

# FORM

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|                                              |          |
|----------------------------------------------|----------|
| <b>1 Data Structure Index</b>                | <b>1</b> |
| 1.1 Data Structures                          | 1        |
| <b>2 File Index</b>                          | <b>5</b> |
| 2.1 File List                                | 5        |
| <b>3 Data Structure Documentation</b>        | <b>7</b> |
| 3.1 AEdge Class Reference                    | 7        |
| 3.1.1 Detailed Description                   | 7        |
| 3.1.2 Constructor & Destructor Documentation | 8        |
| 3.1.2.1 AEdge()                              | 8        |
| 3.1.2.2 ~AEdge()                             | 8        |
| 3.1.3 Field Documentation                    | 8        |
| 3.1.3.1 cand                                 | 8        |
| 3.1.3.2 nodes                                | 8        |
| 3.1.3.3 nlegs                                | 8        |
| 3.1.3.4 ptcl                                 | 8        |
| 3.1.3.5 DUMMYPADDING                         | 9        |
| 3.2 AllGlobals Struct Reference              | 9        |
| 3.2.1 Detailed Description                   | 10       |
| 3.2.2 Field Documentation                    | 10       |
| 3.2.2.1 M                                    | 10       |
| 3.2.2.2 Cc                                   | 10       |
| 3.2.2.3 S                                    | 10       |
| 3.2.2.4 R                                    | 10       |
| 3.2.2.5 N                                    | 11       |
| 3.2.2.6 O                                    | 11       |
| 3.2.2.7 P                                    | 11       |
| 3.2.2.8 T                                    | 11       |
| 3.2.2.9 X                                    | 11       |
| 3.3 ANode Class Reference                    | 11       |
| 3.3.1 Detailed Description                   | 12       |
| 3.3.2 Constructor & Destructor Documentation | 12       |
| 3.3.2.1 ANode()                              | 12       |
| 3.3.2.2 ~ANode()                             | 12       |
| 3.3.3 Member Function Documentation          | 12       |
| 3.3.3.1 newleg()                             | 12       |
| 3.3.4 Field Documentation                    | 13       |
| 3.3.4.1 deg                                  | 13       |
| 3.3.4.2 nlegs                                | 13       |
| 3.3.4.3 anodes                               | 13       |
| 3.3.4.4 aedges                               | 13       |
| 3.3.4.5 aelegs                               | 13       |

|                                              |    |
|----------------------------------------------|----|
| 3.3.4.6 cand                                 | 13 |
| 3.4 ARGBUFFER Struct Reference               | 14 |
| 3.4.1 Detailed Description                   | 14 |
| 3.4.2 Field Documentation                    | 14 |
| 3.4.2.1 buffer                               | 14 |
| 3.4.2.2 dollar                               | 14 |
| 3.4.2.3 size                                 | 15 |
| 3.4.2.4 type                                 | 15 |
| 3.4.2.5 dummy                                | 15 |
| 3.5 Assign Class Reference                   | 15 |
| 3.5.1 Detailed Description                   | 17 |
| 3.5.2 Constructor & Destructor Documentation | 17 |
| 3.5.2.1 Assign()                             | 17 |
| 3.5.2.2 ~Assign()                            | 17 |
| 3.5.3 Member Function Documentation          | 17 |
| 3.5.3.1 prCand()                             | 17 |
| 3.5.3.2 checkAG()                            | 17 |
| 3.5.3.3 assignAllVertices()                  | 18 |
| 3.5.3.4 selectVertex()                       | 18 |
| 3.5.3.5 selectVertexSimp()                   | 18 |
| 3.5.3.6 selectLeg()                          | 18 |
| 3.5.3.7 assignVertex()                       | 18 |
| 3.5.3.8 allAssigned()                        | 18 |
| 3.5.3.9 fromMGraph()                         | 18 |
| 3.5.3.10 addEdge()                           | 19 |
| 3.5.3.11 connect()                           | 19 |
| 3.5.3.12 fillEGraph()                        | 19 |
| 3.5.3.13 reordLeg()                          | 19 |
| 3.5.3.14 getLegParticle()                    | 19 |
| 3.5.3.15 legEdgeParticle()                   | 20 |
| 3.5.3.16 legPart()                           | 20 |
| 3.5.3.17 candPart()                          | 20 |
| 3.5.3.18 selUnAssVertex()                    | 20 |
| 3.5.3.19 selUnAssVertexSimp()                | 20 |
| 3.5.3.20 selUnAssLeg()                       | 20 |
| 3.5.3.21 assignVertex()                      | 21 |
| 3.5.3.22 assignPLeg()                        | 21 |
| 3.5.3.23 isOrdPLeg()                         | 21 |
| 3.5.3.24 detEdge()                           | 21 |
| 3.5.3.25 candPartClassify()                  | 21 |
| 3.5.3.26 updateCandNode()                    | 21 |
| 3.5.3.27 checkOrderCpl()                     | 22 |



|                                              |    |
|----------------------------------------------|----|
| 3.5.3.28 isOrdLegs()                         | 22 |
| 3.5.3.29 isIsomorphic()                      | 22 |
| 3.5.3.30 cmpPermGraph()                      | 22 |
| 3.5.3.31 cmpNodes()                          | 22 |
| 3.5.3.32 edgeSym()                           | 22 |
| 3.5.3.33 saveCouple()                        | 23 |
| 3.5.3.34 restoreCouple()                     | 23 |
| 3.5.3.35 subCouple()                         | 23 |
| 3.5.4 Field Documentation                    | 23 |
| 3.5.4.1 egraph                               | 23 |
| 3.5.4.2 mgraph                               | 23 |
| 3.5.4.3 sproc                                | 23 |
| 3.5.4.4 proc                                 | 23 |
| 3.5.4.5 model                                | 24 |
| 3.5.4.6 opt                                  | 24 |
| 3.5.4.7 astack                               | 24 |
| 3.5.4.8 pnclass                              | 24 |
| 3.5.4.9 orbits                               | 24 |
| 3.5.4.10 nNodes                              | 24 |
| 3.5.4.11 nEdges                              | 24 |
| 3.5.4.12 nExtern                             | 24 |
| 3.5.4.13 nETotal                             | 25 |
| 3.5.4.14 nAGraphs                            | 25 |
| 3.5.4.15 wAGraphs                            | 25 |
| 3.5.4.16 nAOPI                               | 25 |
| 3.5.4.17 wAOPI                               | 25 |
| 3.5.4.18 nodes                               | 25 |
| 3.5.4.19 edges                               | 25 |
| 3.5.4.20 checkpoint0                         | 25 |
| 3.5.4.21 cplleft                             | 26 |
| 3.6 AStack Class Reference                   | 26 |
| 3.6.1 Detailed Description                   | 27 |
| 3.6.2 Constructor & Destructor Documentation | 27 |
| 3.6.2.1 AStack()                             | 27 |
| 3.6.2.2 ~AStack()                            | 27 |
| 3.6.3 Member Function Documentation          | 27 |
| 3.6.3.1 setAGraph()                          | 27 |
| 3.6.3.2 checkPoint()                         | 27 |
| 3.6.3.3 restore()                            | 27 |
| 3.6.3.4 prStack()                            | 28 |
| 3.6.3.5 pushNode()                           | 28 |
| 3.6.3.6 pushEdge()                           | 28 |

|                                               |    |
|-----------------------------------------------|----|
| 3.6.4 Field Documentation                     | 28 |
| 3.6.4.1 agraph                                | 28 |
| 3.6.4.2 nStack                                | 28 |
| 3.6.4.3 nStackP                               | 28 |
| 3.6.4.4 nSize                                 | 28 |
| 3.6.4.5 eStack                                | 29 |
| 3.6.4.6 eStackP                               | 29 |
| 3.6.4.7 eSize                                 | 29 |
| 3.7 bit_field Struct Reference                | 29 |
| 3.7.1 Detailed Description                    | 29 |
| 3.7.2 Field Documentation                     | 30 |
| 3.7.2.1 bit_0                                 | 30 |
| 3.7.2.2 bit_1                                 | 30 |
| 3.7.2.3 bit_2                                 | 30 |
| 3.7.2.4 bit_3                                 | 30 |
| 3.7.2.5 bit_4                                 | 30 |
| 3.7.2.6 bit_5                                 | 30 |
| 3.7.2.7 bit_6                                 | 30 |
| 3.7.2.8 bit_7                                 | 31 |
| 3.8 BrAcKeTiNdEx Struct Reference             | 31 |
| 3.8.1 Detailed Description                    | 31 |
| 3.8.2 Field Documentation                     | 32 |
| 3.8.2.1 start                                 | 32 |
| 3.8.2.2 next                                  | 32 |
| 3.8.2.3 bracket                               | 32 |
| 3.8.2.4 termsinbracket                        | 32 |
| 3.9 BrAcKeTiNfO Struct Reference              | 32 |
| 3.9.1 Detailed Description                    | 33 |
| 3.9.2 Field Documentation                     | 33 |
| 3.9.2.1 indexbuffer                           | 33 |
| 3.9.2.2 bracketbuffer                         | 33 |
| 3.9.2.3 bracketbuffersize                     | 33 |
| 3.9.2.4 indexbuffersize                       | 33 |
| 3.9.2.5 bracketfill                           | 34 |
| 3.9.2.6 indexfill                             | 34 |
| 3.9.2.7 SortType                              | 34 |
| 3.10 BracketInfo Struct Reference             | 34 |
| 3.10.1 Detailed Description                   | 35 |
| 3.10.2 Constructor & Destructor Documentation | 35 |
| 3.10.2.1 BracketInfo()                        | 35 |
| 3.10.3 Member Function Documentation          | 35 |
| 3.10.3.1 operator<()                          | 35 |

|                                             |    |
|---------------------------------------------|----|
| 3.10.4 Field Documentation . . . . .        | 35 |
| 3.10.4.1 pattern . . . . .                  | 35 |
| 3.10.4.2 num_terms . . . . .                | 35 |
| 3.10.4.3 dummy . . . . .                    | 36 |
| 3.10.4.4 p . . . . .                        | 36 |
| 3.11 bufIPstruct Struct Reference . . . . . | 36 |
| 3.11.1 Detailed Description . . . . .       | 37 |
| 3.11.2 Field Documentation . . . . .        | 37 |
| 3.11.2.1 i . . . . .                        | 37 |
| 3.11.2.2 e . . . . .                        | 37 |
| 3.12 C_const Struct Reference . . . . .     | 37 |
| 3.12.1 Detailed Description . . . . .       | 42 |
| 3.12.2 Field Documentation . . . . .        | 42 |
| 3.12.2.1 separators . . . . .               | 42 |
| 3.12.2.2 StoreFileSize . . . . .            | 42 |
| 3.12.2.3 dollarnames . . . . .              | 42 |
| 3.12.2.4 exprnames . . . . .                | 42 |
| 3.12.2.5 varnames . . . . .                 | 42 |
| 3.12.2.6 ChannelList . . . . .              | 43 |
| 3.12.2.7 DubiousList . . . . .              | 43 |
| 3.12.2.8 FunctionList . . . . .             | 43 |
| 3.12.2.9 ExpressionList . . . . .           | 43 |
| 3.12.2.10 IndexList . . . . .               | 43 |
| 3.12.2.11 SetElementList . . . . .          | 43 |
| 3.12.2.12 SetList . . . . .                 | 44 |
| 3.12.2.13 SymbolList . . . . .              | 44 |
| 3.12.2.14 VectorList . . . . .              | 44 |
| 3.12.2.15 PotModDolList . . . . .           | 44 |
| 3.12.2.16 ModOptDolList . . . . .           | 44 |
| 3.12.2.17 TableBaseList . . . . .           | 44 |
| 3.12.2.18 cbufList . . . . .                | 45 |
| 3.12.2.19 AutoSymbolList . . . . .          | 45 |
| 3.12.2.20 AutoIndexList . . . . .           | 45 |
| 3.12.2.21 AutoVectorList . . . . .          | 45 |
| 3.12.2.22 AutoFunctionList . . . . .        | 45 |
| 3.12.2.23 autonames . . . . .               | 45 |
| 3.12.2.24 Symbols . . . . .                 | 45 |
| 3.12.2.25 Indices . . . . .                 | 46 |
| 3.12.2.26 Vectors . . . . .                 | 46 |
| 3.12.2.27 Functions . . . . .               | 46 |
| 3.12.2.28 activenames . . . . .             | 46 |
| 3.12.2.29 Streams . . . . .                 | 46 |

|                                         |    |
|-----------------------------------------|----|
| 3.12.2.30 CurrentStream . . . . .       | 46 |
| 3.12.2.31 SwitchArray . . . . .         | 46 |
| 3.12.2.32 models . . . . .              | 47 |
| 3.12.2.33 SwitchHeap . . . . .          | 47 |
| 3.12.2.34 termstack . . . . .           | 47 |
| 3.12.2.35 termsortstack . . . . .       | 47 |
| 3.12.2.36 cmod . . . . .                | 47 |
| 3.12.2.37 powmod . . . . .              | 47 |
| 3.12.2.38 modpowers . . . . .           | 48 |
| 3.12.2.39 halfmod . . . . .             | 48 |
| 3.12.2.40 ProtoType . . . . .           | 48 |
| 3.12.2.41 WildC . . . . .               | 48 |
| 3.12.2.42 IfHeap . . . . .              | 48 |
| 3.12.2.43 IfCount . . . . .             | 48 |
| 3.12.2.44 IfStack . . . . .             | 48 |
| 3.12.2.45 iBuffer . . . . .             | 49 |
| 3.12.2.46 iPointer . . . . .            | 49 |
| 3.12.2.47 iStop . . . . .               | 49 |
| 3.12.2.48 LabelNames . . . . .          | 49 |
| 3.12.2.49 FixIndices . . . . .          | 49 |
| 3.12.2.50 termsumcheck . . . . .        | 49 |
| 3.12.2.51 WildcardNames . . . . .       | 50 |
| 3.12.2.52 Labels . . . . .              | 50 |
| 3.12.2.53 tokens . . . . .              | 50 |
| 3.12.2.54 toptokens . . . . .           | 50 |
| 3.12.2.55 endoftokens . . . . .         | 50 |
| 3.12.2.56 tokenarglevel . . . . .       | 50 |
| 3.12.2.57 modinverses . . . . .         | 51 |
| 3.12.2.58 Fortran90Kind . . . . .       | 51 |
| 3.12.2.59 MultiBracketBuf . . . . .     | 51 |
| 3.12.2.60 extrasym . . . . .            | 51 |
| 3.12.2.61 doloopstack . . . . .         | 51 |
| 3.12.2.62 doloopnest . . . . .          | 51 |
| 3.12.2.63 CheckpointRunAfter . . . . .  | 51 |
| 3.12.2.64 CheckpointRunBefore . . . . . | 52 |
| 3.12.2.65 IfSumCheck . . . . .          | 52 |
| 3.12.2.66 CommutInSet . . . . .         | 52 |
| 3.12.2.67 TestValue . . . . .           | 52 |
| 3.12.2.68 argstack . . . . .            | 52 |
| 3.12.2.69 insidestack . . . . .         | 52 |
| 3.12.2.70 inexprstack . . . . .         | 52 |
| 3.12.2.71 iBufferSize . . . . .         | 53 |

|                                        |    |
|----------------------------------------|----|
| 3.12.2.72 TransEname . . . . .         | 53 |
| 3.12.2.73 ProcessBucketSize . . . . .  | 53 |
| 3.12.2.74 mProcessBucketSize . . . . . | 53 |
| 3.12.2.75 CModule . . . . .            | 53 |
| 3.12.2.76 ThreadBucketSize . . . . .   | 53 |
| 3.12.2.77 CheckpointStamp . . . . .    | 53 |
| 3.12.2.78 CheckpointInterval . . . . . | 54 |
| 3.12.2.79 cbufnum . . . . .            | 54 |
| 3.12.2.80 AutoDeclareFlag . . . . .    | 54 |
| 3.12.2.81 NoShowInput . . . . .        | 54 |
| 3.12.2.82 ShortStats . . . . .         | 54 |
| 3.12.2.83 compiletype . . . . .        | 54 |
| 3.12.2.84 firstconstindex . . . . .    | 54 |
| 3.12.2.85 insidelfirst . . . . .       | 55 |
| 3.12.2.86 minsidelfirst . . . . .      | 55 |
| 3.12.2.87 wildflag . . . . .           | 55 |
| 3.12.2.88 NumLabels . . . . .          | 55 |
| 3.12.2.89 MaxLabels . . . . .          | 55 |
| 3.12.2.90 IDefDim . . . . .            | 55 |
| 3.12.2.91 IDefDim4 . . . . .           | 55 |
| 3.12.2.92 NumWildcardNames . . . . .   | 55 |
| 3.12.2.93 WildcardBufferSize . . . . . | 56 |
| 3.12.2.94 MaxIf . . . . .              | 56 |
| 3.12.2.95 NumStreams . . . . .         | 56 |
| 3.12.2.96 MaxNumStreams . . . . .      | 56 |
| 3.12.2.97 firstctypemessage . . . . .  | 56 |
| 3.12.2.98 tablecheck . . . . .         | 56 |
| 3.12.2.99 idoption . . . . .           | 56 |
| 3.12.2.100 BottomLevel . . . . .       | 56 |
| 3.12.2.101 CompileLevel . . . . .      | 57 |
| 3.12.2.102 TokensWriteFlag . . . . .   | 57 |
| 3.12.2.103 UnsureDollarMode . . . . .  | 57 |
| 3.12.2.104 outsidefun . . . . .        | 57 |
| 3.12.2.105 funpowers . . . . .         | 57 |
| 3.12.2.106 WarnFlag . . . . .          | 57 |
| 3.12.2.107 StatsFlag . . . . .         | 57 |
| 3.12.2.108 NamesFlag . . . . .         | 57 |
| 3.12.2.109 CodesFlag . . . . .         | 58 |
| 3.12.2.110 SetupFlag . . . . .         | 58 |
| 3.12.2.111 SortType . . . . .          | 58 |
| 3.12.2.112 ISortType . . . . .         | 58 |
| 3.12.2.113 ThreadStats . . . . .       | 58 |

|                                          |    |
|------------------------------------------|----|
| 3.12.2.114 FinalStats . . . . .          | 58 |
| 3.12.2.115 OldParallelStats . . . . .    | 58 |
| 3.12.2.116 ThreadsFlag . . . . .         | 58 |
| 3.12.2.117 ThreadBalancing . . . . .     | 59 |
| 3.12.2.118 ThreadSortFileSynch . . . . . | 59 |
| 3.12.2.119 ProcessStats . . . . .        | 59 |
| 3.12.2.120 BracketNormalize . . . . .    | 59 |
| 3.12.2.121 maxtermlevel . . . . .        | 59 |
| 3.12.2.122 dumnumflag . . . . .          | 59 |
| 3.12.2.123 bracketindexflag . . . . .    | 59 |
| 3.12.2.124 paralleflag . . . . .         | 59 |
| 3.12.2.125 mparalleflag . . . . .        | 60 |
| 3.12.2.126 inparalleflag . . . . .       | 60 |
| 3.12.2.127 partodoflag . . . . .         | 60 |
| 3.12.2.128 properorderflag . . . . .     | 60 |
| 3.12.2.129 vetofilling . . . . .         | 60 |
| 3.12.2.130 tablefilling . . . . .        | 60 |
| 3.12.2.131 vetotablebasefill . . . . .   | 60 |
| 3.12.2.132 exprfillwarning . . . . .     | 60 |
| 3.12.2.133 lhdollarflag . . . . .        | 61 |
| 3.12.2.134 NoCompress . . . . .          | 61 |
| 3.12.2.135 IsFortran90 . . . . .         | 61 |
| 3.12.2.136 MultiBracketLevels . . . . .  | 61 |
| 3.12.2.137 topolynomialflag . . . . .    | 61 |
| 3.12.2.138 ffbufnum . . . . .            | 61 |
| 3.12.2.139 OldFactArgFlag . . . . .      | 61 |
| 3.12.2.140 MemDebugFlag . . . . .        | 61 |
| 3.12.2.141 OldGCDflag . . . . .          | 62 |
| 3.12.2.142 WTimeStatsFlag . . . . .      | 62 |
| 3.12.2.143 SortReallocateFlag . . . . .  | 62 |
| 3.12.2.144 PrintBacktraceFlag . . . . .  | 62 |
| 3.12.2.145 FlintPolyFlag . . . . .       | 62 |
| 3.12.2.146 HumanStatsFlag . . . . .      | 62 |
| 3.12.2.147 doloopstacksize . . . . .     | 62 |
| 3.12.2.148 dolooplevel . . . . .         | 62 |
| 3.12.2.149 CheckpointFlag . . . . .      | 63 |
| 3.12.2.150 SizeCommutelnSet . . . . .    | 63 |
| 3.12.2.151 origin . . . . .              | 63 |
| 3.12.2.152 vectorlikeLHS . . . . .       | 63 |
| 3.12.2.153 nummodels . . . . .           | 63 |
| 3.12.2.154 modelspace . . . . .          | 63 |
| 3.12.2.155 ModelLevel . . . . .          | 63 |

|                                     |    |
|-------------------------------------|----|
| 3.12.2.156 argsumcheck . . . . .    | 64 |
| 3.12.2.157 insidesumcheck . . . . . | 64 |
| 3.12.2.158 inexprsumcheck . . . . . | 64 |
| 3.12.2.159 RepSumCheck . . . . .    | 64 |
| 3.12.2.160 IUniTrace . . . . .      | 64 |
| 3.12.2.161 RepLevel . . . . .       | 64 |
| 3.12.2.162 arglevel . . . . .       | 64 |
| 3.12.2.163 insidelevel . . . . .    | 64 |
| 3.12.2.164 inexprlevel . . . . .    | 65 |
| 3.12.2.165 termlevel . . . . .      | 65 |
| 3.12.2.166 MustTestTable . . . . .  | 65 |
| 3.12.2.167 DumNum . . . . .         | 65 |
| 3.12.2.168 ncmmod . . . . .         | 65 |
| 3.12.2.169 npowmod . . . . .        | 65 |
| 3.12.2.170 modmode . . . . .        | 65 |
| 3.12.2.171 nhalfmod . . . . .       | 65 |
| 3.12.2.172 DirtPow . . . . .        | 66 |
| 3.12.2.173 IUnitTrace . . . . .     | 66 |
| 3.12.2.174 NwildC . . . . .         | 66 |
| 3.12.2.175 ComDefer . . . . .       | 66 |
| 3.12.2.176 CollectFun . . . . .     | 66 |
| 3.12.2.177 AltCollectFun . . . . .  | 66 |
| 3.12.2.178 OutputMode . . . . .     | 66 |
| 3.12.2.179 Cnumpows . . . . .       | 66 |
| 3.12.2.180 OutputSpaces . . . . .   | 67 |
| 3.12.2.181 OutNumberType . . . . .  | 67 |
| 3.12.2.182 DidClean . . . . .       | 67 |
| 3.12.2.183 IfLevel . . . . .        | 67 |
| 3.12.2.184 WhileLevel . . . . .     | 67 |
| 3.12.2.185 SwitchLevel . . . . .    | 67 |
| 3.12.2.186 SwitchInArray . . . . .  | 67 |
| 3.12.2.187 MaxSwitch . . . . .      | 67 |
| 3.12.2.188 LogHandle . . . . .      | 68 |
| 3.12.2.189 LineLength . . . . .     | 68 |
| 3.12.2.190 StoreHandle . . . . .    | 68 |
| 3.12.2.191 HideLevel . . . . .      | 68 |
| 3.12.2.192 IPolyFun . . . . .       | 68 |
| 3.12.2.193 IPolyFunInv . . . . .    | 68 |
| 3.12.2.194 IPolyFunType . . . . .   | 68 |
| 3.12.2.195 IPolyFunExp . . . . .    | 68 |
| 3.12.2.196 IPolyFunVar . . . . .    | 69 |
| 3.12.2.197 IPolyFunPow . . . . .    | 69 |

|                               |    |
|-------------------------------|----|
| 3.12.2.198 SymChangeFlag      | 69 |
| 3.12.2.199 CollectPercentage  | 69 |
| 3.12.2.200 ShortStatsMax      | 69 |
| 3.12.2.201 extrasymbols       | 69 |
| 3.12.2.202 PolyRatFunChanged  | 69 |
| 3.12.2.203 ToBelFactors       | 69 |
| 3.12.2.204 InnerTest          | 70 |
| 3.12.2.205 Commercial         | 70 |
| 3.12.2.206 debugFlags         | 70 |
| 3.13 CbUf Struct Reference    | 70 |
| 3.13.1 Detailed Description   | 71 |
| 3.13.2 Field Documentation    | 71 |
| 3.13.2.1 Buffer               | 71 |
| 3.13.2.2 Top                  | 71 |
| 3.13.2.3 Pointer              | 72 |
| 3.13.2.4 lhs                  | 72 |
| 3.13.2.5 rhs                  | 72 |
| 3.13.2.6 CanCommu             | 72 |
| 3.13.2.7 NumTerms             | 72 |
| 3.13.2.8 numdum               | 73 |
| 3.13.2.9 dimension            | 73 |
| 3.13.2.10 boomlijst           | 73 |
| 3.13.2.11 BufferSize          | 73 |
| 3.13.2.12 numlhs              | 73 |
| 3.13.2.13 numrhs              | 73 |
| 3.13.2.14 maxlhs              | 74 |
| 3.13.2.15 maxrhs              | 74 |
| 3.13.2.16 mnumlhs             | 74 |
| 3.13.2.17 mnumrhs             | 74 |
| 3.13.2.18 numtree             | 74 |
| 3.13.2.19 rootnum             | 74 |
| 3.13.2.20 MaxTreeSize         | 74 |
| 3.14 ChAnNeL Struct Reference | 75 |
| 3.14.1 Detailed Description   | 75 |
| 3.14.2 Field Documentation    | 75 |
| 3.14.2.1 name                 | 75 |
| 3.14.2.2 handle               | 75 |
| 3.15 COST Struct Reference    | 75 |
| 3.15.1 Detailed Description   | 76 |
| 3.15.2 Field Documentation    | 76 |
| 3.15.2.1 add                  | 76 |
| 3.15.2.2 mul                  | 76 |



|                                      |    |
|--------------------------------------|----|
| 3.15.2.3 div                         | 76 |
| 3.15.2.4 pow                         | 76 |
| 3.16 CSEEq Struct Reference          | 76 |
| 3.16.1 Detailed Description          | 76 |
| 3.16.2 Member Function Documentation | 77 |
| 3.16.2.1 operator()                  | 77 |
| 3.17 CSEHash Struct Reference        | 77 |
| 3.17.1 Detailed Description          | 77 |
| 3.17.2 Member Function Documentation | 77 |
| 3.17.2.1 operator()                  | 77 |
| 3.18 dbase Struct Reference          | 78 |
| 3.18.1 Detailed Description          | 78 |
| 3.18.2 Field Documentation           | 78 |
| 3.18.2.1 info                        | 78 |
| 3.18.2.2 mode                        | 79 |
| 3.18.2.3 tablenameessize             | 79 |
| 3.18.2.4 topnumber                   | 79 |
| 3.18.2.5 tablenameefill              | 79 |
| 3.18.2.6 rmode                       | 79 |
| 3.18.2.7 iblocks                     | 79 |
| 3.18.2.8 nblocks                     | 79 |
| 3.18.2.9 handle                      | 79 |
| 3.18.2.10 name                       | 80 |
| 3.18.2.11 fullname                   | 80 |
| 3.18.2.12 tablenames                 | 80 |
| 3.19 DGraph Struct Reference         | 80 |
| 3.19.1 Detailed Description          | 81 |
| 3.19.2 Field Documentation           | 81 |
| 3.19.2.1 nnodes                      | 81 |
| 3.19.2.2 nedges                      | 81 |
| 3.19.2.3 nodes                       | 81 |
| 3.19.2.4 edges                       | 81 |
| 3.20 DICTIONARY Struct Reference     | 81 |
| 3.20.1 Detailed Description          | 82 |
| 3.20.2 Field Documentation           | 82 |
| 3.20.2.1 elements                    | 82 |
| 3.20.2.2 name                        | 82 |
| 3.20.2.3 sizeelements                | 82 |
| 3.20.2.4 numelements                 | 82 |
| 3.20.2.5 numbers                     | 82 |
| 3.20.2.6 variables                   | 83 |
| 3.20.2.7 characters                  | 83 |

|                                                    |    |
|----------------------------------------------------|----|
| 3.20.2.8 funwith . . . . .                         | 83 |
| 3.20.2.9 gnumelements . . . . .                    | 83 |
| 3.20.2.10 ranges . . . . .                         | 83 |
| 3.21 DICTIONARY_ELEMENT Struct Reference . . . . . | 83 |
| 3.21.1 Detailed Description . . . . .              | 83 |
| 3.21.2 Field Documentation . . . . .               | 84 |
| 3.21.2.1 lhs . . . . .                             | 84 |
| 3.21.2.2 rhs . . . . .                             | 84 |
| 3.21.2.3 type . . . . .                            | 84 |
| 3.21.2.4 size . . . . .                            | 84 |
| 3.22 DiStRiBuTe Struct Reference . . . . .         | 84 |
| 3.22.1 Detailed Description . . . . .              | 85 |
| 3.22.2 Field Documentation . . . . .               | 85 |
| 3.22.2.1 obj1 . . . . .                            | 85 |
| 3.22.2.2 obj2 . . . . .                            | 85 |
| 3.22.2.3 out . . . . .                             | 85 |
| 3.22.2.4 sign . . . . .                            | 85 |
| 3.22.2.5 n1 . . . . .                              | 85 |
| 3.22.2.6 n2 . . . . .                              | 85 |
| 3.22.2.7 n . . . . .                               | 86 |
| 3.22.2.8 cycle . . . . .                           | 86 |
| 3.23 DNode Struct Reference . . . . .              | 86 |
| 3.23.1 Detailed Description . . . . .              | 86 |
| 3.23.2 Field Documentation . . . . .               | 86 |
| 3.23.2.1 extloop . . . . .                         | 86 |
| 3.23.2.2 intrct . . . . .                          | 86 |
| 3.24 dollar_buf Struct Reference . . . . .         | 87 |
| 3.24.1 Detailed Description . . . . .              | 87 |
| 3.25 DoLIaRS Struct Reference . . . . .            | 87 |
| 3.25.1 Detailed Description . . . . .              | 87 |
| 3.25.2 Field Documentation . . . . .               | 88 |
| 3.25.2.1 where . . . . .                           | 88 |
| 3.25.2.2 factors . . . . .                         | 88 |
| 3.25.2.3 size . . . . .                            | 88 |
| 3.25.2.4 name . . . . .                            | 88 |
| 3.25.2.5 type . . . . .                            | 88 |
| 3.25.2.6 node . . . . .                            | 88 |
| 3.25.2.7 index . . . . .                           | 88 |
| 3.25.2.8 zero . . . . .                            | 89 |
| 3.25.2.9 numdummies . . . . .                      | 89 |
| 3.25.2.10 nfactors . . . . .                       | 89 |
| 3.26 DoLoOp Struct Reference . . . . .             | 89 |

|                                               |    |
|-----------------------------------------------|----|
| 3.26.1 Detailed Description                   | 90 |
| 3.26.2 Field Documentation                    | 90 |
| 3.26.2.1 p                                    | 90 |
| 3.26.2.2 name                                 | 90 |
| 3.26.2.3 vars                                 | 91 |
| 3.26.2.4 contents                             | 91 |
| 3.26.2.5 dollarname                           | 91 |
| 3.26.2.6 startlinenumber                      | 91 |
| 3.26.2.7 firstnum                             | 91 |
| 3.26.2.8 lastnum                              | 91 |
| 3.26.2.9 incnum                               | 91 |
| 3.26.2.10 type                                | 92 |
| 3.26.2.11 NoShowInput                         | 92 |
| 3.26.2.12 errorsinloop                        | 92 |
| 3.26.2.13 firstloopcall                       | 92 |
| 3.26.2.14 firstdollar                         | 92 |
| 3.26.2.15 lastdollar                          | 92 |
| 3.26.2.16 incdollar                           | 92 |
| 3.26.2.17 NumPreTypes                         | 92 |
| 3.26.2.18 PriefLevel                          | 93 |
| 3.26.2.19 PreSwitchLevel                      | 93 |
| 3.27 DuBiOuS Struct Reference                 | 93 |
| 3.27.1 Detailed Description                   | 93 |
| 3.27.2 Field Documentation                    | 93 |
| 3.27.2.1 name                                 | 93 |
| 3.27.2.2 node                                 | 93 |
| 3.27.2.3 dummy                                | 94 |
| 3.28 ECand Class Reference                    | 94 |
| 3.28.1 Detailed Description                   | 94 |
| 3.28.2 Constructor & Destructor Documentation | 94 |
| 3.28.2.1 ECand()                              | 94 |
| 3.28.2.2 ~ECand()                             | 94 |
| 3.28.3 Member Function Documentation          | 95 |
| 3.28.3.1 prECand()                            | 95 |
| 3.28.4 Field Documentation                    | 95 |
| 3.28.4.1 det                                  | 95 |
| 3.28.4.2 nplist                               | 95 |
| 3.28.4.3 plist                                | 95 |
| 3.29 EEdge Class Reference                    | 95 |
| 3.29.1 Detailed Description                   | 96 |
| 3.29.2 Constructor & Destructor Documentation | 96 |
| 3.29.2.1 EEdge() [1/2]                        | 96 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.29.2.2 EEdge() [2/2]                        | 96  |
| 3.29.2.3 ~EEdge()                             | 97  |
| 3.29.3 Member Function Documentation          | 97  |
| 3.29.3.1 copy()                               | 97  |
| 3.29.3.2 setId()                              | 97  |
| 3.29.3.3 print()                              | 97  |
| 3.29.3.4 setType()                            | 97  |
| 3.29.3.5 setLMom()                            | 97  |
| 3.29.3.6 setEMom()                            | 98  |
| 3.29.4 Field Documentation                    | 98  |
| 3.29.4.1 egraph                               | 98  |
| 3.29.4.2 id                                   | 98  |
| 3.29.4.3 ext                                  | 98  |
| 3.29.4.4 ptcl                                 | 98  |
| 3.29.4.5 deleted                              | 98  |
| 3.29.4.6 nodes                                | 98  |
| 3.29.4.7 nlegs                                | 99  |
| 3.29.4.8 emom                                 | 99  |
| 3.29.4.9 lmom                                 | 99  |
| 3.29.4.10 extMom                              | 99  |
| 3.29.4.11 cut                                 | 99  |
| 3.29.4.12 visited                             | 99  |
| 3.29.4.13 conid                               | 99  |
| 3.29.4.14 edtype                              | 99  |
| 3.29.4.15 opicomp                             | 100 |
| 3.29.4.16 dir                                 | 100 |
| 3.30 EFLine Class Reference                   | 100 |
| 3.30.1 Detailed Description                   | 100 |
| 3.30.2 Constructor & Destructor Documentation | 100 |
| 3.30.2.1 EFLine()                             | 100 |
| 3.30.3 Member Function Documentation          | 101 |
| 3.30.3.1 print()                              | 101 |
| 3.30.4 Field Documentation                    | 101 |
| 3.30.4.1 elist                                | 101 |
| 3.30.4.2 ftype                                | 101 |
| 3.30.4.3 fkind                                | 101 |
| 3.30.4.4 nlist                                | 101 |
| 3.31 EGraph Class Reference                   | 102 |
| 3.31.1 Detailed Description                   | 103 |
| 3.31.2 Constructor & Destructor Documentation | 104 |
| 3.31.2.1 EGraph()                             | 104 |
| 3.31.2.2 ~EGraph()                            | 104 |

|                                      |     |
|--------------------------------------|-----|
| 3.31.3 Member Function Documentation | 104 |
| 3.31.3.1 copy()                      | 104 |
| 3.31.3.2 print()                     | 104 |
| 3.31.3.3 fromDGraph()                | 104 |
| 3.31.3.4 fromMGraph()                | 104 |
| 3.31.3.5 isOptE()                    | 105 |
| 3.31.3.6 setExtern()                 | 105 |
| 3.31.3.7 isExternal()                | 105 |
| 3.31.3.8 isFermion()                 | 105 |
| 3.31.3.9 setExtLoop()                | 105 |
| 3.31.3.10 endSetExtLoop()            | 105 |
| 3.31.3.11 connComp()                 | 106 |
| 3.31.3.12 connVisit()                | 106 |
| 3.31.3.13 biconnE()                  | 106 |
| 3.31.3.14 biinitE()                  | 106 |
| 3.31.3.15 bisearchE()                | 106 |
| 3.31.3.16 findRoot()                 | 106 |
| 3.31.3.17 dirEdge()                  | 107 |
| 3.31.3.18 extMomConsv()              | 107 |
| 3.31.3.19 cmpMom()                   | 107 |
| 3.31.3.20 groupLMom()                | 107 |
| 3.31.3.21 chkMomConsv()              | 107 |
| 3.31.3.22 prFLines()                 | 107 |
| 3.31.3.23 getFLines()                | 108 |
| 3.31.3.24 fltrace()                  | 108 |
| 3.31.3.25 addFLine()                 | 108 |
| 3.31.3.26 legParticle()              | 108 |
| 3.31.4 Field Documentation           | 108 |
| 3.31.4.1 opt                         | 108 |
| 3.31.4.2 model                       | 108 |
| 3.31.4.3 proc                        | 109 |
| 3.31.4.4 sproc                       | 109 |
| 3.31.4.5 mgraph                      | 109 |
| 3.31.4.6 econn                       | 109 |
| 3.31.4.7 nodes                       | 109 |
| 3.31.4.8 edges                       | 109 |
| 3.31.4.9 mld                         | 109 |
| 3.31.4.10 ald                        | 109 |
| 3.31.4.11 sld                        | 110 |
| 3.31.4.12 gSubld                     | 110 |
| 3.31.4.13 assigned                   | 110 |
| 3.31.4.14 fsign                      | 110 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.31.4.15 nsym                                | 110 |
| 3.31.4.16 esym                                | 110 |
| 3.31.4.17 nsym1                               | 110 |
| 3.31.4.18 extperm                             | 110 |
| 3.31.4.19 multp                               | 111 |
| 3.31.4.20 pld                                 | 111 |
| 3.31.4.21 sNodes                              | 111 |
| 3.31.4.22 sEdges                              | 111 |
| 3.31.4.23 sMaxdeg                             | 111 |
| 3.31.4.24 sLoops                              | 111 |
| 3.31.4.25 nNodes                              | 111 |
| 3.31.4.26 nEdges                              | 111 |
| 3.31.4.27 nExtern                             | 112 |
| 3.31.4.28 maxdeg                              | 112 |
| 3.31.4.29 nLoops                              | 112 |
| 3.31.4.30 totalc                              | 112 |
| 3.31.4.31 nopicomp                            | 112 |
| 3.31.4.32 opi2plp                             | 112 |
| 3.31.4.33 nopi2p                              | 112 |
| 3.31.4.34 nadj2ptv                            | 112 |
| 3.31.4.35 DUMMYPADDING                        | 113 |
| 3.31.4.36 bidef                               | 113 |
| 3.31.4.37 bilow                               | 113 |
| 3.31.4.38 extMom                              | 113 |
| 3.31.4.39 bconn                               | 113 |
| 3.31.4.40 bicount                             | 113 |
| 3.31.4.41 loopm                               | 113 |
| 3.31.4.42 opiCount                            | 113 |
| 3.31.4.43 flines                              | 114 |
| 3.31.4.44 nflines                             | 114 |
| 3.31.4.45 DUMMYPADDING1                       | 114 |
| 3.32 ENode Class Reference                    | 114 |
| 3.32.1 Detailed Description                   | 115 |
| 3.32.2 Constructor & Destructor Documentation | 115 |
| 3.32.2.1 ENode() [1/2]                        | 115 |
| 3.32.2.2 ENode() [2/2]                        | 115 |
| 3.32.2.3 ~ENode()                             | 115 |
| 3.32.3 Member Function Documentation          | 116 |
| 3.32.3.1 initAss()                            | 116 |
| 3.32.3.2 setId()                              | 116 |
| 3.32.3.3 copy()                               | 116 |
| 3.32.3.4 setExtern()                          | 116 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.32.3.5 setType()                            | 116 |
| 3.32.3.6 print()                              | 116 |
| 3.32.4 Field Documentation                    | 117 |
| 3.32.4.1 egraph                               | 117 |
| 3.32.4.2 edges                                | 117 |
| 3.32.4.3 id                                   | 117 |
| 3.32.4.4 maxdeg                               | 117 |
| 3.32.4.5 deg                                  | 117 |
| 3.32.4.6 extloop                              | 117 |
| 3.32.4.7 ndtype                               | 117 |
| 3.32.4.8 intrct                               | 118 |
| 3.32.4.9 klow                                 | 118 |
| 3.32.4.10 visited                             | 118 |
| 3.32.4.11 DUMMYPADDING                        | 118 |
| 3.33 EStack Class Reference                   | 118 |
| 3.33.1 Detailed Description                   | 118 |
| 3.33.2 Constructor & Destructor Documentation | 119 |
| 3.33.2.1 EStack()                             | 119 |
| 3.33.2.2 ~EStack()                            | 119 |
| 3.33.3 Member Function Documentation          | 119 |
| 3.33.3.1 print()                              | 119 |
| 3.33.4 Field Documentation                    | 119 |
| 3.33.4.1 edgen                                | 119 |
| 3.33.4.2 det                                  | 119 |
| 3.33.4.3 nplist                               | 119 |
| 3.33.4.4 plist                                | 120 |
| 3.34 ExPrEsSiOn Struct Reference              | 120 |
| 3.34.1 Detailed Description                   | 121 |
| 3.34.2 Field Documentation                    | 121 |
| 3.34.2.1 onfile                               | 121 |
| 3.34.2.2 prototype                            | 121 |
| 3.34.2.3 size                                 | 121 |
| 3.34.2.4 renum                                | 122 |
| 3.34.2.5 bracketinfo                          | 122 |
| 3.34.2.6 newbracketinfo                       | 122 |
| 3.34.2.7 renumlists                           | 122 |
| 3.34.2.8 inmem                                | 122 |
| 3.34.2.9 counter                              | 122 |
| 3.34.2.10 name                                | 122 |
| 3.34.2.11 hidelevel                           | 123 |
| 3.34.2.12 vflags                              | 123 |
| 3.34.2.13 printfldg                           | 123 |

|                                                |     |
|------------------------------------------------|-----|
| 3.34.2.14 status . . . . .                     | 123 |
| 3.34.2.15 replace . . . . .                    | 123 |
| 3.34.2.16 node . . . . .                       | 123 |
| 3.34.2.17 whichbuffer . . . . .                | 123 |
| 3.34.2.18 namesize . . . . .                   | 123 |
| 3.34.2.19 compression . . . . .                | 124 |
| 3.34.2.20 numdummies . . . . .                 | 124 |
| 3.34.2.21 numfactors . . . . .                 | 124 |
| 3.34.2.22 sizeprototype . . . . .              | 124 |
| 3.34.2.23 uflags . . . . .                     | 124 |
| 3.35 FaCdOLaR Struct Reference . . . . .       | 124 |
| 3.35.1 Detailed Description . . . . .          | 124 |
| 3.35.2 Field Documentation . . . . .           | 125 |
| 3.35.2.1 where . . . . .                       | 125 |
| 3.35.2.2 size . . . . .                        | 125 |
| 3.35.2.3 type . . . . .                        | 125 |
| 3.35.2.4 value . . . . .                       | 125 |
| 3.36 factorized_poly Class Reference . . . . . | 125 |
| 3.36.1 Detailed Description . . . . .          | 126 |
| 3.36.2 Member Function Documentation . . . . . | 126 |
| 3.36.2.1 add_factor() . . . . .                | 126 |
| 3.36.2.2 toString() . . . . .                  | 126 |
| 3.36.3 Field Documentation . . . . .           | 126 |
| 3.36.3.1 factor . . . . .                      | 126 |
| 3.36.3.2 power . . . . .                       | 126 |
| 3.37 FGInput Struct Reference . . . . .        | 126 |
| 3.37.1 Detailed Description . . . . .          | 127 |
| 3.37.2 Field Documentation . . . . .           | 127 |
| 3.37.2.1 ninitl . . . . .                      | 127 |
| 3.37.2.2 initln . . . . .                      | 127 |
| 3.37.2.3 initlc . . . . .                      | 127 |
| 3.37.2.4 nfinal . . . . .                      | 127 |
| 3.37.2.5 finaln . . . . .                      | 127 |
| 3.37.2.6 finalc . . . . .                      | 127 |
| 3.37.2.7 coupl . . . . .                       | 128 |
| 3.38 FiLe Struct Reference . . . . .           | 128 |
| 3.38.1 Detailed Description . . . . .          | 129 |
| 3.38.2 Field Documentation . . . . .           | 129 |
| 3.38.2.1 PPosition . . . . .                   | 129 |
| 3.38.2.2 filesize . . . . .                    | 129 |
| 3.38.2.3 PObuffer . . . . .                    | 129 |
| 3.38.2.4 PStop . . . . .                       | 129 |



|                                              |     |
|----------------------------------------------|-----|
| 3.38.2.5 POfill . . . . .                    | 129 |
| 3.38.2.6 POfull . . . . .                    | 129 |
| 3.38.2.7 name . . . . .                      | 130 |
| 3.38.2.8 numblocks . . . . .                 | 130 |
| 3.38.2.9 inbuffer . . . . .                  | 130 |
| 3.38.2.10 PSize . . . . .                    | 130 |
| 3.38.2.11 handle . . . . .                   | 130 |
| 3.38.2.12 active . . . . .                   | 130 |
| 3.39 FiLeDaTa Struct Reference . . . . .     | 131 |
| 3.39.1 Detailed Description . . . . .        | 131 |
| 3.39.2 Field Documentation . . . . .         | 132 |
| 3.39.2.1 Index . . . . .                     | 132 |
| 3.39.2.2 Fill . . . . .                      | 132 |
| 3.39.2.3 Position . . . . .                  | 132 |
| 3.39.2.4 Handle . . . . .                    | 132 |
| 3.39.2.5 dirtyflag . . . . .                 | 132 |
| 3.40 FiLeInDeX Struct Reference . . . . .    | 133 |
| 3.40.1 Detailed Description . . . . .        | 133 |
| 3.40.2 Field Documentation . . . . .         | 134 |
| 3.40.2.1 next . . . . .                      | 134 |
| 3.40.2.2 number . . . . .                    | 134 |
| 3.40.2.3 expression . . . . .                | 134 |
| 3.40.2.4 empty . . . . .                     | 134 |
| 3.41 FixedGlobals Struct Reference . . . . . | 134 |
| 3.41.1 Detailed Description . . . . .        | 135 |
| 3.41.2 Field Documentation . . . . .         | 135 |
| 3.41.2.1 Operation . . . . .                 | 135 |
| 3.41.2.2 OperaFind . . . . .                 | 135 |
| 3.41.2.3 VarType . . . . .                   | 135 |
| 3.41.2.4 ExprStat . . . . .                  | 136 |
| 3.41.2.5 FunNam . . . . .                    | 136 |
| 3.41.2.6 swmes . . . . .                     | 136 |
| 3.41.2.7 fname . . . . .                     | 136 |
| 3.41.2.8 fname2 . . . . .                    | 136 |
| 3.41.2.9 s_one . . . . .                     | 136 |
| 3.41.2.10 fnamebase . . . . .                | 136 |
| 3.41.2.11 fname2base . . . . .               | 136 |
| 3.41.2.12 fnamesize . . . . .                | 137 |
| 3.41.2.13 fname2size . . . . .               | 137 |
| 3.41.2.14 cTable . . . . .                   | 137 |
| 3.42 fixedset Struct Reference . . . . .     | 137 |
| 3.42.1 Detailed Description . . . . .        | 137 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.42.2 Field Documentation                    | 137 |
| 3.42.2.1 name                                 | 137 |
| 3.42.2.2 description                          | 138 |
| 3.42.2.3 type                                 | 138 |
| 3.42.2.4 dimension                            | 138 |
| 3.43 flint::fmpz Class Reference              | 138 |
| 3.43.1 Detailed Description                   | 138 |
| 3.43.2 Constructor & Destructor Documentation | 138 |
| 3.43.2.1 fmpz()                               | 138 |
| 3.43.2.2 ~fmpz()                              | 139 |
| 3.43.3 Member Function Documentation          | 139 |
| 3.43.3.1 print()                              | 139 |
| 3.43.4 Field Documentation                    | 139 |
| 3.43.4.1 d                                    | 139 |
| 3.44 Fraction Class Reference                 | 139 |
| 3.44.1 Detailed Description                   | 140 |
| 3.44.2 Constructor & Destructor Documentation | 140 |
| 3.44.2.1 Fraction() [1/2]                     | 140 |
| 3.44.2.2 Fraction() [2/2]                     | 140 |
| 3.44.3 Member Function Documentation          | 140 |
| 3.44.3.1 print()                              | 140 |
| 3.44.3.2 setValue() [1/2]                     | 140 |
| 3.44.3.3 setValue() [2/2]                     | 140 |
| 3.44.3.4 add() [1/2]                          | 141 |
| 3.44.3.5 add() [2/2]                          | 141 |
| 3.44.3.6 sub()                                | 141 |
| 3.44.3.7 gcd()                                | 141 |
| 3.44.3.8 normal()                             | 141 |
| 3.44.3.9 isEq()                               | 141 |
| 3.44.4 Field Documentation                    | 142 |
| 3.44.4.1 num                                  | 142 |
| 3.44.4.2 den                                  | 142 |
| 3.44.4.3 ratio                                | 142 |
| 3.45 FUN_INFO Struct Reference                | 142 |
| 3.45.1 Detailed Description                   | 142 |
| 3.45.2 Field Documentation                    | 143 |
| 3.45.2.1 location                             | 143 |
| 3.45.2.2 numargs                              | 143 |
| 3.45.2.3 numfunnies                           | 143 |
| 3.45.2.4 numwildcards                         | 143 |
| 3.45.2.5 symmet                               | 143 |
| 3.45.2.6 tensor                               | 143 |

|                                                     |     |
|-----------------------------------------------------|-----|
| 3.45.2.7 commute . . . . .                          | 143 |
| 3.46 FuNcTiOn Struct Reference . . . . .            | 144 |
| 3.46.1 Detailed Description . . . . .               | 144 |
| 3.46.2 Field Documentation . . . . .                | 145 |
| 3.46.2.1 tabl . . . . .                             | 145 |
| 3.46.2.2 symminfo . . . . .                         | 145 |
| 3.46.2.3 name . . . . .                             | 145 |
| 3.46.2.4 commute . . . . .                          | 145 |
| 3.46.2.5 complex . . . . .                          | 145 |
| 3.46.2.6 number . . . . .                           | 146 |
| 3.46.2.7 flags . . . . .                            | 146 |
| 3.46.2.8 spec . . . . .                             | 146 |
| 3.46.2.9 symmetric . . . . .                        | 146 |
| 3.46.2.10 node . . . . .                            | 146 |
| 3.46.2.11 namesize . . . . .                        | 147 |
| 3.46.2.12 dimension . . . . .                       | 147 |
| 3.46.2.13 maxnumargs . . . . .                      | 147 |
| 3.46.2.14 minnumargs . . . . .                      | 147 |
| 3.47 gcd_heuristic_failed Class Reference . . . . . | 147 |
| 3.47.1 Detailed Description . . . . .               | 147 |
| 3.48 HANDLERS Struct Reference . . . . .            | 147 |
| 3.48.1 Detailed Description . . . . .               | 148 |
| 3.48.2 Field Documentation . . . . .                | 148 |
| 3.48.2.1 newlogonly . . . . .                       | 148 |
| 3.48.2.2 newhandle . . . . .                        | 148 |
| 3.48.2.3 oldhandle . . . . .                        | 148 |
| 3.48.2.4 oldlogonly . . . . .                       | 148 |
| 3.48.2.5 oldprinttype . . . . .                     | 148 |
| 3.48.2.6 oldsilent . . . . .                        | 149 |
| 3.49 IInput Struct Reference . . . . .              | 149 |
| 3.49.1 Detailed Description . . . . .               | 149 |
| 3.49.2 Field Documentation . . . . .                | 149 |
| 3.49.2.1 name . . . . .                             | 149 |
| 3.49.2.2 icode . . . . .                            | 149 |
| 3.49.2.3 nplistn . . . . .                          | 149 |
| 3.49.2.4 plistn . . . . .                           | 150 |
| 3.49.2.5 plistc . . . . .                           | 150 |
| 3.49.2.6 cvallist . . . . .                         | 150 |
| 3.50 InDeX Struct Reference . . . . .               | 150 |
| 3.50.1 Detailed Description . . . . .               | 150 |
| 3.50.2 Field Documentation . . . . .                | 150 |
| 3.50.2.1 name . . . . .                             | 150 |

|                                  |     |
|----------------------------------|-----|
| 3.50.2.2 type                    | 151 |
| 3.50.2.3 dimension               | 151 |
| 3.50.2.4 number                  | 151 |
| 3.50.2.5 flags                   | 151 |
| 3.50.2.6 nmin4                   | 151 |
| 3.50.2.7 node                    | 151 |
| 3.50.2.8 namesize                | 151 |
| 3.51 indexblock Struct Reference | 152 |
| 3.51.1 Detailed Description      | 152 |
| 3.51.2 Field Documentation       | 152 |
| 3.51.2.1 flags                   | 152 |
| 3.51.2.2 previousblock           | 152 |
| 3.51.2.3 position                | 152 |
| 3.51.2.4 objects                 | 153 |
| 3.52 InDeXeNtRy Struct Reference | 153 |
| 3.52.1 Detailed Description      | 154 |
| 3.52.2 Field Documentation       | 154 |
| 3.52.2.1 position                | 154 |
| 3.52.2.2 length                  | 154 |
| 3.52.2.3 variables               | 154 |
| 3.52.2.4 CompressSize            | 154 |
| 3.52.2.5 nsymbols                | 154 |
| 3.52.2.6 nindices                | 155 |
| 3.52.2.7 nvectors                | 155 |
| 3.52.2.8 nfunctions              | 155 |
| 3.52.2.9 size                    | 155 |
| 3.52.2.10 name                   | 155 |
| 3.53 iniinfo Struct Reference    | 156 |
| 3.53.1 Detailed Description      | 156 |
| 3.53.2 Field Documentation       | 156 |
| 3.53.2.1 entriesinindex          | 156 |
| 3.53.2.2 numberofindexblocks     | 156 |
| 3.53.2.3 firstindexblock         | 156 |
| 3.53.2.4 lastindexblock          | 156 |
| 3.53.2.5 numberoftables          | 157 |
| 3.53.2.6 numberofnamesblocks     | 157 |
| 3.53.2.7 firstnameblock          | 157 |
| 3.53.2.8 lastnameblock           | 157 |
| 3.54 INSIDEINFO Struct Reference | 157 |
| 3.54.1 Detailed Description      | 158 |
| 3.54.2 Field Documentation       | 158 |
| 3.54.2.1 buffer                  | 158 |

|                                                         |     |
|---------------------------------------------------------|-----|
| 3.54.2.2 oldcompiletype . . . . .                       | 158 |
| 3.54.2.3 oldparallelflag . . . . .                      | 158 |
| 3.54.2.4 oldnumpotmoddollars . . . . .                  | 158 |
| 3.54.2.5 size . . . . .                                 | 158 |
| 3.54.2.6 numdollars . . . . .                           | 158 |
| 3.54.2.7 oldcbuf . . . . .                              | 159 |
| 3.54.2.8 oldrbuf . . . . .                              | 159 |
| 3.54.2.9 inscbuf . . . . .                              | 159 |
| 3.54.2.10 oldcnumlhs . . . . .                          | 159 |
| 3.55 Interaction Class Reference . . . . .              | 159 |
| 3.55.1 Detailed Description . . . . .                   | 160 |
| 3.55.2 Constructor & Destructor Documentation . . . . . | 160 |
| 3.55.2.1 Interaction() . . . . .                        | 160 |
| 3.55.2.2 ~Interaction() . . . . .                       | 160 |
| 3.55.3 Member Function Documentation . . . . .          | 161 |
| 3.55.3.1 prInteraction() . . . . .                      | 161 |
| 3.55.4 Field Documentation . . . . .                    | 161 |
| 3.55.4.1 mdl . . . . .                                  | 161 |
| 3.55.4.2 name . . . . .                                 | 161 |
| 3.55.4.3 plist . . . . .                                | 161 |
| 3.55.4.4 clist . . . . .                                | 161 |
| 3.55.4.5 slist . . . . .                                | 161 |
| 3.55.4.6 id . . . . .                                   | 161 |
| 3.55.4.7 csum . . . . .                                 | 162 |
| 3.55.4.8 nlegs . . . . .                                | 162 |
| 3.55.4.9 loop . . . . .                                 | 162 |
| 3.55.4.10 icode . . . . .                               | 162 |
| 3.55.4.11 nplist . . . . .                              | 162 |
| 3.55.4.12 nclist . . . . .                              | 162 |
| 3.55.4.13 nslist . . . . .                              | 162 |
| 3.56 KEYWORD Struct Reference . . . . .                 | 163 |
| 3.56.1 Detailed Description . . . . .                   | 163 |
| 3.56.2 Field Documentation . . . . .                    | 163 |
| 3.56.2.1 name . . . . .                                 | 163 |
| 3.56.2.2 func . . . . .                                 | 163 |
| 3.56.2.3 type . . . . .                                 | 163 |
| 3.56.2.4 flags . . . . .                                | 163 |
| 3.57 KEYWORDV Struct Reference . . . . .                | 164 |
| 3.57.1 Detailed Description . . . . .                   | 164 |
| 3.57.2 Field Documentation . . . . .                    | 164 |
| 3.57.2.1 name . . . . .                                 | 164 |
| 3.57.2.2 var . . . . .                                  | 164 |

|                                       |     |
|---------------------------------------|-----|
| 3.57.2.3 type                         | 164 |
| 3.57.2.4 flags                        | 164 |
| 3.58 LIST Struct Reference            | 165 |
| 3.58.1 Detailed Description           | 165 |
| 3.58.2 Field Documentation            | 165 |
| 3.58.2.1 lijst                        | 165 |
| 3.58.2.2 message                      | 165 |
| 3.58.2.3 num                          | 165 |
| 3.58.2.4 maxnum                       | 166 |
| 3.58.2.5 size                         | 166 |
| 3.58.2.6 numglobal                    | 166 |
| 3.58.2.7 numtemp                      | 166 |
| 3.58.2.8 numclear                     | 166 |
| 3.59 longMultiStruct Struct Reference | 167 |
| 3.59.1 Detailed Description           | 167 |
| 3.59.2 Field Documentation            | 167 |
| 3.59.2.1 buffer                       | 167 |
| 3.59.2.2 bufpos                       | 167 |
| 3.59.2.3 packpos                      | 167 |
| 3.59.2.4 nPacks                       | 168 |
| 3.59.2.5 lastLen                      | 168 |
| 3.59.2.6 next                         | 168 |
| 3.60 M_const Struct Reference         | 168 |
| 3.60.1 Detailed Description           | 172 |
| 3.60.2 Field Documentation            | 172 |
| 3.60.2.1 zeropos                      | 172 |
| 3.60.2.2 S0                           | 172 |
| 3.60.2.3 gcmmod                       | 172 |
| 3.60.2.4 gpowmod                      | 172 |
| 3.60.2.5 TempDir                      | 173 |
| 3.60.2.6 TempSortDir                  | 173 |
| 3.60.2.7 IncDir                       | 173 |
| 3.60.2.8 InputFileName                | 173 |
| 3.60.2.9 LogFileName                  | 173 |
| 3.60.2.10 OutBuffer                   | 173 |
| 3.60.2.11 Path                        | 173 |
| 3.60.2.12 SetupDir                    | 173 |
| 3.60.2.13 SetupFile                   | 174 |
| 3.60.2.14 gFortran90Kind              | 174 |
| 3.60.2.15 gextrasym                   | 174 |
| 3.60.2.16 ggextrasym                  | 174 |
| 3.60.2.17 oldnumextrasymbols          | 174 |

|                                          |     |
|------------------------------------------|-----|
| 3.60.2.18 SpectatorFiles . . . . .       | 174 |
| 3.60.2.19 MaxTer . . . . .               | 174 |
| 3.60.2.20 CompressSize . . . . .         | 174 |
| 3.60.2.21 ScratSize . . . . .            | 175 |
| 3.60.2.22 HideSize . . . . .             | 175 |
| 3.60.2.23 SizeStoreCache . . . . .       | 175 |
| 3.60.2.24 MaxStreamSize . . . . .        | 175 |
| 3.60.2.25 SIOsize . . . . .              | 175 |
| 3.60.2.26 SLargeSize . . . . .           | 175 |
| 3.60.2.27 SSmallEsize . . . . .          | 175 |
| 3.60.2.28 SSmallSize . . . . .           | 175 |
| 3.60.2.29 STermsInSmall . . . . .        | 176 |
| 3.60.2.30 MaxBracketBufferSize . . . . . | 176 |
| 3.60.2.31 hProcessBucketSize . . . . .   | 176 |
| 3.60.2.32 gProcessBucketSize . . . . .   | 176 |
| 3.60.2.33 shmWinSize . . . . .           | 176 |
| 3.60.2.34 OldChildTime . . . . .         | 176 |
| 3.60.2.35 OldSecTime . . . . .           | 176 |
| 3.60.2.36 OldMilliTime . . . . .         | 176 |
| 3.60.2.37 WorkSize . . . . .             | 177 |
| 3.60.2.38 gThreadBucketSize . . . . .    | 177 |
| 3.60.2.39 ggThreadBucketSize . . . . .   | 177 |
| 3.60.2.40 SumTime . . . . .              | 177 |
| 3.60.2.41 SpectatorSize . . . . .        | 177 |
| 3.60.2.42 TimeLimit . . . . .            | 177 |
| 3.60.2.43 FileOnlyFlag . . . . .         | 177 |
| 3.60.2.44 Interact . . . . .             | 177 |
| 3.60.2.45 MaxParLevel . . . . .          | 178 |
| 3.60.2.46 OutBufSize . . . . .           | 178 |
| 3.60.2.47 SMaxFpatches . . . . .         | 178 |
| 3.60.2.48 SMaxPatches . . . . .          | 178 |
| 3.60.2.49 StdOut . . . . .               | 178 |
| 3.60.2.50 ginsidefirst . . . . .         | 178 |
| 3.60.2.51 gDefDim . . . . .              | 178 |
| 3.60.2.52 gDefDim4 . . . . .             | 178 |
| 3.60.2.53 NumFixedSets . . . . .         | 179 |
| 3.60.2.54 NumFixedFunctions . . . . .    | 179 |
| 3.60.2.55 rbufnum . . . . .              | 179 |
| 3.60.2.56 dbufnum . . . . .              | 179 |
| 3.60.2.57 sbufnum . . . . .              | 179 |
| 3.60.2.58 zbufnum . . . . .              | 179 |
| 3.60.2.59 SkipClears . . . . .           | 179 |

|                                           |     |
|-------------------------------------------|-----|
| 3.60.2.60 gTokensWriteFlag . . . . .      | 179 |
| 3.60.2.61 gfunpowers . . . . .            | 180 |
| 3.60.2.62 gStatsFlag . . . . .            | 180 |
| 3.60.2.63 gNamesFlag . . . . .            | 180 |
| 3.60.2.64 gCodesFlag . . . . .            | 180 |
| 3.60.2.65 gSortType . . . . .             | 180 |
| 3.60.2.66 gproperorderflag . . . . .      | 180 |
| 3.60.2.67 hparallelflag . . . . .         | 180 |
| 3.60.2.68 gparallelflag . . . . .         | 180 |
| 3.60.2.69 totalnumberofthreads . . . . .  | 181 |
| 3.60.2.70 gSizeCommuteInSet . . . . .     | 181 |
| 3.60.2.71 gThreadStats . . . . .          | 181 |
| 3.60.2.72 ggThreadStats . . . . .         | 181 |
| 3.60.2.73 gFinalStats . . . . .           | 181 |
| 3.60.2.74 ggFinalStats . . . . .          | 181 |
| 3.60.2.75 gThreadsFlag . . . . .          | 181 |
| 3.60.2.76 ggThreadsFlag . . . . .         | 181 |
| 3.60.2.77 gThreadBalancing . . . . .      | 182 |
| 3.60.2.78 ggThreadBalancing . . . . .     | 182 |
| 3.60.2.79 gThreadSortFileSynch . . . . .  | 182 |
| 3.60.2.80 ggThreadSortFileSynch . . . . . | 182 |
| 3.60.2.81 gProcessStats . . . . .         | 182 |
| 3.60.2.82 ggProcessStats . . . . .        | 182 |
| 3.60.2.83 gOldParallelStats . . . . .     | 182 |
| 3.60.2.84 ggOldParallelStats . . . . .    | 182 |
| 3.60.2.85 maxFlevels . . . . .            | 183 |
| 3.60.2.86 resetTimeOnClear . . . . .      | 183 |
| 3.60.2.87 gcNumDollars . . . . .          | 183 |
| 3.60.2.88 MultiRun . . . . .              | 183 |
| 3.60.2.89 gNoSpacesInNumbers . . . . .    | 183 |
| 3.60.2.90 ggNoSpacesInNumbers . . . . .   | 183 |
| 3.60.2.91 gIsFortran90 . . . . .          | 183 |
| 3.60.2.92 PrintTotalSize . . . . .        | 183 |
| 3.60.2.93 fbufferSize . . . . .           | 184 |
| 3.60.2.94 gOldFactArgFlag . . . . .       | 184 |
| 3.60.2.95 ggOldFactArgFlag . . . . .      | 184 |
| 3.60.2.96 gnumextrasym . . . . .          | 184 |
| 3.60.2.97 ggnumextrasym . . . . .         | 184 |
| 3.60.2.98 NumSpectatorFiles . . . . .     | 184 |
| 3.60.2.99 SizeForSpectatorFiles . . . . . | 184 |
| 3.60.2.100 gOldGCDflag . . . . .          | 184 |
| 3.60.2.101 ggOldGCDflag . . . . .         | 185 |



|                                       |     |
|---------------------------------------|-----|
| 3.60.2.102 gWTimeStatsFlag . . . . .  | 185 |
| 3.60.2.103 ggWTimeStatsFlag . . . . . | 185 |
| 3.60.2.104 jumpratio . . . . .        | 185 |
| 3.60.2.105 MaxTal . . . . .           | 185 |
| 3.60.2.106 IndDum . . . . .           | 185 |
| 3.60.2.107 DumInd . . . . .           | 185 |
| 3.60.2.108 Willnd . . . . .           | 185 |
| 3.60.2.109 gncmod . . . . .           | 186 |
| 3.60.2.110 gnpowmod . . . . .         | 186 |
| 3.60.2.111 gmodmode . . . . .         | 186 |
| 3.60.2.112 gUnitTrace . . . . .       | 186 |
| 3.60.2.113 gOutputMode . . . . .      | 186 |
| 3.60.2.114 gOutputSpaces . . . . .    | 186 |
| 3.60.2.115 gOutNumberType . . . . .   | 186 |
| 3.60.2.116 gCnumpows . . . . .        | 186 |
| 3.60.2.117 gUniTrace . . . . .        | 187 |
| 3.60.2.118 MaxWildcards . . . . .     | 187 |
| 3.60.2.119 mTraceDum . . . . .        | 187 |
| 3.60.2.120 OffsetIndex . . . . .      | 187 |
| 3.60.2.121 OffsetVector . . . . .     | 187 |
| 3.60.2.122 RepMax . . . . .           | 187 |
| 3.60.2.123 LogType . . . . .          | 187 |
| 3.60.2.124 ggStatsFlag . . . . .      | 187 |
| 3.60.2.125 gLineLength . . . . .      | 188 |
| 3.60.2.126 qError . . . . .           | 188 |
| 3.60.2.127 FortranCont . . . . .      | 188 |
| 3.60.2.128 HoldFlag . . . . .         | 188 |
| 3.60.2.129 Ordering . . . . .         | 188 |
| 3.60.2.130 silent . . . . .           | 188 |
| 3.60.2.131 tracebackflag . . . . .    | 188 |
| 3.60.2.132 expnum . . . . .           | 188 |
| 3.60.2.133 denomnum . . . . .         | 189 |
| 3.60.2.134 facnum . . . . .           | 189 |
| 3.60.2.135 invfacnum . . . . .        | 189 |
| 3.60.2.136 sumnum . . . . .           | 189 |
| 3.60.2.137 sumpnum . . . . .          | 189 |
| 3.60.2.138 OldOrderFlag . . . . .     | 189 |
| 3.60.2.139 termfunnum . . . . .       | 189 |
| 3.60.2.140 matchfunnum . . . . .      | 189 |
| 3.60.2.141 countfunnum . . . . .      | 190 |
| 3.60.2.142 gPolyFun . . . . .         | 190 |
| 3.60.2.143 gPolyFunInv . . . . .      | 190 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.60.2.144 gPolyFunType                       | 190 |
| 3.60.2.145 gPolyFunExp                        | 190 |
| 3.60.2.146 gPolyFunVar                        | 190 |
| 3.60.2.147 gPolyFunPow                        | 190 |
| 3.60.2.148 dollarzero                         | 190 |
| 3.60.2.149 atstartup                          | 191 |
| 3.60.2.150 exitflag                           | 191 |
| 3.60.2.151 NumStoreCaches                     | 191 |
| 3.60.2.152 gIndentSpace                       | 191 |
| 3.60.2.153 ggIndentSpace                      | 191 |
| 3.60.2.154 gShortStatsMax                     | 191 |
| 3.60.2.155 ggShortStatsMax                    | 191 |
| 3.60.2.156 gextrasyms                         | 192 |
| 3.60.2.157 ggextrasyms                        | 192 |
| 3.60.2.158 zerorhs                            | 192 |
| 3.60.2.159 onerhs                             | 192 |
| 3.60.2.160 havesortdir                        | 192 |
| 3.60.2.161 vectorzero                         | 192 |
| 3.60.2.162 ClearStore                         | 192 |
| 3.60.2.163 BracketFactors                     | 192 |
| 3.60.2.164 FromStdin                          | 193 |
| 3.60.2.165 IgnoreDeprecation                  | 193 |
| 3.61 MCBLOCK Class Reference                  | 193 |
| 3.61.1 Detailed Description                   | 193 |
| 3.61.2 Constructor & Destructor Documentation | 193 |
| 3.61.2.1 MCBLOCK()                            | 193 |
| 3.61.2.2 ~MCBLOCK()                           | 194 |
| 3.61.3 Member Function Documentation          | 194 |
| 3.61.3.1 init()                               | 194 |
| 3.61.4 Field Documentation                    | 194 |
| 3.61.4.1 edges                                | 194 |
| 3.61.4.2 nmedges                              | 194 |
| 3.61.4.3 nartps                               | 194 |
| 3.61.4.4 loop                                 | 194 |
| 3.61.4.5 DUMMY_PADDING                        | 195 |
| 3.62 MCBridge Class Reference                 | 195 |
| 3.62.1 Detailed Description                   | 195 |
| 3.62.2 Constructor & Destructor Documentation | 195 |
| 3.62.2.1 MCBridge()                           | 195 |
| 3.62.2.2 ~MCBridge()                          | 195 |
| 3.62.3 Field Documentation                    | 195 |
| 3.62.3.1 nodes                                | 195 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.62.3.2 next                                 | 196 |
| 3.63 MConn Class Reference                    | 196 |
| 3.63.1 Detailed Description                   | 197 |
| 3.63.2 Constructor & Destructor Documentation | 197 |
| 3.63.2.1 MConn()                              | 197 |
| 3.63.2.2 ~MConn()                             | 197 |
| 3.63.3 Member Function Documentation          | 198 |
| 3.63.3.1 init()                               | 198 |
| 3.63.3.2 pushNode()                           | 198 |
| 3.63.3.3 pushEdge()                           | 198 |
| 3.63.3.4 addOPlc()                            | 198 |
| 3.63.3.5 addBridge()                          | 198 |
| 3.63.3.6 addArtic()                           | 198 |
| 3.63.3.7 addBlock()                           | 199 |
| 3.63.3.8 addBlockSelf()                       | 199 |
| 3.63.3.9 print()                              | 199 |
| 3.63.4 Field Documentation                    | 199 |
| 3.63.4.1 opics                                | 199 |
| 3.63.4.2 bridges                              | 199 |
| 3.63.4.3 blocks                               | 199 |
| 3.63.4.4 articuls                             | 199 |
| 3.63.4.5 opisp                                | 200 |
| 3.63.4.6 opistk                               | 200 |
| 3.63.4.7 blksp                                | 200 |
| 3.63.4.8 blkstk                               | 200 |
| 3.63.4.9 snodes                               | 200 |
| 3.63.4.10 sedges                              | 200 |
| 3.63.4.11 nopic                               | 200 |
| 3.63.4.12 nlpopic                             | 200 |
| 3.63.4.13 nctopic                             | 201 |
| 3.63.4.14 nbridges                            | 201 |
| 3.63.4.15 ne0bridges                          | 201 |
| 3.63.4.16 ne1bridges                          | 201 |
| 3.63.4.17 nselfloops                          | 201 |
| 3.63.4.18 nblocks                             | 201 |
| 3.63.4.19 na1blocks                           | 201 |
| 3.63.4.20 narticuls                           | 201 |
| 3.63.4.21 neblocks                            | 202 |
| 3.63.4.22 nopisp                              | 202 |
| 3.63.4.23 opistkptr                           | 202 |
| 3.63.4.24 nblksp                              | 202 |
| 3.63.4.25 blkstkptr                           | 202 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.63.4.26 DUMMYPADDING                        | 202 |
| 3.64 MCOpi Class Reference                    | 202 |
| 3.64.1 Detailed Description                   | 203 |
| 3.64.2 Constructor & Destructor Documentation | 203 |
| 3.64.2.1 MCOpi()                              | 203 |
| 3.64.2.2 ~MCOpi()                             | 203 |
| 3.64.3 Member Function Documentation          | 203 |
| 3.64.3.1 init()                               | 203 |
| 3.64.4 Field Documentation                    | 203 |
| 3.64.4.1 nodes                                | 203 |
| 3.64.4.2 nnodes                               | 204 |
| 3.64.4.3 nlegs                                | 204 |
| 3.64.4.4 next                                 | 204 |
| 3.64.4.5 nedges                               | 204 |
| 3.64.4.6 loop                                 | 204 |
| 3.64.4.7 ctloop                               | 204 |
| 3.65 MGraph Class Reference                   | 205 |
| 3.65.1 Detailed Description                   | 206 |
| 3.65.2 Constructor & Destructor Documentation | 206 |
| 3.65.2.1 MGraph() [1/2]                       | 206 |
| 3.65.2.2 MGraph() [2/2]                       | 207 |
| 3.65.2.3 ~MGraph()                            | 207 |
| 3.65.3 Member Function Documentation          | 207 |
| 3.65.3.1 init()                               | 207 |
| 3.65.3.2 generate()                           | 207 |
| 3.65.3.3 isExternal()                         | 207 |
| 3.65.3.4 printAdjMat()                        | 207 |
| 3.65.3.5 print()                              | 208 |
| 3.65.3.6 isConnected()                        | 208 |
| 3.65.3.7 visit()                              | 208 |
| 3.65.3.8 isIsomorphic()                       | 208 |
| 3.65.3.9 permMat()                            | 208 |
| 3.65.3.10 compMat()                           | 208 |
| 3.65.3.11 refineClass()                       | 209 |
| 3.65.3.12 bisearchME()                        | 209 |
| 3.65.3.13 biconnME()                          | 209 |
| 3.65.3.14 isOptM()                            | 209 |
| 3.65.3.15 connectClass()                      | 209 |
| 3.65.3.16 connectNode()                       | 209 |
| 3.65.3.17 connectLeg()                        | 210 |
| 3.65.3.18 newGraph()                          | 210 |
| 3.65.4 Field Documentation                    | 210 |

|                          |     |
|--------------------------|-----|
| 3.65.4.1 opt             | 210 |
| 3.65.4.2 group           | 210 |
| 3.65.4.3 orbits          | 210 |
| 3.65.4.4 nodes           | 210 |
| 3.65.4.5 mld             | 211 |
| 3.65.4.6 clist           | 211 |
| 3.65.4.7 pld             | 211 |
| 3.65.4.8 nNodes          | 211 |
| 3.65.4.9 nEdges          | 211 |
| 3.65.4.10 nLoops         | 211 |
| 3.65.4.11 nExtern        | 211 |
| 3.65.4.12 nClasses       | 211 |
| 3.65.4.13 mindeg         | 212 |
| 3.65.4.14 maxdeg         | 212 |
| 3.65.4.15 adjMat         | 212 |
| 3.65.4.16 curcl          | 212 |
| 3.65.4.17 egraph         | 212 |
| 3.65.4.18 nsym           | 212 |
| 3.65.4.19 esym           | 212 |
| 3.65.4.20 cDiag          | 212 |
| 3.65.4.21 c1PI           | 213 |
| 3.65.4.22 cNoTadpole     | 213 |
| 3.65.4.23 cNoTadBlock    | 213 |
| 3.65.4.24 c1PINoTadBlock | 213 |
| 3.65.4.25 wscon          | 213 |
| 3.65.4.26 wsopi          | 213 |
| 3.65.4.27 ngen           | 213 |
| 3.65.4.28 ngconn         | 213 |
| 3.65.4.29 nCallRefine    | 214 |
| 3.65.4.30 discardRefine  | 214 |
| 3.65.4.31 discardDisc    | 214 |
| 3.65.4.32 discardIso     | 214 |
| 3.65.4.33 opi            | 214 |
| 3.65.4.34 opiloop        | 214 |
| 3.65.4.35 extself        | 214 |
| 3.65.4.36 selfloop       | 214 |
| 3.65.4.37 tadpole        | 215 |
| 3.65.4.38 tadblock       | 215 |
| 3.65.4.39 block          | 215 |
| 3.65.4.40 DUMMYPADDING1  | 215 |
| 3.65.4.41 mconn          | 215 |
| 3.65.4.42 modmat         | 215 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.65.4.43 bidef                               | 215 |
| 3.65.4.44 bilow                               | 215 |
| 3.65.4.45 bicount                             | 216 |
| 3.65.4.46 DUMMYPADDING2                       | 216 |
| 3.66 MiNmAx Struct Reference                  | 216 |
| 3.66.1 Detailed Description                   | 216 |
| 3.66.2 Field Documentation                    | 216 |
| 3.66.2.1 mini                                 | 216 |
| 3.66.2.2 maxi                                 | 216 |
| 3.66.2.3 size                                 | 217 |
| 3.67 MInput Struct Reference                  | 217 |
| 3.67.1 Detailed Description                   | 217 |
| 3.67.2 Field Documentation                    | 217 |
| 3.67.2.1 name                                 | 217 |
| 3.67.2.2 defpart                              | 217 |
| 3.67.2.3 ncouple                              | 217 |
| 3.67.2.4 cnamlist                             | 218 |
| 3.68 MNode Class Reference                    | 218 |
| 3.68.1 Detailed Description                   | 218 |
| 3.68.2 Constructor & Destructor Documentation | 218 |
| 3.68.2.1 MNode() [1/2]                        | 218 |
| 3.68.2.2 MNode() [2/2]                        | 219 |
| 3.68.3 Field Documentation                    | 219 |
| 3.68.3.1 id                                   | 219 |
| 3.68.3.2 deg                                  | 219 |
| 3.68.3.3 cls                                  | 219 |
| 3.68.3.4 extloop                              | 219 |
| 3.68.3.5 cmindeg                              | 219 |
| 3.68.3.6 cmaxdeg                              | 220 |
| 3.68.3.7 freelg                               | 220 |
| 3.68.3.8 visited                              | 220 |
| 3.69 MNodeClass Class Reference               | 220 |
| 3.69.1 Detailed Description                   | 221 |
| 3.69.2 Constructor & Destructor Documentation | 221 |
| 3.69.2.1 MNodeClass()                         | 221 |
| 3.69.2.2 ~MNodeClass()                        | 221 |
| 3.69.3 Member Function Documentation          | 221 |
| 3.69.3.1 init()                               | 221 |
| 3.69.3.2 copy()                               | 221 |
| 3.69.3.3 clCmp()                              | 221 |
| 3.69.3.4 printMat()                           | 222 |
| 3.69.3.5 mkFlist()                            | 222 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.69.3.6 mkNdCl()                             | 222 |
| 3.69.3.7 mkClMat()                            | 222 |
| 3.69.3.8 incMat()                             | 222 |
| 3.69.3.9 cmpMNCArry()                         | 222 |
| 3.69.3.10 reorder()                           | 223 |
| 3.69.4 Field Documentation                    | 223 |
| 3.69.4.1 clmat                                | 223 |
| 3.69.4.2 clist                                | 223 |
| 3.69.4.3 ndcl                                 | 223 |
| 3.69.4.4 flist                                | 223 |
| 3.69.4.5 clord                                | 223 |
| 3.69.4.6 cmindeg                              | 223 |
| 3.69.4.7 cmaxdeg                              | 224 |
| 3.69.4.8 nNodes                               | 224 |
| 3.69.4.9 nClasses                             | 224 |
| 3.69.4.10 maxdeg                              | 224 |
| 3.69.4.11 flg0                                | 224 |
| 3.69.4.12 flg1                                | 224 |
| 3.69.4.13 flg2                                | 224 |
| 3.70 Model Class Reference                    | 225 |
| 3.70.1 Detailed Description                   | 226 |
| 3.70.2 Constructor & Destructor Documentation | 226 |
| 3.70.2.1 Model()                              | 226 |
| 3.70.2.2 ~Model()                             | 226 |
| 3.70.3 Member Function Documentation          | 226 |
| 3.70.3.1 prModel()                            | 226 |
| 3.70.3.2 addParticle()                        | 226 |
| 3.70.3.3 addParticleEnd()                     | 227 |
| 3.70.3.4 addInteraction()                     | 227 |
| 3.70.3.5 addInteractionEnd()                  | 227 |
| 3.70.3.6 findParticleName()                   | 227 |
| 3.70.3.7 findParticleCode()                   | 227 |
| 3.70.3.8 findInteractionName()                | 227 |
| 3.70.3.9 findInteractionCode()                | 227 |
| 3.70.3.10 particleName()                      | 228 |
| 3.70.3.11 particleCode()                      | 228 |
| 3.70.3.12 normalParticle()                    | 228 |
| 3.70.3.13 antiParticle()                      | 228 |
| 3.70.3.14 allParticles()                      | 228 |
| 3.70.3.15 findMClass()                        | 228 |
| 3.70.3.16 prParticleArray()                   | 228 |
| 3.70.4 Field Documentation                    | 229 |

|                              |     |
|------------------------------|-----|
| 3.70.4.1 name                | 229 |
| 3.70.4.2 cnlist              | 229 |
| 3.70.4.3 ncouple             | 229 |
| 3.70.4.4 nParticles          | 229 |
| 3.70.4.5 particles           | 229 |
| 3.70.4.6 pdef                | 229 |
| 3.70.4.7 nallPart            | 229 |
| 3.70.4.8 allPart             | 230 |
| 3.70.4.9 nintPart            | 230 |
| 3.70.4.10 intPart            | 230 |
| 3.70.4.11 nInteracts         | 230 |
| 3.70.4.12 interacts          | 230 |
| 3.70.4.13 vdef               | 230 |
| 3.70.4.14 maxnlegs           | 230 |
| 3.70.4.15 maxcpl             | 230 |
| 3.70.4.16 maxloop            | 231 |
| 3.70.4.17 cplgcp             | 231 |
| 3.70.4.18 cplglg             | 231 |
| 3.70.4.19 cplgnvl            | 231 |
| 3.70.4.20 cplgvl             | 231 |
| 3.70.4.21 ncplgcp            | 231 |
| 3.70.4.22 defpart            | 231 |
| 3.70.4.23 skipFLine          | 231 |
| 3.70.4.24 DUMMY_PADDING      | 232 |
| 3.71 MoDeL Struct Reference  | 232 |
| 3.71.1 Detailed Description  | 233 |
| 3.71.2 Field Documentation   | 233 |
| 3.71.2.1 vertices            | 233 |
| 3.71.2.2 couplings           | 233 |
| 3.71.2.3 name                | 233 |
| 3.71.2.4 grccmodel           | 233 |
| 3.71.2.5 legcouple           | 233 |
| 3.71.2.6 nparticles          | 233 |
| 3.71.2.7 nvertices           | 234 |
| 3.71.2.8 invertices          | 234 |
| 3.71.2.9 sizevertices        | 234 |
| 3.71.2.10 sizecouplings      | 234 |
| 3.71.2.11 ncouplings         | 234 |
| 3.71.2.12 error              | 234 |
| 3.71.2.13 dummy              | 234 |
| 3.72 MODNUM Struct Reference | 235 |
| 3.72.1 Detailed Description  | 235 |



|                                               |     |
|-----------------------------------------------|-----|
| 3.72.2 Field Documentation                    | 235 |
| 3.72.2.1 a                                    | 235 |
| 3.72.2.2 m                                    | 235 |
| 3.72.2.3 na                                   | 235 |
| 3.72.2.4 nm                                   | 235 |
| 3.73 MoDoPtDoLIaRS Struct Reference           | 236 |
| 3.73.1 Detailed Description                   | 236 |
| 3.73.2 Field Documentation                    | 236 |
| 3.73.2.1 number                               | 236 |
| 3.73.2.2 type                                 | 236 |
| 3.74 monomial_larger Class Reference          | 236 |
| 3.74.1 Detailed Description                   | 236 |
| 3.74.2 Constructor & Destructor Documentation | 236 |
| 3.74.2.1 monomial_larger()                    | 236 |
| 3.74.3 Member Function Documentation          | 237 |
| 3.74.3.1 operator()()                         | 237 |
| 3.75 MORbits Class Reference                  | 237 |
| 3.75.1 Detailed Description                   | 237 |
| 3.75.2 Constructor & Destructor Documentation | 237 |
| 3.75.2.1 MORbits()                            | 237 |
| 3.75.2.2 ~MORbits()                           | 238 |
| 3.75.3 Member Function Documentation          | 238 |
| 3.75.3.1 print()                              | 238 |
| 3.75.3.2 initPerm()                           | 238 |
| 3.75.3.3 fromPerm()                           | 238 |
| 3.75.3.4 toOrbits()                           | 238 |
| 3.75.4 Field Documentation                    | 238 |
| 3.75.4.1 nOrbits                              | 238 |
| 3.75.4.2 nNodes                               | 239 |
| 3.75.4.3 nd2or                                | 239 |
| 3.75.4.4 or2nd                                | 239 |
| 3.75.4.5 flist                                | 239 |
| 3.76 flint::mpoly Class Reference             | 239 |
| 3.76.1 Detailed Description                   | 239 |
| 3.76.2 Constructor & Destructor Documentation | 240 |
| 3.76.2.1 mpoly()                              | 240 |
| 3.76.2.2 ~mpoly()                             | 240 |
| 3.76.3 Member Function Documentation          | 240 |
| 3.76.3.1 print()                              | 240 |
| 3.76.4 Field Documentation                    | 240 |
| 3.76.4.1 d                                    | 240 |
| 3.77 flint::mpoly_ctx Class Reference         | 240 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.77.1 Detailed Description                   | 241 |
| 3.77.2 Constructor & Destructor Documentation | 241 |
| 3.77.2.1 mpoly_ctx()                          | 241 |
| 3.77.2.2 ~mpoly_ctx()                         | 241 |
| 3.77.3 Field Documentation                    | 241 |
| 3.77.3.1 d                                    | 241 |
| 3.78 flint::mpoly_factor Class Reference      | 241 |
| 3.78.1 Detailed Description                   | 241 |
| 3.78.2 Constructor & Destructor Documentation | 242 |
| 3.78.2.1 mpoly_factor()                       | 242 |
| 3.78.2.2 ~mpoly_factor()                      | 242 |
| 3.78.3 Field Documentation                    | 242 |
| 3.78.3.1 d                                    | 242 |
| 3.79 N_const Struct Reference                 | 242 |
| 3.79.1 Detailed Description                   | 245 |
| 3.79.2 Field Documentation                    | 245 |
| 3.79.2.1 OldPosIn                             | 245 |
| 3.79.2.2 OldPosOut                            | 245 |
| 3.79.2.3 theposition                          | 245 |
| 3.79.2.4 EndNest                              | 245 |
| 3.79.2.5 Frozen                               | 245 |
| 3.79.2.6 FullProto                            | 246 |
| 3.79.2.7 cTerm                                | 246 |
| 3.79.2.8 RepPoint                             | 246 |
| 3.79.2.9 WildValue                            | 246 |
| 3.79.2.10 WildStop                            | 246 |
| 3.79.2.11 argaddress                          | 246 |
| 3.79.2.12 RepFunList                          | 246 |
| 3.79.2.13 patstop                             | 246 |
| 3.79.2.14 terstop                             | 247 |
| 3.79.2.15 terstart                            | 247 |
| 3.79.2.16 terfirstcomm                        | 247 |
| 3.79.2.17 DumFound                            | 247 |
| 3.79.2.18 DumPlace                            | 247 |
| 3.79.2.19 DumFunPlace                         | 247 |
| 3.79.2.20 UsedSymbol                          | 247 |
| 3.79.2.21 UsedVector                          | 247 |
| 3.79.2.22 UsedIndex                           | 248 |
| 3.79.2.23 UsedFunction                        | 248 |
| 3.79.2.24 MaskPointer                         | 248 |
| 3.79.2.25 ForFindOnly                         | 248 |
| 3.79.2.26 findTerm                            | 248 |

|                                     |     |
|-------------------------------------|-----|
| 3.79.2.27 findPattern . . . . .     | 248 |
| 3.79.2.28 dummyrenumlist . . . . .  | 248 |
| 3.79.2.29 funargs . . . . .         | 248 |
| 3.79.2.30 funlocs . . . . .         | 249 |
| 3.79.2.31 funinds . . . . .         | 249 |
| 3.79.2.32 NoScrat2 . . . . .        | 249 |
| 3.79.2.33 ReplaceScrat . . . . .    | 249 |
| 3.79.2.34 tracestack . . . . .      | 249 |
| 3.79.2.35 selecttermundo . . . . .  | 249 |
| 3.79.2.36 patternbuffer . . . . .   | 249 |
| 3.79.2.37 termbuffer . . . . .      | 249 |
| 3.79.2.38 PoinScrat . . . . .       | 250 |
| 3.79.2.39 FunScrat . . . . .        | 250 |
| 3.79.2.40 RenumScrat . . . . .      | 250 |
| 3.79.2.41 FunInfo . . . . .         | 250 |
| 3.79.2.42 SplitScrat . . . . .      | 250 |
| 3.79.2.43 SplitScrat1 . . . . .     | 250 |
| 3.79.2.44 FunSorts . . . . .        | 250 |
| 3.79.2.45 SoScratC . . . . .        | 250 |
| 3.79.2.46 listinprint . . . . .     | 251 |
| 3.79.2.47 currentTerm . . . . .     | 251 |
| 3.79.2.48 arglist . . . . .         | 251 |
| 3.79.2.49 tlistbuf . . . . .        | 251 |
| 3.79.2.50 compressSpace . . . . .   | 251 |
| 3.79.2.51 SHcombi . . . . .         | 251 |
| 3.79.2.52 poly_vars . . . . .       | 251 |
| 3.79.2.53 cmod . . . . .            | 251 |
| 3.79.2.54 SHvar . . . . .           | 252 |
| 3.79.2.55 deferskipped . . . . .    | 252 |
| 3.79.2.56 InScrat . . . . .         | 252 |
| 3.79.2.57 SplitScratSize . . . . .  | 252 |
| 3.79.2.58 InScrat1 . . . . .        | 252 |
| 3.79.2.59 SplitScratSize1 . . . . . | 252 |
| 3.79.2.60 ninterms . . . . .        | 252 |
| 3.79.2.61 SHcombisize . . . . .     | 252 |
| 3.79.2.62 NumTotWildArgs . . . . .  | 253 |
| 3.79.2.63 UseFindOnly . . . . .     | 253 |
| 3.79.2.64 UsedOtherFind . . . . .   | 253 |
| 3.79.2.65 ErrorInDollar . . . . .   | 253 |
| 3.79.2.66 numfargs . . . . .        | 253 |
| 3.79.2.67 numflocs . . . . .        | 253 |
| 3.79.2.68 nargs . . . . .           | 253 |

|                               |     |
|-------------------------------|-----|
| 3.79.2.69 tohunt              | 253 |
| 3.79.2.70 numoffuns           | 254 |
| 3.79.2.71 funisize            | 254 |
| 3.79.2.72 RSize               | 254 |
| 3.79.2.73 numtracesctack      | 254 |
| 3.79.2.74 intracestack        | 254 |
| 3.79.2.75 numfuninfo          | 254 |
| 3.79.2.76 NumFunSorts         | 254 |
| 3.79.2.77 MaxFunSorts         | 254 |
| 3.79.2.78 arglistsize         | 255 |
| 3.79.2.79 tlistsize           | 255 |
| 3.79.2.80 filenum             | 255 |
| 3.79.2.81 compressSize        | 255 |
| 3.79.2.82 polysortflag        | 255 |
| 3.79.2.83 nogroundlevel       | 255 |
| 3.79.2.84 subsubveto          | 255 |
| 3.79.2.85 MaxRenumScratch     | 255 |
| 3.79.2.86 oldtype             | 256 |
| 3.79.2.87 oldvalue            | 256 |
| 3.79.2.88 NumWild             | 256 |
| 3.79.2.89 RepFunNum           | 256 |
| 3.79.2.90 DisOrderFlag        | 256 |
| 3.79.2.91 WildDirt            | 256 |
| 3.79.2.92 NumFound            | 256 |
| 3.79.2.93 WildReserve         | 256 |
| 3.79.2.94 TelnFun             | 257 |
| 3.79.2.95 TeSuOut             | 257 |
| 3.79.2.96 WildArgs            | 257 |
| 3.79.2.97 WildEat             | 257 |
| 3.79.2.98 PolyNormFlag        | 257 |
| 3.79.2.99 PolyFunTodo         | 257 |
| 3.79.2.100 sizeselecttermundo | 257 |
| 3.79.2.101 patternbuffersize  | 257 |
| 3.79.2.102 numlistinprint     | 258 |
| 3.79.2.103 ncmod              | 258 |
| 3.79.2.104 ExpectedSign       | 258 |
| 3.79.2.105 SignCheck          | 258 |
| 3.79.2.106 IndDum             | 258 |
| 3.79.2.107 poly_num_vars      | 258 |
| 3.79.2.108 idfunctionflag     | 258 |
| 3.79.2.109 poly_vars_type     | 259 |
| 3.79.2.110 tryterm            | 259 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.80 nameblock Struct Reference               | 259 |
| 3.80.1 Detailed Description                   | 259 |
| 3.80.2 Field Documentation                    | 259 |
| 3.80.2.1 previousblock                        | 259 |
| 3.80.2.2 position                             | 259 |
| 3.80.2.3 names                                | 260 |
| 3.81 NaMeNode Struct Reference                | 260 |
| 3.81.1 Detailed Description                   | 260 |
| 3.81.2 Field Documentation                    | 260 |
| 3.81.2.1 name                                 | 260 |
| 3.81.2.2 parent                               | 260 |
| 3.81.2.3 left                                 | 261 |
| 3.81.2.4 right                                | 261 |
| 3.81.2.5 balance                              | 261 |
| 3.81.2.6 type                                 | 261 |
| 3.81.2.7 number                               | 261 |
| 3.82 NaMeSpAcE Struct Reference               | 262 |
| 3.82.1 Detailed Description                   | 262 |
| 3.82.2 Field Documentation                    | 262 |
| 3.82.2.1 previous                             | 262 |
| 3.82.2.2 next                                 | 262 |
| 3.82.2.3 usernames                            | 263 |
| 3.82.2.4 name                                 | 263 |
| 3.83 NaMeTree Struct Reference                | 263 |
| 3.83.1 Detailed Description                   | 264 |
| 3.83.2 Field Documentation                    | 264 |
| 3.83.2.1 namenode                             | 264 |
| 3.83.2.2 namebuffer                           | 264 |
| 3.83.2.3 nodesize                             | 264 |
| 3.83.2.4 nodefill                             | 264 |
| 3.83.2.5 namesize                             | 264 |
| 3.83.2.6 namefill                             | 265 |
| 3.83.2.7 oldnamefill                          | 265 |
| 3.83.2.8 oldnodefill                          | 265 |
| 3.83.2.9 globalnamefill                       | 265 |
| 3.83.2.10 globalnodefill                      | 265 |
| 3.83.2.11 clearnamefill                       | 265 |
| 3.83.2.12 clearnodefill                       | 266 |
| 3.83.2.13 headnode                            | 266 |
| 3.84 NCand Class Reference                    | 266 |
| 3.84.1 Detailed Description                   | 266 |
| 3.84.2 Constructor & Destructor Documentation | 266 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.84.2.1 NCand()                              | 266 |
| 3.84.2.2 ~NCand()                             | 267 |
| 3.84.3 Member Function Documentation          | 267 |
| 3.84.3.1 prNCand()                            | 267 |
| 3.84.4 Field Documentation                    | 267 |
| 3.84.4.1 deg                                  | 267 |
| 3.84.4.2 st                                   | 267 |
| 3.84.4.3 nilist                               | 267 |
| 3.84.4.4 ilist                                | 267 |
| 3.85 NCInput Struct Reference                 | 268 |
| 3.85.1 Detailed Description                   | 268 |
| 3.85.2 Field Documentation                    | 268 |
| 3.85.2.1 clddeg                               | 268 |
| 3.85.2.2 clnum                                | 268 |
| 3.85.2.3 cltyp                                | 268 |
| 3.85.2.4 cmind                                | 268 |
| 3.85.2.5 cmaxd                                | 268 |
| 3.85.2.6 ptcl                                 | 269 |
| 3.85.2.7 cple                                 | 269 |
| 3.86 NeStInG Struct Reference                 | 269 |
| 3.86.1 Detailed Description                   | 269 |
| 3.86.2 Field Documentation                    | 269 |
| 3.86.2.1 termsize                             | 269 |
| 3.86.2.2 funsize                              | 269 |
| 3.86.2.3 argsize                              | 270 |
| 3.87 NoDe Struct Reference                    | 270 |
| 3.87.1 Detailed Description                   | 270 |
| 3.87.2 Field Documentation                    | 270 |
| 3.87.2.1 left                                 | 270 |
| 3.87.2.2 rght                                 | 271 |
| 3.87.2.3 lloser                               | 271 |
| 3.87.2.4 rloser                               | 271 |
| 3.87.2.5 lsrc                                 | 271 |
| 3.87.2.6 rsrc                                 | 271 |
| 3.88 node Struct Reference                    | 271 |
| 3.88.1 Detailed Description                   | 272 |
| 3.88.2 Constructor & Destructor Documentation | 272 |
| 3.88.2.1 node() [1/2]                         | 272 |
| 3.88.2.2 node() [2/2]                         | 272 |
| 3.88.3 Member Function Documentation          | 272 |
| 3.88.3.1 cmp()                                | 272 |
| 3.88.3.2 operator()()                         | 273 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.88.3.3 calcHash()                           | 273 |
| 3.88.4 Field Documentation                    | 273 |
| 3.88.4.1 data                                 | 273 |
| 3.88.4.2 l                                    | 273 |
| 3.88.4.3 r                                    | 273 |
| 3.88.4.4 sign                                 | 273 |
| 3.88.4.5 hash                                 | 273 |
| 3.89 NodeEq Struct Reference                  | 274 |
| 3.89.1 Detailed Description                   | 274 |
| 3.89.2 Member Function Documentation          | 274 |
| 3.89.2.1 operator()                           | 274 |
| 3.90 NodeHash Struct Reference                | 274 |
| 3.90.1 Detailed Description                   | 274 |
| 3.90.2 Member Function Documentation          | 274 |
| 3.90.2.1 operator()                           | 274 |
| 3.91 NStack Class Reference                   | 275 |
| 3.91.1 Detailed Description                   | 275 |
| 3.91.2 Constructor & Destructor Documentation | 275 |
| 3.91.2.1 NStack()                             | 275 |
| 3.91.2.2 ~NStack()                            | 275 |
| 3.91.3 Member Function Documentation          | 275 |
| 3.91.3.1 print()                              | 275 |
| 3.91.4 Field Documentation                    | 275 |
| 3.91.4.1 noden                                | 275 |
| 3.91.4.2 deg                                  | 276 |
| 3.91.4.3 st                                   | 276 |
| 3.91.4.4 nilist                               | 276 |
| 3.91.4.5 ilist                                | 276 |
| 3.92 O_const Struct Reference                 | 276 |
| 3.92.1 Detailed Description                   | 278 |
| 3.92.2 Field Documentation                    | 278 |
| 3.92.2.1 SaveData                             | 278 |
| 3.92.2.2 SaveHeader                           | 278 |
| 3.92.2.3 OptimizeResult                       | 278 |
| 3.92.2.4 OutputLine                           | 279 |
| 3.92.2.5 OutStop                              | 279 |
| 3.92.2.6 OutFill                              | 279 |
| 3.92.2.7 bracket                              | 279 |
| 3.92.2.8 termbuf                              | 279 |
| 3.92.2.9 tabstring                            | 279 |
| 3.92.2.10 wpos                                | 279 |
| 3.92.2.11 wpoin                               | 279 |

|                                           |     |
|-------------------------------------------|-----|
| 3.92.2.12 DollarOutBuffer . . . . .       | 280 |
| 3.92.2.13 CurBufWrt . . . . .             | 280 |
| 3.92.2.14 FlipWORD . . . . .              | 280 |
| 3.92.2.15 FlipLONG . . . . .              | 280 |
| 3.92.2.16 FlipPOS . . . . .               | 280 |
| 3.92.2.17 FlipPOINTER . . . . .           | 280 |
| 3.92.2.18 ResizeData . . . . .            | 280 |
| 3.92.2.19 ResizeWORD . . . . .            | 280 |
| 3.92.2.20 ResizeNCWORD . . . . .          | 281 |
| 3.92.2.21 ResizeLONG . . . . .            | 281 |
| 3.92.2.22 ResizePOS . . . . .             | 281 |
| 3.92.2.23 ResizePOINTER . . . . .         | 281 |
| 3.92.2.24 CheckPower . . . . .            | 281 |
| 3.92.2.25 RenumberVec . . . . .           | 281 |
| 3.92.2.26 Dictionaries . . . . .          | 281 |
| 3.92.2.27 tensorList . . . . .            | 281 |
| 3.92.2.28 inscheme . . . . .              | 282 |
| 3.92.2.29 NumInBrack . . . . .            | 282 |
| 3.92.2.30 wlen . . . . .                  | 282 |
| 3.92.2.31 DollarOutSizeBuffer . . . . .   | 282 |
| 3.92.2.32 DollarInOutBuffer . . . . .     | 282 |
| 3.92.2.33 Optimize . . . . .              | 282 |
| 3.92.2.34 OutInBuffer . . . . .           | 282 |
| 3.92.2.35 NoSpacesInNumbers . . . . .     | 282 |
| 3.92.2.36 BlockSpaces . . . . .           | 283 |
| 3.92.2.37 CurrentDictionary . . . . .     | 283 |
| 3.92.2.38 SizeDictionaries . . . . .      | 283 |
| 3.92.2.39 NumDictionaries . . . . .       | 283 |
| 3.92.2.40 CurDictNumbers . . . . .        | 283 |
| 3.92.2.41 CurDictVariables . . . . .      | 283 |
| 3.92.2.42 CurDictSpecials . . . . .       | 283 |
| 3.92.2.43 CurDictFunWithArgs . . . . .    | 283 |
| 3.92.2.44 CurDictNumberWarning . . . . .  | 284 |
| 3.92.2.45 CurDictNotInFunctions . . . . . | 284 |
| 3.92.2.46 CurDictInDollars . . . . .      | 284 |
| 3.92.2.47 gNumDictionaries . . . . .      | 284 |
| 3.92.2.48 schemenum . . . . .             | 284 |
| 3.92.2.49 transFlag . . . . .             | 284 |
| 3.92.2.50 powerFlag . . . . .             | 284 |
| 3.92.2.51 mpower . . . . .                | 284 |
| 3.92.2.52 resizeFlag . . . . .            | 285 |
| 3.92.2.53 bufferedInd . . . . .           | 285 |



|                                                |     |
|------------------------------------------------|-----|
| 3.92.2.54 OutSkip . . . . .                    | 285 |
| 3.92.2.55 IsBracket . . . . .                  | 285 |
| 3.92.2.56 InFbrack . . . . .                   | 285 |
| 3.92.2.57 PrintType . . . . .                  | 285 |
| 3.92.2.58 FortFirst . . . . .                  | 285 |
| 3.92.2.59 DoubleFlag . . . . .                 | 285 |
| 3.92.2.60 IndentSpace . . . . .                | 286 |
| 3.92.2.61 FactorMode . . . . .                 | 286 |
| 3.92.2.62 FactorNum . . . . .                  | 286 |
| 3.92.2.63 ErrorBlock . . . . .                 | 286 |
| 3.92.2.64 OptimizationLevel . . . . .          | 286 |
| 3.92.2.65 FortDotChar . . . . .                | 286 |
| 3.93 objects Struct Reference . . . . .        | 287 |
| 3.93.1 Detailed Description . . . . .          | 287 |
| 3.93.2 Field Documentation . . . . .           | 287 |
| 3.93.2.1 position . . . . .                    | 287 |
| 3.93.2.2 size . . . . .                        | 287 |
| 3.93.2.3 date . . . . .                        | 287 |
| 3.93.2.4 tablenumber . . . . .                 | 287 |
| 3.93.2.5 uncompressed . . . . .                | 288 |
| 3.93.2.6 spare1 . . . . .                      | 288 |
| 3.93.2.7 spare2 . . . . .                      | 288 |
| 3.93.2.8 spare3 . . . . .                      | 288 |
| 3.93.2.9 element . . . . .                     | 288 |
| 3.94 OptDef Struct Reference . . . . .         | 288 |
| 3.94.1 Detailed Description . . . . .          | 288 |
| 3.94.2 Field Documentation . . . . .           | 289 |
| 3.94.2.1 name . . . . .                        | 289 |
| 3.94.2.2 mean . . . . .                        | 289 |
| 3.94.2.3 defaultv . . . . .                    | 289 |
| 3.94.2.4 DUMMYPADDING . . . . .                | 289 |
| 3.95 optimization Class Reference . . . . .    | 289 |
| 3.95.1 Detailed Description . . . . .          | 290 |
| 3.95.2 Description . . . . .                   | 290 |
| 3.95.3 Member Function Documentation . . . . . | 290 |
| 3.95.3.1 operator<() . . . . .                 | 290 |
| 3.95.4 Field Documentation . . . . .           | 291 |
| 3.95.4.1 type . . . . .                        | 291 |
| 3.95.4.2 arg1 . . . . .                        | 291 |
| 3.95.4.3 arg2 . . . . .                        | 291 |
| 3.95.4.4 improve . . . . .                     | 291 |
| 3.95.4.5 coeff . . . . .                       | 291 |

|                                                         |     |
|---------------------------------------------------------|-----|
| 3.95.4.6 eqnidxs . . . . .                              | 291 |
| 3.96 OPTIMIZE Struct Reference . . . . .                | 292 |
| 3.96.1 Detailed Description . . . . .                   | 292 |
| 3.96.2 Field Documentation . . . . .                    | 292 |
| 3.96.2.1 fval . . . . .                                 | 292 |
| 3.96.2.2 ival . . . . .                                 | 292 |
| 3.96.2.3 horner . . . . .                               | 293 |
| 3.96.2.4 hornerdirection . . . . .                      | 293 |
| 3.96.2.5 method . . . . .                               | 293 |
| 3.96.2.6 mctstimelimit . . . . .                        | 293 |
| 3.96.2.7 mctsnumexpand . . . . .                        | 293 |
| 3.96.2.8 mctsnumkeep . . . . .                          | 293 |
| 3.96.2.9 mctsnumrepeat . . . . .                        | 293 |
| 3.96.2.10 greedytimelimit . . . . .                     | 293 |
| 3.96.2.11 greedyminnum . . . . .                        | 294 |
| 3.96.2.12 greedymaxperc . . . . .                       | 294 |
| 3.96.2.13 printstats . . . . .                          | 294 |
| 3.96.2.14 debugflags . . . . .                          | 294 |
| 3.96.2.15 schemeflags . . . . .                         | 294 |
| 3.96.2.16 mctsdecaymode . . . . .                       | 294 |
| 3.96.2.17 salter . . . . .                              | 294 |
| 3.96.2.18 spare . . . . .                               | 295 |
| 3.97 OPTIMIZERESULT Struct Reference . . . . .          | 295 |
| 3.97.1 Detailed Description . . . . .                   | 295 |
| 3.97.2 Field Documentation . . . . .                    | 295 |
| 3.97.2.1 code . . . . .                                 | 295 |
| 3.97.2.2 nameofexpr . . . . .                           | 295 |
| 3.97.2.3 codesize . . . . .                             | 296 |
| 3.97.2.4 exprnr . . . . .                               | 296 |
| 3.97.2.5 minvar . . . . .                               | 296 |
| 3.97.2.6 maxvar . . . . .                               | 296 |
| 3.98 Options Class Reference . . . . .                  | 296 |
| 3.98.1 Detailed Description . . . . .                   | 297 |
| 3.98.2 Constructor & Destructor Documentation . . . . . | 297 |
| 3.98.2.1 Options() . . . . .                            | 297 |
| 3.98.2.2 ~Options() . . . . .                           | 298 |
| 3.98.3 Member Function Documentation . . . . .          | 298 |
| 3.98.3.1 setDefaultValue() . . . . .                    | 298 |
| 3.98.3.2 setValue() . . . . .                           | 298 |
| 3.98.3.3 getValue() . . . . .                           | 298 |
| 3.98.3.4 print() . . . . .                              | 298 |
| 3.98.3.5 setOutputF() . . . . .                         | 298 |

|                                               |     |
|-----------------------------------------------|-----|
| 3.98.3.6 setOutputP()                         | 299 |
| 3.98.3.7 printLevel()                         | 299 |
| 3.98.3.8 printModel()                         | 299 |
| 3.98.3.9 setOutMG()                           | 299 |
| 3.98.3.10 setOutAG()                          | 299 |
| 3.98.3.11 setEndMG()                          | 299 |
| 3.98.3.12 setErExit()                         | 300 |
| 3.98.3.13 getDef()                            | 300 |
| 3.98.3.14 begin()                             | 300 |
| 3.98.3.15 end()                               | 300 |
| 3.98.3.16 beginProc()                         | 300 |
| 3.98.3.17 endProc()                           | 300 |
| 3.98.3.18 beginSubProc()                      | 300 |
| 3.98.3.19 endSubProc()                        | 301 |
| 3.98.3.20 newMGraph()                         | 301 |
| 3.98.3.21 newAGraph()                         | 301 |
| 3.98.3.22 outModel()                          | 301 |
| 3.98.4 Field Documentation                    | 301 |
| 3.98.4.1 model                                | 301 |
| 3.98.4.2 proc                                 | 301 |
| 3.98.4.3 sproc                                | 301 |
| 3.98.4.4 out                                  | 302 |
| 3.98.4.5 outmg                                | 302 |
| 3.98.4.6 endmg                                | 302 |
| 3.98.4.7 outag                                | 302 |
| 3.98.4.8 argmg                                | 302 |
| 3.98.4.9 argemg                               | 302 |
| 3.98.4.10 argag                               | 302 |
| 3.98.4.11 values                              | 302 |
| 3.98.4.12 DUMMYPADDING                        | 303 |
| 3.98.4.13 time0                               | 303 |
| 3.98.4.14 time1                               | 303 |
| 3.99 Output Class Reference                   | 303 |
| 3.99.1 Detailed Description                   | 304 |
| 3.99.2 Constructor & Destructor Documentation | 304 |
| 3.99.2.1 Output()                             | 304 |
| 3.99.2.2 ~Output()                            | 305 |
| 3.99.3 Member Function Documentation          | 305 |
| 3.99.3.1 setOutgrf()                          | 305 |
| 3.99.3.2 setOutgrp()                          | 305 |
| 3.99.3.3 outBeginF()                          | 305 |
| 3.99.3.4 outBeginP()                          | 305 |

|                                |     |
|--------------------------------|-----|
| 3.99.3.5 outEndF()             | 305 |
| 3.99.3.6 outEndP()             | 306 |
| 3.99.3.7 outProcBeginF()       | 306 |
| 3.99.3.8 outProcBeginP()       | 306 |
| 3.99.3.9 outProcBegin0()       | 306 |
| 3.99.3.10 outSProcBeginF()     | 306 |
| 3.99.3.11 outSProcBeginP()     | 306 |
| 3.99.3.12 outProcEndF()        | 306 |
| 3.99.3.13 outProcEndP()        | 307 |
| 3.99.3.14 outEGraphF()         | 307 |
| 3.99.3.15 outEGraphP()         | 307 |
| 3.99.3.16 outModelF()          | 307 |
| 3.99.3.17 outModelP()          | 307 |
| 3.99.4 Field Documentation     | 307 |
| 3.99.4.1 opt                   | 307 |
| 3.99.4.2 model                 | 307 |
| 3.99.4.3 proc                  | 308 |
| 3.99.4.4 sproc                 | 308 |
| 3.99.4.5 outgrf                | 308 |
| 3.99.4.6 outgrfp               | 308 |
| 3.99.4.7 outgrp                | 308 |
| 3.99.4.8 outgrpp               | 308 |
| 3.99.4.9 proclid               | 308 |
| 3.99.4.10 outproc              | 309 |
| 3.100 P_const Struct Reference | 309 |
| 3.100.1 Detailed Description   | 310 |
| 3.100.2 Field Documentation    | 311 |
| 3.100.2.1 DollarList           | 311 |
| 3.100.2.2 PreVarList           | 311 |
| 3.100.2.3 LoopList             | 311 |
| 3.100.2.4 ProcList             | 311 |
| 3.100.2.5 inside               | 311 |
| 3.100.2.6 firstnamespace       | 311 |
| 3.100.2.7 lastnamespace        | 311 |
| 3.100.2.8 fullname             | 312 |
| 3.100.2.9 PreSwitchStrings     | 312 |
| 3.100.2.10 preStart            | 312 |
| 3.100.2.11 preStop             | 312 |
| 3.100.2.12 preFill             | 312 |
| 3.100.2.13 procedureExtension  | 312 |
| 3.100.2.14 cprocedureExtension | 312 |
| 3.100.2.15 PreAssignStack      | 312 |

|                                     |     |
|-------------------------------------|-----|
| 3.100.2.16 PelfStack                | 313 |
| 3.100.2.17 PreSwitchModes           | 313 |
| 3.100.2.18 PreTypes                 | 313 |
| 3.100.2.19 StopWatchZero            | 313 |
| 3.100.2.20 InOutBuf                 | 313 |
| 3.100.2.21 pSize                    | 313 |
| 3.100.2.22 PreAssignFlag            | 313 |
| 3.100.2.23 PreContinuation          | 313 |
| 3.100.2.24 PreproFlag               | 314 |
| 3.100.2.25 iBufError                | 314 |
| 3.100.2.26 PreOut                   | 314 |
| 3.100.2.27 PreSwitchLevel           | 314 |
| 3.100.2.28 NumPreSwitchStrings      | 314 |
| 3.100.2.29 MaxPreTypes              | 314 |
| 3.100.2.30 NumPreTypes              | 314 |
| 3.100.2.31 MaxPelfLevel             | 314 |
| 3.100.2.32 PelfLevel                | 315 |
| 3.100.2.33 PreInsideLevel           | 315 |
| 3.100.2.34 DelayPrevar              | 315 |
| 3.100.2.35 AllowDelay               | 315 |
| 3.100.2.36 lhdollarerror            | 315 |
| 3.100.2.37 eat                      | 315 |
| 3.100.2.38 gNumPre                  | 315 |
| 3.100.2.39 PreDebug                 | 315 |
| 3.100.2.40 OpenDictionary           | 316 |
| 3.100.2.41 PreAssignLevel           | 316 |
| 3.100.2.42 MaxPreAssignLevel        | 316 |
| 3.100.2.43 fullnamesize             | 316 |
| 3.100.2.44 DebugFlag                | 316 |
| 3.100.2.45 preError                 | 316 |
| 3.100.2.46 ComChar                  | 316 |
| 3.100.2.47 cComChar                 | 317 |
| 3.101 ParallelVars Struct Reference | 317 |
| 3.101.1 Detailed Description        | 318 |
| 3.101.2 Field Documentation         | 318 |
| 3.101.2.1 slavebuf                  | 318 |
| 3.101.2.2 sbuf                      | 318 |
| 3.101.2.3 rbufs                     | 318 |
| 3.101.2.4 me                        | 318 |
| 3.101.2.5 numtasks                  | 318 |
| 3.101.2.6 parallel                  | 318 |
| 3.101.2.7 rhsInParallel             | 319 |

|                                                          |     |
|----------------------------------------------------------|-----|
| 3.101.2.8 mkSlaveInfile . . . . .                        | 319 |
| 3.101.2.9 exprbufsize . . . . .                          | 319 |
| 3.101.2.10 exprtodo . . . . .                            | 319 |
| 3.101.2.11 log . . . . .                                 | 319 |
| 3.101.2.12 numsbuffs . . . . .                           | 319 |
| 3.101.2.13 numrbuffs . . . . .                           | 319 |
| 3.102 PaRtl Struct Reference . . . . .                   | 320 |
| 3.102.1 Detailed Description . . . . .                   | 320 |
| 3.102.2 Field Documentation . . . . .                    | 320 |
| 3.102.2.1 psize . . . . .                                | 320 |
| 3.102.2.2 args . . . . .                                 | 320 |
| 3.102.2.3 nargs . . . . .                                | 320 |
| 3.102.2.4 nfun . . . . .                                 | 320 |
| 3.102.2.5 numargs . . . . .                              | 321 |
| 3.102.2.6 numpart . . . . .                              | 321 |
| 3.102.2.7 where . . . . .                                | 321 |
| 3.103 PaRtlcLe Struct Reference . . . . .                | 321 |
| 3.103.1 Detailed Description . . . . .                   | 321 |
| 3.103.2 Field Documentation . . . . .                    | 321 |
| 3.103.2.1 number . . . . .                               | 321 |
| 3.103.2.2 spin . . . . .                                 | 322 |
| 3.103.2.3 mass . . . . .                                 | 322 |
| 3.103.2.4 type . . . . .                                 | 322 |
| 3.104 Particle Class Reference . . . . .                 | 322 |
| 3.104.1 Detailed Description . . . . .                   | 323 |
| 3.104.2 Constructor & Destructor Documentation . . . . . | 323 |
| 3.104.2.1 Particle() . . . . .                           | 323 |
| 3.104.2.2 ~Particle() . . . . .                          | 323 |
| 3.104.3 Member Function Documentation . . . . .          | 324 |
| 3.104.3.1 particleName() . . . . .                       | 324 |
| 3.104.3.2 particleCode() . . . . .                       | 324 |
| 3.104.3.3 aparticle() . . . . .                          | 324 |
| 3.104.3.4 prParticle() . . . . .                         | 324 |
| 3.104.3.5 isNeutral() . . . . .                          | 324 |
| 3.104.3.6 typeName() . . . . .                           | 324 |
| 3.104.3.7 typeGName() . . . . .                          | 325 |
| 3.104.4 Field Documentation . . . . .                    | 325 |
| 3.104.4.1 mdl . . . . .                                  | 325 |
| 3.104.4.2 name . . . . .                                 | 325 |
| 3.104.4.3 aname . . . . .                                | 325 |
| 3.104.4.4 id . . . . .                                   | 325 |
| 3.104.4.5 ptype . . . . .                                | 325 |

|                                  |     |
|----------------------------------|-----|
| 3.104.4.6 neutral                | 325 |
| 3.104.4.7 pcode                  | 326 |
| 3.104.4.8 acode                  | 326 |
| 3.104.4.9 cmindeg                | 326 |
| 3.104.4.10 cmaxdeg               | 326 |
| 3.104.4.11 extonly               | 326 |
| 3.105 PeRmUtE Struct Reference   | 326 |
| 3.105.1 Detailed Description     | 327 |
| 3.105.2 Field Documentation      | 327 |
| 3.105.2.1 objects                | 327 |
| 3.105.2.2 sign                   | 327 |
| 3.105.2.3 n                      | 327 |
| 3.105.2.4 cycle                  | 327 |
| 3.106 PeRmUtEp Struct Reference  | 327 |
| 3.106.1 Detailed Description     | 328 |
| 3.106.2 Field Documentation      | 328 |
| 3.106.2.1 objects                | 328 |
| 3.106.2.2 sign                   | 328 |
| 3.106.2.3 n                      | 328 |
| 3.106.2.4 cycle                  | 328 |
| 3.107 PF_BUFFER Struct Reference | 328 |
| 3.107.1 Detailed Description     | 329 |
| 3.107.2 Field Documentation      | 329 |
| 3.107.2.1 buff                   | 329 |
| 3.107.2.2 fill                   | 329 |
| 3.107.2.3 full                   | 329 |
| 3.107.2.4 stop                   | 329 |
| 3.107.2.5 status                 | 330 |
| 3.107.2.6 retstat                | 330 |
| 3.107.2.7 request                | 330 |
| 3.107.2.8 type                   | 330 |
| 3.107.2.9 index                  | 330 |
| 3.107.2.10 tag                   | 330 |
| 3.107.2.11 from                  | 330 |
| 3.107.2.12 numbufs               | 330 |
| 3.107.2.13 active                | 331 |
| 3.108 PGInput Struct Reference   | 331 |
| 3.108.1 Detailed Description     | 331 |
| 3.108.2 Field Documentation      | 331 |
| 3.108.2.1 cdeg                   | 331 |
| 3.108.2.2 ctyp                   | 331 |
| 3.108.2.3 cnum                   | 331 |

|                                                |     |
|------------------------------------------------|-----|
| 3.108.2.4 cple                                 | 332 |
| 3.108.2.5 pname                                | 332 |
| 3.108.2.6 pcode                                | 332 |
| 3.109 PInput Struct Reference                  | 332 |
| 3.109.1 Detailed Description                   | 332 |
| 3.109.2 Field Documentation                    | 332 |
| 3.109.2.1 name                                 | 332 |
| 3.109.2.2 aname                                | 333 |
| 3.109.2.3 pcode                                | 333 |
| 3.109.2.4 acode                                | 333 |
| 3.109.2.5 ptypen                               | 333 |
| 3.109.2.6 ptypec                               | 333 |
| 3.109.2.7 extonly                              | 333 |
| 3.110 PNodeClass Class Reference               | 334 |
| 3.110.1 Detailed Description                   | 334 |
| 3.110.2 Constructor & Destructor Documentation | 335 |
| 3.110.2.1 PNodeClass() [1/2]                   | 335 |
| 3.110.2.2 PNodeClass() [2/2]                   | 335 |
| 3.110.2.3 ~PNodeClass()                        | 335 |
| 3.110.3 Member Function Documentation          | 335 |
| 3.110.3.1 prPNodeClass()                       | 335 |
| 3.110.3.2 prElem()                             | 335 |
| 3.110.4 Field Documentation                    | 336 |
| 3.110.4.1 sproc                                | 336 |
| 3.110.4.2 deg                                  | 336 |
| 3.110.4.3 type                                 | 336 |
| 3.110.4.4 particle                             | 336 |
| 3.110.4.5 couple                               | 336 |
| 3.110.4.6 cmindeg                              | 336 |
| 3.110.4.7 cmaxdeg                              | 336 |
| 3.110.4.8 count                                | 337 |
| 3.110.4.9 cl2nd                                | 337 |
| 3.110.4.10 nd2cl                               | 337 |
| 3.110.4.11 cl2mcl                              | 337 |
| 3.110.4.12 nnodes                              | 337 |
| 3.110.4.13 nclass                              | 337 |
| 3.111 flint::poly Class Reference              | 337 |
| 3.111.1 Detailed Description                   | 338 |
| 3.111.2 Constructor & Destructor Documentation | 338 |
| 3.111.2.1 poly()                               | 338 |
| 3.111.2.2 ~poly()                              | 338 |
| 3.111.3 Member Function Documentation          | 338 |



|                                                |     |
|------------------------------------------------|-----|
| 3.111.3.1 print()                              | 338 |
| 3.111.4 Field Documentation                    | 338 |
| 3.111.4.1 d                                    | 338 |
| 3.112 poly Class Reference                     | 339 |
| 3.112.1 Detailed Description                   | 340 |
| 3.112.2 Constructor & Destructor Documentation | 340 |
| 3.112.2.1 poly() [1/3]                         | 340 |
| 3.112.2.2 poly() [2/3]                         | 341 |
| 3.112.2.3 poly() [3/3]                         | 341 |
| 3.112.2.4 ~poly()                              | 341 |
| 3.112.3 Member Function Documentation          | 341 |
| 3.112.3.1 operator+=()                         | 341 |
| 3.112.3.2 operator-=()                         | 341 |
| 3.112.3.3 operator*=()                         | 341 |
| 3.112.3.4 operator/=( )                        | 342 |
| 3.112.3.5 operator%=( )                        | 342 |
| 3.112.3.6 operator+( )                         | 342 |
| 3.112.3.7 operator-( )                         | 342 |
| 3.112.3.8 operator*( )                         | 342 |
| 3.112.3.9 operator/( )                         | 342 |
| 3.112.3.10 operator%( )                        | 342 |
| 3.112.3.11 operator==( )                       | 343 |
| 3.112.3.12 operator!=( )                       | 343 |
| 3.112.3.13 operator=( )                        | 343 |
| 3.112.3.14 operator[]() [1/2]                  | 343 |
| 3.112.3.15 operator[]() [2/2]                  | 343 |
| 3.112.3.16 termscopy()                         | 343 |
| 3.112.3.17 check_memory()                      | 343 |
| 3.112.3.18 expand_memory()                     | 344 |
| 3.112.3.19 is_zero()                           | 344 |
| 3.112.3.20 is_one()                            | 344 |
| 3.112.3.21 is_integer()                        | 344 |
| 3.112.3.22 is_monomial()                       | 344 |
| 3.112.3.23 is_dense_univariate()               | 344 |
| 3.112.4 Description                            | 344 |
| 3.112.5 Notes                                  | 345 |
| 3.112.5.1 sign()                               | 345 |
| 3.112.5.2 degree()                             | 345 |
| 3.112.5.3 total_degree()                       | 345 |
| 3.112.5.4 first_variable()                     | 345 |
| 3.112.5.5 number_of_terms()                    | 345 |
| 3.112.5.6 all_variables()                      | 345 |

|                                             |     |
|---------------------------------------------|-----|
| 3.112.5.7 integer_lcoeff()                  | 346 |
| 3.112.5.8 lcoeff_univar()                   | 346 |
| 3.112.5.9 lcoeff_multivar()                 | 346 |
| 3.112.5.10 coefficient()                    | 346 |
| 3.112.5.11 derivative()                     | 346 |
| 3.112.5.12 setmod()                         | 346 |
| 3.112.5.13 coefficients_modulo()            | 346 |
| 3.112.5.14 simple_poly() <sup>[1/2]</sup>   | 347 |
| 3.112.5.15 simple_poly() <sup>[2/2]</sup>   | 347 |
| 3.112.5.16 get_variables()                  | 347 |
| 3.112.5.17 argument_to_poly()               | 347 |
| 3.112.5.18 poly_to_argument()               | 347 |
| 3.112.5.19 poly_to_argument_with_den()      | 348 |
| 3.112.5.20 size_of_form_notation()          | 348 |
| 3.112.5.21 size_of_form_notation_with_den() | 348 |
| 3.112.5.22 normalize()                      | 348 |
| 3.112.5.23 from_coefficient_list()          | 348 |
| 3.112.5.24 to_coefficient_list()            | 348 |
| 3.112.5.25 coefficient_list_divmod()        | 349 |
| 3.112.5.26 to_string()                      | 349 |
| 3.112.5.27 monomial_compare()               | 349 |
| 3.112.5.28 last_monomial_index()            | 349 |
| 3.112.5.29 last_monomial()                  | 349 |
| 3.112.5.30 compare_degree_vector()          | 349 |
| 3.112.5.31 degree_vector()                  | 349 |
| 3.112.5.32 add()                            | 350 |
| 3.112.5.33 sub()                            | 350 |
| 3.112.5.34 mul()                            | 350 |
| 3.112.6 Description                         | 350 |
| 3.112.6.1 div()                             | 351 |
| 3.112.6.2 mod()                             | 351 |
| 3.112.6.3 divmod()                          | 351 |
| 3.112.7 Description                         | 351 |
| 3.112.7.1 divides()                         | 352 |
| 3.112.7.2 mul_one_term()                    | 352 |
| 3.112.7.3 mul_univar()                      | 352 |
| 3.112.7.4 mul_heap()                        | 352 |
| 3.112.8 Description                         | 352 |
| 3.112.8.1 divmod_one_term()                 | 353 |
| 3.112.8.2 divmod_univar()                   | 353 |
| 3.112.9 Description                         | 354 |
| 3.112.9.1 divmod_heap()                     | 354 |

|                                                |     |
|------------------------------------------------|-----|
| 3.112.10 Description                           | 354 |
| 3.112.10.1 push_heap()                         | 355 |
| 3.112.10.2 pop_heap()                          | 355 |
| 3.112.11 Field Documentation                   | 356 |
| 3.112.11.1 terms                               | 356 |
| 3.112.11.2 size_of_terms                       | 356 |
| 3.112.11.3 modp                                | 356 |
| 3.112.11.4 modn                                | 356 |
| 3.113 flint::poly_factor Class Reference       | 356 |
| 3.113.1 Detailed Description                   | 356 |
| 3.113.2 Constructor & Destructor Documentation | 357 |
| 3.113.2.1 poly_factor()                        | 357 |
| 3.113.2.2 ~poly_factor()                       | 357 |
| 3.113.3 Field Documentation                    | 357 |
| 3.113.3.1 d                                    | 357 |
| 3.114 POLYMOD Struct Reference                 | 357 |
| 3.114.1 Detailed Description                   | 357 |
| 3.114.2 Field Documentation                    | 358 |
| 3.114.2.1 coefs                                | 358 |
| 3.114.2.2 numsym                               | 358 |
| 3.114.2.3 arraysize                            | 358 |
| 3.114.2.4 polysize                             | 358 |
| 3.114.2.5 modnum                               | 358 |
| 3.115 PoSiTiOn Struct Reference                | 358 |
| 3.115.1 Detailed Description                   | 358 |
| 3.115.2 Field Documentation                    | 359 |
| 3.115.2.1 p1                                   | 359 |
| 3.116 PRELOAD Struct Reference                 | 359 |
| 3.116.1 Detailed Description                   | 359 |
| 3.116.2 Field Documentation                    | 359 |
| 3.116.2.1 buffer                               | 359 |
| 3.116.2.2 size                                 | 359 |
| 3.117 pReVaR Struct Reference                  | 360 |
| 3.117.1 Detailed Description                   | 360 |
| 3.117.2 Field Documentation                    | 360 |
| 3.117.2.1 name                                 | 360 |
| 3.117.2.2 value                                | 360 |
| 3.117.2.3 argnames                             | 360 |
| 3.117.2.4 nargs                                | 361 |
| 3.117.2.5 wildarg                              | 361 |
| 3.118 PROCEDURE Struct Reference               | 361 |
| 3.118.1 Detailed Description                   | 362 |

|                                                |     |
|------------------------------------------------|-----|
| 3.118.2 Field Documentation                    | 362 |
| 3.118.2.1 p                                    | 362 |
| 3.118.2.2 name                                 | 362 |
| 3.118.2.3 loadmode                             | 362 |
| 3.119 Process Class Reference                  | 362 |
| 3.119.1 Detailed Description                   | 363 |
| 3.119.2 Constructor & Destructor Documentation | 364 |
| 3.119.2.1 Process()                            | 364 |
| 3.119.2.2 ~Process()                           | 364 |
| 3.119.3 Member Function Documentation          | 364 |
| 3.119.3.1 prProcess()                          | 364 |
| 3.119.3.2 mkSProcess()                         | 364 |
| 3.119.4 Field Documentation                    | 364 |
| 3.119.4.1 model                                | 364 |
| 3.119.4.2 opt                                  | 365 |
| 3.119.4.3 mgrcount                             | 365 |
| 3.119.4.4 agrcount                             | 365 |
| 3.119.4.5 initlPart                            | 365 |
| 3.119.4.6 finalPart                            | 365 |
| 3.119.4.7 id                                   | 365 |
| 3.119.4.8 ninitl                               | 365 |
| 3.119.4.9 nfinal                               | 365 |
| 3.119.4.10 ctotat                              | 366 |
| 3.119.4.11 nExtern                             | 366 |
| 3.119.4.12 loop                                | 366 |
| 3.119.4.13 maxnlegs                            | 366 |
| 3.119.4.14 clist                               | 366 |
| 3.119.4.15 nSubproc                            | 366 |
| 3.119.4.16 sptbl                               | 366 |
| 3.119.4.17 sproc                               | 366 |
| 3.119.4.18 astack                              | 367 |
| 3.119.4.19 ngraphs                             | 367 |
| 3.119.4.20 nopi                                | 367 |
| 3.119.4.21 wgraphs                             | 367 |
| 3.119.4.22 wopi                                | 367 |
| 3.119.4.23 nMGraphs                            | 367 |
| 3.119.4.24 nMOPI                               | 367 |
| 3.119.4.25 wMGraphs                            | 367 |
| 3.119.4.26 wMOPI                               | 368 |
| 3.119.4.27 nAGraphs                            | 368 |
| 3.119.4.28 nAOPI                               | 368 |
| 3.119.4.29 wAGraphs                            | 368 |

|                                          |     |
|------------------------------------------|-----|
| 3.119.4.30 wAOPI . . . . .               | 368 |
| 3.119.4.31 sec . . . . .                 | 368 |
| 3.120 R_const Struct Reference . . . . . | 369 |
| 3.120.1 Detailed Description . . . . .   | 370 |
| 3.120.2 Field Documentation . . . . .    | 371 |
| 3.120.2.1 StoreData . . . . .            | 371 |
| 3.120.2.2 Fscr . . . . .                 | 371 |
| 3.120.2.3 FoStage4 . . . . .             | 371 |
| 3.120.2.4 DefPosition . . . . .          | 371 |
| 3.120.2.5 infile . . . . .               | 371 |
| 3.120.2.6 outfile . . . . .              | 371 |
| 3.120.2.7 hidefile . . . . .             | 371 |
| 3.120.2.8 CompressBuffer . . . . .       | 372 |
| 3.120.2.9 ComprTop . . . . .             | 372 |
| 3.120.2.10 CompressPointer . . . . .     | 372 |
| 3.120.2.11 CompareRoutine . . . . .      | 372 |
| 3.120.2.12 wranfia . . . . .             | 372 |
| 3.120.2.13 moebiustable . . . . .        | 372 |
| 3.120.2.14 OldTime . . . . .             | 372 |
| 3.120.2.15 InInBuf . . . . .             | 372 |
| 3.120.2.16 InHiBuf . . . . .             | 373 |
| 3.120.2.17 pWorkSize . . . . .           | 373 |
| 3.120.2.18 lWorkSize . . . . .           | 373 |
| 3.120.2.19 posWorkSize . . . . .         | 373 |
| 3.120.2.20 wranfseed . . . . .           | 373 |
| 3.120.2.21 NoCompress . . . . .          | 373 |
| 3.120.2.22 gzipCompress . . . . .        | 373 |
| 3.120.2.23 Cnumlhs . . . . .             | 373 |
| 3.120.2.24 outtohide . . . . .           | 374 |
| 3.120.2.25 wranfcall . . . . .           | 374 |
| 3.120.2.26 wranfnpair1 . . . . .         | 374 |
| 3.120.2.27 wranfnpair2 . . . . .         | 374 |
| 3.120.2.28 GetFile . . . . .             | 374 |
| 3.120.2.29 KeptInHold . . . . .          | 374 |
| 3.120.2.30 BracketOn . . . . .           | 374 |
| 3.120.2.31 MaxBracket . . . . .          | 374 |
| 3.120.2.32 CurDum . . . . .              | 375 |
| 3.120.2.33 DeferFlag . . . . .           | 375 |
| 3.120.2.34 TePos . . . . .               | 375 |
| 3.120.2.35 sLevel . . . . .              | 375 |
| 3.120.2.36 Stage4Name . . . . .          | 375 |
| 3.120.2.37 GetOneFile . . . . .          | 375 |

|                                 |     |
|---------------------------------|-----|
| 3.120.2.38 PolyFun              | 375 |
| 3.120.2.39 PolyFunInv           | 375 |
| 3.120.2.40 PolyFunType          | 376 |
| 3.120.2.41 PolyFunExp           | 376 |
| 3.120.2.42 PolyFunVar           | 376 |
| 3.120.2.43 PolyFunPow           | 376 |
| 3.120.2.44 Eside                | 376 |
| 3.120.2.45 MaxDum               | 376 |
| 3.120.2.46 level                | 376 |
| 3.120.2.47 expchanged           | 376 |
| 3.120.2.48 expflags             | 377 |
| 3.120.2.49 CurExpr              | 377 |
| 3.120.2.50 SortType             | 377 |
| 3.120.2.51 ShortSortCount       | 377 |
| 3.120.2.52 modeloptions         | 377 |
| 3.120.2.53 funoffset            | 377 |
| 3.120.2.54 moebiustablesizes    | 377 |
| 3.121 ReNuMbEr Struct Reference | 378 |
| 3.121.1 Detailed Description    | 378 |
| 3.121.2 Field Documentation     | 378 |
| 3.121.2.1 startposition         | 378 |
| 3.121.2.2 symb                  | 379 |
| 3.121.2.3 indi                  | 379 |
| 3.121.2.4 vect                  | 379 |
| 3.121.2.5 func                  | 379 |
| 3.121.2.6 symnum                | 379 |
| 3.121.2.7 indnum                | 380 |
| 3.121.2.8 vecnum                | 380 |
| 3.121.2.9 funnum                | 380 |
| 3.122 S_const Struct Reference  | 380 |
| 3.122.1 Detailed Description    | 381 |
| 3.122.2 Field Documentation     | 381 |
| 3.122.2.1 MaxExprSize           | 381 |
| 3.122.2.2 OldOnFile             | 381 |
| 3.122.2.3 OldNumFactors         | 381 |
| 3.122.2.4 Oldvflags             | 382 |
| 3.122.2.5 Olduflags             | 382 |
| 3.122.2.6 NumOldOnFile          | 382 |
| 3.122.2.7 NumOldNumFactors      | 382 |
| 3.122.2.8 MultiThreaded         | 382 |
| 3.122.2.9 Balancing             | 382 |
| 3.122.2.10 ExecMode             | 382 |

|                                                          |     |
|----------------------------------------------------------|-----|
| 3.122.2.11 CollectOverFlag . . . . .                     | 383 |
| 3.123 SeTs Struct Reference . . . . .                    | 383 |
| 3.123.1 Detailed Description . . . . .                   | 383 |
| 3.123.2 Field Documentation . . . . .                    | 383 |
| 3.123.2.1 name . . . . .                                 | 383 |
| 3.123.2.2 type . . . . .                                 | 383 |
| 3.123.2.3 first . . . . .                                | 383 |
| 3.123.2.4 last . . . . .                                 | 384 |
| 3.123.2.5 node . . . . .                                 | 384 |
| 3.123.2.6 namesize . . . . .                             | 384 |
| 3.123.2.7 dimension . . . . .                            | 384 |
| 3.123.2.8 flags . . . . .                                | 384 |
| 3.124 SETUPPARAMETERS Struct Reference . . . . .         | 384 |
| 3.124.1 Detailed Description . . . . .                   | 385 |
| 3.124.2 Field Documentation . . . . .                    | 385 |
| 3.124.2.1 parameter . . . . .                            | 385 |
| 3.124.2.2 type . . . . .                                 | 385 |
| 3.124.2.3 flags . . . . .                                | 385 |
| 3.124.2.4 value . . . . .                                | 385 |
| 3.125 SGroup Class Reference . . . . .                   | 385 |
| 3.125.1 Detailed Description . . . . .                   | 386 |
| 3.125.2 Constructor & Destructor Documentation . . . . . | 386 |
| 3.125.2.1 SGroup() . . . . .                             | 386 |
| 3.125.2.2 ~SGroup() . . . . .                            | 386 |
| 3.125.3 Member Function Documentation . . . . .          | 386 |
| 3.125.3.1 print() . . . . .                              | 386 |
| 3.125.3.2 newGroup() . . . . .                           | 387 |
| 3.125.3.3 clearGroup() . . . . .                         | 387 |
| 3.125.3.4 delGroup() . . . . .                           | 387 |
| 3.125.3.5 genNext() . . . . .                            | 387 |
| 3.125.3.6 addGroup() . . . . .                           | 387 |
| 3.125.3.7 nElem() . . . . .                              | 387 |
| 3.125.3.8 nextElem() . . . . .                           | 387 |
| 3.125.4 Field Documentation . . . . .                    | 388 |
| 3.125.4.1 size . . . . .                                 | 388 |
| 3.125.4.2 nelem . . . . .                                | 388 |
| 3.125.4.3 elem . . . . .                                 | 388 |
| 3.125.4.4 nnodes . . . . .                               | 388 |
| 3.125.4.5 neclass . . . . .                              | 388 |
| 3.125.4.6 eclass . . . . .                               | 388 |
| 3.125.4.7 cgen . . . . .                                 | 388 |
| 3.125.4.8 csav . . . . .                                 | 389 |

|                                    |     |
|------------------------------------|-----|
| 3.125.4.9 permg                    | 389 |
| 3.125.4.10 perms                   | 389 |
| 3.125.4.11 pgr                     | 389 |
| 3.125.4.12 pgq                     | 389 |
| 3.125.4.13 psr                     | 389 |
| 3.125.4.14 psq                     | 389 |
| 3.126 SHvariables Struct Reference | 390 |
| 3.126.1 Detailed Description       | 390 |
| 3.126.2 Field Documentation        | 390 |
| 3.126.2.1 outterm                  | 390 |
| 3.126.2.2 outfun                   | 390 |
| 3.126.2.3 incoef                   | 390 |
| 3.126.2.4 stop1                    | 390 |
| 3.126.2.5 stop2                    | 391 |
| 3.126.2.6 ststop1                  | 391 |
| 3.126.2.7 ststop2                  | 391 |
| 3.126.2.8 finishuf                 | 391 |
| 3.126.2.9 do_uffle                 | 391 |
| 3.126.2.10 combilast               | 391 |
| 3.126.2.11 nincoef                 | 391 |
| 3.126.2.12 level                   | 391 |
| 3.126.2.13 thefunction             | 392 |
| 3.126.2.14 option                  | 392 |
| 3.127 sOrT Struct Reference        | 392 |
| 3.127.1 Detailed Description       | 394 |
| 3.127.2 Field Documentation        | 394 |
| 3.127.2.1 file                     | 394 |
| 3.127.2.2 SizeInFile               | 394 |
| 3.127.2.3 lBuffer                  | 394 |
| 3.127.2.4 lTop                     | 394 |
| 3.127.2.5 lFill                    | 394 |
| 3.127.2.6 used                     | 394 |
| 3.127.2.7 sBuffer                  | 395 |
| 3.127.2.8 sTop                     | 395 |
| 3.127.2.9 sTop2                    | 395 |
| 3.127.2.10 sHalf                   | 395 |
| 3.127.2.11 sFill                   | 395 |
| 3.127.2.12 sPointer                | 395 |
| 3.127.2.13 PoinFill                | 395 |
| 3.127.2.14 cBuffer                 | 395 |
| 3.127.2.15 Patches                 | 396 |
| 3.127.2.16 pStop                   | 396 |



|            |                            |     |
|------------|----------------------------|-----|
| 3.127.2.17 | poina                      | 396 |
| 3.127.2.18 | poin2a                     | 396 |
| 3.127.2.19 | ktoi                       | 396 |
| 3.127.2.20 | tree                       | 396 |
| 3.127.2.21 | fPatches                   | 396 |
| 3.127.2.22 | inPatches                  | 396 |
| 3.127.2.23 | fPatchesStop               | 397 |
| 3.127.2.24 | iPatches                   | 397 |
| 3.127.2.25 | f                          | 397 |
| 3.127.2.26 | ff                         | 397 |
| 3.127.2.27 | sTerms                     | 397 |
| 3.127.2.28 | LargeSize                  | 397 |
| 3.127.2.29 | SmallSize                  | 397 |
| 3.127.2.30 | SmallEsize                 | 397 |
| 3.127.2.31 | TermsInSmall               | 398 |
| 3.127.2.32 | Terms2InSmall              | 398 |
| 3.127.2.33 | GenTerms                   | 398 |
| 3.127.2.34 | TermsLeft                  | 398 |
| 3.127.2.35 | GenSpace                   | 398 |
| 3.127.2.36 | SpaceLeft                  | 398 |
| 3.127.2.37 | putinsize                  | 398 |
| 3.127.2.38 | ninterms                   | 398 |
| 3.127.2.39 | MaxPatches                 | 399 |
| 3.127.2.40 | MaxFpatches                | 399 |
| 3.127.2.41 | type                       | 399 |
| 3.127.2.42 | IPatch                     | 399 |
| 3.127.2.43 | fPatchN1                   | 399 |
| 3.127.2.44 | PolyWise                   | 399 |
| 3.127.2.45 | PolyFlag                   | 399 |
| 3.127.2.46 | cBufferSize                | 399 |
| 3.127.2.47 | maxtermsize                | 400 |
| 3.127.2.48 | newmaxtermsize             | 400 |
| 3.127.2.49 | outputmode                 | 400 |
| 3.127.2.50 | stagelevel                 | 400 |
| 3.127.2.51 | fPatchN                    | 400 |
| 3.127.2.52 | inNum                      | 400 |
| 3.127.2.53 | stage4                     | 400 |
| 3.128      | SpecTatoR Struct Reference | 401 |
| 3.128.1    | Detailed Description       | 401 |
| 3.128.2    | Field Documentation        | 401 |
| 3.128.2.1  | position                   | 401 |
| 3.128.2.2  | readpos                    | 402 |

|                                                |     |
|------------------------------------------------|-----|
| 3.128.2.3 fh                                   | 402 |
| 3.128.2.4 name                                 | 402 |
| 3.128.2.5 exprnumber                           | 402 |
| 3.128.2.6 flags                                | 402 |
| 3.129 SProcess Class Reference                 | 402 |
| 3.129.1 Detailed Description                   | 404 |
| 3.129.2 Constructor & Destructor Documentation | 404 |
| 3.129.2.1 SProcess() [1/2]                     | 404 |
| 3.129.2.2 SProcess() [2/2]                     | 404 |
| 3.129.2.3 ~SProcess()                          | 404 |
| 3.129.3 Member Function Documentation          | 404 |
| 3.129.3.1 prSProcess()                         | 404 |
| 3.129.3.2 generate()                           | 405 |
| 3.129.3.3 assign()                             | 405 |
| 3.129.3.4 toMNodeClass()                       | 405 |
| 3.129.3.5 match()                              | 405 |
| 3.129.3.6 endMGraph()                          | 405 |
| 3.129.3.7 endAGraph()                          | 405 |
| 3.129.3.8 resultMGraph()                       | 406 |
| 3.129.3.9 resultAGraph()                       | 406 |
| 3.129.4 Field Documentation                    | 406 |
| 3.129.4.1 model                                | 406 |
| 3.129.4.2 proc                                 | 406 |
| 3.129.4.3 opt                                  | 406 |
| 3.129.4.4 pnclass                              | 406 |
| 3.129.4.5 astack                               | 407 |
| 3.129.4.6 mgraph                               | 407 |
| 3.129.4.7 egraph                               | 407 |
| 3.129.4.8 agraph                               | 407 |
| 3.129.4.9 cl2nd                                | 407 |
| 3.129.4.10 nd2cl                               | 407 |
| 3.129.4.11 nclass                              | 407 |
| 3.129.4.12 ninitl                              | 407 |
| 3.129.4.13 nfinal                              | 408 |
| 3.129.4.14 nvert                               | 408 |
| 3.129.4.15 clist                               | 408 |
| 3.129.4.16 id                                  | 408 |
| 3.129.4.17 loop                                | 408 |
| 3.129.4.18 nNodes                              | 408 |
| 3.129.4.19 nEdges                              | 408 |
| 3.129.4.20 nExtern                             | 408 |
| 3.129.4.21 ncouple                             | 409 |

|                                              |     |
|----------------------------------------------|-----|
| 3.129.4.22 tCouple . . . . .                 | 409 |
| 3.129.4.23 DUMMYPadding . . . . .            | 409 |
| 3.129.4.24 mgrcount . . . . .                | 409 |
| 3.129.4.25 agrcount . . . . .                | 409 |
| 3.129.4.26 extperm . . . . .                 | 409 |
| 3.129.4.27 nMGraphs . . . . .                | 409 |
| 3.129.4.28 nMOPI . . . . .                   | 409 |
| 3.129.4.29 wMGraphs . . . . .                | 410 |
| 3.129.4.30 wMOPI . . . . .                   | 410 |
| 3.129.4.31 nAGraphs . . . . .                | 410 |
| 3.129.4.32 nAOPI . . . . .                   | 410 |
| 3.129.4.33 wAGraphs . . . . .                | 410 |
| 3.129.4.34 wAOPI . . . . .                   | 410 |
| 3.130 StOrEcAcHe Struct Reference . . . . .  | 411 |
| 3.130.1 Detailed Description . . . . .       | 411 |
| 3.130.2 Field Documentation . . . . .        | 411 |
| 3.130.2.1 position . . . . .                 | 411 |
| 3.130.2.2 toppos . . . . .                   | 412 |
| 3.130.2.3 next . . . . .                     | 412 |
| 3.130.2.4 buffer . . . . .                   | 412 |
| 3.131 STOREHEADER Struct Reference . . . . . | 412 |
| 3.131.1 Detailed Description . . . . .       | 413 |
| 3.131.2 Field Documentation . . . . .        | 413 |
| 3.131.2.1 headermark . . . . .               | 413 |
| 3.131.2.2 lenWORD . . . . .                  | 413 |
| 3.131.2.3 lenLONG . . . . .                  | 413 |
| 3.131.2.4 lenPOS . . . . .                   | 413 |
| 3.131.2.5 lenPOINTER . . . . .               | 414 |
| 3.131.2.6 endianness . . . . .               | 414 |
| 3.131.2.7 sSym . . . . .                     | 414 |
| 3.131.2.8 sInd . . . . .                     | 414 |
| 3.131.2.9 sVec . . . . .                     | 414 |
| 3.131.2.10 sFun . . . . .                    | 415 |
| 3.131.2.11 maxpower . . . . .                | 415 |
| 3.131.2.12 wildoffset . . . . .              | 415 |
| 3.131.2.13 revision . . . . .                | 415 |
| 3.131.2.14 reserved . . . . .                | 415 |
| 3.132 Stream Struct Reference . . . . .      | 416 |
| 3.132.1 Detailed Description . . . . .       | 416 |
| 3.132.2 Field Documentation . . . . .        | 416 |
| 3.132.2.1 fileposition . . . . .             | 416 |
| 3.132.2.2 linenumber . . . . .               | 417 |

|                                          |     |
|------------------------------------------|-----|
| 3.132.2.3 prevline . . . . .             | 417 |
| 3.132.2.4 buffer . . . . .               | 417 |
| 3.132.2.5 pointer . . . . .              | 417 |
| 3.132.2.6 top . . . . .                  | 417 |
| 3.132.2.7 FoldName . . . . .             | 417 |
| 3.132.2.8 name . . . . .                 | 418 |
| 3.132.2.9 pname . . . . .                | 418 |
| 3.132.2.10 buffersize . . . . .          | 418 |
| 3.132.2.11 bufferposition . . . . .      | 418 |
| 3.132.2.12 inbuffer . . . . .            | 418 |
| 3.132.2.13 previous . . . . .            | 418 |
| 3.132.2.14 handle . . . . .              | 418 |
| 3.132.2.15 type . . . . .                | 419 |
| 3.132.2.16 prevars . . . . .             | 419 |
| 3.132.2.17 previousNoShowInput . . . . . | 419 |
| 3.132.2.18 eqnum . . . . .               | 419 |
| 3.132.2.19 afterwards . . . . .          | 419 |
| 3.132.2.20 olddelay . . . . .            | 419 |
| 3.132.2.21 oldnoshowinput . . . . .      | 419 |
| 3.132.2.22 isnextchar . . . . .          | 419 |
| 3.132.2.23 nextchar . . . . .            | 420 |
| 3.132.2.24 reserved . . . . .            | 420 |
| 3.133 SuBbUf Struct Reference . . . . .  | 420 |
| 3.133.1 Detailed Description . . . . .   | 420 |
| 3.133.2 Field Documentation . . . . .    | 420 |
| 3.133.2.1 subexpnum . . . . .            | 420 |
| 3.133.2.2 buffernum . . . . .            | 420 |
| 3.134 SWITCH Struct Reference . . . . .  | 421 |
| 3.134.1 Detailed Description . . . . .   | 421 |
| 3.134.2 Field Documentation . . . . .    | 421 |
| 3.134.2.1 table . . . . .                | 421 |
| 3.134.2.2 defaultcase . . . . .          | 422 |
| 3.134.2.3 endswitch . . . . .            | 422 |
| 3.134.2.4 typetable . . . . .            | 422 |
| 3.134.2.5 maxcase . . . . .              | 422 |
| 3.134.2.6 mincase . . . . .              | 422 |
| 3.134.2.7 numcases . . . . .             | 422 |
| 3.134.2.8 tablesize . . . . .            | 422 |
| 3.134.2.9 caseoffset . . . . .           | 422 |
| 3.134.2.10 iflevel . . . . .             | 423 |
| 3.134.2.11 whilelevel . . . . .          | 423 |
| 3.134.2.12 nestingsum . . . . .          | 423 |

|                                    |     |
|------------------------------------|-----|
| 3.134.2.13 padding                 | 423 |
| 3.135 SWITCHTABLE Struct Reference | 423 |
| 3.135.1 Detailed Description       | 423 |
| 3.135.2 Field Documentation        | 423 |
| 3.135.2.1 ncase                    | 423 |
| 3.135.2.2 value                    | 424 |
| 3.135.2.3 compbuffer               | 424 |
| 3.136 SyMbOI Struct Reference      | 424 |
| 3.136.1 Detailed Description       | 424 |
| 3.136.2 Field Documentation        | 424 |
| 3.136.2.1 name                     | 424 |
| 3.136.2.2 minpower                 | 424 |
| 3.136.2.3 maxpower                 | 425 |
| 3.136.2.4 complex                  | 425 |
| 3.136.2.5 number                   | 425 |
| 3.136.2.6 flags                    | 425 |
| 3.136.2.7 node                     | 425 |
| 3.136.2.8 namesize                 | 425 |
| 3.136.2.9 dimension                | 425 |
| 3.137 T_const Struct Reference     | 426 |
| 3.137.1 Detailed Description       | 428 |
| 3.137.2 Field Documentation        | 428 |
| 3.137.2.1 S0                       | 428 |
| 3.137.2.2 SS                       | 428 |
| 3.137.2.3 Nest                     | 428 |
| 3.137.2.4 NestStop                 | 428 |
| 3.137.2.5 NestPoin                 | 429 |
| 3.137.2.6 BrackBuf                 | 429 |
| 3.137.2.7 StoreCache               | 429 |
| 3.137.2.8 StoreCacheAlloc          | 429 |
| 3.137.2.9 pWorkSpace               | 429 |
| 3.137.2.10 IWorkSpace              | 429 |
| 3.137.2.11 posWorkSpace            | 429 |
| 3.137.2.12 WorkSpace               | 429 |
| 3.137.2.13 WorkTop                 | 430 |
| 3.137.2.14 WorkPointer             | 430 |
| 3.137.2.15 RepCount                | 430 |
| 3.137.2.16 RepTop                  | 430 |
| 3.137.2.17 WildArgTaken            | 430 |
| 3.137.2.18 factorials              | 430 |
| 3.137.2.19 small_power_n           | 430 |
| 3.137.2.20 small_power             | 430 |

|                                         |     |
|-----------------------------------------|-----|
| 3.137.2.21 bernoullis . . . . .         | 431 |
| 3.137.2.22 primelist . . . . .          | 431 |
| 3.137.2.23 pfac . . . . .               | 431 |
| 3.137.2.24 pBer . . . . .               | 431 |
| 3.137.2.25 TMaddr . . . . .             | 431 |
| 3.137.2.26 WildMask . . . . .           | 431 |
| 3.137.2.27 previousEfactor . . . . .    | 431 |
| 3.137.2.28 TermMemHeap . . . . .        | 431 |
| 3.137.2.29 NumberMemHeap . . . . .      | 432 |
| 3.137.2.30 CacheNumberMemHeap . . . . . | 432 |
| 3.137.2.31 bracketinfo . . . . .        | 432 |
| 3.137.2.32 ListPoly . . . . .           | 432 |
| 3.137.2.33 ListSymbols . . . . .        | 432 |
| 3.137.2.34 NumMem . . . . .             | 432 |
| 3.137.2.35 TopologiesTerm . . . . .     | 432 |
| 3.137.2.36 TopologiesStart . . . . .    | 432 |
| 3.137.2.37 partitions . . . . .         | 433 |
| 3.137.2.38 sBer . . . . .               | 433 |
| 3.137.2.39 pWorkPointer . . . . .       | 433 |
| 3.137.2.40 IWorkPointer . . . . .       | 433 |
| 3.137.2.41 posWorkPointer . . . . .     | 433 |
| 3.137.2.42 InNumMem . . . . .           | 433 |
| 3.137.2.43 sfact . . . . .              | 433 |
| 3.137.2.44 mfac . . . . .               | 433 |
| 3.137.2.45 ebufnum . . . . .            | 434 |
| 3.137.2.46 fbufnum . . . . .            | 434 |
| 3.137.2.47 allbufnum . . . . .          | 434 |
| 3.137.2.48 aebufnum . . . . .           | 434 |
| 3.137.2.49 idallflag . . . . .          | 434 |
| 3.137.2.50 idallnum . . . . .           | 434 |
| 3.137.2.51 idallmaxnum . . . . .        | 434 |
| 3.137.2.52 WildcardBufferSize . . . . . | 434 |
| 3.137.2.53 TermMemMax . . . . .         | 435 |
| 3.137.2.54 TermMemTop . . . . .         | 435 |
| 3.137.2.55 NumberMemMax . . . . .       | 435 |
| 3.137.2.56 NumberMemTop . . . . .       | 435 |
| 3.137.2.57 CacheNumberMemMax . . . . .  | 435 |
| 3.137.2.58 CacheNumberMemTop . . . . .  | 435 |
| 3.137.2.59 bracketindexflag . . . . .   | 435 |
| 3.137.2.60 optimitimes . . . . .        | 435 |
| 3.137.2.61 ListSymbolsSize . . . . .    | 436 |
| 3.137.2.62 NumListSymbols . . . . .     | 436 |

|                                                |     |
|------------------------------------------------|-----|
| 3.137.2.63 numpoly                             | 436 |
| 3.137.2.64 LeaveNegative                       | 436 |
| 3.137.2.65 TrimPower                           | 436 |
| 3.137.2.66 small_power_maxx                    | 436 |
| 3.137.2.67 small_power_maxn                    | 436 |
| 3.137.2.68 dummysubexp                         | 436 |
| 3.137.2.69 comsym                              | 437 |
| 3.137.2.70 comnum                              | 437 |
| 3.137.2.71 comfun                              | 437 |
| 3.137.2.72 comind                              | 437 |
| 3.137.2.73 MinVecArg                           | 437 |
| 3.137.2.74 FunArg                              | 437 |
| 3.137.2.75 locwildvalue                        | 437 |
| 3.137.2.76 mulpat                              | 437 |
| 3.137.2.77 proexp                              | 438 |
| 3.137.2.78 TMout                               | 438 |
| 3.137.2.79 TMbuff                              | 438 |
| 3.137.2.80 TMdolfac                            | 438 |
| 3.137.2.81 nfac                                | 438 |
| 3.137.2.82 nBer                                | 438 |
| 3.137.2.83 mBer                                | 438 |
| 3.137.2.84 PolyAct                             | 438 |
| 3.137.2.85 RecFlag                             | 439 |
| 3.137.2.86 inprimelist                         | 439 |
| 3.137.2.87 sizeprimelist                       | 439 |
| 3.137.2.88 fromindex                           | 439 |
| 3.137.2.89 setinterntopo                       | 439 |
| 3.137.2.90 setexterntopo                       | 439 |
| 3.137.2.91 TopologiesLevel                     | 439 |
| 3.137.2.92 TopologiesOptions                   | 440 |
| 3.138 T_EEdge Class Reference                  | 440 |
| 3.138.1 Detailed Description                   | 440 |
| 3.138.2 Field Documentation                    | 440 |
| 3.138.2.1 nodes                                | 440 |
| 3.138.2.2 ext                                  | 440 |
| 3.138.2.3 momn                                 | 440 |
| 3.138.2.4 momc                                 | 441 |
| 3.138.2.5 padding                              | 441 |
| 3.139 T_EGraph Class Reference                 | 441 |
| 3.139.1 Detailed Description                   | 442 |
| 3.139.2 Constructor & Destructor Documentation | 442 |
| 3.139.2.1 T_EGraph()                           | 442 |

|                                                |     |
|------------------------------------------------|-----|
| 3.139.2.2 <code>~T_EGraph()</code>             | 442 |
| 3.139.3 Member Function Documentation          | 442 |
| 3.139.3.1 <code>print()</code>                 | 442 |
| 3.139.3.2 <code>init()</code>                  | 443 |
| 3.139.3.3 <code>setExtern()</code>             | 443 |
| 3.139.3.4 <code>endSetExtern()</code>          | 443 |
| 3.139.4 Field Documentation                    | 443 |
| 3.139.4.1 <code>gld</code>                     | 443 |
| 3.139.4.2 <code>pId</code>                     | 443 |
| 3.139.4.3 <code>nNodes</code>                  | 443 |
| 3.139.4.4 <code>nEdges</code>                  | 444 |
| 3.139.4.5 <code>maxdeg</code>                  | 444 |
| 3.139.4.6 <code>nExtern</code>                 | 444 |
| 3.139.4.7 <code>opi</code>                     | 444 |
| 3.139.4.8 <code>nsym</code>                    | 444 |
| 3.139.4.9 <code>esym</code>                    | 444 |
| 3.139.4.10 <code>nodes</code>                  | 444 |
| 3.139.4.11 <code>edges</code>                  | 445 |
| 3.140 <code>T_ENode</code> Class Reference     | 445 |
| 3.140.1 Detailed Description                   | 445 |
| 3.140.2 Field Documentation                    | 445 |
| 3.140.2.1 <code>deg</code>                     | 445 |
| 3.140.2.2 <code>ext</code>                     | 445 |
| 3.140.2.3 <code>edges</code>                   | 445 |
| 3.141 <code>T_MGraph</code> Class Reference    | 446 |
| 3.141.1 Detailed Description                   | 447 |
| 3.141.2 Constructor & Destructor Documentation | 447 |
| 3.141.2.1 <code>T_MGraph()</code>              | 447 |
| 3.141.2.2 <code>~T_MGraph()</code>             | 447 |
| 3.141.3 Member Function Documentation          | 447 |
| 3.141.3.1 <code>generate()</code>              | 447 |
| 3.141.4 Field Documentation                    | 447 |
| 3.141.4.1 <code>pId</code>                     | 447 |
| 3.141.4.2 <code>nNodes</code>                  | 447 |
| 3.141.4.3 <code>nEdges</code>                  | 448 |
| 3.141.4.4 <code>nLoops</code>                  | 448 |
| 3.141.4.5 <code>nodes</code>                   | 448 |
| 3.141.4.6 <code>clist</code>                   | 448 |
| 3.141.4.7 <code>nClasses</code>                | 448 |
| 3.141.4.8 <code>mindeg</code>                  | 448 |
| 3.141.4.9 <code>maxdeg</code>                  | 448 |
| 3.141.4.10 <code>selOPI</code>                 | 448 |



|                                                |     |
|------------------------------------------------|-----|
| 3.141.4.11 ndiag                               | 449 |
| 3.141.4.12 n1PI                                | 449 |
| 3.141.4.13 nBridges                            | 449 |
| 3.141.4.14 c1PI                                | 449 |
| 3.141.4.15 adjMat                              | 449 |
| 3.141.4.16 nsym                                | 449 |
| 3.141.4.17 esym                                | 449 |
| 3.141.4.18 wsum                                | 449 |
| 3.141.4.19 wsopi                               | 450 |
| 3.141.4.20 curcl                               | 450 |
| 3.141.4.21 egraph                              | 450 |
| 3.141.4.22 ngen                                | 450 |
| 3.141.4.23 ngconn                              | 450 |
| 3.141.4.24 nCallRefine                         | 450 |
| 3.141.4.25 discardOrd                          | 450 |
| 3.141.4.26 discardRefine                       | 450 |
| 3.141.4.27 discardDisc                         | 451 |
| 3.141.4.28 discardIso                          | 451 |
| 3.142 T_MNode Class Reference                  | 451 |
| 3.142.1 Detailed Description                   | 451 |
| 3.142.2 Constructor & Destructor Documentation | 451 |
| 3.142.2.1 T_MNode()                            | 451 |
| 3.142.3 Field Documentation                    | 452 |
| 3.142.3.1 id                                   | 452 |
| 3.142.3.2 deg                                  | 452 |
| 3.142.3.3 cls                                  | 452 |
| 3.142.3.4 ext                                  | 452 |
| 3.142.3.5 freelg                               | 452 |
| 3.142.3.6 visited                              | 452 |
| 3.143 T_MNodeClass Class Reference             | 453 |
| 3.143.1 Detailed Description                   | 453 |
| 3.143.2 Constructor & Destructor Documentation | 453 |
| 3.143.2.1 T_MNodeClass()                       | 453 |
| 3.143.2.2 ~T_MNodeClass()                      | 453 |
| 3.143.3 Member Function Documentation          | 454 |
| 3.143.3.1 init()                               | 454 |
| 3.143.3.2 copy()                               | 454 |
| 3.143.3.3 clCmp()                              | 454 |
| 3.143.3.4 printMat()                           | 454 |
| 3.143.3.5 mkFlist()                            | 454 |
| 3.143.3.6 mkNdCI()                             | 454 |
| 3.143.3.7 mkCIMat()                            | 455 |

|                                          |     |
|------------------------------------------|-----|
| 3.143.3.8 incMat()                       | 455 |
| 3.143.3.9 chkOrd()                       | 455 |
| 3.143.3.10 cmpArray()                    | 455 |
| 3.143.4 Field Documentation              | 455 |
| 3.143.4.1 nNodes                         | 455 |
| 3.143.4.2 nClasses                       | 455 |
| 3.143.4.3 clist                          | 456 |
| 3.143.4.4 ndcl                           | 456 |
| 3.143.4.5 clmat                          | 456 |
| 3.143.4.6 flist                          | 456 |
| 3.143.4.7 maxdeg                         | 456 |
| 3.143.4.8 forallignment                  | 456 |
| 3.144 TaBIEbAsE Struct Reference         | 457 |
| 3.144.1 Detailed Description             | 457 |
| 3.144.2 Field Documentation              | 457 |
| 3.144.2.1 fillpoint                      | 457 |
| 3.144.2.2 current                        | 458 |
| 3.144.2.3 name                           | 458 |
| 3.144.2.4 tablenumbers                   | 458 |
| 3.144.2.5 subindex                       | 458 |
| 3.144.2.6 numtables                      | 458 |
| 3.145 TaBIEbAsEsUbInDeX Struct Reference | 458 |
| 3.145.1 Detailed Description             | 459 |
| 3.145.2 Field Documentation              | 459 |
| 3.145.2.1 where                          | 459 |
| 3.145.2.2 size                           | 459 |
| 3.146 TaBIEs Struct Reference            | 459 |
| 3.146.1 Detailed Description             | 460 |
| 3.146.2 Field Documentation              | 460 |
| 3.146.2.1 tablepointers                  | 460 |
| 3.146.2.2 prototype                      | 461 |
| 3.146.2.3 pattern                        | 461 |
| 3.146.2.4 mm                             | 461 |
| 3.146.2.5 flags                          | 461 |
| 3.146.2.6 boomlijst                      | 461 |
| 3.146.2.7 argtail                        | 462 |
| 3.146.2.8 spare                          | 462 |
| 3.146.2.9 buffers                        | 462 |
| 3.146.2.10 totind                        | 462 |
| 3.146.2.11 reserved                      | 462 |
| 3.146.2.12 defined                       | 463 |
| 3.146.2.13 mdefined                      | 463 |

---

|                                           |     |
|-------------------------------------------|-----|
| 3.146.2.14 prototypeSize . . . . .        | 463 |
| 3.146.2.15 numind . . . . .               | 463 |
| 3.146.2.16 bounds . . . . .               | 463 |
| 3.146.2.17 strict . . . . .               | 463 |
| 3.146.2.18 sparse . . . . .               | 464 |
| 3.146.2.19 numtree . . . . .              | 464 |
| 3.146.2.20 rootnum . . . . .              | 464 |
| 3.146.2.21 MaxTreeSize . . . . .          | 464 |
| 3.146.2.22 bufnum . . . . .               | 464 |
| 3.146.2.23 buffersize . . . . .           | 465 |
| 3.146.2.24 buffersfill . . . . .          | 465 |
| 3.146.2.25 tablenum . . . . .             | 465 |
| 3.146.2.26 mode . . . . .                 | 465 |
| 3.146.2.27 numdummies . . . . .           | 465 |
| 3.147 TERMINFO Struct Reference . . . . . | 466 |
| 3.147.1 Detailed Description . . . . .    | 466 |
| 3.147.2 Field Documentation . . . . .     | 466 |
| 3.147.2.1 term . . . . .                  | 466 |
| 3.147.2.2 currentModel . . . . .          | 466 |
| 3.147.2.3 currentMODEL . . . . .          | 466 |
| 3.147.2.4 legcouple . . . . .             | 466 |
| 3.147.2.5 numtopo . . . . .               | 467 |
| 3.147.2.6 numdia . . . . .                | 467 |
| 3.147.2.7 diaoffset . . . . .             | 467 |
| 3.147.2.8 level . . . . .                 | 467 |
| 3.147.2.9 externalset . . . . .           | 467 |
| 3.147.2.10 internalset . . . . .          | 467 |
| 3.147.2.11 numextern . . . . .            | 467 |
| 3.147.2.12 flags . . . . .                | 468 |
| 3.148 ToPoTyPe Struct Reference . . . . . | 468 |
| 3.148.1 Detailed Description . . . . .    | 468 |
| 3.148.2 Field Documentation . . . . .     | 469 |
| 3.148.2.1 vert . . . . .                  | 469 |
| 3.148.2.2 vertmax . . . . .               | 469 |
| 3.148.2.3 opt . . . . .                   | 469 |
| 3.148.2.4 cldeg . . . . .                 | 469 |
| 3.148.2.5 cnum . . . . .                  | 469 |
| 3.148.2.6 clext . . . . .                 | 469 |
| 3.148.2.7 cmind . . . . .                 | 469 |
| 3.148.2.8 cmaxd . . . . .                 | 470 |
| 3.148.2.9 ncl . . . . .                   | 470 |
| 3.148.2.10 nvert [1/2] . . . . .          | 470 |

|                               |     |
|-------------------------------|-----|
| 3.148.2.11 nloops             | 470 |
| 3.148.2.12 nlegs              | 470 |
| 3.148.2.13 npadding           | 470 |
| 3.148.2.14 nvert [2/2]        | 470 |
| 3.148.2.15 sopi               | 471 |
| 3.149 TrAcEn Struct Reference | 471 |
| 3.149.1 Detailed Description  | 471 |
| 3.149.2 Field Documentation   | 471 |
| 3.149.2.1 accu                | 471 |
| 3.149.2.2 accup               | 471 |
| 3.149.2.3 temp                | 472 |
| 3.149.2.4 perm                | 472 |
| 3.149.2.5 inlist              | 472 |
| 3.149.2.6 sgn                 | 472 |
| 3.149.2.7 num                 | 472 |
| 3.149.2.8 level               | 472 |
| 3.149.2.9 factor              | 472 |
| 3.149.2.10 allsign            | 473 |
| 3.150 TrAcEs Struct Reference | 473 |
| 3.150.1 Detailed Description  | 474 |
| 3.150.2 Field Documentation   | 474 |
| 3.150.2.1 accu                | 474 |
| 3.150.2.2 accup               | 474 |
| 3.150.2.3 temp                | 474 |
| 3.150.2.4 perm                | 474 |
| 3.150.2.5 inlist              | 474 |
| 3.150.2.6 nt3                 | 474 |
| 3.150.2.7 nt4                 | 475 |
| 3.150.2.8 j3                  | 475 |
| 3.150.2.9 j4                  | 475 |
| 3.150.2.10 e3                 | 475 |
| 3.150.2.11 e4                 | 475 |
| 3.150.2.12 eers               | 475 |
| 3.150.2.13 mepf               | 475 |
| 3.150.2.14 mdel               | 475 |
| 3.150.2.15 pepf               | 476 |
| 3.150.2.16 pdel               | 476 |
| 3.150.2.17 sgn                | 476 |
| 3.150.2.18 stap               | 476 |
| 3.150.2.19 step1              | 476 |
| 3.150.2.20 kstep              | 476 |
| 3.150.2.21 mdum               | 476 |

|                                                |     |
|------------------------------------------------|-----|
| 3.150.2.22 gamm                                | 476 |
| 3.150.2.23 ad                                  | 477 |
| 3.150.2.24 a3                                  | 477 |
| 3.150.2.25 a4                                  | 477 |
| 3.150.2.26 lc3                                 | 477 |
| 3.150.2.27 lc4                                 | 477 |
| 3.150.2.28 sign1                               | 477 |
| 3.150.2.29 sign2                               | 477 |
| 3.150.2.30 gamma5                              | 477 |
| 3.150.2.31 num                                 | 478 |
| 3.150.2.32 level                               | 478 |
| 3.150.2.33 factor                              | 478 |
| 3.150.2.34 allsign                             | 478 |
| 3.150.2.35 finalstep                           | 478 |
| 3.151 tree Struct Reference                    | 478 |
| 3.151.1 Detailed Description                   | 479 |
| 3.151.2 Field Documentation                    | 479 |
| 3.151.2.1 parent                               | 479 |
| 3.151.2.2 left                                 | 479 |
| 3.151.2.3 right                                | 479 |
| 3.151.2.4 value                                | 479 |
| 3.151.2.5 blnce                                | 480 |
| 3.151.2.6 usage                                | 480 |
| 3.152 tree_node Class Reference                | 480 |
| 3.152.1 Detailed Description                   | 481 |
| 3.152.2 Constructor & Destructor Documentation | 481 |
| 3.152.2.1 tree_node() [1/2]                    | 481 |
| 3.152.2.2 tree_node() [2/2]                    | 481 |
| 3.152.3 Member Function Documentation          | 481 |
| 3.152.3.1 operator=()                          | 481 |
| 3.152.4 Field Documentation                    | 481 |
| 3.152.4.1 childs                               | 481 |
| 3.152.4.2 sum_results                          | 482 |
| 3.152.4.3 num_visits                           | 482 |
| 3.152.4.4 var                                  | 482 |
| 3.152.4.5 finished                             | 482 |
| 3.153 VaRrEnUm Struct Reference                | 482 |
| 3.153.1 Detailed Description                   | 482 |
| 3.153.2 Field Documentation                    | 483 |
| 3.153.2.1 start                                | 483 |
| 3.153.2.2 lo                                   | 483 |
| 3.153.2.3 hi                                   | 483 |

|                                                |            |
|------------------------------------------------|------------|
| 3.154 VeCtOr Struct Reference . . . . .        | 483        |
| 3.154.1 Detailed Description . . . . .         | 484        |
| 3.154.2 Field Documentation . . . . .          | 484        |
| 3.154.2.1 name . . . . .                       | 484        |
| 3.154.2.2 complex . . . . .                    | 484        |
| 3.154.2.3 number . . . . .                     | 484        |
| 3.154.2.4 flags . . . . .                      | 484        |
| 3.154.2.5 node . . . . .                       | 484        |
| 3.154.2.6 namesize . . . . .                   | 484        |
| 3.154.2.7 dimension . . . . .                  | 485        |
| 3.155 VeRtEx Struct Reference . . . . .        | 485        |
| 3.155.1 Detailed Description . . . . .         | 485        |
| 3.155.2 Field Documentation . . . . .          | 485        |
| 3.155.2.1 particles . . . . .                  | 485        |
| 3.155.2.2 couplings . . . . .                  | 486        |
| 3.155.2.3 nparticles . . . . .                 | 486        |
| 3.155.2.4 ncouplings . . . . .                 | 486        |
| 3.155.2.5 type . . . . .                       | 486        |
| 3.155.2.6 error . . . . .                      | 486        |
| 3.155.2.7 externonly . . . . .                 | 486        |
| 3.155.2.8 spare . . . . .                      | 486        |
| 3.156 X_const Struct Reference . . . . .       | 487        |
| 3.156.1 Detailed Description . . . . .         | 487        |
| 3.156.2 Field Documentation . . . . .          | 487        |
| 3.156.2.1 currentPrompt . . . . .              | 487        |
| 3.156.2.2 shellname . . . . .                  | 487        |
| 3.156.2.3 stderrname . . . . .                 | 487        |
| 3.156.2.4 timeout . . . . .                    | 487        |
| 3.156.2.5 killSignal . . . . .                 | 488        |
| 3.156.2.6 killWholeGroup . . . . .             | 488        |
| 3.156.2.7 daemonize . . . . .                  | 488        |
| 3.156.2.8 currentExternalChannel . . . . .     | 488        |
| <b>4 File Documentation . . . . .</b>          | <b>489</b> |
| 4.1 argument.c File Reference . . . . .        | 489        |
| 4.1.1 Detailed Description . . . . .           | 490        |
| 4.1.2 Macro Definition Documentation . . . . . | 490        |
| 4.1.2.1 NEWORDER . . . . .                     | 490        |
| 4.1.3 Function Documentation . . . . .         | 490        |
| 4.1.3.1 execarg() . . . . .                    | 490        |
| 4.1.3.2 execterm() . . . . .                   | 490        |
| 4.1.3.3 ArgumentImplode() . . . . .            | 490        |

|                                      |     |
|--------------------------------------|-----|
| 4.1.3.4 ArgumentExplode()            | 491 |
| 4.1.3.5 ArgFactorize()               | 491 |
| 4.1.3.6 FindArg()                    | 491 |
| 4.1.3.7 InsertArg()                  | 491 |
| 4.1.3.8 CleanupArgCache()            | 492 |
| 4.1.3.9 ArgSymbolMerge()             | 492 |
| 4.1.3.10 ArgDotproductMerge()        | 492 |
| 4.1.3.11 TakeArgContent()            | 493 |
| 4.1.3.12 MakeInteger()               | 493 |
| 4.1.3.13 MakeMod()                   | 494 |
| 4.1.3.14 SortWeights()               | 494 |
| 4.2 argument.c                       | 495 |
| 4.3 bugtool.c File Reference         | 535 |
| 4.3.1 Detailed Description           | 535 |
| 4.3.2 Function Documentation         | 536 |
| 4.3.2.1 ExprStatus()                 | 536 |
| 4.4 bugtool.c                        | 536 |
| 4.5 checkpoint.c File Reference      | 537 |
| 4.5.1 Detailed Description           | 538 |
| 4.5.2 Macro Definition Documentation | 538 |
| 4.5.2.1 CACHED_SNAPSHOT              | 538 |
| 4.5.2.2 CACHE_SIZE                   | 538 |
| 4.5.2.3 R_FREE                       | 539 |
| 4.5.2.4 R_FREE_NAMETREE              | 539 |
| 4.5.2.5 R_FREE_STREAM                | 539 |
| 4.5.2.6 R_SET                        | 539 |
| 4.5.2.7 R_COPY_B                     | 539 |
| 4.5.2.8 S_WRITE_B                    | 540 |
| 4.5.2.9 S_FLUSH_B                    | 540 |
| 4.5.2.10 R_COPY_S                    | 540 |
| 4.5.2.11 S_WRITE_S                   | 540 |
| 4.5.2.12 R_COPY_LIST                 | 540 |
| 4.5.2.13 S_WRITE_LIST                | 541 |
| 4.5.2.14 R_COPY_NAMETREE             | 541 |
| 4.5.2.15 S_WRITE_NAMETREE            | 541 |
| 4.5.2.16 S_WRITE_DOLLAR              | 541 |
| 4.5.2.17 ANNOUNCE                    | 542 |
| 4.5.3 Function Documentation         | 542 |
| 4.5.3.1 CheckRecoveryFile()          | 542 |
| 4.5.3.2 DeleteRecoveryFile()         | 542 |
| 4.5.3.3 RecoveryFilename()           | 542 |
| 4.5.3.4 InitRecovery()               | 543 |

|                               |     |
|-------------------------------|-----|
| 4.5.3.5 fwrite_cached()       | 543 |
| 4.5.3.6 flush_cache()         | 543 |
| 4.5.3.7 DoRecovery()          | 543 |
| 4.5.3.8 DoCheckpoint()        | 544 |
| 4.5.4 Variable Documentation  | 545 |
| 4.5.4.1 cache_buffer          | 545 |
| 4.5.4.2 cache_fill            | 545 |
| 4.6 checkpoint.c              | 545 |
| 4.7 comexpr.c File Reference  | 581 |
| 4.7.1 Detailed Description    | 582 |
| 4.7.2 Function Documentation  | 582 |
| 4.7.2.1 CoLocal()             | 582 |
| 4.7.2.2 CoGlobal()            | 582 |
| 4.7.2.3 CoLocalFactorized()   | 582 |
| 4.7.2.4 CoGlobalFactorized()  | 582 |
| 4.7.2.5 DoExpr()              | 582 |
| 4.7.2.6 ColdOld()             | 583 |
| 4.7.2.7 Cold()                | 583 |
| 4.7.2.8 ColdNew()             | 583 |
| 4.7.2.9 CoDisorder()          | 583 |
| 4.7.2.10 CoMany()             | 583 |
| 4.7.2.11 CoMulti()            | 583 |
| 4.7.2.12 ColfMatch()          | 583 |
| 4.7.2.13 ColfNoMatch()        | 584 |
| 4.7.2.14 CoOnce()             | 584 |
| 4.7.2.15 CoOnly()             | 584 |
| 4.7.2.16 CoSelect()           | 584 |
| 4.7.2.17 ColdExpression()     | 584 |
| 4.7.2.18 CoMultiply()         | 584 |
| 4.7.2.19 CoFill()             | 584 |
| 4.7.2.20 CoFillExpression()   | 585 |
| 4.7.2.21 CoPrintTable()       | 585 |
| 4.7.2.22 CoAssign()           | 585 |
| 4.7.2.23 CoDeallocateTable()  | 585 |
| 4.8 comexpr.c                 | 585 |
| 4.9 compcomm.c File Reference | 609 |
| 4.9.1 Detailed Description    | 612 |
| 4.9.2 Function Documentation  | 612 |
| 4.9.2.1 CoFormat()            | 612 |
| 4.9.2.2 CoCollect()           | 612 |
| 4.9.2.3 setonoff()            | 612 |
| 4.9.2.4 CoCompress()          | 613 |



|                           |     |
|---------------------------|-----|
| 4.9.2.5 CoFlags()         | 613 |
| 4.9.2.6 CoOff()           | 613 |
| 4.9.2.7 CoOn()            | 613 |
| 4.9.2.8 CoInsideFirst()   | 613 |
| 4.9.2.9 CoProperCount()   | 613 |
| 4.9.2.10 CoDelete()       | 613 |
| 4.9.2.11 CoKeep()         | 614 |
| 4.9.2.12 CoFixIndex()     | 614 |
| 4.9.2.13 CoMetric()       | 614 |
| 4.9.2.14 DoPrint()        | 614 |
| 4.9.2.15 CoPrint()        | 614 |
| 4.9.2.16 CoPrintB()       | 614 |
| 4.9.2.17 CoNPrint()       | 614 |
| 4.9.2.18 CoPushHide()     | 615 |
| 4.9.2.19 CoPopHide()      | 615 |
| 4.9.2.20 SetExprCases()   | 615 |
| 4.9.2.21 SetExpr()        | 615 |
| 4.9.2.22 CoDrop()         | 615 |
| 4.9.2.23 CoNoDrop()       | 615 |
| 4.9.2.24 CoSkip()         | 616 |
| 4.9.2.25 CoNoSkip()       | 616 |
| 4.9.2.26 CoHide()         | 616 |
| 4.9.2.27 CoIntoHide()     | 616 |
| 4.9.2.28 CoNoIntoHide()   | 616 |
| 4.9.2.29 CoNoHide()       | 616 |
| 4.9.2.30 CoUnHide()       | 616 |
| 4.9.2.31 CoNoUnHide()     | 617 |
| 4.9.2.32 AddToCom()       | 617 |
| 4.9.2.33 AddComString()   | 617 |
| 4.9.2.34 Add2ComStrings() | 617 |
| 4.9.2.35 CoDiscard()      | 617 |
| 4.9.2.36 CoContract()     | 617 |
| 4.9.2.37 CoGoTo()         | 618 |
| 4.9.2.38 CoLabel()        | 618 |
| 4.9.2.39 DoArgument()     | 618 |
| 4.9.2.40 CoArgument()     | 618 |
| 4.9.2.41 CoEndArgument()  | 618 |
| 4.9.2.42 CoInside()       | 618 |
| 4.9.2.43 CoEndInside()    | 618 |
| 4.9.2.44 CoNormalize()    | 619 |
| 4.9.2.45 CoMakeInteger()  | 619 |
| 4.9.2.46 CoSplitArg()     | 619 |

|                               |     |
|-------------------------------|-----|
| 4.9.2.47 CoSplitFirstArg()    | 619 |
| 4.9.2.48 CoSplitLastArg()     | 619 |
| 4.9.2.49 CoFactArg()          | 619 |
| 4.9.2.50 DoSymmetrize()       | 619 |
| 4.9.2.51 CoSymmetrize()       | 620 |
| 4.9.2.52 CoAntiSymmetrize()   | 620 |
| 4.9.2.53 CoCycleSymmetrize()  | 620 |
| 4.9.2.54 CoRCycleSymmetrize() | 620 |
| 4.9.2.55 CoWrite()            | 620 |
| 4.9.2.56 CoNWrite()           | 620 |
| 4.9.2.57 CoRatio()            | 620 |
| 4.9.2.58 CoRedefine()         | 621 |
| 4.9.2.59 CoRenumber()         | 621 |
| 4.9.2.60 CoSum()              | 621 |
| 4.9.2.61 CoToTensor()         | 621 |
| 4.9.2.62 CoToVector()         | 621 |
| 4.9.2.63 CoTrace4()           | 621 |
| 4.9.2.64 CoTraceN()           | 621 |
| 4.9.2.65 CoChisholm()         | 622 |
| 4.9.2.66 DoChain()            | 622 |
| 4.9.2.67 CoChainin()          | 622 |
| 4.9.2.68 CoChainout()         | 622 |
| 4.9.2.69 CoExit()             | 622 |
| 4.9.2.70 CoInParallel()       | 622 |
| 4.9.2.71 CoNotInParallel()    | 622 |
| 4.9.2.72 DoInParallel()       | 623 |
| 4.9.2.73 CoInExpression()     | 623 |
| 4.9.2.74 CoEndInExpression()  | 623 |
| 4.9.2.75 CoSetExitFlag()      | 623 |
| 4.9.2.76 CoTryReplace()       | 623 |
| 4.9.2.77 CoModulus()          | 623 |
| 4.9.2.78 CoRepeat()           | 623 |
| 4.9.2.79 CoEndRepeat()        | 624 |
| 4.9.2.80 DoBrackets()         | 624 |
| 4.9.2.81 CoBracket()          | 624 |
| 4.9.2.82 CoAntiBracket()      | 624 |
| 4.9.2.83 CoMultiBracket()     | 624 |
| 4.9.2.84 CountComp()          | 624 |
| 4.9.2.85 Colf()               | 624 |
| 4.9.2.86 CoElse()             | 625 |
| 4.9.2.87 CoElseif()           | 625 |
| 4.9.2.88 CoEndIf()            | 625 |

|                                |     |
|--------------------------------|-----|
| 4.9.2.89 CoWhile()             | 625 |
| 4.9.2.90 CoEndWhile()          | 625 |
| 4.9.2.91 DoFindLoop()          | 625 |
| 4.9.2.92 CoFindLoop()          | 625 |
| 4.9.2.93 CoReplaceLoop()       | 626 |
| 4.9.2.94 CoFunPowers()         | 626 |
| 4.9.2.95 CoUnitTrace()         | 626 |
| 4.9.2.96 CoTerm()              | 626 |
| 4.9.2.97 CoEndTerm()           | 626 |
| 4.9.2.98 CoSort()              | 626 |
| 4.9.2.99 CoPolyFun()           | 626 |
| 4.9.2.100 CoPolyRatFun()       | 627 |
| 4.9.2.101 CoMerge()            | 627 |
| 4.9.2.102 CoStuffle()          | 627 |
| 4.9.2.103 CoProcessBucket()    | 627 |
| 4.9.2.104 CoThreadBucket()     | 627 |
| 4.9.2.105 DoArgPlode()         | 627 |
| 4.9.2.106 CoArgExplode()       | 627 |
| 4.9.2.107 CoArgImplode()       | 628 |
| 4.9.2.108 CoClearTable()       | 628 |
| 4.9.2.109 CoDenominators()     | 628 |
| 4.9.2.110 CoDropCoefficient()  | 628 |
| 4.9.2.111 CoDropSymbols()      | 628 |
| 4.9.2.112 CoToPolynomial()     | 628 |
| 4.9.2.113 CoFromPolynomial()   | 628 |
| 4.9.2.114 CoArgToExtraSymbol() | 629 |
| 4.9.2.115 CoExtraSymbols()     | 629 |
| 4.9.2.116 GetIfDollarFactor()  | 629 |
| 4.9.2.117 GetDoParam()         | 629 |
| 4.9.2.118 CoDo()               | 629 |
| 4.9.2.119 CoEndDo()            | 629 |
| 4.9.2.120 CoFactDollar()       | 630 |
| 4.9.2.121 CoFactorize()        | 630 |
| 4.9.2.122 CoNFactorize()       | 630 |
| 4.9.2.123 CoUnFactorize()      | 630 |
| 4.9.2.124 CoNUnFactorize()     | 630 |
| 4.9.2.125 DoFactorize()        | 630 |
| 4.9.2.126 CoOptimizeOption()   | 630 |
| 4.9.2.127 CoPutInside()        | 631 |
| 4.9.2.128 CoAntiPutInside()    | 631 |
| 4.9.2.129 DoPutInside()        | 631 |
| 4.9.2.130 CoSwitch()           | 631 |

|                                       |     |
|---------------------------------------|-----|
| 4.9.2.131 CoCase()                    | 631 |
| 4.9.2.132 CoBreak()                   | 631 |
| 4.9.2.133 CoDefault()                 | 631 |
| 4.9.2.134 CoEndSwitch()               | 632 |
| 4.9.2.135 CoSetUserFlag()             | 632 |
| 4.9.2.136 CoClearUserFlag()           | 632 |
| 4.9.2.137 CoCreateAllLoops()          | 632 |
| 4.9.2.138 CoCreateAllPaths()          | 632 |
| 4.9.2.139 CoCreateAll()               | 632 |
| 4.10 compcomm.c                       | 633 |
| 4.11 compiler.c File Reference        | 722 |
| 4.11.1 Detailed Description           | 723 |
| 4.11.2 Macro Definition Documentation | 723 |
| 4.11.2.1 OPTION0                      | 723 |
| 4.11.2.2 OPTION1                      | 723 |
| 4.11.2.3 OPTION2                      | 724 |
| 4.11.2.4 REDUCESUBEXPBUFFERS          | 724 |
| 4.11.3 Function Documentation         | 724 |
| 4.11.3.1 inictable()                  | 724 |
| 4.11.3.2 findcommand()                | 724 |
| 4.11.3.3 ParenthesesTest()            | 724 |
| 4.11.3.4 SkipAName()                  | 724 |
| 4.11.3.5 IsRHS()                      | 725 |
| 4.11.3.6 IsIdStatement()              | 725 |
| 4.11.3.7 CompileAlgebra()             | 725 |
| 4.11.3.8 CompileStatement()           | 725 |
| 4.11.3.9 TestTables()                 | 725 |
| 4.11.3.10 CompileSubExpressions()     | 725 |
| 4.11.3.11 CodeGenerator()             | 726 |
| 4.11.3.12 CompleteTerm()              | 726 |
| 4.11.3.13 CodeFactors()               | 726 |
| 4.11.3.14 GenerateFactors()           | 726 |
| 4.11.4 Variable Documentation         | 726 |
| 4.11.4.1 alfatable1                   | 726 |
| 4.11.4.2 subexpbuffers                | 726 |
| 4.11.4.3 topsubexpbuffers             | 727 |
| 4.11.4.4 insubexpbuffers              | 727 |
| 4.12 compiler.c                       | 727 |
| 4.13 compress.c File Reference        | 753 |
| 4.13.1 Detailed Description           | 753 |
| 4.14 compress.c                       | 754 |
| 4.15 comtool.c File Reference         | 762 |

|                                       |     |
|---------------------------------------|-----|
| 4.15.1 Detailed Description           | 763 |
| 4.15.2 Function Documentation         | 763 |
| 4.15.2.1 inicbufs()                   | 763 |
| 4.15.2.2 finishcbuf()                 | 763 |
| 4.15.2.3 clearcbuf()                  | 764 |
| 4.15.2.4 DoubleCbuffer()              | 764 |
| 4.15.2.5 AddLHS()                     | 764 |
| 4.15.2.6 AddRHS()                     | 765 |
| 4.15.2.7 AddNtoL()                    | 765 |
| 4.15.2.8 AddNtoC()                    | 766 |
| 4.15.2.9 InsTree()                    | 767 |
| 4.15.2.10 FindTree()                  | 767 |
| 4.15.2.11 RedoTree()                  | 767 |
| 4.15.2.12 ClearTree()                 | 767 |
| 4.15.2.13 IniFbuffer()                | 768 |
| 4.15.2.14 numcommute()                | 768 |
| 4.16 comtool.c                        | 768 |
| 4.17 comtool.h File Reference         | 775 |
| 4.17.1 Detailed Description           | 776 |
| 4.18 comtool.h                        | 776 |
| 4.19 declare.h File Reference         | 777 |
| 4.19.1 Detailed Description           | 799 |
| 4.19.2 Macro Definition Documentation | 799 |
| 4.19.2.1 MaX                          | 799 |
| 4.19.2.2 MiN                          | 800 |
| 4.19.2.3 ABS                          | 800 |
| 4.19.2.4 SGN                          | 800 |
| 4.19.2.5 REDLENG                      | 800 |
| 4.19.2.6 INCLENG                      | 800 |
| 4.19.2.7 GETCOEF                      | 800 |
| 4.19.2.8 GETSTOP                      | 801 |
| 4.19.2.9 StuffAdd                     | 801 |
| 4.19.2.10 EXCHN                       | 801 |
| 4.19.2.11 EXCH                        | 801 |
| 4.19.2.12 TOKENTOLINE                 | 801 |
| 4.19.2.13 UngetFromStream             | 801 |
| 4.19.2.14 AddLineFeed                 | 802 |
| 4.19.2.15 TryRecover                  | 802 |
| 4.19.2.16 UngetChar                   | 802 |
| 4.19.2.17 ParseNumber                 | 802 |
| 4.19.2.18 ParseSign                   | 802 |
| 4.19.2.19 ParseSignedNumber           | 802 |

|                                   |     |
|-----------------------------------|-----|
| 4.19.2.20 NCOPY . . . . .         | 803 |
| 4.19.2.21 NCOPYI . . . . .        | 803 |
| 4.19.2.22 NCOPYB . . . . .        | 803 |
| 4.19.2.23 NCOPYI32 . . . . .      | 803 |
| 4.19.2.24 WCOPY . . . . .         | 803 |
| 4.19.2.25 NeedNumber . . . . .    | 804 |
| 4.19.2.26 SKIPBLANKS . . . . .    | 804 |
| 4.19.2.27 FLUSHCONSOLE . . . . .  | 804 |
| 4.19.2.28 SKIPBRA1 . . . . .      | 804 |
| 4.19.2.29 SKIPBRA2 . . . . .      | 804 |
| 4.19.2.30 SKIPBRA3 . . . . .      | 805 |
| 4.19.2.31 SKIPBRA4 . . . . .      | 805 |
| 4.19.2.32 SKIPBRA5 . . . . .      | 805 |
| 4.19.2.33 CYCLE1 . . . . .        | 805 |
| 4.19.2.34 AddToCB . . . . .       | 805 |
| 4.19.2.35 EXCHINOUT . . . . .     | 806 |
| 4.19.2.36 BACKINOUT . . . . .     | 806 |
| 4.19.2.37 CopyArg . . . . .       | 806 |
| 4.19.2.38 FILLARG . . . . .       | 806 |
| 4.19.2.39 COPYARG . . . . .       | 806 |
| 4.19.2.40 ZEROARG . . . . .       | 807 |
| 4.19.2.41 FILLFUN . . . . .       | 807 |
| 4.19.2.42 COPYFUN . . . . .       | 807 |
| 4.19.2.43 COPYFUN3 . . . . .      | 807 |
| 4.19.2.44 FILLFUN3 . . . . .      | 807 |
| 4.19.2.45 FILLSUB . . . . .       | 807 |
| 4.19.2.46 COPYSUB . . . . .       | 808 |
| 4.19.2.47 FILLEXPR . . . . .      | 808 |
| 4.19.2.48 NEXTARG . . . . .       | 808 |
| 4.19.2.49 COPY1ARG . . . . .      | 808 |
| 4.19.2.50 ZeroFillRange . . . . . | 808 |
| 4.19.2.51 TABLESIZE . . . . .     | 809 |
| 4.19.2.52 WORDDIF . . . . .       | 809 |
| 4.19.2.53 wsizeof . . . . .       | 809 |
| 4.19.2.54 VARNAME . . . . .       | 809 |
| 4.19.2.55 DOLLARNAME . . . . .    | 809 |
| 4.19.2.56 EXPRNAME . . . . .      | 809 |
| 4.19.2.57 PREV . . . . .          | 810 |
| 4.19.2.58 Terminate . . . . .     | 810 |
| 4.19.2.59 SETERROR . . . . .      | 810 |
| 4.19.2.60 DUMMYUSE . . . . .      | 810 |
| 4.19.2.61 ADDPOS . . . . .        | 810 |

|                                          |     |
|------------------------------------------|-----|
| 4.19.2.62 SETBASELENGTH . . . . .        | 810 |
| 4.19.2.63 SETBASEPOSITION . . . . .      | 811 |
| 4.19.2.64 ISEQUALPOSINC . . . . .        | 811 |
| 4.19.2.65 ISGEPOSINC . . . . .           | 811 |
| 4.19.2.66 DIVPOS . . . . .               | 811 |
| 4.19.2.67 MULPOS . . . . .               | 811 |
| 4.19.2.68 DIFPOS . . . . .               | 811 |
| 4.19.2.69 DIFBASE . . . . .              | 812 |
| 4.19.2.70 ADD2POS . . . . .              | 812 |
| 4.19.2.71 PUTZERO . . . . .              | 812 |
| 4.19.2.72 BASEPOSITION . . . . .         | 812 |
| 4.19.2.73 SETSTARTPOS . . . . .          | 812 |
| 4.19.2.74 NOTSTARTPOS . . . . .          | 812 |
| 4.19.2.75 ISMINPOS . . . . .             | 812 |
| 4.19.2.76 ISEQUALPOS . . . . .           | 813 |
| 4.19.2.77 ISNOTEQUALPOS . . . . .        | 813 |
| 4.19.2.78 ISLESSPOS . . . . .            | 813 |
| 4.19.2.79 ISGEPOS . . . . .              | 813 |
| 4.19.2.80 ISNOTZEROPOS . . . . .         | 813 |
| 4.19.2.81 ISZEROPOS . . . . .            | 813 |
| 4.19.2.82 ISPOSPOS . . . . .             | 814 |
| 4.19.2.83 ISNEGPOS . . . . .             | 814 |
| 4.19.2.84 TOLONG . . . . .               | 814 |
| 4.19.2.85 Add2Com . . . . .              | 814 |
| 4.19.2.86 Add3Com . . . . .              | 814 |
| 4.19.2.87 Add4Com . . . . .              | 814 |
| 4.19.2.88 Add5Com . . . . .              | 815 |
| 4.19.2.89 WantAddPointers . . . . .      | 815 |
| 4.19.2.90 WantAddLongs . . . . .         | 815 |
| 4.19.2.91 WantAddPositions . . . . .     | 815 |
| 4.19.2.92 FORM_INLINE . . . . .          | 815 |
| 4.19.2.93 MEMORYMACROS . . . . .         | 816 |
| 4.19.2.94 TermMalloc . . . . .           | 816 |
| 4.19.2.95 NumberMalloc . . . . .         | 816 |
| 4.19.2.96 CacheNumberMalloc . . . . .    | 816 |
| 4.19.2.97 TermFree . . . . .             | 816 |
| 4.19.2.98 NumberFree . . . . .           | 816 |
| 4.19.2.99 CacheNumberFree . . . . .      | 817 |
| 4.19.2.100 NestingChecksum . . . . .     | 817 |
| 4.19.2.101 MesNesting . . . . .          | 817 |
| 4.19.2.102 MarkPolyRatFunDirty . . . . . | 817 |
| 4.19.2.103 PUSHPREASSIGNLEVEL . . . . .  | 817 |

|                                               |     |
|-----------------------------------------------|-----|
| 4.19.2.104 POPPREASSIGNLEVEL . . . . .        | 818 |
| 4.19.2.105 EXTERNLOCK . . . . .               | 818 |
| 4.19.2.106 INILOCK . . . . .                  | 818 |
| 4.19.2.107 LOCK . . . . .                     | 818 |
| 4.19.2.108 UNLOCK . . . . .                   | 818 |
| 4.19.2.109 EXTERNRWLOCK . . . . .             | 818 |
| 4.19.2.110 INIRWLOCK . . . . .                | 819 |
| 4.19.2.111 RWLOCKR . . . . .                  | 819 |
| 4.19.2.112 RWLOCKW . . . . .                  | 819 |
| 4.19.2.113 UNRWLOCK . . . . .                 | 819 |
| 4.19.2.114 MLOCK . . . . .                    | 819 |
| 4.19.2.115 MUNLOCK . . . . .                  | 819 |
| 4.19.2.116 GETIDENTITY . . . . .              | 819 |
| 4.19.2.117 GETBIDENTITY . . . . .             | 820 |
| 4.19.2.118 M_alloc . . . . .                  | 820 |
| 4.19.2.119 CompareTerms . . . . .             | 820 |
| 4.19.2.120 FiniShuffle . . . . .              | 820 |
| 4.19.2.121 DoShtuffle . . . . .               | 820 |
| 4.19.3 Typedef Documentation . . . . .        | 820 |
| 4.19.3.1 WRITEBUFTOEXTCHANNEL . . . . .       | 820 |
| 4.19.3.2 GETCFROMEXTCHANNEL . . . . .         | 820 |
| 4.19.3.3 SETTERMINATORFOREXTCHANNEL . . . . . | 821 |
| 4.19.3.4 SETKILLMODEFOREXTCHANNEL . . . . .   | 821 |
| 4.19.3.5 WRITEFILE . . . . .                  | 821 |
| 4.19.3.6 GETTERM . . . . .                    | 821 |
| 4.19.4 Function Documentation . . . . .       | 821 |
| 4.19.4.1 TELLFILE() . . . . .                 | 821 |
| 4.19.4.2 StartVariables() . . . . .           | 822 |
| 4.19.4.3 CodeToLine() . . . . .               | 822 |
| 4.19.4.4 AddArrayIndex() . . . . .            | 822 |
| 4.19.4.5 FindInIndex() . . . . .              | 823 |
| 4.19.4.6 NextFileIndex() . . . . .            | 823 |
| 4.19.4.7 PasteTerm() . . . . .                | 823 |
| 4.19.4.8 StrCopy() . . . . .                  | 823 |
| 4.19.4.9 WrtPower() . . . . .                 | 824 |
| 4.19.4.10 AccumGCD() . . . . .                | 824 |
| 4.19.4.11 AddArgs() . . . . .                 | 824 |
| 4.19.4.12 AddCoef() . . . . .                 | 824 |
| 4.19.4.13 AddLong() . . . . .                 | 825 |
| 4.19.4.14 AddPLon() . . . . .                 | 825 |
| 4.19.4.15 AddPoly() . . . . .                 | 826 |
| 4.19.4.16 AddRat() . . . . .                  | 827 |



|                                    |     |
|------------------------------------|-----|
| 4.19.4.17 AddToLine()              | 827 |
| 4.19.4.18 AddWild()                | 828 |
| 4.19.4.19 BigLong()                | 828 |
| 4.19.4.20 BinomGen()               | 828 |
| 4.19.4.21 CheckWild()              | 828 |
| 4.19.4.22 Chisholm()               | 828 |
| 4.19.4.23 CleanExpr()              | 829 |
| 4.19.4.24 CleanUp()                | 829 |
| 4.19.4.25 ClearWild()              | 829 |
| 4.19.4.26 CompareFunctions()       | 829 |
| 4.19.4.27 Commute()                | 829 |
| 4.19.4.28 DetCommu()               | 829 |
| 4.19.4.29 DoesCommu()              | 829 |
| 4.19.4.30 CompArg()                | 830 |
| 4.19.4.31 CompCoef()               | 830 |
| 4.19.4.32 CompGroup()              | 830 |
| 4.19.4.33 Compare1()               | 830 |
| 4.19.4.34 CountDo()                | 831 |
| 4.19.4.35 CountFun()               | 831 |
| 4.19.4.36 DimensionSubterm()       | 831 |
| 4.19.4.37 DimensionTerm()          | 832 |
| 4.19.4.38 DimensionExpression()    | 832 |
| 4.19.4.39 Deferred()               | 832 |
| 4.19.4.40 DeleteStore()            | 833 |
| 4.19.4.41 DetCurDum()              | 833 |
| 4.19.4.42 DetVars()                | 833 |
| 4.19.4.43 Distribute()             | 834 |
| 4.19.4.44 DivLong()                | 834 |
| 4.19.4.45 DivRat()                 | 834 |
| 4.19.4.46 Divvy()                  | 834 |
| 4.19.4.47 DoDelta()                | 834 |
| 4.19.4.48 DoDelta3()               | 835 |
| 4.19.4.49 TestPartitions()         | 835 |
| 4.19.4.50 DoPartitions()           | 835 |
| 4.19.4.51 CoCanonicalize()         | 835 |
| 4.19.4.52 DoCanonicalize()         | 835 |
| 4.19.4.53 GenTopologies()          | 835 |
| 4.19.4.54 GenDiagrams()            | 836 |
| 4.19.4.55 DoTopologyCanonicalize() | 836 |
| 4.19.4.56 DoShattering()           | 836 |
| 4.19.4.57 GenerateTopologies()     | 836 |
| 4.19.4.58 DoTableExpansion()       | 836 |

|                                    |     |
|------------------------------------|-----|
| 4.19.4.59 DoDistrib()              | 837 |
| 4.19.4.60 DoShuffle()              | 837 |
| 4.19.4.61 DoPermutations()         | 837 |
| 4.19.4.62 Shuffle()                | 837 |
| 4.19.4.63 FinishShuffle()          | 837 |
| 4.19.4.64 DoStuffle()              | 837 |
| 4.19.4.65 Stuffle()                | 838 |
| 4.19.4.66 FinishStuffle()          | 838 |
| 4.19.4.67 StuffRootAdd()           | 838 |
| 4.19.4.68 TestUse()                | 838 |
| 4.19.4.69 FindTB()                 | 838 |
| 4.19.4.70 CheckTableDeclarations() | 838 |
| 4.19.4.71 Apply()                  | 839 |
| 4.19.4.72 ApplyExec()              | 839 |
| 4.19.4.73 ApplyReset()             | 839 |
| 4.19.4.74 TableReset()             | 839 |
| 4.19.4.75 ReWorkT()                | 839 |
| 4.19.4.76 GetIfDollarNum()         | 839 |
| 4.19.4.77 FindVar()                | 840 |
| 4.19.4.78 DolfStatement()          | 840 |
| 4.19.4.79 DoOnePow()               | 840 |
| 4.19.4.80 DoRevert()               | 841 |
| 4.19.4.81 DoSumF1()                | 841 |
| 4.19.4.82 DoSumF2()                | 841 |
| 4.19.4.83 DoTheta()                | 842 |
| 4.19.4.84 EndSort()                | 842 |
| 4.19.4.85 EntVar()                 | 843 |
| 4.19.4.86 EpfCon()                 | 843 |
| 4.19.4.87 EpfFind()                | 843 |
| 4.19.4.88 EpfGen()                 | 843 |
| 4.19.4.89 EqualArg()               | 843 |
| 4.19.4.90 Factorial()              | 844 |
| 4.19.4.91 Bernoulli()              | 844 |
| 4.19.4.92 FactorIn()               | 844 |
| 4.19.4.93 FactorInExpr()           | 844 |
| 4.19.4.94 FindAll()                | 844 |
| 4.19.4.95 FindMulti()              | 844 |
| 4.19.4.96 FindOnce()               | 845 |
| 4.19.4.97 FindOnly()               | 845 |
| 4.19.4.98 FindRest()               | 845 |
| 4.19.4.99 FindrNumber()            | 845 |
| 4.19.4.100 FiniLine()              | 845 |

|                                 |     |
|---------------------------------|-----|
| 4.19.4.101 FiniTerm()           | 845 |
| 4.19.4.102 FlushOut()           | 846 |
| 4.19.4.103 FunLevel()           | 847 |
| 4.19.4.104 AdjustRenumScratch() | 847 |
| 4.19.4.105 GarbHand()           | 847 |
| 4.19.4.106 GcdLong()            | 847 |
| 4.19.4.107 LcmLong()            | 847 |
| 4.19.4.108 Generator()          | 848 |
| 4.19.4.109 GetBinom()           | 850 |
| 4.19.4.110 GetFromStore()       | 850 |
| 4.19.4.111 GetLong()            | 850 |
| 4.19.4.112 GetMoreTerms()       | 850 |
| 4.19.4.113 GetMoreFromMem()     | 850 |
| 4.19.4.114 GetOneTerm()         | 851 |
| 4.19.4.115 GetTable()           | 851 |
| 4.19.4.116 GetTerm()            | 851 |
| 4.19.4.117 Glue()               | 851 |
| 4.19.4.118 InFunction()         | 851 |
| 4.19.4.119 IniLine()            | 852 |
| 4.19.4.120 IniVars()            | 852 |
| 4.19.4.121 InsertTerm()         | 853 |
| 4.19.4.122 LongToLine()         | 853 |
| 4.19.4.123 MakeDirty()          | 853 |
| 4.19.4.124 MarkDirty()          | 854 |
| 4.19.4.125 PolyFunDirty()       | 854 |
| 4.19.4.126 PolyFunClean()       | 854 |
| 4.19.4.127 MakeModTable()       | 854 |
| 4.19.4.128 MatchE()             | 854 |
| 4.19.4.129 MatchCy()            | 854 |
| 4.19.4.130 FunMatchCy()         | 855 |
| 4.19.4.131 FunMatchSy()         | 855 |
| 4.19.4.132 MatchArgument()      | 855 |
| 4.19.4.133 MatchFunction()      | 855 |
| 4.19.4.134 MergePatches()       | 855 |
| 4.19.4.135 MesCerr()            | 856 |
| 4.19.4.136 MesComp()            | 856 |
| 4.19.4.137 Modulus()            | 857 |
| 4.19.4.138 MoveDummies()        | 857 |
| 4.19.4.139 MulLong()            | 857 |
| 4.19.4.140 MulRat()             | 857 |
| 4.19.4.141 Mully()              | 857 |
| 4.19.4.142 MultDo()             | 858 |

|                               |     |
|-------------------------------|-----|
| 4.19.4.143 NewSort()          | 858 |
| 4.19.4.144 ExtraSymbol()      | 858 |
| 4.19.4.145 Normalize()        | 858 |
| 4.19.4.146 BracketNormalize() | 858 |
| 4.19.4.147 DropCoefficient()  | 859 |
| 4.19.4.148 DropSymbols()      | 859 |
| 4.19.4.149 PutInside()        | 859 |
| 4.19.4.150 OpenTemp()         | 859 |
| 4.19.4.151 Pack()             | 859 |
| 4.19.4.152 PasteFile()        | 859 |
| 4.19.4.153 Permute()          | 860 |
| 4.19.4.154 PermuteP()         | 860 |
| 4.19.4.155 PolyFunMul()       | 860 |
| 4.19.4.156 PopVariables()     | 861 |
| 4.19.4.157 PrepPoly()         | 861 |
| 4.19.4.158 Processor()        | 862 |
| 4.19.4.159 Product()          | 863 |
| 4.19.4.160 PrtLong()          | 863 |
| 4.19.4.161 PrtTerms()         | 864 |
| 4.19.4.162 PrintDeprecation() | 864 |
| 4.19.4.163 PrintRunningTime() | 864 |
| 4.19.4.164 GetRunningTime()   | 864 |
| 4.19.4.165 PutBracket()       | 864 |
| 4.19.4.166 PutIn()            | 864 |
| 4.19.4.167 PutInStore()       | 865 |
| 4.19.4.168 PutOut()           | 865 |
| 4.19.4.169 Quotient()         | 866 |
| 4.19.4.170 RaisPow()          | 866 |
| 4.19.4.171 RaisPowCached()    | 866 |
| 4.19.5 Description            | 866 |
| 4.19.6 Notes                  | 867 |
| 4.19.6.1 RaisPowMod()         | 867 |
| 4.19.6.2 NormalModulus()      | 867 |
| 4.19.6.3 MakeInverses()       | 867 |
| 4.19.6.4 GetModInverses()     | 868 |
| 4.19.6.5 GetLongModInverses() | 868 |
| 4.19.6.6 RatToLine()          | 868 |
| 4.19.6.7 RatioFind()          | 868 |
| 4.19.6.8 RatioGen()           | 869 |
| 4.19.6.9 ReNumber()           | 869 |
| 4.19.6.10 ReadSnum()          | 869 |
| 4.19.6.11 Remain10()          | 869 |

|                               |     |
|-------------------------------|-----|
| 4.19.6.12 Remain4()           | 869 |
| 4.19.6.13 ResetScratch()      | 869 |
| 4.19.6.14 ResolveSet()        | 870 |
| 4.19.6.15 RevertScratch()     | 870 |
| 4.19.6.16 ScanFunctions()     | 870 |
| 4.19.6.17 SeekScratch()       | 870 |
| 4.19.6.18 SetEndScratch()     | 870 |
| 4.19.6.19 SetEndHScratch()    | 870 |
| 4.19.6.20 SetFileIndex()      | 871 |
| 4.19.6.21 Sflush()            | 871 |
| 4.19.6.22 Simplify()          | 871 |
| 4.19.6.23 SortWild()          | 872 |
| 4.19.6.24 LocateBase()        | 872 |
| 4.19.6.25 SplitMerge()        | 872 |
| 4.19.6.26 StoreTerm()         | 873 |
| 4.19.6.27 SubPLon()           | 874 |
| 4.19.6.28 Substitute()        | 874 |
| 4.19.6.29 SymFind()           | 874 |
| 4.19.6.30 SymGen()            | 875 |
| 4.19.6.31 Symmetrize()        | 875 |
| 4.19.6.32 FullSymmetrize()    | 875 |
| 4.19.6.33 TakeModulus()       | 875 |
| 4.19.6.34 TakeNormalModulus() | 875 |
| 4.19.6.35 TalToLine()         | 876 |
| 4.19.6.36 TenVec()            | 876 |
| 4.19.6.37 TenVecFind()        | 876 |
| 4.19.6.38 TermRenumber()      | 876 |
| 4.19.6.39 TestDrop()          | 877 |
| 4.19.6.40 PutInVflags()       | 877 |
| 4.19.6.41 TestMatch()         | 877 |
| 4.19.6.42 TestSub()           | 878 |
| 4.19.6.43 TimeCPU()           | 878 |
| 4.19.6.44 TimeChildren()      | 879 |
| 4.19.6.45 TimeWallClock()     | 879 |
| 4.19.6.46 ToStorage()         | 879 |
| 4.19.6.47 TokenToLine()       | 879 |
| 4.19.6.48 Trace4()            | 879 |
| 4.19.6.49 Trace4Gen()         | 880 |
| 4.19.6.50 Trace4no()          | 880 |
| 4.19.6.51 TraceFind()         | 880 |
| 4.19.6.52 TraceN()            | 880 |
| 4.19.6.53 TraceNgen()         | 880 |

|                             |     |
|-----------------------------|-----|
| 4.19.6.54 TraceNno()        | 880 |
| 4.19.6.55 Traces()          | 881 |
| 4.19.6.56 Trick()           | 881 |
| 4.19.6.57 TryDo()           | 881 |
| 4.19.6.58 UnPack()          | 881 |
| 4.19.6.59 VarStore()        | 881 |
| 4.19.6.60 WildFill()        | 882 |
| 4.19.6.61 WriteAll()        | 882 |
| 4.19.6.62 WriteOne()        | 882 |
| 4.19.6.63 WriteArgument()   | 882 |
| 4.19.6.64 WriteExpression() | 882 |
| 4.19.6.65 WriteInnerTerm()  | 882 |
| 4.19.6.66 WriteLists()      | 883 |
| 4.19.6.67 WriteSetup()      | 883 |
| 4.19.6.68 WriteStats()      | 883 |
| 4.19.6.69 WriteSubTerm()    | 884 |
| 4.19.6.70 WriteTerm()       | 884 |
| 4.19.6.71 execarg()         | 884 |
| 4.19.6.72 execterm()        | 884 |
| 4.19.6.73 SpecialCleanup()  | 884 |
| 4.19.6.74 SetMods()         | 884 |
| 4.19.6.75 UnSetMods()       | 885 |
| 4.19.6.76 DoExecute()       | 885 |
| 4.19.6.77 SetScratch()      | 885 |
| 4.19.6.78 Warning()         | 885 |
| 4.19.6.79 HighWarning()     | 885 |
| 4.19.6.80 SpareTable()      | 885 |
| 4.19.6.81 strDup1()         | 886 |
| 4.19.6.82 Malloc1()         | 886 |
| 4.19.6.83 DoTail()          | 886 |
| 4.19.6.84 OpenInput()       | 886 |
| 4.19.6.85 PutPreVar()       | 886 |
| 4.19.6.86 Error0()          | 887 |
| 4.19.6.87 Error1()          | 887 |
| 4.19.6.88 Error2()          | 887 |
| 4.19.6.89 ReadFromStream()  | 887 |
| 4.19.6.90 GetFromStream()   | 887 |
| 4.19.6.91 LookInStream()    | 888 |
| 4.19.6.92 OpenStream()      | 888 |
| 4.19.6.93 LocateFile()      | 888 |
| 4.19.6.94 CloseStream()     | 888 |
| 4.19.6.95 PositionStream()  | 888 |

|                               |     |
|-------------------------------|-----|
| 4.19.6.96 ReverseStatements() | 888 |
| 4.19.6.97 ProcessOption()     | 889 |
| 4.19.6.98 DoSetups()          | 889 |
| 4.19.6.99 TerminateImpl()     | 889 |
| 4.19.6.100 GetNode()          | 889 |
| 4.19.6.101 AddName()          | 889 |
| 4.19.6.102 GetName()          | 890 |
| 4.19.6.103 GetFunction()      | 890 |
| 4.19.6.104 GetNumber()        | 890 |
| 4.19.6.105 GetLastExprName()  | 890 |
| 4.19.6.106 GetAutoName()      | 890 |
| 4.19.6.107 GetVar()           | 890 |
| 4.19.6.108 MakeDubious()      | 891 |
| 4.19.6.109 GetOName()         | 891 |
| 4.19.6.110 DumpTree()         | 891 |
| 4.19.6.111 DumpNode()         | 891 |
| 4.19.6.112 LinkTree()         | 891 |
| 4.19.6.113 CopyTree()         | 891 |
| 4.19.6.114 CompactifyTree()   | 892 |
| 4.19.6.115 MakeNameTree()     | 892 |
| 4.19.6.116 FreeNameTree()     | 892 |
| 4.19.6.117 AddExpression()    | 892 |
| 4.19.6.118 AddSymbol()        | 892 |
| 4.19.6.119 AddDollar()        | 892 |
| 4.19.6.120 ReplaceDollar()    | 893 |
| 4.19.6.121 DollarRaiseLow()   | 893 |
| 4.19.6.122 AddVector()        | 893 |
| 4.19.6.123 AddDubious()       | 893 |
| 4.19.6.124 AddIndex()         | 893 |
| 4.19.6.125 DoDimension()      | 893 |
| 4.19.6.126 AddFunction()      | 894 |
| 4.19.6.127 CoCommutateInSet() | 894 |
| 4.19.6.128 CoFunction()       | 894 |
| 4.19.6.129 TestName()         | 894 |
| 4.19.6.130 AddSet()           | 894 |
| 4.19.6.131 DoElements()       | 894 |
| 4.19.6.132 DoTempSet()        | 895 |
| 4.19.6.133 NameConflict()     | 895 |
| 4.19.6.134 OpenFile()         | 895 |
| 4.19.6.135 OpenAddFile()      | 895 |
| 4.19.6.136 ReOpenFile()       | 895 |
| 4.19.6.137 CreateFile()       | 895 |

|                               |     |
|-------------------------------|-----|
| 4.19.6.138 CreateLogFile()    | 895 |
| 4.19.6.139 CloseFile()        | 896 |
| 4.19.6.140 CopyFile()         | 896 |
| 4.19.6.141 CreateHandle()     | 896 |
| 4.19.6.142 ReadFile()         | 896 |
| 4.19.6.143 ReadPosFile()      | 896 |
| 4.19.6.144 WriteFileToFile()  | 897 |
| 4.19.6.145 SeekFile()         | 897 |
| 4.19.6.146 TellFile()         | 897 |
| 4.19.6.147 FlushFile()        | 897 |
| 4.19.6.148 GetPosFile()       | 897 |
| 4.19.6.149 SetPosFile()       | 897 |
| 4.19.6.150 SynchFile()        | 898 |
| 4.19.6.151 TruncateFile()     | 898 |
| 4.19.6.152 GetChannel()       | 898 |
| 4.19.6.153 GetAppendChannel() | 898 |
| 4.19.6.154 CloseChannel()     | 898 |
| 4.19.6.155 inictable()        | 898 |
| 4.19.6.156 findcommand()      | 898 |
| 4.19.6.157 inicbufs()         | 899 |
| 4.19.6.158 StartFiles()       | 899 |
| 4.19.6.159 MakeDate()         | 899 |
| 4.19.6.160 PreProcessor()     | 899 |
| 4.19.6.161 FromList()         | 899 |
| 4.19.6.162 FromOList()        | 900 |
| 4.19.6.163 FromVarList()      | 900 |
| 4.19.6.164 DoubleList()       | 900 |
| 4.19.6.165 DoubleLList()      | 900 |
| 4.19.6.166 DoubleBuffer()     | 900 |
| 4.19.6.167 ExpandBuffer()     | 900 |
| 4.19.6.168 iexp()             | 901 |
| 4.19.6.169 IsLikeVector()     | 901 |
| 4.19.6.170 AreArgsEqual()     | 901 |
| 4.19.6.171 CompareArgs()      | 901 |
| 4.19.6.172 SkipField()        | 901 |
| 4.19.6.173 StrCmp()           | 901 |
| 4.19.6.174 StrlCmp()          | 902 |
| 4.19.6.175 StrHlCmp()         | 902 |
| 4.19.6.176 StrlCont()         | 902 |
| 4.19.6.177 CmpArray()         | 902 |
| 4.19.6.178 ConWord()          | 902 |
| 4.19.6.179 StrLen()           | 902 |



|                                |     |
|--------------------------------|-----|
| 4.19.6.180 GetPreVar()         | 903 |
| 4.19.6.181 ToGeneral()         | 903 |
| 4.19.6.182 ToPolyFunGeneral()  | 903 |
| 4.19.6.183 ToFast()            | 903 |
| 4.19.6.184 GetSetupPar()       | 903 |
| 4.19.6.185 RecalcSetups()      | 903 |
| 4.19.6.186 AllocSetups()       | 904 |
| 4.19.6.187 AllocSort()         | 904 |
| 4.19.6.188 AllocSortFileName() | 904 |
| 4.19.6.189 LoadInputFile()     | 904 |
| 4.19.6.190 GetInput()          | 904 |
| 4.19.6.191 ClearPushback()     | 904 |
| 4.19.6.192 GetChar()           | 905 |
| 4.19.6.193 CharOut()           | 905 |
| 4.19.6.194 UnsetAllowDelay()   | 905 |
| 4.19.6.195 PopPreVars()        | 905 |
| 4.19.6.196 IniModule()         | 905 |
| 4.19.6.197 IniSpecialModule()  | 905 |
| 4.19.6.198 ModuleInstruction() | 905 |
| 4.19.6.199 PreProInstruction() | 906 |
| 4.19.6.200 LoadInstruction()   | 906 |
| 4.19.6.201 LoadStatement()     | 906 |
| 4.19.6.202 FindKeyWord()       | 906 |
| 4.19.6.203 FindInKeyWord()     | 906 |
| 4.19.6.204 DoDefine()          | 906 |
| 4.19.6.205 DoRedefine()        | 907 |
| 4.19.6.206 TheDefine()         | 907 |
| 4.19.6.207 TheUndefine()       | 907 |
| 4.19.6.208 ClearMacro()        | 908 |
| 4.19.6.209 DoUndefine()        | 908 |
| 4.19.6.210 DoInclude()         | 908 |
| 4.19.6.211 DoReverseInclude()  | 908 |
| 4.19.6.212 Include()           | 908 |
| 4.19.6.213 DoExternal()        | 908 |
| 4.19.6.214 DoToExternal()      | 908 |
| 4.19.6.215 DoFromExternal()    | 909 |
| 4.19.6.216 DoPrompt()          | 909 |
| 4.19.6.217 DoSetExternal()     | 909 |
| 4.19.6.218 DoSetExternalAttr() | 909 |
| 4.19.6.219 DoRmExternal()      | 909 |
| 4.19.6.220 DoFactDollar()      | 909 |
| 4.19.6.221 GetDollarNumber()   | 909 |

|                                 |     |
|---------------------------------|-----|
| 4.19.6.222 DoSetRandom()        | 910 |
| 4.19.6.223 DoOptimize()         | 910 |
| 4.19.6.224 DoClearOptimize()    | 910 |
| 4.19.6.225 DoSkipExtraSymbols() | 910 |
| 4.19.6.226 DoTimeOutAfter()     | 910 |
| 4.19.6.227 DoMessage()          | 910 |
| 4.19.6.228 DoPreOut()           | 910 |
| 4.19.6.229 DoPreAppend()        | 911 |
| 4.19.6.230 DoPreCreate()        | 911 |
| 4.19.6.231 DoPreAssign()        | 911 |
| 4.19.6.232 DoPreBreak()         | 911 |
| 4.19.6.233 DoPreDefault()       | 911 |
| 4.19.6.234 DoPreSwitch()        | 911 |
| 4.19.6.235 DoPreEndSwitch()     | 911 |
| 4.19.6.236 DoPreCase()          | 912 |
| 4.19.6.237 DoPreShow()          | 912 |
| 4.19.6.238 DoPreExchange()      | 912 |
| 4.19.6.239 DoSystem()           | 912 |
| 4.19.6.240 DoPipe()             | 912 |
| 4.19.6.241 StartPrepro()        | 912 |
| 4.19.6.242 Dolfdef()            | 912 |
| 4.19.6.243 Dolfydef()           | 913 |
| 4.19.6.244 Dolfndef()           | 913 |
| 4.19.6.245 DoElse()             | 913 |
| 4.19.6.246 DoElseif()           | 913 |
| 4.19.6.247 DoEndif()            | 913 |
| 4.19.6.248 DoTerminate()        | 913 |
| 4.19.6.249 Dolf()               | 913 |
| 4.19.6.250 DoCall()             | 914 |
| 4.19.6.251 DoDebug()            | 914 |
| 4.19.6.252 DoContinueDo()       | 914 |
| 4.19.6.253 DoDo()               | 914 |
| 4.19.6.254 DoBreakDo()          | 914 |
| 4.19.6.255 DoEnddo()            | 914 |
| 4.19.6.256 DoEndprocedure()     | 914 |
| 4.19.6.257 DoInside()           | 915 |
| 4.19.6.258 DoEndInside()        | 915 |
| 4.19.6.259 DoProcedure()        | 915 |
| 4.19.6.260 DoPrePrintTimes()    | 915 |
| 4.19.6.261 DoPreWrite()         | 915 |
| 4.19.6.262 DoPreClose()         | 915 |
| 4.19.6.263 DoPreRemove()        | 915 |

|                                  |     |
|----------------------------------|-----|
| 4.19.6.264 DoPreSortReallocate() | 916 |
| 4.19.6.265 DoCommentChar()       | 916 |
| 4.19.6.266 DoPrcExtension()      | 916 |
| 4.19.6.267 DoPreReset()          | 916 |
| 4.19.6.268 WriteString()         | 916 |
| 4.19.6.269 WriteUnfinString()    | 916 |
| 4.19.6.270 AddToString()         | 917 |
| 4.19.6.271 PreCalc()             | 917 |
| 4.19.6.272 PreEval()             | 917 |
| 4.19.6.273 NumToStr()            | 917 |
| 4.19.6.274 PreCmp()              | 917 |
| 4.19.6.275 PreEq()               | 918 |
| 4.19.6.276 pParseObject()        | 918 |
| 4.19.6.277 PriefEval()           | 918 |
| 4.19.6.278 EvalPrief()           | 918 |
| 4.19.6.279 PreLoad()             | 918 |
| 4.19.6.280 PreSkip()             | 919 |
| 4.19.6.281 EndOfToken()          | 919 |
| 4.19.6.282 SetSpecialMode()      | 919 |
| 4.19.6.283 MakeGlobal()          | 919 |
| 4.19.6.284 ExecModule()          | 919 |
| 4.19.6.285 ExecStore()           | 919 |
| 4.19.6.286 FullCleanUp()         | 920 |
| 4.19.6.287 DoExecStatement()     | 920 |
| 4.19.6.288 DoPipeStatement()     | 920 |
| 4.19.6.289 DoPolyfun()           | 920 |
| 4.19.6.290 DoPolyratfun()        | 920 |
| 4.19.6.291 CompileStatement()    | 920 |
| 4.19.6.292 ToToken()             | 920 |
| 4.19.6.293 GetDollar()           | 921 |
| 4.19.6.294 MesWork()             | 921 |
| 4.19.6.295 MesCall()             | 921 |
| 4.19.6.296 NumCopy()             | 921 |
| 4.19.6.297 LongCopy()            | 921 |
| 4.19.6.298 LongLongCopy()        | 921 |
| 4.19.6.299 ReserveTempFiles()    | 922 |
| 4.19.6.300 PrintTerm()           | 922 |
| 4.19.6.301 PrintTermC()          | 922 |
| 4.19.6.302 PrintSubTerm()        | 922 |
| 4.19.6.303 PrintWords()          | 922 |
| 4.19.6.304 PrintSeq()            | 922 |
| 4.19.6.305 ExpandTripleDots()    | 923 |

|                                 |     |
|---------------------------------|-----|
| 4.19.6.306 ComPress()           | 923 |
| 4.19.6.307 StageSort()          | 923 |
| 4.19.6.308 M_free()             | 924 |
| 4.19.6.309 ClearWildcardNames() | 924 |
| 4.19.6.310 AddWildcardName()    | 924 |
| 4.19.6.311 GetWildcardName()    | 924 |
| 4.19.6.312 Globalize()          | 924 |
| 4.19.6.313 ResetVariables()     | 924 |
| 4.19.6.314 AddToPreTypes()      | 924 |
| 4.19.6.315 MessPreNesting()     | 925 |
| 4.19.6.316 GetStreamPosition()  | 925 |
| 4.19.6.317 DoubleCbuffer()      | 925 |
| 4.19.6.318 AddLHS()             | 925 |
| 4.19.6.319 AddRHS()             | 926 |
| 4.19.6.320 AddNtoL()            | 926 |
| 4.19.6.321 AddNtoC()            | 927 |
| 4.19.6.322 DoubleIfBuffers()    | 928 |
| 4.19.6.323 CreateStream()       | 928 |
| 4.19.6.324 setonoff()           | 928 |
| 4.19.6.325 DoPrint()            | 928 |
| 4.19.6.326 SetExpr()            | 928 |
| 4.19.6.327 AddToCom()           | 928 |
| 4.19.6.328 Add2ComStrings()     | 929 |
| 4.19.6.329 DoSymmetrize()       | 929 |
| 4.19.6.330 DoArgument()         | 929 |
| 4.19.6.331 ArgFactorize()       | 929 |
| 4.19.6.332 TakeArgContent()     | 929 |
| 4.19.6.333 MakeInteger()        | 930 |
| 4.19.6.334 MakeMod()            | 930 |
| 4.19.6.335 FindArg()            | 931 |
| 4.19.6.336 InsertArg()          | 931 |
| 4.19.6.337 CleanupArgCache()    | 932 |
| 4.19.6.338 ArgSymbolMerge()     | 932 |
| 4.19.6.339 ArgDotproductMerge() | 932 |
| 4.19.6.340 SortWeights()        | 933 |
| 4.19.6.341 DoBrackets()         | 933 |
| 4.19.6.342 DoPutInside()        | 933 |
| 4.19.6.343 CountComp()          | 933 |
| 4.19.6.344 CoAntiBracket()      | 934 |
| 4.19.6.345 CoAntiSymmetrize()   | 934 |
| 4.19.6.346 DoArgPlode()         | 934 |
| 4.19.6.347 CoArgExplode()       | 934 |

|                                |     |
|--------------------------------|-----|
| 4.19.6.348 CoArgImplode()      | 934 |
| 4.19.6.349 CoArgument()        | 934 |
| 4.19.6.350 CoInside()          | 934 |
| 4.19.6.351 ExecInside()        | 935 |
| 4.19.6.352 CoInExpression()    | 935 |
| 4.19.6.353 CoInParallel()      | 935 |
| 4.19.6.354 CoNotInParallel()   | 935 |
| 4.19.6.355 DoInParallel()      | 935 |
| 4.19.6.356 CoEndInExpression() | 935 |
| 4.19.6.357 CoBracket()         | 935 |
| 4.19.6.358 CoPutInside()       | 936 |
| 4.19.6.359 CoAntiPutInside()   | 936 |
| 4.19.6.360 CoMultiBracket()    | 936 |
| 4.19.6.361 CoCFunction()       | 936 |
| 4.19.6.362 CoCTensor()         | 936 |
| 4.19.6.363 CoCollect()         | 936 |
| 4.19.6.364 CoCompress()        | 936 |
| 4.19.6.365 CoContract()        | 937 |
| 4.19.6.366 CoCycleSymmetrize() | 937 |
| 4.19.6.367 CoDelete()          | 937 |
| 4.19.6.368 CoTableBase()       | 937 |
| 4.19.6.369 CoApply()           | 937 |
| 4.19.6.370 CoDenominators()    | 937 |
| 4.19.6.371 CoDimension()       | 937 |
| 4.19.6.372 CoDiscard()         | 938 |
| 4.19.6.373 CoDisorder()        | 938 |
| 4.19.6.374 CoDrop()            | 938 |
| 4.19.6.375 CoDropCoefficient() | 938 |
| 4.19.6.376 CoDropSymbols()     | 938 |
| 4.19.6.377 CoElse()            | 938 |
| 4.19.6.378 CoElseIf()          | 938 |
| 4.19.6.379 CoEndArgument()     | 939 |
| 4.19.6.380 CoEndInside()       | 939 |
| 4.19.6.381 CoEndIf()           | 939 |
| 4.19.6.382 CoEndRepeat()       | 939 |
| 4.19.6.383 CoEndTerm()         | 939 |
| 4.19.6.384 CoEndWhile()        | 939 |
| 4.19.6.385 CoExit()            | 939 |
| 4.19.6.386 CoFactArg()         | 940 |
| 4.19.6.387 CoFactDollar()      | 940 |
| 4.19.6.388 CoFactorize()       | 940 |
| 4.19.6.389 CoNFactorize()      | 940 |

|                                 |     |
|---------------------------------|-----|
| 4.19.6.390 CoUnFactorize()      | 940 |
| 4.19.6.391 CoNUnFactorize()     | 940 |
| 4.19.6.392 DoFactorize()        | 940 |
| 4.19.6.393 CoFill()             | 941 |
| 4.19.6.394 CoFillExpression()   | 941 |
| 4.19.6.395 CoFixIndex()         | 941 |
| 4.19.6.396 CoFormat()           | 941 |
| 4.19.6.397 CoGlobal()           | 941 |
| 4.19.6.398 CoGlobalFactorized() | 941 |
| 4.19.6.399 CoGoTo()             | 941 |
| 4.19.6.400 Cold()               | 942 |
| 4.19.6.401 ColdNew()            | 942 |
| 4.19.6.402 ColdOld()            | 942 |
| 4.19.6.403 Colf()               | 942 |
| 4.19.6.404 ColfMatch()          | 942 |
| 4.19.6.405 ColfNoMatch()        | 942 |
| 4.19.6.406 ColIndex()           | 942 |
| 4.19.6.407 ColInsideFirst()     | 943 |
| 4.19.6.408 CoKeep()             | 943 |
| 4.19.6.409 CoLabel()            | 943 |
| 4.19.6.410 CoLoad()             | 943 |
| 4.19.6.411 CoLocal()            | 943 |
| 4.19.6.412 CoLocalFactorized()  | 943 |
| 4.19.6.413 CoMany()             | 943 |
| 4.19.6.414 CoMerge()            | 944 |
| 4.19.6.415 CoStuffle()          | 944 |
| 4.19.6.416 CoMetric()           | 944 |
| 4.19.6.417 CoModOption()        | 944 |
| 4.19.6.418 CoModuleOption()     | 944 |
| 4.19.6.419 CoModulus()          | 944 |
| 4.19.6.420 CoMulti()            | 944 |
| 4.19.6.421 CoMultiply()         | 945 |
| 4.19.6.422 CoNFunction()        | 945 |
| 4.19.6.423 CoNPrint()           | 945 |
| 4.19.6.424 CoNTensor()          | 945 |
| 4.19.6.425 CoNWrite()           | 945 |
| 4.19.6.426 CoNoDrop()           | 945 |
| 4.19.6.427 CoNoSkip()           | 945 |
| 4.19.6.428 CoNormalize()        | 946 |
| 4.19.6.429 CoMakeInteger()      | 946 |
| 4.19.6.430 CoFlags()            | 946 |
| 4.19.6.431 CoOff()              | 946 |

|                                 |     |
|---------------------------------|-----|
| 4.19.6.432 CoOn()               | 946 |
| 4.19.6.433 CoOnce()             | 946 |
| 4.19.6.434 CoOnly()             | 946 |
| 4.19.6.435 CoOptimizeOption()   | 947 |
| 4.19.6.436 CoPolyFun()          | 947 |
| 4.19.6.437 CoPolyRatFun()       | 947 |
| 4.19.6.438 CoPrint()            | 947 |
| 4.19.6.439 CoPrintB()           | 947 |
| 4.19.6.440 CoProperCount()      | 947 |
| 4.19.6.441 CoUnitTrace()        | 947 |
| 4.19.6.442 CoRCycleSymmetrize() | 948 |
| 4.19.6.443 CoRatio()            | 948 |
| 4.19.6.444 CoRedefine()         | 948 |
| 4.19.6.445 CoRenummer()         | 948 |
| 4.19.6.446 CoRepeat()           | 948 |
| 4.19.6.447 CoSave()             | 948 |
| 4.19.6.448 CoSelect()           | 948 |
| 4.19.6.449 CoSet()              | 949 |
| 4.19.6.450 CoSetExitFlag()      | 949 |
| 4.19.6.451 CoSkip()             | 949 |
| 4.19.6.452 CoProcessBucket()    | 949 |
| 4.19.6.453 CoPushHide()         | 949 |
| 4.19.6.454 CoPopHide()          | 949 |
| 4.19.6.455 CoHide()             | 949 |
| 4.19.6.456 CoIntoHide()         | 950 |
| 4.19.6.457 CoNoIntoHide()       | 950 |
| 4.19.6.458 CoNoHide()           | 950 |
| 4.19.6.459 CoUnHide()           | 950 |
| 4.19.6.460 CoNoUnHide()         | 950 |
| 4.19.6.461 CoSort()             | 950 |
| 4.19.6.462 CoSplitArg()         | 950 |
| 4.19.6.463 CoSplitFirstArg()    | 951 |
| 4.19.6.464 CoSplitLastArg()     | 951 |
| 4.19.6.465 CoSum()              | 951 |
| 4.19.6.466 CoSymbol()           | 951 |
| 4.19.6.467 CoSymmetrize()       | 951 |
| 4.19.6.468 DoTable()            | 951 |
| 4.19.6.469 CoTable()            | 951 |
| 4.19.6.470 CoTerm()             | 952 |
| 4.19.6.471 CoNTable()           | 952 |
| 4.19.6.472 CoCTable()           | 952 |
| 4.19.6.473 EmptyTable()         | 952 |

|                              |     |
|------------------------------|-----|
| 4.19.6.474 CoToTensor()      | 952 |
| 4.19.6.475 CoToVector()      | 952 |
| 4.19.6.476 CoTrace4()        | 952 |
| 4.19.6.477 CoTraceN()        | 953 |
| 4.19.6.478 CoChisholm()      | 953 |
| 4.19.6.479 CoTransform()     | 953 |
| 4.19.6.480 CoClearTable()    | 953 |
| 4.19.6.481 DoChain()         | 953 |
| 4.19.6.482 CoChainin()       | 953 |
| 4.19.6.483 CoChainout()      | 953 |
| 4.19.6.484 CoTryReplace()    | 954 |
| 4.19.6.485 CoVector()        | 954 |
| 4.19.6.486 CoWhile()         | 954 |
| 4.19.6.487 CoWrite()         | 954 |
| 4.19.6.488 CoAuto()          | 954 |
| 4.19.6.489 CoSwitch()        | 954 |
| 4.19.6.490 CoCase()          | 954 |
| 4.19.6.491 CoBreak()         | 955 |
| 4.19.6.492 CoDefault()       | 955 |
| 4.19.6.493 CoEndSwitch()     | 955 |
| 4.19.6.494 CoTBaddto()       | 955 |
| 4.19.6.495 CoTBaudit()       | 955 |
| 4.19.6.496 CoTbcleanup()     | 955 |
| 4.19.6.497 CoTBcreate()      | 955 |
| 4.19.6.498 CoTBenter()       | 956 |
| 4.19.6.499 CoTBhelp()        | 956 |
| 4.19.6.500 CoTBload()        | 956 |
| 4.19.6.501 CoTBoff()         | 956 |
| 4.19.6.502 CoTBon()          | 956 |
| 4.19.6.503 CoTBopen()        | 956 |
| 4.19.6.504 CoTBreplace()     | 956 |
| 4.19.6.505 CoTBuse()         | 957 |
| 4.19.6.506 CoTestUse()       | 957 |
| 4.19.6.507 CoThreadBucket()  | 957 |
| 4.19.6.508 AddComString()    | 957 |
| 4.19.6.509 CompileAlgebra()  | 957 |
| 4.19.6.510 IsIdStatement()   | 957 |
| 4.19.6.511 IsRHS()           | 958 |
| 4.19.6.512 ParenthesesTest() | 958 |
| 4.19.6.513 tokenize()        | 958 |
| 4.19.6.514 WriteTokens()     | 958 |
| 4.19.6.515 simp1token()      | 958 |



|                                        |     |
|----------------------------------------|-----|
| 4.19.6.516 simpwtoken()                | 958 |
| 4.19.6.517 simp2token()                | 958 |
| 4.19.6.518 simp3atoken()               | 959 |
| 4.19.6.519 simp3btoken()               | 959 |
| 4.19.6.520 simp4token()                | 959 |
| 4.19.6.521 simp5token()                | 959 |
| 4.19.6.522 simp6token()                | 959 |
| 4.19.6.523 SkipAName()                 | 959 |
| 4.19.6.524 TestTables()                | 960 |
| 4.19.6.525 GetLabel()                  | 960 |
| 4.19.6.526 ColdExpression()            | 960 |
| 4.19.6.527 CoAssign()                  | 960 |
| 4.19.6.528 DoExpr()                    | 960 |
| 4.19.6.529 CompileSubExpressions()     | 960 |
| 4.19.6.530 CodeGenerator()             | 961 |
| 4.19.6.531 CompleteTerm()              | 961 |
| 4.19.6.532 CodeFactors()               | 961 |
| 4.19.6.533 GenerateFactors()           | 961 |
| 4.19.6.534 InsTree()                   | 961 |
| 4.19.6.535 FindTree()                  | 961 |
| 4.19.6.536 RedoTree()                  | 962 |
| 4.19.6.537 ClearTree()                 | 962 |
| 4.19.6.538 CatchDollar()               | 962 |
| 4.19.6.539 AssignDollar()              | 962 |
| 4.19.6.540 WriteDollarToBuffer()       | 962 |
| 4.19.6.541 WriteDollarFactorToBuffer() | 962 |
| 4.19.6.542 AddToDollarBuffer()         | 963 |
| 4.19.6.543 PutTermInDollar()           | 963 |
| 4.19.6.544 TermAssign()                | 963 |
| 4.19.6.545 WildDollars()               | 963 |
| 4.19.6.546 numcommute()                | 963 |
| 4.19.6.547 FullRenummer()              | 963 |
| 4.19.6.548 Lus()                       | 964 |
| 4.19.6.549 FindLus()                   | 964 |
| 4.19.6.550 CoReplaceLoop()             | 964 |
| 4.19.6.551 CoFindLoop()                | 964 |
| 4.19.6.552 DoFindLoop()                | 964 |
| 4.19.6.553 CoFunPowers()               | 964 |
| 4.19.6.554 SortTheList()               | 965 |
| 4.19.6.555 MatchIsPossible()           | 965 |
| 4.19.6.556 StudyPattern()              | 965 |
| 4.19.6.557 DolToTensor()               | 965 |

|                                 |     |
|---------------------------------|-----|
| 4.19.6.558 DolToFunction()      | 965 |
| 4.19.6.559 DolToVector()        | 965 |
| 4.19.6.560 DolToNumber()        | 965 |
| 4.19.6.561 DolToSymbol()        | 966 |
| 4.19.6.562 DolToIndex()         | 966 |
| 4.19.6.563 DolToLong()          | 966 |
| 4.19.6.564 DollarFactorize()    | 966 |
| 4.19.6.565 CoPrintTable()       | 966 |
| 4.19.6.566 CoDeallocateTable()  | 966 |
| 4.19.6.567 CleanDollarFactors() | 966 |
| 4.19.6.568 TakeDollarContent()  | 967 |
| 4.19.6.569 MakeDollarInteger()  | 967 |
| 4.19.6.570 MakeDollarMod()      | 968 |
| 4.19.6.571 GetDolNum()          | 968 |
| 4.19.6.572 AddPotModdollar()    | 968 |
| 4.19.6.573 Optimize()           | 969 |
| 4.19.7 Description              | 969 |
| 4.19.7.1 ClearOptimize()        | 970 |
| 4.19.8 Description              | 970 |
| 4.19.8.1 TakeLongRoot()         | 971 |
| 4.19.8.2 TakeRatRoot()          | 971 |
| 4.19.8.3 MakeRational()         | 971 |
| 4.19.8.4 MakeLongRational()     | 971 |
| 4.19.8.5 ClearTableTree()       | 972 |
| 4.19.8.6 InsTableTree()         | 972 |
| 4.19.8.7 RedoTableTree()        | 972 |
| 4.19.8.8 FindTableTree()        | 972 |
| 4.19.8.9 finishcbuf()           | 972 |
| 4.19.8.10 clearcbuf()           | 973 |
| 4.19.8.11 CleanUpSort()         | 973 |
| 4.19.8.12 AllocFileHandle()     | 973 |
| 4.19.8.13 DeAllocFileHandle()   | 973 |
| 4.19.8.14 LowerSortLevel()      | 973 |
| 4.19.8.15 PolyRatFunSpecial()   | 974 |
| 4.19.8.16 SimpleSplitMergeRec() | 974 |
| 4.19.8.17 SimpleSplitMerge()    | 974 |
| 4.19.8.18 BinarySearch()        | 974 |
| 4.19.8.19 InsideDollar()        | 974 |
| 4.19.8.20 DolToTerms()          | 974 |
| 4.19.8.21 EvalDoLoopArg()       | 975 |
| 4.19.8.22 SetExprCases()        | 975 |
| 4.19.8.23 TestSelect()          | 975 |

|                                 |     |
|---------------------------------|-----|
| 4.19.8.24 SubInAll()            | 976 |
| 4.19.8.25 TransferBuffer()      | 976 |
| 4.19.8.26 TakeIDfunction()      | 976 |
| 4.19.8.27 MakeSetupAllocs()     | 976 |
| 4.19.8.28 TryFileSetups()       | 976 |
| 4.19.8.29 ExchangeExpressions() | 976 |
| 4.19.8.30 ExchangeDollars()     | 977 |
| 4.19.8.31 GetFirstBracket()     | 977 |
| 4.19.8.32 GetFirstTerm()        | 977 |
| 4.19.8.33 GetContent()          | 977 |
| 4.19.8.34 CleanupTerm()         | 978 |
| 4.19.8.35 ContentMerge()        | 978 |
| 4.19.8.36 PriefDollarEval()     | 978 |
| 4.19.8.37 TermsInDollar()       | 978 |
| 4.19.8.38 SizeOfDollar()        | 978 |
| 4.19.8.39 TermsInExpression()   | 978 |
| 4.19.8.40 SizeOfExpression()    | 978 |
| 4.19.8.41 TranslateExpression() | 979 |
| 4.19.8.42 IsSetMember()         | 979 |
| 4.19.8.43 IsMultipleOf()        | 979 |
| 4.19.8.44 TwoExprCompare()      | 979 |
| 4.19.8.45 UpdatePositions()     | 979 |
| 4.19.8.46 M_print()             | 979 |
| 4.19.8.47 M_check1()            | 980 |
| 4.19.8.48 PrintTime()           | 980 |
| 4.19.8.49 FindBracket()         | 980 |
| 4.19.8.50 PutBracketInIndex()   | 980 |
| 4.19.8.51 ClearBracketIndex()   | 980 |
| 4.19.8.52 OpenBracketIndex()    | 980 |
| 4.19.8.53 DoNoParallel()        | 980 |
| 4.19.8.54 DoParallel()          | 981 |
| 4.19.8.55 DoModSum()            | 981 |
| 4.19.8.56 DoModMax()            | 981 |
| 4.19.8.57 DoModMin()            | 981 |
| 4.19.8.58 DoModLocal()          | 981 |
| 4.19.8.59 DoModDollar()         | 981 |
| 4.19.8.60 DoProcessBucket()     | 981 |
| 4.19.8.61 DoinParallel()        | 982 |
| 4.19.8.62 DonotinParallel()     | 982 |
| 4.19.8.63 FlipTable()           | 982 |
| 4.19.8.64 ChainIn()             | 982 |
| 4.19.8.65 ChainOut()            | 982 |

|                                |     |
|--------------------------------|-----|
| 4.19.8.66 ArgumentImplode()    | 982 |
| 4.19.8.67 ArgumentExplode()    | 983 |
| 4.19.8.68 DenToFunction()      | 983 |
| 4.19.8.69 HowMany()            | 983 |
| 4.19.8.70 RemoveDollars()      | 983 |
| 4.19.8.71 CountTerms1()        | 983 |
| 4.19.8.72 TermsInBracket()     | 983 |
| 4.19.8.73 Crash()              | 984 |
| 4.19.8.74 str_dup()            | 984 |
| 4.19.8.75 convertblock()       | 984 |
| 4.19.8.76 convertnamesblock()  | 984 |
| 4.19.8.77 convertiniinfo()     | 984 |
| 4.19.8.78 ReadIndex()          | 984 |
| 4.19.8.79 WriteIndexBlock()    | 985 |
| 4.19.8.80 WriteNamesBlock()    | 985 |
| 4.19.8.81 WriteIndex()         | 985 |
| 4.19.8.82 WriteIniInfo()       | 985 |
| 4.19.8.83 ReadIniInfo()        | 985 |
| 4.19.8.84 AddToIndex()         | 985 |
| 4.19.8.85 GetDbase()           | 986 |
| 4.19.8.86 OpenDbase()          | 986 |
| 4.19.8.87 ReadObject()         | 986 |
| 4.19.8.88 ReadijObject()       | 986 |
| 4.19.8.89 ExistsObject()       | 986 |
| 4.19.8.90 DeleteObject()       | 986 |
| 4.19.8.91 WriteObject()        | 987 |
| 4.19.8.92 AddObject()          | 987 |
| 4.19.8.93 NewDbase()           | 987 |
| 4.19.8.94 FreeTableBase()      | 987 |
| 4.19.8.95 ComposeTableNames()  | 987 |
| 4.19.8.96 PutTableNames()      | 987 |
| 4.19.8.97 AddTableName()       | 988 |
| 4.19.8.98 GetTableName()       | 988 |
| 4.19.8.99 FindTableNumber()    | 988 |
| 4.19.8.100 TryEnvironment()    | 988 |
| 4.19.8.101 CopyExpression()    | 988 |
| 4.19.8.102 set_in()            | 988 |
| 4.19.8.103 set_set()           | 989 |
| 4.19.8.104 set_del()           | 989 |
| 4.19.8.105 set_sub()           | 989 |
| 4.19.8.106 DoPreAddSeparator() | 989 |
| 4.19.8.107 DoPreRmSeparator()  | 989 |

|                                         |      |
|-----------------------------------------|------|
| 4.19.8.108 openExternalChannel()        | 989  |
| 4.19.8.109 initPresetExternalChannels() | 990  |
| 4.19.8.110 closeExternalChannel()       | 990  |
| 4.19.8.111 selectExternalChannel()      | 990  |
| 4.19.8.112 getCurrentExternalChannel()  | 990  |
| 4.19.8.113 closeAllExternalChannels()   | 990  |
| 4.19.8.114 defineChannel()              | 990  |
| 4.19.8.115 writeToChannel()             | 991  |
| 4.19.8.116 ReleaseTB()                  | 991  |
| 4.19.8.117 SymbolNormalize()            | 991  |
| 4.19.8.118 TestFunFlag()                | 991  |
| 4.19.8.119 CompareSymbols()             | 991  |
| 4.19.8.120 CompareHSymbols()            | 992  |
| 4.19.8.121 NextPrime()                  | 992  |
| 4.19.8.122 Moebius()                    | 992  |
| 4.19.8.123 wranf()                      | 992  |
| 4.19.8.124 iranf()                      | 992  |
| 4.19.8.125 iniwranf()                   | 993  |
| 4.19.8.126 PreRandom()                  | 993  |
| 4.19.8.127 TreatPolyRatFun()            | 993  |
| 4.19.8.128 ReadSaveHeader()             | 993  |
| 4.19.8.129 ReadSaveIndex()              | 993  |
| 4.19.8.130 ReadSaveExpression()         | 994  |
| 4.19.8.131 ReadSaveTerm32()             | 995  |
| 4.19.8.132 ReadSaveVariables()          | 996  |
| 4.19.8.133 WriteStoreHeader()           | 997  |
| 4.19.8.134 InitRecovery()               | 997  |
| 4.19.8.135 CheckRecoveryFile()          | 997  |
| 4.19.8.136 DeleteRecoveryFile()         | 998  |
| 4.19.8.137 RecoveryFilename()           | 998  |
| 4.19.8.138 DoRecovery()                 | 998  |
| 4.19.8.139 DoCheckpoint()               | 999  |
| 4.19.8.140 NumberMallocAddMemory()      | 1000 |
| 4.19.8.141 CacheNumberMallocAddMemory() | 1000 |
| 4.19.8.142 TermMallocAddMemory()        | 1000 |
| 4.19.8.143 ExprStatus()                 | 1000 |
| 4.19.8.144 iniTools()                   | 1000 |
| 4.19.8.145 TestTerm()                   | 1000 |
| 4.19.8.146 RunTransform()               | 1001 |
| 4.19.8.147 RunEncode()                  | 1001 |
| 4.19.8.148 RunDecode()                  | 1001 |
| 4.19.8.149 RunReplace()                 | 1001 |

|                                 |      |
|---------------------------------|------|
| 4.19.8.150 RunImplode()         | 1002 |
| 4.19.8.151 RunExplode()         | 1002 |
| 4.19.8.152 TestArgNum()         | 1002 |
| 4.19.8.153 PutArgInScratch()    | 1002 |
| 4.19.8.154 ReadRange()          | 1002 |
| 4.19.8.155 FindRange()          | 1002 |
| 4.19.8.156 RunPermute()         | 1003 |
| 4.19.8.157 RunReverse()         | 1003 |
| 4.19.8.158 RunCycle()           | 1003 |
| 4.19.8.159 RunAddArg()          | 1003 |
| 4.19.8.160 RunMulArg()          | 1003 |
| 4.19.8.161 RunIsLyndon()        | 1003 |
| 4.19.8.162 RunToLyndon()        | 1004 |
| 4.19.8.163 RunDropArg()         | 1004 |
| 4.19.8.164 RunSelectArg()       | 1004 |
| 4.19.8.165 RunDedup()           | 1004 |
| 4.19.8.166 RunZtoHArg()         | 1004 |
| 4.19.8.167 RunHtoZArg()         | 1004 |
| 4.19.8.168 NormPolyTerm()       | 1005 |
| 4.19.8.169 ConvertToPoly()      | 1005 |
| 4.19.8.170 LocalConvertToPoly() | 1005 |
| 4.19.8.171 ConvertFromPoly()    | 1005 |
| 4.19.8.172 FindSubterm()        | 1005 |
| 4.19.8.173 FindLocalSubterm()   | 1006 |
| 4.19.8.174 PrintSubtermList()   | 1006 |
| 4.19.8.175 PrintExtraSymbol()   | 1006 |
| 4.19.8.176 FindSubexpression()  | 1006 |
| 4.19.8.177 UpdateMaxSize()      | 1006 |
| 4.19.8.178 CoToPolynomial()     | 1006 |
| 4.19.8.179 CoFromPolynomial()   | 1007 |
| 4.19.8.180 CoArgToExtraSymbol() | 1007 |
| 4.19.8.181 CoExtraSymbols()     | 1007 |
| 4.19.8.182 GetDoParam()         | 1007 |
| 4.19.8.183 GetIfDollarFactor()  | 1007 |
| 4.19.8.184 CoDo()               | 1007 |
| 4.19.8.185 CoEndDo()            | 1008 |
| 4.19.8.186 ExtraSymFun()        | 1008 |
| 4.19.8.187 PruneExtraSymbols()  | 1008 |
| 4.19.8.188 IniFbuffer()         | 1008 |
| 4.19.8.189 GCDfunction()        | 1008 |
| 4.19.8.190 GCDfunction3()       | 1008 |
| 4.19.8.191 ReadPolyRatFun()     | 1009 |

|                                         |      |
|-----------------------------------------|------|
| 4.19.8.192 FromPolyRatFun()             | 1009 |
| 4.19.8.193 PolyDiv()                    | 1009 |
| 4.19.8.194 GCDclean()                   | 1009 |
| 4.19.8.195 TakeSymbolContent()          | 1009 |
| 4.19.8.196 GCDterms()                   | 1010 |
| 4.19.8.197 PutExtraSymbols()            | 1010 |
| 4.19.8.198 TakeExtraSymbols()           | 1010 |
| 4.19.8.199 MultiplyWithTerm()           | 1010 |
| 4.19.8.200 TakeContent()                | 1010 |
| 4.19.8.201 MergeSymbolLists()           | 1011 |
| 4.19.8.202 MergeDotproductLists()       | 1011 |
| 4.19.8.203 CreateExpression()           | 1011 |
| 4.19.8.204 DIVfunction()                | 1011 |
| 4.19.8.205 MULfunc()                    | 1012 |
| 4.19.8.206 ConvertArgument()            | 1012 |
| 4.19.8.207 ExpandRat()                  | 1012 |
| 4.19.8.208 InvPoly()                    | 1012 |
| 4.19.8.209 TestDoLoop()                 | 1012 |
| 4.19.8.210 TestEndDoLoop()              | 1012 |
| 4.19.8.211 poly_gcd()                   | 1013 |
| 4.19.9 Description                      | 1013 |
| 4.19.10 Notes                           | 1013 |
| 4.19.10.1 poly_div()                    | 1013 |
| 4.19.10.2 poly_rem()                    | 1014 |
| 4.19.10.3 poly_inverse()                | 1014 |
| 4.19.10.4 poly_mul()                    | 1014 |
| 4.19.10.5 poly_ratfun_add()             | 1014 |
| 4.19.11 Description                     | 1014 |
| 4.19.12 Notes                           | 1015 |
| 4.19.12.1 poly_ratfun_normalize()       | 1015 |
| 4.19.13 Description                     | 1015 |
| 4.19.14 Notes                           | 1016 |
| 4.19.14.1 poly_factorize_argument()     | 1016 |
| 4.19.15 Description                     | 1016 |
| 4.19.16 Notes                           | 1017 |
| 4.19.16.1 poly_factorize_dollar()       | 1017 |
| 4.19.17 Description                     | 1017 |
| 4.19.18 Notes                           | 1017 |
| 4.19.18.1 poly_factorize_expression()   | 1018 |
| 4.19.19 Description                     | 1018 |
| 4.19.20 Notes                           | 1018 |
| 4.19.20.1 poly_unfactorize_expression() | 1019 |

|                                  |      |
|----------------------------------|------|
| 4.19.21 Description              | 1019 |
| 4.19.22 Notes                    | 1019 |
| 4.19.22.1 poly_free_poly_vars()  | 1020 |
| 4.19.22.2 optimize_print_code()  | 1020 |
| 4.19.23 Description              | 1020 |
| 4.19.23.1 DoPreAdd()             | 1020 |
| 4.19.23.2 DoPreUseDictionary()   | 1020 |
| 4.19.23.3 DoPreCloseDictionary() | 1020 |
| 4.19.23.4 DoPreOpenDictionary()  | 1021 |
| 4.19.23.5 RemoveDictionary()     | 1021 |
| 4.19.23.6 UnSetDictionary()      | 1021 |
| 4.19.23.7 SetDictionaryOptions() | 1021 |
| 4.19.23.8 AddToDictionary()      | 1021 |
| 4.19.23.9 AddDictionary()        | 1021 |
| 4.19.23.10 FindDictionary()      | 1021 |
| 4.19.23.11 IsExponentSign()      | 1022 |
| 4.19.23.12 IsMultiplySign()      | 1022 |
| 4.19.23.13 TransformRational()   | 1022 |
| 4.19.23.14 WriteDictionary()     | 1022 |
| 4.19.23.15 ShrinkDictionary()    | 1022 |
| 4.19.23.16 MultiplyToLine()      | 1022 |
| 4.19.23.17 FindSymbol()          | 1022 |
| 4.19.23.18 FindVector()          | 1023 |
| 4.19.23.19 FindIndex()           | 1023 |
| 4.19.23.20 FindFunction()        | 1023 |
| 4.19.23.21 FindFunWithArgs()     | 1023 |
| 4.19.23.22 FindExtraSymbol()     | 1023 |
| 4.19.23.23 DictToBytes()         | 1023 |
| 4.19.23.24 DictFromBytes()       | 1023 |
| 4.19.23.25 CoCreateSpectator()   | 1024 |
| 4.19.23.26 CoToSpectator()       | 1024 |
| 4.19.23.27 CoRemoveSpectator()   | 1024 |
| 4.19.23.28 CoEmptySpectator()    | 1024 |
| 4.19.23.29 CoCopySpectator()     | 1024 |
| 4.19.23.30 PutInSpectator()      | 1024 |
| 4.19.23.31 ClearSpectators()     | 1024 |
| 4.19.23.32 GetFromSpectator()    | 1025 |
| 4.19.23.33 FlushSpectators()     | 1025 |
| 4.19.23.34 ReadFromScratch()     | 1025 |
| 4.19.23.35 AddToScratch()        | 1025 |
| 4.19.23.36 DoPreAppendPath()     | 1025 |
| 4.19.23.37 DoPrePrependPath()    | 1026 |



|                                   |      |
|-----------------------------------|------|
| 4.19.23.38 DoSwitch()             | 1026 |
| 4.19.23.39 DoEndSwitch()          | 1026 |
| 4.19.23.40 FindCase()             | 1026 |
| 4.19.23.41 SwitchSplitMergeRec()  | 1026 |
| 4.19.23.42 SwitchSplitMerge()     | 1026 |
| 4.19.23.43 DoubleSwitchBuffers()  | 1027 |
| 4.19.23.44 DistrN()               | 1027 |
| 4.19.23.45 DoNamespace()          | 1027 |
| 4.19.23.46 DoEndNamespace()       | 1027 |
| 4.19.23.47 DoUse()                | 1027 |
| 4.19.23.48 SkipName()             | 1027 |
| 4.19.23.49 ConstructName()        | 1028 |
| 4.19.23.50 DoSetUserFlag()        | 1028 |
| 4.19.23.51 DoClearUserFlag()      | 1028 |
| 4.19.23.52 CreateVertex()         | 1028 |
| 4.19.23.53 ReadParticle()         | 1028 |
| 4.19.23.54 CoModel()              | 1028 |
| 4.19.23.55 CoParticle()           | 1029 |
| 4.19.23.56 CoVertex()             | 1029 |
| 4.19.23.57 CoEndModel()           | 1029 |
| 4.19.23.58 LoadModel()            | 1029 |
| 4.19.23.59 CoSetUserFlag()        | 1029 |
| 4.19.23.60 CoClearUserFlag()      | 1029 |
| 4.19.23.61 CoCreateAllLoops()     | 1029 |
| 4.19.23.62 CoCreateAllPaths()     | 1030 |
| 4.19.23.63 CoCreateAll()          | 1030 |
| 4.19.23.64 AllLoops()             | 1030 |
| 4.19.23.65 StartLoops()           | 1030 |
| 4.19.23.66 GenLoops()             | 1030 |
| 4.19.23.67 LoopOutput()           | 1031 |
| 4.19.23.68 AllPaths()             | 1031 |
| 4.19.23.69 GenPaths()             | 1031 |
| 4.19.23.70 PathOutput()           | 1031 |
| 4.20 declare.h                    | 1032 |
| 4.21 diagrams.c File Reference    | 1052 |
| 4.21.1 Detailed Description       | 1052 |
| 4.21.2 Function Documentation     | 1052 |
| 4.21.2.1 CoCanonicalize()         | 1052 |
| 4.21.2.2 DoCanonicalize()         | 1052 |
| 4.21.2.3 DoTopologyCanonicalize() | 1053 |
| 4.21.2.4 DoShattering()           | 1053 |
| 4.22 diagrams.c                   | 1053 |

|                                       |      |
|---------------------------------------|------|
| 4.23 diawrap.cc                       | 1061 |
| 4.24 dict.c File Reference            | 1074 |
| 4.24.1 Detailed Description           | 1075 |
| 4.24.2 Function Documentation         | 1075 |
| 4.24.2.1 TransformRational()          | 1075 |
| 4.24.2.2 IsMultiplySign()             | 1075 |
| 4.24.2.3 IsExponentSign()             | 1075 |
| 4.24.2.4 FindSymbol()                 | 1075 |
| 4.24.2.5 FindVector()                 | 1076 |
| 4.24.2.6 FindIndex()                  | 1076 |
| 4.24.2.7 FindFunction()               | 1076 |
| 4.24.2.8 FindFunWithArgs()            | 1076 |
| 4.24.2.9 FindExtraSymbol()            | 1076 |
| 4.24.2.10 FindDictionary()            | 1076 |
| 4.24.2.11 AddDictionary()             | 1076 |
| 4.24.2.12 AddToDictionary()           | 1077 |
| 4.24.2.13 UseDictionary()             | 1077 |
| 4.24.2.14 SetDictionaryOptions()      | 1077 |
| 4.24.2.15 UnSetDictionary()           | 1077 |
| 4.24.2.16 RemoveDictionary()          | 1077 |
| 4.24.2.17 ShrinkDictionary()          | 1077 |
| 4.24.2.18 DoPreOpenDictionary()       | 1078 |
| 4.24.2.19 DoPreCloseDictionary()      | 1078 |
| 4.24.2.20 DoPreUseDictionary()        | 1078 |
| 4.24.2.21 DoPreAdd()                  | 1078 |
| 4.24.2.22 DictToBytes()               | 1078 |
| 4.24.2.23 DictFromBytes()             | 1078 |
| 4.25 dict.c                           | 1079 |
| 4.26 dollar.c File Reference          | 1092 |
| 4.26.1 Detailed Description           | 1093 |
| 4.26.2 Macro Definition Documentation | 1093 |
| 4.26.2.1 STEP2                        | 1093 |
| 4.26.3 Function Documentation         | 1094 |
| 4.26.3.1 CatchDollar()                | 1094 |
| 4.26.3.2 AssignDollar()               | 1094 |
| 4.26.3.3 WriteDollarToBuffer()        | 1094 |
| 4.26.3.4 WriteDollarFactorToBuffer()  | 1094 |
| 4.26.3.5 AddToDollarBuffer()          | 1094 |
| 4.26.3.6 TermAssign()                 | 1094 |
| 4.26.3.7 PutTermInDollar()            | 1095 |
| 4.26.3.8 WildDollars()                | 1095 |
| 4.26.3.9 DoIToTensor()                | 1095 |

|                                       |      |
|---------------------------------------|------|
| 4.26.3.10 DolToFunction()             | 1095 |
| 4.26.3.11 DolToVector()               | 1095 |
| 4.26.3.12 DolToNumber()               | 1095 |
| 4.26.3.13 DolToSymbol()               | 1095 |
| 4.26.3.14 DolToIndex()                | 1096 |
| 4.26.3.15 DolToTerms()                | 1096 |
| 4.26.3.16 DolToLong()                 | 1096 |
| 4.26.3.17 ExecInside()                | 1096 |
| 4.26.3.18 InsideDollar()              | 1096 |
| 4.26.3.19 ExchangeDollars()           | 1096 |
| 4.26.3.20 TermsInDollar()             | 1096 |
| 4.26.3.21 SizeOfDollar()              | 1097 |
| 4.26.3.22 PriefDollarEval()           | 1097 |
| 4.26.3.23 TranslateExpression()       | 1097 |
| 4.26.3.24 IsSetMember()               | 1097 |
| 4.26.3.25 IsMultipleOf()              | 1097 |
| 4.26.3.26 TwoExprCompare()            | 1097 |
| 4.26.3.27 DollarRaiseLow()            | 1098 |
| 4.26.3.28 EvalDoLoopArg()             | 1098 |
| 4.26.3.29 TestDoLoop()                | 1098 |
| 4.26.3.30 TestEndDoLoop()             | 1099 |
| 4.26.3.31 DollarFactorize()           | 1099 |
| 4.26.3.32 CleanDollarFactors()        | 1099 |
| 4.26.3.33 TakeDollarContent()         | 1099 |
| 4.26.3.34 MakeDollarInteger()         | 1099 |
| 4.26.3.35 MakeDollarMod()             | 1100 |
| 4.26.3.36 GetDolNum()                 | 1101 |
| 4.26.3.37 AddPotModdollar()           | 1101 |
| 4.27 dollar.c                         | 1102 |
| 4.28 evaluate.c                       | 1147 |
| 4.29 execute.c File Reference         | 1156 |
| 4.29.1 Detailed Description           | 1156 |
| 4.29.2 Macro Definition Documentation | 1157 |
| 4.29.2.1 CURRENTBRACKET               | 1157 |
| 4.29.2.2 BRACKETCURRENTEXPR           | 1157 |
| 4.29.2.3 BRACKETOTHEREXPR             | 1157 |
| 4.29.2.4 NOBRACKETACTIVE              | 1157 |
| 4.29.3 Function Documentation         | 1157 |
| 4.29.3.1 CleanExpr()                  | 1157 |
| 4.29.3.2 PopVariables()               | 1157 |
| 4.29.3.3 MakeGlobal()                 | 1157 |
| 4.29.3.4 TestDrop()                   | 1158 |

|                                       |      |
|---------------------------------------|------|
| 4.29.3.5 PutInVflags()                | 1158 |
| 4.29.3.6 DoExecute()                  | 1158 |
| 4.29.3.7 PutBracket()                 | 1158 |
| 4.29.3.8 SpecialCleanup()             | 1158 |
| 4.29.3.9 SetMods()                    | 1158 |
| 4.29.3.10 UnSetMods()                 | 1158 |
| 4.29.3.11 ExchangeExpressions()       | 1159 |
| 4.29.3.12 GetFirstBracket()           | 1159 |
| 4.29.3.13 GetFirstTerm()              | 1159 |
| 4.29.3.14 GetContent()                | 1159 |
| 4.29.3.15 CleanupTerm()               | 1160 |
| 4.29.3.16 ContentMerge()              | 1160 |
| 4.29.3.17 TermsInExpression()         | 1160 |
| 4.29.3.18 SizeOfExpression()          | 1160 |
| 4.29.3.19 UpdatePositions()           | 1160 |
| 4.29.3.20 CountTerms1()               | 1160 |
| 4.29.3.21 TermsInBracket()            | 1160 |
| 4.30 execute.c                        | 1161 |
| 4.31 extcmd.c File Reference          | 1192 |
| 4.31.1 Detailed Description           | 1193 |
| 4.31.2 Function Documentation         | 1193 |
| 4.31.2.1 openExternalChannel()        | 1193 |
| 4.31.2.2 initPresetExternalChannels() | 1193 |
| 4.31.2.3 closeExternalChannel()       | 1193 |
| 4.31.2.4 selectExternalChannel()      | 1193 |
| 4.31.2.5 getCurrentExternalChannel()  | 1194 |
| 4.31.2.6 closeAllExternalChannels()   | 1194 |
| 4.32 extcmd.c                         | 1194 |
| 4.33 factor.c File Reference          | 1213 |
| 4.33.1 Detailed Description           | 1213 |
| 4.33.2 Function Documentation         | 1214 |
| 4.33.2.1 FactorIn()                   | 1214 |
| 4.33.2.2 FactorInExpr()               | 1214 |
| 4.34 factor.c                         | 1214 |
| 4.35 findpat.c File Reference         | 1224 |
| 4.35.1 Detailed Description           | 1225 |
| 4.35.2 Function Documentation         | 1225 |
| 4.35.2.1 FindOnly()                   | 1225 |
| 4.35.2.2 FindOnce()                   | 1225 |
| 4.35.2.3 FindMulti()                  | 1225 |
| 4.35.2.4 FindRest()                   | 1225 |
| 4.36 findpat.c                        | 1226 |

|                                                  |      |
|--------------------------------------------------|------|
| 4.37 flintinterface.cc File Reference . . . . .  | 1242 |
| 4.37.1 Detailed Description . . . . .            | 1242 |
| 4.37.2 Macro Definition Documentation . . . . .  | 1242 |
| 4.37.2.1 IFW . . . . .                           | 1242 |
| 4.38 flintinterface.cc . . . . .                 | 1243 |
| 4.39 flintinterface.h File Reference . . . . .   | 1268 |
| 4.39.1 Detailed Description . . . . .            | 1270 |
| 4.39.2 Typedef Documentation . . . . .           | 1270 |
| 4.39.2.1 var_map_t . . . . .                     | 1270 |
| 4.39.3 Function Documentation . . . . .          | 1270 |
| 4.39.3.1 cleanup() . . . . .                     | 1270 |
| 4.39.3.2 cleanup_master() . . . . .              | 1270 |
| 4.39.3.3 divmod_mpoly() . . . . .                | 1271 |
| 4.39.3.4 divmod_poly() . . . . .                 | 1271 |
| 4.39.3.5 factorize_mpoly() . . . . .             | 1271 |
| 4.39.3.6 factorize_poly() . . . . .              | 1271 |
| 4.39.3.7 form_sort() . . . . .                   | 1271 |
| 4.39.3.8 from_argument_mpoly() . . . . .         | 1272 |
| 4.39.3.9 from_argument_poly() . . . . .          | 1272 |
| 4.39.3.10 fmpz_get_form() . . . . .              | 1272 |
| 4.39.3.11 fmpz_set_form() . . . . .              | 1272 |
| 4.39.3.12 gcd_mpoly() . . . . .                  | 1272 |
| 4.39.3.13 gcd_poly() . . . . .                   | 1273 |
| 4.39.3.14 get_variables() . . . . .              | 1273 |
| 4.39.3.15 inverse_poly() . . . . .               | 1273 |
| 4.39.3.16 mul_mpoly() . . . . .                  | 1273 |
| 4.39.3.17 mul_poly() . . . . .                   | 1273 |
| 4.39.3.18 ratfun_add_mpoly() . . . . .           | 1274 |
| 4.39.3.19 ratfun_add_poly() . . . . .            | 1274 |
| 4.39.3.20 ratfun_normalize_mpoly() . . . . .     | 1274 |
| 4.39.3.21 ratfun_normalize_poly() . . . . .      | 1274 |
| 4.39.3.22 ratfun_read_mpoly() . . . . .          | 1274 |
| 4.39.3.23 ratfun_read_poly() . . . . .           | 1275 |
| 4.39.3.24 startup_init() . . . . .               | 1275 |
| 4.39.3.25 to_argument_mpoly() [1/2] . . . . .    | 1275 |
| 4.39.3.26 to_argument_mpoly() [2/2] . . . . .    | 1275 |
| 4.39.3.27 to_argument_poly() [1/2] . . . . .     | 1275 |
| 4.39.3.28 to_argument_poly() [2/2] . . . . .     | 1276 |
| 4.39.3.29 simplify_fmpz() . . . . .              | 1276 |
| 4.39.3.30 simplify_fmpz_poly() . . . . .         | 1276 |
| 4.39.3.31 fix_sign_fmpz_mpoly_ratfun() . . . . . | 1276 |
| 4.39.3.32 fix_sign_fmpz_poly_ratfun() . . . . .  | 1276 |

|                                                 |      |
|-------------------------------------------------|------|
| 4.40 flintinterface.h . . . . .                 | 1277 |
| 4.41 flintwrap.cc File Reference . . . . .      | 1279 |
| 4.41.1 Detailed Description . . . . .           | 1279 |
| 4.41.2 Function Documentation . . . . .         | 1279 |
| 4.41.2.1 flint_final_cleanup_thread() . . . . . | 1279 |
| 4.41.2.2 flint_final_cleanup_master() . . . . . | 1279 |
| 4.41.2.3 flint_div() . . . . .                  | 1280 |
| 4.41.2.4 flint_factorize_argument() . . . . .   | 1280 |
| 4.41.2.5 flint_factorize_dollar() . . . . .     | 1280 |
| 4.41.2.6 flint_gcd() . . . . .                  | 1280 |
| 4.41.2.7 flint_inverse() . . . . .              | 1280 |
| 4.41.2.8 flint_mul() . . . . .                  | 1280 |
| 4.41.2.9 flint_ratfun_add() . . . . .           | 1281 |
| 4.41.2.10 flint_ratfun_normalize() . . . . .    | 1281 |
| 4.41.2.11 flint_rem() . . . . .                 | 1281 |
| 4.41.2.12 flint_startup_init() . . . . .        | 1281 |
| 4.42 flintwrap.cc . . . . .                     | 1281 |
| 4.43 float.c File Reference . . . . .           | 1285 |
| 4.43.1 Detailed Description . . . . .           | 1286 |
| 4.43.2 Macro Definition Documentation . . . . . | 1286 |
| 4.43.2.1 WITHCUTOFF . . . . .                   | 1286 |
| 4.43.2.2 GMPSREAD . . . . .                     | 1286 |
| 4.43.3 Function Documentation . . . . .         | 1287 |
| 4.43.3.1 Form_mpf_init() . . . . .              | 1287 |
| 4.43.3.2 Form_mpf_clear() . . . . .             | 1287 |
| 4.43.3.3 Form_mpf_set_prec_raw() . . . . .      | 1287 |
| 4.43.3.4 FormtoZ() . . . . .                    | 1287 |
| 4.43.3.5 ZtoForm() . . . . .                    | 1287 |
| 4.43.3.6 FloatToInteger() . . . . .             | 1287 |
| 4.43.3.7 IntegerToFloat() . . . . .             | 1288 |
| 4.43.3.8 FloatToRat() . . . . .                 | 1288 |
| 4.43.3.9 AddFloats() . . . . .                  | 1288 |
| 4.43.3.10 MulFloats() . . . . .                 | 1288 |
| 4.43.3.11 DivFloats() . . . . .                 | 1288 |
| 4.43.3.12 AddRatToFloat() . . . . .             | 1289 |
| 4.43.3.13 MulRatToFloat() . . . . .             | 1289 |
| 4.43.3.14 SimpleDelta() . . . . .               | 1289 |
| 4.43.3.15 SimpleDeltaC() . . . . .              | 1289 |
| 4.43.3.16 SingleTable() . . . . .               | 1289 |
| 4.43.3.17 DoubleTable() . . . . .               | 1290 |
| 4.43.3.18 EndTable() . . . . .                  | 1290 |
| 4.43.3.19 deltaMZV() . . . . .                  | 1290 |

|           |                                |      |
|-----------|--------------------------------|------|
| 4.43.3.20 | deltaEuler()                   | 1290 |
| 4.43.3.21 | deltaEulerC()                  | 1290 |
| 4.43.3.22 | CalculateMZVhalf()             | 1291 |
| 4.43.3.23 | CalculateMZV()                 | 1291 |
| 4.43.3.24 | CalculateEuler()               | 1291 |
| 4.43.3.25 | ExpandMZV()                    | 1291 |
| 4.43.3.26 | ExpandEuler()                  | 1291 |
| 4.43.3.27 | PackFloat()                    | 1291 |
| 4.43.3.28 | UnpackFloat()                  | 1292 |
| 4.43.3.29 | RatToFloat()                   | 1292 |
| 4.43.3.30 | Form_mpf_empty()               | 1292 |
| 4.43.3.31 | TestFloat()                    | 1292 |
| 4.43.3.32 | FloatFunToRat()                | 1292 |
| 4.43.3.33 | ZeroTable()                    | 1292 |
| 4.43.3.34 | ReadFloat()                    | 1293 |
| 4.43.3.35 | CheckFloat()                   | 1293 |
| 4.43.3.36 | SetFloatPrecision()            | 1293 |
| 4.43.3.37 | PrintFloat()                   | 1293 |
| 4.43.3.38 | SetupMZVTables()               | 1293 |
| 4.43.3.39 | SetupMPFTables()               | 1293 |
| 4.43.3.40 | ClearMZVTables()               | 1293 |
| 4.43.3.41 | CoToFloat()                    | 1294 |
| 4.43.3.42 | CoToRat()                      | 1294 |
| 4.43.3.43 | ToFloat()                      | 1294 |
| 4.43.3.44 | ToRat()                        | 1294 |
| 4.43.3.45 | AddWithFloat()                 | 1294 |
| 4.43.3.46 | MergeWithFloat()               | 1294 |
| 4.43.3.47 | EvaluateEuler()                | 1295 |
| 4.43.3.48 | CoEvaluate()                   | 1295 |
| 4.44      | float.c                        | 1295 |
| 4.45      | form3.h File Reference         | 1324 |
| 4.45.1    | Detailed Description           | 1326 |
| 4.45.2    | Macro Definition Documentation | 1326 |
| 4.45.2.1  | MAJORVERSION                   | 1326 |
| 4.45.2.2  | MINORVERSION                   | 1326 |
| 4.45.2.3  | PRODUCTIONDATE                 | 1326 |
| 4.45.2.4  | BETAVERSION                    | 1327 |
| 4.45.2.5  | inline                         | 1327 |
| 4.45.2.6  | NDEBUG                         | 1327 |
| 4.45.2.7  | STATIC_ASSERT                  | 1327 |
| 4.45.2.8  | STATIC_ASSERT__1               | 1327 |
| 4.45.2.9  | STATIC_ASSERT__2               | 1327 |

|                            |      |
|----------------------------|------|
| 4.45.2.10 STATIC_ASSERT__3 | 1327 |
| 4.45.2.11 BITSINWORD       | 1328 |
| 4.45.2.12 BITSINLONG       | 1328 |
| 4.45.2.13 WORD_MIN_VALUE   | 1328 |
| 4.45.2.14 WORD_MAX_VALUE   | 1328 |
| 4.45.2.15 LONG_MIN_VALUE   | 1328 |
| 4.45.2.16 LONG_MAX_VALUE   | 1328 |
| 4.45.2.17 TOPBITONLY       | 1328 |
| 4.45.2.18 TOPLONGBITONLY   | 1328 |
| 4.45.2.19 SPECMASK         | 1329 |
| 4.45.2.20 WILDMASK         | 1329 |
| 4.45.2.21 WORDMASK         | 1329 |
| 4.45.2.22 AWORDMASK        | 1329 |
| 4.45.2.23 FULLMAX          | 1329 |
| 4.45.2.24 MAXPOSITIVE      | 1329 |
| 4.45.2.25 MAXLONG          | 1329 |
| 4.45.2.26 MAXPOSITIVE2     | 1329 |
| 4.45.2.27 MAXPOSITIVE4     | 1330 |
| 4.45.2.28 form_alignof     | 1330 |
| 4.45.2.29 PADDUMMY         | 1330 |
| 4.45.2.30 PADPOSITION      | 1330 |
| 4.45.2.31 PADPOINTER       | 1330 |
| 4.45.2.32 PADLONG          | 1331 |
| 4.45.2.33 PADINT           | 1331 |
| 4.45.2.34 PADWORD          | 1331 |
| 4.45.2.35 WITHSORTBOTS     | 1331 |
| 4.45.2.36 FILES            | 1331 |
| 4.45.2.37 Uopen            | 1332 |
| 4.45.2.38 Uflush           | 1332 |
| 4.45.2.39 Uclose           | 1332 |
| 4.45.2.40 Uread            | 1332 |
| 4.45.2.41 Uwrite           | 1332 |
| 4.45.2.42 Usetbuf          | 1332 |
| 4.45.2.43 Useek            | 1333 |
| 4.45.2.44 Utell            | 1333 |
| 4.45.2.45 Ugetpos          | 1333 |
| 4.45.2.46 Usetpos          | 1333 |
| 4.45.2.47 Usync            | 1333 |
| 4.45.2.48 Utruncate        | 1333 |
| 4.45.2.49 Ustdout          | 1334 |
| 4.45.2.50 MAX_OPEN_FILES   | 1334 |
| 4.45.2.51 bzero            | 1334 |



|                                                 |      |
|-------------------------------------------------|------|
| 4.45.2.52 GetPID . . . . .                      | 1334 |
| 4.45.3 Typedef Documentation . . . . .          | 1334 |
| 4.45.3.1 WORD . . . . .                         | 1334 |
| 4.45.3.2 LONG . . . . .                         | 1334 |
| 4.45.3.3 UWORD . . . . .                        | 1334 |
| 4.45.3.4 ULONG . . . . .                        | 1335 |
| 4.45.3.5 SBYTE . . . . .                        | 1335 |
| 4.45.3.6 UBYTE . . . . .                        | 1335 |
| 4.45.3.7 UINT . . . . .                         | 1335 |
| 4.45.3.8 RLONG . . . . .                        | 1335 |
| 4.45.3.9 MLONG . . . . .                        | 1335 |
| 4.45.3.10 BOOL . . . . .                        | 1335 |
| 4.46 form3.h . . . . .                          | 1336 |
| 4.47 fsizes.h File Reference . . . . .          | 1341 |
| 4.47.1 Detailed Description . . . . .           | 1343 |
| 4.47.2 Macro Definition Documentation . . . . . | 1343 |
| 4.47.2.1 MAXPRENAMESIZE . . . . .               | 1343 |
| 4.47.2.2 MAXENAME . . . . .                     | 1343 |
| 4.47.2.3 MAXSAVEFUNCTION . . . . .              | 1343 |
| 4.47.2.4 MAXPARLEVEL . . . . .                  | 1343 |
| 4.47.2.5 MAXNUMBERSIZE . . . . .                | 1343 |
| 4.47.2.6 MAXREPEAT . . . . .                    | 1343 |
| 4.47.2.7 NORMSIZE . . . . .                     | 1344 |
| 4.47.2.8 INITNODESIZE . . . . .                 | 1344 |
| 4.47.2.9 INITNAMESIZE . . . . .                 | 1344 |
| 4.47.2.10 NUMFIXED . . . . .                    | 1344 |
| 4.47.2.11 MAXNEST . . . . .                     | 1344 |
| 4.47.2.12 MAXMATCH . . . . .                    | 1344 |
| 4.47.2.13 MAXIF . . . . .                       | 1344 |
| 4.47.2.14 SIZEFACS . . . . .                    | 1344 |
| 4.47.2.15 NUMFACS . . . . .                     | 1345 |
| 4.47.2.16 MAXLOOPS . . . . .                    | 1345 |
| 4.47.2.17 MAXLABELS . . . . .                   | 1345 |
| 4.47.2.18 COMMERCIALSIZE . . . . .              | 1345 |
| 4.47.2.19 MAXFLAGS . . . . .                    | 1345 |
| 4.47.2.20 COMPRESSBUFFER . . . . .              | 1345 |
| 4.47.2.21 FORTRANCONTINUATIONLINES . . . . .    | 1345 |
| 4.47.2.22 MAXLEVELS . . . . .                   | 1345 |
| 4.47.2.23 MAXLHS . . . . .                      | 1346 |
| 4.47.2.24 MAXWILDC . . . . .                    | 1346 |
| 4.47.2.25 NUMTABLEENTRIES . . . . .             | 1346 |
| 4.47.2.26 COMPILERBUFFER . . . . .              | 1346 |

|                                                |      |
|------------------------------------------------|------|
| 4.47.2.27 MAXPATCHES . . . . .                 | 1346 |
| 4.47.2.28 MAXFPATCHES . . . . .                | 1346 |
| 4.47.2.29 SORTIOSIZE . . . . .                 | 1346 |
| 4.47.2.30 SSMALLBUFFER . . . . .               | 1346 |
| 4.47.2.31 SSMALLOVERFLOW . . . . .             | 1347 |
| 4.47.2.32 STERMSSMALL . . . . .                | 1347 |
| 4.47.2.33 SLARGEBUFFER . . . . .               | 1347 |
| 4.47.2.34 SMAXPATCHES . . . . .                | 1347 |
| 4.47.2.35 SMAXFPATCHES . . . . .               | 1347 |
| 4.47.2.36 SSORTIOSIZE . . . . .                | 1347 |
| 4.47.2.37 SPECTATORSIZE . . . . .              | 1347 |
| 4.47.2.38 MAXFLEVELS . . . . .                 | 1347 |
| 4.47.2.39 COMPINC . . . . .                    | 1348 |
| 4.47.2.40 MAXNUMSIZE . . . . .                 | 1348 |
| 4.47.2.41 MAXBRACKETBUFFERSIZE . . . . .       | 1348 |
| 4.47.2.42 SFHSIZE . . . . .                    | 1348 |
| 4.47.2.43 DEFAULTPROCESSBUCKETSIZE . . . . .   | 1348 |
| 4.47.2.44 SHMWINSIZE . . . . .                 | 1348 |
| 4.47.2.45 TABLEEXTENSION . . . . .             | 1348 |
| 4.47.2.46 GZIPDEFAULT . . . . .                | 1348 |
| 4.47.2.47 DEFAULTTHREADS . . . . .             | 1349 |
| 4.47.2.48 DEFAULTTHREADBUCKETSIZE . . . . .    | 1349 |
| 4.47.2.49 DEFAULTTHREADLOADBALANCING . . . . . | 1349 |
| 4.47.2.50 THREADSCRATCHSIZE . . . . .          | 1349 |
| 4.47.2.51 THREADSCRATCHOUTSIZE . . . . .       | 1349 |
| 4.47.2.52 NUMSTORECACHES . . . . .             | 1349 |
| 4.47.2.53 SIZESTORECACHE . . . . .             | 1349 |
| 4.47.2.54 INDENTSPACE . . . . .                | 1349 |
| 4.47.2.55 MULTIINDENTSPACE . . . . .           | 1350 |
| 4.47.2.56 MAXMULTIBRACKETLEVELS . . . . .      | 1350 |
| 4.47.2.57 FBUFFERSIZE . . . . .                | 1350 |
| 4.47.2.58 NPAIR1 . . . . .                     | 1350 |
| 4.47.2.59 NPAIR2 . . . . .                     | 1350 |
| 4.47.2.60 MAXLINELENGTH . . . . .              | 1350 |
| 4.47.2.61 MINALLOC . . . . .                   | 1350 |
| 4.47.2.62 JUMPRATIO . . . . .                  | 1350 |
| 4.47.2.63 MAXPARTICLES . . . . .               | 1351 |
| 4.47.2.64 MAXCOUPLINGS . . . . .               | 1351 |
| 4.47.2.65 NUMOPTIONS . . . . .                 | 1351 |
| 4.47.2.66 MAXLEGS . . . . .                    | 1351 |
| 4.48 fsizes.h . . . . .                        | 1351 |
| 4.49 ftypes.h File Reference . . . . .         | 1354 |

|                                                 |      |
|-------------------------------------------------|------|
| 4.49.1 Detailed Description . . . . .           | 1366 |
| 4.49.2 Macro Definition Documentation . . . . . | 1366 |
| 4.49.2.1 WITHOUTERROR . . . . .                 | 1366 |
| 4.49.2.2 WITHERROR . . . . .                    | 1366 |
| 4.49.2.3 FILESTREAM . . . . .                   | 1366 |
| 4.49.2.4 PREVARSTREAM . . . . .                 | 1366 |
| 4.49.2.5 PREREADSTREAM . . . . .                | 1366 |
| 4.49.2.6 PIPESTREAM . . . . .                   | 1367 |
| 4.49.2.7 PRECALCSTREAM . . . . .                | 1367 |
| 4.49.2.8 DOLLARSTREAM . . . . .                 | 1367 |
| 4.49.2.9 PREREADSTREAM2 . . . . .               | 1367 |
| 4.49.2.10 EXTERNALCHANNELSTREAM . . . . .       | 1367 |
| 4.49.2.11 PREREADSTREAM3 . . . . .              | 1367 |
| 4.49.2.12 REVERSEFILESTREAM . . . . .           | 1367 |
| 4.49.2.13 INPUTSTREAM . . . . .                 | 1367 |
| 4.49.2.14 ENDOFSTREAM . . . . .                 | 1368 |
| 4.49.2.15 ENDOFINPUT . . . . .                  | 1368 |
| 4.49.2.16 SUBROUTINEFILE . . . . .              | 1368 |
| 4.49.2.17 PROCEDUREFILE . . . . .               | 1368 |
| 4.49.2.18 HEADERFILE . . . . .                  | 1368 |
| 4.49.2.19 SETUPFILE . . . . .                   | 1368 |
| 4.49.2.20 TABLEBASEFILE . . . . .               | 1368 |
| 4.49.2.21 FIRSTMODULE . . . . .                 | 1368 |
| 4.49.2.22 GLOBALMODULE . . . . .                | 1369 |
| 4.49.2.23 SORTMODULE . . . . .                  | 1369 |
| 4.49.2.24 STOREMODULE . . . . .                 | 1369 |
| 4.49.2.25 CLEARMODULE . . . . .                 | 1369 |
| 4.49.2.26 ENDMODULE . . . . .                   | 1369 |
| 4.49.2.27 POLYFUN . . . . .                     | 1369 |
| 4.49.2.28 NOPARALLEL_DOLLAR . . . . .           | 1369 |
| 4.49.2.29 NOPARALLEL_RHS . . . . .              | 1369 |
| 4.49.2.30 NOPARALLEL_CONVPOLY . . . . .         | 1370 |
| 4.49.2.31 NOPARALLEL_SPECTATOR . . . . .        | 1370 |
| 4.49.2.32 NOPARALLEL_USER . . . . .             | 1370 |
| 4.49.2.33 NOPARALLEL_TBLDOLLAR . . . . .        | 1370 |
| 4.49.2.34 NOPARALLEL_NPROC . . . . .            | 1370 |
| 4.49.2.35 PARALLELFLAG . . . . .                | 1370 |
| 4.49.2.36 PRENOACTION . . . . .                 | 1370 |
| 4.49.2.37 PRERAISEAFTER . . . . .               | 1370 |
| 4.49.2.38 PRELOWERAFTER . . . . .               | 1371 |
| 4.49.2.39 WITHSEMICOLON . . . . .               | 1371 |
| 4.49.2.40 WITHOUTSEMICOLON . . . . .            | 1371 |

|                                           |      |
|-------------------------------------------|------|
| 4.49.2.41 MODULEINSTACK . . . . .         | 1371 |
| 4.49.2.42 EXECUTINGIF . . . . .           | 1371 |
| 4.49.2.43 LOOKINGFORELSE . . . . .        | 1371 |
| 4.49.2.44 LOOKINGFORENDIF . . . . .       | 1371 |
| 4.49.2.45 NEWSTATEMENT . . . . .          | 1371 |
| 4.49.2.46 OLDSTATEMENT . . . . .          | 1372 |
| 4.49.2.47 EXECUTINGPRESWITCH . . . . .    | 1372 |
| 4.49.2.48 SEARCHINGPRECASE . . . . .      | 1372 |
| 4.49.2.49 SEARCHINGPREENDSWITCH . . . . . | 1372 |
| 4.49.2.50 PREPROONLY . . . . .            | 1372 |
| 4.49.2.51 DUMPTOCOMPILER . . . . .        | 1372 |
| 4.49.2.52 DUMPOUTTERMS . . . . .          | 1372 |
| 4.49.2.53 DUMPINTERMS . . . . .           | 1372 |
| 4.49.2.54 DUMPTOSORT . . . . .            | 1373 |
| 4.49.2.55 DUMPTOPARALLEL . . . . .        | 1373 |
| 4.49.2.56 THREADSDEBUG . . . . .          | 1373 |
| 4.49.2.57 ERROROUT . . . . .              | 1373 |
| 4.49.2.58 INPUTOUT . . . . .              | 1373 |
| 4.49.2.59 STATSOUT . . . . .              | 1373 |
| 4.49.2.60 EXPRSOUT . . . . .              | 1373 |
| 4.49.2.61 WRITEOUT . . . . .              | 1373 |
| 4.49.2.62 EXTERNALCHANNELOUT . . . . .    | 1374 |
| 4.49.2.63 NUMERICALLOOP . . . . .         | 1374 |
| 4.49.2.64 LISTEDLOOP . . . . .            | 1374 |
| 4.49.2.65 ONEEXPRESSION . . . . .         | 1374 |
| 4.49.2.66 PRETYPENONE . . . . .           | 1374 |
| 4.49.2.67 PRETYPEIF . . . . .             | 1374 |
| 4.49.2.68 PRETYPEDO . . . . .             | 1374 |
| 4.49.2.69 PRETYPEPROCEDURE . . . . .      | 1374 |
| 4.49.2.70 PRETYPESWITCH . . . . .         | 1375 |
| 4.49.2.71 PRETYPEINSIDE . . . . .         | 1375 |
| 4.49.2.72 DECLARATION . . . . .           | 1375 |
| 4.49.2.73 SPECIFICATION . . . . .         | 1375 |
| 4.49.2.74 DEFINITION . . . . .            | 1375 |
| 4.49.2.75 STATEMENT . . . . .             | 1375 |
| 4.49.2.76 TOOOUTPUT . . . . .             | 1375 |
| 4.49.2.77 ATENDOFMODULE . . . . .         | 1375 |
| 4.49.2.78 MIXED2 . . . . .                | 1376 |
| 4.49.2.79 MIXED . . . . .                 | 1376 |
| 4.49.2.80 NOAUTO . . . . .                | 1376 |
| 4.49.2.81 PARTEST . . . . .               | 1376 |
| 4.49.2.82 WITHAUTO . . . . .              | 1376 |

|                                    |      |
|------------------------------------|------|
| 4.49.2.83 ALLVARIABLES . . . . .   | 1376 |
| 4.49.2.84 SYMBOLONLY . . . . .     | 1376 |
| 4.49.2.85 INDEXONLY . . . . .      | 1376 |
| 4.49.2.86 VECTORONLY . . . . .     | 1377 |
| 4.49.2.87 FUNCTIONONLY . . . . .   | 1377 |
| 4.49.2.88 SETONLY . . . . .        | 1377 |
| 4.49.2.89 EXPRESSIONONLY . . . . . | 1377 |
| 4.49.2.90 CDELETE . . . . .        | 1377 |
| 4.49.2.91 ANYTYPE . . . . .        | 1377 |
| 4.49.2.92 CSYMBOL . . . . .        | 1377 |
| 4.49.2.93 CINDEX . . . . .         | 1377 |
| 4.49.2.94 CVECTOR . . . . .        | 1378 |
| 4.49.2.95 CFUNCTION . . . . .      | 1378 |
| 4.49.2.96 CSET . . . . .           | 1378 |
| 4.49.2.97 CEXPRESSION . . . . .    | 1378 |
| 4.49.2.98 CDOTPRODUCT . . . . .    | 1378 |
| 4.49.2.99 CNUMBER . . . . .        | 1378 |
| 4.49.2.100 CSUBEXP . . . . .       | 1378 |
| 4.49.2.101 CDELTA . . . . .        | 1378 |
| 4.49.2.102 CDOLLAR . . . . .       | 1379 |
| 4.49.2.103 CDUBIOUS . . . . .      | 1379 |
| 4.49.2.104 CRANGE . . . . .        | 1379 |
| 4.49.2.105 CMODEL . . . . .        | 1379 |
| 4.49.2.106 CVECTOR1 . . . . .      | 1379 |
| 4.49.2.107 CDOUBLEDOT . . . . .    | 1379 |
| 4.49.2.108 TSYMBOL . . . . .       | 1379 |
| 4.49.2.109 TINDEX . . . . .        | 1379 |
| 4.49.2.110 TVECTOR . . . . .       | 1380 |
| 4.49.2.111 TFUNCTION . . . . .     | 1380 |
| 4.49.2.112 TSET . . . . .          | 1380 |
| 4.49.2.113 TEXPRESSION . . . . .   | 1380 |
| 4.49.2.114 TDOTPRODUCT . . . . .   | 1380 |
| 4.49.2.115 TNUMBER . . . . .       | 1380 |
| 4.49.2.116 TSUBEXP . . . . .       | 1380 |
| 4.49.2.117 TDELTA . . . . .        | 1380 |
| 4.49.2.118 TDOLLAR . . . . .       | 1381 |
| 4.49.2.119 TDUBIOUS . . . . .      | 1381 |
| 4.49.2.120 LPARENTHESIS . . . . .  | 1381 |
| 4.49.2.121 RPARENTHESIS . . . . .  | 1381 |
| 4.49.2.122 TWILDCARD . . . . .     | 1381 |
| 4.49.2.123 TWILDARG . . . . .      | 1381 |
| 4.49.2.124 TDOT . . . . .          | 1381 |

|                                             |      |
|---------------------------------------------|------|
| 4.49.2.125 LBRACE . . . . .                 | 1381 |
| 4.49.2.126 RBRACE . . . . .                 | 1382 |
| 4.49.2.127 TCOMMA . . . . .                 | 1382 |
| 4.49.2.128 TFUNOPEN . . . . .               | 1382 |
| 4.49.2.129 TFUNCLOSE . . . . .              | 1382 |
| 4.49.2.130 TMULTIPLY . . . . .              | 1382 |
| 4.49.2.131 TDIVIDE . . . . .                | 1382 |
| 4.49.2.132 TPOWER . . . . .                 | 1382 |
| 4.49.2.133 TPLUS . . . . .                  | 1382 |
| 4.49.2.134 TMINUS . . . . .                 | 1383 |
| 4.49.2.135 TNOT . . . . .                   | 1383 |
| 4.49.2.136 TENDOFIT . . . . .               | 1383 |
| 4.49.2.137 TSETOPEN . . . . .               | 1383 |
| 4.49.2.138 TSETCLOSE . . . . .              | 1383 |
| 4.49.2.139 TGENINDEX . . . . .              | 1383 |
| 4.49.2.140 TCONJUGATE . . . . .             | 1383 |
| 4.49.2.141 LRPARENTHESSES . . . . .         | 1383 |
| 4.49.2.142 TNUMBER1 . . . . .               | 1384 |
| 4.49.2.143 TPOWER1 . . . . .                | 1384 |
| 4.49.2.144 TEMPTY . . . . .                 | 1384 |
| 4.49.2.145 TSETNUM . . . . .                | 1384 |
| 4.49.2.146 TSGAMMA . . . . .                | 1384 |
| 4.49.2.147 TSETDOL . . . . .                | 1384 |
| 4.49.2.148 TFLOAT . . . . .                 | 1384 |
| 4.49.2.149 TYPEISFUN . . . . .              | 1384 |
| 4.49.2.150 TYPEISSUB . . . . .              | 1385 |
| 4.49.2.151 TYPEISMYSTERY . . . . .          | 1385 |
| 4.49.2.152 LHSIDEX . . . . .                | 1385 |
| 4.49.2.153 LHSIDE . . . . .                 | 1385 |
| 4.49.2.154 RHSIDE . . . . .                 | 1385 |
| 4.49.2.155 FORTRANMODE . . . . .            | 1385 |
| 4.49.2.156 REDUCEMODE . . . . .             | 1385 |
| 4.49.2.157 MAPLEMODE . . . . .              | 1385 |
| 4.49.2.158 MATHEMATICAMODE . . . . .        | 1386 |
| 4.49.2.159 CMODE . . . . .                  | 1386 |
| 4.49.2.160 VORTRANMODE . . . . .            | 1386 |
| 4.49.2.161 PFORTRANMODE . . . . .           | 1386 |
| 4.49.2.162 DOUBLEFORTRANMODE . . . . .      | 1386 |
| 4.49.2.163 DOUBLEPRECISIONFLAG . . . . .    | 1386 |
| 4.49.2.164 NODOUBLEMASK . . . . .           | 1386 |
| 4.49.2.165 QUADRUPLEFORTRANMODE . . . . .   | 1386 |
| 4.49.2.166 QUADRUPLEPRECISIONFLAG . . . . . | 1387 |

|                                       |      |
|---------------------------------------|------|
| 4.49.2.167 ALLINTEGERDOUBLE . . . . . | 1387 |
| 4.49.2.168 NOQUADMASK . . . . .       | 1387 |
| 4.49.2.169 NORMALFORMAT . . . . .     | 1387 |
| 4.49.2.170 NOSPACEFORMAT . . . . .    | 1387 |
| 4.49.2.171 ISNOTFORTRAN90 . . . . .   | 1387 |
| 4.49.2.172 ISFORTRAN90 . . . . .      | 1387 |
| 4.49.2.173 ALSOREVERSE . . . . .      | 1387 |
| 4.49.2.174 CHISHOLM . . . . .         | 1388 |
| 4.49.2.175 NOTRICK . . . . .          | 1388 |
| 4.49.2.176 SORTLOWFIRST . . . . .     | 1388 |
| 4.49.2.177 SORTHIGHFIRST . . . . .    | 1388 |
| 4.49.2.178 SORTPOWERFIRST . . . . .   | 1388 |
| 4.49.2.179 SORTANTIPOWER . . . . .    | 1388 |
| 4.49.2.180 STATSSPLITMERGE . . . . .  | 1388 |
| 4.49.2.181 STATSMERGETOFILE . . . . . | 1388 |
| 4.49.2.182 STATSPOSTSORT . . . . .    | 1389 |
| 4.49.2.183 NOCHECKLOGTYPE . . . . .   | 1389 |
| 4.49.2.184 CHECKLOGTYPE . . . . .     | 1389 |
| 4.49.2.185 NMIN4SHIFT . . . . .       | 1389 |
| 4.49.2.186 SYMBOL . . . . .           | 1389 |
| 4.49.2.187 DOTPRODUCT . . . . .       | 1389 |
| 4.49.2.188 VECTOR . . . . .           | 1389 |
| 4.49.2.189 INDEX . . . . .            | 1389 |
| 4.49.2.190 EXPRESSION . . . . .       | 1390 |
| 4.49.2.191 SUBEXPRESSION . . . . .    | 1390 |
| 4.49.2.192 DOLLAREXPRESSION . . . . . | 1390 |
| 4.49.2.193 SETSET . . . . .           | 1390 |
| 4.49.2.194 ARGWILD . . . . .          | 1390 |
| 4.49.2.195 MINVECTOR . . . . .        | 1390 |
| 4.49.2.196 SETEXP . . . . .           | 1390 |
| 4.49.2.197 DOLLAREXPR2 . . . . .      | 1390 |
| 4.49.2.198 HAAKJE0 . . . . .          | 1391 |
| 4.49.2.199 FUNCTION . . . . .         | 1391 |
| 4.49.2.200 TMPPOLYFUN . . . . .       | 1391 |
| 4.49.2.201 ARGFIELD . . . . .         | 1391 |
| 4.49.2.202 SNUMBER . . . . .          | 1391 |
| 4.49.2.203 LNUMBER . . . . .          | 1391 |
| 4.49.2.204 HAAKJE . . . . .           | 1391 |
| 4.49.2.205 DELTA . . . . .            | 1391 |
| 4.49.2.206 EXPONENT . . . . .         | 1392 |
| 4.49.2.207 DENOMINATOR . . . . .      | 1392 |
| 4.49.2.208 SETFUNCTION . . . . .      | 1392 |

|                                        |      |
|----------------------------------------|------|
| 4.49.2.209 GAMMA . . . . .             | 1392 |
| 4.49.2.210 GAMMAI . . . . .            | 1392 |
| 4.49.2.211 GAMMAFIVE . . . . .         | 1392 |
| 4.49.2.212 GAMMASIX . . . . .          | 1392 |
| 4.49.2.213 GAMMASEVEN . . . . .        | 1392 |
| 4.49.2.214 SUMF1 . . . . .             | 1393 |
| 4.49.2.215 SUMF2 . . . . .             | 1393 |
| 4.49.2.216 DUMFUN . . . . .            | 1393 |
| 4.49.2.217 REPLACEMENT . . . . .       | 1393 |
| 4.49.2.218 REVERSEFUNCTION . . . . .   | 1393 |
| 4.49.2.219 DISTRIBUTION . . . . .      | 1393 |
| 4.49.2.220 DELTA3 . . . . .            | 1393 |
| 4.49.2.221 DUMMYFUN . . . . .          | 1393 |
| 4.49.2.222 DUMMYTEN . . . . .          | 1394 |
| 4.49.2.223 LEVICIVITA . . . . .        | 1394 |
| 4.49.2.224 FACTORIAL . . . . .         | 1394 |
| 4.49.2.225 INVERSEFACTORIAL . . . . .  | 1394 |
| 4.49.2.226 BINOMIAL . . . . .          | 1394 |
| 4.49.2.227 NUMARGSFUN . . . . .        | 1394 |
| 4.49.2.228 SIGNFUN . . . . .           | 1394 |
| 4.49.2.229 MODFUNCTION . . . . .       | 1394 |
| 4.49.2.230 MOD2FUNCTION . . . . .      | 1395 |
| 4.49.2.231 MINFUNCTION . . . . .       | 1395 |
| 4.49.2.232 MAXFUNCTION . . . . .       | 1395 |
| 4.49.2.233 ABSFUNCTION . . . . .       | 1395 |
| 4.49.2.234 SIGFUNCTION . . . . .       | 1395 |
| 4.49.2.235 INTFUNCTION . . . . .       | 1395 |
| 4.49.2.236 THETA . . . . .             | 1395 |
| 4.49.2.237 THETA2 . . . . .            | 1395 |
| 4.49.2.238 DELTA2 . . . . .            | 1396 |
| 4.49.2.239 DELTAP . . . . .            | 1396 |
| 4.49.2.240 BERNOULLIFUNCTION . . . . . | 1396 |
| 4.49.2.241 COUNTFUNCTION . . . . .     | 1396 |
| 4.49.2.242 MATCHFUNCTION . . . . .     | 1396 |
| 4.49.2.243 PATTERNFUNCTION . . . . .   | 1396 |
| 4.49.2.244 TERMFUNCTION . . . . .      | 1396 |
| 4.49.2.245 CONJUGATION . . . . .       | 1396 |
| 4.49.2.246 ROOTFUNCTION . . . . .      | 1397 |
| 4.49.2.247 TABLEFUNCTION . . . . .     | 1397 |
| 4.49.2.248 FIRSTBRACKET . . . . .      | 1397 |
| 4.49.2.249 TERMSINEXPR . . . . .       | 1397 |
| 4.49.2.250 NUMTERMSFUN . . . . .       | 1397 |



|                                      |      |
|--------------------------------------|------|
| 4.49.2.251 GCDFUNCTION . . . . .     | 1397 |
| 4.49.2.252 DIVFUNCTION . . . . .     | 1397 |
| 4.49.2.253 REMFUNCTION . . . . .     | 1397 |
| 4.49.2.254 MAXPOWEROF . . . . .      | 1398 |
| 4.49.2.255 MINPOWEROF . . . . .      | 1398 |
| 4.49.2.256 TABLESTUB . . . . .       | 1398 |
| 4.49.2.257 FACTORIN . . . . .        | 1398 |
| 4.49.2.258 TERMSINBRACKET . . . . .  | 1398 |
| 4.49.2.259 WILDARGFUN . . . . .      | 1398 |
| 4.49.2.260 SQRTFUNCTION . . . . .    | 1398 |
| 4.49.2.261 LNFUNCTION . . . . .      | 1398 |
| 4.49.2.262 SINFUNCTION . . . . .     | 1399 |
| 4.49.2.263 COSFUNCTION . . . . .     | 1399 |
| 4.49.2.264 TANFUNCTION . . . . .     | 1399 |
| 4.49.2.265 ASINFUNCTION . . . . .    | 1399 |
| 4.49.2.266 ACOSFUNCTION . . . . .    | 1399 |
| 4.49.2.267 ATANFUNCTION . . . . .    | 1399 |
| 4.49.2.268 ATAN2FUNCTION . . . . .   | 1399 |
| 4.49.2.269 SINHFUNCTION . . . . .    | 1399 |
| 4.49.2.270 COSHFUNCTION . . . . .    | 1400 |
| 4.49.2.271 TANHFUNCTION . . . . .    | 1400 |
| 4.49.2.272 ASINHFUNCTION . . . . .   | 1400 |
| 4.49.2.273 ACOSHFUNCTION . . . . .   | 1400 |
| 4.49.2.274 ATANHFUNCTION . . . . .   | 1400 |
| 4.49.2.275 LI2FUNCTION . . . . .     | 1400 |
| 4.49.2.276 LINFUNCTION . . . . .     | 1400 |
| 4.49.2.277 EXTRASYMFUN . . . . .     | 1400 |
| 4.49.2.278 RANDOMFUNCTION . . . . .  | 1401 |
| 4.49.2.279 RANPERM . . . . .         | 1401 |
| 4.49.2.280 NUMFACTORS . . . . .      | 1401 |
| 4.49.2.281 FIRSTTERM . . . . .       | 1401 |
| 4.49.2.282 CONTENTTERM . . . . .     | 1401 |
| 4.49.2.283 PRIMENUMBER . . . . .     | 1401 |
| 4.49.2.284 EXTEUCLIDEAN . . . . .    | 1401 |
| 4.49.2.285 MAKERATIONAL . . . . .    | 1401 |
| 4.49.2.286 INVERSEFUNCTION . . . . . | 1402 |
| 4.49.2.287 IDFUNCTION . . . . .      | 1402 |
| 4.49.2.288 PUTFIRST . . . . .        | 1402 |
| 4.49.2.289 PERMUTATIONS . . . . .    | 1402 |
| 4.49.2.290 PARTITIONS . . . . .      | 1402 |
| 4.49.2.291 MULFUNCTION . . . . .     | 1402 |
| 4.49.2.292 MOEBIUS . . . . .         | 1402 |

|                                         |      |
|-----------------------------------------|------|
| 4.49.2.293 TOPOLOGIES . . . . .         | 1402 |
| 4.49.2.294 DIAGRAMS . . . . .           | 1403 |
| 4.49.2.295 TOPO . . . . .               | 1403 |
| 4.49.2.296 NODEFUNCTION . . . . .       | 1403 |
| 4.49.2.297 EDGE . . . . .               | 1403 |
| 4.49.2.298 SIZEOFFUNCTION . . . . .     | 1403 |
| 4.49.2.299 BLOCK . . . . .              | 1403 |
| 4.49.2.300 ONEPI . . . . .              | 1403 |
| 4.49.2.301 PHI . . . . .                | 1403 |
| 4.49.2.302 MAXBUILTINFUNCTION . . . . . | 1404 |
| 4.49.2.303 FIRSTUSERFUNCTION . . . . .  | 1404 |
| 4.49.2.304 ALLFUNCTIONS . . . . .       | 1404 |
| 4.49.2.305 ALLMZVFUNCTIONS . . . . .    | 1404 |
| 4.49.2.306 ISYMBOL . . . . .            | 1404 |
| 4.49.2.307 PISYMBOL . . . . .           | 1404 |
| 4.49.2.308 COEFFSYMBOL . . . . .        | 1404 |
| 4.49.2.309 NUMERATORSYMBOL . . . . .    | 1404 |
| 4.49.2.310 DENOMINATORSYMBOL . . . . .  | 1405 |
| 4.49.2.311 WILDARGSYMBOL . . . . .      | 1405 |
| 4.49.2.312 DIMENSIONSYMBOL . . . . .    | 1405 |
| 4.49.2.313 FACTORSYMBOL . . . . .       | 1405 |
| 4.49.2.314 SEPARATESYMBOL . . . . .     | 1405 |
| 4.49.2.315 EESYMBOL . . . . .           | 1405 |
| 4.49.2.316 EMSYMBOL . . . . .           | 1405 |
| 4.49.2.317 BUILTINSYMBOLS . . . . .     | 1405 |
| 4.49.2.318 FIRSTUSERSYMBOL . . . . .    | 1406 |
| 4.49.2.319 BUILTININDICES . . . . .     | 1406 |
| 4.49.2.320 BUILTINVECTORS . . . . .     | 1406 |
| 4.49.2.321 BUILTINDOLLARS . . . . .     | 1406 |
| 4.49.2.322 WILDARGVECTOR . . . . .      | 1406 |
| 4.49.2.323 WILDARGINDEX . . . . .       | 1406 |
| 4.49.2.324 TYPEEXPRESSION . . . . .     | 1406 |
| 4.49.2.325 TYPEIDNEW . . . . .          | 1406 |
| 4.49.2.326 TYPEIDOLD . . . . .          | 1407 |
| 4.49.2.327 TYPEOPERATION . . . . .      | 1407 |
| 4.49.2.328 TYPEREPEAT . . . . .         | 1407 |
| 4.49.2.329 TYPEENDREPEAT . . . . .      | 1407 |
| 4.49.2.330 TYPECOUNT . . . . .          | 1407 |
| 4.49.2.331 TYPEMULT . . . . .           | 1407 |
| 4.49.2.332 TYPEGOTO . . . . .           | 1407 |
| 4.49.2.333 TYPEDISCARD . . . . .        | 1407 |
| 4.49.2.334 TYPEIF . . . . .             | 1408 |

|                                        |      |
|----------------------------------------|------|
| 4.49.2.335 TYPEELSE . . . . .          | 1408 |
| 4.49.2.336 TYPEELIF . . . . .          | 1408 |
| 4.49.2.337 TYPEENDIF . . . . .         | 1408 |
| 4.49.2.338 TYPESUM . . . . .           | 1408 |
| 4.49.2.339 TYPECHISHOLM . . . . .      | 1408 |
| 4.49.2.340 TYPEREVERSE . . . . .       | 1408 |
| 4.49.2.341 TYPEARG . . . . .           | 1408 |
| 4.49.2.342 TYPENORM . . . . .          | 1409 |
| 4.49.2.343 TYPENORM2 . . . . .         | 1409 |
| 4.49.2.344 TYPENORM3 . . . . .         | 1409 |
| 4.49.2.345 TYPEEXIT . . . . .          | 1409 |
| 4.49.2.346 TYPESETEXIT . . . . .       | 1409 |
| 4.49.2.347 TYPEPRINT . . . . .         | 1409 |
| 4.49.2.348 TYPEFPRINT . . . . .        | 1409 |
| 4.49.2.349 TYPEREDEFPRE . . . . .      | 1409 |
| 4.49.2.350 TYPESPLITARG . . . . .      | 1410 |
| 4.49.2.351 TYPESPLITARG2 . . . . .     | 1410 |
| 4.49.2.352 TYPEFACTARG . . . . .       | 1410 |
| 4.49.2.353 TYPEFACTARG2 . . . . .      | 1410 |
| 4.49.2.354 TYPETRY . . . . .           | 1410 |
| 4.49.2.355 TYPEASSIGN . . . . .        | 1410 |
| 4.49.2.356 TYPERENUMBER . . . . .      | 1410 |
| 4.49.2.357 TYPESUMFIX . . . . .        | 1410 |
| 4.49.2.358 TYPEFINDLOOP . . . . .      | 1411 |
| 4.49.2.359 TYPEUNRAVEL . . . . .       | 1411 |
| 4.49.2.360 TYPEADJUSTBOUNDS . . . . .  | 1411 |
| 4.49.2.361 TYPEINSIDE . . . . .        | 1411 |
| 4.49.2.362 TYPETERM . . . . .          | 1411 |
| 4.49.2.363 TYPESORT . . . . .          | 1411 |
| 4.49.2.364 TYPEDETCURDUM . . . . .     | 1411 |
| 4.49.2.365 TYPEINEXPRESSION . . . . .  | 1411 |
| 4.49.2.366 TYPESPLITFIRSTARG . . . . . | 1412 |
| 4.49.2.367 TYPESPLITLASTARG . . . . .  | 1412 |
| 4.49.2.368 TYPEMERGE . . . . .         | 1412 |
| 4.49.2.369 TYPETESTUSE . . . . .       | 1412 |
| 4.49.2.370 TYPEAPPLY . . . . .         | 1412 |
| 4.49.2.371 TYPEAPPLYRESET . . . . .    | 1412 |
| 4.49.2.372 TYPECHAININ . . . . .       | 1412 |
| 4.49.2.373 TYPECHAINOUT . . . . .      | 1412 |
| 4.49.2.374 TYPENORM4 . . . . .         | 1413 |
| 4.49.2.375 TYPEFACTOR . . . . .        | 1413 |
| 4.49.2.376 TYPEARGIMPLODE . . . . .    | 1413 |

|                                           |      |
|-------------------------------------------|------|
| 4.49.2.377 TYPEARGEXPLODE . . . . .       | 1413 |
| 4.49.2.378 TYPEDENOMINATORS . . . . .     | 1413 |
| 4.49.2.379 TYPESTUFFLE . . . . .          | 1413 |
| 4.49.2.380 TYPEDROPCOEFFICIENT . . . . .  | 1413 |
| 4.49.2.381 TYPETRANSFORM . . . . .        | 1413 |
| 4.49.2.382 TYPETOPOLYNOMIAL . . . . .     | 1414 |
| 4.49.2.383 TYPEFROMPOLYNOMIAL . . . . .   | 1414 |
| 4.49.2.384 TYPEDOLOOP . . . . .           | 1414 |
| 4.49.2.385 TYPEENDDOLOOP . . . . .        | 1414 |
| 4.49.2.386 TYPEDROPSYMBOLS . . . . .      | 1414 |
| 4.49.2.387 TYPEPUTINSIDE . . . . .        | 1414 |
| 4.49.2.388 TYPETOSPECTATOR . . . . .      | 1414 |
| 4.49.2.389 TYPEARGTOEXTRASymbol . . . . . | 1414 |
| 4.49.2.390 TYPECANONICALIZE . . . . .     | 1415 |
| 4.49.2.391 TYPESWITCH . . . . .           | 1415 |
| 4.49.2.392 TYPEENDSWITCH . . . . .        | 1415 |
| 4.49.2.393 TYPESETUSERFLAG . . . . .      | 1415 |
| 4.49.2.394 TYPECLEARUSERFLAG . . . . .    | 1415 |
| 4.49.2.395 TYPEALLLOOPS . . . . .         | 1415 |
| 4.49.2.396 TYPEALLPATHS . . . . .         | 1415 |
| 4.49.2.397 TAKETRACE . . . . .            | 1415 |
| 4.49.2.398 CONTRACT . . . . .             | 1416 |
| 4.49.2.399 RATIO . . . . .                | 1416 |
| 4.49.2.400 SYMMETRIZE . . . . .           | 1416 |
| 4.49.2.401 TENVEC . . . . .               | 1416 |
| 4.49.2.402 SUMNUM1 . . . . .              | 1416 |
| 4.49.2.403 SUMNUM2 . . . . .              | 1416 |
| 4.49.2.404 WILDDUMMY . . . . .            | 1416 |
| 4.49.2.405 SYMTONUM . . . . .             | 1416 |
| 4.49.2.406 SYMTOSYM . . . . .             | 1417 |
| 4.49.2.407 SYMTOSUB . . . . .             | 1417 |
| 4.49.2.408 VECTOMIN . . . . .             | 1417 |
| 4.49.2.409 VECTOVEC . . . . .             | 1417 |
| 4.49.2.410 VECTOSUB . . . . .             | 1417 |
| 4.49.2.411 INDTOIND . . . . .             | 1417 |
| 4.49.2.412 INDTO SUB . . . . .            | 1417 |
| 4.49.2.413 FUNTOFUN . . . . .             | 1417 |
| 4.49.2.414 ARGTOARG . . . . .             | 1418 |
| 4.49.2.415 ARLTOARL . . . . .             | 1418 |
| 4.49.2.416 EXPTOEXP . . . . .             | 1418 |
| 4.49.2.417 FROMBRAC . . . . .             | 1418 |
| 4.49.2.418 FROMSET . . . . .              | 1418 |

|                                        |      |
|----------------------------------------|------|
| 4.49.2.419 SETTONUM . . . . .          | 1418 |
| 4.49.2.420 WILDCARDS . . . . .         | 1418 |
| 4.49.2.421 SETNUMBER . . . . .         | 1418 |
| 4.49.2.422 LOADDOLLAR . . . . .        | 1419 |
| 4.49.2.423 NUMTONUM . . . . .          | 1419 |
| 4.49.2.424 NUMTOSYM . . . . .          | 1419 |
| 4.49.2.425 NUMTOIND . . . . .          | 1419 |
| 4.49.2.426 NUMTOSUB . . . . .          | 1419 |
| 4.49.2.427 EATTENSOR . . . . .         | 1419 |
| 4.49.2.428 CLEANFLAG . . . . .         | 1419 |
| 4.49.2.429 DIRTYFLAG . . . . .         | 1419 |
| 4.49.2.430 DIRTYSYMFLAG . . . . .      | 1420 |
| 4.49.2.431 MUSTCLEANPRF . . . . .      | 1420 |
| 4.49.2.432 SUBTERMUSED1 . . . . .      | 1420 |
| 4.49.2.433 SUBTERMUSED2 . . . . .      | 1420 |
| 4.49.2.434 ALLDIRTY . . . . .          | 1420 |
| 4.49.2.435 ARGHEAD . . . . .           | 1420 |
| 4.49.2.436 FUNHEAD . . . . .           | 1420 |
| 4.49.2.437 SUBEXPSIZE . . . . .        | 1420 |
| 4.49.2.438 EXPRHEAD . . . . .          | 1421 |
| 4.49.2.439 TYPEARGHEADSIZE . . . . .   | 1421 |
| 4.49.2.440 SKIP . . . . .              | 1421 |
| 4.49.2.441 DROP . . . . .              | 1421 |
| 4.49.2.442 HIDE . . . . .              | 1421 |
| 4.49.2.443 UNHIDE . . . . .            | 1421 |
| 4.49.2.444 INTOHIDE . . . . .          | 1421 |
| 4.49.2.445 LOCALEXPRESSION . . . . .   | 1421 |
| 4.49.2.446 SKIPLEXPRESSION . . . . .   | 1422 |
| 4.49.2.447 DROPLEXPRESSION . . . . .   | 1422 |
| 4.49.2.448 DROPEDEXPRESSION . . . . .  | 1422 |
| 4.49.2.449 GLOBALEXPRESSION . . . . .  | 1422 |
| 4.49.2.450 SKIPGEXPRESSION . . . . .   | 1422 |
| 4.49.2.451 DROPGEXPRESSION . . . . .   | 1422 |
| 4.49.2.452 STOREDEXPRESSION . . . . .  | 1422 |
| 4.49.2.453 HIDDENLEXPRESSION . . . . . | 1422 |
| 4.49.2.454 HIDDENGEXPRESSION . . . . . | 1423 |
| 4.49.2.455 INCEXPRESSION . . . . .     | 1423 |
| 4.49.2.456 HIDELEXPRESSION . . . . .   | 1423 |
| 4.49.2.457 HIDEGEXPRESSION . . . . .   | 1423 |
| 4.49.2.458 DROPHLEXPRESSION . . . . .  | 1423 |
| 4.49.2.459 DROPHGEXPRESSION . . . . .  | 1423 |
| 4.49.2.460 UNHIDELEXPRESSION . . . . . | 1423 |

|                                              |      |
|----------------------------------------------|------|
| 4.49.2.461 UNHIDEGEXPRESSION . . . . .       | 1423 |
| 4.49.2.462 INTOHIDELEXPRESSION . . . . .     | 1424 |
| 4.49.2.463 INTOHIDEGEXPRESSION . . . . .     | 1424 |
| 4.49.2.464 SPECTATOREXPRESSION . . . . .     | 1424 |
| 4.49.2.465 DROPSPECTATOREXPRESSION . . . . . | 1424 |
| 4.49.2.466 SKIPUNHIDELEXPRESSION . . . . .   | 1424 |
| 4.49.2.467 SKIPUNHIDEGEXPRESSION . . . . .   | 1424 |
| 4.49.2.468 PRINTOFF . . . . .                | 1424 |
| 4.49.2.469 PRINTON . . . . .                 | 1424 |
| 4.49.2.470 PRINTCONTENTS . . . . .           | 1425 |
| 4.49.2.471 PRINTCONTENT . . . . .            | 1425 |
| 4.49.2.472 PRINTLFILE . . . . .              | 1425 |
| 4.49.2.473 PRINTONETERM . . . . .            | 1425 |
| 4.49.2.474 PRINTONEFUNCTION . . . . .        | 1425 |
| 4.49.2.475 PRINTALL . . . . .                | 1425 |
| 4.49.2.476 REGULAREXPRESSION . . . . .       | 1425 |
| 4.49.2.477 REDEFINEDEXPRESSION . . . . .     | 1425 |
| 4.49.2.478 NEWLYDEFINEDEXPRESSION . . . . .  | 1426 |
| 4.49.2.479 VERTEXFUNCTION . . . . .          | 1426 |
| 4.49.2.480 GENERALFUNCTION . . . . .         | 1426 |
| 4.49.2.481 TENSORFUNCTION . . . . .          | 1426 |
| 4.49.2.482 GAMMAFUNCTION . . . . .           | 1426 |
| 4.49.2.483 POS_ . . . . .                    | 1426 |
| 4.49.2.484 POS0_ . . . . .                   | 1426 |
| 4.49.2.485 NEG_ . . . . .                    | 1426 |
| 4.49.2.486 NEG0_ . . . . .                   | 1427 |
| 4.49.2.487 EVEN_ . . . . .                   | 1427 |
| 4.49.2.488 ODD_ . . . . .                    | 1427 |
| 4.49.2.489 Z_ . . . . .                      | 1427 |
| 4.49.2.490 SYMBOL_ . . . . .                 | 1427 |
| 4.49.2.491 FIXED_ . . . . .                  | 1427 |
| 4.49.2.492 INDEX_ . . . . .                  | 1427 |
| 4.49.2.493 Q_ . . . . .                      | 1427 |
| 4.49.2.494 DUMMYINDEX_ . . . . .             | 1428 |
| 4.49.2.495 VECTOR_ . . . . .                 | 1428 |
| 4.49.2.496 GAMMA1 . . . . .                  | 1428 |
| 4.49.2.497 GAMMA5 . . . . .                  | 1428 |
| 4.49.2.498 GAMMA6 . . . . .                  | 1428 |
| 4.49.2.499 GAMMA7 . . . . .                  | 1428 |
| 4.49.2.500 FUNNYVEC . . . . .                | 1428 |
| 4.49.2.501 FUNNYWILD . . . . .               | 1428 |
| 4.49.2.502 SUMMEDIND . . . . .               | 1429 |

|                                         |      |
|-----------------------------------------|------|
| 4.49.2.503 NOINDEX . . . . .            | 1429 |
| 4.49.2.504 FUNNYDOLLAR . . . . .        | 1429 |
| 4.49.2.505 EMPTYINDEX . . . . .         | 1429 |
| 4.49.2.506 MINSPEC . . . . .            | 1429 |
| 4.49.2.507 USEDFLAG . . . . .           | 1429 |
| 4.49.2.508 DUMMYFLAG . . . . .          | 1429 |
| 4.49.2.509 MAINSORT . . . . .           | 1429 |
| 4.49.2.510 FUNCTIONSORT . . . . .       | 1430 |
| 4.49.2.511 SUBSORT . . . . .            | 1430 |
| 4.49.2.512 FLOATMODE . . . . .          | 1430 |
| 4.49.2.513 RATIONALMODE . . . . .       | 1430 |
| 4.49.2.514 NUMSPECSETS . . . . .        | 1430 |
| 4.49.2.515 ISZERO . . . . .             | 1430 |
| 4.49.2.516 ISUNMODIFIED . . . . .       | 1430 |
| 4.49.2.517 ISCOMPRESSED . . . . .       | 1430 |
| 4.49.2.518 ISINRHS . . . . .            | 1431 |
| 4.49.2.519 ISFACTORIZED . . . . .       | 1431 |
| 4.49.2.520 TOBEFACTORED . . . . .       | 1431 |
| 4.49.2.521 TOBEUNFACTORED . . . . .     | 1431 |
| 4.49.2.522 KEEPZERO . . . . .           | 1431 |
| 4.49.2.523 VARTYPENONE . . . . .        | 1431 |
| 4.49.2.524 VARTYPECOMPLEX . . . . .     | 1431 |
| 4.49.2.525 VARTYPEIMAGINARY . . . . .   | 1431 |
| 4.49.2.526 VARTYPEROOTOFUNITY . . . . . | 1432 |
| 4.49.2.527 VARTYPEMINUS . . . . .       | 1432 |
| 4.49.2.528 CYCLESYMMETRIC . . . . .     | 1432 |
| 4.49.2.529 RCYCLESYMMETRIC . . . . .    | 1432 |
| 4.49.2.530 SYMMETRIC . . . . .          | 1432 |
| 4.49.2.531 ANTISYMMETRIC . . . . .      | 1432 |
| 4.49.2.532 REVERSEORDER . . . . .       | 1432 |
| 4.49.2.533 SUBMULTI . . . . .           | 1432 |
| 4.49.2.534 SUBONCE . . . . .            | 1433 |
| 4.49.2.535 SUBONLY . . . . .            | 1433 |
| 4.49.2.536 SUBMANY . . . . .            | 1433 |
| 4.49.2.537 SUBVECTOR . . . . .          | 1433 |
| 4.49.2.538 SUBSELECT . . . . .          | 1433 |
| 4.49.2.539 SUBALL . . . . .             | 1433 |
| 4.49.2.540 SUBMASK . . . . .            | 1433 |
| 4.49.2.541 SUBDISORDER . . . . .        | 1433 |
| 4.49.2.542 SUBAFTER . . . . .           | 1434 |
| 4.49.2.543 SUBAFTERNOT . . . . .        | 1434 |
| 4.49.2.544 IDHEAD . . . . .             | 1434 |

|                                     |      |
|-------------------------------------|------|
| 4.49.2.545 DOLLARFLAG . . . . .     | 1434 |
| 4.49.2.546 NORMALIZEFLAG . . . . .  | 1434 |
| 4.49.2.547 GIDENT . . . . .         | 1434 |
| 4.49.2.548 GFIVE . . . . .          | 1434 |
| 4.49.2.549 GPLUS . . . . .          | 1434 |
| 4.49.2.550 GMINUS . . . . .         | 1435 |
| 4.49.2.551 LONGNUMBER . . . . .     | 1435 |
| 4.49.2.552 MATCH . . . . .          | 1435 |
| 4.49.2.553 COEFFI . . . . .         | 1435 |
| 4.49.2.554 SUBEXPR . . . . .        | 1435 |
| 4.49.2.555 MULTIPLEOF . . . . .     | 1435 |
| 4.49.2.556 IFDOLLAR . . . . .       | 1435 |
| 4.49.2.557 IFEXPRESSION . . . . .   | 1435 |
| 4.49.2.558 IFDOLLAREXTRA . . . . .  | 1436 |
| 4.49.2.559 IFISFACTORIZED . . . . . | 1436 |
| 4.49.2.560 IFOCCURS . . . . .       | 1436 |
| 4.49.2.561 IFUSERFLAG . . . . .     | 1436 |
| 4.49.2.562 IFFLOATNUMBER . . . . .  | 1436 |
| 4.49.2.563 GREATER . . . . .        | 1436 |
| 4.49.2.564 GREATEREQUAL . . . . .   | 1436 |
| 4.49.2.565 LESS . . . . .           | 1436 |
| 4.49.2.566 LESSEQUAL . . . . .      | 1437 |
| 4.49.2.567 EQUAL . . . . .          | 1437 |
| 4.49.2.568 NOTEQUAL . . . . .       | 1437 |
| 4.49.2.569 ORCOND . . . . .         | 1437 |
| 4.49.2.570 ANDCOND . . . . .        | 1437 |
| 4.49.2.571 DUMMY . . . . .          | 1437 |
| 4.49.2.572 SORT . . . . .           | 1437 |
| 4.49.2.573 STORE . . . . .          | 1437 |
| 4.49.2.574 END . . . . .            | 1438 |
| 4.49.2.575 GLOBAL . . . . .         | 1438 |
| 4.49.2.576 CLEAR . . . . .          | 1438 |
| 4.49.2.577 VECTBIT . . . . .        | 1438 |
| 4.49.2.578 DOTPBIT . . . . .        | 1438 |
| 4.49.2.579 FUNBIT . . . . .         | 1438 |
| 4.49.2.580 SETBIT . . . . .         | 1438 |
| 4.49.2.581 EXTRAPARAMETER . . . . . | 1438 |
| 4.49.2.582 GENCOMMUTE . . . . .     | 1439 |
| 4.49.2.583 GENNONCOMMUTE . . . . .  | 1439 |
| 4.49.2.584 NAMENOTFOUND . . . . .   | 1439 |
| 4.49.2.585 DOLUNDEFINED . . . . .   | 1439 |
| 4.49.2.586 DOLNUMBER . . . . .      | 1439 |



|                                       |      |
|---------------------------------------|------|
| 4.49.2.587 DOLARGUMENT . . . . .      | 1439 |
| 4.49.2.588 DOLSUBTERM . . . . .       | 1439 |
| 4.49.2.589 DOLTERMS . . . . .         | 1439 |
| 4.49.2.590 DOLWILDARGS . . . . .      | 1440 |
| 4.49.2.591 DOLINDEX . . . . .         | 1440 |
| 4.49.2.592 DOLZERO . . . . .          | 1440 |
| 4.49.2.593 FINDLOOP . . . . .         | 1440 |
| 4.49.2.594 REPLACELOOP . . . . .      | 1440 |
| 4.49.2.595 NOFUNPOWERS . . . . .      | 1440 |
| 4.49.2.596 COMFUNPOWERS . . . . .     | 1440 |
| 4.49.2.597 ALLFUNPOWERS . . . . .     | 1440 |
| 4.49.2.598 PROPERORDERFLAG . . . . .  | 1441 |
| 4.49.2.599 REGULAR . . . . .          | 1441 |
| 4.49.2.600 FINISH . . . . .           | 1441 |
| 4.49.2.601 POLYADD . . . . .          | 1441 |
| 4.49.2.602 POLYSUB . . . . .          | 1441 |
| 4.49.2.603 POLYMUL . . . . .          | 1441 |
| 4.49.2.604 POLYDIV . . . . .          | 1441 |
| 4.49.2.605 POLYREM . . . . .          | 1441 |
| 4.49.2.606 POLYGCD . . . . .          | 1442 |
| 4.49.2.607 POLYINTFAC . . . . .       | 1442 |
| 4.49.2.608 POLYNORM . . . . .         | 1442 |
| 4.49.2.609 MODNONE . . . . .          | 1442 |
| 4.49.2.610 MODSUM . . . . .           | 1442 |
| 4.49.2.611 MODMAX . . . . .           | 1442 |
| 4.49.2.612 MODMIN . . . . .           | 1442 |
| 4.49.2.613 MODLOCAL . . . . .         | 1442 |
| 4.49.2.614 ELEMENTUSED . . . . .      | 1443 |
| 4.49.2.615 ELEMENTLOADED . . . . .    | 1443 |
| 4.49.2.616 POSNEG . . . . .           | 1443 |
| 4.49.2.617 INVERSETABLE . . . . .     | 1443 |
| 4.49.2.618 COEFFICIENTSONLY . . . . . | 1443 |
| 4.49.2.619 ALSOPOWERS . . . . .       | 1443 |
| 4.49.2.620 ALSOFUNARGS . . . . .      | 1443 |
| 4.49.2.621 ALSODOLLARS . . . . .      | 1443 |
| 4.49.2.622 NOINVERSES . . . . .       | 1444 |
| 4.49.2.623 POSITIVEONLY . . . . .     | 1444 |
| 4.49.2.624 UNPACK . . . . .           | 1444 |
| 4.49.2.625 NOUNPACK . . . . .         | 1444 |
| 4.49.2.626 FROMFUNCTION . . . . .     | 1444 |
| 4.49.2.627 VARNAMES . . . . .         | 1444 |
| 4.49.2.628 AUTONAMES . . . . .        | 1444 |

|                                           |      |
|-------------------------------------------|------|
| 4.49.2.629 EXPRNAMES . . . . .            | 1444 |
| 4.49.2.630 DOLLARNAMES . . . . .          | 1445 |
| 4.49.2.631 DUMMYBUFFER . . . . .          | 1445 |
| 4.49.2.632 ALLARGS . . . . .              | 1445 |
| 4.49.2.633 NUMARG . . . . .               | 1445 |
| 4.49.2.634 ARGRANGE . . . . .             | 1445 |
| 4.49.2.635 MAKEARGS . . . . .             | 1445 |
| 4.49.2.636 MAXRANGEINDICATOR . . . . .    | 1445 |
| 4.49.2.637 REPLACEARG . . . . .           | 1445 |
| 4.49.2.638 ENCODEARG . . . . .            | 1446 |
| 4.49.2.639 DECODEARG . . . . .            | 1446 |
| 4.49.2.640 IMplodeARG . . . . .           | 1446 |
| 4.49.2.641 EXPLODEARG . . . . .           | 1446 |
| 4.49.2.642 PERMUTEARG . . . . .           | 1446 |
| 4.49.2.643 REVERSEARG . . . . .           | 1446 |
| 4.49.2.644 CYCLEARG . . . . .             | 1446 |
| 4.49.2.645 ISLYNDON . . . . .             | 1446 |
| 4.49.2.646 ISLYNDONR . . . . .            | 1447 |
| 4.49.2.647 TOLYNDON . . . . .             | 1447 |
| 4.49.2.648 TOLYNDONR . . . . .            | 1447 |
| 4.49.2.649 ADDARG . . . . .               | 1447 |
| 4.49.2.650 MULTIPLYARG . . . . .          | 1447 |
| 4.49.2.651 DROPARG . . . . .              | 1447 |
| 4.49.2.652 SELECTARG . . . . .            | 1447 |
| 4.49.2.653 DEDUPARG . . . . .             | 1447 |
| 4.49.2.654 ZTOHARG . . . . .              | 1448 |
| 4.49.2.655 HTOZARG . . . . .              | 1448 |
| 4.49.2.656 BASECODE . . . . .             | 1448 |
| 4.49.2.657 YESLYNDON . . . . .            | 1448 |
| 4.49.2.658 NOLYNDON . . . . .             | 1448 |
| 4.49.2.659 TOPOLYNOMIALFLAG . . . . .     | 1448 |
| 4.49.2.660 FACTARGFLAG . . . . .          | 1448 |
| 4.49.2.661 OLDFACTARG . . . . .           | 1448 |
| 4.49.2.662 NEWFACTARG . . . . .           | 1449 |
| 4.49.2.663 FROMMODULEOPTION . . . . .     | 1449 |
| 4.49.2.664 FROMPOINTINSTRUCTION . . . . . | 1449 |
| 4.49.2.665 EXTRASYMBOL . . . . .          | 1449 |
| 4.49.2.666 REGULARSYMBOL . . . . .        | 1449 |
| 4.49.2.667 EXPRESSIONNUMBER . . . . .     | 1449 |
| 4.49.2.668 O_NONE . . . . .               | 1449 |
| 4.49.2.669 O_CSE . . . . .                | 1449 |
| 4.49.2.670 O_CSEGREEDY . . . . .          | 1450 |

|                                                   |      |
|---------------------------------------------------|------|
| 4.49.2.671 O_GREEDY . . . . .                     | 1450 |
| 4.49.2.672 O_OCCURRENCE . . . . .                 | 1450 |
| 4.49.2.673 O_MCTS . . . . .                       | 1450 |
| 4.49.2.674 O_SIMULATED_ANNEALING . . . . .        | 1450 |
| 4.49.2.675 O_FORWARD . . . . .                    | 1450 |
| 4.49.2.676 O_BACKWARD . . . . .                   | 1450 |
| 4.49.2.677 O_FORWARDORBACKWARD . . . . .          | 1450 |
| 4.49.2.678 O_FORWARDANDBACKWARD . . . . .         | 1451 |
| 4.49.2.679 OPTHEAD . . . . .                      | 1451 |
| 4.49.2.680 DOALL . . . . .                        | 1451 |
| 4.49.2.681 ONLYFUNCTIONS . . . . .                | 1451 |
| 4.49.2.682 INUSE . . . . .                        | 1451 |
| 4.49.2.683 COULDCOMMUTE . . . . .                 | 1451 |
| 4.49.2.684 DOESNOTCOMMUTE . . . . .               | 1451 |
| 4.49.2.685 DICT_NONNUMBERS . . . . .              | 1451 |
| 4.49.2.686 DICT_INTEGERONLY . . . . .             | 1452 |
| 4.49.2.687 DICT_RATIONALONLY . . . . .            | 1452 |
| 4.49.2.688 DICT_ALLNUMBERS . . . . .              | 1452 |
| 4.49.2.689 DICT_NOVARIABLES . . . . .             | 1452 |
| 4.49.2.690 DICT_DOVARIABLES . . . . .             | 1452 |
| 4.49.2.691 DICT_NOSPECIALS . . . . .              | 1452 |
| 4.49.2.692 DICT_DOSPECIALS . . . . .              | 1452 |
| 4.49.2.693 DICT_NOFUNWITHARGS . . . . .           | 1452 |
| 4.49.2.694 DICT_DOFUNWITHARGS . . . . .           | 1453 |
| 4.49.2.695 DICT_NOTINDOLLARS . . . . .            | 1453 |
| 4.49.2.696 DICT_INDOLLARS . . . . .               | 1453 |
| 4.49.2.697 DICT_INTEGERNUMBER . . . . .           | 1453 |
| 4.49.2.698 DICT_RATIONALNUMBER . . . . .          | 1453 |
| 4.49.2.699 DICT_SYMBOL . . . . .                  | 1453 |
| 4.49.2.700 DICT_VECTOR . . . . .                  | 1453 |
| 4.49.2.701 DICT_INDEX . . . . .                   | 1453 |
| 4.49.2.702 DICT_FUNCTION . . . . .                | 1454 |
| 4.49.2.703 DICT_FUNCTION_WITH_ARGUMENTS . . . . . | 1454 |
| 4.49.2.704 DICT_SPECIALCHARACTER . . . . .        | 1454 |
| 4.49.2.705 DICT_RANGE . . . . .                   | 1454 |
| 4.49.2.706 READSPECTATORFLAG . . . . .            | 1454 |
| 4.49.2.707 GLOBALSPECTATORFLAG . . . . .          | 1454 |
| 4.49.2.708 ORDEREDSET . . . . .                   | 1454 |
| 4.49.2.709 DENSETABLE . . . . .                   | 1454 |
| 4.49.2.710 SPARSETABLE . . . . .                  | 1455 |
| 4.49.2.711 ONEPARTICLEIRREDUCIBLE . . . . .       | 1455 |
| 4.49.2.712 WITHINSERTIONS . . . . .               | 1455 |

|                                       |      |
|---------------------------------------|------|
| 4.49.2.713 NOTADPOLES                 | 1455 |
| 4.49.2.714 WITHSYMMETRIZE             | 1455 |
| 4.49.2.715 TOPOLOGIESONLY             | 1455 |
| 4.49.2.716 NONODES                    | 1455 |
| 4.49.2.717 WITHEDGES                  | 1455 |
| 4.49.2.718 CHECKEXTERN                | 1456 |
| 4.49.2.719 WITHBLOCKS                 | 1456 |
| 4.49.2.720 WITHONEPI                  | 1456 |
| 4.49.2.721 NOSNAILS                   | 1456 |
| 4.49.2.722 NOEXTSELF                  | 1456 |
| 4.49.3 Typedef Documentation          | 1456 |
| 4.49.3.1 PVFUNWP                      | 1456 |
| 4.49.3.2 PVFUNV                       | 1456 |
| 4.49.3.3 CFUN                         | 1457 |
| 4.49.3.4 TFUN                         | 1457 |
| 4.49.3.5 TFUN1                        | 1457 |
| 4.50 ftypes.h                         | 1457 |
| 4.51 function.c File Reference        | 1470 |
| 4.51.1 Detailed Description           | 1470 |
| 4.51.2 Function Documentation         | 1470 |
| 4.51.2.1 MakeDirty()                  | 1470 |
| 4.51.2.2 MarkDirty()                  | 1471 |
| 4.51.2.3 PolyFunDirty()               | 1471 |
| 4.51.2.4 PolyFunClean()               | 1471 |
| 4.51.2.5 Symmetrize()                 | 1471 |
| 4.51.2.6 CompGroup()                  | 1471 |
| 4.51.2.7 FullSymmetrize()             | 1471 |
| 4.51.2.8 SymGen()                     | 1472 |
| 4.51.2.9 SymFind()                    | 1472 |
| 4.51.2.10 ChainIn()                   | 1472 |
| 4.51.2.11 ChainOut()                  | 1472 |
| 4.51.2.12 MatchFunction()             | 1472 |
| 4.51.2.13 ScanFunctions()             | 1472 |
| 4.52 function.c                       | 1473 |
| 4.53 fwin.h File Reference            | 1495 |
| 4.53.1 Detailed Description           | 1495 |
| 4.53.2 Macro Definition Documentation | 1496 |
| 4.53.2.1 LINEFEED                     | 1496 |
| 4.53.2.2 CARRIAGERETURN               | 1496 |
| 4.53.2.3 WITHRETURN                   | 1496 |
| 4.53.2.4 WITHSYSTEM                   | 1496 |
| 4.53.2.5 P_term                       | 1496 |

|                                 |      |
|---------------------------------|------|
| 4.53.2.6 SEPARATOR              | 1496 |
| 4.53.2.7 ALTSEPARATOR           | 1496 |
| 4.53.2.8 PATHSEPARATOR          | 1497 |
| 4.53.2.9 WITH_ENV               | 1497 |
| 4.54 fwin.h                     | 1497 |
| 4.55 gentopo.cc                 | 1497 |
| 4.56 gentopo.h                  | 1516 |
| 4.57 grcc.cc                    | 1518 |
| 4.58 grcc.h                     | 1636 |
| 4.59 grccparam.h                | 1652 |
| 4.60 if.c File Reference        | 1655 |
| 4.60.1 Detailed Description     | 1655 |
| 4.60.2 Function Documentation   | 1655 |
| 4.60.2.1 GetIfDollarNum()       | 1655 |
| 4.60.2.2 FindVar()              | 1655 |
| 4.60.2.3 DoIfStatement()        | 1656 |
| 4.60.2.4 HowMany()              | 1656 |
| 4.60.2.5 DoubleIfBuffers()      | 1656 |
| 4.60.2.6 DoSwitch()             | 1656 |
| 4.60.2.7 DoEndSwitch()          | 1656 |
| 4.60.2.8 FindCase()             | 1656 |
| 4.60.2.9 DoubleSwitchBuffers()  | 1657 |
| 4.60.2.10 SwitchSplitMergeRec() | 1657 |
| 4.60.2.11 SwitchSplitMerge()    | 1657 |
| 4.61 if.c                       | 1657 |
| 4.62 index.c File Reference     | 1671 |
| 4.62.1 Detailed Description     | 1672 |
| 4.62.2 Function Documentation   | 1672 |
| 4.62.2.1 FindBracket()          | 1672 |
| 4.62.2.2 PutBracketInIndex()    | 1672 |
| 4.62.2.3 ClearBracketIndex()    | 1672 |
| 4.62.2.4 OpenBracketIndex()     | 1672 |
| 4.62.2.5 PutInside()            | 1673 |
| 4.63 index.c                    | 1673 |
| 4.64 inivar.h File Reference    | 1681 |
| 4.64.1 Detailed Description     | 1682 |
| 4.64.2 Variable Documentation   | 1682 |
| 4.64.2.1 FG                     | 1682 |
| 4.64.2.2 A                      | 1682 |
| 4.64.2.3 fixedsets              | 1682 |
| 4.64.2.4 BufferForOutput        | 1682 |
| 4.64.2.5 setupfilename          | 1682 |

|                                       |      |
|---------------------------------------|------|
| 4.65 inivar.h                         | 1683 |
| 4.66 lus.c File Reference             | 1686 |
| 4.66.1 Detailed Description           | 1687 |
| 4.66.2 Function Documentation         | 1687 |
| 4.66.2.1 Lus()                        | 1687 |
| 4.66.2.2 FindLus()                    | 1687 |
| 4.66.2.3 SortTheList()                | 1687 |
| 4.66.2.4 AllLoops()                   | 1687 |
| 4.66.2.5 StartLoops()                 | 1688 |
| 4.66.2.6 GenLoops()                   | 1688 |
| 4.66.2.7 LoopOutput()                 | 1688 |
| 4.66.2.8 AllPaths()                   | 1688 |
| 4.66.2.9 GenPaths()                   | 1689 |
| 4.66.2.10 PathOutput()                | 1689 |
| 4.66.2.11 AllOnePI()                  | 1689 |
| 4.66.2.12 RemoveBridges()             | 1689 |
| 4.66.2.13 TakeOneLine()               | 1689 |
| 4.66.2.14 OutputOnePI()               | 1690 |
| 4.67 lus.c                            | 1690 |
| 4.68 mallocprotect.h                  | 1708 |
| 4.69 message.c File Reference         | 1713 |
| 4.69.1 Detailed Description           | 1714 |
| 4.69.2 Function Documentation         | 1714 |
| 4.69.2.1 Error0()                     | 1714 |
| 4.69.2.2 Error1()                     | 1714 |
| 4.69.2.3 Error2()                     | 1714 |
| 4.69.2.4 MesWork()                    | 1714 |
| 4.69.2.5 MesPrint()                   | 1714 |
| 4.69.2.6 Warning()                    | 1715 |
| 4.69.2.7 HighWarning()                | 1715 |
| 4.69.2.8 MesCall()                    | 1715 |
| 4.69.2.9 MesCerr()                    | 1715 |
| 4.69.2.10 MesComp()                   | 1715 |
| 4.69.2.11 PrintTerm()                 | 1715 |
| 4.69.2.12 PrintTermC()                | 1716 |
| 4.69.2.13 PrintSubTerm()              | 1716 |
| 4.69.2.14 PrintWords()                | 1716 |
| 4.69.2.15 PrintSeq()                  | 1716 |
| 4.70 message.c                        | 1716 |
| 4.71 minos.c File Reference           | 1727 |
| 4.71.1 Detailed Description           | 1729 |
| 4.71.2 Macro Definition Documentation | 1729 |

|                                       |      |
|---------------------------------------|------|
| 4.71.2.1 CFD                          | 1729 |
| 4.71.2.2 CTD                          | 1729 |
| 4.71.3 Function Documentation         | 1729 |
| 4.71.3.1 minosread()                  | 1729 |
| 4.71.3.2 minoswrite()                 | 1730 |
| 4.71.3.3 str_dup()                    | 1730 |
| 4.71.3.4 convertblock()               | 1730 |
| 4.71.3.5 convertnamesblock()          | 1730 |
| 4.71.3.6 convertiniinfo()             | 1730 |
| 4.71.3.7 LocateBase()                 | 1730 |
| 4.71.3.8 ReadIndex()                  | 1731 |
| 4.71.3.9 WriteIndexBlock()            | 1731 |
| 4.71.3.10 WriteNamesBlock()           | 1731 |
| 4.71.3.11 WriteIndex()                | 1731 |
| 4.71.3.12 WriteIniInfo()              | 1731 |
| 4.71.3.13 ReadIniInfo()               | 1731 |
| 4.71.3.14 GetDbase()                  | 1732 |
| 4.71.3.15 NewDbase()                  | 1732 |
| 4.71.3.16 FreeTableBase()             | 1732 |
| 4.71.3.17 ComposeTableNames()         | 1732 |
| 4.71.3.18 OpenDbase()                 | 1732 |
| 4.71.3.19 AddTableName()              | 1732 |
| 4.71.3.20 GetTableName()              | 1733 |
| 4.71.3.21 PutTableNames()             | 1733 |
| 4.71.3.22 AddToIndex()                | 1733 |
| 4.71.3.23 AddObject()                 | 1733 |
| 4.71.3.24 FindTableNumber()           | 1733 |
| 4.71.3.25 WriteObject()               | 1733 |
| 4.71.3.26 ReadObject()                | 1734 |
| 4.71.3.27 ReadijObject()              | 1734 |
| 4.71.3.28 ExistsObject()              | 1734 |
| 4.71.3.29 DeleteObject()              | 1734 |
| 4.71.4 Variable Documentation         | 1734 |
| 4.71.4.1 withoutflush                 | 1734 |
| 4.72 minos.c                          | 1735 |
| 4.73 minos.h File Reference           | 1752 |
| 4.73.1 Detailed Description           | 1753 |
| 4.73.2 Macro Definition Documentation | 1753 |
| 4.73.2.1 TODISK                       | 1753 |
| 4.73.2.2 FROMDISK                     | 1753 |
| 4.73.2.3 MDIRTYFLAG                   | 1753 |
| 4.73.2.4 MCLEANFLAG                   | 1754 |

|                                       |      |
|---------------------------------------|------|
| 4.73.2.5 INANDOUT                     | 1754 |
| 4.73.2.6 INPUTONLY                    | 1754 |
| 4.73.2.7 OUTPUTONLY                   | 1754 |
| 4.73.2.8 NOCOMPRESS                   | 1754 |
| 4.73.3 Function Documentation         | 1754 |
| 4.73.3.1 minosread()                  | 1754 |
| 4.73.3.2 minoswrite()                 | 1754 |
| 4.73.4 Variable Documentation         | 1755 |
| 4.73.4.1 withoutflush                 | 1755 |
| 4.74 minos.h                          | 1755 |
| 4.75 model.c                          | 1756 |
| 4.76 module.c File Reference          | 1762 |
| 4.76.1 Detailed Description           | 1762 |
| 4.76.2 Function Documentation         | 1762 |
| 4.76.2.1 ModuleInstruction()          | 1762 |
| 4.76.2.2 CoModuleOption()             | 1763 |
| 4.76.2.3 CoModOption()                | 1763 |
| 4.76.2.4 SetSpecialMode()             | 1763 |
| 4.76.2.5 ExecModule()                 | 1763 |
| 4.76.2.6 ExecStore()                  | 1763 |
| 4.76.2.7 FullCleanUp()                | 1763 |
| 4.76.2.8 DoPolyfun()                  | 1763 |
| 4.76.2.9 DoPolyratfun()               | 1764 |
| 4.76.2.10 DoNoParallel()              | 1764 |
| 4.76.2.11 DoParallel()                | 1764 |
| 4.76.2.12 DoModSum()                  | 1764 |
| 4.76.2.13 DoModMax()                  | 1764 |
| 4.76.2.14 DoModMin()                  | 1764 |
| 4.76.2.15 DoModLocal()                | 1764 |
| 4.76.2.16 DoProcessBucket()           | 1765 |
| 4.76.2.17 DoModDollar()               | 1765 |
| 4.76.2.18 DoinParallel()              | 1765 |
| 4.76.2.19 DonotinParallel()           | 1765 |
| 4.76.2.20 DoExecStatement()           | 1765 |
| 4.76.2.21 DoPipeStatement()           | 1765 |
| 4.77 module.c                         | 1766 |
| 4.78 mpi.c File Reference             | 1775 |
| 4.78.1 Detailed Description           | 1776 |
| 4.78.2 Macro Definition Documentation | 1776 |
| 4.78.2.1 PF_PACKSIZE                  | 1776 |
| 4.78.2.2 MPI_ERRCODE_CHECK            | 1777 |
| 4.78.3 Function Documentation         | 1777 |



|                                       |      |
|---------------------------------------|------|
| 4.78.3.1 PF_RealTime()                | 1777 |
| 4.78.3.2 PF_LibInit()                 | 1777 |
| 4.78.3.3 PF_LibTerminate()            | 1778 |
| 4.78.3.4 PF_Probe()                   | 1778 |
| 4.78.3.5 PF_ISendSbuf()               | 1779 |
| 4.78.3.6 PF_RecvWbuf()                | 1779 |
| 4.78.3.7 PF_IRecvRbuf()               | 1779 |
| 4.78.3.8 PF_WaitRbuf()                | 1780 |
| 4.78.3.9 PF_Bcast()                   | 1780 |
| 4.78.3.10 PF_RawSend()                | 1781 |
| 4.78.3.11 PF_RawRecv()                | 1781 |
| 4.78.3.12 PF_RawProbe()               | 1782 |
| 4.78.3.13 PF_PrintPackBuf()           | 1782 |
| 4.78.3.14 PF_PreparePack()            | 1783 |
| 4.78.3.15 PF_Pack()                   | 1783 |
| 4.78.3.16 PF_Unpack()                 | 1783 |
| 4.78.3.17 PF_PackString()             | 1784 |
| 4.78.3.18 PF_UnpackString()           | 1784 |
| 4.78.3.19 PF_Send()                   | 1785 |
| 4.78.3.20 PF_Receive()                | 1785 |
| 4.78.3.21 PF_Broadcast()              | 1786 |
| 4.78.3.22 PF_PrepareLongSinglePack()  | 1786 |
| 4.78.3.23 PF_LongSinglePack()         | 1786 |
| 4.78.3.24 PF_LongSingleUnpack()       | 1787 |
| 4.78.3.25 PF_LongSingleSend()         | 1787 |
| 4.78.3.26 PF_LongSingleReceive()      | 1788 |
| 4.78.3.27 PF_PrepareLongMultiPack()   | 1788 |
| 4.78.3.28 PF_LongMultiPackImpl()      | 1789 |
| 4.78.3.29 PF_LongMultiUnpackImpl()    | 1789 |
| 4.78.3.30 PF_LongMultiBroadcast()     | 1790 |
| 4.78.4 Variable Documentation         | 1791 |
| 4.78.4.1 PF_maxDollarChunkSize        | 1791 |
| 4.79 mpi.c                            | 1791 |
| 4.80 mpidbg.h File Reference          | 1809 |
| 4.80.1 Detailed Description           | 1810 |
| 4.80.2 Macro Definition Documentation | 1810 |
| 4.80.2.1 MPIDBG_RANK                  | 1810 |
| 4.80.2.2 MPIDBG_Out                   | 1810 |
| 4.80.2.3 MPIDBG_EXTARG                | 1810 |
| 4.80.2.4 MPI_Send                     | 1810 |
| 4.80.2.5 MPI_Recv                     | 1810 |
| 4.80.2.6 MPI_Bsend                    | 1810 |

|                               |      |
|-------------------------------|------|
| 4.80.2.7 MPI_Ssend            | 1811 |
| 4.80.2.8 MPI_Rsend            | 1811 |
| 4.80.2.9 MPI_Isend            | 1811 |
| 4.80.2.10 MPI_Ibsend          | 1811 |
| 4.80.2.11 MPI_Issend          | 1811 |
| 4.80.2.12 MPI_Irsend          | 1811 |
| 4.80.2.13 MPI_Irecv           | 1811 |
| 4.80.2.14 MPI_Wait            | 1812 |
| 4.80.2.15 MPI_Test            | 1812 |
| 4.80.2.16 MPI_Waitany         | 1812 |
| 4.80.2.17 MPI_Testany         | 1812 |
| 4.80.2.18 MPI_Waitall         | 1812 |
| 4.80.2.19 MPI_Testall         | 1812 |
| 4.80.2.20 MPI_Waitsome        | 1812 |
| 4.80.2.21 MPI_Testsome        | 1813 |
| 4.80.2.22 MPI_Iprobe          | 1813 |
| 4.80.2.23 MPI_Probe           | 1813 |
| 4.80.2.24 MPI_Cancel          | 1813 |
| 4.80.2.25 MPI_Test_cancelled  | 1813 |
| 4.80.2.26 MPI_Barrier         | 1813 |
| 4.80.2.27 MPI_Bcast           | 1813 |
| 4.81 mpidbg.h                 | 1814 |
| 4.82 mytime.cc                | 1822 |
| 4.83 mytime.h                 | 1822 |
| 4.84 names.c File Reference   | 1823 |
| 4.84.1 Detailed Description   | 1824 |
| 4.84.2 Function Documentation | 1824 |
| 4.84.2.1 GetNode()            | 1824 |
| 4.84.2.2 AddName()            | 1824 |
| 4.84.2.3 GetName()            | 1825 |
| 4.84.2.4 GetFunction()        | 1825 |
| 4.84.2.5 GetNumber()          | 1825 |
| 4.84.2.6 GetLastExprName()    | 1825 |
| 4.84.2.7 GetOName()           | 1825 |
| 4.84.2.8 GetAutoName()        | 1825 |
| 4.84.2.9 GetVar()             | 1826 |
| 4.84.2.10 EntVar()            | 1826 |
| 4.84.2.11 GetDollar()         | 1826 |
| 4.84.2.12 DumpTree()          | 1826 |
| 4.84.2.13 DumpNode()          | 1826 |
| 4.84.2.14 CompactifyTree()    | 1827 |
| 4.84.2.15 CopyTree()          | 1827 |

|                                |      |
|--------------------------------|------|
| 4.84.2.16 LinkTree()           | 1827 |
| 4.84.2.17 MakeNameTree()       | 1827 |
| 4.84.2.18 FreeNameTree()       | 1827 |
| 4.84.2.19 ClearWildcardNames() | 1827 |
| 4.84.2.20 AddWildcardName()    | 1828 |
| 4.84.2.21 GetWildcardName()    | 1828 |
| 4.84.2.22 AddSymbol()          | 1828 |
| 4.84.2.23 CoSymbol()           | 1828 |
| 4.84.2.24 AddIndex()           | 1828 |
| 4.84.2.25 CoIndex()            | 1828 |
| 4.84.2.26 DoDimension()        | 1829 |
| 4.84.2.27 CoDimension()        | 1829 |
| 4.84.2.28 AddVector()          | 1829 |
| 4.84.2.29 CoVector()           | 1829 |
| 4.84.2.30 AddFunction()        | 1829 |
| 4.84.2.31 CoCommutInSet()      | 1829 |
| 4.84.2.32 CoFunction()         | 1830 |
| 4.84.2.33 CoNFunction()        | 1830 |
| 4.84.2.34 CoCFunction()        | 1830 |
| 4.84.2.35 CoNTensor()          | 1830 |
| 4.84.2.36 CoCTensor()          | 1830 |
| 4.84.2.37 DoTable()            | 1830 |
| 4.84.2.38 CoTable()            | 1831 |
| 4.84.2.39 CoNTable()           | 1831 |
| 4.84.2.40 CoCTable()           | 1831 |
| 4.84.2.41 EmptyTable()         | 1831 |
| 4.84.2.42 AddSet()             | 1831 |
| 4.84.2.43 DoElements()         | 1831 |
| 4.84.2.44 CoSet()              | 1832 |
| 4.84.2.45 DoTempSet()          | 1832 |
| 4.84.2.46 CoAuto()             | 1832 |
| 4.84.2.47 AddDollar()          | 1832 |
| 4.84.2.48 ReplaceDollar()      | 1832 |
| 4.84.2.49 AddDubious()         | 1832 |
| 4.84.2.50 MakeDubious()        | 1833 |
| 4.84.2.51 NameConflict()       | 1833 |
| 4.84.2.52 AddExpression()      | 1833 |
| 4.84.2.53 GetLabel()           | 1833 |
| 4.84.2.54 ResetVariables()     | 1833 |
| 4.84.2.55 RemoveDollars()      | 1833 |
| 4.84.2.56 Globalize()          | 1834 |
| 4.84.2.57 TestName()           | 1834 |

|                                       |      |
|---------------------------------------|------|
| 4.85 names.c                          | 1834 |
| 4.86 normal.c File Reference          | 1872 |
| 4.86.1 Detailed Description           | 1872 |
| 4.86.2 Macro Definition Documentation | 1873 |
| 4.86.2.1 MAXNUMBEROFNONCOMTERMS       | 1873 |
| 4.86.3 Function Documentation         | 1873 |
| 4.86.3.1 CompareFunctions()           | 1873 |
| 4.86.3.2 Commute()                    | 1873 |
| 4.86.3.3 Normalize()                  | 1873 |
| 4.86.3.4 ExtraSymbol()                | 1873 |
| 4.86.3.5 DoTheta()                    | 1873 |
| 4.86.3.6 DoDelta()                    | 1874 |
| 4.86.3.7 DoRevert()                   | 1874 |
| 4.86.3.8 DetCommu()                   | 1874 |
| 4.86.3.9 DoesCommu()                  | 1874 |
| 4.86.3.10 TreatPolyRatFun()           | 1874 |
| 4.86.3.11 DropCoefficient()           | 1874 |
| 4.86.3.12 DropSymbols()               | 1874 |
| 4.86.3.13 SymbolNormalize()           | 1875 |
| 4.86.3.14 TestFunFlag()               | 1875 |
| 4.86.3.15 BracketNormalize()          | 1875 |
| 4.87 normal.c                         | 1875 |
| 4.88 notation.c File Reference        | 1938 |
| 4.88.1 Detailed Description           | 1938 |
| 4.88.2 Function Documentation         | 1938 |
| 4.88.2.1 NormPolyTerm()               | 1938 |
| 4.88.2.2 ConvertToPoly()              | 1939 |
| 4.88.2.3 LocalConvertToPoly()         | 1939 |
| 4.88.2.4 ConvertFromPoly()            | 1939 |
| 4.88.2.5 FindSubterm()                | 1939 |
| 4.88.2.6 FindLocalSubterm()           | 1940 |
| 4.88.2.7 PrintSubtermList()           | 1940 |
| 4.88.2.8 PrintExtraSymbol()           | 1940 |
| 4.88.2.9 FindSubexpression()          | 1940 |
| 4.88.2.10 ExtraSymFun()               | 1940 |
| 4.88.2.11 PruneExtraSymbols()         | 1940 |
| 4.89 notation.c                       | 1941 |
| 4.90 opera.c File Reference           | 1955 |
| 4.90.1 Detailed Description           | 1955 |
| 4.90.2 Function Documentation         | 1955 |
| 4.90.2.1 EpfFind()                    | 1955 |
| 4.90.2.2 EpfCon()                     | 1956 |

|                                  |      |
|----------------------------------|------|
| 4.90.2.3 EpfGen()                | 1956 |
| 4.90.2.4 Trick()                 | 1956 |
| 4.90.2.5 Trace4no()              | 1956 |
| 4.90.2.6 Trace4()                | 1956 |
| 4.90.2.7 Trace4Gen()             | 1957 |
| 4.90.2.8 TraceNno()              | 1957 |
| 4.90.2.9 TraceN()                | 1957 |
| 4.90.2.10 TraceNgen()            | 1957 |
| 4.90.2.11 Traces()               | 1957 |
| 4.90.2.12 TraceFind()            | 1957 |
| 4.90.2.13 Chisholm()             | 1958 |
| 4.90.2.14 TenVecFind()           | 1958 |
| 4.90.2.15 TenVec()               | 1958 |
| 4.91 opera.c                     | 1958 |
| 4.92 optimize.cc File Reference  | 1987 |
| 4.92.1 Detailed Description      | 1989 |
| 4.92.2 Function Documentation    | 1989 |
| 4.92.2.1 print_instr()           | 1989 |
| 4.92.2.2 my_random_shuffle()     | 1989 |
| 4.92.3 Description               | 1989 |
| 4.92.3.1 get_expression()        | 1990 |
| 4.92.4 Description               | 1990 |
| 4.92.4.1 get_brackets()          | 1990 |
| 4.92.5 Description               | 1990 |
| 4.92.5.1 count_operators() [1/2] | 1991 |
| 4.92.6 Description               | 1991 |
| 4.92.6.1 count_operators() [2/2] | 1991 |
| 4.92.7 Description               | 1991 |
| 4.92.7.1 occurrence_order()      | 1991 |
| 4.92.8 Description               | 1991 |
| 4.92.8.1 getpower()              | 1992 |
| 4.92.9 Description               | 1992 |
| 4.92.10 Note                     | 1992 |
| 4.92.11 Description              | 1992 |
| 4.92.11.1 fixarg()               | 1992 |
| 4.92.12 Description              | 1993 |
| 4.92.12.1 GcdLong_fix_args()     | 1993 |
| 4.92.12.2 DivLong_fix_args()     | 1993 |
| 4.92.12.3 build_Horner_tree()    | 1993 |
| 4.92.13 Description              | 1994 |
| 4.92.13.1 term_compare()         | 1994 |
| 4.92.14 Description              | 1994 |

|                                              |      |
|----------------------------------------------|------|
| 4.92.14.1 Horner_tree()                      | 1995 |
| 4.92.15 Description                          | 1995 |
| 4.92.15.1 print_tree()                       | 1995 |
| 4.92.15.2 hash_range()                       | 1995 |
| 4.92.15.3 generate_instructions()            | 1996 |
| 4.92.16 Description                          | 1996 |
| 4.92.17 Implementation details               | 1996 |
| 4.92.17.1 count_operators_cse()              | 1997 |
| 4.92.17.2 buildTree()                        | 1997 |
| 4.92.17.3 count_operators_cse_topdown()      | 1997 |
| 4.92.17.4 simulated_annealing()              | 1997 |
| 4.92.17.5 find_Horner_MCTS_expand_tree()     | 1997 |
| 4.92.17.6 find_Horner_MCTS()                 | 1997 |
| 4.92.18 Description                          | 1998 |
| 4.92.18.1 merge_operators()                  | 1998 |
| 4.92.19 Description                          | 1998 |
| 4.92.20 Implementation details               | 1999 |
| 4.92.20.1 find_optimizations()               | 1999 |
| 4.92.21 Description                          | 1999 |
| 4.92.21.1 do_optimization()                  | 1999 |
| 4.92.22 Description                          | 2000 |
| 4.92.22.1 partial_factorize()                | 2000 |
| 4.92.23 Description                          | 2000 |
| 4.92.23.1 optimize_greedy()                  | 2000 |
| 4.92.24 Description                          | 2001 |
| 4.92.24.1 recycle_variables()                | 2001 |
| 4.92.25 Description                          | 2001 |
| 4.92.26 Implementation details               | 2002 |
| 4.92.26.1 optimize_expression_given_Horner() | 2002 |
| 4.92.27 Description                          | 2002 |
| 4.92.27.1 generate_output()                  | 2003 |
| 4.92.28 Description                          | 2003 |
| 4.92.28.1 generate_expression()              | 2003 |
| 4.92.29 Description                          | 2003 |
| 4.92.29.1 optimize_print_code()              | 2004 |
| 4.92.30 Description                          | 2004 |
| 4.92.30.1 Optimize()                         | 2004 |
| 4.92.31 Description                          | 2005 |
| 4.92.31.1 ClearOptimize()                    | 2006 |
| 4.92.32 Description                          | 2006 |
| 4.92.33 Variable Documentation               | 2007 |
| 4.92.33.1 OPER_ADD                           | 2007 |

|                                                  |      |
|--------------------------------------------------|------|
| 4.92.33.2 OPER_MUL . . . . .                     | 2007 |
| 4.92.33.3 OPER_COMMA . . . . .                   | 2007 |
| 4.92.33.4 optimize_expr . . . . .                | 2007 |
| 4.92.33.5 optimize_best_Horner_schemes . . . . . | 2007 |
| 4.92.33.6 optimize_num_vars . . . . .            | 2008 |
| 4.92.33.7 optimize_best_num_oper . . . . .       | 2008 |
| 4.92.33.8 optimize_best_instr . . . . .          | 2008 |
| 4.92.33.9 optimize_best_vars . . . . .           | 2008 |
| 4.92.33.10 mcts_factorized . . . . .             | 2008 |
| 4.92.33.11 mcts_separated . . . . .              | 2008 |
| 4.92.33.12 mcts_vars . . . . .                   | 2008 |
| 4.92.33.13 mcts_root . . . . .                   | 2008 |
| 4.92.33.14 mcts_expr_score . . . . .             | 2009 |
| 4.92.33.15 mcts_best_schemes . . . . .           | 2009 |
| 4.93 optimize.cc . . . . .                       | 2009 |
| 4.94 parallel.c File Reference . . . . .         | 2061 |
| 4.94.1 Detailed Description . . . . .            | 2063 |
| 4.94.2 Macro Definition Documentation . . . . .  | 2063 |
| 4.94.2.1 PRINTFBUF . . . . .                     | 2063 |
| 4.94.2.2 SWAP . . . . .                          | 2064 |
| 4.94.2.3 PACK_LONG . . . . .                     | 2064 |
| 4.94.2.4 UNPACK_LONG . . . . .                   | 2064 |
| 4.94.2.5 CHECK . . . . .                         | 2064 |
| 4.94.2.6 _CHECK . . . . .                        | 2065 |
| 4.94.2.7 __CHECK . . . . .                       | 2065 |
| 4.94.2.8 DBGOUT . . . . .                        | 2065 |
| 4.94.2.9 DBGOUT_NINTERMS . . . . .               | 2065 |
| 4.94.2.10 PF_STATS_SIZE . . . . .                | 2065 |
| 4.94.2.11 PF_SNDFILEBUFSIZE . . . . .            | 2066 |
| 4.94.2.12 recvBuffer . . . . .                   | 2066 |
| 4.94.3 Typedef Documentation . . . . .           | 2066 |
| 4.94.3.1 NODE . . . . .                          | 2066 |
| 4.94.4 Function Documentation . . . . .          | 2066 |
| 4.94.4.1 PF_RealTime() . . . . .                 | 2066 |
| 4.94.4.2 PF_LibInit() . . . . .                  | 2066 |
| 4.94.4.3 PF_LibTerminate() . . . . .             | 2067 |
| 4.94.4.4 PF_Probe() . . . . .                    | 2067 |
| 4.94.4.5 PF_RecvWbuf() . . . . .                 | 2068 |
| 4.94.4.6 PF_IRecvRbuf() . . . . .                | 2068 |
| 4.94.4.7 PF_WaitRbuf() . . . . .                 | 2068 |
| 4.94.4.8 PF_RawSend() . . . . .                  | 2069 |
| 4.94.4.9 PF_RawRecv() . . . . .                  | 2069 |

|                                          |      |
|------------------------------------------|------|
| 4.94.4.10 PF_RawProbe()                  | 2070 |
| 4.94.4.11 PF_EndSort()                   | 2070 |
| 4.94.4.12 PF_Deferred()                  | 2071 |
| 4.94.4.13 PF_Processor()                 | 2072 |
| 4.94.4.14 PF_Init()                      | 2073 |
| 4.94.4.15 PF_Terminate()                 | 2074 |
| 4.94.4.16 PF_GetSlaveTimes()             | 2075 |
| 4.94.4.17 PF_BroadcastNumber()           | 2075 |
| 4.94.4.18 PF_BroadcastBuffer()           | 2076 |
| 4.94.4.19 PF_BroadcastString()           | 2077 |
| 4.94.4.20 PF_BroadcastPreDollar()        | 2077 |
| 4.94.4.21 PF_CollectModifiedDollars()    | 2078 |
| 4.94.4.22 PF_BroadcastModifiedDollars()  | 2079 |
| 4.94.4.23 PF_BroadcastRedefinedPreVars() | 2080 |
| 4.94.4.24 PF_StoreInsideInfo()           | 2081 |
| 4.94.4.25 PF_RestoreInsideInfo()         | 2081 |
| 4.94.4.26 PF_BroadcastCBuf()             | 2081 |
| 4.94.4.27 PF_BroadcastExpFlags()         | 2082 |
| 4.94.4.28 PF_BroadcastExpr()             | 2082 |
| 4.94.4.29 PF_BroadcastRHS()              | 2083 |
| 4.94.4.30 PF_InParallelProcessor()       | 2083 |
| 4.94.4.31 PF_SendFile()                  | 2083 |
| 4.94.4.32 PF_RecvFile()                  | 2084 |
| 4.94.4.33 PF_MLock()                     | 2085 |
| 4.94.4.34 PF_MUnlock()                   | 2085 |
| 4.94.4.35 PF_WriteFileToFile()           | 2085 |
| 4.94.4.36 PF_FlushStdOutBuffer()         | 2086 |
| 4.94.4.37 PF_FreeErrorMessageBuffers()   | 2086 |
| 4.94.5 Variable Documentation            | 2086 |
| 4.94.5.1 PF                              | 2086 |
| 4.95 parallel.c                          | 2087 |
| 4.96 parallel.h File Reference           | 2135 |
| 4.96.1 Detailed Description              | 2137 |
| 4.96.2 Macro Definition Documentation    | 2137 |
| 4.96.2.1 MASTER                          | 2137 |
| 4.96.2.2 PF_RESET                        | 2137 |
| 4.96.2.3 PF_TIME                         | 2138 |
| 4.96.2.4 PF_TERM_MSGTAG                  | 2138 |
| 4.96.2.5 PF_ENDSORT_MSGTAG               | 2138 |
| 4.96.2.6 PF_DOLLAR_MSGTAG                | 2138 |
| 4.96.2.7 PF_BUFFER_MSGTAG                | 2138 |
| 4.96.2.8 PF_ENDBUFFER_MSGTAG             | 2138 |



|                                                |      |
|------------------------------------------------|------|
| 4.96.2.9 PF_READY_MSGTAG . . . . .             | 2138 |
| 4.96.2.10 PF_DATA_MSGTAG . . . . .             | 2139 |
| 4.96.2.11 PF_EMPTY_MSGTAG . . . . .            | 2139 |
| 4.96.2.12 PF_STDOUT_MSGTAG . . . . .           | 2139 |
| 4.96.2.13 PF_LOG_MSGTAG . . . . .              | 2139 |
| 4.96.2.14 PF_OPT_MCTS_MSGTAG . . . . .         | 2139 |
| 4.96.2.15 PF_OPT_HORNER_MSGTAG . . . . .       | 2139 |
| 4.96.2.16 PF_OPT_COLLECT_MSGTAG . . . . .      | 2139 |
| 4.96.2.17 PF_MISC_MSGTAG . . . . .             | 2139 |
| 4.96.2.18 GNUC_PREREQ . . . . .                | 2140 |
| 4.96.2.19 OMPI_SKIP_MPICXX . . . . .           | 2140 |
| 4.96.2.20 indices . . . . .                    | 2140 |
| 4.96.2.21 PF_ANY_SOURCE . . . . .              | 2140 |
| 4.96.2.22 PF_ANY_MSGTAG . . . . .              | 2140 |
| 4.96.2.23 PF_COMM . . . . .                    | 2140 |
| 4.96.2.24 PF_BYTE . . . . .                    | 2140 |
| 4.96.2.25 PF_INT . . . . .                     | 2141 |
| 4.96.2.26 PF_LongMultiPack . . . . .           | 2141 |
| 4.96.2.27 PF_LongMultiUnpack . . . . .         | 2141 |
| 4.96.3 Function Documentation . . . . .        | 2141 |
| 4.96.3.1 PF_IsendSbuf() . . . . .              | 2141 |
| 4.96.3.2 PF_Bcast() . . . . .                  | 2142 |
| 4.96.3.3 PF_RawSend() . . . . .                | 2142 |
| 4.96.3.4 PF_RawRecv() . . . . .                | 2142 |
| 4.96.3.5 PF_PreparePack() . . . . .            | 2143 |
| 4.96.3.6 PF_Pack() . . . . .                   | 2143 |
| 4.96.3.7 PF_Unpack() . . . . .                 | 2144 |
| 4.96.3.8 PF_PackString() . . . . .             | 2144 |
| 4.96.3.9 PF_UnpackString() . . . . .           | 2145 |
| 4.96.3.10 PF_Send() . . . . .                  | 2145 |
| 4.96.3.11 PF_Receive() . . . . .               | 2146 |
| 4.96.3.12 PF_Broadcast() . . . . .             | 2146 |
| 4.96.3.13 PF_PrepareLongSinglePack() . . . . . | 2147 |
| 4.96.3.14 PF_LongSinglePack() . . . . .        | 2147 |
| 4.96.3.15 PF_LongSingleUnpack() . . . . .      | 2147 |
| 4.96.3.16 PF_LongSingleSend() . . . . .        | 2148 |
| 4.96.3.17 PF_LongSingleReceive() . . . . .     | 2148 |
| 4.96.3.18 PF_PrepareLongMultiPack() . . . . .  | 2149 |
| 4.96.3.19 PF_LongMultiPackImpl() . . . . .     | 2149 |
| 4.96.3.20 PF_LongMultiUnpackImpl() . . . . .   | 2150 |
| 4.96.3.21 PF_LongMultiBroadcast() . . . . .    | 2151 |
| 4.96.3.22 PF_EndSort() . . . . .               | 2151 |

|           |                                |      |
|-----------|--------------------------------|------|
| 4.96.3.23 | PF_Deferred()                  | 2152 |
| 4.96.3.24 | PF_Processor()                 | 2153 |
| 4.96.3.25 | PF_Init()                      | 2154 |
| 4.96.3.26 | PF_Terminate()                 | 2155 |
| 4.96.3.27 | PF_GetSlaveTimes()             | 2156 |
| 4.96.3.28 | PF_BroadcastNumber()           | 2156 |
| 4.96.3.29 | PF_BroadcastBuffer()           | 2157 |
| 4.96.3.30 | PF_BroadcastString()           | 2158 |
| 4.96.3.31 | PF_BroadcastPreDollar()        | 2158 |
| 4.96.3.32 | PF_CollectModifiedDollars()    | 2159 |
| 4.96.3.33 | PF_BroadcastModifiedDollars()  | 2160 |
| 4.96.3.34 | PF_BroadcastRedefinedPreVars() | 2161 |
| 4.96.3.35 | PF_BroadcastCBuf()             | 2162 |
| 4.96.3.36 | PF_BroadcastExpFlags()         | 2162 |
| 4.96.3.37 | PF_StoreInsideInfo()           | 2163 |
| 4.96.3.38 | PF_RestoreInsideInfo()         | 2163 |
| 4.96.3.39 | PF_BroadcastExpr()             | 2163 |
| 4.96.3.40 | PF_BroadcastRHS()              | 2164 |
| 4.96.3.41 | PF_InParallelProcessor()       | 2164 |
| 4.96.3.42 | PF_SendFile()                  | 2164 |
| 4.96.3.43 | PF_RecvFile()                  | 2165 |
| 4.96.3.44 | PF_MLock()                     | 2166 |
| 4.96.3.45 | PF_MUnlock()                   | 2166 |
| 4.96.3.46 | PF_WriteFileToFile()           | 2166 |
| 4.96.3.47 | PF_FlushStdOutBuffer()         | 2167 |
| 4.96.4    | Variable Documentation         | 2167 |
| 4.96.4.1  | PF                             | 2167 |
| 4.96.4.2  | PF_maxDollarChunkSize          | 2167 |
| 4.97      | parallel.h                     | 2168 |
| 4.98      | pattern.c File Reference       | 2171 |
| 4.98.1    | Detailed Description           | 2171 |
| 4.98.2    | Macro Definition Documentation | 2172 |
| 4.98.2.1  | PutInBuffers                   | 2172 |
| 4.98.3    | Function Documentation         | 2172 |
| 4.98.3.1  | TestMatch()                    | 2172 |
| 4.98.3.2  | Substitute()                   | 2173 |
| 4.98.3.3  | FindAll()                      | 2173 |
| 4.98.3.4  | TestSelect()                   | 2173 |
| 4.98.3.5  | SubsInAll()                    | 2174 |
| 4.98.3.6  | TransferBuffer()               | 2174 |
| 4.98.3.7  | TakeIDfunction()               | 2174 |
| 4.99      | pattern.c                      | 2174 |

|                                              |      |
|----------------------------------------------|------|
| 4.100 poly.cc . . . . .                      | 2201 |
| 4.101 poly.h . . . . .                       | 2234 |
| 4.102 polyfact.cc File Reference . . . . .   | 2237 |
| 4.102.1 Detailed Description . . . . .       | 2238 |
| 4.102.2 Function Documentation . . . . .     | 2238 |
| 4.102.2.1 operator<<() [1/2] . . . . .       | 2238 |
| 4.102.2.2 operator<<() [2/2] . . . . .       | 2238 |
| 4.103 polyfact.cc . . . . .                  | 2239 |
| 4.104 polyfact.h . . . . .                   | 2254 |
| 4.105 polygcd.cc File Reference . . . . .    | 2256 |
| 4.105.1 Detailed Description . . . . .       | 2256 |
| 4.105.2 Function Documentation . . . . .     | 2256 |
| 4.105.2.1 gcd_heuristic_possible() . . . . . | 2256 |
| 4.105.3 Description . . . . .                | 2257 |
| 4.105.4 Notes . . . . .                      | 2257 |
| 4.105.4.1 gcd_linear_helper() . . . . .      | 2257 |
| 4.106 polygcd.cc . . . . .                   | 2257 |
| 4.107 polygcd.h . . . . .                    | 2273 |
| 4.108 polywrap.cc File Reference . . . . .   | 2275 |
| 4.108.1 Detailed Description . . . . .       | 2275 |
| 4.108.2 Function Documentation . . . . .     | 2276 |
| 4.108.2.1 poly_determine_modulus() . . . . . | 2276 |
| 4.108.3 Description . . . . .                | 2276 |
| 4.108.4 Notes . . . . .                      | 2276 |
| 4.108.4.1 poly_gcd() . . . . .               | 2276 |
| 4.108.5 Description . . . . .                | 2276 |
| 4.108.6 Notes . . . . .                      | 2277 |
| 4.108.6.1 poly_divmod() . . . . .            | 2277 |
| 4.108.6.2 poly_div() . . . . .               | 2277 |
| 4.108.6.3 poly_rem() . . . . .               | 2277 |
| 4.108.6.4 poly_ratfun_read() . . . . .       | 2278 |
| 4.108.7 Description . . . . .                | 2278 |
| 4.108.8 Notes . . . . .                      | 2278 |
| 4.108.8.1 poly_sort() . . . . .              | 2278 |
| 4.108.9 Description . . . . .                | 2278 |
| 4.108.10 Notes . . . . .                     | 2279 |
| 4.108.10.1 poly_ratfun_add() . . . . .       | 2279 |
| 4.108.11 Description . . . . .               | 2279 |
| 4.108.12 Notes . . . . .                     | 2280 |
| 4.108.12.1 poly_ratfun_normalize() . . . . . | 2280 |
| 4.108.13 Description . . . . .               | 2280 |
| 4.108.14 Notes . . . . .                     | 2281 |

|                                                |      |
|------------------------------------------------|------|
| 4.108.14.1 poly_fix_minus_signs()              | 2281 |
| 4.108.14.2 poly_factorize()                    | 2281 |
| 4.108.15 Description                           | 2282 |
| 4.108.16 Notes                                 | 2282 |
| 4.108.16.1 poly_factorize_argument()           | 2282 |
| 4.108.17 Description                           | 2282 |
| 4.108.18 Notes                                 | 2283 |
| 4.108.18.1 poly_factorize_dollar()             | 2283 |
| 4.108.19 Description                           | 2283 |
| 4.108.20 Notes                                 | 2283 |
| 4.108.20.1 poly_factorize_expression()         | 2284 |
| 4.108.21 Description                           | 2284 |
| 4.108.22 Notes                                 | 2284 |
| 4.108.22.1 poly_unfactorize_expression()       | 2285 |
| 4.108.23 Description                           | 2285 |
| 4.108.24 Notes                                 | 2285 |
| 4.108.24.1 poly_inverse()                      | 2286 |
| 4.108.24.2 poly_mul()                          | 2286 |
| 4.108.24.3 poly_free_poly_vars()               | 2286 |
| 4.108.25 Variable Documentation                | 2286 |
| 4.108.25.1 POLYWRAP_DENOMPOWER_INCREASE_FACTOR | 2286 |
| 4.109 polywrap.cc                              | 2286 |
| 4.110 portsignals.h File Reference             | 2308 |
| 4.110.1 Detailed Description                   | 2309 |
| 4.110.2 Macro Definition Documentation         | 2309 |
| 4.110.2.1 FATAL_SIG_ERROR                      | 2309 |
| 4.110.2.2 NSIG                                 | 2309 |
| 4.110.2.3 SIGSEGV                              | 2309 |
| 4.110.2.4 SIGFPE                               | 2310 |
| 4.110.2.5 SIGILL                               | 2310 |
| 4.110.2.6 SIGEMT                               | 2310 |
| 4.110.2.7 SIGSYS                               | 2310 |
| 4.110.2.8 SIGPIPE                              | 2310 |
| 4.110.2.9 SIGLOST                              | 2310 |
| 4.110.2.10 SIGXCPU                             | 2310 |
| 4.110.2.11 SIGXFSZ                             | 2310 |
| 4.110.2.12 SIGTERM                             | 2311 |
| 4.110.2.13 SIGINT                              | 2311 |
| 4.110.2.14 SIGQUIT                             | 2311 |
| 4.110.2.15 SIGHUP                              | 2311 |
| 4.110.2.16 SIGALRM                             | 2311 |
| 4.110.2.17 SIGVTALRM                           | 2311 |

|                                        |      |
|----------------------------------------|------|
| 4.110.2.18 SIGPROF                     | 2311 |
| 4.111 portsignals.h                    | 2312 |
| 4.112 pre.c File Reference             | 2313 |
| 4.112.1 Detailed Description           | 2315 |
| 4.112.2 Macro Definition Documentation | 2315 |
| 4.112.2.1 STRINGIFY                    | 2315 |
| 4.112.2.2 STRINGIFY__                  | 2316 |
| 4.112.2.3 SKIPBUFSIZE                  | 2316 |
| 4.112.2.4 KILL                         | 2316 |
| 4.112.2.5 KILLALL                      | 2316 |
| 4.112.2.6 DAEMON                       | 2316 |
| 4.112.2.7 SHELL                        | 2316 |
| 4.112.2.8 STDERR                       | 2316 |
| 4.112.2.9 TRUE_EXPR                    | 2317 |
| 4.112.2.10 FALSE_EXPR                  | 2317 |
| 4.112.2.11 NOSHELL                     | 2317 |
| 4.112.2.12 TERMINAL                    | 2317 |
| 4.112.3 Function Documentation         | 2317 |
| 4.112.3.1 GetInput()                   | 2317 |
| 4.112.3.2 ClearPushback()              | 2317 |
| 4.112.3.3 GetChar()                    | 2317 |
| 4.112.3.4 CharOut()                    | 2318 |
| 4.112.3.5 UnsetAllowDelay()            | 2318 |
| 4.112.3.6 GetPreVar()                  | 2318 |
| 4.112.3.7 PutPreVar()                  | 2318 |
| 4.112.3.8 PopPreVars()                 | 2319 |
| 4.112.3.9 IniModule()                  | 2319 |
| 4.112.3.10 IniSpecialModule()          | 2319 |
| 4.112.3.11 PreProcessor()              | 2319 |
| 4.112.3.12 PreProInstruction()         | 2319 |
| 4.112.3.13 LoadInstruction()           | 2319 |
| 4.112.3.14 LoadStatement()             | 2319 |
| 4.112.3.15 ExpandTripleDots()          | 2320 |
| 4.112.3.16 FindKeyWord()               | 2320 |
| 4.112.3.17 FindInKeyWord()             | 2320 |
| 4.112.3.18 TheDefine()                 | 2320 |
| 4.112.3.19 DoCommentChar()             | 2321 |
| 4.112.3.20 DoPreAssign()               | 2321 |
| 4.112.3.21 DoDefine()                  | 2321 |
| 4.112.3.22 DoRedefine()                | 2321 |
| 4.112.3.23 ClearMacro()                | 2321 |
| 4.112.3.24 TheUndefine()               | 2322 |

|                                  |      |
|----------------------------------|------|
| 4.112.3.25 DoUndefine()          | 2322 |
| 4.112.3.26 DoInclude()           | 2322 |
| 4.112.3.27 DoReverseInclude()    | 2322 |
| 4.112.3.28 Include()             | 2322 |
| 4.112.3.29 DoPreExchange()       | 2322 |
| 4.112.3.30 DoCall()              | 2322 |
| 4.112.3.31 DoDebug()             | 2323 |
| 4.112.3.32 DoTerminate()         | 2323 |
| 4.112.3.33 DoContinueDo()        | 2323 |
| 4.112.3.34 DoDo()                | 2323 |
| 4.112.3.35 DoBreakDo()           | 2323 |
| 4.112.3.36 DoElse()              | 2323 |
| 4.112.3.37 DoElseif()            | 2323 |
| 4.112.3.38 DoEnddo()             | 2324 |
| 4.112.3.39 DoEndif()             | 2324 |
| 4.112.3.40 DoEndprocedure()      | 2324 |
| 4.112.3.41 Dolf()                | 2324 |
| 4.112.3.42 Dolfdef()             | 2324 |
| 4.112.3.43 Dolfydef()            | 2324 |
| 4.112.3.44 Dolfndef()            | 2324 |
| 4.112.3.45 DoInside()            | 2325 |
| 4.112.3.46 DoEndInside()         | 2325 |
| 4.112.3.47 DoMessage()           | 2325 |
| 4.112.3.48 DoPipe()              | 2325 |
| 4.112.3.49 DoPrCExtension()      | 2325 |
| 4.112.3.50 DoPreOut()            | 2325 |
| 4.112.3.51 DoPrePrintTimes()     | 2325 |
| 4.112.3.52 DoPreSortReallocate() | 2326 |
| 4.112.3.53 DoPreAppend()         | 2326 |
| 4.112.3.54 DoPreCreate()         | 2326 |
| 4.112.3.55 DoPreRemove()         | 2326 |
| 4.112.3.56 DoPreClose()          | 2326 |
| 4.112.3.57 DoPreWrite()          | 2326 |
| 4.112.3.58 DoProcedure()         | 2326 |
| 4.112.3.59 DoPreBreak()          | 2327 |
| 4.112.3.60 DoPreCase()           | 2327 |
| 4.112.3.61 DoPreDefault()        | 2327 |
| 4.112.3.62 DoPreEndSwitch()      | 2327 |
| 4.112.3.63 DoPreSwitch()         | 2327 |
| 4.112.3.64 DoPreShow()           | 2327 |
| 4.112.3.65 DoSystem()            | 2327 |
| 4.112.3.66 PreLoad()             | 2328 |

|                                 |      |
|---------------------------------|------|
| 4.112.3.67 PreSkip()            | 2328 |
| 4.112.3.68 StartPrepro()        | 2328 |
| 4.112.3.69 EvalPrelf()          | 2328 |
| 4.112.3.70 PrelfEval()          | 2328 |
| 4.112.3.71 PreCmp()             | 2328 |
| 4.112.3.72 PreEq()              | 2329 |
| 4.112.3.73 pParseObject()       | 2329 |
| 4.112.3.74 PreCalc()            | 2329 |
| 4.112.3.75 PreEval()            | 2329 |
| 4.112.3.76 AddToPreTypes()      | 2329 |
| 4.112.3.77 MessPreNesting()     | 2329 |
| 4.112.3.78 DoPreAddSeparator()  | 2330 |
| 4.112.3.79 DoPreRmSeparator()   | 2330 |
| 4.112.3.80 DoExternal()         | 2330 |
| 4.112.3.81 DoPrompt()           | 2330 |
| 4.112.3.82 DoSetExternal()      | 2330 |
| 4.112.3.83 DoSetExternalAttr()  | 2330 |
| 4.112.3.84 DoRmExternal()       | 2330 |
| 4.112.3.85 DoFromExternal()     | 2331 |
| 4.112.3.86 DoToExternal()       | 2331 |
| 4.112.3.87 defineChannel()      | 2331 |
| 4.112.3.88 writeToChannel()     | 2331 |
| 4.112.3.89 DoFactDollar()       | 2331 |
| 4.112.3.90 GetDollarNumber()    | 2331 |
| 4.112.3.91 DoSetRandom()        | 2332 |
| 4.112.3.92 DoOptimize()         | 2332 |
| 4.112.3.93 DoClearOptimize()    | 2332 |
| 4.112.3.94 DoSkipExtraSymbols() | 2332 |
| 4.112.3.95 DoPreReset()         | 2332 |
| 4.112.3.96 DoPreAppendPath()    | 2332 |
| 4.112.3.97 DoPrePrependPath()   | 2333 |
| 4.112.3.98 DoTimeOutAfter()     | 2333 |
| 4.112.3.99 DoNamespace()        | 2333 |
| 4.112.3.100 DoEndNamespace()    | 2333 |
| 4.112.3.101 SkipName()          | 2333 |
| 4.112.3.102 ConstructName()     | 2333 |
| 4.112.3.103 DoUse()             | 2334 |
| 4.112.3.104 UserFlags()         | 2334 |
| 4.112.3.105 DoClearUserFlag()   | 2334 |
| 4.112.3.106 DoSetUserFlag()     | 2334 |
| 4.113 pre.c                     | 2334 |
| 4.114 proces.c File Reference   | 2423 |

|                                        |      |
|----------------------------------------|------|
| 4.114.1 Detailed Description           | 2424 |
| 4.114.2 Macro Definition Documentation | 2424 |
| 4.114.2.1 DONE                         | 2424 |
| 4.114.3 Function Documentation         | 2425 |
| 4.114.3.1 Processor()                  | 2425 |
| 4.114.3.2 TestSub()                    | 2426 |
| 4.114.3.3 InFunction()                 | 2426 |
| 4.114.3.4 InsertTerm()                 | 2427 |
| 4.114.3.5 PasteFile()                  | 2428 |
| 4.114.3.6 PasteTerm()                  | 2429 |
| 4.114.3.7 FiniTerm()                   | 2429 |
| 4.114.3.8 Generator()                  | 2430 |
| 4.114.3.9 DoOnePow()                   | 2431 |
| 4.114.3.10 Deferred()                  | 2432 |
| 4.114.3.11 PrepPoly()                  | 2433 |
| 4.114.3.12 PolyFunMul()                | 2434 |
| 4.114.4 Variable Documentation         | 2435 |
| 4.114.4.1 printscratch                 | 2435 |
| 4.115 proces.c                         | 2435 |
| 4.116 ratio.c File Reference           | 2500 |
| 4.116.1 Detailed Description           | 2501 |
| 4.116.2 Function Documentation         | 2501 |
| 4.116.2.1 RatioFind()                  | 2501 |
| 4.116.2.2 RatioGen()                   | 2501 |
| 4.116.2.3 BinomGen()                   | 2502 |
| 4.116.2.4 DoSumF1()                    | 2502 |
| 4.116.2.5 Glue()                       | 2502 |
| 4.116.2.6 DoSumF2()                    | 2502 |
| 4.116.2.7 GCDfunction()                | 2502 |
| 4.116.2.8 GCDfunction3()               | 2503 |
| 4.116.2.9 PutExtraSymbols()            | 2503 |
| 4.116.2.10 TakeExtraSymbols()          | 2503 |
| 4.116.2.11 MultiplyWithTerm()          | 2503 |
| 4.116.2.12 TakeContent()               | 2503 |
| 4.116.2.13 MergeSymbolLists()          | 2504 |
| 4.116.2.14 MergeDotproductLists()      | 2504 |
| 4.116.2.15 CreateExpression()          | 2504 |
| 4.116.2.16 GCDterms()                  | 2504 |
| 4.116.2.17 ReadPolyRatFun()            | 2505 |
| 4.116.2.18 FromPolyRatFun()            | 2505 |
| 4.116.2.19 TakeSymbolContent()         | 2505 |
| 4.116.2.20 GCDclean()                  | 2505 |



|                                        |      |
|----------------------------------------|------|
| 4.116.2.21 PolyDiv()                   | 2506 |
| 4.116.2.22 DIVfunction()               | 2506 |
| 4.116.2.23 MULfunc()                   | 2506 |
| 4.116.2.24 ConvertArgument()           | 2506 |
| 4.116.2.25 ExpandRat()                 | 2506 |
| 4.116.2.26 InvPoly()                   | 2506 |
| 4.116.3 Variable Documentation         | 2507 |
| 4.116.3.1 divrem                       | 2507 |
| 4.116.3.2 TheErrorMessage              | 2507 |
| 4.117 ratio.c                          | 2507 |
| 4.118 reken.c File Reference           | 2549 |
| 4.118.1 Detailed Description           | 2551 |
| 4.118.2 Macro Definition Documentation | 2551 |
| 4.118.2.1 GCDMAX                       | 2551 |
| 4.118.2.2 NEWTRICK                     | 2551 |
| 4.118.2.3 COPYLONG                     | 2551 |
| 4.118.2.4 WARMUP                       | 2552 |
| 4.118.3 Function Documentation         | 2552 |
| 4.118.3.1 Pack()                       | 2552 |
| 4.118.3.2 UnPack()                     | 2552 |
| 4.118.3.3 Mully()                      | 2552 |
| 4.118.3.4 Divvy()                      | 2552 |
| 4.118.3.5 AddRat()                     | 2553 |
| 4.118.3.6 MulRat()                     | 2553 |
| 4.118.3.7 DivRat()                     | 2553 |
| 4.118.3.8 Simplify()                   | 2553 |
| 4.118.3.9 AccumGCD()                   | 2553 |
| 4.118.3.10 TakeRatRoot()               | 2554 |
| 4.118.3.11 AddLong()                   | 2554 |
| 4.118.3.12 AddPLon()                   | 2554 |
| 4.118.3.13 SubPLon()                   | 2554 |
| 4.118.3.14 MulLong()                   | 2554 |
| 4.118.3.15 BigLong()                   | 2555 |
| 4.118.3.16 DivLong()                   | 2555 |
| 4.118.3.17 RaisPow()                   | 2555 |
| 4.118.3.18 RaisPowCached()             | 2555 |
| 4.118.4 Description                    | 2555 |
| 4.118.5 Notes                          | 2556 |
| 4.118.5.1 RaisPowMod()                 | 2556 |
| 4.118.5.2 NormalModulus()              | 2556 |
| 4.118.5.3 MakeInverses()               | 2556 |
| 4.118.5.4 GetModInverses()             | 2557 |

|                                        |      |
|----------------------------------------|------|
| 4.118.5.5 GetLongModInverses()         | 2557 |
| 4.118.5.6 Product()                    | 2557 |
| 4.118.5.7 Quotient()                   | 2557 |
| 4.118.5.8 Remain10()                   | 2558 |
| 4.118.5.9 Remain4()                    | 2558 |
| 4.118.5.10 PrtLong()                   | 2558 |
| 4.118.5.11 GetLong()                   | 2558 |
| 4.118.5.12 GcdLong()                   | 2558 |
| 4.118.5.13 GetBinom()                  | 2559 |
| 4.118.5.14 LcmLong()                   | 2559 |
| 4.118.5.15 TakeLongRoot()              | 2559 |
| 4.118.5.16 MakeRational()              | 2559 |
| 4.118.5.17 MakeLongRational()          | 2559 |
| 4.118.5.18 CompCoef()                  | 2560 |
| 4.118.5.19 Modulus()                   | 2560 |
| 4.118.5.20 TakeModulus()               | 2560 |
| 4.118.5.21 TakeNormalModulus()         | 2560 |
| 4.118.5.22 MakeModTable()              | 2560 |
| 4.118.5.23 Factorial()                 | 2561 |
| 4.118.5.24 Bernoulli()                 | 2561 |
| 4.118.5.25 NextPrime()                 | 2561 |
| 4.118.5.26 Moebius()                   | 2561 |
| 4.118.5.27 iniwranf()                  | 2561 |
| 4.118.5.28 wranf()                     | 2562 |
| 4.118.5.29 iranf()                     | 2562 |
| 4.118.5.30 PreRandom()                 | 2562 |
| 4.119 reken.c                          | 2562 |
| 4.120 reshuf.c File Reference          | 2608 |
| 4.120.1 Detailed Description           | 2608 |
| 4.120.2 Macro Definition Documentation | 2609 |
| 4.120.2.1 NEWCODE                      | 2609 |
| 4.120.3 Function Documentation         | 2609 |
| 4.120.3.1 ReNumber()                   | 2609 |
| 4.120.3.2 FunLevel()                   | 2609 |
| 4.120.3.3 DetCurDum()                  | 2609 |
| 4.120.3.4 FullRenumber()               | 2609 |
| 4.120.3.5 MoveDummies()                | 2609 |
| 4.120.3.6 AdjustRenumScratch()         | 2610 |
| 4.120.3.7 CountDo()                    | 2610 |
| 4.120.3.8 CountFun()                   | 2610 |
| 4.120.3.9 DimensionSubterm()           | 2610 |
| 4.120.3.10 DimensionTerm()             | 2610 |

|                                        |      |
|----------------------------------------|------|
| 4.120.3.11 DimensionExpression()       | 2610 |
| 4.120.3.12 MultDo()                    | 2611 |
| 4.120.3.13 TryDo()                     | 2611 |
| 4.120.3.14 DoDistrib()                 | 2611 |
| 4.120.3.15 EqualArg()                  | 2611 |
| 4.120.3.16 DoDelta3()                  | 2611 |
| 4.120.3.17 TestPartitions()            | 2611 |
| 4.120.3.18 DoPartitions()              | 2612 |
| 4.120.3.19 DoPermutations()            | 2612 |
| 4.120.3.20 DoShuffle()                 | 2612 |
| 4.120.3.21 Shuffle()                   | 2612 |
| 4.120.3.22 FinishShuffle()             | 2612 |
| 4.120.3.23 DoStuffle()                 | 2612 |
| 4.120.3.24 Stuffle()                   | 2613 |
| 4.120.3.25 FinishStuffle()             | 2613 |
| 4.120.3.26 StuffRootAdd()              | 2613 |
| 4.121 reshuf.c                         | 2613 |
| 4.122 sch.c File Reference             | 2649 |
| 4.122.1 Detailed Description           | 2650 |
| 4.122.2 Macro Definition Documentation | 2650 |
| 4.122.2.1 va_dcl                       | 2650 |
| 4.122.2.2 va_start                     | 2650 |
| 4.122.2.3 va_end                       | 2651 |
| 4.122.2.4 va_arg                       | 2651 |
| 4.122.3 Typedef Documentation          | 2651 |
| 4.122.3.1 va_list                      | 2651 |
| 4.122.4 Function Documentation         | 2651 |
| 4.122.4.1 StrCopy()                    | 2651 |
| 4.122.4.2 AddToLine()                  | 2651 |
| 4.122.4.3 FiniLine()                   | 2651 |
| 4.122.4.4 IniLine()                    | 2652 |
| 4.122.4.5 LongToLine()                 | 2652 |
| 4.122.4.6 RatToLine()                  | 2652 |
| 4.122.4.7 TalToLine()                  | 2652 |
| 4.122.4.8 TokenToLine()                | 2652 |
| 4.122.4.9 CodeToLine()                 | 2652 |
| 4.122.4.10 MultiplyToLine()            | 2653 |
| 4.122.4.11 AddArrayIndex()             | 2653 |
| 4.122.4.12 PrtTerms()                  | 2653 |
| 4.122.4.13 WrtPower()                  | 2653 |
| 4.122.4.14 PrintTime()                 | 2653 |
| 4.122.4.15 WriteLists()                | 2653 |

|                                        |      |
|----------------------------------------|------|
| 4.122.4.16 WriteDictionary()           | 2653 |
| 4.122.4.17 WriteArgument()             | 2654 |
| 4.122.4.18 WriteSubTerm()              | 2654 |
| 4.122.4.19 WriteInnerTerm()            | 2654 |
| 4.122.4.20 WriteTerm()                 | 2654 |
| 4.122.4.21 WriteExpression()           | 2654 |
| 4.122.4.22 WriteAll()                  | 2654 |
| 4.122.4.23 WriteOne()                  | 2655 |
| 4.123 sch.c                            | 2655 |
| 4.124 setfile.c File Reference         | 2688 |
| 4.124.1 Detailed Description           | 2689 |
| 4.124.2 Macro Definition Documentation | 2689 |
| 4.124.2.1 NUMERICALVALUE               | 2689 |
| 4.124.2.2 STRINGVALUE                  | 2689 |
| 4.124.2.3 PATHVALUE                    | 2690 |
| 4.124.2.4 ONOFFVALUE                   | 2690 |
| 4.124.2.5 DEFINEVALUE                  | 2690 |
| 4.124.2.6 SETBUFSIZE                   | 2690 |
| 4.124.3 Function Documentation         | 2690 |
| 4.124.3.1 DoSetups()                   | 2690 |
| 4.124.3.2 ProcessOption()              | 2690 |
| 4.124.3.3 GetSetupPar()                | 2690 |
| 4.124.3.4 RecalcSetups()               | 2691 |
| 4.124.3.5 AllocSetups()                | 2691 |
| 4.124.3.6 WriteSetup()                 | 2691 |
| 4.124.3.7 AllocSort()                  | 2691 |
| 4.124.3.8 AllocSortFileName()          | 2691 |
| 4.124.3.9 AllocFileHandle()            | 2691 |
| 4.124.3.10 DeAllocFileHandle()         | 2692 |
| 4.124.3.11 MakeSetupAllocs()           | 2692 |
| 4.124.3.12 TryFileSetups()             | 2692 |
| 4.124.3.13 TryEnvironment()            | 2692 |
| 4.124.4 Variable Documentation         | 2692 |
| 4.124.4.1 curdirp                      | 2692 |
| 4.124.4.2 cursortdirp                  | 2692 |
| 4.124.4.3 commentchar                  | 2692 |
| 4.124.4.4 dotchar                      | 2693 |
| 4.124.4.5 highfirst                    | 2693 |
| 4.124.4.6 lowfirst                     | 2693 |
| 4.124.4.7 procedureextension           | 2693 |
| 4.124.4.8 setupparameters              | 2693 |
| 4.125 setfile.c                        | 2693 |

|                                        |      |
|----------------------------------------|------|
| 4.126 smart.c File Reference           | 2708 |
| 4.126.1 Detailed Description           | 2708 |
| 4.126.2 Function Documentation         | 2709 |
| 4.126.2.1 StudyPattern()               | 2709 |
| 4.126.2.2 MatchIsPossible()            | 2709 |
| 4.127 smart.c                          | 2709 |
| 4.128 sort.c File Reference            | 2714 |
| 4.128.1 Detailed Description           | 2715 |
| 4.128.2 Macro Definition Documentation | 2715 |
| 4.128.2.1 NEWSPLITMERGE                | 2715 |
| 4.128.2.2 NEWSPLITMERGETIMSORT         | 2715 |
| 4.128.2.3 HUMANSTRLEN                  | 2716 |
| 4.128.2.4 HUMANSUFFLEN                 | 2716 |
| 4.128.2.5 INSLLENGTH                   | 2716 |
| 4.128.3 Function Documentation         | 2716 |
| 4.128.3.1 HumanString()                | 2716 |
| 4.128.3.2 WriteStats()                 | 2716 |
| 4.128.3.3 NewSort()                    | 2717 |
| 4.128.3.4 EndSort()                    | 2717 |
| 4.128.3.5 PutIn()                      | 2718 |
| 4.128.3.6 Sflush()                     | 2719 |
| 4.128.3.7 PutOut()                     | 2719 |
| 4.128.3.8 FlushOut()                   | 2720 |
| 4.128.3.9 AddCoef()                    | 2721 |
| 4.128.3.10 AddPoly()                   | 2721 |
| 4.128.3.11 AddArgs()                   | 2722 |
| 4.128.3.12 Compare1()                  | 2723 |
| 4.128.3.13 CompareSymbols()            | 2724 |
| 4.128.3.14 CompareHSymbols()           | 2724 |
| 4.128.3.15 ComPress()                  | 2724 |
| 4.128.3.16 SplitMerge()                | 2725 |
| 4.128.3.17 GarbHand()                  | 2726 |
| 4.128.3.18 MergePatches()              | 2726 |
| 4.128.3.19 StoreTerm()                 | 2727 |
| 4.128.3.20 StageSort()                 | 2728 |
| 4.128.3.21 SortWild()                  | 2728 |
| 4.128.3.22 CleanUpSort()               | 2729 |
| 4.128.3.23 LowerSortLevel()            | 2729 |
| 4.128.3.24 PolyRatFunSpecial()         | 2729 |
| 4.128.3.25 SimpleSplitMergeRec()       | 2729 |
| 4.128.3.26 SimpleSplitMerge()          | 2729 |
| 4.128.3.27 BinarySearch()              | 2730 |

|                                        |      |
|----------------------------------------|------|
| 4.128.4 Variable Documentation         | 2730 |
| 4.128.4.1 numcompares                  | 2730 |
| 4.128.4.2 toterms                      | 2730 |
| 4.128.4.3 humanTermsSuffix             | 2730 |
| 4.128.4.4 humanBytesSuffix             | 2730 |
| 4.129 sort.c                           | 2731 |
| 4.130 spectator.c File Reference       | 2786 |
| 4.130.1 Detailed Description           | 2786 |
| 4.130.2 Function Documentation         | 2787 |
| 4.130.2.1 CoCreateSpectator()          | 2787 |
| 4.130.2.2 CoToSpectator()              | 2787 |
| 4.130.2.3 CoRemoveSpectator()          | 2787 |
| 4.130.2.4 CoEmptySpectator()           | 2787 |
| 4.130.2.5 PutInSpectator()             | 2787 |
| 4.130.2.6 FlushSpectators()            | 2787 |
| 4.130.2.7 CoCopySpectator()            | 2788 |
| 4.130.2.8 GetFromSpectator()           | 2788 |
| 4.130.2.9 ClearSpectators()            | 2788 |
| 4.131 spectator.c                      | 2788 |
| 4.132 startup.c File Reference         | 2796 |
| 4.132.1 Detailed Description           | 2797 |
| 4.132.2 Macro Definition Documentation | 2797 |
| 4.132.2.1 STRINGIFY                    | 2797 |
| 4.132.2.2 STRINGIFY__                  | 2798 |
| 4.132.2.3 FORMNAME                     | 2798 |
| 4.132.2.4 VERSIONSTR__                 | 2798 |
| 4.132.2.5 VERSIONSTR                   | 2798 |
| 4.132.2.6 TAKEPATH                     | 2798 |
| 4.132.2.7 DEFAULTFNAMELENGTH           | 2798 |
| 4.132.3 Function Documentation         | 2798 |
| 4.132.3.1 DoTail()                     | 2798 |
| 4.132.3.2 OpenInput()                  | 2799 |
| 4.132.3.3 ReserveTempFiles()           | 2799 |
| 4.132.3.4 StartVariables()             | 2799 |
| 4.132.3.5 StartMore()                  | 2799 |
| 4.132.3.6 IniVars()                    | 2800 |
| 4.132.3.7 main()                       | 2800 |
| 4.132.3.8 CleanUp()                    | 2800 |
| 4.132.3.9 TerminateImpl()              | 2800 |
| 4.132.3.10 PrintDeprecation()          | 2800 |
| 4.132.3.11 PrintRunningTime()          | 2801 |
| 4.132.3.12 GetRunningTime()            | 2801 |

|                                        |      |
|----------------------------------------|------|
| 4.132.4 Variable Documentation         | 2801 |
| 4.132.4.1 emptystring                  | 2801 |
| 4.132.4.2 defaulttempfilename          | 2801 |
| 4.133 startup.c                        | 2801 |
| 4.134 store.c File Reference           | 2826 |
| 4.134.1 Detailed Description           | 2827 |
| 4.134.2 Macro Definition Documentation | 2827 |
| 4.134.2.1 SAVEREVISION                 | 2827 |
| 4.134.3 Function Documentation         | 2828 |
| 4.134.3.1 OpenTemp()                   | 2828 |
| 4.134.3.2 SeekScratch()                | 2828 |
| 4.134.3.3 SetEndScratch()              | 2828 |
| 4.134.3.4 SetEndHScratch()             | 2828 |
| 4.134.3.5 SetScratch()                 | 2828 |
| 4.134.3.6 RevertScratch()              | 2828 |
| 4.134.3.7 ResetScratch()               | 2829 |
| 4.134.3.8 ReadFromScratch()            | 2829 |
| 4.134.3.9 AddToScratch()               | 2829 |
| 4.134.3.10 CoSave()                    | 2829 |
| 4.134.3.11 CoLoad()                    | 2829 |
| 4.134.3.12 DeleteStore()               | 2829 |
| 4.134.3.13 PutInStore()                | 2830 |
| 4.134.3.14 GetTerm()                   | 2830 |
| 4.134.3.15 GetOneTerm()                | 2830 |
| 4.134.3.16 GetMoreTerms()              | 2830 |
| 4.134.3.17 GetMoreFromMem()            | 2830 |
| 4.134.3.18 GetFromStore()              | 2830 |
| 4.134.3.19 DetVars()                   | 2831 |
| 4.134.3.20 ToStorage()                 | 2831 |
| 4.134.3.21 NextFileIndex()             | 2831 |
| 4.134.3.22 SetFileIndex()              | 2831 |
| 4.134.3.23 VarStore()                  | 2832 |
| 4.134.3.24 TermRenumber()              | 2832 |
| 4.134.3.25 FindrNumber()               | 2832 |
| 4.134.3.26 FindInIndex()               | 2833 |
| 4.134.3.27 GetTable()                  | 2833 |
| 4.134.3.28 CopyExpression()            | 2833 |
| 4.134.3.29 WriteStoreHeader()          | 2833 |
| 4.134.3.30 ReadSaveHeader()            | 2834 |
| 4.134.3.31 ReadSaveIndex()             | 2834 |
| 4.134.3.32 ReadSaveVariables()         | 2834 |
| 4.134.3.33 ReadSaveTerm32()            | 2835 |

|                                        |      |
|----------------------------------------|------|
| 4.134.3.34 ReadSaveExpression()        | 2836 |
| 4.135 store.c                          | 2837 |
| 4.136 structs.h File Reference         | 2895 |
| 4.136.1 Detailed Description           | 2898 |
| 4.136.2 Macro Definition Documentation | 2898 |
| 4.136.2.1 INFILEINDEX                  | 2898 |
| 4.136.2.2 EMPTYININDEX                 | 2899 |
| 4.136.2.3 PHEAD                        | 2899 |
| 4.136.2.4 PHEAD0                       | 2899 |
| 4.136.2.5 BHEAD                        | 2899 |
| 4.136.2.6 BHEAD0                       | 2899 |
| 4.136.3 Typedef Documentation          | 2899 |
| 4.136.3.1 INDEXENTRY                   | 2899 |
| 4.136.3.2 FILEINDEX                    | 2900 |
| 4.136.3.3 VARRENUM                     | 2900 |
| 4.136.3.4 RENUMBER                     | 2900 |
| 4.136.3.5 NAMENODE                     | 2900 |
| 4.136.3.6 NAMETREE                     | 2900 |
| 4.136.3.7 COMPTREE                     | 2900 |
| 4.136.3.8 TABLES                       | 2901 |
| 4.136.3.9 FUNCTIONS                    | 2901 |
| 4.136.3.10 FILEHANDLE                  | 2901 |
| 4.136.3.11 STREAM                      | 2901 |
| 4.136.3.12 TRACES                      | 2901 |
| 4.136.3.13 TRACEN                      | 2901 |
| 4.136.3.14 PREVAR                      | 2902 |
| 4.136.3.15 DOLOOP                      | 2902 |
| 4.136.3.16 set_of_char                 | 2902 |
| 4.136.3.17 one_byte                    | 2902 |
| 4.136.3.18 FINISHUFFLE                 | 2902 |
| 4.136.3.19 DO_UFFLE                    | 2902 |
| 4.136.3.20 COMPAREDUMMY                | 2902 |
| 4.136.3.21 CBUF                        | 2903 |
| 4.136.3.22 CHANNEL                     | 2903 |
| 4.136.3.23 NESTING                     | 2903 |
| 4.136.3.24 STORECACHE                  | 2903 |
| 4.136.3.25 PERM                        | 2903 |
| 4.136.3.26 PERMP                       | 2903 |
| 4.136.3.27 DISTRIBUTE                  | 2903 |
| 4.136.3.28 PARTI                       | 2904 |
| 4.136.3.29 SORTING                     | 2904 |
| 4.136.3.30 ALLGLOBALS                  | 2904 |



|                                               |      |
|-----------------------------------------------|------|
| 4.136.3.31 WCN . . . . .                      | 2904 |
| 4.136.3.32 WCN2 . . . . .                     | 2904 |
| 4.136.3.33 COMPARE . . . . .                  | 2904 |
| 4.136.3.34 FIXEDGLOBALS . . . . .             | 2904 |
| 4.137 structs.h . . . . .                     | 2905 |
| 4.138 symmetr.c File Reference . . . . .      | 2931 |
| 4.138.1 Detailed Description . . . . .        | 2931 |
| 4.138.2 Function Documentation . . . . .      | 2931 |
| 4.138.2.1 MatchE() . . . . .                  | 2931 |
| 4.138.2.2 Permute() . . . . .                 | 2931 |
| 4.138.2.3 PermuteP() . . . . .                | 2932 |
| 4.138.2.4 Distribute() . . . . .              | 2932 |
| 4.138.2.5 MatchCy() . . . . .                 | 2932 |
| 4.138.2.6 FunMatchCy() . . . . .              | 2932 |
| 4.138.2.7 FunMatchSy() . . . . .              | 2932 |
| 4.138.2.8 MatchArgument() . . . . .           | 2933 |
| 4.139 symmetr.c . . . . .                     | 2933 |
| 4.140 tables.c File Reference . . . . .       | 2960 |
| 4.140.1 Detailed Description . . . . .        | 2961 |
| 4.140.2 Function Documentation . . . . .      | 2961 |
| 4.140.2.1 ClearTableTree() . . . . .          | 2961 |
| 4.140.2.2 InsTableTree() . . . . .            | 2961 |
| 4.140.2.3 RedoTableTree() . . . . .           | 2961 |
| 4.140.2.4 FindTableTree() . . . . .           | 2961 |
| 4.140.2.5 DoTableExpansion() . . . . .        | 2962 |
| 4.140.2.6 CoTableBase() . . . . .             | 2962 |
| 4.140.2.7 FlipTable() . . . . .               | 2962 |
| 4.140.2.8 SpareTable() . . . . .              | 2962 |
| 4.140.2.9 FindTB() . . . . .                  | 2962 |
| 4.140.2.10 CoTBcreate() . . . . .             | 2962 |
| 4.140.2.11 CoTBopen() . . . . .               | 2962 |
| 4.140.2.12 CoTBaddto() . . . . .              | 2963 |
| 4.140.2.13 CoTBenter() . . . . .              | 2963 |
| 4.140.2.14 CoTestUse() . . . . .              | 2963 |
| 4.140.2.15 CheckTableDeclarations() . . . . . | 2963 |
| 4.140.2.16 CoTBload() . . . . .               | 2963 |
| 4.140.2.17 TestUse() . . . . .                | 2963 |
| 4.140.2.18 CoTBaudit() . . . . .              | 2963 |
| 4.140.2.19 CoTBon() . . . . .                 | 2964 |
| 4.140.2.20 CoTBoff() . . . . .                | 2964 |
| 4.140.2.21 CoTBcleanup() . . . . .            | 2964 |
| 4.140.2.22 CoTBreplace() . . . . .            | 2964 |

|                                        |      |
|----------------------------------------|------|
| 4.140.2.23 CoTBuse()                   | 2964 |
| 4.140.2.24 CoApply()                   | 2964 |
| 4.140.2.25 CoTBhelp()                  | 2964 |
| 4.140.2.26 ReWorkT()                   | 2965 |
| 4.140.2.27 Apply()                     | 2965 |
| 4.140.2.28 ApplyExec()                 | 2965 |
| 4.140.2.29 ApplyReset()                | 2965 |
| 4.140.2.30 TableReset()                | 2965 |
| 4.140.2.31 ReleaseTB()                 | 2965 |
| 4.140.3 Variable Documentation         | 2966 |
| 4.140.3.1 helptb                       | 2966 |
| 4.141 tables.c                         | 2966 |
| 4.142 threads.c File Reference         | 2993 |
| 4.142.1 Detailed Description           | 2993 |
| 4.143 threads.c                        | 2993 |
| 4.144 token.c File Reference           | 3047 |
| 4.144.1 Detailed Description           | 3048 |
| 4.144.2 Macro Definition Documentation | 3048 |
| 4.144.2.1 CHECKPOLY                    | 3048 |
| 4.144.3 Function Documentation         | 3048 |
| 4.144.3.1 tokenize()                   | 3048 |
| 4.144.3.2 WriteTokens()                | 3048 |
| 4.144.3.3 simp1token()                 | 3048 |
| 4.144.3.4 simpwtoken()                 | 3049 |
| 4.144.3.5 simp2token()                 | 3049 |
| 4.144.3.6 simp3atoken()                | 3049 |
| 4.144.3.7 simp3btoken()                | 3049 |
| 4.144.3.8 simp4token()                 | 3049 |
| 4.144.3.9 simp5token()                 | 3049 |
| 4.144.3.10 simp6token()                | 3050 |
| 4.144.4 Variable Documentation         | 3050 |
| 4.144.4.1 ttypes                       | 3050 |
| 4.145 token.c                          | 3050 |
| 4.146 tools.c File Reference           | 3075 |
| 4.146.1 Detailed Description           | 3077 |
| 4.146.2 Macro Definition Documentation | 3077 |
| 4.146.2.1 FILLVALUE                    | 3077 |
| 4.146.2.2 TERMMEMSTARTNUM              | 3077 |
| 4.146.2.3 TERMEXTRAWORDS               | 3078 |
| 4.146.2.4 NUMBERMEMSTARTNUM            | 3078 |
| 4.146.2.5 NUMBEREXTRAWORDS             | 3078 |
| 4.146.2.6 DODOUBLE                     | 3078 |

---

|                                          |      |
|------------------------------------------|------|
| 4.146.2.7 DOEXPAND . . . . .             | 3078 |
| 4.146.3 Function Documentation . . . . . | 3079 |
| 4.146.3.1 LoadInputFile() . . . . .      | 3079 |
| 4.146.3.2 ReadFromStream() . . . . .     | 3079 |
| 4.146.3.3 GetFromStream() . . . . .      | 3079 |
| 4.146.3.4 LookInStream() . . . . .       | 3079 |
| 4.146.3.5 OpenStream() . . . . .         | 3079 |
| 4.146.3.6 LocateFile() . . . . .         | 3079 |
| 4.146.3.7 CloseStream() . . . . .        | 3080 |
| 4.146.3.8 CreateStream() . . . . .       | 3080 |
| 4.146.3.9 GetStreamPosition() . . . . .  | 3080 |
| 4.146.3.10 PositionStream() . . . . .    | 3080 |
| 4.146.3.11 ReverseStatements() . . . . . | 3080 |
| 4.146.3.12 StartFiles() . . . . .        | 3080 |
| 4.146.3.13 OpenFile() . . . . .          | 3080 |
| 4.146.3.14 OpenAddFile() . . . . .       | 3081 |
| 4.146.3.15 ReOpenFile() . . . . .        | 3081 |
| 4.146.3.16 CreateFile() . . . . .        | 3081 |
| 4.146.3.17 CreateLogFile() . . . . .     | 3081 |
| 4.146.3.18 CloseFile() . . . . .         | 3081 |
| 4.146.3.19 CopyFile() . . . . .          | 3081 |
| 4.146.3.20 CreateHandle() . . . . .      | 3082 |
| 4.146.3.21 ReadFile() . . . . .          | 3082 |
| 4.146.3.22 ReadPosFile() . . . . .       | 3082 |
| 4.146.3.23 WriteFileToFile() . . . . .   | 3082 |
| 4.146.3.24 SeekFile() . . . . .          | 3082 |
| 4.146.3.25 TellFile() . . . . .          | 3082 |
| 4.146.3.26 TELLFILE() . . . . .          | 3083 |
| 4.146.3.27 FlushFile() . . . . .         | 3083 |
| 4.146.3.28 GetPosFile() . . . . .        | 3083 |
| 4.146.3.29 SetPosFile() . . . . .        | 3083 |
| 4.146.3.30 SynchFile() . . . . .         | 3083 |
| 4.146.3.31 TruncateFile() . . . . .      | 3083 |
| 4.146.3.32 GetChannel() . . . . .        | 3084 |
| 4.146.3.33 GetAppendChannel() . . . . .  | 3084 |
| 4.146.3.34 CloseChannel() . . . . .      | 3084 |
| 4.146.3.35 UpdateMaxSize() . . . . .     | 3084 |
| 4.146.3.36 StrCmp() . . . . .            | 3084 |
| 4.146.3.37 StrlCmp() . . . . .           | 3084 |
| 4.146.3.38 StrHlCmp() . . . . .          | 3085 |
| 4.146.3.39 StrlCont() . . . . .          | 3085 |
| 4.146.3.40 CmpArray() . . . . .          | 3085 |

|                                         |      |
|-----------------------------------------|------|
| 4.146.3.41 ConWord()                    | 3085 |
| 4.146.3.42 StrLen()                     | 3085 |
| 4.146.3.43 NumToStr()                   | 3085 |
| 4.146.3.44 WriteString()                | 3086 |
| 4.146.3.45 WriteUnfinString()           | 3086 |
| 4.146.3.46 AddToString()                | 3086 |
| 4.146.3.47 strDup1()                    | 3086 |
| 4.146.3.48 EndOfToken()                 | 3086 |
| 4.146.3.49 ToToken()                    | 3086 |
| 4.146.3.50 SkipField()                  | 3087 |
| 4.146.3.51 ReadSnum()                   | 3087 |
| 4.146.3.52 NumCopy()                    | 3087 |
| 4.146.3.53 LongCopy()                   | 3087 |
| 4.146.3.54 LongLongCopy()               | 3087 |
| 4.146.3.55 MakeDate()                   | 3087 |
| 4.146.3.56 set_in()                     | 3088 |
| 4.146.3.57 set_set()                    | 3088 |
| 4.146.3.58 set_del()                    | 3088 |
| 4.146.3.59 set_sub()                    | 3088 |
| 4.146.3.60 iniTools()                   | 3088 |
| 4.146.3.61 Malloc1()                    | 3088 |
| 4.146.3.62 M_free()                     | 3089 |
| 4.146.3.63 M_check1()                   | 3089 |
| 4.146.3.64 M_print()                    | 3089 |
| 4.146.3.65 TermMallocAddMemory()        | 3089 |
| 4.146.3.66 NumberMallocAddMemory()      | 3089 |
| 4.146.3.67 CacheNumberMallocAddMemory() | 3089 |
| 4.146.3.68 FromList()                   | 3089 |
| 4.146.3.69 From0List()                  | 3090 |
| 4.146.3.70 FromVarList()                | 3090 |
| 4.146.3.71 DoubleList()                 | 3090 |
| 4.146.3.72 DoubleLList()                | 3090 |
| 4.146.3.73 DoubleBuffer()               | 3090 |
| 4.146.3.74 ExpandBuffer()               | 3090 |
| 4.146.3.75 iexp()                       | 3091 |
| 4.146.3.76 ToGeneral()                  | 3091 |
| 4.146.3.77 ToFast()                     | 3091 |
| 4.146.3.78 ToPolyFunGeneral()           | 3091 |
| 4.146.3.79 IsLikeVector()               | 3091 |
| 4.146.3.80 AreArgsEqual()               | 3091 |
| 4.146.3.81 CompareArgs()                | 3092 |
| 4.146.3.82 CompArg()                    | 3092 |

|                                        |      |
|----------------------------------------|------|
| 4.146.3.83 TimeWallClock()             | 3092 |
| 4.146.3.84 TimeChildren()              | 3092 |
| 4.146.3.85 TimeCPU()                   | 3092 |
| 4.146.3.86 Crash()                     | 3093 |
| 4.146.3.87 TestTerm()                  | 3093 |
| 4.146.3.88 DistrN()                    | 3094 |
| 4.146.4 Variable Documentation         | 3094 |
| 4.146.4.1 filelist                     | 3094 |
| 4.146.4.2 numinfilelist                | 3094 |
| 4.146.4.3 filelistsize                 | 3094 |
| 4.146.4.4 WriteFile                    | 3094 |
| 4.147 tools.c                          | 3095 |
| 4.148 topowrap.cc File Reference       | 3140 |
| 4.148.1 Detailed Description           | 3141 |
| 4.148.2 Macro Definition Documentation | 3141 |
| 4.148.2.1 MAXPOINTS                    | 3141 |
| 4.148.3 Function Documentation         | 3141 |
| 4.148.3.1 GenerateVertices()           | 3141 |
| 4.148.3.2 GenerateTopologies()         | 3141 |
| 4.148.3.3 toForm()                     | 3141 |
| 4.149 topowrap.cc                      | 3142 |
| 4.150 transform.c File Reference       | 3145 |
| 4.150.1 Detailed Description           | 3146 |
| 4.150.2 Function Documentation         | 3146 |
| 4.150.2.1 CoTransform()                | 3146 |
| 4.150.2.2 RunTransform()               | 3146 |
| 4.150.2.3 RunEncode()                  | 3146 |
| 4.150.2.4 RunDecode()                  | 3146 |
| 4.150.2.5 RunReplace()                 | 3147 |
| 4.150.2.6 RunImplode()                 | 3147 |
| 4.150.2.7 RunExplode()                 | 3147 |
| 4.150.2.8 RunPermute()                 | 3147 |
| 4.150.2.9 RunReverse()                 | 3147 |
| 4.150.2.10 RunDedup()                  | 3147 |
| 4.150.2.11 RunCycle()                  | 3148 |
| 4.150.2.12 RunAddArg()                 | 3148 |
| 4.150.2.13 RunMulArg()                 | 3148 |
| 4.150.2.14 RunIsLyndon()               | 3148 |
| 4.150.2.15 RunToLyndon()               | 3148 |
| 4.150.2.16 RunDropArg()                | 3148 |
| 4.150.2.17 RunSelectArg()              | 3149 |
| 4.150.2.18 RunZtoHArg()                | 3149 |

|                                        |      |
|----------------------------------------|------|
| 4.150.2.19 RunHtoZArg()                | 3149 |
| 4.150.2.20 TestArgNum()                | 3149 |
| 4.150.2.21 PutArgInScratch()           | 3149 |
| 4.150.2.22 ReadRange()                 | 3149 |
| 4.150.2.23 FindRange()                 | 3150 |
| 4.151 transform.c                      | 3150 |
| 4.152 unix.h File Reference            | 3191 |
| 4.152.1 Detailed Description           | 3192 |
| 4.152.2 Macro Definition Documentation | 3192 |
| 4.152.2.1 LINEFEED                     | 3192 |
| 4.152.2.2 CARRIAGERETURN               | 3192 |
| 4.152.2.3 WITHPIPE                     | 3192 |
| 4.152.2.4 WITHSYSTEM                   | 3192 |
| 4.152.2.5 WITHEXTERNALCHANNEL          | 3192 |
| 4.152.2.6 TRAP SIGNALS                 | 3192 |
| 4.152.2.7 P_term                       | 3193 |
| 4.152.2.8 SEPARATOR                    | 3193 |
| 4.152.2.9 ALTSEPARATOR                 | 3193 |
| 4.152.2.10 PATHSEPARATOR               | 3193 |
| 4.152.2.11 WITH_ENV                    | 3193 |
| 4.152.2.12 WITH_ALARM                  | 3193 |
| 4.153 unix.h                           | 3194 |
| 4.154 unixfile.c File Reference        | 3194 |
| 4.154.1 Detailed Description           | 3195 |
| 4.155 unixfile.c                       | 3195 |
| 4.156 variable.h File Reference        | 3198 |
| 4.156.1 Detailed Description           | 3199 |
| 4.156.2 Macro Definition Documentation | 3199 |
| 4.156.2.1 chartype                     | 3199 |
| 4.156.2.2 Procedures                   | 3199 |
| 4.156.2.3 NumProcedures                | 3200 |
| 4.156.2.4 MaxProcedures                | 3200 |
| 4.156.2.5 DoLoops                      | 3200 |
| 4.156.2.6 NumDoLoops                   | 3200 |
| 4.156.2.7 MaxDoLoops                   | 3200 |
| 4.156.2.8 PreVar                       | 3200 |
| 4.156.2.9 NumPre                       | 3200 |
| 4.156.2.10 MaxNumPre                   | 3200 |
| 4.156.2.11 SetElements                 | 3201 |
| 4.156.2.12 Sets                        | 3201 |
| 4.156.2.13 functions                   | 3201 |
| 4.156.2.14 indices                     | 3201 |

|                              |      |
|------------------------------|------|
| 4.156.2.15 symbols           | 3201 |
| 4.156.2.16 vectors           | 3201 |
| 4.156.2.17 tablebases        | 3201 |
| 4.156.2.18 NumFunctions      | 3201 |
| 4.156.2.19 NumIndices        | 3202 |
| 4.156.2.20 NumSymbols        | 3202 |
| 4.156.2.21 NumVectors        | 3202 |
| 4.156.2.22 NumSets           | 3202 |
| 4.156.2.23 NumSetElements    | 3202 |
| 4.156.2.24 NumTableBases     | 3202 |
| 4.156.2.25 GlobalFunctions   | 3202 |
| 4.156.2.26 GlobalIndices     | 3202 |
| 4.156.2.27 GlobalSymbols     | 3203 |
| 4.156.2.28 GlobalVectors     | 3203 |
| 4.156.2.29 GlobalSets        | 3203 |
| 4.156.2.30 GlobalSetElements | 3203 |
| 4.156.2.31 cbuf              | 3203 |
| 4.156.2.32 channels          | 3203 |
| 4.156.2.33 NumOutputChannels | 3203 |
| 4.156.2.34 Dollars           | 3203 |
| 4.156.2.35 NumDollars        | 3204 |
| 4.156.2.36 Dubious           | 3204 |
| 4.156.2.37 NumDubious        | 3204 |
| 4.156.2.38 Expressions       | 3204 |
| 4.156.2.39 NumExpressions    | 3204 |
| 4.156.2.40 autofunctions     | 3204 |
| 4.156.2.41 autoindices       | 3204 |
| 4.156.2.42 autosymbols       | 3204 |
| 4.156.2.43 autovectors       | 3205 |
| 4.156.2.44 xsymbol           | 3205 |
| 4.156.2.45 numxsymbol        | 3205 |
| 4.156.2.46 PotModdollars     | 3205 |
| 4.156.2.47 NumPotModdollars  | 3205 |
| 4.156.2.48 ModOptdollars     | 3205 |
| 4.156.2.49 NumModOptdollars  | 3205 |
| 4.156.2.50 BUG               | 3205 |
| 4.156.2.51 AC                | 3206 |
| 4.156.2.52 AM                | 3206 |
| 4.156.2.53 AN                | 3206 |
| 4.156.2.54 AO                | 3206 |
| 4.156.2.55 AP                | 3206 |
| 4.156.2.56 AR                | 3206 |

|                                                  |      |
|--------------------------------------------------|------|
| 4.156.2.57 AS . . . . .                          | 3206 |
| 4.156.2.58 AT . . . . .                          | 3206 |
| 4.156.2.59 AX . . . . .                          | 3207 |
| 4.156.3 Variable Documentation . . . . .         | 3207 |
| 4.156.3.1 WriteFile . . . . .                    | 3207 |
| 4.156.3.2 A . . . . .                            | 3207 |
| 4.156.3.3 FG . . . . .                           | 3207 |
| 4.156.3.4 fixedsets . . . . .                    | 3207 |
| 4.156.3.5 setupfilename . . . . .                | 3207 |
| 4.157 variable.h . . . . .                       | 3208 |
| 4.158 vector.h File Reference . . . . .          | 3209 |
| 4.158.1 Detailed Description . . . . .           | 3211 |
| 4.158.2 Macro Definition Documentation . . . . . | 3211 |
| 4.158.2.1 VectorStruct . . . . .                 | 3211 |
| 4.158.2.2 Vector . . . . .                       | 3211 |
| 4.158.2.3 DeclareVector . . . . .                | 3212 |
| 4.158.2.4 VectorInit . . . . .                   | 3212 |
| 4.158.2.5 VectorFree . . . . .                   | 3212 |
| 4.158.2.6 VectorPtr . . . . .                    | 3213 |
| 4.158.2.7 VectorFront . . . . .                  | 3213 |
| 4.158.2.8 VectorBack . . . . .                   | 3213 |
| 4.158.2.9 VectorSize . . . . .                   | 3214 |
| 4.158.2.10 VectorCapacity . . . . .              | 3214 |
| 4.158.2.11 VectorEmpty . . . . .                 | 3214 |
| 4.158.2.12 VectorClear . . . . .                 | 3215 |
| 4.158.2.13 VectorReserve . . . . .               | 3215 |
| 4.158.2.14 VectorPushBack . . . . .              | 3216 |
| 4.158.2.15 VectorPushBacks . . . . .             | 3216 |
| 4.158.2.16 VectorPopBack . . . . .               | 3216 |
| 4.158.2.17 VectorInsert . . . . .                | 3217 |
| 4.158.2.18 VectorInserts . . . . .               | 3217 |
| 4.158.2.19 VectorErase . . . . .                 | 3218 |
| 4.158.2.20 VectorErases . . . . .                | 3218 |
| 4.159 vector.h . . . . .                         | 3218 |
| 4.160 wildcard.c File Reference . . . . .        | 3222 |
| 4.160.1 Detailed Description . . . . .           | 3222 |
| 4.160.2 Macro Definition Documentation . . . . . | 3222 |
| 4.160.2.1 DEBUG . . . . .                        | 3222 |
| 4.160.3 Function Documentation . . . . .         | 3223 |
| 4.160.3.1 WildFill() . . . . .                   | 3223 |
| 4.160.3.2 ResolveSet() . . . . .                 | 3223 |
| 4.160.3.3 ClearWild() . . . . .                  | 3223 |



---

|                           |      |
|---------------------------|------|
| 4.160.3.4 AddWild()       | 3223 |
| 4.160.3.5 CheckWild()     | 3223 |
| 4.160.3.6 DenToFunction() | 3224 |
| 4.161 wildcard.c          | 3224 |



# Chapter 1

## Data Structure Index

### 1.1 Data Structures

Here are the data structures with brief descriptions:

|                    |     |
|--------------------|-----|
| AEdge              | 7   |
| AllGlobals         | 9   |
| ANode              | 11  |
| ARGBUFFER          | 14  |
| Assign             | 15  |
| AStack             | 26  |
| bit_field          | 29  |
| BrAcKeTiNdEx       | 31  |
| BrAcKeTiNfO        | 32  |
| BracketInfo        | 34  |
| bufIPstruct        | 36  |
| C_const            | 37  |
| CbUf               | 70  |
| ChAnNeL            | 75  |
| COST               | 75  |
| CSEEq              | 76  |
| CSEHash            | 77  |
| dbase              | 78  |
| DGraph             | 80  |
| DICTIONARY         | 81  |
| DICTIONARY_ELEMENT | 83  |
| DiStRiBuTe         | 84  |
| DNode              | 86  |
| dollar_buf         | 87  |
| DoLIArS            | 87  |
| DoLoOp             | 89  |
| DuBiOuS            | 93  |
| ECand              | 94  |
| EEdge              | 95  |
| EFLine             | 100 |
| EGraph             | 102 |
| ENode              | 114 |
| EStack             | 118 |
| ExPrEsSiOn         | 120 |
| FaCdOILaR          | 124 |

|                      |     |
|----------------------|-----|
| factorized_poly      | 125 |
| FGInput              | 126 |
| FiLe                 | 128 |
| FiLeDaTa             | 131 |
| FiLeInDeX            | 133 |
| FixedGlobals         | 134 |
| fixedset             | 137 |
| flint::fmpz          | 138 |
| Fraction             | 139 |
| FUN_INFO             | 142 |
| FuNcTiOn             | 144 |
| gcd_heuristic_failed | 147 |
| HANDLERS             | 147 |
| IInput               | 149 |
| InDeX                | 150 |
| indexblock           | 152 |
| InDeXeNtRy           | 153 |
| iniinfo              | 156 |
| INSIDEINFO           | 157 |
| Interaction          | 159 |
| KEYWORD              | 163 |
| KEYWORDV             | 164 |
| LIST                 | 165 |
| longMultiStruct      | 167 |
| M_const              | 168 |
| MCBlock              | 193 |
| MCBridge             | 195 |
| MConn                | 196 |
| MCOpi                | 202 |
| MGraph               | 205 |
| MiNmAx               | 216 |
| MInput               | 217 |
| MNode                | 218 |
| MNodeClass           | 220 |
| Model                | 225 |
| MoDeL                | 232 |
| MODNUM               | 235 |
| MoDoPtDoLIaRS        | 236 |
| monomial_larger      | 236 |
| MORbits              | 237 |
| flint::mpoly         | 239 |
| flint::mpoly_ctx     | 240 |
| flint::mpoly_factor  | 241 |
| N_const              | 242 |
| nameblock            | 259 |
| NaMeNode             | 260 |
| NaMeSpAcE            | 262 |
| NaMeTree             | 263 |
| NCand                | 266 |
| NCInput              | 268 |
| NeStInG              | 269 |
| NoDe                 | 270 |
| node                 | 271 |
| NodeEq               | 274 |
| NodeHash             | 274 |
| NStack               | 275 |
| O_const              | 276 |
| objects              | 287 |

|                    |     |
|--------------------|-----|
| OptDef             | 288 |
| optimization       | 289 |
| OPTIMIZE           | 292 |
| OPTIMIZERESULT     | 295 |
| Options            | 296 |
| Output             | 303 |
| P_const            | 309 |
| ParallelVars       | 317 |
| PaRtl              | 320 |
| PaRtlcLe           | 321 |
| Particle           | 322 |
| PeRmUtE            | 326 |
| PeRmUtEp           | 327 |
| PF_BUFFER          | 328 |
| PGInput            | 331 |
| PInput             | 332 |
| PNodeClass         | 334 |
| flint::poly        | 337 |
| poly               | 339 |
| flint::poly_factor | 356 |
| POLYMOD            | 357 |
| PoSiTiOn           | 358 |
| PRELOAD            | 359 |
| pReVaR             | 360 |
| PROCEDURE          | 361 |
| Process            | 362 |
| R_const            | 369 |
| ReNuMbEr           | 378 |
| S_const            | 380 |
| SeTs               | 383 |
| SETUPPARAMETERS    | 384 |
| SGroup             | 385 |
| SHvariables        | 390 |
| sOrT               | 392 |
| SpecTatoR          | 401 |
| SProcess           | 402 |
| StOrEcAcHe         | 411 |
| STOREHEADER        | 412 |
| StreaM             | 416 |
| SuBbUf             | 420 |
| SWITCH             | 421 |
| SWITCHTABLE        | 423 |
| SyMbOl             | 424 |
| T_const            | 426 |
| T_EEdge            | 440 |
| T_EGraph           | 441 |
| T_ENode            | 445 |
| T_MGraph           | 446 |
| T_MNode            | 451 |
| T_MNodeClass       | 453 |
| TaBIEbAsE          | 457 |
| TaBIEbAsEsUbInDeX  | 458 |
| TaBIes             | 459 |
| TERMINFO           | 466 |
| ToPoTyPe           | 468 |
| TrAcEn             | 471 |
| TrAcEs             | 473 |
| tree               | 478 |

|                           |     |
|---------------------------|-----|
| <a href="#">tree_node</a> | 480 |
| <a href="#">VaRrEnUm</a>  | 482 |
| <a href="#">VeCtOr</a>    | 483 |
| <a href="#">VeRtEx</a>    | 485 |
| <a href="#">X_const</a>   | 487 |

## Chapter 2

# File Index

### 2.1 File List

Here is a list of all documented files with brief descriptions:

|                                   |      |
|-----------------------------------|------|
| <a href="#">argument.c</a>        | 489  |
| <a href="#">bugtool.c</a>         | 535  |
| <a href="#">checkpoint.c</a>      | 537  |
| <a href="#">comexpr.c</a>         | 581  |
| <a href="#">compcomm.c</a>        | 609  |
| <a href="#">compiler.c</a>        | 722  |
| <a href="#">compress.c</a>        | 753  |
| <a href="#">comtool.c</a>         | 762  |
| <a href="#">comtool.h</a>         | 775  |
| <a href="#">declare.h</a>         | 777  |
| <a href="#">diagrams.c</a>        | 1052 |
| <a href="#">diawrap.cc</a>        | 1061 |
| <a href="#">dict.c</a>            | 1074 |
| <a href="#">dollar.c</a>          | 1092 |
| <a href="#">evaluate.c</a>        | 1147 |
| <a href="#">execute.c</a>         | 1156 |
| <a href="#">extcmd.c</a>          | 1192 |
| <a href="#">factor.c</a>          | 1213 |
| <a href="#">findpat.c</a>         | 1224 |
| <a href="#">flintinterface.cc</a> | 1242 |
| <a href="#">flintinterface.h</a>  | 1268 |
| <a href="#">flintwrap.cc</a>      | 1279 |
| <a href="#">float.c</a>           | 1285 |
| <a href="#">form3.h</a>           | 1324 |
| <a href="#">fsizes.h</a>          | 1341 |
| <a href="#">ftypes.h</a>          | 1354 |
| <a href="#">function.c</a>        | 1470 |
| <a href="#">fwin.h</a>            | 1495 |
| <a href="#">gentopo.cc</a>        | 1497 |
| <a href="#">gentopo.h</a>         | 1516 |
| <a href="#">grcc.cc</a>           | 1518 |
| <a href="#">grcc.h</a>            | 1636 |
| <a href="#">grccparam.h</a>       | 1652 |
| <a href="#">if.c</a>              | 1655 |
| <a href="#">index.c</a>           | 1671 |

|                                 |      |
|---------------------------------|------|
| <a href="#">inivar.h</a>        | 1681 |
| <a href="#">lus.c</a>           | 1686 |
| <a href="#">mallocprotect.h</a> | 1708 |
| <a href="#">message.c</a>       | 1713 |
| <a href="#">minos.c</a>         | 1727 |
| <a href="#">minos.h</a>         | 1752 |
| <a href="#">model.c</a>         | 1756 |
| <a href="#">module.c</a>        | 1762 |
| <a href="#">mpi.c</a>           | 1775 |
| <a href="#">mpidbg.h</a>        | 1809 |
| <a href="#">mytime.cc</a>       | 1822 |
| <a href="#">mytime.h</a>        | 1822 |
| <a href="#">names.c</a>         | 1823 |
| <a href="#">normal.c</a>        | 1872 |
| <a href="#">notation.c</a>      | 1938 |
| <a href="#">opera.c</a>         | 1955 |
| <a href="#">optimize.cc</a>     | 1987 |
| <a href="#">parallel.c</a>      | 2061 |
| <a href="#">parallel.h</a>      | 2135 |
| <a href="#">pattern.c</a>       | 2171 |
| <a href="#">poly.cc</a>         | 2201 |
| <a href="#">poly.h</a>          | 2234 |
| <a href="#">polyfact.cc</a>     | 2237 |
| <a href="#">polyfact.h</a>      | 2254 |
| <a href="#">polygcd.cc</a>      | 2256 |
| <a href="#">polygcd.h</a>       | 2273 |
| <a href="#">polywrap.cc</a>     | 2275 |
| <a href="#">portsignals.h</a>   | 2308 |
| <a href="#">pre.c</a>           | 2313 |
| <a href="#">proces.c</a>        | 2423 |
| <a href="#">ratio.c</a>         | 2500 |
| <a href="#">reken.c</a>         | 2549 |
| <a href="#">reshuf.c</a>        | 2608 |
| <a href="#">sch.c</a>           | 2649 |
| <a href="#">setfile.c</a>       | 2688 |
| <a href="#">smart.c</a>         | 2708 |
| <a href="#">sort.c</a>          | 2714 |
| <a href="#">spectator.c</a>     | 2786 |
| <a href="#">startup.c</a>       | 2796 |
| <a href="#">store.c</a>         | 2826 |
| <a href="#">structs.h</a>       | 2895 |
| <a href="#">symmetr.c</a>       | 2931 |
| <a href="#">tables.c</a>        | 2960 |
| <a href="#">threads.c</a>       | 2993 |
| <a href="#">token.c</a>         | 3047 |
| <a href="#">tools.c</a>         | 3075 |
| <a href="#">topowrap.cc</a>     | 3140 |
| <a href="#">transform.c</a>     | 3145 |
| <a href="#">unix.h</a>          | 3191 |
| <a href="#">unixfile.c</a>      | 3194 |
| <a href="#">variable.h</a>      | 3198 |
| <a href="#">vector.h</a>        | 3209 |
| <a href="#">wildcard.c</a>      | 3222 |

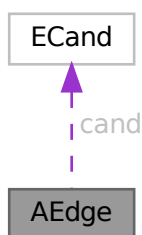


## Chapter 3

# Data Structure Documentation

### 3.1 AEdge Class Reference

Collaboration diagram for AEdge:



#### Public Member Functions

- [AEdge](#) (int n0, int l0, int n1, int l1)

#### Data Fields

- [ECand](#) \* [cand](#)
- int [nodes](#) [2]
- int [nlegs](#) [2]
- int [ptcl](#)
- int [DUMMY\\_PADDING](#)

#### 3.1.1 Detailed Description

Definition at line [1102](#) of file [grcc.h](#).

## 3.1.2 Constructor & Destructor Documentation

### 3.1.2.1 AEdge()

```
AEdge::AEdge (
    int  n0,
    int  l0,
    int  n1,
    int  l1 )
```

Definition at line [7413](#) of file [grcc.cc](#).

### 3.1.2.2 ~AEdge()

```
AEdge::~~AEdge (
    void  )
```

Definition at line [7424](#) of file [grcc.cc](#).

## 3.1.3 Field Documentation

### 3.1.3.1 cand

```
ECand* AEdge::cand
```

Definition at line [1109](#) of file [grcc.h](#).

### 3.1.3.2 nodes

```
int AEdge::nodes[2]
```

Definition at line [1110](#) of file [grcc.h](#).

### 3.1.3.3 nlegs

```
int AEdge::nlegs[2]
```

Definition at line [1111](#) of file [grcc.h](#).

### 3.1.3.4 ptcl

```
int AEdge::ptcl
```

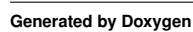
Definition at line [1112](#) of file [grcc.h](#).

```
int AEdge::DUMMYPADDING
```

The documentation for this class was generated from the following files:

- ### 3.2 AllGlobals Struct Reference

Collaboration diagram for AllGlobals:



## Public Member Functions

- **PADPOSITION** (0, 0, 0, 0, sizeof(struct [P\\_const](#))+sizeof(struct [T\\_const](#))+sizeof(struct [X\\_const](#)))

## Data Fields

- struct [M\\_const](#) M
- struct [C\\_const](#) Cc
- struct [S\\_const](#) S
- struct [R\\_const](#) R
- struct [N\\_const](#) N
- struct [O\\_const](#) O
- struct [P\\_const](#) P
- struct [T\\_const](#) T
- struct [X\\_const](#) X

### 3.2.1 Detailed Description

Without pthreads (FORM) the ALLGLOBALS struct has all the global variables

Definition at line [2620](#) of file [structs.h](#).

### 3.2.2 Field Documentation

#### 3.2.2.1 M

```
struct M\_const AllGlobals::M
```

Definition at line [2621](#) of file [structs.h](#).

#### 3.2.2.2 Cc

```
struct C\_const AllGlobals::Cc
```

Definition at line [2622](#) of file [structs.h](#).

#### 3.2.2.3 S

```
struct S\_const AllGlobals::S
```

Definition at line [2623](#) of file [structs.h](#).

#### 3.2.2.4 R

```
struct R\_const AllGlobals::R
```

Definition at line [2624](#) of file [structs.h](#).

### 3.2.2.5 N

```
struct N_const AllGlobals::N
```

Definition at line 2625 of file [structs.h](#).

### 3.2.2.6 O

```
struct O_const AllGlobals::O
```

Definition at line 2626 of file [structs.h](#).

### 3.2.2.7 P

```
struct P_const AllGlobals::P
```

Definition at line 2627 of file [structs.h](#).

### 3.2.2.8 T

```
struct T_const AllGlobals::T
```

Definition at line 2628 of file [structs.h](#).

### 3.2.2.9 X

```
struct X_const AllGlobals::X
```

Definition at line 2629 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.3 ANode Class Reference

Collaboration diagram for ANode:



## Public Member Functions

- [ANode](#) (int dg)
- int [newleg](#) (void)

## Data Fields

- int [deg](#)
- int [nlegs](#)
- int \* [anodes](#)
- int \* [aedges](#)
- int \* [aelegs](#)
- [NCand](#) \* [cand](#)

### 3.3.1 Detailed Description

Definition at line [1078](#) of file [grcc.h](#).

### 3.3.2 Constructor & Destructor Documentation

#### 3.3.2.1 ANode()

```
ANode::ANode (
    int dg )
```

Definition at line [7358](#) of file [grcc.cc](#).

#### 3.3.2.2 ~ANode()

```
ANode::~ANode (
    void )
```

Definition at line [7376](#) of file [grcc.cc](#).

### 3.3.3 Member Function Documentation

#### 3.3.3.1 newleg()

```
int ANode::newleg (
    void )
```

Definition at line [7393](#) of file [grcc.cc](#).

### 3.3.4 Field Documentation

#### 3.3.4.1 deg

```
int ANode::deg
```

Definition at line 1088 of file [grcc.h](#).

#### 3.3.4.2 nlegs

```
int ANode::nlegs
```

Definition at line 1089 of file [grcc.h](#).

#### 3.3.4.3 anodes

```
int* ANode::anodes
```

Definition at line 1090 of file [grcc.h](#).

#### 3.3.4.4 aedges

```
int* ANode::aedges
```

Definition at line 1091 of file [grcc.h](#).

#### 3.3.4.5 aelegs

```
int* ANode::aelegs
```

Definition at line 1092 of file [grcc.h](#).

#### 3.3.4.6 cand

```
NCand* ANode::cand
```

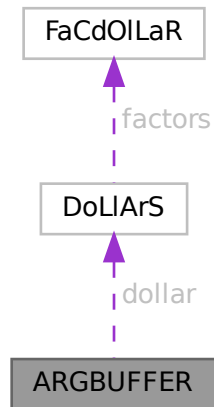
Definition at line 1093 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.4 ARGBUFFER Struct Reference

Collaboration diagram for ARGBUFFER:



### Data Fields

- WORD \* [buffer](#)
- DOLLARS [dollar](#)
- LONG [size](#)
- int [type](#)
- int [dummy](#)

### 3.4.1 Detailed Description

Definition at line [601](#) of file [ratio.c](#).

### 3.4.2 Field Documentation

#### 3.4.2.1 buffer

```
WORD* ARGBUFFER::buffer
```

Definition at line [602](#) of file [ratio.c](#).

#### 3.4.2.2 dollar

```
DOLLARS ARGBUFFER::dollar
```

Definition at line [603](#) of file [ratio.c](#).





## Public Member Functions

- [Assign](#) ([SProcess](#) \*sprc, [MGraph](#) \*mgr, [PNodeClass](#) \*pnc)
- void [prCand](#) (const char \*msg)
- void [checkAG](#) (const char \*msg)
- Bool [assignAllVertices](#) (void)
- Bool [selectVertex](#) (void)
- Bool [selectVertexSimp](#) (int lastv)
- Bool [selectLeg](#) (int v, int lastlg)
- Bool [assignVertex](#) (int v)
- Bool [allAssigned](#) (void)
- Bool [fromMGraph](#) (void)
- void [addEdge](#) (int n0, int n1, int nplist, int \*plist)
- void [connect](#) (int n0, int l0, int eg, int el, int n1, int l1)
- Bool [fillEGraph](#) (int aid, [BigInt](#) nsym, [BigInt](#) esym, [BigInt](#) nsym1)
- int \* [reordLeg](#) (int n, int \*reord, int \*plist, int \*used)
- int [getLegParticle](#) (int n, int ln)
- int [legEdgeParticle](#) (int n, int ln, int pt)
- int [legPart](#) (int v, int lg, int nplst, int \*plst, int \*rlist, const int size)
- int [candPart](#) (int v, int ln, int \*plist, const int size)
- int [selUnAssVertex](#) (void)
- int [selUnAssVertexSimp](#) (int lastv)
- int [selUnAssLeg](#) (int v, int lastlg)
- [NCandSt](#) [assignVertex](#) (int v, int ia)
- Bool [assignPLeg](#) (int n, int ln, int pt)
- Bool [isOrdPLeg](#) (int n, int ln, int pt)
- Bool [detEdge](#) (int e)
- Bool [candPartClassify](#) (int v, int \*npdass, int \*pdass, int \*npuass, int \*puass, const int size)
- Bool [updateCandNode](#) (int v)
- Bool [checkOrderCpl](#) (void)
- Bool [isOrdLegs](#) (void)
- Bool [isIsomorphic](#) ([MNodeClass](#) \*cl, [BigInt](#) \*nsym, [BigInt](#) \*esym, [BigInt](#) \*nsym1)
- int [cmpPermGraph](#) (int \*p, [MNodeClass](#) \*cl)
- int [cmpNodes](#) (int nd0, int nd1, [MNodeClass](#) \*cn)
- [BigInt](#) [edgeSym](#) (void)
- void [saveCouple](#) (int \*sav)
- void [restoreCouple](#) (int \*sav)
- Bool [subCouple](#) (int \*cpl)

## Data Fields

- [EGraph](#) \* egraph
- [MGraph](#) \* mgraph
- [SProcess](#) \* sprc
- [Process](#) \* proc
- [Model](#) \* model
- [Options](#) \* opt
- [AStack](#) \* astack
- [PNodeClass](#) \* pnclass
- [MOrbits](#) \* orbits
- int [nNodes](#)
- int [nEdges](#)
- int [nExtern](#)
- int [nETotal](#)

- [BigInt](#) [nAGraphs](#)
- [Fraction](#) [wAGraphs](#)
- [BigInt](#) [nAOPI](#)
- [Fraction](#) [wAOPI](#)
- [ANode](#) \*\* [nodes](#)
- [AEdge](#) \*\* [edges](#)
- [CheckPt](#) [checkpoint0](#)
- [int](#) [cplleft](#) [[GRCC\\_MAXNCPLG](#)]

### 3.5.1 Detailed Description

Definition at line [1121](#) of file [grcc.h](#).

### 3.5.2 Constructor & Destructor Documentation

#### 3.5.2.1 Assign()

```
Assign::Assign (  
    SProcess * sprc,  
    MGraph * mgr,  
    PNodeClass * pnc )
```

Definition at line [7433](#) of file [grcc.cc](#).

#### 3.5.2.2 ~Assign()

```
Assign::~Assign (  
    void )
```

Definition at line [7533](#) of file [grcc.cc](#).

### 3.5.3 Member Function Documentation

#### 3.5.3.1 prCand()

```
void Assign::prCand (  
    const char * msg )
```

Definition at line [7553](#) of file [grcc.cc](#).

#### 3.5.3.2 checkAG()

```
void Assign::checkAG (  
    const char * msg )
```

Definition at line [7589](#) of file [grcc.cc](#).

### 3.5.3.3 assignAllVertices()

```
Bool Assign::assignAllVertices (
    void )
```

Definition at line 7613 of file [grcc.cc](#).

### 3.5.3.4 selectVertex()

```
Bool Assign::selectVertex (
    void )
```

Definition at line 7686 of file [grcc.cc](#).

### 3.5.3.5 selectVertexSimp()

```
Bool Assign::selectVertexSimp (
    int lastv )
```

Definition at line 7651 of file [grcc.cc](#).

### 3.5.3.6 selectLeg()

```
Bool Assign::selectLeg (
    int v,
    int lastlg )
```

Definition at line 7721 of file [grcc.cc](#).

### 3.5.3.7 assignVertex()

```
Bool Assign::assignVertex (
    int v )
```

Definition at line 7823 of file [grcc.cc](#).

### 3.5.3.8 allAssigned()

```
Bool Assign::allAssigned (
    void )
```

Definition at line 7946 of file [grcc.cc](#).

### 3.5.3.9 fromMGraph()

```
Bool Assign::fromMGraph (
    void )
```

Definition at line 8050 of file [grcc.cc](#).

#### 3.5.3.10 addEdge()

```
void Assign::addEdge (
    int n0,
    int n1,
    int nplist,
    int * plist )
```

Definition at line 8196 of file [grcc.cc](#).

#### 3.5.3.11 connect()

```
void Assign::connect (
    int n0,
    int l0,
    int eg,
    int e1,
    int n1,
    int l1 )
```

Definition at line 8241 of file [grcc.cc](#).

#### 3.5.3.12 fillEGraph()

```
Bool Assign::fillEGraph (
    int aid,
    BigInt nsym,
    BigInt esym,
    BigInt nsym1 )
```

Definition at line 8270 of file [grcc.cc](#).

#### 3.5.3.13 reordLeg()

```
int * Assign::reordLeg (
    int n,
    int * reord,
    int * plist,
    int * used )
```

Definition at line 8417 of file [grcc.cc](#).

#### 3.5.3.14 getLegParticle()

```
int Assign::getLegParticle (
    int n,
    int ln )
```

Definition at line 8492 of file [grcc.cc](#).

### 3.5.3.15 legEdgeParticle()

```
int Assign::legEdgeParticle (
    int n,
    int ln,
    int pt )
```

Definition at line [8519](#) of file [grcc.cc](#).

### 3.5.3.16 legPart()

```
int Assign::legPart (
    int v,
    int lg,
    int nplst,
    int * plst,
    int * rlist,
    const int size )
```

Definition at line [8539](#) of file [grcc.cc](#).

### 3.5.3.17 candPart()

```
int Assign::candPart (
    int v,
    int ln,
    int * plist,
    const int size )
```

Definition at line [8555](#) of file [grcc.cc](#).

### 3.5.3.18 selUnAssVertex()

```
int Assign::selUnAssVertex (
    void )
```

Definition at line [8608](#) of file [grcc.cc](#).

### 3.5.3.19 selUnAssVertexSimp()

```
int Assign::selUnAssVertexSimp (
    int lastv )
```

Definition at line [8588](#) of file [grcc.cc](#).

### 3.5.3.20 selUnAssLeg()

```
int Assign::selUnAssLeg (
    int v,
    int lastlg )
```

Definition at line [8637](#) of file [grcc.cc](#).

### 3.5.3.21 assignVertex()

```
NCandSt Assign::assignVertex (
    int v,
    int ia )
```

Definition at line 8676 of file [grcc.cc](#).

### 3.5.3.22 assignPLeg()

```
Bool Assign::assignPLeg (
    int n,
    int ln,
    int pt )
```

Definition at line 8722 of file [grcc.cc](#).

### 3.5.3.23 isOrdPLeg()

```
Bool Assign::isOrdPLeg (
    int n,
    int ln,
    int pt )
```

Definition at line 8798 of file [grcc.cc](#).

### 3.5.3.24 detEdge()

```
Bool Assign::detEdge (
    int e )
```

Definition at line 8777 of file [grcc.cc](#).

### 3.5.3.25 candPartClassify()

```
Bool Assign::candPartClassify (
    int v,
    int * npdass,
    int * pdass,
    int * npuass,
    int * puass,
    const int size )
```

Definition at line 8842 of file [grcc.cc](#).

### 3.5.3.26 updateCandNode()

```
Bool Assign::updateCandNode (
    int v )
```

Definition at line 8879 of file [grcc.cc](#).

### 3.5.3.27 checkOrderCpl()

```
Bool Assign::checkOrderCpl (
    void )
```

Definition at line 9018 of file [grcc.cc](#).

### 3.5.3.28 isOrdLegs()

```
Bool Assign::isOrdLegs (
    void )
```

Definition at line 9056 of file [grcc.cc](#).

### 3.5.3.29 isIsomorphic()

```
Bool Assign::isIsomorphic (
    MNodeClass * cl,
    BigInt * nsym,
    BigInt * esym,
    BigInt * nsym1 )
```

Definition at line 9094 of file [grcc.cc](#).

### 3.5.3.30 cmpPermGraph()

```
int Assign::cmpPermGraph (
    int * p,
    MNodeClass * cl )
```

Definition at line 9176 of file [grcc.cc](#).

### 3.5.3.31 cmpNodes()

```
int Assign::cmpNodes (
    int nd0,
    int nd1,
    MNodeClass * cn )
```

Definition at line 9274 of file [grcc.cc](#).

### 3.5.3.32 edgeSym()

```
BigInt Assign::edgeSym (
    void )
```

Definition at line 9300 of file [grcc.cc](#).



### 3.5.3.33 saveCouple()

```
void Assign::saveCouple (
    int * sav )
```

Definition at line 7783 of file [grcc.cc](#).

### 3.5.3.34 restoreCouple()

```
void Assign::restoreCouple (
    int * sav )
```

Definition at line 7795 of file [grcc.cc](#).

### 3.5.3.35 subCouple()

```
Bool Assign::subCouple (
    int * cpl )
```

Definition at line 7807 of file [grcc.cc](#).

## 3.5.4 Field Documentation

### 3.5.4.1 egraph

```
EGraph* Assign::egraph
```

Definition at line 1126 of file [grcc.h](#).

### 3.5.4.2 mgraph

```
MGraph* Assign::mgraph
```

Definition at line 1127 of file [grcc.h](#).

### 3.5.4.3 sproc

```
SProcess* Assign::sproc
```

Definition at line 1128 of file [grcc.h](#).

### 3.5.4.4 proc

```
Process* Assign::proc
```

Definition at line 1129 of file [grcc.h](#).

#### 3.5.4.5 model

`Model*` Assign::model

Definition at line 1130 of file [grcc.h](#).

#### 3.5.4.6 opt

`Options*` Assign::opt

Definition at line 1131 of file [grcc.h](#).

#### 3.5.4.7 astack

`AStack*` Assign::astack

Definition at line 1132 of file [grcc.h](#).

#### 3.5.4.8 pnclass

`PNodeClass*` Assign::pnclass

Definition at line 1133 of file [grcc.h](#).

#### 3.5.4.9 orbits

`MOrbits*` Assign::orbits

Definition at line 1134 of file [grcc.h](#).

#### 3.5.4.10 nNodes

`int` Assign::nNodes

Definition at line 1136 of file [grcc.h](#).

#### 3.5.4.11 nEdges

`int` Assign::nEdges

Definition at line 1137 of file [grcc.h](#).

#### 3.5.4.12 nExtern

`int` Assign::nExtern

Definition at line 1138 of file [grcc.h](#).

#### 3.5.4.13 nETotal

```
int Assign::nETotal
```

Definition at line 1139 of file [grcc.h](#).

#### 3.5.4.14 nAGraphs

```
BigInt Assign::nAGraphs
```

Definition at line 1141 of file [grcc.h](#).

#### 3.5.4.15 wAGraphs

```
Fraction Assign::wAGraphs
```

Definition at line 1142 of file [grcc.h](#).

#### 3.5.4.16 nAOPI

```
BigInt Assign::nAOPI
```

Definition at line 1143 of file [grcc.h](#).

#### 3.5.4.17 wAOPI

```
Fraction Assign::wAOPI
```

Definition at line 1144 of file [grcc.h](#).

#### 3.5.4.18 nodes

```
ANode** Assign::nodes
```

Definition at line 1146 of file [grcc.h](#).

#### 3.5.4.19 edges

```
AEdge** Assign::edges
```

Definition at line 1147 of file [grcc.h](#).

#### 3.5.4.20 checkpoint0

```
CheckPt Assign::checkpoint0
```

Definition at line 1149 of file [grcc.h](#).

### 3.5.4.21 cplleft

```
int Assign::cplleft[GRCC_MAXNCPLG]
```

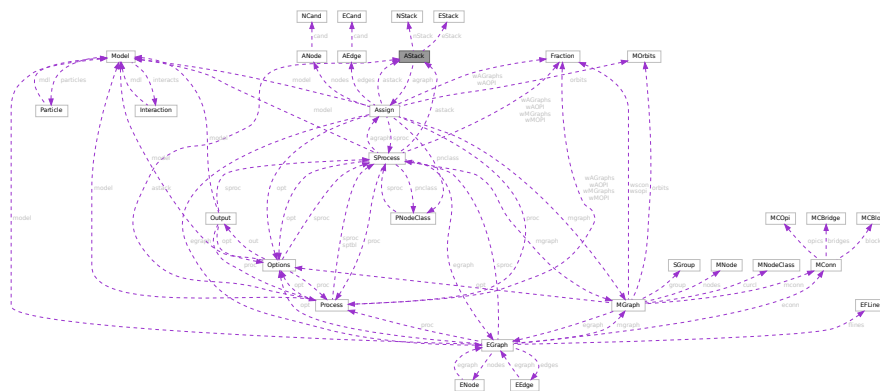
Definition at line 1151 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.6 AStack Class Reference

Collaboration diagram for AStack:



### Public Member Functions

- [AStack](#) (int nSize, int eSize)
- void [setAGraph](#) ([Assign](#) \*ag)
- void [checkPoint](#) (CheckPt sav)
- void [restore](#) (CheckPt sav)
- void [restoreMsg](#) (CheckPt sav, const char \*msg)
- void [prStack](#) (void)
- void [pushNode](#) (int n)
- void [pushEdge](#) (int e)

### Data Fields

- [Assign](#) \* [agraph](#)
- [NStack](#) \*\* [nStack](#)
- int [nStackP](#)
- int [nSize](#)
- [EStack](#) \*\* [eStack](#)
- int [eStackP](#)
- int [eSize](#)

### 3.6.1 Detailed Description

Definition at line 1305 of file [grcc.h](#).

### 3.6.2 Constructor & Destructor Documentation

#### 3.6.2.1 AStack()

```
AStack::AStack (
    int nSize,
    int eSize )
```

Definition at line 9470 of file [grcc.cc](#).

#### 3.6.2.2 ~AStack()

```
AStack::~~AStack (
    void )
```

Definition at line 9502 of file [grcc.cc](#).

### 3.6.3 Member Function Documentation

#### 3.6.3.1 setAGraph()

```
void AStack::setAGraph (
    Assign * ag )
```

Definition at line 9530 of file [grcc.cc](#).

#### 3.6.3.2 checkPoint()

```
void AStack::checkPoint (
    CheckPt sav )
```

Definition at line 9595 of file [grcc.cc](#).

#### 3.6.3.3 restore()

```
void AStack::restore (
    CheckPt sav )
```

Definition at line 9639 of file [grcc.cc](#).

#### 3.6.3.4 prStack()

```
void AStack::prStack (
    void )
```

Definition at line [9662](#) of file [grcc.cc](#).

#### 3.6.3.5 pushNode()

```
void AStack::pushNode (
    int n )
```

Definition at line [9536](#) of file [grcc.cc](#).

#### 3.6.3.6 pushEdge()

```
void AStack::pushEdge (
    int e )
```

Definition at line [9566](#) of file [grcc.cc](#).

### 3.6.4 Field Documentation

#### 3.6.4.1 agraph

```
Assign* AStack::agraph
```

Definition at line [1309](#) of file [grcc.h](#).

#### 3.6.4.2 nStack

```
NStack** AStack::nStack
```

Definition at line [1311](#) of file [grcc.h](#).

#### 3.6.4.3 nStackP

```
int AStack::nStackP
```

Definition at line [1312](#) of file [grcc.h](#).

#### 3.6.4.4 nSize

```
int AStack::nSize
```

Definition at line [1313](#) of file [grcc.h](#).

#### 3.6.4.5 eStack

```
EStack** AStack::eStack
```

Definition at line 1315 of file [grcc.h](#).

#### 3.6.4.6 eStackP

```
int AStack::eStackP
```

Definition at line 1316 of file [grcc.h](#).

#### 3.6.4.7 eSize

```
int AStack::eSize
```

Definition at line 1317 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.7 bit\_field Struct Reference

```
#include <structs.h>
```

### Data Fields

- UBYTE [bit\\_0](#): 1
- UBYTE [bit\\_1](#): 1
- UBYTE [bit\\_2](#): 1
- UBYTE [bit\\_3](#): 1
- UBYTE [bit\\_4](#): 1
- UBYTE [bit\\_5](#): 1
- UBYTE [bit\\_6](#): 1
- UBYTE [bit\\_7](#): 1

### 3.7.1 Detailed Description

The struct [bit\\_field](#) is used by [set\\_in](#), [set\\_set](#), [set\\_del](#) and [set\\_sub](#). They in turn are used in [pre.c](#) to toggle bits that indicate whether a character can be used as a separator of function arguments. This facility is used in the communication with external channels.

Definition at line 927 of file [structs.h](#).

## 3.7.2 Field Documentation

### 3.7.2.1 bit\_0

```
UBYTE bit_field::bit_0
```

Definition at line 928 of file [structs.h](#).

### 3.7.2.2 bit\_1

```
UBYTE bit_field::bit_1
```

Definition at line 929 of file [structs.h](#).

### 3.7.2.3 bit\_2

```
UBYTE bit_field::bit_2
```

Definition at line 930 of file [structs.h](#).

### 3.7.2.4 bit\_3

```
UBYTE bit_field::bit_3
```

Definition at line 931 of file [structs.h](#).

### 3.7.2.5 bit\_4

```
UBYTE bit_field::bit_4
```

Definition at line 932 of file [structs.h](#).

### 3.7.2.6 bit\_5

```
UBYTE bit_field::bit_5
```

Definition at line 933 of file [structs.h](#).

### 3.7.2.7 bit\_6

```
UBYTE bit_field::bit_6
```

Definition at line 934 of file [structs.h](#).



### 3.7.2.8 bit\_7

```
UBYTE bit_field::bit_7
```

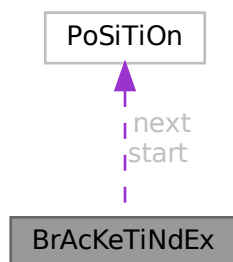
Definition at line 935 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.8 BrAcKeTiNdEx Struct Reference

Collaboration diagram for BrAcKeTiNdEx:



### Public Member Functions

- **PADPOSITION** (0, 2, 0, 0, 0)

### Data Fields

- [POSITION](#) start
- [POSITION](#) next
- LONG [bracket](#)
- LONG [termsinbracket](#)

### 3.8.1 Detailed Description

Definition at line 317 of file [structs.h](#).

### 3.8.2 Field Documentation

#### 3.8.2.1 start

`POSITION BrAcKeTiNdEx::start`

Definition at line 318 of file [structs.h](#).

#### 3.8.2.2 next

`POSITION BrAcKeTiNdEx::next`

Definition at line 319 of file [structs.h](#).

#### 3.8.2.3 bracket

`LONG BrAcKeTiNdEx::bracket`

Definition at line 320 of file [structs.h](#).

#### 3.8.2.4 termsinbracket

`LONG BrAcKeTiNdEx::termsinbracket`

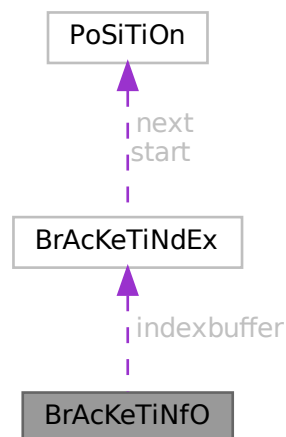
Definition at line 321 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.9 BrAcKeTiNfO Struct Reference

Collaboration diagram for BrAcKeTiNfO:



## Data Fields

- [BRACKETINDEX](#) \* [indexbuffer](#)
- WORD \* [bracketbuffer](#)
- LONG [bracketbuffersize](#)
- LONG [indexbuffersize](#)
- LONG [bracketfill](#)
- LONG [indexfill](#)
- WORD [SortType](#)

### 3.9.1 Detailed Description

Definition at line [329](#) of file [structs.h](#).

### 3.9.2 Field Documentation

#### 3.9.2.1 indexbuffer

```
BRACKETINDEX* BrAcKeTiNfO::indexbuffer
```

[D]

Definition at line [330](#) of file [structs.h](#).

Referenced by [DoRecovery\(\)](#), and [FlushOut\(\)](#).

#### 3.9.2.2 bracketbuffer

```
WORD* BrAcKeTiNfO::bracketbuffer
```

[D]

Definition at line [331](#) of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

#### 3.9.2.3 bracketbuffersize

```
LONG BrAcKeTiNfO::bracketbuffersize
```

Definition at line [332](#) of file [structs.h](#).

#### 3.9.2.4 indexbuffersize

```
LONG BrAcKeTiNfO::indexbuffersize
```

Definition at line [333](#) of file [structs.h](#).

### 3.9.2.5 bracketfill

```
LONG BrAcKeTiNfO::bracketfill
```

Definition at line 334 of file [structs.h](#).

### 3.9.2.6 indexfill

```
LONG BrAcKeTiNfO::indexfill
```

Definition at line 335 of file [structs.h](#).

### 3.9.2.7 SortType

```
WORD BrAcKeTiNfO::SortType
```

The sorting criterium used (like POWERFIRST etc)

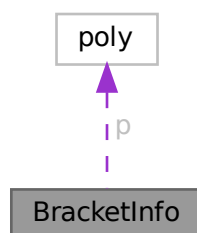
Definition at line 336 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.10 BracketInfo Struct Reference

Collaboration diagram for BracketInfo:



### Public Member Functions

- [BracketInfo](#) (const std::vector< int > &pattern, int num\_terms, const [poly](#) \*p)
- bool [operator<](#) (const [BracketInfo](#) &rhs) const

## Data Fields

- `std::vector< int >` [pattern](#)
- `int` [num\\_terms](#)
- `int` [dummy](#) = 0
- `const` [poly](#) \* [p](#)

### 3.10.1 Detailed Description

Definition at line [1389](#) of file [polygcd.cc](#).

### 3.10.2 Constructor & Destructor Documentation

#### 3.10.2.1 BracketInfo()

```
BracketInfo::BracketInfo (
    const std::vector< int > & pattern,
    int num_terms,
    const poly * p ) [inline]
```

Definition at line [1394](#) of file [polygcd.cc](#).

### 3.10.3 Member Function Documentation

#### 3.10.3.1 operator<()

```
bool BracketInfo::operator< (
    const BracketInfo & rhs ) const [inline]
```

Definition at line [1395](#) of file [polygcd.cc](#).

### 3.10.4 Field Documentation

#### 3.10.4.1 pattern

```
std::vector<int> BracketInfo::pattern
```

Definition at line [1390](#) of file [polygcd.cc](#).

#### 3.10.4.2 num\_terms

```
int BracketInfo::num_terms
```

Definition at line [1391](#) of file [polygcd.cc](#).

### 3.10.4.3 dummy

```
int BracketInfo::dummy = 0
```

Definition at line 1391 of file [polygcd.cc](#).

### 3.10.4.4 p

```
const poly* BracketInfo::p
```

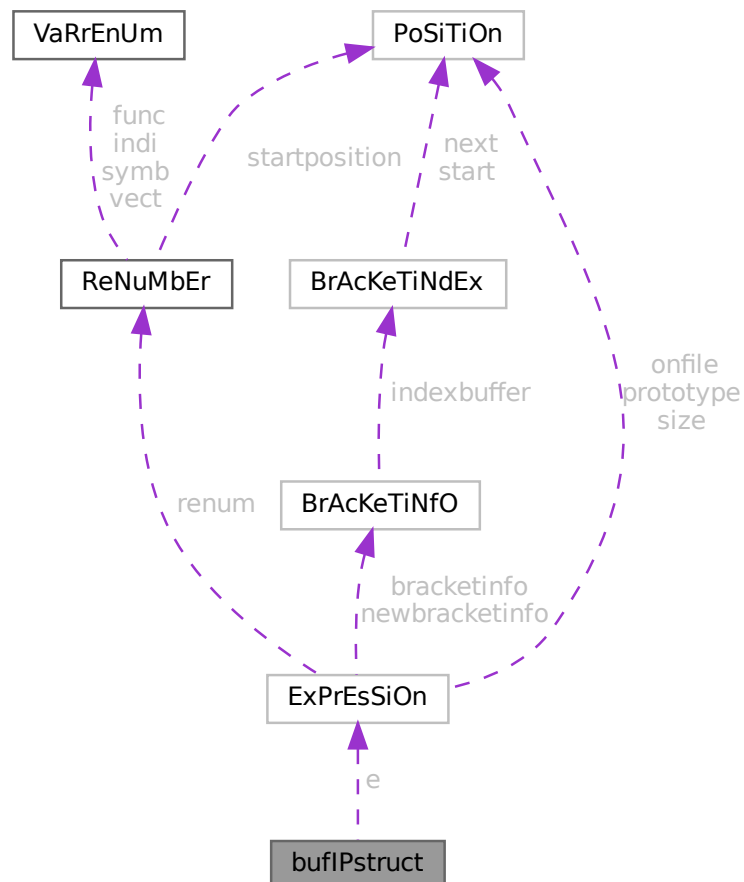
Definition at line 1392 of file [polygcd.cc](#).

The documentation for this struct was generated from the following file:

- [polygcd.cc](#)

## 3.11 bufIPstruct Struct Reference

Collaboration diagram for bufIPstruct:



## Data Fields

- LONG `i`
- struct `ExPrEsSiOn e`

### 3.11.1 Detailed Description

Definition at line 3921 of file [parallel.c](#).

### 3.11.2 Field Documentation

#### 3.11.2.1 `i`

LONG `bufIPstruct::i`

Definition at line 3922 of file [parallel.c](#).

#### 3.11.2.2 `e`

struct `ExPrEsSiOn` `bufIPstruct::e`

Definition at line 3923 of file [parallel.c](#).

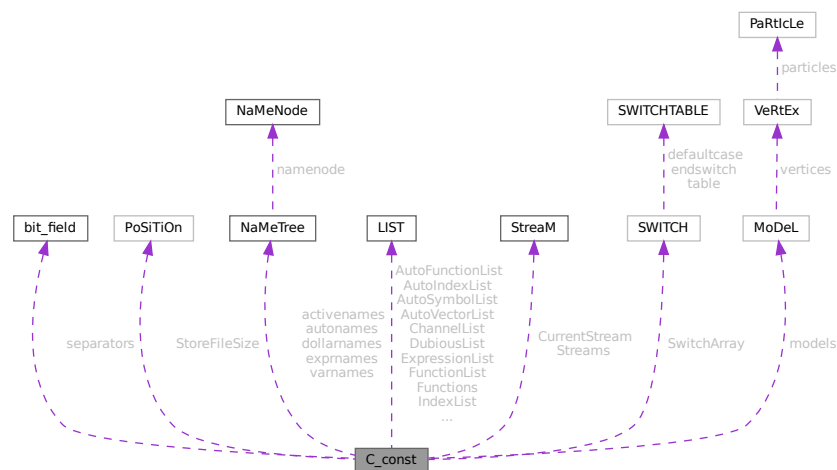
The documentation for this struct was generated from the following file:

- [parallel.c](#)

## 3.12 C\_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for C\_const:



## Public Member Functions

- **PADPOSITION** (48, 8+3 \*MAXNEST, 75, 48+3 \*MAXNEST+MAXREPEAT, COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST) \*17)

## Data Fields

- [set\\_of\\_char](#) separators
- [POSITION](#) StoreFileSize
- [NAMETREE](#) \* [dollarnames](#)
- [NAMETREE](#) \* [exprnames](#)
- [NAMETREE](#) \* [varnames](#)
- [LIST](#) ChannelList
- [LIST](#) DubiousList
- [LIST](#) FunctionList
- [LIST](#) ExpressionList
- [LIST](#) IndexList
- [LIST](#) SetElementList
- [LIST](#) SetList
- [LIST](#) SymbolList
- [LIST](#) VectorList
- [LIST](#) PotModDolList
- [LIST](#) ModOptDolList
- [LIST](#) TableBaseList
- [LIST](#) cbufList
- [LIST](#) AutoSymbolList
- [LIST](#) AutoIndexList
- [LIST](#) AutoVectorList
- [LIST](#) AutoFunctionList
- [NAMETREE](#) \* [autonames](#)
- [LIST](#) \* [Symbols](#)
- [LIST](#) \* [Indices](#)
- [LIST](#) \* [Vectors](#)
- [LIST](#) \* [Functions](#)
- [NAMETREE](#) \*\* [activenames](#)
- [STREAM](#) \* [Streams](#)
- [STREAM](#) \* [CurrentStream](#)
- [SWITCH](#) \* [SwitchArray](#)
- [MODEL](#) \*\* [models](#)
- [WORD](#) \* [SwitchHeap](#)
- [LONG](#) \* [termstack](#)
- [LONG](#) \* [termsortstack](#)
- [UWORD](#) \* [cmod](#)
- [UWORD](#) \* [powmod](#)
- [UWORD](#) \* [modpowers](#)
- [UWORD](#) \* [halfmod](#)
- [WORD](#) \* [ProtoType](#)
- [WORD](#) \* [WildC](#)
- [LONG](#) \* [IfHeap](#)
- [LONG](#) \* [IfCount](#)
- [LONG](#) \* [IfStack](#)
- [UBYTE](#) \* [iBuffer](#)
- [UBYTE](#) \* [iPointer](#)
- [UBYTE](#) \* [iStop](#)



- UBYTE \*\* [LabelNames](#)
- WORD \* [FixIndices](#)
- WORD \* [termsumcheck](#)
- UBYTE \* [WildcardNames](#)
- int \* [Labels](#)
- SBYTE \* [tokens](#)
- SBYTE \* [toptokens](#)
- SBYTE \* [endoftokens](#)
- WORD \* [tokenarglevel](#)
- UWORD \* [modinverses](#)
- UBYTE \* [Fortran90Kind](#)
- WORD \*\* [MultiBracketBuf](#)
- UBYTE \* [extrasym](#)
- WORD \* [doloopstack](#)
- WORD \* [doloopnest](#)
- char \* [CheckpointRunAfter](#)
- char \* [CheckpointRunBefore](#)
- WORD \* [IfSumCheck](#)
- WORD \* [CommutelnSet](#)
- UBYTE \* [TestValue](#)
- LONG [argstack](#) [MAXNEST]
- LONG [insidestack](#) [MAXNEST]
- LONG [inexprstack](#) [MAXNEST]
- LONG [iBufferSize](#)
- LONG [TransEname](#)
- LONG [ProcessBucketSize](#)
- LONG [mProcessBucketSize](#)
- LONG [CModule](#)
- LONG [ThreadBucketSize](#)
- LONG [CheckpointStamp](#)
- LONG [CheckpointInterval](#)
- int [cbufnum](#)
- int [AutoDeclareFlag](#)
- int [NoShowInput](#)
- int [ShortStats](#)
- int [compiletype](#)
- int [firstconstindex](#)
- int [insidefirst](#)
- int [minsidefirst](#)
- int [wildflag](#)
- int [NumLabels](#)
- int [MaxLabels](#)
- int [IDefDim](#)
- int [IDefDim4](#)
- int [NumWildcardNames](#)
- int [WildcardBufferSize](#)
- int [MaxIf](#)
- int [NumStreams](#)
- int [MaxNumStreams](#)
- int [firstctypemessage](#)
- int [tablecheck](#)
- int [idooption](#)
- int [BottomLevel](#)
- int [CompileLevel](#)
- int [TokensWriteFlag](#)

- int [UnsureDollarMode](#)
- int [outsidefun](#)
- int [funpowers](#)
- int [WarnFlag](#)
- int [StatsFlag](#)
- int [NamesFlag](#)
- int [CodesFlag](#)
- int [SetupFlag](#)
- int [SortType](#)
- int [ISortType](#)
- int [ThreadStats](#)
- int [FinalStats](#)
- int [OldParallelStats](#)
- int [ThreadsFlag](#)
- int [ThreadBalancing](#)
- int [ThreadSortFileSynch](#)
- int [ProcessStats](#)
- int [BracketNormalize](#)
- int [maxtermlevel](#)
- int [dumnumflag](#)
- int [bracketindexflag](#)
- int [parallelflag](#)
- int [mparallelflag](#)
- int [inparallelflag](#)
- int [partodoflag](#)
- int [properorderflag](#)
- int [vetofilling](#)
- int [tablefilling](#)
- int [vetotablebasefill](#)
- int [exprfillwarning](#)
- int [lhdollarflag](#)
- int [NoCompress](#)
- int [IsFortran90](#)
- int [MultiBracketLevels](#)
- int [topolynomialflag](#)
- int [ffbufnum](#)
- int [OldFactArgFlag](#)
- int [MemDebugFlag](#)
- int [OldGCDflag](#)
- int [WTimeStatsFlag](#)
- int [SortReallocateFlag](#)
- int [PrintBacktraceFlag](#)
- int [FlintPolyFlag](#)
- int [HumanStatsFlag](#)
- int [doloopstacksize](#)
- int [dolooplevel](#)
- int [CheckpointFlag](#)
- int [SizeCommutelnSet](#)
- int [origin](#)
- int [vectorlikeLHS](#)
- int [nummodels](#)
- int [modelspace](#)
- int [ModelLevel](#)
- WORD [argsumcheck](#) [MAXNEST]
- WORD [insidesumcheck](#) [MAXNEST]

- WORD [inexprsumcheck](#) [MAXNEST]
- WORD [RepSumCheck](#) [MAXREPEAT]
- WORD [IUniTrace](#) [4]
- WORD [RepLevel](#)
- WORD [arglevel](#)
- WORD [insidelevel](#)
- WORD [inexprlevel](#)
- WORD [termlevel](#)
- WORD [MustTestTable](#)
- WORD [DumNum](#)
- WORD [ncmod](#)
- WORD [npowmod](#)
- WORD [modmode](#)
- WORD [nhalfmod](#)
- WORD [DirtPow](#)
- WORD [IUnitTrace](#)
- WORD [NwildC](#)
- WORD [ComDefer](#)
- WORD [CollectFun](#)
- WORD [AltCollectFun](#)
- WORD [OutputMode](#)
- WORD [Cnumpows](#)
- WORD [OutputSpaces](#)
- WORD [OutNumberType](#)
- WORD [DidClean](#)
- WORD [IfLevel](#)
- WORD [WhileLevel](#)
- WORD [SwitchLevel](#)
- WORD [SwitchInArray](#)
- WORD [MaxSwitch](#)
- WORD [LogHandle](#)
- WORD [LineLength](#)
- WORD [StoreHandle](#)
- WORD [HideLevel](#)
- WORD [IPolyFun](#)
- WORD [IPolyFunInv](#)
- WORD [IPolyFunType](#)
- WORD [IPolyFunExp](#)
- WORD [IPolyFunVar](#)
- WORD [IPolyFunPow](#)
- WORD [SymChangeFlag](#)
- WORD [CollectPercentage](#)
- WORD [ShortStatsMax](#)
- WORD [extrasymbols](#)
- WORD [PolyRatFunChanged](#)
- WORD [ToBeInFactors](#)
- WORD [InnerTest](#)
- UBYTE [Commercial](#) [COMMERCIALSIZE+2]
- UBYTE [debugFlags](#) [MAXFLAGS+2]

### 3.12.1 Detailed Description

The `C_const` struct is part of the global data and resides in the `ALLGLOBALS` struct A under the name #C. We see it used with the macro AC as in AC.exprnames. It contains variables that involve the compiler and objects set during compilation.

Definition at line 1697 of file `structs.h`.

### 3.12.2 Field Documentation

#### 3.12.2.1 separators

`set_of_char C_const::separators`

Separators in #call and #do

Definition at line 1698 of file `structs.h`.

#### 3.12.2.2 StoreFileSize

`POSITION C_const::StoreFileSize`

Definition at line 1699 of file `structs.h`.

#### 3.12.2.3 dollarnames

`NAMETREE* C_const::dollarnames`

[D] Names of dollar variables

Definition at line 1700 of file `structs.h`.

#### 3.12.2.4 exprnames

`NAMETREE* C_const::exprnames`

[D] Names of expressions

Definition at line 1701 of file `structs.h`.

#### 3.12.2.5 varnames

`NAMETREE* C_const::varnames`

[D] Names of regular variables

Definition at line 1702 of file `structs.h`.

### 3.12.2.6 ChannelList

`LIST C_const::ChannelList`

Used for the #write statement. Contains [CHANNEL](#)

Definition at line [1703](#) of file [structs.h](#).

### 3.12.2.7 DubiousList

`LIST C_const::DubiousList`

List of dubious variables. Contains DUBIOUSV. If not empty -> no execution

Definition at line [1705](#) of file [structs.h](#).

### 3.12.2.8 FunctionList

`LIST C_const::FunctionList`

List of functions and properties. Contains [FUNCTIONS](#)

Definition at line [1707](#) of file [structs.h](#).

### 3.12.2.9 ExpressionList

`LIST C_const::ExpressionList`

List of expressions, locations etc.

Definition at line [1708](#) of file [structs.h](#).

### 3.12.2.10 IndexList

`LIST C_const::IndexList`

List of indices

Definition at line [1709](#) of file [structs.h](#).

### 3.12.2.11 SetElementList

`LIST C_const::SetElementList`

List of all elements of all sets

Definition at line [1710](#) of file [structs.h](#).

#### 3.12.2.12 SetList

`LIST C_const::SetList`

List of the sets

Definition at line 1711 of file [structs.h](#).

#### 3.12.2.13 SymbolList

`LIST C_const::SymbolList`

List of the symbols and their properties

Definition at line 1712 of file [structs.h](#).

#### 3.12.2.14 VectorList

`LIST C_const::VectorList`

List of the vectors

Definition at line 1713 of file [structs.h](#).

#### 3.12.2.15 PotModDolList

`LIST C_const::PotModDolList`

Potentially changed dollars

Definition at line 1714 of file [structs.h](#).

#### 3.12.2.16 ModOptDolList

`LIST C_const::ModOptDolList`

Module Option Dollars list

Definition at line 1715 of file [structs.h](#).

#### 3.12.2.17 TableBaseList

`LIST C_const::TableBaseList`

TableBase list

Definition at line 1716 of file [structs.h](#).

### 3.12.2.18 cbufList

`LIST C_const::cbufList`

List of compiler buffers

Definition at line 1720 of file [structs.h](#).

### 3.12.2.19 AutoSymbolList

`LIST C_const::AutoSymbolList`

Definition at line 1724 of file [structs.h](#).

### 3.12.2.20 AutoIndexList

`LIST C_const::AutoIndexList`

Definition at line 1725 of file [structs.h](#).

### 3.12.2.21 AutoVectorList

`LIST C_const::AutoVectorList`

Definition at line 1726 of file [structs.h](#).

### 3.12.2.22 AutoFunctionList

`LIST C_const::AutoFunctionList`

Definition at line 1727 of file [structs.h](#).

### 3.12.2.23 autonames

`NAMETREE* C_const::autonames`

[D] Names in autodeclare

Definition at line 1728 of file [structs.h](#).

### 3.12.2.24 Symbols

`LIST* C_const::Symbols`

Definition at line 1730 of file [structs.h](#).

### 3.12.2.25 Indices

`LIST* C_const::Indices`

Definition at line 1732 of file [structs.h](#).

### 3.12.2.26 Vectors

`LIST* C_const::Vectors`

Definition at line 1733 of file [structs.h](#).

### 3.12.2.27 Functions

`LIST* C_const::Functions`

Definition at line 1734 of file [structs.h](#).

### 3.12.2.28 activenames

`NAMETREE** C_const::activenames`

Definition at line 1735 of file [structs.h](#).

### 3.12.2.29 Streams

`STREAM* C_const::Streams`

(C) Pointer for AutoDeclare statement. Points either to varnames or autonames. [D] The input streams.

Definition at line 1737 of file [structs.h](#).

### 3.12.2.30 CurrentStream

`STREAM* C_const::CurrentStream`

(C) The current input stream. Streams are: do loop, file, prevariable. points into Streams memory.

Definition at line 1738 of file [structs.h](#).

### 3.12.2.31 SwitchArray

`SWITCH* C_const::SwitchArray`

Definition at line 1740 of file [structs.h](#).



### 3.12.2.32 models

`MODEL** C_const::models`

Definition at line 1741 of file [structs.h](#).

### 3.12.2.33 SwitchHeap

`WORD* C_const::SwitchHeap`

Definition at line 1742 of file [structs.h](#).

### 3.12.2.34 termstack

`LONG* C_const::termstack`

[D] Last term statement {offset}

Definition at line 1743 of file [structs.h](#).

### 3.12.2.35 termsortstack

`LONG* C_const::termsortstack`

[D] Last sort statement {offset}

Definition at line 1744 of file [structs.h](#).

### 3.12.2.36 cmod

`UWORD* C_const::cmod`

[D] Local setting of modulus. Pointer to value.

Definition at line 1745 of file [structs.h](#).

### 3.12.2.37 powmod

`UWORD* C_const::powmod`

Local setting printing as powers. Points into cmod memory

Definition at line 1746 of file [structs.h](#).

### 3.12.2.38 modpowers

UWORD\* C\_const::modpowers

[D] The conversion table for mod-> powers.

Definition at line 1747 of file [structs.h](#).

### 3.12.2.39 halfmod

UWORD\* C\_const::halfmod

Definition at line 1748 of file [structs.h](#).

### 3.12.2.40 ProtoType

WORD\* C\_const::ProtoType

Definition at line 1749 of file [structs.h](#).

### 3.12.2.41 WildC

WORD\* C\_const::WildC

Definition at line 1750 of file [structs.h](#).

### 3.12.2.42 IfHeap

LONG\* C\_const::IfHeap

[D] Keeps track of where to go in if

Definition at line 1751 of file [structs.h](#).

### 3.12.2.43 IfCount

LONG\* C\_const::IfCount

[D] Keeps track of where to go in if

Definition at line 1752 of file [structs.h](#).

### 3.12.2.44 IfStack

LONG\* C\_const::IfStack

Keeps track of where to go in if. Points into IfHeap-memory

Definition at line 1753 of file [structs.h](#).

#### 3.12.2.45 iBuffer

```
UBYTE* C_const::iBuffer
```

[D] Compiler input buffer

Definition at line 1754 of file [structs.h](#).

#### 3.12.2.46 iPointer

```
UBYTE* C_const::iPointer
```

Running pointer in the compiler input buffer

Definition at line 1755 of file [structs.h](#).

#### 3.12.2.47 iStop

```
UBYTE* C_const::iStop
```

Top of iBuffer

Definition at line 1756 of file [structs.h](#).

#### 3.12.2.48 LabelNames

```
UBYTE** C_const::LabelNames
```

[D] List of names in label statements

Definition at line 1757 of file [structs.h](#).

#### 3.12.2.49 FixIndices

```
WORD* C_const::FixIndices
```

[D] Buffer of fixed indices

Definition at line 1758 of file [structs.h](#).

#### 3.12.2.50 termsumcheck

```
WORD* C_const::termsumcheck
```

[D] Checking of nesting

Definition at line 1759 of file [structs.h](#).

#### 3.12.2.51 WildcardNames

```
UBYTE* C_const::WildcardNames
```

[D] Names of ?a variables

Definition at line 1760 of file [structs.h](#).

#### 3.12.2.52 Labels

```
int* C_const::Labels
```

Label information for during run. Pointer into LabelNames memory.

Definition at line 1761 of file [structs.h](#).

#### 3.12.2.53 tokens

```
SBYTE* C_const::tokens
```

[D] Array with tokens for tokenizer

Definition at line 1762 of file [structs.h](#).

#### 3.12.2.54 toptokens

```
SBYTE* C_const::toptokens
```

Top of tokens

Definition at line 1763 of file [structs.h](#).

#### 3.12.2.55 endoftokens

```
SBYTE* C_const::endoftokens
```

End of the actual tokens

Definition at line 1764 of file [structs.h](#).

#### 3.12.2.56 tokenarglevel

```
WORD* C_const::tokenarglevel
```

[D] Keeps track of function arguments

Definition at line 1765 of file [structs.h](#).

### 3.12.2.57 modinverses

UWORD\* C\_const::modinverses

Definition at line 1766 of file [structs.h](#).

### 3.12.2.58 Fortran90Kind

UBYTE\* C\_const::Fortran90Kind

Definition at line 1767 of file [structs.h](#).

### 3.12.2.59 MultiBracketBuf

WORD\*\* C\_const::MultiBracketBuf

Definition at line 1768 of file [structs.h](#).

### 3.12.2.60 extrasym

UBYTE\* C\_const::extrasym

Definition at line 1769 of file [structs.h](#).

### 3.12.2.61 doloopstack

WORD\* C\_const::doloopstack

Definition at line 1770 of file [structs.h](#).

### 3.12.2.62 doloopnest

WORD\* C\_const::doloopnest

Definition at line 1771 of file [structs.h](#).

### 3.12.2.63 CheckpointRunAfter

char\* C\_const::CheckpointRunAfter

[D] Filename of script to be executed *before* creating the snapshot. =0 if no script shall be executed.

Definition at line 1772 of file [structs.h](#).

#### 3.12.2.64 CheckpointRunBefore

```
char* C_const::CheckpointRunBefore
```

[D] Filename of script to be executed *after* having created the snapshot. =0 if no script shall be executed.

Definition at line 1774 of file [structs.h](#).

#### 3.12.2.65 IfSumCheck

```
WORD* C_const::IfSumCheck
```

[D] Keeps track of if-nesting

Definition at line 1776 of file [structs.h](#).

#### 3.12.2.66 CommuteInSet

```
WORD* C_const::CommuteInSet
```

Definition at line 1777 of file [structs.h](#).

#### 3.12.2.67 TestValue

```
UBYTE* C_const::TestValue
```

Definition at line 1778 of file [structs.h](#).

#### 3.12.2.68 argstack

```
LONG C_const::argstack[MAXNEST]
```

Definition at line 1786 of file [structs.h](#).

#### 3.12.2.69 insidestack

```
LONG C_const::insidestack[MAXNEST]
```

Definition at line 1787 of file [structs.h](#).

#### 3.12.2.70 inexprstack

```
LONG C_const::inexprstack[MAXNEST]
```

Definition at line 1788 of file [structs.h](#).

### 3.12.2.71 iBufferSize

LONG C\_const::iBufferSize

Definition at line 1789 of file [structs.h](#).

### 3.12.2.72 TransName

LONG C\_const::TransName

Definition at line 1790 of file [structs.h](#).

### 3.12.2.73 ProcessBucketSize

LONG C\_const::ProcessBucketSize

Definition at line 1792 of file [structs.h](#).

### 3.12.2.74 mProcessBucketSize

LONG C\_const::mProcessBucketSize

Definition at line 1793 of file [structs.h](#).

### 3.12.2.75 CModule

LONG C\_const::CModule

Definition at line 1794 of file [structs.h](#).

### 3.12.2.76 ThreadBucketSize

LONG C\_const::ThreadBucketSize

Definition at line 1795 of file [structs.h](#).

### 3.12.2.77 CheckpointStamp

LONG C\_const::CheckpointStamp

Timestamp of the last created snapshot (set to Timer(0)).

Definition at line 1796 of file [structs.h](#).

### 3.12.2.78 CheckpointInterval

```
LONG C_const::CheckpointInterval
```

Time interval in milliseconds for snapshots. =0 if snapshots shall be created at the end of every module.

Definition at line 1797 of file [structs.h](#).

### 3.12.2.79 cbufnum

```
int C_const::cbufnum
```

Current compiler buffer

Definition at line 1805 of file [structs.h](#).

### 3.12.2.80 AutoDeclareFlag

```
int C_const::AutoDeclareFlag
```

Definition at line 1806 of file [structs.h](#).

### 3.12.2.81 NoShowInput

```
int C_const::NoShowInput
```

(C) Mode of looking for names. Set to NOAUTO (=0) or WITHAUTO (=2), cf. AutoDeclare statement

Definition at line 1808 of file [structs.h](#).

### 3.12.2.82 ShortStats

```
int C_const::ShortStats
```

Definition at line 1809 of file [structs.h](#).

### 3.12.2.83 compiletype

```
int C_const::compiletype
```

Definition at line 1810 of file [structs.h](#).

### 3.12.2.84 firstconstindex

```
int C_const::firstconstindex
```

Definition at line 1811 of file [structs.h](#).



**3.12.2.85 insidefirst**

```
int C_const::insidefirst
```

Definition at line 1812 of file [structs.h](#).

**3.12.2.86 minsidefirst**

```
int C_const::minsidefirst
```

Definition at line 1813 of file [structs.h](#).

**3.12.2.87 wildflag**

```
int C_const::wildflag
```

Definition at line 1814 of file [structs.h](#).

**3.12.2.88 NumLabels**

```
int C_const::NumLabels
```

Definition at line 1815 of file [structs.h](#).

**3.12.2.89 MaxLabels**

```
int C_const::MaxLabels
```

Definition at line 1816 of file [structs.h](#).

**3.12.2.90 IDefDim**

```
int C_const::lDefDim
```

Definition at line 1817 of file [structs.h](#).

**3.12.2.91 IDefDim4**

```
int C_const::lDefDim4
```

Definition at line 1818 of file [structs.h](#).

**3.12.2.92 NumWildcardNames**

```
int C_const::NumWildcardNames
```

Definition at line 1819 of file [structs.h](#).

### 3.12.2.93 WildcardBufferSize

```
int C_const::WildcardBufferSize
```

Definition at line 1820 of file [structs.h](#).

### 3.12.2.94 MaxIf

```
int C_const::MaxIf
```

Definition at line 1821 of file [structs.h](#).

### 3.12.2.95 NumStreams

```
int C_const::NumStreams
```

Definition at line 1822 of file [structs.h](#).

### 3.12.2.96 MaxNumStreams

```
int C_const::MaxNumStreams
```

Definition at line 1823 of file [structs.h](#).

### 3.12.2.97 firstctypemessage

```
int C_const::firstctypemessage
```

Definition at line 1824 of file [structs.h](#).

### 3.12.2.98 tablecheck

```
int C_const::tablecheck
```

Definition at line 1825 of file [structs.h](#).

### 3.12.2.99 idoption

```
int C_const::idoption
```

Definition at line 1826 of file [structs.h](#).

### 3.12.2.100 BottomLevel

```
int C_const::BottomLevel
```

Definition at line 1827 of file [structs.h](#).

**3.12.2.101 CompileLevel**

```
int C_const::CompileLevel
```

Definition at line 1828 of file [structs.h](#).

**3.12.2.102 TokensWriteFlag**

```
int C_const::TokensWriteFlag
```

Definition at line 1829 of file [structs.h](#).

**3.12.2.103 UnsureDollarMode**

```
int C_const::UnsureDollarMode
```

Definition at line 1830 of file [structs.h](#).

**3.12.2.104 outsidefun**

```
int C_const::outsidefun
```

Definition at line 1831 of file [structs.h](#).

**3.12.2.105 funpowers**

```
int C_const::funpowers
```

Definition at line 1832 of file [structs.h](#).

**3.12.2.106 WarnFlag**

```
int C_const::WarnFlag
```

Definition at line 1833 of file [structs.h](#).

**3.12.2.107 StatsFlag**

```
int C_const::StatsFlag
```

Definition at line 1834 of file [structs.h](#).

**3.12.2.108 NamesFlag**

```
int C_const::NamesFlag
```

Definition at line 1835 of file [structs.h](#).

**3.12.2.109 CodesFlag**

```
int C_const::CodesFlag
```

Definition at line [1836](#) of file [structs.h](#).

**3.12.2.110 SetupFlag**

```
int C_const::SetupFlag
```

Definition at line [1837](#) of file [structs.h](#).

**3.12.2.111 SortType**

```
int C_const::SortType
```

Definition at line [1838](#) of file [structs.h](#).

**3.12.2.112 lSortType**

```
int C_const::lSortType
```

Definition at line [1839](#) of file [structs.h](#).

**3.12.2.113 ThreadStats**

```
int C_const::ThreadStats
```

Definition at line [1840](#) of file [structs.h](#).

**3.12.2.114 FinalStats**

```
int C_const::FinalStats
```

Definition at line [1841](#) of file [structs.h](#).

**3.12.2.115 OldParallelStats**

```
int C_const::OldParallelStats
```

Definition at line [1842](#) of file [structs.h](#).

**3.12.2.116 ThreadsFlag**

```
int C_const::ThreadsFlag
```

Definition at line [1843](#) of file [structs.h](#).

**3.12.2.117 ThreadBalancing**

```
int C_const::ThreadBalancing
```

Definition at line 1844 of file [structs.h](#).

**3.12.2.118 ThreadSortFileSynch**

```
int C_const::ThreadSortFileSynch
```

Definition at line 1845 of file [structs.h](#).

**3.12.2.119 ProcessStats**

```
int C_const::ProcessStats
```

Definition at line 1846 of file [structs.h](#).

**3.12.2.120 BracketNormalize**

```
int C_const::BracketNormalize
```

Definition at line 1847 of file [structs.h](#).

**3.12.2.121 maxtermlevel**

```
int C_const::maxtermlevel
```

Definition at line 1848 of file [structs.h](#).

**3.12.2.122 dumnumflag**

```
int C_const::dumnumflag
```

Definition at line 1849 of file [structs.h](#).

**3.12.2.123 bracketindexflag**

```
int C_const::bracketindexflag
```

Definition at line 1850 of file [structs.h](#).

**3.12.2.124 parallelfag**

```
int C_const::parallelfag
```

Definition at line 1851 of file [structs.h](#).

**3.12.2.125 mparallelflag**

```
int C_const::mparallelflag
```

Definition at line 1852 of file [structs.h](#).

**3.12.2.126 inparallelflag**

```
int C_const::inparallelflag
```

Definition at line 1853 of file [structs.h](#).

**3.12.2.127 partodoflag**

```
int C_const::partodoflag
```

Definition at line 1854 of file [structs.h](#).

**3.12.2.128 properorderflag**

```
int C_const::properorderflag
```

Definition at line 1855 of file [structs.h](#).

**3.12.2.129 vetofilling**

```
int C_const::vetofilling
```

Definition at line 1856 of file [structs.h](#).

**3.12.2.130 tablefilling**

```
int C_const::tablefilling
```

Definition at line 1857 of file [structs.h](#).

**3.12.2.131 vetotablebasefill**

```
int C_const::vetotablebasefill
```

Definition at line 1858 of file [structs.h](#).

**3.12.2.132 exprfillwarning**

```
int C_const::exprfillwarning
```

Definition at line 1859 of file [structs.h](#).

**3.12.2.133 lhdollarflag**

```
int C_const::lhdollarflag
```

Definition at line 1860 of file [structs.h](#).

**3.12.2.134 NoCompress**

```
int C_const::NoCompress
```

Definition at line 1861 of file [structs.h](#).

**3.12.2.135 IsFortran90**

```
int C_const::IsFortran90
```

Definition at line 1862 of file [structs.h](#).

**3.12.2.136 MultiBracketLevels**

```
int C_const::MultiBracketLevels
```

Definition at line 1863 of file [structs.h](#).

**3.12.2.137 topolynomialflag**

```
int C_const::topolynomialflag
```

Definition at line 1864 of file [structs.h](#).

**3.12.2.138 ffbufnum**

```
int C_const::ffbufnum
```

Definition at line 1865 of file [structs.h](#).

**3.12.2.139 OldFactArgFlag**

```
int C_const::OldFactArgFlag
```

Definition at line 1866 of file [structs.h](#).

**3.12.2.140 MemDebugFlag**

```
int C_const::MemDebugFlag
```

Definition at line 1867 of file [structs.h](#).

**3.12.2.141 OldGCDflag**

```
int C_const::OldGCDflag
```

Definition at line 1868 of file [structs.h](#).

**3.12.2.142 WTimeStatsFlag**

```
int C_const::WTimeStatsFlag
```

Definition at line 1869 of file [structs.h](#).

**3.12.2.143 SortReallocateFlag**

```
int C_const::SortReallocateFlag
```

Definition at line 1870 of file [structs.h](#).

**3.12.2.144 PrintBacktraceFlag**

```
int C_const::PrintBacktraceFlag
```

Definition at line 1874 of file [structs.h](#).

**3.12.2.145 FlintPolyFlag**

```
int C_const::FlintPolyFlag
```

Definition at line 1875 of file [structs.h](#).

**3.12.2.146 HumanStatsFlag**

```
int C_const::HumanStatsFlag
```

Definition at line 1876 of file [structs.h](#).

**3.12.2.147 doloopstacksize**

```
int C_const::doloopstacksize
```

Definition at line 1877 of file [structs.h](#).

**3.12.2.148 dolooplevel**

```
int C_const::dolooplevel
```

Definition at line 1878 of file [structs.h](#).



**3.12.2.149 CheckpointFlag**

```
int C_const::CheckpointFlag
```

Tells preprocessor whether checkpoint code must executed. -1 : do recovery from snapshot, set by command line option; 0 : do nothing; 1 : create snapshots, set by On checkpoint statement

Definition at line 1879 of file [structs.h](#).

**3.12.2.150 SizeCommuteInSet**

```
int C_const::SizeCommuteInSet
```

Definition at line 1883 of file [structs.h](#).

**3.12.2.151 origin**

```
int C_const::origin
```

Definition at line 1888 of file [structs.h](#).

**3.12.2.152 vectorlikeLHS**

```
int C_const::vectorlikeLHS
```

Definition at line 1889 of file [structs.h](#).

**3.12.2.153 nummodels**

```
int C_const::nummodels
```

Definition at line 1890 of file [structs.h](#).

**3.12.2.154 modelspace**

```
int C_const::modelspace
```

Definition at line 1891 of file [structs.h](#).

**3.12.2.155 ModelLevel**

```
int C_const::ModelLevel
```

Definition at line 1892 of file [structs.h](#).

**3.12.2.156 argsumcheck**

WORD C\_const::argsumcheck[MAXNEST]

Definition at line 1893 of file [structs.h](#).

**3.12.2.157 insidesumcheck**

WORD C\_const::insidesumcheck[MAXNEST]

Definition at line 1894 of file [structs.h](#).

**3.12.2.158 inexprsumcheck**

WORD C\_const::inexprsumcheck[MAXNEST]

Definition at line 1895 of file [structs.h](#).

**3.12.2.159 RepSumCheck**

WORD C\_const::RepSumCheck[MAXREPEAT]

Definition at line 1896 of file [structs.h](#).

**3.12.2.160 IUniTrace**

WORD C\_const::lUniTrace[4]

Definition at line 1897 of file [structs.h](#).

**3.12.2.161 RepLevel**

WORD C\_const::RepLevel

Definition at line 1898 of file [structs.h](#).

**3.12.2.162 arglevel**

WORD C\_const::arglevel

Definition at line 1899 of file [structs.h](#).

**3.12.2.163 insidelevel**

WORD C\_const::insidelevel

Definition at line 1900 of file [structs.h](#).

**3.12.2.164 inexprlevel**

WORD C\_const::inexprlevel

Definition at line 1901 of file [structs.h](#).

**3.12.2.165 termlevel**

WORD C\_const::termlevel

Definition at line 1902 of file [structs.h](#).

**3.12.2.166 MustTestTable**

WORD C\_const::MustTestTable

Definition at line 1903 of file [structs.h](#).

**3.12.2.167 DumNum**

WORD C\_const::DumNum

Definition at line 1904 of file [structs.h](#).

**3.12.2.168 ncmmod**

WORD C\_const::ncmmod

Definition at line 1905 of file [structs.h](#).

**3.12.2.169 npowmod**

WORD C\_const::npowmod

Definition at line 1906 of file [structs.h](#).

**3.12.2.170 modmode**

WORD C\_const::modmode

Definition at line 1907 of file [structs.h](#).

**3.12.2.171 nhalfmod**

WORD C\_const::nhalfmod

Definition at line 1908 of file [structs.h](#).

### 3.12.2.172 DirtPow

WORD C\_const::DirtPow

Definition at line 1909 of file [structs.h](#).

### 3.12.2.173 lUnitTrace

WORD C\_const::lUnitTrace

Definition at line 1910 of file [structs.h](#).

### 3.12.2.174 NwildC

WORD C\_const::NwildC

Definition at line 1911 of file [structs.h](#).

### 3.12.2.175 ComDefer

WORD C\_const::ComDefer

Definition at line 1912 of file [structs.h](#).

### 3.12.2.176 CollectFun

WORD C\_const::CollectFun

Definition at line 1913 of file [structs.h](#).

### 3.12.2.177 AltCollectFun

WORD C\_const::AltCollectFun

Definition at line 1914 of file [structs.h](#).

### 3.12.2.178 OutputMode

WORD C\_const::OutputMode

Definition at line 1915 of file [structs.h](#).

### 3.12.2.179 Cnumpows

WORD C\_const::Cnumpows

Definition at line 1916 of file [structs.h](#).

### 3.12.2.180 OutputSpaces

WORD C\_const::OutputSpaces

Definition at line 1917 of file [structs.h](#).

### 3.12.2.181 OutNumberType

WORD C\_const::OutNumberType

Definition at line 1918 of file [structs.h](#).

### 3.12.2.182 DidClean

WORD C\_const::DidClean

Definition at line 1919 of file [structs.h](#).

### 3.12.2.183 IfLevel

WORD C\_const::IfLevel

Definition at line 1920 of file [structs.h](#).

### 3.12.2.184 WhileLevel

WORD C\_const::WhileLevel

Definition at line 1921 of file [structs.h](#).

### 3.12.2.185 SwitchLevel

WORD C\_const::SwitchLevel

Definition at line 1922 of file [structs.h](#).

### 3.12.2.186 SwitchInArray

WORD C\_const::SwitchInArray

Definition at line 1923 of file [structs.h](#).

### 3.12.2.187 MaxSwitch

WORD C\_const::MaxSwitch

Definition at line 1924 of file [structs.h](#).

### 3.12.2.188 LogHandle

WORD C\_const::LogHandle

Definition at line 1925 of file [structs.h](#).

### 3.12.2.189 LineLength

WORD C\_const::LineLength

Definition at line 1926 of file [structs.h](#).

### 3.12.2.190 StoreHandle

WORD C\_const::StoreHandle

Definition at line 1927 of file [structs.h](#).

### 3.12.2.191 HideLevel

WORD C\_const::HideLevel

Definition at line 1928 of file [structs.h](#).

### 3.12.2.192 IPolyFun

WORD C\_const::lPolyFun

Definition at line 1929 of file [structs.h](#).

### 3.12.2.193 IPolyFunInv

WORD C\_const::lPolyFunInv

Definition at line 1930 of file [structs.h](#).

### 3.12.2.194 IPolyFunType

WORD C\_const::lPolyFunType

Definition at line 1931 of file [structs.h](#).

### 3.12.2.195 IPolyFunExp

WORD C\_const::lPolyFunExp

Definition at line 1932 of file [structs.h](#).

**3.12.2.196 IPolyFunVar**

WORD C\_const::lPolyFunVar

Definition at line 1933 of file [structs.h](#).

**3.12.2.197 IPolyFunPow**

WORD C\_const::lPolyFunPow

Definition at line 1934 of file [structs.h](#).

**3.12.2.198 SymChangeFlag**

WORD C\_const::SymChangeFlag

Definition at line 1935 of file [structs.h](#).

**3.12.2.199 CollectPercentage**

WORD C\_const::CollectPercentage

Definition at line 1936 of file [structs.h](#).

**3.12.2.200 ShortStatsMax**

WORD C\_const::ShortStatsMax

Definition at line 1937 of file [structs.h](#).

**3.12.2.201 extrasymbols**

WORD C\_const::extrasymbols

Definition at line 1938 of file [structs.h](#).

**3.12.2.202 PolyRatFunChanged**

WORD C\_const::PolyRatFunChanged

Definition at line 1939 of file [structs.h](#).

**3.12.2.203 ToBeInFactors**

WORD C\_const::ToBeInFactors

Definition at line 1940 of file [structs.h](#).

### 3.12.2.204 InnerTest

```
WORD C_const::InnerTest
```

Definition at line 1941 of file [structs.h](#).

### 3.12.2.205 Commercial

```
UBYTE C_const::Commercial[COMMERCIALSIZE+2]
```

Definition at line 1945 of file [structs.h](#).

### 3.12.2.206 debugFlags

```
UBYTE C_const::debugFlags[MAXFLAGS+2]
```

Definition at line 1946 of file [structs.h](#).

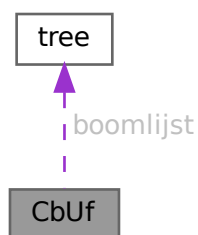
The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.13 CbUf Struct Reference

```
#include <structs.h>
```

Collaboration diagram for CbUf:





## Data Fields

- WORD \* [Buffer](#)
- WORD \* [Top](#)
- WORD \* [Pointer](#)
- WORD \*\* [lhs](#)
- WORD \*\* [rhs](#)
- LONG \* [CanCommu](#)
- LONG \* [NumTerms](#)
- WORD \* [numdum](#)
- WORD \* [dimension](#)
- COMPTREE \* [boomlijst](#)
- LONG [BufferSize](#)
- int [numlhs](#)
- int [numrhs](#)
- int [maxlhs](#)
- int [maxrhs](#)
- int [mnumlhs](#)
- int [mnumrhs](#)
- int [numtree](#)
- int [rootnum](#)
- int [MaxTreeSize](#)

### 3.13.1 Detailed Description

The CBUF struct is used by the compiler. It is a compiler buffer of which since version 3.0 there can be many.

Definition at line [992](#) of file [structs.h](#).

### 3.13.2 Field Documentation

#### 3.13.2.1 Buffer

```
WORD* CbUf::Buffer
```

[D] Size in [BufferSize](#)

Definition at line [993](#) of file [structs.h](#).

Referenced by [CleanupArgCache\(\)](#), [clearcbuf\(\)](#), [DoubleCbuffer\(\)](#), [finishcbuf\(\)](#), [inicbufs\(\)](#), [PF\\_BroadcastCBuf\(\)](#), and [Processor\(\)](#).

#### 3.13.2.2 Top

```
WORD* CbUf::Top
```

pointer to the end of the Buffer memory

Definition at line [994](#) of file [structs.h](#).

Referenced by [AddNtoC\(\)](#), [AddNtoL\(\)](#), [DoubleCbuffer\(\)](#), [finishcbuf\(\)](#), [inicbufs\(\)](#), and [PF\\_BroadcastCBuf\(\)](#).

### 3.13.2.3 Pointer

`WORD* CbUf::Pointer`

pointer into the Buffer memory

Definition at line 995 of file [structs.h](#).

Referenced by [AddLHS\(\)](#), [AddNtoC\(\)](#), [AddNtoL\(\)](#), [AddRHS\(\)](#), [CleanupArgCache\(\)](#), [clearcbuf\(\)](#), [DoubleCbuffer\(\)](#), [finishcbuf\(\)](#), [Generator\(\)](#), [inicbufs\(\)](#), [PF\\_BroadcastCBuf\(\)](#), and [Processor\(\)](#).

### 3.13.2.4 lhs

`WORD** CbUf::lhs`

[D] Size in maxlhs. list of pointers into Buffer.

Definition at line 996 of file [structs.h](#).

Referenced by [AddLHS\(\)](#), [DoubleCbuffer\(\)](#), [finishcbuf\(\)](#), [Generator\(\)](#), [inicbufs\(\)](#), [PF\\_BroadcastCBuf\(\)](#), [Processor\(\)](#), and [TestMatch\(\)](#).

### 3.13.2.5 rhs

`WORD** CbUf::rhs`

[D] Size in maxrhs. list of pointers into Buffer.

Definition at line 997 of file [structs.h](#).

Referenced by [AddRHS\(\)](#), [CleanupArgCache\(\)](#), [clearcbuf\(\)](#), [DoubleCbuffer\(\)](#), [finishcbuf\(\)](#), [Generator\(\)](#), [inicbufs\(\)](#), [IniFbuffer\(\)](#), [PF\\_BroadcastCBuf\(\)](#), and [Processor\(\)](#).

### 3.13.2.6 CanCommu

`LONG* CbUf::CanCommu`

points into rhs memory behind WORD\* area.

Definition at line 998 of file [structs.h](#).

Referenced by [AddRHS\(\)](#), [finishcbuf\(\)](#), [inicbufs\(\)](#), [IniFbuffer\(\)](#), and [PF\\_BroadcastCBuf\(\)](#).

### 3.13.2.7 NumTerms

`LONG* CbUf::NumTerms`

points into rhs memory behind CanCommu area

Definition at line 999 of file [structs.h](#).

Referenced by [AddRHS\(\)](#), [finishcbuf\(\)](#), [inicbufs\(\)](#), [IniFbuffer\(\)](#), and [PF\\_BroadcastCBuf\(\)](#).

### 3.13.2.8 numdum

WORD\* CbUf::numdum

points into rhs memory behind NumTerms

Definition at line 1000 of file [structs.h](#).

Referenced by [AddRHS\(\)](#), [inicbufs\(\)](#), [IniFbuffer\(\)](#), and [PF\\_BroadcastCBuf\(\)](#).

### 3.13.2.9 dimension

WORD\* CbUf::dimension

points into rhs memory behind numdum

Definition at line 1001 of file [structs.h](#).

Referenced by [AddRHS\(\)](#), [inicbufs\(\)](#), [IniFbuffer\(\)](#), and [PF\\_BroadcastCBuf\(\)](#).

### 3.13.2.10 boomlijst

COMPTREE\* CbUf::boomlijst

[D] Number elements in MaxTreeSize

Definition at line 1002 of file [structs.h](#).

Referenced by [CleanupArgCache\(\)](#), [clearcbuf\(\)](#), [finishcbuf\(\)](#), [inicbufs\(\)](#), [IniFbuffer\(\)](#), and [PF\\_BroadcastCBuf\(\)](#).

### 3.13.2.11 BufferSize

LONG CbUf::BufferSize

Number of allocated WORD's in Buffer

Definition at line 1003 of file [structs.h](#).

Referenced by [DoubleCbuffer\(\)](#), [finishcbuf\(\)](#), [inicbufs\(\)](#), and [PF\\_BroadcastCBuf\(\)](#).

### 3.13.2.12 numlhs

int CbUf::numlhs

Definition at line 1004 of file [structs.h](#).

### 3.13.2.13 numrhs

int CbUf::numrhs

Definition at line 1005 of file [structs.h](#).

#### 3.13.2.14 maxlhs

```
int CbUf::maxlhs
```

Definition at line 1006 of file [structs.h](#).

#### 3.13.2.15 maxrhs

```
int CbUf::maxrhs
```

Definition at line 1007 of file [structs.h](#).

#### 3.13.2.16 mnumlhs

```
int CbUf::mnumlhs
```

Definition at line 1008 of file [structs.h](#).

#### 3.13.2.17 mnumrhs

```
int CbUf::mnumrhs
```

Definition at line 1009 of file [structs.h](#).

#### 3.13.2.18 numtree

```
int CbUf::numtree
```

Definition at line 1010 of file [structs.h](#).

#### 3.13.2.19 rootnum

```
int CbUf::rootnum
```

Definition at line 1011 of file [structs.h](#).

#### 3.13.2.20 MaxTreeSize

```
int CbUf::MaxTreeSize
```

Definition at line 1012 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.14 ChAnNeL Struct Reference

```
#include <structs.h>
```

### Data Fields

- char \* [name](#)
- int [handle](#)

### 3.14.1 Detailed Description

When we read input from text files we have to remember not only their handle but also their name. This is needed for error messages. Hence we call such a file a channel and reserve a struct of type [CHANNEL](#) to allow to lay this link.

Definition at line [1023](#) of file [structs.h](#).

### 3.14.2 Field Documentation

#### 3.14.2.1 name

```
char* ChAnNeL::name
```

[D] Name of the channel

Definition at line [1024](#) of file [structs.h](#).

#### 3.14.2.2 handle

```
int ChAnNeL::handle
```

File handle

Definition at line [1025](#) of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.15 COST Struct Reference

### Data Fields

- LONG [add](#)
- LONG [mul](#)
- LONG [div](#)
- LONG [pow](#)

### 3.15.1 Detailed Description

Definition at line [1294](#) of file [structs.h](#).

### 3.15.2 Field Documentation

#### 3.15.2.1 add

```
LONG COST::add
```

Definition at line [1295](#) of file [structs.h](#).

#### 3.15.2.2 mul

```
LONG COST::mul
```

Definition at line [1296](#) of file [structs.h](#).

#### 3.15.2.3 div

```
LONG COST::div
```

Definition at line [1297](#) of file [structs.h](#).

#### 3.15.2.4 pow

```
LONG COST::pow
```

Definition at line [1298](#) of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.16 CSEEq Struct Reference

### Public Member Functions

- bool [operator\(\)](#) (const vector< WORD > &lhs, const vector< WORD > &rhs) const

### 3.16.1 Detailed Description

Definition at line [1060](#) of file [optimize.cc](#).

### 3.16.2 Member Function Documentation

#### 3.16.2.1 operator()

```
bool CSEEq::operator() (
    const vector< WORD > & lhs,
    const vector< WORD > & rhs ) const [inline]
```

Definition at line 1061 of file [optimize.cc](#).

The documentation for this struct was generated from the following file:

- [optimize.cc](#)

## 3.17 CSEHash Struct Reference

### Public Member Functions

- `size_t operator()` (const vector< WORD > &n) const

#### 3.17.1 Detailed Description

Definition at line 1054 of file [optimize.cc](#).

### 3.17.2 Member Function Documentation

#### 3.17.2.1 operator()

```
size_t CSEHash::operator() (
    const vector< WORD > & n ) const [inline]
```

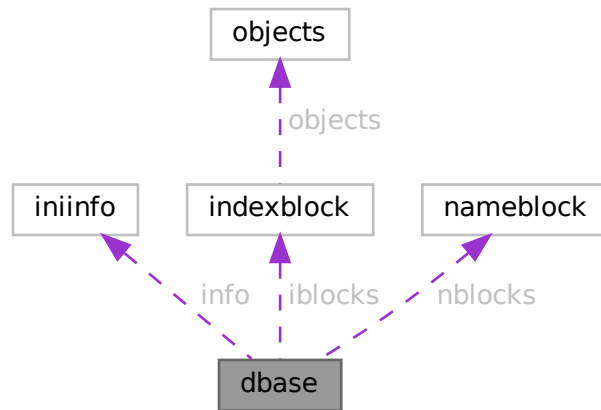
Definition at line 1055 of file [optimize.cc](#).

The documentation for this struct was generated from the following file:

- [optimize.cc](#)

## 3.18 dbase Struct Reference

Collaboration diagram for dbase:



### Data Fields

- `INIINFO` `info`
- `MLONG` `mode`
- `MLONG` `tablenameessize`
- `MLONG` `topnumber`
- `MLONG` `tablenameefill`
- `MLONG` `rwmode`
- `INDEXBLOCK` `** iblocks`
- `NAMESBLOCK` `** nblocks`
- `FILE` `* handle`
- `char *` `name`
- `char *` `fullname`
- `char *` `tablenames`

### 3.18.1 Detailed Description

Definition at line 123 of file `minos.h`.

### 3.18.2 Field Documentation

#### 3.18.2.1 `info`

`INIINFO` `dbase::info`

Definition at line 124 of file `minos.h`.



### 3.18.2.2 mode

MLONG dbase::mode

Definition at line 125 of file [minos.h](#).

### 3.18.2.3 tablenamessize

MLONG dbase::tablenamessize

Definition at line 126 of file [minos.h](#).

### 3.18.2.4 topnumber

MLONG dbase::topnumber

Definition at line 127 of file [minos.h](#).

### 3.18.2.5 tablenameefill

MLONG dbase::tablenameefill

Definition at line 128 of file [minos.h](#).

### 3.18.2.6 rwmode

MLONG dbase::rwmode

Definition at line 129 of file [minos.h](#).

### 3.18.2.7 iblocks

INDEXBLOCK\*\* dbase::iblocks

Definition at line 130 of file [minos.h](#).

### 3.18.2.8 nblocks

NAMESBLOCK\*\* dbase::nblocks

Definition at line 131 of file [minos.h](#).

### 3.18.2.9 handle

FILE\* dbase::handle

Definition at line 132 of file [minos.h](#).

### 3.18.2.10 name

```
char* dbase::name
```

Definition at line 133 of file [minos.h](#).

### 3.18.2.11 fullname

```
char* dbase::fullname
```

Definition at line 134 of file [minos.h](#).

### 3.18.2.12 tablenames

```
char* dbase::tablenames
```

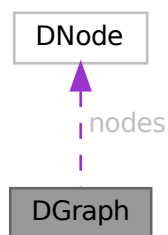
Definition at line 135 of file [minos.h](#).

The documentation for this struct was generated from the following file:

- [minos.h](#)

## 3.19 DGraph Struct Reference

Collaboration diagram for DGraph:



### Data Fields

- int [nnodes](#)
- int [nedges](#)
- [DNode](#) \* [nodes](#)
- [DEdge](#) \* [edges](#)

### 3.19.1 Detailed Description

Definition at line 199 of file [grccparam.h](#).

### 3.19.2 Field Documentation

#### 3.19.2.1 nnodes

```
int DGraph::nnodes
```

Definition at line 200 of file [grccparam.h](#).

#### 3.19.2.2 nedges

```
int DGraph::nedges
```

Definition at line 201 of file [grccparam.h](#).

#### 3.19.2.3 nodes

```
DNode* DGraph::nodes
```

Definition at line 202 of file [grccparam.h](#).

#### 3.19.2.4 edges

```
DEdge* DGraph::edges
```

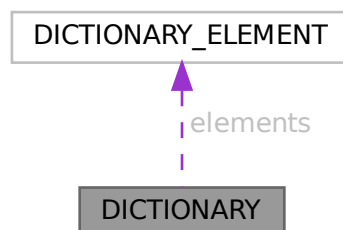
Definition at line 203 of file [grccparam.h](#).

The documentation for this struct was generated from the following file:

- [grccparam.h](#)

## 3.20 DICTIONARY Struct Reference

Collaboration diagram for DICTIONARY:



## Data Fields

- [DICTIONARY\\_ELEMENT](#) \*\* [elements](#)
- UBYTE \* [name](#)
- int [sizeelements](#)
- int [numelements](#)
- int [numbers](#)
- int [variables](#)
- int [characters](#)
- int [funwith](#)
- int [gnumelements](#)
- int [ranges](#)

### 3.20.1 Detailed Description

Definition at line [1361](#) of file [structs.h](#).

### 3.20.2 Field Documentation

#### 3.20.2.1 elements

```
DICTIONARY_ELEMENT** DICTIONARY::elements
```

Definition at line [1362](#) of file [structs.h](#).

#### 3.20.2.2 name

```
UBYTE* DICTIONARY::name
```

Definition at line [1363](#) of file [structs.h](#).

#### 3.20.2.3 sizeelements

```
int DICTIONARY::sizeelements
```

Definition at line [1364](#) of file [structs.h](#).

#### 3.20.2.4 numelements

```
int DICTIONARY::numelements
```

Definition at line [1365](#) of file [structs.h](#).

#### 3.20.2.5 numbers

```
int DICTIONARY::numbers
```

Definition at line [1366](#) of file [structs.h](#).

### 3.20.2.6 variables

```
int DICTIONARY::variables
```

Definition at line 1367 of file [structs.h](#).

### 3.20.2.7 characters

```
int DICTIONARY::characters
```

Definition at line 1368 of file [structs.h](#).

### 3.20.2.8 funwith

```
int DICTIONARY::funwith
```

Definition at line 1369 of file [structs.h](#).

### 3.20.2.9 gnumelements

```
int DICTIONARY::gnumelements
```

Definition at line 1370 of file [structs.h](#).

### 3.20.2.10 ranges

```
int DICTIONARY::ranges
```

Definition at line 1371 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.21 `DICTIONARY_ELEMENT` Struct Reference

### Data Fields

- WORD \* [lhs](#)
- WORD \* [rhs](#)
- int [type](#)
- int [size](#)

### 3.21.1 Detailed Description

Definition at line 1354 of file [structs.h](#).

### 3.21.2 Field Documentation

#### 3.21.2.1 lhs

```
WORD* DICTIONARY_ELEMENT::lhs
```

Definition at line 1355 of file [structs.h](#).

#### 3.21.2.2 rhs

```
WORD* DICTIONARY_ELEMENT::rhs
```

Definition at line 1356 of file [structs.h](#).

#### 3.21.2.3 type

```
int DICTIONARY_ELEMENT::type
```

Definition at line 1357 of file [structs.h](#).

#### 3.21.2.4 size

```
int DICTIONARY_ELEMENT::size
```

Definition at line 1358 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.22 DiStRiBuTe Struct Reference

```
#include <structs.h>
```

### Data Fields

- WORD \* [obj1](#)
- WORD \* [obj2](#)
- WORD \* [out](#)
- WORD [sign](#)
- WORD [n1](#)
- WORD [n2](#)
- WORD [n](#)
- WORD [cycle](#) [MAXMATCH]

### 3.22.1 Detailed Description

The struct DISTRIBUTE is used to help the pattern matcher when matching antisymmetric tensors.

Definition at line 1103 of file [structs.h](#).

### 3.22.2 Field Documentation

#### 3.22.2.1 obj1

```
WORD* DiStRiBuTe::obj1
```

Definition at line 1104 of file [structs.h](#).

#### 3.22.2.2 obj2

```
WORD* DiStRiBuTe::obj2
```

Definition at line 1105 of file [structs.h](#).

#### 3.22.2.3 out

```
WORD* DiStRiBuTe::out
```

Definition at line 1106 of file [structs.h](#).

#### 3.22.2.4 sign

```
WORD DiStRiBuTe::sign
```

Definition at line 1107 of file [structs.h](#).

#### 3.22.2.5 n1

```
WORD DiStRiBuTe::n1
```

Definition at line 1108 of file [structs.h](#).

#### 3.22.2.6 n2

```
WORD DiStRiBuTe::n2
```

Definition at line 1109 of file [structs.h](#).

### 3.22.2.7 n

```
WORD DiStRiBuTe::n
```

Definition at line 1110 of file [structs.h](#).

### 3.22.2.8 cycle

```
WORD DiStRiBuTe::cycle[MAXMATCH]
```

Definition at line 1111 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.23 DNode Struct Reference

### Data Fields

- int [extloop](#)
- int [intrct](#)

### 3.23.1 Detailed Description

Definition at line 192 of file [grccparam.h](#).

### 3.23.2 Field Documentation

#### 3.23.2.1 extloop

```
int DNode::extloop
```

Definition at line 193 of file [grccparam.h](#).

#### 3.23.2.2 intrct

```
int DNode::intrct
```

Definition at line 194 of file [grccparam.h](#).

The documentation for this struct was generated from the following file:

- [grccparam.h](#)



## 3.24 dollar\_buf Struct Reference

### 3.24.1 Detailed Description

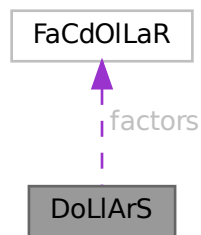
Definition at line 2479 of file [parallel.c](#).

The documentation for this struct was generated from the following file:

- [parallel.c](#)

## 3.25 DoLIaRS Struct Reference

Collaboration diagram for DoLIaRS:



### Data Fields

- WORD \* [where](#)
- [FACDOLLAR](#) \* [factors](#)
- LONG [size](#)
- LONG [name](#)
- WORD [type](#)
- WORD [node](#)
- WORD [index](#)
- WORD [zero](#)
- WORD [numdummies](#)
- WORD [nfactors](#)

### 3.25.1 Detailed Description

Definition at line 530 of file [structs.h](#).

### 3.25.2 Field Documentation

#### 3.25.2.1 where

`WORD* DoLLArS::where`

Definition at line 531 of file [structs.h](#).

#### 3.25.2.2 factors

`FACDOLLAR* DoLLArS::factors`

Definition at line 532 of file [structs.h](#).

#### 3.25.2.3 size

`LONG DoLLArS::size`

Definition at line 537 of file [structs.h](#).

#### 3.25.2.4 name

`LONG DoLLArS::name`

Definition at line 538 of file [structs.h](#).

#### 3.25.2.5 type

`WORD DoLLArS::type`

Definition at line 539 of file [structs.h](#).

#### 3.25.2.6 node

`WORD DoLLArS::node`

Definition at line 540 of file [structs.h](#).

#### 3.25.2.7 index

`WORD DoLLArS::index`

Definition at line 541 of file [structs.h](#).

### 3.25.2.8 zero

```
WORD DoLlArS::zero
```

Definition at line 542 of file [structs.h](#).

### 3.25.2.9 numdummies

```
WORD DoLlArS::numdummies
```

Definition at line 543 of file [structs.h](#).

### 3.25.2.10 nfactors

```
WORD DoLlArS::nfactors
```

Definition at line 544 of file [structs.h](#).

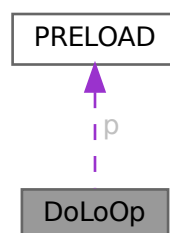
The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.26 DoLoOp Struct Reference

```
#include <structs.h>
```

Collaboration diagram for DoLoOp:



## Data Fields

- [PRELOAD](#) [p](#)
- UBYTE \* [name](#)
- UBYTE \* [vars](#)
- UBYTE \* [contents](#)
- UBYTE \* [dollarname](#)
- LONG [startlinenumber](#)
- LONG [firstnum](#)
- LONG [lastnum](#)
- LONG [incnum](#)
- int [type](#)
- int [NoShowInput](#)
- int [errorsinloop](#)
- int [firstloopcall](#)
- WORD [firstdollar](#)
- WORD [lastdollar](#)
- WORD [incdollar](#)
- WORD [NumPreTypes](#)
- WORD [PrelfLevel](#)
- WORD [PreSwitchLevel](#)

### 3.26.1 Detailed Description

Each preprocessor do loop has a struct of type DOLOOP to keep track of all relevant parameters like where the beginning of the loop is, what the boundaries, increment and value of the loop parameter are, etc. Also we keep the whole loop inside a buffer of type [PRELOAD](#)

Definition at line [897](#) of file [structs.h](#).

### 3.26.2 Field Documentation

#### 3.26.2.1 [p](#)

[PRELOAD](#) [DoLoOp](#)::[p](#)

size, name and buffer

Definition at line [898](#) of file [structs.h](#).

#### 3.26.2.2 [name](#)

UBYTE\* [DoLoOp](#)::[name](#)

pointer into [PRELOAD](#) buffer

Definition at line [899](#) of file [structs.h](#).

### 3.26.2.3 vars

```
UBYTE* DoLoOp::vars
```

Definition at line 900 of file [structs.h](#).

### 3.26.2.4 contents

```
UBYTE* DoLoOp::contents
```

Definition at line 901 of file [structs.h](#).

### 3.26.2.5 dollarname

```
UBYTE* DoLoOp::dollarname
```

For loop over terms in expression. Allocated with Malloc1()

Definition at line 902 of file [structs.h](#).

### 3.26.2.6 startlinenumber

```
LONG DoLoOp::startlinenumber
```

Definition at line 903 of file [structs.h](#).

### 3.26.2.7 firstnum

```
LONG DoLoOp::firstnum
```

Definition at line 904 of file [structs.h](#).

### 3.26.2.8 lastnum

```
LONG DoLoOp::lastnum
```

Definition at line 905 of file [structs.h](#).

### 3.26.2.9 incnum

```
LONG DoLoOp::incnum
```

Definition at line 906 of file [structs.h](#).

### 3.26.2.10 type

```
int DoLoOp::type
```

Definition at line 907 of file [structs.h](#).

### 3.26.2.11 NoShowInput

```
int DoLoOp::NoShowInput
```

Definition at line 908 of file [structs.h](#).

### 3.26.2.12 errorsinloop

```
int DoLoOp::errorsinloop
```

Definition at line 909 of file [structs.h](#).

### 3.26.2.13 firstloopcall

```
int DoLoOp::firstloopcall
```

Definition at line 910 of file [structs.h](#).

### 3.26.2.14 firstdollar

```
WORD DoLoOp::firstdollar
```

Definition at line 911 of file [structs.h](#).

### 3.26.2.15 lastdollar

```
WORD DoLoOp::lastdollar
```

Definition at line 912 of file [structs.h](#).

### 3.26.2.16 incdollar

```
WORD DoLoOp::incdollar
```

Definition at line 913 of file [structs.h](#).

### 3.26.2.17 NumPreTypes

```
WORD DoLoOp::NumPreTypes
```

Definition at line 914 of file [structs.h](#).

### 3.26.2.18 PreIfLevel

WORD DoLoOp::PreIfLevel

Definition at line 915 of file [structs.h](#).

### 3.26.2.19 PreSwitchLevel

WORD DoLoOp::PreSwitchLevel

Definition at line 916 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.27 DuBiOuS Struct Reference

### Data Fields

- LONG [name](#)
- WORD [node](#)
- WORD [dummy](#)

### 3.27.1 Detailed Description

Definition at line 515 of file [structs.h](#).

### 3.27.2 Field Documentation

#### 3.27.2.1 name

LONG DuBiOuS::name

Definition at line 516 of file [structs.h](#).

#### 3.27.2.2 node

WORD DuBiOuS::node

Definition at line 517 of file [structs.h](#).

### 3.27.2.3 dummy

WORD DuBiOuS::dummy

Definition at line 518 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.28 ECand Class Reference

### Public Member Functions

- [ECand](#) (int dt, int nplist, int \*plst)
- void [prECand](#) (const char \*msg)

### Data Fields

- Bool [det](#)
- int [nplist](#)
- int [plist](#) [GRCC\_MAXMPARTICLES2]

### 3.28.1 Detailed Description

Definition at line 1062 of file [grcc.h](#).

### 3.28.2 Constructor & Destructor Documentation

#### 3.28.2.1 ECand()

```
ECand::ECand (  
    int dt,  
    int nplist,  
    int * plst )
```

Definition at line 7325 of file [grcc.cc](#).

#### 3.28.2.2 ~ECand()

```
ECand::~~ECand (  
    void )
```

Definition at line 7345 of file [grcc.cc](#).





## Public Member Functions

- [EEdge](#) ([EGraph](#) \*egrph, int nedges, int nloops)
- void [copy](#) ([EEdge](#) \*ee)
- void [setId](#) ([EGraph](#) \*egrph, const int eid)
- void [print](#) (void)
- void [setType](#) (int typ)
- void [setLMom](#) (int k, int dir)
- void [setEMom](#) (int nedges, int \*extn, int dir)

## Data Fields

- [EGraph](#) \* [egraph](#)
- int [id](#)
- int [ext](#)
- int [ptcl](#)
- int [deleted](#)
- int [nodes](#) [2]
- int [nlegs](#) [2]
- int \* [emom](#)
- int \* [lmom](#)
- int \* [extMom](#)
- Bool [cut](#)
- int [visited](#)
- int [conid](#)
- int [edtype](#)
- int [opicom](#)
- int [dir](#)

### 3.29.1 Detailed Description

Definition at line [538](#) of file [grcc.h](#).

### 3.29.2 Constructor & Destructor Documentation

#### 3.29.2.1 EEdge() [1/2]

```
EEdge::EEdge (
    void )
```

Definition at line [5640](#) of file [grcc.cc](#).

#### 3.29.2.2 EEdge() [2/2]

```
EEdge::EEdge (
    EGraph * egrph,
    int nedges,
    int nloops )
```

Definition at line [5659](#) of file [grcc.cc](#).

### 3.29.2.3 ~EEdge()

```
EEdge::~~EEdge (
    void )
```

Definition at line [5678](#) of file [grcc.cc](#).

## 3.29.3 Member Function Documentation

### 3.29.3.1 copy()

```
void EEdge::copy (
    EEdge * ee )
```

Definition at line [5689](#) of file [grcc.cc](#).

### 3.29.3.2 setId()

```
void EEdge::setId (
    EGraph * egrph,
    const int eid )
```

Definition at line [5701](#) of file [grcc.cc](#).

### 3.29.3.3 print()

```
void EEdge::print (
    void )
```

Definition at line [5737](#) of file [grcc.cc](#).

### 3.29.3.4 setType()

```
void EEdge::setType (
    int typ )
```

Definition at line [5708](#) of file [grcc.cc](#).

### 3.29.3.5 setLMom()

```
void EEdge::setLMom (
    int k,
    int dir )
```

Definition at line [5731](#) of file [grcc.cc](#).

### 3.29.3.6 setEMom()

```
void EEdge::setEMom (
    int nedges,
    int * extn,
    int dir )
```

Definition at line [5719](#) of file [grcc.cc](#).

## 3.29.4 Field Documentation

### 3.29.4.1 egraph

```
EGraph* EEdge::egraph
```

Definition at line [541](#) of file [grcc.h](#).

### 3.29.4.2 id

```
int EEdge::id
```

Definition at line [543](#) of file [grcc.h](#).

### 3.29.4.3 ext

```
int EEdge::ext
```

Definition at line [544](#) of file [grcc.h](#).

### 3.29.4.4 ptcl

```
int EEdge::ptcl
```

Definition at line [545](#) of file [grcc.h](#).

### 3.29.4.5 deleted

```
int EEdge::deleted
```

Definition at line [546](#) of file [grcc.h](#).

### 3.29.4.6 nodes

```
int EEdge::nodes[2]
```

Definition at line [547](#) of file [grcc.h](#).

#### 3.29.4.7 nlegs

```
int EEdge::nlegs[2]
```

Definition at line 548 of file [grcc.h](#).

#### 3.29.4.8 emom

```
int* EEdge::emom
```

Definition at line 551 of file [grcc.h](#).

#### 3.29.4.9 lmom

```
int* EEdge::lmom
```

Definition at line 552 of file [grcc.h](#).

#### 3.29.4.10 extMom

```
int* EEdge::extMom
```

Definition at line 553 of file [grcc.h](#).

#### 3.29.4.11 cut

```
Bool EEdge::cut
```

Definition at line 555 of file [grcc.h](#).

#### 3.29.4.12 visited

```
int EEdge::visited
```

Definition at line 556 of file [grcc.h](#).

#### 3.29.4.13 conid

```
int EEdge::conid
```

Definition at line 557 of file [grcc.h](#).

#### 3.29.4.14 edtype

```
int EEdge::edtype
```

Definition at line 558 of file [grcc.h](#).

#### 3.29.4.15 opicomp

```
int EEdge::opicomp
```

Definition at line 559 of file [grcc.h](#).

#### 3.29.4.16 dir

```
int EEdge::dir
```

Definition at line 560 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.30 EFLine Class Reference

### Public Member Functions

- void [print](#) (const char \*msg)

### Data Fields

- int [elist](#) [GRCC\_MAXNODES]
- FLType [ftype](#)
- int [fkind](#)
- int [nlist](#)

### 3.30.1 Detailed Description

Definition at line 583 of file [grcc.h](#).

## 3.30.2 Constructor & Destructor Documentation

### 3.30.2.1 EFLine()

```
EFLine::EFLine (  
    void )
```

Definition at line 5779 of file [grcc.cc](#).

### 3.30.3 Member Function Documentation

#### 3.30.3.1 print()

```
void EFLine::print (
    const char * msg )
```

Definition at line 5787 of file [grcc.cc](#).

### 3.30.4 Field Documentation

#### 3.30.4.1 elist

```
int EFLine::elist[GRCC_MAXNODES]
```

Definition at line 585 of file [grcc.h](#).

#### 3.30.4.2 ftype

```
FLType EFLine::ftype
```

Definition at line 586 of file [grcc.h](#).

#### 3.30.4.3 fkind

```
int EFLine::fkind
```

Definition at line 587 of file [grcc.h](#).

#### 3.30.4.4 nlist

```
int EFLine::nlist
```

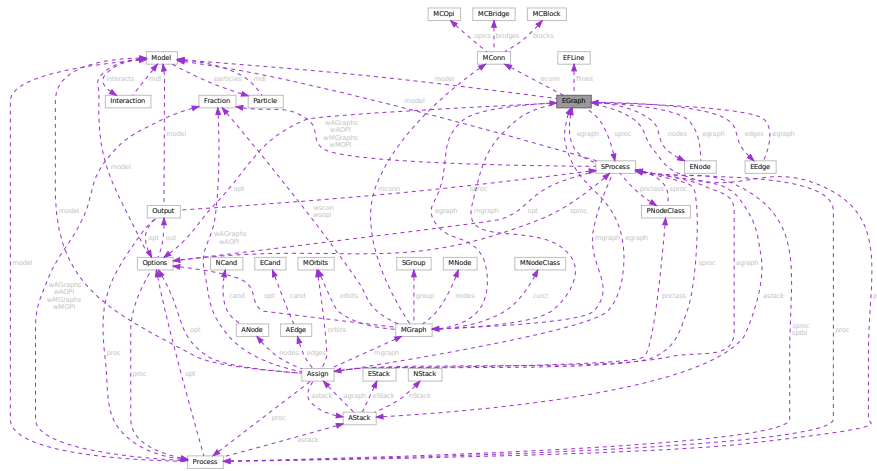
Definition at line 588 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

### 3.31 EGraph Class Reference

Collaboration diagram for EGraph:



#### Public Member Functions

- [EGraph](#) (int nnodes, int nedges, int mxdeg)
- void [copy](#) ([EGraph](#) \*eg)
- void [print](#) (void)
- void [fromDGraph](#) ([DGraph](#) \*dg)
- void [fromMGraph](#) ([MGraph](#) \*mgraph)
- Bool [isOptE](#) (void)
- [ENode](#) \* [setExtern](#) (int n0, int pt, int ndtp)
- Bool [isExternal](#) (int nd)
- Bool [isFermion](#) (int nd)
- void [setExtLoop](#) (int nd, int val)
- void [endSetExtLoop](#) (void)
- int [connComp](#) (void)
- int [connVisit](#) (int nd, int ncc)
- void [biconnE](#) (void)
- void [biinitE](#) (void)
- void [bisearchE](#) (int nd, int \*extlst, int \*intlst, int \*opiext, int \*opiloop)
- int [findRoot](#) (void)
- int [dirEdge](#) (int n, int e)
- void [extMomConsv](#) (void)
- int [cmpMom](#) (int \*lm0, int \*em0, int \*lm1, int \*em1)
- int [groupLMom](#) (int \*grp, int \*ed2gr)
- void [chkMomConsv](#) (void)
- void [prFLines](#) (void)
- void [getFLines](#) (void)
- int [fltrace](#) (int fk, int nd0, int \*fl)
- void [addFLine](#) (const FLType ft, int fk, int nfl, int \*fl)
- int [legParticle](#) (int ed, int lg)



## Data Fields

- [Options](#) \* [opt](#)
- [Model](#) \* [model](#)
- [Process](#) \* [proc](#)
- [SProcess](#) \* [sproc](#)
- [MGraph](#) \* [mgraph](#)
- [MConn](#) \* [econn](#)
- [ENode](#) \*\* [nodes](#)
- [EEdge](#) \*\* [edges](#)
- [BigInt](#) [mld](#)
- [BigInt](#) [ald](#)
- [BigInt](#) [sld](#)
- [BigInt](#) [gSubld](#)
- [Bool](#) [assigned](#)
- [int](#) [fsign](#)
- [BigInt](#) [nsym](#)
- [BigInt](#) [esym](#)
- [BigInt](#) [nsym1](#)
- [BigInt](#) [extperm](#)
- [BigInt](#) [multp](#)
- [int](#) [pld](#)
- [int](#) [sNodes](#)
- [int](#) [sEdges](#)
- [int](#) [sMaxdeg](#)
- [int](#) [sLoops](#)
- [int](#) [nNodes](#)
- [int](#) [nEdges](#)
- [int](#) [nExtern](#)
- [int](#) [maxdeg](#)
- [int](#) [nLoops](#)
- [int](#) [totalc](#)
- [int](#) [nopicomp](#)
- [int](#) [opi2plp](#)
- [int](#) [nopi2p](#)
- [int](#) [nadj2ptv](#)
- [int](#) [DUMMYPADDING](#)
- [int](#) \* [bidef](#)
- [int](#) \* [bilow](#)
- [int](#) \* [extMom](#)
- [int](#) [bconn](#)
- [int](#) [bicount](#)
- [int](#) [loopm](#)
- [int](#) [opiCount](#)
- [EFLine](#) \* [flines](#) [GRCC\_MAXFLINES]
- [int](#) [nflines](#)
- [int](#) [DUMMYPADDING1](#)

### 3.31.1 Detailed Description

Definition at line 595 of file [grcc.h](#).

### 3.31.2 Constructor & Destructor Documentation

#### 3.31.2.1 EGraph()

```
EGraph::EGraph (
    int  nnodes,
    int  nedges,
    int  mxdeg )
```

Definition at line [5812](#) of file [grcc.cc](#).

#### 3.31.2.2 ~EGraph()

```
EGraph::~~EGraph (
    void )
```

Definition at line [5873](#) of file [grcc.cc](#).

### 3.31.3 Member Function Documentation

#### 3.31.3.1 copy()

```
void EGraph::copy (
    EGraph * eg )
```

Definition at line [5904](#) of file [grcc.cc](#).

#### 3.31.3.2 print()

```
void EGraph::print (
    void )
```

Definition at line [6269](#) of file [grcc.cc](#).

#### 3.31.3.3 fromDGraph()

```
void EGraph::fromDGraph (
    DGraph * dg )
```

Definition at line [5969](#) of file [grcc.cc](#).

#### 3.31.3.4 fromMGraph()

```
void EGraph::fromMGraph (
    MGraph * mgraph )
```

Definition at line [6074](#) of file [grcc.cc](#).

### 3.31.3.5 isOptE()

```
Bool EGraph::isOptE (
    void )
```

Definition at line 6440 of file [grcc.cc](#).

### 3.31.3.6 setExtern()

```
ENode * EGraph::setExtern (
    int n0,
    int pt,
    int ndtp )
```

Definition at line 6222 of file [grcc.cc](#).

### 3.31.3.7 isExternal()

```
Bool EGraph::isExternal (
    int nd ) [inline]
```

Definition at line 667 of file [grcc.h](#).

### 3.31.3.8 isFermion()

```
int EGraph::isFermion (
    int nd )
```

Definition at line 7056 of file [grcc.cc](#).

### 3.31.3.9 setExtLoop()

```
void EGraph::setExtLoop (
    int nd,
    int val )
```

Definition at line 5946 of file [grcc.cc](#).

### 3.31.3.10 endSetExtLoop()

```
void EGraph::endSetExtLoop (
    void )
```

Definition at line 5954 of file [grcc.cc](#).

### 3.31.3.11 connComp()

```
int EGraph::connComp (
    void )
```

Definition at line 6586 of file [grcc.cc](#).

### 3.31.3.12 connVisit()

```
int EGraph::connVisit (
    int nd,
    int ncc )
```

Definition at line 6618 of file [grcc.cc](#).

### 3.31.3.13 biconnE()

```
void EGraph::biconnE (
    void )
```

Definition at line 6639 of file [grcc.cc](#).

### 3.31.3.14 biinitE()

```
void EGraph::biinitE (
    void )
```

Definition at line 6708 of file [grcc.cc](#).

### 3.31.3.15 bisearchE()

```
void EGraph::bisearchE (
    int nd,
    int * extlst,
    int * intlst,
    int * opiext,
    int * opiloop )
```

Definition at line 6744 of file [grcc.cc](#).

### 3.31.3.16 findRoot()

```
int EGraph::findRoot (
    void )
```

Definition at line 6537 of file [grcc.cc](#).

**3.31.3.17 dirEdge()**

```
int EGraph::dirEdge (
    int n,
    int e )
```

Definition at line [6347](#) of file [grcc.cc](#).

**3.31.3.18 extMomConsv()**

```
void EGraph::extMomConsv (
    void )
```

Definition at line [6949](#) of file [grcc.cc](#).

**3.31.3.19 cmpMom()**

```
int EGraph::cmpMom (
    int * lm0,
    int * em0,
    int * lm1,
    int * em1 )
```

Definition at line [6357](#) of file [grcc.cc](#).

**3.31.3.20 groupLMom()**

```
int EGraph::groupLMom (
    int * grp,
    int * ed2gr )
```

Definition at line [6407](#) of file [grcc.cc](#).

**3.31.3.21 chkMomConsv()**

```
void EGraph::chkMomConsv (
    void )
```

Definition at line [6977](#) of file [grcc.cc](#).

**3.31.3.22 prFLines()**

```
void EGraph::prFLines (
    void )
```

Definition at line [6334](#) of file [grcc.cc](#).

### 3.31.3.23 getFLines()

```
void EGraph::getFLines (
    void )
```

Definition at line 7154 of file [grcc.cc](#).

### 3.31.3.24 fltrace()

```
int EGraph::fltrace (
    int fk,
    int nd0,
    int * fl )
```

Definition at line 7069 of file [grcc.cc](#).

### 3.31.3.25 addFLine()

```
void EGraph::addFLine (
    const FLType ft,
    int fk,
    int nfl,
    int * fl )
```

Definition at line 7240 of file [grcc.cc](#).

### 3.31.3.26 legParticle()

```
int EGraph::legParticle (
    int ed,
    int lg )
```

Definition at line 7048 of file [grcc.cc](#).

## 3.31.4 Field Documentation

### 3.31.4.1 opt

[Options\\*](#) EGraph::opt

Definition at line 597 of file [grcc.h](#).

### 3.31.4.2 model

[Model\\*](#) EGraph::model

Definition at line 598 of file [grcc.h](#).

#### 3.31.4.3 proc

`Process*` EGraph::proc

Definition at line 599 of file [grcc.h](#).

#### 3.31.4.4 sproc

`SProcess*` EGraph::sproc

Definition at line 600 of file [grcc.h](#).

#### 3.31.4.5 mgraph

`MGraph*` EGraph::mgraph

Definition at line 601 of file [grcc.h](#).

#### 3.31.4.6 econn

`MConn*` EGraph::econn

Definition at line 602 of file [grcc.h](#).

#### 3.31.4.7 nodes

`ENode**` EGraph::nodes

Definition at line 604 of file [grcc.h](#).

#### 3.31.4.8 edges

`EEdge**` EGraph::edges

Definition at line 605 of file [grcc.h](#).

#### 3.31.4.9 mId

`BigInt` EGraph::mId

Definition at line 607 of file [grcc.h](#).

#### 3.31.4.10 aId

`BigInt` EGraph::aId

Definition at line 608 of file [grcc.h](#).

**3.31.4.11 sId**

```
BigInt EGraph::sId
```

Definition at line 609 of file [grcc.h](#).

**3.31.4.12 gSubId**

```
BigInt EGraph::gSubId
```

Definition at line 610 of file [grcc.h](#).

**3.31.4.13 assigned**

```
Bool EGraph::assigned
```

Definition at line 611 of file [grcc.h](#).

**3.31.4.14 fsign**

```
int EGraph::fsign
```

Definition at line 613 of file [grcc.h](#).

**3.31.4.15 nsym**

```
BigInt EGraph::nsym
```

Definition at line 614 of file [grcc.h](#).

**3.31.4.16 esym**

```
BigInt EGraph::esym
```

Definition at line 614 of file [grcc.h](#).

**3.31.4.17 nsym1**

```
BigInt EGraph::nsym1
```

Definition at line 615 of file [grcc.h](#).

**3.31.4.18 extperm**

```
BigInt EGraph::extperm
```

Definition at line 616 of file [grcc.h](#).



#### 3.31.4.19 multp

```
BigInt EGraph::multp
```

Definition at line 617 of file [grcc.h](#).

#### 3.31.4.20 pld

```
int EGraph::pId
```

Definition at line 619 of file [grcc.h](#).

#### 3.31.4.21 sNodes

```
int EGraph::sNodes
```

Definition at line 621 of file [grcc.h](#).

#### 3.31.4.22 sEdges

```
int EGraph::sEdges
```

Definition at line 622 of file [grcc.h](#).

#### 3.31.4.23 sMaxdeg

```
int EGraph::sMaxdeg
```

Definition at line 623 of file [grcc.h](#).

#### 3.31.4.24 sLoops

```
int EGraph::sLoops
```

Definition at line 624 of file [grcc.h](#).

#### 3.31.4.25 nNodes

```
int EGraph::nNodes
```

Definition at line 626 of file [grcc.h](#).

#### 3.31.4.26 nEdges

```
int EGraph::nEdges
```

Definition at line 627 of file [grcc.h](#).

#### 3.31.4.27 nExtern

```
int EGraph::nExtern
```

Definition at line 628 of file [grcc.h](#).

#### 3.31.4.28 maxdeg

```
int EGraph::maxdeg
```

Definition at line 629 of file [grcc.h](#).

#### 3.31.4.29 nLoops

```
int EGraph::nLoops
```

Definition at line 630 of file [grcc.h](#).

#### 3.31.4.30 totalc

```
int EGraph::totalc
```

Definition at line 631 of file [grcc.h](#).

#### 3.31.4.31 nopicomp

```
int EGraph::nopicomp
```

Definition at line 634 of file [grcc.h](#).

#### 3.31.4.32 opi2plp

```
int EGraph::opi2plp
```

Definition at line 635 of file [grcc.h](#).

#### 3.31.4.33 nopi2p

```
int EGraph::nopi2p
```

Definition at line 636 of file [grcc.h](#).

#### 3.31.4.34 nadj2ptv

```
int EGraph::nadj2ptv
```

Definition at line 637 of file [grcc.h](#).

#### 3.31.4.35 DUMMYPADDING

```
int EGraph::DUMMYPADDING
```

Definition at line 639 of file [grcc.h](#).

#### 3.31.4.36 bidef

```
int* EGraph::bidef
```

Definition at line 641 of file [grcc.h](#).

#### 3.31.4.37 bilow

```
int* EGraph::bilow
```

Definition at line 642 of file [grcc.h](#).

#### 3.31.4.38 extMom

```
int* EGraph::extMom
```

Definition at line 643 of file [grcc.h](#).

#### 3.31.4.39 bconn

```
int EGraph::bconn
```

Definition at line 644 of file [grcc.h](#).

#### 3.31.4.40 bicount

```
int EGraph::bicount
```

Definition at line 645 of file [grcc.h](#).

#### 3.31.4.41 loopm

```
int EGraph::loopm
```

Definition at line 646 of file [grcc.h](#).

#### 3.31.4.42 opiCount

```
int EGraph::opiCount
```

Definition at line 647 of file [grcc.h](#).



## Data Fields

- [EGraph](#) \* [egraph](#)
- int \* [edges](#)
- int [id](#)
- int [maxdeg](#)
- int [deg](#)
- int [extloop](#)
- int [ndtype](#)
- int [intrct](#)
- int \* [klow](#)
- int [visited](#)
- int [DUMMYPADDING](#)

### 3.32.1 Detailed Description

Definition at line [504](#) of file [grcc.h](#).

### 3.32.2 Constructor & Destructor Documentation

#### 3.32.2.1 ENode() [1/2]

```
ENode::ENode (  
    void )
```

Definition at line [5526](#) of file [grcc.cc](#).

#### 3.32.2.2 ENode() [2/2]

```
ENode::ENode (  
    EGraph * egrph,  
    int loops,  
    int sdeg )
```

Definition at line [5542](#) of file [grcc.cc](#).

#### 3.32.2.3 ~ENode()

```
ENode::~ENode (  
    void )
```

Definition at line [5562](#) of file [grcc.cc](#).

### 3.32.3 Member Function Documentation

#### 3.32.3.1 `initAss()`

```
void ENode::initAss (
    EGraph * egrph,
    int nid,
    int sdg )
```

Definition at line 5586 of file [grcc.cc](#).

#### 3.32.3.2 `setId()`

```
void ENode::setId (
    EGraph * egrph,
    const int nid )
```

Definition at line 5594 of file [grcc.cc](#).

#### 3.32.3.3 `copy()`

```
void ENode::copy (
    ENode * en )
```

Definition at line 5573 of file [grcc.cc](#).

#### 3.32.3.4 `setExtern()`

```
void ENode::setExtern (
    int typ,
    int pt )
```

Definition at line 5601 of file [grcc.cc](#).

#### 3.32.3.5 `setType()`

```
void ENode::setType (
    int typ )
```

Definition at line 5608 of file [grcc.cc](#).

#### 3.32.3.6 `print()`

```
void ENode::print (
    void )
```

Definition at line 5621 of file [grcc.cc](#).

### 3.32.4 Field Documentation

#### 3.32.4.1 egraph

`EGraph*` `ENode::egraph`

Definition at line 506 of file [grcc.h](#).

#### 3.32.4.2 edges

`int*` `ENode::edges`

Definition at line 507 of file [grcc.h](#).

#### 3.32.4.3 id

`int` `ENode::id`

Definition at line 510 of file [grcc.h](#).

#### 3.32.4.4 maxdeg

`int` `ENode::maxdeg`

Definition at line 511 of file [grcc.h](#).

#### 3.32.4.5 deg

`int` `ENode::deg`

Definition at line 512 of file [grcc.h](#).

#### 3.32.4.6 extloop

`int` `ENode::extloop`

Definition at line 513 of file [grcc.h](#).

#### 3.32.4.7 ndtype

`int` `ENode::ndtype`

Definition at line 514 of file [grcc.h](#).

#### 3.32.4.8 intrct

```
int ENode::intrct
```

Definition at line 515 of file [grcc.h](#).

#### 3.32.4.9 klow

```
int* ENode::klow
```

Definition at line 518 of file [grcc.h](#).

#### 3.32.4.10 visited

```
int ENode::visited
```

Definition at line 519 of file [grcc.h](#).

#### 3.32.4.11 DUMMYPADDING

```
int ENode::DUMMYPADDING
```

Definition at line 521 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.33 EStack Class Reference

### Public Member Functions

- void [print](#) (const char \*msg)

### Data Fields

- int [edgen](#)
- int [det](#)
- int [nplist](#)
- int [plist](#) [GRCC\_MAXMPARTICLES2]

### 3.33.1 Detailed Description

Definition at line 1288 of file [grcc.h](#).



## 3.33.2 Constructor & Destructor Documentation

### 3.33.2.1 EStack()

```
EStack::EStack ( ) [inline]
```

Definition at line 1297 of file [grcc.h](#).

### 3.33.2.2 ~EStack()

```
EStack::~~EStack ( ) [inline]
```

Definition at line 1298 of file [grcc.h](#).

## 3.33.3 Member Function Documentation

### 3.33.3.1 print()

```
void EStack::print (
    const char * msg )
```

Definition at line 9463 of file [grcc.cc](#).

## 3.33.4 Field Documentation

### 3.33.4.1 edgen

```
int EStack::edgen
```

Definition at line 1292 of file [grcc.h](#).

### 3.33.4.2 det

```
int EStack::det
```

Definition at line 1293 of file [grcc.h](#).

### 3.33.4.3 nplist

```
int EStack::nplist
```

Definition at line 1294 of file [grcc.h](#).

### 3.33.4.4 plist

```
int EStack::plist[GRCC_MAXMPARTICLES2]
```

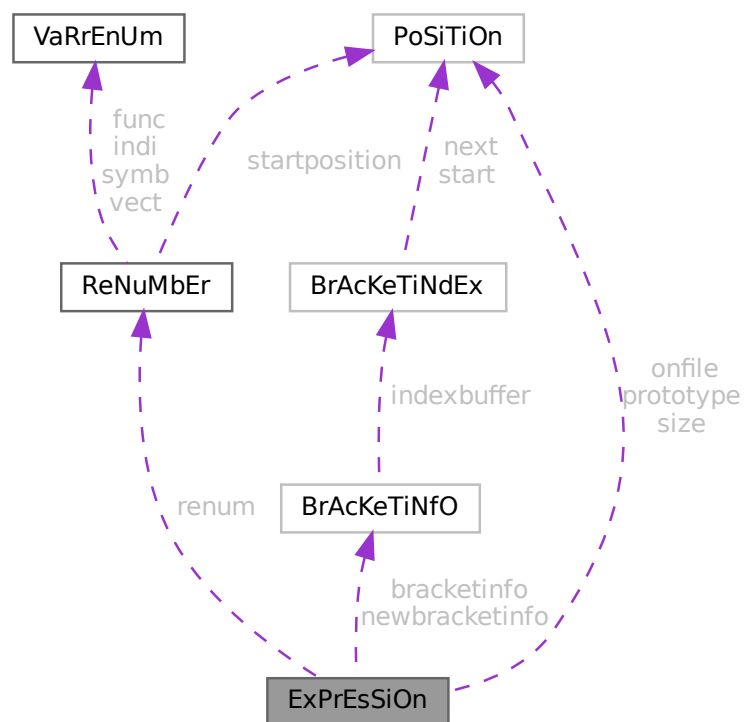
Definition at line 1295 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.34 ExPrEsSiOn Struct Reference

Collaboration diagram for ExPrEsSiOn:



### Public Member Functions

- **PADPOSITION** (5, 2, 0, 13, 0)

## Data Fields

- [POSITION onfile](#)
- [POSITION prototype](#)
- [POSITION size](#)
- [RENUMBER renum](#)
- [BRACKETINFO \\* bracketinfo](#)
- [BRACKETINFO \\* newbracketinfo](#)
- [WORD \\* renumlists](#)
- [WORD \\* inmem](#)
- [LONG counter](#)
- [LONG name](#)
- [WORD hidelevel](#)
- [WORD vflags](#)
- [WORD printflag](#)
- [WORD status](#)
- [WORD replace](#)
- [WORD node](#)
- [WORD whichbuffer](#)
- [WORD namesize](#)
- [WORD compression](#)
- [WORD numdummies](#)
- [WORD numfactors](#)
- [WORD sizeprototype](#)
- [WORD uflags](#)

### 3.34.1 Detailed Description

Definition at line [391](#) of file [structs.h](#).

### 3.34.2 Field Documentation

#### 3.34.2.1 onfile

[POSITION](#) ExPrEsSiOn::onfile

Definition at line [392](#) of file [structs.h](#).

#### 3.34.2.2 prototype

[POSITION](#) ExPrEsSiOn::prototype

Definition at line [393](#) of file [structs.h](#).

#### 3.34.2.3 size

[POSITION](#) ExPrEsSiOn::size

Definition at line [394](#) of file [structs.h](#).

#### 3.34.2.4 renum

`RENUMBER ExPrEsSiOn::renum`

Definition at line 395 of file [structs.h](#).

#### 3.34.2.5 bracketinfo

`BRACKETINFO* ExPrEsSiOn::bracketinfo`

Definition at line 396 of file [structs.h](#).

#### 3.34.2.6 newbracketinfo

`BRACKETINFO* ExPrEsSiOn::newbracketinfo`

Definition at line 397 of file [structs.h](#).

#### 3.34.2.7 renumlists

`WORD* ExPrEsSiOn::renumlists`

Allocated only for threaded version if variables exist, else points to AN.dummyrenumlist

Definition at line 398 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

#### 3.34.2.8 inmem

`WORD* ExPrEsSiOn::inmem`

Definition at line 400 of file [structs.h](#).

#### 3.34.2.9 counter

`LONG ExPrEsSiOn::counter`

Definition at line 401 of file [structs.h](#).

#### 3.34.2.10 name

`LONG ExPrEsSiOn::name`

Definition at line 402 of file [structs.h](#).

#### 3.34.2.11 hidelevel

WORD ExPrEsSiOn::hidelevel

Definition at line 403 of file [structs.h](#).

#### 3.34.2.12 vflags

WORD ExPrEsSiOn::vflags

Definition at line 404 of file [structs.h](#).

#### 3.34.2.13 printflag

WORD ExPrEsSiOn::printflag

Definition at line 405 of file [structs.h](#).

#### 3.34.2.14 status

WORD ExPrEsSiOn::status

Definition at line 406 of file [structs.h](#).

#### 3.34.2.15 replace

WORD ExPrEsSiOn::replace

Definition at line 407 of file [structs.h](#).

#### 3.34.2.16 node

WORD ExPrEsSiOn::node

Definition at line 408 of file [structs.h](#).

#### 3.34.2.17 whichbuffer

WORD ExPrEsSiOn::whichbuffer

Definition at line 409 of file [structs.h](#).

#### 3.34.2.18 namesize

WORD ExPrEsSiOn::namesize

Definition at line 410 of file [structs.h](#).

### 3.34.2.19 compression

WORD ExPrEsSiOn::compression

Definition at line 411 of file [structs.h](#).

### 3.34.2.20 numdummies

WORD ExPrEsSiOn::numdummies

Definition at line 412 of file [structs.h](#).

### 3.34.2.21 numfactors

WORD ExPrEsSiOn::numfactors

Definition at line 413 of file [structs.h](#).

### 3.34.2.22 sizeprototype

WORD ExPrEsSiOn::sizeprototype

Definition at line 414 of file [structs.h](#).

### 3.34.2.23 uflags

WORD ExPrEsSiOn::uflags

Definition at line 415 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.35 FaCdOILaR Struct Reference

### Data Fields

- WORD \* [where](#)
- LONG [size](#)
- WORD [type](#)
- WORD [value](#)

### 3.35.1 Detailed Description

Definition at line 522 of file [structs.h](#).

### 3.35.2 Field Documentation

#### 3.35.2.1 where

WORD\* FaCdOlLaR::where

Definition at line 523 of file [structs.h](#).

#### 3.35.2.2 size

LONG FaCdOlLaR::size

Definition at line 524 of file [structs.h](#).

#### 3.35.2.3 type

WORD FaCdOlLaR::type

Definition at line 525 of file [structs.h](#).

#### 3.35.2.4 value

WORD FaCdOlLaR::value

Definition at line 526 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.36 factorized\_poly Class Reference

### Public Member Functions

- void [add\\_factor](#) (const [poly](#) &f, int p=1)
- const std::string [tostring](#) () const

### Data Fields

- std::vector< [poly](#) > [factor](#)
- std::vector< int > [power](#)

### Friends

- std::ostream & [operator](#)<< (std::ostream &out, const [poly](#) &p)

### 3.36.1 Detailed Description

Definition at line 54 of file [polyfact.h](#).

### 3.36.2 Member Function Documentation

#### 3.36.2.1 `add_factor()`

```
void factorized_poly::add_factor (
    const poly & f,
    int p = 1 )
```

Definition at line 125 of file [polyfact.cc](#).

#### 3.36.2.2 `tostring()`

```
const string factorized_poly::tostring ( ) const
```

Definition at line 59 of file [polyfact.cc](#).

### 3.36.3 Field Documentation

#### 3.36.3.1 `factor`

```
std::vector<poly> factorized_poly::factor
```

Definition at line 59 of file [polyfact.h](#).

#### 3.36.3.2 `power`

```
std::vector<int> factorized_poly::power
```

Definition at line 60 of file [polyfact.h](#).

The documentation for this class was generated from the following files:

- [polyfact.h](#)
- [polyfact.cc](#)

## 3.37 FGInput Struct Reference

### Data Fields

- long [ninitl](#)
- const char \* [initln](#) [GRCC\_MAXNODES]
- int [initlc](#) [GRCC\_MAXNODES]
- long [nfinal](#)
- const char \* [finaln](#) [GRCC\_MAXNODES]
- int [finalc](#) [GRCC\_MAXNODES]
- int [coupl](#) [GRCC\_MAXNCPLG]



### 3.37.1 Detailed Description

Definition at line 165 of file [grccparam.h](#).

### 3.37.2 Field Documentation

#### 3.37.2.1 ninitl

```
long FGInput::ninitl
```

Definition at line 166 of file [grccparam.h](#).

#### 3.37.2.2 initln

```
const char* FGInput::initln[GRCC_MAXNODES]
```

Definition at line 167 of file [grccparam.h](#).

#### 3.37.2.3 initlc

```
int FGInput::initlc[GRCC_MAXNODES]
```

Definition at line 168 of file [grccparam.h](#).

#### 3.37.2.4 nfinal

```
long FGInput::nfinal
```

Definition at line 169 of file [grccparam.h](#).

#### 3.37.2.5 finaln

```
const char* FGInput::finaln[GRCC_MAXNODES]
```

Definition at line 170 of file [grccparam.h](#).

#### 3.37.2.6 finalc

```
int FGInput::finalc[GRCC_MAXNODES]
```

Definition at line 171 of file [grccparam.h](#).

### 3.37.2.7 coupl

```
int FGInput::coupl[GRCC_MAXNCPLG]
```

Definition at line 172 of file [grccparam.h](#).

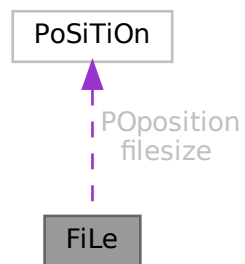
The documentation for this struct was generated from the following file:

- [grccparam.h](#)

## 3.38 FiLe Struct Reference

```
#include <structs.h>
```

Collaboration diagram for FiLe:



### Public Member Functions

- **PADPOSITION** (5, 3, 2, 0, 0)

### Data Fields

- [POSITION](#) [PPosition](#)
- [POSITION](#) [filesize](#)
- WORD \* [PObuffer](#)
- WORD \* [POstop](#)
- WORD \* [POfill](#)
- WORD \* [POfull](#)
- char \* [name](#)
- ULONG [numblocks](#)
- ULONG [inbuffer](#)
- LONG [POsize](#)
- int [handle](#)
- int [active](#)

### 3.38.1 Detailed Description

The type FILEHANDLE is the struct that controls all relevant information of a file, whether it is open or not. The file may even not yet exist. There is a system of caches (PObuffer) and as long as the information to be written still fits inside the cache the file may never be created. There are variables that can store information about different types of files, like scratch files or sort files. Depending on what is available in the system we may also have information about gzip compression (currently sort file only) or locks (TFORM).

Definition at line 682 of file [structs.h](#).

### 3.38.2 Field Documentation

#### 3.38.2.1 PPosition

`POSITION FiLe::PPosition`

Definition at line 683 of file [structs.h](#).

#### 3.38.2.2 filesize

`POSITION FiLe::filesize`

Definition at line 684 of file [structs.h](#).

#### 3.38.2.3 PObuffer

`WORD* FiLe::PObuffer`

Definition at line 685 of file [structs.h](#).

#### 3.38.2.4 PStop

`WORD* FiLe::PStop`

Definition at line 686 of file [structs.h](#).

#### 3.38.2.5 POfill

`WORD* FiLe::POfill`

Definition at line 687 of file [structs.h](#).

#### 3.38.2.6 POfull

`WORD* FiLe::POfull`

Definition at line 688 of file [structs.h](#).

### 3.38.2.7 name

```
char* FiLe::name
```

Definition at line 695 of file [structs.h](#).

### 3.38.2.8 numblocks

```
ULONG FiLe::numblocks
```

Definition at line 700 of file [structs.h](#).

### 3.38.2.9 inbuffer

```
ULONG FiLe::inbuffer
```

Definition at line 701 of file [structs.h](#).

### 3.38.2.10 PSize

```
LONG FiLe::PSize
```

Definition at line 702 of file [structs.h](#).

### 3.38.2.11 handle

```
int FiLe::handle
```

Our own handle. Equal -1 if no file exists.

Definition at line 710 of file [structs.h](#).

Referenced by [DoOnePow\(\)](#), [FlushOut\(\)](#), [GetFirstTerm\(\)](#), [MergePatches\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [PutIn\(\)](#), [PutOut\(\)](#), [Sflush\(\)](#), and [StageSort\(\)](#).

### 3.38.2.12 active

```
int FiLe::active
```

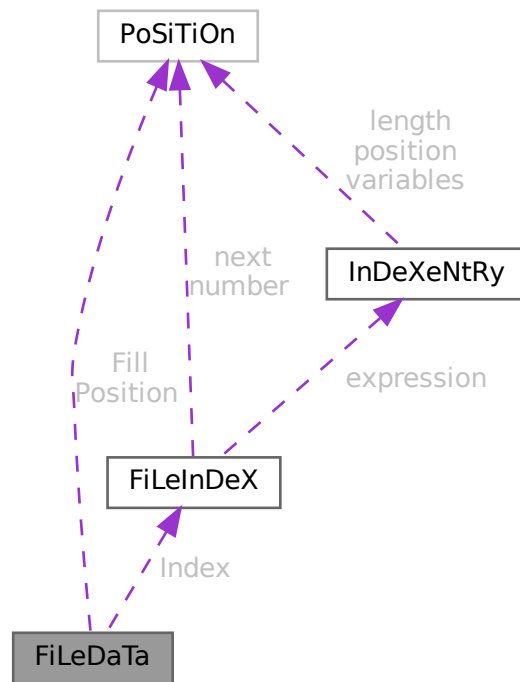
Definition at line 711 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.39 FiLeDaTa Struct Reference

Collaboration diagram for FiLeDaTa:



### Public Member Functions

- **PADPOSITION** (0, 0, 0, 2, 0)

### Data Fields

- [FILEINDEX](#) Index
- [POSITION](#) Fill
- [POSITION](#) Position
- [WORD](#) Handle
- [WORD](#) dirtyflag

### 3.39.1 Detailed Description

Definition at line 151 of file [structs.h](#).

### 3.39.2 Field Documentation

#### 3.39.2.1 Index

`FILEINDEX` `FileDaTa::Index`

Definition at line 152 of file [structs.h](#).

#### 3.39.2.2 Fill

`POSITION` `FileDaTa::Fill`

Definition at line 153 of file [structs.h](#).

#### 3.39.2.3 Position

`POSITION` `FileDaTa::Position`

Definition at line 154 of file [structs.h](#).

#### 3.39.2.4 Handle

`WORD` `FileDaTa::Handle`

Definition at line 155 of file [structs.h](#).

#### 3.39.2.5 dirtyflag

`WORD` `FileDaTa::dirtyflag`

Definition at line 156 of file [structs.h](#).

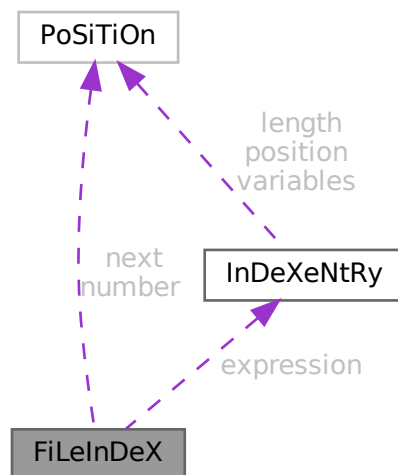
The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.40 FiLeInDeX Struct Reference

```
#include <structs.h>
```

Collaboration diagram for FiLeInDeX:



### Data Fields

- [POSITION](#) next
- [POSITION](#) number
- [INDEXENTRY](#) expression [[INFILEINDEX](#)]
- [SBYTE](#) empty [[EMPTYININDEX](#)]

### 3.40.1 Detailed Description

Defines the structure of a file index in store-files and save-files.

It contains several entries (see struct [InDeXeNtRy](#)) up to a maximum of [INFILEINDEX](#).

The variable number has been made of type [POSITION](#) to avoid padding problems with some types of computers/↔ OS and keep system independence of the .sav files.

This struct is always 512 bytes long.

Definition at line [138](#) of file [structs.h](#).

### 3.40.2 Field Documentation

#### 3.40.2.1 next

`POSITION FileInDeX::next`

Position of next FILEINDEX if any

Definition at line 139 of file [structs.h](#).

Referenced by [ReadSaveIndex\(\)](#).

#### 3.40.2.2 number

`POSITION FileInDeX::number`

Number of used entries in this index

Definition at line 140 of file [structs.h](#).

Referenced by [ReadSaveIndex\(\)](#).

#### 3.40.2.3 expression

`INDEXENTRY FileInDeX::expression[INFILEINDEX]`

File index entries

Definition at line 141 of file [structs.h](#).

#### 3.40.2.4 empty

`SBYTE FileInDeX::empty[EMPTYININDEX]`

Padding to 512 bytes

Definition at line 142 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.41 FixedGlobals Struct Reference

```
#include <structs.h>
```



## Data Fields

- WCN [Operation](#) [8]
- WCN2 [OperaFind](#) [6]
- char \* [VarType](#) [10]
- char \* [ExprStat](#) [21]
- char \* [FunNam](#) [2]
- char \* [swmes](#) [3]
- char \* [fname](#)
- char \* [fname2](#)
- UBYTE \* [s\\_one](#)
- WORD [fnamebase](#)
- WORD [fname2base](#)
- WORD [fnamesize](#)
- WORD [fname2size](#)
- UINT [cTable](#) [256]

### 3.41.1 Detailed Description

The FIXEDGLOBALS struct is an anachronism. It started as the struct with global variables that needed initialization. It contains the elements `Operation` and `OperaFind` which define a very early way of automatically jumping to the proper operation. We find the results of it in parts of the file [opera.c](#). Later operations were treated differently in a more transparent way. We never changed the existing code. The most important part is currently the `cTable` which is used intensively in the compiler.

Definition at line [2674](#) of file [structs.h](#).

### 3.41.2 Field Documentation

#### 3.41.2.1 Operation

```
WCN FixedGlobals::Operation[8]
```

Definition at line [2675](#) of file [structs.h](#).

#### 3.41.2.2 OperaFind

```
WCN2 FixedGlobals::OperaFind[6]
```

Definition at line [2676](#) of file [structs.h](#).

#### 3.41.2.3 VarType

```
char* FixedGlobals::VarType[10]
```

Definition at line [2677](#) of file [structs.h](#).

#### 3.41.2.4 ExprStat

```
char* FixedGlobals::ExprStat[21]
```

Definition at line 2678 of file [structs.h](#).

#### 3.41.2.5 FunNam

```
char* FixedGlobals::FunNam[2]
```

Definition at line 2679 of file [structs.h](#).

#### 3.41.2.6 swmes

```
char* FixedGlobals::swmes[3]
```

Definition at line 2680 of file [structs.h](#).

#### 3.41.2.7 fname

```
char* FixedGlobals::fname
```

Definition at line 2681 of file [structs.h](#).

#### 3.41.2.8 fname2

```
char* FixedGlobals::fname2
```

Definition at line 2682 of file [structs.h](#).

#### 3.41.2.9 s\_one

```
UBYTE* FixedGlobals::s_one
```

Definition at line 2683 of file [structs.h](#).

#### 3.41.2.10 fnamebase

```
WORD FixedGlobals::fnamebase
```

Definition at line 2684 of file [structs.h](#).

#### 3.41.2.11 fname2base

```
WORD FixedGlobals::fname2base
```

Definition at line 2685 of file [structs.h](#).

#### 3.41.2.12 fnamesize

WORD FixedGlobals::fnamesize

Definition at line 2686 of file [structs.h](#).

#### 3.41.2.13 fname2size

WORD FixedGlobals::fname2size

Definition at line 2687 of file [structs.h](#).

#### 3.41.2.14 cTable

UINT FixedGlobals::cTable[256]

Definition at line 2688 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.42 fixedset Struct Reference

### Data Fields

- char \* [name](#)
- char \* [description](#)
- int [type](#)
- int [dimension](#)

### 3.42.1 Detailed Description

Definition at line 571 of file [structs.h](#).

### 3.42.2 Field Documentation

#### 3.42.2.1 name

char\* fixedset::name

Definition at line 572 of file [structs.h](#).

#### 3.42.2.2 description

```
char* fixedset::description
```

Definition at line 573 of file [structs.h](#).

#### 3.42.2.3 type

```
int fixedset::type
```

Definition at line 574 of file [structs.h](#).

#### 3.42.2.4 dimension

```
int fixedset::dimension
```

Definition at line 575 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

### 3.43 flint::fmpz Class Reference

#### Public Member Functions

- void [print](#) (const string &text)

#### Data Fields

- [fmpz\\_t](#) [d](#)

#### 3.43.1 Detailed Description

Definition at line 54 of file [flintinterface.h](#).

#### 3.43.2 Constructor & Destructor Documentation

##### 3.43.2.1 fmpz()

```
flint::fmpz::fmpz ( ) [inline]
```

Definition at line 57 of file [flintinterface.h](#).

### 3.43.2.2 ~fmpz()

```
flint::fmpz::~~fmpz ( ) [inline]
```

Definition at line 58 of file [flintinterface.h](#).

## 3.43.3 Member Function Documentation

### 3.43.3.1 print()

```
void flint::fmpz::print (
    const string & text ) [inline]
```

Definition at line 59 of file [flintinterface.h](#).

## 3.43.4 Field Documentation

### 3.43.4.1 d

```
fmpz_t flint::fmpz::d
```

Definition at line 56 of file [flintinterface.h](#).

The documentation for this class was generated from the following file:

- [flintinterface.h](#)

## 3.44 Fraction Class Reference

### Public Member Functions

- [Fraction](#) (BigInt n, BigInt d)
- void [print](#) (const char \*msg)
- void [setValue](#) (BigInt n, BigInt d)
- void [setValue](#) ([Fraction](#) &f)
- void [add](#) (BigInt n, BigInt d)
- void [add](#) ([Fraction](#) f)
- void [sub](#) ([Fraction](#) f)
- BigInt [gcd](#) (BigInt n0, BigInt n1)
- void [normal](#) (void)
- Bool [isEq](#) ([Fraction](#) f)

### Data Fields

- BigInt [num](#)
- BigInt [den](#)
- double [ratio](#)

### 3.44.1 Detailed Description

Definition at line [182](#) of file [grcc.h](#).

### 3.44.2 Constructor & Destructor Documentation

#### 3.44.2.1 Fraction() [1/2]

```
Fraction::Fraction ( ) [inline]
```

Definition at line [187](#) of file [grcc.h](#).

#### 3.44.2.2 Fraction() [2/2]

```
Fraction::Fraction (
    BigInt n,
    BigInt d )
```

Definition at line [9680](#) of file [grcc.cc](#).

### 3.44.3 Member Function Documentation

#### 3.44.3.1 print()

```
void Fraction::print (
    const char * msg )
```

Definition at line [9691](#) of file [grcc.cc](#).

#### 3.44.3.2 setValue() [1/2]

```
void Fraction::setValue (
    BigInt n,
    BigInt d )
```

Definition at line [9703](#) of file [grcc.cc](#).

#### 3.44.3.3 setValue() [2/2]

```
void Fraction::setValue (
    Fraction & f )
```

Definition at line [9711](#) of file [grcc.cc](#).

**3.44.3.4 add()** [1/2]

```
void Fraction::add (
    BigInt n,
    BigInt d )
```

Definition at line 9756 of file [grcc.cc](#).

**3.44.3.5 add()** [2/2]

```
void Fraction::add (
    Fraction f )
```

Definition at line 9779 of file [grcc.cc](#).

**3.44.3.6 sub()**

```
void Fraction::sub (
    Fraction f )
```

Definition at line 9788 of file [grcc.cc](#).

**3.44.3.7 gcd()**

```
BigInt Fraction::gcd (
    BigInt n0,
    BigInt n1 )
```

Definition at line 9719 of file [grcc.cc](#).

**3.44.3.8 normal()**

```
void Fraction::normal (
    void )
```

Definition at line 9739 of file [grcc.cc](#).

**3.44.3.9 isEq()**

```
Bool Fraction::isEq (
    Fraction f )
```

Definition at line 9797 of file [grcc.cc](#).

### 3.44.4 Field Documentation

#### 3.44.4.1 num

```
BigInt Fraction::num
```

Definition at line 184 of file [grcc.h](#).

#### 3.44.4.2 den

```
BigInt Fraction::den
```

Definition at line 184 of file [grcc.h](#).

#### 3.44.4.3 ratio

```
double Fraction::ratio
```

Definition at line 185 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.45 FUN\_INFO Struct Reference

```
#include <structs.h>
```

### Data Fields

- WORD \* [location](#)
- int [numargs](#)
- int [numfunnies](#)
- int [numwildcards](#)
- int [symmet](#)
- int [tensor](#)
- int [commute](#)

### 3.45.1 Detailed Description

The struct [FUN\\_INFO](#) is used for information about functions in the file [smart.c](#) which is supposed to intelligently look for patterns in complicated wildcard situations involving symmetric functions.

Definition at line 608 of file [structs.h](#).



## 3.45.2 Field Documentation

### 3.45.2.1 location

```
WORD* FUN_INFO::location
```

Definition at line 609 of file [structs.h](#).

### 3.45.2.2 numargs

```
int FUN_INFO::numargs
```

Definition at line 610 of file [structs.h](#).

### 3.45.2.3 numfunnies

```
int FUN_INFO::numfunnies
```

Definition at line 611 of file [structs.h](#).

### 3.45.2.4 numwildcards

```
int FUN_INFO::numwildcards
```

Definition at line 612 of file [structs.h](#).

### 3.45.2.5 symmet

```
int FUN_INFO::symmet
```

Definition at line 613 of file [structs.h](#).

### 3.45.2.6 tensor

```
int FUN_INFO::tensor
```

Definition at line 614 of file [structs.h](#).

### 3.45.2.7 commute

```
int FUN_INFO::commute
```

Definition at line 615 of file [structs.h](#).

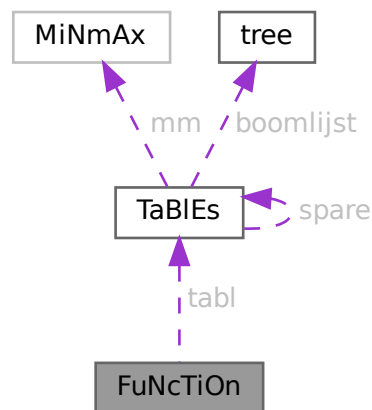
The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.46 FuNcTiOn Struct Reference

```
#include <structs.h>
```

Collaboration diagram for FuNcTiOn:



### Data Fields

- [TABLES](#) `tabl`
- LONG `symminfo`
- LONG `name`
- WORD `commute`
- WORD `complex`
- WORD `number`
- WORD `flags`
- WORD `spec`
- WORD `symmetric`
- WORD `node`
- WORD `namesize`
- WORD `dimension`
- WORD `maxnumargs`
- WORD `minnumargs`

### 3.46.1 Detailed Description

Contains all information about a function. Also used for tables. It is used in the [LIST](#) elements of AC.

Definition at line [477](#) of file [structs.h](#).

## 3.46.2 Field Documentation

### 3.46.2.1 `tabl`

`TABLES FuNcTiOn::tabl`

Used if redefined as table. != 0 if function is a table

Definition at line 478 of file [structs.h](#).

### 3.46.2.2 `symminfo`

`LONG FuNcTiOn::symminfo`

Info regarding symm properties offset in buffer

Definition at line 479 of file [structs.h](#).

### 3.46.2.3 `name`

`LONG FuNcTiOn::name`

Location in namebuffer of [NAMETREE](#)

Definition at line 480 of file [structs.h](#).

Referenced by [StartVariables\(\)](#).

### 3.46.2.4 `commute`

`WORD FuNcTiOn::commute`

Commutation properties

Definition at line 481 of file [structs.h](#).

### 3.46.2.5 `complex`

`WORD FuNcTiOn::complex`

Properties under complex conjugation

Definition at line 482 of file [structs.h](#).

### 3.46.2.6 number

WORD FuNcTiOn::number

Number when stored in file

Definition at line [483](#) of file [structs.h](#).

Referenced by [ReadSaveIndex\(\)](#).

### 3.46.2.7 flags

WORD FuNcTiOn::flags

Used to indicate usage when storing

Definition at line [484](#) of file [structs.h](#).

### 3.46.2.8 spec

WORD FuNcTiOn::spec

Regular, Tensor, etc. See [FunSpecs](#).

Definition at line [485](#) of file [structs.h](#).

Referenced by [ReadSaveVariables\(\)](#).

### 3.46.2.9 symmetric

WORD FuNcTiOn::symmetric

0 if symmetric properties

Definition at line [486](#) of file [structs.h](#).

Referenced by [StartVariables\(\)](#).

### 3.46.2.10 node

WORD FuNcTiOn::node

Location in namenode of [NAMETREE](#)

Definition at line [487](#) of file [structs.h](#).

### 3.46.2.11 namesize

WORD FuNcTiOn::namesize

Length of the name

Definition at line 488 of file [structs.h](#).

### 3.46.2.12 dimension

WORD FuNcTiOn::dimension

Definition at line 489 of file [structs.h](#).

### 3.46.2.13 maxnumargs

WORD FuNcTiOn::maxnumargs

Definition at line 490 of file [structs.h](#).

### 3.46.2.14 minnumargs

WORD FuNcTiOn::minnumargs

Definition at line 491 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.47 gcd\_heuristic\_failed Class Reference

### 3.47.1 Detailed Description

Definition at line 33 of file [polygcd.h](#).

The documentation for this class was generated from the following file:

- [polygcd.h](#)

## 3.48 HANDLERS Struct Reference

```
#include <structs.h>
```

## Data Fields

- WORD [newlogonly](#)
- WORD [newhandle](#)
- WORD [oldhandle](#)
- WORD [oldlogonly](#)
- WORD [oldprinttype](#)
- WORD [oldsilent](#)

### 3.48.1 Detailed Description

The struct [HANDLERS](#) is used in the communication with external channels.

Definition at line [964](#) of file [structs.h](#).

### 3.48.2 Field Documentation

#### 3.48.2.1 newlogonly

```
WORD HANDLERS::newlogonly
```

Definition at line [965](#) of file [structs.h](#).

#### 3.48.2.2 newhandle

```
WORD HANDLERS::newhandle
```

Definition at line [966](#) of file [structs.h](#).

#### 3.48.2.3 oldhandle

```
WORD HANDLERS::oldhandle
```

Definition at line [967](#) of file [structs.h](#).

#### 3.48.2.4 oldlogonly

```
WORD HANDLERS::oldlogonly
```

Definition at line [968](#) of file [structs.h](#).

#### 3.48.2.5 oldprinttype

```
WORD HANDLERS::oldprinttype
```

Definition at line [969](#) of file [structs.h](#).

### 3.48.2.6 oldsilent

```
WORD HANDLERS::oldsilent
```

Definition at line 970 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.49 Input Struct Reference

### Data Fields

- const char \* [name](#)
- int [icode](#)
- int [nplistn](#)
- const char \* [plistn](#) [GRCC\_MAXLEGS]
- int [plistc](#) [GRCC\_MAXLEGS]
- int [cvallist](#) [GRCC\_MAXNCPLG]

### 3.49.1 Detailed Description

Definition at line 137 of file [grccparam.h](#).

### 3.49.2 Field Documentation

#### 3.49.2.1 name

```
const char* IInput::name
```

Definition at line 138 of file [grccparam.h](#).

#### 3.49.2.2 icode

```
int IInput::icode
```

Definition at line 139 of file [grccparam.h](#).

#### 3.49.2.3 nplistn

```
int IInput::nplistn
```

Definition at line 140 of file [grccparam.h](#).

#### 3.49.2.4 plistn

```
const char* IInput::plistn[GRCC_MAXLEGS]
```

Definition at line 141 of file [grccparam.h](#).

#### 3.49.2.5 plistc

```
int IInput::plistc[GRCC_MAXLEGS]
```

Definition at line 142 of file [grccparam.h](#).

#### 3.49.2.6 cvallist

```
int IInput::cvallist[GRCC_MAXNCPLG]
```

Definition at line 143 of file [grccparam.h](#).

The documentation for this struct was generated from the following file:

- [grccparam.h](#)

## 3.50 InDeX Struct Reference

### Data Fields

- LONG [name](#)
- WORD [type](#)
- WORD [dimension](#)
- WORD [number](#)
- WORD [flags](#)
- WORD [nmin4](#)
- WORD [node](#)
- WORD [namesize](#)

### 3.50.1 Detailed Description

Definition at line 445 of file [structs.h](#).

### 3.50.2 Field Documentation

#### 3.50.2.1 name

```
LONG InDeX::name
```

Definition at line 446 of file [structs.h](#).



### 3.50.2.2 type

WORD InDeX::type

Definition at line 447 of file [structs.h](#).

### 3.50.2.3 dimension

WORD InDeX::dimension

Definition at line 448 of file [structs.h](#).

### 3.50.2.4 number

WORD InDeX::number

Definition at line 449 of file [structs.h](#).

### 3.50.2.5 flags

WORD InDeX::flags

Definition at line 450 of file [structs.h](#).

### 3.50.2.6 nmin4

WORD InDeX::nmin4

Definition at line 451 of file [structs.h](#).

### 3.50.2.7 node

WORD InDeX::node

Definition at line 452 of file [structs.h](#).

### 3.50.2.8 namesize

WORD InDeX::namesize

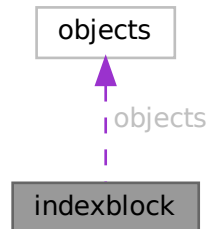
Definition at line 453 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.51 indexblock Struct Reference

Collaboration diagram for indexblock:



### Data Fields

- MLONG [flags](#)
- MLONG [previousblock](#)
- MLONG [position](#)
- OBJECTS [objects](#) [NUMOBJECTS]

### 3.51.1 Detailed Description

Definition at line 110 of file [minos.h](#).

### 3.51.2 Field Documentation

#### 3.51.2.1 flags

MLONG `indexblock::flags`

Definition at line 111 of file [minos.h](#).

#### 3.51.2.2 previousblock

MLONG `indexblock::previousblock`

Definition at line 112 of file [minos.h](#).

#### 3.51.2.3 position

MLONG `indexblock::position`

Definition at line 113 of file [minos.h](#).

### 3.51.2.4 objects

`OBJECTS` `indexblock::objects[NUMOBJECTS]`

Definition at line 114 of file [minos.h](#).

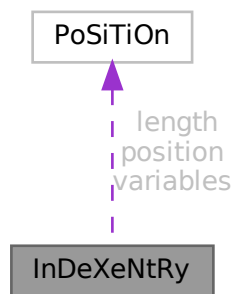
The documentation for this struct was generated from the following file:

- [minos.h](#)

## 3.52 InDeXeNtRy Struct Reference

```
#include <structs.h>
```

Collaboration diagram for InDeXeNtRy:



### Public Member Functions

- **PADPOSITION** (0, 1, 0, 5, MAXENAME+1)

### Data Fields

- [POSITION](#) [position](#)
- [POSITION](#) [length](#)
- [POSITION](#) [variables](#)
- `LONG` [CompressSize](#)
- `WORD` [nsymbols](#)
- `WORD` [nindices](#)
- `WORD` [nvectors](#)
- `WORD` [nfunctions](#)
- `WORD` [size](#)
- `SBYTE` [name](#) [MAXENAME+1]

### 3.52.1 Detailed Description

Defines the structure of an entry in a file index (see struct [FiLeInDeX](#)).

It represents one expression in the file.

Definition at line 100 of file [structs.h](#).

### 3.52.2 Field Documentation

#### 3.52.2.1 position

`POSITION InDeXeNtRy::position`

Position of the expression itself

Definition at line 101 of file [structs.h](#).

#### 3.52.2.2 length

`POSITION InDeXeNtRy::length`

Length of the expression itself

Definition at line 102 of file [structs.h](#).

#### 3.52.2.3 variables

`POSITION InDeXeNtRy::variables`

Position of the list with variables

Definition at line 103 of file [structs.h](#).

#### 3.52.2.4 CompressSize

`LONG InDeXeNtRy::CompressSize`

Size of buffer before compress

Definition at line 104 of file [structs.h](#).

#### 3.52.2.5 nsymbols

`WORD InDeXeNtRy::nsymbols`

Number of symbols in the list

Definition at line 105 of file [structs.h](#).

Referenced by [ReadSaveVariables\(\)](#).

### 3.52.2.6 nindices

WORD InDeXeNtRy::nindices

Number of indices in the list

Definition at line 106 of file [structs.h](#).

Referenced by [ReadSaveVariables\(\)](#).

### 3.52.2.7 nvectors

WORD InDeXeNtRy::nvectors

Number of vectors in the list

Definition at line 107 of file [structs.h](#).

Referenced by [ReadSaveVariables\(\)](#).

### 3.52.2.8 nfunctions

WORD InDeXeNtRy::nfunctions

Number of functions in the list

Definition at line 108 of file [structs.h](#).

Referenced by [ReadSaveVariables\(\)](#).

### 3.52.2.9 size

WORD InDeXeNtRy::size

Size of variables field

Definition at line 109 of file [structs.h](#).

### 3.52.2.10 name

SBYTE InDeXeNtRy::name[MAXENAME+1]

Name of expression

Definition at line 110 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.53 iniinfo Struct Reference

### Data Fields

- MLONG [entriesinindex](#)
- MLONG [numberofindexblocks](#)
- MLONG [firstindexblock](#)
- MLONG [lastindexblock](#)
- MLONG [numberoftables](#)
- MLONG [numberofnamesblocks](#)
- MLONG [firstnameblock](#)
- MLONG [lastnameblock](#)

### 3.53.1 Detailed Description

Definition at line [85](#) of file [minos.h](#).

### 3.53.2 Field Documentation

#### 3.53.2.1 entriesinindex

```
MLONG iniinfo::entriesinindex
```

Definition at line [87](#) of file [minos.h](#).

#### 3.53.2.2 numberofindexblocks

```
MLONG iniinfo::numberofindexblocks
```

Definition at line [88](#) of file [minos.h](#).

#### 3.53.2.3 firstindexblock

```
MLONG iniinfo::firstindexblock
```

Definition at line [89](#) of file [minos.h](#).

#### 3.53.2.4 lastindexblock

```
MLONG iniinfo::lastindexblock
```

Definition at line [90](#) of file [minos.h](#).

### 3.53.2.5 numberoftables

MLONG iniinfo::numberoftables

Definition at line 91 of file [minos.h](#).

### 3.53.2.6 numberofnamesblocks

MLONG iniinfo::numberofnamesblocks

Definition at line 92 of file [minos.h](#).

### 3.53.2.7 firstnameblock

MLONG iniinfo::firstnameblock

Definition at line 93 of file [minos.h](#).

### 3.53.2.8 lastnameblock

MLONG iniinfo::lastnameblock

Definition at line 94 of file [minos.h](#).

The documentation for this struct was generated from the following file:

- [minos.h](#)

## 3.54 INSIDEINFO Struct Reference

```
#include <structs.h>
```

### Data Fields

- WORD \* [buffer](#)
- int [oldcompletype](#)
- int [oldparallelflag](#)
- int [oldnumpotmoddollars](#)
- WORD [size](#)
- WORD [numdollars](#)
- WORD [oldcbuf](#)
- WORD [oldrbuf](#)
- WORD [inscbuf](#)
- WORD [oldcnumlhs](#)

### 3.54.1 Detailed Description

Used for #inside

Definition at line [854](#) of file [structs.h](#).

### 3.54.2 Field Documentation

#### 3.54.2.1 buffer

```
WORD*  INSIDEINFO::buffer
```

Definition at line [855](#) of file [structs.h](#).

#### 3.54.2.2 oldcompiletype

```
int  INSIDEINFO::oldcompiletype
```

Definition at line [856](#) of file [structs.h](#).

#### 3.54.2.3 oldparallelflag

```
int  INSIDEINFO::oldparallelflag
```

Definition at line [857](#) of file [structs.h](#).

#### 3.54.2.4 oldnumpotmoddollars

```
int  INSIDEINFO::oldnumpotmoddollars
```

Definition at line [858](#) of file [structs.h](#).

#### 3.54.2.5 size

```
WORD  INSIDEINFO::size
```

Definition at line [859](#) of file [structs.h](#).

#### 3.54.2.6 numdollars

```
WORD  INSIDEINFO::numdollars
```

Definition at line [860](#) of file [structs.h](#).



### 3.54.2.7 olddbuf

WORD INSIDEINFO::olddbuf

Definition at line 861 of file [structs.h](#).

### 3.54.2.8 oldrbuf

WORD INSIDEINFO::oldrbuf

Definition at line 862 of file [structs.h](#).

### 3.54.2.9 inscbuf

WORD INSIDEINFO::inscbuf

Definition at line 863 of file [structs.h](#).

### 3.54.2.10 oldcnumlhs

WORD INSIDEINFO::oldcnumlhs

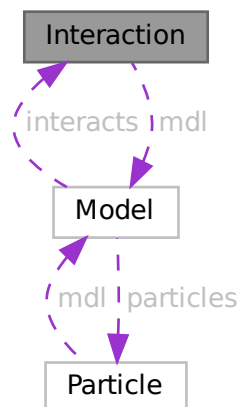
Definition at line 864 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.55 Interaction Class Reference

Collaboration diagram for Interaction:



## Public Member Functions

- [Interaction](#) ([Model](#) \*modl, int iid, const char \*nam, int icode, int \*cpl, int nlgs, int \*plst, int csm, int lp)
- void [prlInteraction](#) (void)

## Data Fields

- [Model](#) \* [mdl](#)
- char \* [name](#)
- int \* [plist](#)
- int \* [clist](#)
- int \* [slist](#)
- int [id](#)
- int [csum](#)
- int [nlegs](#)
- int [loop](#)
- int [icode](#)
- int [nplist](#)
- int [nclist](#)
- int [nslst](#)

### 3.55.1 Detailed Description

Definition at line [236](#) of file [grcc.h](#).

### 3.55.2 Constructor & Destructor Documentation

#### 3.55.2.1 Interaction()

```
Interaction::Interaction (
    Model * modl,
    int iid,
    const char * nam,
    int icode,
    int * cpl,
    int nlgs,
    int * plst,
    int csm,
    int lp )
```

Definition at line [1570](#) of file [grcc.cc](#).

#### 3.55.2.2 ~Interaction()

```
Interaction::~~Interaction (
    void )
```

Definition at line [1691](#) of file [grcc.cc](#).

### 3.55.3 Member Function Documentation

#### 3.55.3.1 prInteraction()

```
void Interaction::prInteraction (
    void )
```

Definition at line 1706 of file [grcc.cc](#).

### 3.55.4 Field Documentation

#### 3.55.4.1 mdl

```
Model* Interaction::mdl
```

Definition at line 238 of file [grcc.h](#).

#### 3.55.4.2 name

```
char* Interaction::name
```

Definition at line 239 of file [grcc.h](#).

#### 3.55.4.3 plist

```
int* Interaction::plist
```

Definition at line 240 of file [grcc.h](#).

#### 3.55.4.4 clist

```
int* Interaction::clist
```

Definition at line 241 of file [grcc.h](#).

#### 3.55.4.5 slist

```
int* Interaction::slist
```

Definition at line 242 of file [grcc.h](#).

#### 3.55.4.6 id

```
int Interaction::id
```

Definition at line 244 of file [grcc.h](#).

#### 3.55.4.7 csum

```
int Interaction::csum
```

Definition at line 245 of file [grcc.h](#).

#### 3.55.4.8 nlegs

```
int Interaction::nlegs
```

Definition at line 246 of file [grcc.h](#).

#### 3.55.4.9 loop

```
int Interaction::loop
```

Definition at line 247 of file [grcc.h](#).

#### 3.55.4.10 icode

```
int Interaction::icode
```

Definition at line 248 of file [grcc.h](#).

#### 3.55.4.11 nplist

```
int Interaction::nplist
```

Definition at line 250 of file [grcc.h](#).

#### 3.55.4.12 nclist

```
int Interaction::nclist
```

Definition at line 251 of file [grcc.h](#).

#### 3.55.4.13 nslist

```
int Interaction::nslist
```

Definition at line 252 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.56 KEYWORD Struct Reference

```
#include <structs.h>
```

### Data Fields

- char \* [name](#)
- TFUN [func](#)
- int [type](#)
- int [flags](#)

### 3.56.1 Detailed Description

The [KEYWORD](#) struct defines names of commands/statements and the routine to be called when they are encountered by the compiler or preprocessor.

Definition at line [222](#) of file [structs.h](#).

### 3.56.2 Field Documentation

#### 3.56.2.1 name

```
char* KEYWORD::name
```

Definition at line [223](#) of file [structs.h](#).

#### 3.56.2.2 func

```
TFUN KEYWORD::func
```

Definition at line [224](#) of file [structs.h](#).

#### 3.56.2.3 type

```
int KEYWORD::type
```

Definition at line [225](#) of file [structs.h](#).

#### 3.56.2.4 flags

```
int KEYWORD::flags
```

Definition at line [226](#) of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.57 KEYWORDV Struct Reference

```
#include <structs.h>
```

### Data Fields

- char \* [name](#)
- int \* [var](#)
- int [type](#)
- int [flags](#)

### 3.57.1 Detailed Description

The [KEYWORDV](#) struct defines names of commands/statements and the variable to be affected when they are encountered by the compiler or preprocessor.

Definition at line [234](#) of file [structs.h](#).

### 3.57.2 Field Documentation

#### 3.57.2.1 name

```
char* KEYWORDV::name
```

Definition at line [235](#) of file [structs.h](#).

#### 3.57.2.2 var

```
int* KEYWORDV::var
```

Definition at line [236](#) of file [structs.h](#).

#### 3.57.2.3 type

```
int KEYWORDV::type
```

Definition at line [237](#) of file [structs.h](#).

#### 3.57.2.4 flags

```
int KEYWORDV::flags
```

Definition at line [238](#) of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.58 LIST Struct Reference

```
#include <structs.h>
```

### Data Fields

- void \* [lijst](#)
- char \* [message](#)
- int [num](#)
- int [maxnum](#)
- int [size](#)
- int [numglobal](#)
- int [numtemp](#)
- int [numclear](#)

### 3.58.1 Detailed Description

Much information is stored in arrays of which we can double the size if the array proves to be too small. Such arrays are controlled by a variable of type [LIST](#). The routines that expand the lists are in the file [tools.c](#)

Definition at line [205](#) of file [structs.h](#).

### 3.58.2 Field Documentation

#### 3.58.2.1 lijst

```
void* LIST::lijst
```

[D] Holds space for "maxnum" elements of size "size" each

Definition at line [206](#) of file [structs.h](#).

#### 3.58.2.2 message

```
char* LIST::message
```

Text for Malloc1 when allocating lijst. Set to constant string.

Definition at line [207](#) of file [structs.h](#).

#### 3.58.2.3 num

```
int LIST::num
```

Number of elements in lijst.

Definition at line [208](#) of file [structs.h](#).

#### 3.58.2.4 maxnum

```
int LIST::maxnum
```

Maximum number of elements in lijst.

Definition at line 209 of file [structs.h](#).

#### 3.58.2.5 size

```
int LIST::size
```

Size of one element in lijst.

Definition at line 210 of file [structs.h](#).

#### 3.58.2.6 numglobal

```
int LIST::numglobal
```

Marker for position when .global is executed.

Definition at line 211 of file [structs.h](#).

#### 3.58.2.7 numtemp

```
int LIST::numtemp
```

At the moment only needed for sets and setstore.

Definition at line 212 of file [structs.h](#).

#### 3.58.2.8 numclear

```
int LIST::numclear
```

Only for the clear instruction.

Definition at line 213 of file [structs.h](#).

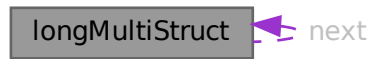
The documentation for this struct was generated from the following file:

- [structs.h](#)



## 3.59 longMultiStruct Struct Reference

Collaboration diagram for longMultiStruct:



### Data Fields

- UBYTE \* [buffer](#)
- int [bufpos](#)
- int [packpos](#)
- int [nPacks](#)
- int [lastLen](#)
- struct [longMultiStruct](#) \* [next](#)

### 3.59.1 Detailed Description

Definition at line [1013](#) of file [mpi.c](#).

### 3.59.2 Field Documentation

#### 3.59.2.1 buffer

```
UBYTE* longMultiStruct::buffer
```

Definition at line [1014](#) of file [mpi.c](#).

#### 3.59.2.2 bufpos

```
int longMultiStruct::bufpos
```

Definition at line [1015](#) of file [mpi.c](#).

#### 3.59.2.3 packpos

```
int longMultiStruct::packpos
```

Definition at line [1016](#) of file [mpi.c](#).

### 3.59.2.4 nPacks

```
int longMultiStruct::nPacks
```

Definition at line 1017 of file [mpi.c](#).

### 3.59.2.5 lastLen

```
int longMultiStruct::lastLen
```

Definition at line 1018 of file [mpi.c](#).

### 3.59.2.6 next

```
struct longMultiStruct* longMultiStruct::next
```

Definition at line 1021 of file [mpi.c](#).

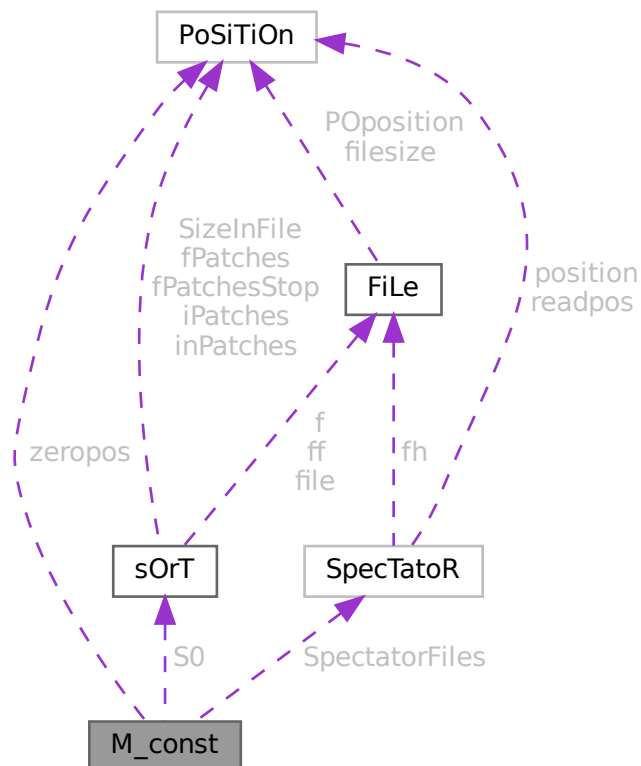
The documentation for this struct was generated from the following file:

- [mpi.c](#)

## 3.60 M\_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for M\_const:



## Public Member Functions

- **PADPOSITION** (17, 28, 62, 84, 2)

## Data Fields

- [POSITION](#) zeropos
- [SORTING](#) \* [S0](#)
- [UWORD](#) \* [gcmmod](#)
- [UWORD](#) \* [gpowmod](#)
- [UBYTE](#) \* [TempDir](#)
- [UBYTE](#) \* [TempSortDir](#)
- [UBYTE](#) \* [IncDir](#)
- [UBYTE](#) \* [InputFileName](#)
- [UBYTE](#) \* [LogFileName](#)
- [UBYTE](#) \* [OutBuffer](#)
- [UBYTE](#) \* [Path](#)
- [UBYTE](#) \* [SetupDir](#)
- [UBYTE](#) \* [SetupFile](#)
- [UBYTE](#) \* [gFortran90Kind](#)
- [UBYTE](#) \* [gextrasym](#)
- [UBYTE](#) \* [ggextrasym](#)
- [UBYTE](#) \* [oldnumextrasymbols](#)
- [SPECTATOR](#) \* [SpectatorFiles](#)
- [LONG](#) [MaxTer](#)
- [LONG](#) [CompressSize](#)
- [LONG](#) [ScratSize](#)
- [LONG](#) [HideSize](#)
- [LONG](#) [SizeStoreCache](#)
- [LONG](#) [MaxStreamSize](#)
- [LONG](#) [SIOsize](#)
- [LONG](#) [SLargeSize](#)
- [LONG](#) [SSmallEsize](#)
- [LONG](#) [SSmallSize](#)
- [LONG](#) [STermsInSmall](#)
- [LONG](#) [MaxBracketBufferSize](#)
- [LONG](#) [hProcessBucketSize](#)
- [LONG](#) [gProcessBucketSize](#)
- [LONG](#) [shmWinSize](#)
- [LONG](#) [OldChildTime](#)
- [LONG](#) [OldSecTime](#)
- [LONG](#) [OldMilliTime](#)
- [LONG](#) [WorkSize](#)
- [LONG](#) [gThreadBucketSize](#)
- [LONG](#) [ggThreadBucketSize](#)
- [LONG](#) [SumTime](#)
- [LONG](#) [SpectatorSize](#)
- [LONG](#) [TimeLimit](#)
- [int](#) [FileOnlyFlag](#)
- [int](#) [Interact](#)
- [int](#) [MaxParLevel](#)
- [int](#) [OutBufSize](#)
- [int](#) [SMaxFpatches](#)
- [int](#) [SMaxPatches](#)

- int StdOut
- int ginsidefirst
- int gDefDim
- int gDefDim4
- int NumFixedSets
- int NumFixedFunctions
- int rbufnum
- int dbufnum
- int sbufnum
- int zbufnum
- int SkipClears
- int gTokensWriteFlag
- int gfunpowers
- int gStatsFlag
- int gNamesFlag
- int gCodesFlag
- int gSortType
- int gproperorderflag
- int hparallelflag
- int gparallelflag
- int totalnumberofthreads
- int gSizeCommuteInSet
- int gThreadStats
- int ggThreadStats
- int gFinalStats
- int ggFinalStats
- int gThreadsFlag
- int ggThreadsFlag
- int gThreadBalancing
- int ggThreadBalancing
- int gThreadSortFileSynch
- int ggThreadSortFileSynch
- int gProcessStats
- int ggProcessStats
- int gOldParallelStats
- int ggOldParallelStats
- int maxFlevels
- int resetTimeOnClear
- int gcNumDollars
- int MultiRun
- int gNoSpacesInNumbers
- int ggNoSpacesInNumbers
- int glsFortran90
- int PrintTotalSize
- int fbufferSize
- int gOldFactArgFlag
- int ggOldFactArgFlag
- int gnumextrasym
- int ggnumextrasym
- int NumSpectatorFiles
- int SizeForSpectatorFiles
- int gOldGCDflag
- int ggOldGCDflag
- int gWTimeStatsFlag
- int ggWTimeStatsFlag

- int [jumpratio](#)
- WORD [MaxTal](#)
- WORD [IndDum](#)
- WORD [DumInd](#)
- WORD [WillInd](#)
- WORD [gncmod](#)
- WORD [gnpowmod](#)
- WORD [gmodmode](#)
- WORD [gUnitTrace](#)
- WORD [gOutputMode](#)
- WORD [gOutputSpaces](#)
- WORD [gOutNumberType](#)
- WORD [gCnumpows](#)
- WORD [gUniTrace](#) [4]
- WORD [MaxWildcards](#)
- WORD [mTraceDum](#)
- WORD [OffsetIndex](#)
- WORD [OffsetVector](#)
- WORD [RepMax](#)
- WORD [LogType](#)
- WORD [ggStatsFlag](#)
- WORD [gLineLength](#)
- WORD [qError](#)
- WORD [FortranCont](#)
- WORD [HoldFlag](#)
- WORD [Ordering](#) [15]
- WORD [silent](#)
- WORD [tracebackflag](#)
- WORD [expnum](#)
- WORD [denomnum](#)
- WORD [facnum](#)
- WORD [invfacnum](#)
- WORD [sumnum](#)
- WORD [sumpnum](#)
- WORD [OldOrderFlag](#)
- WORD [termfunnum](#)
- WORD [matchfunnum](#)
- WORD [countfunnum](#)
- WORD [gPolyFun](#)
- WORD [gPolyFunInv](#)
- WORD [gPolyFunType](#)
- WORD [gPolyFunExp](#)
- WORD [gPolyFunVar](#)
- WORD [gPolyFunPow](#)
- WORD [dollarzero](#)
- WORD [atstartup](#)
- WORD [exitflag](#)
- WORD [NumStoreCaches](#)
- WORD [gIndentSpace](#)
- WORD [ggIndentSpace](#)
- WORD [gShortStatsMax](#)
- WORD [ggShortStatsMax](#)
- WORD [gextrasymbols](#)
- WORD [ggextrasymbols](#)
- WORD [zerorhs](#)

- WORD [onerhs](#)
- WORD [havesortdir](#)
- WORD [vectorzero](#)
- WORD [ClearStore](#)
- WORD [BracketFactors](#) [8]
- BOOL [FromStdin](#)
- BOOL [IgnoreDeprecation](#)

### 3.60.1 Detailed Description

The [M\\_const](#) struct is part of the global data and resides in the [ALLGLOBALS](#) struct A under the name #M. We see it used with the macro AM as in AM.S0. It contains global settings at startup or .clear.

Definition at line [1424](#) of file [structs.h](#).

### 3.60.2 Field Documentation

#### 3.60.2.1 zeropos

```
POSITION M_const::zeropos
```

Definition at line [1425](#) of file [structs.h](#).

#### 3.60.2.2 S0

```
SORTING* M_const::S0
```

[D] The main sort buffer

Definition at line [1426](#) of file [structs.h](#).

#### 3.60.2.3 gcmmod

```
UWORD* M_const::gcmmod
```

Global setting of modulus. Uses AC.cmod's memory

Definition at line [1427](#) of file [structs.h](#).

#### 3.60.2.4 gpowmod

```
UWORD* M_const::gpowmod
```

Global setting printing as powers. Uses AC.cmod's memory

Definition at line [1428](#) of file [structs.h](#).

### 3.60.2.5 TempDir

```
UBYTE* M_const::TempDir
```

Definition at line 1429 of file [structs.h](#).

### 3.60.2.6 TempSortDir

```
UBYTE* M_const::TempSortDir
```

Definition at line 1430 of file [structs.h](#).

### 3.60.2.7 IncDir

```
UBYTE* M_const::IncDir
```

Definition at line 1431 of file [structs.h](#).

### 3.60.2.8 InputFileName

```
UBYTE* M_const::InputFileName
```

Definition at line 1432 of file [structs.h](#).

### 3.60.2.9 LogFileName

```
UBYTE* M_const::LogFileName
```

Definition at line 1433 of file [structs.h](#).

### 3.60.2.10 OutBuffer

```
UBYTE* M_const::OutBuffer
```

Definition at line 1434 of file [structs.h](#).

### 3.60.2.11 Path

```
UBYTE* M_const::Path
```

Definition at line 1435 of file [structs.h](#).

### 3.60.2.12 SetupDir

```
UBYTE* M_const::SetupDir
```

Definition at line 1436 of file [structs.h](#).

### 3.60.2.13 SetupFile

UBYTE\* M\_const::SetupFile

Definition at line 1437 of file [structs.h](#).

### 3.60.2.14 gFortran90Kind

UBYTE\* M\_const::gFortran90Kind

Definition at line 1438 of file [structs.h](#).

### 3.60.2.15 gextrasym

UBYTE\* M\_const::gextrasym

Definition at line 1439 of file [structs.h](#).

### 3.60.2.16 ggextrasym

UBYTE\* M\_const::ggextrasym

Definition at line 1440 of file [structs.h](#).

### 3.60.2.17 oldnumextrasymbols

UBYTE\* M\_const::oldnumextrasymbols

Definition at line 1441 of file [structs.h](#).

### 3.60.2.18 SpectatorFiles

SPECTATOR\* M\_const::SpectatorFiles

Definition at line 1442 of file [structs.h](#).

### 3.60.2.19 MaxTer

LONG M\_const::MaxTer

Definition at line 1450 of file [structs.h](#).

### 3.60.2.20 CompressSize

LONG M\_const::CompressSize

Definition at line 1451 of file [structs.h](#).



### 3.60.2.21 ScratSize

LONG M\_const::SkratSize

Definition at line 1452 of file [structs.h](#).

### 3.60.2.22 HideSize

LONG M\_const::HideSize

Definition at line 1453 of file [structs.h](#).

### 3.60.2.23 SizeStoreCache

LONG M\_const::SizeStoreCache

Definition at line 1454 of file [structs.h](#).

### 3.60.2.24 MaxStreamSize

LONG M\_const::MaxStreamSize

Definition at line 1455 of file [structs.h](#).

### 3.60.2.25 SIOsize

LONG M\_const::SIOsize

Definition at line 1456 of file [structs.h](#).

### 3.60.2.26 SLargeSize

LONG M\_const::SLargeSize

Definition at line 1457 of file [structs.h](#).

### 3.60.2.27 SSmallEsize

LONG M\_const::SSmallEsize

Definition at line 1458 of file [structs.h](#).

### 3.60.2.28 SSmallSize

LONG M\_const::SSmallSize

Definition at line 1459 of file [structs.h](#).

### 3.60.2.29 STermsInSmall

LONG M\_const::STermsInSmall

Definition at line 1460 of file [structs.h](#).

### 3.60.2.30 MaxBracketBufferSize

LONG M\_const::MaxBracketBufferSize

Definition at line 1461 of file [structs.h](#).

### 3.60.2.31 hProcessBucketSize

LONG M\_const::hProcessBucketSize

Definition at line 1462 of file [structs.h](#).

### 3.60.2.32 gProcessBucketSize

LONG M\_const::gProcessBucketSize

Definition at line 1463 of file [structs.h](#).

### 3.60.2.33 shmWinSize

LONG M\_const::shmWinSize

Definition at line 1464 of file [structs.h](#).

### 3.60.2.34 OldChildTime

LONG M\_const::OldChildTime

Definition at line 1465 of file [structs.h](#).

### 3.60.2.35 OldSecTime

LONG M\_const::OldSecTime

Definition at line 1466 of file [structs.h](#).

### 3.60.2.36 OldMilliTime

LONG M\_const::OldMilliTime

Definition at line 1467 of file [structs.h](#).

### 3.60.2.37 WorkSize

LONG M\_const::WorkSize

Definition at line 1468 of file [structs.h](#).

### 3.60.2.38 gThreadBucketSize

LONG M\_const::gThreadBucketSize

Definition at line 1469 of file [structs.h](#).

### 3.60.2.39 ggThreadBucketSize

LONG M\_const::ggThreadBucketSize

Definition at line 1470 of file [structs.h](#).

### 3.60.2.40 SumTime

LONG M\_const::SumTime

Definition at line 1471 of file [structs.h](#).

### 3.60.2.41 SpectatorSize

LONG M\_const::SpectatorSize

Definition at line 1472 of file [structs.h](#).

### 3.60.2.42 TimeLimit

LONG M\_const::TimeLimit

Definition at line 1473 of file [structs.h](#).

### 3.60.2.43 FileOnlyFlag

int M\_const::FileOnlyFlag

Definition at line 1480 of file [structs.h](#).

### 3.60.2.44 Interact

int M\_const::Interact

Definition at line 1481 of file [structs.h](#).

#### 3.60.2.45 MaxParLevel

```
int M_const::MaxParLevel
```

Definition at line 1482 of file [structs.h](#).

#### 3.60.2.46 OutBufSize

```
int M_const::OutBufSize
```

Definition at line 1483 of file [structs.h](#).

#### 3.60.2.47 SMaxFpatches

```
int M_const::SMaxFpatches
```

Definition at line 1484 of file [structs.h](#).

#### 3.60.2.48 SMaxPatches

```
int M_const::SMaxPatches
```

Definition at line 1485 of file [structs.h](#).

#### 3.60.2.49 StdOut

```
int M_const::StdOut
```

Definition at line 1486 of file [structs.h](#).

#### 3.60.2.50 ginsidefirst

```
int M_const::ginsidefirst
```

Definition at line 1487 of file [structs.h](#).

#### 3.60.2.51 gDefDim

```
int M_const::gDefDim
```

Definition at line 1488 of file [structs.h](#).

#### 3.60.2.52 gDefDim4

```
int M_const::gDefDim4
```

Definition at line 1489 of file [structs.h](#).

### 3.60.2.53 NumFixedSets

```
int M_const::NumFixedSets
```

Definition at line 1490 of file [structs.h](#).

### 3.60.2.54 NumFixedFunctions

```
int M_const::NumFixedFunctions
```

Definition at line 1491 of file [structs.h](#).

### 3.60.2.55 rbufnum

```
int M_const::rbufnum
```

Definition at line 1492 of file [structs.h](#).

### 3.60.2.56 dbufnum

```
int M_const::dbufnum
```

Definition at line 1493 of file [structs.h](#).

### 3.60.2.57 sbufnum

```
int M_const::sbufnum
```

Definition at line 1494 of file [structs.h](#).

### 3.60.2.58 zbufnum

```
int M_const::zbufnum
```

Definition at line 1495 of file [structs.h](#).

### 3.60.2.59 SkipClears

```
int M_const::SkipClears
```

Definition at line 1496 of file [structs.h](#).

### 3.60.2.60 gTokensWriteFlag

```
int M_const::gTokensWriteFlag
```

Definition at line 1497 of file [structs.h](#).

**3.60.2.61 gfunpowers**

```
int M_const::gfunpowers
```

Definition at line 1498 of file [structs.h](#).

**3.60.2.62 gStatsFlag**

```
int M_const::gStatsFlag
```

Definition at line 1499 of file [structs.h](#).

**3.60.2.63 gNamesFlag**

```
int M_const::gNamesFlag
```

Definition at line 1500 of file [structs.h](#).

**3.60.2.64 gCodesFlag**

```
int M_const::gCodesFlag
```

Definition at line 1501 of file [structs.h](#).

**3.60.2.65 gSortType**

```
int M_const::gSortType
```

Definition at line 1502 of file [structs.h](#).

**3.60.2.66 gproperorderflag**

```
int M_const::gproperorderflag
```

Definition at line 1503 of file [structs.h](#).

**3.60.2.67 hparallelflag**

```
int M_const::hparallelflag
```

Definition at line 1504 of file [structs.h](#).

**3.60.2.68 gparallelflag**

```
int M_const::gparallelflag
```

Definition at line 1505 of file [structs.h](#).

**3.60.2.69 totalnumberofthreads**

```
int M_const::totalnumberofthreads
```

Definition at line 1506 of file [structs.h](#).

**3.60.2.70 gSizeCommuteInSet**

```
int M_const::gSizeCommuteInSet
```

Definition at line 1507 of file [structs.h](#).

**3.60.2.71 gThreadStats**

```
int M_const::gThreadStats
```

Definition at line 1508 of file [structs.h](#).

**3.60.2.72 ggThreadStats**

```
int M_const::ggThreadStats
```

Definition at line 1509 of file [structs.h](#).

**3.60.2.73 gFinalStats**

```
int M_const::gFinalStats
```

Definition at line 1510 of file [structs.h](#).

**3.60.2.74 ggFinalStats**

```
int M_const::ggFinalStats
```

Definition at line 1511 of file [structs.h](#).

**3.60.2.75 gThreadsFlag**

```
int M_const::gThreadsFlag
```

Definition at line 1512 of file [structs.h](#).

**3.60.2.76 ggThreadsFlag**

```
int M_const::ggThreadsFlag
```

Definition at line 1513 of file [structs.h](#).

**3.60.2.77 gThreadBalancing**

```
int M_const::gThreadBalancing
```

Definition at line 1514 of file [structs.h](#).

**3.60.2.78 ggThreadBalancing**

```
int M_const::ggThreadBalancing
```

Definition at line 1515 of file [structs.h](#).

**3.60.2.79 gThreadSortFileSynch**

```
int M_const::gThreadSortFileSynch
```

Definition at line 1516 of file [structs.h](#).

**3.60.2.80 ggThreadSortFileSynch**

```
int M_const::ggThreadSortFileSynch
```

Definition at line 1517 of file [structs.h](#).

**3.60.2.81 gProcessStats**

```
int M_const::gProcessStats
```

Definition at line 1518 of file [structs.h](#).

**3.60.2.82 ggProcessStats**

```
int M_const::ggProcessStats
```

Definition at line 1519 of file [structs.h](#).

**3.60.2.83 gOldParallelStats**

```
int M_const::gOldParallelStats
```

Definition at line 1520 of file [structs.h](#).

**3.60.2.84 ggOldParallelStats**

```
int M_const::ggOldParallelStats
```

Definition at line 1521 of file [structs.h](#).



**3.60.2.85 maxFlevels**

```
int M_const::maxFlevels
```

Definition at line 1522 of file [structs.h](#).

**3.60.2.86 resetTimeOnClear**

```
int M_const::resetTimeOnClear
```

Definition at line 1523 of file [structs.h](#).

**3.60.2.87 gcNumDollars**

```
int M_const::gcNumDollars
```

Definition at line 1524 of file [structs.h](#).

**3.60.2.88 MultiRun**

```
int M_const::MultiRun
```

Definition at line 1525 of file [structs.h](#).

**3.60.2.89 gNoSpacesInNumbers**

```
int M_const::gNoSpacesInNumbers
```

Definition at line 1526 of file [structs.h](#).

**3.60.2.90 ggNoSpacesInNumbers**

```
int M_const::ggNoSpacesInNumbers
```

Definition at line 1527 of file [structs.h](#).

**3.60.2.91 gIsFortran90**

```
int M_const::gIsFortran90
```

Definition at line 1528 of file [structs.h](#).

**3.60.2.92 PrintTotalSize**

```
int M_const::PrintTotalSize
```

Definition at line 1529 of file [structs.h](#).

### 3.60.2.93 fbufferSize

```
int M_const::fbufferSize
```

Definition at line 1530 of file [structs.h](#).

### 3.60.2.94 gOldFactArgFlag

```
int M_const::gOldFactArgFlag
```

Definition at line 1531 of file [structs.h](#).

### 3.60.2.95 ggOldFactArgFlag

```
int M_const::ggOldFactArgFlag
```

Definition at line 1532 of file [structs.h](#).

### 3.60.2.96 gnumextrasym

```
int M_const::gnumextrasym
```

Definition at line 1533 of file [structs.h](#).

### 3.60.2.97 ggnumextrasym

```
int M_const::ggnumextrasym
```

Definition at line 1534 of file [structs.h](#).

### 3.60.2.98 NumSpectatorFiles

```
int M_const::NumSpectatorFiles
```

Definition at line 1535 of file [structs.h](#).

### 3.60.2.99 SizeForSpectatorFiles

```
int M_const::SizeForSpectatorFiles
```

Definition at line 1536 of file [structs.h](#).

### 3.60.2.100 gOldGCDflag

```
int M_const::gOldGCDflag
```

Definition at line 1537 of file [structs.h](#).

**3.60.2.101 ggOldGCDflag**

```
int M_const::ggOldGCDflag
```

Definition at line 1538 of file [structs.h](#).

**3.60.2.102 gWTimeStatsFlag**

```
int M_const::gWTimeStatsFlag
```

Definition at line 1539 of file [structs.h](#).

**3.60.2.103 ggWTimeStatsFlag**

```
int M_const::ggWTimeStatsFlag
```

Definition at line 1540 of file [structs.h](#).

**3.60.2.104 jumpratio**

```
int M_const::jumpratio
```

Definition at line 1541 of file [structs.h](#).

**3.60.2.105 MaxTal**

```
WORD M_const::MaxTal
```

Definition at line 1542 of file [structs.h](#).

**3.60.2.106 IndDum**

```
WORD M_const::IndDum
```

Definition at line 1543 of file [structs.h](#).

**3.60.2.107 DumInd**

```
WORD M_const::DumInd
```

Definition at line 1544 of file [structs.h](#).

**3.60.2.108 WillInd**

```
WORD M_const::WillInd
```

Definition at line 1545 of file [structs.h](#).

**3.60.2.109 gncmod**

WORD M\_const::gncmod

Definition at line 1546 of file [structs.h](#).

**3.60.2.110 gnpowmod**

WORD M\_const::gnpowmod

Definition at line 1547 of file [structs.h](#).

**3.60.2.111 gmodmode**

WORD M\_const::gmodmode

Definition at line 1548 of file [structs.h](#).

**3.60.2.112 gUnitTrace**

WORD M\_const::gUnitTrace

Definition at line 1549 of file [structs.h](#).

**3.60.2.113 gOutputMode**

WORD M\_const::gOutputMode

Definition at line 1550 of file [structs.h](#).

**3.60.2.114 gOutputSpaces**

WORD M\_const::gOutputSpaces

Definition at line 1551 of file [structs.h](#).

**3.60.2.115 gOutNumberType**

WORD M\_const::gOutNumberType

Definition at line 1552 of file [structs.h](#).

**3.60.2.116 gCnumpows**

WORD M\_const::gCnumpows

Definition at line 1553 of file [structs.h](#).

**3.60.2.117 gUniTrace**

WORD M\_const::gUniTrace[4]

Definition at line 1554 of file [structs.h](#).

**3.60.2.118 MaxWildcards**

WORD M\_const::MaxWildcards

Definition at line 1555 of file [structs.h](#).

**3.60.2.119 mTraceDum**

WORD M\_const::mTraceDum

Definition at line 1556 of file [structs.h](#).

**3.60.2.120 OffsetIndex**

WORD M\_const::OffsetIndex

Definition at line 1557 of file [structs.h](#).

**3.60.2.121 OffsetVector**

WORD M\_const::OffsetVector

Definition at line 1558 of file [structs.h](#).

**3.60.2.122 RepMax**

WORD M\_const::RepMax

Definition at line 1559 of file [structs.h](#).

**3.60.2.123 LogType**

WORD M\_const::LogType

Definition at line 1560 of file [structs.h](#).

**3.60.2.124 ggStatsFlag**

WORD M\_const::ggStatsFlag

Definition at line 1561 of file [structs.h](#).

**3.60.2.125 gLineLength**

WORD M\_const::gLineLength

Definition at line 1562 of file [structs.h](#).

**3.60.2.126 qError**

WORD M\_const::qError

Definition at line 1563 of file [structs.h](#).

**3.60.2.127 FortranCont**

WORD M\_const::FortranCont

Definition at line 1564 of file [structs.h](#).

**3.60.2.128 HoldFlag**

WORD M\_const::HoldFlag

Definition at line 1565 of file [structs.h](#).

**3.60.2.129 Ordering**

WORD M\_const::Ordering[15]

Definition at line 1566 of file [structs.h](#).

**3.60.2.130 silent**

WORD M\_const::silent

Definition at line 1567 of file [structs.h](#).

**3.60.2.131 tracebackflag**

WORD M\_const::tracebackflag

Definition at line 1568 of file [structs.h](#).

**3.60.2.132 expnum**

WORD M\_const::expnum

Definition at line 1569 of file [structs.h](#).

**3.60.2.133 denomnum**

WORD M\_const::denomnum

Definition at line 1570 of file [structs.h](#).

**3.60.2.134 facnum**

WORD M\_const::facnum

Definition at line 1571 of file [structs.h](#).

**3.60.2.135 invfacnum**

WORD M\_const::invfacnum

Definition at line 1572 of file [structs.h](#).

**3.60.2.136 sumnum**

WORD M\_const::sumnum

Definition at line 1573 of file [structs.h](#).

**3.60.2.137 sumpnum**

WORD M\_const::sumpnum

Definition at line 1574 of file [structs.h](#).

**3.60.2.138 OldOrderFlag**

WORD M\_const::OldOrderFlag

Definition at line 1575 of file [structs.h](#).

**3.60.2.139 termfunnum**

WORD M\_const::termfunnum

Definition at line 1576 of file [structs.h](#).

**3.60.2.140 matchfunnum**

WORD M\_const::matchfunnum

Definition at line 1577 of file [structs.h](#).

**3.60.2.141 countfunnum**

WORD M\_const::countfunnum

Definition at line 1578 of file [structs.h](#).

**3.60.2.142 gPolyFun**

WORD M\_const::gPolyFun

Definition at line 1579 of file [structs.h](#).

**3.60.2.143 gPolyFunInv**

WORD M\_const::gPolyFunInv

Definition at line 1580 of file [structs.h](#).

**3.60.2.144 gPolyFunType**

WORD M\_const::gPolyFunType

Definition at line 1581 of file [structs.h](#).

**3.60.2.145 gPolyFunExp**

WORD M\_const::gPolyFunExp

Definition at line 1582 of file [structs.h](#).

**3.60.2.146 gPolyFunVar**

WORD M\_const::gPolyFunVar

Definition at line 1583 of file [structs.h](#).

**3.60.2.147 gPolyFunPow**

WORD M\_const::gPolyFunPow

Definition at line 1584 of file [structs.h](#).

**3.60.2.148 dollarzero**

WORD M\_const::dollarzero

Definition at line 1585 of file [structs.h](#).



**3.60.2.149 atstartup**

WORD M\_const::atstartup

Definition at line 1586 of file [structs.h](#).

**3.60.2.150 exitflag**

WORD M\_const::exitflag

Definition at line 1587 of file [structs.h](#).

**3.60.2.151 NumStoreCaches**

WORD M\_const::NumStoreCaches

Definition at line 1588 of file [structs.h](#).

**3.60.2.152 gIndentSpace**

WORD M\_const::gIndentSpace

Definition at line 1589 of file [structs.h](#).

**3.60.2.153 ggIndentSpace**

WORD M\_const::ggIndentSpace

Definition at line 1590 of file [structs.h](#).

**3.60.2.154 gShortStatsMax**

WORD M\_const::gShortStatsMax

For On FewerStatistics 10;

Definition at line 1591 of file [structs.h](#).

**3.60.2.155 ggShortStatsMax**

WORD M\_const::ggShortStatsMax

For On FewerStatistics 10;

Definition at line 1592 of file [structs.h](#).

**3.60.2.156 gextrasymbols**

WORD M\_const::gextrasymbols

Definition at line 1593 of file [structs.h](#).

**3.60.2.157 ggextrasymbols**

WORD M\_const::ggextrasymbols

Definition at line 1594 of file [structs.h](#).

**3.60.2.158 zerorhs**

WORD M\_const::zerorhs

Definition at line 1595 of file [structs.h](#).

**3.60.2.159 onerhs**

WORD M\_const::onerhs

Definition at line 1596 of file [structs.h](#).

**3.60.2.160 havessortdir**

WORD M\_const::havessortdir

Definition at line 1597 of file [structs.h](#).

**3.60.2.161 vectorzero**

WORD M\_const::vectorzero

Definition at line 1598 of file [structs.h](#).

**3.60.2.162 ClearStore**

WORD M\_const::ClearStore

Definition at line 1599 of file [structs.h](#).

**3.60.2.163 BracketFactors**

WORD M\_const::BracketFactors[8]

Definition at line 1600 of file [structs.h](#).

### 3.60.2.164 FromStdin

```
BOOL M_const::FromStdin
```

Definition at line 1601 of file [structs.h](#).

### 3.60.2.165 IgnoreDeprecation

```
BOOL M_const::IgnoreDeprecation
```

Definition at line 1602 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.61 MCBLOCK Class Reference

### Public Member Functions

- void [init](#) (void)

### Data Fields

- Edge2n \* [edges](#)
- int [nedges](#)
- int [nartps](#)
- int [loop](#)
- int [DUMMYPADDING](#)

### 3.61.1 Detailed Description

Definition at line 933 of file [grcc.h](#).

### 3.61.2 Constructor & Destructor Documentation

#### 3.61.2.1 MCBLOCK()

```
MCBlock::MCBlock (  
    void )
```

Definition at line 5106 of file [grcc.cc](#).

### 3.61.2.2 ~MCBlock()

```
MCBlock::~MCBlock (
    void )
```

Definition at line 5112 of file [grcc.cc](#).

## 3.61.3 Member Function Documentation

### 3.61.3.1 init()

```
void MCBlock::init (
    void )
```

Definition at line 5117 of file [grcc.cc](#).

## 3.61.4 Field Documentation

### 3.61.4.1 edges

```
Edge2n* MCBlock::edges
```

Definition at line 935 of file [grcc.h](#).

### 3.61.4.2 nmedges

```
int MCBlock::nmedges
```

Definition at line 936 of file [grcc.h](#).

### 3.61.4.3 nartps

```
int MCBlock::nartps
```

Definition at line 937 of file [grcc.h](#).

### 3.61.4.4 loop

```
int MCBlock::loop
```

Definition at line 938 of file [grcc.h](#).

#### 3.61.4.5 DUMMYPADDING

```
int MCBlock::DUMMYPADDING
```

Definition at line 940 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.62 MCBridge Class Reference

### Data Fields

- [Edge2n](#) [nodes](#)
- [int](#) [next](#)

### 3.62.1 Detailed Description

Definition at line 921 of file [grcc.h](#).

### 3.62.2 Constructor & Destructor Documentation

#### 3.62.2.1 MCBridge()

```
MCBridge::MCBridge (  
    void )
```

Definition at line 5094 of file [grcc.cc](#).

#### 3.62.2.2 ~MCBridge()

```
MCBridge::~~MCBridge (  
    void )
```

Definition at line 5099 of file [grcc.cc](#).

### 3.62.3 Field Documentation

#### 3.62.3.1 nodes

```
Edge2n MCBridge::nodes
```

Definition at line 923 of file [grcc.h](#).

### 3.62.3.2 next

```
int MCBridge::next
```

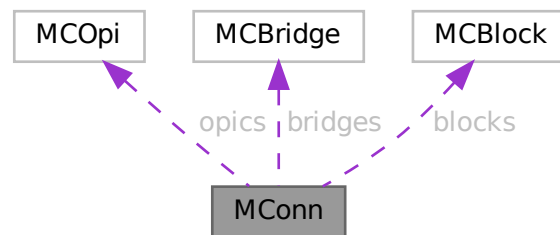
Definition at line 924 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.63 MConn Class Reference

Collaboration diagram for MConn:



### Public Member Functions

- [MConn](#) (int nnod, int nedg)
- void [init](#) (void)
- void [pushNode](#) (int nd)
- void [pushEdge](#) (int n0, int n1)
- void [addOPlc](#) ([MCOpI](#) \*mopi, int stp)
- void [addBridge](#) (int n0, int n1, int nex, int nextot)
- void [addArtic](#) (int nd, int mul)
- void [addBlock](#) ([MCBlock](#) \*eblk, int stp)
- void [addBlockSelf](#) (int nd, int mul)
- void [print](#) (void)

**Data Fields**

- [MCOpi](#) \* [opics](#)
- [MCBridge](#) \* [bridges](#)
- [MCBlock](#) \* [blocks](#)
- int \* [articuls](#)
- int \* [opisp](#)
- int \* [opistk](#)
- [Edge2n](#) \* [blksp](#)
- [Edge2n](#) \* [blkstk](#)
- int [snodes](#)
- int [sedges](#)
- int [nopic](#)
- int [nlpopic](#)
- int [nctopic](#)
- int [nbridges](#)
- int [ne0bridges](#)
- int [ne1bridges](#)
- int [nselfloops](#)
- int [nblocks](#)
- int [na1blocks](#)
- int [narticuls](#)
- int [neblocks](#)
- int [nopisp](#)
- int [opistkptr](#)
- int [nblksp](#)
- int [blkstkptr](#)
- int [DUMMY\\_PADDING](#)

**3.63.1 Detailed Description**

Definition at line [950](#) of file [grcc.h](#).

**3.63.2 Constructor & Destructor Documentation****3.63.2.1 MConn()**

```
MConn::MConn (
    int nnod,
    int nedg )
```

Definition at line [5128](#) of file [grcc.cc](#).

**3.63.2.2 ~MConn()**

```
MConn::~MConn (
    void )
```

Definition at line [5147](#) of file [grcc.cc](#).

### 3.63.3 Member Function Documentation

#### 3.63.3.1 init()

```
void MConn::init (
    void )
```

Definition at line 5161 of file [grcc.cc](#).

#### 3.63.3.2 pushNode()

```
void MConn::pushNode (
    int nd )
```

Definition at line 5189 of file [grcc.cc](#).

#### 3.63.3.3 pushEdge()

```
void MConn::pushEdge (
    int n0,
    int n1 )
```

Definition at line 5198 of file [grcc.cc](#).

#### 3.63.3.4 addOPic()

```
void MConn::addOPic (
    MCOpi * mopi,
    int stp )
```

Definition at line 5214 of file [grcc.cc](#).

#### 3.63.3.5 addBridge()

```
void MConn::addBridge (
    int n0,
    int n1,
    int nex,
    int nextot )
```

Definition at line 5240 of file [grcc.cc](#).

#### 3.63.3.6 addArtic()

```
void MConn::addArtic (
    int nd,
    int mul )
```

Definition at line 5257 of file [grcc.cc](#).



### 3.63.3.7 addBlock()

```
void MConn::addBlock (
    MCBLOCK * eblk,
    int stp )
```

Definition at line 5268 of file [grcc.cc](#).

### 3.63.3.8 addBlockSelf()

```
void MConn::addBlockSelf (
    int nd,
    int mul )
```

Definition at line 5292 of file [grcc.cc](#).

### 3.63.3.9 print()

```
void MConn::print (
    void )
```

Definition at line 5310 of file [grcc.cc](#).

## 3.63.4 Field Documentation

### 3.63.4.1 opics

```
MCOPi* MConn::opics
```

Definition at line 953 of file [grcc.h](#).

### 3.63.4.2 bridges

```
MCBridge* MConn::bridges
```

Definition at line 954 of file [grcc.h](#).

### 3.63.4.3 blocks

```
MCBlock* MConn::blocks
```

Definition at line 955 of file [grcc.h](#).

### 3.63.4.4 articuls

```
int* MConn::articuls
```

Definition at line 956 of file [grcc.h](#).

#### 3.63.4.5 opisp

```
int* MConn::opisp
```

Definition at line 958 of file [grcc.h](#).

#### 3.63.4.6 opistk

```
int* MConn::opistk
```

Definition at line 959 of file [grcc.h](#).

#### 3.63.4.7 blksp

```
Edge2n* MConn::blksp
```

Definition at line 960 of file [grcc.h](#).

#### 3.63.4.8 blkstk

```
Edge2n* MConn::blkstk
```

Definition at line 961 of file [grcc.h](#).

#### 3.63.4.9 snodes

```
int MConn::snodes
```

Definition at line 962 of file [grcc.h](#).

#### 3.63.4.10 sedges

```
int MConn::sedges
```

Definition at line 963 of file [grcc.h](#).

#### 3.63.4.11 nopic

```
int MConn::nopic
```

Definition at line 966 of file [grcc.h](#).

#### 3.63.4.12 nlpopic

```
int MConn::nlpopic
```

Definition at line 967 of file [grcc.h](#).

#### 3.63.4.13 nctopic

```
int MConn::nctopic
```

Definition at line 968 of file [grcc.h](#).

#### 3.63.4.14 nbridges

```
int MConn::nbridges
```

Definition at line 969 of file [grcc.h](#).

#### 3.63.4.15 ne0bridges

```
int MConn::ne0bridges
```

Definition at line 970 of file [grcc.h](#).

#### 3.63.4.16 ne1bridges

```
int MConn::ne1bridges
```

Definition at line 971 of file [grcc.h](#).

#### 3.63.4.17 nselfloops

```
int MConn::nselfloops
```

Definition at line 972 of file [grcc.h](#).

#### 3.63.4.18 nblocks

```
int MConn::nblocks
```

Definition at line 975 of file [grcc.h](#).

#### 3.63.4.19 na1blocks

```
int MConn::na1blocks
```

Definition at line 976 of file [grcc.h](#).

#### 3.63.4.20 narticuls

```
int MConn::narticuls
```

Definition at line 977 of file [grcc.h](#).

#### 3.63.4.21 neblocks

```
int MConn::neblocks
```

Definition at line 978 of file [grcc.h](#).

#### 3.63.4.22 nopisp

```
int MConn::nopisp
```

Definition at line 981 of file [grcc.h](#).

#### 3.63.4.23 opistkptr

```
int MConn::opistkptr
```

Definition at line 982 of file [grcc.h](#).

#### 3.63.4.24 nblksp

```
int MConn::nblksp
```

Definition at line 983 of file [grcc.h](#).

#### 3.63.4.25 blkstkptr

```
int MConn::blkstkptr
```

Definition at line 984 of file [grcc.h](#).

#### 3.63.4.26 DUMMYPADDING

```
int MConn::DUMMYPADDING
```

Definition at line 986 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.64 MCOpi Class Reference

### Public Member Functions

- void [init](#) (void)

## Data Fields

- int \* [nodes](#)
- int [nnodes](#)
- int [nlegs](#)
- int [next](#)
- int [nedges](#)
- int [loop](#)
- int [ctloop](#)

### 3.64.1 Detailed Description

Definition at line [902](#) of file [grcc.h](#).

### 3.64.2 Constructor & Destructor Documentation

#### 3.64.2.1 MCOpI()

```
MCOpI::MCOpI (  
    void )
```

Definition at line [5067](#) of file [grcc.cc](#).

#### 3.64.2.2 ~MCOpI()

```
MCOpI::~~MCOpI (  
    void )
```

Definition at line [5073](#) of file [grcc.cc](#).

### 3.64.3 Member Function Documentation

#### 3.64.3.1 init()

```
void MCOpI::init (  
    void )
```

Definition at line [5079](#) of file [grcc.cc](#).

### 3.64.4 Field Documentation

#### 3.64.4.1 nodes

```
int* MCOpI::nodes
```

Definition at line [904](#) of file [grcc.h](#).

#### 3.64.4.2 nnodes

```
int MCOpi::nnodes
```

Definition at line 905 of file [grcc.h](#).

#### 3.64.4.3 nlegs

```
int MCOpi::nlegs
```

Definition at line 906 of file [grcc.h](#).

#### 3.64.4.4 next

```
int MCOpi::next
```

Definition at line 907 of file [grcc.h](#).

#### 3.64.4.5 nedges

```
int MCOpi::nedges
```

Definition at line 908 of file [grcc.h](#).

#### 3.64.4.6 loop

```
int MCOpi::loop
```

Definition at line 909 of file [grcc.h](#).

#### 3.64.4.7 ctloop

```
int MCOpi::ctloop
```

Definition at line 910 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)



- int [nEdges](#)
- int [nLoops](#)
- int [nExtern](#)
- int [nClasses](#)
- int [mindeg](#)
- int [maxdeg](#)
- int \*\* [adjMat](#)
- [MNodeClass](#) \* [curcl](#)
- [EGraph](#) \* [egraph](#)
- [BigInt](#) [nsym](#)
- [BigInt](#) [esym](#)
- [BigInt](#) [cDiag](#)
- [BigInt](#) [c1PI](#)
- [BigInt](#) [cNoTadpole](#)
- [BigInt](#) [cNoTadBlock](#)
- [BigInt](#) [c1PINoTadBlock](#)
- [Fraction](#) [wscon](#)
- [Fraction](#) [wsopi](#)
- [BigInt](#) [ngen](#)
- [BigInt](#) [ngconn](#)
- [BigInt](#) [nCallRefine](#)
- [BigInt](#) [discardRefine](#)
- [BigInt](#) [discardDisc](#)
- [BigInt](#) [discardIso](#)
- [Bool](#) [opi](#)
- [Bool](#) [opiloop](#)
- [Bool](#) [extself](#)
- [Bool](#) [selfloop](#)
- [Bool](#) [tadpole](#)
- [Bool](#) [tadblock](#)
- [Bool](#) [block](#)
- int [DUMMYPADDING1](#)
- [MConn](#) \* [mconn](#)
- int \*\* [modmat](#)
- int \* [bidef](#)
- int \* [bilow](#)
- int [bicount](#)
- int [DUMMYPADDING2](#)

### 3.65.1 Detailed Description

Definition at line [761](#) of file [grcc.h](#).

### 3.65.2 Constructor & Destructor Documentation

#### 3.65.2.1 MGraph() [1/2]

```
MGraph::MGraph (
    int pid,
    int ncl,
    NCInput * mg,
    Options * opt )
```

Definition at line [3732](#) of file [grcc.cc](#).



### 3.65.2.2 MGraph() [2/2]

```
MGraph::MGraph (
    int pid,
    int ncl,
    int * cldeg,
    int * clnum,
    int * clexl,
    int * cmind,
    int * cmaxd,
    Options * opt )
```

Definition at line 3649 of file [grcc.cc](#).

### 3.65.2.3 ~MGraph()

```
MGraph::~MGraph (
    void )
```

Definition at line 3860 of file [grcc.cc](#).

## 3.65.3 Member Function Documentation

### 3.65.3.1 init()

```
void MGraph::init (
    void )
```

Definition at line 3817 of file [grcc.cc](#).

### 3.65.3.2 generate()

```
BigInt MGraph::generate (
    void )
```

Definition at line 4479 of file [grcc.cc](#).

### 3.65.3.3 isExternal()

```
Bool MGraph::isExternal (
    int nd ) [inline]
```

Definition at line 841 of file [grcc.h](#).

### 3.65.3.4 printAdjMat()

```
void MGraph::printAdjMat (
    MNodeClass * cl )
```

Definition at line 3889 of file [grcc.cc](#).

#### 3.65.3.5 print()

```
void MGraph::print (
    void )
```

Definition at line [3908](#) of file [grcc.cc](#).

#### 3.65.3.6 isConnected()

```
Bool MGraph::isConnected (
    void )
```

Definition at line [3932](#) of file [grcc.cc](#).

#### 3.65.3.7 visit()

```
Bool MGraph::visit (
    int nd )
```

Definition at line [3956](#) of file [grcc.cc](#).

#### 3.65.3.8 isIsomorphic()

```
Bool MGraph::isIsomorphic (
    MNodeClass * cl )
```

Definition at line [3981](#) of file [grcc.cc](#).

#### 3.65.3.9 permMat()

```
void MGraph::permMat (
    int size,
    int * perm,
    int ** mat0,
    int ** mat1 )
```

Definition at line [4039](#) of file [grcc.cc](#).

#### 3.65.3.10 compMat()

```
int MGraph::compMat (
    int size,
    int ** mat0,
    int ** mat1 )
```

Definition at line [4053](#) of file [grcc.cc](#).

### 3.65.3.11 refineClass()

```
MNodeClass * MGraph::refineClass (
    MNodeClass * cl )
```

Definition at line 4071 of file [grcc.cc](#).

### 3.65.3.12 bisearchME()

```
void MGraph::bisearchME (
    int nd,
    int pd,
    int ned,
    MCOpi * mopi,
    MCBLOCK * mblk,
    int * next,
    int * nart )
```

Definition at line 4239 of file [grcc.cc](#).

### 3.65.3.13 biconnME()

```
void MGraph::biconnME (
    void )
```

Definition at line 4153 of file [grcc.cc](#).

### 3.65.3.14 isOptM()

```
Bool MGraph::isOptM (
    void )
```

Definition at line 4731 of file [grcc.cc](#).

### 3.65.3.15 connectClass()

```
void MGraph::connectClass (
    MNodeClass * cl )
```

Definition at line 4505 of file [grcc.cc](#).

### 3.65.3.16 connectNode()

```
void MGraph::connectNode (
    int sc,
    int ss,
    MNodeClass * cl )
```

Definition at line 4537 of file [grcc.cc](#).

### 3.65.3.17 connectLeg()

```
void MGraph::connectLeg (
    int sc,
    int sn,
    int tc,
    int ts,
    MNodeClass * cl )
```

Definition at line [4558](#) of file [grcc.cc](#).

### 3.65.3.18 newGraph()

```
void MGraph::newGraph (
    MNodeClass * cl )
```

Definition at line [4815](#) of file [grcc.cc](#).

## 3.65.4 Field Documentation

### 3.65.4.1 opt

[Options\\*](#) MGraph::opt

Definition at line [766](#) of file [grcc.h](#).

### 3.65.4.2 group

[SGroup\\*](#) MGraph::group

Definition at line [769](#) of file [grcc.h](#).

### 3.65.4.3 orbits

[MOrbits\\*](#) MGraph::orbits

Definition at line [770](#) of file [grcc.h](#).

### 3.65.4.4 nodes

[MNode\\*\\*](#) MGraph::nodes

Definition at line [772](#) of file [grcc.h](#).

#### 3.65.4.5 mId

```
BigInt MGraph::mId
```

Definition at line 773 of file [grcc.h](#).

#### 3.65.4.6 clist

```
int* MGraph::clist
```

Definition at line 774 of file [grcc.h](#).

#### 3.65.4.7 pId

```
int MGraph::pId
```

Definition at line 775 of file [grcc.h](#).

#### 3.65.4.8 nNodes

```
int MGraph::nNodes
```

Definition at line 776 of file [grcc.h](#).

#### 3.65.4.9 nEdges

```
int MGraph::nEdges
```

Definition at line 777 of file [grcc.h](#).

#### 3.65.4.10 nLoops

```
int MGraph::nLoops
```

Definition at line 778 of file [grcc.h](#).

#### 3.65.4.11 nExtern

```
int MGraph::nExtern
```

Definition at line 779 of file [grcc.h](#).

#### 3.65.4.12 nClasses

```
int MGraph::nClasses
```

Definition at line 780 of file [grcc.h](#).

#### 3.65.4.13 mindeg

```
int MGraph::mindeg
```

Definition at line 781 of file [grcc.h](#).

#### 3.65.4.14 maxdeg

```
int MGraph::maxdeg
```

Definition at line 782 of file [grcc.h](#).

#### 3.65.4.15 adjMat

```
int** MGraph::adjMat
```

Definition at line 785 of file [grcc.h](#).

#### 3.65.4.16 curcl

```
MNodeClass* MGraph::curcl
```

Definition at line 786 of file [grcc.h](#).

#### 3.65.4.17 egraph

```
EGraph* MGraph::egraph
```

Definition at line 787 of file [grcc.h](#).

#### 3.65.4.18 nsym

```
BigInt MGraph::nsym
```

Definition at line 788 of file [grcc.h](#).

#### 3.65.4.19 esym

```
BigInt MGraph::esym
```

Definition at line 789 of file [grcc.h](#).

#### 3.65.4.20 cDiag

```
BigInt MGraph::cDiag
```

Definition at line 792 of file [grcc.h](#).

#### 3.65.4.21 c1PI

`BigInt MGraph::c1PI`

Definition at line 793 of file [grcc.h](#).

#### 3.65.4.22 cNoTadpole

`BigInt MGraph::cNoTadpole`

Definition at line 794 of file [grcc.h](#).

#### 3.65.4.23 cNoTadBlock

`BigInt MGraph::cNoTadBlock`

Definition at line 795 of file [grcc.h](#).

#### 3.65.4.24 c1PINoTadBlock

`BigInt MGraph::c1PINoTadBlock`

Definition at line 796 of file [grcc.h](#).

#### 3.65.4.25 wscon

`Fraction MGraph::wscon`

Definition at line 797 of file [grcc.h](#).

#### 3.65.4.26 wsopi

`Fraction MGraph::wsopi`

Definition at line 798 of file [grcc.h](#).

#### 3.65.4.27 ngen

`BigInt MGraph::ngen`

Definition at line 801 of file [grcc.h](#).

#### 3.65.4.28 ngconn

`BigInt MGraph::ngconn`

Definition at line 802 of file [grcc.h](#).

#### 3.65.4.29 nCallRefine

`BigInt MGraph::nCallRefine`

Definition at line 804 of file [grcc.h](#).

#### 3.65.4.30 discardRefine

`BigInt MGraph::discardRefine`

Definition at line 805 of file [grcc.h](#).

#### 3.65.4.31 discardDisc

`BigInt MGraph::discardDisc`

Definition at line 806 of file [grcc.h](#).

#### 3.65.4.32 discardIso

`BigInt MGraph::discardIso`

Definition at line 807 of file [grcc.h](#).

#### 3.65.4.33 opi

`Bool MGraph::opi`

Definition at line 810 of file [grcc.h](#).

#### 3.65.4.34 opiloop

`Bool MGraph::opiloop`

Definition at line 811 of file [grcc.h](#).

#### 3.65.4.35 extself

`Bool MGraph::extself`

Definition at line 812 of file [grcc.h](#).

#### 3.65.4.36 selfloop

`Bool MGraph::selfloop`

Definition at line 813 of file [grcc.h](#).



#### 3.65.4.37 tadpole

```
Bool MGraph::tadpole
```

Definition at line 814 of file [grcc.h](#).

#### 3.65.4.38 tadbblock

```
Bool MGraph::tadbblock
```

Definition at line 815 of file [grcc.h](#).

#### 3.65.4.39 block

```
Bool MGraph::block
```

Definition at line 816 of file [grcc.h](#).

#### 3.65.4.40 DUMMYPADDING1

```
int MGraph::DUMMYPADDING1
```

Definition at line 818 of file [grcc.h](#).

#### 3.65.4.41 mconn

```
MConn* MGraph::mconn
```

Definition at line 821 of file [grcc.h](#).

#### 3.65.4.42 modmat

```
int** MGraph::modmat
```

Definition at line 824 of file [grcc.h](#).

#### 3.65.4.43 bidef

```
int* MGraph::bidef
```

Definition at line 827 of file [grcc.h](#).

#### 3.65.4.44 bilow

```
int* MGraph::bilow
```

Definition at line 828 of file [grcc.h](#).

#### 3.65.4.45 bicount

```
int MGraph::bicount
```

Definition at line 829 of file [grcc.h](#).

#### 3.65.4.46 DUMMYPADDING2

```
int MGraph::DUMMYPADDING2
```

Definition at line 831 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.66 MiNmAx Struct Reference

### Data Fields

- WORD [mini](#)
- WORD [maxi](#)
- WORD [size](#)

### 3.66.1 Detailed Description

Definition at line 307 of file [structs.h](#).

### 3.66.2 Field Documentation

#### 3.66.2.1 mini

```
WORD MiNmAx::mini
```

Minimum value

Definition at line 308 of file [structs.h](#).

#### 3.66.2.2 maxi

```
WORD MiNmAx::maxi
```

Maximum value

Definition at line 309 of file [structs.h](#).

### 3.66.2.3 size

```
WORD MinMax::size
```

Value of one unit in this position.

Definition at line 310 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.67 MInput Struct Reference

### Data Fields

- const char \* [name](#)
- int [defpart](#)
- int [ncouple](#)
- const char \* [cnamlist](#) [GRCC\_MAXNCPLG]

### 3.67.1 Detailed Description

Definition at line 119 of file [grccparam.h](#).

### 3.67.2 Field Documentation

#### 3.67.2.1 name

```
const char* MInput::name
```

Definition at line 120 of file [grccparam.h](#).

#### 3.67.2.2 defpart

```
int MInput::defpart
```

Definition at line 121 of file [grccparam.h](#).

#### 3.67.2.3 ncouple

```
int MInput::ncouple
```

Definition at line 123 of file [grccparam.h](#).

### 3.67.2.4 cnamlist

```
const char* MInput::cnamlist[GRCC_MAXNCPLG]
```

Definition at line 124 of file [grccparam.h](#).

The documentation for this struct was generated from the following file:

- [grccparam.h](#)

## 3.68 MNode Class Reference

### Public Member Functions

- [MNode](#) (int id, int clss, [NCInput](#) \*mgi)
- [MNode](#) (int id, int deg, int extloop, int clss, int cmind, int cmaxd)

### Data Fields

- int [id](#)
- int [deg](#)
- int [clss](#)
- int [extloop](#)
- int [cminddeg](#)
- int [cmaxdeg](#)
- int [freelg](#)
- int [visited](#)

### 3.68.1 Detailed Description

Definition at line 739 of file [grcc.h](#).

### 3.68.2 Constructor & Destructor Documentation

#### 3.68.2.1 MNode() [1/2]

```
MNode::MNode (
    int id,
    int clss,
    NCInput * mgi )
```

Definition at line 3605 of file [grcc.cc](#).

### 3.68.2.2 MNode() [2/2]

```
MNode::MNode (
    int id,
    int deg,
    int extloop,
    int cls,
    int cmin,
    int cmax )
```

Definition at line 3626 of file [grcc.cc](#).

## 3.68.3 Field Documentation

### 3.68.3.1 id

```
int MNode::id
```

Definition at line 741 of file [grcc.h](#).

### 3.68.3.2 deg

```
int MNode::deg
```

Definition at line 742 of file [grcc.h](#).

### 3.68.3.3 cls

```
int MNode::cls
```

Definition at line 743 of file [grcc.h](#).

### 3.68.3.4 extloop

```
int MNode::extloop
```

Definition at line 744 of file [grcc.h](#).

### 3.68.3.5 cmindeg

```
int MNode::cmindeg
```

Definition at line 745 of file [grcc.h](#).

### 3.68.3.6 cmaxdeg

```
int MNode::cmaxdeg
```

Definition at line 746 of file [grcc.h](#).

### 3.68.3.7 freelg

```
int MNode::freelg
```

Definition at line 748 of file [grcc.h](#).

### 3.68.3.8 visited

```
int MNode::visited
```

Definition at line 749 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.69 MNodeClass Class Reference

### Public Member Functions

- [MNodeClass](#) (int nnodes, int nclasses)
- void [init](#) (int \*cl, int mxdeg, int \*\*adjmat)
- void [copy](#) ([MNodeClass](#) \*mnc)
- int [clCmp](#) (int nd0, int nd1, int cn)
- void [printMat](#) (void)
- void [mkFlist](#) (void)
- void [mkNdCl](#) (void)
- void [mkClMat](#) (int \*\*adjmat)
- void [incMat](#) (int nd, int td, int val)
- int [cmpMNCArray](#) (int \*a0, int \*a1, int ma)
- void [reorder](#) ([MGraph](#) \*mg)

### Data Fields

- int [clmat](#) [GRCC\_MAXNODES][GRCC\_MAXNODES]
- int [clist](#) [GRCC\_MAXNODES]
- int [ndcl](#) [GRCC\_MAXNODES]
- int [flist](#) [GRCC\_MAXNODES+1]
- int [clord](#) [GRCC\_MAXNODES]
- int [cmindeg](#) [GRCC\_MAXNODES]
- int [cmaxdeg](#) [GRCC\_MAXNODES]
- int [nNodes](#)
- int [nClasses](#)
- int [maxdeg](#)
- int [flg0](#)
- int [flg1](#)
- int [flg2](#)

### 3.69.1 Detailed Description

Definition at line 865 of file [grcc.h](#).

### 3.69.2 Constructor & Destructor Documentation

#### 3.69.2.1 MNodeClass()

```
MNodeClass::MNodeClass (
    int nnodes,
    int nclasses )
```

Definition at line 3358 of file [grcc.cc](#).

#### 3.69.2.2 ~MNodeClass()

```
MNodeClass::~MNodeClass (
    void )
```

Definition at line 3387 of file [grcc.cc](#).

### 3.69.3 Member Function Documentation

#### 3.69.3.1 init()

```
void MNodeClass::init (
    int * cl,
    int mxdeg,
    int ** adjmat )
```

Definition at line 3394 of file [grcc.cc](#).

#### 3.69.3.2 copy()

```
void MNodeClass::copy (
    MNodeClass * mnc )
```

Definition at line 3418 of file [grcc.cc](#).

#### 3.69.3.3 clCmp()

```
int MNodeClass::clCmp (
    int nd0,
    int nd1,
    int cn )
```

Definition at line 3474 of file [grcc.cc](#).

#### 3.69.3.4 printMat()

```
void MNodeClass::printMat (
    void )
```

Definition at line [3525](#) of file [grcc.cc](#).

#### 3.69.3.5 mkFlist()

```
void MNodeClass::mkFlist (
    void )
```

Definition at line [3442](#) of file [grcc.cc](#).

#### 3.69.3.6 mkNdCl()

```
void MNodeClass::mkNdCl (
    void )
```

Definition at line [3458](#) of file [grcc.cc](#).

#### 3.69.3.7 mkClMat()

```
void MNodeClass::mkClMat (
    int ** adjmat )
```

Definition at line [3496](#) of file [grcc.cc](#).

#### 3.69.3.8 incMat()

```
void MNodeClass::incMat (
    int nd,
    int td,
    int val )
```

Definition at line [3516](#) of file [grcc.cc](#).

#### 3.69.3.9 cmpMNCArray()

```
int MNodeClass::cmpMNCArray (
    int * a0,
    int * a1,
    int ma )
```

Definition at line [3556](#) of file [grcc.cc](#).



### 3.69.3.10 reorder()

```
void MNodeClass::reorder (
    MGraph * mg )
```

Definition at line 3572 of file [grcc.cc](#).

## 3.69.4 Field Documentation

### 3.69.4.1 clmat

```
int MNodeClass::clmat[GRCC_MAXNODES][GRCC_MAXNODES]
```

Definition at line 867 of file [grcc.h](#).

### 3.69.4.2 clist

```
int MNodeClass::clist[GRCC_MAXNODES]
```

Definition at line 868 of file [grcc.h](#).

### 3.69.4.3 ndcl

```
int MNodeClass::ndcl[GRCC_MAXNODES]
```

Definition at line 869 of file [grcc.h](#).

### 3.69.4.4 flist

```
int MNodeClass::flist[GRCC_MAXNODES+1]
```

Definition at line 870 of file [grcc.h](#).

### 3.69.4.5 clord

```
int MNodeClass::clord[GRCC_MAXNODES]
```

Definition at line 871 of file [grcc.h](#).

### 3.69.4.6 cmindeg

```
int MNodeClass::cmindeg[GRCC_MAXNODES]
```

Definition at line 872 of file [grcc.h](#).

#### 3.69.4.7 cmaxdeg

```
int MNodeClass::cmaxdeg[GRCC_MAXNODES]
```

Definition at line 873 of file [grcc.h](#).

#### 3.69.4.8 nNodes

```
int MNodeClass::nNodes
```

Definition at line 874 of file [grcc.h](#).

#### 3.69.4.9 nClasses

```
int MNodeClass::nClasses
```

Definition at line 875 of file [grcc.h](#).

#### 3.69.4.10 maxdeg

```
int MNodeClass::maxdeg
```

Definition at line 876 of file [grcc.h](#).

#### 3.69.4.11 flg0

```
int MNodeClass::flg0
```

Definition at line 878 of file [grcc.h](#).

#### 3.69.4.12 flg1

```
int MNodeClass::flg1
```

Definition at line 879 of file [grcc.h](#).

#### 3.69.4.13 flg2

```
int MNodeClass::flg2
```

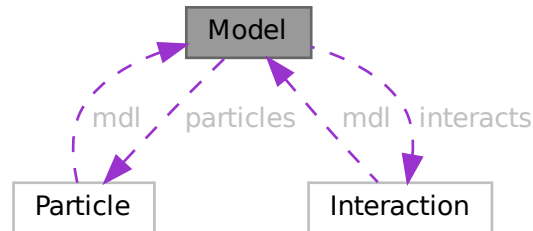
Definition at line 880 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.70 Model Class Reference

Collaboration diagram for Model:



### Public Member Functions

- [Model](#) ([MInput](#) \*minp)
- void [prModel](#) (void)
- void [addParticle](#) ([PInput](#) \*pinp)
- void [addParticleEnd](#) (void)
- void [addInteraction](#) ([IInput](#) \*iinp)
- void [addInteractionEnd](#) (void)
- int [findParticleName](#) (const char \*name)
- int [findParticleCode](#) (int pcd)
- int [findInteractionName](#) (const char \*name)
- int [findInteractionCode](#) (int icd)
- char \* [particleName](#) (int p)
- int [particleCode](#) (int p)
- int [normalParticle](#) (int pt)
- int [antiParticle](#) (int pt)
- int \* [allParticles](#) (int \*len)
- int [findMClass](#) (const int cpl, const int dgr)
- void [prParticleArray](#) (int n, int \*a, const char \*msg)

### Data Fields

- char \* [name](#)
- char \*\* [cnlist](#)
- int [ncouple](#)
- int [nParticles](#)
- [Particle](#) \*\* [particles](#)
- int [pdef](#)
- int [nallPart](#)
- int [allPart](#) [GRCC\_MAXMPARTICLES2]
- int [nintPart](#)
- int [intPart](#) [GRCC\_MAXMPARTICLES2]
- int [nInteracts](#)

- [Interaction](#) \*\* [interacts](#)
- int [vdef](#)
- int [maxnlegs](#)
- int [maxcpl](#)
- int [maxloop](#)
- int \* [cplgcp](#)
- int \* [cplglg](#)
- int \* [cplgnvl](#)
- int \*\* [cplgvl](#)
- int [ncplgcp](#)
- int [defpart](#)
- int [skipFLine](#)
- int [DUMMYPADDING](#)

### 3.70.1 Detailed Description

Definition at line [262](#) of file [grcc.h](#).

### 3.70.2 Constructor & Destructor Documentation

#### 3.70.2.1 Model()

```
Model::Model (
    MInput * minp )
```

Definition at line [1726](#) of file [grcc.cc](#).

#### 3.70.2.2 ~Model()

```
Model::~~Model (
    void )
```

Definition at line [1787](#) of file [grcc.cc](#).

### 3.70.3 Member Function Documentation

#### 3.70.3.1 prModel()

```
void Model::prModel (
    void )
```

Definition at line [1827](#) of file [grcc.cc](#).

#### 3.70.3.2 addParticle()

```
void Model::addParticle (
    PInput * pinp )
```

Definition at line [1876](#) of file [grcc.cc](#).

### 3.70.3.3 addParticleEnd()

```
void Model::addParticleEnd (
    void )
```

Definition at line 1918 of file [grcc.cc](#).

### 3.70.3.4 addInteraction()

```
void Model::addInteraction (
    IInput * iinp )
```

Definition at line 1956 of file [grcc.cc](#).

### 3.70.3.5 addInteractionEnd()

```
void Model::addInteractionEnd (
    void )
```

Definition at line 2084 of file [grcc.cc](#).

### 3.70.3.6 findParticleName()

```
int Model::findParticleName (
    const char * name )
```

Definition at line 2157 of file [grcc.cc](#).

### 3.70.3.7 findParticleCode()

```
int Model::findParticleCode (
    int pcd )
```

Definition at line 2175 of file [grcc.cc](#).

### 3.70.3.8 findInteractionName()

```
int Model::findInteractionName (
    const char * name )
```

Definition at line 2294 of file [grcc.cc](#).

### 3.70.3.9 findInteractionCode()

```
int Model::findInteractionCode (
    int icd )
```

Definition at line 2307 of file [grcc.cc](#).

#### 3.70.3.10 `particleName()`

```
char * Model::particleName (  
    int p )
```

Definition at line [2190](#) of file [grcc.cc](#).

#### 3.70.3.11 `particleCode()`

```
int Model::particleCode (  
    int p )
```

Definition at line [2210](#) of file [grcc.cc](#).

#### 3.70.3.12 `normalParticle()`

```
int Model::normalParticle (  
    int pt )
```

Definition at line [2256](#) of file [grcc.cc](#).

#### 3.70.3.13 `antiParticle()`

```
int Model::antiParticle (  
    int pt )
```

Definition at line [2273](#) of file [grcc.cc](#).

#### 3.70.3.14 `allParticles()`

```
int * Model::allParticles (  
    int * len )
```

Definition at line [2287](#) of file [grcc.cc](#).

#### 3.70.3.15 `findMClass()`

```
int Model::findMClass (  
    const int cpl,  
    const int dgr )
```

Definition at line [2243](#) of file [grcc.cc](#).

#### 3.70.3.16 `prParticleArray()`

```
void Model::prParticleArray (  
    int n,  
    int * a,  
    const char * msg )
```

Definition at line [2228](#) of file [grcc.cc](#).

## 3.70.4 Field Documentation

### 3.70.4.1 name

```
char* Model::name
```

Definition at line 264 of file [grcc.h](#).

### 3.70.4.2 cnlist

```
char** Model::cnlist
```

Definition at line 265 of file [grcc.h](#).

### 3.70.4.3 ncouple

```
int Model::ncouple
```

Definition at line 266 of file [grcc.h](#).

### 3.70.4.4 nParticles

```
int Model::nParticles
```

Definition at line 268 of file [grcc.h](#).

### 3.70.4.5 particles

```
Particle** Model::particles
```

Definition at line 269 of file [grcc.h](#).

### 3.70.4.6 pdef

```
int Model::pdef
```

Definition at line 270 of file [grcc.h](#).

### 3.70.4.7 nallPart

```
int Model::nallPart
```

Definition at line 273 of file [grcc.h](#).

#### 3.70.4.8 allPart

```
int Model::allPart[GRCC_MAXMPARTICLES2]
```

Definition at line 274 of file [grcc.h](#).

#### 3.70.4.9 nintPart

```
int Model::nintPart
```

Definition at line 277 of file [grcc.h](#).

#### 3.70.4.10 intPart

```
int Model::intPart[GRCC_MAXMPARTICLES2]
```

Definition at line 278 of file [grcc.h](#).

#### 3.70.4.11 nInteracts

```
int Model::nInteracts
```

Definition at line 280 of file [grcc.h](#).

#### 3.70.4.12 interacts

```
Interaction** Model::interacts
```

Definition at line 281 of file [grcc.h](#).

#### 3.70.4.13 vdef

```
int Model::vdef
```

Definition at line 282 of file [grcc.h](#).

#### 3.70.4.14 maxnlegs

```
int Model::maxnlegs
```

Definition at line 284 of file [grcc.h](#).

#### 3.70.4.15 maxcpl

```
int Model::maxcpl
```

Definition at line 285 of file [grcc.h](#).



**3.70.4.16 maxloop**

```
int Model::maxloop
```

Definition at line 286 of file [grcc.h](#).

**3.70.4.17 cplgcp**

```
int* Model::cplgcp
```

Definition at line 289 of file [grcc.h](#).

**3.70.4.18 cplglg**

```
int* Model::cplglg
```

Definition at line 290 of file [grcc.h](#).

**3.70.4.19 cplgnvl**

```
int* Model::cplgnvl
```

Definition at line 291 of file [grcc.h](#).

**3.70.4.20 cplgvl**

```
int** Model::cplgvl
```

Definition at line 292 of file [grcc.h](#).

**3.70.4.21 ncplgcp**

```
int Model::ncplgcp
```

Definition at line 293 of file [grcc.h](#).

**3.70.4.22 defpart**

```
int Model::defpart
```

Definition at line 295 of file [grcc.h](#).

**3.70.4.23 skipFLine**

```
int Model::skipFLine
```

Definition at line 296 of file [grcc.h](#).

### 3.70.4.24 DUMMYPADDING

```
int Model::DUMMYPADDING
```

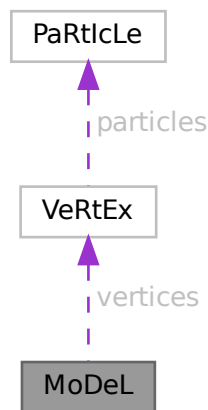
Definition at line 298 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.71 MoDeL Struct Reference

Collaboration diagram for MoDeL:



### Data Fields

- [VERTEX](#) \*\* [vertices](#)
- WORD \* [couplings](#)
- UBYTE \* [name](#)
- void \* [grccmodel](#)
- WORD [legcouple](#) [MAXLEGS+2]
- WORD [nparticles](#)
- WORD [nvertices](#)
- WORD [invertices](#)
- WORD [sizevertices](#)
- WORD [sizecouplings](#)
- WORD [ncouplings](#)
- WORD [error](#)
- WORD [dummy](#)

### 3.71.1 Detailed Description

Definition at line 637 of file [structs.h](#).

### 3.71.2 Field Documentation

#### 3.71.2.1 vertices

```
VERTEX** MoDeL::vertices
```

Definition at line 638 of file [structs.h](#).

#### 3.71.2.2 couplings

```
WORD* MoDeL::couplings
```

Definition at line 639 of file [structs.h](#).

#### 3.71.2.3 name

```
UBYTE* MoDeL::name
```

Definition at line 640 of file [structs.h](#).

#### 3.71.2.4 grccmodel

```
void* MoDeL::grccmodel
```

Definition at line 641 of file [structs.h](#).

#### 3.71.2.5 legcouple

```
WORD MoDeL::legcouple[MAXLEGS+2]
```

Definition at line 642 of file [structs.h](#).

#### 3.71.2.6 nparticles

```
WORD MoDeL::nparticles
```

Definition at line 643 of file [structs.h](#).

### 3.71.2.7 nvertices

WORD MoDeL::nvertices

Definition at line 644 of file [structs.h](#).

### 3.71.2.8 invertices

WORD MoDeL::invertices

Definition at line 645 of file [structs.h](#).

### 3.71.2.9 sizevertices

WORD MoDeL::sizevertices

Definition at line 646 of file [structs.h](#).

### 3.71.2.10 sizecouplings

WORD MoDeL::sizecouplings

Definition at line 647 of file [structs.h](#).

### 3.71.2.11 ncouplings

WORD MoDeL::ncouplings

Definition at line 648 of file [structs.h](#).

### 3.71.2.12 error

WORD MoDeL::error

Definition at line 649 of file [structs.h](#).

### 3.71.2.13 dummy

WORD MoDeL::dummy

Definition at line 650 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.72 MODNUM Struct Reference

### Data Fields

- UWORD \* [a](#)
- UWORD \* [m](#)
- WORD [na](#)
- WORD [nm](#)

### 3.72.1 Detailed Description

Definition at line [1301](#) of file [structs.h](#).

### 3.72.2 Field Documentation

#### 3.72.2.1 [a](#)

UWORD\* MODNUM::a

Definition at line [1302](#) of file [structs.h](#).

#### 3.72.2.2 [m](#)

UWORD\* MODNUM::m

Definition at line [1303](#) of file [structs.h](#).

#### 3.72.2.3 [na](#)

WORD MODNUM::na

Definition at line [1304](#) of file [structs.h](#).

#### 3.72.2.4 [nm](#)

WORD MODNUM::nm

Definition at line [1305](#) of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.73 MoDoPtDoLIaRS Struct Reference

### Data Fields

- WORD [number](#)
- WORD [type](#)

### 3.73.1 Detailed Description

Definition at line [556](#) of file [structs.h](#).

### 3.73.2 Field Documentation

#### 3.73.2.1 number

WORD MoDoPtDoLIaRS::number

Definition at line [560](#) of file [structs.h](#).

#### 3.73.2.2 type

WORD MoDoPtDoLIaRS::type

Definition at line [561](#) of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.74 monomial\_larger Class Reference

### Public Member Functions

- bool [operator\(\)](#) (const WORD \*a, const WORD \*b)

### 3.74.1 Detailed Description

Definition at line [179](#) of file [poly.h](#).

### 3.74.2 Constructor & Destructor Documentation

#### 3.74.2.1 monomial\_larger()

```
monomial_larger::monomial_larger ( ) [inline]
```

Definition at line [184](#) of file [poly.h](#).

### 3.74.3 Member Function Documentation

#### 3.74.3.1 operator()

```
bool monomial_larger::operator() (
    const WORD * a,
    const WORD * b ) [inline]
```

Definition at line 190 of file [poly.h](#).

The documentation for this class was generated from the following file:

- [poly.h](#)

## 3.75 MORbits Class Reference

### Public Member Functions

- void [print](#) (void)
- void [initPerm](#) (int nnodes)
- void [fromPerm](#) (int \*perm)
- void [toOrbits](#) (void)

### Data Fields

- int [nOrbits](#)
- int [nNodes](#)
- int [nd2or](#) [GRCC\_MAXNODES]
- int [or2nd](#) [GRCC\_MAXNODES]
- int [flist](#) [GRCC\_MAXNODES+1]

### 3.75.1 Detailed Description

Definition at line 1005 of file [grcc.h](#).

### 3.75.2 Constructor & Destructor Documentation

#### 3.75.2.1 MORbits()

```
MORbits::MORbits (
    void )
```

Definition at line 4964 of file [grcc.cc](#).

### 3.75.2.2 ~MOrbits()

```
MOrbits::~MOrbits (
    void )
```

Definition at line [4971](#) of file [grcc.cc](#).

## 3.75.3 Member Function Documentation

### 3.75.3.1 print()

```
void MOrbits::print (
    void )
```

Definition at line [4976](#) of file [grcc.cc](#).

### 3.75.3.2 initPerm()

```
void MOrbits::initPerm (
    int nnodes )
```

Definition at line [4989](#) of file [grcc.cc](#).

### 3.75.3.3 fromPerm()

```
void MOrbits::fromPerm (
    int * perm )
```

Definition at line [5000](#) of file [grcc.cc](#).

### 3.75.3.4 toOrbits()

```
void MOrbits::toOrbits (
    void )
```

Definition at line [5039](#) of file [grcc.cc](#).

## 3.75.4 Field Documentation

### 3.75.4.1 nOrbits

```
int MOrbits::nOrbits
```

Definition at line [1017](#) of file [grcc.h](#).



#### 3.75.4.2 nNodes

```
int MORbits::nNodes
```

Definition at line 1018 of file [grcc.h](#).

#### 3.75.4.3 nd2or

```
int MORbits::nd2or[GRCC_MAXNODES]
```

Definition at line 1019 of file [grcc.h](#).

#### 3.75.4.4 or2nd

```
int MORbits::or2nd[GRCC_MAXNODES]
```

Definition at line 1020 of file [grcc.h](#).

#### 3.75.4.5 flist

```
int MORbits::flist[GRCC_MAXNODES+1]
```

Definition at line 1021 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.76 flint::mpoly Class Reference

### Public Member Functions

- [mpoly](#) (fmpz\_mpoly\_ctx\_struct \*ctx\_in)
- void [print](#) (const string &text)

### Data Fields

- fmpz\_mpoly\_t [d](#)

### 3.76.1 Detailed Description

Definition at line 76 of file [flintinterface.h](#).

## 3.76.2 Constructor & Destructor Documentation

### 3.76.2.1 mpoly()

```
flint::mpoly::mpoly (
    fmpz_mpoly_ctx_struct * ctx_in ) [inline], [explicit]
```

Definition at line 81 of file [flintinterface.h](#).

### 3.76.2.2 ~mpoly()

```
flint::mpoly::~~mpoly ( ) [inline]
```

Definition at line 82 of file [flintinterface.h](#).

## 3.76.3 Member Function Documentation

### 3.76.3.1 print()

```
void flint::mpoly::print (
    const string & text ) [inline]
```

Definition at line 83 of file [flintinterface.h](#).

## 3.76.4 Field Documentation

### 3.76.4.1 d

```
fmpz_mpoly_t flint::mpoly::d
```

Definition at line 80 of file [flintinterface.h](#).

The documentation for this class was generated from the following file:

- [flintinterface.h](#)

## 3.77 flint::mpoly\_ctx Class Reference

### Public Member Functions

- [mpoly\\_ctx](#) (int64\_t nvars)

### Data Fields

- [fmpz\\_mpoly\\_ctx\\_t](#) d

### 3.77.1 Detailed Description

Definition at line 99 of file [flintinterface.h](#).

### 3.77.2 Constructor & Destructor Documentation

#### 3.77.2.1 mpoly\_ctx()

```
flint::mpoly_ctx::mpoly_ctx (
    int64_t nvars ) [inline], [explicit]
```

Definition at line 102 of file [flintinterface.h](#).

#### 3.77.2.2 ~mpoly\_ctx()

```
flint::mpoly_ctx::~~mpoly_ctx ( ) [inline]
```

Definition at line 103 of file [flintinterface.h](#).

### 3.77.3 Field Documentation

#### 3.77.3.1 d

```
fmprz_mpoly_ctx_t flint::mpoly_ctx::d
```

Definition at line 101 of file [flintinterface.h](#).

The documentation for this class was generated from the following file:

- [flintinterface.h](#)

## 3.78 flint::mpoly\_factor Class Reference

### Public Member Functions

- [mpoly\\_factor](#) (fmprz\_mpoly\_ctx\_struct \*ctx\_in)

### Data Fields

- fmprz\_mpoly\_factor\_t [d](#)

### 3.78.1 Detailed Description

Definition at line 89 of file [flintinterface.h](#).

### 3.78.2 Constructor & Destructor Documentation

#### 3.78.2.1 `mpoly_factor()`

```
flint::mpoly_factor::mpoly_factor (
    fmpz_mpoly_ctx_struct * ctx_in ) [inline], [explicit]
```

Definition at line 94 of file [flintinterface.h](#).

#### 3.78.2.2 `~mpoly_factor()`

```
flint::mpoly_factor::~~mpoly_factor ( ) [inline]
```

Definition at line 97 of file [flintinterface.h](#).

### 3.78.3 Field Documentation

#### 3.78.3.1 `d`

```
fmpz_mpoly_factor_t flint::mpoly_factor::d
```

Definition at line 93 of file [flintinterface.h](#).

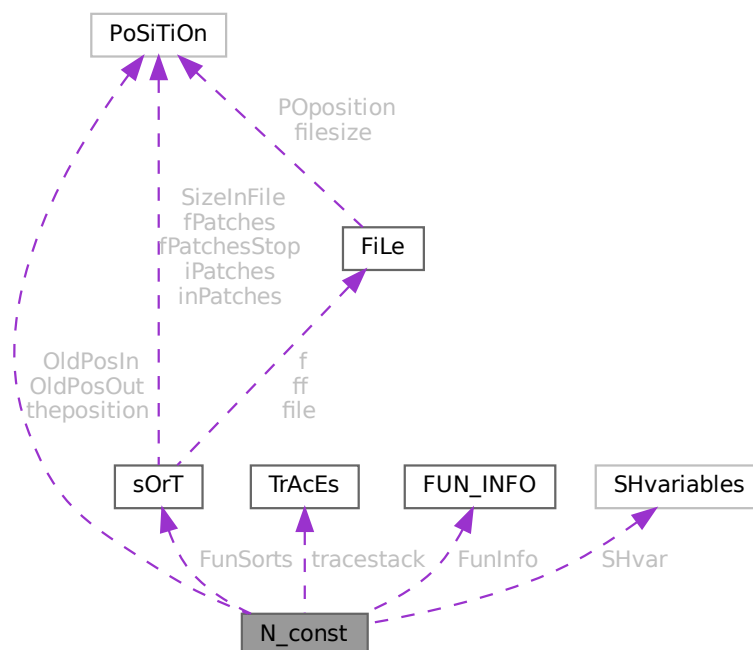
The documentation for this class was generated from the following file:

- [flintinterface.h](#)

## 3.79 `N_const` Struct Reference

```
#include <structs.h>
```

Collaboration diagram for `N_const`:



**Public Member Functions**

- **PADPOSITION** (50, 7, 23, 26, sizeof([SHvariables](#)))

**Data Fields**

- [POSITION](#) OldPosIn
- [POSITION](#) OldPosOut
- [POSITION](#) theposition
- WORD \* [EndNest](#)
- WORD \* [Frozen](#)
- WORD \* [FullProto](#)
- WORD \* [cTerm](#)
- int \* [RepPoint](#)
- WORD \* [WildValue](#)
- WORD \* [WildStop](#)
- WORD \* [argaddress](#)
- WORD \* [RepFunList](#)
- WORD \* [patstop](#)
- WORD \* [terstop](#)
- WORD \* [terstart](#)
- WORD \* [terfirstcomm](#)
- WORD \* [DumFound](#)
- WORD \*\* [DumPlace](#)
- WORD \*\* [DumFunPlace](#)
- WORD \* [UsedSymbol](#)
- WORD \* [UsedVector](#)
- WORD \* [UsedIndex](#)
- WORD \* [UsedFunction](#)
- WORD \* [MaskPointer](#)
- WORD \* [ForFindOnly](#)
- WORD \* [findTerm](#)
- WORD \* [findPattern](#)
- WORD \* [dummyrenumlist](#)
- int \* [funargs](#)
- WORD \*\* [funlocs](#)
- int \* [funinds](#)
- UWORD \* [NoScrat2](#)
- WORD \* [ReplaceScrat](#)
- [TRACES](#) \* [tracestack](#)
- WORD \* [selecttermundo](#)
- WORD \* [patternbuffer](#)
- WORD \* [termbuffer](#)
- WORD \*\* [PoinScratch](#)
- WORD \*\* [FunScratch](#)
- WORD \* [RenumScratch](#)
- [FUN\\_INFO](#) \* [FunInfo](#)
- WORD \*\* [SplitScratch](#)
- WORD \*\* [SplitScratch1](#)
- [SORTING](#) \*\* [FunSorts](#)
- UWORD \* [SoScratC](#)
- WORD \* [listinprint](#)
- WORD \* [currentTerm](#)
- WORD \*\* [arglist](#)

- int \* [tlistbuf](#)
- WORD \* [compressSpace](#)
- UWORD \* [SHcombi](#)
- WORD \* [poly\\_vars](#)
- UWORD \* [cmod](#)
- [SHvariables](#) SHvar
- LONG [deferskipped](#)
- LONG [InScratch](#)
- LONG [SplitScratchSize](#)
- LONG [InScratch1](#)
- LONG [SplitScratchSize1](#)
- LONG [ninterms](#)
- LONG [SHcombisize](#)
- int [NumTotWildArgs](#)
- int [UseFindOnly](#)
- int [UsedOtherFind](#)
- int [ErrorInDollar](#)
- int [numfargs](#)
- int [numflocs](#)
- int [nargs](#)
- int [tohunt](#)
- int [numoffuns](#)
- int [funisize](#)
- int [Rssize](#)
- int [numtracesctack](#)
- int [intracestack](#)
- int [numfuninfo](#)
- int [NumFunSorts](#)
- int [MaxFunSorts](#)
- int [arglistsize](#)
- int [tlistsize](#)
- int [filenum](#)
- int [compressSize](#)
- int [polysortflag](#)
- int [nogroundlevel](#)
- int [subsubveto](#)
- WORD [MaxRenumScratch](#)
- WORD [oldtype](#)
- WORD [oldvalue](#)
- WORD [NumWild](#)
- WORD [RepFunNum](#)
- WORD [DisOrderFlag](#)
- WORD [WildDirt](#)
- WORD [NumFound](#)
- WORD [WildReserve](#)
- WORD [TelInFun](#)
- WORD [TeSuOut](#)
- WORD [WildArgs](#)
- WORD [WildEat](#)
- WORD [PolyNormFlag](#)
- WORD [PolyFunTodo](#)
- WORD [sizeselecttermundo](#)
- WORD [patternbuffersize](#)
- WORD [numlistinprint](#)
- WORD [ncmod](#)

- WORD [ExpectedSign](#)
- WORD [SignCheck](#)
- WORD [IndDum](#)
- WORD [poly\\_num\\_vars](#)
- WORD [idfunctionflag](#)
- WORD [poly\\_vars\\_type](#)
- WORD [tryterm](#)

### 3.79.1 Detailed Description

The [N\\_const](#) struct is part of the global data and resides either in the ALLGLOBALS struct A, or the ALLPRIVATES struct B (TFORM) under the name N We see it used with the macro AN as in AN.RepFunNum It has variables that are private to each thread and are used as temporary storage during the expansion of the terms tree.

Definition at line [2276](#) of file [structs.h](#).

### 3.79.2 Field Documentation

#### 3.79.2.1 OldPosIn

[POSITION](#) [N\\_const::OldPosIn](#)

Definition at line [2277](#) of file [structs.h](#).

#### 3.79.2.2 OldPosOut

[POSITION](#) [N\\_const::OldPosOut](#)

Definition at line [2278](#) of file [structs.h](#).

#### 3.79.2.3 theposition

[POSITION](#) [N\\_const::theposition](#)

Definition at line [2279](#) of file [structs.h](#).

#### 3.79.2.4 EndNest

[WORD\\*](#) [N\\_const::EndNest](#)

Definition at line [2280](#) of file [structs.h](#).

#### 3.79.2.5 Frozen

[WORD\\*](#) [N\\_const::Frozen](#)

Definition at line [2281](#) of file [structs.h](#).

### 3.79.2.6 FullProto

WORD\* N\_const::FullProto

Definition at line 2282 of file [structs.h](#).

### 3.79.2.7 cTerm

WORD\* N\_const::cTerm

Definition at line 2283 of file [structs.h](#).

### 3.79.2.8 RepPoint

int\* N\_const::RepPoint

Definition at line 2284 of file [structs.h](#).

### 3.79.2.9 WildValue

WORD\* N\_const::WildValue

Definition at line 2285 of file [structs.h](#).

### 3.79.2.10 WildStop

WORD\* N\_const::WildStop

Definition at line 2286 of file [structs.h](#).

### 3.79.2.11 argaddress

WORD\* N\_const::argaddress

Definition at line 2287 of file [structs.h](#).

### 3.79.2.12 RepFunList

WORD\* N\_const::RepFunList

Definition at line 2288 of file [structs.h](#).

### 3.79.2.13 patstop

WORD\* N\_const::patstop

Definition at line 2289 of file [structs.h](#).



#### 3.79.2.14 terstop

WORD\* N\_const::terstop

Definition at line 2290 of file [structs.h](#).

#### 3.79.2.15 terstart

WORD\* N\_const::terstart

Definition at line 2291 of file [structs.h](#).

#### 3.79.2.16 terfirstcomm

WORD\* N\_const::terfirstcomm

Definition at line 2292 of file [structs.h](#).

#### 3.79.2.17 DumFound

WORD\* N\_const::DumFound

Definition at line 2293 of file [structs.h](#).

#### 3.79.2.18 DumPlace

WORD\*\* N\_const::DumPlace

Definition at line 2294 of file [structs.h](#).

#### 3.79.2.19 DumFunPlace

WORD\*\* N\_const::DumFunPlace

Definition at line 2295 of file [structs.h](#).

#### 3.79.2.20 UsedSymbol

WORD\* N\_const::UsedSymbol

Definition at line 2296 of file [structs.h](#).

#### 3.79.2.21 UsedVector

WORD\* N\_const::UsedVector

Definition at line 2297 of file [structs.h](#).

### 3.79.2.22 UsedIndex

WORD\* N\_const::UsedIndex

Definition at line 2298 of file [structs.h](#).

### 3.79.2.23 UsedFunction

WORD\* N\_const::UsedFunction

Definition at line 2299 of file [structs.h](#).

### 3.79.2.24 MaskPointer

WORD\* N\_const::MaskPointer

Definition at line 2300 of file [structs.h](#).

### 3.79.2.25 ForFindOnly

WORD\* N\_const::ForFindOnly

Definition at line 2301 of file [structs.h](#).

### 3.79.2.26 findTerm

WORD\* N\_const::findTerm

Definition at line 2302 of file [structs.h](#).

### 3.79.2.27 findPattern

WORD\* N\_const::findPattern

Definition at line 2303 of file [structs.h](#).

### 3.79.2.28 dummyrenumlist

WORD\* N\_const::dummyrenumlist

Definition at line 2308 of file [structs.h](#).

### 3.79.2.29 funargs

int\* N\_const::funargs

Definition at line 2309 of file [structs.h](#).

### 3.79.2.30 funlocs

WORD\*\* N\_const::funlocs

Definition at line 2310 of file [structs.h](#).

### 3.79.2.31 funinds

int\* N\_const::funinds

Definition at line 2311 of file [structs.h](#).

### 3.79.2.32 NoScrat2

UWORD\* N\_const::NoScrat2

Definition at line 2312 of file [structs.h](#).

### 3.79.2.33 ReplaceScrat

WORD\* N\_const::ReplaceScrat

Definition at line 2313 of file [structs.h](#).

### 3.79.2.34 tracestack

TRACES\* N\_const::tracestack

Definition at line 2314 of file [structs.h](#).

### 3.79.2.35 selecttermundo

WORD\* N\_const::selecttermundo

Definition at line 2315 of file [structs.h](#).

### 3.79.2.36 patternbuffer

WORD\* N\_const::patternbuffer

Definition at line 2316 of file [structs.h](#).

### 3.79.2.37 termbuffer

WORD\* N\_const::termbuffer

Definition at line 2317 of file [structs.h](#).

### 3.79.2.38 PoinScratch

WORD\*\* N\_const::PoinScratch

Definition at line 2318 of file [structs.h](#).

### 3.79.2.39 FunScratch

WORD\*\* N\_const::FunScratch

Definition at line 2319 of file [structs.h](#).

### 3.79.2.40 RenumScratch

WORD\* N\_const::RenumScratch

Definition at line 2320 of file [structs.h](#).

### 3.79.2.41 FunInfo

FUN\_INFO\* N\_const::FunInfo

Definition at line 2321 of file [structs.h](#).

### 3.79.2.42 SplitScratch

WORD\*\* N\_const::SplitScratch

Definition at line 2322 of file [structs.h](#).

### 3.79.2.43 SplitScratch1

WORD\*\* N\_const::SplitScratch1

Definition at line 2323 of file [structs.h](#).

### 3.79.2.44 FunSorts

SORTING\*\* N\_const::FunSorts

Definition at line 2324 of file [structs.h](#).

### 3.79.2.45 SoScratC

UWORD\* N\_const::SoScratC

Definition at line 2325 of file [structs.h](#).

#### 3.79.2.46 listinprint

WORD\* N\_const::listinprint

Definition at line 2326 of file [structs.h](#).

#### 3.79.2.47 currentTerm

WORD\* N\_const::currentTerm

Definition at line 2327 of file [structs.h](#).

#### 3.79.2.48 arglist

WORD\*\* N\_const::arglist

Definition at line 2328 of file [structs.h](#).

#### 3.79.2.49 tlistbuf

int\* N\_const::tlistbuf

Definition at line 2329 of file [structs.h](#).

#### 3.79.2.50 compressSpace

WORD\* N\_const::compressSpace

Definition at line 2333 of file [structs.h](#).

#### 3.79.2.51 SHcombi

UWORD\* N\_const::SHcombi

Definition at line 2338 of file [structs.h](#).

#### 3.79.2.52 poly\_vars

WORD\* N\_const::poly\_vars

Definition at line 2339 of file [structs.h](#).

#### 3.79.2.53 cmod

UWORD\* N\_const::cmod

Definition at line 2340 of file [structs.h](#).

#### 3.79.2.54 SHvar

`SHvariables N_const::SHvar`

Definition at line 2341 of file [structs.h](#).

#### 3.79.2.55 deferskipped

`LONG N_const::deferskipped`

Definition at line 2342 of file [structs.h](#).

#### 3.79.2.56 InScratch

`LONG N_const::InScratch`

Definition at line 2343 of file [structs.h](#).

#### 3.79.2.57 SplitScratchSize

`LONG N_const::SplitScratchSize`

Definition at line 2344 of file [structs.h](#).

#### 3.79.2.58 InScratch1

`LONG N_const::InScratch1`

Definition at line 2345 of file [structs.h](#).

#### 3.79.2.59 SplitScratchSize1

`LONG N_const::SplitScratchSize1`

Definition at line 2346 of file [structs.h](#).

#### 3.79.2.60 ninterms

`LONG N_const::ninterms`

Definition at line 2347 of file [structs.h](#).

#### 3.79.2.61 SHcombisize

`LONG N_const::SHcombisize`

Definition at line 2356 of file [structs.h](#).

**3.79.2.62 NumTotWildArgs**

```
int N_const::NumTotWildArgs
```

Definition at line [2357](#) of file [structs.h](#).

**3.79.2.63 UseFindOnly**

```
int N_const::UseFindOnly
```

Definition at line [2358](#) of file [structs.h](#).

**3.79.2.64 UsedOtherFind**

```
int N_const::UsedOtherFind
```

Definition at line [2359](#) of file [structs.h](#).

**3.79.2.65 ErrorInDollar**

```
int N_const::ErrorInDollar
```

Definition at line [2360](#) of file [structs.h](#).

**3.79.2.66 numfargs**

```
int N_const::numfargs
```

Definition at line [2361](#) of file [structs.h](#).

**3.79.2.67 numflocs**

```
int N_const::numflocs
```

Definition at line [2362](#) of file [structs.h](#).

**3.79.2.68 nargs**

```
int N_const::nargs
```

Definition at line [2363](#) of file [structs.h](#).

**3.79.2.69 tohunt**

```
int N_const::tohunt
```

Definition at line [2364](#) of file [structs.h](#).

### 3.79.2.70 numoffuns

```
int N_const::numoffuns
```

Definition at line 2365 of file [structs.h](#).

### 3.79.2.71 funisize

```
int N_const::funisize
```

Definition at line 2366 of file [structs.h](#).

### 3.79.2.72 RSize

```
int N_const::RSize
```

Definition at line 2367 of file [structs.h](#).

### 3.79.2.73 numtracesctack

```
int N_const::numtracesctack
```

Definition at line 2368 of file [structs.h](#).

### 3.79.2.74 intracestack

```
int N_const::intracestack
```

Definition at line 2369 of file [structs.h](#).

### 3.79.2.75 numfuninfo

```
int N_const::numfuninfo
```

Definition at line 2370 of file [structs.h](#).

### 3.79.2.76 NumFunSorts

```
int N_const::NumFunSorts
```

Definition at line 2371 of file [structs.h](#).

### 3.79.2.77 MaxFunSorts

```
int N_const::MaxFunSorts
```

Definition at line 2372 of file [structs.h](#).



**3.79.2.78 arglistsize**

```
int N_const::arglistsize
```

Definition at line [2373](#) of file [structs.h](#).

**3.79.2.79 tlistsize**

```
int N_const::tlistsize
```

Definition at line [2374](#) of file [structs.h](#).

**3.79.2.80 filenum**

```
int N_const::filenum
```

Definition at line [2375](#) of file [structs.h](#).

**3.79.2.81 compressSize**

```
int N_const::compressSize
```

Definition at line [2376](#) of file [structs.h](#).

**3.79.2.82 polysortflag**

```
int N_const::polysortflag
```

Definition at line [2377](#) of file [structs.h](#).

**3.79.2.83 nogroundlevel**

```
int N_const::nogroundlevel
```

Definition at line [2378](#) of file [structs.h](#).

**3.79.2.84 subsubveto**

```
int N_const::subsubveto
```

Definition at line [2379](#) of file [structs.h](#).

**3.79.2.85 MaxRenumScratch**

```
WORD N_const::MaxRenumScratch
```

Definition at line [2380](#) of file [structs.h](#).

### 3.79.2.86 oldtype

WORD N\_const::oldtype

Definition at line 2381 of file [structs.h](#).

### 3.79.2.87 oldvalue

WORD N\_const::oldvalue

Definition at line 2382 of file [structs.h](#).

### 3.79.2.88 NumWild

WORD N\_const::NumWild

Definition at line 2383 of file [structs.h](#).

### 3.79.2.89 RepFunNum

WORD N\_const::RepFunNum

Definition at line 2384 of file [structs.h](#).

### 3.79.2.90 DisOrderFlag

WORD N\_const::DisOrderFlag

Definition at line 2385 of file [structs.h](#).

### 3.79.2.91 WildDirt

WORD N\_const::WildDirt

Definition at line 2386 of file [structs.h](#).

### 3.79.2.92 NumFound

WORD N\_const::NumFound

Definition at line 2387 of file [structs.h](#).

### 3.79.2.93 WildReserve

WORD N\_const::WildReserve

Definition at line 2388 of file [structs.h](#).

#### 3.79.2.94 TeInFun

WORD N\_const::TeInFun

Definition at line 2389 of file [structs.h](#).

#### 3.79.2.95 TeSuOut

WORD N\_const::TeSuOut

Definition at line 2390 of file [structs.h](#).

#### 3.79.2.96 WildArgs

WORD N\_const::WildArgs

Definition at line 2391 of file [structs.h](#).

#### 3.79.2.97 WildEat

WORD N\_const::WildEat

Definition at line 2392 of file [structs.h](#).

#### 3.79.2.98 PolyNormFlag

WORD N\_const::PolyNormFlag

Definition at line 2393 of file [structs.h](#).

#### 3.79.2.99 PolyFunTodo

WORD N\_const::PolyFunTodo

Definition at line 2394 of file [structs.h](#).

#### 3.79.2.100 sizeselecttermundo

WORD N\_const::sizeselecttermundo

Definition at line 2395 of file [structs.h](#).

#### 3.79.2.101 patternbuffersize

WORD N\_const::patternbuffersize

Definition at line 2396 of file [structs.h](#).

**3.79.2.102 numlistinprint**

WORD N\_const::numlistinprint

Definition at line [2397](#) of file [structs.h](#).

**3.79.2.103 ncmmod**

WORD N\_const::ncmmod

Definition at line [2398](#) of file [structs.h](#).

**3.79.2.104 ExpectedSign**

WORD N\_const::ExpectedSign

Definition at line [2399](#) of file [structs.h](#).

**3.79.2.105 SignCheck**

WORD N\_const::SignCheck

Used in pattern matching of antisymmetric functions

Definition at line [2400](#) of file [structs.h](#).

**3.79.2.106 IndDum**

WORD N\_const::IndDum

Used in pattern matching of antisymmetric functions

Definition at line [2401](#) of file [structs.h](#).

**3.79.2.107 poly\_num\_vars**

WORD N\_const::poly\_num\_vars

Definition at line [2402](#) of file [structs.h](#).

**3.79.2.108 idfunctionflag**

WORD N\_const::idfunctionflag

Definition at line [2403](#) of file [structs.h](#).

### 3.79.2.109 poly\_vars\_type

WORD N\_const::poly\_vars\_type

Definition at line 2404 of file [structs.h](#).

### 3.79.2.110 tryterm

WORD N\_const::tryterm

Definition at line 2405 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.80 nameblock Struct Reference

### Data Fields

- MLONG [previousblock](#)
- MLONG [position](#)
- char [names](#) [NAMETABLESIZE]

### 3.80.1 Detailed Description

Definition at line 117 of file [minos.h](#).

### 3.80.2 Field Documentation

#### 3.80.2.1 previousblock

MLONG nameblock::previousblock

Definition at line 118 of file [minos.h](#).

#### 3.80.2.2 position

MLONG nameblock::position

Definition at line 119 of file [minos.h](#).

### 3.80.2.3 names

```
char nameblock::names[NAMETABLESIZE]
```

Definition at line 120 of file [minos.h](#).

The documentation for this struct was generated from the following file:

- [minos.h](#)

## 3.81 NaMeNode Struct Reference

```
#include <structs.h>
```

### Data Fields

- LONG [name](#)
- WORD [parent](#)
- WORD [left](#)
- WORD [right](#)
- WORD [balance](#)
- WORD [type](#)
- WORD [number](#)

### 3.81.1 Detailed Description

The names of variables are kept in an array. Elements of type [NAMENODE](#) define a tree (that is kept balanced) that make it easy and fast to look for variables. See also [NAMETREE](#).

Definition at line 247 of file [structs.h](#).

### 3.81.2 Field Documentation

#### 3.81.2.1 name

```
LONG NaMeNode::name
```

Offset into [NAMETREE::namebuffer](#).

Definition at line 248 of file [structs.h](#).

#### 3.81.2.2 parent

```
WORD NaMeNode::parent
```

== -1 if no parent.

Definition at line 249 of file [structs.h](#).

### 3.81.2.3 left

WORD NaMeNode::left

=-1 if no child.

Definition at line 250 of file [structs.h](#).

### 3.81.2.4 right

WORD NaMeNode::right

=-1 if no child.

Definition at line 251 of file [structs.h](#).

### 3.81.2.5 balance

WORD NaMeNode::balance

Used for the balancing of the tree.

Definition at line 252 of file [structs.h](#).

### 3.81.2.6 type

WORD NaMeNode::type

Type associated with the name. See [compiler types](#).

Definition at line 253 of file [structs.h](#).

### 3.81.2.7 number

WORD NaMeNode::number

Number of variable in [LIST](#)'s like for example [C\\_const::SymbolList](#).

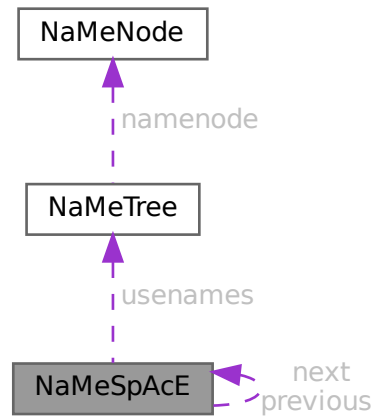
Definition at line 254 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.82 NaMeSpAcE Struct Reference

Collaboration diagram for NaMeSpAcE:



### Data Fields

- struct [NaMeSpAcE](#) \* [previous](#)
- struct [NaMeSpAcE](#) \* [next](#)
- [NAMETREE](#) \* [usernames](#)
- UBYTE \* [name](#)

### 3.82.1 Detailed Description

Definition at line [658](#) of file [structs.h](#).

### 3.82.2 Field Documentation

#### 3.82.2.1 previous

```
struct NaMeSpAcE* NaMeSpAcE::previous
```

Definition at line [659](#) of file [structs.h](#).

#### 3.82.2.2 next

```
struct NaMeSpAcE* NaMeSpAcE::next
```

Definition at line [660](#) of file [structs.h](#).



### 3.82.2.3 usernames

`NAMETREE* NaMeSpAcE::usernames`

Definition at line 661 of file [structs.h](#).

### 3.82.2.4 name

`UBYTE* NaMeSpAcE::name`

Definition at line 662 of file [structs.h](#).

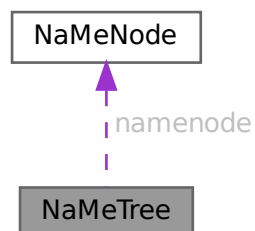
The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.83 NaMeTree Struct Reference

```
#include <structs.h>
```

Collaboration diagram for NaMeTree:



### Data Fields

- `NAMENODE * namenode`
- `UBYTE * namebuffer`
- `LONG nodesize`
- `LONG nodefill`
- `LONG namesize`
- `LONG namefill`
- `LONG oldnamefill`
- `LONG oldnodefill`
- `LONG globalnamefill`
- `LONG globalnodefill`
- `LONG clearnamefill`
- `LONG clearnodefill`
- `WORD headnode`

### 3.83.1 Detailed Description

A struct of type [NAMETREE](#) controls a complete (balanced) tree of names for the compiler. The compiler maintains several of such trees and the system has been set up in such a way that one could define more of them if we ever want to work with local name spaces.

Definition at line [265](#) of file [structs.h](#).

### 3.83.2 Field Documentation

#### 3.83.2.1 namenode

```
NAMENODE* NaMeTree::namenode
```

[D] Vector of [NAMENODE](#)'s. Number of elements is [nodesize](#). =0 if no memory has been allocated.

Definition at line [266](#) of file [structs.h](#).

#### 3.83.2.2 namebuffer

```
UBYTE* NaMeTree::namebuffer
```

[D] Buffer that holds all the name strings referred to by the [NAMENODE](#)'s. Allocation size is [namesize](#). =0 if no memory has been allocated.

Definition at line [268](#) of file [structs.h](#).

#### 3.83.2.3 nodesize

```
LONG NaMeTree::nodesize
```

Maximum number of elements in [namenode](#).

Definition at line [271](#) of file [structs.h](#).

#### 3.83.2.4 nodefill

```
LONG NaMeTree::nodefill
```

Number of currently used nodes in [namenode](#).

Definition at line [272](#) of file [structs.h](#).

#### 3.83.2.5 namesize

```
LONG NaMeTree::namesize
```

Allocation size of [namebuffer](#) in bytes.

Definition at line [273](#) of file [structs.h](#).

### 3.83.2.6 namefill

```
LONG NaMeTree::namefill
```

Number of bytes occupied.

Definition at line 274 of file [structs.h](#).

### 3.83.2.7 oldnamefill

```
LONG NaMeTree::oldnamefill
```

UNUSED

Definition at line 275 of file [structs.h](#).

### 3.83.2.8 oldnodefill

```
LONG NaMeTree::oldnodefill
```

UNUSED

Definition at line 276 of file [structs.h](#).

### 3.83.2.9 globalnamefill

```
LONG NaMeTree::globalnamefill
```

Set by .global statement to the value of [namefill](#). When a .store command is processed, this value will be used to reset namefill.

Definition at line 277 of file [structs.h](#).

### 3.83.2.10 globalnodefill

```
LONG NaMeTree::globalnodefill
```

Same usage as [globalnamefill](#), but for nodefill.

Definition at line 279 of file [structs.h](#).

### 3.83.2.11 clearnamefill

```
LONG NaMeTree::clearnamefill
```

Marks the reset point used by the .clear statement.

Definition at line 280 of file [structs.h](#).

### 3.83.2.12 clearnodefill

```
LONG NaMeTree::clearnodefill
```

Marks the reset point used by the .clear statement.

Definition at line 281 of file [structs.h](#).

### 3.83.2.13 headnode

```
WORD NaMeTree::headnode
```

Offset in [namenode](#) of head node. ==-1 if tree is empty.

Definition at line 282 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.84 NCand Class Reference

### Public Member Functions

- [NCand](#) (const NCandSt sta, const int dega, const int nilst, int \*ilst)
- void [prNCand](#) (const char \*msg)

### Data Fields

- int [deg](#)
- NCandSt [st](#)
- int [nilist](#)
- int [ilst](#) [GRCC\_MAXMINTERACT]

### 3.84.1 Detailed Description

Definition at line 1044 of file [grcc.h](#).

### 3.84.2 Constructor & Destructor Documentation

#### 3.84.2.1 NCand()

```
NCand::NCand (
    const NCandSt sta,
    const int dega,
    const int nilst,
    int * ilst )
```

Definition at line 7285 of file [grcc.cc](#).

### 3.84.2.2 ~NCand()

```
NCand::~NCand (
    void )
```

Definition at line 7312 of file [grcc.cc](#).

## 3.84.3 Member Function Documentation

### 3.84.3.1 prNCand()

```
void NCand::prNCand (
    const char * msg )
```

Definition at line 7317 of file [grcc.cc](#).

## 3.84.4 Field Documentation

### 3.84.4.1 deg

```
int NCand::deg
```

Definition at line 1048 of file [grcc.h](#).

### 3.84.4.2 st

```
NCandSt NCand::st
```

Definition at line 1049 of file [grcc.h](#).

### 3.84.4.3 nilist

```
int NCand::nilist
```

Definition at line 1050 of file [grcc.h](#).

### 3.84.4.4 ilist

```
int NCand::ilist[GRCC_MAXMINTERACT]
```

Definition at line 1051 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.85 NCInput Struct Reference

### Data Fields

- int [cldeg](#)
- int [clnum](#)
- int [cltyp](#)
- int [cmind](#)
- int [cmaxd](#)
- int [ptcl](#)
- int [cple](#)

### 3.85.1 Detailed Description

Definition at line [149](#) of file [grccparam.h](#).

### 3.85.2 Field Documentation

#### 3.85.2.1 cldeg

```
int NCInput::cldeg
```

Definition at line [151](#) of file [grccparam.h](#).

#### 3.85.2.2 clnum

```
int NCInput::clnum
```

Definition at line [152](#) of file [grccparam.h](#).

#### 3.85.2.3 cltyp

```
int NCInput::cltyp
```

Definition at line [153](#) of file [grccparam.h](#).

#### 3.85.2.4 cmind

```
int NCInput::cmind
```

Definition at line [154](#) of file [grccparam.h](#).

#### 3.85.2.5 cmaxd

```
int NCInput::cmaxd
```

Definition at line [155](#) of file [grccparam.h](#).

### 3.85.2.6 ptcl

```
int NCInput::ptcl
```

Definition at line 158 of file [grccparam.h](#).

### 3.85.2.7 cple

```
int NCInput::cple
```

Definition at line 159 of file [grccparam.h](#).

The documentation for this struct was generated from the following file:

- [grccparam.h](#)

## 3.86 NeStInG Struct Reference

```
#include <structs.h>
```

### Data Fields

- WORD \* [termsize](#)
- WORD \* [funsize](#)
- WORD \* [argsize](#)

### 3.86.1 Detailed Description

The NESTING struct is used when we enter the argument of functions and there is the possibility that we have to change something there. Because functions can be nested we have to keep track of all levels of functions in case we have to move the outer layers to make room for a larger function argument.

Definition at line 1053 of file [structs.h](#).

### 3.86.2 Field Documentation

#### 3.86.2.1 termsize

```
WORD* NeStInG::termsize
```

Definition at line 1054 of file [structs.h](#).

#### 3.86.2.2 funsize

```
WORD* NeStInG::funsize
```

Definition at line 1055 of file [structs.h](#).

### 3.86.2.3 argsize

WORD\* NeStInG::argsize

Definition at line 1056 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.87 NoDe Struct Reference

Collaboration diagram for NoDe:



### Data Fields

- struct [NoDe](#) \* [left](#)
- struct [NoDe](#) \* [right](#)
- int [lloser](#)
- int [rloser](#)
- int [lsrc](#)
- int [rsrc](#)

### 3.87.1 Detailed Description

A node for the tree of losers in the final sorting on the master.

Definition at line 265 of file [parallel.c](#).

### 3.87.2 Field Documentation

#### 3.87.2.1 left

struct [NoDe](#)\* NoDe::left

Definition at line 266 of file [parallel.c](#).



### 3.87.2.2 right

```
struct NoDe* NoDe::right
```

Definition at line 267 of file [parallel.c](#).

### 3.87.2.3 lloser

```
int NoDe::lloser
```

Definition at line 268 of file [parallel.c](#).

### 3.87.2.4 rloser

```
int NoDe::rloser
```

Definition at line 269 of file [parallel.c](#).

### 3.87.2.5 lsrc

```
int NoDe::lsrc
```

Definition at line 270 of file [parallel.c](#).

### 3.87.2.6 rsrc

```
int NoDe::rsrc
```

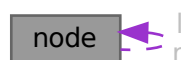
Definition at line 271 of file [parallel.c](#).

The documentation for this struct was generated from the following file:

- [parallel.c](#)

## 3.88 node Struct Reference

Collaboration diagram for node:



## Public Member Functions

- [node](#) (const WORD \*data)
- int [cmp](#) (const struct [node](#) \*rhs) const
- bool [operator\(\)](#) (const struct [node](#) \*lhs, const struct [node](#) \*rhs) const
- void [calcHash](#) ()

## Data Fields

- const WORD \* [data](#)
- struct [node](#) \* [l](#)
- struct [node](#) \* [r](#)
- WORD [sign](#)
- UWORD [hash](#)

### 3.88.1 Detailed Description

Definition at line [1496](#) of file [optimize.cc](#).

### 3.88.2 Constructor & Destructor Documentation

#### 3.88.2.1 [node\(\)](#) [1/2]

```
node::node ( ) [inline]
```

Definition at line [1503](#) of file [optimize.cc](#).

#### 3.88.2.2 [node\(\)](#) [2/2]

```
node::node (
    const WORD * data ) [inline]
```

Definition at line [1504](#) of file [optimize.cc](#).

### 3.88.3 Member Function Documentation

#### 3.88.3.1 [cmp\(\)](#)

```
int node::cmp (
    const struct node * rhs ) const [inline]
```

Definition at line [1508](#) of file [optimize.cc](#).

### 3.88.3.2 operator()

```
bool node::operator() (
    const struct node * lhs,
    const struct node * rhs ) const [inline]
```

Definition at line 1532 of file [optimize.cc](#).

### 3.88.3.3 calcHash()

```
void node::calcHash ( ) [inline]
```

Definition at line 1537 of file [optimize.cc](#).

## 3.88.4 Field Documentation

### 3.88.4.1 data

```
const WORD* node::data
```

Definition at line 1497 of file [optimize.cc](#).

### 3.88.4.2 l

```
struct node* node::l
```

Definition at line 1498 of file [optimize.cc](#).

### 3.88.4.3 r

```
struct node* node::r
```

Definition at line 1499 of file [optimize.cc](#).

### 3.88.4.4 sign

```
WORD node::sign
```

Definition at line 1500 of file [optimize.cc](#).

### 3.88.4.5 hash

```
UWORD node::hash
```

Definition at line 1501 of file [optimize.cc](#).

The documentation for this struct was generated from the following file:

- [optimize.cc](#)

## 3.89 NodeEq Struct Reference

### Public Member Functions

- `bool operator()` (const `NODE` \*lhs, const `NODE` \*rhs) const

### 3.89.1 Detailed Description

Definition at line 1558 of file [optimize.cc](#).

### 3.89.2 Member Function Documentation

#### 3.89.2.1 operator()

```
bool NodeEq::operator() (
    const NODE * lhs,
    const NODE * rhs ) const [inline]
```

Definition at line 1559 of file [optimize.cc](#).

The documentation for this struct was generated from the following file:

- [optimize.cc](#)

## 3.90 NodeHash Struct Reference

### Public Member Functions

- `size_t operator()` (const `NODE` \*n) const

### 3.90.1 Detailed Description

Definition at line 1552 of file [optimize.cc](#).

### 3.90.2 Member Function Documentation

#### 3.90.2.1 operator()

```
size_t NodeHash::operator() (
    const NODE * n ) const [inline]
```

Definition at line 1553 of file [optimize.cc](#).

The documentation for this struct was generated from the following file:

- [optimize.cc](#)

## 3.91 NStack Class Reference

### Public Member Functions

- void [print](#) (const char \*msg)

### Data Fields

- int [noden](#)
- int [deg](#)
- NCandSt [st](#)
- int [nilist](#)
- int [ilist](#) [GRCC\_MAXMINTERACT]

### 3.91.1 Detailed Description

Definition at line [1270](#) of file [grcc.h](#).

### 3.91.2 Constructor & Destructor Documentation

#### 3.91.2.1 NStack()

```
NStack::NStack ( ) [inline]
```

Definition at line [1280](#) of file [grcc.h](#).

#### 3.91.2.2 ~NStack()

```
NStack::~~NStack ( ) [inline]
```

Definition at line [1281](#) of file [grcc.h](#).

### 3.91.3 Member Function Documentation

#### 3.91.3.1 print()

```
void NStack::print (
    const char * msg )
```

Definition at line [9456](#) of file [grcc.cc](#).

### 3.91.4 Field Documentation

#### 3.91.4.1 noden

```
int NStack::noden
```

Definition at line [1274](#) of file [grcc.h](#).

### 3.91.4.2 deg

```
int NStack::deg
```

Definition at line 1275 of file [grcc.h](#).

### 3.91.4.3 st

```
NCandSt NStack::st
```

Definition at line 1276 of file [grcc.h](#).

### 3.91.4.4 nilist

```
int NStack::nilist
```

Definition at line 1277 of file [grcc.h](#).

### 3.91.4.5 ilist

```
int NStack::ilist[GRCC_MAXMINTERACT]
```

Definition at line 1278 of file [grcc.h](#).

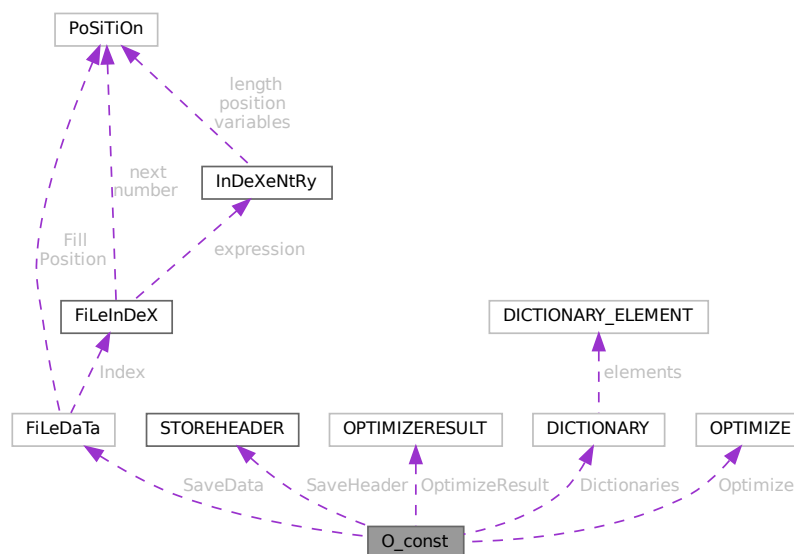
The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.92 O\_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for O\_const:



## Public Member Functions

- **PADPOSITION** (25, 4, 36, 17, 1)

## Data Fields

- FILEDATA SaveData
- STOREHEADER SaveHeader
- OPTIMIZERESULT OptimizeResult
- UBYTE \* OutputLine
- UBYTE \* OutStop
- UBYTE \* OutFill
- WORD \* bracket
- WORD \* termbuf
- WORD \* tabstring
- UBYTE \* wpos
- UBYTE \* wpoin
- UBYTE \* DollarOutBuffer
- UBYTE \* CurBufWrt
- void(\* FlipWORD )(UBYTE \*)
- void(\* FlipLONG )(UBYTE \*)
- void(\* FlipPOS )(UBYTE \*)
- void(\* FlipPOINTER )(UBYTE \*)
- void(\* ResizeData )(UBYTE \*, int, UBYTE \*, int)
- void(\* ResizeWORD )(UBYTE \*, UBYTE \*)
- void(\* ResizeNCWORD )(UBYTE \*, UBYTE \*)
- void(\* ResizeLONG )(UBYTE \*, UBYTE \*)
- void(\* ResizePOS )(UBYTE \*, UBYTE \*)
- void(\* ResizePOINTER )(UBYTE \*, UBYTE \*)
- void(\* CheckPower )(UBYTE \*)
- void(\* RenumVec )(UBYTE \*)
- DICTIONARY \*\* Dictionaries
- UBYTE \* tensorList
- WORD \* inscheme
- LONG NumInBrack
- LONG wlen
- LONG DollarOutSizeBuffer
- LONG DollarInOutBuffer
- OPTIMIZE Optimize
- int OutInBuffer
- int NoSpacesInNumbers
- int BlockSpaces
- int CurrentDictionary
- int SizeDictionaries
- int NumDictionaries
- int CurDictNumbers
- int CurDictVariables
- int CurDictSpecials
- int CurDictFunWithArgs
- int CurDictNumberWarning
- int CurDictNotInFunctions
- int CurDictInDollars
- int gNumDictionaries
- WORD schemenum

- WORD [transFlag](#)
- WORD [powerFlag](#)
- WORD [mpower](#)
- WORD [resizeFlag](#)
- WORD [bufferedInd](#)
- WORD [OutSkip](#)
- WORD [IsBracket](#)
- WORD [InFbrack](#)
- WORD [PrintType](#)
- WORD [FortFirst](#)
- WORD [DoubleFlag](#)
- WORD [IndentSpace](#)
- WORD [FactorMode](#)
- WORD [FactorNum](#)
- WORD [ErrorBlock](#)
- WORD [OptimizationLevel](#)
- UBYTE [FortDotChar](#)

### 3.92.1 Detailed Description

The [O\\_const](#) struct is part of the global data and resides in the ALLGLOBALS struct A under the name O We see it used with the macro AO as in AO.OutputLine It contains variables that involve the writing of text output and save/store files.

Definition at line [2453](#) of file [structs.h](#).

### 3.92.2 Field Documentation

#### 3.92.2.1 SaveData

[FILEDATA](#) [O\\_const::SaveData](#)

Definition at line [2454](#) of file [structs.h](#).

#### 3.92.2.2 SaveHeader

[STOREHEADER](#) [O\\_const::SaveHeader](#)

Definition at line [2455](#) of file [structs.h](#).

#### 3.92.2.3 OptimizeResult

[OPTIMIZERESULT](#) [O\\_const::OptimizeResult](#)

Definition at line [2456](#) of file [structs.h](#).



#### 3.92.2.4 OutputLine

UBYTE\* O\_const::OutputLine

Definition at line 2457 of file [structs.h](#).

#### 3.92.2.5 OutStop

UBYTE\* O\_const::OutStop

Definition at line 2458 of file [structs.h](#).

#### 3.92.2.6 OutFill

UBYTE\* O\_const::OutFill

Definition at line 2459 of file [structs.h](#).

#### 3.92.2.7 bracket

WORD\* O\_const::bracket

Definition at line 2460 of file [structs.h](#).

#### 3.92.2.8 termbuf

WORD\* O\_const::termbuf

Definition at line 2461 of file [structs.h](#).

#### 3.92.2.9 tabstring

WORD\* O\_const::tabstring

Definition at line 2462 of file [structs.h](#).

#### 3.92.2.10 wpos

UBYTE\* O\_const::wpos

Definition at line 2463 of file [structs.h](#).

#### 3.92.2.11 wpoin

UBYTE\* O\_const::wpoin

Definition at line 2464 of file [structs.h](#).

### 3.92.2.12 DollarOutBuffer

```
UBYTE* O_const::DollarOutBuffer
```

Definition at line [2465](#) of file [structs.h](#).

### 3.92.2.13 CurBufWrt

```
UBYTE* O_const::CurBufWrt
```

Definition at line [2466](#) of file [structs.h](#).

### 3.92.2.14 FlipWORD

```
void(* O_const::FlipWORD) (UBYTE *)
```

Definition at line [2467](#) of file [structs.h](#).

### 3.92.2.15 FlipLONG

```
void(* O_const::FlipLONG) (UBYTE *)
```

Definition at line [2468](#) of file [structs.h](#).

### 3.92.2.16 FlipPOS

```
void(* O_const::FlipPOS) (UBYTE *)
```

Definition at line [2469](#) of file [structs.h](#).

### 3.92.2.17 FlipPOINTER

```
void(* O_const::FlipPOINTER) (UBYTE *)
```

Definition at line [2470](#) of file [structs.h](#).

### 3.92.2.18 ResizeData

```
void(* O_const::ResizeData) (UBYTE *, int, UBYTE *, int)
```

Definition at line [2471](#) of file [structs.h](#).

### 3.92.2.19 ResizeWORD

```
void(* O_const::ResizeWORD) (UBYTE *, UBYTE *)
```

Definition at line [2472](#) of file [structs.h](#).

### 3.92.2.20 ResizeNCWORD

```
void(* O_const::ResizeNCWORD) (UBYTE *, UBYTE *)
```

Definition at line 2473 of file [structs.h](#).

### 3.92.2.21 ResizeLONG

```
void(* O_const::ResizeLONG) (UBYTE *, UBYTE *)
```

Definition at line 2474 of file [structs.h](#).

### 3.92.2.22 ResizePOS

```
void(* O_const::ResizePOS) (UBYTE *, UBYTE *)
```

Definition at line 2475 of file [structs.h](#).

### 3.92.2.23 ResizePOINTER

```
void(* O_const::ResizePOINTER) (UBYTE *, UBYTE *)
```

Definition at line 2476 of file [structs.h](#).

### 3.92.2.24 CheckPower

```
void(* O_const::CheckPower) (UBYTE *)
```

Definition at line 2477 of file [structs.h](#).

### 3.92.2.25 RenumberVec

```
void(* O_const::ReNUMBERVec) (UBYTE *)
```

Definition at line 2478 of file [structs.h](#).

### 3.92.2.26 Dictionaries

```
DICTIoNARY** O_const::Dictionaries
```

Definition at line 2479 of file [structs.h](#).

### 3.92.2.27 tensorList

```
UBYTE* O_const::tensorList
```

Definition at line 2480 of file [structs.h](#).

### 3.92.2.28 inscheme

```
WORD* O_const::inscheme
```

Definition at line 2481 of file [structs.h](#).

### 3.92.2.29 NumInBrack

```
LONG O_const::NumInBrack
```

Definition at line 2487 of file [structs.h](#).

### 3.92.2.30 wlen

```
LONG O_const::wlen
```

Definition at line 2488 of file [structs.h](#).

### 3.92.2.31 DollarOutSizeBuffer

```
LONG O_const::DollarOutSizeBuffer
```

Definition at line 2489 of file [structs.h](#).

### 3.92.2.32 DollarInOutBuffer

```
LONG O_const::DollarInOutBuffer
```

Definition at line 2490 of file [structs.h](#).

### 3.92.2.33 Optimize

```
OPTIMIZE O_const::Optimize
```

Definition at line 2495 of file [structs.h](#).

### 3.92.2.34 OutInBuffer

```
int O_const::OutInBuffer
```

Definition at line 2496 of file [structs.h](#).

### 3.92.2.35 NoSpacesInNumbers

```
int O_const::NoSpacesInNumbers
```

Definition at line 2497 of file [structs.h](#).

### 3.92.2.36 BlockSpaces

```
int O_const::BlockSpaces
```

Definition at line 2498 of file [structs.h](#).

### 3.92.2.37 CurrentDictionary

```
int O_const::CurrentDictionary
```

Definition at line 2499 of file [structs.h](#).

### 3.92.2.38 SizeDictionaries

```
int O_const::SizeDictionaries
```

Definition at line 2500 of file [structs.h](#).

### 3.92.2.39 NumDictionaries

```
int O_const::NumDictionaries
```

Definition at line 2501 of file [structs.h](#).

### 3.92.2.40 CurDictNumbers

```
int O_const::CurDictNumbers
```

Definition at line 2502 of file [structs.h](#).

### 3.92.2.41 CurDictVariables

```
int O_const::CurDictVariables
```

Definition at line 2503 of file [structs.h](#).

### 3.92.2.42 CurDictSpecials

```
int O_const::CurDictSpecials
```

Definition at line 2504 of file [structs.h](#).

### 3.92.2.43 CurDictFunWithArgs

```
int O_const::CurDictFunWithArgs
```

Definition at line 2505 of file [structs.h](#).

#### 3.92.2.44 CurDictNumberWarning

```
int O_const::CurDictNumberWarning
```

Definition at line 2506 of file [structs.h](#).

#### 3.92.2.45 CurDictNotInFunctions

```
int O_const::CurDictNotInFunctions
```

Definition at line 2507 of file [structs.h](#).

#### 3.92.2.46 CurDictInDollars

```
int O_const::CurDictInDollars
```

Definition at line 2508 of file [structs.h](#).

#### 3.92.2.47 gNumDictionaries

```
int O_const::gNumDictionaries
```

Definition at line 2509 of file [structs.h](#).

#### 3.92.2.48 schemenum

```
WORD O_const::schemenum
```

Definition at line 2513 of file [structs.h](#).

#### 3.92.2.49 transFlag

```
WORD O_const::transFlag
```

Definition at line 2514 of file [structs.h](#).

#### 3.92.2.50 powerFlag

```
WORD O_const::powerFlag
```

Definition at line 2515 of file [structs.h](#).

#### 3.92.2.51 mpower

```
WORD O_const::mpower
```

Definition at line 2516 of file [structs.h](#).

### 3.92.2.52 resizeFlag

WORD O\_const::resizeFlag

Definition at line 2517 of file [structs.h](#).

### 3.92.2.53 bufferedInd

WORD O\_const::bufferedInd

Definition at line 2518 of file [structs.h](#).

### 3.92.2.54 OutSkip

WORD O\_const::OutSkip

Definition at line 2519 of file [structs.h](#).

### 3.92.2.55 IsBracket

WORD O\_const::IsBracket

Definition at line 2520 of file [structs.h](#).

### 3.92.2.56 InFbrack

WORD O\_const::InFbrack

Definition at line 2521 of file [structs.h](#).

### 3.92.2.57 PrintType

WORD O\_const::PrintType

Definition at line 2522 of file [structs.h](#).

### 3.92.2.58 FortFirst

WORD O\_const::FortFirst

Definition at line 2523 of file [structs.h](#).

### 3.92.2.59 DoubleFlag

WORD O\_const::DoubleFlag

Definition at line 2524 of file [structs.h](#).

### 3.92.2.60 IndentSpace

WORD O\_const::IndentSpace

Definition at line 2525 of file [structs.h](#).

### 3.92.2.61 FactorMode

WORD O\_const::FactorMode

Definition at line 2526 of file [structs.h](#).

### 3.92.2.62 FactorNum

WORD O\_const::FactorNum

Definition at line 2527 of file [structs.h](#).

### 3.92.2.63 ErrorBlock

WORD O\_const::ErrorBlock

Definition at line 2528 of file [structs.h](#).

### 3.92.2.64 OptimizationLevel

WORD O\_const::OptimizationLevel

Definition at line 2529 of file [structs.h](#).

### 3.92.2.65 FortDotChar

UBYTE O\_const::FortDotChar

Definition at line 2530 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)



## 3.93 objects Struct Reference

### Data Fields

- MLONG [position](#)
- MLONG [size](#)
- MLONG [date](#)
- MLONG [tablename](#)
- MLONG [uncompressed](#)
- MLONG [spare1](#)
- MLONG [spare2](#)
- MLONG [spare3](#)
- char [element](#) [ELEMENTSIZE]

### 3.93.1 Detailed Description

Definition at line [97](#) of file [minos.h](#).

### 3.93.2 Field Documentation

#### 3.93.2.1 position

MLONG `objects::position`

Definition at line [99](#) of file [minos.h](#).

#### 3.93.2.2 size

MLONG `objects::size`

Definition at line [100](#) of file [minos.h](#).

#### 3.93.2.3 date

MLONG `objects::date`

Definition at line [101](#) of file [minos.h](#).

#### 3.93.2.4 tablename

MLONG `objects::tablename`

Definition at line [102](#) of file [minos.h](#).

### 3.93.2.5 uncompressed

```
MLONG objects::uncompressed
```

Definition at line 103 of file [minos.h](#).

### 3.93.2.6 spare1

```
MLONG objects::spare1
```

Definition at line 104 of file [minos.h](#).

### 3.93.2.7 spare2

```
MLONG objects::spare2
```

Definition at line 105 of file [minos.h](#).

### 3.93.2.8 spare3

```
MLONG objects::spare3
```

Definition at line 106 of file [minos.h](#).

### 3.93.2.9 element

```
char objects::element[ELEMENTSIZE]
```

Definition at line 107 of file [minos.h](#).

The documentation for this struct was generated from the following file:

- [minos.h](#)

## 3.94 OptDef Struct Reference

### Data Fields

- const char \* [name](#)
- const char \* [mean](#)
- int [defaultv](#)
- int [DUMMY\\_PADDING](#)

### 3.94.1 Detailed Description

Definition at line 101 of file [grccparam.h](#).

### 3.94.2 Field Documentation

#### 3.94.2.1 name

```
const char* OptDef::name
```

Definition at line 102 of file [grccparam.h](#).

#### 3.94.2.2 mean

```
const char* OptDef::mean
```

Definition at line 103 of file [grccparam.h](#).

#### 3.94.2.3 defaultv

```
int OptDef::defaultv
```

Definition at line 104 of file [grccparam.h](#).

#### 3.94.2.4 DUMMYPADDING

```
int OptDef::DUMMYPADDING
```

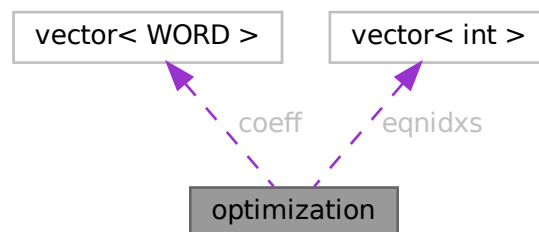
Definition at line 105 of file [grccparam.h](#).

The documentation for this struct was generated from the following file:

- [grccparam.h](#)

## 3.95 optimization Class Reference

Collaboration diagram for optimization:



## Public Member Functions

- bool `operator<` (const `optimization` &a) const

## Data Fields

- int `type`
- int `arg1`
- int `arg2`
- int `improve`
- vector< WORD > `coeff`
- vector< int > `eqnidxs`

### 3.95.1 Detailed Description

class Optimization

### 3.95.2 Description

This object represents an optimization. Its type is a number in the range 0 to 5. Depending on this type, the variables `arg1`, `arg2` and `coeff` indicate:

type==0 : optimization of the form  $x[\text{arg1}] \wedge \text{arg2}$  (coeff=empty) type==1 : optimization of the form  $x[\text{arg1}] * x[\text{arg2}]$  (coeff=empty) type==2 : optimization of the form  $x[\text{arg1}] * \text{coeff}$  (arg2=0) type==3 : optimization of the form  $x[\text{arg1}] + \text{coeff}$  (arg2=0) type==4 : optimization of the form  $x[\text{arg1}] + x[\text{arg2}]$  (coeff=empty) type==5 : optimization of the form  $x[\text{arg1}] - x[\text{arg2}]$  (coeff=empty)

Here, "`x[arg]`" represents a symbol (if positive) or an extrasymbol (if negative). The represented symbol's id is  $\text{ABS}(x[\text{arg}]) - 1$ .

"eqns" is a list of equation, where this optimization can be performed.

"improve" is the total improvement of this optimization.

Definition at line 2670 of file `optimize.cc`.

### 3.95.3 Member Function Documentation

#### 3.95.3.1 `operator<()`

```
bool optimization::operator< (
    const optimization & a ) const [inline]
```

Definition at line 2676 of file `optimize.cc`.

### 3.95.4 Field Documentation

#### 3.95.4.1 type

```
int optimization::type
```

Definition at line 2672 of file [optimize.cc](#).

#### 3.95.4.2 arg1

```
int optimization::arg1
```

Definition at line 2672 of file [optimize.cc](#).

#### 3.95.4.3 arg2

```
int optimization::arg2
```

Definition at line 2672 of file [optimize.cc](#).

#### 3.95.4.4 improve

```
int optimization::improve
```

Definition at line 2672 of file [optimize.cc](#).

#### 3.95.4.5 coeff

```
vector<WORD> optimization::coeff
```

Definition at line 2673 of file [optimize.cc](#).

#### 3.95.4.6 eqnidxs

```
vector<int> optimization::eqnidxs
```

Definition at line 2674 of file [optimize.cc](#).

The documentation for this class was generated from the following file:

- [optimize.cc](#)

## 3.96 OPTIMIZE Struct Reference

### Data Fields

- union {  
    float [fval](#)  
    int [ival](#) [2]  
} **mctsconstant**
- int [horner](#)
- int [hornerdirection](#)
- int [method](#)
- int [mctstimelimit](#)
- int [mctsnumexpand](#)
- int [mctsnumkeep](#)
- int [mctsnumrepeat](#)
- int [greedytimelimit](#)
- int [greedyminnum](#)
- int [greedymaxperc](#)
- int [printstats](#)
- int [debugflags](#)
- int [schemeflags](#)
- int [mctsdecaymode](#)
- int [salter](#)
- union {  
    float [fval](#)  
    int [ival](#) [2]  
} **saMaxT**
- union {  
    float [fval](#)  
    int [ival](#) [2]  
} **saMinT**
- int [spare](#)

### 3.96.1 Detailed Description

Definition at line [1312](#) of file [structs.h](#).

### 3.96.2 Field Documentation

#### 3.96.2.1 fval

```
float OPTIMIZE::fval
```

Definition at line [1314](#) of file [structs.h](#).

#### 3.96.2.2 ival

```
int OPTIMIZE::ival[2]
```

Definition at line [1315](#) of file [structs.h](#).

### 3.96.2.3 horner

```
int OPTIMIZE::horner
```

Definition at line 1317 of file [structs.h](#).

### 3.96.2.4 hornerdirection

```
int OPTIMIZE::hornerdirection
```

Definition at line 1318 of file [structs.h](#).

### 3.96.2.5 method

```
int OPTIMIZE::method
```

Definition at line 1319 of file [structs.h](#).

### 3.96.2.6 mctstimelimit

```
int OPTIMIZE::mctstimelimit
```

Definition at line 1320 of file [structs.h](#).

### 3.96.2.7 mctsnumexpand

```
int OPTIMIZE::mctsnumexpand
```

Definition at line 1321 of file [structs.h](#).

### 3.96.2.8 mctsnumkeep

```
int OPTIMIZE::mctsnumkeep
```

Definition at line 1322 of file [structs.h](#).

### 3.96.2.9 mctsnumrepeat

```
int OPTIMIZE::mctsnumrepeat
```

Definition at line 1323 of file [structs.h](#).

### 3.96.2.10 greedytimelimit

```
int OPTIMIZE::greedytimelimit
```

Definition at line 1324 of file [structs.h](#).

### 3.96.2.11 greedyminnum

```
int OPTIMIZE::greedyminnum
```

Definition at line 1325 of file [structs.h](#).

### 3.96.2.12 greedymaxperc

```
int OPTIMIZE::greedymaxperc
```

Definition at line 1326 of file [structs.h](#).

### 3.96.2.13 printstats

```
int OPTIMIZE::printstats
```

Definition at line 1327 of file [structs.h](#).

### 3.96.2.14 debugflags

```
int OPTIMIZE::debugflags
```

Definition at line 1328 of file [structs.h](#).

### 3.96.2.15 schemeflags

```
int OPTIMIZE::schemeflags
```

Definition at line 1329 of file [structs.h](#).

### 3.96.2.16 mctsdecaymode

```
int OPTIMIZE::mctsdecaymode
```

Definition at line 1330 of file [structs.h](#).

### 3.96.2.17 salter

```
int OPTIMIZE::saIter
```

Definition at line 1331 of file [structs.h](#).



### 3.96.2.18 spare

```
int OPTIMIZE::spare
```

Definition at line 1340 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.97 OPTIMIZERESULT Struct Reference

### Public Member Functions

- **PADPOSITION** (2, 1, 0, 3, 0)

### Data Fields

- WORD \* [code](#)
- UBYTE \* [nameofexpr](#)
- LONG [codesize](#)
- WORD [exprnr](#)
- WORD [minvar](#)
- WORD [maxvar](#)

### 3.97.1 Detailed Description

Definition at line 1343 of file [structs.h](#).

### 3.97.2 Field Documentation

#### 3.97.2.1 code

```
WORD* OPTIMIZERESULT::code
```

Definition at line 1344 of file [structs.h](#).

#### 3.97.2.2 nameofexpr

```
UBYTE* OPTIMIZERESULT::nameofexpr
```

Definition at line 1345 of file [structs.h](#).

### 3.97.2.3 codesize

LONG OPTIMIZERESULT::codesize

Definition at line 1346 of file [structs.h](#).

### 3.97.2.4 exprnr

WORD OPTIMIZERESULT::exprnr

Definition at line 1347 of file [structs.h](#).

### 3.97.2.5 minvar

WORD OPTIMIZERESULT::minvar

Definition at line 1348 of file [structs.h](#).

### 3.97.2.6 maxvar

WORD OPTIMIZERESULT::maxvar

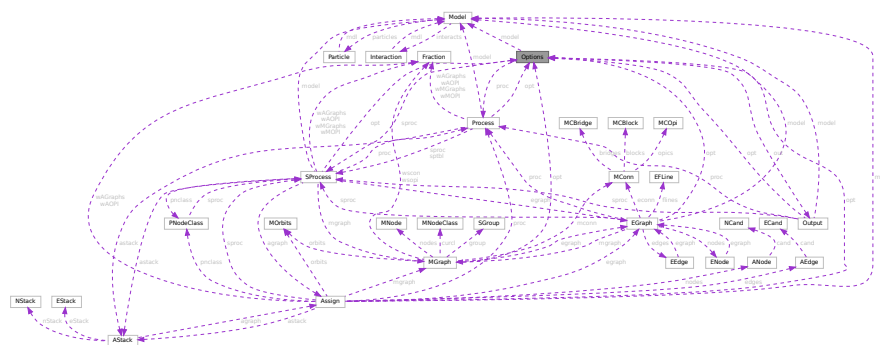
Definition at line 1349 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.98 Options Class Reference

Collaboration diagram for Options:



**Public Member Functions**

- void [setDefault](#)Value (void)
- void [setValue](#) (int ind, int val)
- int [getValue](#) (int ind)
- void [print](#) (void)
- void [setOutputF](#) (Bool outgrf, const char \*fname)
- void [setOutputP](#) (Bool outgrp, const char \*fname)
- void [printLevel](#) (int l)
- void [printModel](#) (void)
- void [setOutMG](#) (OutEGB \*omg, void \*pt)
- void [setOutAG](#) (OutEG \*oag, void \*pt)
- void [setEndMG](#) (OutEGB \*omg, void \*pt)
- void [setErExit](#) (ErExit \*ere, void \*pt)
- const [OptDef](#) \* [getDef](#) (void)
- void [begin](#) ([Model](#) \*mdl)
- void [end](#) (void)
- void [beginProc](#) ([Process](#) \*prc)
- void [endProc](#) (void)
- void [beginSubProc](#) ([SProcess](#) \*sprc)
- void [endSubProc](#) (void)
- void [newMGraph](#) ([MGraph](#) \*mgr)
- void [newAGraph](#) ([EGraph](#) \*egr)
- void [outModel](#) (void)

**Data Fields**

- [Model](#) \* [model](#)
- [Process](#) \* [proc](#)
- [SProcess](#) \* [sprc](#)
- [Output](#) \* [out](#)
- OutEGB \* [outmg](#)
- OutEGB \* [endmg](#)
- OutEG \* [outag](#)
- void \* [argmg](#)
- void \* [argemg](#)
- void \* [argag](#)
- int [values](#) [GRCC\_OPT\_Size]
- int [DUMMY\\_PADDING](#)
- double [time0](#)
- double [time1](#)

**3.98.1 Detailed Description**

Definition at line 88 of file [grcc.h](#).

**3.98.2 Constructor & Destructor Documentation****3.98.2.1 Options()**

```
Options::Options (
    void )
```

Definition at line 117 of file [grcc.cc](#).

### 3.98.2.2 ~Options()

```
Options::~~Options (
    void )
```

Definition at line 149 of file [grcc.cc](#).

## 3.98.3 Member Function Documentation

### 3.98.3.1 setDefaultValue()

```
void Options::setDefaultValue (
    void )
```

Definition at line 161 of file [grcc.cc](#).

### 3.98.3.2 setValue()

```
void Options::setValue (
    int ind,
    int val )
```

Definition at line 212 of file [grcc.cc](#).

### 3.98.3.3 getValue()

```
int Options::getValue (
    int ind )
```

Definition at line 227 of file [grcc.cc](#).

### 3.98.3.4 print()

```
void Options::print (
    void )
```

Definition at line 241 of file [grcc.cc](#).

### 3.98.3.5 setOutputF()

```
void Options::setOutputF (
    Bool outgrf,
    const char * fname )
```

Definition at line 264 of file [grcc.cc](#).

### 3.98.3.6 setOutputP()

```
void Options::setOutputP (
    Bool outgrp,
    const char * fname )
```

Definition at line 271 of file [grcc.cc](#).

### 3.98.3.7 printLevel()

```
void Options::printLevel (
    int l )
```

Definition at line 206 of file [grcc.cc](#).

### 3.98.3.8 printModel()

```
void Options::printModel (
    void )
```

Definition at line 278 of file [grcc.cc](#).

### 3.98.3.9 setOutMG()

```
void Options::setOutMG (
    OutEGB * omg,
    void * pt )
```

Definition at line 178 of file [grcc.cc](#).

### 3.98.3.10 setOutAG()

```
void Options::setOutAG (
    OutEG * oag,
    void * pt )
```

Definition at line 192 of file [grcc.cc](#).

### 3.98.3.11 setEndMG()

```
void Options::setEndMG (
    OutEGB * omg,
    void * pt )
```

Definition at line 185 of file [grcc.cc](#).

### 3.98.3.12 setErExit()

```
void Options::setErExit (
    ErExit * ere,
    void * pt )
```

Definition at line 199 of file [grcc.cc](#).

### 3.98.3.13 getDef()

```
const OptDef * Options::getDef (
    void )
```

Definition at line 257 of file [grcc.cc](#).

### 3.98.3.14 begin()

```
void Options::begin (
    Model * mdl )
```

Definition at line 299 of file [grcc.cc](#).

### 3.98.3.15 end()

```
void Options::end (
    void )
```

Definition at line 319 of file [grcc.cc](#).

### 3.98.3.16 beginProc()

```
void Options::beginProc (
    Process * prc )
```

Definition at line 351 of file [grcc.cc](#).

### 3.98.3.17 endProc()

```
void Options::endProc (
    void )
```

Definition at line 367 of file [grcc.cc](#).

### 3.98.3.18 beginSubProc()

```
void Options::beginSubProc (
    SProcess * sprc )
```

Definition at line 456 of file [grcc.cc](#).

### 3.98.3.19 endSubProc()

```
void Options::endSubProc (
    void )
```

Definition at line 477 of file [grcc.cc](#).

### 3.98.3.20 newMGraph()

```
void Options::newMGraph (
    MGraph * mgr )
```

Definition at line 546 of file [grcc.cc](#).

### 3.98.3.21 newAGraph()

```
void Options::newAGraph (
    EGraph * egr )
```

Definition at line 583 of file [grcc.cc](#).

### 3.98.3.22 outModel()

```
void Options::outModel (
    void )
```

Definition at line 290 of file [grcc.cc](#).

## 3.98.4 Field Documentation

### 3.98.4.1 model

```
Model* Options::model
```

Definition at line 91 of file [grcc.h](#).

### 3.98.4.2 proc

```
Process* Options::proc
```

Definition at line 92 of file [grcc.h](#).

### 3.98.4.3 sproc

```
SProcess* Options::sproc
```

Definition at line 93 of file [grcc.h](#).

#### 3.98.4.4 out

`Output* Options::out`

Definition at line 94 of file [grcc.h](#).

#### 3.98.4.5 outmg

`OutEGB* Options::outmg`

Definition at line 95 of file [grcc.h](#).

#### 3.98.4.6 endmg

`OutEGB* Options::endmg`

Definition at line 96 of file [grcc.h](#).

#### 3.98.4.7 outag

`OutEG* Options::outag`

Definition at line 97 of file [grcc.h](#).

#### 3.98.4.8 argmg

`void* Options::argmg`

Definition at line 98 of file [grcc.h](#).

#### 3.98.4.9 argemg

`void* Options::argemg`

Definition at line 99 of file [grcc.h](#).

#### 3.98.4.10 argag

`void* Options::argag`

Definition at line 100 of file [grcc.h](#).

#### 3.98.4.11 values

`int Options::values[GRCC_OPT_Size]`

Definition at line 103 of file [grcc.h](#).



### 3.98.4.12 DUMMYPADDING

```
int Options::DUMMYPADDING
```

Definition at line 105 of file [grcc.h](#).

### 3.98.4.13 time0

```
double Options::time0
```

Definition at line 108 of file [grcc.h](#).

### 3.98.4.14 time1

```
double Options::time1
```

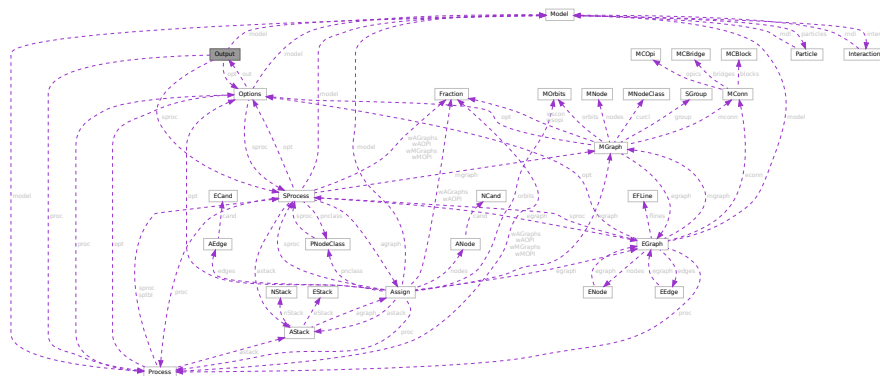
Definition at line 109 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.99 Output Class Reference

Collaboration diagram for Output:



## Public Member Functions

- [Output](#) ([Options](#) \*optn)
- void [setOutgrf](#) (const char \*fname)
- void [setOutgrp](#) (const char \*fname)
- Bool [outBeginF](#) ([Model](#) \*mdl, Bool pr)
- Bool [outBeginP](#) ([Model](#) \*mdl, Bool pr)
- void [outEndF](#) (void)
- void [outEndP](#) (void)
- void [outProcBeginF](#) ([Process](#) \*prc)
- void [outProcBeginP](#) ([Process](#) \*prc)
- void [outProcBegin0](#) (int next, int couple, int loop)
- void [outSProcBeginF](#) ([SProcess](#) \*sprc)
- void [outSProcBeginP](#) ([SProcess](#) \*sprc)
- void [outProcEndF](#) (void)
- void [outProcEndP](#) (void)
- void [outEGraphF](#) ([EGraph](#) \*egraph)
- void [outEGraphP](#) ([EGraph](#) \*egraph)
- void [outModelF](#) (void)
- void [outModelP](#) (void)

## Data Fields

- [Options](#) \* opt
- [Model](#) \* model
- [Process](#) \* proc
- [SProcess](#) \* sprc
- char \* [outgrf](#)
- FILE \* [outgrfp](#)
- char \* [outgrp](#)
- FILE \* [outgrpp](#)
- int [procl](#)
- Bool [outproc](#)

### 3.99.1 Detailed Description

Definition at line 144 of file [grcc.h](#).

### 3.99.2 Constructor & Destructor Documentation

#### 3.99.2.1 Output()

```
Output::Output (
    Options * optn )
```

Definition at line 637 of file [grcc.cc](#).

### 3.99.2.2 ~Output()

```
Output::~~Output (
    void )
```

Definition at line 653 of file [grcc.cc](#).

## 3.99.3 Member Function Documentation

### 3.99.3.1 setOutgrf()

```
void Output::setOutgrf (
    const char * fname )
```

Definition at line 684 of file [grcc.cc](#).

### 3.99.3.2 setOutgrp()

```
void Output::setOutgrp (
    const char * fname )
```

Definition at line 701 of file [grcc.cc](#).

### 3.99.3.3 outBeginF()

```
Bool Output::outBeginF (
    Model * mdl,
    Bool pr )
```

Definition at line 718 of file [grcc.cc](#).

### 3.99.3.4 outBeginP()

```
Bool Output::outBeginP (
    Model * mdl,
    Bool pr )
```

Definition at line 756 of file [grcc.cc](#).

### 3.99.3.5 outEndF()

```
void Output::outEndF (
    void )
```

Definition at line 787 of file [grcc.cc](#).

#### 3.99.3.6 outEndP()

```
void Output::outEndP (
    void )
```

Definition at line 803 of file [grcc.cc](#).

#### 3.99.3.7 outProcBeginF()

```
void Output::outProcBeginF (
    Process * prc )
```

Definition at line 819 of file [grcc.cc](#).

#### 3.99.3.8 outProcBeginP()

```
void Output::outProcBeginP (
    Process * prc )
```

Definition at line 864 of file [grcc.cc](#).

#### 3.99.3.9 outProcBegin0()

```
void Output::outProcBegin0 (
    int next,
    int couple,
    int loop )
```

Definition at line 886 of file [grcc.cc](#).

#### 3.99.3.10 outSProcBeginF()

```
void Output::outSProcBeginF (
    SProcess * sprc )
```

Definition at line 918 of file [grcc.cc](#).

#### 3.99.3.11 outSProcBeginP()

```
void Output::outSProcBeginP (
    SProcess * sprc )
```

Definition at line 973 of file [grcc.cc](#).

#### 3.99.3.12 outProcEndF()

```
void Output::outProcEndF (
    void )
```

Definition at line 993 of file [grcc.cc](#).

### 3.99.3.13 outProcEndP()

```
void Output::outProcEndP (
    void )
```

Definition at line 1012 of file [grcc.cc](#).

### 3.99.3.14 outEGraphF()

```
void Output::outEGraphF (
    EGraph * egraph )
```

Definition at line 1024 of file [grcc.cc](#).

### 3.99.3.15 outEGraphP()

```
void Output::outEGraphP (
    EGraph * egraph )
```

Definition at line 1152 of file [grcc.cc](#).

### 3.99.3.16 outModelF()

```
void Output::outModelF (
    void )
```

Definition at line 1254 of file [grcc.cc](#).

### 3.99.3.17 outModelP()

```
void Output::outModelP (
    void )
```

Definition at line 1335 of file [grcc.cc](#).

## 3.99.4 Field Documentation

### 3.99.4.1 opt

[Options\\*](#) Output::opt

Definition at line 147 of file [grcc.h](#).

### 3.99.4.2 model

[Model\\*](#) Output::model

Definition at line 148 of file [grcc.h](#).

#### 3.99.4.3 proc

`Process*` Output::proc

Definition at line 149 of file [grcc.h](#).

#### 3.99.4.4 sproc

`SProcess*` Output::sproc

Definition at line 150 of file [grcc.h](#).

#### 3.99.4.5 outgrf

`char*` Output::outgrf

Definition at line 151 of file [grcc.h](#).

#### 3.99.4.6 outgrfp

`FILE*` Output::outgrfp

Definition at line 152 of file [grcc.h](#).

#### 3.99.4.7 outgrp

`char*` Output::outgrp

Definition at line 153 of file [grcc.h](#).

#### 3.99.4.8 outgrpp

`FILE*` Output::outgrpp

Definition at line 154 of file [grcc.h](#).

#### 3.99.4.9 proclId

`int` Output::procId

Definition at line 155 of file [grcc.h](#).

## 3.99.4.10 outproc

```
Bool Output::outproc
```

Definition at line 156 of file [grcc.h](#).

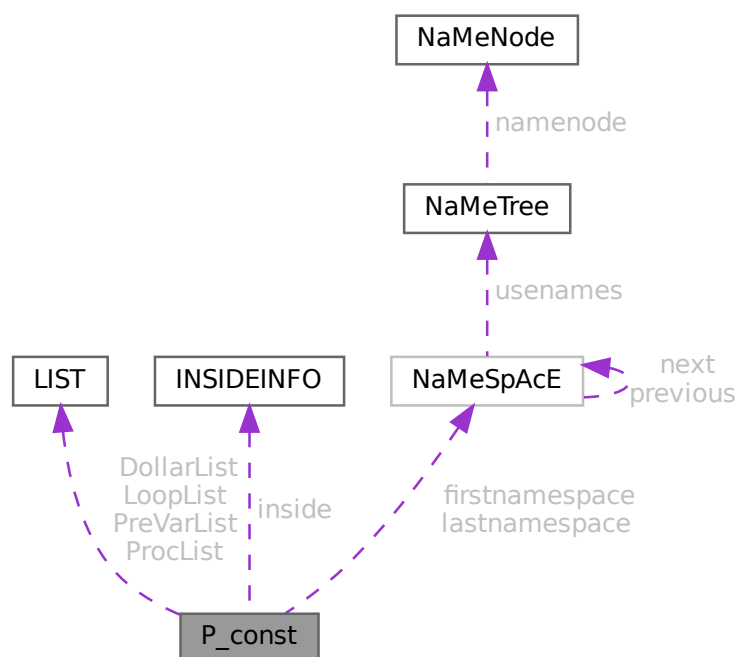
The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.100 P\_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for P\_const:



## Data Fields

- [LIST DollarList](#)
- [LIST PreVarList](#)
- [LIST LoopList](#)
- [LIST ProcList](#)
- [INSIDEINFO inside](#)

- [NAMESPACE](#) \* [firstnamespace](#)
- [NAMESPACE](#) \* [lastnamespace](#)
- [UBYTE](#) \* [fullName](#)
- [UBYTE](#) \*\* [PreSwitchStrings](#)
- [UBYTE](#) \* [preStart](#)
- [UBYTE](#) \* [preStop](#)
- [UBYTE](#) \* [preFill](#)
- [UBYTE](#) \* [procedureExtension](#)
- [UBYTE](#) \* [cprocedureExtension](#)
- [LONG](#) \* [PreAssignStack](#)
- [int](#) \* [PrelfStack](#)
- [int](#) \* [PreSwitchModes](#)
- [int](#) \* [PreTypes](#)
- [LONG](#) [StopWatchZero](#)
- [LONG](#) [InOutBuf](#)
- [LONG](#) [pSize](#)
- [int](#) [PreAssignFlag](#)
- [int](#) [PreContinuation](#)
- [int](#) [PreproFlag](#)
- [int](#) [iBufError](#)
- [int](#) [PreOut](#)
- [int](#) [PreSwitchLevel](#)
- [int](#) [NumPreSwitchStrings](#)
- [int](#) [MaxPreTypes](#)
- [int](#) [NumPreTypes](#)
- [int](#) [MaxPrelfLevel](#)
- [int](#) [PrelfLevel](#)
- [int](#) [PreInsideLevel](#)
- [int](#) [DelayPrevar](#)
- [int](#) [AllowDelay](#)
- [int](#) [lhdollarerror](#)
- [int](#) [eat](#)
- [int](#) [gNumPre](#)
- [int](#) [PreDebug](#)
- [int](#) [OpenDictionary](#)
- [int](#) [PreAssignLevel](#)
- [int](#) [MaxPreAssignLevel](#)
- [int](#) [fullnamesize](#)
- [WORD](#) [DebugFlag](#)
- [WORD](#) [preError](#)
- [UBYTE](#) [ComChar](#)
- [UBYTE](#) [cComChar](#)

### 3.100.1 Detailed Description

The [P\\_const](#) struct is part of the global data and resides in the ALLGLOBALS struct A under the name P We see it used with the macro AP as in AP.InOutBuf It contains objects that have dealings with the preprocessor.

Definition at line [1628](#) of file [structs.h](#).



## 3.100.2 Field Documentation

### 3.100.2.1 DollarList

`LIST P_const::DollarList`

Definition at line 1629 of file [structs.h](#).

### 3.100.2.2 PreVarList

`LIST P_const::PreVarList`

Definition at line 1631 of file [structs.h](#).

### 3.100.2.3 LoopList

`LIST P_const::LoopList`

Definition at line 1633 of file [structs.h](#).

### 3.100.2.4 ProcList

`LIST P_const::ProcList`

Definition at line 1634 of file [structs.h](#).

### 3.100.2.5 inside

`INSIDEINFO P_const::inside`

Definition at line 1635 of file [structs.h](#).

### 3.100.2.6 firstnamespace

`NAMESPACE* P_const::firstnamespace`

Definition at line 1636 of file [structs.h](#).

### 3.100.2.7 lastnamespace

`NAMESPACE* P_const::lastnamespace`

Definition at line 1637 of file [structs.h](#).

### 3.100.2.8 fullname

```
UBYTE* P_const::fullname
```

Definition at line 1638 of file [structs.h](#).

### 3.100.2.9 PreSwitchStrings

```
UBYTE** P_const::PreSwitchStrings
```

Definition at line 1639 of file [structs.h](#).

### 3.100.2.10 preStart

```
UBYTE* P_const::preStart
```

Definition at line 1640 of file [structs.h](#).

### 3.100.2.11 preStop

```
UBYTE* P_const::preStop
```

Definition at line 1641 of file [structs.h](#).

### 3.100.2.12 preFill

```
UBYTE* P_const::preFill
```

Definition at line 1642 of file [structs.h](#).

### 3.100.2.13 procedureExtension

```
UBYTE* P_const::procedureExtension
```

Definition at line 1643 of file [structs.h](#).

### 3.100.2.14 cprocedureExtension

```
UBYTE* P_const::cprocedureExtension
```

Definition at line 1644 of file [structs.h](#).

### 3.100.2.15 PreAssignStack

```
LONG* P_const::PreAssignStack
```

Definition at line 1645 of file [structs.h](#).

**3.100.2.16 PreIfStack**

```
int* P_const::PreIfStack
```

Definition at line 1646 of file [structs.h](#).

**3.100.2.17 PreSwitchModes**

```
int* P_const::PreSwitchModes
```

Definition at line 1647 of file [structs.h](#).

**3.100.2.18 PreTypes**

```
int* P_const::PreTypes
```

Definition at line 1648 of file [structs.h](#).

**3.100.2.19 StopWatchZero**

```
LONG P_const::StopWatchZero
```

Definition at line 1652 of file [structs.h](#).

**3.100.2.20 InOutBuf**

```
LONG P_const::InOutBuf
```

Definition at line 1653 of file [structs.h](#).

**3.100.2.21 pSize**

```
LONG P_const::pSize
```

Definition at line 1654 of file [structs.h](#).

**3.100.2.22 PreAssignFlag**

```
int P_const::PreAssignFlag
```

Definition at line 1655 of file [structs.h](#).

**3.100.2.23 PreContinuation**

```
int P_const::PreContinuation
```

Definition at line 1656 of file [structs.h](#).

**3.100.2.24 PreproFlag**

```
int P_const::PreproFlag
```

Definition at line 1657 of file [structs.h](#).

**3.100.2.25 iBufError**

```
int P_const::iBufError
```

Definition at line 1658 of file [structs.h](#).

**3.100.2.26 PreOut**

```
int P_const::PreOut
```

Definition at line 1659 of file [structs.h](#).

**3.100.2.27 PreSwitchLevel**

```
int P_const::PreSwitchLevel
```

Definition at line 1660 of file [structs.h](#).

**3.100.2.28 NumPreSwitchStrings**

```
int P_const::NumPreSwitchStrings
```

Definition at line 1661 of file [structs.h](#).

**3.100.2.29 MaxPreTypes**

```
int P_const::MaxPreTypes
```

Definition at line 1662 of file [structs.h](#).

**3.100.2.30 NumPreTypes**

```
int P_const::NumPreTypes
```

Definition at line 1663 of file [structs.h](#).

**3.100.2.31 MaxPreIfLevel**

```
int P_const::MaxPreIfLevel
```

Definition at line 1664 of file [structs.h](#).

**3.100.2.32 PreIfLevel**

```
int P_const::PreIfLevel
```

Definition at line 1665 of file [structs.h](#).

**3.100.2.33 PreInsideLevel**

```
int P_const::PreInsideLevel
```

Definition at line 1666 of file [structs.h](#).

**3.100.2.34 DelayPrevar**

```
int P_const::DelayPrevar
```

Definition at line 1667 of file [structs.h](#).

**3.100.2.35 AllowDelay**

```
int P_const::AllowDelay
```

Definition at line 1668 of file [structs.h](#).

**3.100.2.36 lhdollarerror**

```
int P_const::lhdollarerror
```

Definition at line 1669 of file [structs.h](#).

**3.100.2.37 eat**

```
int P_const::eat
```

Definition at line 1670 of file [structs.h](#).

**3.100.2.38 gNumPre**

```
int P_const::gNumPre
```

Definition at line 1671 of file [structs.h](#).

**3.100.2.39 PreDebug**

```
int P_const::PreDebug
```

Definition at line 1672 of file [structs.h](#).

#### 3.100.2.40 OpenDictionary

```
int P_const::OpenDictionary
```

Definition at line 1673 of file [structs.h](#).

#### 3.100.2.41 PreAssignLevel

```
int P_const::PreAssignLevel
```

Definition at line 1674 of file [structs.h](#).

#### 3.100.2.42 MaxPreAssignLevel

```
int P_const::MaxPreAssignLevel
```

Definition at line 1675 of file [structs.h](#).

#### 3.100.2.43 fullnamesize

```
int P_const::fullnamesize
```

Definition at line 1676 of file [structs.h](#).

#### 3.100.2.44 DebugFlag

```
WORD P_const::DebugFlag
```

Definition at line 1677 of file [structs.h](#).

#### 3.100.2.45 preError

```
WORD P_const::preError
```

Definition at line 1678 of file [structs.h](#).

#### 3.100.2.46 ComChar

```
UBYTE P_const::ComChar
```

Definition at line 1679 of file [structs.h](#).

## 3.100.2.47 cComChar

```
UBYTE P_const::cComChar
```

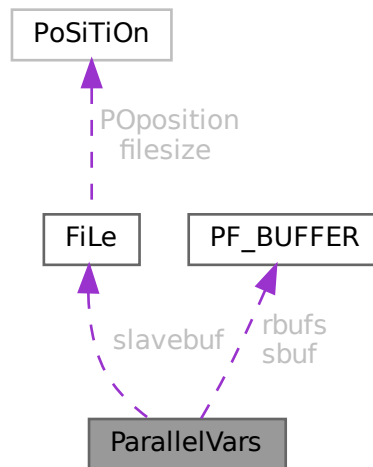
Definition at line 1680 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.101 ParallelVars Struct Reference

Collaboration diagram for ParallelVars:



## Public Member Functions

- **PADPOSITION** (2, 0, 8, 2, 0)

## Data Fields

- [FILEHANDLE](#) [slavebuf](#)
- [PF\\_BUFFER](#) \* [sbuf](#)
- [PF\\_BUFFER](#) \*\* [rbufs](#)
- int [me](#)
- int [numtasks](#)
- int [parallel](#)
- int [rhsInParallel](#)
- int [mkSlaveInfile](#)
- int [exprbufsize](#)
- int [exprtodo](#)
- int [log](#)
- WORD [numsbufs](#)
- WORD [numrbufs](#)

### 3.101.1 Detailed Description

Definition at line 169 of file [parallel.h](#).

### 3.101.2 Field Documentation

#### 3.101.2.1 slavebuf

```
FILEHANDLE ParallelVars::slavebuf
```

Definition at line 170 of file [parallel.h](#).

#### 3.101.2.2 sbuf

```
PF_BUFFER* ParallelVars::sbuf
```

Definition at line 172 of file [parallel.h](#).

#### 3.101.2.3 rbufs

```
PF_BUFFER** ParallelVars::rbufs
```

Definition at line 173 of file [parallel.h](#).

#### 3.101.2.4 me

```
int ParallelVars::me
```

Definition at line 174 of file [parallel.h](#).

#### 3.101.2.5 numtasks

```
int ParallelVars::numtasks
```

Definition at line 175 of file [parallel.h](#).

#### 3.101.2.6 parallel

```
int ParallelVars::parallel
```

Definition at line 176 of file [parallel.h](#).



### 3.101.2.7 rhsInParallel

```
int ParallelVars::rhsInParallel
```

Definition at line 178 of file [parallel.h](#).

### 3.101.2.8 mkSlaveInfile

```
int ParallelVars::mkSlaveInfile
```

Definition at line 179 of file [parallel.h](#).

### 3.101.2.9 exprbufsize

```
int ParallelVars::exprbufsize
```

Definition at line 180 of file [parallel.h](#).

### 3.101.2.10 exptodo

```
int ParallelVars::exptodo
```

Definition at line 181 of file [parallel.h](#).

### 3.101.2.11 log

```
int ParallelVars::log
```

Definition at line 182 of file [parallel.h](#).

### 3.101.2.12 numsbufs

```
WORD ParallelVars::numsbufs
```

Definition at line 183 of file [parallel.h](#).

### 3.101.2.13 numrbufs

```
WORD ParallelVars::numrbufs
```

Definition at line 184 of file [parallel.h](#).

The documentation for this struct was generated from the following file:

- [parallel.h](#)

## 3.102 PaRtl Struct Reference

```
#include <structs.h>
```

### Data Fields

- WORD \* [psize](#)
- WORD \* [args](#)
- WORD \* [nargs](#)
- WORD \* [nfun](#)
- WORD [numargs](#)
- WORD [numpart](#)
- WORD [where](#)

### 3.102.1 Detailed Description

The struct PARTI is used to help determining whether a `partition_` function can be replaced.

Definition at line [1120](#) of file [structs.h](#).

### 3.102.2 Field Documentation

#### 3.102.2.1 psize

```
WORD* PaRtI::psize
```

Definition at line [1121](#) of file [structs.h](#).

#### 3.102.2.2 args

```
WORD* PaRtI::args
```

Definition at line [1122](#) of file [structs.h](#).

#### 3.102.2.3 nargs

```
WORD* PaRtI::nargs
```

Definition at line [1123](#) of file [structs.h](#).

#### 3.102.2.4 nfun

```
WORD* PaRtI::nfun
```

Definition at line [1124](#) of file [structs.h](#).

### 3.102.2.5 numargs

WORD PaRtI::numargs

Definition at line 1125 of file [structs.h](#).

### 3.102.2.6 numpart

WORD PaRtI::numpart

Definition at line 1126 of file [structs.h](#).

### 3.102.2.7 where

WORD PaRtI::where

Definition at line 1127 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.103 PaRtlcLe Struct Reference

### Data Fields

- WORD [number](#)
- WORD [spin](#)
- WORD [mass](#)
- WORD [type](#)

### 3.103.1 Detailed Description

Definition at line 619 of file [structs.h](#).

### 3.103.2 Field Documentation

#### 3.103.2.1 number

WORD PaRtIcLe::number

Definition at line 620 of file [structs.h](#).

### 3.103.2.2 spin

WORD PaRtIcLe::spin

Definition at line 621 of file [structs.h](#).

### 3.103.2.3 mass

WORD PaRtIcLe::mass

Definition at line 622 of file [structs.h](#).

### 3.103.2.4 type

WORD PaRtIcLe::type

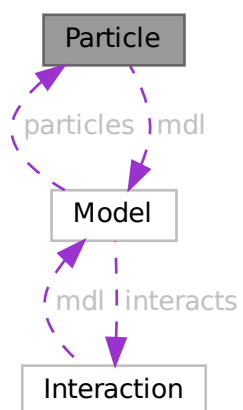
Definition at line 623 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.104 Particle Class Reference

Collaboration diagram for Particle:



## Public Member Functions

- [Particle](#) ([Model](#) \*modl, int pid, [PInput](#) \*pinp)
- char \* [particleName](#) (int p)
- int [particleCode](#) (int p)
- char \* [interactionName](#) (int p)
- char \* [aparticle](#) (void)
- void [prParticle](#) (void)
- int [isNeutral](#) (void)
- const char \* [typeName](#) (void)
- const char \* [typeGName](#) (void)

## Data Fields

- [Model](#) \* [mdl](#)
- char \* [name](#)
- char \* [aname](#)
- int [id](#)
- int [ptype](#)
- int [neutral](#)
- int [pcode](#)
- int [acode](#)
- int [cmindeg](#)
- int [cmaxdeg](#)
- int [extonly](#)

### 3.104.1 Detailed Description

Definition at line 205 of file [grcc.h](#).

### 3.104.2 Constructor & Destructor Documentation

#### 3.104.2.1 Particle()

```
Particle::Particle (  
    Model * modl,  
    int pid,  
    PInput * pinp )
```

Definition at line 1422 of file [grcc.cc](#).

#### 3.104.2.2 ~Particle()

```
Particle::~~Particle (  
    void )
```

Definition at line 1499 of file [grcc.cc](#).

### 3.104.3 Member Function Documentation

#### 3.104.3.1 `particleName()`

```
char * Particle::particleName (  
    int p )
```

Definition at line 1506 of file [grcc.cc](#).

#### 3.104.3.2 `particleCode()`

```
int Particle::particleCode (  
    int p )
```

Definition at line 1516 of file [grcc.cc](#).

#### 3.104.3.3 `aparticle()`

```
char * Particle::aparticle (  
    void )
```

Definition at line 1544 of file [grcc.cc](#).

#### 3.104.3.4 `prParticle()`

```
void Particle::prParticle (  
    void )
```

Definition at line 1556 of file [grcc.cc](#).

#### 3.104.3.5 `isNeutral()`

```
int Particle::isNeutral (  
    void )
```

Definition at line 1526 of file [grcc.cc](#).

#### 3.104.3.6 `typeName()`

```
const char * Particle::typeName (  
    void )
```

Definition at line 1532 of file [grcc.cc](#).

### 3.104.3.7 typeGName()

```
const char * Particle::typeGName (  
    void )
```

Definition at line 1538 of file [grcc.cc](#).

## 3.104.4 Field Documentation

### 3.104.4.1 mdl

```
Model* Particle::mdl
```

Definition at line 207 of file [grcc.h](#).

### 3.104.4.2 name

```
char* Particle::name
```

Definition at line 208 of file [grcc.h](#).

### 3.104.4.3 aname

```
char* Particle::aname
```

Definition at line 209 of file [grcc.h](#).

### 3.104.4.4 id

```
int Particle::id
```

Definition at line 210 of file [grcc.h](#).

### 3.104.4.5 ptype

```
int Particle::ptype
```

Definition at line 211 of file [grcc.h](#).

### 3.104.4.6 neutral

```
int Particle::neutral
```

Definition at line 212 of file [grcc.h](#).

#### 3.104.4.7 pcode

```
int Particle::pcode
```

Definition at line 213 of file [grcc.h](#).

#### 3.104.4.8 acode

```
int Particle::acode
```

Definition at line 214 of file [grcc.h](#).

#### 3.104.4.9 cmindeg

```
int Particle::cmindeg
```

Definition at line 215 of file [grcc.h](#).

#### 3.104.4.10 cmaxdeg

```
int Particle::cmaxdeg
```

Definition at line 216 of file [grcc.h](#).

#### 3.104.4.11 extonly

```
int Particle::extonly
```

Definition at line 218 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

### 3.105 PeRmUtE Struct Reference

```
#include <structs.h>
```

#### Data Fields

- WORD \* [objects](#)
- WORD [sign](#)
- WORD [n](#)
- WORD [cycle](#) [MAXMATCH]



### 3.105.1 Detailed Description

The struct PERM is used to generate all permutations when the pattern matcher has to try to match (anti)symmetric functions.

Definition at line 1078 of file [structs.h](#).

### 3.105.2 Field Documentation

#### 3.105.2.1 objects

```
WORD* PeRmUtE::objects
```

Definition at line 1079 of file [structs.h](#).

#### 3.105.2.2 sign

```
WORD PeRmUtE::sign
```

Definition at line 1080 of file [structs.h](#).

#### 3.105.2.3 n

```
WORD PeRmUtE::n
```

Definition at line 1081 of file [structs.h](#).

#### 3.105.2.4 cycle

```
WORD PeRmUtE::cycle[MAXMATCH]
```

Definition at line 1082 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.106 PeRmUtEp Struct Reference

```
#include <structs.h>
```

### Data Fields

- WORD \*\* [objects](#)
- WORD [sign](#)
- WORD [n](#)
- WORD [cycle](#) [MAXMATCH]

### 3.106.1 Detailed Description

Like struct PERM but works with pointers.

Definition at line 1090 of file [structs.h](#).

### 3.106.2 Field Documentation

#### 3.106.2.1 objects

```
WORD** PeRmUtEp::objects
```

Definition at line 1091 of file [structs.h](#).

#### 3.106.2.2 sign

```
WORD PeRmUtEp::sign
```

Definition at line 1092 of file [structs.h](#).

#### 3.106.2.3 n

```
WORD PeRmUtEp::n
```

Definition at line 1093 of file [structs.h](#).

#### 3.106.2.4 cycle

```
WORD PeRmUtEp::cycle[MAXMATCH]
```

Definition at line 1094 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.107 PF\_BUFFER Struct Reference

```
#include <parallel.h>
```

## Data Fields

- WORD \*\* [buff](#)
- WORD \*\* [fill](#)
- WORD \*\* [full](#)
- WORD \*\* [stop](#)
- MPI\_Status \* [status](#)
- MPI\_Status \* [retstat](#)
- MPI\_Request \* [request](#)
- MPI\_Datatype \* [type](#)
- int \* [index](#)
- int \* [tag](#)
- int \* [from](#)
- int [numbufs](#)
- int [active](#)

### 3.107.1 Detailed Description

A struct for nonblocking, unbuffered send of the sorted terms in the PObuffers back to the master using several "rotating" PObuffers.

Definition at line [147](#) of file [parallel.h](#).

### 3.107.2 Field Documentation

#### 3.107.2.1 buff

```
WORD** PF_BUFFER::buff
```

Definition at line [148](#) of file [parallel.h](#).

#### 3.107.2.2 fill

```
WORD** PF_BUFFER::fill
```

Definition at line [149](#) of file [parallel.h](#).

#### 3.107.2.3 full

```
WORD** PF_BUFFER::full
```

Definition at line [150](#) of file [parallel.h](#).

#### 3.107.2.4 stop

```
WORD** PF_BUFFER::stop
```

Definition at line [151](#) of file [parallel.h](#).

### 3.107.2.5 status

```
MPI_Status* PF_BUFFER::status
```

Definition at line 152 of file [parallel.h](#).

### 3.107.2.6 retstat

```
MPI_Status* PF_BUFFER::retstat
```

Definition at line 153 of file [parallel.h](#).

### 3.107.2.7 request

```
MPI_Request* PF_BUFFER::request
```

Definition at line 154 of file [parallel.h](#).

### 3.107.2.8 type

```
MPI_Datatype* PF_BUFFER::type
```

Definition at line 155 of file [parallel.h](#).

### 3.107.2.9 index

```
int* PF_BUFFER::index
```

Definition at line 156 of file [parallel.h](#).

### 3.107.2.10 tag

```
int* PF_BUFFER::tag
```

Definition at line 157 of file [parallel.h](#).

### 3.107.2.11 from

```
int* PF_BUFFER::from
```

Definition at line 158 of file [parallel.h](#).

### 3.107.2.12 numbufs

```
int PF_BUFFER::numbufs
```

Definition at line 159 of file [parallel.h](#).

### 3.107.2.13 active

```
int PF_BUFFER::active
```

Definition at line 160 of file [parallel.h](#).

The documentation for this struct was generated from the following file:

- [parallel.h](#)

## 3.108 PGInput Struct Reference

### Data Fields

- int [cdeg](#)
- int [ctyp](#)
- int [cnum](#)
- int [cple](#)
- const char \* [pname](#)
- long [pcode](#)

### 3.108.1 Detailed Description

Definition at line 176 of file [grccparam.h](#).

### 3.108.2 Field Documentation

#### 3.108.2.1 cdeg

```
int PGInput::cdeg
```

Definition at line 177 of file [grccparam.h](#).

#### 3.108.2.2 ctyp

```
int PGInput::ctyp
```

Definition at line 178 of file [grccparam.h](#).

#### 3.108.2.3 cnum

```
int PGInput::cnum
```

Definition at line 179 of file [grccparam.h](#).

#### 3.108.2.4 cple

```
int PGInput::cple
```

Definition at line 180 of file [grccparam.h](#).

#### 3.108.2.5 pname

```
const char* PGInput::pname
```

Definition at line 182 of file [grccparam.h](#).

#### 3.108.2.6 pcode

```
long PGInput::pcode
```

Definition at line 184 of file [grccparam.h](#).

The documentation for this struct was generated from the following file:

- [grccparam.h](#)

### 3.109 PInput Struct Reference

#### Data Fields

- const char \* [name](#)
- const char \* [aname](#)
- int [pcode](#)
- int [acode](#)
- const char \* [ptypen](#)
- int [ptypec](#)
- int [extonly](#)

#### 3.109.1 Detailed Description

Definition at line 127 of file [grccparam.h](#).

#### 3.109.2 Field Documentation

##### 3.109.2.1 name

```
const char* PInput::name
```

Definition at line 128 of file [grccparam.h](#).

### 3.109.2.2 aname

```
const char* PInput::aname
```

Definition at line 129 of file [grccparam.h](#).

### 3.109.2.3 pcode

```
int PInput::pcode
```

Definition at line 130 of file [grccparam.h](#).

### 3.109.2.4 acode

```
int PInput::acode
```

Definition at line 131 of file [grccparam.h](#).

### 3.109.2.5 ptypen

```
const char* PInput::ptypen
```

Definition at line 132 of file [grccparam.h](#).

### 3.109.2.6 ptypec

```
int PInput::ptypec
```

Definition at line 133 of file [grccparam.h](#).

### 3.109.2.7 extonly

```
int PInput::extonly
```

Definition at line 134 of file [grccparam.h](#).

The documentation for this struct was generated from the following file:

- [grccparam.h](#)





## 3.110.2 Constructor & Destructor Documentation

### 3.110.2.1 PNodeClass() [1/2]

```
PNodeClass::PNodeClass (
    SProcess * sprc,
    int nnods,
    int nclss,
    NCInput * cls )
```

Definition at line 2325 of file [grcc.cc](#).

### 3.110.2.2 PNodeClass() [2/2]

```
PNodeClass::PNodeClass (
    SProcess * sprc,
    int nnods,
    int nclss,
    int * dgs,
    int * typ,
    int * ptcl,
    int * cpl,
    int * cnt,
    int * cmind,
    int * cmaxd )
```

Definition at line 2399 of file [grcc.cc](#).

### 3.110.2.3 ~PNodeClass()

```
PNodeClass::~~PNodeClass (
    void )
```

Definition at line 2477 of file [grcc.cc](#).

## 3.110.3 Member Function Documentation

### 3.110.3.1 prPNodeClass()

```
void PNodeClass::prPNodeClass (
    void )
```

Definition at line 2492 of file [grcc.cc](#).

### 3.110.3.2 prElem()

```
void PNodeClass::prElem (
    int e )
```

Definition at line 2504 of file [grcc.cc](#).

### 3.110.4 Field Documentation

#### 3.110.4.1 `sproc`

`SProcess* PNodeClass::sproc`

Definition at line 327 of file [grcc.h](#).

#### 3.110.4.2 `deg`

`int* PNodeClass::deg`

Definition at line 328 of file [grcc.h](#).

#### 3.110.4.3 `type`

`int* PNodeClass::type`

Definition at line 329 of file [grcc.h](#).

#### 3.110.4.4 `particle`

`int* PNodeClass::particle`

Definition at line 330 of file [grcc.h](#).

#### 3.110.4.5 `couple`

`int* PNodeClass::couple`

Definition at line 331 of file [grcc.h](#).

#### 3.110.4.6 `cmindeg`

`int* PNodeClass::cmindeg`

Definition at line 332 of file [grcc.h](#).

#### 3.110.4.7 `cmaxdeg`

`int* PNodeClass::cmaxdeg`

Definition at line 333 of file [grcc.h](#).

#### 3.110.4.8 count

```
int* PNodeClass::count
```

Definition at line 334 of file [grcc.h](#).

#### 3.110.4.9 cl2nd

```
int* PNodeClass::cl2nd
```

Definition at line 335 of file [grcc.h](#).

#### 3.110.4.10 nd2cl

```
int* PNodeClass::nd2cl
```

Definition at line 336 of file [grcc.h](#).

#### 3.110.4.11 cl2mcl

```
int* PNodeClass::cl2mcl
```

Definition at line 337 of file [grcc.h](#).

#### 3.110.4.12 nnodes

```
int PNodeClass::nnodes
```

Definition at line 338 of file [grcc.h](#).

#### 3.110.4.13 nclass

```
int PNodeClass::nclass
```

Definition at line 339 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.111 flint::poly Class Reference

### Public Member Functions

- void [print](#) (const string &text)

## Data Fields

- `mpz_poly_t` [d](#)

### 3.111.1 Detailed Description

Definition at line [61](#) of file [flintinterface.h](#).

### 3.111.2 Constructor & Destructor Documentation

#### 3.111.2.1 `poly()`

```
flint::poly::poly ( ) [inline]
```

Definition at line [64](#) of file [flintinterface.h](#).

#### 3.111.2.2 `~poly()`

```
flint::poly::~~poly ( ) [inline]
```

Definition at line [65](#) of file [flintinterface.h](#).

### 3.111.3 Member Function Documentation

#### 3.111.3.1 `print()`

```
void flint::poly::print (
    const string & text ) [inline]
```

Definition at line [66](#) of file [flintinterface.h](#).

### 3.111.4 Field Documentation

#### 3.111.4.1 `d`

```
mpz_poly_t flint::poly::d
```

Definition at line [63](#) of file [flintinterface.h](#).

The documentation for this class was generated from the following file:

- [flintinterface.h](#)

## 3.112 poly Class Reference

### Public Member Functions

- [poly](#) (PHEAD int, WORD=-1, WORD=1)
- [poly](#) (PHEAD const UWORD \*, WORD, WORD=-1, WORD=1)
- [poly](#) (const [poly](#) &, WORD=-1, WORD=1)
- [poly](#) & [operator+=](#) (const [poly](#) &)
- [poly](#) & [operator-=](#) (const [poly](#) &)
- [poly](#) & [operator\\*=](#) (const [poly](#) &)
- [poly](#) & [operator/=](#) (const [poly](#) &)
- [poly](#) & [operator%=>](#) (const [poly](#) &)
- const [poly](#) [operator+](#) (const [poly](#) &) const
- const [poly](#) [operator-](#) (const [poly](#) &) const
- const [poly](#) [operator\\*](#) (const [poly](#) &) const
- const [poly](#) [operator/](#) (const [poly](#) &) const
- const [poly](#) [operator%](#) (const [poly](#) &) const
- bool [operator==](#) (const [poly](#) &) const
- bool [operator!=](#) (const [poly](#) &) const
- [poly](#) & [operator=](#) (const [poly](#) &)
- WORD & [operator\[\]](#) (int)
- const WORD & [operator\[\]](#) (int) const
- void [termscopy](#) (const WORD \*, int, int)
- void [check\\_memory](#) (int)
- void [expand\\_memory](#) (int)
- bool [is\\_zero](#) () const
- bool [is\\_one](#) () const
- bool [is\\_integer](#) () const
- bool [is\\_monomial](#) () const
- int [is\\_dense\\_univariate](#) () const
- int [sign](#) () const
- int [degree](#) (int) const
- int [total\\_degree](#) () const
- int [first\\_variable](#) () const
- int [number\\_of\\_terms](#) () const
- const std::vector< int > [all\\_variables](#) () const
- const [poly](#) [integer\\_lcoeff](#) () const
- const [poly](#) [lcoeff\\_univar](#) (int) const
- const [poly](#) [lcoeff\\_multivar](#) (int) const
- const [poly](#) [coefficient](#) (int, int) const
- const [poly](#) [derivative](#) (int) const
- void [setmod](#) (WORD, WORD=1)
- void [coefficients\\_modulo](#) (UWORD \*, WORD, bool)
- int [size\\_of\\_form\\_notation](#) () const
- int [size\\_of\\_form\\_notation\\_with\\_den](#) (WORD) const
- const [poly](#) & [normalize](#) ()
- const std::string [to\\_string](#) () const
- WORD [last\\_monomial\\_index](#) () const
- WORD \* [last\\_monomial](#) () const
- int [compare\\_degree\\_vector](#) (const [poly](#) &) const
- std::vector< int > [degree\\_vector](#) () const
- int [compare\\_degree\\_vector](#) (const std::vector< int > &) const

## Static Public Member Functions

- static const [poly simple\\_poly](#) (PHEAD int, int=0, int=1, int=0, int=1)
- static const [poly simple\\_poly](#) (PHEAD int, const [poly](#) &, int=1, int=0, int=1)
- static void [get\\_variables](#) (PHEAD const std::vector< WORD \* > &, bool, bool)
- static const [poly argument\\_to\\_poly](#) (PHEAD WORD \*, bool, bool, [poly](#) \*den=NULL)
- static void [poly\\_to\\_argument](#) (const [poly](#) &, WORD \*, bool)
- static void [poly\\_to\\_argument\\_with\\_den](#) (const [poly](#) &, WORD, const UWORD \*, WORD \*, bool)
- static const [poly from\\_coefficient\\_list](#) (PHEAD const std::vector< WORD > &, int, WORD)
- static const std::vector< WORD > [to\\_coefficient\\_list](#) (const [poly](#) &)
- static const std::vector< WORD > [coefficient\\_list\\_divmod](#) (const std::vector< WORD > &, const std::vector< WORD > &, WORD, int)
- static int [monomial\\_compare](#) (PHEAD const WORD \*, const WORD \*)
- static void [add](#) (const [poly](#) &, const [poly](#) &, [poly](#) &)
- static void [sub](#) (const [poly](#) &, const [poly](#) &, [poly](#) &)
- static void [mul](#) (const [poly](#) &, const [poly](#) &, [poly](#) &)
- static void [div](#) (const [poly](#) &, const [poly](#) &, [poly](#) &)
- static void [mod](#) (const [poly](#) &, const [poly](#) &, [poly](#) &)
- static void [divmod](#) (const [poly](#) &, const [poly](#) &, [poly](#) &, [poly](#) &, bool only\_divides)
- static bool [divides](#) (const [poly](#) &, const [poly](#) &)
- static void [mul\\_one\\_term](#) (const [poly](#) &, const [poly](#) &, [poly](#) &)
- static void [mul\\_univar](#) (const [poly](#) &, const [poly](#) &, [poly](#) &, int)
- static void [mul\\_heap](#) (const [poly](#) &, const [poly](#) &, [poly](#) &)
- static void [divmod\\_one\\_term](#) (const [poly](#) &, const [poly](#) &, [poly](#) &, [poly](#) &, bool)
- static void [divmod\\_univar](#) (const [poly](#) &, const [poly](#) &, [poly](#) &, [poly](#) &, int, bool)
- static void [divmod\\_heap](#) (const [poly](#) &, const [poly](#) &, [poly](#) &, [poly](#) &, bool, bool, bool &)
- static void [push\\_heap](#) (PHEAD WORD \*\*, int)
- static void [pop\\_heap](#) (PHEAD WORD \*\*, int)

## Data Fields

- WORD \* [terms](#)
- LONG [size\\_of\\_terms](#)
- WORD [modp](#)
- WORD [modn](#)

### 3.112.1 Detailed Description

Definition at line 53 of file [poly.h](#).

### 3.112.2 Constructor & Destructor Documentation

#### 3.112.2.1 [poly\(\)](#) [1/3]

```
poly::poly (
    PHEAD int a,
    WORD modp = -1,
    WORD modn = 1 )
```

Definition at line 54 of file [poly.cc](#).

### 3.112.2.2 poly() [2/3]

```
poly::poly (
    PHEAD const UWORD * a,
    WORD na,
    WORD modp = -1,
    WORD modn = 1 )
```

Definition at line 83 of file [poly.cc](#).

### 3.112.2.3 poly() [3/3]

```
poly::poly (
    const poly & a,
    WORD modp = -1,
    WORD modn = 1 )
```

Definition at line 111 of file [poly.cc](#).

### 3.112.2.4 ~poly()

```
poly::~~poly ( )
```

Definition at line 133 of file [poly.cc](#).

## 3.112.3 Member Function Documentation

### 3.112.3.1 operator+=(())

```
poly & poly::operator+= (
    const poly & a )
```

Definition at line 1966 of file [poly.cc](#).

### 3.112.3.2 operator-=(())

```
poly & poly::operator-= (
    const poly & a )
```

Definition at line 1967 of file [poly.cc](#).

### 3.112.3.3 operator\*=(())

```
poly & poly::operator*= (
    const poly & a )
```

Definition at line 1968 of file [poly.cc](#).

#### 3.112.3.4 operator/=( )

```
poly & poly::operator/= (
    const poly & a )
```

Definition at line 1969 of file [poly.cc](#).

#### 3.112.3.5 operator%=( )

```
poly & poly::operator%= (
    const poly & a )
```

Definition at line 1970 of file [poly.cc](#).

#### 3.112.3.6 operator+( )

```
const poly poly::operator+ (
    const poly & a ) const
```

Definition at line 1959 of file [poly.cc](#).

#### 3.112.3.7 operator-( )

```
const poly poly::operator- (
    const poly & a ) const
```

Definition at line 1960 of file [poly.cc](#).

#### 3.112.3.8 operator\*( )

```
const poly poly::operator* (
    const poly & a ) const
```

Definition at line 1961 of file [poly.cc](#).

#### 3.112.3.9 operator/( )

```
const poly poly::operator/ (
    const poly & a ) const
```

Definition at line 1962 of file [poly.cc](#).

#### 3.112.3.10 operator%( )

```
const poly poly::operator% (
    const poly & a ) const
```

Definition at line 1963 of file [poly.cc](#).



**3.112.3.11 operator==( )**

```
bool poly::operator== (
    const poly & a ) const
```

Definition at line 1973 of file [poly.cc](#).

**3.112.3.12 operator"!=()"**

```
bool poly::operator!= (
    const poly & a ) const
```

Definition at line 1979 of file [poly.cc](#).

**3.112.3.13 operator=( )**

```
poly & poly::operator= (
    const poly & a )
```

Definition at line 1939 of file [poly.cc](#).

**3.112.3.14 operator[]() [1/2]**

```
WORD & poly::operator[] (
    int i ) [inline]
```

Definition at line 249 of file [poly.h](#).

**3.112.3.15 operator[]() [2/2]**

```
const WORD & poly::operator[] (
    int i ) const [inline]
```

Definition at line 253 of file [poly.h](#).

**3.112.3.16 termscopy()**

```
void poly::termscopy (
    const WORD * source,
    int dest,
    int num ) [inline]
```

Definition at line 260 of file [poly.h](#).

**3.112.3.17 check\_memory()**

```
void poly::check_memory (
    int i ) [inline]
```

Definition at line 242 of file [poly.h](#).

#### 3.112.3.18 `expand_memory()`

```
void poly::expand_memory (
    int i )
```

Definition at line 149 of file [poly.cc](#).

#### 3.112.3.19 `is_zero()`

```
bool poly::is_zero ( ) const
```

Definition at line 2209 of file [poly.cc](#).

#### 3.112.3.20 `is_one()`

```
bool poly::is_one ( ) const
```

Definition at line 2219 of file [poly.cc](#).

#### 3.112.3.21 `is_integer()`

```
bool poly::is_integer ( ) const
```

Definition at line 2239 of file [poly.cc](#).

#### 3.112.3.22 `is_monomial()`

```
bool poly::is_monomial ( ) const
```

Definition at line 2259 of file [poly.cc](#).

#### 3.112.3.23 `is_dense_univariate()`

```
int poly::is_dense_univariate ( ) const
```

Dense univariate detection

### 3.112.4 Description

This method returns whether the polynomial is dense and univariate. The possible return values are:

-2 is not dense univariate -1 is no variables  $n \geq 0$  is univariate in  $n$

### 3.112.5 Notes

A univariate polynomial is considered dense iff more than half of the coefficients `a_0...a_deg` are non-zero.

Definition at line 2284 of file [poly.cc](#).

Referenced by [divmod\(\)](#), and [mul\(\)](#).

#### 3.112.5.1 `sign()`

```
int poly::sign ( ) const
```

Definition at line 2039 of file [poly.cc](#).

#### 3.112.5.2 `degree()`

```
int poly::degree (
    int x ) const
```

Definition at line 2050 of file [poly.cc](#).

#### 3.112.5.3 `total_degree()`

```
int poly::total_degree ( ) const
```

Definition at line 2063 of file [poly.cc](#).

#### 3.112.5.4 `first_variable()`

```
int poly::first_variable ( ) const
```

Definition at line 2000 of file [poly.cc](#).

#### 3.112.5.5 `number_of_terms()`

```
int poly::number_of_terms ( ) const
```

Definition at line 1986 of file [poly.cc](#).

#### 3.112.5.6 `all_variables()`

```
const vector< int > poly::all_variables ( ) const
```

Definition at line 2016 of file [poly.cc](#).

#### 3.112.5.7 integer\_lcoeff()

```
const poly poly::integer_lcoeff ( ) const
```

Definition at line 2083 of file [poly.cc](#).

#### 3.112.5.8 lcoeff\_univar()

```
const poly poly::lcoeff_univar (
    int x ) const
```

Definition at line 2133 of file [poly.cc](#).

#### 3.112.5.9 lcoeff\_multivar()

```
const poly poly::lcoeff_multivar (
    int x ) const
```

Definition at line 2140 of file [poly.cc](#).

#### 3.112.5.10 coefficient()

```
const poly poly::coefficient (
    int x,
    int n ) const
```

Definition at line 2105 of file [poly.cc](#).

#### 3.112.5.11 derivative()

```
const poly poly::derivative (
    int x ) const
```

Definition at line 2172 of file [poly.cc](#).

#### 3.112.5.12 setmod()

```
void poly::setmod (
    WORD _modp,
    WORD _modn = 1 )
```

Definition at line 179 of file [poly.cc](#).

#### 3.112.5.13 coefficients\_modulo()

```
void poly::coefficients_modulo (
    UWORD * a,
    WORD na,
    bool small )
```

Definition at line 215 of file [poly.cc](#).

**3.112.5.14 simple\_poly()** [1/2]

```
const poly poly::simple_poly (
    PHEAD int x,
    int a = 0,
    int b = 1,
    int p = 0,
    int n = 1 ) [static]
```

Definition at line 2317 of file [poly.cc](#).

**3.112.5.15 simple\_poly()** [2/2]

```
const poly poly::simple_poly (
    PHEAD int x,
    const poly & a,
    int b = 1,
    int p = 0,
    int n = 1 ) [static]
```

Definition at line 2356 of file [poly.cc](#).

**3.112.5.16 get\_variables()**

```
void poly::get_variables (
    PHEAD const std::vector< WORD * > & ,
    bool ,
    bool ) [static]
```

Definition at line 2395 of file [poly.cc](#).

**3.112.5.17 argument\_to\_poly()**

```
const poly poly::argument_to_poly (
    PHEAD WORD * e,
    bool with_arghead,
    bool sort_univar,
    poly * den = NULL ) [static]
```

Definition at line 2498 of file [poly.cc](#).

**3.112.5.18 poly\_to\_argument()**

```
void poly::poly_to_argument (
    const poly & a,
    WORD * res,
    bool with_arghead ) [static]
```

Definition at line 2611 of file [poly.cc](#).

#### 3.112.5.19 poly\_to\_argument\_with\_den()

```
void poly::poly_to_argument_with_den (
    const poly & a,
    WORD nden,
    const UWORD * den,
    WORD * res,
    bool with_arghead ) [static]
```

Definition at line 2683 of file [poly.cc](#).

#### 3.112.5.20 size\_of\_form\_notation()

```
int poly::size_of_form_notation ( ) const
```

Definition at line 2766 of file [poly.cc](#).

#### 3.112.5.21 size\_of\_form\_notation\_with\_den()

```
int poly::size_of_form_notation_with_den (
    WORD nden ) const
```

Definition at line 2795 of file [poly.cc](#).

#### 3.112.5.22 normalize()

```
const poly & poly::normalize ( )
```

Definition at line 362 of file [poly.cc](#).

#### 3.112.5.23 from\_coefficient\_list()

```
const poly poly::from_coefficient_list (
    PHEAD const std::vector< WORD > & ,
    int ,
    WORD ) [static]
```

Definition at line 2885 of file [poly.cc](#).

#### 3.112.5.24 to\_coefficient\_list()

```
const vector< WORD > poly::to_coefficient_list (
    const poly & a ) [static]
```

Definition at line 2823 of file [poly.cc](#).

**3.112.5.25 coefficient\_list\_divmod()**

```
const vector< WORD > poly::coefficient_list_divmod (
    const std::vector< WORD > & ,
    const std::vector< WORD > & ,
    WORD ,
    int ) [static]
```

Definition at line 2846 of file [poly.cc](#).

**3.112.5.26 to\_string()**

```
const string poly::to_string ( ) const
```

Definition at line 267 of file [poly.cc](#).

**3.112.5.27 monomial\_compare()**

```
int poly::monomial_compare (
    PHEAD const WORD * a,
    const WORD * b ) [static]
```

Definition at line 350 of file [poly.cc](#).

**3.112.5.28 last\_monomial\_index()**

```
WORD poly::last_monomial_index ( ) const
```

Definition at line 465 of file [poly.cc](#).

**3.112.5.29 last\_monomial()**

```
WORD * poly::last_monomial ( ) const
```

Definition at line 472 of file [poly.cc](#).

**3.112.5.30 compare\_degree\_vector()**

```
int poly::compare_degree_vector (
    const poly & b ) const
```

Definition at line 481 of file [poly.cc](#).

**3.112.5.31 degree\_vector()**

```
std::vector< int > poly::degree_vector ( ) const
```

Definition at line 503 of file [poly.cc](#).

### 3.112.5.32 add()

```
void poly::add (
    const poly & a,
    const poly & b,
    poly & c ) [static]
```

Definition at line 538 of file [poly.cc](#).

### 3.112.5.33 sub()

```
void poly::sub (
    const poly & a,
    const poly & b,
    poly & c ) [static]
```

Definition at line 622 of file [poly.cc](#).

### 3.112.5.34 mul()

```
void poly::mul (
    const poly & a,
    const poly & b,
    poly & c ) [static]
```

Polynomial multiplication

## 3.112.6 Description

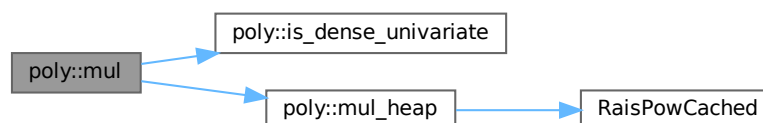
This routine determines which multiplication routine to use for multiplying two polynomials. The logic is as follows:

- If a or b consists of only one term, call `mul_one_term`;
- Otherwise, if both are univariate and dense, call `mul_univar`;
- Otherwise, call `mul_heap`.

Definition at line 1195 of file [poly.cc](#).

References [is\\_dense\\_univariate\(\)](#), and [mul\\_heap\(\)](#).

Here is the call graph for this function:





**3.112.6.1 div()**

```
void poly::div (
    const poly & a,
    const poly & b,
    poly & c ) [static]
```

Definition at line 1915 of file [poly.cc](#).

**3.112.6.2 mod()**

```
void poly::mod (
    const poly & a,
    const poly & b,
    poly & c ) [static]
```

Definition at line 1927 of file [poly.cc](#).

**3.112.6.3 divmod()**

```
void poly::divmod (
    const poly & a,
    const poly & b,
    poly & q,
    poly & r,
    bool only_divides ) [static]
```

Polynomial division

**3.112.7 Description**

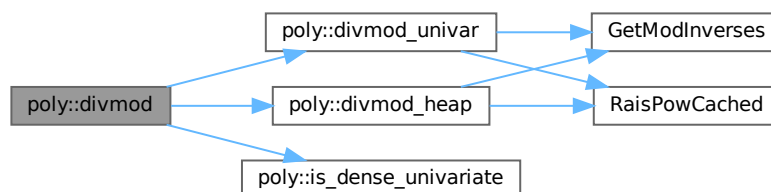
This routine determines which division routine to use for dividing two polynomials. The logic is as follows:

- If b consists of only one term, call `divmod_one_term`;
- Otherwise, if both are univariate and dense, call `divmod_univar`;
- Otherwise, call `divmod_heap`.

Definition at line 1856 of file [poly.cc](#).

References [divmod\\_heap\(\)](#), [divmod\\_univar\(\)](#), and [is\\_dense\\_univariate\(\)](#).

Here is the call graph for this function:



### 3.112.7.1 divides()

```
bool poly::divides (
    const poly & a,
    const poly & b ) [static]
```

Definition at line 1902 of file [poly.cc](#).

### 3.112.7.2 mul\_one\_term()

```
void poly::mul_one_term (
    const poly & a,
    const poly & b,
    poly & c ) [static]
```

Definition at line 755 of file [poly.cc](#).

### 3.112.7.3 mul\_univar()

```
void poly::mul_univar (
    const poly & a,
    const poly & b,
    poly & c,
    int var ) [static]
```

Definition at line 829 of file [poly.cc](#).

### 3.112.7.4 mul\_heap()

```
void poly::mul_heap (
    const poly & a,
    const poly & b,
    poly & c ) [static]
```

Multiplication of polynomials with a heap

## 3.112.8 Description

Multiplies two multivariate polynomials. The next element of the product is efficiently determined by using a heap. If the product of the maximum power in all variables is small, a hash table is used to add equal terms for extra speed.

A heap element h is formatted as follows:

- h[0] = index in a
- h[1] = index in b
- h[2] = hash code (-1 if no hash is used)
- h[3] = length of coefficient with sign
- h[4...4+AN.poly\_num\_vars-1] = powers

- $h[4+AN.poly\_num\_vars...4+h[3]-1]$  = coefficient

Definition at line 961 of file [poly.cc](#).

References [RaisPowCached\(\)](#).

Referenced by [mul\(\)](#).

Here is the call graph for this function:



#### 3.112.8.1 `divmod_one_term()`

```
void poly::divmod_one_term (
    const poly & a,
    const poly & b,
    poly & q,
    poly & r,
    bool only_divides ) [static]
```

Definition at line 1245 of file [poly.cc](#).

#### 3.112.8.2 `divmod_univar()`

```
void poly::divmod_univar (
    const poly & a,
    const poly & b,
    poly & q,
    poly & r,
    int var,
    bool only_divides ) [static]
```

Division of dense univariate polynomials.

### 3.112.9 Description

Divides two dense univariate polynomials. For each power, the method collects all terms that result in that power.

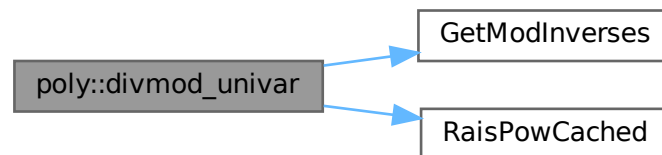
Relevant formula  $[Q=A/B, P=\text{SUM}(p_i \cdot x^i), n=\text{deg}(A), m=\text{deg}(B)]: q_k = [a_{m+k} - \text{SUM}(i=k+1 \dots n-m, b_{m+k-i} \cdot q_i)] / b_m$

Definition at line 1376 of file [poly.cc](#).

References [GetModInverses\(\)](#), and [RaisPowCached\(\)](#).

Referenced by [divmod\(\)](#).

Here is the call graph for this function:



#### 3.112.9.1 divmod\_heap()

```

void poly::divmod_heap (
    const poly & a,
    const poly & b,
    poly & q,
    poly & r,
    bool only_divides,
    bool check_div,
    bool & div_fail ) [static]
  
```

Division of polynomials with a heap

### 3.112.10 Description

Divides two multivariate polynomials. The next element of the quotient/remainder is efficiently determined by using a heap. If the product of the maximum power in all variables is small, a hash table is used to add equal terms for extra speed.

If the input flag `check_div` is set then if the result of any coefficient division results in a non-zero remainder (indicating that division has failed over the integers) the output flag `div_fail` will be set and the division will be terminated early (`q`, `r` will be incorrect). If `check_div` is not set then non-zero remainders from coefficient division will be written into `r`.

A heap element `h` is formatted as follows:

- $h[0]$  = index in  $a$
- $h[1]$  = index in  $b$
- $h[2] = -1$  (no hash is used)
- $h[3]$  = length of coefficient with sign
- $h[4 \dots 4 + \text{AN.poly\_num\_vars} - 1]$  = powers
- $h[4 + \text{AN.poly\_num\_vars} \dots 4 + h[3] - 1]$  = coefficient

Note: the hashing trick as in multiplication cannot be used easily, since there is no tight upperbound on the exponents in the answer.

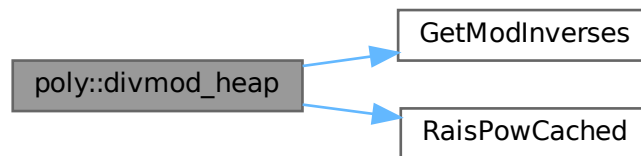
For details, see M. Monagan, "Polynomial Division using Dynamic Array, Heaps, and Packed Exponent Vectors"

Definition at line 1579 of file [poly.cc](#).

References [GetModInverses\(\)](#), and [RaisPowCached\(\)](#).

Referenced by [divmod\(\)](#).

Here is the call graph for this function:



#### 3.112.10.1 `push_heap()`

```
void poly::push_heap (
    PHEAD WORD ** heap,
    int n ) [static]
```

Definition at line 739 of file [poly.cc](#).

#### 3.112.10.2 `pop_heap()`

```
void poly::pop_heap (
    PHEAD WORD ** heap,
    int n ) [static]
```

Definition at line 707 of file [poly.cc](#).

### 3.112.11 Field Documentation

#### 3.112.11.1 terms

WORD\* poly::terms

Definition at line 62 of file [poly.h](#).

#### 3.112.11.2 size\_of\_terms

LONG poly::size\_of\_terms

Definition at line 63 of file [poly.h](#).

#### 3.112.11.3 modp

WORD poly::modp

Definition at line 64 of file [poly.h](#).

#### 3.112.11.4 modn

WORD poly::modn

Definition at line 64 of file [poly.h](#).

The documentation for this class was generated from the following files:

- [poly.h](#)
- [poly.cc](#)

## 3.113 flint::poly\_factor Class Reference

### Data Fields

- [fmpz\\_poly\\_factor\\_t](#) [d](#)

### 3.113.1 Detailed Description

Definition at line 70 of file [flintinterface.h](#).

### 3.113.2 Constructor & Destructor Documentation

#### 3.113.2.1 `poly_factor()`

```
flint::poly_factor::poly_factor ( ) [inline]
```

Definition at line 73 of file [flintinterface.h](#).

#### 3.113.2.2 `~poly_factor()`

```
flint::poly_factor::~~poly_factor ( ) [inline]
```

Definition at line 74 of file [flintinterface.h](#).

### 3.113.3 Field Documentation

#### 3.113.3.1 `d`

```
fmprz_poly_factor_t flint::poly_factor::d
```

Definition at line 72 of file [flintinterface.h](#).

The documentation for this class was generated from the following file:

- [flintinterface.h](#)

## 3.114 POLYMOD Struct Reference

```
#include <structs.h>
```

### Data Fields

- WORD \* [coefs](#)
- WORD [numsym](#)
- WORD [arraysize](#)
- WORD [polysize](#)
- WORD [modnum](#)

### 3.114.1 Detailed Description

The [POLYMOD](#) struct controls one univariate polynomial of which the coefficients have been taken modulus a (prime) number that fits inside a variable of type WORD. The polynomial is stored as an array of coefficients of size WORD.

Definition at line 1268 of file [structs.h](#).

### 3.114.2 Field Documentation

#### 3.114.2.1 `coefs`

`WORD* POLYMOD::coefs`

Definition at line 1269 of file [structs.h](#).

#### 3.114.2.2 `numsym`

`WORD POLYMOD::numsym`

Definition at line 1270 of file [structs.h](#).

#### 3.114.2.3 `arraysize`

`WORD POLYMOD::arraysize`

Definition at line 1271 of file [structs.h](#).

#### 3.114.2.4 `polysize`

`WORD POLYMOD::polysize`

Definition at line 1272 of file [structs.h](#).

#### 3.114.2.5 `modnum`

`WORD POLYMOD::modnum`

Definition at line 1273 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.115 PoSiTiOn Struct Reference

### Data Fields

- `off_t p1`

### 3.115.1 Detailed Description

Definition at line 59 of file [structs.h](#).



## 3.115.2 Field Documentation

### 3.115.2.1 p1

```
off_t PoSiTiOn::p1
```

Definition at line 60 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.116 PRELOAD Struct Reference

```
#include <structs.h>
```

### Data Fields

- UBYTE \* [buffer](#)
- LONG [size](#)

### 3.116.1 Detailed Description

Used by the preprocessor to load the contents of a doloop or a procedure. The struct [PRELOAD](#) is used both in the DOLOOP and [PROCEDURE](#) structs.

Definition at line 873 of file [structs.h](#).

## 3.116.2 Field Documentation

### 3.116.2.1 buffer

```
UBYTE* PRELOAD::buffer
```

Definition at line 874 of file [structs.h](#).

### 3.116.2.2 size

```
LONG PRELOAD::size
```

Definition at line 875 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.117 pReVaR Struct Reference

```
#include <structs.h>
```

### Data Fields

- UBYTE \* [name](#)
- UBYTE \* [value](#)
- UBYTE \* [argnames](#)
- int [nargs](#)
- int [wildarg](#)

### 3.117.1 Detailed Description

An element of the type PREVAR is needed for each preprocessor variable.

Definition at line [841](#) of file [structs.h](#).

### 3.117.2 Field Documentation

#### 3.117.2.1 name

```
UBYTE* pReVaR::name
```

allocated

Definition at line [842](#) of file [structs.h](#).

#### 3.117.2.2 value

```
UBYTE* pReVaR::value
```

points into memory of name

Definition at line [843](#) of file [structs.h](#).

#### 3.117.2.3 argnames

```
UBYTE* pReVaR::argnames
```

names of arguments, zero separated. points into memory of name

Definition at line [844](#) of file [structs.h](#).

### 3.117.2.4 nargs

```
int pReVaR::nargs
```

0 = regular, >= 1: number of macro arguments. total number

Definition at line 845 of file [structs.h](#).

### 3.117.2.5 wildarg

```
int pReVaR::wildarg
```

The number of a potential ?var. If none: 0. wildarg<nargs

Definition at line 846 of file [structs.h](#).

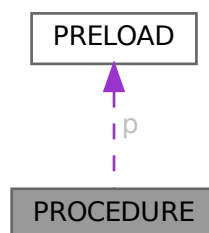
The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.118 PROCEDURE Struct Reference

```
#include <structs.h>
```

Collaboration diagram for PROCEDURE:



### Data Fields

- [PRELOAD](#) p
- UBYTE \* [name](#)
- int [loadmode](#)

### 3.118.1 Detailed Description

An element of the type [PROCEDURE](#) is needed for each procedure in the system.

Definition at line [883](#) of file [structs.h](#).

### 3.118.2 Field Documentation

#### 3.118.2.1 p

[PRELOAD](#) `PROCEDURE::p`

Definition at line [884](#) of file [structs.h](#).

#### 3.118.2.2 name

`UBYTE* PROCEDURE::name`

Definition at line [885](#) of file [structs.h](#).

#### 3.118.2.3 loadmode

`int PROCEDURE::loadmode`

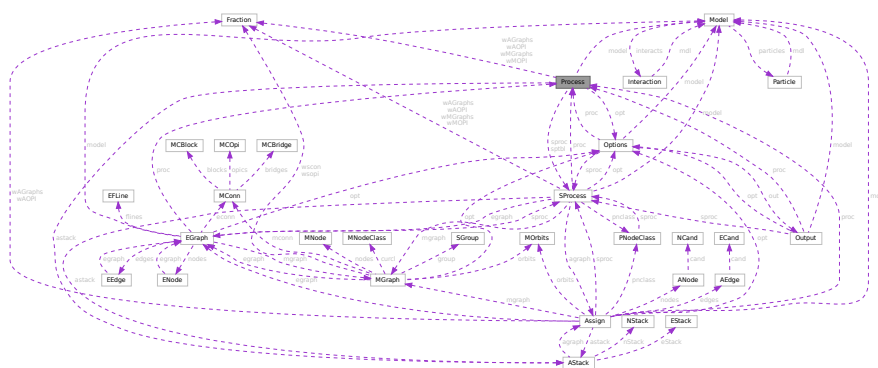
Definition at line [886](#) of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.119 Process Class Reference

Collaboration diagram for Process:



## Public Member Functions

- [Process](#) (int pid, [Model](#) \*model, [Options](#) \*opt, int nin, int \*initlPart, int nfin, int \*finalPart, int \*coupling)
- **Process** (int pid, [Model](#) \*model, [Options](#) \*opt, [FGInput](#) \*fgi)
- void [prProcess](#) (void)
- void [mkSProcess](#) (void)

## Data Fields

- [Model](#) \* [model](#)
- [Options](#) \* [opt](#)
- [BigInt](#) [mgrcount](#)
- [BigInt](#) [agrcount](#)
- int \* [initlPart](#)
- int \* [finalPart](#)
- int [id](#)
- int [ninitl](#)
- int [nfinal](#)
- int [ctotal](#)
- int [nExtern](#)
- int [loop](#)
- int [maxnlegs](#)
- int [clist](#) [GRCC\_MAXNCPLG]
- int [nSubproc](#)
- [SProcess](#) \* [sptbl](#) [GRCC\_MAXSUBPROCS]
- [SProcess](#) \* [sproc](#)
- [AStack](#) \* [astack](#)
- [BigInt](#) [ngraphs](#)
- [BigInt](#) [nopi](#)
- [BigInt](#) [wgraphs](#)
- [BigInt](#) [wopi](#)
- [BigInt](#) [nMGraphs](#)
- [BigInt](#) [nMOPI](#)
- [Fraction](#) [wMGraphs](#)
- [Fraction](#) [wMOPI](#)
- [BigInt](#) [nAGraphs](#)
- [BigInt](#) [nAOPI](#)
- [Fraction](#) [wAGraphs](#)
- [Fraction](#) [wAOPI](#)
- double [sec](#)

### 3.119.1 Detailed Description

Definition at line 424 of file [grcc.h](#).

### 3.119.2 Constructor & Destructor Documentation

#### 3.119.2.1 Process()

```
Process::Process (
    int pid,
    Model * model,
    Options * opt,
    int nin,
    int * initlPart,
    int nfin,
    int * finalPart,
    int * coupling )
```

Definition at line 3036 of file [grcc.cc](#).

#### 3.119.2.2 ~Process()

```
Process::~~Process (
    void )
```

Definition at line 3134 of file [grcc.cc](#).

### 3.119.3 Member Function Documentation

#### 3.119.3.1 prProcess()

```
void Process::prProcess (
    void )
```

Definition at line 3142 of file [grcc.cc](#).

#### 3.119.3.2 mkSProcess()

```
void Process::mkSProcess (
    void )
```

Definition at line 3150 of file [grcc.cc](#).

### 3.119.4 Field Documentation

#### 3.119.4.1 model

```
Model* Process::model
```

Definition at line 426 of file [grcc.h](#).

#### 3.119.4.2 opt

`Options*` `Process::opt`

Definition at line 427 of file [grcc.h](#).

#### 3.119.4.3 mgrcount

`BigInt` `Process::mgrcount`

Definition at line 428 of file [grcc.h](#).

#### 3.119.4.4 agrcount

`BigInt` `Process::agrcount`

Definition at line 429 of file [grcc.h](#).

#### 3.119.4.5 initlPart

`int*` `Process::initlPart`

Definition at line 430 of file [grcc.h](#).

#### 3.119.4.6 finalPart

`int*` `Process::finalPart`

Definition at line 431 of file [grcc.h](#).

#### 3.119.4.7 id

`int` `Process::id`

Definition at line 433 of file [grcc.h](#).

#### 3.119.4.8 ninitl

`int` `Process::ninitl`

Definition at line 434 of file [grcc.h](#).

#### 3.119.4.9 nfinal

`int` `Process::nfinal`

Definition at line 435 of file [grcc.h](#).

#### 3.119.4.10 ctotat

```
int Process::ctotat
```

Definition at line 436 of file [grcc.h](#).

#### 3.119.4.11 nExtern

```
int Process::nExtern
```

Definition at line 437 of file [grcc.h](#).

#### 3.119.4.12 loop

```
int Process::loop
```

Definition at line 438 of file [grcc.h](#).

#### 3.119.4.13 maxnlegs

```
int Process::maxnlegs
```

Definition at line 439 of file [grcc.h](#).

#### 3.119.4.14 clist

```
int Process::clist[GRCC_MAXNCPLG]
```

Definition at line 440 of file [grcc.h](#).

#### 3.119.4.15 nSubproc

```
int Process::nSubproc
```

Definition at line 444 of file [grcc.h](#).

#### 3.119.4.16 sptbl

```
SProcess* Process::sptbl[GRCC_MAXSUBPROCS]
```

Definition at line 445 of file [grcc.h](#).

#### 3.119.4.17 sproc

```
SProcess* Process::sproc
```

Definition at line 446 of file [grcc.h](#).



#### 3.119.4.18 astack

`AStack*` `Process::astack`

Definition at line 449 of file [grcc.h](#).

#### 3.119.4.19 ngraphs

`BigInt` `Process::ngraphs`

Definition at line 452 of file [grcc.h](#).

#### 3.119.4.20 nopi

`BigInt` `Process::nopi`

Definition at line 453 of file [grcc.h](#).

#### 3.119.4.21 wgraphs

`BigInt` `Process::wgraphs`

Definition at line 454 of file [grcc.h](#).

#### 3.119.4.22 wopi

`BigInt` `Process::wopi`

Definition at line 455 of file [grcc.h](#).

#### 3.119.4.23 nMGraphs

`BigInt` `Process::nMGraphs`

Definition at line 458 of file [grcc.h](#).

#### 3.119.4.24 nMOPI

`BigInt` `Process::nMOPI`

Definition at line 459 of file [grcc.h](#).

#### 3.119.4.25 wMGraphs

`Fraction` `Process::wMGraphs`

Definition at line 460 of file [grcc.h](#).

#### 3.119.4.26 wMOPI

`Fraction` `Process::wMOPI`

Definition at line 461 of file [grcc.h](#).

#### 3.119.4.27 nAGraphs

`BigInt` `Process::nAGraphs`

Definition at line 463 of file [grcc.h](#).

#### 3.119.4.28 nAOPI

`BigInt` `Process::nAOPI`

Definition at line 464 of file [grcc.h](#).

#### 3.119.4.29 wAGraphs

`Fraction` `Process::wAGraphs`

Definition at line 465 of file [grcc.h](#).

#### 3.119.4.30 wAOPI

`Fraction` `Process::wAOPI`

Definition at line 466 of file [grcc.h](#).

#### 3.119.4.31 sec

`double` `Process::sec`

Definition at line 468 of file [grcc.h](#).

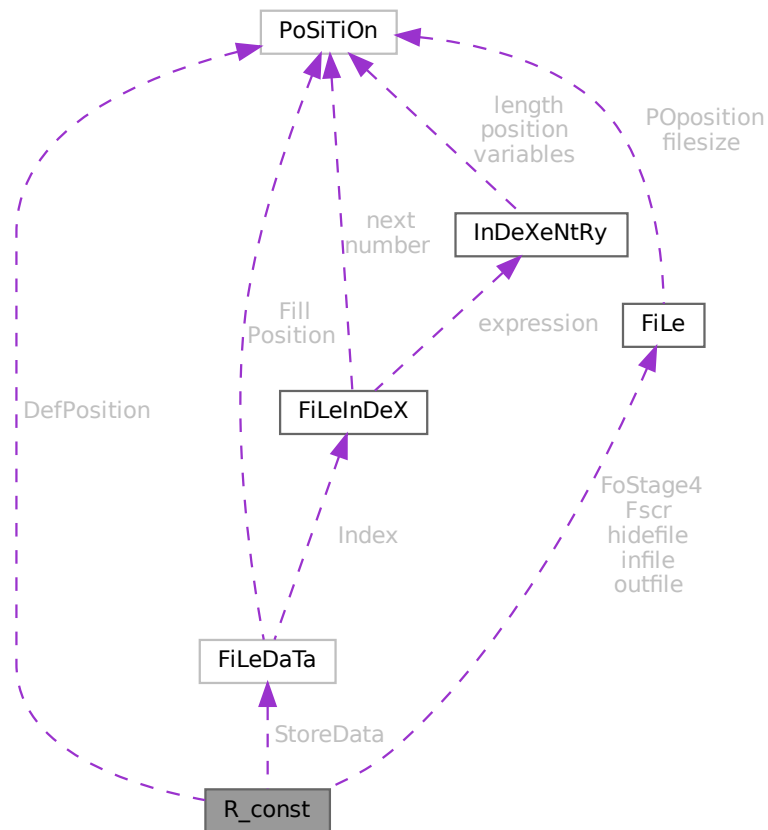
The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.120 R\_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for R\_const:



### Public Member Functions

- **PADPOSITION** (8, 7, 7, 27, 0)

### Data Fields

- [FILEDATA](#) `StoreData`
- [FILEHANDLE](#) `Fscr` [3]
- [FILEHANDLE](#) `FoStage4` [2]
- [POSITION](#) `DefPosition`
- [FILEHANDLE](#) \* `infile`
- [FILEHANDLE](#) \* `outfile`
- [FILEHANDLE](#) \* `hidefile`
- [WORD](#) \* `CompressBuffer`
- [WORD](#) \* `ComprTop`

- WORD \* [CompressPointer](#)
- COMPAREDUMMY [CompareRoutine](#)
- ULONG \* [wranfia](#)
- SBYTE \* [moebiustable](#)
- LONG [OldTime](#)
- LONG [InInBuf](#)
- LONG [InHiBuf](#)
- LONG [pWorkSize](#)
- LONG [IWorkSize](#)
- LONG [posWorkSize](#)
- ULONG [wranfseed](#)
- int [NoCompress](#)
- int [gzipCompress](#)
- int [Cnumlhs](#)
- int [outtohide](#)
- int [wranfcall](#)
- int [wranfnpair1](#)
- int [wranfnpair2](#)
- WORD [GetFile](#)
- WORD [KeptInHold](#)
- WORD [BracketOn](#)
- WORD [MaxBracket](#)
- WORD [CurDum](#)
- WORD [DeferFlag](#)
- WORD [TePos](#)
- WORD [sLevel](#)
- WORD [Stage4Name](#)
- WORD [GetOneFile](#)
- WORD [PolyFun](#)
- WORD [PolyFunInv](#)
- WORD [PolyFunType](#)
- WORD [PolyFunExp](#)
- WORD [PolyFunVar](#)
- WORD [PolyFunPow](#)
- WORD [Eside](#)
- WORD [MaxDum](#)
- WORD [level](#)
- WORD [expchanged](#)
- WORD [expflags](#)
- WORD [CurExpr](#)
- WORD [SortType](#)
- WORD [ShortSortCount](#)
- WORD [modeloptions](#)
- WORD [funoffset](#)
- WORD [moebiustablesizes](#)

### 3.120.1 Detailed Description

The [R\\_const](#) struct is part of the global data and resides either in the ALLGLOBALS struct A, or the ALLPRIVATE struct B (TFORM) under the name R We see it used with the macro AR as in AR.infile It has the variables that define the running environment and that should be transferred with a term in a multithreaded run.

Definition at line [2029](#) of file [structs.h](#).

## 3.120.2 Field Documentation

### 3.120.2.1 StoreData

`FILEDATA R_const::StoreData`

Definition at line 2030 of file [structs.h](#).

### 3.120.2.2 Fscr

`FILEHANDLE R_const::Fscr[3]`

Definition at line 2031 of file [structs.h](#).

### 3.120.2.3 FoStage4

`FILEHANDLE R_const::FoStage4[2]`

Definition at line 2032 of file [structs.h](#).

### 3.120.2.4 DefPosition

`POSITION R_const::DefPosition`

Definition at line 2033 of file [structs.h](#).

### 3.120.2.5 infile

`FILEHANDLE* R_const::infile`

Definition at line 2034 of file [structs.h](#).

### 3.120.2.6 outfile

`FILEHANDLE* R_const::outfile`

Definition at line 2035 of file [structs.h](#).

### 3.120.2.7 hidefile

`FILEHANDLE* R_const::hidefile`

Definition at line 2036 of file [structs.h](#).

### 3.120.2.8 CompressBuffer

WORD\* R\_const::CompressBuffer

Definition at line 2038 of file [structs.h](#).

### 3.120.2.9 ComprTop

WORD\* R\_const::ComprTop

Definition at line 2039 of file [structs.h](#).

### 3.120.2.10 CompressPointer

WORD\* R\_const::CompressPointer

Definition at line 2040 of file [structs.h](#).

### 3.120.2.11 CompareRoutine

COMPAREDUMMY R\_const::CompareRoutine

Definition at line 2041 of file [structs.h](#).

### 3.120.2.12 wranfia

ULONG\* R\_const::wranfia

Definition at line 2042 of file [structs.h](#).

### 3.120.2.13 moebiustable

SBYTE\* R\_const::moebiustable

Definition at line 2043 of file [structs.h](#).

### 3.120.2.14 OldTime

LONG R\_const::OldTime

Definition at line 2045 of file [structs.h](#).

### 3.120.2.15 InInBuf

LONG R\_const::InInBuf

Definition at line 2046 of file [structs.h](#).

**3.120.2.16 InHiBuf**

```
LONG R_const::InHiBuf
```

Definition at line 2047 of file [structs.h](#).

**3.120.2.17 pWorkSize**

```
LONG R_const::pWorkSize
```

Definition at line 2048 of file [structs.h](#).

**3.120.2.18 lWorkSize**

```
LONG R_const::lWorkSize
```

Definition at line 2049 of file [structs.h](#).

**3.120.2.19 posWorkSize**

```
LONG R_const::posWorkSize
```

Definition at line 2050 of file [structs.h](#).

**3.120.2.20 wranfseed**

```
ULONG R_const::wranfseed
```

Definition at line 2051 of file [structs.h](#).

**3.120.2.21 NoCompress**

```
int R_const::NoCompress
```

Definition at line 2052 of file [structs.h](#).

**3.120.2.22 gzipCompress**

```
int R_const::gzipCompress
```

Definition at line 2053 of file [structs.h](#).

**3.120.2.23 Cnumlhs**

```
int R_const::Cnumlhs
```

Definition at line 2054 of file [structs.h](#).

#### 3.120.2.24 outtohide

```
int R_const::outtohide
```

Definition at line 2055 of file [structs.h](#).

#### 3.120.2.25 wranfcall

```
int R_const::wranfcall
```

Definition at line 2059 of file [structs.h](#).

#### 3.120.2.26 wranfnpair1

```
int R_const::wranfnpair1
```

Definition at line 2060 of file [structs.h](#).

#### 3.120.2.27 wranfnpair2

```
int R_const::wranfnpair2
```

Definition at line 2061 of file [structs.h](#).

#### 3.120.2.28 GetFile

```
WORD R_const::GetFile
```

Definition at line 2067 of file [structs.h](#).

#### 3.120.2.29 KeptInHold

```
WORD R_const::KeptInHold
```

Definition at line 2068 of file [structs.h](#).

#### 3.120.2.30 BracketOn

```
WORD R_const::BracketOn
```

Definition at line 2069 of file [structs.h](#).

#### 3.120.2.31 MaxBracket

```
WORD R_const::MaxBracket
```

Definition at line 2070 of file [structs.h](#).



### 3.120.2.32 CurDum

WORD R\_const::CurDum

Definition at line 2071 of file [structs.h](#).

### 3.120.2.33 DeferFlag

WORD R\_const::DeferFlag

Definition at line 2072 of file [structs.h](#).

### 3.120.2.34 TePos

WORD R\_const::TePos

Definition at line 2073 of file [structs.h](#).

### 3.120.2.35 sLevel

WORD R\_const::sLevel

Definition at line 2074 of file [structs.h](#).

### 3.120.2.36 Stage4Name

WORD R\_const::Stage4Name

Definition at line 2075 of file [structs.h](#).

### 3.120.2.37 GetOneFile

WORD R\_const::GetOneFile

Definition at line 2076 of file [structs.h](#).

### 3.120.2.38 PolyFun

WORD R\_const::PolyFun

Definition at line 2077 of file [structs.h](#).

### 3.120.2.39 PolyFunInv

WORD R\_const::PolyFunInv

Definition at line 2078 of file [structs.h](#).

#### 3.120.2.40 PolyFunType

WORD R\_const::PolyFunType

Definition at line 2079 of file [structs.h](#).

#### 3.120.2.41 PolyFunExp

WORD R\_const::PolyFunExp

Definition at line 2080 of file [structs.h](#).

#### 3.120.2.42 PolyFunVar

WORD R\_const::PolyFunVar

Definition at line 2081 of file [structs.h](#).

#### 3.120.2.43 PolyFunPow

WORD R\_const::PolyFunPow

Definition at line 2082 of file [structs.h](#).

#### 3.120.2.44 Eside

WORD R\_const::Eside

Definition at line 2083 of file [structs.h](#).

#### 3.120.2.45 MaxDum

WORD R\_const::MaxDum

Definition at line 2084 of file [structs.h](#).

#### 3.120.2.46 level

WORD R\_const::level

Definition at line 2085 of file [structs.h](#).

#### 3.120.2.47 expchanged

WORD R\_const::expchanged

Definition at line 2086 of file [structs.h](#).

### 3.120.2.48 expflags

WORD R\_const::expflags

Definition at line 2087 of file [structs.h](#).

### 3.120.2.49 CurExpr

WORD R\_const::CurExpr

Definition at line 2088 of file [structs.h](#).

### 3.120.2.50 SortType

WORD R\_const::SortType

Definition at line 2089 of file [structs.h](#).

### 3.120.2.51 ShortSortCount

WORD R\_const::ShortSortCount

Definition at line 2090 of file [structs.h](#).

### 3.120.2.52 modeloptions

WORD R\_const::modeloptions

Definition at line 2091 of file [structs.h](#).

### 3.120.2.53 funoffset

WORD R\_const::funoffset

Definition at line 2092 of file [structs.h](#).

### 3.120.2.54 moebiustablesizesize

WORD R\_const::moebiustablesizesize

Definition at line 2093 of file [structs.h](#).

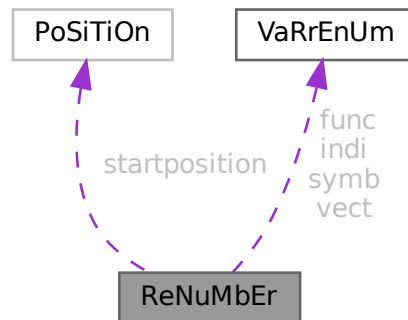
The documentation for this struct was generated from the following file:

- [structs.h](#)

### 3.121 ReNuMbEr Struct Reference

```
#include <structs.h>
```

Collaboration diagram for ReNuMbEr:



#### Public Member Functions

- **PADPOSITION** (4, 0, 0, 0, sizeof([VARRENUM](#)) \*4)

#### Data Fields

- [POSITION](#) startposition
- [VARRENUM](#) symb
- [VARRENUM](#) indi
- [VARRENUM](#) vect
- [VARRENUM](#) func
- WORD \* [symnum](#)
- WORD \* [indnum](#)
- WORD \* [vecnum](#)
- WORD \* [funnum](#)

#### 3.121.1 Detailed Description

Only symb.lo gets dynamically allocated. All other pointers points into this memory.

Definition at line 178 of file [structs.h](#).

#### 3.121.2 Field Documentation

##### 3.121.2.1 startposition

[POSITION](#) ReNuMbEr::startposition

Definition at line 179 of file [structs.h](#).

### 3.121.2.2 symb

`VARRENUM ReNuMbEr::symb`

Symbols

Definition at line 181 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#), [Generator\(\)](#), [GetFirstTerm\(\)](#), and [TermRenumber\(\)](#).

### 3.121.2.3 indi

`VARRENUM ReNuMbEr::indi`

Indices

Definition at line 182 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#), and [TermRenumber\(\)](#).

### 3.121.2.4 vect

`VARRENUM ReNuMbEr::vect`

Vectors

Definition at line 183 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#), and [TermRenumber\(\)](#).

### 3.121.2.5 func

`VARRENUM ReNuMbEr::func`

Functions

Definition at line 184 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#), and [TermRenumber\(\)](#).

### 3.121.2.6 symnum

`WORD* ReNuMbEr::symnum`

Renumbered symbols

Definition at line 186 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#), and [TermRenumber\(\)](#).

### 3.121.2.7 indnum

```
WORD* ReNuMbEr::indnum
```

Renumbered indices

Definition at line 187 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#), and [TermRenumber\(\)](#).

### 3.121.2.8 vecnum

```
WORD* ReNuMbEr::vecnum
```

Renumbered vectors

Definition at line 188 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#), and [TermRenumber\(\)](#).

### 3.121.2.9 funnum

```
WORD* ReNuMbEr::funnum
```

Renumbered functions

Definition at line 189 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#), and [TermRenumber\(\)](#).

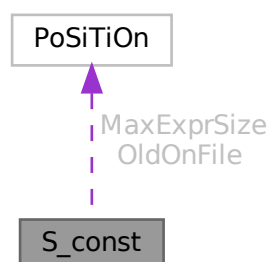
The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.122 S\_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for S\_const:



## Public Member Functions

- **PADPOSITION** (4, 0, 4, 2, 0)

## Data Fields

- [POSITION](#) MaxExprSize
- [POSITION](#) \* OldOnFile
- [WORD](#) \* OldNumFactors
- [WORD](#) \* Oldvflags
- [WORD](#) \* Olduflags
- [int](#) NumOldOnFile
- [int](#) NumOldNumFactors
- [int](#) MultiThreaded
- [int](#) Balancing
- [WORD](#) ExecMode
- [WORD](#) CollectOverFlag

### 3.122.1 Detailed Description

The [S\\_const](#) struct is part of the global data and resides in the ALLGLOBALS struct A under the name S We see it used with the macro AS as in AS.ExecMode It has some variables used by the master in multithreaded runs

Definition at line [1977](#) of file [structs.h](#).

### 3.122.2 Field Documentation

#### 3.122.2.1 MaxExprSize

[POSITION](#) S\_const::MaxExprSize

Definition at line [1978](#) of file [structs.h](#).

#### 3.122.2.2 OldOnFile

[POSITION](#)\* S\_const::OldOnFile

Definition at line [1984](#) of file [structs.h](#).

#### 3.122.2.3 OldNumFactors

[WORD](#)\* S\_const::OldNumFactors

Definition at line [1985](#) of file [structs.h](#).

#### 3.122.2.4 Oldvflags

```
WORD* S_const::Oldvflags
```

Definition at line 1986 of file [structs.h](#).

#### 3.122.2.5 Olduflags

```
WORD* S_const::Olduflags
```

Definition at line 1987 of file [structs.h](#).

#### 3.122.2.6 NumOldOnFile

```
int S_const::NumOldOnFile
```

Definition at line 1991 of file [structs.h](#).

#### 3.122.2.7 NumOldNumFactors

```
int S_const::NumOldNumFactors
```

Definition at line 1992 of file [structs.h](#).

#### 3.122.2.8 MultiThreaded

```
int S_const::MultiThreaded
```

Definition at line 1993 of file [structs.h](#).

#### 3.122.2.9 Balancing

```
int S_const::Balancing
```

Definition at line 2000 of file [structs.h](#).

#### 3.122.2.10 ExecMode

```
WORD S_const::ExecMode
```

Definition at line 2001 of file [structs.h](#).



### 3.122.2.11 CollectOverFlag

WORD S\_const::CollectOverFlag

Definition at line 2003 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.123 SeTs Struct Reference

### Data Fields

- LONG [name](#)
- WORD [type](#)
- WORD [first](#)
- WORD [last](#)
- WORD [node](#)
- WORD [namesize](#)
- WORD [dimension](#)
- WORD [flags](#)

### 3.123.1 Detailed Description

Definition at line 499 of file [structs.h](#).

### 3.123.2 Field Documentation

#### 3.123.2.1 name

LONG SeTs::name

Definition at line 500 of file [structs.h](#).

#### 3.123.2.2 type

WORD SeTs::type

Definition at line 501 of file [structs.h](#).

#### 3.123.2.3 first

WORD SeTs::first

Definition at line 502 of file [structs.h](#).

#### 3.123.2.4 last

WORD SeTs::last

Definition at line 503 of file [structs.h](#).

#### 3.123.2.5 node

WORD SeTs::node

Definition at line 504 of file [structs.h](#).

#### 3.123.2.6 namesize

WORD SeTs::namesize

Definition at line 505 of file [structs.h](#).

#### 3.123.2.7 dimension

WORD SeTs::dimension

Definition at line 506 of file [structs.h](#).

#### 3.123.2.8 flags

WORD SeTs::flags

Definition at line 507 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

### 3.124 SETUPPARAMETERS Struct Reference

```
#include <structs.h>
```

#### Data Fields

- UBYTE \* [parameter](#)
- int [type](#)
- int [flags](#)
- LONG [value](#)

### 3.124.1 Detailed Description

Each setup parameter has one element of the struct [SETUPPARAMETERS](#) assigned to it. By binary search in the array of them we can then locate the proper element by name. We have to assume that two ints make a long and either one or two longs make a pointer. The long before the ints and padding gives a problem in the initialization.

Definition at line [1038](#) of file [structs.h](#).

### 3.124.2 Field Documentation

#### 3.124.2.1 parameter

```
UBYTE* SETUPPARAMETERS::parameter
```

Definition at line [1039](#) of file [structs.h](#).

#### 3.124.2.2 type

```
int SETUPPARAMETERS::type
```

Definition at line [1040](#) of file [structs.h](#).

#### 3.124.2.3 flags

```
int SETUPPARAMETERS::flags
```

Definition at line [1041](#) of file [structs.h](#).

#### 3.124.2.4 value

```
LONG SETUPPARAMETERS::value
```

Definition at line [1042](#) of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.125 SGroup Class Reference

### Public Member Functions

- void [print](#) (void)
- void [newGroup](#) (int nelm, int nclss, int \*clss)
- void [clearGroup](#) (void)
- void [delGroup](#) (void)
- int \* [genNext](#) (void)
- void [addGroup](#) (int \*p)
- BigInt [nElem](#) (void)
- int \* [nextElem](#) (void)

## Data Fields

- BigInt [size](#)
- BigInt [nelem](#)
- int \*\* [elem](#)
- int [nnodes](#)
- int [neclass](#)
- int [eclass](#) [GRCC\_MAXNODES]
- int [cgen](#)
- int [csav](#)
- int [permg](#) [GRCC\_MAXNODES]
- int [perms](#) [GRCC\_MAXNODES]
- int [pgr](#) [GRCC\_MAXNODES]
- int [pgq](#) [GRCC\_MAXNODES]
- int [psr](#) [GRCC\_MAXNODES]
- int [psq](#) [GRCC\_MAXNODES]

### 3.125.1 Detailed Description

Definition at line [700](#) of file [grcc.h](#).

### 3.125.2 Constructor & Destructor Documentation

#### 3.125.2.1 SGroup()

```
SGroup::SGroup (  
    void )
```

Definition at line [5376](#) of file [grcc.cc](#).

#### 3.125.2.2 ~SGroup()

```
SGroup::~~SGroup (  
    void )
```

Definition at line [5385](#) of file [grcc.cc](#).

### 3.125.3 Member Function Documentation

#### 3.125.3.1 print()

```
void SGroup::print (  
    void )
```

Definition at line [5402](#) of file [grcc.cc](#).

### 3.125.3.2 newGroup()

```
void SGroup::newGroup (
    int nelm,
    int nclss,
    int * clss )
```

Definition at line 5417 of file [grcc.cc](#).

### 3.125.3.3 clearGroup()

```
void SGroup::clearGroup (
    void )
```

Definition at line 5442 of file [grcc.cc](#).

### 3.125.3.4 delGroup()

```
void SGroup::delGroup (
    void )
```

Definition at line 5450 of file [grcc.cc](#).

### 3.125.3.5 genNext()

```
int * SGroup::genNext (
    void )
```

Definition at line 5466 of file [grcc.cc](#).

### 3.125.3.6 addGroup()

```
void SGroup::addGroup (
    int * p )
```

Definition at line 5476 of file [grcc.cc](#).

### 3.125.3.7 nElem()

```
BigInt SGroup::nElem (
    void )
```

Definition at line 5493 of file [grcc.cc](#).

### 3.125.3.8 nextElem()

```
int * SGroup::nextElem (
    void )
```

Definition at line 5499 of file [grcc.cc](#).

### 3.125.4 Field Documentation

#### 3.125.4.1 size

```
BigInt SGroup::size
```

Definition at line 702 of file [grcc.h](#).

#### 3.125.4.2 nelem

```
BigInt SGroup::nelem
```

Definition at line 703 of file [grcc.h](#).

#### 3.125.4.3 elem

```
int** SGroup::elem
```

Definition at line 704 of file [grcc.h](#).

#### 3.125.4.4 nnodes

```
int SGroup::nnodes
```

Definition at line 705 of file [grcc.h](#).

#### 3.125.4.5 neclass

```
int SGroup::neclass
```

Definition at line 706 of file [grcc.h](#).

#### 3.125.4.6 eclass

```
int SGroup::eclass[GRCC_MAXNODES]
```

Definition at line 707 of file [grcc.h](#).

#### 3.125.4.7 cgen

```
int SGroup::cgen
```

Definition at line 708 of file [grcc.h](#).

#### 3.125.4.8 csav

```
int SGroup::csav
```

Definition at line 709 of file [grcc.h](#).

#### 3.125.4.9 permg

```
int SGroup::permg[GRCC_MAXNODES]
```

Definition at line 710 of file [grcc.h](#).

#### 3.125.4.10 perms

```
int SGroup::perms[GRCC_MAXNODES]
```

Definition at line 711 of file [grcc.h](#).

#### 3.125.4.11 pgr

```
int SGroup::pgr[GRCC_MAXNODES]
```

Definition at line 713 of file [grcc.h](#).

#### 3.125.4.12 pgq

```
int SGroup::pgq[GRCC_MAXNODES]
```

Definition at line 714 of file [grcc.h](#).

#### 3.125.4.13 psr

```
int SGroup::psr[GRCC_MAXNODES]
```

Definition at line 715 of file [grcc.h](#).

#### 3.125.4.14 psq

```
int SGroup::psq[GRCC_MAXNODES]
```

Definition at line 716 of file [grcc.h](#).

The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.126 SHvariables Struct Reference

### Data Fields

- WORD \* [outterm](#)
- WORD \* [outfun](#)
- WORD \* [incoef](#)
- WORD \* [stop1](#)
- WORD \* [stop2](#)
- WORD \* [ststop1](#)
- WORD \* [ststop2](#)
- FINISHUFFLE [finishuf](#)
- DO\_UFFLE [do\\_uffle](#)
- LONG [combilast](#)
- WORD [nincoef](#)
- WORD [level](#)
- WORD [thefunction](#)
- WORD [option](#)

### 3.126.1 Detailed Description

Definition at line [1276](#) of file [structs.h](#).

### 3.126.2 Field Documentation

#### 3.126.2.1 outterm

```
WORD* SHvariables::outterm
```

Definition at line [1277](#) of file [structs.h](#).

#### 3.126.2.2 outfun

```
WORD* SHvariables::outfun
```

Definition at line [1278](#) of file [structs.h](#).

#### 3.126.2.3 incoef

```
WORD* SHvariables::incoef
```

Definition at line [1279](#) of file [structs.h](#).

#### 3.126.2.4 stop1

```
WORD* SHvariables::stop1
```

Definition at line [1280](#) of file [structs.h](#).



### 3.126.2.5 stop2

WORD\* SHvariables::stop2

Definition at line 1281 of file [structs.h](#).

### 3.126.2.6 ststop1

WORD\* SHvariables::ststop1

Definition at line 1282 of file [structs.h](#).

### 3.126.2.7 ststop2

WORD\* SHvariables::ststop2

Definition at line 1283 of file [structs.h](#).

### 3.126.2.8 finishuf

FINISHUFFLE SHvariables::finishuf

Definition at line 1284 of file [structs.h](#).

### 3.126.2.9 do\_uffle

DO\_UFFLE SHvariables::do\_uffle

Definition at line 1285 of file [structs.h](#).

### 3.126.2.10 combilast

LONG SHvariables::combilast

Definition at line 1286 of file [structs.h](#).

### 3.126.2.11 nincoef

WORD SHvariables::nincoef

Definition at line 1287 of file [structs.h](#).

### 3.126.2.12 level

WORD SHvariables::level

Definition at line 1288 of file [structs.h](#).

### 3.126.2.13 thefunction

WORD SHvariables::thefunction

Definition at line 1289 of file [structs.h](#).

### 3.126.2.14 option

WORD SHvariables::option

Definition at line 1290 of file [structs.h](#).

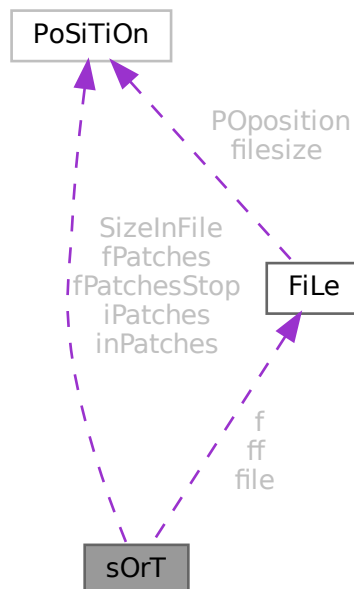
The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.127 sOrT Struct Reference

```
#include <structs.h>
```

Collaboration diagram for sOrT:



### Public Member Functions

- **PADPOSITION** (24, 12, 12, 3, 0)

## Data Fields

- [FILEHANDLE](#) [file](#)
- [POSITION](#) [SizeInFile](#) [3]
- WORD \* [lBuffer](#)
- WORD \* [lTop](#)
- WORD \* [lFill](#)
- WORD \* [used](#)
- WORD \* [sBuffer](#)
- WORD \* [sTop](#)
- WORD \* [sTop2](#)
- WORD \* [sHalf](#)
- WORD \* [sFill](#)
- WORD \*\* [sPointer](#)
- WORD \*\* [PoinFill](#)
- WORD \* [cBuffer](#)
- WORD \*\* [Patches](#)
- WORD \*\* [pStop](#)
- WORD \*\* [poina](#)
- WORD \*\* [poin2a](#)
- WORD \* [ktoi](#)
- WORD \* [tree](#)
- [POSITION](#) \* [fPatches](#)
- [POSITION](#) \* [inPatches](#)
- [POSITION](#) \* [fPatchesStop](#)
- [POSITION](#) \* [iPatches](#)
- [FILEHANDLE](#) \* [f](#)
- [FILEHANDLE](#) \*\* [ff](#)
- LONG [sTerms](#)
- LONG [LargeSize](#)
- LONG [SmallSize](#)
- LONG [SmallEsize](#)
- LONG [TermsInSmall](#)
- LONG [Terms2InSmall](#)
- LONG [GenTerms](#)
- LONG [TermsLeft](#)
- LONG [GenSpace](#)
- LONG [SpaceLeft](#)
- LONG [putinsize](#)
- LONG [ninterms](#)
- int [MaxPatches](#)
- int [MaxFpatches](#)
- int [type](#)
- int [lPatch](#)
- int [fPatchN1](#)
- int [PolyWise](#)
- int [PolyFlag](#)
- int [cBufferSize](#)
- int [maxtermsize](#)
- int [newmaxtermsize](#)
- int [outputmode](#)
- int [stagelevel](#)
- WORD [fPatchN](#)
- WORD [inNum](#)
- WORD [stage4](#)

### 3.127.1 Detailed Description

The struct SORTING is used to control a sort operation. It includes a small and a large buffer and arrays for keeping track of various stages of the (merge) sorts. Each sort level has its own struct and different levels can have different sizes for its arrays. Also different threads have their own set of SORTING structs.

Definition at line 1140 of file [structs.h](#).

### 3.127.2 Field Documentation

#### 3.127.2.1 file

```
FILEHANDLE sOrT::file
```

Definition at line 1141 of file [structs.h](#).

#### 3.127.2.2 SizeInFile

```
POSITION sOrT::SizeInFile[3]
```

Definition at line 1142 of file [structs.h](#).

#### 3.127.2.3 lBuffer

```
WORD* sOrT::lBuffer
```

Definition at line 1143 of file [structs.h](#).

#### 3.127.2.4 lTop

```
WORD* sOrT::lTop
```

Definition at line 1144 of file [structs.h](#).

#### 3.127.2.5 lFill

```
WORD* sOrT::lFill
```

Definition at line 1145 of file [structs.h](#).

#### 3.127.2.6 used

```
WORD* sOrT::used
```

Definition at line 1146 of file [structs.h](#).

### 3.127.2.7 sBuffer

WORD\* sOrT::sBuffer

Definition at line 1147 of file [structs.h](#).

### 3.127.2.8 sTop

WORD\* sOrT::sTop

Definition at line 1148 of file [structs.h](#).

### 3.127.2.9 sTop2

WORD\* sOrT::sTop2

Definition at line 1149 of file [structs.h](#).

### 3.127.2.10 sHalf

WORD\* sOrT::sHalf

Definition at line 1150 of file [structs.h](#).

### 3.127.2.11 sFill

WORD\* sOrT::sFill

Definition at line 1151 of file [structs.h](#).

### 3.127.2.12 sPointer

WORD\*\* sOrT::sPointer

Definition at line 1152 of file [structs.h](#).

### 3.127.2.13 PoinFill

WORD\*\* sOrT::PoinFill

Definition at line 1153 of file [structs.h](#).

### 3.127.2.14 cBuffer

WORD\* sOrT::cBuffer

Definition at line 1154 of file [structs.h](#).

### 3.127.2.15 Patches

```
WORD** sOrT::Patches
```

Definition at line 1155 of file [structs.h](#).

### 3.127.2.16 pStop

```
WORD** sOrT::pStop
```

Definition at line 1156 of file [structs.h](#).

### 3.127.2.17 poina

```
WORD** sOrT::poina
```

Definition at line 1157 of file [structs.h](#).

### 3.127.2.18 poin2a

```
WORD** sOrT::poin2a
```

Definition at line 1158 of file [structs.h](#).

### 3.127.2.19 ktoi

```
WORD* sOrT::ktoi
```

Definition at line 1159 of file [structs.h](#).

### 3.127.2.20 tree

```
WORD* sOrT::tree
```

Definition at line 1160 of file [structs.h](#).

### 3.127.2.21 fPatches

```
POSITION* sOrT::fPatches
```

Definition at line 1166 of file [structs.h](#).

### 3.127.2.22 inPatches

```
POSITION* sOrT::inPatches
```

Definition at line 1167 of file [structs.h](#).

### 3.127.2.23 fPatchesStop

`POSITION*` sOrT::fPatchesStop

Definition at line 1168 of file [structs.h](#).

### 3.127.2.24 iPatches

`POSITION*` sOrT::iPatches

Definition at line 1169 of file [structs.h](#).

### 3.127.2.25 f

`FILEHANDLE*` sOrT::f

Definition at line 1170 of file [structs.h](#).

### 3.127.2.26 ff

`FILEHANDLE**` sOrT::ff

Definition at line 1171 of file [structs.h](#).

### 3.127.2.27 sTerms

`LONG` sOrT::sTerms

Definition at line 1172 of file [structs.h](#).

### 3.127.2.28 LargeSize

`LONG` sOrT::LargeSize

Definition at line 1173 of file [structs.h](#).

### 3.127.2.29 SmallSize

`LONG` sOrT::SmallSize

Definition at line 1174 of file [structs.h](#).

### 3.127.2.30 SmallEsize

`LONG` sOrT::SmallEsize

Definition at line 1175 of file [structs.h](#).

### 3.127.2.31 TermsInSmall

LONG sOrT::TermsInSmall

Definition at line 1176 of file [structs.h](#).

### 3.127.2.32 Terms2InSmall

LONG sOrT::Terms2InSmall

Definition at line 1177 of file [structs.h](#).

### 3.127.2.33 GenTerms

LONG sOrT::GenTerms

Definition at line 1178 of file [structs.h](#).

### 3.127.2.34 TermsLeft

LONG sOrT::TermsLeft

Definition at line 1179 of file [structs.h](#).

### 3.127.2.35 GenSpace

LONG sOrT::GenSpace

Definition at line 1180 of file [structs.h](#).

### 3.127.2.36 SpaceLeft

LONG sOrT::SpaceLeft

Definition at line 1181 of file [structs.h](#).

### 3.127.2.37 putinsize

LONG sOrT::putinsize

Definition at line 1182 of file [structs.h](#).

### 3.127.2.38 ninterms

LONG sOrT::ninterms

Definition at line 1183 of file [structs.h](#).



**3.127.2.39 MaxPatches**

```
int sOrT::MaxPatches
```

Definition at line 1184 of file [structs.h](#).

**3.127.2.40 MaxFpatches**

```
int sOrT::MaxFpatches
```

Definition at line 1185 of file [structs.h](#).

**3.127.2.41 type**

```
int sOrT::type
```

Definition at line 1186 of file [structs.h](#).

**3.127.2.42 lPatch**

```
int sOrT::lPatch
```

Definition at line 1187 of file [structs.h](#).

**3.127.2.43 fPatchN1**

```
int sOrT::fPatchN1
```

Definition at line 1188 of file [structs.h](#).

**3.127.2.44 PolyWise**

```
int sOrT::PolyWise
```

Definition at line 1189 of file [structs.h](#).

**3.127.2.45 PolyFlag**

```
int sOrT::PolyFlag
```

Definition at line 1190 of file [structs.h](#).

**3.127.2.46 cBufferSize**

```
int sOrT::cBufferSize
```

Definition at line 1191 of file [structs.h](#).

**3.127.2.47 maxtermsize**

```
int sOrT::maxtermsize
```

Definition at line 1192 of file [structs.h](#).

**3.127.2.48 newmaxtermsize**

```
int sOrT::newmaxtermsize
```

Definition at line 1193 of file [structs.h](#).

**3.127.2.49 outputmode**

```
int sOrT::outputmode
```

Definition at line 1194 of file [structs.h](#).

**3.127.2.50 stagelevel**

```
int sOrT::stagelevel
```

Definition at line 1195 of file [structs.h](#).

**3.127.2.51 fPatchN**

```
WORD sOrT::fPatchN
```

Definition at line 1196 of file [structs.h](#).

**3.127.2.52 inNum**

```
WORD sOrT::inNum
```

Definition at line 1197 of file [structs.h](#).

**3.127.2.53 stage4**

```
WORD sOrT::stage4
```

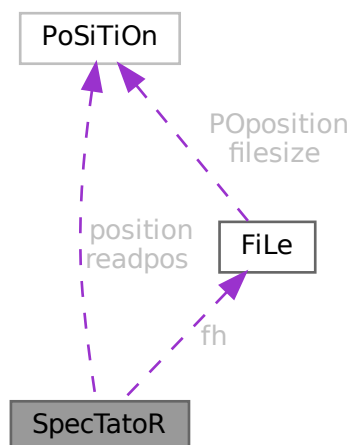
Definition at line 1198 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.128 SpecTatoR Struct Reference

Collaboration diagram for SpecTatoR:



### Public Member Functions

- **PADPOSITION** (2, 0, 0, 2, 0)

### Data Fields

- [POSITION](#) position
- [POSITION](#) readpos
- [FILEHANDLE](#) \* fh
- char \* name
- [WORD](#) exprnumber
- [WORD](#) flags

### 3.128.1 Detailed Description

Definition at line 765 of file [structs.h](#).

### 3.128.2 Field Documentation

#### 3.128.2.1 position

[POSITION](#) SpecTatoR::position

Definition at line 766 of file [structs.h](#).

### 3.128.2.2 readpos

**POSITION** `SpecTatoR::readpos`

Definition at line 767 of file [structs.h](#).

### 3.128.2.3 fh

**FILEHANDLE\*** `SpecTatoR::fh`

Definition at line 768 of file [structs.h](#).

### 3.128.2.4 name

**char\*** `SpecTatoR::name`

Definition at line 769 of file [structs.h](#).

### 3.128.2.5 exprnumber

**WORD** `SpecTatoR::exprnumber`

Definition at line 770 of file [structs.h](#).

### 3.128.2.6 flags

**WORD** `SpecTatoR::flags`

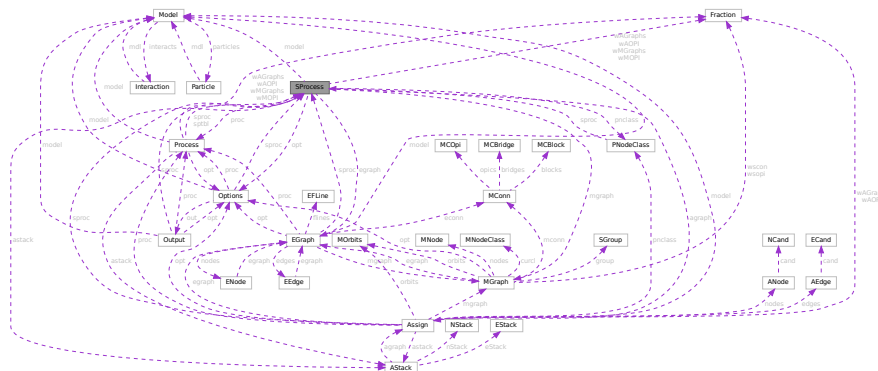
Definition at line 771 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.129 SProcess Class Reference

Collaboration diagram for SProcess:



## Public Member Functions

- [SProcess](#) ([Model](#) \*mdl, [Process](#) \*prc, [Options](#) \*opts, int sid, int \*clst, int ncls, [NCInput](#) \*cls)
- [SProcess](#) ([Model](#) \*mdl, [Process](#) \*prc, [Options](#) \*opts, int sid, int \*clst, int ncls, int \*cdeg, int \*ctyp, int \*ptcl, int \*cpl, int \*cnum, int \*cmind, int \*cmaxd)
- void [prSProcess](#) (void)
- [BigInt](#) [generate](#) (void)
- void [assign](#) ([MGraph](#) \*mgr)
- int [toMNodeClass](#) (int \*ctyp, int \*cldeg, int \*clnum, int \*cmind, int \*cmaxd)
- [PNodeClass](#) \* [match](#) ([MGraph](#) \*mgr)
- void [endMGraph](#) ([MGraph](#) \*mgr)
- void [endAGraph](#) ([EGraph](#) \*egr)
- void [resultMGraph](#) ([BigInt](#) nmgraphs, [Fraction](#) mwsum, [BigInt](#) nmopi, [Fraction](#) mwopi)
- void [resultAGraph](#) ([BigInt](#) nagraphs, [Fraction](#) awsum, [BigInt](#) naopi, [Fraction](#) awopi)

## Data Fields

- [Model](#) \* model
- [Process](#) \* proc
- [Options](#) \* opt
- [PNodeClass](#) \* pnclass
- [AStack](#) \* astack
- [MGraph](#) \* mgraph
- [EGraph](#) \* egraph
- [Assign](#) \* agraph
- int \* cl2nd
- int \* nd2cl
- int nclass
- int ninitl
- int nfinal
- int nvert
- int clist [GRCC\_MAXNCPLG]
- int id
- int loop
- int nNodes
- int nEdges
- int nExtern
- int ncouple
- int tCouple
- int DUMMY\_PADDING
- [BigInt](#) mgrcount
- [BigInt](#) agrcount
- [BigInt](#) extperm
- [BigInt](#) nMGraphs
- [BigInt](#) nMOPI
- [Fraction](#) wMGraphs
- [Fraction](#) wMOPI
- [BigInt](#) nAGraphs
- [BigInt](#) nAOPI
- [Fraction](#) wAGraphs
- [Fraction](#) wAOPI

### 3.129.1 Detailed Description

Definition at line 350 of file [grcc.h](#).

### 3.129.2 Constructor & Destructor Documentation

#### 3.129.2.1 SProcess() [1/2]

```
SProcess::SProcess (
    Model * mdl,
    Process * prc,
    Options * opts,
    int sid,
    int * clst,
    int ncls,
    NCInput * cls )
```

Definition at line 2695 of file [grcc.cc](#).

#### 3.129.2.2 SProcess() [2/2]

```
SProcess::SProcess (
    Model * mdl,
    Process * prc,
    Options * opts,
    int sid,
    int * clst,
    int ncls,
    int * cdeg,
    int * ctyp,
    int * ptcl,
    int * cpl,
    int * cnum,
    int * cmind,
    int * cmaxd )
```

Definition at line 2526 of file [grcc.cc](#).

#### 3.129.2.3 ~SProcess()

```
SProcess::~~SProcess (
    void )
```

Definition at line 2857 of file [grcc.cc](#).

### 3.129.3 Member Function Documentation

#### 3.129.3.1 prSProcess()

```
void SProcess::prSProcess (
    void )
```

Definition at line 2865 of file [grcc.cc](#).

### 3.129.3.2 generate()

```
BigInt SProcess::generate (
    void )
```

Definition at line [2877](#) of file [grcc.cc](#).

### 3.129.3.3 assign()

```
void SProcess::assign (
    MGraph * mgr )
```

Definition at line [2921](#) of file [grcc.cc](#).

### 3.129.3.4 toMNodeClass()

```
int SProcess::toMNodeClass (
    int * ctyp,
    int * cldeg,
    int * clnum,
    int * cmind,
    int * cmaxd )
```

Definition at line [2902](#) of file [grcc.cc](#).

### 3.129.3.5 match()

```
PNodeClass * SProcess::match (
    MGraph * mgr )
```

Definition at line [2938](#) of file [grcc.cc](#).

### 3.129.3.6 endMGraph()

```
void SProcess::endMGraph (
    MGraph * mgr )
```

Definition at line [3017](#) of file [grcc.cc](#).

### 3.129.3.7 endAGraph()

```
void SProcess::endAGraph (
    EGraph * egr )
```

Definition at line [3027](#) of file [grcc.cc](#).

### 3.129.3.8 resultMGraph()

```
void SProcess::resultMGraph (
    BigInt nmgraphs,
    Fraction mwsum,
    BigInt nmopi,
    Fraction mwopi )
```

Definition at line 2999 of file [grcc.cc](#).

### 3.129.3.9 resultAGraph()

```
void SProcess::resultAGraph (
    BigInt nagraphs,
    Fraction awsum,
    BigInt naopi,
    Fraction awopi )
```

Definition at line 3008 of file [grcc.cc](#).

## 3.129.4 Field Documentation

### 3.129.4.1 model

```
Model* SProcess::model
```

Definition at line 357 of file [grcc.h](#).

### 3.129.4.2 proc

```
Process* SProcess::proc
```

Definition at line 358 of file [grcc.h](#).

### 3.129.4.3 opt

```
Options* SProcess::opt
```

Definition at line 359 of file [grcc.h](#).

### 3.129.4.4 pnclass

```
PNodeClass* SProcess::pnclass
```

Definition at line 360 of file [grcc.h](#).



#### 3.129.4.5 astack

`AStack* SProcess::astack`

Definition at line 361 of file [grcc.h](#).

#### 3.129.4.6 mgraph

`MGraph* SProcess::mgraph`

Definition at line 363 of file [grcc.h](#).

#### 3.129.4.7 egraph

`EGraph* SProcess::egraph`

Definition at line 364 of file [grcc.h](#).

#### 3.129.4.8 agraph

`Assign* SProcess::agraph`

Definition at line 365 of file [grcc.h](#).

#### 3.129.4.9 cl2nd

`int* SProcess::cl2nd`

Definition at line 367 of file [grcc.h](#).

#### 3.129.4.10 nd2cl

`int* SProcess::nd2cl`

Definition at line 369 of file [grcc.h](#).

#### 3.129.4.11 nclass

`int SProcess::nclass`

Definition at line 370 of file [grcc.h](#).

#### 3.129.4.12 ninitl

`int SProcess::ninitl`

Definition at line 372 of file [grcc.h](#).

**3.129.4.13 nfinal**

```
int SProcess::nfinal
```

Definition at line 373 of file [grcc.h](#).

**3.129.4.14 nvert**

```
int SProcess::nvert
```

Definition at line 374 of file [grcc.h](#).

**3.129.4.15 clist**

```
int SProcess::clist[GRCC_MAXNCPLG]
```

Definition at line 376 of file [grcc.h](#).

**3.129.4.16 id**

```
int SProcess::id
```

Definition at line 377 of file [grcc.h](#).

**3.129.4.17 loop**

```
int SProcess::loop
```

Definition at line 378 of file [grcc.h](#).

**3.129.4.18 nNodes**

```
int SProcess::nNodes
```

Definition at line 379 of file [grcc.h](#).

**3.129.4.19 nEdges**

```
int SProcess::nEdges
```

Definition at line 380 of file [grcc.h](#).

**3.129.4.20 nExtern**

```
int SProcess::nExtern
```

Definition at line 381 of file [grcc.h](#).

#### 3.129.4.21 ncouple

```
int SProcess::ncouple
```

Definition at line 382 of file [grcc.h](#).

#### 3.129.4.22 tCouple

```
int SProcess::tCouple
```

Definition at line 383 of file [grcc.h](#).

#### 3.129.4.23 DUMMYPADDING

```
int SProcess::DUMMYPADDING
```

Definition at line 385 of file [grcc.h](#).

#### 3.129.4.24 mgrcount

```
BigInt SProcess::mgrcount
```

Definition at line 387 of file [grcc.h](#).

#### 3.129.4.25 agrcount

```
BigInt SProcess::agrcount
```

Definition at line 388 of file [grcc.h](#).

#### 3.129.4.26 extperm

```
BigInt SProcess::extperm
```

Definition at line 389 of file [grcc.h](#).

#### 3.129.4.27 nMGraphs

```
BigInt SProcess::nMGraphs
```

Definition at line 392 of file [grcc.h](#).

#### 3.129.4.28 nMOPI

```
BigInt SProcess::nMOPI
```

Definition at line 393 of file [grcc.h](#).

#### 3.129.4.29 wMGraphs

`Fraction SProcess::wMGraphs`

Definition at line 394 of file [grcc.h](#).

#### 3.129.4.30 wMOPI

`Fraction SProcess::wMOPI`

Definition at line 395 of file [grcc.h](#).

#### 3.129.4.31 nAGraphs

`BigInt SProcess::nAGraphs`

Definition at line 397 of file [grcc.h](#).

#### 3.129.4.32 nAOPI

`BigInt SProcess::nAOPI`

Definition at line 398 of file [grcc.h](#).

#### 3.129.4.33 wAGraphs

`Fraction SProcess::wAGraphs`

Definition at line 399 of file [grcc.h](#).

#### 3.129.4.34 wAOPI

`Fraction SProcess::wAOPI`

Definition at line 400 of file [grcc.h](#).

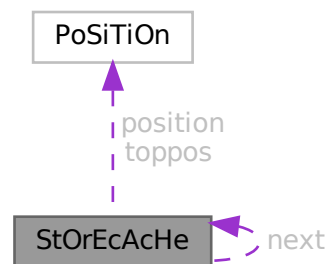
The documentation for this class was generated from the following files:

- [grcc.h](#)
- [grcc.cc](#)

## 3.130 StOrEcAcHe Struct Reference

```
#include <structs.h>
```

Collaboration diagram for StOrEcAcHe:



### Public Member Functions

- **PADPOSITION** (1, 0, 0, 2, 0)

### Data Fields

- **POSITION** position
- **POSITION** toppos
- struct **StOrEcAcHe** \* next
- **WORD** buffer [2]

### 3.130.1 Detailed Description

The struct of type STORECACHE is used by a caching system for reading terms from stored expressions. Each thread should have its own system of caches.

Definition at line 1065 of file [structs.h](#).

### 3.130.2 Field Documentation

#### 3.130.2.1 position

**POSITION** StOrEcAcHe::position

Definition at line 1066 of file [structs.h](#).

### 3.130.2.2 toppos

`POSITION StOrEcAcHe::toppos`

Definition at line 1067 of file [structs.h](#).

### 3.130.2.3 next

`struct StOrEcAcHe* StOrEcAcHe::next`

Definition at line 1068 of file [structs.h](#).

### 3.130.2.4 buffer

`WORD StOrEcAcHe::buffer[2]`

Definition at line 1069 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.131 STOREHEADER Struct Reference

```
#include <structs.h>
```

### Data Fields

- UBYTE [headermark](#) [8]
- UBYTE [lenWORD](#)
- UBYTE [lenLONG](#)
- UBYTE [lenPOS](#)
- UBYTE [lenPOINTER](#)
- UBYTE [endianness](#) [16]
- UBYTE [sSym](#)
- UBYTE [sInd](#)
- UBYTE [sVec](#)
- UBYTE [sFun](#)
- UBYTE [maxpower](#) [16]
- UBYTE [wildoffset](#) [16]
- UBYTE [revision](#)
- UBYTE [reserved](#) [512-8-4-16-4-16-16-1]

### 3.131.1 Detailed Description

Defines the structure of the file header for store-files and save-files.

The first 8 bytes serve as a unique mark to identity save-files that contain such a header. Older versions of FORM don't have this header and will write the POSITION of the next file index (struct [FiLeInDeX](#)) here, which is always different from this pattern.

It is always 512 bytes long.

Definition at line 75 of file [structs.h](#).

### 3.131.2 Field Documentation

#### 3.131.2.1 headermark

```
UBYTE STOREHEADER::headermark[8]
```

Pattern for header identification. Old versions of FORM have a maximum sizeof(POSITION) of 8

Definition at line 76 of file [structs.h](#).

#### 3.131.2.2 lenWORD

```
UBYTE STOREHEADER::lenWORD
```

Number of bytes for WORD

Definition at line 78 of file [structs.h](#).

Referenced by [WriteStoreHeader\(\)](#).

#### 3.131.2.3 lenLONG

```
UBYTE STOREHEADER::lenLONG
```

Number of bytes for LONG

Definition at line 79 of file [structs.h](#).

Referenced by [WriteStoreHeader\(\)](#).

#### 3.131.2.4 lenPOS

```
UBYTE STOREHEADER::lenPOS
```

Number of bytes for POSITION

Definition at line 80 of file [structs.h](#).

Referenced by [WriteStoreHeader\(\)](#).

### 3.131.2.5 lenPOINTER

```
UBYTE STOREHEADER::lenPOINTER
```

Number of bytes for void \*

Definition at line 81 of file [structs.h](#).

Referenced by [WriteStoreHeader\(\)](#).

### 3.131.2.6 endianness

```
UBYTE STOREHEADER::endianness[16]
```

Used to determine endianness, sizeof(int) should be <= 16

Definition at line 82 of file [structs.h](#).

Referenced by [WriteStoreHeader\(\)](#).

### 3.131.2.7 sSym

```
UBYTE STOREHEADER::sSym
```

sizeof(struct SyMbOl)

Definition at line 83 of file [structs.h](#).

Referenced by [WriteStoreHeader\(\)](#).

### 3.131.2.8 sInd

```
UBYTE STOREHEADER::sInd
```

sizeof(struct InDeX)

Definition at line 84 of file [structs.h](#).

Referenced by [WriteStoreHeader\(\)](#).

### 3.131.2.9 sVec

```
UBYTE STOREHEADER::sVec
```

sizeof(struct VeCtOr)

Definition at line 85 of file [structs.h](#).

Referenced by [WriteStoreHeader\(\)](#).



### 3.131.2.10 sFun

```
UBYTE STOREHEADER::sFun
```

sizeof(struct FuNcTiOn)

Definition at line 86 of file [structs.h](#).

Referenced by [WriteStoreHeader\(\)](#).

### 3.131.2.11 maxpower

```
UBYTE STOREHEADER::maxpower[16]
```

Maximum power, see #MAXPOWER

Definition at line 87 of file [structs.h](#).

Referenced by [WriteStoreHeader\(\)](#).

### 3.131.2.12 wildoffset

```
UBYTE STOREHEADER::wildoffset[16]
```

#WILDOFFSET macro

Definition at line 88 of file [structs.h](#).

Referenced by [WriteStoreHeader\(\)](#).

### 3.131.2.13 revision

```
UBYTE STOREHEADER::revision
```

Revision number of save-file system

Definition at line 89 of file [structs.h](#).

### 3.131.2.14 reserved

```
UBYTE STOREHEADER::reserved[512-8-4-16-4-16-16-1]
```

Padding to 512 bytes

Definition at line 90 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.132 Stream Struct Reference

```
#include <structs.h>
```

### Public Member Functions

- **PADPOSITION** (6, 3, 9, 0, 4)

### Data Fields

- off\_t [fileposition](#)
- off\_t [linenumber](#)
- off\_t [prevline](#)
- UBYTE \* [buffer](#)
- UBYTE \* [pointer](#)
- UBYTE \* [top](#)
- UBYTE \* [FoldName](#)
- UBYTE \* [name](#)
- UBYTE \* [pname](#)
- LONG [buffersize](#)
- LONG [bufferposition](#)
- LONG [inbuffer](#)
- int [previous](#)
- int [handle](#)
- int [type](#)
- int [prevars](#)
- int [previousNoShowInput](#)
- int [eqnum](#)
- int [afterwards](#)
- int [olddelay](#)
- int [oldnoshowinput](#)
- UBYTE [isnextchar](#)
- UBYTE [nextchar](#) [2]
- UBYTE [reserved](#)

### 3.132.1 Detailed Description

Input is read from 'streams' which are represented by objects of type STREAM. A stream can be a file, a do-loop, a procedure, the string value of a preprocessor variable ..... When a new stream is opened we have to keep information about where to fall back in the parent stream to allow this to happen even in the middle of reading names etc as would be the case with a`i'b

Definition at line [736](#) of file [structs.h](#).

### 3.132.2 Field Documentation

#### 3.132.2.1 fileposition

```
off_t Stream::fileposition
```

Definition at line [737](#) of file [structs.h](#).

### 3.132.2.2 `linenumber`

```
off_t Stream::linenumber
```

Definition at line [738](#) of file [structs.h](#).

### 3.132.2.3 `prevline`

```
off_t Stream::prevline
```

Definition at line [739](#) of file [structs.h](#).

### 3.132.2.4 `buffer`

```
UBYTE* Stream::buffer
```

[D] Size in buffersize

Definition at line [740](#) of file [structs.h](#).

### 3.132.2.5 `pointer`

```
UBYTE* Stream::pointer
```

pointer into buffer memory

Definition at line [741](#) of file [structs.h](#).

### 3.132.2.6 `top`

```
UBYTE* Stream::top
```

pointer into buffer memory

Definition at line [742](#) of file [structs.h](#).

### 3.132.2.7 `FoldName`

```
UBYTE* Stream::FoldName
```

[D]

Definition at line [743](#) of file [structs.h](#).

### 3.132.2.8 name

```
UBYTE* Stream::name
```

[D]

Definition at line 744 of file [structs.h](#).

### 3.132.2.9 pname

```
UBYTE* Stream::pname
```

for DOLLARSTREAM and PREVARSTREAM it points always to name, else it is undefined

Definition at line 745 of file [structs.h](#).

### 3.132.2.10 buffersize

```
LONG Stream::buffersize
```

Definition at line 747 of file [structs.h](#).

### 3.132.2.11 bufferposition

```
LONG Stream::bufferposition
```

Definition at line 748 of file [structs.h](#).

### 3.132.2.12 inbuffer

```
LONG Stream::inbuffer
```

Definition at line 749 of file [structs.h](#).

### 3.132.2.13 previous

```
int Stream::previous
```

Definition at line 750 of file [structs.h](#).

### 3.132.2.14 handle

```
int Stream::handle
```

Definition at line 751 of file [structs.h](#).

**3.132.2.15 type**

```
int Stream::type
```

Definition at line 752 of file [structs.h](#).

**3.132.2.16 prevars**

```
int Stream::prevars
```

Definition at line 753 of file [structs.h](#).

**3.132.2.17 previousNoShowInput**

```
int Stream::previousNoShowInput
```

Definition at line 754 of file [structs.h](#).

**3.132.2.18 eqnum**

```
int Stream::eqnum
```

Definition at line 755 of file [structs.h](#).

**3.132.2.19 afterwards**

```
int Stream::afterwards
```

Definition at line 756 of file [structs.h](#).

**3.132.2.20 olddelay**

```
int Stream::olddelay
```

Definition at line 757 of file [structs.h](#).

**3.132.2.21 oldnoshowinput**

```
int Stream::oldnoshowinput
```

Definition at line 758 of file [structs.h](#).

**3.132.2.22 isnextchar**

```
UBYTE Stream::isnextchar
```

Definition at line 759 of file [structs.h](#).

### 3.132.2.23 nextchar

```
UBYTE Stream::nextchar[2]
```

Definition at line 760 of file [structs.h](#).

### 3.132.2.24 reserved

```
UBYTE Stream::reserved
```

Definition at line 761 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.133 SuBbUf Struct Reference

### Data Fields

- WORD [subexpnum](#)
- WORD [buffernum](#)

### 3.133.1 Detailed Description

Definition at line 257 of file [compiler.c](#).

### 3.133.2 Field Documentation

#### 3.133.2.1 subexpnum

```
WORD SuBbUf::subexpnum
```

Definition at line 258 of file [compiler.c](#).

#### 3.133.2.2 buffernum

```
WORD SuBbUf::buffernum
```

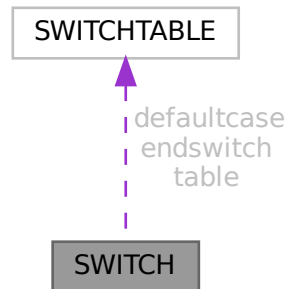
Definition at line 259 of file [compiler.c](#).

The documentation for this struct was generated from the following file:

- [compiler.c](#)

## 3.134 SWITCH Struct Reference

Collaboration diagram for SWITCH:



### Data Fields

- [SWITCHTABLE \\* table](#)
- [SWITCHTABLE defaultcase](#)
- [SWITCHTABLE endswitch](#)
- WORD [typetable](#)
- WORD [maxcase](#)
- WORD [mincase](#)
- WORD [numcases](#)
- WORD [tablesize](#)
- WORD [caseoffset](#)
- WORD [iflevel](#)
- WORD [whilelevel](#)
- WORD [nestingsum](#)
- WORD [padding](#)

### 3.134.1 Detailed Description

Definition at line [1380](#) of file [structs.h](#).

### 3.134.2 Field Documentation

#### 3.134.2.1 table

[SWITCHTABLE\\*](#) [SWITCH::table](#)

Definition at line [1381](#) of file [structs.h](#).

### 3.134.2.2 defaultcase

`SWITCHTABLE SWITCH::defaultcase`

Definition at line 1382 of file [structs.h](#).

### 3.134.2.3 endswitch

`SWITCHTABLE SWITCH::endswitch`

Definition at line 1383 of file [structs.h](#).

### 3.134.2.4 typetable

`WORD SWITCH::typetable`

Definition at line 1384 of file [structs.h](#).

### 3.134.2.5 maxcase

`WORD SWITCH::maxcase`

Definition at line 1385 of file [structs.h](#).

### 3.134.2.6 mincase

`WORD SWITCH::mincase`

Definition at line 1386 of file [structs.h](#).

### 3.134.2.7 numcases

`WORD SWITCH::numcases`

Definition at line 1387 of file [structs.h](#).

### 3.134.2.8 tablesize

`WORD SWITCH::tablesize`

Definition at line 1388 of file [structs.h](#).

### 3.134.2.9 caseoffset

`WORD SWITCH::caseoffset`

Definition at line 1389 of file [structs.h](#).



#### 3.134.2.10 iflevel

WORD SWITCH::iflevel

Definition at line 1390 of file [structs.h](#).

#### 3.134.2.11 whilelevel

WORD SWITCH::whilelevel

Definition at line 1391 of file [structs.h](#).

#### 3.134.2.12 nestingsum

WORD SWITCH::nestingsum

Definition at line 1392 of file [structs.h](#).

#### 3.134.2.13 padding

WORD SWITCH::padding

Definition at line 1393 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.135 SWITCHTABLE Struct Reference

### Data Fields

- WORD [ncase](#)
- WORD [value](#)
- WORD [compbuffer](#)

### 3.135.1 Detailed Description

Definition at line 1374 of file [structs.h](#).

### 3.135.2 Field Documentation

#### 3.135.2.1 ncase

WORD SWITCHTABLE::ncase

Definition at line 1375 of file [structs.h](#).

### 3.135.2.2 value

WORD SWITCHTABLE::value

Definition at line 1376 of file [structs.h](#).

### 3.135.2.3 compbuffer

WORD SWITCHTABLE::compbuffer

Definition at line 1377 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.136 SyMbOl Struct Reference

### Data Fields

- LONG [name](#)
- WORD [minpower](#)
- WORD [maxpower](#)
- WORD [complex](#)
- WORD [number](#)
- WORD [flags](#)
- WORD [node](#)
- WORD [namesize](#)
- WORD [dimension](#)

### 3.136.1 Detailed Description

Definition at line 428 of file [structs.h](#).

### 3.136.2 Field Documentation

#### 3.136.2.1 name

LONG SyMbOl::name

Definition at line 429 of file [structs.h](#).

#### 3.136.2.2 minpower

WORD SyMbOl::minpower

Definition at line 430 of file [structs.h](#).

### 3.136.2.3 maxpower

WORD SyMbOl::maxpower

Definition at line 431 of file [structs.h](#).

### 3.136.2.4 complex

WORD SyMbOl::complex

Definition at line 432 of file [structs.h](#).

### 3.136.2.5 number

WORD SyMbOl::number

Definition at line 433 of file [structs.h](#).

### 3.136.2.6 flags

WORD SyMbOl::flags

Definition at line 434 of file [structs.h](#).

### 3.136.2.7 node

WORD SyMbOl::node

Definition at line 435 of file [structs.h](#).

### 3.136.2.8 namesize

WORD SyMbOl::namesize

Definition at line 436 of file [structs.h](#).

### 3.136.2.9 dimension

WORD SyMbOl::dimension

Definition at line 437 of file [structs.h](#).

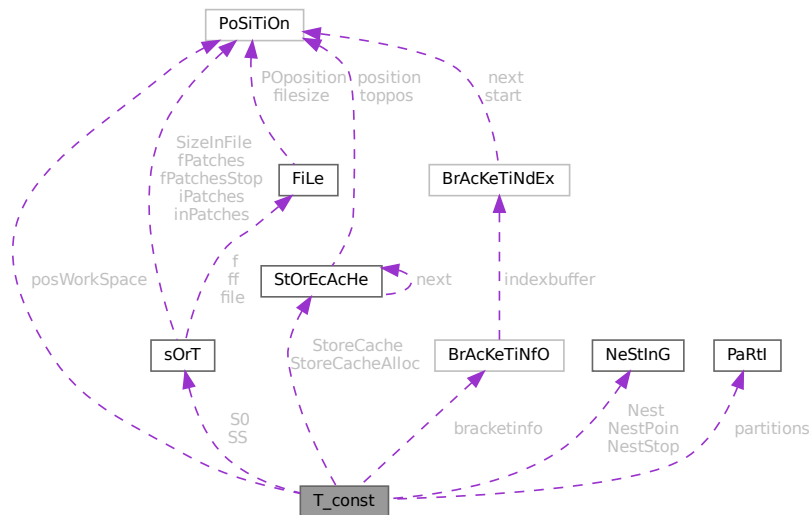
The documentation for this struct was generated from the following file:

- [structs.h](#)

### 3.137 T\_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for T\_const:



#### Data Fields

- [SORTING](#) \* [S0](#)
- [SORTING](#) \* [SS](#)
- [NESTING](#) [Nest](#)
- [NESTING](#) [NestStop](#)
- [NESTING](#) [NestPoin](#)
- [WORD](#) \* [BrackBuf](#)
- [STORECACHE](#) [StoreCache](#)
- [STORECACHE](#) [StoreCacheAlloc](#)
- [WORD](#) \*\* [pWorkspace](#)
- [LONG](#) \* [lWorkspace](#)
- [POSITION](#) \* [posWorkspace](#)
- [WORD](#) \* [Workspace](#)
- [WORD](#) \* [WorkTop](#)
- [WORD](#) \* [WorkPointer](#)
- [int](#) \* [RepCount](#)
- [int](#) \* [RepTop](#)
- [WORD](#) \* [WildArgTaken](#)
- [UWORD](#) \* [factorials](#)
- [WORD](#) \* [small\\_power\\_n](#)
- [UWORD](#) \*\* [small\\_power](#)
- [UWORD](#) \* [bernoullis](#)
- [WORD](#) \* [primelist](#)
- [LONG](#) \* [pfac](#)
- [LONG](#) \* [pBer](#)
- [WORD](#) \* [TMaddr](#)

- WORD \* WildMask
- WORD \* previousEfactor
- WORD \*\* TermMemHeap
- UWORD \*\* NumberMemHeap
- UWORD \*\* CacheNumberMemHeap
- BRACKETINFO \* bracketinfo
- WORD \*\* ListPoly
- WORD \* ListSymbols
- UWORD \* NumMem
- WORD \* TopologiesTerm
- WORD \* TopologiesStart
- PARTI partitions
- LONG sBer
- LONG pWorkPointer
- LONG IWorkPointer
- LONG posWorkPointer
- LONG lnNumMem
- int sfact
- int mfac
- int ebufnum
- int fbufnum
- int allbufnum
- int aebufnum
- int idallflag
- int idallnum
- int idallmaxnum
- int WildcardBufferSize
- int TermMemMax
- int TermMemTop
- int NumberMemMax
- int NumberMemTop
- int CacheNumberMemMax
- int CacheNumberMemTop
- int bracketindexflag
- int optentimes
- int ListSymbolsSize
- int NumListSymbols
- int numpoly
- int LeaveNegative
- int TrimPower
- WORD small\_power\_maxx
- WORD small\_power\_maxn
- WORD dummysubexp [SUBEXPSIZE+4]
- WORD comsym [8]
- WORD comnum [4]
- WORD comfun [FUNHEAD+4]
- WORD comind [7]
- WORD MinVecArg [7+ARGHEAD]
- WORD FunArg [4+ARGHEAD+FUNHEAD]
- WORD locwildvalue [SUBEXPSIZE]
- WORD mulpat [SUBEXPSIZE+5]
- WORD proexp [SUBEXPSIZE+5]
- WORD TMout [40]
- WORD TMbuff
- WORD TMdolfac

- WORD [nfac](#)
- WORD [nBer](#)
- WORD [mBer](#)
- WORD [PolyAct](#)
- WORD [RecFlag](#)
- WORD [inprimelist](#)
- WORD [sizeprimelist](#)
- WORD [fromindex](#)
- WORD [setinterntopo](#)
- WORD [setexterntopo](#)
- WORD [TopologiesLevel](#)
- WORD [TopologiesOptions](#) [2]

### 3.137.1 Detailed Description

The [T\\_const](#) struct is part of the global data and resides either in the ALLGLOBALS struct A, or the ALLPRIVATES struct B (TFORM) under the name T We see it used with the macro AT as in AT.WorkPointer It has variables that are private to each thread, most of which have to be defined at startup.

Definition at line [2121](#) of file [structs.h](#).

### 3.137.2 Field Documentation

#### 3.137.2.1 S0

```
SORTING* T_const::S0
```

Definition at line [2125](#) of file [structs.h](#).

#### 3.137.2.2 SS

```
SORTING* T_const::SS
```

Definition at line [2126](#) of file [structs.h](#).

#### 3.137.2.3 Nest

```
NESTING T_const::Nest
```

Definition at line [2127](#) of file [structs.h](#).

#### 3.137.2.4 NestStop

```
NESTING T_const::NestStop
```

Definition at line [2128](#) of file [structs.h](#).

### 3.137.2.5 NestPoin

`NESTING T_const::NestPoin`

Definition at line 2129 of file [structs.h](#).

### 3.137.2.6 BrackBuf

`WORD* T_const::BrackBuf`

Definition at line 2130 of file [structs.h](#).

### 3.137.2.7 StoreCache

`STORECACHE T_const::StoreCache`

Definition at line 2131 of file [structs.h](#).

### 3.137.2.8 StoreCacheAlloc

`STORECACHE T_const::StoreCacheAlloc`

Definition at line 2132 of file [structs.h](#).

### 3.137.2.9 pWorkSpace

`WORD** T_const::pWorkSpace`

Definition at line 2133 of file [structs.h](#).

### 3.137.2.10 lWorkSpace

`LONG* T_const::lWorkSpace`

Definition at line 2134 of file [structs.h](#).

### 3.137.2.11 posWorkSpace

`POSITION* T_const::posWorkSpace`

Definition at line 2135 of file [structs.h](#).

### 3.137.2.12 WorkSpace

`WORD* T_const::WorkSpace`

Definition at line 2136 of file [structs.h](#).

### 3.137.2.13 WorkTop

WORD\* T\_const::WorkTop

Definition at line 2137 of file [structs.h](#).

### 3.137.2.14 WorkPointer

WORD\* T\_const::WorkPointer

Definition at line 2138 of file [structs.h](#).

### 3.137.2.15 RepCount

int\* T\_const::RepCount

Definition at line 2139 of file [structs.h](#).

### 3.137.2.16 RepTop

int\* T\_const::RepTop

Definition at line 2140 of file [structs.h](#).

### 3.137.2.17 WildArgTaken

WORD\* T\_const::WildArgTaken

Definition at line 2141 of file [structs.h](#).

### 3.137.2.18 factorials

UWORD\* T\_const::factorials

Definition at line 2142 of file [structs.h](#).

### 3.137.2.19 small\_power\_n

WORD\* T\_const::small\_power\_n

Definition at line 2143 of file [structs.h](#).

### 3.137.2.20 small\_power

UWORD\*\* T\_const::small\_power

Definition at line 2144 of file [structs.h](#).



#### 3.137.2.21 bernoullis

UWORD\* T\_const::bernoullis

Definition at line 2145 of file [structs.h](#).

#### 3.137.2.22 primelist

WORD\* T\_const::primelist

Definition at line 2146 of file [structs.h](#).

#### 3.137.2.23 pfac

LONG\* T\_const::pfac

Definition at line 2147 of file [structs.h](#).

#### 3.137.2.24 pBer

LONG\* T\_const::pBer

Definition at line 2148 of file [structs.h](#).

#### 3.137.2.25 TMaddr

WORD\* T\_const::TMaddr

Definition at line 2149 of file [structs.h](#).

#### 3.137.2.26 WildMask

WORD\* T\_const::WildMask

Definition at line 2150 of file [structs.h](#).

#### 3.137.2.27 previousEfactor

WORD\* T\_const::previousEfactor

Definition at line 2151 of file [structs.h](#).

#### 3.137.2.28 TermMemHeap

WORD\*\* T\_const::TermMemHeap

Definition at line 2152 of file [structs.h](#).

### 3.137.2.29 NumberMemHeap

UWORD\*\* T\_const::NumberMemHeap

Definition at line 2153 of file [structs.h](#).

### 3.137.2.30 CacheNumberMemHeap

UWORD\*\* T\_const::CacheNumberMemHeap

Definition at line 2154 of file [structs.h](#).

### 3.137.2.31 bracketinfo

BRACKETINFO\* T\_const::bracketinfo

Definition at line 2155 of file [structs.h](#).

### 3.137.2.32 ListPoly

WORD\*\* T\_const::ListPoly

Definition at line 2156 of file [structs.h](#).

### 3.137.2.33 ListSymbols

WORD\* T\_const::ListSymbols

Definition at line 2157 of file [structs.h](#).

### 3.137.2.34 NumMem

UWORD\* T\_const::NumMem

Definition at line 2158 of file [structs.h](#).

### 3.137.2.35 TopologiesTerm

WORD\* T\_const::TopologiesTerm

Definition at line 2159 of file [structs.h](#).

### 3.137.2.36 TopologiesStart

WORD\* T\_const::TopologiesStart

Definition at line 2160 of file [structs.h](#).

### 3.137.2.37 partitions

`PARTI T_const::partitions`

Definition at line 2169 of file [structs.h](#).

### 3.137.2.38 sBer

`LONG T_const::sBer`

Definition at line 2170 of file [structs.h](#).

### 3.137.2.39 pWorkPointer

`LONG T_const::pWorkPointer`

Definition at line 2171 of file [structs.h](#).

### 3.137.2.40 lWorkPointer

`LONG T_const::lWorkPointer`

Definition at line 2172 of file [structs.h](#).

### 3.137.2.41 posWorkPointer

`LONG T_const::posWorkPointer`

Definition at line 2173 of file [structs.h](#).

### 3.137.2.42 InNumMem

`LONG T_const::InNumMem`

Definition at line 2174 of file [structs.h](#).

### 3.137.2.43 sfact

`int T_const::sfact`

Definition at line 2175 of file [structs.h](#).

### 3.137.2.44 mfac

`int T_const::mfac`

Definition at line 2176 of file [structs.h](#).

**3.137.2.45 ebufnum**

```
int T_const::ebufnum
```

Definition at line 2177 of file [structs.h](#).

**3.137.2.46 fbufnum**

```
int T_const::fbufnum
```

Definition at line 2178 of file [structs.h](#).

**3.137.2.47 allbufnum**

```
int T_const::allbufnum
```

Definition at line 2179 of file [structs.h](#).

**3.137.2.48 aebufnum**

```
int T_const::aebufnum
```

Definition at line 2180 of file [structs.h](#).

**3.137.2.49 idallflag**

```
int T_const::idallflag
```

Definition at line 2181 of file [structs.h](#).

**3.137.2.50 idallnum**

```
int T_const::idallnum
```

Definition at line 2182 of file [structs.h](#).

**3.137.2.51 idallmaxnum**

```
int T_const::idallmaxnum
```

Definition at line 2183 of file [structs.h](#).

**3.137.2.52 WildcardBufferSize**

```
int T_const::WildcardBufferSize
```

Definition at line 2184 of file [structs.h](#).

**3.137.2.53 TermMemMax**

```
int T_const::TermMemMax
```

Definition at line 2193 of file [structs.h](#).

**3.137.2.54 TermMemTop**

```
int T_const::TermMemTop
```

Definition at line 2194 of file [structs.h](#).

**3.137.2.55 NumberMemMax**

```
int T_const::NumberMemMax
```

Definition at line 2195 of file [structs.h](#).

**3.137.2.56 NumberMemTop**

```
int T_const::NumberMemTop
```

Definition at line 2196 of file [structs.h](#).

**3.137.2.57 CacheNumberMemMax**

```
int T_const::CacheNumberMemMax
```

Definition at line 2197 of file [structs.h](#).

**3.137.2.58 CacheNumberMemTop**

```
int T_const::CacheNumberMemTop
```

Definition at line 2198 of file [structs.h](#).

**3.137.2.59 bracketindexflag**

```
int T_const::bracketindexflag
```

Definition at line 2199 of file [structs.h](#).

**3.137.2.60 optentimes**

```
int T_const::optentimes
```

Definition at line 2200 of file [structs.h](#).

**3.137.2.61 ListSymbolsSize**

```
int T_const::ListSymbolsSize
```

Definition at line [2201](#) of file [structs.h](#).

**3.137.2.62 NumListSymbols**

```
int T_const::NumListSymbols
```

Definition at line [2202](#) of file [structs.h](#).

**3.137.2.63 numpoly**

```
int T_const::numpoly
```

Definition at line [2203](#) of file [structs.h](#).

**3.137.2.64 LeaveNegative**

```
int T_const::LeaveNegative
```

Definition at line [2204](#) of file [structs.h](#).

**3.137.2.65 TrimPower**

```
int T_const::TrimPower
```

Definition at line [2205](#) of file [structs.h](#).

**3.137.2.66 small\_power\_maxx**

```
WORD T_const::small_power_maxx
```

Definition at line [2206](#) of file [structs.h](#).

**3.137.2.67 small\_power\_maxn**

```
WORD T_const::small_power_maxn
```

Definition at line [2207](#) of file [structs.h](#).

**3.137.2.68 dummysubexp**

```
WORD T_const::dummysubexp[SUBEXPSIZE+4]
```

Definition at line [2208](#) of file [structs.h](#).

### 3.137.2.69 comsym

```
WORD T_const::comsym[8]
```

Definition at line 2209 of file [structs.h](#).

### 3.137.2.70 comnum

```
WORD T_const::comnum[4]
```

Definition at line 2210 of file [structs.h](#).

### 3.137.2.71 comfun

```
WORD T_const::comfun[FUNHEAD+4]
```

Definition at line 2211 of file [structs.h](#).

### 3.137.2.72 comind

```
WORD T_const::comind[7]
```

Definition at line 2213 of file [structs.h](#).

### 3.137.2.73 MinVecArg

```
WORD T_const::MinVecArg[7+ARGHEAD]
```

Definition at line 2214 of file [structs.h](#).

### 3.137.2.74 FunArg

```
WORD T_const::FunArg[4+ARGHEAD+FUNHEAD]
```

Definition at line 2215 of file [structs.h](#).

### 3.137.2.75 locwildvalue

```
WORD T_const::locwildvalue[SUBEXPSIZE]
```

Definition at line 2216 of file [structs.h](#).

### 3.137.2.76 mulpat

```
WORD T_const::mulpat[SUBEXPSIZE+5]
```

Definition at line 2217 of file [structs.h](#).

### 3.137.2.77 proexp

```
WORD T_const::proexp[SUBEXPSIZE+5]
```

Definition at line 2219 of file [structs.h](#).

### 3.137.2.78 TMout

```
WORD T_const::TMout[40]
```

Definition at line 2220 of file [structs.h](#).

### 3.137.2.79 TMbuff

```
WORD T_const::TMbuff
```

Definition at line 2221 of file [structs.h](#).

### 3.137.2.80 TMdolfac

```
WORD T_const::TMdolfac
```

Definition at line 2222 of file [structs.h](#).

### 3.137.2.81 nfac

```
WORD T_const::nfac
```

Definition at line 2223 of file [structs.h](#).

### 3.137.2.82 nBer

```
WORD T_const::nBer
```

Definition at line 2224 of file [structs.h](#).

### 3.137.2.83 mBer

```
WORD T_const::mBer
```

Definition at line 2225 of file [structs.h](#).

### 3.137.2.84 PolyAct

```
WORD T_const::PolyAct
```

Definition at line 2226 of file [structs.h](#).



**3.137.2.85 RecFlag**

WORD T\_const::RecFlag

Definition at line 2227 of file [structs.h](#).

**3.137.2.86 inprimelist**

WORD T\_const::inprimelist

Definition at line 2228 of file [structs.h](#).

**3.137.2.87 sizeprimelist**

WORD T\_const::sizeprimelist

Definition at line 2229 of file [structs.h](#).

**3.137.2.88 fromindex**

WORD T\_const::fromindex

Definition at line 2230 of file [structs.h](#).

**3.137.2.89 setinterntopo**

WORD T\_const::setinterntopo

Definition at line 2231 of file [structs.h](#).

**3.137.2.90 setexterntopo**

WORD T\_const::setexterntopo

Definition at line 2232 of file [structs.h](#).

**3.137.2.91 TopologiesLevel**

WORD T\_const::TopologiesLevel

Definition at line 2233 of file [structs.h](#).

### 3.137.2.92 TopologiesOptions

WORD T\_const::TopologiesOptions[2]

Definition at line 2234 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.138 T\_EEdge Class Reference

### Data Fields

- int [nodes](#) [2]
- int [ext](#)
- int [momn](#)
- char [momc](#) [2]
- char [padding](#) [6]

### 3.138.1 Detailed Description

Definition at line 24 of file [gentopo.h](#).

### 3.138.2 Field Documentation

#### 3.138.2.1 nodes

int T\_EEdge::nodes[2]

Definition at line 26 of file [gentopo.h](#).

#### 3.138.2.2 ext

int T\_EEdge::ext

Definition at line 27 of file [gentopo.h](#).

#### 3.138.2.3 momn

int T\_EEdge::momn

Definition at line 28 of file [gentopo.h](#).

### 3.138.2.4 momc

```
char T_EEdge::momc[2]
```

Definition at line 29 of file [gentopo.h](#).

### 3.138.2.5 padding

```
char T_EEdge::padding[6]
```

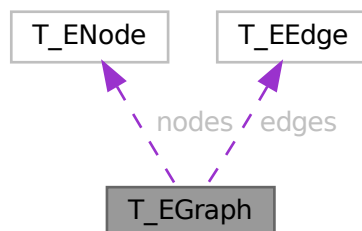
Definition at line 31 of file [gentopo.h](#).

The documentation for this class was generated from the following file:

- [gentopo.h](#)

## 3.139 T\_EGraph Class Reference

Collaboration diagram for T\_EGraph:



### Public Member Functions

- [T\\_EGraph](#) (int nnodes, int nedges, int mxdeg)
- void [print](#) (void)
- void [init](#) (int pid, long gid, int \*\*adjmat, int sopi, BigInt nsym, BigInt esym)
- void [setExtern](#) (int nd, int val)
- void [endSetExtern](#) (void)

## Data Fields

- long [gld](#)
- int [pld](#)
- int [nNodes](#)
- int [nEdges](#)
- int [maxdeg](#)
- int [nExtern](#)
- int [opi](#)
- BigInt [nsym](#)
- BigInt [esym](#)
- [T\\_ENode](#) \* [nodes](#)
- [T\\_EEdge](#) \* [edges](#)

### 3.139.1 Detailed Description

Definition at line [34](#) of file [gentopo.h](#).

### 3.139.2 Constructor & Destructor Documentation

#### 3.139.2.1 T\_EGraph()

```
T_EGraph::T_EGraph (
    int nnodes,
    int nedges,
    int mxdeg )
```

Definition at line [88](#) of file [gentopo.cc](#).

#### 3.139.2.2 ~T\_EGraph()

```
T_EGraph::~T_EGraph ( )
```

Definition at line [109](#) of file [gentopo.cc](#).

### 3.139.3 Member Function Documentation

#### 3.139.3.1 print()

```
void T_EGraph::print (
    void )
```

Definition at line [200](#) of file [gentopo.cc](#).

### 3.139.3.2 init()

```
void T_EGraph::init (
    int pid,
    long gid,
    int ** adjmat,
    int sopi,
    BigInt nsym,
    BigInt esym )
```

Definition at line 124 of file [gentopo.cc](#).

### 3.139.3.3 setExtern()

```
void T_EGraph::setExtern (
    int nd,
    int val )
```

Definition at line 181 of file [gentopo.cc](#).

### 3.139.3.4 endSetExtern()

```
void T_EGraph::endSetExtern (
    void )
```

Definition at line 187 of file [gentopo.cc](#).

## 3.139.4 Field Documentation

### 3.139.4.1 gId

```
long T_EGraph::gId
```

Definition at line 36 of file [gentopo.h](#).

### 3.139.4.2 pId

```
int T_EGraph::pId
```

Definition at line 37 of file [gentopo.h](#).

### 3.139.4.3 nNodes

```
int T_EGraph::nNodes
```

Definition at line 38 of file [gentopo.h](#).

**3.139.4.4 nEdges**

```
int T_EGraph::nEdges
```

Definition at line 39 of file [gentopo.h](#).

**3.139.4.5 maxdeg**

```
int T_EGraph::maxdeg
```

Definition at line 40 of file [gentopo.h](#).

**3.139.4.6 nExtern**

```
int T_EGraph::nExtern
```

Definition at line 41 of file [gentopo.h](#).

**3.139.4.7 opi**

```
int T_EGraph::opi
```

Definition at line 43 of file [gentopo.h](#).

**3.139.4.8 nsym**

```
BigInt T_EGraph::nsym
```

Definition at line 44 of file [gentopo.h](#).

**3.139.4.9 esym**

```
BigInt T_EGraph::esym
```

Definition at line 44 of file [gentopo.h](#).

**3.139.4.10 nodes**

```
T\_ENode\* T_EGraph::nodes
```

Definition at line 46 of file [gentopo.h](#).

### 3.139.4.11 edges

`T_EEdge* T_EGraph::edges`

Definition at line 47 of file [gentopo.h](#).

The documentation for this class was generated from the following files:

- [gentopo.h](#)
- [gentopo.cc](#)

## 3.140 T\_ENode Class Reference

### Data Fields

- `int` [deg](#)
- `int` [ext](#)
- `int *` [edges](#)

### 3.140.1 Detailed Description

Definition at line 17 of file [gentopo.h](#).

### 3.140.2 Field Documentation

#### 3.140.2.1 deg

`int T_ENode::deg`

Definition at line 19 of file [gentopo.h](#).

#### 3.140.2.2 ext

`int T_ENode::ext`

Definition at line 20 of file [gentopo.h](#).

#### 3.140.2.3 edges

`int* T_ENode::edges`

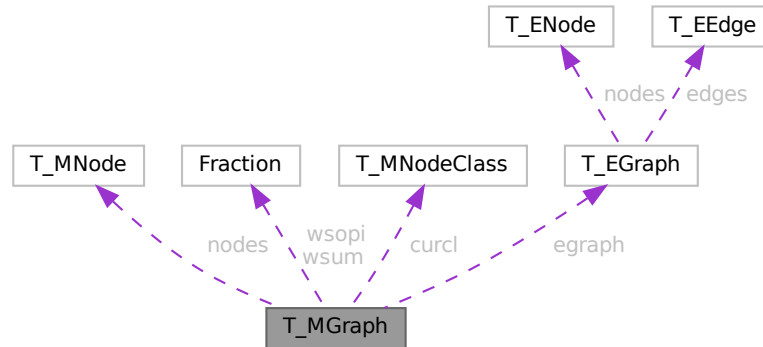
Definition at line 21 of file [gentopo.h](#).

The documentation for this class was generated from the following file:

- [gentopo.h](#)

### 3.141 T\_MGraph Class Reference

Collaboration diagram for T\_MGraph:



#### Public Member Functions

- [T\\_MGraph](#) (int pid, int ncl, int \*cldeg, int \*clnum, int \*clext, int sopi)
- long [generate](#) (void)

#### Data Fields

- int [pId](#)
- int [nNodes](#)
- int [nEdges](#)
- int [nLoops](#)
- [T\\_MNode](#) \*\* [nodes](#)
- int \* [clist](#)
- int [nClasses](#)
- int [mindeg](#)
- int [maxdeg](#)
- int [selOPI](#)
- long [ndiag](#)
- long [n1PI](#)
- int [nBridges](#)
- int [c1PI](#)
- int \*\* [adjMat](#)
- BigInt [nsym](#)
- BigInt [esym](#)
- [Fraction](#) [wsum](#)
- [Fraction](#) [wsopi](#)
- [T\\_MNodeClass](#) \* [curcl](#)
- [T\\_EGraph](#) \* [egraph](#)
- long [ngen](#)
- long [ngconn](#)
- long [nCallRefine](#)
- long [discardOrd](#)
- long [discardRefine](#)
- long [discardDisc](#)
- long [discardIso](#)



### 3.141.1 Detailed Description

Definition at line 84 of file [gentopo.h](#).

### 3.141.2 Constructor & Destructor Documentation

#### 3.141.2.1 T\_MGraph()

```
T_MGraph::T_MGraph (
    int pid,
    int ncl,
    int * cldeg,
    int * clnum,
    int * clext,
    int sopi )
```

Definition at line 473 of file [gentopo.cc](#).

#### 3.141.2.2 ~T\_MGraph()

```
T_MGraph::~T_MGraph ( )
```

Definition at line 572 of file [gentopo.cc](#).

### 3.141.3 Member Function Documentation

#### 3.141.3.1 generate()

```
long T_MGraph::generate (
    void )
```

Definition at line 911 of file [gentopo.cc](#).

### 3.141.4 Field Documentation

#### 3.141.4.1 pld

```
int T_MGraph::pId
```

Definition at line 89 of file [gentopo.h](#).

#### 3.141.4.2 nNodes

```
int T_MGraph::nNodes
```

Definition at line 90 of file [gentopo.h](#).

#### 3.141.4.3 nEdges

```
int T_MGraph::nEdges
```

Definition at line 91 of file [gentopo.h](#).

#### 3.141.4.4 nLoops

```
int T_MGraph::nLoops
```

Definition at line 92 of file [gentopo.h](#).

#### 3.141.4.5 nodes

```
T_MNode** T_MGraph::nodes
```

Definition at line 93 of file [gentopo.h](#).

#### 3.141.4.6 clist

```
int* T_MGraph::clist
```

Definition at line 94 of file [gentopo.h](#).

#### 3.141.4.7 nClasses

```
int T_MGraph::nClasses
```

Definition at line 95 of file [gentopo.h](#).

#### 3.141.4.8 mindeg

```
int T_MGraph::mindeg
```

Definition at line 97 of file [gentopo.h](#).

#### 3.141.4.9 maxdeg

```
int T_MGraph::maxdeg
```

Definition at line 98 of file [gentopo.h](#).

#### 3.141.4.10 selOPI

```
int T_MGraph::selOPI
```

Definition at line 100 of file [gentopo.h](#).

**3.141.4.11 ndiag**

```
long T_MGraph::ndiag
```

Definition at line 103 of file [gentopo.h](#).

**3.141.4.12 n1PI**

```
long T_MGraph::n1PI
```

Definition at line 104 of file [gentopo.h](#).

**3.141.4.13 nBridges**

```
int T_MGraph::nBridges
```

Definition at line 105 of file [gentopo.h](#).

**3.141.4.14 c1PI**

```
int T_MGraph::c1PI
```

Definition at line 108 of file [gentopo.h](#).

**3.141.4.15 adjMat**

```
int** T_MGraph::adjMat
```

Definition at line 109 of file [gentopo.h](#).

**3.141.4.16 nsym**

```
BigInt T_MGraph::nsym
```

Definition at line 110 of file [gentopo.h](#).

**3.141.4.17 esym**

```
BigInt T_MGraph::esym
```

Definition at line 111 of file [gentopo.h](#).

**3.141.4.18 wsum**

```
Fraction T_MGraph::wsum
```

Definition at line 112 of file [gentopo.h](#).

**3.141.4.19 wsopi**

`Fraction T_MGraph::wsopi`

Definition at line 113 of file [gentopo.h](#).

**3.141.4.20 curcl**

`T_MNodeClass* T_MGraph::curcl`

Definition at line 114 of file [gentopo.h](#).

**3.141.4.21 egraph**

`T_EGraph* T_MGraph::egraph`

Definition at line 115 of file [gentopo.h](#).

**3.141.4.22 ngen**

`long T_MGraph::ngen`

Definition at line 118 of file [gentopo.h](#).

**3.141.4.23 ngconn**

`long T_MGraph::ngconn`

Definition at line 119 of file [gentopo.h](#).

**3.141.4.24 nCallRefine**

`long T_MGraph::nCallRefine`

Definition at line 121 of file [gentopo.h](#).

**3.141.4.25 discardOrd**

`long T_MGraph::discardOrd`

Definition at line 122 of file [gentopo.h](#).

**3.141.4.26 discardRefine**

`long T_MGraph::discardRefine`

Definition at line 123 of file [gentopo.h](#).

#### 3.141.4.27 discardDisc

```
long T_MGraph::discardDisc
```

Definition at line 124 of file [gentopo.h](#).

#### 3.141.4.28 discardIso

```
long T_MGraph::discardIso
```

Definition at line 125 of file [gentopo.h](#).

The documentation for this class was generated from the following files:

- [gentopo.h](#)
- [gentopo.cc](#)

## 3.142 T\_MNode Class Reference

### Public Member Functions

- [T\\_MNode](#) (int id, int deg, int ext, int clss)

### Data Fields

- int [id](#)
- int [deg](#)
- int [clss](#)
- int [ext](#)
- int [freelg](#)
- int [visited](#)

### 3.142.1 Detailed Description

Definition at line 61 of file [gentopo.h](#).

### 3.142.2 Constructor & Destructor Documentation

#### 3.142.2.1 T\_MNode()

```
T_MNode::T_MNode (
    int id,
    int deg,
    int ext,
    int clss )
```

Definition at line 237 of file [gentopo.cc](#).

### 3.142.3 Field Documentation

#### 3.142.3.1 id

```
int T_MNode::id
```

Definition at line 63 of file [gentopo.h](#).

#### 3.142.3.2 deg

```
int T_MNode::deg
```

Definition at line 64 of file [gentopo.h](#).

#### 3.142.3.3 clss

```
int T_MNode::clss
```

Definition at line 65 of file [gentopo.h](#).

#### 3.142.3.4 ext

```
int T_MNode::ext
```

Definition at line 66 of file [gentopo.h](#).

#### 3.142.3.5 freeIg

```
int T_MNode::freeIg
```

Definition at line 68 of file [gentopo.h](#).

#### 3.142.3.6 visited

```
int T_MNode::visited
```

Definition at line 69 of file [gentopo.h](#).

The documentation for this class was generated from the following files:

- [gentopo.h](#)
- [gentopo.cc](#)

## 3.143 T\_MNodeClass Class Reference

### Public Member Functions

- [T\\_MNodeClass](#) (int nnodes, int ncl)
- void [init](#) (int \*cl, int mxdeg, int \*\*adjmat)
- void [copy](#) ([T\\_MNodeClass](#) \*mnc)
- int [clCmp](#) (int nd0, int nd1, int cn)
- void [printMat](#) (void)
- void [mkFlist](#) (void)
- void [mkNdCl](#) (void)
- void [mkClMat](#) (int \*\*adjmat)
- void [incMat](#) (int nd, int td, int val)
- Bool [chkOrd](#) (int nd, int ndc, [T\\_MNodeClass](#) \*cl, int \*dtcl)
- int [cmpArray](#) (int \*a0, int \*a1, int ma)

### Data Fields

- int [nNodes](#)
- int [nClasses](#)
- int \* [clist](#)
- int \* [ndcl](#)
- int \*\* [clmat](#)
- int \* [flist](#)
- int [maxdeg](#)
- int [foralignment](#)

### 3.143.1 Detailed Description

Definition at line 249 of file [gentopo.cc](#).

### 3.143.2 Constructor & Destructor Documentation

#### 3.143.2.1 T\_MNodeClass()

```
T_MNodeClass::T_MNodeClass (
    int nnodes,
    int ncl )
```

Definition at line 275 of file [gentopo.cc](#).

#### 3.143.2.2 ~T\_MNodeClass()

```
T_MNodeClass::~T_MNodeClass ( )
```

Definition at line 290 of file [gentopo.cc](#).

### 3.143.3 Member Function Documentation

#### 3.143.3.1 init()

```
void T_MNodeClass::init (
    int * cl,
    int mxdeg,
    int ** adjmat )
```

Definition at line 302 of file [gentopo.cc](#).

#### 3.143.3.2 copy()

```
void T_MNodeClass::copy (
    T_MNodeClass * mnc )
```

Definition at line 315 of file [gentopo.cc](#).

#### 3.143.3.3 clCmp()

```
int T_MNodeClass::clCmp (
    int nd0,
    int nd1,
    int cn )
```

Definition at line 364 of file [gentopo.cc](#).

#### 3.143.3.4 printMat()

```
void T_MNodeClass::printMat (
    void )
```

Definition at line 435 of file [gentopo.cc](#).

#### 3.143.3.5 mkFlist()

```
void T_MNodeClass::mkFlist (
    void )
```

Definition at line 336 of file [gentopo.cc](#).

#### 3.143.3.6 mkNdCl()

```
void T_MNodeClass::mkNdCl (
    void )
```

Definition at line 350 of file [gentopo.cc](#).



### 3.143.3.7 mkClMat()

```
void T_MNodeClass::mkClMat (
    int ** adjmat )
```

Definition at line 384 of file [gentopo.cc](#).

### 3.143.3.8 incMat()

```
void T_MNodeClass::incMat (
    int nd,
    int td,
    int val )
```

Definition at line 398 of file [gentopo.cc](#).

### 3.143.3.9 chkOrd()

```
Bool T_MNodeClass::chkOrd (
    int nd,
    int ndc,
    T_MNodeClass * cl,
    int * dtcl )
```

Definition at line 405 of file [gentopo.cc](#).

### 3.143.3.10 cmpArray()

```
int T_MNodeClass::cmpArray (
    int * a0,
    int * a1,
    int ma )
```

Definition at line 458 of file [gentopo.cc](#).

## 3.143.4 Field Documentation

### 3.143.4.1 nNodes

```
int T_MNodeClass::nNodes
```

Definition at line 251 of file [gentopo.cc](#).

### 3.143.4.2 nClasses

```
int T_MNodeClass::nClasses
```

Definition at line 252 of file [gentopo.cc](#).

**3.143.4.3 clist**

```
int* T_MNodeClass::clist
```

Definition at line [253](#) of file [gentopo.cc](#).

**3.143.4.4 ndcl**

```
int* T_MNodeClass::ndcl
```

Definition at line [254](#) of file [gentopo.cc](#).

**3.143.4.5 clmat**

```
int** T_MNodeClass::clmat
```

Definition at line [255](#) of file [gentopo.cc](#).

**3.143.4.6 flist**

```
int* T_MNodeClass::flist
```

Definition at line [256](#) of file [gentopo.cc](#).

**3.143.4.7 maxdeg**

```
int T_MNodeClass::maxdeg
```

Definition at line [257](#) of file [gentopo.cc](#).

**3.143.4.8 forallignment**

```
int T_MNodeClass::forallignment
```

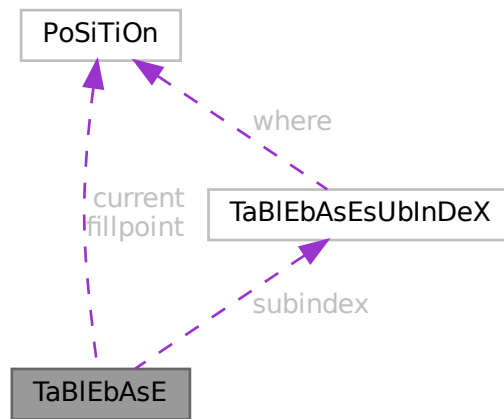
Definition at line [258](#) of file [gentopo.cc](#).

The documentation for this class was generated from the following file:

- [gentopo.cc](#)

## 3.144 TaBIEbAsE Struct Reference

Collaboration diagram for TaBIEbAsE:



### Public Member Functions

- **PADPOSITION** (3, 0, 1, 0, 0)

### Data Fields

- [POSITION](#) fillpoint
- [POSITION](#) current
- `UBYTE * name`
- `int * tablenumbers`
- [TABLEBASESUBINDEX](#) \* [subindex](#)
- `int numtables`

### 3.144.1 Detailed Description

Definition at line 592 of file [structs.h](#).

### 3.144.2 Field Documentation

#### 3.144.2.1 fillpoint

[POSITION](#) `TaBIEbAsE::fillpoint`

Definition at line 593 of file [structs.h](#).

### 3.144.2.2 current

`POSITION TaBlEbAsE::current`

Definition at line 594 of file [structs.h](#).

### 3.144.2.3 name

`UBYTE* TaBlEbAsE::name`

Definition at line 595 of file [structs.h](#).

### 3.144.2.4 tablenumbers

`int* TaBlEbAsE::tablenumbers`

Definition at line 596 of file [structs.h](#).

### 3.144.2.5 subindex

`TABLEBASESUBINDEX* TaBlEbAsE::subindex`

Definition at line 597 of file [structs.h](#).

### 3.144.2.6 numtables

`int TaBlEbAsE::numtables`

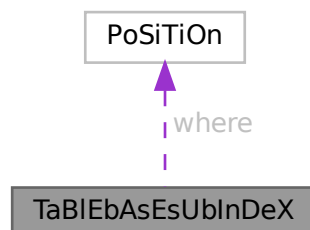
Definition at line 598 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.145 TaBlEbAsEsUbInDeX Struct Reference

Collaboration diagram for TaBlEbAsEsUbInDeX:



## Public Member Functions

- **PADPOSITION** (0, 1, 0, 0, 0)

## Data Fields

- **POSITION** where
- **LONG** size

### 3.145.1 Detailed Description

Definition at line 582 of file [structs.h](#).

### 3.145.2 Field Documentation

#### 3.145.2.1 where

**POSITION** TaBlEbAsEsUbInDeX::where

Definition at line 583 of file [structs.h](#).

#### 3.145.2.2 size

**LONG** TaBlEbAsEsUbInDeX::size

Definition at line 584 of file [structs.h](#).

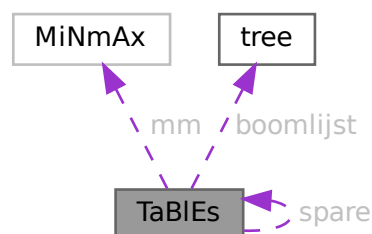
The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.146 TaBlEs Struct Reference

```
#include <structs.h>
```

Collaboration diagram for TaBlEs:



## Data Fields

- WORD \* [tablepointers](#)
- WORD \* [prototype](#)
- WORD \* [pattern](#)
- [MINMAX](#) \* [mm](#)
- WORD \* [flags](#)
- [COMPTREE](#) \* [boomlijst](#)
- UBYTE \* [argtail](#)
- struct [TaBles](#) \* [spare](#)
- WORD \* [buffers](#)
- LONG [totind](#)
- LONG [reserved](#)
- LONG [defined](#)
- LONG [mdefined](#)
- int [prototypeSize](#)
- int [numind](#)
- int [bounds](#)
- int [strict](#)
- int [sparse](#)
- int [numtree](#)
- int [rootnum](#)
- int [MaxTreeSize](#)
- WORD [bufnum](#)
- WORD [bufferssize](#)
- WORD [buffersfill](#)
- WORD [tablenum](#)
- WORD [mode](#)
- WORD [numdummies](#)

### 3.146.1 Detailed Description

[buffers](#), [mm](#), [flags](#), and [prototype](#) are always dynamically allocated, [tablepointers](#) only if needed (=0 if unallocated), [boomlijst](#) and [argtail](#) only for sparse tables.

Allocation is done for both the normal and the stub instance ([spare](#)), except for [prototype](#) and [argtail](#) which share memory.

Definition at line [350](#) of file [structs.h](#).

### 3.146.2 Field Documentation

#### 3.146.2.1 [tablepointers](#)

```
WORD* TaBles::tablepointers
```

[D] Start in [tablepointers](#) table.

Definition at line [351](#) of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.2 prototype

WORD\* TaBlEs::prototype

[D] The wildcard prototyping for arguments

Definition at line 356 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.3 pattern

WORD\* TaBlEs::pattern

The pattern with which to match the arguments

Definition at line 357 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.4 mm

MINMAX\* TaBlEs::mm

[D] Array bounds, dimension by dimension. # elements = numind.

Definition at line 359 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.5 flags

WORD\* TaBlEs::flags

[D] Is element in use ? etc. # elements = numind.

Definition at line 360 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.6 boomlijst

COMPTREE\* TaBlEs::boomlijst

[D] Tree for searching in sparse tables

Definition at line 361 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.7 argtail

```
UBYTE* TaBles::argtail
```

[D] The arguments in characters. Starts for tablebase with parenthesis to indicate tail

Definition at line 362 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.8 spare

```
struct TaBles* TaBles::spare
```

[D] For tablebase. Alternatingly stubs and real

Definition at line 364 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.9 buffers

```
WORD* TaBles::buffers
```

[D] When we use more than one compiler buffer.

Definition at line 365 of file [structs.h](#).

Referenced by [AddRHS\(\)](#), and [DoRecovery\(\)](#).

### 3.146.2.10 totind

```
LONG TaBles::totind
```

Total number requested

Definition at line 366 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.11 reserved

```
LONG TaBles::reserved
```

Total reservation in tablepointers for sparse

Definition at line 367 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).



### 3.146.2.12 defined

```
LONG TaBlEs::defined
```

Number of table elements that are defined

Definition at line 368 of file [structs.h](#).

### 3.146.2.13 mdefined

```
LONG TaBlEs::mdefined
```

Same as defined but after .global

Definition at line 369 of file [structs.h](#).

### 3.146.2.14 prototypeSize

```
int TaBlEs::prototypeSize
```

Size of allocated memory for prototype in bytes.

Definition at line 370 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.15 numind

```
int TaBlEs::numind
```

Number of array indices

Definition at line 371 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.16 bounds

```
int TaBlEs::bounds
```

Array bounds check on/off.

Definition at line 372 of file [structs.h](#).

### 3.146.2.17 strict

```
int TaBlEs::strict
```

>0: all must be defined. <0: undefined not substitute

Definition at line 373 of file [structs.h](#).

### 3.146.2.18 sparse

```
int TaBles::sparse
```

0 --> sparse table

Definition at line 374 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.19 numtree

```
int TaBles::numtree
```

For the tree for sparse tables

Definition at line 375 of file [structs.h](#).

### 3.146.2.20 rootnum

```
int TaBles::rootnum
```

For the tree for sparse tables

Definition at line 376 of file [structs.h](#).

### 3.146.2.21 MaxTreeSize

```
int TaBles::MaxTreeSize
```

For the tree for sparse tables

Definition at line 377 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

### 3.146.2.22 bufnum

```
WORD TaBles::bufnum
```

Each table potentially its own buffer

Definition at line 378 of file [structs.h](#).

Referenced by [AddRHS\(\)](#).

### 3.146.2.23 buffersize

WORD TaBlEs::buffersize

When we use more than one compiler buffer

Definition at line 379 of file [structs.h](#).

Referenced by [AddRHS\(\)](#), and [DoRecovery\(\)](#).

### 3.146.2.24 buffersfill

WORD TaBlEs::buffersfill

When we use more than one compiler buffer

Definition at line 380 of file [structs.h](#).

Referenced by [AddRHS\(\)](#).

### 3.146.2.25 tablenum

WORD TaBlEs::tablenum

For testing of tableuse

Definition at line 381 of file [structs.h](#).

### 3.146.2.26 mode

WORD TaBlEs::mode

0: normal, 1: stub

Definition at line 382 of file [structs.h](#).

### 3.146.2.27 numdummies

WORD TaBlEs::numdummies

Definition at line 383 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.147 TERMINFO Struct Reference

### Data Fields

- WORD \* [term](#)
- void \* [currentModel](#)
- void \* [currentMODEL](#)
- WORD \* [legcouple](#) [MAXLEGS+1]
- LONG [numtopo](#)
- LONG [numdia](#)
- WORD [diaoffset](#)
- WORD [level](#)
- WORD [externalset](#)
- WORD [internalset](#)
- WORD [numextern](#)
- WORD [flags](#)

### 3.147.1 Detailed Description

Definition at line [1396](#) of file [structs.h](#).

### 3.147.2 Field Documentation

#### 3.147.2.1 term

```
WORD* TERMINFO::term
```

Definition at line [1397](#) of file [structs.h](#).

#### 3.147.2.2 currentModel

```
void* TERMINFO::currentModel
```

Definition at line [1398](#) of file [structs.h](#).

#### 3.147.2.3 currentMODEL

```
void* TERMINFO::currentMODEL
```

Definition at line [1399](#) of file [structs.h](#).

#### 3.147.2.4 legcouple

```
WORD* TERMINFO::legcouple[MAXLEGS+1]
```

Definition at line [1400](#) of file [structs.h](#).

### 3.147.2.5 numtopo

LONG TERMINFO::numtopo

Definition at line 1401 of file [structs.h](#).

### 3.147.2.6 numdia

LONG TERMINFO::numdia

Definition at line 1402 of file [structs.h](#).

### 3.147.2.7 diaoffset

WORD TERMINFO::diaoffset

Definition at line 1403 of file [structs.h](#).

### 3.147.2.8 level

WORD TERMINFO::level

Definition at line 1404 of file [structs.h](#).

### 3.147.2.9 externalset

WORD TERMINFO::externalset

Definition at line 1405 of file [structs.h](#).

### 3.147.2.10 internalset

WORD TERMINFO::internalset

Definition at line 1406 of file [structs.h](#).

### 3.147.2.11 numextern

WORD TERMINFO::numextern

Definition at line 1407 of file [structs.h](#).



## 3.148.2 Field Documentation

### 3.148.2.1 vert

WORD \* ToPoTyPe::vert

Definition at line 13 of file [diawrap.cc](#).

### 3.148.2.2 vertmax

WORD \* ToPoTyPe::vertmax

Definition at line 14 of file [diawrap.cc](#).

### 3.148.2.3 opt

Options\* ToPoTyPe::opt

Definition at line 15 of file [diawrap.cc](#).

### 3.148.2.4 cldeg

int ToPoTyPe::cldeg

Definition at line 16 of file [diawrap.cc](#).

### 3.148.2.5 clnum

int ToPoTyPe::clnum

Definition at line 16 of file [diawrap.cc](#).

### 3.148.2.6 clext

int ToPoTyPe::clext

Definition at line 16 of file [diawrap.cc](#).

### 3.148.2.7 cmind

int ToPoTyPe::cmind[MAXLEGS+1]

Definition at line 17 of file [diawrap.cc](#).

### 3.148.2.8 cmaxd

```
int ToPoTyPe::cmaxd[MAXLEGS+1]
```

Definition at line 17 of file [diawrap.cc](#).

### 3.148.2.9 ncl

```
int ToPoTyPe::ncl
```

Definition at line 18 of file [diawrap.cc](#).

### 3.148.2.10 nvert [1/2]

```
int ToPoTyPe::nvert
```

Definition at line 18 of file [diawrap.cc](#).

### 3.148.2.11 nloops

```
int ToPoTyPe::nloops
```

Definition at line 48 of file [topowrap.cc](#).

### 3.148.2.12 nlegs

```
int ToPoTyPe::nlegs
```

Definition at line 48 of file [topowrap.cc](#).

### 3.148.2.13 npadding

```
int ToPoTyPe::npadding
```

Definition at line 48 of file [topowrap.cc](#).

### 3.148.2.14 nvert [2/2]

```
WORD ToPoTyPe::nvert
```

Definition at line 51 of file [topowrap.cc](#).



### 3.148.2.15 sopi

WORD ToPoTyPe::sopi

Definition at line 52 of file [topowrap.cc](#).

The documentation for this struct was generated from the following files:

- [diawrap.cc](#)
- [topowrap.cc](#)

## 3.149 TrAcEn Struct Reference

```
#include <structs.h>
```

### Data Fields

- WORD \* [accu](#)
- WORD \* [accup](#)
- WORD \* [termp](#)
- WORD \* [perm](#)
- WORD \* [inlist](#)
- WORD [sgn](#)
- WORD [num](#)
- WORD [level](#)
- WORD [factor](#)
- WORD [allsign](#)

### 3.149.1 Detailed Description

The struct TRACEN keeps track of the progress during the expansion of a 4-dimensional trace. Each time a term gets generated the expansion tree continues in the next statement. When it returns it has to know where to continue.

Definition at line 822 of file [structs.h](#).

### 3.149.2 Field Documentation

#### 3.149.2.1 accu

WORD\* TrAcEn::accu

Definition at line 823 of file [structs.h](#).

#### 3.149.2.2 accup

WORD\* TrAcEn::accup

Definition at line 824 of file [structs.h](#).

### 3.149.2.3 temp

WORD\* TrAcEn::temp

Definition at line 825 of file [structs.h](#).

### 3.149.2.4 perm

WORD\* TrAcEn::perm

Definition at line 826 of file [structs.h](#).

### 3.149.2.5 inlist

WORD\* TrAcEn::inlist

Definition at line 827 of file [structs.h](#).

### 3.149.2.6 sgn

WORD TrAcEn::sgn

Definition at line 828 of file [structs.h](#).

### 3.149.2.7 num

WORD TrAcEn::num

Definition at line 828 of file [structs.h](#).

### 3.149.2.8 level

WORD TrAcEn::level

Definition at line 828 of file [structs.h](#).

### 3.149.2.9 factor

WORD TrAcEn::factor

Definition at line 828 of file [structs.h](#).

### 3.149.2.10 allsign

WORD TrAcEn::allsign

Definition at line 828 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.150 TrAcEs Struct Reference

```
#include <structs.h>
```

### Data Fields

- WORD \* [accu](#)
- WORD \* [accup](#)
- WORD \* [termp](#)
- WORD \* [perm](#)
- WORD \* [inlist](#)
- WORD \* [nt3](#)
- WORD \* [nt4](#)
- WORD \* [j3](#)
- WORD \* [j4](#)
- WORD \* [e3](#)
- WORD \* [e4](#)
- WORD \* [eers](#)
- WORD \* [mepf](#)
- WORD \* [mdel](#)
- WORD \* [pepf](#)
- WORD \* [pdel](#)
- WORD [sgn](#)
- WORD [stap](#)
- WORD [step1](#)
- WORD [kstep](#)
- WORD [mdum](#)
- WORD [gamm](#)
- WORD [ad](#)
- WORD [a3](#)
- WORD [a4](#)
- WORD [lc3](#)
- WORD [lc4](#)
- WORD [sign1](#)
- WORD [sign2](#)
- WORD [gamma5](#)
- WORD [num](#)
- WORD [level](#)
- WORD [factor](#)
- WORD [allsign](#)
- WORD [finalstep](#)

### 3.150.1 Detailed Description

The struct TRACES keeps track of the progress during the expansion of a 4-dimensional trace. Each time a term gets generated the expansion tree continues in the next statement. When it returns it has to know where to continue. The 4-dimensional traces are more complicated than the n-dimensional traces (see TRACEN) because of the extra tricks that can be used. They are responsible for the shorter final expressions.

Definition at line 789 of file [structs.h](#).

### 3.150.2 Field Documentation

#### 3.150.2.1 `accu`

```
WORD* TrAcEs::accu
```

Definition at line 790 of file [structs.h](#).

#### 3.150.2.2 `accup`

```
WORD* TrAcEs::accup
```

Definition at line 791 of file [structs.h](#).

#### 3.150.2.3 `termp`

```
WORD* TrAcEs::termp
```

Definition at line 792 of file [structs.h](#).

#### 3.150.2.4 `perm`

```
WORD* TrAcEs::perm
```

Definition at line 793 of file [structs.h](#).

#### 3.150.2.5 `inlist`

```
WORD* TrAcEs::inlist
```

Definition at line 794 of file [structs.h](#).

#### 3.150.2.6 `nt3`

```
WORD* TrAcEs::nt3
```

Definition at line 795 of file [structs.h](#).

### 3.150.2.7 nt4

WORD\* TrAcEs::nt4

Definition at line 796 of file [structs.h](#).

### 3.150.2.8 j3

WORD\* TrAcEs::j3

Definition at line 797 of file [structs.h](#).

### 3.150.2.9 j4

WORD\* TrAcEs::j4

Definition at line 798 of file [structs.h](#).

### 3.150.2.10 e3

WORD\* TrAcEs::e3

Definition at line 799 of file [structs.h](#).

### 3.150.2.11 e4

WORD\* TrAcEs::e4

Definition at line 800 of file [structs.h](#).

### 3.150.2.12 eers

WORD\* TrAcEs::eers

Definition at line 801 of file [structs.h](#).

### 3.150.2.13 mepf

WORD\* TrAcEs::mepf

Definition at line 802 of file [structs.h](#).

### 3.150.2.14 mdel

WORD\* TrAcEs::mdel

Definition at line 803 of file [structs.h](#).

**3.150.2.15   pef**

WORD\* TrAcEs::pepf

Definition at line 804 of file [structs.h](#).

**3.150.2.16   pdel**

WORD\* TrAcEs::pdel

Definition at line 805 of file [structs.h](#).

**3.150.2.17   sgn**

WORD TrAcEs::sgn

Definition at line 806 of file [structs.h](#).

**3.150.2.18   stap**

WORD TrAcEs::stap

Definition at line 807 of file [structs.h](#).

**3.150.2.19   step1**

WORD TrAcEs::step1

Definition at line 808 of file [structs.h](#).

**3.150.2.20   kstep**

WORD TrAcEs::kstep

Definition at line 808 of file [structs.h](#).

**3.150.2.21   mdum**

WORD TrAcEs::mdum

Definition at line 808 of file [structs.h](#).

**3.150.2.22   gamm**

WORD TrAcEs::gamm

Definition at line 809 of file [structs.h](#).

**3.150.2.23 ad**

WORD TrAcEs::ad

Definition at line 809 of file [structs.h](#).

**3.150.2.24 a3**

WORD TrAcEs::a3

Definition at line 809 of file [structs.h](#).

**3.150.2.25 a4**

WORD TrAcEs::a4

Definition at line 809 of file [structs.h](#).

**3.150.2.26 lc3**

WORD TrAcEs::lc3

Definition at line 809 of file [structs.h](#).

**3.150.2.27 lc4**

WORD TrAcEs::lc4

Definition at line 809 of file [structs.h](#).

**3.150.2.28 sign1**

WORD TrAcEs::sign1

Definition at line 810 of file [structs.h](#).

**3.150.2.29 sign2**

WORD TrAcEs::sign2

Definition at line 810 of file [structs.h](#).

**3.150.2.30 gamma5**

WORD TrAcEs::gamma5

Definition at line 810 of file [structs.h](#).

### 3.150.2.31 num

WORD TrAcEs::num

Definition at line 810 of file [structs.h](#).

### 3.150.2.32 level

WORD TrAcEs::level

Definition at line 810 of file [structs.h](#).

### 3.150.2.33 factor

WORD TrAcEs::factor

Definition at line 810 of file [structs.h](#).

### 3.150.2.34 allsign

WORD TrAcEs::allsign

Definition at line 810 of file [structs.h](#).

### 3.150.2.35 finalstep

WORD TrAcEs::finalstep

Definition at line 811 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.151 tree Struct Reference

```
#include <structs.h>
```

### Data Fields

- int [parent](#)
- int [left](#)
- int [right](#)
- int [value](#)
- int [blnce](#)
- int [usage](#)



### 3.151.1 Detailed Description

The subexpressions in the compiler are kept track of in a (balanced) tree to reduce the need for subexpressions and hence save much space in large rhs expressions (like when we have xxxxxx occurrences of objects like  $f(x+1, x+1)$  in which each  $x+1$  becomes a subexpression. The struct that controls this tree is COMPTREE.

Definition at line 294 of file [structs.h](#).

### 3.151.2 Field Documentation

#### 3.151.2.1 parent

```
int tree::parent
```

Index of parent

Definition at line 295 of file [structs.h](#).

Referenced by [IniFbuffer\(\)](#).

#### 3.151.2.2 left

```
int tree::left
```

Left child (if not -1)

Definition at line 296 of file [structs.h](#).

Referenced by [IniFbuffer\(\)](#).

#### 3.151.2.3 right

```
int tree::right
```

Right child (if not -1)

Definition at line 297 of file [structs.h](#).

Referenced by [IniFbuffer\(\)](#).

#### 3.151.2.4 value

```
int tree::value
```

The object to be sorted and searched

Definition at line 298 of file [structs.h](#).

Referenced by [IniFbuffer\(\)](#).

### 3.151.2.5 blnce

```
int tree::blnce
```

Balance factor

Definition at line 299 of file [structs.h](#).

Referenced by [IniFbuffer\(\)](#).

### 3.151.2.6 usage

```
int tree::usage
```

Number of uses in some types of trees

Definition at line 300 of file [structs.h](#).

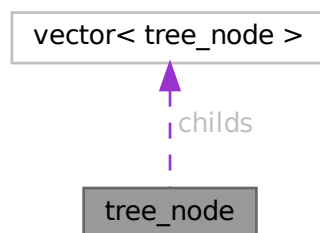
Referenced by [CleanupArgCache\(\)](#), and [IniFbuffer\(\)](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.152 tree\_node Class Reference

Collaboration diagram for tree\_node:



### Public Member Functions

- [tree\\_node](#) (int \_var=0)
- [tree\\_node](#) (const [tree\\_node](#) &other)
- [tree\\_node](#) & [operator=](#) (const [tree\\_node](#) &other)

## Data Fields

- vector< [tree\\_node](#) > [childs](#)
- double [sum\\_results](#)
- int [num\\_visits](#)
- WORD [var](#)
- bool [finished](#)

### 3.152.1 Detailed Description

Definition at line 125 of file [optimize.cc](#).

### 3.152.2 Constructor & Destructor Documentation

#### 3.152.2.1 `tree_node()` [1/2]

```
tree_node::tree_node (  
    int _var = 0 ) [inline]
```

Definition at line 134 of file [optimize.cc](#).

#### 3.152.2.2 `tree_node()` [2/2]

```
tree_node::tree_node (  
    const tree\_node & other ) [inline]
```

Definition at line 137 of file [optimize.cc](#).

### 3.152.3 Member Function Documentation

#### 3.152.3.1 `operator=()`

```
tree\_node & tree_node::operator= (  
    const tree\_node & other ) [inline]
```

Definition at line 144 of file [optimize.cc](#).

### 3.152.4 Field Documentation

#### 3.152.4.1 `childs`

```
vector<tree\_node> tree_node::childs
```

Definition at line 127 of file [optimize.cc](#).

#### 3.152.4.2 `sum_results`

```
double tree_node::sum_results
```

Definition at line 128 of file [optimize.cc](#).

#### 3.152.4.3 `num_visits`

```
int tree_node::num_visits
```

Definition at line 129 of file [optimize.cc](#).

#### 3.152.4.4 `var`

```
WORD tree_node::var
```

Definition at line 130 of file [optimize.cc](#).

#### 3.152.4.5 `finished`

```
bool tree_node::finished
```

Definition at line 131 of file [optimize.cc](#).

The documentation for this class was generated from the following file:

- [optimize.cc](#)

### 3.153 VaRrEnUm Struct Reference

```
#include <structs.h>
```

#### Data Fields

- WORD \* [start](#)
- WORD \* [lo](#)
- WORD \* [hi](#)

#### 3.153.1 Detailed Description

Contains the pointers to an array in which a binary search will be performed.

Definition at line 166 of file [structs.h](#).

### 3.153.2 Field Documentation

#### 3.153.2.1 start

WORD\* VaRrEnUm::start

Start point for search. Points inbetween lo and hi

Definition at line 167 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

#### 3.153.2.2 lo

WORD\* VaRrEnUm::lo

Start of memory area

Definition at line 168 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#), [Generator\(\)](#), and [GetFirstTerm\(\)](#).

#### 3.153.2.3 hi

WORD\* VaRrEnUm::hi

End of memory area

Definition at line 169 of file [structs.h](#).

Referenced by [DoRecovery\(\)](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.154 VeCtOr Struct Reference

### Data Fields

- LONG [name](#)
- WORD [complex](#)
- WORD [number](#)
- WORD [flags](#)
- WORD [node](#)
- WORD [namesize](#)
- WORD [dimension](#)

### 3.154.1 Detailed Description

Definition at line [461](#) of file [structs.h](#).

### 3.154.2 Field Documentation

#### 3.154.2.1 name

```
LONG VeCtOr::name
```

Definition at line [462](#) of file [structs.h](#).

#### 3.154.2.2 complex

```
WORD VeCtOr::complex
```

Definition at line [463](#) of file [structs.h](#).

#### 3.154.2.3 number

```
WORD VeCtOr::number
```

Definition at line [464](#) of file [structs.h](#).

#### 3.154.2.4 flags

```
WORD VeCtOr::flags
```

Definition at line [465](#) of file [structs.h](#).

#### 3.154.2.5 node

```
WORD VeCtOr::node
```

Definition at line [466](#) of file [structs.h](#).

#### 3.154.2.6 namesize

```
WORD VeCtOr::namesize
```

Definition at line [467](#) of file [structs.h](#).

### 3.154.2.7 dimension

WORD VeCtOr::dimension

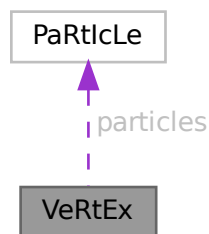
Definition at line 468 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

## 3.155 VeRtEx Struct Reference

Collaboration diagram for VeRtEx:



### Data Fields

- [PARTICLE](#) [particles](#) [MAXPARTICLES]
- WORD [couplings](#) [2 \*MAXCOUPLINGS]
- WORD [nparticles](#)
- WORD [ncouplings](#)
- WORD [type](#)
- WORD [error](#)
- WORD [externonly](#)
- WORD [spare](#)

### 3.155.1 Detailed Description

Definition at line 626 of file [structs.h](#).

### 3.155.2 Field Documentation

#### 3.155.2.1 particles

[PARTICLE](#) VeRtEx::particles [MAXPARTICLES]

Definition at line 627 of file [structs.h](#).

### 3.155.2.2 couplings

WORD VeRtEx::couplings[2 \*MAXCOUPLINGS]

Definition at line 628 of file [structs.h](#).

### 3.155.2.3 nparticles

WORD VeRtEx::nparticles

Definition at line 629 of file [structs.h](#).

### 3.155.2.4 ncouplings

WORD VeRtEx::ncouplings

Definition at line 630 of file [structs.h](#).

### 3.155.2.5 type

WORD VeRtEx::type

Definition at line 631 of file [structs.h](#).

### 3.155.2.6 error

WORD VeRtEx::error

Definition at line 632 of file [structs.h](#).

### 3.155.2.7 externonly

WORD VeRtEx::externonly

Definition at line 633 of file [structs.h](#).

### 3.155.2.8 spare

WORD VeRtEx::spare

Definition at line 634 of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)



## 3.156 X\_const Struct Reference

```
#include <structs.h>
```

### Data Fields

- `UBYTE *` [currentPrompt](#)
- `UBYTE *` [shellname](#)
- `UBYTE *` [stderrname](#)
- `int` [timeout](#)
- `int` [killSignal](#)
- `int` [killWholeGroup](#)
- `int` [daemonize](#)
- `int` [currentExternalChannel](#)

### 3.156.1 Detailed Description

The [X\\_const](#) struct is part of the global data and resides in the ALLGLOBALS struct A under the name X We see it used with the macro AX as in AX.timeout It contains variables that involve communication with external programs

Definition at line [2559](#) of file [structs.h](#).

### 3.156.2 Field Documentation

#### 3.156.2.1 currentPrompt

```
UBYTE* X_const::currentPrompt
```

Definition at line [2560](#) of file [structs.h](#).

#### 3.156.2.2 shellname

```
UBYTE* X_const::shellname
```

Definition at line [2561](#) of file [structs.h](#).

#### 3.156.2.3 stderrname

```
UBYTE* X_const::stderrname
```

Definition at line [2563](#) of file [structs.h](#).

#### 3.156.2.4 timeout

```
int X_const::timeout
```

Definition at line [2565](#) of file [structs.h](#).

### 3.156.2.5 killSignal

```
int X_const::killSignal
```

Definition at line [2568](#) of file [structs.h](#).

### 3.156.2.6 killWholeGroup

```
int X_const::killWholeGroup
```

Definition at line [2569](#) of file [structs.h](#).

### 3.156.2.7 daemonize

```
int X_const::daemonize
```

Definition at line [2571](#) of file [structs.h](#).

### 3.156.2.8 currentExternalChannel

```
int X_const::currentExternalChannel
```

Definition at line [2572](#) of file [structs.h](#).

The documentation for this struct was generated from the following file:

- [structs.h](#)

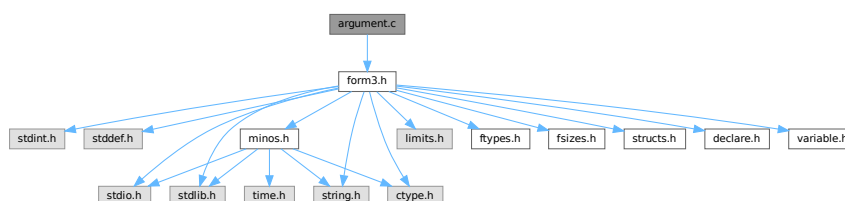
## Chapter 4

# File Documentation

### 4.1 argument.c File Reference

```
#include "form3.h"
```

Include dependency graph for argument.c:



#### Macros

- `#define` [NEWORDER](#)

#### Functions

- int [execarg](#) (PHEAD WORD \*term, WORD level)
- int [execterm](#) (PHEAD WORD \*term, WORD level)
- int [ArgumentImplode](#) (PHEAD WORD \*term, WORD \*thelist)
- int [ArgumentExplode](#) (PHEAD WORD \*term, WORD \*thelist)
- int [ArgFactorize](#) (PHEAD WORD \*argin, WORD \*argout)
- WORD [FindArg](#) (PHEAD WORD \*a)
- int [InsertArg](#) (PHEAD WORD \*argin, WORD \*argout, int par)
- int [CleanupArgCache](#) (PHEAD WORD bufnum)
- int [ArgSymbolMerge](#) (WORD \*t1, WORD \*t2)
- int [ArgDotproductMerge](#) (WORD \*t1, WORD \*t2)
- WORD \* [TakeArgContent](#) (PHEAD WORD \*argin, WORD \*argout)
- WORD \* [MakeInteger](#) (PHEAD WORD \*argin, WORD \*argout, WORD \*argfree)
- WORD \* [MakeMod](#) (PHEAD WORD \*argin, WORD \*argout, WORD \*argfree)
- void [SortWeights](#) (LONG \*weights, LONG \*extraspace, WORD number)

### 4.1.1 Detailed Description

Contains the routines that deal with the execution phase of the argument and related statements (like term)

Definition in file [argument.c](#).

### 4.1.2 Macro Definition Documentation

#### 4.1.2.1 NEWORDER

```
#define NEWORDER
```

Factorizes an argument in general notation (meaning that the first word of the argument is a positive size indicator)  
Input (argin): pointer to the complete argument [Output](#) (argout): Pointer to where the output should be written. This is in the WorkSpace Return value should be negative if anything goes wrong.

The notation of the output should be a string of arguments terminated by the number zero.

Originally we sorted in a way that the constants came last. This gave conflicts with the dollar and expression factorizations (in the expressions we wanted the zero first and then followed by the constants).

Definition at line [2059](#) of file [argument.c](#).

### 4.1.3 Function Documentation

#### 4.1.3.1 `execarg()`

```
int execarg (
    PHEAD WORD * term,
    WORD level )
```

Definition at line [56](#) of file [argument.c](#).

#### 4.1.3.2 `execterm()`

```
int execterm (
    PHEAD WORD * term,
    WORD level )
```

Definition at line [1770](#) of file [argument.c](#).

#### 4.1.3.3 `ArgumentImplode()`

```
int ArgumentImplode (
    PHEAD WORD * term,
    WORD * thelist )
```

Definition at line [1846](#) of file [argument.c](#).

#### 4.1.3.4 ArgumentExplode()

```
int ArgumentExplode (
    PHEAD WORD * term,
    WORD * thelist )
```

Definition at line 1950 of file [argument.c](#).

#### 4.1.3.5 ArgFactorize()

```
int ArgFactorize (
    PHEAD WORD * argin,
    WORD * argout )
```

Definition at line 2061 of file [argument.c](#).

#### 4.1.3.6 FindArg()

```
WORD FindArg (
    PHEAD WORD * a )
```

Looks the argument up in the (workers) table. If it is found the number in the table is returned (plus one to make it positive). If it is not found we look in the compiler provided table. If it is found - the number in the table is returned (minus one to make it negative). If in neither table we return zero.

Definition at line 2514 of file [argument.c](#).

#### 4.1.3.7 InsertArg()

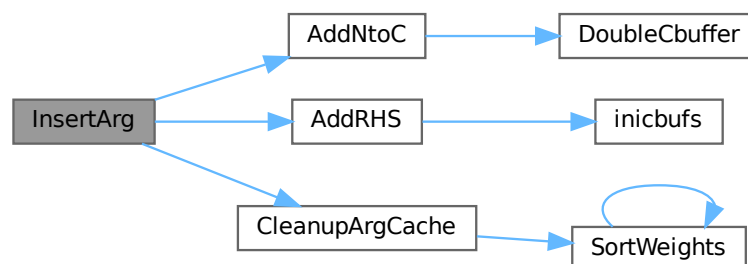
```
int InsertArg (
    PHEAD WORD * argin,
    WORD * argout,
    int par )
```

Inserts the argument into the (workers) table. If the table is too full we eliminate half of it. The eliminated elements are the ones that have not been used most recently, weighted by their total use and age(?). If par == 0 it inserts in the regular factorization cache. If par == 1 it inserts in the cache defined with the FactorCache statement

Definition at line 2538 of file [argument.c](#).

References [AddNtoC\(\)](#), [AddRHS\(\)](#), and [CleanupArgCache\(\)](#).

Here is the call graph for this function:



#### 4.1.3.8 CleanupArgCache()

```
int CleanupArgCache (
    PHEAD WORD bufnum )
```

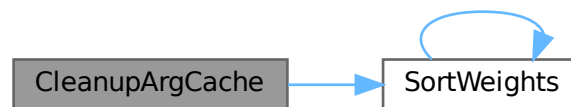
Cleans up the argument factorization cache. We throw half the elements. For a weight of what we want to keep we use the product of usage and the number in the buffer.

Definition at line 2573 of file [argument.c](#).

References [CbUf::boomlijst](#), [CbUf::Buffer](#), [CbUf::Pointer](#), [CbUf::rhs](#), [SortWeights\(\)](#), and [tree::usage](#).

Referenced by [InsertArg\(\)](#).

Here is the call graph for this function:



#### 4.1.3.9 ArgSymbolMerge()

```
int ArgSymbolMerge (
    WORD * t1,
    WORD * t2 )
```

Definition at line 2644 of file [argument.c](#).

#### 4.1.3.10 ArgDotproductMerge()

```
int ArgDotproductMerge (
    WORD * t1,
    WORD * t2 )
```

Definition at line 2699 of file [argument.c](#).

#### 4.1.3.11 TakeArgContent()

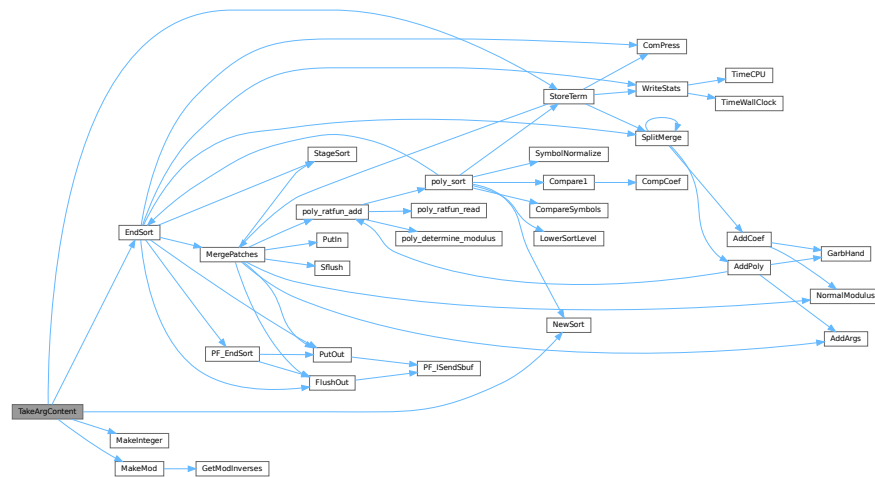
```
WORD * TakeArgContent (
    PHEAD WORD * argin,
    WORD * argout )
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. The common pieces are put in argout as a sequence of arguments. The part with the multiple terms that are now relative prime is put in argfree which is allocated via TermMalloc and is given as the return value. The difference with the old code is that negative powers are always removed. Hence it is as in MakeInteger in which only numerators will be left: now only zero or positive powers will be remaining.

Definition at line 2767 of file [argument.c](#).

References [EndSort\(\)](#), [MakeInteger\(\)](#), [MakeMod\(\)](#), [NewSort\(\)](#), and [StoreTerm\(\)](#).

Here is the call graph for this function:



#### 4.1.3.12 MakeInteger()

```
WORD * MakeInteger (
    PHEAD WORD * argin,
    WORD * argout,
    WORD * argfree )
```

For normalizing everything to integers we have to determine for all elements of this argument the LCM of the denominators and the GCD of the numerators. The input argument is in argin. The number that comes out should go to argout. The new pointer in the argout buffer is the return value. The normalized argument is in argfree.

Definition at line 3313 of file [argument.c](#).

Referenced by [TakeArgContent\(\)](#).

#### 4.1.3.13 MakeMod()

```
WORD * MakeMod (
    PHEAD WORD * argin,
    WORD * argout,
    WORD * argfree )
```

Similar to MakeInteger but now with modulus arithmetic using only a one WORD 'prime'. We make the coefficient of the first term in the argument equal to one. Already the coefficients are taken modulus AN.cmod and AN.ncmod == 1

Definition at line 3484 of file [argument.c](#).

References [GetModInverses\(\)](#).

Referenced by [TakeArgContent\(\)](#).

Here is the call graph for this function:



#### 4.1.3.14 SortWeights()

```
void SortWeights (
    LONG * weights,
    LONG * extraspace,
    WORD number )
```

Sorts an array of LONGS in the same way SplitMerge (in [sort.c](#)) works We use gradual division in two.

Definition at line 3529 of file [argument.c](#).

References [SortWeights\(\)](#).

Referenced by [CleanupArgCache\(\)](#), and [SortWeights\(\)](#).

Here is the call graph for this function:





## 4.2 argument.c

[Go to the documentation of this file.](#)

```

00001
00007 /* #[ License : */
00008 /*
00009  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00010  * When using this file you are requested to refer to the publication
00011  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00012  * This is considered a matter of courtesy as the development was paid
00013  * for by FOM the Dutch physics granting agency and we would like to
00014  * be able to track its scientific use to convince FOM of its value
00015  * for the community.
00016  *
00017  * This file is part of FORM.
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00022  * version.
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00024  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
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00026  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00027  * details.
00028  *
00029  * You should have received a copy of the GNU General Public License along
00030  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00031  */
00032 /* #[ License : */
00033
00034 /*
00035  *#[ include : argument.c
00036  */
00037
00038 #include "form3.h"
00039
00040 /*
00041  *#[ include :
00042  *#[ execarg :
00043
00044  Executes the subset of statements in an argument environment.
00045  The calling routine should be of the type
00046  if ( C->lhs[level][0] == TYPEARG ) {
00047      if ( execarg(term,level) ) goto GenCall;
00048      level = C->lhs[level][2];
00049      goto SkipCount;
00050  }
00051  Note that there will be cases in which extra space is needed.
00052  In addition the compare with C->numlhs isn't very fine, because we
00053  need to insert a different value (C->lhs[level][2]).
00054  */
00055
00056 int execarg(PHEAD WORD *term, WORD level)
00057 {
00058     GETBIDENTITY
00059     WORD *t, *r, *m, *v;
00060     WORD *start, *stop, *rstop, *r1, *r2 = 0, *r3 = 0, *r4, *r5, *r6, *r7, *r8, *r9;
00061     WORD *mm, *mstop, *rnext, *rr, *factor, type, ngcd, nq;
00062     CBUF *C = cbuf+AM.rbufnum, *CC = cbuf+AT.ebufnum;
00063     WORD i, j, k, oldnumlhs = AR.Cnumlhs, count, olddefer = AR.DeferFlag;
00064     int action = 0;
00065     WORD oldnumrhs = CC->numrhs, size, pow, jj;
00066     LONG oldcpointer = CC->Pointer - CC->Buffer, oldppointer = AT.pWorkPointer, lp;
00067     WORD *oldwork = AT.WorkPointer, *oldwork2, scale, renorm;
00068     WORD kLCM = 0, kGCD = 0, kGCD2, kkLCM = 0, jLCM = 0, jGCD, sign = 1;
00069     int ii, didpolyratfun;
00070     UWORD *EAscrat, *GCDBuffer = 0, *GCDBuffer2 = 0, *LCMBuffer = 0, *LCMb = 0, *LCMc = 0;
00071     AT.WorkPointer += *term;
00072     start = C->lhs[level];
00073     AR.Cnumlhs = start[2];
00074     stop = start + start[1];
00075     type = *start;
00076     scale = start[4];
00077     renorm = start[5];
00078     start += TYPEARGHEADSIZE;
00079     /*
00080     *#[ Dollars :
00081     */
00082     if ( renorm && start[1] != 0 ) { /* We have to evaluate $ symbols inside () */
00083         t = start+1; factor = oldwork2 = v = AT.WorkPointer;
00084         i = *t; t++;
00085         *v++ = i+3; i--; NCOPY(v,t,i);
00086         *v++ = 1; *v++ = 1; *v++ = 3;
00087         AT.WorkPointer = v;

```

```

00088     start = t; AR.Eside = LHSIDEX;
00089     NewSort(BHEAD0);
00090     if ( Generator(BHEAD factor,AR.Cnumlhs) ) {
00091         LowerSortLevel();
00092         AT.WorkPointer = oldwork;
00093         return(-1);
00094     }
00095     AT.WorkPointer = v;
00096     if ( EndSort(BHEAD factor,0) < 0 ) {}
00097     if ( *factor && *(factor+*factor) != 0 ) {
00098         MLOCK(ErrorMessageLock);
00099         MesPrint("%$ in () does not evaluate into a single term");
00100         MUNLOCK(ErrorMessageLock);
00101         return(-1);
00102     }
00103     AR.Eside = RHSIDE;
00104     if ( *factor > 0 ) {
00105         v = factor+*factor;
00106         v -= ABS(v[-1]);
00107         *factor = v-factor;
00108     }
00109     AT.WorkPointer = v;
00110 }
00111 else {
00112     if ( *start < 0 ) {
00113         factor = start + 1;
00114         start += -*start;
00115     }
00116     else factor = 0;
00117 }
00118 /*
00119 #] Dollars :
00120 */
00121 t = term;
00122 r = t + *t;
00123 rstop = r - ABS(r[-1]);
00124 t++;
00125 /*
00126 #[ Argument detection : + argument statement
00127 */
00128 /*
00129 Allocate Numbers for MakeInteger here, for re-use in case multiple
00130 functions are treated in the same term.
00131 */
00132 if ( type == TYPENORM4 ) {
00133     GCDBuffer = NumberMalloc("execarg");
00134     GCDBuffer2 = NumberMalloc("execarg");
00135     LCMbuffer = NumberMalloc("execarg");
00136     LCMb = NumberMalloc("execarg"); LCMc = NumberMalloc("execarg");
00137 }
00138 didpolyratfun = 0;
00139 while ( t < rstop ) {
00140     if ( *t >= FUNCTION && functions[*t-FUNCTION].spec <= 0 ) {
00141 /*
00142         We have a function. First count the number of arguments.
00143         Tensors are excluded.
00144 */
00145         count = 0;
00146         v = t;
00147         m = t + FUNHEAD;
00148         r = t + t[1];
00149         while ( m < r ) {
00150             count++;
00151             NEXTARG(m)
00152         }
00153         if ( count <= 0 ) { t += t[1]; continue; }
00154 /*
00155         Now we take the arguments one by one and test for a match
00156 */
00157         for ( i = 1; i <= count; i++ ) {
00158             m = start;
00159             while ( m < stop ) {
00160                 r = m + m[1];
00161                 j = *r++;
00162                 if ( j > 1 ) {
00163                     while ( --j > 0 ) {
00164                         if ( *r == i ) goto RightNum;
00165                         r++;
00166                     }
00167                     m = r;
00168                     continue;
00169                 }
00170 RightNum:
00171                 if ( m[1] == 2 ) {
00172 #ifdef WITHFLOAT
00173                     if ( *t != FLOATFUN || AT.aux_ == 0 || TestFloat(t) == 0 )
00174 #endif

```

```

00175         {
00176             m += 2;
00177             m += *m;
00178             goto HaveToDo;
00179         }
00180     }
00181     else {
00182         r = m + m[1];
00183         m += 2;
00184         while ( m < r ) {
00185             if ( *m == CSET ) {
00186                 r1 = SetElements + Sets[m[1]].first;
00187                 r2 = SetElements + Sets[m[1]].last;
00188                 while ( r1 < r2 ) {
00189                     if ( *r1++ == *t ) goto HaveToDo;
00190                 }
00191             }
00192             else if ( m[1] == *t ) goto HaveToDo;
00193             m += 2;
00194         }
00195     }
00196     m += *m;
00197 }
00198 continue;
00199 HaveToDo:
00200 /*
00201     If we come here we have to do the argument i (first is 1).
00202 */
00203 sign = 1;
00204 action = 1;
00205 if ( *t == AR.PolyFun ) didpolyratfun = 1;
00206 v[2] |= DIRTYFLAG;
00207 r = t + FUNHEAD;
00208 j = i;
00209 while ( --j > 0 ) { NEXTARG(r) }
00210 if ( ( type == TYPESPLITARG ) || ( type == TYPESPLITFIRSTARG )
00211     || ( type == TYPESPLITLASTARG ) ) {
00212     if ( *t > FUNCTION && *r > 0 ) {
00213         WantAddPointers(2);
00214         AT.pWorkSpace[AT.pWorkPointer++] = t;
00215         AT.pWorkSpace[AT.pWorkPointer++] = r;
00216     }
00217     continue;
00218 }
00219 else if ( type == TYPESPLITARG2 ) {
00220     if ( *t > FUNCTION && *r > 0 ) {
00221         WantAddPointers(2);
00222         AT.pWorkSpace[AT.pWorkPointer++] = t;
00223         AT.pWorkSpace[AT.pWorkPointer++] = r;
00224     }
00225     continue;
00226 }
00227 else if ( type == TYPEFACTARG || type == TYPEFACTARG2 ) {
00228     if ( *t > FUNCTION || *t == DENOMINATOR ) {
00229         if ( *r > 0 ) {
00230             mm = r + ARGHEAD; mstop = r + *r;
00231             if ( mm + *mm < mstop ) {
00232                 WantAddPointers(2);
00233                 AT.pWorkSpace[AT.pWorkPointer++] = t;
00234                 AT.pWorkSpace[AT.pWorkPointer++] = r;
00235                 continue;
00236             }
00237             if ( *mm == 1+ABS(mstop[-1]) ) continue;
00238             if ( mstop[-3] != 1 || mstop[-2] != 1
00239                 || mstop[-1] != 3 ) {
00240                 WantAddPointers(2);
00241                 AT.pWorkSpace[AT.pWorkPointer++] = t;
00242                 AT.pWorkSpace[AT.pWorkPointer++] = r;
00243                 continue;
00244             }
00245             GETSTOP(mm,mstop); mm++;
00246             if ( mm + mm[1] < mstop ) {
00247                 WantAddPointers(2);
00248                 AT.pWorkSpace[AT.pWorkPointer++] = t;
00249                 AT.pWorkSpace[AT.pWorkPointer++] = r;
00250                 continue;
00251             }
00252             if ( *mm == SYMBOL && ( mm[1] > 4 ||
00253                 ( mm[3] != 1 && mm[3] != -1 ) ) ) {
00254                 WantAddPointers(2);
00255                 AT.pWorkSpace[AT.pWorkPointer++] = t;
00256                 AT.pWorkSpace[AT.pWorkPointer++] = r;
00257                 continue;
00258             }
00259             else if ( *mm == DOTPRODUCT && ( mm[1] > 5 ||
00260                 ( mm[4] != 1 && mm[4] != -1 ) ) ) {
00261                 WantAddPointers(2);

```

```

00262         AT.pWorkSpace[AT.pWorkPointer++] = t;
00263         AT.pWorkSpace[AT.pWorkPointer++] = r;
00264         continue;
00265     }
00266     else if ( ( *mm == DELTA || *mm == VECTOR )
00267         && mm[1] > 4 ) {
00268         WantAddPointers(2);
00269         AT.pWorkSpace[AT.pWorkPointer++] = t;
00270         AT.pWorkSpace[AT.pWorkPointer++] = r;
00271         continue;
00272     }
00273     }
00274     else if ( factor && *factor == 4 && factor[2] == 1 ) {
00275         WantAddPointers(2);
00276         AT.pWorkSpace[AT.pWorkPointer++] = t;
00277         AT.pWorkSpace[AT.pWorkPointer++] = r;
00278         continue;
00279     }
00280     else if ( factor && *factor == 0
00281     && ( *r == -SNUMBER && r[1] != 1 ) ) {
00282         WantAddPointers(2);
00283         AT.pWorkSpace[AT.pWorkPointer++] = t;
00284         AT.pWorkSpace[AT.pWorkPointer++] = r;
00285         continue;
00286     }
00287     else if ( *r == -MINVECTOR ) {
00288         WantAddPointers(2);
00289         AT.pWorkSpace[AT.pWorkPointer++] = t;
00290         AT.pWorkSpace[AT.pWorkPointer++] = r;
00291         continue;
00292     }
00293     }
00294     continue;
00295 }
00296 else if ( type == TYPENORM || type == TYPENORM2 || type == TYPENORM3 || type ==
TYPENORM4 ) {
00297     if ( *r < 0 ) {
00298         WORD rone;
00299         if ( *r == -MINVECTOR ) { rone = -1; *r = -INDEX; }
00300         else if ( *r != -SNUMBER || r[1] == 1 || r[1] == 0 ) continue;
00301         else { rone = r[1]; r[1] = 1; }
00302     /*
00303     Now we must multiply the general coefficient by r[1]
00304     */
00305     if ( scale && ( factor == 0 || *factor ) ) {
00306         action = 1;
00307         v[2] |= DIRTYFLAG;
00308         if ( rone < 0 ) {
00309             if ( type == TYPENORM3 ) k = 1;
00310             else k = -1;
00311             rone = -rone;
00312         }
00313         else k = 1;
00314         r1 = term + *term;
00315         size = r1[-1];
00316         size = REDLENG(size);
00317         if ( scale > 0 ) {
00318             for ( jj = 0; jj < scale; jj++ ) {
00319                 if ( Mully(BHEAD (UWORD *)rstop,&size,(UWORD *)(&rone),k) )
00320                     goto execargerr;
00321             }
00322         }
00323         else {
00324             for ( jj = 0; jj > scale; jj-- ) {
00325                 if ( Divvy(BHEAD (UWORD *)rstop,&size,(UWORD *)(&rone),k) )
00326                     goto execargerr;
00327             }
00328         }
00329         size = INCLENG(size);
00330         k = size < 0 ? -size: size;
00331         rstop[k-1] = size;
00332         *term = (WORD)(rstop - term) + k;
00333     }
00334     continue;
00335 }
00336 /*
00337 Now we have to find a reference term.
00338 If factor is defined and *factor != 0 we have to
00339 look for the first term that matches the pattern exactly
00340 Otherwise the first term plays this role
00341 If its coefficient is not one,
00342 we must set up a division of the whole argument by
00343 this coefficient, and a multiplication of the term
00344 when the type is not equal to TYPENORM2.
00345 We first multiply the coefficient of the term.
00346 Then we set up the division.
00347

```

```

00348                                     First find the magic term
00349 */
00350                                     if ( type == TYPENORM4 ) {
00351 /*
00352                                     For normalizing everything to integers we have to
00353                                     determine for all elements of this argument the LCM of
00354                                     the denominators and the GCD of the numerators.
00355                                     The buffers have been allocated already.
00356 */
00357                                     r4 = r + *r;
00358                                     r1 = r + ARGHEAD;
00359 /*
00360                                     First take the first term to load up the LCM and the GCD
00361 */
00362                                     r2 = r1 + *r1;
00363                                     j = r2[-1];
00364                                     if ( j < 0 ) sign = -1;
00365                                     r3 = r2 - ABS(j);
00366                                     k = REDLENG(j);
00367                                     if ( k < 0 ) k = -k;
00368                                     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00369                                     for ( kGCD = 0; kGCD < k; kGCD++ ) GCDBuffer[kGCD] = r3[kGCD];
00370                                     k = REDLENG(j);
00371                                     if ( k < 0 ) k = -k;
00372                                     r3 += k;
00373                                     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00374                                     for ( kLCM = 0; kLCM < k; kLCM++ ) LCMbuffer[kLCM] = r3[kLCM];
00375                                     r1 = r2;
00376 /*
00377                                     Now go through the rest of the terms in this argument.
00378 */
00379                                     while ( r1 < r4 ) {
00380                                         r2 = r1 + *r1;
00381                                         j = r2[-1];
00382                                         r3 = r2 - ABS(j);
00383                                         k = REDLENG(j);
00384                                         if ( k < 0 ) k = -k;
00385                                         while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00386                                         if ( ( GCDBuffer[0] == 1 ) && ( kGCD == 1 ) ) {
00387 /*
00388                                             GCD is already 1
00389 */
00390                                             }
00391                                         else if ( ( ( k != 1 ) || ( r3[0] != 1 ) ) ) {
00392                                             if ( GcdLong(BHEAD GCDBuffer,kGCD, (UWORD *)r3,k,GCDBuffer2,&kGCD2) ) {
00393                                                 NumberFree(GCDBuffer,"execarg");
00394                                                 NumberFree(GCDBuffer2,"execarg");
00395                                                 NumberFree(LCMbuffer,"execarg");
00396                                                 NumberFree(LCMb,"execarg"); NumberFree(LCMc,"execarg");
00397                                                 goto execargerr;
00398                                             }
00399                                             kGCD = kGCD2;
00400                                             for ( ii = 0; ii < kGCD; ii++ ) GCDBuffer[ii] = GCDBuffer2[ii];
00401                                         }
00402                                         else {
00403                                             kGCD = 1; GCDBuffer[0] = 1;
00404                                         }
00405                                         k = REDLENG(j);
00406                                         if ( k < 0 ) k = -k;
00407                                         r3 += k;
00408                                         while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00409                                         if ( ( LCMbuffer[0] == 1 ) && ( kLCM == 1 ) ) {
00410                                             for ( kLCM = 0; kLCM < k; kLCM++ )
00411                                                 LCMbuffer[kLCM] = r3[kLCM];
00412                                         }
00413                                         else if ( ( k != 1 ) || ( r3[0] != 1 ) ) {
00414                                             if ( GcdLong(BHEAD LCMbuffer,kLCM, (UWORD *)r3,k,LCMb,&kkLCM) ) {
00415                                                 NumberFree(GCDBuffer,"execarg"); NumberFree(GCDBuffer2,"execarg");
00416                                                 NumberFree(LCMbuffer,"execarg"); NumberFree(LCMb,"execarg");
00417                                                 NumberFree(LCMc,"execarg");
00418                                                 goto execargerr;
00419                                             }
00420                                             DivLong( (UWORD *)r3,k,LCMb,kkLCM,LCMb,&kkLCM,LCMc,&jLCM);
00421                                             Mullong(LCMbuffer,kLCM,LCMb,kkLCM,LCMc,&jLCM);
00422                                             for ( kLCM = 0; kLCM < jLCM; kLCM++ )
00423                                                 LCMbuffer[kLCM] = LCMc[kLCM];
00424                                         }
00425                                         else {} /* LCM doesn't change */
00426                                         r1 = r2;
00427 /*
00428                                     Now put the factor together: GCD/LCM
00429 */
00430                                     r3 = (WORD *) (GCDBuffer);
00431                                     if ( kGCD == kLCM ) {
00432                                         for ( jGCD = 0; jGCD < kGCD; jGCD++ )
00433                                             r3[jGCD+kGCD] = LCMbuffer[jGCD];

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00434         k = kGCD;
00435     }
00436     else if ( kGCD > kLCM ) {
00437         for ( jGCD = 0; jGCD < kLCM; jGCD++ )
00438             r3[jGCD+kGCD] = LCMBuffer[jGCD];
00439         for ( jGCD = kLCM; jGCD < kGCD; jGCD++ )
00440             r3[jGCD+kGCD] = 0;
00441         k = kGCD;
00442     }
00443     else {
00444         for ( jGCD = kGCD; jGCD < kLCM; jGCD++ )
00445             r3[jGCD] = 0;
00446         for ( jGCD = 0; jGCD < kLCM; jGCD++ )
00447             r3[jGCD+kLCM] = LCMBuffer[jGCD];
00448         k = kLCM;
00449     }
00450
00451     j = 2*k+1;
00452 /*
00453     Now we have to correct the overall factor
00454 */
00455     if ( scale && ( factor == 0 || *factor > 0 ) )
00456         goto ScaledVariety;
00457 /*
00458     The if was added 28-nov-2012 to give MakeInteger also
00459     the (0) option.
00460 */
00461     if ( scale && ( factor == 0 || *factor ) ) {
00462         size = term[*term-1];
00463         size = REDLENG(size);
00464         if ( MulRat(BHEAD (UWORD *)rstop,size,(UWORD *)r3,k,
00465             (UWORD *)rstop,&size) ) goto execargerr;
00466         size = INCLENG(size);
00467         k = size < 0 ? -size: size;
00468         rstop[k-1] = size*sign;
00469         *term = (WORD)(rstop - term) + k;
00470     }
00471 }
00472 else {
00473     if ( factor && *factor >= 1 ) {
00474         r4 = r + *r;
00475         r1 = r + ARGHEAD;
00476         while ( r1 < r4 ) {
00477             r2 = r1 + *r1;
00478             r3 = r2 - ABS(r2[-1]);
00479             j = r3 - r1;
00480             r5 = factor;
00481             if ( j != *r5 ) { r1 = r2; continue; }
00482             r5++; r6 = r1+1;
00483             while ( --j > 0 ) {
00484                 if ( *r5 != *r6 ) break;
00485                 r5++; r6++;
00486             }
00487             if ( j > 0 ) { r1 = r2; continue; }
00488             break;
00489         }
00490         if ( r1 >= r4 ) continue;
00491     }
00492     else {
00493         r1 = r + ARGHEAD;
00494         r2 = r1 + *r1;
00495         r3 = r2 - ABS(r2[-1]);
00496     }
00497     if ( *r3 == 1 && r3[1] == 1 ) {
00498         if ( r2[-1] == 3 ) continue;
00499         if ( r2[-1] == -3 && type == TYPENORM3 ) continue;
00500     }
00501     action = 1;
00502     v[2] |= DIRTYFLAG;
00503     j = r2[-1];
00504     k = REDLENG(j);
00505     if ( j < 0 ) j = -j;
00506     if ( type == TYPENORM && scale && ( factor == 0 || *factor ) ) {
00507 /*
00508         Now we correct the overall factor
00509 */
00510     ScaledVariety;;
00511         size = term[*term-1];
00512         size = REDLENG(size);
00513         if ( scale > 0 ) {
00514             for ( jj = 0; jj < scale; jj++ ) {
00515                 if ( MulRat(BHEAD (UWORD *)rstop,size,(UWORD *)r3,k,
00516                     (UWORD *)rstop,&size) ) goto execargerr;
00517             }
00518         }
00519         else {
00520             for ( jj = 0; jj > scale; jj-- ) {

```

```

00521                                     if ( DivRat(BHEAD (UWORD *)rstop,size,(UWORD *)r3,k,
00522                                     (UWORD *)rstop,&size) ) goto execargerr;
00523                                     }
00524                                     }
00525                                     size = INCLENG(size);
00526                                     k = size < 0 ? -size: size;
00527                                     rstop[k-1] = size*sign;
00528                                     *term = (WORD)(rstop - term) + k;
00529                                     }
00530                                     }
00531                                     /*
00532                                     We generate a statement for adapting all terms in the
00533                                     argument successively
00534                                     */
00535                                     r4 = AddRHS(AT.ebufnum,1);
00536                                     while ( (r4+j+12) > CC->Top ) r4 = DoubleCbuffer(AT.ebufnum,r4,3);
00537                                     *r4++ = j+1;
00538                                     i = (j-1)/2; /* was (j-1)*2 ????? 17-oct-2017 */
00539                                     for ( k = 0; k < i; k++ ) *r4++ = r3[i+k];
00540                                     for ( k = 0; k < i; k++ ) *r4++ = r3[k];
00541                                     if ( ( type == TYPENORM3 ) || ( type == TYPENORM4 ) ) *r4++ = j*sign;
00542                                     else *r4++ = r3[j-1];
00543                                     *r4++ = 0;
00544                                     CC->rhs[CC->numrhs+1] = r4;
00545                                     CC->Pointer = r4;
00546                                     AT.mulpat[5] = CC->numrhs;
00547                                     AT.mulpat[7] = AT.ebufnum;
00548                                     }
00549                                     else if ( type == TYPEARGTOEXTRASymbol ) {
00550                                     WORD n;
00551                                     if ( r[0] < 0 ) {
00552                                     /* The argument is in the fast notation. */
00553                                     WORD tmp[Max(9,FUNHEAD+5)];
00554                                     switch ( r[0] ) {
00555                                     case -SNUMBER:
00556                                     if ( r[1] == 0 ) {
00557                                     tmp[0] = 0;
00558                                     }
00559                                     else {
00560                                     tmp[0] = 4;
00561                                     tmp[1] = ABS(r[1]);
00562                                     tmp[2] = 1;
00563                                     tmp[3] = r[1] > 0 ? 3 : -3;
00564                                     tmp[4] = 0;
00565                                     }
00566                                     break;
00567                                     case -SYMBOL:
00568                                     tmp[0] = 8;
00569                                     tmp[1] = SYMBOL;
00570                                     tmp[2] = 4;
00571                                     tmp[3] = r[1];
00572                                     tmp[4] = 1;
00573                                     tmp[5] = 1;
00574                                     tmp[6] = 1;
00575                                     tmp[7] = 3;
00576                                     tmp[8] = 0;
00577                                     break;
00578                                     case -INDEX:
00579                                     case -VECTOR:
00580                                     case -MINVECTOR:
00581                                     tmp[0] = 7;
00582                                     tmp[1] = INDEX;
00583                                     tmp[2] = 3;
00584                                     tmp[3] = r[1];
00585                                     tmp[4] = 1;
00586                                     tmp[5] = 1;
00587                                     tmp[6] = r[0] != -MINVECTOR ? 3 : -3;
00588                                     tmp[7] = 0;
00589                                     break;
00590                                     default:
00591                                     if ( r[0] <= -FUNCTION ) {
00592                                     tmp[0] = FUNHEAD+4;
00593                                     tmp[1] = -r[0];
00594                                     tmp[2] = FUNHEAD;
00595                                     ZeroFillRange(tmp,3,1+FUNHEAD);
00596                                     tmp[FUNHEAD+1] = 1;
00597                                     tmp[FUNHEAD+2] = 1;
00598                                     tmp[FUNHEAD+3] = 3;
00599                                     tmp[FUNHEAD+4] = 0;
00600                                     break;
00601                                     }
00602                                     else {
00603                                     MLOCK(ErrorMessageLock);
00604                                     MesPrint("Unknown fast notation found (TYPEARGTOEXTRASymbol)");
00605                                     MUNLOCK(ErrorMessageLock);
00606                                     return(-1);
00607                                     }

```

```

00608         }
00609         n = FindSubexpression(tmp);
00610     }
00611     else {
00612         /*
00613          * NOTE: writing to r[r[0]] is legal. As long as we work
00614          * in a part of the term, at least the coefficient of
00615          * the term must follow.
00616          */
00617         WORD old_rr0 = r[r[0]];
00618         r[r[0]] = 0; /* zero-terminated */
00619         n = FindSubexpression(r+ARGHEAD);
00620         r[r[0]] = old_rr0;
00621     }
00622     /* Put the new argument in the work space. */
00623     if ( AT.WorkPointer+2 > AT.WorkTop ) {
00624         MLOCK(ErrorMessageLock);
00625         MesWork();
00626         MUNLOCK(ErrorMessageLock);
00627         return(-1);
00628     }
00629     r1 = AT.WorkPointer;
00630     if ( scale ) { /* means "tonumber" */
00631         r1[0] = -SNUMBER;
00632         r1[1] = n;
00633     }
00634     else {
00635         r1[0] = -SYMBOL;
00636         r1[1] = MAXVARIABLES-n;
00637     }
00638     /* We need r2, r3, m and k to shift the data. */
00639     r2 = r + ( r[0] > 0 ? r[0] : r[0] <= -FUNCTION ? 1 : 2 );
00640     r3 = r;
00641     m = r1+ARGHEAD+2;
00642     k = 2;
00643     goto do_shift;
00644 }
00645 r3 = r;
00646 AR.DeferFlag = 0;
00647 if ( *r > 0 ) {
00648     NewSort(BHEAD0);
00649     action = 1;
00650     r2 = r + *r;
00651     r += ARGHEAD;
00652     while ( r < r2 ) { /* Sum over the terms */
00653         m = AT.WorkPointer;
00654         j = *r;
00655         while ( --j >= 0 ) *m++ = *r++;
00656         r1 = AT.WorkPointer;
00657         AT.WorkPointer = m;
00658     } /*
00659     What to do with dummy indices?
00660     */
00661     if ( type == TYPENORM || type == TYPENORM2 || type == TYPENORM3 || type ==
TYPENORM4 ) {
00662         if ( MultDo(BHEAD r1,AT.mulpat) ) goto execargerr;
00663         AT.WorkPointer = r1 + *r1;
00664     }
00665     if ( Generator(BHEAD r1,level) ) goto execargerr;
00666     AT.WorkPointer = r1;
00667 }
00668 }
00669 else {
00670     r2 = r + (( *r <= -FUNCTION ) ? 1:2);
00671     r1 = AT.WorkPointer;
00672     ToGeneral(r,r1,0);
00673     m = r1 + ARGHEAD;
00674     AT.WorkPointer = r1 + *r1;
00675     NewSort(BHEAD0);
00676     action = 1;
00677 } /*
00678 What to do with dummy indices?
00679 */
00680 if ( type == TYPENORM || type == TYPENORM2 || type == TYPENORM3 || type ==
TYPENORM4 ) {
00681     if ( MultDo(BHEAD m,AT.mulpat) ) goto execargerr;
00682     AT.WorkPointer = m + *m;
00683 }
00684 if ( (*m != 0) && Generator(BHEAD m,level) ) goto execargerr;
00685 AT.WorkPointer = r1;
00686 }
00687 if ( EndSort(BHEAD AT.WorkPointer+ARGHEAD,1) < 0 ) goto execargerr;
00688 AR.DeferFlag = olddefer;
00689 /*
00690 Now shift the sorted entity over the old argument.
00691 */
00692 m = AT.WorkPointer+ARGHEAD;

```



```

00693         while ( *m ) m += *m;
00694         k = WORDDIF(m,AT.WorkPointer);
00695         *AT.WorkPointer = k;
00696         AT.WorkPointer[1] = 0;
00697         if ( ToFast(AT.WorkPointer,AT.WorkPointer) ) {
00698             if ( *AT.WorkPointer <= -FUNCTION ) k = 1;
00699             else k = 2;
00700         }
00701 do_shift:
00702         if ( *r3 > 0 ) j = k - *r3;
00703         else if ( *r3 <= -FUNCTION ) j = k - 1;
00704         else j = k - 2;
00705
00706         t[1] += j;
00707         action = 1;
00708         v[2] |= DIRTYFLAG;
00709         if ( j > 0 ) {
00710             r = m + j;
00711             while ( m > AT.WorkPointer ) *--r = *--m;
00712             AT.WorkPointer = r;
00713             m = term + *term;
00714             r = m + j;
00715             while ( m > r2 ) *--r = *--m;
00716         }
00717         else if ( j < 0 ) {
00718             r = r2 + j;
00719             r1 = term + *term;
00720             while ( r2 < r1 ) *r++ = *r2++;
00721         }
00722         r = r3;
00723         m = AT.WorkPointer;
00724         NCOPY(r,m,k);
00725         *term += j;
00726         rstop += j;
00727         CC->numrhs = oldnumrhs;
00728         CC->Pointer = CC->Buffer + oldcpointer;
00729     }
00730 }
00731 t += t[1];
00732 }
00733 /*
00734     If TYPENORM4, we allocated Number buffers before the above while loop. Free them.
00735 */
00736 if ( type == TYPENORM4 ) {
00737     NumberFree(GCdbuffer,"execarg");
00738     NumberFree(GCdbuffer2,"execarg");
00739     NumberFree(LCMbuffer,"execarg");
00740     NumberFree(LCMb,"execarg"); NumberFree(LCMc,"execarg");
00741 }
00742 if ( didpolyratfun ) {
00743     PolyFunDirty(BHEAD term);
00744     didpolyratfun = 0;
00745 }
00746 /*
00747 #] Argument detection :
00748 #[ SplitArg : + varieties
00749 */
00750 if ( ( type == TYPESPLITARG || type == TYPESPLITARG2
00751     || type == TYPESPLITFIRSTARG || type == TYPESPLITLASTARG ) &&
00752     AT.pWorkPointer > oldppointer ) {
00753     t = term+1;
00754     r1 = AT.WorkPointer + 1;
00755     lp = oldppointer;
00756     while ( t < rstop ) {
00757         if ( lp < AT.pWorkPointer && t == AT.pWorkspace[lp] ) {
00758             v = t;
00759             m = t + FUNHEAD;
00760             r = t + t[1];
00761             r2 = r1; while ( t < m ) *r1++ = *t++;
00762             while ( m < r ) {
00763                 t = m;
00764                 NEXTARG(m)
00765                 if ( lp >= AT.pWorkPointer || t != AT.pWorkspace[lp+1] ) {
00766                     if ( *t > 0 ) t[1] = 0;
00767                     while ( t < m ) *r1++ = *t++;
00768                     continue;
00769                 }
00770             }
00771             /*
00772             Now we have a nontrivial argument that should be done.
00773             */
00774             lp += 2;
00775             action = 1;
00776             v[2] |= DIRTYFLAG;
00777             r3 = t + *t;
00778             t += ARGHEAD;
00779             if ( type == TYPESPLITFIRSTARG ) {
00780                 r4 = r1; r5 = t; r7 = oldwork;

```

```

00780         *r1++ = *t + ARGHEAD;
00781     for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
00782     j = 0;
00783     while ( t < r3 ) {
00784         i = *t;
00785         if ( j == 0 ) {
00786             NCOPY(r7,t,i)
00787             j++;
00788         }
00789         else {
00790             NCOPY(r1,t,i)
00791         }
00792     }
00793     *r4 = r1 - r4;
00794     if ( j ) {
00795         if ( ToFast(r4,r4) ) {
00796             r1 = r4;
00797             if ( *r1 > -FUNCTION ) r1++;
00798             r1++;
00799         }
00800         r7 = oldwork;
00801         while ( --j >= 0 ) {
00802             r4 = r1; i = *r7;
00803             *r1++ = i+ARGHEAD; *r1++ = 0;
00804             FILLARG(r1);
00805             NCOPY(r1,r7,i)
00806             if ( ToFast(r4,r4) ) {
00807                 r1 = r4;
00808                 if ( *r1 > -FUNCTION ) r1++;
00809                 r1++;
00810             }
00811         }
00812     }
00813     t = r3;
00814 }
00815 else if ( type == TYPESPLITLASTARG ) {
00816     r4 = r1; r5 = t; r7 = oldwork;
00817     *r1++ = *t + ARGHEAD;
00818     for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
00819     j = 0;
00820     while ( t < r3 ) {
00821         i = *t;
00822         if ( t+i >= r3 ) {
00823             NCOPY(r7,t,i)
00824             j++;
00825         }
00826         else {
00827             NCOPY(r1,t,i)
00828         }
00829     }
00830     *r4 = r1 - r4;
00831     if ( j ) {
00832         if ( ToFast(r4,r4) ) {
00833             r1 = r4;
00834             if ( *r1 > -FUNCTION ) r1++;
00835             r1++;
00836         }
00837         r7 = oldwork;
00838         while ( --j >= 0 ) {
00839             r4 = r1; i = *r7;
00840             *r1++ = i+ARGHEAD; *r1++ = 0;
00841             FILLARG(r1);
00842             NCOPY(r1,r7,i)
00843             if ( ToFast(r4,r4) ) {
00844                 r1 = r4;
00845                 if ( *r1 > -FUNCTION ) r1++;
00846                 r1++;
00847             }
00848         }
00849     }
00850     t = r3;
00851 }
00852 else if ( factor == 0 || ( type == TYPESPLITARG2 && *factor == 0 ) ) {
00853     while ( t < r3 ) {
00854         r4 = r1;
00855         *r1++ = *t + ARGHEAD;
00856         for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
00857         i = *t;
00858         while ( --i >= 0 ) *r1++ = *t++;
00859         if ( ToFast(r4,r4) ) {
00860             r1 = r4;
00861             if ( *r1 > -FUNCTION ) r1++;
00862             r1++;
00863         }
00864     }
00865 }
00866 else if ( type == TYPESPLITARG2 ) {

```

```

00867 /*
00868 Here we better put the pattern matcher at work?
00869 Remember: there are no wildcards.
00870 */
00871 WORD *oRepFunList = AN.RepFunList;
00872 WORD *oWildMask = AT.WildMask, *oWildValue = AN.WildValue;
00873 AN.WildValue = AT.locwildvalue; AT.WildMask = AT.locwildvalue+2;
00874 AN.NumWild = 0;
00875 r4 = r1; r5 = t; r7 = oldwork;
00876 *r1++ = *t + ARGHEAD;
00877 for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
00878 j = 0;
00879 while ( t < r3 ) {
00880     AN.UseFindOnly = 0; oldwork2 = AT.WorkPointer;
00881     AN.RepFunList = r1;
00882     AT.WorkPointer = r1+AN.RepFunNum+2;
00883     i = *t;
00884     if ( FindRest(BHEAD t,factor) &&
00885         ( AN.UsedOtherFind || FindOnce(BHEAD t,factor) ) ) {
00886         NCOPY(r7,t,i)
00887         j++;
00888     }
00889     else if ( factor[0] == FUNHEAD+1 && factor[1] >= FUNCTION ) {
00890         WORD *rr1 = t+1, *rr2 = t+i;
00891         rr2 -= ABS(rr2[-1]);
00892         while ( rr1 < rr2 ) {
00893             if ( *rr1 == factor[1] ) break;
00894             rr1 += rr1[1];
00895         }
00896         if ( rr1 < rr2 ) {
00897             NCOPY(r7,t,i)
00898             j++;
00899         }
00900         else {
00901             NCOPY(r1,t,i)
00902         }
00903     }
00904     else {
00905         NCOPY(r1,t,i)
00906     }
00907     AT.WorkPointer = oldwork2;
00908 }
00909 AN.RepFunList = oRepFunList;
00910 *r4 = r1 - r4;
00911 if ( j ) {
00912     if ( ToFast(r4,r4) ) {
00913         r1 = r4;
00914         if ( *r1 > -FUNCTION ) r1++;
00915         r1++;
00916     }
00917     r7 = oldwork;
00918     while ( --j >= 0 ) {
00919         r4 = r1; i = *r7;
00920         *r1++ = i+ARGHEAD; *r1++ = 0;
00921         FILLARG(r1);
00922         NCOPY(r1,r7,i)
00923         if ( ToFast(r4,r4) ) {
00924             r1 = r4;
00925             if ( *r1 > -FUNCTION ) r1++;
00926             r1++;
00927         }
00928     }
00929 }
00930 t = r3;
00931 AT.WildMask = oWildMask; AN.WildValue = oWildValue;
00932 }
00933 else {
00934 /*
00935 This code deals with splitting off a single term
00936 */
00937 r4 = r1; r5 = t;
00938 *r1++ = *t + ARGHEAD;
00939 for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
00940 j = 0;
00941 while ( t < r3 ) {
00942     r6 = t + *t; r6 -= ABS(r6[-1]);
00943     if ( (r6 - t) == *factor ) {
00944         k = *factor - 1;
00945         for ( ; k > 0; k-- ) {
00946             if ( t[k] != factor[k] ) break;
00947         }
00948         if ( k <= 0 ) {
00949             j = r3 - t; t += *t; continue;
00950         }
00951     }
00952     else if ( (r6 - t) == 1 && *factor == 0 ) {
00953         j = r3 - t; t += *t; continue;

```

```

00954         }
00955         i = *t;
00956         NCOPY(r1,t,i)
00957     }
00958     *r4 = r1 - r4;
00959     if ( j ) {
00960         if ( ToFast(r4,r4) ) {
00961             r1 = r4;
00962             if ( *r1 > -FUNCTION ) r1++;
00963             r1++;
00964         }
00965         t = r3 - j;
00966         r4 = r1;
00967         *r1++ = *t + ARGHEAD;
00968         for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
00969         i = *t;
00970         while ( --i >= 0 ) *r1++ = *t++;
00971         if ( ToFast(r4,r4) ) {
00972             r1 = r4;
00973             if ( *r1 > -FUNCTION ) r1++;
00974             r1++;
00975         }
00976     }
00977     t = r3;
00978 }
00979 }
00980 r2[1] = r1 - r2;
00981 }
00982 else {
00983     r = t + t[1];
00984     while ( t < r ) *r1++ = *t++;
00985 }
00986 }
00987 r = term + *term;
00988 while ( t < r ) *r1++ = *t++;
00989 m = AT.WorkPointer;
00990 i = m[0] = r1 - m;
00991 t = term;
00992 while ( --i >= 0 ) *t++ = *m++;
00993 if ( AT.WorkPointer < m ) AT.WorkPointer = m;
00994 }
00995 /*
00996 #] SplitArg :
00997 #[ FACTARG :
00998 */
00999 if ( ( type == TYPEFACTARG || type == TYPEFACTARG2 ) &&
01000 AT.pWorkPointer > oldppointer ) {
01001     t = term+1;
01002     r1 = AT.WorkPointer + 1;
01003     lp = oldppointer;
01004     while ( t < rstop ) {
01005         if ( lp < AT.pWorkPointer && AT.pWorkspace[lp] == t ) {
01006             v = t;
01007             m = t + FUNHEAD;
01008             r = t + t[1];
01009             r2 = r1; while ( t < m ) *r1++ = *t++;
01010             while ( m < r ) {
01011                 rr = t = m;
01012                 NEXTARG(m)
01013                 if ( lp >= AT.pWorkPointer || AT.pWorkspace[lp+1] != t ) {
01014                     if ( *t > 0 ) t[1] = 0;
01015                     while ( t < m ) *r1++ = *t++;
01016                     continue;
01017                 }
01018             }
01019             /*
01020             Now we have a nontrivial argument that should be studied.
01021             Try to find common factors.
01022             */
01023             lp += 2;
01024             if ( *t < 0 ) {
01025                 if ( factor && ( *factor == 0 && *t == -SNUMBER ) ) {
01026                     *r1++ = *t++;
01027                     if ( *t == 0 ) *r1++ = *t++;
01028                     else { *r1++ = 1; t++; }
01029                     continue;
01030                 }
01031                 else if ( factor && *factor == 4 && factor[2] == 1 ) {
01032                     if ( *t == -SNUMBER ) {
01033                         if ( factor[3] < 0 || t[1] >= 0 ) {
01034                             while ( t < m ) *r1++ = *t++;
01035                         }
01036                     }
01037                     else {
01038                         *r1++ = -SNUMBER; *r1++ = -1;
01039                         *r1++ = *t++; *r1++ = -*t++;
01040                     }
01041                 }
01042             }
01043             else {

```

```

01041             while ( t < m ) *r1++ = *t++;
01042             *r1++ = -SNUMBER; *r1++ = 1;
01043         }
01044         continue;
01045     }
01046     else if ( *t == -MINVECTOR ) {
01047         if ( AC.OldFactArgFlag == NEWFACTARG ) {
01048             *r1++ = -SNUMBER; *r1++ = -1;
01049             *r1++ = -VECTOR; t++; *r1++ = *t++;
01050         }
01051         else {
01052             *r1++ = -VECTOR; t++; *r1++ = *t++;
01053             *r1++ = -SNUMBER; *r1++ = -1;
01054             *r1++ = -SNUMBER; *r1++ = 1;
01055         }
01056         continue;
01057     }
01058 }
01059 /*
01060 Now we have a nontrivial argument
01061 */
01062 r3 = t + *t;
01063 t += ARGHEAD; r5 = t; /* Store starting point */
01064 /* We have terms from r5 to r3 */
01065 if ( r5+r5 == r3 && factor ) { /* One term only */
01066     if ( *factor == 0 ) {
01067         GETSTOP(t,r6);
01068         r9 = r1; *r1++ = 0; *r1++ = 1;
01069         FILLARG(r1);
01070         *r1++ = (r6-t)+3; t++;
01071         while ( t < r6 ) *r1++ = *t++;
01072         *r1++ = 1; *r1++ = 1; *r1++ = 3;
01073         *r9 = r1-r9;
01074         if ( ToFast(r9,r9) ) {
01075             if ( *r9 <= -FUNCTION ) r1 = r9+1;
01076             else r1 = r9+2;
01077         }
01078         t = r3; continue;
01079     }
01080     if ( factor[0] == 4 && factor[2] == 1 ) {
01081         GETSTOP(t,r6);
01082         r7 = r1; *r1++ = (r6-t)+3+ARGHEAD; *r1++ = 0;
01083         FILLARG(r1);
01084         *r1++ = (r6-t)+3; t++;
01085         while ( t < r6 ) *r1++ = *t++;
01086         *r1++ = 1; *r1++ = 1; *r1++ = 3;
01087         if ( ToFast(r7,r7) ) {
01088             if ( *r7 <= -FUNCTION ) r1 = r7+1;
01089             else r1 = r7+2;
01090         }
01091         if ( r3[-1] < 0 && factor[3] > 0 ) {
01092             *r1++ = -SNUMBER; *r1++ = -1;
01093             if ( r3[-1] == -3 && r3[-2] == 1
01094                 && ( r3[-3] & MAXPOSITIVE ) == r3[-3] ) {
01095                 *r1++ = -SNUMBER; *r1++ = r3[-3];
01096             }
01097             else {
01098                 *r1++ = (r3-r6)+1+ARGHEAD;
01099                 *r1++ = 0;
01100                 FILLARG(r1);
01101                 *r1++ = (r3-r6+1);
01102                 while ( t < r3 ) *r1++ = *t++;
01103                 r1[-1] = -r1[-1];
01104             }
01105         }
01106         else {
01107             if ( ( r3[-1] == -3 || r3[-1] == 3 )
01108                 && r3[-2] == 1
01109                 && ( r3[-3] & MAXPOSITIVE ) == r3[-3] ) {
01110                 *r1++ = -SNUMBER; *r1++ = r3[-3];
01111                 if ( r3[-1] < 0 ) r1[-1] = - r1[-1];
01112             }
01113             else {
01114                 *r1++ = (r3-r6)+1+ARGHEAD;
01115                 *r1++ = 0;
01116                 FILLARG(r1);
01117                 *r1++ = (r3-r6+1);
01118                 while ( t < r3 ) *r1++ = *t++;
01119             }
01120         }
01121         t = r3; continue;
01122     }
01123 }
01124 /*
01125 Now we take the first term and look for its pieces
01126 inside the other terms.
01127

```

```

01128         It is at this point that a more general factorization
01129         routine could take over (allowing for writing the output
01130         properly of course).
01131     */
01132     if ( AC.OldFactArgFlag == NEWFACTARG ) {
01133         if ( factor == 0 ) {
01134             WORD *oldworkpointer2 = AT.WorkPointer;
01135             AT.WorkPointer = r1 + AM.MaxTer+FUNHEAD;
01136             if ( ArgFactorize(BHEAD t-ARGHEAD,r1) < 0 ) {
01137                 MesCall("ExecArg");
01138                 return(-1);
01139             }
01140             AT.WorkPointer = oldworkpointer2;
01141             t = r3;
01142             while ( *r1 ) { NEXTARG(r1) }
01143         }
01144         else {
01145             rnext = t + *t;
01146             GETSTOP(t,r6);
01147             t++;
01148             t = r5; pow = 1;
01149             while ( t < r3 ) {
01150                 t += *t; if ( t[-1] > 0 ) { pow = 0; break; }
01151             }
01152         }
01153     /*
01154         We have to add here the code for computing the GCD
01155         and to divide it out.
01156
01157     # Numerical factor :
01158     */
01159     t = r5;
01160     EAscrat = (UWORD *) (TermMalloc("execarg"));
01161     if ( t + *t == r3 ) {
01162         if ( factor == 0 || *factor > 2 ) {
01163             if ( pow > 0 ) {
01164                 *r1++ = -SNUMBER; *r1++ = -1;
01165                 t = r5;
01166                 while ( t < r3 ) {
01167                     t += *t; t[-1] = -t[-1];
01168                 }
01169                 t = rr; *r1++ = *t++; *r1++ = 1; t++;
01170                 COPYARG(r1,t);
01171                 while ( t < m ) *r1++ = *t++;
01172             }
01173         }
01174         else {
01175             GETSTOP(t,r6);
01176             ngcd = t[t[0]-1];
01177             i = abs(ngcd)-1;
01178             while ( --i >= 0 ) EAscrat[i] = r6[i];
01179             t += *t;
01180             while ( t < r3 ) {
01181                 GETSTOP(t,r6);
01182                 i = t[t[0]-1];
01183                 if ( AccumGCD(BHEAD EAscrat,&ngcd, (UWORD *)r6,i) ) goto execargerr;
01184                 if ( ngcd == 3 && EAscrat[0] == 1 && EAscrat[1] == 1 ) break;
01185                 t += *t;
01186             }
01187         }
01188     if ( ngcd != 3 || EAscrat[0] != 1 || EAscrat[1] != 1 )
01189     /*
01190     */
01191     {
01192         if ( pow ) ngcd = -ngcd;
01193         t = r5; r9 = r1; *r1++ = t[-ARGHEAD]; *r1++ = 1;
01194         FILLARG(r1); ngcd = REDLENG(ngcd);
01195         while ( t < r3 ) {
01196             GETSTOP(t,r6);
01197             r7 = t; r8 = r1;
01198             while ( r7 < r6 ) *r1++ = *r7++;
01199             t += *t;
01200             i = REDLENG(t[-1]);
01201             if ( DivRat (BHEAD (UWORD *)r6,i,EAscrat,ngcd, (UWORD *)r1,&nq) ) goto
execargerr;
01202             nq = INCLENG(nq);
01203             i = ABS(nq)-1;
01204             r1 += i; *r1++ = nq; *r8 = r1-r8;
01205         }
01206         *r9 = r1-r9;
01207         ngcd = INCLENG(ngcd);
01208         i = ABS(ngcd)-1;
01209         if ( factor && *factor == 0 ) {}
01210         else if ( ( factor && factor[0] == 4 && factor[2] == 1
&& factor[3] == -3 ) || pow == 0 ) {
01211             r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
01212             FILLARG(r1); *r1++ = i+2;
01213             for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];

```

```

01214         *r1++ = ngcd;
01215         if ( ToFast(r9,r9) ) r1 = r9+2;
01216     }
01217     else if ( factor && factor[0] == 4 && factor[2] == 1
01218 && factor[3] > 0 && pow ) {
01219         if ( ngcd < 0 ) ngcd = -ngcd;
01220         *r1++ = -SNUMBER; *r1++ = -1;
01221         r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
01222         FILLARG(r1); *r1++ = i+2;
01223         for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];
01224         *r1++ = ngcd;
01225         if ( ToFast(r9,r9) ) r1 = r9+2;
01226     }
01227     else {
01228         if ( ngcd < 0 ) ngcd = -ngcd;
01229         if ( pow ) { *r1++ = -SNUMBER; *r1++ = -1; }
01230         if ( ngcd != 3 || EAscrat[0] != 1 || EAscrat[1] != 1 ) {
01231             r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
01232             FILLARG(r1); *r1++ = i+2;
01233             for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];
01234             *r1++ = ngcd;
01235             if ( ToFast(r9,r9) ) r1 = r9+2;
01236         }
01237     }
01238 }
01239 /*
01240 #] Numerical factor :
01241     else {
01242 onetermnew;;
01243
01244         if ( factor == 0 || *factor > 2 ) {
01245             if ( pow > 0 ) {
01246                 *r1++ = -SNUMBER; *r1++ = -1;
01247                 t = r5;
01248                 while ( t < r3 ) {
01249                     t += *t; t[-1] = -t[-1];
01250                 }
01251             }
01252             t = rr; *r1++ = *t++; *r1++ = 1; t++;
01253             COPYARG(r1,t);
01254             while ( t < m ) *r1++ = *t++;
01255         }
01256     }
01257 onetermnew;;
01258 */
01259 }
01260 TermFree(EAscrat,"execarg");
01261 }
01262 }
01263 else { /* AC.OldFactArgFlag is ON */
01264 {
01265     WORD *mnext, ncom;
01266     rnext = t + *t;
01267     GETSTOP(t,r6);
01268     t++;
01269     if ( factor == 0 ) {
01270         while ( t < r6 ) {
01271 /*
01272 #[ SYMBOL :
01273 */
01274         if ( *t == SYMBOL ) {
01275             r7 = t; r8 = t + t[1]; t += 2;
01276             while ( t < r8 ) {
01277                 pow = t[1];
01278                 mm = rnext;
01279                 while ( mm < r3 ) {
01280                     mnext = mm + *mm;
01281                     GETSTOP(mm,mstop); mm++;
01282                     while ( mm < mstop ) {
01283                         if ( *mm != SYMBOL ) mm += mm[1];
01284                         else break;
01285                     }
01286                     if ( *mm == SYMBOL ) {
01287                         mstop = mm + mm[1]; mm += 2;
01288                         while ( *mm != *t && mm < mstop ) mm += 2;
01289                         if ( mm >= mstop ) pow = 0;
01290                         else if ( pow > 0 && mm[1] > 0 ) {
01291                             if ( mm[1] < pow ) pow = mm[1];
01292                         }
01293                         else if ( pow < 0 && mm[1] < 0 ) {
01294                             if ( mm[1] > pow ) pow = mm[1];
01295                         }
01296                         else pow = 0;
01297                     }
01298                     else pow = 0;
01299                     if ( pow == 0 ) break;
01300                     mm = mnext;

```

```

01301                                     }
01302                                     if ( pow == 0 ) { t += 2; continue; }
01303 /*
01304                                     We have a factor
01305 */
01306                                     action = 1; i = pow;
01307                                     if ( i > 0 ) {
01308                                         while ( --i >= 0 ) {
01309                                             *r1++ = -SYMBOL;
01310                                             *r1++ = *t;
01311                                         }
01312                                     }
01313                                     else {
01314                                         while ( i++ < 0 ) {
01315                                             *r1++ = 8 + ARGHEAD;
01316                                             for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
01317                                             *r1++ = 8; *r1++ = SYMBOL;
01318                                             *r1++ = 4; *r1++ = *t; *r1++ = -1;
01319                                             *r1++ = 1; *r1++ = 1; *r1++ = 3;
01320                                         }
01321                                     }
01322 /*
01323                                     Now we have to remove the symbols
01324 */
01325                                     t[1] -= pow;
01326                                     mm = rnnext;
01327                                     while ( mm < r3 ) {
01328                                         mnnext = mm + *mm;
01329                                         GETSTOP(mm,mstop); mm++;
01330                                         while ( mm < mstop ) {
01331                                             if ( *mm != SYMBOL ) mm += mm[1];
01332                                             else break;
01333                                         }
01334                                         mstop = mm + mm[1]; mm += 2;
01335                                         while ( mm < mstop && *mm != *t ) mm += 2;
01336                                         mm[1] -= pow;
01337                                         mm = mnnext;
01338                                     }
01339                                     t += 2;
01340                                     }
01341                                     }
01342 /*
01343                                     #] SYMBOL :
01344                                     #[ DOTPRODUCT :
01345 */
01346                                     else if ( *t == DOTPRODUCT ) {
01347                                         r7 = t; r8 = t + t[1]; t += 2;
01348                                         while ( t < r8 ) {
01349                                             pow = t[2];
01350                                             mm = rnnext;
01351                                             while ( mm < r3 ) {
01352                                                 mnnext = mm + *mm;
01353                                                 GETSTOP(mm,mstop); mm++;
01354                                                 while ( mm < mstop ) {
01355                                                     if ( *mm != DOTPRODUCT ) mm += mm[1];
01356                                                     else break;
01357                                                 }
01358                                                 if ( *mm == DOTPRODUCT ) {
01359                                                     mstop = mm + mm[1]; mm += 2;
01360                                                     while ( ( *mm != *t || mm[1] != t[1] )
01361                                                         && mm < mstop ) mm += 3;
01362                                                     if ( mm >= mstop ) pow = 0;
01363                                                     else if ( pow > 0 && mm[2] > 0 ) {
01364                                                         if ( mm[2] < pow ) pow = mm[2];
01365                                                     }
01366                                                     else if ( pow < 0 && mm[2] < 0 ) {
01367                                                         if ( mm[2] > pow ) pow = mm[2];
01368                                                     }
01369                                                     else pow = 0;
01370                                                 }
01371                                                 else pow = 0;
01372                                                 if ( pow == 0 ) break;
01373                                                 mm = mnnext;
01374                                             }
01375                                         if ( pow == 0 ) { t += 3; continue; }
01376 /*
01377                                     We have a factor
01378 */
01379                                     action = 1; i = pow;
01380                                     if ( i > 0 ) {
01381                                         while ( --i >= 0 ) {
01382                                             *r1++ = 9 + ARGHEAD;
01383                                             for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
01384                                             *r1++ = 9; *r1++ = DOTPRODUCT;
01385                                             *r1++ = 5; *r1++ = *t; *r1++ = t[1]; *r1++ = 1;
01386                                             *r1++ = 1; *r1++ = 1; *r1++ = 3;
01387                                         }

```



```

01388     }
01389     else {
01390         while ( i++ < 0 ) {
01391             *r1++ = 9 + ARGHEAD;
01392             for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
01393             *r1++ = 9; *r1++ = DOTPRODUCT;
01394             *r1++ = 5; *r1++ = *t; *r1++ = t[1]; *r1++ = -1;
01395             *r1++ = 1; *r1++ = 1; *r1++ = 3;
01396         }
01397     }
01398     /*
01399     Now we have to remove the dotproducts
01400     */
01401     t[2] -= pow;
01402     mm = rnext;
01403     while ( mm < r3 ) {
01404         mnext = mm + *mm;
01405         GETSTOP(mm,mstop); mm++;
01406         while ( mm < mstop ) {
01407             if ( *mm != DOTPRODUCT ) mm += mm[1];
01408             else break;
01409         }
01410         mstop = mm + mm[1]; mm += 2;
01411         while ( mm < mstop && ( *mm != *t
01412             || mm[1] != t[1] ) ) mm += 3;
01413         mm[2] -= pow;
01414         mm = mnext;
01415     }
01416     t += 3;
01417 }
01418 }
01419 /*
01420 #] DOTPRODUCT :
01421 #[ DELTA/VECTOR :
01422 */
01423     else if ( *t == DELTA || *t == VECTOR ) {
01424         r7 = t; r8 = t + t[1]; t += 2;
01425         while ( t < r8 ) {
01426             mm = rnext;
01427             pow = 1;
01428             while ( mm < r3 ) {
01429                 mnext = mm + *mm;
01430                 GETSTOP(mm,mstop); mm++;
01431                 while ( mm < mstop ) {
01432                     if ( *mm != *r7 ) mm += mm[1];
01433                     else break;
01434                 }
01435                 if ( *mm == *r7 ) {
01436                     mstop = mm + mm[1]; mm += 2;
01437                     while ( ( *mm != *t || mm[1] != t[1] )
01438                         && mm < mstop ) mm += 2;
01439                     if ( mm >= mstop ) pow = 0;
01440                 }
01441                 else pow = 0;
01442                 if ( pow == 0 ) break;
01443                 mm = mnext;
01444             }
01445             if ( pow == 0 ) { t += 2; continue; }
01446         /*
01447         We have a factor
01448         */
01449         action = 1;
01450         *r1++ = 8 + ARGHEAD;
01451         for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
01452         *r1++ = 8; *r1++ = *r7;
01453         *r1++ = 4; *r1++ = *t; *r1++ = t[1];
01454         *r1++ = 1; *r1++ = 1; *r1++ = 3;
01455     /*
01456     Now we have to remove the delta's/vectors
01457     */
01458     mm = rnext;
01459     while ( mm < r3 ) {
01460         mnext = mm + *mm;
01461         GETSTOP(mm,mstop); mm++;
01462         while ( mm < mstop ) {
01463             if ( *mm != *r7 ) mm += mm[1];
01464             else break;
01465         }
01466         mstop = mm + mm[1]; mm += 2;
01467         while ( mm < mstop && (
01468             *mm != *t || mm[1] != t[1] ) ) mm += 2;
01469         *mm = mm[1] = NOINDEX;
01470         mm = mnext;
01471     }
01472     *t = t[1] = NOINDEX;
01473     t += 2;
01474 }

```

```

01475     }
01476  /*
01477      #] DELTA/VECTOR :
01478      #[ INDEX :
01479  */
01480      else if ( *t == INDEX ) {
01481          r7 = t; r8 = t + t[1]; t += 2;
01482          while ( t < r8 ) {
01483              mm = rnext;
01484              pow = 1;
01485              while ( mm < r3 ) {
01486                  mnext = mm + *mm;
01487                  GETSTOP(mm,mstop); mm++;
01488                  while ( mm < mstop ) {
01489                      if ( *mm != *r7 ) mm += mm[1];
01490                      else break;
01491                  }
01492                  if ( *mm == *r7 ) {
01493                      mstop = mm + mm[1]; mm += 2;
01494                      while ( *mm != *t
01495                          && mm < mstop ) mm++;
01496                      if ( mm >= mstop ) pow = 0;
01497                  }
01498                  else pow = 0;
01499                  if ( pow == 0 ) break;
01500                  mm = mnext;
01501              }
01502              if ( pow == 0 ) { t++; continue; }
01503  /*
01504      We have a factor
01505  */
01506      action = 1;
01507  /*
01508      The next looks like an error.
01509      We should have here a VECTOR or INDEX like object
01510
01511      *r1++ = 7 + ARGHEAD;
01512      for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
01513      *r1++ = 7; *r1++ = *r7;
01514      *r1++ = 3; *r1++ = *t;
01515      *r1++ = 1; *r1++ = 1; *r1++ = 3;
01516
01517      Replace this by: (11-apr-2007)
01518  */
01519      if ( *t < 0 ) { *r1++ = -VECTOR; }
01520      else { *r1++ = -INDEX; }
01521      *r1++ = *t;
01522  /*
01523      Now we have to remove the index
01524  */
01525      *t = NOINDEX;
01526      mm = rnext;
01527      while ( mm < r3 ) {
01528          mnext = mm + *mm;
01529          GETSTOP(mm,mstop); mm++;
01530          while ( mm < mstop ) {
01531              if ( *mm != *r7 ) mm += mm[1];
01532              else break;
01533          }
01534          mstop = mm + mm[1]; mm += 2;
01535          while ( mm < mstop &&
01536              *mm != *t ) mm += 1;
01537          *mm = NOINDEX;
01538          mm = mnext;
01539      }
01540      t += 1;
01541      }
01542      }
01543  /*
01544      #] INDEX :
01545      #[ FUNCTION :
01546  */
01547      else if ( *t >= FUNCTION ) {
01548  /*
01549      In the next code we should actually look inside
01550      the DENOMINATOR or EXPONENT for noncommuting objects
01551  */
01552      if ( *t >= FUNCTION &&
01553          functions[*t-FUNCTION].commute == 0 ) ncom = 0;
01554      else ncom = 1;
01555      if ( ncom ) {
01556          mm = r5 + 1;
01557          while ( mm < t && ( *mm == DUMMYFUN
01558              || *mm == DUMMYTEN ) ) mm += mm[1];
01559          if ( mm < t ) { t += t[1]; continue; }
01560      }
01561      mm = rnext; pow = 1;

```

```

01562         while ( mm < r3 ) {
01563             mnext = mm + *mm;
01564             GETSTOP(mm,mstop); mm++;
01565             while ( mm < mstop ) {
01566                 if ( *mm == *t && mm[1] == t[1] ) {
01567                     for ( i = 2; i < t[1]; i++ ) {
01568                         if ( mm[i] != t[i] ) break;
01569                     }
01570                     if ( i >= t[1] )
01571                         { mm += mm[1]; goto nextmterm; }
01572                 }
01573                 if ( ncom && *mm != DUMMYFUN && *mm != DUMMYTEN )
01574                     { pow = 0; break; }
01575                 mm += mm[1];
01576             }
01577             if ( mm >= mstop ) pow = 0;
01578             if ( pow == 0 ) break;
01579 nextmterm:
01580             mm = mnext;
01581         }
01582         if ( pow == 0 ) { t += t[1]; continue; }
01583
01584         Copy the function
01585
01586         action = 1;
01587         *r1++ = t[1] + 4 + ARGHEAD;
01588         for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
01589         *r1++ = t[1] + 4;
01590         for ( i = 0; i < t[1]; i++ ) *r1++ = t[i];
01591         *r1++ = 1; *r1++ = 1; *r1++ = 3;
01592
01593         Now we have to take out the functions
01594
01595         mm = rnext;
01596         while ( mm < r3 ) {
01597             mnext = mm + *mm;
01598             GETSTOP(mm,mstop); mm++;
01599             while ( mm < mstop ) {
01600                 if ( *mm == *t && mm[1] == t[1] ) {
01601                     for ( i = 2; i < t[1]; i++ ) {
01602                         if ( mm[i] != t[i] ) break;
01603                     }
01604                     if ( i >= t[1] ) {
01605                         if ( functions[*t-FUNCTION].spec > 0 )
01606                             *mm = DUMMYTEN;
01607                         else
01608                             *mm = DUMMYFUN;
01609                         mm += mm[1];
01610                         goto nexttterm;
01611                     }
01612                 }
01613                 mm += mm[1];
01614             }
01615             mm = mnext;
01616         }
01617         if ( functions[*t-FUNCTION].spec > 0 )
01618             *t = DUMMYTEN;
01619         else
01620             *t = DUMMYFUN;
01621         action = 1;
01622         v[2] = DIRTYFLAG;
01623         t += t[1];
01624     }
01625
01626     #] FUNCTION :
01627
01628     else {
01629         t += t[1];
01630     }
01631 }
01632 t = r5; pow = 1;
01633 while ( t < r3 ) {
01634     t += *t; if ( t[-1] > 0 ) { pow = 0; break; }
01635 }
01636
01637 We have to add here the code for computing the GCD
01638 and to divide it out.
01639
01640
01641 #[ Numerical factor :
01642
01643 t = r5;
01644 EAscrat = (UWORD *) (TermMalloc("execarg"));
01645 if ( t + *t == r3 ) goto oneterm;
01646 GETSTOP(t,r6);
01647 ngcd = t[t[0]-1];
01648 i = abs(ngcd)-1;

```

```

01649         while ( --i >= 0 ) EAscrat[i] = r6[i];
01650         t += *t;
01651         while ( t < r3 ) {
01652             GETSTOP(t,r6);
01653             i = t[t[0]-1];
01654             if ( AccumGCD(BHEAD EAscrat,&ngcd,(UWORD *)r6,i) ) goto execargerr;
01655             if ( ngcd == 3 && EAscrat[0] == 1 && EAscrat[1] == 1 ) break;
01656             t += *t;
01657         }
01658         if ( ngcd != 3 || EAscrat[0] != 1 || EAscrat[1] != 1 ) {
01659             if ( pow ) ngcd = -ngcd;
01660             t = r5; r9 = r1; *r1++ = t[-ARGHEAD]; *r1++ = 1;
01661             FILLARG(r1); ngcd = REDLENG(ngcd);
01662             while ( t < r3 ) {
01663                 GETSTOP(t,r6);
01664                 r7 = t; r8 = r1;
01665                 while ( r7 < r6 ) *r1++ = *r7++;
01666                 t += *t;
01667                 i = REDLENG(t[-1]);
01668                 if ( DivRat(BHEAD (UWORD *)r6,i,EAscrat,ngcd,(UWORD *)r1,&nq) ) goto
execargerr;
01669                 nq = INCLENG(nq);
01670                 i = ABS(nq)-1;
01671                 r1 += i; *r1++ = nq; *r8 = r1-r8;
01672             }
01673             *r9 = r1-r9;
01674             ngcd = INCLENG(ngcd);
01675             i = ABS(ngcd)-1;
01676             if ( factor && *factor == 0 ) {}
01677             else if ( ( factor && factor[0] == 4 && factor[2] == 1
01678 && factor[3] == -3 ) || pow == 0 ) {
01679                 r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
01680                 FILLARG(r1); *r1++ = i+2;
01681                 for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];
01682                 *r1++ = ngcd;
01683                 if ( ToFast(r9,r9) ) r1 = r9+2;
01684             }
01685             else if ( factor && factor[0] == 4 && factor[2] == 1
01686 && factor[3] > 0 && pow ) {
01687                 if ( ngcd < 0 ) ngcd = -ngcd;
01688                 *r1++ = -SNUMBER; *r1++ = -1;
01689                 r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
01690                 FILLARG(r1); *r1++ = i+2;
01691                 for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];
01692                 *r1++ = ngcd;
01693                 if ( ToFast(r9,r9) ) r1 = r9+2;
01694             }
01695             else {
01696                 if ( ngcd < 0 ) ngcd = -ngcd;
01697                 if ( pow ) { *r1++ = -SNUMBER; *r1++ = -1; }
01698                 if ( ngcd != 3 || EAscrat[0] != 1 || EAscrat[1] != 1 ) {
01699                     r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
01700                     FILLARG(r1); *r1++ = i+2;
01701                     for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];
01702                     *r1++ = ngcd;
01703                     if ( ToFast(r9,r9) ) r1 = r9+2;
01704                 }
01705             }
01706         }
01707     /*
01708     #] Numerical factor :
01709     */
01710     else {
01711 oneterm;;
01712         if ( factor == 0 || *factor > 2 ) {
01713             if ( pow > 0 ) {
01714                 *r1++ = -SNUMBER; *r1++ = -1;
01715                 t = r5;
01716                 while ( t < r3 ) {
01717                     t += *t; t[-1] = -t[-1];
01718                 }
01719             }
01720             t = rr; *r1++ = *t++; *r1++ = 1; t++;
01721             COPYARG(r1,t);
01722             while ( t < m ) *r1++ = *t++;
01723         }
01724     }
01725     TermFree(EAscrat,"execarg");
01726 }
01727 } /* AC.OldFactArgFlag */
01728 }
01729 /* r1 is fout in ons voorbeeld. */
01730 r2[1] = r1 - r2;
01731 action = 1;
01732 v[2] = DIRTYFLAG;
01733 }
01734 else {

```

```

01735         r = t + t[l];
01736         while ( t < r ) *r1++ = *t++;
01737     }
01738 }
01739 r = term + *term;
01740 while ( t < r ) *r1++ = *t++;
01741 m = AT.WorkPointer;
01742 i = m[0] = r1 - m;
01743 t = term;
01744 while ( --i >= 0 ) *t++ = *m++;
01745 if ( AT.WorkPointer < t ) AT.WorkPointer = t;
01746 }
01747 /*
01748 #] FACTARG :
01749 */
01750 AR.Cnumlhs = oldnumlhs;
01751 if ( action && Normalize(BHEAD term) ) goto execargerr;
01752 AT.WorkPointer = oldwork;
01753 if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
01754 AT.pWorkPointer = oldppointer;
01755 return(action);
01756 execargerr:
01757 AT.WorkPointer = oldwork;
01758 AT.pWorkPointer = oldppointer;
01759 MLOCK(ErrorMessageLock);
01760 MesCall("execarg");
01761 MUNLOCK(ErrorMessageLock);
01762 return(-1);
01763 }
01764
01765 /*
01766 #] execarg :
01767 #[ execterm :
01768 */
01769
01770 int execterm(PHEAD WORD *term, WORD level)
01771 {
01772     GETBIDENTITY
01773     CBUF *C = cbuf+AM.rbufnum;
01774     WORD oldnumlhs = AR.Cnumlhs;
01775     WORD maxisat = C->lhs[level][2];
01776     WORD *buffer1 = 0;
01777     WORD *oldworkpointer = AT.WorkPointer;
01778     WORD *tl, i;
01779     WORD olddeferflag = AR.DeferFlag, tryterm = 0;
01780     AR.DeferFlag = 0;
01781     do {
01782         AR.Cnumlhs = C->lhs[level][3];
01783         NewSort(BHEAD0);
01784         /*
01785          Normally for function arguments we do not use PolyFun/PolyRatFun.
01786          Hence NewSort sets the corresponding variables to zero.
01787          Here we overwrite that.
01788         */
01789         AN.FunSorts[AR.sLevel]->PolyFlag = ( AR.PolyFun != 0 ) ? AR.PolyFunType: 0;
01790         if ( AR.PolyFun == 0 ) { AN.FunSorts[AR.sLevel]->PolyFlag = 0; }
01791         else if ( AR.PolyFunType == 1 ) { AN.FunSorts[AR.sLevel]->PolyFlag = 1; }
01792         else if ( AR.PolyFunType == 2 ) {
01793             if ( AR.PolyFunExp == 2 ) AN.FunSorts[AR.sLevel]->PolyFlag = 1;
01794             else AN.FunSorts[AR.sLevel]->PolyFlag = 2;
01795         }
01796         if ( buffer1 ) {
01797             term = buffer1;
01798             while ( *term ) {
01799                 tl = oldworkpointer;
01800                 i = *term; while ( --i >= 0 ) *tl++ = *term++;
01801                 AT.WorkPointer = tl;
01802                 if ( Generator(BHEAD oldworkpointer,level) ) goto exectermerr;
01803             }
01804         }
01805         else {
01806             if ( Generator(BHEAD term,level) ) goto exectermerr;
01807         }
01808         if ( buffer1 ) {
01809             if ( tryterm ) { TermFree(buffer1,"buffer in sort statement"); tryterm = 0; }
01810             else { M_free((void *)buffer1,"buffer in sort statement"); }
01811             buffer1 = 0;
01812         }
01813         AN.tryterm = 1;
01814         if ( EndSort(BHEAD (WORD *)((void *)&buffer1)),2) < 0 ) goto exectermerr;
01815         tryterm = AN.tryterm; AN.tryterm = 0;
01816         level = AR.Cnumlhs;
01817     } while ( AR.Cnumlhs < maxisat );
01818     AR.Cnumlhs = oldnumlhs;
01819     AR.DeferFlag = olddeferflag;
01820     term = buffer1;
01821     while ( *term ) {

```

```

01822         t1 = oldworkpointer;
01823         i = *term; while ( --i >= 0 ) *t1++ = *term++;
01824         AT.WorkPointer = t1;
01825         if ( Generator(BHEAD oldworkpointer,level) ) goto exectermerr;
01826     }
01827     if ( tryterm ) { TermFree(buffer1,"buffer in term statement"); tryterm = 0; }
01828     else { M_free(buffer1,"buffer in term statement"); }
01829     buffer1 = 0;
01830     AT.WorkPointer = oldworkpointer;
01831     return(0);
01832 exectermerr:
01833     AT.WorkPointer = oldworkpointer;
01834     AR.DeferFlag = olddeferflag;
01835     MLOCK(ErrorMessageLock);
01836     MesCall("execterm");
01837     MUNLOCK(ErrorMessageLock);
01838     return(-1);
01839 }
01840
01841 /*
01842 #] execterm :
01843 #[ ArgumentImplode :
01844 */
01845
01846 int ArgumentImplode(PHEAD WORD *term, WORD *thelist)
01847 {
01848     GETBIDENTITY
01849     WORD *liststart, *liststop, *inlist;
01850     WORD *w, *t, *tend, *tstop, *tt, *ttstop, *ttt, ncount, i;
01851     int action = 0;
01852     liststop = thelist + thelist[1];
01853     liststart = thelist + 2;
01854     t = term;
01855     tend = t + *t;
01856     tstop = tend - ABS(tend[-1]);
01857     t++;
01858     while ( t < tstop ) {
01859         if ( *t >= FUNCTION ) {
01860             inlist = liststart;
01861             while ( inlist < liststop && *inlist != *t ) inlist += inlist[1];
01862             if ( inlist < liststop ) {
01863                 tt = t; ttstop = t + t[1]; w = AT.WorkPointer;
01864                 for ( i = 0; i < FUNHEAD; i++ ) *w++ = *tt++;
01865                 while ( tt < ttstop ) {
01866                     ncount = 0;
01867                     if ( *tt == -SNUMBER && tt[1] == 0 ) {
01868                         ncount = 1; ttt = tt; tt += 2;
01869                         while ( tt < ttstop && *tt == -SNUMBER && tt[1] == 0 ) {
01870                             ncount++; tt += 2;
01871                         }
01872                     }
01873                     if ( ncount > 0 ) {
01874                         if ( tt < ttstop && *tt == -SNUMBER && ( tt[1] == 1 || tt[1] == -1 ) ) {
01875                             *w++ = -SNUMBER;
01876                             *w++ = (ncount+1) * tt[1];
01877                             tt += 2;
01878                             action = 1;
01879                         }
01880                         else if ( ( tt[0] == tt[ARGHEAD] + ARGHEAD )
01881                             && ( ABS(tt[tt[0]-1]) == 3 )
01882                             && ( tt[tt[0]-2] == 1 )
01883                             && ( tt[tt[0]-3] == 1 ) ) { /* Single term with coef +/- 1 */
01884                             i = *tt; NCOPY(w,tt,i)
01885                             w[-3] = ncount+1;
01886                             action = 1;
01887                         }
01888                         else if ( *tt == -SYMBOL ) {
01889                             *w++ = ARGHEAD+8;
01890                             *w++ = 0;
01891                             FILLARG(w)
01892                             *w++ = 8;
01893                             *w++ = SYMBOL;
01894                             *w++ = tt[1];
01895                             *w++ = 1;
01896                             *w++ = ncount+1; *w++ = 1; *w++ = 3;
01897                             tt += 2;
01898                             action = 1;
01899                         }
01900                         else if ( *tt <= -FUNCTION ) {
01901                             *w++ = ARGHEAD+FUNHEAD+4;
01902                             *w++ = 0;
01903                             FILLARG(w)
01904                             *w++ = -*tt++;
01905                             *w++ = FUNHEAD+4;
01906                             FILLFUN(w)
01907                             *w++ = ncount+1; *w++ = 1; *w++ = 3;
01908                             action = 1;

```

```

01909         }
01910         else {
01911             while ( ttt < tt ) *w++ = *ttt++;
01912             if ( tt < ttstop && *tt == -SNUMBER ) {
01913                 *w++ = *tt++; *w++ = *tt++;
01914             }
01915         }
01916     }
01917     else if ( *tt <= -FUNCTION ) {
01918         *w++ = *tt++;
01919     }
01920     else if ( *tt < 0 ) {
01921         *w++ = *tt++;
01922         *w++ = *tt++;
01923     }
01924     else {
01925         i = *tt; NCOPY(w,tt,i)
01926     }
01927 }
01928 AT.WorkPointer[1] = w - AT.WorkPointer;
01929 while ( tt < tend ) *w++ = *tt++;
01930 ttt = AT.WorkPointer; tt = t;
01931 while ( ttt < w ) *tt++ = *ttt++;
01932 term[0] = tt - term;
01933 AT.WorkPointer = tt;
01934 tend = tt; tstop = tt - ABS(tt[-1]);
01935 }
01936 }
01937 t += t[1];
01938 }
01939 if ( action ) {
01940     if ( Normalize(BHEAD term) ) return(-1);
01941 }
01942 return(0);
01943 }
01944
01945 /*
01946 #] ArgumentImplode :
01947 #[ ArgumentExplode :
01948 */
01949
01950 int ArgumentExplode(PHEAD WORD *term, WORD *thelist)
01951 {
01952     GETBIDENTITY
01953     WORD *liststart, *liststop, *inlist, *old;
01954     WORD *w, *t, *tend, *tstop, *tt, *ttstop, *ttt, ncount, i;
01955     int action = 0;
01956     LONG x;
01957     liststop = thelist + thelist[1];
01958     liststart = thelist + 2;
01959     t = term;
01960     tend = t + *t;
01961     tstop = tend - ABS(tend[-1]);
01962     t++;
01963     while ( t < tstop ) {
01964         if ( *t >= FUNCTION ) {
01965             inlist = liststart;
01966             while ( inlist < liststop && *inlist != *t ) inlist += inlist[1];
01967             if ( inlist < liststop ) {
01968                 tt = t; ttstop = t + t[1]; w = AT.WorkPointer;
01969                 for ( i = 0; i < FUNHEAD; i++ ) *w++ = *tt++;
01970                 while ( tt < ttstop ) {
01971                     if ( *tt == -SNUMBER && tt[1] != 0 ) {
01972                         if ( tt[1] < AM.MaxTer/((WORD)sizeof(WORD)*4)
01973                             && tt[1] > -(AM.MaxTer/((WORD)sizeof(WORD)*4))
01974                             && ( tt[1] > 1 || tt[1] < -1 ) ) {
01975                             ncount = ABS(tt[1]);
01976                             while ( ncount > 1 ) {
01977                                 *w++ = -SNUMBER; *w++ = 0; ncount--;
01978                             }
01979                             *w++ = -SNUMBER;
01980                             if ( tt[1] < 0 ) *w++ = -1;
01981                             else *w++ = 1;
01982                             tt += 2;
01983                             action = 1;
01984                         }
01985                     }
01986                     else {
01987                         *w++ = *tt++; *w++ = *tt++;
01988                     }
01989                 }
01990             }
01991             else if ( *tt <= -FUNCTION ) {
01992                 *w++ = *tt++;
01993             }
01994             else if ( *tt < 0 ) {
01995                 *w++ = *tt++;
01996                 *w++ = *tt++;
01997             }
01998         }
01999     }

```

```

01996         else if ( tt[0] == tt[ARGHEAD]+ARGHEAD ) {
01997             ttt = tt + tt[0] - 1;
01998             i = (ABS(ttt[0])-1)/2;
01999             if ( i > 1 ) {
02000 TooMany:
02001                 old = AN.currentTerm;
02002                 AN.currentTerm = term;
02003                 MesPrint("Too many arguments in output of ArgExplode");
02004                 MesPrint("Term = %t");
02005                 AN.currentTerm = old;
02006                 return(-1);
02007             }
02008             if ( ttt[-1] != 1 ) goto NoExplode;
02009             x = ttt[-2];
02010             if ( 2*x > (AT.WorkTop-w)-*term ) goto TooMany;
02011             ncount = x - 1;
02012             while ( ncount > 0 ) {
02013                 *w++ = -SNUMBER; *w++ = 0; ncount--;
02014             }
02015             ttt[-2] = 1;
02016             i = *tt; NCOPY(w,tt,i)
02017             action = 1;
02018         }
02019 NoExplode:
02020         i = *tt; NCOPY(w,tt,i)
02021     }
02022     }
02023     AT.WorkPointer[1] = w - AT.WorkPointer;
02024     while ( tt < tend ) *w++ = *tt++;
02025     ttt = AT.WorkPointer; tt = t;
02026     while ( ttt < w ) *tt++ = *ttt++;
02027     term[0] = tt - term;
02028     AT.WorkPointer = tt;
02029     tend = tt; tstop = tt - ABS(tt[-1]);
02030 }
02031 }
02032 t += t[1];
02033 }
02034 if ( action ) {
02035     if ( Normalize(BHEAD term) ) return(-1);
02036 }
02037 return(0);
02038 }
02039
02040 /*
02041 #] ArgumentExplode :
02042 #[ ArgFactorize :
02043 */
02044 #define NEWORDER
02045
02046 int ArgFactorize(PHEAD WORD *argin, WORD *argout)
02047 {
02048     /*
02049     #[ step 0 : Declarations and initializations
02050     */
02051     WORD *argfree, *argextra, *argcopy, *t, *tstop, *a, *a1, *a2;
02052     #ifndef NEWORDER
02053     WORD *tt;
02054     #endif
02055     WORD startebuf = cbuf[AT.ebufnum].numrhs,oldword;
02056     WORD oldsorttype = AR.SortType, numargs;
02057     int error = 0, action = 0, i, ii, number, sign = 1;
02058
02059     *argout = 0;
02060     /*
02061     #] step 0 :
02062     #[ step 1 : Take care of ordering
02063     */
02064     AR.SortType = SORTHIGHFIRST;
02065     if ( oldsorttype != AR.SortType ) {
02066         NewSort(BHEAD0);
02067         oldword = argin[*argin]; argin[*argin] = 0;
02068         t = argin+ARGHEAD;
02069         while ( *t ) {
02070             tstop = t + *t;
02071             if ( AN.ncmod != 0 ) {
02072                 if ( AN.ncmod != 1 || ( (WORD)AN.cmod[0] < 0 ) ) {
02073                     MLOCK(ErrorMessageLock);
02074                     MesPrint("Factorization modulus a number, greater than a WORD not implemented.");
02075                     MUNLOCK(ErrorMessageLock);
02076                     Terminate(-1);
02077                 }
02078             }
02079             if ( Modulus(t) ) {
02080                 MLOCK(ErrorMessageLock);
02081                 MesCall("ArgFactorize");
02082                 MUNLOCK(ErrorMessageLock);
02083                 Terminate(-1);
02084             }
02085         }
02086     }

```



```

02098         }
02099         if ( !*t ) { t = tstop; continue; }
02100     }
02101     StoreTerm(BHEAD t);
02102     t = tstop;
02103 }
02104 /* par = 1, in case the arg has more than SubTermsInSmall terms */
02105 EndSort (BHEAD argin+ARGHEAD,1);
02106 argin[*argin] = oldword;
02107 }
02108 /*
02109 #] step 1 :
02110 #[ step 2 : take out the 'content'.
02111 */
02112 argfree = TakeArgContent(BHEAD argin,argout);
02113 {
02114     al = argout;
02115     while ( *al ) {
02116         if ( al[0] == -SNUMBER && ( al[1] == 1 || al[1] == -1 ) ) {
02117             if ( al[1] == -1 ) { sign = -sign; al[1] = 1; }
02118             if ( al[2] ) {
02119                 a = t = al+2; while ( *t ) NEXTARG(t);
02120                 i = t - al-2;
02121                 t = al; NCOPY(t,a,i);
02122                 *t = 0;
02123                 continue;
02124             }
02125             else {
02126                 al[0] = 0;
02127             }
02128             break;
02129         }
02130         else if ( al[0] == FUNHEAD+ARGHEAD+4 && al[ARGHEAD] == FUNHEAD+4
02131             && al[*al-1] == 3 && al[*al-2] == 1 && al[*al-3] == 1
02132             && al[ARGHEAD+1] >= FUNCTION ) {
02133             a = t = al+*al; while ( *t ) NEXTARG(t);
02134             i = t - a;
02135             *al = -al[ARGHEAD+1]; t = al+1; NCOPY(t,a,i);
02136             *t = 0;
02137         }
02138         NEXTARG(al);
02139     }
02140 }
02141 if ( argfree == 0 ) {
02142     argfree = argin;
02143 }
02144 else if ( argfree[0] == ( argfree[ARGHEAD]+ARGHEAD ) ) {
02145     Normalize(BHEAD argfree+ARGHEAD);
02146     argfree[0] = argfree[ARGHEAD]+ARGHEAD;
02147     argfree[1] = 0;
02148     if ( ( argfree[0] == ARGHEAD+4 ) && ( argfree[ARGHEAD+3] == 3 )
02149         && ( argfree[ARGHEAD+1] == 1 ) && ( argfree[ARGHEAD+2] == 1 ) ) {
02150         goto return0;
02151     }
02152 }
02153 else {
02154 /*
02155     The way we took out objects is rather brutish. We have to
02156     normalize
02157 */
02158     NewSort(BHEAD0);
02159     t = argfree+ARGHEAD;
02160     while ( *t ) {
02161         tstop = t + *t;
02162         Normalize(BHEAD t);
02163         StoreTerm(BHEAD t);
02164         t = tstop;
02165     }
02166     /* par = 1, in case the arg has more than SubTermsInSmall terms */
02167     EndSort (BHEAD argfree+ARGHEAD,1);
02168     t = argfree+ARGHEAD;
02169     while ( *t ) t += *t;
02170     *argfree = t - argfree;
02171 }
02172 /*
02173 #] step 2 :
02174 #[ step 3 : look whether we have done this one already.
02175 */
02176 if ( ( number = FindArg(BHEAD argfree) ) != 0 ) {
02177     if ( number > 0 ) t = cbuf[AT.fbufnum].rhs[number-1];
02178     else t = cbuf[AC.ffbufnum].rhs[-number-1];
02179 /*
02180     Now position on the result. Remember we have in the cache:
02181         inputarg,0,outputargs,0
02182     t is currently at inputarg. *inputarg is always positive.
02183     in principle this holds also for the arguments in the output
02184     but we take no risks here (in case of future developments).

```

```

02185 */
02186     t += *t; t++;
02187     tstop = t;
02188     ii = 0;
02189     while ( *tstop ) {
02190         if ( *tstop == -SNUMBER && tstop[1] == -1 ) {
02191             sign = -sign; ii += 2;
02192         }
02193         NEXTARG(tstop);
02194     }
02195     a = argout; while ( *a ) NEXTARG(a);
02196 #ifndef NEWORDER
02197     if ( sign == -1 ) { *a++ = -SNUMBER; *a++ = -1; *a = 0; sign = 1; }
02198 #endif
02199     i = tstop - t - ii;
02200     ii = a - argout;
02201     a2 = a; a1 = a + i;
02202     *a1 = 0;
02203     while ( ii > 0 ) { *--a1 = *--a2; ii--; }
02204     a = argout;
02205     while ( *t ) {
02206         if ( *t == -SNUMBER && t[1] == -1 ) { t += 2; }
02207         else { COPY1ARG(a,t) }
02208     }
02209     goto return0;
02210 }
02211 /*
02212 #] step 3 :
02213 #[ step 4 : invoke ConvertToPoly
02214
02215         We make a copy first in case there are no factors
02216 */
02217 argcopy = TermMalloc("argcopy");
02218 for ( i = 0; i <= *argfree; i++ ) argcopy[i] = argfree[i];
02219
02220 tstop = argfree + *argfree;
02221 {
02222     WORD sumcommu = 0;
02223     t = argfree + ARGHEAD;
02224     while ( t < tstop ) {
02225         sumcommu += DoesCommu(t);
02226         t += *t;
02227     }
02228     if ( sumcommu > 1 ) {
02229         MesPrint("ERROR: Cannot factorize an argument with more than one noncommuting object");
02230         Terminate(-1);
02231     }
02232 }
02233 t = argfree + ARGHEAD;
02234
02235 while ( t < tstop ) {
02236     if ( ( t[1] != SYMBOL ) && ( *t != (ABS(t[*t-1])+1) ) ) {
02237         action = 1; break;
02238     }
02239     t += *t;
02240 }
02241 if ( action ) {
02242     t = argfree + ARGHEAD;
02243     argextra = AT.WorkPointer;
02244     NewSort(BHEAD0);
02245     while ( t < tstop ) {
02246         if ( LocalConvertToPoly(BHEAD t,argextra,startebuf,0) < 0 ) {
02247             error = -1;
02248         }
02249     }
02250     AR.SortType = oldsorttype;
02251     TermFree(argcopy,"argcopy");
02252     if ( argfree != argin ) TermFree(argfree,"argfree");
02253     MesCall("ArgFactorize");
02254     Terminate(-1);
02255     return(-1);
02256 }
02257 StoreTerm(BHEAD argextra);
02258 t += *t; argextra += *argextra;
02259 }
02260 /* par = 1, in case the arg has more than SubTermsInSmall terms */
02261 if ( EndSort(BHEAD argfree+ARGHEAD,1) < 0 ) { error = -2; goto getout; }
02262 t = argfree + ARGHEAD;
02263 while ( *t > 0 ) t += *t;
02264 *argfree = t - argfree;
02265 }
02266 /*
02267 #] step 4 :
02268 #[ step 5 : If not in the tables, we have to do this by hard work.
02269 */
02270 a = argout;
02271 while ( *a ) NEXTARG(a);

```

```

02272     if ( poly_factorize_argument(BHEAD argfree,a) < 0 ) {
02273         MesCall("ArgFactorize");
02274         error = -1;
02275     }
02276 /*
02277 #] step 5 :
02278 #[ step 6 : use now ConvertFromPoly
02279
02280         Be careful: there should be more than one argument now.
02281 */
02282     if ( error == 0 && action ) {
02283         a1 = a; NEXTARG(a1);
02284         if ( *a1 != 0 ) {
02285             CBUF *C = cbuf+AC.cbunum;
02286             CBUF *CC = cbuf+AT.ebunum;
02287             WORD *oldworkpointer = AT.WorkPointer;
02288             WORD *argcopy2 = TermMalloc("argcopy2"), *a1, *a2;
02289             a1 = a; a2 = argcopy2;
02290             while ( *a1 ) {
02291                 if ( *a1 < 0 ) {
02292                     if ( *a1 > -FUNCTION ) *a2++ = *a1++;
02293                     *a2++ = *a1++; *a2 = 0;
02294                     continue;
02295                 }
02296                 t = a1 + ARGHEAD;
02297                 tstop = a1 + *a1;
02298                 argextra = AT.WorkPointer;
02299                 NewSort(BHEAD0);
02300                 while ( t < tstop ) {
02301                     if ( ConvertFromPoly(BHEAD t,argextra,numxsymbol,CC->numrhs-startebuf+numxsymbol
02302 ,startebuf-numxsymbol,1) <= 0 ) {
02303                         TermFree(argcopy2,"argcopy2");
02304                         LowerSortLevel();
02305                         error = -3;
02306                         goto getout;
02307                     }
02308                     t += *t;
02309                     AT.WorkPointer = argextra + *argextra;
02310 /*
02311         ConvertFromPoly leaves terms with subexpressions. Hence:
02312 */
02313                     if ( Generator(BHEAD argextra,C->numlhs) ) {
02314                         TermFree(argcopy2,"argcopy2");
02315                         LowerSortLevel();
02316                         error = -4;
02317                         goto getout;
02318                     }
02319                 }
02320                 AT.WorkPointer = oldworkpointer;
02321                 /* par = 1, in case the factor has more than SubTermsInSmall terms */
02322                 if ( EndSort(BHEAD a2+ARGHEAD,1) < 0 ) { error = -5; goto getout; }
02323                 t = a2+ARGHEAD; while ( *t ) t += *t;
02324                 *a2 = t - a2; a2[1] = 0; ZEROARG(a2);
02325                 ToFast(a2,a2); NEXTARG(a2);
02326                 a1 = tstop;
02327             }
02328             i = a2 - argcopy2;
02329             a2 = argcopy2; a1 = a;
02330             NCOPY(a1,a2,i);
02331             *a1 = 0;
02332             TermFree(argcopy2,"argcopy2");
02333 /*
02334         Erase the entries we made temporarily in cbuf[AT.ebunum]
02335 */
02336             CC->numrhs = startebuf;
02337         }
02338         else { /* no factorization. recover the argument from before step 3. */
02339             for ( i = 0; i <= *argcopy; i++ ) a[i] = argcopy[i];
02340         }
02341     }
02342 /*
02343 #] step 6 :
02344 #[ step 7 : Add this one to the tables.
02345
02346         Possibly drop some elements in the tables
02347         when they become too full.
02348 */
02349     if ( error == 0 && AN.ncmod == 0 ) {
02350         if ( InsertArg(BHEAD argcopy,a,0) < 0 ) { error = -1; }
02351     }
02352 /*
02353 #] step 7 :
02354 #[ step 8 : Clean up and return.
02355
02356         Change the order of the arguments in argout and a.
02357         Use argcopy as spare space.
02358 */

```

```

02359     ii = a - argout;
02360     for ( i = 0; i < ii; i++ ) argcopy[i] = argout[i];
02361     al = a;
02362     while ( *al ) {
02363         if ( *al == -SNUMBER && al[1] < 0 ) {
02364             sign = -sign; al[1] = -al[1];
02365             if ( al[1] == 1 ) {
02366                 a2 = al+2; while ( *a2 ) NEXTARG(a2);
02367                 i = a2-al-2; a2 = al+2;
02368                 NCOPY(al,a2,i);
02369                 *al = 0;
02370             }
02371             while ( *al ) NEXTARG(al);
02372             break;
02373         }
02374         else {
02375             if ( *al > 0 && *al == al[ARGHEAD]+ARGHEAD && al[*al-1] < 0 ) {
02376                 al[*al-1] = -al[*al-1]; sign = -sign;
02377             }
02378             if ( *al == ARGHEAD+4 && al[ARGHEAD+1] == 1 && al[ARGHEAD+2] == 1 ) {
02379                 a2 = al+ARGHEAD+4; while ( *a2 ) NEXTARG(a2);
02380                 i = a2-al-ARGHEAD-4; a2 = al+ARGHEAD+4;
02381                 NCOPY(al,a2,i);
02382                 *al = 0;
02383                 break;
02384             }
02385             while ( *al ) NEXTARG(al);
02386             break;
02387         }
02388         NEXTARG(al);
02389     }
02390     i = al - a;
02391     a2 = argout;
02392     NCOPY(a2,a,i);
02393     for ( i = 0; i < ii; i++ ) *a2++ = argcopy[i];
02394 #ifndef NEWORDER
02395     if ( sign == -1 ) { *a2++ = -SNUMBER; *a2++ = -1; sign = 1; }
02396 #endif
02397     *a2 = 0;
02398     TermFree(argcopy,"argcopy");
02399     return 0;
02400     if ( argfree != argin ) TermFree(argfree,"argfree");
02401     if ( oldsorttype != AR.SortType ) {
02402         AR.SortType = oldsorttype;
02403         a = argout;
02404         while ( *a ) {
02405             if ( *a > 0 ) {
02406                 NewSort(BHEAD0);
02407                 oldword = a[*a]; a[*a] = 0;
02408                 t = a+ARGHEAD;
02409                 while ( *t ) {
02410                     tstop = t + *t;
02411                     StoreTerm(BHEAD t);
02412                     t = tstop;
02413                 }
02414                 /* par = 1, in case the factor has more than SubTermsInSmall terms */
02415                 EndSort(BHEAD a+ARGHEAD,1);
02416                 a[*a] = oldword;
02417                 a += *a;
02418             }
02419             else { NEXTARG(a); }
02420         }
02421     }
02422 #ifndef NEWORDER
02423     t = argout; numargs = 0;
02424     while ( *t ) {
02425         tt = t;
02426         NEXTARG(t);
02427         if ( *tt == ABS(t[-1])+1+ARGHEAD && sign == -1 ) { t[-1] = -t[-1]; sign = 1; }
02428         else if ( *tt == -SNUMBER && sign == -1 ) { tt[1] = -tt[1]; sign = 1; }
02429         numargs++;
02430     }
02431     if ( sign == -1 ) {
02432         *t++ = -SNUMBER; *t++ = -1; *t = 0; sign = 1; numargs++;
02433     }
02434 #else
02435 /*
02436     Now we have to sort the arguments
02437     First have the number of 'nontrivial/nonnumerical' arguments
02438     Then make a piece of code like in FullSymmetrize with that number
02439     of arguments to be symmetrized.
02440     Put a function in front
02441     Call the Symmetrize routine
02442 */
02443     t = argout; numargs = 0;
02444     while ( *t && *t != -SNUMBER && ( *t < 0 || ( ABS(t[*t-1]) != *t-1 ) ) ) {
02445         NEXTARG(t);

```

```

02446         numargs++;
02447     }
02448 #endif
02449     if ( numargs > 1 ) {
02450         WORD *Lijst;
02451         WORD x[3];
02452         x[0] = argout[-FUNHEAD];
02453         x[1] = argout[-FUNHEAD+1];
02454         x[2] = argout[-FUNHEAD+2];
02455         while ( *t ) { NEXTARG(t); }
02456         argout[-FUNHEAD] = SQRTFUNCTION;
02457         argout[-FUNHEAD+1] = t-argout+FUNHEAD;
02458         argout[-FUNHEAD+2] = 0;
02459         AT.WorkPointer = t+1;
02460         Lijst = AT.WorkPointer;
02461         for ( i = 0; i < numargs; i++ ) Lijst[i] = i;
02462         AT.WorkPointer += numargs;
02463         error = Symmetrize(BHEAD argout-FUNHEAD,Lijst,numargs,1,SYMMETRIC);
02464         AT.WorkPointer = Lijst;
02465         argout[-FUNHEAD] = x[0];
02466         argout[-FUNHEAD+1] = x[1];
02467         argout[-FUNHEAD+2] = x[2];
02468 #ifdef NEWORDER
02469 /*
02470     Now we have to get a potential numerical argument to the first position
02471 */
02472     tstop = argout; while ( *tstop ) { NEXTARG(tstop); }
02473     t = argout; number = 0;
02474     while ( *t ) {
02475         tt = t; NEXTARG(t);
02476         if ( *tt == -SNUMBER ) {
02477             if ( number == 0 ) break;
02478             x[0] = tt[1];
02479             while ( tt > argout ) { ---t = ---tt; }
02480             argout[0] = -SNUMBER; argout[1] = x[0];
02481             break;
02482         }
02483         else if ( *tt == ABS(t[-1])+1+ARGHEAD ) {
02484             if ( number == 0 ) break;
02485             ii = t - tt;
02486             for ( i = 0; i < ii; i++ ) tstop[i] = tt[i];
02487             while ( tt > argout ) { ---t = ---tt; }
02488             for ( i = 0; i < ii; i++ ) argout[i] = tstop[i];
02489             *tstop = 0;
02490             break;
02491         }
02492         number++;
02493     }
02494 #endif
02495     }
02496 /*
02497     #] step 8 :
02498 */
02499     return(error);
02500 }
02501
02502 /*
02503     #] ArgFactorize :
02504     #[ FindArg :
02505 */
02514 WORD FindArg(PHEAD WORD *a)
02515 {
02516     int number;
02517     if ( AN.ncmod != 0 ) return(0); /* no room for mod stuff */
02518     number = FindTree(AT.fbufnum,a);
02519     if ( number >= 0 ) return(number+1);
02520     number = FindTree(AC.ffbufnum,a);
02521     if ( number >= 0 ) return(-number-1);
02522     return(0);
02523 }
02524
02525 /*
02526     #] FindArg :
02527     #[ InsertArg :
02528 */
02538 int InsertArg(PHEAD WORD *argin, WORD *argout,int par)
02539 {
02540     CBUF *C;
02541     WORD *a, i, bufnum;
02542     if ( par == 0 ) {
02543         bufnum = AT.fbufnum;
02544         C = cbuf+bufnum;
02545         if ( C->numrhs >= (C->maxrhs-2) ) CleanupArgCache(BHEAD AT.fbufnum);
02546     }
02547     else if ( par == 1 ) {
02548         bufnum = AC.ffbufnum;
02549         C = cbuf+bufnum;

```

```

02550     }
02551     else { return(-1); }
02552     AddRHS(bufnum,1);
02553     AddNtoC(bufnum,*argin,argin,1);
02554     AddToCB(C,0)
02555     a = argout; while ( *a ) NEXTARG(a);
02556     i = a - argout;
02557     AddNtoC(bufnum,i,argout,2);
02558     AddToCB(C,0)
02559     return(InsTree(bufnum,C->numrhs));
02560 }
02561
02562 /*
02563 #] InsertArg :
02564 #[ CleanupArgCache :
02565 */
02573 int CleanupArgCache(PHEAD WORD bufnum)
02574 {
02575     CBUF *C = cbuf + bufnum;
02576     COMPTREE *boomlijst = C->boomlijst;
02577     LONG *weights = (LONG *)Malloc1(2*(C->numrhs+1)*sizeof(LONG),"CleanupArgCache");
02578     LONG w, whalf, *extraweights;
02579     WORD *a, *to, *from;
02580     int i,j,k;
02581     for ( i = 1; i <= C->numrhs; i++ ) {
02582         weights[i] = ((LONG)i) * (boomlijst[i].usage);
02583     }
02584     /*
02585         Now sort the weights and determine the halfway weight
02586     */
02587     extraweights = weights+C->numrhs+1;
02588     SortWeights(weights+1,extraweights,C->numrhs);
02589     whalf = weights[C->numrhs/2+1];
02590     /*
02591         We should drop everybody with a weight < whalf.
02592     */
02593     to = C->Buffer;
02594     k = 1;
02595     for ( i = 1; i <= C->numrhs; i++ ) {
02596         from = C->rhs[i]; w = ((LONG)i) * (boomlijst[i].usage);
02597         if ( w >= whalf ) {
02598             if ( i < C->numrhs-1 ) {
02599                 if ( to == from ) {
02600                     to = C->rhs[i+1];
02601                 }
02602                 else {
02603                     j = C->rhs[i+1] - from;
02604                     C->rhs[k] = to;
02605                     NCOPY(to,from,j)
02606                 }
02607             }
02608             else if ( to == from ) {
02609                 to += *to + 1; while ( *to ) NEXTARG(to); to++;
02610             }
02611             else {
02612                 a = from; a += *a+1; while ( *a ) NEXTARG(a); a++;
02613                 j = a - from;
02614                 C->rhs[k] = to;
02615                 NCOPY(to,from,j)
02616             }
02617             weights[k++] = boomlijst[i].usage;
02618         }
02619     }
02620     C->numrhs = --k;
02621     C->Pointer = to;
02622     /*
02623         Next we need to rebuild the tree.
02624         Note that this can probably be done much faster by using the
02625         remains of the old tree !!!!!!!!!!!!!!!!!!!!!
02626     */
02627     ClearTree(AT.fbufnum);
02628     for ( i = 1; i <= k; i++ ) {
02629         InsTree(AT.fbufnum,i);
02630         boomlijst[i].usage = weights[i];
02631     }
02632     /*
02633         And cleanup
02634     */
02635     M_free(weights,"CleanupArgCache");
02636     return(0);
02637 }
02638
02639 /*
02640 #] CleanupArgCache :
02641 #[ ArgSymbolMerge :
02642 */
02643

```

```

02644 int ArgSymbolMerge(WORD *t1, WORD *t2)
02645 {
02646     WORD *t1e = t1+t1[1];
02647     WORD *t2e = t2+t2[1];
02648     WORD *t1a = t1+2;
02649     WORD *t2a = t2+2;
02650     WORD *t3;
02651     while ( t1a < t1e && t2a < t2e ) {
02652         if ( *t1a < *t2a ) {
02653             if ( t1a[1] >= 0 ) {
02654                 t3 = t1a+2;
02655                 while ( t3 < t1e ) { t3[-2] = *t3; t3[-1] = t3[1]; t3 += 2; }
02656                 t1e -= 2;
02657             }
02658             else t1a += 2;
02659         }
02660         else if ( *t1a > *t2a ) {
02661             if ( t2a[1] >= 0 ) t2a += 2;
02662             else {
02663                 t3 = t1e;
02664                 while ( t3 > t1a ) { *t3 = t3[-2]; t3[1] = t3[-1]; t3 -= 2; }
02665                 *t1a++ = *t2a++;
02666                 *t1a++ = *t2a++;
02667                 t1e += 2;
02668             }
02669         }
02670         else {
02671             if ( t2a[1] < t1a[1] ) t1a[1] = t2a[1];
02672             t1a += 2; t2a += 2;
02673         }
02674     }
02675     while ( t2a < t2e ) {
02676         if ( t2a[1] < 0 ) {
02677             *t1a++ = *t2a++;
02678             *t1a++ = *t2a++;
02679         }
02680         else t2a += 2;
02681     }
02682     while ( t1a < t1e ) {
02683         if ( t1a[1] >= 0 ) {
02684             t3 = t1a+2;
02685             while ( t3 < t1e ) { t3[-2] = *t3; t3[-1] = t3[1]; t3 += 2; }
02686             t1e -= 2;
02687         }
02688         else t1a += 2;
02689     }
02690     t1[1] = t1a - t1;
02691     return(0);
02692 }
02693
02694 /*
02695 #] ArgSymbolMerge :
02696 #[ ArgDotproductMerge :
02697 */
02698
02699 int ArgDotproductMerge(WORD *t1, WORD *t2)
02700 {
02701     WORD *t1e = t1+t1[1];
02702     WORD *t2e = t2+t2[1];
02703     WORD *t1a = t1+2;
02704     WORD *t2a = t2+2;
02705     WORD *t3;
02706     while ( t1a < t1e && t2a < t2e ) {
02707         if ( *t1a < *t2a || ( *t1a == *t2a && t1a[1] < t2a[1] ) ) {
02708             if ( t1a[2] >= 0 ) {
02709                 t3 = t1a+3;
02710                 while ( t3 < t1e ) { t3[-3] = *t3; t3[-2] = t3[1]; t3[-1] = t3[2]; t3 += 3; }
02711                 t1e -= 3;
02712             }
02713             else t1a += 3;
02714         }
02715         else if ( *t1a > *t2a || ( *t1a == *t2a && t1a[1] > t2a[1] ) ) {
02716             if ( t2a[2] >= 0 ) t2a += 3;
02717             else {
02718                 t3 = t1e;
02719                 while ( t3 > t1a ) { *t3 = t3[-3]; t3[1] = t3[-2]; t3[2] = t3[-1]; t3 -= 3; }
02720                 *t1a++ = *t2a++;
02721                 *t1a++ = *t2a++;
02722                 *t1a++ = *t2a++;
02723                 t1e += 3;
02724             }
02725         }
02726         else {
02727             if ( t2a[2] < t1a[2] ) t1a[2] = t2a[2];
02728             t1a += 3; t2a += 3;
02729         }
02730     }

```

```

02731     while ( t2a < t2e ) {
02732         if ( t2a[2] < 0 ) {
02733             *t1a++ = *t2a++;
02734             *t1a++ = *t2a++;
02735             *t1a++ = *t2a++;
02736         }
02737         else t2a += 3;
02738     }
02739     while ( t1a < t1e ) {
02740         if ( t1a[2] >= 0 ) {
02741             t3 = t1a+3;
02742             while ( t3 < t1e ) { t3[-3] = *t3; t3[-2] = t3[1]; t3[-1] = t3[2]; t3 += 3; }
02743             t1e -= 3;
02744         }
02745         else t1a += 2;
02746     }
02747     t1[1] = t1a - t1;
02748     return(0);
02749 }
02750
02751 /*
02752  #] ArgDotproductMerge :
02753  #[ TakeArgContent :
02754 */
02767 WORD *TakeArgContent(PHEAD WORD *argin, WORD *argout)
02768 {
02769     GETBIDENTITY
02770     WORD *t, *rnext, *r1, *r2, *r3, *r5, *r6, *r7, *r8, *r9;
02771     WORD pow, *mm, *mnext, *mstop, *argin2 = argin, *argin3 = argin, *argfree;
02772     WORD ncom;
02773     int j, i, act;
02774     r5 = t = argin + ARGHEAD;
02775     r3 = argin + *argin;
02776     rnext = t + *t;
02777     GETSTOP(t,r6);
02778     r1 = argout;
02779     t++;
02780 /*
02781     First pass: arrange everything but the symbols and dotproducts.
02782     They need separate treatment because we have to take out negative
02783     powers.
02784 */
02785     while ( t < r6 ) {
02786 /*
02787         #[ DELTA/VECTOR :
02788 */
02789         if ( *t == DELTA || *t == VECTOR ) {
02790             r7 = t; r8 = t + t[1]; t += 2;
02791             while ( t < r8 ) {
02792                 mm = rnext;
02793                 pow = 1;
02794                 while ( mm < r3 ) {
02795                     mnext = mm + *mm;
02796                     GETSTOP(mm,mstop); mm++;
02797                     while ( mm < mstop ) {
02798                         if ( *mm != *r7 ) mm += mm[1];
02799                         else break;
02800                     }
02801                     if ( *mm == *r7 ) {
02802                         mstop = mm + mm[1]; mm += 2;
02803                         while ( ( *mm != *t || mm[1] != t[1] )
02804                             && mm < mstop ) mm += 2;
02805                         if ( mm >= mstop ) pow = 0;
02806                     }
02807                     else pow = 0;
02808                     if ( pow == 0 ) break;
02809                     mm = mnext;
02810                 }
02811                 if ( pow == 0 ) { t += 2; continue; }
02812 /*
02813                 We have a factor
02814 */
02815                 *r1++ = 8 + ARGHEAD;
02816                 for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
02817                 *r1++ = 8; *r1++ = *r7;
02818                 *r1++ = 4; *r1++ = *t; *r1++ = t[1];
02819                 *r1++ = 1; *r1++ = 1; *r1++ = 3;
02820                 argout = r1;
02821 /*
02822                 Now we have to remove the delta's/vectors
02823 */
02824                 mm = rnext;
02825                 while ( mm < r3 ) {
02826                     mnext = mm + *mm;
02827                     GETSTOP(mm,mstop); mm++;
02828                     while ( mm < mstop ) {
02829                         if ( *mm != *r7 ) mm += mm[1];

```



```

02830         else break;
02831     }
02832     mstop = mm + mm[1]; mm += 2;
02833     while ( mm < mstop && (
02834         *mm != *t || mm[1] != t[1] ) ) mm += 2;
02835     *mm = mm[1] = NOINDEX;
02836     mm = mnext;
02837 }
02838 *t = t[1] = NOINDEX;
02839 t += 2;
02840 }
02841 }
02842 /*
02843 #] DELTA/VECTOR :
02844 #[ INDEX :
02845 */
02846 else if ( *t == INDEX ) {
02847     r7 = t; r8 = t + t[1]; t += 2;
02848     while ( t < r8 ) {
02849         mm = rnext;
02850         pow = 1;
02851         while ( mm < r3 ) {
02852             mnext = mm + *mm;
02853             GETSTOP(mm,mstop); mm++;
02854             while ( mm < mstop ) {
02855                 if ( *mm != *r7 ) mm += mm[1];
02856                 else break;
02857             }
02858             if ( *mm == *r7 ) {
02859                 mstop = mm + mm[1]; mm += 2;
02860                 while ( *mm != *t
02861                     && mm < mstop ) mm++;
02862                 if ( mm >= mstop ) pow = 0;
02863             }
02864             else pow = 0;
02865             if ( pow == 0 ) break;
02866             mm = mnext;
02867         }
02868         if ( pow == 0 ) { t++; continue; }
02869     /*
02870     We have a factor
02871     */
02872     if ( *t < 0 ) { *r1++ = -VECTOR; }
02873     else { *r1++ = -INDEX; }
02874     *r1++ = *t;
02875     argout = r1;
02876     /*
02877     Now we have to remove the index
02878     */
02879     *t = NOINDEX;
02880     mm = rnext;
02881     while ( mm < r3 ) {
02882         mnext = mm + *mm;
02883         GETSTOP(mm,mstop); mm++;
02884         while ( mm < mstop ) {
02885             if ( *mm != *r7 ) mm += mm[1];
02886             else break;
02887         }
02888         mstop = mm + mm[1]; mm += 2;
02889         while ( mm < mstop &&
02890             *mm != *t ) mm += 1;
02891         *mm = NOINDEX;
02892         mm = mnext;
02893     }
02894     t += 1;
02895 }
02896 }
02897 /*
02898 #] INDEX :
02899 #[ FUNCTION :
02900 */
02901 else if ( *t >= FUNCTION ) {
02902     /*
02903     In the next code we should actually look inside
02904     the DENOMINATOR or EXPONENT for noncommuting objects
02905     */
02906     if ( *t >= FUNCTION &&
02907         functions[*t-FUNCTION].commute == 0 ) ncom = 0;
02908     else ncom = 1;
02909     if ( ncom ) {
02910         mm = r5 + 1;
02911         while ( mm < t && ( *mm == DUMMYFUN
02912             || *mm == DUMMYTEN ) ) mm += mm[1];
02913         if ( mm < t ) { t += t[1]; continue; }
02914     }
02915     mm = rnext; pow = 1;
02916     while ( mm < r3 ) {

```

```

02917         mnext = mm + *mm;
02918         GETSTOP(mm,mstop); mm++;
02919         while ( mm < mstop ) {
02920             if ( *mm == *t && mm[1] == t[1] ) {
02921                 for ( i = 2; i < t[1]; i++ ) {
02922                     if ( mm[i] != t[i] ) break;
02923                 }
02924                 if ( i >= t[1] )
02925                     { mm += mm[1]; goto nextmterm; }
02926             }
02927             if ( ncom && *mm != DUMMYFUN && *mm != DUMMYTEN )
02928                 { pow = 0; break; }
02929             mm += mm[1];
02930         }
02931         if ( mm >= mstop ) pow = 0;
02932         if ( pow == 0 ) break;
02933 nextmterm:    mm = mnext;
02934     }
02935     if ( pow == 0 ) { t += t[1]; continue; }
02936 /*
02937     Copy the function
02938 */
02939     *r1++ = t[1] + 4 + ARGHEAD;
02940     for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
02941     *r1++ = t[1] + 4;
02942     for ( i = 0; i < t[1]; i++ ) *r1++ = t[i];
02943     *r1++ = 1; *r1++ = 1; *r1++ = 3;
02944    argout = r1;
02945 /*
02946     Now we have to take out the functions
02947 */
02948     mm = rnext;
02949     while ( mm < r3 ) {
02950         mnext = mm + *mm;
02951         GETSTOP(mm,mstop); mm++;
02952         while ( mm < mstop ) {
02953             if ( *mm == *t && mm[1] == t[1] ) {
02954                 for ( i = 2; i < t[1]; i++ ) {
02955                     if ( mm[i] != t[i] ) break;
02956                 }
02957                 if ( i >= t[1] ) {
02958                     if ( functions[*t-FUNCTION].spec > 0 )
02959                         *mm = DUMMYTEN;
02960                     else
02961                         *mm = DUMMYFUN;
02962                     mm += mm[1];
02963                     goto nextterm;
02964                 }
02965             }
02966             mm += mm[1];
02967         }
02968 nextterm:    mm = mnext;
02969     }
02970     if ( functions[*t-FUNCTION].spec > 0 )
02971         *t = DUMMYTEN;
02972     else
02973         *t = DUMMYFUN;
02974     t += t[1];
02975 }
02976 /*
02977     #] FUNCTION :
02978 */
02979     else {
02980         t += t[1];
02981     }
02982 }
02983 /*
02984     #[ SYMBOL :
02985
02986     Now collect all symbols. We can use the space after r1 as storage
02987 */
02988     t = argin+ARGHEAD;
02989     rnext = t + *t;
02990     r2 = r1;
02991     while ( t < r3 ) {
02992         GETSTOP(t,r6);
02993         t++;
02994         act = 0;
02995         while ( t < r6 ) {
02996             if ( *t == SYMBOL ) {
02997                 act = 1;
02998                 i = t[1];
02999                 NCOPY(r2,t,i)
03000             }
03001             else { t += t[1]; }
03002         }
03003         if ( act == 0 ) {

```

```

03004         *r2++ = SYMBOL; *r2++ = 2;
03005     }
03006     t = rnext; rnext = rnext + *rnext;
03007 }
03008 *r2 = 0;
03009 argin2 = argin;
03010 /*
03011     Now we have a list of all symbols as a sequence of SYMBOL subterms.
03012     Any symbol that is absent in a subterm has power zero.
03013     We now need a list of all minimum powers.
03014     This can be done by subsequent merges.
03015 */
03016 r7 = r1;          /* The first object into which we merge. */
03017 r8 = r7 + r7[1];  /* The object that gets merged into r7. */
03018 while ( *r8 ) {
03019     r2 = r8 + r8[1]; /* Next object */
03020     ArgSymbolMerge(r7,r8);
03021     r8 = r2;
03022 }
03023 /*
03024     Now we have to divide by the object in r7 and take it apart as factors.
03025     The division can be simple if there are no negative powers.
03026 */
03027 if ( r7[1] > 2 ) {
03028     r8 = r7+2;
03029     r2 = r7 + r7[1];
03030     act = 0;
03031     pow = 0;
03032     while ( r8 < r2 ) {
03033         if ( r8[1] < 0 ) { act = 1; pow += -r8[1]*(ARGHEAD+8); }
03034         else { pow += 2*r8[1]; }
03035         r8 += 2;
03036     }
03037 /*
03038     The amount of space we need to move r7 is given in pow
03039 */
03040 if ( act == 0 ) { /* this can be done 'in situ' */
03041     t = argin + ARGHEAD;
03042     while ( t < r3 ) {
03043         rnext = t + *t;
03044         GETSTOP(t,r6);
03045         t++;
03046         while ( t < r6 ) {
03047             if ( *t != SYMBOL ) { t += t[1]; continue; }
03048             r8 = r7+2; r9 = t + t[1]; t += 2;
03049             while ( ( t < r9 ) && ( r8 < r2 ) ) {
03050                 if ( *t == *r8 ) {
03051                     t[1] -= r8[1]; t += 2; r8 += 2;
03052                 }
03053                 else { /* *t must be < than *r8 !!! */
03054                     t += 2;
03055                 }
03056             }
03057             t = r9;
03058         }
03059         t = rnext;
03060     }
03061 /*
03062     And now the factors that go to argout.
03063     First we have to move r7 out of the way.
03064 */
03065 r8 = r7+pow; i = r7[1];
03066 while ( --i >= 0 ) r8[i] = r7[i];
03067 r2 += pow;
03068 r8 += 2;
03069 while ( r8 < r2 ) {
03070     for ( i = 0; i < r8[1]; i++ ) { *r1++ = -SYMBOL; *r1++ = *r8; }
03071     r8 += 2;
03072 }
03073 }
03074 else { /* this needs a new location */
03075     argin2 = TermMalloc("TakeArgContent2");
03076 /*
03077     We have to multiply the inverse of r7 into argin
03078     The answer should go to argin2.
03079 */
03080 r5 = argin2; *r5++ = 0; *r5++ = 0; FILLARG(r5);
03081 t = argin+ARGHEAD;
03082 while ( t < r3 ) {
03083     rnext = t + *t;
03084     GETSTOP(t,r6);
03085     r9 = r5;
03086     *r5++ = *t++ + r7[1];
03087     while ( t < r6 ) *r5++ = *t++;
03088     i = r7[1] - 2; r8 = r7+2;
03089     *r5++ = r7[0]; *r5++ = r7[1];
03090     while ( i > 0 ) { *r5++ = *r8++; *r5++ = -*r8++; i -= 2; }

```

```

03091         while ( t < rnext ) *r5++ = *t++;
03092         Normalize(BHEAD r9);
03093         r5 = r9 + *r9;
03094     }
03095     *r5 = 0;
03096     *argin2 = r5-argin2;
03097 /*
03098     We may have to sort the terms in argin2.
03099 */
03100     NewSort(BHEAD0);
03101     t = argin2+ARGHEAD;
03102     while ( *t ) {
03103         StoreTerm(BHEAD t);
03104         t += *t;
03105     }
03106     t = argin2+ARGHEAD;
03107     if ( EndSort(BHEAD t,0) < 0 ) goto Irreg;
03108     while ( *t ) t += *t;
03109     *argin2 = t - argin2;
03110     r3 = t;
03111 /*
03112     And now the factors that go to argout.
03113     First we have to move r7 out of the way.
03114 */
03115     r8 = r7+pow; i = r7[1];
03116     while ( --i >= 0 ) r8[i] = r7[i];
03117     r2 += pow;
03118     r8 += 2;
03119     while ( r8 < r2 ) {
03120         if ( r8[1] >= 0 ) {
03121             for ( i = 0; i < r8[1]; i++ ) { *r1++ = -SYMBOL; *r1++ = *r8; }
03122         }
03123         else {
03124             for ( i = 0; i < -r8[1]; i++ ) {
03125                 *r1++ = ARGHEAD+8; *r1++ = 0;
03126                 FILLARG(r1);
03127                 *r1++ = 8; *r1++ = SYMBOL; *r1++ = 4; *r1++ = *r8;
03128                 *r1++ = -1; *r1++ = 1; *r1++ = 1; *r1++ = 3;
03129             }
03130         }
03131         r8 += 2;
03132     }
03133     argout = r1;
03134 }
03135 }
03136 /*
03137     #] SYMBOL :
03138     #[ DOTPRODUCT :
03139
03140     Now collect all dotproducts. We can use the space after r1 as storage
03141 */
03142     t = argin2+ARGHEAD;
03143     rnext = t + *t;
03144     r2 = r1;
03145     while ( t < r3 ) {
03146         GETSTOP(t,r6);
03147         t++;
03148         act = 0;
03149         while ( t < r6 ) {
03150             if ( *t == DOTPRODUCT ) {
03151                 act = 1;
03152                 i = t[1];
03153                 NCOPY(r2,t,i)
03154             }
03155             else { t += t[1]; }
03156         }
03157         if ( act == 0 ) {
03158             *r2++ = DOTPRODUCT; *r2++ = 2;
03159         }
03160         t = rnext; rnext = rnext + *rnext;
03161     }
03162     *r2 = 0;
03163     argin3 = argin2;
03164 /*
03165     Now we have a list of all dotproducts as a sequence of DOTPRODUCT
03166     subterms. Any dotproduct that is absent in a subterm has power zero.
03167     We now need a list of all minimum powers.
03168     This can be done by subsequent merges.
03169 */
03170     r7 = r1; /* The first object into which we merge. */
03171     r8 = r7 + r7[1]; /* The object that gets merged into r7. */
03172     while ( *r8 ) {
03173         r2 = r8 + r8[1]; /* Next object */
03174         ArgDotproductMerge(r7,r8);
03175         r8 = r2;
03176     }
03177 /*

```

```

03178      Now we have to divide by the object in r7 and take it apart as factors.
03179      The division can be simple if there are no negative powers.
03180  */
03181  if ( r7[1] > 2 ) {
03182      r8 = r7+2;
03183      r2 = r7 + r7[1];
03184      act = 0;
03185      pow = 0;
03186      while ( r8 < r2 ) {
03187          if ( r8[2] < 0 ) { pow += -r8[2]*(ARGHEAD+9); }
03188          else { pow += r8[2]*(ARGHEAD+9); }
03189          r8 += 3;
03190      }
03191  /*
03192      The amount of space we need to move r7 is given in pow
03193      For dotproducts we always need a new location
03194  */
03195      {
03196          argin3 = TermMalloc("TakeArgContent3");
03197      /*
03198          We have to multiply the inverse of r7 into argin
03199          The answer should go to argin2.
03200  */
03201          r5 = argin3; *r5++ = 0; *r5++ = 0; FILLARG(r5);
03202          t = argin2+ARGHEAD;
03203          while ( t < r3 ) {
03204              rnext = t + *t;
03205              GETSTOP(t,r6);
03206              r9 = r5;
03207              *r5++ = *t++ + r7[1];
03208              while ( t < r6 ) *r5++ = *t++;
03209              i = r7[1] - 2; r8 = r7+2;
03210              *r5++ = r7[0]; *r5++ = r7[1];
03211              while ( i > 0 ) { *r5++ = *r8++; *r5++ = *r8++; *r5++ = -*r8++; i -= 3; }
03212              while ( t < rnext ) *r5++ = *t++;
03213              Normalize(BHEAD r9);
03214              r5 = r9 + *r9;
03215          }
03216          *r5 = 0;
03217          *argin3 = r5-argin3;
03218  /*
03219          We may have to sort the terms in argin3.
03220  */
03221          NewSort (BHEAD0);
03222          t = argin3+ARGHEAD;
03223          while ( *t ) {
03224              StoreTerm(BHEAD t);
03225              t += *t;
03226          }
03227          t = argin3+ARGHEAD;
03228          if ( EndSort(BHEAD t,0) < 0 ) goto Irreg;
03229          while ( *t ) t += *t;
03230          *argin3 = t - argin3;
03231          r3 = t;
03232  /*
03233          And now the factors that go to argout.
03234          First we have to move r7 out of the way.
03235  */
03236          r8 = r7+pow; i = r7[1];
03237          while ( --i >= 0 ) r8[i] = r7[i];
03238          r2 += pow;
03239          r8 += 2;
03240          while ( r8 < r2 ) {
03241              for ( i = ABS(r8[2]); i > 0; i-- ) {
03242                  *r1++ = ARGHEAD+9; *r1++ = 0; FILLARG(r1);
03243                  *r1++ = 9; *r1++ = DOTPRODUCT; *r1++ = 5; *r1++ = *r8;
03244                  *r1++ = r8[1]; *r1++ = r8[2] < 0 ? -1: 1;
03245                  *r1++ = 1; *r1++ = 1; *r1++ = 3;
03246              }
03247              r8 += 3;
03248          }
03249          argout = r1;
03250      }
03251  }
03252  /*
03253      #] DOTPRODUCT :
03254
03255      We have now in argin3 the argument stripped of negative powers and
03256      common factors. The only thing left to deal with is to make the
03257      coefficients integer. For that we have to find the LCM of the denominators
03258      and the GCD of the numerators. And to start with, the sign.
03259      We force the sign of the first term to be positive.
03260  */
03261      t = argin3 + ARGHEAD; pow = 1;
03262      t += *t;
03263      if ( t[-1] < 0 ) {
03264          pow = 0;

```

```

03265         t[-1] = -t[-1];
03266         while ( t < r3 ) {
03267             t += *t; t[-1] = -t[-1];
03268         }
03269     }
03270 /*
03271 Now the GCD of the numerators and the LCM of the denominators:
03272 */
03273     argfree = TermMalloc("TakeArgContent1");
03274     if ( AN.cmod != 0 ) {
03275         r1 = MakeMod(BHEAD argin3,r1,argfree);
03276     }
03277     else {
03278         r1 = MakeInteger(BHEAD argin3,r1,argfree);
03279     }
03280     if ( pow == 0 ) {
03281         *r1++ = -SNUMBER;
03282         *r1++ = -1;
03283     }
03284     *r1 = 0;
03285 /*
03286 Cleanup
03287 */
03288     if ( argin3 != argin2 ) TermFree(argin3,"TakeArgContent3");
03289     if ( argin2 != argin ) TermFree(argin2,"TakeArgContent2");
03290     return(argfree);
03291 Irreg:
03292     MesPrint("Irregularity while sorting argument in TakeArgContent");
03293     if ( argin3 != argin2 ) TermFree(argin3,"TakeArgContent3");
03294     if ( argin2 != argin ) TermFree(argin2,"TakeArgContent2");
03295     Terminate(-1);
03296     return(0);
03297 }
03298
03299 /*
03300 #] TakeArgContent :
03301 #[ MakeInteger :
03302 */
03313 WORD *MakeInteger(PHEAD WORD *argin,WORD *argout,WORD *argfree)
03314 {
03315     UWORD *GCDBuffer, *GCDBuffer2, *LCMbuffer, *LCMb, *LCMc;
03316     WORD *r, *r1, *r2, *r3, *r4, *r5, *rnext, i, k, j;
03317     WORD kGCD, kLCM, kGCD2, kkLCM, jLCM, jGCD;
03318     GCDBuffer = NumberMalloc("MakeInteger");
03319     GCDBuffer2 = NumberMalloc("MakeInteger");
03320     LCMbuffer = NumberMalloc("MakeInteger");
03321     LCMb = NumberMalloc("MakeInteger");
03322     LCMc = NumberMalloc("MakeInteger");
03323     r4 = argin + *argin;
03324     r = argin + ARGHEAD;
03325 /*
03326 First take the first term to load up the LCM and the GCD
03327 */
03328     r2 = r + *r;
03329     j = r2[-1];
03330     r3 = r2 - ABS(j);
03331     k = REDLENG(j);
03332     if ( k < 0 ) k = -k;
03333     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
03334     for ( kGCD = 0; kGCD < k; kGCD++ ) GCDBuffer[kGCD] = r3[kGCD];
03335     k = REDLENG(j);
03336     if ( k < 0 ) k = -k;
03337     r3 += k;
03338     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
03339     for ( kLCM = 0; kLCM < k; kLCM++ ) LCMbuffer[kLCM] = r3[kLCM];
03340     r1 = r2;
03341 /*
03342 Now go through the rest of the terms in this argument.
03343 */
03344     while ( r1 < r4 ) {
03345         r2 = r1 + *r1;
03346         j = r2[-1];
03347         r3 = r2 - ABS(j);
03348         k = REDLENG(j);
03349         if ( k < 0 ) k = -k;
03350         while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
03351         if ( ( ( GCDBuffer[0] == 1 ) && ( kGCD == 1 ) ) ) {
03352 /*
03353             GCD is already 1
03354 */
03355         }
03356         else if ( ( ( k != 1 ) || ( r3[0] != 1 ) ) ) {
03357             if ( GcdLong(BHEAD GCDBuffer,kGCD,(UWORD *)r3,k,GCDBuffer2,&kGCD2) ) {
03358                 NumberFree(GCDBuffer,"MakeInteger");
03359                 NumberFree(GCDBuffer2,"MakeInteger");
03360                 NumberFree(LCMbuffer,"MakeInteger");
03361                 NumberFree(LCMb,"MakeInteger"); NumberFree(LCMc,"MakeInteger");

```

```

03362         goto MakeIntegerErr;
03363     }
03364     kGCD = kGCD2;
03365     for ( i = 0; i < kGCD; i++ ) GCDbuffer[i] = GCDbuffer2[i];
03366 }
03367 else {
03368     kGCD = 1; GCDbuffer[0] = 1;
03369 }
03370 k = REDLENG(j);
03371 if ( k < 0 ) k = -k;
03372 r3 += k;
03373 while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
03374 if ( ( ( LCMbuffer[0] == 1 ) && ( kLCM == 1 ) ) ) {
03375     for ( kLCM = 0; kLCM < k; kLCM++ )
03376         LCMbuffer[kLCM] = r3[kLCM];
03377 }
03378 else if ( ( k != 1 ) || ( r3[0] != 1 ) ) {
03379     if ( GcdLong(BHEAD LCMbuffer, kLCM, (UWORD *)r3, k, LCMb, &kLCM) ) {
03380         NumberFree(GCDbuffer, "MakeInteger"); NumberFree(GCDbuffer2, "MakeInteger");
03381         NumberFree(LCMbuffer, "MakeInteger"); NumberFree(LCMb, "MakeInteger");
03382         NumberFree(LCMc, "MakeInteger");
03383         goto MakeIntegerErr;
03384     }
03385     DivLong((UWORD *)r3, k, LCMb, kLCM, LCMb, &kLCM, LCMc, &jLCM);
03386     MulLong(LCMbuffer, kLCM, LCMb, kLCM, LCMc, &jLCM);
03387     for ( kLCM = 0; kLCM < jLCM; kLCM++ )
03388         LCMbuffer[kLCM] = LCMc[kLCM];
03389 }
03390 else { /* LCM doesn't change */
03391     r1 = r2;
03392 }
03393 /*
03394 Now put the factor together: GCD/LCM
03395 */
03396 r3 = (WORD *) (GCDbuffer);
03397 if ( kGCD == kLCM ) {
03398     for ( jGCD = 0; jGCD < kGCD; jGCD++ )
03399         r3[jGCD+kGCD] = LCMbuffer[jGCD];
03400     k = kGCD;
03401 }
03402 else if ( kGCD > kLCM ) {
03403     for ( jGCD = 0; jGCD < kLCM; jGCD++ )
03404         r3[jGCD+kGCD] = LCMbuffer[jGCD];
03405     for ( jGCD = kLCM; jGCD < kGCD; jGCD++ )
03406         r3[jGCD+kGCD] = 0;
03407     k = kGCD;
03408 }
03409 else {
03410     for ( jGCD = kGCD; jGCD < kLCM; jGCD++ )
03411         r3[jGCD] = 0;
03412     for ( jGCD = 0; jGCD < kLCM; jGCD++ )
03413         r3[jGCD+kLCM] = LCMbuffer[jGCD];
03414     k = kLCM;
03415 }
03416 j = 2*k+1;
03417 /*
03418 Now we have to write this to argout
03419 */
03420 if ( ( j == 3 ) && ( r3[1] == 1 ) && ( (WORD)(r3[0]) > 0 ) ) {
03421     *argout = -SNUMBER;
03422     argout[1] = r3[0];
03423     r1 = argout+2;
03424 }
03425 else {
03426     r1 = argout;
03427     *r1++ = j+1+ARGHEAD; *r1++ = 0; FILLARG(r1);
03428     *r1++ = j+1; r2 = r3;
03429     for ( i = 0; i < k; i++ ) { *r1++ = *r2++; *r1++ = *r2++; }
03430     *r1++ = j;
03431 }
03432 /*
03433 Next we have to take the factor out from the argument.
03434 This cannot be done in location, because the denominator stuff can make
03435 coefficients longer.
03436 */
03437 r2 = argfree + 2; FILLARG(r2)
03438 while ( r < r4 ) {
03439     rnext = r + *r;
03440     j = ABS(rnext[-1]);
03441     r5 = rnext - j;
03442     r3 = r2;
03443     while ( r < r5 ) *r2++ = *r++;
03444     j = (j-1)/2; /* reduced length. Remember, k is the other red length */
03445     if ( DivRat(BHEAD (UWORD *)r5, j, GCDbuffer, k, (UWORD *)r2, &i) ) {
03446         goto MakeIntegerErr;
03447     }
03448     i = 2*i+1;

```

```

03448         r2 = r2 + i;
03449         if ( rnext[-1] < 0 ) r2[-1] = -i;
03450         else                 r2[-1] = i;
03451         *r3 = r2-r3;
03452         r = rnext;
03453     }
03454     *r2 = 0;
03455     argfree[0] = r2-argfree;
03456     argfree[1] = 0;
03457 /*
03458 Cleanup
03459 */
03460     NumberFree(LCMc,"MakeInteger");
03461     NumberFree(LCmb,"MakeInteger");
03462     NumberFree(LCMbuffer,"MakeInteger");
03463     NumberFree(GCdbuffer2,"MakeInteger");
03464     NumberFree(GCdbuffer,"MakeInteger");
03465     return(r1);
03466
03467 MakeIntegerErr:
03468     MesCall("MakeInteger");
03469     Terminate(-1);
03470     return(0);
03471 }
03472
03473 /*
03474 #] MakeInteger :
03475 #[ MakeMod :
03476 */
03484 WORD *MakeMod(PHEAD WORD *argin,WORD *argout,WORD *argfree)
03485 {
03486     WORD *r, *instop, *r1, *m, x, xx, ix, ip;
03487     int i;
03488     r = argin; instop = r + *r; r += ARGHEAD;
03489     x = r[*r-3];
03490     if ( r[*r-1] < 0 ) x += AN.cmod[0];
03491     if ( GetModInverses(x,(WORD)(AN.cmod[0]),&ix,&ip) ) {
03492         Terminate(-1);
03493     }
03494     argout[0] = -SNUMBER;
03495     argout[1] = x;
03496     argout[2] = 0;
03497     r1 = argout+2;
03498 /*
03499 Now we have to multiply all coefficients by ix.
03500 This does not make things longer, but we should keep to the conventions
03501 of MakeInteger.
03502 */
03503     m = argfree + ARGHEAD;
03504     while ( r < instop ) {
03505         xx = r[*r-3]; if ( r[*r-1] < 0 ) xx += AN.cmod[0];
03506         xx = (WORD)((LONG)xx*ix) % AN.cmod[0];
03507         if ( xx != 0 ) {
03508             i = *r; NCOPY(m,r,i);
03509             m[-3] = xx; m[-1] = 3;
03510         }
03511         else { r += *r; }
03512     }
03513     *m = 0;
03514     *argfree = m - argfree;
03515     argfree[1] = 0;
03516     argfree += 2; FILLARG(argfree);
03517     return(r1);
03518 }
03519
03520 /*
03521 #] MakeMod :
03522 #[ SortWeights :
03523 */
03529 void SortWeights(LONG *weights, LONG *extraspace, WORD number)
03530 {
03531     LONG w, *fill, *from1, *from2;
03532     int n1,n2,i;
03533     if ( number >= 4 ) {
03534         n1 = number/2; n2 = number - n1;
03535         SortWeights(weights,extraspace,n1);
03536         SortWeights(weights+n1,extraspace,n2);
03537 /*
03538 We copy the first patch to the extra space. Then we merge
03539 Note that a potential remaining n2 objects are already in place.
03540 */
03541         for ( i = 0; i < n1; i++ ) extraspace[i] = weights[i];
03542         fill = weights; from1 = extraspace; from2 = weights+n1;
03543         while ( n1 > 0 && n2 > 0 ) {
03544             if ( *from1 <= *from2 ) { *fill++ = *from1++; n1--; }
03545             else { *fill++ = *from2++; n2--; }
03546         }

```



```

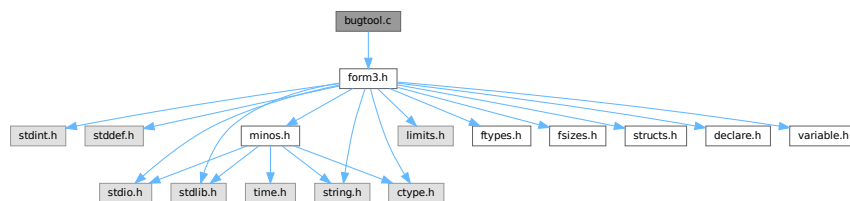
03547     while ( n1 > 0 ) { *fill++ = *from1++; n1--; }
03548   }
03549   /*
03550   Special cases
03551   */
03552   else if ( number == 3 ) { /* 6 permutations of which one is trivial */
03553     if ( weights[0] > weights[1] ) {
03554       if ( weights[1] > weights[2] ) {
03555         w = weights[0]; weights[0] = weights[2]; weights[2] = w;
03556       }
03557       else if ( weights[0] > weights[2] ) {
03558         w = weights[0]; weights[0] = weights[1];
03559         weights[1] = weights[2]; weights[2] = w;
03560       }
03561       else {
03562         w = weights[0]; weights[0] = weights[1]; weights[1] = w;
03563       }
03564     }
03565     else if ( weights[0] > weights[2] ) {
03566       w = weights[0]; weights[0] = weights[2];
03567       weights[2] = weights[1]; weights[1] = w;
03568     }
03569     else if ( weights[1] > weights[2] ) {
03570       w = weights[1]; weights[1] = weights[2]; weights[2] = w;
03571     }
03572   }
03573   else if ( number == 2 ) {
03574     if ( weights[0] > weights[1] ) {
03575       w = weights[0]; weights[0] = weights[1]; weights[1] = w;
03576     }
03577   }
03578   return;
03579 }
03580
03581 /*
03582 #] SortWeights :
03583 */

```

## 4.3 bugtool.c File Reference

#include "form3.h"

Include dependency graph for bugtool.c:



### Functions

- void [ExprStatus](#) (EXPRESSIONS e)

### 4.3.1 Detailed Description

Low level routines for debugging

Definition in file [bugtool.c](#).

## 4.3.2 Function Documentation

### 4.3.2.1 ExprStatus()

```
void ExprStatus (
    EXPRESSIONS e )
```

Definition at line 66 of file [bugtool.c](#).

## 4.4 bugtool.c

[Go to the documentation of this file.](#)

```
00001
00006 /* #[] License : */
00007 /*
00008 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 * When using this file you are requested to refer to the publication
00010 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 * This is considered a matter of courtesy as the development was paid
00012 * for by FOM the Dutch physics granting agency and we would like to
00013 * be able to track its scientific use to convince FOM of its value
00014 * for the community.
00015 *
00016 * This file is part of FORM.
00017 *
00018 * FORM is free software: you can redistribute it and/or modify it under the
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00020 * Foundation, either version 3 of the License, or (at your option) any later
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00022 *
00023 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 * details.
00027 *
00028 * You should have received a copy of the GNU General Public License along
00029 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #[] License : */
00032
00033 /*
00034 #[] Includes :
00035 */
00036
00037 #include "form3.h"
00038
00039 /*
00040 #[] Includes :
00041 #[] ExprStatus :
00042 */
00043
00044 static UBYTE *statusexpr[] = {
00045     (UBYTE *) "LOCALEXPRESSION"
00046     , (UBYTE *) "SKIPLEXPRESSION"
00047     , (UBYTE *) "DROPLEXPRESSION"
00048     , (UBYTE *) "DROPEDEXPRESSION"
00049     , (UBYTE *) "GLOBALEXPRESSION"
00050     , (UBYTE *) "SKIPGEXPRESSION"
00051     , (UBYTE *) "DROPGEXPRESSION"
00052     , (UBYTE *) "UNKNOWN"
00053     , (UBYTE *) "STOREDEXPRESSION"
00054     , (UBYTE *) "HIDDENLEXPRESSION"
00055     , (UBYTE *) "HIDELEXPRESSION"
00056     , (UBYTE *) "DROPHLEXPRESSION"
00057     , (UBYTE *) "UNHIDELEXPRESSION"
00058     , (UBYTE *) "HIDDENGEXPRESSION"
00059     , (UBYTE *) "HIDEGEXPRESSION"
00060     , (UBYTE *) "DROPHGEXPRESSION"
00061     , (UBYTE *) "UNHIDEGEXPRESSION"
00062     , (UBYTE *) "INTOHIDELEXPRESSION"
00063     , (UBYTE *) "INTOHIDEGEXPRESSION"
00064 };
00065
00066 void ExprStatus(EXPRESSIONS e)
00067 {
00068     MesPrint("Expression %s(%d) has status %s(%d,%d). Buffer: %d, Position: %15p",
```

```

00069      AC.exprnames->namebuffer+e->name, (WORD) (e->Expressions),
00070      statusexpr[e->status], e->status, e->hidelevel,
00071      e->whichbuffer, &(e->onfile));
00072  }
00073
00074  /*
00075   #] ExprStatus :
00076   */

```

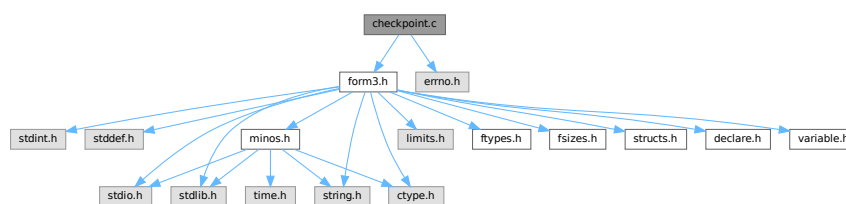
## 4.5 checkpoint.c File Reference

```

#include "form3.h"
#include <errno.h>

```

Include dependency graph for checkpoint.c:



### Macros

- #define [CACHED\\_SNAPSHOT](#)
- #define [CACHE\\_SIZE](#) 4096
- #define [R\\_FREE](#)(ARG) if ( ARG ) M\_free(ARG, #ARG);
- #define [R\\_FREE\\_NAMETREE](#)(ARG)
- #define [R\\_FREE\\_STREAM](#)(ARG)
- #define [R\\_SET](#)(VAR, TYPE) VAR = \*((TYPE\*)p); p = (unsigned char\*)p + sizeof(TYPE);
- #define [R\\_COPY\\_B](#)(VAR, SIZE, CAST)
- #define [S\\_WRITE\\_B](#)(BUF, LEN) if ( fwrite\_cached(BUF, 1, LEN, fd) != (size\_t)(LEN) ) return(\_\_LINE\_\_);
- #define [S\\_FLUSH\\_B](#) if ( flush\_cache(fd) != 1 ) return(\_\_LINE\_\_);
- #define [R\\_COPY\\_S](#)(VAR, CAST)
- #define [S\\_WRITE\\_S](#)(STR)
- #define [R\\_COPY\\_LIST](#)(ARG)
- #define [S\\_WRITE\\_LIST](#)(LST)
- #define [R\\_COPY\\_NAMETREE](#)(ARG)
- #define [S\\_WRITE\\_NAMETREE](#)(ARG)
- #define [S\\_WRITE\\_DOLLAR](#)(ARG)
- #define [ANNOUNCE](#)(str)

### Functions

- int [CheckRecoveryFile](#) (void)
- void [DeleteRecoveryFile](#) (void)
- char \* [RecoveryFilename](#) (void)
- void [InitRecovery](#) (void)
- size\_t [fwrite\\_cached](#) (const void \*ptr, size\_t size, size\_t nmemb, FILE \*fd)
- size\_t [flush\\_cache](#) (FILE \*fd)
- int [DoRecovery](#) (int \*moduletype)
- void [DoCheckpoint](#) (int moduletype)

## Variables

- unsigned char `cache_buffer` [CACHE\_SIZE]
- size\_t `cache_fill` = 0

### 4.5.1 Detailed Description

Contains all functions that deal with the recovery mechanism controlled and activated by the On Checkpoint switch.

The main function are DoCheckpoint, DoRecovery, and DoSnapshot. If the checkpoints are activated DoCheckpoint is called every time a module is finished executing. If the conditions for the creation of a recovery snapshot are met DoCheckpoint calls DoSnapshot. DoRecovery is called once when FORM starts up with the command line argument -R. Most of the other code contains debugging facilities that are only compiled if the macro PRINTDEBUG is defined.

The recovery mechanism is atomic, i.e. only if everything went well, the final recovery file is created (and the older one overwritten) in a single step (copying). If some errors occur, a warning is issued and the program continues without having created a new recovery file. The only situation in which the creation of the recovery data leads to a termination of the running program is if not enough disk or memory space is left.

For ParFORM each slave creates its own recovery file, sends it to the master and then it deletes the recovery file. The master stores all the recovery files and on recovery it feeds these files to the slaves. It is nearly impossible to recover after some MPI fault so ParFORM terminates on any recovery failure.

DoRecovery and DoSnapshot do the loading and saving of the recovery data, respectively. Every change in one functions needs to be accompanied by the appropriate change in the other function. The structure of both functions is quite similar. They handle the relevant global structs one after the other and then care about the copying of the hide and scratch files.

The names of the recovery, scratch and hide files are hard-coded in the variables in fold "filenames and system commands".

If the global structs AM,AP,AC,AR are changed, DoRecovery and DoSnapshot usually also have to be changed. Some structs are read/written as a whole (AP,AC), some are read/written only partly as a selection of their individual elements (AM,AR). If AM or AR have been changed by adding or removing an element that is important for the runtime status, then the reading/writing statements have to be added to or removed from DoRecovery and DoSnapshot. If AP or AC are changed, then for non-pointer variables (in the case of a struct it also means that none of its elements is a pointer) nothing has to be changed in the functions here. If pointers are involved, extra code has to be added (or removed). See the comments of DoRecovery and DoSnapshot.

Definition in file [checkpoint.c](#).

### 4.5.2 Macro Definition Documentation

#### 4.5.2.1 CACHED\_SNAPSHOT

```
#define CACHED_SNAPSHOT
```

Definition at line 1214 of file [checkpoint.c](#).

#### 4.5.2.2 CACHE\_SIZE

```
#define CACHE_SIZE 4096
```

Definition at line 1216 of file [checkpoint.c](#).

#### 4.5.2.3 R\_FREE

```
#define R_FREE(  
    ARG )    if ( ARG ) M_free(ARG, #ARG);
```

Definition at line 1281 of file [checkpoint.c](#).

#### 4.5.2.4 R\_FREE\_NAMETREE

```
#define R_FREE_NAMETREE(  
    ARG )
```

**Value:**

```
R_FREE(ARG->namenode); \  
R_FREE(ARG->namebuffer); \  
R_FREE(ARG);
```

Definition at line 1284 of file [checkpoint.c](#).

#### 4.5.2.5 R\_FREE\_STREAM

```
#define R_FREE_STREAM(  
    ARG )
```

**Value:**

```
R_FREE(ARG.buffer); \  
R_FREE(ARG.FoldName); \  
R_FREE(ARG.name);
```

Definition at line 1289 of file [checkpoint.c](#).

#### 4.5.2.6 R\_SET

```
#define R_SET(  
    VAR,  
    TYPE )    VAR = *((TYPE*)p); p = (unsigned char*)p + sizeof(TYPE);
```

Definition at line 1296 of file [checkpoint.c](#).

#### 4.5.2.7 R\_COPY\_B

```
#define R_COPY_B(  
    VAR,  
    SIZE,  
    CAST )
```

**Value:**

```
VAR = (CAST)Malloc1(SIZE, #VAR); \  
memcpy(VAR, p, SIZE); p = (unsigned char*)p + SIZE;
```

Definition at line 1301 of file [checkpoint.c](#).

#### 4.5.2.8 S\_WRITE\_B

```
#define S_WRITE_B(
    BUF,
    LEN ) if ( fwrite_cached(BUF, 1, LEN, fd) != (size_t) (LEN) ) return(__LINE__);
```

Definition at line 1305 of file [checkpoint.c](#).

#### 4.5.2.9 S\_FLUSH\_B

```
#define S_FLUSH_B if ( flush_cache(fd) != 1 ) return(__LINE__);
```

Definition at line 1308 of file [checkpoint.c](#).

#### 4.5.2.10 R\_COPY\_S

```
#define R_COPY_S(
    VAR,
    CAST )
```

**Value:**

```
if ( VAR ) { \
    VAR = (CAST)Malloc1(strlen(p)+1,"R_COPY_S"); \
    strcpy((char*)VAR, p); p = (unsigned char*)p + strlen(p) + 1; \
}
```

Definition at line 1313 of file [checkpoint.c](#).

#### 4.5.2.11 S\_WRITE\_S

```
#define S_WRITE_S(
    STR )
```

**Value:**

```
if ( STR ) { \
    l = strlen((char*)STR) + 1; \
    if ( fwrite_cached(STR, 1, l, fd) != (size_t)l ) return(__LINE__); \
}
```

Definition at line 1319 of file [checkpoint.c](#).

#### 4.5.2.12 R\_COPY\_LIST

```
#define R_COPY_LIST(
    ARG )
```

**Value:**

```
if ( ARG.maxnum ) { \
    R_COPY_B(ARG.lijst, ARG.size*ARG.maxnum, void*) \
}
```

Definition at line 1327 of file [checkpoint.c](#).

#### 4.5.2.13 S\_WRITE\_LIST

```
#define S_WRITE_LIST(
    LST )
```

**Value:**

```
if ( LST.maxnum ) { \
    S_WRITE_B((char*)LST.lijst, LST.maxnum*LST.size) \
}
```

Definition at line 1332 of file [checkpoint.c](#).

#### 4.5.2.14 R\_COPY\_NAMETREE

```
#define R_COPY_NAMETREE(
    ARG )
```

**Value:**

```
R_COPY_B(ARG, sizeof(NAMETREE), NAMETREE*); \
if ( ARG->namenode ) { \
    R_COPY_B(ARG->namenode, ARG->nodesize*sizeof(NAMENODE), NAMENODE*); \
} \
if ( ARG->namebuffer ) { \
    R_COPY_B(ARG->namebuffer, ARG->namesize, UBYTE*); \
}
```

Definition at line 1339 of file [checkpoint.c](#).

#### 4.5.2.15 S\_WRITE\_NAMETREE

```
#define S_WRITE_NAMETREE(
    ARG )
```

**Value:**

```
S_WRITE_B(ARG, sizeof(NAMETREE)); \
if ( ARG->namenode ) { \
    S_WRITE_B(ARG->namenode, ARG->nodesize*sizeof(struct NaMeNode)); \
} \
if ( ARG->namebuffer ) { \
    S_WRITE_B(ARG->namebuffer, ARG->namesize); \
}
```

Definition at line 1348 of file [checkpoint.c](#).

#### 4.5.2.16 S\_WRITE\_DOLLAR

```
#define S_WRITE_DOLLAR(
    ARG )
```

**Value:**

```
if ( ARG.size && ARG.where && ARG.where != &(AM.dollarzero) ) { \
    S_WRITE_B(ARG.where, ARG.size*sizeof(WORD)) \
}
```

Definition at line 1359 of file [checkpoint.c](#).

#### 4.5.2.17 ANNOUNCE

```
#define ANNOUNCE(  
    str )
```

Definition at line 1370 of file [checkpoint.c](#).

### 4.5.3 Function Documentation

#### 4.5.3.1 CheckRecoveryFile()

```
int CheckRecoveryFile (  
    void )
```

Checks whether a snapshot/recovery file exists. Returns 1 if it exists, 0 otherwise.

Definition at line 278 of file [checkpoint.c](#).

References [RecoveryFilename\(\)](#).

Here is the call graph for this function:



#### 4.5.3.2 DeleteRecoveryFile()

```
void DeleteRecoveryFile (  
    void )
```

Deletes the recovery files. It is called by `CleanUp()` in the case of a successful completion.

Definition at line 333 of file [checkpoint.c](#).

#### 4.5.3.3 RecoveryFilename()

```
char * RecoveryFilename (  
    void )
```

Returns pointer to recovery filename.

Definition at line 364 of file [checkpoint.c](#).

Referenced by [CheckRecoveryFile\(\)](#), and [DoRecovery\(\)](#).



#### 4.5.3.4 InitRecovery()

```
void InitRecovery (
    void )
```

Sets up the strings for the filenames of the recovery files. This functions should only be called once to avoid memory leaks and after AM.TempDir has been initialized.

Definition at line 399 of file [checkpoint.c](#).

#### 4.5.3.5 fwrite\_cached()

```
size_t fwrite_cached (
    const void * ptr,
    size_t size,
    size_t nmemb,
    FILE * fd )
```

Definition at line 1222 of file [checkpoint.c](#).

#### 4.5.3.6 flush\_cache()

```
size_t flush_cache (
    FILE * fd )
```

Definition at line 1247 of file [checkpoint.c](#).

#### 4.5.3.7 DoRecovery()

```
int DoRecovery (
    int * moduletype )
```

Reads from the recovery file and restores all necessary variables and states in FORM, so that the execution can recommence in preprocessor() as if no restart of FORM had occurred.

The recovery file is read into memory as a whole. The pointer p then points into this memory at the next non-processed data. The macros by which variables are restored, like R\_SET, automatically increase p appropriately.

If something goes wrong, the function returns with a non-zero value.

Allocated memory that would be lost when overwriting the global structs with data from the file is freed first. A major part of the code deals with the restoration of pointers. The idiom we use is to memorize the original pointer value (org), allocate new memory and copy the data from the file into this memory, calculate the offset between the old pointer value and the new allocated memory position (ofs), and then correct all affected pointers (+=ofs).

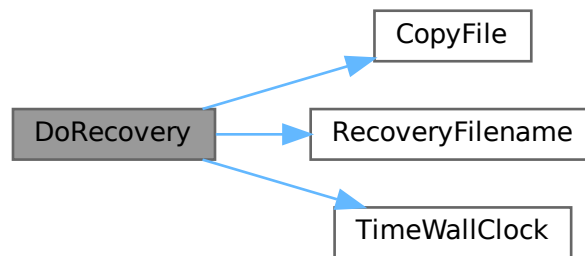
We rely on the fact that several variables (especially in AM) are already assigned the correct values by the startup functions. That means, in principle, that a change in the setup files between snapshot creation and recovery will be noticed.

Definition at line 1402 of file [checkpoint.c](#).

References [TaBIEs::argtail](#), [TaBIEs::boomlijst](#), [BrAcKeTiNfO::bracketbuffer](#), [TaBIEs::buffers](#), [TaBIEs::bufferssize](#), [CopyFile\(\)](#), [TaBIEs::flags](#), [ReNuMbEr::func](#), [ReNuMbEr::funnum](#), [VaRrEnUm::hi](#), [BrAcKeTiNfO::indexbuffer](#),

[ReNuMbEr::indi](#), [ReNuMbEr::indnum](#), [VaRrEnUm::lo](#), [TaBIes::MaxTreeSize](#), [TaBIes::mm](#), [TaBIes::numind](#), [TaBIes::pattern](#), [TaBIes::prototype](#), [TaBIes::prototypeSize](#), [RecoveryFilename\(\)](#), [ExPrEsSiOn::renumlists](#), [TaBIes::reserved](#), [TaBIes::spare](#), [TaBIes::sparse](#), [VaRrEnUm::start](#), [ReNuMbEr::symb](#), [ReNuMbEr::symnum](#), [TaBIes::tablepointers](#), [TimeWallClock\(\)](#), [TaBIes::totind](#), [ReNuMbEr::vecnum](#), and [ReNuMbEr::vect](#).

Here is the call graph for this function:



#### 4.5.3.8 DoCheckpoint()

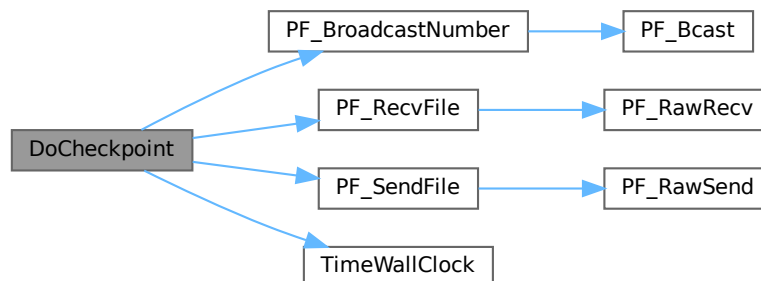
```
void DoCheckpoint (
    int moduletype )
```

Checks whether a snapshot should be done. Calls `DoSnapshot()` to create the snapshot.

Definition at line 3111 of file [checkpoint.c](#).

References [PF\\_BroadcastNumber\(\)](#), [PF\\_RecvFile\(\)](#), [PF\\_SendFile\(\)](#), and [TimeWallClock\(\)](#).

Here is the call graph for this function:



## 4.5.4 Variable Documentation

### 4.5.4.1 cache\_buffer

```
unsigned char cache_buffer[CACHE_SIZE]
```

Definition at line 1219 of file [checkpoint.c](#).

### 4.5.4.2 cache\_fill

```
size_t cache_fill = 0
```

Definition at line 1220 of file [checkpoint.c](#).

## 4.6 checkpoint.c

[Go to the documentation of this file.](#)

```
00001 /*
00002      #[ Explanations :
00003      */
00052 /*
00053      #] Explanations :
00054      #[ License :
00055      *
00056      *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00057      *   When using this file you are requested to refer to the publication
00058      *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00059      *   This is considered a matter of courtesy as the development was paid
00060      *   for by FOM the Dutch physics granting agency and we would like to
00061      *   be able to track its scientific use to convince FOM of its value
00062      *   for the community.
00063      *
00064      *   This file is part of FORM.
00065      *
00066      *   FORM is free software: you can redistribute it and/or modify it under the
00067      *   terms of the GNU General Public License as published by the Free Software
00068      *   Foundation, either version 3 of the License, or (at your option) any later
00069      *   version.
00070      *
00071      *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00072      *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00073      *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00074      *   details.
00075      *
00076      *   You should have received a copy of the GNU General Public License along
00077      *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00078      */
00079 /*
00080      #] License :
00081      #[ Includes :
00082      */
00083
00084 #include "form3.h"
00085
00086 #include <errno.h>
00087
00088 /*
00089 #define PRINTDEBUG
00090 */
00091
00092 /*
00093 #define PRINTTIMEMARKS
00094 */
00095
00096 /*
00097      #] Includes :
00098      #[ filenames and system commands :
00099      */
00100
00104 #ifndef WITHMPI
00105 #define BASENAME_FMT "%c%04dFORMrecv"
```

```

00111 static char BaseName[] = BASENAME_FMT;
00112 #else
00113 static char *BaseName = "FORMrecv";
00114 #endif
00118 static char *recoveryfile = 0;
00124 static char *intermedfile = 0;
00128 static char *sortfile = 0;
00132 static char *hidefile = 0;
00136 static char *storefile = 0;
00137
00142 static int done_snapshot = 0;
00143
00144 #ifdef WITHMPI
00149 static int PF_fmt_pos;
00150
00155 static const char *PF_recoveryfile(char prefix, int id, int intermed)
00156 {
00157     /*
00158      * Assume that InitRecovery() has been already called, namely
00159      * recoveryfile, intermedfile and PF_fmt_pos are already initialized.
00160      */
00161     static char *tmp_recovery = NULL;
00162     static char *tmp_intermed = NULL;
00163     char *tmp, c;
00164     if ( tmp_recovery == NULL ) {
00165         if ( PF.numtasks > 9999 ) { /* see BASENAME_FMT */
00166             MesPrint("Checkpoint: too many number of processors.");
00167             Terminate(-1);
00168         }
00169         tmp_recovery = (char *)Malloc1(strlen(recoveryfile) + strlen(intermedfile) + 2,
"PF_recoveryfile");
00170         tmp_intermed = tmp_recovery + strlen(recoveryfile) + 1;
00171         strcpy(tmp_recovery, recoveryfile);
00172         strcpy(tmp_intermed, intermedfile);
00173     }
00174     tmp = intermed ? tmp_intermed : tmp_recovery;
00175     c = tmp[PF_fmt_pos + 13]; /* The magic number 13 comes from BASENAME_FMT. */
00176     sprintf(tmp + PF_fmt_pos, BASENAME_FMT, prefix, id);
00177     tmp[PF_fmt_pos + 13] = c;
00178     return tmp;
00179 }
00180 #endif
00181
00182 /*
00183  #] filenames and system commands :
00184  #[ CheckRecoveryFile :
00185  */
00186
00191 #ifdef WITHMPI
00192
00198 static int PF_CheckRecoveryFile()
00199 {
00200     int i,ret=0;
00201     FILE *fd;
00202     /* Check if the recovery file for the master exists. */
00203     if ( PF.me == MASTER ) {
00204         if ( (fd = fopen(recoveryfile, "r")) ) {
00205             fclose(fd);
00206             PF_BroadcastNumber(1);
00207         }
00208         else {
00209             PF_BroadcastNumber(0);
00210             return 0;
00211         }
00212     }
00213     else {
00214         if ( !PF_BroadcastNumber(0) )
00215             return 0;
00216     }
00217     /* Now the main part. */
00218     if (PF.me == MASTER){
00219         /*We have to have recovery files for the master and all the slaves:*/
00220         for(i=1; i<PF.numtasks;i++){
00221             const char *tmpnam = PF_recoveryfile('m', i, 0);
00222             if ( (fd = fopen(tmpnam, "r")) )
00223                 fclose(fd);
00224             else
00225                 break;
00226         }/*for(i=0; i<PF.numtasks;i++)*/
00227         if(i!=PF.numtasks){/*some files are absent*/
00228             int j;
00229             /*Send all slaves failure*/
00230             for(j=1; j<PF.numtasks;j++){
00231                 ret=PF_SendFile(j, NULL);
00232                 if(ret<0)
00233                     return(-1);
00234             }

```

```

00235         if(i==0)
00236             return(0);/*Ok, no recovery files at all.*/
00237         /*The master recovery file exists but some slave files are absent*/
00238         MesPrint("The file %s exists but some of the slave recovery files are absent.",
00239                 RecoveryFilename());
00240         return(-1);
00241     }/*if(i!=PF.numtasks)*/
00242     /*All the files are here.*/
00243     /*Send all slaves success and files:*/
00244     for(i=1; i<PF.numtasks;i++){
00245         const char *tmpnam = PF_recoveryfile('m', i, 0);
00246         fd = fopen(tmpnam, "r");
00247         ret=PF_SendFile(i, fd);/*if fd==NULL, PF_SendFile seds to a slave the failure tag*/
00248         if(fd == NULL)
00249             return(-1);
00250         else
00251             fclose(fd);
00252         if(ret<0)
00253             return(-1);
00254     }/*for(i=0; i<PF.numtasks;i++)*/
00255     return(1);
00256 }/*if (PF.me == MASTER)*/
00257 /*Slave:*/
00258 /*Get the answer from the master:*/
00259 fd=fopen(recoveryfile,"w");
00260 if(fd == NULL) {
00261     MesPrint("Failed to open %s in write mode in process %w", recoveryfile);
00262     return(-1);
00263 }
00264 ret=PF_RecvFile(MASTER,fd);
00265 if(ret<0)
00266     return(-1);
00267 fclose(fd);
00268 if(ret==0){
00269     /*Nothing is found by the master*/
00270     remove(recoveryfile);
00271     return(0);
00272 }
00273 /*Recovery file is successfully transferred*/
00274 return(1);
00275 }
00276 #endif
00277
00278 int CheckRecoveryFile(void)
00279 {
00280     int ret = 0;
00281 #ifdef WITHMPI
00282     ret = PF_CheckRecoveryFile();
00283 #else
00284     FILE *fd;
00285     if ( ( fd = fopen(recoveryfile, "r") ) ) {
00286         fclose(fd);
00287         ret = 1;
00288     }
00289 #endif
00290     if ( ret < 0 ){/*In ParFORM CheckRecoveryFile() may return a fatal error.*/
00291         MesPrint("Fail checking recovery file");
00292         Terminate(-1);
00293     }
00294     else if ( ret > 0 ) {
00295         if ( AC.CheckpointFlag != -1 ) {
00296             /* recovery file exists but recovery option is not given */
00297 #ifdef WITHMPI
00298             if ( PF.me == MASTER ) {
00299 #endif
00300                 MesPrint("The recovery file %s exists, but the recovery option -R has not been given!",
00301                         RecoveryFilename());
00302                 MesPrint("FORM will be terminated to avoid unintentional loss of data.");
00303                 MesPrint("Delete the recovery file manually, if you want to start FORM without
00304 recovery.");
00305 #ifdef WITHMPI
00306             }
00307             if(PF.me != MASTER)
00308                 remove(RecoveryFilename());
00309 #endif
00310             Terminate(-1);
00311         }
00312         else {
00313             if ( AC.CheckpointFlag == -1 ) {
00314                 /* recovery option given but recovery file does not exist */
00315 #ifdef WITHMPI
00316                 if ( PF.me == MASTER )
00317 #endif
00318                 MesPrint("Option -R for recovery has been given, but the recovery file %s does not
00319 exist!", RecoveryFilename());
00320                 Terminate(-1);

```

```

00319     }
00320 }
00321 return(ret);
00322 }
00323
00324 /*
00325  #] CheckRecoveryFile :
00326  #[ DeleteRecoveryFile :
00327 */
00328
00333 void DeleteRecoveryFile(void)
00334 {
00335     if ( done_snapshot ) {
00336         remove(recoveryfile);
00337 #ifdef WITHMPI
00338         if( PF.me == MASTER){
00339             int i;
00340             for(i=1; i<PF.numtasks;i++){
00341                 const char *tmpnam = PF_recoveryfile('m', i, 0);
00342                 remove(tmpnam);
00343             }/*for(i=1; i<PF.numtasks;i++)*/
00344             remove(storefile);
00345             remove(sortfile);
00346             remove(hidefile);
00347         }/*if( PF.me == MASTER)*/
00348 #else
00349         remove(storefile);
00350         remove(sortfile);
00351         remove(hidefile);
00352 #endif
00353     }
00354 }
00355
00356 /*
00357  #] DeleteRecoveryFile :
00358  #[ RecoveryFilename :
00359 */
00360
00364 char *RecoveryFilename(void)
00365 {
00366     return(recoveryfile);
00367 }
00368
00369 /*
00370  #] RecoveryFilename :
00371  #[ InitRecovery :
00372 */
00373
00377 static char *InitName(char *str, char *ext)
00378 {
00379     char *s, *d = str;
00380     if ( AM.TempDir ) {
00381         s = (char*)AM.TempDir;
00382         while ( *s ) { *d++ = *s++; }
00383         *d++ = SEPARATOR;
00384     }
00385     s = BaseName;
00386     while ( *s ) { *d++ = *s++; }
00387     *d++ = '.';
00388     s = ext;
00389     while ( *s ) { *d++ = *s++; }
00390     *d++ = 0;
00391     return d;
00392 }
00393
00399 void InitRecovery(void)
00400 {
00401     int lenpath = AM.TempDir ? strlen((char*)AM.TempDir)+1 : 0;
00402 #ifdef WITHMPI
00403     sprintf(BaseName,BASENAME_FMT,(PF.me == MASTER)?'m':'s',PF.me);
00404     /*Now BaseName has a form ?XXXXFORMrecv where ? == 'm' for master and 's' for slave,
00405        XXXX is a zero - padded PF.me*/
00406     PF_fmt_pos = lenpath;
00407 #endif
00408     recoveryfile = (char*)Malloc(5*(lenpath+strlen(BaseName)+4+1),"InitRecovery");
00409     intermedfile = InitName(recoveryfile, "tmp");
00410     sortfile      = InitName(intermedfile, "XXX");
00411     hidefile      = InitName(sortfile, "out");
00412     storefile     = InitName(hidefile, "hid");
00413     InitName(storefile, "str");
00414 }
00415
00416 /*
00417  #] InitRecovery :
00418  #[ Debugging :
00419 */
00420

```

```

00421 #ifdef PRINTDEBUG
00422
00423 void print_BYTE(void *p)
00424 {
00425     UBYTE h = (UBYTE) (*(UBYTE*)p) >> 4;
00426     UBYTE l = (UBYTE) (*(UBYTE*)p) & 0x0F;
00427     if ( h > 9 ) h += 55; else h += 48;
00428     if ( l > 9 ) l += 55; else l += 48;
00429     printf("%c%c ", h, l);
00430 }
00431
00432 static void print_STR(UBYTE *p)
00433 {
00434     if ( p ) {
00435         MesPrint("%s", (char*)p);
00436     }
00437     else {
00438         MesPrint("NULL");
00439     }
00440 }
00441
00442 static void print_WORDB(WORD *buf, WORD *top)
00443 {
00444     LONG size = top-buf;
00445     int i;
00446     while ( size > 0 ) {
00447         if ( size > MAXPOSITIVE ) i = MAXPOSITIVE;
00448         else i = size;
00449         size -= i;
00450         MesPrint("%a",i,buf);
00451         buf += i;
00452     }
00453 }
00454
00455 static void print_VOIDP(void *p, size_t size)
00456 {
00457     int i;
00458     if ( p ) {
00459         while ( size > 0 ) {
00460             if ( size > MAXPOSITIVE ) i = MAXPOSITIVE;
00461             else i = size;
00462             size -= i;
00463             MesPrint("%b",i,(UBYTE *)p);
00464             p = ((UBYTE *)p)+i;
00465         }
00466     }
00467     else {
00468         MesPrint("NULL");
00469     }
00470 }
00471
00472 static void print_CHARS(UBYTE *p, size_t size)
00473 {
00474     int i;
00475     while ( size > 0 ) {
00476         if ( size > MAXPOSITIVE ) i = MAXPOSITIVE;
00477         else i = size;
00478         size -= i;
00479         MesPrint("%C",i,(char *)p);
00480         p += i;
00481     }
00482 }
00483
00484 static void print_WORDV(WORD *p, size_t size)
00485 {
00486     int i;
00487     if ( p ) {
00488         while ( size > 0 ) {
00489             if ( size > MAXPOSITIVE ) i = MAXPOSITIVE;
00490             else i = size;
00491             size -= i;
00492             MesPrint("%a",i,p);
00493             p += i;
00494         }
00495     }
00496     else {
00497         MesPrint("NULL");
00498     }
00499 }
00500
00501 static void print_INTV(int *p, size_t size)
00502 {
00503     int iarray[8];
00504     WORD i = 0;
00505     if ( p ) {
00506         while ( size > 0 ) {
00507             if ( i >= 8 ) {

```

```

00508         MesPrint("%I",i,iarray);
00509         i = 0;
00510     }
00511     iarray[i++] = *p++;
00512     size--;
00513 }
00514 if ( i > 0 ) MesPrint("%I",i,iarray);
00515 }
00516 else {
00517     MesPrint("NULL");
00518 }
00519 }
00520
00521 static void print_LONGV(LONG *p, size_t size)
00522 {
00523     LONG larray[8];
00524     WORD i = 0;
00525     if ( p ) {
00526         while ( size > 0 ) {
00527             if ( i >= 8 ) {
00528                 MesPrint("%I",i,larray);
00529                 i = 0;
00530             }
00531             larray[i++] = *p++;
00532             size--;
00533         }
00534         if ( i > 0 ) MesPrint("%I",i,larray);
00535     }
00536     else {
00537         MesPrint("NULL");
00538     }
00539 }
00540
00541 static void print_PRELOAD(PRELOAD *l)
00542 {
00543     if ( l->size ) {
00544         print_CHARS(l->buffer, l->size);
00545     }
00546     MesPrint("%ld", l->size);
00547 }
00548
00549 static void print_PREVAR(PREVAR *l)
00550 {
00551     MesPrint("%s", l->name);
00552     print_STR(l->value);
00553     if ( l->nargs ) print_STR(l->argnames);
00554     MesPrint("%d", l->nargs);
00555     MesPrint("%d", l->wildarg);
00556 }
00557
00558 static void print_DOLLARS(DOLLARS l)
00559 {
00560     print_VOIDP(l->where, l->size);
00561     MesPrint("%ld", l->size);
00562     MesPrint("%ld", l->name);
00563     MesPrint("%s", AC.dollarnames->namebuffer+l->name);
00564     MesPrint("%d", l->type);
00565     MesPrint("%d", l->node);
00566     MesPrint("%d", l->index);
00567     MesPrint("%d", l->zero);
00568     MesPrint("%d", l->numdummies);
00569     MesPrint("%d", l->nfactors);
00570 }
00571
00572 static void print_LIST(LIST *l)
00573 {
00574     print_VOIDP(l->lijst, l->size);
00575     MesPrint("%s", l->message);
00576     MesPrint("%d", l->num);
00577     MesPrint("%d", l->maxnum);
00578     MesPrint("%d", l->size);
00579     MesPrint("%d", l->numglobal);
00580     MesPrint("%d", l->numtemp);
00581     MesPrint("%d", l->numclear);
00582 }
00583
00584 static void print_DOLOOP(DOLOOP *l)
00585 {
00586     print_PRELOAD(&(l->p));
00587     print_STR(l->name);
00588     if ( l->type != NUMERICALLOOP ) {
00589         print_STR(l->vars);
00590     }
00591     print_STR(l->contents);
00592     if ( l->type != LISTEDLOOP && l->type != NUMERICALLOOP ) {
00593         print_STR(l->dollarname);
00594     }

```



```

00595     MesPrint("%l", l->startlinenumber);
00596     MesPrint("%l", l->firstnum);
00597     MesPrint("%l", l->lastnum);
00598     MesPrint("%l", l->incnum);
00599     MesPrint("%d", l->type);
00600     MesPrint("%d", l->NoShowInput);
00601     MesPrint("%d", l->errorsinloop);
00602     MesPrint("%d", l->firstloopcall);
00603 }
00604
00605 static void print_PROCEDURE(PROCEDURE *l)
00606 {
00607     if ( l->loadmode != 1 ) {
00608         print_PRELOAD(&(l->p));
00609     }
00610     print_STR(l->name);
00611     MesPrint("%d", l->loadmode);
00612 }
00613
00614 static void print_NAMETREE(NAMETREE *t)
00615 {
00616     int i;
00617     for ( i=0; i<t->nodefill; ++i ) {
00618         MesPrint("%l %d %d %d %d %d %d\n", t->namenode[i].name,
00619             t->namenode[i].parent, t->namenode[i].left, t->namenode[i].right,
00620             t->namenode[i].balance, t->namenode[i].type, t->namenode[i].number );
00621     }
00622     print_CHARS(t->namebuffer, t->namefill);
00623     MesPrint("%l", t->namesize);
00624     MesPrint("%l", t->namefill);
00625     MesPrint("%l", t->nodesize);
00626     MesPrint("%l", t->nodefill);
00627     MesPrint("%l", t->oldnamefill);
00628     MesPrint("%l", t->oldnodefill);
00629     MesPrint("%l", t->globalnamefill);
00630     MesPrint("%l", t->globalnodefill);
00631     MesPrint("%l", t->clearnamefill);
00632     MesPrint("%l", t->clearnodefill);
00633     MesPrint("%d", t->headnode);
00634 }
00635
00636 void print_CBUF(CBUF *c)
00637 {
00638     int i;
00639     print_WORDV(c->Buffer, c->BufferSize);
00640     /*
00641     MesPrint("%x", c->Buffer);
00642     MesPrint("%x", c->lhs);
00643     MesPrint("%x", c->rhs);
00644     */
00645     for ( i=0; i<c->numlhs; ++i ) {
00646         if ( c->lhs[i] ) MesPrint("%d", *(c->lhs[i]));
00647     }
00648     for ( i=0; i<c->numrhs; ++i ) {
00649         if ( c->rhs[i] ) MesPrint("%d", *(c->rhs[i]));
00650     }
00651     MesPrint("%l", *c->CanCommu);
00652     MesPrint("%l", *c->NumTerms);
00653     MesPrint("%d", *c->numdum);
00654     for ( i=0; i<c->MaxTreeSize; ++i ) {
00655         MesPrint("%d %d %d %d %d", c->boomlijst[i].parent, c->boomlijst[i].left,
c->boomlijst[i].right,
00656             c->boomlijst[i].value, c->boomlijst[i].blnce);
00657     }
00658 }
00659
00660 static void print_STREAM(STREAM *t)
00661 {
00662     print_CHARS(t->buffer, t->inbuffer);
00663     MesPrint("%l", (LONG)(t->pointer-t->buffer));
00664     print_STR(t->FoldName);
00665     print_STR(t->name);
00666     if ( t->type == PREVARSTREAM || t->type == DOLLARSTREAM ) {
00667         print_STR(t->pname);
00668     }
00669     MesPrint("%l", (LONG)t->fileposition);
00670     MesPrint("%l", (LONG)t->linenumber);
00671     MesPrint("%l", (LONG)t->prevline);
00672     MesPrint("%l", t->buffersize);
00673     MesPrint("%l", t->bufferposition);
00674     MesPrint("%l", t->inbuffer);
00675     MesPrint("%d", t->previous);
00676     MesPrint("%d", t->handle);
00677     switch ( t->type ) {
00678         case FILESTREAM: MesPrint("%d == FILESTREAM", t->type); break;
00679         case PREVARSTREAM: MesPrint("%d == PREVARSTREAM", t->type); break;
00680         case PREREADSTREAM: MesPrint("%d == PREREADSTREAM", t->type); break;

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00681         case PIPESTREAM: MesPrint("%d == PIPESTREAM", t->type); break;
00682         case PRECALCSTREAM: MesPrint("%d == PRECALCSTREAM", t->type); break;
00683         case DOLLARSTREAM: MesPrint("%d == DOLLARSTREAM", t->type); break;
00684         case PREREADSTREAM2: MesPrint("%d == PREREADSTREAM2", t->type); break;
00685         case EXTERNALCHANNELSTREAM: MesPrint("%d == EXTERNALCHANNELSTREAM", t->type); break;
00686         case PREREADSTREAM3: MesPrint("%d == PREREADSTREAM3", t->type); break;
00687         default: MesPrint("%d == UNKNOWN", t->type);
00688     }
00689 }
00690
00691 static void print_M()
00692 {
00693     MesPrint("%%% M_const");
00694     MesPrint("%d", *AM.gcmmod);
00695     MesPrint("%d", *AM.gpowmod);
00696     print_STR(AM.TempDir);
00697     print_STR(AM.TempSortDir);
00698     print_STR(AM.IncDir);
00699     print_STR(AM.InputFileName);
00700     print_STR(AM.LogFileName);
00701     print_STR(AM.OutBuffer);
00702     print_STR(AM.Path);
00703     print_STR(AM.SetupDir);
00704     print_STR(AM.SetupFile);
00705     MesPrint("--MARK 1");
00706     MesPrint("%l", (LONG)BASEPOSITION(AM.zeropos));
00707 #ifdef WITHPTHREADS
00708     MesPrint("%l", AM.ThreadScratSize);
00709     MesPrint("%l", AM.ThreadScratOutSize);
00710 #endif
00711     MesPrint("%l", AM.MaxTer);
00712     MesPrint("%l", AM.CompressSize);
00713     MesPrint("%l", AM.ScratSize);
00714     MesPrint("%l", AM.SizeStoreCache);
00715     MesPrint("%l", AM.MaxStreamSize);
00716     MesPrint("%l", AM.SIOsize);
00717     MesPrint("%l", AM.SLargeSize);
00718     MesPrint("%l", AM.SSmallEsize);
00719     MesPrint("%l", AM.SSmallSize);
00720     MesPrint("--MARK 2");
00721     MesPrint("%l", AM.STermsInSmall);
00722     MesPrint("%l", AM.MaxBracketBufferSize);
00723     MesPrint("%l", AM.hProcessBucketSize);
00724     MesPrint("%l", AM.gProcessBucketSize);
00725     MesPrint("%l", AM.shmWinSize);
00726     MesPrint("%l", AM.OldChildTime);
00727     MesPrint("%l", AM.OldSecTime);
00728     MesPrint("%l", AM.OldMilliTime);
00729     MesPrint("%l", AM.WorkSize);
00730     MesPrint("%l", AM.gThreadBucketSize);
00731     MesPrint("--MARK 3");
00732     MesPrint("%l", AM.ggThreadBucketSize);
00733     MesPrint("%d", AM.FileOnlyFlag);
00734     MesPrint("%d", AM.Interact);
00735     MesPrint("%d", AM.MaxParLevel);
00736     MesPrint("%d", AM.OutBufSize);
00737     MesPrint("%d", AM.SMaxFpatches);
00738     MesPrint("%d", AM.SMaxPatches);
00739     MesPrint("%d", AM.StdOut);
00740     MesPrint("%d", AM.ginsidefirst);
00741     MesPrint("%d", AM.gDefDim);
00742     MesPrint("%d", AM.gDefDim4);
00743     MesPrint("--MARK 4");
00744     MesPrint("%d", AM.NumFixedSets);
00745     MesPrint("%d", AM.NumFixedFunctions);
00746     MesPrint("%d", AM.rbufnum);
00747     MesPrint("%d", AM.dbufnum);
00748     MesPrint("%d", AM.SkipClears);
00749     MesPrint("%d", AM.gfunpowers);
00750     MesPrint("%d", AM.gStatsFlag);
00751     MesPrint("%d", AM.gNamesFlag);
00752     MesPrint("%d", AM.gCodesFlag);
00753     MesPrint("%d", AM.gTokensWriteFlag);
00754     MesPrint("%d", AM.gSortType);
00755     MesPrint("%d", AM.gproperorderflag);
00756     MesPrint("--MARK 5");
00757     MesPrint("%d", AM.hparallelflag);
00758     MesPrint("%d", AM.gparallelflag);
00759     MesPrint("%d", AM.totalnumberofthreads);
00760     MesPrint("%d", AM.gSizeCommuteInSet);
00761     MesPrint("%d", AM.gThreadStats);
00762     MesPrint("%d", AM.ggThreadStats);
00763     MesPrint("%d", AM.gFinalStats);
00764     MesPrint("%d", AM.ggFinalStats);
00765     MesPrint("%d", AM.gThreadsFlag);
00766     MesPrint("%d", AM.ggThreadsFlag);
00767     MesPrint("%d", AM.gThreadBalancing);

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00768     MesPrint("%d", AM.ggThreadBalancing);
00769     MesPrint("%d", AM.gThreadSortFileSynch);
00770     MesPrint("%d", AM.ggThreadSortFileSynch);
00771     MesPrint("%d", AM.gProcessStats);
00772     MesPrint("%d", AM.ggProcessStats);
00773     MesPrint("%d", AM.gOldParallelStats);
00774     MesPrint("%d", AM.ggOldParallelStats);
00775     MesPrint("%d", AM.gWTimeStatsFlag);
00776     MesPrint("%d", AM.ggWTimeStatsFlag);
00777     MesPrint("%d", AM.maxFlevels);
00778     MesPrint("--MARK 6");
00779     MesPrint("%d", AM.resetTimeOnClear);
00780     MesPrint("%d", AM.gcNumDollars);
00781     MesPrint("%d", AM.MultiRun);
00782     MesPrint("%d", AM.gNoSpacesInNumbers);
00783     MesPrint("%d", AM.ggNoSpacesInNumbers);
00784     MesPrint("%d", AM.MaxTal);
00785     MesPrint("%d", AM.IndDum);
00786     MesPrint("%d", AM.DumInd);
00787     MesPrint("%d", AM.WilInd);
00788     MesPrint("%d", AM.gncmod);
00789     MesPrint("%d", AM.gnpowmod);
00790     MesPrint("%d", AM.gmodmode);
00791     MesPrint("--MARK 7");
00792     MesPrint("%d", AM.gUnitTrace);
00793     MesPrint("%d", AM.gOutputMode);
00794     MesPrint("%d", AM.gCnumpows);
00795     MesPrint("%d", AM.gOutputSpaces);
00796     MesPrint("%d", AM.gOutNumberType);
00797     MesPrint("%d %d %d %d", AM.gUniTrace[0], AM.gUniTrace[1], AM.gUniTrace[2], AM.gUniTrace[3]);
00798     MesPrint("%d", AM.MaxWildcards);
00799     MesPrint("%d", AM.mTraceDum);
00800     MesPrint("%d", AM.OffsetIndex);
00801     MesPrint("%d", AM.OffsetVector);
00802     MesPrint("%d", AM.RepMax);
00803     MesPrint("%d", AM.LogType);
00804     MesPrint("%d", AM.ggStatsFlag);
00805     MesPrint("%d", AM.gLineLength);
00806     MesPrint("%d", AM.qError);
00807     MesPrint("--MARK 8");
00808     MesPrint("%d", AM.FortranCont);
00809     MesPrint("%d", AM.HoldFlag);
00810     MesPrint("%d %d %d %d %d", AM.Ordering[0], AM.Ordering[1], AM.Ordering[2], AM.Ordering[3],
AM.Ordering[4]);
00811     MesPrint("%d %d %d %d %d", AM.Ordering[5], AM.Ordering[6], AM.Ordering[7], AM.Ordering[8],
AM.Ordering[9]);
00812     MesPrint("%d %d %d %d %d", AM.Ordering[10], AM.Ordering[11], AM.Ordering[12], AM.Ordering[13],
AM.Ordering[14]);
00813     MesPrint("%d", AM.silent);
00814     MesPrint("%d", AM.tracebackflag);
00815     MesPrint("%d", AM.expnum);
00816     MesPrint("%d", AM.denomnum);
00817     MesPrint("%d", AM.facnum);
00818     MesPrint("%d", AM.invfacnum);
00819     MesPrint("%d", AM.sumnum);
00820     MesPrint("%d", AM.sumpnum);
00821     MesPrint("--MARK 9");
00822     MesPrint("%d", AM.OldOrderFlag);
00823     MesPrint("%d", AM.termfunnum);
00824     MesPrint("%d", AM.matchfunnum);
00825     MesPrint("%d", AM.countfunnum);
00826     MesPrint("%d", AM.gPolyFun);
00827     MesPrint("%d", AM.gPolyFunInv);
00828     MesPrint("%d", AM.gPolyFunType);
00829     MesPrint("%d", AM.gPolyFunExp);
00830     MesPrint("%d", AM.gPolyFunVar);
00831     MesPrint("%d", AM.gPolyFunPow);
00832     MesPrint("--MARK 10");
00833     MesPrint("%d", AM.dollarzero);
00834     MesPrint("%d", AM.atstartup);
00835     MesPrint("%d", AM.exitflag);
00836     MesPrint("%d", AM.NumStoreCaches);
00837     MesPrint("%d", AM.gIndentSpace);
00838     MesPrint("%d", AM.ggIndentSpace);
00839     MesPrint("%d", AM.gShortStatsMax);
00840     MesPrint("%d", AM.ggShortStatsMax);
00841     MesPrint("--MARK 11");
00842     MesPrint("%d", AM.FromStdin);
00843     MesPrint("%d", AM.IgnoreDeprecation);
00844     MesPrint("%%% END M_const");
00845     /* fflush(0); */
00846 }
00847
00848 static void print_P()
00849 {
00850     int i;
00851     MesPrint("%%% P_const");

```

```

00852     print_LIST(&AP.DollarList);
00853     for ( i=0; i<AP.DollarList.num; ++i ) {
00854         print_DOLLARS(&(Dollars[i]));
00855     }
00856     MesPrint("--MARK 1");
00857     print_LIST(&AP.PreVarList);
00858     for ( i=0; i<AP.PreVarList.num; ++i ) {
00859         print_PREVAR(&(PreVar[i]));
00860     }
00861     MesPrint("--MARK 2");
00862     print_LIST(&AP.LoopList);
00863     for ( i=0; i<AP.LoopList.num; ++i ) {
00864         print_DOLOOP(&(DoLoops[i]));
00865     }
00866     MesPrint("--MARK 3");
00867     print_LIST(&AP.ProcList);
00868     for ( i=0; i<AP.ProcList.num; ++i ) {
00869         print_PROCEDURE(&(Procedures[i]));
00870     }
00871     MesPrint("--MARK 4");
00872     for ( i=0; i<=AP.PreSwitchLevel; ++i ) {
00873         print_STR(AP.PreSwitchStrings[i]);
00874     }
00875     MesPrint("%l", AP.preStop-AP.preStart);
00876     if ( AP.preFill ) MesPrint("%l", AP.preFill-AP.preStart);
00877     print_CHARS(AP.preStart, AP.pSize);
00878     MesPrint("%s", AP.procedureExtension);
00879     MesPrint("%s", AP.cprocedureExtension);
00880     print_INTV(AP.PreIfStack, AP.MaxPreIfLevel);
00881     print_INTV(AP.PreSwitchModes, AP.NumPreSwitchStrings+1);
00882     print_INTV(AP.PreTypes, AP.NumPreTypes+1);
00883     MesPrint("%d", AP.PreAssignFlag);
00884     MesPrint("--MARK 5");
00885     MesPrint("%d", AP.PreContinuation);
00886     MesPrint("%l", AP.InOutBuf);
00887     MesPrint("%l", AP.pSize);
00888     MesPrint("%d", AP.PreproFlag);
00889     MesPrint("%d", AP.iBufError);
00890     MesPrint("%d", AP.PreOut);
00891     MesPrint("%d", AP.PreSwitchLevel);
00892     MesPrint("%d", AP.NumPreSwitchStrings);
00893     MesPrint("%d", AP.MaxPreTypes);
00894     MesPrint("--MARK 6");
00895     MesPrint("%d", AP.NumPreTypes);
00896     MesPrint("%d", AP.DelayPrevar);
00897     MesPrint("%d", AP.AllowDelay);
00898     MesPrint("%d", AP.lhdollarerror);
00899     MesPrint("%d", AP.eat);
00900     MesPrint("%d", AP.gNumPre);
00901     MesPrint("%d", AP.PreDebug);
00902     MesPrint("--MARK 7");
00903     MesPrint("%d", AP.DebugFlag);
00904     MesPrint("%d", AP.preError);
00905     MesPrint("%C", 1, &(AP.ComChar));
00906     MesPrint("%C", 1, &(AP.cComChar));
00907     MesPrint("%%%" END P_const");
00908     /* fflush(0); */
00909 }
00910
00911 static void print_C()
00912 {
00913     int i;
00914     UBYTE buf[40], *t;
00915     MesPrint("%%%" C_const");
00916     for ( i=0; i<32; ++i ) {
00917         t = buf;
00918         t = NumCopy((WORD) (AC.separators[i].bit_7), t);
00919         t = NumCopy((WORD) (AC.separators[i].bit_6), t);
00920         t = NumCopy((WORD) (AC.separators[i].bit_5), t);
00921         t = NumCopy((WORD) (AC.separators[i].bit_4), t);
00922         t = NumCopy((WORD) (AC.separators[i].bit_3), t);
00923         t = NumCopy((WORD) (AC.separators[i].bit_2), t);
00924         t = NumCopy((WORD) (AC.separators[i].bit_1), t);
00925         t = NumCopy((WORD) (AC.separators[i].bit_0), t);
00926         MesPrint("%s ", buf);
00927     }
00928     print_NAMETREE(AC.dollarnames);
00929     print_NAMETREE(AC.exprnames);
00930     print_NAMETREE(AC.varnames);
00931     MesPrint("--MARK 1");
00932     print_LIST(&AC.ChannelList);
00933     for ( i=0; i<AC.ChannelList.num; ++i ) {
00934         MesPrint("%s %d", channels[i].name, channels[i].handle);
00935     }
00936     MesPrint("--MARK 2");
00937     print_LIST(&AC.DubiousList);
00938     MesPrint("--MARK 3");

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```

00939     print_LIST(&AC.FunctionList);
00940     for ( i=0; i<AC.FunctionList.num; ++i ) {
00941         if ( functions[i].tabl ) {
00942             }
00943             MesPrint("%1", functions[i].symminfo);
00944             MesPrint("%1", functions[i].name);
00945             MesPrint("%d", functions[i].namesize);
00946         }
00947     MesPrint("--MARK 4");
00948     print_LIST(&AC.ExpressionList);
00949     print_LIST(&AC.IndexList);
00950     print_LIST(&AC.SetElementList);
00951     print_LIST(&AC.SetList);
00952     MesPrint("--MARK 5");
00953     print_LIST(&AC.SymbolList);
00954     print_LIST(&AC.VectorList);
00955     print_LIST(&AC.PotModDolList);
00956     print_LIST(&AC.ModOptDolList);
00957     print_LIST(&AC.TableBaseList);
00958
00959
00960 /*
00961     print_LIST(&AC.cbufList);
00962     for ( i=0; i<AC.cbufList.num; ++i ) {
00963         MesPrint("cbufList.num == %d", i);
00964         print_CBUF(cbuf+i);
00965     }
00966     MesPrint("%d", AC.cbufnum);
00967 */
00968     MesPrint("--MARK 6");
00969
00970     print_LIST(&AC.AutoSymbolList);
00971     print_LIST(&AC.AutoIndexList);
00972     print_LIST(&AC.AutoVectorList);
00973     print_LIST(&AC.AutoFunctionList);
00974
00975     print_NAMETREE(AC.autonames);
00976     MesPrint("--MARK 7");
00977
00978     print_LIST(AC.Symbols);
00979     print_LIST(AC.Indices);
00980     print_LIST(AC.Vectors);
00981     print_LIST(AC.Functions);
00982     MesPrint("--MARK 8");
00983
00984     print_NAMETREE(*AC.activenames);
00985
00986     MesPrint("--MARK 9");
00987
00988     MesPrint("%d", AC.AutoDeclareFlag);
00989
00990     for ( i=0; i<AC.NumStreams; ++i ) {
00991         MesPrint("Stream %d\n", i);
00992         print_STREAM(AC.Streams+i);
00993     }
00994     print_STREAM(AC.CurrentStream);
00995     MesPrint("--MARK 10");
00996
00997     print_LONGV(AC.termstack, AC.maxtermlevel);
00998     print_LONGV(AC.termstack, AC.maxtermlevel);
00999     print_VOIDP(AC.cmod, AM.MaxTal*4*sizeof(UWORD));
01000     print_WORDV((WORD *) (AC.cmod), 1);
01001     print_WORDV((WORD *) (AC.powmod), 1);
01002     print_WORDV((WORD *) AC.modpowers, 1);
01003     print_WORDV((WORD *) AC.halfmod, 1);
01004     MesPrint("--MARK 10-2");
01005     /*
01006     print_WORDV(AC.ProtoType, AC.ProtoType[1]);
01007     print_WORDV(AC.WildC, 1);
01008     */
01009
01010     MesPrint("--MARK 11");
01011     /* IfHeap ... Labels */
01012
01013     print_CHARS((UBYTE *) AC.tokens, AC.toptokens-AC.tokens);
01014     MesPrint("%1", AC.endoftokens-AC.tokens);
01015     print_WORDV(AC.tokenarglevel, AM.MaxParLevel);
01016     print_WORDV((WORD *) AC.modinverses, ABS(AC.ncmod));
01017 #ifdef WITHPTHREADS
01018     print_LONGV(AC.inputnumbers, AC.sizefirstnum+AC.sizefirstnum*sizeof(WORD)/sizeof(LONG));
01019     print_WORDV(AC.pfirstnum, 1);
01020 #endif
01021     MesPrint("--MARK 12");
01022     print_LONGV(AC.argstack, MAXNEST);
01023     print_LONGV(AC.insidestack, MAXNEST);
01024     print_LONGV(AC.inexprstack, MAXNEST);
01025     MesPrint("%1", AC.iBufferSize);

```

```

01026     MesPrint("%l", AC.TransEname);
01027     MesPrint("%l", AC.ProcessBucketSize);
01028     MesPrint("%l", AC.mProcessBucketSize);
01029     MesPrint("%l", AC.CModule);
01030     MesPrint("%l", AC.ThreadBucketSize);
01031     MesPrint("%d", AC.NoShowInput);
01032     MesPrint("%d", AC.ShortStats);
01033     MesPrint("%d", AC.compiletype);
01034     MesPrint("%d", AC.firstconstindex);
01035     MesPrint("%d", AC.insidefirst);
01036     MesPrint("%d", AC.minsidefirst);
01037     MesPrint("%d", AC.wildflag);
01038     MesPrint("%d", AC.NumLabels);
01039     MesPrint("%d", AC.MaxLabels);
01040     MesPrint("--MARK 13");
01041     MesPrint("%d", AC.lDefDim);
01042     MesPrint("%d", AC.lDefDim4);
01043     MesPrint("%d", AC.NumWildcardNames);
01044     MesPrint("%d", AC.WildcardBufferSize);
01045     MesPrint("%d", AC.MaxIf);
01046     MesPrint("%d", AC.NumStreams);
01047     MesPrint("%d", AC.MaxNumStreams);
01048     MesPrint("%d", AC.firstctypemessage);
01049     MesPrint("%d", AC.tablecheck);
01050     MesPrint("%d", AC.idoption);
01051     MesPrint("%d", AC.BottomLevel);
01052     MesPrint("%d", AC.CompileLevel);
01053     MesPrint("%d", AC.TokensWriteFlag);
01054     MesPrint("%d", AC.UnsureDollarMode);
01055     MesPrint("%d", AC.outsidefun);
01056     MesPrint("%d", AC.funpowers);
01057     MesPrint("--MARK 14");
01058     MesPrint("%d", AC.WarnFlag);
01059     MesPrint("%d", AC.StatsFlag);
01060     MesPrint("%d", AC.NamesFlag);
01061     MesPrint("%d", AC.CodesFlag);
01062     MesPrint("%d", AC.TokensWriteFlag);
01063     MesPrint("%d", AC.SetupFlag);
01064     MesPrint("%d", AC.SortType);
01065     MesPrint("%d", AC.lSortType);
01066     MesPrint("%d", AC.ThreadStats);
01067     MesPrint("%d", AC.FinalStats);
01068     MesPrint("%d", AC.ThreadsFlag);
01069     MesPrint("%d", AC.ThreadBalancing);
01070     MesPrint("%d", AC.ThreadSortFileSynch);
01071     MesPrint("%d", AC.ProcessStats);
01072     MesPrint("%d", AC.OldParallelStats);
01073     MesPrint("%d", AC.WTimeStatsFlag);
01074     MesPrint("%d", AC.BraceNormalizer);
01075     MesPrint("%d", AC.maxtermlevel);
01076     MesPrint("%d", AC.dumnumflag);
01077     MesPrint("--MARK 15");
01078     MesPrint("%d", AC.bracketindexflag);
01079     MesPrint("%d", AC.parallelflag);
01080     MesPrint("%d", AC.mparallelflag);
01081     MesPrint("%d", AC.properorderflag);
01082     MesPrint("%d", AC.vetofilling);
01083     MesPrint("%d", AC.tablefilling);
01084     MesPrint("%d", AC.vetotablebasefill);
01085     MesPrint("%d", AC.exprfillwarning);
01086     MesPrint("%d", AC.lhdollarflag);
01087     MesPrint("%d", AC.NoCompress);
01088 #ifdef WITHPTHREADS
01089     MesPrint("%d", AC.numpfirstnum);
01090     MesPrint("%d", AC.sizepfirstnum);
01091 #endif
01092     MesPrint("%d", AC.RepLevel);
01093     MesPrint("%d", AC.arglevel);
01094     MesPrint("%d", AC.insidelevel);
01095     MesPrint("%d", AC.inexprlevel);
01096     MesPrint("%d", AC.termlevel);
01097     MesPrint("--MARK 16");
01098     print_WORDV(AC.argsumcheck, MAXNEST);
01099     print_WORDV(AC.insidesumcheck, MAXNEST);
01100     print_WORDV(AC.inexprsumcheck, MAXNEST);
01101     MesPrint("%d", AC.MustTestTable);
01102     MesPrint("%d", AC.DumNum);
01103     MesPrint("%d", AC.ncmod);
01104     MesPrint("%d", AC.npowmod);
01105     MesPrint("%d", AC.modmode);
01106     MesPrint("%d", AC.nhalfmod);
01107     MesPrint("%d", AC.DirtPow);
01108     MesPrint("%d", AC.lUnitTrace);
01109     MesPrint("%d", AC.NwildC);
01110     MesPrint("%d", AC.ComDefer);
01111     MesPrint("%d", AC.CollectFun);
01112     MesPrint("%d", AC.AltCollectFun);

```

```

01113     MesPrint("--MARK 17");
01114     MesPrint("%d", AC.OutputMode);
01115     MesPrint("%d", AC.Cnumpows);
01116     MesPrint("%d", AC.OutputSpaces);
01117     MesPrint("%d", AC.OutNumberType);
01118     print_WORDV(AC.lUniTrace, 4);
01119     print_WORDV(AC.RepSumCheck, MAXREPEAT);
01120     MesPrint("%d", AC.DidClean);
01121     MesPrint("%d", AC.IfLevel);
01122     MesPrint("%d", AC.WhileLevel);
01123     print_WORDV(AC.IfSumCheck, (AC.MaxIf+1));
01124     MesPrint("%d", AC.LogHandle);
01125     MesPrint("%d", AC.LineLength);
01126     MesPrint("%d", AC.StoreHandle);
01127     MesPrint("%d", AC.HideLevel);
01128     MesPrint("%d", AC.lPolyFun);
01129     MesPrint("%d", AC.lPolyFunInv);
01130     MesPrint("%d", AC.lPolyFunType);
01131     MesPrint("%d", AC.lPolyFunExp);
01132     MesPrint("%d", AC.lPolyFunVar);
01133     MesPrint("%d", AC.lPolyFunPow);
01134     MesPrint("%d", AC.SymChangeFlag);
01135     MesPrint("%d", AC.CollectPercentage);
01136     MesPrint("%d", AC.ShortStatsMax);
01137     MesPrint("--MARK 18");
01138
01139     print_CHARS(AC.Commercial, COMMERCIALSIZE+2);
01140
01141     MesPrint("%", AC.CheckpointFlag);
01142     MesPrint("%l", AC.CheckpointStamp);
01143     print_STR((unsigned char*)AC.CheckpointRunAfter);
01144     print_STR((unsigned char*)AC.CheckpointRunBefore);
01145     MesPrint("%l", AC.CheckpointInterval);
01146
01147     MesPrint("%%% END C_const");
01148     /* fflush(0); */
01149 }
01150
01151 static void print_R()
01152 {
01153     GETIDENTITY
01154     int i;
01155     MesPrint("%%% R_const");
01156     MesPrint("%l", (LONG)(AR.infile-AR.Fscr));
01157     MesPrint("%s", AR.infile->name);
01158     MesPrint("%l", (LONG)(AR.outfile-AR.Fscr));
01159     MesPrint("%s", AR.outfile->name);
01160     MesPrint("%l", AR.hidefile-AR.Fscr);
01161     MesPrint("%s", AR.hidefile->name);
01162     for ( i=0; i<3; ++i ) {
01163         MesPrint("FSCR %d", i);
01164         print_WORDB(AR.Fscr[i].PObuffer, AR.Fscr[i].POfull);
01165     }
01166     /* ... */
01167     MesPrint("%l", AR.OldTime);
01168     MesPrint("%l", AR.InInBuf);
01169     MesPrint("%l", AR.InHiBuf);
01170     MesPrint("%l", AR.pWorkSize);
01171     MesPrint("%l", AR.lWorkSize);
01172     MesPrint("%l", AR.posWorkSize);
01173     MesPrint("%d", AR.NoCompress);
01174     MesPrint("%d", AR.gzipCompress);
01175     MesPrint("%d", AR.Cnumlhs);
01176 #ifdef WITHPTHREADS
01177     MesPrint("%d", AR.exprtodo);
01178 #endif
01179     MesPrint("%d", AR.GetFile);
01180     MesPrint("%d", AR.KeptInHold);
01181     MesPrint("%d", AR.BraceOn);
01182     MesPrint("%d", AR.MaxBracket);
01183     MesPrint("%d", AR.CurDum);
01184     MesPrint("%d", AR.DeferFlag);
01185     MesPrint("%d", AR.TePos);
01186     MesPrint("%d", AR.sLevel);
01187     MesPrint("%d", AR.Stage4Name);
01188     MesPrint("%d", AR.GetOneFile);
01189     MesPrint("%d", AR.PolyFun);
01190     MesPrint("%d", AR.PolyFunInv);
01191     MesPrint("%d", AR.PolyFunType);
01192     MesPrint("%d", AR.PolyFunExp);
01193     MesPrint("%d", AR.PolyFunVar);
01194     MesPrint("%d", AR.PolyFunPow);
01195     MesPrint("%d", AR.Eside);
01196     MesPrint("%d", AR.MaxDum);
01197     MesPrint("%d", AR.level);
01198     MesPrint("%d", AR.expchanged);
01199     MesPrint("%d", AR.expflags);

```

```

01200     MesPrint("%d", AR.CurExpr);
01201     MesPrint("%d", AR.SortType);
01202     MesPrint("%d", AR.ShortSortCount);
01203     MesPrint("%%% END R_const");
01204 /*  fflush(0); */
01205 }
01206
01207 #endif /* ifdef PRINTDEBUG */
01208
01209 /*
01210     #] Debugging :
01211     #[ Cached file operation functions :
01212 */
01213
01214 #define CACHED_SNAPSHOT
01215
01216 #define CACHE_SIZE 4096
01217
01218 #ifndef CACHED_SNAPSHOT
01219 unsigned char cache_buffer[CACHE_SIZE];
01220 size_t cache_fill = 0;
01221
01222 size_t fwrite_cached(const void *ptr, size_t size, size_t nmemb, FILE *fd)
01223 {
01224     size_t fullsize = size*nmemb;
01225     if ( fullsize+cache_fill >= CACHE_SIZE ) {
01226         size_t overlap = CACHE_SIZE-cache_fill;
01227         memcpy(cache_buffer+cache_fill, (unsigned char*)ptr, overlap);
01228         if ( fwrite(cache_buffer, 1, CACHE_SIZE, fd) != CACHE_SIZE ) return 0;
01229         fullsize -= overlap;
01230         if ( fullsize >= CACHE_SIZE ) {
01231             cache_fill = fullsize % CACHE_SIZE;
01232             if ( cache_fill ) memcpy(cache_buffer, (unsigned char*)ptr+overlap+fullsize-cache_fill,
cache_fill);
01233             if ( fwrite((unsigned char*)ptr+overlap, 1, fullsize-cache_fill, fd) !=
fullsize-cache_fill ) return 0;
01234         }
01235         else {
01236             memcpy(cache_buffer, (unsigned char*)ptr+overlap, fullsize);
01237             cache_fill = fullsize;
01238         }
01239     }
01240     else {
01241         memcpy(cache_buffer+cache_fill, (unsigned char*)ptr, fullsize);
01242         cache_fill += fullsize;
01243     }
01244     return nmemb;
01245 }
01246
01247 size_t flush_cache(FILE *fd)
01248 {
01249     if ( cache_fill ) {
01250         size_t retval = fwrite(cache_buffer, 1, cache_fill, fd);
01251         if ( retval != cache_fill ) {
01252             cache_fill = 0;
01253             return 0;
01254         }
01255         cache_fill = 0;
01256     }
01257     return 1;
01258 }
01259 #else
01260 size_t fwrite_cached(const void *ptr, size_t size, size_t nmemb, FILE *fd)
01261 {
01262     return fwrite(ptr, size, nmemb, fd);
01263 }
01264
01265 size_t flush_cache(FILE *fd)
01266 {
01267     DUMMYUSE(fd)
01268     return 1;
01269 }
01270 #endif
01271
01272 /*
01273     #] Cached file operation functions :
01274     #[ Helper Macros :
01275 */
01276
01277 /* some helper macros to streamline the code in DoSnapshot() and DoRecovery() */
01278
01279 /* freeing memory */
01280
01281 #define R_FREE(ARG) \
01282     if ( ARG ) M_free(ARG, #ARG);
01283
01284 #define R_FREE_NAMETREE(ARG) \

```



```

01285     R_FREE(ARG->namenode); \
01286     R_FREE(ARG->namebuffer); \
01287     R_FREE(ARG);
01288
01289 #define R_FREE_STREAM(ARG) \
01290     R_FREE(ARG.buffer); \
01291     R_FREE(ARG.FoldName); \
01292     R_FREE(ARG.name);
01293
01294 /* reading a single variable */
01295
01296 #define R_SET(VAR,TYPE) \
01297     VAR = *((TYPE*)p); p = (unsigned char*)p + sizeof(TYPE);
01298
01299 /* general buffer */
01300
01301 #define R_COPY_B(VAR,SIZE,CAST) \
01302     VAR = (CAST)Malloc1(SIZE,#VAR); \
01303     memcpy(VAR, p, SIZE); p = (unsigned char*)p + SIZE;
01304
01305 #define S_WRITE_B(BUF,LEN) \
01306     if ( fwrite_cached(BUF, 1, LEN, fd) != (size_t)(LEN) ) return(__LINE__);
01307
01308 #define S_FLUSH_B \
01309     if ( flush_cache(fd) != 1 ) return(__LINE__);
01310
01311 /* character strings */
01312
01313 #define R_COPY_S(VAR,CAST) \
01314     if ( VAR ) { \
01315         VAR = (CAST)Malloc1(strlen(p)+1,"R_COPY_S"); \
01316         strcpy((char*)VAR, p); p = (unsigned char*)p + strlen(p) + 1; \
01317     }
01318
01319 #define S_WRITE_S(STR) \
01320     if ( STR ) { \
01321         l = strlen((char*)STR) + 1; \
01322         if ( fwrite_cached(STR, 1, l, fd) != (size_t)l ) return(__LINE__); \
01323     }
01324
01325 /* LIST */
01326
01327 #define R_COPY_LIST(ARG) \
01328     if ( ARG.maxnum ) { \
01329         R_COPY_B(ARG.lijst, ARG.size*ARG.maxnum, void*) \
01330     }
01331
01332 #define S_WRITE_LIST(LST) \
01333     if ( LST.maxnum ) { \
01334         S_WRITE_B((char*)LST.lijst, LST.maxnum*LST.size) \
01335     }
01336
01337 /* NAMETREE */
01338
01339 #define R_COPY_NAMETREE(ARG) \
01340     R_COPY_B(ARG, sizeof(NAMETREE), NAMETREE*); \
01341     if ( ARG->namenode ) { \
01342         R_COPY_B(ARG->namenode, ARG->nodesize*sizeof(NAMENODE), NAMENODE*); \
01343     } \
01344     if ( ARG->namebuffer ) { \
01345         R_COPY_B(ARG->namebuffer, ARG->namesize, UBYTE*); \
01346     }
01347
01348 #define S_WRITE_NAMETREE(ARG) \
01349     S_WRITE_B(ARG, sizeof(NAMETREE)); \
01350     if ( ARG->namenode ) { \
01351         S_WRITE_B(ARG->namenode, ARG->nodesize*sizeof(struct NaMeNode)); \
01352     } \
01353     if ( ARG->namebuffer ) { \
01354         S_WRITE_B(ARG->namebuffer, ARG->namesize); \
01355     }
01356
01357 /* DOLLAR */
01358
01359 #define S_WRITE_DOLLAR(ARG) \
01360     if ( ARG.size && ARG.where && ARG.where != &(AM.dollarzero) ) { \
01361         S_WRITE_B(ARG.where, ARG.size*sizeof(WORD)) \
01362     }
01363
01364 /* Printing time marks with ANNOUNCE macro */
01365
01366 #ifdef PRINTTIMEMARKS
01367     time_t announce_time;
01368     #define ANNOUNCE(str) time(&announce_time); MesPrint("TIMEMARK %s %s", ctime(&announce_time), #str);
01369     #else
01370     #define ANNOUNCE(str)
01371 #endif

```

```

01372
01373 /*
01374 #] Helper Macros :
01375 #[ DoRecovery :
01376 */
01377
01402 int DoRecovery(int *moduletype)
01403 {
01404     GETIDENTITY
01405     FILE *fd;
01406     POSITION pos;
01407     void *buf, *p;
01408     LONG size, l;
01409     int i, j;
01410     UBYTE *org;
01411     char *namebufout, *namebufhide;
01412     LONG ofs;
01413     void *oldAMdollarzero;
01414     LIST PotModDolListBackup;
01415     LIST ModOptDolListBackup;
01416     WORD oldLogHandle;
01417
01418     MesPrint("Recovering ... %"); fflush(0);
01419
01420     if ( !(fd = fopen(recoveryfile, "r")) ) return(__LINE__);
01421
01422     /* load the complete recovery file into a buffer */
01423     if ( fread(&pos, sizeof(POSITION), 1, fd) != 1 ) return(__LINE__);
01424     size = BASEPOSITION(pos) - sizeof(POSITION);
01425     buf = Malloc1(size, "recovery buffer");
01426     if ( fread(buf, size, 1, fd) != 1 ) return(__LINE__);
01427
01428     /* pointer p will go through the buffer in the following */
01429     p = buf;
01430
01431     /* read moduletype */
01432     R_SET(*moduletype, int);
01433
01434     /*#[ AM : */
01435
01436     /* only certain elements will be restored. the rest of AM should have gotten
01437      * the correct values at startup. */
01438
01439     R_SET(AM.hparallelflag, int);
01440     R_SET(AM.gparallelflag, int);
01441     R_SET(AM.gCodesFlag, int);
01442     R_SET(AM.gNamesFlag, int);
01443     R_SET(AM.gStatsFlag, int);
01444     R_SET(AM.gTokensWriteFlag, int);
01445     R_SET(AM.gNoSpacesInNumbers, int);
01446     R_SET(AM.gIndentSpace, WORD);
01447     R_SET(AM.gUnitTrace, WORD);
01448     R_SET(AM.gDefDim, int);
01449     R_SET(AM.gDefDim4, int);
01450     R_SET(AM.gncmod, WORD);
01451     R_SET(AM.gnpowmod, WORD);
01452     R_SET(AM.gmodmode, WORD);
01453     R_SET(AM.gOutputMode, WORD);
01454     R_SET(AM.gCnumpows, WORD);
01455     R_SET(AM.gOutputSpaces, WORD);
01456     R_SET(AM.gOutNumberType, WORD);
01457     R_SET(AM.gfunpowers, int);
01458     R_SET(AM.gPolyFun, WORD);
01459     R_SET(AM.gPolyFunInv, WORD);
01460     R_SET(AM.gPolyFunType, WORD);
01461     R_SET(AM.gPolyFunExp, WORD);
01462     R_SET(AM.gPolyFunVar, WORD);
01463     R_SET(AM.gPolyFunPow, WORD);
01464     R_SET(AM.gProcessBucketSize, LONG);
01465     R_SET(AM.OldChildTime, LONG);
01466     R_SET(AM.OldSecTime, LONG);
01467     R_SET(AM.OldMilliTime, LONG);
01468     R_SET(AM.gproperorderflag, int);
01469     R_SET(AM.gThreadBucketSize, LONG);
01470     R_SET(AM.gSizeCommuteInSet, int);
01471     R_SET(AM.gThreadStats, int);
01472     R_SET(AM.gFinalStats, int);
01473     R_SET(AM.gThreadsFlag, int);
01474     R_SET(AM.gThreadBalancing, int);
01475     R_SET(AM.gThreadSortFileSynch, int);
01476     R_SET(AM.gProcessStats, int);
01477     R_SET(AM.gOldParallelStats, int);
01478     R_SET(AM.gSortType, int);
01479     R_SET(AM.gShortStatsMax, WORD);
01480     R_SET(AM.gIsFortran90, int);
01481     R_SET(oldAMdollarzero, void*);
01482     R_FREE(AM.gFortran90Kind);

```

```

01483     R_SET(AM.gFortran90Kind, UBYTE *);
01484     R_COPY_S(AM.gFortran90Kind, UBYTE *);
01485
01486     R_COPY_S(AM.gextrasy, UBYTE *);
01487     R_COPY_S(AM.ggextrasy, UBYTE *);
01488
01489     R_SET(AM.PrintTotalSize, int);
01490     R_SET(AM.fbufferSize, int);
01491     R_SET(AM.gOldFactArgFlag, int);
01492     R_SET(AM.ggOldFactArgFlag, int);
01493
01494     R_SET(AM.gnumextrasy, int);
01495     R_SET(AM.ggnumextrasy, int);
01496     R_SET(AM.NumSpectatorFiles, int);
01497     R_SET(AM.SizeForSpectatorFiles, int);
01498     R_SET(AM.gOldGCDflag, int);
01499     R_SET(AM.ggOldGCDflag, int);
01500     R_SET(AM.gWTimeStatsFlag, int);
01501
01502     R_FREE(AM.Path);
01503     R_SET(AM.Path, UBYTE *);
01504     R_COPY_S(AM.Path, UBYTE *);
01505
01506     R_SET(AM.FromStdin, BOOL);
01507     R_SET(AM.IgnoreDeprecation, BOOL);
01508
01509 #ifdef PRINTDEBUG
01510     print_M();
01511 #endif
01512
01513     /*#] AM : */
01514     /*#[ AC : */
01515
01516     /* #[ AC free pointers */
01517
01518     /* AC will be overwritten by data from the recovery file, therefore
01519      * dynamically allocated memory must be freed first. */
01520
01521     R_FREE_NAMETREE(AC.dollarnames);
01522     R_FREE_NAMETREE(AC.exprnames);
01523     R_FREE_NAMETREE(AC.varnames);
01524     for ( i=0; i<AC.ChannellList.num; ++i ) {
01525         R_FREE(channels[i].name);
01526     }
01527     R_FREE(AC.ChannellList.lijst);
01528     R_FREE(AC.DubiousList.lijst);
01529     for ( i=0; i<AC.FunctionList.num; ++i ) {
01530         TABLES T = functions[i].tabl;
01531         if ( T ) {
01532             R_FREE(T->buffers);
01533             R_FREE(T->mm);
01534             R_FREE(T->flags);
01535             R_FREE(T->prototype);
01536             R_FREE(T->tablepointers);
01537             if ( T->sparse ) {
01538                 R_FREE(T->boomlijst);
01539                 R_FREE(T->argtail);
01540             }
01541             if ( T->spare ) {
01542                 R_FREE(T->spare->buffers);
01543                 R_FREE(T->spare->mm);
01544                 R_FREE(T->spare->flags);
01545                 R_FREE(T->spare->tablepointers);
01546                 if ( T->spare->sparse ) {
01547                     R_FREE(T->spare->boomlijst);
01548                 }
01549                 R_FREE(T->spare);
01550             }
01551             R_FREE(T);
01552         }
01553     }
01554     R_FREE(AC.FunctionList.lijst);
01555     for ( i=0; i<AC.ExpressionList.num; ++i ) {
01556         if ( Expressions[i].renum ) {
01557             R_FREE(Expressions[i].renum->symb.lo);
01558             R_FREE(Expressions[i].renum);
01559         }
01560         if ( Expressions[i].bracketinfo ) {
01561             R_FREE(Expressions[i].bracketinfo->indexbuffer);
01562             R_FREE(Expressions[i].bracketinfo->bracketbuffer);
01563             R_FREE(Expressions[i].bracketinfo);
01564         }
01565         if ( Expressions[i].newbracketinfo ) {
01566             R_FREE(Expressions[i].newbracketinfo->indexbuffer);
01567             R_FREE(Expressions[i].newbracketinfo->bracketbuffer);
01568             R_FREE(Expressions[i].newbracketinfo);
01569         }

```

```

01570         if ( Expressions[i].renumlists != AN.dummyrenumlist ) {
01571             R_FREE(Expressions[i].renumlists);
01572         }
01573         R_FREE(Expressions[i].inmem);
01574     }
01575     R_FREE(AC.ExpressionList.lijst);
01576     R_FREE(AC.IndexList.lijst);
01577     R_FREE(AC.SetElementList.lijst);
01578     R_FREE(AC.SetList.lijst);
01579     R_FREE(AC.SymbolList.lijst);
01580     R_FREE(AC.VectorList.lijst);
01581     for ( i=0; i<AC.TableBaseList.num; ++i ) {
01582         R_FREE(tablebases[i].iblocks);
01583         R_FREE(tablebases[i].nblocks);
01584         R_FREE(tablebases[i].name);
01585         R_FREE(tablebases[i].fullname);
01586         R_FREE(tablebases[i].tablenames);
01587     }
01588     R_FREE(AC.TableBaseList.lijst);
01589     for ( i=0; i<AC.cbufList.num; ++i ) {
01590         R_FREE(cbuf[i].Buffer);
01591         R_FREE(cbuf[i].lhs);
01592         R_FREE(cbuf[i].rhs);
01593         R_FREE(cbuf[i].boomlijst);
01594     }
01595     R_FREE(AC.cbufList.lijst);
01596     R_FREE(AC.AutoSymbolList.lijst);
01597     R_FREE(AC.AutoIndexList.lijst);
01598     R_FREE(AC.AutoVectorList.lijst);
01599     /* Tables cannot be auto-declared, therefore no extra code here */
01600     R_FREE(AC.AutoFunctionList.lijst);
01601     R_FREE_NAMETREE(AC.autonames);
01602     for ( i=0; i<AC.NumStreams; ++i ) {
01603         R_FREE_STREAM(AC.Streams[i]);
01604     }
01605     R_FREE(AC.Streams);
01606     R_FREE(AC.termstack);
01607     R_FREE(AC.termstortstack);
01608     R_FREE(AC.cmod);
01609     R_FREE(AC.modpowers);
01610     R_FREE(AC.halfmod);
01611     R_FREE(AC.IfHeap);
01612     R_FREE(AC.IfCount);
01613     R_FREE(AC.iBuffer);
01614     for ( i=0; i<AC.NumLabels; ++i ) {
01615         R_FREE(AC.LabelNames[i]);
01616     }
01617     R_FREE(AC.LabelNames);
01618     R_FREE(AC.FixIndices);
01619     R_FREE(AC.termsumcheck);
01620     R_FREE(AC.WildcardNames);
01621     R_FREE(AC.tokens);
01622     R_FREE(AC.tokenarglevel);
01623     R_FREE(AC.modinverses);
01624     R_FREE(AC.Fortran90Kind);
01625 #ifdef WITHPTHREADS
01626     R_FREE(AC.inputnumbers);
01627 #endif
01628     R_FREE(AC.IfSumCheck);
01629     R_FREE(AC.CommuteInSet);
01630     R_FREE(AC.CheckpointRunAfter);
01631     R_FREE(AC.CheckpointRunBefore);
01632
01633     /* #] AC free pointers */
01634
01635     /* backup some lists in order to restore it to the initial setup */
01636     PotModDolListBackup = AC.PotModDolList;
01637     ModOptDolListBackup = AC.ModOptDolList;
01638     oldLogHandle = AC.LogHandle;
01639
01640     /* first we copy AC as a whole and then restore the pointer structures step
01641        by step. */
01642
01643     AC = *((struct C_const*)p); p = (unsigned char*)p + sizeof(struct C_const);
01644
01645     R_COPY_NAMETREE(AC.dollarnames);
01646     R_COPY_NAMETREE(AC.exprnames);
01647     R_COPY_NAMETREE(AC.varnames);
01648
01649     R_COPY_LIST(AC.ChannelList);
01650     for ( i=0; i<AC.ChannelList.num; ++i ) {
01651         R_COPY_S(channels[i].name, char*);
01652         channels[i].handle = ReOpenFile(channels[i].name);
01653     }
01654     AC.ChannelList.message = "channel buffer";
01655
01656     AC.DubiousList.lijst = 0;

```

```

01657     AC.DubiousList.message = "ambiguous variable";
01658     AC.DubiousList.num =
01659     AC.DubiousList.maxnum =
01660     AC.DubiousList.numglobal =
01661     AC.DubiousList.numtemp =
01662     AC.DubiousList.numclear = 0;
01663
01664     R_COPY_LIST(AC.FunctionList);
01665     for ( i=0; i<AC.FunctionList.num; ++i ) {
01666         if ( functions[i].tabl ) {
01667             TABLES tabl;
01668             R_COPY_B(tabl, sizeof(struct TableEs), TABLES);
01669             functions[i].tabl = tabl;
01670             if ( tabl->tablepointers ) {
01671                 if ( tabl->sparse ) {
01672                     R_COPY_B(tabl->tablepointers,
01673                             tabl->reserved*sizeof(WORD)*(tabl->numind+TABLEEXTENSION),
01674                             WORD*);
01675                 }
01676                 else {
01677                     R_COPY_B(tabl->tablepointers,
01678                             TABLEEXTENSION*sizeof(WORD)*(tabl->totind), WORD*);
01679                 }
01680             }
01681             org = (UBYTE*)tabl->prototype;
01682             #ifdef WITHPTHREADS
01683             R_COPY_B(tabl->prototype, tabl->prototypeSize, WORD*);
01684             ofs = (UBYTE*)tabl->prototype - org;
01685             for ( j=0; j<AM.totalnumberofthreads; ++j ) {
01686                 if ( tabl->prototype[j] ) {
01687                     tabl->prototype[j] = (WORD*)((UBYTE*)tabl->prototype[j] + ofs);
01688                 }
01689             }
01690             if ( tabl->pattern ) {
01691                 tabl->pattern = (WORD*)((UBYTE*)tabl->pattern + ofs);
01692                 for ( j=0; j<AM.totalnumberofthreads; ++j ) {
01693                     if ( tabl->pattern[j] ) {
01694                         tabl->pattern[j] = (WORD*)((UBYTE*)tabl->pattern[j] + ofs);
01695                     }
01696                 }
01697             }
01698             #else
01699             ofs = tabl->pattern - tabl->prototype;
01700             R_COPY_B(tabl->prototype, tabl->prototypeSize, WORD*);
01701             if ( tabl->pattern ) {
01702                 tabl->pattern = tabl->prototype + ofs;
01703             }
01704             #endif
01705             R_COPY_B(tabl->mm, tabl->numind*(LONG)sizeof(MINMAX), MINMAX*);
01706             R_COPY_B(tabl->flags, tabl->numind*(LONG)sizeof(WORD), WORD*);
01707             if ( tabl->sparse ) {
01708                 R_COPY_B(tabl->boomlijst, tabl->MaxTreeSize*(LONG)sizeof(COMPTREE), COMPTREE*);
01709                 R_COPY_S(tabl->argtail, UBYTE*);
01710             }
01711             R_COPY_B(tabl->buffers, tabl->buffersize*(LONG)sizeof(WORD), WORD*);
01712             if ( tabl->spare ) {
01713                 TABLES spare;
01714                 R_COPY_B(spare, sizeof(struct TableEs), TABLES);
01715                 tabl->spare = spare;
01716                 if ( spare->tablepointers ) {
01717                     if ( spare->sparse ) {
01718                         R_COPY_B(spare->tablepointers,
01719                                 spare->reserved*sizeof(WORD)*(spare->numind+TABLEEXTENSION),
01720                                 WORD*);
01721                     }
01722                     else {
01723                         R_COPY_B(spare->tablepointers,
01724                                 TABLEEXTENSION*sizeof(WORD)*(spare->totind), WORD*);
01725                     }
01726                 }
01727                 spare->prototype = tabl->prototype;
01728                 spare->pattern = tabl->pattern;
01729                 R_COPY_B(spare->mm, spare->numind*(LONG)sizeof(MINMAX), MINMAX*);
01730                 R_COPY_B(spare->flags, spare->numind*(LONG)sizeof(WORD), WORD*);
01731                 if ( tabl->sparse ) {
01732                     R_COPY_B(spare->boomlijst, spare->MaxTreeSize*(LONG)sizeof(COMPTREE), COMPTREE*);
01733                     spare->argtail = tabl->argtail;
01734                 }
01735                 spare->spare = tabl;
01736                 R_COPY_B(spare->buffers, spare->buffersize*(LONG)sizeof(WORD), WORD*);
01737             }
01738         }
01739     }
01740     AC.FunctionList.message = "function";
01741
01742     R_COPY_LIST(AC.ExpressionList);
01743     for ( i=0; i<AC.ExpressionList.num; ++i ) {

```

```

01744     EXPRESSIONS ex = Expressions + i;
01745     if ( ex->renum ) {
01746         R_COPY_B(ex->renum, sizeof(struct ReNuMbEr), RENUMBER);
01747         org = (UBYTE*)ex->renum->symb.lo;
01748         R_SET(size, size_t);
01749         R_COPY_B(ex->renum->symb.lo, size, WORD*);
01750         ofs = (UBYTE*)ex->renum->symb.lo - org;
01751         ex->renum->symb.start = (WORD*) ((UBYTE*)ex->renum->symb.start + ofs);
01752         ex->renum->symb.hi = (WORD*) ((UBYTE*)ex->renum->symb.hi + ofs);
01753         ex->renum->indi.lo = (WORD*) ((UBYTE*)ex->renum->indi.lo + ofs);
01754         ex->renum->indi.start = (WORD*) ((UBYTE*)ex->renum->indi.start + ofs);
01755         ex->renum->indi.hi = (WORD*) ((UBYTE*)ex->renum->indi.hi + ofs);
01756         ex->renum->vect.lo = (WORD*) ((UBYTE*)ex->renum->vect.lo + ofs);
01757         ex->renum->vect.start = (WORD*) ((UBYTE*)ex->renum->vect.start + ofs);
01758         ex->renum->vect.hi = (WORD*) ((UBYTE*)ex->renum->vect.hi + ofs);
01759         ex->renum->func.lo = (WORD*) ((UBYTE*)ex->renum->func.lo + ofs);
01760         ex->renum->func.start = (WORD*) ((UBYTE*)ex->renum->func.start + ofs);
01761         ex->renum->func.hi = (WORD*) ((UBYTE*)ex->renum->func.hi + ofs);
01762         ex->renum->symnum = (WORD*) ((UBYTE*)ex->renum->symnum + ofs);
01763         ex->renum->indnum = (WORD*) ((UBYTE*)ex->renum->indnum + ofs);
01764         ex->renum->vecnum = (WORD*) ((UBYTE*)ex->renum->vecnum + ofs);
01765         ex->renum->funnum = (WORD*) ((UBYTE*)ex->renum->funnum + ofs);
01766     }
01767     if ( ex->bracketinfo ) {
01768         R_COPY_B(ex->bracketinfo, sizeof(BRACKETINFO), BRACKETINFO*);
01769         R_COPY_B(ex->bracketinfo->indexbuffer,
01770 ex->bracketinfo->indexbuffersize*sizeof(BRACKETINDEX), BRACKETINDEX*);
01771         R_COPY_B(ex->bracketinfo->bracketbuffer, ex->bracketinfo->bracketbuffersize*sizeof(WORD),
01772 WORD*);
01773     }
01774     if ( ex->newbracketinfo ) {
01775         R_COPY_B(ex->newbracketinfo, sizeof(BRACKETINFO), BRACKETINFO*);
01776         R_COPY_B(ex->newbracketinfo->indexbuffer,
01777 ex->newbracketinfo->indexbuffersize*sizeof(BRACKETINDEX), BRACKETINDEX*);
01778         R_COPY_B(ex->newbracketinfo->bracketbuffer,
01779 ex->newbracketinfo->bracketbuffersize*sizeof(WORD), WORD*);
01780     }
01781 #ifdef WITHPTHREADS
01782     ex->renumlists = 0;
01783 #else
01784     ex->renumlists = AN.dummyrenumlist;
01785 #endif
01786     if ( ex->inmem ) {
01787         R_SET(size, size_t);
01788         R_COPY_B(ex->inmem, size, WORD*);
01789     }
01790     AC.ExpressionList.message = "expression";
01791     R_COPY_LIST(AC.IndexList);
01792     AC.IndexList.message = "index";
01793     R_COPY_LIST(AC.SetElementList);
01794     AC.SetElementList.message = "set element";
01795     R_COPY_LIST(AC.SetList);
01796     AC.SetList.message = "set";
01797     R_COPY_LIST(AC.SymbolList);
01798     AC.SymbolList.message = "symbol";
01799     R_COPY_LIST(AC.VectorList);
01800     AC.VectorList.message = "vector";
01801     AC.PotModDolList = PotModDolListBackup;
01802     AC.ModOptDolList = ModOptDolListBackup;
01803     R_COPY_LIST(AC.TableBaseList);
01804     for ( i=0; i<AC.TableBaseList.num; ++i ) {
01805         if ( tablebases[i].iblocks ) {
01806             R_COPY_B(tablebases[i].iblocks,
01807 tablebases[i].info.numberofindexblocks*sizeof(INDEXBLOCK*), INDEXBLOCK**);
01808             for ( j=0; j<tablebases[i].info.numberofindexblocks; ++j ) {
01809                 if ( tablebases[i].iblocks[j] ) {
01810                     R_COPY_B(tablebases[i].iblocks[j], sizeof(INDEXBLOCK), INDEXBLOCK*);
01811                 }
01812             }
01813             if ( tablebases[i].nblocks ) {
01814                 R_COPY_B(tablebases[i].nblocks,
01815 tablebases[i].info.numberofnamesblocks*sizeof(NAMESBLOCK*), NAMESBLOCK**);
01816                 for ( j=0; j<tablebases[i].info.numberofindexblocks; ++j ) {
01817                     if ( tablebases[i].nblocks[j] ) {
01818                         R_COPY_B(tablebases[i].nblocks[j], sizeof(NAMESBLOCK), NAMESBLOCK*);
01819                     }
01820                 }
01821             }
01822             /* reopen file */
01823             if ( ( tablebases[i].handle = fopen(tablebases[i].fullname, "r+b") ) == NULL ) {
01824                 MesPrint("ERROR: Could not reopen tablebase %s!", tablebases[i].name);
01825                 Terminate(-1);
01826             }
01827         }
01828     }

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```

01825     }
01826     R_COPY_S(tablebases[i].name,char*);
01827     R_COPY_S(tablebases[i].fullname,char*);
01828     R_COPY_S(tablebases[i].tablenames,char*);
01829 }
01830 AC.TableBaseList.message = "list of tablebases";
01831
01832 R_COPY_LIST(AC.cbufList);
01833 for ( i=0; i<AC.cbufList.num; ++i ) {
01834     org = (UBYTE*)cbuf[i].Buffer;
01835     R_COPY_B(cbuf[i].Buffer, cbuf[i].BufferSize*sizeof(WORD), WORD*);
01836     ofs = (UBYTE*)cbuf[i].Buffer - org;
01837     cbuf[i].Top = (WORD*)((UBYTE*)cbuf[i].Top + ofs);
01838     cbuf[i].Pointer = (WORD*)((UBYTE*)cbuf[i].Pointer + ofs);
01839     R_COPY_B(cbuf[i].lhs, cbuf[i].maxlhs*(LONG)sizeof(WORD*), WORD**);
01840     for ( j=1; j<=cbuf[i].numlhs; ++j ) {
01841         if ( cbuf[i].lhs[j] ) cbuf[i].lhs[j] = (WORD*)((UBYTE*)cbuf[i].lhs[j] + ofs);
01842     }
01843     org = (UBYTE*)cbuf[i].rhs;
01844     R_COPY_B(cbuf[i].rhs, cbuf[i].maxrhs*(LONG)(sizeof(WORD*)+2*sizeof(LONG)+2*sizeof(WORD)),
WORD**);
01845     for ( j=1; j<=cbuf[i].numrhs; ++j ) {
01846         if ( cbuf[i].rhs[j] ) cbuf[i].rhs[j] = (WORD*)((UBYTE*)cbuf[i].rhs[j] + ofs);
01847     }
01848     ofs = (UBYTE*)cbuf[i].rhs - org;
01849     cbuf[i].CanCommu = (LONG*)((UBYTE*)cbuf[i].CanCommu + ofs);
01850     cbuf[i].NumTerms = (LONG*)((UBYTE*)cbuf[i].NumTerms + ofs);
01851     cbuf[i].numdum = (WORD*)((UBYTE*)cbuf[i].numdum + ofs);
01852     cbuf[i].dimension = (WORD*)((UBYTE*)cbuf[i].dimension + ofs);
01853     if ( cbuf[i].boomlijst ) {
01854         R_COPY_B(cbuf[i].boomlijst, cbuf[i].MaxTreeSize*sizeof(COMPTREE), COMPTREE*);
01855     }
01856 }
01857 AC.cbufList.message = "compiler buffer";
01858
01859 R_COPY_LIST(AC.AutoSymbolList);
01860 AC.AutoSymbolList.message = "autosymbol";
01861 R_COPY_LIST(AC.AutoIndexList);
01862 AC.AutoIndexList.message = "autoindex";
01863 R_COPY_LIST(AC.AutoVectorList);
01864 AC.AutoVectorList.message = "autovector";
01865 R_COPY_LIST(AC.AutoFunctionList);
01866 AC.AutoFunctionList.message = "autofunction";
01867
01868 R_COPY_NAMETREE(AC.autonames);
01869
01870 AC.Symbols = &(AC.SymbolList);
01871 AC.Indices = &(AC.IndexList);
01872 AC.Vectors = &(AC.VectorList);
01873 AC.Functions = &(AC.FunctionList);
01874 AC.activenames = &(AC.varnames);
01875
01876 org = (UBYTE*)AC.Streams;
01877 R_COPY_B(AC.Streams, AC.MaxNumStreams*(LONG)sizeof(STREAM), STREAM*);
01878 for ( i=0; i<AC.NumStreams; ++i ) {
01879     if ( AC.Streams[i].type != FILESTREAM ) {
01880         UBYTE *org2;
01881         org2 = AC.Streams[i].buffer;
01882         if ( AC.Streams[i].inbuffer ) {
01883             R_COPY_B(AC.Streams[i].buffer, AC.Streams[i].inbuffer, UBYTE*);
01884         }
01885         ofs = AC.Streams[i].buffer - org2;
01886         AC.Streams[i].pointer += ofs;
01887         AC.Streams[i].top += ofs;
01888     }
01889     else {
01890         p = (unsigned char*)p + AC.Streams[i].inbuffer;
01891     }
01892     AC.Streams[i].buffersize = AC.Streams[i].inbuffer;
01893     R_COPY_S(AC.Streams[i].FoldName,UBYTE*);
01894     R_COPY_S(AC.Streams[i].name,UBYTE*);
01895     if ( AC.Streams[i].type == PREVARSTREAM || AC.Streams[i].type == DOLLARSTREAM ) {
01896         AC.Streams[i].pname = AC.Streams[i].name;
01897     }
01898     else if ( AC.Streams[i].type == FILESTREAM ) {
01899         UBYTE *org2;
01900         org2 = AC.Streams[i].buffer;
01901         AC.Streams[i].buffer = (UBYTE*)Malloc1(AC.Streams[i].buffersize, "buffer");
01902         ofs = AC.Streams[i].buffer - org2;
01903         AC.Streams[i].pointer += ofs;
01904         AC.Streams[i].top += ofs;
01905
01906         /* open file except for already opened main input file */
01907         if ( i ) {
01908             AC.Streams[i].handle = OpenFile((char *) (AC.Streams[i].name));
01909             if ( AC.Streams[i].handle == -1 ) {
01910                 MesPrint("ERROR: Could not reopen stream %s!",AC.Streams[i].name);

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01911         Terminate(-1);
01912     }
01913 }
01914
01915     PUTZERO(pos);
01916     ADDPOS(pos, AC.Streams[i].bufferposition);
01917     SeekFile(AC.Streams[i].handle, &pos, SEEK_SET);
01918
01919     AC.Streams[i].inbuffer = ReadFile(AC.Streams[i].handle, AC.Streams[i].buffer,
AC.Streams[i].inbuffer);
01920
01921     SETBASEPOSITION(pos, AC.Streams[i].fileposition);
01922     SeekFile(AC.Streams[i].handle, &pos, SEEK_SET);
01923 }
01924 /*
01925  * Ideally, we should check if we have a type PREREADSTREAM, PREREADSTREAM2, and
01926  * PRECALCSTREAM here. If so, we should free element name and point it
01927  * to the name element of the embracing stream's struct. In practice,
01928  * this is undoable without adding new data to STREAM. Since we create
01929  * only a small memory leak here (some few byte for each existing
01930  * stream of these types) and only once when we do a recovery, we
01931  * tolerate this leak and keep it STREAM as it is.
01932  */
01933 }
01934 ofs = (UBYTE*)AC.Streams - org;
01935 AC.CurrentStream = (STREAM*)((UBYTE*)AC.CurrentStream + ofs);
01936
01937 if ( AC.termstack ) {
01938     R_COPY_B(AC.termstack, AC.maxtermlevel*(LONG)sizeof(LONG), LONG*);
01939 }
01940
01941 if ( AC.termstack ) {
01942     R_COPY_B(AC.termstack, AC.maxtermlevel*(LONG)sizeof(LONG), LONG*);
01943 }
01944
01945 /* exception: here we also change values from struct AM */
01946 R_COPY_B(AC.cmod, AM.MaxTal*4*(LONG)sizeof(WORD), WORD*);
01947 AM.gcmmod = AC.cmod + AM.MaxTal;
01948 AC.powmod = AM.gcmmod + AM.MaxTal;
01949 AM.gpowmod = AC.powmod + AM.MaxTal;
01950
01951 AC.modpowers = 0;
01952 AC.halfmod = 0;
01953
01954 /* we don't care about AC.ProtoType/WildC */
01955
01956 if ( AC.IfHeap ) {
01957     ofs = AC.IfStack - AC.IfHeap;
01958     R_COPY_B(AC.IfHeap, (LONG)sizeof(LONG)*(AC.MaxIf+1), LONG*);
01959     AC.IfStack = AC.IfHeap + ofs;
01960     R_COPY_B(AC.IfCount, (LONG)sizeof(LONG)*(AC.MaxIf+1), LONG*);
01961 }
01962
01963 org = AC.iBuffer;
01964 size = AC.iStop - AC.iBuffer + 2;
01965 R_COPY_B(AC.iBuffer, size, UBYTE*);
01966 ofs = AC.iBuffer - org;
01967 AC.iPointer += ofs;
01968 AC.iStop += ofs;
01969
01970 if ( AC.LabelNames ) {
01971     org = (UBYTE*)AC.LabelNames;
01972     R_COPY_B(AC.LabelNames, AC.MaxLabels*(LONG)(sizeof(UBYTE*)+sizeof(WORD)), UBYTE*);
01973     for ( i=0; i<AC.NumLabels; ++i ) {
01974         R_COPY_S(AC.LabelNames[i], UBYTE*);
01975     }
01976     ofs = (UBYTE*)AC.LabelNames - org;
01977     AC.Labels = (int*)((UBYTE*)AC.Labels + ofs);
01978 }
01979
01980 R_COPY_B(AC.FixIndices, AM.OffsetIndex*(LONG)sizeof(WORD), WORD*);
01981
01982 if ( AC.termsumcheck ) {
01983     R_COPY_B(AC.termsumcheck, AC.maxtermlevel*(LONG)sizeof(WORD), WORD*);
01984 }
01985
01986 R_COPY_B(AC.WildcardNames, AC.WildcardBufferSize, UBYTE*);
01987
01988 if ( AC.tokens ) {
01989     size = AC.toptokens - AC.tokens;
01990     if ( size ) {
01991         org = (UBYTE*)AC.tokens;
01992         R_COPY_B(AC.tokens, size, SBYTE*);
01993         ofs = (UBYTE*)AC.tokens - org;
01994         AC.endoftokens += ofs;
01995         AC.toptokens += ofs;
01996     }

```



```

01997         else {
01998             AC.tokens = 0;
01999             AC.endoftokens = AC.tokens;
02000             AC.toptokens = AC.tokens;
02001         }
02002     }
02003
02004     R_COPY_B(AC.tokenarglevel, AM.MaxParLevel*(LONG)sizeof(WORD), WORD*);
02005
02006     AC.modinverses = 0;
02007
02008     R_COPY_S(AC.Fortran90Kind, UBYTE *);
02009
02010 #ifdef WITHPTHREADS
02011     if ( AC.inputnumbers ) {
02012         org = (UBYTE*)AC.inputnumbers;
02013         R_COPY_B(AC.inputnumbers, AC.sizepfirstnum*(LONG)(sizeof(WORD)+sizeof(LONG)), LONG*);
02014         ofs = (UBYTE*)AC.inputnumbers - org;
02015         AC.pfirstnum = (WORD*)((UBYTE*)AC.pfirstnum + ofs);
02016     }
02017     AC.halfmodlock = dummylock;
02018 #endif /* ifdef WITHPTHREADS */
02019
02020     if ( AC.IfSumCheck ) {
02021         R_COPY_B(AC.IfSumCheck, (LONG)sizeof(WORD)*(AC.MaxIf+1), WORD*);
02022     }
02023     if ( AC.CommuteInSet ) {
02024         R_COPY_B(AC.CommuteInSet, (LONG)sizeof(WORD)*(AC.SizeCommuteInSet+1), WORD*);
02025     }
02026
02027     AC.LogHandle = oldLogHandle;
02028
02029     R_COPY_S(AC.CheckpointRunAfter, char*);
02030     R_COPY_S(AC.CheckpointRunBefore, char*);
02031
02032     R_COPY_S(AC.extrasym, UBYTE *);
02033
02034 #ifdef PRINTDEBUG
02035     print_C();
02036 #endif
02037
02038     /*#] AC : */
02039     /*#[ AP : */
02040
02041     /* #[ AP free pointers */
02042
02043     /* AP will be overwritten by data from the recovery file, therefore
02044      * dynamically allocated memory must be freed first. */
02045
02046     for ( i=0; i<AP.DollarList.num; ++i ) {
02047         if ( Dollars[i].size ) {
02048             R_FREE(Dollars[i].where);
02049         }
02050         CleanDollarFactors(Dollars+i);
02051     }
02052     R_FREE(AP.DollarList.lijst);
02053
02054     for ( i=0; i<AP.PreVarList.num; ++i ) {
02055         R_FREE(PreVar[i].name);
02056     }
02057     R_FREE(AP.PreVarList.lijst);
02058
02059     for ( i=0; i<AP.LoopList.num; ++i ) {
02060         R_FREE(DoLoops[i].p.buffer);
02061         if ( DoLoops[i].dollarname ) {
02062             R_FREE(DoLoops[i].dollarname);
02063         }
02064     }
02065     R_FREE(AP.LoopList.lijst);
02066
02067     for ( i=0; i<AP.ProcList.num; ++i ) {
02068         R_FREE(Procedures[i].p.buffer);
02069         R_FREE(Procedures[i].name);
02070     }
02071     R_FREE(AP.ProcList.lijst);
02072
02073     for ( i=1; i<=AP.PreSwitchLevel; ++i ) {
02074         R_FREE(AP.PreSwitchStrings[i]);
02075     }
02076     R_FREE(AP.PreSwitchStrings);
02077     R_FREE(AP.preStart);
02078     R_FREE(AP.procedureExtension);
02079     R_FREE(AP.cprocedureExtension);
02080     R_FREE(AP.PreIfStack);
02081     R_FREE(AP.PreSwitchModes);
02082     R_FREE(AP.PreTypes);
02083

```

```

02084      /* #] AP free pointers */
02085
02086      /* first we copy AP as a whole and then restore the pointer structures step
02087         by step. */
02088
02089      AP = *((struct P_const*)p); p = (unsigned char*)p + sizeof(struct P_const);
02090 #ifdef WITHPTHREADS
02091      AP.PreVarLock = dummylock;
02092 #endif
02093
02094      R_COPY_LIST(AP.DollarList);
02095      for ( i=0; i<AP.DollarList.num; ++i ) {
02096          DOLLARS d = Dollars + i;
02097          size = d->size * sizeof(WORD);
02098          if ( size && d->where && d->where != oldAMDollarzero ) {
02099              R_COPY_B(d->where, size, void*);
02100          }
02101 #ifdef WITHPTHREADS
02102          d->pthreadlockread = dummylock;
02103          d->pthreadlockwrite = dummylock;
02104 #endif
02105          if ( d->nfactors > 1 ) {
02106              R_COPY_B(d->factors, sizeof(FACDOLLAR)*d->nfactors, FACDOLLAR*);
02107              for ( j = 0; j < d->nfactors; j++ ) {
02108                  if ( d->factors[j].size > 0 ) {
02109                      R_COPY_B(d->factors[j].where, sizeof(WORD)*(d->factors[j].size+1), WORD*);
02110                  }
02111              }
02112          }
02113      }
02114      AP.DollarList.message = "$-variable";
02115
02116      R_COPY_LIST(AP.PreVarList);
02117      for ( i=0; i<AP.PreVarList.num; ++i ) {
02118          R_SET(size, size_t);
02119          org = PreVar[i].name;
02120          R_COPY_B(PreVar[i].name, size, UBYTE*);
02121          ofs = PreVar[i].name - org;
02122          if ( PreVar[i].value ) PreVar[i].value += ofs;
02123          if ( PreVar[i].argnames ) PreVar[i].argnames += ofs;
02124      }
02125      AP.PreVarList.message = "PreVariable";
02126
02127      R_COPY_LIST(AP.LoopList);
02128      for ( i=0; i<AP.LoopList.num; ++i ) {
02129          org = DoLoops[i].p.buffer;
02130          R_COPY_B(DoLoops[i].p.buffer, DoLoops[i].p.size, UBYTE*);
02131          ofs = DoLoops[i].p.buffer - org;
02132          if ( DoLoops[i].name ) DoLoops[i].name += ofs;
02133          if ( DoLoops[i].vars ) DoLoops[i].vars += ofs;
02134          if ( DoLoops[i].contents ) DoLoops[i].contents += ofs;
02135          if ( DoLoops[i].type == ONEEXPRESSION ) {
02136              R_COPY_S(DoLoops[i].dollarname, UBYTE*);
02137          }
02138      }
02139      AP.LoopList.message = "doloop";
02140
02141      R_COPY_LIST(AP.ProcList);
02142      for ( i=0; i<AP.ProcList.num; ++i ) {
02143          if ( Procedures[i].p.size ) {
02144              if ( Procedures[i].loadmode != 1 ) {
02145                  R_COPY_B(Procedures[i].p.buffer, Procedures[i].p.size, UBYTE*);
02146              }
02147              else {
02148                  R_SET(j, int);
02149                  Procedures[i].p.buffer = Procedures[j].p.buffer;
02150              }
02151          }
02152          R_COPY_S(Procedures[i].name, UBYTE*);
02153      }
02154      AP.ProcList.message = "procedure";
02155
02156      size = (AP.NumPreSwitchStrings+1)*(LONG)sizeof(UBYTE*);
02157      R_COPY_B(AP.PreSwitchStrings, size, UBYTE*);
02158      for ( i=1; i<=AP.PreSwitchLevel; ++i ) {
02159          R_COPY_S(AP.PreSwitchStrings[i], UBYTE*);
02160      }
02161
02162      org = AP.preStart;
02163      R_COPY_B(AP.preStart, AP.pSize, UBYTE*);
02164      ofs = AP.preStart - org;
02165      if ( AP.preFill ) AP.preFill += ofs;
02166      if ( AP.preStop ) AP.preStop += ofs;
02167
02168      R_COPY_S(AP.procedureExtension, UBYTE*);
02169      R_COPY_S(AP.cprocedureExtension, UBYTE*);
02170

```

```

02171     R_COPY_B(AP.PreAssignStack, AP.MaxPreAssignLevel*(LONG)sizeof(LONG), LONG*);
02172     R_COPY_B(AP.PreIfStack, AP.MaxPreIfLevel*(LONG)sizeof(int), int*);
02173     R_COPY_B(AP.PreSwitchModes, (AP.NumPreSwitchStrings+1)*(LONG)sizeof(int), int*);
02174     R_COPY_B(AP.PreTypes, (AP.MaxPreTypes+1)*(LONG)sizeof(int), int*);
02175
02176 #ifdef PRINTDEBUG
02177     print_P();
02178 #endif
02179
02180     /*#] AP : */
02181     /*#[ AR : */
02182
02183     R_SET(ofs, LONG);
02184     if ( ofs ) {
02185         AR.infile = AR.Fscr+1;
02186         AR.outfile = AR.Fscr;
02187         AR.hidefile = AR.Fscr+2;
02188     }
02189     else {
02190         AR.infile = AR.Fscr;
02191         AR.outfile = AR.Fscr+1;
02192         AR.hidefile = AR.Fscr+2;
02193     }
02194
02195     /* #[ AR free pointers */
02196
02197     /* Parts of AR will be overwritten by data from the recovery file, therefore
02198      * dynamically allocated memory must be freed first. */
02199
02200     namebufout = AR.outfile->name;
02201     namebufhide = AR.hidefile->name;
02202     R_FREE(AR.outfile->PObuffer);
02203 #ifdef WITHZLIB
02204     R_FREE(AR.outfile->zsp);
02205     R_FREE(AR.outfile->ziobuffer);
02206 #endif
02207     namebufhide = AR.hidefile->name;
02208     R_FREE(AR.hidefile->PObuffer);
02209 #ifdef WITHZLIB
02210     R_FREE(AR.hidefile->zsp);
02211     R_FREE(AR.hidefile->ziobuffer);
02212 #endif
02213     /* no files should be opened -> nothing to do with handle */
02214
02215     /* #] AR free pointers */
02216
02217     /* outfile */
02218     R_SET(*AR.outfile, FILEHANDLE);
02219     org = (UBYTE*)AR.outfile->PObuffer;
02220     size = AR.outfile->POfull - AR.outfile->PObuffer;
02221     AR.outfile->PObuffer = (WORD*)Malloc1(AR.outfile->POsize, "PObuffer");
02222     if ( size ) {
02223         memcpy(AR.outfile->PObuffer, p, size*sizeof(WORD));
02224         p = (unsigned char*)p + size*sizeof(WORD);
02225     }
02226     ofs = (UBYTE*)AR.outfile->PObuffer - org;
02227     AR.outfile->PPostop = (WORD*)((UBYTE*)AR.outfile->PPostop + ofs);
02228     AR.outfile->POfill = (WORD*)((UBYTE*)AR.outfile->POfill + ofs);
02229     AR.outfile->POfull = (WORD*)((UBYTE*)AR.outfile->POfull + ofs);
02230     AR.outfile->name = namebufout;
02231 #ifdef WITHPTHREADS
02232     AR.outfile->wPObuffer = AR.outfile->PObuffer;
02233     AR.outfile->wPPostop = AR.outfile->PPostop;
02234     AR.outfile->wPOfill = AR.outfile->POfill;
02235     AR.outfile->wPOfull = AR.outfile->POfull;
02236 #endif
02237 #ifdef WITHZLIB
02238     /* zsp and ziobuffer will be allocated when used */
02239     AR.outfile->zsp = 0;
02240     AR.outfile->ziobuffer = 0;
02241 #endif
02242     /* reopen old outfile */
02243 #ifdef WITHMPI
02244     if (PF.me==MASTER)
02245 #endif
02246     if ( AR.outfile->handle >= 0 ) {
02247         if ( CopyFile(sortfile, AR.outfile->name) ) {
02248             MesPrint("ERROR: Could not copy old output sort file %s!", sortfile);
02249             Terminate(-1);
02250         }
02251         AR.outfile->handle = ReOpenFile(AR.outfile->name);
02252         if ( AR.outfile->handle == -1 ) {
02253             MesPrint("ERROR: Could not reopen output sort file %s!", AR.outfile->name);
02254             Terminate(-1);
02255         }
02256         SeekFile(AR.outfile->handle, &AR.outfile->POposition, SEEK_SET);
02257     }

```

```

02258
02259 /* hidefile */
02260 R_SET(*AR.hidefile, FILEHANDLE);
02261 AR.hidefile->name = namebufhide;
02262 if ( AR.hidefile->PObuffer ) {
02263     org = (UBYTE*)AR.hidefile->PObuffer;
02264     size = AR.hidefile->POfull - AR.hidefile->PObuffer;
02265     AR.hidefile->PObuffer = (WORD*)Malloc1(AR.hidefile->POsize, "PObuffer");
02266     if ( size ) {
02267         memcpy(AR.hidefile->PObuffer, p, size*sizeof(WORD));
02268         p = (unsigned char*)p + size*sizeof(WORD);
02269     }
02270     ofs = (UBYTE*)AR.hidefile->PObuffer - org;
02271     AR.hidefile->POstop = (WORD*)((UBYTE*)AR.hidefile->POstop + ofs);
02272     AR.hidefile->POfill = (WORD*)((UBYTE*)AR.hidefile->POfill + ofs);
02273     AR.hidefile->POfull = (WORD*)((UBYTE*)AR.hidefile->POfull + ofs);
02274 #ifdef WITHPTHREADS
02275     AR.hidefile->wPObuffer = AR.hidefile->PObuffer;
02276     AR.hidefile->wPOstop = AR.hidefile->POstop;
02277     AR.hidefile->wPOfill = AR.hidefile->POfill;
02278     AR.hidefile->wPOfull = AR.hidefile->POfull;
02279 #endif
02280 }
02281 #ifdef WITHZLIB
02282 /* zsp and ziobuffer will be allocated when used */
02283 AR.hidefile->zsp = 0;
02284 AR.hidefile->ziobuffer = 0;
02285 #endif
02286 /* reopen old hidefile */
02287 if ( AR.hidefile->handle >= 0 ) {
02288     if ( CopyFile(hidefile, AR.hidefile->name) ) {
02289         MesPrint("ERROR: Could not copy old hide file %s!",hidefile);
02290         Terminate(-1);
02291     }
02292     AR.hidefile->handle = ReOpenFile(AR.hidefile->name);
02293     if ( AR.hidefile->handle == -1 ) {
02294         MesPrint("ERROR: Could not reopen hide file %s!",AR.hidefile->name);
02295         Terminate(-1);
02296     }
02297     SeekFile(AR.hidefile->handle, &AR.hidefile->POposition, SEEK_SET);
02298 }
02299
02300 /* store file */
02301 R_SET(pos, POSITION);
02302 if ( ISNOTZEROPOS(pos) ) {
02303     CloseFile(AR.StoreData.Handle);
02304     R_SET(AR.StoreData, FILEDATA);
02305     if ( CopyFile(storefile, FG.fname) ) {
02306         MesPrint("ERROR: Could not copy old store file %s!",storefile);
02307         Terminate(-1);
02308     }
02309     AR.StoreData.Handle = (WORD)ReOpenFile(FG.fname);
02310     SeekFile(AR.StoreData.Handle, &AR.StoreData.Position, SEEK_SET);
02311 }
02312
02313 R_SET(AR.DefPosition, POSITION);
02314 R_SET(AR.OldTime, LONG);
02315 R_SET(AR.InInBuf, LONG);
02316 R_SET(AR.InHiBuf, LONG);
02317
02318 R_SET(AR.NoCompress, int);
02319 R_SET(AR.gzipCompress, int);
02320
02321 R_SET(AR.outtohide, int);
02322
02323 R_SET(AR.GetFile, WORD);
02324 R_SET(AR.KeptInHold, WORD);
02325 R_SET(AR.BracketOn, WORD);
02326 R_SET(AR.MaxBracket, WORD);
02327 R_SET(AR.CurDum, WORD);
02328 R_SET(AR.DeferFlag, WORD);
02329 R_SET(AR.TePos, WORD);
02330 R_SET(AR.sLevel, WORD);
02331 R_SET(AR.Stage4Name, WORD);
02332 R_SET(AR.GetOneFile, WORD);
02333 R_SET(AR.PolyFun, WORD);
02334 R_SET(AR.PolyFunInv, WORD);
02335 R_SET(AR.PolyFunType, WORD);
02336 R_SET(AR.PolyFunExp, WORD);
02337 R_SET(AR.PolyFunVar, WORD);
02338 R_SET(AR.PolyFunPow, WORD);
02339 R_SET(AR.Eside, WORD);
02340 R_SET(AR.MaxDum, WORD);
02341 R_SET(AR.level, WORD);
02342 R_SET(AR.expchanged, WORD);
02343 R_SET(AR.expflags, WORD);
02344 R_SET(AR.CurExpr, WORD);

```

```

02345     R_SET(AR.SortType, WORD);
02346     R_SET(AR.ShortSortCount, WORD);
02347
02348     /* this is usually done in Process(), but sometimes FORM does not
02349        end up executing Process() before it uses the AR.CompressPointer,
02350        so we need to explicitly set it here. */
02351     AR.CompressPointer = AR.CompressBuffer;
02352
02353 #ifdef WITHPTHREADS
02354     for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
02355         R_SET(AB[j]->R.wranfnpair1, int);
02356         R_SET(AB[j]->R.wranfnpair2, int);
02357         R_SET(AB[j]->R.wranfcall, int);
02358         R_SET(AB[j]->R.wranfseed, ULONG);
02359         R_SET(AB[j]->R.wranfia, ULONG*);
02360         if ( AB[j]->R.wranfia ) {
02361             R_COPY_B(AB[j]->R.wranfia, sizeof(ULONG)*AB[j]->R.wranfnpair2, ULONG*);
02362         }
02363     }
02364 #else
02365     R_SET(AR.wranfnpair1, int);
02366     R_SET(AR.wranfnpair2, int);
02367     R_SET(AR.wranfcall, int);
02368     R_SET(AR.wranfseed, ULONG);
02369     R_SET(AR.wranfia, ULONG*);
02370     if ( AR.wranfia ) {
02371         R_COPY_B(AR.wranfia, sizeof(ULONG)*AR.wranfnpair2, ULONG*);
02372     }
02373 #endif
02374
02375 #ifdef PRINTDEBUG
02376     print_R();
02377 #endif
02378
02379     /*#] AR : */
02380     /*#[ AO :*/
02381     /*
02382     We copy all non-pointer variables.
02383     */
02384     l = sizeof(A.O) - ((UBYTE *)(&(A.O.NumInBrack))-(UBYTE *)(&A.O));
02385     memcpy(&(A.O.NumInBrack), p, l); p = (unsigned char*)p + l;
02386     /*
02387     Now the variables in OptimizeResult
02388     */
02389     memcpy(&(A.O.OptimizeResult), p, sizeof(OPTIMIZERESULT));
02390     p = (unsigned char*)p + sizeof(OPTIMIZERESULT);
02391
02392     if ( A.O.OptimizeResult.codesize > 0 ) {
02393         R_COPY_B(A.O.OptimizeResult.code, A.O.OptimizeResult.codesize*sizeof(WORD), WORD *);
02394     }
02395     R_COPY_S(A.O.OptimizeResult.nameofexpr, UBYTE *);
02396     /*
02397     And now the dictionaries. We know how many there are. We also know
02398     how many elements the array AO.Dictionaries should have.
02399     */
02400     if ( AO.SizeDictionaries > 0 ) {
02401         AO.Dictionaries = (DICTIONARY **)Malloc1(AO.SizeDictionaries*sizeof(DICTIONARY *),
02402            "Dictionaries");
02403         for ( i = 0; i < AO.NumDictionaries; i++ ) {
02404             R_SET(i, LONG)
02405             AO.Dictionaries[i] = DictFromBytes(p);
02406             p = (char *)p + l;
02407         }
02408     }
02409     /*#] AO :*/
02410 #ifdef WITHMPI
02411     /*#[ PF : */
02412     /*Block*/
02413     int numtasks;
02414     R_SET(numtasks, int);
02415     if(numtasks!=PF.numtasks){
02416         MesPrint("%d number of tasks expected instead of %d; use mpirun -np %d",
02417             numtasks, PF.numtasks, numtasks);
02418         if(PF.me!=MASTER)
02419             remove(RecoveryFilename());
02420         Terminate(-1);
02421     }
02422     /*Block*/
02423     R_SET(PF.rhsInParallel, int);
02424     R_SET(PF.exprbufsize, int);
02425     R_SET(PF.log, int);
02426     /*#] PF : */
02427 #endif
02428
02429 #ifdef WITHPTHREADS
02430     /* read timing information of individual threads */
02431     R_SET(i, int);

```

```

02432     for ( j=1; j<AM.totalnumberofthreads; ++j ) {
02433         /* ... and correcting OldTime */
02434         AB[j]->R.OldTime = -(*(LONG*)p+j));
02435     }
02436     WriteTimerInfo((LONG*)p, (LONG *) ((unsigned char*)p + i*(LONG)sizeof(LONG)));
02437     p = (unsigned char*)p + 2*i*(LONG)sizeof(LONG);
02438 #endif /* ifdef WITHPTHREADS */
02439
02440     if ( fclose(fd) ) return(__LINE__);
02441
02442     M_free(buf, "recovery buffer");
02443
02444     /* cares about data in S_const */
02445     UpdatePositions();
02446     AT.SS = AT.S0;
02447 /*
02448     Set the checkpoint parameter right for the next checkpoint.
02449 */
02450     AC.CheckpointStamp = TimeWallClock(1);
02451
02452     done_snapshot = 1;
02453     MesPrint("done."); fflush(0);
02454
02455     return(0);
02456 }
02457
02458 /*
02459     #] DoRecovery :
02460     #[ DoSnapshot :
02461 */
02462
02463 static int DoSnapshot(int moduletype)
02464 {
02465     GETIDENTITY
02466     FILE *fd;
02467     POSITION pos;
02468     int i, j;
02469     LONG l;
02470     WORD *w;
02471     void *adr;
02472 #ifdef WITHPTHREADS
02473     LONG *longp, *longpp;
02474 #endif /* ifdef WITHPTHREADS */
02475
02476     MesPrint("Saving recovery point ... %"); fflush(0);
02477 #ifdef PRINTTIMEMARKS
02478     MesPrint("\n");
02479 #endif
02480
02481     if ( !(fd = fopen(intermedfile, "wb")) ) return(__LINE__);
02482
02483     /* reserve space in the file for a length field */
02484     if ( fwrite(&pos, 1, sizeof(POSITION), fd) != sizeof(POSITION) ) return(__LINE__);
02485
02486     /* write moduletype */
02487     if ( fwrite(&moduletype, 1, sizeof(int), fd) != sizeof(int) ) return(__LINE__);
02488
02489     /*#[ AM :*/
02490
02491     /* since most values don't change during execution, AM doesn't need to be
02492      * written as a whole. all values will be correctly set when starting up
02493      * anyway. only the exceptions need to be taken care of. see MakeGlobal()
02494      * and PopVariables() in execute.c. */
02495
02496     ANNOUNCE(AM)
02497     S_WRITE_B(&AM.hparallelflag, sizeof(int));
02498     S_WRITE_B(&AM.gparallelflag, sizeof(int));
02499     S_WRITE_B(&AM.gCodesFlag, sizeof(int));
02500     S_WRITE_B(&AM.gNamesFlag, sizeof(int));
02501     S_WRITE_B(&AM.gStatsFlag, sizeof(int));
02502     S_WRITE_B(&AM.gTokensWriteFlag, sizeof(int));
02503     S_WRITE_B(&AM.gNoSpacesInNumbers, sizeof(int));
02504     S_WRITE_B(&AM.gIndentSpace, sizeof(WORD));
02505     S_WRITE_B(&AM.gUnitTrace, sizeof(WORD));
02506     S_WRITE_B(&AM.gDefDim, sizeof(int));
02507     S_WRITE_B(&AM.gDefDim4, sizeof(int));
02508     S_WRITE_B(&AM.gncmod, sizeof(WORD));
02509     S_WRITE_B(&AM.gnpowmod, sizeof(WORD));
02510     S_WRITE_B(&AM.gmodmode, sizeof(WORD));
02511     S_WRITE_B(&AM.gOutputMode, sizeof(WORD));
02512     S_WRITE_B(&AM.gCnumpows, sizeof(WORD));
02513     S_WRITE_B(&AM.gOutputSpaces, sizeof(WORD));
02514     S_WRITE_B(&AM.gOutNumberType, sizeof(WORD));
02515     S_WRITE_B(&AM.gfunpowers, sizeof(int));
02516     S_WRITE_B(&AM.gPolyFun, sizeof(WORD));
02517     S_WRITE_B(&AM.gPolyFunInv, sizeof(WORD));
02518     S_WRITE_B(&AM.gPolyFunType, sizeof(WORD));

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02534     S_WRITE_B(&AM.gPolyFunExp, sizeof(WORD));
02535     S_WRITE_B(&AM.gPolyFunVar, sizeof(WORD));
02536     S_WRITE_B(&AM.gPolyFunPow, sizeof(WORD));
02537     S_WRITE_B(&AM.gProcessBucketSize, sizeof(LONG));
02538     S_WRITE_B(&AM.OldChildTime, sizeof(LONG));
02539     S_WRITE_B(&AM.OldSecTime, sizeof(LONG));
02540     S_WRITE_B(&AM.OldMilliTime, sizeof(LONG));
02541     S_WRITE_B(&AM.gproperorderflag, sizeof(int));
02542     S_WRITE_B(&AM.gThreadBucketSize, sizeof(LONG));
02543     S_WRITE_B(&AM.gSizeCommuteInSet, sizeof(int));
02544     S_WRITE_B(&AM.gThreadStats, sizeof(int));
02545     S_WRITE_B(&AM.gFinalStats, sizeof(int));
02546     S_WRITE_B(&AM.gThreadsFlag, sizeof(int));
02547     S_WRITE_B(&AM.gThreadBalancing, sizeof(int));
02548     S_WRITE_B(&AM.gThreadSortFileSynch, sizeof(int));
02549     S_WRITE_B(&AM.gProcessStats, sizeof(int));
02550     S_WRITE_B(&AM.gOldParallelStats, sizeof(int));
02551     S_WRITE_B(&AM.gSortType, sizeof(int));
02552     S_WRITE_B(&AM.gShortStatsMax, sizeof(WORD));
02553     S_WRITE_B(&AM.gIsFortran90, sizeof(int));
02554     adr = &AM.dollarzero;
02555     S_WRITE_B(adr, sizeof(void*));
02556     S_WRITE_B(&AM.gFortran90Kind, sizeof(UBYTE *));
02557     S_WRITE_S(AM.gFortran90Kind);
02558
02559     S_WRITE_S(AM.gextrasymp);
02560     S_WRITE_S(AM.ggextrasymp);
02561
02562     S_WRITE_B(&AM.PrintTotalSize, sizeof(int));
02563     S_WRITE_B(&AM.fbuffersize, sizeof(int));
02564     S_WRITE_B(&AM.gOldFactArgFlag, sizeof(int));
02565     S_WRITE_B(&AM.ggOldFactArgFlag, sizeof(int));
02566
02567     S_WRITE_B(&AM.gnumextrasymp, sizeof(int));
02568     S_WRITE_B(&AM.ggnumextrasymp, sizeof(int));
02569     S_WRITE_B(&AM.NumSpectatorFiles, sizeof(int));
02570     S_WRITE_B(&AM.SizeForSpectatorFiles, sizeof(int));
02571     S_WRITE_B(&AM.gOldGCDflag, sizeof(int));
02572     S_WRITE_B(&AM.ggOldGCDflag, sizeof(int));
02573     S_WRITE_B(&AM.gWTimeStatsFlag, sizeof(int));
02574
02575     S_WRITE_B(&AM.Path, sizeof(UBYTE *));
02576     S_WRITE_S(AM.Path);
02577
02578     S_WRITE_B(&AM.FromStdin, sizeof(BOOL));
02579     S_WRITE_B(&AM.IgnoreDeprecation, sizeof(BOOL));
02580
02581     /*#] AM :*/
02582     /*#[ AC :*/
02583
02584     /* we write AC as a whole and then write all additional data step by step.
02585      * AC.DubiousList doesn't need to be treated, because it should be empty. */
02586
02587     ANNOUNCE(AC)
02588     S_WRITE_B(&AC, sizeof(struct C_const));
02589
02590     S_WRITE_NAMETREE(AC.dollarnames);
02591     S_WRITE_NAMETREE(AC.exprnames);
02592     S_WRITE_NAMETREE(AC.varnames);
02593
02594     S_WRITE_LIST(AC.ChannelList);
02595     for ( i=0; i<AC.ChannelList.num; ++i ) {
02596         S_WRITE_S(channels[i].name);
02597     }
02598
02599     ANNOUNCE(AC.FunctionList)
02600     S_WRITE_LIST(AC.FunctionList);
02601     for ( i=0; i<AC.FunctionList.num; ++i ) {
02602         /* if the function is a table */
02603         if ( functions[i].tabl ) {
02604             TABLES tabl = functions[i].tabl;
02605             S_WRITE_B(tabl, sizeof(struct TABEs));
02606             if ( tabl->tablepointers ) {
02607                 if ( tabl->sparse ) {
02608                     /* sparse tables. reserved holds number of allocated
02609                      * elements. the size of an element is numind plus
02610                      * TABLEEXTENSION times the size of WORD. */
02611                     S_WRITE_B(tabl->tablepointers,
02612                               tabl->reserved*sizeof(WORD)*(tabl->numind+TABLEEXTENSION));
02613                 }
02614                 else {
02615                     /* matrix like tables. */
02616                     S_WRITE_B(tabl->tablepointers,
02617                               TABLEEXTENSION*sizeof(WORD)*(tabl->totind));
02618                 }
02619             }
02620             S_WRITE_B(tabl->prototype, tabl->prototypeSize);

```

```

02621     S_WRITE_B(tabl->mm, tabl->numind*(LONG)sizeof(MINMAX));
02622     S_WRITE_B(tabl->flags, tabl->numind*(LONG)sizeof(WORD));
02623     if ( tabl->sparse ) {
02624         S_WRITE_B(tabl->boomlijst, tabl->MaxTreeSize*(LONG)sizeof(COMPTREE));
02625         S_WRITE_S(tabl->argtail);
02626     }
02627     S_WRITE_B(tabl->buffers, tabl->bufferssize*(LONG)sizeof(WORD));
02628     if ( tabl->sparse ) {
02629         TABLES spare = tabl->spare;
02630         S_WRITE_B(spare, sizeof(struct TaBLeS));
02631         if ( spare->tablepointers ) {
02632             if ( spare->sparse ) {
02633                 /* sparse tables */
02634                 S_WRITE_B(spare->tablepointers,
02635                     spare->reserved*sizeof(WORD)*(spare->numind+TABLEEXTENSION));
02636             }
02637             else {
02638                 /* matrix like tables */
02639                 S_WRITE_B(spare->tablepointers,
02640                     TABLEEXTENSION*sizeof(WORD)*(spare->totind));
02641             }
02642         }
02643         S_WRITE_B(spare->mm, spare->numind*(LONG)sizeof(MINMAX));
02644         S_WRITE_B(spare->flags, spare->numind*(LONG)sizeof(WORD));
02645         if ( spare->sparse ) {
02646             S_WRITE_B(spare->boomlijst, spare->MaxTreeSize*(LONG)sizeof(COMPTREE));
02647         }
02648         S_WRITE_B(spare->buffers, spare->bufferssize*(LONG)sizeof(WORD));
02649     }
02650 }
02651 }
02652
02653 ANNOUNCE(AC.ExpressionList)
02654 S_WRITE_LIST(AC.ExpressionList);
02655 for ( i=0; i<AC.ExpressionList.num; ++i ) {
02656     EXPRESSIONS ex = Expressions + i;
02657     if ( ex->renum ) {
02658         S_WRITE_B(ex->renum, sizeof(struct ReNuMbEr));
02659         /* there is one dynamically allocated buffer for struct ReNuMbEr and
02660          * symb.lo points to its beginning. the size of the buffer is not
02661          * stored anywhere but we know it is 2*sizeof(WORD)*N, where N is
02662          * the number of all vectors, indices, functions and symbols. since
02663          * funum points into the buffer at a distance 2N-[Number of
02664          * functions] from symb.lo (see GetTable() in store.c), we can
02665          * calculate the buffer size by some pointer arithmetic. the size is
02666          * then written to the file. */
02667         l = ex->renum->funnum - ex->renum->symb.lo;
02668         l += ex->renum->funnum - ex->renum->func.lo;
02669         S_WRITE_B(&l, sizeof(size_t));
02670         S_WRITE_B(ex->renum->symb.lo, l);
02671     }
02672     if ( ex->bracketinfo ) {
02673         S_WRITE_B(ex->bracketinfo, sizeof(BRACKETINFO));
02674         S_WRITE_B(ex->bracketinfo->indexbuffer,
02675             ex->bracketinfo->indexbuffersize*sizeof(BRACKETINDEX));
02676         S_WRITE_B(ex->bracketinfo->bracketbuffer,
02677             ex->bracketinfo->bracketbuffersize*sizeof(WORD));
02678     }
02679     if ( ex->newbracketinfo ) {
02680         S_WRITE_B(ex->newbracketinfo, sizeof(BRACKETINFO));
02681         S_WRITE_B(ex->newbracketinfo->indexbuffer,
02682             ex->newbracketinfo->indexbuffersize*sizeof(BRACKETINDEX));
02683         S_WRITE_B(ex->newbracketinfo->bracketbuffer,
02684             ex->newbracketinfo->bracketbuffersize*sizeof(WORD));
02685     }
02686     /* don't need to write ex->renumlists */
02687     if ( ex->inmem ) {
02688         /* size of the inmem buffer has to be determined. we use the fact
02689          * that the end of an expression is marked by a zero. */
02690         w = ex->inmem;
02691         while ( *w++ );
02692         l = w - ex->inmem;
02693         S_WRITE_B(&l, sizeof(size_t));
02694         S_WRITE_B(ex->inmem, l);
02695     }
02696 }
02697
02698 ANNOUNCE(AC.IndexList)
02699 S_WRITE_LIST(AC.IndexList);
02700 S_WRITE_LIST(AC.SetElementList);
02701 S_WRITE_LIST(AC.SetList);
02702 S_WRITE_LIST(AC.SymbolList);
02703 S_WRITE_LIST(AC.VectorList);
02704
02705 ANNOUNCE(AC.TableBaseList)
02706 S_WRITE_LIST(AC.TableBaseList);
02707 for ( i=0; i<AC.TableBaseList.num; ++i ) {

```



```

02704      /* see struct dbase in minos.h */
02705      if ( tablebases[i].iblocks ) {
02706          S_WRITE_B(tablebases[i].iblocks, tablebases[i].info.numberofindexblocks *
sizeof(INDEXBLOCK*));
02707          for ( j=0; j<tablebases[i].info.numberofindexblocks; ++j ) {
02708              if ( tablebases[i].iblocks[j] ) {
02709                  S_WRITE_B(tablebases[i].iblocks[j], sizeof(INDEXBLOCK));
02710              }
02711          }
02712      }
02713      if ( tablebases[i].nblocks ) {
02714          S_WRITE_B(tablebases[i].nblocks, tablebases[i].info.numberofnamesblocks *
sizeof(NAMESBLOCK*));
02715          for ( j=0; j<tablebases[i].info.numberofnamesblocks; ++j ) {
02716              if ( tablebases[i].nblocks[j] ) {
02717                  S_WRITE_B(tablebases[i].nblocks[j], sizeof(NAMESBLOCK));
02718              }
02719          }
02720      }
02721      S_WRITE_S(tablebases[i].name);
02722      S_WRITE_S(tablebases[i].fullname);
02723      S_WRITE_S(tablebases[i].tablenames);
02724  }
02725
02726  ANNOUNCE(AC.cbufList)
02727  S_WRITE_LIST(AC.cbufList);
02728  for ( i=0; i<AC.cbufList.num; ++i ) {
02729      S_WRITE_B(cbuf[i].Buffer, cbuf[i].BufferSize*sizeof(WORD));
02730      /* see inicbufs in comtool.c */
02731      S_WRITE_B(cbuf[i].lhs, cbuf[i].maxlhs*(LONG)sizeof(WORD*));
02732      S_WRITE_B(cbuf[i].rhs, cbuf[i].maxrhs*(LONG)(sizeof(WORD*)+2*sizeof(LONG)+2*sizeof(WORD)));
02733      if ( cbuf[i].boomlijst ) {
02734          S_WRITE_B(cbuf[i].boomlijst, cbuf[i].MaxTreeSize*(LONG)sizeof(COMPTREE));
02735      }
02736  }
02737
02738  S_WRITE_LIST(AC.AutoSymbolList);
02739  S_WRITE_LIST(AC.AutoIndexList);
02740  S_WRITE_LIST(AC.AutoVectorList);
02741  S_WRITE_LIST(AC.AutoFunctionList);
02742
02743  S_WRITE_NAMETREE(AC.autonames);
02744
02745  ANNOUNCE(AC.Streams)
02746  S_WRITE_B(AC.Streams, AC.MaxNumStreams*(LONG)sizeof(STREAM));
02747  for ( i=0; i<AC.NumStreams; ++i ) {
02748      if ( AC.Streams[i].inbuffer ) {
02749          S_WRITE_B(AC.Streams[i].buffer, AC.Streams[i].inbuffer);
02750      }
02751      S_WRITE_S(AC.Streams[i].FoldName);
02752      S_WRITE_S(AC.Streams[i].name);
02753  }
02754
02755  if ( AC.termstack ) {
02756      S_WRITE_B(AC.termstack, AC.maxtermlevel*(LONG)sizeof(LONG));
02757  }
02758
02759  if ( AC.termstortstack ) {
02760      S_WRITE_B(AC.termstortstack, AC.maxtermlevel*(LONG)sizeof(LONG));
02761  }
02762
02763  S_WRITE_B(AC.cmod, AM.MaxTal*4*(LONG)sizeof(UWORD));
02764
02765  if ( AC.IfHeap ) {
02766      S_WRITE_B(AC.IfHeap, (LONG)sizeof(LONG)*(AC.MaxIf+1));
02767      S_WRITE_B(AC.IfCount, (LONG)sizeof(LONG)*(AC.MaxIf+1));
02768  }
02769
02770  l = AC.iStop - AC.iBuffer + 2;
02771  S_WRITE_B(AC.iBuffer, l);
02772
02773  if ( AC.LabelNames ) {
02774      S_WRITE_B(AC.LabelNames, AC.MaxLabels*(LONG)(sizeof(UBYTE*)+sizeof(WORD)));
02775      for ( i=0; i<AC.NumLabels; ++i ) {
02776          S_WRITE_S(AC.LabelNames[i]);
02777      }
02778  }
02779
02780  S_WRITE_B(AC.FixIndices, AM.OffsetIndex*(LONG)sizeof(WORD));
02781
02782  if ( AC.termsumcheck ) {
02783      S_WRITE_B(AC.termsumcheck, AC.maxtermlevel*(LONG)sizeof(WORD));
02784  }
02785
02786  S_WRITE_B(AC.WildcardNames, AC.WildcardBufferSize);
02787
02788  if ( AC.tokens ) {

```

```

02789         l = AC.toptokens - AC.tokens;
02790         if ( l ) {
02791             S_WRITE_B(AC.tokens, l);
02792         }
02793     }
02794
02795     S_WRITE_B(AC.tokenarglevel, AM.MaxParLevel*(LONG)sizeof(WORD));
02796
02797     S_WRITE_S(AC.Fortran90Kind);
02798
02799 #ifdef WITHPTHREADS
02800     if ( AC.inputnumbers ) {
02801         S_WRITE_B(AC.inputnumbers, AC.sizepfirstnum*(LONG) (sizeof(WORD)+sizeof(LONG)));
02802     }
02803 #endif /* ifdef WITHPTHREADS */
02804
02805     if ( AC.IfSumCheck ) {
02806         S_WRITE_B(AC.IfSumCheck, (LONG)sizeof(WORD)*(AC.MaxIf+1));
02807     }
02808     if ( AC.CommuteInSet ) {
02809         S_WRITE_B(AC.CommuteInSet, (LONG)sizeof(WORD)*(AC.SizeCommuteInSet+1));
02810     }
02811
02812     S_WRITE_S(AC.CheckpointRunAfter);
02813     S_WRITE_S(AC.CheckpointRunBefore);
02814
02815     S_WRITE_S(AC.extrasym);
02816
02817     /*# AC :*/
02818     /*# AP :*/
02819
02820     /* we write AP as a whole and then write all additional data step by step. */
02821
02822     ANNOUNCE(AP)
02823     S_WRITE_B(&AP, sizeof(struct P_const));
02824
02825     S_WRITE_LIST(AP.DollarList);
02826     for ( i=0; i<AP.DollarList.num; ++i ) {
02827         DOLLARS d = Dollars + i;
02828         S_WRITE_DOLLAR(Dollars[i]);
02829         if ( d->nfactors > 1 ) {
02830             S_WRITE_B(&(d->factors),sizeof(FACDOLLAR)*d->nfactors);
02831             for ( j = 0; j < d->nfactors; j++ ) {
02832                 if ( d->factors[j].size > 0 ) {
02833                     S_WRITE_B(&(d->factors[i].where),sizeof(WORD)*(d->factors[j].size+1));
02834                 }
02835             }
02836         }
02837     }
02838
02839     S_WRITE_LIST(AP.PreVarList);
02840     for ( i=0; i<AP.PreVarList.num; ++i ) {
02841         /* there is one dynamically allocated buffer in struct pReVar holding
02842          * the strings name, value and several argnames. the size of the buffer
02843          * can be calculated by adding up their string lengths. */
02844         l = strlen((char*)PreVar[i].name) + 1;
02845         if ( PreVar[i].value ) {
02846             l += strlen((char*)(PreVar[i].name+1)) + 1;
02847         }
02848         for ( j=0; j<PreVar[i].nargs; ++j ) {
02849             l += strlen((char*)(PreVar[i].name+1)) + 1;
02850         }
02851         S_WRITE_B(&l, sizeof(size_t));
02852         S_WRITE_B(PreVar[i].name, l);
02853     }
02854
02855     ANNOUNCE(AP.LoopList)
02856     S_WRITE_LIST(AP.LoopList);
02857     for ( i=0; i<AP.LoopList.num; ++i ) {
02858         S_WRITE_B(DoLoops[i].p.buffer, DoLoops[i].p.size);
02859         if ( DoLoops[i].type == ONEEXPRESSION ) {
02860             /* do loops with an expression keep this expression in a dynamically
02861              * allocated buffer in dollarname */
02862             S_WRITE_S(DoLoops[i].dollarname);
02863         }
02864     }
02865
02866     S_WRITE_LIST(AP.ProcList);
02867     for ( i=0; i<AP.ProcList.num; ++i ) {
02868         if ( Procedures[i].p.size ) {
02869             if ( Procedures[i].loadmode != 1 ) {
02870                 S_WRITE_B(Procedures[i].p.buffer, Procedures[i].p.size);
02871             }
02872             else {
02873                 for ( j=0; j<AP.ProcList.num; ++j ) {
02874                     if ( Procedures[i].p.buffer == Procedures[j].p.buffer ) {
02875                         break;

```

```

02876         }
02877     }
02878     if ( j == AP.ProcList.num ) {
02879         MesPrint("Error writing procedures to recovery file!");
02880     }
02881     S_WRITE_B(&j, sizeof(int));
02882 }
02883 }
02884 S_WRITE_S(Procedures[i].name);
02885 }
02886
02887 S_WRITE_B(AP.PreSwitchStrings, (AP.NumPreSwitchStrings+1)*(LONG)sizeof(UBYTE*));
02888 for ( i=1; i<=AP.PreSwitchLevel; ++i ) {
02889     S_WRITE_S(AP.PreSwitchStrings[i]);
02890 }
02891
02892 S_WRITE_B(AP.preStart, AP.pSize);
02893
02894 S_WRITE_S(AP.procedureExtension);
02895 S_WRITE_S(AP.cprocedureExtension);
02896
02897 S_WRITE_B(AP.PreAssignStack, AP.MaxPreAssignLevel*(LONG)sizeof(LONG));
02898 S_WRITE_B(AP.PreIfStack, AP.MaxPreIfLevel*(LONG)sizeof(int));
02899 S_WRITE_B(AP.PreSwitchModes, (AP.NumPreSwitchStrings+1)*(LONG)sizeof(int));
02900 S_WRITE_B(AP.PreTypes, (AP.MaxPreTypes+1)*(LONG)sizeof(int));
02901
02902 /*#] AP :*/
02903 /*#[ AR :*/
02904
02905 ANNOUNCE(AR)
02906 /* to remember which entry in AR.Fscr corresponds to the infile */
02907 l = AR.infile - AR.Fscr;
02908 S_WRITE_B(&l, sizeof(LONG));
02909
02910 /* write the FILEHANDLES */
02911 S_WRITE_B(AR.outfile, sizeof(FILEHANDLE));
02912 l = AR.outfile->POfull - AR.outfile->PObuffer;
02913 if ( l ) {
02914     S_WRITE_B(AR.outfile->PObuffer, l*sizeof(WORD));
02915 }
02916 S_WRITE_B(AR.hidefile, sizeof(FILEHANDLE));
02917 l = AR.hidefile->POfull - AR.hidefile->PObuffer;
02918 if ( l ) {
02919     S_WRITE_B(AR.hidefile->PObuffer, l*sizeof(WORD));
02920 }
02921
02922 S_WRITE_B(&AR.StoreData.Fill, sizeof(POSITION));
02923 if ( ISNOTZEROPOS(AR.StoreData.Fill) ) {
02924     S_WRITE_B(&AR.StoreData, sizeof(FILEDATA));
02925 }
02926
02927 S_WRITE_B(&AR.DefPosition, sizeof(POSITION));
02928
02929 l = TimeCPU(1); l = -l;
02930 S_WRITE_B(&l, sizeof(LONG));
02931
02932 ANNOUNCE(AR.InInBuf)
02933 S_WRITE_B(&AR.InInBuf, sizeof(LONG));
02934 S_WRITE_B(&AR.InHiBuf, sizeof(LONG));
02935
02936 S_WRITE_B(&AR.NoCompress, sizeof(int));
02937 S_WRITE_B(&AR.zipCompress, sizeof(int));
02938
02939 S_WRITE_B(&AR.outtohide, sizeof(int));
02940
02941 S_WRITE_B(&AR.GetFile, sizeof(WORD));
02942 S_WRITE_B(&AR.KeptInHold, sizeof(WORD));
02943 S_WRITE_B(&AR.BraceOn, sizeof(WORD));
02944 S_WRITE_B(&AR.MaxBracket, sizeof(WORD));
02945 S_WRITE_B(&AR.CurDum, sizeof(WORD));
02946 S_WRITE_B(&AR.DeferFlag, sizeof(WORD));
02947 S_WRITE_B(&AR.TePos, sizeof(WORD));
02948 S_WRITE_B(&AR.sLevel, sizeof(WORD));
02949 S_WRITE_B(&AR.Stage4Name, sizeof(WORD));
02950 S_WRITE_B(&AR.GetOneFile, sizeof(WORD));
02951 S_WRITE_B(&AR.PolyFun, sizeof(WORD));
02952 S_WRITE_B(&AR.PolyFunInv, sizeof(WORD));
02953 S_WRITE_B(&AR.PolyFunType, sizeof(WORD));
02954 S_WRITE_B(&AR.PolyFunExp, sizeof(WORD));
02955 S_WRITE_B(&AR.PolyFunVar, sizeof(WORD));
02956 S_WRITE_B(&AR.PolyFunPow, sizeof(WORD));
02957 S_WRITE_B(&AR.Eside, sizeof(WORD));
02958 S_WRITE_B(&AR.MaxDum, sizeof(WORD));
02959 S_WRITE_B(&AR.level, sizeof(WORD));
02960 S_WRITE_B(&AR.expchanged, sizeof(WORD));
02961 S_WRITE_B(&AR.expflags, sizeof(WORD));
02962 S_WRITE_B(&AR.CurExpr, sizeof(WORD));

```

```

02963     S_WRITE_B(&AR.SortType, sizeof(WORD));
02964     S_WRITE_B(&AR.ShortSortCount, sizeof(WORD));
02965
02966 #ifdef WITHPTHREADS
02967     for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
02968         S_WRITE_B(&(AB[j]->R.wranfnpair1), sizeof(int));
02969         S_WRITE_B(&(AB[j]->R.wranfnpair2), sizeof(int));
02970         S_WRITE_B(&(AB[j]->R.wranfcall), sizeof(int));
02971         S_WRITE_B(&(AB[j]->R.wranfseed), sizeof(ULONG));
02972         S_WRITE_B(&(AB[j]->R.wranfia), sizeof(ULONG *));
02973         if ( AB[j]->R.wranfia ) {
02974             S_WRITE_B(AB[j]->R.wranfia, sizeof(ULONG)*AB[j]->R.wranfnpair2);
02975         }
02976     }
02977 #else
02978     S_WRITE_B(&(AR.wranfnpair1), sizeof(int));
02979     S_WRITE_B(&(AR.wranfnpair2), sizeof(int));
02980     S_WRITE_B(&(AR.wranfcall), sizeof(int));
02981     S_WRITE_B(&(AR.wranfseed), sizeof(ULONG));
02982     S_WRITE_B(&(AR.wranfia), sizeof(ULONG *));
02983     if ( AR.wranfia ) {
02984         S_WRITE_B(AR.wranfia, sizeof(ULONG)*AR.wranfnpair2);
02985     }
02986 #endif
02987
02988     /*#] AR :*/
02989     /*#[ AO :*/
02990 /*
02991     We copy all non-pointer variables.
02992 */
02993     ANNOUNCE(AO)
02994     l = sizeof(A.O) - ((UBYTE *)(&(A.O.NumInBrack)) - (UBYTE *)(&(A.O)));
02995     S_WRITE_B(&(A.O.NumInBrack), l);
02996 /*
02997     Now the variables in OptimizeResult
02998 */
02999     S_WRITE_B(&(A.O.OptimizeResult), sizeof(OPTIMIZERESULT));
03000     if ( A.O.OptimizeResult.codesize > 0 ) {
03001         S_WRITE_B(A.O.OptimizeResult.code, A.O.OptimizeResult.codesize*sizeof(WORD));
03002     }
03003     S_WRITE_S(A.O.OptimizeResult.nameofexpr);
03004 /*
03005     And now the dictionaries.
03006     We write each dictionary to a buffer and get the size of that buffer.
03007     Then we write the size and the buffer.
03008 */
03009     for ( i = 0; i < AO.NumDictionaries; i++ ) {
03010         l = DictToBytes(AO.Dictionaries[i], (UBYTE *) (AT.WorkPointer));
03011         S_WRITE_B(&l, sizeof(LONG));
03012         S_WRITE_B(AT.WorkPointer, l);
03013     }
03014
03015     /*#] AO :*/
03016     /*#[ PF :*/
03017 #ifdef WITHMPI
03018     S_WRITE_B(&PF.numtasks, sizeof(int));
03019     S_WRITE_B(&PF.rhsInParallel, sizeof(int));
03020     S_WRITE_B(&PF.exprbufsize, sizeof(int));
03021     S_WRITE_B(&PF.log, sizeof(int));
03022 #endif
03023     /*#] PF :*/
03024
03025 #ifdef WITHPTHREADS
03026
03027     ANNOUNCE(GetTimerInfo)
03028 /*
03029     write timing information of individual threads
03030 */
03031     i = GetTimerInfo(&longp, &longpp);
03032     S_WRITE_B(&i, sizeof(int));
03033     S_WRITE_B(longp, i*(LONG)sizeof(LONG));
03034     S_WRITE_B(&l, sizeof(int));
03035     S_WRITE_B(longpp, i*(LONG)sizeof(LONG));
03036
03037 #endif
03038
03039     S_FLUSH_B /* because we will call fwrite() directly in the following code */
03040
03041     /* save length of data at the beginning of the file */
03042     ANNOUNCE(file length)
03043     SETBASEPOSITION(pos, (ftell(fd)));
03044     fseek(fd, 0, SEEK_SET);
03045     if ( fwrite(&pos, 1, sizeof(POSITION), fd) != sizeof(POSITION) ) return(__LINE__);
03046     fseek(fd, BASEPOSITION(pos), SEEK_SET);
03047
03048     ANNOUNCE(file close)
03049     if ( fclose(fd) ) return(__LINE__);

```

```

03050 #ifdef WITHMPI
03051     if ( PF.me == MASTER ) {
03052 #endif
03053 /*
03054     copy store file if necessary
03055 */
03056     ANNOUNCE(copy store file)
03057     if ( ISNOTZEROPOS(AR.StoreData.Fill) ) {
03058         if ( CopyFile(FG.fname, storefile) ) return(__LINE__);
03059     }
03060 /*
03061     copy sort file if necessary
03062 */
03063     ANNOUNCE(copy sort file)
03064     if ( AR.outfile->handle >= 0 ) {
03065         if ( CopyFile(AR.outfile->name, sortfile) ) return(__LINE__);
03066     }
03067 /*
03068     copy hide file if necessary
03069 */
03070     ANNOUNCE(copy hide file)
03071     if ( AR.hidefile->handle >= 0 ) {
03072         if ( CopyFile(AR.hidefile->name, hidefile) ) return(__LINE__);
03073     }
03074 #ifdef WITHMPI
03075     }
03076 /*
03077     * For ParFORM, the renaming will be performed after the master got
03078     * all recovery files from the slaves.
03079     */
03080 #else
03081 /*
03082     make the intermediate file the recovery file
03083 */
03084     ANNOUNCE(rename intermediate file)
03085     if ( rename(intermedfile, recoveryfile) ) return(__LINE__);
03086
03087     done_snapshot = 1;
03088
03089     MesPrint("done."); fflush(0);
03090 #endif
03091
03092 #ifdef PRINTDEBUG
03093     print_M();
03094     print_C();
03095     print_P();
03096     print_R();
03097 #endif
03098
03099     return(0);
03100 }
03101
03102 /*
03103     #] DoSnapshot :
03104     #[ DoCheckpoint :
03105 */
03106
03111 void DoCheckpoint(int moduletype)
03112 {
03113     int error;
03114     LONG timestamp = TimeWallClock(1);
03115 #ifdef WITHMPI
03116     if (PF.me == MASTER){
03117 #endif
03118         if ( timestamp - AC.CheckpointStamp >= AC.CheckpointInterval ) {
03119             char argbuf[20];
03120             int retvalue = 0;
03121             if ( AC.CheckpointRunBefore ) {
03122                 size_t l1, l2;
03123                 char *str;
03124                 l1 = strlen(AC.CheckpointRunBefore);
03125                 NumToStr((UBYTE*)argbuf, AC.CModule);
03126                 l2 = strlen(argbuf);
03127                 str = (char*)Malloc1(l1+l2+2, "callbefore");
03128                 strcpy(str, AC.CheckpointRunBefore);
03129                 *(str+l1) = ' ';
03130                 strcpy(str+l1+1, argbuf);
03131                 retvalue = system(str);
03132                 M_free(str, "callbefore");
03133                 if ( retvalue ) {
03134                     MesPrint("Script returned error -> no recovery file will be created.");
03135                 }
03136             }
03137 #ifdef WITHMPI
03138             /* Confirm slaves to make snapshots. */
03139             PF_BroadcastNumber(retvalue == 0);
03140 #endif

```

```

03141         if ( retvalue == 0 ) {
03142             if ( (error = DoSnapshot(moduletype)) ) {
03143                 MesPrint("Error creating recovery files: %d", error);
03144             }
03145 #ifdef WITHMPI
03146         {
03147             int i;
03148             /*get recovery files from slaves:*/
03149             for(i=1; i<PF.numtasks;i++){
03150                 FILE *fd;
03151                 const char *tmpnam = PF_recoveryfile('m', i, 1);
03152                 fd = fopen(tmpnam, "w");
03153                 if(fd == NULL){
03154                     MesPrint("Error opening recovery file for slave %d",i);
03155                     Terminate(-1);
03156                 }/*if(fd == NULL)*/
03157                 retvalue=PF_RecvFile(i,fd);
03158                 if(retvalue<=0){
03159                     MesPrint("Error receiving recovery file from slave %d",i);
03160                     Terminate(-1);
03161                 }/*if(retvalue<=0)*/
03162                 fclose(fd);
03163             }/*for(i=0; i<PF.numtasks;i++)*/
03164             /*
03165              * Make the intermediate files the recovery files.
03166              */
03167             ANNOUNCE(rename intermediate file)
03168             for ( i = 0; i < PF.numtasks; i++ ) {
03169                 const char *src = PF_recoveryfile('m', i, 1);
03170                 const char *dst = PF_recoveryfile('m', i, 0);
03171                 if ( rename(src, dst) ) {
03172                     MesPrint("Error renaming recovery file %s -> %s", src, dst);
03173                 }
03174             }
03175             done_snapshot = 1;
03176             MesPrint("done."); fflush(0);
03177         }
03178 #endif
03179     }
03180     if ( AC.CheckpointRunAfter ) {
03181         size_t l1, l2;
03182         char *str;
03183         l1 = strlen(AC.CheckpointRunAfter);
03184         NumToStr((UBYTE*)argbuf, AC.CModule);
03185         l2 = strlen(argbuf);
03186         str = (char*)Malloc1(l1+l2+2, "callafter");
03187         strcpy(str, AC.CheckpointRunAfter);
03188         *(str+l1) = ' ';
03189         strcpy(str+l1+1, argbuf);
03190         retvalue = system(str);
03191         M_free(str, "callafter");
03192         if ( retvalue ) {
03193             MesPrint("Error calling script after recovery.");
03194         }
03195     }
03196     AC.CheckpointStamp = TimeWallClock(1);
03197 }
03198 #ifdef WITHMPI
03199 else/* timestamp - AC.CheckpointStamp < AC.CheckpointInterval*/
03200 /* The slaves don't need to make snapshots. */
03201 PF_BroadcastNumber(0);
03202 }
03203 }/*if(PF.me == MASTER)*/
03204 else{/*Slave*/
03205     int i;
03206     /* Check if the slave needs to make a snapshot. */
03207     if ( PF_BroadcastNumber(0) ) {
03208         error = DoSnapshot(moduletype);
03209         if(error == 0){
03210             FILE *fd;
03211             /*
03212              * Send the recovery file to the master. Note that no renaming
03213              * has been performed and what we have to send is actually sitting
03214              * in the intermediate file.
03215              */
03216             fd = fopen(intermedfile, "r");
03217             i=PF_SendFile(MASTER, fd);/*if fd=NULL, PF_SendFile seds to a slave the failure tag*/
03218             if(fd == NULL)
03219                 Terminate(-1);
03220             fclose(fd);
03221             if(i<=0)
03222                 Terminate(-1);
03223             /*Now the slave need not the recovery file so remove it:*/
03224             remove(intermedfile);
03225         }
03226         else{
03227             /*send the error tag to the master:*/

```

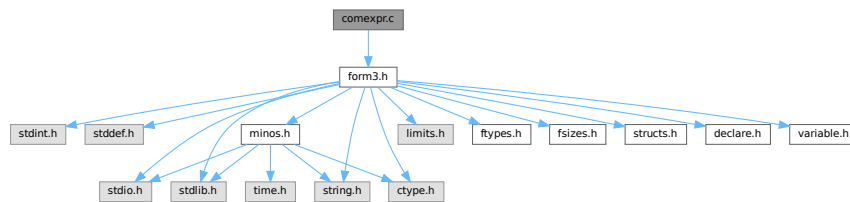
```

03228         PF_SendFile(MASTER, NULL); /*if fd==NULL, PF_SendFile seds to a slave the failure tag*/
03229         Terminate(-1);
03230     }
03231     done_snapshot = 1;
03232     } /*if (tag=PF_DATA_MSGTAG) */
03233     } /*if (PF.me != MASTER) */
03234 #endif
03235 }
03236
03237 /*
03238     #] DoCheckpoint :
03239 */

```

## 4.7 comexpr.c File Reference

#include "form3.h"  
 Include dependency graph for comexpr.c:



### Data Structures

- struct **id\_options**

### Functions

- int **CoLocal** (UBYTE \*inp)
- int **CoGlobal** (UBYTE \*inp)
- int **CoLocalFactorized** (UBYTE \*inp)
- int **CoGlobalFactorized** (UBYTE \*inp)
- int **DoExpr** (UBYTE \*inp, int type, int par)
- int **ColdOld** (UBYTE \*inp)
- int **Cold** (UBYTE \*inp)
- int **ColdNew** (UBYTE \*inp)
- int **CoDisorder** (UBYTE \*inp)
- int **CoMany** (UBYTE \*inp)
- int **CoMulti** (UBYTE \*inp)
- int **ColfMatch** (UBYTE \*inp)
- int **ColfNoMatch** (UBYTE \*inp)
- int **CoOnce** (UBYTE \*inp)
- int **CoOnly** (UBYTE \*inp)
- int **CoSelect** (UBYTE \*inp)
- int **ColdExpression** (UBYTE \*inp, int type)
- int **CoMultiply** (UBYTE \*inp)
- int **CoFill** (UBYTE \*inp)
- int **CoFillExpression** (UBYTE \*inp)
- int **CoPrintTable** (UBYTE \*inp)
- int **CoAssign** (UBYTE \*inp)
- int **CoDeallocateTable** (UBYTE \*inp)

### 4.7.1 Detailed Description

Compiler routines for statements that involve algebraic expressions. These involve definitions, id-statements, the multiply statement and the fill statement.

Definition in file [comexpr.c](#).

### 4.7.2 Function Documentation

#### 4.7.2.1 CoLocal()

```
int CoLocal (
    UBYTE * inp )
```

Definition at line 66 of file [comexpr.c](#).

#### 4.7.2.2 CoGlobal()

```
int CoGlobal (
    UBYTE * inp )
```

Definition at line 73 of file [comexpr.c](#).

#### 4.7.2.3 CoLocalFactorized()

```
int CoLocalFactorized (
    UBYTE * inp )
```

Definition at line 80 of file [comexpr.c](#).

#### 4.7.2.4 CoGlobalFactorized()

```
int CoGlobalFactorized (
    UBYTE * inp )
```

Definition at line 87 of file [comexpr.c](#).

#### 4.7.2.5 DoExpr()

```
int DoExpr (
    UBYTE * inp,
    int type,
    int par )
```

Definition at line 96 of file [comexpr.c](#).



#### 4.7.2.6 ColdOld()

```
int ColdOld (
    UBYTE * inp )
```

Definition at line 384 of file [comexpr.c](#).

#### 4.7.2.7 Cold()

```
int Cold (
    UBYTE * inp )
```

Definition at line 395 of file [comexpr.c](#).

#### 4.7.2.8 ColdNew()

```
int ColdNew (
    UBYTE * inp )
```

Definition at line 406 of file [comexpr.c](#).

#### 4.7.2.9 CoDisorder()

```
int CoDisorder (
    UBYTE * inp )
```

Definition at line 417 of file [comexpr.c](#).

#### 4.7.2.10 CoMany()

```
int CoMany (
    UBYTE * inp )
```

Definition at line 428 of file [comexpr.c](#).

#### 4.7.2.11 CoMulti()

```
int CoMulti (
    UBYTE * inp )
```

Definition at line 439 of file [comexpr.c](#).

#### 4.7.2.12 ColfMatch()

```
int ColfMatch (
    UBYTE * inp )
```

Definition at line 450 of file [comexpr.c](#).

#### 4.7.2.13 ColfNoMatch()

```
int ColfNoMatch (
    UBYTE * inp )
```

Definition at line 461 of file [comexpr.c](#).

#### 4.7.2.14 CoOnce()

```
int CoOnce (
    UBYTE * inp )
```

Definition at line 472 of file [comexpr.c](#).

#### 4.7.2.15 CoOnly()

```
int CoOnly (
    UBYTE * inp )
```

Definition at line 483 of file [comexpr.c](#).

#### 4.7.2.16 CoSelect()

```
int CoSelect (
    UBYTE * inp )
```

Definition at line 494 of file [comexpr.c](#).

#### 4.7.2.17 ColdExpression()

```
int ColdExpression (
    UBYTE * inp,
    int type )
```

Definition at line 507 of file [comexpr.c](#).

#### 4.7.2.18 CoMultiply()

```
int CoMultiply (
    UBYTE * inp )
```

Definition at line 1031 of file [comexpr.c](#).

#### 4.7.2.19 CoFill()

```
int CoFill (
    UBYTE * inp )
```

Definition at line 1070 of file [comexpr.c](#).

#### 4.7.2.20 CoFillExpression()

```
int CoFillExpression (
    UBYTE * inp )
```

Definition at line 1356 of file [comexpr.c](#).

#### 4.7.2.21 CoPrintTable()

```
int CoPrintTable (
    UBYTE * inp )
```

Definition at line 1748 of file [comexpr.c](#).

#### 4.7.2.22 CoAssign()

```
int CoAssign (
    UBYTE * inp )
```

Definition at line 1924 of file [comexpr.c](#).

#### 4.7.2.23 CoDeallocateTable()

```
int CoDeallocateTable (
    UBYTE * inp )
```

Definition at line 1982 of file [comexpr.c](#).

## 4.8 comexpr.c

[Go to the documentation of this file.](#)

```
00001
00008 /* # [ License : */
00009 /*
00010 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00011 * When using this file you are requested to refer to the publication
00012 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00013 * This is considered a matter of courtesy as the development was paid
00014 * for by FOM the Dutch physics granting agency and we would like to
00015 * be able to track its scientific use to convince FOM of its value
00016 * for the community.
00017 *
00018 * This file is part of FORM.
00019 *
00020 * FORM is free software: you can redistribute it and/or modify it under the
00021 * terms of the GNU General Public License as published by the Free Software
00022 * Foundation, either version 3 of the License, or (at your option) any later
00023 * version.
00024 *
00025 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00026 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00027 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00028 * details.
00029 *
00030 * You should have received a copy of the GNU General Public License along
00031 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00032 */
00033 /* # ] License : */
00034
00035 /*
```

```

00036     # [ Includes : compi2.c
00037
00038     File contains most of what has to do with compiling expressions.
00039     Main supporting file: token.c
00040 */
00041
00042 #include "form3.h"
00043
00044 static struct id_options {
00045     UBYTE *name;
00046     int code;
00047     int dummy;
00048 } IdOptions[] = {
00049     { (UBYTE *) "multi",      SUBMULTI      , 0 }
00050     , { (UBYTE *) "many",     SUBMANY       , 0 }
00051     , { (UBYTE *) "only",     SUBONLY       , 0 }
00052     , { (UBYTE *) "once",     SUBONCE       , 0 }
00053     , { (UBYTE *) "ifmatch",  SUBAFTER    , 0 }
00054     , { (UBYTE *) "ifnomatch", SUBAFTERNOT , 0 }
00055     , { (UBYTE *) "ifnotmatch", SUBAFTERNOT , 0 }
00056     , { (UBYTE *) "disorder", SUBDISORDER , 0 }
00057     , { (UBYTE *) "select",   SUBSELECT   , 0 }
00058     , { (UBYTE *) "all",      SUBALL       , 0 }
00059 };
00060
00061 /*
00062 #] Includes :
00063 # [ CoLocal :
00064 */
00065
00066 int CoLocal(UBYTE *inp) { return(DoExpr(inp, LOCALEXPRESSION, 0)); }
00067
00068 /*
00069 #] CoLocal :
00070 # [ CoGlobal :
00071 */
00072
00073 int CoGlobal(UBYTE *inp) { return(DoExpr(inp, GLOBALEXPRESSION, 0)); }
00074
00075 /*
00076 #] CoGlobal :
00077 # [ CoLocalFactorized :
00078 */
00079
00080 int CoLocalFactorized(UBYTE *inp) { return(DoExpr(inp, LOCALEXPRESSION, 1)); }
00081
00082 /*
00083 #] CoLocalFactorized :
00084 # [ CoGlobalFactorized :
00085 */
00086
00087 int CoGlobalFactorized(UBYTE *inp) { return(DoExpr(inp, GLOBALEXPRESSION, 1)); }
00088
00089 /*
00090 #] CoGlobalFactorized :
00091 # [ DoExpr:
00092
00093
00094 */
00095
00096 int DoExpr(UBYTE *inp, int type, int par)
00097 {
00098     GETIDENTITY
00099     int error = 0;
00100     UBYTE *p, *q, c;
00101     WORD *w, i, j = 0, c1, c2, *OldWork = AT.WorkPointer, osize;
00102     WORD jold = 0;
00103     POSITION pos;
00104     while ( *inp == ',' ) inp++;
00105     if ( par ) AC.ToBeInFactors = 1;
00106     else AC.ToBeInFactors = 0;
00107     p = inp;
00108     while ( *p && *p != '=' ) {
00109         if ( *p == '(' ) SKIPBRA4(p)
00110         else if ( *p == '{' ) SKIPBRA5(p)
00111         else if ( *p == '[' ) SKIPBRA1(p)
00112         else p++;
00113     }
00114     if ( *p ) { /* Variety with the = sign */
00115         if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
00116             MesPrint("%Illegal name for expression");
00117             error = 1;
00118             if ( q[-1] == '_' ) {
00119                 while ( FG.cTable[*q] < 2 || *q == '_' ) q++;
00120             }
00121         }
00122         else {

```

```

00123     c = *q; *q = 0;
00124     if ( GetVar(inp,&c1,&c2,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
00125         if ( c1 == CEXPRESSION ) {
00126             if ( Expressions[c2].status == STOREDEXPRESSION ) {
00127                 MesPrint("&Illegal attempt to overwrite a stored expression");
00128                 error = 1;
00129             }
00130             else {
00131                 HighWarning("Expression is replaced by new definition");
00132                 if ( AO.OptimizeResult.nameofexpr != NULL &&
00133                     StrCmp(inp,AO.OptimizeResult.nameofexpr) == 0 ) {
00134                     ClearOptimize();
00135                 }
00136                 if ( Expressions[c2].status != DROPEDEXPRESSION ) {
00137                     w = &(Expressions[c2].status);
00138                     if ( *w == LOCALEXPRESSION || *w == SKIPLEXPRESSION )
00139                         *w = DROPEDEXPRESSION;
00140                     else if ( *w == GLOBALEXPRESSION || *w == SKIPGEXPRESSION )
00141                         *w = DROPGEXPRESSION;
00142                     else if ( *w == HIDDENLEXPRESSION )
00143                         *w = DROPHLEXPRESSION;
00144                     else if ( *w == HIDDENGEXPRESSION )
00145                         *w = DROPHGEXPRESSION;
00146                 }
00147                 AC.TransEname = Expressions[c2].name;
00148                 j = EntVar(CEXPRESSION,0,type,0,0,0);
00149                 Expressions[j].node = Expressions[c2].node;
00150                 Expressions[c2].replace = j;
00151             }
00152         }
00153         else {
00154             MesPrint("&name of expression is also name of a variable");
00155             error = 1;
00156             j = EntVar(CEXPRESSION,inp,type,0,0,0);
00157         }
00158         jold = c2;
00159     }
00160     else {
00161         /*
00162             Here we have to worry about reuse of the expression in the
00163             same module. That will need AS.Oldvflags but that may not
00164             be defined or have the proper value.
00165         */
00166         j = EntVar(CEXPRESSION,inp,type,0,0,0);
00167         jold = j;
00168     }
00169     *q = c;
00170     OldWork = w = AT.WorkPointer;
00171     *w++ = TYPEEXPRESSION;
00172     *w++ = 3+SUBEXPSIZE;
00173     *w++ = j;
00174     AC.ProtoType = w;
00175     AR.CurExpr = j; /* Block expression j */
00176     *w++ = SUBEXPRESSION;
00177     *w++ = SUBEXPSIZE;
00178     *w++ = j;
00179     *w++ = 1;
00180     *w++ = AC.cbunum;
00181     FILLSUB(w)
00182
00183     if ( c == '(' ) {
00184         while ( *q == ',' || *q == '(' ) {
00185             inp = q+1;
00186             if ( ( q = SkipAName(inp) ) == 0 ) {
00187                 MesPrint("&Illegal name for expression argument");
00188                 error = 1;
00189                 q = p - 1;
00190                 break;
00191             }
00192             c = *q; *q = 0;
00193             if ( GetVar(inp,&c1,&c2,ALLVARIABLES,WITHAUTO) < 0 ) c1 = -1;
00194             switch ( c1 ) {
00195                 case CSYMBOL :
00196                     *w++ = SYMTOSYM; *w++ = 4; *w++ = c2; *w++ = 0;
00197                     break;
00198                 case CINDEXT :
00199                     *w++ = INDTOIND; *w++ = 4;
00200                     *w++ = c2 + AM.OffsetIndex; *w++ = 0;
00201                     break;
00202                 case CVECTOR :
00203                     *w++ = VECTOVEC; *w++ = 4;
00204                     *w++ = c2 + AM.OffsetVector; *w++ = 0;
00205                     break;
00206                 case CFUNCTION :
00207                     *w++ = FUNTOFUN; *w++ = 4; *w++ = c2 + FUNCTION; *w++ = 0;
00208                     break;
00209                 default :

```

```

00210             MesPrint("&Illegal expression parameter: %s",inp);
00211             error = 1;
00212             break;
00213         }
00214         *q = c;
00215     }
00216     if ( *q != ')' || q+1 != p ) {
00217         MesPrint("&Illegal use of arguments for expression");
00218         error = 1;
00219     }
00220     AC.ProtoType[1] = w - AC.ProtoType;
00221 }
00222 else if ( c != '=' ) {
00223 /*
00224     The dummy accepted L F := RHS;
00225 */
00226     MesPrint("&Illegal LHS for expression definition");
00227     error = 1;
00228 }
00229 *w++ = 1;
00230 *w++ = 1;
00231 *w++ = 3;
00232 *w++ = 0;
00233 SeekScratch(AR.outfile,&pos);
00234 Expressions[j].counter = 1;
00235 Expressions[j].onfile = pos;
00236 Expressions[j].whichbuffer = 0;
00237 #ifdef PARALLELCODE
00238     Expressions[j].partodo = AC.inparallelfalg;
00239 #endif
00240 OldWork[2] = w - OldWork - 3;
00241 AT.WorkPointer = w;
00242 /*
00243     Writing the expression prototype to disk and to the compiler
00244     buffer is done only after the RHS has been compiled because
00245     we don't know the number of the main level RHS yet.
00246 */
00247 }
00248 inp = p+1;
00249 ClearWildcardNames();
00250 osize = AC.ProtoType[1]; AC.ProtoType[1] = SUBEXPSIZE;
00251 PutInVflags(jold);
00252 if ( ( i = CompileAlgebra(inp,RHSIDE,AC.ProtoType) ) < 0 ) {
00253     AC.ProtoType[1] = osize;
00254     error = 1;
00255 }
00256 else if ( error == 0 ) {
00257     AC.ProtoType[1] = osize;
00258     AC.ProtoType[2] = i;
00259     if ( PutOut(BHEAD OldWork+2,&pos,AR.outfile,0) < 0 ) {
00260         MesPrint("&Cannot create expression");
00261         error = -1;
00262     }
00263     else {
00264         Expressions[j].sizeprototype = OldWork[2];
00265         OldWork[2] = 4+SUBEXPSIZE;
00266         OldWork[4] = SUBEXPSIZE;
00267         OldWork[5] = i;
00268         OldWork[SUBEXPSIZE+3] = 1;
00269         OldWork[SUBEXPSIZE+4] = 1;
00270         OldWork[SUBEXPSIZE+5] = 3;
00271         OldWork[SUBEXPSIZE+6] = 0;
00272         if ( PutOut(BHEAD OldWork+2,&pos,AR.outfile,0) < 0 ) {
00273             || FlushOut(&pos,AR.outfile,0) ) {
00274                 MesPrint("&Cannot create expression");
00275                 error = -1;
00276             }
00277             AR.outfile->POfull = AR.outfile->POfill;
00278         }
00279         OldWork[2] = j;
00280 /*
00281     Seems unnecessary (13-feb-2018)
00282
00283     AddNtoL(OldWork[1],OldWork);
00284 */
00285     AT.WorkPointer = OldWork;
00286     if ( AC.dumnumflag ) Add2Com(TYPEDETCURDUM)
00287 }
00288 AC.ToBeInFactors = 0;
00289 }
00290 else { /* Variety in which expressions change property */
00291 /*
00292     This code got a major revision because it didn't
00293     take hidden expressions into account. (1-jun-2010 JV)
00294 */
00295     do {
00296         if ( ( q = SkipAName(inp) ) == 0 ) {

```

```

00297         MesPrint("&Illegal name(s) for expression(s)");
00298         return(1);
00299     }
00300     c = *q; *q = 0;
00301     if ( GetName(AC.exprnames,inp,&c2,NOAUTO) == NAMENOTFOUND ) {
00302         MesPrint("&%s is not a valid expression",inp);
00303         error = 1;
00304     }
00305     else {
00306         w = &(Expressions[c2].status);
00307         if ( type == LOCALEXPRESSION ) {
00308             switch ( *w ) {
00309                 case GLOBALEXPRESSION:
00310                     *w = LOCALEXPRESSION;
00311                     break;
00312                 case SKIPGEXPRESSION:
00313                     *w = SKIPLEXPRESSION;
00314                     break;
00315                 case DROPGEXPRESSION:
00316                     *w = DROPLEXPRESSION;
00317                     break;
00318                 case HIDDENGEXPRESSION:
00319                     *w = HIDDENLEXPRESSION;
00320                     break;
00321                 case HIDEGEXPRESSION:
00322                     *w = HIDELEXPRESSION;
00323                     break;
00324                 case UNHIDEGEXPRESSION:
00325                     *w = UNHIDELEXPRESSION;
00326                     break;
00327                 case INTOHIDEGEXPRESSION:
00328                     *w = INTOHIDELEXPRESSION;
00329                     break;
00330                 case DROPHGEXPRESSION:
00331                     *w = DROPHLEXPRESSION;
00332                     break;
00333             }
00334         }
00335         else if ( type == GLOBALEXPRESSION ) {
00336             switch ( *w ) {
00337                 case LOCALEXPRESSION:
00338                     *w = GLOBALEXPRESSION;
00339                     break;
00340                 case SKIPLEXPRESSION:
00341                     *w = SKIPGEXPRESSION;
00342                     break;
00343                 case DROPLEXPRESSION:
00344                     *w = DROPGEXPRESSION;
00345                     break;
00346                 case HIDDENLEXPRESSION:
00347                     *w = HIDDENGEXPRESSION;
00348                     break;
00349                 case HIDELEXPRESSION:
00350                     *w = HIDEGEXPRESSION;
00351                     break;
00352                 case UNHIDELEXPRESSION:
00353                     *w = UNHIDEGEXPRESSION;
00354                     break;
00355                 case INTOHIDELEXPRESSION:
00356                     *w = INTOHIDEGEXPRESSION;
00357                     break;
00358                 case DROPHLEXPRESSION:
00359                     *w = DROPHGEXPRESSION;
00360                     break;
00361             }
00362         }
00363         /*
00364         old code
00365         if ( type != LOCALEXPRESSION || *w != STOREDEXPRESSION )
00366             *w = type;
00367         */
00368     }
00369     *q = c; inp = q+1;
00370 } while ( c == ',' );
00371 if ( c ) {
00372     MesPrint("&Illegal object in local or global redefinition");
00373     error = 1;
00374 }
00375 }
00376 return(error);
00377 }
00378
00379 /*
00380 #] DoExpr:
00381 #[ CoIdOld :
00382 */
00383

```

```
00384 int CoIdOld(UBYTE *inp)
00385 {
00386     AC.idoption = 0;
00387     return(CoIdExpression(inp, TYPEIDOLD));
00388 }
00389
00390 /*
00391     #] CoIdOld :
00392     #[ CoId :
00393 */
00394
00395 int CoId(UBYTE *inp)
00396 {
00397     AC.idoption = 0;
00398     return(CoIdExpression(inp, TYPEIDNEW));
00399 }
00400
00401 /*
00402     #] CoId :
00403     #[ CoIdNew :
00404 */
00405
00406 int CoIdNew(UBYTE *inp)
00407 {
00408     AC.idoption = 0;
00409     return(CoIdExpression(inp, TYPEIDNEW));
00410 }
00411
00412 /*
00413     #] CoIdNew :
00414     #[ CoDisorder :
00415 */
00416
00417 int CoDisorder(UBYTE *inp)
00418 {
00419     AC.idoption = SUBDISORDER;
00420     return(CoIdExpression(inp, TYPEIDNEW));
00421 }
00422
00423 /*
00424     #] CoDisorder :
00425     #[ CoMany :
00426 */
00427
00428 int CoMany(UBYTE *inp)
00429 {
00430     AC.idoption = SUBMANY;
00431     return(CoIdExpression(inp, TYPEIDNEW));
00432 }
00433
00434 /*
00435     #] CoMany :
00436     #[ CoMulti :
00437 */
00438
00439 int CoMulti(UBYTE *inp)
00440 {
00441     AC.idoption = SUBMULTI;
00442     return(CoIdExpression(inp, TYPEIDNEW));
00443 }
00444
00445 /*
00446     #] CoMulti :
00447     #[ CoIfMatch :
00448 */
00449
00450 int CoIfMatch(UBYTE *inp)
00451 {
00452     AC.idoption = SUBAFTER;
00453     return(CoIdExpression(inp, TYPEIDNEW));
00454 }
00455
00456 /*
00457     #] CoIfMatch :
00458     #[ CoIfNoMatch :
00459 */
00460
00461 int CoIfNoMatch(UBYTE *inp)
00462 {
00463     AC.idoption = SUBAFTERNOT;
00464     return(CoIdExpression(inp, TYPEIDNEW));
00465 }
00466
00467 /*
00468     #] CoIfNoMatch :
00469     #[ CoOnce :
00470 */
```



```

00471
00472 int CoOnce(UBYTE *inp)
00473 {
00474     AC.idoption = SUBONCE;
00475     return(CoIdExpression(inp,TYPEIDNEW));
00476 }
00477
00478 /*
00479     #] CoOnce :
00480     #[ CoOnly :
00481 */
00482
00483 int CoOnly(UBYTE *inp)
00484 {
00485     AC.idoption = SUBONLY;
00486     return(CoIdExpression(inp,TYPEIDNEW));
00487 }
00488
00489 /*
00490     #] CoOnly :
00491     #[ CoSelect :
00492 */
00493
00494 int CoSelect(UBYTE *inp)
00495 {
00496     AC.idoption = SUBSELECT;
00497     return(CoIdExpression(inp,TYPEIDNEW));
00498 }
00499
00500 /*
00501     #] CoSelect :
00502     #[ CoIdExpression :
00503
00504     First finish dealing with secondary keywords
00505 */
00506
00507 int CoIdExpression(UBYTE *inp, int type)
00508 {
00509     GETIDENTITY
00510     int i, j, idhead, error = 0, MinusSign = 0, opt, retcode;
00511     WORD *w, *s, *m, *mm, *ww, *FirstWork, *OldWork, cl, numsets = 0,
00512         oldnumrhs, *ow, oldEside;
00513     UBYTE *p, *pp, c;
00514     CBUF *C = cbuf+AC.cbufnum;
00515     LONG oldcpointer, x;
00516     FirstWork = OldWork = AT.WorkPointer;
00517 /*
00518     Don't forget to change in StudyPattern if we change/add_to the
00519     following setup.
00520     if ( type == TYPEIF ) idhead = IDHEAD-1;
00521     else
00522 */
00523     idhead = IDHEAD;
00524     AR.CurExpr = -1;
00525     w = AT.WorkPointer;
00526     *w++ = type;
00527     *w++ = idhead + SUBEXPSIZE;
00528     w++;
00529     if ( idhead >= IDHEAD ) *w++ = -1;
00530 #if IDHEAD > 4
00531     for ( i = 4; i < idhead; i++ ) *w++ = 0;
00532 #endif
00533     while ( *inp == ',' ) inp++;
00534     p = inp;
00535     if ( AC.idoption == SUBSELECT ) {
00536         p--;
00537         goto findsets;
00538     }
00539     else if ( ( AC.idoption == SUBAFTER ) || ( AC.idoption == SUBAFTERNOT ) ) {
00540         while ( *p && *p != '=' && *p != ',' ) {
00541             if ( *p == '(' ) SKIPBRA4(p)
00542             else if ( *p == '{' ) SKIPBRA5(p)
00543             else if ( *p == '[' ) SKIPBRA1(p)
00544             else p++;
00545         }
00546         if ( *p == '=' || *inp != '-' || inp[1] != '>' ) {
00547             MesPrint("&Illegal use if if[no]match in id statement");
00548             error = 1; goto AllDone;
00549         }
00550         if ( *p == 0 ) {
00551             MesPrint("&id-statement without = sign");
00552             error = 1; goto AllDone;
00553         }
00554         inp += 2; pp = inp;
00555         goto readlabel;
00556     }
00557     for(;;) {

```

```

00558     while ( *p && *p != '=' && *p != ',' ) {
00559         if ( *p == '(' ) SKIPBRA4(p)
00560         else if ( *p == '{' ) SKIPBRA5(p)
00561         else if ( *p == '[' ) SKIPBRA1(p)
00562         else p++;
00563     }
00564     if ( *p == '=' ) break;
00565     if ( *p == 0 ) {
00566         MesPrint("&id-statement without = sign");
00567         error = 1; goto AllDone;
00568     }
00569     /*
00570     We have either a secondary option or a syntax error
00571     */
00572     pp = inp;
00573     while ( FG.cTable[*pp] == 0 ) pp++;
00574     c = *pp; *pp = 0;
00575     i = sizeof(IdOptions)/sizeof(struct id_options);
00576     while ( --i >= 0 ) {
00577         if ( StrICmp(inp,IdOptions[i].name) == 0 ) break;
00578     }
00579     if ( i < 0 ) {
00580         MesPrint("&Illegal option %s in id-statement",inp);
00581         *pp = c; error = 1; p++; inp = p; continue;
00582     }
00583     opt = IdOptions[i].code;
00584     *pp = c;
00585     inp = pp+1;
00586     switch ( opt ) {
00587         case SUBDISORDER:
00588             if ( pp != p ) goto IllField;
00589             AC.idoption |= SUBDISORDER;
00590             p++; inp = p;
00591             break;
00592         case SUBSELECT:
00593             if ( p != pp ) goto IllField;
00594             if ( ( AC.idoption & SUBMASK ) != 0 ) {
00595                 if ( AC.idoption == SUBMULTI && type == TYPEIF ) {}
00596                 else {
00597                     MesPrint("&Conflicting options in id-statement");
00598                     error = 1;
00599                 }
00600             }
00601     findsets;;
00602     /*
00603     Now we read the sets
00604     */
00605     numsets = 0;
00606     for(;;) {
00607         inp = ++p;
00608         while ( *p && *p != '=' && *p != ',' ) {
00609             if ( *p == '(' ) SKIPBRA4(p)
00610             else if ( *p == '{' ) SKIPBRA5(p)
00611             else if ( *p == '[' ) SKIPBRA1(p)
00612             else p++;
00613         }
00614         if ( *p == '=' ) break;
00615         if ( *p == 0 ) {
00616             MesPrint("&id-statement without = sign");
00617             error = 1; goto AllDone;
00618         }
00619     /*
00620     We have a set at inp.
00621     */
00622     if ( *inp == '{' ) {
00623         if ( p[-1] != '}' ) {
00624             c = *p; *p = 0;
00625             MesPrint("&Illegal temporary set: %s",inp);
00626             error = 1; *p = c;
00627         }
00628         else {
00629             inp++;
00630             c = p[-1]; p[-1] = 0;
00631             c1 = DoTempSet(inp,p-1);
00632             *w++ = c1;
00633             p[-1] = c;
00634             numsets++;
00635             if ( w[-1] < 0 ) error = 1;
00636         }
00637     }
00638     else {
00639         c = *p; *p = 0;
00640         if ( GetName(AC.varnames,inp,&c1,NOAUTO) != CSET ) {
00641             MesPrint("&%s is not a set",inp);
00642             error = 1;
00643         }
00644     }

```

```

00645             if ( c1 < AM.NumFixedSets ) {
00646                 MesPrint("&Built in sets are not allowed in the select option");
00647                 error = 1;
00648             }
00649             else if ( Sets[c1].type == CRANGE ) {
00650                 MesPrint("&Ranged sets are not allowed in the select option");
00651                 error = 1;
00652             }
00653             numsets++;
00654             *w++ = c1;
00655         }
00656         *p = c;
00657     }
00658 }
00659 /*
00660     Now exchange the positions a bit.
00661     Regular stuff at OldWork, numsets sets at FirstWork[idhead]
00662 */
00663 OldWork = w;
00664 for ( i = 0; i < idhead; i++ ) *w++ = FirstWork[i];
00665 AC.idoption = SUBSELECT;
00666 break;
00667 case SUBAFTER:
00668 case SUBAFTERNOT:
00669     if ( type == TYPEIF ) {
00670         MesPrint("&The if[no]match->label option is not allowed in an if statement");
00671         error = 1; goto AllDone;
00672     }
00673     if ( pp[0] != '-' || pp[1] != '>' ) goto IllField;
00674     pp += 2; /* points now at the label */
00675     inp = pp;
00676     AC.idoption |= opt;
00677 readlabel:
00678     while ( FG.cTable[*pp] <= 1 ) pp++;
00679     if ( pp != p ) {
00680         c = *p; *p = 0;
00681         MesPrint("&Illegal label %s in if[no]match option of id-statement",inp);
00682         *p = c; error = 1; inp = p+1; continue;
00683     }
00684     c = *p; *p = 0;
00685     OldWork[3] = GetLabel(inp);
00686     *p++ = c; inp = p;
00687     break;
00688 case SUBALL:
00689     x = 0;
00690     if ( *pp == '(' ) {
00691         if ( FG.cTable[*inp] == 1 ) {
00692             while ( *inp >= '0' && *inp <= '9' ) x = 10*x+*inp++ - '0';
00693         }
00694         else {
00695             pp++;
00696             while ( FG.cTable[*inp] == 0 ) inp++;
00697             c = *inp; *inp = 0;
00698             if ( StrICont(pp,(UBYTE *)"normalize") != 0 ) goto IllOpt;
00699             *inp = c;
00700             OldWork[4] |= NORMALIZEFLAG;
00701         }
00702         if ( *inp != ')' || inp+1 != p ) {
00703             c = *inp; *inp = 0;
00704 IllOpt:
00705             MesPrint("&Illegal ALL option in id-statement: ",pp);
00706             *inp++ = c;
00707             error = 1;
00708             continue;
00709         }
00710         pp = inp;
00711         inp = pp+1;
00712     }
00713 /*
00714     Note that the following statement limits x to
00715 */
00716     if ( x > MAXPOSITIVE ) {
00717         MesPrint("&Requested maximum number of matches %1 in ALL option in id-statement is
greater than %1 ",x,MAXPOSITIVE);
00718         error = 1;
00719     }
00720     OldWork[5] = x;
00721     if ( type != TYPEIDNEW ) {
00722         if ( type == TYPEIDOLD ) {
00723             MesPrint("&Requested ALL option not allowed in idold/also statement.");
00724             error = 1;
00725         }
00726         else if ( type == TYPEIF ) {
00727             MesPrint("&Requested ALL option not allowed in if(match())");
00728             error = 1;
00729         }
00730         else {

```

```

00731             MesPrint("&ALL option only allowed in regular id-statement.");
00732             error = 1;
00733         }
00734     }
00735     p++; inp = p;
00736     AC.idoption = opt;
00737     break;
00738     default:
00739         if ( pp != p ) {
00740 IllField:
00741             c = *p; *p = 0;
00742             MesPrint("&Illegal optionfield %s in id-statement",inp);
00743             *p = c; error = 1; inp = p+1; continue;
00744         }
00745         i = AC.idoption & SUBMASK;
00746         if ( i && i != opt ) {
00747             MesPrint("&Conflicting options in id-statement");
00748             error = 1; continue;
00749         }
00750         else AC.idoption |= opt;
00751         while ( *p == ',' ) p++;
00752         inp = p;
00753         break;
00754     }
00755     if ( ( AC.idoption & SUBMASK ) == 0 ) AC.idoption |= SUBMULTI;
00756     OldWork[2] = AC.idoption;
00757 /*
00758     Now we have a field till the = sign
00759     Now the subexpression prototype
00760 */
00761     AC.ProtoType = w;
00762     *w++ = SUBEXPRESSION;
00763     *w++ = SUBEXPSIZE;
00764     *w++ = C->numrhs+1;
00765     *w++ = 1;
00766     *w++ = AC.cbufnum;
00767     FILLSUB(w)
00768     AC.WildC = w;
00769     AC.NwildC = 0;
00770     AT.WorkPointer = s = w + 4*AM.MaxWildcards + 8;
00771 /*
00772     Now read the LHS
00773 */
00774     ClearWildcardNames();
00775     oldcpointer = AddLHS(AC.cbufnum) - C->Buffer;
00776
00777     *p = 0;
00778     oldnumrhs = C->numrhs;
00779     if ( ( retcode = CompileAlgebra(inp,LHSIDE,AC.ProtoType) ) < 0 ) { error = 1; }
00780     else AC.ProtoType[2] = retcode;
00781     *p = '='; inp = p+1;
00782     AT.WorkPointer = s;
00783     if ( AC.NwildC && SortWild(w,AC.NwildC) ) error = 1;
00784
00785     /* Make the LHS pointers ready */
00786
00787     OldWork[1] = AC.WildC-OldWork;
00788     OldWork[idhead+1] = OldWork[1] - idhead;
00789     w = AC.WildC;
00790     AT.WorkPointer = w;
00791     s = C->rhs[C->numrhs];
00792 /*
00793     Now check whether wildcards get converted to dollars (for PARALLEL)
00794 */
00795     {
00796         WORD *tw, *twstop;
00797         tw = AC.ProtoType; twstop = tw + tw[1]; tw += SUBEXPSIZE;
00798         while ( tw < twstop ) {
00799             if ( *tw == LOADDOLLAR ) {
00800                 AddPotModdollar(tw[2]);
00801             }
00802             tw += tw[1];
00803         }
00804     }
00805 /*
00806     We have the expression in the compiler buffers.
00807     The main level is at lhs[numlhs]
00808     The partial lhs (including ProtoType) is in OldWork (in Workspace)
00809     We need to load the result at w after the prototype
00810     Because these sort routines don't use the Workspace
00811     there should not be a conflict
00812 */
00813     if ( !error && *s == 0 ) {
00814 IllLeft:MesPrint("&Illegal LHS");
00815         AC.lhdollarflag = 0;
00816         return(1);
00817     }

```

```

00818     if ( !error && *(s+s) != 0 ) {
00819         MesPrint("&LHS should be one term only");
00820         return(1);
00821     }
00822     if ( error == 0 ) {
00823         WORD oldpolyfun = AR.PolyFun;
00824         if ( NewSort(BHEAD0) || NewSort(BHEAD0) ) {
00825             if ( !error ) error = 1;
00826             return(error);
00827         }
00828         AN.RepPoint = AT.RepCount + 1;
00829         ow = (WORD *) ((UBYTE *) (AT.WorkPointer)) + AM.MaxTer;
00830         mm = s; ww = ow; i = *mm;
00831         while ( --i >= 0 ) { *ww++ = *mm++; } AT.WorkPointer = ww;
00832         AC.lhdollarflag = 0; oldEside = AR.Eside; AR.Eside = LHSIDE;
00833         AR.Cnumlhs = C->numlhs;
00834         AR.PolyFun = 0;
00835         if ( Generator(BHEAD ow,C->numlhs) ) {
00836             AR.Eside = oldEside;
00837             LowerSortLevel(); LowerSortLevel(); AR.PolyFun = oldpolyfun; goto IllLeft;
00838         }
00839         AR.Eside = oldEside;
00840         AT.WorkPointer = w;
00841         if ( EndSort(BHEAD w,0) < 0 ) { LowerSortLevel(); AR.PolyFun = oldpolyfun; goto IllLeft; }
00842         AR.PolyFun = oldpolyfun;
00843         if ( *w == 0 || *(w+w) != 0 ) {
00844             MesPrint("&LHS must be one term");
00845             AC.lhdollarflag = 0;
00846             return(1);
00847         }
00848         LowerSortLevel();
00849         if ( AC.lhdollarflag ) MarkDirty(w,DIRTYFLAG);
00850     }
00851     AT.WorkPointer = w + *w;
00852     AC.DumNum = 0;
00853 /*
00854     Everything is now after OldWork. We can pop the compilerbuffer.
00855     Next test for illegal things like a coefficient
00856     At this point we have:
00857     w = the term of the LHS
00858 */
00859     C->Pointer = C->Buffer + oldcpointer;
00860     C->numrhs = oldnumrhs;
00861     C->numlhs--;
00862
00863     m = w + *w - 3;
00864     AC.vectorlikeLHS = 0;
00865     if ( !error ) {
00866         if ( m[2] != 3 || m[1] != 1 || *m != 1 ) {
00867             if ( *m == 1 && m[1] == 1 && m[2] == -3 ) {
00868                 MinusSign = 1;
00869             }
00870             else {
00871                 MesPrint("&Coefficient in LHS");
00872                 error = 1;
00873                 AC.DumNum = 0;
00874                 *w -= ABS(m[2])-3;
00875             }
00876         }
00877         if ( *w == 7 && w[1] == INDEX && w[3] < 0 ) {
00878             if ( ( AC.idoption & SUBMASK ) != 0 && ( AC.idoption & SUBMASK ) !=
00879                 SUBMULTI ) {
00880                 MesPrint("&Illegal option for substitution of a vector");
00881                 error = 1;
00882             }
00883             AC.DumNum = AM.IndDum;
00884             OldWork[2] = ( OldWork[2] - ( OldWork[2] & SUBMASK ) ) | SUBVECTOR;
00885             c1 = w[3];
00886             /* We overwrite the LHS */
00887             *w++ = INDTOIND;
00888             *w++ = 4;
00889             *w++ = AC.DumNum + WILDOFFSET;
00890             *w++ = 0;
00891             w[0] = 5;
00892             w[1] = VECTOR;
00893             w[2] = 4;
00894             w[3] = c1;
00895             w[4] = AC.DumNum + WILDOFFSET;
00896             OldWork[idhead+1] = w - OldWork - idhead;
00897             AC.vectorlikeLHS = 1;
00898         }
00899         else {
00900             AC.DumNum = 0;
00901             *w -= 3;
00902             i = OldWork[2] & SUBMASK;
00903             m = w + *w;
00904             if ( i == 0 || i == SUBMULTI ) {

```

```

00905         s = w+1;
00906         while ( s < m ) {
00907             if ( *s == SYMBOL ) {
00908                 j = s[1]/2; s += 2;
00909                 while ( --j >= 0 ) {
00910                     if ( ABS(s[1]) > 2*MAXPOWER ) {
00911                         OldWork[2] = ( OldWork[2] - i ) | SUBONCE;
00912                         break;
00913                     }
00914                     s += 2;
00915                 }
00916                 if ( j >= 0 ) break;
00917             }
00918             else if ( *s == DOTPRODUCT ) {
00919                 j = s[1]/3; s += 2;
00920                 while ( --j >= 0 ) {
00921                     if ( ABS(s[2]) > 2*MAXPOWER ) {
00922                         OldWork[2] = ( OldWork[2] - i ) | SUBONCE;
00923                         break;
00924                     }
00925                     else if ( s[1] >= -(2*WILDOFFSET) || s[0] >= -(2*WILDOFFSET) ) {
00926                         OldWork[2] = ( OldWork[2] - i ) | SUBMANY;
00927                         i = SUBMANY;
00928                     }
00929                     s += 3;
00930                 }
00931                 if ( j >= 0 ) break;
00932             }
00933             else {
00934                 OldWork[2] = ( OldWork[2] - i ) | SUBMANY;
00935                 break;
00936             }
00937         }
00938     }
00939     if ( ( OldWork[2] & SUBMASK ) == 0 ) OldWork[2] |= SUBMULTI;
00940 }
00941 if ( ( OldWork[2] & SUBMASK ) == SUBSELECT ) {
00942     /*
00943         Paste the SETSET information after the pattern.
00944         Important note: We will still get function information for the
00945         smart patternmatching after it. To distinguish them we need to have
00946         that SETSET != m*n+1 in which m is the number of words per function
00947         and n the number of functions. Currently (29-may-1997) m = 4.
00948     */
00949     *m++ = SETSET;
00950     *m++ = numsets+2;
00951     s = FirstWork + idhead;
00952     while ( --numsets >= 0 ) *m++ = *s++;
00953 }
00954 else {
00955     m = w + *w;
00956 }
00957 }
00958 /*
00959     We keep the whole thing in OldWork for the moment.
00960     We still have to add the number of the RHS expression.
00961     There is also some opportunity now to be smart about the pattern.
00962     This is needed for complicated wilddcarding with symmetric functions.
00963     We do this in a special routine during compile time to make sure
00964     that we loose as little time as possible (during running) if there
00965     is no need to be smart.
00966 */
00967 *m++ = 0;
00968 OldWork[1] = m - OldWork;
00969 AC.ProtoType = OldWork+idhead;
00970 if ( !error ) {
00971     if ( StudyPattern(OldWork) ) error = 1;
00972 }
00973 AT.WorkPointer = OldWork + OldWork[1];
00974 if ( AC.lhdollarflag ) OldWork[4] |= DOLLARFLAG;
00975 AC.lhdollarflag = 0;
00976 /*
00977     Test whether the id/idold configuration is fine.
00978 */
00979 if ( type == TYPEIDOLD ) {
00980     WORD ci = C->numlhs;
00981     while ( ci >= 1 ) {
00982         if ( C->lhs[ci][0] == TYPEIDNEW ) {
00983             if ( (C->lhs[ci][2] & SUBMASK) == SUBALL ) {
00984                 MesPrint("&Idold/also cannot follow an id,all statement.");
00985                 error = 1;
00986             }
00987             break;
00988         }
00989         else if ( C->lhs[ci][0] == TYPEDETCURDUM ) { ci--; continue; }
00990         else if ( C->lhs[ci][0] == TYPEIDOLD ) { ci--; continue; }
00991         else ci = 0;

```

```

00992     }
00993     if ( ci < 1 ) {
00994         MesPrint("&Idold/also should follow an id/idnew statement.");
00995         error = 1;
00996     }
00997 }
00998 /*
00999 Now the right hand side.
01000 */
01001 if ( type != TYPEIF ) {
01002     if ( ( retcode = CompileAlgebra(inp,RHSIDE,AC.ProtoType) ) < 0 ) error = 1;
01003     else {
01004         AC.ProtoType[2] = retcode;
01005         AC.DumNum = 0;
01006         if ( MinusSign ) { /* Flip the sign of the RHS */
01007             w = C->rhs[retcode];
01008             while ( *w ) { w += *w; w[-1] = -w[-1]; }
01009         }
01010         if ( AC.dumnumflag ) Add2Com(TYPEDETCURDUM)
01011     }
01012 }
01013 /*
01014 Actual adding happens only now after numrhs insertion
01015 */
01016 if ( !error ) { AddNtoL(OldWork[1],OldWork); }
01017 AllDone:
01018 AC.lhdollarflag = 0;
01019 AT.WorkPointer = FirstWork;
01020 return(error);
01021 }
01022
01023 /*
01024 #] CoIdExpression :
01025 #[ CoMultiply :
01026 */
01027
01028 static WORD mularray[13] = { TYPEMULT, SUBEXPSIZE+3, 0, SUBEXPRESSION,
01029     SUBEXPSIZE, 0, 1, 0, 0, 0, 0, 0, 0, 0 };
01030
01031 int CoMultiply(UBYTE *inp)
01032 {
01033     UBYTE *p;
01034     int error = 0, RetCode;
01035     mularray[2] = 0; /* right multiply is default */
01036     while ( *inp == ',' ) inp++;
01037     /* if ( inp[-1] == '-' || inp[-1] == '+' ) inp--; */
01038     p = SkipField(inp,0);
01039     if ( *p ) {
01040         *p = 0;
01041         if ( StrICont(inp,(UBYTE *)"left") == 0 ) mularray[2] = 1;
01042         else if ( StrICont(inp,(UBYTE *)"right") == 0 ) mularray[2] = 0;
01043         else {
01044             MesPrint("&Illegal option in multiply statement or ; forgotten.");
01045             return(1);
01046         }
01047         *p = ',';
01048         inp = p + 1;
01049     }
01050     ClearWildcardNames();
01051     while ( *inp == ',' ) inp++;
01052     AC.ProtoType = mularray+3;
01053     mularray[7] = AC.cbufnum;
01054     if ( ( RetCode = CompileAlgebra(inp,RHSIDE,AC.ProtoType) ) < 0 ) error = 1;
01055     else {
01056         mularray[5] = RetCode;
01057         AddNtoL(SUBEXPSIZE+3,mularray);
01058         if ( AC.dumnumflag ) Add2Com(TYPEDETCURDUM)
01059     }
01060     return(error);
01061 }
01062
01063 /*
01064 #] CoMultiply :
01065 #[ CoFill :
01066
01067 Special additions for tablebase-like tables added 12-aug-2002
01068 */
01069 int CoFill(UBYTE *inp)
01070 {
01071     GETIDENTITY
01072     WORD error = 0, x, xx, funnum, type, *oldwp = AT.WorkPointer;
01073     int i, oldcbufnum = AC.cbufnum, nofill = 0, numover, redef = 0;
01074     WORD *w, *wold, *Tprototype;
01075     UBYTE *p = inp, c, *inpl;
01076     TABLES T = 0, oldT;
01077     LONG newreservation, sum = 0;

```

```

01079     UBYTE *p1, *p2, *p3, *p4, *fake = 0;
01080     int tablestub = 0;
01081     if ( AC.exprfillwarning == 1 ) AC.exprfillwarning = 0;
01082     /*
01083     Read the name of the function and test that it is in the table.
01084     */
01085     p1 = inp;
01086     if ( ( p = SkipAName(inp) ) == 0 ) return(1);
01087     p2 = p;
01088     c = *p; *p = 0;
01089     if ( ( GetVar(inp,&type,&funnum,CFUNCTION,WITHAUTO) == NAMENOTFOUND )
01090     || ( T = functions[funnum].tabl ) == 0 || ( T->numind > 0 && c != '(' ) ) {
01091         MesPrint("%s should be a table with argument(s)",inp);
01092         *p = c; return(1);
01093     }
01094     oldT = T;
01095     *p++ = c;
01096     if ( T->numind == 0 ) {
01097         if ( c == '(' ) {
01098             if ( *p != ')' ) {
01099                 c = *p; *p = 0;
01100                 MesPrint("%s should be a table without arguments",inp);
01101                 *p = c; return(1);
01102             }
01103             else { p++; }
01104         }
01105         else { p--; }
01106         sum = 0;
01107         p3 = p;
01108         goto andagain;
01109     }
01110     w = oldwp;
01111     if ( T->numind < 0 ) { /* Pick up the first index */
01112         ParseSignedNumber(xx,p);
01113         if ( FG.cTable[p[-1]] != 1 || *p != ',' || xx < 1 || ( xx > ( -T->numind - 1 ) ) ) {
01114             MesPrint("&No valid number of table indices in *-table fill statement.");
01115             return(1);
01116         }
01117         *w++ = xx;
01118         p++;
01119     }
01120     else { xx = T->numind; }
01121     for ( sum = 0, i = 0; i < xx; i++ ) {
01122         ParseSignedNumber(x,p);
01123         if ( FG.cTable[p[-1]] != 1 || ( *p != ',' && *p != ')' ) ) {
01124             MesPrint("&Table arguments in fill statement should be numbers");
01125             return(1);
01126         }
01127         if ( T->sparse ) *w++ = x;
01128         else if ( x < T->mm[i].mini || x > T->mm[i].maxi ) {
01129             MesPrint("&Value %d for argument %d of table out of bounds",x,i+1);
01130             error = 1; nofill = 1;
01131         }
01132         else sum += ( x - T->mm[i].mini ) * T->mm[i].size;
01133         if ( *p == ')' ) break;
01134         p++;
01135     }
01136     p3 = p;
01137     if ( T->numind < 0 ) {
01138         for ( ; i < ABS(T->numind)-1; i++ ) *w++ = 0;
01139         xx = -T->numind;
01140     }
01141     if ( *p != ')' || i < ( xx - 1 ) ) {
01142         MesPrint("&Incorrect number of table arguments in fill statement. Should be %d",
01143             ,T->numind);
01144         error = 1; nofill = 1;
01145     }
01146     AT.WorkPointer = w;
01147     if ( T->sparse == 0 ) sum *= TABLEEXTENSION;
01148     andagain:;
01149     AC.cbunum = T->bunum;
01150     if ( T->sparse ) {
01151         i = FindTableTree(T,oldwp,1);
01152         if ( i >= 0 ) {
01153             sum = i + ABS(T->numind);
01154             if ( tablestub == 0 && ( ( T->sparse & 2 ) == 2 ) && ( T->mode != 0 )
01155             && ( AC.vetotablebasefill == 0 ) ) {
01156                 /*
01157                 This redefinition does not need a new stub
01158                 */
01159                 functions[funnum].tabl = T = T->sparse;
01160                 tablestub = 1;
01161                 goto andagain;
01162             }
01163             redef = 1;
01164             goto redef;
01165         }

```



```

01166     if ( T->totind >= T->reserved ) {
01167         if ( T->reserved == 0 ) newreservation = 20;
01168         else newreservation = T->reserved;
01169         while ( T->totind >= newreservation && newreservation < MAXTABLECOMBUF )
01170             newreservation = 2*newreservation;
01171         if ( newreservation > MAXTABLECOMBUF ) newreservation = MAXTABLECOMBUF;
01172         if ( T->totind >= newreservation ) {
01173             MesPrint("@More than %ld elements in sparse table",MAXTABLECOMBUF);
01174             AC.cbufnum = oldcbufnum;
01175             Terminate(-1);
01176         }
01177         wold = (WORD *)Malloc1(newreservation*sizeof(WORD)*
01178                               (ABS(T->numind)+TABLEEXTENSION),"tablepointers");
01179         for ( i = T->reserved*(ABS(T->numind)+TABLEEXTENSION)-1; i >= 0; i-- )
01180             wold[i] = T->tablepointers[i];
01181         if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
01182         T->tablepointers = wold;
01183         T->reserved = newreservation;
01184     }
01185     w = oldwp;
01186     for ( sum = T->totind*(ABS(T->numind)+TABLEEXTENSION), i = 0; i < ABS(T->numind); i++ ) {
01187         T->tablepointers[sum++] = *w++;
01188     }
01189     InsTableTree(T,T->tablepointers+sum-ABS(T->numind));
01190 #if TABLEEXTENSION == 2
01191     T->tablepointers[sum+TABLEEXTENSION-1] = -1; /* New element! */
01192 #else
01193     T->tablepointers[sum+1] = T->bufnum;
01194     T->tablepointers[sum+2] = -1;
01195     T->tablepointers[sum+3] = -1;
01196     T->tablepointers[sum+4] = 0;
01197     T->tablepointers[sum+5] = 0;
01198 #endif
01199 }
01200     else {
01201         if ( !nofill && T->tablepointers[sum] >= 0 ) {
01202 redef::
01203             if ( AC.vetofilling ) nofill = 1;
01204             else {
01205                 Warning("Table element was already defined. New definition will be used");
01206             }
01207         }
01208 #if TABLEEXTENSION == 2
01209         T->tablepointers[sum+TABLEEXTENSION-1] = -1; /* New element! */
01210 #else
01211         T->tablepointers[sum+1] = T->bufnum;
01212         T->tablepointers[sum+2] = -1;
01213         T->tablepointers[sum+3] = -1;
01214         T->tablepointers[sum+4] = 0;
01215         T->tablepointers[sum+5] = 0;
01216 #endif
01217     }
01218     if ( T->numind ) { p++; }
01219     if ( *p != '=' ) {
01220         MesPrint("&Fill statement misses = sign after the table element");
01221         AC.cbufnum = oldcbufnum;
01222         AT.WorkPointer = oldwp;
01223         functions[funnum].tabl = oldT;
01224         return(1);
01225     }
01226     if ( tablestub == 0 && T->mode == 1 && AC.vetotablebasefill == 0 ) {
01227 /*
01228     Here we construct a righthandside from the indices and the wildcards
01229 */
01230         int numfake;
01231         tablestub = 1;
01232         p4 = T->argtail;
01233         while ( *p4 ) p4++;
01234         numfake = (p4-T->argtail)+(p3-p1)+10;
01235
01236         fake = (UBYTE *)Malloc1(numfake*sizeof(UBYTE),"Fill fake rhs");
01237         p = fake;
01238         *p++ = 't'; *p++ = 'b'; *p++ = 'l'; *p++ = '_'; *p++ = '(';
01239         p4 = p1; while ( p4 < p2 ) *p++ = *p4++; *p++ = ',';
01240         p4 = p2+1; while ( p4 < p3 ) *p++ = *p4++;
01241         if ( T->argtail ) {
01242             p4 = T->argtail + 1;
01243             while ( FG.cTable[*p4] == 1 ) p4++;
01244             while ( *p4 ) {
01245                 if ( *p4 == '?' && p[-1] != ',' ) {
01246                     p4++;
01247                     if ( FG.cTable[*p4] == 0 || *p4 == '$' || *p4 == '[' ) {
01248                         p4 = SkipAName(p4);
01249                         if ( *p4 == '[' ) {
01250                             SKIPBRA1(p4);
01251                         }
01252                     }

```

```

01253             else if ( *p4 == '{' ) {
01254                 SKIPBRA2(p4);
01255             }
01256             else if ( *p4 ) { *p++ = *p4++; continue; }
01257         }
01258         else *p++ = *p4++;
01259     }
01260 }
01261 *p++ = ' ';
01262 *p = 0;
01263 inpl = fake;
01264 }
01265 else {
01266     inpl = ++p;
01267 }
01268 c = 0;
01269 /*
01270 Now we have the indices and p points to the rhs.
01271 */
01272 numover = 0;
01273 AC.tablefilling = funnum;
01274 while ( *inpl ) {
01275     p = SkipField(inpl,0);
01276     c = *p; *p = 0;
01277 #ifdef WITHPTHREADS
01278     Tprototype = T->prototype[0];
01279 #else
01280     Tprototype = T->prototype;
01281 #endif
01282     if ( ( i = CompileAlgebra(inpl,RHSIDE,Tprototype) ) < 0 ) { error = 1; i = 0; }
01283     if ( !nofill ) {
01284         T->tablepointers[sum] = i;
01285         T->tablepointers[sum+1] = T->bufnum;
01286     }
01287     AC.DumNum = 0;
01288     *p = c;
01289     if ( T->sparse || c == 0 ) break;
01290     inpl = ++p;
01291 #if ( TABLEEXTENSION == 2 )
01292     sum++;
01293 #else
01294     sum += 2;
01295 #endif
01296     if ( !nofill && T->tablepointers[sum] >= 0 ) numover++;
01297 #if ( TABLEEXTENSION == 2 )
01298     sum++;
01299 #else
01300     sum += TABLEEXTENSION-2;
01301 #endif
01302 }
01303 if ( AC.exprfillwarning == 1 ) {
01304     AC.exprfillwarning = 2;
01305     Warning("Use of expressions and/or $variables in Fill statements is potentially very
dangerous.");
01306 }
01307 AC.tablefilling = 0;
01308 if ( T->sparse && c != 0 ) {
01309     MesPrint("&In sparse tables one can fill only one element at a time");
01310     error = 1;
01311 }
01312 else if ( numover ) {
01313     if ( numover == 1 )
01314         Warning("one element was overwritten. New definition will be used");
01315     else if ( AC.WarnFlag )
01316         MesPrint("&Warning: %d elements were overwritten. New definitions will be used",numover);
01317 }
01318 if ( T->sparse ) {
01319     if ( redef == 0 ) T->totind++;
01320 }
01321 else T->defined++;
01322 /*
01323 NumSets = AC.SetList.numtemp;
01324 NumSetElements = AC.SetElementList.numtemp;
01325 */
01326 if ( fake ) {
01327     M_free(fake,"Fill fake rhs");
01328     fake = 0;
01329     functions[funnum].tabl = T = T->sparse;
01330     p = p3;
01331     goto andagain;
01332 }
01333 AC.cbunum = oldcbunum;
01334 AC.SymChangeFlag = 1;
01335 AT.WorkPointer = oldwp;
01336 functions[funnum].tabl = oldT;
01337 return(error);
01338 }

```

```

01339
01340 /*
01341 #] CoFill :
01342 #[ CoFillExpression :
01343
01344 Syntax: FillExpression table = expression(x1,...,xn);
01345 The arguments should have been bracketed. Each corresponds to one
01346 of the dimensions of the table. Then the bracket with x1^2*x3^4
01347 will fill the (2,0,4) element of the table (if n=3 of course).
01348 Brackets that don't fit will be skipped. It just gives a warning.
01349
01350 New option (13-jul-2005)
01351 Syntax: FillExpression table = expression(f);
01352 The table indices are arguments of the function f which should
01353 have been bracketed before.
01354 */
01355
01356 int CoFillExpression(UBYTE *inp)
01357 {
01358     GETIDENTITY
01359     UBYTE *p, c;
01360     WORD type, funnum, expnum, symnum, numsym = 0, *oldwork = AT.WorkPointer;
01361     WORD *brackets, *term, brasize, *b, *m, *w, *pw, *tstop, zero = 0;
01362     WORD oldcbuf = AC.cbufnum, curelement = 0;
01363     int weneedit, i, j, numzero, pow, numfirst;
01364     TABLES T = 0;
01365     LONG newreservation, numcommu, sum;
01366     POSITION oldposition;
01367     FILEHANDLE *fi;
01368     CBUF *C;
01369     WORD numdummies;
01370
01371     AN.IndDum = AM.IndDum;
01372     if ( ( p = SkipAName(inp) ) == 0 ) return(1);
01373     c = *p; *p = 0;
01374     if ( ( GetVar(inp,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
01375     || ( T = functions[funnum].tabl ) == 0 ) {
01376         MesPrint("%s should be a previously declared table",inp);
01377         *p = c; return(1);
01378     }
01379     *p++ = c;
01380     if ( T->spare ) T = T->spare;
01381     C = cbuf + T->bufnum;
01382     if ( c != '=' ) {
01383         MesPrint("%sNo = sign in FillExpression statement");
01384         return(1);
01385     }
01386     inp = p;
01387     if ( ( p = SkipAName(inp) ) == 0 ) return(1);
01388     c = *p; *p = 0;
01389     if ( ( type = GetName(AC.exprnames,inp,&expnum,NOAUTO) ) == NAMENOTFOUND
01390     || c != '(' || (
01391         Expressions[expnum].status != LOCALEXPRESSION &&
01392         Expressions[expnum].status != SKIPLEXPRESSION &&
01393         Expressions[expnum].status != DROPLEXPRESSION &&
01394         Expressions[expnum].status != GLOBALEXPRESSION &&
01395         Expressions[expnum].status != SKIPGEXPRESSION &&
01396         Expressions[expnum].status != DROPGEXPRESSION ) ) {
01397         MesPrint("%s should be an active expression with arguments",inp);
01398         *p = c; return(1);
01399     }
01400     if ( Expressions[expnum].inmem ) {
01401         MesPrint("%s cannot be used in a FillExpression statement in the same %n\
01402 module that it has been redefined",inp);
01403         *p = c; return(1);
01404     }
01405     *p++ = c;
01406     while ( *p ) {
01407         inp = p;
01408         if ( ( p = SkipAName(inp) ) == 0 ) return(1);
01409         c = *p; *p = 0;
01410
01411         if ( GetVar(inp,&type,&symnum,-1,NOAUTO) == NAMENOTFOUND ) {
01412             MesPrint("%s should be a previously declared symbol or function",inp);
01413             *p = c; return(1);
01414         }
01415         else if ( type == CSYMBOL ) {
01416             *p++ = c;
01417             *AT.WorkPointer++ = symnum;
01418             numsym++;
01419         }
01420         else if ( type == CFUNCTION ) {
01421             numsym = -1;
01422             *p++ = c;
01423             if ( c != ')' ) {
01424                 MesPrint("%sArgument should be a single function or a list of symbols");
01425                 return(1);

```

```

01426         }
01427         symnum += FUNCTION;
01428         *AT.WorkPointer++ = symnum;
01429     }
01430     else {
01431         MesPrint("%&s should be a previously declared symbol or function",inp);
01432         *p = c; return(1);
01433     }
01434     if ( c == ' ' ) break;
01435     if ( c != ',' ) {
01436         MesPrint("%Illegal separator in FillExpression statement");
01437         goto noway;
01438     }
01439 }
01440 if ( *p ) {
01441     MesPrint("%Illegal end of FillExpression statement");
01442     goto noway;
01443 }
01444 /*
01445     We have the number of the table in funnum.
01446     The number of the expression in expnum, the table struct in T
01447     and either the numbers of the symbols in oldwork (there are numsym of them)
01448     or the number of the function in oldwork (just one and numsym = -1).
01449     We don't sort them!!!!
01450 */
01451 if ( ( numsym > 0 ) && ( ABS(T->numind) != numsym ) ) {
01452     MesPrint("%This table needs %d symbols for its array indices");
01453     goto noway;
01454 }
01455 EXCHINOUT
01456 #ifdef WITHMPI
01457 /*
01458     * The workers can't access to the data of the input expression. We need to
01459     * broadcast it to all the workers.
01460 */
01461 PF_BroadcastExpr(&Expressions[expnum], AR.infile);
01462 if ( PF.me == MASTER ) {
01463     /*
01464         * Restore the file position on the master.
01465     */
01466     POSITION pos;
01467     SetEndScratch(AR.infile, &pos);
01468 }
01469 #endif
01470 fi = AR.infile;
01471 if ( fi->handle >= 0 ) {
01472     PUTZERO(oldposition);
01473     SeekFile(fi->handle,&oldposition,SEEK_CUR);
01474     SetScratch(fi,&(Expressions[expnum].onfile));
01475     if ( ISNEGPOS(Expressions[expnum].onfile) ) {
01476         MesPrint("%File error in FillExpression");
01477         BACKINOUT
01478         goto noway;
01479     }
01480 }
01481 else {
01482 /*
01483     Note: Because everything fits inside memory we never get problems
01484     with excessive file sizes.
01485 */
01486     SETBASEPOSITION(oldposition,(UBYTE *) (fi->POfill)-(UBYTE *) (fi->PObuffer));
01487     fi->POfill = (WORD *) ((UBYTE *) (fi->PObuffer) + BASEPOSITION(Expressions[expnum].onfile));
01488 }
01489 pw = AT.WorkPointer;
01490 if ( numsym < 0 ) { brackets = pw + 1; }
01491 else { brackets = pw + numsym; }
01492 brasize = -1; wenedit = 0; /* stands for we need it */
01493 term = (WORD *) ((UBYTE *) (brackets)) + AM.MaxTer;
01494 AT.WorkPointer = (WORD *) ((UBYTE *) (term)) + AM.MaxTer;
01495 AC.cbufnum = T->bufnum;
01496 AC.tablefilling = funnum;
01497 if ( GetTerm(BHEAD term) > 0 ) { /* Skip prototype */
01498     while ( GetTerm(BHEAD term) > 0 ) {
01499         GETSTOP(term,tstop);
01500         w = m = term + 1;
01501         while ( m < tstop && *m != HAAKJE ) m += m[1];
01502         if ( *m != HAAKJE ) {
01503             MesPrint("%Illegal attempt to put an expression without brackets in a table");
01504             BACKINOUT
01505             goto noway;
01506         }
01507         if ( brasize == m - w ) {
01508             b = brackets;
01509             while ( *b == *w && w < m ) { b++; w++; }
01510             if ( w == m ) { /* Same as current bracket. Copy. */
01511                 if ( wenedit ) {
01512                     m += m[1] - 1;

```

```

01513         *m = *term - (m-term);
01514         AddNtoC(AC.cbufnum,*m,m,3);
01515         numdummies = DetCurDum(BHEAD term) - AM.IndDum;
01516         if ( numdummies > T->numdummies ) T->numdummies = numdummies;
01517     }
01518     continue; /* Next term */
01519 }
01520 }
01521 if ( weneedit ) {
01522     AddNtoC(AC.cbufnum,1,&zero,4); /* Terminate old bracket */
01523     numcommu = numcommute(C->rhs[curelement],&(C->NumTerms[curelement]));
01524     C->CanCommu[curelement] = numcommu;
01525 }
01526 b = brackets; w = term + 1;
01527 if ( numsym < 0 ) pw = oldwork + 1;
01528 else pw = oldwork + numsym;
01529 while ( w < m ) *b++ = *w++;
01530 brasize = b - brackets;
01531 /*
01532 Now compute the element. See whether we need it
01533 */
01534 if ( numsym < 0 ) {
01535     WORD *bb, bnum;
01536     if ( *brackets != symnum || brasize != brackets[1] ) {
01537         weneedit = 0; continue; /* Cannot work! */
01538     }
01539 /*
01540 Now count the number of arguments and whether they are numbers
01541 */
01542 b = brackets + FUNHEAD;
01543 bb = brackets+brackets[1];
01544 i = 0;
01545 if ( T->numind < 0 ) {
01546     bnum = b[1]+1;
01547     if ( bnum > -T->numind ) {
01548         weneedit = 0; continue; /* Cannot work! */
01549     }
01550 }
01551 else bnum = T->numind;
01552 while ( b < bb ) {
01553     if ( *b != -SNUMBER ) break;
01554     i++;
01555     b += 2;
01556 }
01557 if ( b < bb || i != bnum ) {
01558     weneedit = 0; continue; /* Cannot work! */
01559 }
01560 }
01561 else if ( brasize > 0 && ( *brackets != SYMBOL
01562 || brackets[1] < brasize || (brackets[1]-2) > numsym*2 ) ) {
01563     weneedit = 0; continue; /* Cannot work! */
01564 }
01565 numzero = 0; sum = 0;
01566 numfirst = 0;
01567 if ( numsym > 0 ) {
01568     for ( i = 0; i < numsym; i++ ) {
01569         if ( brasize > 0 ) {
01570             b = brackets + 2; j = brackets[1]-2;
01571             while ( j > 0 ) {
01572                 if ( *b == oldwork[i] ) break;
01573                 j -= 2; b += 2;
01574             }
01575             if ( j <= 0 ) { /* it was not there */
01576                 numzero++; pow = 0;
01577                 if ( 2*numzero+brackets[1]-2 > numsym*2 ) {
01578                     weneedit = 0; goto nextterm;
01579                 }
01580             }
01581             else pow = b[1];
01582         }
01583         else pow = 0;
01584         if ( T->sparse ) {
01585             if ( T->numind < 0 ) {
01586                 if ( i == 0 ) {
01587                     numfirst = pow;
01588                     if ( pow > -T->numind ) {
01589                         weneedit = 0; goto nextterm;
01590                     }
01591                 }
01592                 else if ( i > pow ) {
01593                     weneedit = 0; goto nextterm;
01594                 }
01595             }
01596             *pw++ = pow;
01597         }
01598         else if ( pow < T->mm[i].mini || pow > T->mm[i].maxi ) {
01599             weneedit = 0; goto nextterm;

```

```

01600         }
01601         else sum += ( pow - T->mm[i].mini ) * T->mm[i].size;
01602     }
01603 }
01604 else {
01605     WORD xx;
01606     b = brackets + FUNHEAD;
01607     sum = 0;
01608 /*
01609     Now scan the arguments of the function.
01610     We did check already the number and type of the arguments.
01611 */
01612     xx = (brackets[1]-FUNHEAD)/2;
01613     for ( i = 0; i < xx; i++ ) {
01614         pow = b[1];
01615         b += 2;
01616         if ( T->sparse ) {
01617             if ( T->numind < 0 ) {
01618                 if ( i == 0 ) {
01619                     numfirst = pow;
01620                     if ( pow >= -T->numind ) {
01621                         weneedit = 0; goto nextterm;
01622                     }
01623                 }
01624                 *pw++ = pow;
01625             }
01626             else if ( pow < T->mm[i].mini || pow > T->mm[i].maxi ) {
01627                 weneedit = 0; goto nextterm;
01628             }
01629             else sum += ( pow - T->mm[i].mini ) * T->mm[i].size;
01630         }
01631     }
01632     if ( T->numind < 0 ) {
01633         for ( i = numfirst+1; i < -T->numind; i++ ) *pw++ = 0;
01634     }
01635     weneedit = 1;
01636     if ( T->sparse ) {
01637         if ( numsym < 0 ) pw = oldwork + 1;
01638         else pw = oldwork + ABS(T->numind);
01639         i = FindTableTree(T,pw,1);
01640         if ( i >= 0 ) {
01641             sum = i+ABS(T->numind);
01642             /*
01643             Wrong!!!!
01644             C->rhs[T->tablepointers[sum]] = C->Pointer;
01645             */
01646             C->Pointer--; /* Back up over the zero */
01647             goto newentry;
01648         }
01649         if ( T->totind >= T->reserved ) {
01650             if ( T->reserved == 0 ) newreservation = 20;
01651             else newreservation = T->reserved;
01652             /*---Copied from Fill-----*/
01653             while ( T->totind >= newreservation && newreservation < MAXTABLECOMBUF )
01654                 newreservation = 2*newreservation;
01655             if ( newreservation > MAXTABLECOMBUF ) newreservation = MAXTABLECOMBUF;
01656             if ( T->totind >= newreservation ) {
01657                 MesPrint("@More than %ld elements in sparse table",MAXTABLECOMBUF);
01658                 AC.cbufnum = oldcbuf;
01659                 AT.WorkPointer = oldwork;
01660                 Terminate(-1);
01661             }
01662             /*---Copied from Fill-----*/
01663             if ( T->totind >= newreservation ) {
01664                 MesPrint("@More than %ld elements in sparse table",MAXTABLECOMBUF);
01665                 AC.cbufnum = oldcbuf;
01666                 AT.WorkPointer = oldwork;
01667                 Terminate(-1);
01668             }
01669             w = (WORD *)Malloca1(newreservation*sizeof(WORD)*
01670                 (ABS(T->numind)+TABLEEXTENSION),"tablepointers");
01671             for ( i = T->reserved*(ABS(T->numind)+TABLEEXTENSION)-1; i >= 0; i-- )
01672                 w[i] = T->tablepointers[i];
01673             if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
01674             T->tablepointers = w;
01675             T->reserved = newreservation;
01676         }
01677         if ( numsym < 0 ) pw = oldwork + 1;
01678         else pw = oldwork + numsym;
01679         for ( sum = T->totind*(ABS(T->numind)+TABLEEXTENSION), i = 0; i < ABS(T->numind); i++
01680     ) {
01681         T->tablepointers[sum++] = *pw++;
01682     }
01683     InsTableTree(T,T->tablepointers+sum-ABS(T->numind));
01684     (T->totind)++;
01685     #if ( TABLEEXTENSION != 2 )

```

```

01686         else {
01687             sum *= TABLEEXTENSION;
01688         }
01689 #endif
01690 /*
01691     Start a new entry. Copy the element.
01692 */
01693     AddRHS(T->bufnum,0);
01694     T->tablepointers[sum] = C->numrhs;
01695 #if ( TABLEEXTENSION == 2 )
01696     T->tablepointers[sum+TABLEEXTENSION-1] = -1;
01697 #else
01698     T->tablepointers[sum+1] = T->bufnum;
01699     T->tablepointers[sum+2] = -1;
01700     T->tablepointers[sum+3] = -1;
01701     T->tablepointers[sum+4] = 0;
01702     T->tablepointers[sum+5] = 0;
01703 #endif
01704 newentry: if ( *m == HAAKJE ) { m += m[1] - 1; }
01705           else m--;
01706           *m = *term - (m-term);
01707           AddNtoC(AC.cbufnum,*m,m,5);
01708           curelement = T->tablepointers[sum];
01709 nextterm:
01710     }
01711     if ( weneedit ) {
01712         AddNtoC(AC.cbufnum,1,&zero,6); /* Terminate old bracket */
01713         numcommu = numcommute(C->rhs[curelement],&(C->NumTerms[curelement]));
01714         C->CanCommu[curelement] = numcommu;
01715     }
01716 }
01717 if ( fi->handle >= 0 ) {
01718     SetScratch(fi,&(oldposition));
01719 }
01720 else {
01721     fi->POfill = (WORD *) ((UBYTE *) (fi->PObuffer) + BASEPOSITION(oldposition));
01722 }
01723 BACKINOUT
01724 AC.cbufnum = oldcbuf;
01725 AC.tablefilling = 0;
01726 AT.WorkPointer = oldwork;
01727 return(0);
01728 noway:
01729 BACKINOUT
01730 AC.cbufnum = oldcbuf;
01731 AC.tablefilling = 0;
01732 AT.WorkPointer = oldwork;
01733 return(1);
01734 }
01735
01736 /*
01737 #] CoFillExpression :
01738 #[ CoPrintTable :
01739
01740 Syntax
01741     PrintTable [+f] [+s] tablename [>[>] file];
01742     All defined elements are written with individual Fill statements.
01743     If a file is specified, the result is written to file only.
01744     The flags of the print statement apply as much as possible.
01745     We make use of the regular write routines.
01746 */
01747
01748 int CoPrintTable(UBYTE *inp)
01749 {
01750     GETIDENTITY
01751     int fflag = 0, sflag = 0, addflag = 0, error = 0, sum, i, j;
01752     UBYTE *filename, *p, c, buffer[100], *s, *oldoutputline = AO.OutputLine;
01753     WORD type, funnum, *expr, *m, num;
01754     TABLES T = 0;
01755     WORD oldSkip = AO.OutSkip, oldMode = AC.OutputMode, oldHandle = AC.LogHandle;
01756     WORD oldType = AO.PrintType, *oldwork = AT.WorkPointer;
01757     UBYTE *oldFill = AO.OutFill, *oldLine = AO.OutputLine;
01758 #ifdef WITHMPI
01759     if ( PF.me != MASTER ) return 0;
01760 #endif
01761 /*
01762     First the flags
01763 */
01764     while ( *inp == '+' ) {
01765         inp++;
01766         if ( *inp == 'f' || *inp == 'F' ) { fflag = 1; inp++; }
01767         else if ( *inp == 's' || *inp == 'S' ) { sflag = PRINTONETERM; inp++; }
01768         else {
01769             MesPrint("%Illegal + option in PrintTable statement");
01770             error = 1; inp++;
01771         }
01772         while ( *inp != ',' && *inp && *inp != '+' ) {

```

```

01773         if ( !error ) {
01774             if ( *inp ) {
01775                 MesPrint("&Illegal + option in PrintTable statement");
01776                 inp++;
01777             }
01778             else {
01779                 MesPrint("&Unfinished PrintTable statement");
01780                 return(1);
01781             }
01782             error = 1;
01783         }
01784         inp++;
01785     }
01786     if ( *inp == ',' ) inp++;
01787 }
01788 /*
01789 Now the name of the table
01790 */
01791 if ( ( p = SkipAName(inp) ) == 0 ) return(1);
01792 c = *p; *p = 0;
01793 if ( ( GetVar(inp,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
01794 || ( T = functions[funnum].tabl ) == 0 ) {
01795     MesPrint("&%s should be a previously declared table",inp);
01796     *p = c; return(1);
01797 }
01798 if ( T->sparse && T->mode == 1 ) T = T->sparse;
01799 *p++ = c;
01800 /*
01801 Check for a filename. Runs to the end of the statement.
01802 */
01803 filename = 0;
01804 if ( c == '>' ) {
01805     if ( *p == '>' ) { addflag = 1; p++; }
01806     filename = p;
01807 }
01808 else filename = 0;
01809
01810 if ( filename ) {
01811     if ( addflag ) AC.LogHandle = OpenAddFile((char *)filename);
01812     else AC.LogHandle = CreateFile((char *)filename);
01813     if ( AC.LogHandle < 0 ) {
01814         MesPrint("&Cannot open file '%s' properly",filename);
01815         error = 1; goto finally;
01816     }
01817     AO.PrintType = PRINTLFILE;
01818 }
01819 else if ( fflag && AC.LogHandle >= 0 ) {
01820     AO.PrintType = PRINTLFILE;
01821 }
01822 AO.OutFill = AO.OutputLine = (UBYTE *)AT.WorkPointer;
01823 AT.WorkPointer += 2*AC.LineLength;
01824
01825 AO.PrintType |= sflag;
01826 AC.OutputMode = 0;
01827 AO.IsBracket = 0;
01828 AO.OutSkip = 0;
01829 AR.DeferFlag = 0;
01830 AC.outsidefun = 1;
01831 if ( AC.LogHandle == oldHandle ) FiniLine();
01832 AO.OutputLine = AO.OutFill = (UBYTE *)Malloc1(AC.LineLength+20,"PrintTable");
01833 AO.OutStop = AO.OutFill + AC.LineLength;
01834 for ( i = 0; i < T->totind; i++ ) {
01835     if ( !T->sparse && T->tablepointers[i*TABLEEXTENSION] < 0 ) continue;
01836     TokenToLine((UBYTE *)"Fill ");
01837     TokenToLine((UBYTE *) (VARNAME(functions,funnum)));
01838     TokenToLine((UBYTE *) "(");
01839     AO.OutSkip = 3;
01840     if ( T->sparse ) {
01841         sum = i * ( T->numind + TABLEEXTENSION );
01842         for ( j = 0; j < T->numind; j++, sum++ ) {
01843             if ( j > 0 ) TokenToLine((UBYTE *)",");
01844             num = T->tablepointers[sum];
01845             s = buffer; s = NumCopy(num,s);
01846             TokenToLine(buffer);
01847         }
01848         expr = cbuf[T->bufnum].rhs[T->tablepointers[sum]];
01849     }
01850     else {
01851         for ( j = 0; j < T->numind; j++ ) {
01852             if ( j > 0 ) {
01853                 TokenToLine((UBYTE *)",");
01854                 num = T->mm[j].mini + ( i % T->mm[j-1].size ) / T->mm[j].size;
01855             }
01856             else {
01857                 num = T->mm[j].mini + i / T->mm[j].size;
01858             }
01859             s = buffer; s = NumCopy(num,s);

```



```

01860         TokenToLine(buffer);
01861     }
01862     expr = cbuf[T->bufnum].rhs[T->tablepointers[TABLEEXTENSION*i]];
01863 }
01864 TOKENTOLINE(" =",")=");
01865 if ( sflag ) {
01866     FiniLine();
01867     if ( AC.OutputSpaces != NOSPACEFORMAT ) TokenToLine((UBYTE *) " ");
01868 }
01869 m = expr;
01870 /*
01871     WORD lbrac, first;
01872     lbrac = 0; first = 1;
01873     while ( *m ) {
01874         if ( WriteTerm(m,&lbrac,first,1,0) ) {
01875             MesPrint("Error while writing table");
01876             error = 1;
01877             goto finally;
01878         }
01879         first = 0;
01880         m += *m;
01881     }
01882     if ( first ) { TOKENTOLINE(" 0","0") }
01883     else if ( lbrac ) { TOKENTOLINE(")","") }
01884 */
01885     while ( *m ) m += *m;
01886     if ( m > expr ) {
01887         if ( WriteExpression(expr,(LONG)(m-expr)) ) { error = 1; goto finally; }
01888         AO.OutSkip = 0;
01889     }
01890     else {
01891         TokenToLine((UBYTE *) "0");
01892     }
01893     TokenToLine((UBYTE *) ";" );
01894     FiniLine();
01895 }
01896 M_free(AO.OutputLine,"PrintTable");
01897 AO.OutputLine = AO.OutFill = oldoutputline;
01898 /*
01899     Reset the file pointers and parameters if any. Close file if needed.
01900 */
01901 finally:
01902     AO.OutSkip = oldSkip;
01903     AC.OutputMode = oldMode;
01904     AC.LogHandle = oldHandle;
01905     AO.PrintType = oldType;
01906     AO.OutFill = oldFill;
01907     AO.OutputLine = oldLine;
01908     AT.WorkPointer = oldwork;
01909     AC.outsidefun = 0;
01910     return(error);
01911 }
01912
01913 /*
01914     #] CoPrintTable :
01915     #[ CoAssign :
01916
01917     This statement has an easy syntax:
01918         $name = expression
01919 */
01920
01921 static WORD AssignLHS[14] = { TYPEASSIGN, 3+SUBEXPSIZE, 0,
01922                             SUBEXPRESSION, SUBEXPSIZE, 0, 1, 0, 0,0,0,0,0 };
01923
01924 int CoAssign(UBYTE *inp)
01925 {
01926     int error = 0, retcode;
01927     UBYTE *name, c;
01928     WORD number;
01929     if ( *inp != '$' ) {
01930 nolhs: MesPrint("&assign statement should have a dollar variable in the LHS");
01931         return(1);
01932     }
01933     inp++; name = inp;
01934     if ( FG.cTable[*inp] != 0 ) goto nolhs;
01935     while ( FG.cTable[*inp] < 2 ) inp++;
01936     if ( AP.PreAssignFlag == 2 ) {
01937         if ( *inp == '_' ) inp++;
01938     }
01939     if ( ( *inp == ',' && inp[1] != '=' ) && ( *inp != '=' ) ) {
01940         MesPrint("&assign statement should have only a dollar variable in the LHS");
01941         return(1);
01942     }
01943     c = *inp;
01944     *inp = 0;
01945     if ( GetName(AC.dollarnames,name,&number,NOAUTO) == NAMENOTFOUND ) {
01946         number = AddDollar(name,DOLUNDEFINED,0,0);

```

```

01947     }
01948     *inp = c;
01949     if ( c == ',' ) inp++;
01950     *inp++ = '=';
01951     if ( *inp == ',' ) inp++;
01952 /*
01953     Fake a Prototype and read the RHS
01954 */
01955     AssignLHS[7] = AC.cbunum;
01956     retcode = CompileAlgebra(inp,RHSIDE,(AssignLHS+3));
01957     if ( retcode < 0 ) error = 1;
01958     AC.DumNum = 0;
01959 /*
01960     Now add the LHS
01961 */
01962     AssignLHS[2] = number;
01963     AssignLHS[5] = retcode;
01964     AddNtoL(AssignLHS[1],AssignLHS);
01965 /*
01966     Add to the list of potentially modified dollars (for PARALLEL)
01967 */
01968     AddPotModdollar(number);
01969     return(error);
01970 }
01971
01972 /*
01973     #] CoAssign :
01974     #[ CoDeallocateTable :
01975
01976     Syntax: DeallocateTable tablename(s);
01977     Should work only for sparse tables.
01978     Action: Cleans all definitions of elements of a table as if there have
01979            never been any fill statements.
01980 */
01981
01982 int CoDeallocateTable(UBYTE *inp)
01983 {
01984     UBYTE *p, c;
01985     TABLES T = 0;
01986     WORD type, funnum, i;
01987     c = *inp;
01988     while ( c ) {
01989         while ( *inp == ',' ) inp++;
01990         if ( *inp == 0 ) break;
01991         if ( ( p = SkipAName(inp) ) == 0 ) return(1);
01992         c = *p; *p = 0;
01993         if ( ( GetVar(inp,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
01994             || ( T = functions[funnum].tabl ) == 0 ) {
01995             MesPrint("%s should be a previously declared table",inp);
01996             *p = c; return(1);
01997         }
01998         if ( T->sparse == 0 ) {
01999             MesPrint("%s should be a sparse table",inp);
02000             *p = c; return(1);
02001         }
02002         if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
02003         ClearTableTree(T);
02004         for ( i = 0; i < T->buffersfill; i++ ) { /* was <= */
02005             finishcbuf(T->buffers[i]);
02006         }
02007         T->bufnum = inicbufs();
02008         T->buffersfill = 0;
02009         T->buffers[T->buffersfill++] = T->bufnum;
02010         T->tablepointers = 0;
02011         T->boomlijst = 0;
02012         T->totind = 0;
02013         T->reserved = 0;
02014
02015         if ( T->sparse ) {
02016             TABLES TT = T->sparse;
02017             if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
02018             ClearTableTree(TT);
02019             for ( i = 0; i < TT->buffersfill; i++ ) { /* was <= */
02020                 finishcbuf(TT->buffers[i]);
02021             }
02022             TT->bufnum = inicbufs();
02023             TT->buffersfill = 0;
02024             TT->buffers[T->buffersfill++] = T->bufnum;
02025             TT->tablepointers = 0;
02026             TT->boomlijst = 0;
02027             TT->totind = 0;
02028             TT->reserved = 0;
02029         }
02030         *p++ = c;
02031         inp = p;
02032     }
02033     return(0);

```

```

02034 }
02035
02036 /*
02037  #] CoDeallocateTable :
02038  #[ CoFactorCache :
02039 */
02040 /*
02050 int CoFactorCache(UBYTE *inp)
02051 {
02052     Code to be added in due time
02053     We need to read 'expression', get its terms through Generator and sort them.
02054     We store the result in the Workspace in argument notation.
02055     This will be argin.
02056     Then we do the same with the sequence of factors. They formargout.
02057     The whole is put in the buffer with the call
02058         InsertArg(BHEAD argin,argout,1)
02059     return(0);
02060 }
02061 */
02062
02063 /*
02064  #] CoFactorCache :
02065 */

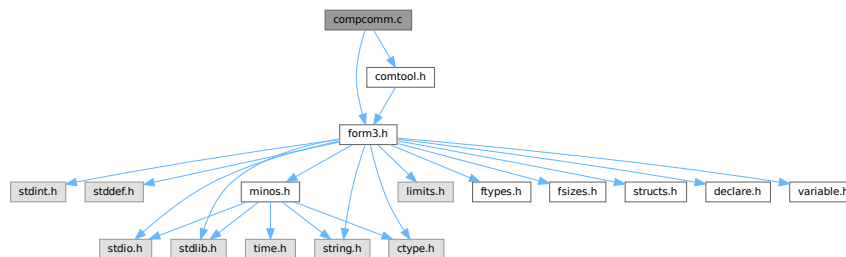
```

## 4.9 compcomm.c File Reference

```
#include "form3.h"
```

```
#include "comtool.h"
```

Include dependency graph for compcomm.c:



### Functions

- int [CoFormat](#) (UBYTE \*s)
- int [CoCollect](#) (UBYTE \*s)
- int [setonoff](#) (UBYTE \*s, int \*flag, int onvalue, int offvalue)
- int [CoCompress](#) (UBYTE \*s)
- int [CoFlags](#) (UBYTE \*s, int value)
- int [CoOff](#) (UBYTE \*s)
- int [CoOn](#) (UBYTE \*s)
- int [CoInsideFirst](#) (UBYTE \*s)
- int [CoProperCount](#) (UBYTE \*s)
- int [CoDelete](#) (UBYTE \*s)
- int [CoKeep](#) (UBYTE \*s)
- int [CoFixIndex](#) (UBYTE \*s)
- int [CoMetric](#) (UBYTE \*s)
- int [DoPrint](#) (UBYTE \*s, int par)
- int [CoPrint](#) (UBYTE \*s)
- int [CoPrintB](#) (UBYTE \*s)

- int [CoNPrint](#) (UBYTE \*s)
- int [CoPushHide](#) (UBYTE \*s)
- int [CoPopHide](#) (UBYTE \*s)
- int [SetExprCases](#) (int par, int setunset, int val)
- int [SetExpr](#) (UBYTE \*s, int setunset, int par)
- int [CoDrop](#) (UBYTE \*s)
- int [CoNoDrop](#) (UBYTE \*s)
- int [CoSkip](#) (UBYTE \*s)
- int [CoNoSkip](#) (UBYTE \*s)
- int [CoHide](#) (UBYTE \*inp)
- int [CoIntoHide](#) (UBYTE \*inp)
- int [CoNoIntoHide](#) (UBYTE \*inp)
- int [CoNoHide](#) (UBYTE \*inp)
- int [CoUnHide](#) (UBYTE \*inp)
- int [CoNoUnHide](#) (UBYTE \*inp)
- void [AddToCom](#) (int n, WORD \*array)
- int [AddComString](#) (int n, WORD \*array, UBYTE \*thestring, int par)
- int [Add2ComStrings](#) (int n, WORD \*array, UBYTE \*string1, UBYTE \*string2)
- int [CoDiscard](#) (UBYTE \*s)
- int [CoContract](#) (UBYTE \*s)
- int [CoGoTo](#) (UBYTE \*inp)
- int [CoLabel](#) (UBYTE \*inp)
- int [DoArgument](#) (UBYTE \*s, int par)
- int [CoArgument](#) (UBYTE \*s)
- int [CoEndArgument](#) (UBYTE \*s)
- int [CoInside](#) (UBYTE \*s)
- int [CoEndInside](#) (UBYTE \*s)
- int [CoNormalize](#) (UBYTE \*s)
- int [CoMakeInteger](#) (UBYTE \*s)
- int [CoSplitArg](#) (UBYTE \*s)
- int [CoSplitFirstArg](#) (UBYTE \*s)
- int [CoSplitLastArg](#) (UBYTE \*s)
- int [CoFactArg](#) (UBYTE \*s)
- int [DoSymmetrize](#) (UBYTE \*s, int par)
- int [CoSymmetrize](#) (UBYTE \*s)
- int [CoAntiSymmetrize](#) (UBYTE \*s)
- int [CoCycleSymmetrize](#) (UBYTE \*s)
- int [CoRCycleSymmetrize](#) (UBYTE \*s)
- int [CoWrite](#) (UBYTE \*s)
- int [CoNWrite](#) (UBYTE \*s)
- int [CoRatio](#) (UBYTE \*s)
- int [CoRedefine](#) (UBYTE \*s)
- int [CoRenumber](#) (UBYTE \*s)
- int [CoSum](#) (UBYTE \*s)
- int [CoToTensor](#) (UBYTE \*s)
- int [CoToVector](#) (UBYTE \*s)
- int [CoTrace4](#) (UBYTE \*s)
- int [CoTraceN](#) (UBYTE \*s)
- int [CoChisholm](#) (UBYTE \*s)
- int [DoChain](#) (UBYTE \*s, int option)
- int [CoChainin](#) (UBYTE \*s)
- int [CoChainout](#) (UBYTE \*s)
- int [CoExit](#) (UBYTE \*s)
- int [CoInParallel](#) (UBYTE \*s)
- int [CoNotInParallel](#) (UBYTE \*s)

- int [DoInParallel](#) (UBYTE \*s, int par)
- int [CoInExpression](#) (UBYTE \*s)
- int [CoEndInExpression](#) (UBYTE \*s)
- int [CoSetExitFlag](#) (UBYTE \*s)
- int [CoTryReplace](#) (UBYTE \*p)
- int [CoModulus](#) (UBYTE \*inp)
- int [CoRepeat](#) (UBYTE \*inp)
- int [CoEndRepeat](#) (UBYTE \*inp)
- int [DoBrackets](#) (UBYTE \*inp, int par)
- int [CoBracket](#) (UBYTE \*inp)
- int [CoAntiBracket](#) (UBYTE \*inp)
- int [CoMultiBracket](#) (UBYTE \*inp)
- WORD \* [CountComp](#) (UBYTE \*inp, WORD \*to)
- int [Colf](#) (UBYTE \*inp)
- int [CoElse](#) (UBYTE \*p)
- int [CoElseif](#) (UBYTE \*inp)
- int [CoEndIf](#) (UBYTE \*inp)
- int [CoWhile](#) (UBYTE \*inp)
- int [CoEndWhile](#) (UBYTE \*inp)
- int [DoFindLoop](#) (UBYTE \*inp, int mode)
- int [CoFindLoop](#) (UBYTE \*inp)
- int [CoReplaceLoop](#) (UBYTE \*inp)
- int [CoFunPowers](#) (UBYTE \*inp)
- int [CoUnitTrace](#) (UBYTE \*s)
- int [CoTerm](#) (UBYTE \*s)
- int [CoEndTerm](#) (UBYTE \*s)
- int [CoSort](#) (UBYTE \*s)
- int [CoPolyFun](#) (UBYTE \*s)
- int [CoPolyRatFun](#) (UBYTE \*s)
- int [CoMerge](#) (UBYTE \*inp)
- int [CoStuffle](#) (UBYTE \*inp)
- int [CoProcessBucket](#) (UBYTE \*s)
- int [CoThreadBucket](#) (UBYTE \*s)
- int [DoArgPlode](#) (UBYTE \*s, int par)
- int [CoArgExplode](#) (UBYTE \*s)
- int [CoArgImplode](#) (UBYTE \*s)
- int [CoClearTable](#) (UBYTE \*s)
- int [CoDenominators](#) (UBYTE \*s)
- int [CoDropCoefficient](#) (UBYTE \*s)
- int [CoDropSymbols](#) (UBYTE \*s)
- int [CoToPolynomial](#) (UBYTE \*inp)
- int [CoFromPolynomial](#) (UBYTE \*inp)
- int [CoArgToExtraSymbol](#) (UBYTE \*s)
- int [CoExtraSymbols](#) (UBYTE \*inp)
- WORD \* [GetIfDollarFactor](#) (UBYTE \*\*inp, WORD \*w)
- UBYTE \* [GetDoParam](#) (UBYTE \*inp, WORD \*\*wp, int par)
- int [CoDo](#) (UBYTE \*inp)
- int [CoEndDo](#) (UBYTE \*inp)
- int [CoFactDollar](#) (UBYTE \*inp)
- int [CoFactorize](#) (UBYTE \*s)
- int [CoNFactorize](#) (UBYTE \*s)
- int [CoUnFactorize](#) (UBYTE \*s)
- int [CoNUnFactorize](#) (UBYTE \*s)
- int [DoFactorize](#) (UBYTE \*s, int par)
- int [CoOptimizeOption](#) (UBYTE \*s)

- int [CoPutInside](#) (UBYTE \*inp)
- int [CoAntiPutInside](#) (UBYTE \*inp)
- int [DoPutInside](#) (UBYTE \*inp, int par)
- int [CoSwitch](#) (UBYTE \*s)
- int [CoCase](#) (UBYTE \*s)
- int [CoBreak](#) (UBYTE \*s)
- int [CoDefault](#) (UBYTE \*s)
- int [CoEndSwitch](#) (UBYTE \*s)
- int [CoSetUserFlag](#) (UBYTE \*s)
- int [CoClearUserFlag](#) (UBYTE \*s)
- int [CoCreateAllLoops](#) (UBYTE \*s)
- int [CoCreateAllPaths](#) (UBYTE \*s)
- int [CoCreateAll](#) (UBYTE \*s)

### 4.9.1 Detailed Description

Compiler routines for most statements that don't involve algebraic expressions. Exceptions: all routines involving declarations are in the file [names.c](#) When making new statements one can add the compiler routines here and have a look whether there is already a routine that is similar. In that case one can make a copy and modify it.

Definition in file [compcomm.c](#).

### 4.9.2 Function Documentation

#### 4.9.2.1 CoFormat()

```
int CoFormat (
    UBYTE * s )
```

Definition at line [153](#) of file [compcomm.c](#).

#### 4.9.2.2 CoCollect()

```
int CoCollect (
    UBYTE * s )
```

Definition at line [428](#) of file [compcomm.c](#).

#### 4.9.2.3 setonoff()

```
int setonoff (
    UBYTE * s,
    int * flag,
    int onvalue,
    int offvalue )
```

Definition at line [490](#) of file [compcomm.c](#).

#### 4.9.2.4 CoCompress()

```
int CoCompress (
    UBYTE * s )
```

Definition at line 506 of file [compcomm.c](#).

#### 4.9.2.5 CoFlags()

```
int CoFlags (
    UBYTE * s,
    int value )
```

Definition at line 555 of file [compcomm.c](#).

#### 4.9.2.6 CoOff()

```
int CoOff (
    UBYTE * s )
```

Definition at line 590 of file [compcomm.c](#).

#### 4.9.2.7 CoOn()

```
int CoOn (
    UBYTE * s )
```

Definition at line 656 of file [compcomm.c](#).

#### 4.9.2.8 CoInsideFirst()

```
int CoInsideFirst (
    UBYTE * s )
```

Definition at line 928 of file [compcomm.c](#).

#### 4.9.2.9 CoProperCount()

```
int CoProperCount (
    UBYTE * s )
```

Definition at line 935 of file [compcomm.c](#).

#### 4.9.2.10 CoDelete()

```
int CoDelete (
    UBYTE * s )
```

Definition at line 942 of file [compcomm.c](#).

#### 4.9.2.11 CoKeep()

```
int CoKeep (
    UBYTE * s )
```

Definition at line 987 of file [compcomm.c](#).

#### 4.9.2.12 CoFixIndex()

```
int CoFixIndex (
    UBYTE * s )
```

Definition at line 999 of file [compcomm.c](#).

#### 4.9.2.13 CoMetric()

```
int CoMetric (
    UBYTE * s )
```

Definition at line 1033 of file [compcomm.c](#).

#### 4.9.2.14 DoPrint()

```
int DoPrint (
    UBYTE * s,
    int par )
```

Definition at line 1041 of file [compcomm.c](#).

#### 4.9.2.15 CoPrint()

```
int CoPrint (
    UBYTE * s )
```

Definition at line 1257 of file [compcomm.c](#).

#### 4.9.2.16 CoPrintB()

```
int CoPrintB (
    UBYTE * s )
```

Definition at line 1264 of file [compcomm.c](#).

#### 4.9.2.17 CoNPrint()

```
int CoNPrint (
    UBYTE * s )
```

Definition at line 1271 of file [compcomm.c](#).



#### 4.9.2.18 CoPushHide()

```
int CoPushHide (
    UBYTE * s )
```

Definition at line 1278 of file [compcomm.c](#).

#### 4.9.2.19 CoPopHide()

```
int CoPopHide (
    UBYTE * s )
```

Definition at line 1322 of file [compcomm.c](#).

#### 4.9.2.20 SetExprCases()

```
int SetExprCases (
    int par,
    int setunset,
    int val )
```

Definition at line 1357 of file [compcomm.c](#).

#### 4.9.2.21 SetExpr()

```
int SetExpr (
    UBYTE * s,
    int setunset,
    int par )
```

Definition at line 1512 of file [compcomm.c](#).

#### 4.9.2.22 CoDrop()

```
int CoDrop (
    UBYTE * s )
```

Definition at line 1557 of file [compcomm.c](#).

#### 4.9.2.23 CoNoDrop()

```
int CoNoDrop (
    UBYTE * s )
```

Definition at line 1564 of file [compcomm.c](#).

#### 4.9.2.24 CoSkip()

```
int CoSkip (
    UBYTE * s )
```

Definition at line 1571 of file [compcomm.c](#).

#### 4.9.2.25 CoNoSkip()

```
int CoNoSkip (
    UBYTE * s )
```

Definition at line 1578 of file [compcomm.c](#).

#### 4.9.2.26 CoHide()

```
int CoHide (
    UBYTE * inp )
```

Definition at line 1585 of file [compcomm.c](#).

#### 4.9.2.27 CoIntoHide()

```
int CoIntoHide (
    UBYTE * inp )
```

Definition at line 1603 of file [compcomm.c](#).

#### 4.9.2.28 CoNoIntoHide()

```
int CoNoIntoHide (
    UBYTE * inp )
```

Definition at line 1621 of file [compcomm.c](#).

#### 4.9.2.29 CoNoHide()

```
int CoNoHide (
    UBYTE * inp )
```

Definition at line 1628 of file [compcomm.c](#).

#### 4.9.2.30 CoUnHide()

```
int CoUnHide (
    UBYTE * inp )
```

Definition at line 1635 of file [compcomm.c](#).

#### 4.9.2.31 CoNoUnHide()

```
int CoNoUnHide (
    UBYTE * inp )
```

Definition at line 1642 of file [compcomm.c](#).

#### 4.9.2.32 AddToCom()

```
void AddToCom (
    int n,
    WORD * array )
```

Definition at line 1649 of file [compcomm.c](#).

#### 4.9.2.33 AddComString()

```
int AddComString (
    int n,
    WORD * array,
    UBYTE * thestring,
    int par )
```

Definition at line 1664 of file [compcomm.c](#).

#### 4.9.2.34 Add2ComStrings()

```
int Add2ComStrings (
    int n,
    WORD * array,
    UBYTE * string1,
    UBYTE * string2 )
```

Definition at line 1723 of file [compcomm.c](#).

#### 4.9.2.35 CoDiscard()

```
int CoDiscard (
    UBYTE * s )
```

Definition at line 1764 of file [compcomm.c](#).

#### 4.9.2.36 CoContract()

```
int CoContract (
    UBYTE * s )
```

Definition at line 1787 of file [compcomm.c](#).

#### 4.9.2.37 CoGoTo()

```
int CoGoTo (
    UBYTE * inp )
```

Definition at line 1816 of file [compcomm.c](#).

#### 4.9.2.38 CoLabel()

```
int CoLabel (
    UBYTE * inp )
```

Definition at line 1835 of file [compcomm.c](#).

#### 4.9.2.39 DoArgument()

```
int DoArgument (
    UBYTE * s,
    int par )
```

Definition at line 1862 of file [compcomm.c](#).

#### 4.9.2.40 CoArgument()

```
int CoArgument (
    UBYTE * s )
```

Definition at line 2096 of file [compcomm.c](#).

#### 4.9.2.41 CoEndArgument()

```
int CoEndArgument (
    UBYTE * s )
```

Definition at line 2103 of file [compcomm.c](#).

#### 4.9.2.42 CoInside()

```
int CoInside (
    UBYTE * s )
```

Definition at line 2129 of file [compcomm.c](#).

#### 4.9.2.43 CoEndInside()

```
int CoEndInside (
    UBYTE * s )
```

Definition at line 2136 of file [compcomm.c](#).

#### 4.9.2.44 CoNormalize()

```
int CoNormalize (
    UBYTE * s )
```

Definition at line 2162 of file [compcomm.c](#).

#### 4.9.2.45 CoMakeInteger()

```
int CoMakeInteger (
    UBYTE * s )
```

Definition at line 2169 of file [compcomm.c](#).

#### 4.9.2.46 CoSplitArg()

```
int CoSplitArg (
    UBYTE * s )
```

Definition at line 2176 of file [compcomm.c](#).

#### 4.9.2.47 CoSplitFirstArg()

```
int CoSplitFirstArg (
    UBYTE * s )
```

Definition at line 2183 of file [compcomm.c](#).

#### 4.9.2.48 CoSplitLastArg()

```
int CoSplitLastArg (
    UBYTE * s )
```

Definition at line 2190 of file [compcomm.c](#).

#### 4.9.2.49 CoFactArg()

```
int CoFactArg (
    UBYTE * s )
```

Definition at line 2197 of file [compcomm.c](#).

#### 4.9.2.50 DoSymmetrize()

```
int DoSymmetrize (
    UBYTE * s,
    int par )
```

Definition at line 2217 of file [compcomm.c](#).

#### 4.9.2.51 CoSymmetrize()

```
int CoSymmetrize (  
    UBYTE * s )
```

Definition at line [2337](#) of file [compcomm.c](#).

#### 4.9.2.52 CoAntiSymmetrize()

```
int CoAntiSymmetrize (  
    UBYTE * s )
```

Definition at line [2344](#) of file [compcomm.c](#).

#### 4.9.2.53 CoCycleSymmetrize()

```
int CoCycleSymmetrize (  
    UBYTE * s )
```

Definition at line [2351](#) of file [compcomm.c](#).

#### 4.9.2.54 CoRCycleSymmetrize()

```
int CoRCycleSymmetrize (  
    UBYTE * s )
```

Definition at line [2358](#) of file [compcomm.c](#).

#### 4.9.2.55 CoWrite()

```
int CoWrite (  
    UBYTE * s )
```

Definition at line [2365](#) of file [compcomm.c](#).

#### 4.9.2.56 CoNWrite()

```
int CoNWrite (  
    UBYTE * s )
```

Definition at line [2390](#) of file [compcomm.c](#).

#### 4.9.2.57 CoRatio()

```
int CoRatio (  
    UBYTE * s )
```

Definition at line [2417](#) of file [compcomm.c](#).

#### 4.9.2.58 CoRedefine()

```
int CoRedefine (
    UBYTE * s )
```

Definition at line [2457](#) of file [compcomm.c](#).

#### 4.9.2.59 CoRenumber()

```
int CoRenumber (
    UBYTE * s )
```

Definition at line [2562](#) of file [compcomm.c](#).

#### 4.9.2.60 CoSum()

```
int CoSum (
    UBYTE * s )
```

Definition at line [2583](#) of file [compcomm.c](#).

#### 4.9.2.61 CoToTensor()

```
int CoToTensor (
    UBYTE * s )
```

Definition at line [2703](#) of file [compcomm.c](#).

#### 4.9.2.62 CoToVector()

```
int CoToVector (
    UBYTE * s )
```

Definition at line [2871](#) of file [compcomm.c](#).

#### 4.9.2.63 CoTrace4()

```
int CoTrace4 (
    UBYTE * s )
```

Definition at line [2941](#) of file [compcomm.c](#).

#### 4.9.2.64 CoTraceN()

```
int CoTraceN (
    UBYTE * s )
```

Definition at line [3028](#) of file [compcomm.c](#).

#### 4.9.2.65 CoChisholm()

```
int CoChisholm (  
    UBYTE * s )
```

Definition at line 3088 of file [compcomm.c](#).

#### 4.9.2.66 DoChain()

```
int DoChain (  
    UBYTE * s,  
    int option )
```

Definition at line 3170 of file [compcomm.c](#).

#### 4.9.2.67 CoChainin()

```
int CoChainin (  
    UBYTE * s )
```

Definition at line 3214 of file [compcomm.c](#).

#### 4.9.2.68 CoChainout()

```
int CoChainout (  
    UBYTE * s )
```

Definition at line 3226 of file [compcomm.c](#).

#### 4.9.2.69 CoExit()

```
int CoExit (  
    UBYTE * s )
```

Definition at line 3236 of file [compcomm.c](#).

#### 4.9.2.70 CoInParallel()

```
int CoInParallel (  
    UBYTE * s )
```

Definition at line 3263 of file [compcomm.c](#).

#### 4.9.2.71 CoNotInParallel()

```
int CoNotInParallel (  
    UBYTE * s )
```

Definition at line 3273 of file [compcomm.c](#).



#### 4.9.2.72 DoInParallel()

```
int DoInParallel (
    UBYTE * s,
    int par )
```

Definition at line 3288 of file [compcomm.c](#).

#### 4.9.2.73 CoInExpression()

```
int CoInExpression (
    UBYTE * s )
```

Definition at line 3357 of file [compcomm.c](#).

#### 4.9.2.74 CoEndInExpression()

```
int CoEndInExpression (
    UBYTE * s )
```

Definition at line 3410 of file [compcomm.c](#).

#### 4.9.2.75 CoSetExitFlag()

```
int CoSetExitFlag (
    UBYTE * s )
```

Definition at line 3436 of file [compcomm.c](#).

#### 4.9.2.76 CoTryReplace()

```
int CoTryReplace (
    UBYTE * p )
```

Definition at line 3450 of file [compcomm.c](#).

#### 4.9.2.77 CoModulus()

```
int CoModulus (
    UBYTE * inp )
```

Definition at line 3553 of file [compcomm.c](#).

#### 4.9.2.78 CoRepeat()

```
int CoRepeat (
    UBYTE * inp )
```

Definition at line 3691 of file [compcomm.c](#).

#### 4.9.2.79 CoEndRepeat()

```
int CoEndRepeat (
    UBYTE * inp )
```

Definition at line 3714 of file [compcomm.c](#).

#### 4.9.2.80 DoBrackets()

```
int DoBrackets (
    UBYTE * inp,
    int par )
```

Definition at line 3752 of file [compcomm.c](#).

#### 4.9.2.81 CoBracket()

```
int CoBracket (
    UBYTE * inp )
```

Definition at line 3879 of file [compcomm.c](#).

#### 4.9.2.82 CoAntiBracket()

```
int CoAntiBracket (
    UBYTE * inp )
```

Definition at line 3887 of file [compcomm.c](#).

#### 4.9.2.83 CoMultiBracket()

```
int CoMultiBracket (
    UBYTE * inp )
```

Definition at line 3898 of file [compcomm.c](#).

#### 4.9.2.84 CountComp()

```
WORD * CountComp (
    UBYTE * inp,
    WORD * to )
```

Definition at line 4030 of file [compcomm.c](#).

#### 4.9.2.85 Colf()

```
int CoIf (
    UBYTE * inp )
```

Definition at line 4220 of file [compcomm.c](#).

#### 4.9.2.86 CoElse()

```
int CoElse (
    UBYTE * p )
```

Definition at line 4862 of file [compcomm.c](#).

#### 4.9.2.87 CoElseIf()

```
int CoElseIf (
    UBYTE * inp )
```

Definition at line 4889 of file [compcomm.c](#).

#### 4.9.2.88 CoEndIf()

```
int CoEndIf (
    UBYTE * inp )
```

Definition at line 4916 of file [compcomm.c](#).

#### 4.9.2.89 CoWhile()

```
int CoWhile (
    UBYTE * inp )
```

Definition at line 4961 of file [compcomm.c](#).

#### 4.9.2.90 CoEndWhile()

```
int CoEndWhile (
    UBYTE * inp )
```

Definition at line 4982 of file [compcomm.c](#).

#### 4.9.2.91 DoFindLoop()

```
int DoFindLoop (
    UBYTE * inp,
    int mode )
```

Definition at line 5010 of file [compcomm.c](#).

#### 4.9.2.92 CoFindLoop()

```
int CoFindLoop (
    UBYTE * inp )
```

Definition at line 5127 of file [compcomm.c](#).

#### 4.9.2.93 CoReplaceLoop()

```
int CoReplaceLoop (
    UBYTE * inp )
```

Definition at line [5135](#) of file [compcomm.c](#).

#### 4.9.2.94 CoFunPowers()

```
int CoFunPowers (
    UBYTE * inp )
```

Definition at line [5155](#) of file [compcomm.c](#).

#### 4.9.2.95 CoUnitTrace()

```
int CoUnitTrace (
    UBYTE * s )
```

Definition at line [5182](#) of file [compcomm.c](#).

#### 4.9.2.96 CoTerm()

```
int CoTerm (
    UBYTE * s )
```

Definition at line [5217](#) of file [compcomm.c](#).

#### 4.9.2.97 CoEndTerm()

```
int CoEndTerm (
    UBYTE * s )
```

Definition at line [5265](#) of file [compcomm.c](#).

#### 4.9.2.98 CoSort()

```
int CoSort (
    UBYTE * s )
```

Definition at line [5292](#) of file [compcomm.c](#).

#### 4.9.2.99 CoPolyFun()

```
int CoPolyFun (
    UBYTE * s )
```

Definition at line [5331](#) of file [compcomm.c](#).

#### 4.9.2.100 CoPolyRatFun()

```
int CoPolyRatFun (
    UBYTE * s )
```

Definition at line 5378 of file [compcomm.c](#).

#### 4.9.2.101 CoMerge()

```
int CoMerge (
    UBYTE * inp )
```

Definition at line 5548 of file [compcomm.c](#).

#### 4.9.2.102 CoStuffle()

```
int CoStuffle (
    UBYTE * inp )
```

Definition at line 5604 of file [compcomm.c](#).

#### 4.9.2.103 CoProcessBucket()

```
int CoProcessBucket (
    UBYTE * s )
```

Definition at line 5659 of file [compcomm.c](#).

#### 4.9.2.104 CoThreadBucket()

```
int CoThreadBucket (
    UBYTE * s )
```

Definition at line 5677 of file [compcomm.c](#).

#### 4.9.2.105 DoArgPlode()

```
int DoArgPlode (
    UBYTE * s,
    int par )
```

Definition at line 5707 of file [compcomm.c](#).

#### 4.9.2.106 CoArgExplode()

```
int CoArgExplode (
    UBYTE * s )
```

Definition at line 5763 of file [compcomm.c](#).

#### 4.9.2.107 CoArgImplode()

```
int CoArgImplode (
    UBYTE * s )
```

Definition at line 5770 of file [compcomm.c](#).

#### 4.9.2.108 CoClearTable()

```
int CoClearTable (
    UBYTE * s )
```

Definition at line 5777 of file [compcomm.c](#).

#### 4.9.2.109 CoDenominators()

```
int CoDenominators (
    UBYTE * s )
```

Definition at line 5859 of file [compcomm.c](#).

#### 4.9.2.110 CoDropCoefficient()

```
int CoDropCoefficient (
    UBYTE * s )
```

Definition at line 5888 of file [compcomm.c](#).

#### 4.9.2.111 CoDropSymbols()

```
int CoDropSymbols (
    UBYTE * s )
```

Definition at line 5902 of file [compcomm.c](#).

#### 4.9.2.112 CoToPolynomial()

```
int CoToPolynomial (
    UBYTE * inp )
```

Definition at line 5925 of file [compcomm.c](#).

#### 4.9.2.113 CoFromPolynomial()

```
int CoFromPolynomial (
    UBYTE * inp )
```

Definition at line 6000 of file [compcomm.c](#).

**4.9.2.114 CoArgToExtraSymbol()**

```
int CoArgToExtraSymbol (
    UBYTE * s )
```

Definition at line [6026](#) of file [compcomm.c](#).

**4.9.2.115 CoExtraSymbols()**

```
int CoExtraSymbols (
    UBYTE * inp )
```

Definition at line [6076](#) of file [compcomm.c](#).

**4.9.2.116 GetIfDollarFactor()**

```
WORD * GetIfDollarFactor (
    UBYTE ** inp,
    WORD * w )
```

Definition at line [6149](#) of file [compcomm.c](#).

**4.9.2.117 GetDoParam()**

```
UBYTE * GetDoParam (
    UBYTE * inp,
    WORD ** wp,
    int par )
```

Definition at line [6207](#) of file [compcomm.c](#).

**4.9.2.118 CoDo()**

```
int CoDo (
    UBYTE * inp )
```

Definition at line [6284](#) of file [compcomm.c](#).

**4.9.2.119 CoEndDo()**

```
int CoEndDo (
    UBYTE * inp )
```

Definition at line [6387](#) of file [compcomm.c](#).

#### 4.9.2.120 CoFactDollar()

```
int CoFactDollar (
    UBYTE * inp )
```

Definition at line 6418 of file [compcomm.c](#).

#### 4.9.2.121 CoFactorize()

```
int CoFactorize (
    UBYTE * s )
```

Definition at line 6447 of file [compcomm.c](#).

#### 4.9.2.122 CoNFactorize()

```
int CoNFactorize (
    UBYTE * s )
```

Definition at line 6454 of file [compcomm.c](#).

#### 4.9.2.123 CoUnFactorize()

```
int CoUnFactorize (
    UBYTE * s )
```

Definition at line 6461 of file [compcomm.c](#).

#### 4.9.2.124 CoNUnFactorize()

```
int CoNUnFactorize (
    UBYTE * s )
```

Definition at line 6468 of file [compcomm.c](#).

#### 4.9.2.125 DoFactorize()

```
int DoFactorize (
    UBYTE * s,
    int par )
```

Definition at line 6475 of file [compcomm.c](#).

#### 4.9.2.126 CoOptimizeOption()

```
int CoOptimizeOption (
    UBYTE * s )
```

Definition at line 6608 of file [compcomm.c](#).



**4.9.2.127 CoPutInside()**

```
int CoPutInside (
    UBYTE * inp )
```

Definition at line 7044 of file [compcomm.c](#).

**4.9.2.128 CoAntiPutInside()**

```
int CoAntiPutInside (
    UBYTE * inp )
```

Definition at line 7045 of file [compcomm.c](#).

**4.9.2.129 DoPutInside()**

```
int DoPutInside (
    UBYTE * inp,
    int par )
```

Definition at line 7047 of file [compcomm.c](#).

**4.9.2.130 CoSwitch()**

```
int CoSwitch (
    UBYTE * s )
```

Definition at line 7167 of file [compcomm.c](#).

**4.9.2.131 CoCase()**

```
int CoCase (
    UBYTE * s )
```

Definition at line 7213 of file [compcomm.c](#).

**4.9.2.132 CoBreak()**

```
int CoBreak (
    UBYTE * s )
```

Definition at line 7263 of file [compcomm.c](#).

**4.9.2.133 CoDefault()**

```
int CoDefault (
    UBYTE * s )
```

Definition at line 7289 of file [compcomm.c](#).

#### 4.9.2.134 CoEndSwitch()

```
int CoEndSwitch (
    UBYTE * s )
```

Definition at line [7316](#) of file [compcomm.c](#).

#### 4.9.2.135 CoSetUserFlag()

```
int CoSetUserFlag (
    UBYTE * s )
```

Definition at line [7384](#) of file [compcomm.c](#).

#### 4.9.2.136 CoClearUserFlag()

```
int CoClearUserFlag (
    UBYTE * s )
```

Definition at line [7412](#) of file [compcomm.c](#).

#### 4.9.2.137 CoCreateAllLoops()

```
int CoCreateAllLoops (
    UBYTE * s )
```

Definition at line [7449](#) of file [compcomm.c](#).

#### 4.9.2.138 CoCreateAllPaths()

```
int CoCreateAllPaths (
    UBYTE * s )
```

Definition at line [7569](#) of file [compcomm.c](#).

#### 4.9.2.139 CoCreateAll()

```
int CoCreateAll (
    UBYTE * s )
```

Definition at line [7697](#) of file [compcomm.c](#).

## 4.10 compcomm.c

[Go to the documentation of this file.](#)

```

00001
00010 /* #[ License : */
00011 /*
00012 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00013 *   When using this file you are requested to refer to the publication
00014 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00015 *   This is considered a matter of courtesy as the development was paid
00016 *   for by FOM the Dutch physics granting agency and we would like to
00017 *   be able to track its scientific use to convince FOM of its value
00018 *   for the community.
00019 *
00020 *   This file is part of FORM.
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00024 *   Foundation, either version 3 of the License, or (at your option) any later
00025 *   version.
00026 *
00027 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00028 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00029 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00030 *   details.
00031 *
00032 *   You should have received a copy of the GNU General Public License along
00033 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00034 */
00035 /* #] License : */
00036 /*
00037 *   #[ includes :
00038 */
00039
00040 #include "form3.h"
00041 #include "comtool.h"
00042 #ifdef WITHFLOAT
00043 #include <gmp.h>
00044 #endif
00045
00046 static KEYWORD formatoptions[] = {
00047     {"allfloat",      (TFUN)0,    ALLINTEGERDOUBLE,    0}
00048     ,{"c",            (TFUN)0,    CMODE,                0}
00049     ,{"doublefortran", (TFUN)0,    DOUBLEFORTRANMODE,  0}
00050     ,{"float",        (TFUN)0,    0,                    2}
00051 #ifdef WITHFLOAT
00052     ,{"floatprecision", (TFUN)0,    0,                    5}
00053 #endif
00054     ,{"fortran",      (TFUN)0,    FORTRANMODE,          0}
00055     ,{"fortran90",    (TFUN)0,    FORTRANMODE,          4}
00056     ,{"maple",        (TFUN)0,    MAPLEMODE,            0}
00057     ,{"mathematica",  (TFUN)0,    MATHEMATICAMODE,      0}
00058     ,{"normal",       (TFUN)0,    NORMALFORMAT,         1}
00059     ,{"nospaces",     (TFUN)0,    NOSPACEFORMAT,        3}
00060     ,{"pfortran",     (TFUN)0,    PFORTRANMODE,         0}
00061     ,{"quadfortran",  (TFUN)0,    QUADRUPLEFORTRANMODE, 0}
00062     ,{"quadruplefortran", (TFUN)0,    QUADRUPLEFORTRANMODE, 0}
00063     ,{"rational",     (TFUN)0,    RATIONALMODE,         1}
00064     ,{"reduce",       (TFUN)0,    REDUCEMODE,           0}
00065     ,{"spaces",       (TFUN)0,    NORMALFORMAT,        3}
00066     ,{"vortran",      (TFUN)0,    VORTRANMODE,          0}
00067 };
00068
00069 static KEYWORD trace4options[] = {
00070     {"contract",      (TFUN)0,    CHISHOLM,             0}
00071     ,{"nocontract",   (TFUN)0,    0,                     CHISHOLM}
00072     ,{"nosymmetrize", (TFUN)0,    0,                     ALSOREVERSE}
00073     ,{"notrick",      (TFUN)0,    NOTRICK,               0}
00074     ,{"symmetrize",   (TFUN)0,    ALSOREVERSE,           0}
00075     ,{"trick",        (TFUN)0,    0,                     NOTRICK}
00076 };
00077
00078 static KEYWORD chisoptions[] = {
00079     {"nosymmetrize", (TFUN)0,    0,                     ALSOREVERSE}
00080     ,{"symmetrize",  (TFUN)0,    ALSOREVERSE,           0}
00081 };
00082
00083 static KEYWORDV writeoptions[] = {
00084     {"stats",         &(AC.StatsFlag), 1, 0}
00085     ,{"statistics",   &(AC.StatsFlag), 1, 0}
00086     ,{"shortstats",   &(AC.ShortStats), 1, 0}
00087     ,{"shortstatistics", &(AC.ShortStats), 1, 0}
00088     ,{"warnings",     &(AC.WarnFlag), 1, 0}
00089     ,{"allwarnings",  &(AC.WarnFlag), 2, 0}
00090     ,{"setup",        &(AC.SetupFlag), 1, 0}

```

```

00091     ,{"names",          &(AC.NamesFlag), 1, 0}
00092     ,{"allnames",       &(AC.NamesFlag), 2, 0}
00093     ,{"codes",          &(AC.CodesFlag), 1, 0}
00094     ,{"highfirst",      &(AC.SortType), SORTHIGHFIRST, SORTLOWFIRST}
00095     ,{"lowfirst",       &(AC.SortType), SORTLOWFIRST, SORTHIGHFIRST}
00096     ,{"powerfirst",     &(AC.SortType), SORTPOWERFIRST, SORTHIGHFIRST}
00097     ,{"tokens",         &(AC.TokensWriteFlag),1, 0}
00098 };
00099
00100 static KEYWORDV onoffoptions[] = {
00101     {"compress",        &(AC.NoCompress), 0, 1}
00102     ,{"checkpoint",     &(AC.CheckpointFlag), 1, 0}
00103     ,{"insidefirst",    &(AC.insidefirst), 1, 0}
00104     ,{"propercount",    &(AC.BottomLevel), 1, 0}
00105     ,{"stats",          &(AC.StatsFlag), 1, 0}
00106     ,{"statistics",     &(AC.StatsFlag), 1, 0}
00107     ,{"shortstats",     &(AC.ShortStats), 1, 0}
00108     ,{"shortstatistics",&(AC.ShortStats), 1, 0}
00109     ,{"names",          &(AC.NamesFlag), 1, 0}
00110     ,{"allnames",       &(AC.NamesFlag), 2, 0}
00111     ,{"warnings",       &(AC.WarnFlag), 1, 0}
00112     ,{"allwarnings",    &(AC.WarnFlag), 2, 0}
00113     ,{"highfirst",      &(AC.SortType), SORTHIGHFIRST, SORTLOWFIRST}
00114     ,{"lowfirst",       &(AC.SortType), SORTLOWFIRST, SORTHIGHFIRST}
00115     ,{"powerfirst",     &(AC.SortType), SORTPOWERFIRST, SORTHIGHFIRST}
00116     ,{"setup",          &(AC.SetupFlag), 1, 0}
00117     ,{"codes",          &(AC.CodesFlag), 1, 0}
00118     ,{"tokens",         &(AC.TokensWriteFlag),1,0}
00119     ,{"properorder",    &(AC.properorderflag),1,0}
00120     ,{"threadloadbalancing",&(AC.ThreadBalancing),1, 0}
00121     ,{"threads",        &(AC.ThreadsFlag),1, 0}
00122     ,{"threadsortfilesynch",&(AC.ThreadSortFileSynch),1, 0}
00123     ,{"threadstats",    &(AC.ThreadStats),1, 0}
00124     ,{"finalstats",     &(AC.FinalStats),1, 0}
00125     ,{"fewerstats",     &(AC.ShortStatsMax), 10, 0}
00126     ,{"fewerstatistics",&(AC.ShortStatsMax), 10, 0}
00127     ,{"processstats",   &(AC.ProcessStats),1, 0}
00128     ,{"oldparallelstats",&(AC.OldParallelStats),1,0}
00129     ,{"parallel",       &(AC.parallelflag),PARALLELFLAG,NOPARALLEL_USER}
00130     ,{"nospacesinnumbers",&(AO.NoSpacesInNumbers),1,0}
00131     ,{"indent",         &(AO.IndentSpace),INDENTSPACE,0}
00132     ,{"totalsize",      &(AM.PrintTotalSize), 1, 0}
00133     ,{"flag",           (int *)&(AC.debugFlags), 1, 0}
00134     ,{"oldfactarg",     &(AC.OldFactArgFlag), 1, 0}
00135     ,{"memdebugflag",   &(AC.MemDebugFlag), 1, 0}
00136     ,{"oldgcd",         &(AC.OldGCDflag), 1, 0}
00137     ,{"innertest",     &(AC.InnerTest), 1, 0}
00138     ,{"wtimestats",    &(AC.WTimeStatsFlag), 1, 0}
00139     ,{"sortreallocate", &(AC.SortReallocateFlag), 1, 0}
00140     ,{"backtrace",     &(AC.PrintBacktraceFlag), 1, 0}
00141     ,{"flint",         &(AC.FlintPolyFlag), 1, 0}
00142     ,{"humanstats",    &(AC.HumanStatsFlag), 1, 0}
00143     ,{"humanstatistics", &(AC.HumanStatsFlag), 1, 0}
00144 };
00145
00146 static WORD one = 1;
00147
00148 /*
00149  #] includes :
00150  #[ CoFormat :
00151  */
00152
00153 int CoFormat (UBYTE *s)
00154 {
00155     int error = 0, x;
00156     KEYWORD *key;
00157     UBYTE *ss;
00158     while ( *s == ' ' || *s == ',' ) s++;
00159     if ( *s == 0 ) {
00160         AC.OutputMode = 72;
00161         AC.OutputSpaces = NORMALFORMAT;
00162         return(error);
00163     }
00164     /*
00165     First the optimization level
00166     */
00167     if ( *s == 'O' || *s == 'o' ) {
00168         if ( ( FG.cTable[s[1]] == 1 ) ||
00169             ( s[1] == '=' && FG.cTable[s[2]] == 1 ) ) {
00170             s++; if ( *s == '=' ) s++;
00171             x = 0;
00172             while ( *s >= '0' && *s <= '9' ) x = 10*x + *s++ - '0';
00173             while ( *s == ',' ) s++;
00174             AO.OptimizationLevel = x;
00175             AO.Optimize.greedytimelimit = 0;
00176             AO.Optimize.mctstimelimit = 0;
00177             AO.Optimize.printstats = 0;

```

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00178     AO.Optimize.debugflags = 0;
00179     AO.Optimize.schemeflags = 0;
00180     AO.Optimize.mctsdecaymode = 1; /* default is decreasing C_p with iteration number */
00181     if ( AO.inscheme ) {
00182         M_free(AO.inscheme,"Horner input scheme");
00183         AO.inscheme = 0; AO.schemenum = 0;
00184     }
00185     switch ( x ) {
00186     case 0:
00187         break;
00188     case 1:
00189         AO.Optimize.mctsconstant.fval = -1.0;
00190         AO.Optimize.horner = O_OCCURRENCE;
00191         AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
00192         AO.Optimize.method = O_CSE;
00193         break;
00194     case 2:
00195         AO.Optimize.horner = O_OCCURRENCE;
00196         AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
00197         AO.Optimize.method = O_GREEDY;
00198         AO.Optimize.greedyminnum = 10;
00199         AO.Optimize.greedymaxperc = 5;
00200         break;
00201     case 3:
00202         AO.Optimize.mctsconstant.fval = 1.0;
00203         AO.Optimize.horner = O_MCTS;
00204         AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
00205         AO.Optimize.method = O_GREEDY;
00206         AO.Optimize.mctsnumexpand = 1000;
00207         AO.Optimize.mctsnumkeep = 10;
00208         AO.Optimize.mctsnumrepeat = 1;
00209         AO.Optimize.greedyminnum = 10;
00210         AO.Optimize.greedymaxperc = 5;
00211         break;
00212     case 4:
00213         AO.Optimize.horner = O_SIMULATED_ANNEALING;
00214         AO.Optimize.saIter = 1000;
00215         AO.Optimize.saMaxT.fval = 2000;
00216         AO.Optimize.saMinT.fval = 1;
00217         break;
00218     default:
00219         error = 1;
00220         MesPrint("&Illegal optimization specification in format statement");
00221         break;
00222     }
00223     if ( error == 0 && *s != 0 && x > 0 ) return(CoOptimizeOption(s));
00224     return(error);
00225 }
00226 #ifndef EXPOPT
00227 { UBYTE c;
00228   ss = s;
00229   while ( FG.cTable[*s] == 0 ) s++;
00230   c = *s; *s = 0;
00231   if ( StrICont(ss,(UBYTE *)"optimize") == 0 ) {
00232       *s = c;
00233       while ( *s == ',' ) s++;
00234       if ( *s == '=' ) s++;
00235       AO.OptimizationLevel = 3;
00236       AO.Optimize.mctsconstant.fval = 1.0;
00237       AO.Optimize.horner = O_MCTS;
00238       AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
00239       AO.Optimize.method = O_GREEDY;
00240       AO.Optimize.mctstimelimit = 0;
00241       AO.Optimize.mctsnumexpand = 1000;
00242       AO.Optimize.mctsnumkeep = 10;
00243       AO.Optimize.mctsnumrepeat = 1;
00244       AO.Optimize.greedytimelimit = 0;
00245       AO.Optimize.greedyminnum = 10;
00246       AO.Optimize.greedymaxperc = 5;
00247       AO.Optimize.printstats = 0;
00248       AO.Optimize.debugflags = 0;
00249       AO.Optimize.schemeflags = 0;
00250       AO.Optimize.mctsdecaymode = 1;
00251       if ( AO.inscheme ) {
00252           M_free(AO.inscheme,"Horner input scheme");
00253           AO.inscheme = 0; AO.schemenum = 0;
00254       }
00255       return(CoOptimizeOption(s));
00256   }
00257   else {
00258       error = 1;
00259       MesPrint("&Illegal optimization specification in format statement");
00260       return(error);
00261   }
00262 }
00263 #endif
00264 }

```

```

00265     else if ( FG.cTable[*s] == 1 ) {
00266         x = 0;
00267         while ( FG.cTable[*s] == 1 ) x = 10*x + *s++ - '0';
00268         if ( x <= 0 || x >= MAXLINELENGTH ) {
00269             error = 1;
00270             MesPrint("&Illegal value for linesize: %d",x);
00271             x = 72;
00272         }
00273         if ( x < 39 ) {
00274             MesPrint(" ... Too small value for linesize corrected to 39");
00275             x = 39;
00276         }
00277         AO.DoubleFlag = 0;
00278     /*
00279     The next line resets the mode to normal. Because the special modes
00280     reset the line length we have a little problem with the special modes
00281     and customized line length. We try to improve by removing the next line
00282     */
00283     /*
00284     AC.OutputMode = 0;
00285     AC.LineLength = x;
00286     if ( *s != 0 ) {
00287         error = 1;
00288         MesPrint("&Illegal linesize field in format statement");
00289     }
00290     else {
00291         key = FindKeyWord(s,formatoptions,
00292             sizeof(formatoptions)/sizeof(KEYWORD));
00293         if ( key ) {
00294             if ( key->type == FORTRANMODE || key->type == PFORTRANMODE || key->type ==
DOUBLEFORTRANMODE
00295                 || key->type == QUADRUPLEFORTRANMODE || key->type == VORTRANMODE ) {
00296                 if ( AC.LineLength > 72 ) AC.LineLength = 72;
00297             }
00298             if ( key->flags == 0 ) {
00299                 if ( key->type == FORTRANMODE || key->type == PFORTRANMODE
00300                     || key->type == DOUBLEFORTRANMODE || key->type == ALLINTEGERDOUBLE
00301                     || key->type == QUADRUPLEFORTRANMODE || key->type == VORTRANMODE ) {
00302                     AC.IsFortran90 = ISNOTFORTRAN90;
00303                     if ( AC.Fortran90Kind ) {
00304                         M_free(AC.Fortran90Kind,"Fortran90 Kind");
00305                         AC.Fortran90Kind = 0;
00306                     }
00307                 }
00308                 if ( ( key->type == ALLINTEGERDOUBLE ) && AO.DoubleFlag != 0 ) {
00309                     AO.DoubleFlag |= 4;
00310                 }
00311                 else {
00312                     AO.DoubleFlag = 0;
00313                     AC.OutputMode = key->type & NODOUBLEMASK;
00314                     if ( ( key->type & DOUBLEPRECISIONFLAG ) != 0 ) {
00315                         AO.DoubleFlag = 1;
00316                     }
00317                     else if ( ( key->type & QUADRUPLEPRECISIONFLAG ) != 0 ) {
00318                         AO.DoubleFlag = 2;
00319                     }
00320                 }
00321             }
00322             else if ( key->flags == 1 ) {
00323                 AC.OutputMode = AC.OutNumberType = key->type;
00324             }
00325             else if ( key->flags == 2 ) {
00326                 while ( FG.cTable[*s] == 0 ) s++;
00327                 if ( *s == 0 ) AC.OutNumberType = 10;
00328                 else if ( *s == ',' ) {
00329                     s++;
00330                     x = 0;
00331                     while ( FG.cTable[*s] == 1 ) x = 10*x + *s++ - '0';
00332                     if ( *s != 0 ) {
00333                         error = 1;
00334                         MesPrint("&Illegal float format specifier");
00335                     }
00336                     else {
00337                         if ( x < 3 ) {
00338                             x = 3;
00339                             MesPrint("& ... float format value corrected to 3");
00340                         }
00341                         if ( x > 100 ) {
00342                             x = 100;
00343                             MesPrint("& ... float format value corrected to 100");
00344                         }
00345                         AC.OutNumberType = x;
00346                     }
00347                 }
00348             }
00349             else if ( key->flags == 3 ) {
00350                 AC.OutputSpaces = key->type;

```

```

00351     }
00352     else if ( key->flags == 4 ) {
00353         AC.IsFortran90 = ISFORTRAN90;
00354         if ( AC.Fortran90Kind ) {
00355             M_free(AC.Fortran90Kind,"Fortran90 Kind");
00356             AC.Fortran90Kind = 0;
00357         }
00358         while ( FG.cTable[*s] <= 1 ) s++;
00359         if ( *s == ',' ) {
00360             s++; ss = s;
00361             while ( *ss && *ss != ',' ) ss++;
00362             if ( *ss == ',' ) {
00363                 MesPrint("&No white space or comma's allowed in Fortran90 option: %s",s);
00364             }
00365             else {
00366                 AC.Fortran90Kind = strdup(s,"Fortran90 Kind");
00367             }
00368         }
00369         AO.DoubleFlag = 0;
00370         AC.OutputMode = key->type & NODOUBLEMASK;
00371     }
00372 #ifdef WITHFLOAT
00373     else if ( key->flags == 5 ) {
00374         /*
00375             Syntax: Format FloatPrecision number;
00376             Format FloatPrecision off;
00377         */
00378         while ( FG.cTable[*s] == 0 ) s++;
00379         while ( *s == ' ' || *s == '\t' || *s == ',' ) s++;
00380         if ( *s == 0 ) {
00381             AO.FloatPrec = 0;
00382         }
00383         else if ( tolower(*s) == 'o' && tolower(s[1]) == 'f'
00384             && tolower(s[2]) == 'f' ) {
00385             ss = s;
00386             s += 3;
00387             while ( *s == ' ' || *s == '\t' || *s == ',' ) s++;
00388             if ( *s ) { s = ss; goto WrongOption; }
00389             AO.FloatPrec = -1;
00390         }
00391         else if ( FG.cTable[*s] == 1 ) {
00392             ss = s;
00393             AO.FloatPrec = 0;
00394             while ( *s <= '9' && *s >= '0' )
00395                 AO.FloatPrec = 10*AO.FloatPrec + (*s++ - '0');
00396             while ( *s == ' ' || *s == '\t' || *s == ',' ) s++;
00397             if ( *s ) { s = ss; goto WrongOption; }
00398         }
00399         else {
00400 WrongOption: MesPrint("&Illegal option in Format FloatPrecision: %s",s);
00401             error = 1;
00402         }
00403     }
00404 #endif
00405 }
00406 else if ( ( *s == 'c' || *s == 'C' ) && ( FG.cTable[s[1]] == 1 ) ) {
00407     UBYTE *ss = s+1;
00408     WORD x = 0;
00409     while ( *ss >= '0' && *ss <= '9' ) x = 10*x + *ss++ - '0';
00410     if ( *ss != 0 ) goto Unknown;
00411     AC.OutputMode = CMODE;
00412     AC.Cnumpows = x;
00413 }
00414 else {
00415 Unknown: MesPrint("&Unknown option: %s",s); error = 1;
00416 }
00417 }
00418 return(error);
00419 }
00420
00421 /*
00422 #] CoFormat :
00423 #[ CoCollect :
00424
00425 Collect,functionname
00426 */
00427
00428 int CoCollect(UBYTE *s)
00429 {
00430     /* -----change 17-feb-2003 Added percentage */
00431     WORD numfun;
00432     int type,x = 0;
00433     UBYTE *t = SkipAName(s), *t1, *t2;
00434     AC.AltCollectFun = 0;
00435     if ( t == 0 ) goto syntaxerror;
00436     t1 = t; while ( *t1 == ',' || *t1 == ' ' || *t1 == '\t' ) t1++;

```

```

00437     *t = 0; t = t1;
00438     if ( *t1 && ( FG.cTable[*t1] == 0 || *t1 == '[' ) ) {
00439         t2 = SkipAName(t1);
00440         if ( t2 == 0 ) goto syntaxerror;
00441         t = t2;
00442         while ( *t == ',' || *t == ' ' || *t == '\\t' ) t++;
00443         *t2 = 0;
00444     }
00445     else t1 = 0;
00446     if ( *t && FG.cTable[*t] == 1 ) {
00447         while ( *t >= '0' && *t <= '9' ) x = 10*x + *t++ - '0';
00448         if ( x > 100 ) x = 100;
00449         while ( *t == ',' || *t == ' ' || *t == '\\t' ) t++;
00450         if ( *t ) goto syntaxerror;
00451     }
00452     else {
00453         if ( *t ) goto syntaxerror;
00454         x = 100;
00455     }
00456     if ( ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
00457     || ( functions[numfun].spec != 0 ) ) {
00458         MesPrint("%s should be a regular function",s);
00459         if ( type < 0 ) {
00460             if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
00461                 AddFunction(s,0,0,0,0,0,-1,-1);
00462         }
00463         return(1);
00464     }
00465     AC.CollectFun = numfun+FUNCTION;
00466     AC.CollectPercentage = (WORD)x;
00467     if ( t1 ) {
00468         if ( ( ( type = GetName(AC.varnames,t1,&numfun,WITHAUTO) ) != CFUNCTION )
00469         || ( functions[numfun].spec != 0 ) ) {
00470             MesPrint("%s should be a regular function",t1);
00471             if ( type < 0 ) {
00472                 if ( GetName(AC.exprnames,t1,&numfun,NOAUTO) == NAMENOTFOUND )
00473                     AddFunction(t1,0,0,0,0,0,-1,-1);
00474             }
00475             return(1);
00476         }
00477         AC.AltCollectFun = numfun+FUNCTION;
00478     }
00479     return(0);
00480 syntaxerror:
00481     MesPrint("&Collect statement needs one or two functions (and a percentage) for its argument(s)");
00482     return(1);
00483 }
00484
00485 /*
00486 #] CoCollect :
00487 #[ setonoff :
00488 */
00489
00490 int setonoff(UBYTE *s, int *flag, int onvalue, int offvalue)
00491 {
00492     if ( StrICmp(s,(UBYTE *)"on") == 0 ) *flag = onvalue;
00493     else if ( StrICmp(s,(UBYTE *)"off") == 0 ) *flag = offvalue;
00494     else {
00495         MesPrint("&Unknown option: %s, on or off expected",s);
00496         return(1);
00497     }
00498     return(0);
00499 }
00500
00501 /*
00502 #] setonoff :
00503 #[ CoCompress :
00504 */
00505
00506 int CoCompress(UBYTE *s)
00507 {
00508     GETIDENTITY
00509     UBYTE *t, c;
00510     if ( StrICmp(s,(UBYTE *)"on") == 0 ) {
00511         AC.NoCompress = 0;
00512         AR.gzipCompress = 0;
00513     }
00514     else if ( StrICmp(s,(UBYTE *)"off") == 0 ) {
00515         AC.NoCompress = 1;
00516         AR.gzipCompress = 0;
00517     }
00518     else {
00519         t = s; while ( FG.cTable[*t] <= 1 ) t++;
00520         c = *t; *t = 0;
00521         if ( StrICmp(s,(UBYTE *)"gzip") == 0 ) {
00522 #ifndef WITHZLIB
00523             Warning("gzip compression not supported on this platform");

```



```

00524 #endif
00525     s = t; *s = c;
00526     if ( *s == 0 ) {
00527         AR.gzipCompress = GZIPDEFAULT; /* Normally should be 6 */
00528         return(0);
00529     }
00530     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
00531     t = s;
00532     if ( FG.cTable[*s] == 1 ) {
00533         AR.gzipCompress = *s - '0';
00534         s++;
00535         while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
00536         if ( *s == 0 ) return(0);
00537     }
00538     MesPrint("&Unknown gzip option: %s, a digit was expected",t);
00539     return(1);
00540
00541 }
00542 else {
00543     MesPrint("&Unknown option: %s, on, off or gzip expected",s);
00544     return(1);
00545 }
00546 }
00547 return(0);
00548 }
00549
00550 /*
00551  #] CoCompress :
00552  #[ CoFlags :
00553  */
00554
00555 int CoFlags(UBYTE *s,int value)
00556 {
00557     int i, error = 0;
00558     if ( *s != ',' ) {
00559         MesPrint("&Proper syntax is: On/Off Flag,number[s];");
00560         error = 1;
00561     }
00562     while ( *s == ',' ) {
00563         do { s++; } while ( *s == ',' );
00564         i = 0;
00565         if ( FG.cTable[*s] != 1 ) {
00566             MesPrint("&Proper syntax is: On/Off Flag,number[s];");
00567             error = 1;
00568             break;
00569         }
00570         while ( FG.cTable[*s] == 1 ) { i = 10*i + *s++ - '0'; }
00571         if ( i <= 0 || i > MAXFLAGS ) {
00572             MesPrint("&The number of a flag in On/Off Flag should be in the range
00573 0-%d", (int)MAXFLAGS);
00574             error = 1;
00575             break;
00576         }
00577         AC.debugFlags[i] = value;
00578     }
00579     if ( *s ) {
00580         MesPrint("&Proper syntax is: On/Off Flag,number[s];");
00581         error = 1;
00582     }
00583     return(error);
00584 }
00585 /*
00586  #] CoFlags :
00587  #[ CoOff :
00588  */
00589
00590 int CoOff(UBYTE *s)
00591 {
00592     GETIDENTITY
00593     UBYTE *t, c;
00594     int i, num = sizeof(onoffoptions)/sizeof(KEYWORD);
00595     for (;;) {
00596         while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
00597         if ( *s == 0 ) return(0);
00598         if ( chartype[*s] != 0 ) {
00599             MesPrint("&Illegal character or option encountered in OFF statement");
00600             return(-1);
00601         }
00602         t = s; while ( chartype[*s] == 0 ) s++;
00603         c = *s; *s = 0;
00604         for ( i = 0; i < num; i++ ) {
00605             if ( StrICont(t, (UBYTE *) (onoffoptions[i].name)) == 0 ) break;
00606         }
00607         if ( i >= num ) {
00608             MesPrint("&Unrecognized option in OFF statement: %s",t);
00609             *s = c; return(-1);

```

```

00610     }
00611     else if ( StrICont(t, (UBYTE *) "compress") == 0 ) {
00612         AR.gzipCompress = 0;
00613     }
00614     else if ( StrICont(t, (UBYTE *) "checkpoint") == 0 ) {
00615         PrintDeprecation("the checkpoint mechanism", "issues/626");
00616         AC.CheckpointInterval = 0;
00617         if ( AC.CheckpointRunBefore ) { free(AC.CheckpointRunBefore); AC.CheckpointRunBefore =
NULL; }
00618         if ( AC.CheckpointRunAfter ) { free(AC.CheckpointRunAfter); AC.CheckpointRunAfter = NULL;
}
00619         if ( AC.NoShowInput == 0 ) MesPrint("Checkpoints deactivated.");
00620     }
00621     else if ( StrICont(t, (UBYTE *) "threads") == 0 ) {
00622         AS.MultiThreaded = 0;
00623     }
00624     else if ( StrICont(t, (UBYTE *) "flag") == 0 ) {
00625         *s = c;
00626         return(CoFlags(s, 0));
00627     }
00628     else if ( StrICont(t, (UBYTE *) "innertest") == 0 ) {
00629         *s = c;
00630         AC.InnerTest = 0;
00631         if ( AC.TestValue ) {
00632             M_free(AC.TestValue, "InnerTest");
00633             AC.TestValue = 0;
00634         }
00635     }
00636     else if ( StrICont(t, (UBYTE *) "sortreallocate") == 0 ) {
00637         if ( AC.SortReallocateFlag == 2 ) {
00638             /* The flag has been set by #sortreallocate, and it was off before. Leave it as 2,
00639              so that the reallocation still happens in the current module. It will be turned
00640              off after the reallocation is done. */
00641             return(0);
00642         }
00643     }
00644     *s = c;
00645     *onoffoptions[i].var = onoffoptions[i].flags;
00646     AR.SortType = AC.SortType;
00647     AC.mparallelfld = AC.parallelfld | AM.hparallelfld;
00648 }
00649 }
00650
00651 /*
00652 #] CoOff :
00653 #[ CoOn :
00654 */
00655
00656 int CoOn(UBYTE *s)
00657 {
00658     GETIDENTITY
00659     UBYTE *t, c;
00660     int i, num = sizeof(onoffoptions)/sizeof(KEYWORD);
00661     LONG interval;
00662     for (;;) {
00663         while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
00664         if ( *s == 0 ) return(0);
00665         if ( chartype[*s] != 0 ) {
00666             MesPrint("&Illegal character or option encountered in ON statement");
00667             return(-1);
00668         }
00669         t = s; while ( chartype[*s] == 0 ) s++;
00670         c = *s; *s = 0;
00671         for ( i = 0; i < num; i++ ) {
00672             if ( StrICont(t, (UBYTE *) (onoffoptions[i].name)) == 0 ) break;
00673         }
00674         if ( i >= num ) {
00675             MesPrint("&Unrecognized option in ON statement: %s", t);
00676             *s = c; return(-1);
00677         }
00678         if ( StrICont(t, (UBYTE *) "backtrace") == 0 ) {
00679             #ifndef ENABLE_BACKTRACE
00680                 Warning("backtrace not supported on this platform");
00681             #endif
00682         }
00683         else if ( StrICont(t, (UBYTE *) "compress") == 0 ) {
00684             AR.gzipCompress = 0;
00685             *s = c;
00686             while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
00687             if ( *s ) {
00688                 t = s;
00689                 while ( FG.cTable[*s] <= 1 ) s++;
00690                 c = *s; *s = 0;
00691                 if ( StrICmp(t, (UBYTE *) "gzip") == 0 ) {
00692                     #ifndef WITHZLIB
00693                         Warning("gzip compression not supported on this platform");
00694                     #endif

```

```

00695 #ifdef WITHZSTD
00696 /* If gzip is specified, turn off zstd compression. zlib still goes via the
    wrapper. */
00697 ZWRAP_useZSTDcompression(0);
00698 #endif
00699 }
00700 else if ( StrICmp(t, (UBYTE *) "zstd") == 0 ) {
00701 #ifdef WITHZSTD
00702     ZWRAP_useZSTDcompression(1);
00703 #else
00704     Warning("zstd compression not supported on this platform");
00705 #endif
00706 }
00707 else {
00708     MesPrint("&Unrecognized option in ON compress statement: %s", t);
00709     return(-1);
00710 }
00711 /* Whether we are using zlib or zstd, accept and use a compression level. */
00712 *s = c;
00713 while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
00714 if ( FG.cTable[*s] == 1 ) {
00715     AR.gzipCompress = *s++ - '0';
00716     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
00717     if ( *s ) {
00718         MesPrint("&Unrecognized option in ON compress gzip/zstd statement: %s", t);
00719         return(-1);
00720     }
00721 }
00722 else if ( *s == 0 ) {
00723     AR.gzipCompress = GZIPDEFAULT;
00724 }
00725 else {
00726     MesPrint("&Unrecognized option in ON compress gzip/zstd statement: %s, single
digit expected", t);
00727     return(-1);
00728 }
00729 }
00730 }
00731 else if ( StrICont(t, (UBYTE *) "checkpoint") == 0 ) {
00732     PrintDeprecation("the checkpoint mechanism", "issues/626");
00733     AC.CheckpointInterval = 0;
00734     if ( AC.CheckpointRunBefore ) { free(AC.CheckpointRunBefore); AC.CheckpointRunBefore =
NULL; }
00735     if ( AC.CheckpointRunAfter ) { free(AC.CheckpointRunAfter); AC.CheckpointRunAfter = NULL;
}
00736 *s = c;
00737 while ( *s ) {
00738     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
00739     if ( FG.cTable[*s] == 1 ) {
00740         interval = 0;
00741         t = s;
00742         do { interval = 10*interval + *s++ - '0'; } while ( FG.cTable[*s] == 1 );
00743         if ( *s == 's' || *s == 'S' ) {
00744             s++;
00745         }
00746         else if ( *s == 'm' || *s == 'M' ) {
00747             interval *= 60; s++;
00748         }
00749         else if ( *s == 'h' || *s == 'H' ) {
00750             interval *= 3600; s++;
00751         }
00752         else if ( *s == 'd' || *s == 'D' ) {
00753             interval *= 86400; s++;
00754         }
00755         if ( *s != ',' && FG.cTable[*s] != 6 && FG.cTable[*s] != 10 ) {
00756             MesPrint("&Unrecognized time interval in ON Checkpoint statement: %s", t);
00757             return(-1);
00758         }
00759         AC.CheckpointInterval = interval * 100; /* in 1/100 of seconds */
00760     }
00761     else if ( FG.cTable[*s] == 0 ) {
00762         int type;
00763         t = s;
00764         while ( FG.cTable[*s] == 0 ) s++;
00765         c = *s; *s = 0;
00766         if ( StrICmp(t, (UBYTE *) "run") == 0 ) {
00767             type = 3;
00768         }
00769         else if ( StrICmp(t, (UBYTE *) "runafter") == 0 ) {
00770             type = 2;
00771         }
00772         else if ( StrICmp(t, (UBYTE *) "runbefore") == 0 ) {
00773             type = 1;
00774         }
00775         else {
00776             MesPrint("&Unrecognized option in ON Checkpoint statement: %s", t);
00777             *s = c; return(-1);

```

```

00778     }
00779     *s = c;
00780     if ( *s != '=' && FG.cTable[*s+1] != 9 ) {
00781         MesPrint("&Unrecognized option in ON Checkpoint statement: %s", t);
00782         return(-1);
00783     }
00784     ++s;
00785     t = ++s;
00786     while ( *s ) {
00787         if ( FG.cTable[*s] == 9 ) {
00788             c = *s; *s = 0;
00789             if ( type & 1 ) {
00790                 if ( AC.CheckpointRunBefore ) {
00791                     free(AC.CheckpointRunBefore); AC.CheckpointRunBefore = NULL;
00792                 }
00793                 if ( s-t > 0 ) {
00794                     AC.CheckpointRunBefore = Malloc1(s-t+1, "AC.CheckpointRunBefore");
00795                     StrCopy(t, (UBYTE*)AC.CheckpointRunBefore);
00796                 }
00797             }
00798             if ( type & 2 ) {
00799                 if ( AC.CheckpointRunAfter ) {
00800                     free(AC.CheckpointRunAfter); AC.CheckpointRunAfter = NULL;
00801                 }
00802                 if ( s-t > 0 ) {
00803                     AC.CheckpointRunAfter = Malloc1(s-t+1, "AC.CheckpointRunAfter");
00804                     StrCopy(t, (UBYTE*)AC.CheckpointRunAfter);
00805                 }
00806             }
00807             *s = c;
00808             break;
00809         }
00810         ++s;
00811     }
00812     if ( FG.cTable[*s] != 9 ) {
00813         MesPrint("&Unrecognized option in ON Checkpoint statement: %s", t);
00814         return(-1);
00815     }
00816     ++s;
00817 }
00818
00819 /*
00820 if ( AC.NoShowInput == 0 ) {
00821     MesPrint("Checkpoints activated.");
00822     if ( AC.CheckpointInterval ) {
00823         MesPrint("-> Minimum saving interval: %1 seconds.", AC.CheckpointInterval/100);
00824     }
00825     else {
00826         MesPrint("-> No minimum saving interval given. Saving after EVERY module.");
00827     }
00828     if ( AC.CheckpointRunBefore ) {
00829         MesPrint("-> Calling script \"%s\" before saving.", AC.CheckpointRunBefore);
00830     }
00831     if ( AC.CheckpointRunAfter ) {
00832         MesPrint("-> Calling script \"%s\" after saving.", AC.CheckpointRunAfter);
00833     }
00834 }
00835 */
00836 }
00837 else if ( StrICont(t, (UBYTE *) "indentSpace") == 0 ) {
00838     *s = c;
00839     while ( *s == ' ' || *s == ',' || *s == '\\t' ) s++;
00840     if ( *s ) {
00841         i = 0;
00842         while ( FG.cTable[*s] == 1 ) { i = 10*i + *s++ - '0'; }
00843         if ( *s ) {
00844             MesPrint("&Unrecognized option in ON IndentSpace statement: %s", t);
00845             return(-1);
00846         }
00847         if ( i > 40 ) {
00848             Warning("IndentSpace parameter adjusted to 40");
00849             i = 40;
00850         }
00851         AO.IndentSpace = i;
00852     }
00853     else {
00854         AO.IndentSpace = AM.ggIndentSpace;
00855     }
00856     return(0);
00857 }
00858 else if ( ( StrICont(t, (UBYTE *) "fewerstats") == 0 ) ||
00859           ( StrICont(t, (UBYTE *) "fewerstatistics") == 0 ) ) {
00860     *s = c;
00861     while ( *s == ' ' || *s == ',' || *s == '\\t' ) s++;
00862     if ( *s ) {
00863         i = 0;
00864         while ( FG.cTable[*s] == 1 ) { i = 10*i + *s++ - '0'; }

```

```

00865         if ( *s ) {
00866             MesPrint("&Unrecognized option in ON FewerStatistics statement: %s",t);
00867             return(-1);
00868         }
00869         if ( i > AM.S0->MaxPatches ) {
00870             if ( AC.WarnFlag )
00871                 MesPrint("&Warning: FewerStatistics parameter greater than MaxPatches(=%d).
Adjusted to %d"
00872                         ,AM.S0->MaxPatches, (AM.S0->MaxPatches+1)/2);
00873             i = (AM.S0->MaxPatches+1)/2;
00874         }
00875         AC.ShortStatsMax = i;
00876     }
00877     else {
00878         AC.ShortStatsMax = 10; /* default value */
00879     }
00880     return(0);
00881 }
00882 else if ( StrICont(t, (UBYTE *) "threads") == 0 ) {
00883     if ( AM.totalnumberofthreads > 1 ) AS.MultiThreaded = 1;
00884 }
00885 else if ( StrICont(t, (UBYTE *) "flag") == 0 ) {
00886     *s = c;
00887     return(CoFlags(s,1));
00888 }
00889 else if ( StrICont(t, (UBYTE *) "innertest") == 0 ) {
00890     UBYTE *t;
00891     *s = c;
00892     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
00893     if ( *s ) {
00894         t = s; while ( *t ) t++;
00895         while ( t[-1] == ' ' || t[-1] == '\t' ) t--;
00896         c = *t; *t = 0;
00897         if ( AC.TestValue ) M_free(AC.TestValue, "InnerTest");
00898         AC.TestValue = strDupl(s, "InnerTest");
00899         *t = c;
00900         s = t;
00901         while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
00902     }
00903     else {
00904         if ( AC.TestValue ) {
00905             M_free(AC.TestValue, "InnerTest");
00906             AC.TestValue = 0;
00907         }
00908     }
00909 }
00910 else if ( StrICont(t, (UBYTE *) "flint") == 0 ) {
00911 #ifndef WITHFLINT
00912     MesPrint("&Warning: FORM was not built with FLINT support.");
00913     MesPrint("Statement has no effect.");
00914 #endif
00915 }
00916 else { *s = c; }
00917 *onoffoptions[i].var = onoffoptions[i].type;
00918 AR.SortType = AC.SortType;
00919 AC.mparallelfalg = AC.parallelfalg | AM.hparallelfalg;
00920 }
00921 }
00922
00923 /*
00924 #] CoOn :
00925 #[ CoInsideFirst :
00926 */
00927
00928 int CoInsideFirst(UBYTE *s) { return(setonoff(s,&AC.insidefirst,1,0)); }
00929
00930 /*
00931 #] CoInsideFirst :
00932 #[ CoProperCount :
00933 */
00934
00935 int CoProperCount(UBYTE *s) { return(setonoff(s,&AC.BottomLevel,1,0)); }
00936
00937 /*
00938 #] CoProperCount :
00939 #[ CoDelete :
00940 */
00941
00942 int CoDelete(UBYTE *s)
00943 {
00944     int error = 0;
00945     if ( StrICmp(s, (UBYTE *) "storage") == 0 ) {
00946         if ( DeleteStore(1) < 0 ) {
00947             MesPrint("&Cannot restart storage file");
00948             error = 1;
00949         }
00950     }
00951 }

```

```

00951     else {
00952         UBYTE *t = s, c;
00953         while ( *t && *t != ',' && *t != '>' ) t++;
00954         c = *t; *t = 0;
00955         if ( ( StrICmp(s, (UBYTE *)"extrasymbols") == 0 )
00956             || ( StrICmp(s, (UBYTE *)"extrasymbol") == 0 ) ) {
00957             WORD x = 0;
00958             /*
00959              * Either deletes all extra symbols or deletes above a given number
00960              */
00961             *t = c; s = t;
00962             if ( *s == '>' ) {
00963                 s++;
00964                 if ( FG.cTable[*s] != 1 ) goto unknown;
00965                 while ( *s <= '9' && *s >= '0' ) x = 10*x + *s++ - '0';
00966                 if ( *s ) goto unknown;
00967             }
00968             else if ( *s ) goto unknown;
00969             if ( x < AM.gnumextrasyms ) x = AM.gnumextrasyms;
00970             PruneExtraSymbols(x);
00971         }
00972     }
00973     *t = c;
00974 unknown:
00975     MesPrint("&Unknown option: %s",s);
00976     error = 1;
00977 }
00978 }
00979 return(error);
00980 }
00981 /*
00982 #] CoDelete :
00983 #[ CoKeep :
00984 */
00985 int CoKeep(UBYTE *s)
00986 {
00987     if ( StrICmp(s, (UBYTE *)"brackets") == 0 ) AC.ComDefer = 1;
00988     else { MesPrint("&Unknown option: '%s'",s); return(1); }
00989     return(0);
00990 }
00991 /*
00992 #] CoKeep :
00993 #[ CoFixIndex :
00994 */
00995 int CoFixIndex(UBYTE *s)
00996 {
00997     int x, y, error = 0;
00998     while ( *s ) {
00999         if ( FG.cTable[*s] != 1 ) {
01000             proper: MesPrint("&Proper syntax is: FixIndex,number:value[,number,value];");
01001             return(1);
01002         }
01003         ParseNumber(x,s)
01004         if ( *s != ':' ) goto proper;
01005         s++;
01006         if ( *s != '-' && *s != '+' && FG.cTable[*s] != 1 ) goto proper;
01007         ParseSignedNumber(y,s)
01008         if ( *s && *s != ',' ) goto proper;
01009         while ( *s == ',' ) s++;
01010         if ( x >= AM.OffsetIndex ) {
01011             MesPrint("&Fixed index out of allowed range. Change ConstIndex in setup file?");
01012             MesPrint("&Current value of ConstIndex = %d",AM.OffsetIndex-1);
01013             error = 1;
01014         }
01015         if ( y != (int)((WORD)y) ) {
01016             MesPrint("&Value of d_(%d,%d) outside range for this computer",x,x);
01017             error = 1;
01018         }
01019         if ( error == 0 ) AC.FixIndices[x] = y;
01020     }
01021     return(error);
01022 }
01023 /*
01024 #] CoFixIndex :
01025 #[ CoMetric :
01026 */
01027 int CoMetric(UBYTE *s)
01028 { DUMMYUSE(s); MesPrint("&The metric statement does not do anything yet"); return(1); }
01029 /*
01030 #] CoMetric :

```

```

01038     #[ DoPrint :
01039     */
01040
01041 int DoPrint(UBYTE *s, int par)
01042 {
01043     int i, error = 0, numdol = 0, type;
01044     WORD handle = -1;
01045     UBYTE *name, c, *t;
01046     EXPRESSIONS e;
01047     WORD numexpr, tofile = 0, *w, par2 = 0;
01048     CBUF *C = cbuf + AC.cbufnum;
01049     while ( *s == ',' ) s++;
01050     if ( ( *s == '+' || *s == '-' ) && ( s[1] == 'f' || s[1] == 'F' ) ) {
01051         t = s + 2; while ( *t == ' ' || *t == ',' ) t++;
01052         if ( *t == '"' ) {
01053             if ( *s == '+' ) { tofile = 1; handle = AC.LogHandle; }
01054             s = t;
01055         }
01056     }
01057     else if ( *s == '<' ) {
01058         UBYTE *filename;
01059         s++; filename = s;
01060         while ( *s && *s != '>' ) s++;
01061         if ( *s == 0 ) {
01062             MesPrint("&Improper filename in print statement");
01063             return(1);
01064         }
01065         *s++ = 0;
01066         tofile = 1;
01067         if ( ( handle = GetChannel((char *)filename,1) ) < 0 ) return(1);
01068         SKIPBLANKS(s) if ( *s == ',' ) s++; SKIPBLANKS(s)
01069         if ( *s == '+' && ( s[1] == 's' || s[1] == 'S' ) ) {
01070             s += 2;
01071             par2 |= PRINTONETERM;
01072             if ( *s == 's' || *s == 'S' ) {
01073                 s++;
01074                 par2 |= PRINTONEFUNCTION;
01075                 if ( *s == 's' || *s == 'S' ) {
01076                     s++;
01077                     par2 |= PRINTALL;
01078                 }
01079             }
01080             SKIPBLANKS(s) if ( *s == ',' ) s++; SKIPBLANKS(s)
01081         }
01082     }
01083     if ( par == PRINTON && *s == '"' ) {
01084         WORD code[3];
01085         if ( tofile == 1 ) code[0] = TYPEFPRINT;
01086         else code[0] = TYPEPRINT;
01087         code[1] = handle;
01088         code[2] = par2;
01089         s++; name = s;
01090         while ( *s && *s != '"' ) {
01091             if ( *s == '\\' ) s++;
01092             if ( *s == '%' && s[1] == '$' ) numdol++;
01093             s++;
01094         }
01095         if ( *s != '"' ) {
01096             MesPrint("&String in print statement should be enclosed in \"");
01097             return(1);
01098         }
01099         *s = 0;
01100         AddComString(3,code,name,1);
01101         *s++ = '"';
01102         while ( *s == ',' ) {
01103             s++;
01104             if ( *s == '$' ) {
01105                 s++; name = s; while ( FG.cTable[*s] <= 1 ) s++;
01106                 c = *s; *s = 0;
01107                 type = GetName(AC.dollarnames,name,&numexpr,NOAUTO);
01108                 if ( type == NAMENOTFOUND ) {
01109                     MesPrint("&$ variable %s not (yet) defined",name);
01110                     error = 1;
01111                 }
01112                 else {
01113                     C->lhs[C->numlhs][1] += 2;
01114                     *(C->Pointer)++ = DOLLAREXPRESSSION;
01115                     *(C->Pointer)++ = numexpr;
01116                     numdol--;
01117                 }
01118             }
01119             else {
01120                 MesPrint("&Illegal object in print statement");
01121                 error = 1;
01122                 return(error);
01123             }
01124             *s = c;

```

```

01125         if ( c == '[' ) {
01126             w = C->Pointer;
01127             s++;
01128             s = GetDoParam(s,&(C->Pointer),-1);
01129             if ( s == 0 ) return(1);
01130             if ( *s != ']' ) {
01131                 MesPrint("&unmatched [] in $ factor");
01132                 return(1);
01133             }
01134             C->lhs[C->numlhs][1] += C->Pointer - w;
01135             s++;
01136         }
01137     }
01138     if ( *s != 0 ) {
01139         MesPrint("&Illegal object in print statement");
01140         error = 1;
01141     }
01142     if ( numdol > 0 ) {
01143         MesPrint("&More $ variables asked for than provided");
01144         error = 1;
01145     }
01146     *(C->Pointer)++ = 0;
01147     return(error);
01148 }
01149 if ( *s == 0 ) { /* All active expressions */
01150 AllExpr:
01151     for ( e = Expressions, i = NumExpressions; i > 0; i--, e++ ) {
01152         if ( e->status == LOCALEXPRESSION || e->status ==
01153             GLOBALEXPRESSION || e->status == UNHIDELEXPRESSION
01154             || e->status == UNHIDEEXPRESSION ) e->printflag = par;
01155     }
01156     return(error);
01157 }
01158 while ( *s ) {
01159     if ( *s == '+' ) {
01160         s++;
01161         if ( tolower(*s) == 'f' ) par |= PRINTLFILE;
01162         else if ( tolower(*s) == 's' ) {
01163             if ( tolower(s[1]) == 's' ) {
01164                 if ( tolower(s[2]) == 's' ) {
01165                     par |= PRINTONEFUNCTION | PRINTONETERM | PRINTALL;
01166                     s++;
01167                 }
01168                 else if ( ( par & 3 ) < 2 ) par |= PRINTONEFUNCTION | PRINTONETERM;
01169                 s++;
01170             }
01171             else {
01172                 if ( ( par & 3 ) < 2 ) par |= PRINTONETERM;
01173             }
01174         }
01175         else {
01176 illeg:
01177             MesPrint("&Illegal option in (n)print statement");
01178             error = 1;
01179         }
01180         s++;
01181         if ( *s == 0 ) goto AllExpr;
01182     }
01183     else if ( *s == '-' ) {
01184         s++;
01185         if ( tolower(*s) == 'f' ) par &= ~PRINTLFILE;
01186         else if ( tolower(*s) == 's' ) {
01187             if ( tolower(s[1]) == 's' ) {
01188                 if ( tolower(s[2]) == 's' ) {
01189                     par &= ~PRINTALL;
01190                     s++;
01191                 }
01192                 else if ( ( par & 3 ) < 2 ) {
01193                     par &= ~PRINTONEFUNCTION;
01194                     par &= ~PRINTALL;
01195                 }
01196                 s++;
01197             }
01198             else {
01199                 if ( ( par & 3 ) < 2 ) {
01200                     par &= ~PRINTONETERM;
01201                     par &= ~PRINTONEFUNCTION;
01202                     par &= ~PRINTALL;
01203                 }
01204             }
01205             else goto illeg;
01206             s++;
01207             if ( *s == 0 ) goto AllExpr;
01208         }
01209     }
01210     else if ( FG.cTable[*s] == 0 || *s == '[' ) {
01211         name = s;
01212         if ( ( s = SkipAName(s) ) == 0 ) {

```



```

01212         MesPrint("&Improper name in (n)print statement");
01213         return(1);
01214     }
01215     c = *s; *s = 0;
01216     if ( ( GetName(AC.exprnames,name,&numexpr,NOAUTO) == CEXPRESSION )
01217         && ( Expressions[numexpr].status == LOCALEXPRESSION
01218         || Expressions[numexpr].status == GLOBALEXPRESSION ) ) {
01219 FoundExpr;;
01220         if ( c == '[' && s[1] == ']' ) {
01221             Expressions[numexpr].printflag = par | PRINTCONTENTS;
01222             *s++ = c; c = *++s;
01223         }
01224         else
01225             Expressions[numexpr].printflag = par;
01226     }
01227     else if ( GetLastExprName(name,&numexpr)
01228         && ( Expressions[numexpr].status == LOCALEXPRESSION
01229         || Expressions[numexpr].status == GLOBALEXPRESSION
01230         || Expressions[numexpr].status == UNHIDELEXPRESSION
01231         || Expressions[numexpr].status == UNHIDEGEXPRESSION
01232         ) ) {
01233         goto FoundExpr;
01234     }
01235     else {
01236         MesPrint("%s is not the name of an active expression",name);
01237         error = 1;
01238     }
01239     *s++ = c;
01240     if ( c == 0 ) return(0);
01241     if ( c == '-' || c == '+' ) s--;
01242 }
01243 else if ( *s == ',' ) s++;
01244 else {
01245     MesPrint("&Illegal object in (n)print statement");
01246     return(1);
01247 }
01248 }
01249 return(0);
01250 }
01251
01252 /*
01253 #] DoPrint :
01254 #[ CoPrint :
01255 */
01256
01257 int CoPrint(UBYTE *s) { return(DoPrint(s,PRINTON)); }
01258
01259 /*
01260 #] CoPrint :
01261 #[ CoPrintB :
01262 */
01263
01264 int CoPrintB(UBYTE *s) { return(DoPrint(s,PRINTCONTENT)); }
01265
01266 /*
01267 #] CoPrintB :
01268 #[ CoNPrint :
01269 */
01270
01271 int CoNPrint(UBYTE *s) { return(DoPrint(s,PRINTOFF)); }
01272
01273 /*
01274 #] CoNPrint :
01275 #[ CoPushHide :
01276 */
01277
01278 int CoPushHide(UBYTE *s)
01279 {
01280     GETIDENTITY
01281     WORD *ScratchBuf;
01282     int i;
01283     if ( AR.Fscr[2].PObuffer == 0 ) {
01284         ScratchBuf = (WORD *)Malloc1(AM.HideSize*sizeof(WORD),"hidesize");
01285         AR.Fscr[2].POsize = AM.HideSize * sizeof(WORD);
01286         AR.Fscr[2].POfull = AR.Fscr[2].POfill = AR.Fscr[2].PObuffer = ScratchBuf;
01287         AR.Fscr[2].POstop = AR.Fscr[2].PObuffer + AM.HideSize;
01288         PUTZERO(AR.Fscr[2].POposition);
01289     }
01290     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
01291     AC.HideLevel += 2;
01292     if ( *s ) {
01293         MesPrint("&PushHide statement should have no arguments");
01294         return(-1);
01295     }
01296     for ( i = 0; i < NumExpressions; i++ ) {
01297         switch ( Expressions[i].status ) {
01298             case DROPLEXPRESSION:

```

```

01299         case SKIPLEXPRESSION:
01300         case LOCALEXPRESSION:
01301             Expressions[i].status = HIDELEXPRESSION;
01302             Expressions[i].hidelevel = AC.HideLevel-1;
01303             break;
01304         case DROPGEXPRESSION:
01305         case SKIPGEXPRESSION:
01306         case GLOBALEXPRESSION:
01307             Expressions[i].status = HIDEGEXPRESSION;
01308             Expressions[i].hidelevel = AC.HideLevel-1;
01309             break;
01310         default:
01311             break;
01312     }
01313 }
01314 return(0);
01315 }
01316
01317 /*
01318 #] CoPushHide :
01319 #[ CoPopHide :
01320 */
01321
01322 int CoPopHide(UBYTE *s)
01323 {
01324     int i;
01325     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
01326     if ( AC.HideLevel <= 0 ) {
01327         MesPrint("&PopHide statement without corresponding PushHide statement");
01328         return(-1);
01329     }
01330     AC.HideLevel -= 2;
01331     if ( *s ) {
01332         MesPrint("&PopHide statement should have no arguments");
01333         return(-1);
01334     }
01335     for ( i = 0; i < NumExpressions; i++ ) {
01336         switch ( Expressions[i].status ) {
01337             case HIDDENLEXPRESSION:
01338                 if ( Expressions[i].hidelevel > AC.HideLevel )
01339                     Expressions[i].status = UNHIDELEXPRESSION;
01340                 break;
01341             case HIDDENGEXPRESSION:
01342                 if ( Expressions[i].hidelevel > AC.HideLevel )
01343                     Expressions[i].status = UNHIDEGEXPRESSION;
01344                 break;
01345             default:
01346                 break;
01347         }
01348     }
01349     return(0);
01350 }
01351
01352 /*
01353 #] CoPopHide :
01354 #[ SetExprCases :
01355 */
01356
01357 int SetExprCases(int par, int setunset, int val)
01358 {
01359     switch ( par ) {
01360         case SKIP:
01361             switch ( val ) {
01362                 case SKIPLEXPRESSION:
01363                     if ( !setunset ) val = LOCALEXPRESSION;
01364                     break;
01365                 case SKIPGEXPRESSION:
01366                     if ( !setunset ) val = GLOBALEXPRESSION;
01367                     break;
01368                 case LOCALEXPRESSION:
01369                     if ( setunset ) val = SKIPLEXPRESSION;
01370                     break;
01371                 case GLOBALEXPRESSION:
01372                     if ( setunset ) val = SKIPGEXPRESSION;
01373                     break;
01374                 case INTOHIDEGEXPRESSION:
01375                 case INTOHIDELEXPRESSION:
01376                 default:
01377                     break;
01378             }
01379             break;
01380         case DROP:
01381             switch ( val ) {
01382                 case SKIPLEXPRESSION:
01383                 case LOCALEXPRESSION:
01384                 case HIDELEXPRESSION:
01385                     if ( setunset ) val = DROPLEXPRESSION;

```

```

01386         break;
01387     case DROPEXPRESSION:
01388         if ( !setunset ) val = LOCALEXPRESSION;
01389         break;
01390     case SKIPGEXPRESSION:
01391     case GLOBALEXPRESSION:
01392     case HIDEGEXPRESSION:
01393         if ( setunset ) val = DROPGEXPRESSION;
01394         break;
01395     case DROPGEXPRESSION:
01396         if ( !setunset ) val = GLOBALEXPRESSION;
01397         break;
01398     case HIDDENLEXPRESSION:
01399     case UNHIDELEXPRESSION:
01400         if ( setunset ) val = DROPHLEXPRESSION;
01401         break;
01402     case HIDDENGEXPRESSION:
01403     case UNHIDEGEXPRESSION:
01404         if ( setunset ) val = DROPHGEXPRESSION;
01405         break;
01406     case DROPHLEXPRESSION:
01407         if ( !setunset ) val = HIDDENLEXPRESSION;
01408         break;
01409     case DROPHGEXPRESSION:
01410         if ( !setunset ) val = HIDDENGEXPRESSION;
01411         break;
01412     case INTOHIDEGEXPRESSION:
01413     case INTOHIDELEXPRESSION:
01414     default:
01415         break;
01416     }
01417     break;
01418 case HIDE:
01419     switch ( val ) {
01420     case DROPEXPRESSION:
01421     case SKIPLEXPRESSION:
01422     case LOCALEXPRESSION:
01423         if ( setunset ) val = HIDELEXPRESSION;
01424         break;
01425     case HIDELEXPRESSION:
01426         if ( !setunset ) val = LOCALEXPRESSION;
01427         break;
01428     case DROPGEXPRESSION:
01429     case SKIPGEXPRESSION:
01430     case GLOBALEXPRESSION:
01431         if ( setunset ) val = HIDEGEXPRESSION;
01432         break;
01433     case HIDEGEXPRESSION:
01434         if ( !setunset ) val = GLOBALEXPRESSION;
01435         break;
01436     case INTOHIDEGEXPRESSION:
01437     case INTOHIDELEXPRESSION:
01438     default:
01439         break;
01440     }
01441     break;
01442 case UNHIDE:
01443     switch ( val ) {
01444     case HIDDENLEXPRESSION:
01445     case DROPHLEXPRESSION:
01446         if ( setunset ) val = UNHIDELEXPRESSION;
01447         break;
01448     case UNHIDELEXPRESSION:
01449         if ( !setunset ) val = HIDDENLEXPRESSION;
01450         break;
01451     case HIDDENGEXPRESSION:
01452     case DROPHGEXPRESSION:
01453         if ( setunset ) val = UNHIDEGEXPRESSION;
01454         break;
01455     case UNHIDEGEXPRESSION:
01456         if ( !setunset ) val = HIDDENGEXPRESSION;
01457         break;
01458     case INTOHIDEGEXPRESSION:
01459     case INTOHIDELEXPRESSION:
01460     default:
01461         break;
01462     }
01463     break;
01464 case INTOHIDE:
01465     switch ( val ) {
01466     case HIDDENLEXPRESSION:
01467     case HIDDENGEXPRESSION:
01468         MesPrint("&Expression is already hidden");
01469         return(-1);
01470     case DROPHLEXPRESSION:
01471     case DROPHGEXPRESSION:
01472     case UNHIDELEXPRESSION:

```

```

01473         case UNHIDEGEXPRESSION:
01474             if ( setunset ) {
01475                 MesPrint("&Cannot unhide/drop and put intohide expression in the same
module");
01476                 return(-1);
01477             }
01478             break;
01479         case LOCALEXPRESSION:
01480         case DROPLEXPRESSION:
01481         case SKIPLEXPRESSION:
01482         case HIDELEXPRESSION:
01483             if ( setunset ) val = INTOHIDELEXPRESSION;
01484             break;
01485         case GLOBALEXPRESSION:
01486         case DROPGEXPRESSION:
01487         case SKIPGEXPRESSION:
01488         case HIDEGEXPRESSION:
01489             if ( setunset ) val = INTOHIDEGEXPRESSION;
01490             break;
01491         case INTOHIDELEXPRESSION:
01492             if ( !setunset ) val = LOCALEXPRESSION;
01493             break;
01494         case INTOHIDEGEXPRESSION:
01495             if ( !setunset ) val = GLOBALEXPRESSION;
01496             break;
01497         default:
01498             break;
01499     }
01500     break;
01501 default:
01502     break;
01503 }
01504 return(val);
01505 }
01506
01507 /*
01508 #] SetExprCases :
01509 #[ SetExpr :
01510 */
01511
01512 int SetExpr(UBYTE *s, int setunset, int par)
01513 {
01514     WORD *w, numexpr;
01515     int error = 0, i;
01516     UBYTE *name, c;
01517     if ( *s == 0 ) {
01518         for ( i = 0; i < NumExpressions; i++ ) {
01519             w = &(Expressions[i].status);
01520             *w = SetExprCases(par,setunset,*w);
01521             if ( *w < 0 ) error = 1;
01522             if ( ( par == HIDE || par == INTOHIDE ) && setunset == 1 )
01523                 Expressions[i].hidelevel = AC.HideLevel;
01524         }
01525         return(0);
01526     }
01527     while ( *s ) {
01528         if ( *s == ',' ) { s++; continue; }
01529         if ( *s == '0' ) { s++; continue; }
01530         name = s;
01531         if ( ( s = SkipAName(s) ) == 0 ) {
01532             MesPrint("&Improper name for an expression: '%s'",name);
01533             return(1);
01534         }
01535         c = *s; *s = 0;
01536         if ( GetName(AC.exprnames,name,&numexpr,NOAUTO) == CEXPRESSION ) {
01537             w = &(Expressions[numexpr].status);
01538             *w = SetExprCases(par,setunset,*w);
01539             if ( *w < 0 ) error = 1;
01540             if ( ( par == HIDE || par == INTOHIDE ) && setunset == 1 )
01541                 Expressions[numexpr].hidelevel = AC.HideLevel;
01542         }
01543         else if ( GetName(AC.varnames,name,&numexpr,NOAUTO) != NAMENOTFOUND ) {
01544             MesPrint("&%s is not an expression",name);
01545             error = 1;
01546         }
01547         *s = c;
01548     }
01549     return(error);
01550 }
01551
01552 /*
01553 #] SetExpr :
01554 #[ CoDrop :
01555 */
01556
01557 int CoDrop(UBYTE *s) { return(SetExpr(s,1,DROP)); }
01558

```

```

01559 /*
01560      #] CoDrop :
01561      #[ CoNoDrop :
01562 */
01563
01564 int CoNoDrop(UBYTE *s) { return(SetExpr(s,0,DROP)); }
01565
01566 /*
01567      #] CoNoDrop :
01568      #[ CoSkip :
01569 */
01570
01571 int CoSkip(UBYTE *s) { return(SetExpr(s,1,SKIP)); }
01572
01573 /*
01574      #] CoSkip :
01575      #[ CoNoSkip :
01576 */
01577
01578 int CoNoSkip(UBYTE *s) { return(SetExpr(s,0,SKIP)); }
01579
01580 /*
01581      #] CoNoSkip :
01582      #[ CoHide :
01583 */
01584
01585 int CoHide(UBYTE *inp) {
01586     GETIDENTITY
01587     WORD *ScratchBuf;
01588     if ( AR.Fscr[2].PObuffer == 0 ) {
01589         ScratchBuf = (WORD *)Malloc1(AM.HideSize*sizeof(WORD),"hidesize");
01590         AR.Fscr[2].POsize = AM.HideSize * sizeof(WORD);
01591         AR.Fscr[2].POfull = AR.Fscr[2].POfill = AR.Fscr[2].PObuffer = ScratchBuf;
01592         AR.Fscr[2].POstop = AR.Fscr[2].PObuffer + AM.HideSize;
01593         PUTZERO(AR.Fscr[2].POposition);
01594     }
01595     return(SetExpr(inp,1,HIDE));
01596 }
01597
01598 /*
01599      #] CoHide :
01600      #[ CoIntoHide :
01601 */
01602
01603 int CoIntoHide(UBYTE *inp) {
01604     GETIDENTITY
01605     WORD *ScratchBuf;
01606     if ( AR.Fscr[2].PObuffer == 0 ) {
01607         ScratchBuf = (WORD *)Malloc1(AM.HideSize*sizeof(WORD),"hidesize");
01608         AR.Fscr[2].POsize = AM.HideSize * sizeof(WORD);
01609         AR.Fscr[2].POfull = AR.Fscr[2].POfill = AR.Fscr[2].PObuffer = ScratchBuf;
01610         AR.Fscr[2].POstop = AR.Fscr[2].PObuffer + AM.HideSize;
01611         PUTZERO(AR.Fscr[2].POposition);
01612     }
01613     return(SetExpr(inp,1,INTOHIDE));
01614 }
01615
01616 /*
01617      #] CoIntoHide :
01618      #[ CoNoIntoHide :
01619 */
01620
01621 int CoNoIntoHide(UBYTE *inp) { return(SetExpr(inp,0,INTOHIDE)); }
01622
01623 /*
01624      #] CoNoIntoHide :
01625      #[ CoNoHide :
01626 */
01627
01628 int CoNoHide(UBYTE *inp) { return(SetExpr(inp,0,HIDE)); }
01629
01630 /*
01631      #] CoNoHide :
01632      #[ CoUnHide :
01633 */
01634
01635 int CoUnHide(UBYTE *inp) { return(SetExpr(inp,1,UNHIDE)); }
01636
01637 /*
01638      #] CoUnHide :
01639      #[ CoNoUnHide :
01640 */
01641
01642 int CoNoUnHide(UBYTE *inp) { return(SetExpr(inp,0,UNHIDE)); }
01643
01644 /*
01645      #] CoNoUnHide :

```

```

01646     #[ AddToCom :
01647 */
01648
01649 void AddToCom(int n, WORD *array)
01650 {
01651     CBUF *C = cbuf+AC.cbunum;
01652 #ifdef COMPUFDEBBUG
01653     MesPrint(" %a",n,array);
01654 #endif
01655     while ( C->Pointer+n >= C->Top ) DoubleCbuffer(AC.cbunum,C->Pointer,18);
01656     while ( --n >= 0 ) *(C->Pointer)++ = *array++;
01657 }
01658
01659 /*
01660 #] AddToCom :
01661 #[ AddComString :
01662 */
01663
01664 int AddComString(int n, WORD *array, UBYTE *thestring, int par)
01665 {
01666     CBUF *C = cbuf+AC.cbunum;
01667     UBYTE *s = thestring, *w;
01668 #ifdef COMPUFDEBBUG
01669     WORD *cc;
01670     UBYTE *ww;
01671 #endif
01672     int i, numchars = 0, size, zeroes;
01673     while ( *s ) {
01674         if ( *s == '\\\\' ) s++;
01675         else if ( par == 1 &&
01676             ( ( *s == '%' && s[1] != 't' && s[1] != 'T' && s[1] != '$' &&
01677               s[1] != 'w' && s[1] != 'W' && s[1] != 'r' && s[1] != 0 ) || *s == '#'
01678             || *s == '@' || *s == '&' ) ) {
01679             numchars++;
01680         }
01681         s++; numchars++;
01682     }
01683     AddLHS(AC.cbunum);
01684     size = numchars/sizeof(WORD)+1;
01685     while ( C->Pointer+size+n+2 >= C->Top ) DoubleCbuffer(AC.cbunum,C->Pointer,19);
01686 #ifdef COMPUFDEBBUG
01687     cc = C->Pointer;
01688 #endif
01689     *(C->Pointer)++ = array[0];
01690     *(C->Pointer)++ = size+n+2;
01691     for ( i = 1; i < n; i++ ) *(C->Pointer)++ = array[i];
01692     *(C->Pointer)++ = size;
01693 #ifdef COMPUFDEBBUG
01694     ww =
01695 #endif
01696     w = (UBYTE *) (C->Pointer);
01697     zeroes = size*sizeof(WORD)-numchars;
01698     s = thestring;
01699     while ( *s ) {
01700         if ( *s == '\\\\' ) s++;
01701         else if ( par == 1 && ( *s == '%' &&
01702             s[1] != 't' && s[1] != 'T' && s[1] != '$' &&
01703             s[1] != 'w' && s[1] != 'W' && s[1] != 'r' && s[1] != 0 ) || *s == '#'
01704             || *s == '@' || *s == '&' ) ) {
01705             *w++ = '%';
01706         }
01707         *w++ = *s++;
01708     }
01709     while ( --zeroes >= 0 ) *w++ = 0;
01710     C->Pointer += size;
01711 #ifdef COMPUFDEBBUG
01712     MesPrint("LH: %a",size+1+n,cc);
01713     MesPrint(" %s",thestring);
01714 #endif
01715     return(0);
01716 }
01717
01718 /*
01719 #] AddComString :
01720 #[ Add2ComStrings :
01721 */
01722
01723 int Add2ComStrings(int n, WORD *array, UBYTE *string1, UBYTE *string2)
01724 {
01725     CBUF *C = cbuf+AC.cbunum;
01726     UBYTE *s1 = string1, *s2 = string2, *w;
01727     int i, num1chars = 0, num2chars = 0, size1, size2, zeroes1, zeroes2;
01728     AddLHS(AC.cbunum);
01729     while ( *s1 ) { s1++; num1chars++; }
01730     size1 = num1chars/sizeof(WORD)+1;
01731     if ( *s2 ) {
01732         while ( *s2 ) { s2++; num2chars++; }

```

```

01733     size2 = num2chars/sizeof(WORD)+1;
01734 }
01735 else size2 = 0;
01736 while ( C->Pointer+size1+size2+n+3 >= C->Top ) DoubleCbuffer (AC.cbufnum,C->Pointer,20);
01737 *(C->Pointer)++ = array[0];
01738 *(C->Pointer)++ = size1+size2+n+3;
01739 for ( i = 1; i < n; i++ ) *(C->Pointer)++ = array[i];
01740 *(C->Pointer)++ = size1;
01741 w = (UBYTE *) (C->Pointer);
01742 zeroes1 = size1*sizeof(WORD)-num1chars;
01743 s1 = string1;
01744 while ( *s1 ) { *w++ = *s1++; }
01745 while ( --zeroes1 >= 0 ) *w++ = 0;
01746 C->Pointer += size1;
01747 *(C->Pointer)++ = size2;
01748 if ( size2 ) {
01749     w = (UBYTE *) (C->Pointer);
01750     zeroes2 = size2*sizeof(WORD)-num2chars;
01751     s2 = string2;
01752     while ( *s2 ) { *w++ = *s2++; }
01753     while ( --zeroes2 >= 0 ) *w++ = 0;
01754     C->Pointer += size2;
01755 }
01756 return(0);
01757 }
01758
01759 /*
01760 #] Add2ComStrings :
01761 #[ CoDiscard :
01762 */
01763
01764 int CoDiscard(UBYTE *s)
01765 {
01766     if ( *s == 0 ) {
01767         Add2Com(TYPEDISCARD)
01768         return(0);
01769     }
01770     MesPrint("&Illegal argument in discard statement: '%s'",s);
01771     return(1);
01772 }
01773
01774 /*
01775 #] CoDiscard :
01776 #[ CoContract :
01777
01778 Syntax:
01779     Contract
01780     Contract:#
01781     Contract #
01782     Contract:##, #
01783 */
01784
01785 static WORD carray[5] = { TYPEOPERATION,5,CONTRACT,0,0 };
01786
01787 int CoContract(UBYTE *s)
01788 {
01789     int x;
01790     if ( *s == ':' ) {
01791         s++;
01792         ParseNumber(x,s)
01793         if ( *s != ',' && *s ) {
01794 proper:         MesPrint("&Illegal number in contract statement");
01795                 return(1);
01796             }
01797             if ( *s ) s++;
01798             carray[4] = x;
01799         }
01800         else carray[4] = 0;
01801         if ( FG.cTable[*s] == 1 ) {
01802             ParseNumber(x,s)
01803             if ( *s ) goto proper;
01804             carray[3] = x;
01805         }
01806         else if ( *s ) goto proper;
01807         else carray[3] = -1;
01808         return(AddNtoL(5,carray));
01809     }
01810
01811 /*
01812 #] CoContract :
01813 #[ CoGoTo :
01814 */
01815
01816 int CoGoTo(UBYTE *inp)
01817 {
01818     UBYTE *s = inp;
01819     int x;

```

```

01820     while ( FG.cTable[*s] <= 1 ) s++;
01821     if ( *s ) {
01822         MesPrint("&Label should be an alpha-numeric string");
01823         return(1);
01824     }
01825     x = GetLabel(inp);
01826     Add3Com(TYPEGOTO,x);
01827     return(0);
01828 }
01829
01830 /*
01831 #] CoGoTo :
01832 #[ CoLabel :
01833 */
01834
01835 int CoLabel(UBYTE *inp)
01836 {
01837     UBYTE *s = inp;
01838     int x;
01839     while ( FG.cTable[*s] <= 1 ) s++;
01840     if ( *s ) {
01841         MesPrint("&Label should be an alpha-numeric string");
01842         return(1);
01843     }
01844     x = GetLabel(inp);
01845     if ( AC.Labels[x] >= 0 ) {
01846         MesPrint("&Label %s defined more than once",AC.LabelNames[x]);
01847         return(1);
01848     }
01849     AC.Labels[x] = cbuf[AC.cbunum].numlhs;
01850     return(0);
01851 }
01852
01853 /*
01854 #] CoLabel :
01855 #[ DoArgument :
01856
01857 Layout:
01858     par,full size,numlhs(+1),par,scale
01859     scale is for normalize
01860 */
01861
01862 int DoArgument(UBYTE *s, int par)
01863 {
01864     GETIDENTITY
01865     UBYTE *name, *t, *v, c;
01866     WORD *oldworkpointer = AT.WorkPointer, *w, *ww, number, *scale;
01867     int error = 0, zeroflag, type, x;
01868     AC.lhdollarflag = 0;
01869     while ( *s == ',' ) s++;
01870     w = AT.WorkPointer;
01871     *w++ = par;
01872     w++;
01873     switch ( par ) {
01874         case TYPEARG:
01875             if ( AC.arglevel >= MAXNEST ) {
01876                 MesPrint("@Nesting of argument statements more than %d levels"
01877                     , (WORD)MAXNEST);
01878                 return(-1);
01879             }
01880             AC.argsumcheck[AC.arglevel] = NestingChecksum();
01881             AC.argstack[AC.arglevel] = cbuf[AC.cbunum].Pointer
01882                 - cbuf[AC.cbunum].Buffer + 2;
01883             AC.arglevel++;
01884             *w++ = cbuf[AC.cbunum].numlhs;
01885             break;
01886         case TYPENORM:
01887         case TYPENORM4:
01888         case TYPESPLITARG:
01889         case TYPESPLITFIRSTARG:
01890         case TYPESPLITLASTARG:
01891         case TYPEFACTARG:
01892         case TYPEARGTOEXTRASymbol:
01893             *w++ = cbuf[AC.cbunum].numlhs+1;
01894             break;
01895     }
01896     *w++ = par;
01897     scale = w;
01898     *w++ = 1;
01899     *w++ = 0;
01900     if ( *s == '^' ) {
01901         s++; ParseSignedNumber(x,s)
01902         while ( *s == ',' ) s++;
01903         *scale = x;
01904     }
01905     if ( *s == '(' ) {
01906         t = s+1; SKIPBRA3(s)      /* We did check the brackets already */

```



```

01907     if ( par == TYPEARG ) {
01908         MesPrint("&Illegal () entry in argument statement");
01909         error = 1; s++; goto skipbracks;
01910     }
01911     else if ( par == TYPESPLITFIRSTARG ) {
01912         MesPrint("&Illegal () entry in splitfirstarg statement");
01913         error = 1; s++; goto skipbracks;
01914     }
01915     else if ( par == TYPESPLITLASTARG ) {
01916         MesPrint("&Illegal () entry in splitlastarg statement");
01917         error = 1; s++; goto skipbracks;
01918     }
01919     v = t;
01920     while ( v < s ) {
01921         if ( *v == '?' ) {
01922             MesPrint("&Wildcarding not allowed in this type of statement");
01923             error = 1; break;
01924         }
01925         v++;
01926     }
01927     v = s++;
01928     if ( *t == '(' && v[-1] == ')' ) {
01929         t++; v--;
01930         if ( par == TYPESPLITARG ) oldworkpointer[0] = TYPESPLITARG2;
01931         else if ( par == TYPEFACTARG ) oldworkpointer[0] = TYPEFACTARG2;
01932         else if ( par == TYPENORM4 ) oldworkpointer[0] = TYPENORM4;
01933         else if ( par == TYPENORM ) {
01934             if ( *t == '-' ) { oldworkpointer[0] = TYPENORM3; t++; }
01935             else { oldworkpointer[0] = TYPENORM2; *scale = 0; }
01936         }
01937     }
01938     if ( error == 0 ) {
01939         CBUF *C = cbuf+AC.cbunum;
01940         WORD oldnumrhs = C->numrhs, oldnumlhs = C->numlhs;
01941         WORD prototype[SUBEXPSIZE+40]; /* Up to 10 nested sums! */
01942         WORD *m, *mm;
01943         int i, retcode;
01944         LONG oldpointer = C->Pointer - C->Buffer;
01945         *v = 0;
01946         prototype[0] = SUBEXPRESSION;
01947         prototype[1] = SUBEXPSIZE;
01948         prototype[2] = C->numrhs+1;
01949         prototype[3] = 1;
01950         prototype[4] = AC.cbunum;
01951         AT.WorkPointer += TYPEARGHEADSIZE+1;
01952         AddLHS(AC.cbunum);
01953         if ( ( retcode = CompileAlgebra(t,LHSIDE,prototype) ) < 0 )
01954             error = 1;
01955         else {
01956             prototype[2] = retcode;
01957             ww = C->lhs[retcode];
01958             AC.lhdollarflag = 0;
01959             if ( *ww == 0 ) {
01960                 *w++ = -2; *w++ = 0;
01961             }
01962             else if ( ww[ww[0]] != 0 ) {
01963                 MesPrint("&There should be only one term between ()");
01964                 error = 1;
01965             }
01966             else if ( NewSort(BHEAD0) ) { if ( !error ) error = 1; }
01967             else if ( NewSort(BHEAD0) ) {
01968                 LowerSortLevel();
01969                 if ( !error ) error = 1;
01970             }
01971             else {
01972                 AN.RepPoint = AT.RepCount + 1;
01973                 m = AT.WorkPointer;
01974                 mm = ww; i = *mm;
01975                 while ( --i >= 0 ) *m++ = *mm++;
01976                 mm = AT.WorkPointer; AT.WorkPointer = m;
01977                 AR.Cnumlhs = C->numlhs;
01978                 if ( Generator(BHEAD mm,C->numlhs) ) {
01979                     LowerSortLevel(); error = 1;
01980                 }
01981                 else if ( EndSort(BHEAD mm,0) < 0 ) {
01982                     error = 1;
01983                     AT.WorkPointer = mm;
01984                 }
01985                 else if ( *mm == 0 ) {
01986                     *w++ = -2; *w++ = 0;
01987                     AT.WorkPointer = mm;
01988                 }
01989                 else if ( mm[mm[0]] != 0 ) {
01990                     error = 1;
01991                     AT.WorkPointer = mm;
01992                 }
01993             }
01994         }
01995     }

```

```

01994         AT.WorkPointer = mm;
01995         m = mm+*mm;
01996         if ( par == TYPEFACTARG ) {
01997             if ( *mm != ABS(m[-1])+1 ) {
01998                 *mm -= ABS(m[-1]); /* Strip coefficient */
01999             }
02000             mm[-1] = -*mm-1; w += *mm+1;
02001         }
02002         else {
02003             *mm -= ABS(m[-1]); /* Strip coefficient */
02004         /*
02005             if ( *mm == 1 ) { *w++ = -2; *w++ = 0; }
02006             else
02007         /*
02008                 { mm[-1] = -*mm-1; w += *mm+1; }
02009             }
02010             oldworkpointer[1] = w - oldworkpointer;
02011         }
02012         LowerSortLevel();
02013     }
02014     oldworkpointer[5] = AC.lhdollarflag;
02015 }
02016 *v = ' ';
02017 C->numrhs = oldnumrhs;
02018 C->numlhs = oldnumlhs;
02019 C->Pointer = C->Buffer + oldpointer;
02020 }
02021 }
02022 skipbracks:
02023     if ( *s == 0 ) { *w++ = 0; *w++ = 2; *w++ = 1; }
02024     else {
02025         do {
02026             if ( *s == ',' ) { s++; continue; }
02027             ww = w; *w++ = 0; w++;
02028             if ( FG.cTable[*s] > 1 && *s != '[' && *s != '{' ) {
02029                 MesPrint("&Illegal parameters in statement");
02030                 error = 1;
02031                 break;
02032             }
02033             while ( FG.cTable[*s] == 0 || *s == '[' || *s == '{' ) {
02034                 if ( *s == '{' ) {
02035                     name = s+1;
02036                     SKIPBRA2(s)
02037                     c = *s; *s = 0;
02038                     number = DoTempSet(name,s);
02039                     name--; *s++ = c; c = *s; *s = 0;
02040                     goto doset;
02041                 }
02042                 else {
02043                     name = s;
02044                     if ( ( s = SkipAName(s) ) == 0 ) {
02045                         MesPrint("&Illegal name '%s'",name);
02046                         return(1);
02047                     }
02048                     c = *s; *s = 0;
02049                     if ( ( type = GetName(AC.varnames,name,&number,WITHAUTO) ) == CSET ) {
02050                         if ( Sets[number].type != CFUNCTION ) goto nofun;
02051                         *w++ = CSET; *w++ = number;
02052                     }
02053                     else if ( type == CFUNCTION ) {
02054                         *w++ = CFUNCTION; *w++ = number + FUNCTION;
02055                     }
02056                     else {
02057                         MesPrint("&%s is not a function or a set of functions",
02058                             name);
02059                         error = 1;
02060                     }
02061                 }
02062                 *s = c;
02063                 while ( *s == ',' ) s++;
02064             }
02065             ww[1] = w - ww;
02066             ww = w; w++; zeroflag = 0;
02067             while ( FG.cTable[*s] == 1 ) {
02068                 ParseNumber(x,s)
02069                 if ( *s && *s != ',' ) {
02070                     MesPrint("&Illegal separator after number");
02071                     error = 1;
02072                     while ( *s && *s != ',' ) s++;
02073                 }
02074                 while ( *s == ',' ) s++;
02075                 if ( x == 0 ) zeroflag = 1;
02076                 if ( !zeroflag ) *w++ = (WORD)x;
02077             }
02078             *ww = w - ww;
02079         } while ( *s );
02080     }

```

```

02081     oldworkpointer[1] = w - oldworkpointer;
02082     if ( par == TYPEARG ) { /* To make sure. The Pointer might move in the future */
02083         AC.argstack[AC.arglevel-1] = cbuf[AC.cbunum].Pointer
02084             - cbuf[AC.cbunum].Buffer + 2;
02085     }
02086     AddNtoL(oldworkpointer[1],oldworkpointer);
02087     AT.WorkPointer = oldworkpointer;
02088     return(error);
02089 }
02090
02091 /*
02092     #] DoArgument :
02093     #[ CoArgument :
02094 */
02095
02096 int CoArgument(UBYTE *s) { return(DoArgument(s,TYPEARG)); }
02097
02098 /*
02099     #] CoArgument :
02100     #[ CoEndArgument :
02101 */
02102
02103 int CoEndArgument(UBYTE *s)
02104 {
02105     CBUF *C = cbuf+AC.cbunum;
02106     while ( *s == ',' ) s++;
02107     if ( *s ) {
02108         MesPrint("%Illegal syntax for EndArgument statement");
02109         return(1);
02110     }
02111     if ( AC.arglevel <= 0 ) {
02112         MesPrint("&EndArgument without corresponding Argument statement");
02113         return(1);
02114     }
02115     AC.arglevel--;
02116     cbuf[AC.cbunum].Buffer[AC.argstack[AC.arglevel]] = C->numlhs;
02117     if ( AC.argsumcheck[AC.arglevel] != NestingChecksum() ) {
02118         MesNesting();
02119         return(1);
02120     }
02121     return(0);
02122 }
02123
02124 /*
02125     #] CoEndArgument :
02126     #[ CoInside :
02127 */
02128
02129 int CoInside(UBYTE *s) { return(ExecInside(s)); }
02130
02131 /*
02132     #] CoInside :
02133     #[ CoEndInside :
02134 */
02135
02136 int CoEndInside(UBYTE *s)
02137 {
02138     CBUF *C = cbuf+AC.cbunum;
02139     while ( *s == ',' ) s++;
02140     if ( *s ) {
02141         MesPrint("%Illegal syntax for EndInside statement");
02142         return(1);
02143     }
02144     if ( AC.insidelevel <= 0 ) {
02145         MesPrint("&EndInside without corresponding Inside statement");
02146         return(1);
02147     }
02148     AC.insidelevel--;
02149     cbuf[AC.cbunum].Buffer[AC.insidestack[AC.insidelevel]] = C->numlhs;
02150     if ( AC.insidesumcheck[AC.insidelevel] != NestingChecksum() ) {
02151         MesNesting();
02152         return(1);
02153     }
02154     return(0);
02155 }
02156
02157 /*
02158     #] CoEndInside :
02159     #[ CoNormalize :
02160 */
02161
02162 int CoNormalize(UBYTE *s) { return(DoArgument(s,TYPENORM)); }
02163
02164 /*
02165     #] CoNormalize :
02166     #[ CoMakeInteger :
02167 */

```

```

02168
02169 int CoMakeInteger(UBYTE *s) { return(DoArgument(s,TYPENORM4)); }
02170
02171 /*
02172 #] CoMakeInteger :
02173 #[ CoSplitArg :
02174 */
02175
02176 int CoSplitArg(UBYTE *s) { return(DoArgument(s,TYPESPLITARG)); }
02177
02178 /*
02179 #] CoSplitArg :
02180 #[ CoSplitFirstArg :
02181 */
02182
02183 int CoSplitFirstArg(UBYTE *s) { return(DoArgument(s,TYPESPLITFIRSTARG)); }
02184
02185 /*
02186 #] CoSplitFirstArg :
02187 #[ CoSplitLastArg :
02188 */
02189
02190 int CoSplitLastArg(UBYTE *s) { return(DoArgument(s,TYPESPLITLASTARG)); }
02191
02192 /*
02193 #] CoSplitLastArg :
02194 #[ CoFactArg :
02195 */
02196
02197 int CoFactArg(UBYTE *s) {
02198     if ( ( AC.topolynomialflag & TOPOLYNOMIALFLAG ) != 0 ) {
02199         MesPrint("&ToPolynomial statement and FactArg statement are not allowed in the same module");
02200         return(1);
02201     }
02202     AC.topolynomialflag |= FACTARGFLAG;
02203     return(DoArgument(s,TYPEFACTARG));
02204 }
02205
02206 /*
02207 #] CoFactArg :
02208 #[ DoSymmetrize :
02209
02210     Syntax:
02211     Symmetrize Fun[:[number]] [Fields]      -> par = 0;
02212     AntiSymmetrize Fun[:[number]] [Fields]  -> par = 1;
02213     CycleSymmetrize Fun[:[number]] [Fields] -> par = 2;
02214     RCycleSymmetrize Fun[:[number]] [Fields]-> par = 3;
02215 */
02216
02217 int DoSymmetrize(UBYTE *s, int par)
02218 {
02219     GETIDENTITY
02220     int extra = 0, error = 0, err, fix, x, groupsize, num, i;
02221     UBYTE *name, c;
02222     WORD funnum, *w, *ww, type;
02223     for(;;) {
02224         name = s;
02225         if ( ( s = SkipAName(s) ) == 0 ) {
02226             MesPrint("&Improper function name");
02227             return(1);
02228         }
02229         c = *s; *s = 0;
02230         if ( c != ',' || ( FG.cTable[s[1]] != 0 && s[1] != '(' ) ) break;
02231         if ( par <= 0 && StrICmp(name,(UBYTE *)"cyclic") == 0 ) extra = 2;
02232         else if ( par <= 0 && StrICmp(name,(UBYTE *)"rcyclic") == 0 ) extra = 6;
02233         else {
02234             MesPrint("&Illegal option: '%s'",name);
02235             error = 1;
02236         }
02237         *s++ = c;
02238     }
02239     if ( ( err = GetVar(name,&type,&funnum,CFUNCTION,WITHAUTO) ) == NAMENOTFOUND ) {
02240         MesPrint("&Undefined function: %s",name);
02241         AddFunction(name,0,0,0,0,0,-1,-1);
02242         *s++ = c;
02243         return(1);
02244     }
02245     funnum += FUNCTION;
02246     if ( err == -1 ) error = 1;
02247     *s = c;
02248     if ( *s == ':' ) {
02249         s++;
02250         if ( *s == ',' || *s == '(' || *s == 0 ) fix = -1;
02251         else if ( FG.cTable[*s] == 1 ) {
02252             ParseNumber(fix,s)
02253             if ( fix == 0 )
02254                 Warning("Restriction to zero arguments removed");

```

```

02255     }
02256     else {
02257         MesPrint("&Illegal character after :");
02258         return(1);
02259     }
02260 }
02261 else fix = 0;
02262 w = AT.WorkPointer;
02263 *w++ = TYPEOPERATION;
02264 w++;
02265 *w++ = SYMMETRIZE;
02266 *w++ = par | extra;
02267 *w++ = funnum;
02268 *w++ = fix;
02269 /*
02270 And now the argument lists. We have either ,#,#,... or (#,#,...,#),(#,...
02271 */
02272 w += 2; ww = w; groupsize = -1;
02273 while ( *s == ',' ) s++;
02274 while ( *s ) {
02275     if ( *s == '(' ) {
02276         s++; num = 0;
02277         while ( *s && *s != ')' ) {
02278             if ( *s == ',' ) { s++; continue; }
02279             if ( FG.cTable[*s] != 1 ) goto illarg;
02280             ParseNumber(x,s)
02281             if ( x <= 0 || ( fix > 0 && x > fix ) ) goto illnum;
02282             num++;
02283             *w++ = x-1;
02284         }
02285         if ( *s == 0 ) {
02286             MesPrint("&Improper termination of statement");
02287             return(1);
02288         }
02289         if ( groupsize < 0 ) groupsize = num;
02290         else if ( groupsize != num ) goto group;
02291         s++;
02292     }
02293     else if ( FG.cTable[*s] == 1 ) {
02294         if ( groupsize < 0 ) groupsize = 1;
02295         else if ( groupsize != 1 ) {
02296             group: MesPrint("&All groups should have the same number of arguments");
02297                 return(1);
02298         }
02299         ParseNumber(x,s)
02300         if ( x <= 0 || ( fix > 0 && x > fix ) ) {
02301             illnum: MesPrint("&Illegal argument number: %d",x);
02302                 return(1);
02303         }
02304         *w++ = x-1;
02305     }
02306     else {
02307         illarg: MesPrint("&Illegal argument");
02308                 return(1);
02309     }
02310     while ( *s == ',' ) s++;
02311 }
02312 /*
02313 Now the completion
02314 */
02315 if ( w == ww ) {
02316     ww[-1] = 1;
02317     ww[-2] = 0;
02318     if ( fix > 0 ) {
02319         for ( i = 0; i < fix; i++ ) *w++ = i;
02320         ww[-2] = fix; /* Bugfix 31-oct-2001. Reported by York Schroeder */
02321     }
02322 }
02323 else {
02324     ww[-1] = groupsize;
02325     ww[-2] = (w-ww)/groupsize;
02326 }
02327 AT.WorkPointer[1] = w - AT.WorkPointer;
02328 AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
02329 return(error);
02330 }
02331
02332 /*
02333 #] DoSymmetrize :
02334 #[ CoSymmetrize :
02335 */
02336
02337 int CoSymmetrize(UBYTE *s) { return(DoSyzmetrize(s,SYMMETRIC)); }
02338
02339 /*
02340 #] CoSymmetrize :
02341 #[ CoAntiSymmetrize :

```

```

02342 */
02343
02344 int CoAntiSymmetrize(UBYTE *s) { return(DoSometrize(s,ANTISYMMETRIC)); }
02345
02346 /*
02347     #] CoAntiSymmetrize :
02348     #[ CoCycleSymmetrize :
02349 */
02350
02351 int CoCycleSymmetrize(UBYTE *s) { return(DoSometrize(s,CYCLESYMMETRIC)); }
02352
02353 /*
02354     #] CoCycleSymmetrize :
02355     #[ CoRCycleSymmetrize :
02356 */
02357
02358 int CoRCycleSymmetrize(UBYTE *s) { return(DoSometrize(s,RCYCLESYMMETRIC)); }
02359
02360 /*
02361     #] CoRCycleSymmetrize :
02362     #[ CoWrite :
02363 */
02364
02365 int CoWrite(UBYTE *s)
02366 {
02367     GETIDENTITY
02368     UBYTE *option;
02369     KEYWORDV *key;
02370     option = s;
02371     if ( ( ( s = SkipAName(s) ) == 0 ) || *s != 0 ) {
02372         MesPrint("&Proper use of write statement is: write option");
02373         return(1);
02374     }
02375     key = (KEYWORDV *)FindInKeyWord(option, (KEYWORD
*)writeoptions,sizeof(writeoptions)/sizeof(KEYWORD));
02376     if ( key == 0 ) {
02377         MesPrint("&Unrecognized option in write statement");
02378         return(1);
02379     }
02380     *key->var = key->type;
02381     AR.SortType = AC.SortType;
02382     return(0);
02383 }
02384
02385 /*
02386     #] CoWrite :
02387     #[ CoNWrite :
02388 */
02389
02390 int CoNWrite(UBYTE *s)
02391 {
02392     GETIDENTITY
02393     UBYTE *option;
02394     KEYWORDV *key;
02395     option = s;
02396     if ( ( ( s = SkipAName(s) ) == 0 ) || *s != 0 ) {
02397         MesPrint("&Proper use of nwrite statement is: nwrite option");
02398         return(1);
02399     }
02400     key = (KEYWORDV *)FindInKeyWord(option, (KEYWORD
*)writeoptions,sizeof(writeoptions)/sizeof(KEYWORD));
02401     if ( key == 0 ) {
02402         MesPrint("&Unrecognized option in nwrite statement");
02403         return(1);
02404     }
02405     *key->var = key->flags;
02406     AR.SortType = AC.SortType;
02407     return(0);
02408 }
02409
02410 /*
02411     #] CoNWrite :
02412     #[ CoRatio :
02413 */
02414
02415 static WORD ratstring[6] = { TYPEOPERATION, 6, RATIO, 0, 0, 0 };
02416
02417 int CoRatio(UBYTE *s)
02418 {
02419     UBYTE c, *t;
02420     int i, type, error = 0;
02421     WORD numsym, *rs;
02422     rs = ratstring+3;
02423     for ( i = 0; i < 3; i++ ) {
02424         if ( *s ) {
02425             t = s;
02426             s = SkipAName(s);

```

```

02427         c = *s; *s = 0;
02428         if ( ( ( type = GetName(AC.varnames,t,&numsym,WITHAUTO) ) != CSYMBOL )
02429             && type != CDUBIOUS ) {
02430             MesPrint("&%s is not a symbol",t);
02431             error = 4;
02432             if ( type < 0 ) numsym = AddSymbol(t,-MAXPOWER,MAXPOWER,0,0);
02433         }
02434         *s = c;
02435         if ( *s == ',' ) s++;
02436     }
02437     else {
02438         if ( error == 0 )
02439             MesPrint("&The ratio statement needs three symbols for its arguments");
02440         error++;
02441         numsym = 0;
02442     }
02443     *rs++ = numsym;
02444 }
02445 AddNtoL(6, ratstring);
02446 return(error);
02447 }
02448
02449 /*
02450 #] CoRatio :
02451 #[ CoRedefine :
02452
02453 We have a preprocessor variable and a (new) value for it.
02454 This value is inside a string that must be stored.
02455 */
02456
02457 int CoRedefine(UBYTE *s)
02458 {
02459     UBYTE *name, c, *args = 0;
02460     int numprevar;
02461     WORD code[2];
02462     name = s;
02463     if ( FG.cTable[*s] || ( s = SkipAName(s) ) == 0 || s[-1] == '_' ) {
02464         MesPrint("&Illegal name for preprocessor variable in redefine statement");
02465         return(1);
02466     }
02467     c = *s; *s = 0;
02468     for ( numprevar = NumPre-1; numprevar >= 0; numprevar-- ) {
02469         if ( StrCmp(name, PreVar[numprevar].name) == 0 ) break;
02470     }
02471     if ( numprevar < 0 ) {
02472         MesPrint("&There is no preprocessor variable with the name '%s'", name);
02473         *s = c;
02474         return(1);
02475     }
02476     *s = c;
02477
02478 /*
02479 The next code worries about arguments.
02480 It is a direct copy of the code in TheDefine in the preprocessor.
02481 */
02482 if ( *s == '(' ) { /* arguments. scan for correctness */
02483     s++; args = s;
02484     for (;;) {
02485         if ( chartype[*s] != 0 ) goto illarg;
02486         s++;
02487         while ( chartype[*s] <= 1 ) s++;
02488         while ( *s == ' ' || *s == '\t' ) s++;
02489         if ( *s == ')' ) break;
02490         if ( *s != ',' ) goto illargs;
02491         s++;
02492         while ( *s == ' ' || *s == '\t' ) s++;
02493     }
02494     *s++ = 0;
02495     while ( *s == ' ' || *s == '\t' ) s++;
02496 }
02497 while ( *s == ',' ) s++;
02498 if ( *s != '"' ) {
02499     MesPrint("&Value for %s should be enclosed in double quotes",
02500             PreVar[numprevar].name);
02501     return(1);
02502 }
02503 s++; name = s; /* actually name points to the new string */
02504 while ( *s && *s != '"' ) { if ( *s == '\\' ) s++; s++; }
02505 if ( *s != '"' ) goto encl;
02506 *s = 0;
02507 code[0] = TYPEREDEFPRE; code[1] = numprevar;
02508
02509 /*
02510 AddComString(2, code, name, 0);
02511
02512 Add2ComStrings(2, code, name, args);
02513 *s = '"';
02514 #ifdef PARALLELLECODE
02515 */

```

```

02514     Now we prepare the input numbering system for pthreads.
02515     We need a list of preprocessor variables that are redefined in this
02516     module.
02517 */
02518 {
02519     int j;
02520     WORD *newpf;
02521     LONG *newin;
02522     for ( j = 0; j < AC.numpfirstnum; j++ ) {
02523         if ( numprevar == AC.pfirstnum[j] ) break;
02524     }
02525     if ( j >= AC.numpfirstnum ) { /* add to list */
02526         if ( j >= AC.sizepfirstnum ) {
02527             if ( AC.sizepfirstnum <= 0 ) { AC.sizepfirstnum = 10; }
02528             else { AC.sizepfirstnum = 2 * AC.sizepfirstnum; }
02529             newin = (LONG *)Malloc1(AC.sizepfirstnum*(sizeof(WORD)+sizeof(LONG)), "AC.pfirstnum");
02530             newpf = (WORD *) (newin+AC.sizepfirstnum);
02531             for ( j = 0; j < AC.numpfirstnum; j++ ) {
02532                 newpf[j] = AC.pfirstnum[j];
02533                 newin[j] = AC.inputnumbers[j];
02534             }
02535             if ( AC.inputnumbers ) M_free(AC.inputnumbers, "AC.pfirstnum");
02536             AC.inputnumbers = newin;
02537             AC.pfirstnum = newpf;
02538         }
02539         AC.pfirstnum[AC.numpfirstnum] = numprevar;
02540         AC.inputnumbers[AC.numpfirstnum] = -1;
02541         AC.numpfirstnum++;
02542     }
02543 }
02544 #endif
02545     return(0);
02546 illarg:;
02547     MesPrint("&Illegally formed name in argument of redefine statement");
02548     return(1);
02549 illargs:;
02550     MesPrint("&Illegally formed arguments in redefine statement");
02551     return(1);
02552 }
02553
02554 /*
02555     #] CoRedefine :
02556     #[ CoRenummer :
02557
02558     renumber      or renumber,0      Only exchanges (n^2 until no improvement)
02559     renumber,1    All permutations (could be slow)
02560 */
02561
02562 int CoRenummer(UBYTE *s)
02563 {
02564     int x;
02565     UBYTE *inp;
02566     while ( *s == ',' ) s++;
02567     inp = s;
02568     if ( *s == 0 ) { x = 0; }
02569     else ParseNumber(x,s)
02570     if ( *s == 0 && x >= 0 && x <= 1 ) {
02571         Add3Com(TYPERENUMBER,x);
02572         return(0);
02573     }
02574     MesPrint("&Illegal argument in Renummer statement: '%s'",inp);
02575     return(1);
02576 }
02577
02578 /*
02579     #] CoRenummer :
02580     #[ CoSum :
02581 */
02582
02583 int CoSum(UBYTE *s)
02584 {
02585     CBUF *C = cbuf+AC.cbufnum;
02586     UBYTE *ss = 0, c, *t;
02587     int error = 0, i = 0, type, x;
02588     WORD numindex,number;
02589     while ( *s ) {
02590         t = s;
02591         if ( *s == '$' ) {
02592             t++; s++; while ( FG.cTable[*s] < 2 ) s++;
02593             c = *s; *s = 0;
02594             if ( ( number = GetDollar(t) ) < 0 ) {
02595                 MesPrint("&Undefined variable %s",t);
02596                 if ( !error ) error = 1;
02597                 number = AddDollar(t,0,0,0);
02598             }
02599             numindex = -number;
02600         }

```



```

02601     else {
02602         if ( ( s = SkipAName(s) ) == 0 ) return(1);
02603         c = *s; *s = 0;
02604         if ( ( ( type = GetOName(AC.exprnames,t,&numindex,NOAUTO) ) != NAMENOTFOUND )
02605         || ( ( type = GetOName(AC.varnames,t,&numindex,WITHAUTO) ) != CINDEX ) ) {
02606             if ( type != NAMENOTFOUND ) error = NameConflict( type,t);
02607             else {
02608                 MesPrint("&%s should have been declared as an index",t);
02609                 error = 1;
02610                 numindex = AddIndex(s,AC.lDefDim,AC.lDefDim4) + AM.OffsetIndex;
02611             }
02612         }
02613     }
02614     Add3Com(TYPESUM,numindex);
02615     i = 3; *s = c;
02616     if ( *s == 0 ) break;
02617     if ( *s != ',' ) {
02618         MesPrint("&Illegal separator between objects in sum statement.");
02619         return(1);
02620     }
02621     s++;
02622     if ( FG.cTable[*s] == 0 || *s == '[' || *s == '$' ) {
02623         while ( FG.cTable[*s] == 0 || *s == '[' || *s == '$' ) {
02624             if ( *s == '$' ) {
02625                 s++;
02626                 ss = t = s;
02627                 while ( FG.cTable[*s] < 2 ) s++;
02628                 c = *s; *s = 0;
02629                 if ( ( number = GetDollar(t) ) < 0 ) {
02630                     MesPrint("&Undefined variable %s",t);
02631                     if ( !error ) error = 1;
02632                     number = AddDollar(t,0,0,0);
02633                 }
02634                 numindex = -number;
02635             }
02636             else {
02637                 ss = t = s;
02638                 if ( ( s = SkipAName(s) ) == 0 ) return(1);
02639                 c = *s; *s = 0;
02640                 if ( ( ( type = GetOName(AC.exprnames,t,&numindex,NOAUTO) ) != NAMENOTFOUND )
02641                 || ( ( type = GetOName(AC.varnames,t,&numindex,WITHAUTO) ) != CINDEX ) ) {
02642                     if ( type != NAMENOTFOUND ) error = NameConflict( type,t);
02643                     else {
02644                         MesPrint("&%s should have been declared as an index",t);
02645                         error = 1;
02646                         numindex = AddIndex(s,AC.lDefDim,AC.lDefDim4) + AM.OffsetIndex;
02647                     }
02648                 }
02649             }
02650             AddToCB(C,numindex)
02651             i++;
02652             C->Pointer[-i+1] = i;
02653             *s = c;
02654             if ( *s == 0 ) return(error);
02655             if ( *s != ',' ) {
02656                 MesPrint("&Illegal separator between objects in sum statement.");
02657                 return(1);
02658             }
02659             s++;
02660         }
02661         if ( FG.cTable[*s] == 1 ) {
02662             C->Pointer[-i+1]--; C->Pointer--; s = ss;
02663         }
02664     }
02665     else if ( FG.cTable[*s] == 1 ) {
02666         while ( FG.cTable[*s] == 1 ) {
02667             t = s;
02668             x = *s++ - '0';
02669             while( FG.cTable[*s] == 1 ) x = 10*x + *s++ - '0';
02670             if ( *s && *s != ',' ) {
02671                 MesPrint("&%s is not a legal fixed index",t);
02672                 return(1);
02673             }
02674             else if ( x >= AM.OffsetIndex ) {
02675                 MesPrint("&%d is too large to be a fixed index",x);
02676                 error = 1;
02677             }
02678             else {
02679                 AddToCB(C,x)
02680                 i++;
02681                 C->Pointer[-i] = TYPESUMFIX;
02682                 C->Pointer[-i+1] = i;
02683             }
02684             if ( *s == 0 ) break;
02685             s++;
02686         }
02687     }

```

```

02688         else {
02689             MesPrint("&Illegal object in sum statement");
02690             error = 1;
02691         }
02692     }
02693     return(error);
02694 }
02695
02696 /*
02697  #] CoSum :
02698  #[ CoToTensor :
02699 */
02700
02701 static WORD cttarray[7] = { TYPEOPERATION,7,TENVEC,0,0,1,0 };
02702
02703 int CoToTensor(UBYTE *s)
02704 {
02705     UBYTE c, *t;
02706     int type, j, nargs, error = 0;
02707     WORD number, dol[2] = { 0, 0 };
02708     cttarray[1] = 6; /* length */
02709     cttarray[3] = 0; /* tensor */
02710     cttarray[4] = 0; /* vector */
02711     cttarray[5] = 1; /* option flags */
02712     /* cttarray[6] = 0; set veto */
02713     /*
02714      Count the number of the arguments. The validity of them is not checked here.
02715     */
02716     nargs = 0;
02717     t = s;
02718     for (;;) {
02719         while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
02720         if ( *s == 0 ) break;
02721         if ( *s == '!' ) {
02722             s++;
02723             if ( *s == '{' ) {
02724                 SKIPBRA2(s)
02725                 s++;
02726             } else {
02727                 if ( ( s = SkipAName(s) ) == 0 ) goto syntax_error;
02728             }
02729         } else {
02730             if ( ( s = SkipAName(s) ) == 0 ) goto syntax_error;
02731         }
02732         nargs++;
02733     }
02734     if ( nargs < 2 ) goto not_enough_arguments;
02735     s = t;
02736     /*
02737      Parse options, which are given as the arguments except the last two.
02738     */
02739     for ( j = 2; j < nargs; j++ ) {
02740         while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
02741         if ( *s == '!' ) {
02742             /*
02743              Handle !set or !{vector,...}. Note: If two or more sets are
02744              specified, then only the last one is used.
02745             */
02746             s++;
02747             cttarray[1] = 7;
02748             cttarray[5] |= 8;
02749             if ( FG.cTable[*s] == 0 || *s == '[' || *s == '_' ) {
02750                 t = s;
02751                 if ( ( s = SkipAName(s) ) == 0 ) goto syntax_error;
02752                 c = *s; *s = 0;
02753                 type = GetName(AC.varnames,t,&number,WITHAUTO);
02754                 if ( type == CVECTOR ) {
02755                     /*
02756                      As written in the manual, "!"p" (without "{}") should work.
02757                     */
02758                     cttarray[6] = DoTempSet(t,s);
02759                     *s = c;
02760                     goto check_tempset;
02761                 }
02762                 else if ( type != CSET ) {
02763                     MesPrint("&%s is not the name of a set or a vector",t);
02764                     error = 1;
02765                 }
02766                 *s = c;
02767                 cttarray[6] = number;
02768             }
02769             else if ( *s == '{' ) {
02770                 t = ++s; SKIPBRA2(s) *s = 0;
02771                 cttarray[6] = DoTempSet(t,s);
02772                 *s++ = '>';
02773             }
02774             check_tempset:
02775             if ( cttarray[6] < 0 ) {

```

```

02775         error = 1;
02776     }
02777     if ( AC.wildflag ) {
02778         MesPrint("&Improper use of wildcard(s) in set specification");
02779         error = 1;
02780     }
02781 }
02782 } else {
02783 /*
02784     Other options.
02785 */
02786     t = s;
02787     if ( ( s = SkipAName(s) ) == 0 ) goto syntax_error;
02788     c = *s; *s = 0;
02789     if ( StrICmp(t, (UBYTE *)"nosquare") == 0 ) cttarray[5] |= 2;
02790     else if ( StrICmp(t, (UBYTE *)"functions") == 0 ) cttarray[5] |= 4;
02791     else {
02792         MesPrint("&Unrecognized option in ToTensor statement: '%s'",t);
02793         *s = c;
02794         return(1);
02795     }
02796     *s = c;
02797 }
02798 }
02799 /*
02800     Now parse a vector and a tensor. The ordering doesn't matter.
02801 */
02802 for ( j = 0; j < 2; j++ ) {
02803     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
02804     t = s;
02805     if ( ( s = SkipAName(s) ) == 0 ) goto syntax_error;
02806     c = *s; *s = 0;
02807     if ( t[0] == '$' ) {
02808         dol[j] = GetDollar(t+1);
02809         if ( dol[j] < 0 ) dol[j] = AddDollar(t+1,DOLUNDEFINED,0,0);
02810     } else {
02811         type = GetName(AC.varnames,t,&number,WITHAUTO);
02812         if ( type == CVECTOR ) {
02813             cttarray[4] = number + AM.OffsetVector;
02814         }
02815         else if ( type == CFUNCTION && ( functions[number].spec > 0 ) ) {
02816             cttarray[3] = number + FUNCTION;
02817         }
02818         else {
02819             MesPrint("&%s is not a vector or a tensor",t);
02820             error = 1;
02821         }
02822     }
02823     *s = c;
02824 }
02825 if ( cttarray[3] == 0 || cttarray[4] == 0 ) {
02826     if ( dol[0] == 0 && dol[1] == 0 ) {
02827         goto not_enough_arguments;
02828     }
02829     else if ( cttarray[3] ) {
02830         if ( dol[1] ) cttarray[4] = dol[1];
02831         else if ( dol[0] ) cttarray[4] = dol[0];
02832         else {
02833             goto not_enough_arguments;
02834         }
02835     }
02836     else if ( cttarray[4] ) {
02837         if ( dol[1] ) cttarray[3] = -dol[1];
02838         else if ( dol[0] ) cttarray[3] = -dol[0];
02839         else {
02840             goto not_enough_arguments;
02841         }
02842     }
02843     else {
02844         if ( dol[0] == 0 || dol[1] == 0 ) {
02845             goto not_enough_arguments;
02846         }
02847         else {
02848             cttarray[3] = -dol[0]; cttarray[4] = dol[1];
02849         }
02850     }
02851 }
02852 AddNtoL(cttarray[1],cttarray);
02853 return(error);
02854
02855 syntax_error:
02856     MesPrint("&Syntax error in ToTensor statement");
02857     return(1);
02858
02859 not_enough_arguments:
02860     MesPrint("&ToTensor statement needs a vector and a tensor");
02861     return(1);

```

```

02862 }
02863
02864 /*
02865     #] CoToTensor :
02866     #[ CoToVector :
02867 */
02868
02869 static WORD ctvarray[6] = { TYPEOPERATION,6,TENVEC,0,0,0 };
02870
02871 int CoToVector(UBYTE *s)
02872 {
02873     UBYTE *t, c;
02874     int j, type, error = 0;
02875     WORD number, dol[2];
02876     dol[0] = dol[1] = 0;
02877     ctvarray[3] = ctvarray[4] = ctvarray[5] = 0;
02878     for ( j = 0; j < 2; j++ ) {
02879         t = s;
02880         if ( ( s = SkipAName(s) ) == 0 ) {
02881 proper:      MesPrint("&Arguments of ToVector statement should be a vector and a tensor");
02882             return(1);
02883         }
02884         c = *s; *s = 0;
02885         if ( *t == '$' ) {
02886             dol[j] = GetDollar(t+1);
02887             if ( dol[j] < 0 ) dol[j] = AddDollar(t+1,DOLUNDEFINED,0,0);
02888         }
02889         else if ( ( type = GetName(AC.varnames,t,&number,WITHAUTO) ) == CVECTOR )
02890             ctvarray[4] = number + AM.OffsetVector;
02891         else if ( type == CFUNCTION && ( functions[number].spec > 0 ) )
02892             ctvarray[3] = number+FUNCTION;
02893         else {
02894             MesPrint("&%s is not a vector or a tensor",t);
02895             error = 1;
02896         }
02897         *s = c; if ( *s && *s != ',' ) goto proper;
02898         if ( *s ) s++;
02899     }
02900     if ( *s != 0 ) goto proper;
02901     if ( ctvarray[3] == 0 || ctvarray[4] == 0 ) {
02902         if ( dol[0] == 0 && dol[1] == 0 ) {
02903             MesPrint("&ToVector statement needs a vector and a tensor");
02904             error = 1;
02905         }
02906         else if ( ctvarray[3] ) {
02907             if ( dol[1] ) ctvarray[4] = dol[1];
02908             else if ( dol[0] ) ctvarray[4] = dol[0];
02909             else {
02910                 MesPrint("&ToVector statement needs a vector and a tensor");
02911                 error = 1;
02912             }
02913         }
02914         else if ( ctvarray[4] ) {
02915             if ( dol[1] ) ctvarray[3] = -dol[1];
02916             else if ( dol[0] ) ctvarray[3] = -dol[0];
02917             else {
02918                 MesPrint("&ToVector statement needs a vector and a tensor");
02919                 error = 1;
02920             }
02921         }
02922         else {
02923             if ( dol[0] == 0 || dol[1] == 0 ) {
02924                 MesPrint("&ToVector statement needs a vector and a tensor");
02925                 error = 1;
02926             }
02927             else {
02928                 ctvarray[3] = -dol[0]; ctvarray[4] = dol[1];
02929             }
02930         }
02931     }
02932     AddNtoL(6,ctvarray);
02933     return(error);
02934 }
02935
02936 /*
02937     #] CoToVector :
02938     #[ CoTrace4 :
02939 */
02940
02941 int CoTrace4(UBYTE *s)
02942 {
02943     int error = 0, type, option = CHISHOLM;
02944     UBYTE *t, c;
02945     WORD numindex, one = 1;
02946     KEYWORD *key;
02947     for (;;) {
02948         t = s;

```

```

02949         if ( FG.cTable[*s] == 1 ) break;
02950         if ( ( s = SkipAName(s) ) == 0 ) {
02951 proper:    MesPrint("&Proper syntax for Trace4 is 'Trace4[,options],index;'");
02952             return(1);
02953         }
02954         if ( *s == 0 ) break;
02955         c = *s; *s = 0;
02956         if ( ( key = FindKeyWord(t,trace4options,
02957             sizeof(trace4options)/sizeof(KEYWORD)) ) == 0 ) break;
02958         else {
02959             option |= key->type;
02960             option &= ~key->flags;
02961         }
02962         if ( ( *s++ = c ) != ',' ) {
02963             MesPrint("&Illegal separator in Trace4 statement");
02964             return(1);
02965         }
02966         if ( *s == 0 ) goto proper;
02967     }
02968     s = t;
02969     if ( FG.cTable[*s] == 1 ) {
02970 retry:
02971         ParseNumber(numindex,s)
02972         if ( *s != 0 ) {
02973             MesPrint("&Last argument of Trace4 should be an index");
02974             return(1);
02975         }
02976         if ( numindex >= AM.OffsetIndex ) {
02977             MesPrint("&fixed index >= %d. Change value of OffsetIndex in setup file"
02978                 ,AM.OffsetIndex);
02979             return(1);
02980         }
02981     }
02982     else if ( *s == '$' ) {
02983         if ( ( type = GetName(AC.dollarnames,s+1,&numindex,NOAUTO) ) == CDOLLAR )
02984             numindex = -numindex;
02985         else {
02986             MesPrint("&%s is undefined",s);
02987             numindex = AddDollar(s+1,DOLINDEX,&one,1);
02988             return(1);
02989         }
02990 tests:   s = SkipAName(s);
02991         if ( *s != 0 ) {
02992             MesPrint("&Trace4 should have a single index or $variable for its argument");
02993             return(1);
02994         }
02995     }
02996     else if ( ( type = GetName(AC.varnames,s,&numindex,WITHAUTO) ) == CINDEX ) {
02997         numindex += AM.OffsetIndex;
02998         goto tests;
02999     }
03000     else if ( type != -1 ) {
03001         if ( type != CDUBIOUS ) {
03002             if ( ( FG.cTable[*s] != 0 ) && ( *s != '[' ) ) {
03003                 if ( *s == '+' && FG.cTable[s[1]] == 1 ) { s++; goto retry; }
03004                 goto proper;
03005             }
03006             NameConflict(type,s);
03007             type = MakeDubious(AC.varnames,s,&numindex);
03008         }
03009         return(1);
03010     }
03011     else {
03012         MesPrint("&%s is not an index",s);
03013         numindex = AddIndex(s,AC.lDefDim,AC.lDefDim4) + AM.OffsetIndex;
03014         return(1);
03015     }
03016     if ( error ) return(error);
03017     if ( ( option & CHISHOLM ) != 0 )
03018         Add4Com(TYPECHISHOLM,numindex,(option & ALSOREVERSE));
03019     Add5Com(TYPEOPERATION,TAKETRACE,4 + (option & NOTRICK),numindex);
03020     return(0);
03021 }
03022
03023 /*
03024 #] CoTrace4 :
03025 #[ CoTraceN :
03026 */
03027
03028 int CoTraceN(UBYTE *s)
03029 {
03030     WORD numindex, one = 1;
03031     int type;
03032     if ( FG.cTable[*s] == 1 ) {
03033 retry:
03034         ParseNumber(numindex,s)
03035         if ( *s != 0 ) {

```

```

03036 proper:      MesPrint("&TraceN should have a single index for its argument");
03037               return(1);
03038           }
03039           if ( numindex >= AM.OffsetIndex ) {
03040               MesPrint("&fixed index >= %d. Change value of OffsetIndex in setup file"
03041                       ,AM.OffsetIndex);
03042               return(1);
03043           }
03044       }
03045       else if ( *s == '$' ) {
03046           if ( ( type = GetName(AC.dollarnames,s+1,&numindex,NOAUTO) ) == CDOLLAR )
03047               numindex = -numindex;
03048           else {
03049               MesPrint("&%s is undefined",s);
03050               numindex = AddDollar(s+1,DOLINDEX,&one,1);
03051               return(1);
03052           }
03053       tests:  s = SkipAName(s);
03054           if ( *s != 0 ) {
03055               MesPrint("&TraceN should have a single index or $variable for its argument");
03056               return(1);
03057           }
03058       }
03059       else if ( ( type = GetName(AC.varnames,s,&numindex,WITHAUTO) ) == CINDEX ) {
03060           numindex += AM.OffsetIndex;
03061           goto tests;
03062       }
03063       else if ( type != -1 ) {
03064           if ( type != CDUBIOUS ) {
03065               if ( ( FG.cTable[*s] != 0 ) && ( *s != '[' ) ) {
03066                   if ( *s == '+' && FG.cTable[s[1]] == 1 ) { s++; goto retry; }
03067                   goto proper;
03068               }
03069               NameConflict(type,s);
03070               type = MakeDubious(AC.varnames,s,&numindex);
03071           }
03072           return(1);
03073       }
03074       else {
03075           MesPrint("&%s is not an index",s);
03076           numindex = AddIndex(s,AC.lDefDim,AC.lDefDim4) + AM.OffsetIndex;
03077           return(1);
03078       }
03079       Add5Com(TYPEOPERATION,TAKETRACE,0,numindex);
03080       return(0);
03081   }
03082   /*
03083   #] CoTraceN :
03084   #[ CoChisholm :
03085   */
03086   /*
03087   int CoChisholm(UBYTE *s)
03088   {
03089       int error = 0, type, option = CHISHOLM;
03090       UBYTE *t, c;
03091       WORD numindex, one = 1;
03092       KEYWORD *key;
03093       for (;;) {
03094           t = s;
03095           if ( FG.cTable[*s] == 1 ) break;
03096           if ( ( s = SkipAName(s) ) == 0 ) {
03097               proper:  MesPrint("&Proper syntax for Chisholm is 'Chisholm[,options],index;'");
03098                       return(1);
03099           }
03100           if ( *s == 0 ) break;
03101           c = *s; *s = 0;
03102           if ( ( key = FindKeyWord(t,chisoptions,
03103                                   sizeof(chisoptions)/sizeof(KEYWORD)) ) == 0 ) break;
03104           else {
03105               option |= key->type;
03106               option &= ~key->flags;
03107           }
03108           if ( ( *s++ = c ) != ',' ) {
03109               MesPrint("&Illegal separator in Chisholm statement");
03110               return(1);
03111           }
03112           if ( *s == 0 ) goto proper;
03113       }
03114       s = t;
03115       if ( FG.cTable[*s] == 1 ) {
03116           ParseNumber(numindex,s)
03117           if ( *s != 0 ) {
03118               MesPrint("&Last argument of Chisholm should be an index");
03119               return(1);
03120           }
03121       }
03122       if ( numindex >= AM.OffsetIndex ) {

```

```

03123         MesPrint("&fixed index >= %d. Change value of OffsetIndex in setup file"
03124         ,AM.OffsetIndex);
03125         return(1);
03126     }
03127 }
03128 else if ( *s == '$' ) {
03129     if ( ( type = GetName(AC.dollarnames,s+1,&numindex,NOAUTO) ) == CDOLLAR )
03130         numindex = -numindex;
03131     else {
03132         MesPrint("&%s is undefined",s);
03133         numindex = AddDollar(s+1,DOLINDEX,&one,1);
03134         return(1);
03135     }
03136 tests: s = SkipAName(s);
03137     if ( *s != 0 ) {
03138         MesPrint("&Chisholm should have a single index or $variable for its argument");
03139         return(1);
03140     }
03141 }
03142 else if ( ( type = GetName(AC.varnames,s,&numindex,WITHAUTO) ) == CINDEX ) {
03143     numindex += AM.OffsetIndex;
03144     goto tests;
03145 }
03146 else if ( type != -1 ) {
03147     if ( type != CDUBIOUS ) {
03148         NameConflict(type,s);
03149         type = MakeDubious(AC.varnames,s,&numindex);
03150     }
03151     return(1);
03152 }
03153 else {
03154     MesPrint("&%s is not an index",s);
03155     numindex = AddIndex(s,AC.lDefDim,AC.lDefDim4) + AM.OffsetIndex;
03156     return(1);
03157 }
03158 if ( error ) return(error);
03159 Add4Com(TYPECHISHOLM,numindex,(option & ALSOREVERSE));
03160 return(0);
03161 }
03162
03163 /*
03164 #] CoChisholm :
03165 #[ DoChain :
03166
03167 Syntax: Chainxx functionname;
03168 */
03169
03170 int DoChain(UBYTE *s, int option)
03171 {
03172     WORD numfunc,type;
03173     if ( *s == '$' ) {
03174         if ( ( type = GetName(AC.dollarnames,s+1,&numfunc,NOAUTO) ) == CDOLLAR )
03175             numfunc = -numfunc;
03176         else {
03177             MesPrint("&%s is undefined",s);
03178             numfunc = AddDollar(s+1,DOLINDEX,&one,1);
03179             return(1);
03180         }
03181 tests: s = SkipAName(s);
03182         if ( *s != 0 ) {
03183             MesPrint("&ChainIn/ChainOut should have a single function or $variable for its argument");
03184             return(1);
03185         }
03186     }
03187     else if ( ( type = GetName(AC.varnames,s,&numfunc,WITHAUTO) ) == CFUNCTION ) {
03188         numfunc += FUNCTION;
03189         goto tests;
03190     }
03191     else if ( type != -1 ) {
03192         if ( type != CDUBIOUS ) {
03193             NameConflict(type,s);
03194             type = MakeDubious(AC.varnames,s,&numfunc);
03195         }
03196         return(1);
03197     }
03198     else {
03199         MesPrint("&%s is not a function",s);
03200         numfunc = AddFunction(s,0,0,0,0,0,-1,-1) + FUNCTION;
03201         return(1);
03202     }
03203     Add3Com(option,numfunc);
03204     return(0);
03205 }
03206
03207 /*
03208 #] DoChain :
03209 #[ CoChainin :

```

```

03210
03211     Syntax: Chainin functionname;
03212 */
03213
03214 int CoChainin(UBYTE *s)
03215 {
03216     return(DoChain(s,TYPECHAININ));
03217 }
03218
03219 /*
03220     #] CoChainin :
03221     #[ CoChainout :
03222
03223     Syntax: Chainout functionname;
03224 */
03225
03226 int CoChainout(UBYTE *s)
03227 {
03228     return(DoChain(s,TYPECHAINOUT));
03229 }
03230
03231 /*
03232     #] CoChainout :
03233     #[ CoExit :
03234 */
03235
03236 int CoExit(UBYTE *s)
03237 {
03238     UBYTE *name;
03239     WORD code = TYPEEXIT;
03240     while ( *s == ',' ) s++;
03241     if ( *s == 0 ) {
03242         Add3Com(TYPEEXIT,0);
03243         return(0);
03244     }
03245     name = s+1;
03246     s++;
03247     while ( *s ) { if ( *s == '\\' ) s++; s++; }
03248     if ( name[-1] != '"' || s[-1] != '"' ) {
03249         MesPrint("&Illegal syntax for exit statement");
03250         return(1);
03251     }
03252     s[-1] = 0;
03253     AddComString(1,&code,name,0);
03254     s[-1] = '"';
03255     return(0);
03256 }
03257
03258 /*
03259     #] CoExit :
03260     #[ CoInParallel :
03261 */
03262
03263 int CoInParallel(UBYTE *s)
03264 {
03265     return(DoInParallel(s,1));
03266 }
03267
03268 /*
03269     #] CoInParallel :
03270     #[ CoNotInParallel :
03271 */
03272
03273 int CoNotInParallel(UBYTE *s)
03274 {
03275     return(DoInParallel(s,0));
03276 }
03277
03278 /*
03279     #] CoNotInParallel :
03280     #[ DoInParallel :
03281
03282     InParallel;
03283     InParallel,names;
03284     NotInParallel;
03285     NotInParallel,names;
03286 */
03287
03288 int DoInParallel(UBYTE *s, int par)
03289 {
03290     #ifdef PARALLELCODE
03291         EXPRESSIONS e;
03292         WORD i;
03293     #endif
03294     WORD number;
03295     UBYTE *t, c;
03296     int error = 0;

```



```

03297 #ifndef WITHPTHREADS
03298     DUMMYUSE(par);
03299 #endif
03300     if ( *s == 0 ) {
03301         AC.inparallelflag = par;
03302 #ifdef PARALLELCODE
03303         for ( i = NumExpressions-1; i >= 0; i-- ) {
03304             e = Expressions+i;
03305             if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
03306                 || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
03307             ) {
03308                 e->partodo = par;
03309             }
03310         }
03311 #endif
03312     }
03313     else {
03314         for(;;) { /* Look for a (comma separated) list of variables */
03315             while ( *s == ',' ) s++;
03316             if ( *s == 0 ) break;
03317             if ( *s == '[' || FG.cTable[*s] == 0 ) {
03318                 t = s;
03319                 if ( ( s = SkipAName(s) ) == 0 ) {
03320                     MesPrint("&Improper name for an expression: '%s'",t);
03321                     return(1);
03322                 }
03323                 c = *s; *s = 0;
03324                 if ( GetName(AC.exprnames,t,&number,NOAUTO) == CEXPRESSION ) {
03325 #ifdef PARALLELCODE
03326                     e = Expressions+number;
03327                     if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
03328                         || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
03329                     ) {
03330                         e->partodo = par;
03331                     }
03332 #endif
03333                 }
03334                 else if ( GetName(AC.varnames,t,&number,NOAUTO) != NAMENOTFOUND ) {
03335                     MesPrint("&%s is not an expression",t);
03336                     error = 1;
03337                 }
03338                 *s = c;
03339             }
03340             else {
03341                 MesPrint("&Illegal object in InExpression statement");
03342                 error = 1;
03343                 while ( *s && *s != ',' ) s++;
03344                 if ( *s == 0 ) break;
03345             }
03346         }
03347     }
03348     }
03349     return(error);
03350 }
03351
03352 /*
03353 #] DoInParallel :
03354 #[ CoInExpression :
03355 */
03356
03357 int CoInExpression(UBYTE *s)
03358 {
03359     GETIDENTITY
03360     UBYTE *t, c;
03361     WORD *w, number;
03362     int error = 0;
03363     w = AT.WorkPointer;
03364     if ( AC.inexprlevel >= MAXNEST ) {
03365         MesPrint("@Nesting of inexpression statements more than %d levels", (WORD)MAXNEST);
03366         return(-1);
03367     }
03368     AC.inexprsumcheck[AC.inexprlevel] = NestingChecksum();
03369     AC.inexprstack[AC.inexprlevel] = cbuf[AC.cbunum].Pointer
03370         - cbuf[AC.cbunum].Buffer + 2;
03371     AC.inexprlevel++;
03372     *w++ = TYPEINEXPRESSION;
03373     w++; w++;
03374     for(;;) { /* Look for a (comma separated) list of variables */
03375         while ( *s == ',' ) s++;
03376         if ( *s == 0 ) break;
03377         if ( *s == '[' || FG.cTable[*s] == 0 ) {
03378             t = s;
03379             if ( ( s = SkipAName(s) ) == 0 ) {
03380                 MesPrint("&Improper name for an expression: '%s'",t);
03381                 return(1);
03382             }
03383             c = *s; *s = 0;

```

```

03384         if ( GetName(AC.exprnames,t,&number,NOAUTO) == CEXPRESSION ) {
03385             *w++ = number;
03386         }
03387         else if ( GetName(AC.varnames,t,&number,NOAUTO) != NAMENOTFOUND ) {
03388             MesPrint("& %s is not an expression",t);
03389             error = 1;
03390         }
03391         *s = c;
03392     }
03393     else {
03394         MesPrint("& Illegal object in InExpression statement");
03395         error = 1;
03396         while ( *s && *s != ',' ) s++;
03397         if ( *s == 0 ) break;
03398     }
03399 }
03400 AT.WorkPointer[1] = w - AT.WorkPointer;
03401 AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
03402 return(error);
03403 }
03404
03405 /*
03406 #] CoInExpression :
03407 #[ CoEndInExpression :
03408 */
03409 int CoEndInExpression(UBYTE *s)
03410 {
03411     CBUF *C = cbuf+AC.cbunum;
03412     while ( *s == ',' ) s++;
03413     if ( *s ) {
03414         MesPrint("& Illegal syntax for EndInExpression statement");
03415         return(1);
03416     }
03417     if ( AC.inexprlevel <= 0 ) {
03418         MesPrint("& EndInExpression without corresponding InExpression statement");
03419         return(1);
03420     }
03421     AC.inexprlevel--;
03422     cbuf[AC.cbunum].Buffer[AC.inexprstack[AC.inexprlevel]] = C->numlhs;
03423     if ( AC.inexprsumcheck[AC.inexprlevel] != NestingChecksum() ) {
03424         MesNesting();
03425         return(1);
03426     }
03427     return(0);
03428 }
03429 }
03430
03431 /*
03432 #] CoEndInExpression :
03433 #[ CoSetExitFlag :
03434 */
03435 int CoSetExitFlag(UBYTE *s)
03436 {
03437     if ( *s ) {
03438         MesPrint("& Illegal syntax for the SetExitFlag statement");
03439         return(1);
03440     }
03441     Add2Com(TYPESETEXIT);
03442     return(0);
03443 }
03444 }
03445
03446 /*
03447 #] CoSetExitFlag :
03448 #[ CoTryReplace :
03449 */
03450 int CoTryReplace(UBYTE *p)
03451 {
03452     GETIDENTITY
03453     UBYTE *name, c;
03454     WORD *w, error = 0, i, which = -1, c1, minvec = 0;
03455     w = AT.WorkPointer;
03456     *w++ = TYPETRY;
03457     *w++ = 3;
03458     *w++ = 0;
03459     *w++ = REPLACEMENT;
03460     *w++ = FUNHEAD;
03461     FILLFUN(w)
03462     /*
03463     Now we have to read a function argument for the replace_ function.
03464     Current arguments that we allow involve only single arguments
03465     that do not expand further. No brackets!
03466     */
03467     while ( *p ) {
03468         /*
03469             No numbers yet
03470         */

```

```

03471     if ( *p == '-' && minvec == 0 && which == (CVECTOR+1) ) {
03472         minvec = 1; p++;
03473     }
03474     if ( *p == '[' || FG.cTable[*p] == 0 ) {
03475         name = p;
03476         if ( ( p = SkipAName(p) ) == 0 ) return(1);
03477         c = *p; *p = 0;
03478         i = GetName(AC.varnames,name,&c1,WITHAUTO);
03479         if ( which >= 0 && i >= 0 && i != CDUBIOUS && which != (i+1) ) {
03480             MesPrint("&Illegal combination of objects in TryReplace");
03481             error = 1;
03482         }
03483         else if ( minvec && i != CVECTOR && i != CDUBIOUS ) {
03484             MesPrint("&Currently a - sign can be used only with a vector in TryReplace");
03485             error = 1;
03486         }
03487         else switch ( i ) {
03488             case CSYMBOL: *w++ = -SYMBOL; *w++ = c1; break;
03489             case CVECTOR:
03490                 if ( minvec ) *w++ = -MINVECTOR;
03491                 else *w++ = -VECTOR;
03492                 *w++ = c1 + AM.OffsetVector;
03493                 minvec = 0;
03494                 break;
03495             case CINDEXT: *w++ = -INDEX; *w++ = c1 + AM.OffsetIndex;
03496                 if ( c1 >= AM.WilInd && c == '?' ) { *p++ = c; c = *p; }
03497                 break;
03498             case CFUNCTION: *w++ = -c1-FUNCTION; break;
03499             case CDUBIOUS: minvec = 0; error = 1; break;
03500             default:
03501                 MesPrint("&Illegal object type in TryReplace: %s",name);
03502                 error = 1;
03503                 i = 0;
03504                 break;
03505         }
03506         if ( which < 0 ) which = i+1;
03507         else which = -1;
03508         *p = c;
03509         if ( *p == ',' ) p++;
03510         continue;
03511     }
03512     else {
03513         MesPrint("&Illegal object in TryReplace");
03514         error = 1;
03515         while ( *p && *p != ',' ) {
03516             if ( *p == '(' ) SKIPBRA3(p)
03517             else if ( *p == '{' ) SKIPBRA2(p)
03518             else if ( *p == '[' ) SKIPBRA1(p)
03519             else p++;
03520         }
03521     }
03522     if ( *p == ',' ) p++;
03523     if ( which < 0 ) which = 0;
03524     else which = -1;
03525 }
03526 if ( which >= 0 ) {
03527     MesPrint("&Odd number of arguments in TryReplace");
03528     error = 1;
03529 }
03530 i = w - AT.WorkPointer;
03531 AT.WorkPointer[1] = i;
03532 AT.WorkPointer[2] = i - 3;
03533 AT.WorkPointer[4] = i - 3;
03534 AddNtoL((int)i,AT.WorkPointer);
03535 return(error);
03536 }
03537
03538 /*
03539 #] CoTryReplace :
03540 #[ CoModulus :
03541
03542 Old syntax: Modulus [-] number [:number]
03543 New syntax: Modulus [option(s)] number
03544 Options are: NoFunctions/CoefficientsOnly/AlsoFunctions
03545             PlusMin/Positive
03546             InverseTable
03547             PrintPowersOf(number)
03548             AlsoPowers/NoPowers
03549             AlsoDollars/NoDollars
03550 Notice: We change the defaults. This may cause problems to some.
03551 */
03552
03553 int CoModulus(UBYTE *inp)
03554 {
03555     GETIDENTITY
03556     int Retval = 0, sign = 1;
03557     UBYTE *p, c;

```

```

03558     while ( *inp == ',' || *inp == ' ' || *inp == '\t' ) inp++;
03559     if ( *inp == 0 ) {
03560 SwitchOff:
03561         if ( AC.modpowers ) M_free(AC.modpowers,"AC.modpowers");
03562         AC.modpowers = 0;
03563         AN.ncmod = AC.ncmod = 0;
03564         if ( AC.halfmod ) M_free(AC.halfmod,"halfmod");
03565         AC.halfmod = 0; AC.nhalfmod = 0;
03566         if ( AC.modinverses ) M_free(AC.modinverses,"modinverses");
03567         AC.modinverses = 0;
03568         AC.modmode = 0;
03569         return(0);
03570     }
03571 #ifdef WITHFLOAT
03572     if ( AT.aux_ != 0 ) {
03573         MesPrint("&Simultaneous use of floating point and modulus arithmetic makes no sense.");
03574         Retval = 1;
03575     }
03576 #endif
03577     AC.modmode = 0;
03578     if ( *inp == '-' ) {
03579         sign = -1;
03580         inp++;
03581     }
03582     else {
03583         while ( FG.cTable[*inp] == 0 ) {
03584             p = inp;
03585             while ( FG.cTable[*inp] == 0 ) inp++;
03586             c = *inp; *inp = 0;
03587             if ( StrICmp(p,(UBYTE *)"nofunctions") == 0 ) {
03588                 AC.modmode &= ~ALSOFUNARGS;
03589             }
03590             else if ( StrICmp(p,(UBYTE *)"alsofunctions") == 0 ) {
03591                 AC.modmode |= ALSOFUNARGS;
03592             }
03593             else if ( StrICmp(p,(UBYTE *)"coefficientsonly") == 0 ) {
03594                 AC.modmode &= ~ALSOFUNARGS;
03595                 AC.modmode &= ~ALSOPOWERS;
03596                 sign = -1;
03597             }
03598             else if ( StrICmp(p,(UBYTE *)"plusmin") == 0 ) {
03599                 AC.modmode |= POSNEG;
03600             }
03601             else if ( StrICmp(p,(UBYTE *)"positive") == 0 ) {
03602                 AC.modmode &= ~POSNEG;
03603             }
03604             else if ( StrICmp(p,(UBYTE *)"inversetable") == 0 ) {
03605                 AC.modmode |= INVERSETABLE;
03606             }
03607             else if ( StrICmp(p,(UBYTE *)"noinversetable") == 0 ) {
03608                 AC.modmode &= ~INVERSETABLE;
03609             }
03610             else if ( StrICmp(p,(UBYTE *)"nodollars") == 0 ) {
03611                 AC.modmode &= ~ALSODOLLARS;
03612             }
03613             else if ( StrICmp(p,(UBYTE *)"alsodollars") == 0 ) {
03614                 AC.modmode |= ALSODOLLARS;
03615             }
03616             else if ( StrICmp(p,(UBYTE *)"printpowersof") == 0 ) {
03617                 *inp = c;
03618                 if ( *inp != '(' ) {
03619 badsyntax:
03620                     MesPrint("&Bad syntax in argument of PrintPowersOf(number) in Modulus statement");
03621                     return(1);
03622                 }
03623                 while ( *inp == ',' || *inp == ' ' || *inp == '\t' ) inp++;
03624                 inp++; p = inp;
03625                 if ( FG.cTable[*inp] != 1 ) goto badsyntax;
03626                 do { inp++; } while ( FG.cTable[*inp] == 1 );
03627                 c = *inp; *inp = 0;
03628                 if ( GetLong(p,(UWORD *)AC.powmod,&AC.npowmod) ) Retval = -1;
03629                 if ( TakeModulus((UWORD *)AC.powmod,&AC.npowmod,AC.cmod,AC.ncmod,NOUNPACK) ) Retval = -1;
03630                 if ( AC.npowmod == 0 ) {
03631                     MesPrint("&Improper value for generator");
03632                     Retval = -1;
03633                 }
03634                 if ( MakeModTable() ) Retval = -1;
03635                 AC.DirtPow = 1;
03636                 *inp = c;
03637                 while ( *inp == ',' || *inp == ' ' || *inp == '\t' ) inp++;
03638                 if ( *inp != '(' ) goto badsyntax;
03639                 inp++;
03640                 c = *inp;
03641             }
03642             else if ( StrICmp(p,(UBYTE *)"alsopowers") == 0 ) {
03643                 AC.modmode |= ALSOPOWERS;
03644                 sign = 1;

```

```

03645     }
03646     else if ( StrICmp(p, (UBYTE *) "nopowers") == 0 ) {
03647         AC.modmode &= ~ALSOPOWERS;
03648         sign = -1;
03649     }
03650     else {
03651         MesPrint("&Unrecognized option %s in Modulus statement", inp);
03652         return(1);
03653     }
03654     *inp = c;
03655     while ( *inp == ',' || *inp == ' ' || *inp == '\t' ) inp++;
03656     if ( *inp == 0 ) {
03657         MesPrint("&Modulus statement with no value!!!");
03658         return(1);
03659     }
03660 }
03661 }
03662 p = inp;
03663 if ( FG.cTable[*inp] != 1 ) {
03664     MesPrint("&Invalid value for modulus:%s", inp);
03665     if ( AC.modpowers ) M_free(AC.modpowers, "AC.modpowers");
03666     AC.modpowers = 0;
03667     AN.ncmod = AC.ncmod = 0;
03668     if ( AC.halfmod ) M_free(AC.halfmod, "halfmod");
03669     AC.halfmod = 0; AC.nhalfmod = 0;
03670     if ( AC.modinverses ) M_free(AC.modinverses, "modinverses");
03671     AC.modinverses = 0;
03672     return(1);
03673 }
03674 do { inp++; } while ( FG.cTable[*inp] == 1 );
03675 c = *inp; *inp = 0;
03676 Retval = GetLong(p, (UWORD *) AC.cmod, &AC.ncmod);
03677 if ( Retval == 0 && AC.ncmod == 0 ) goto SwitchOff;
03678 if ( sign < 0 ) AC.ncmod = -AC.ncmod;
03679 AN.ncmod = AC.ncmod;
03680 if ( ( AC.modmode & INVERSETABLE ) != 0 ) MakeInverses();
03681 if ( AC.halfmod ) M_free(AC.halfmod, "halfmod");
03682 AC.halfmod = 0; AC.nhalfmod = 0;
03683 return(Retval);
03684 }
03685
03686 /*
03687 #] CoModulus :
03688 #[ CoRepeat :
03689 */
03690
03691 int CoRepeat(UBYTE *inp)
03692 {
03693     int error = 0;
03694     AC.RepSumCheck[AC.RepLevel] = NestingChecksum();
03695     AC.RepLevel++;
03696     if ( AC.RepLevel > AM.RepMax ) {
03697         MesPrint("&Too many repeat levels. Maximum is %d", AM.RepMax);
03698         return(1);
03699     }
03700     Add3Com(TYPEREPEAT, -1) /* Means indefinite */
03701     while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
03702     if ( *inp ) {
03703         error = CompileStatement(inp);
03704         if ( CoEndRepeat(inp) ) error = 1;
03705     }
03706     return(error);
03707 }
03708
03709 /*
03710 #] CoRepeat :
03711 #[ CoEndRepeat :
03712 */
03713
03714 int CoEndRepeat(UBYTE *inp)
03715 {
03716     CBUF *C = cbuf+AC.cbunum;
03717     int level, error = 0, repeatlevel = 0;
03718     DUMMYUSE(inp);
03719     AC.RepLevel--;
03720     if ( AC.RepLevel < 0 ) {
03721         MesPrint("&EndRepeat without Repeat");
03722         AC.RepLevel = 0;
03723         return(1);
03724     }
03725     else if ( AC.RepSumCheck[AC.RepLevel] != NestingChecksum() ) {
03726         MesNesting();
03727         error = 1;
03728     }
03729     level = C->numlhs+1;
03730     while ( level > 0 ) {
03731         if ( C->lhs[--level][0] == TYPEREPEAT ) {

```

```

03732         if ( repeatlevel == 0 ) {
03733             Add3Com(TYPEENDREPEAT,level)
03734             return(error);
03735         }
03736         repeatlevel--;
03737     }
03738     else if ( C->lhs[level][0] == TYPEENDREPEAT ) repeatlevel++;
03739 }
03740 return(1);
03741 }
03742
03743 /*
03744 #] CoEndRepeat :
03745 #[ DoBrackets :
03746
03747     Reads in the bracket information.
03748     Storage is in the form of a regular term.
03749     No subterms and arguments are allowed.
03750 */
03751
03752 int DoBrackets(UBYTE *inp, int par)
03753 {
03754     GETIDENTITY
03755     UBYTE *p, *pp, c;
03756     WORD *to, i, type, *w, error = 0;
03757     WORD c1,c2, *WorkSave;
03758     int bflag;
03759     p = inp;
03760     WorkSave = to = AT.WorkPointer;
03761     to++;
03762     if ( AT.BrackBuf == 0 ) {
03763         AR.MaxBracket = 100;
03764         AT.BrackBuf = (WORD *)Malloca(sizeof(WORD)*(AR.MaxBracket+1),"bracket buffer");
03765     }
03766     *AT.BrackBuf = 0;
03767     AR.BracketOn = 0;
03768     AC.bracketindexflag = 0;
03769     AT.bracketindexflag = 0;
03770     if ( *p == '+' || *p == '-' ) p++;
03771     if ( p[-1] == ',' && *p ) p--;
03772     if ( p[-1] == '+' && *p ) { bflag = 1; if ( *p != ',' ) { *--p = ','; } }
03773     else if ( p[-1] == '-' && *p ) { bflag = -1; if ( *p != ',' ) { *--p = ','; } }
03774     else bflag = 0;
03775     while ( *p == ',' ) {
03776 redo: AR.BracketOn++;
03777         while ( *p == ',' ) p++;
03778         if ( *p == 0 ) break;
03779         if ( *p == '0' ) {
03780             p++; while ( *p == '0' ) p++;
03781             continue;
03782         }
03783         inp = pp = p;
03784         p = SkipAName(p);
03785         if ( p == 0 ) return(1);
03786         c = *p;
03787         *p = 0;
03788         type = GetName(AC.varnames,inp,&c1,WITHAUTO);
03789         if ( c == '.' ) {
03790             if ( type == CVECTOR || type == CDUBIOUS ) {
03791                 *p++ = c;
03792                 inp = p;
03793                 p = SkipAName(p);
03794                 if ( p == 0 ) return(1);
03795                 c = *p;
03796                 *p = 0;
03797                 type = GetName(AC.varnames,inp,&c2,WITHAUTO);
03798                 if ( type != CVECTOR && type != CDUBIOUS ) {
03799                     MesPrint("&Not a vector in dotproduct in bracket statement: %s",inp);
03800                     error = 1;
03801                 }
03802                 else type = CDOTPRODUCT;
03803             }
03804             else {
03805                 MesPrint("&Illegal use of . after %s in bracket statement",inp);
03806                 error = 1;
03807                 *p++ = c;
03808                 goto redo;
03809             }
03810         }
03811         switch ( type ) {
03812             case CSYMBOL :
03813                 *to++ = SYMBOL; *to++ = 4; *to++ = c1; *to++ = 1; break;
03814             case CVECTOR :
03815                 *to++ = INDEX; *to++ = 3; *to++ = AM.OffsetVector + c1; break;
03816             case CFUNCTION :
03817                 *to++ = c1+FUNCTION; *to++ = FUNHEAD; *to++ = 0;
03818                 FILLFUN3(to)

```

```

03819         break;
03820     case CDOTPRODUCT :
03821         *to++ = DOTPRODUCT; *to++ = 5; *to++ = c1 + AM.OffsetVector;
03822         *to++ = c2 + AM.OffsetVector; *to++ = 1; break;
03823     case CDELTA :
03824         *to++ = DELTA; *to++ = 4; *to++ = EMPTYINDEX; *to++ = EMPTYINDEX; break;
03825     case CSET :
03826         *to++ = SETSET; *to++ = 4; *to++ = c1; *to++ = Sets[c1].type; break;
03827     default :
03828         MesPrint("&Illegal bracket request for %s",pp);
03829         error = 1; break;
03830     }
03831     *p = c;
03832 }
03833 if ( *p ) {
03834     MesCerr("separator",p);
03835     AC.BracketNormalize = 0;
03836     AT.WorkPointer = WorkSave;
03837     error = 1;
03838     return(error);
03839 }
03840 *to++ = 1; *to++ = 1; *to++ = 3;
03841 *AT.WorkPointer = to - AT.WorkPointer;
03842 AT.WorkPointer = to;
03843 AC.BracketNormalize = 1;
03844 if ( BracketNormalize(BHEAD WorkSave) ) { error = 1; AR.BracketOn = 0; }
03845 else {
03846     w = WorkSave;
03847     if ( *w == 4 || !*w ) { AR.BracketOn = 0; }
03848     else {
03849         i = *(w+w-1);
03850         if ( i < 0 ) i = -i;
03851         *w -= i;
03852         i = *w;
03853         if ( i > AR.MaxBracket ) {
03854             WORD *newbuf;
03855             newbuf = (WORD *)Malloca(sizeof(WORD)*(i+1),"bracket buffer");
03856             AR.MaxBracket = i;
03857             if ( AT.BrackBuf != 0 ) M_free(AT.BrackBuf,"bracket buffer");
03858             AT.BrackBuf = newbuf;
03859         }
03860         to = AT.BrackBuf;
03861         NCOPY(to,w,i);
03862     }
03863 }
03864 AC.BracketNormalize = 0;
03865 if ( par == 1 ) AR.BracketOn = -AR.BracketOn;
03866 if ( error == 0 ) {
03867     AC.bracketindexflag = biflag;
03868     AT.bracketindexflag = biflag;
03869 }
03870 AT.WorkPointer = WorkSave;
03871 return(error);
03872 }
03873
03874 /*
03875 #] DoBrackets :
03876 #[ CoBracket :
03877 */
03878
03879 int CoBracket(UBYTE *inp)
03880 { return(DoBrackets(inp,0)); }
03881
03882 /*
03883 #] CoBracket :
03884 #[ CoAntiBracket :
03885 */
03886
03887 int CoAntiBracket(UBYTE *inp)
03888 { return(DoBrackets(inp,1)); }
03889
03890 /*
03891 #] CoAntiBracket :
03892 #[ CoMultiBracket :
03893
03894 Syntax:
03895     MultiBracket:{A|B} bracketinfo:...:{A|B} bracketinfo;
03896 */
03897
03898 int CoMultiBracket(UBYTE *inp)
03899 {
03900     GETIDENTITY
03901     int i, error = 0, error1, type, num;
03902     UBYTE *s, c;
03903     WORD *to, *from;
03904
03905     if ( *inp != ':' ) {

```

```

03906         MesPrint("&Illegal Multiple Bracket separator: %s",inp);
03907         return(1);
03908     }
03909     inp++;
03910     if ( AC.MultiBracketBuf == 0 ) {
03911         AC.MultiBracketBuf = (WORD **)Malloca1(sizeof(WORD *)*MAXMULTIBRACKETLEVELS,"multi bracket
buffer");
03912         for ( i = 0; i < MAXMULTIBRACKETLEVELS; i++ ) {
03913             AC.MultiBracketBuf[i] = 0;
03914         }
03915     }
03916     else {
03917         for ( i = 0; i < MAXMULTIBRACKETLEVELS; i++ ) {
03918             if ( AC.MultiBracketBuf[i] ) {
03919                 M_free(AC.MultiBracketBuf[i],"bracket buffer i");
03920                 AC.MultiBracketBuf[i] = 0;
03921             }
03922         }
03923         AC.MultiBracketLevels = 0;
03924     }
03925     AC.MultiBracketLevels = 0;
03926     /*
03927         Start with disabling the regular brackets.
03928     */
03929     if ( AT.BrackBuf == 0 ) {
03930         AR.MaxBracket = 100;
03931         AT.BrackBuf = (WORD *)Malloca1(sizeof(WORD)*(AR.MaxBracket+1),"bracket buffer");
03932     }
03933     *AT.BrackBuf = 0;
03934     AR.BracketOn = 0;
03935     AC.bracketindexflag = 0;
03936     AT.bracketindexflag = 0;
03937     /*
03938         Now loop through the various levels, separated by the colons.
03939     */
03940     for ( i = 0; i < MAXMULTIBRACKETLEVELS; i++ ) {
03941         if ( *inp == 0 ) goto RegEnd;
03942         /*
03943             1: skip to ':', determine bracket or antibracket
03944         */
03945         s = inp;
03946         while ( *s && *s != ':' ) {
03947             if ( *s == '[' ) { SKIPBRA1(s) s++; }
03948             else if ( *s == '{' ) { SKIPBRA2(s) s++; }
03949             else s++;
03950         }
03951         c = *s; *s = 0;
03952         if ( StrICont(inp,(UBYTE *)"antibrackets") == 0 ) { type = 1; }
03953         else if ( StrICont(inp,(UBYTE *)"brackets") == 0 ) { type = 0; }
03954         else {
03955             MesPrint("&Illegal (anti)bracket specification in MultiBracket statement");
03956             if ( error == 0 ) error = 1;
03957             goto NextLevel;
03958         }
03959         while ( FG.cTable[*inp] == 0 ) inp++;
03960         if ( *inp != ',' ) {
03961             MesPrint("&Illegal separator after (anti)bracket specification in MultiBracket
statement");
03962             if ( error == 0 ) error = 1;
03963             goto NextLevel;
03964         }
03965         inp++;
03966         /*
03967             2: call DoBrackets.
03968         */
03969         error1 = DoBrackets(inp, type);
03970         if ( error < 0 ) return(error1);
03971         if ( error1 > error ) error = error1;
03972         /*
03973             3: copy bracket information to the multi bracket arrays
03974         */
03975         if ( AR.BracketOn ) {
03976             num = AT.BrackBuf[0];
03977             to = AC.MultiBracketBuf[i] = (WORD *)Malloca1((num+2)*sizeof(WORD),"bracket buffer i");
03978             from = AT.BrackBuf;
03979             *to++ = AR.BracketOn;
03980             NCOPY(to,from,num);
03981             *to = 0;
03982         }
03983         /*
03984             4: set ready for the next level
03985         */
03986     NextLevel:
03987         *s = c; if ( c == ':' ) s++;
03988         inp = s;
03989         *AT.BrackBuf = 0;
03990         AR.BracketOn = 0;

```



```

03991     }
03992     if ( *inp != 0 ) {
03993         MesPrint("&More than %d levels in MultiBracket statement", (WORD)MAXMULTIBRACKETLEVELS);
03994         if ( error == 0 ) error = 1;
03995     }
03996 RegEnd:
03997     AC.MultiBracketLevels = i;
03998     *AT.BrackBuf = 0;
03999     AR.BraceOn = 0;
04000     AC.bracketindexflag = 0;
04001     AT.bracketindexflag = 0;
04002     return(error);
04003 }
04004
04005 /*
04006 #] CoMultiBracket :
04007 #[ CountComp :
04008
04009     This routine reads the count statement. The syntax is:
04010     count minimum,object,size[,object,size]
04011     Objects can be:
04012         symbol
04013         dotproduct
04014         vector
04015         function
04016     Vectors can have the auxiliary flags:
04017         +v +f +d +?setname
04018
04019     Output for the compiler:
04020     TYPECOUNT,size,minimum,objects
04021     with the objects:
04022     SYMBOL,4,number,size
04023     DOTPRODUCT,5,v1,v2,size
04024     FUNCTION,4,number,size
04025     VECTOR,5,number,bits,size or VECTOR,6,number,bits,setnumber,size
04026
04027     Currently only used in the if statement
04028 */
04029
04030 WORD *CountComp(UBYTE *inp, WORD *to)
04031 {
04032     GETIDENTITY
04033     UBYTE *p, c;
04034     WORD *w, mini = 0, type, c1, c2;
04035     int error = 0;
04036     p = inp;
04037     w = to;
04038     AR.Eside = 2;
04039     *w++ = TYPECOUNT;
04040     *w++ = 0;
04041     *w++ = 0;
04042     while ( *p == ',' ) {
04043         p++; inp = p;
04044         if ( *p == '[' || FG.cTable[*p] == 0 ) {
04045             if ( ( p = SkipAName(inp) ) == 0 ) return(0);
04046             c = *p; *p = 0;
04047             type = GetName(AC.varnames,inp,&c1,WITHAUTO);
04048             if ( c == '.' ) {
04049                 if ( type == CVECTOR || type == CDUBIOUS ) {
04050                     *p++ = c;
04051                     inp = p;
04052                     p = SkipAName(p);
04053                     if ( p == 0 ) return(0);
04054                     c = *p;
04055                     *p = 0;
04056                     type = GetName(AC.varnames,inp,&c2,WITHAUTO);
04057                     if ( type != CVECTOR && type != CDUBIOUS ) {
04058                         MesPrint("&Not a vector in dotproduct in if statement: %s",inp);
04059                         error = 1;
04060                     }
04061                     else type = CDOTPRODUCT;
04062                 }
04063                 else {
04064                     MesPrint("&Illegal use of . after %s in if statement",inp);
04065                     if ( type == NAMENOTFOUND )
04066                         MesPrint("&%s is not a properly declared variable",inp);
04067                     error = 1;
04068                     *p++ = c;
04069                     while ( *p && *p != ')' && *p != ',' ) p++;
04070                     if ( *p == ',' && FG.cTable[p[1]] == 1 ) {
04071                         p++;
04072                         while ( *p && *p != ')' && *p != ',' ) p++;
04073                     }
04074                     continue;
04075                 }
04076             }
04077             *p = c;

```

```

04078         switch ( type ) {
04079             case CSYMBOL:
04080                 *w++ = SYMBOL; *w++ = 4; *w++ = c1;
04081 Sgetnum:         if ( *p != ',' ) {
04082                     MesCerr("sequence",p);
04083                     while ( *p && *p != ')' && *p != ',' ) p++;
04084                     error = 1;
04085                 }
04086                 p++; inp = p;
04087                 ParseSignedNumber(mini,p)
04088                 if ( FG.cTable[p[-1]] != 1 || ( *p && *p != ')' && *p != ',' ) ) {
04089                     while ( *p && *p != ')' && *p != ',' ) p++;
04090                     error = 1;
04091                     c = *p; *p = 0;
04092                     MesPrint("&Improper value in count: %s",inp);
04093                     *p = c;
04094                     while ( *p && *p != ')' && *p != ',' ) p++;
04095                 }
04096                 *w++ = mini;
04097                 break;
04098             case CFUNCTION:
04099                 *w++ = FUNCTION; *w++ = 4; *w++ = c1+FUNCTION; goto Sgetnum;
04100             case CDOTPRODUCT:
04101                 *w++ = DOTPRODUCT; *w++ = 5;
04102                 *w++ = c2 + AM.OffsetVector;
04103                 *w++ = c1 + AM.OffsetVector;
04104                 goto Sgetnum;
04105             case CVECTOR:
04106                 *w++ = VECTOR; *w++ = 5;
04107                 *w++ = c1 + AM.OffsetVector;
04108                 if ( *p == ',' ) {
04109                     *w++ = VECTBIT | DOTPBIT | FUNBIT;
04110                     goto Sgetnum;
04111                 }
04112             else if ( *p == '+' ) {
04113                 p++;
04114                 *w = 0;
04115                 while ( *p && *p != ',' ) {
04116                     if ( *p == 'v' || *p == 'V' ) {
04117                         *w |= VECTBIT; p++;
04118                     }
04119                     else if ( *p == 'd' || *p == 'D' ) {
04120                         *w |= DOTPBIT; p++;
04121                     }
04122                     else if ( *p == 'f' || *p == 'F'
04123                             || *p == 't' || *p == 'T' ) {
04124                         *w |= FUNBIT; p++;
04125                     }
04126                     else if ( *p == '?' ) {
04127                         p++; inp = p;
04128                         if ( *p == '{' ) { /* } */
04129                             SKIPBRA2(p)
04130                             if ( p == 0 ) return(0);
04131                             if ( ( c1 = DoTempSet(inp+1,p) ) < 0 ) return(0);
04132                             if ( Sets[c1].type != CFUNCTION ) {
04133                                 MesPrint("&set type conflict: Function expected");
04134                                 return(0);
04135                             }
04136                             type = CSET;
04137                             c = ++p;
04138                         }
04139                         else {
04140                             p = SkipAName(p);
04141                             if ( p == 0 ) return(0);
04142                             c = *p; *p = 0;
04143                             type = GetName(AC.varnames,inp,&c1,WITHAUTO);
04144                         }
04145                         if ( type != CSET && type != CDUBIOUS ) {
04146                             MesPrint("&%s is not a set",inp);
04147                             error = 1;
04148                         }
04149                         w[-2] = 6;
04150                         *w++ |= SETBIT;
04151                         *w++ = c1;
04152                         *p = c;
04153                         goto Sgetnum;
04154                     }
04155                     else {
04156                         MesCerr("specifier for vector",p);
04157                         error = 1;
04158                     }
04159                 }
04160                 w++;
04161                 goto Sgetnum;
04162             }
04163             else {
04164                 MesCerr("specifier for vector",p);

```

```

04165         while ( *p && *p != ')' && *p != ',' ) p++;
04166         error = 1;
04167         *w++ = VECTBIT | DOTPBIT | FUNBIT;
04168         goto Sgetnum;
04169     }
04170     case CDUBIOUS:
04171         goto skipfield;
04172     default:
04173         *p = 0;
04174         MesPrint("%s is not a symbol, function, vector or dotproduct",inp);
04175         error = 1;
04176 skipfield:
04177         while ( *p && *p != ')' && *p != ',' ) p++;
04178         if ( *p && FG.cTable[p[1]] == 1 ) {
04179             p++;
04180             while ( *p && *p != ')' && *p != ',' ) p++;
04181             break;
04182         }
04183     }
04184     else {
04185         MesCerr("name",p);
04186         while ( *p && *p != ',' ) p++;
04187         error = 1;
04188     }
04189 }
04190 to[1] = w-to;
04191 if ( *p == ')' ) p++;
04192 if ( *p ) { MesCerr("end of statement",p); return(0); }
04193 if ( error ) return(0);
04194 return(w);
04195 }
04196
04197 /*
04198 #] CountComp :
04199 #[ CoIf :
04200
04201     Reads the if statement: There must be a pair of parentheses.
04202     Much work is delegated to the routines in compi2 and CountComp.
04203     The goto is kept hanging as it is forward.
04204     The address in which the label must be written is pushed on
04205     the AC.IfStack.
04206
04207     Here we allow statements of the type
04208     if ( condition ) single statement;
04209     compile the if statement.
04210     test character at end
04211     if not ; or )
04212     copy the statement after the proper parenthesis to the
04213     beginning of the AC.iBuffer.
04214     Have it compiled.
04215     generate an endif statement.
04216 */
04217
04218 static UWORD *CIsratC = 0;
04219
04220 int CoIf(UBYTE *inp)
04221 {
04222     GETIDENTITY
04223     int error = 0, level;
04224     WORD *w, *ww, *u, *s, *OldWork, *OldSpace = AT.WorkSpace;
04225     WORD gotexp = 0; /* Indicates whether there can be a condition */
04226     WORD lenpp, lenlev, ncoef, i, number;
04227     UBYTE *p, *pp, *ppp, c;
04228     CBUF *C = cbuf+AC.cbunum;
04229     LONG x;
04230 #ifdef WITHFLOAT
04231     int spec;
04232 #endif
04233     if ( *inp == '(' && inp[1] == ',' ) inp += 2;
04234     else if ( *inp == '(' ) inp++; /* Usually we enter at the bracket */
04235
04236     if ( CIsratC == 0 )
04237         CIsratC = (UWORD *)Malloca1((AM.MaxTal+2)*sizeof(UWORD),"CoIf");
04238     lenpp = 0;
04239     lenlev = 1;
04240     if ( AC.IfLevel >= AC.MaxIf ) DoubleIfBuffers();
04241     AC.IfCount[lenpp++] = 0;
04242 /*
04243     IfStack is used for organizing the 'go to' for the various if levels
04244 */
04245     *AC.IfStack++ = C->Pointer-C->Buffer+2;
04246 /*
04247     IfSumCheck is used to test for illegal nesting of if, argument or repeat.
04248 */
04249     AC.IfSumCheck[AC.IfLevel] = NestingChecksum();
04250     AC.IfLevel++;
04251     w = OldWork = AT.WorkPointer;

```

```

04252     *w++ = TYPEIF;
04253     w += 2;
04254     p = inp;
04255     for(;;) {
04256         inp = p;
04257         level = 0;
04258 ReDo:
04259     if ( FG.cTable[*p] == 1 ) { /* Number */
04260         if ( gotexp == 1 ) { MesCerr("position for ",p); error = 1; }
04261 #ifdef WITHFLOAT
04262         pp = CheckFloat(p,&spec);
04263         if ( pp > p ) { /* Got one */
04264 HaveFloat:
04265             if ( spec == 1 ) { /* is zero */
04266                 *w++ = LONGNUMBER; *w++ = 3; *w++ = 0;
04267             }
04268             else if ( spec == -1 ) {
04269                 MesPrint("&The floating point system has not been started: %s",p);
04270                 if ( !error ) error = 1;
04271             }
04272             else {
04273                 WORD *ow = AT.WorkPointer;
04274                 AT.WorkPointer = w;
04275                 c = *pp; c = 0;
04276                 ReadFloat((SBYTE *)p); /* Is now at AT.WorkPointer */
04277                 *pp = c;
04278                 AT.WorkPointer[0] = IFFLOATNUMBER;
04279                 w = AT.WorkPointer + AT.WorkPointer[1];
04280                 AT.WorkPointer = ow;
04281                 if ( level ) w[FUNHEAD+3] = -w[FUNHEAD+3];
04282             }
04283             goto DoneWithNumber;
04284         }
04285 /*
04286     Notation: Same as FLOATFUN but FLOATFUN replaced by IFFLOATNUMBER.
04287 */
04288 #endif
04289     u = w;
04290     *w++ = LONGNUMBER;
04291 /*
04292     Notation:
04293     LONGNUMBER,size,reducedsize*sign,numerator,denominator
04294     with the length of denominator and numerator equal to reducedsize
04295 */
04296     w += 2;
04297     if ( GetLong(p,(UWORD *)w,&ncoef) ) { ncoef = 1; error = 1; }
04298     w[-1] = ncoef;
04299     while ( FG.cTable[***p] == 1 );
04300     if ( *p == '/' ) {
04301         p++;
04302         if ( FG.cTable[*p] != 1 ) {
04303             MesCerr("sequence",p); error = 1; goto OnlyNum;
04304         }
04305         if ( GetLong(p,CIsratC,&ncoef) ) {
04306             ncoef = 1; error = 1;
04307         }
04308         while ( FG.cTable[***p] == 1 );
04309         if ( ncoef == 0 ) {
04310             MesPrint("&Division by zero!");
04311             error = 1;
04312         }
04313     }
04314     else {
04315         if ( w[-1] != 0 ) {
04316             if ( Simplify(BHEAD (UWORD *)w,(WORD *) (w-1),
04317                 CIsratC,&ncoef) ) error = 1;
04318             else {
04319                 i = w[-1];
04320                 if ( i >= ncoef ) {
04321                     i = w[-1];
04322                     w += i;
04323                     i -= ncoef;
04324                     s = (WORD *)CIsratC;
04325                     NCOPY(w,s,ncoef);
04326                     while ( --i >= 0 ) *w++ = 0;
04327                 }
04328                 else {
04329                     w += i;
04330                     i = ncoef - i;
04331                     while ( --i >= 0 ) *w++ = 0;
04332                     s = (WORD *)CIsratC;
04333                     NCOPY(w,s,ncoef);
04334                 }
04335             }
04336         }
04337     }
04338 }

```

```

04339         else {
04340 OnlyNum:
04341             w += ncoef;
04342             if ( ncoef > 0 ) {
04343                 ncoef--; *w++ = 1;
04344                 while ( --ncoef >= 0 ) *w++ = 0;
04345             }
04346         }
04347         u[1] = WORDDIF(w,u);
04348         u[2] = (u[1] - 3)/2;
04349         if ( level ) u[2] = -u[2];
04350 #ifdef WITHFLOAT
04351 DoneWithNumber:
04352 #endif
04353         gotexp = 1;
04354     }
04355     else if ( *p == '+' ) { p++; goto ReDo; }
04356     else if ( *p == '-' ) { level ^= 1; p++; goto ReDo; }
04357     else if ( *p == 'c' || *p == 'C' ) { /* Count or Coefficient */
04358         if ( gotexp == 1 ) { MesCerr("position for ",p); error = 1; }
04359         while ( FG.cTable[++p] == 0 );
04360         c = *p; *p = 0;
04361         if ( !StrICmp(inp, (UBYTE *) "count" ) ) {
04362             *p = c;
04363             if ( c != '(' ) {
04364                 MesPrint("&no ( after count");
04365                 error = 1;
04366                 goto endofif;
04367             }
04368             inp = p;
04369             SKIPBRA4(p);
04370             c = *p; *p = 0; *inp = ',';
04371             w = CountComp(inp,w);
04372             *p = c; *inp = '(';
04373             if ( w == 0 ) { error = 1; goto endofif; }
04374             gotexp = 1;
04375         }
04376         else if ( ConWord(inp, (UBYTE *) "coefficient" ) && ( p - inp ) > 3 ) {
04377             *w++ = COEFFI;
04378             *w++ = 2;
04379             *p = c;
04380             gotexp = 1;
04381         }
04382         else goto NoGood;
04383         inp = p;
04384     }
04385     else if ( *p == 'm' || *p == 'M' ) { /* match */
04386         if ( gotexp == 1 ) { MesCerr("position for ",p); error = 1; }
04387         while ( !FG.cTable[++p] );
04388         c = *p; *p = 0;
04389         if ( !StrICmp(inp, (UBYTE *) "match" ) ) {
04390             *p = c;
04391             if ( c != '(' ) {
04392                 MesPrint("&no ( after match");
04393                 error = 1;
04394                 goto endofif;
04395             }
04396             p++; inp = p;
04397             SKIPBRA4(p);
04398             *p = '=';
04399         /*
04400          Now we can call the reading of the lhs of an id statement.
04401          This has to be modified in the future.
04402         */
04403         AT.WorkSpace = AT.WorkPointer = w;
04404         ppp = inp;
04405         while ( FG.cTable[*ppp] == 0 && ppp < p ) ppp++;
04406         if ( *ppp == ',' ) AC.idoption = 0;
04407         else AC.idoption = SUBMULTI;
04408         level = CoIdExpression(inp,TYPEIF);
04409         AT.WorkSpace = OldSpace;
04410         AT.WorkPointer = OldWork;
04411         if ( level != 0 ) {
04412             if ( level < 0 ) { error = -1; goto endofif; }
04413             error = 1;
04414         }
04415     /*
04416      If we pop numlhs we are in good shape
04417     */
04418     s = u = C->lhs[C->numlhs];
04419     while ( u < C->Pointer ) *w++ = *u++;
04420     C->numlhs--; C->Pointer = s;
04421     *p++ = ')';
04422     inp = p;
04423     gotexp = 1;
04424 }
04425     else if ( !StrICmp(inp, (UBYTE *) "multipleof" ) ) {

```

```

04426         if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
04427         *p = c;
04428         if ( c != '(' ) {
04429             MesPrint("&no ( after multipleof");
04430             error = 1; goto endofif;
04431         }
04432         p++;
04433         if ( FG.cTable[*p] != 1 ) {
04434             MesPrint("&multipleof needs a short positive integer argument");
04435             error = 1; goto endofif;
04436         }
04437         ParseNumber(x,p)
04438         if ( *p != ')' || x <= 0 || x > MAXPOSITIVE ) goto Nomulof;
04439         p++;
04440         *w++ = MULTIPLEOF; *w++ = 3; *w++ = (WORD)x;
04441         inp = p;
04442         gotexp = 1;
04443     }
04444     else {
04445         MesPrint("&Unrecognized word: %s",inp);
04446         *p = c;
04447         error = 1;
04448         level = 0;
04449         if ( c == '(' ) SKIPBRA4(p)
04450         inp = ++p;
04451         gotexp = 1;
04452     }
04453 }
04454 else if ( *p == 'f' || *p == 'F' ) { /* FindLoop */
04455     if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
04456     while ( FG.cTable[++p] == 0 );
04457     c = *p; *p = 0;
04458     if ( !StrICmp(inp, (UBYTE *) "findloop") ) {
04459         *p = c;
04460         if ( c != '(' ) {
04461             MesPrint("&no ( after findloop");
04462             error = 1;
04463             goto endofif;
04464         }
04465         inp = p;
04466         SKIPBRA4(p);
04467         c = ++p; *p = 0; *inp = ',';
04468         if ( CoFindLoop(inp) ) { error = 1; goto endofif; }
04469         s = u = C->lhs[C->numlhs];
04470         while ( u < C->Pointer ) *w++ = *u++;
04471         C->numlhs--; C->Pointer = s;
04472         *p = c; *inp = '(';
04473         if ( w == 0 ) { error = 1; goto endofif; }
04474         gotexp = 1;
04475     }
04476     else if ( !StrICmp(inp, (UBYTE *) "flag") ) {
04477         UBYTE cc = c, *pppp;
04478         *p = cc;
04479         if ( cc != '(' ) {
04480             MesPrint("&no ( after flag");
04481             error = 1;
04482             goto endofif;
04483         }
04484         inp = p;
04485         SKIPBRA4(p);
04486         cc = ++p; *p = 0; *inp = ','; pppp = p;
04487         ww = w;
04488         *w++ = IFUSERFLAG; *w++ = 0;
04489         while ( *inp ) {
04490             int x = 0;
04491             while ( *inp == ',' ) inp++;
04492             if ( *inp == 0 || *inp == ')' ) break;
04493             while ( *inp >= '0' && *inp <= '9' ) x = 10*x + (*inp++ - '0');
04494             if ( x < 1 || x > BITSINWORD ) {
04495                 MesPrint("&Flag number %d outside the permitted range 1-%d.",BITSINWORD);
04496                 error = 1;
04497             }
04498             *w++ = x-1;
04499         }
04500         ww[1] = w-ww;
04501         p = pppp; *p = cc; *inp = '(';
04502         gotexp = 1;
04503         if ( ww[1] <= 2 ) {
04504             MesPrint("&The userflag condition in the if statement needs arguments.");
04505             error = 1;
04506         }
04507         inp = p;
04508         gotexp = 1;
04509     }
04510     else goto NoGood;
04511 }
04512 else if ( *p == 'e' || *p == 'E' ) { /* Expression */

```

```

04513         if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
04514         while ( FG.cTable[***p] == 0 );
04515         c = *p; *p = 0;
04516         if ( !StrICmp(inp, (UBYTE *)"expression") ) {
04517             *p = c;
04518             if ( c != '(' ) {
04519                 MesPrint("&no ( after expression");
04520                 error = 1;
04521                 goto endofif;
04522             }
04523             p++; ww = w; *w++ = IFEXPRESSION; w++;
04524             while ( *p != ')' ) {
04525                 if ( *p == ',' ) { p++; continue; }
04526                 if ( *p == '[' || FG.cTable[*p] == 0 ) {
04527                     pp = p;
04528                     if ( ( p = SkipAName(p) ) == 0 ) {
04529                         MesPrint("&Improper name for an expression: '%s'",pp);
04530                         error = 1;
04531                         goto endofif;
04532                     }
04533                     c = *p; *p = 0;
04534                     if ( GetName(AC.exprnames,pp,&number,NOAUTO) == CEXPRESSION ) {
04535                         *w++ = number;
04536                     }
04537                     else if ( GetName(AC.varnames,pp,&number,NOAUTO) != NAMENOTFOUND ) {
04538                         MesPrint("&%s is not an expression",pp);
04539                         error = 1;
04540                         *w++ = number;
04541                     }
04542                     *p = c;
04543                 }
04544                 else {
04545                     MesPrint("&Illegal object in Expression in if-statement");
04546                     error = 1;
04547                     while ( *p && *p != ',' && *p != ')' ) p++;
04548                     if ( *p == 0 || *p == ')' ) break;
04549                 }
04550             }
04551             ww[1] = w - ww;
04552             p++;
04553             gotexp = 1;
04554         }
04555         else goto NoGood;
04556         inp = p;
04557     }
04558     else if ( *p == 'i' || *p == 'I' ) { /* IsFactorized */
04559         if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
04560         while ( FG.cTable[***p] == 0 );
04561         c = *p; *p = 0;
04562         if ( !StrICmp(inp, (UBYTE *)"isfactorized") ) {
04563             *p = c;
04564             if ( c != '(' ) { /* No expression means current expression */
04565                 ww = w; *w++ = IFISFACTORIZED; w++;
04566             }
04567             else {
04568                 p++; ww = w; *w++ = IFISFACTORIZED; w++;
04569                 while ( *p != ')' ) {
04570                     if ( *p == ',' ) { p++; continue; }
04571                     if ( *p == '[' || FG.cTable[*p] == 0 ) {
04572                         pp = p;
04573                         if ( ( p = SkipAName(p) ) == 0 ) {
04574                             MesPrint("&Improper name for an expression: '%s'",pp);
04575                             error = 1;
04576                             goto endofif;
04577                         }
04578                         c = *p; *p = 0;
04579                         if ( GetName(AC.exprnames,pp,&number,NOAUTO) == CEXPRESSION ) {
04580                             *w++ = number;
04581                         }
04582                         else if ( GetName(AC.varnames,pp,&number,NOAUTO) != NAMENOTFOUND ) {
04583                             MesPrint("&%s is not an expression",pp);
04584                             error = 1;
04585                             *w++ = number;
04586                         }
04587                         *p = c;
04588                     }
04589                     else {
04590                         MesPrint("&Illegal object in IsFactorized in if-statement");
04591                         error = 1;
04592                         while ( *p && *p != ',' && *p != ')' ) p++;
04593                         if ( *p == 0 || *p == ')' ) break;
04594                     }
04595                 }
04596                 p++;
04597             }
04598             ww[1] = w - ww;
04599             gotexp = 1;

```

```

04600     }
04601     else goto NoGood;
04602     inp = p;
04603 }
04604 else if ( *p == 'o' || *p == 'O' ) { /* Occurs */
04605 /*
04606     Tests whether variables occur inside a term.
04607     At the moment this is done one by one.
04608     If we want to do them in groups we should do the reading
04609     a bit different: each as a variable in a term, and then
04610     use Normalize to get the variables grouped and in order.
04611     That way FindVar (in if.c) can work more efficiently.
04612     Still to be done!!!
04613     TASK: Nice little task for someone to learn.
04614 */
04615     UBYTE cc;
04616     if ( gotexp == 1 ) { MesCerr("position for "),p); error = 1; }
04617     while ( FG.cTable[++p] == 0 );
04618     c = cc = *p; *p = 0;
04619     if ( !StrICmp(inp,(UBYTE *)"occurs") ) {
04620         WORD c1, c2, type;
04621         *p = cc;
04622         if ( cc != '(' ) {
04623             MesPrint("&no ( after occurs");
04624             error = 1;
04625             goto endofif;
04626         }
04627         inp = p;
04628         SKIPBRA4(p);
04629         cc = ++p; *p = 0; *inp = ','; pp = p;
04630         ww = w;
04631         *w++ = IFOCCURS; *w++ = 0;
04632         while ( *inp ) {
04633             while ( *inp == ',' ) inp++;
04634             if ( *inp == 0 || *inp == ')' ) break;
04635 /*
04636             Now read a list of names
04637             We can have symbols, vectors, dotproducts, indices, functions.
04638             There could also be dummy indices and/or extra symbols.
04639 */
04640             if ( *inp == '[' || FG.cTable[*inp] == 0 ) {
04641                 if ( ( p = SkipAName(inp) ) == 0 ) return(0);
04642                 c = *p; *p = 0;
04643                 type = GetName(AC.varnames,inp,&c1,WITHAUTO);
04644                 if ( c == '.' ) {
04645                     if ( type == CVECTOR || type == CDUBIOUS ) {
04646                         *p++ = c;
04647                         inp = p;
04648                         p = SkipAName(p);
04649                         if ( p == 0 ) return(0);
04650                         c = *p;
04651                         *p = 0;
04652                         type = GetName(AC.varnames,inp,&c2,WITHAUTO);
04653                         if ( type != CVECTOR && type != CDUBIOUS ) {
04654                             MesPrint("&Not a vector in dotproduct in if statement: %s",inp);
04655                             error = 1;
04656                         }
04657                         else type = CDOTPRODUCT;
04658                     }
04659                     else {
04660                         MesPrint("&Illegal use of . after %s in if statement",inp);
04661                         if ( type == NAMENOTFOUND )
04662                             MesPrint("&%s is not a properly declared variable",inp);
04663                         error = 1;
04664                         *p++ = c;
04665                         while ( *p && *p != ')' && *p != ',' ) p++;
04666                         if ( *p == ',' && FG.cTable[p[1]] == 1 ) {
04667                             p++;
04668                             while ( *p && *p != ')' && *p != ',' ) p++;
04669                         }
04670                         continue;
04671                     }
04672                 }
04673                 *p = c;
04674                 switch ( type ) {
04675                     case CSYMBOL: /* To worry about extra symbols */
04676                         *w++ = SYMBOL;
04677                         *w++ = c1;
04678                     break;
04679                     case CINDEX:
04680                         *w++ = INDEX;
04681                         *w++ = c1 + AM.OffsetIndex;
04682                     break;
04683                     case CVECTOR:
04684                         *w++ = VECTOR;
04685                         *w++ = c1 + AM.OffsetVector;
04686                     break;

```



```

04687             case CDOTPRODUCT:
04688                 *w++ = DOTPRODUCT;
04689                 *w++ = c1 + AM.OffsetVector;
04690                 *w++ = c2 + AM.OffsetVector;
04691             break;
04692             case CFUNCTION:
04693                 *w++ = FUNCTION;
04694                 *w++ = c1+FUNCTION;
04695             break;
04696             default:
04697                 MesPrint("&Illegal variable %s in occurs condition in if
statement",inp);
04698                 error = 1;
04699                 break;
04700             }
04701             inp = p;
04702         }
04703         else {
04704             MesPrint("&Illegal object %s in occurs condition in if statement",inp);
04705             error = 1;
04706             break;
04707         }
04708     }
04709     ww[1] = w-ww;
04710     p = pp; *p = cc; *inp = '(';
04711     gotexp = 1;
04712     if ( ww[1] <= 2 ) {
04713         MesPrint("&The occurs condition in the if statement needs arguments.");
04714         error = 1;
04715     }
04716 }
04717 else goto NoGood;
04718 inp = p;
04719 }
04720 else if ( *p == '$' ) {
04721     if ( gotexp == 1 ) { MesCerr("position for ",p); error = 1; }
04722     p++; inp = p;
04723     while ( FG.cTable[*p] == 0 || FG.cTable[*p] == 1 ) p++;
04724     c = *p; *p = 0;
04725     if ( ( i = GetDollar(inp) ) < 0 ) {
04726         MesPrint("&undefined dollar expression %s",inp);
04727         error = 1;
04728         i = AddDollar(inp,DOLUNDEFINED,0,0);
04729     }
04730     *p = c;
04731     *w++ = IFDOLLAR; *w++ = 3; *w++ = i;
04732 /*
04733     And then the IFDOLLAREXTRA pieces for [1] [$y] etc
04734 */
04735     if ( *p == '[' ) {
04736         p++;
04737         if ( ( w = GetIfDollarFactor(&p,w) ) == 0 ) {
04738             error = 1;
04739             goto endofif;
04740         }
04741         else if ( *p != ']' ) {
04742             error = 1;
04743             goto endofif;
04744         }
04745         p++;
04746     }
04747     inp = p;
04748     gotexp = 1;
04749 }
04750 #ifdef WITHFLOAT
04751     else if ( *p == '.' ) {
04752         pp = CheckFloat(p,&spec);
04753         if ( pp > p ) goto HaveFloat;
04754     }
04755 #endif
04756     else if ( *p == '(' ) {
04757         if ( gotexp ) {
04758             MesCerr("parenthesis",p);
04759             error = 1;
04760             goto endofif;
04761         }
04762         gotexp = 0;
04763         if ( ++lenlev >= AC.MaxIf ) DoubleIfBuffers();
04764         AC.IfCount[lenpp++] = w-OldWork;
04765         *w++ = SUBEXPR;
04766         w += 2;
04767         p++;
04768     }
04769     else if ( *p == ')' ) {
04770         if ( gotexp == 0 ) { MesCerr("position for ",p); error = 1; }
04771         gotexp = 1;
04772         u = AC.IfCount[--lenpp]+OldWork;

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```

04773         lenlev--;
04774         u[1] = w - u;
04775         if ( lenlev <= 0 ) {          /* End if condition */
04776             AT.WorkSpace = OldSpace;
04777             AT.WorkPointer = OldWork;
04778             AddNtoL(OldWork[1],OldWork);
04779             p++;
04780             if ( *p == ')' ) {
04781                 MesPrint("&unmatched parenthesis in if/while ()");
04782                 error = 1;
04783                 while ( *++p == ')' );
04784             }
04785             if ( *p ) {
04786                 level = CompileStatement(p);
04787                 if ( level ) error = level;
04788                 while ( *p ) p++;
04789                 if ( CoEndIf(p) && error == 0 ) error = 1;
04790             }
04791             return(error);
04792         }
04793         p++;
04794     }
04795     else if ( *p == '>' ) {
04796         if ( gotexp == 0 ) goto NoExp;
04797         if ( p[1] == '=' ) { *w++ = GREATEREQUAL; *w++ = 2; p += 2; }
04798         else                { *w++ = GREATER;      *w++ = 2; p++; }
04799         gotexp = 0;
04800     }
04801     else if ( *p == '<' ) {
04802         if ( gotexp == 0 ) goto NoExp;
04803         if ( p[1] == '=' ) { *w++ = LESSEQUAL; *w++ = 2; p += 2; }
04804         else                { *w++ = LESS;      *w++ = 2; p++; }
04805         gotexp = 0;
04806     }
04807     else if ( *p == '=' ) {
04808         if ( gotexp == 0 ) goto NoExp;
04809         if ( p[1] == '=' ) p++;
04810         *w++ = EQUAL; *w++ = 2; p++;
04811         gotexp = 0;
04812     }
04813     else if ( *p == '!' && p[1] == '=' ) {
04814         if ( gotexp == 0 ) { p++; goto NoExp; }
04815         *w++ = NOTEQUAL; *w++ = 2; p += 2;
04816         gotexp = 0;
04817     }
04818     else if ( *p == '|' && p[1] == '|' ) {
04819         if ( gotexp == 0 ) { p++; goto NoExp; }
04820         *w++ = ORCOND; *w++ = 2; p += 2;
04821         gotexp = 0;
04822     }
04823     else if ( *p == '&' && p[1] == '&' ) {
04824         if ( gotexp == 0 ) {
04825             p++;
04826             p++;
04827             MesCerr("sequence",p);
04828             error = 1;
04829         }
04830         else {
04831             *w++ = ANDCOND; *w++ = 2; p += 2;
04832             gotexp = 0;
04833         }
04834     }
04835     else if ( *p == 0 ) {
04836         MesPrint("&Unmatched parentheses");
04837         error = 1;
04838         goto endofif;
04839     }
04840     else {
04841         if ( FG.cTable[*p] == 0 ) {
04842             WORD ij;
04843             inp = p;
04844             while ( ( ij = FG.cTable[*++p] ) == 0 || ij == 1 );
04845             c = *p; *p = 0;
04846             goto NoGood;
04847         }
04848         MesCerr("sequence",p);
04849         error = 1;
04850         p++;
04851     }
04852 }
04853 endofif;
04854 return(error);
04855 }
04856
04857 /*
04858 #] CoIf :
04859 #[ CoElse :

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```

04860 */
04861
04862 int CoElse(UBYTE *p)
04863 {
04864     int error = 0;
04865     CBUF *C = cbuf+AC.cbunum;
04866     if ( *p != 0 ) {
04867         while ( *p == ',' ) p++;
04868         if ( tolower(*p) == 'i' && tolower(p[1]) == 'f' && p[2] == '(' )
04869             return(CoElseIf(p+2));
04870         MesPrint("&No extra text allowed as part of an else statement");
04871         error = 1;
04872     }
04873     if ( AC.IfLevel <= 0 ) { MesPrint("&else statement without if"); return(1); }
04874     if ( AC.IfSumCheck[AC.IfLevel-1] != NestingChecksum() - 1 ) {
04875         MesNesting();
04876         error = 1;
04877     }
04878     Add3Com(TYPEELSE,AC.IfLevel)
04879     C->Buffer[AC.IfStack[-1]] = C->numlhs;
04880     AC.IfStack[-1] = C->Pointer - C->Buffer - 1;
04881     return(error);
04882 }
04883
04884 /*
04885 #] CoElse :
04886 #[ CoElseIf :
04887 */
04888
04889 int CoElseIf(UBYTE *inp)
04890 {
04891     CBUF *C = cbuf+AC.cbunum;
04892     if ( AC.IfLevel <= 0 ) { MesPrint("&elseif statement without if"); return(1); }
04893     Add3Com(TYPEELSE,-AC.IfLevel)
04894     AC.IfLevel--;
04895     C->Buffer[*--AC.IfStack] = C->numlhs;
04896     return(CoIf(inp));
04897 }
04898
04899 /*
04900 #] CoElseIf :
04901 #[ CoEndIf :
04902
04903     It puts a RHS-level at the position indicated in the AC.IfStack.
04904     This corresponds to the label belonging to a forward goto.
04905     It is the goto that belongs either to the failing condition
04906     of the if (no else statement), or the completion of the
04907     success path (with else statement)
04908     The code is a jump to the next statement. It is there to prevent
04909     problems with
04910     if ( .. )
04911         if ( .. )
04912         endif;
04913     elseif ( .. )
04914 */
04915
04916 int CoEndIf(UBYTE *inp)
04917 {
04918     CBUF *C = cbuf+AC.cbunum;
04919     WORD i = C->numlhs, to, k = -AC.IfLevel;
04920     int error = 0;
04921     while ( *inp == ',' ) inp++;
04922     if ( *inp != 0 ) {
04923         error = 1;
04924         MesPrint("&No extra text allowed as part of an endif/elseif statement");
04925     }
04926     if ( AC.IfLevel <= 0 ) {
04927         MesPrint("&Endif statement without corresponding if"); return(1);
04928     }
04929     AC.IfLevel--;
04930     C->Buffer[*--AC.IfStack] = i+1;
04931     if ( AC.IfSumCheck[AC.IfLevel] != NestingChecksum() ) {
04932         MesNesting();
04933         error = 1;
04934     }
04935     Add3Com(TYPEENDIF,i+1)
04936
04937     /*
04938     Now the search for the TYPEELSE in front of the elseif statements
04939     */
04940     to = C->numlhs;
04941     while ( i > 0 ) {
04942         if ( C->lhs[i][0] == TYPEELSE && C->lhs[i][2] == to ) to = i;
04943         if ( C->lhs[i][0] == TYPEIF ) {
04944             if ( C->lhs[i][2] == to ) {
04945                 i--;
04946                 if ( i <= 0 || C->lhs[i][0] != TYPEELSE
04947                     || C->lhs[i][2] != k ) break;

```

```

04947         C->lhs[i][2] = C->numlhs;
04948         to = i;
04949     }
04950 }
04951     i--;
04952 }
04953     return(error);
04954 }
04955
04956 /*
04957     #] CoEndIf :
04958     #[ CoWhile :
04959 */
04960
04961 int CoWhile(UBYTE *inp)
04962 {
04963     CBUF *C = cbuf+AC.cbunum;
04964     WORD startnum = C->numlhs + 1;
04965     int error;
04966     AC.WhileLevel++;
04967     error = CoIf(inp);
04968     if ( C->numlhs > startnum && C->lhs[startnum][2] == C->numlhs
04969         && C->lhs[C->numlhs][0] == TYPEENDIF ) {
04970         C->lhs[C->numlhs][2] = startnum-1;
04971         AC.WhileLevel--;
04972     }
04973     else C->lhs[startnum][2] = startnum;
04974     return(error);
04975 }
04976
04977 /*
04978     #] CoWhile :
04979     #[ CoEndWhile :
04980 */
04981
04982 int CoEndWhile(UBYTE *inp)
04983 {
04984     int error = 0;
04985     WORD i;
04986     CBUF *C = cbuf+AC.cbunum;
04987     if ( AC.WhileLevel <= 0 ) {
04988         MesPrint("&EndWhile statement without corresponding While"); return(1);
04989     }
04990     AC.WhileLevel--;
04991     i = C->Buffer[AC.IfStack[-1]];
04992     error = CoEndIf(inp);
04993     C->lhs[C->numlhs][2] = i - 1;
04994     return(error);
04995 }
04996
04997 /*
04998     #] CoEndWhile :
04999     #[ DoFindLoop :
05000
05001     Function,arguments=number,loopsize=number,outfun=function,include=index;
05002 */
05003
05004 static char *messfind[] = {
05005     "Findloop(function,arguments=#,loopsize=#|<#)[,include=index]"
05006     ,"Replaceloop,function,arguments=#,loopsize=#|<#),outfun=function[,include=index]"
05007 };
05008 static WORD comfindloop[7] = { TYPEFINDLOOP,7,0,0,0,0,0 };
05009
05010 int DoFindLoop(UBYTE *inp, int mode)
05011 {
05012     UBYTE *s, c;
05013     WORD funnum, nargs = 0, nloop = 0, indexnum = 0, outfun = 0;
05014     int type, aflag, lflag, indflag, outflag, error = 0, sym;
05015     while ( *inp == ',' ) inp++;
05016     if ( ( s = SkipAName(inp) ) == 0 ) {
05017         syntax:
05018         MesPrint("&Proper syntax is:");
05019         MesPrint("%s",messfind[mode]);
05020         return(1);
05021     }
05022     c = *s; *s = 0;
05023     if ( ( ( type = GetName(AC.varnames,inp,&funnum,WITHAUTO) ) == NAMENOTFOUND )
05024         || type != CFUNCTION || ( ( sym = (functions[funnum].symmetric) & ~REVERSEORDER )
05025         != SYMMETRIC && sym != ANTISYMMETRIC ) ) {
05026         MesPrint("&%s should be a (anti)symmetric function or tensor",inp);
05027         error = 1;
05028     }
05029     funnum += FUNCTION;
05030     *s = c; inp = s;
05031     aflag = lflag = indflag = outflag = 0;
05032     while ( *inp == ',' ) {
05033         while ( *inp == ',' ) inp++;

```

```

05034     s = inp;
05035     if ( ( s = SkipAName(inp) ) == 0 ) goto syntax;
05036     c = *s; *s = 0;
05037     if ( StrICont(inp, (UBYTE *)"arguments") == 0 ) {
05038         if ( c != '=' ) goto syntax;
05039         *s++ = c;
05040         NeedNumber(nargs, s, syntax)
05041         aflag++;
05042         inp = s;
05043     }
05044     else if ( StrICont(inp, (UBYTE *)"loopsize") == 0 ) {
05045         if ( c != '=' && c != '<' ) goto syntax;
05046         *s++ = c;
05047         if ( FG.cTable[*s] == 1 ) {
05048             NeedNumber(nloop, s, syntax)
05049             if ( nloop < 2 ) {
05050                 MesPrint("&loopsize should be at least 2");
05051                 error = 1;
05052             }
05053             if ( c == '<' ) nloop = -nloop;
05054         }
05055         else if ( tolower(*s) == 'a' && tolower(s[1]) == 'l'
05056             && tolower(s[2]) == 'l' && FG.cTable[s[3]] > 1 ) {
05057             nloop = -1; s += 3;
05058             if ( c != '=' ) goto syntax;
05059         }
05060         inp = s;
05061         lflag++;
05062     }
05063     else if ( StrICont(inp, (UBYTE *)"include") == 0 ) {
05064         if ( c != '=' ) goto syntax;
05065         *s++ = c;
05066         if ( ( inp = SkipAName(s) ) == 0 ) goto syntax;
05067         c = *inp; *inp = 0;
05068         if ( ( type = GetName(AC.varnames, s, &indexnum, WITHAUTO) ) != CINDEX ) {
05069             MesPrint("&%s is not a proper index", s);
05070             error = 1;
05071         }
05072         else if ( indexnum < WILD OFFSET
05073             && indices[indexnum].dimension == 0 ) {
05074             MesPrint("&%s should be a summable index", s);
05075             error = 1;
05076         }
05077         indexnum += AM.OffsetIndex;
05078         *inp = c;
05079         indflag++;
05080     }
05081     else if ( StrICont(inp, (UBYTE *)"outfun") == 0 ) {
05082         if ( c != '=' ) goto syntax;
05083         *s++ = c;
05084         if ( ( inp = SkipAName(s) ) == 0 ) goto syntax;
05085         c = *inp; *inp = 0;
05086         if ( ( type = GetName(AC.varnames, s, &outfun, WITHAUTO) ) != CFUNCTION ) {
05087             MesPrint("&%s is not a proper function or tensor", s);
05088             error = 1;
05089         }
05090         outfun += FUNCTION;
05091         outflag++;
05092         *inp = c;
05093     }
05094     else {
05095         MesPrint("&Unrecognized option in FindLoop or ReplaceLoop: %s", inp);
05096         error = 1;
05097         *s = c; inp = s;
05098         while ( *inp && *inp != ',' ) inp++;
05099     }
05100 }
05101 if ( *inp != 0 && mode == REPLACELOOP ) goto syntax;
05102 if ( mode == FINDLOOP && outflag > 0 ) {
05103     MesPrint("&outflag option is illegal in FindLoop");
05104     error = 1;
05105 }
05106 if ( mode == REPLACELOOP && outflag == 0 ) goto syntax;
05107 if ( aflag == 0 || lflag == 0 ) goto syntax;
05108 comfindloop[3] = funnum;
05109 comfindloop[4] = nloop;
05110 comfindloop[5] = nargs;
05111 comfindloop[6] = outfun;
05112 comfindloop[1] = 7;
05113 if ( indflag ) {
05114     if ( mode == 0 ) comfindloop[2] = indexnum + 5;
05115     else comfindloop[2] = -indexnum - 5;
05116 }
05117 else comfindloop[2] = mode;
05118 AddNtoL(comfindloop[1], comfindloop);
05119 return(error);
05120 }

```

```

05121
05122 /*
05123     #] DoFindLoop :
05124     #[ CoFindLoop :
05125 */
05126
05127 int CoFindLoop(UBYTE *inp)
05128 { return(DoFindLoop(inp,FINDLOOP)); }
05129
05130 /*
05131     #] CoFindLoop :
05132     #[ CoReplaceLoop :
05133 */
05134
05135 int CoReplaceLoop(UBYTE *inp)
05136 {
05137     int error = DoFindLoop(inp,REPLACELoop);
05138     if ( error ) {
05139         Terminate(-1);
05140     }
05141     return(error);
05142 }
05143
05144 /*
05145     #] CoReplaceLoop :
05146     #[ CoFunPowers :
05147 */
05148
05149 static UBYTE *FunPowOptions[] = {
05150     (UBYTE *) "nofunpowers"
05151     , (UBYTE *) "commutingonly"
05152     , (UBYTE *) "allfunpowers"
05153 };
05154
05155 int CoFunPowers(UBYTE *inp)
05156 {
05157     UBYTE *option, c;
05158     int i, maxoptions = sizeof(FunPowOptions)/sizeof(UBYTE *);
05159     while ( *inp == ',' ) inp++;
05160     option = inp;
05161     inp = SkipAName(inp); c = *inp; *inp = 0;
05162     for ( i = 0; i < maxoptions; i++ ) {
05163         if ( StrICont(option, FunPowOptions[i]) == 0 ) {
05164             if ( c ) {
05165                 *inp = c;
05166                 MesPrint("&Illegal FunPowers statement");
05167                 return(1);
05168             }
05169             AC.funpowers = i;
05170             return(0);
05171         }
05172     }
05173     MesPrint("&Illegal option in FunPowers statement: %s", option);
05174     return(1);
05175 }
05176
05177 /*
05178     #] CoFunPowers :
05179     #[ CoUnitTrace :
05180 */
05181
05182 int CoUnitTrace(UBYTE *s)
05183 {
05184     WORD num;
05185     if ( FG.cTable[*s] == 1 ) {
05186         ParseNumber(num,s)
05187         if ( *s != 0 ) {
05188             nogood: MesPrint("&Value of UnitTrace should be a (positive) number or a symbol");
05189             return(1);
05190         }
05191         AC.lUnitTrace[0] = SNUMBER;
05192         AC.lUnitTrace[2] = num;
05193     }
05194     else {
05195         if ( GetName(AC.varnames,s,&num,WITHAUTO) == CSYMBOL ) {
05196             AC.lUnitTrace[0] = SYMBOL;
05197             AC.lUnitTrace[2] = num;
05198             num = -num;
05199         }
05200         else goto nogood;
05201         s = SkipAName(s);
05202         if ( *s ) goto nogood;
05203     }
05204     AC.lUnitTrace = num;
05205     return(0);
05206 }
05207

```

```

05208 /*
05209  #] CoUnitTrace :
05210  #[ CoTerm :
05211
05212      Note: termstack holds the offset of the term statement in the compiler
05213      buffer. termsortstack holds the offset of the last sort statement
05214      (or the corresponding term statement)
05215  */
05216
05217 int CoTerm(UBYTE *s)
05218 {
05219     GETIDENTITY
05220     WORD *w = AT.WorkPointer;
05221     int error = 0;
05222     while ( *s == ',' ) s++;
05223     if ( *s ) {
05224         MesPrint("&Illegal syntax for Term statement");
05225         return(1);
05226     }
05227     if ( AC.termlevel+1 >= AC.maxtermlevel ) {
05228         if ( AC.maxtermlevel <= 0 ) {
05229             AC.maxtermlevel = 20;
05230             AC.termstack = (LONG *)Malloc1(AC.maxtermlevel*sizeof(LONG), "termstack");
05231             AC.termsortstack = (LONG *)Malloc1(AC.maxtermlevel*sizeof(LONG), "termsortstack");
05232             AC.termsumcheck = (WORD *)Malloc1(AC.maxtermlevel*sizeof(WORD), "termsumcheck");
05233         }
05234         else {
05235             DoubleBuffer((void **)AC.termstack, (void **)AC.termstack+AC.maxtermlevel,
05236                 sizeof(LONG), "doubling termstack");
05237             DoubleBuffer((void **)AC.termsortstack,
05238                 (void **)AC.termsortstack+AC.maxtermlevel,
05239                 sizeof(LONG), "doubling termsortstack");
05240             DoubleBuffer((void **)AC.termsumcheck,
05241                 (void **)AC.termsumcheck+AC.maxtermlevel,
05242                 sizeof(LONG), "doubling termsumcheck");
05243             AC.maxtermlevel *= 2;
05244         }
05245     }
05246     AC.termsumcheck[AC.termlevel] = NestingChecksum();
05247     AC.termstack[AC.termlevel] = cbuf[AC.cbunum].Pointer
05248         - cbuf[AC.cbunum].Buffer + 2;
05249     AC.termsortstack[AC.termlevel] = AC.termstack[AC.termlevel] + 1;
05250     AC.termlevel++;
05251     *w++ = TYPETERM;
05252     w++;
05253     *w++ = cbuf[AC.cbunum].numlhs;
05254     *w++ = cbuf[AC.cbunum].numlhs;
05255     AT.WorkPointer[1] = w - AT.WorkPointer;
05256     AddNtoL(AT.WorkPointer[1], AT.WorkPointer);
05257     return(error);
05258 }
05259
05260 /*
05261  #] CoTerm :
05262  #[ CoEndTerm :
05263  */
05264
05265 int CoEndTerm(UBYTE *s)
05266 {
05267     CBUF *C = cbuf+AC.cbunum;
05268     while ( *s == ',' ) s++;
05269     if ( *s ) {
05270         MesPrint("&Illegal syntax for EndTerm statement");
05271         return(1);
05272     }
05273     if ( AC.termlevel <= 0 ) {
05274         MesPrint("&EndTerm without corresponding Argument statement");
05275         return(1);
05276     }
05277     AC.termlevel--;
05278     cbuf[AC.cbunum].Buffer[AC.termstack[AC.termlevel]] = C->numlhs;
05279     cbuf[AC.cbunum].Buffer[AC.termsortstack[AC.termlevel]] = C->numlhs;
05280     if ( AC.termsumcheck[AC.termlevel] != NestingChecksum() ) {
05281         MesNesting();
05282         return(1);
05283     }
05284     return(0);
05285 }
05286
05287 /*
05288  #] CoEndTerm :
05289  #[ CoSort :
05290  */
05291
05292 int CoSort(UBYTE *s)
05293 {
05294     GETIDENTITY

```

```

05295     WORD *w = AT.WorkPointer;
05296     int error = 0;
05297     while ( *s == ',' ) s++;
05298     if ( *s ) {
05299         MesPrint("&Illegal syntax for Sort statement");
05300         error = 1;
05301     }
05302     if ( AC.termlevel <= 0 ) {
05303         MesPrint("&The Sort statement can only be used inside a term environment");
05304         error = 1;
05305     }
05306     if ( error ) return(error);
05307     *w++ = TYPESORT;
05308     w++;
05309     w++;
05310     cbuf[AC.cbunum].Buffer[AC.termstack[AC.termlevel-1]] =
05311         *w = cbuf[AC.cbunum].numlhs+1;
05312     w++;
05313     AC.termstack[AC.termlevel-1] = cbuf[AC.cbunum].Pointer
05314         - cbuf[AC.cbunum].Buffer + 3;
05315     if ( AC.termcheck[AC.termlevel-1] != NestingChecksum() - 1 ) {
05316         MesNesting();
05317         return(1);
05318     }
05319     AT.WorkPointer[1] = w - AT.WorkPointer;
05320     AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
05321     return(error);
05322 }
05323
05324 /*
05325 #] CoSort :
05326 #[ CoPolyFun :
05327
05328 Collect,functionname
05329 */
05330
05331 int CoPolyFun(UBYTE *s)
05332 {
05333     GETIDENTITY
05334     WORD numfun;
05335     int type, error = 0;
05336     UBYTE *t;
05337     AR.PolyFun = AC.lPolyFun = 0;
05338     AR.PolyFunInv = AC.lPolyFunInv = 0;
05339     AR.PolyFunType = AC.lPolyFunType = 0;
05340     AR.PolyFunExp = AC.lPolyFunExp = 0;
05341     AR.PolyFunVar = AC.lPolyFunVar = 0;
05342     AR.PolyFunPow = AC.lPolyFunPow = 0;
05343     if ( *s == 0 ) { return(0); }
05344     t = SkipAName(s);
05345     if ( t == 0 || *t != 0 ) {
05346         MesPrint("&PolyFun statement needs a single commuting function for its argument");
05347         return(1);
05348     }
05349     if ( ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
05350     || ( functions[numfun].spec != 0 ) || ( functions[numfun].commute != 0 ) ) {
05351         MesPrint("&%s should be a regular commuting function",s);
05352         if ( type < 0 ) {
05353             if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
05354                 AddFunction(s,0,0,0,0,0,-1,-1);
05355         }
05356         error = 1;
05357     }
05358     else {
05359         AR.PolyFun = AC.lPolyFun = numfun+FUNCTION;
05360         AR.PolyFunType = AC.lPolyFunType = 1;
05361     }
05362 #ifdef WITHFLOAT
05363     if ( mpfaux_ != 0 ) {
05364         MesPrint("&Simultaneous use of PolyFun and float_ is not allowed.");
05365         error = 1;
05366     }
05367 #endif
05368     return(error);
05369 }
05370
05371 /*
05372 #] CoPolyFun :
05373 #[ CoPolyRatFun :
05374
05375 PolyRatFun [,functionname[,functionname](option)]
05376 */
05377
05378 int CoPolyRatFun(UBYTE *s)
05379 {
05380     GETIDENTITY
05381     WORD numfun;

```



```

05382     int type, error = 0;
05383     UBYTE *t, c;
05384     AR.PolyFun = AC.lPolyFun = 0;
05385     AR.PolyFunInv = AC.lPolyFunInv = 0;
05386     AR.PolyFunType = AC.lPolyFunType = 0;
05387     AR.PolyFunExp = AC.lPolyFunExp = 0;
05388     AR.PolyFunVar = AC.lPolyFunVar = 0;
05389     AR.PolyFunPow = AC.lPolyFunPow = 0;
05390     if ( *s == 0 ) return(error);
05391     t = SkipAName(s);
05392     if ( t == 0 ) goto NumErr;
05393     c = *t; *t = 0;
05394 #ifdef WITHFLOAT
05395     if ( mpfaux_ != 0 ) {
05396         MesPrint("&Simultaneous use of PolyFun and float_ is not allowed.");
05397         error = 1;
05398     }
05399 #endif
05400     if ( ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
05401 || ( functions[numfun].spec != 0 ) || ( functions[numfun].commute != 0 ) ) {
05402         MesPrint("&%s should be a regular commuting function",s);
05403         if ( type < 0 ) {
05404             if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
05405                 AddFunction(s,0,0,0,0,0,-1,-1);
05406         }
05407         return(1);
05408     }
05409     AR.PolyFun = AC.lPolyFun = numfun+FUNCTION;
05410     AR.PolyFunInv = AC.lPolyFunInv = 0;
05411     AR.PolyFunType = AC.lPolyFunType = 2;
05412     AC.PolyRatFunChanged = 1;
05413     if ( c == 0 ) return(error);
05414     *t = c;
05415     if ( *t == '-' ) { AC.PolyRatFunChanged = 0; t++; }
05416     while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
05417     if ( *t == 0 ) return(error);
05418     if ( *t != '(' ) {
05419         s = t;
05420         t = SkipAName(s);
05421         if ( t == 0 ) goto NumErr;
05422         c = *t; *t = 0;
05423         if ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
05424         || ( functions[numfun].spec != 0 ) || ( functions[numfun].commute != 0 ) ) {
05425             MesPrint("&%s should be a regular commuting function",s);
05426             if ( type < 0 ) {
05427                 if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
05428                     AddFunction(s,0,0,0,0,0,-1,-1);
05429             }
05430             return(1);
05431         }
05432         AR.PolyFunInv = AC.lPolyFunInv = numfun+FUNCTION;
05433         if ( c == 0 ) return(error);
05434         *t = c;
05435         if ( *t == '-' ) { AC.PolyRatFunChanged = 0; t++; }
05436         while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
05437         if ( *t == 0 ) return(error);
05438     }
05439     if ( *t == '(' ) {
05440         t++;
05441         while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
05442     /*
05443         Next we need a keyword like
05444         (divergence,ep)
05445         (expand,ep,maxpow)
05446     */
05447     s = t;
05448     t = SkipAName(s);
05449     if ( t == 0 ) goto NumErr;
05450     c = *t; *t = 0;
05451     if ( ( StrICmp(s,(UBYTE *)"divergence") == 0 )
05452 || ( StrICmp(s,(UBYTE *)"finddivergence") == 0 ) ) {
05453         if ( c != ',' ) {
05454             MesPrint("&Illegal option field in PolyRatFun statement.");
05455             return(1);
05456         }
05457         *t = c;
05458         while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
05459         s = t;
05460         t = SkipAName(s);
05461         if ( t == 0 ) goto NumErr;
05462         c = *t; *t = 0;
05463         if ( ( type = GetName(AC.varnames,s,&AC.lPolyFunVar,WITHAUTO) ) != CSYMBOL ) {
05464             MesPrint("&Illegal symbol %s in option field in PolyRatFun statement.",s);
05465             return(1);
05466         }
05467         *t = c;
05468         while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;

```

```

05469         if ( *t != ')' ) {
05470             MesPrint("&Illegal termination of option in PolyRatFun statement.");
05471             return(1);
05472         }
05473         AR.PolyFunExp = AC.lPolyFunExp = 1;
05474         AR.PolyFunVar = AC.lPolyFunVar;
05475         symbols[AC.lPolyFunVar].minpower = -MAXPOWER;
05476         symbols[AC.lPolyFunVar].maxpower = MAXPOWER;
05477     }
05478     else if ( StrICmp(s, (UBYTE *)"expand") == 0 ) {
05479         WORD x = 0, etype = 2;
05480         if ( c != ',' ) {
05481             MesPrint("&Illegal option field in PolyRatFun statement.");
05482             return(1);
05483         }
05484         *t = c;
05485         while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
05486         s = t;
05487         t = SkipAName(s);
05488         if ( t == 0 ) goto NumErr;
05489         c = *t; *t = 0;
05490         if ( ( type = GetName(AC.varnames, s, &AC.lPolyFunVar, WITHAUTO) ) != CSYMBOL ) {
05491             MesPrint("&Illegal symbol %s in option field in PolyRatFun statement.", s);
05492             return(1);
05493         }
05494         *t = c;
05495         while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
05496         if ( *t > '9' || *t < '0' ) {
05497             MesPrint("&Illegal option field in PolyRatFun statement.");
05498             return(1);
05499         }
05500         while ( *t <= '9' && *t >= '0' ) x = 10*x + *t++ - '0';
05501         while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
05502         if ( *t != ')' ) {
05503             s = t;
05504             t = SkipAName(s);
05505             if ( t == 0 ) goto ParErr;
05506             c = *t; *t = 0;
05507             if ( StrICmp(s, (UBYTE *)"fixed") == 0 ) {
05508                 etype = 3;
05509             }
05510             else if ( StrICmp(s, (UBYTE *)"relative") == 0 ) {
05511                 etype = 2;
05512             }
05513             else {
05514                 MesPrint("&Illegal termination of option in PolyRatFun statement.");
05515                 return(1);
05516             }
05517             *t = c;
05518             while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
05519             if ( *t != ')' ) {
05520                 MesPrint("&Illegal termination of option in PolyRatFun statement.");
05521                 return(1);
05522             }
05523         }
05524         AR.PolyFunExp = AC.lPolyFunExp = etype;
05525         AR.PolyFunVar = AC.lPolyFunVar;
05526         AR.PolyFunPow = AC.lPolyFunPow = x;
05527         symbols[AC.lPolyFunVar].minpower = -MAXPOWER;
05528         symbols[AC.lPolyFunVar].maxpower = MAXPOWER;
05529     }
05530     else {
05531 ParErr: MesPrint("&Illegal option %s in PolyRatFun statement.", s);
05532         return(1);
05533     }
05534     t++;
05535     while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
05536     if ( *t == 0 ) return(error);
05537 }
05538 NumErr:
05539 MesPrint("&PolyRatFun statement needs one or two commuting function(s) for its argument(s)");
05540 return(1);
05541 }
05542
05543 /*
05544 #] CoPolyRatFun :
05545 #[ CoMerge :
05546 */
05547
05548 int CoMerge(UBYTE *inp)
05549 {
05550     UBYTE *s = inp;
05551     int type;
05552     WORD numfunc, option = 0;
05553     if ( tolower(s[0]) == 'o' && tolower(s[1]) == 'n' && tolower(s[2]) == 'c' &&
05554         tolower(s[3]) == 'e' && tolower(s[4]) == ',' ) {
05555         option = 1; s += 5;

```

```

05556     }
05557     else if ( tolower(s[0]) == 'a' && tolower(s[1]) == 'l' && tolower(s[2]) == 'l' &&
05558              tolower(s[3]) == ',' ) {
05559         option = 0; s += 4;
05560     }
05561     if ( *s == '$' ) {
05562         if ( ( type = GetName(AC.dollarnames,s+1,&numfunc,NOAUTO) ) == CDOLLAR )
05563             numfunc = -numfunc;
05564         else {
05565             MesPrint("&%s is undefined",s);
05566             numfunc = AddDollar(s+1,DOLINDEX,&one,1);
05567             return(1);
05568         }
05569     tests: s = SkipAName(s);
05570     if ( *s != 0 ) {
05571         MesPrint("&Merge/shuffle should have a single function or $variable for its argument");
05572         return(1);
05573     }
05574     }
05575     else if ( ( type = GetName(AC.varnames,s,&numfunc,WITHAUTO) ) == CFUNCTION ) {
05576         numfunc += FUNCTION;
05577         goto tests;
05578     }
05579     else if ( type != -1 ) {
05580         if ( type != CDUBIOUS ) {
05581             NameConflict(type,s);
05582             type = MakeDubious(AC.varnames,s,&numfunc);
05583         }
05584         return(1);
05585     }
05586     else {
05587         MesPrint("&%s is not a function",s);
05588         numfunc = AddFunction(s,0,0,0,0,0,-1,-1) + FUNCTION;
05589         return(1);
05590     }
05591     Add4Com(TYPEMERGE,numfunc,option);
05592     return(0);
05593 }
05594
05595 /*
05596  #] CoMerge :
05597  #[ CoStuffle :
05598
05599     Important for future options: The bit, given by 256 (bit 8) is reserved
05600     internally for keeping track of the sign in the number of Stuffle
05601     additions.
05602 */
05603
05604 int CoStuffle(UBYTE *inp)
05605 {
05606     UBYTE *s = inp, *ss, c;
05607     int type;
05608     WORD numfunc, option = 0;
05609     if ( tolower(s[0]) == 'o' && tolower(s[1]) == 'n' && tolower(s[2]) == 'c' &&
05610          tolower(s[3]) == 'e' && tolower(s[4]) == ',' ) {
05611         option = 1; s += 5;
05612     }
05613     else if ( tolower(s[0]) == 'a' && tolower(s[1]) == 'l' && tolower(s[2]) == 'l' &&
05614              tolower(s[3]) == ',' ) {
05615         option = 0; s += 4;
05616     }
05617     ss = SkipAName(s);
05618     c = *ss; *ss = 0;
05619     if ( *s == '$' ) {
05620         if ( ( type = GetName(AC.dollarnames,s+1,&numfunc,NOAUTO) ) == CDOLLAR )
05621             numfunc = -numfunc;
05622         else {
05623             MesPrint("&%s is undefined",s);
05624             numfunc = AddDollar(s+1,DOLINDEX,&one,1);
05625             return(1);
05626         }
05627     tests: *ss = c;
05628     if ( *ss != '+' && *ss != '-' && ss[1] != 0 ) {
05629         MesPrint("&Stuffle should have a single function or $variable for its argument, followed
05630 by either + or -");
05631         return(1);
05632     }
05633     if ( *ss == '-' ) option += 2;
05634     }
05635     else if ( ( type = GetName(AC.varnames,s,&numfunc,WITHAUTO) ) == CFUNCTION ) {
05636         numfunc += FUNCTION;
05637         goto tests;
05638     }
05639     else if ( type != -1 ) {
05640         if ( type != CDUBIOUS ) {
05641             NameConflict(type,s);
05642             type = MakeDubious(AC.varnames,s,&numfunc);

```

```

05642     }
05643     return(1);
05644 }
05645 else {
05646     MesPrint("%s is not a function",s);
05647     numfunc = AddFunction(s,0,0,0,0,0,-1,-1) + FUNCTION;
05648     return(1);
05649 }
05650 Add4Com(TYPESTUFFLE,numfunc,option);
05651 return(0);
05652 }
05653
05654 /*
05655 #] CoStuffle :
05656 #[ CoProcessBucket :
05657 */
05658
05659 int CoProcessBucket(UBYTE *s)
05660 {
05661     LONG x;
05662     while ( *s == ',' || *s == '=' ) s++;
05663     ParseNumber(x,s);
05664     if ( *s && *s != ' ' && *s != '\t' ) {
05665         MesPrint("&Numerical value expected for ProcessBucketSize");
05666         return(1);
05667     }
05668     AC.ProcessBucketSize = x;
05669     return(0);
05670 }
05671
05672 /*
05673 #] CoProcessBucket :
05674 #[ CoThreadBucket :
05675 */
05676
05677 int CoThreadBucket(UBYTE *s)
05678 {
05679     LONG x;
05680     while ( *s == ',' || *s == '=' ) s++;
05681     ParseNumber(x,s);
05682     if ( *s && *s != ' ' && *s != '\t' ) {
05683         MesPrint("&Numerical value expected for ThreadBucketSize");
05684         return(1);
05685     }
05686     if ( x <= 0 ) {
05687         Warning("Negative of zero value not allowed for ThreadBucketSize. Adjusted to 1.");
05688         x = 1;
05689     }
05690     AC.ThreadBucketSize = x;
05691 #ifdef WITHPTHREADS
05692     if ( AS.MultiThreaded ) MakeThreadBuckets(-1,1);
05693 #endif
05694     return(0);
05695 }
05696
05697 /*
05698 #] CoThreadBucket :
05699 #[ DoArgPlode :
05700
05701 Syntax: a list of functions.
05702 If the functions have an argument it must be a function.
05703 In the case f(g) we treat f(g(...)) with g any argument.
05704 (not yet implemented)
05705 */
05706
05707 int DoArgPlode(UBYTE *s, int par)
05708 {
05709     GETIDENTITY
05710     WORD numfunc, type, error = 0, *w, n;
05711     UBYTE *t,c;
05712     int i;
05713     w = AT.WorkPointer;
05714     *w++ = par;
05715     w++;
05716     while ( *s == ',' ) s++;
05717     while ( *s ) {
05718         if ( *s == '$' ) {
05719             MesPrint("&We don't do dollar variables yet in ArgImplode/ArgExplode");
05720             return(1);
05721         }
05722         t = s;
05723         if ( ( s = SkipAName(s) ) == 0 ) return(1);
05724         c = *s; *s = 0;
05725         if ( ( type = GetName(AC.varnames,t,&numfunc,WITHAUTO) ) == CFUNCTION ) {
05726             numfunc += FUNCTION;
05727         }
05728         else if ( type != -1 ) {

```

```

05729         if ( type != CDUBIOUS ) {
05730             NameConflict(type,t);
05731             type = MakeDubious(AC.varnames,t,&numfunc);
05732         }
05733         error = 1;
05734     }
05735     else {
05736         MesPrint("%s is not a function",t);
05737         numfunc = AddFunction(s,0,0,0,0,0,-1,-1) + FUNCTION;
05738         return(1);
05739     }
05740     *s = c;
05741     *w++ = numfunc;
05742     *w++ = FUNHEAD;
05743     #if FUNHEAD > 2
05744     for ( i = 2; i < FUNHEAD; i++ ) *w++ = 0;
05745     #endif
05746     if ( *s && *s != ',' ) {
05747         MesPrint("&Illegal character in ArgImplode/ArgExplode statement: %s",s);
05748         return(1);
05749     }
05750     while ( *s == ',' ) s++;
05751 }
05752 n = w - AT.WorkPointer;
05753 AT.WorkPointer[1] = n;
05754 AddNtoL(n,AT.WorkPointer);
05755 return(error);
05756 }
05757
05758 /*
05759  #] DoArgPlode :
05760  #[ CoArgExplode :
05761  */
05762
05763 int CoArgExplode(UBYTE *s) { return(DoArgPlode(s,TYPEARGEXPLODE)); }
05764
05765 /*
05766  #] CoArgExplode :
05767  #[ CoArgImplode :
05768  */
05769
05770 int CoArgImplode(UBYTE *s) { return(DoArgPlode(s,TYPEARGIMPLODE)); }
05771
05772 /*
05773  #] CoArgImplode :
05774  #[ CoClearTable :
05775  */
05776
05777 int CoClearTable(UBYTE *s)
05778 {
05779     UBYTE c, *t;
05780     int j, type, error = 0;
05781     WORD numfun;
05782     TABLES T, TT;
05783     if ( *s == 0 ) {
05784         MesPrint("&The ClearTable statement needs at least one (table) argument.");
05785         return(1);
05786     }
05787     while ( *s ) {
05788         t = s;
05789         s = SkipAName(s);
05790         c = *s; *s = 0;
05791         if ( ( ( type = GetName(AC.varnames,t,&numfun,WITHAUTO) ) != CFUNCTION )
05792             && type != CDUBIOUS ) {
05793             MesPrint("&%s is not a table",t);
05794             error = 4;
05795             if ( type < 0 ) numfun = AddFunction(t,0,0,0,0,0,-1,-1);
05796             *s = c;
05797             if ( *s == ',' ) s++;
05798             continue;
05799         }
05800     /*
05801         else if ( ( ( T = functions[numfun].tabl ) == 0 )
05802             || ( T->sparse == 0 ) ) goto nofunc;
05803     */
05804     else if ( ( T = functions[numfun].tabl ) == 0 ) goto nofunc;
05805     numfun += FUNCTION;
05806     *s = c;
05807     if ( *s == ',' ) s++;
05808     /*
05809     Now we clear the table.
05810     */
05811     if ( T->sparse ) {
05812         if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
05813         for ( j = 0; j < T->buffersfill; j++ ) { /* was <= */
05814             finishcbuf(T->buffers[j]);
05815         }

```

```

05816         if ( T->buffers ) M_free(T->buffers,"Table buffers");
05817         finishcbuf(T->bufnum);
05818
05819         T->boomlijst = 0;
05820         T->numtree = 0; T->rootnum = 0; T->MaxTreeSize = 0;
05821         T->boomlijst = 0;
05822         T->bufnum = inicbufs();
05823         T->bufferssize = 8;
05824         T->buffers = (WORD *)Malloc1(sizeof(WORD)*T->bufferssize,"Table buffers");
05825         T->buffersfill = 0;
05826         T->buffers[T->buffersfill++] = T->bufnum;
05827
05828         T->totind = 0;           /* At the moment there are this many */
05829         T->reserved = 0;
05830
05831         ClearTableTree(T);
05832
05833         if ( T->spare ) {
05834             if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
05835             T->tablepointers = 0;
05836             TT = T->spare;
05837             if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
05838             for ( j = 0; j < TT->buffersfill; j++ ) {
05839                 finishcbuf(TT->buffers[j]);
05840             }
05841             if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
05842             if ( TT->buffers ) M_free(TT->buffers,"Table buffers");
05843             if ( TT->mm ) M_free(TT->mm,"tableminmax");
05844             if ( TT->flags ) M_free(TT->flags,"tableflags");
05845             M_free(TT,"table");
05846             SpareTable(T);
05847         }
05848     }
05849     else EmptyTable(T);
05850 }
05851 return(error);
05852 }
05853
05854 /*
05855 #] CoClearTable :
05856 #[ CoDenominators :
05857 */
05858
05859 int CoDenominators(UBYTE *s)
05860 {
05861     WORD numfun;
05862     int type;
05863     UBYTE *t = SkipAName(s), *t1;
05864     if ( t == 0 ) goto syntaxerror;
05865     t1 = t; while ( *t1 == ',' || *t1 == ' ' || *t1 == '\t' ) t1++;
05866     if ( *t1 ) goto syntaxerror;
05867     *t = 0;
05868     if ( ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
05869         || ( functions[numfun].spec != 0 ) ) {
05870         if ( type < 0 ) {
05871             if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
05872                 AddFunction(s,0,0,0,0,0,-1,-1);
05873         }
05874         goto syntaxerror;
05875     }
05876     Add3Com(TYPEDENOMINATORS,numfun+FUNCTION);
05877     return(0);
05878 syntaxerror:
05879     MesPrint("&Denominators statement needs one regular function for its argument");
05880     return(1);
05881 }
05882
05883 /*
05884 #] CoDenominators :
05885 #[ CoDropCoefficient :
05886 */
05887
05888 int CoDropCoefficient(UBYTE *s)
05889 {
05890     if ( *s == 0 ) {
05891         Add2Com(TYPEDROPCOEFFICIENT);
05892         return(0);
05893     }
05894     MesPrint("&Illegal argument in DropCoefficient statement: '%s'",s);
05895     return(1);
05896 }
05897 /*
05898 #] CoDropCoefficient :
05899 #[ CoDropSymbols :
05900 */
05901
05902 int CoDropSymbols(UBYTE *s)

```

```

05903 {
05904     if ( *s == 0 ) {
05905         Add2Com(TYPEDROPSYMBOLS)
05906         return(0);
05907     }
05908     MesPrint("Illegal argument in DropSymbols statement: '%s'",s);
05909     return(1);
05910 }
05911 /*
05912 #] CoDropSymbols :
05913 #[ CoToPolynomial :
05914
05915 Converts the current term as much as possible to symbols.
05916 Keeps a list of all objects converted to symbols in AM.sbufnum.
05917 Note that this cannot be executed in parallel because we have only
05918 a single compiler buffer for this. Hence we switch on the noparallel
05919 module option.
05920
05921 Option(s):
05922     OnlyFunctions [,name1][,name2][,...,namem];
05923 */
05924
05925 int CoToPolynomial(UBYTE *inp)
05926 {
05927     int error = 0;
05928     while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
05929     if ( ( AC.topolynomialflag & ~TOPOLYNOMIALFLAG ) != 0 ) {
05930         MesPrint("&ToPolynomial statement and FactArg statement are not allowed in the same module");
05931         return(1);
05932     }
05933     if ( AO.OptimizeResult.code != NULL ) {
05934         MesPrint("&Using ToPolynomial statement when there are still optimization results active.");
05935         MesPrint("&Please use #ClearOptimize instruction first.");
05936         MesPrint("&This will loose the optimized expression.");
05937         return(1);
05938     }
05939     if ( *inp == 0 ) {
05940         Add3Com(TYPETOPOLYNOMIAL,DOALL)
05941     }
05942     else {
05943         int numargs = 0;
05944         WORD *funnums = 0, type, num;
05945         UBYTE *s, c;
05946         s = SkipAName(inp);
05947         if ( s == 0 ) return(1);
05948         c = *s; *s = 0;
05949         if ( StrICmp(inp,(UBYTE *)"onlyfunctions") ) {
05950             MesPrint("&Illegal option %s in ToPolynomial statement",inp);
05951             *s = c;
05952             return(1);
05953         }
05954         *s = c;
05955         inp = s;
05956         while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
05957         s = inp;
05958         while ( *s ) s++;
05959     /*
05960     Get definitely enough space for the numbers of the functions
05961     */
05962     funnums = (WORD *)Malloc1(((LONG)(s-inp)+3)*sizeof(WORD),"ToPolynomial");
05963     while ( *inp ) {
05964         s = SkipAName(inp);
05965         if ( s == 0 ) return(1);
05966         c = *s; *s = 0;
05967         type = GetName(AC.varnames,inp,&num,WITHAUTO);
05968         if ( type != CFUNCTION ) {
05969             MesPrint("&%s is not a function in ToPolynomial statement",inp);
05970             error = 1;
05971         }
05972         funnums[3+numargs++] = num+FUNCTION;
05973         *s = c;
05974         inp = s;
05975         while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
05976     }
05977     funnums[0] = TYPETOPOLYNOMIAL;
05978     funnums[1] = numargs+3;
05979     funnums[2] = ONLYFUNCTIONS;
05980
05981     AddNtoL(numargs+3,funnums);
05982     if ( funnums ) M_free(funnums,"ToPolynomial");
05983 }
05984 AC.topolynomialflag |= TOPOLYNOMIALFLAG;
05985 #ifdef WITHMPI
05986 /* In ParFORM, ToPolynomial has to be executed on the master. */
05987 AC.mparallelflag |= NOPARALLEL_CONVPOLY;
05988 #endif
05989 return(error);

```

```

05990 }
05991
05992 /*
05993  #] CoToPolynomial :
05994  #[ CoFromPolynomial :
05995
05996  Converts the current term as much as possible back from extra symbols
05997  to their original values. Does not look inside functions.
05998 */
05999
06000 int CoFromPolynomial(UBYTE *inp)
06001 {
06002     while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
06003     if ( *inp == 0 ) {
06004         if ( AO.OptimizeResult.code != NULL ) {
06005             MesPrint("&Using FromPolynomial statement when there are still optimization results
active.");
06006             MesPrint("&Please use #ClearOptimize instruction first.");
06007             MesPrint("&This will loose the optimized expression.");
06008             return(1);
06009         }
06010         Add2Com(TYPEFROMPOLYNOMIAL)
06011         return(0);
06012     }
06013     MesPrint("&Illegal argument in FromPolynomial statement: '%s'",inp);
06014     return(1);
06015 }
06016
06017 /*
06018  #] CoFromPolynomial :
06019  #[ CoArgToExtraSymbol :
06020
06021  Converts the specified function arguments into extra symbols.
06022
06023  Syntax: ArgToExtraSymbol [ToNumber] [<argument specifications>]
06024 */
06025
06026 int CoArgToExtraSymbol(UBYTE *s)
06027 {
06028     CBUF *C = cbuf + AC.cbunum;
06029     WORD *lhs;
06030
06031     /* TODO: resolve interference with rational arithmetic. (#138) */
06032     if ( ( AC.topolynomialflag & ~TOPOLYNOMIALFLAG ) != 0 ) {
06033         MesPrint("&ArgToExtraSymbol statement and FactArg statement are not allowed in the same
module");
06034         return(1);
06035     }
06036     if ( AO.OptimizeResult.code != NULL ) {
06037         MesPrint("&Using ArgToExtraSymbol statement when there are still optimization results
active.");
06038         MesPrint("&Please use #ClearOptimize instruction first.");
06039         MesPrint("&This will loose the optimized expression.");
06040         return(1);
06041     }
06042
06043     SkipSpaces(&s);
06044     int tonumber = ConsumeOption(&s, "tonumber");
06045
06046     int ret = DoArgument(s, TYPEARGTOEXTRASymbol);
06047     if ( ret ) return(ret);
06048
06049     /*
06050     * The "scale" parameter is unused. Instead, we put the "tonumber"
06051     * parameter.
06052     */
06053     lhs = C->lhs[C->numlhs];
06054     if ( lhs[4] != 1 ) {
06055         Warning("scale parameter (^n) is ignored in ArgToExtraSymbol");
06056     }
06057     lhs[4] = tonumber;
06058
06059     AC.topolynomialflag |= TOPOLYNOMIALFLAG; /* This flag is also used in ParFORM. */
06060 #ifdef WITHMPI
06061     /*
06062     * In ParFORM, the conversion to extra symbols has to be performed on
06063     * the master.
06064     */
06065     AC.mparallelfldflag |= NOPARALLEL_CONVPOLY;
06066 #endif
06067     return(0);
06068 }
06069
06070
06071 /*
06072  #] CoArgToExtraSymbol :
06073  #[ CoExtraSymbols :

```



```

06074 */
06075
06076 int CoExtraSymbols(UBYTE *inp)
06077 {
06078     UBYTE *arg1, *arg2, c, *s;
06079     WORD i, j, type, number;
06080     while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
06081     if ( FG.cTable[*inp] != 0 ) {
06082         MesPrint("&Illegal argument in ExtraSymbols statement: '%s'",inp);
06083         return(1);
06084     }
06085     arg1 = inp;
06086     while ( FG.cTable[*inp] == 0 ) inp++;
06087     c = *inp; *inp = 0;
06088     if ( ( StrICmp(arg1, (UBYTE *) "array") == 0 )
06089         || ( StrICmp(arg1, (UBYTE *) "vector") == 0 ) ) {
06090         AC.extrasymbols = 1;
06091     }
06092     else if ( StrICmp(arg1, (UBYTE *) "underscore") == 0 ) {
06093         AC.extrasymbols = 0;
06094     }
06095     /*
06096     else if ( StrICmp(arg1, (UBYTE *) "nothing") == 0 ) {
06097         AC.extrasymbols = 2;
06098     }
06099     */
06100     else {
06101         MesPrint("&Illegal keyword in ExtraSymbols statement: '%s'",arg1);
06102         return(1);
06103     }
06104     *inp = c;
06105     while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
06106     if ( FG.cTable[*inp] != 0 ) {
06107         MesPrint("&Illegal argument in ExtraSymbols statement: '%s'",inp);
06108         return(1);
06109     }
06110     arg2 = inp;
06111     while ( FG.cTable[*inp] <= 1 ) inp++;
06112     if ( *inp != 0 ) {
06113         MesPrint("&Illegal end of ExtraSymbols statement: '%s'",inp);
06114         return(1);
06115     }
06116     /*
06117         Now check whether this object has been declared already.
06118         That would not be allowed.
06119     */
06120     if ( AC.extrasymbols == 1 ) {
06121         type = GetName(AC.varnames,arg2,&number,NOAUTO);
06122         if ( type != NAMENOTFOUND ) {
06123             MesPrint("&ExtraSymbols statement: '%s' has already been declared before",arg2);
06124             return(1);
06125         }
06126     }
06127     else if ( AC.extrasymbols == 0 ) {
06128         if ( *arg2 == 'N' ) {
06129             s = arg2+1;
06130             while ( FG.cTable[*s] == 1 ) s++;
06131             if ( *s == 0 ) {
06132                 MesPrint("&ExtraSymbols statement: '%s' creates conflicts with summed indices",arg2);
06133                 return(1);
06134             }
06135         }
06136     }
06137     if ( AC.extrasym ) { M_free(AC.extrasym,"extrasym"); AC.extrasym = 0; }
06138     i = inp - arg2 + 1;
06139     AC.extrasym = (UBYTE *) Malloc1(i*sizeof(UBYTE),"extrasym");
06140     for ( j = 0; j < i; j++ ) AC.extrasym[j] = arg2[j];
06141     return(0);
06142 }
06143
06144 /*
06145 #] CoExtraSymbols :
06146 #[ GetIfDollarFactor :
06147 */
06148
06149 WORD *GetIfDollarFactor(UBYTE **inp, WORD *w)
06150 {
06151     LONG x;
06152     WORD number;
06153     UBYTE *name, c, *s;
06154     s = *inp;
06155     if ( FG.cTable[*s] == 1 ) {
06156         x = 0;
06157         while ( FG.cTable[*s] == 1 ) {
06158             x = 10*x + *s++ - '0';
06159             if ( x >= MAXPOSITIVE ) {
06160                 MesPrint("&Value in dollar factor too large");

```

```

06161         while ( FG.cTable[*s] == 1 ) s++;
06162         *inp = s;
06163         return(0);
06164     }
06165 }
06166 *w++ = IFDOLLAREXTRA;
06167 *w++ = 3;
06168 *w++ = -x-1;
06169 *inp = s;
06170 return(w);
06171 }
06172 if ( *s != '$' ) {
06173     MesPrint("&Factor indicator for $-variable should be a number or a $-variable.");
06174     return(0);
06175 }
06176 s++; name = s;
06177 while ( FG.cTable[*s] < 2 ) s++;
06178 c = *s; *s = 0;
06179 if ( GetName(AC.dollarnames,name,&number,NOAUTO) == NAMENOTFOUND ) {
06180     MesPrint("&dollar in if statement should have been defined previously");
06181     return(0);
06182 }
06183 *s = c;
06184 *w++ = IFDOLLAREXTRA;
06185 *w++ = 3;
06186 *w++ = number;
06187 if ( c == '[' ) {
06188     s++;
06189     *inp = s;
06190     if ( ( w = GetIfDollarFactor(inp,w) ) == 0 ) return(0);
06191     s = *inp;
06192     if ( *s != ']' ) {
06193         MesPrint("&unmatched [] in $ in if statement");
06194         return(0);
06195     }
06196     s++;
06197     *inp = s;
06198 }
06199 return(w);
06200 }
06201
06202 /*
06203 #] GetIfDollarFactor :
06204 #[ GetDoParam :
06205 */
06206
06207 UBYTE *GetDoParam(UBYTE *inp, WORD **wp, int par)
06208 {
06209     LONG x;
06210     WORD number;
06211     UBYTE *name, c;
06212     if ( FG.cTable[*inp] == 1 ) {
06213         x = 0;
06214         while ( *inp >= '0' && *inp <= '9' ) {
06215             x = 10*x + *inp++ - '0';
06216             if ( x > MAXPOSITIVE ) {
06217                 if ( par == -1 ) {
06218                     MesPrint("&Value in dollar factor too large");
06219                 }
06220                 else {
06221                     MesPrint("&Value in do loop boundaries too large");
06222                 }
06223                 while ( FG.cTable[*inp] == 1 ) inp++;
06224                 return(0);
06225             }
06226         }
06227         if ( par > 0 ) {
06228             *(*wp)++ = SNUMBER;
06229             *(*wp)++ = (WORD)x;
06230         }
06231         else {
06232             *(*wp)++ = DOLLAREXPR2;
06233             *(*wp)++ = -((WORD)x)-1;
06234         }
06235         return(inp);
06236     }
06237     if ( *inp != '$' ) {
06238         return(0);
06239     }
06240     inp++; name = inp;
06241     while ( FG.cTable[*inp] < 2 ) inp++;
06242     c = *inp; *inp = 0;
06243     if ( GetName(AC.dollarnames,name,&number,NOAUTO) == NAMENOTFOUND ) {
06244         if ( par == -1 ) {
06245             MesPrint("&dollar in print statement should have been defined previously");
06246         }
06247         else {

```

```

06248         MesPrint("&dollar in do loop boundaries should have been defined previously");
06249     }
06250     return(0);
06251 }
06252 *inp = c;
06253 if ( par > 0 ) {
06254     *(&wp)++ = DOLLAREXPRESSION;
06255     *(&wp)++ = number;
06256 }
06257 else {
06258     *(&wp)++ = DOLLAREXPR2;
06259     *(&wp)++ = number;
06260 }
06261 if ( c == '[' ) {
06262     inp++;
06263     inp = GetDoParam(inp,wp,0);
06264     if ( inp == 0 ) return(0);
06265     if ( *inp != ']' ) {
06266         if ( par == -1 ) {
06267             MesPrint("&unmatched [] in $ in print statement");
06268         }
06269         else {
06270             MesPrint("&unmatched [] in do loop boundaries");
06271         }
06272         return(0);
06273     }
06274     inp++;
06275 }
06276 return(inp);
06277 }
06278
06279 /*
06280 #] GetDoParam :
06281 #[ CoDo :
06282 */
06283
06284 int CoDo(UBYTE *inp)
06285 {
06286     GETIDENTITY
06287     CBUF *C = cbuf+AC.cbunum;
06288     WORD *w, numparam;
06289     int error = 0, i;
06290     UBYTE *name, c;
06291     if ( AC.doloopstack == 0 ) {
06292         AC.doloopstacksize = 20;
06293         AC.doloopstack = (WORD *)Malloc1(AC.doloopstacksize*2*sizeof(WORD),"doloop stack");
06294         AC.doloopnest = AC.doloopstack + AC.doloopstacksize;
06295     }
06296     if ( AC.dolooplevel >= AC.doloopstacksize ) {
06297         WORD *newstack, *newnest, newsize;
06298         newsize = AC.doloopstacksize * 2;
06299         newstack = (WORD *)Malloc1(newsize*2*sizeof(WORD),"doloop stack");
06300         newnest = newstack + newsize;
06301         for ( i = 0; i < newsize; i++ ) {
06302             newstack[i] = AC.doloopstack[i];
06303             newnest[i] = AC.doloopnest[i];
06304         }
06305         M_free(AC.doloopstack,"doloop stack");
06306         AC.doloopstack = newstack;
06307         AC.doloopnest = newnest;
06308         AC.doloopstacksize = newsize;
06309     }
06310     AC.doloopnest[AC.dolooplevel] = NestingChecksum();
06311
06312     w = AT.WorkPointer;
06313     *w++ = TYPEDOLOOP;
06314     w++; /* Space for the length of the statement */
06315
06316     /* Now the $loopvariable
06317     */
06318     while ( *inp == ',' ) inp++;
06319     if ( *inp != '$' ) {
06320         error = 1;
06321         MesPrint("&do loop parameter should be a dollar variable");
06322     }
06323     else {
06324         inp++;
06325         name = inp;
06326         if ( FG.cTable[*inp] != 0 ) {
06327             error = 1;
06328             MesPrint("&illegal name for do loop parameter");
06329         }
06330         while ( FG.cTable[*inp] < 2 ) inp++;
06331         c = *inp; *inp = 0;
06332         if ( GetName(AC.dollarnames,name,&numparam,NOAUTO) == NAMENOTFOUND ) {
06333             numparam = AddDollar(name,DOLUNDEFINED,0,0);
06334         }

```

```

06335         *w++ = numparam;
06336         *inp = c;
06337         AddPotModdollar(numparam);
06338     }
06339     w++; /* space for the level of the enddo statement */
06340     while ( *inp == ',' ) inp++;
06341     if ( *inp != '=' ) goto IllSyntax;
06342     inp++;
06343     while ( *inp == ',' ) inp++;
06344 /*
06345     The start value
06346 */
06347     inp = GetDoParam(inp,&w,1);
06348     if ( inp == 0 || *inp != ',' ) goto IllSyntax;
06349     while ( *inp == ',' ) inp++;
06350 /*
06351     The end value
06352 */
06353     inp = GetDoParam(inp,&w,1);
06354     if ( inp == 0 || ( *inp != 0 && *inp != ',' ) ) goto IllSyntax;
06355 /*
06356     The increment value
06357 */
06358     if ( *inp != ',' ) {
06359         if ( *inp == 0 ) { *w++ = SNUMBER; *w++ = 1; }
06360         else goto IllSyntax;
06361     }
06362     else {
06363         while ( *inp == ',' ) inp++;
06364         inp = GetDoParam(inp,&w,1);
06365     }
06366     if ( inp == 0 || *inp != 0 ) goto IllSyntax;
06367     *w = 0;
06368     AT.WorkPointer[1] = w - AT.WorkPointer;
06369 /*
06370     Put away and set information for placing enddo information.
06371 */
06372     AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
06373     AC.doloopstack[AC.dolooplevel++] = C->numlhs;
06374
06375     return(error);
06376
06377 IllSyntax:
06378     MesPrint("&Illegal syntax for do statement");
06379     return(1);
06380 }
06381
06382 /*
06383 #] CoDo :
06384 #[ CoEndDo :
06385 */
06386
06387 int CoEndDo(UBYTE *inp)
06388 {
06389     CBUF *C = cbuf+AC.cbufnum;
06390     WORD scratch[3];
06391     while ( *inp == ',' ) inp++;
06392     if ( *inp ) {
06393         MesPrint("&Illegal syntax for EndDo statement");
06394         return(1);
06395     }
06396     if ( AC.dolooplevel <= 0 ) {
06397         MesPrint("&EndDo without corresponding Do statement");
06398         return(1);
06399     }
06400     AC.dolooplevel--;
06401     scratch[0] = TYPEENDDOLOOP;
06402     scratch[1] = 3;
06403     scratch[2] = AC.doloopstack[AC.dolooplevel];
06404     AddNtoL(3,scratch);
06405     cbuf[AC.cbufnum].lhs[AC.doloopstack[AC.dolooplevel]][3] = C->numlhs;
06406     if ( AC.doloopnest[AC.dolooplevel] != NestingChecksum() ) {
06407         MesNesting();
06408         return(1);
06409     }
06410     return(0);
06411 }
06412
06413 /*
06414 #] CoEndDo :
06415 #[ CoFactDollar :
06416 */
06417
06418 int CoFactDollar(UBYTE *inp)
06419 {
06420     WORD numdollar;
06421     if ( *inp == '$' ) {

```

```

06422         if ( GetName(AC.dollarnames,inp+1,&numdollar,NOAUTO) != CDOLLAR ) {
06423             MesPrint("%s is undefined",inp);
06424             numdollar = AddDollar(inp+1,DOLINDEX,&one,1);
06425             return(1);
06426         }
06427         inp = SkipAName(inp+1);
06428         if ( *inp != 0 ) {
06429             MesPrint("&FactDollar should have a single $variable for its argument");
06430             return(1);
06431         }
06432         AddPotModdollar(numdollar);
06433     }
06434     else {
06435         MesPrint("%s is not a $-variable",inp);
06436         return(1);
06437     }
06438     Add3Com(TYPEFACTOR,numdollar);
06439     return(0);
06440 }
06441
06442 /*
06443 #] CoFactDollar :
06444 #[ CoFactorize :
06445 */
06446
06447 int CoFactorize(UBYTE *s) { return(DoFactorize(s,1)); }
06448
06449 /*
06450 #] CoFactorize :
06451 #[ CoNFactorize :
06452 */
06453
06454 int CoNFactorize(UBYTE *s) { return(DoFactorize(s,0)); }
06455
06456 /*
06457 #] CoNFactorize :
06458 #[ CoUnFactorize :
06459 */
06460
06461 int CoUnFactorize(UBYTE *s) { return(DoFactorize(s,3)); }
06462
06463 /*
06464 #] CoUnFactorize :
06465 #[ CoNUnFactorize :
06466 */
06467
06468 int CoNUnFactorize(UBYTE *s) { return(DoFactorize(s,2)); }
06469
06470 /*
06471 #] CoNUnFactorize :
06472 #[ DoFactorize :
06473 */
06474
06475 int DoFactorize(UBYTE *s,int par)
06476 {
06477     EXPRESSIONS e;
06478     WORD i;
06479     WORD number;
06480     UBYTE *t, c;
06481     int error = 0, keepzeroflag = 0;
06482     if ( *s == '(' ) {
06483         s++;
06484         while ( *s != ')' && *s ) {
06485             if ( FG.cTable[*s] == 0 ) {
06486                 t = s; while ( FG.cTable[*s] == 0 ) s++;
06487                 c = *s; *s = 0;
06488                 if ( StrICmp((UBYTE *)"keepzero",t) == 0 ) {
06489                     keepzeroflag = 1;
06490                 }
06491                 else {
06492                     MesPrint("&Illegal option in [N][Un]Factorize statement: %s",t);
06493                     error = 1;
06494                 }
06495                 *s = c;
06496             }
06497             while ( *s == ',' ) s++;
06498             if ( *s && *s != ')' && FG.cTable[*s] != 0 ) {
06499                 MesPrint("&Illegal character in option field of [N][Un]Factorize statement");
06500                 error = 1;
06501                 return(error);
06502             }
06503         }
06504         if ( *s ) s++;
06505         while ( *s == ',' || *s == ' ' ) s++;
06506     }
06507     if ( *s == 0 ) {
06508         for ( i = NumExpressions-1; i >= 0; i-- ) {

```

```

06509         e = Expressions+i;
06510         if ( e->replace >= 0 ) {
06511             e = Expressions + e->replace;
06512         }
06513         if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
06514             || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
06515             || e->status == INTOHIDELEXPRESSION || e->status == INTOHIDEGEXPRESSION
06516         ) {
06517             switch ( par ) {
06518                 case 0:
06519                     e->vflags &= ~TOBEFACTORED;
06520                     break;
06521                 case 1:
06522                     e->vflags |= TOBEFACTORED;
06523                     e->vflags &= ~TOBEUNFACTORED;
06524                     break;
06525                 case 2:
06526                     e->vflags &= ~TOBEUNFACTORED;
06527                     break;
06528                 case 3:
06529                     e->vflags |= TOBEUNFACTORED;
06530                     e->vflags &= ~TOBEFACTORED;
06531                     break;
06532             }
06533         }
06534         if ( ( e->vflags & TOBEFACTORED ) != 0 ) {
06535             if ( keepzeroflag ) e->vflags |= KEEPZERO;
06536             else e->vflags &= ~KEEPZERO;
06537         }
06538         else e->vflags &= ~KEEPZERO;
06539     }
06540 }
06541 else {
06542     for(;;) { /* Look for a (comma separated) list of variables */
06543         while ( *s == ',' ) s++;
06544         if ( *s == 0 ) break;
06545         if ( *s == '[' || FG.cTable[*s] == 0 ) {
06546             t = s;
06547             if ( ( s = SkipAName(s) ) == 0 ) {
06548                 MesPrint("&Improper name for an expression: '%s'",t);
06549                 return(1);
06550             }
06551             c = *s; *s = 0;
06552             if ( GetName(AC.exprnames,t,&number,NOAUTO) == CEXPRESSION ) {
06553                 e = Expressions+number;
06554                 if ( e->replace >= 0 ) {
06555                     e = Expressions + e->replace;
06556                 }
06557                 if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
06558                     || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
06559                     || e->status == INTOHIDELEXPRESSION || e->status == INTOHIDEGEXPRESSION
06560                 ) {
06561                     switch ( par ) {
06562                         case 0:
06563                             e->vflags &= ~TOBEFACTORED;
06564                             break;
06565                         case 1:
06566                             e->vflags |= TOBEFACTORED;
06567                             e->vflags &= ~TOBEUNFACTORED;
06568                             break;
06569                         case 2:
06570                             e->vflags &= ~TOBEUNFACTORED;
06571                             break;
06572                         case 3:
06573                             e->vflags |= TOBEUNFACTORED;
06574                             e->vflags &= ~TOBEFACTORED;
06575                             break;
06576                     }
06577                 }
06578                 if ( ( e->vflags & TOBEFACTORED ) != 0 ) {
06579                     if ( keepzeroflag ) e->vflags |= KEEPZERO;
06580                     else e->vflags &= ~KEEPZERO;
06581                 }
06582                 else e->vflags &= ~KEEPZERO;
06583             }
06584             else if ( GetName(AC.varnames,t,&number,NOAUTO) != NAMENOTFOUND ) {
06585                 MesPrint("&%s is not an expression",t);
06586                 error = 1;
06587             }
06588             *s = c;
06589         }
06590     }
06591     else {
06592         MesPrint("&Illegal object in (N)Factorize statement");
06593         error = 1;
06594         while ( *s && *s != ',' ) s++;
06595         if ( *s == 0 ) break;
06596     }

```

```

06596     }
06597
06598     }
06599     return(error);
06600 }
06601
06602 /*
06603  #] DoFactorize :
06604  #[ CoOptimizeOption :
06605
06606 */
06607
06608 int CoOptimizeOption(UBYTE *s)
06609 {
06610     UBYTE *name, *t1, *t2, c1, c2, *value, *u;
06611     int error = 0, x;
06612     double d;
06613     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
06614     while ( *s ) {
06615         name = s; while ( FG.cTable[*s] == 0 ) s++;
06616         t1 = s; c1 = *t1;
06617         while ( *s == ' ' || *s == '\t' ) s++;
06618         if ( *s != '=' ) {
06619             correctuse:
06620                 MesPrint("&Correct use in Format,Optimize statement is Optionname=value");
06621                 error = 1;
06622                 while ( *s == ' ' || *s == ',' || *s == '\t' || *s == '=' ) s++;
06623                 *t1 = c1;
06624                 continue;
06625             }
06626             *t1 = 0;
06627             s++;
06628             while ( *s == ' ' || *s == '\t' ) s++;
06629             if ( *s == 0 ) goto correctuse;
06630             value = s;
06631             while ( FG.cTable[*s] <= 1 || *s== '.' || *s=='*' || *s == '(' || *s == ')' ) {
06632                 if ( *s == '(' ) { SKIPBRA4(s) }
06633                 s++;
06634             }
06635             t2 = s; c2 = *t2;
06636             while ( *s == ' ' || *s == '\t' ) s++;
06637             if ( *s && *s != ',' ) goto correctuse;
06638             if ( *s ) {
06639                 s++;
06640                 while ( *s == ' ' || *s == '\t' ) s++;
06641             }
06642             *t2 = 0;
06643         }
06644         Now we have name=value with name and value zero terminated strings.
06645     }
06646     if ( StrICmp(name,(UBYTE *)"horner") == 0 ) {
06647         if ( StrICmp(value,(UBYTE *)"occurrence") == 0 ) {
06648             AO.Optimize.horner = O_OCCURRENCE;
06649         }
06650         else if ( StrICmp(value,(UBYTE *)"mcts") == 0 ) {
06651             AO.Optimize.horner = O_MCTS;
06652         }
06653         else if ( StrICmp(value,(UBYTE *)"sa") == 0 ) {
06654             AO.Optimize.horner = O_SIMULATED_ANNEALING;
06655         }
06656         else {
06657             AO.Optimize.horner = -1;
06658             MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
06659             error = 1;
06660         }
06661     }
06662     else if ( StrICmp(name,(UBYTE *)"hornerdirection") == 0 ) {
06663         if ( StrICmp(value,(UBYTE *)"forward") == 0 ) {
06664             AO.Optimize.hornerdirection = O_FORWARD;
06665         }
06666         else if ( StrICmp(value,(UBYTE *)"backward") == 0 ) {
06667             AO.Optimize.hornerdirection = O_BACKWARD;
06668         }
06669         else if ( StrICmp(value,(UBYTE *)"forwardorbackward") == 0 ) {
06670             AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
06671         }
06672         else if ( StrICmp(value,(UBYTE *)"forwardandbackward") == 0 ) {
06673             AO.Optimize.hornerdirection = O_FORWARDANDBACKWARD;
06674         }
06675         else {
06676             AO.Optimize.method = -1;
06677             MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
06678             error = 1;
06679         }
06680     }
06681     else if ( StrICmp(name,(UBYTE *)"method") == 0 ) {
06682         if ( StrICmp(value,(UBYTE *)"none") == 0 ) {

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```

06683         AO.Optimize.method = O_NONE;
06684     }
06685     else if ( StrICmp(value, (UBYTE *) "cse") == 0 ) {
06686         AO.Optimize.method = O_CSE;
06687     }
06688     else if ( StrICmp(value, (UBYTE *) "csegreedy") == 0 ) {
06689         AO.Optimize.method = O_CSEGREEDY;
06690     }
06691     else if ( StrICmp(value, (UBYTE *) "greedy") == 0 ) {
06692         AO.Optimize.method = O_GREEDY;
06693     }
06694     else {
06695         AO.Optimize.method = -1;
06696         MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
06697         error = 1;
06698     }
06699 }
06700 else if ( StrICmp(name, (UBYTE *) "timelimit") == 0 ) {
06701     x = 0;
06702     u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06703     if ( *u != 0 ) {
06704         MesPrint("&Option TimeLimit in Format,Optimize statement should be a positive number:
06705 %s",value);
06706         AO.Optimize.mctstimelimit = 0;
06707         AO.Optimize.greedytimelimit = 0;
06708         error = 1;
06709     }
06710     else {
06711         AO.Optimize.mctstimelimit = x/2;
06712         AO.Optimize.greedytimelimit = x/2;
06713     }
06714 }
06715 else if ( StrICmp(name, (UBYTE *) "mctstimelimit") == 0 ) {
06716     x = 0;
06717     u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06718     if ( *u != 0 ) {
06719         MesPrint("&Option MCTSTimeLimit in Format,Optimize statement should be a positive
06720 number: %s",value);
06721         AO.Optimize.mctstimelimit = 0;
06722         error = 1;
06723     }
06724     else {
06725         AO.Optimize.mctstimelimit = x;
06726     }
06727 }
06728 else if ( StrICmp(name, (UBYTE *) "mctsnmexpand") == 0 ) {
06729     int y;
06730     x = 0;
06731     u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06732     if ( *u == '*' || *u == 'x' || *u == 'X' ) {
06733         u++; y = x;
06734         x = 0;
06735         while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06736     }
06737     else { y = 1; }
06738     if ( *u != 0 ) {
06739         MesPrint("&Option MCTSNmExpand in Format,Optimize statement should be a positive
06740 number: %s",value);
06741         AO.Optimize.mctsnmexpand= 0;
06742         AO.Optimize.mctsnmrepeat= 1;
06743         error = 1;
06744     }
06745     else {
06746         AO.Optimize.mctsnmexpand= x;
06747         AO.Optimize.mctsnmrepeat= y;
06748     }
06749 }
06750 else if ( StrICmp(name, (UBYTE *) "mctsnmrepeat") == 0 ) {
06751     x = 0;
06752     u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06753     if ( *u != 0 ) {
06754         MesPrint("&Option MCTSNmExpand in Format,Optimize statement should be a positive
06755 number: %s",value);
06756         AO.Optimize.mctsnmrepeat= 1;
06757         error = 1;
06758     }
06759     else {
06760         AO.Optimize.mctsnmrepeat= x;
06761     }
06762 }
06763 else if ( StrICmp(name, (UBYTE *) "mctsnmkeep") == 0 ) {
06764     x = 0;
06765     u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06766     if ( *u != 0 ) {
06767         MesPrint("&Option MCTSNmKeep in Format,Optimize statement should be a positive
06768 number: %s",value);
06769         AO.Optimize.mctsnmkeep= 0;

```



```

06765         error = 1;
06766     }
06767     else {
06768         AO.Optimize.mctsnumkeep= x;
06769     }
06770 }
06771 else if ( StrICmp(name, (UBYTE *)"mctsconstant") == 0 ) {
06772     d = 0;
06773     if ( sscanf ((char*)value, "%lf", &d) != 1 ) {
06774         MesPrint("&Option MCTSConstant in Format,Optimize statement should be a positive
number: %s",value);
06775         AO.Optimize.mctsconstant.fval = 0;
06776         error = 1;
06777     }
06778     else {
06779         AO.Optimize.mctsconstant.fval = d;
06780     }
06781 }
06782 else if ( StrICmp(name, (UBYTE *)"greedytimelimit") == 0 ) {
06783     x = 0;
06784     u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06785     if ( *u != 0 ) {
06786         MesPrint("&Option GreedyTimeLimit in Format,Optimize statement should be a positive
number: %s",value);
06787         AO.Optimize.greedytimelimit = 0;
06788         error = 1;
06789     }
06790     else {
06791         AO.Optimize.greedytimelimit = x;
06792     }
06793 }
06794 else if ( StrICmp(name, (UBYTE *)"greedyminnum") == 0 ) {
06795     x = 0;
06796     u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06797     if ( *u != 0 ) {
06798         MesPrint("&Option GreedyMinNum in Format,Optimize statement should be a positive
number: %s",value);
06799         AO.Optimize.greedyminnum= 0;
06800         error = 1;
06801     }
06802     else {
06803         AO.Optimize.greedyminnum= x;
06804     }
06805 }
06806 else if ( StrICmp(name, (UBYTE *)"greedymaxperc") == 0 ) {
06807     x = 0;
06808     u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06809     if ( *u != 0 ) {
06810         MesPrint("&Option GreedyMaxPerc in Format,Optimize statement should be a positive
number: %s",value);
06811         AO.Optimize.greedymaxperc= 0;
06812         error = 1;
06813     }
06814     else {
06815         AO.Optimize.greedymaxperc= x;
06816     }
06817 }
06818 else if ( StrICmp(name, (UBYTE *)"stats") == 0 ) {
06819     if ( StrICmp(value, (UBYTE *)"on") == 0 ) {
06820         AO.Optimize.printstats = 1;
06821     }
06822     else if ( StrICmp(value, (UBYTE *)"off") == 0 ) {
06823         AO.Optimize.printstats = 0;
06824     }
06825     else {
06826         AO.Optimize.printstats = 0;
06827         MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
06828         error = 1;
06829     }
06830 }
06831 else if ( StrICmp(name, (UBYTE *)"printscheme") == 0 ) {
06832     if ( StrICmp(value, (UBYTE *)"on") == 0 ) {
06833         AO.Optimize.schemeflags |= 1;
06834     }
06835     else if ( StrICmp(value, (UBYTE *)"off") == 0 ) {
06836         AO.Optimize.schemeflags &= ~1;
06837     }
06838     else {
06839         AO.Optimize.schemeflags &= ~1;
06840         MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
06841         error = 1;
06842     }
06843 }
06844 else if ( StrICmp(name, (UBYTE *)"debugflag") == 0 ) {
06845     /*
06846         This option is for debugging purposes only. Not in the manual!
06847         0x1: Print statements in reverse order.

```

```

06848         0x2: Print the scheme of the variables.
06849  */
06850         x = 0;
06851         u = value;
06852         if ( FG.cTable[*u] == 1 ) {
06853             while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06854             if ( *u != 0 ) {
06855                 MesPrint("&Numerical value for DebugFlag in Format,Optimize statement should be a
nonnegative number: %s",value);
06856                 AO.Optimize.debugflags = 0;
06857                 error = 1;
06858             }
06859             else {
06860                 AO.Optimize.debugflags = x;
06861             }
06862         }
06863         else if ( StrICmp(value,(UBYTE *)"on") == 0 ) {
06864             AO.Optimize.debugflags = 1;
06865         }
06866         else if ( StrICmp(value,(UBYTE *)"off") == 0 ) {
06867             AO.Optimize.debugflags = 0;
06868         }
06869         else {
06870             AO.Optimize.debugflags = 0;
06871             MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
06872             error = 1;
06873         }
06874     }
06875     else if ( StrICmp(name,(UBYTE *)"scheme") == 0 ) {
06876         UBYTE *ss, *s1, c;
06877         WORD type, numsym;
06878         AO.schemenum = 0;
06879         u = value;
06880         if ( *u != '(' ) {
06881             noscheme:
06882                 MesPrint("&Option Scheme in Format,Optimize statement should be an array of names or
integers between (): %s",value);
06883                 error = 1;
06884                 break;
06885         }
06886         u++; ss = u;
06887         while ( *ss == ' ' || *ss == '\t' || *ss == ',' ) ss++;
06888         if ( FG.cTable[*ss] == 0 || *ss == '$' || *ss == '[' ) { /* Name */
06889             s1 = u; SKIPBRA3(s1)
06890             if ( *s1 != ')' ) goto noscheme;
06891             while ( ss < s1 ) { if ( *ss++ == ',' ) AO.schemenum++; }
06892             *ss++ = 0; while ( *ss == ' ' ) ss++;
06893             if ( *ss != 0 ) goto noscheme;
06894             ss = u;
06895             if ( AO.schemenum < 1 ) {
06896                 MesPrint("&Option Scheme in Format,Optimize statement should have at least one
name or number between ()");
06897                 error = 1;
06898                 break;
06899             }
06900             if ( AO.inscheme ) M_free(AO.inscheme,"Horner input scheme");
06901             AO.inscheme = (WORD *)Malloc1((AO.schemenum+1)*sizeof(WORD),"Horner input scheme");
06902             while ( *ss == ' ' || *ss == '\t' || *ss == ',' ) ss++;
06903             AO.schemenum = 0;
06904             for(;;) {
06905                 if ( *ss == 0 ) break;
06906                 s1 = ss; ss = SkipAName(s1); c = *ss; *ss = 0;
06907             }
06908             if ( ss[-1] == '_' ) {
06909                 /*
06910                 Now AC.extrasym followed by a number and _
06911                 */
06912                 UBYTE *u1, *u2;
06913                 u1 = s1; u2 = AC.extrasym;
06914                 while ( *u1 == *u2 ) { u1++; u2++; }
06915                 if ( *u2 == 0 ) { /* Good start */
06916                     numsym = 0;
06917                     while ( *u1 >= '0' && *u1 <= '9' ) numsym = 10*numsym + *u1++ - '0';
06918                     if ( u1 != ss-1 || numsym == 0 || AC.extrasymbols != 0 ) {
06919                         MesPrint("&Improper use of extra symbol in scheme format option");
06920                         goto noscheme;
06921                     }
06922                     numsym = MAXVARIABLES-numsym;
06923                     ss++;
06924                     goto GotTheNumber;
06925                 }
06926             }
06927             else if ( *s1 == '$' ) {
06928                 GETIDENTITY
06929                 int numdollar;
06930                 if ( ( numdollar = GetDollar(s1+1) ) < 0 ) {
06931                     MesPrint("&Undefined variable %s",s1);

```

```

06932         error = 5;
06933     }
06934     else if ( ( numsym = DolToSymbol(BHEAD numdollar) ) < 0 ) {
06935         MesPrint("&$$s does not evaluate to a symbol",s1);
06936         error = 5;
06937     }
06938     *ss = c;
06939     goto GotTheNumber;
06940 }
06941 else if ( c == '(' ) {
06942     if ( StrCmp(s1,AC.extrasym) == 0 ) {
06943         if ( (AC.extrasymbols&1) != 1 ) {
06944             MesPrint("&Improper use of extra symbol in scheme format option");
06945             goto noscheme;
06946         }
06947         *ss++ = c;
06948         numsym = 0;
06949         while ( *ss >= '0' && *ss <= '9' ) numsym = 10*numsym + *ss++ - '0';
06950         if ( *ss != ')' ) {
06951             MesPrint("&Extra symbol should have a number for its argument.");
06952             goto noscheme;
06953         }
06954         numsym = MAXVARIABLES-numsym;
06955         ss++;
06956         goto GotTheNumber;
06957     }
06958 }
06959 type = GetName(AC.varnames,s1,&numsym,WITHAUTO);
06960 if ( ( type != CSYMBOL ) && type != CDUBIOUS ) {
06961     MesPrint("&$$s is not a symbol",s1);
06962     error = 4;
06963     if ( type < 0 ) numsym = AddSymbol(s1,-MAXPOWER,MAXPOWER,0,0);
06964 }
06965 *ss = c;
06966 GotTheNumber:
06967     AO.inscheme[AO.schemenum++] = numsym;
06968     while ( *ss == ' ' || *ss == '\t' || *ss == ',' ) ss++;
06969 }
06970 }
06971 }
06972 else if ( StrICmp(name,(UBYTE *)"mctsdecaymode") == 0 ) {
06973     x = 0;
06974     u = value;
06975     if ( FG.cTable[*u] == 1 ) {
06976         while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06977         if ( *u != 0 ) {
06978             MesPrint("&Option MCTSDecayMode in Format,Optimize statement should be a
nonnegative integer: %s",value);
06979             AO.Optimize.mctsdecaymode = 0;
06980             error = 1;
06981         }
06982         else {
06983             AO.Optimize.mctsdecaymode = x;
06984         }
06985     }
06986     else {
06987         AO.Optimize.mctsdecaymode = 0;
06988         MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
06989         error = 1;
06990     }
06991 }
06992 else if ( StrICmp(name,(UBYTE *)"saiter") == 0 ) {
06993     x = 0;
06994     u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
06995     if ( *u != 0 ) {
06996         MesPrint("&Option SAIter in Format,Optimize statement should be a positive integer:
%s",value);
06997         AO.Optimize.saIter = 0;
06998         error = 1;
06999     }
07000     else {
07001         AO.Optimize.saIter = x;
07002     }
07003 }
07004 else if ( StrICmp(name,(UBYTE *)"samaxt") == 0 ) {
07005     d = 0;
07006     if ( sscanf ((char*)value, "%lf", &d) != 1 ) {
07007         MesPrint("&Option SAMaxT in Format,Optimize statement should be a positive number:
%s",value);
07008         AO.Optimize.saMaxT.fval = 0;
07009         error = 1;
07010     }
07011     else {
07012         AO.Optimize.saMaxT.fval = d;
07013     }
07014 }
07015 else if ( StrICmp(name,(UBYTE *)"samint") == 0 ) {

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```

07016         d = 0;
07017         if ( sscanf ((char*)value, "%lf", &d) != 1 ) {
07018             MesPrint("&Option SAmiT in Format,Optimize statement should be a positive number:
%s",value);
07019             AO.Optimize.saMinT.fval = 0;
07020             error = 1;
07021         }
07022         else {
07023             AO.Optimize.saMinT.fval = d;
07024         }
07025     }
07026     else {
07027         MesPrint("&Unrecognized option name in Format,Optimize statement: %s",name);
07028         error = 1;
07029     }
07030     *t1 = c1; *t2 = c2;
07031 }
07032 return(error);
07033 }
07034
07035 /*
07036 #] CoOptimizeOption :
07037 #[ DoPutInside :
07038
07039 Syntax:
07040 PutIn[side],functionname[,brackets] -> par = 1
07041 AntiPutIn[side],functionname,antibrackets -> par = -1
07042 */
07043
07044 int CoPutInside(UBYTE *inp) { return(DoPutInside(inp,1)); }
07045 int CoAntiPutInside(UBYTE *inp) { return(DoPutInside(inp,-1)); }
07046
07047 int DoPutInside(UBYTE *inp, int par)
07048 {
07049     GETIDENTITY
07050     UBYTE *p, c;
07051     WORD *to, type, c1,c2,funnum, *WorkSave;
07052     int error = 0;
07053     while ( *inp == ' ' || *inp == '\t' || *inp == ',' ) inp++;
07054 /*
07055     First we need the name of a function. (Not a tensor or table!)
07056 */
07057     p = SkipAName(inp);
07058     if ( p == 0 ) return(1);
07059     c = *p; *p = 0;
07060     type = GetName(AC.varnames,inp,&funnum,WITHAUTO);
07061     if ( type != CFUNCTION || functions[funnum].tabl != 0 || functions[funnum].spec ) {
07062         MesPrint("&PutInside/AntiPutInside expects a regular function for its first argument");
07063         MesPrint("&Argument is %s",inp);
07064         error = 1;
07065     }
07066     funnum += FUNCTION;
07067     *p = c;
07068     inp = p;
07069     while ( *inp == ' ' || *inp == '\t' || *inp == ',' ) inp++;
07070     if ( *inp == 0 ) {
07071         if ( par == 1 ) {
07072             WORD tocompiler[4];
07073             tocompiler[0] = TYPEPUTINSIDE;
07074             tocompiler[1] = 4;
07075             tocompiler[2] = 0;
07076             tocompiler[3] = funnum;
07077             AddNtoL(4,tocompiler);
07078         }
07079         else {
07080             MesPrint("&AntiPutInside needs inside information.");
07081             error = 1;
07082         }
07083         return(error);
07084     }
07085     WorkSave = to = AT.WorkPointer;
07086     *to++ = TYPEPUTINSIDE;
07087     *to++ = 4;
07088     *to++ = par;
07089     *to++ = funnum;
07090     to++;
07091     while ( *inp ) {
07092         while ( *inp == ' ' || *inp == '\t' || *inp == ',' ) inp++;
07093         if ( *inp == 0 ) break;
07094         p = SkipAName(inp);
07095         if ( p == 0 ) { error = 1; break; }
07096         c = *p; *p = 0;
07097         type = GetName(AC.varnames,inp,&c1,WITHAUTO);
07098         if ( c == '.' ) {
07099             if ( type == CVECTOR || type == CDUBIOUS ) {
07100                 *p++ = c;
07101                 inp = p;

```

```

07102         p = SkipAName(inp);
07103         if ( p == 0 ) return(1);
07104         c = *p; *p = 0;
07105         type = GetName(AC.varnames,inp,&c2,WITHAUTO);
07106         if ( type != CVECTOR && type != CDUBIOUS ) {
07107             MesPrint("&Not a vector in dotproduct in PutInside/AntiPutInside statement:
%s",inp);
07108             error = 1;
07109         }
07110         else type = CDOTPRODUCT;
07111     }
07112     else {
07113         MesPrint("&Illegal use of . after %s in PutInside/AntiPutInside statement",inp);
07114         error = 1;
07115         *p = c; inp = p;
07116         continue;
07117     }
07118 }
07119 switch ( type ) {
07120     case CSYMBOL :
07121         *to++ = SYMBOL; *to++ = 4; *to++ = c1; *to++ = 1; break;
07122     case CVECTOR :
07123         *to++ = INDEX; *to++ = 3; *to++ = AM.OffsetVector + c1; break;
07124     case CFUNCTION :
07125         *to++ = c1+FUNCTION; *to++ = FUNHEAD; *to++ = 0;
07126         FILLFUN3(to)
07127         break;
07128     case CDOTPRODUCT :
07129         *to++ = DOTPRODUCT; *to++ = 5; *to++ = c1 + AM.OffsetVector;
07130         *to++ = c2 + AM.OffsetVector; *to++ = 1; break;
07131     case CDELTA :
07132         *to++ = DELTA; *to++ = 4; *to++ = EMPTYINDEX; *to++ = EMPTYINDEX; break;
07133     default :
07134         MesPrint("&Illegal variable request for %s in PutInside/AntiPutInside statement",inp);
07135         error = 1; break;
07136 }
07137 *p = c;
07138 inp = p;
07139 }
07140 *to++ = 1; *to++ = 1; *to++ = 3;
07141 AT.WorkPointer[1] = to - AT.WorkPointer;
07142 AT.WorkPointer[4] = AT.WorkPointer[1]-4;
07143 AT.WorkPointer = to;
07144 AC.BraceNormalize = 1;
07145 if ( Normalize(BHEAD WorkSave+4) ) { error = 1; }
07146 else {
07147     WorkSave[1] = WorkSave[4]+4;
07148     to = WorkSave + WorkSave[1] - 1;
07149     c1 = ABS(*to);
07150     WorkSave[1] -= c1;
07151     WorkSave[4] -= c1;
07152     AddNtoL(WorkSave[1],WorkSave);
07153 }
07154 AC.BraceNormalize = 0;
07155 AT.WorkPointer = WorkSave;
07156 return(error);
07157 }
07158
07159 /*
07160 #] DoPutInside :
07161 #[ CoSwitch :
07162
07163 Syntax: Switch $var;
07164 Be careful with illegal nestings with repeat, if, while.
07165 */
07166
07167 int CoSwitch(UBYTE *s)
07168 {
07169     WORD numdollar;
07170     SWITCH *sw;
07171     if ( *s == '$' ) {
07172         if ( GetName(AC.dollarnames,s+1,&numdollar,NOAUTO) != CDOLLAR ) {
07173             MesPrint("&%s is undefined in switch statement",s);
07174             numdollar = AddDollar(s+1,DOLINDEX,&one,1);
07175             return(1);
07176         }
07177         s = SkipAName(s+1);
07178         if ( *s != 0 ) {
07179             MesPrint("&Switch should have a single $variable for its argument");
07180             return(1);
07181         }
07182         /* AddPotModdollar(numdollar); */
07183     }
07184     else {
07185         MesPrint("&%s is not a $-variable in switch statement",s);
07186         return(1);
07187     }

```

```

07188 /*
07189     Now create the switch table. We will add to it each time we run
07190     into a new case. It will all be sorted out the moment we run into
07191     the endswitch statement.
07192 */
07193     AC.SwitchLevel++;
07194     if ( AC.SwitchInArray >= AC.MaxSwitch ) DoubleSwitchBuffers();
07195     AC.SwitchHeap[AC.SwitchLevel] = AC.SwitchInArray;
07196     sw = AC.SwitchArray + AC.SwitchInArray;
07197
07198     sw->iflevel = AC.IfLevel;
07199     sw->whilelevel = AC.WhileLevel;
07200     sw->nestingsum = NestingChecksum();
07201
07202     Add4Com(TYPESWITCH,numdollar,AC.SwitchInArray);
07203
07204     AC.SwitchInArray++;
07205     return(0);
07206 }
07207
07208 /*
07209     #] CoSwitch :
07210     #[ CoCase :
07211 */
07212
07213 int CoCase(UBYTE *s)
07214 {
07215     SWITCH *sw = AC.SwitchArray + AC.SwitchHeap[AC.SwitchLevel];
07216     WORD x = 0, sign = 1;
07217     while ( *s == ',' ) s++;
07218     SKIPBLANKS(s);
07219     while ( *s == '-' || *s == '+' ) {
07220         if ( *s == '-' ) sign = -sign;
07221         s++;
07222     }
07223     while ( FG.cTable[*s] == 1 ) { x = 10*x + *s++ - '0'; }
07224     x = sign*x;
07225
07226     if ( sw->iflevel != AC.IfLevel || sw->whilelevel != AC.WhileLevel
07227         || sw->nestingsum != NestingChecksum() ) {
07228         MesPrint("%Illegal nesting of switch/case/default with
07229 if/while/repeat/loop/argument/term/...");
07230         return(-1);
07231     }
07232
07233     /*
07234     Now add a case to the table with the current 'address'.
07235     */
07236     if ( sw->numcases >= sw->tablesize ) {
07237         int i;
07238         SWITCHTABLE *newtable;
07239         WORD newsize;
07240         if ( sw->tablesize == 0 ) newsize = 10;
07241         else newsize = 2*sw->tablesize;
07242         newtable = (SWITCHTABLE *)Malloc1(newsize*sizeof(SWITCHTABLE),"Switch table");
07243         if ( sw->table ) {
07244             for ( i = 0; i < sw->tablesize; i++ ) newtable[i] = sw->table[i];
07245             M_free(sw->table,"Switch table");
07246         }
07247         sw->table = newtable;
07248         sw->tablesize = newsize;
07249     }
07250     if ( sw->numcases == 0 ) { sw->mincase = sw->maxcase = x; }
07251     else if ( x > sw->maxcase ) sw->maxcase = x;
07252     else if ( x < sw->mincase ) sw->mincase = x;
07253     sw->table[sw->numcases].ncase = x;
07254     sw->table[sw->numcases].value = cbuf[AC.cbunum].numlhs;
07255     sw->table[sw->numcases].compbuffer = AC.cbunum;
07256     sw->numcases++;
07257     return(0);
07258 }
07259
07260 /*
07261     #] CoCase :
07262     #[ CoBreak :
07263 */
07264
07265 int CoBreak(UBYTE *s)
07266 {
07267     /*
07268     This involves a 'postponed' jump to the end. This can be done
07269     in a special routine during execution.
07270     That routine should also pop the switch level.
07271     */
07272     SWITCH *sw = AC.SwitchArray + AC.SwitchHeap[AC.SwitchLevel];
07273     if ( sw->iflevel != AC.IfLevel || sw->whilelevel != AC.WhileLevel
07274         || sw->nestingsum != NestingChecksum() ) {
07275         MesPrint("%Illegal nesting of switch/case/default with

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    if/while/repeat/loop/argument/term/...");
07274     return(-1);
07275 }
07276 if ( *s ) {
07277     MesPrint("&No parameters allowed in Break statement");
07278     return(-1);
07279 }
07280 Add3Com(TYPEENDSWITCH,AC.SwitchHeap[AC.SwitchLevel]);
07281 return(0);
07282 }
07283
07284 /*
07285 #] CoBreak :
07286 #[ CoDefault :
07287 */
07288
07289 int CoDefault (UBYTE *s)
07290 {
07291     /*
07292     A bit like case, except that the address gets stored directly in the
07293     SWITCH struct.
07294     */
07295     SWITCH *sw = AC.SwitchArray + AC.SwitchHeap[AC.SwitchLevel];
07296     if ( sw->iflevel != AC.IfLevel || sw->whilelevel != AC.WhileLevel
07297         || sw->nestingsum != NestingChecksum() ) {
07298         MesPrint("&Illegal nesting of switch/case/default with
07299 if/while/repeat/loop/argument/term/...");
07300         return(-1);
07301     }
07302     if ( *s ) {
07303         MesPrint("&No parameters allowed in Default statement");
07304         return(-1);
07305     }
07306     sw->defaultcase.ncase = 0;
07307     sw->defaultcase.value = cbuf[AC.cbunum].numlhs;
07308     sw->defaultcase.compbuffer = AC.cbunum;
07309     return(0);
07310 }
07311 /*
07312 #] CoDefault :
07313 #[ CoEndSwitch :
07314 */
07315
07316 int CoEndSwitch (UBYTE *s)
07317 {
07318     /*
07319     We store this address in the SWITCH struct.
07320     Next we sort the table by ncase.
07321     Then we decide whether the table is DENSE or SPARSE.
07322     If it is dense we change the allocation and spread the cases is necessary.
07323     Finally we pop levels.
07324     */
07325     SWITCH *sw = AC.SwitchArray + AC.SwitchHeap[AC.SwitchLevel];
07326     WORD i;
07327     WORD totcases = sw->maxcase-sw->mincase+1;
07328     while ( *s == ',' ) s++;
07329     SKIPBLANKS(s)
07330     if ( *s ) {
07331         MesPrint("&No parameters allowed in EndSwitch statement");
07332         return(-1);
07333     }
07334     if ( sw->iflevel != AC.IfLevel || sw->whilelevel != AC.WhileLevel
07335         || sw->nestingsum != NestingChecksum() ) {
07336         MesPrint("&Illegal nesting of switch/case/default with
07337 if/while/repeat/loop/argument/term/...");
07338         return(-1);
07339     }
07340     if ( sw->defaultcase.value == 0 ) CoDefault(s);
07341     if ( totcases > sw->numcases*AM.jumpratio ) { /* The factor is experimental */
07342         sw->caseoffset = 0;
07343         sw->typetable = SPARSETABLE;
07344     }
07345     /*
07346     Now we need to sort sw->table
07347     */
07348     SwitchSplitMerge(sw->table,sw->numcases);
07349 }
07350 else { /* DENSE */
07351     SWITCHTABLE *ntable;
07352     sw->caseoffset = sw->mincase;
07353     sw->typetable = DENSETABLE;
07354     ntable = (SWITCHTABLE *)Malloc1(totcases*sizeof(SWITCHTABLE),"Switch table");
07355     for ( i = 0; i < totcases; i++ ) {
07356         ntable[i].ncase = i+sw->caseoffset;
07357         ntable[i].value = sw->defaultcase.value;
07358         ntable[i].compbuffer = sw->defaultcase.compbuffer;
07359     }

```

```

07358         for ( i = 0; i < sw->numcases; i++ ) {
07359             ntable[sw->table[i].ncase-sw->caseoffset] = sw->table[i];
07360         }
07361         M_free(sw->table,"Switch table");
07362         sw->table = ntable;
07363         sw->numcases = totcases;
07364     }
07365     sw->endswitch.ncase = 0;
07366     sw->endswitch.value = cbuf[AC.cbufnum].numlhs;
07367     sw->endswitch.compbuffer = AC.cbufnum;
07368     if ( sw->defaultcase.value == 0 ) {
07369         sw->defaultcase = sw->endswitch;
07370     }
07371     Add3Com(TYPEENDSWITCH,AC.SwitchHeap[AC.SwitchLevel]);
07372     /*
07373     Now we need to pop.
07374     */
07375     AC.SwitchLevel--;
07376     return(0);
07377 }
07378
07379 /*
07380 #] CoEndSwitch :
07381 #[ CoSetUserFlag :
07382 */
07383
07384 int CoSetUserFlag(UBYTE *s)
07385 {
07386     int error = 0;
07387     while ( *s && ( *s == ',' || *s == ' ' || *s == '\t' ) ) s++;
07388     while ( *s && ( FG.cTable[*s] == 1 ) ) {
07389         int x = 0;
07390         while ( *s && ( FG.cTable[*s] == 1 ) ) x = 10*x+(s++ - '0');
07391         if ( x < 1 || x > BITSINWORD ) {
07392             MesPrint("&Flag number %d outside the permitted range 1-&d.",BITSINWORD);
07393             error = 1;
07394         }
07395         else {
07396             Add3Com(TYPESETUSERFLAG,x-1);
07397         }
07398         while ( *s && ( *s == ',' || *s == ' ' || *s == '\t' ) ) s++;
07399     }
07400     if ( *s ) {
07401         MesPrint("&Illegal character in SetUserFlag statement: %s",s);
07402         error = 1;
07403     }
07404     return(error);
07405 }
07406
07407 /*
07408 #] CoSetUserFlag :
07409 #[ CoClearUserFlag :
07410 */
07411
07412 int CoClearUserFlag(UBYTE *s)
07413 {
07414     int error = 0;
07415     while ( *s && ( *s == ',' || *s == ' ' || *s == '\t' ) ) s++;
07416     while ( *s && ( FG.cTable[*s] == 1 ) ) {
07417         int x = 0;
07418         while ( *s && ( FG.cTable[*s] == 1 ) ) x = 10*x+(s++ - '0');
07419         if ( x < 1 || x > BITSINWORD ) {
07420             MesPrint("&Flag number %d outside the permitted range 1-&d.",BITSINWORD);
07421             error = 1;
07422         }
07423         else {
07424             Add3Com(TYPECLEARUSERFLAG,x);
07425         }
07426         while ( *s && ( *s == ',' || *s == ' ' || *s == '\t' ) ) s++;
07427     }
07428     if ( *s ) {
07429         MesPrint("&Illegal character in SetUserFlag statement: %s",s);
07430         error = 1;
07431     }
07432     return(error);
07433 }
07434
07435 /*
07436 #] CoClearUserFlag :
07437 #[ CoCreateAllLoops :
07438
07439 Syntax:
07440     CoCreateAllLoops,in-function,out-function,type-argument,ifnolooop;
07441 Types allowed:
07442     vector, index, symbol, snumber
07443 ifnolooop:
07444     ifnolooop=0 or ifnolooop=1

```



```

07445     in-function can be a tensor. In that case type can be only vector or index.
07446     out-function can be a tensor. In that case type can be only vector or index.
07447 */
07448
07449 int CoCreateAllLoops(UBYTE *s)
07450 {
07451     GETIDENTITY
07452     UBYTE *inname, *outname, *stype, c;
07453     WORD infun, outfun, x, type, tensorflag, typenum;
07454     WORD *WorkSave, *to;
07455     while ( *s == ',' || *s == ' ' ) s++;
07456     inname = s; s = SkipAName(s);
07457     c = *s; *s = 0;
07458     if ( ( ( type = GetName(AC.varnames,inname,&infun,WITHAUTO) ) != CFUNCTION )
07459     || ( ( functions[infun].spec != 0 ) && ( functions[infun].spec != TENSORFUNCTION ) ) ) {
07460         MesPrint("%s should be a regular function or a tensor.",inname);
07461         if ( type < 0 ) {
07462             if ( GetName(AC.exprnames,s,&infun,NOAUTO) == NAMENOTFOUND )
07463                 AddFunction(s,0,0,0,0,0,-1,-1);
07464         }
07465         return(1);
07466     }
07467     infun += FUNCTION;
07468     *s++ = c;
07469     while ( *s == ',' || *s == ' ' ) s++;
07470     outname = s; s = SkipAName(s);
07471     c = *s; *s = 0;
07472     if ( ( ( type = GetName(AC.varnames,outname,&outfun,WITHAUTO) ) != CFUNCTION )
07473     || ( ( functions[outfun].spec != 0 ) && ( functions[outfun].spec != TENSORFUNCTION ) ) ) {
07474         MesPrint("%s should be a regular function or a tensor.",outname);
07475         if ( type < 0 ) {
07476             if ( GetName(AC.exprnames,s,&outfun,NOAUTO) == NAMENOTFOUND )
07477                 AddFunction(s,0,0,0,0,0,-1,-1);
07478         }
07479         return(1);
07480     }
07481     outfun += FUNCTION;
07482     *s++ = c;
07483     if ( functions[infun].spec == TENSORFUNCTION ||
07484         functions[outfun].spec == TENSORFUNCTION ) tensorflag = 1;
07485     else tensorflag = 0;
07486 /*
07487     Now the type: type=....
07488 */
07489     while ( *s == ',' || *s == ' ' ) s++;
07490     stype = s;
07491     while ( FG.cTable[*s] == 0 ) s++;
07492     c = *s; *s = 0;
07493     if ( StrICmp(stype,(UBYTE *)"type") != 0 || c != '=' ) {
07494         MesPrint("&In CreateAllLoops statement: expected type=vartype.");
07495         return(1);
07496     }
07497     *s++ = c;
07498     stype = s;
07499     while ( FG.cTable[*s] == 0 ) s++;
07500     c = *s; *s = 0;
07501     if ( StrICmp(stype,(UBYTE *)"vector") == 0 ) {
07502         typenum = -VECTOR;
07503     }
07504     else if ( StrICmp(stype,(UBYTE *)"index") == 0 ) {
07505         typenum = -INDEX;
07506     }
07507     else if ( StrICmp(stype,(UBYTE *)"symbol") == 0 ) {
07508         if ( tensorflag ) goto notintensor;
07509         typenum = -SYMBOL;
07510     }
07511     else if ( StrICmp(stype,(UBYTE *)"snumber") == 0 ) {
07512         if ( tensorflag ) goto notintensor;
07513         typenum = -SNUMBER;
07514     }
07515     else {
07516         MesPrint("&Unknown/not allowed variable type in CreateAllLoops: %s",stype);
07517         return(1);
07518     }
07519     *s = c;
07520     while ( *s == ',' || *s == ' ' ) s++;
07521
07522     stype = s;
07523     while ( FG.cTable[*s] == 0 ) s++;
07524     c = *s; *s = 0;
07525     if ( StrICmp(stype,(UBYTE *)"ifnolooop") != 0 || c != '=' ) {
07526         MesPrint("&Unrecognised option in CreateAllLoops statement: %s",stype);
07527         return(1);
07528     }
07529     *s++ = c;
07530     x = -1;
07531     if ( FG.cTable[*s] == 1 ) {

```

```

07532         x = 0;
07533         do { x = 10*x + (*s++-'0'); } while (FG.cTable[*s] == 1);
07534     }
07535     if ( x != 0 && x != 1 ) {
07536         MesPrint("&Only options allowed for ifnolooop are 0 or 1.");
07537         return(1);
07538     }
07539     WorkSave = to = AT.WorkPointer;
07540     *to++ = TYPEALLLOOPS;
07541     *to++ = 6;
07542     *to++ = infun;
07543     *to++ = outfun;
07544     *to++ = typenum;
07545     *to++ = x;
07546
07547     AddNtoL(WorkSave[1],WorkSave);
07548
07549     return(0);
07550 notintensor:
07551     MesPrint("&Variable type not allowed in tensors: %s",type);
07552     return(1);
07553 }
07554
07555 /*
07556 #] CoCreateAllLoops :
07557 #[ CoCreateAllPaths :
07558
07559 Syntax:
07560     CreateAllPaths,end-function,intermediate-function,out-function,type-argument,ifnopath;
07561 Types allowed:
07562     vector, index, symbol, snumber
07563 ifnolooop:
07564     ifnolooop=0 or ifnolooop=1
07565 in-function can be a tensor. In that case type can be only vector or index.
07566 out-function can be a tensor. In that case type can be only vector or index.
07567 */
07568
07569 int CoCreateAllPaths(UBYTE *s)
07570 {
07571     GETIDENTITY
07572     UBYTE *endname,*iname, *outname, *stype, c;
07573     WORD endfun, infun, outfun, x, type, tensorflag, typenum;
07574     WORD *WorkSave, *to;
07575     while ( *s == ',' || *s == ' ' ) s++;
07576     endname = s; s = SkipAName(s);
07577     c = *s; *s = 0;
07578     if ( ( ( type = GetName(AC.varnames,endname,&endfun,WITHAUTO) ) != CFUNCTION )
07579 || ( ( functions[endfun].spec != 0 ) && ( functions[endfun].spec != TENSORFUNCTION ) ) ) {
07580         MesPrint("&%s should be a regular function or a tensor.",endname);
07581         if ( type < 0 ) {
07582             if ( GetName(AC.exprnames,s,&endfun,NOAUTO) == NAMENOTFOUND )
07583                 AddFunction(s,0,0,0,0,0,-1,-1);
07584         }
07585         return(1);
07586     }
07587     endfun += FUNCTION;
07588     *s++ = c;
07589     while ( *s == ',' || *s == ' ' ) s++;
07590     inname = s; s = SkipAName(s);
07591     c = *s; *s = 0;
07592     if ( ( ( type = GetName(AC.varnames,inname,&infun,WITHAUTO) ) != CFUNCTION )
07593 || ( ( functions[infun].spec != 0 ) && ( functions[infun].spec != TENSORFUNCTION ) ) ) {
07594         MesPrint("&%s should be a regular function or a tensor.",iname);
07595         if ( type < 0 ) {
07596             if ( GetName(AC.exprnames,s,&infun,NOAUTO) == NAMENOTFOUND )
07597                 AddFunction(s,0,0,0,0,0,-1,-1);
07598         }
07599         return(1);
07600     }
07601     infun += FUNCTION;
07602     *s++ = c;
07603     while ( *s == ',' || *s == ' ' ) s++;
07604     outname = s; s = SkipAName(s);
07605     c = *s; *s = 0;
07606     if ( ( ( type = GetName(AC.varnames,outname,&outfun,WITHAUTO) ) != CFUNCTION )
07607 || ( ( functions[outfun].spec != 0 ) && ( functions[outfun].spec != TENSORFUNCTION ) ) ) {
07608         MesPrint("&%s should be a regular function or a tensor.",outname);
07609         if ( type < 0 ) {
07610             if ( GetName(AC.exprnames,s,&outfun,NOAUTO) == NAMENOTFOUND )
07611                 AddFunction(s,0,0,0,0,0,-1,-1);
07612         }
07613         return(1);
07614     }
07615     outfun += FUNCTION;
07616     *s++ = c;
07617     if ( functions[infun].spec == TENSORFUNCTION ||
07618         functions[outfun].spec == TENSORFUNCTION ) tensorflag = 1;

```

```

07619     else tensorflag = 0;
07620 /*
07621     Now the type: type=....
07622 */
07623     while ( *s == ',' || *s == ' ' ) s++;
07624     stype = s;
07625     while ( FG.cTable[*s] == 0 ) s++;
07626     c = *s; *s = 0;
07627     if ( StrICmp(stype, (UBYTE *)"type") != 0 || c != '=' ) {
07628         MesPrint("&In CreateAllPaths statement: expected type=vartype.");
07629         return(1);
07630     }
07631     *s++ = c;
07632     stype = s;
07633     while ( FG.cTable[*s] == 0 ) s++;
07634     c = *s; *s = 0;
07635     if ( StrICmp(stype, (UBYTE *)"vector") == 0 ) {
07636         typenum = -VECTOR;
07637     }
07638     else if ( StrICmp(stype, (UBYTE *)"index") == 0 ) {
07639         typenum = -INDEX;
07640     }
07641     else if ( StrICmp(stype, (UBYTE *)"symbol") == 0 ) {
07642         if ( tensorflag ) goto notintensor;
07643         typenum = -SYMBOL;
07644     }
07645     else if ( StrICmp(stype, (UBYTE *)"snumber") == 0 ) {
07646         if ( tensorflag ) goto notintensor;
07647         typenum = -SNUMBER;
07648     }
07649     else {
07650         MesPrint("&Unknown/not allowed variable type in CreateAllPaths: %s", stype);
07651         return(1);
07652     }
07653     *s = c;
07654     while ( *s == ',' || *s == ' ' ) s++;
07655
07656     stype = s;
07657     while ( FG.cTable[*s] == 0 ) s++;
07658     c = *s; *s = 0;
07659     if ( StrICmp(stype, (UBYTE *)"ifnopath") != 0 || c != '=' ) {
07660         MesPrint("&Unrecognised option in CreateAllPaths statement: %s", stype);
07661         return(1);
07662     }
07663     *s++ = c;
07664     x = -1;
07665     if ( FG.cTable[*s] == 1 ) {
07666         x = 0;
07667         do { x = 10*x + (*s++-'0'); } while (FG.cTable[*s] == 1);
07668     }
07669     if ( x != 0 && x != 1 ) {
07670         MesPrint("&Only options allowed for ifnopath are 0 or 1.");
07671         return(1);
07672     }
07673     WorkSave = to = AT.WorkPointer;
07674     *to++ = TYPEALLPATHS;
07675     *to++ = 7;
07676     *to++ = endfun;
07677     *to++ = infun;
07678     *to++ = outfun;
07679     *to++ = typenum;
07680     *to++ = x;
07681
07682     AddNtoL(WorkSave[1], WorkSave);
07683
07684     return(0);
07685 notintensor:
07686     MesPrint("&CreateAllPaths: Variable type not allowed in tensors: %s", stype);
07687     return(1);
07688 }
07689
07690 /*
07691     #] CoCreateAllPaths :
07692     #[ CoCreateAll :
07693
07694     Syntax: subkey, two or three functions, type, ifnone
07695 */
07696 int CoCreateAll(UBYTE *s)
07697 {
07698     UBYTE *subkey;
07699     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
07700     subkey = s;
07701     while ( FG.cTable[*s] == 0 ) s++;
07702     if ( *s != ' ' && *s != ',' && *s != '\t' ) {
07703         MesPrint("&Illegal subkey in CoCreate statement.");
07704         return(1);
07705     }

```

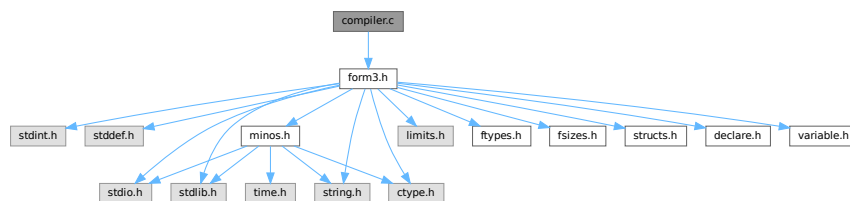
```

07706     }
07707     /* c = *s; */ *s++ = 0;
07708     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
07709     if ( StrICmp(subkey, (UBYTE *) "loops" ) == 0 ) {
07710         return(CoCreateAllLoops(s));
07711     }
07712     else if ( StrICmp(subkey, (UBYTE *) "paths" ) == 0 ) {
07713         return(CoCreateAllPaths(s));
07714     }
07715     /*
07716     else if ( StrICmp(subkey, (UBYTE *) "motics" ) == 0 ) {
07717     }
07718     else if ( StrICmp(subkey, (UBYTE *) "onepi" ) == 0 ) {
07719     }
07720     else if ( StrICmp(subkey, (UBYTE *) "cuts" ) == 0 ) {
07721     }
07722     */
07723     else {
07724         MesPrint("%Illegal subkey in CoCreate statement: %s.", subkey);
07725         return(1);
07726     }
07727 }
07728
07729 /*
07730 #] CoCreateAll :
07731 */

```

## 4.11 compiler.c File Reference

#include "form3.h"  
 Include dependency graph for compiler.c:



### Data Structures

- struct [SuBbUf](#)

### Macros

- #define [OPTION0](#) 1
- #define [OPTION1](#) 2
- #define [OPTION2](#) 3
- #define [REDUCESUBEXPBUFFERS](#)

### Typedefs

- typedef struct [SuBbUf](#) [SUBBUF](#)

## Functions

- void [inictable](#) (void)
- [KEYWORD](#) \* [findcommand](#) (UBYTE \*in)
- int [ParenthesesTest](#) (UBYTE \*sin)
- UBYTE \* [SkipAName](#) (UBYTE \*s)
- UBYTE \* [IsRHS](#) (UBYTE \*s, UBYTE c)
- int [IsIdStatement](#) (UBYTE \*s)
- int [CompileAlgebra](#) (UBYTE \*s, int leftright, WORD \*prototype)
- int [CompileStatement](#) (UBYTE \*in)
- int [TestTables](#) (void)
- int [CompileSubExpressions](#) (SBYTE \*tokens)
- int [CodeGenerator](#) (SBYTE \*tokens)
- int [CompleteTerm](#) (WORD \*term, UWORD \*numer, UWORD \*denom, WORD nnum, WORD nden, int sign)
- int [CodeFactors](#) (SBYTE \*tokens)
- WORD [GenerateFactors](#) (WORD n, WORD inc)

## Variables

- int [alfatable1](#) [27]
- [SUBBUF](#) \* [subexpbuffers](#) = 0
- [SUBBUF](#) \* [topsubexpbuffers](#) = 0
- LONG [insubexpbuffers](#) = 0

### 4.11.1 Detailed Description

The heart of the compiler. It contains the tables of statements. It finds the statements in the tables and calls the proper routines. For algebraic expressions it runs the compilation by first calling the tokenizer, splitting things into subexpressions and generating the code. There is a system for recognizing already existing subexpressions. This economizes on the length of the output.

Note: the compiler of FORM doesn't attempt to normalize the input. Hence  $x+1$  and  $1+x$  are different objects during compilation. Similarly  $(a+b-b)$  will not be simplified to  $(a)$ .

Definition in file [compiler.c](#).

### 4.11.2 Macro Definition Documentation

#### 4.11.2.1 OPTION0

```
#define OPTION0 1
```

Definition at line 253 of file [compiler.c](#).

#### 4.11.2.2 OPTION1

```
#define OPTION1 2
```

Definition at line 254 of file [compiler.c](#).

#### 4.11.2.3 OPTION2

```
#define OPTION2 3
```

Definition at line 255 of file [compiler.c](#).

#### 4.11.2.4 REDUCESUBEXPBUFFERS

```
#define REDUCESUBEXPBUFFERS
```

**Value:**

```
{ if ( (topsubexpbuffers-subexpbuffers) > 256 ) {\
  M_free(subexpbuffers,"subexpbuffers");\
  subexpbuffers = (SUBBUF *)Malloc1(256*sizeof(SUBBUF),"subexpbuffers");\
  topsubexpbuffers = subexpbuffers+256; } insubexpbuffers = 0; }
```

Definition at line 266 of file [compiler.c](#).

### 4.11.3 Function Documentation

#### 4.11.3.1 inictable()

```
void inictable (
    void )
```

Definition at line 308 of file [compiler.c](#).

#### 4.11.3.2 findcommand()

```
KEYWORD * findcommand (
    UBYTE * in )
```

Definition at line 334 of file [compiler.c](#).

#### 4.11.3.3 ParenthesesTest()

```
int ParenthesesTest (
    UBYTE * sin )
```

Definition at line 378 of file [compiler.c](#).

#### 4.11.3.4 SkipAName()

```
UBYTE * SkipAName (
    UBYTE * s )
```

Definition at line 428 of file [compiler.c](#).

#### 4.11.3.5 IsRHS()

```
UBYTE * IsRHS (
    UBYTE * s,
    UBYTE c )
```

Definition at line [456](#) of file [compiler.c](#).

#### 4.11.3.6 IsIdStatement()

```
int IsIdStatement (
    UBYTE * s )
```

Definition at line [502](#) of file [compiler.c](#).

#### 4.11.3.7 CompileAlgebra()

```
int CompileAlgebra (
    UBYTE * s,
    int leftright,
    WORD * prototype )
```

Definition at line [516](#) of file [compiler.c](#).

#### 4.11.3.8 CompileStatement()

```
int CompileStatement (
    UBYTE * in )
```

Definition at line [552](#) of file [compiler.c](#).

#### 4.11.3.9 TestTables()

```
int TestTables (
    void )
```

Definition at line [667](#) of file [compiler.c](#).

#### 4.11.3.10 CompileSubExpressions()

```
int CompileSubExpressions (
    SBYTE * tokens )
```

Definition at line [711](#) of file [compiler.c](#).

#### 4.11.3.11 CodeGenerator()

```
int CodeGenerator (
    SBYTE * tokens )
```

Definition at line 846 of file [compiler.c](#).

#### 4.11.3.12 CompleteTerm()

```
int CompleteTerm (
    WORD * term,
    UWORD * numer,
    UWORD * denom,
    WORD nnum,
    WORD nden,
    int sign )
```

Definition at line 1909 of file [compiler.c](#).

#### 4.11.3.13 CodeFactors()

```
int CodeFactors (
    SBYTE * tokens )
```

Definition at line 1946 of file [compiler.c](#).

#### 4.11.3.14 GenerateFactors()

```
WORD GenerateFactors (
    WORD n,
    WORD inc )
```

Definition at line 2243 of file [compiler.c](#).

### 4.11.4 Variable Documentation

#### 4.11.4.1 alfatable1

```
int alfatable1[27]
```

Definition at line 251 of file [compiler.c](#).

#### 4.11.4.2 subexpbuffers

```
SUBBUF* subexpbuffers = 0
```

Definition at line 262 of file [compiler.c](#).



#### 4.11.4.3 topsubexpbuffers

`SUBBUF* topsubexpbuffers = 0`

Definition at line 263 of file [compiler.c](#).

#### 4.11.4.4 insubexpbuffers

`LONG insubexpbuffers = 0`

Definition at line 264 of file [compiler.c](#).

## 4.12 compiler.c

[Go to the documentation of this file.](#)

```

00001
00015 /* #[] License : */
00016 /*
00017 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00018 *   When using this file you are requested to refer to the publication
00019 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00020 *   This is considered a matter of courtesy as the development was paid
00021 *   for by FOM the Dutch physics granting agency and we would like to
00022 *   be able to track its scientific use to convince FOM of its value
00023 *   for the community.
00024 *
00025 *   This file is part of FORM.
00026 *
00027 *   FORM is free software: you can redistribute it and/or modify it under the
00028 *   terms of the GNU General Public License as published by the Free Software
00029 *   Foundation, either version 3 of the License, or (at your option) any later
00030 *   version.
00031 *
00032 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00033 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00034 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00035 *   details.
00036 *
00037 *   You should have received a copy of the GNU General Public License along
00038 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00039 */
00040 /* #[] License : */
00041 /*
00042 *   #[] includes :
00043 */
00044
00045 #include "form3.h"
00046
00047 /*
00048 *   com1commands are the commands of which only part of the word has to
00049 *   be present. The order is rather important here.
00050 *   com2commands are the commands that must have their whole word match.
00051 *   here we can do a binary search.
00052 *   {{{
00053 */
00054
00055 static KEYWORD com1commands[] = {
00056     {"also", (TFUN)CoIdOld, STATEMENT, PARTEST},
00057     {"abackets", (TFUN)CoAntiBracket, TOOUTPUT, PARTEST},
00058     {"antisymmetrize", (TFUN)CoAntiSymmetrize, STATEMENT, PARTEST},
00059     {"antibrackets", (TFUN)CoAntiBracket, TOOUTPUT, PARTEST},
00060     {"brackets", (TFUN)CoBracket, TOOUTPUT, PARTEST},
00061     {"cfunctions", (TFUN)CoCFunction, DECLARATION, PARTEST|WITHAUTO},
00062     {"commuting", (TFUN)CoCFunction, DECLARATION, PARTEST|WITHAUTO},
00063     {"compress", (TFUN)CoCompress, DECLARATION, PARTEST},
00064     {"ctensors", (TFUN)CoCTensor, DECLARATION, PARTEST|WITHAUTO},
00065     {"cyclesymmetrize", (TFUN)CoCycleSymmetrize, STATEMENT, PARTEST},
00066     {"dimension", (TFUN)CoDimension, DECLARATION, PARTEST},
00067     {"discard", (TFUN)CoDiscard, STATEMENT, PARTEST},
00068     {"functions", (TFUN)CoFunction, DECLARATION, PARTEST|WITHAUTO},
00069     {"format", (TFUN)CoFormat, TOOUTPUT, PARTEST},
00070     {"fixindex", (TFUN)CoFixIndex, DECLARATION, PARTEST},
00071     {"global", (TFUN)CoGlobal, DEFINITION, PARTEST}

```

```

00072     ,{"gfactorized",      (TFUN)CoGlobalFactorized, DEFINITION, PARTEST}
00073     ,{"globalfactorized", (TFUN)CoGlobalFactorized, DEFINITION, PARTEST}
00074     ,{"goto",            (TFUN)CoGoTo,          STATEMENT, PARTEST}
00075     ,{"indexes",         (TFUN)CoIndex,         DECLARATION, PARTEST|WITHAUTO}
00076     ,{"indices",         (TFUN)CoIndex,         DECLARATION, PARTEST|WITHAUTO}
00077     ,{"identify",        (TFUN)CoId,           STATEMENT, PARTEST}
00078     ,{"idnew",           (TFUN)CoIdNew,         STATEMENT, PARTEST}
00079     ,{"idold",           (TFUN)CoIdOld,         STATEMENT, PARTEST}
00080     ,{"local",           (TFUN)CoLocal,         DEFINITION, PARTEST}
00081     ,{"lfactorized",     (TFUN)CoLocalFactorized, DEFINITION, PARTEST}
00082     ,{"localfactorized", (TFUN)CoLocalFactorized, DEFINITION, PARTEST}
00083     ,{"load",            (TFUN)CoLoad,          DECLARATION, PARTEST}
00084     ,{"label",           (TFUN)CoLabel,         STATEMENT, PARTEST}
00085     ,{"modulus",         (TFUN)CoModulus,       DECLARATION, PARTEST}
00086     ,{"multiply",        (TFUN)CoMultiply,      STATEMENT, PARTEST}
00087     ,{"nfunctions",      (TFUN)CoNFunction,     DECLARATION, PARTEST|WITHAUTO}
00088     ,{"nprint",          (TFUN)CoNPrint,        TOOUTPUT, PARTEST}
00089     ,{"ntensors",        (TFUN)CoNTensor,       DECLARATION, PARTEST|WITHAUTO}
00090     ,{"nwrite",          (TFUN)CoNWrite,        DECLARATION, PARTEST}
00091     ,{"print",           (TFUN)CoPrint,         MIXED, 0}
00092     ,{"redefine",        (TFUN)CoRedefine,      STATEMENT, 0}
00093     ,{"rcyclesymmetrize", (TFUN)CoRCycleSymmetrize, STATEMENT, PARTEST}
00094     ,{"symbols",         (TFUN)CoSymbol,        DECLARATION, PARTEST|WITHAUTO}
00095     ,{"save",            (TFUN)CoSave,          DECLARATION, PARTEST}
00096     ,{"symmetrize",      (TFUN)CoSymmetrize,     STATEMENT, PARTEST}
00097     ,{"tensors",        (TFUN)CoCTensor,       DECLARATION, PARTEST|WITHAUTO}
00098     ,{"unittrace",       (TFUN)CoUnitTrace,     DECLARATION, PARTEST}
00099     ,{"vectors",         (TFUN)CoVector,        DECLARATION, PARTEST|WITHAUTO}
00100     ,{"write",           (TFUN)CoWrite,         DECLARATION, PARTEST}
00101 };
00102
00103 static KEYWORD com2commands[] = {
00104     {"antiputinside",    (TFUN)CoAntiPutInside, STATEMENT, PARTEST}
00105     ,{"apply",           (TFUN)CoApply,         STATEMENT, PARTEST}
00106     ,{"aputinside",      (TFUN)CoAntiPutInside, STATEMENT, PARTEST}
00107     ,{"argexplode",      (TFUN)CoArgExplode,     STATEMENT, PARTEST}
00108     ,{"argimplode",      (TFUN)CoArgImplode,     STATEMENT, PARTEST}
00109     ,{"argtoextrasymbol", (TFUN)CoArgToExtraSymbol, STATEMENT, PARTEST}
00110     ,{"argument",        (TFUN)CoArgument,       STATEMENT, PARTEST}
00111     ,{"assign",          (TFUN)CoAssign,         STATEMENT, PARTEST}
00112     ,{"auto",            (TFUN)CoAuto,           DECLARATION, PARTEST}
00113     ,{"autodeclare",     (TFUN)CoAuto,           DECLARATION, PARTEST}
00114     ,{"break",           (TFUN)CoBreak,         STATEMENT, PARTEST}
00115     ,{"canonicalize",    (TFUN)CoCanonicalize,   STATEMENT, PARTEST}
00116     ,{"case",            (TFUN)CoCase,          STATEMENT, PARTEST}
00117     ,{"chainin",         (TFUN)CoChainin,        STATEMENT, PARTEST}
00118     ,{"chainout",        (TFUN)CoChainout,       STATEMENT, PARTEST}
00119     ,{"chisholm",        (TFUN)CoChisholm,      STATEMENT, PARTEST}
00120     ,{"cleartable",      (TFUN)CoClearTable,     DECLARATION, PARTEST}
00121     ,{"clearflag",       (TFUN)CoClearUserFlag,  STATEMENT, PARTEST}
00122     ,{"collect",         (TFUN)CoCollect,        SPECIFICATION, PARTEST}
00123     ,{"commuteinset",    (TFUN)CoCommuteInSet,   DECLARATION, PARTEST}
00124     ,{"contract",        (TFUN)CoContract,       STATEMENT, PARTEST}
00125     ,{"copspectator",    (TFUN)CoCopySpectator,  DEFINITION, PARTEST}
00126     ,{"createall",       (TFUN)CoCreateAll,      STATEMENT, PARTEST}
00127     ,{"createspectator", (TFUN)CoCreateSpectator,  DECLARATION, PARTEST}
00128     ,{"ctable",          (TFUN)CoCTable,        DECLARATION, PARTEST}
00129     ,{"deallocate",      (TFUN)CoDeallocateTable, DECLARATION, PARTEST}
00130     ,{"default",         (TFUN)CoDefault,        STATEMENT, PARTEST}
00131     ,{"delete",          (TFUN)CoDelete,        SPECIFICATION, PARTEST}
00132     ,{"denominators",    (TFUN)CoDenominators,   STATEMENT, PARTEST}
00133     ,{"disorder",        (TFUN)CoDisorder,       STATEMENT, PARTEST}
00134     ,{"do",              (TFUN)CoDo,            STATEMENT, PARTEST}
00135     ,{"drop",            (TFUN)CoDrop,          SPECIFICATION, PARTEST}
00136     ,{"dropcoefficient", (TFUN)CoDropCoefficient,  STATEMENT, PARTEST}
00137     ,{"dropsymbols",     (TFUN)CoDropSymbols,    STATEMENT, PARTEST}
00138     ,{"else",            (TFUN)CoElse,          STATEMENT, PARTEST}
00139     ,{"elseif",          (TFUN)CoElseIf,        STATEMENT, PARTEST}
00140     ,{"emptyspectator",  (TFUN)CoEmptySpectator,  SPECIFICATION, PARTEST}
00141     ,{"endargument",     (TFUN)CoEndArgument,    STATEMENT, PARTEST}
00142     ,{"enddo",           (TFUN)CoEndDo,         STATEMENT, PARTEST}
00143     ,{"endif",           (TFUN)CoEndIf,         STATEMENT, PARTEST}
00144     ,{"endinexpression", (TFUN)CoEndInExpression,  STATEMENT, PARTEST}
00145     ,{"endinside",       (TFUN)CoEndInside,     STATEMENT, PARTEST}
00146     ,{"endmodel",        (TFUN)CoEndModel,       DECLARATION, PARTEST}
00147     ,{"endrepeat",       (TFUN)CoEndRepeat,     STATEMENT, PARTEST}
00148     ,{"endswitch",       (TFUN)CoEndSwitch,     STATEMENT, PARTEST}
00149     ,{"endterm",         (TFUN)CoEndTerm,        STATEMENT, PARTEST}
00150     ,{"endwhile",        (TFUN)CoEndWhile,      STATEMENT, PARTEST}
00151 #ifdef WITHFLOAT
00152     ,{"evaluate",        (TFUN)CoEvaluate,      STATEMENT, PARTEST}
00153 #endif
00154     ,{"exit",            (TFUN)CoExit,          STATEMENT, PARTEST}
00155     ,{"extrasymbols",    (TFUN)CoExtraSymbols,   DECLARATION, PARTEST}
00156     ,{"factarg",         (TFUN)CoFactArg,        STATEMENT, PARTEST}
00157     ,{"factdollar",      (TFUN)CoFactDollar,    STATEMENT, PARTEST}
00158     ,{"factorize",       (TFUN)CoFactorize,     TOOUTPUT, PARTEST}

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00159     ,{"fill", (TFUN) CoFill, DECLARATION, PARTEST}
00160     ,{"fillexpression", (TFUN) CoFillExpression, DECLARATION, PARTEST}
00161     ,{"frompolynomial", (TFUN) CoFromPolynomial, STATEMENT, PARTEST}
00162     ,{"funpowers", (TFUN) CoFunPowers, DECLARATION, PARTEST}
00163     ,{"hide", (TFUN) CoHide, SPECIFICATION, PARTEST}
00164     ,{"if", (TFUN) CoIf, STATEMENT, PARTEST}
00165     ,{"ifmatch", (TFUN) CoIfMatch, STATEMENT, PARTEST}
00166     ,{"ifnomatch", (TFUN) CoIfNoMatch, STATEMENT, PARTEST}
00167     ,{"ifnotmatch", (TFUN) CoIfNoMatch, STATEMENT, PARTEST}
00168     ,{"inexpression", (TFUN) CoInExpression, STATEMENT, PARTEST}
00169     ,{"inparallel", (TFUN) CoInParallel, SPECIFICATION, PARTEST}
00170     ,{"inside", (TFUN) CoInside, STATEMENT, PARTEST}
00171     ,{"insidefirst", (TFUN) CoInsideFirst, DECLARATION, PARTEST}
00172     ,{"intohide", (TFUN) CoIntoHide, SPECIFICATION, PARTEST}
00173     ,{"keep", (TFUN) CoKeep, SPECIFICATION, PARTEST}
00174     ,{"makeinteger", (TFUN) CoMakeInteger, STATEMENT, PARTEST}
00175     ,{"many", (TFUN) CoMany, STATEMENT, PARTEST}
00176     ,{"merge", (TFUN) CoMerge, STATEMENT, PARTEST}
00177     ,{"metric", (TFUN) CoMetric, DECLARATION, PARTEST}
00178     ,{"model", (TFUN) CoModel, DECLARATION, PARTEST}
00179     ,{"moduleoption", (TFUN) CoModuleOption, ATENDOFMODULE, PARTEST}
00180     ,{"multi", (TFUN) CoMulti, STATEMENT, PARTEST}
00181     ,{"multibracket", (TFUN) CoMultiBracket, STATEMENT, PARTEST}
00182     ,{"ndrop", (TFUN) CoNoDrop, SPECIFICATION, PARTEST}
00183     ,{"nfactorize", (TFUN) CoNFactorize, TOOUTPUT, PARTEST}
00184     ,{"nhide", (TFUN) CoNoHide, SPECIFICATION, PARTEST}
00185     ,{"nintohide", (TFUN) CoNoIntoHide, SPECIFICATION, PARTEST}
00186     ,{"normalize", (TFUN) CoNormalize, STATEMENT, PARTEST}
00187     ,{"notinparallel", (TFUN) CoNotInParallel, SPECIFICATION, PARTEST}
00188     ,{"nskip", (TFUN) CoNoSkip, SPECIFICATION, PARTEST}
00189     ,{"ntable", (TFUN) CoNTable, DECLARATION, PARTEST}
00190     ,{"nunfactorize", (TFUN) CoUnFactorize, TOOUTPUT, PARTEST}
00191     ,{"nunhide", (TFUN) CoNoUnHide, SPECIFICATION, PARTEST}
00192     ,{"off", (TFUN) CoOff, DECLARATION, PARTEST}
00193     ,{"on", (TFUN) CoOn, DECLARATION, PARTEST}
00194     ,{"once", (TFUN) CoOnce, STATEMENT, PARTEST}
00195     ,{"only", (TFUN) CoOnly, STATEMENT, PARTEST}
00196     ,{"particle", (TFUN) CoParticle, DECLARATION, PARTEST}
00197     ,{"polyfun", (TFUN) CoPolyFun, DECLARATION, PARTEST}
00198     ,{"polyratfun", (TFUN) CoPolyRatFun, DECLARATION, PARTEST}
00199     ,{"pophide", (TFUN) CoPopHide, SPECIFICATION, PARTEST}
00200     ,{"print[]", (TFUN) CoPrintB, TOOUTPUT, PARTEST}
00201     ,{"printtable", (TFUN) CoPrintTable, MIXED, PARTEST}
00202     ,{"processbucketsize", (TFUN) CoProcessBucket, DECLARATION, PARTEST}
00203     ,{"propercount", (TFUN) CoProperCount, DECLARATION, PARTEST}
00204     ,{"pushhide", (TFUN) CoPushHide, SPECIFICATION, PARTEST}
00205     ,{"putinside", (TFUN) CoPutInside, STATEMENT, PARTEST}
00206     ,{"ratio", (TFUN) CoRatio, STATEMENT, PARTEST}
00207     ,{"removespectator", (TFUN) CoRemoveSpectator, SPECIFICATION, PARTEST}
00208     ,{"renumber", (TFUN) CoRenumber, STATEMENT, PARTEST}
00209     ,{"repeat", (TFUN) CoRepeat, STATEMENT, PARTEST}
00210     ,{"replaceloop", (TFUN) CoReplaceLoop, STATEMENT, PARTEST}
00211     ,{"select", (TFUN) CoSelect, STATEMENT, PARTEST}
00212     ,{"set", (TFUN) CoSet, DECLARATION, PARTEST}
00213     ,{"setexitflag", (TFUN) CoSetExitFlag, STATEMENT, PARTEST}
00214     ,{"setflag", (TFUN) CoSetUserFlag, STATEMENT, PARTEST}
00215     ,{"shuffle", (TFUN) CoMerge, STATEMENT, PARTEST}
00216     ,{"skip", (TFUN) CoSkip, SPECIFICATION, PARTEST}
00217     ,{"sort", (TFUN) CoSort, STATEMENT, PARTEST}
00218     ,{"splitarg", (TFUN) CoSplitArg, STATEMENT, PARTEST}
00219     ,{"splitfirstarg", (TFUN) CoSplitFirstArg, STATEMENT, PARTEST}
00220     ,{"splitlastarg", (TFUN) CoSplitLastArg, STATEMENT, PARTEST}
00221     ,{"shuffle", (TFUN) CoShuffle, STATEMENT, PARTEST}
00222     ,{"sum", (TFUN) CoSum, STATEMENT, PARTEST}
00223     ,{"switch", (TFUN) CoSwitch, STATEMENT, PARTEST}
00224     ,{"table", (TFUN) CoTable, DECLARATION, PARTEST}
00225     ,{"tablebase", (TFUN) CoTableBase, DECLARATION, PARTEST}
00226     ,{"tb", (TFUN) CoTableBase, DECLARATION, PARTEST}
00227     ,{"term", (TFUN) CoTerm, STATEMENT, PARTEST}
00228     ,{"testuse", (TFUN) CoTestUse, STATEMENT, PARTEST}
00229     ,{"threadbucketsize", (TFUN) CoThreadBucket, DECLARATION, PARTEST}
00230 #ifdef WITHFLOAT
00231     ,{"tofloat", (TFUN) CoToFloat, STATEMENT, PARTEST}
00232 #endif
00233     ,{"topolynomial", (TFUN) CoToPolynomial, STATEMENT, PARTEST}
00234 #ifdef WITHFLOAT
00235     ,{"torat", (TFUN) CoToRat, STATEMENT, PARTEST}
00236     ,{"torational", (TFUN) CoToRat, STATEMENT, PARTEST}
00237 #endif
00238     ,{"tospectator", (TFUN) CoToSpectator, STATEMENT, PARTEST}
00239     ,{"totensor", (TFUN) CoToTensor, STATEMENT, PARTEST}
00240     ,{"tovector", (TFUN) CoToVector, STATEMENT, PARTEST}
00241     ,{"trace4", (TFUN) CoTrace4, STATEMENT, PARTEST}
00242     ,{"tracen", (TFUN) CoTraceN, STATEMENT, PARTEST}
00243     ,{"transform", (TFUN) CoTransform, STATEMENT, PARTEST}
00244     ,{"tryreplace", (TFUN) CoTryReplace, STATEMENT, PARTEST}
00245     ,{"unfactorize", (TFUN) CoUnFactorize, TOOUTPUT, PARTEST}

```

```

00246     ,{"unhide",          (TFUN)CoUnHide,          SPECIFICATION,PARTEST}
00247     ,{"vertex",          (TFUN)CoVertex,          DECLARATION,  PARTEST}
00248     ,{"while",           (TFUN)CoWhile,           STATEMENT,    PARTEST}
00249 };
00250
00251 int alfatable1[27];
00252
00253 #define OPTION0 1
00254 #define OPTION1 2
00255 #define OPTION2 3
00256
00257 typedef struct SuBbUf {
00258     WORD      subexpnum;
00259     WORD      buffernum;
00260 } SUBBUF;
00261
00262 SUBBUF *subexpbuffers = 0;
00263 SUBBUF *topsubexpbuffers = 0;
00264 LONG insubexpbuffers = 0;
00265
00266 #define REDUCESUBEXPBUFFERS { if ( (topsubexpbuffers-subexpbuffers) > 256 ) {\
00267     M_free(subexpbuffers,"subexpbuffers");\
00268     subexpbuffers = (SUBBUF *)Malloca1(256*sizeof(SUBBUF),"subexpbuffers");\
00269     topsubexpbuffers = subexpbuffers+256; } insubexpbuffers = 0; }
00270
00271 #if BITSINWORD == 32
00272 #define PUTNUMBER128(t,n) { if ( n >= 2097152 ) { \
00273     *t++ = ((n/128)/128)/128; *t++ = ((n/128)/128)%128; *t++ = (n/128)%128; *t++ = n%128;
00274 } \
00275     else if ( n >= 16384 ) { \
00276     *t++ = n/(128*128); *t++ = (n/128)%128; *t++ = n%128; } \
00277     else if ( n >= 128 ) { *t++ = n/128; *t++ = n%128; } \
00278     else *t++ = n; }
00279 #define PUTNUMBER100(t,n) { if ( n >= 1000000 ) { \
00280     *t++ = ((n/100)/100)/100; *t++ = ((n/100)/100)%100; *t++ = (n/100)%100; *t++ = n%100;
00281 } \
00282     else if ( n >= 10000 ) { \
00283     *t++ = n/10000; *t++ = (n/100)%100; *t++ = n%100; } \
00284     else if ( n >= 100 ) { *t++ = n/100; *t++ = n%100; } \
00285     else *t++ = n; }
00286 #elif BITSINWORD == 16
00287 #define PUTNUMBER128(t,n) { if ( n >= 16384 ) { \
00288     *t++ = n/(128*128); *t++ = (n/128)%128; *t++ = n%128; } \
00289     else if ( n >= 128 ) { *t++ = n/128; *t++ = n%128; } \
00290     else *t++ = n; }
00291 #define PUTNUMBER100(t,n) { if ( n >= 10000 ) { \
00292     *t++ = n/10000; *t++ = (n/100)%100; *t++ = n%100; } \
00293     else if ( n >= 100 ) { *t++ = n/100; *t++ = n%100; } \
00294     else *t++ = n; }
00295 #else
00296 #error Only 64-bit and 32-bit platforms are supported.
00297 #endif
00298
00299 /*
00300 #] includes :
00301 #[ Compiler :
00302     #[ inictable :
00303
00304     Routine sets the table for 1-st characters that allow a faster
00305     start in the search in table 1 which should be sequential.
00306     Search in table 2 can be binary.
00307 */
00308 void inictable(void)
00309 {
00310     KEYWORD *k = com1commands;
00311     int i, j, ksize;
00312     ksize = sizeof(com1commands)/sizeof(KEYWORD);
00313     j = 0;
00314     alfatable1[0] = 0;
00315     for ( i = 0; i < 26; i++ ) {
00316         while ( j < ksize && k[j].name[0] == 'a'+i ) j++;
00317         alfatable1[i+1] = j;
00318     }
00319 }
00320
00321 /*
00322 #] inictable :
00323 #[ findcommand :
00324
00325     Checks whether a command is in the command table.
00326     If so a pointer to the table element is returned.
00327     If not we return 0.
00328     Note that when a command is not in the table, we have
00329     to test whether it is an id command without id. It should
00330     then have the structure pattern = rhs. This should be done

```

```

00331         in the calling routine.
00332 */
00333
00334 KEYWORD *findcommand(UBYTE *in)
00335 {
00336     int hi, med, lo, i;
00337     UBYTE *s, c;
00338     s = in;
00339     while ( FG.cTable[*s] <= 1 ) s++;
00340     if ( s > in && *s == '[' && s[1] == ']' ) s += 2;
00341     if ( *s ) { c = *s; *s = 0; }
00342     else c = 0;
00343 /*
00344     First do a binary search in the second table
00345 */
00346     lo = 0;
00347     hi = sizeof(com2commands)/sizeof(KEYWORD)-1;
00348     do {
00349         med = ( hi + lo ) / 2;
00350         i = StrICmp(in, (UBYTE *)com2commands[med].name);
00351         if ( i == 0 ) { if ( c ) *s = c; return(com2commands+med); }
00352         if ( i < 0 ) hi = med-1;
00353         else lo = med+1;
00354     } while ( hi >= lo );
00355 /*
00356     Now do a 'hashed' search in the first table. It is sequential.
00357 */
00358     i = tolower(*in) - 'a';
00359     med = alfatable1[i];
00360     hi = alfatable1[i+1];
00361     while ( med < hi ) {
00362         if ( StrICont(in, (UBYTE *)com1commands[med].name) == 0 )
00363             { if ( c ) *s = c; return(com1commands+med); }
00364         med++;
00365     }
00366     if ( c ) *s = c;
00367 /*
00368     Unrecognized. Too bad!
00369 */
00370     return(0);
00371 }
00372
00373 /*
00374     #] findcommand :
00375     #[ ParenthesesTest :
00376 */
00377
00378 int ParenthesesTest(UBYTE *sin)
00379 {
00380     WORD L1 = 0, L2 = 0, L3 = 0;
00381     UBYTE *s = sin;
00382     while ( *s ) {
00383         if ( *s == '[' ) L1++;
00384         else if ( *s == ']' ) {
00385             L1--;
00386             if ( L1 < 0 ) { MesPrint("&Unmatched []"); return(1); }
00387         }
00388         s++;
00389     }
00390     if ( L1 > 0 ) { MesPrint("&Unmatched []"); return(1); }
00391     s = sin;
00392     while ( *s ) {
00393         if ( *s == '[' ) SKIPBRA1(s)
00394         else if ( *s == '(' ) { L2++; s++; }
00395         else if ( *s == ')' ) {
00396             L2--; s++;
00397             if ( L2 < 0 ) { MesPrint("&Unmatched ()"); return(1); }
00398         }
00399         else s++;
00400     }
00401     if ( L2 > 0 ) { MesPrint("&Unmatched ()"); return(1); }
00402     s = sin;
00403     while ( *s ) {
00404         if ( *s == '[' ) SKIPBRA1(s)
00405         else if ( *s == '[' ) SKIPBRA4(s)
00406         else if ( *s == '{' ) { L3++; s++; }
00407         else if ( *s == '}' ) {
00408             L3--; s++;
00409             if ( L3 < 0 ) { MesPrint("&Unmatched {}"); return(1); }
00410         }
00411         else s++;
00412     }
00413     if ( L3 > 0 ) { MesPrint("&Unmatched {}"); return(1); }
00414     return(0);
00415 }
00416
00417 /*

```

```

00418         #] ParenthesesTest :
00419         #[ SkipAName :
00420
00421         Skips a name and gives a pointer to the object after the name.
00422         If there is not a proper name, it returns a zero pointer.
00423         In principle the brackets match already, so the `if ( *s == 0 )'
00424         code is not really needed, but you never know how the program
00425         is extended later.
00426     */
00427
00428     UBYTE *SkipAName(UBYTE *s)
00429     {
00430         UBYTE *t = s;
00431         if ( *s == '[' ) {
00432             SKIPBRA1(s)
00433             if ( *s == 0 ) {
00434                 MesPrint("&Illegal name: '%s'",t);
00435                 return(0);
00436             }
00437             s++;
00438         }
00439         else if ( FG.cTable[*s] == 0 || *s == '_' || *s == '$' ) {
00440             if ( *s == '$' ) s++;
00441             while ( FG.cTable[*s] <= 1 ) s++;
00442             if ( *s == '_' ) s++;
00443         }
00444         else {
00445             MesPrint("&Illegal name: '%s'",t);
00446             return(0);
00447         }
00448         return(s);
00449     }
00450
00451     /*
00452         #] SkipAName :
00453         #[ IsRHS :
00454     */
00455
00456     UBYTE *IsRHS(UBYTE *s, UBYTE c)
00457     {
00458         while ( *s && *s != c ) {
00459             if ( *s == '[' ) {
00460                 SKIPBRA1(s);
00461                 if ( *s != ']' ) {
00462                     MesPrint("&Unmatched []");
00463                     return(0);
00464                 }
00465             }
00466             else if ( *s == '{' ) {
00467                 SKIPBRA2(s);
00468                 if ( *s != '}' ) {
00469                     MesPrint("&Unmatched {}");
00470                     return(0);
00471                 }
00472             }
00473             else if ( *s == '(' ) {
00474                 SKIPBRA3(s);
00475                 if ( *s != ')' ) {
00476                     MesPrint("&Unmatched ()");
00477                     return(0);
00478                 }
00479             }
00480             else if ( *s == ')' ) {
00481                 MesPrint("&Unmatched ()");
00482                 return(0);
00483             }
00484             else if ( *s == '}' ) {
00485                 MesPrint("&Unmatched {}");
00486                 return(0);
00487             }
00488             else if ( *s == ']' ) {
00489                 MesPrint("&Unmatched []");
00490                 return(0);
00491             }
00492             s++;
00493         }
00494         return(s);
00495     }
00496
00497     /*
00498         #] IsRHS :
00499         #[ IsIdStatement :
00500     */
00501
00502     int IsIdStatement(UBYTE *s)
00503     {
00504         DUMMYUSE(s);

```

```

00505     return(0);
00506 }
00507
00508 /*
00509     #] IsIdStatement :
00510     #[ CompileAlgebra :
00511
00512     Returns either the number of the main level RHS (>= 0)
00513     or an error code (< 0)
00514 */
00515
00516 int CompileAlgebra(UBYTE *s, int leftright, WORD *prototype)
00517 {
00518     GETIDENTITY
00519     int error;
00520     WORD *oldproto = AC.ProtoType;
00521     AC.ProtoType = prototype;
00522     if ( AC.TokensWriteFlag ) {
00523         MesPrint("To tokenize: %s",s);
00524         error = tokenize(s,leftright);
00525         MesPrint(" The contents of the token buffer are:");
00526         WriteTokens(AC.tokens);
00527     }
00528     else error = tokenize(s,leftright);
00529     if ( error == 0 ) {
00530         AR.Eside = leftright;
00531         AC.CompileLevel = 0;
00532         if ( leftright == LHSIDE ) { AC.DumNum = AR.CurDum = 0; }
00533         error = CompileSubExpressions(AC.tokens);
00534         REDUCESUBEXPBUFFERS
00535     }
00536     else {
00537         AC.ProtoType = oldproto;
00538         return(-1);
00539     }
00540     AC.ProtoType = oldproto;
00541     if ( error < 0 ) return(-1);
00542     else if ( leftright == LHSIDE ) return(cbuf[AC.cbufnum].numlhs);
00543     else return(cbuf[AC.cbufnum].numrhs);
00544 }
00545
00546 /*
00547     #] CompileAlgebra :
00548     #[ CompileStatement :
00549
00550 */
00551
00552 int CompileStatement(UBYTE *in)
00553 {
00554     KEYWORD *k;
00555     UBYTE *s;
00556     int error1 = 0, error2;
00557     /* A.iStatement = */ s = in;
00558     if ( *s == 0 ) return(0);
00559     if ( *s == '$' ) {
00560         k = findcommand((UBYTE *)"assign");
00561     }
00562     else {
00563         if ( ( k = findcommand(s) ) == 0 && IsIdStatement(s) == 0 ) {
00564             MesPrint("&Unrecognized statement %s",s);
00565             return(1);
00566         }
00567         if ( k == 0 ) { /* Id statement without id. Note: id must be in table */
00568             k = com1commands + alfatable1['i'-'a'];
00569             while ( k->name[1] != 'd' || k->name[2] ) k++;
00570         }
00571         else {
00572             while ( FG.cTable[*s] <= 1 ) s++;
00573             if ( s > in && *s == '[' && s[1] == ']' ) s += 2;
00574         }
00575         /*
00576             The next statement is rather mysterious
00577             It is undone in DoPrint and CoMultiply, but it also causes effects
00578             in other (wrong) statements like dimension -4; or Trace4 -1;
00579             The code in pre.c (LoadStatement) has been changed 8-sep-2009
00580             to force a comma after the keyword. This means that the
00581             'mysterious' line is automatically inactive. Hence it is taken out.
00582
00583             if ( *s == '+' || *s == '-' ) s++;
00584         */
00585         if ( *s == ',' ) s++;
00586     }
00587     /*
00588         First the test on the order of the statements.
00589         This is relatively new (2.2c) and may cause some problems with old
00590         programs. Hence the first error message should explain!
00591     */

```

```

00592     if ( AP.PreAssignFlag == 0 && AM.OldOrderFlag == 0 ) {
00593         if ( AP.PreInsideLevel ) {
00594             if ( k->type != STATEMENT && k->type != MIXED ) {
00595                 MesPrint("&Only executable and print statements are allowed in an %#inside/%#endinside
construction");
00596                 return(-1);
00597             }
00598         }
00599     else {
00600         if ( ( AC.compiletype == DECLARATION || AC.compiletype == SPECIFICATION )
00601             && ( k->type == STATEMENT || k->type == DEFINITION || k->type == TOOOUTPUT ) ) {
00602             if ( AC.tablecheck == 0 ) {
00603                 AC.tablecheck = 1;
00604                 if ( TestTables() ) error1 = 1;
00605             }
00606         }
00607         if ( k->type == MIXED ) {
00608             if ( AC.compiletype <= DEFINITION ) {
00609                 AC.compiletype = STATEMENT;
00610             }
00611         }
00612         else if ( k->type == MIXED2 ) {}
00613         else if ( k->type > AC.compiletype ) {
00614             if ( StrCmp((UBYTE *) (k->name), (UBYTE *) "format") != 0 )
00615                 AC.compiletype = k->type;
00616         }
00617         else if ( k->type < AC.compiletype ) {
00618             switch ( k->type ) {
00619                 case DECLARATION:
00620                     MesPrint("&Declaration out of order");
00621                     MesPrint("&  %s",in);
00622                     break;
00623                 case DEFINITION:
00624                     MesPrint("&Definition out of order");
00625                     MesPrint("&  %s",in);
00626                     break;
00627                 case SPECIFICATION:
00628                     MesPrint("&Specification out of order");
00629                     MesPrint("&  %s",in);
00630                     break;
00631                 case STATEMENT:
00632                     MesPrint("&Statement out of order");
00633                     break;
00634                 case TOOOUTPUT:
00635                     MesPrint("&Output control statement out of order");
00636                     MesPrint("&  %s",in);
00637                     break;
00638             }
00639             AC.compiletype = k->type;
00640             if ( AC.firstctypemessage == 0 ) {
00641                 MesPrint("&Proper order inside a module is:");
00642                 MesPrint("Declarations, specifications, definitions, statements, output control
statements");
00643                 AC.firstctypemessage = 1;
00644             }
00645             error1 = 1;
00646         }
00647     }
00648 }
00649 /*
00650 Now we execute the tests that are prescribed by the flags.
00651 */
00652 if ( AC.AutoDeclareFlag && ( ( k->flags & WITHAUTO ) == 0 ) ) {
00653     MesPrint("&Illegal type of auto-declaration");
00654     return(1);
00655 }
00656 if ( ( ( k->flags & PARTEST ) != 0 ) && ParenthesesTest(s) ) return(1);
00657 error2 = (*k->func)(s);
00658 if ( error2 == 0 ) return(error1);
00659 return(error2);
00660 }
00661 /*
00662 #] CompileStatement :
00663 #[ TestTables :
00664 */
00665
00666 int TestTables(void)
00667 {
00668     FUNCTIONS f = functions;
00669     TABLES t;
00670     WORD j;
00671     int error = 0, i;
00672     LONG x;
00673     i = NumFunctions + FUNCTION - MAXBUILTINFUNCTION - 1;
00674     f = f + MAXBUILTINFUNCTION - FUNCTION + 1;
00675     if ( AC.MustTestTable > 0 ) {

```



```

00677     while ( i > 0 ) {
00678         if ( ( t = f->tabl ) != 0 && t->strict > 0 && !t->sparse ) {
00679             for ( x = 0, j = 0; x < t->totind; x++ ) {
00680                 if ( t->tablepointers[TABLEEXTENSION*x] < 0 ) j++;
00681             }
00682             if ( j > 0 ) {
00683                 if ( j > 1 ) {
00684                     MesPrint("&In table %s there are %d unfilled elements",
00685                         AC.varnames->namebuffer+f->name, j);
00686                 }
00687                 else {
00688                     MesPrint("&In table %s there is one unfilled element",
00689                         AC.varnames->namebuffer+f->name);
00690                 }
00691                 error = 1;
00692             }
00693         }
00694         i--; f++;
00695     }
00696     AC.MustTestTable--;
00697 }
00698 return(error);
00699 }
00700
00701 /*
00702     #] TestTables :
00703     #[ CompileSubExpressions :
00704
00705     Now we attack the subexpressions from inside out.
00706     We try to see whether we had any of them already.
00707     We have to worry about adding the wildcard sum parameter
00708     to the prototype.
00709 */
00710
00711 int CompileSubExpressions(SBYTE *tokens)
00712 {
00713     GETIDENTITY
00714     SBYTE *fill = tokens, *s = tokens, *t;
00715     WORD number[MAXNUMSIZE], *oldwork, *w1, *w2;
00716     int level, num, i, sumlevel = 0, sumtype = SYMTOSYM;
00717     int retval, error = 0;
00718 /*
00719     Eliminate all subexpressions. They are marked by LPARENTHESIS, RPARENTHESIS
00720 */
00721     AC.CompileLevel++;
00722     while ( *s != TENDOFIT ) {
00723         if ( *s == TFUNOPEN ) {
00724             if ( fill < s ) *fill = TENDOFIT;
00725             t = fill - 1;
00726             while ( t >= tokens && t[0] >= 0 ) t--;
00727             if ( t >= tokens && *t == TFUNCTION ) {
00728                 t++; i = 0; while ( *t >= 0 ) i = 128*i + *t++;
00729                 if ( i == AM.sumnum || i == AM.sumpnum ) {
00730                     t = s + 1;
00731                     if ( *t == TSYMBOL || *t == TINDEX ) {
00732                         t++; i = 0; while ( *t >= 0 ) i = 128*i + *t++;
00733                         if ( s[1] == TINDEX ) {
00734                             i += AM.OffsetIndex;
00735                             sumtype = INDTOIND;
00736                         }
00737                         else sumtype = SYMTOSYM;
00738                         sumlevel = i;
00739                     }
00740                 }
00741             }
00742             *fill++ = *s++;
00743         }
00744         else if ( *s == TFUNCLOSE ) { sumlevel = 0; *fill++ = *s++; }
00745         else if ( *s == LPARENTHESIS ) {
00746 /*
00747             We must make an exception here.
00748             If the subexpression is just an integer, whatever its length,
00749             we should try to keep it.
00750             This is important when we have a function with an integer
00751             argument. In particular this is relevant for the MZV program.
00752 */
00753             t = s; level = 0;
00754             while ( level >= 0 ) {
00755                 s++;
00756                 if ( *s == LPARENTHESIS ) level++;
00757                 else if ( *s == RPARENTHESIS ) level--;
00758                 else if ( *s == TENDOFIT ) {
00759                     MesPrint("&Unbalanced subexpression parentheses");
00760                     return(-1);
00761                 }
00762             }
00763             t++; *s = TENDOFIT;

```

```

00764         if ( sumlevel > 0 ) { /* Inside sum. Add wildcard to prototype */
00765             oldwork = w1 = AT.WorkPointer;
00766             w2 = AC.ProtoType;
00767             i = w2[1];
00768             while ( --i >= 0 ) *w1++ = *w2++;
00769             oldwork[1] += 4;
00770             *w1++ = sumtype; *w1++ = 4; *w1++ = sumlevel; *w1++ = sumlevel;
00771             w2 = AC.ProtoType; AT.WorkPointer = w1;
00772             AC.ProtoType = oldwork;
00773             num = CompileSubExpressions(t);
00774             AC.ProtoType = w2; AT.WorkPointer = oldwork;
00775         }
00776         else num = CompileSubExpressions(t);
00777         if ( num < 0 ) return(-1);
00778     /*
00779         Note that the subexpression code should always fit.
00780         We had two parentheses and at least two bytes contents.
00781         There cannot be more than 2^21 subexpressions or we get outside
00782         this minimum. Ignoring this might lead to really rare and
00783         hard to find errors, years from now.
00784     */
00785         if ( insubexpbuffers >= MAXSUBEXPRESSIONS ) {
00786             MesPrint("&More than %d subexpressions inside one
expression", (WORD)MAXSUBEXPRESSIONS);
00787             Terminate(-1);
00788         }
00789         if ( subexpbuffers+insubexpbuffers >= topsubexpbuffers ) {
00790             DoubleBuffer((void **) ((void *) (&subexpbuffers))
00791                 , (void **) ((void *) (&topsubexpbuffers)), sizeof(SUBBUF), "subexpbuffers");
00792         }
00793         subexpbuffers[insubexpbuffers].subexpnum = num;
00794         subexpbuffers[insubexpbuffers].buffernum = AC.cbufnum;
00795         num = insubexpbuffers++;
00796         *fill++ = TSUBEXP;
00797         i = 0;
00798         do { number[i++] = num & 0x7F; num >= 7; } while ( num );
00799         while ( --i >= 0 ) *fill++ = (SBYTE)(number[i]);
00800         s++;
00801     }
00802     else if ( *s == EMPTY ) s++;
00803     else *fill++ = *s++;
00804 }
00805 *fill = TENDOFIT;
00806 /*
00807     At this stage there are no more subexpressions.
00808     Hence we can do the basic compilation.
00809 */
00810     if ( AC.CompileLevel == 1 && AC.ToBeInFactors ) {
00811         error = CodeFactors(tokens);
00812     }
00813     AC.CompileLevel--;
00814     retval = CodeGenerator(tokens);
00815     if ( error < 0 ) return(error);
00816     return(retval);
00817 }
00818
00819 /*
00820     #] CompileSubExpressions :
00821     #[ CodeGenerator :
00822
00823     This routine does the real code generation.
00824     It returns the number of the rhs subexpression.
00825     At this point we do not have to worry about subexpressions,
00826     sets, setelements, simple vs complicated function arguments
00827     simple vs complicated powers etc.
00828
00829     The variable 'first' indicates whether we are starting a new term
00830
00831     The major complication are the set elements of type set[n].
00832     We have marked them as TSETNUM,n,Ttype,setnum
00833     They go into
00834     SETSET,size,subterm,relocation list
00835     in which the subterm should be ready to become a regular
00836     subterm in which the sets have been replaced by their element
00837     The relocation list consists of pairs of numbers:
00838     1: offset in the subterm, 2: the symbol n.
00839     Note that such a subterm can be a whole function with its arguments.
00840     We use the variable inset to indicate that we have something going.
00841     The relocation list is collected in the top of the Workspace.
00842 */
00843
00844 static UWORD *CGscrat7 = 0;
00845
00846 int CodeGenerator(SBYTE *tokens)
00847 {
00848     GETIDENTITY
00849     SBYTE *s = tokens, c;

```

```

00850     int i, sign = 1, first = 1, deno = 1, error = 0, minus, n, needarg, numexp, cc;
00851     int base, sumlevel = 0, sumtype = SYMTOSYM, firstsumarg, inset = 0;
00852     int funflag = 0, settype, x1, x2, mulflag = 0;
00853     WORD *t, *v, *r, *term, nnumerator, ndenominator, *oldwork, x3, y, nin;
00854     WORD *w1, *w2, *tsize = 0, *relo = 0;
00855     UWORD *numerator, *denominator, *innum;
00856     CBUF *C;
00857     POSITION position;
00858     WORD TmpProto[SUBEXPSIZE];
00859     /*
00860     #ifdef WITHPTHREADS
00861         RENUMBER renumber;
00862     #endif
00863     */
00864     RENUMBER renumber;
00865     if ( AC.TokensWriteFlag ) WriteTokens(tokens);
00866     if ( CGscrat7 == 0 )
00867         CGscrat7 = (UWORD *)Malloc1((AM.MaxTal+2)*sizeof(WORD),"CodeGenerator");
00868     AddrRHS(AC.cbunum,0);
00869     C = cbuf + AC.cbunum;
00870     numexp = C->numrhs;
00871     C->NumTerms[numexp] = 0;
00872     C->numdum[numexp] = 0;
00873     oldwork = AT.WorkPointer;
00874     numerator = (UWORD *) (AT.WorkPointer);
00875     denominator = numerator + 2*AM.MaxTal;
00876     innum = denominator + 2*AM.MaxTal;
00877     term = (WORD *) (innum + 2*AM.MaxTal);
00878     AT.WorkPointer = term + AM.MaxTer/sizeof(WORD);
00879     if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
00880     cc = 0;
00881     t = term+1;
00882     numerator[0] = denominator[0] = 1;
00883     nnumerator = ndenominator = 1;
00884     while ( *s != TENDOFIT ) {
00885         if ( *s == TPLUS || *s == TMINUS ) {
00886             if ( first || mulflag ) { if ( *s == TMINUS ) sign = -sign; }
00887             else {
00888                 *term = t-term;
00889                 C->NumTerms[numexp]++;
00890                 if ( cc && sign ) C->CanCommu[numexp]++;
00891                 CompleteTerm(term,numerator,denominator,nnumerator,ndenominator,sign);
00892                 first = 1; cc = 0; t = term + 1; deno = 1;
00893                 numerator[0] = denominator[0] = 1;
00894                 nnumerator = ndenominator = 1;
00895                 if ( *s == TMINUS ) sign = -1;
00896                 else sign = 1;
00897             }
00898             s++;
00899         }
00900         else {
00901             mulflag = first = 0; c = *s++;
00902             switch ( c ) {
00903             case TSYMBOL:
00904                 x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
00905                 if ( *s == TWILDCARD ) { s++; x1 += 2*MAXPOWER; }
00906                 *t++ = SYMBOL; *t++ = 4; *t++ = x1;
00907                 if ( inset ) *relo = 2;
00908             TryPower:
00909                 if ( *s == TPOWER ) {
00910                     s++;
00911                     if ( *s == TMINUS ) { s++; deno = -deno; }
00912                     c = *s++;
00913                     base = ( c == TNUMBER ) ? 100: 128;
00914                     x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
00915                     if ( c == TSYMBOL ) {
00916                         if ( *s == TWILDCARD ) s++;
00917                         x2 += 2*MAXPOWER;
00918                     }
00919                     *t++ = deno*x2;
00920                 }
00921                 else *t++ = deno;
00922             fin:
00923                 deno = 1;
00924                 if ( inset ) {
00925                     while ( relo < AT.WorkTop ) *t++ = *relo++;
00926                     inset = 0; tsize[1] = t - tsize;
00927                 }
00928                 break;
00929             case TINDEX:
00930                 x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
00931                 *t++ = INDEX; *t++ = 3;
00932                 if ( *s == TWILDCARD ) { s++; x1 += WILDOFFSET; }
00933                 if ( inset ) { *t++ = x1; *relo = 2; }
00934                 else *t++ = x1 + AM.OffsetIndex;
00935                 if ( t[-1] > AM.IndDum ) {
00936                     x1 = t[-1] - AM.IndDum;
00937                     if ( x1 > C->numdum[numexp] ) C->numdum[numexp] = x1;
00938                 }
00939             }
00940         }
00941     }

```

```

00937         goto fin;
00938     case TGENINDEX:
00939         *t++ = INDEX; *t++ = 3; *t++ = AC.DumNum+WILDOFFSET;
00940         deno = 1;
00941         break;
00942     case TVECTOR:
00943         x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
00944     dovector:
00945         if ( inset == 0 ) x1 += AM.OffsetVector;
00946         if ( *s == TWILDCARD ) { s++; x1 += WILDOFFSET; }
00947         if ( inset ) *relo = 2;
00948         if ( *s == TDOT ) { /* DotProduct ? */
00949             s++;
00950             if ( *s == TSETNUM || *s == TSETDOL ) {
00951                 settype = ( *s == TSETDOL );
00952                 s++; x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
00953                 if ( settype ) x2 = -x2;
00954                 if ( inset == 0 ) {
00955                     tsize = t; *t++ = SETSET; *t++ = 0;
00956                     relo = AT.WorkTop;
00957                 }
00958                 inset += 2;
00959                 *--relo = x2; *--relo = 3;
00960             }
00961             if ( *s != TVECTOR && *s != TDUBIOUS ) {
00962                 MesPrint("&Illegally formed dotproduct");
00963                 error = 1;
00964             }
00965             s++; x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
00966             if ( inset < 2 ) x2 += AM.OffsetVector;
00967             if ( *s == TWILDCARD ) { s++; x2 += WILDOFFSET; }
00968             *t++ = DOTPRODUCT; *t++ = 5; *t++ = x1; *t++ = x2;
00969             goto TryPower;
00970         }
00971     else if ( *s == TFUNOPEN ) {
00972         s++;
00973         if ( *s == TSETNUM || *s == TSETDOL ) {
00974             settype = ( *s == TSETDOL );
00975             s++; x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
00976             if ( settype ) x2 = -x2;
00977             if ( inset == 0 ) {
00978                 tsize = t; *t++ = SETSET; *t++ = 0;
00979                 relo = AT.WorkTop;
00980             }
00981             inset += 2;
00982             *--relo = x2; *--relo = 3;
00983         }
00984         if ( *s == TINDEX || *s == TDUBIOUS ) {
00985             s++;
00986             x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
00987             if ( inset < 2 ) x2 += AM.OffsetIndex;
00988             if ( *s == TWILDCARD ) { s++; x2 += WILDOFFSET; }
00989             *t++ = VECTOR; *t++ = 4; *t++ = x1; *t++ = x2;
00990             if ( t[-1] > AM.IndDum ) {
00991                 x2 = t[-1] - AM.IndDum;
00992                 if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
00993             }
00994         }
00995     else if ( *s == TGENINDEX ) {
00996         *t++ = VECTOR; *t++ = 4; *t++ = x1;
00997         *t++ = AC.DumNum + WILDOFFSET;
00998     }
00999     else if ( *s == TNUMBER || *s == TNUMBER1 ) {
01000         base = ( *s == TNUMBER ) ? 100: 128;
01001         s++;
01002         x2 = 0; while ( *s >= 0 ) { x2 = x2*base + *s++; }
01003         if ( x2 >= AM.OffsetIndex && inset < 2 ) {
01004             MesPrint("&Fixed index in vector greater than %d",
01005                 AM.OffsetIndex);
01006             return(-1);
01007         }
01008         *t++ = VECTOR; *t++ = 4; *t++ = x1; *t++ = x2;
01009     }
01010     else if ( *s == TVECTOR || ( *s == TMINUS && s[1] == TVECTOR ) ) {
01011         if ( *s == TMINUS ) { s++; sign = -sign; }
01012         s++;
01013         x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
01014         if ( inset < 2 ) x2 += AM.OffsetVector;
01015         if ( *s == TWILDCARD ) { s++; x2 += WILDOFFSET; }
01016         *t++ = DOTPRODUCT; *t++ = 5; *t++ = x1; *t++ = x2; *t++ = deno;
01017     }
01018     else {
01019         MesPrint("&Illegal argument for vector");
01020         return(-1);
01021     }
01022     if ( *s != TFUNCLOSE ) {
01023         MesPrint("&Illegal argument for vector");
01024         return(-1);

```

```

01024         }
01025         s++;
01026     }
01027     else {
01028         if ( AC.DumNum ) {
01029             *t++ = VECTOR; *t++ = 4; *t++ = x1;
01030             *t++ = AC.DumNum + WILDOFFSET;
01031         }
01032         else {
01033             *t++ = INDEX; *t++ = 3; *t++ = x1;
01034         }
01035     }
01036     goto fin;
01037 case TDELTA:
01038     if ( *s != TFUNOPEN ) {
01039         MesPrint("&d_ needs two arguments");
01040         error = -1;
01041     }
01042     v = t; *t++ = DELTA; *t++ = 4;
01043     needarg = 2; x3 = x1 = -1;
01044     goto dotensor;
01045 case TFUNCTION:
01046     x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
01047     if ( x1 == AM.sumnum || x1 == AM.sumpnum ) sumlevel = x1;
01048     x1 += FUNCTION;
01049     if ( x1 == FIRSTBRACKET ) {
01050         if ( s[0] == TFUNOPEN && s[1] == TEXPRESSON ) {
01051 doexpr:
01052             s += 2;
01053             *t++ = x1; *t++ = FUNHEAD+2; *t++ = 0;
01054             if ( x1 == AR.PolyFun && AR.PolyFunType == 2 && AR.Eside != LHSIDE )
01055                 t[-1] |= MUSTCLEANPRF;
01056             FILLFUN3(t)
01057             x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
01058             *t++ = -EXPRESSION; *t++ = x2;
01059
01060             /*
01061             The next code is added to facilitate parallel processing
01062             We need to call GetTable here to make sure all processors
01063             have the same numbering of all variables.
01064             */
01065             if ( Expressions[x2].status == STOREDEXPRESSION ) {
01066                 TMproto[0] = EXPRESSION;
01067                 TMproto[1] = SUBEXPSIZE;
01068                 TMproto[2] = x2;
01069                 TMproto[3] = 1;
01070                 { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
01071                 AT.TMaddr = TMproto;
01072                 PUTZERO(position);
01073
01074                 if ( (
01075 #ifdef WITHPTHREADS
01076                     renumber =
01077                     GetTable(x2,&position,0) ) == 0 ) {
01078                         error = 1;
01079                         MesPrint("&Problems getting information about stored expression %s(1)"
01080                             ,EXPRNAME(x2));
01081                     }
01082 #ifdef WITHPTHREADS
01083                     M_free(renumber->symb.lo,"VarSpace");
01084                     M_free(renumber,"Renumber");
01085                 #endif
01086
01087                 if ( ( renumber = GetTable(x2,&position,0) ) == 0 ) {
01088                     error = 1;
01089                     MesPrint("&Problems getting information about stored expression %s(1)"
01090                         ,EXPRNAME(x2));
01091                 }
01092                 if ( renumber->symb.lo != AN.dummyrenumlist )
01093                     M_free(renumber->symb.lo,"VarSpace");
01094                 M_free(renumber,"Renumber");
01095                 AR.StoreData.dirtyflag = 1;
01096             }
01097         if ( *s != TFUNCLOSE ) {
01098             if ( x1 == FIRSTBRACKET )
01099                 MesPrint("&Problems with argument of FirstBracket_");
01100             else if ( x1 == FIRSTTERM )
01101                 MesPrint("&Problems with argument of FirstTerm_");
01102             else if ( x1 == CONTENTTERM )
01103                 MesPrint("&Problems with argument of FirstTerm_");
01104             else if ( x1 == TERMSINEXPR )
01105                 MesPrint("&Problems with argument of TermsIn_");
01106             else if ( x1 == SIZEOFFUNCTION )
01107                 MesPrint("&Problems with argument of SizeOf_");
01108             else if ( x1 == NUMFACTORS )
01109                 MesPrint("&Problems with argument of NumFactors_");
01110             else
01111                 MesPrint("&Problems with argument of FactorIn_");

```

```

01111         error = 1;
01112         while ( *s != TENDOFIT && *s != TFUNCLOSE ) s++;
01113     }
01114     if ( *s == TFUNCLOSE ) s++;
01115     goto fin;
01116 }
01117 }
01118 else if ( x1 == TERMSINEXPR || x1 == SIZEOFFUNCTION || x1 == FACTORIN
01119 || x1 == NUMFACTORS || x1 == FIRSTTERM || x1 == CONTENTTERM ) {
01120     if ( s[0] == TFUNOPEN && s[1] == TEXPRESSON ) goto doexpr;
01121     if ( s[0] == TFUNOPEN && s[1] == TDOLLAR ) {
01122         s += 2;
01123         *t++ = x1; *t++ = FUNHEAD+2; *t++ = 0;
01124         FILLFUN3(t)
01125         x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
01126         *t++ = -DOLLAREXPRESSON; *t++ = x2;
01127         if ( *s != TFUNCLOSE ) {
01128             if ( x1 == TERMSINEXPR )
01129                 MesPrint("&Problems with argument of TermsIn_");
01130             else if ( x1 == SIZEOFFUNCTION )
01131                 MesPrint("&Problems with argument of SizeOf_");
01132             else if ( x1 == NUMFACTORS )
01133                 MesPrint("&Problems with argument of NumFactors_");
01134             else
01135                 MesPrint("&Problems with argument of FactorIn_");
01136             error = 1;
01137             while ( *s != TENDOFIT && *s != TFUNCLOSE ) s++;
01138         }
01139         if ( *s == TFUNCLOSE ) s++;
01140         goto fin;
01141     }
01142 }
01143 x3 = x1;
01144 if ( inset && ( t-tsize == 2 ) ) x1 -= FUNCTION;
01145 if ( *s == TWILDCARD ) { x1 += WILDOFFSET; s++; }
01146 if ( functions[x3-FUNCTION].commute ) cc = 1;
01147 if ( *s != TFUNOPEN ) {
01148     *t++ = x1; *t++ = FUNHEAD; *t++ = 0;
01149     if ( x1 == AR.PolyFun && AR.PolyFunType == 2 && AR.Eside != LHSIDE )
01150         t[-1] |= MUSTCLEANPRF;
01151     FILLFUN3(t) sumlevel = 0; goto fin;
01152 }
01153 v = t; *t++ = x1; *t++ = FUNHEAD; *t++ = DIRTYFLAG;
01154 if ( x1 == AR.PolyFun && AR.PolyFunType == 2 && AR.Eside != LHSIDE )
01155     t[-1] |= MUSTCLEANPRF;
01156 FILLFUN3(t)
01157 needarg = -1;
01158 if ( !inset && functions[x3-FUNCTION].spec >= TENSORFUNCTION ) {
01159 dotensor:
01160     do {
01161         if ( needarg == 0 ) {
01162             if ( x1 >= 0 ) {
01163                 x3 = x1;
01164                 if ( x3 >= FUNCTION+WILDOFFSET ) x3 -= WILDOFFSET;
01165                 MesPrint("&Too many arguments in function %s",
01166                     VARNAME(functions, (x3-FUNCTION)) );
01167             }
01168             else
01169                 MesPrint("&d_ needs exactly two arguments");
01170             error = -1;
01171             needarg--;
01172         }
01173         else if ( needarg > 0 ) needarg--;
01174         s++;
01175         c = *s++;
01176         if ( c == TMINUS && *s == TVECTOR ) { sign = -sign; c = *s++; }
01177         base = ( c == TNUMBER ) ? 100: 128;
01178         x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
01179         if ( *s == TWILDCARD && c != TNUMBER ) { x2 += WILDOFFSET; s++; }
01180         if ( c == TSETNUM || c == TSETDOL ) {
01181             if ( c == TSETDOL ) x2 = -x2;
01182             if ( !inset == 0 ) {
01183                 w1 = t; t += 2; w2 = t;
01184                 while ( w1 > v ) *--w2 = *--w1;
01185                 tsize = v; relo = AT.WorkTop;
01186                 *v++ = SETSET; *v++ = 0;
01187             }
01188             inset = 2; *--relo = x2; *--relo = t - v;
01189             c = *s++;
01190             x2 = 0; while ( *s >= 0 ) x2 = 128*x2 + *s++;
01191             switch ( c ) {
01192                 case TINDEX:
01193                     *t++ = x2;
01194                     if ( t[-1]+AM.OffsetIndex > AM.IndDum ) {
01195                         x2 = t[-1]+AM.OffsetIndex - AM.IndDum;
01196                         if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
01197                     }

```

```

01198         break;
01199     case TVECTOR:
01200         *t++ = x2; break;
01201     case TNUMBER1:
01202         if ( x2 >= 0 && x2 < AM.OffsetIndex ) {
01203             *t++ = x2; break;
01204         }
01205         /* fall through */
01206     default:
01207         MesPrint("&Illegal type of set inside tensor");
01208         error = 1;
01209         *t++ = x2;
01210         break;
01211     }
01212 }
01213 else { switch ( c ) {
01214     case TINDEX:
01215         if ( inset < 2 ) *t++ = x2 + AM.OffsetIndex;
01216         else *t++ = x2;
01217         if ( x2+AM.OffsetIndex > AM.IndDum ) {
01218             x2 = x2+AM.OffsetIndex - AM.IndDum;
01219             if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
01220         }
01221         break;
01222     case TGENINDEX:
01223         *t++ = AC.DumNum + WILDOFFSET;
01224         break;
01225     case TVECTOR:
01226         if ( inset < 2 ) *t++ = x2 + AM.OffsetVector;
01227         else *t++ = x2;
01228         break;
01229     case TWILDARG:
01230         *t++ = FUNNYWILD; *t++ = x2;
01231         v[2] = 0; /*
01232         break;
01233     case TDOLLAR:
01234         *t++ = FUNNYDOLLAR; *t++ = x2;
01235         break;
01236     case TDUBIOUS:
01237         if ( inset < 2 ) *t++ = x2 + AM.OffsetVector;
01238         else *t++ = x2;
01239         break;
01240     case TSGAMMA: /* Special gamma's */
01241         if ( x3 != GAMMA ) {
01242             MesPrint("&5_,6_,7_ can only be used inside g_");
01243             error = -1;
01244         }
01245         *t++ = -x2;
01246         break;
01247     case TNUMBER:
01248     case TNUMBER1:
01249         if ( x2 >= AM.OffsetIndex && inset < 2 ) {
01250             MesPrint("&Value of constant index in tensor too large");
01251             error = -1;
01252         }
01253         *t++ = x2;
01254         break;
01255     default:
01256         MesPrint("&Illegal object in tensor");
01257         error = -1;
01258         break;
01259 }}
01260 if ( inset >= 2 ) inset = 1;
01261 } while ( *s == TCOMMA );
01262 }
01263 else {
01264     dofunction: firstsumarg = 1;
01265     do {
01266         unsigned int ux2;
01267         s++;
01268         c = *s++;
01269         if ( c == TMINUS && ( *s == TVECTOR || *s == TNUMBER
01270             || *s == TNUMBER1 || *s == TSUBEXP ) ) {
01271             minus = 1; c = *s++;
01272         }
01273         else minus = 0;
01274         base = ( c == TNUMBER ) ? 100: 128;
01275         ux2 = 0; while ( *s >= 0 ) { ux2 = base*ux2 + *s++; }
01276         x2 = ux2; /* may cause an implementation-defined behaviour */
01277     } /*
01278     !!!!!!!! What if it does not fit?
01279     */
01280     if ( firstsumarg ) {
01281         firstsumarg = 0;
01282         if ( sumlevel > 0 ) {
01283             if ( c == TSYMBOL ) {
01284                 sumlevel = x2; sumtype = SYMTOSYM;

```

```

01285         }
01286     else if ( c == TINDEX ) {
01287         sumlevel = x2+AM.OffsetIndex; sumtype = INDTOIND;
01288         if ( sumlevel > AM.IndDum ) {
01289             x2 = sumlevel - AM.IndDum;
01290             if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
01291         }
01292     }
01293 }
01294 }
01295 if ( *s == TWILDCARD ) {
01296     if ( c == TSYMBOL ) x2 += 2*MAXPOWER;
01297     else if ( c != TNUMBER ) x2 += WILDOFFSET;
01298     s++;
01299 }
01300 switch ( c ) {
01301 case TSYMBOL:
01302     *t++ = -SYMBOL; *t++ = x2; break;
01303 case TDOLLAR:
01304     *t++ = -DOLLAREXPRESSION; *t++ = x2; break;
01305 case TEXPRESS:
01306     *t++ = -EXPRESSION; *t++ = x2;
01307 /*
01308     The next code is added to facilitate parallel processing
01309     We need to call GetTable here to make sure all processors
01310     have the same numbering of all variables.
01311 */
01312     if ( Expressions[x2].status == STOREDEXPRESSION ) {
01313         TMproto[0] = EXPRESSION;
01314         TMproto[1] = SUBEXPSIZE;
01315         TMproto[2] = x2;
01316         TMproto[3] = 1;
01317         { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
01318         AT.TMaddr = TMproto;
01319         PUTZERO(position);
01320 /*
01321         if ( (
01322             renumber =
01323                 GetTable(x2,&position,0) ) == 0 ) {
01324             error = 1;
01325             MesPrint("&Problems getting information about stored
01326                 ,EXPRNAME(x2));
01327         }
01328         M_free(renumber->symb.lo,"VarSpace");
01329         M_free(renumber,"Renumber");
01330 */
01331         if ( ( renumber = GetTable(x2,&position,0) ) == 0 ) {
01332             error = 1;
01333             MesPrint("&Problems getting information about stored
01334                 ,EXPRNAME(x2));
01335         }
01336         if ( renumber->symb.lo != AN.dummyrenumlist )
01337             M_free(renumber->symb.lo,"VarSpace");
01338         M_free(renumber,"Renumber");
01339         AR.StoreData.dirtyflag = 1;
01340     }
01341     break;
01342 case TINDEX:
01343     *t++ = -INDEX; *t++ = x2 + AM.OffsetIndex;
01344     if ( t[-1] > AM.IndDum ) {
01345         x2 = t[-1] - AM.IndDum;
01346         if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
01347     }
01348     break;
01349 case TGENINDEX:
01350     *t++ = -INDEX; *t++ = AC.DumNum + WILDOFFSET;
01351     break;
01352 case TVECTOR:
01353     if ( minus ) *t++ = -MINVECTOR;
01354     else *t++ = -VECTOR;
01355     *t++ = x2 + AM.OffsetVector;
01356     break;
01357 case TSGAMMA: /* Special gamma's */
01358     MesPrint("&5_,6_,7_ can only be used inside g_");
01359     error = -1;
01360     *t++ = -INDEX;
01361     *t++ = -x2;
01362     break;
01363 case TDUBIOUS:
01364     *t++ = -SYMBOL; *t++ = x2; break;
01365 case TFUNCTION:

```



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01370             *t++ = -x2-FUNCTION;
01371             break;
01372         case TSET:
01373             *t++ = -SETSET;
01374             *t++ = x2;
01375             break;
01376         case TWILDARG:
01377             *t++ = -ARGWILD; *t++ = x2; break;
01378         case TSETDOL:
01379             x2 = -x2;
01380             /* fall through */
01381         case TSETNUM:
01382             if ( inset == 0 ) {
01383                 w1 = t; t += 2; w2 = t;
01384                 while ( w1 > v ) *--w2 = *--w1;
01385                 tsize = v; relo = AT.WorkTop;
01386                 *v++ = SETSET; *v++ = 0;
01387                 inset = 1;
01388             }
01389             *--relo = x2; *--relo = t-v+1;
01390             c = *s++;
01391             x2 = 0; while ( *s >= 0 ) x2 = 128*x2 + *s++;
01392             switch ( c ) {
01393                 case TFUNCTION:
01394                     (*relo)--; *t++ = -x2-1; break;
01395                 case TSYMBOL:
01396                     *t++ = -SYMBOL; *t++ = x2; break;
01397                 case TINDEX:
01398                     *t++ = -INDEX; *t++ = x2;
01399                     if ( x2+AM.OffsetIndex > AM.IndDum ) {
01400                         x2 = x2+AM.OffsetIndex - AM.IndDum;
01401                         if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
01402                     }
01403                     break;
01404                 case TVECTOR:
01405                     *t++ = -VECTOR; *t++ = x2; break;
01406                 case TNUMBER1:
01407                     *t++ = -SNUMBER; *t++ = x2; break;
01408                 default:
01409                     MesPrint("&Internal error 435");
01410                     error = 1;
01411                     *t++ = -SYMBOL; *t++ = x2; break;
01412             }
01413             break;
01414         case TSUBEXP:
01415             w2 = AC.ProtoType; i = w2[1];
01416             w1 = t;
01417             *t++ = i+ARGHEAD+4;
01418             *t++ = 1;
01419             FILLARG(t);
01420             *t++ = i + 4;
01421             while ( --i >= 0 ) *t++ = *w2++;
01422             w1[ARGHEAD+3] = subexpbuffers[x2].subexpnum;
01423             w1[ARGHEAD+5] = subexpbuffers[x2].buffernum;
01424             if ( sumlevel > 0 ) {
01425                 w1[0] += 4;
01426                 w1[ARGHEAD] += 4;
01427                 w1[ARGHEAD+2] += 4;
01428                 *t++ = sumtype; *t++ = 4;
01429                 *t++ = sumlevel; *t++ = sumlevel;
01430             }
01431             *t++ = 1; *t++ = 1;
01432             if ( minus ) *t++ = -3;
01433             else *t++ = 3;
01434             break;
01435         case TNUMBER:
01436         case TNUMBER1:
01437             if ( minus ) x2 = UnsignedToInt(-IntAbs(x2));
01438             *t++ = -SNUMBER;
01439             *t++ = x2;
01440             break;
01441         default:
01442             MesPrint("&Illegal object in function");
01443             error = -1;
01444             break;
01445     }
01446     } while ( *s == TCOMMA );
01447 }
01448 if ( *s != TFUNCLOSE ) {
01449     MesPrint("&Illegal argument field for function. Expected ");
01450     return(-1);
01451 }
01452 s++; sumlevel = 0;
01453 v[1] = t-v;
01454 /*
01455     if ( *v == AM.termfunnum && ( v[1] != FUNHEAD+2 ||
01456         v[FUNHEAD] != -DOLLAREXPRESSION ) ) {

```

```

01457             MesPrint("&The function term_ can only have one argument with a single
01458 $-expression");
01459             error = 1;
01460         }
01461     /*
01462     case TDUBIOUS:
01463         x1 = 0; while ( *s >= 0 ) x1 = 128*x1 + *s++;
01464         if ( *s == TWILDCARD ) s++;
01465         if ( *s == TDOT ) goto dovector;
01466         if ( *s == TFUNOPEN ) {
01467             x1 += FUNCTION;
01468             cc = 1;
01469             v = t; *t++ = x1; *t++ = FUNHEAD; *t++ = DIRTYFLAG;
01470             if ( x1 == AR.PolyFun && AR.PolyFunType == 2 && AR.Eside != LHSIDE )
01471                 t[-1] |= MUSTCLEANPRF;
01472             FILLFUN3(t)
01473             needarg = -1; goto dofunction;
01474         }
01475         *t++ = SYMBOL; *t++ = 4; *t++ = 0;
01476         if ( inset ) *relo = 2;
01477         goto TryPower;
01478     case TSUBEXP:
01479         x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
01480         if ( *s == TPOWER ) {
01481             s++; c = *s++;
01482             base = ( c == TNUMBER ) ? 100: 128;
01483             x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
01484             if ( *s == TWILDCARD ) { x2 += 2*MAXPOWER; s++; }
01485             else if ( c == TSYMBOL ) x2 += 2*MAXPOWER;
01486         }
01487         else x2 = 1;
01488         r = AC.ProtoType; n = r[1] - 5; r += 5;
01489         *t++ = SUBEXPRESSION; *t++ = r[-4];
01490         *t++ = subexpbuffers[x1].subexpnum;
01491         *t++ = x2*deno;
01492         *t++ = subexpbuffers[x1].buffernum;
01493         NCOPY(t,r,n);
01494         if ( cbuf[subexpbuffers[x1].buffernum].CanCommu[subexpbuffers[x1].subexpnum] ) cc = 1;
01495         deno = 1;
01496         break;
01497     case TMULTIPLY:
01498         mulflag = 1;
01499         break;
01500     case TDIVIDE:
01501         mulflag = 1;
01502         deno = -deno;
01503         break;
01504     case TEXPRESSION:
01505         cc = 1;
01506         x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
01507         v = t;
01508         *t++ = EXPRESSION; *t++ = SUBEXPSIZE; *t++ = x1; *t++ = deno;
01509         *t++ = 0; FILLSUB(t)
01510     /*
01511     Here we had some erroneous code before. It should be after
01512     the reading of the parameters as it is now (after 15-jan-2007).
01513     Thomas Hahn noticed this and reported it.
01514     */
01515     if ( *s == TFUNOPEN ) {
01516         do {
01517             s++; c = *s++;
01518             base = ( c == TNUMBER ) ? 100: 128;
01519             x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
01520             switch ( c ) {
01521                 case TSYMBOL:
01522                     *t++ = SYMBOL; *t++ = 4; *t++ = x2; *t++ = 1;
01523                     break;
01524                 case TINDEX:
01525                     *t++ = INDEX; *t++ = 3; *t++ = x2+AM.OffsetIndex;
01526                     if ( t[-1] > AM.IndDum ) {
01527                         x2 = t[-1] - AM.IndDum;
01528                         if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
01529                     }
01530                     break;
01531                 case TVECTOR:
01532                     *t++ = INDEX; *t++ = 3; *t++ = x2+AM.OffsetVector;
01533                     break;
01534                 case TFUNCTION:
01535                     *t++ = x2+FUNCTION; *t++ = 2; break;
01536                 case TNUMBER:
01537                 case TNUMBER1:
01538                     if ( x2 >= AM.OffsetIndex || x2 < 0 ) {
01539                         MesPrint("&Index as argument of expression has illegal value");
01540                         error = -1;
01541                     }
01542                     *t++ = INDEX; *t++ = 3; *t++ = x2; break;

```

```

01543         case TSETDOL:
01544             x2 = -x2;
01545             /* fall through */
01546         case TSETNUM:
01547             if ( inset == 0 ) {
01548                 w1 = t; t += 2; w2 = t;
01549                 while ( w1 > v ) *--w2 = *--w1;
01550                 tsize = v; relo = AT.WorkTop;
01551                 *v++ = SETSET; *v++ = 0;
01552                 inset = 1;
01553             }
01554             *--relo = x2; *--relo = t-v+2;
01555             c = *s++;
01556             x2 = 0; while ( *s >= 0 ) x2 = 128*x2 + *s++;
01557             switch ( c ) {
01558                 case TFUNCTION:
01559                     *relo -= 2; *t++ = -x2-1; break;
01560                 case TSYMBOL:
01561                     *t++ = SYMBOL; *t++ = 4; *t++ = x2; *t++ = 1; break;
01562                 case TINDEX:
01563                     *t++ = INDEX; *t++ = 3; *t++ = x2;
01564                     if ( x2+AM.OffsetIndex > AM.IndDum ) {
01565                         x2 = x2+AM.OffsetIndex - AM.IndDum;
01566                         if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
01567                     }
01568                     break;
01569                 case TVECTOR:
01570                     *t++ = VECTOR; *t++ = 3; *t++ = x2; break;
01571                 case TNUMBER1:
01572                     *t++ = SNUMBER; *t++ = 4; *t++ = x2; *t++ = 1; break;
01573                 default:
01574                     MesPrint("&Internal error 435");
01575                     error = 1;
01576                     *t++ = SYMBOL; *t++ = 4; *t++ = x2; *t++ = 1; break;
01577             }
01578             break;
01579         default:
01580             MesPrint("&Argument of expression can only be symbol, index, vector or
function");
01581             error = -1;
01582             break;
01583     }
01584     } while ( *s == TCOMMA );
01585     if ( *s != TFUNCLOSE ) {
01586         MesPrint("&Illegal object in argument field for expression");
01587         error = -1;
01588         while ( *s != TFUNCLOSE ) s++;
01589     }
01590     s++;
01591 }
01592 r = AC.ProtoType; n = r[1];
01593 if ( n > SUBEXPSIZE ) {
01594     *t++ = WILDCARDS; *t++ = n+2;
01595     NCOPY(t,r,n);
01596 }
01597 /*
01598 Code added for parallel processing.
01599 This is different from the other occurrences to test immediately
01600 for renumbering. Here we have to read the parameters first.
01601 */
01602 if ( Expressions[x1].status == STOREDEXPRESSION ) {
01603     v[1] = t-v;
01604     AT.TMaddr = v;
01605     PUTZERO(position);
01606 /*
01607     if ( (
01608 #ifndef WITHPTHREADS
01609         renumber =
01610 #endif
01611         GetTable(x1,&position,0) ) == 0 ) {
01612         error = 1;
01613         MesPrint("&Problems getting information about stored expression %s(3) ",
01614             EXPRNAME(x1));
01615     }
01616 #ifndef WITHPTHREADS
01617     M_free(renumber->symb.lo,"VarSpace");
01618     M_free(renumber,"Renumber");
01619 #endif
01620 */
01621     if ( ( renumber = GetTable(x1,&position,0) ) == 0 ) {
01622         error = 1;
01623         MesPrint("&Problems getting information about stored expression %s(3) ",
01624             EXPRNAME(x1));
01625     }
01626     if ( renumber->symb.lo != AN.dummyrenumlist )
01627         M_free(renumber->symb.lo,"VarSpace");
01628     M_free(renumber,"Renumber");

```

```

01629         AR.StoreData.dirtyflag = 1;
01630     }
01631     if ( *s == LBRACE ) {
01632         /*
01633             This should be one term that should be inserted
01634             FROMBRAC size+2 ( term )
01635             Because this term should have been translated
01636             already we can copy it from the 'subexpression'
01637         */
01638         s++;
01639         if ( *s != TSUBEXP ) {
01640             MesPrint("&Internal error 23");
01641             Terminate(-1);
01642         }
01643         s++; x2 = 0; while ( *s >= 0 ) { x2 = 128*x2 + *s++; }
01644         r = cbuf[subexpbuffers[x2].buffernum].rhs[subexpbuffers[x2].subexpnum];
01645         *t++ = FROMBRAC; *t++ = *r+2;
01646         n = *r;
01647         NCOPY(t,r,n);
01648         if ( *r != 0 ) {
01649             MesPrint("&Object between [] in expression should be a single term");
01650             error = -1;
01651         }
01652         if ( *s != RBRACE ) {
01653             MesPrint("&Internal error 23b");
01654             Terminate(-1);
01655         }
01656         s++;
01657     }
01658     if ( *s == TPOWER ) {
01659         s++; c = *s++;
01660         base = ( c == TNUMBER ) ? 100: 128;
01661         x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
01662         if ( *s == TWILDCARD || c == TSYMBOL ) { x2 += 2*MAXPOWER; s++; }
01663         v[3] = x2;
01664     }
01665     v[1] = t - v;
01666     deno = 1;
01667     break;
01668 case TNUMBER:
01669     if ( *s == 0 ) {
01670         s++;
01671         if ( *s == TPOWER ) {
01672             s++; if ( *s == TMINUS ) { s++; deno = -deno; }
01673             c = *s++; base = ( c == TNUMBER ) ? 100: 128;
01674             x2 = 0; while ( *s >= 0 ) { x2 = x2*base + *s++; }
01675             if ( x2 == 0 ) {
01676                 error = -1;
01677                 MesPrint("&Encountered 0^0 during compilation");
01678             }
01679             if ( deno < 0 ) {
01680                 error = -1;
01681                 MesPrint("&Division by zero during compilation (0 to the power negative
01682 number)");
01683             }
01684             else if ( deno < 0 ) {
01685                 error = -1;
01686                 MesPrint("&Division by zero during compilation");
01687             }
01688             sign = 0; break; /* term is zero */
01689         }
01690         y = *s++;
01691         if ( *s >= 0 ) { y = 100*y + *s++; }
01692         innum[0] = y; nin = 1;
01693         while ( *s >= 0 ) {
01694             y = *s++; x2 = 100;
01695             if ( *s >= 0 ) { y = 100*y + *s++; x2 = 10000; }
01696             Product(innum,&nin,(WORD)x2);
01697             if ( y ) AddLong(innum,nin,(UWORD *)(&y),(WORD)1,innum,&nin);
01698         }
01699     docoef:
01700     if ( *s == TPOWER ) {
01701         s++; if ( *s == TMINUS ) { s++; deno = -deno; }
01702         c = *s++; base = ( c == TNUMBER ) ? 100: 128;
01703         x2 = 0; while ( *s >= 0 ) { x2 = x2*base + *s++; }
01704         if ( x2 == 0 ) {
01705             innum[0] = 1; nin = 1;
01706         }
01707         else if ( RaisPow(BHEAD innum,&nin,x2) ) {
01708             error = -1; innum[0] = 1; nin = 1;
01709         }
01710     }
01711     if ( deno > 0 ) {
01712         Simplify(BHEAD innum,&nin,denominator,&ndenominator);
01713         for ( i = 0; i < nnumerator; i++ ) CGscrat7[i] = numerator[i];
01714         MulLong(innum,nin,CGscrat7,nnumerator,numerator,&nnumerator);

```

```

01715         }
01716     else if ( deno < 0 ) {
01717         Simplify(BHEAD innum,&nin,numerator,&nnumerator);
01718         for ( i = 0; i < ndenominator; i++ ) CGscrat7[i] = denominator[i];
01719         MulLong(innum,nin,CGscrat7,ndenominator,denominator,&ndenominator);
01720     }
01721     deno = 1;
01722     break;
01723 case TNUMBER1:
01724     if ( *s == 0 ) { s++; sign = 0; break; /* term is zero */ }
01725     y = *s++;
01726     if ( *s >= 0 ) { y = 128*y + *s++; }
01727     if ( inset == 0 ) {
01728         innum[0] = y; nin = 1;
01729         while ( *s >= 0 ) {
01730             y = *s++; x2 = 128;
01731             if ( *s >= 0 ) { y = 128*y + *s++; x2 = 16384; }
01732             Product(innum,&nin,(WORD)x2);
01733             if ( y ) AddLong(innum,nin,(UWORD *)&y,(WORD)1,innum,&nin);
01734         }
01735         goto doccoef;
01736     }
01737     *relo = 2; *t++ = SNUMBER; *t++ = 4; *t++ = y;
01738     goto TryPower;
01739 #ifdef WITHFLOAT
01740 case TFLOAT:
01741     { WORD *w;
01742       s = ReadFloat(s);
01743       i = AT.WorkPointer[1];
01744       w = AT.WorkPointer;
01745       NCOPY(t,w,i);
01746       /*
01747       Power?
01748       */
01749     }
01750     break;
01751 #endif
01752 case TDOLLAR:
01753     {
01754         WORD *powplace;
01755         x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
01756         if ( AR.Eside != LHSIDE ) {
01757             *t++ = SUBEXPRESSION; *t++ = SUBEXPSIZE; *t++ = x1;
01758         }
01759         else {
01760             *t++ = DOLLAREXPRESSION; *t++ = SUBEXPSIZE; *t++ = x1;
01761         }
01762         powplace = t; t++;
01763         *t++ = AM.dbufnum; FILLSUB(t)
01764         /*
01765         Now we have to test for factors of dollars with [ ] and [ [ ] ]
01766         */
01767         if ( *s == LBRACE ) {
01768             int bracelevel = 1;
01769             s++;
01770             while ( bracelevel > 0 ) {
01771                 if ( *s == RBRACE ) {
01772                     bracelevel--; s++;
01773                 }
01774                 else if ( *s == TNUMBER ) {
01775                     s++;
01776                     x2 = 0; while ( *s >= 0 ) { x2 = 100*x2 + *s++; }
01777                     *t++ = DOLLAREXPR2; *t++ = 3; *t++ = -x2-1;
01778 CloseBraces:
01779                     while ( bracelevel > 0 ) {
01780                         if ( *s != RBRACE ) {
01781 ErrorBraces:
01782                             error = -1;
01783                             MesPrint("&Improper use of [] in $-variable.");
01784                             return(error);
01785                         }
01786                         else {
01787                             s++; bracelevel--;
01788                         }
01789                     }
01790                 }
01791                 else if ( *s == TDOLLAR ) {
01792                     s++;
01793                     x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
01794                     *t++ = DOLLAREXPR2; *t++ = 3; *t++ = x1;
01795                     if ( *s == RBRACE ) goto CloseBraces;
01796                     else if ( *s == LBRACE ) {
01797                         s++; bracelevel++;
01798                     }
01799                 }
01800                 else goto ErrorBraces;
01801             }

```

```

01802         }
01803     /*
01804         Finally we can continue with the power
01805     */
01806         if ( *s == TPOWER ) {
01807             s++;
01808             if ( *s == TMINUS ) { s++; deno = -deno; }
01809             c = *s++;
01810             base = ( c == TNUMBER ) ? 100: 128;
01811             x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
01812             if ( c == TSYMBOL ) {
01813                 if ( *s == TWILDCARD ) s++;
01814                 x2 += 2*MAXPOWER;
01815             }
01816             *powplace = deno*x2;
01817         }
01818         else *powplace = deno;
01819         deno = 1;
01820     /*
01821         if ( inset ) {
01822             while ( relo < AT.WorkTop ) *t++ = *relo++;
01823             inset = 0; tsize[1] = t - tsize;
01824         }
01825     */
01826     }
01827     break;
01828 case TSETNUM:
01829     inset = 1; tsize = t; relo = AT.WorkTop;
01830     *t++ = SETSET; *t++ = 0;
01831     x1 = 0; while ( *s >= 0 ) x1 = x1*128 + *s++;
01832     *--relo = x1; *--relo = 0;
01833     break;
01834 case TSETDOL:
01835     inset = 1; tsize = t; relo = AT.WorkTop;
01836     *t++ = SETSET; *t++ = 0;
01837     x1 = 0; while ( *s >= 0 ) x1 = x1*128 + *s++;
01838     *--relo = -x1; *--relo = 0;
01839     break;
01840 case TFUNOPEN:
01841     MesPrint("&Illegal use of function arguments");
01842     error = -1;
01843     funflag = 1;
01844     deno = 1;
01845     break;
01846 case TFUNCLOSE:
01847     if ( funflag == 0 )
01848         MesPrint("&Illegal use of function arguments");
01849     error = -1;
01850     funflag = 0;
01851     deno = 1;
01852     break;
01853 case TSGAMMA:
01854     MesPrint("&Illegal use special gamma symbols 5_, 6_, 7_");
01855     error = -1;
01856     funflag = 0;
01857     deno = 1;
01858     break;
01859 case TCONJUGATE:
01860     MesPrint("&Complex conjugate operator (%#) is not implemented");
01861     error = -1;
01862     deno = 1;
01863     break;
01864 default:
01865     MesPrint("&Internal error in code generator. Unknown object: %d",c);
01866     error = -1;
01867     deno = 1;
01868     break;
01869     }
01870 }
01871 }
01872 if ( mulflag ) {
01873     MesPrint("&Irregular end of statement.");
01874     error = 1;
01875 }
01876 if ( !first && error == 0 ) {
01877     *term = t-term;
01878     C->NumTerms[numexp]++;
01879     if ( cc && sign ) C->CanCommu[numexp]++;
01880     error = CompleteTerm(term,numerator,denominator,nnumerator,ndenominator,sign);
01881 }
01882 AT.WorkPointer = oldwork;
01883 if ( error ) return(-1);
01884 AddToCB(C,0)
01885 if ( AC.CompileLevel > 0 && AR.Eside != LHSIDE ) {
01886     /* See whether we have this one already */
01887     error = InsTree(AC.cbufnum,C->numrhs);
01888     if ( error < (C->numrhs) ) {

```

```

01889         C->Pointer = C->rhs[C->numrhs--];
01890         return(error);
01891     }
01892 }
01893 return(C->numrhs);
01894 OverWork:
01895     MLOCK(ErrorMessageLock);
01896     MesWork();
01897     MUNLOCK(ErrorMessageLock);
01898     return(-1);
01899 }
01900
01901 /*
01902     #] CodeGenerator :
01903     #[ CompleteTerm :
01904
01905     Completes the term
01906     Puts it in the buffer
01907 */
01908
01909 int CompleteTerm(WORD *term, UWORD *numer, UWORD *denom, WORD nnum, WORD nden, int sign)
01910 {
01911     int nsize, i;
01912     WORD *t;
01913     if ( sign == 0 ) return(0);      /* Term is zero */
01914     if ( nnum >= nden ) nsize = nnum;
01915     else nsize = nden;
01916     t = term + *term;
01917     for ( i = 0; i < nnum; i++ ) *t++ = numer[i];
01918     for ( ; i < nsize; i++ ) *t++ = 0;
01919     for ( i = 0; i < nden; i++ ) *t++ = denom[i];
01920     for ( ; i < nsize; i++ ) *t++ = 0;
01921     *t++ = (2*nsize+1)*sign;
01922     *term = t - term;
01923     AddNtoC(AC.cbufnum,*term,term,7);
01924     return(0);
01925 }
01926
01927 /*
01928     #] CompleteTerm :
01929     #[ CodeFactors :
01930
01931     This routine does the part of reading in in terms of factors.
01932     If there is more than one term at this level we have only one
01933     factor. In that case any expression should first be unfactorized.
01934     Then the whole expression gets read as a new subexpression and finally
01935     we generate factor_*subexpression.
01936     If the whole has only multiplications we have factors. Then the
01937     nasty thing is powers of objects and in particular powers of
01938     factorized expressions or dollars.
01939     For a power we generate a new subexpression of the type
01940     1+factor_+...+factor_^(power-1)
01941     with which we multiply.
01942
01943     WE HAVE NOT YET WORRIED ABOUT SETS
01944 */
01945
01946 int CodeFactors(SBYTE *tokens)
01947 {
01948     GETIDENTITY
01949     EXPRESSIONS e = Expressions + AR.CurExpr;
01950     int nfactor = 1, nparenthesis, i, last = 0, error = 0;
01951     SBYTE *t, *startobject, *tt, *sl, *out, *outtokens;
01952     WORD nexpt, subexp = 0, power, pow, x2, powfactor, first;
01953
01954     /* First scan the number of factors
01955     */
01956     t = tokens;
01957     while ( *t != TENDOFIT ) {
01958         if ( *t >= 0 ) { while ( *t >= 0 ) t++; continue; }
01959         if ( *t == LPARENTHESIS || *t == LBRACE || *t == TSETOPEN || *t == TFUNOPEN ) {
01960             nparenthesis = 0; t++;
01961             while ( nparenthesis >= 0 ) {
01962                 if ( *t == LPARENTHESIS || *t == LBRACE || *t == TSETOPEN || *t == TFUNOPEN )
01963                     nparenthesis++;
01964                 else if ( *t == RPARENTHESIS || *t == RBRACE || *t == TSETCLOSE || *t == TFUNCLOSE )
01965                     nparenthesis--;
01966                 t++;
01967             }
01968             else if ( ( *t == TPLUS || *t == TMINUS ) && ( t > tokens )
01969                 && ( t[-1] != TPLUS && t[-1] != TMINUS ) ) {
01970                 if ( t[-1] >= 0 || t[-1] == RPARENTHESIS || t[-1] == RBRACE
01971                     || t[-1] == TSETCLOSE || t[-1] == TFUNCLOSE ) {
01972                     subexp = CodeGenerator(tokens);
01973                     if ( subexp < 0 ) error = -1;

```

```

01974         if ( insubexpbuffers >= MAXSUBEXPRESSIONS ) {
01975             MesPrint("&More than %d subexpressions inside one
expression", (WORD)MAXSUBEXPRESSIONS);
01976             Terminate(-1);
01977         }
01978         if ( subexpbuffers+insubexpbuffers >= topsubexpbuffers ) {
01979             DoubleBuffer((void **) ((void *) (&subexpbuffers))
01980                 , (void **) ((void *) (&topsubexpbuffers)), sizeof(SUBBUF), "subexpbuffers");
01981         }
01982         subexpbuffers[insubexpbuffers].subexpnum = subexp;
01983         subexpbuffers[insubexpbuffers].buffernum = AC.cbufnum;
01984         subexp = insubexpbuffers++;
01985         t = tokens;
01986         *t++ = TSYMBOL; *t++ = FACTORSYMBOL;
01987         *t++ = TMULTIPLY; *t++ = TSUBEXP;
01988         PUTNUMBER128(t, subexp)
01989         *t++ = TENDOFIT;
01990         e->numfactors = 1;
01991         e->vflags |= ISFACTORIZED;
01992         return(subexp);
01993     }
01994 }
01995 else if ( ( *t == TMULTIPLY || *t == TDIVIDE ) && t > tokens ) {
01996     nfactor++;
01997 }
01998 else if ( *t == TEXPRESSON ) {
01999     t++;
02000     nexpt = 0; while ( *t >= 0 ) { nexpt = nexpt*128 + *t++; }
02001     if ( *t == LBACE ) continue;
02002     if ( ( AS.Oldvflags[nexpt] & ISFACTORIZED ) != 0 ) {
02003         nfactor += AS.OldNumFactors[nexpt];
02004     }
02005     else { nfactor++; }
02006     continue;
02007 }
02008 else if ( *t == TDOLLAR ) {
02009     t++;
02010     nexpt = 0; while ( *t >= 0 ) { nexpt = nexpt*128 + *t++; }
02011     if ( *t == LBACE ) continue;
02012     if ( Dollars[nexpt].nfactors > 0 ) {
02013         nfactor += Dollars[nexpt].nfactors;
02014     }
02015     else { nfactor++; }
02016     continue;
02017 }
02018 t++;
02019 }
02020 /*
02021 Now the real pass.
02022 nfactor is a not so reliable measure for the space we need.
02023 */
02024 outtokens = (SBYTE *)Malloc1(((t-tokens)+(nfactor+2)*25)*sizeof(SBYTE), "CodeFactors");
02025 out = outtokens;
02026 t = tokens; first = 1; powfactor = 1;
02027 while ( *t == TPLUS || *t == TMINUS ) { if ( *t == TMINUS ) first = -first; t++; }
02028 if ( first < 0 ) {
02029     *out++ = TMINUS; *out++ = TSYMBOL; *out++ = FACTORSYMBOL;
02030     *out++ = TPOWER; *out++ = TNUMBER; PUTNUMBER100(out, powfactor)
02031     powfactor++;
02032 }
02033 startobject = t; power = 1;
02034 while ( *t != TENDOFIT ) {
02035     if ( *t >= 0 ) { while ( *t >= 0 ) t++; continue; }
02036     if ( *t == LPARENTHESIS || *t == LBACE || *t == TSETOPEN || *t == TFUNOPEN ) {
02037         nparenthesis = 0; t++;
02038         while ( nparenthesis >= 0 ) {
02039             if ( *t == LPARENTHESIS || *t == LBACE || *t == TSETOPEN || *t == TFUNOPEN )
nparenthesis++;
02040             else if ( *t == RPARENTHESIS || *t == RBACE || *t == TSETCLOSE || *t == TFUNCLOSE )
nparenthesis--;
02041             t++;
02042         }
02043         continue;
02044     }
02045     else if ( ( *t == TMULTIPLY || *t == TDIVIDE ) && ( t > tokens ) ) {
02046         if ( t[-1] >= 0 || t[-1] == RPARENTHESIS || t[-1] == RBACE
02047             || t[-1] == TSETCLOSE || t[-1] == TFUNCLOSE ) {
02048 dolast:
02049             if ( startobject ) { /* apparently power is 1 or -1 */
02050                 *out++ = TPLUS;
02051                 if ( power < 0 ) { *out++ = TNUMBER; *out++ = 1; *out++ = TDIVIDE; }
02052                 s1 = startobject;
02053                 while ( s1 < t ) *out++ = *s1++;
02054                 *out++ = TMULTIPLY; *out++ = TSYMBOL; *out++ = FACTORSYMBOL;
02055                 *out++ = TPOWER; *out++ = TNUMBER; PUTNUMBER100(out, powfactor)
02056                 powfactor++;
02057             }

```



```

02058         if ( last ) { startobject = 0; break; }
02059         startobject = t+1;
02060         if ( *t == TDIVIDE ) power = -1;
02061         if ( *t == TMULTIPLY ) power = 1;
02062     }
02063 }
02064 else if ( *t == TPOWER ) {
02065     pow = 1;
02066     tt = t+1;
02067     while ( ( *tt == TMINUS ) || ( *tt == TPLUS ) ) {
02068         if ( *tt == TMINUS ) pow = -pow;
02069         tt++;
02070     }
02071     if ( *tt == TSYMBOL ) {
02072         tt++; while ( *tt >= 0 ) tt++;
02073         t = tt; continue;
02074     }
02075     tt++; x2 = 0; while ( *tt >= 0 ) { x2 = 100*x2 + *tt++; }
02076 /*
02077     We have an object in startobject till t. The power is
02078     power*pow*x2
02079 */
02080     power = power*pow*x2;
02081     if ( power < 0 ) { pow = -power; power = -1; }
02082     else if ( power == 0 ) { t = tt; startobject = tt; continue; }
02083     else { pow = power; power = 1; }
02084     *out++ = TPLUS;
02085     if ( pow > 1 ) {
02086         subexp = GenerateFactors(pow,1);
02087         if ( subexp < 0 ) { error = -1; subexp = 0; }
02088         *out++ = TSUBEXP; PUTNUMBER128(out,subexp);
02089     }
02090     *out++ = TSYMBOL; *out++ = FACTORSYMBOL;
02091     *out++ = TPOWER; *out++ = TNUMBER; PUTNUMBER100(out,powfactor)
02092     powfactor += pow;
02093     if ( power > 0 ) *out++ = TMULTIPLY;
02094     else *out++ = TDIVIDE;
02095     s1 = startobject; while ( s1 < t ) *out++ = *s1++;
02096     startobject = 0; t = tt; continue;
02097 }
02098 else if ( *t == TEXPRESSSION ) {
02099     startobject = t;
02100     t++;
02101     nex = 0; while ( *t >= 0 ) { nex = nex*128 + *t++; }
02102     if ( *t == LBRACE ) continue;
02103     if ( *t == LPARENTHESIS ) {
02104         nparenthesis = 0; t++;
02105         while ( nparenthesis >= 0 ) {
02106             if ( *t == LPARENTHESIS ) nparenthesis++;
02107             else if ( *t == RPARENTHESIS ) nparenthesis--;
02108             t++;
02109         }
02110     }
02111     if ( ( AS.Oldvflags[nex] & ISFACTORIZED ) == 0 ) continue;
02112     if ( *t == TPOWER ) {
02113         pow = 1;
02114         tt = t+1;
02115         while ( ( *tt == TMINUS ) || ( *tt == TPLUS ) ) {
02116             if ( *tt == TMINUS ) pow = -pow;
02117             tt++;
02118         }
02119         if ( *tt != TNUMBER ) {
02120             MesPrint("Internal problems(1) in CodeFactors");
02121             return(-1);
02122         }
02123         tt++; x2 = 0; while ( *tt >= 0 ) { x2 = 100*x2 + *tt++; }
02124 /*
02125         We have an object in startobject till t. The power is
02126         power*pow*x2
02127 */
02128     dopower:
02129         power = power*pow*x2;
02130         if ( power < 0 ) { pow = -power; power = -1; }
02131         else if ( power == 0 ) { t = tt; startobject = tt; continue; }
02132         else { pow = power; power = 1; }
02133         *out++ = TPLUS;
02134         if ( pow > 1 ) {
02135             subexp = GenerateFactors(pow,AS.OldNumFactors[nex]);
02136             if ( subexp < 0 ) { error = -1; subexp = 0; }
02137             *out++ = TSUBEXP; PUTNUMBER128(out,subexp)
02138             *out++ = TMULTIPLY;
02139         }
02140         i = powfactor-1;
02141         if ( i > 0 ) {
02142             *out++ = TSYMBOL; *out++ = FACTORSYMBOL;
02143             if ( i > 1 ) {
02144                 *out++ = TPOWER; *out++ = TNUMBER; PUTNUMBER100(out,i)

```

```

02145         }
02146         *out++ = TMULTIPLY;
02147     }
02148     powfactor += AS.OldNumFactors[nexp]*pow;
02149     s1 = startobject;
02150     while ( s1 < t ) *out++ = *s1++;
02151     startobject = 0; t = tt; continue;
02152 }
02153 else {
02154     tt = t; pow = 1; x2 = 1; goto dopower;
02155 }
02156 }
02157 else if ( *t == TDOLLAR ) {
02158     startobject = t;
02159     t++;
02160     nexp = 0; while ( *t >= 0 ) { nexp = nexp*128 + *t++; }
02161     if ( *t == LBRACE ) continue;
02162     if ( Dollars[nexp].nfactors == 0 ) continue;
02163     if ( *t == TPOWER ) {
02164         pow = 1;
02165         tt = t+1;
02166         while ( ( *tt == TMINUS ) || ( *tt == TPLUS ) ) {
02167             if ( *tt == TMINUS ) pow = -pow;
02168             tt++;
02169         }
02170         if ( *tt != TNUMBER ) {
02171             MesPrint("Internal problems(2) in CodeFactors");
02172             return(-1);
02173         }
02174         tt++; x2 = 0; while ( *tt >= 0 ) { x2 = 100*x2 + *tt++; }
02175     /*
02176         We have an object in startobject till t. The power is
02177         power*pow*x2
02178     */
02179     dopowerd:
02180         power = power*pow*x2;
02181         if ( power < 0 ) { pow = -power; power = -1; }
02182         else if ( power == 0 ) { t = tt; startobject = tt; continue; }
02183         else { pow = power; power = 1; }
02184         if ( pow > 1 ) {
02185             subexp = GenerateFactors(pow,1);
02186             if ( subexp < 0 ) { error = -1; subexp = 0; }
02187         }
02188         for ( i = 1; i <= Dollars[nexp].nfactors; i++ ) {
02189             s1 = startobject; *out++ = TPLUS;
02190             while ( s1 < t ) *out++ = *s1++;
02191             *out++ = LBRACE; *out++ = TNUMBER; PUTNUMBER128(out,i)
02192             *out++ = RBACE;
02193             *out++ = TMULTIPLY;
02194             *out++ = TSYMBOL; *out++ = FACTORSYMBOL;
02195             *out++ = TPOWER; *out++ = TNUMBER; PUTNUMBER100(out,powfactor)
02196             powfactor += pow;
02197             if ( pow > 1 ) {
02198                 *out++ = TSUBEXP; PUTNUMBER128(out,subexp)
02199             }
02200         }
02201         startobject = 0; t = tt; continue;
02202     }
02203     else {
02204         tt = t; pow = 1; x2 = 1; goto dopower;
02205     }
02206 }
02207 t++;
02208 }
02209 if ( last == 0 ) { last = 1; goto dolast; }
02210 *out = TENDOFIT;
02211 e->numfactors = powfactor-1;
02212 e->vflags |= ISFACTORIZED;
02213 subexp = CodeGenerator(outtokens);
02214 if ( subexp < 0 ) error = -1;
02215 if ( insubexpbuffers >= MAXSUBEXPRESSIONS ) {
02216     MesPrint("&More than %d subexpressions inside one expression", (WORD)MAXSUBEXPRESSIONS);
02217     Terminate(-1);
02218 }
02219 if ( subexpbuffers+insubexpbuffers >= topsubexpbuffers ) {
02220     DoubleBuffer((void **)((void *)(&subexpbuffers))
02221         , (void **)((void *)(&topsubexpbuffers)), sizeof(SUBBUF), "subexpbuffers");
02222 }
02223 subexpbuffers[insubexpbuffers].subexpnum = subexp;
02224 subexpbuffers[insubexpbuffers].buffernum = AC.cbunum;
02225 subexp = insubexpbuffers++;
02226 M_free(outtokens, "CodeFactors");
02227 s1 = tokens;
02228 *s1++ = TSUBEXP; PUTNUMBER128(s1,subexp); *s1++ = TENDOFIT;
02229 if ( error < 0 ) return(-1);
02230 else return(subexp);
02231 }

```

```

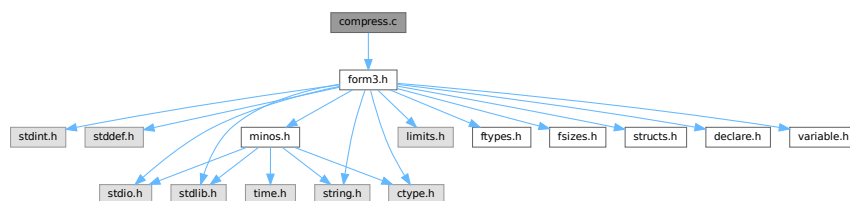
02232
02233 /*
02234     #] CodeFactors :
02235     #[ GenerateFactors :
02236
02237     Generates an expression of the type
02238     1+factor_+factor_^2+...+factor_^(n-1)
02239     (this is if inc=1)
02240     Returns the subexpression pointer of it.
02241 */
02242
02243 WORD GenerateFactors(WORD n,WORD inc)
02244 {
02245     int subexp;
02246     int i, error = 0;
02247     SBYTE *s;
02248     SBYTE *tokenbuffer = (SBYTE *)Malloc1(8*n*sizeof(SBYTE), "GenerateFactors");
02249     s = tokenbuffer;
02250     *s++ = TNUMBER; *s++ = 1;
02251     for ( i = inc; i < n*inc; i += inc ) {
02252         *s++ = TPLUS; *s++ = TSYMBOL; *s++ = FACTORSYMBOL;
02253         if ( i > 1 ) {
02254             *s++ = TPOWER; *s++ = TNUMBER;
02255             PUTNUMBER100(s,i)
02256         }
02257     }
02258     *s++ = TENDOFIT;
02259     subexp = CodeGenerator(tokenbuffer);
02260     if ( subexp < 0 ) error = -1;
02261     M_free(tokenbuffer, "GenerateFactors");
02262     if ( insubexpbuffers >= MAXSUBEXPRESSIONS ) {
02263         MesPrint("&More than %d subexpressions inside one expression", (WORD)MAXSUBEXPRESSIONS);
02264         Terminate(-1);
02265     }
02266     if ( subexpbuffers+insubexpbuffers >= topsubexpbuffers ) {
02267         DoubleBuffer((void **) (&subexpbuffers))
02268         , (void **) (&topsubexpbuffers), sizeof(SUBBUF), "subexpbuffers");
02269     }
02270     subexpbuffers[insubexpbuffers].subexpnum = subexp;
02271     subexpbuffers[insubexpbuffers].buffernum = AC.cbufnum;
02272     subexp = insubexpbuffers++;
02273     if ( error < 0 ) return(error);
02274     return(subexp);
02275 }
02276
02277 /*
02278     #] GenerateFactors :
02279     #] Compiler :
02280 */

```

## 4.13 compress.c File Reference

#include "form3.h"

Include dependency graph for compress.c:



### 4.13.1 Detailed Description

The routines for the use of gzip (de)compression of the information in the sort file.

Definition in file [compress.c](#).

## 4.14 compress.c

[Go to the documentation of this file.](#)

```

00001
00007 /* #[ License : */
00008 /*
00009 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00010 *   When using this file you are requested to refer to the publication
00011 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00012 *   This is considered a matter of courtesy as the development was paid
00013 *   for by FOM the Dutch physics granting agency and we would like to
00014 *   be able to track its scientific use to convince FOM of its value
00015 *   for the community.
00016 *
00017 *   This file is part of FORM.
00018 *
00019 *   FORM is free software: you can redistribute it and/or modify it under the
00020 *   terms of the GNU General Public License as published by the Free Software
00021 *   Foundation, either version 3 of the License, or (at your option) any later
00022 *   version.
00023 *
00024 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00025 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00026 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00027 *   details.
00028 *
00029 *   You should have received a copy of the GNU General Public License along
00030 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00031 */
00032 /* #[ License : */
00033
00034 #include "form3.h"
00035
00036 #ifdef WITHZLIB
00037 /*
00038 #define GZIPDEBUG
00039
00040     Low level routines for dealing with zlib during sorting and handling
00041     the scratch files. Work started 5-sep-2005.
00042     The .sor file handling was more or less completed on 8-sep-2005
00043     The handling of the scratch files still needs some thinking.
00044     Complications are:
00045         gzip compression should be per expression, not per buffer.
00046         No gzip compression for expressions with a bracket index.
00047         Separate decompression buffers for expressions in the rhs.
00048         This last one will involve more buffer work and organization.
00049         Information about compression should be stored for each expr.
00050         (including what method/program was used)
00051     Note: Be careful with compression. By far the most compact method
00052     is the original problem....
00053
00054     #[ Variables :
00055
00056     The following variables are to contain the intermediate buffers
00057     for the inflation of the various patches in the sort file.
00058     There can be up to MaxFpatches (FilePatches in the setup) and hence
00059     we can have that many streams simultaneously. We set this up once
00060     and only when needed.
00061         (in struct A.N or AB[threadnum].N)
00062     Bytef **AN.ziobufnum;
00063     Bytef *AN.ziobuffers;
00064 */
00065
00066 /*
00067     #[ Variables :
00068     #[ SetupOutputGZIP :
00069
00070     Routine prepares a gzip output stream for the given file.
00071 */
00072
00073 int SetupOutputGZIP(FILEHANDLE *f)
00074 {
00075     GETIDENTITY
00076
00077     if ( AT.SS != AT.S0 ) return(0);
00078     if ( AR.NoCompress == 1 ) return(0);
00079     if ( AR.gzipCompress <= 0 ) return(0);
00080
00081     if ( f->ziobuffer == 0 ) {
00082 /*
00083         1: Allocate a struct for the gzip stream:
00084 */
00085         f->zsp = Malloc1(sizeof(z_stream),"output zstream");
00086 /*
00087         2: Allocate the output buffer.

```

```

00088 */
00089     f->zibuffer =
00090         (Bytef *)Malloc1(f->ziosize*sizeof(char),"output zbuffer");
00091     if ( f->zsp == 0 || f->zibuffer == 0 ) {
00092         MLOCK(ErrorMessageLock);
00093         MesCall("SetupOutputGZIP");
00094         MUNLOCK(ErrorMessageLock);
00095         Terminate(-1);
00096     }
00097 }
00098 /*
00099 3: Set the default fields:
00100 */
00101 f->zsp->zalloc = Z_NULL;
00102 f->zsp->zfree  = Z_NULL;
00103 f->zsp->opaque = Z_NULL;
00104 /*
00105 4: Set the output space:
00106 */
00107 f->zsp->next_out = f->zibuffer;
00108 f->zsp->avail_out = f->ziosize;
00109 f->zsp->total_out = 0;
00110 /*
00111 5: Set the input space:
00112 */
00113 f->zsp->next_in = (Bytef *) (f->PObuffer);
00114 f->zsp->avail_in = (Bytef *) (f->POfill) - (Bytef *) (f->PObuffer);
00115 f->zsp->total_in = 0;
00116 /*
00117 6: Initiate the deflation
00118 */
00119 if ( deflateInit(f->zsp,AR.gzipCompress) != Z_OK ) {
00120     MLOCK(ErrorMessageLock);
00121     MesPrint("Error from zlib: %s",f->zsp->msg);
00122     MesCall("SetupOutputGZIP");
00123     MUNLOCK(ErrorMessageLock);
00124     Terminate(-1);
00125 }
00126
00127 return(0);
00128 }
00129
00130 /*
00131 #] SetupOutputGZIP :
00132 #[ PutOutputGZIP :
00133
00134 Routine is called when the PObuffer of f is full.
00135 The contents of it will be compressed and whenever the output buffer
00136 f->zibuffer is full it will be written and the output buffer
00137 will be reset.
00138 Upon exit the input buffer will be cleared.
00139 */
00140
00141 int PutOutputGZIP(FILEHANDLE *f)
00142 {
00143     GETIDENTITY
00144     int zerror;
00145     /*
00146     First set the number of bytes in the input
00147     */
00148     f->zsp->next_in = (Bytef *) (f->PObuffer);
00149     f->zsp->avail_in = (Bytef *) (f->POfill) - (Bytef *) (f->PObuffer);
00150     f->zsp->total_in = 0;
00151
00152     while ( ( zerror = deflate(f->zsp,Z_NO_FLUSH) ) == Z_OK ) {
00153         if ( f->zsp->avail_out == 0 ) {
00154             /*
00155             zibuffer is full. Write the output.
00156             */
00157             #ifdef GZIPDEBUG
00158             {
00159                 char *s = (char *) ((UBYTE *) (f->zibuffer)+f->ziosize);
00160                 MLOCK(ErrorMessageLock);
00161                 MesPrint("%wWriting %l bytes at %10p: %d %d %d %d %d"
00162                     ,f->ziosize,&(f->POposition),s[-5],s[-4],s[-3],s[-2],s[-1]);
00163                 MUNLOCK(ErrorMessageLock);
00164             }
00165             #endif
00166             #ifdef ALLLOCK
00167             LOCK(f->pthreadlock);
00168             #endif
00169             if ( f == AR.hidefile ) {
00170                 LOCK(AS.inputslock);
00171             }
00172             SeekFile(f->handle,&(f->POposition),SEEK_SET);
00173             if ( WriteFile(f->handle,(UBYTE *) (f->zibuffer),f->ziosize)
00174                 != f->ziosize ) {

```

```

00175         if ( f == AR.hidefile ) {
00176             UNLOCK(AS.inputslock);
00177         }
00178 #ifdef ALLLOCK
00179         UNLOCK(f->pthreadlock);
00180 #endif
00181         MLOCK(ErrorMessageLock);
00182         MesPrint("%wWrite error during compressed sort. Disk full?");
00183         MUNLOCK(ErrorMessageLock);
00184         return(-1);
00185     }
00186     if ( f == AR.hidefile ) {
00187         UNLOCK(AS.inputslock);
00188     }
00189 #ifdef ALLLOCK
00190         UNLOCK(f->pthreadlock);
00191 #endif
00192         ADDPOS(f->filesize,f->ziosize);
00193         ADDPOS(f->POposition,f->ziosize);
00194 #ifdef WITHPTHREADS
00195         if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
00196             if ( f->handle >= 0 ) SynchFile(f->handle);
00197         }
00198 #endif
00199 /*
00200     Reset the output
00201 */
00202     f->zsp->next_out = f->ziobuffer;
00203     f->zsp->avail_out = f->ziosize;
00204     f->zsp->total_out = 0;
00205 }
00206 else if ( f->zsp->avail_in == 0 ) {
00207 /*
00208     We compressed everything and it sits in ziobuffer. Finish
00209 */
00210     return(0);
00211 }
00212 else {
00213     MLOCK(ErrorMessageLock);
00214     MesPrint("%w avail_in = %d, avail_out = %d.",f->zsp->avail_in,f->zsp->avail_out);
00215     MUNLOCK(ErrorMessageLock);
00216     break;
00217 }
00218 }
00219 MLOCK(ErrorMessageLock);
00220 MesPrint("%wError in gzip handling of output. zerror = %d",zerror);
00221 MUNLOCK(ErrorMessageLock);
00222 return(-1);
00223 }
00224 /*
00225 #] PutOutputGZIP :
00226 #[ FlushOutputGZIP :
00227
00228 Routine is called to flush a stream. The compression of the input buffer
00229 will be completed and the contents of f->ziobuffer will be written.
00230 Both buffers will be cleared.
00231 */
00232 int FlushOutputGZIP(FILEHANDLE *f)
00233 {
00234     GETIDENTITY
00235     int zerror;
00236 /*
00237     Set the proper parameters
00238 */
00239     f->zsp->next_in = (Bytef *) (f->PObuffer);
00240     f->zsp->avail_in = (Bytef *) (f->POfill) - (Bytef *) (f->PObuffer);
00241     f->zsp->total_in = 0;
00242     while ( ( zerror = deflate(f->zsp,Z_FINISH) ) == Z_OK ) {
00243         if ( f->zsp->avail_out == 0 ) {
00244             /*
00245                 Write the output
00246             */
00247             #ifdef GZIPDEBUG
00248                 MLOCK(ErrorMessageLock);
00249                 MesPrint("%wWriting %l bytes at %l0p",f->ziosize,&(f->POposition));
00250                 MUNLOCK(ErrorMessageLock);
00251             #endif
00252             #ifdef ALLLOCK
00253                 LOCK(f->pthreadlock);
00254             #endif
00255             if ( f == AR.hidefile ) {
00256                 UNLOCK(AS.inputslock);
00257             }
00258             SeekFile(f->handle,&(f->POposition),SEEK_SET);

```

```

00262         if ( WriteFile(f->handle, (UBYTE *) (f->ziobuffer), f->ziosize)
00263             != f->ziosize ) {
00264             if ( f == AR.hidefile ) {
00265                 UNLOCK(AS.inputslock);
00266             }
00267 #ifdef ALLLOCK
00268         UNLOCK(f->pthreadlock);
00269 #endif
00270         MLOCK(ErrorMessageLock);
00271         MesPrint("%wWrite error during compressed sort. Disk full?");
00272         MUNLOCK(ErrorMessageLock);
00273         return(-1);
00274     }
00275     if ( f == AR.hidefile ) {
00276         UNLOCK(AS.inputslock);
00277     }
00278 #ifdef ALLLOCK
00279     UNLOCK(f->pthreadlock);
00280 #endif
00281     ADDPOS(f->filesize, f->ziosize);
00282     ADDPOS(f->POposition, f->ziosize);
00283 #ifdef WITHPTHREADS
00284     if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
00285         if ( f->handle >= 0 ) SynchFile(f->handle);
00286     }
00287 #endif
00288 /*
00289     Reset the output
00290 */
00291     f->zsp->next_out = f->ziobuffer;
00292     f->zsp->avail_out = f->ziosize;
00293     f->zsp->total_out = 0;
00294 }
00295 }
00296 if ( zerror == Z_STREAM_END ) {
00297 /*
00298     Write the output
00299 */
00300 #ifdef GZIPDEBUG
00301     MLOCK(ErrorMessageLock);
00302     MesPrint("%wWriting %l bytes at %l0p", (LONG) (f->zsp->avail_out), &(f->POposition));
00303     MUNLOCK(ErrorMessageLock);
00304 #endif
00305 #ifdef ALLLOCK
00306     LOCK(f->pthreadlock);
00307 #endif
00308     if ( f == AR.hidefile ) {
00309         LOCK(AS.inputslock);
00310     }
00311     SeekFile(f->handle, &(f->POposition), SEEK_SET);
00312     if ( WriteFile(f->handle, (UBYTE *) (f->ziobuffer), f->zsp->total_out)
00313         != (LONG) (f->zsp->total_out) ) {
00314         if ( f == AR.hidefile ) {
00315             UNLOCK(AS.inputslock);
00316         }
00317 #ifdef ALLLOCK
00318         UNLOCK(f->pthreadlock);
00319 #endif
00320         MLOCK(ErrorMessageLock);
00321         MesPrint("%wWrite error during compressed sort. Disk full?");
00322         MUNLOCK(ErrorMessageLock);
00323         return(-1);
00324     }
00325     if ( f == AR.hidefile ) {
00326         LOCK(AS.inputslock);
00327     }
00328 #ifdef ALLLOCK
00329     UNLOCK(f->pthreadlock);
00330 #endif
00331     ADDPOS(f->filesize, f->zsp->total_out);
00332     ADDPOS(f->POposition, f->zsp->total_out);
00333 #ifdef WITHPTHREADS
00334     if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
00335         if ( f->handle >= 0 ) SynchFile(f->handle);
00336     }
00337 #endif
00338 #ifdef GZIPDEBUG
00339     MLOCK(ErrorMessageLock);
00340     { char *s = f->ziobuffer+f->zsp->total_out;
00341       MesPrint("%w    Last bytes written: %d %d %d %d %d", s[-5], s[-4], s[-3], s[-2], s[-1]);
00342     }
00343     MesPrint("%w    Perceived position in FlushOutputGZIP is %l0p", &(f->POposition));
00344     MUNLOCK(ErrorMessageLock);
00345 #endif
00346 /*
00347     Reset the output
00348 */

```

```

00349         f->zsp->next_out = f->ziobuffer;
00350         f->zsp->avail_out = f->ziosize;
00351         f->zsp->total_out = 0;
00352         if ( ( zerror = deflateEnd(f->zsp) ) == Z_OK ) return(0);
00353         MLOCK(ErrorMessageLock);
00354         if ( f->zsp->msg ) {
00355             MesPrint("%wError in finishing gzip handling of output: %s",f->zsp->msg);
00356         }
00357         else {
00358             MesPrint("%wError in finishing gzip handling of output.");
00359         }
00360         MUNLOCK(ErrorMessageLock);
00361     }
00362     else {
00363         MLOCK(ErrorMessageLock);
00364         MesPrint("%wError in gzip handling of output.");
00365         MUNLOCK(ErrorMessageLock);
00366     }
00367     return(-1);
00368 }
00369
00370 /*
00371  #] FlushOutputGZIP :
00372  #[ SetupAllInputGZIP :
00373
00374  Routine prepares all gzip input streams for a merge.
00375
00376  Problem (29-may-2008): If we never use GZIP compression, this routine
00377  will still allocate the array space. This is an enormous amount!
00378  It places an effective restriction on the value of SortIOSize
00379  */
00380
00381 int SetupAllInputGZIP(SORTING *S)
00382 {
00383     GETIDENTITY
00384     int i, NumberOpened = 0;
00385     z_stream zsp;
00386     /*
00387     This code was added 29-may-2008 by JV to prevent further processing if
00388     there is no compression at all (usually).
00389     */
00390     for ( i = 0; i < S->inNum; i++ ) {
00391         if ( S->fpincompressed[i] ) break;
00392     }
00393     if ( i >= S->inNum ) return(0);
00394
00395     if ( S->zsparray == 0 ) {
00396         S->zsparray = (z_stream*)Malloc1(sizeof(z_stream)*S->MaxFpatches,"input zstreams");
00397         if ( S->zsparray == 0 ) {
00398             MLOCK(ErrorMessageLock);
00399             MesCall("SetupAllInputGZIP");
00400             MUNLOCK(ErrorMessageLock);
00401             Terminate(-1);
00402         }
00403     }
00404     /*
00405     We add 128 bytes in the hope that if it can happen that it goes
00406     outside the buffer during decompression, it does not do damage.
00407     */
00408     AN.ziobuffers = (Bytef *)Malloc1(S->MaxFpatches*(S->file.ziosize+128)*sizeof(Bytef),"input raw
00409 buffers");
00410     /*
00411     This seems to be one of the really stupid errors:
00412     We allocate way too much space. Way way way too much.
00413     AN.ziobufnum = (Bytef **)Malloc1(S->MaxFpatches*S->file.ziosize*sizeof(Bytef *),"input raw
00414 pointers");
00415     */
00416     AN.ziobufnum = (Bytef **)Malloc1(S->MaxFpatches*sizeof(Bytef *),"input raw pointers");
00417     if ( AN.ziobuffers == 0 || AN.ziobufnum == 0 ) {
00418         MLOCK(ErrorMessageLock);
00419         MesCall("SetupAllInputGZIP");
00420         MUNLOCK(ErrorMessageLock);
00421         Terminate(-1);
00422     }
00423     for ( i = 0; i < S->MaxFpatches; i++ ) {
00424         AN.ziobufnum[i] = AN.ziobuffers + i * (S->file.ziosize+128);
00425     }
00426     for ( i = 0; i < S->inNum; i++ ) {
00427         #ifdef GZIPDEBUG
00428         MLOCK(ErrorMessageLock);
00429         MesPrint("%wPreparing z-stream %d with compression %d",i,S->fpincompressed[i]);
00430         MUNLOCK(ErrorMessageLock);
00431         #endif
00432         if ( S->fpincompressed[i] ) {
00433             zsp = &(S->zsparray[i]);
00434             /*
00435             1: Set the default fields:

```



```

00434 */
00435     zsp->zalloc = Z_NULL;
00436     zsp->zfree  = Z_NULL;
00437     zsp->opaque = Z_NULL;
00438 /*
00439     2: Set the output space:
00440 */
00441     zsp->next_out = Z_NULL;
00442     zsp->avail_out = 0;
00443     zsp->total_out = 0;
00444 /*
00445     3: Set the input space temporarily:
00446 */
00447     zsp->next_in = Z_NULL;
00448     zsp->avail_in = 0;
00449     zsp->total_in = 0;
00450 /*
00451     4: Initiate the inflation
00452 */
00453     if ( inflateInit(zsp) != Z_OK ) {
00454         MLOCK(ErrorMessageLock);
00455         if ( zsp->msg ) MesPrint("%wError from inflateInit: %s",zsp->msg);
00456         else           MesPrint("%wError from inflateInit");
00457         MesCall("SetupAllInputGZIP");
00458         MUNLOCK(ErrorMessageLock);
00459         Terminate(-1);
00460     }
00461     NumberOpened++;
00462 }
00463 }
00464 return (NumberOpened);
00465 }
00466
00467 /*
00468 #] SetupAllInputGZIP :
00469 #[ FillInputGZIP :
00470
00471 Routine is called when we need new input in the specified buffer.
00472 This buffer is used for the output and we keep reading and uncompressing
00473 input till either this buffer is full or the input stream is finished.
00474 The return value is the number of bytes in the buffer.
00475 */
00476
00477 LONG FillInputGZIP(FILEHANDLE *f, POSITION *position, UBYTE *buffer, LONG buffersize, int numstream)
00478 {
00479     GETIDENTITY
00480     int zerror;
00481     LONG readsize, toread = 0;
00482     SORTING *S = AT.SS;
00483     z_streamp zsp;
00484     POSITION pos;
00485     if ( S->fpincompressed[numstream] ) {
00486         zsp = &(S->zsparray[numstream]);
00487         zsp->next_out = (Bytef *)buffer;
00488         zsp->avail_out = buffersize;
00489         zsp->total_out = 0;
00490         if ( zsp->avail_in == 0 ) {
00491 /*
00492         First loading of the input
00493 */
00494         if ( ISGEPOSINC(S->fPatchesStop[numstream],*position,f->ziosize) ) {
00495             toread = f->ziosize;
00496         }
00497         else {
00498             DIFFPOS(pos,S->fPatchesStop[numstream],*position);
00499             toread = (LONG) (BASEPOSITION(pos));
00500         }
00501         if ( toread > 0 ) {
00502 #ifdef GZIPDEBUG
00503             MLOCK(ErrorMessageLock);
00504             MesPrint("%w-+Reading %l bytes in stream %d at position %l0p; stop at
00505 %l0p",toread,numstream,position,&(S->fPatchesStop[numstream]));
00506             MUNLOCK(ErrorMessageLock);
00507 #endif
00508 #ifdef ALLLOCK
00509             LOCK(f->pthreadslock);
00510 #endif
00511             SeekFile(f->handle,position,SEEK_SET);
00512             readsize = ReadFile(f->handle,(UBYTE *) (AN.ziobufnum[numstream]),toread);
00513             SeekFile(f->handle,position,SEEK_CUR);
00514 #ifdef ALLLOCK
00515             UNLOCK(f->pthreadslock);
00516 #endif
00517 #ifdef GZIPDEBUG
00518             MLOCK(ErrorMessageLock);
00519             { char *s = AN.ziobufnum[numstream]+readsize;
00520               MesPrint("%w read: %l +Last bytes read: %d %d %d %d %d in %s, newpos =

```

```

        %10p",readsize,s[-5],s[-4],s[-3],s[-2],s[-1],f->name,position);
00520     }
00521     MUNLOCK(ErrorMessageLock);
00522 #endif
00523     if ( readsize == 0 ) {
00524         zsp->next_in = AN.ziobufnum[numstream];
00525         zsp->avail_in = f->ziosize;
00526         zsp->total_in = 0;
00527         return(zsp->total_out);
00528     }
00529     if ( readsize < 0 ) {
00530         MLOCK(ErrorMessageLock);
00531         MesPrint("%wFillInputGZIP: Read error during compressed sort.");
00532         MUNLOCK(ErrorMessageLock);
00533         return(-1);
00534     }
00535     ADDPOS(f->filesize,readsize);
00536     ADDPOS(f->POposition,readsize);
00537 /*
00538     Set the input
00539 */
00540     zsp->next_in = AN.ziobufnum[numstream];
00541     zsp->avail_in = readsize;
00542     zsp->total_in = 0;
00543 }
00544 }
00545 if ( toread > 0 || zsp->avail_in ) {
00546     while ( ( zerror = inflate(zsp,Z_NO_FLUSH) ) == Z_OK ) {
00547         if ( zsp->avail_out == 0 ) {
00548 /*
00549             Finish
00550 */
00551             return((LONG) (zsp->total_out));
00552         }
00553         if ( zsp->avail_in == 0 ) {
00554             if ( ISEQUALPOS(S->fPatchesStop[numstream],*position) ) {
00555 /*
00556                 We finished this stream. Try to terminate.
00557 */
00558                 zerror = Z_STREAM_END;
00559                 break;
00560             }
00561             if ( ( zerror = inflate(zsp,Z_SYNC_FLUSH) ) == Z_OK ) {
00562                 return((LONG) (zsp->total_out));
00563             }
00564             else {
00565                 break;
00566             }
00567 /*
00568 */
00569 #ifdef GZIPDEBUG
00570             MLOCK(ErrorMessageLock);
00571             MesPrint("%wClosing stream %d",numstream);
00572 #endif
00573             readsize = zsp->total_out;
00574 #ifdef GZIPDEBUG
00575             if ( readsize > 0 ) {
00576                 WORD *s = (WORD *) (buffer+zsp->total_out);
00577                 MesPrint("%w    Last words: %d %d %d %d %d",s[-5],s[-4],s[-3],s[-2],s[-1]);
00578             }
00579             else {
00580                 MesPrint("%w No words");
00581             }
00582             MUNLOCK(ErrorMessageLock);
00583 #endif
00584             if ( ( zerror = inflateEnd(zsp) ) == Z_OK ) return(readsize);
00585             break;
00586 /*
00587         }
00588 */
00589         Read more input
00590 */
00591 #ifdef GZIPDEBUG
00592         if ( numstream == 0 ) {
00593             MLOCK(ErrorMessageLock);
00594             MesPrint("%wWant to read in stream 0 at position %10p",position);
00595             MUNLOCK(ErrorMessageLock);
00596         }
00597 #endif
00598         if ( ISGEPOSINC(S->fPatchesStop[numstream],*position,f->ziosize) ) {
00599             toread = f->ziosize;
00600         }
00601         else {
00602             DIFPOS(pos,S->fPatchesStop[numstream],*position);
00603             toread = (LONG) (BASEPOSITION(pos));
00604         }
00605 #ifdef GZIPDEBUG

```

```

00606             MLOCK(ErrorMessageLock);
00607             MesPrint("%w--Reading %l bytes in stream %d at position
%10p",toread,numstream,position);
00608             MUNLOCK(ErrorMessageLock);
00609 #endif
00610 #ifdef ALLLOCK
00611             LOCK(f->pthreadlock);
00612 #endif
00613             SeekFile(f->handle,position,SEEK_SET);
00614             readsize = ReadFile(f->handle,(UBYTE *) (AN.ziobufnum[numstream]),toread);
00615             SeekFile(f->handle,position,SEEK_CUR);
00616 #ifdef ALLLOCK
00617             UNLOCK(f->pthreadlock);
00618 #endif
00619 #ifdef GZIPDEBUG
00620             MLOCK(ErrorMessageLock);
00621             { char *s = AN.ziobufnum[numstream]+readsize;
00622               MesPrint("%w Last bytes read: %d %d %d %d %d",s[-5],s[-4],s[-3],s[-2],s[-1]);
00623             }
00624             MUNLOCK(ErrorMessageLock);
00625 #endif
00626             if ( readsize == 0 ) {
00627                 zsp->next_in = AN.ziobufnum[numstream];
00628                 zsp->avail_in = f->ziosize;
00629                 zsp->total_in = 0;
00630                 return(zsp->total_out);
00631             }
00632             if ( readsize < 0 ) {
00633                 MLOCK(ErrorMessageLock);
00634                 MesPrint("%wFillInputGZIP: Read error during compressed sort.");
00635                 MUNLOCK(ErrorMessageLock);
00636                 return(-1);
00637             }
00638             ADDPOS(f->filesize,readsize);
00639             ADDPOS(f->PPosition,readsize);
00640 /*
00641             Reset the input
00642 */
00643             zsp->next_in = AN.ziobufnum[numstream];
00644             zsp->avail_in = readsize;
00645             zsp->total_in = 0;
00646         }
00647         else {
00648             break;
00649         }
00650     }
00651 }
00652 else {
00653     zerror = Z_STREAM_END;
00654     zsp->total_out = 0;
00655 }
00656 #ifdef GZIPDEBUG
00657 MLOCK(ErrorMessageLock);
00658 MesPrint("%w zerror = %d in stream %d. At position %10p",zerror,numstream,position);
00659 MUNLOCK(ErrorMessageLock);
00660 #endif
00661 if ( zerror == Z_STREAM_END ) {
00662 /*
00663     Reset the input
00664 */
00665     zsp->next_in = Z_NULL;
00666     zsp->avail_in = 0;
00667     zsp->total_in = 0;
00668 /*
00669     Make the final call and finish
00670 */
00671 #ifdef GZIPDEBUG
00672 MLOCK(ErrorMessageLock);
00673 MesPrint("%wClosing stream %d",numstream);
00674 #endif
00675     readsize = zsp->total_out;
00676 #ifdef GZIPDEBUG
00677     if ( readsize > 0 ) {
00678         WORD *s = (WORD *) (buffer+zsp->total_out);
00679         MesPrint("%w -Last words: %d %d %d %d %d",s[-5],s[-4],s[-3],s[-2],s[-1]);
00680     }
00681     else {
00682         MesPrint("%w No words");
00683     }
00684     MUNLOCK(ErrorMessageLock);
00685 #endif
00686     if ( zsp->zalloc != Z_NULL ) {
00687         zerror = inflateEnd(zsp);
00688         zsp->zalloc = Z_NULL;
00689     }
00690     if ( zerror == Z_OK || zerror == Z_STREAM_END ) return(readsize);
00691 }

```

```

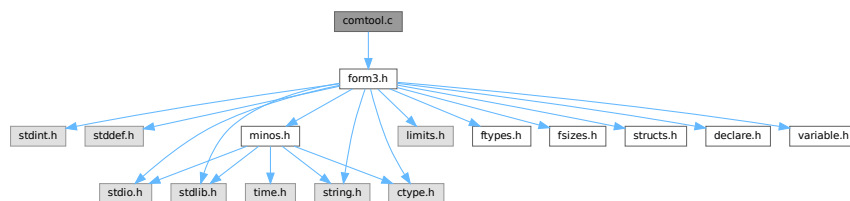
00692
00693     MLOCK(ErrorMessageLock);
00694     MesPrint("%wFillInputGZIP: Error in gzip handling of input. zerror = %d",zerror);
00695     MUNLOCK(ErrorMessageLock);
00696     return(-1);
00697 }
00698 else {
00699 #ifdef GZIPDEBUG
00700     MLOCK(ErrorMessageLock);
00701     MesPrint("%w++Reading %l bytes at position %l0p",buffer,size,position);
00702     MUNLOCK(ErrorMessageLock);
00703 #endif
00704 #ifdef ALLLOCK
00705     LOCK(f->pthreadlock);
00706 #endif
00707     SeekFile(f->handle,position,SEEK_SET);
00708     readsize = ReadFile(f->handle,buffer,buffer_size);
00709     SeekFile(f->handle,position,SEEK_CUR);
00710 #ifdef ALLLOCK
00711     UNLOCK(f->pthreadlock);
00712 #endif
00713     if ( readsize < 0 ) {
00714         MLOCK(ErrorMessageLock);
00715         MesPrint("%wFillInputGZIP: Read error during uncompressed sort.");
00716         MesPrint("%w++Reading %l bytes at position %l0p",buffer,size,position);
00717         MUNLOCK(ErrorMessageLock);
00718     }
00719     return(readsize);
00720 }
00721 }
00722
00723 /*
00724 #] FillInputGZIP :
00725 #[ ClearSortGZIP :
00726 */
00727
00728 void ClearSortGZIP(FILEHANDLE *f)
00729 {
00730     if ( f->zbuffer ) {
00731         M_free(f->zbuffer,"output zbuffer");
00732         M_free(f->zsp,"output zstream");
00733         f->zbuffer = 0;
00734     }
00735 }
00736
00737 /*
00738 #] ClearSortGZIP :
00739 */
00740 #endif

```

## 4.15 comtool.c File Reference

#include "form3.h"

Include dependency graph for comtool.c:



### Functions

- int [inicbufs](#) (void)
- void [finishcbuf](#) (WORD num)
- void [clearcbuf](#) (WORD num)

- WORD \* [DoubleCbuffer](#) (int num, WORD \*w, int par)
- WORD \* [AddLHS](#) (int num)
- WORD \* [AddRHS](#) (int num, int type)
- int [AddNtoL](#) (int n, WORD \*array)
- int [AddNtoC](#) (int bufnum, int n, WORD \*array, int par)
- int [InsTree](#) (int bufnum, int h)
- int [FindTree](#) (int bufnum, WORD \*subexpr)
- void [RedoTree](#) (CBUF \*C, int size)
- void [ClearTree](#) (int i)
- int [IniFbuffer](#) (WORD bufnum)
- LONG [numcommute](#) (WORD \*terms, LONG \*numterms)

### 4.15.1 Detailed Description

Utility routines for the compiler.

Definition in file [comtool.c](#).

### 4.15.2 Function Documentation

#### 4.15.2.1 inicbufs()

```
int inicbufs (
    void )
```

Creates a new compiler buffer and returns its ID number.

#### Returns

The ID number for the new compiler buffer.

Definition at line 47 of file [comtool.c](#).

References [CbUf::boomlijst](#), [CbUf::Buffer](#), [CbUf::BufferSize](#), [CbUf::CanCommu](#), [CbUf::dimension](#), [CbUf::lhs](#), [CbUf::numdum](#), [CbUf::NumTerms](#), [CbUf::Pointer](#), [CbUf::rhs](#), and [CbUf::Top](#).

Referenced by [AddRHS\(\)](#), and [StartVariables\(\)](#).

#### 4.15.2.2 finishcbuf()

```
void finishcbuf (
    WORD num )
```

Frees a compiler buffer.

#### Parameters

|            |                                           |
|------------|-------------------------------------------|
| <i>num</i> | The ID number for the buffer to be freed. |
|------------|-------------------------------------------|

Definition at line 89 of file [comtool.c](#).

References [CbUf::boomlijst](#), [CbUf::Buffer](#), [CbUf::BufferSize](#), [CbUf::CanCommu](#), [CbUf::lhs](#), [CbUf::NumTerms](#), [CbUf::Pointer](#), [CbUf::rhs](#), and [CbUf::Top](#).

Referenced by [PF\\_BroadcastCBuf\(\)](#).

#### 4.15.2.3 clearcbuf()

```
void clearcbuf (
    WORD num )
```

Clears contents in a compiler buffer.

##### Parameters

|            |                                             |
|------------|---------------------------------------------|
| <i>num</i> | The ID number for the buffer to be cleared. |
|------------|---------------------------------------------|

Definition at line 116 of file [comtool.c](#).

References [CbUf::boomlijst](#), [CbUf::Buffer](#), [CbUf::Pointer](#), and [CbUf::rhs](#).

#### 4.15.2.4 DoubleCbuffer()

```
WORD * DoubleCbuffer (
    int num,
    WORD * w,
    int par )
```

Doubles a compiler buffer.

##### Parameters

|            |                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------|
| <i>num</i> | The ID number for the buffer to be doubled.                                                                                |
| <i>w</i>   | The pointer to the end (exclusive) of the current buffer. The contents in the range of [cbuff[num].Buffer,w) will be kept. |

Definition at line 143 of file [comtool.c](#).

References [CbUf::Buffer](#), [CbUf::BufferSize](#), [CbUf::lhs](#), [CbUf::Pointer](#), [CbUf::rhs](#), and [CbUf::Top](#).

Referenced by [AddNtoC\(\)](#), and [AddNtoL\(\)](#).

#### 4.15.2.5 AddLHS()

```
WORD * AddLHS (
    int num )
```

Adds an LHS to a compiler buffer and returns the pointer to a buffer for the new LHS.

**Parameters**

|            |                                                  |
|------------|--------------------------------------------------|
| <i>num</i> | The ID number for the buffer to get another LHS. |
|------------|--------------------------------------------------|

Definition at line 188 of file [comtool.c](#).

References [CbUf::lhs](#), and [CbUf::Pointer](#).

Referenced by [AddNtoL\(\)](#).

**4.15.2.6 AddRHS()**

```
WORD * AddRHS (
    int num,
    int type )
```

Adds an RHS to a compiler buffer and returns the pointer to a buffer for the new RHS.

**Parameters**

|             |                                                   |
|-------------|---------------------------------------------------|
| <i>num</i>  | The ID number for the buffer to get another RHS.  |
| <i>type</i> | If 0, the subexpression tree will be reallocated. |

Definition at line 214 of file [comtool.c](#).

References [TaBIEs::buffers](#), [TaBIEs::buffersfill](#), [TaBIEs::bufferssize](#), [TaBIEs::bufnum](#), [CbUf::CanCommu](#), [CbUf::dimension](#), [inicbufs\(\)](#), [CbUf::numdum](#), [CbUf::NumTerms](#), [CbUf::Pointer](#), and [CbUf::rhs](#).

Referenced by [InsertArg\(\)](#), [StartVariables\(\)](#), and [TestMatch\(\)](#).

Here is the call graph for this function:

**4.15.2.7 AddNtoL()**

```
int AddNtoL (
    int n,
    WORD * array )
```

Adds an LHS with the given data to the current compiler buffer.

**Parameters**

|              |                         |
|--------------|-------------------------|
| <i>n</i>     | The length of the data. |
| <i>array</i> | The data to be added.   |

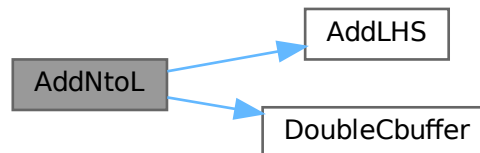
**Returns**

0 if succeeds.

Definition at line 288 of file [comtool.c](#).

References [AddLHS\(\)](#), [DoubleCbuffer\(\)](#), [CbUf::Pointer](#), and [CbUf::Top](#).

Here is the call graph for this function:

**4.15.2.8 AddNtoC()**

```

int AddNtoC (
    int bufnum,
    int n,
    WORD * array,
    int par )
  
```

Adds the given data to the last LHS/RHS in a compiler buffer.

**Parameters**

|               |                                                            |
|---------------|------------------------------------------------------------|
| <i>bufnum</i> | The ID number for the buffer where the data will be added. |
| <i>n</i>      | The length of the data.                                    |
| <i>array</i>  | The data to be added.                                      |

**Returns**

0 if succeeds.

Definition at line 317 of file [comtool.c](#).

References [DoubleCbuffer\(\)](#), [CbUf::Pointer](#), and [CbUf::Top](#).



Referenced by [InsertArg\(\)](#), and [StartVariables\(\)](#).

Here is the call graph for this function:



#### 4.15.2.9 `InsTree()`

```
int InsTree (
    int bufnum,
    int h )
```

Definition at line [355](#) of file [comtool.c](#).

#### 4.15.2.10 `FindTree()`

```
int FindTree (
    int bufnum,
    WORD * subexpr )
```

Definition at line [533](#) of file [comtool.c](#).

#### 4.15.2.11 `RedoTree()`

```
void RedoTree (
    CBUF * C,
    int size )
```

Definition at line [569](#) of file [comtool.c](#).

#### 4.15.2.12 `ClearTree()`

```
void ClearTree (
    int i )
```

Definition at line [588](#) of file [comtool.c](#).

#### 4.15.2.13 IniFbuffer()

```
int IniFbuffer (
    WORD bufnum )
```

Initialize a factorization cache buffer. We set the size of the rhs and boomlijst buffers immediately to their final values.

Definition at line 614 of file [comtool.c](#).

References [tree::blnce](#), [CbUf::boomlijst](#), [CbUf::CanCommu](#), [CbUf::dimension](#), [tree::left](#), [CbUf::numdum](#), [CbUf::NumTerms](#), [tree::parent](#), [CbUf::rhs](#), [tree::right](#), [tree::usage](#), and [tree::value](#).

#### 4.15.2.14 numcommute()

```
LONG numcommute (
    WORD * terms,
    LONG * numterms )
```

Definition at line 659 of file [comtool.c](#).

### 4.16 comtool.c

[Go to the documentation of this file.](#)

```
00001
00005 /* # [ License : */
00006 /*
00007 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00008 * When using this file you are requested to refer to the publication
00009 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010 * This is considered a matter of courtesy as the development was paid
00011 * for by FOM the Dutch physics granting agency and we would like to
00012 * be able to track its scientific use to convince FOM of its value
00013 * for the community.
00014 *
00015 * This file is part of FORM.
00016 *
00017 * FORM is free software: you can redistribute it and/or modify it under the
00018 * terms of the GNU General Public License as published by the Free Software
00019 * Foundation, either version 3 of the License, or (at your option) any later
00020 * version.
00021 *
00022 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00023 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00024 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00025 * details.
00026 *
00027 * You should have received a copy of the GNU General Public License along
00028 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00029 */
00030 /* # [ License : */
00031 /*
00032 * # [ Includes :
00033 */
00034
00035 #include "form3.h"
00036
00037 /*
00038 * #] Includes :
00039 * # [ inicbufs :
00040 */
00041
00047 int inicbufs(void)
00048 {
00049     int i, num = AC.cbufList.num;
00050     CBUF *C = cbuf;
00051     for ( i = 0; i < num; i++, C++ ) {
00052         if ( C->Buffer == 0 ) break;
```

```

00053     }
00054     if ( i >= num ) C = (CBUF *)FromList(&AC.cbuffers);
00055     else num = i;
00056     C->BufferSize = 2000;
00057     C->Buffer = (WORD *)Malloc1(C->BufferSize*sizeof(WORD),"compiler buffer-1");
00058     C->Pointer = C->Buffer;
00059     C->Top = C->Buffer + C->BufferSize;
00060     C->maxlhs = 10;
00061     C->lhs = (WORD **)Malloc1(C->maxlhs*sizeof(WORD *),"compiler buffer-2");
00062     C->numlhs = 0;
00063     C->mnumlhs = 0;
00064     C->maxrhs = 25;
00065     C->rhs = (WORD **)Malloc1(C->maxrhs*(sizeof(WORD *)+2*sizeof(LONG)+2*sizeof(WORD)),"compiler
buffer-3");
00066     C->CanCommu = (LONG *) (C->rhs+C->maxrhs);
00067     C->NumTerms = C->CanCommu+C->maxrhs;
00068     C->numdum = (WORD *) (C->NumTerms+C->maxrhs);
00069     C->dimension = C->numdum + C->maxrhs;
00070     C->numrhs = 0;
00071     C->mnumrhs = 0;
00072     C->rhs[0] = C->rhs[1] = C->Pointer;
00073     C->boomlijst = 0;
00074     RedoTree(C,C->maxrhs);
00075     ClearTree(num);
00076     return(num);
00077 }
00078
00079 /*
00080 #] inicbufs :
00081 #[ finishcbuf :
00082 */
00083
00089 void finishcbuf(WORD num)
00090 {
00091     CBUF *C = cbuf+num;
00092     if ( C->Buffer ) M_free(C->Buffer,"compiler buffer-1");
00093     if ( C->rhs ) M_free(C->rhs,"compiler buffer-3");
00094     if ( C->lhs ) M_free(C->lhs,"compiler buffer-2");
00095     if ( C->boomlijst ) M_free(C->boomlijst,"boomlijst");
00096     C->Top = C->Pointer = C->Buffer = 0;
00097     C->rhs = C->lhs = 0;
00098     C->CanCommu = 0;
00099     C->NumTerms = 0;
00100     C->BufferSize = 0;
00101     C->boomlijst = 0;
00102     C->numlhs = C->numrhs = C->maxlhs = C->maxrhs = C->mnumlhs =
00103     C->mnumrhs = C->numtree = C->rootnum = C->MaxTreeSize = 0;
00104 }
00105
00106 /*
00107 #] finishcbuf :
00108 #[ clearcbuf :
00109 */
00110
00116 void clearcbuf(WORD num)
00117 {
00118     CBUF *C = cbuf+num;
00119     if ( C->boomlijst ) M_free(C->boomlijst,"boomlijst");
00120     C->Pointer = C->Buffer;
00121     C->numrhs = C->numlhs = 0;
00122     C->mnumlhs = 0;
00123     C->boomlijst = 0;
00124     C->mnumrhs = 0;
00125     C->rhs[0] = C->rhs[1] = C->Pointer;
00126     C->numtree = C->rootnum = C->MaxTreeSize = 0;
00127     RedoTree(C,C->maxrhs);
00128     ClearTree(num);
00129 }
00130
00131 /*
00132 #] clearcbuf :
00133 #[ DoubleCbuffer :
00134 */
00135
00143 WORD *DoubleCbuffer(int num, WORD *w,int par)
00144 {
00145     CBUF *C = cbuf + num;
00146     LONG newsize = C->BufferSize*2;
00147     WORD *newbuffer = (WORD *)Malloc1(newsize*sizeof(WORD),"compiler buffer-4");
00148     WORD *wl, *w2;
00149     LONG offset, j, i;
00150     DUMMYUSE(par)
00151     /*
00152     MLOCK(ErrorMessageLock);
00153     MesPrint(" doubleCbuffer: par = %d",par);
00154     MUNLOCK(ErrorMessageLock);
00155     */

```

```

00156     w1 = C->Buffer; w2 = newbuffer;
00157     i = w - w1;
00158     j = i & 7;
00159     while ( --j >= 0 ) *w2++ = *w1++;
00160     i >= 3;
00161     while ( --i >= 0 ) {
00162         *w2++ = *w1++; *w2++ = *w1++; *w2++ = *w1++; *w2++ = *w1++;
00163         *w2++ = *w1++; *w2++ = *w1++; *w2++ = *w1++; *w2++ = *w1++;
00164     }
00165     offset = newbuffer - C->Buffer;
00166     for ( i = 0; i <= C->numlhs; i++ ) C->lhs[i] += offset;
00167     for ( i = 1; i <= C->numrhs; i++ ) C->rhs[i] += offset;
00168     w1 = C->Buffer;
00169     C->Pointer += offset;
00170     C->Top = newbuffer + newsize;
00171     C->BufferSize = newsize;
00172     C->Buffer = newbuffer;
00173     M_free(w1,"DoubleCbuffer");
00174     return(w2);
00175 }
00176
00177 /*
00178     #] DoubleCbuffer :
00179     #[ AddLHS :
00180 */
00181
00182 WORD *AddLHS(int num)
00183 {
00184     CBUF *C = cbuf + num;
00185     C->numlhs++;
00186     if ( C->numlhs >= (C->maxlhs-2) ) {
00187         WORD **ppp = &(C->lhs); /* to avoid compiler warning */
00188         if ( DoubleList((void ***)ppp,&(C->maxlhs),sizeof(WORD *),
00189             "statement lists") ) Terminate(-1);
00190     }
00191     C->lhs[C->numlhs] = C->Pointer;
00192     C->lhs[C->numlhs+1] = 0;
00193     return(C->Pointer);
00194 }
00195
00196 /*
00197     #] AddLHS :
00198     #[ AddRHS :
00199 */
00200
00201 WORD *AddRHS(int num, int type)
00202 {
00203     LONG fullsize, *lold, newsize;
00204     int i;
00205     WORD *old, *wold;
00206     CBUF *C;
00207 restart:;
00208     C = cbuf + num;
00209     if ( C->numrhs >= (C->maxrhs-2) ) {
00210         if ( C->maxrhs == 0 ) newsize = 100;
00211         else newsize = C->maxrhs * 2;
00212         if ( newsize > MAXCOMBUFRHS ) newsize = MAXCOMBUFRHS;
00213         if ( newsize == C->maxrhs ) {
00214             if ( AC.tablefilling ) {
00215                 TABLES T = functions[AC.tablefilling].tabl;
00216                 /*
00217                     We add a compiler buffer, change a few settings and continue.
00218                 */
00219                 if ( T->buffersfill >= T->bufferssize ) {
00220                     int new1 = 2*T->bufferssize;
00221                     WORD *nbufs = (WORD *)Malloc1(new1*sizeof(WORD),"Table compile buffers");
00222                     for ( i = 0; i < T->buffersfill; i++ )
00223                         nbufs[i] = T->buffers[i];
00224                     for ( ; i < new1; i++ ) nbufs[i] = 0;
00225                     M_free(T->buffers,"Table compile buffers");
00226                     T->buffers = nbufs;
00227                     T->bufferssize = new1;
00228                 }
00229                 T->buffers[T->buffersfill++] = T->bufnum = inicbufs();
00230                 AC.cbufnum = num = T->bufnum;
00231                 goto restart;
00232             }
00233             else {
00234                 MesPrint("@Compiler buffer overflow. Try to make modules smaller");
00235                 Terminate(-1);
00236             }
00237         }
00238     }
00239     old = C->rhs;
00240     fullsize = newsize * (sizeof(WORD *) + 2*sizeof(LONG) + 2*sizeof(WORD));
00241     C->rhs = (WORD **)Malloc1(fullsize,"subexpression lists");
00242     for ( i = 0; i < C->maxrhs; i++ ) C->rhs[i] = old[i];
00243     lold = C->CanCommu; C->CanCommu = (LONG *) (C->rhs+newsize);

```

```

00256         for ( i = 0; i < C->maxrhs; i++ ) C->CanCommu[i] = lold[i];
00257         lold = C->NumTerms; C->NumTerms = (LONG *) (C->rhs+2*newsize);
00258         for ( i = 0; i < C->maxrhs; i++ ) C->NumTerms[i] = lold[i];
00259         wold = C->numdum; C->numdum = (WORD *) (C->NumTerms+newsize);
00260         for ( i = 0; i < C->maxrhs; i++ ) C->numdum[i] = wold[i];
00261         wold = C->dimension; C->dimension = (WORD *) (C->numdum+newsize);
00262         for ( i = 0; i < C->maxrhs; i++ ) C->dimension[i] = wold[i];
00263         if ( old ) M_free(old,"subexpression lists");
00264         C->maxrhs = newsize;
00265         if ( type == 0 ) RedoTree(C,C->maxrhs);
00266     }
00267     C->numrhs++;
00268     C->CanCommu[C->numrhs] = 0;
00269     C->NumTerms[C->numrhs] = 0;
00270     C->numdum[C->numrhs] = 0;
00271     C->dimension[C->numrhs] = 0;
00272     C->rhs[C->numrhs] = C->Pointer;
00273     return(C->Pointer);
00274 }
00275
00276 /*
00277  #] AddRHS :
00278  #[ AddNtoL :
00279  */
00280
00281 int AddNtoL(int n, WORD *array)
00282 {
00283     int i;
00284     CBUF *C = cbuf+AC.cbunum;
00285     #ifdef COMPUFDEBBUG
00286     MesPrint("LH: %a",n,array);
00287     #endif
00288     AddLHS(AC.cbunum);
00289     while ( C->Pointer+n >= C->Top ) DoubleCbuffer(AC.cbunum,C->Pointer,1);
00290     for ( i = 0; i < n; i++ ) *(C->Pointer)++ = *array++;
00291     return(0);
00292 }
00293
00294 /*
00295  #] AddNtoL :
00296  #[ AddNtoC :
00297
00298     Commentary: added the bufnum on 14-sep-2010 to make the whole a bit
00299     more flexible (JV). Still to do with AddNtoL.
00300  */
00301
00302 int AddNtoC(int bufnum, int n, WORD *array,int par)
00303 {
00304     int i;
00305     WORD *w;
00306     CBUF *C = cbuf+bufnum;
00307     #ifdef COMPUFDEBBUG
00308     MesPrint("RH: %a",n,array);
00309     #endif
00310     while ( C->Pointer+n+1 >= C->Top ) DoubleCbuffer(bufnum,C->Pointer,50+par);
00311     w = C->Pointer;
00312     for ( i = 0; i < n; i++ ) *w++ = *array++;
00313     C->Pointer = w;
00314     return(0);
00315 }
00316
00317 /*
00318  #] AddNtoC :
00319  #[ InsTree :
00320
00321     Routines for balanced tree searching and insertion.
00322     Compared to Knuth we have a parent link. This minimizes the
00323     number of compares. That is better for anything that is more
00324     complicated than just single numbers.
00325     There are no provisions for removing elements from the tree.
00326     The routines are:
00327     void RedoTree(size) Re-allocates the tree space. There will
00328                        be MaxTreeSize = size elements.
00329     void ClearTree() Prunes the tree down to the root element.
00330     int InsTree(int,int) Searches for the requested element. If not found it
00331                        will allocate a new element, balance the tree if
00332                        necessary and return the called number.
00333                        If it was in the tree, it returns the tree 'value'.
00334
00335     Commentary: added the bufnum on 14-sep-2010 to make the whole a bit
00336     more flexible (JV).
00337  */
00338 static COMPTREE comptreezero = {0,0,0,0,0,0};
00339
00340 int InsTree(int bufnum, int h)
00341 {
00342     CBUF *C = cbuf + bufnum;

```

```

00358     COMPTREE *boomlijst = C->boomlijst, *q = boomlijst + C->rootnum, *p, *s;
00359     WORD *v1, *v2, *v3;
00360     int ip, iq, is;
00361
00362     if ( C->numtree + 1 >= C->MaxTreeSize ) {
00363         if ( C->MaxTreeSize == 0 ) {
00364             COMPTREE *root;
00365             C->MaxTreeSize = 125;
00366             C->boomlijst = (COMPTREE *)Malloc1((C->MaxTreeSize+1)*sizeof(COMPTREE),
00367                 "ClearInsTree");
00368             root = C->boomlijst;
00369             C->numtree = 0;
00370             C->rootnum = 0;
00371             root->left = -1;
00372             root->right = -1;
00373             root->parent = -1;
00374             root->blnce = 0;
00375             root->value = -1;
00376             root->usage = 0;
00377             for ( ip = 1; ip < C->MaxTreeSize; ip++ ) { C->boomlijst[ip] = comptreezero; }
00378         }
00379         else {
00380             is = C->MaxTreeSize * 2;
00381             s = (COMPTREE *)Malloc1((is+1)*sizeof(COMPTREE), "InsTree");
00382             for ( ip = 0; ip < C->MaxTreeSize; ip++ ) { s[ip] = C->boomlijst[ip]; }
00383             for ( ip = C->MaxTreeSize; ip <= is; ip++ ) { s[ip] = comptreezero; }
00384             if ( C->boomlijst ) M_free(C->boomlijst, "InsTree");
00385             C->boomlijst = s;
00386             C->MaxTreeSize = is;
00387         }
00388         boomlijst = C->boomlijst;
00389         q = boomlijst + C->rootnum;
00390     }
00391
00392     if ( q->right == -1 ) { /* First element */
00393         C->numtree++;
00394         s = boomlijst+C->numtree;
00395         q->right = C->numtree;
00396         s->parent = C->rootnum;
00397         s->left = s->right = -1;
00398         s->blnce = 0;
00399         s->value = h;
00400         s->usage = 1;
00401         return(h);
00402     }
00403     ip = q->right;
00404     while ( ip >= 0 ) {
00405         p = boomlijst + ip;
00406         v1 = C->rhs[p->value]; v2 = v3 = C->rhs[h];
00407         while ( *v3 ) v3 += *v3; /* find the 0 that indicates end-of-expr */
00408         while ( *v1 == *v2 && v2 < v3 ) { v1++; v2++; }
00409         if ( *v1 > *v2 ) {
00410             iq = p->right;
00411             if ( iq >= 0 ) { ip = iq; }
00412             else {
00413                 C->numtree++;
00414                 is = C->numtree;
00415                 p->right = is;
00416                 s = boomlijst + is;
00417                 s->parent = ip; s->left = s->right = -1;
00418                 s->blnce = 0; s->value = h; s->usage = 1;
00419                 p->blnce++;
00420                 if ( p->blnce == 0 ) return(h);
00421                 goto balance;
00422             }
00423         }
00424         else if ( *v1 < *v2 ) {
00425             iq = p->left;
00426             if ( iq >= 0 ) { ip = iq; }
00427             else {
00428                 C->numtree++;
00429                 is = C->numtree;
00430                 s = boomlijst+is;
00431                 p->left = is;
00432                 s->parent = ip; s->left = s->right = -1;
00433                 s->blnce = 0; s->value = h; s->usage = 1;
00434                 p->blnce--;
00435                 if ( p->blnce == 0 ) return(h);
00436                 goto balance;
00437             }
00438         }
00439         else {
00440             p->usage++;
00441             return(p->value);
00442         }
00443     }
00444     MesPrint("We vallen uit de boom!");

```

```

00445     Terminate(-1);
00446     return(h);
00447 balance;;
00448     for (;;) {
00449         p = boomlijst + ip;
00450         iq = p->parent;
00451         if ( iq == C->rootnum ) break;
00452         q = boomlijst + iq;
00453         if ( ip == q->left ) q->blnce--;
00454         else q->blnce++;
00455         if ( q->blnce == 0 ) break;
00456         if ( q->blnce == -2 ) {
00457             if ( p->blnce == -1 ) { /* single rotation */
00458                 q->left = p->right;
00459                 p->right = iq;
00460                 p->parent = q->parent;
00461                 q->parent = ip;
00462                 if ( boomlijst[p->parent].left == iq ) boomlijst[p->parent].left = ip;
00463                 else boomlijst[p->parent].right = ip;
00464                 if ( q->left >= 0 ) boomlijst[q->left].parent = iq;
00465                 q->blnce = p->blnce = 0;
00466             }
00467             else { /* double rotation */
00468                 s = boomlijst + is;
00469                 q->left = s->right;
00470                 p->right = s->left;
00471                 s->right = iq;
00472                 s->left = ip;
00473                 if ( p->right >= 0 ) boomlijst[p->right].parent = ip;
00474                 if ( q->left >= 0 ) boomlijst[q->left].parent = iq;
00475                 s->parent = q->parent;
00476                 q->parent = is;
00477                 p->parent = is;
00478                 if ( boomlijst[s->parent].left == iq )
00479                     boomlijst[s->parent].left = is;
00480                 else boomlijst[s->parent].right = is;
00481                 if ( s->blnce > 0 ) { q->blnce = s->blnce = 0; p->blnce = -1; }
00482                 else if ( s->blnce < 0 ) { p->blnce = s->blnce = 0; q->blnce = 1; }
00483                 else { p->blnce = s->blnce = q->blnce = 0; }
00484             }
00485             break;
00486         }
00487         else if ( q->blnce == 2 ) {
00488             if ( p->blnce == 1 ) { /* single rotation */
00489                 q->right = p->left;
00490                 p->left = iq;
00491                 p->parent = q->parent;
00492                 q->parent = ip;
00493                 if ( boomlijst[p->parent].left == iq ) boomlijst[p->parent].left = ip;
00494                 else boomlijst[p->parent].right = ip;
00495                 if ( q->right >= 0 ) boomlijst[q->right].parent = iq;
00496                 q->blnce = p->blnce = 0;
00497             }
00498             else { /* double rotation */
00499                 s = boomlijst + is;
00500                 q->right = s->left;
00501                 p->left = s->right;
00502                 s->left = iq;
00503                 s->right = ip;
00504                 if ( p->left >= 0 ) boomlijst[p->left].parent = ip;
00505                 if ( q->right >= 0 ) boomlijst[q->right].parent = iq;
00506                 s->parent = q->parent;
00507                 q->parent = is;
00508                 p->parent = is;
00509                 if ( boomlijst[s->parent].left == iq ) boomlijst[s->parent].left = is;
00510                 else boomlijst[s->parent].right = is;
00511                 if ( s->blnce < 0 ) { q->blnce = s->blnce = 0; p->blnce = 1; }
00512                 else if ( s->blnce > 0 ) { p->blnce = s->blnce = 0; q->blnce = -1; }
00513                 else { p->blnce = s->blnce = q->blnce = 0; }
00514             }
00515             break;
00516         }
00517         is = ip; ip = iq;
00518     }
00519     return(h);
00520 }
00521
00522 /*
00523  #] InsTree :
00524  #[ FindTree :
00525
00526  Routines for balanced tree searching.
00527  Is like InsTree but without the insertions.
00528  Returns -1 if the element is not in the tree.
00529  The advantage of this routine over InsTree is that this routine
00530  can be run in parallel.
00531 */

```

```

00532
00533 int FindTree(int bufnum, WORD *subexpr)
00534 {
00535     CBUF *C = cbuf + bufnum;
00536     COMPTREE *boomlijst = C->boomlijst, *q = boomlijst + C->rootnum, *p;
00537     WORD *v1, *v2, *v3;
00538     int ip, iq;
00539
00540     ip = q->right;
00541     while ( ip >= 0 ) {
00542         p = boomlijst + ip;
00543         v1 = C->rhs[p->value]; v2 = v3 = subexpr;
00544         while ( *v3 ) v3 += *v3; /* find the 0 that indicates end-of-expr */
00545         while ( *v1 == *v2 && v2 < v3 ) { v1++; v2++; }
00546         if ( *v1 > *v2 ) {
00547             iq = p->right;
00548             if ( iq >= 0 ) { ip = iq; }
00549             else { return(-1); }
00550         }
00551         else if ( *v1 < *v2 ) {
00552             iq = p->left;
00553             if ( iq >= 0 ) { ip = iq; }
00554             else { return(-1); }
00555         }
00556         else {
00557             p->usage++;
00558             return(p->value);
00559         }
00560     }
00561     return(-1);
00562 }
00563
00564 /*
00565 #] FindTree :
00566 #[ RedoTree :
00567 */
00568
00569 void RedoTree(CBUF *C, int size)
00570 {
00571     COMPTREE *newboomlijst;
00572     int i;
00573     newboomlijst = (COMPTREE *)Malloc1((size+1)*sizeof(COMPTREE), "newboomlijst");
00574     if ( C->boomlijst ) {
00575         if ( C->MaxTreeSize > size ) C->MaxTreeSize = size;
00576         for ( i = 0; i < C->MaxTreeSize; i++ ) newboomlijst[i] = C->boomlijst[i];
00577         M_free(C->boomlijst, "boomlijst");
00578     }
00579     C->boomlijst = newboomlijst;
00580     C->MaxTreeSize = size;
00581 }
00582
00583 /*
00584 #] RedoTree :
00585 #[ ClearTree :
00586 */
00587
00588 void ClearTree(int i)
00589 {
00590     CBUF *C = cbuf + i;
00591     COMPTREE *root = C->boomlijst;
00592     if ( root ) {
00593         C->numtree = 0;
00594         C->rootnum = 0;
00595         root->left = -1;
00596         root->right = -1;
00597         root->parent = -1;
00598         root->blnce = 0;
00599         root->value = -1;
00600         root->usage = 0;
00601     }
00602 }
00603
00604 /*
00605 #] ClearTree :
00606 #[ IniFbuffer :
00607 */
00614 int IniFbuffer(WORD bufnum)
00615 {
00616     CBUF *C = cbuf + bufnum;
00617     COMPTREE *root;
00618     int i;
00619     LONG fullsize;
00620     C->maxrhs = AM.fbuffersize;
00621     C->MaxTreeSize = AM.fbuffersize;
00622
00623     /*
00624     * Note that bufnum is a return value of inicbufs(). So C has been already

```



```

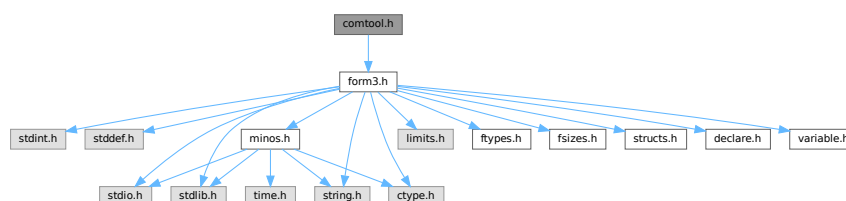
00625     * initialized. (TU 20 Dec 2011)
00626     */
00627     if ( C->boomlijst ) M_free(C->boomlijst, "IniFbuffer-tree");
00628     if ( C->rhs ) M_free(C->rhs, "IniFbuffer-rhs");
00629
00630     C->boomlijst = (COMPTREE *)Malloc1((C->MaxTreeSize+1)*sizeof(COMPTREE),"IniFbuffer-tree");
00631     root = C->boomlijst;
00632     C->numtree = 0;
00633     C->rootnum = 0;
00634     root->left = -1;
00635     root->right = -1;
00636     root->parent = -1;
00637     root->blnce = 0;
00638     root->value = -1;
00639     root->usage = 0;
00640     for ( i = 1; i < C->MaxTreeSize; i++ ) { C->boomlijst[i] = comptreezero; }
00641
00642     fullsize = (C->maxrhs+1) * (sizeof(WORD *) + 2*sizeof(LONG) + 2*sizeof(WORD));
00643     C->rhs = (WORD **)Malloc1(fullsize,"IniFbuffer-rhs");
00644     C->CanCommu = (LONG *) (C->rhs+C->maxrhs);
00645     C->NumTerms = (LONG *) (C->rhs+2*C->maxrhs);
00646     C->numdum = (WORD *) (C->NumTerms+C->maxrhs);
00647     C->dimension = (WORD *) (C->numdum+C->maxrhs);
00648
00649     return(0);
00650 }
00651
00652 /*
00653  #] IniFbuffer :
00654  #[ numcommute :
00655
00656  Returns the number of non-commuting terms in the expression
00657  */
00658
00659 LONG numcommute(WORD *terms, LONG *numterms)
00660 {
00661     LONG num = 0;
00662     WORD *t, *m;
00663     *numterms = 0;
00664     while ( *terms ) {
00665         *numterms += 1;
00666         t = terms + 1;
00667         GETSTOP(terms,m);
00668         while ( t < m ) {
00669             if ( *t >= FUNCTION ) {
00670                 if ( functions[*t-FUNCTION].commute ) { num++; break; }
00671             }
00672             t += t[1];
00673         }
00674         terms = terms + *terms;
00675     }
00676     return(num);
00677 }
00678
00679 /*
00680  #] numcommute :
00681  */

```

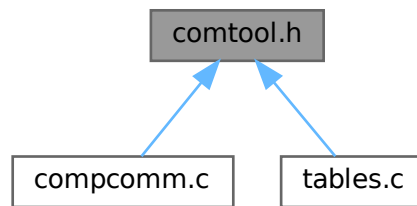
## 4.17 comtool.h File Reference

```
#include "form3.h"
```

Include dependency graph for comtool.h:



This graph shows which files directly or indirectly include this file:



### 4.17.1 Detailed Description

Utility routines for the compiler.

Definition in file [comtool.h](#).

## 4.18 comtool.h

[Go to the documentation of this file.](#)

```

00001
00005 /* #[] License : */
00006 /*
00007  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00008  * When using this file you are requested to refer to the publication
00009  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010  * This is considered a matter of courtesy as the development was paid
00011  * for by FOM the Dutch physics granting agency and we would like to
00012  * be able to track its scientific use to convince FOM of its value
00013  * for the community.
00014  *
00015  * This file is part of FORM.
00016  *
00017  * FORM is free software: you can redistribute it and/or modify it under the
00018  * terms of the GNU General Public License as published by the Free Software
00019  * Foundation, either version 3 of the License, or (at your option) any later
00020  * version.
00021  *
00022  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00023  * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00024  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00025  * details.
00026  *
00027  * You should have received a copy of the GNU General Public License along
00028  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00029  */
00030 /* #[] License : */
00031 #ifndef FORM_COMTOOL_H_
00032 #define FORM_COMTOOL_H_
00033 /*
00034  * #[] Includes :
00035  */
00036 #include "form3.h"
00037
00038 /*
00039  * #[] Includes :
00040  * #[] Inline functions :
00041  */
00042
00043 static inline void SkipSpaces(UBYTE **s)
00051 {
  
```

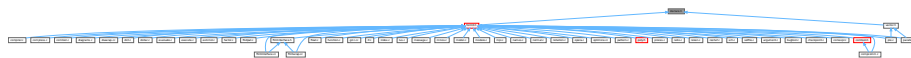
```

00052     const char *p = (const char *)s;
00053     while ( *p == ' ' || *p == ',' || *p == '\\t' ) p++;
00054     *s = (UBYTE *)p;
00055 }
00056
00066 static inline int ConsumeOption(UBYTE **s, const char *opt)
00067 {
00068     const char *p = (const char *)s;
00069     while ( *p && *opt && tolower(*p) == tolower(*opt) ) {
00070         p++;
00071         opt++;
00072     }
00073     /* Check if `opt` ended. */
00074     if ( !*opt ) {
00075         /* Check if `*p` is a word boundary. */
00076         if ( !*p || !(FG.cTable[(unsigned char)*p] == 0 ||
00077             FG.cTable[(unsigned char)*p] == 1 || *p == ' ' ||
00078             *p == '$') ) {
00079             /* Consume the option. Skip the trailing spaces. */
00080             *s = (UBYTE *)p;
00081             SkipSpaces(s);
00082             return(1);
00083         }
00084     }
00085     return(0);
00086 }
00087
00088 /*
00089  #] Inline functions :
00090  */
00091 #endif /* FORM_COMTOOL_H_ */

```

## 4.19 declare.h File Reference

This graph shows which files directly or indirectly include this file:



### Macros

- #define **MaX**(x, y) ((x) > (y) ? (x): (y))
- #define **MiN**(x, y) ((x) < (y) ? (x): (y))
- #define **ABS**(x) ( (x) < 0 ? -(x): (x) )
- #define **SGN**(x) ( (x) > 0 ? 1 : (x) < 0 ? -1 : 0 )
- #define **REDLENG**(x) (((x)<0)?((x)+1):((x)-1))/2)
- #define **INCLENG**(x) (((x)<0)?(((x)\*2)-1):(((x)\*2)+1))
- #define **GETCOEF**(x, y) x += \*x; y = x[-1]; x -= ABS(y); y=REDLENG(y)
- #define **GETSTOP**(x, y) y=x+(\*x)-1; y -= ABS(\*y)-1
- #define **StuffAdd**(x, y) (((x)<0?-1:1)\*y)+((y)<0?-1:1)\*(x))
- #define **EXCHN**(t1, t2, n) { WORD a,i; for(i=0;i<n;i++){a=t1[i];t1[i]=t2[i];t2[i]=a; } }
- #define **EXCH**(x, y) { WORD a = (x); (x) = (y); (y) = a; }
- #define **TOKENTOLINE**(x, y)
- #define **UngetFromStream**(stream, c) ((stream)->nextchar[(stream)->isnextchar++]=c)
- #define **AddLineFeed**(s, n) { (s)[(n)++] = LINEFEED; }
- #define **TryRecover**(x) Terminate(-1)
- #define **UngetChar**(c) { pushbackchar = c; }
- #define **ParseNumber**(x, s) {(x)=0;while(\*(s)>='0'&&\*(s)<='9')(x)=10\*(x)+\*(s)++ -'0';}
- #define **ParseSign**(sgn, s)
- #define **ParseSignedNumber**(x, s)
- #define **NCOPY**(s, t, n) { while ( (n)-- > 0 ) { \*s++ = \*t++; } }
- #define **NCOPYI**(s, t, n) { while ( (n)-- > 0 ) { \*s++ = \*t++; } }

- #define NCOPYB(s, t, n) { while ( (n)-- > 0 ) { \*s++ = \*t++; } }
- #define NCOPYI32(s, t, n) { while ( (n)-- > 0 ) { \*s++ = \*t++; } }
- #define WCOPY(s, t, n) { int nn=n; WORD \*ss=(WORD \*)s, \*tt=(WORD \*)t; while ( (nn)-- > 0 ) { \*ss++ = \*tt++; } }
- #define NeedNumber(x, s, err)
- #define SKIPBLANKS(s) { while ( \*(s) == ' ' || \*(s) == '\t' ) (s)++; }
- #define FLUSHCONSOLE if ( AP.InOutBuf > 0 ) CharOut(LINEFEED)
- #define SKIPBRA1(s)
- #define SKIPBRA2(s)
- #define SKIPBRA3(s)
- #define SKIPBRA4(s)
- #define SKIPBRA5(s)
- #define CYCLE1(t, a, i) { t iX=\*a; WORD jX; for(jX=1;jX<i;jX++)a[jX-1]=a[jX]; a[i-1]=iX; }
- #define AddToCB(c, wx)
- #define EXCHINOUT
- #define BACKINOUT
- #define CopyArg(to, from)
- #define FILLARG(w)
- #define COPYARG(w, t)
- #define ZEROARG(w)
- #define FILLFUN(w)
- #define COPYFUN(w, t)
- #define COPYFUN3(w, t)
- #define FILLFUN3(w)
- #define FILLSUB(w)
- #define COPYSUB(w, ww)
- #define FILLEXPR(w)
- #define NEXTARG(x) if(\*x>0) x += \*x; else if(\*x <= -FUNCTION)x++; else x += 2;
- #define COPY1ARG(s1, t1)
- #define ZeroFillRange(w, begin, end)
- #define TABLESIZE(a, b) (((WORD)sizeof(a))/((WORD)sizeof(b)))
- #define WORDDIF(x, y) (WORD)(x-y)
- #define wsizeof(a) ((WORD)sizeof(a))
- #define VARNAME(type, num) (AC.varnames->namebuffer+type[num].name)
- #define DOLLARNAME(type, num) (AC.dollarnames->namebuffer+type[num].name)
- #define EXPRNAME(num) (AC.exprnames->namebuffer+Expressions[num].name)
- #define PREV(x) prevorder?prevorder:x
- #define Terminate(x) do { TerminateImpl(x, \_\_FILE\_\_, \_\_LINE\_\_, \_\_FUNCTION\_\_); } while(0)
- #define SETERROR(x) { Terminate(-1); return(-1); }
- #define DUMMYUSE(x) (void)(x);
- #define ADDPOS(pp, x) (pp).p1 = ((pp).p1+(LONG)(x))
- #define SETBASELENGTH(ss, x) (ss).p1 = (LONG)(x)
- #define SETBASEPOSITION(pp, x) (pp).p1 = (LONG)(x)
- #define ISEQUALPOSINC(pp1, pp2, x) ( (pp1).p1 == ((pp2).p1+(LONG)(x)) )
- #define ISGEPOSINC(pp1, pp2, x) ( (pp1).p1 >= ((pp2).p1+(LONG)(x)) )
- #define DIVPOS(pp, n) ( (pp).p1/(LONG)(n) )
- #define MULPOS(pp, n) (pp).p1 \*= (LONG)(n)
- #define DIFPOS(ss, pp1, pp2) (ss).p1 = ((pp1).p1-(pp2).p1)
- #define DIFBASE(pp1, pp2) ((pp1).p1-(pp2).p1)
- #define ADD2POS(pp1, pp2) (pp1).p1 += (pp2).p1
- #define PUTZERO(pp) (pp).p1 = 0
- #define BASEPOSITION(pp) ((pp).p1)
- #define SETSTARTPOS(pp) (pp).p1 = -2
- #define NOTSTARTPOS(pp) ( (pp).p1 > -2 )
- #define ISMINPOS(pp) ( (pp).p1 == -1 )

- #define [ISEQUALPOS](#)(pp1, pp2) ( (pp1).p1 == (pp2).p1 )
- #define [ISNOTEQUALPOS](#)(pp1, pp2) ( (pp1).p1 != (pp2).p1 )
- #define [ISLESSPOS](#)(pp1, pp2) ( (pp1).p1 < (pp2).p1 )
- #define [ISGEPOS](#)(pp1, pp2) ( (pp1).p1 >= (pp2).p1 )
- #define [ISNOTZEROPOS](#)(pp) ( (pp).p1 != 0 )
- #define [ISZEROPOS](#)(pp) ( (pp).p1 == 0 )
- #define [ISPOSP](#)(pp) ( (pp).p1 > 0 )
- #define [ISNEGPOS](#)(pp) ( (pp).p1 < 0 )
- #define [TOLONG](#)(x) ((LONG)(x))
- #define [Add2Com](#)(x) { WORD cod[2]; cod[0] = x; cod[1] = 2; [AddNtoL](#)(2,cod); }
- #define [Add3Com](#)(x1, x2) { WORD cod[3]; cod[0] = x1; cod[1] = 3; cod[2] = x2; [AddNtoL](#)(3,cod); }
- #define [Add4Com](#)(x1, x2, x3)
- #define [Add5Com](#)(x1, x2, x3, x4)
- #define [WantAddPointers](#)(x)
- #define [WantAddLongs](#)(x)
- #define [WantAddPositions](#)(x)
- #define [FORM\\_INLINE](#) inline
- #define [MEMORYMACROS](#)
- #define [TermMalloc](#)(x) ( (AT.TermMemTop <= 0) ? TermMallocAddMemory(BHEAD0), AT.TermMemHeap[--AT.TermMemTop]: AT.TermMemHeap[--AT.TermMemTop] )
- #define [NumberMalloc](#)(x) ( (AT.NumberMemTop <= 0) ? NumberMallocAddMemory(BHEAD0), AT.NumberMemHeap[--AT.NumberMemTop]: AT.NumberMemHeap[--AT.NumberMemTop] )
- #define [CacheNumberMalloc](#)(x) ( (AT.CacheNumberMemTop <= 0) ? CacheNumberMallocAddMemory(BHEAD0), AT.CacheNumberMemHeap[--AT.CacheNumberMemTop]: AT.CacheNumberMemHeap[--AT.CacheNumberMemTop] )
- #define [TermFree](#)(TermMem, x) AT.TermMemHeap[AT.TermMemTop++] = (WORD \*) (TermMem)
- #define [NumberFree](#)(NumberMem, x) AT.NumberMemHeap[AT.NumberMemTop++] = (UWORD \*) (NumberMem)
- #define [CacheNumberFree](#)(NumberMem, x) AT.CacheNumberMemHeap[AT.CacheNumberMemTop++] = (UWORD \*) (NumberMem)
- #define [NestingChecksum](#)() (AC.IfLevel + AC.RepLevel + AC.arglevel + AC.insidelevel + AC.termlevel + AC.inexprlevel + AC.dolooplevel + AC.SwitchLevel)
- #define [MesNesting](#)() MesPrint("&Illegal nesting of if, repeat, argument, inside, term, inexpression and do")
- #define [MarkPolyRatFunDirty](#)(T)
- #define [PUSHPREASSIGNLEVEL](#)
- #define [POPPREASSIGNLEVEL](#)
- #define [EXTERNLOCK](#)(x)
- #define [INILOCK](#)(x)
- #define [LOCK](#)(x)
- #define [UNLOCK](#)(x)
- #define [EXTERNRWLOCK](#)(x)
- #define [INIRWLOCK](#)(x)
- #define [RWLOCKR](#)(x)
- #define [RWLOCKW](#)(x)
- #define [UNRWLOCK](#)(x)
- #define [MLOCK](#)(x)
- #define [MUNLOCK](#)(x)
- #define [GETIDENTITY](#)
- #define [GETBIDENTITY](#)
- #define [M\\_alloc](#)(x) malloc((size\_t)(x))
- #define [CompareTerms](#) ((COMPARE)AR.CompareRoutine)
- #define [FiniShuffle](#) AN.SHvar.finishuf
- #define [DoShuffle](#) ((DO\_UFFLE)AN.SHvar.do\_uffle)

## Typedefs

- typedef int(\* [WRITEBUFTOEXTCHANNEL](#)) (char \*, size\_t)
- typedef int(\* [GETCFROMEXTCHANNEL](#)) (void)
- typedef int(\* [SETTERMINATORFOREXTERNALCHANNEL](#)) (char \*)
- typedef int(\* [SETKILLMODEFOREXTERNALCHANNEL](#)) (int, int)
- typedef LONG(\* [WRITEFILE](#)) (int, UBYTE \*, LONG)
- typedef WORD(\* [GETTERM](#)) (PHEAD WORD \*)

## Functions

- void [TELLFILE](#) (int, [POSITION](#) \*)
- void [StartVariables](#) (void)
- void [setSignalHandlers](#) (void)
- UBYTE \* [CodeToLine](#) (WORD, UBYTE \*)
- UBYTE \* [AddArrayIndex](#) (WORD, UBYTE \*)
- [INDEXENTRY](#) \* [FindInIndex](#) (WORD, [FILEDATA](#) \*, WORD, WORD)
- [INDEXENTRY](#) \* [NextFileIndex](#) ([POSITION](#) \*)
- WORD \* [PasteTerm](#) (PHEAD WORD, WORD \*, WORD \*, WORD, WORD)
- UBYTE \* [StrCopy](#) (UBYTE \*, UBYTE \*)
- UBYTE \* [WrtPower](#) (UBYTE \*, WORD)
- int [AccumGCD](#) (PHEAD UWORD \*, WORD \*, UWORD \*, WORD)
- void [AddArgs](#) (PHEAD WORD \*, WORD \*, WORD \*)
- int [AddCoef](#) (PHEAD WORD \*\*, WORD \*\*)
- int [AddLong](#) (UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*)
- int [AddPLon](#) (UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*)
- int [AddPoly](#) (PHEAD WORD \*\*, WORD \*\*)
- int [AddRat](#) (PHEAD UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*)
- void [AddToLine](#) (UBYTE \*)
- int [AddWild](#) (PHEAD WORD, WORD, WORD)
- int [BigLong](#) (UWORD \*, WORD, UWORD \*, WORD)
- int [BinomGen](#) (PHEAD WORD \*, WORD, WORD \*\*, WORD, WORD, WORD, WORD, WORD, UWORD \*, WORD)
- int [CheckWild](#) (PHEAD WORD, WORD, WORD, WORD \*)
- int [Chisholm](#) (PHEAD WORD \*, WORD)
- int [CleanExpr](#) (WORD)
- void [CleanUp](#) (WORD)
- void [ClearWild](#) (PHEAD0)
- int [CompareFunctions](#) (WORD \*, WORD \*)
- int [Commute](#) (WORD \*, WORD \*)
- WORD [DetCommu](#) (WORD \*)
- WORD [DoesCommu](#) (WORD \*)
- int [CompArg](#) (WORD \*, WORD \*)
- WORD [CompCoef](#) (WORD \*, WORD \*)
- int [CompGroup](#) (PHEAD WORD, WORD \*\*, WORD \*, WORD \*, WORD)
- WORD [Compare1](#) (PHEAD WORD \*, WORD \*, WORD)
- WORD [CountDo](#) (WORD \*, WORD \*)
- WORD [CountFun](#) (WORD \*, WORD \*)
- WORD [DimensionSubterm](#) (WORD \*)
- WORD [DimensionTerm](#) (WORD \*)
- WORD [DimensionExpression](#) (PHEAD WORD \*)
- int [Deferred](#) (PHEAD WORD \*, WORD)
- int [DeleteStore](#) (WORD)
- WORD [DetCurDum](#) (PHEAD WORD \*)

- void [DetVars](#) (WORD \*, WORD)
- int [Distribute](#) (DISTRIBUTE \*, WORD)
- int [DivLong](#) (UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*, UWORD \*, WORD \*)
- int [DivRat](#) (PHEAD UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*)
- int [Divvy](#) (PHEAD UWORD \*, WORD \*, UWORD \*, WORD)
- int [DoDelta](#) (WORD \*)
- int [DoDelta3](#) (PHEAD WORD \*, WORD)
- int [TestPartitions](#) (WORD \*, [PARTI](#) \*)
- int [DoPartitions](#) (PHEAD WORD \*, WORD)
- int [CoCanonicalize](#) (UBYTE \*)
- int [DoCanonicalize](#) (PHEAD WORD \*, WORD \*)
- int [GenTopologies](#) (PHEAD WORD \*, WORD)
- int [GenDiagrams](#) (PHEAD WORD \*, WORD)
- int [DoTopologyCanonicalize](#) (PHEAD WORD \*, WORD, WORD, WORD \*)
- int [DoShattering](#) (PHEAD WORD \*, WORD \*, WORD \*, WORD)
- int [GenerateTopologies](#) (PHEAD WORD, WORD, WORD, WORD)
- int [DoTableExpansion](#) (WORD \*, WORD)
- int [DoDistrib](#) (PHEAD WORD \*, WORD)
- int [DoShuffle](#) (WORD \*, WORD, WORD, WORD)
- int [DoPermutations](#) (PHEAD WORD \*, WORD)
- int [Shuffle](#) (WORD \*, WORD \*, WORD \*)
- int [FinishShuffle](#) (WORD \*)
- int [DoStuffle](#) (WORD \*, WORD, WORD, WORD)
- int [Stuffle](#) (WORD \*, WORD \*, WORD \*)
- int [FinishStuffle](#) (WORD \*)
- WORD \* [StuffRootAdd](#) (WORD \*, WORD \*, WORD \*)
- int [TestUse](#) (WORD \*, WORD)
- DBASE \* [FindTB](#) (UBYTE \*)
- int [CheckTableDeclarations](#) (DBASE \*)
- void [Apply](#) (WORD \*, WORD)
- int [ApplyExec](#) (WORD \*, int, WORD)
- void [ApplyReset](#) (WORD)
- void [TableReset](#) (void)
- void [ReWorkT](#) (WORD \*, WORD \*, WORD)
- WORD [GetIfDollarNum](#) (WORD \*, WORD \*)
- int [FindVar](#) (WORD \*, WORD \*)
- int [DolfStatement](#) (PHEAD WORD \*, WORD \*)
- int [DoOnePow](#) (PHEAD WORD \*, WORD, WORD, WORD \*, WORD \*, WORD, WORD \*)
- void [DoRevert](#) (WORD \*, WORD \*)
- int [DoSumF1](#) (PHEAD WORD \*, WORD \*, WORD, WORD)
- int [DoSumF2](#) (PHEAD WORD \*, WORD \*, WORD, WORD)
- int [DoTheta](#) (PHEAD WORD \*)
- LONG [EndSort](#) (PHEAD WORD \*, int)
- int [EntVar](#) (WORD, UBYTE \*, WORD, WORD, WORD, WORD)
- int [EpfCon](#) (PHEAD WORD \*, WORD \*, WORD, WORD)
- int [EpfFind](#) (PHEAD WORD \*, WORD \*)
- WORD [EpfGen](#) (WORD, WORD \*, WORD \*, WORD \*, WORD)
- int [EqualArg](#) (WORD \*, WORD, WORD)
- int [Factorial](#) (PHEAD WORD, UWORD \*, WORD \*)
- int [Bernoulli](#) (WORD, UWORD \*, WORD \*)
- int [FactorIn](#) (PHEAD WORD \*, WORD)
- int [FactorInExpr](#) (PHEAD WORD \*, WORD)
- int [FindAll](#) (PHEAD WORD \*, WORD \*, WORD, WORD \*)
- WORD [FindMulti](#) (PHEAD WORD \*, WORD \*)
- int [FindOnce](#) (PHEAD WORD \*, WORD \*)

- int [FindOnly](#) (PHEAD WORD \*, WORD \*)
- int [FindRest](#) (PHEAD WORD \*, WORD \*)
- void [FindSpecial](#) (WORD \*)
- WORD [FindrNumber](#) (WORD, [VARRENUM](#) \*)
- void [FiniLine](#) (void)
- int [FiniTerm](#) (PHEAD WORD \*, WORD \*, WORD \*, WORD, WORD)
- int [FlushOut](#) ([POSITION](#) \*, [FILEHANDLE](#) \*, int)
- void [FunLevel](#) (PHEAD WORD \*)
- void [AdjustRenumScratch](#) (PHEAD0)
- void [GarbHand](#) (void)
- int [GcdLong](#) (PHEAD UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*)
- int [LcmLong](#) (PHEAD UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*)
- void [GCD](#) (UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*)
- ULONG [GCD2](#) (ULONG, ULONG)
- int [Generator](#) (PHEAD WORD \*, WORD)
- int [GetBinom](#) (UWORD \*, WORD \*, WORD, WORD)
- WORD [GetFromStore](#) (WORD \*, [POSITION](#) \*, [RENUMBER](#), WORD \*, WORD)
- int [GetLong](#) (UBYTE \*, UWORD \*, WORD \*)
- WORD [GetMoreTerms](#) (WORD \*)
- int [GetMoreFromMem](#) (WORD \*, WORD \*\*)
- WORD [GetOneTerm](#) (PHEAD WORD \*, [FILEHANDLE](#) \*, [POSITION](#) \*, int)
- [RENUMBER](#) [GetTable](#) (WORD, [POSITION](#) \*, WORD)
- WORD [GetTerm](#) (PHEAD WORD \*)
- int [Glue](#) (PHEAD WORD \*, WORD \*, WORD \*, WORD)
- int [InFunction](#) (PHEAD WORD \*, WORD \*)
- void [IniLine](#) (WORD)
- void [IniVars](#) (void)
- int [InsertTerm](#) (PHEAD WORD \*, WORD, WORD, WORD \*, WORD \*, WORD)
- void [LongToLine](#) (UWORD \*, WORD)
- int [MakeDirty](#) (WORD \*, WORD \*, WORD)
- void [MarkDirty](#) (WORD \*, WORD)
- void [PolyFunDirty](#) (PHEAD WORD \*)
- void [PolyFunClean](#) (PHEAD WORD \*)
- int [MakeModTable](#) (void)
- int [MatchE](#) (PHEAD WORD \*, WORD \*, WORD \*, WORD)
- int [MatchCy](#) (PHEAD WORD \*, WORD \*, WORD \*, WORD)
- int [FunMatchCy](#) (PHEAD WORD \*, WORD \*, WORD \*, WORD)
- int [FunMatchSy](#) (PHEAD WORD \*, WORD \*, WORD \*, WORD)
- int [MatchArgument](#) (PHEAD WORD \*, WORD \*)
- int [MatchFunction](#) (PHEAD WORD \*, WORD \*, WORD \*)
- int [MergePatches](#) (WORD)
- int [MesCerr](#) (char \*, UBYTE \*)
- int [MesComp](#) (char \*, UBYTE \*, UBYTE \*)
- int [Modulus](#) (WORD \*)
- void [MoveDummies](#) (PHEAD WORD \*, WORD)
- int [MulLong](#) (UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*)
- int [MulRat](#) (PHEAD UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*)
- int [Mully](#) (PHEAD UWORD \*, WORD \*, UWORD \*, WORD)
- int [MultDo](#) (PHEAD WORD \*, WORD \*)
- int [NewSort](#) (PHEAD0)
- int [ExtraSymbol](#) (WORD, WORD, WORD, WORD \*, WORD \*)
- int [Normalize](#) (PHEAD WORD \*)
- int [BracketNormalize](#) (PHEAD WORD \*)
- void [DropCoefficient](#) (PHEAD WORD \*)
- void [DropSymbols](#) (PHEAD WORD \*)



- int [PutInside](#) (PHEAD WORD \*, WORD \*)
- void [OpenTemp](#) (void)
- void [Pack](#) (UWORD \*, WORD \*, UWORD \*, WORD)
- LONG [PasteFile](#) (PHEAD WORD, WORD \*, [POSITION](#) \*, WORD \*\*, [RENUMBER](#), WORD \*, WORD)
- int [Permute](#) ([PERM](#) \*, WORD)
- int [PermuteP](#) ([PERMP](#) \*, WORD)
- int [PolyFunMul](#) (PHEAD WORD \*)
- int [PopVariables](#) (void)
- int [PrepPoly](#) (PHEAD WORD \*, WORD)
- int [Processor](#) (void)
- int [Product](#) (UWORD \*, WORD \*, WORD)
- void [PrtLong](#) (UWORD \*, WORD, UBYTE \*)
- void [PrtTerms](#) (void)
- void [PrintDeprecation](#) (const char \*, const char \*)
- void [PrintRunningTime](#) (void)
- LONG [GetRunningTime](#) (void)
- int [PutBracket](#) (PHEAD WORD \*)
- LONG [PutIn](#) ([FILEHANDLE](#) \*, [POSITION](#) \*, WORD \*, WORD \*\*, int)
- int [PutInStore](#) ([INDEXENTRY](#) \*, WORD)
- WORD [PutOut](#) (PHEAD WORD \*, [POSITION](#) \*, [FILEHANDLE](#) \*, WORD)
- UWORD [Quotient](#) (UWORD \*, WORD \*, WORD)
- int [RaisPow](#) (PHEAD UWORD \*, WORD \*, UWORD)
- void [RaisPowCached](#) (PHEAD WORD, WORD, UWORD \*\*, WORD \*)
- WORD [RaisPowMod](#) (WORD, WORD, WORD)
- int [NormalModulus](#) (UWORD \*, WORD \*)
- int [MakeInverses](#) (void)
- int [GetModInverses](#) (WORD, WORD, WORD \*, WORD \*)
- int [GetLongModInverses](#) (PHEAD UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*, UWORD \*, WORD \*)
- void [RatToLine](#) (UWORD \*, WORD)
- int [RatioFind](#) (PHEAD WORD \*, WORD \*)
- int [RatioGen](#) (PHEAD WORD \*, WORD \*, WORD, WORD)
- WORD [ReNumber](#) (PHEAD WORD \*)
- WORD [ReadSnum](#) (UBYTE \*\*)
- WORD [Remain10](#) (UWORD \*, WORD \*)
- WORD [Remain4](#) (UWORD \*, WORD \*)
- int [ResetScratch](#) (void)
- int [ResolveSet](#) (PHEAD WORD \*, WORD \*, WORD \*)
- int [RevertScratch](#) (void)
- int [ScanFunctions](#) (PHEAD WORD \*, WORD \*, WORD)
- void [SeekScratch](#) ([FILEHANDLE](#) \*, [POSITION](#) \*)
- void [SetEndScratch](#) ([FILEHANDLE](#) \*, [POSITION](#) \*)
- void [SetEndHScratch](#) ([FILEHANDLE](#) \*, [POSITION](#) \*)
- int [SetFileIndex](#) (void)
- int [Sflush](#) ([FILEHANDLE](#) \*)
- int [Simplify](#) (PHEAD UWORD \*, WORD \*, UWORD \*, WORD \*)
- int [SortWild](#) (WORD \*, WORD)
- FILE \* [LocateBase](#) (char \*\*, char \*\*, char \*)
- LONG [SplitMerge](#) (PHEAD WORD \*\*, LONG)
- int [StoreTerm](#) (PHEAD WORD \*)
- void [SubPLon](#) (UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*)
- void [Substitute](#) (PHEAD WORD \*, WORD \*, WORD)
- int [SymFind](#) (PHEAD WORD \*, WORD \*)
- int [SymGen](#) (PHEAD WORD \*, WORD \*, WORD, WORD)
- WORD [Symmetrize](#) (PHEAD WORD \*, WORD \*, WORD, WORD, WORD)

- int [FullSymmetrize](#) (PHEAD WORD \*, int)
- int [TakeModulus](#) (UWORD \*, WORD \*, UWORD \*, WORD, WORD)
- int [TakeNormalModulus](#) (UWORD \*, WORD \*, UWORD \*, WORD, WORD)
- void [TalToLine](#) (UWORD)
- int [TenVec](#) (PHEAD WORD \*, WORD \*, WORD, WORD)
- int [TenVecFind](#) (PHEAD WORD \*, WORD \*)
- int [TermRenummer](#) (WORD \*, [RENUMBER](#), WORD)
- void [TestDrop](#) (void)
- void [PutInVflags](#) (WORD)
- int [TestMatch](#) (PHEAD WORD \*, WORD \*)
- WORD [TestSub](#) (PHEAD WORD \*, WORD)
- LONG [TimeCPU](#) (WORD)
- LONG [TimeChildren](#) (WORD)
- LONG [TimeWallClock](#) (WORD)
- LONG [Timer](#) (int)
- int [GetTimerInfo](#) (LONG \*\*, LONG \*\*)
- void [WriteTimerInfo](#) (LONG \*, LONG \*)
- LONG [GetWorkerTimes](#) (void)
- int [ToStorage](#) ([EXPRESSIONS](#), [POSITION](#) \*)
- void [TokenToLine](#) (UBYTE \*)
- int [Trace4](#) (PHEAD WORD \*, WORD \*, WORD, WORD)
- int [Trace4Gen](#) (PHEAD [TRACES](#) \*, WORD)
- int [Trace4no](#) (WORD, WORD \*, [TRACES](#) \*)
- int [TraceFind](#) (PHEAD WORD \*, WORD \*)
- int [TraceN](#) (PHEAD WORD \*, WORD \*, WORD, WORD)
- int [TraceNgen](#) (PHEAD [TRACES](#) \*, WORD)
- WORD [TraceNno](#) (WORD, WORD \*, [TRACES](#) \*)
- int [Traces](#) (PHEAD WORD \*, WORD \*, WORD, WORD)
- WORD [Trick](#) (WORD \*, [TRACES](#) \*)
- int [TryDo](#) (PHEAD WORD \*, WORD \*, WORD)
- void [UnPack](#) (UWORD \*, WORD, WORD \*, WORD \*)
- int [VarStore](#) (UBYTE \*, WORD, WORD, WORD)
- WORD [WildFill](#) (PHEAD WORD \*, WORD \*, WORD \*)
- int [WriteAll](#) (void)
- int [WriteOne](#) (UBYTE \*, int, int, WORD)
- void [WriteArgument](#) (WORD \*)
- int [WriteExpression](#) (WORD \*, LONG)
- int [WriteInnerTerm](#) (WORD \*, WORD)
- void [WriteLists](#) (void)
- void [WriteSetup](#) (void)
- void [WriteStats](#) ([POSITION](#) \*, WORD, WORD)
- int [WriteSubTerm](#) (WORD \*, WORD)
- int [WriteTerm](#) (WORD \*, WORD \*, WORD, WORD, WORD)
- int [execarg](#) (PHEAD WORD \*, WORD)
- int [execterm](#) (PHEAD WORD \*, WORD)
- void [SpecialCleanup](#) (PHEAD0)
- void [SetMods](#) (void)
- void [UnSetMods](#) (void)
- int [DoExecute](#) (WORD, WORD)
- void [SetScratch](#) ([FILEHANDLE](#) \*, [POSITION](#) \*)
- void [Warning](#) (char \*)
- void [HighWarning](#) (char \*)
- int [SpareTable](#) ([TABLES](#))
- UBYTE \* [strDup1](#) (UBYTE \*, char \*)
- void \* [Malloc1](#) (LONG, const char \*)

- int [DoTail](#) (int, UBYTE \*\*)
- int [OpenInput](#) (void)
- int [PutPreVar](#) (UBYTE \*, UBYTE \*, UBYTE \*, int)
- void [Error0](#) (char \*)
- void [Error1](#) (char \*, UBYTE \*)
- void [Error2](#) (char \*, char \*, UBYTE \*)
- UBYTE [ReadFromStream](#) (STREAM \*)
- UBYTE [GetFromStream](#) (STREAM \*)
- UBYTE [LookInStream](#) (STREAM \*)
- STREAM \* [OpenStream](#) (UBYTE \*, int, int, int)
- int [LocateFile](#) (UBYTE \*\*, int)
- STREAM \* [CloseStream](#) (STREAM \*)
- void [PositionStream](#) (STREAM \*, LONG)
- int [ReverseStatements](#) (STREAM \*)
- int [ProcessOption](#) (UBYTE \*, UBYTE \*, int)
- int [DoSetups](#) (void)
- void [TerminatImpl](#) (int, const char \*, int, const char \*)
- NAMETREE \* [GetNode](#) (NAMETREE \*, UBYTE \*)
- int [AddName](#) (NAMETREE \*, UBYTE \*, WORD, WORD, int \*)
- int [GetName](#) (NAMETREE \*, UBYTE \*, WORD \*, int)
- UBYTE \* [GetFunction](#) (UBYTE \*, WORD \*)
- UBYTE \* [GetNumber](#) (UBYTE \*, WORD \*)
- int [GetLastExprName](#) (UBYTE \*, WORD \*)
- int [GetAutoName](#) (UBYTE \*, WORD \*)
- int [GetVar](#) (UBYTE \*, WORD \*, WORD \*, int, int)
- int [MakeDubious](#) (NAMETREE \*, UBYTE \*, WORD \*)
- int [GetOName](#) (NAMETREE \*, UBYTE \*, WORD \*, int)
- void [DumpTree](#) (NAMETREE \*)
- void [DumpNode](#) (NAMETREE \*, WORD, WORD)
- void [LinkTree](#) (NAMETREE \*, WORD, WORD)
- void [CopyTree](#) (NAMETREE \*, NAMETREE \*, WORD, WORD)
- int [CompactifyTree](#) (NAMETREE \*, WORD)
- NAMETREE \* [MakeNameTree](#) (void)
- void [FreeNameTree](#) (NAMETREE \*)
- int [AddExpression](#) (UBYTE \*, int, int)
- int [AddSymbol](#) (UBYTE \*, int, int, int, int)
- int [AddDollar](#) (UBYTE \*, WORD, WORD \*, LONG)
- int [ReplaceDollar](#) (WORD, WORD, WORD \*, LONG)
- int [DollarRaiseLow](#) (UBYTE \*, LONG)
- int [AddVector](#) (UBYTE \*, int, int)
- int [AddDubious](#) (UBYTE \*)
- int [AddIndex](#) (UBYTE \*, int, int)
- UBYTE \* [DoDimension](#) (UBYTE \*, int \*, int \*)
- int [AddFunction](#) (UBYTE \*, int, int, int, int, int, int, int)
- int [CoCommuteInSet](#) (UBYTE \*)
- int [CoFunction](#) (UBYTE \*, int, int)
- int [TestName](#) (UBYTE \*)
- int [AddSet](#) (UBYTE \*, WORD)
- int [DoElements](#) (UBYTE \*, SETS, UBYTE \*)
- int [DoTempSet](#) (UBYTE \*, UBYTE \*)
- int [NameConflict](#) (int, UBYTE \*)
- int [OpenFile](#) (char \*)
- int [OpenAddFile](#) (char \*)
- int [ReOpenFile](#) (char \*)
- int [CreateFile](#) (char \*)

- int [CreateLogFile](#) (char \*)
- void [CloseFile](#) (int)
- int [CopyFile](#) (char \*, char \*)
- int [CreateHandle](#) (void)
- LONG [ReadFile](#) (int, UBYTE \*, LONG)
- LONG [ReadPosFile](#) (PHEAD [FILEHANDLE](#) \*, UBYTE \*, LONG, [POSITION](#) \*)
- LONG [WriteFileToFile](#) (int, UBYTE \*, LONG)
- void [SeekFile](#) (int, [POSITION](#) \*, int)
- LONG [TellFile](#) (int)
- void [FlushFile](#) (int)
- int [GetPosFile](#) (int, fpos\_t \*)
- int [SetPosFile](#) (int, fpos\_t \*)
- void [SynchFile](#) (int)
- void [TruncateFile](#) (int)
- int [GetChannel](#) (char \*, int)
- int [GetAppendChannel](#) (char \*)
- int [CloseChannel](#) (char \*)
- void [inictable](#) (void)
- [KEYWORD](#) \* [findcommand](#) (UBYTE \*)
- int [inicbufs](#) (void)
- void [StartFiles](#) (void)
- UBYTE \* [MakeDate](#) (void)
- void [PreProcessor](#) (void)
- void \* [FromList](#) ([LIST](#) \*)
- void \* [From0List](#) ([LIST](#) \*)
- void \* [FromVarList](#) ([LIST](#) \*)
- int [DoubleList](#) (void \*\*\*, int \*, int, char \*)
- int [DoubleLList](#) (void \*\*\*, LONG \*, int, char \*)
- void [DoubleBuffer](#) (void \*\*, void \*\*, int, char \*)
- void [ExpandBuffer](#) (void \*\*, LONG \*, int)
- LONG [iexp](#) (LONG, int)
- int [IsLikeVector](#) (WORD \*)
- int [AreArgsEqual](#) (WORD \*, WORD \*)
- int [CompareArgs](#) (WORD \*, WORD \*)
- UBYTE \* [SkipField](#) (UBYTE \*, int)
- int [StrCmp](#) (UBYTE \*, UBYTE \*)
- int [StrLCmp](#) (UBYTE \*, UBYTE \*)
- int [StrHICmp](#) (UBYTE \*, UBYTE \*)
- int [StrlCont](#) (UBYTE \*, UBYTE \*)
- int [CmpArray](#) (WORD \*, WORD \*, WORD)
- int [ConWord](#) (UBYTE \*, UBYTE \*)
- int [StrLen](#) (UBYTE \*)
- UBYTE \* [GetPreVar](#) (UBYTE \*, int)
- void [ToGeneral](#) (WORD \*, WORD \*, WORD)
- WORD [ToPolyFunGeneral](#) (PHEAD WORD \*)
- int [ToFast](#) (WORD \*, WORD \*)
- [SETUPPARAMETERS](#) \* [GetSetupPar](#) (UBYTE \*)
- int [RecalcSetups](#) (void)
- int [AllocSetups](#) (void)
- [SORTING](#) \* [AllocSort](#) (LONG, LONG, LONG, LONG, int, int, LONG, int)
- void [AllocSortFileName](#) ([SORTING](#) \*)
- UBYTE \* [LoadInputFile](#) (UBYTE \*, int)
- UBYTE [GetInput](#) (void)
- void [ClearPushback](#) (void)
- UBYTE [GetChar](#) (int)

- void [CharOut](#) (UBYTE)
- void [UnsetAllowDelay](#) (void)
- void [PopPreVars](#) (int)
- void [IniModule](#) (int)
- void [IniSpecialModule](#) (int)
- int [ModuleInstruction](#) (int \*, int \*)
- int [PreProInstruction](#) (void)
- int [LoadInstruction](#) (int)
- int [LoadStatement](#) (int)
- KEYWORD \* [FindKeyword](#) (UBYTE \*, KEYWORD \*, int)
- KEYWORD \* [FindInKeyword](#) (UBYTE \*, KEYWORD \*, int)
- int [DoDefine](#) (UBYTE \*)
- int [DoRedefine](#) (UBYTE \*)
- int [TheDefine](#) (UBYTE \*, int)
- int [TheUndefine](#) (UBYTE \*)
- int [ClearMacro](#) (UBYTE \*)
- int [DoUndefine](#) (UBYTE \*)
- int [DoInclude](#) (UBYTE \*)
- int [DoReverseInclude](#) (UBYTE \*)
- int [Include](#) (UBYTE \*, int)
- int [DoExternal](#) (UBYTE \*)
- int [DoToExternal](#) (UBYTE \*)
- int [DoFromExternal](#) (UBYTE \*)
- int [DoPrompt](#) (UBYTE \*)
- int [DoSetExternal](#) (UBYTE \*)
- int [DoSetExternalAttr](#) (UBYTE \*)
- int [DoRmExternal](#) (UBYTE \*)
- int [DoFactDollar](#) (UBYTE \*)
- WORD [GetDollarNumber](#) (UBYTE \*\*, DOLLARS)
- int [DoSetRandom](#) (UBYTE \*)
- int [DoOptimize](#) (UBYTE \*)
- int [DoClearOptimize](#) (UBYTE \*)
- int [DoSkipExtraSymbols](#) (UBYTE \*)
- int [DoTimeOutAfter](#) (UBYTE \*)
- int [DoMessage](#) (UBYTE \*)
- int [DoPreOut](#) (UBYTE \*)
- int [DoPreAppend](#) (UBYTE \*)
- int [DoPreCreate](#) (UBYTE \*)
- int [DoPreAssign](#) (UBYTE \*)
- int [DoPreBreak](#) (UBYTE \*)
- int [DoPreDefault](#) (UBYTE \*)
- int [DoPreSwitch](#) (UBYTE \*)
- int [DoPreEndSwitch](#) (UBYTE \*)
- int [DoPreCase](#) (UBYTE \*)
- int [DoPreShow](#) (UBYTE \*)
- int [DoPreExchange](#) (UBYTE \*)
- int [DoSystem](#) (UBYTE \*)
- int [DoPipe](#) (UBYTE \*)
- void [StartPrepro](#) (void)
- int [Dolfdef](#) (UBYTE \*, int)
- int [Dolfydef](#) (UBYTE \*)
- int [Dolfndef](#) (UBYTE \*)
- int [DoElse](#) (UBYTE \*)
- int [DoElseif](#) (UBYTE \*)
- int [DoEndif](#) (UBYTE \*)

- int [DoTerminate](#) (UBYTE \*)
- int [Dolf](#) (UBYTE \*)
- int [DoCall](#) (UBYTE \*)
- int [DoDebug](#) (UBYTE \*)
- int [DoContinueDo](#) (UBYTE \*)
- int [DoDo](#) (UBYTE \*)
- int [DoBreakDo](#) (UBYTE \*)
- int [DoEnddo](#) (UBYTE \*)
- int [DoEndprocedure](#) (UBYTE \*)
- int [DoInside](#) (UBYTE \*)
- int [DoEndInside](#) (UBYTE \*)
- int [DoProcedure](#) (UBYTE \*)
- int [DoPrePrintTimes](#) (UBYTE \*)
- int [DoPreWrite](#) (UBYTE \*)
- int [DoPreClose](#) (UBYTE \*)
- int [DoPreRemove](#) (UBYTE \*)
- int [DoPreSortReallocate](#) (UBYTE \*)
- int [DoCommentChar](#) (UBYTE \*)
- int [DoPrcExtension](#) (UBYTE \*)
- int [DoPreReset](#) (UBYTE \*)
- void [WriteString](#) (int, UBYTE \*, int)
- void [WriteUnfinString](#) (int, UBYTE \*, int)
- UBYTE \* [AddToString](#) (UBYTE \*, UBYTE \*, int)
- UBYTE \* [PreCalc](#) (void)
- UBYTE \* [PreEval](#) (UBYTE \*, LONG \*)
- void [NumToStr](#) (UBYTE \*, LONG)
- int [PreCmp](#) (int, int, UBYTE \*, int, int, UBYTE \*, int)
- int [PreEq](#) (int, int, UBYTE \*, int, int, UBYTE \*, int)
- UBYTE \* [pParseObject](#) (UBYTE \*, int \*, LONG \*)
- UBYTE \* [PrelfEval](#) (UBYTE \*, int \*)
- int [EvalPrelf](#) (UBYTE \*)
- int [PreLoad](#) (PRELOAD \*, UBYTE \*, UBYTE \*, int, char \*)
- int [PreSkip](#) (UBYTE \*, UBYTE \*, int)
- UBYTE \* [EndOfToken](#) (UBYTE \*)
- void [SetSpecialMode](#) (int, int)
- void [MakeGlobal](#) (void)
- int [ExecModule](#) (int)
- int [ExecStore](#) (void)
- void [FullCleanUp](#) (void)
- int [DoExecStatement](#) (void)
- int [DoPipeStatement](#) (void)
- int [DoPolyfun](#) (UBYTE \*)
- int [DoPolyratfun](#) (UBYTE \*)
- int [CompileStatement](#) (UBYTE \*)
- UBYTE \* [ToToken](#) (UBYTE \*)
- int [GetDollar](#) (UBYTE \*)
- int [MesWork](#) (void)
- int **MesPrint** (const char \*,...)
- int [MesCall](#) (char \*)
- UBYTE \* [NumCopy](#) (WORD, UBYTE \*)
- char \* [LongCopy](#) (LONG, char \*)
- char \* [LongLongCopy](#) (off\_t \*, char \*)
- void [ReserveTempFiles](#) (int)
- void [PrintTerm](#) (WORD \*, char \*)
- void [PrintTermC](#) (WORD \*, char \*)

- void [PrintSubTerm](#) (WORD \*, char \*)
- void [PrintWords](#) (WORD \*, LONG)
- void [PrintSeq](#) (WORD \*, char \*)
- int [ExpandTripleDots](#) (int)
- LONG [ComPress](#) (WORD \*\*, LONG \*)
- void [StageSort](#) (FILEHANDLE \*)
- void [M\\_free](#) (void \*, const char \*)
- void [ClearWildcardNames](#) (void)
- int [AddWildcardName](#) (UBYTE \*)
- int [GetWildcardName](#) (UBYTE \*)
- void [Globalize](#) (int)
- void [ResetVariables](#) (int)
- void [AddToPreTypes](#) (int)
- void [MessPreNesting](#) (int)
- LONG [GetStreamPosition](#) (STREAM \*)
- WORD \* [DoubleCbuffer](#) (int, WORD \*, int)
- WORD \* [AddLHS](#) (int)
- WORD \* [AddRHS](#) (int, int)
- int [AddNtoL](#) (int, WORD \*)
- int [AddNtoC](#) (int, int, WORD \*, int)
- void [DoubleIfBuffers](#) (void)
- STREAM \* [CreateStream](#) (UBYTE \*)
- int [setonoff](#) (UBYTE \*, int \*, int, int)
- int [DoPrint](#) (UBYTE \*, int)
- int [SetExpr](#) (UBYTE \*, int, int)
- void [AddToCom](#) (int, WORD \*)
- int [Add2ComStrings](#) (int, WORD \*, UBYTE \*, UBYTE \*)
- int [DoSymmetrize](#) (UBYTE \*, int)
- int [DoArgument](#) (UBYTE \*, int)
- int [ArgFactorize](#) (PHEAD WORD \*, WORD \*)
- WORD \* [TakeArgContent](#) (PHEAD WORD \*, WORD \*)
- WORD \* [MakeInteger](#) (PHEAD WORD \*, WORD \*, WORD \*)
- WORD \* [MakeMod](#) (PHEAD WORD \*, WORD \*, WORD \*)
- WORD [FindArg](#) (PHEAD WORD \*)
- int [InsertArg](#) (PHEAD WORD \*, WORD \*, int)
- int [CleanupArgCache](#) (PHEAD WORD)
- int [ArgSymbolMerge](#) (WORD \*, WORD \*)
- int [ArgDotproductMerge](#) (WORD \*, WORD \*)
- void [SortWeights](#) (LONG \*, LONG \*, WORD)
- int [DoBrackets](#) (UBYTE \*, int)
- int [DoPutInside](#) (UBYTE \*, int)
- WORD \* [CountComp](#) (UBYTE \*, WORD \*)
- int [CoAntiBracket](#) (UBYTE \*)
- int [CoAntiSymmetrize](#) (UBYTE \*)
- int [DoArgPlode](#) (UBYTE \*, int)
- int [CoArgExplode](#) (UBYTE \*)
- int [CoArgImplode](#) (UBYTE \*)
- int [CoArgument](#) (UBYTE \*)
- int [CoInside](#) (UBYTE \*)
- int [ExeclnInside](#) (UBYTE \*)
- int [CoInExpression](#) (UBYTE \*)
- int [CoInParallel](#) (UBYTE \*)
- int [CoNotInParallel](#) (UBYTE \*)
- int [DoInParallel](#) (UBYTE \*, int)
- int [CoEndInExpression](#) (UBYTE \*)

- int [CoBracket](#) (UBYTE \*)
- int [CoPutInside](#) (UBYTE \*)
- int [CoAntiPutInside](#) (UBYTE \*)
- int [CoMultiBracket](#) (UBYTE \*)
- int [CoCFunction](#) (UBYTE \*)
- int [CoCTensor](#) (UBYTE \*)
- int [CoCollect](#) (UBYTE \*)
- int [CoCompress](#) (UBYTE \*)
- int [CoContract](#) (UBYTE \*)
- int [CoCycleSymmetrize](#) (UBYTE \*)
- int [CoDelete](#) (UBYTE \*)
- int [CoTableBase](#) (UBYTE \*)
- int [CoApply](#) (UBYTE \*)
- int [CoDenominators](#) (UBYTE \*)
- int [CoDimension](#) (UBYTE \*)
- int [CoDiscard](#) (UBYTE \*)
- int [CoDisorder](#) (UBYTE \*)
- int [CoDrop](#) (UBYTE \*)
- int [CoDropCoefficient](#) (UBYTE \*)
- int [CoDropSymbols](#) (UBYTE \*)
- int [CoElse](#) (UBYTE \*)
- int [CoElseif](#) (UBYTE \*)
- int [CoEndArgument](#) (UBYTE \*)
- int [CoEndInside](#) (UBYTE \*)
- int [CoEndIf](#) (UBYTE \*)
- int [CoEndRepeat](#) (UBYTE \*)
- int [CoEndTerm](#) (UBYTE \*)
- int [CoEndWhile](#) (UBYTE \*)
- int [CoExit](#) (UBYTE \*)
- int [CoFactArg](#) (UBYTE \*)
- int [CoFactDollar](#) (UBYTE \*)
- int [CoFactorize](#) (UBYTE \*)
- int [CoNFactorize](#) (UBYTE \*)
- int [CoUnFactorize](#) (UBYTE \*)
- int [CoNUnFactorize](#) (UBYTE \*)
- int [DoFactorize](#) (UBYTE \*, int)
- int [CoFill](#) (UBYTE \*)
- int [CoFillExpression](#) (UBYTE \*)
- int [CoFixIndex](#) (UBYTE \*)
- int [CoFormat](#) (UBYTE \*)
- int [CoGlobal](#) (UBYTE \*)
- int [CoGlobalFactorized](#) (UBYTE \*)
- int [CoGoTo](#) (UBYTE \*)
- int [Cold](#) (UBYTE \*)
- int [ColdNew](#) (UBYTE \*)
- int [ColdOld](#) (UBYTE \*)
- int [Colf](#) (UBYTE \*)
- int [ColfMatch](#) (UBYTE \*)
- int [ColfNoMatch](#) (UBYTE \*)
- int [CoIndex](#) (UBYTE \*)
- int [CoInsideFirst](#) (UBYTE \*)
- int [CoKeep](#) (UBYTE \*)
- int [CoLabel](#) (UBYTE \*)
- int [CoLoad](#) (UBYTE \*)
- int [CoLocal](#) (UBYTE \*)



- int [CoLocalFactorized](#) (UBYTE \*)
- int [CoMany](#) (UBYTE \*)
- int [CoMerge](#) (UBYTE \*)
- int [CoShuffle](#) (UBYTE \*)
- int [CoMetric](#) (UBYTE \*)
- int [CoModOption](#) (UBYTE \*)
- int [CoModuleOption](#) (UBYTE \*)
- int [CoModulus](#) (UBYTE \*)
- int [CoMulti](#) (UBYTE \*)
- int [CoMultiply](#) (UBYTE \*)
- int [CoNFunction](#) (UBYTE \*)
- int [CoNPrint](#) (UBYTE \*)
- int [CoNTensor](#) (UBYTE \*)
- int [CoNWrite](#) (UBYTE \*)
- int [CoNoDrop](#) (UBYTE \*)
- int [CoNoSkip](#) (UBYTE \*)
- int [CoNormalize](#) (UBYTE \*)
- int [CoMakeInteger](#) (UBYTE \*)
- int [CoFlags](#) (UBYTE \*, int)
- int [CoOff](#) (UBYTE \*)
- int [CoOn](#) (UBYTE \*)
- int [CoOnce](#) (UBYTE \*)
- int [CoOnly](#) (UBYTE \*)
- int [CoOptimizeOption](#) (UBYTE \*)
- int [CoPolyFun](#) (UBYTE \*)
- int [CoPolyRatFun](#) (UBYTE \*)
- int [CoPrint](#) (UBYTE \*)
- int [CoPrintB](#) (UBYTE \*)
- int [CoProperCount](#) (UBYTE \*)
- int [CoUnitTrace](#) (UBYTE \*)
- int [CoRCycleSymmetrize](#) (UBYTE \*)
- int [CoRatio](#) (UBYTE \*)
- int [CoRedefine](#) (UBYTE \*)
- int [CoRenumber](#) (UBYTE \*)
- int [CoRepeat](#) (UBYTE \*)
- int [CoSave](#) (UBYTE \*)
- int [CoSelect](#) (UBYTE \*)
- int [CoSet](#) (UBYTE \*)
- int [CoSetExitFlag](#) (UBYTE \*)
- int [CoSkip](#) (UBYTE \*)
- int [CoProcessBucket](#) (UBYTE \*)
- int [CoPushHide](#) (UBYTE \*)
- int [CoPopHide](#) (UBYTE \*)
- int [CoHide](#) (UBYTE \*)
- int [CoIntoHide](#) (UBYTE \*)
- int [CoNoIntoHide](#) (UBYTE \*)
- int [CoNoHide](#) (UBYTE \*)
- int [CoUnHide](#) (UBYTE \*)
- int [CoNoUnHide](#) (UBYTE \*)
- int [CoSort](#) (UBYTE \*)
- int [CoSplitArg](#) (UBYTE \*)
- int [CoSplitFirstArg](#) (UBYTE \*)
- int [CoSplitLastArg](#) (UBYTE \*)
- int [CoSum](#) (UBYTE \*)
- int [CoSymbol](#) (UBYTE \*)

- int [CoSymmetrize](#) (UBYTE \*)
- int [DoTable](#) (UBYTE \*, int)
- int [CoTable](#) (UBYTE \*)
- int [CoTerm](#) (UBYTE \*)
- int [CoNTable](#) (UBYTE \*)
- int [CoCTable](#) (UBYTE \*)
- void [EmptyTable](#) ([TABLES](#))
- int [CoToTensor](#) (UBYTE \*)
- int [CoToVector](#) (UBYTE \*)
- int [CoTrace4](#) (UBYTE \*)
- int [CoTraceN](#) (UBYTE \*)
- int [CoChisholm](#) (UBYTE \*)
- int [CoTransform](#) (UBYTE \*)
- int [CoClearTable](#) (UBYTE \*)
- int [DoChain](#) (UBYTE \*, int)
- int [CoChainin](#) (UBYTE \*)
- int [CoChainout](#) (UBYTE \*)
- int [CoTryReplace](#) (UBYTE \*)
- int [CoVector](#) (UBYTE \*)
- int [CoWhile](#) (UBYTE \*)
- int [CoWrite](#) (UBYTE \*)
- int [CoAuto](#) (UBYTE \*)
- int [CoSwitch](#) (UBYTE \*)
- int [CoCase](#) (UBYTE \*)
- int [CoBreak](#) (UBYTE \*)
- int [CoDefault](#) (UBYTE \*)
- int [CoEndSwitch](#) (UBYTE \*)
- int [CoTBaddto](#) (UBYTE \*)
- int [CoTBaudit](#) (UBYTE \*)
- int [CoTbcleanup](#) (UBYTE \*)
- int [CoTbcreate](#) (UBYTE \*)
- int [CoTBenter](#) (UBYTE \*)
- int [CoTBhelp](#) (UBYTE \*)
- int [CoTbload](#) (UBYTE \*)
- int [CoTBoff](#) (UBYTE \*)
- int [CoTBon](#) (UBYTE \*)
- int [CoTBopen](#) (UBYTE \*)
- int [CoTBreplace](#) (UBYTE \*)
- int [CoTBuse](#) (UBYTE \*)
- int [CoTestUse](#) (UBYTE \*)
- int [CoThreadBucket](#) (UBYTE \*)
- int [AddComString](#) (int, WORD \*, UBYTE \*, int)
- int [CompileAlgebra](#) (UBYTE \*, int, WORD \*)
- int [IsIdStatement](#) (UBYTE \*)
- UBYTE \* [IsRHS](#) (UBYTE \*, UBYTE)
- int [ParenthesesTest](#) (UBYTE \*)
- int [tokenize](#) (UBYTE \*, WORD)
- void [WriteTokens](#) (SBYTE \*)
- int [simp1token](#) (SBYTE \*)
- int [simpwtoken](#) (SBYTE \*)
- int [simp2token](#) (SBYTE \*)
- int [simp3atoken](#) (SBYTE \*, int)
- int [simp3btoken](#) (SBYTE \*, int)
- int [simp4token](#) (SBYTE \*)
- int [simp5token](#) (SBYTE \*, int)

- int [simp6token](#) (SBYTE \*, int)
- UBYTE \* [SkipAName](#) (UBYTE \*)
- int [TestTables](#) (void)
- int [GetLabel](#) (UBYTE \*)
- int [ColdExpression](#) (UBYTE \*, int)
- int [CoAssign](#) (UBYTE \*)
- int [DoExpr](#) (UBYTE \*, int, int)
- int [CompileSubExpressions](#) (SBYTE \*)
- int [CodeGenerator](#) (SBYTE \*)
- int [CompleteTerm](#) (WORD \*, UWORD \*, UWORD \*, WORD, WORD, int)
- int [CodeFactors](#) (SBYTE \*s)
- WORD [GenerateFactors](#) (WORD, WORD)
- int [InsTree](#) (int, int)
- int [FindTree](#) (int, WORD \*)
- void [RedoTree](#) (CBUF \*, int)
- void [ClearTree](#) (int)
- int [CatchDollar](#) (int)
- int [AssignDollar](#) (PHEAD WORD \*, WORD)
- UBYTE \* [WriteDollarToBuffer](#) (WORD, WORD)
- UBYTE \* [WriteDollarFactorToBuffer](#) (WORD, WORD, WORD)
- void [AddToDollarBuffer](#) (UBYTE \*)
- int [PutTermInDollar](#) (WORD \*, WORD)
- void [TermAssign](#) (WORD \*)
- void [WildDollars](#) (PHEAD WORD \*)
- LONG [numcommute](#) (WORD \*, LONG \*)
- int [FullRenumber](#) (PHEAD WORD \*, WORD)
- int [Lus](#) (WORD \*, WORD, WORD, WORD, WORD, WORD)
- int [FindLus](#) (int, int, int)
- int [CoReplaceLoop](#) (UBYTE \*)
- int [CoFindLoop](#) (UBYTE \*)
- int [DoFindLoop](#) (UBYTE \*, int)
- int [CoFunPowers](#) (UBYTE \*)
- int [SortTheList](#) (int \*, int)
- int [MatchIsPossible](#) (WORD \*, WORD \*)
- int [StudyPattern](#) (WORD \*)
- WORD [DolToTensor](#) (PHEAD WORD)
- WORD [DolToFunction](#) (PHEAD WORD)
- WORD [DolToVector](#) (PHEAD WORD)
- WORD [DolToNumber](#) (PHEAD WORD)
- WORD [DolToSymbol](#) (PHEAD WORD)
- WORD [DolToIndex](#) (PHEAD WORD)
- LONG [DolToLong](#) (PHEAD WORD)
- int [DollarFactorize](#) (PHEAD WORD)
- int [CoPrintTable](#) (UBYTE \*)
- int [CoDeallocateTable](#) (UBYTE \*)
- void [CleanDollarFactors](#) (DOLLARS)
- WORD \* [TakeDollarContent](#) (PHEAD WORD \*, WORD \*\*)
- WORD \* [MakeDollarInteger](#) (PHEAD WORD \*, WORD \*\*)
- WORD \* [MakeDollarMod](#) (PHEAD WORD \*, WORD \*\*)
- int [GetDolNum](#) (PHEAD WORD \*, WORD \*)
- void [AddPotModdollar](#) (WORD)
- int [Optimize](#) (WORD, int)
- int [ClearOptimize](#) (void)
- int [TakeLongRoot](#) (UWORD \*, WORD \*, WORD)
- int [TakeRatRoot](#) (UWORD \*, WORD \*, WORD)

- int [MakeRational](#) (WORD, WORD, WORD \*, WORD \*)
- int [MakeLongRational](#) (PHEAD UWORD \*, WORD, UWORD \*, WORD, UWORD \*, WORD \*)
- void [ClearTableTree](#) ([TABLES](#))
- int [InsTableTree](#) ([TABLES](#), WORD \*)
- void [RedoTableTree](#) ([TABLES](#), int)
- int [FindTableTree](#) ([TABLES](#), WORD \*, int)
- void [finishcbuf](#) (WORD)
- void [clearcbuf](#) (WORD)
- void [CleanUpSort](#) (int)
- FILEHANDLE \* [AllocFileHandle](#) (WORD, char \*)
- void [DeAllocFileHandle](#) (FILEHANDLE \*)
- void [LowerSortLevel](#) (void)
- WORD \* [PolyRatFunSpecial](#) (PHEAD WORD \*, WORD \*)
- void [SimpleSplitMergeRec](#) (WORD \*, WORD, WORD \*)
- void [SimpleSplitMerge](#) (WORD \*, WORD)
- WORD [BinarySearch](#) (WORD \*, WORD, WORD)
- int [InsideDollar](#) (PHEAD WORD \*, WORD)
- DOLLARS [DoToTerms](#) (PHEAD WORD)
- WORD [EvalDoLoopArg](#) (PHEAD WORD \*, WORD)
- int [SetExprCases](#) (int, int, int)
- int [TestSelect](#) (WORD \*, WORD \*)
- void [SubsInAll](#) (PHEAD0)
- void [TransferBuffer](#) (int, int, int)
- int [TakeIDfunction](#) (PHEAD WORD \*)
- int [MakeSetupAllocs](#) (void)
- int [TryFileSetups](#) (void)
- void [ExchangeExpressions](#) (int, int)
- void [ExchangeDollars](#) (int, int)
- int [GetFirstBracket](#) (WORD \*, int)
- int [GetFirstTerm](#) (WORD \*, int, int)
- int [GetContent](#) (WORD \*, int)
- int [CleanupTerm](#) (WORD \*)
- WORD [ContentMerge](#) (PHEAD WORD \*, WORD \*)
- UBYTE \* [PrelfDollarEval](#) (UBYTE \*, int \*)
- LONG [TermsInDollar](#) (WORD)
- LONG [SizeOfDollar](#) (WORD)
- LONG [TermsInExpression](#) (WORD)
- LONG [SizeOfExpression](#) (WORD)
- WORD \* [TranslateExpression](#) (UBYTE \*)
- int [IsSetMember](#) (WORD \*, WORD)
- int [IsMultipleOf](#) (WORD \*, WORD \*)
- int [TwoExprCompare](#) (WORD \*, WORD \*, int)
- void [UpdatePositions](#) (void)
- void [M\\_check](#) (void)
- void [M\\_print](#) (void)
- void [M\\_check1](#) (void)
- void [PrintTime](#) (UBYTE \*)
- POSITION \* [FindBracket](#) (WORD, WORD \*)
- void [PutBracketInIndex](#) (PHEAD WORD \*, [POSITION](#) \*)
- void [ClearBracketIndex](#) (WORD)
- void [OpenBracketIndex](#) (WORD)
- int [DoNoParallel](#) (UBYTE \*)
- int [DoParallel](#) (UBYTE \*)
- int [DoModSum](#) (UBYTE \*)
- int [DoModMax](#) (UBYTE \*)

- int [DoModMin](#) (UBYTE \*)
- int [DoModLocal](#) (UBYTE \*)
- UBYTE \* [DoModDollar](#) (UBYTE \*, int)
- int [DoProcessBucket](#) (UBYTE \*)
- int [DoinParallel](#) (UBYTE \*)
- int [DonotinParallel](#) (UBYTE \*)
- int [FlipTable](#) (FUNCTIONS, int)
- int [ChainIn](#) (PHEAD WORD \*, WORD)
- int [ChainOut](#) (PHEAD WORD \*, WORD)
- int [ArgumentImplode](#) (PHEAD WORD \*, WORD \*)
- int [ArgumentExplode](#) (PHEAD WORD \*, WORD \*)
- int [DenToFunction](#) (WORD \*, WORD)
- WORD [HowMany](#) (PHEAD WORD \*, WORD \*)
- void [RemoveDollars](#) (void)
- LONG [CountTerms1](#) (PHEAD0)
- LONG [TermsInBracket](#) (PHEAD WORD \*, WORD)
- int [Crash](#) (void)
- char \* [str\\_dup](#) (char \*)
- void [convertblock](#) (INDEXBLOCK \*, INDEXBLOCK \*, int)
- void [convertnamesblock](#) (NAMESBLOCK \*, NAMESBLOCK \*, int)
- void [convertiniinfo](#) (INIINFO \*, INIINFO \*, int)
- int [ReadIndex](#) (DBASE \*)
- int [WriteIndexBlock](#) (DBASE \*, MLONG)
- int [WriteNamesBlock](#) (DBASE \*, MLONG)
- int [WriteIndex](#) (DBASE \*)
- int [WriteIniInfo](#) (DBASE \*)
- int [ReadIniInfo](#) (DBASE \*)
- int [AddToIndex](#) (DBASE \*, MLONG)
- DBASE \* [GetDbase](#) (char \*, MLONG)
- DBASE \* [OpenDbase](#) (char \*)
- char \* [ReadObject](#) (DBASE \*, MLONG, char \*)
- char \* [ReadijObject](#) (DBASE \*, MLONG, MLONG, char \*)
- int [ExistsObject](#) (DBASE \*, MLONG, char \*)
- int [DeleteObject](#) (DBASE \*, MLONG, char \*)
- int [WriteObject](#) (DBASE \*, MLONG, char \*, char \*, MLONG)
- MLONG [AddObject](#) (DBASE \*, MLONG, char \*, char \*)
- DBASE \* [NewDbase](#) (char \*, MLONG)
- void [FreeTableBase](#) (DBASE \*)
- int [ComposeTableNames](#) (DBASE \*)
- int [PutTableNames](#) (DBASE \*)
- MLONG [AddTableName](#) (DBASE \*, char \*, TABLES)
- MLONG [GetTableName](#) (DBASE \*, char \*)
- MLONG [FindTableNumber](#) (DBASE \*, char \*)
- int [TryEnvironment](#) (void)
- int [CopyExpression](#) (FILEHANDLE \*, FILEHANDLE \*)
- int [set\\_in](#) (UBYTE, set\_of\_char)
- one\_byte [set\\_set](#) (UBYTE, set\_of\_char)
- one\_byte [set\\_del](#) (UBYTE, set\_of\_char)
- one\_byte [set\\_sub](#) (set\_of\_char, set\_of\_char, set\_of\_char)
- int [DoPreAddSeparator](#) (UBYTE \*)
- int [DoPreRmSeparator](#) (UBYTE \*)
- int [openExternalChannel](#) (UBYTE \*, int, UBYTE \*, UBYTE \*)
- int [initPresetExternalChannels](#) (UBYTE \*, int)
- int [closeExternalChannel](#) (int)
- int [selectExternalChannel](#) (int)

- int [getCurrentExternalChannel](#) (void)
- void [closeAllExternalChannels](#) (void)
- UBYTE \* [defineChannel](#) (UBYTE \*, [HANDLERS](#) \*)
- int [writeToChannel](#) (int, UBYTE \*, [HANDLERS](#) \*)
- int [writeBufToExtChannelOk](#) (char \*, size\_t)
- int [getcFromExtChannelOk](#) (void)
- int [setKillModeForExternalChannelOk](#) (int, int)
- int [setTerminatorForExternalChannelOk](#) (char \*)
- int [getcFromExtChannelFailure](#) (void)
- int [setKillModeForExternalChannelFailure](#) (int, int)
- int [setTerminatorForExternalChannelFailure](#) (char \*)
- int [writeBufToExtChannelFailure](#) (char \*, size\_t)
- int [ReleaseTB](#) (void)
- int [SymbolNormalize](#) (WORD \*)
- int [TestFunFlag](#) (PHEAD WORD \*)
- WORD [CompareSymbols](#) (PHEAD WORD \*, WORD \*, WORD)
- WORD [CompareHSymbols](#) (PHEAD WORD \*, WORD \*, WORD)
- WORD [NextPrime](#) (PHEAD WORD)
- WORD [Moebius](#) (PHEAD WORD)
- UWORD [wranf](#) (PHEAD0)
- UWORD [iranf](#) (PHEAD UWORD)
- void [iniwranf](#) (PHEAD0)
- UBYTE \* [PreRandom](#) (UBYTE \*)
- WORD \* [PolyNormPoly](#) (PHEAD WORD)
- WORD \* [EvaluateGcd](#) (PHEAD WORD \*)
- int [TreatPolyRatFun](#) (PHEAD WORD \*)
- int [ReadSaveHeader](#) (void)
- LONG [ReadSaveIndex](#) ([FILEINDEX](#) \*)
- LONG [ReadSaveExpression](#) (UBYTE \*, UBYTE \*, LONG \*, LONG \*)
- UBYTE \* [ReadSaveTerm32](#) (UBYTE \*, UBYTE \*, UBYTE \*\*, UBYTE \*, UBYTE \*, int)
- LONG [ReadSaveVariables](#) (UBYTE \*, UBYTE \*, LONG \*, LONG \*, [INDEXENTRY](#) \*, LONG \*)
- LONG [WriteStoreHeader](#) (WORD)
- void [InitRecovery](#) (void)
- int [CheckRecoveryFile](#) (void)
- void [DeleteRecoveryFile](#) (void)
- char \* [RecoveryFilename](#) (void)
- int [DoRecovery](#) (int \*)
- void [DoCheckpoint](#) (int)
- void [NumberMallocAddMemory](#) (PHEAD0)
- void [CacheNumberMallocAddMemory](#) (PHEAD0)
- void [TermMallocAddMemory](#) (PHEAD0)
- void [ExprStatus](#) ([EXPRESSIONS](#))
- void [iniTools](#) (void)
- int [TestTerm](#) (WORD \*)
- int [RunTransform](#) (PHEAD WORD \*term, WORD \*params)
- int [RunEncode](#) (PHEAD WORD \*fun, WORD \*args, WORD \*info)
- int [RunDecode](#) (PHEAD WORD \*fun, WORD \*args, WORD \*info)
- int [RunReplace](#) (PHEAD WORD \*fun, WORD \*args, WORD \*info)
- int [RunImplode](#) (WORD \*fun, WORD \*args)
- int [RunExplode](#) (PHEAD WORD \*fun, WORD \*args)
- int [TestArgNum](#) (int n, int totarg, WORD \*args)
- WORD [PutArgInScratch](#) (WORD \*arg, UWORD \*scrat)
- UBYTE \* [ReadRange](#) (UBYTE \*s, WORD \*out, int par)
- int [FindRange](#) (PHEAD WORD \*, WORD \*, WORD \*, WORD)
- int [RunPermute](#) (PHEAD WORD \*fun, WORD \*args, WORD \*info)

- int [RunReverse](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunCycle](#) (PHEAD WORD \*fun, WORD \*args, WORD \*info)
- int [RunAddArg](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunMulArg](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunIsLyndon](#) (PHEAD WORD \*fun, WORD \*args, int par)
- WORD [RunToLyndon](#) (PHEAD WORD \*fun, WORD \*args, int par)
- int [RunDropArg](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunSelectArg](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunDedup](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunZtoHArg](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunHtoZArg](#) (PHEAD WORD \*fun, WORD \*args)
- int [NormPolyTerm](#) (PHEAD WORD \*)
- WORD [ComparePoly](#) (WORD \*, WORD \*, WORD)
- int [ConvertToPoly](#) (PHEAD WORD \*, WORD \*, WORD \*, WORD)
- int [LocalConvertToPoly](#) (PHEAD WORD \*, WORD \*, WORD, WORD)
- int [ConvertFromPoly](#) (PHEAD WORD \*, WORD \*, WORD, WORD, WORD, WORD)
- int [FindSubterm](#) (WORD \*)
- int [FindLocalSubterm](#) (PHEAD WORD \*, WORD)
- void [PrintSubtermList](#) (int, int)
- void [PrintExtraSymbol](#) (int, WORD \*, int)
- int [FindSubexpression](#) (WORD \*)
- void [UpdateMaxSize](#) (void)
- int [CoToPolynomial](#) (UBYTE \*)
- int [CoFromPolynomial](#) (UBYTE \*)
- int [CoArgToExtraSymbol](#) (UBYTE \*)
- int [CoExtraSymbols](#) (UBYTE \*)
- UBYTE \* [GetDoParam](#) (UBYTE \*, WORD \*\*, int)
- WORD \* [GetIfDollarFactor](#) (UBYTE \*\*, WORD \*)
- int [CoDo](#) (UBYTE \*)
- int [CoEndDo](#) (UBYTE \*)
- int [ExtraSymFun](#) (PHEAD WORD \*, WORD)
- int [PruneExtraSymbols](#) (WORD)
- int [IniFbuffer](#) (WORD)
- void [IniFbufs](#) (void)
- int [GCDfunction](#) (PHEAD WORD \*, WORD)
- WORD \* [GCDfunction3](#) (PHEAD WORD \*, WORD \*)
- int [ReadPolyRatFun](#) (PHEAD WORD \*)
- int [FromPolyRatFun](#) (PHEAD WORD \*, WORD \*\*, WORD \*\*)
- WORD \* [PolyDiv](#) (PHEAD WORD \*, WORD \*, char \*)
- void [GCDclean](#) (PHEAD WORD \*, WORD \*)
- WORD \* [TakeSymbolContent](#) (PHEAD WORD \*, WORD \*)
- int [GCDterms](#) (PHEAD WORD \*, WORD \*, WORD \*)
- WORD \* [PutExtraSymbols](#) (PHEAD WORD \*, WORD, int \*)
- WORD \* [TakeExtraSymbols](#) (PHEAD WORD \*, WORD)
- WORD \* [MultiplyWithTerm](#) (PHEAD WORD \*, WORD \*, WORD)
- WORD \* [TakeContent](#) (PHEAD WORD \*, WORD \*)
- int [MergeSymbolLists](#) (PHEAD WORD \*, WORD \*, int)
- int [MergeDotproductLists](#) (PHEAD WORD \*, WORD \*, int)
- WORD \* [CreateExpression](#) (PHEAD WORD)
- int [DIVfunction](#) (PHEAD WORD \*, WORD, int)
- WORD \* [MULfunc](#) (PHEAD WORD \*, WORD \*)
- WORD \* [ConvertArgument](#) (PHEAD WORD \*, int \*)
- int [ExpandRat](#) (PHEAD WORD \*)
- int [InvPoly](#) (PHEAD WORD \*, WORD, WORD)
- WORD [TestDoLoop](#) (PHEAD WORD \*, WORD)

- WORD [TestEndDoLoop](#) (PHEAD WORD \*, WORD)
- WORD \* [poly\\_gcd](#) (PHEAD WORD \*, WORD \*, WORD)
- WORD \* [poly\\_div](#) (PHEAD WORD \*, WORD \*, WORD)
- WORD \* [poly\\_rem](#) (PHEAD WORD \*, WORD \*, WORD)
- WORD \* [poly\\_inverse](#) (PHEAD WORD \*, WORD \*)
- WORD \* [poly\\_mul](#) (PHEAD WORD \*, WORD \*)
- WORD \* [poly\\_ratfun\\_add](#) (PHEAD WORD \*, WORD \*)
- int [poly\\_ratfun\\_normalize](#) (PHEAD WORD \*)
- int [poly\\_factorize\\_argument](#) (PHEAD WORD \*, WORD \*)
- WORD \* [poly\\_factorize\\_dollar](#) (PHEAD WORD \*)
- int [poly\\_factorize\\_expression](#) (EXPRESSIONS)
- int [poly\\_unfactorize\\_expression](#) (EXPRESSIONS)
- void [poly\\_free\\_poly\\_vars](#) (PHEAD const char \*)
- void [optimize\\_print\\_code](#) (int)
- int [DoPreAdd](#) (UBYTE \*s)
- int [DoPreUseDictionary](#) (UBYTE \*s)
- int [DoPreCloseDictionary](#) (UBYTE \*s)
- int [DoPreOpenDictionary](#) (UBYTE \*s)
- void [RemoveDictionary](#) (DICTIONARY \*dict)
- void [UnSetDictionary](#) (void)
- int [SetDictionaryOptions](#) (UBYTE \*options)
- int [AddToDictionary](#) (DICTIONARY \*dict, UBYTE \*left, UBYTE \*right)
- int [AddDictionary](#) (UBYTE \*name)
- int [FindDictionary](#) (UBYTE \*name)
- UBYTE \* [IsExponentSign](#) (void)
- UBYTE \* [IsMultiplySign](#) (void)
- void [TransformRational](#) (UWORD \*a, WORD na)
- void [WriteDictionary](#) (DICTIONARY \*)
- void [ShrinkDictionary](#) (DICTIONARY \*)
- void [MultiplyToLine](#) (void)
- UBYTE \* [FindSymbol](#) (WORD num)
- UBYTE \* [FindVector](#) (WORD num)
- UBYTE \* [FindIndex](#) (WORD num)
- UBYTE \* [FindFunction](#) (WORD num)
- UBYTE \* [FindFunWithArgs](#) (WORD \*t)
- UBYTE \* [FindExtraSymbol](#) (WORD num)
- LONG [DictToBytes](#) (DICTIONARY \*dict, UBYTE \*buf)
- DICTIONARY \* [DictFromBytes](#) (UBYTE \*buf)
- int [CoCreateSpectator](#) (UBYTE \*inp)
- int [CoToSpectator](#) (UBYTE \*inp)
- int [CoRemoveSpectator](#) (UBYTE \*inp)
- int [CoEmptySpectator](#) (UBYTE \*inp)
- int [CoCopySpectator](#) (UBYTE \*inp)
- int [PutInSpectator](#) (WORD \*, WORD)
- void [ClearSpectators](#) (WORD)
- WORD [GetFromSpectator](#) (WORD \*, WORD)
- void [FlushSpectators](#) (void)
- WORD \* [PreGCD](#) (PHEAD WORD \*, WORD \*, int)
- int [ReadFromScratch](#) (FILEHANDLE \*, POSITION \*, UBYTE \*, POSITION \*)
- int [AddToScratch](#) (FILEHANDLE \*, POSITION \*, UBYTE \*, POSITION \*, int)
- int [DoPreAppendPath](#) (UBYTE \*)
- int [DoPrePrependPath](#) (UBYTE \*)
- int [DoSwitch](#) (PHEAD WORD \*, WORD \*)
- int [DoEndSwitch](#) (PHEAD WORD \*, WORD \*)
- SWITCHTABLE \* [FindCase](#) (WORD, WORD)



- void [SwitchSplitMergeRec](#) ([SWITCHTABLE](#) \*, WORD, [SWITCHTABLE](#) \*)
- void [SwitchSplitMerge](#) ([SWITCHTABLE](#) \*, WORD)
- int [DoubleSwitchBuffers](#) (void)
- int [DistrN](#) (int, int \*, int, int \*)
- int [DoNamespace](#) (UBYTE \*)
- int [DoEndNamespace](#) (UBYTE \*)
- int [DoUse](#) (UBYTE \*)
- UBYTE \* [SkipName](#) (UBYTE \*)
- UBYTE \* [ConstructName](#) (UBYTE \*, UBYTE)
- int [DoSetUserFlag](#) (UBYTE \*)
- int [DoClearUserFlag](#) (UBYTE \*)
- [VERTEX](#) \* [CreateVertex](#) ([MODEL](#) \*)
- UBYTE \* [ReadParticle](#) (UBYTE \*, [VERTEX](#) \*, [MODEL](#) \*, int)
- int [CoModel](#) (UBYTE \*)
- int [CoParticle](#) (UBYTE \*)
- int [CoVertex](#) (UBYTE \*)
- int [CoEndModel](#) (UBYTE \*)
- int [LoadModel](#) ([MODEL](#) \*)
- int [CoSetUserFlag](#) (UBYTE \*)
- int [CoClearUserFlag](#) (UBYTE \*)
- int [CoCreateAllLoops](#) (UBYTE \*)
- int [CoCreateAllPaths](#) (UBYTE \*)
- int [CoCreateAll](#) (UBYTE \*)
- int [AllLoops](#) (PHEAD WORD \*, WORD)
- LONG [StartLoops](#) (PHEAD WORD \*, WORD, LONG, WORD, WORD \*, WORD, WORD \*, WORD)
- LONG [GenLoops](#) (PHEAD WORD \*, WORD, LONG, WORD, WORD \*, WORD, WORD \*, WORD)
- void [LoopOutput](#) (PHEAD WORD \*, WORD, WORD \*, WORD)
- int [AllPaths](#) (PHEAD WORD \*, WORD)
- LONG [GenPaths](#) (PHEAD WORD \*, WORD, LONG, WORD, WORD \*, WORD, WORD \*, WORD)
- void [PathOutput](#) (PHEAD WORD \*, WORD, WORD \*, WORD)

### 4.19.1 Detailed Description

Contains macros and function declarations.

Definition in file [declare.h](#).

### 4.19.2 Macro Definition Documentation

#### 4.19.2.1 MaX

```
#define MaX(
    x,
    y ) ((x) > (y) ? (x) : (y))
```

Definition at line 40 of file [declare.h](#).

#### 4.19.2.2 MiN

```
#define MiN(  
    x,  
    y ) ( (x) < (y) ? (x) : (y) )
```

Definition at line 41 of file [declare.h](#).

#### 4.19.2.3 ABS

```
#define ABS(  
    x ) ( (x) < 0 ? -(x) : (x) )
```

Definition at line 42 of file [declare.h](#).

#### 4.19.2.4 SGN

```
#define SGN(  
    x ) ( (x) > 0 ? 1 : (x) < 0 ? -1 : 0 )
```

Definition at line 43 of file [declare.h](#).

#### 4.19.2.5 REDLENG

```
#define REDLENG(  
    x ) ( ((x)<0) ? ((x)+1) : ((x)-1) ) / 2 )
```

Definition at line 44 of file [declare.h](#).

#### 4.19.2.6 INCLENG

```
#define INCLENG(  
    x ) ( ((x)<0) ? ((x)*2)-1 : ((x)*2)+1 )
```

Definition at line 45 of file [declare.h](#).

#### 4.19.2.7 GETCOEF

```
#define GETCOEF(  
    x,  
    y ) x += *x; y = x[-1]; x -= ABS(y); y=REDLENG(y)
```

Definition at line 46 of file [declare.h](#).

#### 4.19.2.8 GETSTOP

```
#define GETSTOP(  
    x,  
    y ) y=x+(*x)-1;y -= ABS(*y)-1
```

Definition at line 47 of file [declare.h](#).

#### 4.19.2.9 StuffAdd

```
#define StuffAdd(  
    x,  
    y ) ((x)<0?-1:1)*(y)+((y)<0?-1:1)*(x)
```

Definition at line 48 of file [declare.h](#).

#### 4.19.2.10 EXCHN

```
#define EXCHN(  
    t1,  
    t2,  
    n ) { WORD a,i; for(i=0;i<n;i++){a=t1[i];t1[i]=t2[i];t2[i]=a;} }
```

Definition at line 50 of file [declare.h](#).

#### 4.19.2.11 EXCH

```
#define EXCH(  
    x,  
    y ) { WORD a = (x); (x) = (y); (y) = a; }
```

Definition at line 51 of file [declare.h](#).

#### 4.19.2.12 TOKENTOLINE

```
#define TOKENTOLINE(  
    x,  
    y )
```

**Value:**

```
if ( AC.OutputSpaces == NOSPACEFORMAT ) { \  
    TokenToLine((UBYTE *) (y)); } else { TokenToLine((UBYTE *) (x)); }
```

Definition at line 53 of file [declare.h](#).

#### 4.19.2.13 UngetFromStream

```
#define UngetFromStream(  
    stream,  
    c ) ((stream)->nextchar[(stream)->isnextchar++]=c)
```

Definition at line 56 of file [declare.h](#).

**4.19.2.14 AddLineFeed**

```
#define AddLineFeed(
    s,
    n ) { (s)[(n)++] = LINEFEED; }
```

Definition at line 60 of file [declare.h](#).

**4.19.2.15 TryRecover**

```
#define TryRecover(
    x ) Terminate(-1)
```

Definition at line 62 of file [declare.h](#).

**4.19.2.16 UngetChar**

```
#define UngetChar(
    c ) { pushbackchar = c; }
```

Definition at line 63 of file [declare.h](#).

**4.19.2.17 ParseNumber**

```
#define ParseNumber(
    x,
    s ) { (x)=0;while(*(s)>='0'&&*(s)<='9') (x)=10*(x)+*(s)++ -'0'; }
```

Definition at line 64 of file [declare.h](#).

**4.19.2.18 ParseSign**

```
#define ParseSign(
    sgn,
    s )
```

**Value:**

```
{ (sgn)=0;while(*(s)=='-'||*(s)=='+'){\
if ( *(s)++ == '-' ) sgn ^= 1;}}
```

Definition at line 65 of file [declare.h](#).

**4.19.2.19 ParseSignedNumber**

```
#define ParseSignedNumber(
    x,
    s )
```

**Value:**

```
{ int sgn; ParseSign(sgn,s)\
ParseNumber(x,s) if ( sgn ) x = -x; }
```

Definition at line 67 of file [declare.h](#).

#### 4.19.2.20 NCOPY

```
#define NCOPY(  
    s,  
    t,  
    n ) { while ( (n)-- > 0 ) { *s++ = *t++; } }
```

Definition at line 71 of file [declare.h](#).

#### 4.19.2.21 NCOPYI

```
#define NCOPYI(  
    s,  
    t,  
    n ) { while ( (n)-- > 0 ) { *s++ = *t++; } }
```

Definition at line 73 of file [declare.h](#).

#### 4.19.2.22 NCOPYB

```
#define NCOPYB(  
    s,  
    t,  
    n ) { while ( (n)-- > 0 ) { *s++ = *t++; } }
```

Definition at line 74 of file [declare.h](#).

#### 4.19.2.23 NCOPYI32

```
#define NCOPYI32(  
    s,  
    t,  
    n ) { while ( (n)-- > 0 ) { *s++ = *t++; } }
```

Definition at line 75 of file [declare.h](#).

#### 4.19.2.24 WCOPY

```
#define WCOPY(  
    s,  
    t,  
    n ) { int nn=n; WORD *ss=(WORD *)s, *tt=(WORD *)t; while ( (nn)-- > 0 ) { *ss++ =  
*tt++; } }
```

Definition at line 76 of file [declare.h](#).

#### 4.19.2.25 NeedNumber

```
#define NeedNumber(
    x,
    s,
    err )
```

##### Value:

```
{ int sgn = 1;
while ( *s == ' ' || *s == '\t' || *s == '-' || *s == '+' ) {
    if ( *s == '-' ) {sgn = -sgn;} s++; }
if ( chartype[*s] != 1 ) goto err;
ParseNumber(x,s)
if ( sgn < 0 ) {(x) = -(x);} while ( *s == ' ' || *s == '\t' ) s++;
}
```

Definition at line 78 of file [declare.h](#).

#### 4.19.2.26 SKIPBLANKS

```
#define SKIPBLANKS(
    s ) { while ( *(s) == ' ' || *(s) == '\t' ) (s)++; }
```

Definition at line 85 of file [declare.h](#).

#### 4.19.2.27 FLUSHCONSOLE

```
#define FLUSHCONSOLE if ( AP.InOutBuf > 0 ) CharOut(LINEFEED)
```

Definition at line 86 of file [declare.h](#).

#### 4.19.2.28 SKIPBRA1

```
#define SKIPBRA1(
    s )
```

##### Value:

```
{ int lev1=0; s++; while(*s) { if(*s=='{'){lev1++;} \
else {if(*s=='}'&&--lev1<0){break;}} s++;} }
```

Definition at line 88 of file [declare.h](#).

#### 4.19.2.29 SKIPBRA2

```
#define SKIPBRA2(
    s )
```

##### Value:

```
{ int lev2=0; s++; while(*s) { if(*s=='{'){lev2++;} \
else {if(*s=='}'&&--lev2<0){break;}} \
else {if(*s=='{'){SKIPBRA1(s)}} s++;} }
```

Definition at line 90 of file [declare.h](#).

#### 4.19.2.30 SKIPBRA3

```
#define SKIPBRA3(
    s )
```

**Value:**

```
{ int lev3=0; s++; while(*s) { if(*s=='('){lev3++;} \
else {if(*s=='&--lev3<0){break;} \
else {if(*s=='('){SKIPBRA2(s)} \
else {if(*s=='['){SKIPBRA1(s)}}} s++;} }
```

Definition at line 93 of file [declare.h](#).

#### 4.19.2.31 SKIPBRA4

```
#define SKIPBRA4(
    s )
```

**Value:**

```
{ int lev4=0; s++; while(*s) { if(*s=='('){lev4++;} \
else {if(*s=='&--lev4<0){break;} \
else {if(*s=='('){SKIPBRA1(s)}} s++;} }
```

Definition at line 97 of file [declare.h](#).

#### 4.19.2.32 SKIPBRA5

```
#define SKIPBRA5(
    s )
```

**Value:**

```
{ int lev5=0; s++; while(*s) { if(*s=='('){lev5++;} \
else {if(*s=='&--lev5<0){break;} \
else {if(*s=='('){SKIPBRA4(s)} \
else {if(*s=='['){SKIPBRA1(s)}}} s++;} }
```

Definition at line 100 of file [declare.h](#).

#### 4.19.2.33 CYCLE1

```
#define CYCLE1(
    t,
    a,
    i ) {t iX=*a; WORD jX; for(jX=1;jX<i;jX++)a[jX-1]=a[jX]; a[i-1]=iX;}
```

Definition at line 108 of file [declare.h](#).

#### 4.19.2.34 AddToCB

```
#define AddToCB(
    c,
    wx )
```

**Value:**

```
if(c->Pointer>=c->Top) \
{DoubleCbuffer(c-cbuf,c->Pointer,21);} \
*(c->Pointer)++ = wx;
```

Definition at line 110 of file [declare.h](#).

#### 4.19.2.35 EXCHINOUT

```
#define EXCHINOUT
```

**Value:**

```
{ FILEHANDLE *ffFi = AR.outfile; \  
  AR.outfile = AR.infile; AR.infile = ffFi; }
```

Definition at line 114 of file [declare.h](#).

#### 4.19.2.36 BACKINOUT

```
#define BACKINOUT
```

**Value:**

```
{ FILEHANDLE *ffFi = AR.outfile; POSITION posi; \  
  AR.outfile = AR.infile; AR.infile = ffFi; \  
  SetEndScratch(AR.infile,&posi); }
```

Definition at line 116 of file [declare.h](#).

#### 4.19.2.37 CopyArg

```
#define CopyArg(  
    to,  
    from )
```

**Value:**

```
{ if ( *from > 0 ) { int ica = *from; NCOPY(to,from,ica) } \  
  else if ( *from <= -FUNCTION ) *to++ = *from++; \  
  else { *to++ = *from++; *to++ = *from++; } }
```

Definition at line 120 of file [declare.h](#).

#### 4.19.2.38 FILLARG

```
#define FILLARG(  
    w )
```

Definition at line 129 of file [declare.h](#).

#### 4.19.2.39 COPYARG

```
#define COPYARG(  
    w,  
    t )
```

Definition at line 130 of file [declare.h](#).



#### 4.19.2.40 ZEROARG

```
#define ZEROARG(  
    w )
```

Definition at line 131 of file [declare.h](#).

#### 4.19.2.41 FILLFUN

```
#define FILLFUN(  
    w )
```

Definition at line 138 of file [declare.h](#).

#### 4.19.2.42 COPYFUN

```
#define COPYFUN(  
    w,  
    t )
```

Definition at line 139 of file [declare.h](#).

#### 4.19.2.43 COPYFUN3

```
#define COPYFUN3(  
    w,  
    t )
```

Definition at line 146 of file [declare.h](#).

#### 4.19.2.44 FILLFUN3

```
#define FILLFUN3(  
    w )
```

Definition at line 147 of file [declare.h](#).

#### 4.19.2.45 FILLSUB

```
#define FILLSUB(  
    w )
```

Definition at line 154 of file [declare.h](#).

**4.19.2.46 COPYSUB**

```
#define COPYSUB(
    w,
    ww )
```

Definition at line 155 of file [declare.h](#).

**4.19.2.47 FILLEXP**

```
#define FILLEXP(
    w )
```

Definition at line 161 of file [declare.h](#).

**4.19.2.48 NEXTARG**

```
#define NEXTARG(
    x ) if(*x>0) x += *x; else if(*x <= -FUNCTION)x++; else x += 2;
```

Definition at line 164 of file [declare.h](#).

**4.19.2.49 COPY1ARG**

```
#define COPY1ARG(
    sl,
    tl )
```

**Value:**

```
{ int ica; if ( (ica=tl) > 0 ) { NCOPY(sl,tl,ica) } \
else if(*tl<=-FUNCTION){*sl++=*tl++;} else{*sl++=*tl++;*sl++=*tl++;} }
```

Definition at line 165 of file [declare.h](#).

**4.19.2.50 ZeroFillRange**

```
#define ZeroFillRange(
    w,
    begin,
    end )
```

**Value:**

```
do { \
    int tmp_i; \
    for ( tmp_i = begin; tmp_i < end; tmp_i++ ) { (w)[tmp_i] = 0; } \
} while (0)
```

Fills a buffer by zero in the range [begin,end).

**Parameters**

|              |                                                 |
|--------------|-------------------------------------------------|
| <i>w</i>     | The buffer.                                     |
| <i>begin</i> | The index for the beginning of the range.       |
| <i>end</i>   | The index for the end of the range (exclusive). |

Definition at line 175 of file [declare.h](#).

#### 4.19.2.51 TABLESIZE

```
#define TABLESIZE(  
    a,  
    b ) ((WORD)sizeof(a)) / ((WORD)sizeof(b))
```

Definition at line 180 of file [declare.h](#).

#### 4.19.2.52 WORDDIF

```
#define WORDDIF(  
    x,  
    y ) (WORD) (x-y)
```

Definition at line 181 of file [declare.h](#).

#### 4.19.2.53 wsizeof

```
#define wsizeof(  
    a ) ((WORD)sizeof(a))
```

Definition at line 182 of file [declare.h](#).

#### 4.19.2.54 VARNAME

```
#define VARNAME(  
    type,  
    num ) (AC.varnames->namebuffer+type[num].name)
```

Definition at line 183 of file [declare.h](#).

#### 4.19.2.55 DOLLARNAME

```
#define DOLLARNAME(  
    type,  
    num ) (AC.dollarnames->namebuffer+type[num].name)
```

Definition at line 184 of file [declare.h](#).

#### 4.19.2.56 EXPRNAME

```
#define EXPRNAME(  
    num ) (AC.exprnames->namebuffer+Expressions[num].name)
```

Definition at line 185 of file [declare.h](#).

#### 4.19.2.57 PREV

```
#define PREV(  
    x ) prevorder?prevorder:x
```

Definition at line 187 of file [declare.h](#).

#### 4.19.2.58 Terminate

```
#define Terminate(  
    x ) do { TerminateImpl(x, __FILE__, __LINE__, __FUNCTION__); } while(0)
```

Definition at line 189 of file [declare.h](#).

#### 4.19.2.59 SETERROR

```
#define SETERROR(  
    x ) { Terminate(-1); return(-1); }
```

Definition at line 190 of file [declare.h](#).

#### 4.19.2.60 DUMMYUSE

```
#define DUMMYUSE(  
    x ) (void) (x);
```

Definition at line 193 of file [declare.h](#).

#### 4.19.2.61 ADDPOS

```
#define ADDPOS(  
    pp,  
    x ) (pp).p1 = ((pp).p1+(LONG) (x))
```

Definition at line 219 of file [declare.h](#).

#### 4.19.2.62 SETBASELENGTH

```
#define SETBASELENGTH(  
    ss,  
    x ) (ss).p1 = (LONG) (x)
```

Definition at line 220 of file [declare.h](#).

#### 4.19.2.63 SETBASEPOSITION

```
#define SETBASEPOSITION(  
    pp,  
    x ) (pp).p1 = (LONG)(x)
```

Definition at line 221 of file [declare.h](#).

#### 4.19.2.64 ISEQUALPOSINC

```
#define ISEQUALPOSINC(  
    pp1,  
    pp2,  
    x ) ( (pp1).p1 == ((pp2).p1+(LONG)(x)) )
```

Definition at line 222 of file [declare.h](#).

#### 4.19.2.65 ISGEPOSINC

```
#define ISGEPOSINC(  
    pp1,  
    pp2,  
    x ) ( (pp1).p1 >= ((pp2).p1+(LONG)(x)) )
```

Definition at line 223 of file [declare.h](#).

#### 4.19.2.66 DIVPOS

```
#define DIVPOS(  
    pp,  
    n ) ( (pp).p1/(LONG)(n) )
```

Definition at line 224 of file [declare.h](#).

#### 4.19.2.67 MULPOS

```
#define MULPOS(  
    pp,  
    n ) (pp).p1 *= (LONG)(n)
```

Definition at line 225 of file [declare.h](#).

#### 4.19.2.68 DIFPOS

```
#define DIFPOS(  
    ss,  
    pp1,  
    pp2 ) (ss).p1 = ((pp1).p1-(pp2).p1)
```

Definition at line 228 of file [declare.h](#).

#### 4.19.2.69 DIFBASE

```
#define DIFBASE(  
    pp1,  
    pp2 ) ((pp1).p1-(pp2).p1)
```

Definition at line 229 of file [declare.h](#).

#### 4.19.2.70 ADD2POS

```
#define ADD2POS(  
    pp1,  
    pp2 ) (pp1).p1 += (pp2).p1
```

Definition at line 230 of file [declare.h](#).

#### 4.19.2.71 PUTZERO

```
#define PUTZERO(  
    pp ) (pp).p1 = 0
```

Definition at line 231 of file [declare.h](#).

#### 4.19.2.72 BASEPOSITION

```
#define BASEPOSITION(  
    pp ) ((pp).p1)
```

Definition at line 232 of file [declare.h](#).

#### 4.19.2.73 SETSTARTPOS

```
#define SETSTARTPOS(  
    pp ) (pp).p1 = -2
```

Definition at line 233 of file [declare.h](#).

#### 4.19.2.74 NOTSTARTPOS

```
#define NOTSTARTPOS(  
    pp ) ( (pp).p1 > -2 )
```

Definition at line 234 of file [declare.h](#).

#### 4.19.2.75 ISMINPOS

```
#define ISMINPOS(  
    pp ) ( (pp).p1 == -1 )
```

Definition at line 235 of file [declare.h](#).

#### 4.19.2.76 ISEQUALPOS

```
#define ISEQUALPOS(  
    pp1,  
    pp2 ) ( (pp1).p1 == (pp2).p1 )
```

Definition at line 236 of file [declare.h](#).

#### 4.19.2.77 ISNOTEQUALPOS

```
#define ISNOTEQUALPOS(  
    pp1,  
    pp2 ) ( (pp1).p1 != (pp2).p1 )
```

Definition at line 237 of file [declare.h](#).

#### 4.19.2.78 ISLESSPOS

```
#define ISLESSPOS(  
    pp1,  
    pp2 ) ( (pp1).p1 < (pp2).p1 )
```

Definition at line 238 of file [declare.h](#).

#### 4.19.2.79 ISGEPOS

```
#define ISGEPOS(  
    pp1,  
    pp2 ) ( (pp1).p1 >= (pp2).p1 )
```

Definition at line 239 of file [declare.h](#).

#### 4.19.2.80 ISNOTZEROPOS

```
#define ISNOTZEROPOS(  
    pp ) ( (pp).p1 != 0 )
```

Definition at line 240 of file [declare.h](#).

#### 4.19.2.81 ISZEROPOS

```
#define ISZEROPOS(  
    pp ) ( (pp).p1 == 0 )
```

Definition at line 241 of file [declare.h](#).

#### 4.19.2.82 ISPOSPOS

```
#define ISPOSPOS(  
    pp ) ( (pp).p1 > 0 )
```

Definition at line 242 of file [declare.h](#).

#### 4.19.2.83 ISNEGPOS

```
#define ISNEGPOS(  
    pp ) ( (pp).p1 < 0 )
```

Definition at line 243 of file [declare.h](#).

#### 4.19.2.84 TOLONG

```
#define TOLONG(  
    x ) ((LONG)(x))
```

Definition at line 246 of file [declare.h](#).

#### 4.19.2.85 Add2Com

```
#define Add2Com(  
    x ) { WORD cod[2]; cod[0] = x; cod[1] = 2; AddNtoL(2,cod); }
```

Definition at line 248 of file [declare.h](#).

#### 4.19.2.86 Add3Com

```
#define Add3Com(  
    x1,  
    x2 ) { WORD cod[3]; cod[0] = x1; cod[1] = 3; cod[2] = x2; AddNtoL(3,cod); }
```

Definition at line 249 of file [declare.h](#).

#### 4.19.2.87 Add4Com

```
#define Add4Com(  
    x1,  
    x2,  
    x3 )
```

**Value:**

```
{ WORD cod[4]; cod[0] = x1; cod[1] = 4; \  
cod[2] = x2; cod[3] = x3; AddNtoL(4,cod); }
```

Definition at line 250 of file [declare.h](#).



#### 4.19.2.88 Add5Com

```
#define Add5Com(  
    x1,  
    x2,  
    x3,  
    x4 )
```

**Value:**

```
{ WORD cod[5]; cod[0] = x1; cod[1] = 5; \  
  cod[2] = x2; cod[3] = x3; cod[4] = x4; AddNtL(5,cod); }
```

Definition at line 252 of file [declare.h](#).

#### 4.19.2.89 WantAddPointers

```
#define WantAddPointers(  
    x )
```

**Value:**

```
while ((AT.pWorkPointer+(x))>AR.pWorkSize) {WORD **ppp=&AT.pWorkSpace;\br/>  ExpandBuffer((void **)ppp,&AR.pWorkSize,(int) (sizeof(WORD *)));}
```

Definition at line 258 of file [declare.h](#).

#### 4.19.2.90 WantAddLongs

```
#define WantAddLongs(  
    x )
```

**Value:**

```
while ((AT.lWorkPointer+(x))>AR.lWorkSize) {LONG **ppp=&AT.lWorkSpace;\br/>  ExpandBuffer((void **)ppp,&AR.lWorkSize,sizeof(LONG));}
```

Definition at line 260 of file [declare.h](#).

#### 4.19.2.91 WantAddPositions

```
#define WantAddPositions(  
    x )
```

**Value:**

```
while ((AT.posWorkPointer+(x))>AR.posWorkSize) {POSITION **ppp=&AT.posWorkSpace;\br/>  ExpandBuffer((void **)ppp,&AR.posWorkSize,sizeof(POSITION));}
```

Definition at line 262 of file [declare.h](#).

#### 4.19.2.92 FORM\_INLINE

```
#define FORM_INLINE inline
```

Definition at line 266 of file [declare.h](#).

#### 4.19.2.93 MEMORYMACROS

```
#define MEMORYMACROS
```

Definition at line 274 of file [declare.h](#).

#### 4.19.2.94 TermMalloc

```
#define TermMalloc(  
    x ) ( (AT.TermMemTop <= 0 ) ? TermMallocAddMemory(BHEAD0), AT.TermMemHeap[--AT.↵  
TermMemTop]: AT.TermMemHeap[--AT.TermMemTop] )
```

Definition at line 278 of file [declare.h](#).

#### 4.19.2.95 NumberMalloc

```
#define NumberMalloc(  
    x ) ( (AT.NumberMemTop <= 0 ) ? NumberMallocAddMemory(BHEAD0), AT.NumberMem↵  
Heap[--AT.NumberMemTop]: AT.NumberMemHeap[--AT.NumberMemTop] )
```

Definition at line 279 of file [declare.h](#).

#### 4.19.2.96 CacheNumberMalloc

```
#define CacheNumberMalloc(  
    x ) ( (AT.CacheNumberMemTop <= 0 ) ? CacheNumberMallocAddMemory(BHEAD0), AT.↵  
CacheNumberMemHeap[--AT.CacheNumberMemTop]: AT.CacheNumberMemHeap[--AT.CacheNumberMemTop] )
```

Definition at line 280 of file [declare.h](#).

#### 4.19.2.97 TermFree

```
#define TermFree(  
    TermMem,  
    x ) AT.TermMemHeap[AT.TermMemTop++] = (WORD *) (TermMem)
```

Definition at line 281 of file [declare.h](#).

#### 4.19.2.98 NumberFree

```
#define NumberFree(  
    NumberMem,  
    x ) AT.NumberMemHeap[AT.NumberMemTop++] = (UWORD *) (NumberMem)
```

Definition at line 282 of file [declare.h](#).

**4.19.2.99 CacheNumberFree**

```
#define CacheNumberFree(
    NumberMem,
    x ) AT.CacheNumberMemHeap[AT.CacheNumberMemTop++] = (UWORD *) (NumberMem)
```

Definition at line 283 of file [declare.h](#).

**4.19.2.100 NestingChecksum**

```
#define NestingChecksum( ) (AC.IfLevel + AC.RepLevel + AC.arglevel + AC.insidelevel + AC.↵
termlevel + AC.inexprlevel + AC.dolooplevel +AC.SwitchLevel)
```

Definition at line 311 of file [declare.h](#).

**4.19.2.101 MesNesting**

```
#define MesNesting( ) MesPrint("&Illegal nesting of if, repeat, argument, inside, term, inexpression
and do")
```

Definition at line 312 of file [declare.h](#).

**4.19.2.102 MarkPolyRatFunDirty**

```
#define MarkPolyRatFunDirty(
    T )
```

**Value:**

```
{if (*T&&AR.PolyFunType==2) {WORD *TP, *TT; TT=T+*T; TT-=ABS(TT[-1]); \
TP=T+1; while (TP<TT) {if (*TP==AR.PolyFun) {TP[2] |= (DIRTYFLAG|MUSTCLEANPRF); } TP+=TP[1]; }}}}
```

Definition at line 314 of file [declare.h](#).

**4.19.2.103 PUSHPREASSIGNLEVEL**

```
#define PUSHPREASSIGNLEVEL
```

**Value:**

```
AP.PreAssignLevel++; { GETIDENTITY \
if ( AP.PreAssignLevel >= AP.MaxPreAssignLevel ) { int i; \
LONG *ap = (LONG *)Malloca(2*AP.MaxPreAssignLevel*sizeof(LONG *), "PreAssignStack"); \
for ( i = 0; i < AP.MaxPreAssignLevel; i++ ) ap[i] = AP.PreAssignStack[i]; \
M_free(AP.PreAssignStack, "PreAssignStack"); \
AP.MaxPreAssignLevel *= 2; AP.PreAssignStack = ap; \
} \
*AT.WorkPointer++ = AP.PreContinuation; AP.PreContinuation = 0; \
AP.PreAssignStack[AP.PreAssignLevel] = AC.iPointer - AC.iBuffer; }
```

Definition at line 321 of file [declare.h](#).

#### 4.19.2.104 POPPREASSIGNLEVEL

```
#define POPPREASSIGNLEVEL
```

**Value:**

```
if ( AP.PreAssignLevel > 0 ) { GETIDENTITY \
AC.iPointer = AC.iBuffer + AP.PreAssignStack[AP.PreAssignLevel--]; \
AP.PreContinuation = *--AT.WorkPointer; \
*AC.iPointer = 0; }
```

Definition at line 331 of file [declare.h](#).

#### 4.19.2.105 EXTERNLOCK

```
#define EXTERNLOCK(
    x )
```

NOTE: We have replaced LOCK(ErrorMessageLock) and UNLOCK(ErrorMessageLock) by MLOCK(ErrorMessageLock) and MUNLOCK(ErrorMessageLock). They are used for the synchronised output in ParFORM. (TU 28 May 2011)

Definition at line 477 of file [declare.h](#).

#### 4.19.2.106 INILOCK

```
#define INILOCK(
    x )
```

Definition at line 478 of file [declare.h](#).

#### 4.19.2.107 LOCK

```
#define LOCK(
    x )
```

Definition at line 479 of file [declare.h](#).

#### 4.19.2.108 UNLOCK

```
#define UNLOCK(
    x )
```

Definition at line 480 of file [declare.h](#).

#### 4.19.2.109 EXTERNRWLOCK

```
#define EXTERNRWLOCK(
    x )
```

Definition at line 481 of file [declare.h](#).

#### 4.19.2.110 INIRWLOCK

```
#define INIRWLOCK(  
    x )
```

Definition at line 482 of file [declare.h](#).

#### 4.19.2.111 RWLOCKR

```
#define RWLOCKR(  
    x )
```

Definition at line 483 of file [declare.h](#).

#### 4.19.2.112 RWLOCKW

```
#define RWLOCKW(  
    x )
```

Definition at line 484 of file [declare.h](#).

#### 4.19.2.113 UNRWLOCK

```
#define UNRWLOCK(  
    x )
```

Definition at line 485 of file [declare.h](#).

#### 4.19.2.114 MLOCK

```
#define MLOCK(  
    x )
```

Definition at line 490 of file [declare.h](#).

#### 4.19.2.115 MUNLOCK

```
#define MUNLOCK(  
    x )
```

Definition at line 491 of file [declare.h](#).

#### 4.19.2.116 GETIDENTITY

```
#define GETIDENTITY
```

Definition at line 493 of file [declare.h](#).

#### 4.19.2.117 GETBIDENTITY

```
#define GETBIDENTITY
```

Definition at line 494 of file [declare.h](#).

#### 4.19.2.118 M\_alloc

```
#define M_alloc(  
    x ) malloc((size_t)(x))
```

Definition at line 1006 of file [declare.h](#).

#### 4.19.2.119 CompareTerms

```
#define CompareTerms ((COMPARE)AR.CompareRoutine)
```

Definition at line 1467 of file [declare.h](#).

#### 4.19.2.120 FiniShuffle

```
#define FiniShuffle AN.SHvar.finishuf
```

Definition at line 1468 of file [declare.h](#).

#### 4.19.2.121 DoShtuffle

```
#define DoShtuffle ((DO_UFFLE)AN.SHvar.do_uffle)
```

Definition at line 1469 of file [declare.h](#).

### 4.19.3 Typedef Documentation

#### 4.19.3.1 WRITEBUFTOEXTCHANNEL

```
typedef int(* WRITEBUFTOEXTCHANNEL) (char *, size_t)
```

Definition at line 1460 of file [declare.h](#).

#### 4.19.3.2 GETCFROMEXTCHANNEL

```
typedef int(* GETCFROMEXTCHANNEL) (void)
```

Definition at line 1461 of file [declare.h](#).

#### 4.19.3.3 SETTERMINATORFOREXTERNALCHANNEL

```
typedef int (* SETTERMINATORFOREXTERNALCHANNEL) (char *)
```

Definition at line 1462 of file [declare.h](#).

#### 4.19.3.4 SETKILLMODEFOREXTERNALCHANNEL

```
typedef int (* SETKILLMODEFOREXTERNALCHANNEL) (int, int)
```

Definition at line 1463 of file [declare.h](#).

#### 4.19.3.5 WRITEFILE

```
typedef LONG (* WRITEFILE) (int, UBYTE *, LONG)
```

Definition at line 1464 of file [declare.h](#).

#### 4.19.3.6 GETTERM

```
typedef WORD (* GETTERM) (PHEAD WORD *)
```

Definition at line 1465 of file [declare.h](#).

### 4.19.4 Function Documentation

#### 4.19.4.1 TELLFILE()

```
void TELLFILE (  
    int handle,  
    POSITION * position ) [extern]
```

Definition at line 1406 of file [tools.c](#).

#### 4.19.4.2 StartVariables()

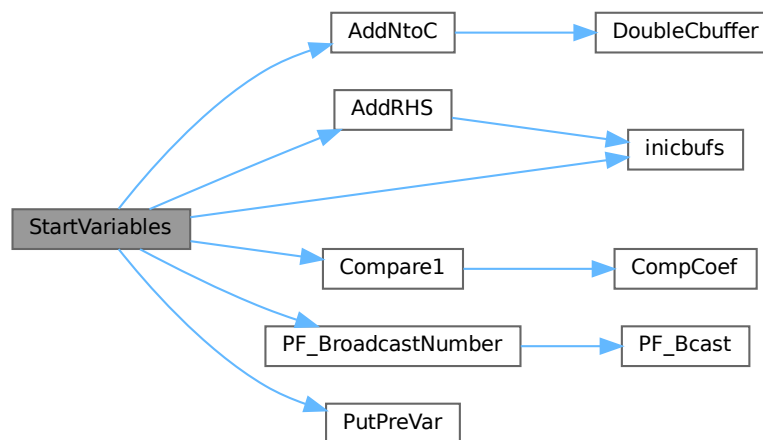
```
void StartVariables (
    void ) [extern]
```

All functions (well, nearly all) are declared here.

Definition at line 905 of file [startup.c](#).

References [AddNtoC\(\)](#), [AddRHS\(\)](#), [Compare1\(\)](#), [inicbufs\(\)](#), [FuNcTiOn::name](#), [PF\\_BroadcastNumber\(\)](#), [PutPreVar\(\)](#), and [FuNcTiOn::symmetric](#).

Here is the call graph for this function:



#### 4.19.4.3 CodeToLine()

```
UBYTE * CodeToLine (
    WORD number,
    UBYTE * Out ) [extern]
```

Definition at line 658 of file [sch.c](#).

#### 4.19.4.4 AddArrayIndex()

```
UBYTE * AddArrayIndex (
    WORD num,
    UBYTE * out ) [extern]
```

Definition at line 696 of file [sch.c](#).



#### 4.19.4.5 FindInIndex()

```
INDEXENTRY * FindInIndex (
    WORD expr,
    FILEDATA * f,
    WORD par,
    WORD mode ) [extern]
```

Definition at line 2664 of file [store.c](#).

#### 4.19.4.6 NextFileIndex()

```
INDEXENTRY * NextFileIndex (
    POSITION * indexpos ) [extern]
```

Definition at line 2257 of file [store.c](#).

#### 4.19.4.7 PasteTerm()

```
WORD * PasteTerm (
    PHEAD WORD number,
    WORD * accum,
    WORD * position,
    WORD times,
    WORD divby ) [extern]
```

Puts the term at position in the accumulator *accum* at position '*number*+1'. if *times* > 0 the coefficient of this term is multiplied by *times*/*divby*.

##### Parameters

|                 |                                                                                   |
|-----------------|-----------------------------------------------------------------------------------|
| <i>number</i>   | The number of term fragments in <i>accum</i> that should be skipped               |
| <i>accum</i>    | The accumulator of term fragments                                                 |
| <i>position</i> | A position in (typically) a compiler buffer from where a (piece of a) term comes. |
| <i>times</i>    | Multiply the result by this                                                       |
| <i>divby</i>    | Divide the result by this.                                                        |

This routine is typically used when we have to replace a (sub)expression pointer by a power of a (sub)expression. This uses mostly a binomial expansion and the new term is the old term multiplied one by one by terms of the new expression. The factors *times* and *divby* keep track of the binomial coefficient. Once this is complete, the routine *FiniTerm* will make the contents of the accumulator into a proper term that still needs to be normalized.

Definition at line 3003 of file [proces.c](#).

Referenced by [Generator\(\)](#).

#### 4.19.4.8 StrCopy()

```
UBYTE * StrCopy (
    UBYTE * from,
    UBYTE * to ) [extern]
```

Definition at line 67 of file [sch.c](#).

#### 4.19.4.9 WrtPower()

```

UBYTE * WrtPower (
    UBYTE * Out,
    WORD Power ) [extern]

```

Definition at line [738](#) of file [sch.c](#).

#### 4.19.4.10 AccumGCD()

```

int AccumGCD (
    PHEAD UWORD * a,
    WORD * na,
    UWORD * b,
    WORD nb ) [extern]

```

Definition at line [648](#) of file [reken.c](#).

#### 4.19.4.11 AddArgs()

```

void AddArgs (
    PHEAD WORD * s1,
    WORD * s2,
    WORD * m ) [extern]

```

Adds the arguments of two occurrences of the PolyFun.

##### Parameters

|           |                                        |
|-----------|----------------------------------------|
| <i>s1</i> | Pointer to the first occurrence.       |
| <i>s2</i> | Pointer to the second occurrence.      |
| <i>m</i>  | Pointer to where the answer should be. |

Definition at line [2327](#) of file [sort.c](#).

Referenced by [AddPoly\(\)](#), and [MergePatches\(\)](#).

#### 4.19.4.12 AddCoef()

```

int AddCoef (
    PHEAD WORD ** ps1,
    WORD ** ps2 ) [extern]

```

Adds the coefficients of the terms \*ps1 and \*ps2. The problem comes when there is not enough space for a new longer coefficient. First a local solution is tried. If this is not successful we need to move terms around. The possibility of a garbage collection should not be ignored, as avoiding this costs very much extra space which is nearly wasted otherwise.

If the return value is zero the terms cancelled.

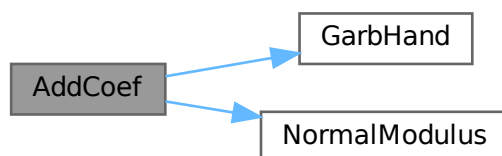
The resulting term is left in \*ps1.

Definition at line 2026 of file [sort.c](#).

References [GarbHand\(\)](#), and [NormalModulus\(\)](#).

Referenced by [SplitMerge\(\)](#).

Here is the call graph for this function:



#### 4.19.4.13 AddLong()

```
int AddLong (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc ) [extern]
```

Definition at line 706 of file [reken.c](#).

#### 4.19.4.14 AddPLon()

```
int AddPLon (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc ) [extern]
```

Definition at line 751 of file [reken.c](#).

#### 4.19.4.15 AddPoly()

```
int AddPoly (
    PHEAD WORD ** ps1,
    WORD ** ps2 ) [extern]
```

Routine should be called when  $S \rightarrow \text{PolyWise} \neq 0$ . It points then to the position of  $AR.PolyFun$  in both terms.

We add the contents of the arguments of the two polynomials. Special attention has to be given to special arguments. We have to reserve a space equal to the size of one term + the size of the argument of the other. The addition has to be done in this routine because not all objects are reentrant.

Newer addition (12-nov-2007). The PolyFun can have two arguments. In that case  $S \rightarrow \text{PolyFlag}$  is 2 and we have to call the routine for adding rational polynomials. We have to be rather careful what happens with: The location of the output The order of the terms in the arguments At first we allow only univariate polynomials in the PolyFun. This restriction will be lifted a.s.a.p.

## Parameters

|            |                                              |
|------------|----------------------------------------------|
| <i>ps1</i> | A pointer to the position of the first term  |
| <i>ps2</i> | A pointer to the position of the second term |

## Returns

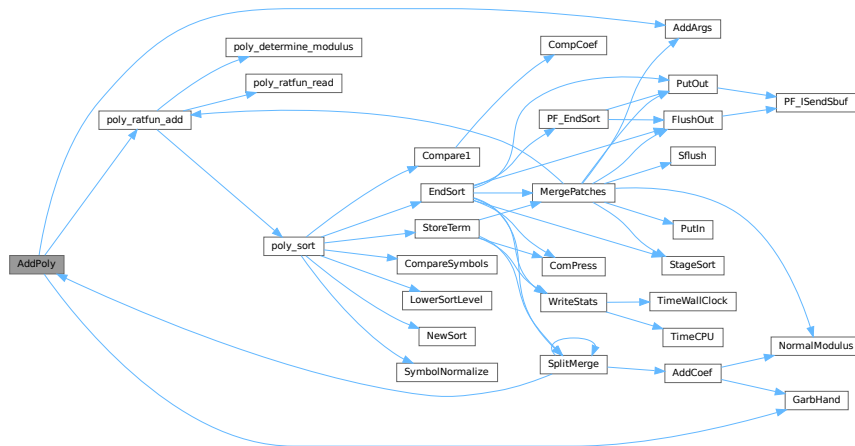
If zero the terms cancel. Otherwise the new term is in \*ps1.

Definition at line 2155 of file [sort.c](#).

References [AddArgs\(\)](#), [GarbHand\(\)](#), and [poly\\_ratfun\\_add\(\)](#).

Referenced by [SplitMerge\(\)](#).

Here is the call graph for this function:



## 4.19.4.16 AddRat()

```
int AddRat (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc ) [extern]
```

Definition at line 210 of file [reken.c](#).

## 4.19.4.17 AddToLine()

```
void AddToLine (
    UBYTE * s ) [extern]
```

Definition at line 82 of file [sch.c](#).

#### 4.19.4.18 AddWild()

```
int AddWild (
    PHEAD WORD,
    WORD type,
    WORD newnumber ) [extern]
```

Definition at line 1544 of file [wildcard.c](#).

#### 4.19.4.19 BigLong()

```
int BigLong (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb ) [extern]
```

Definition at line 945 of file [reken.c](#).

#### 4.19.4.20 BinomGen()

```
int BinomGen (
    PHEAD WORD * term,
    WORD level,
    WORD ** tstops,
    WORD x1,
    WORD x2,
    WORD pow1,
    WORD pow2,
    WORD sign,
    UWORD * coef,
    WORD ncoef ) [extern]
```

Definition at line 320 of file [ratio.c](#).

#### 4.19.4.21 CheckWild()

```
int CheckWild (
    PHEAD WORD,
    WORD type,
    WORD newnumber,
    WORD * newval ) [extern]
```

Definition at line 1791 of file [wildcard.c](#).

#### 4.19.4.22 Chisholm()

```
int Chisholm (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 1966 of file [opera.c](#).

#### 4.19.4.23 CleanExpr()

```
int CleanExpr (
    WORD par ) [extern]
```

Definition at line 47 of file [execute.c](#).

#### 4.19.4.24 CleanUp()

```
void CleanUp (
    WORD par ) [extern]
```

Definition at line 1760 of file [startup.c](#).

#### 4.19.4.25 ClearWild()

```
void ClearWild (
    PHEAD0 ) [extern]
```

Definition at line 1518 of file [wildcard.c](#).

#### 4.19.4.26 CompareFunctions()

```
int CompareFunctions (
    WORD * fleft,
    WORD * fright ) [extern]
```

Definition at line 54 of file [normal.c](#).

#### 4.19.4.27 Commute()

```
int Commute (
    WORD * fleft,
    WORD * fright ) [extern]
```

Definition at line 118 of file [normal.c](#).

#### 4.19.4.28 DetCommu()

```
WORD DetCommu (
    WORD * terms ) [extern]
```

Definition at line 4511 of file [normal.c](#).

#### 4.19.4.29 DoesCommu()

```
WORD DoesCommu (
    WORD * term ) [extern]
```

Definition at line 4570 of file [normal.c](#).

#### 4.19.4.30 CompArg()

```
int CompArg (
    WORD * s1,
    WORD * s2 ) [extern]
```

Definition at line 3226 of file [tools.c](#).

#### 4.19.4.31 CompCoef()

```
WORD CompCoef (
    WORD * term1,
    WORD * term2 ) [extern]
```

Routine takes  $a_1 \bmod m_1$  and  $a_2 \bmod m_2$  and returns  $a \bmod m_1*m_2$  with  
 $a \bmod m_1 = a_1$  and  $a \bmod m_2 = a_2$

Chinese remainder:  $a \bmod (m_1*m_2) = q_1*m_1 + a_1$   $a \bmod (m_1*m_2) = q_2*m_2 + a_2$  Compute  $n_1$  such that  $(n_1*m_1) \bmod m_2$  is one  
 Compute  $n_2$  such that  $(n_2*m_2) \bmod m_1$  is one Then  $(a_1*n_2*m_2 + a_2*n_1*m_1) \bmod (m_1*m_2)$  is  $a \bmod (m_1*m_2)$

Definition at line 3048 of file [reken.c](#).

Referenced by [Compare1\(\)](#).

#### 4.19.4.32 CompGroup()

```
int CompGroup (
    PHEAD WORD,
    WORD ** args,
    WORD * a1,
    WORD * a2,
    WORD num ) [extern]
```

Definition at line 373 of file [function.c](#).

#### 4.19.4.33 Compare1()

```
WORD Compare1 (
    PHEAD WORD * term1,
    WORD * term2,
    WORD level ) [extern]
```

Compares two terms. The answer is: 0 equal ( with exception of the coefficient if level == 0. ) >0 term1 comes first.  
 <0 term2 comes first. Some special precautions may be needed to keep the CompCoef routine from generating overflows, although this is very unlikely in subterms. This routine should not return an error condition.

Originally this routine was called Compare. With the treatment of special polynomials with terms that contain only symbols and the need for extreme speed for the polynomial routines we made a special compare routine and now we store the address of the current compare routine in AR.CompareRoutine and have a macro Compare which makes all existing code work properly and we can just replace the routine on a thread by thread basis (each thread has its own AR struct).



## Parameters

|              |                                                 |
|--------------|-------------------------------------------------|
| <i>term1</i> | First input term                                |
| <i>term2</i> | Second input term                               |
| <i>level</i> | The sorting level (may influence on the result) |

## Returns

0 equal ( with exception of the coefficient if level == 0. ) >0 term1 comes first. <0 term2 comes first.

When there are floating point numbers active (float\_ = FLOATFUN) the presence of one or more float\_ functions is returned in AT.SortFloatMode: 0: no float\_ 1: float\_ in term1 only 2: float\_ in term2 only 3: float\_ in both terms

Definition at line 2620 of file [sort.c](#).

References [CompCoef\(\)](#).

Referenced by [InFunction\(\)](#), [poly\\_sort\(\)](#), and [StartVariables\(\)](#).

Here is the call graph for this function:



## 4.19.4.34 CountDo()

```
WORD CountDo (
    WORD * term,
    WORD * instruct ) [extern]
```

Definition at line 552 of file [reshuf.c](#).

## 4.19.4.35 CountFun()

```
WORD CountFun (
    WORD * term,
    WORD * countfun ) [extern]
```

Definition at line 706 of file [reshuf.c](#).

## 4.19.4.36 DimensionSubterm()

```
WORD DimensionSubterm (
    WORD * subterm ) [extern]
```

Definition at line 867 of file [reshuf.c](#).

**4.19.4.37 DimensionTerm()**

```
WORD DimensionTerm (
    WORD * term ) [extern]
```

Definition at line [968](#) of file [reshuf.c](#).

**4.19.4.38 DimensionExpression()**

```
WORD DimensionExpression (
    PHEAD WORD * expr ) [extern]
```

Definition at line [1000](#) of file [reshuf.c](#).

**4.19.4.39 Deferred()**

```
int Deferred (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Picks up the deferred brackets. These are the bracket contents of which we postpone the reading when we use the 'Keep Brackets' statement. These contents are multiplying the terms just before they are sent to the sorting system. Special attention goes to having it thread-safe We have to lock positioning the file and reading it in a thread specific buffer.

**Parameters**

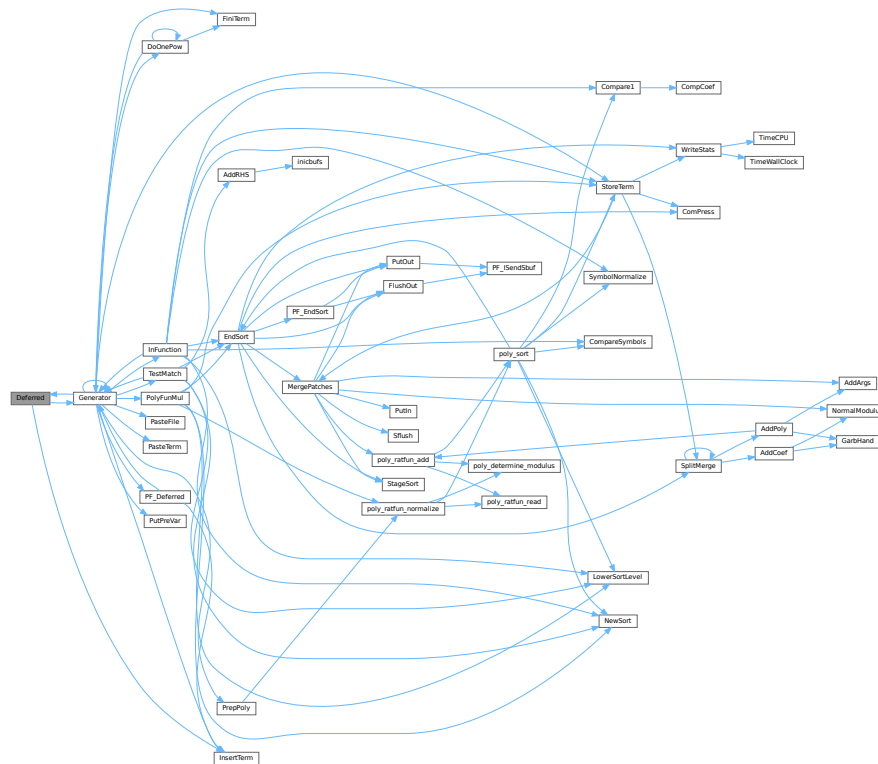
|              |                                                                                                    |
|--------------|----------------------------------------------------------------------------------------------------|
| <i>term</i>  | The term that must be multiplied by the contents of the current bracket                            |
| <i>level</i> | The compiler level. This is needed because after multiplying term by term we call Generator again. |

Definition at line [4846](#) of file [proces.c](#).

References [Generator\(\)](#), and [InsertTerm\(\)](#).

Referenced by [Generator\(\)](#).

Here is the call graph for this function:



#### 4.19.4.40 DeleteStore()

```
int DeleteStore (
    WORD par ) [extern]
```

Definition at line 772 of file [store.c](#).

#### 4.19.4.41 DetCurDum()

```
WORD DetCurDum (
    PHEAD WORD * t ) [extern]
```

Definition at line 252 of file [reshuf.c](#).

#### 4.19.4.42 DetVars()

```
void DetVars (
    WORD * term,
    WORD par ) [extern]
```

Definition at line 1829 of file [store.c](#).

#### 4.19.4.43 Distribute()

```
int Distribute (
    DISTRIBUTE * d,
    WORD first ) [extern]
```

Definition at line 510 of file [symmetr.c](#).

#### 4.19.4.44 DivLong()

```
int DivLong (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc,
    UWORD * d,
    WORD * nd ) [extern]
```

Definition at line 973 of file [reken.c](#).

#### 4.19.4.45 DivRat()

```
int DivRat (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc ) [extern]
```

Definition at line 500 of file [reken.c](#).

#### 4.19.4.46 Divvy()

```
int Divvy (
    PHEAD UWORD * a,
    WORD * na,
    UWORD * b,
    WORD nb ) [extern]
```

Definition at line 172 of file [reken.c](#).

#### 4.19.4.47 DoDelta()

```
int DoDelta (
    WORD * t ) [extern]
```

Definition at line 4356 of file [normal.c](#).

#### 4.19.4.48 DoDelta3()

```
int DoDelta3 (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 1405 of file [reshuf.c](#).

#### 4.19.4.49 TestPartitions()

```
int TestPartitions (
    WORD * tfun,
    PARTI * parti ) [extern]
```

Definition at line 1607 of file [reshuf.c](#).

#### 4.19.4.50 DoPartitions()

```
int DoPartitions (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 1722 of file [reshuf.c](#).

#### 4.19.4.51 CoCanonicalize()

```
int CoCanonicalize (
    UBYTE * s ) [extern]
```

Definition at line 64 of file [diagrams.c](#).

#### 4.19.4.52 DoCanonicalize()

```
int DoCanonicalize (
    PHEAD WORD * term,
    WORD * params ) [extern]
```

Definition at line 185 of file [diagrams.c](#).

#### 4.19.4.53 GenTopologies()

```
int GenTopologies (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 962 of file [diawrap.cc](#).

#### 4.19.4.54 GenDiagrams()

```
int GenDiagrams (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 732 of file [diawrap.cc](#).

#### 4.19.4.55 DoTopologyCanonicalize()

```
int DoTopologyCanonicalize (
    PHEAD WORD * term,
    WORD vert,
    WORD edge,
    WORD * args ) [extern]
```

Definition at line 417 of file [diagrams.c](#).

#### 4.19.4.56 DoShattering()

```
int DoShattering (
    PHEAD WORD * connect,
    WORD * environ,
    WORD * partitions,
    WORD nvert ) [extern]
```

Definition at line 626 of file [diagrams.c](#).

#### 4.19.4.57 GenerateTopologies()

```
int GenerateTopologies (
    PHEAD WORD,
    WORD nlegs,
    WORD setvert,
    WORD setmax ) [extern]
```

Definition at line 123 of file [topowrap.cc](#).

#### 4.19.4.58 DoTableExpansion()

```
int DoTableExpansion (
    WORD * term,
    WORD level ) [extern]
```

Definition at line 334 of file [tables.c](#).

#### 4.19.4.59 DoDistrib()

```
int DoDistrib (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 1119 of file [reshuf.c](#).

#### 4.19.4.60 DoShuffle()

```
int DoShuffle (
    WORD * term,
    WORD level,
    WORD fun,
    WORD option ) [extern]
```

Definition at line 2095 of file [reshuf.c](#).

#### 4.19.4.61 DoPermutations()

```
int DoPermutations (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 2007 of file [reshuf.c](#).

#### 4.19.4.62 Shuffle()

```
int Shuffle (
    WORD * from1,
    WORD * from2,
    WORD * to ) [extern]
```

Definition at line 2229 of file [reshuf.c](#).

#### 4.19.4.63 FinishShuffle()

```
int FinishShuffle (
    WORD * fini ) [extern]
```

Definition at line 2421 of file [reshuf.c](#).

#### 4.19.4.64 DoStuffle()

```
int DoStuffle (
    WORD * term,
    WORD level,
    WORD fun,
    WORD option ) [extern]
```

Definition at line 2487 of file [reshuf.c](#).

#### 4.19.4.65 Stuffle()

```
int Stuffle (
    WORD * from1,
    WORD * from2,
    WORD * to ) [extern]
```

Definition at line 2702 of file [reshuf.c](#).

#### 4.19.4.66 FinishStuffle()

```
int FinishStuffle (
    WORD * fini ) [extern]
```

Definition at line 2829 of file [reshuf.c](#).

#### 4.19.4.67 StuffRootAdd()

```
WORD * StuffRootAdd (
    WORD * t1,
    WORD * t2,
    WORD * to ) [extern]
```

Definition at line 2876 of file [reshuf.c](#).

#### 4.19.4.68 TestUse()

```
int TestUse (
    WORD * term,
    WORD level ) [extern]
```

Definition at line 1414 of file [tables.c](#).

#### 4.19.4.69 FindTB()

```
DBASE * FindTB (
    UBYTE * name ) [extern]
```

Definition at line 734 of file [tables.c](#).

#### 4.19.4.70 CheckTableDeclarations()

```
int CheckTableDeclarations (
    DBASE * d ) [extern]
```

Definition at line 1178 of file [tables.c](#).



#### 4.19.4.71 Apply()

```
void Apply (
    WORD * term,
    WORD level ) [extern]
```

Definition at line 1981 of file [tables.c](#).

#### 4.19.4.72 ApplyExec()

```
int ApplyExec (
    WORD * term,
    int maxtogo,
    WORD level ) [extern]
```

Definition at line 2039 of file [tables.c](#).

#### 4.19.4.73 ApplyReset()

```
void ApplyReset (
    WORD level ) [extern]
```

Definition at line 2256 of file [tables.c](#).

#### 4.19.4.74 TableReset()

```
void TableReset (
    void ) [extern]
```

Definition at line 2291 of file [tables.c](#).

#### 4.19.4.75 ReWorkT()

```
void ReWorkT (
    WORD * term,
    WORD * funs,
    WORD numfuns ) [extern]
```

Definition at line 1900 of file [tables.c](#).

#### 4.19.4.76 GetIfDollarNum()

```
WORD GetIfDollarNum (
    WORD * ifp,
    WORD * ifstop ) [extern]
```

Definition at line 106 of file [if.c](#).

**4.19.4.77 FindVar()**

```
int FindVar (
    WORD * v,
    WORD * term ) [extern]
```

Definition at line 173 of file [if.c](#).

**4.19.4.78 DoIfStatement()**

```
int DoIfStatement (
    PHEAD WORD * ifcode,
    WORD * term ) [extern]
```

Definition at line 280 of file [if.c](#).

**4.19.4.79 DoOnePow()**

```
int DoOnePow (
    PHEAD WORD * term,
    WORD power,
    WORD nexp,
    WORD * accum,
    WORD * aa,
    WORD level,
    WORD * freeze ) [extern]
```

Routine gets one power of an expression in the scratch system. If there are more powers needed there will be a recursion.

No attempt is made to use binomials because we have no information about commuting properties.

There is a searching for the contents of brackets if needed. This searching may be rather slow because of the single links.

**Parameters**

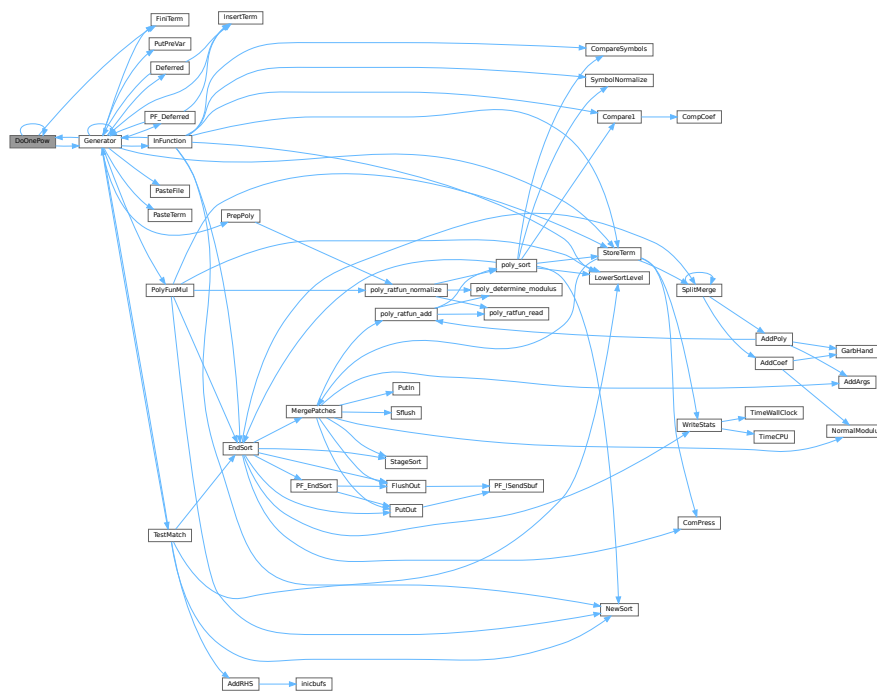
|               |                                                                                                                                      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <i>term</i>   | is the term we are adding to.                                                                                                        |
| <i>power</i>  | is the power of the expression that we need.                                                                                         |
| <i>nexp</i>   | is the number of the expression.                                                                                                     |
| <i>accum</i>  | is the accumulator of terms. It accepts the termfragments that are made into a proper term in FiniTerm                               |
| <i>aa</i>     | points to the start of the entire accumulator. In *aa we store the number of term fragments that are in the accumulator.             |
| <i>level</i>  | is the current depth in the tree of statements. It is needed to continue to the next operation/substitution with each generated term |
| <i>freeze</i> | is the pointer to the bracket information that should be matched.                                                                    |

Definition at line 4625 of file [proces.c](#).

References [DoOnePow\(\)](#), [FiniTerm\(\)](#), [Generator\(\)](#), and [FiLe::handle](#).

Referenced by [DoOnePow\(\)](#), and [Generator\(\)](#).

Here is the call graph for this function:



#### 4.19.4.80 DoRevert()

```
void DoRevert (
    WORD * fun,
    WORD * tmp ) [extern]
```

Definition at line 4423 of file [normal.c](#).

#### 4.19.4.81 DoSumF1()

```
int DoSumF1 (
    PHEAD WORD * term,
    WORD * params,
    WORD * replac,
    WORD level ) [extern]
```

Definition at line 395 of file [ratio.c](#).

#### 4.19.4.82 DoSumF2()

```
int DoSumF2 (
    PHEAD WORD * term,
    WORD * params,
    WORD * replac,
    WORD level ) [extern]
```

Definition at line 520 of file [ratio.c](#).

#### 4.19.4.83 DoTheta()

```
int DoTheta (
    PHEAD WORD * t ) [extern]
```

Definition at line 4259 of file [normal.c](#).

#### 4.19.4.84 EndSort()

```
LONG EndSort (
    PHEAD WORD * buffer,
    int par ) [extern]
```

Finishes a sort. At AR.sLevel == 0 the output is to the regular output stream. When AR.sLevel > 0, the parameter par determines the actual output. The AR.sLevel will be popped. All ongoing stages are finished and if the sortfile is open it is closed. The statistics are printed when AR.sLevel == 0 par == 0 [Output](#) to the buffer. par == 1 Sort for function arguments. The output will be copied into the buffer. It is assumed that this is in the WorkSpace. par == 2 Sort for \$-variable. We return the address of the buffer that contains the output in buffer (treated like WORD \*\*). We first catch the output in a file (unless we can intercept things after the small buffer has been sorted) Then we read from the file into a buffer. Only when par == 0 data compression can be attempted at AT.SS==AT.S0.

##### Parameters

|               |                               |
|---------------|-------------------------------|
| <i>buffer</i> | buffer for output when needed |
| <i>par</i>    | See above                     |

##### Returns

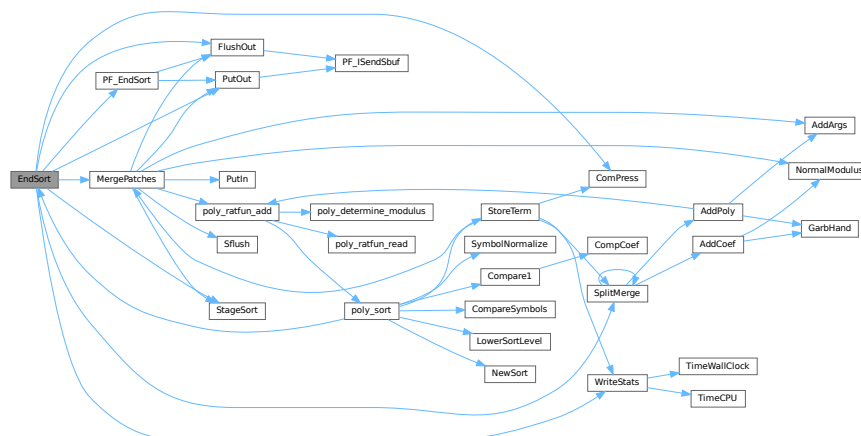
If negative: error. If positive: number of words in output.

Definition at line 733 of file [sort.c](#).

References [ComPress\(\)](#), [FlushOut\(\)](#), [MergePatches\(\)](#), [PF\\_EndSort\(\)](#), [PutOut\(\)](#), [SplitMerge\(\)](#), [StageSort\(\)](#), and [WriteStats\(\)](#).

Referenced by [generate\\_expression\(\)](#), [get\\_expression\(\)](#), [InFunction\(\)](#), [MakeDollarInteger\(\)](#), [MakeDollarMod\(\)](#), [PF\\_CollectModifiedDollars\(\)](#), [PF\\_Processor\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_sort\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [PolyFunMul\(\)](#), [Processor\(\)](#), [TakeArgContent\(\)](#), and [TestMatch\(\)](#).

Here is the call graph for this function:



#### 4.19.4.85 EntVar()

```
int EntVar (
    WORD type,
    UBYTE * name,
    WORD x,
    WORD y,
    WORD z,
    WORD d ) [extern]
```

Definition at line 557 of file [names.c](#).

#### 4.19.4.86 EpfCon()

```
int EpfCon (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level ) [extern]
```

Definition at line 212 of file [opera.c](#).

#### 4.19.4.87 EpfFind()

```
int EpfFind (
    PHEAD WORD * term,
    WORD * params ) [extern]
```

Definition at line 58 of file [opera.c](#).

#### 4.19.4.88 EpfGen()

```
WORD EpfGen (
    WORD number,
    WORD * inlist,
    WORD * kron,
    WORD * perm,
    WORD sgn ) [extern]
```

Definition at line 282 of file [opera.c](#).

#### 4.19.4.89 EqualArg()

```
int EqualArg (
    WORD * parms,
    WORD num1,
    WORD num2 ) [extern]
```

Definition at line 1379 of file [reshuf.c](#).

#### 4.19.4.90 Factorial()

```
int Factorial (
    PHEAD WORD,
    UWORD * a,
    WORD * na ) [extern]
```

Definition at line [3450](#) of file [reken.c](#).

#### 4.19.4.91 Bernoulli()

```
int Bernoulli (
    WORD n,
    UWORD * a,
    WORD * na ) [extern]
```

Definition at line [3530](#) of file [reken.c](#).

#### 4.19.4.92 FactorIn()

```
int FactorIn (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line [46](#) of file [factor.c](#).

#### 4.19.4.93 FactorInExpr()

```
int FactorInExpr (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line [421](#) of file [factor.c](#).

#### 4.19.4.94 FindAll()

```
int FindAll (
    PHEAD WORD * term,
    WORD * pattern,
    WORD level,
    WORD * par ) [extern]
```

Definition at line [1197](#) of file [pattern.c](#).

#### 4.19.4.95 FindMulti()

```
WORD FindMulti (
    PHEAD WORD * term,
    WORD * pattern ) [extern]
```

Definition at line [954](#) of file [findpat.c](#).

#### 4.19.4.96 FindOnce()

```
int FindOnce (
    PHEAD WORD * term,
    WORD * pattern ) [extern]
```

Definition at line 419 of file [findpat.c](#).

#### 4.19.4.97 FindOnly()

```
int FindOnly (
    PHEAD WORD * term,
    WORD * pattern ) [extern]
```

Definition at line 61 of file [findpat.c](#).

#### 4.19.4.98 FindRest()

```
int FindRest (
    PHEAD WORD * term,
    WORD * pattern ) [extern]
```

Definition at line 1070 of file [findpat.c](#).

#### 4.19.4.99 FindrNumber()

```
WORD FindrNumber (
    WORD n,
    VARRENUM * v ) [extern]
```

Definition at line 2587 of file [store.c](#).

#### 4.19.4.100 FiniLine()

```
void FiniLine (
    void ) [extern]
```

Definition at line 182 of file [sch.c](#).

#### 4.19.4.101 FiniTerm()

```
int FiniTerm (
    PHEAD WORD * term,
    WORD * accum,
    WORD * termout,
    WORD number,
    WORD tepos ) [extern]
```

Concatenates the contents of the accumulator into a single legal term, which replaces the subexpression pointer

## Parameters

|                |                                                    |
|----------------|----------------------------------------------------|
| <i>term</i>    | the input term with the (sub)expression subterm    |
| <i>accum</i>   | the accumulator with the term fragments            |
| <i>termout</i> | the location where the output should be written    |
| <i>number</i>  | the number of term fragments in the accumulator    |
| <i>tepos</i>   | the position of the subterm in term to be replaced |

Definition at line 3068 of file [proces.c](#).

Referenced by [DoOnePow\(\)](#), and [Generator\(\)](#).

**4.19.4.102 FlushOut()**

```
int FlushOut (
    POSITION * position,
    FILEHANDLE * fi,
    int compr ) [extern]
```

Completes output to an output file and writes the trailing zero.

## Parameters

|                 |                                                          |
|-----------------|----------------------------------------------------------|
| <i>position</i> | The position in the file after writing                   |
| <i>fi</i>       | The file (or its cache)                                  |
| <i>compr</i>    | Indicates whether there should be compression with gzip. |

## Returns

Regular conventions (OK -> 0).

Definition at line 1812 of file [sort.c](#).

References [FiLe::handle](#), [BrAcKeTiNfO::indexbuffer](#), and [PF\\_ISendSbuf\(\)](#).

Referenced by [EndSort\(\)](#), [MergePatches\(\)](#), [PF\\_EndSort\(\)](#), and [Processor\(\)](#).

Here is the call graph for this function:





#### 4.19.4.103 FunLevel()

```
void FunLevel (
    PHEAD WORD * term ) [extern]
```

Definition at line 130 of file [reshuf.c](#).

#### 4.19.4.104 AdjustRenumScratch()

```
void AdjustRenumScratch (
    PHEAD0 ) [extern]
```

Definition at line 506 of file [reshuf.c](#).

#### 4.19.4.105 GarbHand()

```
void GarbHand (
    void ) [extern]
```

Garbage collection that takes place when the small extension is full and we need to place more terms there. When this is the case there are many holes in the small buffer and the whole can be compactified. The major complication is the buffer for SplitMerge. There are two options for temporary memory: 1: find some buffer that has enough space (maybe in the large buffer). 2: allocate a buffer. Give it back afterwards of course. If the small extension is properly dimensioned this routine should be called very rarely. Most of the time it will be called when the polyfun or polyratfun is active.

Definition at line 3626 of file [sort.c](#).

Referenced by [AddCoef\(\)](#), and [AddPoly\(\)](#).

#### 4.19.4.106 GcdLong()

```
int GcdLong (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc ) [extern]
```

Definition at line 2277 of file [reken.c](#).

#### 4.19.4.107 LcmLong()

```
int LcmLong (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc ) [extern]
```

Definition at line 2702 of file [reken.c](#).

#### 4.19.4.108 Generator()

```
int Generator (
    PHEAD WORD * term,
    WORD level ) [extern]
```

The heart of the program. Here the expansion tree is set up in one giant recursion

## Parameters

|              |                                                        |
|--------------|--------------------------------------------------------|
| <i>term</i>  | the input term. may be overwritten                     |
| <i>level</i> | the level in the compiler buffer (number of statement) |

## Returns

Normal conventions (OK = 0).

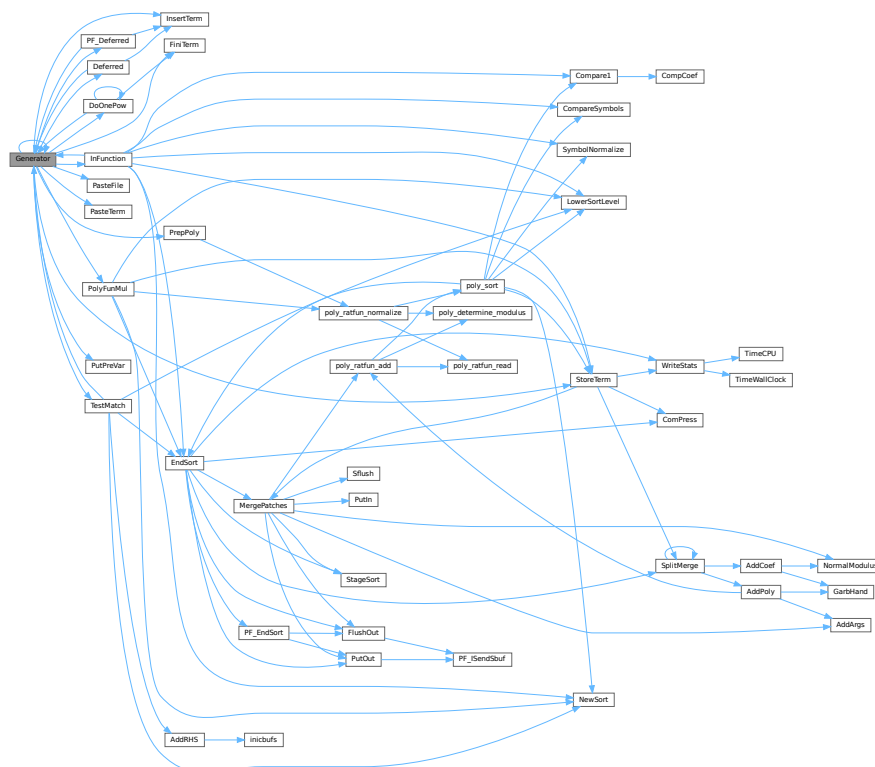
The routine looks first whether there are unsubstituted (sub)expressions. If so, one of them gets inserted term by term and the new term is used in a renewed call to Generator. If there are no (sub)expressions, the term is normalized, the compiler level is raised (next statement) and the program looks what type of statement this is. If this is a special statement it is either treated on the spot or the appropriate routine is called. If it is a substitution, the pattern matcher is called (TestMatch) which tells whether there was a match. If so we need to call TestSub again to test for (sub)expressions. If we run out of levels, the term receives a final treatment for modulus calculus and/or brackets and is then sent off to the sorting routines.

Definition at line 3267 of file [proces.c](#).

References [Deferred\(\)](#), [DoOnePow\(\)](#), [FiniTerm\(\)](#), [Generator\(\)](#), [InFunction\(\)](#), [InsertTerm\(\)](#), [CbUf::lhs](#), [VaRrEnUm::lo](#), [PasteFile\(\)](#), [PasteTerm\(\)](#), [PF\\_Deferred\(\)](#), [CbUf::Pointer](#), [PolyFunMul\(\)](#), [PrepPoly\(\)](#), [PutPreVar\(\)](#), [CbUf::rhs](#), [StoreTerm\(\)](#), [ReNuMbEr::symb](#), and [TestMatch\(\)](#).

Referenced by [Deferred\(\)](#), [DoOnePow\(\)](#), [generate\\_expression\(\)](#), [Generator\(\)](#), [InFunction\(\)](#), [MakeDollarInteger\(\)](#), [MakeDollarMod\(\)](#), [PF\\_CollectModifiedDollars\(\)](#), [PF\\_Deferred\(\)](#), [PF\\_Processor\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [Processor\(\)](#), and [TestMatch\(\)](#).

Here is the call graph for this function:



#### 4.19.4.109 GetBinom()

```
int GetBinom (
    UWORD * a,
    WORD * na,
    WORD i1,
    WORD i2 ) [extern]
```

Definition at line 2668 of file [reken.c](#).

#### 4.19.4.110 GetFromStore()

```
WORD GetFromStore (
    WORD * to,
    POSITION * position,
    RENUMBER renumber,
    WORD * InCompState,
    WORD nexpr ) [extern]
```

Definition at line 1628 of file [store.c](#).

#### 4.19.4.111 GetLong()

```
int GetLong (
    UBYTE * s,
    UWORD * a,
    WORD * na ) [extern]
```

Definition at line 1775 of file [reken.c](#).

#### 4.19.4.112 GetMoreTerms()

```
WORD GetMoreTerms (
    WORD * term ) [extern]
```

Definition at line 1425 of file [store.c](#).

#### 4.19.4.113 GetMoreFromMem()

```
int GetMoreFromMem (
    WORD * term,
    WORD ** tpoin ) [extern]
```

Definition at line 1521 of file [store.c](#).

#### 4.19.4.114 GetOneTerm()

```
WORD GetOneTerm (
    PHEAD WORD * term,
    FILEHANDLE * fi,
    POSITION * pos,
    int par ) [extern]
```

Definition at line 1267 of file [store.c](#).

#### 4.19.4.115 GetTable()

```
RENUMBER GetTable (
    WORD expr,
    POSITION * position,
    WORD mode ) [extern]
```

Definition at line 2825 of file [store.c](#).

#### 4.19.4.116 GetTerm()

```
WORD GetTerm (
    PHEAD WORD * term ) [extern]
```

Definition at line 992 of file [store.c](#).

#### 4.19.4.117 Glue()

```
int Glue (
    PHEAD WORD * term1,
    WORD * term2,
    WORD * sub,
    WORD insert ) [extern]
```

Definition at line 461 of file [ratio.c](#).

#### 4.19.4.118 InFunction()

```
int InFunction (
    PHEAD WORD * term,
    WORD * termout ) [extern]
```

Makes the replacement of the subexpression with the number 'replac' in a function argument. Additional information is passed in some of the AR, AN, AT variables.

##### Parameters

|                |                 |
|----------------|-----------------|
| <i>term</i>    | The input term  |
| <i>termout</i> | The output term |

## Returns

0: everything is fine, Negative: fatal, Positive: error.

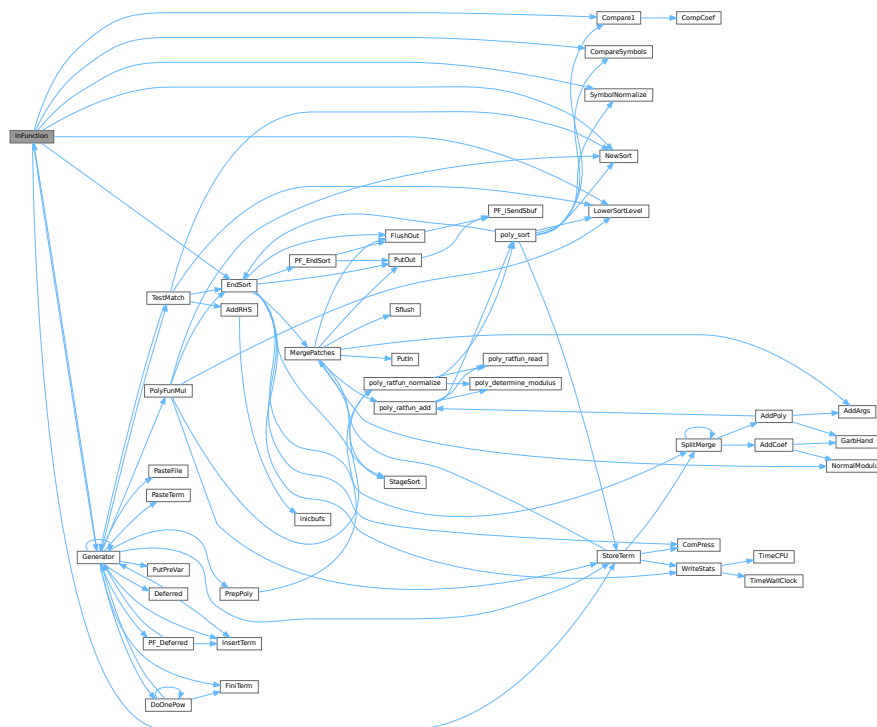
Special attention should be given to nested functions!

Definition at line 2175 of file [proces.c](#).

References [Compare1\(\)](#), [CompareSymbols\(\)](#), [EndSort\(\)](#), [Generator\(\)](#), [LowerSortLevel\(\)](#), [NewSort\(\)](#), [StoreTerm\(\)](#), and [SymbolNormalize\(\)](#).

Referenced by [Generator\(\)](#).

Here is the call graph for this function:



### 4.19.4.119 IniLine()

```
void IniLine (
    WORD extrablank ) [extern]
```

Definition at line 267 of file [sch.c](#).

### 4.19.4.120 IniVars()

```
void IniVars (
    void ) [extern]
```

Definition at line 1354 of file [startup.c](#).

#### 4.19.4.121 InsertTerm()

```
int InsertTerm (
    PHEAD WORD * term,
    WORD replac,
    WORD extractbuff,
    WORD * position,
    WORD * termout,
    WORD tepos ) [extern]
```

Puts the terms 'term' and 'position' together into a single legal term in termout. replac is the number of the subexpression that should be replaced. It must be a positive term. When action is needed in the argument of a function all terms in that argument are dealt with recursively. The subexpression is sorted. Only one subexpression is done at a time this way.

##### Parameters

|                    |                                                             |
|--------------------|-------------------------------------------------------------|
| <i>term</i>        | the input term                                              |
| <i>replac</i>      | number of the subexpression pointer to replace              |
| <i>extractbuff</i> | number of the compiler buffer replac refers to              |
| <i>position</i>    | position from where to take the term in the compiler buffer |
| <i>termout</i>     | the output term                                             |
| <i>tepos</i>       | offset in term where the subexpression is.                  |

##### Returns

Normal conventions (OK = 0).

Definition at line 2745 of file [proces.c](#).

Referenced by [Deferred\(\)](#), [Generator\(\)](#), and [PF\\_Deferred\(\)](#).

#### 4.19.4.122 LongToLine()

```
void LongToLine (
    UWORD * a,
    WORD na ) [extern]
```

Definition at line 303 of file [sch.c](#).

#### 4.19.4.123 MakeDirty()

```
int MakeDirty (
    WORD * term,
    WORD * x,
    WORD par ) [extern]
```

Definition at line 51 of file [function.c](#).

#### 4.19.4.124 MarkDirty()

```
void MarkDirty (
    WORD * term,
    WORD flags ) [extern]
```

Definition at line 97 of file [function.c](#).

#### 4.19.4.125 PolyFunDirty()

```
void PolyFunDirty (
    PHEAD WORD * term ) [extern]
```

Definition at line 139 of file [function.c](#).

#### 4.19.4.126 PolyFunClean()

```
void PolyFunClean (
    PHEAD WORD * term ) [extern]
```

Definition at line 174 of file [function.c](#).

#### 4.19.4.127 MakeModTable()

```
int MakeModTable (
    void ) [extern]
```

Definition at line 3364 of file [reken.c](#).

#### 4.19.4.128 MatchE()

```
int MatchE (
    PHEAD WORD * pattern,
    WORD * fun,
    WORD * inter,
    WORD par ) [extern]
```

Definition at line 56 of file [symmetr.c](#).

#### 4.19.4.129 MatchCy()

```
int MatchCy (
    PHEAD WORD * pattern,
    WORD * fun,
    WORD * inter,
    WORD par ) [extern]
```

Definition at line 571 of file [symmetr.c](#).



**4.19.4.130 FunMatchCy()**

```
int FunMatchCy (
    PHEAD WORD * pattern,
    WORD * fun,
    WORD * inter,
    WORD par ) [extern]
```

Definition at line 1067 of file [symmetr.c](#).

**4.19.4.131 FunMatchSy()**

```
int FunMatchSy (
    PHEAD WORD * pattern,
    WORD * fun,
    WORD * inter,
    WORD par ) [extern]
```

Definition at line 1525 of file [symmetr.c](#).

**4.19.4.132 MatchArgument()**

```
int MatchArgument (
    PHEAD WORD * arg,
    WORD * pat ) [extern]
```

Definition at line 2052 of file [symmetr.c](#).

**4.19.4.133 MatchFunction()**

```
int MatchFunction (
    PHEAD WORD * pattern,
    WORD * interm,
    WORD * wilds ) [extern]
```

Definition at line 826 of file [function.c](#).

**4.19.4.134 MergePatches()**

```
int MergePatches (
    WORD par ) [extern]
```

The general merge routine. Can be used for the large buffer and the file merging. The array S->Patches tells where the patches start S->pStop tells where they end (has to be computed first). The end of a 'line to be merged' is indicated by a zero. If the end is reached without running into a zero or a term runs over the boundary of a patch it is a file merging operation and a new piece from the file is read in.

## Parameters

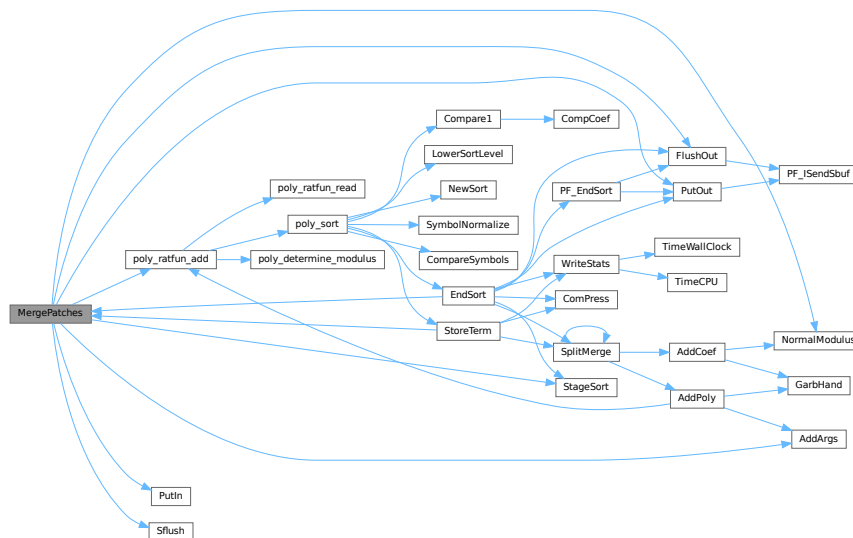
|            |                                                                                                                                                                                                |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>par</i> | If <code>par == 0</code> the sort is for file -> outfile. If <code>par == 1</code> the sort is for large buffer -> sortfile. If <code>par == 2</code> the sort is for large buffer -> outfile. |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Definition at line 3741 of file [sort.c](#).

References [AddArgs\(\)](#), [FlushOut\(\)](#), [FiLe::handle](#), [NormalModulus\(\)](#), [poly\\_ratfun\\_add\(\)](#), [PutIn\(\)](#), [PutOut\(\)](#), [Sflush\(\)](#), and [StageSort\(\)](#).

Referenced by [EndSort\(\)](#), and [StoreTerm\(\)](#).

Here is the call graph for this function:



#### 4.19.4.135 MesCerr()

```
int MesCerr (
    char * s,
    UBYTE * t ) [extern]
```

Definition at line 824 of file [message.c](#).

#### 4.19.4.136 MesComp()

```
int MesComp (
    char * s,
    UBYTE * p,
    UBYTE * q ) [extern]
```

Definition at line 843 of file [message.c](#).

#### 4.19.4.137 Modulus()

```
int Modulus (
    WORD * term ) [extern]
```

Definition at line [3103](#) of file [reken.c](#).

#### 4.19.4.138 MoveDummies()

```
void MoveDummies (
    PHEAD WORD * term,
    WORD shift ) [extern]
```

Definition at line [436](#) of file [reshuf.c](#).

#### 4.19.4.139 MulLong()

```
int MulLong (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc ) [extern]
```

Definition at line [842](#) of file [reken.c](#).

#### 4.19.4.140 MulRat()

```
int MulRat (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc ) [extern]
```

Definition at line [378](#) of file [reken.c](#).

#### 4.19.4.141 Mully()

```
int Mully (
    PHEAD UWORD * a,
    WORD * na,
    UWORD * b,
    WORD nb ) [extern]
```

Definition at line [126](#) of file [reken.c](#).

**4.19.4.142 MultDo()**

```
int MultDo (
    PHEAD WORD * term,
    WORD * pattern ) [extern]
```

Definition at line [1044](#) of file [reshuf.c](#).

**4.19.4.143 NewSort()**

```
int NewSort (
    PHEAD0 ) [extern]
```

Starts a new sort. At the lowest level this is a 'main sort' with the struct according to the parameters in S0. At higher levels this is a sort for functions, subroutines or dollars. We prepare the arrays and structs.

**Returns**

Regular convention (OK -> 0)

Definition at line [643](#) of file [sort.c](#).

Referenced by [generate\\_expression\(\)](#), [get\\_expression\(\)](#), [InFunction\(\)](#), [MakeDollarInteger\(\)](#), [MakeDollarMod\(\)](#), [PF\\_CollectModifiedDollars\(\)](#), [PF\\_Processor\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_sort\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [PolyFunMul\(\)](#), [Processor\(\)](#), [TakeArgContent\(\)](#), and [TestMatch\(\)](#).

**4.19.4.144 ExtraSymbol()**

```
int ExtraSymbol (
    WORD sym,
    WORD pow,
    WORD nsym,
    WORD * ppsym,
    WORD * ncoef ) [extern]
```

Definition at line [4197](#) of file [normal.c](#).

**4.19.4.145 Normalize()**

```
int Normalize (
    PHEAD WORD * term ) [extern]
```

Definition at line [193](#) of file [normal.c](#).

**4.19.4.146 BracketNormalize()**

```
int BracketNormalize (
    PHEAD WORD * term ) [extern]
```

Definition at line [5272](#) of file [normal.c](#).

**4.19.4.147 DropCoefficient()**

```
void DropCoefficient (
    PHEAD WORD * term ) [extern]
```

Definition at line 5097 of file [normal.c](#).

**4.19.4.148 DropSymbols()**

```
void DropSymbols (
    PHEAD WORD * term ) [extern]
```

Definition at line 5115 of file [normal.c](#).

**4.19.4.149 PutInside()**

```
int PutInside (
    PHEAD WORD * term,
    WORD * code ) [extern]
```

Definition at line 609 of file [index.c](#).

**4.19.4.150 OpenTemp()**

```
void OpenTemp (
    void ) [extern]
```

Definition at line 48 of file [store.c](#).

**4.19.4.151 Pack()**

```
void Pack (
    UWORD * a,
    WORD * na,
    UWORD * b,
    WORD nb ) [extern]
```

Definition at line 62 of file [reken.c](#).

**4.19.4.152 PasteFile()**

```
LONG PasteFile (
    PHEAD WORD number,
    WORD * accum,
    POSITION * position,
    WORD ** accfill,
    RENUMBER renumber,
    WORD * freeze,
    WORD nexpr ) [extern]
```

Gets a term from stored expression *expr* and puts it in the accumulator at position number. It returns the length of the term that came from file.

**Parameters**

|                 |                                                                |
|-----------------|----------------------------------------------------------------|
| <i>number</i>   | number of partial terms to skip in accum                       |
| <i>accum</i>    | the accumulator                                                |
| <i>position</i> | file position from where to get the stored term                |
| <i>accfill</i>  | returns tail position in accum                                 |
| <i>renumber</i> | the renumber struct for the variables in the stored expression |
| <i>freeze</i>   | information about if we need only the contents of a bracket    |
| <i>nexpr</i>    | the number of the stored expression                            |

**Returns**

Normal conventions (OK = 0).

Definition at line [2881](#) of file [proces.c](#).

Referenced by [Generator\(\)](#).

**4.19.4.153 Permute()**

```
int Permute (
    PERM * perm,
    WORD first ) [extern]
```

Definition at line [444](#) of file [symmetr.c](#).

**4.19.4.154 PermuteP()**

```
int PermuteP (
    PERMP * perm,
    WORD first ) [extern]
```

Definition at line [478](#) of file [symmetr.c](#).

**4.19.4.155 PolyFunMul()**

```
int PolyFunMul (
    PHEAD WORD * term ) [extern]
```

Multiplies the arguments of multiple occurrences of the polyfun. In this routine we do the original PolyFun with one argument only. The PolyRatFun (PolyFunType = 2) is done in a dedicated routine in the file [polywrap.cc](#) The new result is written over the old result.

**Parameters**

|             |                                                  |
|-------------|--------------------------------------------------|
| <i>term</i> | It contains the input term and later the output. |
|-------------|--------------------------------------------------|

## Returns

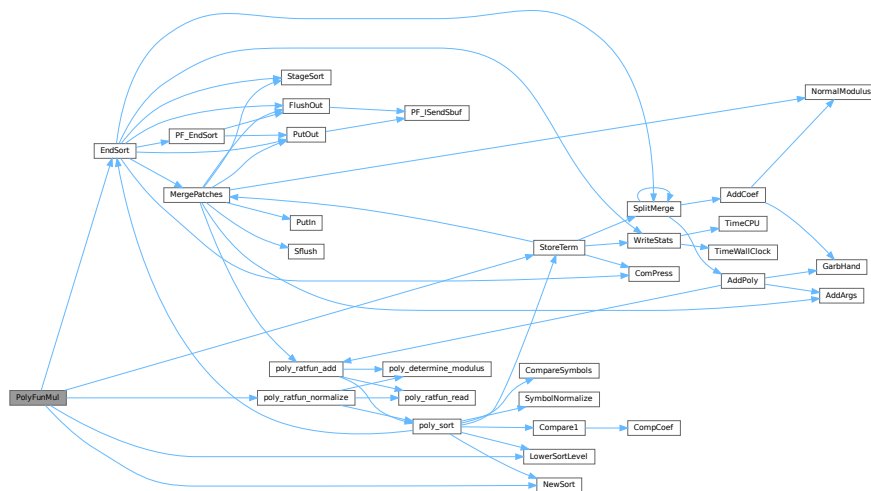
Normal conventions (OK = 0).

Definition at line 5362 of file [proces.c](#).

References [EndSort\(\)](#), [LowerSortLevel\(\)](#), [NewSort\(\)](#), [poly\\_ratfun\\_normalize\(\)](#), and [StoreTerm\(\)](#).

Referenced by [Generator\(\)](#).

Here is the call graph for this function:



## 4.19.4.156 PopVariables()

```
int PopVariables (
    void ) [extern]
```

Definition at line 211 of file [execute.c](#).

## 4.19.4.157 PrepPoly()

```
int PrepPoly (
    PHEAD WORD * term,
    WORD par ) [extern]
```

Routine checks whether the count of function AR.PolyFun is zero or one. If it is one and it has one scalarlike argument the coefficient of the term is pulled inside the argument. If the count is zero a new function is made with the coefficient as its only argument. The function should be placed at its proper position.

When this function is active it places the PolyFun as last object before the coefficient. This is needed because otherwise the compress algorithm has problems in MergePatches.

The bracket routine should also place the PolyFun at a comparable spot. The compression should then stop at the PolyFun. It doesn't really have to stop when writing the final result but this may be too complicated.

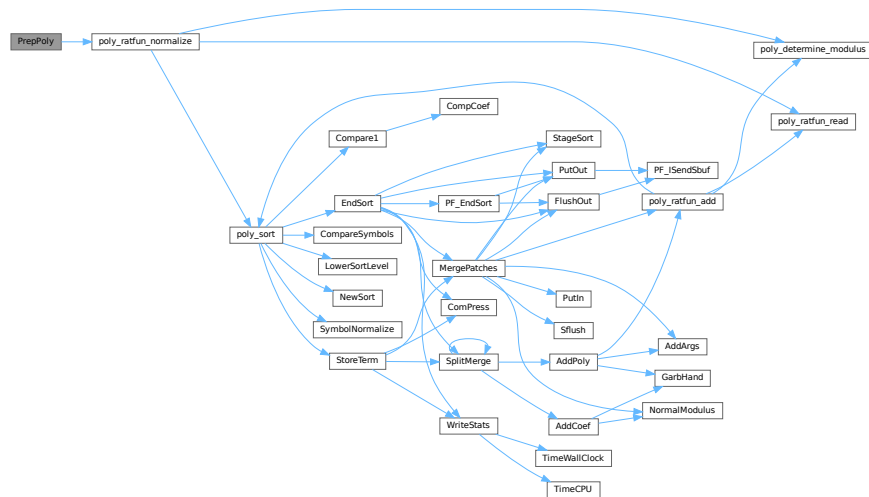
The parameter `par` tells whether we are at groundlevel or inside a function or dollar variable.

Definition at line 4974 of file `proces.c`.

References `poly_ratfun_normalize()`.

Referenced by `Generator()`.

Here is the call graph for this function:



#### 4.19.4.158 Processor()

```
int Processor (
    void ) [extern]
```

This is the central processor. It accepts a stream of Expressions which is accessed by calls to `GetTerm`. The expressions reside either in `AR.infile` or `AR.hidefile`. The definitions of an expression are seen as an id-statement, so the primary Expressions should be written to the system of scratch files as single terms with an expression pointer. Each expression is terminated with a zero and the whole is terminated by two zeroes.

The routine `DoExecute` should determine whether results are to be printed, should revert the scratch I/O directions etc. In principle it is `DoExecute` that calls `Processor`.

#### Returns

if everything OK: 0. Otherwise error. The preprocessor may continue with compilation though. Really fatal errors should return on the spot by calling `Terminate`.

Definition at line 64 of file `proces.c`.

References `CbUf::Buffer`, `EndSort()`, `FlushOut()`, `Generator()`, `CbUf::lhs`, `LowerSortLevel()`, `NewSort()`, `PF_BroadcastRHS()`, `PF_InParallelProcessor()`, `PF_Processor()`, `CbUf::Pointer`, `poly_factorize_expression()`, `poly_unfactorize_expression()`, `PutOut()`, `CbUf::rhs`, and `StoreTerm()`.



```
int Product (
    UWORD * a,
    WORD * na,
    WORD b ) [extern]
```

#### 4.19.4.160 PrtLong()

Definition at line 1713 of file reken.c.

**4.19.4.161 PrtTerms()**

```
void PrtTerms (
    void ) [extern]
```

Definition at line 716 of file [sch.c](#).

**4.19.4.162 PrintDeprecation()**

```
void PrintDeprecation (
    const char * feature,
    const char * issue ) [extern]
```

Prints a deprecation warning for a given feature.

**Parameters**

|                |                                           |
|----------------|-------------------------------------------|
| <i>feature</i> | The name of the deprecated feature.       |
| <i>issue</i>   | The associated issue, e.g., "issues/700". |

Definition at line 2062 of file [startup.c](#).

**4.19.4.163 PrintRunningTime()**

```
void PrintRunningTime (
    void ) [extern]
```

Definition at line 2086 of file [startup.c](#).

**4.19.4.164 GetRunningTime()**

```
LONG GetRunningTime (
    void ) [extern]
```

Definition at line 2127 of file [startup.c](#).

**4.19.4.165 PutBracket()**

```
int PutBracket (
    PHEAD WORD * termin ) [extern]
```

Definition at line 1112 of file [execute.c](#).

**4.19.4.166 PutIn()**

```
LONG PutIn (
    FILEHANDLE * file,
    POSITION * position,
    WORD * buffer,
    WORD ** take,
    int npat ) [extern]
```

Reads a new patch from position in file handle. It is put at buffer, anything after take is moved forward. This would be part of a term that hasn't been used yet. Because of this there should be some space before the start of the buffer

## Parameters

|                 |                                                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <i>file</i>     | The file system from which to read                                                                                                        |
| <i>position</i> | The position from which to read                                                                                                           |
| <i>buffer</i>   | The buffer into which to read                                                                                                             |
| <i>take</i>     | The unused tail should be moved before the buffer                                                                                         |
| <i>npat</i>     | The number of the patch. Is needed if the information was compressed with gzip, because each patch has its own independent gzip encoding. |

Definition at line 1304 of file [sort.c](#).

References [FiLe::handle](#).

Referenced by [MergePatches\(\)](#).

#### 4.19.4.167 PutInStore()

```
int PutInStore (
    INDEXENTRY * ind,
    WORD num ) [extern]
```

Definition at line 873 of file [store.c](#).

#### 4.19.4.168 PutOut()

```
WORD PutOut (
    PHEAD WORD * term,
    POSITION * position,
    FILEHANDLE * fi,
    WORD ncomp ) [extern]
```

Routine writes one term to file handle at position. It returns the new value of the position.

NOTE: For 'final output' we may have to index the brackets. See the struct BRACKETINDEX. We should maintain:  
 1: a list with brackets array with the brackets  
 2: a list of objects of type BRACKETINDEX. It contains array with either pointers or offsets to the list of brackets. starting positions in the file. The index may be tied to a maximum size. In that case we may have to prune the list occasionally.

## Parameters

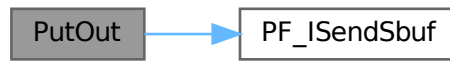
|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| <i>term</i>     | The term to be written                                    |
| <i>position</i> | The position in the file. Afterwards it is updated        |
| <i>fi</i>       | The file (or its cache) to which should be written        |
| <i>ncomp</i>    | Information about what type of compression should be used |

Definition at line 1450 of file [sort.c](#).

References [FiLe::handle](#), and [PF\\_IsSendSbuf\(\)](#).

Referenced by [EndSort\(\)](#), [generate\\_expression\(\)](#), [MergePatches\(\)](#), [PF\\_EndSort\(\)](#), [PF\\_Processor\(\)](#), and [Processor\(\)](#).

Here is the call graph for this function:



#### 4.19.4.169 Quotient()

```
UWORD Quotient (  
    UWORD * a,  
    WORD * na,  
    WORD b ) [extern]
```

Definition at line 1621 of file [reken.c](#).

#### 4.19.4.170 RaisPow()

```
int RaisPow (  
    PHEAD UWORD * a,  
    WORD * na,  
    UWORD b ) [extern]
```

Definition at line 1229 of file [reken.c](#).

#### 4.19.4.171 RaisPowCached()

```
void RaisPowCached (  
    PHEAD WORD x,  
    WORD n,  
    UWORD ** c,  
    WORD * nc ) [extern]
```

Computes power  $x^n$  and caches the value

### 4.19.5 Description

Calculates the power  $x^n$  and stores the results for caching purposes. The pointer c (i.e., the pointer, and not what it points to) is overwritten. What it points to should not be overwritten in the calling function.

### 4.19.6 Notes

- Caching is done in `AT.small_power[]`. This array is extended if necessary.

Definition at line 1297 of file [reken.c](#).

Referenced by [poly::divmod\\_heap\(\)](#), [poly::divmod\\_univar\(\)](#), and [poly::mul\\_heap\(\)](#).

#### 4.19.6.1 RaisPowMod()

```
WORD RaisPowMod (
    WORD x,
    WORD n,
    WORD m ) [extern]
```

Definition at line 1384 of file [reken.c](#).

#### 4.19.6.2 NormalModulus()

```
int NormalModulus (
    UWORD * a,
    WORD * na ) [extern]
```

Brings a modular representation in the range  $-p/2$  to  $+p/2$  The return value tells whether anything was done. Routine made in the general modulus revamp of July 2008 (JV).

Definition at line 1404 of file [reken.c](#).

Referenced by [AddCoef\(\)](#), and [MergePatches\(\)](#).

#### 4.19.6.3 MakeInverses()

```
int MakeInverses (
    void ) [extern]
```

Makes a table of inverses in modular calculus The modulus is in `AC.cmod` and `AC.ncmod` One should notice that the table of inverses can only be made if the modulus fits inside a single FORM word. Otherwise the table lookup becomes too difficult and the table too long.

Definition at line 1441 of file [reken.c](#).

References [GetModInverses\(\)](#).

Here is the call graph for this function:



#### 4.19.6.4 GetModInverses()

```
int GetModInverses (
    WORD m1,
    WORD m2,
    WORD * im1,
    WORD * im2 ) [extern]
```

Input m1 and m2, which are relative prime. determines  $a*m1+b*m2 = 1$  (and 1 is the gcd of m1 and m2) then  $a*m1 = 1 \bmod m2$  and hence  $im1 = a$ . and  $b*m2 = 1 \bmod m1$  and hence  $im2 = b$ . Set  $m1 = 0*m1+1*m2 = a1*m1+b1*m2$   
 $m2 = 1*m1+0*m2 = a2*m1+b2*m2$  If everything is OK, the return value is zero

Definition at line 1477 of file [reken.c](#).

Referenced by [poly::divmod\\_heap\(\)](#), [poly::divmod\\_univar\(\)](#), [MakeDollarMod\(\)](#), [MakeInverses\(\)](#), [MakeMod\(\)](#), [TakeContent\(\)](#), and [TakeSymbolContent\(\)](#).

#### 4.19.6.5 GetLongModInverses()

```
int GetLongModInverses (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * ia,
    WORD * nia,
    UWORD * ib,
    WORD * nib ) [extern]
```

Definition at line 1509 of file [reken.c](#).

#### 4.19.6.6 RatToLine()

```
void RatToLine (
    UWORD * a,
    WORD na ) [extern]
```

Definition at line 337 of file [sch.c](#).

#### 4.19.6.7 RatioFind()

```
int RatioFind (
    PHEAD WORD * term,
    WORD * params ) [extern]
```

Definition at line 62 of file [ratio.c](#).

#### 4.19.6.8 RatioGen()

```
int RatioGen (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level ) [extern]
```

Definition at line 170 of file [ratio.c](#).

#### 4.19.6.9 ReNumber()

```
WORD ReNumber (
    PHEAD WORD * term ) [extern]
```

Definition at line 80 of file [reshuf.c](#).

#### 4.19.6.10 ReadSnum()

```
WORD ReadSnum (
    UBYTE ** p ) [extern]
```

Definition at line 1973 of file [tools.c](#).

#### 4.19.6.11 Remain10()

```
WORD Remain10 (
    UWORD * a,
    WORD * na ) [extern]
```

Definition at line 1660 of file [reken.c](#).

#### 4.19.6.12 Remain4()

```
WORD Remain4 (
    UWORD * a,
    WORD * na ) [extern]
```

Definition at line 1687 of file [reken.c](#).

#### 4.19.6.13 ResetScratch()

```
int ResetScratch (
    void ) [extern]
```

Definition at line 234 of file [store.c](#).

#### 4.19.6.14 ResolveSet()

```
int ResolveSet (
    PHEAD WORD * from,
    WORD * to,
    WORD * subs ) [extern]
```

Definition at line [1361](#) of file [wildcard.c](#).

#### 4.19.6.15 RevertScratch()

```
int RevertScratch (
    void ) [extern]
```

Definition at line [189](#) of file [store.c](#).

#### 4.19.6.16 ScanFunctions()

```
int ScanFunctions (
    PHEAD WORD * inpat,
    WORD * inter,
    WORD par ) [extern]
```

Definition at line [1616](#) of file [function.c](#).

#### 4.19.6.17 SeekScratch()

```
void SeekScratch (
    FILEHANDLE * fi,
    POSITION * pos ) [extern]
```

Definition at line [63](#) of file [store.c](#).

#### 4.19.6.18 SetEndScratch()

```
void SetEndScratch (
    FILEHANDLE * f,
    POSITION * position ) [extern]
```

Definition at line [74](#) of file [store.c](#).

#### 4.19.6.19 SetEndHScratch()

```
void SetEndHScratch (
    FILEHANDLE * f,
    POSITION * position ) [extern]
```

Definition at line [88](#) of file [store.c](#).



#### 4.19.6.20 SetFileIndex()

```
int SetFileIndex (
    void ) [extern]
```

Reads the next file index and puts it into AR.StoreData.Index. TODO

##### Returns

= 0 everything okay, != 0 an error occurred

Definition at line 2320 of file [store.c](#).

References [WriteStoreHeader\(\)](#).

Here is the call graph for this function:



#### 4.19.6.21 Sflush()

```
int Sflush (
    FILEHANDLE * fi ) [extern]
```

Puts the contents of a buffer to output Only to be used when there is a single patch in the large buffer.

##### Parameters

|           |                                                                    |
|-----------|--------------------------------------------------------------------|
| <i>fi</i> | The filesystem (or its cache) to which the patch should be written |
|-----------|--------------------------------------------------------------------|

Definition at line 1364 of file [sort.c](#).

References [FiLe::handle](#).

Referenced by [MergePatches\(\)](#).

#### 4.19.6.22 Simplify()

```
int Simplify (
    PHEAD UWORD * a,
    WORD * na,
    UWORD * b,
    WORD * nb ) [extern]
```

Definition at line 531 of file [reken.c](#).

#### 4.19.6.23 SortWild()

```
int SortWild (
    WORD * w,
    WORD nw ) [extern]
```

Sorts the wildcard entries in the parameter w. Double entries are removed. Full space taken is nw words. Routine serves for the reading of wildcards in the compiler. The entries come in the format: (type,4,number,0) in which the zero is reserved for the future replacement of 'number'.

##### Parameters

|           |                               |
|-----------|-------------------------------|
| <i>w</i>  | buffer with wildcard entries. |
| <i>nw</i> | number of wildcard entries.   |

##### Returns

Normal conventions (OK -> 0)

Definition at line [4737](#) of file [sort.c](#).

#### 4.19.6.24 LocateBase()

```
FILE * LocateBase (
    char ** name,
    char ** newname,
    char * iomode ) [extern]
```

Definition at line [225](#) of file [minos.c](#).

#### 4.19.6.25 SplitMerge()

```
LONG SplitMerge (
    PHEAD WORD ** Pointer,
    LONG number ) [extern]
```

Algorithm by J.A.M.Vermaseren (31-7-1988)

Note that AN.SplitScratch and AN.InScratch are used also in GarbHand

Merge sort in memory. The input is an array of pointers. Sorting is done recursively by dividing the array in two equal parts and calling SplitMerge for each. When the parts are small enough we can do the compare and take the appropriate action. An addition is that we look for 'runs'. Sequences that are already ordered. This happens a lot when there is very little action in a module. This made FORM faster by a few percent.

##### Parameters

|                |                                                  |
|----------------|--------------------------------------------------|
| <i>Pointer</i> | The array of pointers to the terms to be sorted. |
| <i>number</i>  | The number of pointers in Pointer.               |

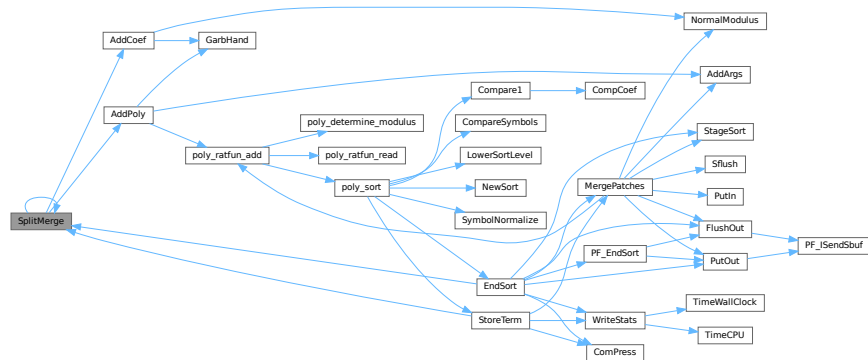
The terms are supposed to be sitting in the small buffer and there is supposed to be an extension to this buffer for when there are two terms that should be added and the result takes more space than each of the original terms. The notation guarantees that the result never needs more space than the sum of the spaces of the original terms.

Definition at line 3346 of file [sort.c](#).

References [AddCoef\(\)](#), [AddPoly\(\)](#), and [SplitMerge\(\)](#).

Referenced by [EndSort\(\)](#), [SplitMerge\(\)](#), and [StoreTerm\(\)](#).

Here is the call graph for this function:



#### 4.19.6.26 StoreTerm()

```
int StoreTerm (
    PHEAD WORD * term ) [extern]
```

The central routine to accept terms, store them and keep things at least partially sorted. A call to EndSort will then complete storing and sorting.

##### Parameters

|             |                       |
|-------------|-----------------------|
| <i>term</i> | The term to be stored |
|-------------|-----------------------|

##### Returns

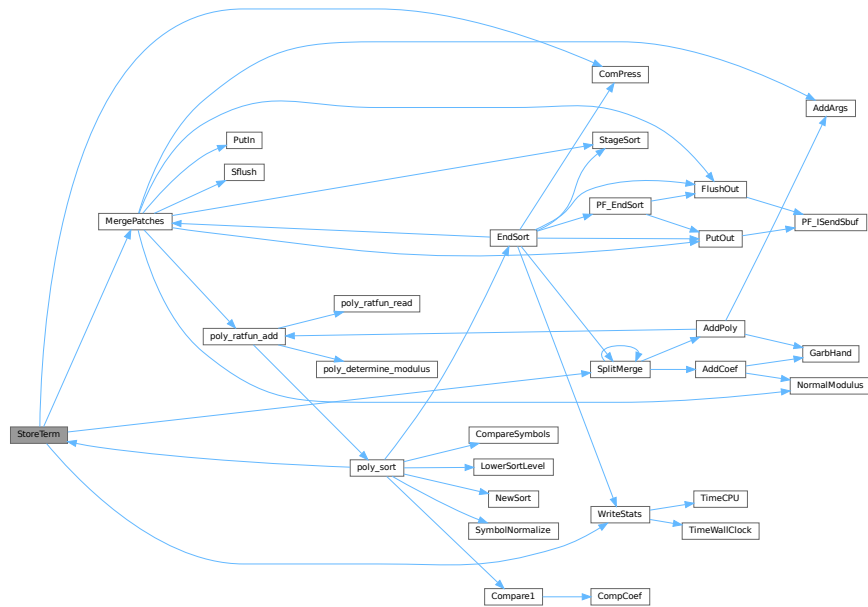
Regular return conventions (OK -> 0)

Definition at line 4524 of file [sort.c](#).

References [ComPress\(\)](#), [MergePatches\(\)](#), [SplitMerge\(\)](#), and [WriteStats\(\)](#).

Referenced by [Generator\(\)](#), [get\\_expression\(\)](#), [InFunction\(\)](#), [PF\\_Processor\(\)](#), [poly\\_sort\(\)](#), [PolyFunMul\(\)](#), [Processor\(\)](#), and [TakeArgContent\(\)](#).

Here is the call graph for this function:



#### 4.19.6.27 SubPLon()

```
void SubPLon (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc ) [extern]
```

Definition at line 804 of file [reken.c](#).

#### 4.19.6.28 Substitute()

```
void Substitute (
    PHEAD WORD * term,
    WORD * pattern,
    WORD power ) [extern]
```

Definition at line 641 of file [pattern.c](#).

#### 4.19.6.29 SymFind()

```
int SymFind (
    PHEAD WORD * term,
    WORD * params ) [extern]
```

Definition at line 633 of file [function.c](#).

#### 4.19.6.30 SymGen()

```
int SymGen (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level ) [extern]
```

Definition at line 518 of file [function.c](#).

#### 4.19.6.31 Symmetrize()

```
WORD Symmetrize (
    PHEAD WORD * func,
    WORD * Lijst,
    WORD ngroups,
    WORD gsize,
    WORD type ) [extern]
```

Definition at line 213 of file [function.c](#).

#### 4.19.6.32 FullSymmetrize()

```
int FullSymmetrize (
    PHEAD WORD * fun,
    int type ) [extern]
```

Definition at line 473 of file [function.c](#).

#### 4.19.6.33 TakeModulus()

```
int TakeModulus (
    UWORD * a,
    WORD * na,
    UWORD * cmodvec,
    WORD ncmod,
    WORD par ) [extern]
```

Definition at line 3150 of file [reken.c](#).

#### 4.19.6.34 TakeNormalModulus()

```
int TakeNormalModulus (
    UWORD * a,
    WORD * na,
    UWORD * c,
    WORD nc,
    WORD par ) [extern]
```

Definition at line 3322 of file [reken.c](#).

**4.19.6.35 TalToLine()**

```
void TalToLine (
    UWORD x ) [extern]
```

Definition at line 522 of file [sch.c](#).

**4.19.6.36 TenVec()**

```
int TenVec (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level ) [extern]
```

Definition at line 2315 of file [opera.c](#).

**4.19.6.37 TenVecFind()**

```
int TenVecFind (
    PHEAD WORD * term,
    WORD * params ) [extern]
```

Definition at line 2181 of file [opera.c](#).

**4.19.6.38 TermRenumber()**

```
int TermRenumber (
    WORD * term,
    RENUMBER renumber,
    WORD nexpr ) [extern]
```

```
!! WORD *memterm=term; static LONG ctrap=0; !!!
```

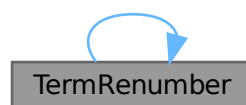
```
!! ctrap++; !!!
```

Definition at line 2427 of file [store.c](#).

References [ReNuMbEr::func](#), [ReNuMbEr::funnum](#), [ReNuMbEr::indi](#), [ReNuMbEr::indnum](#), [ReNuMbEr::symb](#), [ReNuMbEr::symnum](#), [TermRenumber\(\)](#), [ReNuMbEr::vecnum](#), and [ReNuMbEr::vect](#).

Referenced by [TermRenumber\(\)](#).

Here is the call graph for this function:



#### 4.19.6.39 TestDrop()

```
void TestDrop (
    void ) [extern]
```

Definition at line 484 of file [execute.c](#).

#### 4.19.6.40 PutInVflags()

```
void PutInVflags (
    WORD nexpr ) [extern]
```

Definition at line 562 of file [execute.c](#).

#### 4.19.6.41 TestMatch()

```
int TestMatch (
    PHEAD WORD * term,
    WORD * level ) [extern]
```

This routine governs the pattern matching. If it decides that a substitution should be made, this can be either the insertion of a right hand side (C->rhs) or the automatic generation of terms as a result of an operation (like trace). The object to be replaced is removed from term and a subexpression pointer is inserted. If the substitution is made more than once there can be more subexpression pointers. Its number is positive as it corresponds to the level at which the C->rhs can be found in the compiler output. The subexpression pointer contains the wildcard substitution information. The power is found in \*AT.TMout. For operations the subexpression pointer is negative and corresponds to an address in the array AT.TMout. In this array are then the instructions for the routine to be called and its number in the array 'Operations' The format is here: length,functionnumber,length-2 parameters

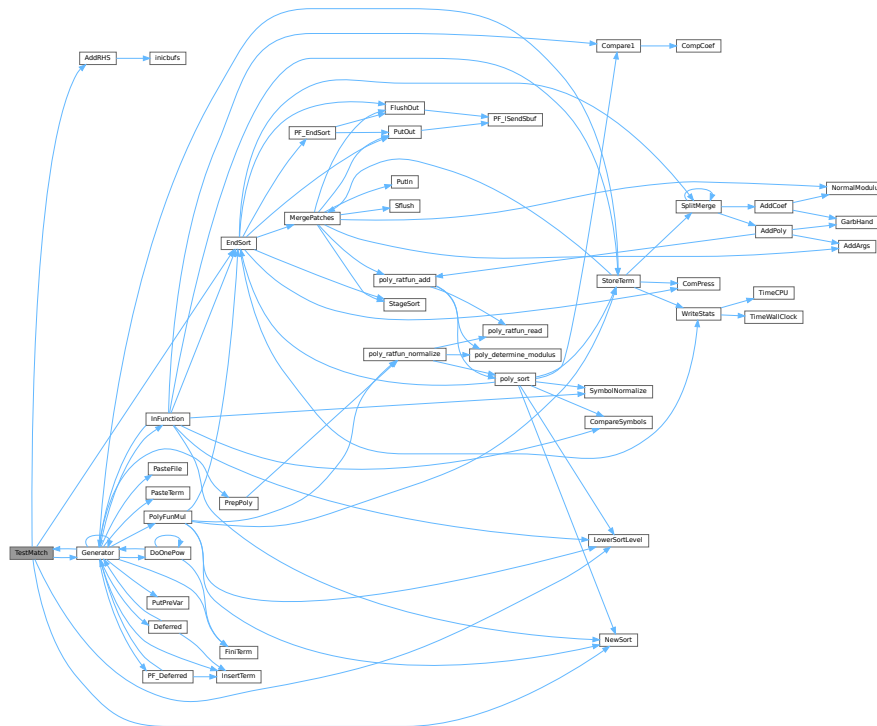
There is a certain complexity wrt repeat levels. Another complication is the poking of the wildcard values in the subexpression prototype in the compiler buffer. This was how things were done in the past with sequential FORM, but with the advent of TFORM this cannot be maintained. Now, for TFORM we make a copy of it. 7-may-2008 (JV): We cannot yet guarantee that this has been done 100% correctly. There are errors that occur in TFORM only and that may indicate problems.

Definition at line 97 of file [pattern.c](#).

References [AddRHS\(\)](#), [EndSort\(\)](#), [Generator\(\)](#), [CbUf::lhs](#), [LowerSortLevel\(\)](#), and [NewSort\(\)](#).

Referenced by [Generator\(\)](#).

Here is the call graph for this function:



#### 4.19.6.42 TestSub()

```
WORD TestSub (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 685 of file [proces.c](#).

#### 4.19.6.43 TimeCPU()

```
LONG TimeCPU (
    WORD par ) [extern]
```

Returns the CPU time.

##### Parameters

|            |                                           |
|------------|-------------------------------------------|
| <i>par</i> | If zero, the CPU time will be reset to 0. |
|------------|-------------------------------------------|

##### Returns

The CPU time in milliseconds.

Definition at line 3470 of file [tools.c](#).

Referenced by [PF\\_GetSlaveTimes\(\)](#), [PF\\_Processor\(\)](#), and [WriteStats\(\)](#).



#### 4.19.6.44 TimeChildren()

```
LONG TimeChildren (
    WORD par ) [extern]
```

Definition at line 3452 of file [tools.c](#).

#### 4.19.6.45 TimeWallClock()

```
LONG TimeWallClock (
    WORD par ) [extern]
```

Returns the wall-clock time.

##### Parameters

|            |                                                  |
|------------|--------------------------------------------------|
| <i>par</i> | If zero, the wall-clock time will be reset to 0. |
|------------|--------------------------------------------------|

##### Returns

The wall-clock time in centiseconds.

Definition at line 3396 of file [tools.c](#).

Referenced by [DoCheckpoint\(\)](#), [DoRecovery\(\)](#), [find\\_Horner\\_MCTS\(\)](#), [optimize\\_expression\\_given\\_Horner\(\)](#), [optimize\\_greedy\(\)](#), and [WriteStats\(\)](#).

#### 4.19.6.46 ToStorage()

```
int ToStorage (
    EXPRESSIONS e,
    POSITION * length ) [extern]
```

Definition at line 2016 of file [store.c](#).

#### 4.19.6.47 TokenToLine()

```
void TokenToLine (
    UBYTE * s ) [extern]
```

Definition at line 551 of file [sch.c](#).

#### 4.19.6.48 Trace4()

```
int Trace4 (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level ) [extern]
```

Definition at line 695 of file [opera.c](#).

#### 4.19.6.49 Trace4Gen()

```
int Trace4Gen (
    PHEAD TRACES * t,
    WORD number ) [extern]
```

Definition at line 858 of file [opera.c](#).

#### 4.19.6.50 Trace4no()

```
int Trace4no (
    WORD number,
    WORD * kron,
    TRACES * t ) [extern]
```

Definition at line 418 of file [opera.c](#).

#### 4.19.6.51 TraceFind()

```
int TraceFind (
    PHEAD WORD * term,
    WORD * params ) [extern]
```

Definition at line 1856 of file [opera.c](#).

#### 4.19.6.52 TraceN()

```
int TraceN (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level ) [extern]
```

Definition at line 1384 of file [opera.c](#).

#### 4.19.6.53 TraceNgen()

```
int TraceNgen (
    PHEAD TRACES * t,
    WORD number ) [extern]
```

Definition at line 1449 of file [opera.c](#).

#### 4.19.6.54 TraceNno()

```
WORD TraceNno (
    WORD number,
    WORD * kron,
    TRACES * t ) [extern]
```

Definition at line 1343 of file [opera.c](#).

#### 4.19.6.55 Traces()

```
int Traces (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level ) [extern]
```

Definition at line 1834 of file [opera.c](#).

#### 4.19.6.56 Trick()

```
WORD Trick (
    WORD * in,
    TRACES * t ) [extern]
```

Definition at line 331 of file [opera.c](#).

#### 4.19.6.57 TryDo()

```
int TryDo (
    PHEAD WORD * term,
    WORD * pattern,
    WORD level ) [extern]
```

Definition at line 1072 of file [reshuf.c](#).

#### 4.19.6.58 UnPack()

```
void UnPack (
    UWORD * a,
    WORD na,
    WORD * denom,
    WORD * numer ) [extern]
```

Definition at line 100 of file [reken.c](#).

#### 4.19.6.59 VarStore()

```
int VarStore (
    UBYTE * s,
    WORD n,
    WORD name,
    WORD namesize ) [extern]
```

Definition at line 2369 of file [store.c](#).

#### 4.19.6.60 WildFill()

```
WORD WildFill (
    PHEAD WORD * to,
    WORD * from,
    WORD * sub ) [extern]
```

Definition at line 65 of file [wildcard.c](#).

#### 4.19.6.61 WriteAll()

```
int WriteAll (
    void ) [extern]
```

Definition at line 2501 of file [sch.c](#).

#### 4.19.6.62 WriteOne()

```
int WriteOne (
    UBYTE * name,
    int alreadyinline,
    int nosemi,
    WORD plus ) [extern]
```

Definition at line 2727 of file [sch.c](#).

#### 4.19.6.63 WriteArgument()

```
void WriteArgument (
    WORD * t ) [extern]
```

Definition at line 1424 of file [sch.c](#).

#### 4.19.6.64 WriteExpression()

```
int WriteExpression (
    WORD * terms,
    LONG ltot ) [extern]
```

Definition at line 2470 of file [sch.c](#).

#### 4.19.6.65 WriteInnerTerm()

```
int WriteInnerTerm (
    WORD * term,
    WORD first ) [extern]
```

Definition at line 1951 of file [sch.c](#).

**4.19.6.66 WriteLists()**

```
void WriteLists (
    void ) [extern]
```

Definition at line 810 of file [sch.c](#).

**4.19.6.67 WriteSetup()**

```
void WriteSetup (
    void ) [extern]
```

Definition at line 811 of file [setfile.c](#).

**4.19.6.68 WriteStats()**

```
void WriteStats (
    POSITION * plspace,
    WORD par,
    WORD checkLogType ) [extern]
```

Writes the statistics.

**Parameters**

|                     |                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <i>plspace</i>      | The size in bytes currently occupied                                                                                                     |
| <i>par</i>          | par = 0 = STATSSPLITMERGE after a splitmerge. par = 1 = STATSMERGETOFILE after merge to sortfile. par = 2 = STATSPOSTSORT after the sort |
| <i>checkLogType</i> | == CHECKLOGTYPE: The output should not go to the log file if AM.LogType is 1.                                                            |

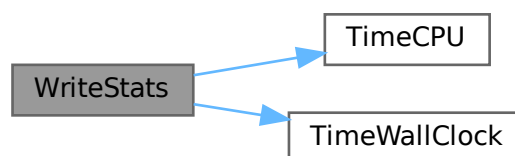
current expression is to be found in AR.CurExpr. terms are in S->TermsLeft. S->GenTerms.

Definition at line 121 of file [sort.c](#).

References [TimeCPU\(\)](#), and [TimeWallClock\(\)](#).

Referenced by [EndSort\(\)](#), [PF\\_Processor\(\)](#), and [StoreTerm\(\)](#).

Here is the call graph for this function:



#### 4.19.6.69 WriteSubTerm()

```
int WriteSubTerm (
    WORD * sterm,
    WORD first ) [extern]
```

Definition at line 1542 of file [sch.c](#).

#### 4.19.6.70 WriteTerm()

```
int WriteTerm (
    WORD * term,
    WORD * lbrac,
    WORD first,
    WORD prtf,
    WORD br ) [extern]
```

Definition at line 2148 of file [sch.c](#).

#### 4.19.6.71 execarg()

```
int execarg (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 56 of file [argument.c](#).

#### 4.19.6.72 execterm()

```
int execterm (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 1770 of file [argument.c](#).

#### 4.19.6.73 SpecialCleanup()

```
void SpecialCleanup (
    PHEAD0 ) [extern]
```

Definition at line 1624 of file [execute.c](#).

#### 4.19.6.74 SetMods()

```
void SetMods (
    void ) [extern]
```

Definition at line 1638 of file [execute.c](#).

#### 4.19.6.75 UnSetMods()

```
void UnSetMods (
    void ) [extern]
```

Definition at line 1656 of file [execute.c](#).

#### 4.19.6.76 DoExecute()

```
int DoExecute (
    WORD par,
    WORD skip ) [extern]
```

Definition at line 616 of file [execute.c](#).

#### 4.19.6.77 SetScratch()

```
void SetScratch (
    FILEHANDLE * f,
    POSITION * position ) [extern]
```

Definition at line 114 of file [store.c](#).

#### 4.19.6.78 Warning()

```
void Warning (
    char * s ) [extern]
```

Definition at line 790 of file [message.c](#).

#### 4.19.6.79 HighWarning()

```
void HighWarning (
    char * s ) [extern]
```

Definition at line 802 of file [message.c](#).

#### 4.19.6.80 SpareTable()

```
int SpareTable (
    TABLES TT ) [extern]
```

Definition at line 694 of file [tables.c](#).

#### 4.19.6.81 strDup1()

```
UBYTE * strDup1 (
    UBYTE * instring,
    char * ifwrong ) [extern]
```

Definition at line 1904 of file [tools.c](#).

#### 4.19.6.82 Malloc1()

```
void * Malloc1 (
    LONG size,
    const char * messageifwrong ) [extern]
```

Definition at line 2222 of file [tools.c](#).

#### 4.19.6.83 DoTail()

```
int DoTail (
    int argc,
    UBYTE ** argv ) [extern]
```

Definition at line 237 of file [startup.c](#).

#### 4.19.6.84 OpenInput()

```
int OpenInput (
    void ) [extern]
```

Definition at line 542 of file [startup.c](#).

#### 4.19.6.85 PutPreVar()

```
int PutPreVar (
    UBYTE * name,
    UBYTE * value,
    UBYTE * args,
    int mode ) [extern]
```

Inserts/Updates a preprocessor variable in the name administration.

##### Parameters

|              |                                                                                                                                                                                                                    |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>name</i>  | Character string with the variable name.                                                                                                                                                                           |
| <i>value</i> | Character string with a possible value. Special case: if this argument is zero, then we have no value. Note: This is different from having an empty argument! This should only occur when the name starts with a ? |
| <i>args</i>  | Character string with possible arguments.                                                                                                                                                                          |
| <i>mode</i>  | =0: always create a new name entry, =1: try to do a redefinition if possible.                                                                                                                                      |



## Returns

Index of used entry in name list.

Definition at line 718 of file [pre.c](#).

Referenced by [ClearOptimize\(\)](#), [Generator\(\)](#), [Optimize\(\)](#), [PF\\_BroadcastRedefinedPreVars\(\)](#), [StartVariables\(\)](#), and [TheDefine\(\)](#).

### 4.19.6.86 Error0()

```
void Error0 (
    char * s ) [extern]
```

Definition at line 56 of file [message.c](#).

### 4.19.6.87 Error1()

```
void Error1 (
    char * s,
    UBYTE * t ) [extern]
```

Definition at line 67 of file [message.c](#).

### 4.19.6.88 Error2()

```
void Error2 (
    char * s1,
    char * s2,
    UBYTE * t ) [extern]
```

Definition at line 78 of file [message.c](#).

### 4.19.6.89 ReadFromStream()

```
UBYTE ReadFromStream (
    STREAM * stream ) [extern]
```

Definition at line 151 of file [tools.c](#).

### 4.19.6.90 GetFromStream()

```
UBYTE GetFromStream (
    STREAM * stream ) [extern]
```

Definition at line 286 of file [tools.c](#).

#### 4.19.6.91 LookInStream()

```
UBYTE LookInStream (
    STREAM * stream ) [extern]
```

Definition at line 309 of file [tools.c](#).

#### 4.19.6.92 OpenStream()

```
STREAM * OpenStream (
    UBYTE * name,
    int type,
    int prevarmode,
    int raiselow ) [extern]
```

Definition at line 321 of file [tools.c](#).

#### 4.19.6.93 LocateFile()

```
int LocateFile (
    UBYTE ** name,
    int type ) [extern]
```

Definition at line 576 of file [tools.c](#).

#### 4.19.6.94 CloseStream()

```
STREAM * CloseStream (
    STREAM * stream ) [extern]
```

Definition at line 663 of file [tools.c](#).

#### 4.19.6.95 PositionStream()

```
void PositionStream (
    STREAM * stream,
    LONG position ) [extern]
```

Definition at line 823 of file [tools.c](#).

#### 4.19.6.96 ReverseStatements()

```
int ReverseStatements (
    STREAM * stream ) [extern]
```

Definition at line 855 of file [tools.c](#).

#### 4.19.6.97 ProcessOption()

```
int ProcessOption (
    UBYTE * s1,
    UBYTE * s2,
    int filetype ) [extern]
```

Definition at line [182](#) of file [setfile.c](#).

#### 4.19.6.98 DoSetups()

```
int DoSetups (
    void ) [extern]
```

Definition at line [132](#) of file [setfile.c](#).

#### 4.19.6.99 TerminateImpl()

```
void TerminateImpl (
    int errorcode,
    const char * file,
    int line,
    const char * function ) [extern]
```

Definition at line [1849](#) of file [startup.c](#).

#### 4.19.6.100 GetNode()

```
NAMENODE * GetNode (
    NAMETREE * nametree,
    UBYTE * name ) [extern]
```

Definition at line [49](#) of file [names.c](#).

#### 4.19.6.101 AddName()

```
int AddName (
    NAMETREE * nametree,
    UBYTE * name,
    WORD type,
    WORD number,
    int * nodenum ) [extern]
```

Definition at line [71](#) of file [names.c](#).

#### 4.19.6.102 GetName()

```
int GetName (
    NAMETREE * nametree,
    UBYTE * namein,
    WORD * number,
    int par ) [extern]
```

Definition at line 252 of file [names.c](#).

#### 4.19.6.103 GetFunction()

```
UBYTE * GetFunction (
    UBYTE * s,
    WORD * funnum ) [extern]
```

Definition at line 342 of file [names.c](#).

#### 4.19.6.104 GetNumber()

```
UBYTE * GetNumber (
    UBYTE * s,
    WORD * num ) [extern]
```

Definition at line 388 of file [names.c](#).

#### 4.19.6.105 GetLastExprName()

```
int GetLastExprName (
    UBYTE * name,
    WORD * number ) [extern]
```

Definition at line 438 of file [names.c](#).

#### 4.19.6.106 GetAutoName()

```
int GetAutoName (
    UBYTE * name,
    WORD * number ) [extern]
```

Definition at line 479 of file [names.c](#).

#### 4.19.6.107 GetVar()

```
int GetVar (
    UBYTE * name,
    WORD * type,
    WORD * number,
    int wantedtype,
    int par ) [extern]
```

Definition at line 524 of file [names.c](#).

#### 4.19.6.108 MakeDubious()

```
int MakeDubious (
    NAMETREE * nametree,
    UBYTE * name,
    WORD * number ) [extern]
```

Definition at line 2693 of file [names.c](#).

#### 4.19.6.109 GetOName()

```
int GetOName (
    NAMETREE * nametree,
    UBYTE * name,
    WORD * number,
    int par ) [extern]
```

Definition at line 460 of file [names.c](#).

#### 4.19.6.110 DumpTree()

```
void DumpTree (
    NAMETREE * nametree ) [extern]
```

Definition at line 602 of file [names.c](#).

#### 4.19.6.111 DumpNode()

```
void DumpNode (
    NAMETREE * nametree,
    WORD node,
    WORD depth ) [extern]
```

Definition at line 615 of file [names.c](#).

#### 4.19.6.112 LinkTree()

```
void LinkTree (
    NAMETREE * tree,
    WORD offset,
    WORD numnodes ) [extern]
```

Definition at line 784 of file [names.c](#).

#### 4.19.6.113 CopyTree()

```
void CopyTree (
    NAMETREE * newtree,
    NAMETREE * oldtree,
    WORD node,
    WORD par ) [extern]
```

Definition at line 696 of file [names.c](#).

#### 4.19.6.114 CompactifyTree()

```
int CompactifyTree (
    NAMETREE * nametree,
    WORD par ) [extern]
```

Definition at line 634 of file [names.c](#).

#### 4.19.6.115 MakeNameTree()

```
NAMETREE * MakeNameTree (
    void ) [extern]
```

Definition at line 815 of file [names.c](#).

#### 4.19.6.116 FreeNameTree()

```
void FreeNameTree (
    NAMETREE * n ) [extern]
```

Definition at line 834 of file [names.c](#).

#### 4.19.6.117 AddExpression()

```
int AddExpression (
    UBYTE * name,
    int x,
    int y ) [extern]
```

Definition at line 2744 of file [names.c](#).

#### 4.19.6.118 AddSymbol()

```
int AddSymbol (
    UBYTE * name,
    int minpow,
    int maxpow,
    int cplx,
    int dim ) [extern]
```

Definition at line 920 of file [names.c](#).

#### 4.19.6.119 AddDollar()

```
int AddDollar (
    UBYTE * name,
    WORD type,
    WORD * start,
    LONG size ) [extern]
```

Definition at line 2602 of file [names.c](#).

**4.19.6.120 ReplaceDollar()**

```
int ReplaceDollar (
    WORD number,
    WORD newtype,
    WORD * newstart,
    LONG newsize ) [extern]
```

Definition at line [2646](#) of file [names.c](#).

**4.19.6.121 DollarRaiseLow()**

```
int DollarRaiseLow (
    UBYTE * name,
    LONG value ) [extern]
```

Definition at line [2552](#) of file [dollar.c](#).

**4.19.6.122 AddVector()**

```
int AddVector (
    UBYTE * name,
    int cplx,
    int dim ) [extern]
```

Definition at line [1244](#) of file [names.c](#).

**4.19.6.123 AddDubious()**

```
int AddDubious (
    UBYTE * name ) [extern]
```

Definition at line [2679](#) of file [names.c](#).

**4.19.6.124 AddIndex()**

```
int AddIndex (
    UBYTE * name,
    int dim,
    int dim4 ) [extern]
```

Definition at line [1088](#) of file [names.c](#).

**4.19.6.125 DoDimension()**

```
UBYTE * DoDimension (
    UBYTE * s,
    int * dim,
    int * dim4 ) [extern]
```

Definition at line [1160](#) of file [names.c](#).

**4.19.6.126 AddFunction()**

```
int AddFunction (
    UBYTE * name,
    int comm,
    int istensor,
    int cplx,
    int symprop,
    int dim,
    int argmax,
    int argmin ) [extern]
```

Definition at line [1326](#) of file [names.c](#).

**4.19.6.127 CoCommuteInSet()**

```
int CoCommuteInSet (
    UBYTE * s ) [extern]
```

Definition at line [1356](#) of file [names.c](#).

**4.19.6.128 CoFunction()**

```
int CoFunction (
    UBYTE * s,
    int comm,
    int istensor ) [extern]
```

Definition at line [1446](#) of file [names.c](#).

**4.19.6.129 TestName()**

```
int TestName (
    UBYTE * name ) [extern]
```

Definition at line [3254](#) of file [names.c](#).

**4.19.6.130 AddSet()**

```
int AddSet (
    UBYTE * name,
    WORD dim ) [extern]
```

Definition at line [2122](#) of file [names.c](#).

**4.19.6.131 DoElements()**

```
int DoElements (
    UBYTE * s,
    SETS set,
    UBYTE * name ) [extern]
```

Definition at line [2155](#) of file [names.c](#).



**4.19.6.132 DoTempSet()**

```
int DoTempSet (
    UBYTE * from,
    UBYTE * to ) [extern]
```

Definition at line [2480](#) of file [names.c](#).

**4.19.6.133 NameConflict()**

```
int NameConflict (
    int type,
    UBYTE * name ) [extern]
```

Definition at line [2728](#) of file [names.c](#).

**4.19.6.134 OpenFile()**

```
int OpenFile (
    char * name ) [extern]
```

Definition at line [983](#) of file [tools.c](#).

**4.19.6.135 OpenAddFile()**

```
int OpenAddFile (
    char * name ) [extern]
```

Definition at line [1002](#) of file [tools.c](#).

**4.19.6.136 ReOpenFile()**

```
int ReOpenFile (
    char * name ) [extern]
```

Definition at line [1023](#) of file [tools.c](#).

**4.19.6.137 CreateFile()**

```
int CreateFile (
    char * name ) [extern]
```

Definition at line [1043](#) of file [tools.c](#).

**4.19.6.138 CreateLogFile()**

```
int CreateLogFile (
    char * name ) [extern]
```

Definition at line [1060](#) of file [tools.c](#).

#### 4.19.6.139 CloseFile()

```
void CloseFile (
    int handle ) [extern]
```

Definition at line [1078](#) of file [tools.c](#).

#### 4.19.6.140 CopyFile()

```
int CopyFile (
    char * source,
    char * dest ) [extern]
```

Copies a file with name *\*source* to a file named *\*dest*. The involved files must not be open. Returns non-zero if an error occurred. Uses if possible the combined large and small sorting buffers as cache.

Definition at line [1101](#) of file [tools.c](#).

Referenced by [DoRecovery\(\)](#).

#### 4.19.6.141 CreateHandle()

```
int CreateHandle (
    void ) [extern]
```

Definition at line [1162](#) of file [tools.c](#).

#### 4.19.6.142 ReadFile()

```
LONG ReadFile (
    int handle,
    UBYTE * buffer,
    LONG size ) [extern]
```

Definition at line [1217](#) of file [tools.c](#).

#### 4.19.6.143 ReadPosFile()

```
LONG ReadPosFile (
    PHEAD FILEHANDLE * fi,
    UBYTE * buffer,
    LONG size,
    POSITION * pos ) [extern]
```

Definition at line [1267](#) of file [tools.c](#).

**4.19.6.144 WriteFileToFile()**

```
LONG WriteFileToFile (
    int handle,
    UBYTE * buffer,
    LONG size ) [extern]
```

Definition at line [1333](#) of file [tools.c](#).

**4.19.6.145 SeekFile()**

```
void SeekFile (
    int handle,
    POSITION * offset,
    int origin ) [extern]
```

Definition at line [1371](#) of file [tools.c](#).

**4.19.6.146 TellFile()**

```
LONG TellFile (
    int handle ) [extern]
```

Definition at line [1396](#) of file [tools.c](#).

**4.19.6.147 FlushFile()**

```
void FlushFile (
    int handle ) [extern]
```

Definition at line [1420](#) of file [tools.c](#).

**4.19.6.148 GetPosFile()**

```
int GetPosFile (
    int handle,
    fpos_t * pospointer ) [extern]
```

Definition at line [1434](#) of file [tools.c](#).

**4.19.6.149 SetPosFile()**

```
int SetPosFile (
    int handle,
    fpos_t * pospointer ) [extern]
```

Definition at line [1448](#) of file [tools.c](#).

#### 4.19.6.150 SynchFile()

```
void SynchFile (
    int handle ) [extern]
```

Definition at line 1468 of file [tools.c](#).

#### 4.19.6.151 TruncateFile()

```
void TruncateFile (
    int handle ) [extern]
```

Definition at line 1490 of file [tools.c](#).

#### 4.19.6.152 GetChannel()

```
int GetChannel (
    char * name,
    int mode ) [extern]
```

Definition at line 1510 of file [tools.c](#).

#### 4.19.6.153 GetAppendChannel()

```
int GetAppendChannel (
    char * name ) [extern]
```

Definition at line 1546 of file [tools.c](#).

#### 4.19.6.154 CloseChannel()

```
int CloseChannel (
    char * name ) [extern]
```

Definition at line 1576 of file [tools.c](#).

#### 4.19.6.155 inictable()

```
void inictable (
    void ) [extern]
```

Definition at line 308 of file [compiler.c](#).

#### 4.19.6.156 findcommand()

```
KEYWORD * findcommand (
    UBYTE * in ) [extern]
```

Definition at line 334 of file [compiler.c](#).

#### 4.19.6.157 inicbufs()

```
int inicbufs (
    void ) [extern]
```

Creates a new compiler buffer and returns its ID number.

##### Returns

The ID number for the new compiler buffer.

Definition at line 47 of file [comtool.c](#).

References [CbUf::boomlijst](#), [CbUf::Buffer](#), [CbUf::BufferSize](#), [CbUf::CanCommu](#), [CbUf::dimension](#), [CbUf::lhs](#), [CbUf::numdum](#), [CbUf::NumTerms](#), [CbUf::Pointer](#), [CbUf::rhs](#), and [CbUf::Top](#).

Referenced by [AddRHS\(\)](#), and [StartVariables\(\)](#).

#### 4.19.6.158 StartFiles()

```
void StartFiles (
    void ) [extern]
```

Definition at line 956 of file [tools.c](#).

#### 4.19.6.159 MakeDate()

```
UBYTE * MakeDate (
    void ) [extern]
```

Definition at line 2090 of file [tools.c](#).

#### 4.19.6.160 PreProcessor()

```
void PreProcessor (
    void ) [extern]
```

Definition at line 947 of file [pre.c](#).

#### 4.19.6.161 FromList()

```
void * FromList (
    LIST * L ) [extern]
```

Definition at line 2699 of file [tools.c](#).

#### 4.19.6.162 From0List()

```
void * From0List (
    LIST * L ) [extern]
```

Definition at line 2725 of file [tools.c](#).

#### 4.19.6.163 FromVarList()

```
void * FromVarList (
    LIST * L ) [extern]
```

Definition at line 2754 of file [tools.c](#).

#### 4.19.6.164 DoubleList()

```
int DoubleList (
    void *** lijst,
    int * oldsize,
    int objectsize,
    char * nameoftype ) [extern]
```

Definition at line 2796 of file [tools.c](#).

#### 4.19.6.165 DoubleLList()

```
int DoubleLList (
    void *** lijst,
    LONG * oldsize,
    int objectsize,
    char * nameoftype ) [extern]
```

Definition at line 2848 of file [tools.c](#).

#### 4.19.6.166 DoubleBuffer()

```
void DoubleBuffer (
    void ** start,
    void ** stop,
    int size,
    char * text ) [extern]
```

Definition at line 2896 of file [tools.c](#).

#### 4.19.6.167 ExpandBuffer()

```
void ExpandBuffer (
    void ** buffer,
    LONG * oldsize,
    int type ) [extern]
```

Definition at line 2921 of file [tools.c](#).

**4.19.6.168 iexp()**

```
LONG iexp (
    LONG x,
    int p ) [extern]
```

Definition at line 2945 of file [tools.c](#).

**4.19.6.169 IsLikeVector()**

```
int IsLikeVector (
    WORD * arg ) [extern]
```

Definition at line 3148 of file [tools.c](#).

**4.19.6.170 AreArgsEqual()**

```
int AreArgsEqual (
    WORD * arg1,
    WORD * arg2 ) [extern]
```

Definition at line 3176 of file [tools.c](#).

**4.19.6.171 CompareArgs()**

```
int CompareArgs (
    WORD * arg1,
    WORD * arg2 ) [extern]
```

Definition at line 3195 of file [tools.c](#).

**4.19.6.172 SkipField()**

```
UBYTE * SkipField (
    UBYTE * s,
    int level ) [extern]
```

Definition at line 1946 of file [tools.c](#).

**4.19.6.173 StrCmp()**

```
int StrCmp (
    UBYTE * s1,
    UBYTE * s2 ) [extern]
```

Definition at line 1700 of file [tools.c](#).

#### 4.19.6.174 StrICmp()

```
int StrICmp (
    UBYTE * s1,
    UBYTE * s2 ) [extern]
```

Definition at line 1711 of file [tools.c](#).

#### 4.19.6.175 StrHICmp()

```
int StrHICmp (
    UBYTE * s1,
    UBYTE * s2 ) [extern]
```

Definition at line 1722 of file [tools.c](#).

#### 4.19.6.176 StrICont()

```
int StrICont (
    UBYTE * s1,
    UBYTE * s2 ) [extern]
```

Definition at line 1733 of file [tools.c](#).

#### 4.19.6.177 CmpArray()

```
int CmpArray (
    WORD * t1,
    WORD * t2,
    WORD n ) [extern]
```

Definition at line 1745 of file [tools.c](#).

#### 4.19.6.178 ConWord()

```
int ConWord (
    UBYTE * s1,
    UBYTE * s2 ) [extern]
```

Definition at line 1759 of file [tools.c](#).

#### 4.19.6.179 StrLen()

```
int StrLen (
    UBYTE * s ) [extern]
```

Definition at line 1771 of file [tools.c](#).



**4.19.6.180 GetPreVar()**

```
UBYTE * GetPreVar (
    UBYTE * name,
    int flag ) [extern]
```

Definition at line [536](#) of file [pre.c](#).

**4.19.6.181 ToGeneral()**

```
void ToGeneral (
    WORD * r,
    WORD * m,
    WORD par ) [extern]
```

Definition at line [2976](#) of file [tools.c](#).

**4.19.6.182 ToPolyFunGeneral()**

```
WORD ToPolyFunGeneral (
    PHEAD WORD * term ) [extern]
```

Definition at line [3073](#) of file [tools.c](#).

**4.19.6.183 ToFast()**

```
int ToFast (
    WORD * r,
    WORD * m ) [extern]
```

Definition at line [3020](#) of file [tools.c](#).

**4.19.6.184 GetSetupPar()**

```
SETUPPARAMETERS * GetSetupPar (
    UBYTE * s ) [extern]
```

Definition at line [337](#) of file [setfile.c](#).

**4.19.6.185 RecalcSetups()**

```
int RecalcSetups (
    void ) [extern]
```

Definition at line [357](#) of file [setfile.c](#).

#### 4.19.6.186 AllocSetups()

```
int AllocSetups (
    void ) [extern]
```

Definition at line 404 of file [setfile.c](#).

#### 4.19.6.187 AllocSort()

```
SORTING * AllocSort (
    LONG inLargeSize,
    LONG inSmallSize,
    LONG inSmallEsize,
    LONG inTermsInSmall,
    int inMaxPatches,
    int inMaxFpatches,
    LONG inIOSize,
    int level ) [extern]
```

Definition at line 869 of file [setfile.c](#).

#### 4.19.6.188 AllocSortFileName()

```
void AllocSortFileName (
    SORTING * sort ) [extern]
```

Definition at line 1019 of file [setfile.c](#).

#### 4.19.6.189 LoadInputFile()

```
UBYTE * LoadInputFile (
    UBYTE * filename,
    int type ) [extern]
```

Definition at line 112 of file [tools.c](#).

#### 4.19.6.190 GetInput()

```
UBYTE GetInput (
    void ) [extern]
```

Definition at line 132 of file [pre.c](#).

#### 4.19.6.191 ClearPushback()

```
void ClearPushback (
    void ) [extern]
```

Definition at line 161 of file [pre.c](#).

#### 4.19.6.192 GetChar()

```
UBYTE GetChar (
    int level ) [extern]
```

Definition at line 179 of file [pre.c](#).

#### 4.19.6.193 CharOut()

```
void CharOut (
    UBYTE c ) [extern]
```

Definition at line 489 of file [pre.c](#).

#### 4.19.6.194 UnsetAllowDelay()

```
void UnsetAllowDelay (
    void ) [extern]
```

Definition at line 510 of file [pre.c](#).

#### 4.19.6.195 PopPreVars()

```
void PopPreVars (
    int tonumber ) [extern]
```

Definition at line 820 of file [pre.c](#).

#### 4.19.6.196 IniModule()

```
void IniModule (
    int type ) [extern]
```

Definition at line 835 of file [pre.c](#).

#### 4.19.6.197 IniSpecialModule()

```
void IniSpecialModule (
    int type ) [extern]
```

Definition at line 937 of file [pre.c](#).

#### 4.19.6.198 ModuleInstruction()

```
int ModuleInstruction (
    int * moduletype,
    int * specialtype ) [extern]
```

Definition at line 76 of file [module.c](#).

#### 4.19.6.199 PreProInstruction()

```
int PreProInstruction (
    void ) [extern]
```

Definition at line 1166 of file [pre.c](#).

#### 4.19.6.200 LoadInstruction()

```
int LoadInstruction (
    int mode ) [extern]
```

Definition at line 1254 of file [pre.c](#).

#### 4.19.6.201 LoadStatement()

```
int LoadStatement (
    int type ) [extern]
```

Definition at line 1457 of file [pre.c](#).

#### 4.19.6.202 FindKeyWord()

```
KEYWORD * FindKeyWord (
    UBYTE * theword,
    KEYWORD * table,
    int size ) [extern]
```

Definition at line 1963 of file [pre.c](#).

#### 4.19.6.203 FindInKeyWord()

```
KEYWORD * FindInKeyWord (
    UBYTE * theword,
    KEYWORD * table,
    int size ) [extern]
```

Definition at line 1994 of file [pre.c](#).

#### 4.19.6.204 DoDefine()

```
int DoDefine (
    UBYTE * s ) [extern]
```

Definition at line 2159 of file [pre.c](#).

#### 4.19.6.205 DoRedefine()

```
int DoRedefine (
    UBYTE * s ) [extern]
```

Definition at line 2169 of file [pre.c](#).

#### 4.19.6.206 TheDefine()

```
int TheDefine (
    UBYTE * s,
    int mode ) [extern]
```

Preprocessor assignment. Possible arguments and values are treated and the new preprocessor variable is put into the name administration.

##### Parameters

|             |                                                                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>s</i>    | Pointer to the character string following the preprocessor command.                                                                                 |
| <i>mode</i> | Bitmask. 0-bit clear: always create a new name entry, 0-bit set: try to redefine an existing name, 1-bit set: ignore preprocessor if/switch status. |

##### Returns

zero: no errors, negative number: errors.

Definition at line 2024 of file [pre.c](#).

References [PutPreVar\(\)](#).

Here is the call graph for this function:



#### 4.19.6.207 TheUndefine()

```
int TheUndefine (
    UBYTE * name ) [extern]
```

Definition at line 2209 of file [pre.c](#).

#### 4.19.6.208 ClearMacro()

```
int ClearMacro (
    UBYTE * name ) [extern]
```

Definition at line 2181 of file [pre.c](#).

#### 4.19.6.209 DoUndefine()

```
int DoUndefine (
    UBYTE * s ) [extern]
```

Definition at line 2263 of file [pre.c](#).

#### 4.19.6.210 DoInclude()

```
int DoInclude (
    UBYTE * s ) [extern]
```

Definition at line 2315 of file [pre.c](#).

#### 4.19.6.211 DoReverseInclude()

```
int DoReverseInclude (
    UBYTE * s ) [extern]
```

Definition at line 2322 of file [pre.c](#).

#### 4.19.6.212 Include()

```
int Include (
    UBYTE * s,
    int type ) [extern]
```

Definition at line 2329 of file [pre.c](#).

#### 4.19.6.213 DoExternal()

```
int DoExternal (
    UBYTE * s ) [extern]
```

Definition at line 5424 of file [pre.c](#).

#### 4.19.6.214 DoToExternal()

```
int DoToExternal (
    UBYTE * s ) [extern]
```

Definition at line 5986 of file [pre.c](#).

**4.19.6.215 DoFromExternal()**

```
int DoFromExternal (
    UBYTE * s ) [extern]
```

Definition at line 5791 of file [pre.c](#).

**4.19.6.216 DoPrompt()**

```
int DoPrompt (
    UBYTE * s ) [extern]
```

Definition at line 5506 of file [pre.c](#).

**4.19.6.217 DoSetExternal()**

```
int DoSetExternal (
    UBYTE * s ) [extern]
```

Definition at line 5539 of file [pre.c](#).

**4.19.6.218 DoSetExternalAttr()**

```
int DoSetExternalAttr (
    UBYTE * s ) [extern]
```

Definition at line 5609 of file [pre.c](#).

**4.19.6.219 DoRmExternal()**

```
int DoRmExternal (
    UBYTE * s ) [extern]
```

Definition at line 5726 of file [pre.c](#).

**4.19.6.220 DoFactDollar()**

```
int DoFactDollar (
    UBYTE * s ) [extern]
```

Definition at line 6633 of file [pre.c](#).

**4.19.6.221 GetDollarNumber()**

```
WORD GetDollarNumber (
    UBYTE ** inp,
    DOLLARS d ) [extern]
```

Definition at line 6670 of file [pre.c](#).

**4.19.6.222 DoSetRandom()**

```
int DoSetRandom (
    UBYTE * s ) [extern]
```

Definition at line [6760](#) of file [pre.c](#).

**4.19.6.223 DoOptimize()**

```
int DoOptimize (
    UBYTE * s ) [extern]
```

Definition at line [6803](#) of file [pre.c](#).

**4.19.6.224 DoClearOptimize()**

```
int DoClearOptimize (
    UBYTE * s ) [extern]
```

Definition at line [6936](#) of file [pre.c](#).

**4.19.6.225 DoSkipExtraSymbols()**

```
int DoSkipExtraSymbols (
    UBYTE * s ) [extern]
```

Definition at line [6956](#) of file [pre.c](#).

**4.19.6.226 DoTimeOutAfter()**

```
int DoTimeOutAfter (
    UBYTE * s ) [extern]
```

Definition at line [7182](#) of file [pre.c](#).

**4.19.6.227 DoMessage()**

```
int DoMessage (
    UBYTE * s ) [extern]
```

Definition at line [3676](#) of file [pre.c](#).

**4.19.6.228 DoPreOut()**

```
int DoPreOut (
    UBYTE * s ) [extern]
```

Definition at line [3747](#) of file [pre.c](#).



**4.19.6.229 DoPreAppend()**

```
int DoPreAppend (  
    UBYTE * s ) [extern]
```

Definition at line [3804](#) of file [pre.c](#).

**4.19.6.230 DoPreCreate()**

```
int DoPreCreate (  
    UBYTE * s ) [extern]
```

Definition at line [3847](#) of file [pre.c](#).

**4.19.6.231 DoPreAssign()**

```
int DoPreAssign (  
    UBYTE * s ) [extern]
```

Definition at line [2128](#) of file [pre.c](#).

**4.19.6.232 DoPreBreak()**

```
int DoPreBreak (  
    UBYTE * s ) [extern]
```

Definition at line [4047](#) of file [pre.c](#).

**4.19.6.233 DoPreDefault()**

```
int DoPreDefault (  
    UBYTE * s ) [extern]
```

Definition at line [4110](#) of file [pre.c](#).

**4.19.6.234 DoPreSwitch()**

```
int DoPreSwitch (  
    UBYTE * s ) [extern]
```

Definition at line [4161](#) of file [pre.c](#).

**4.19.6.235 DoPreEndSwitch()**

```
int DoPreEndSwitch (  
    UBYTE * s ) [extern]
```

Definition at line [4134](#) of file [pre.c](#).

**4.19.6.236 DoPreCase()**

```
int DoPreCase (
    UBYTE * s ) [extern]
```

Definition at line 4071 of file [pre.c](#).

**4.19.6.237 DoPreShow()**

```
int DoPreShow (
    UBYTE * s ) [extern]
```

Definition at line 4215 of file [pre.c](#).

**4.19.6.238 DoPreExchange()**

```
int DoPreExchange (
    UBYTE * s ) [extern]
```

Definition at line 2490 of file [pre.c](#).

**4.19.6.239 DoSystem()**

```
int DoSystem (
    UBYTE * s ) [extern]
```

Definition at line 4254 of file [pre.c](#).

**4.19.6.240 DoPipe()**

```
int DoPipe (
    UBYTE * s ) [extern]
```

Definition at line 3690 of file [pre.c](#).

**4.19.6.241 StartPrepro()**

```
void StartPrepro (
    void ) [extern]
```

Definition at line 4432 of file [pre.c](#).

**4.19.6.242 Dolfdef()**

```
int DoIfdef (
    UBYTE * s,
    int par ) [extern]
```

Definition at line 3391 of file [pre.c](#).

**4.19.6.243 DoIfydef()**

```
int DoIfydef (
    UBYTE * s ) [extern]
```

Definition at line [3416](#) of file [pre.c](#).

**4.19.6.244 DoIfndef()**

```
int DoIfndef (
    UBYTE * s ) [extern]
```

Definition at line [3426](#) of file [pre.c](#).

**4.19.6.245 DoElse()**

```
int DoElse (
    UBYTE * s ) [extern]
```

Definition at line [3095](#) of file [pre.c](#).

**4.19.6.246 DoElseif()**

```
int DoElseif (
    UBYTE * s ) [extern]
```

Definition at line [3132](#) of file [pre.c](#).

**4.19.6.247 DoEndif()**

```
int DoEndif (
    UBYTE * s ) [extern]
```

Definition at line [3313](#) of file [pre.c](#).

**4.19.6.248 DoTerminate()**

```
int DoTerminate (
    UBYTE * s ) [extern]
```

Definition at line [2757](#) of file [pre.c](#).

**4.19.6.249 DoIf()**

```
int DoIf (
    UBYTE * s ) [extern]
```

Definition at line [3367](#) of file [pre.c](#).

#### 4.19.6.250 DoCall()

```
int DoCall (
    UBYTE * s ) [extern]
```

Definition at line [2551](#) of file [pre.c](#).

#### 4.19.6.251 DoDebug()

```
int DoDebug (
    UBYTE * s ) [extern]
```

Definition at line [2720](#) of file [pre.c](#).

#### 4.19.6.252 DoContinueDo()

```
int DoContinueDo (
    UBYTE * s ) [extern]
```

Definition at line [2780](#) of file [pre.c](#).

#### 4.19.6.253 DoDo()

```
int DoDo (
    UBYTE * s ) [extern]
```

Definition at line [2811](#) of file [pre.c](#).

#### 4.19.6.254 DoBreakDo()

```
int DoBreakDo (
    UBYTE * s ) [extern]
```

Definition at line [3026](#) of file [pre.c](#).

#### 4.19.6.255 DoEnddo()

```
int DoEnddo (
    UBYTE * s ) [extern]
```

Definition at line [3167](#) of file [pre.c](#).

#### 4.19.6.256 DoEndprocedure()

```
int DoEndprocedure (
    UBYTE * s ) [extern]
```

Definition at line [3341](#) of file [pre.c](#).

**4.19.6.257 DoInside()**

```
int DoInside (
    UBYTE * s ) [extern]
```

Definition at line [3451](#) of file [pre.c](#).

**4.19.6.258 DoEndInside()**

```
int DoEndInside (
    UBYTE * s ) [extern]
```

Definition at line [3544](#) of file [pre.c](#).

**4.19.6.259 DoProcedure()**

```
int DoProcedure (
    UBYTE * s ) [extern]
```

Definition at line [4005](#) of file [pre.c](#).

**4.19.6.260 DoPrePrintTimes()**

```
int DoPrePrintTimes (
    UBYTE * s ) [extern]
```

Definition at line [3770](#) of file [pre.c](#).

**4.19.6.261 DoPreWrite()**

```
int DoPreWrite (
    UBYTE * s ) [extern]
```

Definition at line [3964](#) of file [pre.c](#).

**4.19.6.262 DoPreClose()**

```
int DoPreClose (
    UBYTE * s ) [extern]
```

Definition at line [3920](#) of file [pre.c](#).

**4.19.6.263 DoPreRemove()**

```
int DoPreRemove (
    UBYTE * s ) [extern]
```

Definition at line [3887](#) of file [pre.c](#).

#### 4.19.6.264 DoPreSortReallocate()

```
int DoPreSortReallocate (
    UBYTE * s ) [extern]
```

Definition at line [3784](#) of file [pre.c](#).

#### 4.19.6.265 DoCommentChar()

```
int DoCommentChar (
    UBYTE * s ) [extern]
```

Definition at line [2097](#) of file [pre.c](#).

#### 4.19.6.266 DoPrcExtension()

```
int DoPrcExtension (
    UBYTE * s ) [extern]
```

Definition at line [3713](#) of file [pre.c](#).

#### 4.19.6.267 DoPreReset()

```
int DoPreReset (
    UBYTE * s ) [extern]
```

Definition at line [6995](#) of file [pre.c](#).

#### 4.19.6.268 WriteString()

```
void WriteString (
    int type,
    UBYTE * str,
    int num ) [extern]
```

Definition at line [1807](#) of file [tools.c](#).

#### 4.19.6.269 WriteUnfinString()

```
void WriteUnfinString (
    int type,
    UBYTE * str,
    int num ) [extern]
```

Definition at line [1841](#) of file [tools.c](#).

**4.19.6.270 AddToString()**

```
UBYTE * AddToString (
    UBYTE * outstring,
    UBYTE * extrastring,
    int par ) [extern]
```

Definition at line [1866](#) of file [tools.c](#).

**4.19.6.271 PreCalc()**

```
UBYTE * PreCalc (
    void ) [extern]
```

Definition at line [5113](#) of file [pre.c](#).

**4.19.6.272 PreEval()**

```
UBYTE * PreEval (
    UBYTE * s,
    LONG * x ) [extern]
```

Definition at line [5191](#) of file [pre.c](#).

**4.19.6.273 NumToStr()**

```
void NumToStr (
    UBYTE * s,
    LONG x ) [extern]
```

Definition at line [1783](#) of file [tools.c](#).

**4.19.6.274 PreCmp()**

```
int PreCmp (
    int type,
    int val,
    UBYTE * t,
    int type2,
    int val2,
    UBYTE * t2,
    int cmpop ) [extern]
```

Definition at line [4624](#) of file [pre.c](#).

**4.19.6.275 PreEq()**

```
int PreEq (
    int type,
    int val,
    UBYTE * t,
    int type2,
    int val2,
    UBYTE * t2,
    int egop ) [extern]
```

Definition at line [4648](#) of file [pre.c](#).

**4.19.6.276 pParseObject()**

```
UBYTE * pParseObject (
    UBYTE * s,
    int * type,
    LONG * val2 ) [extern]
```

Definition at line [4677](#) of file [pre.c](#).

**4.19.6.277 PreIfEval()**

```
UBYTE * PreIfEval (
    UBYTE * s,
    int * value ) [extern]
```

Definition at line [4495](#) of file [pre.c](#).

**4.19.6.278 EvalPreIf()**

```
int EvalPreIf (
    UBYTE * s ) [extern]
```

Definition at line [4460](#) of file [pre.c](#).

**4.19.6.279 PreLoad()**

```
int PreLoad (
    PRELOAD * p,
    UBYTE * start,
    UBYTE * stop,
    int mode,
    char * message ) [extern]
```

Definition at line [4294](#) of file [pre.c](#).



**4.19.6.280 PreSkip()**

```
int PreSkip (
    UBYTE * start,
    UBYTE * stop,
    int mode ) [extern]
```

Definition at line [4377](#) of file [pre.c](#).

**4.19.6.281 EndOfToken()**

```
UBYTE * EndOfToken (
    UBYTE * s ) [extern]
```

Definition at line [1919](#) of file [tools.c](#).

**4.19.6.282 SetSpecialMode()**

```
void SetSpecialMode (
    int moduletype,
    int specialtype ) [extern]
```

Definition at line [257](#) of file [module.c](#).

**4.19.6.283 MakeGlobal()**

```
void MakeGlobal (
    void ) [extern]
```

Definition at line [383](#) of file [execute.c](#).

**4.19.6.284 ExecModule()**

```
int ExecModule (
    int moduletype ) [extern]
```

Definition at line [274](#) of file [module.c](#).

**4.19.6.285 ExecStore()**

```
int ExecStore (
    void ) [extern]
```

Definition at line [299](#) of file [module.c](#).

#### 4.19.6.286 FullCleanUp()

```
void FullCleanUp (  
    void ) [extern]
```

Definition at line [314](#) of file [module.c](#).

#### 4.19.6.287 DoExecStatement()

```
int DoExecStatement (  
    void ) [extern]
```

Definition at line [780](#) of file [module.c](#).

#### 4.19.6.288 DoPipeStatement()

```
int DoPipeStatement (  
    void ) [extern]
```

Definition at line [797](#) of file [module.c](#).

#### 4.19.6.289 DoPolyfun()

```
int DoPolyfun (  
    UBYTE * s ) [extern]
```

Definition at line [395](#) of file [module.c](#).

#### 4.19.6.290 DoPolyratfun()

```
int DoPolyratfun (  
    UBYTE * s ) [extern]
```

Definition at line [458](#) of file [module.c](#).

#### 4.19.6.291 CompileStatement()

```
int CompileStatement (  
    UBYTE * in ) [extern]
```

Definition at line [552](#) of file [compiler.c](#).

#### 4.19.6.292 ToToken()

```
UBYTE * ToToken (  
    UBYTE * s ) [extern]
```

Definition at line [1931](#) of file [tools.c](#).

**4.19.6.293 GetDollar()**

```
int GetDollar (
    UBYTE * name ) [extern]
```

Definition at line [590](#) of file [names.c](#).

**4.19.6.294 MesWork()**

```
int MesWork (
    void ) [extern]
```

Definition at line [89](#) of file [message.c](#).

**4.19.6.295 MesCall()**

```
int MesCall (
    char * s ) [extern]
```

Definition at line [814](#) of file [message.c](#).

**4.19.6.296 NumCopy()**

```
UBYTE * NumCopy (
    WORD y,
    UBYTE * to ) [extern]
```

Definition at line [1997](#) of file [tools.c](#).

**4.19.6.297 LongCopy()**

```
char * LongCopy (
    LONG y,
    char * to ) [extern]
```

Definition at line [2024](#) of file [tools.c](#).

**4.19.6.298 LongLongCopy()**

```
char * LongLongCopy (
    off_t * y,
    char * to ) [extern]
```

Definition at line [2051](#) of file [tools.c](#).

#### 4.19.6.299 ReserveTempFiles()

```
void ReserveTempFiles (
    int par ) [extern]
```

Definition at line [666](#) of file [startup.c](#).

#### 4.19.6.300 PrintTerm()

```
void PrintTerm (
    WORD * term,
    char * where ) [extern]
```

Definition at line [857](#) of file [message.c](#).

#### 4.19.6.301 PrintTermC()

```
void PrintTermC (
    WORD * term,
    char * where ) [extern]
```

Definition at line [887](#) of file [message.c](#).

#### 4.19.6.302 PrintSubTerm()

```
void PrintSubTerm (
    WORD * term,
    char * where ) [extern]
```

Definition at line [921](#) of file [message.c](#).

#### 4.19.6.303 PrintWords()

```
void PrintWords (
    WORD * buffer,
    LONG number ) [extern]
```

Definition at line [943](#) of file [message.c](#).

#### 4.19.6.304 PrintSeq()

```
void PrintSeq (
    WORD * a,
    char * text ) [extern]
```

Definition at line [961](#) of file [message.c](#).

**4.19.6.305 ExpandTripleDots()**

```
int ExpandTripleDots (
    int par ) [extern]
```

Definition at line 1595 of file [pre.c](#).

**4.19.6.306 ComPress()**

```
LONG ComPress (
    WORD ** ss,
    LONG * n ) [extern]
```

Gets a list of pointers to terms and compresses the terms. In *n* it collects the number of terms and the return value of the function is the space that is occupied.

We have to pay some special attention to the compression of terms with a PolyFun. This PolyFun should occur only straight before the coefficient, so we can use the same trick as for the coefficient to sabotage compression of this object (Replace in the history the function pointer by zero. This is safe, because terms that would be identical otherwise would have been added).

**Parameters**

|           |                                              |
|-----------|----------------------------------------------|
| <i>ss</i> | Array of pointers to terms to be compressed. |
| <i>n</i>  | Number of pointers in <i>ss</i> .            |

**Returns**

Total number of words needed for the compressed result.

Definition at line 3180 of file [sort.c](#).

Referenced by [EndSort\(\)](#), and [StoreTerm\(\)](#).

**4.19.6.307 StageSort()**

```
void StageSort (
    FILEHANDLE * fout ) [extern]
```

Prepares a stage 4 or higher sort. Stage 4 sorts occur when the sort file contains more patches than can be merged in one pass.

Definition at line 4638 of file [sort.c](#).

References [FiLe::handle](#).

Referenced by [EndSort\(\)](#), and [MergePatches\(\)](#).

**4.19.6.308 M\_free()**

```
void M_free (
    void * x,
    const char * where ) [extern]
```

Definition at line [2302](#) of file [tools.c](#).

**4.19.6.309 ClearWildcardNames()**

```
void ClearWildcardNames (
    void ) [extern]
```

Definition at line [849](#) of file [names.c](#).

**4.19.6.310 AddWildcardName()**

```
int AddWildcardName (
    UBYTE * name ) [extern]
```

Definition at line [854](#) of file [names.c](#).

**4.19.6.311 GetWildcardName()**

```
int GetWildcardName (
    UBYTE * name ) [extern]
```

Definition at line [898](#) of file [names.c](#).

**4.19.6.312 Globalize()**

```
void Globalize (
    int par ) [extern]
```

Definition at line [3155](#) of file [names.c](#).

**4.19.6.313 ResetVariables()**

```
void ResetVariables (
    int par ) [extern]
```

Definition at line [2832](#) of file [names.c](#).

**4.19.6.314 AddToPreTypes()**

```
void AddToPreTypes (
    int type ) [extern]
```

Definition at line [5341](#) of file [pre.c](#).

#### 4.19.6.315 MessPreNesting()

```
void MessPreNesting (
    int par ) [extern]
```

Definition at line 5359 of file [pre.c](#).

#### 4.19.6.316 GetStreamPosition()

```
LONG GetStreamPosition (
    STREAM * stream ) [extern]
```

Definition at line 813 of file [tools.c](#).

#### 4.19.6.317 DoubleCbuffer()

```
WORD * DoubleCbuffer (
    int num,
    WORD * w,
    int par ) [extern]
```

Doubles a compiler buffer.

##### Parameters

|            |                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------|
| <i>num</i> | The ID number for the buffer to be doubled.                                                                                |
| <i>w</i>   | The pointer to the end (exclusive) of the current buffer. The contents in the range of [cbuff[num].Buffer,w) will be kept. |

Definition at line 143 of file [comtool.c](#).

References [CbUf::Buffer](#), [CbUf::BufferSize](#), [CbUf::lhs](#), [CbUf::Pointer](#), [CbUf::rhs](#), and [CbUf::Top](#).

Referenced by [AddNtoC\(\)](#), and [AddNtoL\(\)](#).

#### 4.19.6.318 AddLHS()

```
WORD * AddLHS (
    int num ) [extern]
```

Adds an LHS to a compiler buffer and returns the pointer to a buffer for the new LHS.

##### Parameters

|            |                                                  |
|------------|--------------------------------------------------|
| <i>num</i> | The ID number for the buffer to get another LHS. |
|------------|--------------------------------------------------|

Definition at line 188 of file [comtool.c](#).

References [CbUf::lhs](#), and [CbUf::Pointer](#).

Referenced by [AddNtoL\(\)](#).

#### 4.19.6.319 AddRHS()

```
WORD * AddRHS (
    int num,
    int type ) [extern]
```

Adds an RHS to a compiler buffer and returns the pointer to a buffer for the new RHS.

##### Parameters

|             |                                                   |
|-------------|---------------------------------------------------|
| <i>num</i>  | The ID number for the buffer to get another RHS.  |
| <i>type</i> | If 0, the subexpression tree will be reallocated. |

Definition at line 214 of file [comtool.c](#).

References [TaBIEs::buffers](#), [TaBIEs::buffersfill](#), [TaBIEs::bufferssize](#), [TaBIEs::bufnum](#), [CbUf::CanCommu](#), [CbUf::dimension](#), [inicbufs\(\)](#), [CbUf::numdum](#), [CbUf::NumTerms](#), [CbUf::Pointer](#), and [CbUf::rhs](#).

Referenced by [InsertArg\(\)](#), [StartVariables\(\)](#), and [TestMatch\(\)](#).

Here is the call graph for this function:



#### 4.19.6.320 AddNtoL()

```
int AddNtoL (
    int n,
    WORD * array ) [extern]
```

Adds an LHS with the given data to the current compiler buffer.

##### Parameters

|              |                         |
|--------------|-------------------------|
| <i>n</i>     | The length of the data. |
| <i>array</i> | The data to be added.   |

##### Returns

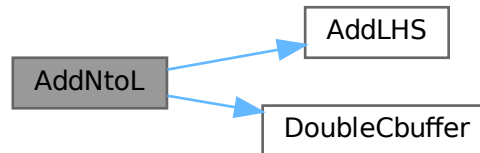
0 if succeeds.



Definition at line 288 of file [comtool.c](#).

References [AddLHS\(\)](#), [DoubleCbuffer\(\)](#), [CbUf::Pointer](#), and [CbUf::Top](#).

Here is the call graph for this function:



#### 4.19.6.321 AddNtoC()

```

int AddNtoC (
    int bufnum,
    int n,
    WORD * array,
    int par ) [extern]
  
```

Adds the given data to the last LHS/RHS in a compiler buffer.

##### Parameters

|               |                                                            |
|---------------|------------------------------------------------------------|
| <i>bufnum</i> | The ID number for the buffer where the data will be added. |
| <i>n</i>      | The length of the data.                                    |
| <i>array</i>  | The data to be added.                                      |

##### Returns

0 if succeeds.

Definition at line 317 of file [comtool.c](#).

References [DoubleCbuffer\(\)](#), [CbUf::Pointer](#), and [CbUf::Top](#).

Referenced by [InsertArg\(\)](#), and [StartVariables\(\)](#).

Here is the call graph for this function:



#### 4.19.6.322 DoubleIfBuffers()

```
void DoubleIfBuffers (
    void ) [extern]
```

Definition at line [1033](#) of file [if.c](#).

#### 4.19.6.323 CreateStream()

```
STREAM * CreateStream (
    UBYTE * where ) [extern]
```

Definition at line [782](#) of file [tools.c](#).

#### 4.19.6.324 setonoff()

```
int setonoff (
    UBYTE * s,
    int * flag,
    int onvalue,
    int offvalue ) [extern]
```

Definition at line [490](#) of file [compcomm.c](#).

#### 4.19.6.325 DoPrint()

```
int DoPrint (
    UBYTE * s,
    int par ) [extern]
```

Definition at line [1041](#) of file [compcomm.c](#).

#### 4.19.6.326 SetExpr()

```
int SetExpr (
    UBYTE * s,
    int setunset,
    int par ) [extern]
```

Definition at line [1512](#) of file [compcomm.c](#).

#### 4.19.6.327 AddToCom()

```
void AddToCom (
    int n,
    WORD * array ) [extern]
```

Definition at line [1649](#) of file [compcomm.c](#).

**4.19.6.328 Add2ComStrings()**

```
int Add2ComStrings (
    int n,
    WORD * array,
    UBYTE * string1,
    UBYTE * string2 ) [extern]
```

Definition at line 1723 of file [compcomm.c](#).

**4.19.6.329 DoSymmetrize()**

```
int DoSymmetrize (
    UBYTE * s,
    int par ) [extern]
```

Definition at line 2217 of file [compcomm.c](#).

**4.19.6.330 DoArgument()**

```
int DoArgument (
    UBYTE * s,
    int par ) [extern]
```

Definition at line 1862 of file [compcomm.c](#).

**4.19.6.331 ArgFactorize()**

```
int ArgFactorize (
    PHEAD WORD * argin,
    WORD * argout ) [extern]
```

Definition at line 2061 of file [argument.c](#).

**4.19.6.332 TakeArgContent()**

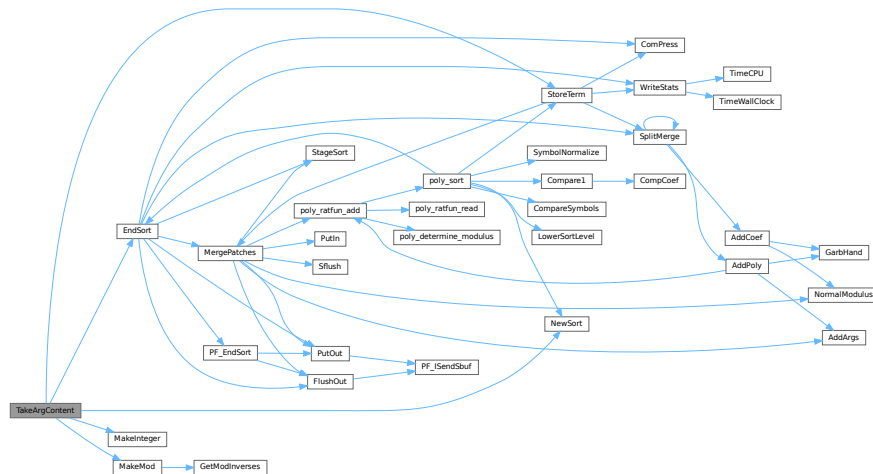
```
WORD * TakeArgContent (
    PHEAD WORD * argin,
    WORD * argout ) [extern]
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. The common pieces are put in argout as a sequence of arguments. The part with the multiple terms that are now relative prime is put in argfree which is allocated via TermMalloc and is given as the return value. The difference with the old code is that negative powers are always removed. Hence it is as in MakeInteger in which only numerators will be left: now only zero or positive powers will be remaining.

Definition at line 2767 of file [argument.c](#).

References [EndSort\(\)](#), [MakeInteger\(\)](#), [MakeMod\(\)](#), [NewSort\(\)](#), and [StoreTerm\(\)](#).

Here is the call graph for this function:



#### 4.19.6.333 MakeInteger()

```
WORD * MakeInteger (
    PHEAD WORD * argin,
    WORD * argout,
    WORD * argfree ) [extern]
```

For normalizing everything to integers we have to determine for all elements of this argument the LCM of the denominators and the GCD of the numerators. The input argument is in argin. The number that comes out should go to argout. The new pointer in the argout buffer is the return value. The normalized argument is in argfree.

Definition at line 3313 of file [argument.c](#).

Referenced by [TakeArgContent\(\)](#).

#### 4.19.6.334 MakeMod()

```
WORD * MakeMod (
    PHEAD WORD * argin,
    WORD * argout,
    WORD * argfree ) [extern]
```

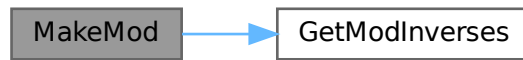
Similar to MakeInteger but now with modulus arithmetic using only a one WORD 'prime'. We make the coefficient of the first term in the argument equal to one. Already the coefficients are taken modulus AN.cmod and AN.ncmod == 1

Definition at line 3484 of file [argument.c](#).

References [GetModInverses\(\)](#).

Referenced by [TakeArgContent\(\)](#).

Here is the call graph for this function:



#### 4.19.6.335 FindArg()

```
WORD FindArg (  
    PHEAD WORD * a ) [extern]
```

Looks the argument up in the (workers) table. If it is found the number in the table is returned (plus one to make it positive). If it is not found we look in the compiler provided table. If it is found - the number in the table is returned (minus one to make it negative). If in neither table we return zero.

Definition at line 2514 of file [argument.c](#).

#### 4.19.6.336 InsertArg()

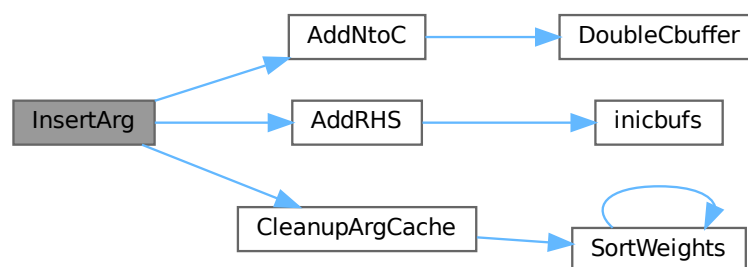
```
int InsertArg (  
    PHEAD WORD * argin,  
    WORD * argout,  
    int par ) [extern]
```

Inserts the argument into the (workers) table. If the table is too full we eliminate half of it. The eliminated elements are the ones that have not been used most recently, weighted by their total use and age(?). If par == 0 it inserts in the regular factorization cache If par == 1 it inserts in the cache defined with the FactorCache statement

Definition at line 2538 of file [argument.c](#).

References [AddNtoC\(\)](#), [AddRHS\(\)](#), and [CleanupArgCache\(\)](#).

Here is the call graph for this function:



#### 4.19.6.337 CleanupArgCache()

```
int CleanupArgCache (
    PHEAD WORD bufnum ) [extern]
```

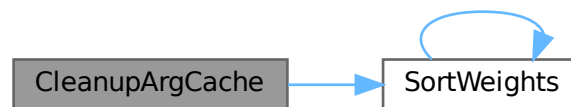
Cleans up the argument factorization cache. We throw half the elements. For a weight of what we want to keep we use the product of usage and the number in the buffer.

Definition at line 2573 of file [argument.c](#).

References [CbUf::boomlijst](#), [CbUf::Buffer](#), [CbUf::Pointer](#), [CbUf::rhs](#), [SortWeights\(\)](#), and [tree::usage](#).

Referenced by [InsertArg\(\)](#).

Here is the call graph for this function:



#### 4.19.6.338 ArgSymbolMerge()

```
int ArgSymbolMerge (
    WORD * t1,
    WORD * t2 ) [extern]
```

Definition at line 2644 of file [argument.c](#).

#### 4.19.6.339 ArgDotproductMerge()

```
int ArgDotproductMerge (
    WORD * t1,
    WORD * t2 ) [extern]
```

Definition at line 2699 of file [argument.c](#).

#### 4.19.6.340 SortWeights()

```
void SortWeights (
    LONG * weights,
    LONG * extraspace,
    WORD number ) [extern]
```

Sorts an array of LONGS in the same way SplitMerge (in [sort.c](#)) works We use gradual division in two.

Definition at line [3529](#) of file [argument.c](#).

References [SortWeights\(\)](#).

Referenced by [CleanupArgCache\(\)](#), and [SortWeights\(\)](#).

Here is the call graph for this function:



#### 4.19.6.341 DoBrackets()

```
int DoBrackets (
    UBYTE * inp,
    int par ) [extern]
```

Definition at line [3752](#) of file [compcomm.c](#).

#### 4.19.6.342 DoPutInside()

```
int DoPutInside (
    UBYTE * inp,
    int par ) [extern]
```

Definition at line [7047](#) of file [compcomm.c](#).

#### 4.19.6.343 CountComp()

```
WORD * CountComp (
    UBYTE * inp,
    WORD * to ) [extern]
```

Definition at line [4030](#) of file [compcomm.c](#).

#### 4.19.6.344 CoAntiBracket()

```
int CoAntiBracket (
    UBYTE * inp ) [extern]
```

Definition at line 3887 of file [compcomm.c](#).

#### 4.19.6.345 CoAntiSymmetrize()

```
int CoAntiSymmetrize (
    UBYTE * s ) [extern]
```

Definition at line 2344 of file [compcomm.c](#).

#### 4.19.6.346 DoArgPlode()

```
int DoArgPlode (
    UBYTE * s,
    int par ) [extern]
```

Definition at line 5707 of file [compcomm.c](#).

#### 4.19.6.347 CoArgExplode()

```
int CoArgExplode (
    UBYTE * s ) [extern]
```

Definition at line 5763 of file [compcomm.c](#).

#### 4.19.6.348 CoArgImplode()

```
int CoArgImplode (
    UBYTE * s ) [extern]
```

Definition at line 5770 of file [compcomm.c](#).

#### 4.19.6.349 CoArgument()

```
int CoArgument (
    UBYTE * s ) [extern]
```

Definition at line 2096 of file [compcomm.c](#).

#### 4.19.6.350 CoInside()

```
int CoInside (
    UBYTE * s ) [extern]
```

Definition at line 2129 of file [compcomm.c](#).



**4.19.6.351 ExecInside()**

```
int ExecInside (
    UBYTE * s ) [extern]
```

Definition at line 1681 of file [dollar.c](#).

**4.19.6.352 CoInExpression()**

```
int CoInExpression (
    UBYTE * s ) [extern]
```

Definition at line 3357 of file [compcomm.c](#).

**4.19.6.353 CoInParallel()**

```
int CoInParallel (
    UBYTE * s ) [extern]
```

Definition at line 3263 of file [compcomm.c](#).

**4.19.6.354 CoNotInParallel()**

```
int CoNotInParallel (
    UBYTE * s ) [extern]
```

Definition at line 3273 of file [compcomm.c](#).

**4.19.6.355 DoInParallel()**

```
int DoInParallel (
    UBYTE * s,
    int par ) [extern]
```

Definition at line 3288 of file [compcomm.c](#).

**4.19.6.356 CoEndInExpression()**

```
int CoEndInExpression (
    UBYTE * s ) [extern]
```

Definition at line 3410 of file [compcomm.c](#).

**4.19.6.357 CoBracket()**

```
int CoBracket (
    UBYTE * inp ) [extern]
```

Definition at line 3879 of file [compcomm.c](#).

#### 4.19.6.358 CoPutInside()

```
int CoPutInside (
    UBYTE * inp ) [extern]
```

Definition at line [7044](#) of file [compcomm.c](#).

#### 4.19.6.359 CoAntiPutInside()

```
int CoAntiPutInside (
    UBYTE * inp ) [extern]
```

Definition at line [7045](#) of file [compcomm.c](#).

#### 4.19.6.360 CoMultiBracket()

```
int CoMultiBracket (
    UBYTE * inp ) [extern]
```

Definition at line [3898](#) of file [compcomm.c](#).

#### 4.19.6.361 CoCFunction()

```
int CoCFunction (
    UBYTE * s ) [extern]
```

Definition at line [1625](#) of file [names.c](#).

#### 4.19.6.362 CoCTensor()

```
int CoCTensor (
    UBYTE * s ) [extern]
```

Definition at line [1627](#) of file [names.c](#).

#### 4.19.6.363 CoCollect()

```
int CoCollect (
    UBYTE * s ) [extern]
```

Definition at line [428](#) of file [compcomm.c](#).

#### 4.19.6.364 CoCompress()

```
int CoCompress (
    UBYTE * s ) [extern]
```

Definition at line [506](#) of file [compcomm.c](#).

**4.19.6.365 CoContract()**

```
int CoContract (
    UBYTE * s ) [extern]
```

Definition at line 1787 of file [compcomm.c](#).

**4.19.6.366 CoCycleSymmetrize()**

```
int CoCycleSymmetrize (
    UBYTE * s ) [extern]
```

Definition at line 2351 of file [compcomm.c](#).

**4.19.6.367 CoDelete()**

```
int CoDelete (
    UBYTE * s ) [extern]
```

Definition at line 942 of file [compcomm.c](#).

**4.19.6.368 CoTableBase()**

```
int CoTableBase (
    UBYTE * s ) [extern]
```

Definition at line 627 of file [tables.c](#).

**4.19.6.369 CoApply()**

```
int CoApply (
    UBYTE * s ) [extern]
```

Definition at line 1797 of file [tables.c](#).

**4.19.6.370 CoDenominators()**

```
int CoDenominators (
    UBYTE * s ) [extern]
```

Definition at line 5859 of file [compcomm.c](#).

**4.19.6.371 CoDimension()**

```
int CoDimension (
    UBYTE * s ) [extern]
```

Definition at line 1226 of file [names.c](#).

**4.19.6.372 CoDiscard()**

```
int CoDiscard (
    UBYTE * s ) [extern]
```

Definition at line 1764 of file [compcomm.c](#).

**4.19.6.373 CoDisorder()**

```
int CoDisorder (
    UBYTE * inp ) [extern]
```

Definition at line 417 of file [comexpr.c](#).

**4.19.6.374 CoDrop()**

```
int CoDrop (
    UBYTE * s ) [extern]
```

Definition at line 1557 of file [compcomm.c](#).

**4.19.6.375 CoDropCoefficient()**

```
int CoDropCoefficient (
    UBYTE * s ) [extern]
```

Definition at line 5888 of file [compcomm.c](#).

**4.19.6.376 CoDropSymbols()**

```
int CoDropSymbols (
    UBYTE * s ) [extern]
```

Definition at line 5902 of file [compcomm.c](#).

**4.19.6.377 CoElse()**

```
int CoElse (
    UBYTE * p ) [extern]
```

Definition at line 4862 of file [compcomm.c](#).

**4.19.6.378 CoElseIf()**

```
int CoElseIf (
    UBYTE * inp ) [extern]
```

Definition at line 4889 of file [compcomm.c](#).

**4.19.6.379 CoEndArgument()**

```
int CoEndArgument (
    UBYTE * s ) [extern]
```

Definition at line [2103](#) of file [compcomm.c](#).

**4.19.6.380 CoEndInside()**

```
int CoEndInside (
    UBYTE * s ) [extern]
```

Definition at line [2136](#) of file [compcomm.c](#).

**4.19.6.381 CoEndIf()**

```
int CoEndIf (
    UBYTE * inp ) [extern]
```

Definition at line [4916](#) of file [compcomm.c](#).

**4.19.6.382 CoEndRepeat()**

```
int CoEndRepeat (
    UBYTE * inp ) [extern]
```

Definition at line [3714](#) of file [compcomm.c](#).

**4.19.6.383 CoEndTerm()**

```
int CoEndTerm (
    UBYTE * s ) [extern]
```

Definition at line [5265](#) of file [compcomm.c](#).

**4.19.6.384 CoEndWhile()**

```
int CoEndWhile (
    UBYTE * inp ) [extern]
```

Definition at line [4982](#) of file [compcomm.c](#).

**4.19.6.385 CoExit()**

```
int CoExit (
    UBYTE * s ) [extern]
```

Definition at line [3236](#) of file [compcomm.c](#).

**4.19.6.386 CoFactArg()**

```
int CoFactArg (  
    UBYTE * s ) [extern]
```

Definition at line [2197](#) of file [compcomm.c](#).

**4.19.6.387 CoFactDollar()**

```
int CoFactDollar (  
    UBYTE * inp ) [extern]
```

Definition at line [6418](#) of file [compcomm.c](#).

**4.19.6.388 CoFactorize()**

```
int CoFactorize (  
    UBYTE * s ) [extern]
```

Definition at line [6447](#) of file [compcomm.c](#).

**4.19.6.389 CoNFactorize()**

```
int CoNFactorize (  
    UBYTE * s ) [extern]
```

Definition at line [6454](#) of file [compcomm.c](#).

**4.19.6.390 CoUnFactorize()**

```
int CoUnFactorize (  
    UBYTE * s ) [extern]
```

Definition at line [6461](#) of file [compcomm.c](#).

**4.19.6.391 CoNUnFactorize()**

```
int CoNUnFactorize (  
    UBYTE * s ) [extern]
```

Definition at line [6468](#) of file [compcomm.c](#).

**4.19.6.392 DoFactorize()**

```
int DoFactorize (  
    UBYTE * s,  
    int par ) [extern]
```

Definition at line [6475](#) of file [compcomm.c](#).

**4.19.6.393 CoFill()**

```
int CoFill (
    UBYTE * inp ) [extern]
```

Definition at line [1070](#) of file [comexpr.c](#).

**4.19.6.394 CoFillExpression()**

```
int CoFillExpression (
    UBYTE * inp ) [extern]
```

Definition at line [1356](#) of file [comexpr.c](#).

**4.19.6.395 CoFixIndex()**

```
int CoFixIndex (
    UBYTE * s ) [extern]
```

Definition at line [999](#) of file [compcomm.c](#).

**4.19.6.396 CoFormat()**

```
int CoFormat (
    UBYTE * s ) [extern]
```

Definition at line [153](#) of file [compcomm.c](#).

**4.19.6.397 CoGlobal()**

```
int CoGlobal (
    UBYTE * inp ) [extern]
```

Definition at line [73](#) of file [comexpr.c](#).

**4.19.6.398 CoGlobalFactorized()**

```
int CoGlobalFactorized (
    UBYTE * inp ) [extern]
```

Definition at line [87](#) of file [comexpr.c](#).

**4.19.6.399 CoGoTo()**

```
int CoGoTo (
    UBYTE * inp ) [extern]
```

Definition at line [1816](#) of file [compcomm.c](#).

#### 4.19.6.400 Cold()

```
int CoId (
    UBYTE * inp ) [extern]
```

Definition at line [395](#) of file [comexpr.c](#).

#### 4.19.6.401 ColdNew()

```
int CoIdNew (
    UBYTE * inp ) [extern]
```

Definition at line [406](#) of file [comexpr.c](#).

#### 4.19.6.402 ColdOld()

```
int CoIdOld (
    UBYTE * inp ) [extern]
```

Definition at line [384](#) of file [comexpr.c](#).

#### 4.19.6.403 Colf()

```
int CoIf (
    UBYTE * inp ) [extern]
```

Definition at line [4220](#) of file [compcomm.c](#).

#### 4.19.6.404 ColfMatch()

```
int CoIfMatch (
    UBYTE * inp ) [extern]
```

Definition at line [450](#) of file [comexpr.c](#).

#### 4.19.6.405 ColfNoMatch()

```
int CoIfNoMatch (
    UBYTE * inp ) [extern]
```

Definition at line [461](#) of file [comexpr.c](#).

#### 4.19.6.406 CoIndex()

```
int CoIndex (
    UBYTE * s ) [extern]
```

Definition at line [1112](#) of file [names.c](#).



**4.19.6.407 CoInsideFirst()**

```
int CoInsideFirst (
    UBYTE * s ) [extern]
```

Definition at line 928 of file [compcomm.c](#).

**4.19.6.408 CoKeep()**

```
int CoKeep (
    UBYTE * s ) [extern]
```

Definition at line 987 of file [compcomm.c](#).

**4.19.6.409 CoLabel()**

```
int CoLabel (
    UBYTE * inp ) [extern]
```

Definition at line 1835 of file [compcomm.c](#).

**4.19.6.410 CoLoad()**

```
int CoLoad (
    UBYTE * inp ) [extern]
```

Definition at line 572 of file [store.c](#).

**4.19.6.411 CoLocal()**

```
int CoLocal (
    UBYTE * inp ) [extern]
```

Definition at line 66 of file [comexpr.c](#).

**4.19.6.412 CoLocalFactorized()**

```
int CoLocalFactorized (
    UBYTE * inp ) [extern]
```

Definition at line 80 of file [comexpr.c](#).

**4.19.6.413 CoMany()**

```
int CoMany (
    UBYTE * inp ) [extern]
```

Definition at line 428 of file [comexpr.c](#).

**4.19.6.414 CoMerge()**

```
int CoMerge (
    UBYTE * inp ) [extern]
```

Definition at line [5548](#) of file [compcomm.c](#).

**4.19.6.415 CoStuffle()**

```
int CoStuffle (
    UBYTE * inp ) [extern]
```

Definition at line [5604](#) of file [compcomm.c](#).

**4.19.6.416 CoMetric()**

```
int CoMetric (
    UBYTE * s ) [extern]
```

Definition at line [1033](#) of file [compcomm.c](#).

**4.19.6.417 CoModOption()**

```
int CoModOption (
    UBYTE * s ) [extern]
```

Definition at line [212](#) of file [module.c](#).

**4.19.6.418 CoModuleOption()**

```
int CoModuleOption (
    UBYTE * s ) [extern]
```

Definition at line [143](#) of file [module.c](#).

**4.19.6.419 CoModulus()**

```
int CoModulus (
    UBYTE * inp ) [extern]
```

Definition at line [3553](#) of file [compcomm.c](#).

**4.19.6.420 CoMulti()**

```
int CoMulti (
    UBYTE * inp ) [extern]
```

Definition at line [439](#) of file [comexpr.c](#).

**4.19.6.421 CoMultiply()**

```
int CoMultiply (
    UBYTE * inp ) [extern]
```

Definition at line [1031](#) of file [comexpr.c](#).

**4.19.6.422 CoNFunction()**

```
int CoNFunction (
    UBYTE * s ) [extern]
```

Definition at line [1624](#) of file [names.c](#).

**4.19.6.423 CoNPrint()**

```
int CoNPrint (
    UBYTE * s ) [extern]
```

Definition at line [1271](#) of file [compcomm.c](#).

**4.19.6.424 CoNTensor()**

```
int CoNTensor (
    UBYTE * s ) [extern]
```

Definition at line [1626](#) of file [names.c](#).

**4.19.6.425 CoNWrite()**

```
int CoNWrite (
    UBYTE * s ) [extern]
```

Definition at line [2390](#) of file [compcomm.c](#).

**4.19.6.426 CoNoDrop()**

```
int CoNoDrop (
    UBYTE * s ) [extern]
```

Definition at line [1564](#) of file [compcomm.c](#).

**4.19.6.427 CoNoSkip()**

```
int CoNoSkip (
    UBYTE * s ) [extern]
```

Definition at line [1578](#) of file [compcomm.c](#).

**4.19.6.428 CoNormalize()**

```
int CoNormalize (
    UBYTE * s ) [extern]
```

Definition at line [2162](#) of file [compcomm.c](#).

**4.19.6.429 CoMakeInteger()**

```
int CoMakeInteger (
    UBYTE * s ) [extern]
```

Definition at line [2169](#) of file [compcomm.c](#).

**4.19.6.430 CoFlags()**

```
int CoFlags (
    UBYTE * s,
    int value ) [extern]
```

Definition at line [555](#) of file [compcomm.c](#).

**4.19.6.431 CoOff()**

```
int CoOff (
    UBYTE * s ) [extern]
```

Definition at line [590](#) of file [compcomm.c](#).

**4.19.6.432 CoOn()**

```
int CoOn (
    UBYTE * s ) [extern]
```

Definition at line [656](#) of file [compcomm.c](#).

**4.19.6.433 CoOnce()**

```
int CoOnce (
    UBYTE * inp ) [extern]
```

Definition at line [472](#) of file [comexpr.c](#).

**4.19.6.434 CoOnly()**

```
int CoOnly (
    UBYTE * inp ) [extern]
```

Definition at line [483](#) of file [comexpr.c](#).

**4.19.6.435 CoOptimizeOption()**

```
int CoOptimizeOption (
    UBYTE * s ) [extern]
```

Definition at line 6608 of file [compcomm.c](#).

**4.19.6.436 CoPolyFun()**

```
int CoPolyFun (
    UBYTE * s ) [extern]
```

Definition at line 5331 of file [compcomm.c](#).

**4.19.6.437 CoPolyRatFun()**

```
int CoPolyRatFun (
    UBYTE * s ) [extern]
```

Definition at line 5378 of file [compcomm.c](#).

**4.19.6.438 CoPrint()**

```
int CoPrint (
    UBYTE * s ) [extern]
```

Definition at line 1257 of file [compcomm.c](#).

**4.19.6.439 CoPrintB()**

```
int CoPrintB (
    UBYTE * s ) [extern]
```

Definition at line 1264 of file [compcomm.c](#).

**4.19.6.440 CoProperCount()**

```
int CoProperCount (
    UBYTE * s ) [extern]
```

Definition at line 935 of file [compcomm.c](#).

**4.19.6.441 CoUnitTrace()**

```
int CoUnitTrace (
    UBYTE * s ) [extern]
```

Definition at line 5182 of file [compcomm.c](#).

#### 4.19.6.442 CoRCycleSymmetrize()

```
int CoRCycleSymmetrize (
    UBYTE * s ) [extern]
```

Definition at line [2358](#) of file [compcomm.c](#).

#### 4.19.6.443 CoRatio()

```
int CoRatio (
    UBYTE * s ) [extern]
```

Definition at line [2417](#) of file [compcomm.c](#).

#### 4.19.6.444 CoRedefine()

```
int CoRedefine (
    UBYTE * s ) [extern]
```

Definition at line [2457](#) of file [compcomm.c](#).

#### 4.19.6.445 CoRenumber()

```
int CoRenumber (
    UBYTE * s ) [extern]
```

Definition at line [2562](#) of file [compcomm.c](#).

#### 4.19.6.446 CoRepeat()

```
int CoRepeat (
    UBYTE * inp ) [extern]
```

Definition at line [3691](#) of file [compcomm.c](#).

#### 4.19.6.447 CoSave()

```
int CoSave (
    UBYTE * inp ) [extern]
```

Definition at line [362](#) of file [store.c](#).

#### 4.19.6.448 CoSelect()

```
int CoSelect (
    UBYTE * inp ) [extern]
```

Definition at line [494](#) of file [comexpr.c](#).

**4.19.6.449 CoSet()**

```
int CoSet (
    UBYTE * s ) [extern]
```

Definition at line [2355](#) of file [names.c](#).

**4.19.6.450 CoSetExitFlag()**

```
int CoSetExitFlag (
    UBYTE * s ) [extern]
```

Definition at line [3436](#) of file [compcomm.c](#).

**4.19.6.451 CoSkip()**

```
int CoSkip (
    UBYTE * s ) [extern]
```

Definition at line [1571](#) of file [compcomm.c](#).

**4.19.6.452 CoProcessBucket()**

```
int CoProcessBucket (
    UBYTE * s ) [extern]
```

Definition at line [5659](#) of file [compcomm.c](#).

**4.19.6.453 CoPushHide()**

```
int CoPushHide (
    UBYTE * s ) [extern]
```

Definition at line [1278](#) of file [compcomm.c](#).

**4.19.6.454 CoPopHide()**

```
int CoPopHide (
    UBYTE * s ) [extern]
```

Definition at line [1322](#) of file [compcomm.c](#).

**4.19.6.455 CoHide()**

```
int CoHide (
    UBYTE * inp ) [extern]
```

Definition at line [1585](#) of file [compcomm.c](#).

**4.19.6.456 CoIntoHide()**

```
int CoIntoHide (
    UBYTE * inp ) [extern]
```

Definition at line [1603](#) of file [compcomm.c](#).

**4.19.6.457 CoNoIntoHide()**

```
int CoNoIntoHide (
    UBYTE * inp ) [extern]
```

Definition at line [1621](#) of file [compcomm.c](#).

**4.19.6.458 CoNoHide()**

```
int CoNoHide (
    UBYTE * inp ) [extern]
```

Definition at line [1628](#) of file [compcomm.c](#).

**4.19.6.459 CoUnHide()**

```
int CoUnHide (
    UBYTE * inp ) [extern]
```

Definition at line [1635](#) of file [compcomm.c](#).

**4.19.6.460 CoNoUnHide()**

```
int CoNoUnHide (
    UBYTE * inp ) [extern]
```

Definition at line [1642](#) of file [compcomm.c](#).

**4.19.6.461 CoSort()**

```
int CoSort (
    UBYTE * s ) [extern]
```

Definition at line [5292](#) of file [compcomm.c](#).

**4.19.6.462 CoSplitArg()**

```
int CoSplitArg (
    UBYTE * s ) [extern]
```

Definition at line [2176](#) of file [compcomm.c](#).



**4.19.6.463 CoSplitFirstArg()**

```
int CoSplitFirstArg (  
    UBYTE * s ) [extern]
```

Definition at line 2183 of file [compcomm.c](#).

**4.19.6.464 CoSplitLastArg()**

```
int CoSplitLastArg (  
    UBYTE * s ) [extern]
```

Definition at line 2190 of file [compcomm.c](#).

**4.19.6.465 CoSum()**

```
int CoSum (  
    UBYTE * s ) [extern]
```

Definition at line 2583 of file [compcomm.c](#).

**4.19.6.466 CoSymbol()**

```
int CoSymbol (  
    UBYTE * s ) [extern]
```

Definition at line 946 of file [names.c](#).

**4.19.6.467 CoSymmetrize()**

```
int CoSymmetrize (  
    UBYTE * s ) [extern]
```

Definition at line 2337 of file [compcomm.c](#).

**4.19.6.468 DoTable()**

```
int DoTable (  
    UBYTE * s,  
    int par ) [extern]
```

Definition at line 1666 of file [names.c](#).

**4.19.6.469 CoTable()**

```
int CoTable (  
    UBYTE * s ) [extern]
```

Definition at line 2048 of file [names.c](#).

**4.19.6.470 CoTerm()**

```
int CoTerm (
    UBYTE * s ) [extern]
```

Definition at line 5217 of file [compcomm.c](#).

**4.19.6.471 CoNTable()**

```
int CoNTable (
    UBYTE * s ) [extern]
```

Definition at line 2058 of file [names.c](#).

**4.19.6.472 CoCTable()**

```
int CoCTable (
    UBYTE * s ) [extern]
```

Definition at line 2068 of file [names.c](#).

**4.19.6.473 EmptyTable()**

```
void EmptyTable (
    TABLES T ) [extern]
```

Definition at line 2078 of file [names.c](#).

**4.19.6.474 CoToTensor()**

```
int CoToTensor (
    UBYTE * s ) [extern]
```

Definition at line 2703 of file [compcomm.c](#).

**4.19.6.475 CoToVector()**

```
int CoToVector (
    UBYTE * s ) [extern]
```

Definition at line 2871 of file [compcomm.c](#).

**4.19.6.476 CoTrace4()**

```
int CoTrace4 (
    UBYTE * s ) [extern]
```

Definition at line 2941 of file [compcomm.c](#).

**4.19.6.477 CoTraceN()**

```
int CoTraceN (  
    UBYTE * s ) [extern]
```

Definition at line 3028 of file [compcomm.c](#).

**4.19.6.478 CoChisholm()**

```
int CoChisholm (  
    UBYTE * s ) [extern]
```

Definition at line 3088 of file [compcomm.c](#).

**4.19.6.479 CoTransform()**

```
int CoTransform (  
    UBYTE * in ) [extern]
```

Definition at line 87 of file [transform.c](#).

**4.19.6.480 CoClearTable()**

```
int CoClearTable (  
    UBYTE * s ) [extern]
```

Definition at line 5777 of file [compcomm.c](#).

**4.19.6.481 DoChain()**

```
int DoChain (  
    UBYTE * s,  
    int option ) [extern]
```

Definition at line 3170 of file [compcomm.c](#).

**4.19.6.482 CoChainin()**

```
int CoChainin (  
    UBYTE * s ) [extern]
```

Definition at line 3214 of file [compcomm.c](#).

**4.19.6.483 CoChainout()**

```
int CoChainout (  
    UBYTE * s ) [extern]
```

Definition at line 3226 of file [compcomm.c](#).

**4.19.6.484 CoTryReplace()**

```
int CoTryReplace (
    UBYTE * p ) [extern]
```

Definition at line [3450](#) of file [compcomm.c](#).

**4.19.6.485 CoVector()**

```
int CoVector (
    UBYTE * s ) [extern]
```

Definition at line [1267](#) of file [names.c](#).

**4.19.6.486 CoWhile()**

```
int CoWhile (
    UBYTE * inp ) [extern]
```

Definition at line [4961](#) of file [compcomm.c](#).

**4.19.6.487 CoWrite()**

```
int CoWrite (
    UBYTE * s ) [extern]
```

Definition at line [2365](#) of file [compcomm.c](#).

**4.19.6.488 CoAuto()**

```
int CoAuto (
    UBYTE * inp ) [extern]
```

Definition at line [2572](#) of file [names.c](#).

**4.19.6.489 CoSwitch()**

```
int CoSwitch (
    UBYTE * s ) [extern]
```

Definition at line [7167](#) of file [compcomm.c](#).

**4.19.6.490 CoCase()**

```
int CoCase (
    UBYTE * s ) [extern]
```

Definition at line [7213](#) of file [compcomm.c](#).

**4.19.6.491 CoBreak()**

```
int CoBreak (
    UBYTE * s ) [extern]
```

Definition at line 7263 of file [compcomm.c](#).

**4.19.6.492 CoDefault()**

```
int CoDefault (
    UBYTE * s ) [extern]
```

Definition at line 7289 of file [compcomm.c](#).

**4.19.6.493 CoEndSwitch()**

```
int CoEndSwitch (
    UBYTE * s ) [extern]
```

Definition at line 7316 of file [compcomm.c](#).

**4.19.6.494 CoTBaddto()**

```
int CoTBaddto (
    UBYTE * s ) [extern]
```

Definition at line 801 of file [tables.c](#).

**4.19.6.495 CoTBaudit()**

```
int CoTBaudit (
    UBYTE * s ) [extern]
```

Definition at line 1474 of file [tables.c](#).

**4.19.6.496 CoTbcleanup()**

```
int CoTbcleanup (
    UBYTE * s ) [extern]
```

Definition at line 1576 of file [tables.c](#).

**4.19.6.497 CoTBcreate()**

```
int CoTBcreate (
    UBYTE * s ) [extern]
```

Definition at line 756 of file [tables.c](#).

**4.19.6.498 CoTBenter()**

```
int CoTBenter (
    UBYTE * s ) [extern]
```

Definition at line 974 of file [tables.c](#).

**4.19.6.499 CoTBhelp()**

```
int CoTBhelp (
    UBYTE * s ) [extern]
```

Definition at line 1884 of file [tables.c](#).

**4.19.6.500 CoTBload()**

```
int CoTBload (
    UBYTE * ss ) [extern]
```

Definition at line 1252 of file [tables.c](#).

**4.19.6.501 CoTBoff()**

```
int CoTBoff (
    UBYTE * s ) [extern]
```

Definition at line 1545 of file [tables.c](#).

**4.19.6.502 CoTBon()**

```
int CoTBon (
    UBYTE * s ) [extern]
```

Definition at line 1514 of file [tables.c](#).

**4.19.6.503 CoTBopen()**

```
int CoTBopen (
    UBYTE * s ) [extern]
```

Definition at line 772 of file [tables.c](#).

**4.19.6.504 CoTBreplace()**

```
int CoTBreplace (
    UBYTE * s ) [extern]
```

Definition at line 1588 of file [tables.c](#).

**4.19.6.505 CoTBuse()**

```
int CoTBuse (
    UBYTE * s ) [extern]
```

Definition at line 1603 of file [tables.c](#).

**4.19.6.506 CoTestUse()**

```
int CoTestUse (
    UBYTE * s ) [extern]
```

Definition at line 1133 of file [tables.c](#).

**4.19.6.507 CoThreadBucket()**

```
int CoThreadBucket (
    UBYTE * s ) [extern]
```

Definition at line 5677 of file [compcomm.c](#).

**4.19.6.508 AddComString()**

```
int AddComString (
    int n,
    WORD * array,
    UBYTE * thestring,
    int par ) [extern]
```

Definition at line 1664 of file [compcomm.c](#).

**4.19.6.509 CompileAlgebra()**

```
int CompileAlgebra (
    UBYTE * s,
    int leftright,
    WORD * prototype ) [extern]
```

Definition at line 516 of file [compiler.c](#).

**4.19.6.510 IsIdStatement()**

```
int IsIdStatement (
    UBYTE * s ) [extern]
```

Definition at line 502 of file [compiler.c](#).

#### 4.19.6.511 IsRHS()

```
UBYTE * IsRHS (
    UBYTE * s,
    UBYTE c ) [extern]
```

Definition at line 456 of file [compiler.c](#).

#### 4.19.6.512 ParenthesesTest()

```
int ParenthesesTest (
    UBYTE * sin ) [extern]
```

Definition at line 378 of file [compiler.c](#).

#### 4.19.6.513 tokenize()

```
int tokenize (
    UBYTE * in,
    WORD leftright ) [extern]
```

Definition at line 58 of file [token.c](#).

#### 4.19.6.514 WriteTokens()

```
void WriteTokens (
    SBYTE * in ) [extern]
```

Definition at line 657 of file [token.c](#).

#### 4.19.6.515 simp1token()

```
int simp1token (
    SBYTE * s ) [extern]
```

Definition at line 705 of file [token.c](#).

#### 4.19.6.516 simpwtoken()

```
int simpwtoken (
    SBYTE * s ) [extern]
```

Definition at line 794 of file [token.c](#).

#### 4.19.6.517 simp2token()

```
int simp2token (
    SBYTE * s ) [extern]
```

Definition at line 948 of file [token.c](#).



**4.19.6.518 simp3atoken()**

```
int simp3atoken (
    SBYTE * s,
    int mode ) [extern]
```

Definition at line 1196 of file [token.c](#).

**4.19.6.519 simp3btoken()**

```
int simp3btoken (
    SBYTE * s,
    int mode ) [extern]
```

Definition at line 1437 of file [token.c](#).

**4.19.6.520 simp4token()**

```
int simp4token (
    SBYTE * s ) [extern]
```

Definition at line 1822 of file [token.c](#).

**4.19.6.521 simp5token()**

```
int simp5token (
    SBYTE * s,
    int mode ) [extern]
```

Definition at line 1982 of file [token.c](#).

**4.19.6.522 simp6token()**

```
int simp6token (
    SBYTE * tokens,
    int mode ) [extern]
```

Definition at line 2028 of file [token.c](#).

**4.19.6.523 SkipAName()**

```
UBYTE * SkipAName (
    UBYTE * s ) [extern]
```

Definition at line 428 of file [compiler.c](#).

#### 4.19.6.524 TestTables()

```
int TestTables (
    void ) [extern]
```

Definition at line 667 of file [compiler.c](#).

#### 4.19.6.525 GetLabel()

```
int GetLabel (
    UBYTE * name ) [extern]
```

Definition at line 2787 of file [names.c](#).

#### 4.19.6.526 ColdExpression()

```
int CoIdExpression (
    UBYTE * inp,
    int type ) [extern]
```

Definition at line 507 of file [comexpr.c](#).

#### 4.19.6.527 CoAssign()

```
int CoAssign (
    UBYTE * inp ) [extern]
```

Definition at line 1924 of file [comexpr.c](#).

#### 4.19.6.528 DoExpr()

```
int DoExpr (
    UBYTE * inp,
    int type,
    int par ) [extern]
```

Definition at line 96 of file [comexpr.c](#).

#### 4.19.6.529 CompileSubExpressions()

```
int CompileSubExpressions (
    SBYTE * tokens ) [extern]
```

Definition at line 711 of file [compiler.c](#).

#### 4.19.6.530 CodeGenerator()

```
int CodeGenerator (
    SBYTE * tokens ) [extern]
```

Definition at line 846 of file [compiler.c](#).

#### 4.19.6.531 CompleteTerm()

```
int CompleteTerm (
    WORD * term,
    UWORD * numer,
    UWORD * denom,
    WORD nnum,
    WORD nden,
    int sign ) [extern]
```

Definition at line 1909 of file [compiler.c](#).

#### 4.19.6.532 CodeFactors()

```
int CodeFactors (
    SBYTE * s ) [extern]
```

Definition at line 1946 of file [compiler.c](#).

#### 4.19.6.533 GenerateFactors()

```
WORD GenerateFactors (
    WORD n,
    WORD inc ) [extern]
```

Definition at line 2243 of file [compiler.c](#).

#### 4.19.6.534 InsTree()

```
int InsTree (
    int bufnum,
    int h ) [extern]
```

Definition at line 355 of file [comtool.c](#).

#### 4.19.6.535 FindTree()

```
int FindTree (
    int bufnum,
    WORD * subexpr ) [extern]
```

Definition at line 533 of file [comtool.c](#).

**4.19.6.536 RedoTree()**

```
void RedoTree (
    CBUF * C,
    int size ) [extern]
```

Definition at line 569 of file [comtool.c](#).

**4.19.6.537 ClearTree()**

```
void ClearTree (
    int i ) [extern]
```

Definition at line 588 of file [comtool.c](#).

**4.19.6.538 CatchDollar()**

```
int CatchDollar (
    int par ) [extern]
```

Definition at line 54 of file [dollar.c](#).

**4.19.6.539 AssignDollar()**

```
int AssignDollar (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 209 of file [dollar.c](#).

**4.19.6.540 WriteDollarToBuffer()**

```
UBYTE * WriteDollarToBuffer (
    WORD numdollar,
    WORD par ) [extern]
```

Definition at line 596 of file [dollar.c](#).

**4.19.6.541 WriteDollarFactorToBuffer()**

```
UBYTE * WriteDollarFactorToBuffer (
    WORD numdollar,
    WORD numfac,
    WORD par ) [extern]
```

Definition at line 687 of file [dollar.c](#).

**4.19.6.542 AddToDollarBuffer()**

```
void AddToDollarBuffer (
    UBYTE * s ) [extern]
```

Definition at line 762 of file [dollar.c](#).

**4.19.6.543 PutTermInDollar()**

```
int PutTermInDollar (
    WORD * term,
    WORD numdollar ) [extern]
```

Definition at line 863 of file [dollar.c](#).

**4.19.6.544 TermAssign()**

```
void TermAssign (
    WORD * term ) [extern]
```

Definition at line 803 of file [dollar.c](#).

**4.19.6.545 WildDollars()**

```
void WildDollars (
    PHEAD WORD * term ) [extern]
```

Definition at line 891 of file [dollar.c](#).

**4.19.6.546 numcommute()**

```
LONG numcommute (
    WORD * terms,
    LONG * numterms ) [extern]
```

Definition at line 659 of file [comtool.c](#).

**4.19.6.547 FullRenumber()**

```
int FullRenumber (
    PHEAD WORD * term,
    WORD par ) [extern]
```

Definition at line 324 of file [reshuf.c](#).

**4.19.6.548 Lus()**

```
int Lus (
    WORD * term,
    WORD funnum,
    WORD loopsize,
    WORD numargs,
    WORD outfun,
    WORD mode ) [extern]
```

Definition at line 57 of file [lus.c](#).

**4.19.6.549 FindLus()**

```
int FindLus (
    int from,
    int level,
    int openindex ) [extern]
```

Definition at line 515 of file [lus.c](#).

**4.19.6.550 CoReplaceLoop()**

```
int CoReplaceLoop (
    UBYTE * inp ) [extern]
```

Definition at line 5135 of file [compcomm.c](#).

**4.19.6.551 CoFindLoop()**

```
int CoFindLoop (
    UBYTE * inp ) [extern]
```

Definition at line 5127 of file [compcomm.c](#).

**4.19.6.552 DoFindLoop()**

```
int DoFindLoop (
    UBYTE * inp,
    int mode ) [extern]
```

Definition at line 5010 of file [compcomm.c](#).

**4.19.6.553 CoFunPowers()**

```
int CoFunPowers (
    UBYTE * inp ) [extern]
```

Definition at line 5155 of file [compcomm.c](#).

**4.19.6.554 SortTheList()**

```
int SortTheList (
    int * slist,
    int num ) [extern]
```

Definition at line 578 of file [lus.c](#).

**4.19.6.555 MatchIsPossible()**

```
int MatchIsPossible (
    WORD * pattern,
    WORD * term ) [extern]
```

Definition at line 299 of file [smart.c](#).

**4.19.6.556 StudyPattern()**

```
int StudyPattern (
    WORD * lhs ) [extern]
```

Definition at line 62 of file [smart.c](#).

**4.19.6.557 DoIToTensor()**

```
WORD DoIToTensor (
    PHEAD WORD ) [extern]
```

Definition at line 1082 of file [dollar.c](#).

**4.19.6.558 DoIToFunction()**

```
WORD DoIToFunction (
    PHEAD WORD ) [extern]
```

Definition at line 1143 of file [dollar.c](#).

**4.19.6.559 DoIToVector()**

```
WORD DoIToVector (
    PHEAD WORD ) [extern]
```

Definition at line 1200 of file [dollar.c](#).

**4.19.6.560 DoIToNumber()**

```
WORD DoIToNumber (
    PHEAD WORD ) [extern]
```

Definition at line 1264 of file [dollar.c](#).

#### 4.19.6.561 DoIToSymbol()

```
WORD DoIToSymbol (
    PHEAD WORD ) [extern]
```

Definition at line 1323 of file [dollar.c](#).

#### 4.19.6.562 DoIToIndex()

```
WORD DoIToIndex (
    PHEAD WORD ) [extern]
```

Definition at line 1377 of file [dollar.c](#).

#### 4.19.6.563 DoIToLong()

```
LONG DoIToLong (
    PHEAD WORD ) [extern]
```

Definition at line 1606 of file [dollar.c](#).

#### 4.19.6.564 DollarFactorize()

```
int DollarFactorize (
    PHEAD WORD ) [extern]
```

Definition at line 2955 of file [dollar.c](#).

#### 4.19.6.565 CoPrintTable()

```
int CoPrintTable (
    UBYTE * inp ) [extern]
```

Definition at line 1748 of file [comexpr.c](#).

#### 4.19.6.566 CoDeallocateTable()

```
int CoDeallocateTable (
    UBYTE * inp ) [extern]
```

Definition at line 1982 of file [comexpr.c](#).

#### 4.19.6.567 CleanDollarFactors()

```
void CleanDollarFactors (
    DOLLARS d ) [extern]
```

Definition at line 3554 of file [dollar.c](#).



```
WORD * TakeDollarContent (
    PHEAD WORD * dollarbuffer,
    WORD ** factor ) [extern]
```

#### 4.19.6.569 MakeDollarInteger()

For normalizing everything to integers we have to determine for all elements of this argument the LCM of the denominators and the GCD of the numerators. The input argument is in `bufin`. The number that comes out is the return value. The normalized argument is in `bufout`.

References [EndSort\(\)](#), [Generator\(\)](#), and [NewSort\(\)](#).



## Parameters

|                  |                                           |
|------------------|-------------------------------------------|
| <i>numdollar</i> | The index of the \$-variable to be added. |
|------------------|-------------------------------------------|

Definition at line 3959 of file [dollar.c](#).

**4.19.6.573 Optimize()**

```
int Optimize (
    WORD exprnr,
    int do_print ) [extern]
```

Optimization of expression

**4.19.7 Description**

This method takes an input expression and generates optimized code to calculate its value. The following methods are called to do so:

- (1) `get_expression` : to read to expression
- (2) `get_brackets` : find brackets for simultaneous optimization
- (3) `occurrence_order` or `find_Horner_MCTS` : to determine (the) Horner scheme(s) to use; this depends on AO.↔  
`optimize.horner`
- (4) `optimize_expression_given_Horner` : to do the optimizations for each Horner scheme; this method does either CSE or greedy optimizations dependings on AO.`optimize.method`
- (5) `generate_output` : to format the output in Form notation and store it in a buffer
- (6a) `optimize_print_code` : to print the expression (for "Print") or (6b) `generate_expression` : to modify the expression (for "#Optimize")

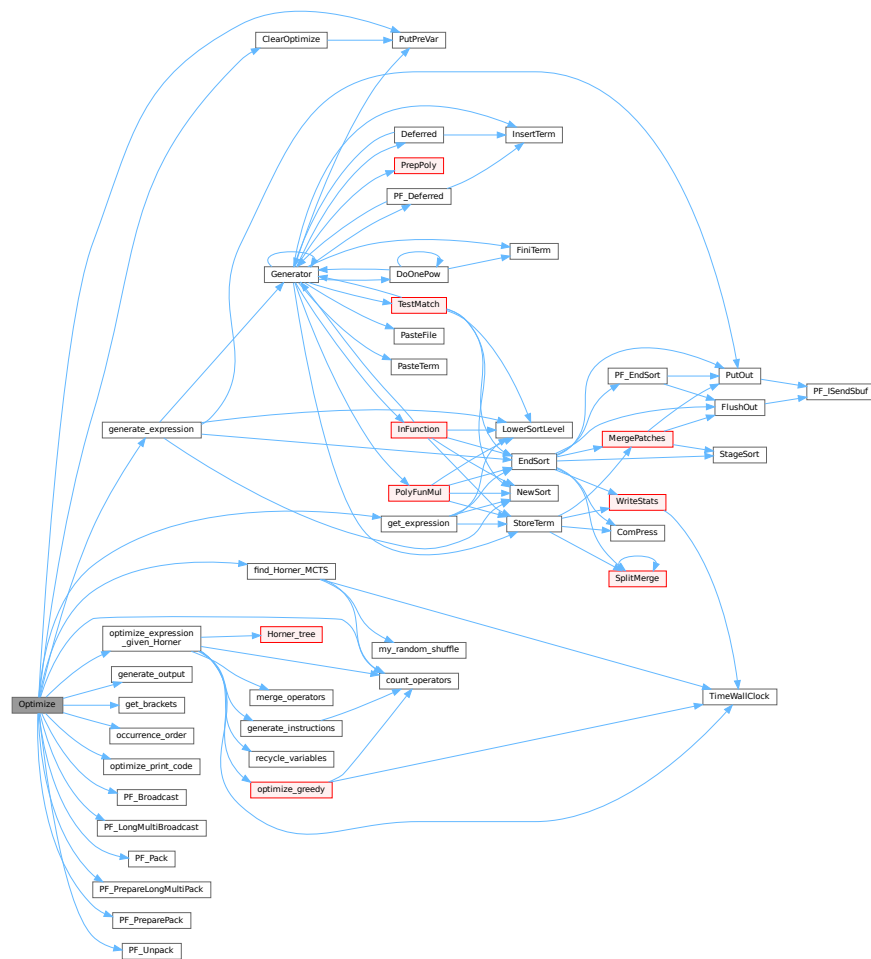
On ParFORM, all the processes must call this function at the same time. Then

- (1) Because only the master can access to the expression to be optimized, the master broadcast the expression to all the slaves after reading the expression (PF\_`get_expression`).
- (2) `get_brackets` reads `optimize_expr` as the input and it works also on the slaves. We leave it although the bracket information is not needed on the slaves (used in (5) on the master).
- (3) and (4) `find_Horner_MCTS` and `optimize_expression_given_Horner` are parallelized.
- (5), (6a) and (6b) are needed only on the master.

Definition at line 4638 of file [optimize.cc](#).

References [ClearOptimize\(\)](#), [count\\_operators\(\)](#), [find\\_Horner\\_MCTS\(\)](#), [generate\\_expression\(\)](#), [generate\\_output\(\)](#), [get\\_brackets\(\)](#), [get\\_expression\(\)](#), [occurrence\\_order\(\)](#), [optimize\\_expression\\_given\\_Horner\(\)](#), [optimize\\_print\\_code\(\)](#), [PF\\_Broadcast\(\)](#), [PF\\_LongMultiBroadcast\(\)](#), [PF\\_Pack\(\)](#), [PF\\_PrepareLongMultiPack\(\)](#), [PF\\_PreparePack\(\)](#), [PF\\_Unpack\(\)](#), and [PutPreVar\(\)](#).

Here is the call graph for this function:



#### 4.19.7.1 ClearOptimize()

```
int ClearOptimize (
    void ) [extern]
```

Optimization of expression

#### 4.19.8 Description

Clears the buffers that were used for optimization output. Clears the expression from the buffers (marks it to be dropped). Note: we need to use the expression by its name, because numbers may change if we drop other expressions between when we do the optimizations and clear the results (in [execute.c](#)). Also this is not 100% safe, because we could overwrite the optimized expression. But that can be done only in a Local or Global statement and hence we only have to test there that we might have to call ClearOptimize first. (in file [comexpr.c](#))

Definition at line 4975 of file [optimize.cc](#).

References [PutPreVar\(\)](#).

Referenced by [Optimize\(\)](#).

Here is the call graph for this function:



#### 4.19.8.1 TakeLongRoot()

```
int TakeLongRoot (
    UWORD * a,
    WORD * n,
    WORD power ) [extern]
```

Definition at line [2736](#) of file [reken.c](#).

#### 4.19.8.2 TakeRatRoot()

```
int TakeRatRoot (
    UWORD * a,
    WORD * n,
    WORD power ) [extern]
```

Definition at line [681](#) of file [reken.c](#).

#### 4.19.8.3 MakeRational()

```
int MakeRational (
    WORD a,
    WORD m,
    WORD * b,
    WORD * c ) [extern]
```

Definition at line [2856](#) of file [reken.c](#).

#### 4.19.8.4 MakeLongRational()

```
int MakeLongRational (
    PHEAD UWORD * a,
    WORD na,
    UWORD * m,
    WORD nm,
    UWORD * b,
    WORD * nb ) [extern]
```

Definition at line [2904](#) of file [reken.c](#).

**4.19.8.5 ClearTableTree()**

```
void ClearTableTree (
    TABLES T ) [extern]
```

Definition at line 72 of file [tables.c](#).

**4.19.8.6 InsTableTree()**

```
int InsTableTree (
    TABLES T,
    WORD * tp ) [extern]
```

Definition at line 103 of file [tables.c](#).

**4.19.8.7 RedoTableTree()**

```
void RedoTableTree (
    TABLES T,
    int newsize ) [extern]
```

Definition at line 266 of file [tables.c](#).

**4.19.8.8 FindTableTree()**

```
int FindTableTree (
    TABLES T,
    WORD * tp,
    int inc ) [extern]
```

Definition at line 291 of file [tables.c](#).

**4.19.8.9 finishcbuf()**

```
void finishcbuf (
    WORD num ) [extern]
```

Frees a compiler buffer.

**Parameters**

|            |                                           |
|------------|-------------------------------------------|
| <i>num</i> | The ID number for the buffer to be freed. |
|------------|-------------------------------------------|

Definition at line 89 of file [comtool.c](#).

References [CbUf::boomlijst](#), [CbUf::Buffer](#), [CbUf::BufferSize](#), [CbUf::CanCommu](#), [CbUf::lhs](#), [CbUf::NumTerms](#), [CbUf::Pointer](#), [CbUf::rhs](#), and [CbUf::Top](#).

Referenced by [PF\\_BroadcastCBuf\(\)](#).

#### 4.19.8.10 clearcbuf()

```
void clearcbuf (
    WORD num ) [extern]
```

Clears contents in a compiler buffer.

##### Parameters

|            |                                             |
|------------|---------------------------------------------|
| <i>num</i> | The ID number for the buffer to be cleared. |
|------------|---------------------------------------------|

Definition at line 116 of file [comtool.c](#).

References [CbUf::boomlijst](#), [CbUf::Buffer](#), [CbUf::Pointer](#), and [CbUf::rhs](#).

#### 4.19.8.11 CleanUpSort()

```
void CleanUpSort (
    int num ) [extern]
```

Partially or completely frees function sort buffers.

Definition at line 4830 of file [sort.c](#).

#### 4.19.8.12 AllocFileHandle()

```
FILEHANDLE * AllocFileHandle (
    WORD par,
    char * name ) [extern]
```

Definition at line 1044 of file [setfile.c](#).

#### 4.19.8.13 DeAllocFileHandle()

```
void DeAllocFileHandle (
    FILEHANDLE * fh ) [extern]
```

Definition at line 1105 of file [setfile.c](#).

#### 4.19.8.14 LowerSortLevel()

```
void LowerSortLevel (
    void ) [extern]
```

Lowers the level in the sort system.

Definition at line 4930 of file [sort.c](#).

Referenced by [generate\\_expression\(\)](#), [get\\_expression\(\)](#), [InFunction\(\)](#), [PF\\_CollectModifiedDollars\(\)](#), [PF\\_Processor\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_sort\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [PolyFunMul\(\)](#), [Processor\(\)](#), and [TestMatch\(\)](#).

#### 4.19.8.15 PolyRatFunSpecial()

```
WORD * PolyRatFunSpecial (
    PHEAD WORD * t1,
    WORD * t2 ) [extern]
```

Definition at line [4947](#) of file [sort.c](#).

#### 4.19.8.16 SimpleSplitMergeRec()

```
void SimpleSplitMergeRec (
    WORD * array,
    WORD num,
    WORD * auxarray ) [extern]
```

Definition at line [5049](#) of file [sort.c](#).

#### 4.19.8.17 SimpleSplitMerge()

```
void SimpleSplitMerge (
    WORD * array,
    WORD num ) [extern]
```

Definition at line [5077](#) of file [sort.c](#).

#### 4.19.8.18 BinarySearch()

```
WORD BinarySearch (
    WORD * array,
    WORD num,
    WORD x ) [extern]
```

Definition at line [5095](#) of file [sort.c](#).

#### 4.19.8.19 InsideDollar()

```
int InsideDollar (
    PHEAD WORD * ll,
    WORD level ) [extern]
```

Definition at line [1746](#) of file [dollar.c](#).

#### 4.19.8.20 DolToTerms()

```
DOLLARS DolToTerms (
    PHEAD WORD ) [extern]
```

Definition at line [1458](#) of file [dollar.c](#).



**4.19.8.21 EvalDoLoopArg()**

```
WORD EvalDoLoopArg (
    PHEAD WORD * arg,
    WORD par ) [extern]
```

Evaluates one argument of a do loop. Such an argument is constructed from SNUMBERS DOLLAREXPRES-  
SIONs and possibly DOLLAREXPR2s which indicate factors of the preceding dollar. Hence we have SNUM-  
BER,num DOLLAREXPRESSSION,numdollar DOLLAREXPRESSSION,numdollar,DOLLAREXPR2,numfactor DOL-  
LAREXPRESSSION,numdollar,DOLLAREXPR2,numfactor,DOLLAREXPR2,numfactor etc. Because we have a do-  
loop at every stage we should have a number. The notation in DOLLAREXPR2 is that  $\geq 0$  is number of yet another  
dollar and  $< 0$  is  $-n-1$  with  $n$  the array element or zero. The return value is the (short) number. The routine works its  
way through the list in a recursive manner.

Definition at line 2651 of file [dollar.c](#).

References [EvalDoLoopArg\(\)](#).

Referenced by [EvalDoLoopArg\(\)](#).

Here is the call graph for this function:

**4.19.8.22 SetExprCases()**

```
int SetExprCases (
    int par,
    int setunset,
    int val ) [extern]
```

Definition at line 1357 of file [compcomm.c](#).

**4.19.8.23 TestSelect()**

```
int TestSelect (
    WORD * term,
    WORD * setp ) [extern]
```

Definition at line 1748 of file [pattern.c](#).

#### 4.19.8.24 SubsInAll()

```
void SubsInAll (
    PHEAD0 ) [extern]
```

Definition at line 1982 of file [pattern.c](#).

#### 4.19.8.25 TransferBuffer()

```
void TransferBuffer (
    int from,
    int to,
    int spectator ) [extern]
```

Definition at line 2143 of file [pattern.c](#).

#### 4.19.8.26 TakeIDfunction()

```
int TakeIDfunction (
    PHEAD WORD * term ) [extern]
```

Definition at line 2213 of file [pattern.c](#).

#### 4.19.8.27 MakeSetupAllocs()

```
int MakeSetupAllocs (
    void ) [extern]
```

Definition at line 1122 of file [setfile.c](#).

#### 4.19.8.28 TryFileSetups()

```
int TryFileSetups (
    void ) [extern]
```

Definition at line 1142 of file [setfile.c](#).

#### 4.19.8.29 ExchangeExpressions()

```
void ExchangeExpressions (
    int num1,
    int num2 ) [extern]
```

Definition at line 1671 of file [execute.c](#).

#### 4.19.8.30 ExchangeDollars()

```
void ExchangeDollars (
    int num1,
    int num2 ) [extern]
```

Definition at line 1849 of file [dollar.c](#).

#### 4.19.8.31 GetFirstBracket()

```
int GetFirstBracket (
    WORD * term,
    int num ) [extern]
```

Definition at line 1745 of file [execute.c](#).

#### 4.19.8.32 GetFirstTerm()

```
int GetFirstTerm (
    WORD * term,
    int num,
    int pre ) [extern]
```

Gets the first term of an expression. Routine should be thread safe.

##### Parameters

|     |             |                                                                                                            |
|-----|-------------|------------------------------------------------------------------------------------------------------------|
| out | <i>term</i> | Pointer to the buffer where the term should be written.                                                    |
| in  | <i>num</i>  | The expression number from which to get the term.                                                          |
| in  | <i>pre</i>  | Denotes a pre-processor call (1) or not (0). Used to determine the correct source file for the expression. |

##### Returns

First WORD of term, a negative value denotes an error.

Definition at line 1864 of file [execute.c](#).

References [FiLe::handle](#), [VaRrEnUm::lo](#), and [ReNuMbEr::symb](#).

#### 4.19.8.33 GetContent()

```
int GetContent (
    WORD * content,
    int num ) [extern]
```

Definition at line 1972 of file [execute.c](#).

#### 4.19.8.34 CleanupTerm()

```
int CleanupTerm (
    WORD * term ) [extern]
```

Definition at line 2089 of file [execute.c](#).

#### 4.19.8.35 ContentMerge()

```
WORD ContentMerge (
    PHEAD WORD * content,
    WORD * term ) [extern]
```

Definition at line 2113 of file [execute.c](#).

#### 4.19.8.36 PreIfDollarEval()

```
UBYTE * PreIfDollarEval (
    UBYTE * s,
    int * value ) [extern]
```

Definition at line 1978 of file [dollar.c](#).

#### 4.19.8.37 TermsInDollar()

```
LONG TermsInDollar (
    WORD num ) [extern]
```

Definition at line 1867 of file [dollar.c](#).

#### 4.19.8.38 SizeOfDollar()

```
LONG SizeOfDollar (
    WORD num ) [extern]
```

Definition at line 1916 of file [dollar.c](#).

#### 4.19.8.39 TermsInExpression()

```
LONG TermsInExpression (
    WORD num ) [extern]
```

Definition at line 2377 of file [execute.c](#).

#### 4.19.8.40 SizeOfExpression()

```
LONG SizeOfExpression (
    WORD num ) [extern]
```

Definition at line 2389 of file [execute.c](#).

#### 4.19.8.41 TranslateExpression()

```
WORD * TranslateExpression (
    UBYTE * s ) [extern]
```

Definition at line 2160 of file [dollar.c](#).

#### 4.19.8.42 IsSetMember()

```
int IsSetMember (
    WORD * buffer,
    WORD numset ) [extern]
```

Definition at line 2217 of file [dollar.c](#).

#### 4.19.8.43 IsMultipleOf()

```
int IsMultipleOf (
    WORD * buf1,
    WORD * buf2 ) [extern]
```

Definition at line 2398 of file [dollar.c](#).

#### 4.19.8.44 TwoExprCompare()

```
int TwoExprCompare (
    WORD * buf1,
    WORD * buf2,
    int oprtr ) [extern]
```

Definition at line 2473 of file [dollar.c](#).

#### 4.19.8.45 UpdatePositions()

```
void UpdatePositions (
    void ) [extern]
```

Definition at line 2401 of file [execute.c](#).

#### 4.19.8.46 M\_print()

```
void M_print (
    void ) [extern]
```

Definition at line 2436 of file [tools.c](#).

#### 4.19.8.47 M\_check1()

```
void M_check1 (
    void ) [extern]
```

Definition at line 2435 of file [tools.c](#).

#### 4.19.8.48 PrintTime()

```
void PrintTime (
    UBYTE * mess ) [extern]
```

Definition at line 784 of file [sch.c](#).

#### 4.19.8.49 FindBracket()

```
POSITION * FindBracket (
    WORD nexp,
    WORD * bracket ) [extern]
```

Definition at line 65 of file [index.c](#).

#### 4.19.8.50 PutBracketInIndex()

```
void PutBracketInIndex (
    PHEAD WORD * term,
    POSITION * newpos ) [extern]
```

Definition at line 331 of file [index.c](#).

#### 4.19.8.51 ClearBracketIndex()

```
void ClearBracketIndex (
    WORD numexp ) [extern]
```

Definition at line 546 of file [index.c](#).

#### 4.19.8.52 OpenBracketIndex()

```
void OpenBracketIndex (
    WORD nexpr ) [extern]
```

Definition at line 578 of file [index.c](#).

#### 4.19.8.53 DoNoParallel()

```
int DoNoParallel (
    UBYTE * s ) [extern]
```

Definition at line 529 of file [module.c](#).

#### 4.19.8.54 DoParallel()

```
int DoParallel (
    UBYTE * s ) [extern]
```

Definition at line 544 of file [module.c](#).

#### 4.19.8.55 DoModSum()

```
int DoModSum (
    UBYTE * s ) [extern]
```

Definition at line 559 of file [module.c](#).

#### 4.19.8.56 DoModMax()

```
int DoModMax (
    UBYTE * s ) [extern]
```

Definition at line 579 of file [module.c](#).

#### 4.19.8.57 DoModMin()

```
int DoModMin (
    UBYTE * s ) [extern]
```

Definition at line 599 of file [module.c](#).

#### 4.19.8.58 DoModLocal()

```
int DoModLocal (
    UBYTE * s ) [extern]
```

Definition at line 619 of file [module.c](#).

#### 4.19.8.59 DoModDollar()

```
UBYTE * DoModDollar (
    UBYTE * s,
    int type ) [extern]
```

Definition at line 657 of file [module.c](#).

#### 4.19.8.60 DoProcessBucket()

```
int DoProcessBucket (
    UBYTE * s ) [extern]
```

Definition at line 639 of file [module.c](#).

#### 4.19.8.61 DoinParallel()

```
int DoinParallel (
    UBYTE * s ) [extern]
```

Definition at line 758 of file [module.c](#).

#### 4.19.8.62 DonotinParallel()

```
int DonotinParallel (
    UBYTE * s ) [extern]
```

Definition at line 768 of file [module.c](#).

#### 4.19.8.63 FlipTable()

```
int FlipTable (
    FUNCTIONS f,
    int type ) [extern]
```

Definition at line 671 of file [tables.c](#).

#### 4.19.8.64 ChainIn()

```
int ChainIn (
    PHEAD WORD * term,
    WORD funnum ) [extern]
```

Definition at line 691 of file [function.c](#).

#### 4.19.8.65 ChainOut()

```
int ChainOut (
    PHEAD WORD * term,
    WORD funnum ) [extern]
```

Definition at line 748 of file [function.c](#).

#### 4.19.8.66 ArgumentImplode()

```
int ArgumentImplode (
    PHEAD WORD * term,
    WORD * thelist ) [extern]
```

Definition at line 1846 of file [argument.c](#).



#### 4.19.8.67 ArgumentExplode()

```
int ArgumentExplode (
    PHEAD WORD * term,
    WORD * thelist ) [extern]
```

Definition at line 1950 of file [argument.c](#).

#### 4.19.8.68 DenToFunction()

```
int DenToFunction (
    WORD * term,
    WORD numfun ) [extern]
```

Definition at line 2557 of file [wildcard.c](#).

#### 4.19.8.69 HowMany()

```
WORD HowMany (
    PHEAD WORD * ifcode,
    WORD * term ) [extern]
```

Definition at line 840 of file [if.c](#).

#### 4.19.8.70 RemoveDollars()

```
void RemoveDollars (
    void ) [extern]
```

Definition at line 3128 of file [names.c](#).

#### 4.19.8.71 CountTerms1()

```
LONG CountTerms1 (
    PHEAD0 ) [extern]
```

Definition at line 2462 of file [execute.c](#).

#### 4.19.8.72 TermsInBracket()

```
LONG TermsInBracket (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 2570 of file [execute.c](#).

#### 4.19.8.73 Crash()

```
int Crash (
    void ) [extern]
```

Definition at line 3753 of file [tools.c](#).

#### 4.19.8.74 str\_dup()

```
char * str_dup (
    char * str ) [extern]
```

Definition at line 101 of file [minos.c](#).

#### 4.19.8.75 convertblock()

```
void convertblock (
    INDEXBLOCK * in,
    INDEXBLOCK * out,
    int mode ) [extern]
```

Definition at line 120 of file [minos.c](#).

#### 4.19.8.76 convertnamesblock()

```
void convertnamesblock (
    NAMESBLOCK * in,
    NAMESBLOCK * out,
    int mode ) [extern]
```

Definition at line 171 of file [minos.c](#).

#### 4.19.8.77 convertiniinfo()

```
void convertiniinfo (
    INIINFO * in,
    INIINFO * out,
    int mode ) [extern]
```

Definition at line 197 of file [minos.c](#).

#### 4.19.8.78 ReadIndex()

```
int ReadIndex (
    DBASE * d ) [extern]
```

Definition at line 288 of file [minos.c](#).

#### 4.19.8.79 WriteIndexBlock()

```
int WriteIndexBlock (
    DBASE * d,
    MLONG num ) [extern]
```

Definition at line 399 of file [minos.c](#).

#### 4.19.8.80 WriteNamesBlock()

```
int WriteNamesBlock (
    DBASE * d,
    MLONG num ) [extern]
```

Definition at line 420 of file [minos.c](#).

#### 4.19.8.81 WriteIndex()

```
int WriteIndex (
    DBASE * d ) [extern]
```

Definition at line 443 of file [minos.c](#).

#### 4.19.8.82 WriteIniInfo()

```
int WriteIniInfo (
    DBASE * d ) [extern]
```

Definition at line 506 of file [minos.c](#).

#### 4.19.8.83 ReadIniInfo()

```
int ReadIniInfo (
    DBASE * d ) [extern]
```

Definition at line 523 of file [minos.c](#).

#### 4.19.8.84 AddToIndex()

```
int AddToIndex (
    DBASE * d,
    MLONG number ) [extern]
```

Definition at line 1065 of file [minos.c](#).

#### 4.19.8.85 GetDbase()

```
DBASE * GetDbase (
    char * filename,
    MLONG rwmode ) [extern]
```

Definition at line 546 of file [minos.c](#).

#### 4.19.8.86 OpenDbase()

```
DBASE * OpenDbase (
    char * filename ) [extern]
```

Definition at line 820 of file [minos.c](#).

#### 4.19.8.87 ReadObject()

```
char * ReadObject (
    DBASE * d,
    MLONG tablenumber,
    char * arguments ) [extern]
```

Definition at line 1294 of file [minos.c](#).

#### 4.19.8.88 ReadijObject()

```
char * ReadijObject (
    DBASE * d,
    MLONG i,
    MLONG j,
    char * arguments ) [extern]
```

Definition at line 1375 of file [minos.c](#).

#### 4.19.8.89 ExistsObject()

```
int ExistsObject (
    DBASE * d,
    MLONG tablenumber,
    char * arguments ) [extern]
```

Definition at line 1427 of file [minos.c](#).

#### 4.19.8.90 DeleteObject()

```
int DeleteObject (
    DBASE * d,
    MLONG tablenumber,
    char * arguments ) [extern]
```

Definition at line 1459 of file [minos.c](#).

#### 4.19.8.91 WriteObject()

```
int WriteObject (
    DBASE * d,
    MLONG tablename,
    char * arguments,
    char * rhs,
    MLONG number ) [extern]
```

Definition at line 1194 of file [minos.c](#).

#### 4.19.8.92 AddObject()

```
MLONG AddObject (
    DBASE * d,
    MLONG tablename,
    char * arguments,
    char * rhs ) [extern]
```

Definition at line 1152 of file [minos.c](#).

#### 4.19.8.93 NewDbase()

```
DBASE * NewDbase (
    char * name,
    MLONG number ) [extern]
```

Definition at line 602 of file [minos.c](#).

#### 4.19.8.94 FreeTableBase()

```
void FreeTableBase (
    DBASE * d ) [extern]
```

Definition at line 739 of file [minos.c](#).

#### 4.19.8.95 ComposeTableNames()

```
int ComposeTableNames (
    DBASE * d ) [extern]
```

Definition at line 771 of file [minos.c](#).

#### 4.19.8.96 PutTableNames()

```
int PutTableNames (
    DBASE * d ) [extern]
```

Definition at line 969 of file [minos.c](#).

#### 4.19.8.97 AddTableName()

```
MLONG AddTableName (
    DBASE * d,
    char * name,
    TABLES T ) [extern]
```

Definition at line [854](#) of file [minos.c](#).

#### 4.19.8.98 GetTableName()

```
MLONG GetTableName (
    DBASE * d,
    char * name ) [extern]
```

Definition at line [937](#) of file [minos.c](#).

#### 4.19.8.99 FindTableNumber()

```
MLONG FindTableNumber (
    DBASE * d,
    char * name ) [extern]
```

Definition at line [1166](#) of file [minos.c](#).

#### 4.19.8.100 TryEnvironment()

```
int TryEnvironment (
    void ) [extern]
```

Definition at line [1228](#) of file [setfile.c](#).

#### 4.19.8.101 CopyExpression()

```
int CopyExpression (
    FILEHANDLE * from,
    FILEHANDLE * to ) [extern]
```

Definition at line [3307](#) of file [store.c](#).

#### 4.19.8.102 set\_in()

```
int set_in (
    UBYTE ch,
    set_of_char set ) [extern]
```

Definition at line [2112](#) of file [tools.c](#).

**4.19.8.103 set\_set()**

```
one_byte set_set (
    UBYTE ch,
    set_of_char set ) [extern]
```

Definition at line 2132 of file [tools.c](#).

**4.19.8.104 set\_del()**

```
one_byte set_del (
    UBYTE ch,
    set_of_char set ) [extern]
```

Definition at line 2153 of file [tools.c](#).

**4.19.8.105 set\_sub()**

```
one_byte set_sub (
    set_of_char set,
    set_of_char set1,
    set_of_char set2 ) [extern]
```

Definition at line 2175 of file [tools.c](#).

**4.19.8.106 DoPreAddSeparator()**

```
int DoPreAddSeparator (
    UBYTE * s ) [extern]
```

Definition at line 5382 of file [pre.c](#).

**4.19.8.107 DoPreRmSeparator()**

```
int DoPreRmSeparator (
    UBYTE * s ) [extern]
```

Definition at line 5407 of file [pre.c](#).

**4.19.8.108 openExternalChannel()**

```
int openExternalChannel (
    UBYTE * cmd,
    int daemonize,
    UBYTE * shellname,
    UBYTE * stderrname ) [extern]
```

Definition at line 230 of file [extcmd.c](#).

**4.19.8.109 initPresetExternalChannels()**

```
int initPresetExternalChannels (
    UBYTE * theline,
    int thetimeout ) [extern]
```

Definition at line [232](#) of file [extcmd.c](#).

**4.19.8.110 closeExternalChannel()**

```
int closeExternalChannel (
    int n ) [extern]
```

Definition at line [233](#) of file [extcmd.c](#).

**4.19.8.111 selectExternalChannel()**

```
int selectExternalChannel (
    int n ) [extern]
```

Definition at line [234](#) of file [extcmd.c](#).

**4.19.8.112 getCurrentExternalChannel()**

```
int getCurrentExternalChannel (
    void ) [extern]
```

Definition at line [235](#) of file [extcmd.c](#).

**4.19.8.113 closeAllExternalChannels()**

```
void closeAllExternalChannels (
    void ) [extern]
```

Definition at line [236](#) of file [extcmd.c](#).

**4.19.8.114 defineChannel()**

```
UBYTE * defineChannel (
    UBYTE * s,
    HANDLERS * h ) [extern]
```

Definition at line [6033](#) of file [pre.c](#).



**4.19.8.115 writeToChannel()**

```
int writeToChannel (
    int wtype,
    UBYTE * s,
    HANDLERS * h ) [extern]
```

Definition at line 6068 of file [pre.c](#).

**4.19.8.116 ReleaseTB()**

```
int ReleaseTB (
    void ) [extern]
```

Definition at line 2317 of file [tables.c](#).

**4.19.8.117 SymbolNormalize()**

```
int SymbolNormalize (
    WORD * term ) [extern]
```

Routine normalizes terms that contain only symbols. Regular minimum and maximum properties are ignored.

We check whether there are negative powers in the output. This is not allowed.

Definition at line 5145 of file [normal.c](#).

Referenced by [InFunction\(\)](#), and [poly\\_sort\(\)](#).

**4.19.8.118 TestFunFlag()**

```
int TestFunFlag (
    PHEAD WORD * tfun ) [extern]
```

Definition at line 5241 of file [normal.c](#).

**4.19.8.119 CompareSymbols()**

```
WORD CompareSymbols (
    PHEAD WORD * term1,
    WORD * term2,
    WORD par ) [extern]
```

Compares the terms, based on the value of AN.polysortflag. If term1 < term2 the return value is -1 If term1 > term2 the return value is 1 If term1 = term2 the return value is 0 The coefficients may differ. The terms contain only a single subterm of type SYMBOL. If AN.polysortflag = 0 it is a 'regular' compare. If AN.polysortflag = 1 the sum of the powers is more important par is a dummy parameter to make the parameter field identical to that of Compare1 which is the regular compare routine in [sort.c](#)

Definition at line 3084 of file [sort.c](#).

Referenced by [InFunction\(\)](#), and [poly\\_sort\(\)](#).

**4.19.8.120 CompareHSymbols()**

```
WORD CompareHSymbols (
    PHEAD WORD * term1,
    WORD * term2,
    WORD par ) [extern]
```

Compares terms that can have only SYMBOL and HAAKJE subterms. If  $term1 < term2$  the return value is -1 If  $term1 > term2$  the return value is 1 If  $term1 = term2$  the return value is 0 par is a dummy parameter to make the parameter field identical to that of Compare1 which is the regular compare routine in [sort.c](#)

Definition at line [3127](#) of file [sort.c](#).

**4.19.8.121 NextPrime()**

```
WORD NextPrime (
    PHEAD WORD num ) [extern]
```

Gives the next prime number in the list of prime numbers.

If the list isn't long enough we expand it. For ease in ParForm and because these lists shouldn't be very big we let each worker keep its own list.

The list is cut off at MAXPOWER, because we don't want to get into trouble that the power of a variable gets larger than the prime number.

Definition at line [3665](#) of file [reken.c](#).

**4.19.8.122 Moebius()**

```
WORD Moebius (
    PHEAD WORD ) [extern]
```

Definition at line [3729](#) of file [reken.c](#).

**4.19.8.123 wranf()**

```
UWORD wranf (
    PHEAD0 ) [extern]
```

Definition at line [3869](#) of file [reken.c](#).

**4.19.8.124 iranf()**

```
UWORD iranf (
    PHEAD UWORD ) [extern]
```

Definition at line [3888](#) of file [reken.c](#).

**4.19.8.125 iniwranf()**

```
void iniwranf (
    PHEAD0 ) [extern]
```

Definition at line 3820 of file [reken.c](#).

**4.19.8.126 PreRandom()**

```
UBYTE * PreRandom (
    UBYTE * s ) [extern]
```

Definition at line 3913 of file [reken.c](#).

**4.19.8.127 TreatPolyRatFun()**

```
int TreatPolyRatFun (
    PHEAD WORD * prf ) [extern]
```

Definition at line 4961 of file [normal.c](#).

**4.19.8.128 ReadSaveHeader()**

```
int ReadSaveHeader (
    void ) [extern]
```

Reads the header in the save file and sets function pointers and flags according to the information found there. Must be called before any other ReadSave... function.

Currently works only for the exchange between 32bit and 64bit machines (WORD size must be 2 or 4 bytes)!

It is called by CoLoad().

**Returns**

= 0 everything okay, != 0 an error occurred

Definition at line 4057 of file [store.c](#).

**4.19.8.129 ReadSaveIndex()**

```
LONG ReadSaveIndex (
    FILEINDEX * fileind ) [extern]
```

Reads a FILEINDEX from the open save file specified by AO.SaveData.Handle. Translations for adjusting endianness and data sizes are done if necessary.

Depends on the assumption that sizeof(FILEINDEX) is the same everywhere. If FILEINDEX or INDEXENTRY change, then this functions has to be adjusted.

Called by CoLoad() and FindInIndex().

## Parameters

|                |                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------|
| <i>fileind</i> | contains the read FILEINDEX after successful return. must point to allocated, big enough memory. |
|----------------|--------------------------------------------------------------------------------------------------|

## Returns

= 0 everything okay, != 0 an error occurred

Definition at line 4175 of file [store.c](#).

References [INFILEINDEX](#), [FiLeInDeX::next](#), [FiLeInDeX::number](#), and [FuNcTiOn::number](#).

**4.19.8.130 ReadSaveExpression()**

```
LONG ReadSaveExpression (
    UBYTE * buffer,
    UBYTE * top,
    LONG * size,
    LONG * outsize ) [extern]
```

Reads an expression from the open file specified by AO.SaveData.Handle. The endianness flip and a resizing without renumbering is done in this function. Thereafter the buffer consists of chunks with a uniform maximal word size (32bit at the moment). The actual renumbering is then done by calling the function [ReadSaveTerm32\(\)](#). The result is returned in *buffer*.

If the translation at some point doesn't fit into the buffer anymore, the function returns and must be called again. In any case *size* returns the number of successfully read bytes, *outsize* returns the number of successfully written bytes, and the file will be positioned at the next byte after the successfully read data.

It is called by PutInStore().

## Parameters

|                |                                                  |
|----------------|--------------------------------------------------|
| <i>buffer</i>  | output buffer, holds the (translated) expression |
| <i>top</i>     | end of buffer                                    |
| <i>size</i>    | number of read bytes                             |
| <i>outsize</i> | number of written bytes                          |

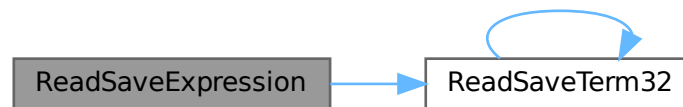
## Returns

= 0 everything okay, != 0 an error occurred

Definition at line 5138 of file [store.c](#).

References [ReadSaveTerm32\(\)](#).

Here is the call graph for this function:



#### 4.19.8.131 ReadSaveTerm32()

```

UBYTE * ReadSaveTerm32 (
    UBYTE * bin,
    UBYTE * binend,
    UBYTE ** bout,
    UBYTE * boutend,
    UBYTE * top,
    int terminbuf ) [extern]
  
```

Reads a single term from the given buffer at *bin* and write the translated term back to this buffer at *bout*.

[ReadSaveTerm32\(\)](#) is currently the only instantiation of a `ReadSaveTerm`-function. It only deals with data that already has the correct endianness and that is resized to 32bit words but without being renumbered or translated in any other way. It uses the compress buffer `AR.CompressBuffer`.

The function is reentrant in order to cope with nested function arguments. It is called by [ReadSaveExpression\(\)](#) and itself.

The *return value* indicates the position in the input buffer up to which the data has already been successfully processed. The parameter *bout* returns the corresponding position in the output buffer.

##### Parameters

|                  |                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <i>bin</i>       | start of the input buffer                                                                                                       |
| <i>binend</i>    | end of the input buffer                                                                                                         |
| <i>bout</i>      | as input points to the beginning of the output buffer, as output points behind the already translated data in the output buffer |
| <i>boutend</i>   | end of already decompressed data in output buffer                                                                               |
| <i>top</i>       | end of output buffer                                                                                                            |
| <i>terminbuf</i> | flag whether decompressed data is already in the output buffer. used in recursive calls                                         |

##### Returns

pointer to the next unprocessed data in the input buffer

Definition at line 4728 of file [store.c](#).

References [ReadSaveTerm32\(\)](#).

Referenced by [ReadSaveExpression\(\)](#), and [ReadSaveTerm32\(\)](#).

Here is the call graph for this function:



#### 4.19.8.132 ReadSaveVariables()

```

LONG ReadSaveVariables (
    UBYTE * buffer,
    UBYTE * top,
    LONG * size,
    LONG * outsize,
    INDEXENTRY * ind,
    LONG * stage ) [extern]
  
```

Reads the variables from the open file specified by AO.SaveData.Handle. It reads the *\*size* bytes and writes them to the *\*buffer*. It is called by PutInStore().

If translation is necessary, the data might shrink or grow in size, then *\*size* is adjusted so that the reading and writing fits into the memory from the buffer to the top. The actual number of read bytes is returned in *\*size*, the number of written bytes is returned in *\*outsize*.

If the *\*size* is smaller than the actual size of the variables, this function will be called several times and needs to remember the current position in the variable structure. The parameter *stage* does this job. When [ReadSaveVariables\(\)](#) is called for the first time, this parameter should have the value -1.

The parameter *ind* is used to get the number of variables.

##### Parameters

|                |                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------|
| <i>buffer</i>  | read variables are written into this allocated memory                                                                      |
| <i>top</i>     | upper end of allocated memory                                                                                              |
| <i>size</i>    | number of bytes to read. might return a smaller number of read bytes if translation was necessary                          |
| <i>outsize</i> | if translation has be done, outsize contains the number of written bytes                                                   |
| <i>ind</i>     | pointer of INDEXENTRY for the current expression. read-only                                                                |
| <i>stage</i>   | should be -1 for the first call, will be increased by ReadSaveVariables to memorize the position in the variable structure |

##### Returns

= 0 everything okay, != 0 an error occurred

Definition at line 4361 of file [store.c](#).

References [InDeXeNtRy::nfunctions](#), [InDeXeNtRy::nindices](#), [InDeXeNtRy::nsymbols](#), [InDeXeNtRy::nvectors](#), and [FuNcTiOn::spec](#).

#### 4.19.8.133 WriteStoreHeader()

```
LONG WriteStoreHeader (
    WORD handle ) [extern]
```

Writes header with information about system architecture and FORM revision to an open store file.

Called by [SetFileIndex\(\)](#).

##### Parameters

|               |                                                     |
|---------------|-----------------------------------------------------|
| <i>handle</i> | specifies open file to which header will be written |
|---------------|-----------------------------------------------------|

##### Returns

= 0 everything okay, != 0 an error occurred

Definition at line 3964 of file [store.c](#).

References [STOREHEADER::endianness](#), [STOREHEADER::lenLONG](#), [STOREHEADER::lenPOINTER](#), [STOREHEADER::lenPOS](#), [STOREHEADER::lenWORD](#), [STOREHEADER::maxpower](#), [STOREHEADER::sFun](#), [STOREHEADER::sInd](#), [STOREHEADER::sSym](#), [STOREHEADER::sVec](#), and [STOREHEADER::wildoffset](#).

Referenced by [SetFileIndex\(\)](#).

#### 4.19.8.134 InitRecovery()

```
void InitRecovery (
    void ) [extern]
```

Sets up the strings for the filenames of the recovery files. This functions should only be called once to avoid memory leaks and after AM.TempDir has been initialized.

Definition at line 399 of file [checkpoint.c](#).

#### 4.19.8.135 CheckRecoveryFile()

```
int CheckRecoveryFile (
    void ) [extern]
```

Checks whether a snapshot/recovery file exists. Returns 1 if it exists, 0 otherwise.

Definition at line 278 of file [checkpoint.c](#).

References [RecoveryFilename\(\)](#).

Here is the call graph for this function:



#### 4.19.8.136 DeleteRecoveryFile()

```
void DeleteRecoveryFile (
    void ) [extern]
```

Deletes the recovery files. It is called by `CleanUp()` in the case of a successful completion.

Definition at line 333 of file [checkpoint.c](#).

#### 4.19.8.137 RecoveryFilename()

```
char * RecoveryFilename (
    void ) [extern]
```

Returns pointer to recovery filename.

Definition at line 364 of file [checkpoint.c](#).

Referenced by [CheckRecoveryFile\(\)](#), and [DoRecovery\(\)](#).

#### 4.19.8.138 DoRecovery()

```
int DoRecovery (
    int * moduletype ) [extern]
```

Reads from the recovery file and restores all necessary variables and states in FORM, so that the execution can recommence in preprocessor() as if no restart of FORM had occurred.

The recovery file is read into memory as a whole. The pointer `p` then points into this memory at the next non-processed data. The macros by which variables are restored, like `R_SET`, automatically increase `p` appropriately.

If something goes wrong, the function returns with a non-zero value.

Allocated memory that would be lost when overwriting the global structs with data from the file is freed first. A major part of the code deals with the restoration of pointers. The idiom we use is to memorize the original pointer value (`org`), allocate new memory and copy the data from the file into this memory, calculate the offset between the old pointer value and the new allocated memory position (`ofs`), and then correct all affected pointers (`+=ofs`).

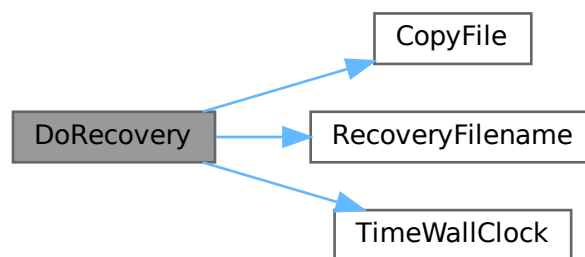


We rely on the fact that several variables (especially in AM) are already assigned the correct values by the startup functions. That means, in principle, that a change in the setup files between snapshot creation and recovery will be noticed.

Definition at line 1402 of file [checkpoint.c](#).

References [TaBlEs::argtail](#), [TaBlEs::boomlijst](#), [BrAcKeTiNfO::bracketbuffer](#), [TaBlEs::buffers](#), [TaBlEs::buffersize](#), [CopyFile\(\)](#), [TaBlEs::flags](#), [ReNuMbEr::func](#), [ReNuMbEr::funnum](#), [VaRrEnUm::hi](#), [BrAcKeTiNfO::indexbuffer](#), [ReNuMbEr::indi](#), [ReNuMbEr::indnum](#), [VaRrEnUm::lo](#), [TaBlEs::MaxTreeSize](#), [TaBlEs::mm](#), [TaBlEs::numind](#), [TaBlEs::pattern](#), [TaBlEs::prototype](#), [TaBlEs::prototypeSize](#), [RecoveryFilename\(\)](#), [ExPrEsSiOn::renumlists](#), [TaBlEs::reserved](#), [TaBlEs::spare](#), [TaBlEs::sparse](#), [VaRrEnUm::start](#), [ReNuMbEr::symb](#), [ReNuMbEr::symnum](#), [TaBlEs::tablepointers](#), [TimeWallClock\(\)](#), [TaBlEs::totind](#), [ReNuMbEr::vecnum](#), and [ReNuMbEr::vect](#).

Here is the call graph for this function:



#### 4.19.8.139 DoCheckpoint()

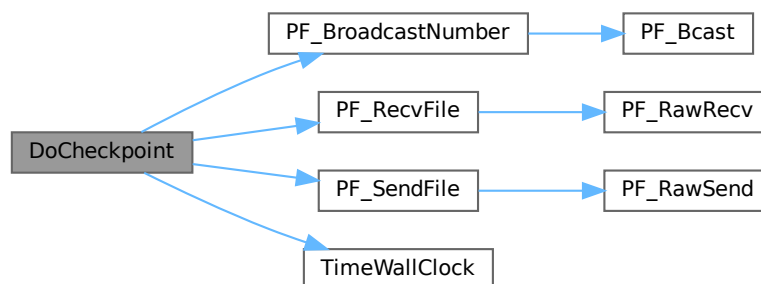
```
void DoCheckpoint (
    int moduletype ) [extern]
```

Checks whether a snapshot should be done. Calls `DoSnapshot()` to create the snapshot.

Definition at line 3111 of file [checkpoint.c](#).

References [PF\\_BroadcastNumber\(\)](#), [PF\\_RecvFile\(\)](#), [PF\\_SendFile\(\)](#), and [TimeWallClock\(\)](#).

Here is the call graph for this function:



#### 4.19.8.140 NumberMallocAddMemory()

```
void NumberMallocAddMemory (
    PHEAD0 ) [extern]
```

Definition at line 2573 of file [tools.c](#).

#### 4.19.8.141 CacheNumberMallocAddMemory()

```
void CacheNumberMallocAddMemory (
    PHEAD0 ) [extern]
```

Definition at line 2647 of file [tools.c](#).

#### 4.19.8.142 TermMallocAddMemory()

```
void TermMallocAddMemory (
    PHEAD0 ) [extern]
```

Definition at line 2469 of file [tools.c](#).

#### 4.19.8.143 ExprStatus()

```
void ExprStatus (
    EXPRESSIONS e ) [extern]
```

Definition at line 66 of file [bugtool.c](#).

#### 4.19.8.144 iniTools()

```
void iniTools (
    void ) [extern]
```

Definition at line 2201 of file [tools.c](#).

#### 4.19.8.145 TestTerm()

```
int TestTerm (
    WORD * term ) [extern]
```

Tests the consistency of the term. Returns 0 when the term is OK. Any nonzero value is trouble. In the current version the testing isn't 100% complete. For instance, we don't check the validity of the symbols nor do we check the range of their powers. Etc. This should be extended when the need is there.

##### Parameters

|             |                       |
|-------------|-----------------------|
| <i>term</i> | the term to be tested |
|-------------|-----------------------|

Definition at line 3781 of file [tools.c](#).

References [TestTerm\(\)](#).

Referenced by [TestTerm\(\)](#).

Here is the call graph for this function:



#### 4.19.8.146 RunTransform()

```
int RunTransform (
    PHEAD WORD * term,
    WORD * params ) [extern]
```

Definition at line 732 of file [transform.c](#).

#### 4.19.8.147 RunEncode()

```
int RunEncode (
    PHEAD WORD * fun,
    WORD * args,
    WORD * info ) [extern]
```

Definition at line 982 of file [transform.c](#).

#### 4.19.8.148 RunDecode()

```
int RunDecode (
    PHEAD WORD * fun,
    WORD * args,
    WORD * info ) [extern]
```

Definition at line 1169 of file [transform.c](#).

#### 4.19.8.149 RunReplace()

```
int RunReplace (
    PHEAD WORD * fun,
    WORD * args,
    WORD * info ) [extern]
```

Definition at line 1339 of file [transform.c](#).

#### 4.19.8.150 RunImplode()

```
int RunImplode (
    WORD * fun,
    WORD * args ) [extern]
```

Definition at line 1817 of file [transform.c](#).

#### 4.19.8.151 RunExplode()

```
int RunExplode (
    PHEAD WORD * fun,
    WORD * args ) [extern]
```

Definition at line 2018 of file [transform.c](#).

#### 4.19.8.152 TestArgNum()

```
int TestArgNum (
    int n,
    int totarg,
    WORD * args ) [extern]
```

Definition at line 3283 of file [transform.c](#).

#### 4.19.8.153 PutArgInScratch()

```
WORD PutArgInScratch (
    WORD * arg,
    UWORD * scrat ) [extern]
```

Definition at line 3352 of file [transform.c](#).

#### 4.19.8.154 ReadRange()

```
UBYTE * ReadRange (
    UBYTE * s,
    WORD * out,
    int par ) [extern]
```

Definition at line 3388 of file [transform.c](#).

#### 4.19.8.155 FindRange()

```
int FindRange (
    PHEAD WORD * args,
    WORD * arg1,
    WORD * arg2,
    WORD totarg ) [extern]
```

Definition at line 3548 of file [transform.c](#).

**4.19.8.156 RunPermute()**

```
int RunPermute (
    PHEAD WORD * fun,
    WORD * args,
    WORD * info ) [extern]
```

Definition at line [2132](#) of file [transform.c](#).

**4.19.8.157 RunReverse()**

```
int RunReverse (
    PHEAD WORD * fun,
    WORD * args ) [extern]
```

Definition at line [2359](#) of file [transform.c](#).

**4.19.8.158 RunCycle()**

```
int RunCycle (
    PHEAD WORD * fun,
    WORD * args,
    WORD * info ) [extern]
```

Definition at line [2535](#) of file [transform.c](#).

**4.19.8.159 RunAddArg()**

```
int RunAddArg (
    PHEAD WORD * fun,
    WORD * args ) [extern]
```

Definition at line [2687](#) of file [transform.c](#).

**4.19.8.160 RunMulArg()**

```
int RunMulArg (
    PHEAD WORD * fun,
    WORD * args ) [extern]
```

Definition at line [2776](#) of file [transform.c](#).

**4.19.8.161 RunIsLyndon()**

```
int RunIsLyndon (
    PHEAD WORD * fun,
    WORD * args,
    int par ) [extern]
```

Definition at line [2917](#) of file [transform.c](#).

**4.19.8.162 RunToLyndon()**

```
WORD RunToLyndon (  
    PHEAD WORD * fun,  
    WORD * args,  
    int par ) [extern]
```

Definition at line [2995](#) of file [transform.c](#).

**4.19.8.163 RunDropArg()**

```
int RunDropArg (  
    PHEAD WORD * fun,  
    WORD * args ) [extern]
```

Definition at line [3107](#) of file [transform.c](#).

**4.19.8.164 RunSelectArg()**

```
int RunSelectArg (  
    PHEAD WORD * fun,  
    WORD * args ) [extern]
```

Definition at line [3133](#) of file [transform.c](#).

**4.19.8.165 RunDedup()**

```
int RunDedup (  
    PHEAD WORD * fun,  
    WORD * args ) [extern]
```

Definition at line [2446](#) of file [transform.c](#).

**4.19.8.166 RunZtoHArg()**

```
int RunZtoHArg (  
    PHEAD WORD * fun,  
    WORD * args ) [extern]
```

Definition at line [3161](#) of file [transform.c](#).

**4.19.8.167 RunHtoZArg()**

```
int RunHtoZArg (  
    PHEAD WORD * fun,  
    WORD * args ) [extern]
```

Definition at line [3217](#) of file [transform.c](#).

#### 4.19.8.168 NormPolyTerm()

```
int NormPolyTerm (
    PHEAD WORD * term ) [extern]
```

Definition at line 53 of file [notation.c](#).

#### 4.19.8.169 ConvertToPoly()

```
int ConvertToPoly (
    PHEAD WORD * term,
    WORD * outterm,
    WORD * comlist,
    WORD par ) [extern]
```

Definition at line 307 of file [notation.c](#).

#### 4.19.8.170 LocalConvertToPoly()

```
int LocalConvertToPoly (
    PHEAD WORD * term,
    WORD * outterm,
    WORD startebuf,
    WORD par ) [extern]
```

Converts a generic term to polynomial notation in which there are only symbols and brackets. During conversion there will be only symbols. Brackets are stripped. Objects that need 'translation' are put inside a special compiler buffer and represented by a symbol. The numbering of the extra symbols is down from the maximum. In principle there can be a problem when running into the already assigned ones. This uses the FindTree for searching in the global tree and then looks further in the AT.ebufnum. This allows fully parallel processing. Hence we need no locks. Cannot be used in the same module as ConvertToPoly.

Definition at line 510 of file [notation.c](#).

Referenced by [poly\\_factorize\\_expression\(\)](#).

#### 4.19.8.171 ConvertFromPoly()

```
int ConvertFromPoly (
    PHEAD WORD * term,
    WORD * outterm,
    WORD from,
    WORD to,
    WORD offset,
    WORD par ) [extern]
```

Definition at line 660 of file [notation.c](#).

#### 4.19.8.172 FindSubterm()

```
int FindSubterm (
    WORD * subterm ) [extern]
```

Definition at line 773 of file [notation.c](#).

**4.19.8.173 FindLocalSubterm()**

```
int FindLocalSubterm (
    PHEAD WORD * subterm,
    WORD startebuf ) [extern]
```

Definition at line [843](#) of file [notation.c](#).

**4.19.8.174 PrintSubtermList()**

```
void PrintSubtermList (
    int from,
    int to ) [extern]
```

Definition at line [897](#) of file [notation.c](#).

**4.19.8.175 PrintExtraSymbol()**

```
void PrintExtraSymbol (
    int num,
    WORD * terms,
    int par ) [extern]
```

Definition at line [1009](#) of file [notation.c](#).

**4.19.8.176 FindSubexpression()**

```
int FindSubexpression (
    WORD * subexpr ) [extern]
```

Definition at line [1096](#) of file [notation.c](#).

**4.19.8.177 UpdateMaxSize()**

```
void UpdateMaxSize (
    void ) [extern]
```

Definition at line [1610](#) of file [tools.c](#).

**4.19.8.178 CoToPolynomial()**

```
int CoToPolynomial (
    UBYTE * inp ) [extern]
```

Definition at line [5925](#) of file [compcomm.c](#).



**4.19.8.179 CoFromPolynomial()**

```
int CoFromPolynomial (
    UBYTE * inp ) [extern]
```

Definition at line [6000](#) of file [compcomm.c](#).

**4.19.8.180 CoArgToExtraSymbol()**

```
int CoArgToExtraSymbol (
    UBYTE * s ) [extern]
```

Definition at line [6026](#) of file [compcomm.c](#).

**4.19.8.181 CoExtraSymbols()**

```
int CoExtraSymbols (
    UBYTE * inp ) [extern]
```

Definition at line [6076](#) of file [compcomm.c](#).

**4.19.8.182 GetDoParam()**

```
UBYTE * GetDoParam (
    UBYTE * inp,
    WORD ** wp,
    int par ) [extern]
```

Definition at line [6207](#) of file [compcomm.c](#).

**4.19.8.183 GetIfDollarFactor()**

```
WORD * GetIfDollarFactor (
    UBYTE ** inp,
    WORD * w ) [extern]
```

Definition at line [6149](#) of file [compcomm.c](#).

**4.19.8.184 CoDo()**

```
int CoDo (
    UBYTE * inp ) [extern]
```

Definition at line [6284](#) of file [compcomm.c](#).

#### 4.19.8.185 CoEndDo()

```
int CoEndDo (
    UBYTE * inp ) [extern]
```

Definition at line 6387 of file [compcomm.c](#).

#### 4.19.8.186 ExtraSymFun()

```
int ExtraSymFun (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 1149 of file [notation.c](#).

#### 4.19.8.187 PruneExtraSymbols()

```
int PruneExtraSymbols (
    WORD downto ) [extern]
```

Definition at line 1217 of file [notation.c](#).

#### 4.19.8.188 IniFbuffer()

```
int IniFbuffer (
    WORD bufnum ) [extern]
```

Initialize a factorization cache buffer. We set the size of the rhs and boomlijst buffers immediately to their final values.

Definition at line 614 of file [comtool.c](#).

References [tree::blnce](#), [CbUf::boomlijst](#), [CbUf::CanCommu](#), [CbUf::dimension](#), [tree::left](#), [CbUf::numdum](#), [CbUf::NumTerms](#), [tree::parent](#), [CbUf::rhs](#), [tree::right](#), [tree::usage](#), and [tree::value](#).

#### 4.19.8.189 GCDfunction()

```
int GCDfunction (
    PHEAD WORD * term,
    WORD level ) [extern]
```

Definition at line 609 of file [ratio.c](#).

#### 4.19.8.190 GCDfunction3()

```
WORD * GCDfunction3 (
    PHEAD WORD * in1,
    WORD * in2 ) [extern]
```

Definition at line 1141 of file [ratio.c](#).

**4.19.8.191 ReadPolyRatFun()**

```
int ReadPolyRatFun (
    PHEAD WORD * term ) [extern]
```

Definition at line 2264 of file [ratio.c](#).

**4.19.8.192 FromPolyRatFun()**

```
int FromPolyRatFun (
    PHEAD WORD * fun,
    WORD ** numout,
    WORD ** denout ) [extern]
```

Definition at line 2383 of file [ratio.c](#).

**4.19.8.193 PolyDiv()**

```
WORD * PolyDiv (
    PHEAD WORD * a,
    WORD * b,
    char * text ) [extern]
```

Definition at line 2741 of file [ratio.c](#).

**4.19.8.194 GCDclean()**

```
void GCDclean (
    PHEAD WORD * num,
    WORD * den ) [extern]
```

Definition at line 2644 of file [ratio.c](#).

**4.19.8.195 TakeSymbolContent()**

```
WORD * TakeSymbolContent (
    PHEAD WORD * in,
    WORD * term ) [extern]
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. We allow only symbols as this code is used for the polyratfun only. We have a special routine, because the generic TakeContent does too much work and speed is at a premium here. Input: in is the input expression as a sequence of terms. **Output:** term: the content return value: the contentfree expression. it is in new allocation, made by TermMalloc. (should be in a TermMalloc space?)

Definition at line 2434 of file [ratio.c](#).

References [GetModInverses\(\)](#).

Here is the call graph for this function:



#### 4.19.8.196 GCDterms()

```
int GCDterms (
    PHEAD WORD * term1,
    WORD * term2,
    WORD * termout ) [extern]
```

Definition at line 2076 of file [ratio.c](#).

#### 4.19.8.197 PutExtraSymbols()

```
WORD * PutExtraSymbols (
    PHEAD WORD * in,
    WORD startebuf,
    int * actionflag ) [extern]
```

Definition at line 1244 of file [ratio.c](#).

#### 4.19.8.198 TakeExtraSymbols()

```
WORD * TakeExtraSymbols (
    PHEAD WORD * in,
    WORD startebuf ) [extern]
```

Definition at line 1273 of file [ratio.c](#).

#### 4.19.8.199 MultiplyWithTerm()

```
WORD * MultiplyWithTerm (
    PHEAD WORD * in,
    WORD * term,
    WORD par ) [extern]
```

Definition at line 1312 of file [ratio.c](#).

#### 4.19.8.200 TakeContent()

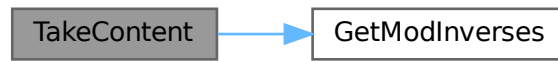
```
WORD * TakeContent (
    PHEAD WORD * in,
    WORD * term ) [extern]
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. Here the input is a sequence of terms in 'in' and the answer is a content-free sequence of terms. This sequence has been allocated by the Malloc1 routine in a call to EndSort, unless the expression was already content-free. In that case the input pointer is returned. The content is returned in term. This is supposed to be a separate allocation, made by TermMalloc in the calling routine.

Definition at line 1376 of file [ratio.c](#).

References [GetModInverses\(\)](#).

Here is the call graph for this function:



#### 4.19.8.201 MergeSymbolLists()

```
int MergeSymbolLists (  
    PHEAD WORD * old,  
    WORD * extra,  
    int par ) [extern]
```

Definition at line 1808 of file [ratio.c](#).

#### 4.19.8.202 MergeDotproductLists()

```
int MergeDotproductLists (  
    PHEAD WORD * old,  
    WORD * extra,  
    int par ) [extern]
```

Definition at line 1927 of file [ratio.c](#).

#### 4.19.8.203 CreateExpression()

```
WORD * CreateExpression (  
    PHEAD WORD ) [extern]
```

Definition at line 2020 of file [ratio.c](#).

#### 4.19.8.204 DIVfunction()

```
int DIVfunction (  
    PHEAD WORD * term,  
    WORD level,  
    int par ) [extern]
```

Definition at line 2788 of file [ratio.c](#).

#### 4.19.8.205 MULfunc()

```
WORD * MULfunc (
    PHEAD WORD * p1,
    WORD * p2 ) [extern]
```

Definition at line [2957](#) of file [ratio.c](#).

#### 4.19.8.206 ConvertArgument()

```
WORD * ConvertArgument (
    PHEAD WORD * arg,
    int * type ) [extern]
```

Definition at line [3018](#) of file [ratio.c](#).

#### 4.19.8.207 ExpandRat()

```
int ExpandRat (
    PHEAD WORD * fun ) [extern]
```

Definition at line [3113](#) of file [ratio.c](#).

#### 4.19.8.208 InvPoly()

```
int InvPoly (
    PHEAD WORD * inpoly,
    WORD maxpow,
    WORD sym ) [extern]
```

Definition at line [3592](#) of file [ratio.c](#).

#### 4.19.8.209 TestDoLoop()

```
WORD TestDoLoop (
    PHEAD WORD * lhsbuf,
    WORD level ) [extern]
```

Definition at line [2762](#) of file [dollar.c](#).

#### 4.19.8.210 TestEndDoLoop()

```
WORD TestEndDoLoop (
    PHEAD WORD * lhsbuf,
    WORD level ) [extern]
```

Definition at line [2840](#) of file [dollar.c](#).

#### 4.19.8.211 poly\_gcd()

```
WORD * poly_gcd (
    PHEAD WORD * a,
    WORD * b,
    WORD fit ) [extern]
```

Polynomial gcd

### 4.19.9 Description

This method calculates the greatest common divisor of two polynomials, given by two zero-terminated Form-style term lists.

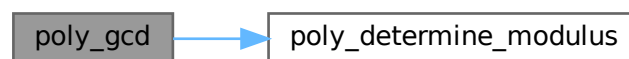
### 4.19.10 Notes

- The result is written at newly allocated memory
- Called from [ratio.c](#)
- Calls `polygcd::gcd`

Definition at line [124](#) of file [polywrap.cc](#).

References [poly\\_determine\\_modulus\(\)](#).

Here is the call graph for this function:



#### 4.19.10.1 poly\_div()

```
WORD * poly_div (
    PHEAD WORD * a,
    WORD * b,
    WORD fit ) [extern]
```

Definition at line [435](#) of file [polywrap.cc](#).

#### 4.19.10.2 poly\_rem()

```
WORD * poly_rem (
    PHEAD WORD * a,
    WORD * b,
    WORD fit ) [extern]
```

Definition at line 463 of file [polywrap.cc](#).

#### 4.19.10.3 poly\_inverse()

```
WORD * poly_inverse (
    PHEAD WORD * arga,
    WORD * argb ) [extern]
```

Definition at line 1743 of file [polywrap.cc](#).

#### 4.19.10.4 poly\_mul()

```
WORD * poly_mul (
    PHEAD WORD * a,
    WORD * b ) [extern]
```

Definition at line 1948 of file [polywrap.cc](#).

#### 4.19.10.5 poly\_ratfun\_add()

```
WORD * poly_ratfun_add (
    PHEAD WORD * t1,
    WORD * t2 ) [extern]
```

Addition of PolyRatFuns

### 4.19.11 Description

This method gets two pointers to polyratfuns with up to two arguments each and calculates the sum.



### 4.19.12 Notes

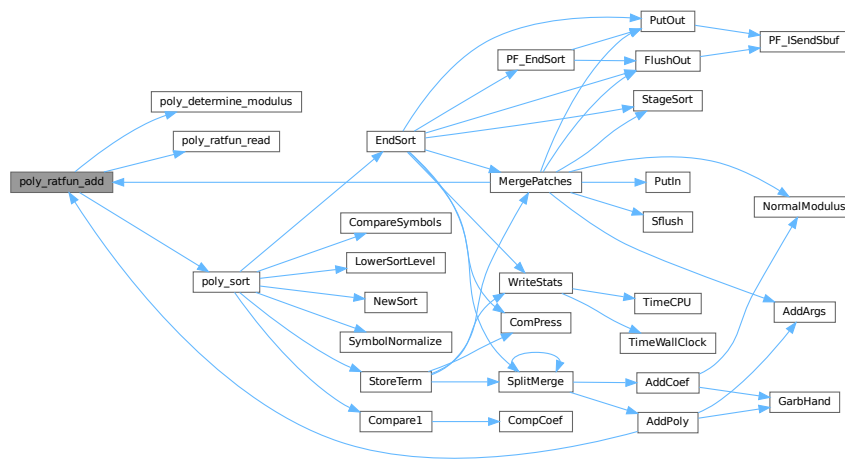
- The result is written at the workpointer
- Called from [sort.c](#) and [threads.c](#)
- Calls `poly::operators` and `polygcd::gcd`

Definition at line 633 of file [polywrap.cc](#).

References [poly\\_determine\\_modulus\(\)](#), [poly\\_ratfun\\_read\(\)](#), and [poly\\_sort\(\)](#).

Referenced by [AddPoly\(\)](#), and [MergePatches\(\)](#).

Here is the call graph for this function:



#### 4.19.12.1 poly\_ratfun\_normalize()

```
int poly_ratfun_normalize (
    PHEAD WORD * term ) [extern]
```

Multiplication/normalization of PolyRatFuns

### 4.19.13 Description

This method searches a term for multiple polyratfuns and multiplies their contents. The result is properly normalized. Normalization also works for terms with a single polyratfun.

#### 4.19.14 Notes

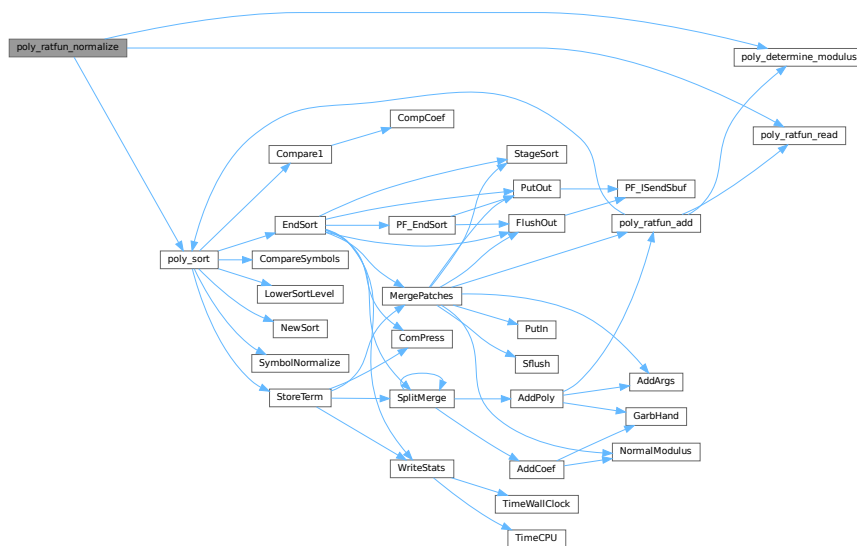
- The result overwrites the original term
- Called from [proces.c](#)
- Calls `poly::operators` and `polygcd::gcd`

Definition at line 769 of file [polywrap.cc](#).

References [poly\\_determine\\_modulus\(\)](#), [poly\\_ratfun\\_read\(\)](#), and [poly\\_sort\(\)](#).

Referenced by [PolyFunMul\(\)](#), and [PrepPoly\(\)](#).

Here is the call graph for this function:



##### 4.19.14.1 poly\_factorize\_argument()

```

int poly_factorize_argument (
    PHEAD WORD * argin,
    WORD * argout ) [extern]
  
```

Factorization of function arguments

#### 4.19.15 Description

This method factorizes the Form-style argument `argin`.

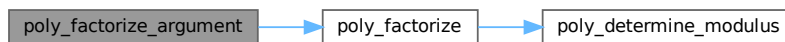
### 4.19.16 Notes

- The result is written at argout
- Called from [argument.c](#)
- Calls `poly_factorize`

Definition at line 1112 of file [polywrap.cc](#).

References [poly\\_factorize\(\)](#).

Here is the call graph for this function:



#### 4.19.16.1 poly\_factorize\_dollar()

```
WORD * poly_factorize_dollar (
    PHEAD WORD * argin ) [extern]
```

Factorization of dollar variables

### 4.19.17 Description

This method factorizes a dollar variable.

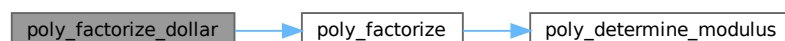
### 4.19.18 Notes

- The result is written at newly allocated memory.
- Called from [dollar.c](#)
- Calls `poly_factorize`

Definition at line 1146 of file [polywrap.cc](#).

References [poly\\_factorize\(\)](#).

Here is the call graph for this function:





### 4.19.20.1 poly\_unfactorize\_expression()

```
int poly_unfactorize_expression (
    EXPRESSIONS expr ) [extern]
```

Unfactorization of expressions

### 4.19.21 Description

This method expands a factorized expression.

### 4.19.22 Notes

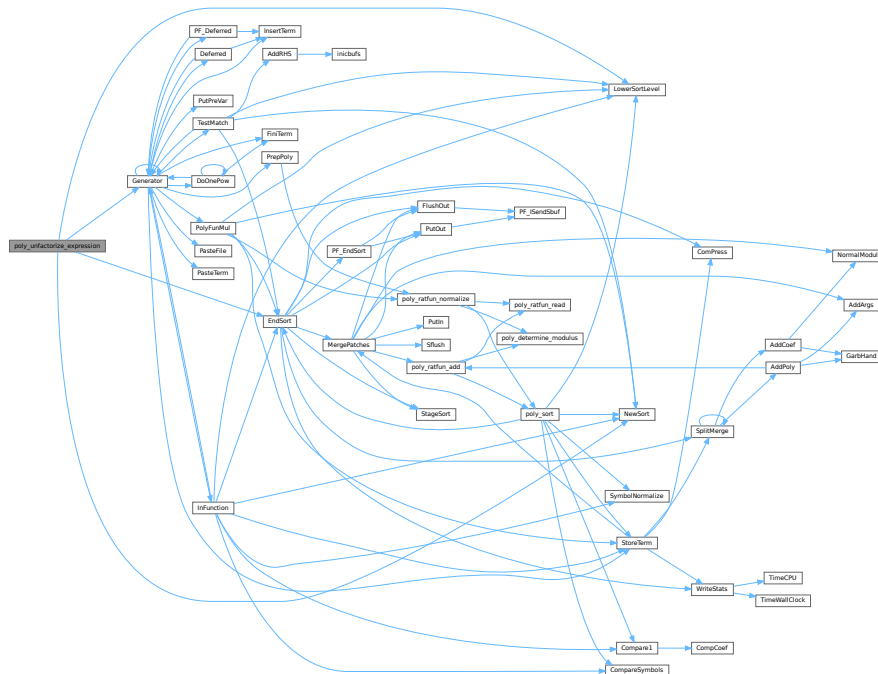
- The result overwrites the input expression
- Called from [proces.c](#)

Definition at line 1535 of file [polywrap.cc](#).

References [EndSort\(\)](#), [Generator\(\)](#), [FiLe::handle](#), [LowerSortLevel\(\)](#), and [NewSort\(\)](#).

Referenced by [PF\\_Processor\(\)](#), and [Processor\(\)](#).

Here is the call graph for this function:



#### 4.19.22.1 poly\_free\_poly\_vars()

```
void poly_free_poly_vars (
    PHEAD const char * text ) [extern]
```

Definition at line [2014](#) of file [polywrap.cc](#).

#### 4.19.22.2 optimize\_print\_code()

```
void optimize_print_code (
    int print_expr ) [extern]
```

Print optimized code

### 4.19.23 Description

This method prints the optimized code via "PrintExtraSymbol". Depending on the flag, the original expression is printed (for "Print") or not (for "#Optimize / #write "O").

Definition at line [4525](#) of file [optimize.cc](#).

Referenced by [Optimize\(\)](#).

#### 4.19.23.1 DoPreAdd()

```
int DoPreAdd (
    UBYTE * s ) [extern]
```

Definition at line [1067](#) of file [dict.c](#).

#### 4.19.23.2 DoPreUseDictionary()

```
int DoPreUseDictionary (
    UBYTE * s ) [extern]
```

Definition at line [1022](#) of file [dict.c](#).

#### 4.19.23.3 DoPreCloseDictionary()

```
int DoPreCloseDictionary (
    UBYTE * s ) [extern]
```

Definition at line [998](#) of file [dict.c](#).

#### 4.19.23.4 DoPreOpenDictionary()

```
int DoPreOpenDictionary (
    UBYTE * s ) [extern]
```

Definition at line 959 of file [dict.c](#).

#### 4.19.23.5 RemoveDictionary()

```
void RemoveDictionary (
    DICTIONARY * dict ) [extern]
```

Definition at line 902 of file [dict.c](#).

#### 4.19.23.6 UnSetDictionary()

```
void UnSetDictionary (
    void ) [extern]
```

Definition at line 883 of file [dict.c](#).

#### 4.19.23.7 SetDictionaryOptions()

```
int SetDictionaryOptions (
    UBYTE * options ) [extern]
```

Definition at line 790 of file [dict.c](#).

#### 4.19.23.8 AddToDictionary()

```
int AddToDictionary (
    DICTIONARY * dict,
    UBYTE * left,
    UBYTE * right ) [extern]
```

Definition at line 491 of file [dict.c](#).

#### 4.19.23.9 AddDictionary()

```
int AddDictionary (
    UBYTE * name ) [extern]
```

Definition at line 449 of file [dict.c](#).

#### 4.19.23.10 FindDictionary()

```
int FindDictionary (
    UBYTE * name ) [extern]
```

Definition at line 434 of file [dict.c](#).

#### 4.19.23.11 IsExponentSign()

```
UBYTE * IsExponentSign (
    void ) [extern]
```

Definition at line 238 of file [dict.c](#).

#### 4.19.23.12 IsMultiplySign()

```
UBYTE * IsMultiplySign (
    void ) [extern]
```

Definition at line 218 of file [dict.c](#).

#### 4.19.23.13 TransformRational()

```
void TransformRational (
    UWORD * a,
    WORD na ) [extern]
```

Definition at line 71 of file [dict.c](#).

#### 4.19.23.14 WriteDictionary()

```
void WriteDictionary (
    DICTIONARY * dict ) [extern]
```

Definition at line 1307 of file [sch.c](#).

#### 4.19.23.15 ShrinkDictionary()

```
void ShrinkDictionary (
    DICTIONARY * dict ) [extern]
```

Definition at line 945 of file [dict.c](#).

#### 4.19.23.16 MultiplyToLine()

```
void MultiplyToLine (
    void ) [extern]
```

Definition at line 671 of file [sch.c](#).

#### 4.19.23.17 FindSymbol()

```
UBYTE * FindSymbol (
    WORD num ) [extern]
```

Definition at line 258 of file [dict.c](#).



#### 4.19.23.18 FindVector()

```
UBYTE * FindVector (
    WORD num ) [extern]
```

Definition at line 279 of file [dict.c](#).

#### 4.19.23.19 FindIndex()

```
UBYTE * FindIndex (
    WORD num ) [extern]
```

Definition at line 301 of file [dict.c](#).

#### 4.19.23.20 FindFunction()

```
UBYTE * FindFunction (
    WORD num ) [extern]
```

Definition at line 323 of file [dict.c](#).

#### 4.19.23.21 FindFunWithArgs()

```
UBYTE * FindFunWithArgs (
    WORD * t ) [extern]
```

Definition at line 345 of file [dict.c](#).

#### 4.19.23.22 FindExtraSymbol()

```
UBYTE * FindExtraSymbol (
    WORD num ) [extern]
```

Definition at line 377 of file [dict.c](#).

#### 4.19.23.23 DictToBytes()

```
LONG DictToBytes (
    DICTIONARY * dict,
    UBYTE * buf ) [extern]
```

Definition at line 1119 of file [dict.c](#).

#### 4.19.23.24 DictFromBytes()

```
DICTIONARY * DictFromBytes (
    UBYTE * buf ) [extern]
```

Definition at line 1152 of file [dict.c](#).

#### 4.19.23.25 CoCreateSpectator()

```
int CoCreateSpectator (
    UBYTE * inp ) [extern]
```

Definition at line 89 of file [spectator.c](#).

#### 4.19.23.26 CoToSpectator()

```
int CoToSpectator (
    UBYTE * inp ) [extern]
```

Definition at line 181 of file [spectator.c](#).

#### 4.19.23.27 CoRemoveSpectator()

```
int CoRemoveSpectator (
    UBYTE * inp ) [extern]
```

Definition at line 233 of file [spectator.c](#).

#### 4.19.23.28 CoEmptySpectator()

```
int CoEmptySpectator (
    UBYTE * inp ) [extern]
```

Definition at line 293 of file [spectator.c](#).

#### 4.19.23.29 CoCopySpectator()

```
int CoCopySpectator (
    UBYTE * inp ) [extern]
```

Definition at line 459 of file [spectator.c](#).

#### 4.19.23.30 PutInSpectator()

```
int PutInSpectator (
    WORD * term,
    WORD specnum ) [extern]
```

Definition at line 345 of file [spectator.c](#).

#### 4.19.23.31 ClearSpectators()

```
void ClearSpectators (
    WORD par ) [extern]
```

Definition at line 675 of file [spectator.c](#).

#### 4.19.23.32 GetFromSpectator()

```
WORD GetFromSpectator (
    WORD * term,
    WORD specnum ) [extern]
```

Definition at line 601 of file [spectator.c](#).

#### 4.19.23.33 FlushSpectators()

```
void FlushSpectators (
    void ) [extern]
```

Definition at line 402 of file [spectator.c](#).

#### 4.19.23.34 ReadFromScratch()

```
int ReadFromScratch (
    FILEHANDLE * fi,
    POSITION * pos,
    UBYTE * buffer,
    POSITION * length ) [extern]
```

Definition at line 272 of file [store.c](#).

#### 4.19.23.35 AddToScratch()

```
int AddToScratch (
    FILEHANDLE * fi,
    POSITION * pos,
    UBYTE * buffer,
    POSITION * length,
    int withflush ) [extern]
```

Definition at line 299 of file [store.c](#).

#### 4.19.23.36 DoPreAppendPath()

```
int DoPreAppendPath (
    UBYTE * s ) [extern]
```

Appends the given path (absolute or relative to the current file directory) to the FORM path.

Syntax: `#appendpath <path>`

Definition at line 7153 of file [pre.c](#).

#### 4.19.23.37 DoPrePrependPath()

```
int DoPrePrependPath (
    UBYTE * s ) [extern]
```

Prepends the given path (absolute or relative to the current file directory) to the FORM path.

Syntax: #prependpath <path>

Definition at line 7170 of file [pre.c](#).

#### 4.19.23.38 DoSwitch()

```
int DoSwitch (
    PHEAD WORD * term,
    WORD * lhs ) [extern]
```

Definition at line 1070 of file [if.c](#).

#### 4.19.23.39 DoEndSwitch()

```
int DoEndSwitch (
    PHEAD WORD * term,
    WORD * lhs ) [extern]
```

Definition at line 1086 of file [if.c](#).

#### 4.19.23.40 FindCase()

```
SWITCHTABLE * FindCase (
    WORD nswitch,
    WORD ncase ) [extern]
```

Definition at line 1097 of file [if.c](#).

#### 4.19.23.41 SwitchSplitMergeRec()

```
void SwitchSplitMergeRec (
    SWITCHTABLE * array,
    WORD num,
    SWITCHTABLE * auxarray ) [extern]
```

Definition at line 1184 of file [if.c](#).

#### 4.19.23.42 SwitchSplitMerge()

```
void SwitchSplitMerge (
    SWITCHTABLE * array,
    WORD num ) [extern]
```

Definition at line 1213 of file [if.c](#).

#### 4.19.23.43 DoubleSwitchBuffers()

```
int DoubleSwitchBuffers (
    void ) [extern]
```

Definition at line 1135 of file [if.c](#).

#### 4.19.23.44 DistrN()

```
int DistrN (
    int n,
    int * cpl,
    int ncpl,
    int * scratch ) [extern]
```

Definition at line 3967 of file [tools.c](#).

#### 4.19.23.45 DoNamespace()

```
int DoNamespace (
    UBYTE * s ) [extern]
```

Definition at line 7234 of file [pre.c](#).

#### 4.19.23.46 DoEndNamespace()

```
int DoEndNamespace (
    UBYTE * s ) [extern]
```

Definition at line 7282 of file [pre.c](#).

#### 4.19.23.47 DoUse()

```
int DoUse (
    UBYTE * s ) [extern]
```

Definition at line 7497 of file [pre.c](#).

#### 4.19.23.48 SkipName()

```
UBYTE * SkipName (
    UBYTE * s ) [extern]
```

Definition at line 7305 of file [pre.c](#).

#### 4.19.23.49 ConstructName()

```
UBYTE * ConstructName (
    UBYTE * s,
    UBYTE type ) [extern]
```

Definition at line 7405 of file [pre.c](#).

#### 4.19.23.50 DoSetUserFlag()

```
int DoSetUserFlag (
    UBYTE * s ) [extern]
```

Definition at line 7662 of file [pre.c](#).

#### 4.19.23.51 DoClearUserFlag()

```
int DoClearUserFlag (
    UBYTE * s ) [extern]
```

Definition at line 7652 of file [pre.c](#).

#### 4.19.23.52 CreateVertex()

```
VERTEX * CreateVertex (
    MODEL * m )
```

Definition at line 46 of file [model.c](#).

#### 4.19.23.53 ReadParticle()

```
UBYTE * ReadParticle (
    UBYTE * s,
    VERTEX * v,
    MODEL * m,
    int par )
```

Definition at line 76 of file [model.c](#).

#### 4.19.23.54 CoModel()

```
int CoModel (
    UBYTE * s )
```

Definition at line 119 of file [model.c](#).

#### 4.19.23.55 CoParticle()

```
int CoParticle (
    UBYTE * s )
```

Definition at line 194 of file [model.c](#).

#### 4.19.23.56 CoVertex()

```
int CoVertex (
    UBYTE * s )
```

Definition at line 276 of file [model.c](#).

#### 4.19.23.57 CoEndModel()

```
int CoEndModel (
    UBYTE * s )
```

Definition at line 379 of file [model.c](#).

#### 4.19.23.58 LoadModel()

```
int LoadModel (
    MODEL * m )
```

Definition at line 27 of file [diawrap.cc](#).

#### 4.19.23.59 CoSetUserFlag()

```
int CoSetUserFlag (
    UBYTE * s )
```

Definition at line 7384 of file [compcomm.c](#).

#### 4.19.23.60 CoClearUserFlag()

```
int CoClearUserFlag (
    UBYTE * s )
```

Definition at line 7412 of file [compcomm.c](#).

#### 4.19.23.61 CoCreateAllLoops()

```
int CoCreateAllLoops (
    UBYTE * s )
```

Definition at line 7449 of file [compcomm.c](#).

#### 4.19.23.62 CoCreateAllPaths()

```
int CoCreateAllPaths (
    UBYTE * s )
```

Definition at line [7569](#) of file [compcomm.c](#).

#### 4.19.23.63 CoCreateAll()

```
int CoCreateAll (
    UBYTE * s )
```

Definition at line [7697](#) of file [compcomm.c](#).

#### 4.19.23.64 AllLoops()

```
int AllLoops (
    PHEAD WORD * term,
    WORD level )
```

Definition at line [650](#) of file [lus.c](#).

#### 4.19.23.65 StartLoops()

```
LONG StartLoops (
    PHEAD WORD * term,
    WORD level,
    LONG vert,
    WORD nvert,
    WORD * arglist,
    WORD nargs,
    WORD * loop,
    WORD nloop )
```

Definition at line [894](#) of file [lus.c](#).

#### 4.19.23.66 GenLoops()

```
LONG GenLoops (
    PHEAD WORD * term,
    WORD level,
    LONG vert,
    WORD nvert,
    WORD * arglist,
    WORD nargs,
    WORD * loop,
    WORD nloop )
```

Definition at line [980](#) of file [lus.c](#).



#### 4.19.23.67 LoopOutput()

```
void LoopOutput (
    PHEAD WORD * term,
    WORD level,
    WORD * loop,
    WORD nloop )
```

Definition at line [1059](#) of file [lus.c](#).

#### 4.19.23.68 AllPaths()

```
int AllPaths (
    PHEAD WORD * term,
    WORD level )
```

Definition at line [1123](#) of file [lus.c](#).

#### 4.19.23.69 GenPaths()

```
LONG GenPaths (
    PHEAD WORD * term,
    WORD level,
    LONG vert,
    WORD nvert,
    WORD * arglist,
    WORD nargs,
    WORD * path,
    WORD npath )
```

Definition at line [1413](#) of file [lus.c](#).

#### 4.19.23.70 PathOutput()

```
void PathOutput (
    PHEAD WORD * term,
    WORD level,
    WORD * path,
    WORD npath )
```

Definition at line [1486](#) of file [lus.c](#).

## 4.20 declare.h

[Go to the documentation of this file.](#)

```

00001 #ifndef __FDECLARE__
00002 #define __FDECLARE__
00003
00009 /* #[] License : */
00010 /*
00011  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00012  * When using this file you are requested to refer to the publication
00013  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00014  * This is considered a matter of courtesy as the development was paid
00015  * for by FOM the Dutch physics granting agency and we would like to
00016  * be able to track its scientific use to convince FOM of its value
00017  * for the community.
00018  *
00019  * This file is part of FORM.
00020  *
00021  * FORM is free software: you can redistribute it and/or modify it under the
00022  * terms of the GNU General Public License as published by the Free Software
00023  * Foundation, either version 3 of the License, or (at your option) any later
00024  * version.
00025  *
00026  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00027  * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00028  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00029  * details.
00030  *
00031  * You should have received a copy of the GNU General Public License along
00032  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00033  */
00034 /* #[] License : */
00035
00036 /*
00037  #[] Macro's :
00038 */
00039
00040 #define MaX(x,y) ((x) > (y) ? (x) : (y))
00041 #define MiN(x,y) ((x) < (y) ? (x) : (y))
00042 #define ABS(x) ( (x) < 0 ? -(x) : (x) )
00043 #define SGN(x) ( (x) > 0 ? 1 : (x) < 0 ? -1 : 0 )
00044 #define REDLENG(x) (((x)<0)?((x)+1):((x)-1))/2)
00045 #define INCLENG(x) (((x)<0)?((x)*2)-1:((x)*2)+1))
00046 #define GETCOEF(x,y) x += *x;y = x[-1];x -= ABS(y);y=REDLENG(y)
00047 #define GETSTOP(x,y) y=x+(*x)-1;y -= ABS(*y)-1
00048 #define StuffAdd(x,y) ((x)<0?-1:1)* (y)+((y)<0?-1:1)* (x))
00049
00050 #define EXCHN(t1,t2,n) { WORD a,i; for(i=0;i<n;i++){a=t1[i];t1[i]=t2[i];t2[i]=a;} }
00051 #define EXCH(x,y) { WORD a = (x); (x) = (y); (y) = a; }
00052
00053 #define TOKENTOLINE(x,y) if ( AC.OutputSpaces == NOSPACEFORMAT ) { \
00054     TokenToLine((UBYTE *) (y)); } else { TokenToLine((UBYTE *) (x)); }
00055
00056 #define UngetFromStream(stream,c) ((stream)->nextchar[(stream)->isnextchar++]=c)
00057 #ifdef WITHRETURN
00058 #define AddLineFeed(s,n) { (s)[(n)++] = CARRIAGERETURN; (s)[(n)++] = LINEFEED; }
00059 #else
00060 #define AddLineFeed(s,n) { (s)[(n)++] = LINEFEED; }
00061 #endif
00062 #define TryRecover(x) Terminate(-1)
00063 #define UngetChar(c) { pushbackchar = c; }
00064 #define ParseNumber(x,s) { (x)=0;while(*(s)>='0'&&*(s)<='9')(x)=10*(x)+*(s)++ -'0';}
00065 #define ParseSign(sgn,s) { (sgn)=0;while(*(s)=='-'||*(s)=='+'){ \
00066     if ( *(s)++ == '-' ) sgn ^= 1;}}
00067 #define ParseSignedNumber(x,s) { int sgn; ParseSign(sgn,s)\
00068     ParseNumber(x,s) if ( sgn ) x = -x; }
00069
00070 /* (n) is necessary here, since the macro is sometimes passed dereferenced pointers for n */
00071 #define NCOPY(s,t,n) { while ( (n)-- > 0 ) { *s++ = *t++; } }
00072 /*#define NCOPY(s,t,n) { memcpy(s,t,n*sizeof(WORD)); s+=n; t+=n; n = -1; }*/
00073 #define NCOPYI(s,t,n) { while ( (n)-- > 0 ) { *s++ = *t++; } }
00074 #define NCOPYB(s,t,n) { while ( (n)-- > 0 ) { *s++ = *t++; } }
00075 #define NCOPYI32(s,t,n) { while ( (n)-- > 0 ) { *s++ = *t++; } }
00076 #define WCOPY(s,t,n) { int nn=n; WORD *ss=(WORD *)s, *tt=(WORD *)t; while ( (nn)-- > 0 ) { *ss++ =
    *tt++; } }
00077
00078 #define NeedNumber(x,s,err) { int sgn = 1;
00079     while ( *s == ' ' || *s == '\t' || *s == '-' || *s == '+' ) {
00080         if ( *s == '-' ) {sgn = -sgn;} s++; }
00081         if ( chartype[*s] != 1 ) goto err;
00082         ParseNumber(x,s)
00083         if ( sgn < 0 ) {(x) = -(x);} while ( *s == ' ' || *s == '\t' ) s++; \
00084     }
00085 #define SKIPBLANKS(s) { while ( *(s) == ' ' || *(s) == '\t' ) (s)++; }
00086 #define FLUSHCONSOLE if ( AP.InOutBuf > 0 ) CharOut(LINEFEED)

```

```

00087
00088 #define SKIPBRA1(s) { int lev1=0; s++; while(*s) { if(*s=='{') {lev1++;} \
00089 else {if(*s=='}' && --lev1<0) {break;} } s++; } }
00090 #define SKIPBRA2(s) { int lev2=0; s++; while(*s) { if(*s=='{') {lev2++;} \
00091 else {if(*s=='}' && --lev2<0) {break;} } \
00092 else {if(*s=='[') {SKIPBRA1(s);} } s++; } }
00093 #define SKIPBRA3(s) { int lev3=0; s++; while(*s) { if(*s=='(') {lev3++;} \
00094 else {if(*s==')' && --lev3<0) {break;} } \
00095 else {if(*s=='{') {SKIPBRA2(s);} \
00096 else {if(*s=='[') {SKIPBRA1(s);} } } s++; } }
00097 #define SKIPBRA4(s) { int lev4=0; s++; while(*s) { if(*s=='(') {lev4++;} \
00098 else {if(*s==')' && --lev4<0) {break;} } \
00099 else {if(*s=='{') {SKIPBRA1(s);} } s++; } }
00100 #define SKIPBRA5(s) { int lev5=0; s++; while(*s) { if(*s=='{') {lev5++;} \
00101 else {if(*s=='}' && --lev5<0) {break;} } \
00102 else {if(*s=='(') {SKIPBRA4(s);} \
00103 else {if(*s=='[') {SKIPBRA1(s);} } } s++; } }
00104
00105 /*
00106 #define CYCLE1(a,i) {WORD iX,jX; iX=*a; for(jX=1;jX<i;jX++)a[jX-1]=a[jX]; a[i-1]=iX;}
00107 */
00108 #define CYCLE1(t,a,i) {t iX=*a; WORD jX; for(jX=1;jX<i;jX++)a[jX-1]=a[jX]; a[i-1]=iX;}
00109
00110 #define AddToCB(c,wx) if(c->Pointer>=c->Top) \
00111 {DoubleCbuffer(c-cbuf,c->Pointer,21);} \
00112 *(c->Pointer)++ = wx;
00113
00114 #define EXCHINOUT { FILEHANDLE *ffFi = AR.outfile; \
00115 AR.outfile = AR.infile; AR.infile = ffFi; }
00116 #define BACKINOUT { FILEHANDLE *ffFi = AR.outfile; POSITION posi; \
00117 AR.outfile = AR.infile; AR.infile = ffFi; \
00118 SetEndScratch(AR.infile,&posi); }
00119
00120 #define CopyArg(to,from) { if ( *from > 0 ) { int ica = *from; NCOPY(to,from,ica) } \
00121 else if ( *from <= -FUNCTION ) *to++ = *from++; \
00122 else { *to++ = *from++; *to++ = *from++; } }
00123
00124 #if ARGHEAD > 2
00125 #define FILLARG(w) { int i = ARGHEAD-2; while ( --i >= 0 ) *w++ = 0; }
00126 #define COPYARG(w,t) { int i = ARGHEAD-2; while ( --i >= 0 ) *w++ = *t++; }
00127 #define ZEROARG(w) { int i; for ( i = 2; i < ARGHEAD; i++ ) w[i] = 0; }
00128 #else
00129 #define FILLARG(w)
00130 #define COPYARG(w,t)
00131 #define ZEROARG(w)
00132 #endif
00133
00134 #if FUNHEAD > 2
00135 #define FILLFUN(w) { *w++ = 0; FILLFUN3(w) }
00136 #define COPYFUN(w,t) { *w++ = *t++; COPYFUN3(w,t) }
00137 #else
00138 #define FILLFUN(w)
00139 #define COPYFUN(w,t)
00140 #endif
00141
00142 #if FUNHEAD > 3
00143 #define FILLFUN3(w) { int ie = FUNHEAD-3; while ( --ie >= 0 ) *w++ = 0; }
00144 #define COPYFUN3(w,t) { int ie = FUNHEAD-3; while ( --ie >= 0 ) *w++ = *t++; }
00145 #else
00146 #define COPYFUN3(w,t)
00147 #define FILLFUN3(w)
00148 #endif
00149
00150 #if SUBEXPSIZE > 5
00151 #define FILLSUB(w) { int ie = SUBEXPSIZE-5; while ( --ie >= 0 ) *w++ = 0; }
00152 #define COPYSUB(w,ww) { int ie = SUBEXPSIZE-5; while ( --ie >= 0 ) *w++ = *ww++; }
00153 #else
00154 #define FILLSUB(w)
00155 #define COPYSUB(w,ww)
00156 #endif
00157
00158 #if EXPRHEAD > 4
00159 #define FILLEXPR(w) { int ie = EXPRHEAD-4; while ( --ie >= 0 ) *w++ = 0; }
00160 #else
00161 #define FILLEXPR(w)
00162 #endif
00163
00164 #define NEXTARG(x) if(*x>0) x += *x; else if(*x <= -FUNCTION)x++; else x += 2;
00165 #define COPY1ARG(sl,t1) { int ica; if ( (ica==t1) > 0 ) { NCOPY(sl,t1,ica) } \
00166 else if(*t1<=-FUNCTION){*sl++=*t1++;} else{*sl++=*t1++;*sl++=*t1++;} }
00167
00175 #define ZeroFillRange(w,begin,end) do { \
00176 int tmp_i; \
00177 for ( tmp_i = begin; tmp_i < end; tmp_i++ ) { (w)[tmp_i] = 0; } \
00178 } while (0)
00179
00180 #define TABLESIZE(a,b) (((WORD)sizeof(a))/((WORD)sizeof(b)))

```

```

00181 #define WORDDIFF(x,y) (WORD)(x-y)
00182 #define wsizeof(a) ((WORD)sizeof(a))
00183 #define VARNAME(type,num) (AC.varnames->namebuffer+type[num].name)
00184 #define DOLLARNAME(type,num) (AC.dollarnames->namebuffer+type[num].name)
00185 #define EXPRNAME(num) (AC.exprnames->namebuffer+Expressions[num].name)
00186
00187 #define PREV(x) prevorder?prevorder:x
00188
00189 #define Terminate(x) do { TerminateImpl(x, __FILE__, __LINE__, __FUNCTION__); } while(0)
00190 #define SETERROR(x) { Terminate(-1); return(-1); }
00191
00192 /* use this macro to avoid the unused parameter warning */
00193 #define DUMMYUSE(x) (void)(x);
00194
00195 #ifndef _FILE_OFFSET_BITS
00196 #if _FILE_OFFSET_BITS==64
00197 /*:[19mar2004 mt]*/
00198
00199 #define ADDPOS(pp,x) (pp).p1 = ((pp).p1+(off_t)(x))
00200 #define SETBASELENGTH(ss,x) (ss).p1 = (off_t)(x)
00201 #define SETBASEPOSITION(pp,x) (pp).p1 = (off_t)(x)
00202 #define ISEQUALPOSINC(pp1,pp2,x) ( (pp1).p1 == ((pp2).p1+(off_t)(x)) )
00203 #define ISGEPOSINC(pp1,pp2,x) ( (pp1).p1 >= ((pp2).p1+(off_t)(x)) )
00204 #define DIVPOS(pp,n) ( (pp).p1/(off_t)(n) )
00205 #define MULPOS(pp,n) (pp).p1 *= (off_t)(n)
00206
00207 #else
00208
00209 #define ADDPOS(pp,x) (pp).p1 = ((pp).p1+(x))
00210 #define SETBASELENGTH(ss,x) (ss).p1 = (x)
00211 #define SETBASEPOSITION(pp,x) (pp).p1 = (x)
00212 #define ISEQUALPOSINC(pp1,pp2,x) ( (pp1).p1 == ((pp2).p1+(LONG)(x)) )
00213 #define ISGEPOSINC(pp1,pp2,x) ( (pp1).p1 >= ((pp2).p1+(LONG)(x)) )
00214 #define DIVPOS(pp,n) ( (pp).p1/(n) )
00215 #define MULPOS(pp,n) (pp).p1 *= (n)
00216 #endif
00217 #else
00218
00219 #define ADDPOS(pp,x) (pp).p1 = ((pp).p1+(LONG)(x))
00220 #define SETBASELENGTH(ss,x) (ss).p1 = (LONG)(x)
00221 #define SETBASEPOSITION(pp,x) (pp).p1 = (LONG)(x)
00222 #define ISEQUALPOSINC(pp1,pp2,x) ( (pp1).p1 == ((pp2).p1+(LONG)(x)) )
00223 #define ISGEPOSINC(pp1,pp2,x) ( (pp1).p1 >= ((pp2).p1+(LONG)(x)) )
00224 #define DIVPOS(pp,n) ( (pp).p1/(LONG)(n) )
00225 #define MULPOS(pp,n) (pp).p1 *= (LONG)(n)
00226
00227 #endif
00228 #define DIFPOS(ss,pp1,pp2) (ss).p1 = ((pp1).p1-(pp2).p1)
00229 #define DIFBASE(pp1,pp2) ((pp1).p1-(pp2).p1)
00230 #define ADD2POS(pp1,pp2) (pp1).p1 += (pp2).p1
00231 #define PUTZERO(pp) (pp).p1 = 0
00232 #define BASEPOSITION(pp) ((pp).p1)
00233 #define SETSTARTPOS(pp) (pp).p1 = -2
00234 #define NOTSTARTPOS(pp) ( (pp).p1 > -2 )
00235 #define ISMINPOS(pp) ( (pp).p1 == -1 )
00236 #define ISEQUALPOS(pp1,pp2) ( (pp1).p1 == (pp2).p1 )
00237 #define ISNOTEQUALPOS(pp1,pp2) ( (pp1).p1 != (pp2).p1 )
00238 #define ISLESSPOS(pp1,pp2) ( (pp1).p1 < (pp2).p1 )
00239 #define ISGEPOS(pp1,pp2) ( (pp1).p1 >= (pp2).p1 )
00240 #define ISNOTZEROPOS(pp) ( (pp).p1 != 0 )
00241 #define ISZEROPOS(pp) ( (pp).p1 == 0 )
00242 #define ISPOSPOS(pp) ( (pp).p1 > 0 )
00243 #define ISNEGPOS(pp) ( (pp).p1 < 0 )
00244 extern void TELLFILE(int,POSITION *);
00245
00246 #define TOLONG(x) ((LONG)(x))
00247
00248 #define Add2Com(x) { WORD cod[2]; cod[0] = x; cod[1] = 2; AddNtoL(2,cod); }
00249 #define Add3Com(x1,x2) { WORD cod[3]; cod[0] = x1; cod[1] = 3; cod[2] = x2; AddNtoL(3,cod); }
00250 #define Add4Com(x1,x2,x3) { WORD cod[4]; cod[0] = x1; cod[1] = 4; \
00251     cod[2] = x2; cod[3] = x3; AddNtoL(4,cod); }
00252 #define Add5Com(x1,x2,x3,x4) { WORD cod[5]; cod[0] = x1; cod[1] = 5; \
00253     cod[2] = x2; cod[3] = x3; cod[4] = x4; AddNtoL(5,cod); }
00254
00255 /*
00256     The temporary variable ppp is to avoid a compiler warning about strict aliasing
00257 */
00258 #define WantAddPointers(x) while((AT.pWorkPointer+(x))>AR.pWorkSize){WORD ***ppp=&AT.pWorkSpace;\
00259     ExpandBuffer((void **)ppp,&AR.pWorkSize,(int)(sizeof(WORD *)));}
00260 #define WantAddLongs(x) while((AT.lWorkPointer+(x))>AR.lWorkSize){LONG **ppp=&AT.lWorkSpace;\
00261     ExpandBuffer((void **)ppp,&AR.lWorkSize,sizeof(LONG));}
00262 #define WantAddPositions(x) while((AT.posWorkPointer+(x))>AR.posWorkSize){POSITION
00263     **ppp=&AT.posWorkSpace;\
00264     ExpandBuffer((void **)ppp,&AR.posWorkSize,sizeof(POSITION));}
00265
00266 /* inline in form3.h (or config.h). */
00267 #define FORM_INLINE inline

```

```

00267
00268 /*
00269     Macro's for memory management. This can be done by routines, but that
00270     would be slower. Inline routines could do this, but we don't want to
00271     leave this to the friendliness of the compiler(s).
00272     The routines can be found in the file tools.c
00273 */
00274 #define MEMORYMACROS
00275
00276 #ifndef MEMORYMACROS
00277
00278 #define TermMalloc(x) ( (AT.TermMemTop <= 0) ? TermMallocAddMemory(BHEAD0),
AT.TermMemHeap[--AT.TermMemTop]: AT.TermMemHeap[--AT.TermMemTop] )
00279 #define NumberMalloc(x) ( (AT.NumberMemTop <= 0) ? NumberMallocAddMemory(BHEAD0),
AT.NumberMemHeap[--AT.NumberMemTop]: AT.NumberMemHeap[--AT.NumberMemTop] )
00280 #define CacheNumberMalloc(x) ( (AT.CacheNumberMemTop <= 0) ? CacheNumberMallocAddMemory(BHEAD0),
AT.CacheNumberMemHeap[--AT.CacheNumberMemTop]: AT.CacheNumberMemHeap[--AT.CacheNumberMemTop] )
00281 #define TermFree(TermMem,x) AT.TermMemHeap[AT.TermMemTop++] = (WORD *) (TermMem)
00282 #define NumberFree(NumberMem,x) AT.NumberMemHeap[AT.NumberMemTop++] = (UWORD *) (NumberMem)
00283 #define CacheNumberFree(NumberMem,x) AT.CacheNumberMemHeap[AT.CacheNumberMemTop++] = (UWORD
*) (NumberMem)
00284
00285 #else
00286
00287 #define TermMalloc(x) TermMalloc2(BHEAD (char *) (x))
00288 #define NumberMalloc(x) NumberMalloc2(BHEAD (char *) (x))
00289 #define CacheNumberMalloc(x) CacheNumberMalloc2(BHEAD (char *) (x))
00290 #define TermFree(x,y) TermFree2(BHEAD (WORD *) (x), (char *) (y))
00291 #define NumberFree(x,y) NumberFree2(BHEAD (UWORD *) (x), (char *) (y))
00292 #define CacheNumberFree(x,y) CacheNumberFree2(BHEAD (UWORD *) (x), (char *) (y))
00293
00294 #endif
00295
00296 /*
00297 * Macros for checking nesting levels in the compiler, used as follows:
00298 *
00299 *   AC.IfSumCheck[AC.IfLevel] = NestingChecksum();
00300 *   AC.IfLevel++;
00301 *
00302 *   AC.IfLevel--;
00303 *   if ( AC.IfSumCheck[AC.IfLevel] != NestingChecksum() ) {
00304 *       MesNesting();
00305 *   }
00306 *
00307 * Note that NestingChecksum() also contains AC.IfLevel and so in this case
00308 * using increment/decrement operators on it in the left-hand side may be
00309 * confusing.
00310 */
00311 #define NestingChecksum() (AC.IfLevel + AC.RepLevel + AC.arglevel + AC.insidelevel + AC.termlevel +
AC.inexprlevel + AC.dolooplevel + AC.SwitchLevel)
00312 #define MesNesting() MesPrint("&Illegal nesting of if, repeat, argument, inside, term, inexpression
and do")
00313
00314 #define MarkPolyRatFunDirty(T) {if (*T&&AR.PolyFunType==2) {WORD *TP,*TT;TT=T+*T;TT-=ABS(TT[-1]);\
TP=T+1;while (TP<TT) {if (*TP==AR.PolyFun) {TP[2]|=(DIRTYFLAG|MUSTCLEANPRF);}TP+=TP[1];}}
00315
00316
00317 /*
00318     Macros for nesting input levels for #name = ...; assign instructions.
00319     Note that the level should never go below zero.
00320 */
00321 #define PUSHPREASSIGNLEVEL AP.PreAssignLevel++; { GETIDENTITY \
00322     if ( AP.PreAssignLevel >= AP.MaxPreAssignLevel ) { int i; \
00323     LONG *ap = (LONG *) Malloc1(2*AP.MaxPreAssignLevel*sizeof(LONG *),"PreAssignStack"); \
00324     for ( i = 0; i < AP.MaxPreAssignLevel; i++ ) ap[i] = AP.PreAssignStack[i]; \
00325     M_free(AP.PreAssignStack,"PreAssignStack"); \
00326     AP.MaxPreAssignLevel *= 2; AP.PreAssignStack = ap; \
00327     } \
00328     *AT.WorkPointer++ = AP.PreContinuation; AP.PreContinuation = 0; \
00329     AP.PreAssignStack[AP.PreAssignLevel] = AC.iPointer - AC.iBuffer; }
00330
00331 #define POPPREASSIGNLEVEL if ( AP.PreAssignLevel > 0 ) { GETIDENTITY \
00332     AC.iPointer = AC.iBuffer + AP.PreAssignStack[AP.PreAssignLevel--]; \
00333     AP.PreContinuation = *--AT.WorkPointer; \
00334     *AC.iPointer = 0; }
00335
00336 #ifndef WITHFLOAT
00337 /*
00338     The following macro's are needed to avoid problems with the compilers
00339     and gmp.h. For the C++ files we need the Form include files to be inside
00340     and extern C {} environment, but then the structs.h needs gmp.h to
00341     recognise the mpf_t datatype. This causes no end of problems.
00342     Hence we collect the sensitive objects as (void *) and cast them to
00343     something usable with the macro's below. This way the gmp.h file
00344     needs to be included only in a very limited number of .c files.
00345 */
00346 #define mpftab1 ((mpf_t *) (AT.mpf_tab1))
00347 #define mpftab2 ((mpf_t *) (AT.mpf_tab2))

```

```

00348 #define mpfaux_ ((mpf_t *) (AT.aux_))
00349 #define aux1 ((mpf_t *) (AT.aux_)) [0]
00350 #define aux2 ((mpf_t *) (AT.aux_)) [1]
00351 #define aux3 ((mpf_t *) (AT.aux_)) [2]
00352 #define aux4 ((mpf_t *) (AT.aux_)) [3]
00353 #define aux5 ((mpf_t *) (AT.aux_)) [4]
00354 #define auxjm ((mpf_t *) (AT.aux_)) [5]
00355 #define auxjjm ((mpf_t *) (AT.aux_)) [6]
00356 #define auxsum ((mpf_t *) (AT.aux_)) [7]
00357
00358 #define mpfdelta1 ((mpf_t *) (AS.delta_1)) [0]
00359
00360 #endif
00361
00362
00363 /*
00364     MesPrint("P-level popped to %d with %d", AP.PreAssignLevel, (WORD) (AC.iPointer - AC.iBuffer));
00365
00366     #] Macro's :
00367     #[ Inline functions :
00368 */
00369
00370 /*
00371 * The following three functions give the unsigned absolute value of a signed
00372 * integer even for the most negative integer. This is beyond the scope of
00373 * the standard abs() function and its family, whose return-values are signed.
00374 * In short, we should not use the unary minus operator with signed numbers
00375 * unless we are sure that there are no integer overflows. Instead, we rely on
00376 * two well-defined operations: (i) signed-to-unsigned conversion and
00377 * (ii) unary minus of unsigned operands.
00378 *
00379 * See also:
00380 *   https://stackoverflow.com/a/4536188 (Unary minus and signed-to-unsigned conversion)
00381 *   https://stackoverflow.com/q/8026694 (C: unary minus operator behavior with unsigned operands)
00382 *   https://stackoverflow.com/q/1610947 (Why does stdlib.h's abs() family of functions return a
signed value?)
00383 *   https://blog.regehr.org/archives/226 (A Guide to Undefined Behavior in C and C++, Part 2)
00384 */
00385 static inline unsigned int IntAbs(int x)
00386 {
00387     if ( x >= 0 ) return (unsigned int)x;
00388     return(-(unsigned int)x);
00389 }
00390
00391 static inline UWORD WordAbs(WORD x)
00392 {
00393     if ( x >= 0 ) return (UWORD)x;
00394     return(-(UWORD)x);
00395 }
00396
00397 static inline ULONG LongAbs(LONG x)
00398 {
00399     if ( x >= 0 ) return (ULONG)x;
00400     return(-(ULONG)x);
00401 }
00402
00403 /*
00404 * The following functions provide portable unsigned-to-signed conversions
00405 * (to avoid the implementation-defined behaviour), which is expected to be
00406 * optimized to a no-op.
00407 *
00408 * See also:
00409 *   https://stackoverflow.com/a/13208789 (Efficient unsigned-to-signed cast avoiding
implementation-defined behavior)
00410 */
00411 static inline int UnsignedToInt(unsigned int x)
00412 {
00413     extern void TerminateImpl(int, const char*, int, const char*);
00414     if ( x <= INT_MAX ) return (int)x;
00415     if ( x >= (unsigned int)INT_MIN )
00416         return((int) (x - (unsigned int)INT_MIN) + INT_MIN);
00417     Terminate(1);
00418     return(0);
00419 }
00420
00421 static inline WORD UWordToWord(UWORD x)
00422 {
00423     extern void TerminateImpl(int, const char*, int, const char*);
00424     if ( x <= WORD_MAX_VALUE ) return (WORD)x;
00425     if ( x >= (UWORD)WORD_MIN_VALUE )
00426         return((WORD) (x - (UWORD)WORD_MIN_VALUE) + WORD_MIN_VALUE);
00427     Terminate(1);
00428     return(0);
00429 }
00430
00431 static inline LONG ULongToLong(ULONG x)
00432 {

```

```

00433     extern void TerminateImpl(int, const char*, int, const char*);
00434     if ( x <= LONG_MAX_VALUE ) return((LONG)x);
00435     if ( x >= (ULONG)LONG_MIN_VALUE )
00436         return((LONG)(x - (ULONG)LONG_MIN_VALUE) + LONG_MIN_VALUE);
00437     Terminate(1);
00438     return(0);
00439 }
00440
00441 /*
00442     #] Inline functions :
00443     #[ Thread objects :
00444 */
00445
00452 #ifdef WITHPTHREADS
00453
00454 #define EXTERNLOCK(x) extern pthread_mutex_t x;
00455 #define INILOCK(x) pthread_mutex_t x = PTHREAD_MUTEX_INITIALIZER;
00456 #define EXTERNRWLOCK(x) extern pthread_rwlock_t x;
00457 #define INIRWLOCK(x) pthread_rwlock_t x = PTHREAD_RWLOCK_INITIALIZER;
00458 #ifdef DEBUGGINGLOCKS
00459 #include <asm/errno.h>
00460 #define LOCK(x) while ( pthread_mutex_trylock(&(x)) == EBUSY ) {}
00461 #define RWLOCKR(x) while ( pthread_rwlock_tryrdlock(&(x)) == EBUSY ) {}
00462 #define RWLOCKW(x) while ( pthread_rwlock_trywrlock(&(x)) == EBUSY ) {}
00463 #else
00464 #define LOCK(x) pthread_mutex_lock(&(x))
00465 #define RWLOCKR(x) pthread_rwlock_rdlock(&(x))
00466 #define RWLOCKW(x) pthread_rwlock_wrlock(&(x))
00467 #endif
00468 #define UNLOCK(x) pthread_mutex_unlock(&(x))
00469 #define UNRWLOCK(x) pthread_rwlock_unlock(&(x))
00470 #define MLOCK(x) LOCK(x)
00471 #define MUNLOCK(x) UNLOCK(x)
00472
00473 #define GETBIDENTITY
00474 #define GETIDENTITY int identity = WhoAmI(); ALLPRIVATES *B = AB[identity];
00475 #else
00476
00477 #define EXTERNLOCK(x)
00478 #define INILOCK(x)
00479 #define LOCK(x)
00480 #define UNLOCK(x)
00481 #define EXTERNRWLOCK(x)
00482 #define INIRWLOCK(x)
00483 #define RWLOCKR(x)
00484 #define RWLOCKW(x)
00485 #define UNRWLOCK(x)
00486 #ifdef WITHMPI
00487 #define MLOCK(x) do { if ( PF.me != MASTER ) PF_MLock(); } while (0)
00488 #define MUNLOCK(x) do { if ( PF.me != MASTER ) PF_MUnlock(); } while (0)
00489 #else
00490 #define MLOCK(x)
00491 #define MUNLOCK(x)
00492 #endif
00493 #define GETIDENTITY
00494 #define GETBIDENTITY
00495
00496 #endif
00497
00498 /*
00499     #] Thread objects :
00500     #[ Declarations :
00501 */
00502
00503 #ifdef TERMMALLOCDEBUG
00504 extern WORD **DebugHeap1, **DebugHeap2;
00505 #endif
00506
00511 extern void StartVariables(void);
00512 extern void setSignalHandlers(void);
00513 extern UBYTE *CodeToLine(WORD, UBYTE *);
00514 extern UBYTE *AddArrayIndex(WORD, UBYTE *);
00515 extern INDEXENTRY *FindInIndex(WORD, FILEDATA *, WORD, WORD);
00516 extern INDEXENTRY *NextFileIndex(POSITION *);
00517 extern WORD *PasteTerm(PHEAD WORD, WORD *, WORD *, WORD, WORD);
00518 extern UBYTE *StrCopy(UBYTE *, UBYTE *);
00519 extern UBYTE *WrtPower(UBYTE *, WORD);
00520 extern int AccumGCD(PHEAD UWORD *, WORD *, UWORD *, WORD);
00521 extern void AddArgs(PHEAD WORD *, WORD *, WORD *);
00522 extern int AddCoef(PHEAD WORD **, WORD **);
00523 extern int AddLong(UWORD *, WORD, UWORD *, WORD, UWORD *, WORD *);
00524 extern int AddPLon(UWORD *, WORD, UWORD *, WORD, UWORD *, WORD *);
00525 extern int AddPoly(PHEAD WORD **, WORD **);
00526 extern int AddRat(PHEAD UWORD *, WORD, UWORD *, WORD, UWORD *, WORD *);
00527 extern void AddToLine(UBYTE *);
00528 extern int AddWild(PHEAD WORD, WORD, WORD);
00529 extern int BigLong(UWORD *, WORD, UWORD *, WORD);

```

```

00530 extern int      BinomGen(PHEAD WORD *,WORD,WORD **,WORD,WORD,WORD,WORD,WORD,UWORD *,WORD);
00531 extern int      CheckWild(PHEAD WORD,WORD,WORD,WORD *);
00532 extern int      Chisholm(PHEAD WORD *,WORD);
00533 extern int      CleanExpr(WORD);
00534 extern void     CleanUp(WORD);
00535 extern void     ClearWild(PHEAD0);
00536 extern int      CompareFunctions(WORD *,WORD *);
00537 extern int      Commute(WORD *,WORD *);
00538 extern WORD     DetCommu(WORD *);
00539 extern WORD     DoesCommu(WORD *);
00540 extern int      CompArg(WORD *,WORD *);
00541 extern WORD     CompCoef(WORD *,WORD *);
00542 extern int      CompGroup(PHEAD WORD,WORD **,WORD *,WORD *,WORD);
00543 extern WORD     Compare1(PHEAD WORD *,WORD *,WORD);
00544 extern WORD     CountDo(WORD *,WORD *);
00545 extern WORD     CountFun(WORD *,WORD *);
00546 extern WORD     DimensionSubterm(WORD *);
00547 extern WORD     DimensionTerm(WORD *);
00548 extern WORD     DimensionExpression(PHEAD WORD *);
00549 extern int      Deferred(PHEAD WORD *,WORD);
00550 extern int      DeleteStore(WORD);
00551 extern WORD     DetCurDum(PHEAD WORD *);
00552 extern void     DetVars(WORD *,WORD);
00553 extern int      Distribute(DISTRIBUTE *,WORD);
00554 extern int      DivLong(UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *,UWORD *,WORD *);
00555 extern int      DivRat(PHEAD UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *);
00556 extern int      Divvy(PHEAD UWORD *,WORD *,UWORD *,WORD);
00557 extern int      DoDelta(WORD *);
00558 extern int      DoDelta3(PHEAD WORD *,WORD);
00559 extern int      TestPartitions(WORD *,PARTI *);
00560 extern int      DoPartitions(PHEAD WORD *,WORD);
00561 extern int      CoCanonicalize(UBYTE *);
00562 extern int      DoCanonicalize(PHEAD WORD *,WORD *);
00563 extern int      GenTopologies(PHEAD WORD *,WORD);
00564 extern int      GenDiagrams(PHEAD WORD *,WORD);
00565 extern int      DoTopologyCanonicalize(PHEAD WORD *,WORD,WORD,WORD *);
00566 extern int      DoShattering(PHEAD WORD *,WORD *,WORD *,WORD);
00567 extern int      GenerateTopologies(PHEAD WORD,WORD,WORD,WORD);
00568 extern int      DoTableExpansion(WORD *,WORD);
00569 extern int      DoDistrib(PHEAD WORD *,WORD);
00570 extern int      DoShuffle(WORD *,WORD,WORD,WORD);
00571 extern int      DoPermutations(PHEAD WORD *,WORD);
00572 extern int      Shuffle(WORD *,WORD *,WORD *);
00573 extern int      FinishShuffle(WORD *);
00574 extern int      DoStuffle(WORD *,WORD,WORD,WORD);
00575 extern int      Stuffle(WORD *,WORD *,WORD *);
00576 extern int      FinishStuffle(WORD *);
00577 extern WORD     *StuffRootAdd(WORD *,WORD *,WORD *);
00578 extern int      TestUse(WORD *,WORD);
00579 extern DBASE     *FindTB(UBYTE *);
00580 extern int      CheckTableDeclarations(DBASE *);
00581 extern void     Apply(WORD *,WORD);
00582 extern int      ApplyExec(WORD *,int,WORD);
00583 extern void     ApplyReset(WORD);
00584 extern void     TableReset(void);
00585 extern void     ReWorkT(WORD *,WORD *,WORD);
00586 extern WORD     GetIfDollarNum(WORD *,WORD *);
00587 extern int      FindVar(WORD *,WORD *);
00588 extern int      DoIfStatement(PHEAD WORD *,WORD *);
00589 extern int      DoOnePow(PHEAD WORD *,WORD,WORD,WORD *,WORD *,WORD,WORD *);
00590 extern void     DoRevert(WORD *,WORD *);
00591 extern int      DoSumF1(PHEAD WORD *,WORD *,WORD,WORD);
00592 extern int      DoSumF2(PHEAD WORD *,WORD *,WORD,WORD);
00593 extern int      DoTheta(PHEAD WORD *);
00594 extern LONG     EndSort(PHEAD WORD *,int);
00595 extern int      EntVar(WORD,UBYTE *,WORD,WORD,WORD,WORD);
00596 extern int      EpfCon(PHEAD WORD *,WORD *,WORD,WORD);
00597 extern int      EpfFind(PHEAD WORD *,WORD *);
00598 extern WORD     EpfGen(WORD,WORD *,WORD *,WORD *,WORD);
00599 extern int      EqualArg(WORD *,WORD,WORD);
00600 extern int      Factorial(PHEAD WORD,UWORD *,WORD *);
00601 extern int      Bernoulli(WORD,UWORD *,WORD *);
00602 extern int      FactorIn(PHEAD WORD *,WORD);
00603 extern int      FactorInExpr(PHEAD WORD *,WORD);
00604 extern int      FindAll(PHEAD WORD *,WORD *,WORD,WORD *);
00605 extern WORD     FindMulti(PHEAD WORD *,WORD *);
00606 extern int      FindOnce(PHEAD WORD *,WORD *);
00607 extern int      FindOnly(PHEAD WORD *,WORD *);
00608 extern int      FindRest(PHEAD WORD *,WORD *);
00609 extern void     FindSpecial(WORD *);
00610 extern WORD     FindrNumber(WORD,VARENUM *);
00611 extern void     FiniLine(void);
00612 extern int      FiniTerm(PHEAD WORD *,WORD *,WORD *,WORD,WORD);
00613 extern int      FlushOut(POSITION *,FILEHANDLE *,int);
00614 extern void     FunLevel(PHEAD WORD *);
00615 extern void     AdjustRenumScratch(PHEAD0);
00616 extern void     GarbHand(void);

```



```

00617 extern int      GcdLong (PHEAD UWORD *, WORD, UWORD *, WORD, UWORD *, WORD *);
00618 extern int      LcmLong (PHEAD UWORD *, WORD, UWORD *, WORD, UWORD *, WORD *);
00619 extern void      GCD (UWORD *, WORD, UWORD *, WORD, UWORD *, WORD *);
00620 extern ULONG     GCD2 (ULONG, ULONG);
00621 extern int      Generator (PHEAD WORD *, WORD);
00622 extern int      GetBinom (UWORD *, WORD *, WORD, WORD);
00623 extern WORD     GetFromStore (WORD *, POSITION *, RENUMBER, WORD *, WORD);
00624 extern int      GetLong (UBYTE *, UWORD *, WORD *);
00625 extern WORD     GetMoreTerms (WORD *);
00626 extern int      GetMoreFromMem (WORD *, WORD **);
00627 extern WORD     GetOneTerm (PHEAD WORD *, FILEHANDLE *, POSITION *, int);
00628 extern RENUMBER GetTable (WORD, POSITION *, WORD);
00629 extern WORD     GetTerm (PHEAD WORD *);
00630 extern int      Glue (PHEAD WORD *, WORD *, WORD *, WORD);
00631 extern int      InFunction (PHEAD WORD *, WORD *);
00632 extern void     IniLine (WORD);
00633 extern void     IniVars (void);
00634 extern int      InsertTerm (PHEAD WORD *, WORD, WORD, WORD *, WORD *, WORD);
00635 extern void     LongToLine (UWORD *, WORD);
00636 extern int      MakeDirty (WORD *, WORD *, WORD);
00637 extern void     MarkDirty (WORD *, WORD);
00638 extern void     PolyFunDirty (PHEAD WORD *);
00639 extern void     PolyFunClean (PHEAD WORD *);
00640 extern int      MakeModTable (void);
00641 extern int      MatchE (PHEAD WORD *, WORD *, WORD *, WORD);
00642 extern int      MatchCy (PHEAD WORD *, WORD *, WORD *, WORD);
00643 extern int      FunMatchCy (PHEAD WORD *, WORD *, WORD *, WORD);
00644 extern int      FunMatchSy (PHEAD WORD *, WORD *, WORD *, WORD);
00645 extern int      MatchArgument (PHEAD WORD *, WORD *);
00646 extern int      MatchFunction (PHEAD WORD *, WORD *, WORD *);
00647 extern int      MergePatches (WORD);
00648 extern int      MesCerr (char *, UBYTE *);
00649 extern int      MesComp (char *, UBYTE *, UBYTE *);
00650 extern int      Modulus (WORD *);
00651 extern void     MoveDummies (PHEAD WORD *, WORD);
00652 extern int      MulLong (UWORD *, WORD, UWORD *, WORD, UWORD *, WORD *);
00653 extern int      MulRat (PHEAD UWORD *, WORD, UWORD *, WORD, UWORD *, WORD *);
00654 extern int      Mully (PHEAD UWORD *, WORD *, UWORD *, WORD);
00655 extern int      MultDo (PHEAD WORD *, WORD *);
00656 extern int      NewSort (PHEAD0);
00657 extern int      ExtraSymbol (WORD, WORD, WORD, WORD *, WORD *);
00658 extern int      Normalize (PHEAD WORD *);
00659 extern int      BracketNormalize (PHEAD WORD *);
00660 extern void     DropCoefficient (PHEAD WORD *);
00661 extern void     DropSymbols (PHEAD WORD *);
00662 extern int      PutInside (PHEAD WORD *, WORD *);
00663 extern void     OpenTemp (void);
00664 extern void     Pack (UWORD *, WORD *, UWORD *, WORD *);
00665 extern LONG     PasteFile (PHEAD WORD, WORD *, POSITION *, WORD **, RENUMBER, WORD *, WORD);
00666 extern int      Permute (PERM *, WORD);
00667 extern int      PermuteP (PERMP *, WORD);
00668 extern int      PolyFunMul (PHEAD WORD *);
00669 extern int      PopVariables (void);
00670 extern int      PrepPoly (PHEAD WORD *, WORD);
00671 extern int      Processor (void);
00672 extern int      Product (UWORD *, WORD *, WORD);
00673 extern void     PrtLong (UWORD *, WORD, UBYTE *);
00674 extern void     PrtTerms (void);
00675 extern void     PrintDeprecation (const char *, const char *);
00676 extern void     PrintRunningTime (void);
00677 extern LONG     GetRunningTime (void);
00678 extern int      PutBracket (PHEAD WORD *);
00679 extern LONG     PutIn (FILEHANDLE *, POSITION *, WORD *, WORD **, int);
00680 extern int      PutInStore (INDEXENTRY *, WORD);
00681 extern WORD     PutOut (PHEAD WORD *, POSITION *, FILEHANDLE *, WORD);
00682 extern UWORD    Quotient (UWORD *, WORD *, WORD);
00683 extern int      RaisPow (PHEAD UWORD *, WORD *, UWORD);
00684 extern void     RaisPowCached (PHEAD WORD, WORD, UWORD **, WORD *);
00685 extern WORD     RaisPowMod (WORD, WORD, WORD);
00686 extern int      NormalModulus (UWORD *, WORD *);
00687 extern int      MakeInverses (void);
00688 extern int      GetModInverses (WORD, WORD, WORD *, WORD *);
00689 extern int      GetLongModInverses (PHEAD UWORD *, WORD, UWORD *, WORD, UWORD *, WORD *, UWORD *, WORD
*);
00690 extern void     RatToLine (UWORD *, WORD);
00691 extern int      RatioFind (PHEAD WORD *, WORD *);
00692 extern int      RatioGen (PHEAD WORD *, WORD *, WORD, WORD);
00693 extern WORD     ReNumber (PHEAD WORD *);
00694 extern WORD     ReadSnum (UBYTE **);
00695 extern WORD     Remain10 (UWORD *, WORD *);
00696 extern WORD     Remain4 (UWORD *, WORD *);
00697 extern int      ResetScratch (void);
00698 extern int      ResolveSet (PHEAD WORD *, WORD *, WORD *);
00699 extern int      RevertScratch (void);
00700 extern int      ScanFunctions (PHEAD WORD *, WORD *, WORD);
00701 extern void     SeekScratch (FILEHANDLE *, POSITION *);
00702 extern void     SetEndScratch (FILEHANDLE *, POSITION *);

```

```

00703 extern void SetEndHScratch(FILEHANDLE *, POSITION *);
00704 extern int SetFileIndex(void);
00705 extern int SFlush(FILEHANDLE *);
00706 extern int Simplify(PHEAD UWORD *, WORD *, UWORD *, WORD *);
00707 extern int SortWild(WORD *, WORD);
00708 extern FILE *LocateBase(char **, char **, char *);
00709 extern LONG SplitMerge(PHEAD WORD **, LONG);
00710 extern int StoreTerm(PHEAD WORD *);
00711 extern void SubPLon(UWORD *, WORD, UWORD *, WORD, UWORD *, WORD *);
00712 extern void Substitute(PHEAD WORD *, WORD *, WORD);
00713 extern int SymFind(PHEAD WORD *, WORD *);
00714 extern int SymGen(PHEAD WORD *, WORD *, WORD, WORD);
00715 extern WORD Symmetrize(PHEAD WORD *, WORD *, WORD, WORD, WORD);
00716 extern int FullSymmetrize(PHEAD WORD *, int);
00717 extern int TakeModulus(UWORD *, WORD *, UWORD *, WORD, WORD);
00718 extern int TakeNormalModulus(UWORD *, WORD *, UWORD *, WORD, WORD);
00719 extern void TalToLine(UWORD);
00720 extern int TenVec(PHEAD WORD *, WORD *, WORD, WORD);
00721 extern int TenVecFind(PHEAD WORD *, WORD *);
00722 extern int TermRenumber(WORD *, RENUMBER, WORD);
00723 extern void TestDrop(void);
00724 extern void PutInVflags(WORD);
00725 extern int TestMatch(PHEAD WORD *, WORD *);
00726 extern WORD TestSub(PHEAD WORD *, WORD);
00727 extern LONG TimeCPU(WORD);
00728 extern LONG TimeChildren(WORD);
00729 extern LONG TimeWallClock(WORD);
00730 extern LONG Timer(int);
00731 extern int GetTimerInfo(LONG **, LONG **);
00732 extern void WriteTimerInfo(LONG *, LONG *);
00733 extern LONG GetWorkerTimes(void);
00734 extern int ToStorage(EXPRESSIONS, POSITION *);
00735 extern void TokenToLine(UBYTE *);
00736 extern int Trace4(PHEAD WORD *, WORD *, WORD, WORD);
00737 extern int Trace4Gen(PHEAD TRACES *, WORD);
00738 extern int Trace4no(WORD, WORD *, TRACES *);
00739 extern int TraceFind(PHEAD WORD *, WORD *);
00740 extern int TraceN(PHEAD WORD *, WORD *, WORD, WORD);
00741 extern int TraceNgen(PHEAD TRACES *, WORD);
00742 extern WORD TraceNno(WORD, WORD *, TRACES *);
00743 extern int Traces(PHEAD WORD *, WORD *, WORD, WORD);
00744 extern WORD Trick(WORD *, TRACES *);
00745 extern int TryDo(PHEAD WORD *, WORD *, WORD);
00746 extern void UnPack(UWORD *, WORD, WORD *, WORD *);
00747 extern int VarStore(UBYTE *, WORD, WORD, WORD);
00748 extern WORD WildFill(PHEAD WORD *, WORD *, WORD *);
00749 extern int WriteAll(void);
00750 extern int WriteOne(UBYTE *, int, int, WORD);
00751 extern void WriteArgument(WORD *);
00752 extern int WriteExpression(WORD *, LONG);
00753 extern int WriteInnerTerm(WORD *, WORD);
00754 extern void WriteLists(void);
00755 extern void WriteSetup(void);
00756 extern void WriteStats(POSITION *, WORD, WORD);
00757 extern int WriteSubTerm(WORD *, WORD);
00758 extern int WriteTerm(WORD *, WORD *, WORD, WORD, WORD);
00759 extern int execarg(PHEAD WORD *, WORD);
00760 extern int execterm(PHEAD WORD *, WORD);
00761 extern void SpecialCleanup(PHEAD0);
00762 extern void SetMods(void);
00763 extern void UnSetMods(void);
00764
00765 /*-----*/
00766
00767 extern int DoExecute(WORD, WORD);
00768 extern void SetScratch(FILEHANDLE *, POSITION *);
00769 extern void Warning(char *);
00770 extern void HighWarning(char *);
00771 extern int SpareTable(TABLES);
00772
00773 extern UBYTE *strDup1(UBYTE *, char *);
00774 extern void *Mallocl1(LONG, const char *);
00775 extern int DoTail(int, UBYTE **);
00776 extern int OpenInput(void);
00777 extern int PutPreVar(UBYTE *, UBYTE *, UBYTE *, int);
00778 extern void Error0(char *);
00779 extern void Error1(char *, UBYTE *);
00780 extern void Error2(char *, char *, UBYTE *);
00781 extern UBYTE ReadFromStream(STREAM *);
00782 extern UBYTE GetFromStream(STREAM *);
00783 extern UBYTE LookInStream(STREAM *);
00784 extern STREAM *OpenStream(UBYTE *, int, int, int);
00785 extern int LocateFile(UBYTE **, int);
00786 extern STREAM *CloseStream(STREAM *);
00787 extern void PositionStream(STREAM *, LONG);
00788 extern int ReverseStatements(STREAM *);
00789 extern int ProcessOption(UBYTE *, UBYTE *, int);

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```

00790 extern int      DoSetups(void);
00791 extern void      TerminateImpl(int, const char *,int, const char *);
00792 extern NAMENODE *GetNode(NAMETREE *,UBYTE *);
00793 extern int      AddName(NAMETREE *,UBYTE *,WORD,WORD,int *);
00794 extern int      GetName(NAMETREE *,UBYTE *,WORD *,int);
00795 extern UBYTE    *GetFunction(UBYTE *,WORD *);
00796 extern UBYTE    *GetNumber(UBYTE *,WORD *);
00797 extern int      GetLastExprName(UBYTE *,WORD *);
00798 extern int      GetAutoName(UBYTE *,WORD *);
00799 extern int      GetVar(UBYTE *,WORD *,WORD *,int,int);
00800 extern int      MakeDubious(NAMETREE *,UBYTE *,WORD *);
00801 extern int      GetOName(NAMETREE *,UBYTE *,WORD *,int);
00802 extern void      DumpTree(NAMETREE *);
00803 extern void      DumpNode(NAMETREE *,WORD,WORD);
00804 extern void      LinkTree(NAMETREE *,WORD,WORD);
00805 extern void      CopyTree(NAMETREE *,NAMETREE *,WORD,WORD);
00806 extern int      CompactifyTree(NAMETREE *,WORD);
00807 extern NAMETREE *MakeNameTree(void);
00808 extern void      FreeNameTree(NAMETREE *);
00809 extern int      AddExpression(UBYTE *,int,int);
00810 extern int      AddSymbol(UBYTE *,int,int,int,int);
00811 extern int      AddDollar(UBYTE *,WORD,WORD *,LONG);
00812 extern int      ReplaceDollar(WORD,WORD,WORD *,LONG);
00813 extern int      DollarRaiseLow(UBYTE *,LONG);
00814 extern int      AddVector(UBYTE *,int,int);
00815 extern int      AddDubious(UBYTE *);
00816 extern int      AddIndex(UBYTE *,int,int);
00817 extern UBYTE    *DoDimension(UBYTE *,int *,int *);
00818 extern int      AddFunction(UBYTE *,int,int,int,int,int,int,int);
00819 extern int      CoCommuteInSet(UBYTE *);
00820 extern int      CoFunction(UBYTE *,int,int);
00821 extern int      TestName(UBYTE *);
00822 extern int      AddSet(UBYTE *,WORD);
00823 extern int      DoElements(UBYTE *,SETS,UBYTE *);
00824 extern int      DoTempSet(UBYTE *,UBYTE *);
00825 extern int      NameConflict(int,UBYTE *);
00826 extern int      OpenFile(char *);
00827 extern int      OpenAddFile(char *);
00828 extern int      ReOpenFile(char *);
00829 extern int      CreateFile(char *);
00830 extern int      CreateLogFile(char *);
00831 extern void      CloseFile(int);
00832 extern int      CopyFile(char *, char *);
00833 extern int      CreateHandle(void);
00834 extern LONG      ReadFile(int,UBYTE *,LONG);
00835 extern LONG      ReadPosFile(PHEAD FILEHANDLE *,UBYTE *,LONG,POSITION *);
00836 extern LONG      WriteFileToFile(int,UBYTE *,LONG);
00837 extern void      SeekFile(int,POSITION *,int);
00838 extern LONG      TellFile(int);
00839 extern void      FlushFile(int);
00840 extern int      GetPosFile(int,fpos_t *);
00841 extern int      SetPosFile(int,fpos_t *);
00842 extern void      SynchFile(int);
00843 extern void      TruncateFile(int);
00844 extern int      GetChannel(char *,int);
00845 extern int      GetAppendChannel(char *);
00846 extern int      CloseChannel(char *);
00847 extern void      inictable(void);
00848 extern KEYWORD   *findcommand(UBYTE *);
00849 extern int      inicbufs(void);
00850 extern void      StartFiles(void);
00851 extern UBYTE    *MakeDate(void);
00852 extern void      PreProcessor(void);
00853 extern void      *FromList(LIST *);
00854 extern void      *From0List(LIST *);
00855 extern void      *FromVarList(LIST *);
00856 extern int      DoubleList(void ***,int *,int,char *);
00857 extern int      DoubleLList(void ***,LONG *,int,char *);
00858 extern void      DoubleBuffer(void **,void **,int,char *);
00859 extern void      ExpandBuffer(void **,LONG *,int);
00860 extern LONG      iexp(LONG,int);
00861 extern int      IsLikeVector(WORD *);
00862 extern int      AreArgsEqual(WORD *,WORD *);
00863 extern int      CompareArgs(WORD *,WORD *);
00864 extern UBYTE    *SkipField(UBYTE *,int);
00865 extern int      StrCmp(UBYTE *,UBYTE *);
00866 extern int      StrICmp(UBYTE *,UBYTE *);
00867 extern int      StrHICmp(UBYTE *,UBYTE *);
00868 extern int      StrICont(UBYTE *,UBYTE *);
00869 extern int      CmpArray(WORD *,WORD *,WORD);
00870 extern int      ConWord(UBYTE *,UBYTE *);
00871 extern int      StrLen(UBYTE *);
00872 extern UBYTE    *GetPreVar(UBYTE *,int);
00873 extern void      ToGeneral(WORD *,WORD *,WORD);
00874 extern WORD      ToPolyFunGeneral(PHEAD WORD *);
00875 extern int      ToFast(WORD *,WORD *);
00876 extern SETUPPARAMETERS *GetSetupPar(UBYTE *);

```

```
00877 extern int RecalcSetups(void);
00878 extern int AllocSetups(void);
00879 extern SORTING *AllocSort (LONG, LONG, LONG, LONG, int, int, LONG, int);
00880 extern void AllocSortFileName (SORTING *);
00881 extern UBYTE *LoadInputFile (UBYTE *, int);
00882 extern UBYTE GetInput (void);
00883 extern void ClearPushback (void);
00884 extern UBYTE GetChar (int);
00885 extern void CharOut (UBYTE);
00886 extern void UnsetAllowDelay (void);
00887 extern void PopPreVars (int);
00888 extern void IniModule (int);
00889 extern void IniSpecialModule (int);
00890 extern int ModuleInstruction (int *, int *);
00891 extern int PreProInstruction (void);
00892 extern int LoadInstruction (int);
00893 extern int LoadStatement (int);
00894 extern KEYWORD *FindKeyWord (UBYTE *, KEYWORD *, int);
00895 extern KEYWORD *FindInKeyWord (UBYTE *, KEYWORD *, int);
00896 extern int DoDefine (UBYTE *);
00897 extern int DoRedefine (UBYTE *);
00898 extern int TheDefine (UBYTE *, int);
00899 extern int TheUndefine (UBYTE *);
00900 extern int ClearMacro (UBYTE *);
00901 extern int DoUndefine (UBYTE *);
00902 extern int DoInclude (UBYTE *);
00903 extern int DoReverseInclude (UBYTE *);
00904 extern int Include (UBYTE *, int);
00905 /*[14apr2004 mt]:*/
00906 extern int DoExternal (UBYTE *);
00907 extern int DoToExternal (UBYTE *);
00908 extern int DoFromExternal (UBYTE *);
00909 extern int DoPrompt (UBYTE *);
00910 extern int DoSetExternal (UBYTE *);
00911 /*[10may2006 mt]:*/
00912 extern int DoSetExternalAttr (UBYTE *);
00913 /*:[10may2006 mt]:*/
00914 extern int DoRmExternal (UBYTE *);
00915 /*:[14apr2004 mt]:*/
00916 extern int DoFactDollar (UBYTE *);
00917 extern WORD GetDollarNumber (UBYTE **, DOLLARS);
00918 extern int DoSetRandom (UBYTE *);
00919 extern int DoOptimize (UBYTE *);
00920 extern int DoClearOptimize (UBYTE *);
00921 extern int DoSkipExtraSymbols (UBYTE *);
00922 extern int DoTimeOutAfter (UBYTE *);
00923 extern int DoMessage (UBYTE *);
00924 extern int DoPreOut (UBYTE *);
00925 extern int DoPreAppend (UBYTE *);
00926 extern int DoPreCreate (UBYTE *);
00927 extern int DoPreAssign (UBYTE *);
00928 extern int DoPreBreak (UBYTE *);
00929 extern int DoPreDefault (UBYTE *);
00930 extern int DoPreSwitch (UBYTE *);
00931 extern int DoPreEndSwitch (UBYTE *);
00932 extern int DoPreCase (UBYTE *);
00933 extern int DoPreShow (UBYTE *);
00934 extern int DoPreExchange (UBYTE *);
00935 extern int DoSystem (UBYTE *);
00936 extern int DoPipe (UBYTE *);
00937 extern void StartPrepro (void);
00938 extern int DoIfdef (UBYTE *, int);
00939 extern int DoIfydef (UBYTE *);
00940 extern int DoIfndef (UBYTE *);
00941 extern int DoElse (UBYTE *);
00942 extern int DoElseif (UBYTE *);
00943 extern int DoEndif (UBYTE *);
00944 extern int DoTerminate (UBYTE *);
00945 extern int DoIf (UBYTE *);
00946 extern int DoCall (UBYTE *);
00947 extern int DoDebug (UBYTE *);
00948 extern int DoContinueDo (UBYTE *);
00949 extern int DoDo (UBYTE *);
00950 extern int DoBreakDo (UBYTE *);
00951 extern int DoEnddo (UBYTE *);
00952 extern int DoEndprocedure (UBYTE *);
00953 extern int DoInside (UBYTE *);
00954 extern int DoEndInside (UBYTE *);
00955 extern int DoProcedure (UBYTE *);
00956 extern int DoPrePrintTimes (UBYTE *);
00957 extern int DoPreWrite (UBYTE *);
00958 extern int DoPreClose (UBYTE *);
00959 extern int DoPreRemove (UBYTE *);
00960 extern int DoPreSortReallocate (UBYTE *);
00961 extern int DoCommentChar (UBYTE *);
00962 extern int DoPrcExtension (UBYTE *);
00963 extern int DoPreReset (UBYTE *);
```

```

00964 extern void    WriteString(int, UBYTE *, int);
00965 extern void    WriteUnfinString(int, UBYTE *, int);
00966 extern UBYTE *AddToString(UBYTE *, UBYTE *, int);
00967 extern UBYTE *PreCalc(void);
00968 extern UBYTE *PreEval(UBYTE *, LONG *);
00969 extern void    NumToStr(UBYTE *, LONG);
00970 extern int     PreCmp(int, int, UBYTE *, int, int, UBYTE *, int);
00971 extern int     PreEq(int, int, UBYTE *, int, int, UBYTE *, int);
00972 extern UBYTE *pParseObject(UBYTE *, int *, LONG *);
00973 extern UBYTE *PreIfEval(UBYTE *, int *);
00974 extern int     EvalPreIf(UBYTE *);
00975 extern int     PreLoad(PRELOAD *, UBYTE *, UBYTE *, int, char *);
00976 extern int     PreSkip(UBYTE *, UBYTE *, int);
00977 extern UBYTE *EndOfToken(UBYTE *);
00978 extern void    SetSpecialMode(int, int);
00979 extern void    MakeGlobal(void);
00980 extern int     ExecModule(int);
00981 extern int     ExecStore(void);
00982 extern void    FullCleanUp(void);
00983 extern int     DoExecStatement(void);
00984 extern int     DoPipeStatement(void);
00985 extern int     DoPolyfun(UBYTE *);
00986 extern int     DoPolyratfun(UBYTE *);
00987 extern int     CompileStatement(UBYTE *);
00988 extern UBYTE *ToToken(UBYTE *);
00989 extern int     GetDollar(UBYTE *);
00990 extern int     MesWork(void);
00991 extern int     MesPrint(const char *, ...);
00992 extern int     MesCall(char *);
00993 extern UBYTE *NumCopy(WORD, UBYTE *);
00994 extern char *LongCopy(LONG, char *);
00995 extern char *LongLongCopy(off_t *, char *);
00996 extern void    ReserveTempFiles(int);
00997 extern void    PrintTerm(WORD *, char *);
00998 extern void    PrintTermC(WORD *, char *);
00999 extern void    PrintSubTerm(WORD *, char *);
01000 extern void    PrintWords(WORD *, LONG);
01001 extern void    PrintSeq(WORD *, char *);
01002 extern int     ExpandTripleDots(int);
01003 extern LONG    ComPress(WORD **, LONG *);
01004 extern void    StageSort(FILEHANDLE *);
01005
01006 #define M_alloc(x)      malloc((size_t)(x))
01007
01008 extern void    M_free(void *, const char *);
01009 extern void    ClearWildcardNames(void);
01010 extern int     AddWildcardName(UBYTE *);
01011 extern int     GetWildcardName(UBYTE *);
01012 extern void    Globalize(int);
01013 extern void    ResetVariables(int);
01014 extern void    AddToPreTypes(int);
01015 extern void    MessPreNesting(int);
01016 extern LONG    GetStreamPosition(STREAM *);
01017 extern WORD *DoubleCbuffer(int, WORD *, int);
01018 extern WORD *AddLHS(int);
01019 extern WORD *AddRHS(int, int);
01020 extern int     AddNtoL(int, WORD *);
01021 extern int     AddNtoC(int, int, WORD *, int);
01022 extern void    DoubleIfBuffers(void);
01023 extern STREAM *CreateStream(UBYTE *);
01024
01025 extern int     setonoff(UBYTE *, int *, int, int);
01026 extern int     DoPrint(UBYTE *, int);
01027 extern int     SetExpr(UBYTE *, int, int);
01028 extern void    AddToCom(int, WORD *);
01029 extern int     Add2ComStrings(int, WORD *, UBYTE *, UBYTE *);
01030 extern int     DoSymmetrize(UBYTE *, int);
01031 extern int     DoArgument(UBYTE *, int);
01032 extern int     ArgFactorize(PHEAD WORD *, WORD *);
01033 extern WORD *TakeArgContent(PHEAD WORD *, WORD *);
01034 extern WORD *MakeInteger(PHEAD WORD *, WORD *, WORD *);
01035 extern WORD *MakeMod(PHEAD WORD *, WORD *, WORD *);
01036 extern WORD *FindArg(PHEAD WORD *);
01037 extern int     InsertArg(PHEAD WORD *, WORD *, int);
01038 extern int     CleanupArgCache(PHEAD WORD *);
01039 extern int     ArgSymbolMerge(WORD *, WORD *);
01040 extern int     ArgDotproductMerge(WORD *, WORD *);
01041 extern void    SortWeights(LONG *, LONG *, WORD);
01042 extern int     DoBrackets(UBYTE *, int);
01043 extern int     DoPutInside(UBYTE *, int);
01044 extern WORD *CountComp(UBYTE *, WORD *);
01045 extern int     CoAntiBracket(UBYTE *);
01046 extern int     CoAntiSymmetrize(UBYTE *);
01047 extern int     DoArgPlode(UBYTE *, int);
01048 extern int     CoArgExplode(UBYTE *);
01049 extern int     CoArgImplode(UBYTE *);
01050 extern int     CoArgument(UBYTE *);

```

```

01051 extern int      CoInside(UBYTE *);
01052 extern int      ExecInside(UBYTE *);
01053 extern int      CoInExpression(UBYTE *);
01054 extern int      CoInParallel(UBYTE *);
01055 extern int      CoNotInParallel(UBYTE *);
01056 extern int      DoInParallel(UBYTE *,int);
01057 extern int      CoEndInExpression(UBYTE *);
01058 extern int      CoBracket(UBYTE *);
01059 extern int      CoPutInside(UBYTE *);
01060 extern int      CoAntiPutInside(UBYTE *);
01061 extern int      CoMultiBracket(UBYTE *);
01062 extern int      CoCFunction(UBYTE *);
01063 extern int      CoCTensor(UBYTE *);
01064 extern int      CoCollect(UBYTE *);
01065 extern int      CoCompress(UBYTE *);
01066 extern int      CoContract(UBYTE *);
01067 extern int      CoCycleSymmetrize(UBYTE *);
01068 extern int      CoDelete(UBYTE *);
01069 extern int      CoTableBase(UBYTE *);
01070 extern int      CoApply(UBYTE *);
01071 extern int      CoDenominators(UBYTE *);
01072 extern int      CoDimension(UBYTE *);
01073 extern int      CoDiscard(UBYTE *);
01074 extern int      CoDisorder(UBYTE *);
01075 extern int      CoDrop(UBYTE *);
01076 extern int      CoDropCoefficient(UBYTE *);
01077 extern int      CoDropSymbols(UBYTE *);
01078 extern int      CoElse(UBYTE *);
01079 extern int      CoElseIf(UBYTE *);
01080 extern int      CoEndArgument(UBYTE *);
01081 extern int      CoEndInside(UBYTE *);
01082 extern int      CoEndIf(UBYTE *);
01083 extern int      CoEndRepeat(UBYTE *);
01084 extern int      CoEndTerm(UBYTE *);
01085 extern int      CoEndWhile(UBYTE *);
01086 extern int      CoExit(UBYTE *);
01087 extern int      CoFactArg(UBYTE *);
01088 extern int      CoFactDollar(UBYTE *);
01089 extern int      CoFactorize(UBYTE *);
01090 extern int      CoNFactorize(UBYTE *);
01091 extern int      CoUnFactorize(UBYTE *);
01092 extern int      CoNUnFactorize(UBYTE *);
01093 extern int      DoFactorize(UBYTE *,int);
01094 extern int      CoFill(UBYTE *);
01095 extern int      CoFillExpression(UBYTE *);
01096 extern int      CoFixIndex(UBYTE *);
01097 extern int      CoFormat(UBYTE *);
01098 extern int      CoGlobal(UBYTE *);
01099 extern int      CoGlobalFactorized(UBYTE *);
01100 extern int      CoGoTo(UBYTE *);
01101 extern int      CoId(UBYTE *);
01102 extern int      CoIdNew(UBYTE *);
01103 extern int      CoIdOld(UBYTE *);
01104 extern int      CoIf(UBYTE *);
01105 extern int      CoIfMatch(UBYTE *);
01106 extern int      CoIfNoMatch(UBYTE *);
01107 extern int      CoIndex(UBYTE *);
01108 extern int      CoInsideFirst(UBYTE *);
01109 extern int      CoKeep(UBYTE *);
01110 extern int      CoLabel(UBYTE *);
01111 extern int      CoLoad(UBYTE *);
01112 extern int      CoLocal(UBYTE *);
01113 extern int      CoLocalFactorized(UBYTE *);
01114 extern int      CoMany(UBYTE *);
01115 extern int      CoMerge(UBYTE *);
01116 extern int      CoShuffle(UBYTE *);
01117 extern int      CoMetric(UBYTE *);
01118 extern int      CoModOption(UBYTE *);
01119 extern int      CoModuleOption(UBYTE *);
01120 extern int      CoModulus(UBYTE *);
01121 extern int      CoMulti(UBYTE *);
01122 extern int      CoMultiply(UBYTE *);
01123 extern int      CoNFunction(UBYTE *);
01124 extern int      CoNPrint(UBYTE *);
01125 extern int      CoNTensor(UBYTE *);
01126 extern int      CoNWrite(UBYTE *);
01127 extern int      CoNoDrop(UBYTE *);
01128 extern int      CoNoSkip(UBYTE *);
01129 extern int      CoNormalize(UBYTE *);
01130 extern int      CoMakeInteger(UBYTE *);
01131 extern int      CoFlags(UBYTE *,int);
01132 extern int      CoOff(UBYTE *);
01133 extern int      CoOn(UBYTE *);
01134 extern int      CoOnce(UBYTE *);
01135 extern int      CoOnly(UBYTE *);
01136 extern int      CoOptimizeOption(UBYTE *);
01137 extern int      CoPolyFun(UBYTE *);

```



```

01138 extern int      CoPolyRatFun(UBYTE *);
01139 extern int      CoPrint(UBYTE *);
01140 extern int      CoPrintB(UBYTE *);
01141 extern int      CoProperCount(UBYTE *);
01142 extern int      CoUnitTrace(UBYTE *);
01143 extern int      CoRCycleSymmetrize(UBYTE *);
01144 extern int      CoRatio(UBYTE *);
01145 extern int      CoRedefine(UBYTE *);
01146 extern int      CoRenumber(UBYTE *);
01147 extern int      CoRepeat(UBYTE *);
01148 extern int      CoSave(UBYTE *);
01149 extern int      CoSelect(UBYTE *);
01150 extern int      CoSet(UBYTE *);
01151 extern int      CoSetExitFlag(UBYTE *);
01152 extern int      CoSkip(UBYTE *);
01153 extern int      CoProcessBucket(UBYTE *);
01154 extern int      CoPushHide(UBYTE *);
01155 extern int      CoPopHide(UBYTE *);
01156 extern int      CoHide(UBYTE *);
01157 extern int      CoIntoHide(UBYTE *);
01158 extern int      CoNoIntoHide(UBYTE *);
01159 extern int      CoNoHide(UBYTE *);
01160 extern int      CoUnHide(UBYTE *);
01161 extern int      CoNoUnHide(UBYTE *);
01162 extern int      CoSort(UBYTE *);
01163 extern int      CoSplitArg(UBYTE *);
01164 extern int      CoSplitFirstArg(UBYTE *);
01165 extern int      CoSplitLastArg(UBYTE *);
01166 extern int      CoSum(UBYTE *);
01167 extern int      CoSymbol(UBYTE *);
01168 extern int      CoSymmetrize(UBYTE *);
01169 extern int      DoTable(UBYTE *,int);
01170 extern int      CoTable(UBYTE *);
01171 extern int      CoTerm(UBYTE *);
01172 extern int      CoNTable(UBYTE *);
01173 extern int      CoCTable(UBYTE *);
01174 extern void     EmptyTable(TABLES);
01175 extern int      CoToTensor(UBYTE *);
01176 extern int      CoToVector(UBYTE *);
01177 extern int      CoTrace4(UBYTE *);
01178 extern int      CoTraceN(UBYTE *);
01179 extern int      CoChisholm(UBYTE *);
01180 extern int      CoTransform(UBYTE *);
01181 extern int      CoClearTable(UBYTE *);
01182 extern int      DoChain(UBYTE *,int);
01183 extern int      CoChainin(UBYTE *);
01184 extern int      CoChainout(UBYTE *);
01185 extern int      CoTryReplace(UBYTE *);
01186 extern int      CoVector(UBYTE *);
01187 extern int      CoWhile(UBYTE *);
01188 extern int      CoWrite(UBYTE *);
01189 extern int      CoAuto(UBYTE *);
01190 extern int      CoSwitch(UBYTE *);
01191 extern int      CoCase(UBYTE *);
01192 extern int      CoBreak(UBYTE *);
01193 extern int      CoDefault(UBYTE *);
01194 extern int      CoEndSwitch(UBYTE *);
01195 extern int      CoTBaddto(UBYTE *);
01196 extern int      CoTBaudit(UBYTE *);
01197 extern int      CoTbcleanup(UBYTE *);
01198 extern int      CoTBcreate(UBYTE *);
01199 extern int      CoTBenter(UBYTE *);
01200 extern int      CoTBhelp(UBYTE *);
01201 extern int      CoTBload(UBYTE *);
01202 extern int      CoTBoff(UBYTE *);
01203 extern int      CoTBon(UBYTE *);
01204 extern int      CoTBopen(UBYTE *);
01205 extern int      CoTBreplace(UBYTE *);
01206 extern int      CoTBuse(UBYTE *);
01207 extern int      CoTestUse(UBYTE *);
01208 extern int      CoThreadBucket(UBYTE *);
01209 extern int      AddComString(int,WORD *,UBYTE *,int);
01210 extern int      CompileAlgebra(UBYTE *,int,WORD *);
01211 extern int      IsIdStatement(UBYTE *);
01212 extern UBYTE    *IsRHS(UBYTE *,UBYTE);
01213 extern int      ParenthesesTest(UBYTE *);
01214 extern int      tokenize(UBYTE *,WORD);
01215 extern void     WriteTokens(SBYTE *);
01216 extern int      simpltoken(SBYTE *);
01217 extern int      simpwtoken(SBYTE *);
01218 extern int      simp2token(SBYTE *);
01219 extern int      simp3atoken(SBYTE *,int);
01220 extern int      simp3btoken(SBYTE *,int);
01221 extern int      simp4token(SBYTE *);
01222 extern int      simp5token(SBYTE *,int);
01223 extern int      simp6token(SBYTE *,int);
01224 extern UBYTE    *SkipAName(UBYTE *);

```

```

01225 extern int      TestTables(void);
01226 extern int      GetLabel(UBYTE *);
01227 extern int      CoIdExpression(UBYTE *,int);
01228 extern int      CoAssign(UBYTE *);
01229 extern int      DoExpr(UBYTE *,int,int);
01230 extern int      CompileSubExpressions(SBYTE *);
01231 extern int      CodeGenerator(SBYTE *);
01232 extern int      CompleteTerm(WORD *,UWORD *,UWORD *,WORD,WORD,int);
01233 extern int      CodeFactors(SBYTE *s);
01234 extern WORD     GenerateFactors(WORD,WORD);
01235 extern int      InsTree(int,int);
01236 extern int      FindTree(int,WORD *);
01237 extern void     RedoTree(CBUF *,int);
01238 extern void     ClearTree(int);
01239 extern int      CatchDollar(int);
01240 extern int      AssignDollar(PHEAD WORD *,WORD);
01241 extern UBYTE     *WriteDollarToBuffer(WORD,WORD);
01242 extern UBYTE     *WriteDollarFactorToBuffer(WORD,WORD,WORD);
01243 extern void     AddToDollarBuffer(UBYTE *);
01244 extern int      PutTermInDollar(WORD *,WORD);
01245 extern void     TermAssign(WORD *);
01246 extern void     WildDollars(PHEAD WORD *);
01247 extern LONG     numcommute(WORD *,LONG *);
01248 extern int      FullRenummer(PHEAD WORD *,WORD);
01249 extern int      Lus(WORD *,WORD,WORD,WORD,WORD,WORD);
01250 extern int      FindLus(int,int,int);
01251 extern int      CoReplaceLoop(UBYTE *);
01252 extern int      CoFindLoop(UBYTE *);
01253 extern int      DoFindLoop(UBYTE *,int);
01254 extern int      CoFunPowers(UBYTE *);
01255 extern int      SortTheList(int *,int);
01256 extern int      MatchIsPossible(WORD *,WORD *);
01257 extern int      StudyPattern(WORD *);
01258 extern WORD     DolToTensor(PHEAD WORD);
01259 extern WORD     DolToFunction(PHEAD WORD);
01260 extern WORD     DolToVector(PHEAD WORD);
01261 extern WORD     DolToNumber(PHEAD WORD);
01262 extern WORD     DolToSymbol(PHEAD WORD);
01263 extern WORD     DolToIndex(PHEAD WORD);
01264 extern LONG     DolToLong(PHEAD WORD);
01265 extern int      DollarFactorize(PHEAD WORD);
01266 extern int      CoPrintTable(UBYTE *);
01267 extern int      CoDeallocateTable(UBYTE *);
01268 extern void     CleanDollarFactors(DOLLARS);
01269 extern WORD     *TakeDollarContent(PHEAD WORD *,WORD **);
01270 extern WORD     *MakeDollarInteger(PHEAD WORD *,WORD **);
01271 extern WORD     *MakeDollarMod(PHEAD WORD *,WORD **);
01272 extern int      GetDolNum(PHEAD WORD *, WORD *);
01273 extern void     AddPotModdollar(WORD);
01274
01275 extern int      Optimize(WORD, int);
01276 extern int      ClearOptimize(void);
01277 extern int      TakeLongRoot(UWORD *,WORD *,WORD);
01278 extern int      TakeRatRoot(UWORD *,WORD *,WORD);
01279 extern int      MakeRational(WORD ,WORD , WORD *, WORD *);
01280 extern int      MakeLongRational(PHEAD UWORD *,WORD ,UWORD *,WORD ,UWORD *,WORD *);
01281 extern void     ClearTableTree(TABLES);
01282 extern int      InsTableTree(TABLES,WORD *);
01283 extern void     RedoTableTree(TABLES,int);
01284 extern int      FindTableTree(TABLES,WORD *,int);
01285 extern void     finishcbuf(WORD);
01286 extern void     clearcbuf(WORD);
01287 extern void     CleanupSort(int);
01288 extern FILEHANDLE *AllocFileHandle(WORD,char *);
01289 extern void     DeAllocFileHandle(FILEHANDLE *);
01290 extern void     LowerSortLevel(void);
01291 extern WORD     *PolyRatFunSpecial(PHEAD WORD *, WORD *);
01292 extern void     SimpleSplitMergeRec(WORD *,WORD,WORD *);
01293 extern void     SimpleSplitMerge(WORD *,WORD);
01294 extern WORD     BinarySearch(WORD *,WORD,WORD);
01295 extern int      InsideDollar(PHEAD WORD *,WORD);
01296 extern DOLLARS  DolToTerms(PHEAD WORD);
01297 extern WORD     EvalDoLoopArg(PHEAD WORD *,WORD);
01298 extern int      SetExprCases(int,int,int);
01299 extern int      TestSelect(WORD *,WORD *);
01300 extern void     SubsInAll(PHEAD0);
01301 extern void     TransferBuffer(int,int,int);
01302 extern int      TakeIDfunction(PHEAD WORD *);
01303 extern int      MakeSetupAllocs(void);
01304 extern int      TryFileSetups(void);
01305 extern void     ExchangeExpressions(int,int);
01306 extern void     ExchangeDollars(int,int);
01307 extern int      GetFirstBracket(WORD *,int);
01308 extern int      GetFirstTerm(WORD *,int,int);
01309 extern int      GetContent(WORD *,int);
01310 extern int      CleanupTerm(WORD *);
01311 extern WORD     ContentMerge(PHEAD WORD *,WORD *);

```



```

01312 extern UBYTE *PreIfDollarEval(UBYTE *,int *);
01313 extern LONG TermsInDollar(WORD);
01314 extern LONG SizeOfDollar(WORD);
01315 extern LONG TermsInExpression(WORD);
01316 extern LONG SizeOfExpression(WORD);
01317 extern WORD *TranslateExpression(UBYTE *);
01318 extern int IsSetMember(WORD *,WORD);
01319 extern int IsMultipleOf(WORD *,WORD *);
01320 extern int TwoExprCompare(WORD *,WORD *,int);
01321 extern void UpdatePositions(void);
01322 extern void M_check(void);
01323 extern void M_print(void);
01324 extern void M_check1(void);
01325 extern void PrintTime(UBYTE *);
01326
01327 extern POSITION *FindBracket(WORD,WORD *);
01328 extern void PutBracketInIndex(PHEAD WORD *,POSITION *);
01329 extern void ClearBracketIndex(WORD);
01330 extern void OpenBracketIndex(WORD);
01331 extern int DoNoParallel(UBYTE *);
01332 extern int DoParallel(UBYTE *);
01333 extern int DoModSum(UBYTE *);
01334 extern int DoModMax(UBYTE *);
01335 extern int DoModMin(UBYTE *);
01336 extern int DoModLocal(UBYTE *);
01337 extern UBYTE *DoModDollar(UBYTE *,int);
01338 extern int DoProcessBucket(UBYTE *);
01339 extern int DoinParallel(UBYTE *);
01340 extern int DonotinParallel(UBYTE *);
01341
01342 extern int FlipTable(FUNCTIONS,int);
01343 extern int ChainIn(PHEAD WORD *,WORD);
01344 extern int ChainOut(PHEAD WORD *,WORD);
01345 extern int ArgumentImplode(PHEAD WORD *,WORD *);
01346 extern int ArgumentExplode(PHEAD WORD *,WORD *);
01347 extern int DenToFunction(WORD *,WORD);
01348
01349 extern WORD HowMany(PHEAD WORD *,WORD *);
01350 extern void RemoveDollars(void);
01351 extern LONG CountTerms1(PHEAD);
01352 extern LONG TermsInBracket(PHEAD WORD *,WORD);
01353 extern int Crash(void);
01354
01355 extern char *str_dup(char *);
01356 extern void convertblock(INDEXBLOCK *,INDEXBLOCK *,int);
01357 extern void convertnamesblock(NAMESBLOCK *,NAMESBLOCK *,int);
01358 extern void convertiniinfo(INIINFO *,INIINFO *,int);
01359 extern int ReadIndex(DBASE *);
01360 extern int WriteIndexBlock(DBASE *,MLONG);
01361 extern int WriteNamesBlock(DBASE *,MLONG);
01362 extern int WriteIndex(DBASE *);
01363 extern int WriteIniInfo(DBASE *);
01364 extern int ReadIniInfo(DBASE *);
01365 extern int AddToIndex(DBASE *,MLONG);
01366 extern DBASE *GetDbase(char *,MLONG);
01367 extern DBASE *OpenDbase(char *);
01368 extern char *ReadObject(DBASE *,MLONG,char *);
01369 extern char *ReadObjObject(DBASE *,MLONG,MLONG,char *);
01370 extern int ExistsObject(DBASE *,MLONG,char *);
01371 extern int DeleteObject(DBASE *,MLONG,char *);
01372 extern int WriteObject(DBASE *,MLONG,char *,char *,MLONG);
01373 extern MLONG AddObject(DBASE *,MLONG,char *,char *);
01374 extern DBASE *NewDbase(char *,MLONG);
01375 extern void FreeTableBase(DBASE *);
01376 extern int ComposeTableName(DBASE *);
01377 extern int PutTableName(DBASE *);
01378 extern MLONG AddTableName(DBASE *,char *,TABLES);
01379 extern MLONG GetTableName(DBASE *,char *);
01380 extern MLONG FindTableNumber(DBASE *,char *);
01381 extern int TryEnvironment(void);
01382
01383 #ifdef WITHZLIB
01384 extern int SetupOutputGZIP(FILEHANDLE *);
01385 extern int PutOutputGZIP(FILEHANDLE *);
01386 extern int FlushOutputGZIP(FILEHANDLE *);
01387 extern int SetupAllInputGZIP(SORTING *);
01388 extern LONG FillInputGZIP(FILEHANDLE *,POSITION *,UBYTE *,LONG,int);
01389 extern void ClearSortGZIP(FILEHANDLE *f);
01390 #endif
01391
01392 #ifdef WITHPTHREADS
01393 extern void BeginIdentities(void);
01394 extern int WhoAmI(void);
01395 extern int StartAllThreads(int);
01396 extern void StartHandleLock(void);
01397 extern void TerminateAllThreads(void);
01398 extern int GetAvailableThread(void);

```

```

01399 extern int      ConditionalGetAvailableThread(void);
01400 extern int      BalanceRunThread(PHEAD int,WORD *,WORD);
01401 extern void     WakeupThread(int,int);
01402 extern int      MasterWait(void);
01403 extern int      InParallelProcessor(void);
01404 extern int      ThreadsProcessor(EXPRESSIONS,WORD,WORD);
01405 extern int      MasterMerge(void);
01406 extern int      PutToMaster(PHEAD WORD *);
01407 extern void     SetWorkerFiles(void);
01408 extern int      MakeThreadBuckets(int,int);
01409 extern int      SendOneBucket(int);
01410 extern int      LoadOneThread(int,int,THREADBUCKET *,int);
01411 extern void     *RunSortBot(void *);
01412 extern void     MasterWaitAllSortBots(void);
01413 extern int      SortBotMerge(PHEAD0);
01414 extern int      SortBotOut(PHEAD WORD *);
01415 extern void     DefineSortBotTree(void);
01416 extern int      SortBotMasterMerge(void);
01417 extern int      SortBotWait(int);
01418 extern void     StartIdentity(void);
01419 extern void     FinishIdentity(void *);
01420 extern int      SetIdentity(int *);
01421 extern ALLPRIVATES *InitializeOneThread(int);
01422 extern void     FinalizeOneThread(int);
01423 extern void     ClearAllThreads(void);
01424 extern void     *RunThread(void *);
01425 extern void     IAmAvailable(int);
01426 extern int      ThreadWait(int);
01427 extern int      ThreadClaimedBlock(int);
01428 extern int      GetThread(int);
01429 extern int      UpdateOneThread(int);
01430 extern void     MasterWaitAll(void);
01431 extern void     MasterWaitAllBlocks(void);
01432 extern int      MasterWaitThread(int);
01433 extern void     WakeupMasterFromThread(int,int);
01434 extern int      LoadReadjusted(void);
01435 extern int      IniSortBlocks(int);
01436 extern int      UpdateSortBlocks(int);
01437 extern int      TreatIndexEntry(PHEAD LONG);
01438 extern WORD     GetTerm2(PHEAD WORD *);
01439 extern void     SetHideFiles(void);
01440
01441 #endif
01442
01443 extern int      CopyExpression(FILEHANDLE *,FILEHANDLE *);
01444
01445 extern int      set_in(UBYTE, set_of_char);
01446 extern one_byte set_set(UBYTE, set_of_char);
01447 extern one_byte set_del(UBYTE, set_of_char);
01448 extern one_byte set_sub(set_of_char, set_of_char, set_of_char);
01449 extern int      DoPreAddSeparator(UBYTE *);
01450 extern int      DoPreRmSeparator(UBYTE *);
01451
01452 /*See the file extcmd.c*/
01453 extern int      openExternalChannel(UBYTE *,int,UBYTE *,UBYTE *);
01454 extern int      initPresetExternalChannels(UBYTE *, int);
01455 extern int      closeExternalChannel(int);
01456 extern int      selectExternalChannel(int);
01457 extern int      getCurrentExternalChannel(void);
01458 extern void     closeAllExternalChannels(void);
01459
01460 typedef int      (*WRITEBUFTOEXTCHANNEL)(char *,size_t);
01461 typedef int      (*GETCFROMEXTCHANNEL)(void);
01462 typedef int      (*SETTERMINATORFOREXTCHANNEL)(char *);
01463 typedef int      (*SETKILLMODEFOREXTCHANNEL)(int,int);
01464 typedef LONG      (*WRITEFILE)(int,UBYTE *,LONG);
01465 typedef WORD      (*GETTERM)(PHEAD WORD *);
01466
01467 #define CompareTerms ((COMPARE)AR.CompareRoutine)
01468 #define FiniShuffle AN.SHvar.finishuf
01469 #define DoShtuffle ((DO_UFFLE)AN.SHvar.do_uffle)
01470
01471 extern UBYTE *defineChannel(UBYTE*, HANDLERS*);
01472 extern int      writeToChannel(int,UBYTE *,HANDLERS*);
01473 #ifdef WITHEXTCHANNEL
01474 extern LONG      WriteToExternalChannel(int,UBYTE *,LONG);
01475 #endif
01476 extern int      writeBufToExtChannelOk(char *,size_t);
01477 extern int      getcFromExtChannelOk(void);
01478 extern int      setKillModeForExternalChannelOk(int,int);
01479 extern int      setTerminatorForExternalChannelOk(char *);
01480 extern int      getcFromExtChannelFailure(void);
01481 extern int      setKillModeForExternalChannelFailure(int,int);
01482 extern int      setTerminatorForExternalChannelFailure(char *);
01483 extern int      writeBufToExtChannelFailure(char *,size_t);
01484
01485 extern int      ReleaseTB(void);

```

```

01486
01487 extern int      SymbolNormalize(WORD *);
01488 extern int      TestFunFlag(PHEAD WORD *);
01489 extern WORD     CompareSymbols(PHEAD WORD *,WORD *,WORD);
01490 extern WORD     CompareHSymbols(PHEAD WORD *,WORD *,WORD);
01491 extern WORD     NextPrime(PHEAD WORD);
01492 extern WORD     Moebius(PHEAD WORD);
01493 extern UWORD    wranf(PHEAD0);
01494 extern UWORD    iranf(PHEAD UWORD);
01495 extern void     iniwranf(PHEAD0);
01496 extern UBYTE    *PreRandom(UBYTE *);
01497
01498 extern WORD     *PolyNormPoly(PHEAD WORD);
01499 extern WORD     *EvaluateGcd(PHEAD WORD *);
01500 extern int      TreatPolyRatFun(PHEAD WORD *);
01501
01502 extern int      ReadSaveHeader(void);
01503 extern LONG     ReadSaveIndex(FILEINDEX *);
01504 extern LONG     ReadSaveExpression(UBYTE *,UBYTE *,LONG *,LONG *);
01505 extern UBYTE    *ReadSaveTerm32(UBYTE *,UBYTE *,UBYTE **,UBYTE *,UBYTE *,int);
01506 extern LONG     ReadSaveVariables(UBYTE *,UBYTE *,LONG *,LONG *,INDEXENTRY *,LONG *);
01507 extern LONG     WriteStoreHeader(WORD);
01508
01509 extern void     InitRecovery(void);
01510 extern int      CheckRecoveryFile(void);
01511 extern void     DeleteRecoveryFile(void);
01512 extern char     *RecoveryFilename(void);
01513 extern int      DoRecovery(int *);
01514 extern void     DoCheckpoint(int);
01515
01516 extern void     NumberMallocAddMemory(PHEAD0);
01517 extern void     CacheNumberMallocAddMemory(PHEAD0);
01518 extern void     TermMallocAddMemory(PHEAD0);
01519 #ifndef MEMORYMACROS
01520 extern WORD     *TermMalloc2(PHEAD char *text);
01521 extern void     TermFree2(PHEAD WORD *term,char *text);
01522 extern UWORD    *NumberMalloc2(PHEAD char *text);
01523 extern UWORD    *CacheNumberMalloc2(PHEAD char *text);
01524 extern void     NumberFree2(PHEAD UWORD *NumberMem,char *text);
01525 extern void     CacheNumberFree2(PHEAD UWORD *NumberMem,char *text);
01526 #endif
01527
01528 extern void     ExprStatus(EXPRESSIONS);
01529 extern void     iniTools(void);
01530 extern int      TestTerm(WORD *);
01531
01532 extern int      RunTransform(PHEAD WORD *term, WORD *params);
01533 extern int      RunEncode(PHEAD WORD *fun, WORD *args, WORD *info);
01534 extern int      RunDecode(PHEAD WORD *fun, WORD *args, WORD *info);
01535 extern int      RunReplace(PHEAD WORD *fun, WORD *args, WORD *info);
01536 extern int      RunImplode(WORD *fun, WORD *args);
01537 extern int      RunExplode(PHEAD WORD *fun, WORD *args);
01538 extern int      TestArgNum(int n, int totarg, WORD *args);
01539 extern WORD     PutArgInScratch(WORD *arg,UWORD *scrat);
01540 extern UBYTE    *ReadRange(UBYTE *s, WORD *out, int par);
01541 extern int      FindRange(PHEAD WORD *,WORD *,WORD *,WORD);
01542 extern int      RunPermute(PHEAD WORD *fun, WORD *args, WORD *info);
01543 extern int      RunReverse(PHEAD WORD *fun, WORD *args);
01544 extern int      RunCycle(PHEAD WORD *fun, WORD *args, WORD *info);
01545 extern int      RunAddArg(PHEAD WORD *fun, WORD *args);
01546 extern int      RunMulArg(PHEAD WORD *fun, WORD *args);
01547 extern int      RunIsLyndon(PHEAD WORD *fun, WORD *args, int par);
01548 extern WORD     RunToLyndon(PHEAD WORD *fun, WORD *args, int par);
01549 extern int      RunDropArg(PHEAD WORD *fun, WORD *args);
01550 extern int      RunSelectArg(PHEAD WORD *fun, WORD *args);
01551 extern int      RunDedup(PHEAD WORD *fun, WORD *args);
01552 extern int      RunZtoHArg(PHEAD WORD *fun, WORD *args);
01553 extern int      RunHtoZArg(PHEAD WORD *fun, WORD *args);
01554
01555 extern int      NormPolyTerm(PHEAD WORD *);
01556 extern WORD     ComparePoly(WORD *, WORD *, WORD);
01557 extern int      ConvertToPoly(PHEAD WORD *, WORD *,WORD *,WORD);
01558 extern int      LocalConvertToPoly(PHEAD WORD *, WORD *, WORD,WORD);
01559 extern int      ConvertFromPoly(PHEAD WORD *, WORD *, WORD, WORD, WORD, WORD);
01560 extern int      FindSubterm(WORD *);
01561 extern int      FindLocalSubterm(PHEAD WORD *, WORD);
01562 extern void     PrintSubtermList(int,int);
01563 extern void     PrintExtraSymbol(int,WORD *,int);
01564 extern int      FindSubexpression(WORD *);
01565
01566 extern void     UpdateMaxSize(void);
01567
01568 extern int      CoToPolynomial(UBYTE *);
01569 extern int      CoFromPolynomial(UBYTE *);
01570 extern int      CoArgToExtraSymbol(UBYTE *);
01571 extern int      CoExtraSymbols(UBYTE *);
01572 extern UBYTE    *GetDoParam(UBYTE *, WORD **, int);

```

```

01573 extern WORD *GetIfDollarFactor(UBYTE **, WORD *);
01574 extern int CoDo(UBYTE *);
01575 extern int CoEndDo(UBYTE *);
01576 extern int ExtraSymFun(PHEAD WORD *,WORD);
01577 extern int PruneExtraSymbols(WORD);
01578 extern int IniFbuffer(WORD);
01579 extern void IniFbufs(void);
01580 extern int GCDfunction(PHEAD WORD *,WORD);
01581 extern WORD *GCDfunction3(PHEAD WORD *,WORD *);
01582 extern int ReadPolyRatFun(PHEAD WORD *);
01583 extern int FromPolyRatFun(PHEAD WORD *, WORD **, WORD **);
01584 extern WORD *PolyDiv(PHEAD WORD *,WORD *,char *);
01585 extern void GCDclean(PHEAD WORD *, WORD *);
01586 extern WORD *TakeSymbolContent(PHEAD WORD *,WORD *);
01587 extern int GCDterms(PHEAD WORD *,WORD *,WORD *);
01588 extern WORD *PutExtraSymbols(PHEAD WORD *,WORD,int *);
01589 extern WORD *TakeExtraSymbols(PHEAD WORD *,WORD);
01590 extern WORD *MultiplyWithTerm(PHEAD WORD *, WORD *,WORD);
01591 extern WORD *TakeContent(PHEAD WORD *, WORD *);
01592 extern int MergeSymbolLists(PHEAD WORD *, WORD *, int);
01593 extern int MergeDotproductLists(PHEAD WORD *, WORD *, int);
01594 extern WORD *CreateExpression(PHEAD WORD);
01595 extern int DIVfunction(PHEAD WORD *,WORD,int);
01596 extern WORD *MULfunc(PHEAD WORD *, WORD *);
01597 extern WORD *ConvertArgument(PHEAD WORD *,int *);
01598 extern int ExpandRat(PHEAD WORD *);
01599 extern int InvPoly(PHEAD WORD *,WORD,WORD);
01600 extern WORD TestDoLoop(PHEAD WORD *,WORD);
01601 extern WORD TestEndDoLoop(PHEAD WORD *,WORD);
01602
01603 extern WORD *poly_gcd(PHEAD WORD *, WORD *, WORD);
01604 extern WORD *poly_div(PHEAD WORD *, WORD *, WORD);
01605 extern WORD *poly_rem(PHEAD WORD *, WORD *, WORD);
01606 extern WORD *poly_inverse(PHEAD WORD *, WORD *);
01607 extern WORD *poly_mul(PHEAD WORD *, WORD *);
01608 extern WORD *poly_ratfun_add(PHEAD WORD *, WORD *);
01609 extern int poly_ratfun_normalize(PHEAD WORD *);
01610 extern int poly_factorize_argument(PHEAD WORD *, WORD *);
01611 extern WORD *poly_factorize_dollar(PHEAD WORD *);
01612 extern int poly_factorize_expression(EXPRESSIONS);
01613 extern int poly_unfactorize_expression(EXPRESSIONS);
01614 extern void poly_free_poly_vars(PHEAD const char *);
01615
01616 #ifdef WITHFLINT
01617 extern void flint_final_cleanup_thread(void);
01618 extern void flint_final_cleanup_master(void);
01619 extern WORD* flint_div(PHEAD WORD *, WORD *, const WORD);
01620 extern int flint_factorize_argument(PHEAD WORD *, WORD *);
01621 extern WORD* flint_factorize_dollar(PHEAD WORD *);
01622 extern WORD* flint_gcd(PHEAD WORD *, WORD *, const WORD);
01623 extern WORD* flint_inverse(PHEAD WORD *, WORD *);
01624 extern WORD* flint_mul(PHEAD WORD *, WORD *);
01625 extern WORD* flint_ratfun_add(PHEAD WORD *, WORD *);
01626 extern int flint_ratfun_normalize(PHEAD WORD *);
01627 extern WORD* flint_rem(PHEAD WORD *, WORD *, const WORD);
01628 extern void flint_startup_init(void);
01629 #endif
01630
01631 extern void optimize_print_code (int);
01632
01633 #ifdef WITHPTHREADS
01634 extern void find_Horner_MCTS_expand_tree(void);
01635 extern void find_Horner_MCTS_expand_tree_threaded(void);
01636 extern void optimize_expression_given_Horner(void);
01637 extern void optimize_expression_given_Horner_threaded(void);
01638 #endif
01639
01640 extern int DoPreAdd(UBYTE *s);
01641 extern int DoPreUseDictionary(UBYTE *s);
01642 extern int DoPreCloseDictionary(UBYTE *s);
01643 extern int DoPreOpenDictionary(UBYTE *s);
01644 extern void RemoveDictionary(DICTIONARY *dict);
01645 extern void UnSetDictionary(void);
01646 extern int SetDictionaryOptions(UBYTE *options);
01647 extern int AddToDictionary(DICTIONARY *dict,UBYTE *left,UBYTE *right);
01648 extern int AddDictionary(UBYTE *name);
01649 extern int FindDictionary(UBYTE *name);
01650 extern UBYTE *IsExponentSign(void);
01651 extern UBYTE *IsMultiplySign(void);
01652 extern void TransformRational(UWORD *a, WORD na);
01653 extern void WriteDictionary(DICTIONARY *);
01654 extern void ShrinkDictionary(DICTIONARY *);
01655 extern void MultiplyToLine(void);
01656 extern UBYTE *FindSymbol(WORD num);
01657 extern UBYTE *FindVector(WORD num);
01658 extern UBYTE *FindIndex(WORD num);
01659 extern UBYTE *FindFunction(WORD num);

```

```

01660 extern UBYTE *FindFunWithArgs(WORD *t);
01661 extern UBYTE *FindExtraSymbol(WORD num);
01662 extern LONG DictToBytes(DICTIONARY *dict, UBYTE *buf);
01663 extern DICTIONARY *DictFromBytes(UBYTE *buf);
01664 extern int CoCreateSpectator(UBYTE *inp);
01665 extern int CoToSpectator(UBYTE *inp);
01666 extern int CoRemoveSpectator(UBYTE *inp);
01667 extern int CoEmptySpectator(UBYTE *inp);
01668 extern int CoCopySpectator(UBYTE *inp);
01669 extern int PutInSpectator(WORD *, WORD);
01670 extern void ClearSpectators(WORD);
01671 extern WORD GetFromSpectator(WORD *, WORD);
01672 extern void FlushSpectators(void);
01673
01674 extern WORD *PreGCD(PHEAD WORD *, WORD *, int);
01675 extern int InvPoly(PHEAD WORD *, WORD, WORD);
01676
01677 extern int ReadFromScratch(FILEHANDLE *, POSITION *, UBYTE *, POSITION *);
01678 extern int AddToScratch(FILEHANDLE *, POSITION *, UBYTE *, POSITION *, int);
01679
01680 extern int DoPreAppendPath(UBYTE *);
01681 extern int DoPrePrependPath(UBYTE *);
01682
01683 extern int DoSwitch(PHEAD WORD *, WORD *);
01684 extern int DoEndSwitch(PHEAD WORD *, WORD *);
01685 extern SWITCHTABLE *FindCase(WORD, WORD);
01686 extern void SwitchSplitMergeRec(SWITCHTABLE *, WORD, SWITCHTABLE *);
01687 extern void SwitchSplitMerge(SWITCHTABLE *, WORD);
01688 extern int DoubleSwitchBuffers(void);
01689
01690 extern int DistrN(int, int *, int, int *);
01691
01692 extern int DoNamespace(UBYTE *);
01693 extern int DoEndNamespace(UBYTE *);
01694 extern int DoUse(UBYTE *);
01695 extern UBYTE *SkipName(UBYTE *);
01696 extern UBYTE *ConstructName(UBYTE *, UBYTE);
01697 extern int DoSetUserFlag(UBYTE *);
01698 extern int DoClearUserFlag(UBYTE *);
01699
01700 VERTEX *CreateVertex(MODEL *);
01701 UBYTE *ReadParticle(UBYTE *, VERTEX *, MODEL *, int);
01702 int CoModel(UBYTE *);
01703 int CoParticle(UBYTE *);
01704 int CoVertex(UBYTE *);
01705 int CoEndModel(UBYTE *);
01706 int LoadModel(MODEL *);
01707
01708 int CoSetUserFlag(UBYTE *);
01709 int CoClearUserFlag(UBYTE *);
01710 int CoCreateAllLoops(UBYTE *);
01711 int CoCreateAllPaths(UBYTE *);
01712 int CoCreateAll(UBYTE *);
01713
01714 int AllLoops(PHEAD WORD *, WORD);
01715 LONG StartLoops(PHEAD WORD *, WORD, LONG, WORD, WORD *, WORD, WORD *, WORD);
01716 LONG GenLoops(PHEAD WORD *, WORD, LONG, WORD, WORD *, WORD, WORD *, WORD);
01717 void LoopOutput(PHEAD WORD *, WORD, WORD *, WORD);
01718 int AllPaths(PHEAD WORD *, WORD);
01719 LONG GenPaths(PHEAD WORD *, WORD, LONG, WORD, WORD *, WORD, WORD *, WORD);
01720 void PathOutput(PHEAD WORD *, WORD, WORD *, WORD);
01721
01722 #ifdef WITHFLOAT
01723 int DoStartFloat(UBYTE *);
01724 int DoEndFloat(UBYTE *);
01725 int SetFloatPrecision(WORD);
01726 void SetupMZVTables(void);
01727 void SetupMPFTTables(void);
01728 void ClearMZVTables(void);
01729 int EvaluateEuler(PHEAD WORD *, WORD, WORD);
01730 int EvaluateSqrt(PHEAD WORD *, WORD, WORD);
01731 int CoEvaluate(UBYTE *);
01732 int PrintFloat(WORD *fun, int numdigits);
01733 int CoToRat(UBYTE *);
01734 int CoToFloat(UBYTE *);
01735 int ToRat(PHEAD WORD *, WORD);
01736 int ToFloat(PHEAD WORD *, WORD);
01737 WORD FloatFunToRat(PHEAD UWORD *, WORD *);
01738 int AddWithFloat(PHEAD WORD **, WORD **);
01739 int MergeWithFloat(PHEAD WORD **, WORD **);
01740 void ClearfFloat(void);
01741 int TestFloat(WORD *);
01742 SBYTE *ReadFloat(SBYTE *);
01743 UBYTE *CheckFloat(UBYTE *, int *);
01744 void SetfFloatPrecision(LONG);
01745 int EvaluateFun(PHEAD WORD *, WORD, WORD *);
01746 #endif

```

```

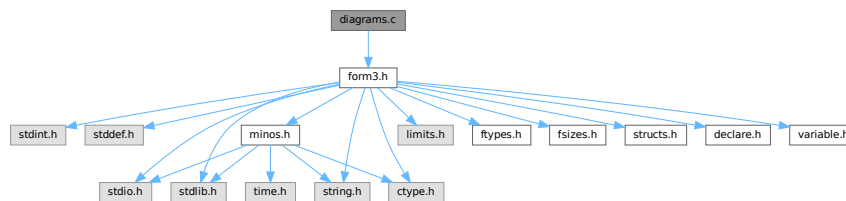
01747
01748 /*
01749     #] Declarations :
01750 */
01751 #endif

```

## 4.21 diagrams.c File Reference

```
#include "form3.h"
```

Include dependency graph for diagrams.c:



### Functions

- int [CoCanonicalize](#) (UBYTE \*s)
- int [DoCanonicalize](#) (PHEAD WORD \*term, WORD \*params)
- int [DoTopologyCanonicalize](#) (PHEAD WORD \*term, WORD vert, WORD edge, WORD \*args)
- int [DoShattering](#) (PHEAD WORD \*connect, WORD \*environ, WORD \*partitions, WORD nvert)

### 4.21.1 Detailed Description

Contains the wrapper routines for diagram manipulations.

Definition in file [diagrams.c](#).

### 4.21.2 Function Documentation

#### 4.21.2.1 CoCanonicalize()

```

int CoCanonicalize (
    UBYTE * s )

```

Definition at line 64 of file [diagrams.c](#).

#### 4.21.2.2 DoCanonicalize()

```

int DoCanonicalize (
    PHEAD WORD * term,
    WORD * params )

```

Definition at line 185 of file [diagrams.c](#).

### 4.21.2.3 DoTopologyCanonicalize()

```
int DoTopologyCanonicalize (
    PHEAD WORD * term,
    WORD vert,
    WORD edge,
    WORD * args )
```

Definition at line 417 of file [diagrams.c](#).

### 4.21.2.4 DoShattering()

```
int DoShattering (
    PHEAD WORD * connect,
    WORD * environ,
    WORD * partitions,
    WORD nvert )
```

Definition at line 626 of file [diagrams.c](#).

## 4.22 diagrams.c

[Go to the documentation of this file.](#)

```
00001
00005 /* #[] License : */
00006 /*
00007 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00008 * When using this file you are requested to refer to the publication
00009 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010 * This is considered a matter of courtesy as the development was paid
00011 * for by FOM the Dutch physics granting agency and we would like to
00012 * be able to track its scientific use to convince FOM of its value
00013 * for the community.
00014 *
00015 * This file is part of FORM.
00016 *
00017 * FORM is free software: you can redistribute it and/or modify it under the
00018 * terms of the GNU General Public License as published by the Free Software
00019 * Foundation, either version 3 of the License, or (at your option) any later
00020 * version.
00021 *
00022 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00023 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00024 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00025 * details.
00026 *
00027 * You should have received a copy of the GNU General Public License along
00028 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00029 */
00030 /* #[] License : */
00031 /*
00032 * #[] Includes : diagrams.c
00033 */
00034 #include "form3.h"
00035
00036 static WORD one = 1;
00037
00038 /*
00039 * #[] Includes :
00040 * #[] CoCanonicalize :
00041
00042 Syntax:
00043 Canonicalize,mainoption,...
00044 With the main options currently:
00045 Canonicalize,topology,vertexfunction,edgefunction,OutDollar,extraoptions;
00046 Canonicalize,polynomial,InDollar,set_or_setname_or_dollar,OutDollar,extraoptions;
00047 The vertex function needs to have the format (assume it is called vx):
00048 vx(p1,p2,-p3) or vx(-p1,p2,p3,-p4) etc.
00049
```

```

00050     The external lines have a vertex with only one line.
00051     All momenta that form connections should be unique.
00052     In principle the - signs are not relevant for the topology, but they
00053     may exist already in the remaining part of the diagram. They are also
00054     part of the canonical form.
00055     The edge function can be used in different ways, depending on the options.
00056     The extraoption(s) should be nonnegative integers or $variables that evaluate
00057     into nonnegative integers (integers less than 2^31).
00058     The Indollar variable contains the polynomial to be canonicalized.
00059     The OutDollar variable should be the name of a $-variable (as in $out) which
00060     will be filled with a replace_ function. The canonicalization can then
00061     be executed in the whole term with the    Multiply $out;    command.
00062 */
00063
00064 int CoCanonicalize(UBYTE *s)
00065 {
00066     WORD args[10], *a, num;
00067     UBYTE *t, c;
00068     args[0] = TYPECANONICALIZE;
00069     a = args+2;
00070     while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
00071     t = s; while ( FG.cTable[*s] == 0 ) s++;
00072     c = *s; *s = 0;
00073     if ( StrICmp(t,(UBYTE *)("topology")) == 0 ) {
00074         *s = c; *a++ = 0;
00075         while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
00076         s = GetFunction(s,a);
00077         while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
00078         s = GetFunction(s,a+1);
00079         if ( *a == 0 || a[1] == 0 ) return(1);
00080         a += 2;
00081     }
00082     else if ( StrICmp(t,(UBYTE *)("polynomial")) == 0 ) {
00083         *s = c; *a++ = 1;
00084         while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
00085     /*
00086         Now get the name of the input $-variable
00087     */
00088         if ( *s != '$' ) {
00089             MesPrint("&Canonicalize statement needs a $-variable for its input.");
00090             return(1);
00091         }
00092         s++; t = s; while ( FG.cTable[*s] < 2 ) s++;
00093         c = *s; *s = 0;
00094         if ( GetName(AC.dollarnames,t,&num,NOAUTO) == CDOLLAR ) *a++ = num;
00095         else { *a++ = AddDollar(t,DOLINDEX,&one,1); }
00096         *s = c;
00097     /*
00098         Now the set
00099     */
00100         if ( *s == '{' ) {
00101             t = s+1; SKIPBRA2(s)
00102             c = *s; *s = 0;
00103             *a++ = DoTempSet(t,s);
00104             *s++ = c;
00105         }
00106         else if ( FG.cTable[*s] == 0 || *s == '[' ) {
00107             t = s;
00108             if ( ( s = SkipAName(s) ) == 0 ) {
00109                 MesPrint("&Illegal name for set in Canonicalize statement: %s",t);
00110                 return(1);
00111             }
00112             c = *s; *s = 0;
00113             if ( GetName(AC.varnames,t,a,WITHAUTO) == CSET ) {
00114                 if ( Sets[*a].type != CSYMBOL ) {
00115                     MesPrint("&In Canonicalize: %s is not a set of symbols.",t);
00116                     return(1);
00117                 }
00118             }
00119             else {
00120                 MesPrint("&In Canonicalize: %s is not a set.",t);
00121                 return(1);
00122             }
00123             *s = c; a++;
00124         }
00125         else if ( *s == '$' ) {
00126             s++; t = s; while ( FG.cTable[*s] < 2 ) s++;
00127             c = *s; *s = 0;
00128             if ( GetName(AC.dollarnames,t,&num,NOAUTO) == CDOLLAR ) *a++ = -num-2;
00129             else {
00130                 MesPrint("&In Canonicalize: %s is undefined.",t-1);
00131                 return(1);
00132             }
00133             *s = c;
00134         }
00135         else {
00136             MesPrint("&In Canonicalize: Illegal third(=set) argument.");

```



```

00137         return(1);
00138     }
00139 }
00140 else {
00141     MesPrint("&Unrecognized option in Canonicalize statement: %s",t);
00142     return(1);
00143 }
00144 /*
00145 Now get the name of the output $-variable
00146 */
00147 while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
00148 if ( *s != '$' ) {
00149     MesPrint("&Canonicalize statement needs a $-variable for its output.");
00150     return(1);
00151 }
00152 s++; t = s; while ( FG.cTable[*s] < 2 ) s++;
00153 c = *s; *s = 0;
00154 if ( GetName(AC.dollarnames,t,&num,NOAUTO) == CDOLLAR ) *a++ = num;
00155 else { *a++ = AddDollar(t,DOLINDEX,&one,1); }
00156 *s = c;
00157 /*
00158 Now the options. At the moment we just do one of them.
00159 (the first extra option is relevant to determine the use of the edge function)
00160 In the future we may have to be more flexible.
00161 */
00162 a[0] = 0; /* default value */
00163 while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
00164 if ( *s != 0 ) {
00165     s = GetNumber(s,a);
00166     if ( *a == -1 ) return(1);
00167     while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
00168     a++;
00169 }
00170 /*
00171 Now complete the args string and put it in the compiler buffer
00172 */
00173 args[1] = a-args;
00174 AddNtoL(args[1],args);
00175 return(0);
00176 }
00177 /*
00178 #] CoCanonicalize :
00179 #[ DoCanonicalize :
00180
00181 Does the canonicalization. The output term overwrites the input term.
00182 */
00183 int DoCanonicalize(PHEAD WORD *term, WORD *params)
00184 {
00185     WORD args[10];
00186     int i;
00187     /*
00188 First check whether we need to expand dollars;
00189 */
00190 for ( i = 0; i < params[1]; i++ ) args[i] = params[i];
00191 if ( args[2] == 0 ) { /* topology */
00192     for ( i = 3; i < 5; i++ ) {
00193         if ( args[i] < 0 ) { /* This is a dollar */
00194             args[i] = DolToFunction(BHEAD -args[i]-2);
00195             if ( args[i] == 0 ) {
00196                 MLOCK(ErrorMessageLock);
00197                 MesPrint("Value of $-variable in Canonicalize statement should be a function.");
00198                 MUNLOCK(ErrorMessageLock);
00199                 Terminate(-1);
00200             }
00201         }
00202     }
00203 }
00204 for ( i = 6; i < args[1]; i++ ) { /* Extra options */
00205     if ( args[i] < 0 ) { /* This is a dollar */
00206         args[i] = DolToNumber(BHEAD -args[i]-2);
00207         if ( args[i] < 0 ) {
00208             MLOCK(ErrorMessageLock);
00209             MesPrint("Value of $-variable in Canonicalize statement should be a nonnegative
00210 number < %l.",(LONG)MAXPOSITIVE);
00211             MUNLOCK(ErrorMessageLock);
00212             Terminate(-1);
00213         }
00214     }
00215 }
00216 switch ( args[6] ) {
00217     case 1: { /* pass the vertex and edge functions. */
00218         WORD *tstop, *t, *tedge, *te;
00219         tstop = term + *term; tstop -= ABS(tstop[-1]);
00220         t = term+1;
00221         tedge = AT.WorkPointer; te = tedge+1;
00222         while ( t < tstop ) {

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```

00223         if ( *t != args[3] && *t != args[4] ) { t += t[1]; continue; }
00224         for ( i = 0; i < t[1]; i++ ) te[i] = t[i];
00225         te += t[1]; t += t[1];
00226     }
00227     *te++ = 1; *te++ = 1; *te++ = 3;
00228     tedge[0] = te - tedge;
00229     AT.WorkPointer = te;
00230 /*
00231     DoVertexCanonicalize(BHEAD term, tedge, args[3], args[4], args[5], args[6]);
00232 */
00233     AT.WorkPointer = tedge;
00234 } break;
00235 case 2: { /* pass the edge functions only */
00236     WORD *tstop, *t, *tedge, *te;
00237     tstop = term + *term; tstop -= ABS(tstop[-1]);
00238     t = term + 1;
00239     tedge = AT.WorkPointer; te = tedge + 1;
00240     while ( t < tstop ) {
00241         if ( *t != args[4] ) { t += t[1]; continue; }
00242         for ( i = 0; i < t[1]; i++ ) te[i] = t[i];
00243         te += t[1]; t += t[1];
00244     }
00245     *te++ = 1; *te++ = 1; *te++ = 3;
00246     tedge[0] = te - tedge;
00247     AT.WorkPointer = te;
00248 /*
00249     DoEdgeCanonicalize(BHEAD term, tedge, args[5]);
00250 */
00251     AT.WorkPointer = tedge;
00252 } break;
00253 default: {
00254     DoTopologyCanonicalize(BHEAD term, args[3], args[4], args+5);
00255 } break;
00256 }
00257 /*
00258     Call here the topology canonicalization
00259     We will have the arguments:
00260     args[3]: The function used as vertex function
00261     args[4]: The function used as edge function
00262     args[5]: The number of the dollar to be used for the output
00263     args[6]: Potentially other options (like saying how to use args[4]).
00264     term:    The term in which the topology resides.
00265 */
00266 }
00267 else if ( args[2] == 1 ) { /* polynomial */
00268     WORD *symlist, *nsymlist;
00269     for ( i = 6; i < args[1]; i++ ) { /* Extra options */
00270         if ( args[i] < 0 ) { /* This is a dollar */
00271             args[i] = DolToNumber(BHEAD -args[i]-2);
00272             if ( args[i] < 0 ) {
00273                 MLOCK(ErrorMessageLock);
00274                 MesPrint("Value of $-variable in Canonicalize statement should be a nonnegative
00275 number < %1.", (LONG)MAXPOSITIVE);
00276                 MUNLOCK(ErrorMessageLock);
00277                 Terminate(-1);
00278             }
00279         }
00280     }
00281 /*
00282     Now we sort out the set. We create a pointer to the list of set
00283     elements, and we determine the number of elements in the set.
00284 */
00285     symlist = AT.WorkPointer;
00286     if ( args[4] < -1 ) { /* Dollar that should expand into a list of symbols */
00287         DOLLARS d = Dollars - args[4] - 2;
00288         WORD *ds, *insym;
00289         if ( d->type != DOLWILDARGS ) {
00290 NoWildArg:
00291             MLOCK(ErrorMessageLock);
00292             MesPrint("Value of $-variable in Canonicalize statement should be a argument wildcard
00293 of symbol arguments.");
00294             MUNLOCK(ErrorMessageLock);
00295             Terminate(-1);
00296         }
00297         insym = symlist; ds = d->where+1;
00298         while ( *ds ) {
00299             if ( *ds != -SYMBOL ) goto NoWildArg;
00300             *insym++ = ds[1];
00301             ds += 2;
00302         }
00303         nsymlist = insym - symlist;
00304     } else { /*if ( args[4] >= 0 ) */
00305         WORD *ss, *sy, n;
00306         ss = (WORD *) (AC.SetElementList.lijst) + Sets[args[4]].first;
00307         nsymlist = n = Sets[args[4]].last - Sets[args[4]].first;

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```

00308         sy = symlist = AT.WorkPointer;
00309         NCOPY(sy,ss,n);
00310     }
00311     AT.WorkPointer = symlist+nsymlist;
00312  /*
00313     Call here the polynomial canonicalization
00314     We will have the arguments:
00315     args[3]: The number of the dollar to be used for the input
00316     symlist: an array of symbols
00317     nsymlist: the number of symbols in symlist
00318     args[5]: The number of the dollar to be used for the output
00319     args[6]: Potentially other options.
00320  */
00321  /*
00322     DoPolynomialCanonicalize(BHEAD args[3],symlist,nsymlist,args[5],args[6]);
00323  */
00324     AT.WorkPointer = symlist;
00325 }
00326 return(0);
00327 }
00328
00329  /*
00330  #] DoCanonicalize :
00331  #[ GenTopologies :
00332
00333     This function has the syntax
00334         topologies_(nloops,      Number of loops
00335         nlegs,                    Number of legs
00336         setvertexsizes,          A set which tells which vertices are allowed like {3,4}.
00337         set_extmomenta,          The name of a set with the external momenta
00338         set_intmomenta           The name of a set with the internal momenta
00339         [,options])
00340     The output will be using the built in functions vertex_ and edge_.
00341
00342     The test for whether this function can be evaluated is in TestSub (inside
00343     file proces.c) (search for the string TOPOLOGIES).
00344     This passes the code -15 in AN.TeInFun to Generator, which then calls
00345     the GenTopologies routine.
00346  */
00347
00348  #ifdef OLDTPO
00349
00350  int GenTopologies(PHEAD WORD *term,WORD level)
00351  {
00352      WORD *t1, *ttl, *tstop, *t, *tt;
00353      WORD *oldworkpointer = AT.WorkPointer;
00354      WORD option1 = 0, option2 = 0, setoption = -1;
00355      int retval;
00356  /*
00357
00358     We have to go through the testing procedure again, because there could
00359     be more than one topologies_ function and not all have to be expandable.
00360  */
00361      tstop = term+*term; tstop -= ABS(tstop[-1]);
00362      tt = term+1;
00363      while ( tt < tstop ) {
00364          t = tt; tt = t+t[1];
00365          if ( *t != TOPOLOGIES ) continue;
00366          tt = t + t[1]; t1 = t + FUNHEAD;
00367          if ( t1+10 > tt || *t1 != -SNUMBER || t1[1] < 0 ||          /* loops */
00368              t1[2] != -SNUMBER || ( t1[3] < 0 && t1[3] != -2 ) || /* legs */
00369              t1[4] != -SETSET || Sets[t1[5]].type != CNUMBER ||    /* set vertices */
00370              t1[6] != -SETSET || Sets[t1[7]].type != CVECTOR ||    /* outvectors */
00371              t1[8] != -SETSET || Sets[t1[9]].type != CVECTOR ) continue;
00372          ttl = t1 + 10;
00373          if ( ttl+2 <= tt && ttl[0] == -SETSET ) {
00374              if ( Sets[t1[5]].last-Sets[t1[5]].first !=
00375                  Sets[ttl[1]].last-Sets[ttl[1]].first ) continue;
00376              setoption = ttl[1]; ttl += 2;
00377          }
00378          if ( ttl+2 <= tt && ttl[0] == -SNUMBER ) { option1 = ttl[1]; ttl += 2; }
00379          if ( ttl+2 <= tt && ttl[0] == -SNUMBER ) { option2 = ttl[1]; ttl += 2; }
00380          AT.setinterntopo = t1[9];
00381          AT.setexterntopo = t1[7];
00382          AT.TopologiesTerm = term;
00383          AT.TopologiesStart = t;
00384          AT.TopologiesLevel = level;
00385          AT.TopologiesOptions[0] = option1;
00386          AT.TopologiesOptions[1] = option2;
00387          retval = GenerateTopologies(BHEAD t1[1],t1[3],t1[5],setoption);
00388          AT.WorkPointer = oldworkpointer;
00389          return(retval);
00390      }
00391      MLOCK(ErrorMessageLock);
00392      MesPrint("Internal error: topologies_ function not encountered.");
00393      MUNLOCK(ErrorMessageLock);
00394      return(-1);

```

```

00395
00396 }
00397
00398 #endif
00399
00400 /*
00401  #] GenTopologies :
00402  #[ DoTopologyCanonicalize :
00403
00404  term: The term
00405  vert: the vertex function
00406  edge: the edge function
00407  args[0]: the number of the output dollar
00408  args[1]: options
00409  return value should be zero if all is correct.
00410
00411  The external lines connect to an 'external vertex' which has only one line.
00412  The vertices are of a type: vertex_(p1,p2,-p3)*vertex_(p3,p4,p5) etc.
00413  The edges indicate noninteger powers of the lines:
00414  edge_(p1,2) means 1/p1.p1^(2*ep)
00415 */
00416
00417 int DoTopologyCanonicalize(PHEAD WORD *term,WORD vert,WORD edge,WORD *args)
00418 {
00419     int nvert = 0, nvert2, i, ii, jj, flipnames = 0, nparts/*, level, num */;
00420     WORD *tstop, *t, *tt, *tend, *td;
00421     WORD *oldworkpointer = AT.WorkPointer;
00422     WORD *termcopy = TermMalloc("TopologyCanonize1");
00423     WORD *vet= TermMalloc("TopologyCanonize2");
00424     WORD *partition, *environ, *connect, *pparts, *p;
00425 /*
00426     WORD *pparts;
00427 */
00428     WORD momenta[150],flips[50],nmomenta = 0, nflips = 0;
00429 /*
00430     Step one: the vertices should get a number. We copy the term for this.
00431     We need a high number for the vertex function to make sure
00432     that it comes after the edge function in the sorting.
00433 */
00434     if ( args[0] < args[1] ) { flipnames = 1; }
00435     tend = term + *term; tend -= ABS(tend[-1]); t = term+1; tt = termcopy+1;
00436     while ( t < tend ) {
00437         if ( *t == vert ) {
00438             for ( i = FUNHEAD; i < t[1]; i += 2 ) {
00439                 if ( t[i] == -VECTOR || ( t[i] == -INDEX && t[i+1] < 0 ) ) {
00440                     momenta[nmomenta++] = -VECTOR;
00441                     momenta[nmomenta++] = t[i+1];
00442                 }
00443                 else if ( t[i] == -MINVECTOR ) {
00444                     momenta[nmomenta++] = -MINVECTOR;
00445                     momenta[nmomenta++] = t[i+1];
00446                 }
00447                 else goto notgoodvertex;
00448                 momenta[nmomenta++] = nvert;
00449             }
00450             ii = FUNHEAD; i = t[1]-FUNHEAD;
00451             NCOPY(tt,t,ii)
00452             if ( flipnames ) tt[-FUNHEAD] = edge;
00453             tt[-FUNHEAD+1] += 2;
00454             *tt++ = -CNUMBER; *tt++ = nvert++;
00455         }
00456         else if ( *t == edge && flipnames ) {
00457             i = t[1] - 1; *tt++ = vert; t++;
00458         }
00459         else {
00460 notgoodvertex:
00461             i = t[1];
00462         }
00463         NCOPY(tt,t,i)
00464     }
00465     while ( t < tend ) *tt++ = *t++;
00466     termcopy[0] = tt - termcopy;
00467     if ( flipnames ) EXCH(edge,vert)
00468     nvert2 = nvert*nvert;
00469 /*
00470     Sort the momenta. Keep the sign order.
00471 */
00472     for ( i = 0; i < nmomenta-3; i+=3 ) {
00473         jj = i;
00474         while ( jj >= 0 && momenta[jj+4] > momenta[jj+1] ) {
00475             EXCH(momenta[jj+5],momenta[jj+2])
00476             EXCH(momenta[jj+4],momenta[jj+1])
00477             EXCH(momenta[jj+3],momenta[jj])
00478             jj -= 3;
00479         }
00480     }
00481 /*

```

```

00482      Step two: make now the edge functions in the proper notation.
00483  */
00484      t = vet+1;
00485      for ( i = 0; i < nmomenta; i += 6 ) {
00486          if ( momenta[i] == -VECTOR && momenta[i+3] == -MINVECTOR
00487              && momenta[i+1] == momenta[i+4] ) {
00488              }
00489          else if ( momenta[i] == -MINVECTOR && momenta[i+3] == -VECTOR
00490                  && momenta[i+1] == momenta[i+4] ) {
00491              flips[nflips++] = momenta[i+1];
00492              DUMMYUSE(flips[nflips-1]);
00493          }
00494          else { /* something wrong with the momenta */
00495              MLOCK(ErrorMessageLock);
00496              MesPrint("No momentum conservation or wrong momenta in Canonicalize statement");
00497              MUNLOCK(ErrorMessageLock);
00498              Terminate(-1);
00499          }
00500          *t++ = EDGE; *t++ = FUNHEAD+10; FILLFUN(t)
00501          *t++ = -SNUMBER; *t++ = momenta[i+2];
00502          *t++ = -SNUMBER; *t++ = momenta[i+5];
00503          *t++ = -VECTOR; *t++ = momenta[i+1];
00504          *t++ = -SNUMBER; *t++ = 0; /* provisional power/color, multiple of ep */
00505          *t++ = -SNUMBER; *t++ = 0; /* provisional power/color, integer */
00506      }
00507      tend = t;
00508      *t++ = 1; *t++ = 1; *t++ = 3; vet[0] = t-vet; *t = 0;
00509  /*
00510      Now the powers of the denominators
00511  */
00512      tstop = termcopy+termcopy; tstop -= ABS(tstop[-1]); td = termcopy+1;
00513      while ( td < tstop ) {
00514          if ( *td == edge && td[1] == FUNHEAD+4 ) {
00515              if ( td[FUNHEAD+2] == -SNUMBER && ( td[FUNHEAD] == -VECTOR || td[FUNHEAD] == -INDEX
00516                  || td[FUNHEAD] == -MINVECTOR ) ) {}
00517              else {
00518                  MLOCK(ErrorMessageLock);
00519                  MesPrint("Illegal argument in edge function in Canonicalize statement");
00520                  MUNLOCK(ErrorMessageLock);
00521                  Terminate(-1);
00522              }
00523              tt = vet+1;
00524              while ( tt < tend ) {
00525                  if ( tt[FUNHEAD+5] == td[FUNHEAD+1] ) { tt[FUNHEAD+7] = td[FUNHEAD+3]; break; }
00526                  tt += tt[1];
00527              }
00528          }
00529          else if ( *td == DOTPRODUCT ) break;
00530          td += td[1];
00531      }
00532      if ( td < tstop ) {
00533          tt = vet+1;
00534          while ( tt < tend ) {
00535              /*
00536                  tt[FUNHEAD+5] is a vector. We look for dotproducts with twice tt[FUNHEAD+5]
00537              */
00538              for ( i = 2; i < td[1]; i += 3 ) {
00539                  if ( td[i] == tt[FUNHEAD+5] && td[i+1] == tt[FUNHEAD+5] ) {
00540                      tt[FUNHEAD+9] = td[i+2];
00541                      break;
00542                  }
00543              }
00544              tt += tt[1];
00545          }
00546      }
00547      Normalize(BHEAD vet);
00548  /*
00549      Now we have a term `vet' in the proper notation and we can start.
00550      To keep track of the shattering we use an array of 2*nvert.
00551      each entry is Number,marker
00552      When the marker is zero, the vertices are in the same partition.
00553      For the environment we need a matrix that is nvert x nvert
00554      At the same time we keep the connectivity matrix, because that will
00555      save much time later.
00556      The partitions are stored in a matrix as well. This allows them to be
00557      treated as a stack. The entries are separated by 0 if they belong to
00558      the same part, and by a 1 when they belong to different parts.
00559  */
00560      partition = AT.WorkPointer; AT.WorkPointer += 2*nvert2;
00561      for ( i = 0; i < nvert; i++ ) { partition[2*i] = i; partition[2*i+1] = 0; }
00562      partition[2*i-1] = -1; /* end of the first part which is currently all vertices */
00563      nparts = 1;
00564
00565      connect = AT.WorkPointer; AT.WorkPointer += nvert2;
00566      for ( i = 0; i < nvert2; i++ ) connect[i] = 0;
00567      tstop = vet+vet; tstop -= ABS(tstop[-1]); t = vet+1;
00568      while ( t < tstop ) {

```

```

00569         if ( *t == EDGE ) {
00570             connect[t[FUNHEAD+1]*nvert+t[FUNHEAD+3]]++;
00571             connect[t[FUNHEAD+3]*nvert+t[FUNHEAD+1]]++;
00572         }
00573         t += t[1];
00574     }
00575     for ( i = 0; i < nvert; i++ ) {
00576         MesPrint("connectivity: %d -- %a",i,nvert,connect+i*nvert);
00577     }
00578     /*
00579     Create the environment matrix and sort it.
00580     */
00581     environ = AT.WorkPointer; AT.WorkPointer += nvert2;
00582     /*
00583     And now the refinement process starts.
00584     */
00585     WantAddPointers(nvert+1);
00586     for ( i = 0; i < nvert2; i++ ) environ[i] = 0;
00587     /* level = 0; */
00588     pparts = partition;
00589     while ( nparts < nvert ) {
00590         nparts = DoShattering(BHEAD connect,environ,pparts,nvert);
00591         if ( nparts < nvert ) { /* raise level and make a copy and split a part */
00592             p = pparts + 2*nvert;
00593             /* level++; */
00594             for ( i = 0; i < 2*nvert; i++ ) p[i] = pparts[i];
00595             for ( ii = 0; ii < 2*nvert; ii += 2 ) {
00596                 if ( p[ii+1] == 0 ) { /* found a part with more than one */
00597                     /* num = 2; */ i = ii+2;
00598                     while ( p[ii+1] == 0 ) { /* num++; */ i += 2; }
00599                     p[ii+1] = -1; pparts = p;
00600                     break;
00601                 }
00602             }
00603         }
00604     }
00605 }
00606 MesPrint("partition: %d -- %a",nparts,2*nvert,pparts);
00607
00608 }
00609 /*
00610 Just for now
00611 */
00612 PutTermInDollar(vet,args[0]);
00613
00614
00615 TermFree(vet,"TopologyCanonize2");
00616 TermFree(termcopy,"TopologyCanonize1");
00617 AT.WorkPointer = oldworkpointer;
00618 return(0);
00619 }
00620
00621 /*
00622 #] DoTopologyCanonicalize :
00623 #[ DoShattering :
00624 */
00625
00626 int DoShattering(PHEAD WORD *connect, WORD *environ, WORD *partitions, WORD nvert)
00627 {
00628     int nparts, i, j, ii, jj, iii, jjj, newmarker;
00629     WORD **p = AT.pWorkSpace + AT.pWorkPointer, *part, *endpart;
00630     WORD *poin1, *poin2;
00631     #ifdef SHATBUG
00632     MesPrint("Entering DoShattering. partitions = %a",2*nvert,partitions);
00633     #endif
00634     restart:
00635     /*
00636     Determine the number of parts
00637     p will be an array with pointers to the parts.
00638     We made space for this array in the calling routine and because this
00639     routine is not calling any other routines we do not need to raise
00640     the pointer in this stack (AT.pWorkPointer).
00641     */
00642     nparts = 0; newmarker = 0;
00643     part = partitions; endpart = part + 2*nvert;
00644     p[0] = part;
00645     while ( part < endpart ) {
00646         if ( part[1] != 0 ) { p[++nparts] = part+2; }
00647         part += 2;
00648     }
00649     for ( i = 0; i < nparts; i++ )
00650         AT.WorkPointer[i] = (p[i+1]-p[i])/2;
00651     #ifdef SHATBUG
00652     MesPrint("DoShattering: calculated the pointers");
00653     MesPrint("DoShattering: sizes: %a",nparts,AT.WorkPointer);
00654     MesPrint("DoShattering: p[0]: %a, p[1]: %a",6,p[0],6,p[1]);
00655     #endif

```

```

00656     for ( i = 0; i < nparts; i++ ) {
00657         if ( AT.WorkPointer[i] > 1 ) {
00658             for ( j = 0; j < nparts; j++ ) {
00659                 /*
00660                  Shatter part i wrt part j.
00661                  if there is action, go to restart.
00662                 */
00663                 for ( ii = 0; ii < AT.WorkPointer[i]; ii++ ) {
00664                     for ( jj = 0; jj < AT.WorkPointer[j]; jj++ ) {
00665                         environ[ii*AT.WorkPointer[j]+jj] += connect[p[i][2*ii]*nvert+p[j][2*jj]];
00666                     }
00667                 }
00668 #ifdef SHATBUG
00669                 for ( ii = 0; ii < AT.WorkPointer[i]; ii++ ) {
00670                     MesPrint("Environ(%d,%d): %a",i,j,AT.WorkPointer[j],environ+ii*AT.WorkPointer[j]);
00671                 }
00672 #endif
00673                 /*
00674                  Sort the rows internally, then sort the rows wrt each other
00675                  and finally place new markers. If a new marker, we restart.
00676                  Don't forget to clean up the environ array.
00677                 */
00678                 for ( ii = 0; ii < nvert; ii++ ) {
00679                     poin1 = environ+ii*AT.WorkPointer[j];
00680                     for ( jj = 0; jj < AT.WorkPointer[j]-1; jj++ ) {
00681                         jjj = jj;
00682                         while ( jjj >= 0 && poin1[jjj+1] > poin1[jjj] ) {
00683                             EXCH(poin1[jjj+1],poin1[jjj])
00684                             jjj--;
00685                         }
00686                     }
00687                 }
00688 #ifdef SHATBUG
00689                 for ( ii = 0; ii < AT.WorkPointer[i]; ii++ ) {
00690                     MesPrint("environ(%d,%d): %a",i,j,AT.WorkPointer[j],environ+ii*AT.WorkPointer[j]);
00691                 }
00692 #endif
00693                 for ( ii = 0; ii < AT.WorkPointer[i]-1; ii++ ) {
00694                     poin2 = environ+ii*AT.WorkPointer[j]; poin1 = poin2+AT.WorkPointer[j];
00695                     iii = ii;
00696                     while ( iii >= 0 && ( CmpArray(poin1,poin2,AT.WorkPointer[j]) < 0 ) ) {
00697                         EXCHN(poin2,poin1,AT.WorkPointer[j])
00698                         EXCH(p[i][2*iii+2],p[i][2*iii])
00699                         iii--; poin1 = poin2; poin2 = poin1-AT.WorkPointer[j];
00700                     }
00701                 }
00702 #ifdef SHATBUG
00703                 for ( ii = 0; ii < AT.WorkPointer[i]; ii++ ) {
00704                     MesPrint("environ(%d,%d): %a",i,j,AT.WorkPointer[j],environ+ii*AT.WorkPointer[j]);
00705                 }
00706                 MesPrint("partitions = %a",2*nvert,partitions);
00707 #endif
00708                 for ( ii = 0; ii < AT.WorkPointer[i]-1; ii++ ) {
00709                     poin2 = environ+ii*AT.WorkPointer[j]; poin1 = poin2+AT.WorkPointer[j];
00710                     if ( CmpArray(poin1,poin2,AT.WorkPointer[j]) == 0 ) continue;
00711                     p[i][2*ii+1] = -1; nparts++; newmarker++;
00712                 }
00713 #ifdef SHATBUG
00714                 MesPrint("partitions = %a",2*nvert,partitions);
00715 #endif
00716                 /*
00717                  Clear environ. This is probably faster than just clearing the whole array.
00718                  Maybe in the future a test could be done on nvert to decide how to clear.
00719                 */
00720                 for ( ii = 0; ii < AT.WorkPointer[i]; ii++ ) {
00721                     for ( jj = 0; jj < AT.WorkPointer[j]; jj++ ) {
00722                         environ[ii*AT.WorkPointer[j]+jj] = 0;
00723                     }
00724                 }
00725                 if ( newmarker ) { goto restart; }
00726             }
00727         }
00728     }
00729     return(nparts);
00730 }
00731
00732 /*
00733 #] DoShattering :
00734 */

```

## 4.23 diawrap.cc

```
00001 // #[ Includes : diawrap.cc
```

```

00002
00003 extern "C" {
00004 #include "form3.h"
00005 }
00006
00007 #include "grccparam.h"
00008 #include "grcc.h"
00009
00010 #define MAXPOINTS 120
00011
00012 typedef struct ToPoTyPe {
00013     WORD *vert;
00014     WORD *vertmax;
00015     Options *opt;
00016     int cldeg[MAXPOINTS], clnum[MAXPOINTS], clex[1][MAXPOINTS];
00017     int cmin[1][MAXLEGS+1], cmax[1][MAXLEGS+1];
00018     int ncl, nvert;
00019 } TOPOTYPE;
00020
00021 static void ProcessDiagram(EGraph *eg, void *ti);
00022 static int processVertex(TOPOTYPE *TopoInf, int pointsremaining, int level);
00023
00024 // #] Includes :
00025 // #[ LoadModel :
00026
00027 int LoadModel(MODEL *m)
00028 {
00029 //
00030 // First check whether there was a model already.
00031 // In the Kaneko setup there can be only one at the same time.
00032 // Hence if there was one we need to remove it first.
00033 // Unless of course, it was already the model we want.
00034 //
00035 if ( m->grccmodel != NULL ) return(0);
00036
00037 int i,j,k;
00038 //
00039 // First the information that goes into the Model struct.
00040 // Note that new Model takes over (does not copy) minp.cnamlist.
00041 //
00042 MInput minp;
00043 PInput pinp;
00044 IInput iinp;
00045 if ( m->ncouplings > GRCC_MAXNCPLG ) {
00046     MesPrint("Too many coupling constants in model. Current limit is %d.",(WORD)GRCC_MAXNCPLG);
00047     MesPrint("Suggestion: recompile Form with a larger value for GRCC_MAXNCPLG");
00048     return(-1);
00049 }
00050 minp.defpart = GRCC_DEFBYCODE;
00051 minp.name = (char *) (m->name);
00052 minp.ncouple = m->ncouplings;
00053 for ( i = 0; i < GRCC_MAXNCPLG; i++ ) minp.cnamlist[i] = NULL;
00054 for ( i = 0; i < minp.ncouple; i++ )
00055     minp.cnamlist[i] = (char *) (VARNAME(symbols,m->couplings[i]));
00056 Model *mdl = new Model(&minp);
00057 //
00058 // Now the particles
00059 //
00060 for ( i = 0; i < m->nparticles; i++ ) {
00061     if ( minp.defpart == GRCC_DEFBYCODE ) {
00062         pinp.name = NULL;
00063         pinp.aname = NULL;
00064         pinp.pcode = m->vertices[i]->particles[0].number;
00065         pinp.acode = m->vertices[i]->particles[1].number;
00066         switch ( m->vertices[i]->particles[0].spin ) {
00067             case 1:
00068                 pinp.ptypec = GRCC_PT_Scalar;
00069                 break;
00070             case -1:
00071                 pinp.ptypec = GRCC_PT_Ghost;
00072                 break;
00073             case 2:
00074                 if ( m->vertices[i]->particles[0].type == 0 )
00075                     pinp.ptypec = GRCC_PT_Majorana;
00076                 else
00077                     pinp.ptypec = GRCC_PT_Undef;
00078                 break;
00079             case -2:
00080                 if ( m->vertices[i]->particles[0].type == 0 )
00081                     pinp.ptypec = GRCC_PT_Majorana;
00082                 else
00083                     pinp.ptypec = GRCC_PT_Dirac;
00084                 break;
00085             case 3:
00086                 pinp.ptypec = GRCC_PT_Vector;
00087                 break;
00088             default:

```



```

00089         pinp.ptypec = GRCC_PT_Undef;
00090         break;
00091     }
00092 }
00093 else {
00094     pinp.name = (char *) (VARNAME(functions,m->vertices[i]->particles[0].number));
00095     pinp.aname = (char *) (VARNAME(functions,m->vertices[i]->particles[1].number));
00096     switch ( m->vertices[i]->particles[0].spin ) {
00097         case 1:
00098             pinp.ptypen = "scalar";
00099             break;
00100         case -1:
00101             pinp.ptypen = "ghost";
00102             break;
00103         case 2:
00104             if ( m->vertices[i]->particles[0].type == 0 )
00105                 pinp.ptypen = "majorana";
00106             else
00107                 pinp.ptypen = "undef";
00108             break;
00109         case -2:
00110             if ( m->vertices[i]->particles[0].type == 0 )
00111                 pinp.ptypen = "majorana";
00112             else
00113                 pinp.ptypen = "dirac";
00114             break;
00115         case 3:
00116             pinp.ptypen = "vector";
00117             break;
00118         default:
00119             pinp.ptypen = "undef";
00120             break;
00121     }
00122 }
00123 pinp.extonly = m->vertices[i]->externonly;
00124 mdl->addParticle(&pinp);
00125 }
00126 mdl->addParticleEnd();
00127 //
00128 // Now the vertices
00129 //
00130 for ( i = m->nparticles; i < m->invertices; i++ ) {
00131     VERTEX *v = m->vertices[i];
00132     if ( minp.defpart == GRCC_DEFBYCODE ) {
00133         iinp.icode = NODEFUNCTION+i;
00134         iinp.name = NULL;
00135     }
00136     else {
00137         iinp.name = "node_";
00138     }
00139     iinp.nplistn = v->nparticles;
00140     for ( j = 0; j < iinp.nplistn; j++ ) {
00141         if ( minp.defpart == GRCC_DEFBYCODE ) {
00142             iinp.plistc[j] = v->particles[j].number;
00143         }
00144         else {
00145             iinp.plistn[j] = (char *) (VARNAME(functions,v->particles[j].number));
00146         }
00147     }
00148     //
00149     // We need a properly ordered list of coupling constants
00150     // The ordered list is in m->couplings.
00151     // For each vertex they are in v->couplings.
00152     //
00153     iinp.cvallist = (int *) Malloc1(m->ncouplings*sizeof(int),"couplings");
00154     for ( j = 0; j < m->ncouplings; j++ ) {
00155         iinp.cvallist[j] = 0;
00156         for ( k = 0; k < v->ncouplings; k += 2 ) {
00157             if ( v->couplings[k] == m->couplings[j] ) {
00158                 iinp.cvallist[j] = v->couplings[k+1];
00159                 break;
00160             }
00161         }
00162     }
00163 }
00164 mdl->addInteraction(&iinp);
00165 }
00166 mdl->addInteractionEnd();
00167 m->grccmodel = (void *)mdl;
00168 return(0);
00169 }
00170
00171 // #] LoadModel :
00172 // #[ ConvertParticle :
00173
00174 int ConvertParticle(Model *model,int formnum)
00175 {

```

```

00176 //
00177 // Returns the grcc number of the particle, because grcc does not convert
00178 //
00179 int i;
00180 for ( i = 0; i < model->nParticles; i++ ) {
00181     if ( model->particles[i]->pcode == formnum ) { return(i); }
00182     else if ( model->particles[i]->acode == formnum ) { return(-i); }
00183 }
00184 MesPrint("Particle %d not found in model %s",formnum,model->name);
00185 Terminate(-1);
00186 return(0);
00187 }
00188
00189 // #] ConvertParticle :
00190 // #[ ReConvertParticle :
00191
00192 int ReConvertParticle(Model *model,int grccnum)
00193 {
00194 //
00195 // Returns the grcc number of the particle, because grcc does not convert
00196 //
00197 if ( grccnum < 0 ) { return(model->particles[-grccnum]->acode); }
00198 else { return(model->particles[grccnum]->pcode); }
00199 }
00200
00201 // #] ReConvertParticle :
00202 // #[ numParticle :
00203
00204 int numParticle(MODEL *m,WORD n)
00205 {
00206     int i;
00207     for ( i = 0; i < m->nparticles; i++ ) {
00208         if ( m->vertices[i]->particles[0].number == n ) return(i);
00209         if ( m->vertices[i]->particles[1].number == n ) return(i);
00210     }
00211     MesPrint("numParticle: particle %d not found in model",n);
00212     Terminate(-1);
00213     return(-1);
00214 }
00215
00216 // #] numParticle :
00217 // #[ ProcessDiagram :
00218
00219 void ProcessDiagram(EGraph *eg, void *ti)
00220 {
00221 //
00222 // This is the routine that gets a complete diagram and passes it on
00223 // to Form (Generator) for further algebraic manipulations.
00224 // The term is picked up from AT.diaterm and a new term is constructed
00225 // in the workspace.
00226 //
00227 GETIDENTITY
00228 TERMINFO *info = (TERMINFO *)ti;
00229
00230 if ( ( info->flags & TOPOLOGIESONLY ) == TOPOLOGIESONLY ) return;
00231
00232 WORD *term = info->term, *newterm, *oldworkpointer = AT.WorkPointer;
00233 WORD *tdia = term + info->diaoffset;
00234 WORD *tail = tdia + tdia[1];
00235 WORD *tend = term + *term;
00236 WORD *fill, *startfill, *cfill, *afill;
00237 int i, j, intr;
00238 Model *model = (Model *)info->currentModel;
00239 MODEL *m = (MODEL *)info->currentMODEL;
00240 int numlegs, vect, edge;
00241
00242 newterm = term + *term;
00243 for ( i = 1; i < info->diaoffset; i++ ) newterm[i] = term[i];
00244 fill = newterm + info->diaoffset;
00245 //
00246 // Now get the nodes
00247 //
00248 if ( ( info->flags & NONODES ) == 0 ) {
00249     for ( i = 0; i < eg->nNodes; i++ ) {
00250 //
00251 //         node_(number,coupling,particle_1(momentum_1),...,particle_n(momentum_n))
00252 //
00253         numlegs = eg->nodes[i]->deg;
00254         startfill = fill;
00255         *fill++ = NODEFUNCTION;
00256         *fill++ = 0;
00257         FILLFUN(fill)
00258         *fill++ = -SNUMBER; *fill++ = i+1;
00259 //
00260 //         Now we put the coupling constants. This is done inside the
00261 //         function for when we want to work with counterterms.
00262 //

```

```

00263     if ( !eg->isExternal(i) ) {
00264         afill = fill; *fill++ = 0; *fill++ = 1; FILLARG(fill)
00265         cfill = fill; *fill++ = 0;
00266         intr = eg->nodes[i]->intrct;
00267         for ( j = 0; j < model->interacts[intr]->nclist; j++ ) {
00268             if ( model->interacts[intr]->clist[j] != 0 ) {
00269                 *fill++ = SYMBOL; *fill++ = 4;
00270                 *fill++ = m->couplings[j];
00271                 *fill++ = model->interacts[intr]->clist[j];
00272             }
00273         }
00274         *fill++ = 1; *fill++ = 1; *fill++ = 3;
00275         *cfill = fill - cfill;
00276         *afill = fill - afill;
00277     }
00278     else {
00279         *fill++ = -SNUMBER;
00280         *fill++ = 1;
00281     }
00282     //
00283     // Now the particles and their momenta.
00284     //
00285     for ( j = 0; j < numlegs; j++ ) {
00286         *fill++ = ARGHEAD+FUNHEAD+6;
00287         *fill++ = 0;
00288         FILLARG(fill)
00289         edge = eg->nodes[i]->edges[j];
00290         vect = ABS(edge)-1;
00291         *fill++ = 6+FUNHEAD;
00292         int a;
00293         if ( edge < 0 ) { a = ReConvertParticle(model,-eg->edges[vect]->ptcl); }
00294         else { a = ReConvertParticle(model,eg->edges[vect]->ptcl); }
00295         *fill++ = a;
00296         *fill++ = FUNHEAD+2;
00297         FILLFUN(fill)
00298         *fill++ = edge < 0 ? -MINVECTOR: -VECTOR;
00299         if ( numlegs == 1 || vect < info->numextern ) { // Look up in set of external momenta
00300             *fill++ = SetElements[Sets[info->externalset].first+vect];
00301         }
00302         else { // Look up in set of internal momenta set
00303             *fill++ = SetElements[Sets[info->internalset].first+(vect-eg->nExtern)];
00304         }
00305         *fill++ = 1; *fill++ = 1; *fill++ = 3;
00306     }
00307     startfill[1] = fill-startfill;
00308 }
00309 }
00310 if ( ( info->flags & WITHEGES ) == WITHEGES ) {
00311     for ( i = 0; i < eg->nEdges; i++ ) {
00312         int n1 = eg->edges[i]->nodes[0];
00313         int n2 = eg->edges[i]->nodes[1];
00314         // int l1 = eg->edges[i]->nlegs[0];
00315         // int l2 = eg->edges[i]->nlegs[1];
00316
00317         startfill = fill;
00318         *fill++ = EDGE;
00319         *fill++ = 0;
00320         FILLFUN(fill)
00321         *fill++ = -SNUMBER; *fill++ = i+1; // number of the edge
00322         //
00323         *fill++ = ARGHEAD+FUNHEAD+6;
00324         *fill++ = 0;
00325         FILLARG(fill)
00326         *fill++ = 6+FUNHEAD;
00327         int a = ReConvertParticle(model,eg->edges[i]->ptcl);
00328         *fill++ = a;
00329         *fill++ = FUNHEAD+2;
00330         FILLFUN(fill)
00331         *fill++ = -VECTOR;
00332         // Look up in set of internal momenta set
00333         if ( i < info->numextern ) { // Look up in set of external momenta
00334             *fill++ = SetElements[Sets[info->externalset].first+i];
00335         }
00336         else { // Look up in set of internal momenta set
00337             *fill++ = SetElements[Sets[info->internalset].first+(i-eg->nExtern)];
00338         }
00339         *fill++ = 1; *fill++ = 1; *fill++ = 3;
00340         //
00341         *fill++ = -SNUMBER; *fill++ = n1+1; // number of the node from
00342         *fill++ = -SNUMBER; *fill++ = n2+1; // number of the node to
00343         startfill[1] = fill - startfill;
00344     }
00345 }
00346 if ( ( info->flags & WITHBLOCKS ) == WITHBLOCKS ) {
00347     for ( i = 0; i < eg->econn->nblocks; i++ ) {
00348         startfill = fill;
00349         *fill++ = BLOCK;

```

```

00350         *fill++ = 0;
00351         FILLFUN(fill);
00352         *fill++ = -SNUMBER;
00353         *fill++ = i+1;
00354         *fill++ = -SNUMBER;
00355         *fill++ = eg->econn->blocks[i].loop;
00356 //
00357 //         Now we have to make a list of all nodes inside this block
00358 //
00359         int bnodes[GRCC_MAXNODES], k;
00360         WORD *argfill = fill, *funfill;
00361         *fill++ = 0; *fill++ = 0; FILLARG(fill)
00362         *fill++ = 0;
00363         for ( k = 0; k < GRCC_MAXNODES; k++ ) bnodes[k] = 0;
00364         for ( k = 0; k < eg->econn->blocks[i].nmedges; k++ ) {
00365             bnodes[eg->econn->blocks[i].edges[k][0]] = 1;
00366             bnodes[eg->econn->blocks[i].edges[k][1]] = 1;
00367         }
00368         for ( k = 0; k < GRCC_MAXNODES; k++ ) {
00369             if ( bnodes[k] == 0 ) continue;
00370 //
00371 //         Now we put the node inside this argument.
00372 //
00373             funfill = fill;
00374             *fill++ = NODEFUNCTION;
00375             *fill++ = 0;
00376             FILLFUN(fill)
00377             *fill++ = -SNUMBER; *fill++ = k+1;
00378             numlegs = eg->nodes[k]->deg;
00379             for ( j = 0; j < numlegs; j++ ) {
00380                 edge = eg->nodes[k]->edges[j];
00381                 vect = ABS(edge)-1;
00382                 *fill++ = -VECTOR;
00383                 if ( numlegs == 1 || vect < info->numextern ) { // Look up in set of external
momenta
00384                     *fill++ = SetElements[Sets[info->externalset].first+vect];
00385                 }
00386                 else { // Look up in set of internal momenta set
00387                     *fill++ = SetElements[Sets[info->internalset].first+(vect-info->numextern)];
00388                 }
00389             }
00390             funfill[1] = fill-funfill;
00391         }
00392         *fill++ = 1; *fill++ = 1; *fill++ = 3;
00393         *argfill = fill - argfill;
00394         argfill[ARGHEAD] = argfill[0] - ARGHEAD;
00395         startfill[1] = fill-startfill;
00396     }
00397 }
00398 if ( ( info->flags & WITHONEPI ) == WITHONEPI ) {
00399     for ( i = 0; i < eg->econn->nopic; i++ ) {
00400         startfill = fill;
00401         *fill++ = ONEPI;
00402         *fill++ = 0;
00403         FILLFUN(fill);
00404         *fill++ = -SNUMBER;
00405         *fill++ = i+1;
00406         *fill++ = -ONEPI;
00407         for ( j = 0; j < eg->econn->opics[i].nnodes; j++ ) {
00408             *fill++ = -SNUMBER;
00409             *fill++ = eg->econn->opics[i].nodes[j]+1;
00410         }
00411         startfill[1] = fill-startfill;
00412     }
00413 }
00414 //
00415 // Topology counter. We have exaggerated a bit with the eye on the far future.
00416 //
00417 if ( info->numtopo-1 < MAXPOSITIVE ) {
00418     *fill++ = TOPO; *fill++ = FUNHEAD+2; FILLFUN(fill)
00419     *fill++ = -SNUMBER; *fill++ = (WORD)(info->numtopo-1);
00420 }
00421 else if ( info->numtopo-1 < FULLMAX-1 ) {
00422     *fill++ = TOPO; *fill++ = FUNHEAD+ARGHEAD+4; FILLFUN(fill)
00423     *fill++ = ARGHEAD+4; *fill++ = 0; FILLARG(fill)
00424     *fill++ = 4;
00425     *fill++ = (WORD)((info->numtopo-1) & WORDMASK);
00426     *fill++ = 1; *fill++ = 3;
00427 }
00428 else { // for now: science fiction
00429     *fill++ = TOPO; *fill++ = FUNHEAD+ARGHEAD+6; FILLFUN(fill)
00430     *fill++ = ARGHEAD+6; *fill++ = 0; FILLARG(fill)
00431     *fill++ = 6; *fill++ = (WORD)((info->numtopo-1) » BITSINWORD);
00432     *fill++ = (WORD)((info->numtopo-1) & WORDMASK);
00433     *fill++ = 0; *fill++ = 1; *fill++ = 5;
00434 }
00435 //

```

```

00436 // Symmetry factors. We let Normalize do the multiplication.
00437 //
00438 if ( eg->nsym != 1 ) {
00439     *fill++ = SNUMBER; *fill++ = 4; *fill++ = (WORD)eg->nsym; *fill++ = -1;
00440 }
00441 if ( eg->esym != 1 ) {
00442     *fill++ = SNUMBER; *fill++ = 4; *fill++ = (WORD)eg->esym; *fill++ = -1;
00443 }
00444 if ( eg->extperm != 1 ) {
00445     *fill++ = SNUMBER; *fill++ = 4; *fill++ = (WORD)eg->extperm; *fill++ = 1;
00446 }
00447 //
00448 // finish it off
00449 //
00450 while ( tail < tend ) *fill++ = *tail++;
00451 if ( eg->fsign < 0 ) fill[-1] = -fill[-1];
00452 *newterm = fill - newterm;
00453 AT.WorkPointer = fill;
00454
00455 // MesPrint("<> %a",newterm[0],newterm);
00456
00457 Generator(BHEAD newterm,info->level);
00458 AT.WorkPointer = oldworkpointer;
00459 }
00460
00461 // #] ProcessDiagram :
00462 // #[ fendMG :
00463
00464 Bool fendMG(EGraph *eg, void *ti)
00465 {
00466     DUMMYUSE(eg);
00467     DUMMYUSE(ti);
00468     return True;
00469 }
00470
00471 // #] fendMG :
00472 // #[ ProcessTopology :
00473
00474 Bool ProcessTopology(EGraph *eg, void *ti)
00475 {
00476     //
00477     // This routine is called for each new topology.
00478     // New convention: return True; generate the corresponding diagrams if needed
00479     // return False; skip diagram generation (when asked for).
00480     //
00481     TERMINFO *info = (TERMINFO *)ti;
00482 #ifdef WITHEARLYVETO
00483     if ( ( ( info->flags & CHECKEXTERN ) == CHECKEXTERN ) && info->currentMODEL != NULL ) {
00484         int i, j;
00485         int numlegs, vect, edge;
00486         for ( i = 0; i < eg->nNodes; i++ ) {
00487             if ( eg->isExternal(i) ) continue;
00488             numlegs = eg->nodes[i]->deg;
00489             for ( j = 0; j < numlegs; j++ ) {
00490                 edge = eg->nodes[i]->edges[j];
00491                 vect = ABS(edge)-1;
00492                 if ( vect < info->numextern && info->legcouple[vect][numlegs] == 0 ) {
00493
00494                     // This cannot be.
00495
00496                     info->numtopo++;
00497                     return False;
00498                 }
00499             }
00500         }
00501     }
00502 #endif
00503     if ( ( info->flags & TOPOLOGIESONLY ) == 0 ) {
00504         info->numtopo++;
00505         return True;
00506     }
00507 //
00508 // Now we are just generating topologies.
00509 //
00510 GETIDENTITY
00511 WORD *term = info->term, *newterm, *oldworkpointer = AT.WorkPointer;
00512 WORD *tdia = term + info->diaoffset;
00513 WORD *tail = tdia + tdia[1];
00514 WORD *tend = term + *term;
00515 WORD *fill, *startfill;
00516 Model *model = (Model *)info->currentModel;
00517 MODEL *m = (MODEL *)info->currentMODEL;
00518 int i, j;
00519 int numlegs, vect, edge;
00520
00521 newterm = term + *term;
00522 for ( i = 1; i < info->diaoffset; i++ ) newterm[i] = term[i];

```

```

00523     fill = newterm + info->diaoffset;
00524 //
00525 // Now get the nodes
00526 //
00527 for ( i = 0; i < eg->nNodes; i++ ) {
00528 //
00529 //     node_(number,momentum_1,...,momentum_n)
00530 //
00531     numlegs = eg->nodes[i]->deg;
00532     startfill = fill;
00533     *fill++ = NODEFUNCTION;
00534     *fill++ = 0;
00535     FILLFUN(fill)
00536     *fill++ = -SNUMBER; *fill++ = i+1;
00537     if ( model != NULL && m != NULL ) {
00538         if ( !eg->isExternal(i) ) {
00539             WORD *afill = fill; *fill++ = 0; *fill++ = 1; FILLARG(fill)
00540             WORD *cfill = fill; *fill++ = 0;
00541             int cpl = 2*eg->nodes[i]->extloop+eg->nodes[i]->deg-2;
00542             *fill++ = SYMBOL; *fill++ = 4;
00543             *fill++ = m->couplings[0];
00544             *fill++ = cpl;
00545             *fill++ = 1; *fill++ = 1; *fill++ = 3;
00546             *cfill = fill - cfill;
00547             *afill = fill - afill;
00548             if ( *afill == ARGHEAD+8 && afill[ARGHEAD+4] == 1 ) {
00549                 fill = afill; *fill++ = -SYMBOL; *fill++ = afill[ARGHEAD+3];
00550             }
00551         }
00552         else {
00553             *fill++ = -SNUMBER;
00554             *fill++ = 1;
00555         }
00556     }
00557 //
00558 // Now the momenta.
00559 //
00560 for ( j = 0; j < numlegs; j++ ) {
00561     edge = eg->nodes[i]->edges[j];
00562     vect = ABS(edge)-1;
00563     *fill++ = edge < 0 ? -MINVECTOR: -VECTOR;
00564     if ( numlegs == 1 || vect < info->numextern ) { // Look up in set of external momenta
00565         *fill++ = SetElements[Sets[info->externalset].first+vect];
00566     }
00567     else { // Look up in set of internal momenta set
00568         *fill++ = SetElements[Sets[info->internalset].first+(vect-info->numextern)];
00569     }
00570 }
00571 startfill[1] = fill-startfill;
00572 }
00573 if ( ( info->flags & WITHEGES ) == WITHEGES ) {
00574     for ( i = 0; i < eg->nEdges; i++ ) {
00575         int n1 = eg->edges[i]->nodes[0];
00576         int n2 = eg->edges[i]->nodes[1];
00577 //         int l1 = eg->edges[i]->nlegs[0];
00578 //         int l2 = eg->edges[i]->nlegs[1];
00579         startfill = fill;
00580         *fill++ = EDGE;
00581         *fill++ = 0;
00582         FILLFUN(fill)
00583         *fill++ = -SNUMBER; *fill++ = i+1; // number of the edge
00584 //
00585         *fill++ = -VECTOR;
00586         if ( i < info->numextern ) { // Look up in set of external momenta
00587             *fill++ = SetElements[Sets[info->externalset].first+i];
00588         }
00589         else { // Look up in set of internal momenta set
00590             *fill++ = SetElements[Sets[info->internalset].first+(i-eg->nExtern)];
00591         }
00592 //
00593         *fill++ = -SNUMBER; *fill++ = n1+1; // number of the node from
00594         *fill++ = -SNUMBER; *fill++ = n2+1; // number of the node to
00595         startfill[1] = fill - startfill;
00596     }
00597 }
00598 //
00599 // Block information
00600 //
00601 if ( ( info->flags & WITHBLOCKS ) == WITHBLOCKS ) {
00602     for ( i = 0; i < eg->econn->nblocks; i++ ) {
00603         startfill = fill;
00604         *fill++ = BLOCK;
00605         *fill++ = 0;
00606         FILLFUN(fill);
00607         *fill++ = -SNUMBER;
00608         *fill++ = i+1;
00609         *fill++ = -SNUMBER;

```

```

00610         *fill++ = eg->econn->blocks[i].loop;
00611 //
00612 //     Now we have to make a list of all nodes inside this block
00613 //
00614     int bnodes[GRCC_MAXNODES], k;
00615     WORD *argfill = fill, *funfill;
00616     *fill++ = 0; *fill++ = 0; FILLARG(fill)
00617     *fill++ = 0;
00618     for ( k = 0; k < GRCC_MAXNODES; k++ ) bnodes[k] = 0;
00619     for ( k = 0; k < eg->econn->blocks[i].nedges; k++ ) {
00620         bnodes[eg->econn->blocks[i].edges[k][0]] = 1;
00621         bnodes[eg->econn->blocks[i].edges[k][1]] = 1;
00622     }
00623     for ( k = 0; k < GRCC_MAXNODES; k++ ) {
00624         if ( bnodes[k] == 0 ) continue;
00625 //
00626 //     Now we put the node inside this argument.
00627 //
00628         funfill = fill;
00629         *fill++ = NODEFUNCTION;
00630         *fill++ = 0;
00631         FILLFUN(fill)
00632         *fill++ = -SNUMBER; *fill++ = k+1;
00633         numlegs = eg->nodes[k]->deg;
00634         for ( j = 0; j < numlegs; j++ ) {
00635             edge = eg->nodes[k]->edges[j];
00636             vect = ABS(edge)-1;
00637             *fill++ = -VECTOR;
00638             if ( numlegs == 1 || vect < info->numextern ) { // Look up in set of external
momenta
00639                 *fill++ = SetElements(Sets[info->externalset].first+vect);
00640             }
00641             else { // Look up in set of internal momenta set
00642                 *fill++ = SetElements(Sets[info->internalset].first+(vect-info->numextern));
00643             }
00644         }
00645         funfill[1] = fill-funfill;
00646     }
00647     *fill++ = 1; *fill++ = 1; *fill++ = 3;
00648     *argfill = fill - argfill;
00649     argfill[ARGHEAD] = argfill[0] - ARGHEAD;
00650     startfill[1] = fill-startfill;
00651 }
00652
00653 //     if ( eg->econn->narticuls > 0 ) {
00654 //         startfill = fill;
00655 //         *fill++ = BLOCK;
00656 //         *fill++ = 0;
00657 //         FILLFUN(fill);
00658 //         for ( i = 0; i < eg->econn->snodes; i++ ) {
00659 //             if ( eg->econn->articuls[i] != 0 ) {
00660 //                 *fill++ = -SNUMBER;
00661 //                 *fill++ = i+1;
00662 //             }
00663 //         }
00664 //         startfill[1] = fill-startfill;
00665 //     }
00666 }
00667 if ( ( info->flags & WITHONEPI ) == WITHONEPI ) {
00668     for ( i = 0; i < eg->econn->nopic; i++ ) {
00669         startfill = fill;
00670         *fill++ = ONEPI;
00671         *fill++ = 0;
00672         FILLFUN(fill);
00673         *fill++ = -SNUMBER;
00674         *fill++ = i+1;
00675         *fill++ = -ONEPI;
00676         for ( j = 0; j < eg->econn->opics[i].nnodes; j++ ) {
00677             *fill++ = -SNUMBER;
00678             *fill++ = eg->econn->opics[i].nodes[j]+1;
00679         }
00680         startfill[1] = fill-startfill;
00681     }
00682 }
00683 //
00684 //     Topology counter. We have exaggerated a bit with the eye on the far future.
00685 //
00686     if ( info->numtopo < MAXPOSITIVE ) {
00687         *fill++ = TOPO; *fill++ = FUNHEAD+2; FILLFUN(fill)
00688         *fill++ = -SNUMBER; *fill++ = (WORD)(info->numtopo);
00689     }
00690     else if ( info->numtopo < FULLMAX-1 ) {
00691         *fill++ = TOPO; *fill++ = FUNHEAD+ARGHEAD+4; FILLFUN(fill)
00692         *fill++ = ARGHEAD+4; *fill++ = 0; FILLARG(fill)
00693         *fill++ = 4;
00694         *fill++ = (WORD)(info->numtopo & WORDMASK);
00695         *fill++ = 1; *fill++ = 3;

```

```

00696     }
00697     else { // for now: science fiction
00698         *fill++ = TOPO; *fill++ = FUNHEAD+ARGHEAD+6; FILLFUN(fill)
00699         *fill++ = ARGHEAD+6; *fill++ = 0; FILLARG(fill)
00700         *fill++ = 6; *fill++ = (WORD)(info->numtopo » BITSINWORD);
00701         *fill++ = (WORD)(info->numtopo & WORDMASK);
00702         *fill++ = 0; *fill++ = 1; *fill++ = 5;
00703     }
00704 //
00705 // Symmetry factors. We let Normalize do the multiplication.
00706 //
00707     if ( eg->nsym != 1 ) {
00708         *fill++ = SNUMBER; *fill++ = 4; *fill++ = (WORD)eg->nsym; *fill++ = -1;
00709     }
00710     if ( eg->esym != 1 ) {
00711         *fill++ = SNUMBER; *fill++ = 4; *fill++ = (WORD)eg->esym; *fill++ = -1;
00712     }
00713 //
00714 // finish it off
00715 //
00716     while ( tail < tend ) *fill++ = *tail++;
00717     if ( eg->fsign < 0 ) fill[-1] = -fill[-1];
00718     *newterm = fill - newterm;
00719     AT.WorkPointer = fill;
00720
00721 //MesPrint("<> %a",*newterm,newterm);
00722
00723     Generator(BHEAD newterm,info->level);
00724     AT.WorkPointer = oldworkpointer;
00725     info->numtopo++;
00726     return False;
00727 }
00728
00729 // #] ProcessTopology :
00730 // #[ GenDiagrams :
00731
00732 int GenDiagrams(PHEAD WORD *term, WORD level)
00733 {
00734     Model *model;
00735     MODEL *m;
00736     Options *opt;
00737     Process *proc;
00738     int pid = 1, x;
00739     int babble = 0; // Later we may set this at the FORM code level
00740     TERMINFO info;
00741     WORD inset,outset,*coupl,setnum,optionnumber = 0;
00742     int i, j, cpl[GRCC_MAXNCPLG];
00743     int ninitl, initlPart[GRCC_MAXLEGS], nfinal, finalPart[GRCC_MAXLEGS];
00744     for ( i = 0; i < GRCC_MAXNCPLG; i++ ) cpl[i] = 0;
00745 //
00746 // Here we create an object of type Option and load it up.
00747 // Next we run the diagram generation on it.
00748 //
00749     info.term = term;
00750     info.level = level;
00751     info.diaoffset = AR.funoffset;
00752     info.externalset = term[info.diaoffset+FUNHEAD+7];
00753     info.internalset = term[info.diaoffset+FUNHEAD+9];
00754     info.flags = 0;
00755     inset = term[info.diaoffset+FUNHEAD+3];
00756     outset = term[info.diaoffset+FUNHEAD+5];
00757     coupl = term + info.diaoffset + FUNHEAD + 10;
00758     if ( *coupl < 0 ) {
00759         if ( term[info.diaoffset+1] > FUNHEAD + 12 ) {
00760             optionnumber = term[info.diaoffset+FUNHEAD+13];
00761         }
00762     }
00763     else {
00764         if ( term[info.diaoffset+1] > *coupl+FUNHEAD+10 )
00765             optionnumber = term[info.diaoffset+*coupl+FUNHEAD+11];
00766     }
00767     setnum = term[info.diaoffset+FUNHEAD+1];
00768
00769     m = AC.models[SetElements[Sets[setnum].first]];
00770     LoadModel(m);
00771     model = (Model *)m->grccmodel;
00772
00773     info.currentModel = (void *)model;
00774     info.currentMODEL = (void *)m;
00775     info.numdia = 0;
00776     info.numtopo = 1;
00777     info.flags = optionnumber;
00778
00779     opt = new Options();
00780
00781     opt->setOutAG(ProcessDiagram, &info);
00782     opt->setOutMG(ProcessTopology, &info);

```



```

00783 // opt->setEndMG(fendMG, &info);
00784
00785 opt->values[GRCC_OPT_1PI] = ( optionnumber & ONEPARTICLEIRREDUCIBLE ) == ONEPARTICLEIRREDUCIBLE;
00786 opt->values[GRCC_OPT_NoTadpole] = ( optionnumber & NOTADPOLES ) == NOTADPOLES;
00787 //
00788 // Next are snails:
00789 //
00790 opt->values[GRCC_OPT_No1PtBlock] = ( optionnumber & NOTADPOLES ) == NOTADPOLES;
00791 //
00792 if ( ( optionnumber & WITHINSERTIONS ) == WITHINSERTIONS ) {
00793     opt->values[GRCC_OPT_No2PtL1PI] = True;
00794     opt->values[GRCC_OPT_NoAdj2PtV] = True;
00795     opt->values[GRCC_OPT_No2PtL1PI] = True;
00796 }
00797 else {
00798     opt->values[GRCC_OPT_NoAdj2PtV] = True;
00799 }
00800 opt->values[GRCC_OPT_SymmInitial] = ( optionnumber & WITHSYMMETRIZE ) == WITHSYMMETRIZE;
00801 opt->values[GRCC_OPT_SymmFinal] = ( optionnumber & WITHSYMMETRIZE ) == WITHSYMMETRIZE;
00802
00803 // opt->values[GRCC_OPT_Block] = ( optionnumber & WITHBLOCKS ) == WITHBLOCKS;
00804
00805 opt->setOutputF(False, "");
00806 opt->setOutputP(False, "");
00807 opt->printLevel(babble);
00808
00809 // opt->values[GRCC_OPT_Step] = GRCC_AGraph;
00810
00811 // Load the various arrays.
00812
00813 ninitl = Sets[inset].last - Sets[inset].first;
00814 for ( i = 0; i < ninitl; i++ ) {
00815     x = SetElements[Sets[inset].first+i];
00816     initlPart[i] = ConvertParticle(model,x);
00817     info.legcouple[i] = m->vertices[numParticle(m,x)]->couplings;
00818 }
00819 nfinal = Sets[outset].last - Sets[outset].first;
00820 for ( i = 0; i < nfinal; i++ ) {
00821     x = SetElements[Sets[outset].first+i];
00822     finalPart[i] = ConvertParticle(model,x);
00823     info.legcouple[i+ninitl] = m->vertices[numParticle(m,x)]->couplings;
00824 }
00825 info.numextern = ninitl + nfinal;
00826 for ( i = 2; i <= MAXLEGS; i++ ) {
00827     if ( m->legcouple[i] == 1 ) {
00828         for ( j = 0; j < info.numextern; j++ ) {
00829             if ( info.legcouple[j][i] == 0 ) { info.flags |= CHECKEXTERN; goto Go_on; }
00830         }
00831     }
00832 }
00833 Go_on;;
00834 //
00835 // Now we have to sort out the coupling constants.
00836 // The argument at coupl can be of type -SNUMBER, -SYMBOL or generic
00837 // It has however already be tested for syntax.
00838 // Note that one cannot have 1 for the coupling constants.
00839 // In that case one should select 0 loops or something equivalent.
00840 //
00841 if ( *coupl == -SNUMBER ) { // Number of loops
00842 //
00843 // This is the complicated case.
00844 // We have to compute the number of coupling constants and then
00845 // generate diagrams for all combinations with the proper power.
00846 //
00847 int nc = coupl[1]*2 + ninitl + nfinal - 2;
00848 int *scratch = (int *)Malloca(nc*sizeof(int), "DistrN");
00849 scratch[0] = -2; // indicating startup cq first call.
00850
00851 if ( ( info.flags & TOPOLOGIESONLY ) == 0 ) {
00852     while ( DistrN(nc, cpl, m->ncouplings, scratch) ) {
00853         proc = new Process(pid, model, opt,
00854                             ninitl, initlPart, nfinal, finalPart, cpl);
00855         delete proc;
00856         info.numtopo = 1;
00857     }
00858 }
00859 else {
00860     cpl[0] = nc;
00861     proc = new Process(pid, model, opt,
00862                         ninitl, initlPart, nfinal, finalPart, cpl);
00863     delete proc;
00864 }
00865 M_free(scratch, "DistrN");
00866 opt->end();
00867 delete opt;
00868 return(0);
00869 }

```

```

00870     else if ( *coupl == -SYMBOL ) { // Just a single power of one constant
00871         for ( i = 0; i < m->ncouplings; i++ ) {
00872             if ( m->couplings[i] == coupl[1] ) {
00873                 cpl[i] = 1;
00874                 break;
00875             }
00876         }
00877     }
00878     else { // One term with powers of coupling constants
00879         WORD *t, *tstop;
00880         t = coupl + ARGHEAD+3;
00881         tstop = coupl+*coupl; tstop -= ABS(tstop[-1]);
00882         while ( t < tstop ) {
00883             for ( i = 0; i < m->ncouplings; i++ ) {
00884                 if ( m->couplings[i] == *t ) {
00885                     cpl[i] = t[1];
00886                     break;
00887                 }
00888             }
00889             t += 2;
00890         }
00891     }
00892     /*
00893     And now the generation:
00894     */
00895     proc = new Process(pid, model, opt,
00896                       ninitl, initlPart, nfinal, finalPart, cpl);
00897     opt->end();
00898     delete proc;
00899     delete opt;
00900     return(0);
00901 }
00902
00903 // #] GenDiagrams :
00904 // #[ processVertex :
00905
00906 // Routine is to be used recursively to work its way through a list
00907 // of possible vertices. The array of vertices is in TopoInf->vert
00908 // with TopoInf->nvert the number of possible vertices.
00909 // Currently we allow in TopoInf->vert only vertices with 3 or more edges.
00910 //
00911 // We work with a point system. Each n-point vertex contributes n-2 points.
00912 // When all points are assigned, we can call mgraph->generate().
00913 //
00914 // The number of vertices of a given number of edges is stored in
00915 // TopoInf->cnum[...] but the loop that determines how many there are
00916 // may be limited by the corresponding element in TopoInf->vertmax[level]
00917
00918 int processVertex(TOPOTYPE *TopoInf, int pointsremaining, int level)
00919 {
00920     int i, j;
00921
00922     for ( i = pointsremaining, j = 0; i >= 0; i -= TopoInf->vert[level]-2, j++ ) {
00923         if ( TopoInf->vertmax && TopoInf->vertmax[level] >= 0
00924             && j > TopoInf->vertmax[level] ) break;
00925         if ( i == 0 ) { // We got one!
00926             TopoInf->cldeg[TopoInf->ncl] = TopoInf->vert[level];
00927             TopoInf->cnum[TopoInf->ncl] = j;
00928             TopoInf->clxt[TopoInf->ncl] = 0;
00929             TopoInf->ncl++;
00930
00931             MGraph *mgraph = new MGraph(1, TopoInf->ncl, TopoInf->cldeg,
00932                                         TopoInf->cnum, TopoInf->clxt,
00933                                         TopoInf->cmind, TopoInf->cmaxd, TopoInf->opt);
00934
00935             mgraph->generate();
00936
00937             delete mgraph;
00938
00939             TopoInf->ncl--;
00940
00941             break;
00942         }
00943         if ( level < TopoInf->nvert-1 ) {
00944             if ( j > 0 ) {
00945                 TopoInf->cldeg[TopoInf->ncl] = TopoInf->vert[level];
00946                 TopoInf->cnum[TopoInf->ncl] = j;
00947                 TopoInf->clxt[TopoInf->ncl] = 0;
00948                 TopoInf->ncl++;
00949             }
00950             if ( processVertex(TopoInf,i,level+1) < 0 ) return(-1);
00951             if ( j > 0 ) { TopoInf->ncl--; }
00952         }
00953     }
00954     return(0);
00955 }
00956

```

```

00957 // #] processVertex :
00958 // #[ GenTopologies :
00959
00960 #define TOPO_MAXVERT 10
00961
00962 int GenTopologies(PHEAD WORD *term, WORD level)
00963 {
00964     Options *opt = new Options;
00965     int nlegs, nloops, i, identical;
00966     TERMINFO info;
00967     WORD *t, *t1, *tstop;
00968     TOPOTYPE TopoInf;
00969     SETS s;
00970 //
00971     info.term = term;
00972     info.level = level;
00973     info.diaoffset = AR.funoffset;
00974     info.flags = 0;
00975
00976     t = term + info.diaoffset; // the function
00977     t1 = t + FUNHEAD;          // its arguments
00978     tstop = t + t[1];
00979
00980     info.externalset = t1[7];
00981     info.internalset = t1[9];
00982
00983     s = &(Sets[t1[5]]);
00984     TopoInf.nvert = s->last - s->first;
00985     TopoInf.vert = &(SetElements[s->first]);
00986
00987     nloops = t1[1];
00988     nlegs = t1[3];
00989
00990     info.numextern = nlegs;
00991
00992     for ( i = 0; i <= MAXLEGS; i++ ) { TopoInf.cmind[i] = TopoInf.cmaxd[i] = 0; }
00993
00994     t1 += 10;
00995     if ( t1 < tstop && t1[0] == -SETSET ) {
00996         TopoInf.vertmax = &(SetElements[Sets[t1[1]].first]);
00997         t1 += 2;
00998     }
00999     else TopoInf.vertmax = NULL;
01000
01001     info.flags |= TOPOLOGIESONLY; // this is the topologies_ function after all.
01002     if ( t1 < tstop && t1[0] == -SNUMBER ) {
01003         if ( ( t1[1] & NONODES ) == NONODES ) info.flags |= NONODES;
01004         if ( ( t1[1] & WITHEGES ) == WITHEGES ) info.flags |= WITHEGES;
01005         if ( ( t1[1] & WITHBLOCKS ) == WITHBLOCKS ) info.flags |= WITHBLOCKS;
01006         if ( ( t1[1] & WITHONEPI ) == WITHONEPI ) info.flags |= WITHONEPI;
01007         opt->values[GRCC_OPT_1PI] = ( t1[1] & ONEPARTICLEIRREDUCIBLE ) == ONEPARTICLEIRREDUCIBLE;
01008 //         opt->values[GRCC_OPT_NoTadpole] = ( t1[1] & NOTADPOLES ) == NOTADPOLES;
01009         opt->values[GRCC_OPT_NoTadpole] = ( t1[1] & NOSNAILS ) == NOSNAILS;
01010         opt->values[GRCC_OPT_No1PtBlock] = ( t1[1] & NOTADPOLES ) == NOTADPOLES;
01011         opt->values[GRCC_OPT_NoExtSelf] = ( t1[1] & NOEXTSELF ) == NOEXTSELF;
01012
01013         if ( ( t1[1] & WITHINSERTIONS ) == WITHINSERTIONS ) {
01014             opt->values[GRCC_OPT_No2PtL1PI] = True;
01015             opt->values[GRCC_OPT_NoAdj2PtV] = True;
01016             opt->values[GRCC_OPT_No2PtL1PI] = True;
01017         }
01018         opt->values[GRCC_OPT_SymmInitial] = ( t1[1] & WITHSYMMETRIZE ) == WITHSYMMETRIZE;
01019     }
01020
01021     info.numdia = 0;
01022     info.numtopo = 1;
01023
01024     opt->setOutAG(ProcessDiagram, &info);
01025     opt->setOutMG(ProcessTopology, &info);
01026 //
01027 // Now we should sum over all possible vertices and run MGraph for
01028 // each combination. This is done by recursion in the processVertex routine
01029 // First load up the relevant arrays.
01030 //
01031
01032 // First the external nodes.
01033
01034     if ( nlegs == -2 ) {
01035         nlegs = 2;
01036         identical = 1;
01037     }
01038     for ( i = 0; i < nlegs; i++ ) {
01039         TopoInf.cldeg[i] = 1; TopoInf.clnum[i] = 1; TopoInf.clex[i] = -1;
01040     }
01041     int points = 2*nloops-2+nlegs;
01042
01043     if ( identical == 1 ) { /* Only propagator topologies..... */

```

```

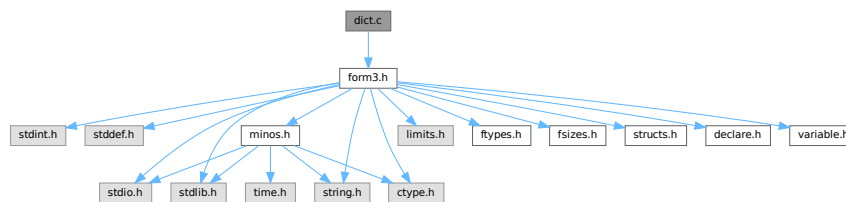
01044         nlegs = 1;
01045         TopoInf.clnum[0] = 2;
01046     }
01047     TopoInf.ncl = nlegs;
01048     TopoInf.opt = opt;
01049
01050     if ( points >= MAXPOINTS ) {
01051         MLOCK(ErrorMessageLock);
01052         MesPrint("GenTopologies: %d loops and %d legs considered excessive",nloops,nlegs);
01053         MUNLOCK(ErrorMessageLock);
01054         Terminate(-1);
01055     }
01056     if ( processVertex(&TopoInf,points,0) != 0 ) {
01057         MLOCK(ErrorMessageLock);
01058         MesPrint("Called from GenTopologies with %d loops and %d legs",nloops,nlegs);
01059         MUNLOCK(ErrorMessageLock);
01060         Terminate(-1);
01061     }
01062     delete opt;
01063     return(0);
01064 }
01065
01066 // #] GenTopologies :
01067

```

## 4.24 dict.c File Reference

```
#include "form3.h"
```

Include dependency graph for dict.c:



### Functions

- void [TransformRational](#) (UWORD \*a, WORD na)
- UBYTE \* [IsMultiplySign](#) (void)
- UBYTE \* [IsExponentSign](#) (void)
- UBYTE \* [FindSymbol](#) (WORD num)
- UBYTE \* [FindVector](#) (WORD num)
- UBYTE \* [FindIndex](#) (WORD num)
- UBYTE \* [FindFunction](#) (WORD num)
- UBYTE \* [FindFunWithArgs](#) (WORD \*t)
- UBYTE \* [FindExtraSymbol](#) (WORD num)
- int [FindDictionary](#) (UBYTE \*name)
- int [AddDictionary](#) (UBYTE \*name)
- int [AddToDictionary](#) (DICTIONARY \*dict, UBYTE \*left, UBYTE \*right)
- int [UseDictionary](#) (UBYTE \*name, UBYTE \*options)
- int [SetDictionaryOptions](#) (UBYTE \*options)
- void [UnSetDictionary](#) (void)
- void [RemoveDictionary](#) (DICTIONARY \*dict)
- void [ShrinkDictionary](#) (DICTIONARY \*dict)
- int [DoPreOpenDictionary](#) (UBYTE \*s)

- int [DoPreCloseDictionary](#) (UBYTE \*s)
- int [DoPreUseDictionary](#) (UBYTE \*s)
- int [DoPreAdd](#) (UBYTE \*s)
- LONG [DictToBytes](#) (DICTIONARY \*dict, UBYTE \*buf)
- DICTIONARY \* [DictFromBytes](#) (UBYTE \*buf)

### 4.24.1 Detailed Description

Contains the code pertaining to dictionaries. Commands are: #opendictionary name #closedictionary #selectdictionary name <options>. There can be several dictionaries, but only one can be active. Defining elements is done with #add object: "replacement". Replacements are strings when a dictionary is for output translation. Objects can be 1: a number (rational) 2: a variable 3: \* ^ 4: a function with arguments.

Definition in file [dict.c](#).

### 4.24.2 Function Documentation

#### 4.24.2.1 TransformRational()

```
void TransformRational (
    UWORD * a,
    WORD na )
```

Definition at line 71 of file [dict.c](#).

#### 4.24.2.2 IsMultiplySign()

```
UBYTE * IsMultiplySign (
    void )
```

Definition at line 218 of file [dict.c](#).

#### 4.24.2.3 IsExponentSign()

```
UBYTE * IsExponentSign (
    void )
```

Definition at line 238 of file [dict.c](#).

#### 4.24.2.4 FindSymbol()

```
UBYTE * FindSymbol (
    WORD num )
```

Definition at line 258 of file [dict.c](#).

#### 4.24.2.5 FindVector()

```
UBYTE * FindVector (
    WORD num )
```

Definition at line 279 of file [dict.c](#).

#### 4.24.2.6 FindIndex()

```
UBYTE * FindIndex (
    WORD num )
```

Definition at line 301 of file [dict.c](#).

#### 4.24.2.7 FindFunction()

```
UBYTE * FindFunction (
    WORD num )
```

Definition at line 323 of file [dict.c](#).

#### 4.24.2.8 FindFunWithArgs()

```
UBYTE * FindFunWithArgs (
    WORD * t )
```

Definition at line 345 of file [dict.c](#).

#### 4.24.2.9 FindExtraSymbol()

```
UBYTE * FindExtraSymbol (
    WORD num )
```

Definition at line 377 of file [dict.c](#).

#### 4.24.2.10 FindDictionary()

```
int FindDictionary (
    UBYTE * name )
```

Definition at line 434 of file [dict.c](#).

#### 4.24.2.11 AddDictionary()

```
int AddDictionary (
    UBYTE * name )
```

Definition at line 449 of file [dict.c](#).

#### 4.24.2.12 AddToDictionary()

```
int AddToDictionary (
    DICTIONARY * dict,
    UBYTE * left,
    UBYTE * right )
```

Definition at line [491](#) of file [dict.c](#).

#### 4.24.2.13 UseDictionary()

```
int UseDictionary (
    UBYTE * name,
    UBYTE * options )
```

Definition at line [766](#) of file [dict.c](#).

#### 4.24.2.14 SetDictionaryOptions()

```
int SetDictionaryOptions (
    UBYTE * options )
```

Definition at line [790](#) of file [dict.c](#).

#### 4.24.2.15 UnSetDictionary()

```
void UnSetDictionary (
    void )
```

Definition at line [883](#) of file [dict.c](#).

#### 4.24.2.16 RemoveDictionary()

```
void RemoveDictionary (
    DICTIONARY * dict )
```

Definition at line [902](#) of file [dict.c](#).

#### 4.24.2.17 ShrinkDictionary()

```
void ShrinkDictionary (
    DICTIONARY * dict )
```

Definition at line [945](#) of file [dict.c](#).

#### 4.24.2.18 DoPreOpenDictionary()

```
int DoPreOpenDictionary (
    UBYTE * s )
```

Definition at line 959 of file [dict.c](#).

#### 4.24.2.19 DoPreCloseDictionary()

```
int DoPreCloseDictionary (
    UBYTE * s )
```

Definition at line 998 of file [dict.c](#).

#### 4.24.2.20 DoPreUseDictionary()

```
int DoPreUseDictionary (
    UBYTE * s )
```

Definition at line 1022 of file [dict.c](#).

#### 4.24.2.21 DoPreAdd()

```
int DoPreAdd (
    UBYTE * s )
```

Definition at line 1067 of file [dict.c](#).

#### 4.24.2.22 DictToBytes()

```
LONG DictToBytes (
    DICTIONARY * dict,
    UBYTE * buf )
```

Definition at line 1119 of file [dict.c](#).

#### 4.24.2.23 DictFromBytes()

```
DICTIONARY * DictFromBytes (
    UBYTE * buf )
```

Definition at line 1152 of file [dict.c](#).



## 4.25 dict.c

[Go to the documentation of this file.](#)

```

00001
00018 /* #[ License : */
00019 /*
00020 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00021 *   When using this file you are requested to refer to the publication
00022 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00023 *   This is considered a matter of courtesy as the development was paid
00024 *   for by FOM the Dutch physics granting agency and we would like to
00025 *   be able to track its scientific use to convince FOM of its value
00026 *   for the community.
00027 *
00028 *   This file is part of FORM.
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00030 *   FORM is free software: you can redistribute it and/or modify it under the
00031 *   terms of the GNU General Public License as published by the Free Software
00032 *   Foundation, either version 3 of the License, or (at your option) any later
00033 *   version.
00034 *
00035 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00036 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00037 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00038 *   details.
00039 *
00040 *   You should have received a copy of the GNU General Public License along
00041 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00042 */
00043 /* #[ License : */
00044 /*
00045     #[ Includes : ratio.c
00046
00047     Data setup:
00048         AO.Dictionary          Array of pointers to DICTIONARY
00049         AO.NumDictionaries
00050         AO.SizeDictionaries
00051         AO.CurrentDictionary
00052         AO.CurDictNumbers
00053         AO.CurDictVariables
00054         AO.CurDictSpecials
00055         AP.OpenDictionary
00056 */
00057
00058 #include "form3.h"
00059
00060 /*
00061     #[ Includes :
00062     #[ TransformRational:
00063
00064     Tries to transform the rational number a according to the rules of
00065     the current dictionary. Whatever cannot be translated goes to the
00066     regular output.
00067     Options for AO.CurDictNumbers are:
00068         DICT_ALLNUMBERS, DICT_RATIONALONLY, DICT_INTEGERONLY, DICT_NONUMBERS
00069 */
00070
00071 void TransformRational(UWORD *a, WORD na)
00072 {
00073     DICTIONARY *dict;
00074     WORD i, j, nb, i1, i2; UWORD *b;
00075     if ( AO.CurrentDictionary <= 0 ) goto NoAction;
00076     dict = AO.Dictionaries[AO.CurrentDictionary-1];
00077     if ( na < 0 ) na = -na;
00078     switch ( AO.CurDictNumbers ) {
00079     case DICT_NONUMBERS:
00080         goto NoAction;
00081     case DICT_INTEGERONLY:
00082         if ( a[na] != 1 ) goto NoAction;
00083         if ( na > 1 ) {
00084             for ( i = 1; i < na; i++ ) {
00085                 if ( a[na+i] != 0 ) goto NoAction;
00086             }
00087         }
00088     Numeratoronly:
00089         for ( i = dict->numelements-1; i >= 0; i-- ) {
00090             if ( dict->elements[i]->type == DICT_INTEGERNUMBER ) {
00091                 if ( dict->elements[i]->size == na ) {
00092                     for ( j = 0; j < na; j++ ) {
00093                         if ( (UWORD)(dict->elements[i]->lhs[j]) != a[j] ) break;
00094                     }
00095                     if ( j == na ) { /* Got it */
00096                         TokenToLine( (UBYTE *) (dict->elements[i]->rhs));
00097                         return;
00098                     }
00099                 }
00100             }
00101         }
00102     }
00103     NoAction:
00104     }
00105 }

```

```

00099         }
00100     }
00101 }
00102 goto NotFound;
00103 case DICT_RATIONALONLY:
00104     nb = 2*na;
00105     for ( i = dict->numelements-1; i >= 0; i-- ) {
00106         if ( dict->elements[i]->type == DICT_RATIONALNUMBER ) {
00107             if ( dict->elements[i]->size == nb+2 ) {
00108                 for ( j = 0; j < nb; j++ ) {
00109                     if ( (UWORD)(dict->elements[i]->lhs[j+1]) != a[j] ) break;
00110                 }
00111                 if ( j == nb ) { /* Got it */
00112                     TokenToLine((UBYTE *) (dict->elements[i]->rhs));
00113                     return;
00114                 }
00115             }
00116         }
00117     }
00118     goto NotFound;
00119 case DICT_ALLNUMBERS:
00120 /*
00121     First fish for rationals
00122 */
00123     nb = 2*na;
00124     for ( i = dict->numelements-1; i >= 0; i-- ) {
00125         if ( dict->elements[i]->type == DICT_RATIONALNUMBER ) {
00126             if ( dict->elements[i]->size == nb+2 ) {
00127                 for ( j = 0; j < nb; j++ ) {
00128                     if ( (UWORD)(dict->elements[i]->lhs[j+1]) != a[j] ) break;
00129                 }
00130                 if ( j == nb ) { /* Got it */
00131                     TokenToLine((UBYTE *) (dict->elements[i]->rhs));
00132                     return;
00133                 }
00134             }
00135         }
00136     }
00137 /*
00138     Now look for element[j1]/element[j2]
00139 */
00140     nb = na; b = a+na;
00141     while ( b[nb-1] == 0 ) nb--;
00142     if ( nb == 1 && b[0] == 1 ) goto Numeratoronly;
00143     while ( a[na-1] == 0 ) na--;
00144     for ( i1 = dict->numelements-1; i1 >= 0; i1-- ) {
00145         if ( dict->elements[i1]->type == DICT_INTEGERNUMBER ) {
00146             if ( dict->elements[i1]->size == na ) {
00147                 for ( j = 0; j < na; j++ ) {
00148                     if ( (UWORD)(dict->elements[i1]->lhs[j]) != a[j] ) break;
00149                 }
00150                 if ( j == na ) break;
00151             }
00152         }
00153     }
00154     for ( i2 = dict->numelements-1; i2 >= 0; i2-- ) {
00155         if ( dict->elements[i2]->type == DICT_INTEGERNUMBER ) {
00156             if ( dict->elements[i2]->size == nb ) {
00157                 for ( j = 0; j < nb; j++ ) {
00158                     if ( (UWORD)(dict->elements[i2]->lhs[j]) != b[j] ) break;
00159                 }
00160                 if ( j == nb ) break;
00161             }
00162         }
00163     }
00164     if ( i1 < 0 ) {
00165         if ( i2 < 0 ) goto NotFound;
00166         else { /* number/replacement[i2] */
00167             LongToLine(a,na);
00168             if ( na > 1 || ( AO.DoubleFlag & 4 ) == 4 ) {
00169                 if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
00170                     || AC.OutputMode == CMODE ) {
00171                     if ( ( AO.DoubleFlag & 2 ) == 2 ) { AddToLine((UBYTE *)".Q0/"); }
00172                     else if ( ( AO.DoubleFlag & 1 ) == 1 ) { AddToLine((UBYTE *)".D0/"); }
00173                     else { AddToLine((UBYTE *)"/"); }
00174                 }
00175             }
00176             else AddToLine((UBYTE *)("/"));
00177             TokenToLine((UBYTE *) (dict->elements[i2]->rhs));
00178         }
00179     }
00180     else if ( i2 < 0 ) { /* replacement[i1]/number */
00181         TokenToLine((UBYTE *) (dict->elements[i1]->rhs));
00182         AddToLine((UBYTE *)("/"));
00183         LongToLine((UWORD *) (b),nb);
00184         if ( nb > 1 || ( AO.DoubleFlag & 4 ) == 4 ) {
00185             if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE

```

```

00186         || AC.OutputMode == CMODE ) {
00187             if ( ( AO.DoubleFlag & 2 ) == 2 ) { AddToLine((UBYTE *)".Q0"); }
00188             else if ( ( AO.DoubleFlag & 1 ) == 1 ) { AddToLine((UBYTE *)".D0"); }
00189         }
00190     }
00191 }
00192 else { /* replacement[i1]/replacement[i2] */
00193     TokenToLine((UBYTE *) (dict->elements[i1]->rhs));
00194     AddToLine((UBYTE *) ("/"));
00195     TokenToLine((UBYTE *) (dict->elements[i2]->rhs));
00196 }
00197 break;
00198 default:
00199     MesPrint("Illegal code in TransformRational: %d",AO.CurDictNumbers);
00200     Terminate(-1);
00201 }
00202 return;
00203 NotFound:
00204 if ( na != 1 || a[1] != 1 ) {
00205     if ( AO.CurDictNumberWarning ) {
00206         MesPrint("»»»»»Could not translate coefficient with dictionary %s«««««",dict->name);
00207     }
00208 NoAction:
00209     RatToLine(a,na);
00210     return;
00211 }
00212
00213 /*
00214     #] TransformRational:
00215     #[ IsMultiplySign:
00216 */
00217 UBYTE *IsMultiplySign(void)
00218 {
00219     DICTIONARY *dict;
00220     int i;
00221     if ( AO.CurrentDictionary <= 0 ) return(0);
00222     dict = AO.Dictionaries[AO.CurrentDictionary-1];
00223     if ( dict->characters == 0 ) return(0);
00224     for ( i = dict->numelements-1; i >= 0; i-- ) {
00225         if ( ( dict->elements[i]->type == DICT_SPECIALCHARACTER )
00226             && ( dict->elements[i]->lhs[0] == (WORD)(' *' ) ) )
00227             return((UBYTE *) (dict->elements[i]->rhs));
00228     }
00229     return(0);
00230 }
00231
00232 /*
00233     #] IsMultiplySign:
00234     #[ IsExponentSign:
00235 */
00236 UBYTE *IsExponentSign(void)
00237 {
00238     DICTIONARY *dict;
00239     int i;
00240     if ( AO.CurrentDictionary <= 0 ) return(0);
00241     dict = AO.Dictionaries[AO.CurrentDictionary-1];
00242     if ( dict->characters == 0 ) return(0);
00243     for ( i = dict->numelements-1; i >= 0; i-- ) {
00244         if ( ( dict->elements[i]->type == DICT_SPECIALCHARACTER )
00245             && ( dict->elements[i]->lhs[0] == (WORD)(' ^' ) ) )
00246             return((UBYTE *) (dict->elements[i]->rhs));
00247     }
00248     return(0);
00249 }
00250
00251 /*
00252     #] IsExponentSign:
00253     #[ FindSymbol :
00254 */
00255 UBYTE *FindSymbol(WORD num)
00256 {
00257     if ( AO.CurrentDictionary > 0 ) {
00258         DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
00259         int i;
00260         if ( dict->variables > 0 && AO.CurDictVariables == DICT_DOVARIABLES ) {
00261             for ( i = dict->numelements-1; i >= 0; i-- ) {
00262                 if ( dict->elements[i]->type == DICT_SYMBOL &&
00263                     dict->elements[i]->lhs[0] == num )
00264                     return((UBYTE *) (dict->elements[i]->rhs));
00265             }
00266         }
00267     }
00268     return(VARNAME(symbols,num));
00269 }
00270
00271 }
00272
00273 }

```

```

00273
00274 /*
00275     #] FindSymbol :
00276     #[ FindVector :
00277 */
00278
00279 UBYTE *FindVector(WORD num)
00280 {
00281     if ( AO.CurrentDictionary > 0 ) {
00282         DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
00283         int i;
00284         if ( dict->variables > 0 && AO.CurDictVariables == DICT_DOVARIABLES ) {
00285             for ( i = dict->numelements-1; i >= 0; i-- ) {
00286                 if ( dict->elements[i]->type == DICT_VECTOR &&
00287                     dict->elements[i]->lhs[0] == num )
00288                     return((UBYTE *) (dict->elements[i]->rhs));
00289             }
00290         }
00291     }
00292     num -= AM.OffsetVector;
00293     return (VARNAME(vectors,num));
00294 }
00295
00296 /*
00297     #] FindVector :
00298     #[ FindIndex :
00299 */
00300
00301 UBYTE *FindIndex(WORD num)
00302 {
00303     if ( AO.CurrentDictionary > 0 ) {
00304         DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
00305         int i;
00306         if ( dict->variables > 0 && AO.CurDictVariables == DICT_DOVARIABLES ) {
00307             for ( i = dict->numelements-1; i >= 0; i-- ) {
00308                 if ( dict->elements[i]->type == DICT_INDEX &&
00309                     dict->elements[i]->lhs[0] == num )
00310                     return((UBYTE *) (dict->elements[i]->rhs));
00311             }
00312         }
00313     }
00314     num -= AM.OffsetIndex;
00315     return (VARNAME(indices,num));
00316 }
00317
00318 /*
00319     #] FindIndex :
00320     #[ FindFunction :
00321 */
00322
00323 UBYTE *FindFunction(WORD num)
00324 {
00325     if ( AO.CurrentDictionary > 0 ) {
00326         DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
00327         int i;
00328         if ( dict->variables > 0 && AO.CurDictVariables == DICT_DOVARIABLES ) {
00329             for ( i = dict->numelements-1; i >= 0; i-- ) {
00330                 if ( dict->elements[i]->type == DICT_FUNCTION &&
00331                     dict->elements[i]->lhs[0] == num )
00332                     return((UBYTE *) (dict->elements[i]->rhs));
00333             }
00334         }
00335     }
00336     num -= FUNCTION;
00337     return (VARNAME(functions,num));
00338 }
00339
00340 /*
00341     #] FindFunction :
00342     #[ FindFunWithArgs :
00343 */
00344
00345 UBYTE *FindFunWithArgs(WORD *t)
00346 {
00347     if ( AO.CurrentDictionary > 0 ) {
00348         DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
00349         int i, j;
00350         if ( dict->funwith > 0
00351             && AO.CurDictFunWithArgs == DICT_DOFUNWITHARGS ) {
00352             for ( i = dict->numelements-1; i >= 0; i-- ) {
00353                 if ( dict->elements[i]->type == DICT_FUNCTION_WITH_ARGUMENTS &&
00354                     (WORD)(dict->elements[i]->lhs[0]) == t[0] &&
00355                     (WORD)(dict->elements[i]->lhs[1]) == t[1] ) {
00356                     for ( j = 2; j < t[1]; j++ ) {
00357                         if ( (WORD)(dict->elements[i]->lhs[j]) != t[j] ) break;
00358                     }
00359                     if ( j >= t[1] ) return((UBYTE *) (dict->elements[i]->rhs));

```

```

00360     }
00361     }
00362     }
00363 }
00364 return(0);
00365 }
00366
00367 /*
00368 #] FindFunWithArgs :
00369 #[ FindExtraSymbol :
00370
00371     The extra symbol is constructed in the Workspace. This way we do not
00372     have to worry about Malloc and freeing the object later.
00373     The input value num is already the number of the extra symbol.
00374     We do NOT need num = MAXVARIABLES-num;
00375 */
00376
00377 UBYTE *FindExtraSymbol(WORD num)
00378 {
00379     GETIDENTITY;
00380     UBYTE *out = (UBYTE *) (AT.WorkPointer);
00381     *out = 0;
00382     if ( AO.CurrentDictionary > 0 ) {
00383         DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
00384         int i;
00385         if ( dict->ranges > 0 && AO.CurDictVariables == DICT_DOVARIABLES ) {
00386             for ( i = dict->numelements-1; i >= 0; i-- ) {
00387                 if ( dict->elements[i]->type == DICT_RANGE
00388                     && num >= dict->elements[i]->lhs[0]
00389                     && num <= dict->elements[i]->lhs[1] ) {
00390                     /*
00391                         Now we have to translate the rhs
00392                         %# gives the number
00393                         %@ gives the number as its position in the range
00394                     */
00395                     UBYTE *r = (UBYTE *) (dict->elements[i]->rhs);
00396                     while ( *r ) {
00397                         if ( *r == (UBYTE) '%' && ( r[1] == (UBYTE) '#'
00398                             || r[1] == (UBYTE) '@' ) ) {
00399                             if ( r[1] == (UBYTE) '#' ) {
00400                                 out = NumCopy(num,out);
00401                             }
00402                             else {
00403                                 out = NumCopy(num-dict->elements[i]->lhs[0]+1,out);
00404                             }
00405                             r += 2;
00406                         }
00407                         else {
00408                             *out++ = *r++;
00409                         }
00410                     }
00411                     *out = 0;
00412                     return((UBYTE *) (AT.WorkPointer));
00413                 }
00414             }
00415         }
00416     }
00417
00418     out = StrCopy((UBYTE *) AC.extrasym,out);
00419     if ( AC.extrasymbols == 0 ) {
00420         out = NumCopy(num,out);
00421         out = StrCopy((UBYTE *) "_",out);
00422     }
00423     else if ( AC.extrasymbols == 1 ) {
00424         out = AddArrayIndex(num,out);
00425     }
00426     return((UBYTE *) (AT.WorkPointer));
00427 }
00428
00429 /*
00430 #] FindExtraSymbol :
00431 #[ FindDictionary :
00432 */
00433
00434 int FindDictionary(UBYTE *name)
00435 {
00436     int i;
00437     for ( i = 0; i < AO.NumDictionaries; i++ ) {
00438         if ( StrCmp(AO.Dictionaries[i]->name,name) == 0 )
00439             return(i+1);
00440     }
00441     return(0);
00442 }
00443
00444 /*
00445 #] FindDictionary :
00446 #[ AddDictionary :

```

```

00447 */
00448
00449 int AddDictionary(UBYTE *name)
00450 {
00451     DICTIONARY *dict;
00452     /*
00453     First make space for the pointer in the list.
00454     */
00455     if ( AO.NumDictionaries >= AO.SizeDictionaries-1 ) {
00456         DICTIONARY **d;
00457         int i;
00458         if ( AO.SizeDictionaries <= 0 ) AO.SizeDictionaries = 10;
00459         else AO.SizeDictionaries = 2*AO.SizeDictionaries;
00460         d = (DICTIONARY **)Malloc1(AO.SizeDictionaries*sizeof(DICTIONARY *),"Dictionaries");
00461         for ( i = 0; i < AO.NumDictionaries; i++ ) d[i] = AO.Dictionaries[i];
00462         if ( AO.Dictionaries != 0 ) M_free(AO.Dictionaries,"Dictionaries");
00463         AO.Dictionaries = d;
00464     }
00465     /*
00466     Now create an empty dictionary.
00467     */
00468     dict = (DICTIONARY *)Malloc1(sizeof(DICTIONARY),"Dictionary");
00469     AO.Dictionaries[AO.NumDictionaries++] = dict;
00470     dict->elements = 0;
00471     dict->name = strDup1(name,"DictionaryName");
00472     dict->sizeelements = 0;
00473     dict->numelements = 0;
00474     dict->numbers = 0;
00475     dict->variables = 0;
00476     dict->characters = 0;
00477     dict->funwith = 0;
00478     dict->gnumelements = 0;
00479     dict->ranges = 0;
00480
00481     return(AO.NumDictionaries);
00482 }
00483
00484 /*
00485 #] AddDictionary :
00486 #[ AddToDictionary :
00487
00488     To be called from #add left:right
00489 */
00490
00491 int AddToDictionary(DICTIONARY *dict,UBYTE *left,UBYTE *right)
00492 {
00493     GETIDENTITY
00494     CBUF *C = cbuf+AC.cbunum;
00495     WORD *w = AT.WorkPointer;
00496     WORD *OldWork = AT.WorkPointer;
00497     WORD *s, oldnumrhs = C->numrhs, oldnumlhs = C->numlhs;
00498     WORD *ow, *ww, *mm, oldEside, *where = 0, type, number, range[3];
00499     LONG oldcpointer;
00500     int error = 0, sizelhs, sizerrhs, i, retcode;
00501     UBYTE *r;
00502     DICTIONARY_ELEMENT *new;
00503     WORD power = (WORD)('^'), times = (WORD)('*');
00504     if ( ( left[0] == '^' && left[1] == 0 )
00505         || ( left[0] == '*' && left[1] == '*' && left[2] == 0 ) ) {
00506         type = DICT_SPECIALCHARACTER;
00507         number = 1;
00508         where = &power;
00509         goto TestDouble;
00510     }
00511     else if ( left[0] == '*' && left[1] == 0 ) {
00512         type = DICT_SPECIALCHARACTER;
00513         number = 1;
00514         where = &times;
00515         goto TestDouble;
00516     }
00517     else if ( left[0] == '(' ) { /* range of extra symbols */
00518         WORD x1 = 0, x2 = 0;
00519         r = left+1;
00520         while ( FG.cTable[*r] == 1 ) x1 = 10*x1 + *r++ - '0';
00521         if ( *r == ',' ) {
00522             r++;
00523             while ( FG.cTable[*r] == 1 ) x2 = 10*x2 + *r++ - '0';
00524         }
00525         else x2 = x1;
00526         number = 2;
00527         if ( *r != ')' ) {
00528             MesPrint("&Illegal range specification in LHS of %#add instruction.");
00529             return(1);
00530         }
00531         type = DICT_RANGE;
00532         if ( x1 <= 0 || x2 <= 0 || x1 > x2 ) {
00533             MesPrint("&Illegal range in LHS of %#add instruction.");

```

```

00534         return(1);
00535     }
00536     range[0] = x1;
00537     range[1] = x2;
00538     range[2] = 0;
00539     where = range;
00540     goto TestDouble;
00541 }
00542 /*
00543 Translate the left part. Determine type.
00544 We follow the code in ColdExpression and then veto what we do not like.
00545 Just make sure to pop what needs to be popped in the compiler buffer.
00546 */
00547 AC.ProtoType = w;
00548 *w++ = SUBEXPRESSION;
00549 *w++ = SUBEXPSIZE;
00550 *w++ = C->numrhs+1;
00551 *w++ = 1;
00552 *w++ = AC.cbufnum;
00553 FILLSUB(w)
00554 AC.WildC = w;
00555 AC.NwildC = 0;
00556 AT.WorkPointer = s = w + 4*AM.MaxWildcards + 8;
00557 /*
00558 Now read the LHS
00559 */
00560 oldcpointer = AddLHS(AC.cbufnum) - C->Buffer;
00561
00562 if ( ( retcode = CompileAlgebra(left,LHSIDE,AC.ProtoType) ) < 0 ) { error = 1; }
00563 else AC.ProtoType[2] = retcode;
00564 AT.WorkPointer = s;
00565 if ( AC.NwildC && SortWild(w,AC.NwildC) ) error = 1;
00566
00567 OldWork[1] = AC.WildC-OldWork;
00568 w = AC.WildC;
00569 AT.WorkPointer = w;
00570 s = C->rhs[C->numrhs];
00571 /*
00572 We have the expression in the compiler buffers.
00573 The main level is at lhs[numlhs]
00574 The partial lhs (including ProtoType) is in OldWork (in Workspace)
00575 We need to load the result at w after the prototype
00576 Because these sort routines don't use the Workspace
00577 there should not be a conflict
00578 */
00579 if ( !error && *s == 0 ) {
00580 IllLeft:MesPrint("&Illegal LHS in dictionary");
00581 AC.lhdollarflag = 0;
00582 return(1);
00583 }
00584 if ( !error && *(s+s) != 0 ) {
00585 MesPrint("&LHS in dictionary should be one term only");
00586 return(1);
00587 }
00588 if ( error == 0 ) {
00589 if ( NewSort(BHEAD0) || NewSort(BHEAD0) ) {
00590 if ( !error ) error = 1;
00591 return(error);
00592 }
00593 AN.RepPoint = AT.RepCount + 1;
00594 ow = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
00595 mm = s; ww = ow; i = *mm;
00596 while ( --i >= 0 ) { *ww++ = *mm++; } AT.WorkPointer = ww;
00597 AC.lhdollarflag = 0; oldEside = AR.Eside; AR.Eside = LHSIDE;
00598 AR.Cnumlhs = C->numlhs;
00599 if ( Generator(BHEAD ow,C->numlhs) ) {
00600 AR.Eside = oldEside;
00601 LowerSortLevel(); LowerSortLevel(); goto IllLeft;
00602 }
00603 AR.Eside = oldEside;
00604 AT.WorkPointer = w;
00605 if ( EndSort(BHEAD w,0) < 0 ) { LowerSortLevel(); goto IllLeft; }
00606 if ( *w == 0 || *(w+w) != 0 ) {
00607 MesPrint("&LHS must be one term");
00608 AC.lhdollarflag = 0;
00609 return(1);
00610 }
00611 LowerSortLevel();
00612 }
00613 AT.WorkPointer = w + *w;
00614 AC.DumNum = 0;
00615 /*
00616 Everything is now after OldWork. We can pop the compilerbuffer.
00617 Next test for illegal things like a coefficient
00618 At this point we have:
00619 w = the term of the LHS
00620 */

```

```

00621     C->Pointer = C->Buffer + oldcpointer;
00622     C->numrhs = oldnumrhs;
00623     C->numlhs = oldnumlhs;
00624     AC.lhdollarflag = 0;
00625     /*
00626     Test for undesirables.
00627         1: wildcards
00628         2: sign
00629         3: more than one term
00630         4: composite terms
00631     */
00632     if ( AC.ProtoType[1] != SUBEXPSIZE ) {
00633         MesPrint("& Currently no wildcards allowed in dictionaries.");
00634         return(1);
00635     }
00636     if ( w[w[0]-1] < 0 ) {
00637         MesPrint("& Currently no sign allowed in dictionaries.");
00638         return(1);
00639     }
00640     if ( w[w[0]] != 0 ) {
00641         MesPrint("& More than one term in dictionary element.");
00642         return(1);
00643     }
00644     if ( w[0] == w[w[0]-1]+1 ) { /* Only coefficient */
00645         WORD *number, *denom;
00646         WORD nsize, dsize;
00647         nsize = dsize = (w[w[0]-1]-1)/2;
00648         number = w+1;
00649         denom = number+nsize;
00650         while ( number[nsize-1] == 0 ) nsize--;
00651         while ( denom[dsize-1] == 0 ) dsize--;
00652         if ( dsize == 1 && denom[0] == 1 ) {
00653             type = DICT_INTEGERNUMBER;
00654             number = nsize;
00655             where = number;
00656         }
00657         else {
00658             type = DICT_RATIONALNUMBER;
00659             number = w[0];
00660             where = w;
00661         }
00662     }
00663     else {
00664         s = w + w[0]-1;
00665         if ( s[0] != 3 || s[-1] != 1 || s[-2] != 1 ) {
00666 Compositeness:;
00667             MesPrint("& Currently no composite objects allowed in dictionaries.");
00668             return(1);
00669         }
00670         if ( w[0] != w[2]+4 ) goto Compositeness;
00671         s = w+1;
00672         switch ( *s ) {
00673             case SYMBOL:
00674                 if ( s[1] != 4 || s[3] != 1 ) goto Compositeness;
00675                 type = DICT_SYMBOL;
00676                 number = 1;
00677                 where = s+2;
00678                 break;
00679             case INDEX:
00680                 if ( s[1] != 3 ) goto Compositeness;
00681                 if ( s[2] < 0 ) type = DICT_VECTOR;
00682                 else type = DICT_INDEX;
00683                 number = 1;
00684                 where = s+2;
00685                 break;
00686             default:
00687                 if ( *s < FUNCTION ) {
00688                     MesPrint("& Illegal object in dictionary.");
00689                     return(1);
00690                 }
00691                 if ( s[1] == FUNHEAD ) {
00692                     type = DICT_FUNCTION;
00693                     number = 1;
00694                     where = s;
00695                     break;
00696                 }
00697                 else {
00698                     type = DICT_FUNCTION_WITH_ARGUMENTS;
00699                     number = s[1];
00700                     where = s;
00701                 }
00702                 break;
00703         }
00704     }
00705 TestDouble:;
00706     /*
00707     Create a new element

```



```

00708 */
00709     if ( dict->numelements >= dict->sizeelements ) {
00710         DICTIONARY_ELEMENT **d;
00711         if ( dict->sizeelements <= 0 ) dict->sizeelements = 10;
00712         else dict->sizeelements *= 2;
00713         d = (DICTIONARY_ELEMENT **)Malloc1(
00714             sizeof(DICTIONARY_ELEMENT *)*dict->sizeelements,"Dictionary elements");
00715         for ( i = 0; i < dict->numelements; i++ )
00716             d[i] = dict->elements[i];
00717         if ( dict->elements ) M_free(dict->elements,"Dictionary elements");
00718         dict->elements = d;
00719     }
00720     sizelhs = number+1;
00721     sizerhs = 1; r = right; while ( *r++ ) sizerhs++;
00722     sizerhs = (sizerhs+sizeof(WORD)-1)/sizeof(WORD)+1;
00723     new = (DICTIONARY_ELEMENT *)Malloc1(sizeof(DICTIONARY_ELEMENT)
00724         +sizeof(WORD)*(sizelhs+sizerhs),"Dictionary element");
00725     new->lhs = (WORD *) (new+1);
00726     new->rhs = new->lhs+sizelhs;
00727     new->type = type;
00728     new->size = number;
00729     for ( i = 0; i < number; i++ ) new->lhs[i] = where[i];
00730     new->lhs[i] = 0;
00731     r = (UBYTE *) (new->rhs);
00732     while ( *right ) {
00733         if ( *right == '\\' && ( right[1] == '\'' || right[1] == '\"' ) ) right++;
00734         *r++ = *right++;
00735     }
00736     *r = 0;
00737
00738     dict->elements[dict->numelements++] = new;
00739
00740     switch ( type ) {
00741         case DICT_INTEGERNUMBER:
00742         case DICT_RATIONALNUMBER:
00743             dict->numbers++; break;
00744         case DICT_SYMBOL:
00745         case DICT_VECTOR:
00746         case DICT_INDEX:
00747         case DICT_FUNCTION:
00748             dict->variables++; break;
00749         case DICT_FUNCTION_WITH_ARGUMENTS:
00750             dict->funwith++; break;
00751         case DICT_SPECIALCHARACTER:
00752             dict->characters++; break;
00753         case DICT_RANGE:
00754             dict->ranges++; break;
00755     }
00756
00757     AT.WorkPointer = OldWork;
00758     return(0);
00759 }
00760
00761 /*
00762 #] AddToDictionary :
00763 #[ UseDictionary :
00764 */
00765
00766 int UseDictionary(UBYTE *name,UBYTE *options)
00767 {
00768     int i;
00769     for ( i = 0; i < AO.NumDictionaries; i++ ) {
00770         if ( StrCmp(AO.Dictionaries[i]->name,name) == 0 ) {
00771             AO.CurrentDictionary = i+1;
00772             if ( SetDictionaryOptions(options) < 0 ) {
00773                 AO.CurrentDictionary = 0;
00774                 return(-1);
00775             }
00776             else { /* Now test whether what is requested is really there? */
00777                 return(0);
00778             }
00779         }
00780     }
00781     MesPrint("@There is no dictionary with the name %s",name);
00782     exit(-1);
00783 }
00784
00785 /*
00786 #] UseDictionary :
00787 #[ SetDictionaryOptions :
00788 */
00789
00790 int SetDictionaryOptions(UBYTE *options)
00791 {
00792     UBYTE *opt, *s, c;
00793     int retval = 0;
00794     s = options;

```

```

00795     AO.CurDictNumbers = DICT_ALLNUMBERS;
00796     AO.CurDictVariables = DICT_DOVARIABLES;
00797     AO.CurDictSpecials = DICT_DOSPECIALS;
00798     AO.CurDictFunWithArgs = DICT_DOFUNWITHARGS;
00799     AO.CurDictNumberWarning = 0;
00800     AO.CurDictNotInFunctions= 0;
00801     AO.CurDictInDollars = DICT_NOTINDOLLARS;
00802     while ( *s ) {
00803         opt = s;
00804         while ( *s && *s != ',' && *s != ' ' ) s++;
00805         c = *s; *s = 0;
00806         if ( opt[0] == '$' && opt[1] == 0 ) {
00807             AO.CurDictInDollars = DICT_INDOLLARS;
00808         }
00809         else if ( StrICmp(opt,(UBYTE *)"nonumbers") == 0 ) {
00810             AO.CurDictNumbers = DICT_NONUMBERS;
00811         }
00812         else if ( StrICmp(opt,(UBYTE *)"integersonly") == 0 ) {
00813             AO.CurDictNumbers = DICT_INTEGERONLY;
00814         }
00815         else if ( StrICmp(opt,(UBYTE *)"rationalonly") == 0 ) {
00816             AO.CurDictNumbers = DICT_RATIONALONLY;
00817         }
00818         else if ( StrICmp(opt,(UBYTE *)"allnumbers") == 0 ) {
00819             AO.CurDictNumbers = DICT_ALLNUMBERS;
00820         }
00821         else if ( StrICmp(opt,(UBYTE *)"novariables") == 0 ) {
00822             AO.CurDictVariables = DICT_NOVARIABLES;
00823         }
00824         else if ( StrICmp(opt,(UBYTE *)"numberonly") == 0 ) {
00825             AO.CurDictNumbers = DICT_ALLNUMBERS;
00826             AO.CurDictVariables = DICT_NOVARIABLES;
00827             AO.CurDictSpecials = DICT_NOSPECIALS;
00828             AO.CurDictFunWithArgs = DICT_NOFUNWITHARGS;
00829         }
00830         else if ( StrICmp(opt,(UBYTE *)"variablesonly") == 0 ) {
00831             AO.CurDictNumbers = DICT_NONUMBERS;
00832             AO.CurDictVariables = DICT_DOVARIABLES;
00833             AO.CurDictSpecials = DICT_NOSPECIALS;
00834             AO.CurDictFunWithArgs = DICT_NOFUNWITHARGS;
00835         }
00836         else if ( StrICmp(opt,(UBYTE *)"nospecials") == 0 ) {
00837             AO.CurDictSpecials = DICT_NOSPECIALS;
00838         }
00839         else if ( StrICmp(opt,(UBYTE *)"specialonly") == 0 ) {
00840             AO.CurDictNumbers = DICT_NONUMBERS;
00841             AO.CurDictVariables = DICT_NOVARIABLES;
00842             AO.CurDictSpecials = DICT_DOSPECIALS;
00843             AO.CurDictFunWithArgs = DICT_NOFUNWITHARGS;
00844         }
00845         else if ( StrICmp(opt,(UBYTE *)"nofunwithargs") == 0 ) {
00846             AO.CurDictFunWithArgs = DICT_NOFUNWITHARGS;
00847         }
00848         else if ( StrICmp(opt,(UBYTE *)"funwithargsonly") == 0 ) {
00849             AO.CurDictNumbers = DICT_NONUMBERS;
00850             AO.CurDictVariables = DICT_NOVARIABLES;
00851             AO.CurDictSpecials = DICT_NOSPECIALS;
00852             AO.CurDictFunWithArgs = DICT_DOFUNWITHARGS;
00853         }
00854         else if ( StrICmp(opt,(UBYTE *)"warnings") == 0
00855             || StrICmp(opt,(UBYTE *)"warning") == 0 ) {
00856             AO.CurDictNumberWarning = 1;
00857         }
00858         else if ( StrICmp(opt,(UBYTE *)"nowarnings") == 0
00859             || StrICmp(opt,(UBYTE *)"nowarning") == 0 ) {
00860             AO.CurDictNumberWarning = 0;
00861         }
00862         else if ( StrICmp(opt,(UBYTE *)"infunctions") == 0 ) {
00863             AO.CurDictNotInFunctions= 0;
00864         }
00865         else if ( StrICmp(opt,(UBYTE *)"notinfunctions") == 0 ) {
00866             AO.CurDictNotInFunctions= 1;
00867         }
00868         else {
00869             MesPrint("@ Unrecognized option in %#SetDictionary: %s",opt);
00870             retval = -1;
00871         }
00872         *s = c;
00873         if ( c == ',' ) s++;
00874     }
00875     return(retval);
00876 }
00877
00878 /*
00879 #] SetDictionaryOptions :
00880 #[ UnSetDictionary :
00881 */

```

```

00882
00883 void UnSetDictionary(void)
00884 {
00885     AO.CurrentDictionary = 0;
00886     AO.CurDictNumbers = -1;
00887     AO.CurDictVariables = -1;
00888     AO.CurDictSpecials = -1;
00889     AO.CurDictFunWithArgs = -1;
00890     AO.CurDictFunWithArgs = -1;
00891     AO.CurDictNumberWarning = -1;
00892     AO.CurDictNotInFunctions= -1;
00893 }
00894
00895 /*
00896 #] UnSetDictionary :
00897 #[ RemoveDictionary :
00898
00899     Mostly needed for .clear
00900 */
00901
00902 void RemoveDictionary(DICTIONARY *dict)
00903 {
00904     int i;
00905     if ( dict == 0 ) return;
00906     for ( i = 0; i < AO.NumDictionaries; i++ ) {
00907         if ( AO.Dictionaries[i] == dict ) {
00908             for ( i++; i < AO.NumDictionaries; i++ ) {
00909                 AO.Dictionaries[i-1] = AO.Dictionaries[i];
00910             }
00911             AO.NumDictionaries--;
00912             goto removeit;
00913         }
00914     }
00915     MesPrint("@ Dictionary not found in RemoveDictionary");
00916     exit(-1);
00917 removeit:;
00918     for ( i = 0; i < dict->numelements; i++ )
00919         M_free(dict->elements[i], "Dictionary element");
00920     for ( i = 0; i < dict->numelements; i++ ) dict->elements[i] = 0;
00921     if ( dict->elements ) M_free(dict->elements, "Dictionary elements");
00922     if ( dict->name ) {
00923         M_free(dict->name, "DictionaryName");
00924         dict->name = 0;
00925     }
00926     dict->sizeelements = 0;
00927     dict->numelements = 0;
00928     dict->numbers = 0;
00929     dict->variables = 0;
00930     dict->characters = 0;
00931     dict->funwith = 0;
00932     dict->gnumelements = 0;
00933     dict->ranges = 0;
00934 }
00935
00936 /*
00937 #] RemoveDictionary :
00938 #[ ShrinkDictionary :
00939
00940     To be called after a .store to restore the dictionary to the state
00941     it had at the last .global
00942     We do not make the elements array shorter.
00943 */
00944
00945 void ShrinkDictionary(DICTIONARY *dict)
00946 {
00947     while ( dict->numelements > dict->gnumelements ) {
00948         dict->numelements--;
00949         M_free(dict->elements[dict->numelements], "Dictionary element");
00950         dict->elements[dict->numelements] = 0;
00951     }
00952 }
00953
00954 /*
00955 #] ShrinkDictionary :
00956 #[ DoPreOpenDictionary :
00957 */
00958
00959 int DoPreOpenDictionary(UBYTE *s)
00960 {
00961     UBYTE *name;
00962     int dict;
00963     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
00964     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
00965     while ( *s == ' ' ) s++;
00966
00967     name = s; s = SkipAName(s);
00968     if ( *s != 0 && *s != ';' ) {

```

```

00969         MesPrint("@proper syntax is #opendictionary name");
00970         return(-1);
00971     }
00972     *s = 0;
00973
00974     if ( AP.OpenDictionary > 0 ) {
00975         MesPrint("@you cannot nest #opendictionary instructions");
00976         MesPrint("@dictionary %s is open already",
00977             AO.Dictionaries[AP.OpenDictionary-1]->name);
00978         return(-1);
00979     }
00980     if ( AO.CurrentDictionary > 0 ) {
00981         MesPrint("@before opening a dictionary you have to first close the selected dictionary");
00982         return(-1);
00983     }
00984     /*
00985     Do we have this dictionary already?
00986     */
00987     dict = FindDictionary(name);
00988     if ( dict == 0 ) dict = AddDictionary(name);
00989     AP.OpenDictionary = dict;
00990     return(0);
00991 }
00992
00993 /*
00994 #] DoPreOpenDictionary :
00995 #[ DoPreCloseDictionary :
00996 */
00997 int DoPreCloseDictionary(UBYTE *s)
00998 {
01000     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
01001     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
01002     while ( *s == ' ' ) s++;
01003
01004     if ( AP.OpenDictionary == 0 && AO.CurrentDictionary == 0 ) {
01005         MesPrint("@you have neither an open, nor a selected dictionary");
01006         return(-1);
01007     }
01008
01009     AP.OpenDictionary = 0;
01010     AO.CurrentDictionary = 0;
01011
01012     AO.CurDictNotInFunctions = 0;
01013
01014     return(0);
01015 }
01016
01017 /*
01018 #] DoPreCloseDictionary :
01019 #[ DoPreUseDictionary :
01020 */
01021 int DoPreUseDictionary(UBYTE *s)
01022 {
01023     UBYTE *options, c, *ss, *sss, *name;
01024     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
01025     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
01026     while ( *s == ' ' ) s++;
01027
01028     if ( AP.OpenDictionary > 0 ) {
01029         MesPrint("@before selecting a dictionary you have to first close the open dictionary");
01030         return(-1);
01031     }
01032
01033     name = s; s = SkipAName(s);
01034     ss = s; while ( *s && *s != ' (' ) s++;
01035     c = *ss; *ss = 0;
01036     if ( c == 0 ) {
01037         options = ss;
01038     }
01039     else {
01040         options = s+1; SKIPBRA3(s)
01041         if ( *s != ')' ) {
01042             MesPrint("@Irregular end of %#UseDictionary instruction");
01043             return(-1);
01044         }
01045         sss = s;
01046         s++; while ( *s == ' ' || *s == '\t' || *s == ';' ) s++;
01047         *sss = 0;
01048         if ( *s ) {
01049             MesPrint("@Irregular end of %#UseDictionary instruction");
01050             return(-1);
01051         }
01052     }
01053 }
01054 return(UseDictionary(name,options));
01055 }

```

```

01056
01057 /*
01058     #] DoPreUseDictionary :
01059     #[ DoPreAdd :
01060
01061     Syntax:
01062         #add left :right
01063         #add left : "right"
01064     Adds to the currently open dictionary
01065 */
01066
01067 int DoPreAdd(UBYTE *s)
01068 {
01069     UBYTE *left, *right;
01070
01071     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
01072     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
01073     while ( *s == ' ' ) s++;
01074
01075     if ( AP.OpenDictionary == 0 ) {
01076         MesPrint("@there is no open dictionary to add to");
01077         return(-1);
01078     }
01079     /*
01080     Scan to the : and mark the left and right parts.
01081     */
01082     left = s;
01083     while ( *s && *s != ':' ) {
01084         if ( *s == '[' ) { SKIPBRA1(s) s++; }
01085         else if ( *s == '{' ) { SKIPBRA2(s) s++; }
01086         else if ( *s == '(' ) { SKIPBRA3(s) s++; }
01087         else if ( *s == ']' || *s == '}' || *s == ')' ) {
01088             MesPrint("@unmatched brackets in #add instruction");
01089             return(-1);
01090         }
01091         else s++;
01092     }
01093     if ( *s == 0 ) {
01094         MesPrint("@Missing : in #add instruction");
01095         return(-1);
01096     }
01097     *s++ = 0;
01098     right = s;
01099     while ( *s == ' ' || *s == '\t' ) s++;
01100     if ( *s == '"' && s[1] ) {
01101         right = s+1;
01102         s = s+2;
01103         while ( *s ) s++;
01104         while ( s[-1] != '"' ) s--;
01105         if ( s <= right ) {
01106             MesPrint("@Irregular use of double quotes in #add instruction");
01107             return(-1);
01108         }
01109         s[-1] = 0;
01110     }
01111     return(AddToDictionary(AO.Dictionaries[AP.OpenDictionary-1],left,right));
01112 }
01113
01114 /*
01115     #] DoPreAdd :
01116     #[ DictToBytes :
01117 */
01118
01119 LONG DictToBytes(DICTIONARY *dict,UBYTE *buf)
01120 {
01121     int numelements = dict->numelements, sizeelement, i, j, x;
01122     UBYTE *s1, *s2 = buf;
01123     DICTIONARY_ELEMENT *e;
01124     /*
01125     First copy the struct
01126     */
01127     s1 = (UBYTE *)dict; j = sizeof(DICTIONARY);
01128     NCOPY(s2,s1,j)
01129     /*
01130     Now the elements. Put a size indicator in front of each of them.
01131     */
01132     for ( i = 0; i < numelements; i++ ) {
01133         e = dict->elements[i];
01134         sizeelement = sizeof(DICTIONARY_ELEMENT)+(e->size+1)*sizeof(WORD);
01135         s1 = (UBYTE *)e->rhs; x = 0;
01136         while ( *s1 ) { s1++; x++; }
01137         x /= sizeof(WORD);
01138         sizeelement += (x+1) * sizeof(WORD);
01139         s1 = (UBYTE *)(&sizeelement); j = sizeof(WORD); NCOPY(s2,s1,j)
01140         s1 = (UBYTE *)e; j = sizeof(DICTIONARY_ELEMENT); NCOPY(s2,s1,j)
01141         s1 = (UBYTE *)e->lhs; j = (e->size+1)*(sizeof(WORD)); NCOPY(s2,s1,j)
01142         s1 = (UBYTE *)e->rhs; j = (x+1)*(sizeof(WORD)); NCOPY(s2,s1,j)

```

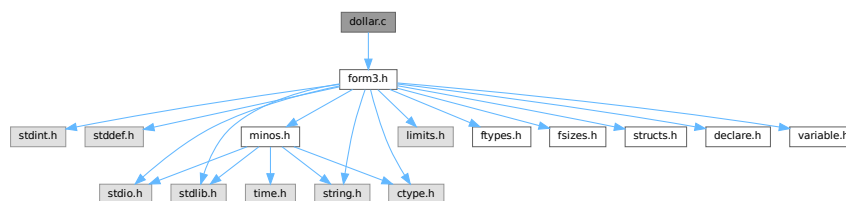
```

01143     }
01144     return(s2-buf);
01145 }
01146
01147 /*
01148  #] DictToBytes :
01149  #[ DictFromBytes :
01150 */
01151
01152 DICTIONARY *DictFromBytes(UBYTE *buf)
01153 {
01154     DICTIONARY *dict = Malloc1(sizeof(DICTIONARY), "Dictionary");
01155     UBYTE *s1, *s2;
01156     int i, j, sizeelement;
01157     DICTIONARY_ELEMENT *e;
01158     /*
01159     First read the dictionary itself
01160     */
01161     s1 = buf;
01162     s2 = (UBYTE *)dict; j = sizeof(DICTIONARY); NCOPY(s2, s1, j)
01163     /*
01164     Allocate the elements array:
01165     */
01166     dict->elements = (DICTIONARY_ELEMENT **)Malloc1(
01167         sizeof(DICTIONARY_ELEMENT *)*dict->sizeelements, "dictionary elements");
01168     for ( i = 0; i < dict->numelements; i++ ) {
01169         s2 = (UBYTE *)(&sizeelement); j = sizeof(WORD); NCOPY(s2, s1, j)
01170         e = (DICTIONARY_ELEMENT *)Malloc1(sizeelement*sizeof(UBYTE), "dictionary element");
01171         dict->elements[i] = e;
01172         j = sizeelement; s2 = (UBYTE *)e; NCOPY(s2, s1, j)
01173         e->lhs = (WORD *) (e+1);
01174         e->rhs = e->lhs + e->size+1;
01175     }
01176     return(dict);
01177 }
01178
01179 /*
01180  #] DictFromBytes :
01181 */

```

## 4.26 dollar.c File Reference

#include "form3.h"  
 Include dependency graph for dollar.c:



### Macros

- #define [STEP2](#)

### Functions

- int [CatchDollar](#) (int par)
- int [AssignDollar](#) (PHEAD WORD \*term, WORD level)
- UBYTE \* [WriteDollarToBuffer](#) (WORD numdollar, WORD par)
- UBYTE \* [WriteDollarFactorToBuffer](#) (WORD numdollar, WORD numfac, WORD par)

- void [AddToDollarBuffer](#) (UBYTE \*s)
- void [TermAssign](#) (WORD \*term)
- int [PutTermInDollar](#) (WORD \*term, WORD numdollar)
- void [WildDollars](#) (PHEAD WORD \*term)
- WORD [DolToTensor](#) (PHEAD WORD numdollar)
- WORD [DolToFunction](#) (PHEAD WORD numdollar)
- WORD [DolToVector](#) (PHEAD WORD numdollar)
- WORD [DolToNumber](#) (PHEAD WORD numdollar)
- WORD [DolToSymbol](#) (PHEAD WORD numdollar)
- WORD [DolToIndex](#) (PHEAD WORD numdollar)
- [DOLLARS DolToTerms](#) (PHEAD WORD numdollar)
- LONG [DolToLong](#) (PHEAD WORD numdollar)
- int [ExecInside](#) (UBYTE \*s)
- int [InsideDollar](#) (PHEAD WORD \*ll, WORD level)
- void [ExchangeDollars](#) (int num1, int num2)
- LONG [TermsInDollar](#) (WORD num)
- LONG [SizeOfDollar](#) (WORD num)
- UBYTE \* [PrelfDollarEval](#) (UBYTE \*s, int \*value)
- WORD \* [TranslateExpression](#) (UBYTE \*s)
- int [IsSetMember](#) (WORD \*buffer, WORD numset)
- int [IsMultipleOf](#) (WORD \*buf1, WORD \*buf2)
- int [TwoExprCompare](#) (WORD \*buf1, WORD \*buf2, int oprtr)
- int [DollarRaiseLow](#) (UBYTE \*name, LONG value)
- WORD [EvalDoLoopArg](#) (PHEAD WORD \*arg, WORD par)
- WORD [TestDoLoop](#) (PHEAD WORD \*lhsbuf, WORD level)
- WORD [TestEndDoLoop](#) (PHEAD WORD \*lhsbuf, WORD level)
- int [DollarFactorize](#) (PHEAD WORD numdollar)
- void [CleanDollarFactors](#) ([DOLLARS](#) d)
- WORD \* [TakeDollarContent](#) (PHEAD WORD \*dollarbuffer, WORD \*\*factor)
- WORD \* [MakeDollarInteger](#) (PHEAD WORD \*bufin, WORD \*\*bufout)
- WORD \* [MakeDollarMod](#) (PHEAD WORD \*buffer, WORD \*\*bufout)
- int [GetDolNum](#) (PHEAD WORD \*t, WORD \*tstop)
- void [AddPotModdollar](#) (WORD numdollar)

### 4.26.1 Detailed Description

The routines that deal with the dollar variables. The name administration is to be found in the file [names.c](#)

Definition in file [dollar.c](#).

### 4.26.2 Macro Definition Documentation

#### 4.26.2.1 STEP2

```
#define STEP2
```

Factors a dollar expression. Notation: d->nfactors becomes nonzero. if the number of factors is one, we leave d->factors zero. Otherwise factors is an array of pointers to the factors. These are pointers of the type FAC-DOLLAR. fd->where pointer to contents in term notation fd->size size of the buffer fd->where points to fd->type DOLNUMBER or DOLTERMS fd->value value if type is DOLNUMBER and it fits in a WORD.

Definition at line 2953 of file [dollar.c](#).

## 4.26.3 Function Documentation

### 4.26.3.1 CatchDollar()

```
int CatchDollar (
    int par )
```

Definition at line 54 of file [dollar.c](#).

### 4.26.3.2 AssignDollar()

```
int AssignDollar (
    PHEAD WORD * term,
    WORD level )
```

Definition at line 209 of file [dollar.c](#).

### 4.26.3.3 WriteDollarToBuffer()

```
UBYTE * WriteDollarToBuffer (
    WORD numdollar,
    WORD par )
```

Definition at line 596 of file [dollar.c](#).

### 4.26.3.4 WriteDollarFactorToBuffer()

```
UBYTE * WriteDollarFactorToBuffer (
    WORD numdollar,
    WORD numfac,
    WORD par )
```

Definition at line 687 of file [dollar.c](#).

### 4.26.3.5 AddToDollarBuffer()

```
void AddToDollarBuffer (
    UBYTE * s )
```

Definition at line 762 of file [dollar.c](#).

### 4.26.3.6 TermAssign()

```
void TermAssign (
    WORD * term )
```

Definition at line 803 of file [dollar.c](#).



#### 4.26.3.7 PutTermInDollar()

```
int PutTermInDollar (
    WORD * term,
    WORD numdollar )
```

Definition at line 863 of file [dollar.c](#).

#### 4.26.3.8 WildDollars()

```
void WildDollars (
    PHEAD WORD * term )
```

Definition at line 891 of file [dollar.c](#).

#### 4.26.3.9 DolToTensor()

```
WORD DolToTensor (
    PHEAD WORD numdollar )
```

Definition at line 1082 of file [dollar.c](#).

#### 4.26.3.10 DolToFunction()

```
WORD DolToFunction (
    PHEAD WORD numdollar )
```

Definition at line 1143 of file [dollar.c](#).

#### 4.26.3.11 DolToVector()

```
WORD DolToVector (
    PHEAD WORD numdollar )
```

Definition at line 1200 of file [dollar.c](#).

#### 4.26.3.12 DolToNumber()

```
WORD DolToNumber (
    PHEAD WORD numdollar )
```

Definition at line 1264 of file [dollar.c](#).

#### 4.26.3.13 DolToSymbol()

```
WORD DolToSymbol (
    PHEAD WORD numdollar )
```

Definition at line 1323 of file [dollar.c](#).

#### 4.26.3.14 DolToIndex()

```
WORD DolToIndex (
    PHEAD WORD numdollar )
```

Definition at line 1377 of file [dollar.c](#).

#### 4.26.3.15 DolToTerms()

```
DOLLARS DolToTerms (
    PHEAD WORD numdollar )
```

Definition at line 1458 of file [dollar.c](#).

#### 4.26.3.16 DolToLong()

```
LONG DolToLong (
    PHEAD WORD numdollar )
```

Definition at line 1606 of file [dollar.c](#).

#### 4.26.3.17 ExecInside()

```
int ExecInside (
    UBYTE * s )
```

Definition at line 1681 of file [dollar.c](#).

#### 4.26.3.18 InsideDollar()

```
int InsideDollar (
    PHEAD WORD * ll,
    WORD level )
```

Definition at line 1746 of file [dollar.c](#).

#### 4.26.3.19 ExchangeDollars()

```
void ExchangeDollars (
    int num1,
    int num2 )
```

Definition at line 1849 of file [dollar.c](#).

#### 4.26.3.20 TermsInDollar()

```
LONG TermsInDollar (
    WORD num )
```

Definition at line 1867 of file [dollar.c](#).

#### 4.26.3.21 SizeOfDollar()

```
LONG SizeOfDollar (
    WORD num )
```

Definition at line 1916 of file [dollar.c](#).

#### 4.26.3.22 PreIfDollarEval()

```
UBYTE * PreIfDollarEval (
    UBYTE * s,
    int * value )
```

Definition at line 1978 of file [dollar.c](#).

#### 4.26.3.23 TranslateExpression()

```
WORD * TranslateExpression (
    UBYTE * s )
```

Definition at line 2160 of file [dollar.c](#).

#### 4.26.3.24 IsSetMember()

```
int IsSetMember (
    WORD * buffer,
    WORD numset )
```

Definition at line 2217 of file [dollar.c](#).

#### 4.26.3.25 IsMultipleOf()

```
int IsMultipleOf (
    WORD * buf1,
    WORD * buf2 )
```

Definition at line 2398 of file [dollar.c](#).

#### 4.26.3.26 TwoExprCompare()

```
int TwoExprCompare (
    WORD * buf1,
    WORD * buf2,
    int oprtr )
```

Definition at line 2473 of file [dollar.c](#).

#### 4.26.3.27 DollarRaiseLow()

```
int DollarRaiseLow (
    UBYTE * name,
    LONG value )
```

Definition at line 2552 of file [dollar.c](#).

#### 4.26.3.28 EvalDoLoopArg()

```
WORD EvalDoLoopArg (
    PHEAD WORD * arg,
    WORD par )
```

Evaluates one argument of a do loop. Such an argument is constructed from SNUMBERS DOLLAREXPRES-  
SIONs and possibly DOLLAREXPR2s which indicate factors of the preceding dollar. Hence we have SNUM-  
BER,num DOLLAREXPRESSSION,numdollar DOLLAREXPRESSSION,numdollar,DOLLAREXPR2,numfactor DOL-  
LAREXPRESSSION,numdollar,DOLLAREXPR2,numfactor,DOLLAREXPR2,numfactor etc. Because we have a do-  
loop at every stage we should have a number. The notation in DOLLAREXPR2 is that  $\geq 0$  is number of yet another  
dollar and  $< 0$  is  $-n-1$  with  $n$  the array element or zero. The return value is the (short) number. The routine works its  
way through the list in a recursive manner.

Definition at line 2651 of file [dollar.c](#).

References [EvalDoLoopArg\(\)](#).

Referenced by [EvalDoLoopArg\(\)](#).

Here is the call graph for this function:



#### 4.26.3.29 TestDoLoop()

```
WORD TestDoLoop (
    PHEAD WORD * lhsbuf,
    WORD level )
```

Definition at line 2762 of file [dollar.c](#).

#### 4.26.3.30 TestEndDoLoop()

```
WORD TestEndDoLoop (
    PHEAD WORD * lhsbuf,
    WORD level )
```

Definition at line [2840](#) of file [dollar.c](#).

#### 4.26.3.31 DollarFactorize()

```
int DollarFactorize (
    PHEAD WORD numdollar )
```

Definition at line [2955](#) of file [dollar.c](#).

#### 4.26.3.32 CleanDollarFactors()

```
void CleanDollarFactors (
    DOLLARS d )
```

Definition at line [3554](#) of file [dollar.c](#).

#### 4.26.3.33 TakeDollarContent()

```
WORD * TakeDollarContent (
    PHEAD WORD * dollarbuffer,
    WORD ** factor )
```

Definition at line [3575](#) of file [dollar.c](#).

#### 4.26.3.34 MakeDollarInteger()

```
WORD * MakeDollarInteger (
    PHEAD WORD * bufin,
    WORD ** bufout )
```

For normalizing everything to integers we have to determine for all elements of this argument the LCM of the denominators and the GCD of the numerators. The input argument is in bufin. The number that comes out is the return value. The normalized argument is in bufout.

Definition at line [3627](#) of file [dollar.c](#).

References [EndSort\(\)](#), [Generator\(\)](#), and [NewSort\(\)](#).





## 4.27 dollar.c

[Go to the documentation of this file.](#)

```

00001
00006 /* #[] License : */
00007 /*
00008 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 *   When using this file you are requested to refer to the publication
00010 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 *   This is considered a matter of courtesy as the development was paid
00012 *   for by FOM the Dutch physics granting agency and we would like to
00013 *   be able to track its scientific use to convince FOM of its value
00014 *   for the community.
00015 *
00016 *   This file is part of FORM.
00017 *
00018 *   FORM is free software: you can redistribute it and/or modify it under the
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00022 *
00023 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 *   details.
00027 *
00028 *   You should have received a copy of the GNU General Public License along
00029 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #[] License : */
00032 /*
00033 *   #[] Includes :
00034 */
00035
00036 #include "form3.h"
00037
00038 /* EXTERNLOCK(dummylock) */
00039
00040 static UBYTE underscore[2] = {'_',0};
00041
00042 /*
00043 *   #[] Includes :
00044 *   #[] CatchDollar :
00045
00046 *   Works out a dollar expression during compile type.
00047 *   Steals it from the buffer and puts it in an assignment.
00048 *   At the moment we should keep this inside the small buffer.
00049 *   Later with more sort buffers we can do this better.
00050 *   Par == 0 : regular assignment
00051 *   par == -1: after error. Just make zero for now.
00052 */
00053
00054 int CatchDollar(int par)
00055 {
00056     GETIDENTITY
00057     CBUF *C = cbuf + AC.cbufnum;
00058     int error = 0, numterms = 0, numdollar, resetmods = 0;
00059     LONG newsize, retval;
00060     WORD *w, *t, n, nsize, *oldwork = AT.WorkPointer, *dbuffer;
00061     WORD oldncmod = AN.ncmod;
00062     DOLLARS d;
00063     if ( AN.ncmod && ( ( AC.modmode & ALSODOLLARS ) == 0 ) ) AN.ncmod = 0;
00064     if ( AN.ncmod && AN.cmod == 0 ) { SetMods(); resetmods = 1; }
00065
00066     numdollar = C->lhs[C->numlhs][2];
00067
00068     d = Dollars+numdollar;
00069     if ( par == -1 ) {
00070         d->type = DOLUNDEFINED;
00071         cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
00072         cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
00073         if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"$-buffer old");
00074         d->size = 0; d->where = &(AM.dollarzero);
00075         cbuf[AM.dbufnum].rhs[numdollar] = d->where;
00076         AN.ncmod = oldncmod;
00077         if ( resetmods ) UnSetMods();
00078         return(0);
00079     }
00080 #ifdef WITHMPI
00081 /*
00082 *   The problem here is that only the master can make an assignment
00083 *   like $a=g; where g is an expression: only the master has an access to
00084 *   the expression. So, in cases where the RHS contains expression names,
00085 *   only the master invokes Generator() and then broadcasts the result to
00086 *   the all slaves.

```



```

00087      * Broadcasting must be performed immediately; one cannot postpone it
00088      * to the end of the module because the dollar variable is visible
00089      * in the current module. For the same reason, this should be done
00090      * regardless of on/off parallel status.
00091      * If the RHS does not contain any expression names, it can be processed
00092      * in each slave.
00093      */
00094      if ( PF.me == MASTER || !AC.RhsExprInModuleFlag ) {
00095 #endif
00096
00097      EXCHINOUT
00098
00099      if ( NewSort(BHEAD0) ) { if ( !error ) error = 1; goto onerror; }
00100      if ( NewSort(BHEAD0) ) {
00101          LowerSortLevel();
00102          if ( !error ) error = 1;
00103          goto onerror;
00104      }
00105      AN.RepPoint = AT.RepCount + 1;
00106      w = C->rhs[C->lhs[C->numlhs][5]];
00107      while ( *w ) {
00108          n = *w; t = oldwork;
00109          NCOPY(t,w,n);
00110          AT.WorkPointer = t;
00111          AR.Cnumlhs = C->numlhs;
00112          if ( Generator(BHEAD oldwork,C->numlhs) ) { error = 1; break; }
00113      }
00114      AT.WorkPointer = oldwork;
00115      AN.tryterm = 0; /* for now */
00116      dbuffer = 0;
00117      if ( ( retval = EndSort(BHEAD (WORD *)((void *)&dbuffer)),2) < 0 ) { error = 1; }
00118      LowerSortLevel();
00119      if ( retval <= 1 || dbuffer == 0 ) {
00120          d->type = DOLZERO;
00121          if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"$-buffer old");
00122          d->size = 0; d->where = &(AM.dollarzero);
00123          cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
00124          cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
00125          goto docopy2;
00126      }
00127      w = dbuffer;
00128      if ( error == 0 )
00129          while ( *w ) { w += *w; numterms++; }
00130      else
00131          goto onerror;
00132      newsize = (w-dbuffer)+1;
00133 #ifdef WITHMPI
00134      }
00135      if ( AC.RhsExprInModuleFlag )
00136          /* PF_BroadcastPreDollar allocates dbuffer for slaves! */
00137          if ( (error = PF_BroadcastPreDollar(&dbuffer, &newsize, &numterms)) != 0 )
00138              goto onerror;
00139 #endif
00140      if ( newsize < MINALLOC ) newsize = MINALLOC;
00141      newsize = (newsize+7)/8*8;
00142      if ( numterms == 0 ) {
00143          d->type = DOLZERO;
00144          goto docopy;
00145      }
00146      else if ( numterms == 1 ) {
00147          t = dbuffer;
00148          n = *t;
00149          nsize = t[n-1];
00150          if ( nsize < 0 ) { nsize = -nsize; }
00151          if ( nsize == (n-1) ) { /* numerical */
00152              nsize = (nsize-1)/2;
00153              w = t + 1 + nsize;
00154              if ( *w != 1 ) goto doterms;
00155              w++; while ( w < ( t + n - 1 ) ) { if ( *w ) break; w++; }
00156              if ( w < ( t + n - 1 ) ) goto doterms;
00157              d->type = DOLNUMBER;
00158              goto docopy;
00159          }
00160          else if ( n == 7 && t[6] == 3 && t[5] == 1 && t[4] == 1
00161                  && t[1] == INDEX && t[2] == 3 ) {
00162              d->type = DOLINDEX;
00163              d->index = t[3];
00164              goto docopy;
00165          }
00166          else goto doterms;
00167      }
00168      else {
00169      doterms:;
00170          d->type = DOLTERMS;
00171          cbuf[AM.dbufnum].CanCommu[numdollar] = numcommute(dbuffer,
00172                  &(cbuf[AM.dbufnum].NumTerms[numdollar]));
00173      docopy:;

```

```

00174         if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"$-buffer old");
00175         d->size = newsize; d->where = dbuffer;
00176 docopy2;;
00177         cbuf[AM.dbufnum].rhs[numdollar] = d->where;
00178     }
00179     if ( C->Pointer > C->rhs[C->numrhs] ) C->Pointer = C->rhs[C->numrhs];
00180     C->numlhs--; C->numrhs--;
00181 onerror:
00182 #ifdef WITHMPI
00183     if ( PF.me == MASTER || !AC.RhsExprInModuleFlag )
00184 #endif
00185     BACKINOUT
00186     AN.ncmod = oldncmod;
00187     if ( resetmods ) UnSetMods();
00188     return(error);
00189 }
00190
00191 /*
00192  #] CatchDollar :
00193  #[ AssignDollar :
00194
00195  To be called from Generator. Assigns an expression to a $ variable.
00196  This one is slightly different from CatchDollar.
00197  We have no easy buffer this time.
00198  We will have to hack our way using what we normally use for functions.
00199
00200  Note that in the threaded case we trust the user. That means that
00201  we are not going to recheck whether there is a maximum, minimum or sum.
00202  If the user says it is like that, we treat it like that.
00203  We only check that in this centralized version MODLOCAL isn't used.
00204
00205  In a later stage dtype could be used for actually checking MODMAX
00206  and MODMIN cases.
00207 */
00208
00209 int AssignDollar(PHEAD WORD *term, WORD level)
00210 {
00211     GETBIDENTITY
00212     CBUF *C = cbuf+AM.rbufnum;
00213     int numterms = 0, numdollar = C->lhs[level][2];
00214     LONG newsize;
00215     DOLLARS d = Dollars + numdollar;
00216     WORD *w, *t, n, nsize, *rh = cbuf[C->lhs[level][7]].rhs[C->lhs[level][5]];
00217     WORD *ss, *ww;
00218     WORD olddefer, oldcompress, oldncmod = AN.ncmod;
00219 #ifdef WITHPTHREADS
00220     int nummodopt, dtype = -1, dw;
00221     WORD numvalue;
00222     if ( AN.ncmod && ( ( AC.modmode & ALSODOLLARS ) == 0 ) ) AN.ncmod = 0;
00223     if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
00224 /*
00225         Here we come only when the module runs with more than one thread.
00226         This must be a variable with a special module option.
00227         For the multi-threaded version we only allow MODSUM, MODMAX and MODMIN.
00228 */
00229         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
00230             if ( numdollar == ModOptdollars[nummodopt].number ) break;
00231         }
00232         if ( nummodopt >= NumModOptdollars ) {
00233             MLOCK(ErrorMessageLock);
00234             MesPrint("Illegal attempt to change $-variable in multi-threaded module %1",AC.CModule);
00235             MUNLOCK(ErrorMessageLock);
00236             Terminate(-1);
00237         }
00238         dtype = ModOptdollars[nummodopt].type;
00239         if ( dtype == MODLOCAL ) {
00240             d = ModOptdollars[nummodopt].dstruct+AT.identity;
00241         }
00242     }
00243 #endif
00244     DUMMYUSE(term);
00245     w = rh;
00246 /*
00247     First some shortcuts
00248 */
00249     if ( *w == 0 ) {
00250 /*
00251         #[ Thread version : Zero case
00252 */
00253 #ifdef WITHPTHREADS
00254         if ( dtype > 0 ) {
00255 /*             LOCK(d->pthreadlockwrite); */
00256             LOCK(d->pthreadlockread);
00257 NewValIsZero;;
00258             switch ( d->type ) {
00259                 case DOLZERO: goto NoChangeZero;
00260                 case DOLNUMBER:

```

```

00261         case DOLTERMS:
00262             if ( ( dw = d->where[0] ) > 0 && d->where[dw] != 0 ) {
00263                 break; /* was not a single number. Trust the user */
00264             }
00265             if ( dtype == MODMAX && d->where[dw-1] >= 0 ) goto NoChangeZero;
00266             if ( dtype == MODMIN && d->where[dw-1] <= 0 ) goto NoChangeZero;
00267             break;
00268         default:
00269             numvalue = DolToNumber(BHEAD numdollar);
00270             if ( AN.ErrorInDollar != 0 ) break;
00271             if ( dtype == MODMAX && numvalue >= 0 ) goto NoChangeZero;
00272             if ( dtype == MODMIN && numvalue <= 0 ) goto NoChangeZero;
00273             break;
00274     }
00275     d->type = DOLZERO;
00276     d->where[0] = 0;
00277     cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
00278     cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
00279 NoChangeZero:;
00280     CleanDollarFactors(d);
00281     /* UNLOCK(d->pthreadlockwrite); */
00282     UNLOCK(d->pthreadlockread);
00283     AN.ncmod = oldncmod;
00284     return(0);
00285 }
00286 #endif
00287 /*
00288     #] Thread version :
00289 */
00290     d->type = DOLZERO;
00291     d->where[0] = 0;
00292     cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
00293     cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
00294     CleanDollarFactors(d);
00295     AN.ncmod = oldncmod;
00296     return(0);
00297 }
00298 else if ( *w == 4 && w[4] == 0 && w[2] == 1 ) {
00299     /*
00300         #[ Thread version : New value is 'single precision'
00301     */
00302     #ifdef WITHPTHREADS
00303         if ( dtype > 0 ) {
00304             /* LOCK(d->pthreadlockwrite); */
00305             LOCK(d->pthreadlockread);
00306             if ( d->size < MINALLOC ) {
00307                 WORD oldsize, *oldwhere, i;
00308                 oldsize = d->size; oldwhere = d->where;
00309                 d->size = MINALLOC;
00310                 d->where = (WORD *)Malloc1(d->size*sizeof(WORD), "dollar contents");
00311                 cbuf[AM.dbufnum].rhs[numdollar] = d->where;
00312                 if ( oldsize > 0 ) {
00313                     for ( i = 0; i < oldsize; i++ ) d->where[i] = oldwhere[i];
00314                 }
00315                 else d->where[0] = 0;
00316                 if ( oldwhere && oldwhere != &(AM.dollarzero) ) M_free(oldwhere, "dollar contents");
00317             }
00318             switch ( d->type ) {
00319                 case DOLZERO:
00320 HandleDolZero:;
00321                     if ( dtype == MODMAX && w[3] <= 0 ) goto NoChangeOne;
00322                     if ( dtype == MODMIN && w[3] >= 0 ) goto NoChangeOne;
00323                     break;
00324                     case DOLNUMBER:
00325                     case DOLTERMS:
00326                         if ( ( dw = d->where[0] ) > 0 && d->where[dw] != 0 ) {
00327                             break; /* was not a single number. Trust the user */
00328                         }
00329                         if ( dtype == MODMAX && CompCoef(d->where,w) >= 0 ) goto NoChangeOne;
00330                         if ( dtype == MODMIN && CompCoef(d->where,w) <= 0 ) goto NoChangeOne;
00331                         break;
00332                     default:
00333                         {
00334                             /*
00335                                 Note that we convert the type for the next time around.
00336                             */
00337                             WORD extraterm[4];
00338                             numvalue = DolToNumber(BHEAD numdollar);
00339                             if ( AN.ErrorInDollar != 0 ) break;
00340                             if ( numvalue == 0 ) {
00341                                 d->type = DOLZERO;
00342                                 d->where[0] = 0;
00343                                 cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
00344                                 cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
00345                                 goto HandleDolZero;
00346                             }
00347                             d->where[0] = extraterm[0] = 4;

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00348         d->where[1] = extraterm[1] = ABS(numvalue);
00349         d->where[2] = extraterm[2] = 1;
00350         d->where[3] = extraterm[3] = numvalue > 0 ? 3 : -3;
00351         d->where[4] = 0;
00352         d->type = DOLNUMBER;
00353         if ( dtype == MODMAX && CompCoef(extraterm,w) >= 0 ) goto NoChangeOne;
00354         if ( dtype == MODMIN && CompCoef(extraterm,w) <= 0 ) goto NoChangeOne;
00355         break;
00356     }
00357 }
00358 d->where[0] = w[0];
00359 d->where[1] = w[1];
00360 d->where[2] = w[2];
00361 d->where[3] = w[3];
00362 d->where[4] = 0;
00363 d->type = DOLNUMBER;
00364 cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
00365 cbuf[AM.dbufnum].NumTerms[numdollar] = 1;
00366 NoChangeOne;;
00367 CleanDollarFactors(d);
00368 /* UNLOCK(d->pthreadswlockwrite); */
00369 UNLOCK(d->pthreadswlockread);
00370 AN.ncmod = oldncmod;
00371 return(0);
00372 }
00373 #endif
00374 /*
00375 #] Thread version :
00376 */
00377 if ( d->size < MINALLOC ) {
00378     if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollar contents");
00379     d->size = MINALLOC;
00380     d->where = (WORD *)Malloc1(d->size*sizeof(WORD),"dollar contents");
00381     cbuf[AM.dbufnum].rhs[numdollar] = d->where;
00382 }
00383 d->where[0] = w[0];
00384 d->where[1] = w[1];
00385 d->where[2] = w[2];
00386 d->where[3] = w[3];
00387 d->where[4] = 0;
00388 d->type = DOLNUMBER;
00389 cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
00390 cbuf[AM.dbufnum].NumTerms[numdollar] = 1;
00391 CleanDollarFactors(d);
00392 AN.ncmod = oldncmod;
00393 return(0);
00394 }
00395 /*
00396 Now the real evaluation.
00397 In the case of threads and MODSUM this requires an immediate lock.
00398 Otherwise the lock could be placed later.
00399 */
00400 #ifdef WITHPTHREADS
00401 if ( dtype == MODSUM ) {
00402     /* LOCK(d->pthreadswlockwrite); */
00403     LOCK(d->pthreadswlockread);
00404 }
00405 #endif
00406 CleanDollarFactors(d);
00407 /*
00408 The following case cannot occur. We treated it already
00409
00410 if ( *w == 0 ) {
00411     ss = 0; numterms = 0; newsize = 0;
00412     olddefer = AR.DeferFlag; AR.DeferFlag = 0;
00413     oldcompress = AR.NoCompress; AR.NoCompress = 1;
00414 }
00415 else
00416 */
00417 {
00418     /*
00419     New value is an expression that has to be evaluated first
00420     This is all generic. It won't foliate due to the sort level
00421     */
00422     if ( NewSort(BHEAD0) ) {
00423         AN.ncmod = oldncmod;
00424         return(1);
00425     }
00426     olddefer = AR.DeferFlag; AR.DeferFlag = 0;
00427     oldcompress = AR.NoCompress; AR.NoCompress = 1;
00428     while ( *w ) {
00429         n = *w; t = ww = AT.WorkPointer;
00430         NCOPY(t,w,n);
00431         AT.WorkPointer = t;
00432         if ( Generator(BHEAD ww,AR.Cnumlhs) ) {
00433             AT.WorkPointer = ww;
00434             LowerSortLevel();

```

```

00435         AR.DeferFlag = olddefer;
00436         AN.ncmod = oldncmod;
00437         return(1);
00438     }
00439     AT.WorkPointer = ww;
00440 }
00441 AN.tryterm = 0; /* for now */
00442 if ( ( newsize = EndSort(BHEAD (WORD *) ((void *) (&ss)), 2) ) < 0 ) {
00443     AN.ncmod = oldncmod;
00444     return(1);
00445 }
00446 numterms = 0; t = ss; while ( *t ) { numterms++; t += *t; }
00447 }
00448 #ifdef WITHPTHREADS
00449     if ( dtype != MODSUM ) {
00450         /* LOCK(d->pthreadlockwrite); */
00451         LOCK(d->pthreadlockread);
00452     }
00453 #endif
00454     if ( numterms == 0 ) {
00455         /*
00456             the new value evaluates to zero
00457         */
00458         #ifdef WITHPTHREADS
00459             if ( dtype == MODMAX || dtype == MODMIN ) {
00460                 if ( ss ) { M_free(ss, "Sort of $"); ss = 0; }
00461                 AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
00462                 goto NewValIsZero;
00463             }
00464             else
00465         #endif
00466         {
00467             if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where, "dollar contents");
00468             d->where = &(AM.dollarzero);
00469             d->size = 0;
00470             cbuf[AM.dbufnum].rhs[numdollar] = 0;
00471             cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
00472             cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
00473             d->type = DOLZERO;
00474         }
00475         if ( ss ) { M_free(ss, "Sort of $"); ss = 0; }
00476     }
00477     else {
00478         /*
00479             #! Thread version :
00480         */
00481         #ifdef WITHPTHREADS
00482             if ( dtype == MODMAX || dtype == MODMIN ) {
00483                 if ( numterms == 1 && ( *ss-1 == ABS(ss[*ss-1]) ) ) { /* is number */
00484                     switch ( d->type ) {
00485                         case DOLZERO:
00486                             HandleDolZero1;
00487                             if ( dtype == MODMAX && ss[*ss-1] > 0 ) break;
00488                             if ( dtype == MODMIN && ss[*ss-1] < 0 ) break;
00489                             if ( ss ) { M_free(ss, "Sort of $"); ss = 0; }
00490                             AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
00491                             goto NoChange;
00492                         case DOLTERMS:
00493                         case DOLNUMBER:
00494                             if ( ( dw = d->where[0] ) > 0 && d->where[dw] != 0 ) break;
00495                             if ( dtype == MODMAX && CompCoef(ss, d->where) > 0 ) break;
00496                             if ( dtype == MODMIN && CompCoef(ss, d->where) < 0 ) break;
00497                             if ( ss ) { M_free(ss, "Sort of $"); ss = 0; }
00498                             AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
00499                             goto NoChange;
00500                         default: {
00501                             WORD extraterm[4];
00502                             numvalue = DolToNumber(BHEAD numdollar);
00503                             if ( AN.ErrorInDollar != 0 ) break;
00504                             if ( numvalue == 0 ) {
00505                                 d->type = DOLZERO;
00506                                 d->where[0] = 0;
00507                                 cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
00508                                 cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
00509                                 goto HandleDolZero1;
00510                             }
00511                             d->where[0] = extraterm[0] = 4;
00512                             d->where[1] = extraterm[1] = ABS(numvalue);
00513                             d->where[2] = extraterm[2] = 1;
00514                             d->where[3] = extraterm[3] = numvalue > 0 ? 3 : -3;
00515                             d->where[4] = 0;
00516                             d->type = DOLNUMBER;
00517                             if ( dtype == MODMAX && CompCoef(ss, extraterm) > 0 ) break;
00518                             if ( dtype == MODMIN && CompCoef(ss, extraterm) < 0 ) break;
00519                             if ( ss ) { M_free(ss, "Sort of $"); ss = 0; }
00520                             AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
00521                             goto NoChange;

```

```

00522     }
00523     }
00524     }
00525     else {
00526         if ( ss ) { M_free(ss,"Sort of $"); ss = 0; }
00527         AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
00528         goto NoChange;
00529     }
00530 }
00531 #endif
00532 /*
00533 #] Thread version :
00534 */
00535     d->type = DOLTERMS;
00536     if ( d->where && d->where != &(AM.dollarzero) ) { M_free(d->where,"dollar contents"); d->where
= 0; }
00537     d->size = newsize + 1;
00538     d->where = ss;
00539     cbuf[AM.dbufnum].rhs[numdollar] = w = d->where;
00540 }
00541     AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
00542 /*
00543     Now find the special cases
00544 */
00545     if ( numterms == 0 ) {
00546         d->type = DOLZERO;
00547     }
00548     else if ( numterms == 1 ) {
00549         t = d->where;
00550         n = *t;
00551         nsize = t[n-1];
00552         if ( nsize < 0 ) { nsize = -nsize; }
00553         if ( nsize == (n-1) ) {
00554             nsize = (nsize-1)/2;
00555             w = t + 1 + nsize;
00556             if ( *w == 1 ) {
00557                 w++; while ( w < ( t + n - 1 ) ) { if ( *w ) break; w++; }
00558                 if ( w >= ( t + n - 1 ) ) d->type = DOLNUMBER;
00559             }
00560         }
00561         else if ( n == 7 && t[6] == 3 && t[5] == 1 && t[4] == 1
&& t[1] == INDEX && t[2] == 3 ) {
00562             d->type = DOLINDEX;
00563             d->index = t[3];
00564         }
00565     }
00566 }
00567     if ( d->type == DOLTERMS ) {
00568         cbuf[AM.dbufnum].CanCommu[numdollar] = numcommute(d->where,
&cbuf[AM.dbufnum].NumTerms[numdollar]);
00569     }
00570     else {
00571         cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
00572         cbuf[AM.dbufnum].NumTerms[numdollar] = 1;
00573     }
00574 }
00575 #ifdef WITHPTHREADS
00576 NoChange;;
00577 /* UNLOCK(d->pthreadlockwrite); */
00578 UNLOCK(d->pthreadlockread);
00579 #endif
00580     AN.ncmod = oldncmod;
00581     return(0);
00582 }
00583
00584 /*
00585 #] AssignDollar :
00586 #[ WriteDollarToBuffer :
00587
00588     Takes the numbered dollar expression and writes it to output.
00589     We catch however the output in a buffer and return its address.
00590     This routine is needed when we need a text representation of
00591     a dollar expression like for the construction '$name' in the preprocessor.
00592     If par==0 we leave the current printing mode.
00593     If par==1 we insist on normal mode
00594 */
00595
00596 UBYTE *WriteDollarToBuffer(WORD numdollar, WORD par)
00597 {
00598     DOLLARS d = Dollars+numdollar;
00599     UBYTE *s, *oldcurbufwrt = AO.CurBufWrt;
00600     WORD *t, lbrac = 0, first = 0, arg[2], oldOutputMode = AC.OutputMode;
00601     WORD oldinfrack = AO.InFbrack;
00602     int error = 0;
00603     int dict = AO.CurrentDictionary;
00604
00605     AO.DollarOutSizeBuffer = 32;
00606     AO.DollarOutBuffer = (UBYTE *)Malloc1(AO.DollarOutSizeBuffer,"DollarOutBuffer");
00607     AO.DollarInOutBuffer = 1;

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```

00608     AO.PrintType = 1;
00609     AO.InFbrack = 0;
00610     s = AO.DollarOutBuffer;
00611     *s = 0;
00612     if ( par > 0 && AO.CurDictInDollars == 0 ) {
00613         AC.OutputMode = NORMALFORMAT;
00614         AO.CurrentDictionary = 0;
00615     }
00616     else {
00617         AO.CurBufWrt = (UBYTE *)underscore;
00618     }
00619     AO.OutInBuffer = 1;
00620     switch ( d->type ) {
00621     case DOLARGUMENT:
00622         WriteArgument(d->where);
00623         break;
00624     case DOLSUBTERM:
00625         WriteSubTerm(d->where,1);
00626         break;
00627     case DOLNUMBER:
00628     case DOLTERMS:
00629         t = d->where;
00630         while ( *t ) {
00631             if ( WriteTerm(t,&lbrac,first,PRINTON,0) ) {
00632                 error = 1; break;
00633             }
00634             t += *t;
00635         }
00636         break;
00637     case DOLWILDARGS:
00638         t = d->where+1;
00639         while ( *t ) {
00640             WriteArgument(t);
00641             NEXTARG(t)
00642             if ( *t ) TokenToLine((UBYTE *)(","));
00643         }
00644         break;
00645     case DOLINDEX:
00646         arg[0] = -INDEX; arg[1] = d->index;
00647         WriteArgument(arg);
00648         break;
00649     case DOLZERO:
00650         *s++ = '0'; *s = 0;
00651         AO.DollarInOutBuffer = 1;
00652         break;
00653     case DOLUNDEFINED:
00654         *s = 0;
00655         AO.DollarInOutBuffer = 1;
00656         break;
00657     }
00658     AC.OutputMode = oldOutputMode;
00659     AO.OutInBuffer = 0;
00660     AO.InFbrack = oldinfbrack;
00661     AO.CurBufWrt = oldcurbufwrt;
00662     AO.CurrentDictionary = dict;
00663     if ( error ) {
00664         MLOCK(ErrorMessageLock);
00665         MesPrint("&Illegal dollar object for writing");
00666         MUNLOCK(ErrorMessageLock);
00667         M_free(AO.DollarOutBuffer,"DollarOutBuffer");
00668         AO.DollarOutBuffer = 0;
00669         AO.DollarOutSizeBuffer = 0;
00670         return(0);
00671     }
00672     return(AO.DollarOutBuffer);
00673 }
00674
00675 /*
00676 #] WriteDollarToBuffer :
00677 #[ WriteDollarFactorToBuffer :
00678
00679 Takes the numbered dollar expression and writes it to output.
00680 We catch however the output in a buffer and return its address.
00681 This routine is needed when we need a text representation of
00682 a dollar expression like for the construction '$name' in the preprocessor.
00683 If par==0 we leave the current printing mode.
00684 If par==1 we insist on normal mode
00685 */
00686
00687 UBYTE *WriteDollarFactorToBuffer(WORD numdollar, WORD numfac, WORD par)
00688 {
00689     DOLLARS d = Dollars+numdollar;
00690     UBYTE *s, *oldcurbufwrt = AO.CurBufWrt;
00691     WORD *t, lbrac = 0, first = 0, n[5], oldOutputMode = AC.OutputMode;
00692     WORD oldinfbrack = AO.InFbrack;
00693     int error = 0;
00694     int dict = AO.CurrentDictionary;

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```

00695
00696     if ( numfac > d->nfactors || numfac < 0 ) {
00697         MLOCK(ErrorMessageLock);
00698         MesPrint("&Illegal factor number for this dollar variable: %d",numfac);
00699         MesPrint("&There are %d factors",d->nfactors);
00700         MUNLOCK(ErrorMessageLock);
00701         return(0);
00702     }
00703
00704     AO.DollarOutSizeBuffer = 32;
00705     AO.DollarOutBuffer = (UBYTE *)Malloc1(AO.DollarOutSizeBuffer,"DollarOutBuffer");
00706     AO.DollarInOutBuffer = 1;
00707     AO.PrintType = 1;
00708     AO.InFbrack = 0;
00709     s = AO.DollarOutBuffer;
00710     *s = 0;
00711     if ( par > 0 ) {
00712         AC.OutputMode = NORMALFORMAT;
00713         AO.CurrentDictionary = 0;
00714     }
00715     else {
00716         AO.CurBufWrt = (UBYTE *)underscore;
00717     }
00718     AO.OutInBuffer = 1;
00719     if ( numfac == 0 ) { /* write the number d->nfactors */
00720         n[0] = 4; n[1] = d->nfactors; n[2] = 1; n[3] = 3; n[4] = 0; t = n;
00721     }
00722     else if ( numfac == 1 && d->factors == 0 ) { /* Here d->factors is zero and d->where is fine */
00723         t = d->where;
00724     }
00725     else if ( d->factors[numfac-1].where == 0 ) { /* write the value */
00726         if ( d->factors[numfac-1].value < 0 ) {
00727             n[0] = 4; n[1] = -d->factors[numfac-1].value; n[2] = 1; n[3] = -3; n[4] = 0; t = n;
00728         }
00729         else {
00730             n[0] = 4; n[1] = d->factors[numfac-1].value; n[2] = 1; n[3] = 3; n[4] = 0; t = n;
00731         }
00732     }
00733     else { t = d->factors[numfac-1].where; }
00734     while ( *t ) {
00735         if ( WriteTerm(t,&lbrac,first,PRINTON,0) ) {
00736             error = 1; break;
00737         }
00738         t += *t;
00739     }
00740     AC.OutputMode = oldOutputMode;
00741     AO.OutInBuffer = 0;
00742     AO.InFbrack = oldinfbrack;
00743     AO.CurBufWrt = oldcurbufwrt;
00744     AO.CurrentDictionary = dict;
00745     if ( error ) {
00746         MLOCK(ErrorMessageLock);
00747         MesPrint("&Illegal dollar object for writing");
00748         MUNLOCK(ErrorMessageLock);
00749         M_free(AO.DollarOutBuffer,"DollarOutBuffer");
00750         AO.DollarOutBuffer = 0;
00751         AO.DollarOutSizeBuffer = 0;
00752         return(0);
00753     }
00754     return(AO.DollarOutBuffer);
00755 }
00756
00757 /*
00758 #] WriteDollarFactorToBuffer :
00759 #[ AddToDollarBuffer :
00760 */
00761
00762 void AddToDollarBuffer(UBYTE *s)
00763 {
00764     int i;
00765     UBYTE *t = s, *u, *newdob;
00766     LONG j;
00767     while ( *t ) { t++; }
00768     i = t - s;
00769     while ( i + AO.DollarInOutBuffer >= AO.DollarOutSizeBuffer ) {
00770         j = AO.DollarInOutBuffer;
00771         AO.DollarOutSizeBuffer *= 2;
00772         t = AO.DollarOutBuffer;
00773         newdob = (UBYTE *)Malloc1(AO.DollarOutSizeBuffer,"DollarOutBuffer");
00774         u = newdob;
00775         while ( --j >= 0 ) *u++ = *t++;
00776         M_free(AO.DollarOutBuffer,"DollarOutBuffer");
00777         AO.DollarOutBuffer = newdob;
00778     }
00779     t = AO.DollarOutBuffer + AO.DollarInOutBuffer-1;
00780     while ( t == AO.DollarOutBuffer && ( *s == '+' || *s == ' ' ) ) s++;
00781     i = 0;

```



```

00782     if ( AO.CurrentDictionary == 0 ) {
00783         while ( *s ) {
00784             if ( *s == ' ' ) { s++; continue; }
00785             *t++ = *s++; i++;
00786         }
00787     }
00788     else {
00789         while ( *s ) { *t++ = *s++; i++; }
00790     }
00791     *t = 0;
00792     AO.DollarInOutBuffer += i;
00793 }
00794
00795 /*
00796 #] AddToDollarBuffer :
00797 #[ TermAssign :
00798
00799 This routine is called from a piece of code in Normalize that has been
00800 commented out.
00801 */
00802
00803 void TermAssign(WORD *term)
00804 {
00805     DOLLARS d;
00806     WORD *t, *tstop, *astop, *w, *m;
00807     WORD i, newsize;
00808     for (;;) {
00809         astop = term + *term;
00810         tstop = astop - ABS(astop[-1]);
00811         t = term + 1;
00812         while ( t < tstop ) {
00813             if ( *t == AM.termfunnum && t[1] == FUNHEAD+2
00814                 && t[FUNHEAD] == -DOLLAREXPRESSION ) {
00815                 d = Dollars + t[FUNHEAD+1];
00816                 newsize = *term - FUNHEAD - 1;
00817                 if ( newsize < MINALLOC ) newsize = MINALLOC;
00818                 newsize = (newsize+7)/8*8;
00819                 if ( d->size > 2*newsize && d->size > 1000 ) {
00820                     if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollar
00821 contents");
00822                     d->size = 0;
00823                     d->where = &(AM.dollarzero);
00824                 }
00825                 if ( d->size < newsize ) {
00826                     if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollar
00827 contents");
00828                     d->size = newsize;
00829                     d->where = (WORD *)Malloc1(newsize*sizeof(WORD),"dollar contents");
00830                 }
00831                 cbuf[AM.dbufnum].rhs[t[FUNHEAD+1]] = w = d->where;
00832                 m = term;
00833                 while ( m < t ) *w++ = *m++;
00834                 m += t[1];
00835                 while ( m < tstop ) {
00836                     if ( *m == AM.termfunnum && m[1] == FUNHEAD+2
00837                         && m[FUNHEAD] == -DOLLAREXPRESSION ) { m += m[1]; }
00838                     else {
00839                         i = m[1];
00840                         while ( --i >= 0 ) *w++ = *m++;
00841                     }
00842                 }
00843                 while ( m < astop ) *w++ = *m++;
00844                 *(d->where) = w - d->where;
00845                 *w = 0;
00846                 d->type = DOLTERMS;
00847                 w = t; m = t + t[1];
00848                 while ( m < astop ) *w++ = *m++;
00849                 *term = w - term;
00850                 break;
00851             }
00852             t += t[1];
00853         }
00854         if ( t >= tstop ) return;
00855     }
00856 }
00857
00858 /*
00859 #] TermAssign :
00860 #[ PutTermInDollar :
00861
00862 We assume here that the dollar is local.
00863 */
00864
00865 int PutTermInDollar(WORD *term, WORD numdollar)
00866 {
00867     DOLLARS d = Dollars+numdollar;
00868     WORD i;

```

```

00867     if ( term == 0 || *term == 0 ) {
00868         d->type = DOLZERO;
00869         return(0);
00870     }
00871     if ( d->size < *term || d->size > 2*term[0] || d->where == 0 ) {
00872         if ( d->size > 0 && d->where ) {
00873             M_free(d->where,"dollar contents");
00874         }
00875         d->where = Malloc1((term[0]+1)*sizeof(WORD),"dollar contents");
00876         d->size = term[0]+1;
00877     }
00878     d->type = DOLTERMS;
00879     for ( i = 0; i < term[0]; i++ ) d->where[i] = term[i];
00880     d->where[i] = 0;
00881     return(0);
00882 }
00883
00884 /*
00885 #] PutTermInDollar :
00886 #[ WildDollars :
00887
00888     Note that we cannot upload wildcards into dollar variables when WITHPTHREADS.
00889 */
00890
00891 void WildDollars(PHEAD WORD *term)
00892 {
00893     GETBIDENTITY
00894     DOLLARS d;
00895     WORD *m, *t, *w, *ww, *orig = 0, *wildvalue, *wildstop;
00896     int numdollar;
00897     LONG weneed, i;
00898     #ifdef WITHPTHREADS
00899         int dtype = -1;
00900     #endif
00901     if ( term == 0 ) {
00902         m = wildvalue = AN.WildValue;
00903         wildstop = AN.WildStop;
00904     }
00905     else {
00906         ww = term + *term; ww -= ABS(ww[-1]); w = term+1;
00907         while ( w < ww && *w != SUBEXPRESSION ) w += w[1];
00908         if ( w >= ww ) return;
00909         wildstop = w + w[1];
00910         w += SUBEXPSIZE;
00911         wildvalue = m = w;
00912     }
00913     while ( m < wildstop ) {
00914         if ( *m != LOADDOLLAR ) { m += m[1]; continue; }
00915         t = m - 4;
00916         while ( *t == LOADDOLLAR || *t == FROMSET || *t == SETTONUM ) t -= 4;
00917         if ( t < wildvalue ) {
00918             MLOCK(ErrorMessageLock);
00919             MesPrint("&Serious bug in wildcard prototype. Found in WildDollars");
00920             MUNLOCK(ErrorMessageLock);
00921             Terminate(-1);
00922         }
00923         numdollar = m[2];
00924         d = Dollars + numdollar;
00925         #ifdef WITHPTHREADS
00926         {
00927             int nummodopt;
00928             dtype = -1;
00929             if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
00930                 for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
00931                     if ( numdollar == ModOptdollars[nummodopt].number ) break;
00932                 }
00933                 if ( nummodopt < NumModOptdollars ) {
00934                     dtype = ModOptdollars[nummodopt].type;
00935                     if ( dtype == MODLOCAL ) {
00936                         d = ModOptdollars[nummodopt].dstruct+AT.identity;
00937                     }
00938                     else {
00939                         MLOCK(ErrorMessageLock);
00940                         MesPrint("&Illegal attempt to use $-variable %s in module %1",
00941                             DOLLARNAME(Dollars,numdollar),AC.CModule);
00942                         MUNLOCK(ErrorMessageLock);
00943                         Terminate(-1);
00944                     }
00945                 }
00946             }
00947         }
00948     #endif
00949     /*
00950         The value of this wildcard goes into our $-variable
00951         First compute the space we need.
00952     */
00953     switch ( *t ) {

```

```

00954         case SYMTONUM:
00955             weneed = 5;
00956             break;
00957         case SYMTOSYM:
00958             weneed = 9;
00959             break;
00960         case SYMTOSUB:
00961         case VECTOSUB:
00962         case INDTOSUB:
00963             orig = cbuf[AT.ebufnum].rhs[t[3]];
00964             w = orig; while ( *w ) w += *w;
00965             weneed = w - orig + 1;
00966             break;
00967         case VECTOMIN:
00968         case VECTOVEC:
00969         case INDTOIND:
00970             weneed = 8;
00971             break;
00972         case FUNTOFUN:
00973             weneed = FUNHEAD+5;
00974             break;
00975         case ARGTOARG:
00976             orig = cbuf[AT.ebufnum].rhs[t[3]];
00977             if ( *orig > 0 ) weneed = *orig+2;
00978             else {
00979                 w = orig+1; while ( *w ) { NEXTARG(w) }
00980                 weneed = w - orig + 1;
00981             }
00982             break;
00983         default:
00984             weneed = MINALLOC;
00985             break;
00986     }
00987     if ( weneed < MINALLOC ) weneed = MINALLOC;
00988     weneed = ((weneed+7)/8)*8;
00989     if ( d->size > 2*weneed && d->size > 1000 ) {
00990         if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollarspace");
00991         d->where = &(AM.dollarzero);
00992         d->size = 0;
00993     }
00994     if ( d->size < weneed ) {
00995         if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollarspace");
00996         d->where = (WORD *)Malloc1(weneed*sizeof(WORD),"dollarspace");
00997         d->size = weneed;
00998     }
00999 /*
01000     It is not clear what the following code does for TFORM
01001
01002     if ( dtype != MODLOCAL ) {
01003 */
01004         cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
01005         cbuf[AM.dbufnum].NumTerms[numdollar] = 1;
01006 /*
01007         cbuf[AM.dbufnum].rhs[numdollar] = d->where; */
01008 /*
01009         cbuf[AM.dbufnum].rhs[numdollar] = (WORD *) (1);
01010
01011     }
01012     Now load up the value of the wildcard in compiler buffer format
01013 */
01014     w = d->where;
01015     d->type = DOLTERMS;
01016     switch ( *t ) {
01017         case SYMTONUM:
01018             d->where[0] = 4; d->where[2] = 1;
01019             if ( t[3] >= 0 ) { d->where[1] = t[3]; d->where[3] = 3; }
01020             else { d->where[1] = -t[3]; d->where[3] = -3; }
01021             if ( t[3] == 0 ) { d->type = DOLZERO; d->where[0] = 0; }
01022             else { d->type = DOLNUMBER; d->where[4] = 0; }
01023             break;
01024         case SYMTOSYM:
01025             *w++ = 8;
01026             *w++ = SYMBOL;
01027             *w++ = 4;
01028             *w++ = t[3];
01029             *w++ = 1;
01030             *w++ = 1;
01031             *w++ = 1;
01032             *w++ = 3;
01033             *w = 0;
01034             break;
01035         case SYMTOSUB:
01036         case VECTOSUB:
01037         case INDTOSUB:
01038             while ( *orig ) {
01039                 i = *orig; while ( --i >= 0 ) *w++ = *orig++;
01040             }
01041             *w = 0;
01042     }
01043 /*

```

```

01041             And then we have to fix up CanCommu
01042  */
01043             break;
01044         case VECTOMIN:
01045             *w++ = 7; *w++ = INDEX; *w++ = 3; *w++ = t[3];
01046             *w++ = 1; *w++ = 1; *w++ = -3; *w = 0;
01047             break;
01048         case VECTOVEC:
01049             *w++ = 7; *w++ = INDEX; *w++ = 3; *w++ = t[3];
01050             *w++ = 1; *w++ = 1; *w++ = 3; *w = 0;
01051             break;
01052         case INDTOIND:
01053             d->type = DOLINDEX; d->index = t[3]; *w = 0;
01054             break;
01055         case FUNTOFUN:
01056             *w++ = FUNHEAD+4; *w++ = t[3]; *w++ = FUNHEAD;
01057             FILLFUN(w)
01058             *w++ = 1; *w++ = 1; *w++ = 3; *w = 0;
01059             break;
01060         case ARGTOARG:
01061             if ( *orig > 0 ) ww = orig + *orig + 1;
01062             else {
01063                 ww = orig+1; while ( *ww ) { NEXTARG(ww) }
01064             }
01065             while ( orig < ww ) *w++ = *orig++;
01066             *w = 0;
01067             d->type = DOLWILDARGS;
01068             break;
01069         default:
01070             d->type = DOLUNDEFINED;
01071             break;
01072     }
01073     m += m[1];
01074 }
01075 }
01076
01077 /*
01078  #] WildDollars :
01079  #[ DolToTensor :      with LOCK
01080  */
01081
01082 WORD DolToTensor(PHEAD WORD numdollar)
01083 {
01084     GETBIDENTITY
01085     DOLLARS d = Dollars + numdollar;
01086     WORD retval;
01087     #ifdef WITHPTHREADS
01088     int nummodopt, dtype = -1;
01089     if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
01090         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01091             if ( numdollar == ModOptdollars[nummodopt].number ) break;
01092         }
01093         if ( nummodopt < NumModOptdollars ) {
01094             dtype = ModOptdollars[nummodopt].type;
01095             if ( dtype == MODLOCAL ) {
01096                 d = ModOptdollars[nummodopt].dstruct+AT.identity;
01097             }
01098             else {
01099                 LOCK(d->pthreadlockread);
01100             }
01101         }
01102     }
01103     #endif
01104     AN.ErrorInDollar = 0;
01105     if ( d->type == DOLTERMS && d->where[0] == FUNHEAD+4 &&
01106         d->where[FUNHEAD+4] == 0 && d->where[FUNHEAD+3] == 3 &&
01107         d->where[FUNHEAD+2] == 1 && d->where[FUNHEAD+1] == 1 &&
01108         d->where[1] >= FUNCTION && d->where[1] < FUNCTION+WILDOFFSET
01109         && functions[d->where[1]-FUNCTION].spec >= TENSORFUNCTION ) {
01110         retval = d->where[1];
01111     }
01112     else if ( d->type == DOLARGUMENT &&
01113         d->where[0] <= -FUNCTION && d->where[0] > -FUNCTION-WILDOFFSET
01114         && functions[-d->where[0]-FUNCTION].spec >= TENSORFUNCTION ) {
01115         retval = -d->where[0];
01116     }
01117     else if ( d->type == DOLWILDARGS && d->where[0] == 0
01118         && d->where[1] <= -FUNCTION && d->where[1] > -FUNCTION-WILDOFFSET
01119         && d->where[2] == 0
01120         && functions[-d->where[1]-FUNCTION].spec >= TENSORFUNCTION ) {
01121         retval = -d->where[1];
01122     }
01123     else if ( d->type == DOLSUBTERM &&
01124         d->where[0] >= FUNCTION && d->where[0] < FUNCTION+WILDOFFSET
01125         && functions[d->where[0]-FUNCTION].spec >= TENSORFUNCTION ) {
01126         retval = d->where[0];
01127     }

```

```

01128     else {
01129         AN.ErrorInDollar = 1;
01130         retval = 0;
01131     }
01132 #ifdef WITHPTHREADS
01133     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
01134 #endif
01135     return(retval);
01136 }
01137
01138 /*
01139 #] DolToTensor :
01140 #[ DolToFunction : with LOCK
01141 */
01142
01143 WORD DolToFunction(PHEAD WORD numdollar)
01144 {
01145     GETBIDENTITY
01146     DOLLARS d = Dollars + numdollar;
01147     WORD retval;
01148 #ifdef WITHPTHREADS
01149     int nummodopt, dtype = -1;
01150     if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
01151         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01152             if ( numdollar == ModOptdollars[nummodopt].number ) break;
01153         }
01154         if ( nummodopt < NumModOptdollars ) {
01155             dtype = ModOptdollars[nummodopt].type;
01156             if ( dtype == MODLOCAL ) {
01157                 d = ModOptdollars[nummodopt].dstruct+AT.identity;
01158             }
01159             else {
01160                 LOCK(d->pthreadlockread);
01161             }
01162         }
01163     }
01164 #endif
01165     AN.ErrorInDollar = 0;
01166     if ( d->type == DOLTERMS && d->where[0] == FUNHEAD+4 &&
01167         d->where[FUNHEAD+4] == 0 && d->where[FUNHEAD+3] == 3 &&
01168         d->where[FUNHEAD+2] == 1 && d->where[FUNHEAD+1] == 1 &&
01169         d->where[1] >= FUNCTION && d->where[1] < FUNCTION+WILDOFFSET ) {
01170         retval = d->where[1];
01171     }
01172     else if ( d->type == DOLARGUMENT &&
01173         d->where[0] <= -FUNCTION && d->where[0] > -FUNCTION-WILDOFFSET ) {
01174         retval = -d->where[0];
01175     }
01176     else if ( d->type == DOLWILDARGS && d->where[0] == 0
01177         && d->where[1] <= -FUNCTION && d->where[1] > -FUNCTION-WILDOFFSET
01178         && d->where[2] == 0 ) {
01179         retval = -d->where[1];
01180     }
01181     else if ( d->type == DOLSUBTERM &&
01182         d->where[0] >= FUNCTION && d->where[0] < FUNCTION+WILDOFFSET ) {
01183         retval = d->where[0];
01184     }
01185     else {
01186         AN.ErrorInDollar = 1;
01187         retval = 0;
01188     }
01189 #ifdef WITHPTHREADS
01190     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
01191 #endif
01192     return(retval);
01193 }
01194
01195 /*
01196 #] DolToFunction :
01197 #[ DolToVector : with LOCK
01198 */
01199
01200 WORD DolToVector(PHEAD WORD numdollar)
01201 {
01202     GETBIDENTITY
01203     DOLLARS d = Dollars + numdollar;
01204     WORD retval;
01205 #ifdef WITHPTHREADS
01206     int nummodopt, dtype = -1;
01207     if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
01208         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01209             if ( numdollar == ModOptdollars[nummodopt].number ) break;
01210         }
01211         if ( nummodopt < NumModOptdollars ) {
01212             dtype = ModOptdollars[nummodopt].type;
01213             if ( dtype == MODLOCAL ) {
01214                 d = ModOptdollars[nummodopt].dstruct+AT.identity;

```

```

01215         }
01216         else {
01217             LOCK(d->pthreadlockread);
01218         }
01219     }
01220 }
01221 #endif
01222 AN.ErrorInDollar = 0;
01223 if ( d->type == DOLINDEX && d->index < 0 ) {
01224     retval = d->index;
01225 }
01226 else if ( d->type == DOLARGUMENT && ( d->where[0] == -VECTOR
01227 || d->where[0] == -MINVECTOR ) ) {
01228     retval = d->where[1];
01229 }
01230 else if ( d->type == DOLSUBTERM && d->where[0] == INDEX
01231 && d->where[1] == 3 && d->where[2] < 0 ) {
01232     retval = d->where[2];
01233 }
01234 else if ( d->type == DOLTERMS && d->where[0] == 7 &&
01235 d->where[7] == 0 && d->where[6] == 3 &&
01236 d->where[5] == 1 && d->where[4] == 1 &&
01237 d->where[1] >= INDEX && d->where[3] < 0 ) {
01238     retval = d->where[3];
01239 }
01240 else if ( d->type == DOLWILDARGS && d->where[0] == 0
01241 && ( d->where[1] == -VECTOR || d->where[1] == -MINVECTOR )
01242 && d->where[3] == 0 ) {
01243     retval = d->where[2];
01244 }
01245 else if ( d->type == DOLWILDARGS && d->where[0] == 1
01246 && d->where[1] < 0 ) {
01247     retval = d->where[1];
01248 }
01249 else {
01250     AN.ErrorInDollar = 1;
01251     retval = 0;
01252 }
01253 #ifdef WITHPTHREADS
01254 if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
01255 #endif
01256 return(retval);
01257 }
01258
01259 /*
01260 #] DolToVector :
01261 #[ DolToNumber :
01262 */
01263
01264 WORD DolToNumber(PHEAD WORD numdollar)
01265 {
01266     GETBIDENTITY
01267     DOLLARS d = Dollars + numdollar;
01268 #ifdef WITHPTHREADS
01269     int nummodopt, dtype = -1;
01270     if ( AS.MultiThreaded && ( AC.mparallelfld == PARALLELFIELD ) ) {
01271         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01272             if ( numdollar == ModOptdollars[nummodopt].number ) break;
01273         }
01274         if ( nummodopt < NumModOptdollars ) {
01275             dtype = ModOptdollars[nummodopt].type;
01276             if ( dtype == MODLOCAL ) {
01277                 d = ModOptdollars[nummodopt].dstruct+AT.identity;
01278             }
01279         }
01280     }
01281 #endif
01282 AN.ErrorInDollar = 0;
01283 if ( ( d->type == DOLTERMS || d->type == DOLNUMBER )
01284 && d->where[0] == 4 &&
01285 d->where[4] == 0 && ( d->where[3] == 3 || d->where[3] == -3 )
01286 && d->where[2] == 1 && ( d->where[1] & TOPBITONLY ) == 0 ) {
01287     if ( d->where[3] > 0 ) return(d->where[1]);
01288     else return(-d->where[1]);
01289 }
01290 else if ( d->type == DOLARGUMENT && d->where[0] == -SNUMBER ) {
01291     return(d->where[1]);
01292 }
01293 else if ( d->type == DOLARGUMENT && d->where[0] == -INDEX
01294 && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
01295     return(d->where[1]);
01296 }
01297 else if ( d->type == DOLZERO ) return(0);
01298 else if ( d->type == DOLWILDARGS && d->where[0] == 0
01299 && d->where[1] == -SNUMBER && d->where[3] == 0 ) {
01300     return(d->where[2]);
01301 }

```

```

01302     else if ( d->type == DOLINDEX && d->index >= 0 && d->index < AM.OffsetIndex ) {
01303         return(d->index);
01304     }
01305     else if ( d->type == DOLWILDARGS && d->where[0] == 1
01306 && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
01307         return(d->where[1]);
01308     }
01309     else if ( d->type == DOLWILDARGS && d->where[0] == 0
01310 && d->where[1] == -INDEX && d->where[3] == 0 && d->where[2] >= 0
01311 && d->where[2] < AM.OffsetIndex ) {
01312         return(d->where[2]);
01313     }
01314     AN.ErrorInDollar = 1;
01315     return(0);
01316 }
01317
01318 /*
01319 #] DolToNumber :
01320 #[ DolToSymbol :      with LOCK
01321 */
01322
01323 WORD DolToSymbol(PHEAD WORD numdollar)
01324 {
01325     GETBIDENTITY
01326     DOLLARS d = Dollars + numdollar;
01327     WORD retval;
01328 #ifdef WITHPTHREADS
01329     int nummodopt, dtype = -1;
01330     if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
01331         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01332             if ( numdollar == ModOptdollars[nummodopt].number ) break;
01333         }
01334         if ( nummodopt < NumModOptdollars ) {
01335             dtype = ModOptdollars[nummodopt].type;
01336             if ( dtype == MODLOCAL ) {
01337                 d = ModOptdollars[nummodopt].dstruct+AT.identity;
01338             }
01339             else {
01340                 LOCK(d->pthreadlockread);
01341             }
01342         }
01343     }
01344 #endif
01345     AN.ErrorInDollar = 0;
01346     if ( d->type == DOLTERMS && d->where[0] == 8 &&
01347 d->where[8] == 0 && d->where[7] == 3 && d->where[6] == 1
01348 && d->where[5] == 1 && d->where[4] == 1 && d->where[1] == SYMBOL ) {
01349         retval = d->where[3];
01350     }
01351     else if ( d->type == DOLARGUMENT && d->where[0] == -SYMBOL ) {
01352         retval = d->where[1];
01353     }
01354     else if ( d->type == DOLSUBTERM && d->where[0] == SYMBOL
01355 && d->where[1] == 4 && d->where[3] == 1 ) {
01356         retval = d->where[2];
01357     }
01358     else if ( d->type == DOLWILDARGS && d->where[0] == 0
01359 && d->where[1] == -SYMBOL && d->where[3] == 0 ) {
01360         retval = d->where[2];
01361     }
01362     else {
01363         AN.ErrorInDollar = 1;
01364         retval = -1;
01365     }
01366 #ifdef WITHPTHREADS
01367     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
01368 #endif
01369     return(retval);
01370 }
01371
01372 /*
01373 #] DolToSymbol :
01374 #[ DolToIndex :      with LOCK
01375 */
01376
01377 WORD DolToIndex(PHEAD WORD numdollar)
01378 {
01379     GETBIDENTITY
01380     DOLLARS d = Dollars + numdollar;
01381     WORD retval;
01382 #ifdef WITHPTHREADS
01383     int nummodopt, dtype = -1;
01384     if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
01385         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01386             if ( numdollar == ModOptdollars[nummodopt].number ) break;
01387         }
01388         if ( nummodopt < NumModOptdollars ) {

```

```

01389         dtype = ModOptdollars[nummodopt].type;
01390         if ( dtype == MODLOCAL ) {
01391             d = ModOptdollars[nummodopt].dstruct+AT.identity;
01392         }
01393         else {
01394             LOCK(d->pthreadlockread);
01395         }
01396     }
01397 }
01398 #endif
01399 AN.ErrorInDollar = 0;
01400 if ( d->type == DOLTERMS && d->where[0] == 7 &&
01401     d->where[7] == 0 && d->where[6] == 3 && d->where[5] == 1
01402     && d->where[4] == 1 && d->where[1] == INDEX && d->where[3] >= 0 ) {
01403     retval = d->where[3];
01404 }
01405 else if ( d->type == DOLARGUMENT && d->where[0] == -SNUMBER
01406     && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
01407     retval = d->where[1];
01408 }
01409 else if ( d->type == DOLARGUMENT && d->where[0] == -INDEX
01410     && d->where[1] >= 0 ) {
01411     retval = d->where[1];
01412 }
01413 else if ( d->type == DOLZERO ) return(0);
01414 else if ( d->type == DOLWILDARGS && d->where[0] == 0
01415     && d->where[1] == -SNUMBER && d->where[3] == 0 && d->where[2] >= 0
01416     && d->where[2] < AM.OffsetIndex ) {
01417     retval = d->where[2];
01418 }
01419 else if ( d->type == DOLINDEX && d->index >= 0 ) {
01420     retval = d->index;
01421 }
01422 else if ( d->type == DOLNUMBER && d->where[0] == 4 && d->where[2] == 1
01423     && d->where[3] == 3 && d->where[4] == 0 && d->where[1] < AM.OffsetIndex ) {
01424     retval = d->where[1];
01425 }
01426 else if ( d->type == DOLWILDARGS && d->where[0] == 1
01427     && d->where[1] >= 0 ) {
01428     retval = d->where[1];
01429 }
01430 else if ( d->type == DOLSUBTERM && d->where[0] == INDEX
01431     && d->where[1] == 3 && d->where[2] >= 0 ) {
01432     retval = d->where[2];
01433 }
01434 else if ( d->type == DOLWILDARGS && d->where[0] == 0
01435     && d->where[1] == -INDEX && d->where[3] == 0 && d->where[2] >= 0 ) {
01436     retval = d->where[2];
01437 }
01438 else {
01439     AN.ErrorInDollar = 1;
01440     retval = 0;
01441 }
01442 #ifdef WITHPTHREADS
01443 if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
01444 #endif
01445 return(retval);
01446 }
01447
01448 /*
01449 #] DolToIndex :
01450 #[ DolToTerms :
01451
01452 Returns a struct of type DOLLARS which contains a copy of the
01453 original dollar variable, provided it can be expressed in terms of
01454 an expression (type = DOLTERMS). Otherwise it returns zero.
01455 The dollar is expressed in terms in the buffer "where"
01456 */
01457
01458 DOLLARS DolToTerms(PHEAD WORD numdollar)
01459 {
01460     GETBIDENTITY
01461     LONG size;
01462     DOLLARS d = Dollars + numdollar, newd;
01463     WORD *t, *w, i;
01464 #ifdef WITHPTHREADS
01465     int nummodopt, dtype = -1;
01466     if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
01467         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01468             if ( numdollar == ModOptdollars[nummodopt].number ) break;
01469         }
01470         if ( nummodopt < NumModOptdollars ) {
01471             dtype = ModOptdollars[nummodopt].type;
01472             if ( dtype == MODLOCAL ) {
01473                 d = ModOptdollars[nummodopt].dstruct+AT.identity;
01474             }
01475         }

```



```

01476     }
01477 #endif
01478     AN.ErrorInDollar = 0;
01479     switch ( d->type ) {
01480     case DOLARGUMENT:
01481         t = d->where;
01482         if ( t[0] < 0 ) {
01483     ShortArgument:
01484             w = AT.WorkPointer;
01485             if ( t[0] <= -FUNCTION ) {
01486                 *w++ = FUNHEAD+4; *w++ = -t[0];
01487                 *w++ = FUNHEAD; FILLFUN(w)
01488                 *w++ = 1; *w++ = 1; *w++ = 3;
01489             }
01490             else if ( t[0] == -SYMBOL ) {
01491                 *w++ = 8; *w++ = SYMBOL; *w++ = 4; *w++ = t[1];
01492                 *w++ = 1; *w++ = 1; *w++ = 1; *w++ = 3;
01493             }
01494             else if ( t[0] == -VECTOR || t[0] == -INDEX ) {
01495                 *w++ = 7; *w++ = INDEX; *w++ = 3; *w++ = t[1];
01496                 *w++ = 1; *w++ = 1; *w++ = 3;
01497             }
01498             else if ( t[0] == -MINVECTOR ) {
01499                 *w++ = 7; *w++ = INDEX; *w++ = 3; *w++ = t[1];
01500                 *w++ = 1; *w++ = 1; *w++ = -3;
01501             }
01502             else if ( t[0] == -SNUMBER ) {
01503                 *w++ = 4;
01504                 if ( t[1] < 0 ) {
01505                     *w++ = -t[1]; *w++ = 1; *w++ = -3;
01506                 }
01507                 else {
01508                     *w++ = t[1]; *w++ = 1; *w++ = 3;
01509                 }
01510             }
01511             *w = 0; size = w - AT.WorkPointer;
01512             w = AT.WorkPointer;
01513             break;
01514         }
01515         /* fall through */
01516     case DOLNUMBER:
01517     case DOLTERMS:
01518         t = d->where;
01519         while ( *t ) t += *t;
01520         size = t - d->where;
01521         w = d->where;
01522         break;
01523     case DOLSUBTERM:
01524         w = AT.WorkPointer;
01525         size = d->where[1];
01526         *w++ = size+4; t = d->where; NCOPY(w,t,size)
01527         *w++ = 1; *w++ = 1; *w++ = 3;
01528         w = AT.WorkPointer; size = d->where[1]+4;
01529         break;
01530     case DOLINDEX:
01531         w = AT.WorkPointer;
01532         *w++ = 7; *w++ = INDEX; *w++ = 3; *w++ = d->index;
01533         *w++ = 1; *w++ = 1; *w++ = 3; *w = 0;
01534         w = AT.WorkPointer; size = 7;
01535         break;
01536     case DOLWILDARGS:
01537     /*
01538         In some cases we can make a copy
01539     */
01540         t = d->where+1;
01541         if ( *t == 0 ) return(0);
01542         NEXTARG(t);
01543         if ( *t ) { /* More than one argument in here */
01544             MLOCK(ErrorMessageLock);
01545             MesPrint("Trying to convert a $ with an argument field into an expression");
01546             MUNLOCK(ErrorMessageLock);
01547             Terminate(-1);
01548         }
01549     /*
01550         Now we have a single argument
01551     */
01552         t = d->where+1;
01553         if ( *t < 0 ) goto ShortArgument;
01554         size = *t - ARGHEAD;
01555         w = t + ARGHEAD;
01556         break;
01557     case DOLUNDEFINED:
01558         MLOCK(ErrorMessageLock);
01559         MesPrint("Trying to use an undefined $ in an expression");
01560         MUNLOCK(ErrorMessageLock);
01561         Terminate(-1);
01562         /* fall through */

```

```

01563         case DOLZERO:
01564             if ( d->where ) { d->where[0] = 0; }
01565             else d->where = &(AM.dollarzero);
01566             size = 0;
01567             w = d->where;
01568             break;
01569         default:
01570             return(0);
01571     }
01572     newd = (DOLLARS)Malloc1(sizeof(struct DoLLaRS)+(size+1)*sizeof(WORD),
01573         "Copy of dollar variable");
01574     t = (WORD *) (newd+1);
01575     newd->where = t;
01576     newd->name = d->name;
01577     newd->node = d->node;
01578     newd->type = DOLTERMS;
01579     newd->size = size;
01580     newd->numdummies = d->numdummies;
01581 #ifdef WITHPTHREADS
01582     newd->pthreadlockread = dummylock;
01583     newd->pthreadlockwrite = dummylock;
01584 #endif
01585     size++;
01586     NCOPY(t,w,size);
01587     newd->nfactors = d->nfactors;
01588     if ( d->nfactors > 1 ) {
01589         newd->factors = (FACDOLLAR *)Malloc1(d->nfactors*sizeof(FACDOLLAR),"Dollar factors");
01590         for ( i = 0; i < d->nfactors; i++ ) {
01591             newd->factors[i].where = 0;
01592             newd->factors[i].size = 0;
01593             newd->factors[i].type = DOLUNDEFINED;
01594             newd->factors[i].value = d->factors[i].value;
01595         }
01596     }
01597     else { newd->factors = 0; }
01598     return(newd);
01599 }
01600
01601 /*
01602  #] DolToTerms :
01603  #[ DolToLong :
01604  */
01605
01606 LONG DolToLong(PHEAD WORD numdollar)
01607 {
01608     GETBIDENTITY
01609     DOLLARS d = Dollars + numdollar;
01610     LONG x;
01611 #ifdef WITHPTHREADS
01612     int nummodopt, dtype = -1;
01613     if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
01614         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01615             if ( numdollar == ModOptdollars[nummodopt].number ) break;
01616         }
01617         if ( nummodopt < NumModOptdollars ) {
01618             dtype = ModOptdollars[nummodopt].type;
01619             if ( dtype == MODLOCAL ) {
01620                 d = ModOptdollars[nummodopt].dstruct+AT.identity;
01621             }
01622         }
01623     }
01624 #endif
01625     AN.ErrorInDollar = 0;
01626     if ( ( d->type == DOLTERMS || d->type == DOLNUMBER )
01627         && d->where[0] == 4 &&
01628         d->where[4] == 0 && ( d->where[3] == 3 || d->where[3] == -3 )
01629         && d->where[2] == 1 && ( d->where[1] & TOPBITONLY ) == 0 ) {
01630         x = d->where[1];
01631         if ( d->where[3] > 0 ) return(x);
01632         else return(-x);
01633     }
01634     else if ( ( d->type == DOLTERMS || d->type == DOLNUMBER )
01635         && d->where[0] == 6 &&
01636         d->where[6] == 0 && ( d->where[5] == 5 || d->where[5] == -5 )
01637         && d->where[3] == 1 && d->where[4] == 1 && ( d->where[2] & TOPBITONLY ) == 0 ) {
01638         x = d->where[1] + ( (LONG) (d->where[2]) << BITSINWORD );
01639         if ( d->where[5] > 0 ) return(x);
01640         else return(-x);
01641     }
01642     else if ( d->type == DOLARGUMENT && d->where[0] == -SNUMBER ) {
01643         x = d->where[1];
01644         return(x);
01645     }
01646     else if ( d->type == DOLARGUMENT && d->where[0] == -INDEX
01647         && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
01648         x = d->where[1];
01649         return(x);

```

```

01650     }
01651     else if ( d->type == DOLZERO ) return(0);
01652     else if ( d->type == DOLWILDARGS && d->where[0] == 0
01653     && d->where[1] == -SNUMBER && d->where[3] == 0 ) {
01654         x = d->where[2];
01655         return(x);
01656     }
01657     else if ( d->type == DOLINDEX && d->index >= 0 && d->index < AM.OffsetIndex ) {
01658         x = d->index;
01659         return(x);
01660     }
01661     else if ( d->type == DOLWILDARGS && d->where[0] == 1
01662     && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
01663         x = d->where[1];
01664         return(x);
01665     }
01666     else if ( d->type == DOLWILDARGS && d->where[0] == 0
01667     && d->where[1] == -INDEX && d->where[3] == 0 && d->where[2] >= 0
01668     && d->where[2] < AM.OffsetIndex ) {
01669         x = d->where[2];
01670         return(x);
01671     }
01672     AN.ErrorInDollar = 1;
01673     return(0);
01674 }
01675
01676 /*
01677  #] DolToLong :
01678  #[ ExecInside :
01679  */
01680
01681 int ExecInside(UBYTE *s)
01682 {
01683     GETIDENTITY
01684     UBYTE *t, c;
01685     WORD *w, number;
01686     int error = 0;
01687     w = AT.WorkPointer;
01688     if ( AC.insidelevel >= MAXNEST ) {
01689         MLOCK(ErrorMessageLock);
01690         MesPrint("@Nesting of inside statements more than %d levels", (WORD)MAXNEST);
01691         MUNLOCK(ErrorMessageLock);
01692         return(-1);
01693     }
01694     AC.insidesumcheck[AC.insidelevel] = NestingChecksum();
01695     AC.insidestack[AC.insidelevel] = cbuf[AC.cbunum].Pointer
01696         - cbuf[AC.cbunum].Buffer + 2;
01697     AC.insidelevel++;
01698     *w++ = TYPEINSIDE;
01699     w++; w++;
01700     for(;;) { /* Look for a (comma separated) list of dollar variables */
01701         while ( *s == ',' ) s++;
01702         if ( *s == 0 ) break;
01703         if ( *s == '$' ) {
01704             s++; t = s;
01705             if ( FG.cTable[*s] != 0 ) {
01706                 MLOCK(ErrorMessageLock);
01707                 MesPrint("Illegal name for $ variable: %s", s-1);
01708                 MUNLOCK(ErrorMessageLock);
01709                 goto skipdol;
01710             }
01711             while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
01712             c = *s; *s = 0;
01713             if ( ( number = GetDollar(t) ) < 0 ) {
01714                 number = AddDollar(t, 0, 0, 0);
01715             }
01716             *s = c;
01717             *w++ = number;
01718             AddPotModdollar(number);
01719         }
01720         else {
01721             MLOCK(ErrorMessageLock);
01722             MesPrint("&Illegal object in Inside statement");
01723             MUNLOCK(ErrorMessageLock);
01724             skipdol: error = 1;
01725             while ( *s && *s != ',' && s[1] != '$' ) s++;
01726             if ( *s == 0 ) break;
01727         }
01728     }
01729     AT.WorkPointer[1] = w - AT.WorkPointer;
01730     AddNtoL(AT.WorkPointer[1], AT.WorkPointer);
01731     return(error);
01732 }
01733
01734 /*
01735  #] ExecInside :
01736  #[ InsideDollar :

```

```

01737
01738     Execution part of Inside $a;
01739     We have to take the variables one by one and then
01740     convert them into proper terms and call Generator for the proper levels.
01741     The conversion copies the whole dollar into a new buffer, making us
01742     insensitive to redefinitions of $a inside the Inside.
01743     In the end we sort and redefine $a.
01744 */
01745
01746 int InsideDollar(PHEAD WORD *ll, WORD level)
01747 {
01748     GETBIDENTITY
01749     int numvar = (int)(ll[1]-3), j, error = 0;
01750     WORD numdol, *oldcterm, *oldwork = AT.WorkPointer, olddefer, *r, *m;
01751     WORD oldnumlhs, *dbuffer;
01752     DOLLARS d, newd;
01753     oldcterm = AN.cTerm; AN.cTerm = 0;
01754     oldnumlhs = AR.Cnumlhs; AR.Cnumlhs = ll[2];
01755     ll += 3;
01756     olddefer = AR.DeferFlag;
01757     AR.DeferFlag = 0;
01758     while ( --numvar >= 0 ) {
01759         numdol = *ll++;
01760         d = Dollars + numdol;
01761         {
01762             #ifdef WITHPTHREADS
01763             int nummodopt, dtype = -1;
01764             if ( AS.MultiThreaded && ( AC.mparallelfld == PARALLELFLAG ) ) {
01765                 for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01766                     if ( numdol == ModOptdollars[nummodopt].number ) break;
01767                 }
01768                 if ( nummodopt < NumModOptdollars ) {
01769                     dtype = ModOptdollars[nummodopt].type;
01770                     if ( dtype == MODLOCAL ) {
01771                         d = ModOptdollars[nummodopt].dstruct+AT.identity;
01772                     }
01773                     else {
01774                         LOCK(d->pthreadlockwrite); /*
01775                         LOCK(d->pthreadlockread);
01776                     }
01777                 }
01778             }
01779             #endif
01780             newd = DolToTerms(BHEAD numdol);
01781             if ( newd == 0 ) {
01782                 continue;
01783             }
01784             if ( newd->where[0] == 0 ) {
01785                 // DolToTerms potentially allocates memory. Free it.
01786                 // The free below is inside the while loop.
01787                 if ( newd->factors ) M_free(newd->factors,"Dollar factors");
01788                 M_free(newd,"Copy of dollar variable");
01789                 continue;
01790             }
01791             r = newd->where;
01792             NewSort(BHEAD0);
01793             while ( *r ) { /* Sum over the terms */
01794                 m = AT.WorkPointer;
01795                 j = *r;
01796                 while ( --j >= 0 ) *m++ = *r++;
01797                 AT.WorkPointer = m;
01798             } /*
01799             What to do with dummy indices?
01800             */
01801             if ( Generator(BHEAD oldwork,level) ) {
01802                 LowerSortLevel();
01803                 error = -1; goto idcall;
01804             }
01805             AT.WorkPointer = oldwork;
01806         }
01807         AN.tryterm = 0; /* for now */
01808         if ( EndSort(BHEAD (WORD *)((void *)(&dbuffer)),2) < 0 ) { error = 1; break; }
01809         if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"old buffer of dollar");
01810         d->where = dbuffer;
01811         if ( dbuffer == 0 || *dbuffer == 0 ) {
01812             d->type = DOLZERO;
01813             if ( dbuffer ) M_free(dbuffer,"buffer of dollar");
01814             d->where = &(AM.dollarzero); d->size = 0;
01815         }
01816         else {
01817             d->type = DOLTERMS;
01818             r = d->where; while ( *r ) r += *r;
01819             d->size = (r-d->where)+1;
01820         }
01821         /* cbuf[AM.dbufnum].rhs[numdol] = d->where; */
01822         cbuf[AM.dbufnum].rhs[numdol] = (WORD *) (1);
01823         /*

```

```

01824         Now we have a little cleaning up to do
01825     */
01826 #ifdef WITHPTHREADS
01827     if ( dtype > 0 && dtype != MODLOCAL ) {
01828     /*         UNLOCK(d->pthreadlockwrite); */
01829         UNLOCK(d->pthreadlockread);
01830     }
01831 #endif
01832     if ( newd->factors ) M_free(newd->factors,"Dollar factors");
01833     M_free(newd,"Copy of dollar variable");
01834 }
01835 }
01836 idcall::
01837     AR.Cnumlhs = oldnumlhs;
01838     AR.DeferFlag = olddefer;
01839     AN.cTerm = oldcTerm;
01840     AT.WorkPointer = oldwork;
01841     return(error);
01842 }
01843
01844 /*
01845     #] InsideDollar :
01846     #[ ExchangeDollars :
01847 */
01848
01849 void ExchangeDollars(int num1, int num2)
01850 {
01851     DOLLARS d1, d2;
01852     WORD node1, node2;
01853     LONG nam;
01854     d1 = Dollars + num1; node1 = d1->node;
01855     d2 = Dollars + num2; node2 = d2->node;
01856     nam = d1->name; d1->name = d2->name; d2->name = nam;
01857     d1->node = node2; d2->node = node1;
01858     AC.dollarnames->namenode[node1].number = num2;
01859     AC.dollarnames->namenode[node2].number = num1;
01860 }
01861
01862 /*
01863     #] ExchangeDollars :
01864     #[ TermsInDollar :
01865 */
01866
01867 LONG TermsInDollar(WORD num)
01868 {
01869     GETIDENTITY
01870     DOLLARS d = Dollars + num;
01871     WORD *t;
01872     LONG n;
01873 #ifdef WITHPTHREADS
01874     int nummodopt, dtype = -1;
01875     if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELF ) ) {
01876         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01877             if ( num == ModOptdollars[nummodopt].number ) break;
01878         }
01879         if ( nummodopt < NumModOptdollars ) {
01880             dtype = ModOptdollars[nummodopt].type;
01881             if ( dtype == MODLOCAL ) {
01882                 d = ModOptdollars[nummodopt].dstruct+AT.identity;
01883             }
01884             else {
01885                 LOCK(d->pthreadlockread);
01886             }
01887         }
01888     }
01889 #endif
01890     if ( d->type == DOLTERMS ) {
01891         n = 0;
01892         t = d->where;
01893         while ( *t ) { t += *t; n++; }
01894     }
01895     else if ( d->type == DOLWILDARGS ) {
01896         n = 0;
01897         if ( d->where[0] == 0 ) {
01898             t = d->where+1;
01899             while ( *t != 0 ) { NEXTARG(t); n++; }
01900         }
01901         else if ( d->where[0] == 1 ) n = 1;
01902     }
01903     else if ( d->type == DOLZERO ) n = 0;
01904     else n = 1;
01905 #ifdef WITHPTHREADS
01906     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
01907 #endif
01908     return(n);
01909 }
01910

```

```

01911 /*
01912     #] TermsInDollar :
01913     #[ SizeOfDollar :
01914 */
01915
01916 LONG SizeOfDollar(WORD num)
01917 {
01918     GETIDENTITY
01919     DOLLARS d = Dollars + num;
01920     WORD *t;
01921     LONG n;
01922 #ifndef WITHPTHREADS
01923     int nummodopt, dtype = -1;
01924     if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
01925         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01926             if ( num == ModOptdollars[nummodopt].number ) break;
01927         }
01928         if ( nummodopt < NumModOptdollars ) {
01929             dtype = ModOptdollars[nummodopt].type;
01930             if ( dtype == MODLOCAL ) {
01931                 d = ModOptdollars[nummodopt].dstruct+AT.identity;
01932             }
01933             else {
01934                 LOCK(d->pthreadlockread);
01935             }
01936         }
01937     }
01938 #endif
01939     if ( d->type == DOLTERMS ) {
01940         t = d->where;
01941         while ( *t ) t += *t;
01942         t++;
01943         n = (LONG)(t - d->where);
01944     }
01945     else if ( d->type == DOLWILDARGS ) {
01946         n = 0;
01947         if ( d->where[0] == 0 ) {
01948             t = d->where+1;
01949             while ( *t != 0 ) { NEXTARG(t); n++; }
01950             t++;
01951             n = (LONG)(t - d->where);
01952         }
01953         else if ( d->where[0] == 1 ) n = 1;
01954     }
01955     else if ( d->type == DOLZERO ) n = 0;
01956     else n = 1;
01957 #ifdef WITHPTHREADS
01958     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
01959 #endif
01960     return(n);
01961 }
01962
01963 /*
01964     #] SizeOfDollar :
01965     #[ PreIfDollarEval :
01966
01967     Routine is invoked in #if etc after $( is encountered.
01968     $(expr1 operator expr2) makes compares between expressions,
01969     $(expr1 operator _keyword) makes compares between expressions,
01970     interpreted as expressions. We are here mainly looking at $variables.
01971     First we look for the operator:
01972         >, <, ==, >=, <=, != : < means that it comes before.
01973     _keywords can be:
01974         _set(setname)    (does the expr belong to the set (only with == or !=))
01975         _productof(expr)
01976 */
01977
01978 UBYTE *PreIfDollarEval(UBYTE *s, int *value)
01979 {
01980     GETIDENTITY
01981     UBYTE *s1,*s2,*s3,*s4,*s5,*t,c,c1,c2,c3;
01982     int oprtr, type;
01983     WORD *buf1 = 0, *buf2 = 0, numset, *oldwork = AT.WorkPointer;
01984     EXCHINOUT
01985 /*
01986     Find the three composing objects (epxression, operator, expression or keyw
01987 */
01988     while ( *s == ' ' || *s == '\t' || *s == '\n' || *s == '\r' ) s++;
01989     s1 = t = s;
01990     while ( *t != '=' && *t != '!' && *t != '>' && *t != '<' ) {
01991         if ( *t == '[' ) { SKIPBRA1(t) }
01992         else if ( *t == '{' ) { SKIPBRA2(t) }
01993         else if ( *t == '(' ) { SKIPBRA3(t) }
01994         else if ( *t == ']' || *t == '}' || *t == ')' ) {
01995             MLOCK(ErrorMessageLock);
01996             MesPrint("@Improper bracketting in #if");
01997             MUNLOCK(ErrorMessageLock);

```

```

01998         goto onerror;
01999     }
02000     t++;
02001 }
02002 s2 = t;
02003 while ( *t == '=' || *t == '!' || *t == '>' || *t == '<' ) t++;
02004 s3 = t;
02005 while ( *t && *t != ')' ) {
02006     if ( *t == '[' ) { SKIPBRA1(t) }
02007     else if ( *t == '{' ) { SKIPBRA2(t) }
02008     else if ( *t == '(' ) { SKIPBRA3(t) }
02009     else if ( *t == ']' || *t == '}' ) {
02010         MLOCK(ErrorMessageLock);
02011         MesPrint("@Improper brackets in #if");
02012         MUNLOCK(ErrorMessageLock);
02013         goto onerror;
02014     }
02015     t++;
02016 }
02017 if ( *t == 0 ) {
02018     MLOCK(ErrorMessageLock);
02019     MesPrint("@Missing ) to match $( in #if");
02020     MUNLOCK(ErrorMessageLock);
02021     goto onerror;
02022 }
02023 s4 = t; c2 = *s4; *s4 = 0;
02024 if ( s2+2 < s3 || s2 == s3 ) {
02025     IllOp;;
02026     MLOCK(ErrorMessageLock);
02027     MesPrint("@Illegal operator in $( option of #if");
02028     MUNLOCK(ErrorMessageLock);
02029     goto onerror;
02030 }
02031 if ( s2+1 == s3 ) {
02032     if ( *s2 == '=' ) oprtr = EQUAL;
02033     else if ( *s2 == '>' ) oprtr = GREATER;
02034     else if ( *s2 == '<' ) oprtr = LESS;
02035     else goto IllOp;
02036 }
02037 else if ( *s2 == '!' && s2[1] == '=' ) oprtr = NOTEQUAL;
02038 else if ( *s2 == '=' && s2[1] == '=' ) oprtr = EQUAL;
02039 else if ( *s2 == '<' && s2[1] == '=' ) oprtr = LESSEQUAL;
02040 else if ( *s2 == '>' && s2[1] == '=' ) oprtr = GREATEREQUAL;
02041 else goto IllOp;
02042 c1 = *s2; *s2 = 0;
02043 /*
02044     The two expressions are now zero terminated
02045     Look for the special keywords
02046 */
02047 while ( *s3 == ' ' || *s3 == '\t' || *s3 == '\n' || *s3 == '\r' ) s3++;
02048 t = s3;
02049 while ( chartype[*t] == 0 ) t++;
02050 if ( *t == '_' ) {
02051     t++; c = *t; *t = 0;
02052     if ( StrICmp(s3, (UBYTE *) "set_") == 0 ) {
02053         if ( oprtr != EQUAL && oprtr != NOTEQUAL ) {
02054             ImpOp;;
02055             MLOCK(ErrorMessageLock);
02056             MesPrint("@Improper operator for special keyword in $( ) option");
02057             MUNLOCK(ErrorMessageLock);
02058             goto onerror;
02059         }
02060         type = 1;
02061     }
02062     else if ( StrICmp(s3, (UBYTE *) "multipleof_") == 0 ) {
02063         if ( oprtr != EQUAL && oprtr != NOTEQUAL ) goto ImpOp;
02064         type = 2;
02065     }
02066     /*
02067     else if ( StrICmp(s3, (UBYTE *) "productof_") == 0 ) {
02068         if ( oprtr != EQUAL && oprtr != NOTEQUAL ) goto ImpOp;
02069         type = 3;
02070     }
02071 */
02072     else type = 0;
02073 }
02074 else { type = 0; c = *t; }
02075 if ( type > 0 ) {
02076     *t++ = c; s3 = t; s5 = s4-1;
02077     while ( *s5 != ')' ) {
02078         if ( *s5 == ' ' || *s5 == '\t' || *s5 == '\n' || *s5 == '\r' ) s5--;
02079         else {
02080             MLOCK(ErrorMessageLock);
02081             MesPrint("@Improper use of special keyword in $( ) option");
02082             MUNLOCK(ErrorMessageLock);
02083             goto onerror;
02084         }

```

```

02085     }
02086     c3 = *s5; *s5 = 0;
02087 }
02088 else { c3 = c2; s5 = s4; }
02089 /*
02090 Expand the first expression.
02091 */
02092 if ( ( buf1 = TranslateExpression(s1) ) == 0 ) {
02093     AT.WorkPointer = oldwork;
02094     goto onerror;
02095 }
02096 if ( type == 1 ) { /* determine the set */
02097     if ( *s3 == '{' ) {
02098         t = s3+1;
02099         SKIPBRA2(s3)
02100         numset = DoTempSet(t,s3);
02101         s3++;
02102         if ( numset < 0 ) {
02103 noset::
02104             MLOCK(ErrorMessageLock);
02105             MesPrint("@Argument of set_ is not a valid set");
02106             MUNLOCK(ErrorMessageLock);
02107             goto onerror;
02108         }
02109     }
02110     else {
02111         t = s3;
02112         while ( FG.cTable[*s3] == 0 || FG.cTable[*s3] == 1
02113             || *s3 == '_' ) s3++;
02114         c = *s3; *s3 = 0;
02115         if ( GetName(AC.varnames,t,&numset,NOAUTO) != CSET ) {
02116             *s3 = c; goto noset;
02117         }
02118         *s3 = c;
02119     }
02120     while ( *s3 == ' ' || *s3 == '\t' || *s3 == '\n' || *s3 == '\r' ) s3++;
02121     if ( s3 != s5 ) goto noset;
02122     *value = IsSetMember(buf1,numset);
02123     if ( oprtr == NOTEQUAL ) *value ^= 1;
02124 }
02125 else {
02126     if ( ( buf2 = TranslateExpression(s3) ) == 0 ) goto onerror;
02127 }
02128 if ( type == 0 ) {
02129     *value = TwoExprCompare(buf1,buf2,oprtr);
02130 }
02131 else if ( type == 2 ) {
02132     *value = IsMultipleOf(buf1,buf2);
02133     if ( oprtr == NOTEQUAL ) *value ^= 1;
02134 }
02135 /*
02136 else if ( type == 3 ) {
02137     *value = IsProductOf(buf1,buf2);
02138     if ( oprtr == NOTEQUAL ) *value ^= 1;
02139 }
02140 */
02141 if ( buf1 ) M_free(buf1,"Buffer in $()");
02142 if ( buf2 ) M_free(buf2,"Buffer in $()");
02143 *s5 = c3; *s4++ = c2; *s2 = c1;
02144 AT.WorkPointer = oldwork;
02145 BACKINOUT
02146 return(s4);
02147 onerror:
02148 if ( buf1 ) M_free(buf1,"Buffer in $()");
02149 if ( buf2 ) M_free(buf2,"Buffer in $()");
02150 AT.WorkPointer = oldwork;
02151 BACKINOUT
02152 return(0);
02153 }
02154
02155 /*
02156 #] PreIfDollarEval :
02157 #[ TranslateExpression :
02158 */
02159
02160 WORD *TranslateExpression(UBYTE *s)
02161 {
02162     GETIDENTITY
02163     CBUF *C = cbuf+AC.cbunum;
02164     WORD oldnumrhs = C->numrhs;
02165     LONG oldcpointer = C->Pointer - C->Buffer;
02166     WORD *w = AT.WorkPointer;
02167     WORD retcode, oldEside;
02168     WORD *outbuffer;
02169     *w++ = SUBEXPSize + 4;
02170     AC.ProtoType = w;
02171     *w++ = SUBEXPRESSION;

```



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02172     *w++ = SUBEXPSIZE;
02173     *w++ = C->numrhs+1;
02174     *w++ = 1;
02175     *w++ = AC.cbufnum;
02176     FILLSUB(w)
02177     *w++ = 1; *w++ = 1; *w++ = 3; *w++ = 0;
02178     AT.WorkPointer = w;
02179     if ( ( retcode = CompileAlgebra(s,RHSIDE,AC.ProtoType) ) < 0 ) {
02180         MLOCK(ErrorMessageLock);
02181         MesPrint("@Error translating first expression in $( ) option");
02182         MUNLOCK(ErrorMessageLock);
02183         return(0);
02184     }
02185     else { AC.ProtoType[2] = retcode; }
02186 /*
02187     Evaluate this expression
02188 */
02189     if ( NewSort(BHEAD0) || NewSort(BHEAD0) ) { return(0); }
02190     AN.RepPoint = AT.RepCount + 1;
02191     oldEside = AR.Eside; AR.Eside = RHSIDE;
02192     AR.Cnumlhs = C->numlhs;
02193     if ( Generator(BHEAD AC.ProtoType-1,C->numlhs) ) {
02194         AR.Eside = oldEside;
02195         LowerSortLevel(); LowerSortLevel(); return(0);
02196     }
02197     AR.Eside = oldEside;
02198     AT.WorkPointer = w;
02199     AN.tryterm = 0; /* for now */
02200     if ( EndSort(BHEAD (WORD *)((void *)&outbuffer)),2) < 0 ) { LowerSortLevel(); return(0); }
02201     LowerSortLevel();
02202     C->Pointer = C->Buffer + oldcpointer;
02203     C->numrhs = oldnumrhs;
02204     AT.WorkPointer = AC.ProtoType - 1;
02205     return(outbuffer);
02206 }
02207
02208 /*
02209     #] TranslateExpression :
02210     #[ IsSetMember :
02211
02212     Checks whether the expression in the buffer can be seen as an element
02213     of the given set.
02214     For the special sets: if more than one term: no match!!!
02215 */
02216
02217 int IsSetMember(WORD *buffer, WORD numset)
02218 {
02219     WORD *t = buffer, *tt, num, csize, num1;
02220     WORD bufterm[4];
02221     int i, j, type;
02222     if ( numset < AM.NumFixedSets ) {
02223         if ( t[*t] != 0 ) return(0); /* More than one term */
02224         if ( *t == 0 ) {
02225             if ( numset == POS0_ || numset == NEG0_ || numset == EVEN_
02226                 || numset == Z_ || numset == Q_ ) return(1);
02227             else return(0);
02228         }
02229         if ( numset == SYMBOL_ ) {
02230             if ( *t == 8 && t[1] == SYMBOL && t[7] == 3 && t[6] == 1
02231                 && t[5] == 1 && t[4] == 1 ) return(1);
02232             else return(0);
02233         }
02234         if ( numset == INDEX_ ) {
02235             if ( *t == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
02236                 && t[4] == 1 && t[3] > 0 ) return(1);
02237             if ( *t == 4 && t[3] == 3 && t[2] == 1 && t[1] < AM.OffsetIndex )
02238                 return(1);
02239             return(0);
02240         }
02241         if ( numset == FIXED_ ) {
02242             if ( *t == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
02243                 && t[4] == 1 && t[3] > 0 && t[3] < AM.OffsetIndex ) return(1);
02244             if ( *t == 4 && t[3] == 3 && t[2] == 1 && t[1] < AM.OffsetIndex )
02245                 return(1);
02246             return(0);
02247         }
02248         if ( numset == DUMMYINDEX_ ) {
02249             if ( *t == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
02250                 && t[4] == 1 && t[3] >= AM.IndDum && t[3] < AM.IndDum+MAXDUMMIES ) return(1);
02251             if ( *t == 4 && t[3] == 3 && t[2] == 1
02252                 && t[1] >= AM.IndDum && t[1] < AM.IndDum+MAXDUMMIES ) return(1);
02253             return(0);
02254         }
02255         if ( numset == VECTOR_ ) {
02256             if ( *t == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
02257                 && t[4] == 1 && t[3] < (AM.OffsetVector+WILDOFFSET) && t[3] >= AM.OffsetVector )
02258                 return(1);

```

```

02258         return(0);
02259     }
02260     tt = t + *t - 1;
02261     if ( ABS(tt[0]) != *t-1 ) return(0);
02262     if ( numset == Q_ ) return(1);
02263     if ( numset == POS_ || numset == POS0_ ) return(tt[0]>0);
02264     else if ( numset == NEG_ || numset == NEG0_ ) return(tt[0]<0);
02265     i = (ABS(tt[0])-1)/2;
02266     tt -= i;
02267     if ( tt[0] != 1 ) return(0);
02268     for ( j = 1; j < i; j++ ) { if ( tt[j] != 0 ) return(0); }
02269     if ( numset == Z_ ) return(1);
02270     if ( numset == ODD_ ) return(t[1]&1);
02271     if ( numset == EVEN_ ) return(1-(t[1]&1));
02272     return(0);
02273 }
02274 if ( t[*t] != 0 ) return(0); /* More than one term */
02275 type = Sets[numset].type;
02276 switch ( type ) {
02277     case CSYMBOL:
02278         if ( t[0] == 8 && t[1] == SYMBOL && t[7] == 3 && t[6] == 1
02279             && t[5] == 1 && t[4] == 1 ) {
02280             num = t[3];
02281         }
02282         else if ( t[0] == 4 && t[2] == 1 && t[1] <= MAXPOWER ) {
02283             num = t[1];
02284             if ( t[3] < 0 ) num = -num;
02285             num += 2*MAXPOWER;
02286         }
02287         else return(0);
02288         break;
02289     case CVECTOR:
02290         if ( t[0] == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
02291             && t[4] == 1 && t[3] < 0 ) {
02292             num = t[3];
02293         }
02294         else return(0);
02295         break;
02296     case CINDEX:
02297         if ( t[0] == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
02298             && t[4] == 1 && t[3] > 0 ) {
02299             num = t[3];
02300         }
02301         else if ( t[0] == 4 && t[3] == 3 && t[2] == 1 && t[1] < AM.OffsetIndex ) {
02302             num = t[1];
02303         }
02304         else return(0);
02305         break;
02306     case CFUNCTION:
02307         if ( t[0] == 4+FUNHEAD && t[3+FUNHEAD] == 3 && t[2+FUNHEAD] == 1
02308             && t[1+FUNHEAD] == 1 && t[1] >= FUNCTION ) {
02309             num = t[1];
02310         }
02311         else return(0);
02312         break;
02313     case CNUMBER:
02314         if ( t[0] == 4 && t[2] == 1 && t[1] <= AM.OffsetIndex && t[3] == 3 ) {
02315             num = t[1];
02316         }
02317         else return(0);
02318         break;
02319     case CRANGE:
02320         csize = t[t[0]-1];
02321         csize = ABS(csize);
02322         if ( csize != t[0]-1 ) return(0);
02323         if ( Sets[numset].first < 3*MAXPOWER ) {
02324             num1 = num = Sets[numset].first;
02325             if ( num >= MAXPOWER ) num -= 2*MAXPOWER;
02326             if ( num == 0 ) {
02327                 if ( num1 < MAXPOWER ) {
02328                     if ( t[t[0]-1] >= 0 ) return(0);
02329                 }
02330                 else if ( t[t[0]-1] > 0 ) return(0);
02331             }
02332             else {
02333                 bufterm[0] = 4; bufterm[1] = ABS(num);
02334                 bufterm[2] = 1;
02335                 if ( num < 0 ) bufterm[3] = -3;
02336                 else bufterm[3] = 3;
02337                 num = CompCoef(t,bufterm);
02338                 if ( num1 < MAXPOWER ) {
02339                     if ( num >= 0 ) return(0);
02340                 }
02341                 else if ( num > 0 ) return(0);
02342             }
02343         }
02344         if ( Sets[numset].last > -3*MAXPOWER ) {

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```

02345         num1 = num = Sets[numset].last;
02346         if ( num <= -MAXPOWER ) num += 2*MAXPOWER;
02347         if ( num == 0 ) {
02348             if ( num1 > -MAXPOWER ) {
02349                 if ( t[t[0]-1] <= 0 ) return(0);
02350             }
02351             else if ( t[t[0]-1] < 0 ) return(0);
02352         }
02353         else {
02354             bufterm[0] = 4; bufterm[1] = ABS(num);
02355             bufterm[2] = 1;
02356             if ( num < 0 ) bufterm[3] = -3;
02357             else bufterm[3] = 3;
02358             num = CompCoeef(t,bufterm);
02359             if ( num1 > -MAXPOWER ) {
02360                 if ( num <= 0 ) return(0);
02361             }
02362             else if ( num < 0 ) return(0);
02363         }
02364     }
02365     return(1);
02366     break;
02367     default: return(0);
02368 }
02369 t = SetElements + Sets[numset].first;
02370 tt = SetElements + Sets[numset].last;
02371 do {
02372     if ( num == *t ) return(1);
02373     t++;
02374 } while ( t < tt );
02375 return(0);
02376 }
02377
02378 /*
02379 #] IsSetMember :
02380 #[ IsProductOf :
02381
02382 Checks whether the expression in buf1 is a single term multiple of
02383 the expression in buf2.
02384
02385 int IsProductOf(WORD *buf1, WORD *buf2)
02386 {
02387     return(0);
02388 }
02389
02390
02391 #] IsProductOf :
02392 #[ IsMultipleOf :
02393
02394 Checks whether the expression in buf1 is a numerical multiple of
02395 the expression in buf2.
02396 */
02397
02398 int IsMultipleOf(WORD *buf1, WORD *buf2)
02399 {
02400     GETIDENTITY
02401     LONG num1, num2;
02402     WORD *t1, *t2, *m1, *m2, *r1, *r2, nc1, nc2, ni1, ni2;
02403     UWORD *IfScrat1, *IfScrat2;
02404     int i, j;
02405     if ( *buf1 == 0 && *buf2 == 0 ) return(1);
02406     /*
02407     First count terms
02408     */
02409     t1 = buf1; t2 = buf2; num1 = 0; num2 = 0;
02410     while ( *t1 ) { t1 += *t1; num1++; }
02411     while ( *t2 ) { t2 += *t2; num2++; }
02412     if ( num1 != num2 ) return(0);
02413     /*
02414     Test similarity of terms. Difference up to a number.
02415     */
02416     t1 = buf1; t2 = buf2;
02417     while ( *t1 ) {
02418         m1 = t1+1; m2 = t2+1; t1 += *t1; t2 += *t2;
02419         r1 = t1 - ABS(t1[-1]); r2 = t2 - ABS(t2[-1]);
02420         if ( r1-m1 != r2-m2 ) return(0);
02421         while ( m1 < r1 ) {
02422             if ( *m1 != *m2 ) return(0);
02423             m1++; m2++;
02424         }
02425     }
02426     /*
02427     Now we have to test the constant factor
02428     */
02429     IfScrat1 = (UWORD *) (TermMalloc("IsMultipleOf")); IfScrat2 = (UWORD
02430 *) (TermMalloc("IsMultipleOf"));
02431     t1 = buf1; t2 = buf2;

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```

02431     t1 += *t1; t2 += *t2;
02432     if ( *t1 == 0 && *t2 == 0 ) return(1);
02433     r1 = t1 - ABS(t1[-1]); r2 = t2 - ABS(t2[-1]);
02434     nc1 = REDLENG(t1[-1]); nc2 = REDLENG(t2[-1]);
02435     if ( DivRat(BHEAD (UWORD *)r1,nc1,(UWORD *)r2,nc2,IfScrat1,&ni1) ) {
02436         MLOCK(ErrorMessageLock);
02437         MesPrint("@Called from MultipleOf in $( )");
02438         MUNLOCK(ErrorMessageLock);
02439         TermFree(IfScrat1,"IsMultipleOf"); TermFree(IfScrat2,"IsMultipleOf");
02440         Terminate(-1);
02441     }
02442     while ( *t1 ) {
02443         t1 += *t1; t2 += *t2;
02444         r1 = t1 - ABS(t1[-1]); r2 = t2 - ABS(t2[-1]);
02445         nc1 = REDLENG(t1[-1]); nc2 = REDLENG(t2[-1]);
02446         if ( DivRat(BHEAD (UWORD *)r1,nc1,(UWORD *)r2,nc2,IfScrat2,&ni2) ) {
02447             MLOCK(ErrorMessageLock);
02448             MesPrint("@Called from MultipleOf in $( )");
02449             MUNLOCK(ErrorMessageLock);
02450             TermFree(IfScrat1,"IsMultipleOf"); TermFree(IfScrat2,"IsMultipleOf");
02451             Terminate(-1);
02452         }
02453         if ( ni1 != ni2 ) return(0);
02454         i = 2*ABS(ni1);
02455         for ( j = 0; j < i; j++ ) {
02456             if ( IfScrat1[j] != IfScrat2[j] ) {
02457                 TermFree(IfScrat1,"IsMultipleOf"); TermFree(IfScrat2,"IsMultipleOf");
02458                 return(0);
02459             }
02460         }
02461     }
02462     TermFree(IfScrat1,"IsMultipleOf"); TermFree(IfScrat2,"IsMultipleOf");
02463     return(1);
02464 }
02465
02466 /*
02467 #] IsMultipleOf :
02468 #[ TwoExprCompare :
02469
02470 Compares the expressions in buf1 and buf2 according to oprtr
02471 */
02472
02473 int TwoExprCompare(WORD *buf1, WORD *buf2, int oprtr)
02474 {
02475     GETIDENTITY
02476     WORD *t1, *t2, cond;
02477     t1 = buf1; t2 = buf2;
02478     while ( *t1 && *t2 ) {
02479         cond = CompareTerms(BHEAD t1,t2,1);
02480         if ( cond != 0 ) {
02481             if ( cond > 0 ) { /* t1 comes first */
02482                 switch ( oprtr ) { /* t1 is less */
02483                     case EQUAL: return(0);
02484                     case NOTEQUAL: return(1);
02485                     case GREATEREQUAL: return(0);
02486                     case GREATER: return(0);
02487                     case LESS: return(1);
02488                     case LESSEQUAL: return(1);
02489                 }
02490             }
02491             else {
02492                 switch ( oprtr ) {
02493                     case EQUAL: return(0);
02494                     case NOTEQUAL: return(1);
02495                     case GREATEREQUAL: return(1);
02496                     case GREATER: return(1);
02497                     case LESS: return(0);
02498                     case LESSEQUAL: return(0);
02499                 }
02500             }
02501         }
02502         t1 += *t1; t2 += *t2;
02503     }
02504     if ( *t1 == *t2 ) { /* They are equal */
02505         switch ( oprtr ) {
02506             case EQUAL: return(1);
02507             case NOTEQUAL: return(0);
02508             case GREATEREQUAL: return(1);
02509             case GREATER: return(0);
02510             case LESS: return(0);
02511             case LESSEQUAL: return(1);
02512         }
02513     }
02514     else if ( *t1 ) { /* t1 is greater */
02515         switch ( oprtr ) {
02516             case EQUAL: return(0);
02517             case NOTEQUAL: return(1);

```

```

02518         case GREATEREQUAL: return(1);
02519         case GREATER: return(1);
02520         case LESS: return(0);
02521         case LESSEQUAL: return(0);
02522     }
02523 }
02524 else {
02525     switch ( oprtr ) { /* t1 is less */
02526         case EQUAL: return(0);
02527         case NOTEQUAL: return(1);
02528         case GREATEREQUAL: return(0);
02529         case GREATER: return(0);
02530         case LESS: return(1);
02531         case LESSEQUAL: return(1);
02532     }
02533 }
02534 MLOCK(ErrorMessageLock);
02535 MesPrint("@Internal problems with operator in $( )");
02536 MUNLOCK(ErrorMessageLock);
02537 Terminate(-1);
02538 return(0);
02539 }
02540
02541 /*
02542  #] TwoExprCompare :
02543  #[ DollarRaiseLow :
02544
02545  Raises or lowers the numerical value of a dollar variable
02546  Not to be used in parallel.
02547 */
02548
02549 static UWORD *dscrat = 0;
02550 static WORD ndscrat;
02551
02552 int DollarRaiseLow(UBYTE *name, LONG value)
02553 {
02554     GETIDENTITY
02555     int num;
02556     DOLLARS d;
02557     int sgn = 1;
02558     WORD lnum[4], nnum, *t1, *t2, i;
02559     UBYTE *s, c;
02560     s = name; while ( *s ) s++;
02561     if ( s[-1] == '-' && s[-2] == '-' && s > name+2 ) s -= 2;
02562     else if ( s[-1] == '+' && s[-2] == '+' && s > name+2 ) s -= 2;
02563     c = *s; *s = 0;
02564     num = GetDollar(name);
02565     *s = c;
02566     d = Dollars + num;
02567     if ( value < 0 ) { value = -value; sgn = -1; }
02568     if ( d->type == DOLZERO ) {
02569         if ( d->where ) M_free(d->where, "DollarRaiseLow");
02570         d->size = MINALLOCS;
02571         d->where = (WORD *)Malloc1(d->size*sizeof(WORD), "DollarRaiseLow");
02572         if ( ( value & AWORDMASK ) != 0 ) {
02573             d->where[0] = 6; d->where[1] = value » BITSINWORD;
02574             d->where[2] = (WORD)value; d->where[3] = 1; d->where[4] = 0;
02575             d->where[5] = 5*sgn; d->where[6] = 0;
02576             d->type = DOLTERMS;
02577         }
02578     }
02579     else {
02580         d->where[0] = 4; d->where[1] = (WORD)value; d->where[2] = 1;
02581         d->where[3] = 3*sgn; d->where[4] = 0;
02582         d->type = DOLNUMBER;
02583     }
02584     else if ( d->type == DOLNUMBER || ( d->type == DOLTERMS
02585     && d->where[d->where[0]] == 0
02586     && d->where[0] == ABS(d->where[d->where[0]-1])+1 ) ) {
02587         if ( ( value & AWORDMASK ) != 0 ) {
02588             lnum[0] = value » BITSINWORD;
02589             lnum[1] = (WORD)value; lnum[2] = 1; lnum[3] = 0;
02590             nnum = 2*sgn;
02591         }
02592         else {
02593             lnum[0] = (WORD)value; lnum[1] = 1; nnum = sgn;
02594         }
02595         i = d->where[d->where[0]-1];
02596         i = REDLENG(i);
02597         if ( dscrat == 0 ) {
02598             dscrat = (UWORD *)Malloc1((AM.MaxTal+2)*sizeof(UWORD), "DollarRaiseLow");
02599         }
02600         if ( AddRat(BHEAD (UWORD *) (d->where+1), i,
02601         (UWORD *)lnum, nnum, dscrat, &ndscrat) ) {
02602             MLOCK(ErrorMessageLock);
02603             MesCall("DollarRaiseLow");
02604             MUNLOCK(ErrorMessageLock);

```

```

02605         Terminate(-1);
02606     }
02607     ndscrat = INCLENG(ndscrat);
02608     i = ABS(ndscrat);
02609     if ( i == 0 ) {
02610         M_free(d->where,"DollarRaiseLow");
02611         d->where = 0;
02612         d->type = DOLZERO;
02613         d->size = 0;
02614         return(0);
02615     }
02616     if ( i+2 > d->size ) {
02617         M_free(d->where,"DollarRaiseLow");
02618         d->size = i+2;
02619         if ( d->size < MINALLOC ) d->size = MINALLOC;
02620         d->size = ((d->size+7)/8)*8;
02621         d->where = (WORD *)Malloc1(d->size*sizeof(WORD),"DollarRaiseLow");
02622     }
02623     t1 = d->where; *t1++ = i+1; t2 = (WORD *)dscrat;
02624     while ( --i > 0 ) *t1++ = *t2++;
02625     *t1++ = ndscrat; *t1 = 0;
02626     d->type = DOLTERMS;
02627 }
02628 return(0);
02629 }
02630
02631 /*
02632 #] DollarRaiseLow :
02633 #[ EvalDoLoopArg :
02634 */
02651 WORD EvalDoLoopArg(PHEAD WORD *arg, WORD par)
02652 {
02653     WORD num, type, *td;
02654     DOLLARS d;
02655     if ( *arg == SNUMBER ) return(arg[1]);
02656     if ( *arg == DOLLAREXPR2 && arg[1] < 0 ) return(-arg[1]-1);
02657     d = Dollars + arg[1];
02658 #ifdef WITHPTHREADS
02659     {
02660         int nummodopt, dtype = -1;
02661         if ( AS.MultiThreaded && ( AC.mparallelfalg == PARALLELFLAG ) ) {
02662             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
02663                 if ( arg[1] == ModOptdollars[nummodopt].number ) break;
02664             }
02665             if ( nummodopt < NumModOptdollars ) {
02666                 dtype = ModOptdollars[nummodopt].type;
02667                 if ( dtype == MODLOCAL ) {
02668                     d = ModOptdollars[nummodopt].dstruct+AT.identity;
02669                 }
02670             }
02671         }
02672     }
02673 #endif
02674     if ( *arg == DOLLAREXPRESSION ) {
02675         if ( arg[2] != DOLLAREXPR2 ) { /* end of chain */
02676             endofchain:
02677                 type = d->type;
02678                 if ( type == DOLZERO ) {}
02679                 else if ( type == DOLNUMBER ) {
02680                     td = d->where;
02681                     if ( ( td[0] != 4 ) || ( (td[1]&SPECMASK) != 0 ) || ( td[2] != 1 ) ) {
02682                         MLOCK(ErrorMessageLock);
02683                         if ( par == -1 ) {
02684                             MesPrint("$-variable is not a short number in print statement");
02685                         }
02686                         else {
02687                             MesPrint("$-variable is not a short number in do loop");
02688                         }
02689                         MUNLOCK(ErrorMessageLock);
02690                         Terminate(-1);
02691                     }
02692                     return( td[3] > 0 ? td[1]: -td[1] );
02693                 }
02694                 else {
02695                     MLOCK(ErrorMessageLock);
02696                     if ( par == -1 ) {
02697                         MesPrint("$-variable is not a number in print statement");
02698                     }
02699                     else {
02700                         MesPrint("$-variable is not a number in do loop");
02701                     }
02702                     MUNLOCK(ErrorMessageLock);
02703                     Terminate(-1);
02704                 }
02705                 return(0);
02706             }
02707             num = EvalDoLoopArg(BHEAD arg+2,par);

```

```

02708     }
02709     else if ( *arg == DOLLAREXP2 ) {
02710         if ( arg[1] < 0 ) { num = -arg[1]-1; }
02711         else if ( arg[2] != DOLLAREXP2 && par == -1 ) {
02712             goto endofchain;
02713         }
02714         else { num = EvalDoLoopArg(BHEAD arg+2,par); }
02715     }
02716     else {
02717         MLOCK(ErrorMessageLock);
02718         if ( par == -1 ) {
02719             MesPrint("Invalid $-variable in print statement");
02720         }
02721         else {
02722             MesPrint("Invalid $-variable in do loop");
02723         }
02724         MUNLOCK(ErrorMessageLock);
02725         Terminate(-1);
02726         return(0);
02727     }
02728     if ( num == 0 ) return(d->nfactors);
02729     if ( num > d->nfactors || num < 1 ) {
02730         MLOCK(ErrorMessageLock);
02731         if ( par == -1 ) {
02732             MesPrint("Not a valid factor number for $-variable in print statement");
02733         }
02734         else {
02735             MesPrint("Not a valid factor number for $-variable in do loop");
02736         }
02737         MUNLOCK(ErrorMessageLock);
02738         Terminate(-1);
02739         return(0);
02740     }
02741     if ( d->factors[num].type == DOLNUMBER )
02742         return(d->factors[num].value);
02743     else { /* If correct, type can only be DOLNUMBER or DOLTERMS */
02744         MLOCK(ErrorMessageLock);
02745         if ( par == -1 ) {
02746             MesPrint("$-variable in print statement is not a number");
02747         }
02748         else {
02749             MesPrint("$-variable in do loop is not a number");
02750         }
02751         MUNLOCK(ErrorMessageLock);
02752         Terminate(-1);
02753         return(0);
02754     }
02755 }
02756
02757 /*
02758 #] EvalDoLoopArg :
02759 #[ TestDoLoop :
02760 */
02761
02762 WORD TestDoLoop(PHEAD WORD *lhsbuf, WORD level)
02763 {
02764     GETBIDENTITY
02765     WORD start,finish,incr;
02766     WORD *h;
02767     DOLLARS d;
02768     h = lhsbuf + 4; /* address of the start value */
02769     start = EvalDoLoopArg(BHEAD h,0);
02770     while ( ( *h == DOLLAREXP2 || *h == DOLLAREXP2 )
02771         && ( h[2] == DOLLAREXP2 ) ) h += 2;
02772     h += 2;
02773     finish = EvalDoLoopArg(BHEAD h,0);
02774     while ( ( *h == DOLLAREXP2 || *h == DOLLAREXP2 )
02775         && ( h[2] == DOLLAREXP2 ) ) h += 2;
02776     h += 2;
02777     incr = EvalDoLoopArg(BHEAD h,0);
02778
02779     if ( ( finish == start ) || ( finish > start && incr > 0 )
02780         || ( finish < start && incr < 0 ) ) {}
02781     else { level = lhsbuf[3]; } /* skips the loop */
02782 /*
02783 Put start in the dollar variable indicated by lhsbuf[2]
02784 */
02785     d = Dollars + lhsbuf[2];
02786 #ifdef WITHPTHREADS
02787     {
02788         int nummodopt, dtype = -1;
02789         if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
02790             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
02791                 if ( lhsbuf[2] == ModOptdollars[nummodopt].number ) break;
02792             }
02793             if ( nummodopt < NumModOptdollars ) {
02794                 dtype = ModOptdollars[nummodopt].type;

```

```

02795         if ( dtype == MODLOCAL ) {
02796             d = ModOptdollars[nummodopt].dstruct+AT.identity;
02797         }
02798     }
02799 }
02800 }
02801 #endif
02802
02803     if ( d->size < MINALLOC ) {
02804         if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollar contents");
02805         d->size = MINALLOC;
02806         d->where = (WORD *)Malloc1(d->size*sizeof(WORD),"dollar contents");
02807     }
02808     if ( start > 0 ) {
02809         d->where[0] = 4;
02810         d->where[1] = start;
02811         d->where[2] = 1;
02812         d->where[3] = 3;
02813         d->where[4] = 0;
02814         d->type = DOLNUMBER;
02815     }
02816     else if ( start < 0 ) {
02817         d->where[0] = 4;
02818         d->where[1] = -start;
02819         d->where[2] = 1;
02820         d->where[3] = -3;
02821         d->where[4] = 0;
02822         d->type = DOLNUMBER;
02823     }
02824     else
02825         d->type = DOLZERO;
02826
02827     if ( d == Dollars + lhsbuf[2] ) {
02828         cbuf[AM.dbufnum].CanCommu[lhsbuf[2]] = 0;
02829         cbuf[AM.dbufnum].NumTerms[lhsbuf[2]] = 1;
02830         cbuf[AM.dbufnum].rhs[lhsbuf[2]] = d->where;
02831     }
02832     return(level);
02833 }
02834
02835 /*
02836 #] TestDoLoop :
02837 #[ TestEndDoLoop :
02838 */
02839
02840 WORD TestEndDoLoop(PHEAD WORD *lhsbuf, WORD level)
02841 {
02842     GETBIDENTITY
02843     WORD start,finish,incr,value;
02844     WORD *h;
02845     DOLLARS d;
02846     h = lhsbuf + 4; /* address of the start value */
02847     start = EvalDoLoopArg(BHEAD h,0);
02848     while ( ( *h == DOLLAREXPRESSION || *h == DOLLAREXP2 )
02849         && ( h[2] == DOLLAREXP2 ) ) h += 2;
02850     h += 2;
02851     finish = EvalDoLoopArg(BHEAD h,0);
02852     while ( ( *h == DOLLAREXPRESSION || *h == DOLLAREXP2 )
02853         && ( h[2] == DOLLAREXP2 ) ) h += 2;
02854     h += 2;
02855     incr = EvalDoLoopArg(BHEAD h,0);
02856
02857     if ( ( finish == start ) || ( finish > start && incr > 0 )
02858         || ( finish < start && incr < 0 ) ) {}
02859     else { level = lhsbuf[3]; } /* skips the loop */
02860 /*
02861 Put start in the dollar variable indicated by lhsbuf[2]
02862 */
02863     d = Dollars + lhsbuf[2];
02864 #ifdef WITHPTHREADS
02865     {
02866         int nummodopt, dtype = -1;
02867         if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
02868             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
02869                 if ( lhsbuf[2] == ModOptdollars[nummodopt].number ) break;
02870             }
02871             if ( nummodopt < NumModOptdollars ) {
02872                 dtype = ModOptdollars[nummodopt].type;
02873                 if ( dtype == MODLOCAL ) {
02874                     d = ModOptdollars[nummodopt].dstruct+AT.identity;
02875                 }
02876             }
02877         }
02878     }
02879 #endif
02880 /*
02881 Get the value

```



```

02882 */
02883     if ( d->type == DOLZERO ) {
02884         value = 0;
02885     }
02886     else if ( ( d->type == DOLNUMBER || d->type == DOLTERMS )
02887         && ( d->where[4] == 0 ) && ( d->where[0] == 4 )
02888         && ( d->where[1] > 0 ) && ( d->where[2] == 1 ) ) {
02889         value = ( d->where[3] < 0 ) ? -d->where[1]: d->where[1];
02890     }
02891     else {
02892         MLOCK(ErrorMessageLock);
02893         MesPrint("Wrong type of object in do loop parameter");
02894         MUNLOCK(ErrorMessageLock);
02895         Terminate(-1);
02896         return(level);
02897     }
02898     value += incr;
02899     if ( ( finish > start && value <= finish ) ||
02900         ( finish < start && value >= finish ) ||
02901         ( finish == start && value == finish ) ) {}
02902     else level = lhsbuf[3];
02903
02904     if ( d->size < MINALLOC ) {
02905         if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollar contents");
02906         d->size = MINALLOC;
02907         d->where = (WORD *)Malloc1(d->size*sizeof(WORD),"dollar contents");
02908     }
02909     if ( value > 0 ) {
02910         d->where[0] = 4;
02911         d->where[1] = value;
02912         d->where[2] = 1;
02913         d->where[3] = 3;
02914         d->where[4] = 0;
02915         d->type = DOLNUMBER;
02916     }
02917     else if ( start < 0 ) {
02918         d->where[0] = 4;
02919         d->where[1] = -value;
02920         d->where[2] = 1;
02921         d->where[3] = -3;
02922         d->where[4] = 0;
02923         d->type = DOLNUMBER;
02924     }
02925     else
02926         d->type = DOLZERO;
02927
02928     if ( d == Dollars + lhsbuf[2] ) {
02929         cbuf[AM.dbufnum].CanCommu[lhsbuf[2]] = 0;
02930         cbuf[AM.dbufnum].NumTerms[lhsbuf[2]] = 1;
02931         cbuf[AM.dbufnum].rhs[lhsbuf[2]] = d->where;
02932     }
02933     return(level);
02934 }
02935
02936 /*
02937     #] TestEndDoLoop :
02938     #[ DollarFactorize :
02939 */
02940 /* #define STEP2 */
02941 #define STEP2
02942
02943 int DollarFactorize(PHEAD WORD numdollar)
02944 {
02945     GETBIDENTITY
02946     DOLLARS d = Dollars + numdollar;
02947     CBUF *C, *CC;
02948     WORD *oldworkpointer;
02949     WORD *buf1, *t, *term, *buf1content, *buf2, *termextra;
02950     WORD *buf3, *argextra;
02951 #ifdef STEP2
02952     WORD *tstop, pow, *r;
02953 #endif
02954     int i, j, jj, action = 0, sign = 1;
02955     LONG insize, ii;
02956     WORD startebuf = cbuf[AT.ebufnum].numrhs;
02957     WORD nfactors, factorsincontent, extrafactor = 0;
02958     WORD oldsorttype = AR.SortType;
02959 #ifdef WITHPTHREADS
02960     int nummodopt, dtype;
02961     dtype = -1;
02962     if ( AS.MultiThreaded ) {
02963         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
02964             if ( numdollar == ModOptdollars[nummodopt].number ) break;
02965         }
02966         if ( nummodopt < NumModOptdollars ) {
02967             dtype = ModOptdollars[nummodopt].type;
02968         }
02969     }
02970 #endif

```

```

02981         if ( dtype == MODLOCAL ) {
02982             d = ModOptdollars[nummodopt].dstruct+AT.identity;
02983         }
02984         else {
02985             LOCK(d->pthreadlockread);
02986         }
02987     }
02988 }
02989 #endif
02990 CleanDollarFactors(d);
02991 #ifdef WITHPTHREADS
02992     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
02993 #endif
02994     if ( d->type != DOLTERMS ) { /* only one term */
02995         if ( d->type != DOLZERO ) d->nfactors = 1;
02996         return(0);
02997     }
02998     if ( d->where[d->where[0]] == 0 ) { /* only one term. easy */
02999     }
03000 /*
03001     Here should come the code for the factorization
03002     We copied the routine ArgFactorize in argument.c and changed the
03003     memory management completely. For the actual factorization it
03004     calls WORD *DoFactorizeDollar(PHEAD WORD *expr) which allocates
03005     space for the answer. Notation:
03006     term,...,term,0,term,...,term,0,term,...,term,0,0
03007
03008     #[ Step 1: sort the terms properly and/or make copy --> buf1,insize
03009 */
03010     term = d->where;
03011     AR.SortType = SORTHIGHFIRST;
03012     if ( oldsorttype != AR.SortType ) {
03013         NewSort(BHEAD0);
03014         while ( *term ) {
03015             t = term + *term;
03016             if ( AN.ncmod != 0 ) {
03017                 if ( AN.ncmod != 1 || ( (WORD)AN.cmod[0] < 0 ) ) {
03018                     AR.SortType = oldsorttype;
03019                     MLOCK(ErrorMessageLock);
03020                     MesPrint("Factorization modulus a number, greater than a WORD not implemented.");
03021                     MUNLOCK(ErrorMessageLock);
03022                     Terminate(-1);
03023                 }
03024                 if ( Modulus(term) ) {
03025                     AR.SortType = oldsorttype;
03026                     MLOCK(ErrorMessageLock);
03027                     MesCall("DollarFactorize");
03028                     MUNLOCK(ErrorMessageLock);
03029                     Terminate(-1);
03030                 }
03031                 if ( !*term ) { term = t; continue; }
03032             }
03033             StoreTerm(BHEAD term);
03034             term = t;
03035         }
03036         AN.tryterm = 0; /* for now */
03037         EndSort(BHEAD (WORD *)((void *)(&buf1)),2);
03038         t = buf1; while ( *t ) t += *t;
03039         insize = t - buf1;
03040     }
03041     else {
03042         t = term; while ( *t ) t += *t;
03043         ii = insize = t - term;
03044         buf1 = (WORD *)Malloc1((insize+1)*sizeof(WORD),"DollarFactorize-1");
03045         t = buf1;
03046         NCOPY(t,term,ii);
03047         *t++ = 0;
03048     }
03049 /*
03050     #] Step 1:
03051     #[ Step 2: take out the 'content'.
03052 */
03053 #ifdef STEP2
03054     buf1content = TermMalloc("DollarContent");
03055     AN.tryterm = -1;
03056     if ( ( buf2 = TakeContent(BHEAD buf1,buf1content) ) == 0 ) {
03057         AN.tryterm = 0;
03058         TermFree(buf1content,"DollarContent");
03059         M_free(buf1,"DollarFactorize-1");
03060         AR.SortType = oldsorttype;
03061         MLOCK(ErrorMessageLock);
03062         MesCall("DollarFactorize");
03063         MUNLOCK(ErrorMessageLock);
03064         Terminate(-1);
03065         return(1);
03066     }
03067     else if ( ( buf1content[0] == 4 ) && ( buf1content[1] == 1 ) &&

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```

03068         ( buf1content[2] == 1 ) && ( buf1content[3] == 3 ) ) { /* Nothing happened */
03069     AN.tryterm = 0;
03070     if ( buf2 != buf1 ) {
03071         M_free(buf2,"DollarFactorize-2");
03072         buf2 = buf1;
03073     }
03074     factorsincontent = 0;
03075 }
03076 else {
03077 /*
03078     The way we took out objects is rather brutish. We have to normalize
03079 */
03080     AN.tryterm = 0;
03081     if ( buf2 != buf1 ) M_free(buf1,"DollarFactorize-1");
03082     buf1 = buf2;
03083     t = buf1; while ( *t ) t += *t;
03084     insize = t - buf1;
03085 /*
03086     Now analyse how many factors there are in the content
03087 */
03088     factorsincontent = 0;
03089     term = buf1content;
03090     tstop = term + *term;
03091     if ( tstop[-1] < 0 ) factorsincontent++;
03092     if ( ABS(tstop[-1]) == 3 && tstop[-2] == 1 && tstop[-3] == 1 ) {
03093         tstop -= ABS(tstop[-1]);
03094     }
03095     else {
03096         factorsincontent++;
03097         tstop -= ABS(tstop[-1]);
03098     }
03099     term++;
03100     while ( term < tstop ) {
03101         switch ( *term ) {
03102             case SYMBOL:
03103                 t = term+2; i = (term[1]-2)/2;
03104                 while ( i > 0 ) {
03105                     factorsincontent += ABS(t[1]);
03106                     i--; t += 2;
03107                 }
03108                 break;
03109             case DOTPRODUCT:
03110                 t = term+2; i = (term[1]-2)/3;
03111                 while ( i > 0 ) {
03112                     factorsincontent += ABS(t[2]);
03113                     i--; t += 3;
03114                 }
03115                 break;
03116             case VECTOR:
03117             case DELTA:
03118                 factorsincontent += (term[1]-2)/2;
03119                 break;
03120             case INDEX:
03121                 factorsincontent += term[1]-2;
03122                 break;
03123             default:
03124                 if ( *term >= FUNCTION ) factorsincontent++;
03125                 break;
03126         }
03127         term += term[1];
03128     }
03129 }
03130 #else
03131     factorsincontent = 0;
03132     buf1content = 0;
03133 #endif
03134 /*
03135     #] Step 2: take out the 'content'.
03136     #[ Step 3: ConvertToPoly
03137         if there are objects that are not SYMBOLs,
03138         invoke ConvertToPoly
03139         We keep the original in buf1 in case there are no factors
03140 */
03141     t = buf1;
03142     while ( *t ) {
03143         if ( ( t[1] != SYMBOL ) && ( *t != (ABS(t[*t-1])+1) ) ) {
03144             action = 1; break;
03145         }
03146         t += *t;
03147     }
03148     if ( DetCommu(buf1) > 1 ) {
03149         MesPrint("Cannot factorize a $-expression with more than one noncommuting object");
03150         AR.SortType = oldsorttype;
03151         M_free(buf1,"DollarFactorize-2");
03152         if ( buf1content ) TermFree(buf1content,"DollarContent");
03153         MesCall("DollarFactorize");
03154         Terminate(-1);

```

```

03155         return(-1);
03156     }
03157     if ( action ) {
03158         t = buf1;
03159         termextra = AT.WorkPointer;
03160         NewSort(BHEAD0);
03161         NewSort(BHEAD0);
03162         while ( *t ) {
03163             if ( LocalConvertToPoly(BHEAD t,termextra,startebuf,0) < 0 ) {
03164 getout:
03165                 AR.SortType = oldsorttype;
03166                 M_free(buf1,"DollarFactorize-2");
03167                 if ( buf1content ) TermFree(buf1content,"DollarContent");
03168                 MesCall("DollarFactorize");
03169                 Terminate(-1);
03170                 return(-1);
03171             }
03172             StoreTerm(BHEAD termextra);
03173             t += *t;
03174         }
03175         AN.tryterm = 0; /* for now */
03176         if ( EndSort(BHEAD (WORD *)((void *)(&buf2)),2) < 0 ) { goto getout; }
03177         LowerSortLevel();
03178         t = buf2; while ( *t > 0 ) t += *t;
03179     }
03180     else {
03181         buf2 = buf1;
03182     }
03183     /*
03184         #] Step 3: ConvertToPoly
03185         #[ Step 4: Now the hard work.
03186     */
03187     if ( ( buf3 = poly_factorize_dollar(BHEAD buf2) ) == 0 ) {
03188         MesCall("DollarFactorize");
03189         AR.SortType = oldsorttype;
03190         if ( buf2 != buf1 && buf2 ) M_free(buf2,"DollarFactorize-3");
03191         M_free(buf1,"DollarFactorize-3");
03192         if ( buf1content ) TermFree(buf1content,"DollarContent");
03193         Terminate(-1);
03194         return(-1);
03195     }
03196     if ( buf2 != buf1 && buf2 ) {
03197         M_free(buf2,"DollarFactorize-3");
03198         buf2 = 0;
03199     }
03200     term = buf3;
03201     AR.SortType = oldsorttype;
03202     /*
03203     Count the factors and strip a factor -1
03204     */
03205     nfactors = 0;
03206     while ( *term ) {
03207 #ifdef STEP2
03208         if ( *term == 4 && term[4] == 0 && term[3] == -3 && term[2] == 1
03209             && term[1] == 1 ) {
03210             WORD *ttl, *tt2, *ttstop;
03211             sign = -sign;
03212             ttl = term; tt2 = term + *term + 1;
03213             ttstop = tt2;
03214             while ( *ttstop ) {
03215                 while ( *ttstop ) ttstop += *ttstop;
03216                 ttstop++;
03217             }
03218             while ( tt2 < ttstop ) *ttl++ = *tt2++;
03219             *ttl = 0;
03220             factorsincontent++;
03221             extrafactor++;
03222         }
03223     else
03224 #endif
03225     {
03226         term += *term;
03227         while ( *term ) { term += *term; }
03228         nfactors++; term++;
03229     }
03230 }
03231 /*
03232 We have now:
03233 buf1: the original before ConvertToPoly for if only one factor
03234 buf3: the factored expression with nfactors factors
03235
03236 #] Step 4:
03237 #[ Step 5: ConvertFromPoly
03238     If ConvertToPoly was used, use now ConvertFromPoly
03239     Be careful: there should be more than one factor now.
03240 */
03241 #ifdef WITHPTHREADS

```

```

03242     if ( dtype > 0 && dtype != MODLOCAL ) { LOCK(d->pthreadlockread); }
03243 #endif
03244     if ( nfactors == 1 && extrafactor == 0 ) { /* we can use the buf1 contents */
03245         if ( factorsincontent == 0 ) {
03246             d->nfactors = 1;
03247 #ifdef WITHPTHREADS
03248             if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
03249 #endif
03250 /*
03251     We used here (before 3-sep-2015) the original and did not make
03252     provisions for having a factors struct, figuring that all info
03253     is identical to the full dollar. This makes things too
03254     complicated at later stages.
03255 */
03256     d->factors = (FACDOLLAR *)Malloc1(sizeof(FACDOLLAR),"factors in dollar");
03257     term = buf1; while ( *term ) term += *term;
03258     d->factors[0].size = i = term - buf1;
03259     d->factors[0].where = t = (WORD *)Malloc1(sizeof(WORD)*(i+1),"DollarFactorize-5");
03260     term = buf1; NCOPY(t,term,i); *t = 0;
03261     AR.SortType = oldsorttype;
03262     M_free(buf3,"DollarFactorize-4");
03263     if ( buf2 != buf1 && buf2 ) M_free(buf2,"DollarFactorize-4");
03264     M_free(buf1,"DollarFactorize-4");
03265     if ( buf1content ) TermFree(buf1content,"DollarContent");
03266     return(0);
03267 }
03268     else {
03269         d->factors = (FACDOLLAR *)Malloc1(sizeof(FACDOLLAR)*(nfactors+factorsincontent),"factors
in dollar");
03270         term = buf1; while ( *term ) term += *term;
03271         d->factors[0].size = i = term - buf1;
03272         d->factors[0].where = t = (WORD *)Malloc1(sizeof(WORD)*(i+1),"DollarFactorize-5");
03273         term = buf1; NCOPY(t,term,i); *t = 0;
03274         M_free(buf3,"DollarFactorize-4");
03275         buf3 = 0;
03276         if ( buf2 != buf1 && buf2 ) {
03277             M_free(buf2,"DollarFactorize-4");
03278             buf2 = 0;
03279         }
03280     }
03281 }
03282     else if ( action ) {
03283         C = cbuf+AC.cbunum;
03284         CC = cbuf+AT.ebunum;
03285         oldworkpointer = AT.WorkPointer;
03286         d->factors = (FACDOLLAR *)Malloc1(sizeof(FACDOLLAR)*(nfactors+factorsincontent),"factors in
dollar");
03287         term = buf3;
03288         for ( i = 0; i < nfactors; i++ ) {
03289             argextra = AT.WorkPointer;
03290             NewSort(BHEAD0);
03291             NewSort(BHEAD0);
03292             while ( *term ) {
03293                 if ( ConvertFromPoly(BHEAD term,argextra,numxsymbol,CC->numrhs-startebuf+numxsymbol
,startebuf-numxsymbol,1) <= 0 ) {
03294                     LowerSortLevel();
03295                     AR.SortType = oldsorttype;
03296 getout2:    M_free(d->factors,"factors in dollar");
03297                     d->factors = 0;
03298 #ifdef WITHPTHREADS
03299                     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
03300 #endif
03301                     M_free(buf3,"DollarFactorize-4");
03302                     if ( buf2 != buf1 && buf2 ) M_free(buf2,"DollarFactorize-4");
03303                     M_free(buf1,"DollarFactorize-4");
03304                     if ( buf1content ) TermFree(buf1content,"DollarContent");
03305                     return(-3);
03306                 }
03307                 AT.WorkPointer = argextra + *argextra;
03308 /*
03309     ConvertFromPoly leaves terms with subexpressions. Hence:
03310 */
03311             if ( Generator(BHEAD argextra,C->numlhs+1) ) {
03312                 goto getout2;
03313             }
03314             term += *term;
03315         }
03316         term++;
03317         AT.WorkPointer = oldworkpointer;
03318         AN.tryterm = 0; /* for now */
03319         EndSort(BHEAD (WORD *)((void *)(&(d->factors[i].where))),2);
03320         LowerSortLevel();
03321         d->factors[i].type = DOLTERMS;
03322         t = d->factors[i].where;
03323         while ( *t ) t += *t;
03324         d->factors[i].size = t - d->factors[i].where;
03325     }
03326 }

```

```

03327         CC->numrhs = startebuf;
03328     }
03329     else {
03330         C = cbuf+AC.cbunum;
03331         oldworkpointer = AT.WorkPointer;
03332         d->factors = (FACDOLLAR *)Malloc1(sizeof(FACDOLLAR)*(nfactors+factorsincontent),"factors in
dollar");
03333         term = buf3;
03334         for ( i = 0; i < nfactors; i++ ) {
03335             NewSort(BHEAD0);
03336             while ( *term ) {
03337                 argextra = oldworkpointer;
03338                 j = *term;
03339                 NCOPY(argextra,term,j)
03340                 AT.WorkPointer = argextra;
03341                 if ( Generator(BHEAD oldworkpointer,C->numlhs+1) ) {
03342                     goto getout2;
03343                 }
03344             }
03345             term++;
03346             AT.WorkPointer = oldworkpointer;
03347             AN.tryterm = 0; /* for now */
03348             EndSort(BHEAD (WORD *)((void *)(&(d->factors[i].where))),2);
03349             d->factors[i].type = DOLTERMS;
03350             t = d->factors[i].where;
03351             while ( *t ) t += *t;
03352             d->factors[i].size = t - d->factors[i].where;
03353         }
03354     }
03355     d->nfactors = nfactors + factorsincontent;
03356     /*
03357         #] Step 5: ConvertFromPoly
03358         #[ Step 6: The factors of the content
03359     */
03360     if ( buf3 ) M_free(buf3,"DollarFactorize-5");
03361     if ( buf2 != buf1 && buf2 ) M_free(buf2,"DollarFactorize-5");
03362     M_free(buf1,"DollarFactorize-5");
03363     j = nfactors;
03364 #ifdef STEP2
03365     term = buf1content;
03366     tstop = term + *term;
03367     if ( tstop[-1] < 0 ) { tstop[-1] = -tstop[-1]; sign = -sign; }
03368     tstop -= tstop[-1];
03369     term++;
03370     while ( term < tstop ) {
03371         switch ( *term ) {
03372             case SYMBOL:
03373                 t = term+2; i = (term[1]-2)/2;
03374                 while ( i > 0 ) {
03375                     if ( t[1] < 0 ) { t[1] = -t[1]; pow = -1; }
03376                     else { pow = 1; }
03377                     for ( jj = 0; jj < t[1]; jj++ ) {
03378                         r = d->factors[j].where = (WORD *)Malloc1(9*sizeof(WORD),"factor");
03379                         r[0] = 8; r[1] = SYMBOL; r[2] = 4; r[3] = *t; r[4] = pow;
03380                         r[5] = 1; r[6] = 1; r[7] = 3; r[8] = 0;
03381                         d->factors[j].type = DOLTERMS;
03382                         d->factors[j].size = 8;
03383                         j++;
03384                     }
03385                     i--; t += 2;
03386                 }
03387                 break;
03388             case DOTPRODUCT:
03389                 t = term+2; i = (term[1]-2)/3;
03390                 while ( i > 0 ) {
03391                     if ( t[2] < 0 ) { t[2] = -t[2]; pow = -1; }
03392                     else { pow = 1; }
03393                     for ( jj = 0; jj < t[2]; jj++ ) {
03394                         r = d->factors[j].where = (WORD *)Malloc1(10*sizeof(WORD),"factor");
03395                         r[0] = 9; r[1] = DOTPRODUCT; r[2] = 5; r[3] = t[0]; r[4] = t[1];
03396                         r[5] = pow; r[6] = 1; r[7] = 1; r[8] = 3; r[9] = 0;
03397                         d->factors[j].type = DOLTERMS;
03398                         d->factors[j].size = 9;
03399                         j++;
03400                     }
03401                     i--; t += 3;
03402                 }
03403                 break;
03404             case VECTOR:
03405             case DELTA:
03406                 t = term+2; i = (term[1]-2)/2;
03407                 while ( i > 0 ) {
03408                     for ( jj = 0; jj < t[1]; jj++ ) {
03409                         r = d->factors[j].where = (WORD *)Malloc1(9*sizeof(WORD),"factor");
03410                         r[0] = 8; r[1] = *term; r[2] = 4; r[3] = *t; r[4] = t[1];
03411                         r[5] = 1; r[6] = 1; r[7] = 3; r[8] = 0;
03412                         d->factors[j].type = DOLTERMS;

```

```

03413             d->factors[j].size = 8;
03414             j++;
03415         }
03416         i--; t += 2;
03417     }
03418     break;
03419 case INDEX:
03420     t = term+2; i = term[1]-2;
03421     while ( i > 0 ) {
03422         for ( jj = 0; jj < t[1]; jj++ ) {
03423             r = d->factors[j].where = (WORD *)Malloca1(8*sizeof(WORD), "factor");
03424             r[0] = 7; r[1] = *term; r[2] = 3; r[3] = *t;
03425             r[4] = 1; r[5] = 1; r[6] = 3; r[7] = 0;
03426             d->factors[j].type = DOLTERMS;
03427             d->factors[j].size = 7;
03428             j++;
03429         }
03430         i--; t++;
03431     }
03432     break;
03433 default:
03434     if ( *term >= FUNCTION ) {
03435         r = d->factors[j].where = (WORD *)Malloca1((term[1]+5)*sizeof(WORD), "factor");
03436         *r++ = d->factors[j].size = term[1]+4;
03437         for ( jj = 0; jj < t[1]; jj++ ) *r++ = term[jj];
03438         *r++ = 1; *r++ = 1; *r++ = 3; *r = 0;
03439         j++;
03440     }
03441     break;
03442 }
03443 term += term[1];
03444 }
03445 #endif
03446 /*
03447     #] Step 6:
03448     #[ Step 7: Numerical factors
03449 */
03450 #ifdef STEP2
03451     term = buf1content;
03452     tstop = term + *term;
03453     if ( tstop[-1] == 3 && tstop[-2] == 1 && tstop[-3] == 1 ) {}
03454     else if ( tstop[-1] == 3 && tstop[-2] == 1 && (UWORD)(tstop[-3]) <= MAXPOSITIVE ) {
03455         d->factors[j].where = 0;
03456         d->factors[j].size = 0;
03457         d->factors[j].type = DOLNUMBER;
03458         d->factors[j].value = sign*tstop[-3];
03459         sign = 1;
03460         j++;
03461     }
03462     else {
03463         d->factors[j].where = r = (WORD *)Malloca1((tstop[-1]+2)*sizeof(WORD), "numfactor");
03464         d->factors[j].size = tstop[-1]+1;
03465         d->factors[j].type = DOLTERMS;
03466         d->factors[j].value = 0;
03467         i = tstop[-1];
03468         t = tstop - i;
03469         *r++ = tstop[-1]+1;
03470         NCOPY(r,t,i);
03471         *r = 0;
03472         if ( sign < 0 ) {
03473             r = d->factors[j].where;
03474             while ( *r ) {
03475                 r += *r; r[-1] = -r[-1];
03476             }
03477             sign = 1;
03478         }
03479         j++;
03480     }
03481 #endif
03482     if ( sign < 0 ) { /* Note that this guy should come first */
03483         for ( jj = j; jj > 0; jj-- ) {
03484             d->factors[jj] = d->factors[jj-1];
03485         }
03486         d->factors[0].where = 0;
03487         d->factors[0].size = 0;
03488         d->factors[0].type = DOLNUMBER;
03489         d->factors[0].value = -1;
03490         j++;
03491     }
03492     d->nfactors = j;
03493     if ( buf1content ) TermFree(buf1content, "DollarContent");
03494 /*
03495     #] Step 7:
03496     #[ Step 8: Sorting the factors
03497
03498     There are d->nfactors factors. Look which ones have a 'where'
03499     Sort them by bubble sort

```

```

03500 */
03501     if ( d->nfactors > 1 ) {
03502         WORD ***fac, j1, j2, k, ret, *s1, *s2, *s3;
03503         LONG **facsize, x;
03504         facsize = (LONG **)Malloc1((sizeof(WORD **)+sizeof(LONG *))d->nfactors,"SortDollarFactors");
03505         fac = (WORD **)(facsize+d->nfactors);
03506         k = 0;
03507         for ( j = 0; j < d->nfactors; j++ ) {
03508             if ( d->factors[j].where ) {
03509                 fac[k] = &(d->factors[j].where);
03510                 facsize[k] = &(d->factors[j].size);
03511                 k++;
03512             }
03513         }
03514         if ( k > 1 ) {
03515             for ( j = 1; j < k; j++ ) { /* bubble sort */
03516                 j1 = j; j2 = j1-1;
03517 nextj1:;
03518                 s1 = *(fac[j1]); s2 = *(fac[j2]);
03519                 while ( *s1 && *s2 ) {
03520                     if ( ( ret = CompareTerms(BHEAD s2, s1, (WORD)2) ) == 0 ) {
03521                         s1 += *s1; s2 += *s2;
03522                     }
03523                     else if ( ret > 0 ) goto nextj;
03524                     else {
03525     exch:
03526                         s3 = *(fac[j1]); *(fac[j1]) = *(fac[j2]); *(fac[j2]) = s3;
03527                         x = *(facsize[j1]); *(facsize[j1]) = *(facsize[j2]); *(facsize[j2]) = x;
03528                         j1--; j2--;
03529                         if ( j1 > 0 ) goto nextj1;
03530                         goto nextj;
03531                     }
03532                 }
03533                 if ( *s1 ) goto nextj;
03534                 if ( *s2 ) goto exch;
03535 nextj:;
03536             }
03537         }
03538         M_free(facsize,"SortDollarFactors");
03539     }
03540     /*
03541         #] Step 8:
03542     */
03543     #ifdef WITHPTHREADS
03544     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
03545     #endif
03546     return(0);
03547 }
03548
03549 /*
03550     #] DollarFactorize :
03551     #[ CleanDollarFactors :
03552 */
03553
03554 void CleanDollarFactors(DOLLARS d)
03555 {
03556     int i;
03557     if ( d->nfactors > 1 ) {
03558         for ( i = 0; i < d->nfactors; i++ ) {
03559             if ( d->factors[i].where )
03560                 M_free(d->factors[i].where,"dollar factors");
03561         }
03562     }
03563     if ( d->factors ) {
03564         M_free(d->factors,"dollar factors");
03565         d->factors = 0;
03566     }
03567     d->nfactors = 0;
03568 }
03569
03570 /*
03571     #] CleanDollarFactors :
03572     #[ TakeDollarContent :
03573 */
03574
03575 WORD *TakeDollarContent(PHEAD WORD *dollarbuffer, WORD **factor)
03576 {
03577     WORD *remain, *t;
03578     int pow;
03579     /*
03580     We force the sign of the first term to be positive.
03581     */
03582     t = dollarbuffer; pow = 1;
03583     t += *t;
03584     if ( t[-1] < 0 ) {
03585         pow = 0;
03586         t[-1] = -t[-1];

```



```

03587         while ( *t ) {
03588             t += *t; t[-1] = -t[-1];
03589         }
03590     }
03591     /*
03592     Now the GCD of the numerators and the LCM of the denominators:
03593     */
03594     if ( AN.cmod != 0 ) {
03595         if ( ( *factor = MakeDollarMod(BHEAD dollarbuffer,&remain) ) == 0 ) {
03596             Terminate(-1);
03597         }
03598         if ( pow == 0 ) {
03599             (*factor)[**factor-1] = -(*factor)[**factor-1];
03600             (*factor)[**factor-1] += AN.cmod[0];
03601         }
03602     }
03603     else {
03604         if ( ( *factor = MakeDollarInteger(BHEAD dollarbuffer,&remain) ) == 0 ) {
03605             Terminate(-1);
03606         }
03607         if ( pow == 0 ) {
03608             (*factor)[**factor-1] = -(*factor)[**factor-1];
03609         }
03610     }
03611     return(remain);
03612 }
03613
03614 /*
03615 #] TakeDollarContent :
03616 #[ MakeDollarInteger :
03617 */
03627 WORD *MakeDollarInteger(PHEAD WORD *bufin,WORD **bufout)
03628 {
03629     GETBIDENTITY
03630     UWORD *GCDBuffer, *GCDBuffer2, *LCMbuffer, *LCMb, *LCMc;
03631     WORD *r, *r1, *r2, *r3, *rnext, i, k, j, *oldworkpointer, *factor;
03632     WORD kGCD, kLCM, kGCD2, kkLCM, jLCM, jGCD;
03633     CBUF *C = cbuf+AC.cbunum;
03634
03635     GCDBuffer = NumberMalloc("MakeDollarInteger");
03636     GCDBuffer2 = NumberMalloc("MakeDollarInteger");
03637     LCMbuffer = NumberMalloc("MakeDollarInteger");
03638     LCMb = NumberMalloc("MakeDollarInteger");
03639     LCMc = NumberMalloc("MakeDollarInteger");
03640     r = bufin;
03641     /*
03642     First take the first term to load up the LCM and the GCD
03643     */
03644     r2 = r + *r;
03645     j = r2[-1];
03646     r3 = r2 - ABS(j);
03647     k = REDLENG(j);
03648     if ( k < 0 ) k = -k;
03649     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
03650     for ( kGCD = 0; kGCD < k; kGCD++ ) GCDBuffer[kGCD] = r3[kGCD];
03651     k = REDLENG(j);
03652     if ( k < 0 ) k = -k;
03653     r3 += k;
03654     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
03655     for ( kLCM = 0; kLCM < k; kLCM++ ) LCMbuffer[kLCM] = r3[kLCM];
03656     r1 = r2;
03657     /*
03658     Now go through the rest of the terms in this argument.
03659     */
03660     while ( *r1 ) {
03661         r2 = r1 + *r1;
03662         j = r2[-1];
03663         r3 = r2 - ABS(j);
03664         k = REDLENG(j);
03665         if ( k < 0 ) k = -k;
03666         while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
03667         if ( ( ( GCDBuffer[0] == 1 ) && ( kGCD == 1 ) ) ) {
03668             /*
03669             GCD is already 1
03670             */
03671         }
03672         else if ( ( ( k != 1 ) || ( r3[0] != 1 ) ) ) {
03673             if ( GcdLong(BHEAD GCDBuffer,kGCD,(UWORD *)r3,k,GCDBuffer2,&kGCD2) ) {
03674                 goto MakeDollarIntegerErr;
03675             }
03676             kGCD = kGCD2;
03677             for ( i = 0; i < kGCD; i++ ) GCDBuffer[i] = GCDBuffer2[i];
03678         }
03679         else {
03680             kGCD = 1; GCDBuffer[0] = 1;
03681         }
03682         k = REDLENG(j);

```

```

03683         if ( k < 0 ) k = -k;
03684         r3 += k;
03685         while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
03686         if ( ( ( LCMbuffer[0] == 1 ) && ( kLCM == 1 ) ) ) {
03687             for ( kLCM = 0; kLCM < k; kLCM++ )
03688                 LCMbuffer[kLCM] = r3[kLCM];
03689         }
03690         else if ( ( k != 1 ) || ( r3[0] != 1 ) ) {
03691             if ( GcdLong(BHEAD LCMbuffer, kLCM, (UWORD *)r3, k, LCMb, &kLCM) ) {
03692                 goto MakeDollarIntegerErr;
03693             }
03694             DivLong( (UWORD *)r3, k, LCMb, kLCM, LCMb, &kLCM, LCMc, &jLCM );
03695             Mullong(LCMbuffer, kLCM, LCMb, kLCM, LCMc, &jLCM);
03696             for ( kLCM = 0; kLCM < jLCM; kLCM++ )
03697                 LCMbuffer[kLCM] = LCMc[kLCM];
03698         }
03699         else {} /* LCM doesn't change */
03700         r1 = r2;
03701     }
03702     /*
03703     Now put the factor together: GCD/LCM
03704     */
03705     r3 = (WORD *) (GCDbuffer);
03706     if ( kgcd == kLCM ) {
03707         for ( jgcd = 0; jgcd < kgcd; jgcd++ )
03708             r3[jgcd+kgcd] = LCMbuffer[jgcd];
03709         k = kgcd;
03710     }
03711     else if ( kgcd > kLCM ) {
03712         for ( jgcd = 0; jgcd < kLCM; jgcd++ )
03713             r3[jgcd+kgcd] = LCMbuffer[jgcd];
03714         for ( jgcd = kLCM; jgcd < kgcd; jgcd++ )
03715             r3[jgcd+kgcd] = 0;
03716         k = kgcd;
03717     }
03718     else {
03719         for ( jgcd = kgcd; jgcd < kLCM; jgcd++ )
03720             r3[jgcd] = 0;
03721         for ( jgcd = 0; jgcd < kLCM; jgcd++ )
03722             r3[jgcd+kLCM] = LCMbuffer[jgcd];
03723         k = kLCM;
03724     }
03725     j = 2*k+1;
03726     /*
03727     Now we have to write this to factor
03728     */
03729     factor = r1 = (WORD *) Malloc1((j+2)*sizeof(WORD), "MakeDollarInteger");
03730     *r1++ = j+1; r2 = r3;
03731     for ( i = 0; i < k; i++ ) { *r1++ = *r2++; *r1++ = *r2++; }
03732     *r1++ = j;
03733     *r1 = 0;
03734     /*
03735     Next we have to take the factor out from the argument.
03736     This cannot be done in location, because the denominator stuff can make
03737     coefficients longer.
03738
03739     We do this via a sort because the things may be jumbled any way and we
03740     do not know in advance how much space we need.
03741     */
03742     NewSort(BHEAD0);
03743     r = bufin;
03744     oldworkpointer = AT.WorkPointer;
03745     while ( *r ) {
03746         rnext = r + *r;
03747         j = ABS(rnext[-1]);
03748         r3 = rnext - j;
03749         r2 = oldworkpointer;
03750         while ( r < r3 ) *r2++ = *r++;
03751         j = (j-1)/2; /* reduced length. Remember, k is the other red length */
03752         if ( DivRat(BHEAD (UWORD *)r3, j, GCDbuffer, k, (UWORD *)r2, &i) ) {
03753             goto MakeDollarIntegerErr;
03754         }
03755         i = 2*i+1;
03756         r2 = r2 + i;
03757         if ( rnext[-1] < 0 ) r2[-1] = -i;
03758         else r2[-1] = i;
03759         *oldworkpointer = r2-oldworkpointer;
03760         AT.WorkPointer = r2;
03761         if ( Generator(BHEAD oldworkpointer, C->numlhs) ) {
03762             goto MakeDollarIntegerErr;
03763         }
03764         r = rnext;
03765     }
03766     AT.WorkPointer = oldworkpointer;
03767     AN.tryterm = 0; /* for now */
03768     EndSort(BHEAD (WORD *)bufout, 2);
03769     /*

```

```

03770 Cleanup
03771 */
03772 NumberFree(LCMC,"MakeDollarInteger");
03773 NumberFree(LCMb,"MakeDollarInteger");
03774 NumberFree(LCMbuffer,"MakeDollarInteger");
03775 NumberFree(GCDBuffer2,"MakeDollarInteger");
03776 NumberFree(GCDBuffer,"MakeDollarInteger");
03777 return(factor);
03778
03779 MakeDollarIntegerErr:
03780 NumberFree(LCMC,"MakeDollarInteger");
03781 NumberFree(LCMb,"MakeDollarInteger");
03782 NumberFree(LCMbuffer,"MakeDollarInteger");
03783 NumberFree(GCDBuffer2,"MakeDollarInteger");
03784 NumberFree(GCDBuffer,"MakeDollarInteger");
03785 MesCall("MakeDollarInteger");
03786 Terminate(-1);
03787 return(0);
03788 }
03789
03790 /*
03791 #] MakeDollarInteger :
03792 #[ MakeDollarMod :
03793 */
03801 WORD *MakeDollarMod(PHEAD WORD *buffer, WORD **bufout)
03802 {
03803     GETBIDENTITY
03804     WORD *r, *r1, x, xx, ix, ip;
03805     WORD *factor, *oldworkpointer;
03806     int i;
03807     CBUF *C = cbuf+AC.cbunum;
03808     r = buffer;
03809     x = r[*r-3];
03810     if ( r[*r-1] < 0 ) x += AN.cmod[0];
03811     if ( GetModInverses(x,(WORD)(AN.cmod[0]),&ix,&ip) ) {
03812         Terminate(-1);
03813     }
03814     factor = (WORD *)Malloca(5*sizeof(WORD),"MakeDollarMod");
03815     factor[0] = 4; factor[1] = x; factor[2] = 1; factor[3] = 3; factor[4] = 0;
03816 /*
03817 Now we have to multiply all coefficients by ix.
03818 This does not make things longer, but we should keep to the conventions
03819 of MakeDollarInteger.
03820 */
03821 NewSort(BHEAD0);
03822 r = buffer;
03823 oldworkpointer = AT.WorkPointer;
03824 while ( *r ) {
03825     r1 = oldworkpointer; i = *r;
03826     NCOPY(r1,r,i);
03827     xx = r1[-3]; if ( r1[-1] < 0 ) xx += AN.cmod[0];
03828     r1[-1] = (WORD)((LONG)xx*ix) % AN.cmod[0];
03829     *r1 = 0; AT.WorkPointer = r1;
03830     if ( Generator(BHEAD oldworkpointer,C->numlhs) ) {
03831         Terminate(-1);
03832     }
03833 }
03834 AT.WorkPointer = oldworkpointer;
03835 AN.tryterm = 0; /* for now */
03836 EndSort(BHEAD (WORD *)bufout,2);
03837 return(factor);
03838 }
03839 /*
03840 #] MakeDollarMod :
03841 #[ GetDolNum :
03842
03843 Evaluates a chain of DOLLAREXPR2 into a number
03844 */
03845
03846 int GetDolNum(PHEAD WORD *t, WORD *tstop)
03847 {
03848     DOLLARS d;
03849     WORD num, *w;
03850     if ( t+3 < tstop && t[3] == DOLLAREXPR2 ) {
03851         d = Dollars + t[2];
03852 #ifdef WITHPTHREADS
03853     {
03854         int nummodopt, dtype;
03855         dtype = -1;
03856         if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
03857             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
03858                 if ( t[2] == ModOptdollars[nummodopt].number ) break;
03859             }
03860             if ( nummodopt < NumModOptdollars ) {
03861                 dtype = ModOptdollars[nummodopt].type;
03862                 if ( dtype == MODLOCAL ) {
03863                     d = ModOptdollars[nummodopt].dstruct+AT.identity;

```

```

03864         }
03865     else {
03866         MLOCK(ErrorMessageLock);
03867         MesPrint("&Illegal attempt to use $-variable %s in module %1",
03868             DOLLARNAME(Dollars,t[2]),AC.CModule);
03869         MUNLOCK(ErrorMessageLock);
03870         Terminate(-1);
03871     }
03872 }
03873 }
03874 }
03875 #endif
03876 if ( d->factors == 0 ) {
03877     MLOCK(ErrorMessageLock);
03878     MesPrint("Attempt to use a factor of an unfactored $-variable");
03879     MUNLOCK(ErrorMessageLock);
03880     Terminate(-1);
03881 }
03882 num = GetDolNum(BHEAD t+t[1],tstop);
03883 if ( num == 0 ) return(d->nfactors);
03884 if ( num > d->nfactors ) {
03885     MLOCK(ErrorMessageLock);
03886     MesPrint("Attempt to use an nonexistent factor %d of a $-variable",num);
03887     MUNLOCK(ErrorMessageLock);
03888     Terminate(-1);
03889 }
03890 w = d->factors[num-1].where;
03891 if ( w == 0 ) return(d->factors[num-1].value);
03892 if ( w[0] == 4 && w[4] == 0 && w[3] == 3 && w[2] == 1 && w[1] > 0
03893     && w[1] < MAXPOSITIVE ) return(w[1]);
03894 else {
03895     MLOCK(ErrorMessageLock);
03896     MesPrint("Illegal type of factor number of a $-variable");
03897     MUNLOCK(ErrorMessageLock);
03898     Terminate(-1);
03899 }
03900 }
03901 else if ( t[2] < 0 ) {
03902     return(-t[2]-1);
03903 }
03904 else {
03905     d = Dollars + t[2];
03906 #ifdef WITHPTHREADS
03907     {
03908         int nummodopt, dtype;
03909         dtype = -1;
03910         if ( AS.MultiThreaded && ( AC.mparallelfalg == PARALLELFALG ) ) {
03911             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
03912                 if ( t[2] == ModOptdollars[nummodopt].number ) break;
03913             }
03914             if ( nummodopt < NumModOptdollars ) {
03915                 dtype = ModOptdollars[nummodopt].type;
03916                 if ( dtype == MODLOCAL ) {
03917                     d = ModOptdollars[nummodopt].dstruct+AT.identity;
03918                 }
03919                 else {
03920                     MLOCK(ErrorMessageLock);
03921                     MesPrint("&Illegal attempt to use $-variable %s in module %1",
03922                         DOLLARNAME(Dollars,t[2]),AC.CModule);
03923                     MUNLOCK(ErrorMessageLock);
03924                     Terminate(-1);
03925                 }
03926             }
03927         }
03928     }
03929 #endif
03930 if ( d->type == DOLZERO ) return(0);
03931 if ( d->type == DOLTERMS || d->type == DOLNUMBER ) {
03932     if ( d->where[0] == 4 && d->where[4] == 0 && d->where[3] == 3
03933         && d->where[2] == 1 && d->where[1] > 0
03934         && d->where[1] < MAXPOSITIVE ) return(d->where[1]);
03935     MLOCK(ErrorMessageLock);
03936     MesPrint("Attempt to use an nonexistent factor of a $-variable");
03937     MUNLOCK(ErrorMessageLock);
03938     Terminate(-1);
03939 }
03940 MLOCK(ErrorMessageLock);
03941 MesPrint("Illegal type of factor number of a $-variable");
03942 MUNLOCK(ErrorMessageLock);
03943 Terminate(-1);
03944 }
03945 return(0);
03946 }
03947 /*
03948 #] GetDolNum :
03949 #[ AddPotModdollar :

```

```

03951 */
03952
03959 void AddPotModdollar (WORD numdollar)
03960 {
03961     int i, n = NumPotModdollars;
03962     for ( i = 0; i < n; i++ ) {
03963         if ( numdollar == PotModdollars[i] ) break;
03964     }
03965     if ( i >= n ) {
03966         *(WORD *)FromList (&AC.PotModDolList) = numdollar;
03967     }
03968 }
03969
03970 /*
03971     #] AddPotModdollar :
03972 */

```

## 4.28 evaluate.c

```

00001 /*
00002     #[ includes :
00003 */
00004
00005 #include "form3.h"
00006 #include <stdarg.h>
00007 #include <gmp.h>
00008 #include <mpfr.h>
00009
00010 #define EXTRAPRECISION 4
00011
00012 #define RND MPFR_RNDN
00013
00014 #define auxr1 ((mpfr_t *) (AT.auxr_)) [0]
00015 #define auxr2 ((mpfr_t *) (AT.auxr_)) [1]
00016 #define auxr3 ((mpfr_t *) (AT.auxr_)) [2]
00017 #define auxr4 ((mpfr_t *) (AT.auxr_)) [3]
00018 #define auxr5 ((mpfr_t *) (AT.auxr_)) [4]
00019
00020 mpfr_t ln2;
00021
00022 int PackFloat (WORD *,mpfr_t);
00023 int UnpackFloat (mpfr_t,WORD *);
00024 void RatToFloat (mpfr_t, UWORD *, int);
00025 void FormtoZ (mpz_t,UWORD *,WORD);
00026
00027 /*
00028     #] includes :
00029     #[ mpfr_ :
00030         #[ IntegerToFloatr :
00031
00032             Converts a Form long integer to a mpfr_ float of default size.
00033             We assume that sizeof(unsigned long int) = 2*sizeof(UWORD).
00034 */
00035
00036 void IntegerToFloatr (mpfr_t result, UWORD *formlong, int longsize)
00037 {
00038     mpz_t z;
00039     mpz_init (z);
00040     FormtoZ (z,formlong,longsize);
00041     mpfr_set_z (result,z,RND);
00042     mpz_clear (z);
00043 }
00044
00045 /*
00046     #] IntegerToFloatr :
00047     #[ RatToFloatr :
00048
00049         Converts a Form rational to a gmp float of default size.
00050 */
00051
00052 void RatToFloatr (mpfr_t result, UWORD *formrat, int ratsize)
00053 {
00054     GETIDENTITY
00055     int nnum, nden;
00056     UWORD *num, *den;
00057     int sgn = 0;
00058     if ( ratsize < 0 ) { ratsize = -ratsize; sgn = 1; }
00059     nnum = nden = (ratsize-1)/2;
00060     num = formrat; den = formrat+nnum;
00061     while ( nnum > 1 && num[nnum-1] == 0 ) { nnum--; }
00062     while ( nden > 1 && den[nden-1] == 0 ) { nden--; }
00063     IntegerToFloatr (auxr4,num,nnum);
00064     IntegerToFloatr (auxr5,den,nden);

```

```

00065     mpfr_div(result,auxr4,auxr5,RND);
00066     if ( sgn > 0 ) mpfr_neg(result,result,RND);
00067 }
00068
00069 /*
00070     #] RatToFloat :
00071     #[ SetFloatPrecision :
00072
00073     The AC.DefaultPrecision is in bits.
00074     prec is in limbs.
00075     fprec is NOT in bits for mpfr_ which is slightly less than in mpf_
00076     We also set the auxr1,auxr2,auxr3,auxr4,auxr5 variables.
00077 */
00078
00079 void SetFloatPrecision(LONG prec)
00080 {
00081     /*
00082         mpfr_prec_t fprec = prec * mp_bits_per_limb - EXTRAPRECISION;
00083     */
00084     mpfr_prec_t fprec = prec + 1;
00085     mpfr_set_default_prec(fprec);
00086 #ifdef WITHPTHREADS
00087     int totnum = AM.totalnumberofthreads, id;
00088     mpfr_t *a;
00089 #ifdef WITHSORTBOTS
00090     totnum = MaX(2*AM.totalnumberofthreads-3,AM.totalnumberofthreads);
00091 #endif
00092     if ( AB[0]->T.auxr_ ) {
00093         for ( id = 0; id < totnum; id++ ) {
00094             a = (mpfr_t *)AB[id]->T.auxr_;
00095             mpfr_clears(a[0],a[1],a[2],a[3],a[4],(mpfr_ptr)0);
00096             M_free(AB[id]->T.auxr_,"AB[id]->T.auxr_");
00097             AB[id]->T.auxr_ = 0;
00098         }
00099     }
00100     for ( id = 0; id < totnum; id++ ) {
00101         AB[id]->T.auxr_ = (void *)Malloc1(sizeof(mpfr_t)*5,"AB[id]->T.auxr_");
00102         a = (mpfr_t *)AB[id]->T.auxr_;
00103     /*
00104         We work here with a[0] etc because the aux1 etc contain B which
00105         in the current routine would be AB[0] only
00106     */
00107         mpfr_inits2(fprec,a[0],a[1],a[2],a[3],a[4],(mpfr_ptr)0);
00108     }
00109 #else
00110     if ( AT.auxr_ ) {
00111         mpfr_clears(auxr1,auxr2,auxr3,auxr4,auxr5,(mpfr_ptr)0);
00112         M_free(AT.auxr_,"AT.auxr_");
00113         AT.auxr_ = 0;
00114     }
00115     AT.auxr_ = (void *)Malloc1(sizeof(mpfr_t)*5,"AT.auxr_");
00116     mpfr_inits2(fprec,auxr1,auxr2,auxr3,auxr4,auxr5,(mpfr_ptr)0);
00117 #endif
00118 }
00119
00120 /*
00121     #] SetFloatPrecision :
00122     #[ ClearFloat :
00123 */
00124
00125 void ClearFloat(void)
00126 {
00127 #ifdef WITHPTHREADS
00128     int totnum = AM.totalnumberofthreads, id;
00129     mpfr_t *a;
00130 #ifdef WITHSORTBOTS
00131     totnum = MaX(2*AM.totalnumberofthreads-3,AM.totalnumberofthreads);
00132 #endif
00133     if ( AB[0]->T.auxr_ ) {
00134         for ( id = 0; id < totnum; id++ ) {
00135             a = (mpfr_t *)AB[id]->T.auxr_;
00136             mpfr_clears(a[0],a[1],a[2],a[3],a[4],(mpfr_ptr)0);
00137             M_free(AB[id]->T.auxr_,"AB[id]->T.auxr_");
00138             AB[id]->T.auxr_ = 0;
00139         }
00140     /*
00141         M_free(AB[0]->T.auxr_,"AB[0]->T.auxr_");
00142         AB[0]->T.auxr_ = 0;
00143     */
00144     }
00145 #else
00146     if ( AT.auxr_ ) {
00147         mpfr_clears(auxr1,auxr2,auxr3,auxr4,auxr5,(mpfr_ptr)0);
00148         M_free(AT.auxr_,"AT.auxr_");
00149         AT.auxr_ = 0;
00150     }
00151 #endif

```

```

00152 /*
00153     if ( AS.delta_1 ) {
00154         mpf_clear(ln2);
00155         AS.delta_1 = 0;
00156     }
00157 */
00158 }
00159
00160 /*
00161     #] ClearfFloat :
00162     #[ GetFloatArgument :
00163
00164     Convert an argument to an mpfr float if possible.
00165     Return value: 0: was converted successfully.
00166                 -1: could not convert.
00167     Note: arguments with more than one term should be evaluated first
00168     inside an Argument environment. If there is still more than one
00169     term remaining we get the code: "could not convert".
00170     par tells which argument we need. If it is negative it should also
00171     be the last argument.
00172 */
00173
00174 int GetFloatArgument(PHEAD mpfr_t f_out,WORD *fun,int par)
00175 {
00176     WORD *term, *tn, *t, *tstop, first, ncoef, *arg, *argp, abspar;
00177     arg = fun+FUNHEAD;
00178     abspar = ABS(par);
00179     while ( arg < fun+fun[1] && abspar > 1 ) { abspar--; NEXTARG(arg) }
00180     if ( arg >= fun+fun[1] || abspar!= 1 ) return(-1);
00181     if ( par < 0 ) {
00182         argp = arg; NEXTARG(argp); if ( argp != fun+fun[1] ) return(-1);
00183     }
00184     if ( *arg < 0 ) {
00185         if ( *arg == -SNUMBER ) {
00186             mpfr_set_si(f_out,(LONG)(arg[1]),RND);
00187         }
00188         else if ( *arg == -SYMBOL && ( arg[1] == PISYMBOL ) ) {
00189             mpfr_const_pi(f_out,RND);
00190         }
00191         else if ( *arg == -INDEX && arg[1] >= 0 && arg[1] < AM.OffsetIndex ) {
00192             mpfr_set_ui(f_out,(ULONG)(arg[1]),RND);
00193         }
00194         else { return(-1); }
00195         return(0);
00196     }
00197     else if ( arg[0] != arg[ARGHEAD]+ARGHEAD ) { /* more than one term */
00198         return(-1);
00199     }
00200     term = arg+ARGHEAD;
00201     tn = term+*term;
00202     tstop = tn-ABS(tn[-1]);
00203     t = term+1;
00204     first = 1;
00205     while ( t <= tstop ) {
00206         if ( t == tstop ) { /* Fraction */
00207             if ( first ) RatToFloat(f_out,(UWORD *)tstop,tn[-1]);
00208             else {
00209                 RatToFloat(auxr4,(UWORD *)tstop,tn[-1]);
00210                 mpfr_mul(f_out,f_out,auxr4,RND);
00211             }
00212             return(0);
00213         }
00214         else if ( *t == FLOATFUN ) {
00215             if ( t+t[1] != tstop ) return(-1);
00216             if ( UnpackFloat(aux5,t) < 0 ) return(-1);
00217             if ( first ) {
00218                 mpfr_set_f(f_out,aux5,RND);
00219                 first = 0;
00220             }
00221             else {
00222                 mpfr_set_f(auxr4,aux5,RND);
00223                 mpfr_mul(f_out,f_out,auxr4,RND);
00224             }
00225             first = 0;
00226             if ( tn[-1] < 0 ) { /* change sign */
00227                 ncoef = -tn[-1];
00228                 mpfr_neg(f_out,f_out,RND);
00229             }
00230             else ncoef = tn[-1];
00231             if ( ncoef == 3 && tn[-2] == 1 && tn[-3] == 1 ) return(0);
00232         }
00233         /*
00234         If the argument was properly normalized we are not supposed to come here.
00235         */
00236         MLOCK(ErrorMessageLock);
00237         MesPrint("Unnormalized argument in GetFloatArgument: %a",*term,term);
00238         MUNLOCK(ErrorMessageLock);
00239         Terminate(-1);

```

```

00239     }
00240     else if ( t[0] == SYMBOL && t[1] == 4 && t[2] == PISYMBOL && t[3] == 1 ) {
00241         if ( first ) {
00242             mpfr_const_pi(f_out,RND);
00243             first = 0;
00244         }
00245         else {
00246             mpfr_const_pi(auxr5,RND);
00247             mpfr_mul(f_out,f_out,auxr5,RND);
00248         }
00249     }
00250     else { /* This we do not / cannot do */
00251         return(-1);
00252     }
00253     t += t[1];
00254 }
00255 return(0);
00256 }
00257
00258 /*
00259     #] GetFloatArgument :
00260     #[ GetPiArgument :
00261
00262     Tests for sin,cos,tan whether the argument is a simple
00263     multiple of pi or can be reduced to such.
00264     Return value: -1: the answer is no.
00265     0-23: we have 0, 1/12*pi, ..., 23/12*pi_
00266     With success most values can be worked out by simple means.
00267 */
00268
00269 int GetPiArgument(PHEAD WORD *arg)
00270 {
00271     UWORD *coef, *co, *tco, twelve, *number, *denom, *c, *d;
00272     WORD i, ii, i2, iflip, nc, nd, *t, *tn, *tstop;
00273     int rem;
00274     /*
00275     One: determine whether there is a rational coefficient and a pi_:
00276     */
00277     if ( *arg == -SNUMBER && arg[1] == 0 ) return(0);
00278     if ( *arg == -SYMBOL && arg[1] == PISYMBOL ) return(12);
00279     if ( *arg < 0 ) return(-1);
00280     if ( arg[ARGHEAD] != *arg-ARGHEAD ) return(-1);
00281     t = arg+ARGHEAD+1;
00282     tn = arg+arg; tstop = tn - ABS(tn[-1]);
00283     if ( *t != SYMBOL || t[1] != 4 || t[2] != PISYMBOL || t[3] != 1
00284         || t+t[1] != tstop ) return(-1);
00285     /*
00286     The denominator must be a divisor of 12
00287     1: copy the coefficient
00288     */
00289     co = coef = NumberMalloc("GetPiArgument");
00290     tco = (UWORD *)tstop;
00291     i = tn[-1]; if ( i < 0 ) { i = -i; iflip = 1; } else iflip = 0;
00292     ii = (i-1)/2;
00293     twelve = 12;
00294     NCOPY(co,tco,i);
00295     co = coef;
00296     Mully(BHEAD co,&ii,&twelve,1);
00297     /*
00298     Now the denominator should be 1
00299     */
00300     i = ii; i2 = i-1;
00301     numer = co; denom = numer + i;
00302     if ( i > 1 ) {
00303         while ( i2 > 0 ) {
00304             if ( denom[i2--] != 0 ) {
00305                 NumberFree(co,"GetPiArgument");
00306                 return(-1);
00307             }
00308         }
00309     }
00310     if ( *denom != 1 ) {
00311         NumberFree(co,"GetPiArgument");
00312         return(-1);
00313     }
00314     /*
00315     Now we need the numerator modulus 24.
00316     */
00317     if ( i == 1 ) {
00318         rem = *numer % 24;
00319     }
00320     else {
00321         c = NumberMalloc("GetPiArgument");
00322         d = NumberMalloc("GetPiArgument");
00323         twelve *= 2;
00324         DivLong(numer,i,&twelve,1,c,&nc,d,&nd);
00325         rem = *d % 24;

```



```

00326         NumberFree(d,"GetPiArgument");
00327         NumberFree(c,"GetPiArgument");
00328     }
00329     NumberFree(co,"GetPiArgument");
00330     if ( iflip && rem != 0 ) rem = 24-rem;
00331     return(rem);
00332 }
00333
00334 /*
00335     #] GetPiArgument :
00336     #[ EvaluateFun :
00337
00338     What we need to do is:
00339     1: look for a function to be treated.
00340     2: make sure its argument is treatable.
00341     3: call the proper mpfr_ function.
00342     4: accumulate the result.
00343     For some functions we have to insert 'smart' shortcuts as is
00344     the case with sin_(pi_) or sin_(pi_/6) of sqrt(4/9) etc.
00345     Otherwise we may have to insert a value for pi_ first.
00346     There are several types of arguments:
00347     a: (short) integers.
00348     b: rationals.
00349     c: floats.
00350     We accumulate the result(s) in auxr2. The argument comes in aux5
00351     and a result in auxr3 which then gets multiplied into auxr2.
00352     In the end we have to combine auxr2 with whatever coefficient
00353     existed already.
00354
00355     When the float system is started we need for aux only 3 variables
00356     per thread. auxr1-auxr3. This should be done separately.
00357     The main problem is the conversion of mpfr_t to float_ and/or mpf_t
00358     and back.
00359 */
00360
00361 int EvaluateFun(PHEAD WORD *term, WORD level, WORD *pars)
00362 {
00363     WORD *t, *tstop, *tt, *tnext, *newterm, i;
00364     WORD *oldworkpointer = AT.WorkPointer, nsize, nsgn, nsgn2;
00365     int retval = 0, first = 1, pimul;
00366
00367     tstop = term + *term; tstop -= ABS(tstop[-1]);
00368     if ( AT.WorkPointer < term+*term ) AT.WorkPointer = term + *term;
00369     t = term+1;
00370     mpfr_set_ui(auxr2,1L,RND);
00371     while ( t < tstop ) {
00372         if ( pars[2] == *t ) { /* have to do this one if possible */
00373             TestArgument:
00374             /*
00375                 There must be a single argument, except for the AGM function
00376             */
00377             tnext = t+t[1]; tt = t+FUNHEAD; NEXTARG(tt);
00378             if( *t == SYMBOL ) {
00379                 for ( WORD* ti = t+2; ti < t+t[1]; ti+=2 ) {
00380                     if( ( *ti == PISYMBOL || *ti == EESYMBOL || *ti == EMSYMBOL )
00381                         && ( pars[2] == ALLFUNCTIONS || pars[3] == *ti ) ) {
00382                         if ( *ti == PISYMBOL )
00383                             mpfr_const_pi(auxr3,RND);
00384                         else if ( *ti == EESYMBOL ) {
00385                             mpfr_set_ui(auxr3,1,RND);
00386                             mpfr_exp(auxr3,auxr3,RND);
00387                         }
00388                         else if ( *ti == EMSYMBOL )
00389                             mpfr_const_euler(auxr3,RND);
00390                         if ( ti[1] != 1 )
00391                             mpfr_pow_si(auxr3,auxr3,ti[1],RND);
00392                         mpfr_mul(auxr2,auxr2,auxr3,RND);
00393                         ti[1] = 0;
00394                     }
00395                 }
00396                 first = 0;
00397                 goto nextfun;
00398             }
00399             if ( tt != tnext && *t != AGMFUNCTION ) goto nextfun;
00400             if ( *t == SINFUNCTION ) {
00401                 pimul = GetPiArgument(BHEAD t+FUNHEAD);
00402                 if ( pimul >= 0 && pimul < 24 ) {
00403                     if ( pimul == 0 || pimul == 12 ) goto getout;
00404                     if ( pimul > 12 ) { pimul = 24-pimul; nsgn = -1; }
00405                     else nsgn = 1;
00406                     if ( pimul > 6 ) pimul = 12-pimul;
00407                     switch ( pimul ) {
00408                         case 1:
00409                             label1:
00410                                 mpfr_sqrt_ui(auxr3,3L,RND);
00411                                 mpfr_sub_ui(auxr3,auxr3,1L,RND);
00412

```

```

00413             mpfr_sqrt_ui (auxr1, 2L, RND);
00414             mpfr_div_ui (auxr1, auxr1, 4L, RND);
00415             mpfr_mul (auxr3, auxr3, auxr1, RND);
00416             if ( nsgn < 0 ) mpfr_neg (auxr3, auxr3, RND);
00417             mpfr_mul (auxr2, auxr2, auxr3, RND);
00418             break;
00419         case 2:
00420 label12:
00421             mpfr_div_ui (auxr2, auxr2, 2L, RND);
00422             if ( nsgn < 0 ) mpfr_neg (auxr2, auxr2, RND);
00423             break;
00424         case 3:
00425 label13:
00426             mpfr_sqrt_ui (auxr3, 2L, RND);
00427             mpfr_div_ui (auxr3, auxr3, 2L, RND);
00428             if ( nsgn < 0 ) mpfr_neg (auxr3, auxr3, RND);
00429             mpfr_mul (auxr2, auxr2, auxr3, RND);
00430             break;
00431         case 4:
00432 label14:
00433             mpfr_sqrt_ui (auxr3, 3L, RND);
00434             mpfr_div_ui (auxr3, auxr3, 2L, RND);
00435             if ( nsgn < 0 ) mpfr_neg (auxr3, auxr3, RND);
00436             mpfr_mul (auxr2, auxr2, auxr3, RND);
00437             break;
00438         case 5:
00439 label15:
00440             mpfr_sqrt_ui (auxr3, 3L, RND);
00441             mpfr_add_ui (auxr3, auxr3, 1L, RND);
00442             mpfr_sqrt_ui (auxr1, 2L, RND);
00443             mpfr_div_ui (auxr1, auxr1, 4L, RND);
00444             mpfr_mul (auxr3, auxr3, auxr1, RND);
00445             if ( nsgn < 0 ) mpfr_neg (auxr3, auxr3, RND);
00446             mpfr_mul (auxr2, auxr2, auxr3, RND);
00447             break;
00448         case 6:
00449 label16:
00450             if ( nsgn < 0 ) mpfr_neg (auxr2, auxr2, RND);
00451             break;
00452     }
00453     *t = 0; first = 0;
00454     goto nextfun;
00455 }
00456 }
00457 else if ( *t == COSFUNCTION ) {
00458     pimul = GetPiArgument (BHEAD t+FUNHEAD);
00459     if ( pimul >= 0 && pimul < 24 ) {
00460         if ( pimul > 12 ) pimul = 24-pimul;
00461         if ( pimul > 6 ) { pimul = 12-pimul; nsgn = -1; }
00462         else nsgn = 1;
00463         if ( pimul == 6 ) goto getout;
00464         switch ( pimul ) {
00465             case 0: goto label6;
00466             case 1: goto label5;
00467             case 2: goto label4;
00468             case 3: goto label3;
00469             case 4: goto label2;
00470             case 5: goto label1;
00471         }
00472     }
00473 }
00474 else if ( *t == TANFUNCTION ) {
00475     pimul = GetPiArgument (BHEAD t+FUNHEAD);
00476     if ( pimul >= 0 && pimul < 24 ) {
00477         if ( pimul == 6 || pimul == 18 ) goto nextfun;
00478         if ( pimul == 0 || pimul == 12 ) goto getout;
00479         if ( pimul > 12 ) { pimul = 24-pimul; nsgn = -1; }
00480         else nsgn = 1;
00481         if ( pimul > 6 ) { pimul = 12-pimul; nsgn = -nsgn; }
00482         switch ( pimul ) {
00483             case 1:
00484                 mpfr_sqrt_ui (auxr3, 3L, RND);
00485                 mpfr_sub_ui (auxr3, auxr3, 2L, RND);
00486                 if ( nsgn > 0 ) mpfr_neg (auxr3, auxr3, RND);
00487                 mpfr_mul (auxr2, auxr2, auxr3, RND);
00488                 break;
00489             case 2:
00490                 mpfr_sqrt_ui (auxr3, 3L, RND);
00491                 mpfr_div_ui (auxr3, auxr3, 3L, RND);
00492                 if ( nsgn < 0 ) mpfr_neg (auxr3, auxr3, RND);
00493                 mpfr_mul (auxr2, auxr2, auxr3, RND);
00494                 break;
00495             case 3:
00496                 break;
00497             case 4:
00498                 mpfr_sqrt_ui (auxr3, 3L, RND);
00499                 if ( nsgn < 0 ) mpfr_neg (auxr3, auxr3, RND);

```

```

00500             mpfr_mul(auxr2,auxr2,auxr3,RND);
00501             break;
00502         case 5:
00503             mpfr_sqrt_ui(auxr3,3L,RND);
00504             mpfr_add_ui(auxr3,auxr3,2L,RND);
00505             if ( nsgn < 0 ) mpfr_neg(auxr3,auxr3,RND);
00506             mpfr_mul(auxr2,auxr2,auxr3,RND);
00507             break;
00508     }
00509     *t = 0; first = 0;
00510     goto nextfun;
00511 }
00512 }
00513
00514 if ( *t == AGMFUNCTION ) {
00515     if ( GetFloatArgument(BHEAD auxr1,t,1) < 0 ) goto nextfun;
00516     if ( GetFloatArgument(BHEAD auxr3,t,-2) < 0 ) goto nextfun;
00517 }
00518 else if ( GetFloatArgument(BHEAD auxr1,t,-1) < 0 ) goto nextfun;
00519 nsgn = mpfr_sgn(auxr1);
00520
00521 switch ( *t ) {
00522     case SQRTFUNCTION:
00523         if ( nsgn < 0 ) goto nextfun;
00524         if ( nsgn == 0 ) goto getout;
00525         else mpfr_sqrt(auxr3,auxr1,RND);
00526         mpfr_mul(auxr2,auxr2,auxr3,RND);
00527         break;
00528     case LNFUNCTION:
00529         if ( nsgn <= 0 ) goto nextfun;
00530         if ( mpfr_cmp_ui(auxr1,1L) == 0 ) goto getout;
00531         else mpfr_log(auxr3,auxr1,RND);
00532         mpfr_mul(auxr2,auxr2,auxr3,RND);
00533         break;
00534     case LI2FUNCTION: /* should be between -1 and +1 */
00535         if ( nsgn == 0 ) goto getout;
00536         mpfr_abs(auxr3,auxr1,RND);
00537         if ( mpfr_cmp_ui(auxr3,1L) > 0 ) goto nextfun;
00538         mpfr_li2(auxr3,auxr1,RND);
00539         mpfr_mul(auxr2,auxr2,auxr3,RND);
00540         break;
00541     case GAMMAFUN:
00542 /*
00543      We cannot do this when the argument is a non-positive integer
00544 */
00545         if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] <= 0 ) goto nextfun;
00546         if ( t[FUNHEAD] == t[1]-FUNHEAD && ABS(t[t[1]-1]) ==
00547             t[FUNHEAD]-ARGHEAD-1 && t[t[1]-1] < 0 ) {
00548             nsize = (-t[t[1]-1]-1)/2;
00549             if ( t[t[1]-1-nsize] == 1 ) {
00550                 for ( i = 1; i < nsize; i++ ) {
00551                     if ( t[t[1]-1-nsize+i] != 0 ) break;
00552                 }
00553                 if ( i >= nsize ) goto nextfun;
00554             }
00555             mpfr_gamma(auxr3,auxr1,RND);
00556             mpfr_mul(auxr2,auxr2,auxr3,RND);
00557             break;
00558         case AGMFUNCTION:
00559             nsgn = mpfr_sgn(auxr1);
00560             nsgn2 = mpfr_sgn(auxr3);
00561             if ( nsgn == 0 || nsgn2 == 0 ) goto getout;
00562             if ( nsgn < 0 || nsgn2 < 0 ) goto nextfun;
00563             mpfr_agm(auxr3,auxr1,auxr3,RND);
00564             mpfr_mul(auxr2,auxr2,auxr3,RND);
00565             break;
00566         case SINHFUNCTION:
00567             if ( nsgn == 0 ) goto getout;
00568             mpfr_sinh(auxr3,auxr1,RND);
00569             mpfr_mul(auxr2,auxr2,auxr3,RND);
00570             break;
00571         case COSHFUNCTION:
00572             mpfr_cosh(auxr3,auxr1,RND);
00573             mpfr_mul(auxr2,auxr2,auxr3,RND);
00574             break;
00575         case TANHFUNCTION:
00576             if ( nsgn == 0 ) goto getout;
00577             mpfr_tanh(auxr3,auxr1,RND);
00578             mpfr_mul(auxr2,auxr2,auxr3,RND);
00579             break;
00580         case ASINHFUNCTION:
00581             if ( nsgn == 0 ) goto getout;
00582             mpfr_asinh(auxr3,auxr1,RND);
00583             mpfr_mul(auxr2,auxr2,auxr3,RND);
00584             break;
00585         case ACOSHFUNCTION:

```

```

00587         if ( mpfr_cmp_ui(auxr1,1L) < 0 ) goto nextfun;
00588         if ( mpfr_cmp_ui(auxr1,1L) == 0 ) goto getout;
00589         mpfr_acosh(auxr3,auxr1,RND);
00590         mpfr_mul(auxr2,auxr2,auxr3,RND);
00591     break;
00592     case ATANHFUNCTION:
00593         if ( nsgn == 0 ) goto getout;
00594         mpfr_abs(auxr3,auxr1,RND);
00595         if ( mpfr_cmp_ui(auxr3,1L) >= 0 ) goto nextfun;
00596         mpfr_atanh(auxr3,auxr1,RND);
00597         mpfr_mul(auxr2,auxr2,auxr3,RND);
00598     break;
00599     case ASINFUNCTION:
00600         if ( nsgn == 0 ) goto getout;
00601         mpfr_abs(auxr3,auxr1,RND);
00602         if ( mpfr_cmp_ui(auxr3,1L) > 0 ) goto nextfun;
00603         mpfr_asin(auxr3,auxr1,RND);
00604         mpfr_mul(auxr2,auxr2,auxr3,RND);
00605     break;
00606     case ACOSFUNCTION:
00607         if ( mpfr_cmp_ui(auxr1,1L) == 0 ) goto getout;
00608         mpfr_abs(auxr3,auxr1,RND);
00609         if ( mpfr_cmp_ui(auxr3,1L) > 0 ) goto nextfun;
00610         mpfr_acos(auxr3,auxr1,RND);
00611         mpfr_mul(auxr2,auxr2,auxr3,RND);
00612     break;
00613     case ATANFUNCTION:
00614         if ( nsgn == 0 ) goto getout;
00615         mpfr_atan(auxr3,auxr1,RND);
00616         mpfr_mul(auxr2,auxr2,auxr3,RND);
00617     break;
00618     case SINFUNCTION:
00619         mpfr_sin(auxr3,auxr1,RND);
00620         mpfr_mul(auxr2,auxr2,auxr3,RND);
00621     break;
00622     case COSFUNCTION:
00623         mpfr_cos(auxr3,auxr1,RND);
00624         mpfr_mul(auxr2,auxr2,auxr3,RND);
00625     break;
00626     case TANFUNCTION:
00627         mpfr_tan(auxr3,auxr1,RND);
00628         mpfr_mul(auxr2,auxr2,auxr3,RND);
00629     break;
00630     default:
00631         goto nextfun;
00632     break;
00633 }
00634 first = 0;
00635 *t = 0;
00636 goto nextfun;
00637 }
00638 else if ( pars[2] == ALLFUNCTIONS ) {
00639     /*
00640     Now we have to test whether this is one of our functions
00641     */
00642     if ( t[1] == FUNHEAD ) goto nextfun;
00643     switch ( *t ) {
00644     case SQRTFUNCTION:
00645     case LNFUNCTION:
00646     case LI2FUNCTION:
00647     case LINFUNCTION:
00648     case ASINFUNCTION:
00649     case ACOSFUNCTION:
00650     case ATANFUNCTION:
00651     case SINHFUNCTION:
00652     case COSHFUNCTION:
00653     case TANHFUNCTION:
00654     case ASINHFUNCTION:
00655     case ACOSHFUNCTION:
00656     case ATANHFUNCTION:
00657     case SINFUNCTION:
00658     case COSFUNCTION:
00659     case TANFUNCTION:
00660     case AGMFUNCTION:
00661     case SYMBOL:
00662         goto TestArgument;
00663     case MZV:
00664     case EULER:
00665     case MZVHALF:
00666         goto nextfun;
00667     default:
00668         MLOCK(ErrorMessageLock);
00669         MesPrint("Function in evaluate statement not yet implemented.");
00670         MUNLOCK(ErrorMessageLock);
00671         break;
00672     }
00673 }

```

```

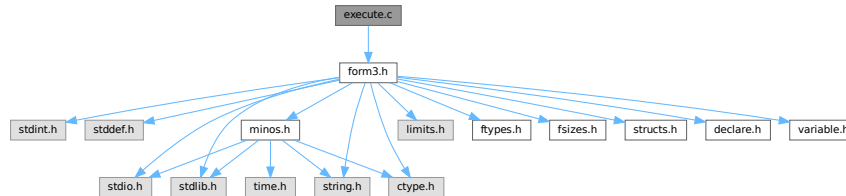
00674         else goto nextfun;
00675 nextfun:
00676     t += t[1];
00677 }
00678 if ( first == 1 ) return(Generator(BHEAD term,level));
00679 mpfr_get_f(aux4,auxr2,RND);
00680 /*
00681 Step 3:
00682 Now the regular coefficient, if it is not 1/l.
00683 We have two cases: size +- 3, or bigger.
00684 */
00685
00686 nsize = term[*term-1];
00687 if ( nsize < 0 ) { nsize = -nsize; nsgn = -1; }
00688 else nsgn = 1;
00689 if ( aux4->_mp_size < 0 ) {
00690     aux4->_mp_size = -aux4->_mp_size;
00691     nsgn = -nsgn;
00692 }
00693
00694 if ( nsize == 3 ) {
00695     if ( tstop[0] != 1 ) {
00696         mpfr_mul_ui(aux4,aux4,(ULONG)((UWORD)tstop[0]));
00697     }
00698     if ( tstop[1] != 1 ) {
00699         mpfr_div_ui(aux4,aux4,(ULONG)((UWORD)tstop[1]));
00700     }
00701 }
00702 else {
00703     RatToFloat(aux5,(UWORD *)tstop,nsize);
00704     mpfr_mul(aux4,aux4,aux5);
00705 }
00706 /*
00707 Now we have to locate possible other float_ functions.
00708 Note possible incompatibilities between the mpfr and mpfr formats.
00709 */
00710 t = term+1;
00711 while ( t < tstop ) {
00712     if ( *t == FLOATFUN ) {
00713         UnpackFloat(aux5,t);
00714         mpfr_mul(aux4,aux4,aux5);
00715     }
00716     t += t[1];
00717 }
00718 /*
00719 Now we should compose the new term in the Workspace.
00720 */
00721 t = term+1;
00722 newterm = AT.WorkPointer;
00723 tt = newterm+1;
00724 while ( t < tstop ) {
00725     if ( *t == 0 || *t == FLOATFUN ) t += t[1];
00726     else {
00727         i = t[1]; NCOPY(tt,t,i);
00728     }
00729 }
00730 PackFloat(tt,aux4);
00731 tt += tt[1];
00732 *tt++ = 1; *tt++ = 1; *tt++ = 3*nsgn;
00733 *newterm = tt-newterm;
00734 AT.WorkPointer = tt;
00735 retval = Generator(BHEAD newterm,level);
00736 getout:
00737 AT.WorkPointer = oldworkpointer;
00738 return(retval);
00739 }
00740
00741 /*
00742 #] EvaluateFun :
00743 #] mpfr_ :
00744 */

```

## 4.29 execute.c File Reference

```
#include "form3.h"
```

Include dependency graph for execute.c:



### Macros

- `#define` [CURRENTBRACKET](#) 1
- `#define` [BRACKETCURRENTEXPR](#) 2
- `#define` [BRACKETOTHEREXPR](#) 3
- `#define` [NOBRACKETACTIVE](#) 4

### Functions

- `int` [CleanExpr](#) (WORD par)
- `int` [PopVariables](#) (void)
- `void` [MakeGlobal](#) (void)
- `void` [TestDrop](#) (void)
- `void` [PutInVflags](#) (WORD nexpr)
- `int` [DoExecute](#) (WORD par, WORD skip)
- `int` [PutBracket](#) (PHEAD WORD \*termin)
- `void` [SpecialCleanup](#) (PHEAD0)
- `void` [SetMods](#) (void)
- `void` [UnSetMods](#) (void)
- `void` [ExchangeExpressions](#) (int num1, int num2)
- `int` [GetFirstBracket](#) (WORD \*term, int num)
- `int` [GetFirstTerm](#) (WORD \*term, int num, int pre)
- `int` [GetContent](#) (WORD \*content, int num)
- `int` [CleanupTerm](#) (WORD \*term)
- `WORD` [ContentMerge](#) (PHEAD WORD \*content, WORD \*term)
- `LONG` [TermsInExpression](#) (WORD num)
- `LONG` [SizeOfExpression](#) (WORD num)
- `void` [UpdatePositions](#) (void)
- `LONG` [CountTerms1](#) (PHEAD0)
- `LONG` [TermsInBracket](#) (PHEAD WORD \*term, WORD level)

### 4.29.1 Detailed Description

The routines that start the execution phase of a module. It also contains the routines for placing the bracket subterm.

Definition in file [execute.c](#).

## 4.29.2 Macro Definition Documentation

### 4.29.2.1 CURRENTBRACKET

```
#define CURRENTBRACKET 1
```

Definition at line 2565 of file [execute.c](#).

### 4.29.2.2 BRACKETCURRENTEXPR

```
#define BRACKETCURRENTEXPR 2
```

Definition at line 2566 of file [execute.c](#).

### 4.29.2.3 BRACKETOTHEREXPR

```
#define BRACKETOTHEREXPR 3
```

Definition at line 2567 of file [execute.c](#).

### 4.29.2.4 NOBRACKETACTIVE

```
#define NOBRACKETACTIVE 4
```

Definition at line 2568 of file [execute.c](#).

## 4.29.3 Function Documentation

### 4.29.3.1 CleanExpr()

```
int CleanExpr (  
    WORD par )
```

Definition at line 47 of file [execute.c](#).

### 4.29.3.2 PopVariables()

```
int PopVariables (  
    void )
```

Definition at line 211 of file [execute.c](#).

### 4.29.3.3 MakeGlobal()

```
void MakeGlobal (  
    void )
```

Definition at line 383 of file [execute.c](#).

#### 4.29.3.4 TestDrop()

```
void TestDrop (
    void )
```

Definition at line [484](#) of file [execute.c](#).

#### 4.29.3.5 PutInVflags()

```
void PutInVflags (
    WORD nexpr )
```

Definition at line [562](#) of file [execute.c](#).

#### 4.29.3.6 DoExecute()

```
int DoExecute (
    WORD par,
    WORD skip )
```

Definition at line [616](#) of file [execute.c](#).

#### 4.29.3.7 PutBracket()

```
int PutBracket (
    PHEAD WORD * termin )
```

Definition at line [1112](#) of file [execute.c](#).

#### 4.29.3.8 SpecialCleanup()

```
void SpecialCleanup (
    PHEAD0 )
```

Definition at line [1624](#) of file [execute.c](#).

#### 4.29.3.9 SetMods()

```
void SetMods (
    void )
```

Definition at line [1638](#) of file [execute.c](#).

#### 4.29.3.10 UnSetMods()

```
void UnSetMods (
    void )
```

Definition at line [1656](#) of file [execute.c](#).



#### 4.29.3.11 ExchangeExpressions()

```
void ExchangeExpressions (
    int num1,
    int num2 )
```

Definition at line 1671 of file [execute.c](#).

#### 4.29.3.12 GetFirstBracket()

```
int GetFirstBracket (
    WORD * term,
    int num )
```

Definition at line 1745 of file [execute.c](#).

#### 4.29.3.13 GetFirstTerm()

```
int GetFirstTerm (
    WORD * term,
    int num,
    int pre )
```

Gets the first term of an expression. Routine should be thread safe.

##### Parameters

|     |             |                                                                                                            |
|-----|-------------|------------------------------------------------------------------------------------------------------------|
| out | <i>term</i> | Pointer to the buffer where the term should be written.                                                    |
| in  | <i>num</i>  | The expression number from which to get the term.                                                          |
| in  | <i>pre</i>  | Denotes a pre-processor call (1) or not (0). Used to determine the correct source file for the expression. |

##### Returns

First WORD of term, a negative value denotes an error.

Definition at line 1864 of file [execute.c](#).

References [FiLe::handle](#), [VaRrEnUm::lo](#), and [ReNuMbEr::symb](#).

#### 4.29.3.14 GetContent()

```
int GetContent (
    WORD * content,
    int num )
```

Definition at line 1972 of file [execute.c](#).

#### 4.29.3.15 CleanupTerm()

```
int CleanupTerm (
    WORD * term )
```

Definition at line 2089 of file [execute.c](#).

#### 4.29.3.16 ContentMerge()

```
WORD ContentMerge (
    PHEAD WORD * content,
    WORD * term )
```

Definition at line 2113 of file [execute.c](#).

#### 4.29.3.17 TermsInExpression()

```
LONG TermsInExpression (
    WORD num )
```

Definition at line 2377 of file [execute.c](#).

#### 4.29.3.18 SizeOfExpression()

```
LONG SizeOfExpression (
    WORD num )
```

Definition at line 2389 of file [execute.c](#).

#### 4.29.3.19 UpdatePositions()

```
void UpdatePositions (
    void )
```

Definition at line 2401 of file [execute.c](#).

#### 4.29.3.20 CountTerms1()

```
LONG CountTerms1 (
    PHEAD0 )
```

Definition at line 2462 of file [execute.c](#).

#### 4.29.3.21 TermsInBracket()

```
LONG TermsInBracket (
    PHEAD WORD * term,
    WORD level )
```

Definition at line 2570 of file [execute.c](#).

## 4.30 execute.c

[Go to the documentation of this file.](#)

```

00001
00006 /* #[ License : */
00007 /*
00008 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 *   When using this file you are requested to refer to the publication
00010 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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00012 *   for by FOM the Dutch physics granting agency and we would like to
00013 *   be able to track its scientific use to convince FOM of its value
00014 *   for the community.
00015 *
00016 *   This file is part of FORM.
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00022 *
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00026 *   details.
00027 *
00028 *   You should have received a copy of the GNU General Public License along
00029 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #[ License : */
00032 /*
00033     #[ Includes : execute.c
00034 */
00035
00036 #include "form3.h"
00037
00038 /*
00039     #[ Includes :
00040     #[ DoExecute :
00041         #[ CleanExpr :
00042
00043         par == 1 after .store or .clear
00044         par == 0 after .sort
00045 */
00046
00047 int CleanExpr(WORD par)
00048 {
00049     GETIDENTITY
00050     WORD j, n, i;
00051     POSITION length;
00052     EXPRESSIONS e_in, e_out, e;
00053     int numhid = 0;
00054     NAMENODE *node;
00055     n = NumExpressions;
00056     j = 0;
00057     e_in = e_out = Expressions;
00058     if ( n > 0 ) { do {
00059         e_in->vflags &= ~( TOBEFACTORED | TOBEUNFACTORED );
00060         if ( par ) {
00061             if ( e_in->renumlists ) {
00062                 if ( e_in->renumlists != AN.dummyrenumlist )
00063                     M_free(e_in->renumlists, "Renumlist");
00064                 e_in->renumlists = 0;
00065             }
00066             if ( e_in->renum ) {
00067                 M_free(e_in->renum, "Renum"); e_in->renum = 0;
00068             }
00069         }
00070         if ( e_in->status == HIDDENLEXPRESSION
00071             || e_in->status == HIDDENEXPRESSION ) numhid++;
00072         switch ( e_in->status ) {
00073             case SPECTATOREXPRESSION:
00074             case LOCALEXPRESSION:
00075             case HIDDENLEXPRESSION:
00076                 if ( par ) {
00077                     AC.exprnames->namenode[e_in->node].type = CDELETE;
00078                     AC.DidClean = 1;
00079                     if ( e_in->status != HIDDENLEXPRESSION )
00080                         ClearBracketIndex(e_in->Expressions);
00081                     break;
00082                 }
00083                 /* fall through */
00084             case GLOBALEXPRESSION:
00085             case HIDDENEXPRESSION:
00086                 if ( par ) {

```

```

00087 #ifdef WITHMPI
00088     /*
00089      * Broadcast the global expression from the master to the all workers.
00090      */
00091     if ( PF_BroadcastExpr(e_in, e_in->status == HIDDENEXPRESSION ? AR.hidefile :
AR.outfile) ) return -1;
00092     if ( PF.me == MASTER ) {
00093 #endif
00094         e = e_in;
00095         i = n-1;
00096         while ( --i >= 0 ) {
00097             e++;
00098             if ( e_in->status == HIDDENEXPRESSION ) {
00099                 if ( e->status == HIDDENEXPRESSION
00100                     || e->status == HIDDENLEXPRESSION ) break;
00101             }
00102             else {
00103                 if ( e->status == GLOBALEXPRESSION
00104                     || e->status == LOCALEXPRESSION ) break;
00105             }
00106         }
00107 #ifdef WITHMPI
00108     }
00109     else {
00110         /*
00111          * On the slaves, the broadcast expression is sitting at the end of the file.
00112          */
00113         e = e_in;
00114         i = -1;
00115     }
00116 #endif
00117     if ( i >= 0 ) {
00118         DIFPOS(length,e->onfile,e_in->onfile);
00119     }
00120     else {
00121         FILEHANDLE *f = e_in->status == HIDDENEXPRESSION ? AR.hidefile : AR.outfile;
00122         if ( f->handle < 0 ) {
00123             SETBASELENGTH(length,TOLONG(f->Pofull)
00124                 - TOLONG(f->Pobuffer)
00125                 - BASEPOSITION(e_in->onfile));
00126         }
00127         else {
00128             SeekFile(f->handle,&(f->filesize),SEEK_SET);
00129             DIFPOS(length,f->filesize,e_in->onfile);
00130         }
00131     }
00132     if ( ToStorage(e_in,&length) ) {
00133         return (MesCall("CleanExpr"));
00134     }
00135     e_in->status = STOREDEXPRESSION;
00136     if ( e_in->status != HIDDENEXPRESSION )
00137         ClearBracketIndex(e_in->Expressions);
00138 }
00139 /* fall through */
00140 case SKIPLEXPRESSION:
00141 case DROPLEXPRESSION:
00142 case DROPHLEXPRESSION:
00143 case DROPGEXPRESSION:
00144 case DROPHGEXPRESSION:
00145 case STOREDEXPRESSION:
00146 case DROPSPECTATOREXPRESSION:
00147     if ( e_out != e_in ) {
00148         node = AC.exprnames->namenode + e_in->node;
00149         node->number = e_out - Expressions;
00150
00151         e_out->onfile = e_in->onfile;
00152         e_out->size = e_in->size;
00153         e_out->printflag = 0;
00154         if ( par ) e_out->status = STOREDEXPRESSION;
00155         else e_out->status = e_in->status;
00156         e_out->name = e_in->name;
00157         e_out->node = e_in->node;
00158         e_out->renum = e_in->renum;
00159         e_out->renumlists = e_in->renumlists;
00160         e_out->counter = e_in->counter;
00161         e_out->hidelevel = e_in->hidelevel;
00162         e_out->inmem = e_in->inmem;
00163         e_out->bracketinfo = e_in->bracketinfo;
00164         e_out->newbracketinfo = e_in->newbracketinfo;
00165         e_out->numdummies = e_in->numdummies;
00166         e_out->numfactors = e_in->numfactors;
00167         e_out->vflags = e_in->vflags;
00168         e_out->uflags = e_in->uflags;
00169         e_out->sizeprototype = e_in->sizeprototype;
00170     }
00171 #ifdef PARALLELCODE
00172     e_out->partodo = 0;

```

```

00173 #endif
00174         e_out++;
00175         j++;
00176         break;
00177     case DROPEXPRESSION:
00178         break;
00179     default:
00180         AC.exprnames->namenode[e_in->node].type = CDELETE;
00181         AC.DidClean = 1;
00182         break;
00183     }
00184     e_in++;
00185 } while ( --n > 0 ); }
00186 UpdateMaxSize();
00187 NumExpressions = j;
00188 if ( numhid == 0 && AR.hidefile->PObuffer ) {
00189     if ( AR.hidefile->handle >= 0 ) {
00190         CloseFile(AR.hidefile->handle);
00191         remove(AR.hidefile->name);
00192         AR.hidefile->handle = -1;
00193     }
00194     AR.hidefile->POfull =
00195     AR.hidefile->POfill = AR.hidefile->PObuffer;
00196     PUTZERO(AR.hidefile->POposition);
00197 }
00198 FlushSpectators();
00199 return(0);
00200 }
00201
00202 /*
00203     #] CleanExpr :
00204     #[ PopVariables :
00205
00206     Pops the local variables from the tables.
00207     The Expressions are reprocessed and their tables are compactified.
00208
00209 */
00210
00211 int PopVariables(void)
00212 {
00213     GETIDENTITY
00214     WORD i, j;
00215     int retval;
00216     UBYTE *s;
00217
00218     retval = CleanExpr(1);
00219     ResetVariables(1);
00220
00221     if ( AC.DidClean ) CompactifyTree(AC.exprnames,EXPRNAMES);
00222
00223     AC.CodesFlag = AM.gCodesFlag;
00224     AC.NamesFlag = AM.gNamesFlag;
00225     AC.StatsFlag = AM.gStatsFlag;
00226 #ifdef WITHFLOAT
00227     AC.MaxWeight = AM.gMaxWeight;
00228     AC.DefaultPrecision = AM.gDefaultPrecision;
00229 #endif
00230     AC.OldFactArgFlag = AM.gOldFactArgFlag;
00231     AC.TokensWriteFlag = AM.gTokensWriteFlag;
00232     AC.extrasymbols = AM.gextrasymbols;
00233     if ( AC.extrasym ) { M_free(AC.extrasym,"extrasym"); AC.extrasym = 0; }
00234     i = 1; s = AM.gextrasym; while ( *s ) { s++; i++; }
00235     AC.extrasym = (UBYTE *)Malloc1(i*sizeof(UBYTE),"extrasym");
00236     for ( j = 0; j < i; j++ ) AC.extrasym[j] = AM.gextrasym[j];
00237     AO.NoSpacesInNumbers = AM.gNoSpacesInNumbers;
00238     AC.IndentSpace = AM.gIndentSpace;
00239     AC.lUnitTrace = AM.gUnitTrace;
00240     AC.lDefDim = AM.gDefDim;
00241     AC.lDefDim4 = AM.gDefDim4;
00242     if ( AC.halfmod ) {
00243         if ( AC.ncmod == AM.gncmod && AC.modmode == AM.gmodmode ) {
00244             j = ABS(AC.ncmod);
00245             while ( --j >= 0 ) {
00246                 if ( AC.cmod[j] != AM.gcmod[j] ) break;
00247             }
00248             if ( j >= 0 ) {
00249                 M_free(AC.halfmod,"halfmod");
00250                 AC.halfmod = 0; AC.nhalfmod = 0;
00251             }
00252         }
00253         else {
00254             M_free(AC.halfmod,"halfmod");
00255             AC.halfmod = 0; AC.nhalfmod = 0;
00256         }
00257     }
00258     if ( AC.modinverses ) {
00259         if ( AC.ncmod == AM.gncmod && AC.modmode == AM.gmodmode ) {

```

```

00260         j = ABS(AC.ncmod);
00261         while ( --j >= 0 ) {
00262             if ( AC.cmod[j] != AM.gcmmod[j] ) break;
00263         }
00264         if ( j >= 0 ) {
00265             M_free(AC.modinverses,"modinverses");
00266             AC.modinverses = 0;
00267         }
00268     }
00269     else {
00270         M_free(AC.modinverses,"modinverses");
00271         AC.modinverses = 0;
00272     }
00273 }
00274 AN.ncmod = AC.ncmod = AM.gncmod;
00275 AC.npowmod = AM.gnpowmod;
00276 AC.modmode = AM.gmodmode;
00277 if ( ( ( AC.modmode & INVERSETABLE ) != 0 ) && ( AC.modinverses == 0 ) )
00278     MakeInverses();
00279 AC.funpowers = AM.gfunpowers;
00280 AC.lPolyFun = AM.gPolyFun;
00281 AC.lPolyFunInv = AM.gPolyFunInv;
00282 AC.lPolyFunType = AM.gPolyFunType;
00283 AC.lPolyFunExp = AM.gPolyFunExp;
00284 AR.PolyFunVar = AC.lPolyFunVar = AM.gPolyFunVar;
00285 AC.lPolyFunPow = AM.gPolyFunPow;
00286 AC.parallelflag = AM.gparallelflag;
00287 AC.ProcessBucketSize = AC.mProcessBucketSize = AM.gProcessBucketSize;
00288 AC.properorderflag = AM.gproperorderflag;
00289 AC.ThreadBucketSize = AM.gThreadBucketSize;
00290 AC.ThreadStats = AM.gThreadStats;
00291 AC.FinalStats = AM.gFinalStats;
00292 AC.OldGCDflag = AM.gOldGCDflag;
00293 AC.WTimeStatsFlag = AM.gWTimeStatsFlag;
00294 AC.ThreadsFlag = AM.gThreadsFlag;
00295 AC.ThreadBalancing = AM.gThreadBalancing;
00296 AC.ThreadSortFileSynch = AM.gThreadSortFileSynch;
00297 AC.ProcessStats = AM.gProcessStats;
00298 AC.OldParallelStats = AM.gOldParallelStats;
00299 AC.IsFortran90 = AM.gIsFortran90;
00300 AC.SizeCommuteInSet = AM.gSizeCommuteInSet;
00301 PruneExtraSymbols(AM.gnumextrasyms);
00302
00303 if ( AC.Fortran90Kind ) {
00304     M_free(AC.Fortran90Kind,"Fortran90 Kind");
00305     AC.Fortran90Kind = 0;
00306 }
00307 if ( AM.gFortran90Kind ) {
00308     AC.Fortran90Kind = strdup(AM.gFortran90Kind,"Fortran90 Kind");
00309 }
00310 if ( AC.ThreadsFlag && AM.totalnumberofthreads > 1 ) AS.MultiThreaded = 1;
00311 {
00312     UWORD *p, *m;
00313     p = AM.gcmmod;
00314     m = AC.cmod;
00315     j = ABS(AC.ncmod);
00316     NCOPY(m,p,j);
00317     p = AM.gpowmod;
00318     m = AC.powmod;
00319     j = AC.npowmod;
00320     NCOPY(m,p,j);
00321     if ( AC.DirtPow ) {
00322         if ( MakeModTable() ) {
00323             MesPrint("===No printing in powers of generator");
00324         }
00325         AC.DirtPow = 0;
00326     }
00327 }
00328 {
00329     WORD *p, *m;
00330     p = AM.gUniTrace;
00331     m = AC.lUniTrace;
00332     j = 4;
00333     NCOPY(m,p,j);
00334 }
00335 AC.Cnumpows = AM.gCnumpows;
00336 AC.OutputMode = AM.gOutputMode;
00337 AC.OutputSpaces = AM.gOutputSpaces;
00338 AC.OutNumberType = AM.gOutNumberType;
00339 AR.SortType = AC.SortType = AM.gSortType;
00340 AC.ShortStatsMax = AM.gShortStatsMax;
00341 /*
00342 Now we have to clean up the commutation properties
00343 */
00344 for ( i = 0; i < NumFunctions; i++ ) functions[i].flags &= ~COULDCOMMUTE;
00345 if ( AC.CommuteInSet ) {
00346     WORD *g, *gg;

```

```

00347     g = AC.CommuteInSet;
00348     while ( *g ) {
00349         gg = g+1; g += *g;
00350         while ( gg < g ) {
00351             if ( *gg <= GAMMASEVEN && *gg >= GAMMA ) {
00352                 functions[GAMMA-FUNCTION].flags |= COULDCOMMUTE;
00353                 functions[GAMMAI-FUNCTION].flags |= COULDCOMMUTE;
00354                 functions[GAMMAFIVE-FUNCTION].flags |= COULDCOMMUTE;
00355                 functions[GAMMASIX-FUNCTION].flags |= COULDCOMMUTE;
00356                 functions[GAMMASEVEN-FUNCTION].flags |= COULDCOMMUTE;
00357             }
00358             else {
00359                 functions[*gg-FUNCTION].flags |= COULDCOMMUTE;
00360             }
00361         }
00362     }
00363 }
00364 /*
00365 Clean up the dictionaries.
00366 */
00367 for ( i = AO.NumDictionaries-1; i >= AO.gNumDictionaries; i-- ) {
00368     RemoveDictionary(AO.Dictionaries[i]);
00369     M_free(AO.Dictionaries[i], "Dictionary");
00370 }
00371 for( ; i >= 0; i-- ) {
00372     ShrinkDictionary(AO.Dictionaries[i]);
00373 }
00374 AO.NumDictionaries = AO.gNumDictionaries;
00375 return(retval);
00376 }
00377
00378 /*
00379     #] PopVariables :
00380     #[ MakeGlobal :
00381 */
00382
00383 void MakeGlobal(void)
00384 {
00385     WORD i, j, *pp, *mm;
00386     UWORD *p, *m;
00387     UBYTE *s;
00388     Globalize(0);
00389
00390     AM.gCodesFlag = AC.CodesFlag;
00391     AM.gNamesFlag = AC.NamesFlag;
00392     AM.gStatsFlag = AC.StatsFlag;
00393     #ifdef WITHFLOAT
00394     AM.gMaxWeight = AC.MaxWeight;
00395     AM.gDefaultPrecision = AC.DefaultPrecision;
00396     #endif
00397     AM.gOldFactArgFlag = AC.OldFactArgFlag;
00398     AM.gextrasyms = AC.extrasymbols;
00399     if ( AM.gextrasym ) { M_free(AM.gextrasym, "extrasym"); AM.gextrasym = 0; }
00400     i = 1; s = AC.extrasym; while ( *s ) { s++; i++; }
00401     AM.gextrasym = (UBYTE *)Malloc1(i*sizeof(UBYTE), "extrasym");
00402     for ( j = 0; j < i; j++ ) AM.gextrasym[j] = AC.extrasym[j];
00403     AM.gTokensWriteFlag = AC.TokensWriteFlag;
00404     AM.gNoSpacesInNumbers = AO.NoSpacesInNumbers;
00405     AM.gIndentSpace = AO.IndentSpace;
00406     AM.gUnitTrace = AC.lUnitTrace;
00407     AM.gDefDim = AC.lDefDim;
00408     AM.gDefDim4 = AC.lDefDim4;
00409     AM.gncmod = AC.ncmod;
00410     AM.gnpowmod = AC.npowmod;
00411     AM.gmodmode = AC.modmode;
00412     AM.gCnumpows = AC.Cnumpows;
00413     AM.gOutputMode = AC.OutputMode;
00414     AM.gOutputSpaces = AC.OutputSpaces;
00415     AM.gOutNumberType = AC.OutNumberType;
00416     AM.gfunpowers = AC.funpowers;
00417     AM.gPolyFun = AC.lPolyFun;
00418     AM.gPolyFunInv = AC.lPolyFunInv;
00419     AM.gPolyFunType = AC.lPolyFunType;
00420     AM.gPolyFunExp = AC.lPolyFunExp;
00421     AM.gPolyFunVar = AC.lPolyFunVar;
00422     AM.gPolyFunPow = AC.lPolyFunPow;
00423     AM.gparallelflag = AC.parallelflag;
00424     AM.gProcessBucketSize = AC.ProcessBucketSize;
00425     AM.gproperorderflag = AC.properorderflag;
00426     AM.gThreadBucketSize = AC.ThreadBucketSize;
00427     AM.gThreadStats = AC.ThreadStats;
00428     AM.gFinalStats = AC.FinalStats;
00429     AM.gOldGCDflag = AC.OldGCDflag;
00430     AM.gWTimeStatsFlag = AC.WTimeStatsFlag;
00431     AM.gThreadsFlag = AC.ThreadsFlag;
00432     AM.gThreadBalancing = AC.ThreadBalancing;
00433     AM.gThreadSortFileSynch = AC.ThreadSortFileSynch;

```

```

00434     AM.gProcessStats = AC.ProcessStats;
00435     AM.gOldParallelStats = AC.OldParallelStats;
00436     AM.gIsFortran90 = AC.IsFortran90;
00437     AM.gSizeCommuteInSet = AC.SizeCommuteInSet;
00438     AM.gnumextrasyms = (cbuf+AM.sbufnum)->numrhs;
00439     if ( AM.gFortran90Kind ) {
00440         M_free(AM.gFortran90Kind,"Fortran 90 Kind");
00441         AM.gFortran90Kind = 0;
00442     }
00443     if ( AC.Fortran90Kind ) {
00444         AM.gFortran90Kind = strDup1(AC.Fortran90Kind,"Fortran 90 Kind");
00445     }
00446     p = AM.gcmmod;
00447     m = AC.cmod;
00448     i = ABS(AC.ncmod);
00449     NCOPY(p,m,i);
00450     p = AM.gpowmod;
00451     m = AC.powmod;
00452     i = AC.npowmod;
00453     NCOPY(p,m,i);
00454     pp = AM.gUniTrace;
00455     mm = AC.lUniTrace;
00456     i = 4;
00457     NCOPY(pp,mm,i);
00458     AM.gSortType = AC.SortType;
00459     AM.gShortStatsMax = AC.ShortStatsMax;
00460
00461     if ( AO.CurrentDictionary > 0 || AP.OpenDictionary > 0 ) {
00462         Warning("You cannot have an open or selected dictionary at a .global. Dictionary closed.");
00463         AP.OpenDictionary = 0;
00464         AO.CurrentDictionary = 0;
00465     }
00466
00467     AO.gNumDictionaries = AO.NumDictionaries;
00468     for ( i = 0; i < AO.NumDictionaries; i++ ) {
00469         AO.Dictionaries[i]->gnumelements = AO.Dictionaries[i]->numelements;
00470     }
00471     if ( AM.NumSpectatorFiles > 0 ) {
00472         for ( i = 0; i < AM.SizeForSpectatorFiles; i++ ) {
00473             if ( AM.SpectatorFiles[i].name != 0 )
00474                 AM.SpectatorFiles[i].flags |= GLOBALSPECTATORFLAG;
00475         }
00476     }
00477 }
00478
00479 /*
00480     #] MakeGlobal :
00481     #[ TestDrop :
00482 */
00483
00484 void TestDrop(void)
00485 {
00486     EXPRESSIONS e;
00487     WORD j;
00488     for ( j = 0, e = Expressions; j < NumExpressions; j++, e++ ) {
00489         switch ( e->status ) {
00490             case SKIPLEXPRESSION:
00491                 e->status = LOCALEXPRESSION;
00492                 break;
00493             case UNHIDELEXPRESSION:
00494                 e->status = LOCALEXPRESSION;
00495                 ClearBracketIndex(j);
00496                 e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
00497                 break;
00498             case HIDELEXPRESSION:
00499                 e->status = HIDDENLEXPRESSION;
00500                 break;
00501             case SKIPGEXPRESSION:
00502                 e->status = GLOBALEXPRESSION;
00503                 break;
00504             case UNHIDEGEXPRESSION:
00505                 e->status = GLOBALEXPRESSION;
00506                 ClearBracketIndex(j);
00507                 e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
00508                 break;
00509             case HIDEGEXPRESSION:
00510                 e->status = HIDDENGEXPRESSION;
00511                 break;
00512             case DROPLEXPRESSION:
00513             case DROPGEXPRESSION:
00514             case DROPHLEXPRESSION:
00515             case DROPHGEXPRESSION:
00516             case DROPSPECTATOREXPRESSION:
00517                 e->status = DROPPDEXPRESSION;
00518                 ClearBracketIndex(j);
00519                 e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
00520                 if ( e->replace >= 0 ) {

```



```

00521         Expressions[e->replace].replace = REGULAREXPRESSION;
00522         AC.exprnames->namenode[e->node].number = e->replace;
00523         e->replace = REGULAREXPRESSION;
00524     }
00525     else {
00526         AC.exprnames->namenode[e->node].type = CDELETE;
00527         AC.DidClean = 1;
00528     }
00529     break;
00530 case LOCALEXPRESSION:
00531 case GLOBALEXPRESSION:
00532     ClearBracketIndex(j);
00533     e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
00534     break;
00535 case HIDDENLEXPRESSION:
00536 case HIDDENGEXPRESSION:
00537     break;
00538 case INTOHIDELEXPRESSION:
00539     ClearBracketIndex(j);
00540     e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
00541     e->status = HIDDENLEXPRESSION;
00542     break;
00543 case INTOHIDEGEXPRESSION:
00544     ClearBracketIndex(j);
00545     e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
00546     e->status = HIDDENGEXPRESSION;
00547     break;
00548 default:
00549     ClearBracketIndex(j);
00550     e->bracketinfo = 0;
00551     break;
00552 }
00553 if ( e->replace == NEWLYDEFINEDEXPRESSION ) e->replace = REGULAREXPRESSION;
00554 }
00555 }
00556
00557 /*
00558     #] TestDrop :
00559     #[ PutInVflags :
00560 */
00561 void PutInVflags(WORD nexpr)
00562 {
00563     EXPRESSIONS e = Expressions + nexpr;
00564     POSITION *old;
00565     WORD *oldw;
00566     int i;
00567 restart:;
00568     if ( AS.OldOnFile == 0 ) {
00569         AS.NumOldOnFile = 20;
00570         AS.OldOnFile = (POSITION *)Malloc1(AS.NumOldOnFile*sizeof(POSITION),"file pointers");
00571     }
00572     else if ( nexpr >= AS.NumOldOnFile ) {
00573         old = AS.OldOnFile;
00574         AS.OldOnFile = (POSITION *)Malloc1(2*AS.NumOldOnFile*sizeof(POSITION),"file pointers");
00575         for ( i = 0; i < AS.NumOldOnFile; i++ ) AS.OldOnFile[i] = old[i];
00576         AS.NumOldOnFile = 2*AS.NumOldOnFile;
00577         M_free(old,"process file pointers");
00578     }
00579     if ( AS.OldNumFactors == 0 ) {
00580         AS.NumOldNumFactors = 20;
00581         AS.OldNumFactors = (WORD *)Malloc1(AS.NumOldNumFactors*sizeof(WORD),"numfactors pointers");
00582         AS.Oldvflags = (WORD *)Malloc1(AS.NumOldNumFactors*sizeof(WORD),"vflags pointers");
00583         AS.Olduflags = (WORD *)Malloc1(AS.NumOldNumFactors*sizeof(WORD),"uflags pointers");
00584     }
00585     else if ( nexpr >= AS.NumOldNumFactors ) {
00586         oldw = AS.OldNumFactors;
00587         AS.OldNumFactors = (WORD *)Malloc1(2*AS.NumOldNumFactors*sizeof(WORD),"numfactors pointers");
00588         for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.OldNumFactors[i] = oldw[i];
00589         M_free(oldw,"numfactors pointers");
00590         oldw = AS.Oldvflags;
00591         AS.Oldvflags = (WORD *)Malloc1(2*AS.NumOldNumFactors*sizeof(WORD),"vflags pointers");
00592         for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.Oldvflags[i] = oldw[i];
00593         M_free(oldw,"vflags pointers");
00594         oldw = AS.Olduflags;
00595         AS.Olduflags = (WORD *)Malloc1(2*AS.NumOldNumFactors*sizeof(WORD),"uflags pointers");
00596         for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.Olduflags[i] = oldw[i];
00597         M_free(oldw,"uflags pointers");
00598         AS.NumOldNumFactors = 2*AS.NumOldNumFactors;
00599     }
00600 }
00601 /*
00602     The next is needed when we Load a .sav file with lots of expressions.
00603 */
00604 if ( nexpr >= AS.NumOldOnFile || nexpr >= AS.NumOldNumFactors ) goto restart;
00605 AS.OldOnFile[nexpr] = e->onfile;
00606 AS.OldNumFactors[nexpr] = e->numfactors;
00607 AS.Oldvflags[nexpr] = e->vflags;

```

```

00608     AS.Olduflags[nexpr] = e->uflags;
00609 }
00610
00611 /*
00612     #] PutInVflags :
00613     #[ DoExecute :
00614 */
00615
00616 int DoExecute(WORD par, WORD skip)
00617 {
00618     GETIDENTITY
00619     int RetCode = 0;
00620     int i, oldmultithreaded = AS.MultiThreaded;
00621 #ifndef PARALLELCODE
00622     int j;
00623 #endif
00624
00625     SpecialCleanup(BHEAD0);
00626     if ( skip ) goto skipexec;
00627     if ( AC.IfLevel > 0 ) {
00628         MesPrint(" %d endif statement(s) missing",AC.IfLevel);
00629         RetCode = 1;
00630     }
00631     if ( AC.WhileLevel > 0 ) {
00632         MesPrint(" %d endwhile statement(s) missing",AC.WhileLevel);
00633         RetCode = 1;
00634     }
00635     if ( AC.arglevel > 0 ) {
00636         MesPrint(" %d endargument statement(s) missing",AC.arglevel);
00637         RetCode = 1;
00638     }
00639     if ( AC.termlevel > 0 ) {
00640         MesPrint(" %d endterm statement(s) missing",AC.termlevel);
00641         RetCode = 1;
00642     }
00643     if ( AC.insidelevel > 0 ) {
00644         MesPrint(" %d endinside statement(s) missing",AC.insidelevel);
00645         RetCode = 1;
00646     }
00647     if ( AC.inexprlevel > 0 ) {
00648         MesPrint(" %d endinexpression statement(s) missing",AC.inexprlevel);
00649         RetCode = 1;
00650     }
00651     if ( AC.NumLabels > 0 ) {
00652         for ( i = 0; i < AC.NumLabels; i++ ) {
00653             if ( AC.Labels[i] < 0 ) {
00654                 MesPrint(" -->Label %s missing",AC.LabelNames[i]);
00655                 RetCode = 1;
00656             }
00657         }
00658     }
00659     if ( AC.SwitchLevel > 0 ) {
00660         MesPrint(" %d endswitch statement(s) missing",AC.SwitchLevel);
00661         RetCode = 1;
00662     }
00663     if ( AC.dolooplevel > 0 ) {
00664         MesPrint(" %d enddo statement(s) missing",AC.dolooplevel);
00665         RetCode = 1;
00666     }
00667     if ( AP.OpenDictionary > 0 ) {
00668         MesPrint(" Dictionary %s has not been closed.",
00669             AO.Dictionaries[AP.OpenDictionary-1]->name);
00670         AP.OpenDictionary = 0;
00671         RetCode = 1;
00672     }
00673     if ( RetCode ) return(RetCode);
00674     AR.Cnumlhs = cbuf[AM.rbufnum].numlhs;
00675
00676     if ( ( AS.ExecMode = par ) == GLOBALMODULE ) AS.ExecMode = 0;
00677 #ifndef PARALLELCODE
00678 /*
00679     Now check whether we have either the regular parallel flag or the
00680     mparallel flag set.
00681     Next check whether any of the expressions has partodo set.
00682     If any of the above we need to check what the dollar status is.
00683 */
00684     AC.partodoflag = -1;
00685     if ( NumPotModdollars >= 0 ) {
00686         for ( i = 0; i < NumExpressions; i++ ) {
00687             if ( Expressions[i].partodo ) { AC.partodoflag = 1; break; }
00688         }
00689     }
00690 #ifdef WITHMPI
00691     if ( AC.partodoflag > 0 && PF.numtasks < 3 ) {
00692         AC.partodoflag = 0;
00693     }
00694 #endif

```

```

00695     if ( AC.partodoflag > 0 || ( NumPotModdollars > 0 && AC.mparallelflag == PARALLELFLAG ) ) {
00696         if ( NumPotModdollars > NumModOptdollars ) {
00697             AC.mparallelflag |= NOPARALLEL_DOLLAR;
00698 #ifdef WITHPTHREADS
00699             AS.MultiThreaded = 0;
00700 #endif
00701             AC.partodoflag = 0;
00702         }
00703         else {
00704             for ( i = 0; i < NumPotModdollars; i++ ) {
00705                 for ( j = 0; j < NumModOptdollars; j++ ) {
00706                     if ( PotModdollars[i] == ModOptdollars[j].number ) break;
00707                     if ( j >= NumModOptdollars ) {
00708                         AC.mparallelflag |= NOPARALLEL_DOLLAR;
00709 #ifdef WITHPTHREADS
00710                         AS.MultiThreaded = 0;
00711 #endif
00712                         AC.partodoflag = 0;
00713                         break;
00714                     }
00715                     switch ( ModOptdollars[j].type ) {
00716                         case MODSUM:
00717                         case MODMAX:
00718                         case MODMIN:
00719                         case MODLOCAL:
00720                             break;
00721                         default:
00722                             AC.mparallelflag |= NOPARALLEL_DOLLAR;
00723                             AS.MultiThreaded = 0;
00724                             AC.partodoflag = 0;
00725                             break;
00726                     }
00727                 }
00728             }
00729         }
00730         else if ( ( AC.mparallelflag & NOPARALLEL_USER ) != 0 ) {
00731 #ifdef WITHPTHREADS
00732             AS.MultiThreaded = 0;
00733 #endif
00734             AC.partodoflag = 0;
00735         }
00736         if ( AC.partodoflag == 0 ) {
00737             for ( i = 0; i < NumExpressions; i++ ) {
00738                 Expressions[i].partodo = 0;
00739             }
00740         }
00741         else if ( AC.partodoflag == -1 ) {
00742             AC.partodoflag = 0;
00743         }
00744 #endif
00745 #ifdef WITHMPI
00746 /*
00747  * Check RHS expressions.
00748  */
00749 if ( AC.RhsExprInModuleFlag && (AC.mparallelflag == PARALLELFLAG || AC.partodoflag) ) {
00750     if ( PF.rhsInParallel ) {
00751         PF.mkSlaveInfile=1;
00752         if (PF.me != MASTER) {
00753             PF.slavebuf.PObuffer=(WORD *)Malloc1(AM.ScratSize*sizeof(WORD),"PF inbuf");
00754             PF.slavebuf.POsize=AM.ScratSize*sizeof(WORD);
00755             PF.slavebuf.POfull = PF.slavebuf.POfill = PF.slavebuf.PObuffer;
00756             PF.slavebuf.POstop= PF.slavebuf.PObuffer+AM.ScratSize;
00757             PUTZERO(PF.slavebuf.POposition);
00758             /*if (PF.me != MASTER)*/
00759         }
00760         else {
00761             AC.mparallelflag |= NOPARALLEL_RHS;
00762             AC.partodoflag = 0;
00763             for ( i = 0; i < NumExpressions; i++ ) {
00764                 Expressions[i].partodo = 0;
00765             }
00766         }
00767     }
00768 /*
00769  * Set $-variables with MODSUM to zero on the slaves.
00770  */
00771 if ( (AC.mparallelflag == PARALLELFLAG || AC.partodoflag) && PF.me != MASTER ) {
00772     for ( i = 0; i < NumModOptdollars; i++ ) {
00773         if ( ModOptdollars[i].type == MODSUM ) {
00774             DOLLARS d = Dollars + ModOptdollars[i].number;
00775             d->type = DOLZERO;
00776             if ( d->where && d->where != &AM.dollarzero ) M_free(d->where, "old content of
dollar");
00777             d->where = &AM.dollarzero;
00778             d->size = 0;
00779             CleanDollarFactors(d);
00780         }

```

```

00781     }
00782 }
00783 #endif
00784 AR.SortType = AC.SortType;
00785 #ifdef WITHMPI
00786     if ( PF.me == MASTER )
00787 #endif
00788 {
00789     if ( AC.SetupFlag ) WriteSetup();
00790     if ( AC.NamesFlag || AC.CodesFlag ) WriteLists();
00791 }
00792 if ( par == GLOBALMODULE ) MakeGlobal();
00793 if ( RevertScratch() ) return(-1);
00794 if ( AC.ncmod ) SetMods();
00795 /*
00796     Warn if the module has to run in sequential mode due to some problems.
00797 */
00798 #ifdef WITHMPI
00799     if ( PF.me == MASTER )
00800 #endif
00801 {
00802     if ( !AC.ThreadsFlag || AC.mparallelfld & NOPARALLEL_USER ) {
00803         /* The user switched off the parallel execution explicitly. */
00804     }
00805     else if ( AC.mparallelfld & NOPARALLEL_DOLLAR ) {
00806         if ( AC.WarnFlag >= 1 ) { /* Warning */
00807             int i, j, k, n;
00808             UBYTE *s, *s1;
00809             s = strDup1((UBYTE *)"", "NOPARALLEL_DOLLAR s");
00810             n = 0;
00811             j = NumPotModdollars;
00812             for ( i = 0; i < j; i++ ) {
00813                 for ( k = 0; k < NumModOptdollars; k++ )
00814                     if ( ModOptdollars[k].number == PotModdollars[i] ) break;
00815                 if ( k >= NumModOptdollars ) {
00816                     /* global $-variable */
00817                     if ( n > 0 )
00818                         s = AddToString(s, (UBYTE *)", ", 0);
00819                     s = AddToString(s, (UBYTE *)"$", 0);
00820                     s = AddToString(s, DOLLARNAME(Dollars, PotModdollars[i]), 0);
00821                     n++;
00822                 }
00823             }
00824             s1 = strDup1((UBYTE *)"This module is forced to run in sequential mode due to
00825 $-variable", "NOPARALLEL_DOLLAR s1");
00826             if ( n != 1 )
00827                 s1 = AddToString(s1, (UBYTE *)"s", 0);
00828             s1 = AddToString(s1, (UBYTE *)": ", 0);
00829             s1 = AddToString(s1, s, 0);
00830             Warning((char *)s1);
00831             M_free(s, "NOPARALLEL_DOLLAR s");
00832             M_free(s1, "NOPARALLEL_DOLLAR s1");
00833         }
00834         else if ( AC.mparallelfld & NOPARALLEL_RHS ) {
00835             HighWarning("This module is forced to run in sequential mode due to RHS expression
00836 names");
00837         }
00838         else if ( AC.mparallelfld & NOPARALLEL_CONVPOLY ) {
00839             HighWarning("This module is forced to run in sequential mode due to conversion to extra
00840 symbols");
00841         }
00842         else if ( AC.mparallelfld & NOPARALLEL_SPECTATOR ) {
00843             HighWarning("This module is forced to run in sequential mode due to
00844 tospectator/copspectator");
00845         }
00846         else if ( AC.mparallelfld & NOPARALLEL_TBLDOLLAR ) {
00847             HighWarning("This module is forced to run in sequential mode due to $-variable assignments
00848 in tables");
00849         }
00850         else if ( AC.mparallelfld & NOPARALLEL_NPROC ) {
00851             HighWarning("This module is forced to run in sequential mode because there is only one
00852 processor");
00853         }
00854     }
00855 }
00856 /*
00857     Now the actual execution
00858 */
00859 #ifdef WITHMPI
00860     /*
00861         * Turn on AS.printflag to print runtime errors occurring on slaves.
00862         */
00863     AS.printflag = 1;
00864 #endif
00865 if ( AP.preError == 0 && ( Processor() || WriteAll() ) ) RetCode = -1;
00866 #ifdef WITHMPI
00867     AS.printflag = 0;

```

```

00862 #endif
00863 /*
00864     That was it. Next is cleanup.
00865 */
00866     if ( AC.ncmod ) UnSetMods();
00867     AS.MultiThreaded = oldmultithreaded;
00868     TableReset();
00869
00870 /*[28sep2005 mt]:*/
00871 #ifdef WITHMPI
00872     /* Combine and then broadcast modified dollar variables. */
00873     if ( NumPotModdollars > 0 ) {
00874         RetCode = PF_CollectModifiedDollars();
00875         if ( RetCode ) return RetCode;
00876         RetCode = PF_BroadcastModifiedDollars();
00877         if ( RetCode ) return RetCode;
00878     }
00879     /* Broadcast the list of objects converted to symbols in AM.sbufnum. */
00880     if ( AC.topolynomialflag & TOPOLYNOMIALFLAG ) {
00881         RetCode = PF_BroadcastCBuf(AM.sbufnum);
00882         if ( RetCode ) return RetCode;
00883     }
00884     /*
00885     * Broadcast AR.expflags, which may be used on the slaves in the next module
00886     * via ZERO_ or UNCHANGED_. It also broadcasts several flags of each expression.
00887     */
00888     RetCode = PF_BroadcastExpFlags();
00889     if ( RetCode ) return RetCode;
00890     /*
00891     * Clean the hide file on the slaves, which was used for RHS expressions
00892     * broadcast from the master at the beginning of the module.
00893     */
00894     if ( PF.me != MASTER && AR.hidefile->PObuffer ) {
00895         if ( AR.hidefile->handle >= 0 ) {
00896             CloseFile(AR.hidefile->handle);
00897             AR.hidefile->handle = -1;
00898             remove(AR.hidefile->name);
00899         }
00900         AR.hidefile->POfull = AR.hidefile->POfill = AR.hidefile->PObuffer;
00901         PUTZERO(AR.hidefile->POposition);
00902     }
00903 #endif
00904 #ifdef WITHPTHREADS
00905     for ( j = 0; j < NumModOptdollars; j++ ) {
00906         if ( ModOptdollars[j].dstruct ) {
00907             /*
00908                 First clean up dollar values.
00909             */
00910             for ( i = 0; i < AM.totalnumberofthreads; i++ ) {
00911                 if ( ModOptdollars[j].dstruct[i].size > 0 ) {
00912                     CleanDollarFactors(&(ModOptdollars[j].dstruct[i]));
00913                     M_free(ModOptdollars[j].dstruct[i].where,"Local dollar value");
00914                 }
00915             }
00916             /*
00917                 Now clean up the whole array.
00918             */
00919             M_free(ModOptdollars[j].dstruct,"Local DOLLARS");
00920             ModOptdollars[j].dstruct = 0;
00921         }
00922     }
00923 #endif
00924 /*:[28sep2005 mt]:*/
00925
00926 /*
00927     @@@@@@@@@@@@@@@@@@
00928     Now follows the code to invalidate caches for all objects in the
00929     PotModdollars. There are NumPotModdollars of them and PotModdollars
00930     is an array of WORD.
00931 */
00932 /*
00933     Cleanup:
00934 */
00935 #ifdef JV_IS_WRONG
00936 /*
00937     Giving back this memory gives way too much activity with Malloc1
00938     Better to keep it and just put the number of used objects to zero (JV)
00939     If you put the lijst equal to NULL, please also make maxnum = 0
00940 */
00941     if ( ModOptdollars ) M_free(ModOptdollars, "ModOptdollars pointer");
00942     if ( PotModdollars ) M_free(PotModdollars, "PotModdollars pointer");
00943
00944     /* ModOptdollars changed to AC.ModOptDolList.lijst because AIX C compiler complained. MF
00945     30/07/2003. */
00946     AC.ModOptDolList.lijst = NULL;
00947     /* PotModdollars changed to AC.PotModDolList.lijst because AIX C compiler complained. MF
00948     30/07/2003. */

```

```

00947     AC.PotModDolList.lijst = NULL;
00948 #endif
00949     NumPotModdollars = 0;
00950     NumModOptdollars = 0;
00951
00952 skipexec:
00953 /*
00954     Clean up the switch information.
00955     We keep the switch array and heap.
00956 */
00957 if ( AC.SwitchInArray > 0 ) {
00958     for ( i = 0; i < AC.SwitchInArray; i++ ) {
00959         SWITCH *sw = AC.SwitchArray + i;
00960         if ( sw->table ) M_free(sw->table,"Switch table");
00961         sw->table = 0;
00962         sw->defaultcase.ncase = 0;
00963         sw->defaultcase.value = 0;
00964         sw->defaultcase.compbuffer = 0;
00965         sw->endswitch.ncase = 0;
00966         sw->endswitch.value = 0;
00967         sw->endswitch.compbuffer = 0;
00968         sw->typetable = 0;
00969         sw->maxcase = 0;
00970         sw->mincase = 0;
00971         sw->numcases = 0;
00972         sw->tablesize = 0;
00973         sw->caseoffset = 0;
00974         sw->iflevel = 0;
00975         sw->whilelevel = 0;
00976         sw->nestingsum = 0;
00977     }
00978     AC.SwitchInArray = 0;
00979     AC.SwitchLevel = 0;
00980 }
00981 #ifdef PARALLELCODE
00982     AC.numpfirstnum = 0;
00983 #endif
00984     AC.DidClean = 0;
00985     AC.PolyRatFunChanged = 0;
00986     TestDrop();
00987     if ( par == STOREMODULE || par == CLEARMODULE ) {
00988         ClearOptimize();
00989         if ( par == STOREMODULE && PopVariables() ) RetCode = -1;
00990         if ( AR.infile->handle >= 0 ) {
00991             CloseFile(AR.infile->handle);
00992             remove(AR.infile->name);
00993             AR.infile->handle = -1;
00994         }
00995         AR.infile->POfill = AR.infile->PObuffer;
00996         PUTZERO(AR.infile->POposition);
00997         AR.infile->POfull = AR.infile->PObuffer;
00998         if ( AR.outfile->handle >= 0 ) {
00999             CloseFile(AR.outfile->handle);
01000             remove(AR.outfile->name);
01001             AR.outfile->handle = -1;
01002         }
01003         AR.outfile->POfull =
01004         AR.outfile->POfill = AR.outfile->PObuffer;
01005         PUTZERO(AR.outfile->POposition);
01006         if ( AR.hidefile->handle >= 0 ) {
01007             CloseFile(AR.hidefile->handle);
01008             remove(AR.hidefile->name);
01009             AR.hidefile->handle = -1;
01010         }
01011         AR.hidefile->POfull =
01012         AR.hidefile->POfill = AR.hidefile->PObuffer;
01013         PUTZERO(AR.hidefile->POposition);
01014         AC.HideLevel = 0;
01015         if ( par == CLEARMODULE ) {
01016             if ( DeleteStore(0) < 0 ) {
01017                 MesPrint("Cannot restart the storage file");
01018                 RetCode = -1;
01019             }
01020             else RetCode = 0;
01021             CleanUp(1);
01022             ResetVariables(2);
01023             AM.gProcessBucketSize = AM.hProcessBucketSize;
01024             AM.gparallelflag = PARALLELFLAG;
01025             AM.gnumextrasyms = AM.ggnumextrasyms;
01026             PruneExtraSymbols(AM.ggnumextrasyms);
01027             IniVars();
01028         }
01029         ClearSpectators(par);
01030     }
01031     else {
01032         if ( CleanExpr(0) ) RetCode = -1;
01033         if ( AC.DidClean ) CompactifyTree(AC.exprnames,EXPRNAMES);

```

```

01034     ResetVariables(0);
01035     CleanupSort(-1);
01036 }
01037 clearcbuf(AC.cbunum);
01038 if ( AC.MultiBracketBuf != 0 ) {
01039     for ( i = 0; i < MAXMULTIBRACKETLEVELS; i++ ) {
01040         if ( AC.MultiBracketBuf[i] ) {
01041             M_free(AC.MultiBracketBuf[i], "bracket buffer i");
01042             AC.MultiBracketBuf[i] = 0;
01043         }
01044     }
01045     AC.MultiBracketLevels = 0;
01046     M_free(AC.MultiBracketBuf, "multi bracket buffer");
01047     AC.MultiBracketBuf = 0;
01048 }
01049
01050 if ( AC.SortReallocateFlag ) {
01051     /* Reallocate the sort buffers to reduce resident set usage */
01052     /* AT.SS is the same as AT.S0 here */
01053     SORTING* S = AT.S0;
01054     M_free(S->lBuffer, "SortReallocate lBuffer+sBuffer");
01055     S->lBuffer = Malloc1(sizeof(*(S->lBuffer))*(S->LargeSize+S->SmallSize), "SortReallocate
lBuffer+sBuffer");
01056     S->lTop = S->lBuffer+S->LargeSize;
01057     S->sBuffer = S->lTop;
01058     if ( S->LargeSize == 0 ) { S->lBuffer = 0; S->lTop = 0; }
01059     S->sTop = S->sBuffer + S->SmallSize;
01060     S->sTop2 = S->sBuffer + S->SmallSize;
01061     S->sHalf = S->sBuffer + (LONG)((S->SmallSize+S->SmallSize)>>1);
01062
01063 #ifdef WITHPTHREADS
01064     /* We have to re-set the pointers into master lBuffer in the SortBlocks */
01065     UpdateSortBlocks(AM.totalnumberofthreads-1);
01066
01067     /* The SortBots do not have a real sort buffer to reallocate. */
01068     /* AB[0] has been reallocated above already. */
01069     for ( i = 1; i < AM.totalnumberofthreads; i++ ) {
01070         SORTING* S = AB[i]->T.S0;
01071         M_free(S->lBuffer, "SortReallocate lBuffer+sBuffer");
01072         S->lBuffer = Malloc1(sizeof(*(S->lBuffer))*(S->LargeSize+S->SmallSize), "SortReallocate
lBuffer+sBuffer");
01073         S->lTop = S->lBuffer+S->LargeSize;
01074         S->sBuffer = S->lTop;
01075         if ( S->LargeSize == 0 ) { S->lBuffer = 0; S->lTop = 0; }
01076         S->sTop = S->sBuffer + S->SmallSize;
01077         S->sTop2 = S->sBuffer + S->SmallSize;
01078         S->sHalf = S->sBuffer + (LONG)((S->SmallSize+S->SmallSize)>>1);
01079     }
01080 #endif
01081 }
01082 if ( AC.SortReallocateFlag == 2 ) {
01083     /* The Flag was set for a single module by the preprocessor #sortreallocate,
01084        so turn it off again. */
01085     AC.SortReallocateFlag = 0;
01086 }
01087
01088 return(RetCode);
01089 }
01090
01091 /*
01092     #] DoExecute :
01093     #[ PutBracket :
01094
01095     Routine uses the bracket info to split a term into two pieces:
01096     1: the part outside the bracket, and
01097     2: the part inside the bracket.
01098     These parts are separated by a subterm of type HAAKJE.
01099     This subterm looks like: HAAKJE,3,level
01100     The level is used for nestings of brackets. The print routines
01101     cannot handle this yet (31-Mar-1988).
01102
01103     The Bracket selector is in AT.BrackBuf in the form of a regular term,
01104     but without coefficient.
01105     When AR.BracketOn < 0 we have a so-called antibracket. The main effect
01106     is an exchange of the inner and outer part and where the coefficient goes.
01107
01108     Routine recoded to facilitate b p1,p2; etc for dotproducts and tensors
01109     15-oct-1991
01110 */
01111
01112 int PutBracket(PHEAD WORD *termin)
01113 {
01114     GETBIDENTITY
01115     WORD *t, *t1, *b, i, j, *lastfun;
01116     WORD *t2, *s1, *s2;
01117     WORD *bStop, *bb, *bf, *tStop;
01118     WORD *term1, *term2, *m1, *m2, *tStopa;

```

```

01119     WORD *bbb = 0, *bind, *binst = 0, bwild = 0, *bss = 0, *bns = 0, bset = 0;
01120     term1 = AT.WorkPointer+1;
01121     term2 = (WORD *)(((UBYTE *) (term1)) + AM.MaxTer);
01122     if ( ( (WORD *)(((UBYTE *) (term2)) + AM.MaxTer) ) > AT.WorkTop ) return(MesWork());
01123     if ( AR.BracketOn < 0 ) {
01124         t2 = term1; t1 = term2;      /* AntiBracket */
01125     }
01126     else {
01127         t1 = term1; t2 = term2;      /* Regular bracket */
01128     }
01129     b = AT.BrackBuf; bStop = b+b; b++;
01130     while ( b < bStop ) {
01131         if ( *b == INDEX ) { bwild = 1; bbb = b+2; binst = b + b[1]; }
01132         if ( *b == SETSET ) { bset = 1; bss = b+2; bns = b + b[1]; }
01133         b += b[1];
01134     }
01135
01136     t = termin; tStopa = t + *t; i = *(t + *t -1); i = ABS(i);
01137     if ( AR.PolyFun && AT.PolyAct ) tStop = termin + AT.PolyAct;
01138 #ifdef WITHFLOAT
01139     else if ( AT.FloatPos ) tStop = termin + AT.FloatPos;
01140 #endif
01141     else tStop = tStopa - i;
01142     t++;
01143     if ( AR.BracketOn < 0 ) {
01144         lastfun = 0;
01145         while ( t < tStop && *t >= FUNCTION
01146             && functions[*t-FUNCTION].commute ) {
01147             b = AT.BrackBuf+1;
01148             while ( b < bStop ) {
01149                 if ( *b == *t ) {
01150                     lastfun = t;
01151                     while ( t < tStop && *t >= FUNCTION
01152                         && functions[*t-FUNCTION].commute ) t += t[1];
01153                     goto NextNcom1;
01154                 }
01155                 b += b[1];
01156             }
01157             if ( bset ) {
01158                 b = bss;
01159                 while ( b < bns ) {
01160                     if ( b[1] == CFUNCTION ) { /* Set of functions */
01161                         SETS set = Sets+b[0]; WORD i;
01162                         for ( i = set->first; i < set->last; i++ ) {
01163                             if ( SetElements[i] == *t ) {
01164                                 lastfun = t;
01165                                 while ( t < tStop && *t >= FUNCTION
01166                                     && functions[*t-FUNCTION].commute ) t += t[1];
01167                                 goto NextNcom1;
01168                             }
01169                         }
01170                     }
01171                     b += 2;
01172                 }
01173             }
01174             if ( bwild && *t >= FUNCTION && functions[*t-FUNCTION].spec ) {
01175                 s1 = t + t[1];
01176                 s2 = t + FUNHEAD;
01177                 while ( s2 < s1 ) {
01178                     bind = bbb;
01179                     while ( bind < binst ) {
01180                         if ( *bind == *s2 ) {
01181                             lastfun = t;
01182                             while ( t < tStop && *t >= FUNCTION
01183                                 && functions[*t-FUNCTION].commute ) t += t[1];
01184                             goto NextNcom1;
01185                         }
01186                         bind++;
01187                     }
01188                     s2++;
01189                 }
01190             }
01191             t += t[1];
01192         }
01193     NextNcom1:
01194         s1 = termin + 1;
01195         if ( lastfun ) {
01196             while ( s1 < lastfun ) *t2++ = *s1++;
01197             while ( s1 < t ) *t1++ = *s1++;
01198         }
01199         else {
01200             while ( s1 < t ) *t2++ = *s1++;
01201         }
01202     }
01203 }
01204 else {
01205     lastfun = t;

```



```

01206     while ( t < tStop && *t >= FUNCTION
01207         && functions[*t-FUNCTION].commute ) {
01208         b = AT.BrackBuf+1;
01209         while ( b < bStop ) {
01210             if ( *b == *t ) { lastfun = t + t[1]; goto NextNcom; }
01211             b += b[1];
01212         }
01213         if ( bset ) {
01214             b = bss;
01215             while ( b < bns ) {
01216                 if ( b[1] == CFUNCTION ) { /* Set of functions */
01217                     SETS set = Sets+b[0]; WORD i;
01218                     for ( i = set->first; i < set->last; i++ ) {
01219                         if ( SetElements[i] == *t ) {
01220                             lastfun = t + t[1];
01221                             goto NextNcom;
01222                         }
01223                     }
01224                 }
01225                 b += 2;
01226             }
01227         }
01228         if ( bwild && *t >= FUNCTION && functions[*t-FUNCTION].spec ) {
01229             s1 = t + t[1];
01230             s2 = t + FUNHEAD;
01231             while ( s2 < s1 ) {
01232                 bind = bbb;
01233                 while ( bind < binst ) {
01234                     if ( *bind == *s2 ) { lastfun = t + t[1]; goto NextNcom; }
01235                     bind++;
01236                 }
01237                 s2++;
01238             }
01239         }
01240     NextNcom:
01241         t += t[1];
01242     }
01243     s1 = termin + 1;
01244     while ( s1 < lastfun ) *t1++ = *s1++;
01245     while ( s1 < t ) *t2++ = *s1++;
01246 }
01247 /*
01248     Now we have only commuting functions left. Move the b pointer to them.
01249 */
01250 b = AT.BrackBuf + 1;
01251 while ( b < bStop && *b >= FUNCTION
01252     && ( *b < FUNCTION || functions[*b-FUNCTION].commute ) ) {
01253     b += b[1];
01254 }
01255 bf = b;
01256
01257 while ( t < tStop && ( bf < bStop || bwild || bset ) ) {
01258     b = bf;
01259     while ( b < bStop && *b != *t ) { b += b[1]; }
01260     i = t[1];
01261     if ( *t >= FUNCTION ) { /* We are in function territory */
01262         if ( b < bStop && *b == *t ) goto FunBrac;
01263         if ( bset ) {
01264             b = bss;
01265             while ( b < bns ) {
01266                 if ( b[1] == CFUNCTION ) { /* Set of functions */
01267                     SETS set = Sets+b[0]; WORD i;
01268                     for ( i = set->first; i < set->last; i++ ) {
01269                         if ( SetElements[i] == *t ) goto FunBrac;
01270                     }
01271                 }
01272                 b += 2;
01273             }
01274         }
01275         if ( bwild && *t >= FUNCTION && functions[*t-FUNCTION].spec ) {
01276             s1 = t + t[1];
01277             s2 = t + FUNHEAD;
01278             while ( s2 < s1 ) {
01279                 bind = bbb;
01280                 while ( bind < binst ) {
01281                     if ( *bind == *s2 ) goto FunBrac;
01282                     bind++;
01283                 }
01284                 s2++;
01285             }
01286         }
01287         NCOPY(t2,t,i);
01288         continue;
01289     FunBrac:
01290         NCOPY(t1,t,i);
01291         continue;
01292 }
01293 /*

```

```

01293     We have left: DELTA, INDEX, VECTOR, DOTPRODUCT, SYMBOL
01294 */
01295     if ( *t == DELTA ) {
01296         if ( b < bStop && *b == DELTA ) {
01297             b += b[1];
01298             NCOPY(t1,t,i);
01299         }
01300         else { NCOPY(t2,t,i); }
01301     }
01302     else if ( *t == INDEX ) {
01303         if ( bwild ) {
01304             m1 = t1; m2 = t2;
01305             *t1++ = *t; t1++; *t2++ = *t; t2++;
01306             bind = bbb;
01307             j = t[1] - 2;
01308             t += 2;
01309             while ( --j >= 0 ) {
01310                 while ( *bind < *t && bind < binst ) bind++;
01311                 if ( *bind == *t && bind < binst ) {
01312                     *t1++ = *t++;
01313                 }
01314                 else if ( bset ) {
01315                     WORD *b3 = bss;
01316                     while ( b3 < bns ) {
01317                         if ( b3[1] == CVECTOR ) {
01318                             SETS set = Sets+b3[0]; WORD i;
01319                             for ( i = set->first; i < set->last; i++ ) {
01320                                 if ( SetElements[i] == *t ) {
01321                                     *t1++ = *t++;
01322                                     goto nextind;
01323                                 }
01324                             }
01325                         }
01326                         b3 += 2;
01327                     }
01328                     *t2++ = *t++;
01329                 }
01330                 else *t2++ = *t++;
01331 nextind:;
01332             }
01333             m1[1] = WORDDIF(t1,m1);
01334             if ( m1[1] == 2 ) t1 = m1;
01335             m2[1] = WORDDIF(t2,m2);
01336             if ( m2[1] == 2 ) t2 = m2;
01337         }
01338         else if ( bset ) {
01339             m1 = t1; m2 = t2;
01340             *t1++ = *t; t1++; *t2++ = *t; t2++;
01341             j = t[1] - 2;
01342             t += 2;
01343             while ( --j >= 0 ) {
01344                 WORD *b3 = bss;
01345                 while ( b3 < bns ) {
01346                     if ( b3[1] == CVECTOR ) {
01347                         SETS set = Sets+b3[0]; WORD i;
01348                         for ( i = set->first; i < set->last; i++ ) {
01349                             if ( SetElements[i] == *t ) {
01350                                 *t1++ = *t++;
01351                                 goto nextind2;
01352                             }
01353                         }
01354                     }
01355                     b3 += 2;
01356                 }
01357                 *t2++ = *t++;
01358 nextind2:;
01359             }
01360             m1[1] = WORDDIF(t1,m1);
01361             if ( m1[1] == 2 ) t1 = m1;
01362             m2[1] = WORDDIF(t2,m2);
01363             if ( m2[1] == 2 ) t2 = m2;
01364         }
01365         else {
01366             NCOPY(t2,t,i);
01367         }
01368     }
01369     else if ( *t == VECTOR ) {
01370         if ( ( b < bStop && *b == VECTOR ) || bwild ) {
01371             if ( b < bStop && *b == VECTOR ) {
01372                 bb = b + b[1]; b += 2;
01373             }
01374             else bb = b;
01375             j = t[1] - 2;
01376             m1 = t1; m2 = t2; *t1++ = *t; *t2++ = *t; t1++; t2++; t += 2;
01377             while ( j > 0 ) {
01378                 j -= 2;
01379                 while ( b < bb && ( *b < *t ||

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```

01380         ( *b == *t && b[1] < t[1] ) ) ) b += 2;
01381     if ( b < bb && ( *t == *b && t[1] == b[1] ) ) {
01382         *t1++ = *t++; *t1++ = *t++; goto nextvec;
01383     }
01384     else if ( bwild ) {
01385         bind = bbb;
01386         while ( bind < binst ) {
01387             if ( *t == *bind || t[1] == *bind ) {
01388                 *t1++ = *t++; *t1++ = *t++;
01389                 goto nextvec;
01390             }
01391             bind++;
01392         }
01393     }
01394     if ( bset ) {
01395         WORD *b3 = bss;
01396         while ( b3 < bns ) {
01397             if ( b3[1] == CVECTOR ) {
01398                 SETS set = Sets+b3[0]; WORD i;
01399                 for ( i = set->first; i < set->last; i++ ) {
01400                     if ( SetElements[i] == *t ) {
01401                         *t1++ = *t++; *t1++ = *t++;
01402                         goto nextvec;
01403                     }
01404                 }
01405             }
01406             b3 += 2;
01407         }
01408     }
01409     *t2++ = *t++; *t2++ = *t++;
01410 nextvec:;
01411     }
01412     m1[1] = WORDDIF(t1,m1);
01413     if ( m1[1] == 2 ) t1 = m1;
01414     m2[1] = WORDDIF(t2,m2);
01415     if ( m2[1] == 2 ) t2 = m2;
01416 }
01417 else if ( bset ) {
01418     m1 = t1; *t1++ = *t; t1++;
01419     m2 = t2; *t2++ = *t; t2++;
01420     s2 = t + i; t += 2;
01421     while ( t < s2 ) {
01422         WORD *b3 = bss;
01423         while ( b3 < bns ) {
01424             if ( b3[1] == CVECTOR ) {
01425                 SETS set = Sets+b3[0]; WORD i;
01426                 for ( i = set->first; i < set->last; i++ ) {
01427                     if ( SetElements[i] == *t ) {
01428                         *t1++ = *t++; *t1++ = *t++;
01429                         goto nextvec2;
01430                     }
01431                 }
01432             }
01433             b3 += 2;
01434         }
01435         *t2++ = *t++; *t2++ = *t++;
01436 nextvec2:;
01437     }
01438     m1[1] = WORDDIF(t1,m1);
01439     if ( m1[1] == 2 ) t1 = m1;
01440     m2[1] = WORDDIF(t2,m2);
01441     if ( m2[1] == 2 ) t2 = m2;
01442 }
01443 else {
01444     NCOPY(t2,t,i);
01445 }
01446 }
01447 else if ( *t == DOTPRODUCT ) {
01448     if ( ( b < bStop && *b == *t ) || bwild ) {
01449         m1 = t1; *t1++ = *t; t1++;
01450         m2 = t2; *t2++ = *t; t2++;
01451         if ( b >= bStop || *b != *t ) { bb = b; s1 = b; }
01452         else {
01453             s1 = b + b[1]; bb = b + 2;
01454         }
01455         s2 = t + i; t += 2;
01456         while ( t < s2 && ( bb < s1 || bwild || bset ) ) {
01457             while ( bb < s1 && ( *bb < *t ||
01458                 ( *bb == *t && bb[1] < t[1] ) ) ) bb += 3;
01459             if ( bb < s1 && *bb == *t && bb[1] == t[1] ) {
01460                 *t1++ = *t++; *t1++ = *t++; *t1++ = *t++; bb += 3;
01461                 goto nextdot;
01462             }
01463             else if ( bwild ) {
01464                 bind = bbb;
01465                 while ( bind < binst ) {
01466                     if ( *bind == *t || *bind == t[1] ) {

```

```

01467             *t1++ = *t++; *t1++ = *t++; *t1++ = *t++;
01468             goto nextdot;
01469         }
01470         bind++;
01471     }
01472 }
01473 if ( bset ) {
01474     WORD *b3 = bss;
01475     while ( b3 < bns ) {
01476         if ( b3[1] == CVECTOR ) {
01477             SETS set = Sets+b3[0]; WORD i;
01478             for ( i = set->first; i < set->last; i++ ) {
01479                 if ( SetElements[i] == *t || SetElements[i] == t[1] ) {
01480                     *t1++ = *t++; *t1++ = *t++; *t1++ = *t++;
01481                     goto nextdot;
01482                 }
01483             }
01484         }
01485         b3 += 2;
01486     }
01487 }
01488 *t2++ = *t++; *t2++ = *t++; *t2++ = *t++;
01489 nextdot:;
01490 }
01491 while ( t < s2 ) *t2++ = *t++;
01492 m1[1] = WORDDIF(t1,m1);
01493 if ( m1[1] == 2 ) t1 = m1;
01494 m2[1] = WORDDIF(t2,m2);
01495 if ( m2[1] == 2 ) t2 = m2;
01496 }
01497 else if ( bset ) {
01498     m1 = t1; *t1++ = *t; t1++;
01499     m2 = t2; *t2++ = *t; t2++;
01500     s2 = t + i; t += 2;
01501     while ( t < s2 ) {
01502         WORD *b3 = bss;
01503         while ( b3 < bns ) {
01504             if ( b3[1] == CVECTOR ) {
01505                 SETS set = Sets+b3[0]; WORD i;
01506                 for ( i = set->first; i < set->last; i++ ) {
01507                     if ( SetElements[i] == *t || SetElements[i] == t[1] ) {
01508                         *t1++ = *t++; *t1++ = *t++; *t1++ = *t++;
01509                         goto nextdot2;
01510                     }
01511                 }
01512             }
01513             b3 += 2;
01514         }
01515         *t2++ = *t++; *t2++ = *t++; *t2++ = *t++;
01516 nextdot2:;
01517     }
01518     m1[1] = WORDDIF(t1,m1);
01519     if ( m1[1] == 2 ) t1 = m1;
01520     m2[1] = WORDDIF(t2,m2);
01521     if ( m2[1] == 2 ) t2 = m2;
01522 }
01523 else { NCOPY(t2,t,i); }
01524 }
01525 else if ( *t == SYMBOL ) {
01526     if ( b < bStop && *b == *t ) {
01527         m1 = t1; *t1++ = *t; t1++;
01528         m2 = t2; *t2++ = *t; t2++;
01529         s1 = b + b[1]; bb = b+2;
01530         s2 = t + i; t += 2;
01531         while ( bb < s1 && t < s2 ) {
01532             while ( bb < s1 && *bb < *t ) bb += 2;
01533             if ( bb >= s1 ) {
01534                 if ( bset ) goto TrySymbolSet;
01535                 break;
01536             }
01537             if ( *bb == *t ) { *t1++ = *t++; *t1++ = *t++; }
01538             else if ( bset ) {
01539                 WORD *bbb;
01540 TrySymbolSet:
01541                 bbb = bss;
01542                 while ( bbb < bns ) {
01543                     if ( bbb[1] == CSYMBOL ) { /* Set of symbols */
01544                         SETS set = Sets+bbb[0]; WORD i;
01545                         for ( i = set->first; i < set->last; i++ ) {
01546                             if ( SetElements[i] == *t ) {
01547                                 *t1++ = *t++; *t1++ = *t++;
01548                                 goto NextSymbol;
01549                             }
01550                         }
01551                     }
01552                     bbb += 2;
01553                 }

```

```

01554             *t2++ = *t++; *t2++ = *t++;
01555         }
01556         else { *t2++ = *t++; *t2++ = *t++; }
01557 NextSymbol;;
01558     }
01559     while ( t < s2 ) *t2++ = *t++;
01560     m1[1] = WORDDIF(t1,m1);
01561     if ( m1[1] == 2 ) t1 = m1;
01562     m2[1] = WORDDIF(t2,m2);
01563     if ( m2[1] == 2 ) t2 = m2;
01564 }
01565 else if ( bset ) {
01566     WORD *bbb;
01567     m1 = t1; *t1++ = *t; t1++;
01568     m2 = t2; *t2++ = *t; t2++;
01569     s2 = t + i; t += 2;
01570     while ( t < s2 ) {
01571         bbb = bss;
01572         while ( bbb < bns ) {
01573             if ( bbb[1] == CSYMBOL ) { /* Set of symbols */
01574                 SETS set = Sets+bbb[0]; WORD i;
01575                 for ( i = set->first; i < set->last; i++ ) {
01576                     if ( SetElements[i] == *t ) {
01577                         *t1++ = *t++; *t1++ = *t++;
01578                         goto NextSymbol2;
01579                     }
01580                 }
01581             }
01582             bbb += 2;
01583         }
01584         *t2++ = *t++; *t2++ = *t++;
01585 NextSymbol2;;
01586     }
01587     m1[1] = WORDDIF(t1,m1);
01588     if ( m1[1] == 2 ) t1 = m1;
01589     m2[1] = WORDDIF(t2,m2);
01590     if ( m2[1] == 2 ) t2 = m2;
01591 }
01592 else { NCOPY(t2,t,i); }
01593 }
01594 else {
01595     NCOPY(t2,t,i);
01596 }
01597 }
01598 if ( ( i = WORDDIF(tStop,t) ) > 0 ) NCOPY(t2,t,i);
01599 if ( AR.BracketOn < 0 ) {
01600     s1 = t1; t1 = t2; t2 = s1;
01601 }
01602 do { *t2++ = *t++; } while ( t < (WORD *)tStopa );
01603 t = AT.WorkPointer;
01604 i = WORDDIF(t1,term1);
01605 *t++ = 4 + i + WORDDIF(t2,term2);
01606 t += i;
01607 *t++ = HAAKJE;
01608 *t++ = 3;
01609 *t++ = 0; /* This feature won't be used for a while */
01610 i = WORDDIF(t2,term2);
01611 t1 = term2;
01612 if ( i > 0 ) NCOPY(t,t1,i);
01613
01614 AT.WorkPointer = t;
01615
01616 return(0);
01617 }
01618
01619 /*
01620     #] PutBracket :
01621     #[ SpecialCleanup :
01622 */
01623
01624 void SpecialCleanup(PHEAD0)
01625 {
01626     GETBIDENTITY
01627     if ( AT.previousEfactor ) M_free(AT.previousEfactor,"Efactor cache");
01628     AT.previousEfactor = 0;
01629 }
01630
01631 /*
01632     #] SpecialCleanup :
01633     #[ SetMods :
01634 */
01635
01636 #ifndef WITHPTHREADS
01637 void SetMods(void)
01638 {
01639     int i, n;

```

```

01641     if ( AN.cmod != 0 ) M_free(AN.cmod,"AN.cmod");
01642     n = ABS(AN.ncmod);
01643     AN.cmod = (UWORD *)Malloc1(sizeof(WORD)*n,"AN.cmod");
01644     for ( i = 0; i < n; i++ ) AN.cmod[i] = AC.cmod[i];
01645 }
01646
01647 #endif
01648
01649 /*
01650     #] SetMods :
01651     #[ UnSetMods :
01652 */
01653
01654 #ifndef WITHPTHREADS
01655
01656 void UnSetMods(void)
01657 {
01658     if ( AN.cmod != 0 ) M_free(AN.cmod,"AN.cmod");
01659     AN.cmod = 0;
01660 }
01661
01662 #endif
01663
01664 /*
01665     #] UnSetMods :
01666     #] DoExecute :
01667     #[ Expressions :
01668     #[ ExchangeExpressions :
01669 */
01670
01671 void ExchangeExpressions(int num1, int num2)
01672 {
01673     GETIDENTITY
01674     WORD node1, node2, namesize, TMproto[SUBEXPSIZE];
01675     INDEXENTRY *ind;
01676     EXPRESSIONS e1, e2;
01677     LONG a;
01678     SBYTE *s1, *s2;
01679     int i;
01680     e1 = Expressions + num1;
01681     e2 = Expressions + num2;
01682     node1 = e1->node;
01683     node2 = e2->node;
01684     AC.exprnames->namenode[node1].number = num2;
01685     AC.exprnames->namenode[node2].number = num1;
01686     a = e1->name; e1->name = e2->name; e2->name = a;
01687     namesize = e1->namesize; e1->namesize = e2->namesize; e2->namesize = namesize;
01688     e1->node = node2;
01689     e2->node = node1;
01690     if ( e1->status == STOREDEXPRESSION ) {
01691     /*
01692         Find the name in the index and replace by the new name
01693     */
01694         TMproto[0] = EXPRESSION;
01695         TMproto[1] = SUBEXPSIZE;
01696         TMproto[2] = num1;
01697         TMproto[3] = 1;
01698         { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
01699         AT.TMaddr = TMproto;
01700         ind = FindInIndex(num1,&AR.StoreData,0,0);
01701         s1 = (SBYTE *) (AC.exprnames->namebuffer+e1->name);
01702         i = e1->namesize;
01703         s2 = ind->name;
01704         NCOPY(s2,s1,i);
01705         *s2 = 0;
01706         SeekFile(AR.StoreData.Handle,&(e1->onfile),SEEK_SET);
01707         if ( WriteFile(AR.StoreData.Handle,(UBYTE *)ind,
01708             (LONG)(sizeof(INDEXENTRY)) != sizeof(INDEXENTRY) ) {
01709             MesPrint("File error while exchanging expressions");
01710             Terminate(-1);
01711         }
01712         FlushFile(AR.StoreData.Handle);
01713     }
01714     if ( e2->status == STOREDEXPRESSION ) {
01715     /*
01716         Find the name in the index and replace by the new name
01717     */
01718         TMproto[0] = EXPRESSION;
01719         TMproto[1] = SUBEXPSIZE;
01720         TMproto[2] = num2;
01721         TMproto[3] = 1;
01722         { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
01723         AT.TMaddr = TMproto;
01724         ind = FindInIndex(num1,&AR.StoreData,0,0);
01725         s1 = (SBYTE *) (AC.exprnames->namebuffer+e2->name);
01726         i = e2->namesize;
01727         s2 = ind->name;

```

```

01728     NCOPY(s2,s1,i);
01729     *s2 = 0;
01730     SeekFile(AR.StoreData.Handle,&(e2->onfile),SEEK_SET);
01731     if ( WriteFile(AR.StoreData.Handle,(UBYTE *)ind,
01732 (LONG)(sizeof(INDEXENTRY)) != sizeof(INDEXENTRY) ) {
01733         MesPrint("File error while exchanging expressions");
01734         Terminate(-1);
01735     }
01736     FlushFile(AR.StoreData.Handle);
01737 }
01738 }
01739
01740 /*
01741     #] ExchangeExpressions :
01742     #[ GetFirstBracket :
01743 */
01744
01745 int GetFirstBracket(WORD *term, int num)
01746 {
01747     /*
01748         Gets the first bracket of the expression 'num'
01749         Puts it in term. If no brackets the answer is one.
01750         Routine should be thread-safe
01751     */
01752     GETIDENTITY
01753     POSITION position, oldposition;
01754     RENUMBER renumber;
01755     FILEHANDLE *fi;
01756     WORD type, *oldcomppointer, oldonefile, numword;
01757     WORD *t, *tstop;
01758
01759     oldcomppointer = AR.CompressPointer;
01760     type = Expressions[num].status;
01761     if ( type == STOREDEXPRESSION ) {
01762         WORD TMproto[SUBEXPSIZE];
01763         TMproto[0] = EXPRESSION;
01764         TMproto[1] = SUBEXPSIZE;
01765         TMproto[2] = num;
01766         TMproto[3] = 1;
01767         { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
01768         AT.TMaddr = TMproto;
01769         PUTZERO(position);
01770         if ( ( renumber = GetTable(num,&position,0) ) == 0 ) {
01771             MesCall("GetFirstBracket");
01772             SETERROR(-1)
01773         }
01774         if ( GetFromStore(term,&position,renumber,&numword,num) < 0 ) {
01775             MesCall("GetFirstBracket");
01776             SETERROR(-1)
01777         }
01778     /*
01779     #ifdef WITHPTHREADS
01780     */
01781         if ( renumber->symb.lo != AN.dummyrenumlist )
01782             M_free(renumber->symb.lo,"VarSpace");
01783         M_free(renumber,"Renummer");
01784     /*
01785     #endif
01786     */
01787     }
01788     else { /* Active expression */
01789         oldonefile = AR.GetOneFile;
01790         if ( type == HIDDENLEXPRESSION || type == HIDDENLEXPRESSION ) {
01791             AR.GetOneFile = 2; fi = AR.hidefile;
01792         }
01793         else {
01794             AR.GetOneFile = 0; fi = AR.infile;
01795         }
01796         if ( fi->handle >= 0 ) {
01797             PUTZERO(oldposition);
01798         /*
01799             SeekFile(fi->handle,&oldposition,SEEK_CUR);
01800         */
01801         }
01802         else {
01803             SETBASEPOSITION(oldposition,fi->POfill-fi->PObuffer);
01804         }
01805         position = AS.OldOnFile[num];
01806         if ( GetOneTerm(BHEAD term,fi,&position,1) < 0
01807         || ( GetOneTerm(BHEAD term,fi,&position,1) < 0 ) ) {
01808             MLOCK(ErrorMessageLock);
01809             MesCall("GetFirstBracket");
01810             MUNLOCK(ErrorMessageLock);
01811             SETERROR(-1)
01812         }
01813         if ( fi->handle >= 0 ) {
01814         /*

```

```

01815         SeekFile(fi->handle,&oldposition,SEEK_SET);
01816         if ( ISNEGPOS(oldposition) ) {
01817             MLOCK(ErrorMessageLock);
01818             MesPrint("File error");
01819             MUNLOCK(ErrorMessageLock);
01820             SETERROR(-1)
01821         }
01822     /*
01823     }
01824     else {
01825         fi->POfill = fi->PObuffer+BASEPOSITION(oldposition);
01826     }
01827     AR.GetOneFile = oldonefile;
01828 }
01829 AR.CompressPointer = oldcomppointer;
01830 if ( *term ) {
01831     tstop = term + *term; tstop -= ABS(tstop[-1]);
01832     t = term + 1;
01833     while ( t < tstop ) {
01834         if ( *t == HAAKJE ) break;
01835         t += t[1];
01836     }
01837     if ( t >= tstop ) {
01838         term[0] = 4; term[1] = 1; term[2] = 1; term[3] = 3;
01839     }
01840     else {
01841         *t++ = 1; *t++ = 1; *t++ = 3; *term = t - term;
01842     }
01843 }
01844 else {
01845     term[0] = 4; term[1] = 1; term[2] = 1; term[3] = 3;
01846 }
01847 return(*term);
01848 }
01849
01850 /*
01851     #] GetFirstBracket :
01852     #[ GetFirstTerm :
01853 */
01854
01864 int GetFirstTerm(WORD *term, int num, int pre)
01865 {
01866     GETIDENTITY
01867     POSITION position, oldposition;
01868     RENUMBER renumber;
01869     FILEHANDLE *fi;
01870     WORD type, *oldcomppointer, oldonefile, numword;
01871
01872     oldcomppointer = AR.CompressPointer;
01873     type = Expressions[num].status;
01874     if ( type == STOREDEXPRESSION ) {
01875         WORD TMproto[SUBEXPSIZE];
01876         TMproto[0] = EXPRESSION;
01877         TMproto[1] = SUBEXPSIZE;
01878         TMproto[2] = num;
01879         TMproto[3] = 1;
01880         { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
01881         AT.TMaddr = TMproto;
01882         PUTZERO(position);
01883         if ( ( renumber = GetTable(num,&position,0) ) == 0 ) {
01884             MesCall("GetFirstTerm");
01885             SETERROR(-1)
01886         }
01887         if ( GetFromStore(term,&position,renumber,&numword,num) < 0 ) {
01888             MesCall("GetFirstTerm");
01889             SETERROR(-1)
01890         }
01891     /*
01892     #ifdef WITHPTHREADS
01893     */
01894         if ( renumber->symb.lo != AN.dummyrenumlist )
01895             M_free(renumber->symb.lo,"VarSpace");
01896         M_free(renumber,"Renummer");
01897     /*
01898     #endif
01899     */
01900     }
01901     else { /* Active expression */
01902         oldonefile = AR.GetOneFile;
01903         if ( type == HIDDENLEXPRESSION || type == HIDDENEXPRESSION ) {
01904             AR.GetOneFile = 2; fi = AR.hidefile;
01905         }
01906         else {
01907             AR.GetOneFile = 0;
01908             if ( Expressions[num].replace == NEWLYDEFINEDEXPRESSION ) {
01909                 /* During execution, if the expression has already been processed it
01910                  will be in the outfile. If it has not, the usage is illegal according

```



```

01911         to the manual, though no error is given. */
01912         if ( pre == 0 ) { fi = AR.outfile; }
01913         /* During preprocessing, the expression certainly has not been processed
01914         yet. Print an error and terminate. */
01915         else {
01916             MesPrint("&isnumerical: expression is not yet defined!");
01917             SETERROR(-1);
01918         }
01919     }
01920     else {
01921         /* During execution, we should use the definition as stored at the end
01922         of the previous module. This is in the infile. */
01923         if ( pre == 0 ) { fi = AR.infile; }
01924         /* During preprocessing, this function is called before the RevertScratch
01925         at the beginning of this module's execution. Thus the expressions are
01926         in the outfile of the previous module. */
01927         else { fi = AR.outfile; }
01928     }
01929 }
01930 if ( fi->handle >= 0 ) {
01931     PUTZERO(oldposition);
01932     /*
01933         SeekFile(fi->handle,&oldposition,SEEK_CUR);
01934     */
01935 }
01936 else {
01937     SETBASEPOSITION(oldposition,fi->POfill-fi->PObuffer);
01938 }
01939 position = AS.OldOnFile[num];
01940 if ( GetOneTerm(BHEAD term,fi,&position,1) < 0
01941 || ( GetOneTerm(BHEAD term,fi,&position,1) < 0 ) ) {
01942     MLOCK(ErrorMessageLock);
01943     MesCall("GetFirstTerm");
01944     MUNLOCK(ErrorMessageLock);
01945     SETERROR(-1)
01946 }
01947 if ( fi->handle >= 0 ) {
01948     /*
01949         SeekFile(fi->handle,&oldposition,SEEK_SET);
01950         if ( ISNEGPOS(oldposition) ) {
01951             MLOCK(ErrorMessageLock);
01952             MesPrint("File error");
01953             MUNLOCK(ErrorMessageLock);
01954             SETERROR(-1)
01955         }
01956     */
01957 }
01958 else {
01959     fi->POfill = fi->PObuffer+BASEPOSITION(oldposition);
01960 }
01961 AR.GetOneFile = oldonefile;
01962 }
01963 AR.CompressPointer = oldcomppointer;
01964 return(*term);
01965 }
01966 /*
01967     #] GetFirstTerm :
01968     #[ GetContent :
01969 */
01970
01971 int GetContent(WORD *content, int num)
01972 {
01973     /*
01974         Gets the content of the expression 'num'
01975         Puts it in content.
01976         Routine should be thread-safe
01977         The content is defined as the term that will make the expression 'num'
01978         with integer coefficients, no GCD and all common factors taken out,
01979         all negative powers removed when we divide the expression by this
01980         content.
01981     */
01982     /*
01983         GETIDENTITY
01984         POSITION position, oldposition;
01985         RENUMBER renumber;
01986         FILEHANDLE *fi;
01987         WORD type, *oldcomppointer, oldonefile, numword, *term, i;
01988         WORD *cbuffer = TermMalloc("GetContent");
01989         WORD *oldworkpointer = AT.WorkPointer;
01990
01991         oldcomppointer = AR.CompressPointer;
01992         type = Expressions[num].status;
01993         if ( type == STOREDEXPRESSION ) {
01994             WORD TMproto[SUBEXPSIZE];
01995             TMproto[0] = EXPRESSION;
01996             TMproto[1] = SUBEXPSIZE;
01997             TMproto[2] = num;

```

```

01998     TMproto[3] = 1;
01999     { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
02000     AT.TMaddr = TMproto;
02001     PUTZERO(position);
02002     if ( ( renumber = GetTable(num,&position,0) ) == 0 ) goto CalledFrom;
02003     if ( GetFromStore(cbuffer,&position,renumber,&numword,num) < 0 ) goto CalledFrom;
02004     for(;;) {
02005         term = oldworkpointer;
02006         AR.CompressPointer = oldcomppointer;
02007         if ( GetFromStore(term,&position,renumber,&numword,num) < 0 ) goto CalledFrom;
02008         if ( *term == 0 ) break;
02009     /*
02010         'merge' the two terms
02011     */
02012         if ( ContentMerge(BHEAD cbuffer,term) < 0 ) goto CalledFrom;
02013     }
02014     /*
02015     #ifdef WITHPTHREADS
02016     */
02017     if ( renumber->symb.lo != AN.dummyrenumlist )
02018         M_free(renumber->symb.lo,"VarSpace");
02019     M_free(renumber,"Renumber");
02020     /*
02021     #endif
02022     */
02023     }
02024     else { /* Active expression */
02025         oldonefile = AR.GetOneFile;
02026         if ( type == HIDDENLEXPRESSION || type == HIDDENGEXPRESSION ) {
02027             AR.GetOneFile = 2; fi = AR.hidefile;
02028         }
02029         else {
02030             AR.GetOneFile = 0;
02031             if ( Expressions[num].replace == NEWLYDEFINEDEXPRESSSION )
02032                 fi = AR.outfile;
02033             else fi = AR.infile;
02034         }
02035         if ( fi->handle >= 0 ) {
02036             PUTZERO(oldposition);
02037         /*
02038             SeekFile(fi->handle,&oldposition,SEEK_CUR);
02039         */
02040         }
02041         else {
02042             SETBASEPOSITION(oldposition,fi->POfill-fi->PObuffer);
02043         }
02044         position = AS.OldOnFile[num];
02045         if ( GetOneTerm(BHEAD cbuffer,fi,&position,1) < 0 ) goto CalledFrom;
02046         AR.CompressPointer = oldcomppointer;
02047         if ( GetOneTerm(BHEAD cbuffer,fi,&position,1) < 0 ) goto CalledFrom;
02048     /*
02049         Now go through the terms. For each term we have to test whether
02050         what is in cbuffer is also in that term. If not, we have to remove
02051         it from cbuffer. Additionally we have to accumulate the GCD of the
02052         numerators and the LCM of the denominators. This is all done in the
02053         routine ContentMerge.
02054     */
02055     for(;;) {
02056         term = oldworkpointer;
02057         AR.CompressPointer = oldcomppointer;
02058         if ( GetOneTerm(BHEAD term,fi,&position,1) < 0 ) goto CalledFrom;
02059         if ( *term == 0 ) break;
02060     /*
02061         'merge' the two terms
02062     */
02063         if ( ContentMerge(BHEAD cbuffer,term) < 0 ) goto CalledFrom;
02064     }
02065     if ( fi->handle < 0 ) {
02066         fi->POfill = fi->PObuffer+BASEPOSITION(oldposition);
02067     }
02068     AR.GetOneFile = oldonefile;
02069     }
02070     AR.CompressPointer = oldcomppointer;
02071     for ( i = 0; i < *cbuffer; i++ ) content[i] = cbuffer[i];
02072     TermFree(cbuffer,"GetContent");
02073     AT.WorkPointer = oldworkpointer;
02074     return(*content);
02075 CalledFrom:
02076     MLOCK(ErrorMessageLock);
02077     MesCall("GetContent");
02078     MUNLOCK(ErrorMessageLock);
02079     SETERROR(-1)
02080 }
02081 /*
02082 #] GetContent :
02083 #[ CleanupTerm :
02084

```

```

02085
02086         Removes noncommuting objects from the term
02087     */
02088
02089     int CleanupTerm(WORD *term)
02090     {
02091         WORD *tstop, *t, *tfill, *tt;
02092         GETSTOP(term, tstop);
02093         t = term+1;
02094         while ( t < tstop ) {
02095             if ( *t >= FUNCTION && ( functions[*t-FUNCTION].commute || *t == DENOMINATOR ) ) {
02096                 tfill = t; tt = t + t[1]; tstop = term + *term;
02097                 while ( tt < tstop ) *tfill++ = *tt++;
02098                 *term = tfill - term;
02099                 tstop -= ABS(tfill[-1]);
02100             }
02101             else {
02102                 t += t[1];
02103             }
02104         }
02105         return(0);
02106     }
02107
02108     /*
02109         #] CleanupTerm :
02110         #[ ContentMerge :
02111     */
02112
02113     WORD ContentMerge(PHEAD WORD *content, WORD *term)
02114     {
02115         GETBIDENTITY
02116         WORD *cstop, csize, crsize, sign = 1, numsize, densize, i, tnsz, tdsz;
02117         UWORD *num, *den, *tnum, *tden;
02118         WORD *outfill, *outb = TermMalloc("ContentMerge"), *ct;
02119         WORD *t, *tstop, tsz, trsz, *told;
02120         WORD *t1, *t2, *c1, *c2, i1, i2, *out1;
02121         WORD didsymbol = 0, diddotp = 0, tfirst;
02122         cstop = content + *content;
02123         csize = cstop[-1];
02124         if ( csize < 0 ) { sign = -sign; csize = -csize; }
02125         cstop -= csize;
02126         numsize = densize = crsize = (csize-1)/2;
02127         num = NumberMalloc("ContentMerge");
02128         den = NumberMalloc("ContentMerge");
02129         for ( i = 0; i < numsize; i++ ) num[i] = (UWORD) (cstop[i]);
02130         for ( i = 0; i < densize; i++ ) den[i] = (UWORD) (cstop[i+crsize]);
02131         while ( num[numsize-1] == 0 ) numsize--;
02132         while ( den[densize-1] == 0 ) densize--;
02133     /*
02134         First we do the coefficient
02135     */
02136         tstop = term + *term;
02137         tsz = tstop[-1];
02138         if ( tsz < 0 ) tsz = -tsz;
02139     /* else { sign = 1; } */
02140         tstop = tstop - tsz;
02141         tnsz = tdsz = trsz = (tsz-1)/2;
02142         tnum = (UWORD *)tstop; tden = (UWORD *) (tstop + trsz);
02143         while ( tnum[tnsz-1] == 0 ) tnsz--;
02144         while ( tden[tdsz-1] == 0 ) tdsz--;
02145         GcdLong(BHEAD num, numsize, tnum, tnsz, num, &numsize);
02146         if ( LcmLong(BHEAD den, densize, tden, tdsz, den, &densize) ) goto CalledFrom;
02147         outfill = outb + 1;
02148         ct = content + 1;
02149         t = term + 1;
02150         while ( ct < cstop ) {
02151             switch ( *ct ) {
02152                 case SYMBOL:
02153                     didsymbol = 1;
02154                     t = term+1;
02155                     while ( t < tstop && *t != *ct ) t += t[1];
02156                     if ( t >= tstop ) break;
02157                     t1 = t+2; t2 = t+t[1];
02158                     c1 = ct+2; c2 = ct+ct[1];
02159                     out1 = outfill; *outfill++ = *ct; outfill++;
02160                     while ( c1 < c2 && t1 < t2 ) {
02161                         if ( *c1 == *t1 ) {
02162                             if ( t1[1] <= c1[1] ) {
02163                                 *outfill++ = *t1++; *outfill++ = *t1++;
02164                                 c1 += 2;
02165                             }
02166                             else {
02167                                 *outfill++ = *c1++; *outfill++ = *c1++;
02168                                 t1 += 2;
02169                             }
02170                         }
02171                     }
02172                     else if ( *c1 < *t1 ) {

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```

02172         if ( c1[1] < 0 ) {
02173             *outfill++ = *c1++; *outfill++ = *c1++;
02174         }
02175         else { c1 += 2; }
02176     }
02177     else {
02178         if ( t1[1] < 0 ) {
02179             *outfill++ = *t1++; *outfill++ = *t1++;
02180         }
02181         else t1 += 2;
02182     }
02183 }
02184 while ( c1 < c2 ) {
02185     if ( c1[1] < 0 ) { *outfill++ = c1[0]; *outfill++ = c1[1]; }
02186     c1 += 2;
02187 }
02188 while ( t1 < t2 ) {
02189     if ( t1[1] < 0 ) { *outfill++ = t1[0]; *outfill++ = t1[1]; }
02190     t1 += 2;
02191 }
02192 out1[1] = outfill - out1;
02193 if ( out1[1] == 2 ) outfill = out1;
02194 break;
02195 case DOTPRODUCT:
02196     diddotp = 1;
02197     t = term+1;
02198     while ( t < tstop && *t != *ct ) t += t[1];
02199     if ( t >= tstop ) break;
02200     t1 = t+2; t2 = t+t[1];
02201     c1 = ct+2; c2 = ct+ct[1];
02202     out1 = outfill; *outfill++ = *ct; outfill++;
02203     while ( c1 < c2 && t1 < t2 ) {
02204         if ( *c1 == *t1 && c1[1] == t1[1] ) {
02205             if ( t1[2] <= c1[2] ) {
02206                 *outfill++ = *t1++; *outfill++ = *t1++; *outfill++ = *t1++;
02207                 c1 += 3;
02208             }
02209             else {
02210                 *outfill++ = *c1++; *outfill++ = *c1++; *outfill++ = *c1++;
02211                 t1 += 3;
02212             }
02213         }
02214         else if ( *c1 < *t1 || ( *c1 == *t1 && c1[1] < t1[1] ) ) {
02215             if ( c1[2] < 0 ) {
02216                 *outfill++ = *c1++; *outfill++ = *c1++; *outfill++ = *c1++;
02217             }
02218             else { c1 += 3; }
02219         }
02220         else {
02221             if ( t1[2] < 0 ) {
02222                 *outfill++ = *t1++; *outfill++ = *t1++; *outfill++ = *t1++;
02223             }
02224             else t1 += 3;
02225         }
02226     }
02227     while ( c1 < c2 ) {
02228         if ( c1[2] < 0 ) { *outfill++ = c1[0]; *outfill++ = c1[1]; *outfill++ = c1[1]; }
02229         c1 += 3;
02230     }
02231     while ( t1 < t2 ) {
02232         if ( t1[2] < 0 ) { *outfill++ = t1[0]; *outfill++ = t1[1]; *outfill++ = t1[1]; }
02233         t1 += 3;
02234     }
02235     out1[1] = outfill - out1;
02236     if ( out1[1] == 2 ) outfill = out1;
02237     break;
02238 case INDEX:
02239     t = term+1;
02240     while ( t < tstop && *t != *ct ) t += t[1];
02241     if ( t >= tstop ) break;
02242     t1 = t+2; t2 = t+t[1];
02243     c1 = ct+2; c2 = ct+ct[1];
02244     out1 = outfill; *outfill++ = *ct; outfill++;
02245     while ( c1 < c2 && t1 < t2 ) {
02246         if ( *c1 == *t1 ) {
02247             *outfill++ = *c1++;
02248             t1 += 1;
02249         }
02250         else if ( *c1 < *t1 ) { c1 += 1; }
02251         else { t1 += 1; }
02252     }
02253     out1[1] = outfill - out1;
02254     if ( out1[1] == 2 ) outfill = out1;
02255     break;
02256 case VECTOR:
02257 case DELTA:
02258     t = term+1;

```

```

02259         while ( t < tstop && *t != *ct ) t += t[1];
02260         if ( t >= tstop ) break;
02261         t1 = t+2; t2 = t+t[1];
02262         c1 = ct+2; c2 = ct+ct[1];
02263         out1 = outfill; *outfill++ = *ct; outfill++;
02264         while ( c1 < c2 && t1 < t2 ) {
02265             if ( *c1 == *t1 && c1[1] && t1[1] ) {
02266                 *outfill++ = *c1++; *outfill++ = *c1++;
02267                 t1 += 2;
02268             }
02269             else if ( *c1 < *t1 || ( *c1 == *t1 && c1[1] < t1[1] ) ) {
02270                 c1 += 2;
02271             }
02272             else {
02273                 t1 += 2;
02274             }
02275         }
02276         out1[1] = outfill - out1;
02277         if ( out1[1] == 2 ) outfill = out1;
02278         break;
02279     case GAMMA:
02280     default:          /* Functions */
02281         told = t;
02282         t = term+1;
02283         while ( t < tstop ) {
02284             if ( *t != *ct ) { t += t[1]; continue; }
02285             if ( ct[1] != t[1] ) { t += t[1]; continue; }
02286             if ( ct[2] != t[2] ) { t += t[1]; continue; }
02287             t1 = t; t2 = ct; i1 = t1[1]; i2 = t2[1];
02288             while ( i1 > 0 ) {
02289                 if ( *t1 != *t2 ) break;
02290                 t1++; t2++; i1--;
02291             }
02292             if ( i1 != 0 ) { t += t[1]; continue; }
02293             t1 = t;
02294             for ( i = 0; i < i2; i++ ) { *outfill++ = *t++; }
02295             /*
02296              Mark as 'used'. The flags must be different!
02297             */
02298             t1[2] |= SUBTERMUSED1;
02299             ct[2] |= SUBTERMUSED2;
02300             t = told;
02301             break;
02302         }
02303         break;
02304     }
02305     ct += ct[1];
02306 }
02307 if ( diddotp == 0 ) {
02308     t = term+1; while ( t < tstop && *t != DOTPRODUCT ) t += t[1];
02309     if ( t < tstop ) { /* now we need the negative powers */
02310         tfirst = 1; told = outfill;
02311         for ( i = 2; i < t[1]; i += 3 ) {
02312             if ( t[i+2] < 0 ) {
02313                 if ( tfirst ) { *outfill++ = DOTPRODUCT; *outfill++ = 0; tfirst = 0; }
02314                 *outfill++ = t[i]; *outfill++ = t[i+1]; *outfill++ = t[i+2];
02315             }
02316         }
02317         if ( outfill > told ) told[1] = outfill-told;
02318     }
02319 }
02320 if ( didsymbol == 0 ) {
02321     t = term+1; while ( t < tstop && *t != SYMBOL ) t += t[1];
02322     if ( t < tstop ) { /* now we need the negative powers */
02323         tfirst = 1; told = outfill;
02324         for ( i = 2; i < t[1]; i += 2 ) {
02325             if ( t[i+1] < 0 ) {
02326                 if ( tfirst ) { *outfill++ = SYMBOL; *outfill++ = 0; tfirst = 0; }
02327                 *outfill++ = t[i]; *outfill++ = t[i+1];
02328             }
02329         }
02330         if ( outfill > told ) told[1] = outfill-told;
02331     }
02332 }
02333 /*
02334 Now put the coefficient back.
02335 */
02336 if ( numsize < densize ) {
02337     for ( i = numsize; i < densize; i++ ) num[i] = 0;
02338     numsize = densize;
02339 }
02340 else if ( densize < numsize ) {
02341     for ( i = densize; i < numsize; i++ ) den[i] = 0;
02342     densize = numsize;
02343 }
02344 for ( i = 0; i < numsize; i++ ) *outfill++ = num[i];
02345 for ( i = 0; i < densize; i++ ) *outfill++ = den[i];

```

```

02346     csize = numsize+densize+1;
02347     if ( sign < 0 ) csize = -csize;
02348     *outfill++ = csize;
02349     *outb = outfill-outb;
02350     NumberFree(den,"ContentMerge");
02351     NumberFree(num,"ContentMerge");
02352     for ( i = 0; i < *outb; i++ ) content[i] = outb[i];
02353     TermFree(outb,"ContentMerge");
02354 /*
02355     Now we have to 'restore' the term to its original.
02356     We do not restore the content, because if anything was used the
02357     new content overwrites the old. 6-mar-2018 JV
02358 */
02359     t = term + 1;
02360     while ( t < tstop ) {
02361         if ( *t >= FUNCTION ) t[2] &= ~SUBTERMUSED1;
02362         t += t[1];
02363     }
02364     return(*content);
02365 CalledFrom:
02366     MLOCK(ErrorMessageLock);
02367     MesCall("GetContent");
02368     MUNLOCK(ErrorMessageLock);
02369     SETERROR(-1)
02370 }
02371
02372 /*
02373     #] ContentMerge :
02374     #[ TermsInExpression :
02375 */
02376
02377 LONG TermsInExpression(WORD num)
02378 {
02379     LONG x = Expressions[num].counter;
02380     if ( x >= 0 ) return(x);
02381     return(-1);
02382 }
02383
02384 /*
02385     #] TermsInExpression :
02386     #[ SizeOfExpression :
02387 */
02388
02389 LONG SizeOfExpression(WORD num)
02390 {
02391     LONG x = (LONG) (DIVPOS(Expressions[num].size,sizeof(WORD)));
02392     if ( x >= 0 ) return(x);
02393     return(-1);
02394 }
02395
02396 /*
02397     #] SizeOfExpression :
02398     #[ UpdatePositions :
02399 */
02400
02401 void UpdatePositions(void)
02402 {
02403     EXPRESSIONS e = Expressions;
02404     POSITION *oldw;
02405     WORD *oldw;
02406     int i;
02407     if ( NumExpressions > 0 &&
02408         ( AS.OldOnFile == 0 || AS.NumOldOnFile < NumExpressions ) ) {
02409         if ( AS.OldOnFile ) {
02410             oldw = AS.OldOnFile;
02411             AS.OldOnFile = (POSITION *)Malloc1(NumExpressions*sizeof(POSITION),"file pointers");
02412             for ( i = 0; i < AS.NumOldOnFile; i++ ) AS.OldOnFile[i] = oldw[i];
02413             AS.NumOldOnFile = NumExpressions;
02414             M_free(oldw,"process file pointers");
02415         }
02416         else {
02417             AS.OldOnFile = (POSITION *)Malloc1(NumExpressions*sizeof(POSITION),"file pointers");
02418             AS.NumOldOnFile = NumExpressions;
02419         }
02420     }
02421     if ( NumExpressions > 0 &&
02422         ( AS.OldNumFactors == 0 || AS.NumOldNumFactors < NumExpressions ) ) {
02423         if ( AS.OldNumFactors ) {
02424             oldw = AS.OldNumFactors;
02425             AS.OldNumFactors = (WORD *)Malloc1(NumExpressions*sizeof(WORD),"numfactors pointers");
02426             for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.OldNumFactors[i] = oldw[i];
02427             M_free(oldw,"numfactors pointers");
02428             oldw = AS.Oldvflags;
02429             AS.Oldvflags = (WORD *)Malloc1(NumExpressions*sizeof(WORD),"vflags pointers");
02430             for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.Oldvflags[i] = oldw[i];
02431             AS.NumOldNumFactors = NumExpressions;
02432             M_free(oldw,"vflags pointers");

```

```

02433         oldw = AS.Olduflags;
02434         AS.Olduflags = (WORD *)Malloca1(NumExpressions*sizeof(WORD),"uflags pointers");
02435         for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.Olduflags[i] = oldw[i];
02436         AS.NumOldNumFactors = NumExpressions;
02437         M_free(oldw,"uflags pointers");
02438     }
02439     else {
02440         AS.OldNumFactors = (WORD *)Malloca1(NumExpressions*sizeof(WORD),"numfactors pointers");
02441         AS.Oldvflags = (WORD *)Malloca1(NumExpressions*sizeof(WORD),"vflags pointers");
02442         AS.Olduflags = (WORD *)Malloca1(NumExpressions*sizeof(WORD),"uflags pointers");
02443         AS.NumOldNumFactors = NumExpressions;
02444     }
02445 }
02446 for ( i = 0; i < NumExpressions; i++ ) {
02447     AS.OldOnFile[i] = e[i].onfile;
02448     AS.OldNumFactors[i] = e[i].numfactors;
02449     AS.Oldvflags[i] = e[i].vflags;
02450     AS.Olduflags[i] = e[i].uflags;
02451 }
02452 }
02453
02454 /*
02455     #] UpdatePositions :
02456     #[ CountTerms1 :      LONG CountTerms1()
02457
02458     Counts the terms in the current deferred bracket
02459     Is mainly an adaptation of the routine Deferred in proces.c
02460 */
02461 LONG CountTerms1(PHEAD0)
02462 {
02463     GETBIDENTITY
02464     POSITION oldposition, startposition;
02465     WORD *t, *m, *mstop, decr, i, *oldwork, retval;
02466     WORD *oldipointer = AR.CompressPointer;
02467     WORD oldGetOneFile = AR.GetOneFile, olddeferflag = AR.DeferFlag;
02468     LONG numterms = 0;
02469     AR.GetOneFile = 1;
02470     oldwork = AT.WorkPointer;
02471     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
02472     AR.DeferFlag = 0;
02473     startposition = AR.DefPosition;
02474
02475     /*
02476         Store old position
02477     */
02478     if ( AR.infile->handle >= 0 ) {
02479         PUTZERO(oldposition);
02480     /*
02481         SeekFile(AR.infile->handle,&oldposition,SEEK_CUR);
02482     */
02483     }
02484     else {
02485         SETBASEPOSITION(oldposition,AR.infile->POfill-AR.infile->PObuffer);
02486         AR.infile->POfill = (WORD *)(((UBYTE *) (AR.infile->PObuffer)
02487             +BASEPOSITION(startposition)));
02488     }
02489     /*
02490         Look in the CompressBuffer where the bracket contents start
02491     */
02492     t = m = AR.CompressBuffer;
02493     t += *t;
02494     mstop = t - ABS(t[-1]);
02495     m++;
02496     while ( *m != HAAKJE && m < mstop ) m += m[1];
02497     if ( m >= mstop ) { /* No deferred action! */
02498         numterms = 1;
02499         AR.DeferFlag = olddeferflag;
02500         AT.WorkPointer = oldwork;
02501         AR.GetOneFile = oldGetOneFile;
02502         return(numterms);
02503     }
02504     mstop = m + m[1];
02505     decr = WORDDIF(mstop,AR.CompressBuffer)-1;
02506
02507     m = AR.CompressBuffer;
02508     t = AR.CompressPointer;
02509     i = *m;
02510     NCOPY(t,m,i);
02511     AR.TePos = 0;
02512     AN.TeSuOut = 0;
02513     /*
02514         Status:
02515         First bracket content starts at mstop.
02516         Next term starts at startposition.
02517         Decompression information is in AR.CompressPointer.
02518         The outside of the bracket runs from AR.CompressBuffer+1 to mstop.
02519     */

```

```

02520     AR.CompressPointer = oldipointer;
02521     for(;;) {
02522         numterms++;
02523         retval = GetOneTerm(BHEAD AT.WorkPointer,AR.infile,&startposition,0);
02524         if ( retval >= 0 ) AR.CompressPointer = oldipointer;
02525         if ( retval <= 0 ) break;
02526         t = AR.CompressPointer;
02527         if ( *t < (1 + decr + ABS(*(t+t-1))) ) break;
02528         t++;
02529         m = AR.CompressBuffer+1;
02530         while ( m < mstop ) {
02531             if ( *m != *t ) goto Thatsit;
02532             m++; t++;
02533         }
02534     }
02535 Thatsit:;
02536 /*
02537     Finished. Reposition the file, restore information and return.
02538 */
02539     AT.WorkPointer = oldwork;
02540     if ( AR.infile->handle >= 0 ) {
02541 /*
02542         SeekFile(AR.infile->handle,&oldposition,SEEK_SET);
02543 */
02544     }
02545     else {
02546         AR.infile->POfill = AR.infile->PObuffer + BASEPOSITION(oldposition);
02547     }
02548     AR.DeferFlag = olddeferflag;
02549     AR.GetOneFile = oldGetOneFile;
02550     return(numterms);
02551 }
02552
02553 /*
02554     #] CountTerms1 :
02555     #[ TermsInBracket :    LONG TermsInBracket(term,level)
02556
02557     The function TermsInBracket_()
02558     Syntax:
02559     TermsInBracket_() : The current bracket in a Keep Brackets
02560     TermsInBracket_(bracket) : This bracket in the current expression
02561     TermsInBracket_(expression,bracket) : This bracket in the given expression
02562     All other specifications don't have any effect.
02563 */
02564
02565 #define CURRENTBRACKET 1
02566 #define BRACKETCURRENTEXPR 2
02567 #define BRACKETOTHEREXPR 3
02568 #define NOBRACKETACTIVE 4
02569
02570 LONG TermsInBracket(PHEAD WORD *term, WORD level)
02571 {
02572     WORD *t, *tstop, *b, *tt, *n1, *n2;
02573     int type = 0, i, num;
02574     LONG numterms = 0;
02575     WORD *bracketbuffer = AT.WorkPointer;
02576     t = term; GETSTOP(t,tstop);
02577     t++; b = bracketbuffer;
02578     while ( t < tstop ) {
02579         if ( *t != TERMSINBRACKET ) { t += t[1]; continue; }
02580         if ( t[1] == FUNHEAD || (
02581             t[1] == FUNHEAD+2
02582             && t[FUNHEAD] == -SNUMBER
02583             && t[FUNHEAD+1] == 0
02584         ) ) {
02585             if ( AC.ComDefer == 0 ) {
02586                 type = NOBRACKETACTIVE;
02587             }
02588             else {
02589                 type = CURRENTBRACKET;
02590             }
02591             *b = 0;
02592             break;
02593         }
02594         if ( t[FUNHEAD] == -EXPRESSION ) {
02595             if ( t[FUNHEAD+2] < 0 ) {
02596                 if ( ( t[FUNHEAD+2] <= -FUNCTION ) && ( t[1] == FUNHEAD+3 ) ) {
02597                     type = BRACKETOTHEREXPR;
02598                     *b++ = FUNHEAD+4; *b++ = -t[FUNHEAD+2]; *b++ = FUNHEAD;
02599                     for ( i = 2; i < FUNHEAD; i++ ) *b++ = 0;
02600                     *b++ = 1; *b++ = 1; *b++ = 3;
02601                     break;
02602                 }
02603                 else if ( ( t[FUNHEAD+2] > -FUNCTION ) && ( t[1] == FUNHEAD+4 ) ) {
02604                     type = BRACKETOTHEREXPR;
02605                     tt = t + FUNHEAD+2;
02606                     switch ( *tt ) {

```



```

02607         case -SYMBOL:
02608             *b++ = 8; *b++ = SYMBOL; *b++ = 4; *b++ = tt[1];
02609             *b++ = 1; *b++ = 1; *b++ = 1; *b++ = 3;
02610             break;
02611         case -SNUMBER:
02612             if ( tt[1] == 1 ) {
02613                 *b++ = 4; *b++ = 1; *b++ = 1; *b++ = 3;
02614             }
02615             else goto IllBraReq;
02616             break;
02617         default:
02618             goto IllBraReq;
02619     }
02620     break;
02621 }
02622 }
02623 else if ( ( t[FUNHEAD+2] == (t[1]-FUNHEAD-2) ) &&
02624 ( t[FUNHEAD+2+ARGHEAD] == (t[FUNHEAD+2]-ARGHEAD) ) ) {
02625     type = BRACKETOTHEREXPR;
02626     tt = t + FUNHEAD + ARGHEAD; num = *tt;
02627     for ( i = 0; i < num; i++ ) *b++ = *tt++;
02628     break;
02629 }
02630 }
02631 else {
02632     if ( t[FUNHEAD] < 0 ) {
02633         if ( ( t[FUNHEAD] <= -FUNCTION ) && ( t[1] == FUNHEAD+1 ) ) {
02634             type = BRACKETCURRENTEXPR;
02635             *b++ = FUNHEAD+4; *b++ = -t[FUNHEAD+2]; *b++ = FUNHEAD;
02636             for ( i = 2; i < FUNHEAD; i++ ) *b++ = 0;
02637             *b++ = 1; *b++ = 1; *b++ = 3; *b = 0;
02638             break;
02639         }
02640         else if ( ( t[FUNHEAD] > -FUNCTION ) && ( t[1] == FUNHEAD+2 ) ) {
02641             type = BRACKETCURRENTEXPR;
02642             tt = t + FUNHEAD+2;
02643             switch ( *tt ) {
02644                 case -SYMBOL:
02645                     *b++ = 8; *b++ = SYMBOL; *b++ = 4; *b++ = tt[1];
02646                     *b++ = 1; *b++ = 1; *b++ = 1; *b++ = 3;
02647                     break;
02648                 case -SNUMBER:
02649                     if ( tt[1] == 1 ) {
02650                         *b++ = 4; *b++ = 1; *b++ = 1; *b++ = 3;
02651                     }
02652                     else goto IllBraReq;
02653                     break;
02654                 default:
02655                     goto IllBraReq;
02656             }
02657             break;
02658         }
02659     }
02660     else if ( ( t[FUNHEAD] == (t[1]-FUNHEAD) ) &&
02661 ( t[FUNHEAD+ARGHEAD] == (t[FUNHEAD]-ARGHEAD) ) ) {
02662         type = BRACKETCURRENTEXPR;
02663         tt = t + FUNHEAD + ARGHEAD; num = *tt;
02664         for ( i = 0; i < num; i++ ) *b++ = *tt++;
02665         break;
02666     }
02667     else {
02668         IllBraReq;
02669         MLOCK(ErrorMessageLock);
02670         MesPrint("Illegal bracket request in termsinbracket_ function.");
02671         MUNLOCK(ErrorMessageLock);
02672         Terminate(-1);
02673     }
02674 }
02675 t += t[1];
02676 }
02677 AT.WorkPointer = b;
02678 if ( AT.WorkPointer + *term + 4 > AT.WorkTop ) {
02679     MLOCK(ErrorMessageLock);
02680     MesWork();
02681     MesPrint("Called from termsinbracket_ function.");
02682     MUNLOCK(ErrorMessageLock);
02683     return(-1);
02684 }
02685 /*
02686 We are now in the position to look for the bracket
02687 */
02688 switch ( type ) {
02689     case CURRENTBRACKET:
02690 /*
02691     The code here should be rather similar to when we pick up
02692     the contents of the bracket. In our case we only count the
02693     terms though.

```

```

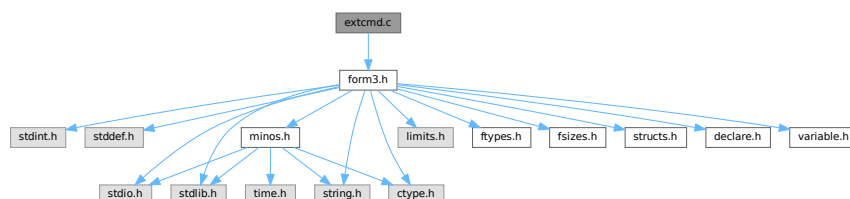
02694 */
02695         numterms = CountTerms1(BHEAD0);
02696         break;
02697     case BRACKETCURRENTEXPR:
02698     /*
02699         Not implemented yet.
02700     */
02701         MLOCK(ErrorMessageLock);
02702         MesPrint("termsinbracket_ function currently only handles Keep Brackets.");
02703         MUNLOCK(ErrorMessageLock);
02704         return(-1);
02705     case BRACKETOTHEREXPR:
02706         MLOCK(ErrorMessageLock);
02707         MesPrint("termsinbracket_ function currently only handles Keep Brackets.");
02708         MUNLOCK(ErrorMessageLock);
02709         return(-1);
02710     case NOBRACKETACTIVE:
02711         numterms = 1;
02712         break;
02713 }
02714 /*
02715 Now we have the number in numterms. We replace the function by it.
02716 */
02717 n1 = term; n2 = AT.WorkPointer; tstop = n1 + *n1;
02718 while ( n1 < t ) *n2++ = *n1++;
02719 i = numterms » BITSINWORD;
02720 if ( i == 0 ) {
02721     *n2++ = LNUMBER; *n2++ = 4; *n2++ = 1; *n2++ = (WORD)(numterms & WORDMASK);
02722 }
02723 else {
02724     *n2++ = LNUMBER; *n2++ = 5; *n2++ = 2;
02725     *n2++ = (WORD)(numterms & WORDMASK); *n2++ = i;
02726 }
02727 n1 += n1[1];
02728 while ( n1 < tstop ) *n2++ = *n1++;
02729 AT.WorkPointer[0] = n2 - AT.WorkPointer;
02730 AT.WorkPointer = n2;
02731 if ( Generator(BHEAD n1,level) < 0 ) {
02732     AT.WorkPointer = bracketbuffer;
02733     MLOCK(ErrorMessageLock);
02734     MesPrint("Called from termsinbracket_ function.");
02735     MUNLOCK(ErrorMessageLock);
02736     return(-1);
02737 }
02738 /*
02739 Finished. Reset things and return.
02740 */
02741 AT.WorkPointer = bracketbuffer;
02742 return(numterms);
02743 }
02744 /*
02745 #] TermsInBracket :      LONG TermsInBracket(term,level)
02746 #] Expressions :
02747 */

```

## 4.31 extcmd.c File Reference

#include "form3.h"

Include dependency graph for extcmd.c:



### Functions

- int [openExternalChannel](#) (UBYTE \*cmd, int daemonize, UBYTE \*shellname, UBYTE \*stderrname)

- int [initPresetExternalChannels](#) (UBYTE \*theline, int thetimeout)
- int [closeExternalChannel](#) (int n)
- int [selectExternalChannel](#) (int n)
- int [getCurrentExternalChannel](#) (void)
- void [closeAllExternalChannels](#) (void)

### 4.31.1 Detailed Description

The system that takes care of communication with external programs.

Definition in file [extcmd.c](#).

### 4.31.2 Function Documentation

#### 4.31.2.1 openExternalChannel()

```
int openExternalChannel (
    UBYTE * cmd,
    int daemonize,
    UBYTE * shellname,
    UBYTE * stderrname )
```

Definition at line [230](#) of file [extcmd.c](#).

#### 4.31.2.2 initPresetExternalChannels()

```
int initPresetExternalChannels (
    UBYTE * theline,
    int thetimeout )
```

Definition at line [232](#) of file [extcmd.c](#).

#### 4.31.2.3 closeExternalChannel()

```
int closeExternalChannel (
    int n )
```

Definition at line [233](#) of file [extcmd.c](#).

#### 4.31.2.4 selectExternalChannel()

```
int selectExternalChannel (
    int n )
```

Definition at line [234](#) of file [extcmd.c](#).

#### 4.31.2.5 `getCurrentExternalChannel()`

```
int getCurrentExternalChannel (
    void )
```

Definition at line 235 of file `extcmd.c`.

#### 4.31.2.6 `closeAllExternalChannels()`

```
void closeAllExternalChannels (
    void )
```

Definition at line 236 of file `extcmd.c`.

### 4.32 `extcmd.c`

[Go to the documentation of this file.](#)

```
00001
00005 /* #[] License : */
00006 /*
00007 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00008 *   When using this file you are requested to refer to the publication
00009 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010 *   This is considered a matter of courtesy as the development was paid
00011 *   for by FOM the Dutch physics granting agency and we would like to
00012 *   be able to track its scientific use to convince FOM of its value
00013 *   for the community.
00014 *
00015 *   This file is part of FORM.
00016 *
00017 *   FORM is free software: you can redistribute it and/or modify it under the
00018 *   terms of the GNU General Public License as published by the Free Software
00019 *   Foundation, either version 3 of the License, or (at your option) any later
00020 *   version.
00021 *
00022 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00023 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00024 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00025 *   details.
00026 *
00027 *   You should have received a copy of the GNU General Public License along
00028 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00029 */
00030 /* #[] License : */
00031 /*
00032    #[] Documentation :
00033
00034    This module is written by M.Tentyukov as a part of implementation of
00035    interaction between FORM and external processes, first release
00036    09.04.2004. A part of this code is copied from the DIANA project, which was
00037    written by M. Tentyukov and published under the GPL version 2 as
00038    published by the Free Software Foundation.
00039
00040    This file is completely re-written by M.Tentyukov in May 2006.
00041    Since the interface was changed, the public function were changed,
00042    also. A new public functions were added: initPresetExternalChannels()
00043    (see comments just before this function in the present file) and
00044    setKillModeForExternalChannel (a pointer, not a function).
00045
00046    If a macro WITHEXTERNALCHANNEL is not defined, all public functions
00047    are stubs returning failure.
00048
00049    The idea is to start an external command swallowing
00050    its stdin and stdout. This can be done by means of the function
00051    int openExternalChannel(cmd,daemonize,shellname,stderrname), where
00052    cmd is a command to run,
00053    daemonize: if !=0 then start the command in the "daemon" mode,
00054    shellname: if !=NULL, execute the command in a subshell,
00055    stderrname: if != NULL, redirect stderr of the command to this file.
00056    The function returns some small positive integer number (the
00057    descriptor of a newly created external channel), or -1 on failure.
00058
```

```

00059     After the command is started, it becomes a _current_ opened external
00060     channel. The buffer can be sent to its stdin by a function
00061     int writeBufToExtChannel(buf, n)
00062     (here buf is a pointer to the buffer, n is the length in bytes; the
00063     function returns 0 in success, or -1 on failure),
00064     or one character can be read from its stdout by
00065     means of the function
00066     int getcFromExtChannel().
00067
00068     The latter returns the
00069     character casted to integer, or something <0. This can be -2 (if there
00070     is no current external channel) or EOF, if the external program closes
00071     its stdout, or if the external program outputs a string coinciding
00072     with a _terminator_.
00073
00074     By default, the terminator if an empty line. For the current external
00075     channel it can be set by means of the function
00076     int setTerminatorForExternalChannel(newterminator).
00077     The function returns 0 in success, or !0 if something is wrong (no
00078     current channel, too long terminator).
00079
00080     After getcFromExtChannel() returns EOF, the current channel becomes
00081     undefined. Any further attempts to read information by
00082     getcFromExtChannel() result in -2. To set (re-set) a current channel,
00083     the function
00084     int selectExternalChannel(n)
00085     can be used. This function accepts the valid external channel
00086     descriptor (returned by openExternalChannel) and returns the
00087     descriptor of a previous current channel (0, if there was no current
00088     channel, or -1, if the external channel descriptor is invalid).
00089     If n == 0, the function undefine the current external channel.
00090
00091     The function
00092     int closeExternalChannel(n)
00093     destroys the opened external channel with the descriptor n. It returns
00094     0 in success, or -1 on failure. If the corresponding external channel
00095     was the current one, the current channel becomes undefined. If n==0,
00096     the function closes the current external channel.
00097
00098     The function
00099     int getCurrentExternalChannel(void)
00100     returns the descriptor if the current external channel, or 0 , if
00101     there is no current external channel.
00102
00103     The function
00104     void closeAllExternalChannels(void)
00105
00106     destroys all opened external channels.
00107
00108     List of all public functions:
00109     int openExternalChannel(UBYTE *cmd,int daemonize,UBYTE *shellname, UBYTE * stderrname);
00110     int initPresetExternalChannels(UBYTE *theline, int thetimeout);
00111     int setTerminatorForExternalChannel(char *newterminator);
00112     int setKillModeForExternalChannel(int signum, int sentToWholeGroup);
00113     int closeExternalChannel(int n);
00114     int selectExternalChannel(int n);
00115     int writeBufToExtChannel(char *buf,int n);
00116     int getcFromExtChannel(void);
00117     int getCurrentExternalChannel(void);
00118     void closeAllExternalChannels(void);
00119
00120     ATTENTION!
00121
00122     Four of them:
00123     1 setTerminatorForExternalChannel
00124     2 setKillModeForExternalChannel
00125     3 writeBufToExtChannel
00126     4 getcFromExtChannel
00127
00128     are NOT functions, but variables (pointers) of a corrsponding type.
00129     They are initialised by proper values to avoid repeated error checking.
00130
00131     All public functions are independent of realization hidden in this module.
00132     All other functions may have a returned type/parameters type local w.r.t.
00133     this module; they are not declared outside of this file.
00134
00135     #] Documentation :
00136     #[ Selftest initializing:
00137  */
00138
00139  /*
00140  Uncomment to get a self-consistent program:
00141  #define SELFTEST 1
00142  */
00143
00144
00145  #ifndef SELFTEST

```

```

00146 #define WITHEXTERNALCHANNEL 1
00147 #ifdef _MSC_VER
00148 #define FORM_INLINE __inline
00149 #else
00150 #define FORM_INLINE inline
00151 #endif
00152
00153 /*
00154     From form3.h:
00155 */
00156 typedef unsigned char UBYTE;
00157
00158 /*The following variables should be defined in variable.h:*/
00159 extern int (*writeBufToExtChannel)(char *buffer, size_t n);
00160 extern int (*getcFromExtChannel)();
00161 extern int (*setTerminatorForExternalChannel)(char *buffer);
00162 extern int (*setKillModeForExternalChannel)(int signum, int sentToWholeGroup);
00163
00164 #else /*ifdef SELFTEST*/
00165 #include "form3.h"
00166 #endif /*ifdef SELFTEST ... else*/
00167 /*
00168 pid_t getExternalChannelPid(void);
00169 */
00170 /*
00171     #] Selftest initializing:
00172     #[ Includes :
00173 */
00174 #ifdef WITHEXTERNALCHANNEL
00175 #include <stdio.h>
00176 #ifndef _MSC_VER
00177 #include <unistd.h>
00178 #endif
00179 #include <fcntl.h>
00180 #include <sys/types.h>
00181 #ifndef _MSC_VER
00182 #include <sys/time.h>
00183 #include <sys/wait.h>
00184 #endif
00185 #include <errno.h>
00186 #include <signal.h>
00187 #include <limits.h>
00188 /*
00189     #] Includes :
00190     #[ FailureFunctions:
00191 */
00192
00193 /*Non-initialized variant of public functions:*/
00194 int writeBufToExtChannelFailure(char *buf, size_t count)
00195 {
00196     DUMMYUSE(buf); DUMMYUSE(count);
00197     return(-1);
00198 } /*writeBufToExtChannelFailure*/
00199
00200 int setTerminatorForExternalChannelFailure(char *newTerminator)
00201 {
00202     DUMMYUSE(newTerminator);
00203     return(-1);
00204 } /*setTerminatorForExternalChannelFailure*/
00205
00206 int setKillModeForExternalChannelFailure(int signum, int sentToWholeGroup)
00207 {
00208     DUMMYUSE(signum); DUMMYUSE(sentToWholeGroup);
00209     return(-1);
00210 } /*setKillModeForExternalChannelFailure*/
00211
00212 int getcFromExtChannelFailure(void)
00213 {
00214     return(-2);
00215 } /*getcFromExtChannelFailure*/
00216
00217 int (*writeBufToExtChannel)(char *buffer, size_t n) = &writeBufToExtChannelFailure;
00218 int (*setTerminatorForExternalChannel)(char *buffer) =
00219     &setTerminatorForExternalChannelFailure;
00220 int (*setKillModeForExternalChannel)(int signum, int sentToWholeGroup) =
00221     &setKillModeForExternalChannelFailure;
00222 int (*getcFromExtChannel)(void) = &getcFromExtChannelFailure;
00223 #endif
00224 /*
00225     #] FailureFunctions:
00226     #[ Stubs :
00227 */
00228 #ifndef WITHEXTERNALCHANNEL
00229 /*Stubs for public functions:*/
00230 int openExternalChannel(UBYTE *cmd, int daemonize, UBYTE *shellname, UBYTE *stderrname)
00231 { DUMMYUSE(cmd); DUMMYUSE(daemonize); DUMMYUSE(shellname); DUMMYUSE(stderrname); return(-1); };
00232 int initPresetExternalChannels(UBYTE *theline, int thetimeout) { DUMMYUSE(theline);

```

```

        DUMMYUSE(thetimeout); return(-1); };
00233 int closeExternalChannel(int n) { DUMMYUSE(n); return(-1); };
00234 int selectExternalChannel(int n) { DUMMYUSE(n); return(-1); };
00235 int getCurrentExternalChannel(void) { return(0); };
00236 void closeAllExternalChannels(void) {};
00237 #else /*ifndef WITHEXTERNALCHANNEL*/
00238 /*
00239     #] Stubs :
00240     #[ Local types :
00241 */
00242 /*First argument for the function signal:*/
00243 #ifndef INTSIGHANDLER
00244 typedef void (*mysighandler_t)(int);
00245 #else
00246 /* Sometimes, this nonsense may occurs:*/
00247 /*typedef int (*mysighandler_t)(int);*/
00248 #endif
00249
00250 /*Input IO buffer size increment -- each time the buffer
00251 is expired it will be increased by this value (in bytes):*/
00252 #define DELTA_EXT_BUF 128
00253
00254 /*Re-allocatable array containing External Channel
00255 handlers increased each time by this value:*/
00256 #define DELTA_EXT_LIST 8
00257
00258 /*How many times I/O routines may attempt to continue their work
00259 in some failures:*/
00260 #define MAX_FAILS_IO 2
00261
00262 /*The external channel handler structure:*/
00263 typedef struct ExternalChannel {
00264     pid_t pid; /*PID of the external process*/
00265     pid_t gpid; /*process group ID of the external process.
00266                 If <=0, not used, if >0, the kill signals
00267                 is sent to the whole group */
00268     FILE *frec; /*stdout of the external process*/
00269     char *INbuf; /*External channel buffer*/
00270     char *IBfill; /*Position in INbuf from which the next input character will be read*/
00271     char *IBfull; /*End of read INbuf*/
00272     char *IBstop; /*end of allocated space for INbuf*/
00273     char *terminator; /* Terminator - when extern. program outputs ONLY this string,
00274                       it is assumed that the answer is ready, and getcFromExtChannel
00275                       returns EOF. Should not be longer then the minimal buffer!*/
00276     /*Info fields, not changable after creating a channel:*/
00277     char *cmd; /*the command*/
00278     char *shellname;
00279     char *stderrname; /*filename to redirect stderr, or NULL*/
00280     int fsend; /*stdin of the external process*/
00281     int killSignal; /*signal to kill*/
00282     int daemonize; /*0 --neither setsid nor daemonize, !=0 -- full daemonization*/
00283     PADPOINTER(0,3,0,0);
00284 } EXTHANDLE;
00285
00286
00287 static EXTHANDLE *externalChannelsList=0;
00288 /*Here integers are better than pointers: */
00289 static int externalChannelsListStop=0;
00290 static int externalChannelsListFill=0;
00291
00292 /*"current" external channel:*/
00293 static EXTHANDLE *externalChannelsListTop=0;
00294 /*
00295     #] Local types :
00296     #[ Selftest functions :
00297 */
00298 #ifdef SELFTEST
00299
00300 /*For malloc prototype:*/
00301 #include <stdlib.h>
00302
00303 /*StrLen, Malloc1, M_free and strDupl are defined in tools.c -- here only emulation:*/
00304 int StrLen(char *pattern)
00305 {
00306     register char *p=(char*)pattern;
00307     while(*p)p++;
00308     return((int) ((p-(char*)pattern)) );
00309 } /*StrLen*/
00310 void *Malloc1(int l, char *c)
00311 {
00312     return(malloc(l));
00313 }
00314 void M_free(void *p,char *c)
00315 {
00316     return(free(p));
00317 }
00318

```

```

00319 char *strDupl(UBYTE *instring, char *ifwrong)
00320 {
00321     UBYTE *s = instring, *to;
00322     while ( *s ) s++;
00323     to = s = (UBYTE *)Malloc1((s-instring)+1,ifwrong);
00324     while ( *instring ) *to++ = *instring++;
00325     *to = 0;
00326     return(s);
00327 }
00328
00329 /*PutPreVar from pre.c -- just ths stub:*/
00330 int PutPreVar(UBYTE *a,UBYTE *b,UBYTE *c,int i)
00331 {
00332     return(0);
00333 }
00334
00335 #endif
00336 /*
00337     #] Selftest functions :
00338     #[ Local functions :
00339 */
00340
00341 /*Initialize one cell of handler:*/
00342 static FORM_INLINE void extHandlerInit(EXTHANDLE *h)
00343 {
00344     h->pid=-1;
00345     h->gpip=-1;
00346     h->fsend=0;
00347     h->killSignal=SIGKILL;
00348     h->daemonize=1;
00349     h->frec=NULL;
00350     h->INbuf=h->IBfill=h->IBfull=h->IBstop=
00351     h->terminator=h->cmd=h->shellname=h->stderrname=NULL;
00352 }/*extHandlerInit*/
00353
00354 /* Copies each field of handler:*/
00355 static FORM_INLINE void extHandlerSwallowCopy(EXTHANDLE *to, EXTHANDLE *from)
00356 {
00357     to->pid=from->pid;
00358     to->gpip=from->gpip;
00359     to->fsend=from->fsend;
00360     to->killSignal=from->killSignal;
00361     to->daemonize=from->daemonize;
00362     to->frec=from->frec;
00363     to->INbuf=from->INbuf;
00364     to->IBfill=from->IBfill;
00365     to->IBfull=from->IBfull;
00366     to->IBstop=from->IBstop;
00367     to->terminator=from->terminator;
00368     to->cmd=from->cmd;
00369     to->shellname=from->shellname;
00370     to->stderrname=from->stderrname;
00371 }/*extHandlerSwallow*/
00372
00373 /*Allocates memory for fields of handler which have no fixed
00374 storage size and initializes some fields:*/
00375 static FORM_INLINE void
00376 extHandlerAlloc(EXTHANDLE *h, char *cmd, char *shellname, char *stderrname)
00377 {
00378     h->IBfill=h->IBfull=h->INbuf=
00379     Malloc1(DELTA_EXT_BUF,"External channel buffer");
00380     h->IBstop=h->INbuf+DELTA_EXT_BUF;
00381     /*Initialize a terminator:*/
00382     *(h->terminator=Malloc1(DELTA_EXT_BUF,"External channel terminator"))='\n';
00383     (h->terminator)[1]='\0';/*By default the terminator is '\n'*/
00384     /*Deep copy all strings:*/
00385     if(cmd!=NULL)
00386         h->cmd=(char *)strDupl((UBYTE *)cmd,"External channel command");
00387     else/*cmd cannot be NULL! If this is NULL then force it to be something special*/
00388         h->cmd=(char *)strDupl((UBYTE *)"/","External channel command");
00389     if(shellname!=NULL)
00390         h->shellname=
00391         (char *)strDupl((UBYTE *)shellname,"External channel shell name");
00392     if(stderrname!=NULL)
00393         h->stderrname=
00394         (char *)strDupl((UBYTE *)stderrname,"External channel stderr name");
00395 }/*extHandlerAlloc*/
00396
00397 /*Disallocates dynamically allocated fields of a handler:*/
00398 static FORM_INLINE void extHandlerFree(EXTHANDLE *h)
00399 {
00400     if(h->stderrname) M_free(h->stderrname,"External channel stderr name");
00401     if(h->shellname) M_free(h->shellname,"External channel shell name");
00402     if(h->cmd) M_free(h->cmd,"External channel command");
00403     if(h->terminator)M_free(h->terminator,"External channel terminator");
00404     if(h->INbuf)M_free(h->INbuf,"External channel buffer");
00405     extHandlerInit(h);

```



```

00406 */*extHandlerFree*/
00407 /* Closes all descriptors, kills the external process, frees all internal fields,
00408 BUT does NOT free the main container:*/
00409 static void destroyExternalChannel(EXTHANDLE *h)
00410 {
00411     /*Note, this function works in parallel mode correctly, see comments below.*/
00412     /*Note, for slaves in a parallel mode h->pid == 0:*/
00413     if( (h->pid > 0) && (h->killSignal > 0) ){
00414         int chstatus;
00415         if( h->gpuid > 0)
00416             chstatus=kill(-h->gpuid,h->killSignal);
00417         else
00418             chstatus=kill(h->pid,h->killSignal);
00419         if(chstatus==0)
00420             /*If the process will not be killed by this signal, FORM hangs up here!*/
00421             waitpid(h->pid, &chstatus, 0);
00422     }/*if( (h->pid > 0) && (h->killSignal > 0) )*/
00423     /*Note, for slaves in a parallel mode h->frec == h->fsend == 0:*/
00424     if(h->frec) fclose(h->frec);
00425     if( h->fsend > 0) close(h->fsend);
00426     extHandlerFree(h);
00427     /*Does not do "free(h)"!*/
00428 }/*destroyExternalChannel*/
00429
00430 /*Wrapper to the read() syscall, to handle possible interrupts by unblocked signals:*/
00431 static FORM_INLINE ssize_t read2b(int fd, char *buf, size_t count)
00432 {
00433     ssize_t res;
00434     if( (res=read(fd,buf,count)) <1 )/*EOF or read is interrupted by a signal?:*/
00435         while( (errno == EINTR)&&(res <1) )
00436             /*The call was interrupted by a signal before any data was read, try again:*/
00437             res=read(fd,buf,count);
00438     return (res);
00439 }/*read2b*/
00440
00441 /*Wrapper to the write() syscall, to handle possible interrupts by unblocked signals:*/
00442 static FORM_INLINE ssize_t writeFromb(int fd, char *buf, size_t count)
00443 {
00444     ssize_t res;
00445     if( (res=write(fd,buf,count)) <1 )/*Is write interrupted by a signal?:*/
00446         while( (errno == EINTR)&&(res <1) )
00447             /*The call was interrupted by a signal before any data was written, try again:*/
00448             res=write(fd,buf,count);
00449     return (res);
00450 }/*writeFromb*/
00451
00452 /* Read one (binary) PID from the file descriptor fd:*/
00453 static FORM_INLINE pid_t readpid(int fd)
00454 {
00455     pid_t tmp;
00456     if(read2b(fd, (char*)&tmp,sizeof(pid_t))!=sizeof(pid_t))
00457         return (pid_t)-1;
00458     return tmp;
00459 }/*readpid*/
00460
00461 /* Writeone (binary) PID to the file descriptor fd:*/
00462 static FORM_INLINE pid_t writepid(int fd, pid_t thepid)
00463 {
00464     if(writeFromb(fd, (char*)&thepid,sizeof(pid_t))!=sizeof(pid_t))
00465         return (pid_t)-1;
00466     return (pid_t)0;
00467 }/*writepid*/
00468
00469 /*Writes exactly count bytes from the buffer buf into the descriptor fd, independently on
00470 nonblocked signals and the MPU/buffer hits. Returns 0 or -1:
00471 */
00472 static FORM_INLINE int writexactly(int fd, char *buf, size_t count)
00473 {
00474     ssize_t i;
00475     int j=0,n=0;
00476     for(;;){
00477         if( (i=writeFromb(fd, buf+j, count-j)) < 0 ) return(-1);
00478         j+=i;
00479         if ( ((size_t)j) == count ) break;
00480         if(i==0)n++;
00481         else n=0;
00482         if(n>MAX_FAILS_IO)return (-1);
00483     }/*for(;;)*/
00484     return (0);
00485 }/*writexactly*/
00486
00487 /* Set the FD_CLOEXEC flag of desc if value is nonzero,

```

```

00493 or clear the flag if value is 0.
00494 Return 0 on success, or -1 on error with errno set. */
00495 static int set_cloexec_flag(int desc, int value)
00496 {
00497     int oldflags = fcntl (desc, F_GETFD, 0);
00498     /* If reading the flags failed, return error indication now.*/
00499     if (oldflags < 0)
00500         return (oldflags);
00501     /* Set just the flag we want to set. */
00502     if (value != 0)
00503         oldflags |= FD_CLOEXEC;
00504     else
00505         oldflags &= ~FD_CLOEXEC;
00506     /* Store modified flag word in the descriptor. */
00507     return (fcntl(desc, F_SETFD, oldflags));
00508 }/*set_cloexec_flag*/
00509
00510 /* Adds the integer fd to the array fifo of length top+1 so that
00511 the array is ascendantly ordered. It is supposed that all 0 -- top-1
00512 elements in the array are already ordered:*/
00513 static void pushDescriptor(int *fifo, int top, int fd)
00514 {
00515     if ( top == 0 ) {
00516         fifo[top] = fd;
00517     } else {
00518         int ins=top-1;
00519         if( fifo[ins]<=fd )
00520             fifo[top]=fd;
00521         else{
00522             /*Find the position:*/
00523             while( (ins>=0)&&(fifo[ins]>fd) )ins--;
00524             /*Move all elements starting from the position to the right:*/
00525             for(ins++;top>ins; top--)
00526                 fifo[top]=fifo[top-1];
00527             /*Put the element:*/
00528             fifo[ins]=fd;
00529         }
00530     }
00531 }/*pushDescriptor*/
00532
00533 /*Close all descriptors greater or equal than startFrom except those
00534 listed in the ascendantly ordered array usedFd of length top:*/
00535 static FORM_INLINE void closeAllDescriptors(int startFrom, int *usedFd, int top)
00536 {
00537     int n,maxfd;
00538     for(n=0;n<top; n++){
00539         maxfd=usedFd[n];
00540         for(;startFrom<maxfd;startFrom++)/*Close all less than maxfd*/
00541             close(startFrom);
00542         startFrom++;/*skip maxfd*/
00543     }/*for(;startFrom<maxfd;startFrom++)*/
00544     /*Close all the rest:*/
00545     maxfd=sysconf(_SC_OPEN_MAX);
00546     for(;startFrom<maxfd;startFrom++)
00547         close(startFrom);
00548 }/*closeAllDescriptors*/
00549
00550 typedef int L_APIPE[2];
00551 /*Closes both pipe descriptors if not -1:*/
00552 static void closepipe(L_APIPE *thepipe)
00553 {
00554     if( (*thepipe)[0] != -1) close ((*thepipe)[0]);
00555     if( (*thepipe)[1] != -1) close ((*thepipe)[1]);
00556 }/*closepipe*/
00557
00558 /*Parses the cmd line like "sh -c myprg", passes each option to the
00559 corresponding element of argv, ends agrv by NULL. Returns the
00560 number of stored argv elements, or -1 if fails:*/
00561 static FORM_INLINE int parseline(char **argv, char *cmd)
00562 {
00563     int n=0;
00564     while(*cmd != '\0'){
00565         for(; (*cmd <= ' ') && (*cmd != '\0') ;cmd++);
00566         if(*cmd != '\0'){
00567             argv[n]=cmd;
00568             while(++cmd > ' ');
00569             if(*cmd != '\0')
00570                 *cmd++ = '\0';
00571             n++;
00572         }/*if(*cmd != '\0')*/
00573     }/*while(*cmd != '\0')*/
00574     argv[n]=NULL;
00575     if(n==0)return -1;
00576     return n;
00577 }/*parseline*/
00578
00579 /*Reads positive decimal number (not bigger than maxnum)

```

```

00580     from the string and returns it;
00581     the pointer *b is set to the next non-converted character:*/
00582 static LONG str2i(char *str, char **b, LONG maxnum)
00583 {
00584     LONG n=0;
00585     /*Eat trailing spaces:*/
00586     while(*str<=' ')if(*str++ == '\0')return(-1);
00587     (*b)=str;
00588     while (*str>='0'&&*str<='9')
00589         if( (n=10*n + *str++ - '0')>maxnum )
00590             return(-1);
00591     if((*b)==str)/*No single number!*/
00592         return(-1);
00593     (*b)=str;
00594     return(n);
00595 }
00596
00597 /*Converts long integer to a decimal representation.
00598 For portability reasons we cannot use LongCopy from tools.c
00599 since theoretically LONG may be smaller than pid_t:*/
00600 static char *l2s(LONG x, char *to)
00601 {
00602     char *s;
00603     int i = 0, j;
00604     s = to;
00605     do { *s++ = (x % 10)+'0'; i++; } while ( ( x /= 10 ) != 0 );
00606     *s-- = '\0';
00607     j = ( i - 1 ) >> 1;
00608     while ( j >= 0 ) {
00609         i = to[j]; to[j] = s[-j]; s[-j] = (char)i; j--;
00610     }
00611     return(s+1);
00612 }
00613
00614 /*like strcat() but returns the pointer to the end of the
00615 resulting string:*/
00616 static FORM_INLINE char *addStr(char *to, char *from)
00617 {
00618     while( (*to++ = *from++) != '\0' );
00619     return(to-1);
00620 }/*addStr*/
00621
00622
00623 /*Try to write (atomically) short buffer (of length count) to fd.
00624 timeout is a timeout in millisecs. Returns number of written bytes or -1:*/
00625 static FORM_INLINE ssize_t writeSome(int fd, char *buf, size_t count, int timeout)
00626 {
00627     ssize_t res = 0;
00628     fd_set wfds;
00629     struct timeval tv;
00630     int nrep=5; /*five attempts it interrupted by a non-blocking signal*/
00631     int flags = fcntl(fd, F_GETFL, 0);
00632
00633     /*Add O_NONBLOCK:*/
00634     fcntl(fd, F_SETFL, flags | O_NONBLOCK);
00635     /* important -- in order to avoid blocking of short receiver buffer*/
00636
00637     do{
00638         FD_ZERO(&wfds);
00639         FD_SET(fd, &wfds);
00640         /* Wait up to timeout. */
00641
00642         tv.tv_sec =timeout /1000;
00643         tv.tv_usec = (timeout % 1000)*1000;
00644         nrep--;
00645
00646         switch(select(fd+1, NULL, &wfds, NULL, &tv)){
00647             case -1:
00648                 if((nrep == 0)|| ( errno != EINTR) ){
00649                     perror("select()");
00650                     res=-1;
00651                     nrep=0;
00652                 }/*else -- A non blocked signal was caught, just repeat*/
00653                 break;
00654             case 0:/*timeout*/
00655                 res=-1;
00656                 nrep=0;
00657                 break;
00658             default:
00659                 if( (res=write(fd,buf,count)) <0 )/*Signal?*/
00660                     while( (errno == EINTR)&&(res <0) )
00661                         res=write(fd,buf,count);
00662                 nrep=0;
00663             }/*switch*/
00664         }while(nrep);
00665
00666     }/*restore the flags:*/

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```

00667     fcntl(fd,F_SETFL, flags);
00668     return (res);
00669 }/*writeSome*/
00670
00671 /*Try to read short buffer (of length not more than count)
00672 from fd. timeout is a timeout in millisecs. Returns number
00673 of written bytes or -1: */
00674 static FORM_INLINE ssize_t readSome(int fd, char *buf, size_t count, int timeout)
00675 {
00676     ssize_t res = 0;
00677     fd_set rfd;
00678     struct timeval tv;
00679     int nrep=5;/*five attempts it interrupted by a non-blocking signal*/
00680
00681     do{
00682         FD_ZERO(&rfd);
00683         FD_SET(fd, &rfd);
00684         /* Wait up to timeout. */
00685
00686         tv.tv_sec = timeout/1000;
00687         tv.tv_usec = (timeout % 1000)*1000;
00688         nrep--;
00689
00690         switch(select(fd+1, &rfd, NULL, NULL, &tv)){
00691             case -1:
00692                 if((nrep == 0)|| (errno != EINTR) ){
00693                     perror("select()");
00694                     res=-1;
00695                     nrep=0;
00696                 }/*else -- A non blocked signal was caught, just repeat*/
00697                 break;
00698             case 0:/*timeout*/
00699                 res=-1;
00700                 nrep=0;
00701                 break;
00702             default:
00703                 if( (res=read(fd,buf,count)) <0 )/*Signal?*/
00704                     while( (errno == EINTR)&&(res <0) )
00705                         res=read(fd,buf,count);
00706                 nrep=0;
00707             }/*switch*/
00708         }while(nrep);
00709         return (res);
00710 }/*readSome*/
00711
00712 /*
00713 #] Local functions :
00714 #[ Ok functions:
00715 */
00716
00717 /*Copies (deep copy) newTerminator to thehandler->terminator. Returns 0 if
00718 newTerminator fits to the buffer, or !0 if it does not fit. ATT! In the
00719 latter case thehandler->terminator is NOT '\0' terminated! */
00720 int setTerminatorForExternalChannelOk(char *newTerminator)
00721 {
00722     int i=DELTA_EXT_BUF;
00723     /*
00724     No problems with externalChannelsListTop are
00725     possible since this function may be invoked only
00726     when the current channel is defined and externalChannelsListTop
00727     is set properly
00728     */
00729     char *t=externalChannelsListTop->terminator;
00730
00731     for( ; i>1; i--)
00732         if( (*t++ = *newTerminator++)=='\0' )
00733             break;
00734     /*Add trailing '\n', if absent:*/
00735     if( (i == DELTA_EXT_BUF)/*newTerminator == '\0'*/
00736         || (*t-2)!='\n' )
00737         {
00738             *(t-1)='\n';*t='\0';
00739         }
00740     return(i==1);
00741 }/*setTerminatorForExternalChannelOk*/
00742
00743 /*Interface to change handler fields "killSignal" and "gpId"*/
00744 int setKillModeForExternalChannelOk(int signum, int sentToWholeGroup)
00745 {
00746     if(signum<0)
00747         return(-1);
00748     /*
00749     No problems with externalChannelsListTop are
00750     possible since this function may be invoked only
00751     when the current channel is defined and externalChannelsListTop
00752     is set properly
00753     */
00754     externalChannelsListTop->killSignal=signum;

```

```

00754     if(sentToWholeGroup){/*gpid must be >0*/
00755         if(externalChannelsListTop->gpid <= 0)
00756             externalChannelsListTop->gpid=-externalChannelsListTop->gpid;
00757     }else{/*gpid must be <=0*/
00758         if(externalChannelsListTop->gpid>0)
00759             externalChannelsListTop->gpid=-externalChannelsListTop->gpid;
00760     }
00761     return(0);
00762 }/*setKillModeForExternalChannelOk*/
00763
00764 /*
00765     #[] getcFromExtChannelOk
00766 */
00767 /*Returns one character from the external channel. If the input is expired,
00768 returns EOF. If the external process is finished completely, the function closes
00769 the channel (and returns EOF). If the external process was finished, the function
00770 returns EOF:*/
00771 int getcFromExtChannelOk(void)
00772 {
00773     mysighandler_t oldPIPE = 0;
00774     EXTHANDLE *h;
00775     int ret;
00776
00777     if (externalChannelsListTop->IBfill < externalChannelsListTop->IBfull)
00778         /*in buffer*/
00779         return( *(externalChannelsListTop->IBfill++) );
00780     /*else -- the buffer is empty*/
00781     ret=EOF;
00782     h= externalChannelsListTop;
00783 #ifdef WITHMPI
00784     if ( PF.me == MASTER ){
00785 #endif
00786         /* Temporary ignore this signal:*/
00787         /* if compiler fails here, try to change the definition of
00788            mysighandler_t on the beginning of this file
00789            (just define INTSIGHANDLER).*/
00790         oldPIPE=signal(SIGPIPE,SIG_IGN);
00791 #ifdef WITHMPI
00792         if( fgets(h->INbuf,h->IBstop - h->INbuf, h->frec) == 0 )/*Fail! EOF?*/
00793             *(h->INbuf)='\0';/*Empty line may not appear here!*/
00794     #else
00795         if( ( fgets(h->INbuf,h->IBstop - h->INbuf, h->frec) == 0)/*Fail! EOF?*/
00796             || ( *(h->INbuf) == '\0')/*Empty line? This shouldn't be!*/
00797         ){
00798             closeExternalChannel(externalChannelsListTop-externalChannelsList+1);
00799             /*Note, this code is only for the sequential mode! */
00800             goto getcFromExtChannelReady;
00801             /*Here we assume that fgets is never interrupted by singals*/
00802         }/*if( fgets(h->INbuf,h->IBstop - h->INbuf, h->frec) == 0 )*/
00803     #endif
00804 #ifdef WITHMPI
00805     }/*if ( PF.me == MASTER */
00806
00807     /*Master broadcasts result to slaves, slaves read it from the master:*/
00808     if( PF_BroadcastString((UBYTE *)h->INbuf) ){/*Fail!*/
00809         MesPrint("Fail broadcasting external channel results");
00810         Terminate(-1);
00811     }/*if( PF_BroadcastString((UBYTE *)h->INbuf) )*/
00812
00813     if( *(h->INbuf) == '\0')/*Empty line? This shouldn't be!*/
00814         closeExternalChannel(externalChannelsListTop-externalChannelsList+1);
00815     goto getcFromExtChannelReady;
00816     }/*if( *(h->INbuf) == '\0')*/
00817 #endif
00818
00819     /*Block*/
00820     char *t=h->terminator;
00821     /*Move IBfull to the end of read line and compare the line with the terminator.
00822     Note, by construction the terminator fits to the first read line, see
00823     the function setTerminatorForExternalChannel.*/
00824     for(h->IBfull=h->INbuf; *(h->IBfull)!='\0'; (h->IBfull)++)
00825         if( *t== *(h->IBfull) )
00826             t++;
00827         else
00828             break;/*not a terminator*/
00829     /*Continue moving IBfull to the end of read line:*/
00830     while( *(h->IBfull)!='\0' ) (h->IBfull)++;
00831
00832     if( (t-h->terminator) == (h->IBfull-h->INbuf) ){
00833         /*Terminator!*/
00834         /*Reset the channel*/
00835         h->IBfull=h->IBfill=h->INbuf;
00836         externalChannelsListTop=0;/*Undefine the current channel*/
00837         writeBufToExtChannel=&writeBufToExtChannelFailure;
00838         getcFromExtChannel=&getcFromExtChannelFailure;
00839         setTerminatorForExternalChannel=&setTerminatorForExternalChannelFailure;
00840         setKillModeForExternalChannel=&setKillModeForExternalChannelFailure;

```

```

00841         goto getcFromExtChannelReady;
00842     }/*if(t == (h->IBfull-h->INbuf) )*/
00843 }/*Block*/
00844 /*Does the buffer have enough capacity?*/
00845 while( *(h->IBfull - 1) != '\n' ){/*Buffer is not enough!*/
00846     /*Extend the buffer:*/
00847     int l= (h->IBstop - h->INbuf)+DELTA_EXT_BUF;
00848     char *newbuf=Malloc1(l,"External channel buffer");
00849     /*We wouldn't like to use realloc.*/
00850     /*Copy the buffer:*/
00851     char *n=newbuf,*o=h->INbuf;
00852     while( (*n++ = *o++)!='\0' );
00853     /*Att! The order of the following operators is important!*/
00854     h->IBfull= newbuf+(h->IBfull-h->INbuf);
00855     M_free(h->INbuf,"External channel buffer");
00856     h->INbuf = newbuf;
00857     h->IBstop = h->INbuf+l;
00858 #ifdef WITHMPI
00859     if ( PF.me == MASTER ){
00860         (h->IBfull)[1]='\0';/*Will mark (h->IBfull)[1] as '!' for failure*/
00861         if( fgets(h->IBfull,h->IBstop - h->IBfull, h->frec) == 0 ){
00862             /*EOF! No trailing '\n'?*/
00863             /*Mark:*/
00864             (h->IBfull)[0]='\0';
00865             (h->IBfull)[1]='!';
00866             (h->IBfull)[2]='\0';
00867             /*The string "\0!\0" is used as an image of NULL.*/
00868         }/*if( fgets(h->IBfull,h->IBstop - h->IBfull, h->frec) == 0 )*/
00869     }/*if ( PF.me == MASTER )*/
00870
00871     /*Master broadcasts results to slaves, slaves read it from the master:*/
00872     if( PF_BroadcastString((UBYTE *)h->IBfull) ){/*Fail!*/
00873         MesPrint("Fail broadcasting external channel results");
00874         Terminate(-1);
00875     }/*if( PF_BroadcastString(h->IBfull) )*/
00876
00877     /*The string "\0!\0" is used as the image of NULL.*/
00878     if(
00879         ( (h->IBfull)[0]=='\0' )
00880         &&( (h->IBfull)[1]!='!' )
00881         &&( (h->IBfull)[2]=='\0' )
00882     ){/*EOF! No trailing '\n'?*/
00883         break;
00884     }else
00885         if( fgets(h->IBfull,h->IBstop - h->IBfull, h->frec) == 0 )
00886             /*EOF! No trailing '\n'?*/
00887             break;
00888     #endif
00889     while( *(h->IBfull)!='\0' ) (h->IBfull)++;
00890 }/*while( *(h->IBfull - 1) != '\n' )*/
00891 /*In h->INbuf we have a fresh string.*/
00892 ret=*(h->IBfill=h->INbuf);
00893 h->IBfill++;/*Next time a new, isn't it?*/
00894
00895 getcFromExtChannelReady:
00896 #ifdef WITHMPI
00897     if ( PF.me == MASTER ){
00898     #endif
00899         signal(SIGPIPE,oldPIPE);
00900     #ifdef WITHMPI
00901     }/*if ( PF.me == MASTER )*/
00902     #endif
00903     return(ret);
00904 }/*getcFromExtChannelOk*/
00905 /*
00906     #] getcFromExtChannelOk
00907 */
00908 /*Writes exactly count bytes from the buffer buf to the external channel thehandler
00909 Returns 0 (on success) or -1:
00910 */
00911 int writeBufToExtChannelOk(char *buf, size_t count)
00912 {
00913
00914 int ret;
00915 mysighandler_t oldPIPE;
00916 #ifdef WITHMPI
00917     /*Only master communicates with the external program:*/
00918     if ( PF.me == MASTER ){
00919     #endif
00920         /* Temporary ignore this signal:*/
00921         /* if compiler fails here, try to change the definition of
00922         mysighandler_t on the beginning of this file
00923         (just define INTSIGHANDLER)*/
00924         oldPIPE=signal(SIGPIPE,SIG_IGN);
00925         ret=writeexactly( externalChannelsListTop->fsend, buf, count);
00926         signal(SIGPIPE,oldPIPE);
00927     #ifdef WITHMPI

```

```

00928     }else{
00929         /*Do not wait the master status: this would be too slow!*/
00930         ret=0;
00931     }
00932 #endif
00933     return(ret);
00934 }/*writeBufToExtChannel*/
00935 /*
00936     #] Ok functions:
00937     #[ do_run_cmd :
00938 */
00939 /*The function returns PID of the started command*/
00940 static FORM_INLINE pid_t do_run_cmd(
00941     int *fdsend,
00942     int *fdreceive,
00943     int *gpid, /*returns group process ID*/
00944     int ttymode,
00945     /*
00946         &8 - daemonizeing
00947         &16 - setsid()*/
00948     char *cmd,
00949     char *argv[],
00950     char *stderrname
00951 )
00952 {
00953     int fdin[2]={-1,-1}, fdout[2]={-1,-1}, fdsig[2]={-1,-1};
00954     /*initialised by -1 for possible rollback at failure, see closepipe() above*/
00955
00956     pid_t childpid,fatherchildpid = (pid_t)0;
00957     mysighandler_t oldPIPE=NULL;
00958
00959     if(
00960         (pipe(fdsig)!=0)/*This pipe will be used by a child to tell the father if fail.*/
00961         || (pipe(fdin)!=0)
00962         || (pipe(fdout)!=0)
00963     )goto fail_do_run_cmd;
00964
00965     if((childpid = fork()) == -1){
00966         perror("fork");
00967         goto fail_do_run_cmd;
00968     }/*if((childpid = fork()) == -1)*/
00969
00970     if(childpid == 0){/*Child.*/
00971         int fifo[3], top=0;
00972         /*
00973             To be thread safely we can't rely on ascendant order of opened
00974             file descriptors. So we put each of descriptor we have to
00975             preserve into the array fifo.
00976             Note, in _this_ process there are no any threads but descriptors
00977             were created in frame of the parent process which may have
00978             multiple threads.
00979         */
00980         /*Mark descriptors which will NOT be closed:*/
00981         pushDescriptor(fifo,top++,fdsig[1]);
00982         pushDescriptor(fifo,top++,fdin[0]);
00983         pushDescriptor(fifo,top++,fdout[1]);
00984         /*Close all except stdin, stdout, stderr and placed into fifo:*/
00985         closeAllDescriptors(3,fifo, top);
00986
00987         /*Now reopen stdin and stdout.*/
00988         /*thread-safety is not a problem here since there are no any threads up to now:*/
00989         if(
00990             (close(0) == -1 )||/* Use fdin as stdin :*/
00991             (dup(fdin[0]) == -1 )||
00992             (close(1)==-1)||/* Use fdout as stdout:*/
00993             (dup(fdout[1]) == -1 )
00994         )
00995         {/*Fail!*/
00996             /*Signal to parent:*/
00997             writepid(fdsig[1],(pid_t)-2);
00998             _exit(1);
00999         }
01000
01001         if(stderrname != NULL){
01002             if(
01003                 (close(2) != 0 )||
01004                 (open(stderrname,O_WRONLY)<0)
01005             )
01006             {/*Fail!*/
01007                 writepid(fdsig[1],(pid_t)-2);
01008                 _exit(1);
01009             }
01010         }/*if(stderrname != NULL)*/
01011
01012         if( ttymode & 16 )/* create a session and sets the process group ID */
01013             setsid();
01014     }

```

```

01015      /* */
01016      if(set_cloexec_flag (fdsig[1], 1)!=0){/*Error?*/
01017          /*Signal to parent:*/
01018          writepid(fdsig[1],(pid_t)-2);
01019          _exit(1);
01020      }/*if(set_cloexec_flag (fdsig[1], 1)!=0)*/
01021
01022      if( ttymode & 8 ){/*Daemonize*/
01023          int fdsig2[2];/*To check exec() success*/
01024          if(
01025              pipe(fdsig2)||
01026              (set_cloexec_flag (fdsig2[1], 1)!=0)
01027          )
01028          {/*Error?*/
01029              /*Signal to parent:*/
01030              writepid(fdsig[1],(pid_t)-2);
01031              _exit(1);
01032          }
01033          set_cloexec_flag (fdsig2[0], 1);
01034          switch(childpid=fork()){
01035              case 0:/*grandchild*/
01036                  /*Execute external command:*/
01037                  execvp(cmd, argv);
01038                  /* Control can reach this point only on error!*/
01039                  writepid(fdsig2[1],(pid_t)-2);
01040                  break;
01041              case -1:
01042                  /* Control can reach this point only on error!*/
01043                  /*Inform the father about the failure*/
01044                  writepid(fdsig[1],(pid_t)-2);
01045                  _exit(1);/*The child, just exit, not return*/
01046              default:/*Son of his father*/
01047                  close(fdsig2[1]);
01048                  /*Ignore SIGPIPE (up to the end of the process):*/
01049                  signal(SIGPIPE,SIG_IGN);
01050
01051                  /*Wait on read() while the grandchild close the pipe
01052                     (on success) or send -2 (if exec() fails).*/
01053                  /*There are two possibilities:
01054                     -1 -- this is ok, the pipe was closed on exec,
01055                     the program was successfully executed;
01056                     -2 -- something is wrong, exec failed since the
01057                     grandchild sends -2 after exec.
01058                  */
01059                  if( readpid(fdsig2[0]) != (pid_t)-1 )/*something is wrong*/
01060                      writepid(fdsig[1],(pid_t)-1);
01061                  else/*ok, send PID of the grandchild to the father:*/
01062                      writepid(fdsig[1],childpid);
01063                  /*Die and free the life space for the grandchild:*/
01064                  _exit(0);/*The child, just exit, not return*/
01065          }/*switch(childpid=fork())*/
01066      }else{/*if( ttymode & 8 )*/
01067          execvp(cmd, argv);
01068          /* Control can reach this point only on error!*/
01069          writepid(fdsig[1],(pid_t)-2);
01070          _exit(2);/*The child, just exit, not return*/
01071      }/*if( ttymode & 8 )...else*/
01072  }else{/* The (grand)father*/
01073      close(fdsig[1]);
01074      /*To prevent closing fdsig in rollback:*/
01075      fdsig[1]=-1;
01076      close(fdin[0]);
01077      close(fdout[1]);
01078      *fdsend = fdin[1];
01079      *fdreceive = fdout[0];
01080
01081      /*Get the process group ID.*/
01082      /*Avoid to use getpgid() which is non-standard.*/
01083      if( ttymode & 16){/*setsid() was invoked, the child is a group leader:*/
01084          *gpid=childpid;
01085      }else/*the child belongs to the same process group as the this process:*/
01086          *gpid=getpgrp();/*if compiler fails here, try getpgrp(0) instead!*/
01087      /*
01088      Rationale: getpgrp conform to POSIX.1 while 4.3BSD provides a
01089      getpgrp() function that returns the process group ID for a
01090      specified process.
01091      */
01092
01093      /* Temporary ignore this signal:*/
01094      /* if compiler fails here, try to change the definition of
01095      mysighandler_t on the beginning of this file
01096      (just define INTSIGHANDLER)*/
01097      oldPIPE=signal(SIGPIPE,SIG_IGN);
01098
01099      if( ttymode & 8 ){/*Daemonize*/
01100          /*Read the grandchild PID from the son.*/
01101          fatherchildpid=childpid;

```



```

01102         if( (childpid=readpid(fdsig[0]))<0 ){
01103             /*Daemonization process fails for some reasons!*/
01104             childpid=fatherchildpid; /*for rollback*/
01105             goto fail_do_run_cmd;
01106         }
01107     }else{
01108         /*fdsig[1] should be closed on exec and this read operation
01109         must fail on success:*/
01110         if( readpid(fdsig[0])!= (pid_t)-1 )
01111             goto fail_do_run_cmd;
01112         }/*if( ttymode & 8 ) ... else*/
01113     }/*if(childpid == 0)...else*/
01114
01115     /*Here can be ONLY the father*/
01116     close(fdsig[0]);
01117     /*To prevent closing fdsig in rollback after goto fail_flnk_do_runcmd:*/
01118     fdsig[0]=-1;
01119
01120     if( ttymode & 8 ){/*Daemonize*/
01121         int i;
01122         /*Wait while the father of a grandchild dies:*/
01123         waitpid(fatherchildpid,&i,0);
01124     }
01125
01126     /*Restore the signal:*/
01127     signal(SIGPIPE,oldPIPE);
01128
01129     return(childpid);
01130
01131     fail_do_run_cmd:
01132         closepipe(&fdout);
01133         closepipe(&fdin);
01134         closepipe(&fdsig);
01135         return((pid_t)-1);
01136 }/*do_run_cmd*/
01137 /*
01138 #] do_run_cmd :
01139 #[ run_cmd :
01140 */
01141 /*Starts the command cmd (directly, if shellpath is NULL, or in a subshell),
01142 swallowing its stdin and stdout;
01143 stderr will be re-directed to stderrname (if !=NULL). Returns PID of
01144 the started process. Stdin will be available as fdsend, and stdout
01145 will be available as fdreceive:*/
01146 static FORM_INLINE pid_t run_cmd(char *cmd,
01147                                 int *fdsend,
01148                                 int *fdreceive,
01149                                 int *gpid,
01150                                 int daemonize,
01151                                 char *shellpath,
01152                                 char *stderrname )
01153 {
01154     char **argv;
01155     pid_t thepid;
01156
01157     cmd=(char*)strDup1( (UBYTE*)cmd, "run_cmd: cmd");/*detouch cmd*/
01158
01159
01160     /* Prepare arguments for execvp:*/
01161     if(shellpath != NULL){/*Run in a subshell.*/
01162         int nopt;
01163         /*Allocate space which is definitely enough:*/
01164         argv=Malloc1(StrLen( (UBYTE*)shellpath)*sizeof(char*)+2,"run_cmd:argv");
01165         shellpath=(char*)strDup1( (UBYTE*)shellpath, "run_cmd: shellpath");/*detouch shellpath*/
01166         /*Parse a shell (e.g., "/bin/sh -c"):*/
01167         nopt=parseline(argv, shellpath);
01168         /* and add the command as a shell argument:*/
01169         argv[nopt]=cmd;
01170         argv[nopt+1]=NULL;
01171     }else{/*Run the command directly:*/
01172         /*Allocate space which is definitely enough:*/
01173         argv=Malloc1(StrLen( (UBYTE*)cmd)*sizeof(char*)+1,"run_cmd:argv");
01174         parseline(argv, cmd);
01175     }
01176
01177     thepid=do_run_cmd(
01178         fdsend,
01179         fdreceive,
01180         gpid,
01181         (daemonize)?(8|16):0,
01182         argv[0],
01183         argv,
01184         stderrname
01185     );
01186
01187     M_free(argv,"run_cmd:argv");
01188     if(shellpath)

```

```

01189     M_free(shellpath, "run_cmd:argv");
01190     M_free(cmd, "run_cmd: cmd");
01191
01192     return(theppid);
01193 }/*run_cmd*/
01194 /*
01195     #] run_cmd :
01196     #[ createExternalChannel :
01197 */
01198 /*The structure to pass parameters to createExternalChannel
01199 and openExternalChannel in case of preset channel (instead of
01200 shellname):*/
01201 typedef struct{
01202     int fdin;
01203     int fdout;
01204     pid_t theppid;
01205 }ECINFOSTRUCT;
01206
01207 /* Creates a new external channel starting the command cmd (if cmd !=NULL)
01208 or using information from (ECINFOSTRUCT *)shellname, if cmd ==NULL:*/
01209 static FORM_INLINE void *createExternalChannel(
01210     EXTHANDLE *h,
01211     char *cmd, /*Command to run or NULL*/
01212     /*0 --neither setsid nor daemonize, !=0 -- full daemonization:*/
01213     int daemonize,
01214     char *shellname,/* The shell (like "/bin/sh -c") or NULL*/
01215     char *stderrname/*filename to redirect stderr or NULL*/
01216 )
01217 {
01218     int fdreceive=0;
01219     int gpid = 0;
01220     ECINFOSTRUCT *psetInfo;
01221 #ifdef WITHMPI
01222     char statusbuf[2]={'\0','\0'};/*'\0' if run_cmd returns ok, '!' otherwise.*/
01223 #endif
01224     extHandlerInit(h);
01225
01226     h->pid=0;
01227
01228     if( cmd==NULL ){/*Instead of starting a new command, use preset channel:*/
01229         psetInfo=(ECINFOSTRUCT *)shellname;
01230         shellname=NULL;
01231         h->killSignal=0;
01232         h->daemonize=0;
01233     }
01234
01235     /*Create a channel:*/
01236 #ifdef WITHMPI
01237     if ( PF.me == MASTER ){
01238 #endif
01239         if(cmd!=NULL)
01240             h->pid=run_cmd (cmd, &(h->fsend),
01241                             &fdreceive,&gpid,daemonize,shellname,stderrname);
01242         else{
01243             gpid=-psetInfo->theppid;
01244             h->pid=psetInfo->theppid;
01245             h->fsend=psetInfo->fdout;
01246             fdreceive=psetInfo->fdin;
01247         }
01248 #ifdef WITHMPI
01249         if(h->pid<0)
01250             statusbuf[0]='!';/*Broadcast fail to slaves*/
01251     }
01252     /*else: Keep h->pid = 0 and h->fsend = 0 for slaves in parallel mode!*/
01253
01254     /*Master broadcasts status to slaves, slaves read it from the master:*/
01255     if( PF_BroadcastString((UBYTE *)statusbuf) ){/*Fail!*/
01256         h->pid=-1;
01257     }else if( statusbuf[0]!='!')/*Master fails*/
01258         h->pid=-1;
01259 #endif
01260
01261     if(h->pid<0)goto createExternalChannelFails;
01262 #ifdef WITHMPI
01263     if ( PF.me == MASTER ){
01264 #endif
01265         h->gpid=gpid;
01266         /*Open stdout of a newly created program as FILE* :*/
01267         if( (h->frec=fdopen(fdreceive,"r")) == 0 )goto createExternalChannelFails;
01268
01269 #ifdef WITHMPI
01270     }
01271 #endif
01272     /*Initialize buffers:*/
01273     extHandlerAlloc(h,cmd,shellname,stderrname);
01274     return(h);
01275     /*Something is wrong?*/

```

```

01276     createExternalChannelFails:
01277
01278         destroyExternalChannel(h);
01279         return(NULL);
01280 }/*createExternalChannel*/
01281
01282 /*
01283  #] createExternalChannel :
01284  #[ openExternalChannel :
01285  */
01286 int openExternalChannel(UBYTE *cmd, int daemonize, UBYTE *shellname, UBYTE *stderrname)
01287 {
01288     EXTHANDLE *h=externalChannelsListTop;
01289     int i=0;
01290
01291     for (externalChannelsListTop=0;i<externalChannelsListFill;i++)
01292         if (externalChannelsList[i].cmd==0) {
01293             externalChannelsListTop=externalChannelsList+i;
01294             break;
01295         }/*if (externalChannelsList[i].cmd!=0)*/
01296
01297     if (externalChannelsListTop==0) {
01298         /*No free cells, create the new one:*/
01299         if (externalChannelsListFill == externalChannelsListStop) {
01300             /*The list is full, increase and reallocate it:*/
01301             EXTHANDLE *newbuf=Malloc1(
01302                 (externalChannelsListStop+DELTA_EXT_LIST)*sizeof(EXTHANDLE),
01303                 "External channel list");
01304             for (i=0;i<externalChannelsListFill;i++)
01305                 extHandlerSwallowCopy(newbuf+i,externalChannelsList+i);
01306             if (externalChannelsList!=0)
01307                 M_free(externalChannelsList,"External channel list");
01308             for (;i<externalChannelsListStop;i++)
01309                 extHandlerInit(newbuf+i);
01310             externalChannelsList=newbuf;
01311         }/*if (externalChannelsListFill == externalChannelsListStop)*/
01312         externalChannelsListTop=externalChannelsList+externalChannelsListFill;
01313         externalChannelsListFill++;
01314     }/*if (externalChannelsListTop==0)*/
01315
01316     if (createExternalChannel(
01317         externalChannelsListTop,
01318         (char*) cmd,
01319         daemonize,
01320         (char*) shellname,
01321         (char*) stderrname
01322         ) != NULL) {
01323         writeBufToExtChannel=&writeBufToExtChannelOk;
01324         getcFromExtChannel=&getcFromExtChannelOk;
01325         setTerminatorForExternalChannel=&setTerminatorForExternalChannelOk;
01326         setKillModeForExternalChannel=&setKillModeForExternalChannelOk;
01327         return(externalChannelsListTop-externalChannelsList+1);
01328     }
01329     /*else - failure!*/
01330     externalChannelsListTop=h;
01331     return(-1);
01332 }/*openExternalChannel*/
01333
01334 /*
01335  #] openExternalChannel :
01336  #[ initPresetExternalChannels :
01337  */
01338 /*Just simple wrapper to invoke openExternalChannel()
01339 from initPresetExternalChannels():*/
01340 static FORM_INLINE int openPresetExternalChannel(int fdin, int fdout, pid_t theppid)
01341 {
01342     ECINFOSTRUCT inf;
01343     inf.fdin=fdin; inf.fdout=fdout; inf.theppid=theppid;
01344     return( openExternalChannel(NULL,0,(UBYTE *)&inf,NULL) );
01345 }/*openPresetExternalChannel*/
01346
01347 #define PIDTXTSIZE 23
01348 #define BOTHPIDSIZE 45
01349
01350 #ifndef LONG_MAX
01351 #define LONG_MAX 0x7FFFFFFFL
01352 #endif
01353
01354 /*There is a possibility to start FORM from another program providing
01355 one (or more) communication channels. These channels will be
01356 visible from a FORM program as ``pre-opened'' external channels existing
01357 after FORM starts.
01358 Before starting FORM, the parent application must create one or more
01359 pairs of pipes The read-only descriptor of the first pipe in the pair
01360 and the write-only descriptor of the second pipe must be passed to
01361 FORM as an argument of a command line option -pipe in ASCII decimal
01362 format. The argument of the option is a comma-separated list of pairs
01363 r#,w# where r# is a read-only descriptor and w# is a write-only
01364 descriptor. Alternatively, the environment variable FORM_PIPES can be

```

```

01363 used.
01364 The following function expects as the first argument
01365 this comma-separated list of the descriptor pairs and tries to
01366 initialize each of channel during the timeout milliseconds:*/
01367
01368 int initPresetExternalChannels(UBYTE *theline, int thetimeout)
01369 {
01370     int i, nchannels = 0;
01371     int pidln;
01372     char pidtxt[PIDTXTSIZE], /*The length of FORM PID in pidtxt*/
01373         chdescriptor[PIDTXTSIZE], /*64/Log_2[10] = 19.3, this is enough for any integer*/
01374         bothpidtxt[BOTHPIDSIZE], /*"#,\n\0"*/
01375         *c,*b = 0;
01376     int pip[2];
01377     pid_t ppid;
01378     if ( theline == NULL ) return(-1);
01379     /*Put into pidtxt PID\n:*/
01380     c = l2s((LONG)getpid(),pidtxt);
01381     *c++='\n';
01382     *c = '\0';
01383     pidln = c-pidtxt;
01384
01385     do {
01386         pip[0] = (int)str2i((char*)theline,&c,0xFFFF);
01387         if( ( pip[0] < 0 ) || ( *c != ',' ) ) goto presetFails;
01388
01389         theline = (UBYTE*)c + 1;
01390         pip[1] = (int)str2i((char*)theline,&c,0xFFFF);
01391         if ( (pip[1] < 0 ) || ( ( *c != ',' ) && ( *c != '\0' ) ) ) goto presetFails;
01392         theline = (UBYTE *)c + 1;
01393         /*Now we have two descriptors.
01394            According to the protocol, FORM must send to external channel
01395            it's PID with added '\n' and read back two comma-separated
01396            decimals with added '\n'. The first must be repeated FORM PID,
01397            the second must be the parent PID
01398            */
01399         if ( writeSome(pip[1],pidtxt,pidln,thetimeout) != pidln ) goto presetFails;
01400         i = readSome(pip[0],bothpidtxt,BOTHPIDSIZE,thetimeout);
01401         if( ( i < 4 ) /*at least 1,2\n*/
01402             || ( i == BOTHPIDSIZE ) /*No space for trailing '\0'*/
01403         ) goto presetFails;
01404         /*read the FORM PID:*/
01405         ppid = (pid_t)str2i(bothpidtxt,&b,getpid());
01406         if( ( *b != ',' ) || ( ppid != getpid() ) ) goto presetFails;
01407         /*read the parent PID:*/
01408         /*The problem is that we do not know the the real type of pid_t.
01409            But long should be enough. On obsolete systems (when LONG_MAX
01410            is not defined) we assume pid_t is 32-bit integer.
01411            This can lead to problem with portability: */
01412         ppid = (pid_t)str2i(b+1,&b,LONG_MAX);
01413         if ( (*b != '\n') || ( ppid < 2 ) ) goto presetFails;
01414         i = openPresetExternalChannel(pip[0],pip[1],ppid);
01415         if ( i < 0 ) goto presetFails;
01416         nchannels++;
01417         /*Now use bothpidtxt as a buffer for preprovar, the space is enough:*/
01418         /*"PIPE#" where # is ne order number of the channel:*/
01419         b = l2s(nchannels,addStr(bothpidtxt,"PIPE"));
01420         *b = '\n';
01421         b[1] = '\0';
01422         *l2s(i,chdescriptor) = '\0';
01423         PutPreVar((UBYTE*)bothpidtxt, (UBYTE*)chdescriptor,0,0);
01424     } while ( *c != '\0' );
01425     /*Export preprovar "PIPES_":*/
01426     *l2s(nchannels,chdescriptor)='\0';
01427     PutPreVar((UBYTE*)"PIPES_", (UBYTE*)chdescriptor,0,0);
01428
01429     /*success:*/
01430     return (nchannels);
01431
01432 presetFails:
01433     /*Here we assume the descriptors the beginning of the list!*/
01434     for(i=0; i<nchannels; i++)
01435         destroyExternalChannel(externalChannelsList+i);
01436     return(-1);
01437 } /*initPresetExternalChannels*/
01438 /*
01439 #] initPresetExternalChannels :
01440 #[ selectExternalChannel :
01441 */
01442 /*
01443 Accepts the valid external channel descriptor (returned by
01444 openExternalChannel) and returns the descriptor of a previous current
01445 channel (0, if there was no current channel, or -1, if the external
01446 channel descriptor is invalid). If n == 0, the function undefine the
01447 current external channel:
01448 */
01449 int selectExternalChannel(int n)

```

```

01450 {
01451     int ret=0;
01452     if (externalChannelsListTop!=0)
01453         ret=externalChannelsListTop-externalChannelsList+1;
01454
01455     if (--n<0) {
01456         if (n!=-1)
01457             return(-1);
01458         externalChannelsListTop=0;
01459         writeBufToExtChannel=&writeBufToExtChannelFailure;
01460         getcFromExtChannel=&getcFromExtChannelFailure;
01461         setTerminatorForExternalChannel=&setTerminatorForExternalChannelFailure;
01462         setKillModeForExternalChannel=&setKillModeForExternalChannelFailure;
01463         return(ret);
01464     }
01465     if (
01466         (n>=externalChannelsListFill) ||
01467         (externalChannelsList[n].cmd==0)
01468     )
01469         return(-1);
01470
01471     externalChannelsListTop=externalChannelsList+n;
01472     writeBufToExtChannel=&writeBufToExtChannelOk;
01473     getcFromExtChannel=&getcFromExtChannelOk;
01474     setTerminatorForExternalChannel=&setTerminatorForExternalChannelOk;
01475     setKillModeForExternalChannel=&setKillModeForExternalChannelOk;
01476     return(ret);
01477 } /*selectExternalChannel*/
01478 /*
01479     #] selectExternalChannel :
01480     #[ closeExternalChannel :
01481 */
01482 /*
01483 */
01484 Destroys the opened external channel with the descriptor n. It returns
01485 0 in success, or -1 on failure. If the corresponding external channel
01486 was the current one, the current channel becomes undefined. If n==0,
01487 the function closes the current external channel.
01488 */
01489 int closeExternalChannel(int n)
01490 {
01491     if (n==0)
01492         n=externalChannelsListTop-externalChannelsList;
01493     else
01494         n--; /*Count from 0*/
01495
01496     if (
01497         (n<0) ||
01498         (n>=externalChannelsListFill) ||
01499         (externalChannelsList[n].cmd==0)
01500     ) /*No such a channel*/
01501         return(-1);
01502
01503     destroyExternalChannel(externalChannelsList+n);
01504     /*If the current external channel was destroyed, undefine current channel:*/
01505     if (externalChannelsListTop==externalChannelsList+n) {
01506         externalChannelsListTop=NULL;
01507         writeBufToExtChannel=&writeBufToExtChannelFailure;
01508         getcFromExtChannel=&getcFromExtChannelFailure;
01509         setTerminatorForExternalChannel=&setTerminatorForExternalChannelFailure;
01510         setKillModeForExternalChannel=&setKillModeForExternalChannelFailure;
01511     } /*if (externalChannelsListTop==externalChannelsList+n)*/
01512     return(0);
01513 } /*closeExternalChannel*/
01514 /*
01515     #] closeExternalChannel :
01516     #[ closeAllExternalChannels :
01517 */
01518 void closeAllExternalChannels(void)
01519 {
01520     int i;
01521     for (i=0; i<externalChannelsListFill; i++)
01522         destroyExternalChannel(externalChannelsList+i);
01523     externalChannelsListFill=externalChannelsListStop=0;
01524     externalChannelsListTop=NULL;
01525
01526     writeBufToExtChannel=&writeBufToExtChannelFailure;
01527     getcFromExtChannel=&getcFromExtChannelFailure;
01528     setTerminatorForExternalChannel=&setTerminatorForExternalChannelFailure;
01529     setKillModeForExternalChannel=&setKillModeForExternalChannelFailure;
01530
01531     if (externalChannelsList!=NULL) {
01532         M_free(externalChannelsList, "External channel list");
01533         externalChannelsList=NULL;
01534     }
01535 } /*closeAllExternalChannels*/
01536 /*

```

```

01537     #] closeAllExternalChannels :
01538     #[ getExternalChannelPid :
01539 */
01540 #ifdef SELFTEST
01541 pid_t getExternalChannelPid(void)
01542 {
01543     if(externalChannelsListTop!=0)
01544         return(externalChannelsListTop ->pid);
01545     return(-1);
01546 }/*getExternalChannelPid*/
01547 #endif
01548 /*
01549     #] getExternalChannelPid :
01550     #[ getCurrentExternalChannel :
01551 */
01552 int getCurrentExternalChannel(void)
01553 {
01554     if ( externalChannelsListTop != 0 )
01555         return(externalChannelsListTop-externalChannelsList+1);
01556     return(0);
01557 }/*getCurrentExternalChannel*/
01558 /*
01559     #] getCurrentExternalChannel :
01560     #[ Selftest main :
01561 */
01562 #ifdef SELFTEST
01563 /*
01564     This is the example of how all these public functions may be used:
01565 */
01566 char buf[1024];
01567 char buf2[1024];
01568 void help(void)
01569 {
01570     printf("String started with a special letter is a command\n");
01571     printf("Known commands are:\n");
01572     printf("H or ? -- this help\n");
01573     printf("Nn<command> -- start a new command\n");
01574     printf("S<command> -- start a new command in a subshell,daemon,stderr>/dev/null\n");
01575     printf("C# -- destroy channel #\n");
01576     printf("R# -- set a new cahhel(number#) as a current one\n");
01577     printf("K#1 #2 -- set signal for kill and kill mode (0 or !=0)\n");
01578     printf("    ^d to quit\n");
01579 }/*help*/
01580 int main (void)
01581 {
01582     int i, j, k,last;
01583     long long sum = 0;
01584     /*openExternalChannel(UBYTE *cmd, int daemonize, UBYTE *shellname, UBYTE *stderrname)*/
01585     help();
01586     printf("Initial channel:%d\n",last=openExternalChannel((UBYTE*)"cat",0,NULL,NULL));
01587     if( ( i = setTerminatorForExternalChannel("qu") ) != 0 ) return 1;
01588     printf("Terminator is 'qu'\n");
01589     while ( fgets(buf, 1024, stdin) != NULL ) {
01590         if ( *buf == 'N' ) {
01591             printf("New channel:%d\n",j=openExternalChannel((UBYTE*)buf+1,0,NULL,NULL));
01592             continue;
01593         }
01594         else if ( *buf == 'C' ) {
01595             int n;
01596             sscanf(buf+1,"%d",&n);
01597             printf("Destroy last channel:%d\n",closeExternalChannel(n));
01598             continue;
01599         }
01600         else if ( *buf == 'R' ) {
01601             int n = 0;
01602             sscanf(buf+1,"%d",&n);
01603             printf("Reopen channel %d:%d\n",n,selectExternalChannel(n));
01604             continue;
01605         }else if( *buf == 'K' ) {
01606             int n=0,g = 0;
01607             sscanf(buf+1,"%d %d",&n,&g);
01608             printf("setKillMode %d\n",setKillModeForExternalChannel(n,g));
01609             continue;
01610         }else if( *buf == 'S' ) {
01611             printf("New channel with sh&d&stderr:%d\n",

```

```

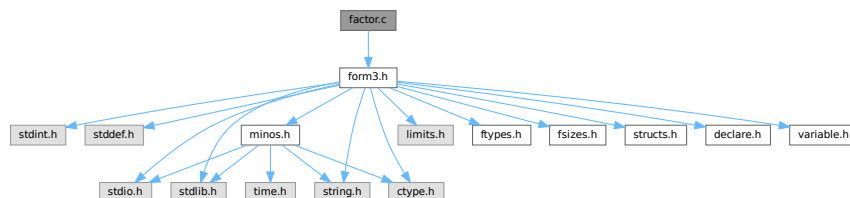
01624         j=openExternalChannel((UBYTE*)buf+1,1,(UBYTE*)" /bin/sh -c", (UBYTE*)" /dev/null");
01625         continue;
01626     }
01627     else if( ( *buf == 'H' ) || ( *buf == '?' ) ){
01628         help();
01629         continue;
01630     }
01631
01632     writeBufToExtChannel(buf,k=StrLen(buf));
01633     sum += k;
01634     for ( j = 0; ( i = getcFromExtChannel() ) != '\n'; j++) {
01635         if ( i == EOF ) {
01636             printf("EOF!\n");
01637             break;
01638         }
01639         buf2[j] = i;
01640     }
01641     buf2[j] = '\0';
01642     printf("I got:'%s'; pid=%d\n",buf2,getExternalChannelPid());
01643 }
01644 printf("Total:%lld bytes\n",sum);
01645 closeAllExternalChannels();
01646 return 0;
01647 }
01648 #endif /*ifdef SELFTEST*/
01649 /*
01650     #] Selftest main :
01651 */
01652
01653 #endif /*ifndef WITHEXTERNALCHANNEL ... else*/

```

## 4.33 factor.c File Reference

```
#include "form3.h"
```

Include dependency graph for factor.c:



### Functions

- int [FactorIn](#) (PHEAD WORD \*term, WORD level)
- int [FactorInExpr](#) (PHEAD WORD \*term, WORD level)

#### 4.33.1 Detailed Description

The routines for finding (one term) factors in dollars and expressions.

Definition in file [factor.c](#).

## 4.33.2 Function Documentation

### 4.33.2.1 FactorIn()

```
int FactorIn (
    PHEAD WORD * term,
    WORD level )
```

Definition at line 46 of file [factor.c](#).

### 4.33.2.2 FactorInExpr()

```
int FactorInExpr (
    PHEAD WORD * term,
    WORD level )
```

Definition at line 421 of file [factor.c](#).

## 4.34 factor.c

[Go to the documentation of this file.](#)

```
00001
00005 /* #[] License : */
00006 /*
00007  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00008  * When using this file you are requested to refer to the publication
00009  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010  * This is considered a matter of courtesy as the development was paid
00011  * for by FOM the Dutch physics granting agency and we would like to
00012  * be able to track its scientific use to convince FOM of its value
00013  * for the community.
00014  *
00015  * This file is part of FORM.
00016  *
00017  * FORM is free software: you can redistribute it and/or modify it under the
00018  * terms of the GNU General Public License as published by the Free Software
00019  * Foundation, either version 3 of the License, or (at your option) any later
00020  * version.
00021  *
00022  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00023  * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00024  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00025  * details.
00026  *
00027  * You should have received a copy of the GNU General Public License along
00028  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00029  */
00030 /* #[] License : */
00031 /*
00032  * #[] Includes : factor.c
00033  */
00034 #include "form3.h"
00035
00036 /*
00037  * #[] Includes :
00038  * #[] FactorIn :
00039  *
00040  * This routine tests for a factor in a dollar expression.
00041  *
00042  * Note that unlike with regular active or hidden expressions we cannot
00043  * add memory as easily as dollars are rather volatile.
00044  */
00045
00046 int FactorIn(PHEAD WORD *term, WORD level)
00047 {
00048     GETBIDENTITY
00049     WORD *t, *tstop, *m, *mm, *oldwork, *mstop, *n1, *n2, *n3, *n4, *n1stop, *n2stop;
00050     WORD *r1, *r2, *r3, *r4, j, k, kGCD, kGCD2, kLCM, jGCD, kkLCM, jLCM, size;
00051     UWORD *GCdbuffer, *GCdbuffer2, *LCMbuffer, *LCMb, *LCMc;
```



```

00052     int fromwhere = 0, i;
00053     DOLLARS d;
00054     t = term; GETSTOP(t,tstop); t++;
00055     while ( ( t < tstop ) && ( *t != FACTORIN || ( ( *t == FACTORIN )
00056         && ( t[FUNHEAD] != -DOLLAREXPRESSON || t[1] != FUNHEAD+2 ) ) ) ) t += t[1];
00057     if ( t >= tstop ) {
00058         MLOCK(ErrorMessageLock);
00059         MesPrint("Internal error. Could not find proper factorin_ function.");
00060         MUNLOCK(ErrorMessageLock);
00061         return(-1);
00062     }
00063     oldwork = AT.WorkPointer;
00064     d = Dollars + t[FUNHEAD+1];
00065     #ifdef WITHPTHREADS
00066     {
00067         int nummodopt, dtype = -1;
00068         if ( AS.MultiThreaded ) {
00069             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
00070                 if ( t[FUNHEAD+1] == ModOptdollars[nummodopt].number ) break;
00071             }
00072             if ( nummodopt < NumModOptdollars ) {
00073                 dtype = ModOptdollars[nummodopt].type;
00074                 if ( dtype == MODLOCAL ) {
00075                     d = ModOptdollars[nummodopt].dstruct+AT.identity;
00076                 }
00077             }
00078         }
00079     }
00080     #endif
00081     if ( d->type == DOLTERMS ) {
00082         fromwhere = 1;
00083     }
00084     else if ( ( d = DolToTerms(BHEAD t[FUNHEAD+1]) ) == 0 ) {
00085         /*
00086          * The variable cannot convert to an expression
00087          * We replace the function by 1.
00088          */
00089         m = oldwork; n1 = term;
00090         while ( n1 < t ) *m++ = *n1++;
00091         n1 = t + t[1]; tstop = term + *term;
00092         while ( n1 < tstop ) *m++ = *n1++;
00093         *oldwork = m - oldwork;
00094         AT.WorkPointer = m;
00095         if ( Generator(BHEAD oldwork,level) ) return(-1);
00096         AT.WorkPointer = oldwork;
00097         return(0);
00098     }
00099     if ( d->where[0] == 0 ) {
00100         if ( fromwhere == 0 ) {
00101             if ( d->factors ) M_free(d->factors,"Dollar factors");
00102             M_free(d,"Dollar in FactorIn_");
00103         }
00104         return(0);
00105     }
00106     /*
00107     Now we have an expression in d->where. Find the symbolic factor that
00108     divides the expression and the numerical factor that makes all
00109     coefficients integer.
00110     For the symbolic factor we make a copy of the first term, and then
00111     go through all terms, scratching in the copy the objects that do not
00112     occur in the terms.
00113     */
00114     m = oldwork;
00115     mm = d->where;
00116     k = *mm - ABS((mm[*mm-1]));
00117     for ( j = 0; j < k; j++ ) *m++ = *mm++;
00118     mstop = m;
00119     *oldwork = k;
00120     /*
00121     The copy is in place. Now search through the terms. Start at the second term
00122     */
00123     mm = d->where + d->where[0];
00124     while ( *mm ) {
00125         m = oldwork+1;
00126         r2 = mm+*mm;
00127         r2 -= ABS(r2[-1]);
00128         r1 = mm+1;
00129         while ( m < mstop ) {
00130             while ( r1 < r2 ) {
00131                 if ( *r1 != *m ) {
00132                     r1 += r1[1]; continue;
00133                 }
00134             }
00135             /*
00136             Now the various cases
00137             */
00138             #[ SYMBOL :

```

```

00139 */
00140         if ( *m == SYMBOL ) {
00141             n1 = m+2; n1stop = m+m[1];
00142             n2stop = r1+r1[1];
00143             while ( n1 < n1stop ) {
00144                 n2 = r1+2;
00145                 while ( n2 < n2stop ) {
00146                     if ( *n1 != *n2 ) { n2 += 2; continue; }
00147                     if ( n1[1] > 0 ) {
00148                         if ( n2[1] < 0 ) { n2 += 2; continue; }
00149                         if ( n2[1] < n1[1] ) n1[1] = n2[1];
00150                     }
00151                     else {
00152                         if ( n2[1] > 0 ) { n2 += 2; continue; }
00153                         if ( n2[1] > n1[1] ) n1[1] = n2[1];
00154                     }
00155                     break;
00156                 }
00157                 if ( n2 >= n2stop ) { /* scratch symbol */
00158                     if ( m[1] == 4 ) goto scratch;
00159                     m[1] -= 2;
00160                     n3 = n1; n4 = n1+2;
00161                     while ( n4 < mstop ) *n3++ = *n4++;
00162                     *oldwork = n3 - oldwork;
00163                     mstop -= 2; n1stop -= 2;
00164                     continue;
00165                 }
00166                 n1 += 2;
00167             }
00168             break;
00169         }
00170 /*
00171 #] SYMBOL :
00172 #[ DOTPRODUCT :
00173 */
00174         else if ( *m == DOTPRODUCT ) {
00175             n1 = m+2; n1stop = m+m[1];
00176             n2stop = r1+r1[1];
00177             while ( n1 < n1stop ) {
00178                 n2 = r1+2;
00179                 while ( n2 < n2stop ) {
00180                     if ( *n1 != *n2 || n1[1] != n2[1] ) { n2 += 3; continue; }
00181                     if ( n1[2] > 0 ) {
00182                         if ( n2[2] < 0 ) { n2 += 3; continue; }
00183                         if ( n2[2] < n1[2] ) n1[2] = n2[2];
00184                     }
00185                     else {
00186                         if ( n2[2] > 0 ) { n2 += 3; continue; }
00187                         if ( n2[2] > n1[2] ) n1[2] = n2[2];
00188                     }
00189                     break;
00190                 }
00191                 if ( n2 >= n2stop ) { /* scratch symbol */
00192                     if ( m[1] == 5 ) goto scratch;
00193                     m[1] -= 3;
00194                     n3 = n1; n4 = n1+3;
00195                     while ( n4 < mstop ) *n3++ = *n4++;
00196                     *oldwork = n3 - oldwork;
00197                     mstop -= 3; n1stop -= 3;
00198                     continue;
00199                 }
00200                 n1 += 3;
00201             }
00202             break;
00203         }
00204 /*
00205 #] DOTPRODUCT :
00206 #[ VECTOR :
00207 */
00208         else if ( *m == VECTOR ) {
00209             /*
00210              Here we have to be careful if there is more than
00211              one of the same
00212              */
00213             n1 = m+2; n1stop = m+m[1];
00214             n2 = r1+2; n2stop = r1+r1[1];
00215             while ( n1 < n1stop ) {
00216                 while ( n2 < n2stop ) {
00217                     if ( *n1 == *n2 && n1[1] == n2[1] ) {
00218                         n2 += 2; goto nextn1;
00219                     }
00220                     n2 += 2;
00221                 }
00222                 if ( n2 >= n2stop ) { /* scratch symbol */
00223                     if ( m[1] == 4 ) goto scratch;
00224                     m[1] -= 2;
00225                     n3 = n1; n4 = n1+2;

```

```

00226             while ( n4 < mstop ) *n3++ = *n4++;
00227             *oldwork = n3 - oldwork;
00228             mstop -= 2; n1stop -= 2;
00229             continue;
00230         }
00231         n2 = r1+2;
00232 nextn1:      n1 += 2;
00233     }
00234     break;
00235 }
00236 /*
00237 #] VECTOR :
00238 #[ REMAINDER :
00239 */
00240     else {
00241 /*
00242         Again: watch for multiple occurrences of the same object
00243 */
00244         if ( m[1] != r1[1] ) { r1 += r1[1]; continue; }
00245         for ( j = 2; j < m[1]; j++ ) {
00246             if ( m[j] != r1[j] ) break;
00247         }
00248         if ( j < m[1] ) { r1 += r1[1]; continue; }
00249         r1 += r1[1]; /* to restart at the next potential match */
00250         goto nextm; /* match */
00251     }
00252 /*
00253 #] REMAINDER :
00254 */
00255     }
00256     if ( r1 >= r2 ) { /* no factor! */
00257 scratch:;
00258         r3 = m + m[1]; r4 = m;
00259         while ( r3 < mstop ) *r4++ = *r3++;
00260         *oldwork = r4 - oldwork;
00261         if ( *oldwork == 1 ) goto nofactor;
00262         mstop = r4;
00263         r1 = mm + 1;
00264         continue;
00265     }
00266     r1 = mm + 1;
00267 nextm:      m += m[1];
00268     }
00269     mm = mm + *mm;
00270 }
00271
00272 nofactor:;
00273 /*
00274     For the coefficient we have to determine the LCM of the denominators
00275     and the GCD of the numerators.
00276 */
00277     GCDBuffer = NumberMalloc("FactorIn"); GCDBuffer2 = NumberMalloc("FactorIn");
00278     LCMbuffer = NumberMalloc("FactorIn"); LCMb = NumberMalloc("FactorIn"); LCMc =
00279     NumberMalloc("FactorIn");
00280     r1 = d->where;
00281 /*
00282     First take the first term to load up the LCM and the GCD
00283 */
00284     r2 = r1 + *r1;
00285     j = r2[-1];
00286     r3 = r2 - ABS(j);
00287     k = REDLENG(j);
00288     if ( k < 0 ) k = -k;
00289     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00290     for ( kGCD = 0; kGCD < k; kGCD++ ) GCDBuffer[kGCD] = r3[kGCD];
00291     k = REDLENG(j);
00292     if ( k < 0 ) k = -k;
00293     r3 += k;
00294     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00295     for ( kLCM = 0; kLCM < k; kLCM++ ) LCMbuffer[kLCM] = r3[kLCM];
00296     r1 = r2;
00297 /*
00298     Now go through the rest of the terms in this dollar buffer.
00299 */
00300     while ( *r1 ) {
00301         r2 = r1 + *r1;
00302         j = r2[-1];
00303         r3 = r2 - ABS(j);
00304         k = REDLENG(j);
00305         if ( k < 0 ) k = -k;
00306         while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00307         if ( ( GCDBuffer[0] == 1 ) && ( kGCD == 1 ) ) {
00308             GCD is already 1
00309         }
00310     }
00311     else if ( ( k != 1 ) || ( r3[0] != 1 ) ) {

```

```

00312         if ( GcdLong(BHEAD GCDbuffer,kGCD, (UWORD *)r3,k,GCDbuffer2,&kGCD2) ) {
00313             goto onerror;
00314         }
00315         kGCD = kGCD2;
00316         for ( i = 0; i < kGCD; i++ ) GCDbuffer[i] = GCDbuffer2[i];
00317     }
00318     else {
00319         kGCD = 1; GCDbuffer[0] = 1;
00320     }
00321     k = REDLENG(j);
00322     if ( k < 0 ) k = -k;
00323     r3 += k;
00324     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00325     if ( ( LCMbuffer[0] == 1 ) && ( kLCM == 1 ) ) {
00326         for ( kLCM = 0; kLCM < k; kLCM++ )
00327             LCMbuffer[kLCM] = r3[kLCM];
00328     }
00329     else if ( ( k != 1 ) || ( r3[0] != 1 ) ) {
00330         if ( GcdLong(BHEAD LCMbuffer,kLCM, (UWORD *)r3,k,LCMb,&kLCM) ) {
00331             goto onerror;
00332         }
00333         DivLong((UWORD *)r3,k,LCMb,kLCM,LCMb,&kLCM,LCMc,&jLCM);
00334         Mullong(LCMbuffer,kLCM,LCMb,kLCM,LCMc,&jLCM);
00335         for ( kLCM = 0; kLCM < jLCM; kLCM++ )
00336             LCMbuffer[kLCM] = LCMc[kLCM];
00337     }
00338     else {} /* LCM doesn't change */
00339     r1 = r2;
00340 }
00341 /*
00342 Now put the factor together: GCD/LCM
00343 */
00344 r3 = (WORD *) (GCDbuffer);
00345 if ( kGCD == kLCM ) {
00346     for ( jGCD = 0; jGCD < kGCD; jGCD++ )
00347         r3[jGCD+kGCD] = LCMbuffer[jGCD];
00348     k = kGCD;
00349 }
00350 else if ( kGCD > kLCM ) {
00351     for ( jGCD = 0; jGCD < kLCM; jGCD++ )
00352         r3[jGCD+kGCD] = LCMbuffer[jGCD];
00353     for ( jGCD = kLCM; jGCD < kGCD; jGCD++ )
00354         r3[jGCD+kGCD] = 0;
00355     k = kGCD;
00356 }
00357 else {
00358     for ( jGCD = kGCD; jGCD < kLCM; jGCD++ )
00359         r3[jGCD] = 0;
00360     for ( jGCD = 0; jGCD < kLCM; jGCD++ )
00361         r3[jGCD+kLCM] = LCMbuffer[jGCD];
00362     k = kLCM;
00363 }
00364 j = 2*k+1;
00365 mm = m = oldwork + oldwork[0];
00366 /*
00367 Now compose the new term
00368 */
00369 n1 = term;
00370 while ( n1 < t ) *m++ = *n1++;
00371 n1 += n1[1];
00372 n2 = oldwork+1;
00373 while ( n2 < mm ) *m++ = *n2++;
00374 while ( n1 < tstop ) *m++ = *n1++;
00375 /*
00376 And the coefficient
00377 */
00378 size = term[*term-1];
00379 size = REDLENG(size);
00380 if ( MulRat(BHEAD (UWORD *)tstop,size, (UWORD *)r3,k,
00381             (UWORD *)m,&size) ) goto onerror;
00382 size = INCLENG(size);
00383 k = size < 0 ? -size: size;
00384 m[k-1] = size;
00385 m += k;
00386 *mm = (WORD) (m - mm);
00387 AT.WorkPointer = m;
00388 if ( Generator(BHEAD mm,level) ) goto onerror;
00389 AT.WorkPointer = oldwork;
00390 if ( fromwhere == 0 ) {
00391     if ( d->factors ) M_free(d->factors,"Dollar factors");
00392     M_free(d,"Dollar in FactorIn");
00393 }
00394 NumberFree(GCDbuffer,"FactorIn"); NumberFree(GCDbuffer2,"FactorIn");
00395 NumberFree(LCMbuffer,"FactorIn"); NumberFree(LCMb,"FactorIn"); NumberFree(LCMc,"FactorIn");
00396 return(0);
00397 onerror:
00398     AT.WorkPointer = oldwork;

```

```

00399     MLOCK(ErrorMessageLock);
00400     MesCall("FactorIn");
00401     MUNLOCK(ErrorMessageLock);
00402     NumberFree(GCDBuffer,"FactorIn"); NumberFree(GCDBuffer2,"FactorIn");
00403     NumberFree(LCMbuffer,"FactorIn"); NumberFree(LCMB,"FactorIn"); NumberFree(LCMc,"FactorIn");
00404     return(-1);
00405 }
00406
00407 /*
00408  #] FactorIn :
00409  #[ FactorInExpr :
00410
00411     This routine tests for a factor in an active or hidden expression.
00412
00413     The factor from the last call is stored in a cache.
00414     Main problem here is whether the cache is global or private to each thread.
00415     A global cache gives most likely the same traffic jam for the computation
00416     as a local cache. The local cache may stay valid longer.
00417     In the future we may make it such that threads can look at the cache
00418     of the others, or even whether a certain factor is under construction.
00419 */
00420
00421 int FactorInExpr(PHEAD WORD *term, WORD level)
00422 {
00423     GETBIDENTITY
00424     WORD *t, *tstop, *m, *oldwork, *mstop, *n1, *n2, *n3, *n4, *n1stop, *n2stop;
00425     WORD *r1, *r2, *r3, *r4, j, k, kGCD, kGCD2, kLCM, jGCD, kkLCM, jLCM, size, sign;
00426     WORD *newterm, expr = 0;
00427     WORD olddeferflag = AR.DeferFlag, oldgetfile = AR.GetFile, oldhold = AR.KeptInHold;
00428     WORD newgetfile, newhold;
00429     int i;
00430     EXPRESSIONS e;
00431     FILEHANDLE *file = 0;
00432     POSITION position, oldposition, startposition;
00433     WORD *oldcpointer = AR.CompressPointer;
00434     UWORD *GCDBuffer, *GCDBuffer2, *LCMBuffer, *LCMB, *LCMc;
00435     GCDBuffer = NumberMalloc("FactorInExpr"); GCDBuffer2 = NumberMalloc("FactorInExpr");
00436     LCMbuffer = NumberMalloc("FactorInExpr"); LCMB = NumberMalloc("FactorInExpr"); LCMc =
00437     NumberMalloc("FactorInExpr");
00438     t = term; GETSTOP(t,tstop); t++;
00439     while ( t < tstop ) {
00440         if ( *t == FACTORIN && t[1] == FUNHEAD+2 && t[FUNHEAD] == -EXPRESSION ) {
00441             expr = t[FUNHEAD+1];
00442             break;
00443         }
00444         t += t[1];
00445     }
00446     if ( t >= tstop ) {
00447         MLOCK(ErrorMessageLock);
00448         MesPrint("Internal error. Could not find proper factorin_ function.");
00449         MUNLOCK(ErrorMessageLock);
00450         NumberFree(GCDBuffer,"FactorInExpr"); NumberFree(GCDBuffer2,"FactorInExpr");
00451         NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMB,"FactorInExpr");
00452         NumberFree(LCMc,"FactorInExpr");
00453         return(-1);
00454     }
00455     oldwork = AT.WorkPointer;
00456     if ( AT.previousEfactor && ( expr == AT.previousEfactor[0] ) ) {
00457         /*
00458            We have a hit in the cache. Construct the new term.
00459            At the moment AT.previousEfactor[1] is reserved for future flags
00460         */
00461         goto PutTheFactor;
00462     }
00463     /*
00464        No hit. We have to do the work. We start with constructing the factor
00465        in the WorkSpace. Later we will move it to the cache.
00466        Finally we will jump to PutTheFactor.
00467     */
00468     e = Expressions + expr;
00469     switch ( e->status ) {
00470         case LOCALEXPRESSION:
00471         case SKIPLEXPRESSION:
00472         case DROPLEXPRESSION:
00473         case GLOBALEXPRESSION:
00474         case SKIPGEXPRESSION:
00475         case DROPGEXPRESSION:
00476         /*
00477            Expression is to be found in the input Scratch file.
00478            Set the file handle and the position.
00479            The rest is done by GetTerm.
00480         */
00481         newhold = 0;
00482         newgetfile = 1;
00483         file = AR.infile;
00484         break;
00485         case HIDDENLEXPRESSION:

```

```

00484         case HIDDENEXPRESSION:
00485         case HIDELEXPRESSION:
00486         case HIDEGEXPRESSION:
00487         case DROPHLEXPRESSION:
00488         case DROPHGEXPRESSION:
00489         case UNHIDELEXPRESSION:
00490         case UNHIDEGEXPRESSION:
00491     /*
00492         Expression is to be found in the hide Scratch file.
00493         Set the file handle and the position.
00494         The rest is done by GetTerm.
00495     */
00496         newhold = 0;
00497         newgetfile = 2;
00498         file = AR.hidefile;
00499         break;
00500     case STOREDEXPRESSION:
00501     /*
00502         This is an 'illegal' case
00503     */
00504         MLOCK(ErrorMessageLock);
00505         MesPrint("Error: factorin_ cannot determine factors in stored expressions.");
00506         MUNLOCK(ErrorMessageLock);
00507         NumberFree(GCDBuffer, "FactorInExpr"); NumberFree(GCDBuffer2, "FactorInExpr");
00508         NumberFree(LCMBuffer, "FactorInExpr"); NumberFree(LCMB, "FactorInExpr");
00509         NumberFree(LCMc, "FactorInExpr");
00510         return(-1);
00511     case DROPEDEXPRESSION:
00512     /*
00513         We replace the function by 1.
00514     */
00515         m = oldwork; n1 = term;
00516         while ( n1 < t ) *m++ = *n1++;
00517         n1 = t + t[1]; tstop = term + *term;
00518         while ( n1 < tstop ) *m++ = *n1++;
00519         *oldwork = m - oldwork;
00520         AT.WorkPointer = m;
00521         if ( Generator(BHEAD oldwork, level) ) {
00522             NumberFree(GCDBuffer, "FactorInExpr"); NumberFree(GCDBuffer2, "FactorInExpr");
00523             NumberFree(LCMBuffer, "FactorInExpr"); NumberFree(LCMB, "FactorInExpr");
00524             NumberFree(LCMc, "FactorInExpr");
00525             return(-1);
00526         }
00527         AT.WorkPointer = oldwork;
00528         NumberFree(GCDBuffer, "FactorInExpr"); NumberFree(GCDBuffer2, "FactorInExpr");
00529         NumberFree(LCMBuffer, "FactorInExpr"); NumberFree(LCMB, "FactorInExpr");
00530         NumberFree(LCMc, "FactorInExpr");
00531         return(0);
00532     default:
00533         MLOCK(ErrorMessageLock);
00534         MesPrint("Error: Illegal expression in factorinexpr.");
00535         MUNLOCK(ErrorMessageLock);
00536         NumberFree(GCDBuffer, "FactorInExpr"); NumberFree(GCDBuffer2, "FactorInExpr");
00537         NumberFree(LCMBuffer, "FactorInExpr"); NumberFree(LCMB, "FactorInExpr");
00538         NumberFree(LCMc, "FactorInExpr");
00539         return(-1);
00540     }
00541     /*
00542     Before we start with the file we set the buffers for the coefficient
00543     For the coefficient we have to determine the LCM of the denominators
00544     and the GCD of the numerators.
00545     */
00546     position = AS.OldOnFile[expr];
00547     AR.DeferFlag = 0; AR.KeptInHold = newhold; AR.GetFile = newgetfile;
00548     SeekScratch(file, &oldposition);
00549     SetScratch(file, &position);
00550     if ( GetTerm(BHEAD oldwork) <= 0 ) {
00551         MLOCK(ErrorMessageLock);
00552         MesPrint("(5) Expression %d has problems in scratchfile", expr);
00553         MUNLOCK(ErrorMessageLock);
00554         NumberFree(GCDBuffer, "FactorInExpr"); NumberFree(GCDBuffer2, "FactorInExpr");
00555         NumberFree(LCMBuffer, "FactorInExpr"); NumberFree(LCMB, "FactorInExpr");
00556         NumberFree(LCMc, "FactorInExpr");
00557         return(-1);
00558     }
00559     SeekScratch(file, &startposition);
00560     SeekScratch(file, &position);
00561     /*
00562     Load the first term in the workspace
00563     */
00564     if ( GetTerm(BHEAD oldwork) == 0 ) {
00565         SetScratch(file, &oldposition); /* We still need this until Processor is clean */
00566         AR.DeferFlag = olddeferflag;
00567         oldwork[0] = 4; oldwork[1] = 1; oldwork[2] = 1; oldwork[3] = 3;
00568         goto Complete;
00569     }
00570     SeekScratch(file, &position);

```

```

00566     AR.DeferFlag = olddeferflag; AR.KeptInHold = oldhold; AR.GetFile = oldgetfile;
00567
00568     r2 = m = oldwork + *oldwork;
00569     j = m[-1];
00570     m -= ABS(j);
00571     *oldwork = (WORD)(m-oldwork);
00572     AT.WorkPointer = newterm = mstop = m;
00573 /*
00574     Now take the coefficient of the first term to load up the LCM and the GCD
00575 */
00576     r3 = m;
00577     k = REDLENG(j);
00578     if ( k < 0 ) { k = -k; sign = -1; }
00579     else { sign = 1; }
00580     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00581     for ( kGCD = 0; kGCD < k; kGCD++ ) GCDbuffer[kGCD] = r3[kGCD];
00582     k = REDLENG(j);
00583     if ( k < 0 ) k = -k;
00584     r3 += k;
00585     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00586     for ( kLCM = 0; kLCM < k; kLCM++ ) LCMbuffer[kLCM] = r3[kLCM];
00587 /*
00588     The copy and the coefficient are in place. Now search through the terms.
00589 */
00590     for (;;) {
00591         AR.DeferFlag = 0; AR.KeptInHold = newhold; AR.GetFile = newgetfile;
00592         SetScratch(file,&position);
00593         size = GetTerm(BHEAD newterm);
00594         SeekScratch(file,&position);
00595         AR.DeferFlag = olddeferflag; AR.KeptInHold = oldhold; AR.GetFile = oldgetfile;
00596         if ( size == 0 ) break;
00597         m = oldwork+1;
00598         r2 = newterm + *newterm;
00599         r2 -= ABS(r2[-1]);
00600         r1 = newterm+1;
00601         while ( m < mstop ) {
00602             while ( r1 < r2 ) {
00603                 if ( *r1 != *m ) {
00604                     r1 += r1[1]; continue;
00605                 }
00606 /*
00607                 Now the various cases
00608                 #[ SYMBOL :
00609 */
00610                 if ( *m == SYMBOL ) {
00611                     n1 = m+2; n1stop = m+m[1];
00612                     n2stop = r1+r1[1];
00613                     while ( n1 < n1stop ) {
00614                         n2 = r1+2;
00615                         while ( n2 < n2stop ) {
00616                             if ( *n1 != *n2 ) { n2 += 2; continue; }
00617                             if ( n1[1] > 0 ) {
00618                                 if ( n2[1] < 0 ) { n2 += 2; continue; }
00619                                 if ( n2[1] < n1[1] ) n1[1] = n2[1];
00620                             }
00621                             else {
00622                                 if ( n2[1] > 0 ) { n2 += 2; continue; }
00623                                 if ( n2[1] > n1[1] ) n1[1] = n2[1];
00624                             }
00625                             break;
00626                         }
00627                     if ( n2 >= n2stop ) { /* scratch symbol */
00628                         if ( m[1] == 4 ) goto scratch;
00629                         m[1] -= 2;
00630                         n3 = n1; n4 = n1+2;
00631                         while ( n4 < mstop ) *n3++ = *n4++;
00632                         *oldwork = n3 - oldwork;
00633                         mstop -= 2; n1stop -= 2;
00634                         continue;
00635                     }
00636                     n1 += 2;
00637                 }
00638                 break;
00639             }
00640 /*
00641             #[ SYMBOL :
00642             #[ DOTPRODUCT :
00643 */
00644             else if ( *m == DOTPRODUCT ) {
00645                 n1 = m+2; n1stop = m+m[1];
00646                 n2stop = r1+r1[1];
00647                 while ( n1 < n1stop ) {
00648                     n2 = r1+2;
00649                     while ( n2 < n2stop ) {
00650                         if ( *n1 != *n2 || n1[1] != n2[1] ) { n2 += 3; continue; }
00651                         if ( n1[2] > 0 ) {
00652                             if ( n2[2] < 0 ) { n2 += 3; continue; }

```

```

00653             if ( n2[2] < n1[2] ) n1[2] = n2[2];
00654         }
00655         else {
00656             if ( n2[2] > 0 ) { n2 += 3; continue; }
00657             if ( n2[2] > n1[2] ) n1[2] = n2[2];
00658         }
00659         break;
00660     }
00661     if ( n2 >= n2stop ) { /* scratch dotproduct */
00662         if ( m[1] == 5 ) goto scratch;
00663         m[1] -= 3;
00664         n3 = n1; n4 = n1+3;
00665         while ( n4 < mstop ) *n3++ = *n4++;
00666         *oldwork = n3 - oldwork;
00667         mstop -= 3; n1stop -= 3;
00668         continue;
00669     }
00670     n1 += 3;
00671 }
00672 break;
00673 }
00674 /*
00675 #] DOTPRODUCT :
00676 #[ VECTOR :
00677 */
00678     else if ( *m == VECTOR ) {
00679 /*
00680         Here we have to be careful if there is more than
00681         one of the same
00682 */
00683         n1 = m+2; n1stop = m+m[1];
00684         n2 = r1+2; n2stop = r1+r1[1];
00685         while ( n1 < n1stop ) {
00686             while ( n2 < n2stop ) {
00687                 if ( *n1 == *n2 && n1[1] == n2[1] ) {
00688                     n2 += 2; goto nextn1;
00689                 }
00690                 n2 += 2;
00691             }
00692             if ( n2 >= n2stop ) { /* scratch vector */
00693                 if ( m[1] == 4 ) goto scratch;
00694                 m[1] -= 2;
00695                 n3 = n1; n4 = n1+2;
00696                 while ( n4 < mstop ) *n3++ = *n4++;
00697                 *oldwork = n3 - oldwork;
00698                 mstop -= 2; n1stop -= 2;
00699                 continue;
00700             }
00701             n2 = r1+2;
00702             n1 += 2;
00703         }
00704         break;
00705     }
00706 /*
00707 #] VECTOR :
00708 #[ REMAINDER :
00709 */
00710     else {
00711 /*
00712         Again: watch for multiple occurrences of the same object
00713 */
00714         if ( m[1] != r1[1] ) { r1 += r1[1]; continue; }
00715         for ( j = 2; j < m[1]; j++ ) {
00716             if ( m[j] != r1[j] ) break;
00717         }
00718         if ( j < m[1] ) { r1 += r1[1]; continue; }
00719         r1 += r1[1]; /* to restart at the next potential match */
00720         goto nextm; /* match */
00721     }
00722 /*
00723 #] REMAINDER :
00724 */
00725 }
00726 if ( r1 >= r2 ) { /* no factor! */
00727 scratch:;
00728     r3 = m + m[1]; r4 = m;
00729     while ( r3 < mstop ) *r4++ = *r3++;
00730     *oldwork = r4 - oldwork;
00731     if ( *oldwork == 1 ) goto nofactor;
00732     mstop = r4;
00733     r1 = newterm + 1;
00734     continue;
00735 }
00736 r1 = newterm + 1;
00737 nextm: m += m[1];
00738 }
00739 nofactor:;

```



```

00740 /*
00741     Now the coefficient
00742 */
00743     r2 = newterm + *newterm;
00744     j = r2[-1];
00745     r3 = r2 - ABS(j);
00746     k = REDLENG(j);
00747     if ( k < 0 ) k = -k;
00748     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00749     if ( ( ( GCDbuffer[0] == 1 ) && ( kGCD == 1 ) ) ) {
00750 /*
00751         GCD is already 1
00752 */
00753     }
00754     else if ( ( ( k != 1 ) || ( r3[0] != 1 ) ) ) {
00755         if ( GcdLong(BHEAD GCDbuffer,kGCD, (UWORD *)r3,k,GCDbuffer2,&kGCD2) ) {
00756             goto onerror;
00757         }
00758         kGCD = kGCD2;
00759         for ( i = 0; i < kGCD; i++ ) GCDbuffer[i] = GCDbuffer2[i];
00760     }
00761     else {
00762         kGCD = 1; GCDbuffer[0] = 1;
00763     }
00764     k = REDLENG(j);
00765     if ( k < 0 ) k = -k;
00766     r3 += k;
00767     while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
00768     if ( ( ( LCMbuffer[0] == 1 ) && ( kLCM == 1 ) ) ) {
00769         for ( kLCM = 0; kLCM < k; kLCM++ )
00770             LCMbuffer[kLCM] = r3[kLCM];
00771     }
00772     else if ( ( k != 1 ) || ( r3[0] != 1 ) ) {
00773         if ( GcdLong(BHEAD LCMbuffer,kLCM, (UWORD *)r3,k,LCMb,&kLCM) ) {
00774             goto onerror;
00775         }
00776         DivLong( (UWORD *)r3,k,LCMb,kLCM,LCMb,&kLCM,LCMc,&jLCM);
00777         Mullong(LCMbuffer,kLCM,LCMb,kLCM,LCMc,&jLCM);
00778         for ( kLCM = 0; kLCM < jLCM; kLCM++ )
00779             LCMbuffer[kLCM] = LCMc[kLCM];
00780     }
00781     else {} /* LCM doesn't change */
00782 }
00783 SetScratch(file,&oldposition); /* Needed until Processor is thread safe */
00784 AR.DeferFlag = olddeferflag;
00785 /*
00786     Now put the term together in oldwork: GCD/LCM
00787     We have already the algebraic contents there.
00788 */
00789     r3 = (WORD *) (GCDbuffer);
00790     r4 = (WORD *) (LCMbuffer);
00791     r1 = oldwork + *oldwork;
00792     if ( kGCD == kLCM ) {
00793         for ( jGCD = 0; jGCD < kGCD; jGCD++ ) *r1++ = *r3++;
00794         for ( jGCD = 0; jGCD < kGCD; jGCD++ ) *r1++ = *r4++;
00795         k = 2*kGCD+1;
00796     }
00797     else if ( kGCD > kLCM ) {
00798         for ( jGCD = 0; jGCD < kGCD; jGCD++ ) *r1++ = *r3++;
00799         for ( jGCD = 0; jGCD < kLCM; jGCD++ ) *r1++ = *r4++;
00800         for ( jGCD = kLCM; jGCD < kGCD; jGCD++ ) *r1++ = 0;
00801         k = 2*kGCD+1;
00802     }
00803     else {
00804         for ( jGCD = 0; jGCD < kGCD; jGCD++ ) *r1++ = *r3++;
00805         for ( jGCD = kGCD; jGCD < kLCM; jGCD++ ) *r1++ = 0;
00806         for ( jGCD = 0; jGCD < kLCM; jGCD++ ) *r1++ = *r4++;
00807         k = 2*kLCM+1;
00808     }
00809     if ( sign < 0 ) *r1++ = -k;
00810     else *r1++ = k;
00811     *oldwork = (WORD) (r1-oldwork);
00812 /*
00813     Now put the new term in the cache
00814 */
00815 Complete:;
00816     if ( AT.previousEfactor ) M_free(AT.previousEfactor,"Efactor cache");
00817     AT.previousEfactor = (WORD *)Malloc1((*oldwork+2)*sizeof(WORD),"Efactor cache");
00818     AT.previousEfactor[0] = expr;
00819     r1 = oldwork; r2 = AT.previousEfactor + 2; k = *oldwork;
00820     NCOPY(r2,r1,k)
00821     AT.previousEfactor[1] = 0;
00822 /*
00823     Now we construct the new term in the workspace.
00824 */
00825 PutTheFactor:;
00826     if ( AT.WorkPointer + AT.previousEfactor[2] >= AT.WorkTop ) {

```

```

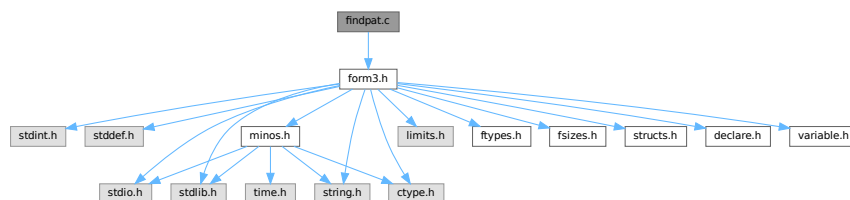
00827         MLOCK(ErrorMessageLock);
00828         MesWork();
00829         MesPrint("Called from factorin_");
00830         MUNLOCK(ErrorMessageLock);
00831         NumberFree(GCDBuffer,"FactorInExpr"); NumberFree(GCDBuffer2,"FactorInExpr");
00832         NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr");
        NumberFree(LCMc,"FactorInExpr");
00833         return(-1);
00834     }
00835     n1 = oldwork; n2 = term; while ( n2 < t ) *n1++ = *n2++;
00836     n2 = AT.previousEfactor+2; GETSTOP(n2,n2stop); n3 = n2 + *n2;
00837     n2++; while ( n2 < n2stop ) *n1++ = *n2++;
00838     n2 = t + t[1]; while ( n2 < tstop ) *n1++ = *n2++;
00839     size = term[*term-1];
00840     size = REDLENG(size);
00841     k = n3[-1]; k = REDLENG(k);
00842     if ( MulRat(BHEAD (UWORD *)tstop,size,(UWORD *)n2stop,k,
00843               (UWORD *)n1,&size) ) goto onerror;
00844     size = INCLENG(size);
00845     k = size < 0 ? -size: size;
00846     n1 += k; n1[-1] = size;
00847     *oldwork = n1 - oldwork;
00848     AT.WorkPointer = n1;
00849     if ( Generator(BHEAD oldwork,level) ) {
00850         NumberFree(GCDBuffer,"FactorInExpr"); NumberFree(GCDBuffer2,"FactorInExpr");
00851         NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr");
        NumberFree(LCMc,"FactorInExpr");
00852         return(-1);
00853     }
00854     AT.WorkPointer = oldwork;
00855     AR.CompressPointer = oldcpointer;
00856     NumberFree(GCDBuffer,"FactorInExpr"); NumberFree(GCDBuffer2,"FactorInExpr");
00857     NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr");
        NumberFree(LCMc,"FactorInExpr");
00858     return(0);
00859 onerror:
00860     AT.WorkPointer = oldwork;
00861     AR.CompressPointer = oldcpointer;
00862     MLOCK(ErrorMessageLock);
00863     MesCall("FactorInExpr");
00864     MUNLOCK(ErrorMessageLock);
00865     NumberFree(GCDBuffer,"FactorInExpr"); NumberFree(GCDBuffer2,"FactorInExpr");
00866     NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr");
        NumberFree(LCMc,"FactorInExpr");
00867     return(-1);
00868 }
00869
00870 /*
00871 #] FactorInExpr :
00872 */

```

## 4.35 findpat.c File Reference

#include "form3.h"

Include dependency graph for findpat.c:



### Functions

- int [FindOnly](#) (PHEAD WORD \*term, WORD \*pattern)
- int [FindOnce](#) (PHEAD WORD \*term, WORD \*pattern)
- WORD [FindMulti](#) (PHEAD WORD \*term, WORD \*pattern)
- int [FindRest](#) (PHEAD WORD \*term, WORD \*pattern)

### 4.35.1 Detailed Description

Pattern matching of symbols and dotproducts. There are various routines because of the options in the id-statements like once, only, multi and many. These are among the oldest routines in FORM and that can be noticed, because the interplay with the function matching is not complete. When we match functions and halfway we fail we can backtrack properly. With the symbols, the dotproducts and the vectors (in [pattern.c](#)) there is no proper backtracking. Hence the routines here need still quite some work or may even have to be rewritten.

Definition in file [findpat.c](#).

### 4.35.2 Function Documentation

#### 4.35.2.1 FindOnly()

```
int FindOnly (
    PHEAD WORD * term,
    WORD * pattern )
```

Definition at line 61 of file [findpat.c](#).

#### 4.35.2.2 FindOnce()

```
int FindOnce (
    PHEAD WORD * term,
    WORD * pattern )
```

Definition at line 419 of file [findpat.c](#).

#### 4.35.2.3 FindMulti()

```
WORD FindMulti (
    PHEAD WORD * term,
    WORD * pattern )
```

Definition at line 954 of file [findpat.c](#).

#### 4.35.2.4 FindRest()

```
int FindRest (
    PHEAD WORD * term,
    WORD * pattern )
```

Definition at line 1070 of file [findpat.c](#).

## 4.36 findpat.c

[Go to the documentation of this file.](#)

```

00001
00013 /* #[ License : */
00014 /*
00015 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00016 *   When using this file you are requested to refer to the publication
00017 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00018 *   This is considered a matter of courtesy as the development was paid
00019 *   for by FOM the Dutch physics granting agency and we would like to
00020 *   be able to track its scientific use to convince FOM of its value
00021 *   for the community.
00022 *
00023 *   This file is part of FORM.
00024 *
00025 *   FORM is free software: you can redistribute it and/or modify it under the
00026 *   terms of the GNU General Public License as published by the Free Software
00027 *   Foundation, either version 3 of the License, or (at your option) any later
00028 *   version.
00029 *
00030 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00031 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00032 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00033 *   details.
00034 *
00035 *   You should have received a copy of the GNU General Public License along
00036 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00037 */
00038 /* #[ License : */
00039 /*
00040     #[ Includes : findpat.c
00041 */
00042 #include "form3.h"
00043
00045 /*
00046     #[ Includes :
00047     #[ Patterns :
00048         #[ FindOnly :          WORD FindOnly(term,pattern)
00049
00050     The current version doesn't scan function arguments yet. 10-Apr-1988
00051
00052     This routine searches for an exact match. This means in particular:
00053     1: x^# must match exactly.
00054     2: x^n? must have a single value for n that cannot be adapted.
00055
00056     When setp != 0 it points to a collection of sets
00057     A match can occur only if no object will be left that belongs
00058     to any of these sets.
00059 */
00060
00061 int FindOnly(PHEAD WORD *term, WORD *pattern)
00062 {
00063     GETBIDENTITY
00064     WORD *t, *m;
00065     WORD *tstop, *mstop;
00066     WORD *xstop, *ystop, *setp = AN.ForFindOnly;
00067     WORD n, nt, *p, nq;
00068     WORD older[NORMSIZE], *q, newval1, newval2, newval3;
00069     AN.UsedOtherFind = 0;
00070     m = pattern;
00071     mstop = m + *m;
00072     m++;
00073     t = term;
00074     t += *term - 1;
00075     tstop = t - ABS(*t) + 1;
00076     t = term;
00077     t++;
00078     while ( t < tstop && *t > DOTPRODUCT ) t += t[1];
00079     while ( m < mstop && *m > DOTPRODUCT ) m += m[1];
00080     if ( m < mstop ) { do {
00081 /*
00082         #[ SYMBOLS :
00083 */
00084         if ( *m == SYMBOL ) {
00085             ystop = m + m[1];
00086             m += 2;
00087             n = 0;
00088             p = older;
00089             if ( t < tstop ) while ( *t != SYMBOL ) {
00090                 t += t[1];
00091                 if ( t >= tstop ) {
00092 OnlyZer1:
00093                     do {

```

```

00094         if ( *m >= 2*MAXPOWER ) return(0);
00095         if ( m[1] >= 2*MAXPOWER ) nt = m[1];
00096         else if ( m[1] <= -2*MAXPOWER ) nt = -m[1];
00097         else return(0);
00098         nt -= 2*MAXPOWER;
00099         if ( CheckWild(BHEAD nt, SYMTONUM, 0, &newval3) ) return(0);
00100         AddWild(BHEAD nt, SYMTONUM, 0);
00101         m += 2;
00102     } while ( m < ystop );
00103     goto EndLoop;
00104 }
00105 }
00106 else goto OnlyZer1;
00107 xstop = t + t[1];
00108 t += 2;
00109 do {
00110     if ( *m == *t && t < xstop ) {
00111         if ( m[1] == t[1] ) { m += 2; t += 2; }
00112         else if ( m[1] >= 2*MAXPOWER ) {
00113             nt = t[1];
00114             nq = m[1];
00115             goto OnlyL2;
00116         }
00117         else if ( m[1] <= -2*MAXPOWER ) {
00118             nt = -t[1];
00119             nq = -m[1];
00120             nq -= 2*MAXPOWER;
00121             if ( CheckWild(BHEAD nq, SYMTONUM, nt, &newval3) ) return(0);
00122             AddWild(BHEAD nq, SYMTONUM, nt);
00123             m += 2;
00124             t += 2;
00125         }
00126         else {
00127             *p++ = *t++; *p++ = *t++; n += 2;
00128         }
00129     }
00130     else if ( *m >= 2*MAXPOWER ) {
00131         while ( t < xstop ) { *p++ = *t++; *p++ = *t++; n += 2; }
00132         nq = n;
00133         p = older;
00134         while ( nq > 0 ) {
00135             if ( !CheckWild(BHEAD *m-2*MAXPOWER, SYMTOSYM, *p, &newval1) ) {
00136                 if ( m[1] == p[1] ) {
00137                     AddWild(BHEAD *m-2*MAXPOWER, SYMTOSYM, newval1);
00138                     break;
00139                 }
00140                 else if ( m[1] >= 2*MAXPOWER && m[1] != *m ) {
00141                     if ( !CheckWild(BHEAD m[1]-2*MAXPOWER, SYMTONUM, p[1], &newval3) ) {
00142                         AddWild(BHEAD m[1]-2*MAXPOWER, SYMTONUM, p[1]);
00143                         AddWild(BHEAD *m-2*MAXPOWER, SYMTOSYM, newval1);
00144                         break;
00145                     }
00146                 }
00147                 else if ( m[1] <= -2*MAXPOWER && m[1] != -(*m) ) {
00148                     if ( !CheckWild(BHEAD -m[1]-2*MAXPOWER, SYMTONUM, -p[1], &newval3) ) {
00149                         AddWild(BHEAD -m[1]-2*MAXPOWER, SYMTONUM, -p[1]);
00150                         AddWild(BHEAD *m-2*MAXPOWER, SYMTOSYM, newval1);
00151                         break;
00152                     }
00153                 }
00154             }
00155             nq -= 2;
00156             p += 2;
00157         }
00158         if ( nq <= 0 ) return(0);
00159         nq -= 2;
00160         n -= 2;
00161         q = p + 2;
00162         while ( --nq >= 0 ) *p++ = *q++;
00163         m += 2;
00164     }
00165     else {
00166         if ( t >= xstop || *m < *t ) {
00167             if ( m[1] >= 2*MAXPOWER ) nt = m[1];
00168             else if ( m[1] <= -2*MAXPOWER ) nt = -m[1];
00169             else return(0);
00170             nt -= 2*MAXPOWER;
00171             if ( CheckWild(BHEAD nt, SYMTONUM, 0, &newval3) ) return(0);
00172             AddWild(BHEAD nt, SYMTONUM, 0);
00173             m += 2;
00174         }
00175         else {
00176             *p++ = *t++; *p++ = *t++; n += 2;
00177         }
00178     }
00179 } while ( m < ystop );
00180 if ( setp ) {

```

```

00181         while ( t < xstop ) { *p++ = *t++; *p++ = *t++; n+= 2; }
00182         p = older;
00183         while ( n > 0 ) {
00184             nq = setp[1] - 2;
00185             m = setp + 2;
00186             while ( --nq >= 0 ) {
00187                 if ( Sets[*m].type != CSYMBOL ) { m++; continue; }
00188                 t = SetElements + Sets[*m].first;
00189                 tstop = SetElements + Sets[*m].last;
00190                 while ( t < tstop ) {
00191                     if ( *t++ == *p ) return(0);
00192                 }
00193                 m++;
00194             }
00195             n -= 2;
00196             p += 2;
00197         }
00198     }
00199     return(1);
00200 }
00201 /*
00202     #] SYMBOLS :
00203     #[ DOTPRODUCTS :
00204 */
00205     else if ( *m == DOTPRODUCT ) {
00206         ystop = m + m[1];
00207         m += 2;
00208         n = 0;
00209         p = older;
00210         if ( t < tstop ) {
00211             if ( *t < DOTPRODUCT ) goto OnlyZer2;
00212             while ( *t > DOTPRODUCT ) {
00213                 t += t[1];
00214                 if ( t >= tstop || *t < DOTPRODUCT ) {
00215 OnlyZer2:
00216                     do {
00217                         if ( *m >= (AM.OffsetVector+WILDOFFSET)
00218                             || m[1] >= (AM.OffsetVector+WILDOFFSET) ) return(0);
00219                         if ( m[2] >= 2*MAXPOWER ) nq = m[2];
00220                         else if ( m[2] <= -2*MAXPOWER ) nq = -m[2];
00221                         else return(0);
00222                         nq -= 2*MAXPOWER;
00223                         if ( CheckWild(BHEAD nq, SYMTONUM, 0, &newval3) ) return(0);
00224                         AddWild(BHEAD nq, SYMTONUM, 0);
00225                         m += 3;
00226                     } while ( m < ystop );
00227                     goto EndLoop;
00228                 }
00229             }
00230         }
00231         else goto OnlyZer2;
00232         xstop = t + t[1];
00233         t += 2;
00234         do {
00235             if ( *m == *t && m[1] == t[1] && t < xstop ) {
00236                 if ( t[2] != m[2] ) {
00237                     if ( m[2] >= 2*MAXPOWER ) {
00238                         nq = m[2];
00239                         nt = t[2];
00240                     }
00241                     else if ( m[2] <= -2*MAXPOWER ) {
00242                         nq = -m[2];
00243                         nt = -t[2];
00244                     }
00245                     else return(0);
00246                     nq -= 2*MAXPOWER;
00247                     if ( CheckWild(BHEAD nq, SYMTONUM, nt, &newval3) ) return(0);
00248                     AddWild(BHEAD nq, SYMTONUM, nt);
00249                 }
00250                 t += 3; m += 3;
00251             }
00252             else if ( *m >= (AM.OffsetVector+WILDOFFSET) ) {
00253                 while ( t < xstop ) {
00254                     *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
00255                 }
00256                 nq = n;
00257                 p = older;
00258                 while ( nq > 0 ) {
00259                     if ( *m == m[1] ) {
00260                         if ( *p != p[1] ) goto NextInDot;
00261                     }
00262                     if ( !CheckWild(BHEAD *m-WILDOFFSET, VECTOVEC, *p, &newval1) &&
00263                         !CheckWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, p[1], &newval2) ) {
00264                         if ( p[2] == m[2] ) {
00265 OnlyL9:
00266                             AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newval2);
00267                             AddWild(BHEAD *m-WILDOFFSET, VECTOVEC, newval1);
00268                             break;

```

```

00268         }
00269         if ( m[2] >= 2*MAXPOWER ) {
00270             if ( !CheckWild(BHEAD m[2]-2*MAXPOWER, SYMTONUM, p[2], &newval3) ) {
00271                 AddWild(BHEAD m[2]-2*MAXPOWER, SYMTONUM, newval3);
00272                 goto OnlyL9;
00273             }
00274         }
00275         else if ( m[2] <= -2*MAXPOWER ) {
00276             if ( !CheckWild(BHEAD -m[2]-2*MAXPOWER, SYMTONUM, -p[2], &newval3) ) {
00277                 AddWild(BHEAD -m[2]-2*MAXPOWER, SYMTONUM, -p[2]);
00278                 goto OnlyL9;
00279             }
00280         }
00281     }
00282     if ( !CheckWild(BHEAD *m-WILDOFFSET, VECTOVEC, p[1], &newval1) &&
00283         !CheckWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, *p, &newval2) ) {
00284         if ( p[2] == m[2] ) {
00285             AddWild(BHEAD *m-WILDOFFSET, VECTOVEC, newval1);
00286             AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newval2);
00287             break;
00288         }
00289         if ( m[2] >= 2*MAXPOWER ) {
00290             if ( !CheckWild(BHEAD m[2]-2*MAXPOWER, SYMTONUM, p[2], &newval3) ) {
00291                 AddWild(BHEAD m[2]-2*MAXPOWER, SYMTONUM, p[2]);
00292                 goto OnlyL10;
00293             }
00294         }
00295         else if ( m[2] <= -2*MAXPOWER ) {
00296             if ( !CheckWild(BHEAD -m[2]-2*MAXPOWER, SYMTONUM, -p[2], &newval3) ) {
00297                 AddWild(BHEAD -m[2]-2*MAXPOWER, SYMTONUM, -p[2]);
00298                 goto OnlyL10;
00299             }
00300         }
00301     }
00302 NextInDot:
00303     p += 3; nq -= 3;
00304 }
00305 if ( nq <= 0 ) return(0);
00306 q = p+3;
00307 nq -= 3;
00308 n = 3;
00309 while ( --nq >= 0 ) *p++ = *q++;
00310 m += 3;
00311 }
00312 else if ( m[1] >= (AM.OffsetVector+WILDOFFSET) ) {
00313     while ( *m >= *t && t < xstop ) {
00314         *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
00315     }
00316     nq = n;
00317     p = older;
00318     while ( nq > 0 ) {
00319         if ( *m == *p && !CheckWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, p[1], &newval1) ) {
00320             if ( p[2] == m[2] ) {
00321                 AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newval1);
00322                 break;
00323             }
00324             else if ( m[2] >= 2*MAXPOWER ) {
00325                 if ( !CheckWild(BHEAD m[2]-2*MAXPOWER, SYMTONUM, p[2], &newval3) ) {
00326                     AddWild(BHEAD m[2]-2*MAXPOWER, SYMTONUM, p[2]);
00327                     AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newval1);
00328                     break;
00329                 }
00330             }
00331             else if ( m[2] <= -2*MAXPOWER ) {
00332                 if ( !CheckWild(BHEAD -m[2]-2*MAXPOWER, SYMTONUM, -p[2], &newval3) ) {
00333                     AddWild(BHEAD -m[2]-2*MAXPOWER, SYMTONUM, -p[2]);
00334                     AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newval1);
00335                     break;
00336                 }
00337             }
00338         }
00339         if ( *m == p[1] && !CheckWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, *p, &newval1) ) {
00340             if ( p[2] == m[2] ) {
00341                 AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newval1);
00342                 break;
00343             }
00344             if ( m[2] >= 2*MAXPOWER ) {
00345                 if ( !CheckWild(BHEAD m[2]-2*MAXPOWER, SYMTONUM, p[2], &newval3) ) {
00346                     AddWild(BHEAD m[2]-2*MAXPOWER, SYMTONUM, p[2]);
00347                     AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newval1);
00348                     break;
00349                 }
00350             }
00351             else if ( m[2] <= -2*MAXPOWER ) {
00352                 if ( !CheckWild(BHEAD -m[2]-2*MAXPOWER, SYMTONUM, -p[2], &newval3) ) {
00353                     AddWild(BHEAD -m[2]-2*MAXPOWER, SYMTONUM, -p[2]);
00354                     AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newval1);

```

```

00355             break;
00356         }
00357     }
00358 }
00359     p += 3; nq -= 3;
00360 }
00361     if ( nq <= 0 ) return(0);
00362     q = p+3;
00363     nq -= 3;
00364     n -= 3;
00365     while ( --nq >= 0 ) *p++ = *q++;
00366     m += 3;
00367 }
00368     else {
00369         if ( t >= xstop || *m < *t || ( *m == *t && m[1] < t[1] ) ) {
00370             if ( m[2] > 2*MAXPOWER ) nt = m[2];
00371             else if ( m[2] <= -2*MAXPOWER ) nt = -m[2];
00372             else return(0);
00373             nt -= 2*MAXPOWER;
00374             if ( CheckWild(BHEAD nt, SYMTONUM, 0, &newval3) ) return(0);
00375             AddWild(BHEAD nt, SYMTONUM, 0);
00376             m += 3;
00377         }
00378         else {
00379             *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
00380         }
00381     }
00382     } while ( m < ystop );
00383     t = xstop;
00384 }
00385 /*
00386     #] DOTPRODUCTS :
00387 */
00388     else {
00389         MLOCK(ErrorMessageLock);
00390         MesPrint("Error in pattern(1)");
00391         MUNLOCK(ErrorMessageLock);
00392         Terminate(-1);
00393     }
00394 EndLoop;;
00395     } while ( m < mstop ); }
00396     if ( setp ) {
00397 /*
00398         while ( t < tstop && *t > SYMBOL ) t += t[1];
00399         if ( t < tstop && setp[1] > 2 ) return(0);
00400 */
00401         /* There were nonempty sets */
00402         /* Empty sets are rejected by the compiler */
00403     }
00404     return(1);
00405 }
00406
00407 /*
00408     #] FindOnly :
00409     #[ FindOnce :          WORD FindOnce(term,pattern)
00410
00411     Searches for a single match in term. The difference with multi
00412     lies mainly in the fact that here functions may occur.
00413     The functions have not been implemented yet. (10-Apr-1988)
00414     Wildcard powers are adjustable. The value closer to zero is taken.
00415     Positive and negative gives (o surprise) zero.
00416
00417 */
00418
00419 int FindOnce(PHEAD WORD *term, WORD *pattern)
00420 {
00421     GETBIDENTITY
00422     WORD *t, *m;
00423     WORD *tstop, *mstop;
00424     WORD *xstop, *ystop;
00425     WORD n, nt, *p, nq, mt, ch;
00426     WORD older[2*NORMSIZE], *q, newval1, newval2, newval3;
00427     AN.UsedOtherFind = 0;
00428     m = pattern;
00429     mstop = m + *m;
00430     m++;
00431     t = term;
00432     t += *term - 1;
00433     tstop = t - ABS(*t) + 1;
00434     t = term;
00435     t++;
00436     while ( t < tstop && *t > DOTPRODUCT ) t += t[1];
00437     while ( m < mstop && *m > DOTPRODUCT ) m += m[1];
00438     if ( m < mstop ) { do {
00439 /*
00440         #] SYMBOLS :
00441 */

```



```

00442         if ( *m == SYMBOL ) {
00443             ystop = m + m[1];
00444             m += 2;
00445             n = 0;
00446             p = older;
00447             if ( t < tstop ) while ( *t != SYMBOL ) {
00448                 t += t[1];
00449                 if ( t >= tstop ) {
00450 TryZero:
00451                     do {
00452                         if ( *m >= 2*MAXPOWER ) return(0);
00453                         if ( m[1] >= 2*MAXPOWER ) nt = m[1];
00454                         else if ( m[1] <= -2*MAXPOWER ) nt = -m[1];
00455                         else return(0);
00456                         nt -= 2*MAXPOWER;
00457                         if ( ( ch = CheckWild(BHEAD nt, SYMTONUM, 0, &newval3) ) != 0 ) {
00458                             if ( ch > 1 ) return(0);
00459                             if ( AN.oldtype != SYMTONUM ) return(0);
00460                             if ( *AN.MaskPointer == 2 ) return(0);
00461                         }
00462                         AddWild(BHEAD nt, SYMTONUM, 0);
00463                         m += 2;
00464                     } while ( m < ystop );
00465                     goto EndLoop;
00466                 }
00467             }
00468             else goto TryZero;
00469             xstop = t + t[1];
00470             t += 2;
00471             do {
00472                 if ( *m == *t && t < xstop ) {
00473                     nt = t[1];
00474                     mt = m[1];
00475                     if ( ( mt > 0 && mt <= nt ) ||
00476                         ( mt < 0 && mt >= nt ) ) { m += 2; t += 2; }
00477                     else if ( mt >= 2*MAXPOWER ) goto OnceL2;
00478                     else if ( mt <= -2*MAXPOWER ) {
00479                         nt = -nt;
00480                         mt = -mt;
00481 OnceL2:
00482                         if ( ( ch = CheckWild(BHEAD mt, SYMTONUM, nt, &newval3) ) != 0 ) {
00483                             if ( ch > 1 ) return(0);
00484                             if ( AN.oldtype != SYMTONUM ) return(0);
00485                             if ( AN.oldvalue <= 0 ) {
00486                                 if ( nt < AN.oldvalue ) nt = AN.oldvalue;
00487                                 else {
00488                                     if ( *AN.MaskPointer == 2 ) return(0);
00489                                     if ( nt > 0 ) nt = 0;
00490                                 }
00491                             }
00492                             if ( AN.oldvalue >= 0 ) {
00493                                 if ( nt > AN.oldvalue ) nt = AN.oldvalue;
00494                                 else {
00495                                     if ( *AN.MaskPointer == 2 ) return(0);
00496                                     if ( nt < 0 ) nt = 0;
00497                                 }
00498                             }
00499                         }
00500                         AddWild(BHEAD mt, SYMTONUM, nt);
00501                         m += 2;
00502                         t += 2;
00503                     }
00504                     else {
00505                         *p++ = *t++; *p++ = *t++; n += 2;
00506                     }
00507                 }
00508                 else if ( *m >= 2*MAXPOWER ) {
00509                     while ( t < xstop ) { *p++ = *t++; *p++ = *t++; n += 2; }
00510                     nq = n;
00511                     p = older;
00512                     while ( nq > 0 ) {
00513                         nt = p[1];
00514                         if ( !CheckWild(BHEAD *m-2*MAXPOWER, SYMTOSYM, *p, &newval1) ) {
00515                             mt = m[1];
00516                             if ( ( mt > 0 && mt <= nt ) ||
00517                                 ( mt < 0 && mt >= nt ) ) {
00518                                 AddWild(BHEAD *m-2*MAXPOWER, SYMTOSYM, newval1);
00519                                 break;
00520                             }
00521                         }
00522                         else if ( mt >= 2*MAXPOWER && mt != *m ) {
00523                             mt -= 2*MAXPOWER;
00524                             if ( ( ch = CheckWild(BHEAD mt, SYMTONUM, nt, &newval3) ) != 0 ) {
00525                                 if ( ch > 1 ) return(0);
00526                                 if ( AN.oldtype == SYMTONUM ) {
00527                                     if ( AN.oldvalue >= 0 ) {
00528                                         if ( nt > AN.oldvalue ) nt = AN.oldvalue;
00529                                         else {

```

```

00529                                     if ( *AN.MaskPointer == 2 ) return(0);
00530                                     if ( nt < 0 ) nt = 0;
00531                                     }
00532                                     }
00533                                     else {
00534                                         if ( nt < AN.oldvalue ) nt = AN.oldvalue;
00535                                         else {
00536                                             if ( *AN.MaskPointer == 2 ) return(0);
00537                                             if ( nt > 0 ) nt = 0;
00538                                         }
00539                                     }
00540                                     AddWild(BHEAD mt, SYMTONUM, nt);
00541                                     AddWild(BHEAD *m-2*MAXPOWER, SYMTOSYM, newval1);
00542                                     break;
00543                                 }
00544                             }
00545                             else {
00546                                 AddWild(BHEAD mt, SYMTONUM, nt);
00547                                 AddWild(BHEAD *m-2*MAXPOWER, SYMTOSYM, newval1);
00548                                 break;
00549                             }
00550                         }
00551                         else if ( mt <= -2*MAXPOWER && mt != -( *m ) ) {
00552                             nt = -nt;
00553                             mt = -mt;
00554                             goto OnceL4a;
00555                         }
00556                     }
00557                     nq -= 2;
00558                     p += 2;
00559                 }
00560                 if ( nq <= 0 ) return(0);
00561                 nq -= 2;
00562                 n -= 2;
00563                 q = p + 2;
00564                 while ( --nq >= 0 ) *p++ = *q++;
00565                 m += 2;
00566             }
00567             else {
00568                 if ( t >= xstop || *m < *t ) {
00569                     if ( m[1] >= 2*MAXPOWER ) nt = m[1];
00570                     else if ( m[1] <= -2*MAXPOWER ) nt = -m[1];
00571                     else return(0);
00572                     nt -= 2*MAXPOWER;
00573                     if ( ( ch = CheckWild(BHEAD nt, SYMTONUM, 0, &newval3) ) != 0 ) {
00574                         if ( ch > 1 ) return(0);
00575                         if ( AN.oldtype != SYMTONUM ) return(0);
00576                         if ( *AN.MaskPointer == 2 ) return(0);
00577                     }
00578                     AddWild(BHEAD nt, SYMTONUM, 0);
00579                     m += 2;
00580                 }
00581                 else {
00582                     *p++ = *t++; *p++ = *t++; n += 2;
00583                 }
00584             }
00585         } while ( m < ystop );
00586     }
00587 /*
00588     #] SYMBOLS :
00589     #[ DOTPRODUCTS :
00590 */
00591     else if ( *m == DOTPRODUCT ) {
00592         ystop = m + m[1];
00593         m += 2;
00594         n = 0;
00595         p = older;
00596         if ( t < tstop ) {
00597             if ( *t < DOTPRODUCT ) goto OnceOp;
00598             while ( *t > DOTPRODUCT ) {
00599                 t += t[1];
00600                 if ( t >= tstop || *t < DOTPRODUCT ) {
00601 OnceOp:
00602                     do {
00603                         if ( *m >= (AM.OffsetVector+WILDOFFSET)
00604                             || m[1] >= (AM.OffsetVector+WILDOFFSET) ) return(0);
00605                         if ( m[2] >= 2*MAXPOWER ) {
00606                             nq = m[2] - 2*MAXPOWER;
00607                         }
00608                         else if ( m[2] <= -2*MAXPOWER ) {
00609                             nq = -m[2] - 2*MAXPOWER;
00610                         }
00611                         else return(0);
00612                         if ( CheckWild(BHEAD nq, SYMTONUM, (WORD)0, &newval3) ) {
00613                             if ( AN.oldtype != SYMTONUM ) return(0);
00614                             if ( *AN.MaskPointer == 2 ) return(0);
00615                         }

```

```

00616             AddWild(BHEAD nq, SYMTONUM, (WORD) 0);
00617             m += 3;
00618         } while ( m < ystop );
00619         goto EndLoop;
00620     }
00621 }
00622 }
00623 else goto OnceOp;
00624 xstop = t + t[1];
00625 t += 2;
00626 do {
00627     if ( *m == *t && m[1] == t[1] && t < xstop ) {
00628         nt = t[2];
00629         mt = m[2];
00630         /*
00631             if ( ( nt > 0 && nt < mt ) ||
00632                 ( nt < 0 && nt > mt ) ) {
00633                 if ( mt <= -2*MAXPOWER ) {
00634                     mt = -mt;
00635                     nt = -nt;
00636                 }
00637                 else if ( mt < 2*MAXPOWER ) return(0);
00638                 mt -= 2*MAXPOWER;
00639                 if ( CheckWild(BHEAD mt, SYMTONUM, nt, &newval3) ) {
00640                     if ( AN.oldtype != SYMTONUM ) return(0);
00641                     if ( AN.oldvalue <= 0 ) {
00642                         if ( nt < AN.oldvalue ) nt = AN.oldvalue;
00643                     } else {
00644                         if ( *AN.MaskPointer == 2 ) return(0);
00645                         if ( nt > 0 ) nt = 0;
00646                     }
00647                 }
00648                 if ( AN.oldvalue >= 0 ) {
00649                     if ( nt > AN.oldvalue ) nt = AN.oldvalue;
00650                 } else {
00651                     if ( *AN.MaskPointer == 2 ) return(0);
00652                     if ( nt < 0 ) nt = 0;
00653                 }
00654             }
00655         }
00656         AddWild(BHEAD mt, SYMTONUM, nt);
00657         m += 3; t += 3;
00658     }
00659     else if ( ( nt > 0 && nt >= mt && mt > -2*MAXPOWER )
00660         || ( nt < 0 && nt <= mt && mt < 2*MAXPOWER ) ) {
00661         m += 3; t += 3;
00662     }
00663     /*
00664         if ( ( mt > 0 && mt <= nt ) ||
00665             ( mt < 0 && mt >= nt ) ) { m += 3; t += 3; }
00666     else if ( mt >= 2*MAXPOWER ) goto OnceL7;
00667     else if ( mt <= -2*MAXPOWER ) {
00668         nt = -nt;
00669         mt = -mt;
00670     OnceL7:
00671         if ( CheckWild(BHEAD mt, SYMTONUM, nt, &newval3) ) {
00672             if ( nt > AN.oldvalue ) return(0);
00673             if ( AN.oldvalue <= 0 ) {
00674                 if ( nt < AN.oldvalue ) nt = AN.oldvalue;
00675             } else {
00676                 if ( *AN.MaskPointer == 2 ) return(0);
00677                 if ( nt > 0 ) nt = 0;
00678             }
00679         }
00680         if ( AN.oldvalue >= 0 ) {
00681             if ( nt > AN.oldvalue ) nt = AN.oldvalue;
00682         } else {
00683             if ( *AN.MaskPointer == 2 ) return(0);
00684             if ( nt < 0 ) nt = 0;
00685         }
00686     }
00687 }
00688 AddWild(BHEAD mt, SYMTONUM, nt);
00689 m += 3;
00690 t += 3;
00691 }
00692 else {
00693     *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
00694 }
00695 }
00696 else if ( *m >= (AM.OffsetVector+WILDOFFSET) ) {
00697     while ( t < xstop ) {
00698         *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
00699     }
00700     nq = n;
00701     p = older;
00702     while ( nq > 0 ) {

```

```

00703         if ( *m == m[1] ) {
00704             if ( *p != p[1] ) goto NextInDot;
00705         }
00706     if ( !CheckWild(BHEAD *m-WILDOFFSET, VECTOVEC, *p, &newval1) &&
00707         !CheckWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, p[1], &newval2) ) {
00708         nt = p[2];
00709         mt = m[2];
00710         if ( ( mt > 0 && nt >= mt ) ||
00711             ( mt < 0 && nt <= mt ) ) {
00712             OnceL9:
00713                 AddWild(BHEAD *m-WILDOFFSET, VECTOVEC, newval1);
00714                 AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newval2);
00715                 break;
00716         }
00717         OnceL9a:
00718         if ( mt >= 2*MAXPOWER ) {
00719             mt -= 2*MAXPOWER;
00720             if ( CheckWild(BHEAD mt, SYMTONUM, nt, &newval3) ) {
00721                 if ( AN.oldtype == SYMTONUM ) {
00722                     if ( AN.oldvalue >= 0 ) {
00723                         if ( nt > AN.oldvalue ) nt = AN.oldvalue;
00724                         else {
00725                             if ( *AN.MaskPointer == 2 ) return(0);
00726                             if ( nt < 0 ) nt = 0;
00727                         }
00728                     }
00729                     else {
00730                         if ( nt < AN.oldvalue ) nt = AN.oldvalue;
00731                         else {
00732                             if ( *AN.MaskPointer == 2 ) return(0);
00733                             if ( nt > 0 ) nt = 0;
00734                         }
00735                     }
00736                     AddWild(BHEAD mt, SYMTONUM, nt);
00737                     goto OnceL9;
00738                 }
00739             }
00740             else {
00741                 AddWild(BHEAD mt, SYMTONUM, nt);
00742                 goto OnceL9;
00743             }
00744         }
00745         else if ( mt <= -2*MAXPOWER ) {
00746             mt = -mt;
00747             nt = -nt;
00748             goto OnceL9a;
00749         }
00750     }
00751     if ( !CheckWild(BHEAD *m-WILDOFFSET, VECTOVEC, p[1], &newval1) &&
00752         !CheckWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, *p, &newval2) ) {
00753         nt = p[2];
00754         mt = m[2];
00755         if ( ( mt > 0 && nt >= mt ) ||
00756             ( mt < 0 && nt <= mt ) ) {
00757             OnceL10:
00758                 AddWild(BHEAD *m-WILDOFFSET, VECTOVEC, newval1);
00759                 AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newval2);
00760                 break;
00761         }
00762         OnceL10a:
00763         if ( mt >= 2*MAXPOWER ) {
00764             mt -= 2*MAXPOWER;
00765             if ( CheckWild(BHEAD mt, SYMTONUM, nt, &newval3) ) {
00766                 if ( AN.oldtype == SYMTONUM ) {
00767                     if ( AN.oldvalue >= 0 ) {
00768                         if ( nt > AN.oldvalue ) nt = AN.oldvalue;
00769                         else {
00770                             if ( *AN.MaskPointer == 2 ) return(0);
00771                             if ( nt < 0 ) nt = 0;
00772                         }
00773                     }
00774                     else {
00775                         if ( nt < AN.oldvalue ) nt = AN.oldvalue;
00776                         else {
00777                             if ( *AN.MaskPointer == 2 ) return(0);
00778                             if ( nt > 0 ) nt = 0;
00779                         }
00780                     }
00781                     AddWild(BHEAD mt, SYMTONUM, nt);
00782                     goto OnceL10;
00783                 }
00784             }
00785             else {
00786                 AddWild(BHEAD mt, SYMTONUM, nt);
00787                 goto OnceL10;
00788             }
00789         }
00790         else if ( mt <= -2*MAXPOWER ) {
00791             mt = -mt;
00792             nt = -nt;
00793             goto OnceL10a;
00794         }
00795     }

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```

00790         }
00791     }
00792 NextInDot:
00793     p += 3; nq -= 3;
00794 }
00795 if ( nq <= 0 ) return(0);
00796 else {
00797     q = p+3;
00798     nq -= 3;
00799     n -= 3;
00800     while ( --nq >= 0 ) *p++ = *q++;
00801 }
00802 m += 3;
00803 }
00804 else if ( m[1] >= (AM.OffsetVector+WILDOFFSET) ) {
00805     while ( *m >= *t && t < xstop ) {
00806         *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
00807     }
00808     nq = n;
00809     p = older;
00810     while ( nq > 0 ) {
00811         if ( *m == *p && !CheckWild(BHEAD m[1]-WILDOFFSET,
00812             VECTOVEC,p[1],&newval1) ) {
00813             nt = p[2];
00814             mt = m[2];
00815             if ( ( mt > 0 && nt >= mt ) ||
00816                 ( mt < 0 && nt <= mt ) ) {
00817                 AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
00818                 break;
00819             }
00820             else if ( mt >= 2*MAXPOWER ) {
00821                 mt -= 2*MAXPOWER;
00822                 if ( CheckWild(BHEAD mt,SYMTONUM,nt,&newval3) ) {
00823                     if ( AN.oldtype == SYMTONUM ) {
00824                         if ( AN.oldvalue >= 0 ) {
00825                             if ( nt > AN.oldvalue ) nt = AN.oldvalue;
00826                             else {
00827                                 if ( *AN.MaskPointer == 2 ) return(0);
00828                                 if ( nt < 0 ) nt = 0;
00829                             }
00830                         }
00831                         else {
00832                             if ( nt < AN.oldvalue ) nt = AN.oldvalue;
00833                             else {
00834                                 if ( *AN.MaskPointer == 2 ) return(0);
00835                                 if ( nt > 0 ) nt = 0;
00836                             }
00837                         }
00838                         AddWild(BHEAD mt,SYMTONUM,nt);
00839                         AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
00840                         break;
00841                     }
00842                 }
00843             else {
00844                 AddWild(BHEAD mt,SYMTONUM,nt);
00845                 AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
00846                 break;
00847             }
00848         }
00849         else if ( mt <= -2*MAXPOWER ) {
00850             mt = -mt;
00851             nt = -nt;
00852             goto OnceL7a;
00853         }
00854     }
00855     if ( *m == p[1] && !CheckWild(BHEAD m[1]-WILDOFFSET,
00856         VECTOVEC,*p,&newval1) ) {
00857         nt = p[2];
00858         mt = m[2];
00859         if ( ( mt > 0 && nt >= mt ) ||
00860             ( mt < 0 && nt <= mt ) ) {
00861             AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
00862             break;
00863         }
00864         if ( mt >= 2*MAXPOWER ) {
00865             mt -= 2*MAXPOWER;
00866             if ( CheckWild(BHEAD mt,SYMTONUM,nt,&newval3) ) {
00867                 if ( AN.oldtype == SYMTONUM ) {
00868                     if ( AN.oldvalue >= 0 ) {
00869                         if ( nt > AN.oldvalue ) nt = AN.oldvalue;
00870                         else {
00871                             if ( *AN.MaskPointer == 2 ) return(0);
00872                             if ( nt < 0 ) nt = 0;
00873                         }
00874                     }
00875                     else {
00876                         if ( nt < AN.oldvalue ) nt = AN.oldvalue;

```

```

00877                                     else {
00878                                         if ( *AN.MaskPointer == 2 ) return(0);
00879                                         if ( nt > 0 ) nt = 0;
00880                                     }
00881                                 }
00882                                 AddWild(BHEAD mt, SYMTONUM, nt);
00883                                 AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newvall);
00884                                 break;
00885                             }
00886                         }
00887                         else {
00888                             AddWild(BHEAD mt, SYMTONUM, nt);
00889                             AddWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, newvall);
00890                             break;
00891                         }
00892                     }
00893                     else if ( mt < -2*MAXPOWER ) {
00894                         mt = -mt;
00895                         nt = -nt;
00896                         goto OnceL8a;
00897                     }
00898                 }
00899                 p += 3; nq -= 3;
00900             }
00901             if ( nq <= 0 ) return(0);
00902             q = p+3;
00903             nq -= 3;
00904             n -= 3;
00905             while ( --nq >= 0 ) *p++ = *q++;
00906             m += 3;
00907         }
00908         else {
00909             if ( t >= xstop || *m < *t || ( *m == *t && m[1] < t[1] ) ) {
00910                 if ( m[2] >= 2*MAXPOWER ) nt = m[2];
00911                 else if ( m[2] <= -2*MAXPOWER ) nt = -m[2];
00912                 else return(0);
00913                 nt -= 2*MAXPOWER;
00914                 if ( CheckWild(BHEAD nt, SYMTONUM, 0, &newval3) ) {
00915                     if ( AN.oldtype != SYMTONUM ) return(0);
00916                     if ( *AN.MaskPointer == 2 ) return(0);
00917                 }
00918                 AddWild(BHEAD nt, SYMTONUM, 0);
00919                 m += 3;
00920             }
00921             else {
00922                 *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
00923             }
00924         }
00925     } while ( m < ystop );
00926     t = xstop;
00927 }
00928 /*
00929     #] DOTPRODUCTS :
00930 */
00931     else {
00932         MLOCK(ErrorMessageLock);
00933         MesPrint("Error in pattern(2)");
00934         MUNLOCK(ErrorMessageLock);
00935         Terminate(-1);
00936     }
00937 EndLoop;;
00938     } while ( m < mstop ); }
00939     else {
00940         return(-1);
00941     }
00942     return(1);
00943 }
00944
00945 /*
00946     #] FindOnce :
00947     #[ FindMulti :          WORD FindMulti(term, pattern)
00948
00949     Note that multi cannot deal with wildcards. Those patterns revert
00950     to many which gives subsequent calls to once.
00951 */
00952 /*
00953
00954 WORD FindMulti(PHEAD WORD *term, WORD *pattern)
00955 {
00956     GETBIDENTITY
00957     WORD *t, *m, *p;
00958     WORD *tstop, *mstop;
00959     WORD *xstop, *ystop;
00960     WORD mt, power, n, nq;
00961     WORD older[2*NORMSIZE], *q, newvall;
00962     AN.UsedOtherFind = 0;
00963     m = pattern;

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00964     mstop = m + *m;
00965     m++;
00966     t = term;
00967     t += *term - 1;
00968     tstop = t - ABS(*t) + 1;
00969     t = term;
00970     t++;
00971     while ( t < tstop && *t > DOTPRODUCT ) t += t[1];
00972     while ( m < mstop && *m > DOTPRODUCT ) m += m[1];
00973     power = -1; /* No power yet */
00974     if ( m < mstop ) { do {
00975 /*
00976         #] SYMBOLS :
00977 */
00978         if ( *m == SYMBOL ) {
00979             ystop = m + m[1];
00980             m += 2;
00981             if ( t >= tstop ) return(0);
00982             while ( *t != SYMBOL ) { t += t[1]; if ( t >= tstop ) return(0); }
00983             xstop = t + t[1];
00984             t += 2;
00985             p = older;
00986             n = 0;
00987             do {
00988                 if ( *m >= 2*MAXPOWER ) {
00989                     while ( t < xstop ) { *p++ = *t++; *p++ = *t++; n += 2; }
00990                     nq = n;
00991                     p = older;
00992                     while ( nq > 0 ) {
00993                         if ( !CheckWild(BHEAD *m-2*MAXPOWER, SYMTOSYM, *p, &newvall) ) {
00994                             mt = p[1]/m[1];
00995                             if ( mt > 0 ) {
00996                                 if ( power < 0 || mt < power ) power = mt;
00997                                 AddWild(BHEAD *m-2*MAXPOWER, SYMTOSYM, newvall);
00998                                 break;
00999                             }
01000                         }
01001                         nq -= 2;
01002                         p += 2;
01003                     }
01004                     if ( nq <= 0 ) return(0);
01005                     nq -= 2;
01006                     n -= 2;
01007                     q = p + 2;
01008                     while ( --nq >= 0 ) *p++ = *q++;
01009                     m += 2;
01010                 }
01011                 else if ( t >= xstop ) return(0);
01012                 else if ( *m == *t ) {
01013                     if ( ( mt = t[1]/m[1] ) <= 0 ) return(0);
01014                     if ( power < 0 || mt < power ) power = mt;
01015                     m += 2;
01016                     t += 2;
01017                 }
01018                 else if ( *m < *t ) return(0);
01019                 else { *p++ = *t++; *p++ = *t++; n += 2; }
01020             } while ( m < ystop );
01021         }
01022 /*
01023         #] SYMBOLS :
01024         #] DOTPRODUCTS :
01025 */
01026         else if ( *m == DOTPRODUCT ) {
01027             ystop = m + m[1];
01028             m += 2;
01029             if ( t >= tstop ) return(0);
01030             while ( *t != DOTPRODUCT ) { t += t[1]; if ( t >= tstop ) return(0); }
01031             xstop = t + t[1];
01032             t += 2;
01033             do {
01034                 if ( t >= xstop ) return(0);
01035                 if ( *t == *m ) {
01036                     if ( t[1] == m[1] ) {
01037                         if ( ( mt = t[2]/m[2] ) <= 0 ) return(0);
01038                         if ( power < 0 || mt < power ) power = mt;
01039                         m += 3;
01040                     }
01041                     else if ( t[1] > m[1] ) return(0);
01042                 }
01043                 else if ( *t > *m ) return(0);
01044                 t += 3;
01045             } while ( m < ystop );
01046             t = xstop;
01047         }
01048 /*
01049         #] DOTPRODUCTS :
01050 */

```

```

01051         else {
01052             MLOCK(ErrorMessageLock);
01053             MesPrint("Error in pattern(3)");
01054             MUNLOCK(ErrorMessageLock);
01055             Terminate(-1);
01056         }
01057     } while ( m < mstop ); }
01058     if ( power < 0 ) power = 0;
01059     return(power);
01060 }
01061
01062 /*
01063     #] FindMulti :
01064     #[ FindRest :          WORD FindRest(term,pattern)
01065
01066     This routine scans for anything but dotproducts and symbols.
01067 */
01068
01069 int FindRest(PHEAD WORD *term, WORD *pattern)
01070 {
01071     GETBIDENTITY
01072     WORD *t, *m, *tt, wild, regular;
01073     WORD *tstop, *mstop;
01074     WORD *xstop, *ystop;
01075     WORD n, *p, nq;
01076     WORD older[NORMSIZE], *q, newval1, newval2;
01077     int i, ntw;
01078     AN.UsedOtherFind = 0;
01079     AN.findTerm = term; AN.findPattern = pattern;
01080     m = AN.WildValue;
01081     i = (m[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
01082     ntw = 0;
01083     while ( i > 0 ) {
01084         if ( m[0] == ARGTOARG ) ntw++;
01085         m += m[1];
01086         i--;
01087     }
01088     t = term;
01089     t += *term - 1;
01090     tstop = t - ABS(*t) + 1;
01091     t = term;
01092     t++; p = t;
01093     while ( t < tstop && *t > DOTPRODUCT ) t += t[1];
01094     tstop = t;
01095     t = p;
01096     m = pattern;
01097     mstop = m + *m;
01098     m++;
01099     p = m;
01100     while ( m < mstop && *m > DOTPRODUCT ) m += m[1];
01101     mstop = m;
01102     m = p;
01103     if ( m < mstop ) {
01104         do {
01105             /*
01106                 #[ FUNCTIONS :
01107             */
01108             if ( *m >= FUNCTION ) {
01109                 if ( *mstop > 5 && !MatchIsPossible(pattern,term) ) return(0);
01110                 ystop = m;
01111                 n = 0;
01112                 do {
01113                     m += m[1]; n++;
01114                 } while ( m < mstop && *m >= FUNCTION );
01115                 AT.WorkPointer += n;
01116                 while ( t < tstop && *t == SUBEXPRESSION ) t += t[1];
01117                 tt = xstop = t;
01118                 nq = 0;
01119                 while ( t < tstop && ( *t >= FUNCTION || *t == SUBEXPRESSION ) ) {
01120                     if ( *t != SUBEXPRESSION ) {
01121                         nq++;
01122                         if ( functions[*t-FUNCTION].commute ) tt = t + t[1];
01123                     }
01124                     t += t[1];
01125                 }
01126                 if ( nq < n ) return(0);
01127                 AN.terstart = term;
01128                 AN.terstop = t;
01129                 AN.terfirstcomm = tt;
01130                 AN.patstop = m;
01131                 AN.NumTotWildArgs = ntw;
01132                 if ( !ScanFunctions(BHEAD ystop,xstop,0) ) return(0);
01133             }
01134             /*
01135                 #] FUNCTIONS :
01136                 #[ VECTORS :
01137             */

```



```

01138 */
01139     else if ( *m == VECTOR ) {
01140         while ( t < tstop && *t != VECTOR ) t += t[1];
01141         if ( t >= tstop ) return(0);
01142         xstop = t + t[1];
01143         ystop = m + m[1];
01144         t += 2;
01145         m += 2;
01146         n = 0;
01147         p = older;
01148         do {
01149             if ( *m == *t && m[1] == t[1] && t < xstop ) {
01150                 m += 2; t += 2;
01151             }
01152             else if ( *m >= (AM.OffsetVector+WILDOFFSET) ) {
01153                 if ( t < xstop ) {
01154                     p = older + n;
01155                     do { *p++ = *t++; n++; } while ( t < xstop );
01156                 }
01157                 p = older;
01158                 nq = n;
01159                 if ( ( m[1] < (AM.OffsetIndex+WILDOFFSET) )
01160                     || ( m[1] >= (AM.OffsetIndex+2*WILDOFFSET) ) ) {
01161                     while ( nq > 0 ) {
01162                         if ( m[1] == p[1] ) {
01163                             if ( !CheckWild(BHEAD *m-WILDOFFSET, VECTOVEC, *p, &newval1) ) {
01164 RestL11: AddWild(BHEAD *m-WILDOFFSET, VECTOVEC, newval1);
01165                             break;
01166                         }
01167                     }
01168                     p += 2;
01169                     nq -= 2;
01170                 }
01171             }
01172             else { /* Double wildcard */
01173                 while ( nq > 0 ) {
01174                     if ( !CheckWild(BHEAD *m-WILDOFFSET, VECTOVEC, *p, &newval1) &&
01175                         !CheckWild(BHEAD m[1]-WILDOFFSET, INDTOIND, p[1], &newval2) ) {
01176                         AddWild(BHEAD m[1]-WILDOFFSET, INDTOIND, newval2);
01177                         goto RestL11;
01178                     }
01179                     p += 2;
01180                     nq -= 2;
01181                 }
01182             }
01183             if ( nq > 0 ) {
01184                 nq -= 2; q = p + 2; n -= 2;
01185                 while ( --nq >= 0 ) *p++ = *q++;
01186             }
01187             else return(0);
01188             m += 2;
01189         }
01190         else if ( ( *m <= *t )
01191             && ( m[1] >= (AM.OffsetIndex + WILDOFFSET) )
01192             && ( m[1] < (AM.OffsetIndex + 2*WILDOFFSET) ) ) {
01193             if ( *m == *t && t < xstop ) {
01194                 p = older;
01195                 p += n;
01196                 *p++ = *t++;
01197                 *p++ = *t++;
01198                 n += 2;
01199             }
01200             p = older;
01201             nq = n;
01202             while ( nq > 0 ) {
01203                 if ( *m == *p ) {
01204                     if ( !CheckWild(BHEAD m[1]-WILDOFFSET, INDTOIND, p[1], &newval1) ) {
01205                         AddWild(BHEAD m[1]-WILDOFFSET, INDTOIND, newval1);
01206                         break;
01207                     }
01208                 }
01209                 p += 2;
01210                 nq -= 2;
01211             }
01212             if ( nq > 0 ) {
01213                 nq -= 2; q = p + 2; n -= 2;
01214                 while ( --nq >= 0 ) *p++ = *q++;
01215             }
01216             else return(0);
01217             m += 2;
01218         }
01219         else {
01220             if ( t >= xstop ) return(0);
01221             *p++ = *t++; *p++ = *t++; n += 2;
01222         }
01223     } while ( m < ystop );
01224 }

```

```

01225 /*
01226         #] VECTORS :
01227         #[ INDICES :
01228 */
01229     else if ( *m == INDEX ) {
01230 /*
01231         This needs only to say that there is a match, after matching
01232         a 'wildcard'. This has to be prepared in TestMatch. The C->rhs
01233         should provide the replacement inside the prototype!
01234         Next question: id,p=q/2+r/2
01235 */
01236         while ( *t != INDEX ) { t += t[1]; if ( t >= tstop ) return(0); }
01237         xstop = t + t[1];
01238         ystop = m + m[1];
01239         t += 2;
01240         m += 2;
01241         n = 0;
01242         p = older;
01243         do {
01244             if ( *m == *t && t < xstop && m < ystop ) {
01245                 t++; m++;
01246             }
01247             else if ( ( *m >= (AM.OffsetIndex+WILDOFFSET) )
01248                 && ( *m < (AM.OffsetIndex+2*WILDOFFSET) ) ) {
01249                 while ( t < xstop ) {
01250                     *p++ = *t++; n++;
01251                 }
01252                 if ( !n ) return(0);
01253                 nq = n;
01254                 q = older;
01255                 do {
01256                     if ( !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*q,&newvall) ) {
01257                         AddWild(BHEAD *m-WILDOFFSET,INDTOIND,newvall);
01258                         break;
01259                     }
01260                     q++;
01261                     nq--;
01262                 } while ( nq > 0 );
01263                 if ( nq <= 0 ) return (0);
01264                 n--;
01265                 nq--;
01266                 p = q + 1;
01267                 while ( nq > 0 ) { *q++ = *p++; nq--; }
01268                 p--;
01269                 m++;
01270             }
01271             else if ( ( *m >= (AM.OffsetVector+WILDOFFSET) )
01272                 && ( *m < (AM.OffsetVector+2*WILDOFFSET) ) ) {
01273                 while ( t < xstop ) {
01274                     *p++ = *t++; n++;
01275                 }
01276                 if ( !n ) return(0);
01277                 nq = n;
01278                 q = older;
01279                 do {
01280                     if ( !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*q,&newvall) ) {
01281                         AddWild(BHEAD *m-WILDOFFSET,VECTOVEC,newvall);
01282                         break;
01283                     }
01284                     q++;
01285                     nq--;
01286                 } while ( nq > 0 );
01287                 if ( nq <= 0 ) return (0);
01288                 n--;
01289                 nq--;
01290                 p = q + 1;
01291                 while ( nq > 0 ) { *q++ = *p++; nq--; }
01292                 p--;
01293                 m++;
01294             }
01295             else {
01296                 if ( t >= xstop ) return(0);
01297                 *p++ = *t++; n++;
01298             }
01299         } while ( m < ystop );
01300
01301 /*
01302         return(0);
01303 */
01304     }
01305 /*
01306         #] INDICES :
01307         #[ DELTAS :
01308 */
01309     else if ( *m == DELTA ) {
01310         while ( *t != DELTA ) { t += t[1]; if ( t >= tstop ) return(0); }
01311         xstop = t + t[1];

```

```

01312     ystop = m + m[1];
01313     t += 2;
01314     m += 2;
01315     n = 0;
01316     p = older;
01317     do {
01318         if ( *t == *m && t[1] == m[1] && t < xstop ) {
01319             m += 2;
01320             t += 2;
01321         }
01322         else if ( ( *m >= (AM.OffsetIndex+WILDOFFSET) )
01323             && ( *m < (AM.OffsetIndex+2*WILDOFFSET) )
01324             && ( m[1] >= (AM.OffsetIndex+WILDOFFSET) )
01325             && ( m[1] < (AM.OffsetIndex+2*WILDOFFSET) ) ) { /* Two dummies */
01326             while ( t < xstop ) {
01327                 *p++ = *t++; *p++ = *t++; n += 2;
01328             }
01329             if ( !n ) return(0);
01330             nq = n;
01331             q = older;
01332             do {
01333                 if ( !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*q,&newval1) &&
01334                     !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,q[1],&newval2) ) {
01335                     AddWild(BHEAD *m-WILDOFFSET,INDTOIND,newval1);
01336                     AddWild(BHEAD m[1]-WILDOFFSET,INDTOIND,newval2);
01337                     break;
01338                 }
01339                 if ( !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,q[1],&newval1) &&
01340                     !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,*q,&newval2) ) {
01341                     AddWild(BHEAD *m-WILDOFFSET,INDTOIND,newval1);
01342                     AddWild(BHEAD m[1]-WILDOFFSET,INDTOIND,newval2);
01343                     break;
01344                 }
01345                 q += 2;
01346                 nq -= 2;
01347             } while ( nq > 0 );
01348             if ( nq <= 0 ) return(0);
01349             n -= 2;
01350             nq -= 2;
01351             p = q + 2;
01352             while ( nq > 0 ) { *q++ = *p++; nq--; }
01353             p -= 2;
01354             m += 2;
01355         }
01356         else if ( ( m[1] >= (AM.OffsetIndex+WILDOFFSET) )
01357             && ( m[1] < (AM.OffsetIndex+2*WILDOFFSET) ) ) {
01358             wild = m[1]; regular = *m;
01359 OneWild:
01360             while ( ( regular == *t || regular == t[1] ) && t < xstop ) {
01361                 *p++ = *t++; *p++ = *t++; n += 2;
01362             }
01363             if ( !n ) return(0);
01364             nq = n;
01365             q = older;
01366             do {
01367                 if ( regular == *q && !CheckWild(BHEAD wild-WILDOFFSET,INDTOIND,q[1],&newval1)
01368                     AddWild(BHEAD wild-WILDOFFSET,INDTOIND,newval1);
01369                     break;
01370                 }
01371                 if ( regular == q[1] && !CheckWild(BHEAD wild-WILDOFFSET,INDTOIND,*q,&newval1)
01372                     AddWild(BHEAD wild-WILDOFFSET,INDTOIND,newval1);
01373                     break;
01374                 }
01375                 q += 2;
01376                 nq -= 2;
01377             } while ( nq > 0 );
01378             if ( nq <= 0 ) return(0);
01379             n -= 2;
01380             nq -= 2;
01381             p = q + 2;
01382             while ( nq > 0 ) { *q++ = *p++; nq--; }
01383             p -= 2;
01384             m += 2;
01385         }
01386         else if ( ( *m >= (AM.OffsetIndex+WILDOFFSET) )
01387             && ( *m < (AM.OffsetIndex+2*WILDOFFSET) ) ) {
01388             wild = *m; regular = m[1];
01389             goto OneWild;
01390         }
01391         else {
01392             if ( t >= tstop || *m < *t || ( *m == *t && m[1] < t[1] ) )
01393                 return(0);
01394             *p++ = *t++; *p++ = *t++; n += 2;
01395         }
01396     } while ( m < ystop );

```



## 4.38 flintinterface.cc

[Go to the documentation of this file.](#)

```

00001
00007 #ifndef HAVE_CONFIG_H
00008 #ifndef CONFIG_H_INCLUDED
00009 #define CONFIG_H_INCLUDED
00010 #include <config.h>
00011 #endif
00012 #endif
00013
00014 extern "C" {
00015 #include "form3.h"
00016 }
00017
00018 #include "flintinterface.h"
00019
00020 /*
00021  *# [ Types
00022  * Use fixed-size integer types (int32_t, uint32_t, int64_t, uint64_t) except when we refer
00023  * specifically to FORM term data, when we use WORD. This code has only been tested on 64bit
00024  * systems where WORDs are int32_t, and some functions (fmpz_get_form, fmpz_set_form) require this
00025  * to be the case. Enforce this:
00026  */
00027 static_assert(sizeof(WORD) == sizeof(int32_t), "flint interface expects 32bit WORD");
00028 static_assert(sizeof(WORD) == sizeof(uint32_t), "flint interface expects 32bit WORD");
00029 static_assert(BITSINWORD == 32, "flint interface expects 32bit WORD");
00030 static_assert(sizeof(ulong) == sizeof(uint64_t), "flint interface expects ulong is uint64_t");
00031 static_assert(sizeof(slong) == sizeof(int64_t), "flint interface expects slong is int64_t");
00032
00033 /*
00034  * Flint functions take arguments or return values which may be "slong" or "ulong" in its
00035  * documentation, and these are int64_t and uint64_t respectively.
00036  */
00037 #] Types
00038 */
00039
00040 /*
00041  *# [ flint::cleanup :
00042  */
00043 void flint::cleanup(void) {
00044     flint_cleanup();
00045 }
00046 /*
00047  *#] flint::cleanup :
00048  *# [ flint::cleanup_master :
00049  */
00050 void flint::cleanup_master(void) {
00051     flint_cleanup_master();
00052 }
00053 /*
00054  *#] flint::cleanup_master :
00055  *# [ flint::divmod_mpoly :
00056  */
00057 /*
00058 WORD* flint::divmod_mpoly(PHEAD const WORD *a, const WORD *b, const bool return_rem,
00059 const WORD must_fit_term, const var_map_t &var_map) {
00060
00061     flint::mpoly_ctx ctx(var_map.size());
00062     flint::mpoly pa(ctx.d), pb(ctx.d), denpa(ctx.d), denpb(ctx.d);
00063
00064     flint::from_argument_mpoly(pa.d, denpa.d, a, false, var_map, ctx.d);
00065     flint::from_argument_mpoly(pb.d, denpb.d, b, false, var_map, ctx.d);
00066
00067     // The input won't have any symbols with negative powers, but there may be rational
00068     // coefficients. Verify this:
00069     if ( fmpz_mpoly_is_fmpz(denpa.d, ctx.d) != 1 ) {
00070         MLOCK(ErrorMessageLock);
00071         MesPrint("flint::divmod_mpoly: error: denpa is non-constant");
00072         MUNLOCK(ErrorMessageLock);
00073         Terminate(-1);
00074     }
00075     if ( fmpz_mpoly_is_fmpz(denpb.d, ctx.d) != 1 ) {
00076         MLOCK(ErrorMessageLock);
00077         MesPrint("flint::divmod_mpoly: error: denpb is non-constant");
00078         MUNLOCK(ErrorMessageLock);
00079         Terminate(-1);
00080     }
00081
00082     flint::fmpz scale;
00083     flint::mpoly div(ctx.d), rem(ctx.d);
00084
00085     fmpz_mpoly_quasidivrem(scale.d, div.d, rem.d, pa.d, pb.d, ctx.d);
00086
00087

```

```

00088 // The quotient must be multiplied by the denominator of the divisor
00089 fmpz_mpoly_mul(div.d, div.d, denpb.d, ctx.d);
00090
00091 // The overall denominator of both div and rem is given by scale*denpa. This we will pass to
00092 // to_argument_mpoly's "denscale" argument which performs the division in the result. We have
00093 // already checked denpa is just an fmpz.
00094 fmpz_mul(scale.d, scale.d, fmpz_mpoly_term_coeff_ref(denpa.d, 0, ctx.d));
00095
00096
00097 // Determine the size of the output by passing write = false.
00098 const bool with_arghead = false;
00099 bool write = false;
00100 const uint64_t prev_size = 0;
00101 const uint64_t out_size = return_rem ?
00102     (uint64_t)flint::to_argument_mpoly(BHEAD NULL, with_arghead, must_fit_term, write, prev_size,
00103         rem.d, var_map, ctx.d, scale.d)
00104     :
00105     (uint64_t)flint::to_argument_mpoly(BHEAD NULL, with_arghead, must_fit_term, write, prev_size,
00106         div.d, var_map, ctx.d, scale.d)
00107     ;
00108 WORD* res = (WORD *)Malloc1(sizeof(WORD)*out_size, "flint::divrem_mpoly");
00109
00110
00111 // Write out the result
00112 write = true;
00113 if ( return_rem ) {
00114     (uint64_t)flint::to_argument_mpoly(BHEAD res, with_arghead, must_fit_term, write, prev_size,
00115         rem.d, var_map, ctx.d, scale.d);
00116 }
00117 else {
00118     (uint64_t)flint::to_argument_mpoly(BHEAD res, with_arghead, must_fit_term, write, prev_size,
00119         div.d, var_map, ctx.d, scale.d);
00120 }
00121
00122 return res;
00123 }
00124 /*
00125 #] flint::divmod_mpoly :
00126 #[ flint::divmod_poly :
00127 */
00128 WORD* flint::divmod_poly(PHEAD const WORD *a, const WORD *b, const bool return_rem,
00129     const WORD must_fit_term, const var_map_t &var_map) {
00130
00131     flint::poly pa, pb, denpa, denpb;
00132
00133     flint::from_argument_poly(pa.d, denpa.d, a, false);
00134     flint::from_argument_poly(pb.d, denpb.d, b, false);
00135
00136     // The input won't have any symbols with negative powers, but there may be rational
00137     // coefficients. Verify this:
00138     if ( fmpz_poly_length(denpa.d) != 1 ) {
00139         MLOCK(ErrorMessageLock);
00140         MesPrint("flint::divmod_poly: error: denpa is non-constant");
00141         MUNLOCK(ErrorMessageLock);
00142         Terminate(-1);
00143     }
00144     if ( fmpz_poly_length(denpb.d) != 1 ) {
00145         MLOCK(ErrorMessageLock);
00146         MesPrint("flint::divmod_poly: error: denpb is non-constant");
00147         MUNLOCK(ErrorMessageLock);
00148         Terminate(-1);
00149     }
00150
00151     flint::fmpz scale;
00152     uint64_t scale_power = 0;
00153     flint::poly div, rem;
00154
00155     // Get the leading coefficient of pb. fmpz_poly_pseudo_divrem returns only the scaling power.
00156     fmpz_poly_get_coeff_fmpz(scale.d, pb.d, fmpz_poly_degree(pb.d));
00157
00158     fmpz_poly_pseudo_divrem(div.d, rem.d, (ulong*)&scale_power, pa.d, pb.d);
00159
00160     // The quotient must be multiplied by the denominator of the divisor
00161     fmpz_poly_mul(div.d, div.d, denpb.d);
00162
00163     // The overall denominator of both div and rem is given by scale^scale_power*denpa. This we will
00164     // pass to to_argument_poly's "denscale" argument which performs the division in the result. We
00165     // have already checked denpa is just an fmpz.
00166     fmpz_pow_ui(scale.d, scale.d, scale_power);
00167     fmpz_mul(scale.d, scale.d, fmpz_poly_get_coeff_ptr(denpa.d, 0));
00168
00169
00170     // Determine the size of the output by passing write = false.
00171     const bool with_arghead = false;
00172     bool write = false;
00173     const uint64_t prev_size = 0;
00174     const uint64_t out_size = return_rem ?

```

```

00175         (uint64_t)flint::to_argument_poly(BHEAD NULL, with_arghead, must_fit_term, write, prev_size,
00176         rem.d, var_map, scale.d)
00177     :
00178     (uint64_t)flint::to_argument_poly(BHEAD NULL, with_arghead, must_fit_term, write, prev_size,
00179     div.d, var_map, scale.d)
00180 ;
00181 WORD* res = (WORD *)Malloca(sizeof(WORD)*out_size, "flint::divrem_poly");
00182
00183
00184 // Write out the result
00185 write = true;
00186 if ( return_rem ) {
00187     (uint64_t)flint::to_argument_poly(BHEAD res, with_arghead, must_fit_term, write, prev_size,
00188     rem.d, var_map, scale.d);
00189 }
00190 else {
00191     (uint64_t)flint::to_argument_poly(BHEAD res, with_arghead, must_fit_term, write, prev_size,
00192     div.d, var_map, scale.d);
00193 }
00194
00195 return res;
00196 }
00197 /*
00198 #] flint::divmod_poly :
00199
00200 #[ flint::factorize_mpoly :
00201 */
00202 WORD* flint::factorize_mpoly(PHEAD const WORD *argin, WORD *argout, const bool with_arghead,
00203 const bool is_fun_arg, const var_map_t &var_map) {
00204
00205     flint::mpoly_ctx ctx(var_map.size());
00206     flint::mpoly arg(ctx.d, den(ctx.d), base(ctx.d));
00207
00208     flint::from_argument_mpoly(arg.d, den.d, argin, with_arghead, var_map, ctx.d);
00209     // The denominator must be 1:
00210     if ( fmpz_mpoly_is_one(den.d, ctx.d) != 1 ) {
00211         MLOCK(ErrorMessageLock);
00212         MesPrint("flint::factorize_mpoly error: den != 1");
00213         MUNLOCK(ErrorMessageLock);
00214         Terminate(-1);
00215     }
00216
00217
00218     // Now we can factor the mpoly:
00219     flint::mpoly_factor arg_fac(ctx.d);
00220     fmpz_mpoly_factor(arg_fac.d, arg.d, ctx.d);
00221     const int64_t num_factors = fmpz_mpoly_factor_length(arg_fac.d, ctx.d);
00222
00223     flint::fmpz overall_constant;
00224     fmpz_mpoly_factor_get_constant_fmpz(overall_constant.d, arg_fac.d, ctx.d);
00225     // FORM should always have taken the overall constant out in the content. Thus this overall
00226     // constant factor should be +- 1 here. Verify this:
00227     if ( ! ( fmpz_equal_si(overall_constant.d, 1) || fmpz_equal_si(overall_constant.d, -1) ) ) {
00228         MLOCK(ErrorMessageLock);
00229         MesPrint("flint::factorize_mpoly error: overall constant factor != +-1");
00230         MUNLOCK(ErrorMessageLock);
00231         Terminate(-1);
00232     }
00233
00234     // Construct the output. If argout is not NULL, we write the result there.
00235     // Otherwise, allocate memory.
00236     // The output is zero-terminated list of factors. If with_arghead, each has an arghead which
00237     // contains its size. Otherwise, the factors are zero separated.
00238
00239     // We only need to determine the size of the output if we are allocating memory, but we need to
00240     // loop through the factors to fix their signs anyway. Do both together in one loop:
00241
00242     // Initially 1, for the final trailing 0.
00243     uint64_t output_size = 1;
00244
00245     // For finding the highest symbol, in FORM's lexicographic ordering
00246     var_map_t var_map_inv;
00247     for ( auto x: var_map ) {
00248         var_map_inv[x.second] = x.first;
00249     }
00250
00251     // Store whether we should flip the factor sign in the output:
00252     vector<int32_t> base_sign(num_factors, 0);
00253
00254     for ( int64_t i = 0; i < num_factors; i++ ) {
00255         fmpz_mpoly_factor_get_base(base.d, arg_fac.d, i, ctx.d);
00256         const int64_t exponent = fmpz_mpoly_factor_get_exp_si(arg_fac.d, i, ctx.d);
00257
00258         // poly_factorize makes sure the highest power of the "highest symbol" (in FORM's
00259         // lexicographic ordering) has a positive coefficient. Check this, update overall_constant
00260         // of the factorization if necessary.
00261         // Store the sign per factor, so that we can flip the signs in the output without re-checking

```

```

00262         // the individual terms again.
00263         uint32_t max_var = 0; // FORM symbols start at 20, 0 is a good initial value.
00264         int32_t max_pow = -1;
00265         vector<int64_t> base_term_exponents(var_map.size(), 0);
00266
00267         for ( int64_t j = 0; j < fmpz_mpoly_length(base.d, ctx.d); j++ ) {
00268             fmpz_mpoly_get_term_exp_si((slong*)base_term_exponents.data(), base.d, (slong)j, ctx.d);
00269
00270             for ( size_t k = 0; k < var_map.size(); k++ ) {
00271                 if ( base_term_exponents[k] > 0 && ( var_map_inv.at(k) > max_var ||
00272                     ( var_map_inv.at(k) == max_var && base_term_exponents[k] > max_pow ) ) ) {
00273
00274                     max_var = var_map_inv.at(k);
00275                     max_pow = base_term_exponents[k];
00276                     base_sign[i] = fmpz_sgn(fmpz_mpoly_term_coeff_ref(base.d, j, ctx.d));
00277                 }
00278             }
00279         }
00280         // If this base's sign will be flipped an odd number of times, there is a contribution to
00281         // the overall sign of the whole factorization:
00282         if ( ( base_sign[i] == -1 ) && ( exponent % 2 == 1 ) ) {
00283             fmpz_neg(overall_constant.d, overall_constant.d);
00284         }
00285
00286         // Now determine the output size of the factor, if we are allocating the memory
00287         if ( argout == NULL ) {
00288             const bool write = false;
00289             for ( int64_t j = 0; j < exponent; j++ ) {
00290                 output_size += (uint64_t)flint::to_argument_mpoly(BHEAD NULL, with_arghead,
00291                     is_fun_arg, write, 0, base.d, var_map, ctx.d);
00292             }
00293         }
00294     }
00295     if ( fmpz_sgn(overall_constant.d) == -1 && argout == NULL ) {
00296         // Add space for a fast-notation number or a normal-notation number and zero separator
00297         output_size += with_arghead ? 2 : 4+1;
00298     }
00299
00300     // Now make the allocation if necessary:
00301     if ( argout == NULL ) {
00302         argout = (WORD*)Malloc1(sizeof(WORD)*output_size, "flint::factorize_mpoly");
00303     }
00304
00305     // And now comes the actual output:
00306     WORD* old_argout = argout;
00307
00308     // If the overall sign is negative, first write a full-notation -1. It will be absorbed into the
00309     // overall factor in the content by the caller.
00310     if ( fmpz_sgn(overall_constant.d) == -1 ) {
00311         if ( with_arghead ) {
00312             // poly writes in fast notation in this case. Fast notation is expected by the caller, to
00313             // properly merge it with the overall factor of the content.
00314             *argout++ = -SNUMBER;
00315             *argout++ = -1;
00316         }
00317         else {
00318             *argout++ = 4; // term size
00319             *argout++ = 1; // numerator
00320             *argout++ = 1; // denominator
00321             *argout++ = -3; // coeff size, negative number
00322             *argout++ = 0; // factor separator
00323         }
00324     }
00325 }
00326
00327 for ( int64_t i = 0; i < num_factors; i++ ) {
00328     fmpz_mpoly_factor_get_base(base.d, arg_fac.d, i, ctx.d);
00329     const int64_t exponent = fmpz_mpoly_factor_get_exp_si(arg_fac.d, i, ctx.d);
00330
00331     if ( base_sign[i] == -1 ) {
00332         fmpz_mpoly_neg(base.d, base.d, ctx.d);
00333     }
00334
00335     const bool write = true;
00336     for ( int64_t j = 0; j < exponent; j++ ) {
00337         argout += flint::to_argument_mpoly(BHEAD argout, with_arghead, is_fun_arg, write,
00338             argout-old_argout, base.d, var_map, ctx.d);
00339     }
00340 }
00341 // Final trailing zero to denote the end of the factors.
00342 *argout++ = 0;
00343
00344 return old_argout;
00345 }
00346 /*
00347 #] flint::factorize_mpoly :
00348 #[ flint::factorize_poly :

```



```

00349 */
00350 WORD* flint::factorize_poly(PHEAD const WORD *argin, WORD *argout, const bool with_arghead,
00351     const bool is_fun_arg, const var_map_t &var_map) {
00352     flint::poly arg, den;
00353
00354     flint::from_argument_poly(arg.d, den.d, argin, with_arghead);
00355     // The denominator must be 1:
00356     if ( fmpz_poly_is_one(den.d) != 1 ) {
00357         MLOCK(ErrorMessageLock);
00358         MesPrint("flint::factorize_poly error: den != 1");
00359         MUNLOCK(ErrorMessageLock);
00360         Terminate(-1);
00361     }
00362
00363     // Now we can factor the poly:
00364     flint::poly_factor arg_fac;
00365     fmpz_poly_factor(arg_fac.d, arg.d);
00366     // fmpz_poly_factor_t lacks some convenience functions which fmpz_mpoly_factor_t has.
00367     // I have worked out how to get the factors by looking at how fmpz_poly_factor_print works.
00368     const long num_factors = (arg_fac.d)->num;
00369
00370     // Construct the output. If argout is not NULL, we write the result there.
00371     // Otherwise, allocate memory.
00372     // The output is zero-terminated list of factors. If with_arghead, each has an arghead which
00373     // contains its size. Otherwise, the factors are zero separated.
00374     if ( argout == NULL ) {
00375         // First we need to determine the size of the output. This is the same procedure as the
00376         // loop below, but we don't write anything in to_argument_poly (write arg: false).
00377         // Initially 1, for the final trailing 0.
00378         uint64_t output_size = 1;
00379
00380         for ( long i = 0; i < num_factors; i++ ) {
00381             fmpz_poly_struct* base = (arg_fac.d)->p + i;
00382
00383             const long exponent = (arg_fac.d)->exp[i];
00384
00385             const bool write = false;
00386             for ( long j = 0; j < exponent; j++ ) {
00387                 output_size += (uint64_t)flint::to_argument_poly(BHEAD NULL, with_arghead,
00388                     is_fun_arg, write, 0, base, var_map);
00389             }
00390         }
00391
00392         argout = (WORD*)Malloca1(sizeof(WORD)*output_size, "flint::factorize_poly");
00393     }
00394
00395     WORD* old_argout = argout;
00396
00397     for ( long i = 0; i < num_factors; i++ ) {
00398         fmpz_poly_struct* base = (arg_fac.d)->p + i;
00399
00400         const long exponent = (arg_fac.d)->exp[i];
00401
00402         const bool write = true;
00403         for ( long j = 0; j < exponent; j++ ) {
00404             argout += flint::to_argument_poly(BHEAD argout, with_arghead, is_fun_arg, write,
00405                 argout-old_argout, base, var_map);
00406         }
00407     }
00408     *argout = 0;
00409
00410     return old_argout;
00411 }
00412
00413 /*
00414 #] flint::factorize_poly :
00415 #[ flint::form_sort :
00416 */
00417 // Sort terms using form's sorting routines. Uses a custom (faster) compare routine, since here
00418 // only symbols can appear.
00419 // This is a modified poly_sort from polywrap.cc.
00420 void flint::form_sort(PHEAD WORD *terms) {
00421     if ( terms[0] < 0 ) {
00422         // Fast notation, there is nothing to do
00423         return;
00424     }
00425
00426     const WORD oldsorttype = AR.SortType;
00427     AR.SortType = SORTHIGHFIRST;
00428
00429     const WORD in_size = terms[0] - ARGHEAD;
00430     WORD out_size;

```

```

00436     if ( NewSort(BHEAD0) ) {
00437         Terminate(-1);
00438     }
00439     AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);
00440
00441     // Make sure the symbols are in the right order within the terms
00442     for ( WORD i = ARGHEAD; i < terms[0]; i += terms[i] ) {
00443         if ( SymbolNormalize(terms+i) < 0 || StoreTerm(BHEAD terms+i) ) {
00444             AR.SortType = oldsorttype;
00445             AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
00446             LowerSortLevel();
00447             Terminate(-1);
00448         }
00449     }
00450
00451     if ( ( out_size = EndSort(BHEAD terms+ARGHEAD, 1) ) < 0 ) {
00452         AR.SortType = oldsorttype;
00453         AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
00454         Terminate(-1);
00455     }
00456
00457     // Check the final size
00458     if ( in_size != out_size ) {
00459         MLOCK(ErrorMessageLock);
00460         MesPrint("flint::form_sort: error: unexpected sorted arg length change %d->%d", in_size,
00461             out_size);
00462         MUNLOCK(ErrorMessageLock);
00463         Terminate(-1);
00464     }
00465
00466     AR.SortType = oldsorttype;
00467     AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
00468     terms[1] = 0; // set dirty flag to zero
00469 }
00470 /*
00471 #] flint::form_sort :
00472
00473 #[ flint::from_argument_mpoly :
00474 */
00475 // Convert a FORM argument (or 0-terminated list of terms: with_arghead == false) to a
00476 // (multi-variate) fmpz_mpoly_t poly. The "denominator" is return in denpoly, and contains the
00477 // overall negative-power numeric and symbolic factor.
00478 uint64_t flint::from_argument_mpoly(fmpz_mpoly_t poly, fmpz_mpoly_t denpoly, const WORD *args,
00479     const bool with_arghead, const var_map_t &var_map, const fmpz_mpoly_ctx_t ctx) {
00480
00481     // Some callers re-use their poly, denpoly to avoid calling init/clear unnecessarily.
00482     // Make sure they are 0 to begin.
00483     fmpz_mpoly_set_si(poly, 0, ctx);
00484     fmpz_mpoly_set_si(denpoly, 0, ctx);
00485
00486     // First check for "fast notation" arguments:
00487     if ( *args == -SNUMBER ) {
00488         fmpz_mpoly_set_si(poly, *(args+1), ctx);
00489         fmpz_mpoly_set_si(denpoly, 1, ctx);
00490         return 2;
00491     }
00492
00493     if ( *args == -SYMBOL ) {
00494         // A "fast notation" SYMBOL has a power and coefficient of 1:
00495         vector<uint64_t> exponents(var_map.size(), 0);
00496         exponents[var_map.at(*(args+1))] = 1;
00497
00498         fmpz_mpoly_set_coeff_ui_ui(poly, (ulong)1, (ulong*)exponents.data(), ctx);
00499         fmpz_mpoly_set_ui(denpoly, (ulong)1, ctx);
00500         return 2;
00501     }
00502
00503
00504     // Now we can iterate through the terms of the argument. If we have
00505     // an ARGHEAD, we already know where to terminate. Otherwise we'll have
00506     // to loop until the terminating 0.
00507     const WORD* arg_stop = with_arghead ? args+args[0] : (WORD*)UINT64_MAX;
00508     uint64_t arg_size = 0;
00509     if ( with_arghead ) {
00510         arg_size = args[0];
00511         args += ARGHEAD;
00512     }
00513
00514
00515     // Search for numerical or symbol denominators to create "denpoly".
00516     flint::fmpz den_coeff, tmp;
00517     fmpz_set_si(den_coeff, 1);
00518     vector<uint64_t> neg_exponents(var_map.size(), 0);
00519
00520     for ( const WORD* term = args; term < arg_stop; term += term[0] ) {
00521         const WORD* term_stop = term+term[0];
00522         const WORD coeff_size = (term_stop)[-1];

```

```

00523     const WORD* symbol_stop = term_stop - ABS(coeff_size);
00524     const WORD* t = term;
00525
00526     t++;
00527     if ( t == symbol_stop ) {
00528         // Just a number, no symbols
00529     }
00530     else {
00531         t++; // this entry is SYMBOL
00532         t++; // this entry just has the size of the symbol array, but we can use symbol_stop
00533
00534         for ( const WORD* s = t; s < symbol_stop; s += 2 ) {
00535             if ( *(s+1) < 0 ) {
00536                 neg_exponents[var_map.at(*s)] =
00537                     MaX(neg_exponents[var_map.at(*s)], (uint64_t)(-(*s+1))) );
00538             }
00539         }
00540     }
00541
00542     // Now check for a denominator in the coefficient:
00543     if ( *(symbol_stop+ABS(coeff_size/2)) != 1 ) {
00544         flint::fmpz_set_form(tmp.d, (UWORD*)(symbol_stop+ABS(coeff_size/2)), ABS(coeff_size/2));
00545         // Record the LCM of the coefficient denominators:
00546         fmpz_lcm(den_coeff.d, den_coeff.d, tmp.d);
00547     }
00548
00549     if ( !with_arghead && *term_stop == 0 ) {
00550         // + 1 for the terminating 0
00551         arg_size = term_stop - args + 1;
00552         break;
00553     }
00554 }
00555
00556 // Assemble denpoly.
00557 fmpz_mpoly_set_coeff_fmpz_ui(denpoly, den_coeff.d, (ulong*)neg_exponents.data(), ctx);
00558
00559 // For the term coefficients
00560 flint::fmpz coeff;
00561
00562 for ( const WORD* term = args; term < arg_stop; term += term[0] ) {
00563     const WORD* term_stop = term+term[0];
00564     const WORD coeff_size = (term_stop)[-1];
00565     const WORD* symbol_stop = term_stop - ABS(coeff_size);
00566     const WORD* t = term;
00567
00568     vector<uint64_t> exponents(var_map.size(), 0);
00569
00570     t++; // skip over the total size entry
00571     if ( t == symbol_stop ) {
00572         // Just a number, no symbols
00573     }
00574     else {
00575         t++; // this entry is SYMBOL
00576         t++; // this entry just has the size of the symbol array, but we can use symbol_stop
00577         for ( const WORD* s = t; s < symbol_stop; s += 2 ) {
00578             exponents[var_map.at(*s)] = *(s+1);
00579         }
00580     }
00581
00582     // Now read the coefficient
00583     flint::fmpz_set_form(coeff.d, (UWORD*)symbol_stop, coeff_size/2);
00584
00585     // Multiply by denominator LCM
00586     fmpz_mul(coeff.d, coeff.d, den_coeff.d);
00587
00588     // Shift by neg_exponents
00589     for ( size_t i = 0; i < var_map.size(); i++ ) {
00590         exponents[i] += neg_exponents[i];
00591     }
00592
00593     // Read the denominator if there is one, and divide it out of the coefficient
00594     if ( *(symbol_stop+ABS(coeff_size/2)) != 1 ) {
00595         flint::fmpz_set_form(tmp.d, (UWORD*)(symbol_stop+ABS(coeff_size/2)), ABS(coeff_size/2));
00596         // By construction, this is an exact division
00597         fmpz_divexact(coeff.d, coeff.d, tmp.d);
00598     }
00599
00600     // Push the term to the mpoly, remember to sort when finished! This is much faster than using
00601     // fmpz_mpoly_set_coeff_fmpz_ui when the terms arrive in the "wrong order".
00602     fmpz_mpoly_push_term_fmpz_ui(poly, coeff.d, (ulong*)exponents.data(), ctx);
00603
00604     if ( !with_arghead && *term_stop == 0 ) {
00605         break;
00606     }
00607 }
00608
00609 // And now sort the mpoly

```

```

00610     fmpz_mpoly_sort_terms(poly, ctx);
00611
00612     return arg_size;
00613 }
00614 /*
00615  #] flint::from_argument_mpoly :
00616  #[ flint::from_argument_poly :
00617  */
00618 // Convert a FORM argument (or 0-terminated list of terms: with_arghead == false) to a
00619 // (uni-variate) fmpz_poly_t poly. The "denominator" is return in denpoly, and contains the
00620 // overall negative-power numeric and symbolic factor.
00621 uint64_t flint::from_argument_poly(fmpz_poly_t poly, fmpz_poly_t denpoly, const WORD *args,
00622     const bool with_arghead) {
00623
00624     // Some callers re-use their poly, denpoly to avoid calling init/clear unnecessarily.
00625     // Make sure they are 0 to begin.
00626     fmpz_poly_set_si(poly, 0);
00627     fmpz_poly_set_si(denpoly, 0);
00628
00629     // First check for "fast notation" arguments:
00630     if ( *args == -SNUMBER ) {
00631         fmpz_poly_set_si(poly, *(args+1));
00632         fmpz_poly_set_si(denpoly, 1);
00633         return 2;
00634     }
00635
00636     if ( *args == -SYMBOL ) {
00637         // A "fast notation" SYMBOL has a power and coefficient of 1:
00638         fmpz_poly_set_coeff_si(poly, 1, 1);
00639         fmpz_poly_set_si(denpoly, 1);
00640         return 2;
00641     }
00642
00643     // Now we can iterate through the terms of the argument. If we have
00644     // an ARGHEAD, we already know where to terminate. Otherwise we'll have
00645     // to loop until the terminating 0.
00646     const WORD* arg_stop = with_arghead ? args+args[0] : (WORD*)UINT64_MAX;
00647     uint64_t arg_size = 0;
00648     if ( with_arghead ) {
00649         arg_size = args[0];
00650         args += ARGHEAD;
00651     }
00652
00653     // Search for numerical or symbol denominators to create "denpoly".
00654     flint::fmpz den_coeff, tmp;
00655     fmpz_set_si(den_coeff, 1);
00656     uint64_t neg_exponent = 0;
00657
00658     for ( const WORD* term = args; term < arg_stop; term += term[0] ) {
00659
00660         const WORD* term_stop = term+term[0];
00661         const WORD coeff_size = (term_stop)[-1];
00662         const WORD* symbol_stop = term_stop - ABS(coeff_size);
00663         const WORD* t = term;
00664
00665         t++; // skip over the total size entry
00666         if ( t == symbol_stop ) {
00667             // Just a number, no symbols
00668         }
00669         else {
00670             t++; // this entry is SYMBOL
00671             t++; // this entry is the size of the symbol array
00672             t++; // this is the first (and only) symbol code
00673             if ( *t < 0 ) {
00674                 neg_exponent = MAX(neg_exponent, (uint64_t)(-(*t)) );
00675             }
00676         }
00677
00678         // Now check for a denominator in the coefficient:
00679         if ( *(symbol_stop+ABS(coeff_size/2)) != 1 ) {
00680             flint::fmpz_set_form(tmp.d, (UWORD*)(symbol_stop+ABS(coeff_size/2)), ABS(coeff_size/2));
00681             // Record the LCM of the coefficient denominators:
00682             fmpz_lcm(den_coeff.d, den_coeff.d, tmp.d);
00683         }
00684
00685         if ( *term_stop == 0 ) {
00686             // + 1 for the terminating 0
00687             arg_size = term_stop - args + 1;
00688             break;
00689         }
00690     }
00691     // Assemble denpoly.
00692     fmpz_poly_set_coeff_fmpz(denpoly, neg_exponent, den_coeff.d);
00693
00694     // For the term coefficients
00695     flint::fmpz coeff;

```

```

00697
00698     for ( const WORD* term = args; term < arg_stop; term += term[0] ) {
00699
00700         const WORD* term_stop = term+term[0];
00701         const WORD  coeff_size = (term_stop)[-1];
00702         const WORD* symbol_stop = term_stop - ABS(coeff_size);
00703         const WORD* t = term;
00704
00705         uint64_t exponent = 0;
00706
00707         t++; // skip over the total size entry
00708         if ( t == symbol_stop ) {
00709             // Just a number, no symbols
00710         }
00711         else {
00712             t++; // this entry is SYMBOL
00713             t++; // this entry is the size of the symbol array
00714             t++; // this is the first (and only) symbol code
00715             exponent = *t++;
00716         }
00717
00718         // Now read the coefficient
00719         flint::fmpz_set_form(coeff.d, (UWORD*)symbol_stop, coeff_size/2);
00720
00721         // Multiply by denominator LCM
00722         fmpz_mul(coeff.d, coeff.d, den_coeff.d);
00723
00724         // Shift by neg_exponent
00725         exponent += neg_exponent;
00726
00727         // Read the denominator if there is one, and divide it out of the coefficient
00728         if ( *(symbol_stop+ABS(coeff_size/2)) != 1 ) {
00729             flint::fmpz_set_form(tmp.d, (UWORD*)(symbol_stop+ABS(coeff_size/2)), ABS(coeff_size/2));
00730             // By construction, this is an exact division
00731             fmpz_divexact(coeff.d, coeff.d, tmp.d);
00732         }
00733
00734         // Add the term to the poly
00735         fmpz_poly_set_coeff_fmpz(poly, exponent, coeff.d);
00736
00737         if ( *term_stop == 0 ) {
00738             break;
00739         }
00740     }
00741
00742     return arg_size;
00743 }
00744 /*
00745 #] flint::from_argument_poly :
00746
00747 #[ flint::fmpz_get_form :
00748 */
00749 // Write FORM's long integer representation of an fmpz at a, and put the number of WORDs at na.
00750 // na carries the sign of the integer.
00751 WORD flint::fmpz_get_form(fmpz_t z, WORD *a) {
00752
00753     WORD na = 0;
00754     const int32_t sgn = fmpz_sgn(z);
00755     if ( sgn == -1 ) {
00756         fmpz_neg(z, z);
00757     }
00758     const int64_t nlimbs = fmpz_size(z);
00759
00760     // This works but is UB?
00761     //fmpz_get_ui_array(reinterpret_cast<uint64_t*>(a), nlimbs, z);
00762
00763     // Use fixed-size functions to get limb data where possible. These probably cover most real
00764     // cases.
00765     if ( nlimbs == 1 ) {
00766         const uint64_t limb = fmpz_get_ui(z);
00767         a[0] = (WORD)(limb & 0xFFFFFFFF);
00768         na++;
00769         a[1] = (WORD)(limb >> BITSINWORD);
00770         if ( a[1] != 0 ) {
00771             na++;
00772         }
00773     }
00774     else if ( nlimbs == 2 ) {
00775         uint64_t limb_hi = 0, limb_lo = 0;
00776         fmpz_get_uivi((ulong*)&limb_hi, (ulong*)&limb_lo, z);
00777         a[0] = (WORD)(limb_lo & 0xFFFFFFFF);
00778         na++;
00779         a[1] = (WORD)(limb_lo >> BITSINWORD);
00780         na++;
00781         a[2] = (WORD)(limb_hi & 0xFFFFFFFF);
00782         na++;
00783         a[3] = (WORD)(limb_hi >> BITSINWORD);

```

```

00784         if ( a[3] != 0 ) {
00785             na++;
00786         }
00787     }
00788     else {
00789         vector<uint64_t> limb_data(nlimbs, 0);
00790         fmpz_get_ui_array((ulong*)limb_data.data(), (ulong)nlimbs, z);
00791         for ( long i = 0; i < nlimbs; i++ ) {
00792             a[2*i] = (WORD)(limb_data[i] & 0xFFFFFFFF);
00793             na++;
00794             a[2*i+1] = (WORD)(limb_data[i] >> BITSINWORD);
00795             if ( a[2*i+1] != 0 || i < (nlimbs-1) ) {
00796                 // The final limb might fit in a single 32bit WORD. Only
00797                 // increment na if the final WORD is non zero.
00798                 na++;
00799             }
00800         }
00801     }
00802
00803     // And now put the sign in the number of limbs
00804     if ( sgn == -1 ) {
00805         na = -na;
00806     }
00807
00808     return na;
00809 }
00810 /*
00811 #] flint::fmpz_get_form :
00812 #[ flint::fmpz_set_form :
00813 */
00814 // Create an fmpz directly from FORM's long integer representation. fmpz uses 64bit unsigned limbs,
00815 // but FORM uses 32bit UWORDS on 64bit architectures so we can't use fmpz_set_ui_array directly.
00816 void flint::fmpz_set_form(fmpz_t z, UWORD *a, WORD na) {
00817
00818     if ( na == 0 ) {
00819         fmpz_zero(z);
00820         return;
00821     }
00822
00823     // Negative na represents a negative number
00824     int32_t sgn = 1;
00825     if ( na < 0 ) {
00826         sgn = -1;
00827         na = -na;
00828     }
00829
00830     // Remove padding. FORM stores numerators and denominators with equal numbers of limbs but we
00831     // don't need to do this within the fmpz. It is not necessary to do this really, the fmpz
00832     // doesn't add zero limbs unnecessarily, but we might be able to avoid creating the limb_data
00833     // array below.
00834     while ( a[na-1] == 0 ) {
00835         na--;
00836     }
00837
00838     // If the number fits in fixed-size fmpz_set functions, we don't need to use additional memory
00839     // to convert to uint64_t. These probably cover most real cases.
00840     if ( na == 1 ) {
00841         fmpz_set_ui(z, (uint64_t)a[0]);
00842     }
00843     else if ( na == 2 ) {
00844         fmpz_set_ui(z, (((uint64_t)a[1])<<BITSINWORD) + (uint64_t)a[0]);
00845     }
00846     else if ( na == 3 ) {
00847         fmpz_set_uiui(z, (uint64_t)a[2], (((uint64_t)a[1])<<BITSINWORD) + (uint64_t)a[0]);
00848     }
00849     else if ( na == 4 ) {
00850         fmpz_set_uiui(z, (((uint64_t)a[3])<<BITSINWORD) + (uint64_t)a[2],
00851             (((uint64_t)a[1])<<BITSINWORD) + (uint64_t)a[0]);
00852     }
00853     else {
00854         const int32_t nlimbs = (na+1)/2;
00855         vector<uint64_t> limb_data(nlimbs, 0);
00856         for ( int32_t i = 0; i < nlimbs; i++ ) {
00857             if ( 2*i+1 <= na-1 ) {
00858                 limb_data[i] = (uint64_t)a[2*i] + (((uint64_t)a[2*i+1])<<BITSINWORD);
00859             }
00860             else {
00861                 limb_data[i] = (uint64_t)a[2*i];
00862             }
00863         }
00864         fmpz_set_ui_array(z, (ulong*)limb_data.data(), nlimbs);
00865     }
00866
00867     // Finally set the sign.
00868     if ( sgn == -1 ) {
00869         fmpz_neg(z, z);
00870     }

```

```

00871
00872     return;
00873 }
00874 /*
00875 #] flint::fmpz_set_form :
00876
00877 #[ flint::gcd_mpoly :
00878 */
00879 // Return a pointer to a buffer containing the GCD of the 0-terminated term lists at a and b.
00880 // If must_fit_term, this should be a TermMalloc buffer. Otherwise Malloc1 the buffer.
00881 // For multi-variate cases.
00882 WORD* flint::gcd_mpoly(PHEAD const WORD *a, const WORD *b, const WORD must_fit_term,
00883     const var_map_t &var_map) {
00884
00885     flint::mpoly_ctx ctx(var_map.size());
00886     flint::mpoly pa(ctx.d), pb(ctx.d), denpa(ctx.d), denpb(ctx.d), gcd(ctx.d);
00887
00888     flint::from_argument_mpoly(pa.d, denpa.d, a, false, var_map, ctx.d);
00889     flint::from_argument_mpoly(pb.d, denpb.d, b, false, var_map, ctx.d);
00890
00891     // denpa, denpb must be 1:
00892     if ( fmpz_mpoly_is_one(denpa.d, ctx.d) != 1 ) {
00893         MLOCK(ErrorMessageLock);
00894         MesPrint("flint::gcd_mpoly: error: denpa != 1");
00895         MUNLOCK(ErrorMessageLock);
00896         Terminate(-1);
00897     }
00898     if ( fmpz_mpoly_is_one(denpb.d, ctx.d) != 1 ) {
00899         MLOCK(ErrorMessageLock);
00900         MesPrint("flint::gcd_mpoly: error: denpb != 1");
00901         MUNLOCK(ErrorMessageLock);
00902         Terminate(-1);
00903     }
00904
00905     // poly returns pa if pa == pb, regardless of the lcoeff sign
00906     if ( fmpz_mpoly_equal(pa.d, pb.d, ctx.d) ) {
00907         fmpz_mpoly_set(gcd.d, pa.d, ctx.d);
00908     }
00909     else {
00910         // We need some gymnastics to have the same sign conventions as the poly class. It takes the
00911         // integer or univar content out of a,b, with the convention that the content sign matches
00912         // the lcoeff sign. Since FORM has already taken out the content, we are left with +-1. In
00913         // Flint, the content always has a positive sign so here we should find +1. Check this:
00914         flint::mpoly tmp(ctx.d);
00915         fmpz_mpoly_term_content(tmp.d, pa.d, ctx.d);
00916         if ( fmpz_mpoly_is_one(tmp.d, ctx.d) != 1 ) {
00917             MLOCK(ErrorMessageLock);
00918             MesPrint("flint::gcd_mpoly: error: content of 1st arg != 1");
00919             MUNLOCK(ErrorMessageLock);
00920             Terminate(-1);
00921         }
00922         fmpz_mpoly_term_content(tmp.d, pb.d, ctx.d);
00923         if ( fmpz_mpoly_is_one(tmp.d, ctx.d) != 1 ) {
00924             MLOCK(ErrorMessageLock);
00925             MesPrint("flint::gcd_mpoly: error: content of 2nd arg != 1");
00926             MUNLOCK(ErrorMessageLock);
00927             Terminate(-1);
00928         }
00929
00930         // The poly class now divides the content out of a,b so that they have a positive lcoeff.
00931         // Then it multiplies the final gcd (which is given a positive lcoeff also) by
00932         // gcd(cont a, cont b). There it has gcd(1,1) = gcd(-1,1) = gcd(1,-1) = 1, and
00933         // gcd(-1,-1) = -1 (because of the pa==pb early return). So: if both input polys have a
00934         // negative lcoeff, we will flip the sign in the final result.
00935         bool flip_sign = 0;
00936         if ( ( fmpz_sgn(fmpz_mpoly_term_coeff_ref(pa.d, 0, ctx.d)) == -1 ) &&
00937             ( fmpz_sgn(fmpz_mpoly_term_coeff_ref(pb.d, 0, ctx.d)) == -1 ) ) {
00938             flip_sign = 1;
00939         }
00940
00941         fmpz_mpoly_gcd(gcd.d, pa.d, pb.d, ctx.d);
00942         if ( flip_sign ) {
00943             fmpz_mpoly_neg(gcd.d, gcd.d, ctx.d);
00944         }
00945     }
00946
00947     // This is freed by the caller
00948     WORD *res;
00949     if ( must_fit_term ) {
00950         res = TermMalloc("flint::gcd_mpoly");
00951     }
00952     else {
00953         // Determine the size of the GCD by passing write = false.
00954         const bool with_arghead = false;
00955         const bool write = false;
00956         const uint64_t prev_size = 0;
00957         const uint64_t gcd_size = (uint64_t)flint::to_argument_mpoly(BHEAD NULL,

```

```

00958         with_arghead, must_fit_term, write, prev_size, gcd.d, var_map, ctx.d);
00959
00960     res = (WORD *)Malloca(sizeof(WORD)*gcd_size, "flint::gcd_mpoly");
00961 }
00962
00963     const bool with_arghead = false;
00964     const bool write = true;
00965     const uint64_t prev_size = 0;
00966     flint::to_argument_mpoly(BHEAD res, with_arghead, must_fit_term, write, prev_size, gcd.d,
00967         var_map, ctx.d);
00968
00969     return res;
00970 }
00971 /*
00972 #] flint::gcd_mpoly :
00973 #[ flint::gcd_poly :
00974 */
00975 // Return a pointer to a buffer containing the GCD of the 0-terminated term lists at a and b.
00976 // If must_fit_term, this should be a TermMalloc buffer. Otherwise Malloca the buffer.
00977 // For uni-variate cases.
00978 WORD* flint::gcd_poly(PHEAD const WORD *a, const WORD *b, const WORD must_fit_term,
00979     const var_map_t &var_map) {
00980
00981     flint::poly pa, pb, denpa, denpb, gcd;
00982
00983     flint::from_argument_poly(pa.d, denpa.d, a, false);
00984     flint::from_argument_poly(pb.d, denpb.d, b, false);
00985
00986     // denpa, denpb must be 1:
00987     if ( fmpz_poly_is_one(denpa.d) != 1 ) {
00988         MLOCK(ErrorMessageLock);
00989         MesPrint("flint::gcd_poly: error: denpa != 1");
00990         MUNLOCK(ErrorMessageLock);
00991         Terminate(-1);
00992     }
00993     if ( fmpz_poly_is_one(denpb.d) != 1 ) {
00994         MLOCK(ErrorMessageLock);
00995         MesPrint("flint::gcd_poly: error: denpb != 1");
00996         MUNLOCK(ErrorMessageLock);
00997         Terminate(-1);
00998     }
00999
01000     // poly returns pa if pa == pb, regardless of the lcoeff sign
01001     if ( fmpz_poly_equal(pa.d, pb.d) ) {
01002         fmpz_poly_set(gcd.d, pa.d);
01003     }
01004     else {
01005         // Here, we don't have to make any sign flips like the mpoly case, because poly's
01006         // integer_gcd(1,1) = integer_gcd(-1,1) = integer_gcd(1,-1) = integer_gcd(-1,-1) = +1.
01007         // Still, verify that the content is 1:
01008         flint::fmpz tmp;
01009         fmpz_poly_content(tmp.d, pa.d);
01010         if ( fmpz_is_one(tmp.d) != 1 ) {
01011             MLOCK(ErrorMessageLock);
01012             MesPrint("flint::gcd_poly: error: content of 1st arg != 1");
01013             MUNLOCK(ErrorMessageLock);
01014             Terminate(-1);
01015         }
01016         fmpz_poly_content(tmp.d, pb.d);
01017         if ( fmpz_is_one(tmp.d) != 1 ) {
01018             MLOCK(ErrorMessageLock);
01019             MesPrint("flint::gcd_poly: error: content of 2nd arg != 1");
01020             MUNLOCK(ErrorMessageLock);
01021             Terminate(-1);
01022         }
01023
01024         fmpz_poly_gcd(gcd.d, pa.d, pb.d);
01025     }
01026
01027     // This is freed by the caller
01028     WORD *res;
01029     if ( must_fit_term ) {
01030         res = TermMalloc("flint::gcd_poly");
01031     }
01032     else {
01033         // Determine the size of the GCD by passing write = false.
01034         const bool with_arghead = false;
01035         const bool write = false;
01036         const uint64_t prev_size = 0;
01037         const uint64_t gcd_size = (uint64_t)flint::to_argument_poly(BHEAD NULL,
01038             with_arghead, must_fit_term, write, prev_size, gcd.d, var_map);
01039
01040         res = (WORD *)Malloca(sizeof(WORD)*gcd_size, "flint::gcd_poly");
01041     }
01042
01043     const bool with_arghead = false;
01044     const uint64_t prev_size = 0;

```



```

01045     const bool write = true;
01046     flint::to_argument_poly(BHEAD res, with_arghead, must_fit_term, write, prev_size, gcd.d,
01047         var_map);
01048
01049     return res;
01050 }
01051 /*
01052 #] flint::gcd_poly :
01053
01054 #[ flint::get_variables :
01055 */
01056 // Get the list of symbols which appear in the vector of expressions. These are polyratfun
01057 // numerators, denominators or expressions from calls to gcd_ etc. Return this list as a map
01058 // between indices and symbol codes.
01059 // TODO FACTORSYMBOL last?
01060 flint::var_map_t flint::get_variables(const vector<WORD*> &es, const bool with_arghead,
01061     const bool sort_vars) {
01062
01063     int32_t num_vars = 0;
01064     // To be used if we sort by highest degree, as the polu code does.
01065     vector<int32_t> degrees;
01066     var_map_t var_map;
01067
01068     // extract all variables
01069     for ( size_t ei = 0; ei < es.size(); ei++ ) {
01070         WORD *e = es[ei];
01071
01072         // fast notation
01073         if ( *e == -SNUMBER ) {
01074             }
01075         else if ( *e == -SYMBOL ) {
01076             if ( !var_map.count(e[1]) ) {
01077                 var_map[e[1]] = num_vars++;
01078                 degrees.push_back(1);
01079             }
01080         }
01081         // JD: Here we need to check for non-symbol/number terms in fast notation.
01082         else if ( *e < 0 ) {
01083             MLOCK(ErrorMessageLock);
01084             MesPrint("ERROR: polynomials and polyratfuns must contain symbols only");
01085             MUNLOCK(ErrorMessageLock);
01086             Terminate(1);
01087         }
01088         else {
01089             for ( WORD i = with_arghead ? ARGHEAD:0; with_arghead ? i < e[0]:e[i] != 0; i += e[i] ) {
01090                 if ( i+1 < i+e[i]-ABS(e[i+e[i]-1]) && e[i+1] != SYMBOL ) {
01091                     MLOCK(ErrorMessageLock);
01092                     MesPrint("ERROR: polynomials and polyratfuns must contain symbols only");
01093                     MUNLOCK(ErrorMessageLock);
01094                     Terminate(1);
01095                 }
01096
01097                 for ( WORD j = i+3; j<i+e[i]-ABS(e[i+e[i]-1]); j += 2 ) {
01098                     if ( !var_map.count(e[j]) ) {
01099                         var_map[e[j]] = num_vars++;
01100                         degrees.push_back(e[j+1]);
01101                     }
01102                     else {
01103                         degrees[var_map[e[j]]] = MaX(degrees[var_map[e[j]]], e[j+1]);
01104                     }
01105                 }
01106             }
01107         }
01108     }
01109
01110     if ( sort_vars ) {
01111         // bubble sort variables in decreasing order of degree
01112         // (this seems better for factorization)
01113         for ( size_t i = 0; i < var_map.size(); i++ ) {
01114             for ( size_t j = 0; j+1 < var_map.size(); j++ ) {
01115                 if ( degrees[j] < degrees[j+1] ) {
01116                     swap(degrees[j], degrees[j+1]);
01117
01118                     // Find the map keys associated with the values we want to swap
01119                     uint32_t j0 = 0;
01120                     uint32_t j1 = 0;
01121                     for ( auto x: var_map ) {
01122                         if ( x.second == j ) {
01123                             j0 = x.first;
01124                         }
01125                         else if ( x.second == j+1 ) {
01126                             j1 = x.first;
01127                         }
01128                     }
01129                     swap(var_map.at(j0), var_map.at(j1));
01130                 }
01131             }
01132         }
01133     }

```

```

01132     }
01133 }
01134 // Otherwise, sort lexicographically in FORM's ordering
01135 else {
01136     for ( size_t i = 0; i < var_map.size(); i++ ) {
01137         for ( size_t j = 0; j+1 < var_map.size(); j++ ) {
01138             uint32_t j0 = 0;
01139             uint32_t j1 = 0;
01140             for ( auto x: var_map ) {
01141                 if ( x.second == j ) {
01142                     j0 = x.first;
01143                 }
01144                 else if ( x.second == j+1 ) {
01145                     j1 = x.first;
01146                 }
01147             }
01148             if ( j0 > j1 ) {
01149                 swap(var_map.at(j0), var_map.at(j1));
01150             }
01151         }
01152     }
01153 }
01154
01155 return var_map;
01156 }
01157 /*
01158 #] flint::get_variables :
01159
01160 #[ flint::inverse_poly :
01161 */
01162 WORD* flint::inverse_poly(PHEAD const WORD *a, const WORD *b, const var_map_t &var_map) {
01163
01164     flint::poly pa, pb, denpa, denpb;
01165
01166     flint::from_argument_poly(pa.d, denpa.d, a, false);
01167     flint::from_argument_poly(pb.d, denpb.d, b, false);
01168
01169     // fmpz_poly_xgcd is undefined if the content of pa and pb are not 1. Take the content out.
01170     // fmpz_poly_content gives the non-negative content and fmpz_poly_primitive normalizes to a
01171     // non-negative lcoeff, so we need to add the sign to the content if the polys have a negative
01172     // lcoeff. We don't need to keep the content of pb, it is a numerical multiple of the modulus.
01173     flint::fmpz content_a, resultant;
01174     flint::poly inverse, tmp;
01175     fmpz_poly_content(content_a.d, pa.d);
01176     if ( fmpz_sgn(fmpz_poly_lead(pa.d)) == -1 ) {
01177         fmpz_neg(content_a.d, content_a.d);
01178     }
01179     fmpz_poly_primitive_part(pa.d, pa.d);
01180     fmpz_poly_primitive_part(pb.d, pb.d);
01181
01182     // Special cases:
01183     // Possibly strange that we give 1 for inverse_(x1,1) but here we take MMA's convention.
01184     if ( fmpz_poly_is_one(pa.d) && fmpz_poly_is_one(pb.d) ) {
01185         fmpz_poly_one(inverse.d);
01186         fmpz_one(resultant.d);
01187     }
01188     else if ( fmpz_poly_is_one(pb.d) ) {
01189         fmpz_poly_zero(inverse.d);
01190         fmpz_one(resultant.d);
01191     }
01192     else {
01193         // Now use the extended Euclidean algorithm to find inverse, resultant, tmp of the Bezout
01194         // identity: inverse*pa + tmp*pb = resultant. Then inverse/resultant is the multiplicative
01195         // inverse of pa mod pb. We'll divide by resultant in the denscale argument of
01196         to_argument_poly.
01197         fmpz_poly_xgcd(resultant.d, inverse.d, tmp.d, pa.d, pb.d);
01198
01199         // If the resultant is zero, the inverse does not exist:
01200         if ( fmpz_is_zero(resultant.d) ) {
01201             MLOCK(ErrorMessageLock);
01202             MesPrint("flint::inverse_poly error: inverse does not exist");
01203             MUNLOCK(ErrorMessageLock);
01204             Terminate(-1);
01205         }
01206
01207         // Multiply inverse by denpa. denpb is a numerical multiple of the modulus, so doesn't matter.
01208         // We also need to divide by content_a, which we do in the denscale argument of to_argument_poly
01209         // by multiplying resultant by content_a here.
01210         fmpz_poly_mul(inverse.d, inverse.d, denpa.d);
01211         fmpz_mul(resultant.d, resultant.d, content_a.d);
01212
01213
01214     WORD* res;
01215     // First determine the result size, and malloc. The result should have no arghead. Here we use
01216     // the "scale" argument of to_argument_poly since resultant might not be 1.
01217     const bool with_arghead = false;

```

```

01218     bool write = false;
01219     const bool must_fit_term = false;
01220     const uint64_t prev_size = 0;
01221     const uint64_t res_size = (uint64_t)flint::to_argument_poly(BHEAD NULL,
01222         with_arghead, must_fit_term, write, prev_size, inverse.d, var_map, resultant.d);
01223     res = (WORD*)Malloca(sizeof(WORD)*res_size, "flint::inverse_poly");
01224
01225     write = true;
01226     flint::to_argument_poly(BHEAD res, with_arghead, must_fit_term, write, prev_size, inverse.d,
01227         var_map, resultant.d);
01228
01229     return res;
01230 }
01231 /*
01232 #] flint::inverse_poly :
01233
01234 #[ flint::mul_mpoly :
01235 */
01236 // Return a pointer to a buffer containing the product of the 0-terminated term lists at a and b.
01237 // For multi-variate cases.
01238 WORD* flint::mul_mpoly(PHEAD const WORD *a, const WORD *b, const var_map_t &var_map) {
01239
01240     flint::mpoly_ctx ctx(var_map.size());
01241     flint::mpoly pa(ctx.d), pb(ctx.d), denpa(ctx.d), denpb(ctx.d);
01242
01243     flint::from_argument_mpoly(pa.d, denpa.d, a, false, var_map, ctx.d);
01244     flint::from_argument_mpoly(pb.d, denpb.d, b, false, var_map, ctx.d);
01245
01246     // denpa, denpb must be integers. Negative symbol powers have been converted to extra symbols.
01247     if ( fmpz_mpoly_is_fmpz(denpa.d, ctx.d) != 1 ) {
01248         MLOCK(ErrorMessageLock);
01249         MesPrint("flint::mul_mpoly: error: denpa is non-constant");
01250         MUNLOCK(ErrorMessageLock);
01251         Terminate(-1);
01252     }
01253     if ( fmpz_mpoly_is_fmpz(denpb.d, ctx.d) != 1 ) {
01254         MLOCK(ErrorMessageLock);
01255         MesPrint("flint::mul_mpoly: error: denpb is non-constant");
01256         MUNLOCK(ErrorMessageLock);
01257         Terminate(-1);
01258     }
01259
01260     // Multiply numerators, store result in pa
01261     fmpz_mpoly_mul(pa.d, pa.d, pb.d, ctx.d);
01262     // Multiply denominators, store result in denpa, and convert to an fmpz:
01263     fmpz_mpoly_mul(denpa.d, denpa.d, denpb.d, ctx.d);
01264     flint::fmpz den;
01265     fmpz_mpoly_get_fmpz(den.d, denpa.d, ctx.d);
01266
01267     WORD* res;
01268     // First determine the result size, and malloc. The result should have no arghead. Here we use
01269     // the "scale" argument of to_argument_mpoly since den might not be 1.
01270     const bool with_arghead = false;
01271     bool write = false;
01272     const bool must_fit_term = false;
01273     const uint64_t prev_size = 0;
01274     const uint64_t mul_size = (uint64_t)flint::to_argument_mpoly(BHEAD NULL,
01275         with_arghead, must_fit_term, write, prev_size, pa.d, var_map, ctx.d, den.d);
01276     res = (WORD*)Malloca(sizeof(WORD)*mul_size, "flint::mul_mpoly");
01277
01278     write = true;
01279     flint::to_argument_mpoly(BHEAD res, with_arghead, must_fit_term, write, prev_size, pa.d,
01280         var_map, ctx.d, den.d);
01281
01282     return res;
01283 }
01284 /*
01285 #] flint::mul_mpoly :
01286 #[ flint::mul_poly :
01287 */
01288 // Return a pointer to a buffer containing the product of the 0-terminated term lists at a and b.
01289 // For uni-variate cases.
01290 WORD* flint::mul_poly(PHEAD const WORD *a, const WORD *b, const var_map_t &var_map) {
01291
01292     flint::poly pa, pb, denpa, denpb;
01293
01294     flint::from_argument_poly(pa.d, denpa.d, a, false);
01295     flint::from_argument_poly(pb.d, denpb.d, b, false);
01296
01297     // denpa, denpb must be integers. Negative symbol powers have been converted to extra symbols.
01298     if ( fmpz_poly_degree(denpa.d) != 0 ) {
01299         MLOCK(ErrorMessageLock);
01300         MesPrint("flint::mul_poly: error: denpa is non-constant");
01301         MUNLOCK(ErrorMessageLock);
01302         Terminate(-1);
01303     }
01304     if ( fmpz_poly_degree(denpb.d) != 0 ) {

```

```

01305         MLOCK(ErrorMessageLock);
01306         MesPrint("flint::mul_poly: error: denpb is non-constant");
01307         MUNLOCK(ErrorMessageLock);
01308         Terminate(-1);
01309     }
01310
01311     // Multiply numerators, store result in pa
01312     fmpz_poly_mul(pa.d, pa.d, pb.d);
01313     // Multiply denominators, store result in denpa, and convert to an fmpz:
01314     fmpz_poly_mul(denpa.d, denpa.d, denpb.d);
01315     flint::fmpz den;
01316     fmpz_poly_get_coeff_fmpz(den.d, denpa.d, 0);
01317
01318     WORD* res;
01319     // First determine the result size, and malloc. The result should have no arghead. Here we use
01320     // the "scale" argument of to_argument_poly since den might not be 1.
01321     const bool with_arghead = false;
01322     bool write = false;
01323     const bool must_fit_term = false;
01324     const uint64_t prev_size = 0;
01325     const uint64_t mul_size = (uint64_t)flint::to_argument_poly(BHEAD NULL,
01326         with_arghead, must_fit_term, write, prev_size, pa.d, var_map, den.d);
01327     res = (WORD*)Malloca(sizeof(WORD)*mul_size, "flint::mul_poly");
01328
01329     write = true;
01330     flint::to_argument_poly(BHEAD res, with_arghead, must_fit_term, write, prev_size, pa.d,
01331         var_map, den.d);
01332
01333     return res;
01334 }
01335 /*
01336 #] flint::mul_poly :
01337
01338 #[ flint::ratfun_add_mpoly :
01339 */
01340 // Add the multi-variate FORM rational polynomials at t1 and t2. The result is written at out.
01341 void flint::ratfun_add_mpoly(PHEAD const WORD *t1, const WORD *t2, WORD *out,
01342     const var_map_t &var_map) {
01343
01344     flint::mpoly_ctx ctx(var_map.size());
01345     flint::mpoly gcd(ctx.d, num1(ctx.d), den1(ctx.d), num2(ctx.d), den2(ctx.d));
01346
01347     flint::ratfun_read_mpoly(t1, num1.d, den1.d, var_map, ctx.d);
01348     flint::ratfun_read_mpoly(t2, num2.d, den2.d, var_map, ctx.d);
01349
01350     if ( fmpz_mpoly_cmp(den1.d, den2.d, ctx.d) != 0 ) {
01351         fmpz_mpoly_gcd_cofactors(gcd.d, den1.d, den2.d, den1.d, den2.d, ctx.d);
01352
01353         fmpz_mpoly_mul(num1.d, num1.d, den2.d, ctx.d);
01354         fmpz_mpoly_mul(num2.d, num2.d, den1.d, ctx.d);
01355
01356         fmpz_mpoly_add(num1.d, num1.d, num2.d, ctx.d);
01357         fmpz_mpoly_mul(den1.d, den1.d, den2.d, ctx.d);
01358         fmpz_mpoly_mul(den1.d, den1.d, gcd.d, ctx.d);
01359     }
01360     else {
01361         fmpz_mpoly_add(num1.d, num1.d, num2.d, ctx.d);
01362     }
01363
01364     // Finally divide out any common factors between the resulting num1, den1:
01365     fmpz_mpoly_gcd_cofactors(gcd.d, num1.d, den1.d, num1.d, den1.d, ctx.d);
01366
01367     flint::util::fix_sign_fmpz_mpoly_ratfun(num1.d, den1.d, ctx.d);
01368
01369     // Result in FORM notation:
01370     *out++ = AR.PolyFun;
01371     WORD* args_size = out++;
01372     WORD* args_flag = out++;
01373     *args_flag = 0; // clean prf
01374     FILLFUN3(out); // Remainder of funhead, if it is larger than 3
01375
01376     const bool with_arghead = true;
01377     const bool must_fit_term = true;
01378     const bool write = true;
01379     // prev_size + 4, to account for final term size and coeff of "1/1"
01380     out += flint::to_argument_mpoly(BHEAD out, with_arghead, must_fit_term, write, out-args_size+4,
01381         num1.d, var_map, ctx.d);
01382     out += flint::to_argument_mpoly(BHEAD out, with_arghead, must_fit_term, write, out-args_size+4,
01383         den1.d, var_map, ctx.d);
01384
01385     *args_size = out - args_size + 1; // The +1 is to include the function ID
01386     AT.WorkPointer = out;
01387 }
01388 /*
01389 #] flint::ratfun_add_mpoly :
01390 #[ flint::ratfun_add_poly :
01391 */

```

```

01392 // Add the uni-variate FORM rational polynomials at t1 and t2. The result is written at out.
01393 void flint::ratfun_add_poly(PHEAD const WORD *t1, const WORD *t2, WORD *out,
01394     const var_map_t &var_map) {
01395
01396     flint::poly gcd, num1, den1, num2, den2;
01397
01398     flint::ratfun_read_poly(t1, num1.d, den1.d);
01399     flint::ratfun_read_poly(t2, num2.d, den2.d);
01400
01401     if ( fmpz_poly_equal(den1.d, den2.d) == 0 ) {
01402         flint::util::simplify_fmpz_poly(den1.d, den2.d, gcd.d);
01403
01404         fmpz_poly_mul(num1.d, num1.d, den2.d);
01405         fmpz_poly_mul(num2.d, num2.d, den1.d);
01406
01407         fmpz_poly_add(num1.d, num1.d, num2.d);
01408         fmpz_poly_mul(den1.d, den1.d, den2.d);
01409         fmpz_poly_mul(den1.d, den1.d, gcd.d);
01410     }
01411     else {
01412         fmpz_poly_add(num1.d, num1.d, num2.d);
01413     }
01414
01415     // Finally divide out any common factors between the resulting num1, den1:
01416     flint::util::simplify_fmpz_poly(num1.d, den1.d, gcd.d);
01417
01418     flint::util::fix_sign_fmpz_poly_ratfun(num1.d, den1.d);
01419
01420     // Result in FORM notation:
01421     *out++ = AR.PolyFun;
01422     WORD* args_size = out++;
01423     WORD* args_flag = out++;
01424     *args_flag = 0; // clean prf
01425     FILLFUN3(out); // Remainder of funhead, if it is larger than 3
01426
01427     const bool with_arghead = true;
01428     const bool must_fit_term = true;
01429     const bool write = true;
01430     // prev_size + 4, to account for final term size and coeff of "1/1"
01431     out += flint::to_argument_poly(BHEAD out, with_arghead, must_fit_term, write, out-args_size+4,
01432         num1.d, var_map);
01433     out += flint::to_argument_poly(BHEAD out, with_arghead, must_fit_term, write, out-args_size+4,
01434         den1.d, var_map);
01435
01436     *args_size = out - args_size + 1; // The +1 is to include the function ID
01437     AT.WorkPointer = out;
01438 }
01439 /*
01440 #] flint::ratfun_add_poly :
01441
01442 #[ flint::ratfun_normalize_mpoly :
01443 */
01444 // Multiply and simplify occurrences of the multi-variate FORM rational polynomials found in term.
01445 // The final term is written in place, with the rational polynomial at the end.
01446 void flint::ratfun_normalize_mpoly(PHEAD WORD *term, const var_map_t &var_map) {
01447
01448     // The length of the coefficient
01449     const WORD ncoeff = (term + *term)[-1];
01450     // The end of the term data, before the coefficient:
01451     const WORD tstop = term + *term - ABS(ncoeff);
01452
01453     flint::mpoly_ctx ctx(var_map.size());
01454     flint::mpoly num1(ctx.d), den1(ctx.d), num2(ctx.d), den2(ctx.d), gcd(ctx.d);
01455
01456     // Start with "trivial" polynomials, and multiply in the num and den of the prf which appear.
01457     flint::fmpz tmpNum, tmpDen;
01458     flint::fmpz_set_form(tmpNum.d, (UWORD*)tstop, ncoeff/2);
01459     flint::fmpz_set_form(tmpDen.d, (UWORD*)tstop+ABS(ncoeff/2), ABS(ncoeff/2));
01460     fmpz_mpoly_set_fmpz(num1.d, tmpNum.d, ctx.d);
01461     fmpz_mpoly_set_fmpz(den1.d, tmpDen.d, ctx.d);
01462
01463     // Loop over the occurrences of PolyFun in the term, and multiply in to num1, den1.
01464     // s tracks where we are writing the non-PolyFun term data. The final PolyFun will
01465     // go at the end.
01466     WORD* term_size = term;
01467     WORD* s = term + 1;
01468     for ( WORD *t = term + 1; t < tstop; ) {
01469         if ( *t == AR.PolyFun ) {
01470             flint::ratfun_read_mpoly(t, num2.d, den2.d, var_map, ctx.d);
01471
01472             // get gcd of num1,den2 and num2,den1 and then assemble
01473             fmpz_mpoly_gcd_cofactors(gcd.d, num1.d, den2.d, num1.d, den2.d, ctx.d);
01474             fmpz_mpoly_gcd_cofactors(gcd.d, num2.d, den1.d, num2.d, den1.d, ctx.d);
01475
01476             fmpz_mpoly_mul(num1.d, num1.d, num2.d, ctx.d);
01477             fmpz_mpoly_mul(den1.d, den1.d, den2.d, ctx.d);
01478

```

```

01479         t += t[1];
01480     }
01481
01482     else {
01483         // Not a PolyFun, just copy or skip over
01484         WORD i = t[1];
01485         if ( s != t ) { NCOPY(s,t,i); }
01486         else { t += i; s += i; }
01487     }
01488 }
01489
01490 flint::util::fix_sign_fmpz_poly_ratfun(num1.d, den1.d, ctx.d);
01491
01492 // Result in FORM notation:
01493 WORD* out = s;
01494 *out++ = AR.PolyFun;
01495 WORD* args_size = out++;
01496 WORD* args_flag = out++;
01497 *args_flag &= ~MUSTCLEANPRF;
01498
01499 const bool with_arghead = true;
01500 const bool must_fit_term = true;
01501 const bool write = true;
01502 out += flint::to_argument_mpoly(BHEAD out, with_arghead, must_fit_term, write, out-term_size,
01503     num1.d, var_map, ctx.d);
01504 out += flint::to_argument_mpoly(BHEAD out, with_arghead, must_fit_term, write, out-term_size,
01505     den1.d, var_map, ctx.d);
01506
01507 *args_size = out - args_size + 1; // The +1 is to include the function ID
01508
01509 // +3 for the coefficient of 1/l, which is added after the check
01510 if ( sizeof(WORD)*(out-term_size+3) > (size_t)AM.MaxTer ) {
01511     MLOCK(ErrorMessageLock);
01512     MesPrint("flint::ratfun_normalize: output exceeds MaxTermSize");
01513     MUNLOCK(ErrorMessageLock);
01514     Terminate(-1);
01515 }
01516
01517 *out++ = 1;
01518 *out++ = 1;
01519 *out++ = 3; // the term's coefficient is now 1/l
01520
01521 *term_size = out - term_size;
01522 }
01523 /*
01524 #] flint::ratfun_normalize_mpoly :
01525 #[ flint::ratfun_normalize_poly :
01526 */
01527 // Multiply and simplify occurrences of the uni-variate FORM rational polynomials found in term.
01528 // The final term is written in place, with the rational polynomial at the end.
01529 void flint::ratfun_normalize_poly(PHEAD WORD *term, const var_map_t &var_map) {
01530
01531     // The length of the coefficient
01532     const WORD ncoeff = (term + *term)[-1];
01533     // The end of the term data, before the coefficient:
01534     const WORD *tstop = term + *term - ABS(ncoeff);
01535
01536     flint::poly num1, den1, num2, den2, gcd;
01537
01538     // Start with "trivial" polynomials, and multiply in the num and den of the prf which appear.
01539     flint::fmpz tmpNum, tmpDen;
01540     flint::fmpz_set_form(tmpNum.d, (UWORD*)tstop, ncoeff/2);
01541     flint::fmpz_set_form(tmpDen.d, (UWORD*)tstop+ABS(ncoeff/2), ABS(ncoeff/2));
01542     fmpz_poly_set_fmpz(num1.d, tmpNum.d);
01543     fmpz_poly_set_fmpz(den1.d, tmpDen.d);
01544
01545     // Loop over the occurrences of PolyFun in the term, and multiply in to num1, den1.
01546     // s tracks where we are writing the non-PolyFun term data. The final PolyFun will
01547     // go at the end.
01548     WORD* term_size = term;
01549     WORD* s = term + 1;
01550     for ( WORD *t = term + 1; t < tstop; ) {
01551         if ( *t == AR.PolyFun ) {
01552             flint::ratfun_read_poly(t, num2.d, den2.d);
01553
01554             // get gcd of num1,den2 and num2,den1 and then assemble
01555             flint::util::simplify_fmpz_poly(num1.d, den2.d, gcd.d);
01556             flint::util::simplify_fmpz_poly(num2.d, den1.d, gcd.d);
01557
01558             fmpz_poly_mul(num1.d, num1.d, num2.d);
01559             fmpz_poly_mul(den1.d, den1.d, den2.d);
01560
01561             t += t[1];
01562         }
01563     }
01564     else {
01565         // Not a PolyFun, just copy or skip over

```

```

01566         WORD i = t[1];
01567         if ( s != t ) { NCOPY(s,t,i); }
01568         else { t += i; s += i; }
01569     }
01570 }
01571
01572 flint::util::fix_sign_fmpz_poly_ratfun(num1.d, den1.d);
01573
01574 // Result in FORM notation:
01575 WORD* out = s;
01576 *out++ = AR.PolyFun;
01577 WORD* args_size = out++;
01578 WORD* args_flag = out++;
01579 *args_flag &= ~MUSTCLEANPRF;
01580
01581 const bool with_arghead = true;
01582 const bool must_fit_term = true;
01583 const bool write = true;
01584 out += flint::to_argument_poly(BHEAD out, with_arghead, must_fit_term, write, out-term_size,
01585     num1.d, var_map);
01586 out += flint::to_argument_poly(BHEAD out, with_arghead, must_fit_term, write, out-term_size,
01587     den1.d, var_map);
01588
01589 *args_size = out - args_size + 1; // The +1 is to include the function ID
01590
01591 // +3 for the coefficient of 1/1, which is added after the check
01592 if ( sizeof(WORD)*(out-term_size+3) > (size_t)AM.MaxTer ) {
01593     MLOCK(ErrorMessageLock);
01594     MesPrint("flint::ratfun_normalize: output exceeds MaxTermSize");
01595     MUNLOCK(ErrorMessageLock);
01596     Terminate(-1);
01597 }
01598
01599 *out++ = 1;
01600 *out++ = 1;
01601 *out++ = 3; // the term's coefficient is now 1/1
01602
01603 *term_size = out - term_size;
01604 }
01605 /*
01606 #] flint::ratfun_normalize_poly :
01607
01608 #[ flint::ratfun_read_mpoly :
01609 */
01610 // Read the multi-variate FORM rational polynomial at a and create fmpz_mpoly_t numerator and
01611 // denominator.
01612 void flint::ratfun_read_mpoly(const WORD *a, fmpz_mpoly_t num, fmpz_mpoly_t den,
01613     const var_map_t &var_map, fmpz_mpoly_ctx_t ctx) {
01614
01615     // The end of the arguments:
01616     const WORD* arg_stop = a+a[1];
01617
01618     const bool must_normalize = (a[2] & MUSTCLEANPRF) != 0;
01619
01620     a += FUNHEAD;
01621     if ( a >= arg_stop ) {
01622         MLOCK(ErrorMessageLock);
01623         MesPrint("ERROR: PolyRatFun cannot have zero arguments");
01624         MUNLOCK(ErrorMessageLock);
01625         Terminate(-1);
01626     }
01627
01628     // Polys to collect the "den of the num" and "den of the den".
01629     // Input can arrive like this when enabling the PolyRatFun or moving things into it.
01630     flint::mpoly den_num(ctx), den_den(ctx);
01631
01632     // Read the numerator
01633     flint::from_argument_mpoly(num, den_num.d, a, true, var_map, ctx);
01634     NEXTARG(a);
01635
01636     if ( a < arg_stop ) {
01637         // Read the denominator
01638         flint::from_argument_mpoly(den, den_den.d, a, true, var_map, ctx);
01639         NEXTARG(a);
01640     }
01641     else {
01642         // The denominator is 1
01643         MLOCK(ErrorMessageLock);
01644         MesPrint("implement this");
01645         MUNLOCK(ErrorMessageLock);
01646         Terminate(-1);
01647     }
01648     if ( a < arg_stop ) {
01649         MLOCK(ErrorMessageLock);
01650         MesPrint("ERROR: PolyRatFun cannot have more than two arguments");
01651         MUNLOCK(ErrorMessageLock);
01652         Terminate(-1);

```

```

01653     }
01654
01655     // Multiply the num by den_den and den by den_num:
01656     fmpz_mpoly_mul(num, num, den_den.d, ctx);
01657     fmpz_mpoly_mul(den, den, den_num.d, ctx);
01658
01659     if ( must_normalize ) {
01660         flint::mpoly gcd(ctx);
01661         fmpz_mpoly_gcd_cofactors(gcd.d, num, den, num, den, ctx);
01662     }
01663 }
01664 /*
01665 #] flint::ratfun_read_mpoly :
01666 #[ flint::ratfun_read_poly :
01667 */
01668 // Read the uni-variate FORM rational polynomial at a and create fmpz_mpoly_t numerator and
01669 // denominator.
01670 void flint::ratfun_read_poly(const WORD *a, fmpz_poly_t num, fmpz_poly_t den) {
01671
01672     // The end of the arguments:
01673     const WORD* arg_stop = a+a[1];
01674
01675     const bool must_normalize = (a[2] & MUSTCLEANPRF) != 0;
01676
01677     a += FUNHEAD;
01678     if ( a >= arg_stop ) {
01679         MLOCK(ErrorMessageLock);
01680         MesPrint("ERROR: PolyRatFun cannot have zero arguments");
01681         MUNLOCK(ErrorMessageLock);
01682         Terminate(-1);
01683     }
01684
01685     // Polys to collect the "den of the num" and "den of the den".
01686     // Input can arrive like this when enabling the PolyRatFun or moving things into it.
01687     flint::poly den_num, den_den;
01688
01689     // Read the numerator
01690     flint::from_argument_poly(num, den_num.d, a, true);
01691     NEXTARG(a);
01692
01693     if ( a < arg_stop ) {
01694         // Read the denominator
01695         flint::from_argument_poly(den, den_den.d, a, true);
01696         NEXTARG(a);
01697     }
01698     else {
01699         // The denominator is 1
01700         MLOCK(ErrorMessageLock);
01701         MesPrint("implement this");
01702         MUNLOCK(ErrorMessageLock);
01703         Terminate(-1);
01704     }
01705     if ( a < arg_stop ) {
01706         MLOCK(ErrorMessageLock);
01707         MesPrint("ERROR: PolyRatFun cannot have more than two arguments");
01708         MUNLOCK(ErrorMessageLock);
01709         Terminate(-1);
01710     }
01711
01712     // Multiply the num by den_den and den by den_num:
01713     fmpz_poly_mul(num, num, den_den.d);
01714     fmpz_poly_mul(den, den, den_num.d);
01715
01716     if ( must_normalize ) {
01717         flint::poly gcd;
01718         flint::util::simplify_fmpz_poly(num, den, gcd.d);
01719     }
01720 }
01721 /*
01722 #] flint::ratfun_read_poly :
01723 #[ flint::startup_init :
01724 */
01725 //
01726 // The purpose of this function is it work around threading issues in flint, reported in
01727 // https://github.com/flintlib/flint/issues/1652 and fixed in
01728 // https://github.com/flintlib/flint/pull/1658 . This implies versions prior to 3.1.0,
01729 // but at least 3.0.0 (when gr was introduced).
01730 void flint::startup_init(void) {
01731
01732     #if __FLINT_RELEASE >= 30000
01733         // Here we initialize some gr contexts so that their method tables are populated. Crashes have
01734         // otherwise been observed due to these two contexts in particular.
01735         fmpz_t dummy_fmpz;
01736         fmpz_init(dummy_fmpz);
01737         fmpz_set_si(dummy_fmpz, 19);
01738
01739         gr_ctx_t dummy_gr_ctx_fmpz_mod;

```



```

01740     gr_ctx_init_fmpz_mod(dummy_gr_ctx_fmpz_mod, dummy_fmpz);
01741     gr_ctx_clear(dummy_gr_ctx_fmpz_mod);
01742
01743     gr_ctx_t dummy_gr_ctx_nmod;
01744     gr_ctx_init_nmod(dummy_gr_ctx_nmod, 19);
01745     gr_ctx_clear(dummy_gr_ctx_nmod);
01746
01747     fmpz_clear(dummy_fmpz);
01748 #endif
01749 }
01750 /*
01751  #] flint::startup_init :
01752
01753  #[ flint::to_argument_mpoly :
01754  */
01755 // Convert a fmpz_mpoly_t to a FORM argument (or 0-terminated list of terms: with_arghead==false).
01756 // If the caller is building an output term, prev_size contains the size of the term so far, to
01757 // check that the output fits in AM.MaxTer if must_fit_term.
01758 // All coefficients will be divided by denscale (which might just be 1).
01759 // If write is false, we never write to out but only track the total would-be size. This lets this
01760 // function be repurposed as a "size of FORM notation" function without duplicating the code.
01761 #define IFW(x) { if ( write ) {x;} }
01762 uint64_t flint::to_argument_mpoly(PHEAD WORD *out, const bool with_arghead,
01763     const bool must_fit_term, const bool write, const uint64_t prev_size, const fmpz_mpoly_t poly,
01764     const var_map_t &var_map, const fmpz_mpoly_ctx_t ctx, const fmpz_t denscale) {
01765
01766     // out is modified later, keep the pointer at entry
01767     const WORD* out_entry = out;
01768
01769     // Track the total size written. We could do this with pointer differences, but if
01770     // write == false we don't write to or move out to be able to find the size that way.
01771     uint64_t ws = 0;
01772
01773     // Check there is at least space for ARGHEAD WORDs (the arghead or fast-notation number/symbol)
01774     if ( write && must_fit_term && (sizeof(WORD)*(prev_size + ARGHEAD) > (size_t)AM.MaxTer) ) {
01775         MLOCK(ErrorMessageLock);
01776         MesPrint("flint::to_argument_mpoly: output exceeds MaxTermSize");
01777         MUNLOCK(ErrorMessageLock);
01778         Terminate(-1);
01779     }
01780
01781     // Create the inverse of var_map, so we don't have to search it for each symbol written
01782     var_map_t var_map_inv;
01783     for ( auto x: var_map ) {
01784         var_map_inv[x.second] = x.first;
01785     }
01786
01787     vector<int64_t> exponents(var_map.size());
01788     const int64_t n_terms = fmpz_mpoly_length(poly, ctx);
01789
01790     if ( n_terms == 0 ) {
01791         if ( with_arghead ) {
01792             IFW(*out++ = -SNUMBER); ws++;
01793             IFW(*out++ = 0); ws++;
01794             return ws;
01795         }
01796         else {
01797             IFW(*out++ = 0); ws++;
01798             return ws;
01799         }
01800     }
01801
01802     // For dividing out denscale
01803     flint::fmpz coeff, den, gcd;
01804
01805     // The mpoly might be constant or a single symbol with coeff 1. Use fast notation if possible.
01806     if ( with_arghead && n_terms == 1 ) {
01807
01808         if ( fmpz_mpoly_is_fmpz(poly, ctx) ) {
01809             // The mpoly is constant. Use fast notation if the number is an integer and small enough:
01810
01811             fmpz_mpoly_get_term_coeff_fmpz(coeff.d, poly, 0, ctx);
01812             fmpz_set(den.d, denscale);
01813             flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);
01814
01815             if ( fmpz_is_one(den.d) && fmpz_fits_si(coeff.d) ) {
01816                 const int64_t fast_coeff = fmpz_get_si(coeff.d);
01817                 // While ">=", could work here, FORM does not use fast notation for INT_MIN
01818                 if ( fast_coeff > INT32_MIN && fast_coeff <= INT32_MAX ) {
01819                     IFW(*out++ = -SNUMBER); ws++;
01820                     IFW(*out++ = (WORD)fast_coeff); ws++;
01821                     return ws;
01822                 }
01823             }
01824         }
01825     }
01826     else {

```

```

01827     fmpz_mpoly_get_term_coeff_fmpz(coeff.d, poly, 0, ctx);
01828     fmpz_set(den.d, denscale);
01829     flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);
01830
01831     if ( fmpz_is_one(coeff.d) && fmpz_is_one(den.d) ) {
01832         // The coefficient is one. Now check the symbol powers:
01833
01834         fmpz_mpoly_get_term_exp_si((slong*)exponents.data(), poly, 0, ctx);
01835         int64_t use_fast = 0;
01836         uint32_t fast_symbol = 0;
01837
01838         for ( size_t i = 0; i < var_map.size(); i++ ) {
01839             if ( exponents[i] == 1 ) fast_symbol = var_map_inv[i];
01840             use_fast += exponents[i];
01841         }
01842
01843         // use_fast has collected the total degree. If it is 1, then fast_symbol holds the
code
01844         if ( use_fast == 1 ) {
01845             IFW(*out++ = -SYMBOL); ws++;
01846             IFW(*out++ = fast_symbol); ws++;
01847             return ws;
01848         }
01849     }
01850 }
01851 }
01852
01853 WORD *tmp_coeff = (WORD *)NumberMalloc("flint::to_argument_mpoly");
01854 WORD *tmp_den = (WORD *)NumberMalloc("flint::to_argument_mpoly");
01855
01856 WORD* arg_size = 0;
01857 WORD* arg_flag = 0;
01858 if ( with_arghead ) {
01859     IFW(arg_size = out++); ws++; // total arg size
01860     IFW(arg_flag = out++); ws++;
01861     IFW(*arg_flag = 0); // clean argument
01862 }
01863
01864 for ( int64_t i = 0; i < n_terms; i++ ) {
01865     fmpz_mpoly_get_term_exp_si((slong*)exponents.data(), poly, i, ctx);
01866
01867     fmpz_mpoly_get_term_coeff_fmpz(coeff.d, poly, i, ctx);
01868     fmpz_set(den.d, denscale);
01869     flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);
01870
01871     uint32_t num_symbols = 0;
01872     for ( size_t j = 0; j < var_map.size(); j++ ) {
01873         if ( exponents[j] != 0 ) { num_symbols += 1; }
01874     }
01875
01876     // Convert the coefficient, write in temporary space
01877     const WORD num_size = flint::fmpz_get_form(coeff.d, tmp_coeff);
01878     const WORD den_size = flint::fmpz_get_form(den.d, tmp_den);
01879     const WORD coeff_size = [num_size, den_size] () -> WORD {
01880         WORD size = ABS(num_size) > ABS(den_size) ? ABS(num_size) : ABS(den_size);
01881         return size * SGN(num_size) * SGN(den_size);
01882     }();
01883
01884     // Now we have the number of symbols and the coeff size, we can determine the output size.
01885     // Check it fits if necessary: term size, num,den of "coeff_size", +1 for total coeff size
01886     uint64_t current_size = prev_size + ws + 1 + 2*ABS(coeff_size) + 1;
01887     if ( num_symbols ) {
01888         // symbols header, code,power of each symbol:
01889         current_size += 2 + 2*num_symbols;
01890     }
01891     if ( write && must_fit_term && (sizeof(WORD)*current_size > (size_t)AM.MaxTer) ) {
01892         MLOCK(ErrorMessageLock);
01893         MesPrint("flint::to_argument_mpoly: output exceeds MaxTermSize");
01894         MUNLOCK(ErrorMessageLock);
01895         Terminate(-1);
01896     }
01897
01898     WORD* term_size = 0;
01899     IFW(term_size = out++); ws++;
01900     if ( num_symbols ) {
01901         IFW(*out++ = SYMBOL); ws++;
01902         WORD* symbol_size = 0;
01903         IFW(symbol_size = out++); ws++;
01904         IFW(*symbol_size = 2);
01905
01906         for ( size_t j = 0; j < var_map.size(); j++ ) {
01907             if ( exponents[j] != 0 ) {
01908                 IFW(*out++ = var_map_inv[j]); ws++;
01909                 IFW(*out++ = exponents[j]); ws++;
01910                 IFW(*symbol_size += 2);
01911             }
01912         }

```

```

01913         }
01914     }
01915 }
01916
01917 // Copy numerator
01918 for ( WORD j = 0; j < ABS(num_size); j++ ) {
01919     IFW(*out++ = tmp_coeff[j]); ws++;
01920 }
01921 for ( WORD j = ABS(num_size); j < ABS(coeff_size); j++ ) {
01922     IFW(*out++ = 0); ws++;
01923 }
01924 // Copy denominator
01925 for ( WORD j = 0; j < ABS(den_size); j++ ) {
01926     IFW(*out++ = tmp_den[j]); ws++;
01927 }
01928 for ( WORD j = ABS(den_size); j < ABS(coeff_size); j++ ) {
01929     IFW(*out++ = 0); ws++;
01930 }
01931
01932 IFW(*out = 2*ABS(coeff_size) + 1); // the size of the coefficient
01933 IFW(if ( coeff_size < 0 ) { *out = -(*out); });
01934 IFW(out++); ws++;
01935
01936 IFW(*term_size = out - term_size);
01937 }
01938
01939 if ( with_arghead ) {
01940     IFW(*arg_size = out - arg_size);
01941     if ( write ) {
01942         // Sort into form highfirst ordering
01943         flint::form_sort(BHEAD (WORD*) (out_entry));
01944     }
01945 }
01946 else {
01947     // with no arghead, we write a terminating zero
01948     IFW(*out++ = 0); ws++;
01949 }
01950
01951 NumberFree(tmp_coeff, "flint::to_argument_mpoly");
01952 NumberFree(tmp_den, "flint::to_argument_mpoly");
01953
01954 return ws;
01955 }
01956
01957 // If no denscale argument is supplied, just set it to 1 and call the usual function
01958 uint64_t flint::to_argument_mpoly(PHEAD WORD *out, const bool with_arghead,
01959 const bool must_fit_term, const bool write, const uint64_t prev_size, const fmpz_mpoly_t poly,
01960 const var_map_t &var_map, const fmpz_mpoly_ctx_t ctx) {
01961     flint::fmpz tmp;
01962     fmpz_set_ui(tmp.d, 1);
01963
01964     uint64_t ret = flint::to_argument_mpoly(BHEAD out, with_arghead, must_fit_term, write,
01965 prev_size, poly, var_map, ctx, tmp.d);
01966
01967     return ret;
01968 }
01969
01970 /*
01971 #] flint::to_argument_mpoly :
01972 #[ flint::to_argument_poly :
01973 */
01974 // Convert a fmpz_poly_t to a FORM argument (or 0-terminated list of terms: with_arghead==false).
01975 // If the caller is building an output term, prev_size contains the size of the term so far, to
01976 // check that the output fits in AM.MaxTer if must_fit_term.
01977 // All coefficients will be divided by denscale (which might just be 1).
01978 // If write is false, we never write to out but only track the total word-size. This lets this
01979 // function be repurposed as a "size of FORM notation" function without duplicating the code.
01980 uint64_t flint::to_argument_poly(PHEAD WORD *out, const bool with_arghead,
01981 const bool must_fit_term, const bool write, const uint64_t prev_size, const fmpz_poly_t poly,
01982 const var_map_t &var_map, const fmpz_t denscale) {
01983
01984     // Track the total size written. We could do this with pointer differences, but if
01985     // write == false we don't write to or move out to be able to find the size that way.
01986     uint64_t ws = 0;
01987
01988     // Check there is at least space for ARGHEAD WORDs (the arghead or fast-notation number/symbol)
01989     if ( ( write && must_fit_term && (sizeof(WORD)*(prev_size + ARGHEAD) > (size_t)AM.MaxTer) ) ) {
01990         MLOCK(ErrorMessageLock);
01991         MesPrint("flint::to_argument_poly: output exceeds MaxTermSize");
01992         MUNLOCK(ErrorMessageLock);
01993         Terminate(-1);
01994     }
01995
01996     // Create the inverse of var_map, so we don't have to search it for each symbol written
01997     var_map_t var_map_inv;
01998     for ( auto x: var_map ) {
01999         var_map_inv[x.second] = x.first;

```

```

02000     }
02001
02002     const int64_t n_terms = fmpz_poly_length(poly);
02003
02004     // The poly is zero
02005     if ( n_terms == 0 ) {
02006         if ( with_arghead ) {
02007             IFW(*out++ = -SNUMBER); ws++;
02008             IFW(*out++ = 0); ws++;
02009             return ws;
02010         }
02011         else {
02012             IFW(*out++ = 0); ws++;
02013             return ws;
02014         }
02015     }
02016
02017     // For dividing out denscale
02018     flint::fmpz coeff, den, gcd;
02019
02020     // The poly is constant, use fast notation if the coefficient is integer and small enough
02021     if ( with_arghead && n_terms == 1 ) {
02022
02023         fmpz_poly_get_coeff_fmpz(coeff.d, poly, 0);
02024         fmpz_set(den.d, denscale);
02025         flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);
02026
02027         if ( fmpz_is_one(den.d) && fmpz_fits_si(coeff.d) ) {
02028             const long fast_coeff = fmpz_get_si(coeff.d);
02029             // While ">=", could work here, FORM does not use fast notation for INT_MIN
02030             if ( fast_coeff > INT_MIN && fast_coeff <= INT_MAX ) {
02031                 IFW(*out++ = -SNUMBER); ws++;
02032                 IFW(*out++ = (WORD)fast_coeff); ws++;
02033                 return ws;
02034             }
02035         }
02036     }
02037
02038     // The poly might be a single symbol with coeff 1, use fast notation if so.
02039     if ( with_arghead && n_terms == 2 ) {
02040         if ( fmpz_is_zero(fmpz_poly_get_coeff_ptr(poly, 0)) ) {
02041             // The constant term is zero
02042
02043             fmpz_poly_get_coeff_fmpz(coeff.d, poly, 1);
02044             fmpz_set(den.d, denscale);
02045             flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);
02046
02047             if ( fmpz_is_one(coeff.d) && fmpz_is_one(den.d) ) {
02048                 // Single symbol with coeff 1. Use fast notation:
02049                 IFW(*out++ = -SYMBOL); ws++;
02050                 IFW(*out++ = var_map_inv[0]); ws++;
02051                 return ws;
02052             }
02053         }
02054     }
02055
02056     WORD *tmp_coeff = (WORD *)NumberMalloc("flint::to_argument_poly");
02057     WORD *tmp_den = (WORD *)NumberMalloc("flint::to_argument_mpoly");
02058
02059     WORD* arg_size = 0;
02060     WORD* arg_flag = 0;
02061     if ( with_arghead ) {
02062         IFW(arg_size = out++); ws++; // total arg size
02063         IFW(arg_flag = out++); ws++;
02064         IFW(*arg_flag = 0); // clean argument
02065     }
02066
02067     // In reverse, since we want a "highfirst" output
02068     for ( int64_t i = n_terms-1; i >= 0; i-- ) {
02069
02070         // fmpz_poly is dense, there might be many zero coefficients:
02071         if ( !fmpz_is_zero(fmpz_poly_get_coeff_ptr(poly, i)) ) {
02072
02073             fmpz_poly_get_coeff_fmpz(coeff.d, poly, i);
02074             fmpz_set(den.d, denscale);
02075             flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);
02076
02077             // Convert the coefficient, write in temporary space
02078             const WORD num_size = flint::fmpz_get_form(coeff.d, tmp_coeff);
02079             const WORD den_size = flint::fmpz_get_form(den.d, tmp_den);
02080             const WORD coeff_size = [num_size, den_size] () -> WORD {
02081                 WORD size = ABS(num_size) > ABS(den_size) ? ABS(num_size) : ABS(den_size);
02082                 return size * SGN(num_size) * SGN(den_size);
02083             }();
02084
02085             // Now we have the coeff size, we can determine the output size
02086             // Check it fits if necessary: symbol code, power, num, den of "coeff_size",

```

```

02087         // +1 for total coeff size
02088         uint64_t current_size = prev_size + ws + 1 + 2*ABS(coeff_size) + 1;
02089         if ( i > 0 ) {
02090             // and also symbols header, code, power of the symbol
02091             current_size += 4;
02092         }
02093         if ( write && must_fit_term && (sizeof(WORD)*current_size > (size_t)AM.MaxTer) ) {
02094             MLOCK(ErrorMessageLock);
02095             MesPrint("flint::to_argument_poly: output exceeds MaxTermSize");
02096             MUNLOCK(ErrorMessageLock);
02097             Terminate(-1);
02098         }
02099
02100         WORD* term_size = 0;
02101         IFW(term_size = out++); ws++;
02102
02103         if ( i > 0 ) {
02104             IFW(*out++ = SYMBOL); ws++;
02105             IFW(*out++ = 4); ws++; // The symbol array size, it is univariate
02106             IFW(*out++ = var_map_inv[0]); ws++;
02107             IFW(*out++ = i); ws++;
02108         }
02109
02110         // Copy numerator
02111         for ( WORD j = 0; j < ABS(num_size); j++ ) {
02112             IFW(*out++ = tmp_coeff[j]); ws++;
02113         }
02114         for ( WORD j = ABS(num_size); j < ABS(coeff_size); j++ ) {
02115             IFW(*out++ = 0); ws++;
02116         }
02117         // Copy denominator
02118         for ( WORD j = 0; j < ABS(den_size); j++ ) {
02119             IFW(*out++ = tmp_den[j]); ws++;
02120         }
02121         for ( WORD j = ABS(den_size); j < ABS(coeff_size); j++ ) {
02122             IFW(*out++ = 0); ws++;
02123         }
02124
02125         IFW(*out = 2*ABS(coeff_size) + 1); // the size of the coefficient
02126         IFW(if (coeff_size < 0) { *out = -(*out); });
02127         IFW(out++); ws++;
02128
02129         IFW(*term_size = out - term_size);
02130     }
02131 }
02132
02133 if ( with_arghead ) {
02134     IFW(*arg_size = out - arg_size);
02135 }
02136 else {
02137     // with no arghead, we write a terminating zero
02138     IFW(*out++ = 0); ws++;
02139 }
02140
02141 NumberFree(tmp_coeff, "flint::to_argument_poly");
02142 NumberFree(tmp_den, "flint::to_argument_poly");
02143
02144 return ws;
02145 }
02146
02147 // If no denscale argument is supplied, just set it to 1 and call the usual function
02148 uint64_t flint::to_argument_poly(PHEAD WORD *out, const bool with_arghead,
02149 const bool must_fit_term, const bool write, const uint64_t prev_size, const fmpz_poly_t poly,
02150 const var_map_t &var_map) {
02151     fmpz_t tmp;
02152     fmpz_set_ui(tmp.d, 1);
02153
02154     uint64_t ret = flint::to_argument_poly(BHEAD out, with_arghead, must_fit_term, write, prev_size,
02155 poly, var_map, tmp.d);
02156
02157     return ret;
02158 }
02159
02160 /*
02161 #] flint::to_argument_poly :
02162 */
02163 // Utility functions
02164 /*
02165 #[ flint::util::simplify_fmpz :
02166 */
02167 // Divide the GCD out of num and den
02168 inline void flint::util::simplify_fmpz(fmpz_t num, fmpz_t den, fmpz_t gcd) {
02169     fmpz_gcd(gcd, num, den);
02170     if ( !fmpz_is_one(gcd) ) {
02171         fmpz_divexact(num, num, gcd);
02172         fmpz_divexact(den, den, gcd);
02173     }

```

```

02174         fmpz_divexact(den, den, gcd);
02175     }
02176 }
02177 /*
02178 #] flint::util::simplify_fmpz :
02179 #[ flint::util::simplify_fmpz_poly :
02180 */
02181 // Divide the GCD out of num and den
02182 inline void flint::util::simplify_fmpz_poly(fmpz_poly_t num, fmpz_poly_t den, fmpz_poly_t gcd) {
02183     fmpz_poly_gcd(gcd, num, den);
02184     if ( !fmpz_poly_is_one(gcd) ) {
02185         #if __FLINT_RELEASE >= 30100
02186             // This should be faster than fmpz_poly_div, see https://github.com/flintlib/flint/pull/1766
02187             fmpz_poly_divexact(num, num, gcd);
02188             fmpz_poly_divexact(den, den, gcd);
02189         #else
02190             fmpz_poly_div(num, num, gcd);
02191             fmpz_poly_div(den, den, gcd);
02192         #endif
02193     }
02194 }
02195 /*
02196 #] flint::util::simplify_fmpz_poly :
02197 #[ flint::util::fix_sign_fmpz_mpoly_ratfun :
02198 */
02199 inline void flint::util::fix_sign_fmpz_mpoly_ratfun(fmpz_mpoly_t num, fmpz_mpoly_t den,
02200     const fmpz_mpoly_ctx_t ctx) {
02201     // Fix sign to align with poly: the leading denominator term should have a positive coeff
02202     if ( fmpz_sgn(fmpz_mpoly_term_coeff_ref(den, 0, ctx)) == -1 ) {
02203         fmpz_mpoly_neg(num, num, ctx);
02204         fmpz_mpoly_neg(den, den, ctx);
02205     }
02206 }
02207 /*
02208 #] flint::util::fix_sign_fmpz_mpoly_ratfun :
02209 #[ flint::util::fix_sign_fmpz_poly_ratfun :
02210 */
02211 inline void flint::util::fix_sign_fmpz_poly_ratfun(fmpz_poly_t num, fmpz_poly_t den) {
02212     // Fix sign to align with poly: the leading denominator term should have a positive coeff
02213     if ( fmpz_sgn(fmpz_poly_get_coeff_ptr(den, fmpz_poly_degree(den))) == -1 ) {
02214         fmpz_poly_neg(num, num);
02215         fmpz_poly_neg(den, den);
02216     }
02217 }
02218 /*
02219 #] flint::util::fix_sign_fmpz_poly_ratfun :
02220 */

```

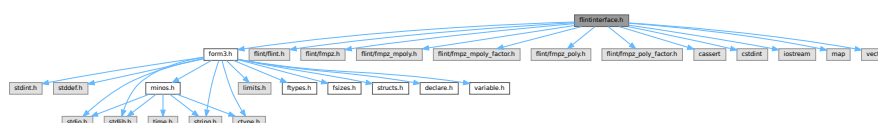
## 4.39 flintinterface.h File Reference

```

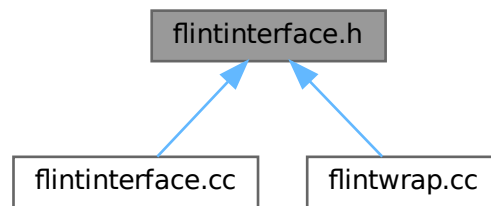
#include "form3.h"
#include <flint/flint.h>
#include <flint/fmpz.h>
#include <flint/fmpz_mpoly.h>
#include <flint/fmpz_mpoly_factor.h>
#include <flint/fmpz_poly.h>
#include <flint/fmpz_poly_factor.h>
#include <cassert>
#include <cstdint>
#include <iostream>
#include <map>
#include <vector>

```

Include dependency graph for flintinterface.h:



This graph shows which files directly or indirectly include this file:



## Data Structures

- class [flint::fmpz](#)
- class [flint::poly](#)
- class [flint::poly\\_factor](#)
- class [flint::mpoly](#)
- class [flint::mpoly\\_factor](#)
- class [flint::mpoly\\_ctx](#)

## Typedefs

- typedef `std::map< uint32_t, uint32_t >` [flint::var\\_map\\_t](#)

## Functions

- void [flint::cleanup](#) (void)
- void [flint::cleanup\\_master](#) (void)
- WORD \* [flint::divmod\\_mpoly](#) (PHEAD const WORD \*, const WORD \*, const bool, const WORD, const var\_map\_t &)
- WORD \* [flint::divmod\\_poly](#) (PHEAD const WORD \*, const WORD \*, const bool, const WORD, const var\_map\_t &)
- WORD \* [flint::factorize\\_mpoly](#) (PHEAD const WORD \*, WORD \*, const bool, const bool, const var\_map\_t &)
- WORD \* [flint::factorize\\_poly](#) (PHEAD const WORD \*, WORD \*, const bool, const bool, const var\_map\_t &)
- void [flint::form\\_sort](#) (PHEAD WORD \*)
- uint64\_t [flint::from\\_argument\\_mpoly](#) (fmpz\_mpoly\_t, fmpz\_mpoly\_t, const WORD \*, const bool, const var\_map\_t &, const fmpz\_mpoly\_ctx\_t)
- uint64\_t [flint::from\\_argument\\_poly](#) (fmpz\_poly\_t, fmpz\_poly\_t, const WORD \*, const bool)
- WORD [flint::fmpz\\_get\\_form](#) (fmpz\_t, WORD \*)
- void [flint::fmpz\\_set\\_form](#) (fmpz\_t, UWORD \*, WORD)
- WORD \* [flint::gcd\\_mpoly](#) (PHEAD const WORD \*, const WORD \*, const WORD, const var\_map\_t &)
- WORD \* [flint::gcd\\_poly](#) (PHEAD const WORD \*, const WORD \*, const WORD, const var\_map\_t &)
- var\_map\_t [flint::get\\_variables](#) (const vector< WORD \* > &, const bool, const bool)
- WORD \* [flint::inverse\\_poly](#) (PHEAD const WORD \*, const WORD \*, const var\_map\_t &)
- WORD \* [flint::mul\\_mpoly](#) (PHEAD const WORD \*, const WORD \*, const var\_map\_t &)
- WORD \* [flint::mul\\_poly](#) (PHEAD const WORD \*, const WORD \*, const var\_map\_t &)

- void [flint::ratfun\\_add\\_mpoly](#) (PHEAD const WORD \*, const WORD \*, WORD \*, const var\_map\_t &)
- void [flint::ratfun\\_add\\_poly](#) (PHEAD const WORD \*, const WORD \*, WORD \*, const var\_map\_t &)
- void [flint::ratfun\\_normalize\\_mpoly](#) (PHEAD WORD \*, const var\_map\_t &)
- void [flint::ratfun\\_normalize\\_poly](#) (PHEAD WORD \*, const var\_map\_t &)
- void [flint::ratfun\\_read\\_mpoly](#) (const WORD \*, fmpz\_mpoly\_t, fmpz\_mpoly\_t, const var\_map\_t &, fmpz\_↵mpoly\_ctx\_t)
- void [flint::ratfun\\_read\\_poly](#) (const WORD \*, fmpz\_poly\_t, fmpz\_poly\_t)
- void [flint::startup\\_init](#) (void)
- uint64\_t [flint::to\\_argument\\_mpoly](#) (PHEAD WORD \*, const bool, const bool, const bool, const uint64\_t, const fmpz\_mpoly\_t, const var\_map\_t &, const fmpz\_mpoly\_ctx\_t)
- uint64\_t [flint::to\\_argument\\_mpoly](#) (PHEAD WORD \*, const bool, const bool, const bool, const uint64\_t, const fmpz\_mpoly\_t, const var\_map\_t &, const fmpz\_mpoly\_ctx\_t, const fmpz\_t)
- uint64\_t [flint::to\\_argument\\_poly](#) (PHEAD WORD \*, const bool, const bool, const bool, const uint64\_t, const fmpz\_poly\_t, const var\_map\_t &)
- uint64\_t [flint::to\\_argument\\_poly](#) (PHEAD WORD \*, const bool, const bool, const bool, const uint64\_t, const fmpz\_poly\_t, const var\_map\_t &, const fmpz\_t)
- void [flint::util::simplify\\_fmpz](#) (fmpz\_t, fmpz\_t, fmpz\_t)
- void [flint::util::simplify\\_fmpz\\_poly](#) (fmpz\_poly\_t, fmpz\_poly\_t, fmpz\_poly\_t)
- void [flint::util::fix\\_sign\\_fmpz\\_mpoly\\_ratfun](#) (fmpz\_mpoly\_t, fmpz\_mpoly\_t, const fmpz\_mpoly\_ctx\_t)
- void [flint::util::fix\\_sign\\_fmpz\\_poly\\_ratfun](#) (fmpz\_poly\_t, fmpz\_poly\_t)

### 4.39.1 Detailed Description

Prototypes for functions in [flintinterface.cc](#)

Definition in file [flintinterface.h](#).

### 4.39.2 Typedef Documentation

#### 4.39.2.1 var\_map\_t

```
typedef std::map<uint32_t,uint32_t> flint::var_map_t
```

Definition at line 50 of file [flintinterface.h](#).

### 4.39.3 Function Documentation

#### 4.39.3.1 cleanup()

```
void flint::cleanup (
    void )
```

Definition at line 43 of file [flintinterface.cc](#).

#### 4.39.3.2 cleanup\_master()

```
void flint::cleanup_master (
    void )
```

Definition at line 50 of file [flintinterface.cc](#).



#### 4.39.3.3 divmod\_mpoly()

```
WORD * flint::divmod_mpoly (
    PHEAD const WORD * a,
    const WORD * b,
    const bool return_rem,
    const WORD must_fit_term,
    const var_map_t & var_map )
```

Definition at line 58 of file [flintinterface.cc](#).

#### 4.39.3.4 divmod\_poly()

```
WORD * flint::divmod_poly (
    PHEAD const WORD * a,
    const WORD * b,
    const bool return_rem,
    const WORD must_fit_term,
    const var_map_t & var_map )
```

Definition at line 128 of file [flintinterface.cc](#).

#### 4.39.3.5 factorize\_mpoly()

```
WORD * flint::factorize_mpoly (
    PHEAD const WORD * argin,
    WORD * argout,
    const bool with_arghead,
    const bool is_fun_arg,
    const var_map_t & var_map )
```

Definition at line 202 of file [flintinterface.cc](#).

#### 4.39.3.6 factorize\_poly()

```
WORD * flint::factorize_poly (
    PHEAD const WORD * argin,
    WORD * argout,
    const bool with_arghead,
    const bool is_fun_arg,
    const var_map_t & var_map )
```

Definition at line 350 of file [flintinterface.cc](#).

#### 4.39.3.7 form\_sort()

```
void flint::form_sort (
    PHEAD WORD * terms )
```

Definition at line 423 of file [flintinterface.cc](#).

#### 4.39.3.8 from\_argument\_mpoly()

```
uint64_t flint::from_argument_mpoly (
    fmpz_mpoly_t poly,
    fmpz_mpoly_t denpoly,
    const WORD * args,
    const bool with_arghead,
    const var_map_t & var_map,
    const fmpz_mpoly_ctx_t ctx )
```

Definition at line 478 of file [flintinterface.cc](#).

#### 4.39.3.9 from\_argument\_poly()

```
uint64_t flint::from_argument_poly (
    fmpz_poly_t poly,
    fmpz_poly_t denpoly,
    const WORD * args,
    const bool with_arghead )
```

Definition at line 621 of file [flintinterface.cc](#).

#### 4.39.3.10 fmpz\_get\_form()

```
WORD flint::fmpz_get_form (
    fmpz_t z,
    WORD * a )
```

Definition at line 751 of file [flintinterface.cc](#).

#### 4.39.3.11 fmpz\_set\_form()

```
void flint::fmpz_set_form (
    fmpz_t z,
    UWORD * a,
    WORD na )
```

Definition at line 816 of file [flintinterface.cc](#).

#### 4.39.3.12 gcd\_mpoly()

```
WORD * flint::gcd_mpoly (
    PHEAD const WORD * a,
    const WORD * b,
    const WORD must_fit_term,
    const var_map_t & var_map )
```

Definition at line 882 of file [flintinterface.cc](#).

#### 4.39.3.13 gcd\_poly()

```
WORD * flint::gcd_poly (
    PHEAD const WORD * a,
    const WORD * b,
    const WORD must_fit_term,
    const var_map_t & var_map )
```

Definition at line 978 of file [flintinterface.cc](#).

#### 4.39.3.14 get\_variables()

```
flint::var_map_t flint::get_variables (
    const vector< WORD * > & es,
    const bool with_arghead,
    const bool sort_vars )
```

Definition at line 1060 of file [flintinterface.cc](#).

#### 4.39.3.15 inverse\_poly()

```
WORD * flint::inverse_poly (
    PHEAD const WORD * a,
    const WORD * b,
    const var_map_t & var_map )
```

Definition at line 1162 of file [flintinterface.cc](#).

#### 4.39.3.16 mul\_mpoly()

```
WORD * flint::mul_mpoly (
    PHEAD const WORD * a,
    const WORD * b,
    const var_map_t & var_map )
```

Definition at line 1238 of file [flintinterface.cc](#).

#### 4.39.3.17 mul\_poly()

```
WORD * flint::mul_poly (
    PHEAD const WORD * a,
    const WORD * b,
    const var_map_t & var_map )
```

Definition at line 1290 of file [flintinterface.cc](#).

#### 4.39.3.18 `ratfun_add_mpoly()`

```
void flint::ratfun_add_mpoly (
    PHEAD const WORD * t1,
    const WORD * t2,
    WORD * out,
    const var_map_t & var_map )
```

Definition at line [1341](#) of file [flintinterface.cc](#).

#### 4.39.3.19 `ratfun_add_poly()`

```
void flint::ratfun_add_poly (
    PHEAD const WORD * t1,
    const WORD * t2,
    WORD * out,
    const var_map_t & var_map )
```

Definition at line [1393](#) of file [flintinterface.cc](#).

#### 4.39.3.20 `ratfun_normalize_mpoly()`

```
void flint::ratfun_normalize_mpoly (
    PHEAD WORD * term,
    const var_map_t & var_map )
```

Definition at line [1446](#) of file [flintinterface.cc](#).

#### 4.39.3.21 `ratfun_normalize_poly()`

```
void flint::ratfun_normalize_poly (
    PHEAD WORD * term,
    const var_map_t & var_map )
```

Definition at line [1529](#) of file [flintinterface.cc](#).

#### 4.39.3.22 `ratfun_read_mpoly()`

```
void flint::ratfun_read_mpoly (
    const WORD * a,
    fmpz_mpoly_t num,
    fmpz_mpoly_t den,
    const var_map_t & var_map,
    fmpz_mpoly_ctx_t ctx )
```

Definition at line [1612](#) of file [flintinterface.cc](#).

#### 4.39.3.23 ratfun\_read\_poly()

```
void flint::ratfun_read_poly (
    const WORD * a,
    fmpz_poly_t num,
    fmpz_poly_t den )
```

Definition at line 1670 of file [flintinterface.cc](#).

#### 4.39.3.24 startup\_init()

```
void flint::startup_init (
    void )
```

Definition at line 1730 of file [flintinterface.cc](#).

#### 4.39.3.25 to\_argument\_mpoly() [1/2]

```
uint64_t flint::to_argument_mpoly (
    PHEAD WORD * out,
    const bool with_arghead,
    const bool must_fit_term,
    const bool write,
    const uint64_t prev_size,
    const fmpz_mpoly_t poly,
    const var_map_t & var_map,
    const fmpz_mpoly_ctx_t ctx )
```

Definition at line 1958 of file [flintinterface.cc](#).

#### 4.39.3.26 to\_argument\_mpoly() [2/2]

```
uint64_t flint::to_argument_mpoly (
    PHEAD WORD * out,
    const bool with_arghead,
    const bool must_fit_term,
    const bool write,
    const uint64_t prev_size,
    const fmpz_mpoly_t poly,
    const var_map_t & var_map,
    const fmpz_mpoly_ctx_t ctx,
    const fmpz_t denscale )
```

Definition at line 1762 of file [flintinterface.cc](#).

#### 4.39.3.27 to\_argument\_poly() [1/2]

```
uint64_t flint::to_argument_poly (
    PHEAD WORD * out,
    const bool with_arghead,
    const bool must_fit_term,
    const bool write,
    const uint64_t prev_size,
    const fmpz_poly_t poly,
    const var_map_t & var_map )
```

Definition at line 2149 of file [flintinterface.cc](#).

#### 4.39.3.28 to\_argument\_poly() [2/2]

```
uint64_t flint::to_argument_poly (
    PHEAD WORD * out,
    const bool with_arghead,
    const bool must_fit_term,
    const bool write,
    const uint64_t prev_size,
    const fmpz_poly_t poly,
    const var_map_t & var_map,
    const fmpz_t denscale )
```

Definition at line 1980 of file [flintinterface.cc](#).

#### 4.39.3.29 simplify\_fmpz()

```
void flint::util::simplify_fmpz (
    fmpz_t num,
    fmpz_t den,
    fmpz_t gcd ) [inline]
```

Definition at line 2170 of file [flintinterface.cc](#).

#### 4.39.3.30 simplify\_fmpz\_poly()

```
void flint::util::simplify_fmpz_poly (
    fmpz_poly_t num,
    fmpz_poly_t den,
    fmpz_poly_t gcd ) [inline]
```

Definition at line 2182 of file [flintinterface.cc](#).

#### 4.39.3.31 fix\_sign\_fmpz\_mpoly\_ratfun()

```
void flint::util::fix_sign_fmpz_mpoly_ratfun (
    fmpz_mpoly_t num,
    fmpz_mpoly_t den,
    const fmpz_mpoly_ctx_t ctx ) [inline]
```

Definition at line 2199 of file [flintinterface.cc](#).

#### 4.39.3.32 fix\_sign\_fmpz\_poly\_ratfun()

```
void flint::util::fix_sign_fmpz_poly_ratfun (
    fmpz_poly_t num,
    fmpz_poly_t den ) [inline]
```

Definition at line 2211 of file [flintinterface.cc](#).

## 4.40 flintinterface.h

[Go to the documentation of this file.](#)

```

00001 #pragma once
00007 extern "C" {
00008 #include "form3.h"
00009 }
00010
00011 #if defined(WINDOWS)
00012 // flint.h defines WORD(xx), which conflicts with the one defined in form3.h.
00013 #undef WORD
00014 #endif
00015
00016 #include <flint/flint.h>
00017 #if __FLINT_RELEASE >= 30000
00018 #include <flint/gr.h>
00019 #endif
00020 #include <flint/fmpz.h>
00021 #include <flint/fmpz_mpoly.h>
00022 #include <flint/fmpz_mpoly_factor.h>
00023 #include <flint/fmpz_poly.h>
00024 #include <flint/fmpz_poly_factor.h>
00025
00026 #if defined(WINDOWS)
00027 // Redefine WORD here to match form3.h.
00028 #undef WORD
00029 #define WORD FORM_WORD
00030 #endif
00031
00032 #include <cassert>
00033 #include <cstdint>
00034 #include <iostream>
00035 #include <map>
00036 #include <vector>
00037
00038
00039 // The bits of std that are needed:
00040 using std::cout;
00041 using std::endl;
00042 using std::map;
00043 using std::swap;
00044 using std::string;
00045 using std::vector;
00046
00047
00048 namespace flint {
00049
00050     typedef std::map<uint32_t, uint32_t> var_map_t;
00051
00052     // Small wrappers around the flint structs to enable RAII init and clear. "d" represents the
00053     // data, and this member will be passed to flint functions.
00054     class fmpz {
00055     public:
00056         fmpz_t d;
00057         fmpz() { fmpz_init(d); }
00058         ~fmpz() { fmpz_clear(d); }
00059         void print(const string& text) { cout << text; fmpz_print(d); cout << endl; }
00060     };
00061     class poly {
00062     public:
00063         fmpz_poly_t d;
00064         poly() { fmpz_poly_init(d); }
00065         ~poly() { fmpz_poly_clear(d); }
00066         void print(const string& text) {
00067             cout << text; fmpz_poly_print_pretty(d, "x"); cout << endl;
00068         }
00069     };
00070     class poly_factor {
00071     public:
00072         fmpz_poly_factor_t d;
00073         poly_factor() { fmpz_poly_factor_init(d); }
00074         ~poly_factor() { fmpz_poly_factor_clear(d); }
00075     };
00076     class mpoly {
00077     private:
00078         fmpz_mpoly_ctx_struct *ctx; // We need to keep a copy of the context pointer for clearing.
00079     public:
00080         fmpz_mpoly_t d;
00081         explicit mpoly(fmpz_mpoly_ctx_struct *ctx_in) : ctx(ctx_in) { fmpz_mpoly_init(d, ctx); }
00082         ~mpoly() { fmpz_mpoly_clear(d, ctx); }
00083         void print(const string& text) {
00084             cout << text;
00085             fmpz_mpoly_print_pretty(d, 0, ctx);
00086             cout << endl;
00087         }
00088     };
00089
00090 }

```

```

00088     };
00089     class mpoly_factor {
00090     private:
00091         fmpz_mpoly_ctx_struct *ctx; // We need to keep a copy of the context pointer for clearing.
00092     public:
00093         fmpz_mpoly_factor_t d;
00094         explicit mpoly_factor(fmpz_mpoly_ctx_struct *ctx_in) : ctx(ctx_in) {
00095             fmpz_mpoly_factor_init(d, ctx);
00096         }
00097         ~mpoly_factor() { fmpz_mpoly_factor_clear(d, ctx); }
00098     };
00099     class mpoly_ctx {
00100     public:
00101         fmpz_mpoly_ctx_t d;
00102         explicit mpoly_ctx(int64_t nvars) { fmpz_mpoly_ctx_init(d, nvars, ORD_LEX); }
00103         ~mpoly_ctx() { fmpz_mpoly_ctx_clear(d); }
00104     };
00105
00106     void cleanup(void);
00107     void cleanup_master(void);
00108
00109     WORD* divmod_mpoly(PHEAD const WORD *, const WORD *, const bool, const WORD, const var_map_t &);
00110     WORD* divmod_poly(PHEAD const WORD *, const WORD *, const bool, const WORD, const var_map_t &);
00111
00112     WORD* factorize_mpoly(PHEAD const WORD *, WORD *, const bool, const bool, const var_map_t &);
00113     WORD* factorize_poly(PHEAD const WORD *, WORD *, const bool, const bool, const var_map_t &);
00114
00115     void form_sort(PHEAD WORD *);
00116
00117     uint64_t from_argument_mpoly(fmpz_mpoly_t, fmpz_mpoly_t, const WORD *, const bool,
00118         const var_map_t &, const fmpz_mpoly_ctx_t);
00119     uint64_t from_argument_poly(fmpz_poly_t, fmpz_poly_t, const WORD *, const bool);
00120
00121     WORD fmpz_get_form(fmpz_t, WORD *);
00122     void fmpz_set_form(fmpz_t, UWORD *, WORD);
00123
00124     WORD* gcd_mpoly(PHEAD const WORD *, const WORD *, const WORD, const var_map_t &);
00125     WORD* gcd_poly(PHEAD const WORD *, const WORD *, const WORD, const var_map_t &);
00126
00127     var_map_t get_variables(const vector <WORD *> &, const bool, const bool);
00128
00129     WORD* inverse_poly(PHEAD const WORD *, const WORD *, const var_map_t &);
00130
00131     WORD* mul_mpoly(PHEAD const WORD *, const WORD *, const var_map_t &);
00132     WORD* mul_poly(PHEAD const WORD *, const WORD *, const var_map_t &);
00133
00134     void ratfun_add_mpoly(PHEAD const WORD *, const WORD *, WORD *, const var_map_t &);
00135     void ratfun_add_poly(PHEAD const WORD *, const WORD *, WORD *, const var_map_t &);
00136
00137     void ratfun_normalize_mpoly(PHEAD WORD *, const var_map_t &);
00138     void ratfun_normalize_poly(PHEAD WORD *, const var_map_t &);
00139
00140     void ratfun_read_mpoly(const WORD *, fmpz_mpoly_t, fmpz_mpoly_t, const var_map_t &,
00141         fmpz_mpoly_ctx_t);
00142     void ratfun_read_poly(const WORD *, fmpz_poly_t, fmpz_poly_t);
00143
00144     void startup_init(void);
00145
00146     uint64_t to_argument_mpoly(PHEAD WORD *, const bool, const bool, const bool, const uint64_t,
00147         const fmpz_mpoly_t, const var_map_t &, const fmpz_mpoly_ctx_t);
00148     uint64_t to_argument_mpoly(PHEAD WORD *, const bool, const bool, const bool, const uint64_t,
00149         const fmpz_mpoly_t, const var_map_t &, const fmpz_mpoly_ctx_t, const fmpz_t);
00150     uint64_t to_argument_poly(PHEAD WORD *, const bool, const bool, const bool, const uint64_t,
00151         const fmpz_poly_t, const var_map_t &);
00152     uint64_t to_argument_poly(PHEAD WORD *, const bool, const bool, const bool, const uint64_t,
00153         const fmpz_poly_t, const var_map_t &, const fmpz_t);
00154
00155     namespace util {
00156
00157         void simplify_fmpz(fmpz_t, fmpz_t, fmpz_t);
00158         void simplify_fmpz_poly(fmpz_poly_t, fmpz_poly_t, fmpz_poly_t);
00159
00160         void fix_sign_fmpz_mpoly_ratfun(fmpz_mpoly_t, fmpz_mpoly_t, const fmpz_mpoly_ctx_t);
00161         void fix_sign_fmpz_poly_ratfun(fmpz_poly_t, fmpz_poly_t);
00162
00163     }
00164
00165 }
00166 }

```





#### 4.41.2.3 flint\_div()

```
WORD * flint_div (
    PHEAD WORD * a,
    WORD * b,
    const WORD must_fit_term )
```

Definition at line 30 of file [flintwrap.cc](#).

#### 4.41.2.4 flint\_factorize\_argument()

```
int flint_factorize_argument (
    PHEAD WORD * argin,
    WORD * argout )
```

Definition at line 51 of file [flintwrap.cc](#).

#### 4.41.2.5 flint\_factorize\_dollar()

```
WORD * flint_factorize_dollar (
    PHEAD WORD * argin )
```

Definition at line 72 of file [flintwrap.cc](#).

#### 4.41.2.6 flint\_gcd()

```
WORD * flint_gcd (
    PHEAD WORD * a,
    WORD * b,
    const WORD must_fit_term )
```

Definition at line 91 of file [flintwrap.cc](#).

#### 4.41.2.7 flint\_inverse()

```
WORD * flint_inverse (
    PHEAD WORD * a,
    WORD * b )
```

Definition at line 111 of file [flintwrap.cc](#).

#### 4.41.2.8 flint\_mul()

```
WORD * flint_mul (
    PHEAD WORD * a,
    WORD * b )
```

Definition at line 131 of file [flintwrap.cc](#).

#### 4.41.2.9 flint\_ratfun\_add()

```
WORD * flint_ratfun_add (
    PHEAD WORD * t1,
    WORD * t2 )
```

Definition at line 149 of file [flintwrap.cc](#).

#### 4.41.2.10 flint\_ratfun\_normalize()

```
int flint_ratfun_normalize (
    PHEAD WORD * term )
```

Definition at line 187 of file [flintwrap.cc](#).

#### 4.41.2.11 flint\_rem()

```
WORD * flint_rem (
    PHEAD WORD * a,
    WORD * b,
    const WORD must_fit_term )
```

Definition at line 259 of file [flintwrap.cc](#).

#### 4.41.2.12 flint\_startup\_init()

```
void flint_startup_init (
    void )
```

Definition at line 280 of file [flintwrap.cc](#).

## 4.42 flintwrap.cc

[Go to the documentation of this file.](#)

```
00001
00006 extern "C" {
00007 #include "form3.h"
00008 }
00009
00010 #include "flintinterface.h"
00011
00012
00013 /*
00014     #[ flint_final_cleanup_thread :
00015 */
00016 void flint_final_cleanup_thread(void) {
00017     flint::cleanup();
00018 }
00019 /*
00020     #] flint_final_cleanup_thread :
00021     #[ flint_final_cleanup_master :
00022 */
00023 void flint_final_cleanup_master(void) {
00024     flint::cleanup_master();
00025 }
00026 /*
00027     #] flint_final_cleanup_master :
00028     #[ flint_div :
```

```

00029 */
00030 WORD* flint_div(PHEAD WORD *a, WORD *b, const WORD must_fit_term) {
00031     // Extract expressions
00032     vector<WORD *> e;
00033     e.push_back(a);
00034     e.push_back(b);
00035     const bool with_arghead = false;
00036     const bool sort_vars = false;
00037     const flint::var_map_t var_map = flint::get_variables(e, with_arghead, sort_vars);
00038
00039     const bool return_rem = false;
00040     if ( var_map.size() > 1 ) {
00041         return flint::divmod_mpoly(BHEAD a, b, return_rem, must_fit_term, var_map);
00042     }
00043     else {
00044         return flint::divmod_poly(BHEAD a, b, return_rem, must_fit_term, var_map);
00045     }
00046 }
00047 /*
00048 #] flint_div :
00049 #[ flint_factorize_argument :
00050 */
00051 int flint_factorize_argument(PHEAD WORD *argin, WORD *argout) {
00052
00053     const bool with_arghead = true;
00054     const bool sort_vars = true;
00055     const flint::var_map_t var_map = flint::get_variables(vector<WORD*>(1,argin), with_arghead,
00056         sort_vars);
00057
00058     const bool is_fun_arg = true;
00059     if ( var_map.size() > 1 ) {
00060         flint::factorize_mpoly(BHEAD argin, argout, with_arghead, is_fun_arg, var_map);
00061     }
00062     else {
00063         flint::factorize_poly(BHEAD argin, argout, with_arghead, is_fun_arg, var_map);
00064     }
00065
00066     return 0;
00067 }
00068 /*
00069 #] flint_factorize_argument :
00070 #[ flint_factorize_dollar :
00071 */
00072 WORD* flint_factorize_dollar(PHEAD WORD *argin) {
00073
00074     const bool with_arghead = false;
00075     const bool sort_vars = true;
00076     const flint::var_map_t var_map = flint::get_variables(vector<WORD*>(1,argin), with_arghead,
00077         sort_vars);
00078
00079     const bool is_fun_arg = false;
00080     if ( var_map.size() > 1 ) {
00081         return flint::factorize_mpoly(BHEAD argin, NULL, with_arghead, is_fun_arg, var_map);
00082     }
00083     else {
00084         return flint::factorize_poly(BHEAD argin, NULL, with_arghead, is_fun_arg, var_map);
00085     }
00086 }
00087 /*
00088 #] flint_factorize_dollar :
00089 #[ flint_gcd :
00090 */
00091 WORD* flint_gcd(PHEAD WORD *a, WORD *b, const WORD must_fit_term) {
00092     // Extract expressions
00093     vector<WORD *> e;
00094     e.push_back(a);
00095     e.push_back(b);
00096     const bool with_arghead = false;
00097     const bool sort_vars = true;
00098     const flint::var_map_t var_map = flint::get_variables(e, with_arghead, sort_vars);
00099
00100     if ( var_map.size() > 1 ) {
00101         return flint::gcd_mpoly(BHEAD a, b, must_fit_term, var_map);
00102     }
00103     else {
00104         return flint::gcd_poly(BHEAD a, b, must_fit_term, var_map);
00105     }
00106 }
00107 /*
00108 #] flint_gcd :
00109 #[ flint_inverse :
00110 */
00111 WORD* flint_inverse(PHEAD WORD *a, WORD *b) {
00112     // Extract expressions
00113     vector<WORD *> e;
00114     e.push_back(a);
00115     e.push_back(b);

```

```

00116     const flint::var_map_t var_map = flint::get_variables(e, false, false);
00117
00118     if ( var_map.size() > 1 ) {
00119         MLOCK(ErrorMessageLock);
00120         MesPrint("flint_inverse: error: only univariate polynomials are supported.");
00121         MUNLOCK(ErrorMessageLock);
00122         Terminate(-1);
00123     }
00124
00125     return flint::inverse_poly(BHEAD a, b, var_map);
00126 }
00127 /*
00128 #] flint_inverse :
00129 #[ flint_mul :
00130 */
00131 WORD* flint_mul(PHEAD WORD *a, WORD *b) {
00132     // Extract expressions
00133     vector<WORD *> e;
00134     e.push_back(a);
00135     e.push_back(b);
00136     const flint::var_map_t var_map = flint::get_variables(e, false, false);
00137
00138     if ( var_map.size() > 1 ) {
00139         return flint::mul_mpoly(BHEAD a, b, var_map);
00140     }
00141     else {
00142         return flint::mul_poly(BHEAD a, b, var_map);
00143     }
00144 }
00145 /*
00146 #] flint_mul :
00147 #[ flint_ratfun_add :
00148 */
00149 WORD* flint_ratfun_add(PHEAD WORD *t1, WORD *t2) {
00150
00151     if ( AR.PolyFunExp == 1 ) {
00152         MLOCK(ErrorMessageLock);
00153         MesPrint("flint_ratfun_add: PolyFunExp unimplemented.");
00154         MUNLOCK(ErrorMessageLock);
00155         Terminate(-1);
00156     }
00157
00158     WORD *oldworkpointer = AT.WorkPointer;
00159
00160     // Extract expressions: the num and den of both prf
00161     vector<WORD *> e;
00162     for (WORD *t=t1+FUNHEAD; t<t1+t1[1];) {
00163         e.push_back(t);
00164         NEXTARG(t);
00165     }
00166     for (WORD *t=t2+FUNHEAD; t<t2+t2[1];) {
00167         e.push_back(t);
00168         NEXTARG(t);
00169     }
00170     const bool with_arghead = true;
00171     const bool sort_vars = true;
00172     const flint::var_map_t var_map = flint::get_variables(e, with_arghead, sort_vars);
00173
00174     if ( var_map.size() > 1 ) {
00175         flint::ratfun_add_mpoly(BHEAD t1, t2, oldworkpointer, var_map);
00176     }
00177     else {
00178         flint::ratfun_add_poly(BHEAD t1, t2, oldworkpointer, var_map);
00179     }
00180
00181     return oldworkpointer;
00182 }
00183 /*
00184 #] flint_ratfun_add :
00185 #[ flint_ratfun_normalize :
00186 */
00187 int flint_ratfun_normalize(PHEAD WORD *term) {
00188
00189     // The length of the coefficient
00190     const WORD ncoeff = (term + *term)[-1];
00191     // The end of the term data, before the coefficient:
00192     const WORD *tstop = term + *term - ABS(ncoeff);
00193
00194     // Search the term for multiple PolyFun or one dirty one.
00195     unsigned num_polyratfun = 0;
00196     for (WORD *t = term+1; t < tstop; t += t[1]) {
00197         if (*t == AR.PolyFun) {
00198             // Found one!
00199             num_polyratfun++;
00200             if ((t[2] & MUSTCLEANPRF) != 0) {
00201                 // Dirty, increment again to force normalisation for single PolyFun
00202                 num_polyratfun++;

```

```

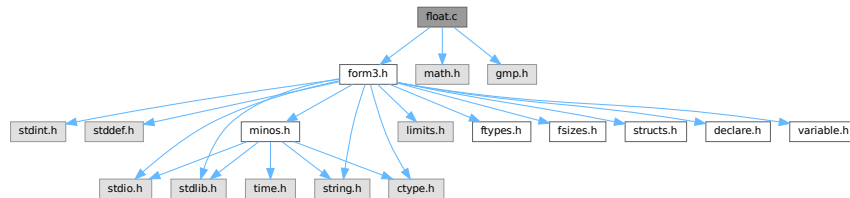
00203         }
00204         if (num_polyratfun > 1) {
00205             // We're not counting all occurrences, just determining if there
00206             // is something to be done.
00207             break;
00208         }
00209     }
00210 }
00211 if (num_polyratfun <= 1) {
00212     // There is nothing to do, return early
00213     return 0;
00214 }
00215
00216 // When there are polyratfun with only one argument: rename them temporarily to TMPPOLYFUN.
00217 for (WORD *t = term+1; t < tstop; t += t[1]) {
00218     if (*t == AR.PolyFun && (t[1] == FUNHEAD+t[FUNHEAD] || t[1] == FUNHEAD+2 ) ) {
00219         *t = TMPPOLYFUN;
00220     }
00221 }
00222
00223 // Extract all variables in the polyfuns
00224 // Collect pointers to each relevant argument
00225 vector<WORD *> e;
00226 for (WORD *t=term+1; t<tstop; t+=t[1]) {
00227     if (*t == AR.PolyFun) {
00228         for (WORD *t2 = t+FUNHEAD; t2<t+t[1];) {
00229             e.push_back(t2);
00230             NEXTARG(t2);
00231         }
00232     }
00233 }
00234 }
00235 const bool with_arghead = true;
00236 const bool sort_vars = true;
00237 const flint::var_map_t var_map = flint::get_variables(e, with_arghead, sort_vars);
00238
00239 if ( var_map.size() > 1 ) {
00240     flint::ratfun_normalize_mpoly(BHEAD term, var_map);
00241 }
00242 else {
00243     flint::ratfun_normalize_poly(BHEAD term, var_map);
00244 }
00245
00246 // Undo renaming of single-argument PolyFun
00247 const WORD *new_tstop = term + *term - ABS((term + *term)[-1]);
00248 for (WORD *t=term+1; t<new_tstop; t+=t[1]) {
00249     if (*t == TMPPOLYFUN ) *t = AR.PolyFun;
00250 }
00251
00252 return 0;
00253 }
00254
00255 /*
00256 #] flint_ratfun_normalize :
00257 #[ flint_rem :
00258 */
00259 WORD* flint_rem(PHEAD WORD *a, WORD *b, const WORD must_fit_term) {
00260     // Extract expressions
00261     vector<WORD *> e;
00262     e.push_back(a);
00263     e.push_back(b);
00264     const bool with_arghead = false;
00265     const bool sort_vars = false;
00266     const flint::var_map_t var_map = flint::get_variables(e, with_arghead, sort_vars);
00267
00268     const bool return_rem = true;
00269     if ( var_map.size() > 1 ) {
00270         return flint::divmod_mpoly(BHEAD a, b, return_rem, must_fit_term, var_map);
00271     }
00272     else {
00273         return flint::divmod_poly(BHEAD a, b, return_rem, must_fit_term, var_map);
00274     }
00275 }
00276
00277 /*
00278 #] flint_rem :
00279 #[ flint_startup_init :
00280 */
00281 void flint_startup_init(void) {
00282     flint::startup_init();
00283 }
00284
00285 /*
00286 #] flint_startup_init :
00287 */

```

## 4.43 float.c File Reference

```
#include "form3.h"
#include <math.h>
#include <gmp.h>
```

Include dependency graph for float.c:



### Macros

- `#define WITHCUTOFF`
- `#define GMPSPREAD (GMP_LIMB_BITS/BITSINWORD)`

### Functions

- void [Form\\_mpf\\_init](#) (mpf\_t t)
- void [Form\\_mpf\\_clear](#) (mpf\_t t)
- void [Form\\_mpf\\_set\\_prec\\_raw](#) (mpf\_t t, ULONG newprec)
- void [FormtoZ](#) (mpz\_t z, UWORD \*a, WORD na)
- void [ZtoForm](#) (UWORD \*a, WORD \*na, mpz\_t z)
- long [FloatToInteger](#) (UWORD \*out, mpf\_t floatin, long \*bitsused)
- void [IntegerToFloat](#) (mpf\_t result, UWORD \*formlong, int longsize)
- int [FloatToRat](#) (UWORD \*ratout, WORD \*nratout, mpf\_t floatin)
- int [AddFloats](#) (PHEAD WORD \*fun3, WORD \*fun1, WORD \*fun2)
- int [MulFloats](#) (PHEAD WORD \*fun3, WORD \*fun1, WORD \*fun2)
- int [DivFloats](#) (PHEAD WORD \*fun3, WORD \*fun1, WORD \*fun2)
- int [AddRatToFloat](#) (PHEAD WORD \*outfun, WORD \*infun, UWORD \*formrat, WORD nrat)
- int [MulRatToFloat](#) (PHEAD WORD \*outfun, WORD \*infun, UWORD \*formrat, WORD nrat)
- void [SimpleDelta](#) (mpf\_t sum, int m)
- void [SimpleDeltaC](#) (mpf\_t sum, int m)
- void [SingleTable](#) (mpf\_t \*tabl, int N, int m, int pow)
- void [DoubleTable](#) (mpf\_t \*tabout, mpf\_t \*tabin, int N, int m, int pow)
- void [EndTable](#) (mpf\_t sum, mpf\_t \*tabin, int N, int m, int pow)
- void [deltaMZV](#) (mpf\_t, int \*, int)
- void [deltaEuler](#) (mpf\_t, int \*, int)
- void [deltaEulerC](#) (mpf\_t, int \*, int)
- void [CalculateMZVhalf](#) (mpf\_t, int \*, int)
- void [CalculateMZV](#) (mpf\_t, int \*, int)
- void [CalculateEuler](#) (mpf\_t, int \*, int)
- int [ExpandMZV](#) (WORD \*term, WORD level)
- int [ExpandEuler](#) (WORD \*term, WORD level)
- int [PackFloat](#) (WORD \*, mpf\_t)
- int [UnpackFloat](#) (mpf\_t, WORD \*)

- void [RatToFloat](#) (mpf\_t result, UWORD \*format, int ratsize)
- void [Form\\_mpf\\_empty](#) (mpf\_t t)
- int [TestFloat](#) (WORD \*fun)
- WORD [FloatFunToRat](#) (PHEAD UWORD \*ratout, WORD \*in)
- void [ZeroTable](#) (mpf\_t \*tab, int N)
- SBYTE \* [ReadFloat](#) (SBYTE \*s)
- UBYTE \* [CheckFloat](#) (UBYTE \*ss, int \*spec)
- int [SetFloatPrecision](#) (WORD prec)
- int [PrintFloat](#) (WORD \*fun, int numdigits)
- void [SetupMZVTables](#) (void)
- void [SetupMPFTables](#) (void)
- void [ClearMZVTables](#) (void)
- int [CoToFloat](#) (UBYTE \*s)
- int [CoToRat](#) (UBYTE \*s)
- int [ToFloat](#) (PHEAD WORD \*term, WORD level)
- int [ToRat](#) (PHEAD WORD \*term, WORD level)
- int [AddWithFloat](#) (PHEAD WORD \*\*ps1, WORD \*\*ps2)
- int [MergeWithFloat](#) (PHEAD WORD \*\*interm1, WORD \*\*interm2)
- int [EvaluateEuler](#) (PHEAD WORD \*term, WORD level, WORD par)
- int [CoEvaluate](#) (UBYTE \*s)

#### 4.43.1 Detailed Description

This file contains numerical routines that deal with floating point numbers. We use the GMP for arbitrary precision floating point arithmetic. The functions of the type sin, cos, ln, sqrt etc are handled by the mpfr library. The reason that for MZV's and the general notation we use the GMP (mpf\_) is because the contents of the structs have been frozen and can be used for storage in a Form function float\_. With mpfr\_ this is not safely possible. All mpfr\_ related code is in the file [evaluate.c](#).

Definition in file [float.c](#).

#### 4.43.2 Macro Definition Documentation

##### 4.43.2.1 WITHCUTOFF

```
#define WITHCUTOFF
```

Definition at line 45 of file [float.c](#).

##### 4.43.2.2 GMPSPREAD

```
#define GMPSPREAD (GMP_LIMB_BITS/BITSINWORD)
```

Definition at line 47 of file [float.c](#).



### 4.43.3 Function Documentation

#### 4.43.3.1 Form\_mpf\_init()

```
void Form_mpf_init (
    mpf_t t )
```

Definition at line 176 of file [float.c](#).

#### 4.43.3.2 Form\_mpf\_clear()

```
void Form_mpf_clear (
    mpf_t t )
```

Definition at line 200 of file [float.c](#).

#### 4.43.3.3 Form\_mpf\_set\_prec\_raw()

```
void Form_mpf_set_prec_raw (
    mpf_t t,
    ULONG newprec )
```

Definition at line 226 of file [float.c](#).

#### 4.43.3.4 FormtoZ()

```
void FormtoZ (
    mpz_t z,
    UWORD * a,
    WORD na )
```

Definition at line 511 of file [float.c](#).

#### 4.43.3.5 ZtoForm()

```
void ZtoForm (
    UWORD * a,
    WORD * na,
    mpz_t z )
```

Definition at line 555 of file [float.c](#).

#### 4.43.3.6 FloatToInteger()

```
long FloatToInteger (
    UWORD * out,
    mpf_t floatin,
    long * bitsused )
```

Definition at line 581 of file [float.c](#).

#### 4.43.3.7 IntegerToFloat()

```
void IntegerToFloat (
    mpf_t result,
    UWORD * formlong,
    int longsize )
```

Definition at line 604 of file [float.c](#).

#### 4.43.3.8 FloatToRat()

```
int FloatToRat (
    UWORD * ratout,
    WORD * nratout,
    mpf_t floatin )
```

Definition at line 668 of file [float.c](#).

#### 4.43.3.9 AddFloats()

```
int AddFloats (
    PHEAD WORD * fun3,
    WORD * fun1,
    WORD * fun2 )
```

Definition at line 978 of file [float.c](#).

#### 4.43.3.10 MulFloats()

```
int MulFloats (
    PHEAD WORD * fun3,
    WORD * fun1,
    WORD * fun2 )
```

Definition at line 994 of file [float.c](#).

#### 4.43.3.11 DivFloats()

```
int DivFloats (
    PHEAD WORD * fun3,
    WORD * fun1,
    WORD * fun2 )
```

Definition at line 1010 of file [float.c](#).

#### 4.43.3.12 AddRatToFloat()

```
int AddRatToFloat (
    PHEAD WORD * outfun,
    WORD * infun,
    UWORD * formrat,
    WORD nrat )
```

Definition at line 1028 of file [float.c](#).

#### 4.43.3.13 MulRatToFloat()

```
int MulRatToFloat (
    PHEAD WORD * outfun,
    WORD * infun,
    UWORD * formrat,
    WORD nrat )
```

Definition at line 1047 of file [float.c](#).

#### 4.43.3.14 SimpleDelta()

```
void SimpleDelta (
    mpf_t sum,
    int m )
```

Definition at line 1681 of file [float.c](#).

#### 4.43.3.15 SimpleDeltaC()

```
void SimpleDeltaC (
    mpf_t sum,
    int m )
```

Definition at line 1728 of file [float.c](#).

#### 4.43.3.16 SingleTable()

```
void SingleTable (
    mpf_t * tabl,
    int N,
    int m,
    int pow )
```

Definition at line 1775 of file [float.c](#).

#### 4.43.3.17 DoubleTable()

```
void DoubleTable (
    mpf_t * tabout,
    mpf_t * tabin,
    int N,
    int m,
    int pow )
```

Definition at line 1827 of file [float.c](#).

#### 4.43.3.18 EndTable()

```
void EndTable (
    mpf_t sum,
    mpf_t * tabin,
    int N,
    int m,
    int pow )
```

Definition at line 1878 of file [float.c](#).

#### 4.43.3.19 deltaMZV()

```
void deltaMZV (
    mpf_t result,
    int * indexes,
    int depth )
```

Definition at line 1923 of file [float.c](#).

#### 4.43.3.20 deltaEuler()

```
void deltaEuler (
    mpf_t result,
    int * indexes,
    int depth )
```

Definition at line 1990 of file [float.c](#).

#### 4.43.3.21 deltaEulerC()

```
void deltaEulerC (
    mpf_t result,
    int * indexes,
    int depth )
```

Definition at line 2046 of file [float.c](#).

#### 4.43.3.22 CalculateMZVhalf()

```
void CalculateMZVhalf (
    mpf_t result,
    int * indexes,
    int depth )
```

Definition at line 2119 of file [float.c](#).

#### 4.43.3.23 CalculateMZV()

```
void CalculateMZV (
    mpf_t result,
    int * indexes,
    int depth )
```

Definition at line 2140 of file [float.c](#).

#### 4.43.3.24 CalculateEuler()

```
void CalculateEuler (
    mpf_t result,
    int * Zindexes,
    int depth )
```

Definition at line 2205 of file [float.c](#).

#### 4.43.3.25 ExpandMZV()

```
int ExpandMZV (
    WORD * term,
    WORD level )
```

Definition at line 2281 of file [float.c](#).

#### 4.43.3.26 ExpandEuler()

```
int ExpandEuler (
    WORD * term,
    WORD level )
```

Definition at line 2293 of file [float.c](#).

#### 4.43.3.27 PackFloat()

```
int PackFloat (
    WORD * fun,
    mpf_t infloat )
```

Definition at line 270 of file [float.c](#).

#### 4.43.3.28 UnpackFloat()

```
int UnpackFloat (
    mpf_t outfloat,
    WORD * fun )
```

Definition at line [364](#) of file [float.c](#).

#### 4.43.3.29 RatToFloat()

```
void RatToFloat (
    mpf_t result,
    UWORD * formrat,
    int ratsize )
```

Definition at line [620](#) of file [float.c](#).

#### 4.43.3.30 Form\_mpf\_empty()

```
void Form_mpf_empty (
    mpf_t t )
```

Definition at line [213](#) of file [float.c](#).

#### 4.43.3.31 TestFloat()

```
int TestFloat (
    WORD * fun )
```

Definition at line [452](#) of file [float.c](#).

#### 4.43.3.32 FloatFunToRat()

```
WORD FloatFunToRat (
    PHEAD UWORD * ratout,
    WORD * in )
```

Definition at line [642](#) of file [float.c](#).

#### 4.43.3.33 ZeroTable()

```
void ZeroTable (
    mpf_t * tab,
    int N )
```

Definition at line [804](#) of file [float.c](#).

#### 4.43.3.34 ReadFloat()

```
SBYTE * ReadFloat (
    SBYTE * s )
```

Definition at line 822 of file [float.c](#).

#### 4.43.3.35 CheckFloat()

```
UBYTE * CheckFloat (
    UBYTE * ss,
    int * spec )
```

Definition at line 848 of file [float.c](#).

#### 4.43.3.36 SetFloatPrecision()

```
int SetFloatPrecision (
    WORD prec )
```

Definition at line 911 of file [float.c](#).

#### 4.43.3.37 PrintFloat()

```
int PrintFloat (
    WORD * fun,
    int numdigits )
```

Definition at line 940 of file [float.c](#).

#### 4.43.3.38 SetupMZVTables()

```
void SetupMZVTables (
    void )
```

Definition at line 1064 of file [float.c](#).

#### 4.43.3.39 SetupMPFTables()

```
void SetupMPFTables (
    void )
```

Definition at line 1161 of file [float.c](#).

#### 4.43.3.40 ClearMZVTables()

```
void ClearMZVTables (
    void )
```

Definition at line 1215 of file [float.c](#).

#### 4.43.3.41 CoToFloat()

```
int CoToFloat (
    UBYTE * s )
```

Definition at line 1292 of file [float.c](#).

#### 4.43.3.42 CoToRat()

```
int CoToRat (
    UBYTE * s )
```

Definition at line 1314 of file [float.c](#).

#### 4.43.3.43 ToFloat()

```
int ToFloat (
    PHEAD WORD * term,
    WORD level )
```

Definition at line 1338 of file [float.c](#).

#### 4.43.3.44 ToRat()

```
int ToRat (
    PHEAD WORD * term,
    WORD level )
```

Definition at line 1370 of file [float.c](#).

#### 4.43.3.45 AddWithFloat()

```
int AddWithFloat (
    PHEAD WORD ** ps1,
    WORD ** ps2 )
```

Definition at line 1439 of file [float.c](#).

#### 4.43.3.46 MergeWithFloat()

```
int MergeWithFloat (
    PHEAD WORD ** interm1,
    WORD ** interm2 )
```

Definition at line 1560 of file [float.c](#).



**4.43.3.47 EvaluateEuler()**

```
int EvaluateEuler (
    PHEAD WORD * term,
    WORD level,
    WORD par )
```

Definition at line [2321](#) of file [float.c](#).

**4.43.3.48 CoEvaluate()**

```
int CoEvaluate (
    UBYTE * s )
```

Definition at line [2439](#) of file [float.c](#).

**4.44 float.c**

[Go to the documentation of this file.](#)

```
00001
00011 /* # License : */
00012 /*
00013 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00014 * When using this file you are requested to refer to the publication
00015 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00016 * This is considered a matter of courtesy as the development was paid
00017 * for by FOM the Dutch physics granting agency and we would like to
00018 * be able to track its scientific use to convince FOM of its value
00019 * for the community.
00020 *
00021 * This file is part of FORM.
00022 *
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00025 * Foundation, either version 3 of the License, or (at your option) any later
00026 * version.
00027 *
00028 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00029 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00030 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00031 * details.
00032 *
00033 * You should have received a copy of the GNU General Public License along
00034 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00035 */
00036 /* # License : */
00037 /*
00038 # Includes : float.c
00039 */
00040
00041 #include "form3.h"
00042 #include <math.h>
00043 #include <gmp.h>
00044
00045 #define WITHCUTOFF
00046
00047 #define GMPSPREAD (GMP_LIMB_BITS/BITSINWORD)
00048
00049 void Form_mpf_init(mpf_t t);
00050 void Form_mpf_clear(mpf_t t);
00051 void Form_mpf_set_prec_raw(mpf_t t,ULONG newprec);
00052 void FormtoZ(mpz_t z,UWORD *a,WORD na);
00053 void ZtoForm(UWORD *a,WORD *na,mpz_t z);
00054 long FloatToInteger(UWORD *out, mpf_t floatin, long *bitsused);
00055 void IntegerToFloat(mpf_t result, UWORD *formlong, int longsize);
00056 int FloatToRat(UWORD *ratout, WORD *nratout, mpf_t floatin);
00057 int AddFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2);
00058 int MulFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2);
00059 int DivFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2);
00060 int AddRatToFloat(PHEAD WORD *outfun, WORD *infun, UWORD *formrat, WORD nrat);
00061 int MulRatToFloat(PHEAD WORD *outfun, WORD *infun, UWORD *formrat, WORD nrat);
```

```

00062 void SimpleDelta(mpf_t sum, int m);
00063 void SimpleDeltaC(mpf_t sum, int m);
00064 void SingleTable(mpf_t *tabl, int N, int m, int pow);
00065 void DoubleTable(mpf_t *tabout, mpf_t *tabin, int N, int m, int pow);
00066 void EndTable(mpf_t sum, mpf_t *tabin, int N, int m, int pow);
00067 void deltaMZV(mpf_t, int *, int);
00068 void deltaEuler(mpf_t, int *, int);
00069 void deltaEulerC(mpf_t, int *, int);
00070 void CalculateMZVhalf(mpf_t, int *, int);
00071 void CalculateMZV(mpf_t, int *, int);
00072 void CalculateEuler(mpf_t, int *, int);
00073 int ExpandMZV(WORD *term, WORD level);
00074 int ExpandEuler(WORD *term, WORD level);
00075 int PackFloat(WORD *, mpf_t);
00076 int UnpackFloat(mpf_t, WORD *);
00077 void RatToFloat(mpf_t result, UWORD *format, int ratsize);
00078
00079 /*
00080  #] Includes :
00081  #[ Some GMP float info :
00082
00083  From gmp.h:
00084
00085  typedef struct
00086  {
00087      int _mp_prec;      Max precision, in number of `mp_limb_t's.
00088                        Set by mpf_init and modified by
00089                        mpf_set_prec. The area pointed to by the
00090                        _mp_d field contains `prec' + 1 limbs.
00091      int _mp_size;      abs(_mp_size) is the number of limbs the
00092                        last field points to. If _mp_size is
00093                        negative this is a negative number.
00094      mp_exp_t _mp_exp;  Exponent, in the base of `mp_limb_t'.
00095      mp_limb_t *_mp_d;  Pointer to the limbs.
00096  } __mpf_struct;
00097
00098  typedef __mpf_struct mpf_t[1];
00099
00100  Currently:
00101      sizeof(int) = 4
00102      sizeof(mpf_t) = 24
00103      sizeof(mp_limb_t) = 8,
00104      sizeof(mp_exp_t) = 8,
00105      sizeof a pointer is 8.
00106  If any of this changes in the future we would have to change the
00107  PackFloat and UnpackFloat routines.
00108
00109  Our float_ function is packed with the 4 arguments:
00110      -SNUMBER _mp_prec
00111      -SNUMBER _mp_size
00112      exponent which can be -SNUMBER or a regular term
00113      the limbs as n/1 in regular term format, or just an -SNUMBER.
00114  During normalization the sign is taken away from the second argument
00115  and put as the sign of the complete term. This is easier for the
00116  WriteInnerTerm routine in sch.c
00117
00118  Wildcarding of functions excludes matches with float_
00119  Float_ always ends up inside the bracket, just like poly(rat)fun.
00120
00121  From mpfr.h
00122
00123  The formats here are different. This has, among others, to do with
00124  the rounding. The result is that we need different aux variables
00125  when working with mpfr and we need a different conversion routine
00126  to float. It also means that we need to treat the mzv_ etc functions
00127  completely separately. For now we try to develop that in the file
00128  Evaluate.c. At a later stage we may merge the two files.
00129  Originally we had the sqrt in float.c but it seems better to put
00130  it in Evaluate.c
00131
00132  #] Some GMP float info :
00133  #[ Low Level :
00134
00135      In the low level routines we interact directly with the content
00136      of the GMP structs. This can be done safely, because their
00137      layout is documented. We pay particular attention to the init
00138      and clear routines, because they involve malloc/free calls.
00139
00140      #[ Explanations :
00141
00142      We need a number of facilities inside Form to deal with floating point
00143      numbers, taking into account that they are only meant for sporadic use.
00144      1: We need a special function float_ who's arguments contain a limb
00145         representation of the mpf_t of the gmp.
00146      2: Some functions may return a floating point number. This is then in
00147         the form of an occurrence of the function float_.
00148      3: We need a print routine to print this float_.

```

```

00149      4: We need routines for conversions:
00150          a: (long) int to float.
00151          b: rat to float.
00152          c: float to rat (if possible)
00153          d: float to integer (with rounding toward zero).
00154      5: We need calculational operations for add, sub, mul, div, exp.
00155      6: We need service routines to pack and unpack the function float_.
00156      7: The coefficient should be pulled inside float_.
00157      8: Float_ cannot occur inside PolyRatFun.
00158      9: We need to be able to treat float_ as the coefficient during sorting.
00159     10: For now, we need the functions mzv_ and eulerf_ for the evaluation
00160          of mzv's and euler sums.
00161     11: We need a setting for the default precision.
00162     12: We need a special flag for the dirty field of arguments to avoid
00163          the normalization of the argument with the limbs.
00164          Alternatively, the float_ should be not a regular function, but
00165          get a status < FUNCTION. Just like SNUMBER, LNUMBER etc.
00166
00167     Note that we can keep this thread-safe. The only routine that is not
00168     thread-safe is the print routine, but that is in accordance with all
00169     other print routines in Form. When called from MesPrint it needs to
00170     be protected by a lock, as is done standard inside Form.
00171
00172          #] Explanations :
00173          #[ Form_mpf_init :
00174  */
00175
00176 void Form_mpf_init(mpf_t t)
00177 {
00178     mp_limb_t *d;
00179     int i, prec;
00180     if ( t->_mp_d ) { M_free(t->_mp_d,"Form_mpf_init"); t->_mp_d = 0; }
00181     prec = (AC.DefaultPrecision + 8*sizeof(mp_limb_t)-1)/(8*sizeof(mp_limb_t)) + 1;
00182     t->_mp_prec = prec;
00183     t->_mp_size = 0;
00184     t->_mp_exp = 0;
00185     d = (mp_limb_t *)Mallc1(prec*sizeof(mp_limb_t),"Form_mpf_init");
00186     if ( d == 0 ) {
00187         MesPrint("Fatal error in Mallc1 call in Form_mpf_init. Trying to allocate %ld bytes.",
00188             prec*sizeof(mp_limb_t));
00189         Terminate(-1);
00190     }
00191     t->_mp_d = d;
00192     for ( i = 0; i < prec; i++ ) d[i] = 0;
00193 }
00194
00195  */
00196          #] Form_mpf_init :
00197          #[ Form_mpf_clear :
00198  */
00199
00200 void Form_mpf_clear(mpf_t t)
00201 {
00202     if ( t->_mp_d ) { M_free(t->_mp_d,"Form_mpf_init"); t->_mp_d = 0; }
00203     t->_mp_prec = 0;
00204     t->_mp_size = 0;
00205     t->_mp_exp = 0;
00206 }
00207
00208  */
00209          #] Form_mpf_clear :
00210          #[ Form_mpf_empty :
00211  */
00212
00213 void Form_mpf_empty(mpf_t t)
00214 {
00215     int i;
00216     for ( i = 0; i < t->_mp_prec; i++ ) t->_mp_d[i] = 0;
00217     t->_mp_size = 0;
00218     t->_mp_exp = 0;
00219 }
00220
00221  */
00222          #] Form_mpf_empty :
00223          #[ Form_mpf_set_prec_raw :
00224  */
00225
00226 void Form_mpf_set_prec_raw(mpf_t t,ULONG newprec)
00227 {
00228     ULONG newpr = (newprec + 8*sizeof(mp_limb_t)-1)/(8*sizeof(mp_limb_t)) + 1;
00229     if ( newpr <= (ULONG)(t->_mp_prec) ) {
00230         int i, used = ABS(t->_mp_size) ;
00231         if ( newpr > (ULONG)used ) {
00232             for ( i = used; i < (int)newpr; i++ ) t->_mp_d[i] = 0;
00233         }
00234         if ( t->_mp_size < 0 ) newpr = -newpr;
00235         t->_mp_size = newpr;

```

```

00236     }
00237     else {
00238  /*
00239         We can choose here between "forget about it" or making a new malloc.
00240         If the program is well designed, we should never get here though.
00241         For now we choose the "forget it" option.
00242  */
00243         MesPrint("Trying too big a precision %ld in Form_mpf_set_prec_raw.", newprec);
00244         MesPrint("Old maximal precision is %ld.",
00245             (size_t) (t->_mp_prec-1)*sizeof(mp_limb_t)*8);
00246         Terminate(-1);
00247     }
00248 }
00249
00250 /*
00251     #] Form_mpf_set_prec_raw :
00252     #[ PackFloat :
00253
00254     Puts an object of type mpf_t inside a function float_
00255     float_(prec, nlimbs, exp, limbs);
00256     Complication: the limbs and the exponent are long's. That means
00257     two times the size of a Form (U)WORD.
00258     To save space we could split the limbs in two because Form stores
00259     coefficients as rationals with equal space for numerator and denominator.
00260     Hence for each limb we could put the most significant UWORD in the
00261     numerator and the least significant limb in the denominator.
00262     This gives a problem with the compilation and subsequent potential
00263     normalization of the 'fraction'. Hence for now we take the wasteful
00264     approach. At a later stage we can try to veto normalization of FLOATFUN.
00265     Caution: gmp/fmp is little endian and Form is big endian.
00266
00267     Zero is represented by zero limbs (and zero exponent).
00268  */
00269
00270 int PackFloat(WORD *fun, mpf_t infloat)
00271 {
00272     WORD *t, nlimbs, num, nnum;
00273     mp_limb_t *d = infloat->_mp_d; /* Pointer to the limbs. */
00274     int i;
00275     long e = infloat->_mp_exp;
00276
00277     t = fun;
00278     *t++ = FLOATFUN;
00279     t++;
00280     FILLFUN(t);
00281     *t++ = -SNUMBER;
00282     *t++ = infloat->_mp_prec;
00283     *t++ = -SNUMBER;
00284     *t++ = infloat->_mp_size;
00285  /*
00286     Now the exponent which is a signed long
00287  */
00288     if ( e < 0 ) {
00289         e = -e;
00290         if ( ( e > (BITSINWORD-1) ) == 0 ) {
00291             *t++ = -SNUMBER; *t++ = -e;
00292         }
00293         else {
00294             *t++ = ARGHEAD+6; *t++ = 0; FILLARG(t);
00295             *t++ = 6;
00296             *t++ = (UWORD)e;
00297             *t++ = (UWORD)(e > BITSINWORD);
00298             *t++ = 1; *t++ = 0; *t++ = -5;
00299         }
00300     }
00301     else {
00302         if ( ( e > (BITSINWORD-1) ) == 0 ) {
00303             *t++ = -SNUMBER; *t++ = e;
00304         }
00305         else {
00306             *t++ = ARGHEAD+6; *t++ = 0; FILLARG(t);
00307             *t++ = 6;
00308             *t++ = (UWORD)e;
00309             *t++ = (UWORD)(e > BITSINWORD);
00310             *t++ = 1; *t++ = 0; *t++ = 5;
00311         }
00312     }
00313  /*
00314     and now the limbs. Note that the number of active limbs could be less
00315     than prec+1 in which case we copy the smaller number.
00316  */
00317     nlimbs = infloat->_mp_size < 0 ? -infloat->_mp_size: infloat->_mp_size;
00318     if ( nlimbs == 0 ) {
00319     }
00320     else if ( nlimbs == 1 && (ULONG)(*d) < ((ULONG)1)«(BITSINWORD-1) ) {
00321         *t++ = -SNUMBER;
00322         *t++ = (UWORD)(*d);

```

```

00323     }
00324     else {
00325         if ( d[nlimbs-1] < ((ULONG)1)«BITSINWORD ) {
00326             num = 2*nlimbs-1; /* number of UWORDS needed */
00327         }
00328         else {
00329             num = 2*nlimbs; /* number of UWORDS needed */
00330         }
00331         nnum = num;
00332         *t++ = 2*num+2+ARGHEAD;
00333         *t++ = 0;
00334         FILLARG(t);
00335         *t++ = 2*num+2;
00336         while ( num > 1 ) {
00337             *t++ = (UWORD) (*d); *t++ = (UWORD) (((ULONG) (*d))«BITSINWORD); d++;
00338             num -= 2;
00339         }
00340         if ( num == 1 ) { *t++ = (UWORD) (*d); }
00341         *t++ = 1;
00342         for ( i = 1; i < nnum; i++ ) *t++ = 0;
00343         *t++ = 2*nnum+1; /* the sign is hidden in infloat->_mp_size */
00344     }
00345     fun[1] = t-fun;
00346     return(fun[1]);
00347 }
00348
00349 /*
00350     #] PackFloat :
00351     #[ UnpackFloat :
00352
00353     Takes the arguments of a function float_ and puts it inside an mpf_t.
00354     For notation see commentary for PutInFloat.
00355
00356     We should make a new allocation for the limbs if the precision exceeds
00357     the present precision of outfloat, or when outfloat has not been
00358     initialized yet.
00359
00360     The return value is -1 if the contents of the FLOATFUN cannot be converted.
00361     Otherwise the return value is the number of limbs.
00362 */
00363 int UnpackFloat(mpf_t outfloat, WORD *fun)
00364 {
00365     UWORD *t;
00366     WORD *f;
00367     int num, i;
00368     mp_limb_t *d;
00369
00370     /*
00371     Very first step: check whether we can use float_:
00372     */
00373     GETIDENTITY
00374     if ( AT.aux_ == 0 ) {
00375         MesPrint("Illegal attempt at using a float_ function without proper startup.");
00376         MesPrint("Please use %#StartFloat <options> first.");
00377         Terminate(-1);
00378     }
00379     /*
00380     Check first the number and types of the arguments
00381     There should be 4. -SNUMBER, -SNUMBER, -SNUMBER or a long number.
00382     + the limbs, either -SNUMBER or Long number in the form of a Form rational.
00383     */
00384     if ( TestFloat(fun) == 0 ) goto Incorrect;
00385     f = fun + FUNHEAD + 2;
00386     if ( ABS(f[1]) > f[-1]+1 ) goto Incorrect;
00387     if ( f[1] > outfloat->_mp_prec+1
00388         || outfloat->_mp_d == 0 ) {
00389         /*
00390             We need to make a new allocation.
00391             Unfortunately we cannot use Malloc1 because that is not
00392             recognised by gmp and hence we cannot clear with mpf_clear.
00393             Maybe we can do something about it by making our own
00394             mpf_init and mpf_clear?
00395         */
00396         if ( outfloat->_mp_d != 0 ) free(outfloat->_mp_d);
00397         outfloat->_mp_d = (mp_limb_t *) malloc((f[1]+1)*sizeof(mp_limb_t));
00398     }
00399     num = f[1];
00400     outfloat->_mp_size = num;
00401     if ( num < 0 ) { num = -num; }
00402     f += 2;
00403     if ( *f == -SNUMBER ) {
00404         outfloat->_mp_exp = (mp_exp_t) (f[1]);
00405         f += 2;
00406     }
00407     else {
00408         f += ARGHEAD+6;
00409         if ( f[-1] == -5 ) {

```

```

00410         outfloat->_mp_exp =
00411             - (mp_exp_t) (((ULONG) (f[-4])) << BITSINWORD) + f[-5]);
00412     }
00413     else if ( f[-1] == 5 ) {
00414         outfloat->_mp_exp =
00415             (mp_exp_t) (((ULONG) (f[-4])) << BITSINWORD) + f[-5]);
00416     }
00417 }
00418 /*
00419 And now the limbs if needed
00420 */
00421 d = outfloat->_mp_d;
00422 if ( outfloat->_mp_size == 0 ) {
00423     for ( i = 0; i < outfloat->_mp_prec; i++ ) *d++ = 0;
00424     return(0);
00425 }
00426 else if ( *f == -SNUMBER ) {
00427     *d++ = (mp_limb_t) f[1];
00428     for ( i = 0; i < outfloat->_mp_prec; i++ ) *d++ = 0;
00429     return(0);
00430 }
00431 num = (*f-ARGHEAD-2)/2; /* 2*number of limbs in the argument */
00432 t = (UWORD *) (f+ARGHEAD+1);
00433 while ( num > 1 ) {
00434     *d++ = (mp_limb_t) (((ULONG) (t[1])) << BITSINWORD) + t[0]);
00435     t += 2; num -= 2;
00436 }
00437 if ( num == 1 ) *d++ = (mp_limb_t) ((UWORD) (*t));
00438 for ( i = d-outfloat->_mp_d; i <= outfloat->_mp_prec; i++ ) *d++ = 0;
00439 return(0);
00440 Incorrect:
00441     return(-1);
00442 }
00443
00444 /*
00445     #] UnpackFloat :
00446     #[ TestFloat :
00447
00448     Tells whether the function (with its arguments) is a legal float_.
00449     We assume, as in the whole float library that one limb is 2 WORDs.
00450 */
00451
00452 int TestFloat (WORD *fun)
00453 {
00454     WORD *fstop, *f, num, nnum, i;
00455     fstop = fun+fun[1];
00456     f = fun + FUNHEAD;
00457     num = 0;
00458 /*
00459 Count arguments
00460 */
00461 while ( f < fstop ) { num++; NEXTARG(f); }
00462 if ( num != 4 ) return(0);
00463 f = fun + FUNHEAD;
00464 if ( *f != -SNUMBER ) return(0);
00465 if ( f[1] < 0 ) return(0);
00466 f += 2;
00467 if ( *f != -SNUMBER ) return(0);
00468 num = ABS(f[1]); /* number of limbs */
00469 f += 2;
00470 if ( *f == -SNUMBER ) { f += 2; }
00471 else {
00472     if ( *f != ARGHEAD+6 ) return(0);
00473     if ( ABS(f[ARGHEAD+5]) != 5 ) return(0);
00474     if ( f[ARGHEAD+3] != 1 ) return(0);
00475     if ( f[ARGHEAD+4] != 0 ) return(0);
00476     f += *f;
00477 }
00478 if ( num == 0 ) return(1);
00479 if ( *f == -SNUMBER ) { if ( num != 1 ) return(0); }
00480 else {
00481     nnum = (ABS(f[*f-1])-1)/2;
00482     if ( ( *f != 4*num+2+ARGHEAD ) && ( *f != 4*num+ARGHEAD ) ) return(0);
00483     if ( ( nnum != 2*num ) && ( nnum != 2*num-1 ) ) return(0);
00484     f += ARGHEAD+nnum+1;
00485     if ( f[0] != 1 ) return(0);
00486     for ( i = 1; i < nnum; i++ ) {
00487         if ( f[i] != 0 ) return(0);
00488     }
00489 }
00490 return(1);
00491 }
00492
00493 /*
00494     #] TestFloat :
00495     #[ FormtoZ :
00496

```

```

00497     Converts a Form long integer to a GMP long integer.
00498
00499     typedef struct
00500     {
00501         int _mp_alloc;    Number of *limbs* allocated and pointed
00502                          to by the _mp_d field.
00503         int _mp_size;    abs(_mp_size) is the number of limbs the
00504                          last field points to. If _mp_size is
00505                          negative this is a negative number.
00506         mp_limb_t *_mp_d; Pointer to the limbs.
00507     } __mpz_struct;
00508     typedef __mpz_struct mpz_t[1];
00509 */
00510
00511 void FormtoZ(mpz_t z, UWORD *a, WORD na)
00512 {
00513     int nlimbs, sgn = 1;
00514     mp_limb_t *d;
00515     UWORD *b = a;
00516     WORD nb = na;
00517
00518     if ( nb == 0 ) {
00519         z->_mp_size = 0;
00520         z->_mp_d[0] = 0;
00521         return;
00522     }
00523     if ( nb < 0 ) { sgn = -1; nb = -nb; }
00524     nlimbs = (nb+1)/2;
00525     if ( mpz_cmp_si(z, 0L) <= 1 ) {
00526         mpz_set_ui(z, 100000000);
00527     }
00528     if ( z->_mp_alloc <= nlimbs ) {
00529         mpz_t zz;
00530         mpz_init(zz);
00531         while ( z->_mp_alloc <= nlimbs ) {
00532             mpz_mul(zz, z, z);
00533             mpz_set(z, zz);
00534         }
00535         mpz_clear(zz);
00536     }
00537     z->_mp_size = sgn*nlimbs;
00538     d = z->_mp_d;
00539     while ( nb > 1 ) {
00540         *d++ = ((mp_limb_t) (b[1])) << BITSINWORD + (mp_limb_t) (b[0]);
00541         b += 2; nb -= 2;
00542     }
00543     if ( nb == 1 ) { *d = (mp_limb_t) (*b); }
00544 }
00545
00546 /*
00547     #] FormtoZ :
00548     #[ ZtoForm :
00549
00550     Converts a GMP long integer to a Form long integer.
00551     Mainly an exercise of going from little endian ULONGs.
00552     to big endian UWORDS
00553 */
00554
00555 void ZtoForm(UWORD *a, WORD *na, mpz_t z)
00556 {
00557     mp_limb_t *d = z->_mp_d;
00558     int nlimbs = ABS(z->_mp_size), i;
00559     UWORD j;
00560     if ( nlimbs == 0 ) { *na = 0; return; }
00561     *na = 0;
00562     for ( i = 0; i < nlimbs-1; i++ ) {
00563         *a++ = (UWORD) (*d);
00564         *a++ = (UWORD) ((*d++) << BITSINWORD);
00565         *na += 2;
00566     }
00567     *na += 1;
00568     *a++ = (UWORD) (*d);
00569     j = (UWORD) ((*d) << BITSINWORD);
00570     if ( j != 0 ) { *a = j; *na += 1; }
00571     if ( z->_mp_size < 0 ) *na = -*na;
00572 }
00573
00574 /*
00575     #] ZtoForm :
00576     #[ FloatToInteger :
00577
00578     Converts a GMP float to a long Form integer.
00579 */
00580
00581 long FloatToInteger(UWORD *out, mpf_t floatin, long *bitsused)
00582 {
00583     mpz_t z;

```

```

00584     WORD nout, nx;
00585     UWORD x;
00586     mpz_init(z);
00587     mpz_set_f(z, floatin);
00588     ZtoForm(out, &nout, z);
00589     mpz_clear(z);
00590     x = out[0]; nx = 0;
00591     while ( x ) { nx++; x >>= 1; }
00592     *bitsused = (nout-1)*BITSINWORD + nx;
00593     return(nout);
00594 }
00595
00596 /*
00597     #] FloatToInteger :
00598     #[ IntegerToFloat :
00599
00600     Converts a Form long integer to a gmp float of default size.
00601     We assume that sizeof(unsigned long int) = 2*sizeof(UWORD).
00602 */
00603
00604 void IntegerToFloat(mpf_t result, UWORD *formlong, int longsize)
00605 {
00606     mpz_t z;
00607     mpz_init(z);
00608     FormtoZ(z, formlong, longsize);
00609     mpf_set_z(result, z);
00610     mpz_clear(z);
00611 }
00612
00613 /*
00614     #] IntegerToFloat :
00615     #[ RatToFloat :
00616
00617     Converts a Form rational to a gmp float of default size.
00618 */
00619
00620 void RatToFloat(mpf_t result, UWORD *formrat, int ratsize)
00621 {
00622     GETIDENTITY
00623     int nnum, nden;
00624     UWORD *num, *den;
00625     int sgn = 0;
00626     if ( ratsize < 0 ) { ratsize = -ratsize; sgn = 1; }
00627     nnum = nden = (ratsize-1)/2;
00628     num = formrat; den = formrat+nnum;
00629     while ( num[nnum-1] == 0 ) { nnum--; }
00630     while ( den[nden-1] == 0 ) { nden--; }
00631     IntegerToFloat(aux2, num, nnum);
00632     IntegerToFloat(aux3, den, nden);
00633     mpf_div(result, aux2, aux3);
00634     if ( sgn > 0 ) mpf_neg(result, result);
00635 }
00636
00637 /*
00638     #] RatToFloat :
00639     #[ FloatFunToRat :
00640 */
00641
00642 WORD FloatFunToRat(PHEAD UWORD *ratout, WORD *in)
00643 {
00644     WORD oldin = in[0], nratout;
00645     in[0] = FLOATFUN;
00646     UnpackFloat(aux4, in);
00647     FloatToRat(ratout, &nratout, aux4);
00648     in[0] = oldin;
00649     return(nratout);
00650 }
00651
00652 /*
00653     #] FloatFunToRat :
00654     #[ FloatToRat :
00655
00656     Converts a gmp float to a Form rational.
00657     The algorithm we use is 'repeated fractions' of the style
00658         n0+1/(n1+1/(n2+1/(n3+....)))
00659     The moment we get an n that is very large we should have the proper fraction
00660     Main problem: what if the sequence keeps going?
00661                 (there are always rounding errors)
00662     We need a cutoff with an error condition.
00663     Basically the product n0*n1*n2*... is an underlimit on the number of
00664     digits in the answer. We can put a limit on this.
00665     A return value of zero means that everything went fine.
00666 */
00667
00668 int FloatToRat(UWORD *ratout, WORD *nratout, mpf_t floatin)
00669 {
00670     GETIDENTITY

```



```

00671     WORD *out = AT.WorkPointer;
00672     WORD s; /* the sign. */
00673     UWORD *a, *b, *c, *d, *mc;
00674     WORD na, nb, nc, nd, i;
00675     int nout;
00676     LONG oldpWorkPointer = AT.pWorkPointer;
00677     long bitsused = 0, totalbitsused = 0, totalbits = AC.DefaultPrecision;
00678     int retval = 0, startnul;
00679     WantAddPointers(AC.DefaultPrecision); /* Horrible overkill */
00680     AT.pWorkSpace[AT.pWorkPointer++] = out;
00681     a = NumberMalloc("FloatToRat");
00682     b = NumberMalloc("FloatToRat");
00683
00684     s = mpf_sgn(floatin);
00685     if ( s < 0 ) mpf_neg(floatin, floatin);
00686
00687     Form_mpf_empty(aux1);
00688     Form_mpf_empty(aux2);
00689     Form_mpf_empty(aux3);
00690
00691     mpf_trunc(aux1, floatin); /* This should now be an integer */
00692     mpf_sub(aux2, floatin, aux1); /* and this >= 0 */
00693     if ( mpf_sgn(aux1) == 0 ) { *out++ = 0; startnul = 1; }
00694     else {
00695         nout = FloatToInteger((UWORD *)out, aux1, &totalbitsused);
00696         out += nout;
00697         startnul = 0;
00698     }
00699     AT.pWorkSpace[AT.pWorkPointer++] = out;
00700     if ( mpf_sgn(aux2) ) {
00701         for(;;) {
00702             mpf_ui_div(aux3, 1, aux2);
00703             mpf_trunc(aux1, aux3);
00704             mpf_sub(aux2, aux3, aux1);
00705             if ( mpf_sgn(aux1) == 0 ) { *out++ = 0; }
00706             else {
00707                 nout = FloatToInteger((UWORD *)out, aux1, &bitsused);
00708                 out += nout;
00709                 totalbitsused += bitsused;
00710             }
00711             if ( bitsused > (totalbits-totalbitsused)/2 ) { break; }
00712             if ( mpf_sgn(aux2) == 0 ) {
00713                 /*if ( startnul == 1 )*/ AT.pWorkSpace[AT.pWorkPointer++] = out;
00714                 break;
00715             }
00716             AT.pWorkSpace[AT.pWorkPointer++] = out;
00717         }
00718     /*
00719     At this point we have the function with the repeated fraction.
00720     Now we should work out the fraction. Form code would be:
00721     repeat id dum_(?a,x1?,x2?) = dum_(?a,x1+1/x2);
00722     id dum_(x?) = x;
00723     We have to work backwards. This is why we made a list of pointers
00724     in AT.pWorkSpace
00725
00726     Now we need the long integers a and b.
00727     Note that by construction there should never be a GCD!
00728 */
00729     out = (WORD *) (AT.pWorkSpace[--AT.pWorkPointer]);
00730     na = 1; a[0] = 1;
00731     c = (UWORD *) (AT.pWorkSpace[--AT.pWorkPointer]);
00732     nc = nb = ((UWORD *)out)-c;
00733     if ( nc > 10 ) {
00734         mc = c;
00735         c = (UWORD *) (AT.pWorkSpace[--AT.pWorkPointer]);
00736         nc = nb = ((UWORD *)mc)-c;
00737     }
00738     for ( i = 0; i < nb; i++ ) b[i] = c[i];
00739     mc = c = NumberMalloc("FloatToRat");
00740     while ( AT.pWorkPointer > oldpWorkPointer ) {
00741         d = (UWORD *) (AT.pWorkSpace[--AT.pWorkPointer]);
00742         nd = (UWORD *) (AT.pWorkSpace[AT.pWorkPointer+1])-d; /* This is the x1 in the formula */
00743         if ( nd == 1 && *d == 0 ) break;
00744         c = a; a = b; b = c; nc = na; na = nb; nb = nc; /* 1/x2 */
00745         MullLong(d, nd, a, na, mc, &nc);
00746         AddLong(mc, nc, b, nb, b, &nb);
00747     }
00748     NumberFree(mc, "FloatToRat");
00749     if ( startnul == 0 ) {
00750         c = a; a = b; b = c; nc = na; na = nb; nb = nc;
00751     }
00752 }
00753 else {
00754     c = (UWORD *) (AT.pWorkSpace[oldpWorkPointer]);
00755     na = (UWORD *)out - c;
00756     for ( i = 0; i < na; i++ ) a[i] = c[i];
00757     b[0] = 1; nb = 1;

```

```

00758     }
00759  /*
00760  Now we have the fraction in a/b. Create the rational and add the sign.
00761  */
00762     d = ratout;
00763     if ( na == nb ) {
00764         *nratout = (2*na+1)*s;
00765         for ( i = 0; i < na; i++ ) *d++ = a[i];
00766         for ( i = 0; i < nb; i++ ) *d++ = b[i];
00767     }
00768     else if ( na < nb ) {
00769         *nratout = (2*nb+1)*s;
00770         for ( i = 0; i < na; i++ ) *d++ = a[i];
00771         for ( i = 0; i < nb; i++ ) *d++ = 0;
00772         for ( i = 0; i < nb; i++ ) *d++ = b[i];
00773     }
00774     else {
00775         *nratout = (2*na+1)*s;
00776         for ( i = 0; i < na; i++ ) *d++ = a[i];
00777         for ( i = 0; i < nb; i++ ) *d++ = b[i];
00778         for ( i = 0; i < na; i++ ) *d++ = 0;
00779     }
00780     *d = (UWORD)(*nratout);
00781     NumberFree(b,"FloatToRat");
00782     NumberFree(a,"FloatToRat");
00783     AT.pWorkPointer = oldpWorkPointer;
00784  /*
00785  Just for check we go back to float
00786  */
00787     if ( s < 0 ) mpf_neg(floatin,floatin);
00788  /*
00789     {
00790         WORD n = *ratout;
00791         RatToFloat(aux1, ratout, n);
00792         mpf_sub(aux2, floatin, aux1);
00793         gmp_printf("Diff is %.*Fe\n", 40, aux2);
00794     }
00795  */
00796     return(retval);
00797 }
00798
00799  /*
00800     #] FloatToRat :
00801     #[ ZeroTable :
00802  */
00803
00804 void ZeroTable(mpf_t *tab, int N)
00805 {
00806     int i, j;
00807     for ( i = 0; i < N; i++ ) {
00808         for ( j = 0; j < tab[i]->_mp_prec; j++ ) tab[i]->_mp_d[j] = 0;
00809     }
00810 }
00811
00812  /*
00813     #] ZeroTable :
00814     #[ ReadFloat :
00815
00816     Used by the compiler. It reads a floating point number and
00817     outputs it at AT.WorkPointer as if it were a float_ function
00818     with its arguments.
00819     The call enters with s[-1] == TFLOAT.
00820  */
00821
00822 SBYTE *ReadFloat(SBYTE *s)
00823 {
00824     GETIDENTITY
00825     SBYTE *ss, c;
00826     ss = s;
00827     while ( *ss > 0 ) ss++;
00828     c = *ss; *ss = 0;
00829     gmp_sscanf((char *)s, "%Ff", aux1);
00830     if ( AT.WorkPointer > AT.WorkTop ) {
00831         MLOCK(ErrorMessageLock);
00832         MesWork();
00833         MUNLOCK(ErrorMessageLock);
00834         Terminate(-1);
00835     }
00836     PackFloat(AT.WorkPointer, aux1);
00837     *ss = c;
00838     return(ss);
00839 }
00840
00841  /*
00842     #] ReadFloat :
00843     #[ CheckFloat :
00844

```

```

00845     For the tokenizer. Tests whether the input string can be a float.
00846  */
00847
00848 UBYTE *CheckFloat(UBYTE *ss, int *spec)
00849 {
00850     GETIDENTITY
00851     UBYTE *s = ss;
00852     int zero = 1, gotdot = 0;
00853     while ( FG.cTable[s[-1]] == 1 ) s--;
00854     *spec = 0;
00855     if ( FG.cTable[*s] == 1 ) {
00856         while ( *s == '0' ) s++;
00857         if ( FG.cTable[*s] == 1 ) {
00858             s++;
00859             while ( FG.cTable[*s] == 1 ) s++;
00860             zero = 0;
00861         }
00862         if ( *s == '.' ) { goto dot; }
00863     }
00864     else if ( *s == '.' ) {
00865 dot:
00866         gotdot = 1;
00867         s++;
00868         if ( FG.cTable[*s] != 1 && zero == 1 ) return(ss);
00869         while ( *s == '0' ) s++;
00870         if ( FG.cTable[*s] == 1 ) {
00871             s++;
00872             while ( FG.cTable[*s] == 1 ) s++;
00873             zero = 0;
00874         }
00875     }
00876     else return(ss);
00877  /*
00878     Now we have had the mantissa part, which may be zero.
00879     Check for an exponent.
00880  */
00881     if ( *s == 'e' || *s == 'E' ) {
00882         s++;
00883         if ( *s == '-' || *s == '+' ) s++;
00884         if ( FG.cTable[*s] == 1 ) {
00885             s++;
00886             while ( FG.cTable[*s] == 1 ) s++;
00887         }
00888         else { return(ss); }
00889     }
00890     else if ( gotdot == 0 ) return(ss);
00891     if ( AT.aux_ == 0 ) { /* no floating point system */
00892         *spec = -1;
00893         return(s);
00894     }
00895     if ( zero ) *spec = 1;
00896     return(s);
00897 }
00898
00899  /*
00900     #] CheckFloat :
00901     #] Low Level :
00902     #[ Float Routines :
00903     #[ SetFloatPrecision :
00904
00905     We set the default precision of the floats and allocate
00906     space for an output string if we want to write the float.
00907     Space needed: exponent: up to 12 chars.
00908     mantissa 2+10*prec/33 + a little bit extra.
00909  */
00910
00911 int SetFloatPrecision(WORD prec)
00912 {
00913     if ( prec <= 0 ) {
00914         MesPrint("&Illegal value for number of bits for floating point constants: %d",prec);
00915         return(-1);
00916     }
00917     else {
00918         AC.DefaultPrecision = prec;
00919         if ( AO.floatspace != 0 ) M_free(AO.floatspace,"floatspace");
00920         AO.floatsize = ((10*prec)/33+20)*sizeof(char);
00921         AO.floatspace = (UBYTE *)Malloc1(AO.floatsize,"floatspace");
00922         mpf_set_default_prec(prec);
00923         return(0);
00924     }
00925 }
00926
00927  /*
00928     #] SetFloatPrecision :
00929     #[ PrintFloat :
00930
00931     Print the float_ function with its arguments as a floating point

```

```

00932     number with numdigits digits.
00933     If however we use rawfloat as a format option it prints it as a
00934     regular Form function and it should never come here because the
00935     print routines in sch.c should intercept it.
00936     The buffer AO.floatspace is allocated when the precision of the
00937     floats is set in the routine SetupM2VTables.
00938 */
00939
00940 int PrintFloat(WORD *fun,int numdigits)
00941 {
00942     GETIDENTITY
00943     int n = 0;
00944     int prec = (AC.DefaultPrecision-AC.MaxWeight-1)*log10(2.0);
00945     if ( numdigits > prec || numdigits == 0 ) {
00946         numdigits = prec;
00947     }
00948 /*
00949     GMP's gmp_snprintf always prints a non-zero number before the decimal point, so we
00950     ask for one digit less.
00951 */
00952     if ( UnpackFloat(aux4,fun) == 0 )
00953         n = gmp_snprintf((char *) (AO.floatspace),AO.floatsize,"%.*Fe",numdigits-1,aux4);
00954     if ( numdigits == prec ) {
00955         UBYTE *s1, *s2;
00956         int n1, n2 = n;
00957         s1 = AO.floatspace+n;
00958         while ( s1 > AO.floatspace && s1[-1] != 'e'
00959             && s1[-1] != 'E' && s1[-1] != '.' ) { s1--; n2--; }
00960         if ( s1 > AO.floatspace && s1[-1] != '.' ) {
00961             s1--; n2--;
00962             s2 = s1; n1 = n2;
00963             while ( s1[-1] == '0' ) { s1--; n1--; }
00964             if ( s1[-1] == '.' ) { s1++; n1++; }
00965             while ( n2 < n ) { *s1++ = *s2++; n2++; }
00966             n -= (n2-n1);
00967             *s1 = 0;
00968         }
00969     }
00970     return(n);
00971 }
00972
00973 /*
00974     #] PrintFloat :
00975     #[ AddFloats :
00976 */
00977
00978 int AddFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2)
00979 {
00980     int retval = 0;
00981     if ( UnpackFloat(aux1,fun1) == 0 && UnpackFloat(aux2,fun2) == 0 ) {
00982         mpf_add(aux1,aux1,aux2);
00983         PackFloat(fun3,aux1);
00984     }
00985     else { retval = -1; }
00986     return(retval);
00987 }
00988
00989 /*
00990     #] AddFloats :
00991     #[ MulFloats :
00992 */
00993
00994 int MulFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2)
00995 {
00996     int retval = 0;
00997     if ( UnpackFloat(aux1,fun1) == 0 && UnpackFloat(aux2,fun2) == 0 ) {
00998         mpf_mul(aux1,aux1,aux2);
00999         PackFloat(fun3,aux1);
01000     }
01001     else { retval = -1; }
01002     return(retval);
01003 }
01004
01005 /*
01006     #] MulFloats :
01007     #[ DivFloats :
01008 */
01009
01010 int DivFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2)
01011 {
01012     int retval = 0;
01013     if ( UnpackFloat(aux1,fun1) == 0 && UnpackFloat(aux2,fun2) == 0 ) {
01014         mpf_div(aux1,aux1,aux2);
01015         PackFloat(fun3,aux1);
01016     }
01017     else { retval = -1; }
01018     return(retval);

```

```

01019 }
01020
01021 /*
01022     #] DivFloats :
01023     #[ AddRatToFloat :
01024
01025     Note: this can be optimized, because often the rat is rather simple.
01026 */
01027
01028 int AddRatToFloat(PHEAD WORD *outfun, WORD *infun, UWORD *formrat, WORD nrat)
01029 {
01030     int retval = 0;
01031     if ( UnpackFloat(aux2,infun) == 0 ) {
01032         RatToFloat(aux1,formrat,nrat);
01033         mpf_add(aux2,aux2,aux1);
01034         PackFloat(outfun,aux2);
01035     }
01036     else retval = -1;
01037     return(retval);
01038 }
01039
01040 /*
01041     #] AddRatToFloat :
01042     #[ MulRatToFloat :
01043
01044     Note: this can be optimized, because often the rat is rather simple.
01045 */
01046
01047 int MulRatToFloat(PHEAD WORD *outfun, WORD *infun, UWORD *formrat, WORD nrat)
01048 {
01049     int retval = 0;
01050     if ( UnpackFloat(aux2,infun) == 0 ) {
01051         RatToFloat(aux1,formrat,nrat);
01052         mpf_mul(aux2,aux2,aux1);
01053         PackFloat(outfun,aux2);
01054     }
01055     else retval = -1;
01056     return(retval);
01057 }
01058
01059 /*
01060     #] MulRatToFloat :
01061     #[ SetupMZVTables :
01062 */
01063
01064 void SetupMZVTables(void)
01065 {
01066     /*
01067     Sets up a table of N+1 mpf_t floats with variable precision.
01068     It assumes that each next element needs one bit less.
01069     Initializes all of them.
01070     We take some extra space, to have one limb overflow everywhere.
01071     Because the depth of the MZV's can get close to the weight
01072     and each deeper sum goes one higher, we make the tablesize a bit bigger.
01073     This may not be needed if we fiddle with the sum boundaries.
01074     */
01075     #ifdef WITHPTHREADS
01076         int i, Nw, id, totnum;
01077         size_t N;
01078         mpf_t *a;
01079         Nw = AC.DefaultPrecision;
01080         N = (size_t)Nw;
01081         totnum = AM.totalnumberofthreads;
01082         for ( id = 0; id < totnum; id++ ) {
01083             if ( AB[id]->T.mpf_tab1 ) {
01084                 a = (mpf_t *)AB[id]->T.mpf_tab1;
01085                 for ( i = 0; i <=Nw; i++ ) {
01086                     mpf_clear(a[i]);
01087                 }
01088                 M_free(AB[id]->T.mpf_tab1,"mpftab1");
01089             }
01090             AB[id]->T.mpf_tab1 = (void *)Malloca1((N+2)*sizeof(mpf_t),"mpftab1");
01091             a = (mpf_t *)AB[id]->T.mpf_tab1;
01092             for ( i = 0; i <=Nw; i++ ) {
01093                 /*
01094                 As explained in the comment above, we could make this variable precision
01095                 using mpf_init2.
01096                 */
01097                 mpf_init(a[i]);
01098             }
01099             if ( AB[id]->T.mpf_tab2 ) {
01100                 a = (mpf_t *)AB[id]->T.mpf_tab2;
01101                 for ( i = 0; i <=Nw; i++ ) {
01102                     mpf_clear(a[i]);
01103                 }
01104                 M_free(AB[id]->T.mpf_tab2,"mpftab2");
01105             }

```

```

01106         AB[id]->T.mpf_tab2 = (void *)Malloc1((N+2)*sizeof(mpf_t), "mpftab2");
01107         a = (mpf_t *)AB[id]->T.mpf_tab2;
01108         for ( i = 0; i <=Nw; i++ ) {
01109             mpf_init(a[i]);
01110         }
01111     }
01112 #else
01113     int i, Nw;
01114     size_t N;
01115     Nw = AC.DefaultPrecision;
01116     N = (size_t)Nw;
01117     if ( AT.mpf_tab1 ) {
01118         for ( i = 0; i <= Nw; i++ ) {
01119             mpf_clear(mpftab1[i]);
01120         }
01121         M_free(AT.mpf_tab1, "mpftab1");
01122     }
01123     AT.mpf_tab1 = (void *)Malloc1((N+2)*sizeof(mpf_t), "mpftab1");
01124     for ( i = 0; i <= Nw; i++ ) {
01125         /*
01126         As explained in the comment above, we could make this variable precision
01127         using mpf_init2.
01128         */
01129         mpf_init(mpftab1[i]);
01130     }
01131     if ( AT.mpf_tab2 ) {
01132         for ( i = 0; i <= Nw; i++ ) {
01133             mpf_clear(mpftab2[i]);
01134         }
01135         M_free(AT.mpf_tab2, "mpftab2");
01136     }
01137     AT.mpf_tab2 = (void *)Malloc1((N+2)*sizeof(mpf_t), "mpftab2");
01138     for ( i = 0; i <= Nw; i++ ) {
01139         mpf_init(mpftab2[i]);
01140     }
01141 #endif
01142     if ( AS.delta_1 ) {
01143         mpf_clear(mpfdelta1);
01144         M_free(AS.delta_1, "delta1");
01145     }
01146     AS.delta_1 = (void *)Malloc1(sizeof(mpf_t), "delta1");
01147     mpf_init(mpfdelta1);
01148     SimpleDelta(mpfdelta1, 1); /* this can speed up things. delta1 = ln(2) */
01149     /*
01150     Finally the character buffer for printing
01151     if ( AO.floatspace ) M_free(AO.floatspace, "floatspace");
01152     AO.floatspace = (UBYTE *)Malloc1(((10*AC.DefaultPrecision)/33+40)*sizeof(UBYTE), "floatspace");
01153     */
01154 }
01155
01156 /*
01157     #] SetupMZVTables :
01158     #[ SetupMPFTables :
01159     */
01160
01161 void SetupMPFTables(void)
01162 {
01163 #ifdef WITHPTHREADS
01164     int id, totnum;
01165     mpf_t *a;
01166     /*
01167     Now the aux variables
01168     */
01169 #ifdef WITHSORTBOTS
01170     totnum = MaX(2*AM.totalnumberofthreads-3, AM.totalnumberofthreads);
01171 #endif
01172     for ( id = 0; id < totnum; id++ ) {
01173         if ( AB[id]->T.aux_ ) {
01174             /*
01175             We work here with a[0] etc because the aux1 etc contain B which
01176             in the current routine would be AB[0] only
01177             */
01178             a = (mpf_t *)AB[id]->T.aux_;
01179             mpf_clears(a[0], a[1], a[2], a[3], a[4], a[5], a[6], a[7], (mpf_ptr)0);
01180             M_free(AB[id]->T.aux_, "AB[id]->T.aux_");
01181         }
01182         AB[id]->T.aux_ = (void *)Malloc1(sizeof(mpf_t)*8, "AB[id]->T.aux_");
01183         a = (mpf_t *)AB[id]->T.aux_;
01184         mpf_inits(a[0], a[1], a[2], a[3], a[4], a[5], a[6], a[7], (mpf_ptr)0);
01185         if ( AB[id]->T.indi1 ) M_free(AB[id]->T.indi1, "indi1");
01186         AB[id]->T.indi1 = (int *)Malloc1(sizeof(int)*AC.MaxWeight*2, "indi1");
01187         AB[id]->T.indi2 = AB[id]->T.indi1 + AC.MaxWeight;
01188     }
01189 #else
01190     /*
01191     Now the aux variables
01192     */

```

```

01193     if ( AT.aux_ ) {
01194         mpf_clears(aux1,aux2,aux3,aux4,aux5,auxjm,auxjjm,auxsum, (mpf_ptr)0);
01195         M_free(AT.aux_, "AT.aux");
01196     }
01197     AT.aux_ = (void *)Malloca(sizeof(mpf_t)*8, "AT.aux_");
01198     mpf_inits(aux1,aux2,aux3,aux4,aux5,auxjm,auxjjm,auxsum, (mpf_ptr)0);
01199     if ( AT.indi1 ) M_free(AT.indi1, "indi1");
01200     AT.indi1 = (int *)Malloca(sizeof(int)*AC.MaxWeight*2, "indi1");
01201     AT.indi2 = AT.indi1 + AC.MaxWeight;
01202 #endif
01203 /*
01204     Finally the character buffer for printing
01205     if ( AO.floatspace ) M_free(AO.floatspace, "floatspace");
01206     AO.floatspace = (UBYTE *)Malloca(((10*AC.DefaultPrecision)/33+40)*sizeof(UBYTE), "floatspace");
01207 */
01208 }
01209
01210 /*
01211     #] SetupMPFTables :
01212     #[ ClearMZVTables :
01213 */
01214
01215 void ClearMZVTables(void)
01216 {
01217 #ifdef WITHPTHREADS
01218     int i, id, totnum;
01219     mpf_t *a;
01220     totnum = AM.totalnumberofthreads;
01221     for ( id = 0; id < totnum; id++ ) {
01222         if ( AB[id]->T.mpf_tab1 ) {
01223             /*
01224                 We work here with a[0] etc because the aux1, mpftabl etc contain B
01225                 which in the current routine would be AB[0] only
01226             */
01227             a = (mpf_t *)AB[id]->T.mpf_tab1;
01228             for ( i = 0; i <= AC.DefaultPrecision; i++ ) {
01229                 mpf_clear(a[i]);
01230             }
01231             M_free(AB[id]->T.mpf_tab1, "mpftabl");
01232             AB[id]->T.mpf_tab1 = 0;
01233         }
01234         if ( AB[id]->T.mpf_tab2 ) {
01235             a = (mpf_t *)AB[id]->T.mpf_tab2;
01236             for ( i = 0; i <= AC.DefaultPrecision; i++ ) {
01237                 mpf_clear(a[i]);
01238             }
01239             M_free(AB[id]->T.mpf_tab2, "mpftab2");
01240             AB[id]->T.mpf_tab2 = 0;
01241         }
01242     }
01243 #endif
01244     totnum = MaX(2*AM.totalnumberofthreads-3, AM.totalnumberofthreads);
01245 #endif
01246     for ( id = 0; id < totnum; id++ ) {
01247         if ( AB[id]->T.aux_ ) {
01248             a = (mpf_t *)AB[id]->T.aux_;
01249             mpf_clears(a[0],a[1],a[2],a[3],a[4],a[5],a[6],a[7], (mpf_ptr)0);
01250             M_free(AB[id]->T.aux_, "AB[id]->T.aux_");
01251             AB[id]->T.aux_ = 0;
01252         }
01253         if ( AB[id]->T.indi1 ) { M_free(AB[id]->T.indi1, "indi1"); AB[id]->T.indi1 = 0; }
01254     }
01255 #else
01256     int i;
01257     if ( AT.mpf_tab1 ) {
01258         for ( i = 0; i <= AC.DefaultPrecision; i++ ) {
01259             mpf_clear(mpftabl[i]);
01260         }
01261         M_free(AT.mpf_tab1, "mpftabl");
01262         AT.mpf_tab1 = 0;
01263     }
01264     if ( AT.mpf_tab2 ) {
01265         for ( i = 0; i <= AC.DefaultPrecision; i++ ) {
01266             mpf_clear(mpftab2[i]);
01267         }
01268         M_free(AT.mpf_tab2, "mpftab2");
01269         AT.mpf_tab2 = 0;
01270     }
01271     if ( AT.aux_ ) {
01272         mpf_clears(aux1,aux2,aux3,aux4,aux5,auxjm,auxjjm,auxsum, (mpf_ptr)0);
01273         M_free(AT.aux_, "AT.aux_");
01274         AT.aux_ = 0;
01275     }
01276     if ( AT.indi1 ) { M_free(AT.indi1, "indi1"); AT.indi1 = 0; }
01277 #endif
01278     if ( AO.floatspace ) { M_free(AO.floatspace, "floatspace"); AO.floatspace = 0; }
01279     AO.floatsize = 0; }

```

```

01280     if ( AS.delta_1 ) {
01281         mpf_clear(mpfdelta1);
01282         M_free(AS.delta_1,"delta1");
01283         AS.delta_1 = 0;
01284     }
01285 }
01286
01287 /*
01288     #] ClearMZVTables :
01289     #[ CoToFloat :
01290 */
01291
01292 int CoToFloat (UBYTE *s)
01293 {
01294     GETIDENTITY
01295     if ( AT.aux_ == 0 ) {
01296         MesPrint("&Illegal attempt to convert to float_ without activating floating point numbers.");
01297         MesPrint("&Forgotten %#startfloat instruction?");
01298         return(1);
01299     }
01300     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
01301     if ( *s ) {
01302         MesPrint("&Illegal argument(s) in ToFloat statement: '%s'",s);
01303         return(1);
01304     }
01305     Add2Com(TYPETOFLOAT);
01306     return(0);
01307 }
01308
01309 /*
01310     #] CoToFloat :
01311     #[ CoToRat :
01312 */
01313
01314 int CoToRat (UBYTE *s)
01315 {
01316     GETIDENTITY
01317     if ( AT.aux_ == 0 ) {
01318         MesPrint("&Illegal attempt to convert from float_ without activating floating point
01319 numbers.");
01320         MesPrint("&Forgotten %#startfloat instruction?");
01321         return(1);
01322     }
01323     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
01324     if ( *s ) {
01325         MesPrint("&Illegal argument(s) in ToRat statement: '%s'",s);
01326         return(1);
01327     }
01328     Add2Com(TYPETORAT);
01329     return(0);
01330 }
01331
01332 /*
01333     #] CoToRat :
01334     #[ ToFloat :
01335
01336     Converts the coefficient to floating point if it is still a rat.
01337 */
01338 int ToFloat(PHEAD WORD *term, WORD level)
01339 {
01340     WORD *t, *tstop, nsize, nsign = 3;
01341     t = term+*term;
01342     nsize = ABS(t[-1]);
01343     tstop = t - nsize;
01344     if ( t[-1] < 0 ) { nsign = -nsign; }
01345     if ( nsize == 3 && t[-2] == 1 && t[-3] == 1 ) { /* Could be float */
01346         t = term + 1;
01347         while ( t < tstop ) {
01348             if ( ( *t == FLOATFUN ) && ( t+t[1] == tstop ) ) {
01349                 return(Generator(BHEAD term,level));
01350             }
01351             t += t[1];
01352         }
01353     }
01354     RatToFloat(aux4, (UWORD *)tstop,nsize);
01355     PackFloat(tstop,aux4);
01356     tstop += tstop[1];
01357     *tstop++ = 1; *tstop++ = 1; *tstop++ = nsign;
01358     *term = tstop - term;
01359     AT.WorkPointer = tstop;
01360     return(Generator(BHEAD term,level));
01361 }
01362
01363 /*
01364     #] ToFloat :
01365     #[ ToRat :

```



```

01366
01367     Tries to convert the coefficient to rational if it is still a float.
01368 */
01369
01370 int ToRat(PHEAD WORD *term, WORD level)
01371 {
01372     WORD *tstop, *t, nsize, nsign, ncoef;
01373     /*
01374     1: find the float which should be at the end.
01375     */
01376     tstop = term + *term; nsize = ABS(tstop[-1]);
01377     nsign = tstop[-1] < 0 ? -1: 1; tstop -= nsize;
01378     t = term+1;
01379     while ( t < tstop ) {
01380         if ( *t == FLOATFUN && t + t[1] == tstop && TestFloat(t) &&
01381             nsize == 3 && tstop[0] == 1 && tstop[1] == 1 ) break;
01382         t += t[1];
01383     }
01384     if ( t < tstop ) {
01385     /*
01386         Now t points at the float_ function and everything is correct.
01387         The result can go straight over the float_ function.
01388     */
01389         UnpackFloat(aux4,t);
01390         if ( FloatToRat((UWORD *)t,&ncoef,aux4) == 0 ) {
01391             t += ABS(ncoef);
01392             t[-1] = ncoef*nsign;
01393             *term = t - term;
01394         }
01395     }
01396     return(Generator(BHEAD term,level));
01397 }
01398
01399 /*
01400     #] ToRat :
01401     #] Float Routines :
01402     #[ Sorting :
01403
01404     We need a method to see whether two terms need addition that could
01405     involve floating point numbers.
01406     1: if PolyWise is active, we do not need anything, because
01407         Poly(Rat)Fun and floating point are mutually exclusive.
01408     2: This means that there should be only interference in AddCoef.
01409         and the AddRat parts in MergePatches, MasterMerge, SortBotMerge
01410         and PF_GetLoser.
01411     The compare routine(s) should recognise the float_ and give off
01412     a code (SortFloatMode) inside the AT struct:
01413     0: no float_
01414     1: float_ in term1 only
01415     2: float_ in term2 only
01416     3: float_ in both terms
01417     Be careful: we have several compare routines, because various codes
01418     use their own routines and we just set a variable with its address.
01419     Currently we have Compare1, CompareSymbols and CompareHSymbols.
01420     Only Compare1 should be active for this. The others should make sure
01421     that the proper variable is always zero.
01422     Remember: the sign of the coefficient is in the last word of the term.
01423
01424     We need two routines:
01425     1: AddWithFloat for SplitMerge which sorts by pointer.
01426     2: MergeWithFloat for MergePatches etc which keeps terms as much
01427         as possible in their location.
01428     The routines start out the same, because AT.SortFloatMode, set in
01429     Compare1, tells more or less what should be done.
01430     The difference is in where we leave the result.
01431
01432     In the future we may want to add a feature that makes the result
01433     zero if the sum comes within a certain distance of the numerical
01434     accuracy, like for instance 5% of the number of bits.
01435
01436     #[ AddWithFloat :
01437 */
01438
01439 int AddWithFloat(PHEAD WORD **ps1, WORD **ps2)
01440 {
01441     GETBIDENTITY
01442     SORTING *S = AT.SS;
01443     WORD *coef1, *coef2, size1, size2, *fun1, *fun2, *fun3;
01444     WORD *s1,*s2,sign3,j,jj, *t1, *t2, i;
01445     s1 = *ps1;
01446     s2 = *ps2;
01447     coef1 = s1+s1; size1 = coef1[-1]; coef1 -= ABS(coef1[-1]);
01448     coef2 = s2+s2; size2 = coef2[-1]; coef2 -= ABS(coef2[-1]);
01449     if ( AT.SortFloatMode == 3 ) {
01450         fun1 = s1+1; while ( fun1 < coef1 && fun1[0] != FLOATFUN ) fun1 += fun1[1];
01451         fun2 = s2+1; while ( fun2 < coef2 && fun2[0] != FLOATFUN ) fun2 += fun2[1];
01452         UnpackFloat(aux1,fun1);

```

```

01453         if ( size1 < 0 ) mpf_neg(aux1,aux1);
01454         UnpackFloat(aux2,fun2);
01455         if ( size2 < 0 ) mpf_neg(aux2,aux2);
01456     }
01457     else if ( AT.SortFloatMode == 1 ) {
01458         fun1 = s1+1; while ( fun1 < coef1 && fun1[0] != FLOATFUN ) fun1 += fun1[1];
01459         UnpackFloat(aux1,fun1);
01460         if ( size1 < 0 ) mpf_neg(aux1,aux1);
01461         fun2 = coef2;
01462         RatToFloat(aux2,(UWORD *)coef2,size2);
01463     }
01464     else if ( AT.SortFloatMode == 2 ) {
01465         fun2 = s2+1; while ( fun2 < coef2 && fun2[0] != FLOATFUN ) fun2 += fun2[1];
01466         fun1 = coef1;
01467         RatToFloat(aux1,(UWORD *)coef1,size1);
01468         UnpackFloat(aux2,fun2);
01469         if ( size2 < 0 ) mpf_neg(aux2,aux2);
01470     }
01471     else {
01472         MLOCK(ErrorMessageLock);
01473         MesPrint("Illegal value %d for AT.SortFloatMode in AddWithFloat.",AT.SortFloatMode);
01474         MUNLOCK(ErrorMessageLock);
01475         Terminate(-1);
01476         return(0);
01477     }
01478     mpf_add(aux3,aux1,aux2);
01479     sign3 = mpf_sgn(aux3);
01480     if ( sign3 == 0 ) { /* May be rare! */
01481         *ps1 = 0; *ps2 = 0; AT.SortFloatMode = 0; return(0);
01482     }
01483     if ( sign3 < 0 ) mpf_neg(aux3,aux3);
01484     fun3 = TermMalloc("AddWithFloat");
01485     PackFloat(fun3,aux3);
01486     /*
01487     Now we have to calculate where the sumterm fits.
01488     If we are sloppy, we can be faster, but run the risk to need the
01489     extension space, even when it is not needed. At slightly lower speed
01490     (ie first creating the result in scribble space) we are better off.
01491     This is why we use TermMalloc.
01492
01493     The new term will have a rational coefficient of 1,1,+-.3.
01494     The size will be (fun1 or fun2 - term) + fun3 +3;
01495     */
01496     if ( AT.SortFloatMode == 3 ) {
01497         if ( fun1[1] == fun3[1] ) {
01498 Over1:
01499             i = fun3[1]; t1 = fun1; t2 = fun3; NCOPY(t1,t2,i);
01500             *t1++ = 1; *t1++ = 1; *t1++ = sign3 < 0 ? -3: 3;
01501             *s1 = t1-s1; goto Finished;
01502         }
01503         else if ( fun2[1] == fun3[1] ) {
01504 Over2:
01505             i = fun3[1]; t1 = fun2; t2 = fun3; NCOPY(t1,t2,i);
01506             *t1++ = 1; *t1++ = 1; *t1++ = sign3 < 0 ? -3: 3;
01507             *s2 = t1-s2; *ps1 = s2; goto Finished;
01508         }
01509         else if ( fun1[1] >= fun3[1] ) goto Over1;
01510         else if ( fun2[1] >= fun3[1] ) goto Over2;
01511     }
01512     else if ( AT.SortFloatMode == 1 ) {
01513         if ( fun1[1] >= fun3[1] ) goto Over1;
01514         else if ( fun3[1]+3 <= ABS(size2) ) goto Over2;
01515     }
01516     else if ( AT.SortFloatMode == 2 ) {
01517         if ( fun2[1] >= fun3[1] ) goto Over2;
01518         else if ( fun3[1]+3 <= ABS(size1) ) goto Over1;
01519     }
01520     /*
01521     Does not fit. Go to extension space.
01522     */
01523     jj = fun1-s1;
01524     j = jj+fun3[1]+3; /* space needed */
01525     if ( (S->sFill + j) >= S->sTop2 ) {
01526         GarbHand();
01527         s1 = *ps1; /* new position of the term after the garbage collection */
01528         fun1 = s1+jj;
01529     }
01530     t1 = S->sFill;
01531     for ( i = 0; i < jj; i++ ) *t1++ = s1[i];
01532     i = fun3[1]; s1 = fun3; NCOPY(t1,s1,i);
01533     *t1++ = 1; *t1++ = 1; *t1++ = sign3 < 0 ? -3: 3;
01534     *ps1 = S->sFill;
01535     **ps1 = t1-*ps1;
01536     S->sFill = t1;
01537 Finished:
01538     *ps2 = 0;
01539     TermFree(fun3,"AddWithFloat");

```

```

01540     AT.SortFloatMode = 0;
01541     if ( **ps1 > AM.MaxTer/((LONG)(sizeof(WORD))) ) {
01542         MLOCK(ErrorMessageLock);
01543         MesPrint("Term too complex after addition in sort. MaxTermSize = %10l",
01544             AM.MaxTer/sizeof(WORD));
01545         MUNLOCK(ErrorMessageLock);
01546         Terminate(-1);
01547     }
01548     return(1);
01549 }
01550
01551 /*
01552     #] AddWithFloat :
01553     #[ MergeWithFloat :
01554
01555     Note that we always park the result on top of term1.
01556     This makes life easier, because the original AddRat in MergePatches
01557     does this as well.
01558 */
01559
01560 int MergeWithFloat(PHEAD WORD **interm1, WORD **interm2)
01561 {
01562     GETBIDENTITY
01563     WORD *coef1, *coef2, size1, size2, *fun1, *fun2, *fun3, *tt;
01564     WORD sign3, j, jj, *t1, *t2, i, *term1 = *interm1, *term2 = *interm2;
01565     int retval = 0;
01566     coef1 = term1+*term1; size1 = coef1[-1]; coef1 -= ABS(size1);
01567     coef2 = term2+*term2; size2 = coef2[-1]; coef2 -= ABS(size2);
01568     if ( AT.SortFloatMode == 3 ) {
01569         fun1 = term1+1; while ( fun1 < coef1 && fun1[0] != FLOATFUN ) fun1 += fun1[1];
01570         fun2 = term2+1; while ( fun2 < coef2 && fun2[0] != FLOATFUN ) fun2 += fun2[1];
01571         UnpackFloat(aux1, fun1);
01572         if ( size1 < 0 ) mpf_neg(aux1, aux1);
01573         UnpackFloat(aux2, fun2);
01574         if ( size2 < 0 ) mpf_neg(aux2, aux2);
01575     }
01576     else if ( AT.SortFloatMode == 1 ) {
01577         fun1 = term1+1; while ( fun1 < coef1 && fun1[0] != FLOATFUN ) fun1 += fun1[1];
01578         UnpackFloat(aux1, fun1);
01579         if ( size1 < 0 ) mpf_neg(aux1, aux1);
01580         fun2 = coef2;
01581         RatToFloat(aux2, (UWORD *)coef2, size2);
01582     }
01583     else if ( AT.SortFloatMode == 2 ) {
01584         fun2 = term2+1; while ( fun2 < coef2 && fun2[0] != FLOATFUN ) fun2 += fun2[1];
01585         fun1 = coef1;
01586         RatToFloat(aux1, (UWORD *)coef1, size1);
01587         UnpackFloat(aux2, fun2);
01588         if ( size2 < 0 ) mpf_neg(aux2, aux2);
01589     }
01590     else {
01591         MLOCK(ErrorMessageLock);
01592         MesPrint("Illegal value %d for AT.SortFloatMode in MergeWithFloat.", AT.SortFloatMode);
01593         MUNLOCK(ErrorMessageLock);
01594         Terminate(-1);
01595         return(0);
01596     }
01597     mpf_add(aux3, aux1, aux2);
01598     sign3 = mpf_sgn(aux3);
01599     if ( sign3 == 0 ) { /* May be very rare! */
01600         AT.SortFloatMode = 0; return(0);
01601     }
01602     /*
01603     Now check whether we can park the result on top of one of the input terms.
01604     */
01605     if ( sign3 < 0 ) mpf_neg(aux3, aux3);
01606     fun3 = TermMalloc("MergeWithFloat");
01607     PackFloat(fun3, aux3);
01608     if ( AT.SortFloatMode == 3 ) {
01609         if ( fun1[1] + ABS(size1) == fun3[1] + 3 ) {
01610             OnTopOf1: t1 = fun3; t2 = fun1;
01611                 for ( i = 0; i < fun3[1]; i++ ) *t2++ = *t1++;
01612                 *t2++ = 1; *t2++ = 1; *t2++ = sign3 < 0 ? -3: 3;
01613                 retval = 1;
01614             }
01615         else if ( fun1[1] + ABS(size1) > fun3[1] + 3 ) {
01616             Shift1: t2 = term1 + *term1; tt = t2;
01617                 *--t2 = sign3 < 0 ? -3: 3; *--t2 = 1; *--t2 = 1;
01618                 t1 = fun3 + fun3[1]; for ( i = 0; i < fun3[1]; i++ ) *--t2 = *--t1;
01619                 t1 = fun1;
01620                 while ( t1 > term1 ) *--t2 = *--t1;
01621                 *t2 = tt-t2; term1 = t2;
01622                 retval = 1;
01623             }
01624         else { /* Here we have to move term1 to the left to make room. */
01625             Over1: jj = fun3[1]-fun1[1]+3-ABS(size1); /* This is positive */
01626             t2 = term1-jj; t1 = term1;

```

```

01627         while ( t1 < fun1 ) *t2++ = *t1++;
01628         term1 -= jj;
01629         *term1 += jj;
01630         for ( i = 0; i < fun3[1]; i++ ) *t2++ = fun3[i];
01631         *t2++ = 1; *t2++ = 1; *t2++ = sign3 < 0 ? -3: 3;
01632         retval = 1;
01633     }
01634 }
01635 else if ( AT.SortFloatMode == 1 ) {
01636     if ( fun1[1] + ABS(size1) == fun3[1] + 3 ) goto OnTopOf1;
01637     else if ( fun1[1] + ABS(size1) > fun3[1] + 3 ) goto Shift1;
01638     else goto Over1;
01639 }
01640 else { /* Can only be 2, based on previous tests */
01641     if ( fun3[1] + 3 == ABS(size1) ) {
01642         t2 = coef1; t1 = fun3;
01643         for ( i = 0; i < fun3[1]; i++ ) *t2++ = *t1++;
01644         *t2++ = 1; *t2++ = 1; *t2++ = sign3 < 0 ? -3: 3;
01645         retval = 1;
01646     }
01647     else if ( fun3[1] + 3 < ABS(size1) ) {
01648         j = ABS(size1) - fun3[1] - 3;
01649         t2 = term1 + *term1; tt = t2;
01650         *--t2 = sign3 < 0 ? -3: 3; *--t2 = 1; *--t2 = 1;
01651         t2 -= fun3[1]; t1 = t2-j;
01652         while ( t2 > term1 ) *--t2 = *--t1;
01653         *t2 = tt-t2; term1 = t2;
01654         retval = 1;
01655     }
01656     else goto Over1;
01657 }
01658 *interm1 = term1;
01659 TermFree(fun3,"MergeWithFloat");
01660 AT.SortFloatMode = 0;
01661 return(retval);
01662 }
01663
01664 /*
01665     #] MergeWithFloat :
01666     #] Sorting :
01667     #[ MZV :
01668
01669     The functions here allow the arbitrary precision calculation
01670     of MZV's and Euler sums.
01671     They are called when the functions mzv_ and/or euler_ are used
01672     and the evaluate statement is applied.
01673     The output is in a float_ function.
01674     The expand statement tries to express the functions in terms of a basis.
01675     The bases are the 'standard basis for the euler sums and the
01676     pushdown bases from the datamine, unless otherwise specified.
01677
01678     #[ SimpleDelta :
01679 */
01680
01681 void SimpleDelta(mpf_t sum, int m)
01682 {
01683     long s = 1;
01684     long prec = AC.DefaultPrecision;
01685     unsigned long xprec = (unsigned long)prec, x;
01686     int j, jmax, n;
01687     mpf_t jm, jjm;
01688     mpf_init(jm); mpf_init(jjm);
01689     if ( m < 0 ) { s = -1; m = -m; }
01690
01691     /*
01692     We will loop until 1/2^j/j^m is smaller than the default precision.
01693     Just running to prec is however overkill, specially when m is large.
01694     We try to estimate a better value.
01695     jmax = prec - ((log2(prec)-1)*m).
01696     Hence we need the leading bit of prec.
01697     We are still overshooting a bit.
01698     */
01699     n = 0; x = xprec;
01700     while ( x ) { x >>= 1; n++; }
01701     /*
01702     We have now n = floor(log2(x))+1.
01703     */
01704     n--;
01705     jmax = (int)((int)xprec - (n-1)*m);
01706     mpf_set_ui(sum,0);
01707     for ( j = 1; j <= jmax; j++ ) {
01708 #ifdef WITHCUTOFF
01709         xprec--;
01710         mpf_set_prec_raw(jm,xprec);
01711         mpf_set_prec_raw(jjm,xprec);
01712 #endif
01713         mpf_set_ui(jm,1L);

```

```

01714     mpf_div_ui(jjm,jm,(unsigned long)j);
01715     mpf_pow_ui(jm,jjm,(unsigned long)m);
01716     mpf_div_2exp(jjm,jm,(unsigned long)j);
01717     if ( s == -1 && j%2 == 1 ) mpf_sub(sum,sum,jjm);
01718     else mpf_add(sum,sum,jjm);
01719 }
01720 mpf_clear(jjm); mpf_clear(jm);
01721 }
01722
01723 /*
01724     #] SimpleDelta :
01725     #[ SimpleDeltaC :
01726 */
01727
01728 void SimpleDeltaC(mpf_t sum, int m)
01729 {
01730     long s = 1;
01731     long prec = AC.DefaultPrecision;
01732     unsigned long xprec = (unsigned long)prec, x;
01733     int j, jmax, n;
01734     mpf_t jm,jjm;
01735     mpf_init(jm); mpf_init(jjm);
01736     if ( m < 0 ) { s = -1; m = -m; }
01737 /*
01738     We will loop until 1/2^j/j^m is smaller than the default precision.
01739     Just running to prec is however overkill, specially when m is large.
01740     We try to estimate a better value.
01741     jmax = prec - ((log2(prec)-1)*m).
01742     Hence we need the leading bit of prec.
01743     We are still overshooting a bit.
01744 */
01745     n = 0; x = xprec;
01746     while ( x ) { x >= 1; n++; }
01747 /*
01748     We have now n = floor(log2(x))+1.
01749 */
01750     n--;
01751     jmax = (int)((int)xprec - (n-1)*m);
01752     if ( s < 0 ) jmax /= 2;
01753     mpf_set_si(sum,0L);
01754     for ( j = 1; j <= jmax; j++ ) {
01755 #ifdef WITHCUTOFF
01756         xprec--;
01757         mpf_set_prec_raw(jm,xprec);
01758         mpf_set_prec_raw(jjm,xprec);
01759 #endif
01760         mpf_set_ui(jm,1L);
01761         mpf_div_ui(jjm,jm,(unsigned long)j);
01762         mpf_pow_ui(jm,jjm,(unsigned long)m);
01763         if ( s == -1 ) mpf_div_2exp(jjm,jm,2*(unsigned long)j);
01764         else mpf_div_2exp(jjm,jm,(unsigned long)j);
01765         mpf_add(sum,sum,jjm);
01766     }
01767     mpf_clear(jjm); mpf_clear(jm);
01768 }
01769
01770 /*
01771     #] SimpleDeltaC :
01772     #[ SingleTable :
01773 */
01774
01775 void SingleTable(mpf_t *tabl, int N, int m, int pow)
01776 {
01777 /*
01778     Creates a table T(1,...,N) with
01779         T(i) = sum_(j,i,N,[sign_(j)]/2^j/j^m)
01780     To make this table we have two options:
01781     1: run the sum backwards with the potential problem that the
01782        precision is difficult to manage.
01783     2: run the sum forwards. Take sum_(j,1,i-1,...) and later subtract.
01784     When doing Euler sums we may need also 1/4^j
01785     pow: 1 -> 1/2^j
01786         2 -> 1/4^j
01787 */
01788     GETIDENTITY
01789     long s = 1;
01790     int j;
01791 #ifdef WITHCUTOFF
01792     long prec = mpf_get_default_prec();
01793 #endif
01794     mpf_t jm,jjm;
01795     mpf_init(jm); mpf_init(jjm);
01796     if ( pow < 1 || pow > 2 ) {
01797         printf("Wrong parameter pow in SingleTable: %d\n",pow);
01798         exit(-1);
01799     }
01800     if ( m < 0 ) { m = -m; s = -1; }

```

```

01801     mpf_set_si(auxsum, 0L);
01802     for ( j = N; j > 0; j-- ) {
01803 #ifdef WITHCUTOFF
01804         mpf_set_prec_raw(jm, prec-j);
01805         mpf_set_prec_raw(jjm, prec-j);
01806 #endif
01807         mpf_set_ui(jm, 1L);
01808         mpf_div_ui(jjm, jm, (unsigned long)j);
01809         mpf_pow_ui(jm, jjm, (unsigned long)m);
01810         mpf_div_2exp(jjm, jm, pow*(unsigned long)j);
01811         if ( pow == 2 ) mpf_add(auxsum, auxsum, jjm);
01812         else if ( s == -1 && j%2 == 1 ) mpf_sub(auxsum, auxsum, jjm);
01813         else mpf_add(auxsum, auxsum, jjm);
01814 /*
01815         And now copy auxsum to tablelement j
01816 */
01817         mpf_set(tabl[j], auxsum);
01818     }
01819     mpf_clear(jjm); mpf_clear(jm);
01820 }
01821
01822 /*
01823     #] SingleTable :
01824     #[ DoubleTable :
01825 */
01826
01827 void DoubleTable(mpf_t *tabout, mpf_t *tabin, int N, int m, int pow)
01828 {
01829     GETIDENTITY
01830     long s = 1;
01831 #ifdef WITHCUTOFF
01832     long prec = mpf_get_default_prec();
01833 #endif
01834     int j;
01835     mpf_t jm, jjm;
01836     mpf_init(jm); mpf_init(jjm);
01837     if ( pow < -1 || pow > 2 ) {
01838         printf("Wrong parameter pow in SingleTable: %d\n", pow);
01839         exit(-1);
01840     }
01841     if ( m < 0 ) { m = -m; s = -1; }
01842     mpf_set_ui(auxsum, 0L);
01843     for ( j = N; j > 0; j-- ) {
01844 #ifdef WITHCUTOFF
01845         mpf_set_prec_raw(jm, prec-j);
01846         mpf_set_prec_raw(jjm, prec-j);
01847 #endif
01848         mpf_set_ui(jm, 1L);
01849         mpf_div_ui(jjm, jm, (unsigned long)j);
01850         mpf_pow_ui(jm, jjm, (unsigned long)m);
01851         if ( pow == -1 ) {
01852             mpf_mul_2exp(jjm, jm, (unsigned long)j);
01853             mpf_mul(jm, jjm, tabin[j+1]);
01854         }
01855         else if ( pow > 0 ) {
01856             mpf_div_2exp(jjm, jm, pow*(unsigned long)j);
01857             mpf_mul(jm, jjm, tabin[j+1]);
01858         }
01859         else {
01860             mpf_mul(jm, jm, tabin[j+1]);
01861         }
01862         if ( pow == 2 ) mpf_add(auxsum, auxsum, jm);
01863         else if ( s == -1 && j%2 == 1 ) mpf_sub(auxsum, auxsum, jm);
01864         else mpf_add(auxsum, auxsum, jm);
01865 /*
01866         And now copy auxsum to tablelement j
01867 */
01868         mpf_set(tabout[j], auxsum);
01869     }
01870     mpf_clear(jjm); mpf_clear(jm);
01871 }
01872
01873 /*
01874     #] DoubleTable :
01875     #[ EndTable :
01876 */
01877
01878 void EndTable(mpf_t sum, mpf_t *tabin, int N, int m, int pow)
01879 {
01880     long s = 1;
01881 #ifdef WITHCUTOFF
01882     long prec = mpf_get_default_prec();
01883 #endif
01884     int j;
01885     mpf_t jm, jjm;
01886     mpf_init(jm); mpf_init(jjm);
01887     if ( pow < -1 || pow > 2 ) {

```

```

01888     printf("Wrong parameter pow in SingleTable: %d\n",pow);
01889     exit(-1);
01890 }
01891 if ( m < 0 ) { m = -m; s = -1; }
01892 mpf_set_si(sum,0L);
01893 for ( j = N; j > 0; j-- ) {
01894 #ifdef WITHCUTOFF
01895     mpf_set_prec_raw(jm,prec-j);
01896     mpf_set_prec_raw(jjm,prec-j);
01897 #endif
01898     mpf_set_ui(jm,1L);
01899     mpf_div_ui(jjm,jm,(unsigned long)j);
01900     mpf_pow_ui(jm,jjm,(unsigned long)m);
01901     if ( pow == -1 ) {
01902         mpf_mul_2exp(jjm,jm,(unsigned long)j);
01903         mpf_mul(jm,jjm,tabin[j+1]);
01904     }
01905     else if ( pow > 0 ) {
01906         mpf_div_2exp(jjm,jm,pow*(unsigned long)j);
01907         mpf_mul(jm,jjm,tabin[j+1]);
01908     }
01909     else {
01910         mpf_mul(jm,jm,tabin[j+1]);
01911     }
01912     if ( s == -1 && j%2 == 1 ) mpf_sub(sum,sum,jm);
01913     else mpf_add(sum,sum,jm);
01914 }
01915 mpf_clear(jjm); mpf_clear(jm);
01916 }
01917
01918 /*
01919     #] EndTable :
01920     #[ deltaMZV :
01921 */
01922
01923 void deltaMZV(mpf_t result, int *indexes, int depth)
01924 {
01925     GETIDENTITY
01926     /*
01927     Because all sums go roughly like 1/2^j we need about depth steps.
01928     */
01929     /* MesPrint("deltaMZV(%a)",depth,indexes); */
01930     if ( depth == 1 ) {
01931         if ( indexes[0] == 1 ) {
01932             mpf_set(result,mpfdelta1);
01933             return;
01934         }
01935         SimpleDelta(result,indexes[0]);
01936     }
01937     else if ( depth == 2 ) {
01938         if ( indexes[0] == 1 && indexes[1] == 1 ) {
01939             mpf_pow_ui(result,mpfdelta1,2);
01940             mpf_div_2exp(result,result,1);
01941         }
01942         else {
01943             SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+1,indexes[0],1);
01944             EndTable(result,mpftab1,AC.DefaultPrecision-AC.MaxWeight,indexes[1],0);
01945         }
01946     }
01947     else if ( depth > 2 ) {
01948         mpf_t *mpftab3;
01949         int d;
01950         /*
01951         Check first whether this is a power of delta1 = ln(2)
01952         */
01953         for ( d = 0; d < depth; d++ ) {
01954             if ( indexes[d] != 1 ) break;
01955         }
01956         if ( d == depth ) { /* divide by fac_(depth) */
01957             mpf_pow_ui(result,mpfdelta1,depth);
01958             for ( d = 2; d <= depth; d++ ) {
01959                 mpf_div_ui(result,result,(unsigned long)d);
01960             }
01961         }
01962         else {
01963             d = depth-1;
01964             SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+d,*indexes,1);
01965             d--; indexes++;
01966             for(;;) {
01967                 DoubleTable(mpftab2,mpftab1,AC.DefaultPrecision-AC.MaxWeight+d,*indexes,0);
01968                 d--; indexes++;
01969                 if ( d == 0 ) {
01970                     EndTable(result,mpftab2,AC.DefaultPrecision-AC.MaxWeight,*indexes,0);
01971                     break;
01972                 }
01973                 mpftab3 = (mpf_t *)AT.mpf_tab1; AT.mpf_tab1 = AT.mpf_tab2;
01974                 AT.mpf_tab2 = (void *)mpftab3;

```

```

01975     }
01976     }
01977 }
01978 else if ( depth == 0 ) {
01979     mpf_set_ui(result,1L);
01980 }
01981 }
01982
01983 /*
01984     #] deltaMZV :
01985     #[ deltaEuler :
01986
01987     Regular Euler delta with - signs, but everywhere 1/2^j
01988 */
01989
01990 void deltaEuler(mpf_t result, int *indexes, int depth)
01991 {
01992     GETIDENTITY
01993     int m;
01994 #ifdef DEBUG
01995     int i;
01996     printf(" deltaEuler(");
01997     for ( i = 0; i < depth; i++ ) {
01998         if ( i != 0 ) printf(",");
01999         printf("%d",indexes[i]);
02000     }
02001     printf(") = ");
02002 #endif
02003     if ( depth == 1 ) {
02004         SimpleDelta(result,indexes[0]);
02005     }
02006     else if ( depth == 2 ) {
02007         SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+1,indexes[0],1);
02008         m = indexes[1]; if ( indexes[0] < 0 ) m = -m;
02009         EndTable(result,mpftab1,AC.DefaultPrecision-AC.MaxWeight,m,0);
02010     }
02011     else if ( depth > 2 ) {
02012         int d;
02013         mpf_t *mpftab3;
02014         d = depth-1;
02015         SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+d,*indexes,1);
02016         d--; indexes++;
02017         m = *indexes; if ( indexes[-1] < 0 ) m = -m;
02018         for(;;) {
02019             DoubleTable(mpftab2,mpftab1,AC.DefaultPrecision-AC.MaxWeight+d,m,0);
02020             d--; indexes++;
02021             m = *indexes; if ( indexes[-1] < 0 ) m = -m;
02022             if ( d == 0 ) {
02023                 EndTable(result,mpftab2,AC.DefaultPrecision-AC.MaxWeight,m,0);
02024                 break;
02025             }
02026             mpftab3 = (mpf_t *)AT.mpf_tab1; AT.mpf_tab1 = AT.mpf_tab2;
02027             AT.mpf_tab2 = (void *)mpftab3;
02028         }
02029     }
02030     else if ( depth == 0 ) {
02031         mpf_set_ui(result,1L);
02032     }
02033 #ifdef DEBUG
02034     gmp_printf("%.256e\n",40,result);
02035 #endif
02036 }
02037
02038 /*
02039     #] deltaEuler :
02040     #[ deltaEulerC :
02041
02042     Conjugate Euler delta without - signs, but possibly 1/4^j
02043     When there is a - in the string we have 1/4.
02044 */
02045
02046 void deltaEulerC(mpf_t result, int *indexes, int depth)
02047 {
02048     GETIDENTITY
02049     int m;
02050 #ifdef DEBUG
02051     int i;
02052     printf(" deltaEulerC(");
02053     for ( i = 0; i < depth; i++ ) {
02054         if ( i != 0 ) printf(",");
02055         printf("%d",indexes[i]);
02056     }
02057     printf(") = ");
02058 #endif
02059     mpf_set_ui(result,0);
02060     if ( depth == 1 ) {
02061         if ( indexes[0] == 0 ) {

```



```

02062         printf("Illegal index in depth=1 deltaEulerC: %d\n", indexes[0]);
02063     }
02064     if ( indexes[0] < 0 ) SimpleDeltaC(result, indexes[0]);
02065     else SimpleDelta(result, indexes[0]);
02066 }
02067 else if ( depth == 2 ) {
02068     int par;
02069     m = indexes[0];
02070     if ( m < 0 ) SingleTable(mpftab1, AC.DefaultPrecision-AC.MaxWeight+depth, -m, 2);
02071     else SingleTable(mpftab1, AC.DefaultPrecision-AC.MaxWeight+depth, m, 1);
02072     m = indexes[1];
02073     if ( m < 0 ) { m = -m; par = indexes[0] < 0 ? 0: 1; }
02074     else { par = indexes[0] < 0 ? -1: 0; }
02075     EndTable(result, mpftab1, AC.DefaultPrecision-AC.MaxWeight, m, par);
02076 }
02077 else if ( depth > 2 ) {
02078     int d, par;
02079     mpf_t *mpftab3;
02080     d = depth-1;
02081     m = indexes[0];
02082     ZeroTable(mpftab1, AC.DefaultPrecision+1);
02083     if ( m < 0 ) SingleTable(mpftab1, AC.DefaultPrecision-AC.MaxWeight+d+1, -m, 2);
02084     else SingleTable(mpftab1, AC.DefaultPrecision-AC.MaxWeight+d+1, m, 1);
02085     d--; indexes++; m = indexes[0];
02086     if ( m < 0 ) { m = -m; par = indexes[-1] < 0 ? 0: 1; }
02087     else { par = indexes[-1] < 0 ? -1: 0; }
02088     for(;;) {
02089         ZeroTable(mpftab2, AC.DefaultPrecision-AC.MaxWeight+d);
02090         DoubleTable(mpftab2, mpftab1, AC.DefaultPrecision-AC.MaxWeight+d, m, par);
02091         d--; indexes++; m = indexes[0];
02092         if ( m < 0 ) { m = -m; par = indexes[-1] < 0 ? 0: 1; }
02093         else { par = indexes[-1] < 0 ? -1: 0; }
02094         if ( d == 0 ) {
02095             mpf_set_ui(result, 0);
02096             EndTable(result, mpftab2, AC.DefaultPrecision-AC.MaxWeight, m, par);
02097             break;
02098         }
02099         mpftab3 = (mpf_t *)AT.mpf_tab1; AT.mpf_tab1 = AT.mpf_tab2;
02100         AT.mpf_tab2 = (void *)mpftab3;
02101     }
02102 }
02103 else if ( depth == 0 ) {
02104     mpf_set_ui(result, 1L);
02105 }
02106 #ifdef DEBUG
02107 gmp_printf("%.Fe\n", 40, result);
02108 #endif
02109 }
02110
02111 /*
02112     #] deltaEulerC :
02113     #[ CalculateMZVhalf :
02114
02115     This routine is mainly for support when large numbers of
02116     MZV's have to be calculated at the same time.
02117 */
02118
02119 void CalculateMZVhalf(mpf_t result, int *indexes, int depth)
02120 {
02121     int i;
02122     if ( depth < 0 ) {
02123         printf("Illegal depth in CalculateMZVhalf: %d\n", depth);
02124         exit(-1);
02125     }
02126     for ( i = 0; i < depth; i++ ) {
02127         if ( indexes[i] <= 0 ) {
02128             printf("Illegal index[%d] in CalculateMZVhalf: %d\n", i, indexes[i]);
02129             exit(-1);
02130         }
02131     }
02132     deltaMZV(result, indexes, depth);
02133 }
02134
02135 /*
02136     #] CalculateMZVhalf :
02137     #[ CalculateMZV :
02138 */
02139
02140 void CalculateMZV(mpf_t result, int *indexes, int depth)
02141 {
02142     GETIDENTITY
02143     int num1, num2 = 0, i, j = 0;
02144     if ( depth < 0 ) {
02145         printf("Illegal depth in CalculateMZV: %d\n", depth);
02146         exit(-1);
02147     }
02148     if ( indexes[0] == 1 ) {

```

```

02149         printf("Divergent MZV in CalculateMZV\n");
02150         exit(-1);
02151     }
02152     /* MesPrint("calculateMZV(%a)",depth,indexes); */
02153     for ( i = 0; i < depth; i++ ) {
02154         if ( indexes[i] <= 0 ) {
02155             printf("Illegal index[%d] in CalculateMZV: %d\n",i,indexes[i]);
02156             exit(-1);
02157         }
02158         AT.indi1[i] = indexes[i];
02159     }
02160     num1 = depth; i = 0;
02161     deltaMZV(result,indexes,depth);
02162     for(;;) {
02163         if ( AT.indi1[0] > 1 ) {
02164             AT.indi1[0] -= 1;
02165             for ( j = num2; j > 0; j-- ) AT.indi2[j] = AT.indi2[j-1];
02166             AT.indi2[0] = 1; num2++;
02167         }
02168         else {
02169             AT.indi2[0] += 1;
02170             for ( j = 1; j < num1; j++ ) AT.indi1[j-1] = AT.indi1[j];
02171             num1--;
02172             if ( num1 == 0 ) break;
02173         }
02174         deltaMZV(aux1,AT.indi1,num1);
02175         deltaMZV(aux2,AT.indi2,num2);
02176         mpf_mul(aux3,aux1,aux2);
02177         mpf_add(result,result,aux3);
02178     }
02179     deltaMZV(aux3,AT.indi2,num2);
02180     mpf_add(result,result,aux3);
02181 }
02182
02183 /*
02184     #] CalculateMZV :
02185     #[ CalculateEuler :
02186
02187     There is a problem of notation here.
02188     Basically there are two notations.
02189     1:  $Z(m_1,m_2,m_3) = \sum_{j_3=1}^{\infty} \frac{\text{sign}_{-}(m_3)}{j_3^{\text{abs}_{-}(m_3)}} * \sum_{j_2=1}^{\infty} \frac{\text{sign}_{-}(m_2)}{j_2^{\text{abs}_{-}(m_2)}} * \sum_{j_1=1}^{\infty} \frac{\text{sign}_{-}(m_1)}{j_1^{\text{abs}_{-}(m_1)}}$ 
02190
02191     See Broadhurst,1996
02192     2:  $L(m_1,m_2,m_3) = \sum_{j_1=1}^{\infty} \frac{\text{sign}_{-}(m_1)}{j_1^{\text{abs}_{-}(m_1)}} * \sum_{j_2=1}^{\infty} \frac{\text{sign}_{-}(m_2)}{j_2^{\text{abs}_{-}(m_2)}} * \sum_{j_3=1}^{\infty} \frac{\text{sign}_{-}(m_3)}{(j_2+j_3)^{\text{abs}_{-}(m_2)}} * \frac{1}{(j_1+j_2+j_3)^{\text{abs}_{-}(m_1)}}$ 
02193
02194     See Borwein, Bradley, Broadhurst and Lisonek, 1999
02195     The L corresponds with the H of the datamine up to  $\text{sign}_{-}(m_1*m_2*m_3)$ 
02196     The algorithm follows 2, but the more common notation is 1.
02197     Hence we start with a conversion.
02198 */
02205 void CalculateEuler(mpf_t result, int *Zindexes, int depth)
02206 {
02207     GETIDENTITY
02208     int s1, num1, num2, i, j;
02209
02210     int *indexes = (int *) (AT.WorkPointer);
02211     for ( i = 0; i < depth; i++ ) indexes[i] = Zindexes[i];
02212     for ( i = 0; i < depth-1; i++ ) {
02213         if ( Zindexes[i] < 0 ) {
02214             for ( j = i+1; j < depth; j++ ) indexes[j] = -indexes[j];
02215         }
02216     }
02217
02218     if ( depth < 0 ) {
02219         printf("Illegal depth in CalculateEuler: %d\n",depth);
02220         exit(-1);
02221     }
02222     if ( indexes[0] == 1 ) {
02223         printf("Divergent Euler sum in CalculateEuler\n");
02224         exit(-1);
02225     }
02226     for ( i = 0, j = 0; i < depth; i++ ) {
02227         if ( indexes[i] == 0 ) {
02228             printf("Illegal index[%d] in CalculateEuler: %d\n",i,indexes[i]);
02229             exit(-1);
02230         }
02231         if ( indexes[i] < 0 ) j = 1;
02232         AT.indi1[i] = indexes[i];
02233     }
02234     if ( j == 0 ) {
02235         CalculateMZV(result,indexes,depth);

```

```

02236     return;
02237 }
02238 num1 = depth; AT.indi2[0] = 0; num2 = 0;
02239 deltaEuler(result,AT.indi1,depth);
02240 s1 = 0;
02241 for(;;) {
02242     s1++;
02243     if ( AT.indi1[0] > 1 ) {
02244         AT.indi1[0] -= 1;
02245         for ( j = num2; j > 0; j-- ) AT.indi2[j] = AT.indi2[j-1];
02246         AT.indi2[0] = 1; num2++;
02247     }
02248     else if ( AT.indi1[0] < -1 ) {
02249         AT.indi1[0] += 1;
02250         for ( j = num2; j > 0; j-- ) AT.indi2[j] = AT.indi2[j-1];
02251         AT.indi2[0] = 1; num2++;
02252     }
02253     else if ( AT.indi1[0] == -1 ) {
02254         for ( j = num2; j > 0; j-- ) AT.indi2[j] = AT.indi2[j-1];
02255         AT.indi2[0] = -1; num2++;
02256         for ( j = 1; j < num1; j++ ) AT.indi1[j-1] = AT.indi1[j];
02257         num1--;
02258     }
02259     else {
02260         AT.indi2[0] = AT.indi2[0] < 0 ? AT.indi2[0]-1: AT.indi2[0]+1;
02261         for ( j = 1; j < num1; j++ ) AT.indi1[j-1] = AT.indi1[j];
02262         num1--;
02263     }
02264     if ( num1 == 0 ) break;
02265     deltaEuler(aux1,AT.indi1,num1);
02266     deltaEulerC(aux2,AT.indi2,num2);
02267     mpf_mul(aux3,aux1,aux2);
02268     if ( (depth+num1+num2+s1)%2 == 0 ) mpf_add(result,result,aux3);
02269     else mpf_sub(result,result,aux3);
02270 }
02271 deltaEulerC(aux3,AT.indi2,num2);
02272 if ( (depth+num2+s1)%2 == 0 ) mpf_add(result,result,aux3);
02273 else mpf_sub(result,result,aux3);
02274 }
02275
02276 /*
02277     #] CalculateEuler :
02278     #[ ExpandMZV :
02279 */
02280
02281 int ExpandMZV(WORD *term, WORD level)
02282 {
02283     DUMMYUSE(term);
02284     DUMMYUSE(level);
02285     return(0);
02286 }
02287
02288 /*
02289     #] ExpandMZV :
02290     #[ ExpandEuler :
02291 */
02292
02293 int ExpandEuler(WORD *term, WORD level)
02294 {
02295     DUMMYUSE(term);
02296     DUMMYUSE(level);
02297     return(0);
02298 }
02299
02300 /*
02301     #] ExpandEuler :
02302     #[ EvaluateEuler :
02303
02304     There are various steps here:
02305     1: locate an Euler function.
02306     2: evaluate it into a float.
02307     3: convert the coefficient to a float and multiply.
02308     4: Put the new term together.
02309     5: Restart to see whether there are more Euler functions.
02310     The compiler should check that there is no conflict between
02311     the evaluate command and a potential polyfun or polyratfun.
02312     Yet, when reading in expressions there can be a conflict.
02313     Hence the Normalize routine should check as well.
02314
02315     par == MZV: MZV
02316     par == EULER: Euler
02317     par == MZVHALF: MZV sums in 1/2 rather than 1. -> deltaMZV.
02318     par == ALLMZVFUNCTIONS: all of the above.
02319 */
02320
02321 int EvaluateEuler(PHEAD WORD *term, WORD level, WORD par)
02322 {

```

```

02323     WORD *t, *tstop, *tt, *tnext, sumweight, depth, first = 1, *newterm, i;
02324     WORD *oldworkpointer = AT.WorkPointer, nsize, *indexes;
02325     int retval;
02326     tstop = term + *term; tstop -= ABS(tstop[-1]);
02327     if ( AT.WorkPointer < term+*term ) AT.WorkPointer = term + *term;
02328 /*
02329     Step 1. We also need to verify that the argument we find is legal
02330 */
02331     t = term+1;
02332     while ( t < tstop ) {
02333         if ( ( *t == par ) || ( ( par == ALLMZVFUNCTIONS ) && (
02334             *t == MZV || *t == EULER || *t == MZVHALF ) ) ) {
02335             /* all arguments should be small integers */
02336             indexes = AT.WorkPointer;
02337             sumweight = 0; depth = 0;
02338             tnext = t + t[1]; tt = t + FUNHEAD;
02339             while ( tt < tnext ) {
02340                 if ( *tt != -SNUMBER ) goto nextfun;
02341                 if ( tt[1] == 0 ) goto nextfun;
02342                 indexes[depth] = tt[1];
02343                 sumweight += ABS(tt[1]); depth++;
02344                 tt += 2;
02345             }
02346             /* euler sum without arguments, i.e. mzv_(), euler_() or mzvhalf_() */
02347             if ( depth == 0 ) goto nextfun;
02348             if ( sumweight > AC.MaxWeight ) {
02349                 MesPrint("Error: Weight of Euler/MZV sum greater than %d",sumweight);
02350                 MesPrint("Please increase MaxWeight in form.set.");
02351                 Terminate(-1);
02352             }
02353 /*
02354             Step 2: evaluate:
02355 */
02356             AT.WorkPointer += depth;
02357             if ( first ) {
02358                 if ( *t == MZV ) CalculateMZV(aux4,indexes,depth);
02359                 else if ( *t == EULER ) CalculateEuler(aux4,indexes,depth);
02360                 else if ( *t == MZVHALF ) CalculateMZVhalf(aux4,indexes,depth);
02361                 first = 0;
02362             }
02363             else {
02364                 if ( *t == MZV ) CalculateMZV(aux5,indexes,depth);
02365                 else if ( *t == EULER ) CalculateEuler(aux5,indexes,depth);
02366                 else if ( *t == MZVHALF ) CalculateMZVhalf(aux5,indexes,depth);
02367                 mpf_mul(aux4,aux4,aux5);
02368             }
02369             *t = 0;
02370         }
02371     nextfun:
02372         t += t[1];
02373     }
02374     if ( first == 1 ) return(Generator(BHEAD term,level));
02375 /*
02376     Step 3:
02377     Now the regular coefficient, if it is not 1/1.
02378     We have two cases: size +- 3, or bigger.
02379 */
02380     nsize = term[*term-1];
02381     if ( nsize < 0 ) {
02382         mpf_neg(aux4,aux4);
02383         nsize = -nsize;
02384     }
02385     if ( nsize == 3 ) {
02386         if ( tstop[0] != 1 ) {
02387             mpf_mul_ui(aux4,aux4, (ULONG) ((UWORD)tstop[0]));
02388         }
02389         if ( tstop[1] != 1 ) {
02390             mpf_div_ui(aux4,aux4, (ULONG) ((UWORD)tstop[1]));
02391         }
02392     }
02393     else {
02394         RatToFloat(aux5, (UWORD *)tstop,nsize);
02395         mpf_mul(aux4,aux4,aux5);
02396     }
02397 /*
02398     Now we have to locate possible other float_ functions.
02399 */
02400     t = term+1;
02401     while ( t < tstop ) {
02402         if ( *t == FLOATFUN ) {
02403             UnpackFloat(aux5,t);
02404             mpf_mul(aux4,aux4,aux5);
02405         }
02406         t += t[1];
02407     }
02408 /*
02409     Now we should compose the new term in the Workspace.

```

```

02410 */
02411     t = term+1;
02412     newterm = AT.WorkPointer;
02413     tt = newterm+1;
02414     while ( t < tstop ) {
02415         if ( *t == 0 || *t == FLOATFUN ) t += t[1];
02416         else {
02417             i = t[1]; NCOPY(tt,t,i);
02418         }
02419     }
02420     PackFloat(tt,aux4);
02421     tt += tt[1];
02422     *tt++ = 1; *tt++ = 1; *tt++ = 3;
02423     *newterm = tt-newterm;
02424     AT.WorkPointer = tt;
02425     retval = Generator(BHEAD newterm,level);
02426     AT.WorkPointer = oldworkpointer;
02427     return(retval);
02428 }
02429
02430 /*
02431     #] EvaluateEuler :
02432     #] MZV :
02433     #[ Functions :
02434     #[ CoEvaluate :
02435
02436         Current subkeys: mzv_, euler_ and sqrt_.
02437 */
02438
02439 int CoEvaluate(UBYTE *s)
02440 {
02441     GETIDENTITY
02442     UBYTE *subkey, c;
02443     WORD numfun, type;
02444     int error = 0;
02445     if ( AT.aux_ == 0 ) {
02446         MesPrint("&Illegal attempt to evaluate a function without activating floating point
numbers.");
02447         MesPrint("&Forgotten %#startfloat instruction?");
02448         return(1);
02449     }
02450     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
02451     if ( *s == 0 ) {
02452         /*
02453             No subkey, which means all functions.
02454         */
02455         Add3Com(TYPEEVALUATE,ALLFUNCTIONS);
02456         /*
02457             The MZV, EULER and MZVHALF are done separately
02458         */
02459         Add3Com(TYPEEVALUATE,ALLMZVFUNCTIONS);
02460         return(0);
02461     }
02462     while ( *s ) {
02463         subkey = s;
02464         while ( FG.cTable[*s] == 0 ) s++;
02465         if ( *s == '2' ) s++; /* cases li2_ and atan2_ */
02466         if ( *s == '-' ) s++;
02467         c = *s; *s = 0;
02468         /*
02469             We still need provisions for pi_ and possibly other constants.
02470         */
02471         if ( ( ( type = GetName(AC.varnames,subkey,&numfun,NOAUTO) ) != CFUNCTION )
|| ( functions[numfun].spec != 0 ) ) {
02472             if ( type == CSYMBOL ) {
02473                 Add4Com(TYPEEVALUATE,SYMBOL,numfun);
02474                 break;
02475             }
02476             /*
02477                 This cannot work.
02478             */
02479             MesPrint("&%s should be a built in function that can be evaluated numerically.",s);
02480             return(1);
02481         }
02482         else {
02483             switch ( numfun+FUNCTION ) {
02484                 case MZV:
02485                 case EULER:
02486                 case MZVHALF:
02487                     /*
02488                         The following functions are treated in evaluate.c
02489                     */
02490                     case SQRTFUNCTION:
02491                     case LNFUNCTION:
02492                     case SINFUNCTION:
02493                     case COSFUNCTION:

```

```

02496         case TANFUNCTION:
02497         case ASINFUNCTION:
02498         case ACOSFUNCTION:
02499         case ATANFUNCTION:
02500         case ATAN2FUNCTION:
02501         case SINHFUNCTION:
02502         case COSHFUNCTION:
02503         case TANHFUNCTION:
02504         case ASINHFUNCTION:
02505         case ACOSHFUNCTION:
02506         case ATANHFUNCTION:
02507         case LI2FUNCTION:
02508         case LINFUNCTION:
02509         case AGMFUNCTION:
02510         case GAMMAFUN:
02511     /*
02512         At a later stage we can add more functions from mpfr here
02513         mpfr_(number,arg(s))
02514     */
02515         Add3Com(TYPEEVALUATE,numfun+FUNCTION);
02516         break;
02517     default:
02518         MesPrint("%s should be a built in function that can be evaluated numerically.",s);
02519         error = 1;
02520         break;
02521     }
02522 }
02523 *s = c;
02524 while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
02525 }
02526 return(error);
02527 }
02528
02529 /*
02530     #] CoEvaluate :
02531     #] Functions :
02532 */

```

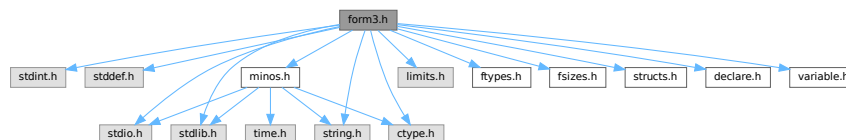
## 4.45 form3.h File Reference

```

#include <stdint.h>
#include <stddef.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <limits.h>
#include "ftypes.h"
#include "fsizes.h"
#include "minos.h"
#include "structs.h"
#include "declare.h"
#include "variable.h"

```

Include dependency graph for form3.h:



## Macros

- #define MAJORVERSION 5

- #define `MINORVERSION` 0
- #define `PRODUCTIONDATE` "8-nov-2022"
- #define `BETAVERSION`
- #define `inline`
- #define `NDEBUG`
- #define `STATIC_ASSERT`(condition) `STATIC_ASSERT__1(condition, __LINE__)`
- #define `STATIC_ASSERT__1`(X, L) `STATIC_ASSERT__2(X,L)`
- #define `STATIC_ASSERT__2`(X, L) `STATIC_ASSERT__3(X,L)`
- #define `STATIC_ASSERT__3`(X, L) `typedef char static_assertion_failed_##L[(!!(X))*2-1]`
- #define `BITSINWORD` 32
- #define `BITSINLONG` 64
- #define `WORD_MIN_VALUE` `INT32_MIN`
- #define `WORD_MAX_VALUE` `INT32_MAX`
- #define `LONG_MIN_VALUE` `INT64_MIN`
- #define `LONG_MAX_VALUE` `INT64_MAX`
- #define `TOPBITONLY` `((ULONG)1 << (BITSINWORD - 1)) /* 0x00008000UL */`
- #define `TOPLONGBITONLY` `((ULONG)1 << (BITSINLONG - 1)) /* 0x80000000UL */`
- #define `SPECMASK` `((UWORD)1 << (BITSINWORD - 1)) /* 0x8000U */`
- #define `WILDMASK` `((UWORD)1 << (BITSINWORD - 2)) /* 0x4000U */`
- #define `WORDMASK` `((ULONG)FULLMAX - 1) /* 0x0000FFFFUL */`
- #define `AWORDMASK` `(WORDMASK << BITSINWORD) /* 0xFFFF0000UL */`
- #define `FULLMAX` `((LONG)1 << BITSINWORD) /* 0x00010000L */`
- #define `MAXPOSITIVE` `((LONG)(TOPBITONLY - 1)) /* 0x00007FFFL */`
- #define `MAXLONG` `((LONG)(TOPLONGBITONLY - 1)) /* 0x7FFFFFFFL */`
- #define `MAXPOSITIVE2` `(MAXPOSITIVE / 2) /* 0x00003FFFL */`
- #define `MAXPOSITIVE4` `(MAXPOSITIVE / 4) /* 0x00001FFFL */`
- #define `form_alignof`(type) `offsetof(struct { char c_; type x_; }, x_)`
- #define `PADDUMMY`(type, size) `UBYTE d_u_m_m_y[form_alignof(type) - ((size) & (form_alignof(type) - 1))]`
- #define `PADPOSITION`(ptr\_, long\_, int\_, word\_, byte\_)
- #define `PADPOINTER`(long\_, int\_, word\_, byte\_)
- #define `PADLONG`(int\_, word\_, byte\_)
- #define `PADINT`(word\_, byte\_)
- #define `PADWORD`(byte\_)
- #define `WITHSORTBOTS`
- #define `FILES` `FILE`
- #define `Uopen`(x, y) `fopen(x,y)`
- #define `Uflush`(x) `fflush(x)`
- #define `Uclose`(x) `fclose(x)`
- #define `Uread`(x, y, z, u) `fread(x,y,z,u)`
- #define `Uwrite`(x, y, z, u) `fwrite(x,y,z,u)`
- #define `Usetbuf`(x, y) `setbuf(x,y)`
- #define `Useek`(x, y, z) `fseek(x,y,z)`
- #define `Utell`(x) `ftell(x)`
- #define `Ugetpos`(x, y) `fgetpos(x,y)`
- #define `Usetpos`(x, y) `fsetpos(x,y)`
- #define `Usync`(x) `fflush(x)`
- #define `Utruncate`(x) `_chsize(_fileno(x),0)`
- #define `Ustdout` `stdout`
- #define `MAX_OPEN_FILES` `FOPEN_MAX`
- #define `bzero`(b, len) `(memset((b), 0, (len)), (void)0)`
- #define `GetPID`() `((LONG)GetCurrentProcessId())`

## Typedefs

- typedef int32\_t [WORD](#)
- typedef int64\_t [LONG](#)
- typedef uint32\_t [UWORD](#)
- typedef uint64\_t [ULONG](#)
- typedef signed char [SBYTE](#)
- typedef unsigned char [UBYTE](#)
- typedef unsigned int [UINT](#)
- typedef ULONG [RLONG](#)
- typedef int64\_t [MLONG](#)
- typedef char [BOOL](#)

## Functions

- **STATIC\_ASSERT** (sizeof(WORD) \*8==BITSINWORD)
- **STATIC\_ASSERT** (sizeof(LONG) \*8==BITSINLONG)
- **STATIC\_ASSERT** (sizeof(LONG) >=sizeof(int \*))
- **STATIC\_ASSERT** (sizeof(int \*) >=sizeof(int))
- **STATIC\_ASSERT** (sizeof(int) >=sizeof(WORD))
- **STATIC\_ASSERT** (sizeof(WORD) >=sizeof(char))
- **STATIC\_ASSERT** (sizeof(char)==1)
- **STATIC\_ASSERT** (sizeof(off\_t) >=sizeof(LONG))

### 4.45.1 Detailed Description

Contains critical defines for the compilation process Also contains the inclusion of all necessary header files. There are also some system dependencies concerning file functions.

Definition in file [form3.h](#).

### 4.45.2 Macro Definition Documentation

#### 4.45.2.1 MAJORVERSION

```
#define MAJORVERSION 5
```

Definition at line [47](#) of file [form3.h](#).

#### 4.45.2.2 MINORVERSION

```
#define MINORVERSION 0
```

Definition at line [48](#) of file [form3.h](#).

#### 4.45.2.3 PRODUCTIONDATE

```
#define PRODUCTIONDATE "8-nov-2022"
```

Definition at line [53](#) of file [form3.h](#).



#### 4.45.2.4 BETAVERSION

```
#define BETAVERSION
```

Definition at line 57 of file [form3.h](#).

#### 4.45.2.5 inline

```
#define inline
```

Definition at line 120 of file [form3.h](#).

#### 4.45.2.6 NDEBUG

```
#define NDEBUG
```

Definition at line 147 of file [form3.h](#).

#### 4.45.2.7 STATIC\_ASSERT

```
#define STATIC_ASSERT(  
    condition ) STATIC_ASSERT__1(condition, __LINE__)
```

Definition at line 158 of file [form3.h](#).

#### 4.45.2.8 STATIC\_ASSERT\_\_1

```
#define STATIC_ASSERT__1(  
    X,  
    L ) STATIC_ASSERT__2(X, L)
```

Definition at line 159 of file [form3.h](#).

#### 4.45.2.9 STATIC\_ASSERT\_\_2

```
#define STATIC_ASSERT__2(  
    X,  
    L ) STATIC_ASSERT__3(X, L)
```

Definition at line 160 of file [form3.h](#).

#### 4.45.2.10 STATIC\_ASSERT\_\_3

```
#define STATIC_ASSERT__3(  
    X,  
    L ) typedef char static_assertion_failed_##L[(!!(X))*2-1]
```

Definition at line 161 of file [form3.h](#).

#### 4.45.2.11 BITSINWORD

```
#define BITSINWORD 32
```

Definition at line 200 of file [form3.h](#).

#### 4.45.2.12 BITSINLONG

```
#define BITSINLONG 64
```

Definition at line 201 of file [form3.h](#).

#### 4.45.2.13 WORD\_MIN\_VALUE

```
#define WORD_MIN_VALUE INT32_MIN
```

Definition at line 202 of file [form3.h](#).

#### 4.45.2.14 WORD\_MAX\_VALUE

```
#define WORD_MAX_VALUE INT32_MAX
```

Definition at line 203 of file [form3.h](#).

#### 4.45.2.15 LONG\_MIN\_VALUE

```
#define LONG_MIN_VALUE INT64_MIN
```

Definition at line 204 of file [form3.h](#).

#### 4.45.2.16 LONG\_MAX\_VALUE

```
#define LONG_MAX_VALUE INT64_MAX
```

Definition at line 205 of file [form3.h](#).

#### 4.45.2.17 TOPBITONLY

```
#define TOPBITONLY ((ULONG)1 << (BITSINWORD - 1)) /* 0x00008000UL */
```

Definition at line 242 of file [form3.h](#).

#### 4.45.2.18 TOPLONGBITONLY

```
#define TOPLONGBITONLY ((ULONG)1 << (BITSINLONG - 1)) /* 0x80000000UL */
```

Definition at line 243 of file [form3.h](#).

#### 4.45.2.19 SPECMASK

```
#define SPECMASK ((UWORD)1 << (BITSINWORD - 1)) /* 0x8000U */
```

Definition at line 244 of file [form3.h](#).

#### 4.45.2.20 WILDMASK

```
#define WILDMASK ((UWORD)1 << (BITSINWORD - 2)) /* 0x4000U */
```

Definition at line 245 of file [form3.h](#).

#### 4.45.2.21 WORDMASK

```
#define WORDMASK ((ULONG)FULLMAX - 1) /* 0x0000FFFFUL */
```

Definition at line 246 of file [form3.h](#).

#### 4.45.2.22 AWORDMASK

```
#define AWORDMASK (WORDMASK << BITSINWORD) /* 0xFFFF0000UL */
```

Definition at line 247 of file [form3.h](#).

#### 4.45.2.23 FULLMAX

```
#define FULLMAX ((LONG)1 << BITSINWORD) /* 0x00010000L */
```

Definition at line 248 of file [form3.h](#).

#### 4.45.2.24 MAXPOSITIVE

```
#define MAXPOSITIVE ((LONG)(TOPBITONLY - 1)) /* 0x00007FFFL */
```

Definition at line 249 of file [form3.h](#).

#### 4.45.2.25 MAXLONG

```
#define MAXLONG ((LONG)(TOPLONGBITONLY - 1)) /* 0x7FFFFFFFL */
```

Definition at line 250 of file [form3.h](#).

#### 4.45.2.26 MAXPOSITIVE2

```
#define MAXPOSITIVE2 (MAXPOSITIVE / 2) /* 0x00003FFFL */
```

Definition at line 251 of file [form3.h](#).

#### 4.45.2.27 MAXPOSITIVE4

```
#define MAXPOSITIVE4 (MAXPOSITIVE / 4) /* 0x00001FFFL */
```

Definition at line 252 of file [form3.h](#).

#### 4.45.2.28 form\_alignof

```
#define form_alignof(  
    type ) offsetof(struct { char c_; type x_; }, x_)
```

Definition at line 268 of file [form3.h](#).

#### 4.45.2.29 PADDUMMY

```
#define PADDUMMY(  
    type,  
    size ) UBYTE d_u_m_m_y[form_alignof(type) - ((size) & (form_alignof(type) -  
1)]]
```

Definition at line 323 of file [form3.h](#).

#### 4.45.2.30 PADPOSITION

```
#define PADPOSITION(  
    ptr_,  
    long_,  
    int_,  
    word_,  
    byte_ )
```

**Value:**

```
PADDUMMY(off_t, \  
+ sizeof(int *) * (ptr_) \  
+ sizeof(LONG) * (long_) \  
+ sizeof(int) * (int_) \  
+ sizeof(WORD) * (word_) \  
+ sizeof(UBYTE) * (byte_) \  
)
```

Definition at line 325 of file [form3.h](#).

#### 4.45.2.31 PADPOINTER

```
#define PADPOINTER(  
    long_,  
    int_,  
    word_,  
    byte_ )
```

**Value:**

```
PADDUMMY(int *, \  
+ sizeof(LONG) * (long_) \  
+ sizeof(int) * (int_) \  
+ sizeof(WORD) * (word_) \  
+ sizeof(UBYTE) * (byte_) \  
)
```

Definition at line 333 of file [form3.h](#).

#### 4.45.2.32 PADLONG

```
#define PADLONG(  
    int_,  
    word_,  
    byte_ )
```

**Value:**

```
PADDUMMY (LONG, \  
+ sizeof(int) * (int_) \  
+ sizeof(WORD) * (word_) \  
+ sizeof(UBYTE) * (byte_) \  
)
```

Definition at line 340 of file [form3.h](#).

#### 4.45.2.33 PADINT

```
#define PADINT(  
    word_,  
    byte_ )
```

**Value:**

```
PADDUMMY (int, \  
+ sizeof(WORD) * (word_) \  
+ sizeof(UBYTE) * (byte_) \  
)
```

Definition at line 346 of file [form3.h](#).

#### 4.45.2.34 PADWORD

```
#define PADWORD(  
    byte_ )
```

**Value:**

```
PADDUMMY (WORD, \  
+ sizeof(UBYTE) * (byte_) \  
)
```

Definition at line 351 of file [form3.h](#).

#### 4.45.2.35 WITHSORTBOTS

```
#define WITHSORTBOTS
```

Definition at line 361 of file [form3.h](#).

#### 4.45.2.36 FILES

```
#define FILES FILE
```

Definition at line 445 of file [form3.h](#).

#### 4.45.2.37 Uopen

```
#define Uopen(  
    x,  
    y ) fopen(x,y)
```

Definition at line 446 of file [form3.h](#).

#### 4.45.2.38 Uflush

```
#define Uflush(  
    x ) fflush(x)
```

Definition at line 447 of file [form3.h](#).

#### 4.45.2.39 Uclose

```
#define Uclose(  
    x ) fclose(x)
```

Definition at line 448 of file [form3.h](#).

#### 4.45.2.40 Uread

```
#define Uread(  
    x,  
    y,  
    z,  
    u ) fread(x,y,z,u)
```

Definition at line 449 of file [form3.h](#).

#### 4.45.2.41 Uwrite

```
#define Uwrite(  
    x,  
    y,  
    z,  
    u ) fwrite(x,y,z,u)
```

Definition at line 450 of file [form3.h](#).

#### 4.45.2.42 Usetbuf

```
#define Usetbuf(  
    x,  
    y ) setbuf(x,y)
```

Definition at line 451 of file [form3.h](#).

#### 4.45.2.43 Useek

```
#define Useek(  
    x,  
    y,  
    z ) fseek(x,y,z)
```

Definition at line 452 of file [form3.h](#).

#### 4.45.2.44 Utell

```
#define Utell(  
    x ) ftell(x)
```

Definition at line 453 of file [form3.h](#).

#### 4.45.2.45 Ugetpos

```
#define Ugetpos(  
    x,  
    y ) fgetpos(x,y)
```

Definition at line 454 of file [form3.h](#).

#### 4.45.2.46 Usetpos

```
#define Usetpos(  
    x,  
    y ) fsetpos(x,y)
```

Definition at line 455 of file [form3.h](#).

#### 4.45.2.47 Usync

```
#define Usync(  
    x ) fflush(x)
```

Definition at line 456 of file [form3.h](#).

#### 4.45.2.48 Utruncate

```
#define Utruncate(  
    x ) _chsize(_fileno(x),0)
```

Definition at line 457 of file [form3.h](#).

#### 4.45.2.49 Ustdout

```
#define Ustdout stdout
```

Definition at line 458 of file [form3.h](#).

#### 4.45.2.50 MAX\_OPEN\_FILES

```
#define MAX_OPEN_FILES FOPEN_MAX
```

Definition at line 459 of file [form3.h](#).

#### 4.45.2.51 bzero

```
#define bzero(  
    b,  
    len ) (memset((b), 0, (len)), (void)0)
```

Definition at line 460 of file [form3.h](#).

#### 4.45.2.52 GetPID

```
#define GetPID( ) ((LONG)GetCurrentProcessId())
```

Definition at line 461 of file [form3.h](#).

### 4.45.3 Typedef Documentation

#### 4.45.3.1 WORD

```
typedef int32_t WORD
```

Definition at line 196 of file [form3.h](#).

#### 4.45.3.2 LONG

```
typedef int64_t LONG
```

Definition at line 197 of file [form3.h](#).

#### 4.45.3.3 UWORD

```
typedef uint32_t UWORD
```

Definition at line 198 of file [form3.h](#).



#### 4.45.3.4 ULONG

```
typedef uint64_t ULONG
```

Definition at line 199 of file [form3.h](#).

#### 4.45.3.5 SBYTE

```
typedef signed char SBYTE
```

Definition at line 231 of file [form3.h](#).

#### 4.45.3.6 UBYTE

```
typedef unsigned char UBYTE
```

Definition at line 232 of file [form3.h](#).

#### 4.45.3.7 UINT

```
typedef unsigned int UINT
```

Definition at line 233 of file [form3.h](#).

#### 4.45.3.8 RLONG

```
typedef ULONG RLONG
```

Definition at line 234 of file [form3.h](#).

#### 4.45.3.9 MLONG

```
typedef int64_t MLONG
```

Definition at line 235 of file [form3.h](#).

#### 4.45.3.10 BOOL

```
typedef char BOOL
```

Definition at line 240 of file [form3.h](#).

## 4.46 form3.h

[Go to the documentation of this file.](#)

```

00001
00008 /* #[ License : */
00009 /*
00010 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00011 *   When using this file you are requested to refer to the publication
00012 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00013 *   This is considered a matter of courtesy as the development was paid
00014 *   for by FOM the Dutch physics granting agency and we would like to
00015 *   be able to track its scientific use to convince FOM of its value
00016 *   for the community.
00017 *
00018 *   This file is part of FORM.
00019 *
00020 *   FORM is free software: you can redistribute it and/or modify it under the
00021 *   terms of the GNU General Public License as published by the Free Software
00022 *   Foundation, either version 3 of the License, or (at your option) any later
00023 *   version.
00024 *
00025 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00026 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00027 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00028 *   details.
00029 *
00030 *   You should have received a copy of the GNU General Public License along
00031 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00032 */
00033 /* #] License : */
00034
00035 #ifndef __FORM3H__
00036 #define __FORM3H__
00037
00038 #ifdef HAVE_CONFIG_H
00039
00040 #ifndef CONFIG_H_INCLUDED
00041 #define CONFIG_H_INCLUDED
00042 #include <config.h>
00043 #endif
00044
00045 #else /* HAVE_CONFIG_H */
00046
00047 #define MAJORVERSION 5
00048 #define MINORVERSION 0
00049
00050 #ifdef __DATE__
00051 #define PRODUCTIONDATE __DATE__
00052 #else
00053 #define PRODUCTIONDATE "8-nov-2022"
00054 #endif
00055
00056 /*#undef BETAVERSION */
00057 #define BETAVERSION
00058
00059 #ifdef LINUX32
00060 #define UNIX
00061 #define LINUX
00062 #define _FILE_OFFSET_BITS 64
00063 #define WITHZLIB
00064 #define WITHGMP
00065 #define WITHPOSIXCLOCK
00066 #endif
00067
00068 #ifdef LINUX64
00069 #define UNIX
00070 #define LINUX
00071 #define WITHZLIB
00072 #define WITHGMP
00073 #define WITHPOSIXCLOCK
00074 #define WITHFLOAT
00075 #endif
00076
00077 #ifdef APPLE32
00078 #define UNIX
00079 #define _FILE_OFFSET_BITS 64
00080 #define WITHZLIB
00081 #endif
00082
00083 #ifdef APPLE64
00084 #define UNIX
00085 #define WITHZLIB
00086 #define WITHGMP
00087 #define WITHPOSIXCLOCK
00088 #define WITHFLOAT

```

```

00089 #define HAVE_UNORDERED_MAP
00090 #define HAVE_UNORDERED_SET
00091 #endif
00092
00093 #ifdef CYGWIN32
00094 #define UNIX
00095 #endif
00096
00097 #ifdef _MSC_VER
00098 #define WINDOWS
00099 #define _CRT_SECURE_NO_WARNINGS
00100 #endif
00101
00102 /*
00103  * We must not define WITHPOSIXCLOCK in compiling the sequential FORM or ParFORM.
00104  */
00105 #if !defined(WITHPTHREADS) && defined(WITHPOSIXCLOCK)
00106 #undef WITHPOSIXCLOCK
00107 #endif
00108
00109 #if !defined(__cplusplus) && !defined(inline)
00110 #if (defined(__STDC_VERSION__) && __STDC_VERSION__ >= 199901L) || (defined(__GNUC__) &&
    !defined(__STRICT_ANSI__))
00111 /* "inline" is available. */
00112 #elif defined(__GNUC__)
00113 /* GNU C compiler has "__inline__". */
00114 #define inline __inline__
00115 #elif defined(_MSC_VER)
00116 /* Microsoft C compiler has "__inline". */
00117 #define inline __inline
00118 #else
00119 /* Inline functions may be not supported. Define "inline" to be empty. */
00120 #define inline
00121 #endif
00122 #endif
00123
00124 #endif /* HAVE_CONFIG_H */
00125
00126 /* Workaround for MSVC. */
00127 #if defined(_MSC_VER)
00128 /*
00129  * Old versions of MSVC didn't support C99 function `snprintf`, which is used
00130  * in poly.cc. On the other hand, macroizing `snprintf` gives a fatal error
00131  * with MSVC >= 2015.
00132  */
00133 #if _MSC_VER < 1900
00134 #define snprintf _snprintf
00135 #endif
00136 #endif
00137
00138 /*
00139  * Translate our dialect "DEBUGGING" to the standard "NDEBUG".
00140  */
00141 #ifdef DEBUGGING
00142 #ifdef NDEBUG
00143 #undef NDEBUG
00144 #endif
00145 #else
00146 #ifndef NDEBUG
00147 #define NDEBUG
00148 #endif
00149 #endif
00150
00151 /*
00152  * STATIC_ASSERT(condition) will fail to be compiled if the given
00153  * condition is false.
00154  */
00155 /*
00156 #define STATIC_ASSERT(condition)
00157 */
00158 #define STATIC_ASSERT(condition) STATIC_ASSERT__1(condition, __LINE__)
00159 #define STATIC_ASSERT__1(X, L) STATIC_ASSERT__2(X, L)
00160 #define STATIC_ASSERT__2(X, L) STATIC_ASSERT__3(X, L)
00161 #define STATIC_ASSERT__3(X, L) \
00162     typedef char static_assertion_failed_##L[(!!(X))*2-1]
00163
00164 /*
00165  * UNIX or WINDOWS must be defined.
00166  */
00167 #if defined(UNIX)
00168 #define mBSD
00169 #define ANSI
00170 #elif defined(WINDOWS)
00171 #define ANSI
00172 #define WIN32_LEAN_AND_MEAN
00173 #include <windows.h>
00174 #include <io.h>

```

```

00175 #include <fcntl.h>
00176 /* Undefine/rename conflicted symbols. */
00177 #undef MAXLONG /* WinNT.h */
00178 #define WORD FORM_WORD /* WinDef.h */
00179 #define LONG FORM_LONG /* WinNT.h */
00180 #define ULONG FORM_ULONG /* WinDef.h */
00181 #define BOOL FORM_BOOL /* WinDef.h */
00182 #undef CreateFile /* WinBase.h */
00183 #undef CopyFile /* WinBase.h */
00184 #define OpenFile FORM_OpenFile /* WinBase.h */
00185 #define ReOpenFile FORM_ReOpenFile /* WinBase.h */
00186 #define ReadFile FORM_ReadFile /* WinBase.h */
00187 #define WriteFile FORM_WriteFile /* WinBase.h */
00188 #define DeleteObject FORM_DeleteObject /* WinGDI.h */
00189 #else
00190 #error UNIX or WINDOWS must be defined!
00191 #endif
00192
00193 #include <stdint.h>
00194
00195 #if UINTPTR_MAX == UINT64_MAX
00196     typedef int32_t WORD;
00197     typedef int64_t LONG;
00198     typedef uint32_t UWORD;
00199     typedef uint64_t ULONG;
00200     #define BITSINWORD 32
00201     #define BITSINLONG 64
00202     #define WORD_MIN_VALUE INT32_MIN
00203     #define WORD_MAX_VALUE INT32_MAX
00204     #define LONG_MIN_VALUE INT64_MIN
00205     #define LONG_MAX_VALUE INT64_MAX
00206 #elif UINTPTR_MAX == UINT32_MAX
00207     typedef int16_t WORD;
00208     typedef int32_t LONG;
00209     typedef uint16_t UWORD;
00210     typedef uint32_t ULONG;
00211     #define BITSINWORD 16
00212     #define BITSINLONG 32
00213     #define WORD_MIN_VALUE INT16_MIN
00214     #define WORD_MAX_VALUE INT16_MAX
00215     #define LONG_MIN_VALUE INT32_MIN
00216     #define LONG_MAX_VALUE INT32_MAX
00217 #else
00218     #error Can not detect if this is a 32-bit or 64-bit platform.
00219 #endif
00220
00221 STATIC_ASSERT(sizeof(WORD) * 8 == BITSINWORD);
00222 STATIC_ASSERT(sizeof(LONG) * 8 == BITSINLONG);
00223 STATIC_ASSERT(sizeof(WORD) * 2 == sizeof(LONG));
00224 STATIC_ASSERT(sizeof(LONG) >= sizeof(int *));
00225 STATIC_ASSERT(sizeof(LONG) >= sizeof(int *));
00226 STATIC_ASSERT(sizeof(int *) >= sizeof(int));
00227 STATIC_ASSERT(sizeof(int) >= sizeof(WORD));
00228 STATIC_ASSERT(sizeof(WORD) >= sizeof(char));
00229 STATIC_ASSERT(sizeof(char) == 1);
00230
00231 typedef signed char SBYTE;
00232 typedef unsigned char UBYTE;
00233 typedef unsigned int UINT;
00234 typedef ULONG RLONG; /* Used in reken.c. */
00235 typedef int64_t MLONG; /* See commentary in minos.h. */
00236 /*
00237  * NOTE: we don't use the standard _Bool (or C++ bool) because its size is
00238  * implementation-dependent and messes up the traditional PADXXX macros.
00239  */
00240 typedef char BOOL;
00241
00242 /* E.g. in 32-bits */
00243 #define TOPBITONLY ((ULONG)1 << (BITSINWORD - 1)) /* 0x00008000UL */
00244 #define TOPLONGBITONLY ((ULONG)1 << (BITSINLONG - 1)) /* 0x80000000UL */
00245 #define SPECMASK ((UWORD)1 << (BITSINWORD - 1)) /* 0x8000U */
00246 #define WILDMASK ((UWORD)1 << (BITSINWORD - 2)) /* 0x4000U */
00247 #define WORDMASK ((ULONG)FULLMAX - 1) /* 0x0000FFFFUL */
00248 #define AWORDMASK (WORDMASK << BITSINWORD) /* 0xFFFF0000UL */
00249 #define FULLMAX ((LONG)1 << BITSINWORD) /* 0x00010000L */
00250 #define MAXPOSITIVE ((LONG)(TOPBITONLY - 1)) /* 0x00007FFF */
00251 #define MAXLONG ((LONG)(TOPLONGBITONLY - 1)) /* 0x7FFFFFFF */
00252 #define MAXPOSITIVE2 (MAXPOSITIVE / 2) /* 0x00003FFF */
00253 #define MAXPOSITIVE4 (MAXPOSITIVE / 4) /* 0x00001FFF */
00254 /*
00255  * form_alignof(type) returns the number of bytes used in the alignment of
00256  * the type.
00257  */
00258 #if !defined(form_alignof)
00259 #if defined(__GNUC__)
00260 /* GNU C compiler has "__alignof__". */
00261 #define form_alignof(type) __alignof__(type)

```

```

00262 #elif defined(_MSC_VER)
00263 /* Microsoft C compiler has "__alignof". */
00264 #define form_alignof(type) __alignof(type)
00265 #elif !defined(__cplusplus)
00266 /* Generic case in C. */
00267 #include <stddef.h>
00268 #define form_alignof(type) offsetof(struct { char c_; type x_; }, x_)
00269 #else
00270 /* Generic case in C++, at least works with a POD struct. */
00271 #include <cstdint>
00272 namespace alignof_impl_ {
00273 template<typename T> struct calc {
00274     struct X { char c_; T x_; };
00275     enum { value = offsetof(X, x_) };
00276 };
00277 }
00278 #define form_alignof(type) alignof_impl_::calc<type>::value
00279 #endif
00280 #endif
00281
00282 /*
00283  * Macros to be inserted at the end of a structure to align the whole structure.
00284  *
00285  * In the currently available systems,
00286  *   sizeof(POSITION) >= sizeof(pointers) == sizeof(LONG) >= sizeof(int)
00287  *   >= sizeof(WORD) >= sizeof(UBYTE) = 1.
00288  * (POSITION is defined in struct.h and contains only an off_t variable.)
00289  * Thus, if we put members of a structure in this order and use those macros,
00290  * then we can align the data without relying on extra paddings added by
00291  * the compiler. For example,
00292  *   typedef struct {
00293  *       int *a;
00294  *       LONG b;
00295  *       WORD c[2];
00296  *       UBYTE d;
00297  *       PADPOINTER(1,0,2,1);
00298  *   } A;
00299  *   typedef struct {
00300  *       POSITION p;
00301  *       A a; // aligned same as pointers
00302  *       int *b;
00303  *       LONG c;
00304  *       UBYTE d;
00305  *       PADPOSITION(1,1,0,0,1+sizeof(A));
00306  *   } B;
00307  * The cost for the use of those PADXXX macros is a padding (>= 1 byte) will
00308  * be always inserted even in the case that no padding is actually needed.
00309  *
00310  * Numbers for the arguments have to be calculated manually and so very
00311  * error-prone. Be careful!
00312  *
00313  * Note that there is a 32-bit system in which off_t is aligned on 8-byte
00314  * boundary, (e.g., Cygwin with large file support), but still the above
00315  * inequalities are satisfied.
00316  *
00317  * The legendary story of these macros--they fixed some problems in ancient
00318  * times when compilers were unreliable and didn't know how to correctly compute
00319  * structure paddings--has been handed down, though nowadays there are only
00320  * disadvantages for them in practice (ancient compilers most likely can't
00321  * compile C99 and C++98+TR1 sources anyway).
00322  */
00323 #define PADDDUMMY(type, size) \
00324     UBYTE d_u_m_m_y[form_alignof(type) - ((size) & (form_alignof(type) - 1))]
00325 #define PADPOSITION(ptr_, long_, int_, word_, byte_) \
00326     PADDDUMMY(off_t, \
00327         + sizeof(int) * (ptr_) \
00328         + sizeof(LONG) * (long_) \
00329         + sizeof(int) * (int_) \
00330         + sizeof(WORD) * (word_) \
00331         + sizeof(UBYTE) * (byte_) \
00332     )
00333 #define PADPOINTER(long_, int_, word_, byte_) \
00334     PADDDUMMY(int *, \
00335         + sizeof(LONG) * (long_) \
00336         + sizeof(int) * (int_) \
00337         + sizeof(WORD) * (word_) \
00338         + sizeof(UBYTE) * (byte_) \
00339     )
00340 #define PADLONG(int_, word_, byte_) \
00341     PADDDUMMY(LONG, \
00342         + sizeof(int) * (int_) \
00343         + sizeof(WORD) * (word_) \
00344         + sizeof(UBYTE) * (byte_) \
00345     )
00346 #define PADINT(word_, byte_) \
00347     PADDDUMMY(int, \
00348         + sizeof(WORD) * (word_) \

```

```

00349         + sizeof(UBYTE) * (byte_) \
00350     )
00351 #define PADWORD(byte_) \
00352     PADDUMMY(WORD, \
00353         + sizeof(UBYTE) * (byte_) \
00354     )
00355
00356 /*
00357 #define WITHPCOUNTER
00358 #define DEBUGGINGLOCKS
00359 #define WITHSTATS
00360 */
00361 #define WITHSORTBOTS
00362
00363 #include <stdio.h>
00364 #include <stdlib.h>
00365 #include <string.h>
00366 #include <ctype.h>
00367 #include <limits.h>
00368 #ifdef ANSI
00369 #include <stdarg.h>
00370 #include <time.h>
00371 #endif
00372 #ifdef WINDOWS
00373 #include "fwin.h"
00374 #endif
00375 #ifdef UNIX
00376 #include <unistd.h>
00377 #include <time.h>
00378 #include <fcntl.h>
00379 #include <sys/file.h>
00380 #include "unix.h"
00381 #endif
00382 #ifdef WITHZLIB
00383 #ifdef WITHZSTD
00384 #include <zstd_zlibwrapper.h>
00385 #else
00386 #include <zlib.h>
00387 #endif
00388 #endif
00389 #ifdef WITHPTHREADS
00390 #include <pthread.h>
00391 #endif
00392
00393 /*
00394     PARALLELCODE indicates code that is common for TFORM and ParFORM but
00395     should not be there for sequential FORM.
00396 */
00397 #if defined(WITHMPI) || defined(WITHPTHREADS)
00398 #define PARALLELCODE
00399 #endif
00400
00401 #include "ftypes.h"
00402 #include "fsizes.h"
00403 #include "minos.h"
00404 #include "structs.h"
00405 #include "declare.h"
00406 #include "variable.h"
00407
00408 STATIC_ASSERT(sizeof(off_t) >= sizeof(LONG));
00409
00410 /*
00411  * The interface to file routines for UNIX or non-UNIX (Windows).
00412  */
00413 #ifdef UNIX
00414
00415 #define UFILES
00416 typedef struct FileS {
00417     int descriptor;
00418 } FILES;
00419 extern FILES *Uopen(char *, char *);
00420 extern int Uclose(FILES *);
00421 extern size_t Uread(char *, size_t, size_t, FILES *);
00422 extern size_t Uwrite(char *, size_t, size_t, FILES *);
00423 extern int Useek(FILES *, off_t, int);
00424 extern off_t Utell(FILES *);
00425 extern void Uflush(FILES *);
00426 extern int Ugetpos(FILES *, fpos_t *);
00427 extern int Usetpos(FILES *, fpos_t *);
00428 extern void Usetbuf(FILES *, char *);
00429 #define Usync(f) fsync(f->descriptor)
00430 #define Utruncate(f) { \
00431     if ( ftruncate(f->descriptor, 0) ) { \
00432         MLOCK(ErrorMessageLock); \
00433         MesPrint("Utruncate failed"); \
00434         MUNLOCK(ErrorMessageLock); \
00435         /* Calling Terminate() here may cause an infinite loop due to CleanUpSort(). */ \

```

```

00436         /* Terminate(-1); */ \
00437     } \
00438 }
00439 extern FILES *Ustdout;
00440 #define MAX_OPEN_FILES getdtablesize()
00441 #define GetPID() ((LONG)getpid())
00442
00443 #else /* UNIX */
00444
00445 #define FILES FILE
00446 #define Uopen(x,y) fopen(x,y)
00447 #define Uflush(x) fflush(x)
00448 #define Uclose(x) fclose(x)
00449 #define Uread(x,y,z,u) fread(x,y,z,u)
00450 #define Uwrite(x,y,z,u) fwrite(x,y,z,u)
00451 #define Usetbuf(x,y) setbuf(x,y)
00452 #define Useek(x,y,z) fseek(x,y,z)
00453 #define Utell(x) ftell(x)
00454 #define Ugetpos(x,y) fgetpos(x,y)
00455 #define Usetpos(x,y) fsetpos(x,y)
00456 #define Usync(x) fflush(x)
00457 #define Utruncate(x) _chsize(_fileno(x),0)
00458 #define Ustdout stdout
00459 #define MAX_OPEN_FILES FOPEN_MAX
00460 #define bzero(b,len) (memset((b), 0, (len)), (void)0)
00461 #define GetPID() ((LONG)GetCurrentProcessId())
00462
00463 #endif /* UNIX */
00464
00465 /*
00466  * Some system may implement the POSIX "environ" variable as a macro, e.g.,
00467  * https://github.com/mingw-w64/mingw-w64/blob/v10.0.0/mingw-w64-headers/crt/stdlib.h#L704
00468  * which breaks the definition of DoShattering() in diagrams.c that uses
00469  * "environ" as a formal parameter. Because FORM doesn't use the POSIX "environ"
00470  * or its variant anyway, we can just undefine it.
00471  */
00472 #ifdef environ
00473 #undef environ
00474 #endif
00475
00476 #ifdef WITHMPI
00477 #include "parallel.h"
00478 #endif
00479
00480 #endif /* __FORM3H__ */

```

## 4.47 fsizes.h File Reference

This graph shows which files directly or indirectly include this file:



### Macros

- #define [MAXPRENAMESIZE](#) 128
- #define [MAXENAME](#) 16
- #define [MAXSAVEFUNCTION](#) 16384
- #define [MAXPARLEVEL](#) 100
- #define [MAXNUMBERSIZE](#) 200
- #define [MAXREPEAT](#) 100
- #define [NORMSIZE](#) 1000
- #define [INITNODESIZE](#) 10
- #define [INITNAMESIZE](#) 100
- #define [NUMFIXED](#) 128
- #define [MAXNEST](#) 100
- #define [MAXMATCH](#) 30

- #define MAXIF 20
- #define SIZEFACS 640L
- #define NUMFACS 50
- #define MAXLOOPS 30
- #define MAXLABELS 20
- #define COMMERCIALSIZE 24
- #define MAXFLAGS 16
- #define COMPRESSBUFFER 90000
- #define FORTRANCONTINUATIONLINES 15
- #define MAXLEVELS 2000
- #define MAXLHS 400
- #define MAXWILDC 100
- #define NUMTABLEENTRIES 1000
- #define COMPILERBUFFER 20000
- #define MAXPATCHES 256
- #define MAXFPATCHES 256
- #define SORTIOSIZE 200000L
- #define SSMALLBUFFER 2560016L
- #define SSMALLOVERFLOW 3840032L
- #define STERMSSMALL 10000L
- #define SLARGEBUFFER 26880512L
- #define SMAXPATCHES 64
- #define SMAXFPATCHES 64
- #define SSORTIOSIZE 32768L
- #define SPECTATORSIZE 1048576L
- #define MAXFLEVELS 30
- #define COMPINC 2
- #define MAXNUMSIZE 10
- #define MAXBRACKETBUFFERSIZE 200000
- #define SFHSIZE 40
- #define DEFAULTPROCESSBUCKETSIZE 1000
- #define SHMWINSIZE 65536L
- #define TABLEEXTENSION 6
- #define GZIPDEFAULT 6
- #define DEFAULTTTHREADS 0
- #define DEFAULTTTHREADBUCKETSIZE 500
- #define DEFAULTTTHREADLOADBALANCING 1
- #define THREADSCRATCHSIZE 100000L
- #define THREADSCRATCHOUTSIZE 2500000L
- #define NUMSTORECACHES 4
- #define SIZESTORECACHE 32768
- #define INDENTSPACE 3
- #define MULTIINDENTSPACE 1
- #define MAXMULTIBRACKETLEVELS 25
- #define FBUFFERSIZE 1026
- #define NPAIR1 38
- #define NPAIR2 89
- #define MAXLINELENGTH 256
- #define MINALLOC 32
- #define JUMPRATIO 4
- #define MAXPARTICLES 20
- #define MAXCOUPLINGS 20
- #define NUMOPTIONS 20
- #define MAXLEGS 20



### 4.47.1 Detailed Description

The definition the default values for certain buffer sizes etc.

Definition in file [fsizes.h](#).

### 4.47.2 Macro Definition Documentation

#### 4.47.2.1 MAXPRENAMESIZE

```
#define MAXPRENAMESIZE 128
```

Definition at line 36 of file [fsizes.h](#).

#### 4.47.2.2 MAXENAME

```
#define MAXENAME 16
```

Definition at line 73 of file [fsizes.h](#).

#### 4.47.2.3 MAXSAVEFUNCTION

```
#define MAXSAVEFUNCTION 16384
```

Definition at line 74 of file [fsizes.h](#).

#### 4.47.2.4 MAXPARLEVEL

```
#define MAXPARLEVEL 100
```

Definition at line 76 of file [fsizes.h](#).

#### 4.47.2.5 MAXNUMBERSIZE

```
#define MAXNUMBERSIZE 200
```

Definition at line 77 of file [fsizes.h](#).

#### 4.47.2.6 MAXREPEAT

```
#define MAXREPEAT 100
```

Definition at line 79 of file [fsizes.h](#).

#### 4.47.2.7 NORMSIZE

```
#define NORMSIZE 1000
```

Definition at line 80 of file [fsizes.h](#).

#### 4.47.2.8 INITNODESIZE

```
#define INITNODESIZE 10
```

Definition at line 82 of file [fsizes.h](#).

#### 4.47.2.9 INITNAME SIZE

```
#define INITNAME SIZE 100
```

Definition at line 83 of file [fsizes.h](#).

#### 4.47.2.10 NUMFIXED

```
#define NUMFIXED 128
```

Definition at line 85 of file [fsizes.h](#).

#### 4.47.2.11 MAXNEST

```
#define MAXNEST 100
```

Definition at line 86 of file [fsizes.h](#).

#### 4.47.2.12 MAXMATCH

```
#define MAXMATCH 30
```

Definition at line 87 of file [fsizes.h](#).

#### 4.47.2.13 MAXIF

```
#define MAXIF 20
```

Definition at line 88 of file [fsizes.h](#).

#### 4.47.2.14 SIZEFACS

```
#define SIZEFACS 640L
```

Definition at line 89 of file [fsizes.h](#).

#### 4.47.2.15 NUMFACS

```
#define NUMFACS 50
```

Definition at line 90 of file [fsizes.h](#).

#### 4.47.2.16 MAXLOOPS

```
#define MAXLOOPS 30
```

Definition at line 91 of file [fsizes.h](#).

#### 4.47.2.17 MAXLABELS

```
#define MAXLABELS 20
```

Definition at line 92 of file [fsizes.h](#).

#### 4.47.2.18 COMMERCIALSIZE

```
#define COMMERCIALSIZE 24
```

Definition at line 93 of file [fsizes.h](#).

#### 4.47.2.19 MAXFLAGS

```
#define MAXFLAGS 16
```

Definition at line 94 of file [fsizes.h](#).

#### 4.47.2.20 COMPRESSBUFFER

```
#define COMPRESSBUFFER 90000
```

Definition at line 99 of file [fsizes.h](#).

#### 4.47.2.21 FORTRANCONTINUATIONLINES

```
#define FORTRANCONTINUATIONLINES 15
```

Definition at line 100 of file [fsizes.h](#).

#### 4.47.2.22 MAXLEVELS

```
#define MAXLEVELS 2000
```

Definition at line 101 of file [fsizes.h](#).

#### 4.47.2.23 MAXLHS

```
#define MAXLHS 400
```

Definition at line 102 of file [fsizes.h](#).

#### 4.47.2.24 MAXWILDC

```
#define MAXWILDC 100
```

Definition at line 103 of file [fsizes.h](#).

#### 4.47.2.25 NUMTABLEENTRIES

```
#define NUMTABLEENTRIES 1000
```

Definition at line 104 of file [fsizes.h](#).

#### 4.47.2.26 COMPILERBUFFER

```
#define COMPILERBUFFER 20000
```

Definition at line 105 of file [fsizes.h](#).

#### 4.47.2.27 MAXPATCHES

```
#define MAXPATCHES 256
```

Definition at line 131 of file [fsizes.h](#).

#### 4.47.2.28 MAXFPATCHES

```
#define MAXFPATCHES 256
```

Definition at line 132 of file [fsizes.h](#).

#### 4.47.2.29 SORTIOSIZE

```
#define SORTIOSIZE 200000L
```

Definition at line 133 of file [fsizes.h](#).

#### 4.47.2.30 SMALLBUFFER

```
#define SMALLBUFFER 2560016L
```

Definition at line 135 of file [fsizes.h](#).

#### 4.47.2.31 SSMALLOVERFLOW

```
#define SSMALLOVERFLOW 3840032L
```

Definition at line 136 of file [fsizes.h](#).

#### 4.47.2.32 STERMSSMALL

```
#define STERMSSMALL 10000L
```

Definition at line 137 of file [fsizes.h](#).

#### 4.47.2.33 SLARGEBUFFER

```
#define SLARGEBUFFER 26880512L
```

Definition at line 138 of file [fsizes.h](#).

#### 4.47.2.34 SMAXPATCHES

```
#define SMAXPATCHES 64
```

Definition at line 139 of file [fsizes.h](#).

#### 4.47.2.35 SMAXFPATCHES

```
#define SMAXFPATCHES 64
```

Definition at line 140 of file [fsizes.h](#).

#### 4.47.2.36 SSORTIOSIZE

```
#define SSORTIOSIZE 32768L
```

Definition at line 141 of file [fsizes.h](#).

#### 4.47.2.37 SPECTATORSIZE

```
#define SPECTATORSIZE 1048576L
```

Definition at line 143 of file [fsizes.h](#).

#### 4.47.2.38 MAXFLEVELS

```
#define MAXFLEVELS 30
```

Definition at line 145 of file [fsizes.h](#).

#### 4.47.2.39 COMPINC

```
#define COMPINC 2
```

Definition at line 147 of file [fsizes.h](#).

#### 4.47.2.40 MAXNUMSIZE

```
#define MAXNUMSIZE 10
```

Definition at line 149 of file [fsizes.h](#).

#### 4.47.2.41 MAXBRACKETBUFFERSIZE

```
#define MAXBRACKETBUFFERSIZE 200000
```

Definition at line 151 of file [fsizes.h](#).

#### 4.47.2.42 SFHSIZE

```
#define SFHSIZE 40
```

Definition at line 153 of file [fsizes.h](#).

#### 4.47.2.43 DEFAULTPROCESSBUCKETSIZE

```
#define DEFAULTPROCESSBUCKETSIZE 1000
```

Definition at line 155 of file [fsizes.h](#).

#### 4.47.2.44 SHMWINSIZE

```
#define SHMWINSIZE 65536L
```

Definition at line 156 of file [fsizes.h](#).

#### 4.47.2.45 TABLEEXTENSION

```
#define TABLEEXTENSION 6
```

Definition at line 158 of file [fsizes.h](#).

#### 4.47.2.46 GZIPDEFAULT

```
#define GZIPDEFAULT 6
```

Definition at line 160 of file [fsizes.h](#).

#### 4.47.2.47 DEFAULTTHREADS

```
#define DEFAULTTHREADS 0
```

Definition at line 161 of file [fsizes.h](#).

#### 4.47.2.48 DEFAULTTHREADBUCKETSIZE

```
#define DEFAULTTHREADBUCKETSIZE 500
```

Definition at line 162 of file [fsizes.h](#).

#### 4.47.2.49 DEFAULTTHREADLOADBALANCING

```
#define DEFAULTTHREADLOADBALANCING 1
```

Definition at line 163 of file [fsizes.h](#).

#### 4.47.2.50 THREADSCRATCHSIZE

```
#define THREADSCRATCHSIZE 100000L
```

Definition at line 164 of file [fsizes.h](#).

#### 4.47.2.51 THREADSCRATCHOUTSIZE

```
#define THREADSCRATCHOUTSIZE 2500000L
```

Definition at line 165 of file [fsizes.h](#).

#### 4.47.2.52 NUMSTORECACHES

```
#define NUMSTORECACHES 4
```

Definition at line 177 of file [fsizes.h](#).

#### 4.47.2.53 SIZESTORECACHE

```
#define SIZESTORECACHE 32768
```

Definition at line 178 of file [fsizes.h](#).

#### 4.47.2.54 INDENTSPACE

```
#define INDENTSPACE 3
```

Definition at line 180 of file [fsizes.h](#).

#### 4.47.2.55 MULTIINDENTSPACE

```
#define MULTIINDENTSPACE 1
```

Definition at line 182 of file [fsizes.h](#).

#### 4.47.2.56 MAXMULTIBRACKETLEVELS

```
#define MAXMULTIBRACKETLEVELS 25
```

Definition at line 183 of file [fsizes.h](#).

#### 4.47.2.57 FBUFFERSIZE

```
#define FBUFFERSIZE 1026
```

Definition at line 185 of file [fsizes.h](#).

#### 4.47.2.58 NPAIR1

```
#define NPAIR1 38
```

Definition at line 189 of file [fsizes.h](#).

#### 4.47.2.59 NPAIR2

```
#define NPAIR2 89
```

Definition at line 190 of file [fsizes.h](#).

#### 4.47.2.60 MAXLINELENGTH

```
#define MAXLINELENGTH 256
```

Definition at line 192 of file [fsizes.h](#).

#### 4.47.2.61 MINALLOC

```
#define MINALLOC 32
```

Definition at line 193 of file [fsizes.h](#).

#### 4.47.2.62 JUMPRATIO

```
#define JUMPRATIO 4
```

Definition at line 195 of file [fsizes.h](#).



#### 4.47.2.63 MAXPARTICLES

```
#define MAXPARTICLES 20
```

Definition at line 199 of file [fsizes.h](#).

#### 4.47.2.64 MAXCOUPLINGS

```
#define MAXCOUPLINGS 20
```

Definition at line 200 of file [fsizes.h](#).

#### 4.47.2.65 NUMOPTIONS

```
#define NUMOPTIONS 20
```

Definition at line 201 of file [fsizes.h](#).

#### 4.47.2.66 MAXLEGS

```
#define MAXLEGS 20
```

Definition at line 202 of file [fsizes.h](#).

## 4.48 fsizes.h

[Go to the documentation of this file.](#)

```
00001
00006 /* #[ License : */
00007 /*
00008 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 * When using this file you are requested to refer to the publication
00010 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 * This is considered a matter of courtesy as the development was paid
00012 * for by FOM the Dutch physics granting agency and we would like to
00013 * be able to track its scientific use to convince FOM of its value
00014 * for the community.
00015 *
00016 * This file is part of FORM.
00017 *
00018 * FORM is free software: you can redistribute it and/or modify it under the
00019 * terms of the GNU General Public License as published by the Free Software
00020 * Foundation, either version 3 of the License, or (at your option) any later
00021 * version.
00022 *
00023 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 * details.
00027 *
00028 * You should have received a copy of the GNU General Public License along
00029 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #[ License : */
00032
00033 /*
00034 * First the fixed variables
00035 */
00036 #define MAXPRENAMESIZE 128
00037 /*
00038 * The following variables are default sizes. They can be changed
00039 * into values read from the setup file
```

```

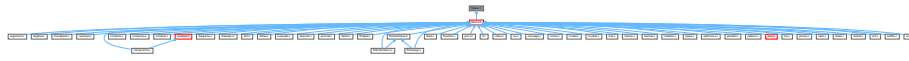
00040
00041     Remark (21-dec-2008 JV): WILDOFFSET*3 should be larger than WILDMASK!!!!
00042     old value was WILDOFFSET 200000100
00043     be careful with old .sav files!!!
00044 */
00045 #if BITSINWORD == 32
00046     #define MAXPOWER 500000000
00047     #define MAXVARIABLES 200000050
00048     #define MAXDOLLARVARIABLES 1000000000L
00049     #define WILDOFFSET 400000100
00050     #define MAXINNAMETREE 2000000000
00051     #define MAXDUMMIES 100000000
00052     #define WORKBUFFER 40000000
00053     #define MAXTER 40000
00054     #define HALFMAX 0x10000
00055     #define MAXSUBEXPRESSIONS 0x1FFFFFFF
00056     #define MAXFILESTREAMSIZE 1024
00057 #elif BITSINWORD == 16
00058     #define MAXPOWER 10000
00059     #define MAXVARIABLES 8050
00060     #define MAXDOLLARVARIABLES 32000
00061     #define WILDOFFSET 6100
00062     #define MAXINNAMETREE 32768
00063     #define MAXDUMMIES 1000
00064     #define WORKBUFFER 10000000
00065     #define MAXTER 10000
00066     #define HALFMAX 0x100
00067     #define MAXSUBEXPRESSIONS 0x3FFF
00068     #define MAXFILESTREAMSIZE 1048576
00069 #else
00070     #error Only 64-bit and 32-bit platforms are supported.
00071 #endif
00072
00073 #define MAXENAME 16
00074 #define MAXSAVEFUNCTION 16384
00075
00076 #define MAXPARLEVEL 100
00077 #define MAXNUMBERSIZE 200
00078
00079 #define MAXREPEAT 100
00080 #define NORMSIZE 1000
00081
00082 #define INITNODESIZE 10
00083 #define INITNAMESIZE 100
00084
00085 #define NUMFIXED 128
00086 #define MAXNEST 100
00087 #define MAXMATCH 30
00088 #define MAXIF 20
00089 #define SIZEFACS 640L
00090 #define NUMFACS 50
00091 #define MAXLOOPS 30
00092 #define MAXLABELS 20
00093 #define COMMERCIALSIZE 24
00094 #define MAXFLAGS 16
00095 /*
00096     The next quantities should still be eliminated from the program
00097     This should be together with changes in setfile!
00098 */
00099 #define COMPRESSBUFFER 90000
00100 #define FORTRANCONTINUATIONLINES 15
00101 #define MAXLEVELS 2000
00102 #define MAXLHS 400
00103 #define MAXWILD 100
00104 #define NUMTABLEENTRIES 1000
00105 #define COMPILERBUFFER 20000
00106
00107 #if BITSINWORD == 32
00108     #ifdef WITHPTHREADS
00109         #define SMALLBUFFER 300000000L
00110         #define SMALLOVERFLOW 600000000L
00111         #define TERMSSMALL 30000000L
00112         #define LARGEBUFFER 1500000000L
00113         #define SCRATCHSIZE 500000000L
00114     #else
00115         #define SMALLBUFFER 150000000L
00116         #define SMALLOVERFLOW 300000000L
00117         #define TERMSSMALL 20000000L
00118         #define LARGEBUFFER 800000000L
00119         #define SCRATCHSIZE 500000000L
00120     #endif
00121 #elif BITSINWORD == 16
00122     #define SMALLBUFFER 100000000L
00123     #define SMALLOVERFLOW 200000000L
00124     #define TERMSSMALL 1000000L
00125     #define LARGEBUFFER 500000000L
00126     #define SCRATCHSIZE 500000000L

```

```
00127 #else
00128     #error Only 64-bit and 32-bit platforms are supported.
00129 #endif
00130
00131 #define MAXPATCHES 256
00132 #define MAXFPATCHES 256
00133 #define SORTIOSIZE 200000L
00134
00135 #define SSMALLBUFFER 2560016L
00136 #define SSMAILOVERFLOW 3840032L
00137 #define STERMSSMALL 10000L
00138 #define SLARGEBUFFER 26880512L
00139 #define SMAXPATCHES 64
00140 #define SMAXFPATCHES 64
00141 #define SSORTIOSIZE 32768L
00142
00143 #define SPECTATORSIZE 1048576L
00144
00145 #define MAXFLEVELS 30
00146
00147 #define COMPINC 2
00148
00149 #define MAXNUMSIZE 10
00150
00151 #define MAXBRACKETBUFFERSIZE 200000
00152
00153 #define SFHSIZE 40
00154
00155 #define DEFAULTPROCESSBUCKETSIZE 1000
00156 #define SHMWINSIZE 65536L
00157
00158 #define TABLEEXTENSION 6
00159
00160 #define GZIPDEFAULT 6
00161 #define DEFAULTTTHREADS 0
00162 #define DEFAULTTTHREADBUCKETSIZE 500
00163 #define DEFAULTTTHREADLOADBALANCING 1
00164 #define THREADSCRATCHSIZE 100000L
00165 #define THREADSCRATCHOUTSIZE 2500000L
00166
00167 #if BITSINWORD == 32
00168     #define MAXTABLECOMBUF 100000000000L
00169     #define MAXCOMBUFRHS 1000000000L
00170 #elif BITSINWORD == 16
00171     #define MAXTABLECOMBUF 1000000L
00172     #define MAXCOMBUFRHS 32500L
00173 #else
00174     #error Only 64-bit and 32-bit platforms are supported.
00175 #endif
00176
00177 #define NUMSTORECACHES 4
00178 #define SIZESTORECACHE 32768
00179
00180 #define INDENTSPACE 3
00181
00182 #define MULTIINDENTSPACE 1
00183 #define MAXMULTIBRACKETLEVELS 25
00184
00185 #define FBUFFERSIZE 1026
00186 /*
00187     For the random number generator (see commentary there)
00188 */
00189 #define NPAIR1 38
00190 #define NPAIR2 89
00191
00192 #define MAXLINELENGTH 256
00193 #define MINALLOC 32
00194
00195 #define JUMPRATIO 4
00196 /*
00197     Note: MAXCOUPLINGS should be at least MAXPARTICLES/2+1
00198 */
00199 #define MAXPARTICLES 20
00200 #define MAXCOUPLINGS 20
00201 #define NUMOPTIONS 20
00202 #define MAXLEGS 20
00203
00204 #ifdef WITHFLOAT
00205 #define MAXWEIGHT 0
00206 #define DEFAULTPRECISION 1000
00207 #endif
```

## 4.49 ftypes.h File Reference

This graph shows which files directly or indirectly include this file:



### Macros

- #define [WITHOUTERROR](#) 0
- #define [WITHERROR](#) 1
- #define [FILESTREAM](#) 0
- #define [PREVARSTREAM](#) 1
- #define [PREREADSTREAM](#) 2
- #define [PIPESTREAM](#) 3
- #define [PRECALCSTREAM](#) 4
- #define [DOLLARSTREAM](#) 5
- #define [PREREADSTREAM2](#) 6
- #define [EXTERNALCHANNELSTREAM](#) 7
- #define [PREREADSTREAM3](#) 8
- #define [REVERSEFILESTREAM](#) 9
- #define [INPUTSTREAM](#) 10
- #define [ENDOFSTREAM](#) 0xFF
- #define [ENDOFINPUT](#) 0xFF
- #define [SUBROUTINEFILE](#) 0
- #define [PROCEDUREFILE](#) 1
- #define [HEADERFILE](#) 2
- #define [SETUPFILE](#) 3
- #define [TABLEBASEFILE](#) 4
- #define [FIRSTMODULE](#) -1
- #define [GLOBALMODULE](#) 0
- #define [SORTMODULE](#) 1
- #define [STOREMODULE](#) 2
- #define [CLEARMODULE](#) 3
- #define [ENDMODULE](#) 4
- #define [POLYFUN](#) 0
- #define [NOPARALLEL\\_DOLLAR](#) 0x0001
- #define [NOPARALLEL\\_RHS](#) 0x0002
- #define [NOPARALLEL\\_CONVPOLY](#) 0x0004
- #define [NOPARALLEL\\_SPECTATOR](#) 0x0008
- #define [NOPARALLEL\\_USER](#) 0x0010
- #define [NOPARALLEL\\_TBLDOLLAR](#) 0x0100
- #define [NOPARALLEL\\_NPROC](#) 0x0200
- #define [PARALLELFLAG](#) 0x0000
- #define [PRENOACTION](#) 0
- #define [PRERAISEAFTER](#) 1
- #define [PRELOWERAFTER](#) 2
- #define [WITHSEMICOLON](#) 0
- #define [WITHOUTSEMICOLON](#) 1
- #define [MODULEINSTACK](#) 8
- #define [EXECUTINGIF](#) 0
- #define [LOOKINGFORELSE](#) 1

- #define [LOOKINGFORENDIF](#) 2
- #define [NEWSTATEMENT](#) 1
- #define [OLDSTATEMENT](#) 0
- #define [EXECUTINGPRESWITCH](#) 0
- #define [SEARCHINGPRECASE](#) 1
- #define [SEARCHINGPREENDSWITCH](#) 2
- #define [PREPROONLY](#) 1
- #define [DUMPTOCOMPILER](#) 2
- #define [DUMPOUTTERMS](#) 4
- #define [DUMPINTERMS](#) 8
- #define [DUMPTOSORT](#) 16
- #define [DUMPTOPARALLEL](#) 32
- #define [THREADSDEBUG](#) 64
- #define [ERROROUT](#) 0
- #define [INPUTOUT](#) 1
- #define [STATSOUT](#) 2
- #define [EXPRSOUT](#) 3
- #define [WRITEOUT](#) 4
- #define [EXTERNALCHANNELOUT](#) 5
- #define [NUMERICALLOOP](#) 0
- #define [LISTEDLOOP](#) 1
- #define [ONEEXPRESSION](#) 2
- #define [PRETYPENONE](#) 0
- #define [PRETYPEIF](#) 1
- #define [PRETYPEDO](#) 2
- #define [PRETYPEPROCEDURE](#) 3
- #define [PRETYPESWITCH](#) 4
- #define [PRETYPEINSIDE](#) 5
- #define [DECLARATION](#) 1
- #define [SPECIFICATION](#) 2
- #define [DEFINITION](#) 3
- #define [STATEMENT](#) 4
- #define [TOOOUTPUT](#) 5
- #define [ATENDOFMODULE](#) 6
- #define [MIXED2](#) 8
- #define [MIXED](#) 9
- #define [NOAUTO](#) 0
- #define [PARTEST](#) 1
- #define [WITHAUTO](#) 2
- #define [ALLVARIABLES](#) -1
- #define [SYMBOLONLY](#) 1
- #define [INDEXONLY](#) 2
- #define [VECTORONLY](#) 4
- #define [FUNCTIONONLY](#) 8
- #define [SETONLY](#) 16
- #define [EXPRESSIONONLY](#) 32

#### **Defines: compiler types**

*Type of variable found by the compiler.*

- #define [CDELETE](#) -1
- #define [ANYTYPE](#) -1
- #define [CSYMBOL](#) 0
- #define [CINDEX](#) 1
- #define [CVECTOR](#) 2

- #define CFUNCTION 3
- #define CSET 4
- #define CEXPRESSION 5
- #define CDOTPRODUCT 6
- #define CNUMBER 7
- #define CSUBEXP 8
- #define CDELTA 9
- #define CDOLLAR 10
- #define CDUBIOUS 11
- #define CRANGE 12
- #define CMODEL 13
- #define CVECTOR1 21
- #define CDOUBLEDOT 22
- #define TSYMBOL -1
- #define TINDEX -2
- #define TVECTOR -3
- #define TFUNCTION -4
- #define TSET -5
- #define TEXPRESSSION -6
- #define TDOTPRODUCT -7
- #define TNUMBER -8
- #define TSUBEXP -9
- #define TDELTA -10
- #define TDOLLAR -11
- #define TDUBIOUS -12
- #define LPARENTHESIS -13
- #define RPARENTHESIS -14
- #define TWILDCARD -15
- #define TWILDARG -16
- #define TDOT -17
- #define LBRACE -18
- #define RBRACE -19
- #define TCOMMA -20
- #define TFUNOPEN -21
- #define TFUNCLOSE -22
- #define TMULTIPLY -23
- #define TDIVIDE -24
- #define TPOWER -25
- #define TPLUS -26
- #define TMINUS -27
- #define TNOT -28
- #define TENDOFIT -29
- #define TSETOPEN -30
- #define TSETCLOSE -31
- #define TGENINDEX -32
- #define TCONJUGATE -33
- #define LRPARENTHESSES -34
- #define TNUMBER1 -35
- #define TPOWER1 -36
- #define TEMPTY -37
- #define TSETNUM -38
- #define TSGAMMA -39
- #define TSETDOL -40
- #define TFLOAT -41
- #define TYPEISFUN 0
- #define TYPEISSUB 1
- #define TYPEISMYSTERY -1
- #define LHSIDEX 2
- #define LHSIDE 1
- #define RHSIDE 0
- #define FORTRANMODE 1
- #define REDUCEMODE 2
- #define MAPLEMODE 3
- #define MATHEMATICAMODE 4

- #define [CMODE](#) 5
- #define [VORTRANMODE](#) 6
- #define [PFORTRANMODE](#) 7
- #define [DOUBLEFORTRANMODE](#) 33
- #define [DOUBLEPRECISIONFLAG](#) 32
- #define [NODOUBLEMASK](#) 31
- #define [QUADRUPLEFORTRANMODE](#) 65
- #define [QUADRUPLEPRECISIONFLAG](#) 64
- #define [ALLINTEGERDOUBLE](#) 128
- #define [NOQUADMASK](#) 63
- #define [NORMALFORMAT](#) 0
- #define [NOSPACEFORMAT](#) 1
- #define [ISNOTFORTRAN90](#) 0
- #define [ISFORTRAN90](#) 1
- #define [ALSOREVERSE](#) 1
- #define [CHISHOLM](#) 2
- #define [NOTRICK](#) 16
- #define [SORTLOWFIRST](#) 0
- #define [SORTHIGHFIRST](#) 1
- #define [SORTPOWERFIRST](#) 2
- #define [SORTANTIPOWER](#) 3
- #define [STATSSPLITMERGE](#) 0
- #define [STATSMERGETOFILE](#) 1
- #define [STATSPOSTSORT](#) 2
- #define [NOCHECKLOGTYPE](#) 0
- #define [CHECKLOGTYPE](#) 1
- #define [NMIN4SHIFT](#) 4
- #define [SYMBOL](#) 1
- #define [DOTPRODUCT](#) 2
- #define [VECTOR](#) 3
- #define [INDEX](#) 4
- #define [EXPRESSION](#) 5
- #define [SUBEXPRESSION](#) 6
- #define [DOLLAREXPRESSION](#) 7
- #define [SETSET](#) 8
- #define [ARGWILD](#) 9
- #define [MINVECTOR](#) 10
- #define [SETEXP](#) 11
- #define [DOLLAREXP2](#) 12
- #define [HAAKJE0](#) 9
- #define [FUNCTION](#) 20
- #define [TMPPOLYFUN](#) 14
- #define [ARGFIELD](#) 15
- #define [SNUMBER](#) 16
- #define [LNUMBER](#) 17
- #define [HAAKJE](#) 18
- #define [DELTA](#) 19
- #define [EXPONENT](#) 20
- #define [DENOMINATOR](#) 21
- #define [SETFUNCTION](#) 22
- #define [GAMMA](#) 23
- #define [GAMMAI](#) 24
- #define [GAMMAFIVE](#) 25
- #define [GAMMASIX](#) 26
- #define [GAMMASEVEN](#) 27
- #define [SUMF1](#) 28
- #define [SUMF2](#) 29
- #define [DUMFUN](#) 30
- #define [REPLACEMENT](#) 31
- #define [REVERSEFUNCTION](#) 32
- #define [DISTRIBUTION](#) 33
- #define [DELTA3](#) 34
- #define [DUMMYFUN](#) 35
- #define [DUMMYTEN](#) 36

- `#define LEVICIVITA` 37
- `#define FACTORIAL` 38
- `#define INVERSEFACTORIAL` 39
- `#define BINOMIAL` 40
- `#define NUMARGSFUN` 41
- `#define SIGNFUN` 42
- `#define MODFUNCTION` 43
- `#define MOD2FUNCTION` 44
- `#define MINFUNCTION` 45
- `#define MAXFUNCTION` 46
- `#define ABSFUNCTION` 47
- `#define SIGFUNCTION` 48
- `#define INTFUNCTION` 49
- `#define THETA` 50
- `#define THETA2` 51
- `#define DELTA2` 52
- `#define DELTAP` 53
- `#define BERNOULLIFUNCTION` 54
- `#define COUNTFUNCTION` 55
- `#define MATCHFUNCTION` 56
- `#define PATTERNFUNCTION` 57
- `#define TERMFUNCTION` 58
- `#define CONJUGATION` 59
- `#define ROOTFUNCTION` 60
- `#define TABLEFUNCTION` 61
- `#define FIRSTBRACKET` 62
- `#define TERMSINEXPR` 63
- `#define NUMTERMSFUN` 64
- `#define GCDFUNCTION` 65
- `#define DIVFUNCTION` 66
- `#define REMFUNCTION` 67
- `#define MAXPOWEROF` 68
- `#define MINPOWEROF` 69
- `#define TABLESTUB` 70
- `#define FACTORIN` 71
- `#define TERMSINBRACKET` 72
- `#define WILDARGFUN` 73
- `#define SQRTFUNCTION` 74
- `#define LNFUNTION` 75
- `#define SINFUNCTION` 76
- `#define COSFUNCTION` 77
- `#define TANFUNCTION` 78
- `#define ASINFUNCTION` 79
- `#define ACOSFUNCTION` 80
- `#define ATANFUNCTION` 81
- `#define ATAN2FUNCTION` 82
- `#define SINHFUNTION` 83
- `#define COSHFUNTION` 84
- `#define TANHFUNCTION` 85
- `#define ASINHFUNTION` 86
- `#define ACOSHFUNTION` 87
- `#define ATANHFUNTION` 88
- `#define LI2FUNCTION` 89
- `#define LINFUNTION` 90
- `#define EXTRASYMFUN` 91
- `#define RANDOMFUNCTION` 92
- `#define RANPERM` 93
- `#define NUMFACTORS` 94
- `#define FIRSTTERM` 95
- `#define CONTENTTERM` 96
- `#define PRIMENUMBER` 97
- `#define EXTEUCLIDEAN` 98
- `#define MAKERATIONAL` 99
- `#define INVERSEFUNCTION` 100



- #define [IDFUNCTION](#) 101
- #define [PUTFIRST](#) 102
- #define [PERMUTATIONS](#) 103
- #define [PARTITIONS](#) 104
- #define [MULFUNCTION](#) 105
- #define [MOEBIUS](#) 106
- #define [TOPOLOGIES](#) 107
- #define [DIAGRAMS](#) 108
- #define [TOPO](#) 109
- #define [NODEFUNCTION](#) 110
- #define [EDGE](#) 111
- #define [SIZEOFFUNCTION](#) 112
- #define [BLOCK](#) 113
- #define [ONEPI](#) 114
- #define [PHI](#) 115
- #define [MAXBUILTINFUNCTION](#) 115
- #define [FIRSTUSERFUNCTION](#) 150
- #define [ALLFUNCTIONS](#) -1
- #define [ALLMZVFUNCTIONS](#) -2
- #define [ISYMBOL](#) 0
- #define [PISYMBOL](#) 1
- #define [COEFFSYMBOL](#) 2
- #define [NUMERATORSYMBOL](#) 3
- #define [DENOMINATORSYMBOL](#) 4
- #define [WILDARGSYMBOL](#) 5
- #define [DIMENSIONSYMBOL](#) 6
- #define [FACTORSYMBOL](#) 7
- #define [SEPARATESYMBOL](#) 8
- #define [EESYMBOL](#) 9
- #define [EMSYMBOL](#) 10
- #define [BUILTINSYMBOLS](#) 11
- #define [FIRSTUSERSYMBOL](#) 20
- #define [BUILTININDICES](#) 1
- #define [BUILTINVECTORS](#) 1
- #define [BUILTINDOLLARS](#) 1
- #define [WILDARGVECTOR](#) 0
- #define [WILDARGINDEX](#) 0
- #define [TYPEEXPRESSION](#) 0
- #define [TYPEIDNEW](#) 1
- #define [TYPEIDOLD](#) 2
- #define [TYPEOPERATION](#) 3
- #define [TYPEREPEAT](#) 4
- #define [TYPEENDREPEAT](#) 5
- #define [TYPECOUNT](#) 20
- #define [TYPEMULT](#) 21
- #define [TYPEGOTO](#) 22
- #define [TYPEDISCARD](#) 23
- #define [TYPEIF](#) 24
- #define [TYPEELSE](#) 25
- #define [TYPEELIF](#) 26
- #define [TYPEENDIF](#) 27
- #define [TYPESUM](#) 28
- #define [TYPECHISHOLM](#) 29
- #define [TYPEREVERSE](#) 30
- #define [TYPEARG](#) 31
- #define [TYPENORM](#) 32
- #define [TYPENORM2](#) 33
- #define [TYPENORM3](#) 34
- #define [TYPEEXIT](#) 35
- #define [TYPESETEXIT](#) 36
- #define [TYPEPRINT](#) 37
- #define [TYPEFPRINT](#) 38
- #define [TYPEREDEFPRE](#) 39
- #define [TYPESPLITARG](#) 40

- #define [TYPESPLITARG2](#) 41
- #define [TYPEFACTARG](#) 42
- #define [TYPEFACTARG2](#) 43
- #define [TYPETRY](#) 44
- #define [TYPEASSIGN](#) 45
- #define [TYPERENUMBER](#) 46
- #define [TYPESUMFIX](#) 47
- #define [TYPEFINDLOOP](#) 48
- #define [TYPEUNRAVEL](#) 49
- #define [TYPEADJUSTBOUNDS](#) 50
- #define [TYPEINSIDE](#) 51
- #define [TYPETERM](#) 52
- #define [TYPESORT](#) 53
- #define [TYPEDETCURDUM](#) 54
- #define [TYPEINEXPRESSION](#) 55
- #define [TYPESPLITFIRSTARG](#) 56
- #define [TYPESPLITLASTARG](#) 57
- #define [TYPEMERGE](#) 58
- #define [TYPETESTUSE](#) 59
- #define [TYPEAPPLY](#) 60
- #define [TYPEAPPLYRESET](#) 61
- #define [TYPECHAININ](#) 62
- #define [TYPECHAINOUT](#) 63
- #define [TYPENORM4](#) 64
- #define [TYPEFACTOR](#) 65
- #define [TYPEARGIMPLODE](#) 66
- #define [TYPEARGEXPLODE](#) 67
- #define [TYPEDENOMINATORS](#) 68
- #define [TYPESTUFFLE](#) 69
- #define [TYPEDROPCOEFFICIENT](#) 70
- #define [TYPETRANSFORM](#) 71
- #define [TYPETOPOLYNOMIAL](#) 72
- #define [TYPEFROMPOLYNOMIAL](#) 73
- #define [TYPEDOLOOP](#) 74
- #define [TYPEENDDOLOOP](#) 75
- #define [TYPEDROPSYMBOLS](#) 76
- #define [TYPEPUTINSIDE](#) 77
- #define [TYPETOSPECTATOR](#) 78
- #define [TYPEARGTOEXTRASymbol](#) 79
- #define [TYPECANONICALIZE](#) 80
- #define [TYPESWITCH](#) 81
- #define [TYPEENDSWITCH](#) 82
- #define [TYPESETUSERFLAG](#) 83
- #define [TYPECLEARUSERFLAG](#) 84
- #define [TYPEALLLOOPS](#) 85
- #define [TYPEALLPATHS](#) 86
- #define [TAKETRACE](#) 1
- #define [CONTRACT](#) 2
- #define [RATIO](#) 3
- #define [SYMMETRIZE](#) 4
- #define [TENVEC](#) 5
- #define [SUMNUM1](#) 6
- #define [SUMNUM2](#) 7
- #define [WILDDUMMY](#) 0
- #define [SYMTONUM](#) 1
- #define [SYMTOSYM](#) 2
- #define [SYMTOSUB](#) 3
- #define [VECTOMIN](#) 4
- #define [VECTOVEC](#) 5
- #define [VECTOSUB](#) 6
- #define [INDTOIND](#) 7
- #define [INDTOSUB](#) 8
- #define [FUNTOFUN](#) 9
- #define [ARGTOARG](#) 10

- #define [ARLTOARL](#) 11
- #define [EXPTOEXP](#) 12
- #define [FROMBRAC](#) 13
- #define [FROMSET](#) 14
- #define [SETTONUM](#) 15
- #define [WILDCARDS](#) 16
- #define [SETNUMBER](#) 17
- #define [LOADDOLLAR](#) 18
- #define [NUMTONUM](#) 20
- #define [NUMTOSYM](#) 21
- #define [NUMTOIND](#) 22
- #define [NUMTOSUB](#) 23
- #define [EATTENSOR](#) 0x2000
- #define [CLEANFLAG](#) 0
- #define [DIRTYFLAG](#) 1
- #define [DIRTSYMSYMLAG](#) 2
- #define [MUSTCLEANPRF](#) 4
- #define [SUBTERMUSED1](#) 8
- #define [SUBTERMUSED2](#) 16
- #define [ALLDIRTY](#) (DIRTYFLAG|DIRTSYMSYMLAG)
- #define [ARGHEAD](#) 2
- #define [FUNHEAD](#) 3
- #define [SUBEXPSIZE](#) 5
- #define [EXPRHEAD](#) 5
- #define [TYPEARGHEADSIZE](#) 6
- #define [SKIP](#) 1
- #define [DROP](#) 2
- #define [HIDE](#) 3
- #define [UNHIDE](#) 4
- #define [INTOHIDE](#) 5
- #define [LOCALEXPRESSION](#) 0
- #define [SKIPLEXPRESSION](#) 1
- #define [DROPLEXPRESSION](#) 2
- #define [DROPPEDEXPRESSION](#) 3
- #define [GLOBALEXPRESSION](#) 4
- #define [SKIPGEXPRESSION](#) 5
- #define [DROPGEXPRESSION](#) 6
- #define [STOREDEXPRESSION](#) 8
- #define [HIDDENLEXPRESSION](#) 9
- #define [HIDDENGEXPRESSION](#) 13
- #define [INCEXPRESSION](#) 9
- #define [HIDELEXPRESSION](#) 10
- #define [HIDEGEXPRESSION](#) 14
- #define [DROPHLEXPRESSION](#) 11
- #define [DROPHGEXPRESSION](#) 15
- #define [UNHIDELEXPRESSION](#) 12
- #define [UNHIDEGEXPRESSION](#) 16
- #define [INTOHIDELEXPRESSION](#) 17
- #define [INTOHIDEGEXPRESSION](#) 18
- #define [SPECTATOREXPRESSION](#) 19
- #define [DROPSPECTATOREXPRESSION](#) 20
- #define [SKIPUNHIDELEXPRESSION](#) 21
- #define [SKIPUNHIDEGEXPRESSION](#) 22
- #define [PRINTOFF](#) 0
- #define [PRINTON](#) 1
- #define [PRINTCONTENTS](#) 2
- #define [PRINTCONTENT](#) 3
- #define [PRINTLFILE](#) 4
- #define [PRINTONETERM](#) 8
- #define [PRINTONEFUNCTION](#) 16
- #define [PRINTALL](#) 32
- #define [REGULAREXPRESSION](#) -1
- #define [REDEFINEDEXPRESSION](#) -2
- #define [NEWLYDEFINEDEXPRESSION](#) -3

**Defines: function specs***Function specifications.*

- #define VERTEXFUNCTION -1
- #define GENERALFUNCTION 0
- #define TENSORFUNCTION 2
- #define GAMMAFUNCTION 4
- #define POS\_ 0 /\* integer > 0 \*/
- #define POS0\_ 1 /\* integer >= 0 \*/
- #define NEG\_ 2 /\* integer < 0 \*/
- #define NEG0\_ 3 /\* integer <= 0 \*/
- #define EVEN\_ 4 /\* integer (even) \*/
- #define ODD\_ 5 /\* integer (odd) \*/
- #define Z\_ 6 /\* integer \*/
- #define SYMBOL\_ 7 /\* symbol only \*/
- #define FIXED\_ 8 /\* fixed index \*/
- #define INDEX\_ 9 /\* index only \*/
- #define Q\_ 10 /\* rational \*/
- #define DUMMYINDEX\_ 11 /\* dummy index only \*/
- #define VECTOR\_ 12 /\* vector only \*/
- #define GAMMA1 0
- #define GAMMA5 -1
- #define GAMMA6 -2
- #define GAMMA7 -3
- #define FUNNYVEC -4
- #define FUNNYWILD -5
- #define SUMMEDIND -6
- #define NOINDEX -7
- #define FUNNYDOLLAR -8
- #define EMPTYINDEX -9
- #define MINSPEC -10
- #define USEDFLAG 2
- #define DUMMYFLAG 1
- #define MAINSORT 0
- #define FUNCTIONSORT 1
- #define SUBSORT 2
- #define FLOATMODE 1
- #define RATIONALMODE 0
- #define NUMSPECSETS 10
- #define ISZERO 1
- #define ISUNMODIFIED 2
- #define ISCOMPRESSED 4
- #define ISINRHS 8
- #define ISFACTORIZED 16
- #define TOBEFACTORED 32
- #define TOBEUNFACTORED 64
- #define KEEPZERO 128
- #define VARTYPENONE 0
- #define VARTYPECOMPLEX 1
- #define VARTYPEIMAGINARY 2
- #define VARTYPEROOTOFUNITY 4
- #define VARTYPEMINUS 8
- #define CYCLESYMMETRIC 1
- #define RCYCLESYMMETRIC 2
- #define SYMMETRIC 3
- #define ANTISYMMETRIC 4
- #define REVERSEORDER 256
- #define SUBMULTI 1
- #define SUBONCE 2
- #define SUBONLY 3
- #define SUBMANY 4
- #define SUBVECTOR 5
- #define SUBSELECT 6
- #define SUBALL 7

- #define SUBMASK 15
- #define SUBDISORDER 16
- #define SUBAFTER 32
- #define SUBAFTERNOT 64
- #define IDHEAD 6
- #define DOLLARFLAG 1
- #define NORMALIZEFLAG 2
- #define GIDENT 1
- #define GFIVE 4
- #define GPLUS 3
- #define GMINUS 2
- #define LONGNUMBER 1
- #define MATCH 2
- #define COEFFI 3
- #define SUBEXPR 4
- #define MULTIPLEOF 5
- #define IFDOLLAR 6
- #define IFEXPRESSION 7
- #define IFDOLLAREXTRA 8
- #define IFISFACTORIZED 9
- #define IFOCCURS 10
- #define IFUSERFLAG 11
- #define IFFLOATNUMBER 12
- #define GREATER 0
- #define GREATEREQUAL 1
- #define LESS 2
- #define LESSEQUAL 3
- #define EQUAL 4
- #define NOTEQUAL 5
- #define ORCOND 6
- #define ANDCOND 7
- #define DUMMY 1
- #define SORT 1
- #define STORE 2
- #define END 3
- #define GLOBAL 4
- #define CLEAR 5
- #define VECTBIT 1
- #define DOTPBIT 2
- #define FUNBIT 4
- #define SETBIT 8
- #define EXTRAPARAMETER 0x4000
- #define GENCOMMUTE 0
- #define GENNONCOMMUTE 0x2000
- #define NAMENOTFOUND -9
- #define DOLUNDEFINED 0
- #define DOLNUMBER 1
- #define DOLARGUMENT 2
- #define DOLSUBTERM 3
- #define DOLTERMS 4
- #define DOLWILDARGS 5
- #define DOLINDEX 6
- #define DOLZERO 7
- #define FINDLOOP 0
- #define REPLACELOOP 1
- #define NOFUNPOWERS 0
- #define COMFUNPOWERS 1
- #define ALLFUNPOWERS 2
- #define PROPERORDERFLAG 0
- #define REGULAR 0
- #define FINISH 1
- #define POLYADD 1
- #define POLYSUB 2
- #define POLYMUL 3

- #define POLYDIV 4
- #define POLYREM 5
- #define POLYGCD 6
- #define POLYINTFAC 7
- #define POLYNORM 8
- #define MODNONE 0
- #define MODSUM 1
- #define MODMAX 2
- #define MODMIN 3
- #define MODLOCAL 4
- #define ELEMENTUSED 1
- #define ELEMENTLOADED 2
- #define POSNEG 0x1
- #define INVERSETABLE 0x2
- #define COEFFICIENTSONLY 0x4
- #define ALSOPOWERS 0x8
- #define ALSOFUNARGS 0x10
- #define ALSODOLLARS 0x20
- #define NOINVERSES 0x40
- #define POSITIVEONLY 0
- #define UNPACK 0x80
- #define NOUNPACK 0
- #define FROMFUNCTION 0x100
- #define VARNAMES 0
- #define AUTONAMES 1
- #define EXPRNAMES 2
- #define DOLLARNAMES 3
- #define DUMMYBUFFER 1
- #define ALLARGS 1
- #define NUMARG 2
- #define ARGRANGE 3
- #define MAKEARGS 4
- #define MAXRANGEINDICATOR 4
- #define REPLACEARG 5
- #define ENCODEARG 6
- #define DECODEARG 7
- #define IMplodeARG 8
- #define EXPLODEARG 9
- #define PERMUTEARG 10
- #define REVERSEARG 11
- #define CYCLEARG 12
- #define ISLYNDON 13
- #define ISLYNDONR 14
- #define TOLYNDON 15
- #define TOLYNDONR 16
- #define ADDARG 17
- #define MULTIPLYARG 18
- #define DROPARG 19
- #define SELECTARG 20
- #define DEDUPARG 21
- #define ZTOHARG 22
- #define HTOZARG 23
- #define BASECODE 1
- #define YESLYNDON 1
- #define NOLYNDON 2
- #define TOPOLYNOMIALFLAG 1
- #define FACTARGFLAG 2
- #define OLDFACTARG 1
- #define NEWFACTARG 0
- #define FROMMODULEOPTION 0
- #define FROMPOINTINSTRUCTION 1
- #define EXTRASYMBOL 0
- #define REGULARSYMBOL 1
- #define EXPRESSIONNUMBER 2

- #define `O_NONE` 0
- #define `O_CSE` 1
- #define `O_CSEGREEDY` 2
- #define `O_GREEDY` 3
- #define `O_OCCURRENCE` 0
- #define `O_MCTS` 1
- #define `O_SIMULATED_ANNEALING` 2
- #define `O_FORWARD` 0
- #define `O_BACKWARD` 1
- #define `O_FORWARDORBACKWARD` 2
- #define `O_FORWARDANDBACKWARD` 3
- #define `OPTHEAD` 3
- #define `DOALL` 1
- #define `ONLYFUNCTIONS` 2
- #define `INUSE` 1
- #define `COULDCOMMUTE` 2
- #define `DOESNOTCOMMUTE` 4
- #define `DICT_NONUMBERS` 0
- #define `DICT_INTEGERONLY` 1
- #define `DICT_RATIONALONLY` 2
- #define `DICT_ALLNUMBERS` 3
- #define `DICT_NOVARIABLES` 0
- #define `DICT_DOVARIABLES` 1
- #define `DICT_NOSPECIALS` 0
- #define `DICT_DOSPECIALS` 1
- #define `DICT_NOFUNWITHARGS` 0
- #define `DICT_DOFUNWITHARGS` 1
- #define `DICT_NOTINDOLLARS` 0
- #define `DICT_INDOLLARS` 1
- #define `DICT_INTEGERNUMBER` 1
- #define `DICT_RATIONALNUMBER` 2
- #define `DICT_SYMBOL` 3
- #define `DICT_VECTOR` 4
- #define `DICT_INDEX` 5
- #define `DICT_FUNCTION` 6
- #define `DICT_FUNCTION_WITH_ARGUMENTS` 7
- #define `DICT_SPECIALCHARACTER` 8
- #define `DICT_RANGE` 9
- #define `READSPECTATORFLAG` 3
- #define `GLOBALSPECTATORFLAG` 1
- #define `ORDEREDSET` 1
- #define `DENSETABLE` 1
- #define `SPARSETABLE` 0
- #define `ONEPARTICLEIRREDUCIBLE` 1
- #define `WITHINSERTIONS` 2
- #define `NOTADPOLES` 4
- #define `WITHSYMMETRIZE` 8
- #define `TOPOLOGIESONLY` 16
- #define `NONODES` 32
- #define `WITHEDGES` 64
- #define `CHECKEXTERN` 128
- #define `WITHBLOCKS` 256
- #define `WITHONEPI` 512
- #define `NOSNAILS` 1024
- #define `NOEXTSELF` 2048

## Typedefs

- typedef void(\* `PVFUNWP`) ()
- typedef void(\* `PVFUNV`) ()
- typedef int(\* `CFUN`) ()
- typedef int(\* `TFUN`) ()
- typedef int(\* `TFUN1`) ()

### 4.49.1 Detailed Description

Contains the definitions of many internal codes. Rather than using numbers directly we do this by defines, making it much easier to change things. Changing things is sometimes also a good way of testing the code.

Definition in file [ftypes.h](#).

### 4.49.2 Macro Definition Documentation

#### 4.49.2.1 WITHOUTERROR

```
#define WITHOUTERROR 0
```

The next macros were introduced when TFORM was programmed. In the case of workers, each worker may need some private data. These can in principle be accessed by some posix calls but that is unnecessarily slow. The passing of a pointer to the complete data struct with private data will be much faster. And anyway, there would have to be a macro that either makes the posix call (TFORM) or doesn't (FORM). The solution by having macro's that either pass the pointer (TFORM) or don't pass it (FORM) is seen as the best solution.

In the declarations and the calling of the functions we have to use the PHEAD or the BHEAD macro, respectively, if the pointer is to be passed. These macro's contain the comma as well. Hence we need special macro's if there are no other arguments. These are called PHEAD0 and BHEAD0.

Definition at line 51 of file [ftypes.h](#).

#### 4.49.2.2 WITHERROR

```
#define WITHERROR 1
```

Definition at line 52 of file [ftypes.h](#).

#### 4.49.2.3 FILESTREAM

```
#define FILESTREAM 0
```

Definition at line 58 of file [ftypes.h](#).

#### 4.49.2.4 PREVARSTREAM

```
#define PREVARSTREAM 1
```

Definition at line 59 of file [ftypes.h](#).

#### 4.49.2.5 PREREADSTREAM

```
#define PREREADSTREAM 2
```

Definition at line 60 of file [ftypes.h](#).



#### 4.49.2.6 PIPESTREAM

```
#define PIPESTREAM 3
```

Definition at line 61 of file [ftypes.h](#).

#### 4.49.2.7 PRECALCSTREAM

```
#define PRECALCSTREAM 4
```

Definition at line 62 of file [ftypes.h](#).

#### 4.49.2.8 DOLLARSTREAM

```
#define DOLLARSTREAM 5
```

Definition at line 63 of file [ftypes.h](#).

#### 4.49.2.9 PREREADSTREAM2

```
#define PREREADSTREAM2 6
```

Definition at line 64 of file [ftypes.h](#).

#### 4.49.2.10 EXTERNALCHANNELSTREAM

```
#define EXTERNALCHANNELSTREAM 7
```

Definition at line 65 of file [ftypes.h](#).

#### 4.49.2.11 PREREADSTREAM3

```
#define PREREADSTREAM3 8
```

Definition at line 66 of file [ftypes.h](#).

#### 4.49.2.12 REVERSEFILESTREAM

```
#define REVERSEFILESTREAM 9
```

Definition at line 67 of file [ftypes.h](#).

#### 4.49.2.13 INPUTSTREAM

```
#define INPUTSTREAM 10
```

Definition at line 68 of file [ftypes.h](#).

#### 4.49.2.14 ENDOFSTREAM

```
#define ENDOFSTREAM 0xFF
```

Definition at line 70 of file [ftypes.h](#).

#### 4.49.2.15 ENDOFINPUT

```
#define ENDOFINPUT 0xFF
```

Definition at line 71 of file [ftypes.h](#).

#### 4.49.2.16 SUBROUTINEFILE

```
#define SUBROUTINEFILE 0
```

Definition at line 77 of file [ftypes.h](#).

#### 4.49.2.17 PROCEDUREFILE

```
#define PROCEDUREFILE 1
```

Definition at line 78 of file [ftypes.h](#).

#### 4.49.2.18 HEADERFILE

```
#define HEADERFILE 2
```

Definition at line 79 of file [ftypes.h](#).

#### 4.49.2.19 SETUPFILE

```
#define SETUPFILE 3
```

Definition at line 80 of file [ftypes.h](#).

#### 4.49.2.20 TABLEBASEFILE

```
#define TABLEBASEFILE 4
```

Definition at line 81 of file [ftypes.h](#).

#### 4.49.2.21 FIRSTMODULE

```
#define FIRSTMODULE -1
```

Definition at line 87 of file [ftypes.h](#).

#### 4.49.2.22 GLOBALMODULE

```
#define GLOBALMODULE 0
```

Definition at line 88 of file [ftypes.h](#).

#### 4.49.2.23 SORTMODULE

```
#define SORTMODULE 1
```

Definition at line 89 of file [ftypes.h](#).

#### 4.49.2.24 STOREMODULE

```
#define STOREMODULE 2
```

Definition at line 90 of file [ftypes.h](#).

#### 4.49.2.25 CLEARMODULE

```
#define CLEARMODULE 3
```

Definition at line 91 of file [ftypes.h](#).

#### 4.49.2.26 ENDMODULE

```
#define ENDMODULE 4
```

Definition at line 92 of file [ftypes.h](#).

#### 4.49.2.27 POLYFUN

```
#define POLYFUN 0
```

Definition at line 94 of file [ftypes.h](#).

#### 4.49.2.28 NOPARALLEL\_DOLLAR

```
#define NOPARALLEL_DOLLAR 0x0001
```

Definition at line 96 of file [ftypes.h](#).

#### 4.49.2.29 NOPARALLEL\_RHS

```
#define NOPARALLEL_RHS 0x0002
```

Definition at line 97 of file [ftypes.h](#).

#### 4.49.2.30 NOPARALLEL\_CONVPOLY

```
#define NOPARALLEL_CONVPOLY 0x0004
```

Definition at line 98 of file [ftypes.h](#).

#### 4.49.2.31 NOPARALLEL\_SPECTATOR

```
#define NOPARALLEL_SPECTATOR 0x0008
```

Definition at line 99 of file [ftypes.h](#).

#### 4.49.2.32 NOPARALLEL\_USER

```
#define NOPARALLEL_USER 0x0010
```

Definition at line 100 of file [ftypes.h](#).

#### 4.49.2.33 NOPARALLEL\_TBLDOLLAR

```
#define NOPARALLEL_TBLDOLLAR 0x0100
```

Definition at line 101 of file [ftypes.h](#).

#### 4.49.2.34 NOPARALLEL\_NPROC

```
#define NOPARALLEL_NPROC 0x0200
```

Definition at line 102 of file [ftypes.h](#).

#### 4.49.2.35 PARALLELFLAG

```
#define PARALLELFLAG 0x0000
```

Definition at line 103 of file [ftypes.h](#).

#### 4.49.2.36 PRENOACTION

```
#define PRENOACTION 0
```

Definition at line 105 of file [ftypes.h](#).

#### 4.49.2.37 PRERAISEAFTER

```
#define PRERAISEAFTER 1
```

Definition at line 106 of file [ftypes.h](#).

#### 4.49.2.38 PRELOWERAFTER

```
#define PRELOWERAFTER 2
```

Definition at line 107 of file [ftypes.h](#).

#### 4.49.2.39 WITHSEMICOLON

```
#define WITHSEMICOLON 0
```

Definition at line 114 of file [ftypes.h](#).

#### 4.49.2.40 WITHOUTSEMICOLON

```
#define WITHOUTSEMICOLON 1
```

Definition at line 115 of file [ftypes.h](#).

#### 4.49.2.41 MODULEINSTACK

```
#define MODULEINSTACK 8
```

Definition at line 116 of file [ftypes.h](#).

#### 4.49.2.42 EXECUTINGIF

```
#define EXECUTINGIF 0
```

Definition at line 117 of file [ftypes.h](#).

#### 4.49.2.43 LOOKINGFORELSE

```
#define LOOKINGFORELSE 1
```

Definition at line 118 of file [ftypes.h](#).

#### 4.49.2.44 LOOKINGFORENDIF

```
#define LOOKINGFORENDIF 2
```

Definition at line 119 of file [ftypes.h](#).

#### 4.49.2.45 NEWSTATEMENT

```
#define NEWSTATEMENT 1
```

Definition at line 120 of file [ftypes.h](#).

#### 4.49.2.46 OLDSTATEMENT

```
#define OLDSTATEMENT 0
```

Definition at line 121 of file [ftypes.h](#).

#### 4.49.2.47 EXECUTINGPRESWITCH

```
#define EXECUTINGPRESWITCH 0
```

Definition at line 123 of file [ftypes.h](#).

#### 4.49.2.48 SEARCHINGPRECASE

```
#define SEARCHINGPRECASE 1
```

Definition at line 124 of file [ftypes.h](#).

#### 4.49.2.49 SEARCHINGPREENDSWITCH

```
#define SEARCHINGPREENDSWITCH 2
```

Definition at line 125 of file [ftypes.h](#).

#### 4.49.2.50 PREPROONLY

```
#define PREPROONLY 1
```

Definition at line 127 of file [ftypes.h](#).

#### 4.49.2.51 DUMPTOCOMPILER

```
#define DUMPTOCOMPILER 2
```

Definition at line 128 of file [ftypes.h](#).

#### 4.49.2.52 DUMPOUTTERMS

```
#define DUMPOUTTERMS 4
```

Definition at line 129 of file [ftypes.h](#).

#### 4.49.2.53 DUMPINTERMS

```
#define DUMPINTERMS 8
```

Definition at line 130 of file [ftypes.h](#).

#### 4.49.2.54 DUMPTOSORT

```
#define DUMPTOSORT 16
```

Definition at line 131 of file [ftypes.h](#).

#### 4.49.2.55 DUMPTOPARALLEL

```
#define DUMPTOPARALLEL 32
```

Definition at line 132 of file [ftypes.h](#).

#### 4.49.2.56 THREADSDEBUG

```
#define THREADSDEBUG 64
```

Definition at line 133 of file [ftypes.h](#).

#### 4.49.2.57 ERROROUT

```
#define ERROROUT 0
```

Definition at line 135 of file [ftypes.h](#).

#### 4.49.2.58 INPUTOUT

```
#define INPUTOUT 1
```

Definition at line 136 of file [ftypes.h](#).

#### 4.49.2.59 STATSOUT

```
#define STATSOUT 2
```

Definition at line 137 of file [ftypes.h](#).

#### 4.49.2.60 EXPRSOUT

```
#define EXPRSOUT 3
```

Definition at line 138 of file [ftypes.h](#).

#### 4.49.2.61 WRITEOUT

```
#define WRITEOUT 4
```

Definition at line 139 of file [ftypes.h](#).

#### 4.49.2.62 EXTERNALCHANNELOUT

```
#define EXTERNALCHANNELOUT 5
```

Definition at line 141 of file [ftypes.h](#).

#### 4.49.2.63 NUMERICALLOOP

```
#define NUMERICALLOOP 0
```

Definition at line 143 of file [ftypes.h](#).

#### 4.49.2.64 LISTEDLOOP

```
#define LISTEDLOOP 1
```

Definition at line 144 of file [ftypes.h](#).

#### 4.49.2.65 ONEEXPRESSION

```
#define ONEEXPRESSION 2
```

Definition at line 145 of file [ftypes.h](#).

#### 4.49.2.66 PRETYPENONE

```
#define PRETYPENONE 0
```

Definition at line 147 of file [ftypes.h](#).

#### 4.49.2.67 PRETYPEIF

```
#define PRETYPEIF 1
```

Definition at line 148 of file [ftypes.h](#).

#### 4.49.2.68 PRETYPEDO

```
#define PRETYPEDO 2
```

Definition at line 149 of file [ftypes.h](#).

#### 4.49.2.69 PRETYPEPROCEDURE

```
#define PRETYPEPROCEDURE 3
```

Definition at line 150 of file [ftypes.h](#).



#### 4.49.2.70 PRETYPESWITCH

```
#define PRETYPESWITCH 4
```

Definition at line 151 of file [ftypes.h](#).

#### 4.49.2.71 PRETYPEINSIDE

```
#define PRETYPEINSIDE 5
```

Definition at line 152 of file [ftypes.h](#).

#### 4.49.2.72 DECLARATION

```
#define DECLARATION 1
```

Definition at line 158 of file [ftypes.h](#).

#### 4.49.2.73 SPECIFICATION

```
#define SPECIFICATION 2
```

Definition at line 159 of file [ftypes.h](#).

#### 4.49.2.74 DEFINITION

```
#define DEFINITION 3
```

Definition at line 160 of file [ftypes.h](#).

#### 4.49.2.75 STATEMENT

```
#define STATEMENT 4
```

Definition at line 161 of file [ftypes.h](#).

#### 4.49.2.76 TOOOUTPUT

```
#define TOOOUTPUT 5
```

Definition at line 162 of file [ftypes.h](#).

#### 4.49.2.77 ATENDOFMODULE

```
#define ATENDOFMODULE 6
```

Definition at line 163 of file [ftypes.h](#).

**4.49.2.78 MIXED2**

```
#define MIXED2 8
```

Definition at line 164 of file [ftypes.h](#).

**4.49.2.79 MIXED**

```
#define MIXED 9
```

Definition at line 165 of file [ftypes.h](#).

**4.49.2.80 NOAUTO**

```
#define NOAUTO 0
```

Definition at line 191 of file [ftypes.h](#).

**4.49.2.81 PARTEST**

```
#define PARTEST 1
```

Definition at line 192 of file [ftypes.h](#).

**4.49.2.82 WITHAUTO**

```
#define WITHAUTO 2
```

Definition at line 193 of file [ftypes.h](#).

**4.49.2.83 ALLVARIABLES**

```
#define ALLVARIABLES -1
```

Definition at line 195 of file [ftypes.h](#).

**4.49.2.84 SYMBOLONLY**

```
#define SYMBOLONLY 1
```

Definition at line 196 of file [ftypes.h](#).

**4.49.2.85 INDEXONLY**

```
#define INDEXONLY 2
```

Definition at line 197 of file [ftypes.h](#).

#### 4.49.2.86 VECTORONLY

```
#define VECTORONLY 4
```

Definition at line 198 of file [ftypes.h](#).

#### 4.49.2.87 FUNCTIONONLY

```
#define FUNCTIONONLY 8
```

Definition at line 199 of file [ftypes.h](#).

#### 4.49.2.88 SETONLY

```
#define SETONLY 16
```

Definition at line 200 of file [ftypes.h](#).

#### 4.49.2.89 EXPRESSIONONLY

```
#define EXPRESSIONONLY 32
```

Definition at line 201 of file [ftypes.h](#).

#### 4.49.2.90 CDELETE

```
#define CDELETE -1
```

Definition at line 212 of file [ftypes.h](#).

#### 4.49.2.91 ANYTYPE

```
#define ANYTYPE -1
```

Definition at line 213 of file [ftypes.h](#).

#### 4.49.2.92 CSYMBOL

```
#define CSYMBOL 0
```

Definition at line 214 of file [ftypes.h](#).

#### 4.49.2.93 CINDEX

```
#define CINDEX 1
```

Definition at line 215 of file [ftypes.h](#).

#### 4.49.2.94 CVECTOR

```
#define CVECTOR 2
```

Definition at line 216 of file [ftypes.h](#).

#### 4.49.2.95 CFUNCTION

```
#define CFUNCTION 3
```

Definition at line 217 of file [ftypes.h](#).

#### 4.49.2.96 CSET

```
#define CSET 4
```

Definition at line 218 of file [ftypes.h](#).

#### 4.49.2.97 CEXPRESSION

```
#define CEXPRESSION 5
```

Definition at line 219 of file [ftypes.h](#).

#### 4.49.2.98 CDOTPRODUCT

```
#define CDOTPRODUCT 6
```

Definition at line 220 of file [ftypes.h](#).

#### 4.49.2.99 CNUMBER

```
#define CNUMBER 7
```

Definition at line 221 of file [ftypes.h](#).

#### 4.49.2.100 CSUBEXP

```
#define CSUBEXP 8
```

Definition at line 222 of file [ftypes.h](#).

#### 4.49.2.101 CDELTA

```
#define CDELTA 9
```

Definition at line 223 of file [ftypes.h](#).

#### 4.49.2.102 CDOLLAR

```
#define CDOLLAR 10
```

Definition at line 224 of file [ftypes.h](#).

#### 4.49.2.103 CDUBIOUS

```
#define CDUBIOUS 11
```

Definition at line 225 of file [ftypes.h](#).

#### 4.49.2.104 CRANGE

```
#define CRANGE 12
```

Definition at line 226 of file [ftypes.h](#).

#### 4.49.2.105 CMODEL

```
#define CMODEL 13
```

Definition at line 227 of file [ftypes.h](#).

#### 4.49.2.106 CVECTOR1

```
#define CVECTOR1 21
```

Definition at line 228 of file [ftypes.h](#).

#### 4.49.2.107 CDOUBLEDOT

```
#define CDOUBLEDOT 22
```

Definition at line 229 of file [ftypes.h](#).

#### 4.49.2.108 TSYMBOL

```
#define TSYMBOL -1
```

Definition at line 237 of file [ftypes.h](#).

#### 4.49.2.109 TINDEX

```
#define TINDEX -2
```

Definition at line 238 of file [ftypes.h](#).

**4.49.2.110 TVECTOR**

```
#define TVECTOR -3
```

Definition at line 239 of file [ftypes.h](#).

**4.49.2.111 TFUNCTION**

```
#define TFUNCTION -4
```

Definition at line 240 of file [ftypes.h](#).

**4.49.2.112 TSET**

```
#define TSET -5
```

Definition at line 241 of file [ftypes.h](#).

**4.49.2.113 TEXPRESSON**

```
#define TEXPRESSON -6
```

Definition at line 242 of file [ftypes.h](#).

**4.49.2.114 TDOTPRODUCT**

```
#define TDOTPRODUCT -7
```

Definition at line 243 of file [ftypes.h](#).

**4.49.2.115 TNUMBER**

```
#define TNUMBER -8
```

Definition at line 244 of file [ftypes.h](#).

**4.49.2.116 TSUBEXP**

```
#define TSUBEXP -9
```

Definition at line 245 of file [ftypes.h](#).

**4.49.2.117 TDELTA**

```
#define TDELTA -10
```

Definition at line 246 of file [ftypes.h](#).

**4.49.2.118 TDOLLAR**

```
#define TDOLLAR -11
```

Definition at line 247 of file [ftypes.h](#).

**4.49.2.119 TDUBIOUS**

```
#define TDUBIOUS -12
```

Definition at line 248 of file [ftypes.h](#).

**4.49.2.120 LPARENTHESIS**

```
#define LPARENTHESIS -13
```

Definition at line 249 of file [ftypes.h](#).

**4.49.2.121 RPARENTHESIS**

```
#define RPARENTHESIS -14
```

Definition at line 250 of file [ftypes.h](#).

**4.49.2.122 TWILDCARD**

```
#define TWILDCARD -15
```

Definition at line 251 of file [ftypes.h](#).

**4.49.2.123 TWILDARG**

```
#define TWILDARG -16
```

Definition at line 252 of file [ftypes.h](#).

**4.49.2.124 TDOT**

```
#define TDOT -17
```

Definition at line 253 of file [ftypes.h](#).

**4.49.2.125 LBRACE**

```
#define LBRACE -18
```

Definition at line 254 of file [ftypes.h](#).

**4.49.2.126 RBRACE**

```
#define RBRACE -19
```

Definition at line 255 of file [ftypes.h](#).

**4.49.2.127 TCOMMA**

```
#define TCOMMA -20
```

Definition at line 256 of file [ftypes.h](#).

**4.49.2.128 TFUNOPEN**

```
#define TFUNOPEN -21
```

Definition at line 257 of file [ftypes.h](#).

**4.49.2.129 TFUNCLOSE**

```
#define TFUNCLOSE -22
```

Definition at line 258 of file [ftypes.h](#).

**4.49.2.130 TMULTIPLY**

```
#define TMULTIPLY -23
```

Definition at line 259 of file [ftypes.h](#).

**4.49.2.131 TDIVIDE**

```
#define TDIVIDE -24
```

Definition at line 260 of file [ftypes.h](#).

**4.49.2.132 TPOWER**

```
#define TPOWER -25
```

Definition at line 261 of file [ftypes.h](#).

**4.49.2.133 TPLUS**

```
#define TPLUS -26
```

Definition at line 262 of file [ftypes.h](#).



#### 4.49.2.134 TMINUS

```
#define TMINUS -27
```

Definition at line 263 of file [ftypes.h](#).

#### 4.49.2.135 TNOT

```
#define TNOT -28
```

Definition at line 264 of file [ftypes.h](#).

#### 4.49.2.136 TENDOFIT

```
#define TENDOFIT -29
```

Definition at line 265 of file [ftypes.h](#).

#### 4.49.2.137 TSETOPEN

```
#define TSETOPEN -30
```

Definition at line 266 of file [ftypes.h](#).

#### 4.49.2.138 TSETCLOSE

```
#define TSETCLOSE -31
```

Definition at line 267 of file [ftypes.h](#).

#### 4.49.2.139 TGENINDEX

```
#define TGENINDEX -32
```

Definition at line 268 of file [ftypes.h](#).

#### 4.49.2.140 TCONJUGATE

```
#define TCONJUGATE -33
```

Definition at line 269 of file [ftypes.h](#).

#### 4.49.2.141 LRPARENTHESSES

```
#define LRPARENTHESSES -34
```

Definition at line 270 of file [ftypes.h](#).

**4.49.2.142 TNUMBER1**

```
#define TNUMBER1 -35
```

Definition at line 271 of file [ftypes.h](#).

**4.49.2.143 TPOWER1**

```
#define TPOWER1 -36
```

Definition at line 272 of file [ftypes.h](#).

**4.49.2.144 TEMPTY**

```
#define TEMPTY -37
```

Definition at line 273 of file [ftypes.h](#).

**4.49.2.145 TSETNUM**

```
#define TSETNUM -38
```

Definition at line 274 of file [ftypes.h](#).

**4.49.2.146 TSGAMMA**

```
#define TSGAMMA -39
```

Definition at line 275 of file [ftypes.h](#).

**4.49.2.147 TSETDOL**

```
#define TSETDOL -40
```

Definition at line 276 of file [ftypes.h](#).

**4.49.2.148 TFLOAT**

```
#define TFLOAT -41
```

Definition at line 277 of file [ftypes.h](#).

**4.49.2.149 TYPEISFUN**

```
#define TYPEISFUN 0
```

Definition at line 279 of file [ftypes.h](#).

**4.49.2.150 TYPEISSUB**

```
#define TYPEISSUB 1
```

Definition at line 280 of file [ftypes.h](#).

**4.49.2.151 TYPEISMYSTERY**

```
#define TYPEISMYSTERY -1
```

Definition at line 281 of file [ftypes.h](#).

**4.49.2.152 LHSIDEX**

```
#define LHSIDEX 2
```

Definition at line 283 of file [ftypes.h](#).

**4.49.2.153 LHSIDE**

```
#define LHSIDE 1
```

Definition at line 284 of file [ftypes.h](#).

**4.49.2.154 RHSIDE**

```
#define RHSIDE 0
```

Definition at line 285 of file [ftypes.h](#).

**4.49.2.155 FORTRANMODE**

```
#define FORTRANMODE 1
```

Definition at line 291 of file [ftypes.h](#).

**4.49.2.156 REDUCEMODE**

```
#define REDUCEMODE 2
```

Definition at line 292 of file [ftypes.h](#).

**4.49.2.157 MAPLEMODE**

```
#define MAPLEMODE 3
```

Definition at line 293 of file [ftypes.h](#).

**4.49.2.158 MATHEMATICAMODE**

```
#define MATHEMATICAMODE 4
```

Definition at line 294 of file [ftypes.h](#).

**4.49.2.159 CMODE**

```
#define CMODE 5
```

Definition at line 295 of file [ftypes.h](#).

**4.49.2.160 VORTRANMODE**

```
#define VORTRANMODE 6
```

Definition at line 296 of file [ftypes.h](#).

**4.49.2.161 PFORTRANMODE**

```
#define PFORTRANMODE 7
```

Definition at line 297 of file [ftypes.h](#).

**4.49.2.162 DOUBLEFORTRANMODE**

```
#define DOUBLEFORTRANMODE 33
```

Definition at line 298 of file [ftypes.h](#).

**4.49.2.163 DOUBLEPRECISIONFLAG**

```
#define DOUBLEPRECISIONFLAG 32
```

Definition at line 299 of file [ftypes.h](#).

**4.49.2.164 NODOUBLEMASK**

```
#define NODOUBLEMASK 31
```

Definition at line 300 of file [ftypes.h](#).

**4.49.2.165 QUADRUPLEFORTRANMODE**

```
#define QUADRUPLEFORTRANMODE 65
```

Definition at line 301 of file [ftypes.h](#).

**4.49.2.166 QUADRUPLERECISIONFLAG**

```
#define QUADRUPLERECISIONFLAG 64
```

Definition at line 302 of file [ftypes.h](#).

**4.49.2.167 ALLINTEGERDOUBLE**

```
#define ALLINTEGERDOUBLE 128
```

Definition at line 303 of file [ftypes.h](#).

**4.49.2.168 NOQUADMASK**

```
#define NOQUADMASK 63
```

Definition at line 304 of file [ftypes.h](#).

**4.49.2.169 NORMALFORMAT**

```
#define NORMALFORMAT 0
```

Definition at line 305 of file [ftypes.h](#).

**4.49.2.170 NOSPACEFORMAT**

```
#define NOSPACEFORMAT 1
```

Definition at line 306 of file [ftypes.h](#).

**4.49.2.171 ISNOTFORTRAN90**

```
#define ISNOTFORTRAN90 0
```

Definition at line 308 of file [ftypes.h](#).

**4.49.2.172 ISFORTRAN90**

```
#define ISFORTRAN90 1
```

Definition at line 309 of file [ftypes.h](#).

**4.49.2.173 ALSOREVERSE**

```
#define ALSOREVERSE 1
```

Definition at line 311 of file [ftypes.h](#).

**4.49.2.174 CHISHOLM**

```
#define CHISHOLM 2
```

Definition at line 312 of file [ftypes.h](#).

**4.49.2.175 NOTRICK**

```
#define NOTRICK 16
```

Definition at line 313 of file [ftypes.h](#).

**4.49.2.176 SORTLOWFIRST**

```
#define SORTLOWFIRST 0
```

Definition at line 315 of file [ftypes.h](#).

**4.49.2.177 SORTHIGHFIRST**

```
#define SORTHIGHFIRST 1
```

Definition at line 316 of file [ftypes.h](#).

**4.49.2.178 SORTPOWERFIRST**

```
#define SORTPOWERFIRST 2
```

Definition at line 317 of file [ftypes.h](#).

**4.49.2.179 SORTANTIPOWER**

```
#define SORTANTIPOWER 3
```

Definition at line 318 of file [ftypes.h](#).

**4.49.2.180 STATSSPLITMERGE**

```
#define STATSSPLITMERGE 0
```

Definition at line 324 of file [ftypes.h](#).

**4.49.2.181 STATSMERGETOFILE**

```
#define STATSMERGETOFILE 1
```

Definition at line 325 of file [ftypes.h](#).

**4.49.2.182 STATPOSTSORT**

```
#define STATPOSTSORT 2
```

Definition at line 326 of file [ftypes.h](#).

**4.49.2.183 NOCHECKLOGTYPE**

```
#define NOCHECKLOGTYPE 0
```

Definition at line 329 of file [ftypes.h](#).

**4.49.2.184 CHECKLOGTYPE**

```
#define CHECKLOGTYPE 1
```

Definition at line 330 of file [ftypes.h](#).

**4.49.2.185 NMIN4SHIFT**

```
#define NMIN4SHIFT 4
```

Definition at line 332 of file [ftypes.h](#).

**4.49.2.186 SYMBOL**

```
#define SYMBOL 1
```

Definition at line 346 of file [ftypes.h](#).

**4.49.2.187 DOTPRODUCT**

```
#define DOTPRODUCT 2
```

Definition at line 347 of file [ftypes.h](#).

**4.49.2.188 VECTOR**

```
#define VECTOR 3
```

Definition at line 348 of file [ftypes.h](#).

**4.49.2.189 INDEX**

```
#define INDEX 4
```

Definition at line 349 of file [ftypes.h](#).

**4.49.2.190 EXPRESSION**

```
#define EXPRESSION 5
```

Definition at line 350 of file [ftypes.h](#).

**4.49.2.191 SUBEXPRESSION**

```
#define SUBEXPRESSION 6
```

Definition at line 351 of file [ftypes.h](#).

**4.49.2.192 DOLLAREXPRESSION**

```
#define DOLLAREXPRESSION 7
```

Definition at line 352 of file [ftypes.h](#).

**4.49.2.193 SETSET**

```
#define SETSET 8
```

Definition at line 353 of file [ftypes.h](#).

**4.49.2.194 ARGWILD**

```
#define ARGWILD 9
```

Definition at line 354 of file [ftypes.h](#).

**4.49.2.195 MINVECTOR**

```
#define MINVECTOR 10
```

Definition at line 355 of file [ftypes.h](#).

**4.49.2.196 SETEXP**

```
#define SETEXP 11
```

Definition at line 356 of file [ftypes.h](#).

**4.49.2.197 DOLLAREXPR2**

```
#define DOLLAREXPR2 12
```

Definition at line 357 of file [ftypes.h](#).



**4.49.2.198 HAAKJE0**

```
#define HAAKJE0 9
```

Definition at line 358 of file [ftypes.h](#).

**4.49.2.199 FUNCTION**

```
#define FUNCTION 20
```

Definition at line 359 of file [ftypes.h](#).

**4.49.2.200 TMPPOLYFUN**

```
#define TMPPOLYFUN 14
```

Definition at line 361 of file [ftypes.h](#).

**4.49.2.201 ARGFIELD**

```
#define ARGFIELD 15
```

Definition at line 362 of file [ftypes.h](#).

**4.49.2.202 SNUMBER**

```
#define SNUMBER 16
```

Definition at line 363 of file [ftypes.h](#).

**4.49.2.203 LNUMBER**

```
#define LNUMBER 17
```

Definition at line 364 of file [ftypes.h](#).

**4.49.2.204 HAAKJE**

```
#define HAAKJE 18
```

Definition at line 365 of file [ftypes.h](#).

**4.49.2.205 DELTA**

```
#define DELTA 19
```

Definition at line 366 of file [ftypes.h](#).

#### 4.49.2.206 EXPONENT

```
#define EXPONENT 20
```

Definition at line 367 of file [ftypes.h](#).

#### 4.49.2.207 DENOMINATOR

```
#define DENOMINATOR 21
```

Definition at line 368 of file [ftypes.h](#).

#### 4.49.2.208 SETFUNCTION

```
#define SETFUNCTION 22
```

Definition at line 369 of file [ftypes.h](#).

#### 4.49.2.209 GAMMA

```
#define GAMMA 23
```

Definition at line 370 of file [ftypes.h](#).

#### 4.49.2.210 GAMMAI

```
#define GAMMAI 24
```

Definition at line 371 of file [ftypes.h](#).

#### 4.49.2.211 GAMMAFIVE

```
#define GAMMAFIVE 25
```

Definition at line 372 of file [ftypes.h](#).

#### 4.49.2.212 GAMMASIX

```
#define GAMMASIX 26
```

Definition at line 373 of file [ftypes.h](#).

#### 4.49.2.213 GAMMASEVEN

```
#define GAMMASEVEN 27
```

Definition at line 374 of file [ftypes.h](#).

**4.49.2.214 SUMF1**

```
#define SUMF1 28
```

Definition at line 375 of file [ftypes.h](#).

**4.49.2.215 SUMF2**

```
#define SUMF2 29
```

Definition at line 376 of file [ftypes.h](#).

**4.49.2.216 DUMFUN**

```
#define DUMFUN 30
```

Definition at line 377 of file [ftypes.h](#).

**4.49.2.217 REPLACEMENT**

```
#define REPLACEMENT 31
```

Definition at line 378 of file [ftypes.h](#).

**4.49.2.218 REVERSEFUNCTION**

```
#define REVERSEFUNCTION 32
```

Definition at line 379 of file [ftypes.h](#).

**4.49.2.219 DISTRIBUTION**

```
#define DISTRIBUTION 33
```

Definition at line 380 of file [ftypes.h](#).

**4.49.2.220 DELTA3**

```
#define DELTA3 34
```

Definition at line 381 of file [ftypes.h](#).

**4.49.2.221 DUMMYFUN**

```
#define DUMMYFUN 35
```

Definition at line 382 of file [ftypes.h](#).

**4.49.2.222 DUMMYTEN**

```
#define DUMMYTEN 36
```

Definition at line 383 of file [ftypes.h](#).

**4.49.2.223 LEVICIVITA**

```
#define LEVICIVITA 37
```

Definition at line 384 of file [ftypes.h](#).

**4.49.2.224 FACTORIAL**

```
#define FACTORIAL 38
```

Definition at line 385 of file [ftypes.h](#).

**4.49.2.225 INVERSEFACTORIAL**

```
#define INVERSEFACTORIAL 39
```

Definition at line 386 of file [ftypes.h](#).

**4.49.2.226 BINOMIAL**

```
#define BINOMIAL 40
```

Definition at line 387 of file [ftypes.h](#).

**4.49.2.227 NUMARGSFUN**

```
#define NUMARGSFUN 41
```

Definition at line 388 of file [ftypes.h](#).

**4.49.2.228 SIGNFUN**

```
#define SIGNFUN 42
```

Definition at line 389 of file [ftypes.h](#).

**4.49.2.229 MODFUNCTION**

```
#define MODFUNCTION 43
```

Definition at line 390 of file [ftypes.h](#).

#### 4.49.2.230 MOD2FUNCTION

```
#define MOD2FUNCTION 44
```

Definition at line 391 of file [ftypes.h](#).

#### 4.49.2.231 MINFUNCTION

```
#define MINFUNCTION 45
```

Definition at line 392 of file [ftypes.h](#).

#### 4.49.2.232 MAXFUNCTION

```
#define MAXFUNCTION 46
```

Definition at line 393 of file [ftypes.h](#).

#### 4.49.2.233 ABSFUNCTION

```
#define ABSFUNCTION 47
```

Definition at line 394 of file [ftypes.h](#).

#### 4.49.2.234 SIGFUNCTION

```
#define SIGFUNCTION 48
```

Definition at line 395 of file [ftypes.h](#).

#### 4.49.2.235 INTFUNCTION

```
#define INTFUNCTION 49
```

Definition at line 396 of file [ftypes.h](#).

#### 4.49.2.236 THETA

```
#define THETA 50
```

Definition at line 397 of file [ftypes.h](#).

#### 4.49.2.237 THETA2

```
#define THETA2 51
```

Definition at line 398 of file [ftypes.h](#).

**4.49.2.238 DELTA2**

```
#define DELTA2 52
```

Definition at line 399 of file [ftypes.h](#).

**4.49.2.239 DELTAP**

```
#define DELTAP 53
```

Definition at line 400 of file [ftypes.h](#).

**4.49.2.240 BERNOULLIFUNCTION**

```
#define BERNOULLIFUNCTION 54
```

Definition at line 401 of file [ftypes.h](#).

**4.49.2.241 COUNTFUNCTION**

```
#define COUNTFUNCTION 55
```

Definition at line 402 of file [ftypes.h](#).

**4.49.2.242 MATCHFUNCTION**

```
#define MATCHFUNCTION 56
```

Definition at line 403 of file [ftypes.h](#).

**4.49.2.243 PATTERNFUNCTION**

```
#define PATTERNFUNCTION 57
```

Definition at line 404 of file [ftypes.h](#).

**4.49.2.244 TERMFUNCTION**

```
#define TERMFUNCTION 58
```

Definition at line 405 of file [ftypes.h](#).

**4.49.2.245 CONJUGATION**

```
#define CONJUGATION 59
```

Definition at line 406 of file [ftypes.h](#).

**4.49.2.246 ROOTFUNCTION**

```
#define ROOTFUNCTION 60
```

Definition at line 407 of file [ftypes.h](#).

**4.49.2.247 TABLEFUNCTION**

```
#define TABLEFUNCTION 61
```

Definition at line 408 of file [ftypes.h](#).

**4.49.2.248 FIRSTBRACKET**

```
#define FIRSTBRACKET 62
```

Definition at line 409 of file [ftypes.h](#).

**4.49.2.249 TERMSINEXPR**

```
#define TERMSINEXPR 63
```

Definition at line 410 of file [ftypes.h](#).

**4.49.2.250 NUMTERMSFUN**

```
#define NUMTERMSFUN 64
```

Definition at line 411 of file [ftypes.h](#).

**4.49.2.251 GCDFUNCTION**

```
#define GCDFUNCTION 65
```

Definition at line 412 of file [ftypes.h](#).

**4.49.2.252 DIVFUNCTION**

```
#define DIVFUNCTION 66
```

Definition at line 413 of file [ftypes.h](#).

**4.49.2.253 REMFUNCTION**

```
#define REMFUNCTION 67
```

Definition at line 414 of file [ftypes.h](#).

**4.49.2.254 MAXPOWEROF**

```
#define MAXPOWEROF 68
```

Definition at line 415 of file [ftypes.h](#).

**4.49.2.255 MINPOWEROF**

```
#define MINPOWEROF 69
```

Definition at line 416 of file [ftypes.h](#).

**4.49.2.256 TABLESTUB**

```
#define TABLESTUB 70
```

Definition at line 417 of file [ftypes.h](#).

**4.49.2.257 FACTORIN**

```
#define FACTORIN 71
```

Definition at line 418 of file [ftypes.h](#).

**4.49.2.258 TERMSINBRACKET**

```
#define TERMSINBRACKET 72
```

Definition at line 419 of file [ftypes.h](#).

**4.49.2.259 WILDARGFUN**

```
#define WILDARGFUN 73
```

Definition at line 420 of file [ftypes.h](#).

**4.49.2.260 SQRTFUNCTION**

```
#define SQRTFUNCTION 74
```

Definition at line 428 of file [ftypes.h](#).

**4.49.2.261 LNFUNCTION**

```
#define LNFUNCTION 75
```

Definition at line 429 of file [ftypes.h](#).



**4.49.2.262 SINFUNCTION**

```
#define SINFUNCTION 76
```

Definition at line 430 of file [ftypes.h](#).

**4.49.2.263 COSFUNCTION**

```
#define COSFUNCTION 77
```

Definition at line 431 of file [ftypes.h](#).

**4.49.2.264 TANFUNCTION**

```
#define TANFUNCTION 78
```

Definition at line 432 of file [ftypes.h](#).

**4.49.2.265 ASINFUNCTION**

```
#define ASINFUNCTION 79
```

Definition at line 433 of file [ftypes.h](#).

**4.49.2.266 ACOSFUNCTION**

```
#define ACOSFUNCTION 80
```

Definition at line 434 of file [ftypes.h](#).

**4.49.2.267 ATANFUNCTION**

```
#define ATANFUNCTION 81
```

Definition at line 435 of file [ftypes.h](#).

**4.49.2.268 ATAN2FUNCTION**

```
#define ATAN2FUNCTION 82
```

Definition at line 436 of file [ftypes.h](#).

**4.49.2.269 SINHFUNCTION**

```
#define SINHFUNCTION 83
```

Definition at line 437 of file [ftypes.h](#).

#### 4.49.2.270 COSHFUNCTION

```
#define COSHFUNCTION 84
```

Definition at line 438 of file [ftypes.h](#).

#### 4.49.2.271 TANHFUNCTION

```
#define TANHFUNCTION 85
```

Definition at line 439 of file [ftypes.h](#).

#### 4.49.2.272 ASINHFUNCTION

```
#define ASINHFUNCTION 86
```

Definition at line 440 of file [ftypes.h](#).

#### 4.49.2.273 ACOSHFUNCTION

```
#define ACOSHFUNCTION 87
```

Definition at line 441 of file [ftypes.h](#).

#### 4.49.2.274 ATANHFUNCTION

```
#define ATANHFUNCTION 88
```

Definition at line 442 of file [ftypes.h](#).

#### 4.49.2.275 LI2FUNCTION

```
#define LI2FUNCTION 89
```

Definition at line 443 of file [ftypes.h](#).

#### 4.49.2.276 LINFUNCTION

```
#define LINFUNCTION 90
```

Definition at line 444 of file [ftypes.h](#).

#### 4.49.2.277 EXTRASYMFUN

```
#define EXTRASYMFUN 91
```

Definition at line 446 of file [ftypes.h](#).

**4.49.2.278 RANDOMFUNCTION**

```
#define RANDOMFUNCTION 92
```

Definition at line 447 of file [ftypes.h](#).

**4.49.2.279 RANPERM**

```
#define RANPERM 93
```

Definition at line 448 of file [ftypes.h](#).

**4.49.2.280 NUMFACTORS**

```
#define NUMFACTORS 94
```

Definition at line 449 of file [ftypes.h](#).

**4.49.2.281 FIRSTTERM**

```
#define FIRSTTERM 95
```

Definition at line 450 of file [ftypes.h](#).

**4.49.2.282 CONTENTTERM**

```
#define CONTENTTERM 96
```

Definition at line 451 of file [ftypes.h](#).

**4.49.2.283 PRIMENUMBER**

```
#define PRIMENUMBER 97
```

Definition at line 452 of file [ftypes.h](#).

**4.49.2.284 EXTEUCLIDEAN**

```
#define EXTEUCLIDEAN 98
```

Definition at line 453 of file [ftypes.h](#).

**4.49.2.285 MAKERATIONAL**

```
#define MAKERATIONAL 99
```

Definition at line 454 of file [ftypes.h](#).

#### 4.49.2.286 INVERSEFUNCTION

```
#define INVERSEFUNCTION 100
```

Definition at line 455 of file [ftypes.h](#).

#### 4.49.2.287 IDFUNCTION

```
#define IDFUNCTION 101
```

Definition at line 456 of file [ftypes.h](#).

#### 4.49.2.288 PUTFIRST

```
#define PUTFIRST 102
```

Definition at line 457 of file [ftypes.h](#).

#### 4.49.2.289 PERMUTATIONS

```
#define PERMUTATIONS 103
```

Definition at line 458 of file [ftypes.h](#).

#### 4.49.2.290 PARTITIONS

```
#define PARTITIONS 104
```

Definition at line 459 of file [ftypes.h](#).

#### 4.49.2.291 MULFUNCTION

```
#define MULFUNCTION 105
```

Definition at line 460 of file [ftypes.h](#).

#### 4.49.2.292 MOEBIUS

```
#define MOEBIUS 106
```

Definition at line 461 of file [ftypes.h](#).

#### 4.49.2.293 TOPOLOGIES

```
#define TOPOLOGIES 107
```

Definition at line 462 of file [ftypes.h](#).

#### 4.49.2.294 DIAGRAMS

```
#define DIAGRAMS 108
```

Definition at line 463 of file [ftypes.h](#).

#### 4.49.2.295 TOPO

```
#define TOPO 109
```

Definition at line 464 of file [ftypes.h](#).

#### 4.49.2.296 NODEFUNCTION

```
#define NODEFUNCTION 110
```

Definition at line 465 of file [ftypes.h](#).

#### 4.49.2.297 EDGE

```
#define EDGE 111
```

Definition at line 466 of file [ftypes.h](#).

#### 4.49.2.298 SIZEOFFUNCTION

```
#define SIZEOFFUNCTION 112
```

Definition at line 467 of file [ftypes.h](#).

#### 4.49.2.299 BLOCK

```
#define BLOCK 113
```

Definition at line 468 of file [ftypes.h](#).

#### 4.49.2.300 ONEPI

```
#define ONEPI 114
```

Definition at line 469 of file [ftypes.h](#).

#### 4.49.2.301 PHI

```
#define PHI 115
```

Definition at line 470 of file [ftypes.h](#).

#### 4.49.2.302 MAXBUILTINFUNCTION

```
#define MAXBUILTINFUNCTION 115
```

Definition at line 482 of file [ftypes.h](#).

#### 4.49.2.303 FIRSTUSERFUNCTION

```
#define FIRSTUSERFUNCTION 150
```

Definition at line 485 of file [ftypes.h](#).

#### 4.49.2.304 ALLFUNCTIONS

```
#define ALLFUNCTIONS -1
```

Definition at line 487 of file [ftypes.h](#).

#### 4.49.2.305 ALLMZVFUNCTIONS

```
#define ALLMZVFUNCTIONS -2
```

Definition at line 488 of file [ftypes.h](#).

#### 4.49.2.306 ISYMBOL

```
#define ISYMBOL 0
```

Definition at line 496 of file [ftypes.h](#).

#### 4.49.2.307 PISYMBOL

```
#define PISYMBOL 1
```

Definition at line 497 of file [ftypes.h](#).

#### 4.49.2.308 COEFFSYMBOL

```
#define COEFFSYMBOL 2
```

Definition at line 498 of file [ftypes.h](#).

#### 4.49.2.309 NUMERATORSYMBOL

```
#define NUMERATORSYMBOL 3
```

Definition at line 499 of file [ftypes.h](#).

**4.49.2.310 DENOMINATORSYMBOL**

```
#define DENOMINATORSYMBOL 4
```

Definition at line 500 of file [ftypes.h](#).

**4.49.2.311 WILDARGSYMBOL**

```
#define WILDARGSYMBOL 5
```

Definition at line 501 of file [ftypes.h](#).

**4.49.2.312 DIMENSIONSYPMBOL**

```
#define DIMENSIONSYPMBOL 6
```

Definition at line 502 of file [ftypes.h](#).

**4.49.2.313 FACTORSYMBOL**

```
#define FACTORSYMBOL 7
```

Definition at line 503 of file [ftypes.h](#).

**4.49.2.314 SEPARATESYMBOL**

```
#define SEPARATESYMBOL 8
```

Definition at line 504 of file [ftypes.h](#).

**4.49.2.315 EESYMBOL**

```
#define EESYMBOL 9
```

Definition at line 505 of file [ftypes.h](#).

**4.49.2.316 EMSYMBOL**

```
#define EMSYMBOL 10
```

Definition at line 506 of file [ftypes.h](#).

**4.49.2.317 BUILTINSYMBOLS**

```
#define BUILTINSYMBOLS 11
```

Definition at line 508 of file [ftypes.h](#).

#### 4.49.2.318 FIRSTUSERSYMBOL

```
#define FIRSTUSERSYMBOL 20
```

Definition at line 509 of file [ftypes.h](#).

#### 4.49.2.319 BUILTININDICES

```
#define BUILTININDICES 1
```

Definition at line 511 of file [ftypes.h](#).

#### 4.49.2.320 BUILTINVECTORS

```
#define BUILTINVECTORS 1
```

Definition at line 512 of file [ftypes.h](#).

#### 4.49.2.321 BUILTINDOLLARS

```
#define BUILTINDOLLARS 1
```

Definition at line 513 of file [ftypes.h](#).

#### 4.49.2.322 WILDARGVECTOR

```
#define WILDARGVECTOR 0
```

Definition at line 515 of file [ftypes.h](#).

#### 4.49.2.323 WILDARGINDEX

```
#define WILDARGINDEX 0
```

Definition at line 516 of file [ftypes.h](#).

#### 4.49.2.324 TYPEEXPRESSION

```
#define TYPEEXPRESSION 0
```

Definition at line 527 of file [ftypes.h](#).

#### 4.49.2.325 TYPEIDNEW

```
#define TYPEIDNEW 1
```

Definition at line 528 of file [ftypes.h](#).



**4.49.2.326 TYPEIDOLD**

```
#define TYPEIDOLD 2
```

Definition at line 529 of file [ftypes.h](#).

**4.49.2.327 TYPEOPERATION**

```
#define TYPEOPERATION 3
```

Definition at line 530 of file [ftypes.h](#).

**4.49.2.328 TYPEEREPEAT**

```
#define TYPEEREPEAT 4
```

Definition at line 531 of file [ftypes.h](#).

**4.49.2.329 TYPEENDREPEAT**

```
#define TYPEENDREPEAT 5
```

Definition at line 532 of file [ftypes.h](#).

**4.49.2.330 TYPECOUNT**

```
#define TYPECOUNT 20
```

Definition at line 536 of file [ftypes.h](#).

**4.49.2.331 TYPEMULT**

```
#define TYPEMULT 21
```

Definition at line 537 of file [ftypes.h](#).

**4.49.2.332 TYPEGOTO**

```
#define TYPEGOTO 22
```

Definition at line 538 of file [ftypes.h](#).

**4.49.2.333 TYPEDISCARD**

```
#define TYPEDISCARD 23
```

Definition at line 539 of file [ftypes.h](#).

#### 4.49.2.334 TYPEIF

```
#define TYPEIF 24
```

Definition at line 540 of file [ftypes.h](#).

#### 4.49.2.335 TYPEELSE

```
#define TYPEELSE 25
```

Definition at line 541 of file [ftypes.h](#).

#### 4.49.2.336 TYPEELIF

```
#define TYPEELIF 26
```

Definition at line 542 of file [ftypes.h](#).

#### 4.49.2.337 TYPEENDIF

```
#define TYPEENDIF 27
```

Definition at line 543 of file [ftypes.h](#).

#### 4.49.2.338 TYPESUM

```
#define TYPESUM 28
```

Definition at line 544 of file [ftypes.h](#).

#### 4.49.2.339 TYPECHISHOLM

```
#define TYPECHISHOLM 29
```

Definition at line 545 of file [ftypes.h](#).

#### 4.49.2.340 TPEREVERSE

```
#define TPEREVERSE 30
```

Definition at line 546 of file [ftypes.h](#).

#### 4.49.2.341 TYPEARG

```
#define TYPEARG 31
```

Definition at line 547 of file [ftypes.h](#).

**4.49.2.342 TYPENORM**

```
#define TYPENORM 32
```

Definition at line 548 of file [ftypes.h](#).

**4.49.2.343 TYPENORM2**

```
#define TYPENORM2 33
```

Definition at line 549 of file [ftypes.h](#).

**4.49.2.344 TYPENORM3**

```
#define TYPENORM3 34
```

Definition at line 550 of file [ftypes.h](#).

**4.49.2.345 TYPEEXIT**

```
#define TYPEEXIT 35
```

Definition at line 551 of file [ftypes.h](#).

**4.49.2.346 TYPESETEXIT**

```
#define TYPESETEXIT 36
```

Definition at line 552 of file [ftypes.h](#).

**4.49.2.347 TYPEPRINT**

```
#define TYPEPRINT 37
```

Definition at line 553 of file [ftypes.h](#).

**4.49.2.348 TYPEFPRINT**

```
#define TYPEFPRINT 38
```

Definition at line 554 of file [ftypes.h](#).

**4.49.2.349 TPEREDEFPRE**

```
#define TPEREDEFPRE 39
```

Definition at line 555 of file [ftypes.h](#).

**4.49.2.350 TYPESPLITARG**

```
#define TYPESPLITARG 40
```

Definition at line 556 of file [ftypes.h](#).

**4.49.2.351 TYPESPLITARG2**

```
#define TYPESPLITARG2 41
```

Definition at line 557 of file [ftypes.h](#).

**4.49.2.352 TYPEFACTARG**

```
#define TYPEFACTARG 42
```

Definition at line 558 of file [ftypes.h](#).

**4.49.2.353 TYPEFACTARG2**

```
#define TYPEFACTARG2 43
```

Definition at line 559 of file [ftypes.h](#).

**4.49.2.354 TYPETRY**

```
#define TYPETRY 44
```

Definition at line 560 of file [ftypes.h](#).

**4.49.2.355 TYPEASSIGN**

```
#define TYPEASSIGN 45
```

Definition at line 561 of file [ftypes.h](#).

**4.49.2.356 TYPERENUMBER**

```
#define TYPERENUMBER 46
```

Definition at line 562 of file [ftypes.h](#).

**4.49.2.357 TYPESUMFIX**

```
#define TYPESUMFIX 47
```

Definition at line 563 of file [ftypes.h](#).

**4.49.2.358 TYPEFINDLOOP**

```
#define TYPEFINDLOOP 48
```

Definition at line 564 of file [ftypes.h](#).

**4.49.2.359 TYPEUNRAVEL**

```
#define TYPEUNRAVEL 49
```

Definition at line 565 of file [ftypes.h](#).

**4.49.2.360 TYPEADJUSTBOUNDS**

```
#define TYPEADJUSTBOUNDS 50
```

Definition at line 566 of file [ftypes.h](#).

**4.49.2.361 TYPEINSIDE**

```
#define TYPEINSIDE 51
```

Definition at line 567 of file [ftypes.h](#).

**4.49.2.362 TYPETERM**

```
#define TYPETERM 52
```

Definition at line 568 of file [ftypes.h](#).

**4.49.2.363 TYPESORT**

```
#define TYPESORT 53
```

Definition at line 569 of file [ftypes.h](#).

**4.49.2.364 TYPEDETCURDUM**

```
#define TYPEDETCURDUM 54
```

Definition at line 570 of file [ftypes.h](#).

**4.49.2.365 TYPEINEXPRESSION**

```
#define TYPEINEXPRESSION 55
```

Definition at line 571 of file [ftypes.h](#).

**4.49.2.366 TYPESPLITFIRSTARG**

```
#define TYPESPLITFIRSTARG 56
```

Definition at line 572 of file [ftypes.h](#).

**4.49.2.367 TYPESPLITLASTARG**

```
#define TYPESPLITLASTARG 57
```

Definition at line 573 of file [ftypes.h](#).

**4.49.2.368 TYPEMERGE**

```
#define TYPEMERGE 58
```

Definition at line 574 of file [ftypes.h](#).

**4.49.2.369 TYPETESTUSE**

```
#define TYPETESTUSE 59
```

Definition at line 575 of file [ftypes.h](#).

**4.49.2.370 TYPEAPPLY**

```
#define TYPEAPPLY 60
```

Definition at line 576 of file [ftypes.h](#).

**4.49.2.371 TYPEAPPLYRESET**

```
#define TYPEAPPLYRESET 61
```

Definition at line 577 of file [ftypes.h](#).

**4.49.2.372 TYPECHAININ**

```
#define TYPECHAININ 62
```

Definition at line 578 of file [ftypes.h](#).

**4.49.2.373 TYPECHAINOUT**

```
#define TYPECHAINOUT 63
```

Definition at line 579 of file [ftypes.h](#).

**4.49.2.374 TYPENORM4**

```
#define TYPENORM4 64
```

Definition at line 580 of file [ftypes.h](#).

**4.49.2.375 TYPEFACTOR**

```
#define TYPEFACTOR 65
```

Definition at line 581 of file [ftypes.h](#).

**4.49.2.376 TYPEARGIMPLODE**

```
#define TYPEARGIMPLODE 66
```

Definition at line 582 of file [ftypes.h](#).

**4.49.2.377 TYPEARGEXPLODE**

```
#define TYPEARGEXPLODE 67
```

Definition at line 583 of file [ftypes.h](#).

**4.49.2.378 TYPEDENOMINATORS**

```
#define TYPEDENOMINATORS 68
```

Definition at line 584 of file [ftypes.h](#).

**4.49.2.379 TYPESTUFFLE**

```
#define TYPESTUFFLE 69
```

Definition at line 585 of file [ftypes.h](#).

**4.49.2.380 TYPEDROPCOEFFICIENT**

```
#define TYPEDROPCOEFFICIENT 70
```

Definition at line 586 of file [ftypes.h](#).

**4.49.2.381 TYPETRANSFORM**

```
#define TYPETRANSFORM 71
```

Definition at line 587 of file [ftypes.h](#).

**4.49.2.382    TYPETOPOLYNOMIAL**

```
#define TYPETOPOLYNOMIAL 72
```

Definition at line 588 of file [ftypes.h](#).

**4.49.2.383    TYPEFROMPOLYNOMIAL**

```
#define TYPEFROMPOLYNOMIAL 73
```

Definition at line 589 of file [ftypes.h](#).

**4.49.2.384    TYPEDOLOOP**

```
#define TYPEDOLOOP 74
```

Definition at line 590 of file [ftypes.h](#).

**4.49.2.385    TYPEENDDOLOOP**

```
#define TYPEENDDOLOOP 75
```

Definition at line 591 of file [ftypes.h](#).

**4.49.2.386    TYPEDROPSYMBOLS**

```
#define TYPEDROPSYMBOLS 76
```

Definition at line 592 of file [ftypes.h](#).

**4.49.2.387    TYPEPUTINSIDE**

```
#define TYPEPUTINSIDE 77
```

Definition at line 593 of file [ftypes.h](#).

**4.49.2.388    TYPETOSPECTATOR**

```
#define TYPETOSPECTATOR 78
```

Definition at line 594 of file [ftypes.h](#).

**4.49.2.389    TYPEARGTOEXTRASymbol**

```
#define TYPEARGTOEXTRASymbol 79
```

Definition at line 595 of file [ftypes.h](#).



**4.49.2.390 TYPECANONICALIZE**

```
#define TYPECANONICALIZE 80
```

Definition at line 596 of file [ftypes.h](#).

**4.49.2.391 TYPESWITCH**

```
#define TYPESWITCH 81
```

Definition at line 597 of file [ftypes.h](#).

**4.49.2.392 TYPEENDSWITCH**

```
#define TYPEENDSWITCH 82
```

Definition at line 598 of file [ftypes.h](#).

**4.49.2.393 TYPESETUSERFLAG**

```
#define TYPESETUSERFLAG 83
```

Definition at line 599 of file [ftypes.h](#).

**4.49.2.394 TYPECLEARUSERFLAG**

```
#define TYPECLEARUSERFLAG 84
```

Definition at line 600 of file [ftypes.h](#).

**4.49.2.395 TYPEALLLOOPS**

```
#define TYPEALLLOOPS 85
```

Definition at line 601 of file [ftypes.h](#).

**4.49.2.396 TYPEALLPATHS**

```
#define TYPEALLPATHS 86
```

Definition at line 602 of file [ftypes.h](#).

**4.49.2.397 TAKETRACE**

```
#define TAKETRACE 1
```

Definition at line 614 of file [ftypes.h](#).

**4.49.2.398 CONTRACT**

```
#define CONTRACT 2
```

Definition at line 615 of file [ftypes.h](#).

**4.49.2.399 RATIO**

```
#define RATIO 3
```

Definition at line 616 of file [ftypes.h](#).

**4.49.2.400 SYMMETRIZE**

```
#define SYMMETRIZE 4
```

Definition at line 617 of file [ftypes.h](#).

**4.49.2.401 TENVEC**

```
#define TENVEC 5
```

Definition at line 618 of file [ftypes.h](#).

**4.49.2.402 SUMNUM1**

```
#define SUMNUM1 6
```

Definition at line 619 of file [ftypes.h](#).

**4.49.2.403 SUMNUM2**

```
#define SUMNUM2 7
```

Definition at line 620 of file [ftypes.h](#).

**4.49.2.404 WILDDUMMY**

```
#define WILDDUMMY 0
```

Definition at line 626 of file [ftypes.h](#).

**4.49.2.405 SYMTONUM**

```
#define SYMTONUM 1
```

Definition at line 627 of file [ftypes.h](#).

**4.49.2.406 SYMTOSYM**

```
#define SYMTOSYM 2
```

Definition at line 628 of file [ftypes.h](#).

**4.49.2.407 SYMTOSUB**

```
#define SYMTOSUB 3
```

Definition at line 629 of file [ftypes.h](#).

**4.49.2.408 VECTOMIN**

```
#define VECTOMIN 4
```

Definition at line 630 of file [ftypes.h](#).

**4.49.2.409 VECTOVEC**

```
#define VECTOVEC 5
```

Definition at line 631 of file [ftypes.h](#).

**4.49.2.410 VECTOSUB**

```
#define VECTOSUB 6
```

Definition at line 632 of file [ftypes.h](#).

**4.49.2.411 INDTOIND**

```
#define INDTOIND 7
```

Definition at line 633 of file [ftypes.h](#).

**4.49.2.412 INDTOSUB**

```
#define INDTOSUB 8
```

Definition at line 634 of file [ftypes.h](#).

**4.49.2.413 FUNTOFUN**

```
#define FUNTOFUN 9
```

Definition at line 635 of file [ftypes.h](#).

**4.49.2.414 ARGTOARG**

```
#define ARGTOARG 10
```

Definition at line 636 of file [ftypes.h](#).

**4.49.2.415 ARLTOARL**

```
#define ARLTOARL 11
```

Definition at line 637 of file [ftypes.h](#).

**4.49.2.416 EXPTOEXP**

```
#define EXPTOEXP 12
```

Definition at line 638 of file [ftypes.h](#).

**4.49.2.417 FROMBRAC**

```
#define FROMBRAC 13
```

Definition at line 639 of file [ftypes.h](#).

**4.49.2.418 FROMSET**

```
#define FROMSET 14
```

Definition at line 640 of file [ftypes.h](#).

**4.49.2.419 SETTONUM**

```
#define SETTONUM 15
```

Definition at line 641 of file [ftypes.h](#).

**4.49.2.420 WILDCARDS**

```
#define WILDCARDS 16
```

Definition at line 642 of file [ftypes.h](#).

**4.49.2.421 SETNUMBER**

```
#define SETNUMBER 17
```

Definition at line 643 of file [ftypes.h](#).

**4.49.2.422 LOADDOLLAR**

```
#define LOADDOLLAR 18
```

Definition at line 644 of file [ftypes.h](#).

**4.49.2.423 NUMTONUM**

```
#define NUMTONUM 20
```

Definition at line 648 of file [ftypes.h](#).

**4.49.2.424 NUMTOSYM**

```
#define NUMTOSYM 21
```

Definition at line 649 of file [ftypes.h](#).

**4.49.2.425 NUMTOIND**

```
#define NUMTOIND 22
```

Definition at line 650 of file [ftypes.h](#).

**4.49.2.426 NUMTOSUB**

```
#define NUMTOSUB 23
```

Definition at line 651 of file [ftypes.h](#).

**4.49.2.427 EATTENSOR**

```
#define EATTENSOR 0x2000
```

Definition at line 655 of file [ftypes.h](#).

**4.49.2.428 CLEANFLAG**

```
#define CLEANFLAG 0
```

Definition at line 661 of file [ftypes.h](#).

**4.49.2.429 DIRTYFLAG**

```
#define DIRTYFLAG 1
```

Definition at line 662 of file [ftypes.h](#).

**4.49.2.430 DIRTYSYMFLAG**

```
#define DIRTYSYMFLAG 2
```

Definition at line 663 of file [ftypes.h](#).

**4.49.2.431 MUSTCLEANPRF**

```
#define MUSTCLEANPRF 4
```

Definition at line 664 of file [ftypes.h](#).

**4.49.2.432 SUBTERMUSED1**

```
#define SUBTERMUSED1 8
```

Definition at line 665 of file [ftypes.h](#).

**4.49.2.433 SUBTERMUSED2**

```
#define SUBTERMUSED2 16
```

Definition at line 666 of file [ftypes.h](#).

**4.49.2.434 ALLDIRTY**

```
#define ALLDIRTY (DIRTYFLAG|DIRTYSYMFLAG)
```

Definition at line 667 of file [ftypes.h](#).

**4.49.2.435 ARGHEAD**

```
#define ARGHEAD 2
```

Definition at line 669 of file [ftypes.h](#).

**4.49.2.436 FUNHEAD**

```
#define FUNHEAD 3
```

Definition at line 670 of file [ftypes.h](#).

**4.49.2.437 SUBEXPSIZE**

```
#define SUBEXPSIZE 5
```

Definition at line 671 of file [ftypes.h](#).

**4.49.2.438   EXPRHEAD**

```
#define EXPRHEAD 5
```

Definition at line 672 of file [ftypes.h](#).

**4.49.2.439   TYPEARGHEADSIZE**

```
#define TYPEARGHEADSIZE 6
```

Definition at line 673 of file [ftypes.h](#).

**4.49.2.440   SKIP**

```
#define SKIP 1
```

Definition at line 680 of file [ftypes.h](#).

**4.49.2.441   DROP**

```
#define DROP 2
```

Definition at line 681 of file [ftypes.h](#).

**4.49.2.442   HIDE**

```
#define HIDE 3
```

Definition at line 682 of file [ftypes.h](#).

**4.49.2.443   UNHIDE**

```
#define UNHIDE 4
```

Definition at line 683 of file [ftypes.h](#).

**4.49.2.444   INTOHIDE**

```
#define INTOHIDE 5
```

Definition at line 684 of file [ftypes.h](#).

**4.49.2.445   LOCALEXPRESSION**

```
#define LOCALEXPRESSION 0
```

Definition at line 690 of file [ftypes.h](#).

**4.49.2.446 SKIPLEXPRESSION**

```
#define SKIPLEXPRESSION 1
```

Definition at line 691 of file [ftypes.h](#).

**4.49.2.447 DROPLEXPRESSION**

```
#define DROPLEXPRESSION 2
```

Definition at line 692 of file [ftypes.h](#).

**4.49.2.448 DROPEDEXPRESSION**

```
#define DROPEDEXPRESSION 3
```

Definition at line 693 of file [ftypes.h](#).

**4.49.2.449 GLOBALEXPRESSION**

```
#define GLOBALEXPRESSION 4
```

Definition at line 694 of file [ftypes.h](#).

**4.49.2.450 SKIPGEXPRESSION**

```
#define SKIPGEXPRESSION 5
```

Definition at line 695 of file [ftypes.h](#).

**4.49.2.451 DROPGEXPRESSION**

```
#define DROPGEXPRESSION 6
```

Definition at line 696 of file [ftypes.h](#).

**4.49.2.452 STOREDEXPRESSION**

```
#define STOREDEXPRESSION 8
```

Definition at line 697 of file [ftypes.h](#).

**4.49.2.453 HIDDENLEXPRESSION**

```
#define HIDDENLEXPRESSION 9
```

Definition at line 698 of file [ftypes.h](#).



**4.49.2.454 HIDDENGEXPRESSION**

```
#define HIDDENGEXPRESSION 13
```

Definition at line 699 of file [ftypes.h](#).

**4.49.2.455 INCEXPRESSION**

```
#define INCEXPRESSION 9
```

Definition at line 700 of file [ftypes.h](#).

**4.49.2.456 HIDELEXPRESSION**

```
#define HIDELEXPRESSION 10
```

Definition at line 701 of file [ftypes.h](#).

**4.49.2.457 HIDEGEXPRESSION**

```
#define HIDEGEXPRESSION 14
```

Definition at line 702 of file [ftypes.h](#).

**4.49.2.458 DROPHLEXPRESSION**

```
#define DROPHLEXPRESSION 11
```

Definition at line 703 of file [ftypes.h](#).

**4.49.2.459 DROPHGEXPRESSION**

```
#define DROPHGEXPRESSION 15
```

Definition at line 704 of file [ftypes.h](#).

**4.49.2.460 UNHIDELEXPRESSION**

```
#define UNHIDELEXPRESSION 12
```

Definition at line 705 of file [ftypes.h](#).

**4.49.2.461 UNHIDEGEXPRESSION**

```
#define UNHIDEGEXPRESSION 16
```

Definition at line 706 of file [ftypes.h](#).

#### 4.49.2.462 INTOHIDELEXPRESSION

```
#define INTOHIDELEXPRESSION 17
```

Definition at line 707 of file [ftypes.h](#).

#### 4.49.2.463 INTOHIDEGEXPRESSION

```
#define INTOHIDEGEXPRESSION 18
```

Definition at line 708 of file [ftypes.h](#).

#### 4.49.2.464 SPECTATOREXPRESSION

```
#define SPECTATOREXPRESSION 19
```

Definition at line 709 of file [ftypes.h](#).

#### 4.49.2.465 DROPSPECTATOREXPRESSION

```
#define DROPSPECTATOREXPRESSION 20
```

Definition at line 710 of file [ftypes.h](#).

#### 4.49.2.466 SKIPUNHIDELEXPRESSION

```
#define SKIPUNHIDELEXPRESSION 21
```

Definition at line 711 of file [ftypes.h](#).

#### 4.49.2.467 SKIPUNHIDEGEXPRESSION

```
#define SKIPUNHIDEGEXPRESSION 22
```

Definition at line 712 of file [ftypes.h](#).

#### 4.49.2.468 PRINTOFF

```
#define PRINTOFF 0
```

Definition at line 714 of file [ftypes.h](#).

#### 4.49.2.469 PRINTON

```
#define PRINTON 1
```

Definition at line 715 of file [ftypes.h](#).

**4.49.2.470 PRINTCONTENTS**

```
#define PRINTCONTENTS 2
```

Definition at line 716 of file [ftypes.h](#).

**4.49.2.471 PRINTCONTENT**

```
#define PRINTCONTENT 3
```

Definition at line 717 of file [ftypes.h](#).

**4.49.2.472 PRINTLFILE**

```
#define PRINTLFILE 4
```

Definition at line 718 of file [ftypes.h](#).

**4.49.2.473 PRINTONETERM**

```
#define PRINTONETERM 8
```

Definition at line 719 of file [ftypes.h](#).

**4.49.2.474 PRINTONEFUNCTION**

```
#define PRINTONEFUNCTION 16
```

Definition at line 720 of file [ftypes.h](#).

**4.49.2.475 PRINTALL**

```
#define PRINTALL 32
```

Definition at line 721 of file [ftypes.h](#).

**4.49.2.476 REGULAREXPRESSION**

```
#define REGULAREXPRESSION -1
```

Definition at line 727 of file [ftypes.h](#).

**4.49.2.477 REDEFINEDEXPRESSION**

```
#define REDEFINEDEXPRESSION -2
```

Definition at line 728 of file [ftypes.h](#).

**4.49.2.478 NEWLYDEFINEDEXPRESSION**

```
#define NEWLYDEFINEDEXPRESSION -3
```

Definition at line 729 of file [ftypes.h](#).

**4.49.2.479 VERTEXFUNCTION**

```
#define VERTEXFUNCTION -1
```

Definition at line 738 of file [ftypes.h](#).

**4.49.2.480 GENERALFUNCTION**

```
#define GENERALFUNCTION 0
```

Definition at line 739 of file [ftypes.h](#).

**4.49.2.481 TENSORFUNCTION**

```
#define TENSORFUNCTION 2
```

Definition at line 740 of file [ftypes.h](#).

**4.49.2.482 GAMMAFUNCTION**

```
#define GAMMAFUNCTION 4
```

Definition at line 741 of file [ftypes.h](#).

**4.49.2.483 POS\_**

```
#define POS_ 0 /* integer > 0 */
```

Definition at line 748 of file [ftypes.h](#).

**4.49.2.484 POS0\_**

```
#define POS0_ 1 /* integer >= 0 */
```

Definition at line 749 of file [ftypes.h](#).

**4.49.2.485 NEG\_**

```
#define NEG_ 2 /* integer < 0 */
```

Definition at line 750 of file [ftypes.h](#).

**4.49.2.486 NEG0\_**

```
#define NEG0_ 3 /* integer <= 0 */
```

Definition at line 751 of file [ftypes.h](#).

**4.49.2.487 EVEN\_**

```
#define EVEN_ 4 /* integer (even) */
```

Definition at line 752 of file [ftypes.h](#).

**4.49.2.488 ODD\_**

```
#define ODD_ 5 /* integer (odd) */
```

Definition at line 753 of file [ftypes.h](#).

**4.49.2.489 Z\_**

```
#define Z_ 6 /* integer */
```

Definition at line 754 of file [ftypes.h](#).

**4.49.2.490 SYMBOL\_**

```
#define SYMBOL_ 7 /* symbol only */
```

Definition at line 755 of file [ftypes.h](#).

**4.49.2.491 FIXED\_**

```
#define FIXED_ 8 /* fixed index */
```

Definition at line 756 of file [ftypes.h](#).

**4.49.2.492 INDEX\_**

```
#define INDEX_ 9 /* index only */
```

Definition at line 757 of file [ftypes.h](#).

**4.49.2.493 Q\_**

```
#define Q_ 10 /* rational */
```

Definition at line 758 of file [ftypes.h](#).

**4.49.2.494 DUMMYINDEX\_**

```
#define DUMMYINDEX_ 11 /* dummy index only */
```

Definition at line 759 of file [ftypes.h](#).

**4.49.2.495 VECTOR\_**

```
#define VECTOR_ 12 /* vector only */
```

Definition at line 760 of file [ftypes.h](#).

**4.49.2.496 GAMMA1**

```
#define GAMMA1 0
```

Definition at line 766 of file [ftypes.h](#).

**4.49.2.497 GAMMA5**

```
#define GAMMA5 -1
```

Definition at line 767 of file [ftypes.h](#).

**4.49.2.498 GAMMA6**

```
#define GAMMA6 -2
```

Definition at line 768 of file [ftypes.h](#).

**4.49.2.499 GAMMA7**

```
#define GAMMA7 -3
```

Definition at line 769 of file [ftypes.h](#).

**4.49.2.500 FUNNYVEC**

```
#define FUNNYVEC -4
```

Definition at line 770 of file [ftypes.h](#).

**4.49.2.501 FUNNYWILD**

```
#define FUNNYWILD -5
```

Definition at line 771 of file [ftypes.h](#).

**4.49.2.502 SUMMEDIND**

```
#define SUMMEDIND -6
```

Definition at line 772 of file [ftypes.h](#).

**4.49.2.503 NOINDEX**

```
#define NOINDEX -7
```

Definition at line 773 of file [ftypes.h](#).

**4.49.2.504 FUNNYDOLLAR**

```
#define FUNNYDOLLAR -8
```

Definition at line 774 of file [ftypes.h](#).

**4.49.2.505 EMPTYINDEX**

```
#define EMPTYINDEX -9
```

Definition at line 775 of file [ftypes.h](#).

**4.49.2.506 MINSPEC**

```
#define MINSPEC -10
```

Definition at line 781 of file [ftypes.h](#).

**4.49.2.507 USEDFLAG**

```
#define USEDFLAG 2
```

Definition at line 783 of file [ftypes.h](#).

**4.49.2.508 DUMMYFLAG**

```
#define DUMMYFLAG 1
```

Definition at line 784 of file [ftypes.h](#).

**4.49.2.509 MAINSORT**

```
#define MAINSORT 0
```

Definition at line 786 of file [ftypes.h](#).

**4.49.2.510 FUNCTIONSORT**

```
#define FUNCTIONSORT 1
```

Definition at line 787 of file [ftypes.h](#).

**4.49.2.511 SUBSORT**

```
#define SUBSORT 2
```

Definition at line 788 of file [ftypes.h](#).

**4.49.2.512 FLOATMODE**

```
#define FLOATMODE 1
```

Definition at line 790 of file [ftypes.h](#).

**4.49.2.513 RATIONALMODE**

```
#define RATIONALMODE 0
```

Definition at line 791 of file [ftypes.h](#).

**4.49.2.514 NUMSPECSETS**

```
#define NUMSPECSETS 10
```

Definition at line 793 of file [ftypes.h](#).

**4.49.2.515 ISZERO**

```
#define ISZERO 1
```

Definition at line 795 of file [ftypes.h](#).

**4.49.2.516 ISUNMODIFIED**

```
#define ISUNMODIFIED 2
```

Definition at line 796 of file [ftypes.h](#).

**4.49.2.517 ISCOMPRESSED**

```
#define ISCOMPRESSED 4
```

Definition at line 797 of file [ftypes.h](#).



**4.49.2.518 ISINRHS**

```
#define ISINRHS 8
```

Definition at line 798 of file [ftypes.h](#).

**4.49.2.519 ISFACTORIZED**

```
#define ISFACTORIZED 16
```

Definition at line 799 of file [ftypes.h](#).

**4.49.2.520 TOBEFACTORED**

```
#define TOBEFACTORED 32
```

Definition at line 800 of file [ftypes.h](#).

**4.49.2.521 TOBEUNFACTORED**

```
#define TOBEUNFACTORED 64
```

Definition at line 801 of file [ftypes.h](#).

**4.49.2.522 KEEPZERO**

```
#define KEEPZERO 128
```

Definition at line 802 of file [ftypes.h](#).

**4.49.2.523 VARTYPENONE**

```
#define VARTYPENONE 0
```

Definition at line 804 of file [ftypes.h](#).

**4.49.2.524 VARTYPECOMPLEX**

```
#define VARTYPECOMPLEX 1
```

Definition at line 805 of file [ftypes.h](#).

**4.49.2.525 VARTYPEIMAGINARY**

```
#define VARTYPEIMAGINARY 2
```

Definition at line 806 of file [ftypes.h](#).

**4.49.2.526 VARTYPEROOTOFUNITY**

```
#define VARTYPEROOTOFUNITY 4
```

Definition at line 807 of file [ftypes.h](#).

**4.49.2.527 VARTYPEMINUS**

```
#define VARTYPEMINUS 8
```

Definition at line 808 of file [ftypes.h](#).

**4.49.2.528 CYCLESYMMETRIC**

```
#define CYCLESYMMETRIC 1
```

Definition at line 809 of file [ftypes.h](#).

**4.49.2.529 RCYCLESYMMETRIC**

```
#define RCYCLESYMMETRIC 2
```

Definition at line 810 of file [ftypes.h](#).

**4.49.2.530 SYMMETRIC**

```
#define SYMMETRIC 3
```

Definition at line 811 of file [ftypes.h](#).

**4.49.2.531 ANTISYMMETRIC**

```
#define ANTISYMMETRIC 4
```

Definition at line 812 of file [ftypes.h](#).

**4.49.2.532 REVERSEORDER**

```
#define REVERSEORDER 256
```

Definition at line 813 of file [ftypes.h](#).

**4.49.2.533 SUBMULTI**

```
#define SUBMULTI 1
```

Definition at line 819 of file [ftypes.h](#).

**4.49.2.534 SUBONCE**

```
#define SUBONCE 2
```

Definition at line 820 of file [ftypes.h](#).

**4.49.2.535 SUBONLY**

```
#define SUBONLY 3
```

Definition at line 821 of file [ftypes.h](#).

**4.49.2.536 SUBMANY**

```
#define SUBMANY 4
```

Definition at line 822 of file [ftypes.h](#).

**4.49.2.537 SUBVECTOR**

```
#define SUBVECTOR 5
```

Definition at line 823 of file [ftypes.h](#).

**4.49.2.538 SUBSELECT**

```
#define SUBSELECT 6
```

Definition at line 824 of file [ftypes.h](#).

**4.49.2.539 SUBALL**

```
#define SUBALL 7
```

Definition at line 825 of file [ftypes.h](#).

**4.49.2.540 SUBMASK**

```
#define SUBMASK 15
```

Definition at line 826 of file [ftypes.h](#).

**4.49.2.541 SUBDISORDER**

```
#define SUBDISORDER 16
```

Definition at line 827 of file [ftypes.h](#).

**4.49.2.542 SUBAFTER**

```
#define SUBAFTER 32
```

Definition at line 828 of file [ftypes.h](#).

**4.49.2.543 SUBAFTERNOT**

```
#define SUBAFTERNOT 64
```

Definition at line 829 of file [ftypes.h](#).

**4.49.2.544 IDHEAD**

```
#define IDHEAD 6
```

Definition at line 831 of file [ftypes.h](#).

**4.49.2.545 DOLLARFLAG**

```
#define DOLLARFLAG 1
```

Definition at line 833 of file [ftypes.h](#).

**4.49.2.546 NORMALIZEFLAG**

```
#define NORMALIZEFLAG 2
```

Definition at line 834 of file [ftypes.h](#).

**4.49.2.547 GIDENT**

```
#define GIDENT 1
```

Definition at line 836 of file [ftypes.h](#).

**4.49.2.548 GFIVE**

```
#define GFIVE 4
```

Definition at line 837 of file [ftypes.h](#).

**4.49.2.549 GPLUS**

```
#define GPLUS 3
```

Definition at line 838 of file [ftypes.h](#).

**4.49.2.550 GMINUS**

```
#define GMINUS 2
```

Definition at line 839 of file [ftypes.h](#).

**4.49.2.551 LONGNUMBER**

```
#define LONGNUMBER 1
```

Definition at line 845 of file [ftypes.h](#).

**4.49.2.552 MATCH**

```
#define MATCH 2
```

Definition at line 846 of file [ftypes.h](#).

**4.49.2.553 COEFFI**

```
#define COEFFI 3
```

Definition at line 847 of file [ftypes.h](#).

**4.49.2.554 SUBEXPR**

```
#define SUBEXPR 4
```

Definition at line 848 of file [ftypes.h](#).

**4.49.2.555 MULTIPLEOF**

```
#define MULTIPLEOF 5
```

Definition at line 849 of file [ftypes.h](#).

**4.49.2.556 IFDOLLAR**

```
#define IFDOLLAR 6
```

Definition at line 850 of file [ftypes.h](#).

**4.49.2.557 IFEXPRESSION**

```
#define IFEXPRESSION 7
```

Definition at line 851 of file [ftypes.h](#).

**4.49.2.558 IFDOLLAREXTRA**

```
#define IFDOLLAREXTRA 8
```

Definition at line 852 of file [ftypes.h](#).

**4.49.2.559 IFISFACTORIZED**

```
#define IFISFACTORIZED 9
```

Definition at line 853 of file [ftypes.h](#).

**4.49.2.560 IFOCCURS**

```
#define IFOCCURS 10
```

Definition at line 854 of file [ftypes.h](#).

**4.49.2.561 IFUSERFLAG**

```
#define IFUSERFLAG 11
```

Definition at line 855 of file [ftypes.h](#).

**4.49.2.562 IFFLOATNUMBER**

```
#define IFFLOATNUMBER 12
```

Definition at line 856 of file [ftypes.h](#).

**4.49.2.563 GREATER**

```
#define GREATER 0
```

Definition at line 857 of file [ftypes.h](#).

**4.49.2.564 GREATEREQUAL**

```
#define GREATEREQUAL 1
```

Definition at line 858 of file [ftypes.h](#).

**4.49.2.565 LESS**

```
#define LESS 2
```

Definition at line 859 of file [ftypes.h](#).

**4.49.2.566 LESSEQUAL**

```
#define LESSEQUAL 3
```

Definition at line 860 of file [ftypes.h](#).

**4.49.2.567 EQUAL**

```
#define EQUAL 4
```

Definition at line 861 of file [ftypes.h](#).

**4.49.2.568 NOTEQUAL**

```
#define NOTEQUAL 5
```

Definition at line 862 of file [ftypes.h](#).

**4.49.2.569 ORCOND**

```
#define ORCOND 6
```

Definition at line 863 of file [ftypes.h](#).

**4.49.2.570 ANDCOND**

```
#define ANDCOND 7
```

Definition at line 864 of file [ftypes.h](#).

**4.49.2.571 DUMMY**

```
#define DUMMY 1
```

Definition at line 865 of file [ftypes.h](#).

**4.49.2.572 SORT**

```
#define SORT 1
```

Definition at line 866 of file [ftypes.h](#).

**4.49.2.573 STORE**

```
#define STORE 2
```

Definition at line 867 of file [ftypes.h](#).

**4.49.2.574 END**

```
#define END 3
```

Definition at line 868 of file [ftypes.h](#).

**4.49.2.575 GLOBAL**

```
#define GLOBAL 4
```

Definition at line 869 of file [ftypes.h](#).

**4.49.2.576 CLEAR**

```
#define CLEAR 5
```

Definition at line 870 of file [ftypes.h](#).

**4.49.2.577 VECTBIT**

```
#define VECTBIT 1
```

Definition at line 872 of file [ftypes.h](#).

**4.49.2.578 DOTPBIT**

```
#define DOTPBIT 2
```

Definition at line 873 of file [ftypes.h](#).

**4.49.2.579 FUNBIT**

```
#define FUNBIT 4
```

Definition at line 874 of file [ftypes.h](#).

**4.49.2.580 SETBIT**

```
#define SETBIT 8
```

Definition at line 875 of file [ftypes.h](#).

**4.49.2.581 EXTRAPARAMETER**

```
#define EXTRAPARAMETER 0x4000
```

Definition at line 877 of file [ftypes.h](#).



**4.49.2.582 GENCOMMUTE**

```
#define GENCOMMUTE 0
```

Definition at line 878 of file [ftypes.h](#).

**4.49.2.583 GENNONCOMMUTE**

```
#define GENNONCOMMUTE 0x2000
```

Definition at line 879 of file [ftypes.h](#).

**4.49.2.584 NAMENOTFOUND**

```
#define NAMENOTFOUND -9
```

Definition at line 881 of file [ftypes.h](#).

**4.49.2.585 DOLUNDEFINED**

```
#define DOLUNDEFINED 0
```

Definition at line 887 of file [ftypes.h](#).

**4.49.2.586 DOLNUMBER**

```
#define DOLNUMBER 1
```

Definition at line 888 of file [ftypes.h](#).

**4.49.2.587 DOLARGUMENT**

```
#define DOLARGUMENT 2
```

Definition at line 889 of file [ftypes.h](#).

**4.49.2.588 DOLSUBTERM**

```
#define DOLSUBTERM 3
```

Definition at line 890 of file [ftypes.h](#).

**4.49.2.589 DOLTERMS**

```
#define DOLTERMS 4
```

Definition at line 891 of file [ftypes.h](#).

**4.49.2.590 DOLWILDARGS**

```
#define DOLWILDARGS 5
```

Definition at line 892 of file [ftypes.h](#).

**4.49.2.591 DOLINDEX**

```
#define DOLINDEX 6
```

Definition at line 893 of file [ftypes.h](#).

**4.49.2.592 DOLZERO**

```
#define DOLZERO 7
```

Definition at line 894 of file [ftypes.h](#).

**4.49.2.593 FINDLOOP**

```
#define FINDLOOP 0
```

Definition at line 896 of file [ftypes.h](#).

**4.49.2.594 REPLACELOOP**

```
#define REPLACELOOP 1
```

Definition at line 897 of file [ftypes.h](#).

**4.49.2.595 NOFUNPOWERS**

```
#define NOFUNPOWERS 0
```

Definition at line 899 of file [ftypes.h](#).

**4.49.2.596 COMFUNPOWERS**

```
#define COMFUNPOWERS 1
```

Definition at line 900 of file [ftypes.h](#).

**4.49.2.597 ALLFUNPOWERS**

```
#define ALLFUNPOWERS 2
```

Definition at line 901 of file [ftypes.h](#).

**4.49.2.598 PROPERORDERFLAG**

```
#define PROPERORDERFLAG 0
```

Definition at line 903 of file [ftypes.h](#).

**4.49.2.599 REGULAR**

```
#define REGULAR 0
```

Definition at line 905 of file [ftypes.h](#).

**4.49.2.600 FINISH**

```
#define FINISH 1
```

Definition at line 906 of file [ftypes.h](#).

**4.49.2.601 POLYADD**

```
#define POLYADD 1
```

Definition at line 908 of file [ftypes.h](#).

**4.49.2.602 POLYSUB**

```
#define POLYSUB 2
```

Definition at line 909 of file [ftypes.h](#).

**4.49.2.603 POLYMUL**

```
#define POLYMUL 3
```

Definition at line 910 of file [ftypes.h](#).

**4.49.2.604 POLYDIV**

```
#define POLYDIV 4
```

Definition at line 911 of file [ftypes.h](#).

**4.49.2.605 POLYREM**

```
#define POLYREM 5
```

Definition at line 912 of file [ftypes.h](#).

**4.49.2.606 POLYGCD**

```
#define POLYGCD 6
```

Definition at line 913 of file [ftypes.h](#).

**4.49.2.607 POLYINTFAC**

```
#define POLYINTFAC 7
```

Definition at line 914 of file [ftypes.h](#).

**4.49.2.608 POLYNORM**

```
#define POLYNORM 8
```

Definition at line 915 of file [ftypes.h](#).

**4.49.2.609 MODNONE**

```
#define MODNONE 0
```

Definition at line 917 of file [ftypes.h](#).

**4.49.2.610 MODSUM**

```
#define MODSUM 1
```

Definition at line 918 of file [ftypes.h](#).

**4.49.2.611 MODMAX**

```
#define MODMAX 2
```

Definition at line 919 of file [ftypes.h](#).

**4.49.2.612 MODMIN**

```
#define MODMIN 3
```

Definition at line 920 of file [ftypes.h](#).

**4.49.2.613 MODLOCAL**

```
#define MODLOCAL 4
```

Definition at line 921 of file [ftypes.h](#).

**4.49.2.614 ELEMENTUSED**

```
#define ELEMENTUSED 1
```

Definition at line 923 of file [ftypes.h](#).

**4.49.2.615 ELEMENTLOADED**

```
#define ELEMENTLOADED 2
```

Definition at line 924 of file [ftypes.h](#).

**4.49.2.616 POSNEG**

```
#define POSNEG 0x1
```

Definition at line 929 of file [ftypes.h](#).

**4.49.2.617 INVERSETABLE**

```
#define INVERSETABLE 0x2
```

Definition at line 930 of file [ftypes.h](#).

**4.49.2.618 COEFFICIENTSONLY**

```
#define COEFFICIENTSONLY 0x4
```

Definition at line 931 of file [ftypes.h](#).

**4.49.2.619 ALSOPOWERS**

```
#define ALSOPOWERS 0x8
```

Definition at line 932 of file [ftypes.h](#).

**4.49.2.620 ALSOFUNARGS**

```
#define ALSOFUNARGS 0x10
```

Definition at line 933 of file [ftypes.h](#).

**4.49.2.621 ALSODOLLARS**

```
#define ALSODOLLARS 0x20
```

Definition at line 934 of file [ftypes.h](#).

**4.49.2.622 NOINVERSES**

```
#define NOINVERSES 0x40
```

Definition at line 935 of file [ftypes.h](#).

**4.49.2.623 POSITIVEONLY**

```
#define POSITIVEONLY 0
```

Definition at line 937 of file [ftypes.h](#).

**4.49.2.624 UNPACK**

```
#define UNPACK 0x80
```

Definition at line 938 of file [ftypes.h](#).

**4.49.2.625 NOUNPACK**

```
#define NOUNPACK 0
```

Definition at line 939 of file [ftypes.h](#).

**4.49.2.626 FROMFUNCTION**

```
#define FROMFUNCTION 0x100
```

Definition at line 940 of file [ftypes.h](#).

**4.49.2.627 VARNames**

```
#define VARNames 0
```

Definition at line 942 of file [ftypes.h](#).

**4.49.2.628 AUTONAMES**

```
#define AUTONAMES 1
```

Definition at line 943 of file [ftypes.h](#).

**4.49.2.629 EXPRNames**

```
#define EXPRNames 2
```

Definition at line 944 of file [ftypes.h](#).

**4.49.2.630 DOLLARNAMES**

```
#define DOLLARNAMES 3
```

Definition at line 945 of file [ftypes.h](#).

**4.49.2.631 DUMMYBUFFER**

```
#define DUMMYBUFFER 1
```

Definition at line 1000 of file [ftypes.h](#).

**4.49.2.632 ALLARGS**

```
#define ALLARGS 1
```

Definition at line 1002 of file [ftypes.h](#).

**4.49.2.633 NUMARG**

```
#define NUMARG 2
```

Definition at line 1003 of file [ftypes.h](#).

**4.49.2.634 ARGRANGE**

```
#define ARGRANGE 3
```

Definition at line 1004 of file [ftypes.h](#).

**4.49.2.635 MAKEARGS**

```
#define MAKEARGS 4
```

Definition at line 1005 of file [ftypes.h](#).

**4.49.2.636 MAXRANGEINDICATOR**

```
#define MAXRANGEINDICATOR 4
```

Definition at line 1006 of file [ftypes.h](#).

**4.49.2.637 REPLACEARG**

```
#define REPLACEARG 5
```

Definition at line 1007 of file [ftypes.h](#).

**4.49.2.638 ENCODEARG**

```
#define ENCODEARG 6
```

Definition at line 1008 of file [ftypes.h](#).

**4.49.2.639 DECODEARG**

```
#define DECODEARG 7
```

Definition at line 1009 of file [ftypes.h](#).

**4.49.2.640 IMplodeARG**

```
#define IMplodeARG 8
```

Definition at line 1010 of file [ftypes.h](#).

**4.49.2.641 EXPLODEARG**

```
#define EXPLODEARG 9
```

Definition at line 1011 of file [ftypes.h](#).

**4.49.2.642 PERMUTEARG**

```
#define PERMUTEARG 10
```

Definition at line 1012 of file [ftypes.h](#).

**4.49.2.643 REVERSEARG**

```
#define REVERSEARG 11
```

Definition at line 1013 of file [ftypes.h](#).

**4.49.2.644 CYCLEARG**

```
#define CYCLEARG 12
```

Definition at line 1014 of file [ftypes.h](#).

**4.49.2.645 ISLYNDON**

```
#define ISLYNDON 13
```

Definition at line 1015 of file [ftypes.h](#).



**4.49.2.646 ISLYNDONR**

```
#define ISLYNDONR 14
```

Definition at line 1016 of file [ftypes.h](#).

**4.49.2.647 TOLYNDON**

```
#define TOLYNDON 15
```

Definition at line 1017 of file [ftypes.h](#).

**4.49.2.648 TOLYNDONR**

```
#define TOLYNDONR 16
```

Definition at line 1018 of file [ftypes.h](#).

**4.49.2.649 ADDARG**

```
#define ADDARG 17
```

Definition at line 1019 of file [ftypes.h](#).

**4.49.2.650 MULTIPLYARG**

```
#define MULTIPLYARG 18
```

Definition at line 1020 of file [ftypes.h](#).

**4.49.2.651 DROPARG**

```
#define DROPARG 19
```

Definition at line 1021 of file [ftypes.h](#).

**4.49.2.652 SELECTARG**

```
#define SELECTARG 20
```

Definition at line 1022 of file [ftypes.h](#).

**4.49.2.653 DEDUPARG**

```
#define DEDUPARG 21
```

Definition at line 1023 of file [ftypes.h](#).

**4.49.2.654 ZTOHARG**

```
#define ZTOHARG 22
```

Definition at line [1024](#) of file [ftypes.h](#).

**4.49.2.655 HTOZARG**

```
#define HTOZARG 23
```

Definition at line [1025](#) of file [ftypes.h](#).

**4.49.2.656 BASECODE**

```
#define BASECODE 1
```

Definition at line [1033](#) of file [ftypes.h](#).

**4.49.2.657 YESLYNDON**

```
#define YESLYNDON 1
```

Definition at line [1034](#) of file [ftypes.h](#).

**4.49.2.658 NOLYNDON**

```
#define NOLYNDON 2
```

Definition at line [1035](#) of file [ftypes.h](#).

**4.49.2.659 TOPOLYNOMIALFLAG**

```
#define TOPOLYNOMIALFLAG 1
```

Definition at line [1037](#) of file [ftypes.h](#).

**4.49.2.660 FACTARGFLAG**

```
#define FACTARGFLAG 2
```

Definition at line [1038](#) of file [ftypes.h](#).

**4.49.2.661 OLDFACTARG**

```
#define OLDFACTARG 1
```

Definition at line [1040](#) of file [ftypes.h](#).

**4.49.2.662 NEWFACTARG**

```
#define NEWFACTARG 0
```

Definition at line 1041 of file [ftypes.h](#).

**4.49.2.663 FROMMODULEOPTION**

```
#define FROMMODULEOPTION 0
```

Definition at line 1043 of file [ftypes.h](#).

**4.49.2.664 FROMPOINTINSTRUCTION**

```
#define FROMPOINTINSTRUCTION 1
```

Definition at line 1044 of file [ftypes.h](#).

**4.49.2.665 EXTRASYMBOL**

```
#define EXTRASYMBOL 0
```

Definition at line 1046 of file [ftypes.h](#).

**4.49.2.666 REGULARSYMBOL**

```
#define REGULARSYMBOL 1
```

Definition at line 1047 of file [ftypes.h](#).

**4.49.2.667 EXPRESSIONNUMBER**

```
#define EXPRESSIONNUMBER 2
```

Definition at line 1048 of file [ftypes.h](#).

**4.49.2.668 O\_NONE**

```
#define O_NONE 0
```

Definition at line 1050 of file [ftypes.h](#).

**4.49.2.669 O\_CSE**

```
#define O_CSE 1
```

Definition at line 1051 of file [ftypes.h](#).

**4.49.2.670 O\_CSEGREEDY**

```
#define O_CSEGREEDY 2
```

Definition at line 1052 of file [ftypes.h](#).

**4.49.2.671 O\_GREEDY**

```
#define O_GREEDY 3
```

Definition at line 1053 of file [ftypes.h](#).

**4.49.2.672 O\_OCCURRENCE**

```
#define O_OCCURRENCE 0
```

Definition at line 1055 of file [ftypes.h](#).

**4.49.2.673 O\_MCTS**

```
#define O_MCTS 1
```

Definition at line 1056 of file [ftypes.h](#).

**4.49.2.674 O\_SIMULATED\_ANNEALING**

```
#define O_SIMULATED_ANNEALING 2
```

Definition at line 1057 of file [ftypes.h](#).

**4.49.2.675 O\_FORWARD**

```
#define O_FORWARD 0
```

Definition at line 1059 of file [ftypes.h](#).

**4.49.2.676 O\_BACKWARD**

```
#define O_BACKWARD 1
```

Definition at line 1060 of file [ftypes.h](#).

**4.49.2.677 O\_FORWARDORBACKWARD**

```
#define O_FORWARDORBACKWARD 2
```

Definition at line 1061 of file [ftypes.h](#).

**4.49.2.678 O\_FORWARDANDBACKWARD**

```
#define O_FORWARDANDBACKWARD 3
```

Definition at line 1062 of file [ftypes.h](#).

**4.49.2.679 OPTHEAD**

```
#define OPTHEAD 3
```

Definition at line 1064 of file [ftypes.h](#).

**4.49.2.680 DOALL**

```
#define DOALL 1
```

Definition at line 1065 of file [ftypes.h](#).

**4.49.2.681 ONLYFUNCTIONS**

```
#define ONLYFUNCTIONS 2
```

Definition at line 1066 of file [ftypes.h](#).

**4.49.2.682 INUSE**

```
#define INUSE 1
```

Definition at line 1068 of file [ftypes.h](#).

**4.49.2.683 COULDCOMMUTE**

```
#define COULDCOMMUTE 2
```

Definition at line 1069 of file [ftypes.h](#).

**4.49.2.684 DOESNOTCOMMUTE**

```
#define DOESNOTCOMMUTE 4
```

Definition at line 1070 of file [ftypes.h](#).

**4.49.2.685 DICT\_NONUMBERS**

```
#define DICT_NONUMBERS 0
```

Definition at line 1072 of file [ftypes.h](#).

**4.49.2.686 DICT\_INTEGERONLY**

```
#define DICT_INTEGERONLY 1
```

Definition at line 1073 of file [ftypes.h](#).

**4.49.2.687 DICT\_RATIONALONLY**

```
#define DICT_RATIONALONLY 2
```

Definition at line 1074 of file [ftypes.h](#).

**4.49.2.688 DICT\_ALLNUMBERS**

```
#define DICT_ALLNUMBERS 3
```

Definition at line 1075 of file [ftypes.h](#).

**4.49.2.689 DICT\_NOVARIABLES**

```
#define DICT_NOVARIABLES 0
```

Definition at line 1076 of file [ftypes.h](#).

**4.49.2.690 DICT\_DOVARIABLES**

```
#define DICT_DOVARIABLES 1
```

Definition at line 1077 of file [ftypes.h](#).

**4.49.2.691 DICT\_NOSPECIALS**

```
#define DICT_NOSPECIALS 0
```

Definition at line 1078 of file [ftypes.h](#).

**4.49.2.692 DICT\_DOSPECIALS**

```
#define DICT_DOSPECIALS 1
```

Definition at line 1079 of file [ftypes.h](#).

**4.49.2.693 DICT\_NOFUNWITHARGS**

```
#define DICT_NOFUNWITHARGS 0
```

Definition at line 1080 of file [ftypes.h](#).

**4.49.2.694 DICT\_DOFUNWITHARGS**

```
#define DICT_DOFUNWITHARGS 1
```

Definition at line 1081 of file [ftypes.h](#).

**4.49.2.695 DICT\_NOTINDOLLARS**

```
#define DICT_NOTINDOLLARS 0
```

Definition at line 1082 of file [ftypes.h](#).

**4.49.2.696 DICT\_INDOLLARS**

```
#define DICT_INDOLLARS 1
```

Definition at line 1083 of file [ftypes.h](#).

**4.49.2.697 DICT\_INTEGERNUMBER**

```
#define DICT_INTEGERNUMBER 1
```

Definition at line 1085 of file [ftypes.h](#).

**4.49.2.698 DICT\_RATIONALNUMBER**

```
#define DICT_RATIONALNUMBER 2
```

Definition at line 1086 of file [ftypes.h](#).

**4.49.2.699 DICT\_SYMBOL**

```
#define DICT_SYMBOL 3
```

Definition at line 1087 of file [ftypes.h](#).

**4.49.2.700 DICT\_VECTOR**

```
#define DICT_VECTOR 4
```

Definition at line 1088 of file [ftypes.h](#).

**4.49.2.701 DICT\_INDEX**

```
#define DICT_INDEX 5
```

Definition at line 1089 of file [ftypes.h](#).

#### 4.49.2.702 DICT\_FUNCTION

```
#define DICT_FUNCTION 6
```

Definition at line 1090 of file [ftypes.h](#).

#### 4.49.2.703 DICT\_FUNCTION\_WITH\_ARGUMENTS

```
#define DICT_FUNCTION_WITH_ARGUMENTS 7
```

Definition at line 1091 of file [ftypes.h](#).

#### 4.49.2.704 DICT\_SPECIALCHARACTER

```
#define DICT_SPECIALCHARACTER 8
```

Definition at line 1092 of file [ftypes.h](#).

#### 4.49.2.705 DICT\_RANGE

```
#define DICT_RANGE 9
```

Definition at line 1093 of file [ftypes.h](#).

#### 4.49.2.706 READSPECTATORFLAG

```
#define READSPECTATORFLAG 3
```

Definition at line 1095 of file [ftypes.h](#).

#### 4.49.2.707 GLOBALSPECTATORFLAG

```
#define GLOBALSPECTATORFLAG 1
```

Definition at line 1096 of file [ftypes.h](#).

#### 4.49.2.708 ORDEREDSET

```
#define ORDEREDSET 1
```

Definition at line 1098 of file [ftypes.h](#).

#### 4.49.2.709 DENSETABLE

```
#define DENSETABLE 1
```

Definition at line 1100 of file [ftypes.h](#).



**4.49.2.710 SPARSETABLE**

```
#define SPARSETABLE 0
```

Definition at line 1101 of file [ftypes.h](#).

**4.49.2.711 ONEPARTICLEIRREDUCIBLE**

```
#define ONEPARTICLEIRREDUCIBLE 1
```

Definition at line 1103 of file [ftypes.h](#).

**4.49.2.712 WITHINSERTIONS**

```
#define WITHINSERTIONS 2
```

Definition at line 1104 of file [ftypes.h](#).

**4.49.2.713 NOTADPOLES**

```
#define NOTADPOLES 4
```

Definition at line 1105 of file [ftypes.h](#).

**4.49.2.714 WITHSYMMETRIZE**

```
#define WITHSYMMETRIZE 8
```

Definition at line 1106 of file [ftypes.h](#).

**4.49.2.715 TOPOLOGIESONLY**

```
#define TOPOLOGIESONLY 16
```

Definition at line 1107 of file [ftypes.h](#).

**4.49.2.716 NONODES**

```
#define NONODES 32
```

Definition at line 1108 of file [ftypes.h](#).

**4.49.2.717 WITHEDGES**

```
#define WITHEDGES 64
```

Definition at line 1109 of file [ftypes.h](#).

#### 4.49.2.718 CHECKEXTERN

```
#define CHECKEXTERN 128
```

Definition at line 1110 of file [ftypes.h](#).

#### 4.49.2.719 WITHBLOCKS

```
#define WITHBLOCKS 256
```

Definition at line 1111 of file [ftypes.h](#).

#### 4.49.2.720 WITHONEPI

```
#define WITHONEPI 512
```

Definition at line 1112 of file [ftypes.h](#).

#### 4.49.2.721 NOSNAILS

```
#define NOSNAILS 1024
```

Definition at line 1113 of file [ftypes.h](#).

#### 4.49.2.722 NOEXTSELF

```
#define NOEXTSELF 2048
```

Definition at line 1114 of file [ftypes.h](#).

### 4.49.3 Typedef Documentation

#### 4.49.3.1 PVFUNWP

```
typedef void(* PVFUNWP) ()
```

Definition at line 183 of file [ftypes.h](#).

#### 4.49.3.2 PVFUNV

```
typedef void(* PVFUNV) ()
```

Definition at line 184 of file [ftypes.h](#).

#### 4.49.3.3 CFUN

```
typedef int(* CFUN) ()
```

Definition at line 185 of file [ftypes.h](#).

#### 4.49.3.4 TFUN

```
typedef int(* TFUN) ()
```

Definition at line 186 of file [ftypes.h](#).

#### 4.49.3.5 TFUN1

```
typedef int(* TFUN1) ()
```

Definition at line 187 of file [ftypes.h](#).

## 4.50 ftypes.h

[Go to the documentation of this file.](#)

```
00001
00009 /* #[ License : */
00010 /*
00011 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00012 *   When using this file you are requested to refer to the publication
00013 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00014 *   This is considered a matter of courtesy as the development was paid
00015 *   for by FOM the Dutch physics granting agency and we would like to
00016 *   be able to track its scientific use to convince FOM of its value
00017 *   for the community.
00018 *
00019 *   This file is part of FORM.
00020 *
00021 *   FORM is free software: you can redistribute it and/or modify it under the
00022 *   terms of the GNU General Public License as published by the Free Software
00023 *   Foundation, either version 3 of the License, or (at your option) any later
00024 *   version.
00025 *
00026 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00027 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00028 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00029 *   details.
00030 *
00031 *   You should have received a copy of the GNU General Public License along
00032 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00033 */
00034 /* #[ License : */
00035
00051 #define WITHOUTERROR 0
00052 #define WITHERROR 1
00053
00054 /*
00055     The various streams. (look also in tools.c)
00056 */
00057
00058 #define FILESTREAM 0
00059 #define PREVARSTREAM 1
00060 #define PREREADSTREAM 2
00061 #define PIPESTREAM 3
00062 #define PRECALCSTREAM 4
00063 #define DOLLARSTREAM 5
00064 #define PREREADSTREAM2 6
00065 #define EXTERNALCHANNELSTREAM 7
00066 #define PREREADSTREAM3 8
00067 #define REVERSEFILESTREAM 9
00068 #define INPUTSTREAM 10
```

```

00069
00070 #define ENDOFSTREAM 0xFF
00071 #define ENDOFINPUT 0xFF
00072
00073 /*
00074     Types of files
00075 */
00076
00077 #define SUBROUTINEFILE 0
00078 #define PROCEDUREFILE 1
00079 #define HEADERFILE 2
00080 #define SETUPFILE 3
00081 #define TABLEBASEFILE 4
00082
00083 /*
00084     Types of modules
00085 */
00086
00087 #define FIRSTMODULE -1
00088 #define GLOBALMODULE 0
00089 #define SORTMODULE 1
00090 #define STOREMODULE 2
00091 #define CLEARMODULE 3
00092 #define ENDMODULE 4
00093
00094 #define POLYFUN 0
00095
00096 #define NOPARALLEL_DOLLAR      0x0001
00097 #define NOPARALLEL_RHS        0x0002
00098 #define NOPARALLEL_CONVPOLY   0x0004
00099 #define NOPARALLEL_SPECTATOR  0x0008
00100 #define NOPARALLEL_USER       0x0010
00101 #define NOPARALLEL_TBLDOLLAR  0x0100
00102 #define NOPARALLEL_NPROC      0x0200
00103 #define PARALLELFLAG          0x0000
00104
00105 #define PRENOACTION 0
00106 #define PRERAISEAFTER 1
00107 #define PRELOWERAFTER 2
00108 /*
00109 #define ELIUMOD 1
00110 #define ELIZMOD 2
00111 #define SKIUMOD 3
00112 #define SKIZMOD 4
00113 */
00114 #define WITHSEMICOLON 0
00115 #define WITHOUTSEMICOLON 1
00116 #define MODULEINSTACK 8
00117 #define EXECUTINGIF 0
00118 #define LOOKINGFORELSE 1
00119 #define LOOKINGFORENDIF 2
00120 #define NEWSTATEMENT 1
00121 #define OLDSTATEMENT 0
00122
00123 #define EXECUTINGPRESWITCH 0
00124 #define SEARCHINGPRECASE 1
00125 #define SEARCHINGPREENDSWITCH 2
00126
00127 #define PREPROONLY 1
00128 #define DUMPTOCOMPILER 2
00129 #define DUMPOUTTERMS 4
00130 #define DUMPINTERMS 8
00131 #define DUMPTOSORT 16
00132 #define DUMPTOPARALLEL 32
00133 #define THREADSDEBUG 64
00134
00135 #define ERROROUT 0
00136 #define INPUTOUT 1
00137 #define STATSOUT 2
00138 #define EXPRSOUT 3
00139 #define WRITEOUT 4
00140
00141 #define EXTERNALCHANNELOUT 5
00142
00143 #define NUMERICALLOOP 0
00144 #define LISTEDLOOP 1
00145 #define ONEEXPRESSION 2
00146
00147 #define PRETYPENONE 0
00148 #define PRETYPEIF 1
00149 #define PRETYPEDO 2
00150 #define PRETYPEPROCEDURE 3
00151 #define PRETYPESWITCH 4
00152 #define PRETYPEINSIDE 5
00153
00154 /*
00155     Type of statement. Used to make sure that the statements are in proper order

```

```
00156 */
00157
00158 #define DECLARATION 1
00159 #define SPECIFICATION 2
00160 #define DEFINITION 3
00161 #define STATEMENT 4
00162 #define TOOUTPUT 5
00163 #define ATENDOFMODULE 6
00164 #define MIXED2 8
00165 #define MIXED 9
00166
00167 /*
00168     The typedefs are to allow the compilers to do better error checking.
00169 */
00170
00171 #ifdef ANSI
00172     typedef void (*PVFUNWP)(WORD *);
00173 #ifdef INTELCOMPILER
00174     typedef void (*PVFUNV)();
00175     typedef int (*CFUN)();
00176 #else
00177     typedef void (*PVFUNV)(void);
00178     typedef int (*CFUN)(void);
00179 #endif
00180     typedef int (*TFUN)(UBYTE *);
00181     typedef int (*TFUN1)(UBYTE *,int);
00182 #else
00183     typedef void (*PVFUNWP)();
00184     typedef void (*PVFUNV)();
00185     typedef int (*CFUN)();
00186     typedef int (*TFUN)();
00187     typedef int (*TFUN1)();
00188 #endif
00189
00190
00191 #define NOAUTO 0
00192 #define PARTEST 1
00193 #define WITHAUTO 2
00194
00195 #define ALLVARIABLES -1
00196 #define SYMBOLONLY 1
00197 #define INDEXONLY 2
00198 #define VECTORONLY 4
00199 #define FUNCTIONONLY 8
00200 #define SETONLY 16
00201 #define EXPRESSIONONLY 32
00202
00203
00211
00212 #define CDELETE -1
00213 #define ANYTYPE -1
00214 #define CSYMBOL 0
00215 #define CINDEX 1
00216 #define CVECTOR 2
00217 #define CFUNCTION 3
00218 #define CSET 4
00219 #define CEXPRESSION 5
00220 #define CDOTPRODUCT 6
00221 #define CNUMBER 7
00222 #define CSUBEXP 8
00223 #define CDELTA 9
00224 #define CDOLLAR 10
00225 #define CDUBIOUS 11
00226 #define CRANGE 12
00227 #define CMODEL 13
00228 #define CVECTOR1 21
00229 #define CDOUBLEDOT 22
00230
00233 /*
00234     Types of tokens in the tokenizer.
00235 */
00236
00237 #define TSYMBOL -1
00238 #define TINDEX -2
00239 #define TVECTOR -3
00240 #define TFUNCTION -4
00241 #define TSET -5
00242 #define TEXPRESSION -6
00243 #define TDOTPRODUCT -7
00244 #define TNUMBER -8
00245 #define TSUBEXP -9
00246 #define TDELTA -10
00247 #define TDOLLAR -11
00248 #define TDUBIOUS -12
00249 #define LPARENTHESIS -13
00250 #define RPARENTHESIS -14
00251 #define TWILDCARD -15
```

```

00252 #define TWILDARG -16
00253 #define TDOT -17
00254 #define LBRACE -18
00255 #define RBRACE -19
00256 #define TCOMMA -20
00257 #define TFUNOPEN -21
00258 #define TFUNCLOSE -22
00259 #define TMULTIPLY -23
00260 #define TDIVIDE -24
00261 #define TPOWER -25
00262 #define TPLUS -26
00263 #define TMINUS -27
00264 #define TNOT -28
00265 #define TENDOFIT -29
00266 #define TSETOPEN -30
00267 #define TSETCLOSE -31
00268 #define TGENINDEX -32
00269 #define TCONJUGATE -33
00270 #define LRPARENTHESSES -34
00271 #define TNUMBER1 -35
00272 #define TPOWER1 -36
00273 #define EMPTY -37
00274 #define TSETNUM -38
00275 #define TSGAMMA -39
00276 #define TSETDOL -40
00277 #define TFLOAT -41
00278
00279 #define TYPEISFUN 0
00280 #define TYPEISSUB 1
00281 #define TYPEISMYSTERY -1
00282
00283 #define LHSIDEX 2
00284 #define LHSIDE 1
00285 #define RHSIDE 0
00286
00287 /*
00288     Output modes
00289 */
00290
00291 #define FORTRANMODE 1
00292 #define REDUCEMODE 2
00293 #define MAPLEMODE 3
00294 #define MATHEMATICAMODE 4
00295 #define CMODE 5
00296 #define VORTRANMODE 6
00297 #define PFORTRANMODE 7
00298 #define DOUBLEFORTRANMODE 33
00299 #define DOUBLEPRECISIONFLAG 32
00300 #define NODOUBLEMASK 31
00301 #define QUADRUPLEFORTRANMODE 65
00302 #define QUADRUPLEPRECISIONFLAG 64
00303 #define ALLINTEGERDOUBLE 128
00304 #define NOQUADMASK 63
00305 #define NORMALFORMAT 0
00306 #define NOSPACEFORMAT 1
00307
00308 #define ISNOTFORTRAN90 0
00309 #define ISFORTRAN90 1
00310
00311 #define ALSOREVERSE 1
00312 #define CHISHOLM 2
00313 #define NOTRICK 16
00314
00315 #define SORTLOWFIRST 0
00316 #define SORTHIGHFIRST 1
00317 #define SORTPOWERFIRST 2
00318 #define SORTANTIPOWER 3
00319
00320 /* These control the output of WriteStats. The different sort
00321  * types have differently formatted stats. These variables should
00322  * not have arbitrary values since they are also used as array
00323  * indices by WriteStats. */
00324 #define STATSSPLITMERGE 0
00325 #define STATSMERGETOFILE 1
00326 #define STATSPOSTSORT 2
00327 /* CHECKLOGTYPE implies that statistics will not be printed in
00328  * the log file if AM.LogType = 1 (when running with -ll or -F). */
00329 #define NOCHECKLOGTYPE 0
00330 #define CHECKLOGTYPE 1
00331
00332 #define NMIN4SHIFT 4
00333 /*
00334     The next are the main codes.
00335     Note: SETSET is not allowed to be 4*n+1
00336     We use those codes in CoIdExpression for function information
00337     after the pattern. Because SETSET also stands there we have to
00338     be careful!!

```

```
00339     Don't forget MAXBUILTINFUNCTION when adding codes!
00340     The object FUNCTION is at the start of the functions that are in regular
00341     notation. Anything below it is in special notation.
00342
00343     Remark: HAAKJE0 is for compression purposes and should only occur
00344     at moments that ARGWILD cannot occur.
00345 */
00346 #define SYMBOL 1
00347 #define DOTPRODUCT 2
00348 #define VECTOR 3
00349 #define INDEX 4
00350 #define EXPRESSION 5
00351 #define SUBEXPRESSION 6
00352 #define DOLLAREXPRESSION 7
00353 #define SETSET 8
00354 #define ARGWILD 9
00355 #define MINVECTOR 10
00356 #define SETEXP 11
00357 #define DOLLAREXPR2 12
00358 #define HAAKJE0 9
00359 #define FUNCTION 20
00360
00361 #define TMPPOLYFUN 14
00362 #define ARGFIELD 15
00363 #define SNUMBER 16
00364 #define LNUMBER 17
00365 #define HAAKJE 18
00366 #define DELTA 19
00367 #define EXPONENT 20
00368 #define DENOMINATOR 21
00369 #define SETFUNCTION 22
00370 #define GAMMA 23
00371 #define GAMMAI 24
00372 #define GAMMAFIVE 25
00373 #define GAMMASIX 26
00374 #define GAMMASEVEN 27
00375 #define SUMF1 28
00376 #define SUMF2 29
00377 #define DUMFUN 30
00378 #define REPLACEMENT 31
00379 #define REVERSEFUNCTION 32
00380 #define DISTRIBUTION 33
00381 #define DELTA3 34
00382 #define DUMMYFUN 35
00383 #define DUMMYTEN 36
00384 #define LEVICIVITA 37
00385 #define FACTORIAL 38
00386 #define INVERSEFACTORIAL 39
00387 #define BINOMIAL 40
00388 #define NUMARGSFUN 41
00389 #define SIGNFUN 42
00390 #define MODFUNCTION 43
00391 #define MOD2FUNCTION 44
00392 #define MINFUNCTION 45
00393 #define MAXFUNCTION 46
00394 #define ABSFUNCTION 47
00395 #define SIGFUNCTION 48
00396 #define INTFUNCTION 49
00397 #define THETA 50
00398 #define THETA2 51
00399 #define DELTA2 52
00400 #define DELTAP 53
00401 #define BERNOULLIFUNCTION 54
00402 #define COUNTFUNCTION 55
00403 #define MATCHFUNCTION 56
00404 #define PATTERNFUNCTION 57
00405 #define TERMFUNCTION 58
00406 #define CONJUGATION 59
00407 #define ROOTFUNCTION 60
00408 #define TABLEFUNCTION 61
00409 #define FIRSTBRACKET 62
00410 #define TERMSINEXPR 63
00411 #define NUMTERMSFUN 64
00412 #define GCDFUNCTION 65
00413 #define DIVFUNCTION 66
00414 #define REMFUNCTION 67
00415 #define MAXPOWEROF 68
00416 #define MINPOWEROF 69
00417 #define TABLESTUB 70
00418 #define FACTORIN 71
00419 #define TERMSINBRACKET 72
00420 #define WILDARGFUN 73
00421 /*
00422     In the past we would add new functions here and raise the numbers
00423     on the special reserved names. This is impractical in the light of
00424     the .sav files. The .sav files need a mechanism that contains the
00425     value of MAXBUILTINFUNCTION at the moment of writing. This allows
```

```

00426     form corrections if this value has changed in the mean time.
00427 */
00428 #define SQRTFUNCTION 74
00429 #define LNFUNCTION 75
00430 #define SINFUNCTION 76
00431 #define COSFUNCTION 77
00432 #define TANFUNCTION 78
00433 #define ASINFUNCTION 79
00434 #define ACOSFUNCTION 80
00435 #define ATANFUNCTION 81
00436 #define ATAN2FUNCTION 82
00437 #define SINHFUNCTION 83
00438 #define COSHFUNCTION 84
00439 #define TANHFUNCTION 85
00440 #define ASINHFUNCTION 86
00441 #define ACOSHFUNCTION 87
00442 #define ATANHFUNCTION 88
00443 #define LI2FUNCTION 89
00444 #define LINFUNCTION 90
00445
00446 #define EXTRASYMFUN 91
00447 #define RANDOMFUNCTION 92
00448 #define RANPERM 93
00449 #define NUMFACTORS 94
00450 #define FIRSTTERM 95
00451 #define CONTENTTERM 96
00452 #define PRIMENUMBER 97
00453 #define EXTEUCLIDEAN 98
00454 #define MAKERATIONAL 99
00455 #define INVERSEFUNCTION 100
00456 #define IDFUNCTION 101
00457 #define PUTFIRST 102
00458 #define PERMUTATIONS 103
00459 #define PARTITIONS 104
00460 #define MULFUNCTION 105
00461 #define MOEBIUS 106
00462 #define TOPOLOGIES 107
00463 #define DIAGRAMS 108
00464 #define TOPO 109
00465 #define NODEFUNCTION 110
00466 #define EDGE 111
00467 #define SIZEOFFUNCTION 112
00468 #define BLOCK 113
00469 #define ONEPI 114
00470 #define PHI 115
00471 #ifdef WITHFLOAT
00472 #define FLOATFUN 116
00473 #define TOFLOAT 117
00474 #define TORAT 118
00475 #define MZV 119
00476 #define EULER 120
00477 #define MZVHALF 121
00478 #define AGMFUNCTION 122
00479 #define GAMMAFUN 123
00480 #define MAXBUILTINFUNCTION 123
00481 #else
00482 #define MAXBUILTINFUNCTION 115
00483 #endif
00484
00485 #define FIRSTUSERFUNCTION 150
00486
00487 #define ALLFUNCTIONS -1
00488 #define ALLMZVFUNCTIONS -2
00489
00490 /*
00491     Note: if we add a builtin table we have to look also inside names.c
00492     in the routine Globalize because there we assume there does not exist
00493     such an object
00494 */
00495
00496 #define ISYMBOL 0
00497 #define PISYMBOL 1
00498 #define COEFFSYMBOL 2
00499 #define NUMERATORSYMBOL 3
00500 #define DENOMINATORSYMBOL 4
00501 #define WILDARGSYMBOL 5
00502 #define DIMENSIONSYMBOL 6
00503 #define FACTORSYMBOL 7
00504 #define SEPARATESYMBOL 8
00505 #define EESYMBOL 9
00506 #define EMSYMBOL 10
00507
00508 #define BUILTINSYMBOLS 11
00509 #define FIRSTUSERSYMBOL 20
00510
00511 #define BUILTININDICES 1
00512 #define BUILTINVECTORS 1

```



```
00513 #define BUILTINDOLLARS 1
00514
00515 #define WILDARGVECTOR 0
00516 #define WILDARGINDEX 0
00517
00518 /*
00519  The objects that have a name that starts with TYPE are codes of statements
00520  made by the compiler. Each statement starts with such a code, followed by
00521  its size. For how most of these statements are used can be seen in the
00522  Generator function in the file proces.c
00523  TYPEOPERATION is an anachronism that remains used only for the statements
00524  that are executed in the file opera.c (like traces and contractions).
00525 */
00526
00527 #define TYPEEXPRESSION 0
00528 #define TYPEIDNEW 1
00529 #define TYPEIDOLD 2
00530 #define TYPEOPERATION 3
00531 #define TYPEPERPEAT 4
00532 #define TYPEENDREPEAT 5
00533 /*
00534  The next counts must be higher than the ones before
00535 */
00536 #define TYPECOUNT 20
00537 #define TYPEMULT 21
00538 #define TYPEGOTO 22
00539 #define TYPEDISCARD 23
00540 #define TYPEIF 24
00541 #define TYPEELSE 25
00542 #define TYPEELIF 26
00543 #define TYPEENDIF 27
00544 #define TYPESUM 28
00545 #define TYPECHISHOLM 29
00546 #define TYPEREVERSE 30
00547 #define TYPEARG 31
00548 #define TYPENORM 32
00549 #define TYPENORM2 33
00550 #define TYPENORM3 34
00551 #define TYPEEXIT 35
00552 #define TYPESETEXIT 36
00553 #define TYPEPRINT 37
00554 #define TYPEFFPRINT 38
00555 #define TYPEDEFPRE 39
00556 #define TYPESPLITARG 40
00557 #define TYPESPLITARG2 41
00558 #define TYPEFACTARG 42
00559 #define TYPEFACTARG2 43
00560 #define TYPETRY 44
00561 #define TYPEASSIGN 45
00562 #define TYPERENUMBER 46
00563 #define TYPESUMFIX 47
00564 #define TYPEFINDLOOP 48
00565 #define TYPEUNRAVEL 49
00566 #define TYPEADJUSTBOUNDS 50
00567 #define TYPEINSIDE 51
00568 #define TYPETERM 52
00569 #define TYPESORT 53
00570 #define TYPEDETCURDUM 54
00571 #define TYPEINEXPRESSION 55
00572 #define TYPESPLITFIRSTARG 56
00573 #define TYPESPLITLASTARG 57
00574 #define TYPEMERGE 58
00575 #define TYPETESTUSE 59
00576 #define TYPEAPPLY 60
00577 #define TYPEAPPLYRESET 61
00578 #define TYPECHAININ 62
00579 #define TYPECHAINOUT 63
00580 #define TYPENORM4 64
00581 #define TYPEFACTOR 65
00582 #define TYPEARGIMPLODE 66
00583 #define TYPEARGEXPLODE 67
00584 #define TYPEDENOMINATORS 68
00585 #define TYPESTUFFLE 69
00586 #define TYPEDROPCOEFFICIENT 70
00587 #define TYPETRANSFORM 71
00588 #define TYPETOPOLYNOMIAL 72
00589 #define TYPEFROMPOLYNOMIAL 73
00590 #define TYPEDOLOOP 74
00591 #define TYPEENDDOLOOP 75
00592 #define TYPEDROPSYMBOLS 76
00593 #define TYPEPUTINSIDE 77
00594 #define TYPETOSPECTATOR 78
00595 #define TYPEARGTOEXTRASymbol 79
00596 #define TYPECANONICALIZE 80
00597 #define TYPESWITCH 81
00598 #define TYPEENDSWITCH 82
00599 #define TYPESETUSERFLAG 83
```

```
00600 #define TYPECLEARUSERFLAG 84
00601 #define TYPEALLLOOPS 85
00602 #define TYPEALLPATHS 86
00603 #ifdef WITHFLOAT
00604 #define TYPEEVALUATE 87
00605 #define TYPEEXPAND 88
00606 #define TYPETOFLOAT 89
00607 #define TYPETORAT 90
00608 #endif
00609
00610 /*
00611     The codes for the 'operations' that are part of TYPEOPERATION.
00612 */
00613
00614 #define TAKETRACE 1
00615 #define CONTRACT 2
00616 #define RATIO 3
00617 #define SYMMETRIZE 4
00618 #define TENVEC 5
00619 #define SUMNUM1 6
00620 #define SUMNUM2 7
00621
00622 /*
00623     The various types of wildcards.
00624 */
00625
00626 #define WILDDUMMY 0
00627 #define SYMTONUM 1
00628 #define SYMTOSYM 2
00629 #define SYMTOSUB 3
00630 #define VECTOMIN 4
00631 #define VECTOVEC 5
00632 #define VECTOSUB 6
00633 #define INDTOIND 7
00634 #define INDTOSUB 8
00635 #define FUNTOFUN 9
00636 #define ARGTOARG 10
00637 #define ARLTOARL 11
00638 #define EXPTOEXP 12
00639 #define FROMBRAC 13
00640 #define FROMSET 14
00641 #define SETTONUM 15
00642 #define WILDCARDS 16
00643 #define SETNUMBER 17
00644 #define LOADDOLLAR 18
00645 /*
00646     Some new types of wildcards that hold only for function arguments.
00647 */
00648 #define NUMTONUM 20
00649 #define NUMTOSYM 21
00650 #define NUMTOIND 22
00651 #define NUMTOSUB 23
00652 /*
00653     Flag or-ed with ARGTOARG to give the number of arguments.
00654 */
00655 #define EATTENSOR 0x2000
00656
00657 /*
00658     Dirty flags (introduced when functions got a field with a dirty flag)
00659 */
00660
00661 #define CLEANFLAG 0
00662 #define DIRTYFLAG 1
00663 #define DIRTYSYMFLAG 2
00664 #define MUSTCLEANPRF 4
00665 #define SUBTERMUSED1 8
00666 #define SUBTERMUSED2 16
00667 #define ALLDIRTY (DIRTYFLAG|DIRTYSYMFLAG)
00668
00669 #define ARGHEAD 2
00670 #define FUNHEAD 3
00671 #define SUBEXPSIZE 5
00672 #define EXPRHEAD 5
00673 #define TYPEARGHEADSIZE 6
00674
00675 /*
00676     Actions to be taken with expressions. They are marked with these objects
00677     during compilation.
00678 */
00679
00680 #define SKIP 1
00681 #define DROP 2
00682 #define HIDE 3
00683 #define UNHIDE 4
00684 #define INTOHIDE 5
00685
00686 /*
```

```

00687     Types of expressions
00688 */
00689
00690 #define LOCALEXPRESSION 0
00691 #define SKIPLEXPRESSION 1
00692 #define DROPLEXPRESSION 2
00693 #define DROPPEDEXPRESSION 3
00694 #define GLOBALEXPRESSION 4
00695 #define SKIPGEXPRESSION 5
00696 #define DROPGEXPRESSION 6
00697 #define STOREDEXPRESSION 8
00698 #define HIDDENLEXPRESSION 9
00699 #define HIDDENGEXPRESSION 13
00700 #define INCEXPRESSION 9
00701 #define HIDELEXPRESSION 10
00702 #define HIDEGEXPRESSION 14
00703 #define DROPHLEXPRESSION 11
00704 #define DROPHGEXPRESSION 15
00705 #define UNHIDELEXPRESSION 12
00706 #define UNHIDEGEXPRESSION 16
00707 #define INTOHIDELEXPRESSION 17
00708 #define INTOHIDEGEXPRESSION 18
00709 #define SPECTATOREXPRESSION 19
00710 #define DROPSPECTATOREXPRESSION 20
00711 #define SKIPUNHIDELEXPRESSION 21
00712 #define SKIPUNHIDEGEXPRESSION 22
00713
00714 #define PRINTOFF 0
00715 #define PRINON 1
00716 #define PRINTCONTENTS 2
00717 #define PRINTCONTENT 3
00718 #define PRINTLFILE 4
00719 #define PRINONETERM 8
00720 #define PRINTONEFUNCTION 16
00721 #define PRINTALL 32
00722
00723 /*
00724     Special codes for the replace variable in the EXPRESSIONS struct
00725 */
00726
00727 #define REGULAREXPRESSION -1
00728 #define REDEFINEDEXPRESSION -2
00729 #define NEWLYDEFINEDEXPRESSION -3
00730
00738 #define VERTEXFUNCTION -1
00739 #define GENERALFUNCTION 0
00740 #define TENSORFUNCTION 2
00741 #define GAMMAFUNCTION 4
00744 /*
00745     Special sets
00746 */
00747
00748 #define POS_      0    /* integer > 0 */
00749 #define POS0_     1    /* integer >= 0 */
00750 #define NEG_      2    /* integer < 0 */
00751 #define NEG0_     3    /* integer <= 0 */
00752 #define EVEN_     4    /* integer (even) */
00753 #define ODD_      5    /* integer (odd) */
00754 #define Z_        6    /* integer */
00755 #define SYMBOL_   7    /* symbol only */
00756 #define FIXED_    8    /* fixed index */
00757 #define INDEX_    9    /* index only */
00758 #define Q_        10   /* rational */
00759 #define DUMMYINDEX_ 11 /* dummy index only */
00760 #define VECTOR_   12   /* vector only */
00761
00762 /*
00763     Special indices.
00764 */
00765
00766 #define GAMMA1 0
00767 #define GAMMA5 -1
00768 #define GAMMA6 -2
00769 #define GAMMA7 -3
00770 #define FUNNYVEC -4
00771 #define FUNNYWILD -5
00772 #define SUMMEDIND -6
00773 #define NOINDEX -7
00774 #define FUNNYDOLLAR -8
00775 #define EMPTYINDEX -9
00776
00777 /*
00778     The next one should be less than all of the above special indices.
00779 */
00780
00781 #define MINSPEC -10
00782

```

```
00783 #define USEDFLAG 2
00784 #define DUMMYFLAG 1
00785
00786 #define MAINSORT 0
00787 #define FUNCTIONSORT 1
00788 #define SUBSORT 2
00789
00790 #define FLOATMODE 1
00791 #define RATIONALMODE 0
00792
00793 #define NUMSPECSETS 10
00794
00795 #define ISZERO 1
00796 #define ISUNMODIFIED 2
00797 #define ISCOMPRESSED 4
00798 #define ISINRHS 8
00799 #define ISFACTORIZED 16
00800 #define TOBEFACTORED 32
00801 #define TOBEUNFACTORED 64
00802 #define KEEPZERO 128
00803
00804 #define VARTYPENONE 0
00805 #define VARTYPECOMPLEX 1
00806 #define VARTYPEIMAGINARY 2
00807 #define VARTYPEROOTOFUNITY 4
00808 #define VARTYPEMINUS 8
00809 #define CYCLESYMMETRIC 1
00810 #define RCYCLESYMMETRIC 2
00811 #define SYMMETRIC 3
00812 #define ANTISYMMETRIC 4
00813 #define REVERSEORDER 256
00814
00815 /*
00816     Types of id statements (substitutions)
00817 */
00818
00819 #define SUBMULTI 1
00820 #define SUBONCE 2
00821 #define SUBONLY 3
00822 #define SUBMANY 4
00823 #define SUBVECTOR 5
00824 #define SUBSELECT 6
00825 #define SUBALL 7
00826 #define SUBMASK 15
00827 #define SUBDISORDER 16
00828 #define SUBAFTER 32
00829 #define SUBAFTERNOT 64
00830
00831 #define IDHEAD 6
00832
00833 #define DOLLARFLAG 1
00834 #define NORMALIZEFLAG 2
00835
00836 #define GIDENT 1
00837 #define GFIVE 4
00838 #define GPLUS 3
00839 #define GMINUS 2
00840
00841 /*
00842     Types of objects inside an if clause.
00843 */
00844
00845 #define LONGNUMBER 1
00846 #define MATCH 2
00847 #define COEFFI 3
00848 #define SUBEXPR 4
00849 #define MULTIPLEOF 5
00850 #define IFDOLLAR 6
00851 #define IFEXPRESSION 7
00852 #define IFDOLLAREXTRA 8
00853 #define IFISFACTORIZED 9
00854 #define IFOCCURS 10
00855 #define IFUSERFLAG 11
00856 #define IFFLOATNUMBER 12
00857 #define GREATER 0
00858 #define GREATEREQUAL 1
00859 #define LESS 2
00860 #define LESSEQUAL 3
00861 #define EQUAL 4
00862 #define NOTEQUAL 5
00863 #define ORCOND 6
00864 #define ANDCOND 7
00865 #define DUMMY 1
00866 #define SORT 1
00867 #define STORE 2
00868 #define END 3
00869 #define GLOBAL 4
```

```

00870 #define CLEAR 5
00871
00872 #define VECTBIT 1
00873 #define DOTPBIT 2
00874 #define FUNBIT 4
00875 #define SETBIT 8
00876
00877 #define EXTRAPARAMETER 0x4000
00878 #define GENCOMMUTE 0
00879 #define GENNONCOMMUTE 0x2000
00880
00881 #define NAMENOTFOUND -9
00882
00883 /*
00884     Types of dollar expressions.
00885 */
00886
00887 #define DOLUNDEFINED 0
00888 #define DOLNUMBER 1
00889 #define DOLARGUMENT 2
00890 #define DOLSUBTERM 3
00891 #define DOLTERMS 4
00892 #define DOLWILDARGS 5
00893 #define DOLINDEX 6
00894 #define DOLZERO 7
00895
00896 #define FINDLOOP 0
00897 #define REPLACELOOP 1
00898
00899 #define NOFUNPOWERS 0
00900 #define COMFUNPOWERS 1
00901 #define ALLFUNPOWERS 2
00902
00903 #define PROPERORDERFLAG 0
00904
00905 #define REGULAR 0
00906 #define FINISH 1
00907
00908 #define POLYADD 1
00909 #define POLYSUB 2
00910 #define POLYMUL 3
00911 #define POLYDIV 4
00912 #define POLYREM 5
00913 #define POLYGCD 6
00914 #define POLYINTFAC 7
00915 #define POLYNORM 8
00916
00917 #define MODNONE 0
00918 #define MODSUM 1
00919 #define MODMAX 2
00920 #define MODMIN 3
00921 #define MODLOCAL 4
00922
00923 #define ELEMENTUSED 1
00924 #define ELEMENTLOADED 2
00925 /*
00926     Variables for the modulus statement, flags in AC.modmode
00927     For explanation, see CoModulus
00928 */
00929 #define POSNEG 0x1
00930 #define INVERSETABLE 0x2
00931 #define COEFFICIENTSONLY 0x4
00932 #define ALSOPOWERS 0x8
00933 #define ALSOFUNARGS 0x10
00934 #define ALSODOLLARS 0x20
00935 #define NOINVERSES 0x40
00936
00937 #define POSITIVEONLY 0
00938 #define UNPACK 0x80
00939 #define NOUNPACK 0
00940 #define FROMFUNCTION 0x100
00941
00942 #define VARNAMES 0
00943 #define AUTONAMES 1
00944 #define EXPRNAMES 2
00945 #define DOLLARNAMES 3
00946
00947 #ifdef WITHPTHREADS
00948 /*
00949     Signals that the workers have to react to
00950 */
00951
00952 #define TERMINATETHREAD -1
00953 #define STARTNEWEXPRESSION 1
00954 #define LOWESTLEVELGENERATION 2
00955 #define FINISHEXPRESSION 3
00956 #define CLEANUPEXPRESSION 4

```

```

00957 #define HIGHERLEVELGENERATION 5
00958 #define STARTNEWMODULE 6
00959 #define CLAIMOUTPUT 7
00960 #define FINISHEXPRESSION2 8
00961 #define INISORTBOT 7
00962 #define RUNSORTBOT 8
00963 #define DOONEEXPRESSION 9
00964 #define DOBRACKETS 10
00965 #define CLEARCLOCK 11
00966 #define MCTSEXPANDTREE 12
00967 #define OPTIMIZEEXPRESSION 13
00968
00969 #define MASTERBUFFERISFULL 1
00970
00971 /*
00972     Bucket states
00973 */
00974
00975 #define BUCKETFREE 1
00976 #define BUCKETINUSE 0
00977 #define BUCKETCOMINGFREE 2
00978 #define BUCKETFILLED -1
00979 #define BUCKETATEND -2
00980 #define BUCKETTERMINATED 3
00981 #define BUCKETRELEASED 4
00982
00983 #define NUMBEROFBLOCKSINSORT 10
00984 #define MINIMUMNUMBEROFTERMS 10
00985
00986 #define BUCKETDOINGTERM 1
00987 #define BUCKETASSIGNED -1
00988 #define BUCKETTOBERELEASED -2
00989 #define BUCKETPREPARINGTERM 0
00990
00991 #define BUCKETDOINGTERMS 0
00992 #define BUCKETDOINGBRACKET 1
00993 #endif
00994
00995 /*
00996     The next variable is because there is some use of cbufnum that is
00997     probably irrelevant. We use here DUMMYBUFNUM instead of AC.cbufnum
00998     just in case we run into trouble later.
00999 */
01000 #define DUMMYBUFFER 1
01001
01002 #define ALLARGS 1
01003 #define NUMARG 2
01004 #define ARGGRANGE 3
01005 #define MAKEARGS 4
01006 #define MAXRANGEINDICATOR 4
01007 #define REPLACEARG 5
01008 #define ENCODEARG 6
01009 #define DECODEARG 7
01010 #define IMplodeARG 8
01011 #define EXPLODEARG 9
01012 #define PERMUTEARG 10
01013 #define REVERSEARG 11
01014 #define CYCLEARG 12
01015 #define ISLYNDON 13
01016 #define ISLYNDONR 14
01017 #define TOLYNDON 15
01018 #define TOLYNDONR 16
01019 #define ADDARG 17
01020 #define MULTIPLYARG 18
01021 #define DROPARG 19
01022 #define SELECTARG 20
01023 #define DEDUPARG 21
01024 #define ZTOHARG 22
01025 #define HTOZARG 23
01026 /*
01027 #define ZTOSARG 24
01028 #define STOZARG 25
01029 #define HTOSARG 26
01030 #define STOHARG 27
01031 */
01032
01033 #define BASECODE 1
01034 #define YESLYNDON 1
01035 #define NOLYNDON 2
01036
01037 #define TOPOLYNOMIALFLAG 1
01038 #define FACTARGFLAG 2
01039
01040 #define OLDFACTARG 1
01041 #define NEWFACTARG 0
01042
01043 #define FROMMODULEOPTION 0

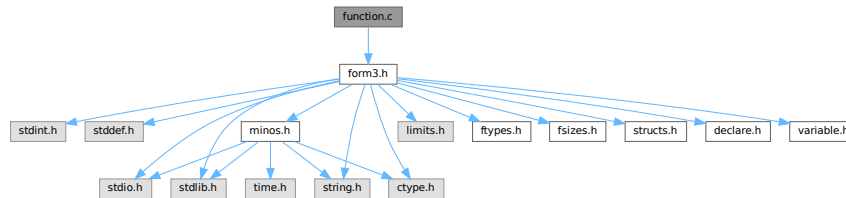
```

```
01044 #define FROMPOINTINSTRUCTION 1
01045
01046 #define EXTRASYMBOL 0
01047 #define REGULARSYMBOL 1
01048 #define EXPRESSIONNUMBER 2
01049
01050 #define O_NONE 0
01051 #define O_CSE 1
01052 #define O_CSEGREEDY 2
01053 #define O_GREEDY 3
01054
01055 #define O_OCCURRENCE 0
01056 #define O_MCTS 1
01057 #define O_SIMULATED_ANNEALING 2
01058
01059 #define O_FORWARD 0
01060 #define O_BACKWARD 1
01061 #define O_FORWARDORBACKWARD 2
01062 #define O_FORWARDANDBACKWARD 3
01063
01064 #define OPTHEAD 3
01065 #define DOALL 1
01066 #define ONLYFUNCTIONS 2
01067
01068 #define INUSE 1
01069 #define COULDCOMMUTE 2
01070 #define DOESNOTCOMMUTE 4
01071
01072 #define DICT_NONUMBERS 0
01073 #define DICT_INTEGERONLY 1
01074 #define DICT_RATIONALONLY 2
01075 #define DICT_ALLNUMBERS 3
01076 #define DICT_NOVARIABLES 0
01077 #define DICT_DOVARIABLES 1
01078 #define DICT_NOSPECIALS 0
01079 #define DICT_DOSPECIALS 1
01080 #define DICT_NOFUNWITHARGS 0
01081 #define DICT_DOFUNWITHARGS 1
01082 #define DICT_NOTINDOLLARS 0
01083 #define DICT_INDOLLARS 1
01084
01085 #define DICT_INTEGERNUMBER 1
01086 #define DICT_RATIONALNUMBER 2
01087 #define DICT_SYMBOL 3
01088 #define DICT_VECTOR 4
01089 #define DICT_INDEX 5
01090 #define DICT_FUNCTION 6
01091 #define DICT_FUNCTION_WITH_ARGUMENTS 7
01092 #define DICT_SPECIALCHARACTER 8
01093 #define DICT_RANGE 9
01094
01095 #define READSPECTATORFLAG 3
01096 #define GLOBALSPECTATORFLAG 1
01097
01098 #define ORDEREDSET 1
01099
01100 #define DENSETABLE 1
01101 #define SPARSETABLE 0
01102
01103 #define ONEPARTICLEIRREDUCIBLE 1
01104 #define WITHINSERTIONS 2
01105 #define NOTADPOLES 4
01106 #define WITHSYMMETRIZE 8
01107 #define TOPOLOGIESONLY 16
01108 #define NONODES 32
01109 #define WITHEDGES 64
01110 #define CHECKEXTERN 128
01111 #define WITHBLOCKS 256
01112 #define WITHONEPI 512
01113 #define NOSNAILS 1024
01114 #define NOEXTSELF 2048
01115
```

## 4.51 function.c File Reference

```
#include "form3.h"
```

Include dependency graph for function.c:



### Functions

- int [MakeDirty](#) (WORD \*term, WORD \*x, WORD par)
- void [MarkDirty](#) (WORD \*term, WORD flags)
- void [PolyFunDirty](#) (PHEAD WORD \*term)
- void [PolyFunClean](#) (PHEAD WORD \*term)
- WORD [Symmetrize](#) (PHEAD WORD \*func, WORD \*Lijst, WORD ngroups, WORD gsize, WORD type)
- int [CompGroup](#) (PHEAD WORD type, WORD \*\*args, WORD \*a1, WORD \*a2, WORD num)
- int [FullSymmetrize](#) (PHEAD WORD \*fun, int type)
- int [SymGen](#) (PHEAD WORD \*term, WORD \*params, WORD num, WORD level)
- int [SymFind](#) (PHEAD WORD \*term, WORD \*params)
- int [ChainIn](#) (PHEAD WORD \*term, WORD funnum)
- int [ChainOut](#) (PHEAD WORD \*term, WORD funnum)
- int [MatchFunction](#) (PHEAD WORD \*pattern, WORD \*interm, WORD \*wilds)
- int [ScanFunctions](#) (PHEAD WORD \*inpat, WORD \*inter, WORD par)

### 4.51.1 Detailed Description

The file with the central routines for the pattern matching of functions and their arguments. The file also contains the routines for the execution of the Symmetrize statement and its variations (like antisymmetrize etc).

Definition in file [function.c](#).

### 4.51.2 Function Documentation

#### 4.51.2.1 MakeDirty()

```
int MakeDirty (
    WORD * term,
    WORD * x,
    WORD par )
```

Definition at line 51 of file [function.c](#).



#### 4.51.2.2 MarkDirty()

```
void MarkDirty (
    WORD * term,
    WORD flags )
```

Definition at line 97 of file [function.c](#).

#### 4.51.2.3 PolyFunDirty()

```
void PolyFunDirty (
    PHEAD WORD * term )
```

Definition at line 139 of file [function.c](#).

#### 4.51.2.4 PolyFunClean()

```
void PolyFunClean (
    PHEAD WORD * term )
```

Definition at line 174 of file [function.c](#).

#### 4.51.2.5 Symmetrize()

```
WORD Symmetrize (
    PHEAD WORD * func,
    WORD * Lijst,
    WORD ngroups,
    WORD gsize,
    WORD type )
```

Definition at line 213 of file [function.c](#).

#### 4.51.2.6 CompGroup()

```
int CompGroup (
    PHEAD WORD type,
    WORD ** args,
    WORD * a1,
    WORD * a2,
    WORD num )
```

Definition at line 373 of file [function.c](#).

#### 4.51.2.7 FullSymmetrize()

```
int FullSymmetrize (
    PHEAD WORD * fun,
    int type )
```

Definition at line 473 of file [function.c](#).

#### 4.51.2.8 SymGen()

```
int SymGen (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level )
```

Definition at line 518 of file [function.c](#).

#### 4.51.2.9 SymFind()

```
int SymFind (
    PHEAD WORD * term,
    WORD * params )
```

Definition at line 633 of file [function.c](#).

#### 4.51.2.10 ChainIn()

```
int ChainIn (
    PHEAD WORD * term,
    WORD funnum )
```

Definition at line 691 of file [function.c](#).

#### 4.51.2.11 ChainOut()

```
int ChainOut (
    PHEAD WORD * term,
    WORD funnum )
```

Definition at line 748 of file [function.c](#).

#### 4.51.2.12 MatchFunction()

```
int MatchFunction (
    PHEAD WORD * pattern,
    WORD * interm,
    WORD * wilds )
```

Definition at line 826 of file [function.c](#).

#### 4.51.2.13 ScanFunctions()

```
int ScanFunctions (
    PHEAD WORD * inpat,
    WORD * inter,
    WORD par )
```

Definition at line 1616 of file [function.c](#).

## 4.52 function.c

[Go to the documentation of this file.](#)

```

00001
00008 /* #[ License : */
00009 /*
00010 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00011 *   When using this file you are requested to refer to the publication
00012 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00013 *   This is considered a matter of courtesy as the development was paid
00014 *   for by FOM the Dutch physics granting agency and we would like to
00015 *   be able to track its scientific use to convince FOM of its value
00016 *   for the community.
00017 *
00018 *   This file is part of FORM.
00019 *
00020 *   FORM is free software: you can redistribute it and/or modify it under the
00021 *   terms of the GNU General Public License as published by the Free Software
00022 *   Foundation, either version 3 of the License, or (at your option) any later
00023 *   version.
00024 *
00025 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00026 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00027 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00028 *   details.
00029 *
00030 *   You should have received a copy of the GNU General Public License along
00031 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00032 */
00033 /* #[ License : */
00034 /*
00035     #[ Includes : function.c
00036 */
00037
00038 #include "form3.h"
00039
00040 /*
00041     #[ Includes :
00042     #[ Utilities :
00043         #[ MakeDirty :
00044
00045         Routine finds the function with the address x in it
00046         and mark all arguments that contain x as dirty.
00047         if par == 0 term is a full term, else term is the start of a
00048         function
00049 */
00050
00051 int MakeDirty(WORD *term, WORD *x, WORD par)
00052 {
00053     WORD *next, *n;
00054     if ( !par ) {
00055         next = term; next += *term;
00056         next -= ABS(next[-1]);
00057         term++;
00058         if ( x < term ) return(0);
00059         if ( x >= next ) return(0);
00060         while ( term < next ) {
00061             n = term + term[1];
00062             if ( x < n ) break;
00063             term = n;
00064         }
00065         /* next = n; */
00066     }
00067     else {
00068         next = term + term[1];
00069         if ( x < term || x >= next ) return(0);
00070     }
00071     if ( *term < FUNCTION ) return(0);
00072     if ( functions[*term-FUNCTION].spec >= TENSORFUNCTION ) return(0);
00073     term += FUNHEAD;
00074     if ( x < term ) return(0);
00075     next = term; NEXTARG(next)
00076     while ( x >= next ) { term = next; NEXTARG(next) }
00077     if ( *term < 0 ) return(0);
00078     term[1] = 1;
00079     term += ARGHEAD;
00080     if ( x < term ) return(1);
00081     next = term + *term;
00082     while ( x >= next ) { term = next; next += *next; }
00083     MakeDirty(term,x,0);
00084     return(1);
00085 }
00086
00087 /*
00088     #[ MakeDirty :

```

```

00089      #] MarkDirty :
00090
00091      Routine marks all functions dirty with the given flags.
00092      Is to be used when there is a possibility that symmetrization
00093      properties of functions may have changed. In that case we play
00094      it safe.
00095  */
00096
00097 void MarkDirty(WORD *term, WORD flags)
00098 {
00099     WORD *t, *r, *m, *tstop;
00100     GETSTOP(term,tstop);
00101     t = term+1;
00102     while ( t < tstop ) {
00103         if ( *t < FUNCTION ) { t += t[1]; continue; }
00104         t[2] |= flags;
00105         if ( *t < FUNCTION+WILDOFFSET && functions[*t-FUNCTION].spec > 0 ) {
00106             t += t[1]; continue;
00107         }
00108         if ( *t >= FUNCTION+WILDOFFSET && functions[*t-FUNCTION-WILDOFFSET].spec > 0 ) {
00109             t += t[1]; continue;
00110         }
00111         r = t + FUNHEAD;
00112         t += t[1];
00113         while ( r < t ) {
00114             if ( *r <= 0 ) {
00115                 if ( *r <= -FUNCTION ) r++;
00116                 else r += 2;
00117                 continue;
00118             }
00119             r[1] |= DIRTYFLAG;
00120             m = r + ARGHEAD;
00121             r += *r;
00122             while ( m < r ) {
00123                 MarkDirty(m,flags);
00124                 m += *m;
00125             }
00126         }
00127     }
00128 }
00129
00130 /*
00131      #] MarkDirty :
00132      #] PolyFunDirty :
00133
00134      Routine marks the PolyFun or the PolyRatFun dirty.
00135      This is used when there is modular calculus and the modulus
00136      has changed for the current module.
00137  */
00138
00139 void PolyFunDirty(PHEAD WORD *term)
00140 {
00141     GETBIDENTITY
00142     WORD *t, *tstop, *endarg;
00143     tstop = term + *term;
00144     tstop -= ABS(tstop[-1]);
00145     t = term+1;
00146     while ( t < tstop ) {
00147         if ( *t == AR.PolyFun ) {
00148             if ( AR.PolyFunType == 2 ) t[2] |= MUSTCLEANPRF;
00149             endarg = t + t[1];
00150             t[2] |= DIRTYFLAG;
00151             t += FUNHEAD;
00152             while ( t < endarg ) {
00153                 if ( *t > 0 ) {
00154                     t[1] |= DIRTYFLAG;
00155                 }
00156                 NEXTARG(t);
00157             }
00158         }
00159         else {
00160             t += t[1];
00161         }
00162     }
00163 }
00164
00165 /*
00166      #] PolyFunDirty :
00167      #] PolyFunClean :
00168
00169      Routine marks the PolyFun or the PolyRatFun clean.
00170      This is used when there is modular calculus and the modulus
00171      has changed for the current module.
00172  */
00173
00174 void PolyFunClean(PHEAD WORD *term)
00175 {

```

```

00176     GETBIDENTITY
00177     WORD *t, *tstop;
00178     tstop = term + *term;
00179     tstop -= ABS(tstop[-1]);
00180     t = term+1;
00181     while ( t < tstop ) {
00182         if ( *t == AR.PolyFun ) {
00183             t[2] &= ~MUSTCLEANPRF;
00184         }
00185         t += t[1];
00186     }
00187 }
00188
00189 /*
00190     #] PolyFunClean :
00191     #[ Symmetrize :
00192
00193     (Anti)Symmetrizes the arguments of a function.
00194     Nlist tells of how many arguments are involved.
00195     Nlist == 0      All arguments must be sorted.
00196     Nlist > 0      Arguments mentioned are to be sorted, rest skipped.
00197     type = SYMMETRIC      Full symmetrization
00198     type = ANTISYMMETRIC: Full symmetrization
00199     type = CYCLESYMMETRIC: Cyclic
00200     type = RCYCLESYMMETRIC: Cyclic or reverse
00201     Return value: OR of:
00202         0 even, 1 odd
00203         2 equal groups
00204         4 there was a permutation.
00205
00206     The information in Lijst tells what grouping is to be applied.
00207     The information is:
00208     ngroups number of groups
00209     gsize size of groups
00210     Lijst[0].... The groups.
00211 */
00212
00213 WORD Symmetrize(PHEAD WORD *func, WORD *Lijst, WORD ngroups, WORD gsize,
00214                WORD type)
00215 {
00216     GETBIDENTITY
00217     WORD **args,**arg,nargs;
00218     WORD *to, *r, *fstop;
00219     WORD i, j, k, ff, exch, nexch, neq;
00220     WORD *a1, *a2, *a3;
00221     WORD reverseorder;
00222     if ( ( type & REVERSEORDER ) != 0 ) reverseorder = -1;
00223     else reverseorder = 1;
00224     type &= ~REVERSEORDER;
00225
00226     ff = ( *func > FUNCTION ) ? functions[*func-FUNCTION].spec: 0;
00227
00228     if ( 2*func[1] > AN.arglistsize ) {
00229         if ( AN.arglist ) M_free(AN.arglist,"Symmetrize");
00230         AN.arglistsize = 2*func[1] + 8;
00231         AN.arglist = (WORD **)Malloca(AN.arglistsize*sizeof(WORD *),"Symmetrize");
00232     }
00233     arg = args = AN.arglist;
00234     to = AT.WorkPointer;
00235     r = func;
00236     fstop = r + r[1];
00237     r += FUNHEAD;
00238     nargs = 0;
00239     while ( r < fstop ) { /* Make list of arguments */
00240         *arg++ = r;
00241         nargs++;
00242         if ( ff ) {
00243             if ( *r == FUNNYWILD ) r++;
00244             r++;
00245         }
00246         else { NEXTARG(r); }
00247     }
00248     exch = 0;
00249     nexch = 0;
00250     neq = 0;
00251     a1 = Lijst;
00252     if ( type == SYMMETRIC || type == ANTISYMMETRIC ) {
00253         for ( i = 1; i < ngroups; i++ ) {
00254             a3 = a2 = a1 + gsize;
00255             k = reverseorder*CompGroup(BHEAD ff,args,a1,a2,gsize);
00256             if ( k < 0 ) {
00257                 j = i-1;
00258                 for(;;) {
00259                     for ( k = 0; k < gsize; k++ ) {
00260                         r = args[a1[k]]; args[a1[k]] = args[a2[k]]; args[a2[k]] = r;
00261                     }
00262                     exch ^= 1;

```

```

00263         nexch = 4;
00264         if ( j <= 0 ) break;
00265         a1 -= gsize;
00266         a2 -= gsize;
00267         k = reverseorder*CompGroup(BHEAD ff,args,a1,a2,gsize);
00268         if ( k == 0 ) neq = 2;
00269         if ( k >= 0 ) break;
00270         j--;
00271     }
00272 }
00273 else if ( k == 0 ) neq = 2;
00274 a1 = a3;
00275 }
00276 }
00277 else if ( type == CYCLESYMMETRIC || type == RCYCLESYMMETRIC ) {
00278     WORD rev = 0, jmin = 0, ii, iimin;
00279 recycle:
00280     for ( j = 1; j < ngroups; j++ ) {
00281         for ( i = 0; i < ngroups; i++ ) {
00282             iimin = jmin + i;
00283             if ( iimin >= ngroups ) iimin -= ngroups;
00284             ii = j + i;
00285             if ( ii >= ngroups ) ii -= ngroups;
00286             k = reverseorder*CompGroup(BHEAD ff,args,Lijst+gsize*iimin,Lijst+gsize*ii,gsize);
00287             if ( k > 0 ) break;
00288             if ( k < 0 ) { jmin = j; nexch = 4; break; }
00289         }
00290     }
00291     if ( type == RCYCLESYMMETRIC && rev == 0 && ngroups > 1 ) {
00292         for ( j = 0; j < ngroups; j++ ) {
00293             for ( i = 0; i < ngroups; i++ ) {
00294                 iimin = jmin + i;
00295                 if ( iimin >= ngroups ) iimin -= ngroups;
00296                 ii = j - i;
00297                 if ( ii < 0 ) ii += ngroups;
00298                 k = reverseorder*CompGroup(BHEAD ff,args,Lijst+gsize*iimin,Lijst+gsize*ii,gsize);
00299                 if ( k > 0 ) break;
00300                 if ( k < 0 ) {
00301                     nexch = 4;
00302                     jmin = 0;
00303                     a1 = Lijst;
00304                     a2 = Lijst + gsize * (ngroups-1);
00305                     while ( a2 > a1 ) {
00306                         for ( k = 0; k < gsize; k++ ) {
00307                             r = args[a1[k]];
00308                             args[a1[k]] = args[a2[k]];
00309                             args[a2[k]] = r;
00310                         }
00311                         a1 += gsize; a2 -= gsize;
00312                     }
00313                     rev = 1;
00314                     goto recycle;
00315                 }
00316             }
00317         }
00318     }
00319     if ( jmin != 0 ) {
00320         arg = AN.arglist + func[1];
00321         a1 = Lijst + gsize * jmin;
00322         k = gsize * ngroups;
00323         a2 = Lijst + k;
00324         for ( i = 0; i < k; i++ ) {
00325             if ( a1 >= a2 ) a1 = Lijst;
00326             *arg++ = args[*a1++];
00327         }
00328         arg = AN.arglist + func[1];
00329         a1 = Lijst;
00330         for ( i = 0; i < k; i++ ) args[*a1++] = *arg++;
00331     }
00332 }
00333 r = func;
00334 i = FUNHEAD;
00335 NCOPY(to,r,i);
00336 for ( i = 0; i < nargs; i++ ) {
00337     if ( ff ) {
00338         if ( *(args[i]) == FUNNYWILD ) {
00339             *to++ = *(args[i]);
00340             *to++ = args[i][1];
00341         }
00342         else *to++ = *(args[i]);
00343     }
00344     else if ( ( j = *args[i] ) < 0 ) {
00345         *to++ = j;
00346         if ( j > -FUNCTION ) *to++ = args[i][1];
00347     }
00348     else {
00349         r = args[i];

```

```

00350         NCOPY(to,r,j);
00351     }
00352 }
00353 i = func[l];
00354 to = func;
00355 r = AT.WorkPointer;
00356 NCOPY(to,r,i);
00357 return ( exch | nexch | neq );
00358 }
00359
00360 /*
00361     #] Symmetrize :
00362     #[ CompGroup :
00363
00364     Routine compares two groups of arguments
00365     The arguments are in args[a1[i]] and args[a2[i]]
00366     for i = 0 to num
00367     type indicates the type of function.
00368     return value: -1 if there should be an exchange
00369     0 if they are equal
00370     1 if they are OK.
00371 */
00372
00373 int CompGroup(PHEAD WORD type, WORD **args, WORD *a1, WORD *a2, WORD num)
00374 {
00375     GETBIDENTITY
00376     WORD *t1, *t2, i1, i2, n, k;
00377
00378     for ( n = 0; n < num; n++ ) {
00379         t1 = args[a1[n]]; t2 = args[a2[n]];
00380         if ( type >= TENSORFUNCTION ) {
00381             if ( AR.Eside == LHSIDE || AR.Eside == LHSIDEX ) {
00382                 if ( *t1 == FUNNYWILD ) {
00383                     if ( *t2 == FUNNYWILD ) {
00384                         if ( t1[1] < t2[1] ) return(1);
00385                         if ( t1[1] > t2[1] ) return(-1);
00386                     }
00387                     return(-1);
00388                 }
00389                 else if ( *t2 == FUNNYWILD ) {
00390                     return(1);
00391                 }
00392                 else {
00393                     if ( *t1 < *t2 ) return(1);
00394                     if ( *t1 > *t2 ) return(-1);
00395                 }
00396             }
00397             else {
00398                 if ( *t1 < *t2 ) return(1);
00399                 if ( *t1 > *t2 ) return(-1);
00400             }
00401         }
00402         else if ( type == 0 ) {
00403             if ( AC.properorderflag ) {
00404                 k = CompArg(t1,t2);
00405                 if ( k < 0 ) return(1);
00406                 if ( k > 0 ) return(-1);
00407                 NEXTARG(t1)
00408                 NEXTARG(t2)
00409             }
00410             else {
00411                 if ( *t1 > 0 ) {
00412                     i1 = *t1 - ARGHEAD - 1;
00413                     t1 += ARGHEAD + 1;
00414                     if ( *t2 > 0 ) {
00415                         i2 = *t2 - ARGHEAD - 1;
00416                         t2 += ARGHEAD + 1;
00417                         while ( i1 > 0 && i2 > 0 ) {
00418                             if ( *t1 > *t2 ) return(-1);
00419                             else if ( *t1 < *t2 ) return(1);
00420                             i1--; i2--; t1++; t2++;
00421                         }
00422                         if ( i1 > 0 ) return(-1);
00423                         else if ( i2 > 0 ) return(1);
00424                     }
00425                 }
00426                 /*
00427                     This seems to be a bug. Reported by Aneesh Monahar, 28-sep-2005
00428                     else return(1);
00429                 */
00430                 else return(-1);
00431             }
00432         }
00433         else if ( *t2 > 0 ) return(1);
00434         else {
00435             if ( *t1 != *t2 ) {
00436                 if ( *t1 <= -FUNCTION && *t2 <= -FUNCTION ) {
00437                     if ( *t1 < *t2 ) return(-1);
00438                     return(1);
00439                 }

```

```

00437         }
00438         else {
00439             if ( *t1 < *t2 ) return(1);
00440             return(-1);
00441         }
00442     }
00443     if ( *t1 > -FUNCTION ) {
00444         if ( t1[1] != t2[1] ) {
00445             if ( t1[1] < t2[1] ) return(1);
00446             return(-1);
00447         }
00448     }
00449 }
00450 }
00451 }
00452 }
00453 return(0);
00454 }
00455
00456 /*
00457     #] CompGroup :
00458     #[ FullSymmetrize :
00459
00460     Relay function for Normalize to execute a full symmetrization
00461     of a function fun. It hooks into Symmetrize according to the
00462     calling conventions for it.
00463     type = 0: Symmetrize
00464     type = 1: AntiSymmetrize
00465     type = 2: CycleSymmetrize
00466     type = 3: RCycleSymmetrize
00467     Return values:
00468     bit 0: odd permutation
00469     bit 1: identical arguments
00470     bit 2: there was a permutation.
00471 */
00472
00473 int FullSymmetrize(PHEAD WORD *fun, int type)
00474 {
00475     GETBIDENTITY
00476     WORD *Lijst, count = 0;
00477     WORD *t, *funstop, i;
00478     int retval;
00479
00480     if ( functions[*fun-FUNCTION].spec > 0 ) {
00481         count = fun[1] - FUNHEAD;
00482         for ( i = fun[1]-1; i >= FUNHEAD; i-- ) {
00483             if ( fun[i] == FUNNYWILD ) count--;
00484         }
00485     }
00486     else {
00487         funstop = fun + fun[1];
00488         t = fun + FUNHEAD;
00489         while ( t < funstop ) { count++; NEXTARG(t) }
00490     }
00491     if ( count < 2 ) {
00492         fun[2] &= ~DIRTYSYMFLAG;
00493         return(0);
00494     }
00495     Lijst = AT.WorkPointer;
00496     for ( i = 0; i < count; i++ ) Lijst[i] = i;
00497     AT.WorkPointer += count;
00498     retval = Symmetrize(BHEAD fun, Lijst, count, 1, type);
00499     fun[2] &= ~DIRTYSYMFLAG;
00500     AT.WorkPointer = Lijst;
00501     return(retval);
00502 }
00503
00504 /*
00505     #] FullSymmetrize :
00506     #[ SymGen :
00507
00508     Routine does the outer work in the symmetrization.
00509     It locates the function(s) and loads up the parameters.
00510     It also studies the result.
00511
00512     if params[4] = -1 and no extra -> all
00513         extra -> strip groups with elements too large
00514         0 -> if group with element too large: nofun
00515         >0 -> must have right number of arguments
00516 */
00517
00518 int SymGen(PHEAD WORD *term, WORD *params, WORD num, WORD level)
00519 {
00520     GETBIDENTITY
00521     WORD *t, *r, *m;
00522     WORD i, j, k, c1, c2, ngroup;
00523     WORD *rstop, Nlist, *inLijst, *Lijst, sign = 1, sumch = 0, count;

```



```

00524     DUMMYUSE(num);
00525     c1 = params[3];          /* function number */
00526     c2 = FUNCTION + WILDOFFSET;
00527     Nlist = params[4];
00528     if ( Nlist < 0 ) Nlist = 0;
00529     else Nlist = params[0] - 7;
00530     t = term;
00531     m = t + *t;
00532     m -= ABS(m[-1]);
00533     t++;
00534     while ( t < m ) {
00535         if ( *t == c1 || c1 > c2 ) {      /* Candidate function */
00536             if ( *t >= FUNCTION && functions[*t-FUNCTION].spec
00537                 >= TENSORFUNCTION ) {
00538                 count = t[1] - FUNHEAD;
00539             }
00540             else {
00541                 count = 0;
00542                 r = t;
00543                 rstop = t + t[1];
00544                 r += FUNHEAD;
00545                 while ( r < rstop ) { count++; NEXTARG(r) }
00546             }
00547             if ( ( j = params[4] ) > 0 && j != count ) goto NextFun;
00548             if ( j == 0 ) {
00549                 inLijst = params+7;
00550                 for ( i = 0; i < Nlist; i++ )
00551                     if ( inLijst[i] > count-1 ) goto NextFun;
00552             }
00553
00554             if ( Nlist > (params[0] - 7) ) Nlist = params[0] - 7;
00555             Lijst = AT.WorkPointer;
00556             inLijst = params + 7;
00557             ngroup = params[5];
00558             if ( Nlist > 0 && j < 0 ) {
00559                 k = 0;
00560                 for ( i = 0; i < ngroup; i++ ) {
00561                     for ( j = 0; j < params[6]; j++ ) {
00562                         if ( inLijst[j] > count+1 ) {
00563                             inLijst += params[6];
00564                             goto NextGroup;
00565                         }
00566                     }
00567                     j = params[6];
00568                     NCOPY(Lijst,inLijst,j);
00569                     k++;
00570 NextGroup:;
00571                 }
00572                 if ( k <= 1 ) goto NextFun;
00573                 ngroup = k;
00574                 inLijst = AT.WorkPointer;
00575                 AT.WorkPointer = Lijst;
00576                 Lijst = inLijst;
00577             }
00578             else if ( Nlist == 0 ) {
00579                 for ( i = 0; i < count; i++ ) Lijst[i] = i;
00580                 AT.WorkPointer += count;
00581                 ngroup = count;
00582             }
00583             else {
00584                 for ( i = 0; i < Nlist; i++ ) Lijst[i] = inLijst[i];
00585                 AT.WorkPointer += Nlist;
00586             }
00587             j = Symmetrize(BHEAD t,Lijst,ngroup,params[6],params[2]);
00588             AT.WorkPointer = Lijst;
00589             if ( params[2] == 4 ) { /* antisymmetric */
00590                 if ( ( j & 1 ) != 0 ) sign = -sign;
00591                 if ( ( j & 2 ) != 0 ) return(0); /* equal arguments */
00592             }
00593             if ( ( j & 4 ) != 0 ) sumch++;
00594             t[2] &= ~DIRTYSYMFLAG;
00595         }
00596 NextFun:
00597         t += t[1];
00598     }
00599     if ( sign < 0 ) {
00600         t = term;
00601         t += *t - 1;
00602         *t = -*t;
00603     }
00604     if ( sumch ) {
00605         if ( Normalize(BHEAD term) ) {
00606             MLOCK(ErrorMessageLock);
00607             MesCall("SymGen");
00608             MUNLOCK(ErrorMessageLock);
00609             return(-1);
00610         }

```

```

00611         if ( !*term ) return(0);
00612         *AN.RepPoint = 1;
00613         AR.expchanged = 1;
00614         if ( AR.CurDum > AM.IndDum && AR.sLevel <= 0 ) ReNumber(BHEAD term);
00615     }
00616     return(Generator(BHEAD term,level));
00617 }
00618
00619 /*
00620     #] SymGen :
00621     #[ SymFind :
00622
00623     There is a certain amount of double work here, as this routine
00624     finds the function to be treated, while the SymGen routine has
00625     to find it again. Note however that this way things remain
00626     uniform and simple. Moreover this avoids problems with actions
00627     on more than one function simultaneously.
00628     Output in AT.TMout:
00629     Number,sym/anti,fun,lenpar,ngroups,gsize,fields
00630 */
00631
00632 int SymFind(PHEAD WORD *term, WORD *params)
00633 {
00634     GETBIDENTITY
00635     WORD *t, *r, *m;
00636     WORD j, c1, c2, count;
00637     WORD *rstop;
00638     c1 = params[4]; /* function number */
00639     c2 = FUNCTION + WILDOFFSET;
00640     t = term;
00641     m = t + *t;
00642     m -= ABS(m[-1]);
00643     t++;
00644     while ( t < m ) {
00645         if ( *t == c1 || c1 > c2 ) { /* Candidate function */
00646             if ( *t >= FUNCTION && functions[*t-FUNCTION].spec
00647                 >= TENSORFUNCTION ) { count = t[1] - FUNHEAD; }
00648             else {
00649                 count = 0;
00650                 r = t;
00651                 rstop = t + t[1];
00652                 r += FUNHEAD;
00653                 while ( r < rstop ) { count++; NEXTARG(r) }
00654             }
00655             if ( ( j = params[5] ) > 0 && j != count ) goto NextFun;
00656             if ( j == 0 ) {
00657                 r = params + 8;
00658                 rstop = params + params[1];
00659                 while ( r < rstop ) {
00660                     if ( *r > count + 1 ) goto NextFun;
00661                     r++;
00662                 }
00663             }
00664             t = AT.TMout;
00665             r = params;
00666             j = r[1] - 1;
00667             *t++ = j;
00668             *t++ = SYMMETRIZE;
00669             r += 3;
00670             j--;
00671             NCOPY(t,r,j);
00672             return(1);
00673         }
00674     }
00675     NextFun:
00676     t += t[1];
00677 }
00678 return(0);
00679 }
00680
00681 /*
00682     #] SymFind :
00683     #[ ChainIn :
00684
00685     Equivalent to repeat id f(?a)*f(?b) = f(?a,?b);
00686
00687     This one always takes less space.
00688 */
00689
00690 int ChainIn(PHEAD WORD *term, WORD funnum)
00691 {
00692     GETBIDENTITY
00693     WORD *t, *tend, *m, *tt, *ts;
00694     int action, normFlag = 0;
00695     if ( funnum < 0 ) { /* Dollar to be expanded */
00696         funnum = DolToFunction(BHEAD -funnum);
00697     }

```

```

00698         if ( AN.ErrorInDollar || funnum <= 0 ) {
00699             MLOCK(ErrorMessageLock);
00700             MesPrint("Dollar variable does not evaluate to function in ChainIn statement");
00701             MUNLOCK(ErrorMessageLock);
00702             return(-1);
00703         }
00704     }
00705     do {
00706         action = 0;
00707         tend = term+*term;
00708         tend -= ABS(tend[-1]);
00709         t = term+1;
00710         while ( t < tend ) {
00711             if ( *t != funnum ) { t += t[1]; continue; }
00712             m = t;
00713             t += t[1];
00714             tt = t;
00715             if ( t >= tend || *t != funnum ) continue;
00716             action = 1;
00717             normFlag = 1;
00718             while ( t < tend && *t == funnum ) {
00719                 ts = t + t[1];
00720                 t += FUNHEAD;
00721                 while ( t < ts ) *tt++ = *t++;
00722             }
00723             m[1] = tt - m;
00724             ts = term + *term;
00725             while ( t < ts ) *tt++ = *t++;
00726             *term = tt - term;
00727             break;
00728         }
00729     } while ( action );
00730
00731     if ( normFlag ) {
00732         /* We need to check the newly-constructed arguments w.r.t symmetry properties */
00733         MarkDirty(term, DIRTSYMFFLAG);
00734         AT.WorkPointer = term + *term;
00735         Normalize(BHEAD term);
00736     }
00737
00738     return(0);
00739 }
00740
00741 /*
00742     #] ChainIn :
00743     #[ ChainOut :
00744
00745     Equivalent to repeat id f(x1?,x2?,?a) = f(x1)*f(x2,?a);
00746 */
00747
00748 int ChainOut(PHEAD WORD *term, WORD funnum)
00749 {
00750     GETBIDENTITY
00751     WORD *t, *tend, *tt, *ts, *w, *ws;
00752     int flag = 0, i;
00753     if ( funnum < 0 ) { /* Dollar to be expanded */
00754         funnum = DolToFunction(BHEAD -funnum);
00755         if ( AN.ErrorInDollar || funnum <= 0 ) {
00756             MLOCK(ErrorMessageLock);
00757             MesPrint("Dollar variable does not evaluate to function in ChainOut statement");
00758             MUNLOCK(ErrorMessageLock);
00759             return(-1);
00760         }
00761     }
00762     tend = term+*term;
00763     if ( AT.WorkPointer < tend ) AT.WorkPointer = tend;
00764     tend -= ABS(tend[-1]);
00765     t = term+1; tt = term; w = AT.WorkPointer;
00766     while ( t < tend ) {
00767         if ( *t != funnum || t[1] == FUNHEAD ) { t += t[1]; continue; }
00768         flag = 1;
00769         while ( tt < t ) *w++ = *tt++;
00770         ts = t + t[1];
00771         t += FUNHEAD;
00772         while ( t < ts ) {
00773             ws = w;
00774             for ( i = 0; i < FUNHEAD; i++ ) *w++ = tt[i];
00775             if ( functions[*tt-FUNCTION].spec >= TENSORFUNCTION ) {
00776                 *w++ = *tt++;
00777             }
00778             else if ( *t < 0 ) {
00779                 if ( *t <= -FUNCTION ) *w++ = *tt++;
00780                 else { *w++ = *t++; *w++ = *t++; }
00781             }
00782             else {
00783                 i = *t; NCOPY(w,t,i);
00784             }

```

```

00785         ws[l] = w - ws;
00786     }
00787     tt = t;
00788 }
00789 if ( flag == 1 ) {
00790     ts = term + *term;
00791     while ( tt < ts ) *w++ = *tt++;
00792     *AT.WorkPointer = w - AT.WorkPointer;
00793     t = term; w = AT.WorkPointer; i = *w;
00794     NCOPY(t,w,i)
00795     AT.WorkPointer = term + *term;
00796     Normalize(BHEAD term);
00797 }
00798 return(0);
00799 }
00800
00801 /*
00802     #] ChainOut :
00803     #] Utilities :
00804     #[ Patterns :
00805         #[ MatchFunction :          WORD MatchFunction(pattern,interm,wilds)
00806
00807         The routine assumes that the function numbers are the same.
00808         The contents are compared and a possible wildcard assignment
00809         is made. Note that it may be necessary to use a wildcard
00810         assignment stack to do things right.
00811         The routine can become arbitrarily complicated as there is
00812         no end to the possible wilddarding.
00813         Examples:
00814         - a: No wilddarding -> straight match
00815         - b: Individual arguments (object -> object)
00816         - c: whole arguments (object to subexpression)
00817         - d: any argumentlist
00818         - e: part of an argument (object inside subexpression)
00819
00820         The ones with a minus sign in front have been implemented.
00821
00822         Note: the argument wilds allows backtracking when multiple
00823         ?a,?b give a match that later turns out to be useless.
00824 */
00825
00826 int MatchFunction(PHEAD WORD *pattern, WORD *interm, WORD *wilds)
00827 {
00828     GETBIDENTITY
00829     WORD *m, *t, *r, i;
00830     WORD *mstop = 0, *tstop = 0;
00831     WORD *argmstop, *argtstop;
00832     WORD *mtrmstop, *ttrmstop;
00833     WORD *msubstop, *mnxtsub;
00834     WORD msizcoef, mcount, tcount, newvalue, j;
00835     WORD *oldm, *oldt;
00836     WORD *OldWork, numofwildarg;
00837     WORD nwstore, tobeeaten, reservevalue = 0, resernum = 0, withwild;
00838     WORD *wildargtaken;
00839     CBUF *C = cbuf+AT.ebufnum;
00840     int ntw = AN.NumTotWildArgs;
00841     LONG oldcpointer = C->Pointer - C->Buffer;
00842 /*
00843     Test first for a straight match
00844 */
00845     AN.RepFunList[AN.RepFunNum+1] = 0;
00846     if ( *wilds == 0 ) {
00847         m = pattern; t = interm;
00848
00849         if ( *m != *t ) {
00850             if ( *m < (FUNCTION + WILDOFFSET) ) return(0);
00851             if ( *t < FUNCTION ) return(0);
00852             if ( functions[*t-FUNCTION].spec !=
00853                 functions[*m-FUNCTION-WILDOFFSET].spec ) return(0);
00854         }
00855         i = m[1];
00856         if ( *m >= (FUNCTION + WILDOFFSET) ) { i--; m++; t++; }
00857         do { if ( *m++ != *t++ ) break; } while ( --i > 0 );
00858         if ( i <= 0 ) { /* Arguments match */
00859             if ( AN.SignCheck && AN.ExpectedSign ) return(0);
00860             i = *pattern - WILDOFFSET;
00861             if ( i >= FUNCTION ) {
00862                 if ( *interm != GAMMA
00863 #ifdef WITHFLOAT
00864                     && ( *interm != FLOATFUN )
00865 #endif
00866                     && !CheckWild(BHEAD i,FUNTOFUN,*interm,&newvalue) ) {
00867                 AddWild(BHEAD i,FUNTOFUN,newvalue);
00868                 return(1);
00869             }
00870         }
00871         return(0);

```

```

00872         }
00873         else return(1);
00874     }
00875 }
00876 /*
00877  Store the current Wildcard assignments
00878 */
00879 t = wildargtaken = OldWork = AT.WorkPointer;
00880 t += ntw;
00881 m = AN.WildValue;
00882 nwstore = i = (m[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
00883 if ( i > 0 ) {
00884     r = AT.WildMask;
00885     do {
00886         *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
00887     } while ( --i > 0 );
00888     *t++ = C->numrhs;
00889 }
00890 if ( t >= AT.WorkTop ) {
00891     MLOCK(ErrorMessageLock);
00892     MesWork();
00893     MUNLOCK(ErrorMessageLock);
00894     Terminate(-1);
00895 }
00896 AT.WorkPointer = t;
00897
00898 if ( *wilds ) {
00899     if ( *wilds == 1 ) goto endoloop;
00900     else goto enloop;          /* tensors = 2 */
00901 }
00902 m = pattern; t = interm;
00903 /*
00904  Single out the specials
00905 */
00906 if ( *t == GAMMA ) {
00907     /*
00908      #[ GAMMA :
00909
00910      For the gamma's we need to do two things:
00911      a: Find that there is a match
00912      b: Find where the match occurs in the string
00913      This last thing cannot be stored in the current conventions,
00914      but once the wildcard assignments have been made it is much
00915      easier to find it back.
00916      Alternative: replace the function number in the term temporarily
00917      by the offset inside the string. This makes things maybe easier.
00918     */
00919     if ( *m != GAMMA ) goto NoCaseB;
00920     i = t[1] - m[1];
00921     if ( m[1] == FUNHEAD+1 ) {
00922         if ( i ) goto NoCaseB;
00923         if ( m[FUNHEAD] < (AM.OffsetIndex+WILDOFFSET) ||
00924             t[FUNHEAD] >= (AM.OffsetIndex+WILDOFFSET) ) goto NoCaseB;
00925
00926         if ( CheckWild(BHEAD m[FUNHEAD]-WILDOFFSET,INDTOIND,t[FUNHEAD],&newvalue) ) goto NoCaseB;
00927         AddWild(BHEAD m[FUNHEAD]-WILDOFFSET,INDTOIND,newvalue);
00928
00929         AT.WorkPointer = OldWork;
00930         if ( AN.SignCheck && AN.ExpectedSign ) return(0);
00931         return(1);          /* m was eaten. we have a match! */
00932     }
00933     if ( i < 0 ) goto NoCaseB; /* Pattern longer than target */
00934     mstop = m + m[1];
00935     tstop = t + t[1];
00936     m += FUNHEAD; t += FUNHEAD;
00937     if ( *m >= (AM.OffsetIndex+WILDOFFSET) && *t < (AM.OffsetIndex+WILDOFFSET) ) {
00938         if ( CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*t,&newvalue) ) goto NoCaseB;
00939         reservevalue = newvalue;
00940         withwild = 1;
00941         resernum = *m-WILDOFFSET;
00942         AddWild(BHEAD *m-WILDOFFSET,INDTOIND,newvalue);
00943     }
00944     else if ( *m != *t ) goto NoCaseB;
00945     else withwild = 0;
00946     m++; t++;
00947     oldm = m; argtstop = oldt = t;
00948     j = 0;          /* No wildcard assignments yet */
00949     while ( i >= 0 ) {
00950         if ( *m == *t ) {
00951             WithGamma:
00952             m++; t++;
00953             if ( m >= mstop ) {
00954                 if ( t < tstop && mstop < AN.patstop ) {
00955                     WORD k;
00956                     mnextsub = pattern + pattern[1];
00957                     k = *mnextsub;
00958                     while ( k == GAMMA && mnextsub[FUNHEAD]

```

```

00959             mnextsub += mnextsub[1];
00960             if ( mnextsub >= AN.patstop ) goto FullOK;
00961             k = *mnextsub;
00962         }
00963         if ( k >= FUNCTION ) {
00964             if ( k > (FUNCTION + WILDOFFSET) ) k -= WILDOFFSET;
00965             if ( functions[k-FUNCTION].commute ) goto NoGamma;
00966         }
00967     }
00968 FullOK:    if ( AN.SignCheck && AN.ExpectedSign ) goto NoGamma;
00969            AN.RepFunList[AN.RepFunNum+1] = WORDDIF(olddt, argtstop);
00970            return(1);
00971        }
00972        if ( t >= tstop ) goto NoCaseB;
00973    }
00974    else if ( *m >= (AM.OffsetIndex+WILDOFFSET)
00975    && *m < (AM.OffsetIndex + (WILDOFFSET+1)) && ( *t >= 0 ||
00976    *t < MINSPEC ) ) { /* Wildcard index */
00977        if ( !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*t,&newvalue) ) {
00978            AddWild(BHEAD *m-WILDOFFSET,INDTOIND,newvalue);
00979            j = 1;
00980            goto WithGamma;
00981        }
00982        else goto NoGamma;
00983    }
00984    else if ( *m < MINSPEC && *m >= (AM.OffsetVector+WILDOFFSET)
00985    && *t < MINSPEC ) { /* Wildcard vector */
00986        if ( !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*t,&newvalue) ) {
00987            AddWild(BHEAD *m-WILDOFFSET,VECTOVEC,newvalue);
00988            j = 1;
00989            goto WithGamma;
00990        }
00991        else goto NoGamma;
00992    }
00993    else {
00994 NoGamma:
00995        if ( j ) { /* Undo wildcards */
00996            m = AN.WildValue;
00997            t = OldWork + AN.NumTotWildArgs; r = AT.WildMask; j = nwstore;
00998            if ( j > 0 ) {
00999                do {
01000                    *m++ = *t++; *m++ = *t++;
01001                    *m++ = *t++; *m++ = *t++; *r++ = *t++;
01002                } while ( --j > 0 );
01003                C->numrhs = *t++;
01004                C->Pointer = C->Buffer + oldcpointer;
01005            }
01006            j = 0;
01007        }
01008        m = oldm; t = ++olddt; i--;
01009        if ( withwild ) {
01010            AddWild(BHEAD resernum,INDTOIND,reservevalue);
01011        }
01012    }
01013 }
01014 goto NoCaseB;
01015 /*
01016 #] GAMMA :
01017 #[ Tensors :
01018 */
01019 }
01020 else if ( *t >= FUNCTION && functions[*t-FUNCTION].spec >= TENSORFUNCTION ) {
01021     mstop = m + m[1];
01022     tstop = t + t[1];
01023     mcount = 0;
01024     m += FUNHEAD;
01025     t += FUNHEAD;
01026     AN.WildArgs = 0;
01027     tcount = WORDDIF(tstop,t);
01028     while ( m < mstop ) {
01029         if ( *m == FUNNYWILD ) { m++; AN.WildArgs++; }
01030         m++; mcount++;
01031     }
01032     tobeeaten = tcount - mcount + AN.WildArgs;
01033     if ( tobeeaten ) {
01034         if ( tobeeaten < 0 || AN.WildArgs == 0 ) {
01035             AT.WorkPointer = OldWork;
01036             return(0); /* Cannot match */
01037         }
01038     }
01039     AT.WildArgTaken[0] = AN.WildEat = tobeeaten;
01040     for ( i = 1; i < AN.WildArgs; i++ ) AT.WildArgTaken[i] = 0;
01041 toploop:
01042     numofwildarg = 0;
01043
01044     m = pattern; t = interm;
01045     mstop = m + m[1];

```

```

01046     if ( *m != *t ) {
01047         i = *m - WILDOFFSET;
01048         if ( CheckWild(BHEAD i,FUNTOFUN,*t,&newvalue) ) goto NoCaseB;
01049         AddWild(BHEAD i,FUNTOFUN,newvalue);
01050     }
01051     m += FUNHEAD;
01052     t += FUNHEAD;
01053     while ( m < mstop ) {
01054         /*
01055             First test for an exact match
01056         */
01057         if ( *m == *t ) { m++; t++; continue; }
01058         /*
01059             No exact match. Try ARGWILD
01060         */
01061         AN.argaddress = t;
01062         if ( *m == FUNNYWILD ) {
01063             tobeeaten = AT.WildArgTaken[numofwildarg++];
01064             if ( CheckWild(BHEAD m[1],ARGTOARG|EATTENSOR,tobeaten,t) ) goto endloop;
01065             AddWild(BHEAD m[1],ARGTOARG|EATTENSOR,tobeaten);
01066             m += 2;
01067             t += tobeeaten;
01068             continue;
01069         }
01070         /*
01071             Now the various cases:
01072         */
01073         i = *m;
01074         if ( i < MINSPEC ) {
01075             if ( *t != i ) {
01076                 if ( *t >= MINSPEC ) goto endloop;
01077                 i -= WILDOFFSET;
01078                 if ( i < AM.OffsetVector ) goto endloop;
01079                 if ( CheckWild(BHEAD i,VECTOVEC,*t,&newvalue) )
01080                     goto endloop;
01081                 AddWild(BHEAD i,VECTOVEC,newvalue);
01082             }
01083         }
01084         else if ( i >= AM.OffsetIndex ) { /* Index */
01085             if ( i < ( AM.OffsetIndex + WILDOFFSET ) ) goto endloop;
01086             if ( i >= ( AM.OffsetIndex + (WILDOFFSET«1) ) ) {
01087                 /* Summed over index */
01088                 goto endloop; /* For the moment */
01089             }
01090             i -= WILDOFFSET;
01091             if ( CheckWild(BHEAD i,INDTOIND,*t,&newvalue) )
01092                 goto endloop; /* Assignment not allowed */
01093             AddWild(BHEAD i,INDTOIND,newvalue);
01094         }
01095         else goto endloop;
01096         m++; t++;
01097     }
01098     if ( AN.SignCheck && AN.ExpectedSign ) goto endloop;
01099     AT.WorkPointer = OldWork;
01100     if ( AN.WildArgs > 1 ) *wilds = 2;
01101     return(1); /* m was eaten. we have a match! */
01102
01103 endloop:;
01104 /*
01105     restore the current Wildcard assignments
01106 */
01107     i = nwstore;
01108     if ( i > 0 ) {
01109         m = AN.WildValue;
01110         t = OldWork + ntw; r = AT.WildMask;
01111         do {
01112             *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01113             while ( --i > 0 );
01114             C->numrhs = *t++;
01115             C->Pointer = C->Buffer + oldcpointer;
01116         }
01117     endloop:;
01118     i = AN.WildArgs - 1;
01119     if ( i <= 0 ) {
01120         AT.WorkPointer = OldWork;
01121         return(0);
01122     }
01123     while ( --i >= 0 ) {
01124         if ( AT.WildArgTaken[i] == 0 ) {
01125             if ( i == 0 ) {
01126                 AT.WorkPointer = OldWork;
01127                 *wilds = 0;
01128                 return(0);
01129             }
01130         }
01131         else {
01132             (AT.WildArgTaken[i])--;

```

```

01133         numofwildarg = 0;
01134         for ( j = 0; j <= i; j++ ) {
01135             numofwildarg += AT.WildArgTaken[j];
01136         }
01137         AT.WildArgTaken[j] = AN.WildEat-numofwildarg;
01138         for ( j++; j < AN.WildArgs; j++ ) AT.WildArgTaken[j] = 0;
01139         break;
01140     }
01141 }
01142 goto toploop;
01143 /*
01144 #] Tensors :
01145 */
01146 }
01147 /*
01148 Count the number of arguments. Either equal or an argument wildcard.
01149 */
01150 mstop = m + m[1];
01151 tstop = t + t[1];
01152 mcount = 0; tcount = 0;
01153 m += FUNHEAD; t += FUNHEAD;
01154 while ( t < tstop ) { tcount++; NEXTARG(t) }
01155 AN.WildArgs = 0;
01156 while ( m < mstop ) {
01157     mcount++;
01158     if ( *m == -ARGWILD ) AN.WildArgs++;
01159     NEXTARG(m)
01160 }
01161 tobeeaten = tcount - mcount + AN.WildArgs;
01162 if ( tobeeaten ) {
01163     if ( tobeeaten < 0 || AN.WildArgs == 0 ) {
01164         AT.WorkPointer = OldWork;
01165         return(0); /* Cannot match */
01166     }
01167 }
01168 /*
01169 Set up the array AT.WildArgTaken for the number of arguments that each
01170 wildarg eats.
01171 */
01172 AT.WildArgTaken[0] = AN.WildEat = tobeeaten;
01173 for ( i = 1; i < AN.WildArgs; i++ ) AT.WildArgTaken[i] = 0;
01174 topofloop:
01175 numofwildarg = 0;
01176 /*
01177 Test for single wildcard object/argument
01178 */
01179 m = pattern; t = interm;
01180 if ( *m != *t ) {
01181     i = *m - WILDOFFSET;
01182     if ( CheckWild(BHEAD i,FUNTOFUN,*t,&newvalue) ) goto NoCaseB;
01183     AddWild(BHEAD i,FUNTOFUN,newvalue);
01184 }
01185 mstop = m + m[1];
01186 /* tstop = t + t[1]; */
01187 m += FUNHEAD;
01188 t += FUNHEAD;
01189 while ( m < mstop ) {
01190     argmstop = oldm = m;
01191     argtstop = oldt = t;
01192     NEXTARG(argmstop)
01193     NEXTARG(argtstop)
01194     if ( t == tstop ) { /* This concerns a very rare bug */
01195         if ( *m == -ARGWILD ) goto ArgAll;
01196         goto endofloop;
01197     }
01198     if ( *m < 0 && *t < 0 ) {
01199         if ( *t <= -FUNCTION ) {
01200             if ( *t == *m ) {}
01201             else if ( *m <= -FUNCTION-WILDOFFSET
01202                 && functions[-*t-FUNCTION].spec
01203                 == functions[-*m-FUNCTION-WILDOFFSET].spec ) {
01204                 i = -*m - WILDOFFSET;
01205                 if ( CheckWild(BHEAD i,FUNTOFUN,*t,&newvalue) ) goto endofloop;
01206                 AddWild(BHEAD i,FUNTOFUN,newvalue);
01207             }
01208             else if ( *m == -SYMBOL && m[1] >= 2*MAXPOWER ) {
01209                 i = m[1] - 2*MAXPOWER;
01210                 AN.argaddress = AT.FunArg;
01211                 AT.FunArg[ARGHEAD+1] = -*t;
01212                 if ( CheckWild(BHEAD i,SYMTOSUB,1,AN.argaddress) ) goto endofloop;
01213                 AddWild(BHEAD i,SYMTOSUB,0);
01214             }
01215             else if ( *m == -ARGWILD ) {
01216 ArgAll:
01217                 i = AT.WildArgTaken[numofwildarg++];
01218                 AN.argaddress = t;
01219                 if ( CheckWild(BHEAD m[1],ARGTOARG,i,t) ) goto endofloop;
01219                 AddWild(BHEAD m[1],ARGTOARG,i);

```



```

01220 /*          m += 2; */
01221          while ( --i >= 0 ) { NEXTARG(t) }
01222          argtstop = t;
01223      }
01224      else goto endofloop;
01225  }
01226  else if ( *t == *m ) {
01227      if ( t[1] == m[1] ) {}
01228      else if ( *t == -SYMBOL ) {
01229          j = SYMTOSYM;
01230  SymAll:
01231          if ( ( i = m[1] - 2*MAXPOWER ) < 0 ) goto endofloop;
01232          if ( CheckWild(BHEAD i,j,t[1],&newvalue) ) goto endofloop;
01233          AddWild(BHEAD i,j,newvalue);
01234      }
01235      else if ( *t == -INDEX ) {
01236  IndAll:
01237          i = m[1] - WILDOFFSET;
01238          if ( i < AM.OffsetIndex || i >= WILDOFFSET+AM.OffsetIndex )
01239              goto endofloop;
01240          /* We kill the summed over indices here */
01241          if ( CheckWild(BHEAD i,INDTOIND,t[1],&newvalue) ) goto endofloop;
01242          AddWild(BHEAD i,INDTOIND,newvalue);
01243      }
01244      else if ( *t == -VECTOR || *t == -MINVECTOR ) {
01245          i = m[1] - WILDOFFSET;
01246          if ( i < AM.OffsetVector ) goto endofloop;
01247          if ( CheckWild(BHEAD i,VECTOVEC,t[1],&newvalue) ) goto endofloop;
01248          AddWild(BHEAD i,VECTOVEC,newvalue);
01249      }
01250      else goto endofloop;
01251  }
01252  else if ( *m == -ARGWILD ) goto ArgAll;
01253  else if ( *m == -INDEX && m[1] >= AM.OffsetIndex+WILDOFFSET
01254  && m[1] < AM.OffsetIndex+(WILDOFFSET<1) ) {
01255      if ( *t == -VECTOR ) goto IndAll;
01256      if ( *t == -SNUMBER && t[1] >= 0 && t[1] < AM.OffsetIndex ) goto IndAll;
01257      if ( *t == -MINVECTOR ) {
01258          i = m[1] - WILDOFFSET;
01259          AN.argaddress = AT.MinVecArg;
01260          AT.MinVecArg[ARGHEAD+3] = t[1];
01261          if ( CheckWild(BHEAD i,INDTOSUB,1,AN.argaddress) ) goto endofloop;
01262          AddWild(BHEAD i,INDTOSUB,(WORD)0);
01263      }
01264      else goto endofloop;
01265  }
01266  else if ( *m == -SYMBOL && m[1] >= 2*MAXPOWER && *t == -SNUMBER ) {
01267      j = SYMTONUM;
01268      goto SymAll;
01269  }
01270  else if ( *m == -VECTOR && *t == -MINVECTOR &&
01271  ( i = m[1] - WILDOFFSET ) >= AM.OffsetVector ) {
01272  /*
01273  =====
01274      AN.argaddress = AT.MinVecArg;
01275      AT.MinVecArg[ARGHEAD+3] = t[1];
01276      if ( CheckWild(BHEAD i,VECTOSUB,1,AN.argaddress) ) goto endofloop;
01277      AddWild(BHEAD i,VECTOSUB,(WORD)0);
01278  =====
01279  */
01280      if ( CheckWild(BHEAD i,VECTOMIN,t[1],&newvalue) ) goto endofloop;
01281      AddWild(BHEAD i,VECTOMIN,newvalue);
01282  }
01283  }
01284  else if ( *m == -MINVECTOR && *t == -VECTOR &&
01285  ( i = m[1] - WILDOFFSET ) >= AM.OffsetVector ) {
01286  /*
01287  =====
01288      AN.argaddress = AT.MinVecArg;
01289      AT.MinVecArg[ARGHEAD+3] = t[1];
01290      if ( CheckWild(BHEAD i,VECTOSUB,1,AN.argaddress) ) goto endofloop;
01291      AddWild(BHEAD i,VECTOSUB,(WORD)0);
01292  =====
01293  */
01294      if ( CheckWild(BHEAD i,VECTOMIN,t[1],&newvalue) ) goto endofloop;
01295      AddWild(BHEAD i,VECTOMIN,newvalue);
01296  }
01297  }
01298  else if ( *t <= -FUNCTION && *m > 0 ) {
01299      if ( ( m[ARGHEAD]+ARGHEAD == *m ) && m[*m-1] == 3
01300      && m[*m-2] == 1 && m[*m-3] == 1 && m[ARGHEAD+1] >= FUNCTION
01301      && m[ARGHEAD+2] == *m-ARGHEAD-4 ) { /* Check for f(?a) etc */
01302          WORD *mmmst, *mmm;
01303          if ( m[ARGHEAD+1] >= FUNCTION+WILDOFFSET ) {
01304              i = *m - WILDOFFSET; /*
01305              i = m[ARGHEAD+1] - WILDOFFSET;
01306              if ( CheckWild(BHEAD i,FUNTOFUN,-*t,&newvalue) ) goto endofloop;

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01307         AddWild(BHEAD i,FUNTOFUN,newvalue);
01308     }
01309     else if ( m[ARGHEAD+1] != -*t ) goto endofloop;
01310 /*
01311         Only arguments allowed are ?a etc.
01312 */
01313     mmmst = m+*m-3;
01314     mmm = m + ARGHEAD + FUNHEAD + 1;
01315     while ( mmm < mmmst ) {
01316         if ( *mmm != -ARGWILD ) goto endofloop;
01317         i = 0;
01318         AN.argaddress = t;
01319         if ( CheckWild(BHEAD mmm[1],ARGTOARG,i,t) ) goto endofloop;
01320         AddWild(BHEAD mmm[1],ARGTOARG,i);
01321         mmm += 2;
01322     }
01323 }
01324 else goto endofloop;
01325 }
01326 else if ( *m < 0 && *t > 0 ) {
01327     if ( *m == -SYMBOL ) { /* SYMTOSUB */
01328         if ( m[1] < 2*MAXPOWER ) goto endofloop;
01329         i = m[1] - 2*MAXPOWER;
01330         AN.argaddress = t;
01331         if ( CheckWild(BHEAD i,SYMTOSUB,1,AN.argaddress) ) goto endofloop;
01332         AddWild(BHEAD i,SYMTOSUB,0);
01333     }
01334     else if ( *m == -VECTOR ) {
01335         if ( ( i = m[1] - WILDOFFSET ) < AM.OffsetVector )
01336             goto endofloop;
01337         AN.argaddress = t;
01338         if ( CheckWild(BHEAD i,VECTOSUB,1,t) ) goto endofloop;
01339         AddWild(BHEAD i,VECTOSUB,(WORD)0);
01340     }
01341     else if ( *m == -INDEX ) {
01342         if ( ( i = m[1] - WILDOFFSET ) < AM.OffsetIndex ) goto endofloop;
01343         if ( i >= AM.OffsetIndex + WILDOFFSET ) goto endofloop;
01344         AN.argaddress = t;
01345         if ( CheckWild(BHEAD i,INDTOSUB,1,AN.argaddress) ) goto endofloop;
01346         AddWild(BHEAD i,INDTOSUB,(WORD)0);
01347     }
01348     else if ( *m == -ARGWILD ) goto ArgAll;
01349     else goto endofloop;
01350 }
01351 else if ( *m > 0 && *t > 0 ) {
01352     WORD ii = *t-*m;
01353     i = *m;
01354     do { if ( *m++ != *t++ ) break; } while ( --i > 0 );
01355     if ( i == 1 && ii == 0 ) { /* sign difference */
01356         goto endofloop;
01357     }
01358     else if ( i > 0 ) {
01359         WORD *cto, *cfrom, *csav, ci;
01360         WORD oRepFunNum;
01361         WORD *oRepFunList;
01362         WORD *oterstart,*oterstop,*opatstop;
01363         WORD oExpectedSign;
01364         WORD wildargs, wildeat;
01365 /*
01366         Not an exact match here.
01367         We have to hope that the pattern contains a composite wildcard.
01368 */
01369         m = oldm; t = oldt;
01370         m += ARGHEAD; t += ARGHEAD; /* Point at (first?) term */
01371         mtrmstop = m + *m;
01372         ttrmstop = t + *t;
01373         if ( mtrmstop < argmstop ) goto endofloop; /* More than one term */
01374         msizcoef = mtrmstop[-1];
01375         if ( msizcoef < 0 ) msizcoef = -msizcoef;
01376         msubstop = mtrmstop - msizcoef;
01377         m++;
01378         if ( m >= msubstop ) goto endofloop; /* Only coefficient */
01379 /*
01380         Here we have a composite term. It can match provided it
01381         matches the entire argument. This argument must be a
01382         single term also and the coefficients should match
01383         (more or less).
01384         The matching takes:
01385         1: Match the functions etc. Nothing can be left.
01386         2: Match dotproducts and symbols. ONLY must match
01387            and nothing may be left.
01388         For safety it is best to take the term out and put it
01389         in workspace.
01390 */
01391         if ( argtstop > ttrmstop ) goto endofloop;
01392         m--;
01393     }

```

```

01394         oterstart = AN.terstart;
01395         oterstop = AN.terstop;
01396         opatstop = AN.patstop;
01397         oRepFunList = AN.RepFunList;
01398         oRepFunNum = AN.RepFunNum;
01399         AN.RepFunNum = 0;
01400         AN.RepFunList = AT.WorkPointer;
01401         AT.WorkPointer = (WORD *) (((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
01402         if ( AT.WorkPointer+*t+5 > AT.WorkTop ) {
01403             MLOCK(ErrorMessageLock);
01404             MesWork();
01405             MUNLOCK(ErrorMessageLock);
01406             return(-1);
01407         }
01408         csav = cto = AT.WorkPointer;
01409         cfrom = t;
01410         ci = *t;
01411         while ( --ci >= 0 ) *cto++ = *cfrom++;
01412         AT.WorkPointer = cto;
01413         ci = msizcoef;
01414         cfrom = mtrmstop;
01415         --ci;
01416         if ( abs(*--cfrom) != abs(*--cto) ) {
01417             AT.WorkPointer = csav;
01418             AN.RepFunList = oRepFunList;
01419             AN.RepFunNum = oRepFunNum;
01420             AN.terstart = oterstart;
01421             AN.terstop = oterstop;
01422             AN.patstop = opatstop;
01423             goto endofloop;
01424         }
01425         i = (*cfrom != *cto) ? 1 : 0; /* buffer AN.ExpectedSign until we are beyond the goto
*/
01426         while ( --ci >= 0 ) {
01427             if ( *--cfrom != *--cto ) {
01428                 AT.WorkPointer = csav;
01429                 AN.RepFunList = oRepFunList;
01430                 AN.RepFunNum = oRepFunNum;
01431                 AN.terstart = oterstart;
01432                 AN.terstop = oterstop;
01433                 AN.patstop = opatstop;
01434                 goto endofloop;
01435             }
01436         }
01437         oExpectedSign = AN.ExpectedSign; /* buffer AN.ExpectedSign until we are beyond
FindRest/FindOnly */
01438         AN.ExpectedSign = i;
01439         *m -= msizcoef;
01440         wildargs = AN.WildArgs;
01441         wildeat = AN.WildEat;
01442         for ( i = 0; i < wildargs; i++ ) wildargtaken[i] = AT.WildArgTaken[i];
01443         AN.ForFindOnly = 0; AN.UseFindOnly = 1;
01444         AN.nogroundlevel++;
01445         if ( FindRest(BHEAD csav,m) && ( AN.UsedOtherFind || FindOnly(BHEAD csav,m) ) ) {}
01446         else {
01447             nomatch:
01448                 *m += msizcoef;
01449                 AT.WorkPointer = csav;
01450                 AN.RepFunList = oRepFunList;
01451                 AN.RepFunNum = oRepFunNum;
01452                 AN.terstart = oterstart;
01453                 AN.terstop = oterstop;
01454                 AN.patstop = opatstop;
01455                 AN.WildArgs = wildargs;
01456                 AN.WildEat = wildeat;
01457                 AN.ExpectedSign = oExpectedSign;
01458                 AN.nogroundlevel--;
01459                 for ( i = 0; i < wildargs; i++ ) AT.WildArgTaken[i] = wildargtaken[i];
01460                 goto endofloop;
01461             }
01462         /*
*/
01463         if ( *m == 1 || m[1] < FUNCTION || functions[m[1]-FUNCTION].spec >= TENSORFUNCTION ) {
01464             if ( *m == 1 || m[1] < FUNCTION ) {
01465                 if ( AN.ExpectedSign ) goto nomatch;
01466             }
01467             else {
01468                 if ( m[1] > FUNCTION + WILDOFFSET ) {
01469                     if ( functions[m[1]-FUNCTION-WILDOFFSET].spec >= TENSORFUNCTION ) {
01470                         if ( AN.ExpectedSign != AN.RepFunList[AN.RepFunNum-1] ) goto nomatch;
01471                     }
01472                 }
01473                 else {
01474                     if ( AN.ExpectedSign != AN.RepFunList[AN.RepFunNum-1] ) goto nomatch;
01475                 }
01476                 if ( functions[m[1]-FUNCTION].spec >= TENSORFUNCTION ) {
01477                     if ( AN.ExpectedSign != AN.RepFunList[AN.RepFunNum-1] ) goto nomatch;
01478                 }
01479             }
01480         }

```

```

01478 */
01479         }
01480     }
01481     AN.nogroundlevel--;
01482     AN.ExpectedSign = oExpectedSign;
01483     AN.WildArgs = wildargs;
01484     AN.WildEat = wildeat;
01485     for ( i = 0; i < wildargs; i++ ) AT.WildArgTaken[i] = wildargtaken[i];
01486     Substitute(BHEAD csav,m,1);
01487     cto = csav;
01488     cfrom = cto + *cto - msizcoef;
01489     cto++;
01490     *m += msizcoef;
01491     AT.WorkPointer = csav;
01492     AN.RepFunList = oRepFunList;
01493     AN.RepFunNum = oRepFunNum;
01494     AN.terstart = oterstart;
01495     AN.terstop = oterstop;
01496     AN.patstop = opatstop;
01497     if ( *cto != SUBEXPRESSION ) goto endofloop;
01498     cto += cto[1];
01499     if ( cto < cfrom ) goto endofloop;
01500 }
01501 }
01502 else goto endofloop;
01503
01504 t = argtstop; /* Next argument */
01505 m = argmstop;
01506 }
01507 if ( AN.SignCheck && AN.ExpectedSign ) goto endofloop;
01508 AT.WorkPointer = OldWork;
01509 if ( AN.WildArgs > 1 ) *wilds = 1;
01510 if ( AN.SignCheck && AN.ExpectedSign ) return(0);
01511 return(1); /* m was eaten. we have a match! */
01512
01513 endofloop;;
01514 /*
01515     restore the current Wildcard assignments
01516 */
01517 i = nwstore;
01518 if ( i > 0 ) {
01519     m = AN.WildValue;
01520     t = OldWork + ntw; r = AT.WildMask;
01521     do {
01522         *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01523     } while ( --i > 0 );
01524     C->numrhs = *t++;
01525     C->Pointer = C->Buffer + oldcpointer;
01526 }
01527
01528 endolop;;
01529 i = AN.WildArgs-1;
01530 if ( i <= 0 ) {
01531     AT.WorkPointer = OldWork;
01532     return(0);
01533 }
01534 while ( --i >= 0 ) {
01535     if ( AT.WildArgTaken[i] == 0 ) {
01536         if ( i == 0 ) {
01537             AT.WorkPointer = OldWork;
01538             return(0);
01539         }
01540     }
01541     else {
01542         (AT.WildArgTaken[i])--;
01543         numofwildarg = 0;
01544         for ( j = 0; j <= i; j++ ) {
01545             numofwildarg += AT.WildArgTaken[j];
01546         }
01547         AT.WildArgTaken[j] = AN.WildEat-numofwildarg;
01548         /* ----> bug to be replaced in other source code */
01549         for ( j++; j < AN.WildArgs; j++ ) AT.WildArgTaken[j] = 0;
01550         break;
01551     }
01552 }
01553 goto topofloop;
01554 NoCaseB:
01555 /*
01556     Restore the old Wildcard assignments
01557 */
01558 i = nwstore;
01559 if ( i > 0 ) {
01560     m = AN.WildValue;
01561     t = OldWork + ntw; r = AT.WildMask;
01562     do {
01563         *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01564     } while ( --i > 0 );

```

```

01565         C->numrhs = *t++;
01566         C->Pointer = C->Buffer + oldcpointer;
01567     }
01568     AT.WorkPointer = OldWork;
01569     return(0);      /* no match */
01570 }
01571
01572 /*
01573     #] MatchFunction :
01574     #[ ScanFunctions :          WORD ScanFunctions(inpat,inter,par)
01575
01576     Finds in which functions to look for a match.
01577     inpat is the start of the pattern still to be matched.
01578     inter is the start of the term still to be matched.
01579     par gives information about commutativity.
01580         par = 0: nothing special
01581         par = 1: regular noncommuting function
01582         par = 2: GAMMA function
01583
01584     AN.patstop: end of the functions field in the search pattern
01585     AN.terstop: end of the functions field in the target pattern
01586     AN.terstart: address of entire term;
01587
01588     The actual matching of the functions and their arguments is done
01589     in a number of different routines. Mainly MatchFunction when there
01590     are no symmetry properties.
01591     Also: MatchE
01592           MatchCy
01593           FunMatchSy
01594           FunMatchCy
01595
01596     The main problem here is backtracking, ie continuing with wildcard
01597     possibilities when a first assignment doesn't work.
01598     Important note: this was completely forgotten in the symmetric
01599     functions till 6-jan-2009. As of the moment this still has to
01600     be fixed.  ?????21-mar-2023????? Is this still unfixed?????
01601
01602     Functions inside functions can cause problems when antisymmetric
01603     functions are involved. The sign of the term may be at stake.
01604     At the lowest level this is no problem but in f(-fas(n2,n1)) this
01605     plays a role. Next is when we have a product of functions inside
01606     an argument. The strategy must be that we test the sign only at the
01607     last function. Hence, when inpat+inpat[1] >= AN.patstop.
01608     We might relax that to the last antisymmetric function at a later stage.
01609
01610     New scheme to be implemented for non-commuting objects:
01611     When we are matching a second (or higher) function, any match can only
01612     be directly after the last matched non-commuting function or a commuting
01613     function. This will take care of whatever happens in MatchE etc.
01614 */
01615
01616 int ScanFunctions(PHEAD WORD *inpat, WORD *inter, WORD par)
01617 {
01618     GETBIDENTITY
01619     WORD i, *m, *t, *r, sym, psym;
01620     WORD *newpat, *newter, *instart, *oinpat = 0, *ointer = 0;
01621     WORD nwstore, offset, *OldWork, SetStop = 0, oRepFunNum = AN.RepFunNum;
01622     WORD wilds, wildargs = 0, wildeat = 0, *wildargtaken;
01623     WORD *Oterfirstcomm = AN.terfirstcomm;
01624     CBUF *C = cbuf+AT.ebufnum;
01625     int ntw = AN.NumTotWildArgs;
01626     LONG oldcpointer = C->Pointer - C->Buffer;
01627     WORD oldSignCheck = AN.SignCheck;
01628     instart = inter;
01629
01630     /*
01631     Only active for the last function in the pattern.
01632     The actual test on the sign is in MatchFunction or the symmetric functions
01633     */
01634     if ( AN.nogroundlevel ) {
01635         AN.SignCheck = ( inpat + inpat[1] >= AN.patstop ) ? 1 : 0;
01636     }
01637     else {
01638         AN.SignCheck = 0;
01639     }
01640
01641     /*
01642     Store the current Wildcard assignments
01643     */
01644     t = wildargtaken = OldWork = AT.WorkPointer;
01645     t += ntw;
01646     m = AN.WildValue;
01647     nwstore = i = (m[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
01648     if ( i > 0 ) {
01649         r = AT.WildMask;
01650         do {
01651             *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
01652         } while ( --i > 0 );
01653         *t++ = C->numrhs;

```

```

01652     }
01653     if ( t >= AT.WorkTop ) {
01654         MLOCK(ErrorMessageLock);
01655         MesWork();
01656         MUNLOCK(ErrorMessageLock);
01657         Terminate(-1);
01658     }
01659     AT.WorkPointer = t;
01660     do {
01661 #ifndef NEWCOMMUTE
01662 /*
01663      Find an eligible unsubstituted function
01664 */
01665      if ( AN.RepFunNum > 0 ) {
01666 /*
01667          First try a non-commuting function, just after the last
01668          substituted non-commuting function.
01669 */
01670          if ( *inter >= FUNCTION && functions[*inter-FUNCTION].commute ) {
01671              do {
01672                  offset = WORDDIF(inter,AN.terstart);
01673                  for ( i = 0; i < AN.RepFunNum; i += 2 ) {
01674                      if ( AN.RepFunList[i] >= offset ) break;
01675                  }
01676                  if ( i >= AN.RepFunNum ) break;
01677                  inter += inter[1];
01678              } while ( inter < AN.terfirstcomm );
01679              if ( inter < AN.terfirstcomm ) { /* Check that it is directly after */
01680                  for ( i = 0; i < AN.RepFunNum; i += 2 ) {
01681                      if ( functions[AN.terstart[AN.RepFunList[i]]-FUNCTION].commute
01682                        && AN.RepFunList[i]+AN.terstart[AN.RepFunList[i]+1] == offset ) break;
01683                  }
01684                  if ( i < AN.RepFunNum ) goto trythis;
01685              }
01686              inter = AN.terfirstcomm;
01687          }
01688 /*
01689          Now try one of the commuting functions
01690 */
01691          while ( inter < AN.terstop ) {
01692              offset = WORDDIF(inter,AN.terstart);
01693              for ( i = 0; i < AN.RepFunNum; i += 2 ) {
01694                  if ( AN.RepFunList[i] == offset ) break;
01695              }
01696              if ( i >= AN.RepFunNum ) break;
01697              inter += inter[1];
01698          }
01699          if ( inter >= AN.terstop ) goto Failure;
01700 trythis:;
01701     }
01702     else {
01703 /*
01704      The first function can be anywhere. We have no problems.
01705 */
01706      offset = WORDDIF(inter,AN.terstart);
01707     }
01708 #else
01709 /* first find an unsubstituted function */
01710     do {
01711         offset = WORDDIF(inter,AN.terstart);
01712         for ( i = 0; i < AN.RepFunNum; i += 2 ) {
01713             if ( AN.RepFunList[i] == offset ) break;
01714         }
01715         if ( i >= AN.RepFunNum ) break;
01716         inter += inter[1];
01717     } while ( inter < AN.terstop );
01718     if ( inter >= AN.terstop ) goto Failure;
01719 #endif
01720     wilds = 0;
01721     /* We found one */
01722     if ( *inter >= FUNCTION && *input >= FUNCTION ) {
01723         if ( *input == *inter || *input >= FUNCTION + WILDOFFSET ) {
01724 /*
01725          if ( inter[1] == FUNHEAD ) goto rewild;
01726 */
01727         if ( functions[*inter-FUNCTION].spec >= TENSORFUNCTION
01728           && (*inter == *input ||
01729             functions[*input-FUNCTION-WILDOFFSET].spec >= TENSORFUNCTION ) ) {
01730             sym = functions[*inter-FUNCTION].symmetric & ~REVERSEORDER;
01731             if ( *input == *inter ) psym = sym;
01732             else psym = functions[*input-FUNCTION-WILDOFFSET].symmetric & ~REVERSEORDER;
01733             if ( sym == ANTISYMMETRIC || sym == SYMMETRIC
01734               || psym == SYMMETRIC || psym == ANTISYMMETRIC ) {
01735                 if ( sym == ANTISYMMETRIC && psym == SYMMETRIC ) goto rewild;
01736                 if ( sym == SYMMETRIC && psym == ANTISYMMETRIC ) goto rewild;
01737 /*
01738          Special function call for (anti)symmetric tensors

```

```

01739 */
01740         if ( MatchE(BHEAD inpat,inter,instart,par) ) goto OnSuccess;
01741     }
01742     else if ( sym == CYCLESYMMETRIC || sym == RCYCLESYMMETRIC
01743 || psym == CYCLESYMMETRIC || psym == RCYCLESYMMETRIC ) {
01744 /*
01745         Special function call for (r)cyclic tensors
01746 */
01747         if ( MatchCy(BHEAD inpat,inter,instart,par) ) goto OnSuccess;
01748     }
01749     else goto rewild;
01750 }
01751 else if ( functions[*inter-FUNCTION].spec <= 0
01752 && ( *inter == *inpat ||
01753 functions[*inpat-FUNCTION-WILDOFFSET].spec <= 0 ) ) {
01754     sym = functions[*inter-FUNCTION].symmetric & ~REVERSEORDER;
01755     if ( *inpat == *inter ) psym = sym;
01756     else psym = functions[*inpat-FUNCTION-WILDOFFSET].symmetric & ~REVERSEORDER;
01757     if ( psym == SYMMETRIC || sym == SYMMETRIC
01758 /*
01759         The next statement was commented out. Why????
01760         Werkt nog niet. Teken wordt nog niet bijgehouden.
01761         5-nov-2001
01762 */
01763 || psym == ANTISYMMETRIC || sym == ANTISYMMETRIC
01764 ) {
01765         if ( sym == ANTISYMMETRIC && psym == SYMMETRIC ) goto rewild;
01766         if ( sym == SYMMETRIC && psym == ANTISYMMETRIC ) goto rewild;
01767         if ( FunMatchSy(BHEAD inpat,inter,instart,par) ) goto OnSuccess;
01768     }
01769     else
01770         if ( sym == CYCLESYMMETRIC || sym == RCYCLESYMMETRIC
01771 || psym == CYCLESYMMETRIC || psym == RCYCLESYMMETRIC ) {
01772             if ( FunMatchCy(BHEAD inpat,inter,instart,par) ) goto OnSuccess;
01773         }
01774         else goto rewild;
01775     }
01776     else goto rewild;
01777     AN.terfirstcomm = Oterfirstcomm;
01778 }
01779 else if ( par > 0 ) { SetStop = 1; goto maybenext; }
01780 }
01781 else {
01782 rewild:
01783     AN.terfirstcomm = Oterfirstcomm;
01784     if ( *inter != SUBEXPRESSION && MatchFunction(BHEAD inpat,inter,&wilds) ) {
01785         AN.terfirstcomm = Oterfirstcomm;
01786         if ( wilds ) {
01787 /*
01788             Store wildcards to continue in MatchFunction if the current
01789             wildcards do not work out.
01790 */
01791             wildargs = AN.WildArgs;
01792             wildeat = AN.WildEat;
01793             for ( i = 0; i < wildargs; i++ ) wildargtaken[i] = AT.WildArgTaken[i];
01794             oinpat = inpat; ointer = inter;
01795         }
01796         if ( par && *inter == GAMMA && AN.RepFunList[AN.RepFunNum+1] ) {
01797             SetStop = 1; goto NoMat;
01798         }
01799         if ( par == 2 ) {
01800             if ( *inter < FUNCTION || functions[*inter-FUNCTION].commute ) {
01801                 goto NoMat;
01802             }
01803             par = 1;
01804         }
01805         AN.RepFunList[AN.RepFunNum] = offset;
01806         AN.RepFunNum += 2;
01807         newpat = inpat + inpat[1];
01808         if ( newpat >= AN.patstop ) {
01809             if ( AN.UseFindOnly == 0 ) {
01810                 if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
01811                     AN.UsedOtherFind = 1;
01812                     goto OnSuccess;
01813                 }
01814                 AN.RepFunNum -= 2;
01815                 goto NoMat;
01816             }
01817             goto OnSuccess;
01818         }
01819         if ( *inter < FUNCTION || functions[*inter-FUNCTION].commute ) {
01820             newer = inter + inter[1];
01821             if ( newer >= AN.terstop ) goto Failure;
01822             if ( *inter == GAMMA && inpat[1] <
01823 inter[1] - AN.RepFunList[AN.RepFunNum-1] ) {
01824                 if ( ScanFunctions(BHEAD newpat,newer,2) ) goto OnSuccess;
01825                 AN.terfirstcomm = Oterfirstcomm;

```

```

01826         }
01827     else if ( *newter == SUBEXPRESSION ) {}
01828     else if ( functions[*inter-FUNCTION].commute ) {
01829         if ( ScanFunctions(BHEAD newpat,newter,1) ) goto OnSuccess;
01830         AN.terfirstcomm = Oterfirstcomm;
01831         if ( ( *newpat < (FUNCTION+WILDOFFSET)
01832             && ( functions[*newpat-FUNCTION].commute == 0 ) ) ||
01833             ( *newpat >= (FUNCTION+WILDOFFSET)
01834             && ( functions[*newpat-FUNCTION-WILDOFFSET].commute == 0 ) ) ) {
01835             newter = AN.terfirstcomm;
01836             if ( newter < AN.terstop && ScanFunctions(BHEAD newpat,newter,1) ) goto
OnSuccess;
01837         }
01838     }
01839     else {
01840         if ( ScanFunctions(BHEAD newpat,instart,1) ) goto OnSuccess;
01841         AN.terfirstcomm = Oterfirstcomm;
01842     }
01843     SetStop = par;
01844 }
01845 else {
01846 /*
01847     Shouldn't this be newpat instead of inpat?????
01848 */
01849     if ( par && inter > instart && ( ( *newpat < (FUNCTION+WILDOFFSET)
01850     && functions[*newpat-FUNCTION].commute ) ||
01851     ( *newpat >= (FUNCTION+WILDOFFSET)
01852     && functions[*newpat-FUNCTION-WILDOFFSET].commute ) ) ) {
01853         SetStop = 1;
01854     }
01855     else {
01856         newter = instart;
01857         if ( ScanFunctions(BHEAD newpat,newter,par) ) goto OnSuccess;
01858         AN.terfirstcomm = Oterfirstcomm;
01859     }
01860 }
01861 /*
01862     Restore the old Wildcard assignments
01863 */
01864 NoMat:
01865     i = nwstore;
01866     if ( i > 0 ) {
01867         m = AN.WildValue;
01868         t = OldWork + ntw; r = AT.WildMask;
01869         do {
01870             *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01871         } while ( --i > 0 );
01872         C->numrhs = *t++;
01873         C->Pointer = C->Buffer + oldcpointer;
01874     }
01875 /*
01876     AN.RepFunNum -= 2; */
01877     AN.RepFunNum = oRepFunNum;
01878     if ( wilds ) {
01879         inter = ointer; inpat = oinpat;
01880         AN.WildArgs = wildargs;
01881         AN.WildEat = wildeat;
01882         for ( i = 0; i < wildargs; i++ ) AT.WildArgTaken[i] = wildargtaken[i];
01883         goto rewild;
01884     }
01885     if ( SetStop ) break;
01886 }
01887 else if ( par ) {
01888 maybenext:
01889     if ( *inpat < (FUNCTION+WILDOFFSET) ) {
01890         if ( *inpat < FUNCTION ||
01891             functions[*inpat-FUNCTION].commute ) break;
01892     }
01893     else {
01894         if ( functions[*inpat-FUNCTION-WILDOFFSET].commute ) break;
01895     }
01896     inter += inter[1];
01897 } while ( inter < AN.terstop );
01898 Failure:
01899     AN.SignCheck = oldSignCheck;
01900     AT.WorkPointer = OldWork;
01901     return(0);
01902 OnSuccess:
01903     if ( AT.idallflag && AN.nogroundlevel <= 0 ) {
01904         if ( AT.idallmaxnum > 0 && AT.idallnum >= AT.idallmaxnum ) {
01905             AN.terfirstcomm = Oterfirstcomm;
01906             AN.SignCheck = oldSignCheck;
01907             AT.WorkPointer = OldWork;
01908             return(0);
01909         }
01910         SubsInAll(BHEAD0);
01911         AT.idallnum++;

```



```

01912         if ( AT.idallmaxnum == 0 || AT.idallnum < AT.idallmaxnum ) goto NoMat;
01913     }
01914     AN.terfirstcomm = Oterfirstcomm;
01915     AN.SignCheck = oldSignCheck;
01916 /*
01917     Now the disorder test
01918 */
01919     if ( AN.DisOrderFlag && AN.RepFunNum >= 4 ) {
01920         WORD k, kk;
01921         for ( i = 2; i < AN.RepFunNum; i += 2 ) {
01922 /*
01923     -----> We still have to copy the code from Normalize wrt properorderflag
01924 */
01925             m = AN.terstart + AN.RepFunList[i-2];
01926             t = AN.terstart + AN.RepFunList[i];
01927             if ( *m != *t ) {
01928                 if ( *m > *t ) continue;
01929                 goto doesmatch;
01930             }
01931             if ( *m >= FUNCTION && functions[*m-FUNCTION].spec >=
01932                 TENSORFUNCTION ) {
01933                 k = m[1] - FUNHEAD;
01934                 kk = t[1] - FUNHEAD;
01935                 m += FUNHEAD;
01936                 t += FUNHEAD;
01937             }
01938             else {
01939                 k = m[1] - FUNHEAD;
01940                 kk = t[1] - FUNHEAD;
01941                 m += FUNHEAD;
01942                 t += FUNHEAD;
01943             }
01944             while ( k > 0 && kk > 0 ) {
01945                 if ( *m < *t ) goto NextFor;
01946                 else if ( *m++ > *t++ ) goto doesmatch;
01947                 k--; kk--;
01948             }
01949             if ( k > 0 ) goto doesmatch;
01950 NextFor;;
01951         }
01952         SetStop = 1;
01953         goto NoMat;
01954     }
01955 doesmatch:
01956     AT.WorkPointer = OldWork;
01957     return(1);
01958 }
01959
01960 /*
01961     #] ScanFunctions :
01962     #] Patterns :
01963 */

```

## 4.53 fwin.h File Reference

### Macros

- #define [LINEFEED](#) '\n'
- #define [CARRIAGEReturn](#) 0x0D
- #define [WITHRETURN](#)
- #define [WITHSYSTEM](#)
- #define [P\\_term](#)(code) exit((int)(code<0?-code:code))
- #define [SEPARATOR](#) '\\'
- #define [ALTSEPARATOR](#) '/'
- #define [PATHSEPARATOR](#) ';'
- #define [WITH\\_ENV](#)

### 4.53.1 Detailed Description

Settings for Windows computers.

Definition in file [fwin.h](#).

## 4.53.2 Macro Definition Documentation

### 4.53.2.1 LINEFEED

```
#define LINEFEED '\n'
```

Definition at line 33 of file [fwin.h](#).

### 4.53.2.2 CARRIAGERETURN

```
#define CARRIAGERETURN 0x0D
```

Definition at line 34 of file [fwin.h](#).

### 4.53.2.3 WITHRETURN

```
#define WITHRETURN
```

Definition at line 35 of file [fwin.h](#).

### 4.53.2.4 WITHSYSTEM

```
#define WITHSYSTEM
```

Definition at line 37 of file [fwin.h](#).

### 4.53.2.5 P\_term

```
#define P_term(  
    code ) exit((int) (code<0?-code:code))
```

Definition at line 39 of file [fwin.h](#).

### 4.53.2.6 SEPARATOR

```
#define SEPARATOR '\\'
```

Definition at line 41 of file [fwin.h](#).

### 4.53.2.7 ALTSEPARATOR

```
#define ALTSEPARATOR '/'
```

Definition at line 42 of file [fwin.h](#).

### 4.53.2.8 PATHSEPARATOR

```
#define PATHSEPARATOR ';'
```

Definition at line 43 of file [fwin.h](#).

### 4.53.2.9 WITH\_ENV

```
#define WITH_ENV
```

Definition at line 44 of file [fwin.h](#).

## 4.54 fwin.h

[Go to the documentation of this file.](#)

```
00001
00006 /* #[ License : */
00007 /*
00008 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 *   When using this file you are requested to refer to the publication
00010 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 *   This is considered a matter of courtesy as the development was paid
00012 *   for by FOM the Dutch physics granting agency and we would like to
00013 *   be able to track its scientific use to convince FOM of its value
00014 *   for the community.
00015 *
00016 *   This file is part of FORM.
00017 *
00018 *   FORM is free software: you can redistribute it and/or modify it under the
00019 *   terms of the GNU General Public License as published by the Free Software
00020 *   Foundation, either version 3 of the License, or (at your option) any later
00021 *   version.
00022 *
00023 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 *   details.
00027 *
00028 *   You should have received a copy of the GNU General Public License along
00029 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #[ License : */
00032
00033 #define LINEFEED '\n'
00034 #define CARRIAGERETURN 0x0D
00035 #define WITHRETURN
00036
00037 #define WITHSYSTEM
00038
00039 #define P_term(code)    exit((int)(code<0?-code:code))
00040
00041 #define SEPARATOR '\\\
00042 #define ALTSEPARATOR '/'
00043 #define PATHSEPARATOR ';'
00044 #define WITH_ENV
```

## 4.55 gentopo.cc

```
00001 // { ( [
00002
00003 #ifdef HAVE_CONFIG_H
00004 #ifndef CONFIG_H_INCLUDED
00005 #define CONFIG_H_INCLUDED
00006 #include <config.h>
00007 #endif
00008 #endif
00009
00010 #include <iostream>
00011 #include <iomanip>
```

```

00012 #include <sstream>
00013 #include <cstdio>
00014 #include <stdlib.h>
00015
00016 #include "grcc.h"
00017 #include "gentopo.h"
00018
00019 #define DUMMYUSE(x) (void)(x)
00020
00021 using namespace std;
00022
00023 // The next limitation is imposed by the fact that the latest compilers
00024 // can give warnings on declarations like "int dtc11[nNodes]"
00025 // With a bit of overkill there should be no real problems.
00026 #define MAXNODES 100
00027 #define MAXNCASSES 100
00028
00029 // Generate scalar connected Feynman graphs.
00030
00031 //=====
00032
00033 typedef int Bool;
00034 const int T_True = 1;
00035 const int T_False = 0;
00036
00037 // compile options
00038 const int CHECK = T_True;
00039 //const int MONITOR = T_True;
00040 const int MONITOR = T_False;
00041
00042 const int OPTPRINT = T_False;
00043
00044 const int DEBUG0 = T_False;
00045 //const int DEBUG1 = T_False;
00046 const int DEBUG = T_False;
00047
00048 // for debugging memory use
00049 #define DEBUGM T_False
00050
00051 //=====
00052 class T_MGraph;
00053 class T_EGraph;
00054
00055 #define Extern
00056 #define Global
00057 #define Local
00058
00059 // External functions
00060 Extern void toForm(T_EGraph *egraph);
00061 Extern int countPhiCon(int ex, int lp, int v4);
00062
00063 // Temporal definition: they should be replaced by FROM functions.
00064 Local BigInt factorial(int n);
00065 Local BigInt ipow(int n, int p);
00066
00067 // Local functions
00068 Local int nextPerm(int nelem, int nclass, int *cl, int *r, int *q, int *p, int count);
00069
00070 Local void erEnd(const char *msg);
00071 Local int *newArray(int size, int val);
00072 Local void deletArray(int *a);
00073 Local void copyArray(int size, int *a0, int *a1);
00074 Local int **newMat(int n0, int n1, int val);
00075 Local void deleteMat(int **m, int n0, int n1);
00076 Local void printArray(int n, int *p);
00077 Local void printMat(int n0, int n1, int **m);
00078
00079 #if DEBUGM
00080 static int countNMC = 0;
00081 static int countEG = 0;
00082 static int countMG = 0;
00083 #endif
00084
00085 //=====
00086 // class T_EGraph
00087
00088 T_EGraph::T_EGraph(int nnodes, int nedges, int mxdeg)
00089 {
00090     int j;
00091
00092     nNodes = nnodes;
00093     nEdges = nedges;
00094     maxdeg = mxdeg; // maximum value of degree of nodes
00095     nExtern = 0;
00096
00097     nodes = new T_ENode[nNodes];
00098     edges = new T_EEdge[nEdges+1];

```

```

00099     for (j = 0; j < nNodes; j++) {
00100         nodes[j].deg = 0;
00101         nodes[j].edges = new int[maxdeg];
00102     }
00103     #if DEBUGM
00104     printf("+++ new T_EGraph %d\n", ++countEG);
00105     if(countEG > 100) { exit(1); }
00106     #endif
00107 }
00108
00109 T_EGraph::~T_EGraph()
00110 {
00111     int j;
00112
00113     for (j = 0; j < nNodes; j++) {
00114         delete[] nodes[j].edges;
00115     }
00116     delete[] nodes;
00117     delete[] edges;
00118     #if DEBUGM
00119     printf("+++ delete T_EGraph %d\n", countEG--);
00120     #endif
00121 }
00122
00123 // construct T_EGraph from adjacency matrix
00124 void T_EGraph::init(int pid, long gid, int **adjmat, Bool sopi, BigInt nsm, BigInt esm)
00125 {
00126     int n0, n1, ed, e, eext, eint;
00127     // Bool ok;
00128
00129     pId = pid;
00130     gId = gid;
00131     opi = sopi;
00132     nsym = nsm;
00133     esym = esm;
00134
00135     for (n0 = 0; n0 < nNodes; n0++) {
00136         nodes[n0].deg = 0;
00137     }
00138     ed = 1;
00139     for (n0 = 0; n0 < nNodes; n0++) {
00140         for (e = 0; e < adjmat[n0][n0]/2; e++, ed++) {
00141             edges[ed].nodes[0] = n0;
00142             edges[ed].nodes[1] = n0;
00143             nodes[n0].edges[nodes[n0].deg++] = - ed;
00144             nodes[n0].edges[nodes[n0].deg++] = ed;
00145             edges[ed].ext = nodes[n0].ext;
00146         }
00147         for (n1 = n0+1; n1 < nNodes; n1++) {
00148             for (e = 0; e < adjmat[n0][n1]; e++, ed++) {
00149                 edges[ed].nodes[0] = n0;
00150                 edges[ed].nodes[1] = n1;
00151                 nodes[n0].edges[nodes[n0].deg++] = - ed;
00152                 nodes[n1].edges[nodes[n1].deg++] = ed;
00153                 edges[ed].ext = (nodes[n0].ext || nodes[n1].ext);
00154             }
00155         }
00156     }
00157     if (CHECK) {
00158         if (ed != nEdges+1) {
00159             printf("*** T_EGraph::init: ed=%d != nEdges=%d\n", ed, nEdges);
00160             erEnd("*** T_EGraph::init: illegal connection\n");
00161         }
00162     }
00163
00164     // name of momenta
00165     eext = 1;
00166     eint = 1;
00167     for (ed = 1; ed <= nEdges; ed++) {
00168         if (edges[ed].ext) {
00169             edges[ed].momn = eext++;
00170             edges[ed].momc[0] = 'Q';
00171             edges[ed].momc[1] = ((char)0);
00172         } else {
00173             edges[ed].momn = eint++;
00174             edges[ed].momc[0] = 'P';
00175             edges[ed].momc[1] = ((char)0);
00176         }
00177     }
00178 }
00179
00180 // set external particle to node 'nd'
00181 void T_EGraph::setExtern(int nd, Bool val)
00182 {
00183     nodes[nd].ext = val;
00184 }
00185

```

```

00186 // end of calling 'setExtern'
00187 void T_EGraph::endSetExtern(void)
00188 {
00189     int n;
00190
00191     nExtern = 0;
00192     for (n = 0; n < nNodes; n++) {
00193         if (nodes[n].ext) {
00194             nExtern++;
00195         }
00196     }
00197 }
00198
00199 // print the T_EGraph
00200 void T_EGraph::print()
00201 {
00202     int nd, lg, ed, nlp;
00203
00204     nlp = nEdges - nNodes + 1;
00205     printf("\n");
00206     printf("T_EGraph: pId=%d gId=%ld nExtern=%d nLoops=%d nNodes=%d nEdges=%d\n",
00207           pId, gId, nExtern, nlp, nNodes, nEdges);
00208     printf("      sym = (%ld * %ld) maxdeg=%d\n", nsym, esym, maxdeg);
00209     printf("  Nodes\n");
00210     for (nd = 0; nd < nNodes; nd++) {
00211         if (nodes[nd].ext) {
00212             printf("    %2d Extern ", nd);
00213         } else {
00214             printf("    %2d Vertex ", nd);
00215         }
00216         printf("deg=%d [", nodes[nd].deg);
00217         for (lg = 0; lg < nodes[nd].deg; lg++) {
00218             printf(" %3d", nodes[nd].edges[lg]);
00219         }
00220         printf("]\n");
00221     }
00222     printf("  Edges\n");
00223     for (ed = 1; ed <= nEdges; ed++) {
00224         if (edges[ed].ext) {
00225             printf("    %2d Extern ", ed);
00226         } else {
00227             printf("    %2d Intern ", ed);
00228         }
00229         printf("%s%d", (edges[ed].ext)?"Q":"p", edges[ed].momn);
00230         printf(" [%3d %3d]\n", edges[ed].nodes[1], edges[ed].nodes[0]);
00231     }
00232 }
00233
00234 //=====
00235 // class T_MNode : nodes in T_MGraph
00236
00237 T_MNode::T_MNode(int vid, int vdeg, Bool vext, int vclss)
00238 {
00239     id      = vid;      // id of the node
00240     deg     = vdeg;     // degree of the node
00241     freelg  = vdeg;     // number of free legs
00242     clss    = vclss;    // class to which the node belongs
00243     ext     = vext;     // external node or not
00244 }
00245
00246 //=====
00247 // class of node-classes for T_MGraph
00248 //
00249 class T_MNodeClass {
00250 public:
00251     int    nNodes;           // the number of nodes
00252     int    nClasses;        // the number of classes
00253     int    *clist;          // the number of nodes in each class
00254     int    *ndcl;           // node --> class
00255     int    **clmat;         // matrix used for classification
00256     int    *flist;          // the first node in each class
00257     int    maxdeg;          // maximal value of degree(node)
00258     int    forallignment;
00259
00260     T_MNodeClass(int nnodes, int ncl);
00261     ~T_MNodeClass();
00262     void init(int *cl, int mxdeg, int **adjmat);
00263     void copy(T_MNodeClass* mnc);
00264     int  clCmp(int nd0, int nd1, int cn);
00265     void printMat(void);
00266
00267     void mkFlist(void);
00268     void mkNdCl(void);
00269     void mkClMat(int **adjmat);
00270     void incMat(int nd, int td, int val);
00271     Bool chkOrd(int nd, int ndc, T_MNodeClass *cl, int *dtcl);
00272     int  cmpArray(int *a0, int *a1, int ma);

```

```

00273 };
00274
00275 T_MNodeClass::T_MNodeClass(int nnodes, int ncl)
00276 {
00277     nNodes    = nnodes;
00278     nClasses   = ncl;
00279     clist      = new int[nClasses];
00280     ndcl       = new int[nNodes];
00281     clmat      = newMat(nNodes, nClasses, 0);
00282     flist      = new int[nClasses+1];
00283
00284 #if DEBUGM
00285     printf("+++ new    T_MNodeClass %d\n", ++countNMC);
00286     if(countNMC > 100) { exit(1); }
00287 #endif
00288 }
00289
00290 T_MNodeClass::~T_MNodeClass()
00291 {
00292     delete[] clist;
00293     delete[] ndcl;
00294     deleteMat(clmat, nNodes, nClasses);
00295     delete[] flist;
00296
00297 #if DEBUGM
00298     printf("+++ delete T_MNodeClass %d\n", countNMC--);
00299 #endif
00300 }
00301
00302 void T_MNodeClass::init(int *cl, int mxdeg, int **adjmat)
00303 {
00304     int j;
00305
00306     for (j = 0; j < nClasses; j++) {
00307         clist[j] = cl[j];
00308     }
00309     maxdeg    = mxdeg;
00310     mkNdCl();
00311     mkClMat(adjmat);
00312     mkFlist();
00313 }
00314
00315 void T_MNodeClass::copy(T_MNodeClass* mnc)
00316 {
00317     int j, k;
00318
00319     for (k = 0; k < nClasses; k++) {
00320         clist[k] = mnc->clist[k];
00321     }
00322     maxdeg    = mnc->maxdeg;
00323     for (j = 0; j < nNodes; j++) {
00324         ndcl[j] = mnc->ndcl[j];
00325         for (k = 0; k < nClasses; k++) {
00326             clmat[j][k] = mnc->clmat[j][k];
00327         }
00328     }
00329     for (k = 0; k < nClasses+1; k++) {
00330         flist[k] = mnc->flist[k];
00331     }
00332 }
00333
00334 // Construct flist
00335 //   The set of nodes in class 'cl' is [flist[cl],...,flist[cl+1]-1]
00336 void T_MNodeClass::mkFlist(void)
00337 {
00338     int j, f;
00339
00340     f = 0;
00341     for (j = 0; j < nClasses; j++) {
00342         flist[j] = f;
00343         f += clist[j];
00344     }
00345     flist[nClasses] = f;
00346 }
00347
00348 // Construct ndcl
00349 //   ndcl[nd] = (the class id in which node 'nd' belongs)
00350 void T_MNodeClass::mkNdCl(void)
00351 {
00352     int c, k;
00353     int nd = 0;
00354
00355     for (c = 0; c < nClasses; c++) {
00356         for (k = 0; k < clist[c]; k++) {
00357             ndcl[nd++] = c;
00358         }
00359     }

```

```

00360 }
00361
00362 // Comparison of two nodes 'nd0' and 'nd1'
00363 // Ordering is lexicographic (class, connection configuration)
00364 int T_MNodeClass::clCmp(int nd0, int nd1, int cn)
00365 {
00366
00367     // Whether two nodes are in a same class or not.
00368     int cmp = ndcl[nd0] - ndcl[nd1];
00369     if (cmp != 0) {
00370         return cmp;
00371     }
00372
00373     // Sign '-' signifies the reverse ordering
00374     cmp = - cmpArray(clmat[nd0], clmat[nd1], cn);
00375     if (cmp != 0) {
00376         return cmp;
00377     }
00378     // for particles ???
00379     return cmp;
00380 }
00381
00382 // Construct a matrix 'clmat[nd][tc]' which is the number
00383 // of edges connecting 'nd' and all nodes in class 'tc'.
00384 void T_MNodeClass::mkClMat(int **adjmat)
00385 {
00386     int nd, td, tc;
00387
00388     for (nd = 0; nd < nNodes; nd++) {
00389         for (td = 0; td < nNodes; td++) {
00390             tc = ndcl[td]; // another node
00391             clmat[nd][tc] += adjmat[nd][td]; // the number of edges 'nd'--'td'
00392         }
00393     }
00394 }
00395
00396 // Increase the number of edges by 'val' between 'nd' and 'td'.
00397 void T_MNodeClass::incMat(int nd, int td, int val)
00398 {
00399     int tdc = ndcl[td]; // class of 'td'
00400     clmat[nd][tdc] += val; // modify matrix 'clmat'.
00401 }
00402
00403 // Check whether the configuration satisfies the ordering condition or not.
00404 Bool T_MNodeClass::chkOrd(int nd, int ndc, T_MNodeClass *cl, int *dtcl)
00405 {
00406     Bool tcl[MAXNCCLASSES];
00407     // Bool tcl[cl->nClasses];
00408     int tn, tc, mxn, cmp, n;
00409     DUMMYUSE(nd);
00410
00411     for (tc = 0; tc < cl->nClasses; tc++) {
00412         tcl[tc] = T_False;
00413     }
00414     for (tn = 0; tn < nNodes; tn++) {
00415         if (dtcl[tn] == 0) {
00416             tcl[cl->ndcl[tn]] = T_False;
00417         }
00418     }
00419     for (tc = 0; tc < cl->nClasses; tc++) {
00420         if (tcl[tc] && ndc != tc) {
00421             mxn = flist[tc+1];
00422             for (n = flist[tc]+1; n < mxn; n++) {
00423                 cmp = - clmat[n-1][tc] + clmat[n][tc];
00424                 if (cmp > 0) {
00425                     return T_False;
00426                 }
00427             }
00428         }
00429     }
00430     return T_True;
00431 }
00432
00433 // Print configuration matrix.
00434 void T_MNodeClass::printMat(void)
00435 {
00436     int j1, j2;
00437
00438     cout << endl;
00439
00440     // the first line
00441     cout << setw(2) << "nd" << ": " << setw(2) << "cl: ";
00442     for (j2 = 0; j2 < nClasses; j2++) {
00443         cout << setw(2) << j2 << " ";
00444     }
00445     cout << endl;

```



```

00447
00448 // print raw
00449 for (j1 = 0; j1 < nNodes; j1++) {
00450     cout << setw(2) << j1 << ": " << setw(2) << ndcl[j1] << ": [";
00451     for (j2 = 0; j2 < nClasses; j2++) {
00452         cout << " " << setw(2) << clmat[j1][j2];
00453     }
00454     cout << "]" << endl;
00455 }
00456 }
00457
00458 int T_MNodeClass::cmpArray(int *a0, int *a1, int ma)
00459 {
00460     for (int j = 0; j < ma; j++) {
00461         if (a0[j] < a1[j]) {
00462             return -1;
00463         } else if (a0[j] > a1[j]) {
00464             return 1;
00465         }
00466     }
00467     return 0;
00468 }
00469
00470 //=====
00471 // class T_MGraph : scalar graph expressed by matrix form
00472
00473 T_MGraph::T_MGraph(int pid, int ncl, int *cldeg, int *clnum, int *clex, Bool sopi)
00474 {
00475     int nn, ne, j, k;
00476
00477     // initial conditions
00478     nClasses = ncl;
00479     clist = new int[ncl];
00480
00481     mindeg = -1;
00482     maxdeg = -1;
00483     ne = 0;
00484     nn = 0;
00485     for (j = 0; j < nClasses; j++) {
00486         clist[j] = clnum[j];
00487         nn += clnum[j];
00488         ne += cldeg[j]*clnum[j];
00489         if (mindeg < 0) {
00490             mindeg = cldeg[j];
00491         } else {
00492             mindeg = min(mindeg, cldeg[j]);
00493         }
00494         if (maxdeg < 0) {
00495             maxdeg = cldeg[j];
00496         } else {
00497             maxdeg = max(maxdeg, cldeg[j]);
00498         }
00499     }
00500     if (ne % 2 != 0) {
00501         printf("Sum of degrees are not even\n");
00502         for (j = 0; j < nClasses; j++) {
00503             printf("class %2d: %2d %2d %2d\n",
00504                 j, cldeg[j], clnum[j], clex[j]);
00505         }
00506         erEnd("illegal degrees of nodes");
00507     }
00508     pId = pid;
00509     nNodes = nn;
00510     nodes = new T_MNode*[nNodes];
00511
00512     selOPI = sopi;
00513
00514     nEdges = ne / 2;
00515     nLoops = nEdges - nNodes + 1;
00516
00517     egraph = new T_EGraph(nNodes, nEdges, maxdeg);
00518     nn = 0;
00519     for (j = 0; j < nClasses; j++) {
00520         for (k = 0; k < clist[j]; k++, nn++) {
00521             nodes[nn] = new T_MNode(nn, cldeg[j], clex[j], j);
00522             egraph->setExtern(nn, clex[j]);
00523         }
00524     }
00525     egraph->endSetExtern();
00526
00527     // generated set of graphs
00528     ndiag = 0;
00529     n1PI = 0;
00530
00531     // the current graph
00532     adjMat = newMat(nNodes, nNodes, 0);
00533     nsym = ToBigInt(0);

```

```

00534     esym      = ToBigInt(0);
00535     clPI      = 0;
00536     wsum      = ToFraction(0, 1);
00537     wsopi     = ToFraction(0, 1);
00538
00539     // current node classification
00540     curcl      = new T_MNodeClass(nNodes, nClasses);
00541
00542     // measures of efficiency
00543     ngen       = 0;
00544     ngconn     = 0;
00545
00546     if (MONITOR) {
00547         nCallRefine    = 0;
00548         discardOrd     = 0;
00549         discardRefine  = 0;
00550         discardDisc    = 0;
00551         discardIso     = 0;
00552     }
00553
00554     // work space for isIsomorphic
00555     modmat = newMat(nNodes, nNodes, 0);
00556     permp  = newArray(nNodes, 0);
00557     permq  = newArray(nNodes, 0);
00558     permr  = newArray(nNodes, 0);
00559
00560     // work space for bisearchM
00561     bidef  = newArray(nNodes, 0);
00562     bilow  = newArray(nNodes, 0);
00563     bicount = 0;
00564     DUMMYUSE(padding);
00565
00566 #if DEBUGM
00567     printf("+++ new    T_MGraph %d\n", ++countMG);
00568     if(countEG > 100) { exit(1); }
00569 #endif
00570 }
00571
00572 T_MGraph::~T_MGraph()
00573 {
00574     int j;
00575
00576     deletArray(bilow);
00577     deletArray(bidef);
00578     deletArray(permr);
00579     deletArray(permq);
00580     deletArray(permp);
00581
00582     deleteMat(modmat, nNodes, nNodes);
00583     delete curcl;
00584     deleteMat(adjMat, nNodes, nNodes);
00585     delete egraph;
00586     for (j = 0; j < nNodes; j++) {
00587         delete nodes[j];
00588     }
00589     delete[] nodes;
00590     delete[] clist;
00591
00592 #if DEBUGM
00593     printf("+++ delete T_MGraph %d\n", countMG++);
00594 #endif
00595 }
00596
00597 void T_MGraph::printAdjMat(T_MNodeClass *cl)
00598 {
00599     int j1, j2;
00600
00601     cout << "      ";
00602     for (j2 = 0; j2 < nNodes; j2++) {
00603         cout << " " << setw(2) << j2;
00604     }
00605     cout << endl;
00606     for (j1 = 0; j1 < nNodes; j1++) {
00607         cout << setw(2) << j1 << ": [";
00608         for (j2 = 0; j2 < nNodes; j2++) {
00609             cout << " " << setw(2) << adjMat[j1][j2];
00610         }
00611         cout << "]" << cl->ndcl[j1] << endl;
00612     }
00613 }
00614 }
00615
00616 //-----
00617 // Check graph can be a connected one.
00618 // If a connected component without free leg is not the whole graph then
00619 // return T_False, otherwise return T_True.
00620 Bool T_MGraph::isConnected(void)

```

```

00621 {
00622     int j, n, nv;
00623
00624     for (j = 0; j < nNodes; j++) {
00625         nodes[j]->visited = -1;
00626     }
00627     if (visit(0)) {
00628         return T_True;
00629     }
00630     nv = 0;
00631     for (n = 0; n < nNodes; n++) {
00632         if (nodes[n]->visited >= 0) {
00633             nv++;
00634         }
00635     }
00636     return (nv == nNodes);
00637 }
00638
00639 // Visiting connected node used for 'isConnected'
00640 // If child nodes has free legs, then this function returns T_True.
00641 // otherwise it returns T_False.
00642 Bool T_MGraph::visit(int nd)
00643 {
00644     int td;
00645
00646     // This node has free legs.
00647     if (nodes[nd]->freelg > 0) {
00648         return T_True;
00649     }
00650     nodes[nd]->visited = 0;
00651     for (td = 0; td < nNodes; td++) {
00652         if ((adjMat[nd][td] > 0) and (nodes[td]->visited < 0)) {
00653             if (visit(td)) {
00654                 return T_True;
00655             }
00656         }
00657     }
00658     // all the child nodes has no free legs.
00659     return T_False;
00660 }
00661
00662 // Check whether the current graph is the Representative of a isomorphic class.
00663 // nsym = symmetry factor by the permutation of nodes.
00664 // esym = symmetry factor by the permutation of edge.
00665 // If this graph is not a representative, then returns T_False.
00666 Bool T_MGraph::isIsomorphic(T_MNodeClass *cl)
00667 {
00668     int j1, j2, cmp, count, nself;
00669
00670     nsym = ToBigInt(0);
00671     esym = ToBigInt(1);
00672
00673     count = 0;
00674     while (T_True) {
00675         count = nextPerm(nNodes, cl->nClasses, cl->clist,
00676             permr, permq, permp, count);
00677
00678         if (count < 0) {
00679             // calculate permutations of edges
00680             esym = ToBigInt(1);
00681             for (j1 = 0; j1 < nNodes; j1++) {
00682                 if (adjMat[j1][j1] > 0) {
00683                     nself = adjMat[j1][j1]/2;
00684                     esym *= factorial(nself);
00685                     esym *= ipow(2, nself);
00686                 }
00687                 for (j2 = j1+1; j2 < nNodes; j2++) {
00688                     if (adjMat[j1][j2] > 0) {
00689                         esym *= factorial(adjMat[j1][j2]);
00690                     }
00691                 }
00692             }
00693             return T_True;
00694         }
00695
00696         permMat(nNodes, permp, adjMat, modmat);
00697         cmp = compMat(nNodes, adjMat, modmat);
00698         if (cmp < 0) {
00699             return T_False;
00700         } else if (cmp == 0) {
00701             // save permutation ???
00702             nsym = nsym + ToBigInt(1);
00703         }
00704     }
00705 }
00706
00707 // apply permutation to matrix 'mat0' and obtain 'mat1'

```

```

00708 void T_MGraph::permMat(int size, int *perm, int **mat0, int **mat1)
00709 {
00710     int j1, j2;
00711     for (j1 = 0; j1 < size; j1++) {
00712         for (j2 = 0; j2 < size; j2++) {
00713             mat1[j1][j2] = mat0[perm[j1]][perm[j2]];
00714         }
00715     }
00716 }
00717 }
00718
00719 // comparison of matrix
00720 int T_MGraph::compMat(int size, int **mat0, int **mat1)
00721 {
00722     int j1, j2, cmp;
00723     for (j1 = 0; j1 < size; j1++) {
00724         for (j2 = 0; j2 < size; j2++) {
00725             cmp = mat0[j1][j2] - mat1[j1][j2];
00726             if (cmp != 0) {
00727                 return cmp;
00728             }
00729         }
00730     }
00731     return 0;
00732 }
00733 }
00734
00735 // Refine the classification
00736 // cl : the current 'T_MNodeClass' object
00737 // cn : the class number
00738 // Returns (the new class number corresponds to 'cn') if OK, or
00739 // 'None' if ordering condition is not satisfied
00740 T_MNodeClass *T_MGraph::refineClass(T_MNodeClass *cl, int cn)
00741 {
00742     T_MNodeClass *ccl = cl;
00743     int ccn = cn;
00744     T_MNodeClass *ncl = NULL;
00745     int ucl[MAXNODES];
00746     int nucl, nce;
00747     int td, cmp;
00748
00749     nucl = 0;
00750     while (ccl->nClasses != nucl) { // repeat refinement.
00751         nce = 0;
00752         nucl = 0;
00753         for (td = 1; td < nNodes; td++) {
00754             // 'td' is the next node and the current node is 'td-1'.
00755             // Count up the number of the elements in the current class
00756             // corresponding to the current node
00757             nce++;
00758             cmp = ccl->clCmp(td-1, td, ccn);
00759
00760             // the ordering condition is not satisfied.
00761             if (cmp > 0) {
00762                 if (DEBUG) {
00763                     cout << "refine: cls = " << ccl->clist << endl;
00764                     ccl->printMat();
00765                     cout << "clmat" << endl;
00766                     ccl->printMat();
00767                     cout << "refine: discard: cls = " << ccl->clist
00768                         << "ucl = " << ucl << endl;
00769                 }
00770                 if (ccl != cl) {
00771                     delete ccl;
00772                 }
00773                 return NULL;
00774             }
00775             else if (cmp < 0) {
00776                 // 'td' is in the next class to the current node.
00777                 ucl[nucl++] = nce; // close the current class
00778
00779                 // start new class
00780                 nce = 0;
00781             }
00782             // nothing to do for the case of 'cmp == 0'.
00783         }
00784     }
00785
00786     // close array 'ucl'.
00787     ucl[nucl++] = nce + 1;
00788
00789     // class is not modified
00790     if (nucl == ccl->nClasses) {
00791         ncl = ccl;
00792         break;
00793     }
00794 }

```

```

00795         // inconsistent
00796     } else if (nucl < ccl->nClasses) {
00797         erEnd("refineClasses : smaller number of classes");
00798
00799         // preparation of the next repetition.
00800     } else {
00801         if (cn == ccl->nClasses) {
00802             td = nNodes;
00803         } else {
00804             td = ccl->flist[cn+1]-1;
00805         }
00806         if (ccl != cl) {
00807             delete ccl;
00808         }
00809         ccl = new T_MNodeClass(nNodes, nucl);
00810         ccl->init(ucl, maxdeg, adjMat);
00811         if (td == nNodes) {
00812             ccn = ccl->nClasses;
00813         } else {
00814             ccn = ccl->ndcl[td];
00815         }
00816         nucl = 0;
00817     }
00818 }
00819
00820 if (DEBUG) {
00821     cout << "refine: ucl = " << ucl << endl;
00822     cout << "refine: ncl = " << ncl->clist << endl;
00823     ncl->printMat();
00824 }
00825
00826 return ncl;
00827 }
00828
00829 // Search biconnected component
00830 // visit : pd --> nd --> td
00831 // ne : the number of edges between pd and nd.
00832 void T_MGraph::bisearchM(int nd, int pd, int ne)
00833 {
00834     int td;
00835
00836     bidef[nd] = bicount;
00837     bilow[nd] = bicount;
00838     bicount++;
00839
00840     for (td = 0; td < nNodes; td++) {
00841         if (nodes[td]->ext) { // ignore external node
00842             continue;
00843         }
00844         else if (td == nd) { // pd --> nd --> nd
00845             continue;
00846         }
00847         else if (td == pd) { // pd --> nd --> pd
00848             if (ne > 1) {
00849                 bilow[nd] = min(bilow[nd], bidef[pd]);
00850             }
00851         }
00852         else if (adjMat[td][nd] < 1) { // td is not adjacent to nd
00853             continue;
00854         }
00855         else if (bidef[td] >= 0) { // back edge
00856             bilow[nd] = min(bilow[nd], bidef[td]);
00857         }
00858         // new node
00859     } else {
00860         bisearchM(td, nd, adjMat[td][nd]);
00861
00862         // ordinary case
00863         if (bilow[td] > bidef[nd]) {
00864             nBridges++;
00865         }
00866         bilow[nd] = min(bilow[nd], bilow[td]);
00867     }
00868 }
00869 }
00870
00871 // nd is the starting point and not an external line
00872 // if (pd < 0 && (!nodes[nd]->ext || nodes[nd]->deg > 1)) {
00873 // if (pd < 0 && !nodes[nd]->ext && nodes[nd]->deg == 1) {
00874 if (pd < 0 && nodes[nd]->deg != 1) {
00875     // nBridges += nodes[nd]->deg;
00876     nBridges++;
00877 }
00878 }
00879
00880 // Count the number of lPI components.
00881 // The algorithm is described in

```

```

00882 // A.V. Aho, J.E. Hopcroft and J.D. Ullman
00883 // 'The Design and Analysis of Computer Algorithms', Chap. 5
00884 // 1974, Addison-Wesley.
00885
00886 int T_MGraph::count1PI(void)
00887 {
00888     int j;
00889
00890     if (nLoops < 0) {
00891         return 1;
00892     }
00893
00894     // initialization
00895     bicount = 0;
00896     nBridges = 0;
00897     for (j = 0; j < nNodes; j++) {
00898         bidef[j] = -1;
00899         bilow[j] = -1;
00900     }
00901
00902     bisearchM(0, -1, 0);
00903
00904     return nBridges;
00905 }
00906
00907 //-----
00908 // Generate graphs
00909 // the generation process starts from 'connectClass'.
00910
00911 long T_MGraph::generate(void)
00912 {
00913     T_MNodeClass *cl;
00914     int dscl[MAXNODES];
00915     int n;
00916
00917     for (n = 0; n < nNodes; n++) {
00918         dscl[n] = T_False;
00919     }
00920
00921     // Initial classification of nodes.
00922     cl = new T_MNodeClass(nNodes, nClasses);
00923     cl->init(clist, maxdeg, adjMat);
00924     connectClass(cl, dscl);
00925
00926     // Print the result.
00927
00928     if (MONITOR) {
00929         cout << endl;
00930         cout << endl;
00931         cout << "* Total " << ndiag << " Graphs.";
00932         cout << "(" << n1PI << " 1PI)";
00933         // cout << " wsum = " << wsum << "(" << wsopi << "1PI)" << endl;
00934         cout << endl;
00935     }
00936
00937     if (MONITOR) {
00938         cout << "* refine: " << nCallRefine << endl;
00939         cout << "* discard for ordering: " << discardOrd << endl;
00940         cout << "* discard for refinement: " << discardRefine << endl;
00941         cout << "* discard for disconnected: " << discardDisc << endl;
00942         cout << "* discard for duplication: " << discardIso << endl;
00943     }
00944     delete cl;
00945
00946     return ndiag;
00947 }
00948
00949 int T_MGraph::findNextCl(T_MNodeClass *cl, int *dscl)
00950 {
00951     int mine, cr, c, n, me;
00952
00953     mine = -1;
00954     cr = -1;
00955     for (c = 0; c < cl->nClasses; c++) {
00956         n = cl->flist[c];
00957         if (!dscl[n]) {
00958             me = nodes[n]->freelg;
00959             if (me > 0) {
00960                 if (nodes[n]->freelg < nodes[n]->deg) {
00961                     return c;
00962                 }
00963                 if (mine < 0 || mine > me) {
00964                     mine = me;
00965                     cr = c;
00966                 }
00967             }
00968         }
00969     }

```

```

00969     }
00970     return cr;
00971 }
00972
00973 int T_MGraph::findNextTC1(T_MNodeClass *cl, int *dtcl)
00974 {
00975     int c, n, mine, cr, me;
00976
00977     mine = -1;
00978     cr = -1;
00979     for (c = 0; c < cl->nClasses; c++) {
00980         for (n = cl->flist[c]; n < cl->flist[c+1]; n++) {
00981             if (!dtcl[n]) {
00982                 me = nodes[n]->freelg;
00983                 if (me > 0) {
00984                     if (nodes[n]->freelg < nodes[n]->deg) {
00985                         return c;
00986                     }
00987                     if (mine < 0 || mine > me) {
00988                         mine = me;
00989                         cr = c;
00990                     }
00991                 }
00992             }
00993         }
00994     }
00995     return cr;
00996 }
00997
00998 // Connect nodes in a class to others
00999 void T_MGraph::connectClass(T_MNodeClass *cl, int *dscl)
01000 {
01001     int sc, sn;
01002     T_MNodeClass *xcl;
01003
01004     if (DEBUG0) {
01005         printf("connectClass:begin:");
01006         printf(" dscl="); printArray(nNodes, dscl);
01007         printf("\n");
01008     }
01009     xcl = refineClass(cl, cl->nClasses);
01010
01011     if (xcl == NULL) {
01012         if (MONITOR) {
01013             discardRefine++;
01014         }
01015     } else {
01016         sc = findNextCl(xcl, dscl);
01017         if (sc < 0) {
01018             newGraph(cl);
01019         } else {
01020             sn = xcl->flist[sc];
01021             connectNode(sc, sn, xcl, dscl);
01022         }
01023     }
01024     if (xcl != cl && xcl != NULL) {
01025         delete xcl;
01026     }
01027     if (DEBUG0) {
01028         printf("connectClass:end\n");
01029     }
01030 }
01031
01032 //-----
01033 void T_MGraph::connectNode(int sc, int ss, T_MNodeClass *cl, int *dscl)
01034 {
01035     int sn;
01036     int dtcl[MAXNODES];
01037
01038     if (DEBUG0) {
01039         printf("connectNode:begin:(%d,%d)", sc, ss);
01040         printf(" dscl="); printArray(nNodes, dscl);
01041         printf("\n");
01042     }
01043     if (ss >= cl->flist[sc+1]) {
01044         connectClass(cl, dscl);
01045         if (DEBUG0) {
01046             printf("connectNode:endl\n");
01047         }
01048         return;
01049     }
01050     copyArray(nNodes, dscl, dtcl);
01051     for (sn = ss; sn < cl->flist[sc+1]; sn++) {
01052         if (!dscl[sn]) {
01053             connectLeg(sc, sn, sc, sn, cl, dscl, dtcl);
01054             if (DEBUG0) {

```

```

01056         printf("connectNode:end2\n");
01057     }
01058     return;
01059 }
01060 }
01061 if (DEBUG0) {
01062     printf("connectNode:end3\n");
01063 }
01064 }
01065
01066 // Add one connection between two legs.
01067 // 1. select another node to connect
01068 // 2. determine multiplicity of the connection
01069 //
01070 // Arguments
01071 //   cn : the current class
01072 //   nd : the current node to be connected.
01073 //   nextnd : {nextnd, ...} is the possible target node of the connection.
01074 //   cl : the current node class
01075
01076 void T_MGraph::connectLeg(int sc, int sn, int tc, int ts, T_MNodeClass *cl, int *dscl, int* dtcl)
01077 {
01078     int tn, maxself, nc2, nc, maxcon, ts1, wc, ncm;
01079     int dtcl1[MAXNODES];
01080
01081     if (DEBUG0) {
01082         printf("connectLeg:begin:(%d,%d,%d,%d)", sc, sn, tc, ts);
01083         printf(" dscl="); printArray(nNodes, dscl);
01084         printf(" dtcl="); printArray(nNodes, dtcl);
01085         printf("\n");
01086         printAdjMat(cl);
01087     }
01088
01089     if (sn >= cl->flist[sc+1]) {
01090         erEnd("*** connectLeg : illegal control");
01091         return;
01092     }
01093     // There remains no free legs in the node 'sn' : move to next node.
01094     } else if (nodes[sn]->freelg < 1) {
01095
01096         if (!isConnected()) {
01097             if (DEBUG0) {
01098                 printf("connectLeg:disconnected\n");
01099             }
01100             if (MONITOR) {
01101                 discardDisc++;
01102             }
01103         } else {
01104             if (DEBUG0) {
01105                 printf("connectLeg: call conNode\n");
01106             }
01107             dscl[sn] = T_True;
01108
01109             // next node in the current class.
01110             connectNode(sc, sn+1, cl, dscl);
01111
01112             dscl[sn] = T_False;
01113         }
01114
01115         // connect a free leg of the current node 'sn'.
01116     } else {
01117         copyArray(nNodes, dtcl, dtcl1);
01118
01119         ts1 = ts;
01120         for (wc = 0; wc < nNodes; wc++) {
01121             if (ts1 >= cl->flist[tc+1]) {
01122                 tc = findNextTCl(cl, dtcl1);
01123                 if (DEBUG0) {
01124                     printf("connectLeg:1:tc=%d\n", tc);
01125                 }
01126                 if (tc < 0) {
01127                     if (DEBUG0) {
01128                         printf("connectLeg:end1:(%d,%d,%d,%d)\n", sc, sn, tc, ts);
01129                     }
01130                     return;
01131                 }
01132                 if (tc != sc && !cl->chkOrd(sn, sc, cl, dtcl)) {
01133                     if (MONITOR) {
01134                         discardOrd++;
01135                     }
01136                     if (DEBUG0) {
01137                         printf("connectLeg:end2:(%d,%d,%d,%d)\n", sc, sn, tc, ts);
01138                     }
01139                     return;
01140                 }
01141             }
01142         }

```



```

01143     tsl = -1;
01144
01145     // repeat for all possible target nodes
01146     // for all tn in tc
01147     if (DEBUG0) {
01148         printf("connectLeg:2:tc=%d, fl:%d-->%d\n", tc, cl->flist[tc], cl->flist[tc+1]);
01149     }
01150     for (tn = cl->flist[tc]; tn < cl->flist[tc+1]; tn++) {
01151         if (DEBUG0) {
01152             printf("connectLeg:2:%d=>try %d\n", sn, tn);
01153         }
01154
01155         if (dtcll[tn]) {
01156             if (DEBUG0) {
01157                 printf("connectLeg:3\n");
01158             }
01159             continue;
01160         }
01161         dtcll[tn] = T_True;
01162
01163         if (sc == tc && sn > tn) {
01164             if (DEBUG0) {
01165                 printf("connectLeg:4\n");
01166             }
01167             continue;
01168
01169             // self-loop
01170         } else if (sn == tn) {
01171             if (nNodes > 1) {
01172                 // there are two or more nodes in the graph :
01173                 // avoid disconnected graph
01174                 maxself = min((nodes[sn]->freelg)/2, (nodes[sn]->deg-1)/2);
01175             } else {
01176                 // there is only one node the graph.
01177                 maxself = nodes[sn]->freelg/2;
01178             }
01179
01180             // If we can assume no tadpole, the following line can be used.
01181             if (selOPI and nNodes > 2) {
01182                 maxself = min((nodes[sn]->deg-2)/2, maxself);
01183             }
01184
01185             // vary the number of connection.
01186             for (nc2 = maxself; nc2 > 0; nc2--) {
01187                 // if nNodes > 1 and ndg == nc2:
01188                 // // disconnected
01189                 // continue
01190                 nc = 2*nc2;
01191                 if (DEBUG0) {
01192                     printf("connectLeg: call conLeg: (same) %d=>%d(%d)\n", sn, tn, nc);
01193                 }
01194                 ncm = 5*nc;
01195                 adjMat[sn][sn] = nc;
01196                 nodes[sn]->freelg -= nc;
01197                 cl->incMat(sn, tn, ncm);
01198                 tsl = tn + 1;
01199
01200                 // next connection
01201                 connectLeg(sc, sn, tc, tsl, cl, dscl, dtcll);
01202
01203                 // restore the configuration
01204                 cl->incMat(tn, sn, -ncm);
01205                 adjMat[sn][sn] = 0;
01206                 nodes[sn]->freelg += nc;
01207                 if (DEBUG0) {
01208                     printf("connectLeg: ret conLeg: (same) %d=>%d(%d)\n", sn, tn, nc);
01209                 }
01210             }
01211
01212             // connections between different nodes.
01213         } else {
01214             // maximum possible connection number
01215             maxcon = min(nodes[sn]->freelg, nodes[tn]->freelg);
01216
01217             // avoid disconnected graphs.
01218             if (nNodes > 2 && nodes[sn]->deg == nodes[tn]->deg) {
01219                 maxcon = min(maxcon, nodes[sn]->deg-1);
01220             }
01221
01222             if (CHECK) {
01223                 if ((adjMat[sn][tn] != 0) || (adjMat[sn][tn] != adjMat[tn][sn])) {
01224                     printf("*** inconsistent connection: sn=%d, tn=%d", sn, tn);
01225                     printf(" dscl="); printArray(nNodes, dscl);
01226                     printf(" dtcl="); printArray(nNodes, dtcl);
01227                     printf("\n");
01228                     printAdjMat(cl);
01229                     erEnd("*** inconsistent connection ");

```

```

01230         }
01231     }
01232
01233     // vary number of connections
01234     for (nc = maxcon; nc > 0; nc--) {
01235         if (DEBUG0) {
01236             printf("connectLeg: call conLeg: (diff) %d=>%d(%d)", sn, tn, nc);
01237             printf(" dtcl="); printArray(nNodes, dtcl);
01238             printf("\n");
01239         }
01240         adjMat[sn][tn] = nc;
01241         adjMat[tn][sn] = nc;
01242         nodes[sn]->freelg -= nc;
01243         nodes[tn]->freelg -= nc;
01244         cl->incMat(sn, tn, nc);
01245         cl->incMat(tn, sn, nc);
01246         ts1 = tn + 1;
01247
01248         // next connection
01249         connectLeg(sc, sn, tc, ts1, cl, dscl, dtcl1);
01250
01251         // restore configuration
01252         cl->incMat(sn, tn, -nc);
01253         cl->incMat(tn, sn, -nc);
01254         adjMat[sn][tn] = 0;
01255         adjMat[tn][sn] = 0;
01256         nodes[sn]->freelg += nc;
01257         nodes[tn]->freelg += nc;
01258         if (DEBUG0) {
01259             printf("connectLeg: ret conLeg: (diff) %d=>%d(%d)\n", sn, tn, nc);
01260             printf(" dtcl="); printArray(nNodes, dtcl);
01261             printAdjMat(cl);
01262             printf("\n");
01263         }
01264     }
01265 }
01266 }
01267 if (ts1 < 0) {
01268     ts1 = cl->flist[tc+1];
01269 }
01270 } // for wc
01271 }
01272 if (DEBUG0) {
01273     printf("connectLeg:end3: (%d,%d,%d,%d)\n", sc, sn, tc, ts);
01274 }
01275 }
01276
01277 // A new candidate diagram is obtained.
01278 // It may be a disconnected graph
01279 void T_MGraph::newGraph(T_MNodeClass *cl)
01280 {
01281     int connected;
01282     T_MNodeClass *xcl;
01283     Bool sopi;
01284
01285     ngen++;
01286     // printf("newGraph: %d\n", ngen);
01287
01288     // refine class and check ordering condition
01289     xcl = refineClass(cl, cl->nClasses);
01290
01291     if (xcl == NULL) {
01292         // printf("newGraph: fail refine\n");
01293         if (MONITOR) {
01294             discardRefine++;
01295         }
01296
01297         // check whether this is a connected graph or not.
01298     } else {
01299         connected = isConnected();
01300         if (!connected) {
01301             // printf("newGraph: not connected\n");
01302             if (DEBUG0) {
01303                 cout << "+++ disconnected graph" << ngen << endl;
01304                 xcl->printMat();
01305             }
01306
01307             if (MONITOR) {
01308                 discardDisc++;
01309             }
01310
01311             // connected graph : check isomorphism of the graph
01312         } else {
01313             ngconn++;
01314             if (!isIsomorphic(xcl)) {
01315                 // printf("newGraph: not isomorphic\n");
01316                 if (DEBUG) {

```

```

01317         cout << "+++ duplicated graph" << ngen << ngconn << endl;
01318         xcl->printMat();
01319     }
01320
01321     if (MONITOR) {
01322         discardIso++;
01323     }
01324
01325     // We got a new connected and a unique representation of a class.
01326 } else {
01327
01328     clPI = count1PI();
01329     if (!selOPI || clPI == 1) {
01330         wsum = wsum + ToFraction(1, nsym*esym);
01331         if (clPI == 1) {
01332             wsopi = wsopi + ToFraction(1, nsym*esym);
01333         }
01334         curcl->copy(xcl);
01335         ndiag++;
01336         sopi = (clPI == 1);
01337         if (sopi) {
01338             nlPI++;
01339         }
01340
01341         if (OPTPRINT) {
01342             cout << endl;
01343             cout << "Graph : " << ndiag
01344                  << "(" << ngen << ")"
01345                  << " lPI comp. = " << clPI
01346                  << " sym. factor = (" << nsym << "*" << esym << ")"
01347                  << endl;
01348             printAdjMat(cl);
01349             // cl->printMat();
01350             if (MONITOR) {
01351                 cout << "refine: " << nCallRefine << endl;
01352                 cout << "discard for ordering: " << discardOrd << endl;
01353                 cout << "discard for refinement: " << discardRefine << endl;
01354                 cout << "discard for disconnected: " << discardDisc << endl;
01355                 cout << "discard for duplication: " << discardIso << endl;
01356             }
01357         }
01358
01359         // go to next step
01360         egraph->init(pid, ndiag, adjMat, sopi, nsym, esym);
01361         toForm(egraph);
01362     }
01363 }
01364 }
01365 }
01366 // printf("in newGraph cl=%p, xcl=%p\n", cl, xcl);
01367 if (xcl != cl && xcl != NULL) {
01368     delete xcl;
01369 }
01370 }
01371
01372
01373 // go to next step
01374 // 1. convert data format which treat the edges as objects.
01375 // 2. definition of loop momenta to the edges.
01376 // 3. analysis of loop structure in a graph.
01377 // 4. assignment of particles to edges and vertices to nodes
01378 // 5. recalculate symmetry factor
01379
01380 // Permutation
01381 Local int nextPerm(int nelem, int nclass, int *cl, int *r, int *q, int *p, int count)
01382 {
01383     int j, k, n, e, t;
01384     Bool b;
01385
01386     for (j = 0; j < nelem; j++) {
01387         p[j] = j;
01388     }
01389     if (count < 1) {
01390         for (j = 0; j < nelem; j++) {
01391             q[j] = 0;
01392             r[j] = 0;
01393         }
01394         j = 0;
01395         for (k = 0; k < nclass; k++) {
01396             n = cl[k];
01397             for (e = 0; e < n; e++) {
01398                 r[j] = n - e - 1;
01399                 j++;
01400             }
01401         }
01402         if (j != nelem) {
01403             erEnd("*** inconsistent # elements");

```

```

01404     }
01405     return 1;
01406 }
01407 b = T_False;
01408 for (j = nelem-1; j >= 0; j--) {
01409     if (q[j] < r[j]) {
01410         for (k = j+1; k < nelem; k++) {
01411             q[k] = 0;
01412         }
01413         q[j]++;
01414         b = T_True;
01415         break;
01416     }
01417 }
01418 if (!b) {
01419     return (-count);
01420 }
01421
01422 for (j = 0; j < nelem; j++) {
01423     k = j + q[j];
01424     t = p[j];
01425     p[j] = p[k];
01426     p[k] = t;
01427 }
01428 return count + 1;
01429 }
01430
01431 Local BigInt factorial(int n)
01432 /* return (n < 1) ? 1 : n!
01433 */
01434 {
01435     int r, j;
01436
01437     r = 1;
01438     for (j = 2; j <= n; j++) {
01439         r *= j;
01440     }
01441     return r;
01442 }
01443
01444 Local BigInt ipow(int n, int p)
01445 {
01446     int r, j;
01447     DUMMYUSE(p);
01448
01449     r = 1;
01450     for (j = 2; j <= n; j++) {
01451         r *= n;
01452     }
01453     return r;
01454 }
01455
01456
01457 Local void erEnd(const char *msg)
01458 {
01459     printf("*** Error : %s\n", msg);
01460     exit(1);
01461 }
01462
01463 /* memory allocation */
01464 Local int *newArray(int size, int val)
01465 {
01466     int *a, j;
01467
01468     a = new int[size];
01469     for (j = 0; j < size; j++) {
01470         a[j] = val;
01471     }
01472     return a;
01473 }
01474
01475 Local void deletArray(int *a)
01476 {
01477     delete[] a;
01478 }
01479
01480 Local void copyArray(int size, int *a0, int *a1)
01481 {
01482     int j;
01483     for (j = 0; j < size; j++) {
01484         a1[j] = a0[j];
01485     }
01486 }
01487
01488 Local int **newMat(int n0, int n1, int val)
01489 {
01490     int **m, j;

```

```

01491
01492     m = new int*[n0];
01493     for (j = 0; j < n0; j++) {
01494         m[j] = newArray(n1, val);
01495     }
01496     return m;
01497 }
01498
01499 Local void deleteMat(int **m, int n0, int n1)
01500 {
01501     int j;
01502     DUMMYUSE(n1);
01503
01504     for (j = 0; j < n0; j++) {
01505         deletArray(m[j]);
01506     }
01507     delete[] m;
01508 }
01509
01510 //=====
01511 // Functions for testing the program.
01512 //
01513 //-----
01514 // Testing function nextPerm
01515 //
01516 Local void printArray(int n, int *p)
01517 {
01518     int j;
01519
01520     printf("[");
01521     for (j = 0; j < n; j++) {
01522         printf(" %2d", p[j]);
01523     }
01524     printf("]");
01525 }
01526
01527 Local void printMat(int n0, int n1, int **m)
01528 {
01529     int j;
01530
01531     for (j = 0; j < n0; j++) {
01532         printArray(n1, m[j]);
01533         printf("\n");
01534     }
01535 }
01536
01537 Global void testPerm()
01538 {
01539     int nelelem, nperm, nclist, n, count;
01540     int *p, *q, *r;
01541     int clist[] = {1, 2, 2, 3};
01542
01543     nclist = sizeof(clist)/sizeof(int);
01544     nelelem = 0;
01545     nperm = 1;
01546     for (n = 0; n < nclist; n++) {
01547         nelelem += clist[n];
01548         nperm *= factorial(clist[n]);
01549     }
01550
01551     printf("+++ clist = (%d) ", nclist);
01552     printArray(nclist, clist);
01553     printf("\n");
01554     printf("+++ nelelem = %d, nperm = %d\n", nelelem, nperm);
01555
01556     p = new int[nelelem];
01557     q = new int[nelelem];
01558     r = new int[nelelem];
01559     count = 0;
01560     while (T_True) {
01561         count = nextPerm(nelelem, nclist, clist, r, q, p, count);
01562         if (count < 0) {
01563             count = -count;
01564             break;
01565         }
01566         printf("%4d:", count);
01567         printArray(nelelem, p);
01568         printf("\n");
01569
01570         if (count > nperm) {
01571             break;
01572         }
01573     }
01574     if (count != nperm) {
01575         printf("*** %d != %d\n", count, nperm);
01576     }
01577     delete[] p;

```

```

01578     delete[] q;
01579     delete[] r;
01580 }
01581
01582 // } } ]
01583

```

## 4.56 gentopo.h

```

00001 #pragma once
00002
00003 // Temporal definition of big integer
00004 #define BigInt long
00005 #define ToBigInt(x) ((BigInt) (x))
00006 //
00007 // Temporal definition of ratio of two big integers
00008 #define Fraction double
00009 #define ToFraction(x, y) (((double) (x))/((double) (y)))
00010
00011 //=====
00012 // Output to FROM
00013
00014 //=====
00015 // class of nodes for T_EGraph
00016
00017 class T_ENode {
00018 public:
00019     int deg;
00020     int ext;
00021     int *edges;
00022 };
00023
00024 class T_EEdge {
00025 public:
00026     int nodes[2];
00027     int ext;
00028     int momn;
00029     char momc[2]; // no more used
00030     // momentum is printed like: ("%s%d", (enode.ext)?"Q":"p", enode.momn)
00031     char padding[6];
00032 };
00033
00034 class T_EGraph {
00035 public:
00036     long gId;
00037     int pId;
00038     int nNodes;
00039     int nEdges;
00040     int maxdeg;
00041     int nExtern;
00042
00043     int opi;
00044     BigInt nsym, esym;
00045
00046     T_ENode *nodes;
00047     T_EEdge *edges;
00048
00049     T_EGraph(int nnodes, int nedges, int mxdeg);
00050     ~T_EGraph();
00051     void print(void);
00052     void init(int pid, long gid, int **adjmat, int sopi, BigInt nsym, BigInt esym);
00053
00054     void setExtern(int nd, int val);
00055     void endSetExtern(void);
00056 };
00057
00058 //=====
00059 // class of nodes for T_MGraph
00060
00061 class T_MNode {
00062 public:
00063     int id; // node id
00064     int deg; // degree(node) = the number of legs
00065     int cls; // initial class number in which the node belongs
00066     int ext;
00067
00068     int freelg; // the number of free legs
00069     int visited;
00070
00071     T_MNode(int id, int deg, int ext, int cls);
00072 };
00073
00074 //=====

```

```

00075 // class of node-classes for T_MGraph
00076 class T_MNodeClass;
00077
00078 //=====
00079 // class of scalar graph expressed by matrix form
00080 //
00081 // Input : the classified set of nodes.
00082 // Output : control passed to 'T_EGraph(self)'
00083
00084 class T_MGraph {
00085
00086 public:
00087
00088     // initial conditions
00089     int pId;           // process/subprocess ID
00090     int nNodes;        // the number of nodes
00091     int nEdges;        // the number of edges
00092     int nLoops;        // the number of loops
00093     T_MNode **nodes;   // table of T_MNode object
00094     int *clist;        // list of initial classes
00095     int nClasses;      // the number of initial classes
00096     // int ndcl;        // node --> initial class number
00097     int mindeg;        // minimum value of degree of nodes
00098     int maxdeg;        // maximum value of degree of nodes
00099
00100     int selOPI;        // flag to select lPI graphs
00101
00102     // generated set of graphs
00103     long ndiag;        // the total number of generated graphs
00104     long nlPI;         // the total number of lPI graphs
00105     int nBridges;      // the number of bridges
00106
00107     // the current graph
00108     int clPI;          // the number of lPI components
00109     int **adjMat;       // adjacency matrix
00110     BigInt nsym;        // symmetry factor from nodes
00111     BigInt esym;        // symmetry factor from edges
00112     Fraction wsum;      // weighted sum of graphs
00113     Fraction wsopi;     // weighted sum of lPI graphs
00114     T_MNodeClass *curcl; // the current 'T_MNodeClass' object
00115     T_EGraph *egraph;
00116
00117     // measures of efficiency
00118     long ngen;          // generated graph before check
00119     long ngconn;        // generated connected graph before check
00120
00121     long nCallRefine;
00122     long discardOrd;
00123     long discardRefine;
00124     long discardDisc;
00125     long discardIso;
00126
00127     // functions */
00128     T_MGraph(int pid, int ncl, int *cldeg, int *clnum, int *clext, int sopi);
00129     ~T_MGraph();
00130     long generate(void);
00131
00132 private:
00133     // work space for isomorphism
00134     int **modmat;       // permutated adjacency matrix
00135     int *permp;
00136     int *permq;
00137     int *permr;
00138
00139     // work space for biconnected component
00140     int *bidef;
00141     int *bilow;
00142     int bicount;
00143     int padding;
00144
00145     void printAdjMat(T_MNodeClass *cl);
00146     int isConnected(void);
00147     int visit(int nd);
00148     int isIsomorphic(T_MNodeClass *cl);
00149     void permMat(int size, int *perm, int **mat0, int **mat1);
00150     int compMat(int size, int **mat0, int **mat1);
00151     T_MNodeClass *refineClass(T_MNodeClass *cl, int cn);
00152     void bisearchM(int nd, int pd, int ne);
00153     int countlPI(void);
00154
00155     int findNextCl(T_MNodeClass *cl, int *dscl);
00156     int findNextTCl(T_MNodeClass *cl, int *dcl);
00157     void connectClass(T_MNodeClass *cl, int *dscl);
00158     void connectNode(int sc, int ss, T_MNodeClass *cl, int *dscl);
00159     void connectLeg(int sc, int sn, int tc, int ts, T_MNodeClass *cl, int *dscl, int* dtcl);
00160     void newGraph(T_MNodeClass *cl);
00161

```

```
00162 };
```

## 4.57 grcc.cc

```
00001 // { ( [
00002 //*****
00003 // grcc.cc
00004
00005 #ifndef NOFORM
00006 extern "C" {
00007 #include "form3.h"
00008 }
00009 #endif
00010
00011 #include <iostream>
00012 #include <iomanip>
00013 #include <sstream>
00014 #include <cstdio>
00015 #include <stdlib.h>
00016 #include <string.h>
00017 #include <time.h>
00018
00019 // #define DEBUG9
00020
00021 // optimization : best combination depends on process by process
00022 #define SIMPSEL
00023
00024 // optimization : lower and upper bound of deg of target node of connection.
00025 #define MINMAXLEG
00026
00027 // optimization : optimization with respect to 'extonly' particles
00028 #define OPTEXTONLY
00029
00030 // check unique interaction by name or by code
00031 #define UNIQINTR
00032
00033 // possible extensions of the program
00034 // #define ORBITS
00035
00036 // reverse direction of an edge of agraph when anti-particle flows on it.
00037 // In this case the direction of edges will be different
00038 // from ones of original mgraph.
00039 // #define EDGEORDER
00040
00041 //-----
00042 #include "grcc.h"
00043
00044 #ifdef GRCC_NAMESPACE
00045 using namespace Grcc;
00046 #endif
00047
00048 //-----
00049 // Macro functions
00050 #define CLWIGHTD(x) (5*(x))
00051 #define CLWIGHTO(x) (3*(x)-2)
00052
00053 //-----
00054 // Static variables
00055
00056 static OptDef optDef[] = {
00057 {"Step", "Generate particle assigned graphs", GRCC_AGraph, 0},
00058 {"Outgrf", "Output to file (out.grf)", False, 0},
00059 {"Outgrp", "Output to file (out.grp)", False, 0},
00060 {"lPI", "Only lPI graphs", True, 0},
00061 {"NoSelfLoop", "Exclude graphs with loops consist of 1 edge", True, 0},
00062 {"NoTadpole", "Exclude graphs with tadpoles (2 edge connected)", True, 0},
00063 {"No1PtBlock", "Exclude graphs with tadpole blocks (2 node connected)", False, 0},
00064 {"No2PtLlPI", "Exclude graphs with 2-point subgraphs", False, 0},
00065 {"NoExtSelf", "Exclude graphs with 2-pt subgraphs at ext. particles", False, 0},
00066 {"NoAdj2PtV", "Exclude graphs with an edge connecting 2-pt vertices", False, 0},
00067 {"Block", "Exclude graphs with more than one block", False, 0},
00068 {"SymmInitial", "Symmetrize initial particles", False, 0},
00069 {"SymmFinal", "Symmetrize final particles", False, 0},
00070 };
00071
00072 static int nOptDef = sizeof(optDef)/sizeof(OptDef);
00073
00074 static int prlevel = 2;
00075 static ErExit *erExit = NULL;
00076 static void *erExitArg = NULL;
00077
00078 //*****
00079 // Utility functions
```



```

00080 //=====
00081
00082 static void      erEnd(const char *msg);
00083 static Bool      nextPart(int nelem, int nclist, int *clist, int *nl, int *r);
00084 static void      prillist(int n, const int *a, const char *msg);
00085 static int       *newArray(int size, int val);
00086 static int       *deleteArray(int *a);
00087 static int       **newMat(int n0, int n1, int val);
00088 static int       **deleteMat(int **m, int n0);
00089 static void      prIntArray(int n, int *p, const char *msg);
00090 static void      prIntArrayErr(int n, int *p, const char *msg);
00091 static int       *intdup(int n, int *v);
00092 static int       *delintdup(int *v);
00093 static void      bsort(int n, int *ord, int *a);
00094 static void      sorti(int n, int *a);
00095 static int       sortb(int n, int *a);
00096 static int       toSList(int n, int *a);
00097 static int       intSetAdd(int n, int *a, int v, const int size);
00098 static int       intSListAdd(int n, int *a, int v, const int size);
00099 static int       cmpArray(int na, int *a, int nb, int *b);
00100 static Bool      leqArray(int n, int *a, int *b);
00101 static void      prMomStr(int mom, const char *ms, int mn);
00102 static void      listDiff(int na, int *a, int nb, int *b, int *np, int *p, int *nq, int *q, int *nr, int
    *r);
00103 static Bool      isSubSList(int na, int *a, int nb, int *b);
00104 static int       subtrSet(int na, int *a, int nb, int *b, int *c, int size);
00105 static int       nextPerm(int nelem, int nclass, int *cl, int *r, int *q, int *p, int count);
00106 static BigInt    ipow(int n, int p);
00107 static BigInt    factorial(int n);
00108
00109 // for debugging
00110 #ifdef CHECK
00111 static Bool      isIn(int n, int *a, int v);
00112 #endif
00113
00114 //=====
00115 // class Options
00116 //-----
00117 Options::Options(void)
00118 {
00119     if (nOptDef != GRCC_OPT_Size) {
00120         fprintf(GRCC_Stderr, "*** Options: inconsistent default values\n");
00121         exit(1);
00122     }
00123     model = NULL;
00124     proc = NULL;
00125     sprc = NULL;
00126
00127     outmg = NULL;
00128     endmg = NULL;
00129     outag = NULL;
00130
00131     argmg = NULL;
00132     argemg = NULL;
00133     argag = NULL;
00134
00135     setDefaultValue();
00136
00137     // for output
00138     out = new Output(this);
00139
00140     // subrpocess
00141     sprc = NULL;
00142
00143     // measuring time
00144     time0 = 0.0;
00145     time1 = 0.0;
00146 }
00147
00148 //-----
00149 Options::~Options(void)
00150 {
00151     outmg = NULL;
00152     outag = NULL;
00153
00154     argmg = NULL;
00155     argag = NULL;
00156
00157     delete out;
00158 }
00159
00160 //-----
00161 void Options::setDefaultValue(void)
00162 {
00163     int j;
00164
00165     // default values

```

```

00166     for (j = 0; j < GRCC_OPT_Size; j++) {
00167         values[j] = optDef[j].defaultv;
00168     }
00169     values[GRCC_OPT_Step] = GRCC_AGraph;
00170
00171     // print level
00172     prlevel = 1;
00173 }
00174
00175
00176
00177 //-----
00178 void Options::setOutMG(OutEGB *omg, void *pt)
00179 {
00180     outmg = omg;
00181     argmg = pt;
00182 }
00183
00184 //-----
00185 void Options::setEndMG(OutEGB *emg, void *pt)
00186 {
00187     endmg = emg;
00188     argemg = pt;
00189 }
00190
00191 //-----
00192 void Options::setOutAG(OutEG *oag, void *pt)
00193 {
00194     outag = oag;
00195     argag = pt;
00196 }
00197
00198 //-----
00199 void Options::setErExit(ErExit *ere, void *erearg)
00200 {
00201     erExit = ere;
00202     erExitArg = erearg;
00203 }
00204
00205 //-----
00206 void Options::printLevel(int l)
00207 {
00208     prlevel = l;
00209 }
00210
00211 //-----
00212 void Options::setValue(int ind, int val)
00213 {
00214     if (0 <= ind && ind < GRCC_OPT_Size) {
00215         values[ind] = val;
00216     } else {
00217         if (prlevel > 0) {
00218             fprintf(GRCC_Stderr, "*** Options::setValue : invalid index=%d ",
00219                 ind);
00220             fprintf(GRCC_Stderr, "(val=%d)\n", val);
00221         }
00222         erEnd("Options::setValue : invalid index");
00223     }
00224 }
00225
00226 //-----
00227 int Options::getValue(int ind)
00228 {
00229     if (0 <= ind && ind < GRCC_OPT_Size) {
00230         return values[ind];
00231     } else {
00232         if (prlevel > 0) {
00233             fprintf(GRCC_Stderr, "*** Options::getValue : invalid index=%d\n", ind);
00234         }
00235         erEnd("Options::getValue : invalid index");
00236     }
00237     return -1;
00238 }
00239
00240 //-----
00241 void Options::print(void)
00242 {
00243     int j;
00244
00245     printf("Options\n");
00246     printf("+++ GRCC_OPT_Size=%d, print level=%d: ",
00247         GRCC_OPT_Size, prlevel);
00248     printf("symbol = value (default)\n");
00249     for (j=0; j < GRCC_OPT_Size; j++) {
00250         printf("    %4d GRCC_OPT_%-15s = %2d (%2d)\n",
00251             j, optDef[j].name, values[j], optDef[j].defaultv);
00252     }

```

```

00253     printf("    outgrf=%s, outgrp=%s\n", out->outgrf, out->outgrp);
00254 }
00255
00256 //-----
00257 const OptDef *Options::getDef(void)
00258 {
00259     return optDef;
00260 }
00261
00262 //-----
00263 void Options::setOutputF(Bool outf, const char *fname)
00264 {
00265     values[GRCC_OPT_Outgrf] = outf;
00266     out->setOutgrf(fname);
00267 }
00268
00269 //-----
00270 void Options::setOutputP(Bool outp, const char *fname)
00271 {
00272     values[GRCC_OPT_Outgrp] = outp;
00273     out->setOutgrp(fname);
00274 }
00275
00276 //-----
00277 void Options::printModel(void)
00278 {
00279     if (prlevel > 0) {
00280         if (model != NULL) {
00281             model->prModel();
00282         } else {
00283             printf("*** model is not defined\n");
00284         }
00285     }
00286 }
00287
00288 //-----
00289 void Options::outModel(void)
00290 {
00291     if (out != NULL && model != NULL) {
00292         out->outModelF();
00293         out->outModelP();
00294     }
00295 }
00296
00297 //-----
00298 void Options::begin(Model *mdl)
00299 {
00300     model = mdl;
00301     if (out != NULL) {
00302         if (values[GRCC_OPT_Outgrf]) {
00303             out->outBeginF(model, values[GRCC_OPT_Outgrf]);
00304             if (model != NULL) {
00305                 out->outModelF();
00306             }
00307         }
00308         if (values[GRCC_OPT_Outgrp]) {
00309             out->outBeginP(model, values[GRCC_OPT_Outgrp]);
00310             if (model != NULL) {
00311                 out->outModelP();
00312             }
00313         }
00314     }
00315 }
00316
00317 //-----
00318 void Options::end(void)
00319 {
00320     if (prlevel > 0) {
00321         printf("Optimization: ");
00322         #ifdef SIMPSEL
00323             printf("SIMPSEL=1 ");
00324         #else
00325             printf("SIMPSEL=0 ");
00326         #endif
00327         #ifdef MINMAXLEG
00328             printf("MINMAXLEG=1 ");
00329         #else
00330             printf("MINMAXLEG=0 ");
00331         #endif
00332         #ifdef OPTEXTONLY
00333             printf("OPTEXTONLY=1 ");
00334         #else
00335             printf("OPTEXTONLY=0 ");
00336         #endif
00337         printf("\n");
00338     }
00339 }

```

```

00340
00341     if (out != NULL && values[GRCC_OPT_Outgrf]) {
00342         out->outEndF();
00343     }
00344     if (out != NULL && values[GRCC_OPT_Outgrp]) {
00345         out->outEndP();
00346     }
00347     model = NULL;
00348 }
00349 //
00350 //-----
00351 void Options::beginProc(Process *proc)
00352 {
00353     proc = proc;
00354     if (out != NULL && values[GRCC_OPT_Outgrf]) {
00355         out->outProcBeginF(proc);
00356     }
00357     if (out != NULL && values[GRCC_OPT_Outgrp]) {
00358         out->outProcBeginP(proc);
00359     }
00360     if (proc != NULL) {
00361         proc->mgrcount = 0;
00362         proc->agrcount = 0;
00363     }
00364 }
00365
00366 //-----
00367 void Options::endProc(void)
00368 {
00369     int k;
00370
00371     if (proc == NULL) {
00372         if (out != NULL && values[GRCC_OPT_Outgrf]) {
00373             out->outProcEndF();
00374         }
00375         if (out != NULL && values[GRCC_OPT_Outgrp]) {
00376             out->outProcEndP();
00377         }
00378         return;
00379     }
00380     if (prlevel > 0) {
00381         printf("\n");
00382         printf("+++ Proc %d: ext=%d, loop=%d, ",
00383             proc->id, proc->nExtern, proc->loop);
00384         if (model != NULL) {
00385             printf("order=");
00386             prIntArray(model->ncouple, proc->clist, ": ");
00387             model->prParticleArray(proc->ninitl, proc->initlPart, "-->");
00388             model->prParticleArray(proc->nfinal, proc->finalPart, "");
00389         }
00390         printf(" (%8.2f sec)\n", proc->sec);
00391
00392         printf("    Proc    %d: Total M-Graphs=%ld, M-Graphs=",
00393             proc->id, proc->nMGraphs);
00394         proc->wMGraphs.print(" (Conn)\n");
00395
00396         printf("    Proc    %d: Total M-Graphs=%ld, M-Graphs=",
00397             proc->id, proc->nMOPI);
00398         proc->wMOPI.print(" (lPI)\n");
00399
00400         printf("    Proc    %d: Total A-Graphs=%ld, A-Graphs=",
00401             proc->id, proc->nAGraphs);
00402         proc->wAGraphs.print(" (Conn)\n");
00403
00404         printf("    Proc    %d: Total A-Graphs=%ld, A-Graphs=",
00405             proc->id, proc->nAOPI);
00406         proc->wAOPI.print(" (lPI)\n");
00407
00408         printf("# { %d,{", proc->ninitl);
00409         for (k = 0; k < proc->ninitl; k++) {
00410             if (k != 0) {
00411                 printf(", ");
00412             }
00413             if (proc->model != NULL) {
00414                 printf("\\"%s\"", proc->model->particleName(proc->initlPart[k]));
00415             } else {
00416                 printf("%d", proc->initlPart[k]);
00417             }
00418         }
00419         printf("}, %d,{", proc->nfinal);
00420         for (k = 0; k < proc->nfinal; k++) {
00421             if (k != 0) {
00422                 printf(", ");
00423             }
00424             if (proc->model != NULL) {
00425                 printf("\\"%s\"", proc->model->particleName(proc->finalPart[k]));
00426

```

```

00427         } else {
00428             printf("%d", proc->initlPart[k]);
00429         }
00430     }
00431     printf(", ");
00432     if (model != NULL) {
00433         printf("{");
00434         for (k = 0; k < model->ncouple; k++) {
00435             if (k != 0) {
00436                 printf(", ");
00437             }
00438             printf("%d", proc->clist[k]);
00439         }
00440         printf("}");
00441     }
00442     printf("},{%6ldL,%6ldL,%3ldL, -1.0, %4.2f},\n",
00443           proc->nAOPI, proc->wAOPI.num, proc->wAOPI.den, proc->sec);
00444 }
00445
00446 if (out != NULL && values[GRCC_OPT_Outgrf]) {
00447     out->outProcEndF();
00448 }
00449 if (out != NULL && values[GRCC_OPT_Outgrp]) {
00450     out->outProcEndP();
00451 }
00452 proc = NULL;
00453 }
00454
00455 //-----
00456 void Options::beginSubProc(SProcess *sprc)
00457 {
00458     sprc = sprc;
00459
00460     // 'beginProc' is called before
00461     if (out != NULL && values[GRCC_OPT_Outgrf]) {
00462         out->outSProcBeginF(sprc);
00463     }
00464     if (out != NULL && values[GRCC_OPT_Outgrp]) {
00465         out->outSProcBeginP(sprc);
00466     }
00467     if (proc != NULL) {
00468         return;
00469     }
00470     if (sprc != NULL) {
00471         sprc->mgrcount = 0;
00472         sprc->agrcount = 0;
00473     }
00474 }
00475
00476 //-----
00477 void Options::endSubProc(void)
00478 {
00479     MGraph *mgraph;
00480
00481     if (sprc == NULL) {
00482         if (out != NULL && values[GRCC_OPT_Outgrf]) {
00483             out->outProcEndF();
00484         }
00485         if (out != NULL && values[GRCC_OPT_Outgrp]) {
00486             out->outProcEndP();
00487         }
00488         return;
00489     }
00490     mgraph = sprc->mgraph;
00491     if (sprc != NULL) {
00492         sprc->resultMGraph(mgraph->cDiag, mgraph->wscon,
00493                           mgraph->c1PI, mgraph->wsopi);
00494     }
00495
00496     if (prlevel > 1) {
00497         printf("\n");
00498         printf("+++ Subproc %d: ext=%d, loop=%d, nodes=%d, edges=%d\n",
00499               sprc->id, sprc->nExtern, sprc->loop,
00500               sprc->nNodes, sprc->nEdges);
00501
00502         printf("    Subproc %d: Total M-Graphs=%ld, M-Wsum=",
00503               sprc->id, sprc->nMGraphs);
00504         sprc->wMGraphs.print(" (Conn)\n");
00505
00506         printf("    Subproc %d: Total M-Graphs=%ld, M-Wsum=",
00507               sprc->id, sprc->nMOPI);
00508         sprc->wMOPI.print(" (1PI)\n");
00509
00510         printf("    Subproc %d: Total A-Graphs=%ld, A-Wsum=",
00511               sprc->id, sprc->nAGraphs);
00512         sprc->wAGraphs.print(" (Conn)\n");
00513     }

```

```

00514         printf("    Subproc %d: Total A-Graphs=%ld, A-Wsum=",
00515                sproc->id, sproc->nAOPI);
00516         sproc->wAOPI.print(" (1PI)\n");
00517     }
00518     if (prlevel > 0) {
00519         printf("\n");
00520         printf("* Total %ld MGraphs; %ld 1PI", mgraph->cDiag, mgraph->c1PI);
00521         printf(" wscon = ");
00522         mgraph->wscon.print(" ( ");
00523         mgraph->wsopi.print(" 1PI)\n");
00524     }
00525 #ifdef MONITOR
00526     printf("* refine: %ld\n", mgraph->nCallRefine);
00527     printf("* discarded for refinement: %ld\n", mgraph->discardRefine);
00528     printf("* discarded for disconnected: %ld\n", mgraph->discardDisc);
00529     printf("* discarded for duplication: %ld\n", mgraph->discardIso);
00530     printf("* discarded by outMG: %ld\n", mgraph->discardMG);
00531 #endif
00532 }
00533
00534 if (proc != NULL) {
00535     return;
00536 }
00537 if (out != NULL && values[GRCC_OPT_Outgrf]) {
00538     out->outProcEndF();
00539 }
00540 if (out != NULL && values[GRCC_OPT_Outgrp]) {
00541     out->outProcEndP();
00542 }
00543 }
00544
00545 //-----
00546 void Options::newMGraph(MGraph *mgr)
00547 {
00548     MGraph *mgraph = mgr;
00549
00550 #ifdef DEBUG
00551     if (proc != NULL) {
00552         printf("+++ New EGraph (MG): %ld\n", proc->mgrcount);
00553     } else if (sproc != NULL) {
00554         printf("+++ New EGraph (MG): %ld\n", sproc->mgrcount);
00555     }
00556 #endif
00557 if (proc != NULL) {
00558     proc->mgrcount++;
00559     mgr->egraph->mId = proc->mgrcount;
00560 } else if (sproc != NULL) {
00561     sproc->mgrcount++;
00562     mgr->egraph->mId = sproc->mgrcount;
00563 }
00564
00565 if (out != NULL
    && values[GRCC_OPT_Step] == GRCC_MGraph) {
00566     if (values[GRCC_OPT_Outgrf]) {
00567         out->outEGraphF(mgraph->egraph);
00568     }
00569     if (values[GRCC_OPT_Outgrp]) {
00570         out->outEGraphP(mgraph->egraph);
00571     }
00572 }
00573
00574 if (sproc != NULL) {
00575     sproc->endMGraph(mgr);
00576 }
00577 if (endmg != NULL) {
00578     (*endmg)(mgr->egraph, argemg);
00579 }
00580 }
00581
00582 //-----
00583 void Options::newAGraph(EGraph *egraph)
00584 {
00585     Fraction sf, zr;
00586
00587 #ifdef DEBUG
00588     if (proc != NULL) {
00589         printf("+++ New EGraph (AG): %ld (%ld)\n",
00590                proc->agrcount, proc->mgrcount);
00591     } else if (sproc != NULL) {
00592         printf("+++ New EGraph (AG): %ld (%ld)\n",
00593                sproc->agrcount, sproc->mgrcount);
00594     }
00595 #endif
00596 if (proc != NULL) {
00597     proc->agrcount++;
00598     egraph->sId = proc->agrcount;
00599 } else if (sproc != NULL) {
00600     sproc->agrcount++;

```

```

00601     egraph->sId = sproc->agrcount;
00602 }
00603
00604 if (sproc != NULL) {
00605     zr = Fraction(0, 1);
00606     if ((egraph->nsym)*(egraph->esym) != 0) {
00607         sf = Fraction(egraph->extperm, egraph->nsym*egraph->esym);
00608     } else {
00609         sf = zr;
00610     }
00611     if (egraph->mgraph->opi) {
00612         sproc->resultAGraph(1, sf, 1, sf);
00613     } else {
00614         sproc->resultAGraph(1, sf, 0, zr);
00615     }
00616 }
00617
00618 if (out != NULL && values[GRCC_OPT_Outgrf]) {
00619     out->outEGraphF(egraph);
00620 }
00621 if (out != NULL && values[GRCC_OPT_Outgrp]) {
00622     out->outEGraphP(egraph);
00623 }
00624 if (outag != NULL) {
00625 #ifdef DEBUG1
00626     printf("call outag\n");
00627     egraph->model->prModel();
00628     egraph->print();
00629 #endif
00630     (*outag)(egraph, argag);
00631 }
00632
00633 sproc->endAGraph(egraph);
00634 }
00635
00636 //=====
00637 Output::Output(Options *optn)
00638 {
00639     opt      = optn;
00640     model    = NULL;
00641     proc     = NULL;
00642     sproc    = NULL;
00643     outgrfp  = NULL;
00644     outgrpp  = NULL;
00645     procId   = -1;
00646     outgrf   = NULL;
00647     outgrp   = NULL;
00648     outproc  = False;
00649 }
00650 }
00651
00652 //-----
00653 Output::~Output(void)
00654 {
00655     if (outgrfp != NULL) {
00656         if (prlevel > 0) {
00657             fprintf(GRCC_Stderr, "*** file has not been closed : \"%s\"\n",
00658                 outgrf);
00659         }
00660         fclose(outgrfp);
00661         outgrfp = NULL;
00662         erEnd("file has not been closed");
00663     }
00664     if (outgrf != NULL) {
00665         free(outgrf);
00666         outgrf = NULL;
00667     }
00668     if (outgrpp != NULL) {
00669         if (prlevel > 0) {
00670             fprintf(GRCC_Stderr, "*** file has not been closed : \"%s\"\n",
00671                 outgrp);
00672         }
00673         fclose(outgrpp);
00674         outgrpp = NULL;
00675         erEnd("file has not been closed");
00676     }
00677     if (outgrp != NULL) {
00678         free(outgrp);
00679         outgrp = NULL;
00680     }
00681 }
00682
00683 //-----
00684 void Output::setOutgrf(const char *fname)
00685 {
00686     if (outgrf != NULL && strcmp(outgrf, fname) == 0) {
00687         return;

```

```

00688     }
00689
00690     if (outgrf != NULL) {
00691         free(outgrf);
00692     }
00693     if (fname == NULL || strlen(fname) < 1) {
00694         outgrf = NULL;
00695     } else {
00696         outgrf = strdup(fname);
00697     }
00698 }
00699
00700 //-----
00701 void Output::setOutgrp(const char *fname)
00702 {
00703     if (outgrp != NULL && strcmp(outgrp, fname) == 0) {
00704         return;
00705     }
00706
00707     if (outgrp != NULL) {
00708         free(outgrp);
00709     }
00710     if (fname == NULL || strlen(fname) < 1) {
00711         outgrp = NULL;
00712     } else {
00713         outgrp = strdup(fname);
00714     }
00715 }
00716
00717 //-----
00718 Bool Output::outBeginF(Model *mdl, Bool pr)
00719 {
00720     model = mdl;
00721
00722     if (outgrf == NULL || strlen(outgrf) < 1) {
00723         outgrfp = NULL;
00724         return True;
00725     } else if (outgrfp != NULL) {
00726         if (prlevel > 0) {
00727             fprintf(GRCC_Stderr, "*** outBegin: \"%s\" is already opened\n",
00728                 outgrf);
00729         }
00730         erEnd("outBegin: file is already opened\n");
00731     }
00732
00733     if (!pr) {
00734         return True;
00735     }
00736     if ((outgrfp = fopen(outgrf, "w")) == NULL) {
00737         if (prlevel > 0) {
00738             fprintf(GRCC_Stderr, "*** outBegin: cannot open %s\n", outgrf);
00739         }
00740         return False;
00741     }
00742     fprintf(outgrfp, "%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%\n");
00743     fprintf(outgrfp, "% Generated by 'grcc'\n");
00744     fprintf(outgrfp, "Version={2,2,0,0};\n");
00745     if (model != NULL) {
00746         fprintf(outgrfp, "Model=\"%s.mdl\";\n", model->name);
00747     } else {
00748         fprintf(outgrfp, "Model=\"%s.mdl\";\n", "Phi34.mdl");
00749     }
00750
00751     return True;
00752 }
00753
00754 //-----
00755 Bool Output::outBeginP(Model *mdl, Bool pr)
00756 {
00757     model = mdl;
00758
00759     if (outgrp == NULL || strlen(outgrp) < 1) {
00760         outgrpp = NULL;
00761         return True;
00762     } else if (outgrpp != NULL) {
00763         if (prlevel > 0) {
00764             fprintf(GRCC_Stderr, "*** outBegin: \"%s\" is already opened\n",
00765                 outgrp);
00766         }
00767         erEnd("outBegin: file is already opened\n");
00768     }
00769
00770     if (!pr) {
00771         return True;
00772     }
00773
00774     if ((outgrpp = fopen(outgrp, "w")) == NULL) {

```



```

00775         if (prlevel > 0) {
00776             fprintf(GRCC_Stderr, "*** outBegin: cannot open %s\n", outgrp);
00777         }
00778         return False;
00779     }
00780     fprintf(outgrpp, "#####\n");
00781     fprintf(outgrpp, "# Generated by 'grcc'\n");
00782
00783     return True;
00784 }
00785
00786 //-----
00787 void Output::outEndF(void)
00788 {
00789     if (outgrfp == NULL) {
00790         return;
00791     }
00792     fprintf(outgrfp, "#####\n");
00793     fprintf(outgrfp, "End=1;\n");
00794     fprintf(outgrfp, "#####\n");
00795     fclose(outgrfp);
00796     outgrfp = NULL;
00797
00798     free(outgrfp);
00799     outgrfp = NULL;
00800 }
00801
00802 //-----
00803 void Output::outEndP(void)
00804 {
00805     if (outgrpp == NULL) {
00806         return;
00807     }
00808     fprintf(outgrpp, "#####\n");
00809     fprintf(outgrpp, "# End\n");
00810     fprintf(outgrpp, "#####\n");
00811     fclose(outgrpp);
00812     outgrpp = NULL;
00813
00814     free(outgrpp);
00815     outgrpp = NULL;
00816 }
00817
00818 //-----
00819 void Output::outProcBeginF(Process *prc)
00820 {
00821     int k, ex;
00822
00823     if (prc == NULL) {
00824         return;
00825     }
00826
00827     proc = prc;
00828     sproc = NULL;
00829     procId = proc->id;
00830
00831     if (outgrfp == NULL) {
00832         return;
00833     }
00834     if (outproc) {
00835         return;
00836     }
00837     outproc = True;
00838
00839     fprintf(outgrfp, "#####\n");
00840     fprintf(outgrfp, "Process=%d;\n", proc->id);
00841     fprintf(outgrfp, "External=%d;\n", proc->ninitl+proc->nfinal);
00842     for (k = 0, ex = 0; k < proc->ninitl; k++, ex++) {
00843         fprintf(outgrfp, "%4d= initial %s;\n", ex,
00844             proc->model->particleName(proc->initlPart[k]));
00845     }
00846     for (k = 0; k < proc->nfinal; k++, ex++) {
00847         fprintf(outgrfp, "%4d= final %s;\n", ex,
00848             proc->model->particleName(proc->finalPart[k]));
00849     }
00850     fprintf(outgrfp, "End;\n");
00851     for (k = 0; k < proc->model->ncouple; k++) {
00852         fprintf(outgrfp, "%s=%d; ", proc->model->cnlist[k], proc->clist[k]);
00853     }
00854     fprintf(outgrfp, "Loop=%d;\n", proc->loop);
00855     fprintf(outgrfp, "OPI=%s;\n", (opt->values[GRCC_OPT_1PI] > 0 ? "Yes" : "No"));
00856     fprintf(outgrfp, "Assign=%s;\n", (opt->values[GRCC_OPT_Step]==GRCC_AGraph ? "Yes" : "No"));
00857     fprintf(outgrfp, "%s Options : dummy values\n");
00858     fprintf(outgrfp, "Expand=Yes; ExpMin=0; ExpMax=-1;\n");
00859     fprintf(outgrfp, "Block=No; Tadpole=Yes; Extself=Yes;\n");
00860     fprintf(outgrfp, "Selfe=Yes; Countert=No; AnyCT=No; Undefp=No;\n");
00861 }

```

```

00862
00863 //-----
00864 void Output::outProcBeginP(Process *prc)
00865 {
00866     if (prc == NULL) {
00867         return;
00868     }
00869
00870     proc    = prc;
00871     sproc   = NULL;
00872     procId  = proc->id;
00873
00874     if (outgrpp == NULL) {
00875         return;
00876     }
00877     if (outproc) {
00878         return;
00879     }
00880     outproc = True;
00881
00882     fprintf(outgrfp, "# Process=%d;\n", proc->id);
00883 }
00884
00885 //-----
00886 void Output::outProcBegin0(int next, int couple, int loop)
00887 {
00888     int k, ex;
00889
00890     procId = 0;
00891
00892     if (outgrfp == NULL) {
00893         return;
00894     }
00895     if (outproc) {
00896         return;
00897     }
00898     outproc = True;
00899
00900     fprintf(outgrfp, "*****\n");
00901     fprintf(outgrfp, "Process=%d;\n", procId);
00902     fprintf(outgrfp, "External=%d;\n", next);
00903     for (k = 0, ex = 0; k < next; k++, ex++) {
00904         fprintf(outgrfp, "%4d= initial undef;\n", ex);
00905     }
00906     fprintf(outgrfp, "End;\n");
00907     fprintf(outgrfp, "GRCC_PHI=%d; ", couple);
00908     fprintf(outgrfp, "Loop=%d;\n", loop);
00909     fprintf(outgrfp, "OPI=%s;\n", (opt->values[GRCC_OPT_1PI] > 0? "Yes" : "No"));
00910     fprintf(outgrfp, "Assign=No;\n");
00911     fprintf(outgrfp, "% Options : dummy values\n");
00912     fprintf(outgrfp, "Expand=Yes; ExpMin=0; ExpMax=-1;\n");
00913     fprintf(outgrfp, "Block=No; Tadpole=Yes; Extself=Yes;\n");
00914     fprintf(outgrfp, "Self=Yes; Countert=No; AnyCT=No; Undefp=No;\n");
00915 }
00916
00917 //-----
00918 void Output::outSProcBeginF(SProcess *sprc)
00919 {
00920     int j, k, ex;
00921     PNodeClass *pnc;
00922
00923     if (sprc == NULL) {
00924         return;
00925     }
00926
00927     sprc    = sprc;
00928     procId  = sprc->id;
00929     pnc     = sprc->pnclass;
00930
00931     if (outgrfp == NULL) {
00932         return;
00933     }
00934     if (outproc) {
00935         return;
00936     }
00937     outproc = True;
00938     fprintf(outgrfp, "*****\n");
00939     fprintf(outgrfp, "Process=%d;\n", sprc->id);
00940     fprintf(outgrfp, "External=%d;\n", sprc->nExtern);
00941
00942     ex = 0;
00943     for (j = 0; j < pnc->nclass; j++) {
00944         if (pnc->type[j] == GRCC_AT_Initial) {
00945             for (k = 0; k < pnc->count[j]; k++) {
00946                 fprintf(outgrfp, "%4d= initial %s;\n", ex,
00947                     model->particleName(pnc->particle[j]));
00948                 ex++;

```

```

00949     }
00950     } else if (pnc->type[j] == GRCC_AT_Final) {
00951         for (k = 0; k < pnc->count[j]; k++) {
00952             fprintf(outgrfp, "%4d= final %s;\n", ex,
00953                 model->particleName(-pnc->particle[j]));
00954             ex++;
00955         }
00956     }
00957 }
00958
00959 fprintf(outgrfp, "End;\n");
00960 for (k = 0; k < sproc->model->ncouple; k++) {
00961     fprintf(outgrfp, "%s=%d; ", sproc->model->cnlist[k], sproc->clist[k]);
00962 }
00963 fprintf(outgrfp, "Loop=%d;\n", sproc->loop);
00964 fprintf(outgrfp, "OPI=%s;\n", (opt->values[GRCC_OPT_1PI] > 0 ? "Yes" : "No"));
00965 fprintf(outgrfp, "Assign=%s;\n", (opt->values[GRCC_OPT_Step]==GRCC_AGraph ? "Yes" : "No"));
00966 fprintf(outgrfp, "%% Options : dummy values\n");
00967 fprintf(outgrfp, "Expand=Yes; ExpMin=0; ExpMax=-1;\n");
00968 fprintf(outgrfp, "Block=No; Tadpole=Yes; Extself=Yes;\n");
00969 fprintf(outgrfp, "Selfe=Yes; Countert=No; AnyCT=No; Undefp=No;\n");
00970 }
00971
00972 //-----
00973 void Output::outSProcBeginP(SProcess *sproc)
00974 {
00975     if (sproc == NULL) {
00976         return;
00977     }
00978
00979     sproc = sproc;
00980     procId = sproc->id;
00981
00982     if (outgrpp == NULL) {
00983         return;
00984     }
00985     if (outproc) {
00986         return;
00987     }
00988     outproc = True;
00989     fprintf(outgrpp, "# SProcess=%d;\n", sproc->id);
00990 }
00991
00992 //-----
00993 void Output::outProcEndF(void)
00994 {
00995     if (outgrfp == NULL) {
00996         return;
00997     }
00998     if (proc != NULL) {
00999         fprintf(outgrfp, "%%-----\n");
01000         fprintf(outgrfp, "Pend=%d;\n", proc->id);
01001     } else if (sproc != NULL) {
01002         fprintf(outgrfp, "%%-----\n");
01003         fprintf(outgrfp, "Pend=%d;\n", sproc->id);
01004     } else {
01005         fprintf(outgrfp, "%%-----\n");
01006         fprintf(outgrfp, "Pend=1;\n");
01007     }
01008     fflush(outgrfp);
01009 }
01010
01011 //-----
01012 void Output::outProcEndP(void)
01013 {
01014     if (outgrpp == NULL) {
01015         return;
01016     }
01017     if (proc != NULL) {
01018         fprintf(outgrpp, "# Pend=%d;\n", proc->id);
01019     }
01020     fflush(outgrpp);
01021 }
01022
01023 //-----
01024 void Output::outEGraphF(EGraph *egraph)
01025 {
01026     int nd, lg, ptcl, ed, intr, j, k, loop;
01027     Model *mdl = egraph->model;
01028     Bool popt;
01029     EFLine *fl;
01030
01031 #ifdef DEBUG9
01032     if (egraph->mgraph != NULL) {
01033         printf("outEGraph: sId=%ld\n", egraph->mId);
01034         // egraph->mgraph->print();
01035         egraph->mgraph->mconn->print();

```

```

01036     }
01037 #endif
01038     if (outgrfp == NULL) {
01039         return;
01040     }
01041     fprintf(outgrfp, "%s-----\n");
01042     if (egraph->assigned) {
01043         fprintf(outgrfp, "Graph=%ld;\n", egraph->sId);
01044         fprintf(outgrfp, "%s AGraph=%ld;\n", egraph->aId);
01045     } else {
01046         fprintf(outgrfp, "Graph=%ld;\n", egraph->mId);
01047     }
01048     fprintf(outgrfp, "Gtype=%ld;\n", egraph->mId);
01049     if (egraph->assigned) {
01050         fprintf(outgrfp, "Sfactor=%ld;\n",
01051             egraph->nsym * egraph->esym * egraph->fsign);
01052     } else {
01053         fprintf(outgrfp, "Sfactor=%ld;\n", egraph->nsym * egraph->esym);
01054     }
01055     fprintf(outgrfp, "%s ExtPerm=%ld; NSym=%ld; ESym=%ld; NSym1=%ld; "
01056         " Multp=%ld;\n",
01057         egraph->extperm, egraph->nsym, egraph->esym, egraph->nsym1,
01058         egraph->multp);
01059
01060     fprintf(outgrfp, "Vertex=%d;\n", egraph->nNodes - egraph->nExtern);
01061
01062     for (nd = 0; nd < egraph->nNodes; nd++) {
01063         fprintf(outgrfp, "%4d", nd);
01064         if (mdl != NULL && egraph->assigned && !egraph->isExternal(nd)) {
01065             intr = egraph->nodes[nd]->intrct;
01066             loop = mdl->interacts[intr]->loop;
01067             popt = False;
01068
01069             // not tree vertex
01070             if (loop > 0) {
01071                 fprintf(outgrfp, "[loop=%d", loop);
01072                 popt = True;
01073             }
01074
01075             // multiple coupling constants
01076             if (mdl->ncouple > 1) {
01077                 if (popt) {
01078                     fprintf(outgrfp, ";order={");
01079                 } else {
01080                     fprintf(outgrfp, "[order={");
01081                     popt = True;
01082                 }
01083                 for (j = 0; j < mdl->interacts[intr]->nclist; j++) {
01084                     if (j != 0) {
01085                         fprintf(outgrfp, ",");
01086                     }
01087                     fprintf(outgrfp, "%d", mdl->interacts[intr]->clist[j]);
01088                 }
01089                 fprintf(outgrfp, "}");
01090             }
01091             if (popt) {
01092                 fprintf(outgrfp, "]");
01093             }
01094         } else if (!egraph->isExternal(nd)) {
01095             loop = egraph->nodes[nd]->extloop;
01096             // not tree vertex
01097             if (loop > 0) {
01098                 fprintf(outgrfp, "[loop=%d", loop);
01099             }
01100         }
01101         fprintf(outgrfp, "={");
01102
01103         // list of legs
01104         for (lg = 0; lg < egraph->nodes[nd]->deg; lg++) {
01105             if (lg != 0) {
01106                 fprintf(outgrfp, ", ");
01107             }
01108             ed = V2Iedge(egraph->nodes[nd]->edges[lg]);
01109             fprintf(outgrfp, "%4d", egraph->nodes[nd]->edges[lg]);
01110             ptcl = egraph->edges[ed]->ptcl;
01111             if (mdl != NULL && ptcl != 0 && egraph->assigned) {
01112                 if (egraph->nodes[nd]->edges[lg] < 0) {
01113                     ptcl = mdl->normalParticle(-ptcl);
01114                 } else {
01115                     ptcl = mdl->normalParticle(ptcl);
01116                 }
01117                 fprintf(outgrfp, "[%s]", mdl->particleName(ptcl));
01118             } else {
01119                 fprintf(outgrfp, "[undef]");
01120             }
01121         }
01122         fprintf(outgrfp, "};\n");

```

```

01123     }
01124     // print Fermion line as comment line
01125     if (egraph->nflines >= 0) {
01126         fprintf(outgrfp, "%s FLines=%d; FSign=%d; sId=%ld;\n",
01127             egraph->nflines, egraph->fsign, egraph->sId);
01128     }
01129     for (j = 0; j < egraph->nflines; j++) {
01130         fl = egraph->flines[j];
01131         fprintf(outgrfp, "%s %4d", j);
01132         if (fl->ftype == FL_Open) {
01133             fprintf(outgrfp, "[Open]=[");
01134         } else if (fl->ftype == FL_Closed) {
01135             fprintf(outgrfp, "[Loop]=[");
01136         } else {
01137             fprintf(outgrfp, " ?%d", fl->ftype);
01138         }
01139         for (k = 0; k < fl->nlist; k++) {
01140             if (k != 0) {
01141                 fprintf(outgrfp, ", ");
01142             }
01143             fprintf(outgrfp, "%d", Abs(fl->elist[k]));
01144         }
01145         fprintf(outgrfp, "];\n");
01146     }
01147     fprintf(outgrfp, "Vend;\n");
01148     fprintf(outgrfp, "Gend;\n");
01149 }
01150
01151 //-----
01152 void Output::outEGraphP(EGraph *eg)
01153 {
01154     int      j, nd, lg, ed, nin, nfi;
01155     char      nl = '\n';
01156     ENode     *node;
01157     static char undef[] = "Undef";
01158     char      *s;
01159
01160     // model
01161     fprintf(outgrpp, "egraph[%ld] = {%c", eg->sId-1, nl);
01162     if (model==NULL) {
01163         fprintf(outgrpp, "  \"Model\": [%\"Undef\", 1],%c", nl);
01164     } else {
01165         fprintf(outgrpp, "  \"Model\": [%\"%s\", %d],%c",
01166             model->name, model->ncouple, nl);
01167     }
01168
01169     // process
01170     nin = nfi = 0;
01171     for (j = 0; j < eg->nNodes; j++) {
01172         if (eg->isExternal(j)) {
01173             if (eg->nodes[j]->extloop == GRCC_AT_Final) {
01174                 nfi++;
01175             } else {
01176                 nin++;
01177             }
01178         }
01179     }
01180     fprintf(outgrpp, "  \"Process\": [[%d, %d], %d, [",
01181         nin, nfi, eg->nLoops);
01182     if (proc != NULL && model != NULL) {
01183         for (j = 0; j < model->ncouple; j++) {
01184             if (j != 0) {
01185                 fprintf(outgrpp, ", ");
01186             }
01187             fprintf(outgrpp, "%d", proc->clist[j]);
01188         }
01189     }
01190     fprintf(outgrpp, "]],%c", nl);
01191
01192     // option of the process
01193     fprintf(outgrpp, "  \"Opt\": [");
01194     if (eg->opt != NULL) {
01195         for (j = 0; j < GRCC_OPT_Size; j++) {
01196             if (j != 0) {
01197                 fprintf(outgrpp, ", ");
01198             }
01199             fprintf(outgrpp, "%d", eg->opt->values[j]);
01200         }
01201     }
01202     fprintf(outgrpp, "],%c", nl);
01203
01204     // graph
01205     fprintf(outgrpp, "  \"Id\": [%ld, %ld, %ld],%c",
01206         eg->mId, eg->aId, eg->sId, nl);
01207     fprintf(outgrpp, "  \"GStat\": [%d, %d, %d],%c",
01208         eg->assigned, eg->nNodes, eg->nEdges, nl);
01209     fprintf(outgrpp, "  \"Sym\": [%d, %ld, %ld, %ld, %ld],%c",

```

```

01210         eg->fsign, eg->nsym, eg->esym, eg->nsym1, eg->multp, nl);
01211
01212 // nodes
01213 fprintf(outgrpp, "  \Nodes\": [%c", nl);
01214 for (nd = 0; nd < eg->nNodes; nd++) {
01215     node = eg->nodes[nd];
01216     if (model==NULL || !eg->assigned) {
01217         s = undef;
01218     } else if (eg->isExternal(nd)) {
01219         s = model->particleName(node->intrct);
01220     } else {
01221         s = model->interacts[node->intrct]->name;
01222     }
01223     fprintf(outgrpp, "      [%d, \"%s\", %d, [",
01224         nd, s, node->extloop);
01225     for (lg = 0; lg < eg->nodes[nd]->deg; lg++) {
01226         if (lg != 0) {
01227             fprintf(outgrpp, ", ");
01228         }
01229         fprintf(outgrpp, "%d", eg->nodes[nd]->edges[lg]);
01230     }
01231     fprintf(outgrpp, "],%c", nl);
01232 }
01233 fprintf(outgrpp, " ],%c", nl);
01234
01235 // edges
01236 fprintf(outgrpp, "  \Edges\": [%c", nl);
01237 for (ed = 0; ed < eg->nEdges; ed++) {
01238     if (model != NULL) {
01239         s = model->particleName(eg->edges[ed]->ptcl);
01240         fprintf(outgrpp, "      [%d, \"%s\", [%d, %d]],%c",
01241             ed+1, s, eg->edges[ed]->nodes[0], eg->edges[ed]->nodes[1], nl);
01242     } else {
01243         fprintf(outgrpp, "      [%d, \"Undef\", [%d, %d]],%c",
01244             ed+1, eg->edges[ed]->nodes[0], eg->edges[ed]->nodes[1], nl);
01245     }
01246 }
01247
01248 fprintf(outgrpp, " ],%c", nl);
01249
01250 fprintf(outgrpp, "};\n");
01251 }
01252
01253 //-----
01254 void Output::outModelF(void)
01255 {
01256     char      *mdlfn;
01257     FILE      *mdlfp;
01258     Particle   *pt;
01259     Interaction *vt;
01260     int        j, k, l, ln;
01261     const char *ptn;
01262
01263     if (model == NULL) {
01264         if (prlevel > 0) {
01265             fprintf(GRCC_Stderr, "*** Output::outModel : model is not defined\n");
01266         }
01267         return;
01268     }
01269     ln = strlen(model->name)+5;
01270     mdlfn = new char[ln];
01271     snprintf(mdlfn, ln, "%s.mdl", model->name);
01272
01273     if ((mdlfp = fopen(mdlfn, "w")) == NULL) {
01274         if (prlevel > 0) {
01275             fprintf(GRCC_Stderr, "*** Output::outModel : cannot open \"%s\\n",
01276                 mdlfn);
01277         }
01278         return;
01279     }
01280
01281     for (l = 0; l < 60; l++) { putc(' ', mdlfp); } putc('\n', mdlfp);
01282     fprintf(mdlfp, "%s File \"%s\\n", mdlfn);
01283     fprintf(mdlfp, "%s Generated by graph generater\n");
01284     fprintf(mdlfp, " Version={2,2,0};\n");
01285
01286 // coupling constants
01287 fprintf(mdlfp, " Order={");
01288 for (j = 0; j < model->ncouple; j++) {
01289     if (j != 0) {
01290         fprintf(mdlfp, ", ");
01291     }
01292     fprintf(mdlfp, "%s", model->cnlist[j]);
01293 }
01294 fprintf(mdlfp, "};\n");
01295
01296 for (l = 0; l < 60; l++) { putc(' ', mdlfp); } putc('\n', mdlfp);

```

```
01297 fprintf(mdlfp, "% particles\n");
01298 for (l = 0; l < 60; l++) { putc('\n', mdlfp); } putc('\n', mdlfp);
01299 for (j = 1; j < model->nParticles; j++) {
01300     pt = model->particles[j];
01301     fprintf(mdlfp, " Particle=%s; Antiparticle=%s;\n",
01302             pt->name, pt->aaname);
01303     ptn = pt->typeGName();
01304     fprintf(mdlfp, " PType=%s; Charge=0; Color=1; Mass=0; Width=0;\n", ptn);
01305     fprintf(mdlfp, " PCODE=%d; Massless; \n", pt->pcode);
01306     fprintf(mdlfp, " Pend;\n");
01307     fprintf(mdlfp, "%\n");
01308 }
01309 for (l = 0; l < 60; l++) { putc('\n', mdlfp); } putc('\n', mdlfp);
01310 fprintf(mdlfp, "% interactions\n");
01311 for (l = 0; l < 60; l++) { putc('\n', mdlfp); } putc('\n', mdlfp);
01312 for (j = 0; j < model->nInteracts; j++) {
01313     vt = model->interacts[j];
01314     fprintf(mdlfp, " Vertex={");
01315     for (k = 0; k < vt->nplist; k++) {
01316         if (k != 0) {
01317             fprintf(mdlfp, ", ");
01318         }
01319         fprintf(mdlfp, "%s", model->particleName(vt->plist[k]));
01320     }
01321     fprintf(mdlfp, "}");
01322     for (k = 0; k < model->ncouple; k++) {
01323         fprintf(mdlfp, "%s=d;", model->cclist[k], vt->cllist[k]);
01324     }
01325     fprintf(mdlfp, "FNAME=%s; Vend;\n", vt->name);
01326 }
01327 for (l = 0; l < 60; l++) { putc('\n', mdlfp); } putc('\n', mdlfp);
01328 fprintf(mdlfp, " Mend;\n");
01329 for (l = 0; l < 60; l++) { putc('\n', mdlfp); } putc('\n', mdlfp);
01330 fclose(mdlfp);
01331 delete[] mdlnf;
01332 }
01333 //-----
01334 void Output::outModelP(void)
01335 {
01336     char *mdlfn;
01337     FILE *mdlfp;
01338     Particle *pt;
01339     Interaction *vt;
01340     int j, k, l, ln;
01341     const char *ptn;
01342 
01343     if (model == NULL) {
01344         if (prlevel > 0) {
01345             fprintf(GRCC_Stderr, "*** Output::outModel : ");
01346             fprintf(GRCC_Stderr, "model is not defined\n");
01347         }
01348         return;
01349     }
01350     ln = strlen(model->name)+5;
01351     mdlfn = new char[ln];
01352     snprintf(mdlfn, ln, "%s.mdp", model->name);
01353 
01354     if ((mdlfp = fopen(mdlfn, "w")) == NULL) {
01355         if (prlevel > 0) {
01356             fprintf(GRCC_Stderr, "*** Output::outModel : cannot open \"%s\"\n",
01357                     mdlfn);
01358         }
01359         return;
01360     }
01361 
01362     for (l = 0; l < 60; l++) { putc('#', mdlfp); } putc('\n', mdlfp);
01363     fprintf(mdlfp, "# File \"%s\"\n", mdlfn);
01364     fprintf(mdlfp, "# Generated by graph generater grcc\n");
01365 
01366     // coupling constants
01367     fprintf(mdlfp, "Model={\n");
01368     fprintf(mdlfp, " \"Model\": [");
01369     fprintf(mdlfp, "\"%s\", %d, [", model->name, model->ncouple);
01370     for (j = 0; j < model->ncouple; j++) {
01371         if (j != 0) {
01372             fprintf(mdlfp, ", ");
01373         }
01374         fprintf(mdlfp, "%s", model->cclist[j]);
01375     }
01376     fprintf(mdlfp, "]\n");
01377 
01378     fprintf(mdlfp, " \"Particles\": [\n");
01379     for (j = 1; j < model->nParticles; j++) {
01380         pt = model->particles[j];
01381         ptn = pt->typeGName();
01382         fprintf(mdlfp, " [%s\", \"%s\", \"%s\", \n",
```

```

01384         pt->name, pt->aname, ptn);
01385     }
01386     fprintf(mdlfp, "  \\"Interactions\\": {\n");
01387     for (j = 0; j < model->nInteracts; j++) {
01388         vt = model->interacts[j];
01389         fprintf(mdlfp, "    [\"%s\", %d, [", vt->name, vt->nplist);
01390         for (k = 0; k < vt->nplist; k++) {
01391             if (k != 0) {
01392                 fprintf(mdlfp, ", ");
01393             }
01394             fprintf(mdlfp, "%s", model->particleName(vt->plist[k]));
01395         }
01396         fprintf(mdlfp, "], ");
01397         for (k = 0; k < model->ncouple; k++) {
01398             if (k != 0) {
01399                 fprintf(mdlfp, ", ");
01400             }
01401             fprintf(mdlfp, "%d", vt->clist[k]);
01402         }
01403         fprintf(mdlfp, "],\n");
01404     }
01405     fprintf(mdlfp, "  ],\n");
01406     fprintf(mdlfp, "],\n");
01407     fclose(mdlfp);
01408     delete[] mdlfn;
01409 }
01410
01411 //*****
01412 // model.cc
01413
01414 //=====
01415
01416 static const char *ptypenames[] = GRCC_PT_Table;
01417 static const char *pgtynames[] = GRCC_PT_GTable;
01418
01419 //=====
01420 // class Particle
01421 //-----
01422 Particle::Particle(Model *mdl, int pid, PInput *pinp)
01423 {
01424     static char buff[100];
01425     static int nbuff=100;
01426     int j;
01427
01428     mdl = mdl; // the model
01429     id = pid; // id of this particle
01430     if (pinp->name == NULL || strlen(pinp->name) < 1) {
01431         if (pinp->pcode != 0) {
01432             snprintf(buff, nbuff, "pa%d", pinp->pcode);
01433             name = strdup(buff);
01434         } else {
01435             snprintf(buff, nbuff, "pi%d", pid);
01436             name = strdup(buff);
01437         }
01438     } else {
01439         name = strdup(pinp->name); // the name of the particle
01440     }
01441     if (pinp->aname == NULL) {
01442         if (pinp->acode != 0) {
01443             snprintf(buff, nbuff, "ap%d", Abs(pinp->acode));
01444             aname = strdup(buff);
01445         } else {
01446             snprintf(buff, nbuff, "ai%d", pid);
01447             aname = strdup(buff);
01448         }
01449     } else {
01450         aname = strdup(pinp->aname); // the name of the anti-particle
01451     }
01452     pcode = pinp->pcode;
01453     acode = pinp->acode;
01454     if (Abs(mdl->defpart) == GRCC_DEFBYCODE) {
01455         neutral = (pcode == acode);
01456     } else if (Abs(mdl->defpart) == GRCC_DEFBYNAME) {
01457         neutral = ((strcmp(name, aname)==0) ? True : False);
01458     }
01459
01460     // convert ptype string to its code.
01461     if (Abs(mdl->defpart) == GRCC_DEFBYCODE) {
01462         if (pinp->ptypc < GRCC_PT_Undef || pinp->ptypc > GRCC_PT_Size) {
01463             if (prlevel > 0) {
01464                 fprintf(GRCC_Stderr, "*** particle type is not defined: %d\n",
01465                     pinp->ptypc);
01466             }
01467             erEnd("particle type is not defined (illegal code)");
01468         }
01469         ptype = pinp->ptypc;
01470     } else if (Abs(mdl->defpart) == GRCC_DEFBYNAME) {

```



```

01471         if (pinp->ptypen == NULL) {
01472             erEnd("particle type is not defined (NULL)");
01473         }
01474         ptype = -1;
01475         for (j = 0; j < GRCC_PT_Size; j++) {
01476             if (strcmp(pinp->ptypen, ptypenames[j]) == 0) {
01477                 ptype = j;
01478                 break;
01479             }
01480         }
01481         if (ptype < 0) {
01482             if (prlevel > 0) {
01483                 fprintf(GRCC_Stderr, "*** particle type \"%s\" is not defined\n",
01484                     pinp->ptypen);
01485             }
01486             erEnd("particle type is not defined (name)");
01487         }
01488     }
01489     if (ptype == GRCC_PT_Undef && ptype != 0) {
01490         erEnd("illegal ptype");
01491     }
01492
01493     extonly = pinp->extonly;
01494     cmindeg = 0;
01495     cmaxdeg = 0;
01496 }
01497
01498 //-----
01499 Particle::~Particle(void)
01500 {
01501     free(aname);
01502     free(name);
01503 }
01504
01505 //-----
01506 char *Particle::particleName(int p)
01507 {
01508     if (p < 0) {
01509         return aname;
01510     } else {
01511         return name;
01512     }
01513 }
01514
01515 //-----
01516 int Particle::particleCode(int p)
01517 {
01518     if (p < 0) {
01519         return acode;
01520     } else {
01521         return pcode;
01522     }
01523 }
01524
01525 //-----
01526 int Particle::isNeutral(void)
01527 {
01528     return neutral;
01529 }
01530
01531 //-----
01532 const char *Particle::typeName(void)
01533 {
01534     return ptypenames[ptype];
01535 }
01536
01537 //-----
01538 const char *Particle::typeGName(void)
01539 {
01540     return pgtypenames[ptype];
01541 }
01542
01543 //-----
01544 char *Particle::aparticle(void)
01545 {
01546     static char prtcl[] = "Particle";
01547
01548     if (isNeutral()) {
01549         return prtcl;
01550     } else {
01551         return aname;
01552     }
01553 }
01554
01555 //-----
01556 void Particle::prParticle(void)
01557 {

```

```

01558     if (isNeutral()) {
01559         printf("pid=%d, name=\"%s\", real_field, ", id, name);
01560     } else {
01561         printf("pid=%d, name=\"%s\", aname=\"%s\", ", id, name, aname);
01562     }
01563     printf("ptype=%s, pcode=%d, acode=%d, extonly=%d, cdeg=(%d,%d)\n",
01564           ptypenames[ptype], pcode, acode, extonly, cmindeg, cmaxdeg);
01565 }
01566
01567 //=====
01568 // class Interaction
01569 //-----
01570 Interaction::Interaction(Model *mdl, int iid, const char *nam, int icd, int *cpl, int nlgs, int
    *plst, int csm, int lp)
01571 {
01572     static char buff[100];
01573     static int  nbuff=100;
01574     Particle    *p;
01575     int         j, ndir, sdir, jdir, nmaj, jmaj, ngho, sgho, jgho, ptcl;
01576     Bool        ok;
01577
01578     mdl    = mdl;           // the model
01579     id     = iid;           // id of this interaction
01580     icode  = icd;           // id of this interaction
01581     csum   = csm;           // the total orders of coupling constants
01582     nlegs  = nlgs;         // the size of plist[]
01583     loop   = lp;           // the number of loops
01584
01585     if (nam == NULL || strlen(nam) < 1) {
01586         snprintf(buff, nbuff, "vt%d", icd);
01587         name = strdup(buff);
01588     } else {
01589         name = strdup(nam); // the name of the interaction
01590     }
01591     nclist = mdl->ncouple;
01592     clist  = new int[nclist];
01593     icode  = icd;
01594     for (j = 0; j < mdl->ncouple; j++) {
01595         clist[j] = cpl[j];
01596     }
01597     plist  = new int[nlegs];
01598     nplist = nlegs;
01599     for (j = 0; j < nlegs; j++) {
01600         plist[j] = plst[j];
01601     }
01602     nslist = 0;
01603     slist  = NULL;
01604
01605     // check fermion number conservation
01606     ndir = 0;
01607     sdir = 0;
01608     jdir = 0;
01609     nmaj = 0;
01610     jmaj = 0;
01611     ngho = 0;
01612     sgho = 0;
01613     jgho = 0;
01614     ok = True;
01615     for (j = 0; j < nplist; j++) {
01616         if (plist[j] > 0) {
01617             p = mdl->particles[plist[j]];
01618             ptcl = 1;
01619         } else {
01620             p = mdl->particles[-plist[j]];
01621             ptcl = -1;
01622         }
01623
01624         if (p->ptype == GRCC_PT_Dirac) {
01625             ndir++;
01626             if (ptcl > 0) {
01627                 sdir++;
01628             } else {
01629                 sdir--;
01630             }
01631             if (Abs(sdir) == 1) {
01632                 jdir = j;
01633             } else if ((Abs(sdir) == 0 && j != jdir + 1) || Abs(sdir) > 1) {
01634                 fprintf(GRCC_Stderr, "*** Interaction: Dirac and anti-Dirac should "
01635                       "be arranged in pairs in an interaction.\n");
01636                 ok = False;
01637             }
01638         } else if (p->ptype == GRCC_PT_Ghost) {
01639             ngho++;
01640             if (ptcl > 0) {
01641                 sgho++;
01642             } else {
01643                 sgho--;
01644             }
01645         }
01646     }

```

```

01644         }
01645         if (Abs(sgho) == 1) {
01646             jgho = j;
01647         } else if ((Abs(sgho) == 0 && j != jgho + 1) || Abs(sgho) > 1) {
01648             fprintf(GRCC_Stderr, "*** Interaction: Ghost and anti-Ghost should "
01649                 "be arranged in pairs in an interaction.\n");
01650             ok = False;
01651         }
01652     } else if (p->ptype == GRCC_PT_Majorana) {
01653         nmaj++;
01654         if (nmaj % 2 == 1) {
01655             jmaj = j;
01656         } else if (j != jmaj + 1) {
01657             fprintf(GRCC_Stderr, "*** Interaction: two Majoranas should "
01658                 "be arranged in pairs in an interaction.\n");
01659             ok = False;
01660         }
01661     }
01662 }
01663 if (sdir != 0) {
01664     fprintf(GRCC_Stderr, "*** Interaction: Dirac number is not conserved\n");
01665     ok = False;
01666 }
01667 if (sgho != 0) {
01668     fprintf(GRCC_Stderr, "*** Interaction: Ghost number is not conserved\n");
01669     ok = False;
01670 }
01671 if (nmaj % 2 != 0) {
01672     fprintf(GRCC_Stderr, "*** Interaction: odd number of Majorana particles\n");
01673     ok = False;
01674 }
01675 if (ndir + nmaj + ngho > 2) {
01676     fprintf(GRCC_Stderr, "+++ Interaction: more than 2 Dirac/Majorana/Ghost "
01677         "particles are interacting.\n");
01678     fprintf(GRCC_Stderr, "    Interaction has %d Dirac, %d Ghost and %d Majorana "
01679         "particles.\n", ndir, nmaj, ngho);
01680     fprintf(GRCC_Stderr, "    Sign factors related to fermions will not "
01681         "be calculated correctly.\n");
01682     mdl->skipFLine = True;
01683 }
01684 if (!ok) {
01685     prInteraction();
01686     erEnd("illegal Dirac/Majorana/Ghost interaction");
01687 }
01688 }
01689
01690 //-----
01691 Interaction::~Interaction(void)
01692 {
01693     if (slist != NULL) {
01694         delete[] slist;
01695     }
01696     if (plist != NULL) {
01697         delete[] plist;
01698     }
01699     if (clist != NULL) {
01700         delete[] clist;
01701     }
01702     free(name);
01703 }
01704
01705 //-----
01706 void Interaction::prInteraction(void)
01707 {
01708     int j;
01709
01710     printf("vid=%d, icode=%d, name=\"%s\", loop=%d, csum=%d, cpl=",
01711         id, icode, name, loop, csum);
01712     prIntArray(mdl->ncouple, clist, "", legs=0);
01713     for (j = 0; j < nplist; j++) {
01714         if (j != 0) {
01715             printf(", ");
01716         }
01717         printf("%s", mdl->particleName(plist[j]));
01718     }
01719     printf("];\n");
01720 }
01721
01722 //=====
01723 // class Model
01724 //-----
01725 Model::Model(MInput *minp)
01726 {
01727     static PInput pundef = {"undef", "undef", 0, 0, "undef", 0, 0};
01728     int j;
01729 }
01730

```

```

01731     if (minp->name == NULL || strlen(minp->name) < 1) {
01732         erEnd("model should have a name");
01733     }
01734     name      = strdup(minp->name);
01735     ncouple   = minp->ncouple;
01736     if (ncouple >= GRCC_MAXNCPLG) {
01737         erEnd("too many coupling constants in the model (GRCC_MAXNCPLG)");
01738     }
01739     // cnlist = minp->cnamlist;
01740     cnlist = new char*[ncouple];
01741     for (j = 0; j < ncouple; j++) {
01742         cnlist[j] = strdup(minp->cnamlist[j]);
01743     }
01744
01745     nParticles = 0;
01746     particles  = new Particle*[GRCC_MAXMPARTICLES];
01747     for (j = 0; j < GRCC_MAXMPARTICLES; j++) {
01748         particles[j] = NULL;
01749     }
01750     pdef = False;
01751
01752     nInteracts = 0;
01753     interacts  = new Interaction*[GRCC_MAXMINTERACT];
01754     for (j = 0; j < GRCC_MAXMINTERACT; j++) {
01755         interacts[j] = NULL;
01756     }
01757     vdef      = False;
01758
01759     if (particles == NULL || interacts == NULL) {
01760         erEnd("memory allocation failed");
01761     }
01762
01763 #ifdef UNIQINTR
01764     if (minp->defpart != GRCC_DEFBYNAME
01765         && minp->defpart != GRCC_DEFBYCODE) {
01766         erEnd("illegal value of defpart");
01767     }
01768 #else
01769     if (Abs(minp->defpart) != GRCC_DEFBYNAME
01770         && Abs(minp->defpart) != GRCC_DEFBYCODE) {
01771         erEnd("illegal value of defpart");
01772     }
01773 #endif
01774     defpart = minp->defpart;
01775
01776     maxnlegs = 0;
01777     maxcpl   = 0;
01778     maxloop  = 0;
01779     ncplgcp  = 0;
01780     skipFLine = False;
01781
01782     addParticle(&pundef);
01783 }
01784
01785 //-----
01786 Model::~Model(void)
01787 {
01788     int j;
01789
01790     for (j = ncplgcp-1; j >= 0; j--) {
01791         cplgvl[j] = delintdup(cplgvl[j]);
01792     }
01793     delete[] cplgvl;
01794     cplgnvl = delintdup(cplgnvl);
01795     cplglg  = delintdup(cplglg);
01796     cplgcp  = delintdup(cplgcp);
01797
01798     for (j = GRCC_MAXMINTERACT-1; j >= 0; j--) {
01799         if (interacts[j] != NULL) {
01800             interacts[j]->slist = delintdup(interacts[j]->slist);
01801             delete interacts[j];
01802             interacts[j] = NULL;
01803         }
01804     }
01805     delete[] interacts;
01806     interacts = NULL;
01807
01808     for (j = GRCC_MAXMPARTICLES-1; j >= 0; j--) {
01809         if (particles[j] != NULL) {
01810             delete particles[j];
01811             particles[j] = NULL;
01812         }
01813     }
01814     delete[] particles;
01815     particles = NULL;
01816 }
01817

```

```

01818     for (j=ncouple-1; j>=0; j--) {
01819         free(cnlist[j]);
01820     }
01821     delete[] cnlist;
01822     free(name);
01823 }
01824 }
01825
01826 //-----
01827 void Model::prModel(void)
01828 {
01829     static char hdr[] = "#####\n";
01830     static char hdl[] = "#-----\n";
01831     int j;
01832
01833     printf("%s", hdr);
01834     printf("Model=\"%s\\", " ", name);
01835     printf("coupling=[");
01836     for (j = 0; j < ncouple; j++) {
01837         if (j != 0) {
01838             printf(", ");
01839         }
01840         printf("%s\\", cnlist[j]);
01841     }
01842     printf("];\n");
01843     printf("%s", hdl);
01844     printf("Particles=%d;\n", nParticles);
01845     for (j = 1; j < nParticles; j++) {
01846         particles[j]->prParticle();
01847     }
01848     printf("EndParticle;\n");
01849     printf("allPart (%d) = [", nallPart);
01850     for (j = 0; j < nallPart; j++) {
01851         if (j != 0) {
01852             printf(", ");
01853         }
01854         printf("%d", allPart[j]);
01855     }
01856     printf("]\n");
01857
01858     printf("%s", hdl);
01859     printf("Interactions=%d;\n", nInteracts);
01860     for (j = 0; j < nInteracts; j++) {
01861         interacts[j]->prInteraction();
01862     }
01863     printf("EndInteraction;\n");
01864     printf("%s", hdl);
01865     printf("InteractionTable;\n");
01866     printf("# %d class in (total coupling, degree)\n", ncplgcp);
01867     for (j = 0; j < ncplgcp; j++) {
01868         printf(" class=%d: cp=%d, lg=%d, vl=", j, cplgcp[j], cplglg[j]);
01869         prIntArray(cplgnvl[j], cplgvl[j], "\n");
01870     }
01871     printf("EndInteractionTable;\n");
01872     printf("%s", hdr);
01873 }
01874
01875 //-----
01876 void Model::addParticle(PInput *pinp)
01877 {
01878     int nid, aid, pid, pcd, acd;
01879
01880     if (nParticles >= GRCC_MAXMPARTICLES) {
01881         erEnd("particle table overflow (GRCC_MAXMPARTICLES)");
01882     }
01883
01884     // uniqueness check
01885     if (Abs(defpart) == GRCC_DEFBYCODE) {
01886         pcd = findParticleCode(pinp->pcode);
01887         acd = findParticleCode(pinp->acode);
01888         if (pcd > 0 || acd > 0) {
01889             if (prlevel > 0) {
01890                 fprintf(GRCC_Stderr, "*** particle code [%d, %d] ",
01891                     pinp->pcode, pinp->acode);
01892                 fprintf(GRCC_Stderr, "is already used.\n");
01893             }
01894             erEnd("particle code is already defined");
01895         }
01896     } else if (Abs(defpart) == GRCC_DEFBYNAME) {
01897         if (pinp->name == NULL || pinp->aname == NULL ||
01898             strlen(pinp->name) < 1 || strlen(pinp->aname) < 1) {
01899             erEnd("no name of particle or anti-particle.");
01900         }
01901         nid = findParticleName(pinp->name);
01902         aid = findParticleName(pinp->aname);
01903         if (nid > 0 || aid > 0) {
01904             if (prlevel > 0) {

```

```

01905             fprintf(GRCC_Stderr, "*** particle name [%s, %s] ",
01906                     pinp->name, pinp->aname);
01907             fprintf(GRCC_Stderr, "is already used.\n");
01908         }
01909         erEnd("particle name is already defined.");
01910     }
01911 }
01912
01913 pid = nParticles++;
01914 particles[pid] = new Particle(this, pid, pinp);
01915 }
01916
01917 //-----
01918 void Model::addParticleEnd(void)
01919 {
01920     int p, ap;
01921
01922     pdef = True;
01923
01924     #ifdef OPTEXTONLY
01925         nallPart = 0;
01926         for (p = nParticles-1; p >= 1; p--) {
01927             if (!particles[p]->extonly) {
01928                 ap = antiParticle(p);
01929                 if (ap != p) {
01930                     allPart[nallPart++] = ap;
01931                 }
01932             }
01933         }
01934         for (p = 1; p < nParticles; p++) {
01935             if (!particles[p]->extonly) {
01936                 allPart[nallPart++] = p;
01937             }
01938         }
01939         sorti(nallPart, allPart);
01940     #else
01941         nallPart = 0;
01942         for (p = nParticles-1; p >= 1; p--) {
01943             ap = antiParticle(p);
01944             if (ap != p) {
01945                 allPart[nallPart++] = ap;
01946             }
01947         }
01948         for (p = 1; p < nParticles; p++) {
01949             allPart[nallPart++] = p;
01950         }
01951         sorti(nallPart, allPart);
01952     #endif
01953 }
01954
01955 //-----
01956 void Model::addInteraction(IInput *iinp)
01957 {
01958     int vid, nlegs, j, k, c;
01959     int csum, lp2, lp;
01960     int plist[GRCC_MAXLEGS];
01961
01962     if (!pdef) {
01963         erEnd("call 'addParticleEnd' before 'addInteraction'");
01964     }
01965     if (nInteracts >= GRCC_MAXMINTERACT) {
01966         erEnd("too many interactions in the model (GRCC_MAXMINTERACT)");
01967     }
01968     nlegs = iinp->nplistn;
01969     if (nlegs >= GRCC_MAXLEGS) {
01970         erEnd("too many legs in an interaction (GRCC_MAXLEGS)");
01971     }
01972
01973     // check identifier
01974     if (defpart == GRCC_DEFBYCODE) {
01975         if (iinp->icode < 0) {
01976             if (prlevel > 0) {
01977                 fprintf(GRCC_Stderr, "*** vertex code %d should be positive\n",
01978                         iinp->icode);
01979             }
01980             erEnd("vertex code should be positive");
01981         } else {
01982             vid = findInteractionCode(iinp->icode);
01983             if (vid >= 0) {
01984                 if (prlevel > 0) {
01985                     fprintf(GRCC_Stderr, "*** vertex code %d is already used\n",
01986                             iinp->icode);
01987                 }
01988                 erEnd("vertex code is already used");
01989             }
01990         }
01991     } else if (defpart == GRCC_DEFBYNAME) {

```

```

01992         if (iinp->name == NULL || strlen(iinp->name) < 1) {
01993             erEnd("no name of vertex\n");
01994         }
01995         vid = findInteractionName(iinp->name);
01996         if (vid >= 0) {
01997             if (prlevel > 0) {
01998                 fprintf(GRCC_Stderr, "*** vertex name %s is already used\n",
01999                     iinp->name);
02000                 erEnd("vertex name is already used");
02001             }
02002         }
02003     } else {
02004 #ifdef UNIQINTR
02005         // defpart ==< 0
02006         erEnd("illegal value of defpart");
02007 #else
02008         // don't care about duplicated name/code of interaction
02009         ;
02010 #endif
02011     }
02012     vid = nInteracts++;
02013
02014     for (j = 0; j < nlegs; j++) {
02015         if (Abs(defpart) == GRCC_DEFBYCODE) {
02016             plist[j] = findParticleCode(iinp->plistc[j]);
02017             if (plist[j] == 0) {
02018                 if (prlevel > 0) {
02019                     fprintf(GRCC_Stderr, "*** particle code %d ",
02020                         iinp->plistc[j]);
02021                     fprintf(GRCC_Stderr, "is not defined\n");
02022                 }
02023                 erEnd("particle is not defined (code)");
02024             }
02025         } else if (Abs(defpart) == GRCC_DEFBYNAME) {
02026             plist[j] = findParticleName(iinp->plistn[j]);
02027             if (plist[j] == 0) {
02028                 if (prlevel > 0) {
02029                     fprintf(GRCC_Stderr, "*** particle name %s ",
02030                         iinp->plistn[j]);
02031                     fprintf(GRCC_Stderr, "is not defined\n");
02032                 }
02033                 erEnd("particle is not defined (name)");
02034             }
02035         }
02036     }
02037
02038     csum = 0;
02039     for (j = 0; j < ncouple; j++) {
02040         c = iinp->cvallist[j];
02041         if (c < 0) {
02042             if (prlevel > 0) {
02043                 fprintf(GRCC_Stderr, "*** illegal value of c-constants \n");
02044                 for (k = 0; k < ncouple; k++) {
02045                     fprintf(GRCC_Stderr, "      %s = %d\n",
02046                         cnlist[k], iinp->cvallist[k]);
02047                 }
02048             }
02049             erEnd("illegal value of c-constants");
02050         }
02051         csum += c;
02052     }
02053     if (csum < 0) {
02054         if (prlevel > 0) {
02055             fprintf(GRCC_Stderr, "*** illegal total coupling-constants %d\n",
02056                 csum);
02057             for (k = 0; k < ncouple; k++) {
02058                 fprintf(GRCC_Stderr, "      %s = %d\n",
02059                     cnlist[k], iinp->cvallist[k]);
02060             }
02061         }
02062         erEnd("illegal value of c-constants");
02063     }
02064
02065     // assume : csum = nlegs - 2 + 2*loop
02066     lp2 = csum - nlegs + 2;
02067     if (lp2 % 2 != 0 || lp2 < 0) {
02068         if (prlevel > 0) {
02069             fprintf(GRCC_Stderr, "*** illegal coupling const : ");
02070             fprintf(GRCC_Stderr, "nlegs - 2 + 2*loop: 2*loop = %d\n", lp2);
02071         }
02072     }
02073     lp = lp2/2;
02074
02075     maxnlegs = Max(maxnlegs, nlegs);
02076     maxloop = Max(maxloop, lp);
02077     maxcpl = Max(maxcpl, csum);
02078

```

```

02079     interacts[vid] = new Interaction(this, vid, iinp->name, iinp->icode,
02080                                     iinp->cvallist, nlegs, plist, csum, lp);
02081 }
02082
02083 //-----
02084 void Model::addInteractionEnd(void)
02085 {
02086     int lst[GRCC_MAXMINTERACT];
02087     int tcp[GRCC_MAXMINTERACT], tlg[GRCC_MAXMINTERACT];
02088     int tnvl[GRCC_MAXMINTERACT], *tv1[GRCC_MAXMINTERACT];
02089     Interaction *vt;
02090     int cp, lg, j, k, p;
02091
02092     vdef = True;
02093     ncplgcp = 0;
02094     for (cp = 0; cp <= maxcpl; cp++) {
02095
02096         for (lg = 0; lg <= maxnlegs; lg++) {
02097             k = 0;
02098             for (j = 0; j < nInteracts; j++) {
02099                 vt = interacts[j];
02100                 if (vt->csum == cp && vt->nlegs == lg) {
02101                     lst[k++] = j;
02102                 }
02103             }
02104             if (k > 0) {
02105                 tcp[ncplgcp] = cp;
02106                 tlg[ncplgcp] = lg;
02107                 tnvl[ncplgcp] = k;
02108                 tv1[ncplgcp] = intdup(k, lst);
02109                 ncplgcp++;
02110             }
02111         }
02112     }
02113     cplgcp = intdup(ncplgcp, tcp);
02114     cplglg = intdup(ncplgcp, tlg);
02115     cplgnvl = intdup(ncplgcp, tnvl);
02116     cplgvl = new int*[ncplgcp];
02117     for (j = 0; j < ncplgcp; j++) {
02118         if (cplgnvl[j] > 0) {
02119             cplgvl[j] = tv1[j];
02120         } else {
02121             cplgvl[j] = NULL;
02122         }
02123     }
02124
02125     for (j = 0; j < nInteracts; j++) {
02126         vt = interacts[j];
02127         if (vt->slist != NULL) {
02128             erEnd("addInteractionEnd: vt->nslist != NULL");
02129         }
02130         vt->slist = intdup(vt->nplist, vt->plist);
02131         vt->nslist = toSList(vt->nplist, vt->slist);
02132     }
02133
02134     for (p = 0; p < nParticles; p++) {
02135         particles[p]->cmindeg = 0;
02136         particles[p]->cmaxdeg = 0;
02137     }
02138     for (j = 0; j < nInteracts; j++) {
02139         vt = interacts[j];
02140         for (k = 0; k < vt->nplist; k++) {
02141             p = abs(vt->plist[k]);
02142             if (particles[p]->cmindeg == 0) {
02143                 particles[p]->cmindeg = vt->nlegs;
02144                 particles[p]->cmaxdeg = vt->nlegs;
02145             } else {
02146                 particles[p]->cmindeg = Min(particles[p]->cmindeg, vt->nlegs);
02147                 particles[p]->cmaxdeg = Max(particles[p]->cmaxdeg, vt->nlegs);
02148             }
02149         }
02150     }
02151 #ifdef DEBUG
02152     prModel();
02153 #endif
02154 }
02155
02156 //-----
02157 int Model::findParticleName(const char *name)
02158 {
02159     int j;
02160
02161     for (j = 0; j < nParticles; j++) {
02162         if (strcmp(name, particles[j]->name) == 0) {
02163             return j;
02164         }
02165     }

```



```

02166     for (j = 0; j < nParticles; j++) {
02167         if (strcmp(name, particles[j]->aname) == 0) {
02168             return -j;
02169         }
02170     }
02171     return 0;
02172 }
02173
02174 //-----
02175 int Model::findParticleCode(int pcd)
02176 {
02177     int j;
02178     for (j = 0; j < nParticles; j++) {
02179         if (pcd == particles[j]->pcode) {
02180             return j;
02181         } else if (pcd == particles[j]->acode) {
02182             return -j;
02183         }
02184     }
02185     return 0;
02186 }
02187
02188 //-----
02189 char *Model::particleName(int p)
02190 {
02191     int q;
02192     if (Abs(p) >= nParticles) {
02193         if (prlevel > 0) {
02194             fprintf(GRCC_Stderr, "\n*** Model::particleName: ");
02195             fprintf(GRCC_Stderr, "illegal particle id=%d\n", p);
02196         }
02197         erEnd("Model::particleName: illegal particle");
02198     }
02199     if (p < 0) {
02200         q = -p;
02201     } else {
02202         q = p;
02203     }
02204     return particles[q]->particleName(p);
02205 }
02206
02207 //-----
02208 int Model::particleCode(int p)
02209 {
02210     int q;
02211     if (Abs(p) >= nParticles) {
02212         fprintf(GRCC_Stderr, "\n*** Model::particleCode: illegal particle id=%d\n",
02213             p);
02214         erEnd("Model::particleCode: illegal particle");
02215     }
02216     if (p < 0) {
02217         q = -p;
02218     } else {
02219         q = p;
02220     }
02221     return particles[q]->particleCode(p);
02222 }
02223
02224 //-----
02225 void Model::prParticleArray(int n, int *a, const char *msg)
02226 {
02227     int j;
02228     printf("[");
02229     for (j = 0; j < n; j++) {
02230         if (j != 0) {
02231             printf(", ");
02232         }
02233         printf("%s", particleName(a[j]));
02234     }
02235     printf("]%s", msg);
02236 }
02237
02238 //-----
02239 int Model::findMClass(const int cpl, const int dgr)
02240 {
02241     int j;
02242     for (j = 0; j < ncplgcp; j++) {
02243         if (cplgcp[j] == cpl && cplglg[j] == dgr) {
02244             return j;
02245         }
02246     }
02247     return -1;
02248 }

```

```

02253 }
02254
02255 //-----
02256 int Model::normalParticle(int pt)
02257 {
02258     int ptc = Abs(pt);
02259     Particle *p;
02260
02261     if (ptc >= nParticles) {
02262         return 0;
02263     }
02264     p = particles[ptc];
02265     if (p->isNeutral()) {
02266         return ptc;
02267     } else {
02268         return pt;
02269     }
02270 }
02271
02272 //-----
02273 int Model::antiParticle(int pt)
02274 {
02275     int ptc = Abs(pt);
02276     Particle *p;
02277
02278     p = particles[ptc];
02279     if (p->isNeutral()) {
02280         return ptc;
02281     } else {
02282         return -pt;
02283     }
02284 }
02285
02286 //-----
02287 int *Model::allParticles(int *len)
02288 {
02289     *len = nallPart;
02290     return allPart;
02291 }
02292
02293 //-----
02294 int Model::findInteractionName(const char *name)
02295 {
02296     int j;
02297
02298     for (j = 0; j < nInteracts; j++) {
02299         if (strcmp(name, interacts[j]->name) == 0) {
02300             return j;
02301         }
02302     }
02303     return -1;
02304 }
02305
02306 //-----
02307 int Model::findInteractionCode(int icd)
02308 {
02309     int j;
02310
02311     for (j = 0; j < nInteracts; j++) {
02312         if (icd == interacts[j]->icode) {
02313             return j;
02314         }
02315     }
02316     return -1;
02317 }
02318
02319 //*****
02320 // proc.cc
02321
02322 //=====
02323 // class PNodeClass
02324 //-----
02325 PNodeClass::PNodeClass(SProcess *spc, int nnods, int nclss, NCInput *cls)
02326 {
02327     // Create a class of nodes
02328     // nnods : # nodes
02329     // nclss : # classes
02330     // cls   : table of class inputs
02331     //
02332     int j, k, nn, lp2;
02333     Bool ok = True;
02334
02335     sproc = spc;
02336     nnods = nnods;
02337     nclss = nclss;
02338     deg = new int[nclss];
02339     type = new int[nclss];

```

```

02340     particle = new int[nclass];
02341     count    = new int[nclass];
02342     couple   = new int[nclass];
02343     cmindeg  = new int[nclass];
02344     cmaxdeg  = new int[nclass];
02345     for (j = 0; j < nclass; j++) {
02346         deg[j]      = cls[j].cldeg;
02347         count[j]    = cls[j].clnum;
02348         type[j]     = cls[j].cltyp;
02349         cmindeg[j]  = cls[j].cmind;
02350         cmaxdeg[j]  = cls[j].cmaxd;
02351         particle[j] = cls[j].ptcl;
02352         couple[j]   = cls[j].cple;
02353         lp2 = (couple[j]-deg[j]+2);
02354         if (!isATEXternal(type[j]) && (lp2 % 2 != 0 || lp2 < 0)) {
02355             if (prlevel > 0) {
02356                 fprintf(GRCC_Stderr, "*** PNodeClass: illegal loop: "
02357                     ": 2*loop=cpl[%d](%d)-deg[%d](%d)+2 =2*loop = %d\n",
02358                     j, couple[j], j, deg[j], lp2);
02359                 for (k = 0; k < nclss; k++) {
02360                     fprintf(GRCC_Stderr, "k=%d, deg=%d, typ=%d, ptcl=%d, ",
02361                         k, deg[k], type[k], particle[k]);
02362                     fprintf(GRCC_Stderr, "cpl=%d, cnt=%d\n",
02363                         couple[k], count[k]);
02364                 }
02365             }
02366             ok = False;
02367         }
02368     }
02369     if (!ok) {
02370         erEnd("PNodeClass: illegal loop");
02371     }
02372     cl2nd = new int[nclass+1];
02373     nd2cl = new int[nnodes];
02374     cl2mcl = new int[nnodes];
02375
02376     nn = 0;
02377     for (j = 0; j < nclass; j++) {
02378         cl2nd[j] = nn;
02379         for (k = 0; k < count[j]; k++, nn++) {
02380             nd2cl[nn] = j;
02381         }
02382         if (isATEXternal(type[j])) {
02383             cl2mcl[j] = -1;
02384         } else {
02385             cl2mcl[j] = spc->model->findMClass(couple[j], deg[j]);
02386             if (cl2mcl[j] < 0) {
02387                 if (prlevel > 0) {
02388                     fprintf(GRCC_Stderr, "*** PNodeClass : no vertex : ");
02389                     fprintf(GRCC_Stderr, "coupling=%d, degree=%d\n",
02390                         couple[j], deg[j]);
02391                 }
02392                 erEnd("PNodeClass : no vertex");
02393             }
02394         }
02395     }
02396     cl2nd[nclass] = nnodes;
02397 }
02398 //-----
02399 PNodeClass::PNodeClass(SProcess *spc, int nnods, int nclss, int *dgs, int *typ, int *ptcl, int *cpl,
02400     int *cnt, int *cmind, int *cmaxd)
02401 {
02402     // Create a class of nodes
02403     // nnodes : # nodes
02404     // nclss : # classes
02405     // dgs : degree of a node, which is common to the vertices in a class
02406     // typ : initial/final/vertex
02407     // ptcl : particle code or list of possible interactions
02408     // cpl : values of coupling constants.
02409     // cnt : the number of nodes in this class
02410
02411     int j, k, nn, lp2;
02412     Bool ok = True;
02413
02414     sproc = spc;
02415     nnodes = nnods;
02416     nclass = nclss;
02417     deg = new int[nclass];
02418     type = new int[nclass];
02419     particle = new int[nclass];
02420     count = new int[nclass];
02421     couple = new int[nclass];
02422     cmindeg = new int[nclass];
02423     cmaxdeg = new int[nclass];
02424     for (j = 0; j < nclass; j++) {
02425         deg[j] = dgs[j];
02426         type[j] = typ[j];

```

```

02426     particle[j] = ptcl[j];
02427     count[j]    = cnt[j];
02428     couple[j]   = cpl[j];
02429     cmindeg[j]  = cmindeg[j];
02430     cmaxdeg[j]  = cmaxdeg[j];
02431     lp2 = (cpl[j]-dgs[j]+2);
02432     if (!isATEXternal(type[j]) && (lp2 % 2 != 0 || lp2 < 0)) {
02433         if (prlevel > 0) {
02434             fprintf(GRCC_Stderr, "*** PNodeClass: illegal loop: "
02435                 ": 2*loop=cpl[%d](%d)-deg[%d](%d)+2 =2*loop = %d\n",
02436                 j, cpl[j], j, dgs[j], lp2);
02437             for (k = 0; k < nclss; k++) {
02438                 fprintf(GRCC_Stderr, "k=%d, dgs=%d, typ=%d, ptcl=%d, ",
02439                     k, dgs[k], typ[k], ptcl[k]);
02440                 fprintf(GRCC_Stderr, "cpl=%d, cnt=%d\n", cpl[k], cnt[k]);
02441             }
02442         }
02443         ok = False;
02444     }
02445 }
02446 if (!ok) {
02447     erEnd("PNodeClass: illegal loop");
02448 }
02449 cl2nd = new int[nclass+1];
02450 nd2cl = new int[nnodes];
02451 cl2mcl = new int[nnodes];
02452
02453 nn = 0;
02454 for (j = 0; j < nclass; j++) {
02455     cl2nd[j] = nn;
02456     for (k = 0; k < count[j]; k++, nn++) {
02457         nd2cl[nn] = j;
02458     }
02459     if (isATEXternal(type[j])) {
02460         cl2mcl[j] = -1;
02461     } else {
02462         cl2mcl[j] = spc->model->findMClass(couple[j], deg[j]);
02463         if (cl2mcl[j] < 0) {
02464             if (prlevel > 0) {
02465                 fprintf(GRCC_Stderr, "*** PNodeClass : no vertex : ");
02466                 fprintf(GRCC_Stderr, "coupling=%d, degree=%d\n",
02467                     couple[j], deg[j]);
02468             }
02469             erEnd("PNodeClass : no vertex");
02470         }
02471     }
02472 }
02473 cl2nd[nclass] = nnodes;
02474 }
02475
02476 //-----
02477 PNodeClass::~PNodeClass(void)
02478 {
02479     delete[] cl2mcl;
02480     delete[] nd2cl;
02481     delete[] cl2nd;
02482     delete[] cmaxdeg;
02483     delete[] cmindeg;
02484     delete[] couple;
02485     delete[] count;
02486     delete[] particle;
02487     delete[] type;
02488     delete[] deg;
02489 }
02490
02491 //-----
02492 void PNodeClass::prPNodeClass(void)
02493 {
02494     int j;
02495
02496     printf("+++ PNodeClass: nclass=%d, nnodes=%d\n", nclass, nnodes);
02497     for (j = 0; j < nclass; j++) {
02498         prElem(j);
02499     }
02500     printf("\n");
02501 }
02502
02503 //-----
02504 void PNodeClass::prElem(int j)
02505 {
02506     int tp;
02507
02508     printf("%3d: %-7s(%2d), ", j, GRCC_AT_NdStr(type[j]), type[j]);
02509     printf("deg=%d, count=%d, nodes[%d--%d], couple=%d, cmindeg=%d, cmaxdeg=%d",
02510         deg[j], count[j], cl2nd[j], cl2nd[j+1], couple[j], cmindeg[j], cmaxdeg[j]);
02511     tp = type[j];
02512     if (isATEXternal(tp)) {

```

```

02513         if (sproc->model != NULL) {
02514             printf(" ptcl=%s ", sproc->model->particleName(particle[j]));
02515         } else {
02516             printf(" ptcl=%d ", particle[j]);
02517         }
02518     }
02519     printf("\n");
02520 }
02521 }
02522
02523 //=====
02524 // class SProcess
02525 //-----
02526 SProcess::SProcess(Model *mdl, Process *prc, Options *opts, int sid, int *clst, int ncls, int *cdeg,
02527 int *ctyp, int *ptcl, int *cpl, int *cnum, int *cmind, int *cmaxd)
02528 {
02529     // Construct a Subprocess object
02530     // mdl          : model
02531     // prc          : process (may be NULL)
02532     // opts        : options
02533     // sid         : id of the sprocess
02534     // ncls        : the number of classes
02535     // cdeg[ncls]  : the table of degrees of nodes.
02536     // ctyp[ncls]  : the table of nodes in the classes
02537     // ptcl[ncls]  : particle(External)/interaction code(Internal)
02538     // cpl[ncls]   : the table of total order of coupling constants.
02539     // cnum[ncls]  : the table of nodes in the classes
02540     // cmind[ncls] : the table of min(deg of connectabl vertex)
02541     // cmaxd[ncls] : the table of max(deg of connectabl vertex)
02542
02543     int j, cp, lp2, ndeg, nvrt, tcp10;
02544     bool ok;
02545
02546     if (ncls < 1) {
02547         fprintf(GRCC_Stderr, "*** SProcess::SProcess: no class %d\n", ncls);
02548         erEnd("SProcess::SProcess: no Class");
02549     }
02550     id = sid;
02551     model = mdl;
02552     proc = prc;
02553     opt = opts;
02554     nclass = ncls;
02555
02556     nNodes = 0;
02557     nEdges = 0;
02558     nExtern = 0;
02559     tCouple = 0;
02560     loop = 0;
02561     mgraph = NULL;
02562     egraph = NULL;
02563     astack = NULL;
02564
02565     mgrcount = 0;
02566     agrcount = 0;
02567     extperm = 1;
02568
02569     // the results of the graph generation
02570     nMGraphs = 0; // the number of generated M-graphs
02571     nMOPI = 0; // the number of 1PI M-graphs
02572     wMGraphs.setValue(0,1); // the weighted sum of M-graphs
02573     wMOPI.setValue(0,1); // the weighted sum of 1PI M-graphs
02574
02575     nAGraphs = 0; // the number of generated A-graphs
02576     nAOPI = 0; // the number of 1PI A-graphs
02577     wAGraphs.setValue(0,1); // the weighted sum of A-graphs
02578     wAOPI.setValue(0,1); // the weighted sum of 1PI A-graphs
02579
02580     // check input
02581     nNodes = 0;
02582     nExtern = 0;
02583
02584     ndeg = 0;
02585     nvrt = 0;
02586     ok = True;
02587
02588     tcp10 = 0;
02589     for (j = 0; j < model->ncouple; j++) {
02590         clist[j] = clst[j];
02591         tcp10 += clist[j];
02592     }
02593     for (j = 0; j < nclass; j++) {
02594         cp = cpl[j];
02595         if (cp > 0) {
02596             lp2 = cpl[j] - cdeg[j] + 2;
02597             if (lp2 % 2 != 0 || lp2 < 0) {
02598                 if (prlevel > 0) {
02599                     fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal loop:"

```

```

02599             "class %d, cp=%d, deg=%d, lp2=%d\n",
02600             j, cp, cdeg[j], lp2);
02601         }
02602         ok = False;
02603     }
02604 }
02605 tCouple += cp*cnum[j];
02606 ndeg += cdeg[j]*cnum[j];
02607
02608 if (!isATEternal(ctyp[j])) {
02609     nvrt += cnum[j];
02610 } else if (isATEternal(ctyp[j])) {
02611     if (model->normalParticle(ptcl[j]) == 0) {
02612         if (prlevel > 0) {
02613             fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
02614             fprintf(GRCC_Stderr, "illegal particle code: ");
02615             fprintf(GRCC_Stderr, "code: class %d, ptcl=%d\n", j, ptcl[j]);
02616         }
02617         ok = False;
02618     } else {
02619         nExtern += cnum[j];
02620     }
02621     extperm *= factorial(cnum[j]);
02622
02623 } else {
02624     if (prlevel > 0) {
02625         fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal type: ");
02626         fprintf(GRCC_Stderr, "class %d, type=%d\n", j, ctyp[j]);
02627     }
02628     ok = False;
02629 }
02630
02631 if (tcpl0 != tCouple) {
02632     if (prlevel > 0) {
02633         fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
02634         fprintf(GRCC_Stderr, "illegal coupling constants:");
02635         fprintf(GRCC_Stderr, " %d != %d\n", tcpl0, tCouple);
02636     }
02637     ok = False;
02638 }
02639 if (ndeg == nExtern) {
02640     if (prlevel > 0) {
02641         fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
02642         fprintf(GRCC_Stderr, "no vertices: ");
02643         fprintf(GRCC_Stderr, "nExtern=%d, ndeg=%d\n", nExtern, ndeg);
02644     }
02645     ok = False;
02646 }
02647 if (ndeg % 2 == 0) {
02648     nEdges = ndeg/2;
02649 } else {
02650     if (prlevel > 0) {
02651         fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
02652         fprintf(GRCC_Stderr, "illegal total deg = %d (not even)\n", ndeg);
02653     }
02654     ok = False;
02655 }
02656 nNodes = nvrt + nExtern;
02657 if (nNodes >= GRCC_MAXNODES) {
02658     if (prlevel > 0) {
02659         fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
02660         fprintf(GRCC_Stderr, "too many nodes = %d\n", nNodes);
02661         fprintf(GRCC_Stderr, "nExtern=%d, nvrt=%d (GRCC_MAXNODES)\n",
02662             nExtern, nvrt);
02663     }
02664     ok = False;
02665 }
02666 if (!ok) {
02667     prSProcess();
02668     erEnd("SProcess::SProcess: illegal input");
02669 }
02670 if (proc == NULL) {
02671     opt->beginProc(NULL);
02672 }
02673
02674 lp2 = tCouple - nExtern + 2;
02675 if (lp2 % 2 != 0 || lp2 < 0) {
02676     if (prlevel > 0) {
02677         fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal loop: "
02678             "tCouple=%d, nExtern=%d, 2*loop=%d\n",
02679             tCouple, nExtern, lp2);
02680     }
02681     ok = False;
02682 }
02683 loop = lp2/2;
02684
02685 // stack for assignment

```

```

02686     if (opt->values[GRCC_OPT_Step] == GRCC_AGraph) {
02687         astack = new AStack(GRCC_MAXNSTACK, GRCC_MAXESTACK);
02688     }
02689
02690     // save to PNodeClass object
02691     pnclass = new PNodeClass(this, nNodes, nclass, cdeg, ctyp, ptcl, cpl, cnum, cmind, cmaxd);
02692 }
02693
02694 //-----
02695 SProcess::SProcess(Model *mdl, Process *prc, Options *opts, int sid, int *clst, int ncls, NCInput
    *cls)
02696 {
02697     // Construct a Subprocess object
02698     //   mdl      : model
02699     //   prc      : process (may be NULL)
02700     //   opts     : options
02701     //   sid     : id of the sprocess
02702     //   ncls    : the number of classes
02703     //   cls     : input data of classes
02704
02705     int j, cp, lp2, ndeg, nvrt, tcpl0;
02706     bool ok;
02707
02708     if (ncls < 1) {
02709         fprintf(GRCC_Stderr, "*** SProcess::SProcess: no class %d\n", ncls);
02710         erEnd("SProcess::SProcess: no Class");
02711     }
02712     id      = sid;
02713     model   = mdl;
02714     proc    = prc;
02715     opt     = opts;
02716     nclass  = ncls;
02717
02718     nNodes  = 0;
02719     nEdges  = 0;
02720     nExtern = 0;
02721     tCouple = 0;
02722     loop    = 0;
02723     mgraph  = NULL;
02724     egraph  = NULL;
02725     astack  = NULL;
02726
02727     mgrcount = 0;
02728     agrcount = 0;
02729     extperm  = 1;
02730
02731     // the results of the graph generation
02732     nMGraphs = 0; // the number of generated M-graphs
02733     nMOPI    = 0; // the number of LPI M-graphs
02734     wMGraphs.setValue(0,1); // the weighted sum of M-graphs
02735     wMOPI.setValue(0,1); // the weighted sum of LPI M-graphs
02736
02737     nAGraphs = 0; // the number of generated A-graphs
02738     nAOPI    = 0; // the number of LPI A-graphs
02739     wAGraphs.setValue(0,1); // the weighted sum of A-graphs
02740     wAOPI.setValue(0,1); // the weighted sum of LPI A-graphs
02741
02742     // check input
02743     nNodes = 0;
02744     nExtern = 0;
02745
02746     ndeg    = 0;
02747     nvrt    = 0;
02748     ok = True;
02749
02750     tcpl0 = 0;
02751     for (j = 0; j < model->ncouple; j++) {
02752         clist[j] = clst[j];
02753         tcpl0 += clist[j];
02754     }
02755     for (j = 0; j < nclass; j++) {
02756         cp = cls[j].cple;
02757         if (cp > 0) {
02758             lp2 = cls[j].cple - cls[j].cldeg + 2;
02759             if (lp2 % 2 != 0 || lp2 < 0) {
02760                 if (prlevel > 0) {
02761                     fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal loop: "
02762                         "class %d, cp=%d, deg=%d, lp2=%d\n",
02763                         j, cp, cls[j].cldeg, lp2);
02764                 }
02765                 ok = False;
02766             }
02767         }
02768         tCouple += cp*cls[j].clnum;
02769         ndeg    += cls[j].cldeg*cls[j].clnum;
02770
02771         if (!isATEXternal(cls[j].cltyp)) {

```

```

02772         nvrt += cls[j].clnum;
02773     } else if (isATExternal(cls[j].cltyp)) {
02774         if (model->normalParticle(cls[j].ptcl) == 0) {
02775             if (prlevel > 0) {
02776                 fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
02777                 fprintf(GRCC_Stderr, "illegal particle code: ");
02778                 fprintf(GRCC_Stderr, "code: class %d, ptcl=%d\n", j, cls[j].ptcl);
02779             }
02780             ok = False;
02781         } else {
02782             nExtern += cls[j].clnum;
02783         }
02784         extperm *= factorial(cls[j].clnum);
02785     } else {
02786         if (prlevel > 0) {
02787             fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal type: ");
02788             fprintf(GRCC_Stderr, "class %d, type=%d\n", j, cls[j].cltyp);
02789         }
02790         ok = False;
02791     }
02792 }
02793
02794 if (tcpl0 != tCouple) {
02795     if (prlevel > 0) {
02796         fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
02797         fprintf(GRCC_Stderr, "illegal coupling constants:");
02798         fprintf(GRCC_Stderr, " %d != %d\n", tcpl0, tCouple);
02799     }
02800     ok = False;
02801 }
02802 if (ndeg == nExtern) {
02803     if (prlevel > 0) {
02804         fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
02805         fprintf(GRCC_Stderr, "no vertices: ");
02806         fprintf(GRCC_Stderr, "nExtern=%d, ndeg=%d\n", nExtern, ndeg);
02807     }
02808     ok = False;
02809 }
02810 if (ndeg % 2 == 0) {
02811     nEdges = ndeg/2;
02812 } else {
02813     if (prlevel > 0) {
02814         fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
02815         fprintf(GRCC_Stderr, "illegal total deg = %d (not even)\n", ndeg);
02816     }
02817     ok = False;
02818 }
02819 nNodes = nvrt + nExtern;
02820 if (nNodes >= GRCC_MAXNODES) {
02821     if (prlevel > 0) {
02822         fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
02823         fprintf(GRCC_Stderr, "too many nodes = %d\n", nNodes);
02824         fprintf(GRCC_Stderr, "nExtern=%d, nvrt=%d (GRCC_MAXNODES)\n",
02825             nExtern, nvrt);
02826     }
02827     ok = False;
02828 }
02829 if (!ok) {
02830     erEnd("SProcess::SProcess: illegal input");
02831 }
02832 if (proc == NULL) {
02833     opt->beginProc(NULL);
02834 }
02835
02836 lp2 = tCouple - nExtern + 2;
02837 if (lp2 % 2 != 0 || lp2 < 0) {
02838     if (prlevel > 0) {
02839         fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal loop: "
02840             "tCouple=%d, nExtern=%d, 2*loop=%d\n",
02841             tCouple, nExtern, lp2);
02842     }
02843     ok = False;
02844 }
02845 loop = lp2/2;
02846
02847 // stack for assignment
02848 if (opt->values[GRCC_OPT_Step] == GRCC_AGraph) {
02849     astack = new AStack(GRCC_MAXNSTACK, GRCC_MAXESTACK);
02850 }
02851
02852 // save to PNodeClass object
02853 pnclass = new PNodeClass(this, nNodes, nclass, cls);
02854 }
02855
02856 //-----
02857 SProcess::~SProcess(void)
02858 {

```



```

02859     delete pnclass;
02860     delete astack;
02861     delete mgraph;
02862 }
02863
02864 //-----
02865 void SProcess::prSProcess(void)
02866 {
02867     printf("\n");
02868     printf("+++ Subprocess %d, class=%d\n", id, nclass);
02869     if (pnclass != NULL) {
02870         pnclass->prPNodeClass();
02871     } else {
02872         printf(" pnclass = NULL\n");
02873     }
02874 }
02875
02876 //-----
02877 BigInt SProcess::generate(void)
02878 {
02879     // Entry point of Feynman graph generation
02880     int cldeg[GRCC_MAXNODES], clnum[GRCC_MAXNODES], cltyp[GRCC_MAXNODES];
02881     int cmind[GRCC_MAXNODES], cmaxd[GRCC_MAXNODES];
02882     int ncl;
02883     BigInt ng;
02884
02885     ncl = toMNodeClass(cltyp, cldeg, clnum, cmind, cmaxd);
02886
02887     opt->beginSubProc(this);
02888
02889     mgraph = new MGraph(id, ncl, cldeg, clnum, cltyp, cmind, cmaxd, opt);
02890
02891     ng = mgraph->generate();
02892
02893     opt->endSubProc();
02894
02895     delete mgraph;
02896     mgraph = NULL;
02897
02898     return ng;
02899 }
02900
02901 //-----
02902 int SProcess::toMNodeClass(int *cltyp, int *cldeg, int *clnum, int *cmind, int *cmaxd)
02903 {
02904     int j;
02905
02906     for (j = 0; j < nclass; j++) {
02907         if (isATEXternal(pnclass->type[j])) {
02908             cltyp[j] = -1;
02909         } else {
02910             cltyp[j] = (pnclass->couple[j]-pnclass->deg[j]+2)/2;
02911         }
02912         cldeg[j] = pnclass->deg[j];
02913         clnum[j] = pnclass->count[j];
02914         cmind[j] = pnclass->cminddeg[j];
02915         cmaxd[j] = pnclass->cmaxdeg[j];
02916     }
02917     return nclass;
02918 }
02919
02920 //-----
02921 void SProcess::assign(MGraph *mgr)
02922 {
02923     PNodeClass *pnc;
02924
02925     pnc = match(mgr);
02926
02927     agraph = new Assign(this, mgr, pnc);
02928
02929 #ifdef CHECK
02930     agraph->checkAG("SProcess::assign");
02931 #endif
02932     delete pnc;
02933     delete agraph;
02934     agraph = NULL;
02935 }
02936
02937 //-----
02938 PNodeClass *SProcess::match(MGraph *mgr)
02939 {
02940     PNodeClass *pnc = NULL;
02941
02942     int typ[GRCC_MAXNODES], dgs[GRCC_MAXNODES];
02943     int cnt[GRCC_MAXNODES], ptcl[GRCC_MAXNODES];
02944     int cpl[GRCC_MAXNODES];
02945     int n2m[GRCC_MAXNODES];

```

```

02946     int cmind[GRCC_MAXNODES], cmaxd[GRCC_MAXNODES];
02947     int j, k, l, nd, mc, mcl, mclss;
02948
02949     if (mgr->nNodes != nNodes) {
02950         if (prlevel > 0) {
02951             fprintf(GRCC_Stderr, "*** SProcess::match: different nNodes=%d != %d\n",
02952                 nNodes, mgr->nNodes);
02953         }
02954         erEnd("SProcess::match: different nNodes");
02955     }
02956
02957     mcl = toMNodeClass(typ, dgs, cnt, cmind, cmaxd);
02958
02959     nd = 0;
02960     for (j = 0; j < mcl; j++) {
02961         for (k = 0; k < cnt[j]; k++, nd++) {
02962             n2m[nd] = j;
02963         }
02964     }
02965
02966     nd = 0;
02967     mclss = mgr->curcl->nClasses;
02968     for (j = 0; j < mclss; j++) {
02969         for (k = 0; k < mgr->curcl->clist[j]; k++, nd++) {
02970             mc = mgr->nodes[nd]->clss;
02971             if (mc != n2m[nd]) {
02972                 if (prlevel > 0) {
02973                     fprintf(GRCC_Stderr, "*** SProcess::match: ");
02974                     fprintf(GRCC_Stderr, "inconsistent class\n");
02975                     for (l = 0; l < nNodes; l++) {
02976                         fprintf(GRCC_Stderr, "    %d: %d, %d\n",
02977                             j, mgr->nodes[l]->clss, n2m[l]);
02978                     }
02979                 }
02980                 erEnd("SProcess::match: inconsistent class");
02981             }
02982         }
02983     }
02984     for (j = 0; j < mcl; j++) {
02985         dgs[j] = pnclass->deg[j];
02986         typ[j] = pnclass->type[j];
02987         ptcl[j] = pnclass->particle[j];
02988         cnt[j] = pnclass->count[j];
02989         cpl[j] = pnclass->couple[j];
02990         cmind[j] = pnclass->cminddeg[j];
02991         cmaxd[j] = pnclass->cmaxdeg[j];
02992     }
02993     pnc = new PNodeClass(this, nNodes, mcl, dgs, typ, ptcl, cpl, cnt, cmind, cmaxd);
02994
02995     return pnc;
02996 }
02997
02998 //-----
02999 void SProcess::resultMGraph(BigInt nmgraphs, Fraction mwsum, BigInt nmopi, Fraction mwopi)
03000 {
03001     nMGraphs += nmgraphs;
03002     nMOPI += nmopi;
03003     wMGraphs.add(mwsum);
03004     wMOPI.add(mwopi);
03005 }
03006
03007 //-----
03008 void SProcess::resultAGraph(BigInt nmgraphs, Fraction mwsum, BigInt nmopi, Fraction mwopi)
03009 {
03010     nAGraphs += nmgraphs;
03011     nAOPI += nmopi;
03012     wAGraphs.add(mwsum);
03013     wAOPI.add(mwopi);
03014 }
03015
03016 //-----
03017 void SProcess::endMGraph(MGraph *mgr)
03018 {
03019     egraph = mgr->egraph;
03020
03021     if (opt->values[GRCC_OPT_Step] != GRCC_MGraph) {
03022         assign(mgr);
03023     }
03024 }
03025
03026 //-----
03027 void SProcess::endAGraph(EGraph *egr)
03028 {
03029     egraph = egr;
03030 }
03031
03032

```

```

03033 //=====
03034 // class Process
03035 //-----
03036 Process::Process(int pid, Model *modl, Options *optn, int nini, int *intlPrt, int nfin, int *finlPrt,
    int *coupling)
03037 {
03038     // Define a process and construct a set of sprocesses
03039     // model      : Model object
03040     // opt        : Option object
03041     // initlPart  : list of particle-id of initlPart particles
03042     // finalPart  : list of particle-id of final particles
03043     // coupling   : list of coupling constants
03044
03045     int j, lp2;
03046     Bool ok;
03047
03048     id      = pid;
03049     model   = modl;
03050     opt     = optn;
03051     ninitl  = nini;
03052     nfinal  = nfin;
03053     initlPart = intdup(ninitl, intlPrt);
03054     finalPart = intdup(nfinal, finlPrt);
03055
03056     // count total number of coupling constants.
03057     cttotal = 0;
03058     for (j = 0; j < GRCC_MAXNCPLG; j++) {
03059         if (j < model->ncouple) {
03060             clist[j] = coupling[j];
03061             cttotal += coupling[j];
03062         } else {
03063             clist[j] = 0;
03064         }
03065     }
03066     if (cttotal < 1) {
03067         erEnd("Process: coupling constant = 0");
03068     }
03069
03070     nExtern = 0;
03071     loop    = -1;
03072     maxnlegs = 0;
03073     mgrcount = 0;
03074     agrcount = 0;
03075
03076     // table of sprocesses
03077     nSubproc = 0;
03078     sproc    = NULL;
03079
03080     // the results of the graph generation
03081     nMGraphs = 0; // the number of generated M-graphs
03082     nMOPI    = 0; // the number of 1PI M-graphs
03083     wMGraphs.setValue(0,1); // the weighted sum of M-graphs
03084     wMOPI.setValue(0,1); // the weighted sum of 1PI M-graphs
03085
03086     nAGraphs = 0; // the number of generated A-graphs
03087     nAOPI    = 0; // the number of 1PI A-graphs
03088     wAGraphs.setValue(0,1); // the weighted sum of A-graphs
03089     wAOPI.setValue(0,1); // the weighted sum of 1PI A-graphs
03090
03091     ok = True;
03092
03093     // numbers
03094     nExtern = ninitl + nfinal;
03095     maxnlegs = Max(nExtern, model->maxnlegs);
03096
03097     // count loops
03098     lp2 = cttotal - nExtern + 2;
03099     if (lp2 % 2 != 0) {
03100         if (prlevel > 0) {
03101             fprintf(GRCC_Stderr, "*** cannot generate : 2*loop is odd : "
03102                 "2*loop = %d, cttotal=%d, nExtern=%d\n",
03103                 lp2, cttotal, nExtern);
03104         }
03105         ok = False;
03106     } else if (lp2 < 0) {
03107         if (prlevel > 0) {
03108             fprintf(GRCC_Stderr, "*** cannot make a connected graph : "
03109                 "2*loop=%d, cttotal=%d, nExtern=%d\n",
03110                 lp2, cttotal, nExtern);
03111         }
03112         ok = False;
03113     }
03114
03115     loop = lp2 / 2;
03116
03117     if (!ok) {
03118         if (prlevel > 0) {

```

```

03119         fprintf(GRCC_Stderr, "*** Process: illegal input: lp2 = %d\n", lp2);
03120     }
03121     erEnd("Process: illegal input");
03122 }
03123
03124 #ifdef DEBUG
03125     // print message
03126     prProcess();
03127 #endif
03128
03129     // construct sprocesses
03130     mkSProcess();
03131 }
03132
03133 //-----
03134 Process::~Process(void)
03135 {
03136     initlPart = delintdup(initlPart);
03137     finalPart = delintdup(finalPart);
03138     delete sproc;
03139 }
03140
03141 //-----
03142 void Process::prProcess(void)
03143 {
03144     printf("+++ process options : OPI = %d, (Step) = %d, coupling=%d: ",
03145         opt->values[GRCC_OPT_1PI], opt->values[GRCC_OPT_Step], ctototal);
03146     prIntArray(model->ncouple, clist, "\n");
03147 }
03148
03149 //-----
03150 void Process::mkSProcess(void)
03151 {
03152     // Construct sprocesses
03153     // Each sprocess is a set of nodes whose degree and order of
03154     // coupling constants are determined.
03155     // Only the total coupling constants are considered here even if
03156     // two or more coupling constants are defined in the model
03157
03158     NCInput cls[GRCC_MAXMINTERACT];
03159     int r, j, k, lin2, nvtx, nleg, nc, n0;
03160     int nl[GRCC_MAXMINTERACT];
03161     double proct0 = 0, proct1 = 0;
03162
03163     // output file to start process
03164     opt->beginProc(this);
03165
03166     // starting time
03167     Second(&proct0);
03168
03169     // generate all possible number of (nl[i]),
03170     // \sum_i nl[i]*cplgcp[i] = (total number of coupling constants)
03171
03172     r = -1;
03173     nSubproc = 0;
03174     ngraphs = 0;
03175     nopi = 0;
03176     wgraphs = 0;
03177     wopi = 0;
03178
03179     // for partitions of (leg, order)
03180
03181     while (nextPart(ctototal, model->ncplgcp, model->cplgcp, nl, &r)) {
03182         #ifdef DEBUG
03183             printf("nextPart:ctototal=%d, nc=%d, c=", ctototal, model->ncplgcp);
03184             prIntArray(model->ncplgcp, model->cplgcp, ", nl=");
03185             prIntArray(model->ncplgcp, nl, "");
03186             printf(", r=%d\n", r);
03187         #endif
03188
03189         // count the total number of vertices and legs.
03190         nvtx = 0;
03191         nleg = 0;
03192         for (j = 0; j < model->ncplgcp; j++) {
03193             nvtx += nl[j];
03194             nleg += nl[j]*model->cplglg[j];
03195         }
03196
03197         // count loops
03198         lin2 = nExtern + nleg;
03199         if (lin2 % 2 != 0) {
03200             continue;
03201         }
03202         loop = lin2/2 - nExtern - nvtx + 1;
03203         if (loop < 0) {
03204             continue;
03205         }
03206     }

```

```

03206     nc = 0;
03207     if (opt->values[GRCC_OPT_SymmInitial]) {
03208         n0 = nc;
03209         for (j = 0; j < ninitl; j++) {
03210             for (k = n0; k < nc; k++) {
03211                 if (cls[k].ptcl == initlPart[j]) {
03212                     cls[k].clnum++;
03213                     break;
03214                 }
03215             }
03216             if (k >= nc) {
03217                 cls[nc].ptcl = initlPart[j];
03218                 cls[nc].clnum = 1;
03219                 cls[nc].cple = 0;
03220                 cls[nc].cldeg = 1;
03221                 cls[nc].cldeg = 1;
03222                 cls[nc].cltyp = GRCC_AT_Initial;
03223                 cls[nc].ptcl = initlPart[j];
03224                 cls[nc].cmind
03225                     = model->particles[Abs(cls[nc].ptcl)]->cminddeg;
03226                 cls[nc].cmaxd
03227                     = model->particles[Abs(cls[nc].ptcl)]->cmaxdeg;
03228                 if (opt->values[GRCC_OPT_NoExtSelf] > 0 && nExtern > 2) {
03229                     cls[nc].cmind = Max(cls[nc].cmind, 3);
03230                 }
03231                 nc++;
03232             }
03233         }
03234     } else {
03235         for (j = 0; j < ninitl; j++) {
03236             cls[nc].ptcl = initlPart[j];
03237             cls[nc].clnum = 1;
03238             cls[nc].cple = 0;
03239             cls[nc].cldeg = 1;
03240             cls[nc].cltyp = GRCC_AT_Initial;
03241             cls[nc].cmind = model->particles[Abs(cls[nc].ptcl)]->cminddeg;
03242             cls[nc].cmaxd = model->particles[Abs(cls[nc].ptcl)]->cmaxdeg;
03243             if (opt->values[GRCC_OPT_NoExtSelf] > 0 && nExtern > 2) {
03244                 cls[nc].cmind = Max(cls[nc].cmind, 3);
03245             }
03246             nc++;
03247         }
03248     }
03249     if (opt->values[GRCC_OPT_SymmFinal]) {
03250         n0 = nc;
03251         for (j = 0; j < nfinal; j++) {
03252             for (k = n0; k < nc; k++) {
03253                 if (cls[k].ptcl == model->antiParticle(finalPart[j])) {
03254                     cls[k].clnum++;
03255                     break;
03256                 }
03257             }
03258             if (k >= nc) {
03259                 cls[nc].ptcl = model->antiParticle(finalPart[j]);
03260                 cls[nc].clnum = 1;
03261                 cls[nc].cple = 0;
03262                 cls[nc].cldeg = 1;
03263                 cls[nc].cltyp = GRCC_AT_Final;
03264                 cls[nc].cmind
03265                     = model->particles[Abs(cls[nc].ptcl)]->cminddeg;
03266                 cls[nc].cmaxd
03267                     = model->particles[Abs(cls[nc].ptcl)]->cmaxdeg;
03268                 if (opt->values[GRCC_OPT_NoExtSelf] > 0 && nExtern > 2) {
03269                     cls[nc].cmind = Max(cls[nc].cmind, 3);
03270                 }
03271                 nc++;
03272             }
03273         }
03274     } else {
03275         for (j = 0; j < nfinal; j++) {
03276             cls[nc].ptcl = model->antiParticle(finalPart[j]);
03277             cls[nc].cldeg = 1;
03278             cls[nc].cple = 0;
03279             cls[nc].clnum = 1;
03280             cls[nc].cltyp = GRCC_AT_Final;
03281             cls[nc].cmind = model->particles[Abs(cls[nc].ptcl)]->cminddeg;
03282             cls[nc].cmaxd = model->particles[Abs(cls[nc].ptcl)]->cmaxdeg;
03283             if (opt->values[GRCC_OPT_NoExtSelf] > 0 && nExtern > 2) {
03284                 cls[nc].cmind = Max(cls[nc].cmind, 3);
03285             }
03286             nc++;
03287         }
03288     }
03289     for (j = 0; j < model->ncplgcp; j++) {
03290         if (nl[j] > 0) {
03291             cls[nc].ptcl = 0;
03292             cls[nc].cple = model->cplgcp[j];

```

```

03293         cls[nc].cldeg = model->cplglg[j];
03294         cls[nc].clnum = nl[j];
03295         // number of loops
03296         cls[nc].cltyp = (model->cplgcp[j] - model->cplglg[j] + 2)/2;
03297         cls[nc].cmind = 0;
03298         cls[nc].cmaxd = 0;
03299         nc++;
03300     }
03301 }
03302 #ifdef DEBUG1
03303     for (j = 0; j < nc; j++) {
03304         printf("%d/%d: deg=%d, typ=%d, ptcl=%d, cple=%d, num=%d\n",
03305             j, nc, cls[j].cldeg, cls[j].cltyp, cls[j].ptcl,
03306             cls[j].cple, cls[j].clnum);
03307     }
03308 #endif
03309 // create a sprocess ??? to be rewritten
03310 sproc = new SProcess(model, this, opt, nSubproc, clist, nc, cls);
03311
03312 // list of sprocesses
03313 if (nSubproc >= GRCC_MAXSUBPROCS) {
03314     erEnd("Subclass: too many sprocesses (GRCC_MAXSUBPROCS)");
03315 }
03316 sptbl[nSubproc++] = sproc;
03317
03318 #ifdef DEBUG
03319     // print sprocesses
03320     sproc->prSProcess();
03321 #endif
03322
03323 // generate M-graphs
03324 sproc->generate();
03325
03326 // summation of the numbers and weighted numbers of graphs
03327 nMGraphs += sproc->nMGraphs;
03328 nMOPI += sproc->nMOPI;
03329 wMGraphs.add(sproc->wMGraphs);
03330 wMOPI.add(sproc->wMOPI);
03331
03332 nAGraphs += sproc->nAGraphs;
03333 nAOPI += sproc->nAOPI;
03334 wAGraphs.add(sproc->wAGraphs);
03335 wAOPI.add(sproc->wAOPI);
03336
03337 if (nAGraphs > 0) {
03338     ngraphs = nAGraphs;
03339 } else {
03340     ngraphs = nAGraphs;
03341 }
03342 delete sproc;
03343 sproc = NULL;
03344 }
03345
03346 // ending time
03347 Second(&proct1);
03348 sec = proct1-proct0;
03349
03350 // output file to finish process
03351 opt->endProc();
03352 }
03353
03354 //*****
03355 // mgraph.cc
03356 //=====
03357 //-----
03358 MNodeClass::MNodeClass(int nnodes, int nclasses)
03359 {
03360     // Input
03361     // nnodes : possible maximal number of nodes
03362     // nclasses : possible maximal number of classes
03363
03364     int j, k;
03365
03366     nNodes = nnodes;
03367     nClasses = nclasses;
03368     maxdeg = 0;
03369     for (j = 0; j < nNodes; j++) {
03370         for (k = 0; k < nNodes; k++) {
03371             clmat[j][k] = 0;
03372         }
03373         clist[j] = 0;
03374         ndcl[j] = 0;
03375         flist[j] = 0;
03376         clord[j] = 0;
03377         cmindeg[j] = 0;
03378         cmaxdeg[j] = 0;
03379     }

```

```

03380     flist[nNodes] = 0;
03381     flg0 = 0;
03382     flg1 = 0;
03383     flg2 = 0;
03384 }
03385
03386 //-----
03387 MNodeClass::~MNodeClass(void)
03388 {
03389 }
03390
03391 //=====
03392 // class MNodeClass
03393 //-----
03394 void MNodeClass::init(int *cl, int mxdeg, int **adjmat)
03395 {
03396     // initialize the object
03397     // Argument
03398     //   cl[j]           : list of number of nodes in the j-th class
03399     //   mxdeg          : maximum degree to nodes
03400     //   adjmat[j][k]   : adjacency matrix
03401
03402     int j;
03403
03404     for (j = 0; j < nClasses; j++) {
03405         clist[j] = cl[j];
03406         clord[j] = j;
03407     }
03408     maxdeg = mxdeg;
03409     mkNdCl();
03410     mkClMat(adjmat);
03411     mkFlist();
03412     flg0 = 0;
03413     flg1 = 0;
03414     flg2 = 0;
03415 }
03416
03417 //-----
03418 void MNodeClass::copy(MNodeClass* mnc)
03419 {
03420     // copy MNodeClass 'mnc' to this object
03421
03422     int j, k;
03423
03424     nClasses = mnc->nClasses;
03425     for (k = 0; k < nClasses; k++) {
03426         clist[k] = mnc->clist[k];
03427     }
03428     maxdeg = mnc->maxdeg;
03429     for (j = 0; j < nNodes; j++) {
03430         ndcl[j] = mnc->ndcl[j];
03431         clord[j] = mnc->clord[j];
03432         for (k = 0; k < nClasses; k++) {
03433             clmat[j][k] = mnc->clmat[j][k];
03434         }
03435     }
03436     for (k = 0; k < nClasses+1; k++) {
03437         flist[k] = mnc->flist[k];
03438     }
03439 }
03440
03441 //-----
03442 void MNodeClass::mkFlist(void)
03443 {
03444     // Construct flist
03445     //   The set of nodes in class 'c' is [flist[c],...,flist[c+1]-1]
03446
03447     int j, f;
03448
03449     f = 0;
03450     for (j = 0; j < nClasses; j++) {
03451         flist[j] = f;
03452         f += clist[j];
03453     }
03454     flist[nClasses] = f;
03455 }
03456
03457 //-----
03458 void MNodeClass::mkNdCl(void)
03459 {
03460     // Construct ndcl
03461     //   ndcl[nd] = (the class id in which node 'nd' belongs)
03462
03463     int c, k;
03464     int nd = 0;
03465
03466     for (c = 0; c < nClasses; c++) {

```

```

03467         for (k = 0; k < clist[c]; k++) {
03468             ndcl[nd++] = c;
03469         }
03470     }
03471 }
03472
03473 //-----
03474 int MNodeClass::clCmp(int nd0, int nd1, int cn)
03475 {
03476     // Comparison of two nodes 'nd0' and 'nd1'
03477     // in lexicographic ordering of [class, connection configuration]
03478
03479     int cmp;
03480     // Wether two nodes are in a same class or not.
03481
03482     cmp = ndcl[nd0] - ndcl[nd1];
03483     if (cmp != 0) {
03484         return cmp;
03485     }
03486
03487     // Sign '-' signifies the reverse ordering
03488     cmp = - cmpMNCArray(clmat[nd0], clmat[nd1], cn);
03489     if (cmp != 0) {
03490         return cmp;
03491     }
03492     return cmp;
03493 }
03494
03495 //-----
03496 void MNodeClass::mkClMat(int **adjmat)
03497 {
03498     // Construct a matrix 'clmat[nd][tc]' which is the number
03499     // of edges connecting 'nd' and all nodes in class 'tc'.
03500
03501     int nd, td, tc;
03502
03503     for (nd = 0; nd < nNodes; nd++) {
03504         for (td = 0; td < nNodes; td++) {
03505             tc = ndcl[td]; // another node
03506             if (nd == td) {
03507                 clmat[nd][tc] += CLWIGHTD(adjmat[nd][td]);
03508             } else {
03509                 clmat[nd][tc] += CLWIGHTO(adjmat[nd][td]);
03510             }
03511         }
03512     }
03513 }
03514
03515 //-----
03516 void MNodeClass::incMat(int nd, int td, int val)
03517 {
03518     // Increase the number of edges by 'val' between 'nd' and 'td'.
03519
03520     int tdc = ndcl[td]; // class of 'td'
03521     clmat[nd][tdc] += val; // modify matrix 'clmat'.
03522 }
03523
03524 //-----
03525 void MNodeClass::printMat(void)
03526 {
03527     // Print configuration matrix.
03528
03529     int j1, j2;
03530
03531     printf("\n");
03532
03533     printf("nClasses=%d", nClasses);
03534     printf(" clord="); prIntArray(nClasses, clord, "");
03535     printf(" flist="); prIntArray(nClasses, flist, "\n");
03536     printf("flg = (%d, %d, %d)\n", flg0, flg1, flg2);
03537
03538     // the first line
03539     printf("nd: cl:  ");
03540     for (j2 = 0; j2 < nClasses; j2++) {
03541         printf("%2d ", j2);
03542     }
03543     printf("\n");
03544
03545     // print raw
03546     for (j1 = 0; j1 < nNodes; j1++) {
03547         printf("%2d; %2d: [", j1, ndcl[j1]);
03548         for (j2 = 0; j2 < nClasses; j2++) {
03549             printf(" %2d", clmat[j1][j2]);
03550         }
03551         printf("]\n");
03552     }
03553 }

```



```

03554
03555 //-----
03556 int MNodeClass::cmpMNCArray(int *a0, int *a1, int ma)
03557 {
03558     int j, k;
03559
03560     for (k = 0; k < ma; k++) {
03561         j = clord[k];
03562         if (a0[j] < a1[j]) {
03563             return -1;
03564         } else if (a0[j] > a1[j]) {
03565             return 1;
03566         }
03567     }
03568     return 0;
03569 }
03570
03571 //-----
03572 void MNodeClass::reorder(MGraph *mg)
03573 {
03574     int flg[GRCC_MAXNODES];
03575     int co, cn;
03576     int f0, f1, f2;
03577
03578     f0 = 0;
03579     f1 = 0;
03580     f2 = 0;
03581     for (co = 0; co < nClasses; co++) {
03582         cn = flist[co];
03583         if (mg->nodes[cn]->freelg < 1) {
03584             flg[co] = 0;
03585             f0++;
03586         } else if (mg->nodes[cn]->freelg < mg->nodes[cn]->deg) {
03587             flg[co] = mg->nodes[cn]->deg + maxdeg*clist[co];
03588             f1++;
03589         } else {
03590             flg[co] = maxdeg*(mg->nodes[cn]->deg + maxdeg*clist[co]);
03591             f2++;
03592         }
03593     }
03594     if (f0 < nClasses) {
03595         bsort(nClasses, clord, flg);
03596     }
03597     flg0 = f0;
03598     flg1 = f0 + f1;
03599     flg2 = f0 + f1 + f2;
03600 }
03601
03602 //=====
03603 // class MNode : nodes in MGraph
03604 //-----
03605 MNode::MNode(int vid, int vclss, NCInput *mgi)
03606 {
03607     // Arguments
03608     // vid : identifier of the vertex
03609     // vdeg : degree of the node
03610     // vextlp : external node (-1) or looped vertex
03611     // vclss : class of the node
03612
03613     id = vid; // id of the node
03614     clss = vclss; // class to which the node belongs
03615     deg = mgi->cldeg; // degree of the node
03616     freelg = mgi->cldeg; // number of free legs
03617     extloop = mgi->cltyp; // external node or not
03618     cmindeg = mgi->cmind; // min(deg of connectable vertex)
03619     cmaxdeg = mgi->cmaxd; // max(deg of connectable vertex)
03620 #ifdef DEBUG3
03621     printf("MNode:id=%d, freelg=%d, cmindeg=%d, cmaxd=%d\n",
03622         id, freelg, cmindeg, cmaxdeg);
03623 #endif
03624 }
03625 //-----
03626 MNode::MNode(int vid, int vdeg, int vextlp, int vclss, int cmin, int cmax)
03627 {
03628     // Arguments
03629     // vid : identifier of the vertex
03630     // vdeg : degree of the node
03631     // vextlp : external node (-1) or looped vertex
03632     // vclss : class of the node
03633
03634     id = vid; // id of the node
03635     deg = vdeg; // degree of the node
03636     freelg = vdeg; // number of free legs
03637     clss = vclss; // class to which the node belongs
03638     extloop = vextlp; // external node or not
03639     cmindeg = cmin; // min(deg of connectable vertex)
03640     cmaxdeg = cmax; // max(deg of connectable vertex)

```

```

03641 #ifdef DEBUG3
03642     printf("MNode:id=%d, freelg=%d, cmind=%d, cmaxd=%d\n", id, freelg, cmindeg, cmaxdeg);
03643 #endif
03644 }
03645
03646 //=====
03647 // class MGraph : scalar graph expressed by matrix form
03648 //-----
03649 MGraph::MGraph(int pid, int ncl, int *cldeg, int *clnum, int *cltyp, int *cmind, int *cmaxd, Options
*opts)
03650 {
03651     int nn, ne, j, k;
03652
03653 #ifdef DEBUGM
03654     printf("MGraph::MGraph(pid=%d, ncl=%d)\n", pid, ncl);
03655     for (j = 0; j < ncl; j++){
03656         printf("%d: deg=%d, num=%d, typ=%d, mind=%d, maxd=%d\n",
03657             j, cldeg[j], clnum[j], cltyp[j], cmindeg[j], cmaxdeg[j]);
03658     }
03659     opts->print();
03660 #endif
03661
03662     // initial conditions
03663     nClasses = ncl;
03664     clist = new int[ncl];
03665     opt = opts;
03666     mId = -1;
03667
03668     mindeg = -1;
03669     maxdeg = -1;
03670     ne = 0;
03671     nn = 0;
03672     for (j = 0; j < nClasses; j++) {
03673         clist[j] = clnum[j];
03674         nn += clnum[j];
03675         ne += cldeg[j]*clnum[j];
03676         if (mindeg < 0) {
03677             mindeg = cldeg[j];
03678         } else {
03679             mindeg = Min(mindeg, cldeg[j]);
03680         }
03681         if (maxdeg < 0) {
03682             maxdeg = cldeg[j];
03683         } else {
03684             maxdeg = Max(maxdeg, cldeg[j]);
03685         }
03686     }
03687     if (ne % 2 != 0) {
03688         if (prlevel > 0) {
03689             fprintf(GRCC_Stderr, "Sum of degrees are not even\n");
03690             for (j = 0; j < nClasses; j++) {
03691                 fprintf(GRCC_Stderr, "class %2d: %2d %2d %2d\n",
03692                     j, cldeg[j], clnum[j], cltyp[j]);
03693             }
03694         }
03695         erEnd("illegal degrees of nodes");
03696     }
03697     pid = pid;
03698     nNodes = nn;
03699     nodes = new MNode*[nNodes];
03700     group = new SGroup();
03701 #ifdef ORBITS
03702     orbits = new MORbits();
03703 #endif
03704
03705     nEdges = ne / 2;
03706     nLoops = nEdges - nNodes + 1;
03707
03708     egraph = new EGraph(nNodes, nEdges, maxdeg);
03709     nn = 0;
03710     nExtern = 0;
03711 #ifdef DEBUG3
03712     printf("MGraph::MGraph: nClasses=%d\n", nClasses);
03713 #endif
03714     for (j = 0; j < nClasses; j++) {
03715 #ifdef DEBUG3
03716         printf("cmind[%d]=%d, cmaxd[%d]=%d\n", j, cmindeg[j], j, cmaxdeg[j]);
03717 #endif
03718         for (k = 0; k < clist[j]; k++, nn++) {
03719             nodes[nn] = new MNode(nn, cldeg[j], cltyp[j], j, cmindeg[j], cmaxdeg[j]);
03720             egraph->setExtLoop(nn, cltyp[j]);
03721             if (cltyp[j] < 0) {
03722                 nExtern++;
03723             }
03724         }
03725     }
03726     egraph->endSetExtLoop();

```

```

03727
03728     init();
03729 }
03730
03731 //-----
03732 MGraph::MGraph(int pid, int ncl, NCInput *mgi, Options *opts)
03733 {
03734     int nn, ne, j, k;
03735
03736 #ifdef DEBUGM
03737     printf("MGraph::MGraph(pid=%d, ncl=%d)\n", pid, ncl);
03738     for (j = 0; j < ncl; j++){
03739         printf("%d: deg=%d, num=%d, typ=%d, mind=%d, maxd=%d\n",
03740             j, mgi[j].cldeg, mgi[j].clnum, mgi[j].cltyp,
03741             mgi[j].cmind, mgi[j].cmaxd);
03742     }
03743     opts->print();
03744 #endif
03745
03746     // initial conditions
03747     nClasses = ncl;
03748     clist     = new int[ncl];
03749     opt       = opts;
03750     mId       = -1;
03751
03752     mindeg    = -1;
03753     maxdeg    = -1;
03754     ne = 0;
03755     nn = 0;
03756     for (j = 0; j < nClasses; j++) {
03757         clist[j] = mgi[j].clnum;
03758         nn += mgi[j].clnum;
03759         ne += mgi[j].cldeg*mgi[j].clnum;
03760         if (mindeg < 0) {
03761             mindeg = mgi[j].cldeg;
03762         } else {
03763             mindeg = Min(mindeg, mgi[j].cldeg);
03764         }
03765         if (maxdeg < 0) {
03766             maxdeg = mgi[j].cldeg;
03767         } else {
03768             maxdeg = Max(maxdeg, mgi[j].cldeg);
03769         }
03770     }
03771     if (ne % 2 != 0) {
03772         if (prlevel > 0) {
03773             fprintf(GRCC_Stderr, "Sum of degrees are not even\n");
03774             for (j = 0; j < nClasses; j++) {
03775                 fprintf(GRCC_Stderr, "class %2d: %2d %2d %2d\n",
03776                     j, mgi[j].cldeg, mgi[j].clnum, mgi[j].cltyp);
03777             }
03778         }
03779         erEnd("illegal degrees of nodes");
03780     }
03781     pid = pid;
03782     nNodes = nn;
03783     nodes = new MNode*[nNodes];
03784     group = new SGroup();
03785 #ifdef ORBITS
03786     orbits = new MORbits();
03787 #endif
03788
03789     nEdges = ne / 2;
03790     nLoops = nEdges - nNodes + 1;
03791
03792     egraph = new EGraph(nNodes, nEdges, maxdeg);
03793     nn = 0;
03794     nExtern = 0;
03795 #ifdef DEBUG3
03796     printf("MGraph::MGraph: nClasses=%d\n", nClasses);
03797 #endif
03798     for (j = 0; j < nClasses; j++) {
03799 #ifdef DEBUG3
03800         printf("cmind[%d]=%d, cmaxd[%d]=%d\n",
03801             j, mgi[j].cmind, j, mgi[j].cmaxd);
03802 #endif
03803         for (k = 0; k < clist[j]; k++, nn++) {
03804             nodes[nn] = new MNode(nn, j, mgi+j);
03805             egraph->setExtLoop(nn, mgi[j].cltyp);
03806             if (mgi[j].cltyp < 0) {
03807                 nExtern++;
03808             }
03809         }
03810     }
03811     egraph->endSetExtLoop();
03812
03813     init();

```

```

03814 }
03815
03816 //-----
03817 void MGraph::init(void)
03818 {
03819     // generated set of graphs
03820     cDiag      = 0;
03821     c1PI       = 0;
03822     cNoTadpole = 0;
03823     cNoTadBlock = 0;
03824     c1PINoTadBlock = 0;
03825
03826     // the current graph
03827     adjMat      = newMat(nNodes, nNodes, 0);
03828     nsym         = ToBigInt(0);
03829     esym         = ToBigInt(0);
03830     wscon        = Fraction(0, 1);
03831     wsopi        = Fraction(0, 1);
03832
03833     // current node classification
03834     curcl        = new MNodeClass(nNodes, nClasses);
03835
03836     // table of n-edge connected components
03837     mconn        = new MConn(nNodes, nEdges);
03838
03839     // measures of efficiency
03840     ngen         = 0;
03841     ngconn       = 0;
03842
03843     #ifdef MONITOR
03844         nCallRefine = 0;
03845         discardRefine = 0;
03846         discardDisc = 0;
03847         discardIso = 0;
03848     #endif
03849
03850     // work space for isIsomorphic
03851     modmat = newMat(nNodes, nNodes, 0);
03852
03853     // work space for bisearchM
03854     bidef = newArray(nNodes, 0);
03855     bilow = newArray(nNodes, 0);
03856     bicount = 0;
03857 }
03858
03859 //-----
03860 MGraph::~MGraph(void)
03861 {
03862     int j;
03863
03864     // group->delGroup();
03865
03866     bilow = deleteArray(bilow);
03867     bidef = deleteArray(bidef);
03868
03869     modmat = deleteMat(modmat, nNodes);
03870     delete mconn;
03871     delete curcl;
03872     adjMat = deleteMat(adjMat, nNodes);
03873     #ifdef ORBITS
03874         delete orbits;
03875     #endif
03876     delete egraph;
03877     delete group;
03878     for (j = 0; j < nNodes; j++) {
03879         delete nodes[j];
03880         nodes[j] = NULL;
03881     }
03882     delete[] nodes;
03883     delete[] clist;
03884 }
03885
03886 //-----
03887 void MGraph::printAdjMat(MNodeClass *cl)
03888 {
03889     int j1, j2;
03890
03891     printf("      ");
03892     for (j2 = 0; j2 < nNodes; j2++) {
03893         printf(" %2d", j2);
03894     }
03895     printf("\n");
03896     for (j1 = 0; j1 < nNodes; j1++) {
03897         printf("%2d : [", j1);
03898         for (j2 = 0; j2 < nNodes; j2++) {

```

```

03901         printf(" %2d", adjMat[j1][j2]);
03902     }
03903     printf("] %2d\n", cl->ndcl[j1]);
03904 }
03905 }
03906
03907 //-----
03908 void MGraph::print(void)
03909 {
03910     int j;
03911
03912     printf("MGraph: pId=%d, cDiag=%ld, cPI=%ld\n", pId, cDiag, cPI);
03913     printf(" nNodes=%d, nEdges=%d, nExtern=%d, nLoops=%d "
03914            "mindeg=%d, maxdeg=%d, sym=(%ld, %ld)\n",
03915            nNodes, nEdges, nExtern, nLoops, mindeg, maxdeg, nsym, esym);
03916     printf(" Nodes=%d\n", nNodes);
03917     for (j = 0; j < nNodes; j++) {
03918         printf("      %2d: id=%d, deg=%d, clss=%d, extloop=%d, ",
03919                j, nodes[j]->id, nodes[j]->deg, nodes[j]->clss,
03920                nodes[j]->extloop);
03921         printf("mind=%d, maxd=%d, freelg=%d\n",
03922                nodes[j]->cmindeg, nodes[j]->cmaxdeg, nodes[j]->freelg);
03923     }
03924
03925     curcl->printMat();
03926     printf("\n");
03927
03928     printAdjMat(curcl);
03929 }
03930
03931 //-----
03932 Bool MGraph::isConnected(void)
03933 {
03934     // Check graph can be a connected one.
03935     // If a connected component without free leg is not the whole graph then
03936     // return False, otherwise return True.
03937
03938     int j, n, nv;
03939
03940     for (j = 0; j < nNodes; j++) {
03941         nodes[j]->visited = -1;
03942     }
03943     if (visit(0)) {
03944         return True;
03945     }
03946     nv = 0;
03947     for (n = 0; n < nNodes; n++) {
03948         if (nodes[n]->visited >= 0) {
03949             nv++;
03950         }
03951     }
03952     return (nv == nNodes);
03953 }
03954
03955 //-----
03956 Bool MGraph::visit(int nd)
03957 {
03958     // Visiting connected node used for 'isConnected'
03959     // If child nodes has free legs, then this function returns True.
03960     // otherwise it returns False.
03961
03962     int td;
03963
03964     // This node has free legs.
03965     if (nodes[nd]->freelg > 0) {
03966         return True;
03967     }
03968     nodes[nd]->visited = 0;
03969     for (td = 0; td < nNodes; td++) {
03970         if ((adjMat[nd][td] > 0) && (nodes[td]->visited < 0)) {
03971             if (visit(td)) {
03972                 return True;
03973             }
03974         }
03975     }
03976     // all the child nodes has no free legs.
03977     return False;
03978 }
03979
03980 //-----
03981 Bool MGraph::isIsomorphic(MNodeClass *cl)
03982 {
03983     // Check whether the current graph is the Representative
03984     // of a isomorphic class.
03985     // nsym = symmetry factor by the permutation of nodes.
03986     // esym = symmetry factor by the permutation of edge.
03987     // If this graph is not a representative, then returns False.

```

```

03988
03989     int j1, j2, cmp, nself;
03990     int *perm;
03991
03992     nsym = ToBigInt(0);
03993     esym = ToBigInt(1);
03994
03995     group->newGroup(nNodes, cl->nClasses, cl->clist);
03996
03997 #ifdef ORBITS
03998     orbits->initPerm(nNodes);
03999 #endif
04000
04001     while (True) {
04002         perm = group->genNext();
04003
04004         if (perm == NULL) {
04005             // calculate permutations of edges
04006             esym = ToBigInt(1);
04007             for (j1 = 0; j1 < nNodes; j1++) {
04008                 if (adjMat[j1][j1] > 0) {
04009                     nself = adjMat[j1][j1]/2;
04010                     esym *= factorial(nself);
04011                     esym *= ipow(2, nself);
04012                 }
04013                 for (j2 = j1+1; j2 < nNodes; j2++) {
04014                     if (adjMat[j1][j2] > 0) {
04015                         esym *= factorial(adjMat[j1][j2]);
04016                     }
04017                 }
04018             }
04019             return True;
04020         }
04021
04022         permMat(nNodes, perm, adjMat, modmat);
04023         cmp = compMat(nNodes, adjMat, modmat);
04024         if (cmp < 0) {
04025             return False;
04026         } else if (cmp == 0) {
04027             // if ngroup < 0, symmetry group is $S_{-group}$.
04028             group->addGroup(perm);
04029
04030             nsym = nsym + ToBigInt(1);
04031 #ifdef ORBITS
04032             orbits->fromPerm(perm);
04033 #endif
04034         }
04035     }
04036 }
04037
04038 //-----
04039 void MGraph::permMat(int size, int *perm, int **mat0, int **mat1)
04040 {
04041     // apply permutation to matrix 'mat0' and obtain 'mat1'
04042
04043     int j1, j2;
04044
04045     for (j1 = 0; j1 < size; j1++) {
04046         for (j2 = 0; j2 < size; j2++) {
04047             mat1[j1][j2] = mat0[perm[j1]][perm[j2]];
04048         }
04049     }
04050 }
04051
04052 //-----
04053 int MGraph::compMat(int size, int **mat0, int **mat1)
04054 {
04055     // comparison of matrix
04056
04057     int j1, j2, cmp;
04058
04059     for (j1 = 0; j1 < size; j1++) {
04060         for (j2 = 0; j2 < size; j2++) {
04061             cmp = mat0[j1][j2] - mat1[j1][j2];
04062             if (cmp != 0) {
04063                 return cmp;
04064             }
04065         }
04066     }
04067     return 0;
04068 }
04069
04070 //-----
04071 MNodeClass *MGraph::refineClass(MNodeClass *cl)
04072 {
04073     // Refine the classification
04074     // cl : the current 'MNodeClass' object

```

```

04075 // cn : the class number
04076 // Returns (the new class number corresponds to 'cn') if OK, or
04077 // 'None' if ordering condition is not satisfied
04078
04079 MNodeClass *ccl = cl;
04080 MNodeClass *xcl = NULL;
04081 MNodeClass *ncl = NULL;
04082 int ucl[GRCC_MAXNODES];
04083 int ccn = cl->nClasses;
04084 int nucl, nce;
04085 int td, cmp;
04086 int nccl;
04087
04088 #ifdef MONITOR
04089 nCallRefine++;
04090 #endif
04091 nucl = 0;
04092 while (ccl->nClasses != nucl) { // repeat refinement.
04093     nce = 0;
04094     nucl = 0;
04095     for (td = 1; td < nNodes; td++) {
04096         // 'td' is the next node and the current node is 'td-1'.
04097         // Count up the number of the elements in the current class
04098         // corresponding to the current node
04099         nce++;
04100         cmp = ccl->clCmp(td-1, td, ccn);
04101
04102         // the ordering condition is not satisfied.
04103         if (cmp > 0) {
04104             if (ccl != cl) {
04105                 delete ccl;
04106             }
04107             return NULL;
04108
04109         } else if (cmp < 0) {
04110             // 'td' is in the next class to the current node.
04111             ucl[nucl++] = nce; // close the current class
04112
04113             // start new class
04114             nce = 0;
04115         }
04116         // nothing to do for the case of 'cmp == 0'.
04117     }
04118
04119     // close array 'ucl'.
04120     ucl[nucl++] = nce + 1;
04121
04122 #ifdef CHECK
04123     // inconsistent
04124     if (nucl < ccl->nClasses) {
04125         erEnd("refineClasses : smaller number of classes");
04126     }
04127 #endif
04128
04129     // preparation of the next repetition.
04130     xcl = new MNodeClass(nNodes, nucl);
04131     xcl->init(ucl, maxdeg, adjMat);
04132     ccn = xcl->nClasses;
04133
04134     nccl = ccl->nClasses;
04135
04136     if (ccl != cl) {
04137         delete ccl;
04138     }
04139     ccl = xcl;
04140     if (ccl->nClasses == nccl) {
04141         ccl->reorder(this);
04142         return ccl;
04143     }
04144
04145     nucl = 0;
04146 }
04147
04148 return ncl;
04149 }
04150
04151 //-----
04152 void MGraph::biconnME(void)
04153 {
04154     // Count the number of lPI components.
04155
04156     int j, root, vr, next, nart;
04157     MCOpi mopi;
04158     MCBLOCK mblk;
04159
04160     // initialization

```

```

04162     bicount      = 0;
04163
04164     mconn->init();
04165
04166     for (j = 0; j < nNodes; j++) {
04167         bidef[j] = -1;
04168         bilow[j] = -1;
04169     }
04170
04171     // find a root for the root of bisearch
04172     // root must be an external node
04173     // or an vertex when there are not external nodes.
04174     root = -1;
04175     vr = -1;
04176     for (j = 0; j < nNodes; j++) {
04177         if (bidef[j] < 0) {
04178             if (isExternal(j)) {
04179                 root = j;
04180                 break;
04181             } else if (vr < 0) {
04182                 vr = j;
04183             }
04184         }
04185     }
04186     // case (2)
04187     if (root < 0) {
04188         root = vr;
04189     }
04190
04191     if (root >= 0) {
04192         bisearchME(root, -1, 0, &mopi, &mblk, &next, &nart);
04193
04194         mconn->nlpopic = 0;
04195         mconn->nctopic = 0;
04196         for (j = 0; j < mconn->nopic; j++) {
04197             if (mconn->opics[j].loop > 0) {
04198                 mconn->nlpopic++;
04199             } else if (mconn->opics[j].ctloop > 0) {
04200                 mconn->nctopic++;
04201             }
04202         }
04203
04204         mconn->ne0bridges = 0;
04205         mconn->nelbridges = 0;
04206         for (j = 0; j < mconn->nbridges; j++) {
04207             if (mconn->bridges[j].next == 0) {
04208                 mconn->ne0bridges++;
04209             }
04210             if (mconn->bridges[j].next == 1) {
04211                 mconn->nelbridges++;
04212             }
04213         }
04214
04215         mconn->nselfloops = 0;
04216         for (j = 0; j < nNodes; j++) {
04217             mconn->nselfloops += adjMat[j][j]/2;
04218         }
04219
04220         mconn->nalblocks = 0;
04221         for (j = 0; j < mconn->nblocks; j++) {
04222             if (mconn->blocks[j].nartps == 1) {
04223                 mconn->nalblocks++;
04224             }
04225         }
04226         mconn->neblocks = mconn->nblocks + mconn->nctopic;
04227
04228         // no vertex.
04229     } else {
04230         // no vertices
04231         // Connected ==> only when ex=2, loop=0
04232         mconn->nopic = 0;
04233         mconn->ne0bridges = 0;
04234         mconn->nelbridges = 0;
04235     }
04236 }
04237
04238 //-----
04239 void MGraph::bisearchME(int nd, int pd, int ned, MCOpI *mopi, MCBloC *mblk, int *next, int *nart)
04240 {
04241     // Search biconnected component
04242     // visit : pd --> nd --> td
04243     // ned : the number of edges between pd and nd.
04244     // Output
04245     //
04246     //      |
04247     //      x-----+
04248     //      |       |

```



```

04249      //          ----x pd |
04250      //          |         |
04251      //          x----x nd | n art. pints.
04252      //          |  /\  |
04253      //          |  /\  |
04254      // m art p. x-x | x-x
04255      //          n1 x n3
04256      //          n2
04257      //
04258      // output of bisearchME(n1, nd, ...) ==> npart = m
04259      // output of bisearchME(n2, nd, ...) ==> npart = 1
04260      // output of bisearchME(n3, nd, ...) ==> npart = n
04261      //
04262      // output of bisearchME(nd, pd, ...) ==> npart = n+1
04263      // n1 => block of (m+1) articulation points
04264      // n2 => block of 2 articulation points
04265      // n3 => no block
04266      //
04267      Bool newv;
04268      int td, td0, lp, conn, next1, nart1, nart2, nvisit, ndart;
04269      int opit, opit0, blkt;
04270      MCOpi mopi1;
04271      MCBlock mblk1;
04272
04273      mopi->init();
04274      mblk->init();
04275
04276      *next = 0; // # external below 'nd' inclusive
04277      *nart = 0; // # articulation points 'nd' inclusive
04278      nart1 = 0; // # articulation points below 'nd'.
04279      nart2 = 0; // # 'nd' is articulation point then 1
04280      ndart = 0; // # the number of blocks attached to 'nd'
04281
04282      newv = (bidef[nd] < 0);
04283
04284      bidef[nd] = bicount;
04285      bilow[nd] = bicount;
04286      bicount++;
04287
04288      opit0 = mconn->opistkptr;
04289
04290      if (pd < 0) {
04291          mconn->pushNode(nd);
04292      }
04293
04294      // external node and not the root
04295      if (isExternal(nd)) {
04296          if (pd >= 0) {
04297              mopi->next = 1;
04298              mopi->nlegs = 1;
04299              mblk->nartps = 1;
04300              *next = 1;
04301              *nart = 1;
04302              return;
04303          }
04304      }
04305
04306      nvisit = 0;
04307      td0 = -1;
04308      for (td = 0; td < nNodes; td++) {
04309          conn = adjMat[nd][td];
04310
04311          // td is not adjacent to nd
04312          if (conn < 1) {
04313              continue;
04314          }
04315          nvisit++;
04316          td0 = td;
04317
04318          // external node
04319          if (isExternal(td) && bidef[td] < 0) {
04320              mopi->nlegs++;
04321              mopi->next++;
04322              (*next)++;
04323              ndart++;
04324              mconn->addArtic(nd, 1);
04325              mblk->nartps++;
04326              nart2 = 1;
04327              if (newv) {
04328                  mconn->pushNode(td);
04329              }
04330
04331              // self-loop : pd --> nd --> nd
04332          } else if (td == nd) {
04333              lp = conn/2;
04334              mopi->loop += lp;
04335              mopi->nedges += lp;

```

```

04336         ndart += lp;
04337         mconn->addArtic(nd, lp);
04338         mconn->addBlockSelf(nd, lp);
04339         mblk->nartps++;
04340         nart2 = 1;
04341
04342
04343         // back to the parent : pd --> nd --> pd, pd is a vertex
04344         // Back edges when the connection is multi-edges (ned > 1)
04345     } else if (td == pd) {
04346         if (ned > 1) {
04347             bilow[nd] = Min(bilow[nd], bidef[pd]);
04348             mopi->loop += ned - 1;
04349             mblk->loop += ned - 1;
04350         } else {
04351             // cancel this visit
04352             nvisit--;
04353         }
04354
04355         // back edge :td is already visited
04356     } else if (bidef[td] >= 0) {
04357         bilow[nd] = Min(bilow[nd], bidef[td]);
04358         if (bidef[nd] >= bidef[td]) {
04359             mopi->loop += conn;
04360             mopi->nedges += conn;
04361             mblk->loop += conn;
04362             mconn->pushEdge(td, nd);
04363         }
04364
04365         // new node
04366     } else if (bidef[td] < 0) {
04367         // stack pointer
04368         if (pd < 0 && isExternal(nd)) {
04369             opit = opit0;
04370         } else {
04371             opit = mconn->opistkptr;
04372         }
04373         blkt = mconn->blkstkptr;
04374         mblk1.init();
04375
04376         mconn->pushEdge(nd, td);
04377
04378         // visit child
04379         bisearchME(td, nd, adjMat[td][nd], &mopil, &mblk1, &next1, &nart1);
04380
04381         // articulation point of bridge
04382         if (bilow[td] > bidef[nd]) {
04383             // new OPI component (not including 'td')
04384             mopil.nlegs++;
04385
04386             mconn->addOPIC(&mopil, opit);
04387             mopil.init();
04388             mopil.nlegs = 1;
04389             if (pd >= 0 || !isExternal(nd)) {
04390                 // opi
04391                 mconn->addBridge(nd, td, next1, nExtern);
04392
04393                 // block
04394                 ndart++;
04395                 mconn->addArtic(nd, 1);
04396                 // discard nparts from nodes below td
04397                 mblk1.nartps = 2;
04398                 mconn->addBlock(&mblk1, blkt);
04399                 mblk1.init();
04400                 mblk1.nartps = 1;
04401
04402                 nart1 = 0;
04403                 nart2 = 1; // nd is an articulation point
04404             }
04405         } else if (bilow[td] == bidef[nd]) {
04406             // articulation point of a block
04407             mopil.nedges += conn;
04408             if (pd >= 0 || !isExternal(nd)) {
04409                 ndart++;
04410                 mconn->addArtic(nd, 1);
04411                 mblk1.nartps++;
04412                 mconn->addBlock(&mblk1, blkt);
04413                 mblk1.init();
04414                 mblk1.nartps = 1;
04415                 nart1 = 0;
04416                 nart2 = 1; // nd is an articulation point
04417             }
04418         }
04419     } else {
04420         // inside a block
04421     }

```

```

04423         mopil.nedges += conn;
04424     }
04425
04426     bilow[nd] = Min(bilow[nd], bilow[td]);
04427     mopi->nlegs += mopil.nlegs;
04428     mopi->next += mopil.next;
04429     mopi->loop += mopil.loop;
04430     mopi->nedges += mopil.nedges;
04431     mopi->ctloop += mopil.ctloop;
04432
04433     mblk->nartps += mblk1.nartps;
04434     mblk->loop += mblk1.loop;
04435
04436     *next += next1;
04437     *nart += nart1;
04438 }
04439
04440 } // end of for
04441
04442 // no connection except for 'pd'==>'nd'. Then 'nd' is an art. point
04443 if (nvisit == 0) {
04444     nart2 = 1;
04445 }
04446 *nart += nart2;
04447
04448 // add # loop inside counter terms
04449 if (newv) {
04450     if (pd >= 0) {
04451         mconn->pushNode(nd);
04452     }
04453     mopi->ctloop += Max(0, nodes[nd]->extloop);
04454 }
04455
04456 // 'nd' is the root node
04457 if (pd < 0) {
04458     if (isExternal(nd)) {
04459         if (td0 >= 0) {
04460             mconn->addArtic(td0, 1);
04461         }
04462         if (mconn->nopic > 0) {
04463             (mconn->opics[mconn->nopic-1].next)++;
04464         }
04465     } else {
04466         mconn->addOPic(mopi, opit0);
04467         if (ndart == 1) {
04468             // 'nd' has only 1 child
04469             // the root is not really an art. point
04470             (*nart)--;
04471             mconn->addArtic(nd, -1);
04472             (mconn->blocks[mconn->nblocks-1].nartps)--;
04473         }
04474     }
04475 }
04476 }
04477
04478 //-----
04479 BigInt MGraph::generate(void)
04480 {
04481     // Generate graphs
04482     // the generation process starts from 'connectClass'.
04483
04484     MNodeClass *cl;
04485
04486 #ifdef DEBUGM
04487     printf("MGraph::generate:\n");
04488     // opt->printLevel(2);
04489 #endif
04490     // Initial classification of nodes.
04491     cl = new MNodeClass(nNodes, nClasses);
04492     cl->init(clist, maxdeg, adjMat);
04493     cl->reorder(this);
04494     connectClass(cl);
04495
04496     delete cl;
04497
04498 #ifdef DEBUGM
04499     printf("MGraph::generate:end:%ld\n", cDiag);
04500 #endif
04501     return cDiag;
04502 }
04503
04504 //-----
04505 void MGraph::connectClass(MNodeClass *cl)
04506 {
04507     // Connect nodes in a class to others
04508
04509     int sc, sn;

```

```

04510     MNodeClass *xcl;
04511
04512     xcl = refineClass(cl);
04513
04514     #ifdef DEBUG3
04515         printf("connectClass\n");
04516         print();
04517     #endif
04518     if (xcl == NULL) {
04519     #ifdef MONITOR
04520         discardRefine++;
04521     #endif
04522     } else {
04523         if (xcl->flg0 >= xcl->nClasses) {
04524             newGraph(xcl);
04525         } else {
04526             sc = xcl->clord[xcl->flg0];
04527             sn = xcl->flist[sc];
04528             connectNode(xcl->flg0, sn, xcl);
04529         }
04530     }
04531     if (xcl != cl && xcl != NULL) {
04532         delete xcl;
04533     }
04534 }
04535
04536 //-----
04537 void MGraph::connectNode(int so, int ss, MNodeClass *cl)
04538 {
04539     int sc, sn;
04540
04541     sc = cl->clord[so];
04542     if (ss >= cl->flist[sc+1]) {
04543         connectClass(cl);
04544         return;
04545     }
04546
04547     #ifdef DEBUG3
04548         \printf("connectNode\n");
04549         print();
04550     #endif
04551     for (sn = ss; sn < cl->flist[sc+1]; sn++) {
04552         connectLeg(so, sn, so, sn, cl);
04553         return;
04554     }
04555 }
04556
04557 //-----
04558 void MGraph::connectLeg(int so, int sn, int to, int ts, MNodeClass *cl)
04559 {
04560     // Add one connection between two legs.
04561     // 1. select another node to connect
04562     // 2. determine multiplicity of the connection
04563
04564     int sc, tc, tn, maxself, nc2, nc, maxcon, tsl, ncm, tol;
04565
04566     sc = cl->clord[so];
04567
04568     #ifdef CHECK
04569     if (sn >= cl->flist[sc+1]) {
04570         erEnd("connectLeg : illegal control");
04571         return;
04572     }
04573     #endif
04574     #ifdef DEBUG3
04575         printf("connectLeg:0\n");
04576         print();
04577     #endif
04578
04579     // There remains no free legs in the node 'sn' : move to next node.
04580     if (nodes[sn]->freelg < 1) {
04581
04582         if (!isConnected()) {
04583         #ifdef MONITOR
04584             discardDisc++;
04585         #endif
04586         } else {
04587         #ifdef DEBUG3
04588             printf("call connectNode:1\n");
04589         #endif
04590             // next node in the current class.
04591             connectNode(so, sn+1, cl);
04592         }
04593         return;
04594     }
04595
04596     #ifdef DEBUG3

```

```

04597     printf("connectLeg:2\n");
04598     print();
04599 #endif
04600     // connect a free leg of the current node 'sn'.
04601     for (tol = to; tol < cl->nClasses; tol++) {
04602         tc = cl->clord[tol];
04603         if (tol == to) {
04604             tsl = ts;
04605         } else {
04606             tsl = cl->flist[tc];
04607         }
04608 #ifdef MINMAXLEG
04609         if (tsl >= nNodes) {
04610             continue;
04611         }
04612 #   ifdef DEBUG3
04613         printf("connectLeg:0: sn=%d, tsl=%d, ?(%d <= %d <= %d)\n",
04614             sn, tsl, nodes[sn]->cmindeg, nodes[tsl]->deg, nodes[sn]->cmaxdeg);
04615         printf("connectLeg:0: tsl=%d, sn=%d, ?(%d <= %d <= %d)\n",
04616             tsl, sn, nodes[tsl]->cmindeg, nodes[sn]->deg, nodes[tsl]->cmaxdeg);
04617 #   endif
04618         if ((nodes[sn]->cmindeg > 0 && nodes[tsl]->deg < nodes[sn]->cmindeg)
04619             || (nodes[sn]->cmaxdeg > 0 && nodes[tsl]->deg > nodes[sn]->cmaxdeg)
04620             || (nodes[tsl]->cmindeg > 0 && nodes[sn]->deg < nodes[tsl]->cmindeg)
04621             || (nodes[tsl]->cmaxdeg > 0 && nodes[sn]->deg > nodes[tsl]->cmaxdeg)) {
04622 #   ifdef DEBUG3
04623             printf("connectLeg:1: sn=%d, tsl=%d, !(%d <= %d <= %d)\n",
04624                 sn, tsl, nodes[sn]->cmindeg, nodes[tsl]->deg, nodes[sn]->cmaxdeg);
04625             printf("connectLeg:2: tsl=%d, sn=%d, !(%d <= %d <= %d)\n",
04626                 tsl, sn, nodes[tsl]->cmindeg, nodes[sn]->deg, nodes[tsl]->cmaxdeg);
04627 #   endif
04628             continue;
04629         }
04630 #ifdef DEBUG3
04631         printf("connectLeg:3: sn=%d, tsl=%d\n", sn, tsl);
04632 #endif
04633 #endif
04634         for (tn = tsl; tn < cl->flist[tc+1]; tn++) {
04635             if (sc == tc && sn > tn) {
04636                 continue;
04637             } else if (nodes[tn]->freelg < 1) {
04638                 continue;
04639             }
04640             // self-loop
04641             } else if (sn == tn) {
04642                 if (opt->values[GRCC_OPT_NoSelfLoop] > 0) {
04643                     continue;
04644                 }
04645             }
04646             if (nNodes > 1) {
04647                 // there are two or more nodes in the graph :
04648                 // avoid disconnected graph
04649                 maxself = Min((nodes[sn]->freelg)/2, (nodes[sn]->deg-1)/2);
04650             } else {
04651                 // there is only one node the graph.
04652                 maxself = nodes[sn]->freelg/2;
04653             }
04654             // If we can assume no tadpole, the following line can be used.
04655             if ((opt->values[GRCC_OPT_1PI] > 0
04656                 || opt->values[GRCC_OPT_NoTadpole] > 0) && nExtern > 2) {
04657                 maxself = Min((nodes[sn]->deg-2)/2, maxself);
04658             }
04659             // for the number of connection.
04660             for (nc2 = maxself; nc2 > 0; nc2--) {
04661                 nc = 2*nc2;
04662                 ncm = CLWIGHTD(nc);
04663                 adjMat[sn][sn] = nc;
04664                 nodes[sn]->freelg -= nc;
04665                 cl->incMat(sn, tn, ncm);
04666             }
04667             // next connection
04668             #ifdef DEBUG3
04669             printf("call connectLeg:1: %d--%d\n", sn, sn);
04670             #endif
04671             connectLeg(so, sn, tol, tn+1, cl);
04672             // restore the configuration
04673             cl->incMat(tn, sn, -ncm);
04674             adjMat[sn][sn] = 0;
04675             nodes[sn]->freelg += nc;
04676         }
04677     }
04678     // connections between different nodes.
04679     } else {
04680         // maximum possible connection number

```

```

04684         maxcon = Min(nodes[sn]->freelg, nodes[tn]->freelg);
04685
04686         // avoid disconnected graphs.
04687         if (nNodes > 2 && nodes[sn]->deg == nodes[tn]->deg) {
04688             maxcon = Min(maxcon, nodes[sn]->deg-1);
04689         }
04690
04691 #ifdef CHECK
04692         if ((adjMat[sn][tn] != 0) ||
04693             (adjMat[sn][tn] != adjMat[tn][sn])) {
04694             printf("*** inconsistent connection: sn=%d, tn=%d",
04695                 sn, tn);
04696             printAdjMat(cl);
04697             erEnd("inconsistent connection ");
04698         }
04699 #endif
04700
04701         // vary number of connections
04702         for (nc = maxcon; nc > 0; nc--) {
04703             ncm = CLWIGHTO(nc);
04704             adjMat[sn][tn] = nc;
04705             adjMat[tn][sn] = nc;
04706             nodes[sn]->freelg -= nc;
04707             nodes[tn]->freelg -= nc;
04708             cl->incMat(sn, tn, ncm);
04709             cl->incMat(tn, sn, ncm);
04710
04711             // next connection
04712 #ifdef DEBUG3
04713             printf("call connectLeg:2: %d--%d\n", sn, tn);
04714 #endif
04715             connectLeg(so, sn, tol, tn+1, cl);
04716
04717             // restore configuration
04718             cl->incMat(sn, tn, -ncm);
04719             cl->incMat(tn, sn, -ncm);
04720             adjMat[sn][tn] = 0;
04721             adjMat[tn][sn] = 0;
04722             nodes[sn]->freelg += nc;
04723             nodes[tn]->freelg += nc;
04724         }
04725     }
04726 }
04727 }
04728 }
04729
04730 //-----
04731 Bool MGraph::isOptM(void)
04732 {
04733     Bool ok = True;
04734
04735     opi      = (mconn->nopic == 1);
04736     opiloop  = (mconn->nlpopic <= 1);
04737     tadpole  = (mconn->ne0bridges >= ((nExtern == 1) ? 0 : 1));
04738     selfloop = (mconn->nselfloops > 0);
04739     tadblock = (mconn->nalblocks > 0);
04740     block    = (mconn->neblocks == 1);
04741     extself  = (mconn->nelbridges > 0);
04742
04743     if (opt->values[GRCC_OPT_1PI] > 0) {
04744         ok = ok && opi;
04745 #ifdef DEBUGM
04746         printf("isOptM: 1PI:%d nopic=%d\n", ok, mconn->nopic);
04747 #endif
04748     } else if (opt->values[GRCC_OPT_1PI] < 0) {
04749         ok = ok && !opi;
04750 #ifdef DEBUGM
04751         printf("isOptM: -1PI:%d nopic=%d\n", ok, mconn->nopic);
04752 #endif
04753     }
04754     if (opt->values[GRCC_OPT_NoExtSelf] > 0) {
04755         ok = ok && !extself;
04756 #ifdef DEBUGM
04757         printf("isOptM: NoExtSelf:%d, extself=%d\n", ok, extself);
04758 #endif
04759     } else if (opt->values[GRCC_OPT_NoExtSelf] < 0) {
04760         ok = ok && extself;
04761 #ifdef DEBUGM
04762         printf("isOptM: -NoExtSelf:%d, extself=%d\n", ok, extself);
04763 #endif
04764     }
04765     if (opt->values[GRCC_OPT_NoTadpole] > 0) {
04766         ok = ok && !tadpole;
04767 #ifdef DEBUGM
04768         printf("isOptM: NoTadPole:%d, tadpole=%d\n", ok, tadpole);
04769 #endif
04770     } else if (opt->values[GRCC_OPT_NoTadpole] < 0) {

```

```

04771         ok = ok && tadpole;
04772 #ifdef DEBUGM
04773     printf("isOptM: -NoTadPole:%d\n", ok);
04774 #endif
04775     }
04776     if (opt->values[GRCC_OPT_NoSelfLoop] > 0) {
04777 #ifdef DEBUGM
04778     printf("isOptM: NoSelfLoop:%d\n", ok);
04779 #endif
04780     } else if (opt->values[GRCC_OPT_NoSelfLoop] < 0) {
04781         ok = ok && selfloop;
04782 #ifdef DEBUGM
04783     printf("isOptM: -NoSelfLoop:%d\n", ok);
04784 #endif
04785     }
04786     if (opt->values[GRCC_OPT_No1PtBlock] > 0) {
04787         ok = ok && !tadblock;
04788 #ifdef DEBUGM
04789     printf("isOptM: No1PtBlock:%d\n", ok);
04790 #endif
04791     } else if (opt->values[GRCC_OPT_No1PtBlock] < 0) {
04792         ok = ok && tadblock;
04793 #ifdef DEBUGM
04794     printf("isOptM: -NoSelfLoop:%d\n", ok);
04795 #endif
04796     }
04797     if (opt->values[GRCC_OPT_Block] > 0) {
04798         ok = ok && block;
04799 #ifdef DEBUGM
04800     printf("isOptM: Block:%d\n", ok);
04801 #endif
04802     } else if (opt->values[GRCC_OPT_Block] < 0) {
04803         ok = ok && !block;
04804 #ifdef DEBUGM
04805     printf("isOptM: -Block:%d\n", ok);
04806 #endif
04807     }
04808 #ifdef DEBUGM
04809     printf("isOptM:%d\n", ok);
04810 #endif
04811     return ok;
04812 }
04813
04814 //-----
04815 void MGraph::newGraph(MNodeClass *cl)
04816 {
04817     // A new candidate diagram is obtained.
04818     // It may be a disconnected graph
04819     //
04820     // Things to be done in the steps followed by this function.
04821     // 1. convert data format which treat the edges as objects.
04822     // 2. definition of loop momenta to the edges.
04823     // 3. analysis of loop structure in a graph.
04824     // 4. assignment of particles to edges and vertices to nodes
04825     // 5. recalculate symmetry factor
04826
04827     int connected;
04828     MNodeClass *xcl;
04829     Bool ok;
04830
04831     ngen++;
04832
04833     // refine class and check ordering condition
04834     xcl = refineClass(cl);
04835     if (xcl == NULL) {
04836 #ifdef MONITOR
04837         discardRefine++;
04838 #endif
04839     }
04840     // check whether this is a connected graph or not.
04841     } else {
04842         connected = isConnected();
04843         if (!connected) {
04844 #ifdef MONITOR
04845             discardDisc++;
04846 #endif
04847         }
04848         // connected graph : check isomorphism of the graph
04849     } else {
04850         ngconn++;
04851         if (!isIsomorphic(xcl)) {
04852 #ifdef MONITOR
04853             discardIso++;
04854 #endif
04855         }
04856         // We got a new connected and a unique representation of a class.
04857     } else {

```

```

04858         biconnME();
04859         if (isOptM()) {
04860             egraph->fromMGraph(this);
04861             if (egraph->isOptE()) {
04862                 curcl->copy(xcl);
04863 #ifdef ORBITS
04864                 orbits->toOrbits();
04865 #endif
04866                 ok = True;
04867                 if (opt->outmg != NULL) {
04868                     if (opt->proc != NULL) {
04869                         egraph->mId = opt->proc->mgrcount + 1;
04870                         egraph->sId = opt->proc->agrcount + 1;
04871                     } else if (opt->sproc != NULL) {
04872                         egraph->mId = opt->sproc->mgrcount + 1;
04873                         egraph->sId = opt->sproc->agrcount + 1;
04874                     } else {
04875                         egraph->mId = cDiag;
04876                     }
04877 #ifdef DEBUG1
04878                     printf("call outmg\n");
04879                     egraph->model->prModel();
04880                     opt->print();
04881                     egraph->print();
04882 #endif
04883                     ok = (*(opt->outmg))(egraph, opt->argmg);
04884 #ifdef DEBUG
04885                     printf("ok=%d\n", ok);
04886 #endif
04887                 }
04888             }
04889             if (ok) {
04890                 // # generated graphs
04891                 cDiag++;
04892                 wscon.add(1, nsym*esym);
04893             }
04894             // # generated lPI graphs
04895             if (opi) {
04896                 wsopi.add(1, nsym*esym);
04897                 clPI++;
04898             }
04899             // # no tadpoles
04900             if (!tadpole) {
04901                 cNoTadpole++;
04902             }
04903             // # no tadblocks
04904             if (!tadblock) {
04905                 cNoTadBlock++;
04906                 if (opi) {
04907                     clPINoTadBlock++;
04908                 }
04909             }
04910         }
04911 #ifdef MONITOR
04912         printf("\n");
04913         printf("Graph : %ld (%ld) lPIComp=%d lPILoop=%d",
04914             cDiag, ngen, nlPIComps, nlPILoop);
04915         printf(" sym. factor = (%ld*%ld)\n", nsym, esym);
04916         printAdjMat(cl);
04917         // cl->printMat();
04918 #endif
04919 #ifdef ORBITS
04920         orbits->print();
04921 #endif
04922 #ifdef DEBUGM
04923         printf("refine: %ld\n",
04924             nCallRefine);
04925         printf("discarded for refinement: %ld\n",
04926             discardRefine);
04927         printf("discarded for disconnected: %ld\n",
04928             discardDisc);
04929         printf("discarded for duplication: %ld\n",
04930             discardIso);
04931 #endif
04932 #endif
04933         // go to next step
04934 #ifdef DEBUGM
04935         printf("newGraph:%ld:accepted\n", ngen);
04936 #endif
04937         opt->newMGraph(this);
04938     } else {
04939 #ifdef DEBUGM
04940         printf("deleted by setOutMG-function\n");
04941 #endif
04942     }
04943 } else {
04944 #ifdef DEBUGM

```



```

04945                 printf("newGraph:%ld:discardOptE\n", ngen);
04946 #endif
04947             }
04948         } else {
04949 #ifdef DEBUGM
04950             printf("newGraph:%ld:discardOptM\n", ngen);
04951 #endif
04952         }
04953     }
04954 }
04955 }
04956 if (xcl != cl && xcl != NULL) {
04957     delete xcl;
04958 }
04959 }
04960
04961 //=====
04962 // class MOrbits: orbits of nodes
04963 //-----
04964 MOrbits::MOrbits(void)
04965 {
04966     nOrbits = 0;
04967     nNodes  = 0;
04968 }
04969
04970 //-----
04971 MOrbits::~MOrbits(void)
04972 {
04973 }
04974
04975 //-----
04976 void MOrbits::print(void)
04977 {
04978     int c;
04979
04980     printf("Orbits : nOrbits=%d: nd2or=", nOrbits);
04981     prIntArray(nNodes, nd2or, "\n");
04982     for (c = 0; c < nOrbits; c++) {
04983         printf("      %2d: (%2d -- %2d) :", c, flist[c], flist[c+1]-1);
04984         prIntArray(flist[c+1]-flist[c], or2nd+flist[c], "\n");
04985     }
04986 }
04987
04988 //-----
04989 void MOrbits::initPerm(int nnodes)
04990 {
04991     int j;
04992
04993     nNodes = nnodes;
04994     for (j = 0; j < nNodes; j++) {
04995         nd2or[j] = j;
04996     }
04997 }
04998
04999 //-----
05000 void MOrbits::fromPerm(int *perm)
05001 {
05002     // update the orbits of nodes by the permutation
05003
05004     int j, j1, k, l;
05005
05006     for (j = 0; j < nNodes; j++) {
05007         if (nd2or[j] != j) {
05008             // not the first element of an old orbit
05009             continue;
05010         }
05011         // merge orbits to orbit 'j'
05012         k = j;
05013         for (j1 = 0; j1 < nNodes && (k = perm[k]) != j; j1++) {
05014             if (nd2or[k] == j) {
05015                 // 'k' is already in 'j'
05016                 ;
05017             } else {
05018                 // old orbit 'k' had several elements: merge to orbit 'j'
05019                 for (l = 0; l < nNodes; l++) {
05020                     if (nd2or[l] == k) {
05021                         nd2or[l] = j;
05022                     }
05023                 }
05024             }
05025         }
05026 #ifdef CHECK
05027         if (j1 >= nNodes) {
05028             printf("*** fromPerm: illegal control: j=%d, k=%d\n", j, k);
05029             printf("perm=");
05030             prIntArray(nNodes, perm, " nd2or=");
05031             prIntArray(nNodes, nd2or, "\n");

```

```

05032         erEnd("fromPerm: illegal control");
05033     }
05034 #endif
05035 }
05036 }
05037
05038 //-----
05039 void MORbits::toOrbits(void)
05040 {
05041     // fill elements from 'nd2or'
05042     int nd, nel, k, lor;
05043
05044     lor = -1;
05045     nel = 0;
05046     nOrbits = 0;
05047     for (nd = 0; nd < nNodes; nd++) {
05048         if (nd2or[nd] > lor) {
05049             // lowest element of a new orbit
05050             lor = nd2or[nd];
05051             flist[nOrbits++] = nel;
05052             for (k = nd; k < nNodes; k++) {
05053                 if (nd2or[k] == lor) {
05054                     or2nd[nel++] = k;
05055                 }
05056             }
05057         }
05058     }
05059     flist[nOrbits] = nNodes;
05060 }
05061
05062 //*****
05063 // n-edge connected components
05064 //=====
05065 // class MCOp: one n-edge connected component
05066 //-----
05067 MCOp::MCOp(void)
05068 {
05069     init();
05070 }
05071
05072 //-----
05073 MCOp::~MCOp(void)
05074 {
05075     nodes = NULL;
05076 }
05077
05078 //-----
05079 void MCOp::init(void)
05080 {
05081     nlegs = 0;
05082     nedges = 0;
05083     nnodes = 0;
05084     next = 0;
05085     loop = 0;
05086     ctloop = 0;
05087     nnodes = 0;
05088     nodes = NULL;
05089 }
05090
05091 //=====
05092 // class MCBridge
05093 //-----
05094 MCBridge::MCBridge(void)
05095 {
05096 }
05097
05098 //-----
05099 MCBridge::~MCBridge(void)
05100 {
05101 }
05102
05103 //=====
05104 // class MCBlock
05105 //-----
05106 MCBlock::MCBlock(void)
05107 {
05108     init();
05109 }
05110
05111 //-----
05112 MCBlock::~MCBlock(void)
05113 {
05114 }
05115
05116 //-----
05117 void MCBlock::init(void)
05118 {

```

```

05119     edges    = NULL;
05120     nmedges  = 0;
05121     nartps   = 0;
05122     loop     = 0;
05123 }
05124
05125 //=====
05126 // class MConn: table of n-edge connected components
05127 //-----
05128 MConn::MConn(int nnod, int nedg)
05129 {
05130     snodes = nnod;
05131     sedges = nedg;
05132
05133     opics   = new MCOp[snodes];
05134     bridges = new MCBridge[sedges];
05135     blocks  = new MCBlock[sedges];
05136     articuls = new int[snodes];
05137
05138     // workspace
05139     opisp    = new int[snodes];
05140     opistk   = new int[snodes];
05141     blksp    = new Edge2n[sedges];
05142     blkstk   = new Edge2n[sedges];
05143     init();
05144 }
05145
05146 //-----
05147 MConn::~MConn(void)
05148 {
05149     delete[] blkstk;
05150     delete[] blksp;
05151     delete[] opistk;
05152     delete[] opisp;
05153
05154     delete[] articuls;
05155     delete[] blocks;
05156     delete[] bridges;
05157     delete[] opics;
05158 }
05159
05160 //-----
05161 void MConn::init(void)
05162 {
05163     int j;
05164
05165     nopic      = 0;
05166     nlpopic    = 0;
05167     nctopic    = 0;
05168     nbridges   = 0;
05169     ne0bridges = 0;
05170     nelbridges = 0;
05171     nselfloops = 0;
05172
05173     nblocks    = 0;
05174     nalblocks  = 0;
05175     narticuls  = 0;
05176     neblocks   = 0;
05177
05178     opistkptra = 0;
05179     nopisp     = 0;
05180     blkstkptr  = 0;
05181     nblksp     = 0;
05182
05183     for (j = 0; j < snodes; j++) {
05184         articuls[j] = 0;
05185     }
05186 }
05187
05188 //-----
05189 void MConn::pushNode(int nd)
05190 {
05191     if (opistkptra >= snodes) {
05192         erEnd("MConn::pushNode: opistkptra >= snodes");
05193     }
05194     opistk[opistkptra++] = nd;
05195 }
05196
05197 //-----
05198 void MConn::pushEdge(int n0, int n1)
05199 {
05200     if (blkstkptr >= sedges) {
05201         erEnd("MConn::pushEdge: blkstkptr >= sedges");
05202     }
05203     if (n0 <= n1) {
05204         blkstk[blkstkptr][0] = n0;
05205         blkstk[blkstkptr][1] = n1;

```

```

05206     } else {
05207         blkstk[blkstkptr][0] = n1;
05208         blkstk[blkstkptr][1] = n0;
05209     }
05210     blkstkptr++;
05211 }
05212
05213 //-----
05214 void MConn::addOPic(MCOpi *mopi, int stp)
05215 {
05216     int j, nn;
05217
05218     if (nopic >= snodes) {
05219         erEnd("MConn::addOPic: table overflow");
05220     }
05221
05222     nn = opistkptr - stp;
05223     for (j = 0; j < nn; j++) {
05224         opisp[nopisp+j] = opistk[stp+j];
05225     }
05226     opics[nopic].nnodes = nn;
05227     opics[nopic].nlegs = mopi->nlegs;
05228     opics[nopic].nedges = mopi->nedges;
05229     opics[nopic].next = mopi->next;
05230     opics[nopic].loop = mopi->loop;
05231     opics[nopic].ctloop = mopi->ctloop;
05232     opics[nopic].nodes = opisp + nopisp;
05233     nopisp += nn;
05234     opistkptr = stp;
05235
05236     nopic++;
05237 }
05238
05239 //-----
05240 void MConn::addBridge(int n0, int n1, int nex, int nextot)
05241 {
05242     if (nbridges >= sedges) {
05243         erEnd("MConn::addBridge: table overflow");
05244     }
05245     if (n0 <= n1) {
05246         bridges[nbridges].nodes[0] = n0;
05247         bridges[nbridges].nodes[1] = n1;
05248     } else {
05249         bridges[nbridges].nodes[0] = n1;
05250         bridges[nbridges].nodes[1] = n0;
05251     }
05252     bridges[nbridges].next = Min(nex, nextot-nex);
05253     nbridges++;
05254 }
05255
05256 //-----
05257 void MConn::addArtic(int nd, int mul)
05258 {
05259     articuls[nd] += mul;
05260     if (articuls[nd] == mul) {
05261         narticuls++;
05262     } else if (articuls[nd] == 0) {
05263         narticuls--;
05264     }
05265 }
05266
05267 //-----
05268 void MConn::addBlock(MCBlock *mblk, int stp)
05269 {
05270     int j, nn;
05271
05272     if (nopic >= snodes) {
05273         erEnd("MConn::addOPic: table overflow");
05274     }
05275
05276     nn = blkstkptr - stp;
05277     for (j = 0; j < nn; j++) {
05278         blksp[nblksp+j][0] = blkstk[stp+j][0];
05279         blksp[nblksp+j][1] = blkstk[stp+j][1];
05280     }
05281     blocks[nblocks].edges = blksp + nblksp;
05282     blocks[nblocks].nmedges = nn;
05283     blocks[nblocks].nartps = mblk->nartps;
05284     blocks[nblocks].loop = mblk->loop;
05285     nblksp += nn;
05286     blkstkptr = stp;
05287
05288     nblocks++;
05289 }
05290
05291 //-----
05292 void MConn::addBlockSelf(int nd, int mult)

```

```

05293 {
05294     MCBlock mblk;
05295     int j, stp;
05296
05297     mblk.edges = NULL;
05298     mblk.nmedges = 0;
05299     mblk.nartps = 1;
05300     mblk.loop = 1;
05301
05302     for (j = 0; j < mult; j++) {
05303         stp = blkstkptr;
05304         pushEdge(nd, nd);
05305         addBlock(&mblk, stp);
05306     }
05307 }
05308
05309 //-----
05310 void MConn::print(void)
05311 {
05312     int j, k;
05313
05314     printf("+++ MConn object: snodes=%d, sedges=%d\n", snodes, sedges);
05315     printf("  nopic=%d, nlpopic=%d, nctopic=%d, nbridges=%d, "
05316           "ne0bridges=%d, nelbridges=%d\n",
05317           nopic, nlpopic, nctopic, nbridges, ne0bridges, nelbridges);
05318     printf("  nblocks=%d, neblocks=%d, nalblocks=%d, narticuls=%d, nselfloop=%d\n",
05319           nblocks, neblocks, nalblocks, narticuls, nselfloops);
05320
05321     printf("  lPI components (%d)\n", nopic);
05322     for (j = 0; j < nopic; j++) {
05323         printf("    %d: nleg=%d, nnodes=%d, nedge=%d, ",
05324               j, opics[j].nlegs, opics[j].nnodes, opics[j].nedges);
05325         printf("next=%d, loop=%d, ctlp=%d: [",
05326               opics[j].next, opics[j].loop, opics[j].ctloop);
05327         for (k = 0; k < opics[j].nnodes; k++) {
05328             printf(" %d", opics[j].nodes[k]);
05329         }
05330         printf("]\n");
05331     }
05332
05333     printf("  bridges (%d)\n", nbridges);
05334     if (nbridges > 0) {
05335         printf("    ");
05336         for (j = 0; j < nbridges; j++) {
05337             printf("(%d,%d)[ex=%d] ",
05338                   bridges[j].nodes[0], bridges[j].nodes[1], bridges[j].next);
05339         }
05340         printf("\n");
05341     }
05342
05343     printf("  blocks (%d)\n", nblocks);
05344     for (j = 0; j < nblocks; j++) {
05345         printf("    %d: nmedges=%d, nartps=%d, loop=%d: [",
05346               j, blocks[j].nmedges, blocks[j].nartps, blocks[j].loop);
05347         for (k = 0; k < blocks[j].nmedges; k++) {
05348             printf(" (%d,%d)", blocks[j].edges[k][0], blocks[j].edges[k][1]);
05349         }
05350         printf("]\n");
05351     }
05352
05353     printf("  articulation points (%d)\n", narticuls);
05354     if (narticuls > 0) {
05355         printf("    ");
05356         for (j = 0; j < snodes; j++) {
05357             if (articuls[j] != 0) {
05358 #if 0
05359                 printf("%d(%d) ", j, articuls[j]);
05360 #else
05361                 printf("%d ", j);
05362 #endif
05363             }
05364         }
05365         printf("\n");
05366     }
05367 }
05368
05369 //*****
05370 // sgroup.cc
05371 //=====
05372 // compile options
05373 //=====
05374 // table of group elements
05375 //-----
05376 SGroup::SGroup(void)
05377 {
05378     // should be called before graph generation
05379     nnodes = -1;

```

```

05380     nelem = 0;
05381     elem = NULL;
05382 }
05383
05384 //-----
05385 SGroup::~SGroup(void)
05386 {
05387     int j;
05388
05389     if (elem != NULL) {
05390         for (j = GRCC_MAXGROUP-1; j >= 0; j--) {
05391             if (elem[j] != NULL) {
05392                 delete[] elem[j];
05393                 elem[j] = NULL;
05394             }
05395         }
05396         delete[] elem;
05397         elem = NULL;
05398     }
05399 }
05400
05401 //-----
05402 void SGroup::print(void)
05403 {
05404     int j;
05405
05406     printf("SGroup : nnodes = %d, nelem=%ld, cgen=%d, csav=%d\n",
05407           nnodes, nelem, cgen, csav);
05408     printf("  eclass =");
05409     prIntArray(necclass, eclass, "\n");
05410     for (j = 0; j < nelem; j++) {
05411         printf("    %4d: ", j);
05412         prIntArray(nnodes, elem[j], "\n");
05413     }
05414 }
05415
05416 //-----
05417 void SGroup::newGroup(int nnd, int nclss, int *clss)
05418 {
05419     int j;
05420
05421     clearGroup();
05422     nnodes = nnd;
05423     necclass = nclss;
05424
05425     if (necclass > GRCC_MAXNODES) {
05426         erEnd("newGroup : too many classes (GRCC_MAXNODES)");
05427     }
05428     for (j = 0; j < necclass; j++) {
05429         eclass[j] = clss[j];
05430     }
05431
05432     nelem = 0;
05433     if (elem == NULL) {
05434         elem = new int*[GRCC_MAXGROUP];
05435         for (j = 0; j < GRCC_MAXGROUP; j++) {
05436             elem[j] = NULL;
05437         }
05438     }
05439 }
05440
05441 //-----
05442 void SGroup::clearGroup(void)
05443 {
05444     nelem = 0;
05445     csav = -1;
05446     cgen = -1;
05447 }
05448
05449 //-----
05450 void SGroup::delGroup(void)
05451 {
05452     int j;
05453
05454     for (j = 0; j < GRCC_MAXGROUP; j++) {
05455         if (elem[j] != NULL) {
05456             delete[] elem[j];
05457             elem[j] = NULL;
05458         }
05459     }
05460     delete[] elem;
05461     elem = NULL;
05462     clearGroup();
05463 }
05464
05465 //-----
05466 int *SGroup::genNext(void)

```

```

05467 {
05468     cgen = nextPerm(nnodes, necclass, eclass, pgr, pgq, permg, cgen);
05469     if (cgen < 0) {
05470         return NULL;
05471     }
05472     return permg;
05473 }
05474
05475 //-----
05476 void SGroup::addGroup(int *p)
05477 {
05478     int j;
05479
05480     if (nelem >= GRCC_MAXGROUP) {
05481         erEnd("addGroup : too many elements (GRCC_MAXGROUP)");
05482     }
05483     if (elem[nelem] == NULL) {
05484         elem[nelem] = new int[GRCC_MAXNODES];
05485     }
05486     for (j = 0; j < nnodes; j++) {
05487         elem[nelem][j] = p[j];
05488     }
05489     nelem++;
05490 }
05491
05492 //-----
05493 BigInt SGroup::nElem(void)
05494 {
05495     return nelem;
05496 }
05497
05498 //-----
05499 int *SGroup::nextElem(void)
05500 {
05501     if (nelem < 0) {
05502         cgen = nextPerm(nelem, necclass, eclass,
05503             psr, psq, perms, cgen);
05504         return perms;
05505     }
05506     if (cgen < 0) {
05507         cgen = 0;
05508     }
05509     if (cgen >= nelem) {
05510         return NULL;
05511     } else {
05512         return elem[cgen++];
05513     }
05514 }
05515
05516 //*****
05517 // egraph.cc
05518 // Generate scalar connected Feynman graphs.
05519 //=====
05520 static const char *GRCC_ND_names[] = GRCC_ND_NAMES;
05521 static const char *GRCC_ED_names[] = GRCC_ED_NAMES;
05522
05523 //=====
05524 // class ENode
05525 //-----
05526 ENode::ENode(void)
05527 {
05528     id      = -1;
05529     egraph  = NULL;
05530     edges   = NULL;
05531     maxdeg  = 0;
05532     deg     = 0;
05533
05534     klow    = NULL;
05535     extloop = 0;
05536     ndtype  = GRCC_ND_Undef;
05537
05538     intrct  = -1;
05539 }
05540
05541 //-----
05542 ENode::ENode(EGraph *egrph, int loops, int sdeg)
05543 {
05544     {
05545         id      = -1;
05546         egraph  = egrph;
05547         edges   = NULL;
05548         maxdeg  = sdeg;
05549         deg     = 0;
05550
05551         klow    = NULL;
05552         extloop = 0;
05553         ndtype  = GRCC_ND_Undef;
05554     }
05555 }

```

```

05554
05555     intrct = -1;
05556
05557     edges = new int[sdeg];
05558     klow = new int[loops+1];
05559 }
05560
05561 //-----
05562 ENode::~ENode(void)
05563 {
05564     if (klow != NULL) {
05565         delete[] klow;
05566         delete[] edges;
05567         klow = NULL;
05568         edges = NULL;
05569     }
05570 }
05571
05572 //-----
05573 void ENode::copy(ENode *en)
05574 {
05575     int d;
05576
05577     deg = en->deg;
05578     extloop = en->extloop;
05579     ndtype = en->ndtype;
05580     for (d = 0; d < deg; d++) {
05581         edges[d] = en->edges[d];
05582     }
05583 }
05584
05585 //-----
05586 void ENode::initAss(EGraph *egrph, int nid, int sdg)
05587 {
05588     id = nid;
05589     egraph = egrph;
05590     maxdeg = sdg;
05591 }
05592
05593 //-----
05594 void ENode::setId(EGraph *egrph, const int nid)
05595 {
05596     id = nid;
05597     egraph = egrph;
05598 }
05599
05600 //-----
05601 void ENode::setExtern(int typ, int pt)
05602 {
05603     ndtype = typ;
05604     intrct = pt;
05605 }
05606
05607 //-----
05608 void ENode::setType(int typ)
05609 {
05610     #ifdef CHECK
05611         if (ndtype != GRCC_ND_Undef && ndtype != typ) {
05612             fprintf(stderr, "*** ndtype is already defined : old=%d, new = %d\n",
05613                 ndtype, typ);
05614             erEnd("ndtype is already defined");
05615         }
05616     #endif
05617     ndtype = typ;
05618 }
05619
05620 //-----
05621 void ENode::print(void)
05622 {
05623     int j;
05624
05625     printf("Enode %d deg=%d, extl=%2d, intr=%d ",
05626         id, deg, extloop, intrct);
05627     if (egrph->bicount > 0) {
05628         printf("(%-8s) ", GRCC_ND_names[ndtype]);
05629     }
05630     printf("edge=[");
05631     for (j = 0; j < deg; j++) {
05632         printf(" %2d", edges[j]);
05633     }
05634     printf("]\n");
05635 }
05636
05637 //=====
05638 // class EEdge
05639 //-----
05640 EEdge::EEdge(void)

```



```

05641 {
05642     id      = -1;
05643     egraph  = NULL;
05644     nodes[0] = -1;
05645     nodes[1] = -1;
05646     // nlegs[0] = -1;
05647     // nlegs[1] = -1;
05648     ptcl    = GRCC_PT_Undef;
05649     deleted  = False;
05650
05651     edtype   = GRCC_ED_Undef;
05652     cut      = False;
05653
05654     emom = NULL;
05655     lmom = NULL;
05656 }
05657
05658 //-----
05659 EEdge::EEdge(EGraph *egrph, int nedges, int nloops)
05660 {
05661     id      = -1;
05662     egraph  = egrph;
05663     nodes[0] = -1;
05664     nodes[1] = -1;
05665     nlegs[0] = -1;
05666     nlegs[1] = -1;
05667     ptcl    = GRCC_PT_Undef;
05668     deleted  = False;
05669
05670     edtype   = GRCC_ED_Undef;
05671     cut      = False;
05672
05673     emom = new int[nedges];
05674     lmom = new int[nloops];
05675 }
05676
05677 //-----
05678 EEdge::~EEdge(void)
05679 {
05680     if (lmom != NULL) {
05681         delete[] lmom;
05682     }
05683     if (emom != NULL) {
05684         delete[] emom;
05685     }
05686 }
05687
05688 //-----
05689 void EEdge::copy(EEdge *ee)
05690 {
05691     nodes[0] = ee->nodes[0];
05692     nodes[1] = ee->nodes[1];
05693     id      = ee->id;
05694     ext     = ee->ext;
05695     dir     = ee->dir;
05696     edtype  = ee->edtype;
05697     cut     = ee->cut;
05698 }
05699
05700 //-----
05701 void EEdge::setId(EGraph *egrph, const int eid)
05702 {
05703     id      = eid;
05704     egraph  = egrph;
05705 }
05706
05707 //-----
05708 void EEdge::setType(int typ)
05709 {
05710     #ifdef CHECK
05711         if (edtype != GRCC_ED_Undef) {
05712             erEnd("edtype is already defined");
05713         }
05714     #endif
05715     edtype = typ;
05716 }
05717
05718 //-----
05719 void EEdge::setEMom(int nedges, int *em, int dir)
05720 {
05721     int e;
05722
05723     for (e = 0; e < nedges; e++) {
05724         if (em[e] != 0) {
05725             emom[e] = dir;
05726         }
05727     }

```

```

05728 }
05729
05730 //-----
05731 void EEdge::setLMom(int k, int dir)
05732 {
05733     lmom[k] = dir;
05734 }
05735
05736 //-----
05737 void EEdge::print(void)
05738 {
05739     int zero, n, k;
05740
05741     printf("Edge %2d ext=%d ptcl=%2d ", id, ext, ptcl);
05742     if (egraph == NULL || !egraph->assigned) {
05743         printf("[%d, %d]", nodes[0], nodes[1]);
05744     } else {
05745         printf("[(%d,%d), (%d,%d)]", nodes[0], nlegs[0], nodes[1], nlegs[1]);
05746     }
05747
05748     if (egraph != NULL && egraph->bicount > 0) {
05749         if (cut) {
05750             printf(" %-9s ", "Cut");
05751         } else {
05752             printf(" %-9s ", GRCC_ED_names[edtype]);
05753         }
05754         printf("c%2d: ", opicomp);
05755         printf("%s%d = ", (ext)?"Q":"p", id);
05756         zero = True;
05757         for (n = 0; n < egraph->nEdges; n++) {
05758             if (emom[n] != 0) {
05759                 prMomStr(emom[n], "Q", n);
05760                 zero = False;
05761             }
05762         }
05763         for (k = 0; k < egraph->nLoops; k++) {
05764             if (lmom[k] != 0) {
05765                 prMomStr(lmom[k], "k", k);
05766                 zero = False;
05767             }
05768         }
05769         if (zero) {
05770             printf(" 0");
05771         }
05772     }
05773     printf("\n");
05774 }
05775
05776 //=====
05777 // class EFLine
05778 //-----
05779 EFLine::EFLine(void)
05780 {
05781     ftype = FL_Open;
05782     fkind = 0;
05783     nlist = 0;
05784 }
05785
05786 //-----
05787 void EFLine::print(const char *msg)
05788 {
05789     if (ftype == FL_Open) {
05790         printf(" Open");
05791     } else if (ftype == FL_Closed) {
05792         printf(" Loop");
05793     } else {
05794         printf(" ?%d", ftype);
05795     }
05796     if (fkind == GRCC_PT_Dirac) {
05797         printf(" Dirac");
05798     } else if (fkind == GRCC_PT_Majorana) {
05799         printf(" Major");
05800     } else if (fkind == GRCC_PT_Ghost) {
05801         printf(" Ghost");
05802     } else {
05803         printf(" ?%d", fkind);
05804     }
05805     printf(" len=%d ", nlist);
05806     prIntArray(nlist, elist, msg);
05807 }
05808
05809 //=====
05810 // class EGraph
05811 //-----
05812 EGraph::EGraph(int nnodes, int nedges, int mxdeg)
05813 {
05814     // Arguments

```

```

05815 // nnodes : the number of nodes
05816 // nedges : the number of edges
05817 // mxdeg : the maximum number of degrees of nodes
05818
05819 int j;
05820
05821 opt      = NULL;
05822 mgraph   = NULL;
05823 econn    = NULL;
05824
05825 sNodes   = nnodes;
05826 sEdges   = nedges;
05827 sMaxdeg  = mxdeg;
05828 sLoops   = sEdges - sNodes + 1; // assume connected graph
05829
05830 pId      = -1;
05831 mId      = -1;
05832 aId      = -1;
05833 sId      = 0;
05834 gSubId   = -1;
05835 assigned = False;
05836
05837 nNodes   = sNodes;
05838 nEdges   = sEdges;
05839 nLoops   = 0;
05840 nExtern  = 0;
05841
05842 nsym     = 1;
05843 esym     = 1;
05844 extperm  = 1;
05845 nsym1    = 1;
05846 multp    = 1;
05847 totalc   = 0;
05848
05849 nodes = new ENode*[sNodes];
05850 for (j = 0; j < sNodes; j++) {
05851     nodes[j] = new ENode(this, sNodes, sMaxdeg);
05852 }
05853 edges = new EEdge*[sEdges];
05854 for (j = 0; j < sEdges; j++) {
05855     edges[j] = new EEdge(this, sEdges, sLoops);
05856 }
05857
05858 bidef    = new int[sNodes];
05859 bilow    = new int[sNodes];
05860 bicount  = -1;
05861
05862 extMom   = new int[sEdges+1];
05863
05864 fsign    = 1;
05865 nflines  = -1;
05866 for (j = 0; j < GRCC_MAXFLINES; j++) {
05867     flines[j] = NULL;
05868 }
05869
05870 }
05871
05872 //-----
05873 EGraph::~EGraph(void)
05874 {
05875     int j;
05876
05877     for (j = GRCC_MAXFLINES-1; j >= 0; j--) {
05878         if (flines[j] != NULL) {
05879             delete flines[j];
05880             flines[j] = NULL;
05881         }
05882     }
05883
05884     delete[] extMom;
05885
05886     delete[] bilow;
05887     delete[] bidef;
05888
05889     for (j = 0; j < sEdges; j++) {
05890         delete edges[j];
05891         edges[j] = NULL;
05892     }
05893     delete[] edges;
05894
05895     for (j = 0; j < sNodes; j++) {
05896         delete nodes[j];
05897         nodes[j] = NULL;
05898     }
05899     delete[] nodes;
05900
05901 }

```

```

05902
05903 //-----
05904 void EGraph::copy(EGraph *eg)
05905 {
05906     int nd, ed;
05907
05908 #ifdef CHECK
05909     if (sNodes < eg->nNodes || sEdges < eg->nEdges ||
05910         sMaxdeg < eg->sMaxdeg || sLoops < eg->nLoops) {
05911         erEnd("EGraph::copy: sizes are too small");
05912     }
05913 #endif
05914     model = eg->model;
05915     proc = eg->proc;
05916     sproc = eg->sproc;
05917     mgraph = eg->mgraph;
05918     opt = mgraph->opt;
05919     econn = mgraph->mconn;
05920
05921     pId = eg->pId;
05922     mId = eg->mId;
05923     gSubId = eg->gSubId;
05924     nNodes = eg->nNodes;
05925     nEdges = eg->nEdges;
05926     nExtern = eg->nExtern;
05927     nLoops = eg->nLoops;
05928
05929     for (nd = 0; nd < nNodes; nd++) {
05930         nodes[nd]->copy(eg->nodes[nd]);
05931     }
05932     for (ed = 0; ed < nEdges; ed++) {
05933         edges[ed]->copy(eg->edges[ed]);
05934     }
05935
05936     nsym = eg->nsym;
05937     esym = eg->esym;
05938     extperm = eg->extperm;
05939     nsym1 = eg->nsym1;
05940     multp = eg->multp;
05941
05942     bicount = -1;
05943 }
05944
05945 //-----
05946 void EGraph::setExtLoop(int nd, int val)
05947 {
05948     // set the node 'nd' being an external node (-1) or a looped vertex
05949
05950     nodes[nd]->extloop = val;
05951 }
05952
05953 //-----
05954 void EGraph::endSetExtLoop(void)
05955 {
05956     // end of calling 'setExtLoop'.
05957
05958     int n;
05959
05960     nExtern = 0;
05961     for (n = 0; n < nNodes; n++) {
05962         if (isExternal(n)) {
05963             nExtern++;
05964         }
05965     }
05966 }
05967
05968 //-----
05969 void EGraph::fromDGraph(DGraph *dg)
05970 {
05971     int deg[GRCC_MAXNODES];
05972     int maxdeg, nextern, nconn, tc;
05973     int n0, nl, e;
05974
05975     if (dg->nnodes > GRCC_MAXNODES) {
05976         fprintf(GRCC_Stderr, "*** too many nodes\n");
05977         exit(1);
05978     }
05979     if (dg->nedges > GRCC_MAXEDGES) {
05980         fprintf(GRCC_Stderr, "*** too many edges\n");
05981         exit(1);
05982     }
05983
05984     for (e = 0; e < dg->nedges; e++) {
05985         if (dg->edges[e][0] >= dg->nnodes || dg->edges[e][0] >= dg->nnodes) {
05986             fprintf(GRCC_Stderr, "*** undefined node:");
05987             fprintf(GRCC_Stderr, "edge[%d] = {%d, %d}\n",
05988                 e, dg->edges[e][0], dg->edges[e][0]);

```

```

05989         exit(1);
05990     }
05991 }
05992
05993 for (n0 = 0; n0 < dg->nnodes; n0++) {
05994     deg[n0] = 0;
05995 }
05996 for (e = 0; e < dg->nedges; e++) {
05997     deg[dg->edges[e][0]]++;
05998     deg[dg->edges[e][1]]++;
05999 }
06000 maxdeg = 0;
06001 nextern = 0;
06002 for (n0 = 0; n0 < dg->nnodes; n0++) {
06003     maxdeg = Max(maxdeg, deg[n0]);
06004     if (deg[n0] == 1) {
06005         nextern++;
06006     } if (deg[n0] < 0) {
06007         fprintf(GRCC_Stderr, "+++ node %d is isolated\n", n0);
06008     }
06009 }
06010
06011 mgraph = NULL;
06012 model = NULL;
06013 proc = NULL;
06014 sproc = NULL;
06015 opt = NULL;
06016
06017 nNodes = dg->nnodes;
06018 nEdges = dg->nedges;
06019 nLoops = 0;
06020 nExtern = nextern;
06021
06022 pId = 0;
06023 mId = 0;
06024 aId = 0;
06025 gSubId = 0;
06026 nsym = 1;
06027 nsym1 = 1;
06028 esym = 1;
06029 extperm = 1;
06030 multp = 1;
06031 // maxdeg = maxdeg;
06032
06033 for (n0 = 0; n0 < dg->nnodes; n0++) {
06034     nodes[n0]->deg = 0;
06035     nodes[n0]->extloop = dg->nodes[n0].extloop;
06036 }
06037 for (e = 0; e < dg->nedges; e++) {
06038     n0 = dg->edges[e][0];
06039     n1 = dg->edges[e][1];
06040     edges[e]->nodes[0] = n0;
06041     edges[e]->nodes[1] = n1;
06042     edges[e]->ext = (isExternal(n0) || isExternal(n1));
06043
06044     nodes[n0]->edges[nodes[n0]->deg++] = I2Vedge(e, -1);
06045     nodes[n1]->edges[nodes[n1]->deg++] = I2Vedge(e, +1);
06046 }
06047 for (n0 = 0; n0 < dg->nnodes; n0++) {
06048     nodes[n0]->setId(this, n0);
06049     nodes[n0]->intrct = dg->nodes[n0].intrct;
06050 }
06051
06052 for (e = 0; e < nEdges; e++) {
06053     edges[e]->setId(this, e);
06054 }
06055 tc = 0;
06056 for (n0 = 0; n0 < nNodes; n0++) {
06057     if (isExternal(n0)) {
06058         setExtern(n0, 1, GRCC_ND_Initial);
06059     } else {
06060         tc += 2*nodes[n0]->extloop + nodes[n0]->deg - 2;
06061     }
06062 }
06063 totalc = tc;
06064
06065 // get the number of connected components
06066 nconn = connComp();
06067 nLoops = nEdges - nNodes + nconn;
06068
06069 bicount = -1;
06070 }
06071
06072
06073 //-----
06074 void EGraph::fromMGraph(MGraph *mg)
06075 {

```

```

06076 // construct EGraph from adjacency matrix.
06077 // This function should be called after
06078 // EGraph(), setExtLoop() and endSetExtLoop().
06079
06080 int n0, n1, ed, e;
06081 int j, n1, nf;
06082 PNodeClass *pnc;
06083 int k;
06084
06085 #ifndef CHECK
06086 if (sNodes < mg->nNodes || sEdges < mg->nEdges || sMaxdeg < mg->maxdeg) {
06087     erEnd("EGraph::fromMGraph: sizes are too small");
06088 }
06089 if (sLoops < mg->nLoops) {
06090     printf("too small sLoops : %d < %d\n", sLoops, mg->nLoops);
06091     erEnd("too small sLoops");
06092 }
06093 #endif
06094 mgraph = mg;
06095 sproc = mgraph->opt->sproc;
06096 if (sproc == NULL) {
06097     proc = NULL;
06098     model = NULL;
06099 } else {
06100     proc = sproc->proc;
06101     model = sproc->model;
06102 }
06103 opt = mgraph->opt;
06104 econn = mgraph->mconn;
06105
06106 nNodes = mg->nNodes;
06107 nEdges = mg->nEdges;
06108 nLoops = mg->nLoops;
06109 nExtern = mg->nExtern;
06110 pId = mg->pId;
06111 mId = mg->cDiag;
06112 aId = -1;
06113 if (opt->proc != NULL) {
06114     mId = mg->opt->proc->mgrcount;
06115     sId = mg->opt->proc->agrcount;
06116 } else if (mg->opt->sproc != NULL) {
06117     mId = mg->opt->sproc->mgrcount;
06118     sId = mg->opt->sproc->agrcount;
06119 } else {
06120     mId = mg->cDiag;
06121     sId = mg->cDiag;
06122 }
06123 gSubId = 0;
06124 nsym = mg->nsym;
06125 esym = mg->esym;
06126 extperm = 1;
06127 nsym1 = nsym;
06128 multp = 1;
06129 maxdeg = mg->maxdeg;
06130
06131 totalc = 0;
06132 if (proc != NULL) {
06133     for (j = 0; j < proc->model->ncouple; j++) {
06134         totalc += proc->clist[j];
06135     }
06136 }
06137 for (ed = 0; ed < nEdges; ed++) {
06138     edges[ed]->edtype = GRCC_ED_Undef;
06139 }
06140
06141 ed = 0;
06142 for (n0 = 0; n0 < nNodes; n0++) {
06143     nodes[n0]->deg = 0;
06144 }
06145 for (n0 = 0; n0 < nNodes; n0++) {
06146     for (e = 0; e < mg->adjMat[n0][n0]/2; e++, ed++) {
06147         edges[ed]->nodes[0] = n0;
06148         edges[ed]->nodes[1] = n0;
06149         nodes[n0]->edges[nodes[n0]->deg++] = I2Vedge(ed, -1);
06150         nodes[n0]->edges[nodes[n0]->deg++] = I2Vedge(ed, +1);
06151         edges[ed]->ext = isExternal(n0);
06152     }
06153     for (n1 = n0+1; n1 < nNodes; n1++) {
06154         for (e = 0; e < mg->adjMat[n0][n1]; e++, ed++) {
06155             edges[ed]->nodes[0] = n0;
06156             edges[ed]->nodes[1] = n1;
06157             nodes[n0]->edges[nodes[n0]->deg++] = I2Vedge(ed, -1);
06158             nodes[n1]->edges[nodes[n1]->deg++] = I2Vedge(ed, +1);
06159             edges[ed]->ext = (isExternal(n0) || isExternal(n1));
06160         }
06161     }
06162 }

```

```

06163 #ifdef CHECK
06164     if (ed != nEdges) {
06165         printf("*** EGraph::init: ed=%d != nEdges=%d\n",
06166             ed, nEdges+1);
06167         erEnd("EGraph::init: illegal connection");
06168     }
06169 #endif
06170
06171     for (j = 0; j < nNodes; j++) {
06172         nodes[j]->setId(this, j);
06173         nodes[j]->intrct = mg->nodes[j]->extloop;
06174     }
06175
06176     for (j = 0; j < nEdges; j++) {
06177         edges[j]->setId(this, j);
06178     }
06179
06180     ni = 0;
06181     nf = 0;
06182     if (proc != NULL) {
06183         ni = proc->ninitl;
06184         for (j = 0; j < ni; j++) {
06185             setExtern(j, proc->initlPart[j], GRCC_ND_Initial);
06186         }
06187
06188         nf = proc->nfinal;
06189         for (j = 0; j < nf; j++) {
06190             setExtern(j+ni, proc->finalPart[j], GRCC_ND_Final);
06191         }
06192         nExtern = ni + nf;
06193     } else if (sproc != NULL) {
06194         pnc = sproc->pnclass;
06195         for (j = 0; j < sproc->pnclass->nclass; j++) {
06196             if (pnc->type[j] == GRCC_AT_Initial || pnc->type[j] == GRCC_AT_External) {
06197                 for (k = pnc->cl2nd[j]; k < pnc->cl2nd[j+1]; k++) {
06198                     setExtern(j, k, GRCC_ND_Initial);
06199                     ni++;
06200                 }
06201             } else if (sproc->pnclass->type[j] == GRCC_AT_Final) {
06202                 for (k = pnc->cl2nd[j]; k < pnc->cl2nd[j+1]; k++) {
06203                     setExtern(j, k, GRCC_ND_Final);
06204                     nf++;
06205                 }
06206             }
06207         }
06208         nExtern = ni + nf;
06209     }
06210
06211     totalc = 0;
06212     for (j = 0; j < nNodes; j++) {
06213         if (!isExternal(j)) {
06214             totalc += 2*nodes[j]->extloop + nodes[j]->deg - 2;
06215         }
06216     }
06217
06218     bicount = -1;
06219 }
06220
06221 //-----
06222 ENode *EGraph::setExtern(int n0, int pt, int ndtyp)
06223 {
06224     ENode *nd;
06225     int npt, ept, eg;
06226
06227     nd = nodes[n0];
06228     if (nd->deg != 1) {
06229         if (prlevel > 0) {
06230             fprintf(GRCC_Stderr, "*** illegal external particle %d, %d, %d\n",
06231                 n0, pt, ndtyp);
06232         }
06233         erEnd("illegal external particle");
06234     }
06235
06236     // particle comes into the node
06237     if (ndtyp == GRCC_ND_Initial) {
06238         npt = pt;
06239     } else if (ndtyp == GRCC_ND_Final) {
06240         npt = -pt;
06241     } else {
06242         npt = 0;
06243         if (prlevel > 0) {
06244             fprintf(GRCC_Stderr, "*** illegal type of an external particle : "
06245                 "node = %d, particle = %d, type = %d", n0, pt, ndtyp);
06246         }
06247         erEnd("illegal type of an external particle");
06248     }
06249 }

```

```

06250 // edge attached to the external node
06251 eg = n0;
06252
06253 // particle flows on e from leg=0 to 1.
06254 if (model != NULL) {
06255     npt = model->normalParticle(npt);
06256     ept = model->normalParticle(-npt);
06257 } else {
06258     npt = 0;
06259     ept = 0;
06260 }
06261
06262 nd->setExtern(ndtyp, npt);
06263 edges[eg]->ptcl = ept;
06264
06265 return nd;
06266 }
06267
06268 //-----
06269 void EGraph::print(void)
06270 {
06271     // print the EGraph
06272
06273     int nd, ed, nlp;
06274
06275     nlp = nEdges - nNodes + 1;
06276     printf("\nEGraph\n");
06277     printf("    pId=%d, gSubId=%ld, mId=%ld, aId=%ld, sId=%ld\n",
06278            pId, gSubId, mId, aId, sId);
06279     printf("    sNodes=%d, sEdges=%d, sMaxdeg=%d, sLoops=%d\n",
06280            sNodes, sEdges, sMaxdeg, sLoops);
06281     printf("    nNodes=%d, nEdges=%d, nExtern=%d, nLoops=%d, totalc=%d\n",
06282            nNodes, nEdges, nExtern, nlp, totalc);
06283     printf("    ");
06284     if (model == NULL) {
06285         printf("model=NULL,");
06286     } else {
06287         printf("model=\"%s\",", model->name);
06288     }
06289     if (proc == NULL) {
06290         printf("proc=NULL,");
06291     } else {
06292         printf("proc=%d,", proc->id);
06293     }
06294     if (sproc == NULL) {
06295         printf("sproc=NULL,");
06296     } else {
06297         printf("sproc=%d,", sproc->id);
06298     }
06299     printf("\n");
06300     printf("    assigned=%d, sym = (%ld * %ld) ",
06301            assigned, nsym, esym);
06302     printf("extperm=%ld, nsym1=%ld, multp=%ld\n",
06303            extperm, nsym1, multp);
06304
06305     printf(" Nodes\n");
06306     for (nd = 0; nd < nNodes; nd++) {
06307         if (isExternal(nd)) {
06308             printf("    %2d Extern ", nd);
06309         } else {
06310             printf("    %2d Vertex ", nd);
06311         }
06312         nodes[nd]->print();
06313     }
06314     printf(" Edges\n");
06315     for (ed = 0; ed < nEdges; ed++) {
06316         if (edges[ed]->ext) {
06317             printf("    %2d Extern ", ed);
06318         } else {
06319             printf("    %2d Intern ", ed);
06320         }
06321         edges[ed]->print();
06322     }
06323     if (bicount > 0) {
06324         printf(" Biconn: nopicomp=%d, nopi2p=%d, opi2plp=%d, nadj2ptv=%d\n",
06325                nopicomp, nopi2p, opi2plp, nadj2ptv);
06326     }
06327     printf("\n");
06328
06329     if (nflines > 0) {
06330         prFLines();
06331     }
06332 }
06333 //-----
06334 void EGraph::prFLines(void)
06335 {
06336     int j;

```



```

06337
06338     printf("   Fermion lines %d, sign=%d (mId=%ld, aId=%ld)\n",
06339           nflines, fsign, mId, aId);
06340     for (j = 0; j < nflines; j++) {
06341         printf("%4d ", j);
06342         flines[j]->print("\n");
06343     }
06344 }
06345
06346 //-----
06347 int EGraph::dirEdge(int n, int e)
06348 {
06349     if (nodes[n]->edges[e] > 0) {
06350         return 1;
06351     } else {
06352         return -1;
06353     }
06354 }
06355
06356 //-----
06357 int EGraph::cmpMom(int *lm0, int *em0, int *lm1, int *em1)
06358 {
06359     int lp, ed, cmp, sgn = 0;
06360
06361     for (lp = 0; lp < nLoops; lp++) {
06362         if (lm0[lp] != 0) {
06363             if (lm1[lp] == 0) {
06364                 return 1;
06365             } else if (sgn == 0) {
06366                 sgn = lm0[lp]*lm1[lp];    // cancel first non-zero elements
06367 #ifdef CHECK
06368                 if (Abs(sgn) != 1) {
06369                     erEnd("cmpMom : illegal element of lmom");
06370                 }
06371 #endif
06372             } else {
06373                 cmp = lm0[lp] - sgn*lm1[lp];
06374                 if (cmp != 0) {
06375                     return cmp;
06376                 }
06377             }
06378         } else if (lm1[lp] != 0) {    // lm0[lp] == 0
06379             return -1;
06380         }
06381     }
06382     for (ed = 0; ed < nEdges; ed++) {
06383         if (em0[ed] != 0) {
06384             if (em1[ed] == 0) {
06385                 return 1;
06386             } else if (sgn == 0) {
06387                 sgn = em0[ed]*em1[ed];    // cancel first non-zero elements
06388 #ifdef CHECK
06389                 if (Abs(sgn) != 1) {
06390                     erEnd("cmpMom : illegal element of emom");
06391                 }
06392 #endif
06393             } else {
06394                 cmp = em0[ed] - sgn*em1[ed];
06395                 if (cmp != 0) {
06396                     return cmp;
06397                 }
06398             }
06399         } else if (em1[ed] != 0) {    // em0[ed] == 0
06400             return -1;
06401         }
06402     }
06403     return 0;
06404 }
06405
06406 //-----
06407 int EGraph::groupLMom(int *grp, int *ed2gr)
06408 {
06409     // grp[g] : the number of elements in the group g (in [0,...,(ngrp-1)])
06410     // ed2gr[ed] : group number of edge ed.
06411
06412     int ed0, ed1, cmp, ngrp = -1;
06413
06414     for (ed0 = 0; ed0 < nEdges; ed0++) {
06415         ed2gr[ed0] = -1;
06416         grp[ed0] = 0;
06417     }
06418
06419     for (ed0 = 0; ed0 < nEdges; ed0++) {
06420         if (ed2gr[ed0] < 0) {
06421             ngrp++;
06422             ed2gr[ed0] = ngrp;
06423             grp[ngrp]++;

```

```

06424         for (ed1 = ed0+1; ed1 < nEdges; ed1++) {
06425             cmp = cmpMom(edges[ed0]->lmom, edges[ed0]->emom,
06426                     edges[ed1]->lmom, edges[ed1]->emom);
06427             if (cmp == 0) {
06428                 ed2gr[ed1] = ngrp;
06429                 grp[ngrp]++;
06430             }
06431         }
06432     }
06433 }
06434 ngrp++;
06435
06436 return ngrp;
06437 }
06438
06439 //-----
06440 Bool EGraph::isOptE(void)
06441 {
06442     EGraph edupv = EGraph(sNodes, sEdges, sMaxdeg);
06443     EGraph *edup = &edupv;
06444     int grp[GRCC_MAXEDGES], ed2gr[GRCC_MAXEDGES];
06445     int g, ed, ngrp;
06446     Bool ok;
06447     int minopi2p;
06448
06449     if (opt->values[GRCC_OPT_No2PtL1PI] == 0
06450         && opt->values[GRCC_OPT_NoAdj2PtV] == 0) {
06451         return True;
06452     }
06453
06454     for (ed = 0; ed < nEdges; ed++) {
06455         edges[ed]->cut = False;
06456     }
06457
06458     biconnE();
06459
06460     if (opt->values[GRCC_OPT_NoAdj2PtV] > 0) {
06461         if (nadj2ptv > 0) {
06462             return False;
06463         }
06464     } else if (opt->values[GRCC_OPT_NoAdj2PtV] < 0) {
06465         if (nadj2ptv < 1) {
06466             return False;
06467         }
06468     }
06469     if (opt->values[GRCC_OPT_No2PtL1PI] == 0) {
06470         return True;
06471     }
06472
06473     if (nExtern == 2 && nopicomp == 1) {
06474         minopi2p = 2;
06475     } else {
06476         minopi2p = 1;
06477     }
06478
06479     if (opi2plp > 0 && nopi2p >= minopi2p) {
06480         if (opt->values[GRCC_OPT_No2PtL1PI] > 0) {
06481             return False;
06482         } else if (opt->values[GRCC_OPT_No2PtL1PI] < 0) {
06483             return True;
06484         }
06485     }
06486
06487     edup->copy(this);
06488     ngrp = groupLMom(grp, ed2gr);
06489
06490     for (g = 0; g < ngrp; g++) {
06491         ok = True;
06492         if (grp[g] < 2) {
06493             continue;
06494         }
06495         for (ed = 0; ed < nEdges; ed++) {
06496             if (ed2gr[ed] == g) {
06497                 if (edges[ed]->ext) {
06498                     ok = False;
06499                     break;
06500                 }
06501                 edup->edges[ed]->cut = True;
06502             } else {
06503                 edup->edges[ed]->cut = False;
06504             }
06505         }
06506         if (ok) {
06507             edup->gSubId++;
06508             edup->biconnE();
06509
06510             if (edup->nExtern == 2 && edup->nopicomp == 1) {

```

```

06511         minopi2p = 2;
06512     } else {
06513         minopi2p = 1;
06514     }
06515
06516     if (edup->opi2plp > 0 && edup->nopi2p >= minopi2p) {
06517         if (opt->values[GRCC_OPT_No2PtL1PI] > 0) {
06518             return False;
06519         } else if (opt->values[GRCC_OPT_No2PtL1PI] < 0) {
06520             return True;
06521         }
06522     }
06523 }
06524 }
06525
06526 if (opt->values[GRCC_OPT_No2PtL1PI] > 0) {
06527     return True;
06528 } else if (opt->values[GRCC_OPT_No2PtL1PI] < 0) {
06529     return False;
06530 } else {
06531     erEnd("isOptE: illegal control");
06532     return False;
06533 }
06534 }
06535
06536 //-----
06537 int EGraph::findRoot(void)
06538 {
06539     int root, vr, er, e, nd0, nd1;
06540
06541     root = -1;
06542     vr = -1;
06543     er = -1;
06544     for (e = 0; e < nEdges; e++) {
06545         if (edges[e]->cut || edges[e]->edtype == GRCC_ED_Deleted) {
06546             continue;
06547         }
06548         if (edges[e]->visited) {
06549             continue;
06550         }
06551         if (root < 0) {
06552             if (edges[e]->ext) {
06553                 nd0 = edges[e]->nodes[0];
06554                 nd1 = edges[e]->nodes[1];
06555                 if (isExternal(nd0) && !isExternal(nd1)) {
06556                     root = nd1;
06557                     break;
06558                 } else if (!isExternal(nd0) && isExternal(nd1)) {
06559                     root = nd0;
06560                     break;
06561                 }
06562             }
06563         }
06564         if (er < 0) {
06565             er = edges[e]->nodes[0];
06566         }
06567         if (vr < 0) {
06568             vr = edges[e]->nodes[0];
06569         }
06570     }
06571     // if root >= 0 : vertex adjacent to an external node
06572     if (root < 0) {
06573         // if vr >= 0 : a vertex
06574         if (vr >= 0) {
06575             root = vr;
06576         }
06577         // no vertex, only external nodes;
06578     } else {
06579         root = er;
06580     }
06581 }
06582 return root;
06583 }
06584
06585 //-----
06586 int EGraph::connComp(void)
06587 {
06588     // Count the number of connected component
06589     // If a connected component without free leg is not the whole graph then
06590     // return False, otherwise return True.
06591
06592     int j, ncc, nelem;
06593
06594     for (j = 0; j < nNodes; j++) {
06595         nodes[j]->visited = -1;
06596     }
06597     ncc = 0;    // # connected components

```

```

06598     nelem = 0;    // # visited nodes
06599     while (nelem < nNodes) {
06600         for (j = 0; j < nNodes; j++) {
06601             if (nodes[j]->visited < 0) {
06602                 break;
06603             }
06604         }
06605         if (j >= nNodes) {
06606             break;
06607         }
06608         nelem += connVisit(j, ncc);
06609         ncc++;
06610     }
06611     if (nelem != nNodes) {
06612         erEnd("EGraph::connComp: illegal control");
06613     }
06614     return ncc;
06615 }
06616
06617 //-----
06618 int EGraph::connVisit(int nd, int ncc)
06619 {
06620     // Visiting connected node used for 'connComp'
06621
06622     int e, en, nn, j, nelem;
06623
06624     nelem = 1;
06625     nodes[nd]->visited = ncc;
06626     for (e = 0; e < nodes[nd]->deg; e++) {
06627         en = V2Iedge(nodes[nd]->edges[e]);
06628         for (j = 0; j < 2; j++) {
06629             nn = edges[en]->nodes[j];
06630             if (nodes[nn]->visited < 0) {
06631                 nelem += connVisit(nn, ncc);
06632             }
06633         }
06634     }
06635     return nelem;
06636 }
06637
06638 //-----
06639 void EGraph::biconnE(void)
06640 {
06641     int e, ie, root;
06642     int extlst[GRCC_MAXEDGES], intlst[GRCC_MAXEDGES];
06643     int opiext, opiloop;
06644
06645     nadj2ptv = 0;
06646     for (e = 0; e < nEdges; e++) {
06647         if (nodes[edges[e]->nodes[0]]->deg == 2 &&
06648             nodes[edges[e]->nodes[1]]->deg == 2 &&
06649             edges[e]->nodes[0] != edges[e]->nodes[1]) {
06650             nadj2ptv++;
06651         }
06652     }
06653
06654     biinitE();    // initialization
06655     bconn = 0;    // the number of connected components
06656
06657     opiext = 0;
06658     opiloop = 0;
06659
06660     for (e = 0; e < nEdges; e++) {
06661         // root of a spanning tree to be searched.
06662         root = findRoot();
06663         if (root < 0) {
06664             // no root : end of search
06665             break;
06666         }
06667
06668         // recursive search
06669         bisearchE(root, extlst, intlst, &opiext, &opiloop);
06670
06671         // opi2plp = Max(opi2plp, opiloop);    // ???
06672
06673         // adjust the #external for the root
06674         if (isExternal(root)) {
06675             opiext++;
06676         }
06677
06678         // 2 point function
06679         if (opiext == 2) {
06680             opi2plp = Max(opi2plp, opiloop);
06681             nopi2p++;
06682         }
06683         for (ie = 0; ie < nEdges; ie++) {
06684             if (intlst[ie]) {

```

```

06685         edges[ie]->opicomp = nopicomp;
06686     }
06687 }
06688     nopicomp++;
06689
06690     bconn++;    // the number of connected component
06691 }
06692
06693 // momentum conservation of external particles
06694 extMomConsv();
06695
06696 #ifdef CHECK
06697     chkMomConsv();
06698 #endif
06699 #ifdef DEBUG
06700     printf("biconnE:opiext=%d, opiloop=%d, opi2plp=%d, nopi2p=%d, nopicomp=%d, bconn=%d\n",
06701         opiext, opiloop, opi2plp, nopi2p, nopicomp, bconn);
06702 #endif
06703
06704     return;
06705 }
06706
06707 //-----
06708 void EGraph::biinitE(void)
06709 {
06710     int n, j, e;
06711
06712     bicount = 0;                // counter of visiting node
06713     loopm   = 0;                // # of independent loop momentum
06714
06715     nopicomp = 0;
06716     opi2plp  = 0;
06717     nopi2p   = 0;
06718
06719     for (n = 0; n < nNodes; n++) {
06720         bidef[n] = -1;
06721         bilow[n] = -1;
06722         nodes[n]->ndtype = GRCC_ND_Undef;
06723         for (j = 0; j < nLoops; j++) {
06724             nodes[n]->klow[j] = -1;
06725         }
06726     }
06727
06728     for (e = 0; e < nEdges; e++) {
06729         // edges[e]->cut is defined before
06730         edges[e]->visited = False;
06731         edges[e]->conid   = -1;                // connected component
06732         edges[e]->edtype  = GRCC_ED_Undef;
06733         edges[e]->opicomp = -1;
06734         for (j = 0; j < nEdges; j++) {
06735             edges[e]->emom[j] = 0;            // coefficients of ext. mom.
06736         }
06737         for (j = 0; j < nLoops; j++) {
06738             edges[e]->lmom[j] = 0;            // coefficients of loop mom.
06739         }
06740     }
06741 }
06742
06743 //-----
06744 void EGraph::bisearchE(int nd, int *extlst, int *intlst, int *opiext, int *opiloop)
06745 {
06746     // Search in the spanning tree below 'nd'.
06747     // Arguments:
06748     //   nd      : input   : node to be visited
06749     //   extlst  : output  : set of external lines in the current lPI comp.
06750     //   intlst  : output  : set of internal lines in the current lPI comp.
06751     //   opiext  : output  : no. external lines of the current lPI component
06752     //   opiloop : output  : no. loops of the current lPI component
06753     //
06754     // EGraph variables
06755     //   bicount   : counter of visiting node
06756     //   loopm     : no. of independent loop momentum
06757     //   nopicomp  : no. of OPI components
06758     //   opi2plp   : maximum number loops among 2-point looped OPI components
06759     //   nopi2p    : no. of 2-point OPI components
06760     //   edges[ie]->opicomp : OPI component no. of the edge
06761     //
06762     int k, e, ie, ed, td, dir, j;
06763     int opiext1, opiloop1;
06764     int extlst1[GRCC_MAXEDGES]; // set of external nodes below
06765     int intlst1[GRCC_MAXEDGES]; // set of vertices in the lPI component
06766
06767     (*opiext) = 0;
06768     (*opiloop) = 0;
06769
06770     // visit node 'nd'
06771

```

```

06772     bidef[nd] = bicount;
06773     bilow[nd] = bicount;
06774     for (k = 0; k < nLoops; k++) {
06775         nodes[nd]->klow[k] = bicount;
06776     }
06777     bicount++;
06778     for (j = 0; j < nEdges; j++) {
06779         extlst[j] = False;
06780         intlst[j] = False;
06781     }
06782
06783     // go to children : nd --> ed --> td
06784     for (e = 0; e < nodes[nd]->deg; e++) {
06785         ed = V2Iedge(nodes[nd]->edges[e]);
06786
06787         // check this edge is already visited
06788         if (edges[ed]->edtype == GRCC_ED_Deleted) {
06789             // permanently deleted edge
06790             continue;
06791         } else if (edges[ed]->visited) {
06792             // already visited (backward visit)
06793             continue;
06794         } else if (edges[ed]->cut) {
06795             // cut edge : treat as nd is an external
06796             (*opiext)++;
06797             nodes[nd]->setType(GRCC_ND_CPoint);
06798             continue;
06799         } else if (!isExternal(nd) && edges[ed]->ext) {
06800             // external
06801             edges[ed]->setType(GRCC_ED_Extern);
06802             edges[ed]->visited = True;
06803             edges[ed]->emom[ed] = 1;
06804             extlst[ed] = True;
06805             (*opiext)++;
06806             nodes[nd]->setType(GRCC_ND_CPoint);
06807             continue;
06808         }
06809
06810         // mark visited
06811         edges[ed]->visited = True;
06812
06813         // momentum is assigned on the backward move. (nd) 1 --> 0 (td)
06814         td = edges[ed]->nodes[0];
06815         dir = 1;
06816         if (td == nd) {
06817             td = edges[ed]->nodes[1];
06818             dir = -1;
06819         }
06820
06821         // biconnected component
06822         if (bidef[td] >= 0) {
06823             // a back edge is found. Already visited. Don't go further
06824             bilow[nd] = Min(bilow[nd], bilow[td]);
06825             edges[ed]->setType(GRCC_ED_Back);
06826             intlst[ed] = True;
06827             (*opiloop)++;
06828
06829             // create new loop momentum
06830             k = loopm++;
06831             nodes[nd]->klow[k] = Min(nodes[nd]->klow[k], nodes[td]->klow[k]);
06832             edges[ed]->setLMom(k, dir);
06833
06834             // self-loop
06835             if (td == nd) {
06836                 nodes[nd]->setType(GRCC_ND_CPoint);
06837             }
06838
06839         } else {
06840             // not a back edge
06841             // go down further
06842             bisearchE(td, extlst1, intlst1, &opiext1, &opiloop1);
06843
06844             // the set of external particles
06845             for (j = 0; j < nEdges; j++) {
06846                 extlst[j] = extlst[j] || extlst1[j];
06847             }
06848
06849             // loop momenta
06850             for (k = 0; k < nLoops; k++) {
06851                 if (nodes[td]->klow[k] <= nodes[nd]->klow[k]) {
06852                     // inside loop 'k'
06853                     edges[ed]->setLMom(k, dir);
06854                 }
06855                 nodes[nd]->klow[k] = Min(nodes[nd]->klow[k], nodes[td]->klow[k]);
06856             }
06857
06858             // cut point ?

```

```

06859         if (bilow[td] >= bidef[nd]) {
06860             // a cut point is found
06861             if (nodes[nd]->ndtype == GRCC_ND_Undef) {
06862                 nodes[nd]->setType(GRCC_ND_CPoint);
06863 #ifndef CHECK
06864             } else if (nodes[nd]->ndtype != GRCC_ND_CPoint) {
06865                 if (prlevel > 0) {
06866                     fprintf(GRCC_Stderr, "bisearch: node %d is a cut point ",
06867                             nd);
06868                     fprintf(GRCC_Stderr, "(not undef %d)\n",
06869                             nodes[nd]->ndtype);
06870                 }
06871 #endif
06872             }
06873         } // cut point
06874
06875         // bridge ?
06876         if (bilow[td] > bidef[nd]) {
06877             // a bridge is found
06878             if (edges[ed]->edtype == GRCC_ED_Undef) {
06879                 edges[ed]->setType(GRCC_ED_Bridge);
06880                 nodes[td]->setType(GRCC_ND_CPoint);
06881 #ifndef CHECK
06882             } else if (edges[ed]->edtype != GRCC_ED_Bridge) {
06883                 if (prlevel > 0) {
06884                     fprintf(GRCC_Stderr, "bisearch: edges %d is a bridge ", ed);
06885                     fprintf(GRCC_Stderr, "(not undef %d)\n", edges[ed]->edtype);
06886                 }
06887 #endif
06888             }
06889
06890             if (!edges[ed]->ext) {
06891                 // internal bridge
06892                 opiextl++; // from bridge
06893                 for (ie = 0; ie < nEdges; ie++) {
06894                     if (intlstl[ie]) {
06895                         // OPI component No.
06896                         edges[ie]->opicomp = nopicomp;
06897                         intlstl[ie] = False;
06898                     }
06899                 }
06900                 // 2pt OPI component
06901                 if (opiextl == 2) {
06902                     opi2plp = Max(opi2plp, opiloop1);
06903
06904                     // the number of 2pt looped OPI components
06905                     nopi2p++;
06906                 }
06907                 // the number of OPI components
06908                 nopicomp++;
06909             }
06910             opiextl = 1; // from bridge
06911             opiloop1 = 0;
06912         } else {
06913             // not a bridge
06914
06915             bilow[nd] = Min(bilow[nd], bilow[td]);
06916
06917             if (edges[ed]->edtype == GRCC_ED_Undef) {
06918                 edges[ed]->setType(GRCC_ED_Inloop);
06919 #ifndef CHECK
06920             } else if (edges[ed]->edtype != GRCC_ED_Inloop) {
06921                 if (prlevel > 0) {
06922                     fprintf(GRCC_Stderr, "bisearch: ");
06923                     fprintf(GRCC_Stderr, "edges %d is not undef (%d)\n",
06924                             ed, edges[ed]->edtype);
06925                 }
06926 #endif
06927             }
06928             intlstl[ed] = True;
06929         }
06930     }
06931
06932     // merge to the current variables
06933     for (ie = 0; ie < nEdges; ie++) {
06934         intlstl[ie] = intlstl[ie] || intlstl[ie];
06935     }
06936     (*opiext) += opiextl;
06937     (*opiloop) += opiloop1;
06938
06939     // linear combination of external momenta
06940     edges[ed]->setEMom(nEdges, extlstl, dir);
06941
06942     } // end of a child
06943 } // for (ed)
06944
06945 return;

```

```

06946 }
06947
06948 //-----
06949 void EGraph::extMomConsrv(void)
06950 {
06951     int e, ex, lex, rs;
06952
06953     lex = -1;
06954     for (e = 0; e < nEdges; e++) {
06955         if (edges[e]->ext) {
06956             lex = e;
06957             extMom[e] = edges[e]->emom[e];
06958         } else {
06959             extMom[e] = 0;
06960         }
06961     }
06962     // eliminate momentum of the last external particle
06963     if (lex < 0) {
06964         return;
06965     }
06966     for (e = 0; e < nEdges; e++) {
06967         rs = edges[e]->emom[lex];
06968         if (rs != 0) {
06969             for (ex = 0; ex < nEdges; ex++) {
06970                 edges[e]->emom[ex] -= rs*extMom[ex];
06971             }
06972         }
06973     }
06974 }
06975
06976 //-----
06977 void EGraph::chkMomConsrv(void)
06978 {
06979     // check momentum conservation
06980
06981     int esum[GRCC_MAXEDGES];
06982     int lsum[GRCC_MAXNODES];
06983     Bool ok, okn;
06984     int n, ex, lk, ej, e, dir;
06985
06986     if (bicount < 1) {
06987         return;
06988     }
06989     ok = True;
06990     for (n = 0; n < nNodes; n++) {
06991         if (isExternal(n)) {
06992             continue;
06993         }
06994         for (ex = 0; ex < nEdges; ex++) {
06995             esum[ex] = 0;
06996         }
06997         for (lk = 0; lk < nLoops; lk++) {
06998             lsum[lk] = 0;
06999         }
07000         for (ej = 0; ej < nodes[n]->deg; ej++) {
07001             e = V2Iedge(nodes[n]->edges[ej]);
07002             if (edges[e]->nodes[0] == edges[e]->nodes[0]) {
07003                 continue;
07004             }
07005             dir = dirEdge(n, ej);
07006             for (ex = 0; ex < nEdges; ex++) {
07007                 if (edges[e]->ext) {
07008                     esum[ex] += dir*edges[e]->emom[ex];
07009                 }
07010             }
07011             for (lk = 0; lk < nLoops; lk++) {
07012                 lsum[lk] += dir*edges[e]->lmom[lk];
07013             }
07014         }
07015
07016         okn = True;
07017         for (ex = 0; ex < nEdges; ex++) {
07018             if (esum[ex] != 0) {
07019                 okn = False;
07020                 fprintf(GRCC_Stderr, "chkMomConsrv:n=%d, esum[%d]=%d\n",
07021                     n, ex, esum[ex]);
07022             }
07023         }
07024
07025         for (lk = 0; lk < nLoops; lk++) {
07026             if (lsum[lk] != 0) {
07027                 okn = False;
07028                 fprintf(GRCC_Stderr, "chkMomConsrv:n=%d, lsum[%d]=%d\n",
07029                     n, lk, lsum[lk]);
07030             }
07031         }
07032     }

```



```

07033
07034         if (!okn) {
07035             ok = False;
07036             fprintf(GRCC_Stderr, "*** Violation of momentum conservation ");
07037             fprintf(GRCC_Stderr, "at node =%d\n",n);
07038         }
07039     }
07040
07041     if (!ok) {
07042         print();
07043         erEnd("inconsistent momentum");
07044     }
07045 }
07046
07047 //-----
07048 int EGraph::legParticle(int ed, int el)
07049 {
07050     // outgoing particle from (edge, leg) = (ed, el)
07051
07052     return model->normalParticle((2*el-1)*edges[ed]->ptcl);
07053 }
07054
07055 //-----
07056 int EGraph::isFermion(int ed)
07057 {
07058     // return if particle on the edge ed follows Fermi statistics or not.
07059
07060     int ptcl, ptype;
07061
07062     ptcl = edges[ed]->ptcl;
07063     ptype = model->particles[Abs(ptcl)]->ptype;
07064
07065     return (ptype == GRCC_PT_Dirac || ptype == GRCC_PT_Majorana || ptype == GRCC_PT_Ghost);
07066 }
07067
07068 //-----
07069 int EGraph::fltrace(int fk, int nd0, int *fl)
07070 {
07071     int nfl, k, i, nd, nl, e, ed, el, fgcnt, fkind, lk;
07072
07073     nfl = 1;
07074     for (k = 0; k < nEdges; k++) {
07075         if (k >= nfl) {
07076             printf("*** fltrace:illegal contorl: k=%d, nEdges=%d\n", k, nEdges);
07077             break;
07078         }
07079         e = fl[k];
07080         ed = V2Iedge(e);
07081         el = V2Ileg(e);
07082 #ifdef CHECK
07083         if (ed > nEdges) {
07084             fprintf(GRCC_Stderr, "*** fltrace: ed=%d > nEdges=%d, fl=",
07085                 ed, nEdges);
07086             prIntArray(nNodes, fl, "\n");
07087             erEnd("fltrace: illegal fl");
07088         }
07089 #endif
07090         nd = edges[ed]->nodes[el];
07091         nl = edges[ed]->nlegs[el];
07092
07093         // end of the fline
07094         if (isExternal(nd) || nd == nd0) {
07095             fgcnt = 1;
07096             break;
07097         }
07098         fgcnt = 0;
07099         ed = 0;
07100         lk = 0;
07101         for (i = 0; i < nodes[nd]->deg; i++) {
07102             e = nodes[nd]->edges[i];
07103             ed = V2Iedge(e);
07104             fkind = model->particles[Abs(edges[ed]->ptcl)]->ptype;
07105             if (fkind == fk) {
07106                 if (lk == 1) {
07107                     lk = -1;
07108                 } else {
07109                     lk = 1;
07110                 }
07111             } else {
07112                 lk = 0;
07113             }
07114             if (!edges[ed]->visited) {
07115                 if (!isFermion(ed)) {
07116                     edges[ed]->visited = True;
07117                 } else {
07118                     if (fkind == fk) {
07119                         // i should be the neighbor of nl

```

```

07120                                     // (nl, i) or (i, nl) should be a pair of fkind
07121                                     if ((lk == 1 && nl == i+1) || (lk == -1 && nl == i-1)) {
07122                                         edges[ed]->visited = True;
07123                                         fgcnt++;
07124                                         if (fgcnt == 1) {
07125                                             fl[nfl++] = - e;
07126                                             break;
07127                                         }
07128                                     }
07129                                 }
07130                             }
07131                         }
07132                     }
07133                 if (fgcnt == 0) {
07134                     printf("*** fline: Fermion number is not conserved\n");
07135                     printf("      nd=%d, e=%d, ed=%d, fgcnt=%d\n",
07136                           nd, e, ed, fgcnt);
07137                     // erEnd("fline: Fermion number is not conserved");
07138                 } else if (fgcnt > 1) {
07139                     printf("+++ fline: more than two fermions: check fsign\n");
07140                     printf("      nd=%d, e=%d, ed=%d, fgcnt=%d\n",
07141                           nd, e, ed, fgcnt);
07142                 }
07143             }
07144             if (k >= nNodes || k >= nfl) {
07145                 printf("*** fline: illegal control\n");
07146                 printf("      nfl=%d, ", nfl);
07147                 prIntArray(nfl, fl, "\n");
07148                 erEnd("fline: illegal control");
07149             }
07150             return nfl;
07151         }
07152
07153         //-----
07154         void EGraph::getFLines(void)
07155         {
07156             int fl[GRCC_MAXNODES];
07157             int nextn, exto[GRCC_MAXNODES];
07158             int e, ed, nd, floop, nswap, nfl, fkind, el, ptcl;
07159
07160             nflines = 0;
07161             for (ed = 0; ed < nEdges; ed++) {
07162                 edges[ed]->visited = False;
07163             }
07164
07165             nextn = 0;
07166             for (ed = 0; ed < nEdges; ed++) {
07167                 if (edges[ed]->visited) {
07168                     continue;
07169                 }
07170                 if (!edges[ed]->ext) {
07171                     continue;
07172                 }
07173                 el = 0;
07174                 nd = edges[ed]->nodes[el];
07175                 if (!isExternal(nd)) {
07176                     el = 1;
07177                     nd = edges[ed]->nodes[el];
07178                     if (!isExternal(nd)) {
07179                         erEnd("*** illegal control");
07180                     }
07181                 }
07182                 if (!isFermion(ed)) {
07183                     continue;
07184                 }
07185                 ptcl = legParticle(ed, 1-el);
07186                 if (ptcl < 0) {
07187                     continue;
07188                 }
07189
07190                 e = I2Vedge(ed, el);
07191                 fl[0] = - e;
07192                 nfl = 1;
07193                 fkind = model->particles[Abs(edges[ed]->ptcl)]->ptype;
07194                 edges[ed]->visited = True;
07195
07196                 nfl = fltrace(fkind, nd, fl);
07197                 addFLine(FL_Open, fkind, nfl, fl);
07198                 e = fl[nfl-1];
07199                 exto[nextn++] = edges[V2Iedge(e)]->nodes[V2Ileg(e)];
07200             }
07201
07202             // closed Fermion line
07203             floop = 0;
07204             for (ed = 0; ed < nEdges; ed++) {
07205                 if (edges[ed]->visited) {
07206                     continue;

```

```

07207     }
07208     if (!isFermion(ed)) {
07209         continue;
07210     }
07211     el = 0;
07212     ptcl = legParticle(ed, 1-el);
07213     if (ptcl < 0) {
07214         el = 1;
07215     }
07216     nd = edges[ed]->nodes[el];
07217     e = I2Vedge(ed, 1-el);
07218     fl[0] = e;
07219     nfl = 1;
07220     edges[ed]->visited = True;
07221
07222     fkind = model->particles[Abs(edges[ed]->ptcl)]->ptype;
07223     nfl = fltrace(fkind, nd, fl);
07224     addFLine(FL_Closed, fkind, nfl, fl);
07225     floop++;
07226 }
07227 if (nextn > 0) {
07228     nswap = sortb(nextn, exto);
07229 } else {
07230     nswap = 0;
07231 }
07232 if ((floop+nswap) % 2 == 0) {
07233     fsign = 1;
07234 } else {
07235     fsign = -1;
07236 }
07237 }
07238
07239 //-----
07240 void EGraph::addFLine(const FLType ft, int fk, int nfl, int *fl)
07241 {
07242     int j;
07243
07244     if (nflines >= GRCC_MAXFLINES) {
07245         erEnd("too many Fermion lines (GRCC_MAXEDGES)");
07246     }
07247     if (flines[nflines] == NULL) {
07248         flines[nflines] = new EFLine();
07249     }
07250     flines[nflines]->ftype = ft;
07251     flines[nflines]->fkind = fk;
07252     flines[nflines]->nlist = nfl;
07253     for (j = 0; j < nfl; j++) {
07254         flines[nflines]->elist[j] = fl[j];
07255     }
07256     nflines++;
07257 }
07258
07259 //*****
07260 // assign.cc
07261 //=====
07262 // Assign particles to the edges and interactions to nodes.
07263 // Completed graph data is saved in the form of EGraph.
07264
07265 // method : selection of assignable node
07266
07267 #ifdef DEBUGM
07268 static int nordleg = 0;
07269 static int nopleg = 0;
07270 static int niso = 0;
07271 static int nisol = 0;
07272 static int nisol1 = 0;
07273 static int nisol11 = 0;
07274 static int nisol12 = 0;
07275 static int nisol2 = 0;
07276 static int nisol3 = 0;
07277 static int nisol4 = 0;
07278 static int niso2 = 0;
07279 static int nivord = 0;
07280 static int nextonly = 0;
07281 #endif
07282
07283 //=====
07284 // class NCand
07285 NCand::NCand(const NCandSt sta, const int dega, const int nilst, int *ilst)
07286 {
07287     // candidate list of interactions attached to a vertex.
07288
07289     int j;
07290
07291     deg = dega;
07292     st = sta;
07293     nilist = nilst;

```

```

07294     for (j = 0; j < nilist; j++) {
07295         ilist[j] = ilst[j];
07296     }
07297
07298 #ifdef CHECK
07299     if (st == AS_Assigned) {
07300         if (nilist != 1) {
07301             erEnd("illegal assignment");
07302         }
07303     } else if (st == AS_AssExt) {
07304         if (nilist != 1) {
07305             erEnd("illegal assignment");
07306         }
07307     }
07308 #endif
07309 }
07310
07311 //-----
07312 NCand::~NCand(void)
07313 {
07314 }
07315
07316 //-----
07317 void NCand::prNCand(const char* msg)
07318 {
07319     printf("%d %d ", st, deg);
07320     prIntArray(nilist, ilist, msg);
07321 }
07322
07323 //=====
07324 // class ECand
07325 ECand::ECand(int dt, int nplst, int *plst)
07326 {
07327     int j;
07328
07329     det = dt;
07330     nplist = nplst;
07331     for (j = 0; j < nplist; j++) {
07332         plist[j] = plst[j];
07333     }
07334
07335     if (det && nplist != 1) {
07336         if (prlevel > 0) {
07337             fprintf(GRCC_Stderr, "*** ECand : len(plist) != 1 : det=%d ", det);
07338             prIntArrayErr(nplist, plist, "\n");
07339         }
07340         erEnd("ECand : len(plist) != 1");
07341     }
07342 }
07343
07344 //-----
07345 ECand::~ECand(void)
07346 {
07347 }
07348
07349 //-----
07350 void ECand::prECand(const char *msg)
07351 {
07352     prIntArray(nplist, plist, "");
07353     printf(" (det=%d)%s", det, msg);
07354 }
07355
07356 //=====
07357 // class ANode
07358 ANode::ANode(int dg)
07359 {
07360     int j;
07361
07362     deg = dg;
07363     nlegs = 0;
07364     anodes = new int[deg];
07365     aedges = new int[deg];
07366     aelegs = new int[deg];
07367     cand = NULL;
07368     for (j = 0; j < deg; j++) {
07369         anodes[j] = -1;
07370         aedges[j] = -1;
07371         aelegs[j] = -1;
07372     }
07373 }
07374
07375 //-----
07376 ANode::~ANode(void)
07377 {
07378     if (cand != NULL) {
07379         delete cand;
07380         cand = NULL;

```

```

07381     }
07382     if (aelegs != NULL) {
07383         delete[] aelegs;
07384         delete[] aedges;
07385         delete[] anodes;
07386         aelegs = NULL;
07387         aedges = NULL;
07388         anodes = NULL;
07389     }
07390 }
07391
07392 //-----
07393 int ANode::newleg(void)
07394 {
07395     // add a slot for a leg
07396
07397     int lg = nlegs;
07398
07399     nlegs++;
07400 #ifdef CHECK
07401     if (nlegs > deg) {
07402         fprintf(GRCC_Stderr, "*** ANode::newleg : nlegs = %d > deg = %d\n",
07403             nlegs, deg);
07404         erEnd("ANode::newleg : nlegs > deg");
07405     }
07406 #endif
07407
07408     return lg;
07409 }
07410
07411 //=====
07412 // class AEdge
07413 AEdge::AEdge(int n0, int l0, int n1, int l1)
07414 {
07415     nodes[0] = n0;        // node of 0 side of the edge
07416     nodes[1] = n1;        // node of 1 side of the edge
07417     nlegs[0] = l0;        // leg of the node of 0 side of the edge
07418     nlegs[1] = l1;        // leg of the node of 1 side of the edge
07419     ptcl = GRCC_PT_Undef;  // particle defined in the model
07420     cand = NULL;          // candidate
07421 }
07422
07423 //-----
07424 AEdge::~AEdge(void)
07425 {
07426     if (cand != NULL) {
07427         delete cand;
07428     }
07429 }
07430
07431 //=====
07432 // class Assign
07433 Assign::Assign(SProcess *sprc, MGraph *mgr, PNodeClass *pnc)
07434 {
07435     Bool ok;
07436     int j;
07437
07438     if (sprc == NULL) {
07439         erEnd("Assign: sprc == NULL");
07440     }
07441
07442     // pointers to related objects
07443     sprc = sprc;
07444     model = sprc->model;
07445     opt = sprc->opt;
07446     mgr = mgr;
07447     pnclass = pnc;
07448
07449     egraph = mgr->egraph;
07450     if (egraph == NULL) {
07451         erEnd("Assign: egraph == NULL");
07452     }
07453
07454     astack = sprc->astack;
07455     if (astack == NULL) {
07456         erEnd("Assign: astack == NULL");
07457     }
07458
07459 #ifdef DEBUG
07460     if (pnclass == NULL) {
07461         printf("+++ Assign::Assign : pnclass = NULL\n");
07462     } else {
07463         printf("+++ Assign::Assign : pnclass : \n");
07464         pnclass->prPNodeClass();
07465     }
07466 #endif
07467

```

```

07468     nNodes      = mgraph->nNodes;
07469     nEdges      = mgraph->nEdges;
07470     nExtern     = sproc->nExtern;
07471     nETotal     = 0;           // total number of edges
07472
07473     // counters of graphs
07474     nAGraphs    = 0;
07475     wAGraphs    = Fraction(0,1);
07476     nAOPI       = 0;
07477     wAOPI       = Fraction(0,1);
07478
07479     nodes       = new ANode*[nNodes];
07480     edges       = new AEdge*[nEdges];
07481
07482     for (j = 0; j < nNodes; j++) {
07483         nodes[j] = NULL;
07484     }
07485     for (j = 0; j < nEdges; j++) {
07486         edges[j] = NULL;
07487     }
07488     for (j = 0; j < model->ncouple; j++) {
07489         cplleft[j] = sproc->clist[j];
07490     }
07491
07492     // initialize astack
07493     astack->setAGraph(this);
07494     astack->checkPoint(checkpoint0);
07495
07496     // import from mgraph
07497     ok = fromMGraph();
07498 #ifdef CHECK
07499     checkAG("Assign::Assign");
07500 #endif
07501
07502     if (ok) {
07503         // for debugging
07504 #ifdef DEBUG0
07505         printf("+++ End of initialization : candidate:\n");
07506         prCand("init");
07507         printf("\n");
07508 #endif
07509
07510         // start assignment
07511         assignAllVertices();
07512 #ifdef DEBBUGM
07513         printf("mId=%ld, sId=%ld, ordleg=%d, ivord=%d, extonly=%d ",
07514             egraph->mId, egraph->sId, nordleg, nivord, nextonly);
07515         printf("niso=%d [%d(%d {&d, %d}, %d, %d, %d) %d]\n",
07516             niso, niso1, niso11, niso111, niso112, niso12, niso13, niso14,
07517             niso2);
07518 #endif
07519     } else {
07520         // cannot assign
07521     }
07522
07523     // restore from the stack
07524 #ifdef CHECK
07525     astack->restoreMsg(checkpoint0, "Assign");
07526 #else
07527     astack->restore(checkpoint0);
07528 #endif
07529 }
07530
07531 //=====
07532 Assign::~Assign(void)
07533 {
07534     int j;
07535
07536     for (j = 0; j < nEdges; j++) {
07537         delete edges[j];
07538         edges[j] = NULL;
07539     }
07540     delete[] edges;
07541     edges = NULL;
07542
07543     for (j = 0; j < nNodes; j++) {
07544         delete nodes[j];
07545         nodes[j] = NULL;
07546     }
07547     delete[] nodes;
07548     nodes = NULL;
07549 }
07550
07551 //-----
07552 void Assign::prCand(const char *msg)
07553 {

```

```

07555 // Print candidate table
07556 int n, e, ne;
07557 AEdge *ed;
07558
07559 printf("\n");
07560 printf("+++ Candidate list: %s\n", msg);
07561 printf(" Nodes %d\n", nNodes);
07562 for (n = 0; n < nNodes; n++) {
07563     printf("%d: edges=", n);
07564     prIntArray(nodes[n]->deg, nodes[n]->aedges, ": cand=");
07565     if (nodes[n]->cand == NULL) {
07566         printf("NULL\n");
07567     } else {
07568         nodes[n]->cand->prNCand("\n");
07569     }
07570 }
07571 ne = Min(nEdges, nETotal);
07572 printf(" Edges %d\n", ne);
07573 for (e = 0; e < ne; e++) {
07574     ed = edges[e];
07575     if (ed == NULL) {
07576         printf("NULL_Edge\n");
07577     } else {
07578         printf("%d: %d->%d: cand=", e, ed->nodes[0], ed->nodes[1]);
07579         if (edges[e]->cand == NULL) {
07580             printf("NULL\n");
07581         } else {
07582             edges[e]->cand->prECand("\n");
07583         }
07584     }
07585 }
07586 }
07587
07588 //=====
07589 void Assign::checkAG(const char *msg)
07590 {
07591     int n, lg;
07592     Bool ok = True;
07593
07594     for (n = 0; n < nNodes; n++) {
07595         for (lg = 0; lg < nodes[n]->deg; lg++) {
07596             if (nodes[n]->aedges[lg] < 0) {
07597                 printf("*** checkAG:%s: n=%d, lg=%d, aedges=%d\n",
07598                     msg, n, lg, nodes[n]->aedges[lg]);
07599                 ok = False;
07600             }
07601         }
07602     }
07603     if (!ok) {
07604         prCand("checkAG");
07605         erEnd("checkAG: failed");
07606     }
07607 }
07608
07609 //=====
07610 // control of the process of particle/interaction assignment
07611
07612 //-----
07613 Bool Assign::assignAllVertices(void)
07614 {
07615     // Entry point of the assignment procedure
07616     Bool ok;
07617
07618 #ifdef CHECK
07619     checkCand("assignAllVertices:1");
07620 #endif
07621
07622 #ifdef DEBUG0
07623     printf("+++ particle assignment for '%ld'\n", mgraph->mId);
07624 #endif
07625
07626     // start main part
07627 #ifdef SIMPSEL
07628     ok = selectVertexSimp(-1);
07629 #else
07630     ok = selectVertex();
07631 #endif
07632
07633 #ifdef DEBUG0
07634     printf("\n");
07635     printf("+++ Total %ld assigned graphs for '%ld'\n",
07636         nAGraphs, mgraph->mId);
07637     printf("result: %ld ", nAGraphs);
07638     wAGraphs.print(" ");
07639     printf("%ld ", nAOPI);
07640     wAOPI.print("\n");
07641 #endif

```

```

07642
07643 #ifndef CHECK
07644     checkCand("assignAllVertices:2");
07645 #endif
07646
07647     return ok;
07648 }
07649
07650 //-----
07651 Bool Assign::selectVertexSimp(int lastv)
07652 {
07653     // Select a vertex for assignment of particles to legs
07654     //
07655     // Argument
07656     // lastv : previously assigned vertex; Nothing when lastv<0.
07657
07658     Bool ok;
07659     int v;
07660
07661 #ifndef CHECK
07662     checkCand("selectVertexSimp");
07663 #endif
07664
07665     // find a vertex to which interaction is assigned
07666     v = selUnAssVertexSimp(lastv);
07667
07668     // no more vertex
07669     if (v < 0) {
07670
07671         ok = allAssigned();
07672         if (ok) {
07673             opt->newAGraph(egraph);
07674         }
07675
07676         return ok;
07677     }
07678
07679     // select a leg and assign a particle to it
07680     ok = selectLeg(v, -1);
07681
07682     return ok;
07683 }
07684
07685 //-----
07686 Bool Assign::selectVertex(void)
07687 {
07688     // Select a vertex for assignment of particles to legs
07689     //
07690     // Argument
07691     // lastv : previously assigned vertex; Nothing when lastv<0.
07692
07693     Bool ok;
07694     int v;
07695
07696 #ifndef CHECK
07697     checkCand("selectVertex");
07698 #endif
07699
07700     // find a vertex to which interaction is assigned
07701     v = selUnAssVertex();
07702
07703     // no more vertex
07704     if (v < 0) {
07705
07706         ok = allAssigned();
07707         if (ok) {
07708             opt->newAGraph(egraph);
07709         }
07710
07711         return ok;
07712     }
07713
07714     // select a leg and assign a particle to it
07715     ok = selectLeg(v, -1);
07716
07717     return ok;
07718 }
07719
07720 //-----
07721 Bool Assign::selectLeg(int v, int lastlg)
07722 {
07723     // Select a leg of vertex 'v' for assignment of particle,
07724     // where the last assigned leg was 'lastlg'.
07725
07726     int pt, ln, j;
07727     Bool ok;
07728     CheckPt sav, sav0;

```



```

07729     int      nplist, plist[GRCC_MAXMPARTICLES2];
07730
07731 #ifdef CHECK
07732     checkNode(v, "selectLeg:0");
07733     checkCand("selectLeg");
07734 #endif
07735
07736     // find a leg to be assigned
07737     ln = selUnAssLeg(v, lastlg);
07738
07739     // no more assignable leg for the current node
07740     if (ln < 0) {
07741         // move to next vertex
07742         ok = assignVertex(v);
07743
07744         return ok;
07745     }
07746
07747     // make the list of possible particles, incoming to the vertex
07748     astack->checkPoint(sav0);
07749     nplist = candPart(v, ln, plist, GRCC_MAXMPARTICLES2);
07750
07751     // no candidate
07752     if (nplist < 1) {
07753         return False;
07754     }
07755
07756     // assign each of possible particle to the leg 'lg'.
07757     astack->checkPoint(sav);
07758     for (j = 0; j < nplist; j++) {
07759         pt = plist[j];
07760
07761         // try to assign 'pt' to (v, ln)
07762         if(assignPLeg(v, ln, pt)) {
07763             // move to next leg
07764             selectLeg(v, ln);
07765         }
07766
07767 #ifdef CHECK
07768         astack->restoreMsg(sav, "selectLeg2");
07769 #else
07770         astack->restore(sav);
07771 #endif
07772     }
07773
07774 #ifdef CHECK
07775     astack->restoreMsg(sav0, "selectLeg2");
07776 #else
07777     astack->restore(sav0);
07778 #endif
07779     return True;
07780 }
07781
07782 //-----
07783 void Assign::saveCouple(int *sav)
07784 {
07785     int j;
07786
07787     if (model->ncouple > 1) {
07788         for (j = 0; j < model->ncouple; j++) {
07789             sav[j] = cplleft[j];
07790         }
07791     }
07792 }
07793
07794 //-----
07795 void Assign::restoreCouple(int *sav)
07796 {
07797     int j;
07798
07799     if (model->ncouple > 1) {
07800         for (j = 0; j < model->ncouple; j++) {
07801             cplleft[j] = sav[j];
07802         }
07803     }
07804 }
07805
07806 //-----
07807 Bool Assign::subCouple(int *cpl)
07808 {
07809     int j;
07810
07811     if (model->ncouple > 1) {
07812         for (j = 0; j < model->ncouple; j++) {
07813             cplleft[j] -= cpl[j];
07814             if (cplleft[j] < 0) {
07815                 return False;

```

```

07816         }
07817     }
07818 }
07819 return True;
07820 }
07821
07822 //-----
07823 Bool Assign::assignVertex(int v)
07824 {
07825     // Assign interactions to vertex 'v'
07826
07827     Bool        done, ok;
07828     CheckPt     sav, savl;
07829     NCand       *nc;
07830     int         ia, n, j, cl;
07831     NCandSt     vst;
07832     Bool        ok1;
07833     int         savc0[GRCC_MAXNCPLG], savc1[GRCC_MAXNCPLG];
07834
07835 #ifdef CHECK
07836     checkCand("assignVertex");
07837 #endif
07838
07839     // save the configuration
07840     astack->checkPoint(sav);
07841     saveCouple(savc0);
07842
07843     // assign to all vertices with only one candidate
07844     done = False;
07845     for (n = 0; n < nNodes; n++) {
07846
07847         // check if vertex
07848         cl = pnclass->nd2cl[n];
07849         if (!isATEXternal(pnclass->type[cl])) {
07850
07851             nc = nodes[n]->cand;
07852             if (nc->st == AS_UnAssLegs && nc->nilist == 1) {
07853
07854                 // bind interaction to the vertex
07855                 ia = nc->ilist[0];
07856                 vst = assignIVertex(n, ia);
07857                 if (vst == AS_Impossible) {
07858                     // discard the current configuration
07859 #ifdef CHECK
07860                     astack->restoreMsg(sav, "assignVertex");
07861 #else
07862                     astack->restore(sav);
07863 #endif
07864                     restoreCouple(savc0);
07865                     return False;
07866                 } else if (vst == AS_Assigned) {
07867                     if (!subCouple(model->interacts[ia]->clist)) {
07868 #ifdef CHECK
07869                         astack->restoreMsg(sav, "assignVertex");
07870 #else
07871                         astack->restore(sav);
07872 #endif
07873                         restoreCouple(savc0);
07874                         return False;
07875                     }
07876                 }
07877             }
07878             if (n == v) {
07879                 done = (vst == AS_Assigned);
07880             }
07881         }
07882     }
07883 }
07884
07885 // uniquely determined
07886 ok = True;
07887 if (done) {
07888     // move to the next vertex
07889 #ifdef SIMPSEL
07890     ok = selectVertexSimp(v);
07891 #else
07892     ok = selectVertex();
07893 #endif
07894
07895 #ifdef CHECK
07896     astack->restoreMsg(sav, "assignVertex");
07897 #else
07898     astack->restore(sav);
07899 #endif
07900     restoreCouple(savc0);
07901
07902     return ok;

```

```

07903     }
07904
07905     // there are still several possibilities.
07906
07907     astack->checkPoint(sav1);
07908     saveCouple(savc1);
07909     for (j = 0; j < nodes[v]->cand->nlist; j++) {
07910         ia = nodes[v]->cand->ilist[j];
07911
07912         // bind interaction to the vertex
07913         ok1 = True;
07914         vst = assignIVertex(v, ia);
07915         if (vst == AS_Assigned) {
07916             ok1 = subCouple(model->interacts[ia]->clist);
07917         }
07918         if (ok1 && (vst == AS_Assigned || vst == AS_Assigned0)) {
07919             // move to the next vertex
07920 #ifdef SIMPSEL
07921             ok = selectVertexSimp(v);
07922 #else
07923             ok = selectVertex();
07924 #endif
07925         }
07926
07927 #ifdef CHECK
07928         astack->restoreMsg(sav1, "assignVertex");
07929 #else
07930         astack->restore(sav1);
07931 #endif
07932         restoreCouple(savc1);
07933     }
07934
07935 #ifdef CHECK
07936     astack->restoreMsg(sav, "assignVertex");
07937 #else
07938     astack->restore(sav);
07939 #endif
07940     restoreCouple(savc0);
07941
07942     return ok;
07943 }
07944
07945 //-----
07946 Bool Assign::allAssigned(void)
07947 {
07948     // Assignment of particles and interactions is finished.
07949     // Check isomorphism etc. and count symmetry factor
07950     // The sign from Fermi statistics is calculated in fillEGraph
07951
07952     BigInt nsym, esym, nsym1;
07953     MNodeClass *cl;
07954     Bool ok = True;
07955 #ifdef OPTEXTONLY
07956 #else
07957     Bool ext;
07958     int e, p;
07959 #endif
07960
07961 #ifdef CHECK
07962     checkCand("allAssigned");
07963 #endif
07964
07965     // check order of coupling constants
07966 #ifdef CHECK
07967     if (!checkOrderCpl()) {
07968         erEnd("allAssigned: checkOrderCpl = False");
07969     }
07970 #endif
07971
07972     // check duplication by violating ordering condition
07973     if (!isOrdLegs()) {
07974 #ifdef DEBUGM
07975         nordleg++;
07976 #endif
07977         return False;
07978     }
07979
07980     // classification of nodes
07981     cl = mgraph->curcl;
07982
07983     // check isomorphism and calculate sfactor
07984     nsym = 0;
07985     esym = 0;
07986
07987     ok = isIsomorphic(cl, &nsym, &esym, &nsym1);
07988     if (!ok || nsym < 1 || esym < 1) {
07989 #ifdef DEBUGM

```

```

07990         niso++;
07991 #endif
07992         return False;
07993     }
07994
07995     //-----
07996     // Now we got a new agraph.
07997
07998     // Update counters
07999     nAGraphs++;
08000     wAGraphs.add(1,nsym*esym);
08001     if (mgraph->opi) {
08002         nAOPI++;
08003         wAOPI.add(1,nsym*esym);
08004     }
08005
08006 #ifdef DEBUG1
08007     for (int j = 0; j < model->ncouple; j++) {
08008         if (cplleft[j] != 0) {
08009             printf("nAGraphs=%ld: 0 != cplleft =", nAGraphs);
08010             prIntArray(model->ncouple, cplleft, "\n");
08011             break;
08012         }
08013     }
08014 #endif
08015
08016 #ifdef CHECK
08017     checkAG("allAssigned");
08018 #endif
08019     // fill information to resulting Egraph
08020     fillEGraph(nAGraphs, nsym, esym, nsym1);
08021
08022 #ifdef OPTEXTONLY
08023 #else
08024     // check extonly particles
08025     for (e = 0; e < nEdges; e++) {
08026         p = Abs(egraph->edges[e]->ptcl);
08027         ext = egraph->edges[e]->ext;
08028         if (!ext) {
08029             if (model->particles[p]->extonly) {
08030 #ifdef DEBUGM
08031                 nextonly++;
08032 #endif
08033                 return False;
08034             }
08035         }
08036     }
08037 #endif
08038
08039 #ifdef DEBUG0
08040     printf("Assigned graph = %ld, sym = (%ld, %ld) ", nAGraphs, nsym, esym);
08041     prCand("allAssigned ");
08042 #endif
08043
08044     return True;
08045 }
08046
08047 //=====
08048 // Interface to MGraph and EGraph
08049 //-----
08050 Bool Assign::fromMGraph(void)
08051 {
08052     // Import from MGraph
08053
08054     int    npall, *pall;
08055     int    np[1];
08056     Bool   ok;
08057     int    j, k, n, nn, nl, deg, nc, ptcl, cl, typ, mcl, cl1, typ1;
08058     int    lcn, ia;
08059     int    nilist, ilist[GRCC_MAXMINTERACT];
08060     NCandSt st;
08061
08062     // create ANodes
08063     for (j = 0; j < nNodes; j++) {
08064         nodes[j] = new ANode(mgraph->nodes[j]->deg);
08065     }
08066
08067     // initialization of edge table
08068     nETotal = 0;
08069
08070     // List of all particles
08071     pall = model->allParticles(&npall);
08072
08073     // add nodes
08074     for (n = 0; n < mgraph->nNodes; n++) {
08075         cl = pnclass->nd2cl[n];
08076         typ = pnclass->type[cl];

```

```

08077
08078 // external particle
08079 if (isATEXternal(typ)) {
08080
08081 #ifdef CHECK
08082     if (mgraph->nodes[n]->deg != 1) {
08083         printf("*** assign:fromMGraph : "
08084             "external but deg[%d] = %d != 1, type=%d\n",
08085             n, mgraph->nodes[n]->deg, typ);
08086         mgraph->print();
08087         erEnd("assign:fromMGraph : illegal external particle");
08088     }
08089 #endif
08090     np[0] = pnclass->particle[c1];
08091     nodes[n]->cand = new NCand(AS_AssExt, 1, 1, np);
08092     for (n1 = 0; n1 < mgraph->nNodes; n1++) {
08093         nc = mgraph->adjMat[n][n1];
08094         for (k = 0; k < nc; k++) {
08095             addEdge(n, n1, 1, np);
08096         }
08097     }
08098
08099 // vertex
08100 } else {
08101
08102     // candidates of vertices
08103     deg = mgraph->nodes[n]->deg;
08104     st = AS_UnAssLegs;
08105     c1 = sproc->pnclass->nd2c1[n]; // class in the process
08106     mcl = sproc->pnclass->c12mcl[c1]; // class in the model
08107
08108     if (nodes[n]->cand != NULL) {
08109         delete nodes[n]->cand;
08110     }
08111     lcn = model->ncouple;
08112
08113     // multiple coupling constants are defined in the model.
08114     if (lcn > 1) {
08115         nilist = 0;
08116         for (j = 0; j < model->cplgnvl[mcl]; j++) {
08117             ia = model->cplgvl[mcl][j];
08118
08119             // select by coupling constants
08120             if (leqArray(lcn, model->interacts[ia]->c1list, cplleft)) {
08121                 ilist[nilist++] = ia;
08122             }
08123         }
08124         nodes[n]->cand = new NCand(st, deg, nilist, ilist);
08125
08126     // there is only one coupling constant
08127     } else {
08128         nodes[n]->cand =
08129             new NCand(st, deg, model->cplgnvl[mcl], model->cplgvl[mcl]);
08130     }
08131
08132     // vertices
08133     for (n1 = n; n1 < mgraph->nNodes; n1++) {
08134         c11 = pnclass->nd2c1[n1];
08135         typ1 = pnclass->type[c11];
08136         if (!isATEXternal(typ1)) {
08137             nc = mgraph->adjMat[n][n1];
08138             if (n == n1) {
08139                 nc = nc/2;
08140             }
08141             for (k = 0; k < nc; k++) {
08142                 addEdge(n, n1, npall, pall);
08143             }
08144         }
08145     }
08146 }
08147 }
08148 #ifdef CHECK
08149 if (nETotal != nEdges) {
08150     printf("*** Assign::fromMGraph nETotal=%d != nEdges=%d\n",
08151         nETotal, nEdges);
08152     erEnd("Assign::fromMGraph nETotal= != nEdges");
08153 }
08154 #endif
08155
08156 // assign particle of an edge next to an external particle
08157 for (n = 0; n < mgraph->nNodes; n++) {
08158     c1 = pnclass->nd2c1[n];
08159     typ = pnclass->type[c1];
08160     if (isATEXternal(typ)) {
08161
08162         c1 = pnclass->nd2c1[n];
08163         ptcl = pnclass->particle[c1];

```

```

08164
08165         // particle ptcl flows in to the node and
08166         // particle (- ptcl) flows in from the edge.
08167
08168 #ifdef CHECK
08169         if (ptcl == 0) {
08170             erEnd("fromMGraph: ptcl=0");
08171         }
08172 #endif
08173         ok = assignPLeg(n, 0, - ptcl);
08174         if (!ok) {
08175             // impossible config
08176             return False;
08177         }
08178         nn = nodes[n]->anodes[0];
08179         ok = updateCandNode(nn);
08180         if (!ok) {
08181             // impossible config
08182             return False;
08183         }
08184     }
08185 }
08186
08187 #ifdef CHECK
08188     checkCand("fromMGraph");
08189     checkAG("fromMGraph");
08190 #endif
08191
08192     return True;
08193 }
08194
08195 //-----
08196 void Assign::addEdge(int n0, int n1, int nplist, int *plist)
08197 {
08198     // add an edge and set connection information
08199     // n0 -- (new edge) -- n1, with candidates 'plist'
08200
08201     int lg0, lg1, eid;
08202     AEdge *aed;
08203
08204 #ifdef CHECK
08205     if (n0 >= nNodes || n1 >= nNodes) {
08206         printf("*** Assign::addEdge : undefined nodes %d: [%d, %d]",
08207             nETotal, n0, n1);
08208         erEnd("Assign::addEdge : undefined nodes");
08209     }
08210 #endif
08211
08212     // legs to be connected
08213     lg0 = nodes[n0]->newleg();
08214     lg1 = nodes[n1]->newleg();
08215
08216     // create edge
08217 #ifdef CHECK
08218     if (nETotal >= nEdges) {
08219         erEnd("too many edges");
08220     }
08221 #endif
08222     aed = new AEdge(n0, lg0, n1, lg1);
08223     aed->cand = new ECand(False, nplist, plist);
08224
08225 #ifdef CHECK
08226     if (edges[nETotal] != NULL) {
08227         erEnd("edges[nETotal] != NULL");
08228     }
08229 #endif
08230 #endif
08231
08232     // register to the edge table
08233     eid = nETotal;
08234     edges[nETotal++] = aed;
08235
08236     // connect edge and nodes
08237     connect(n0, lg0, eid, 0, n1, lg1);
08238 }
08239
08240 //-----
08241 void Assign::connect(int n0, int l0, int eg, int el, int n1, int l1)
08242 {
08243     // Connect (n0, l0) -- (eg, el) -- (n1)
08244
08245     ANode *nd0, *nd1;
08246     AEdge *ed;
08247     int eo;
08248
08249     nd0 = nodes[n0];
08250     nd1 = nodes[n1];

```

```

08251     ed = edges[eg];
08252     eo = 1 - el;
08253
08254     nd0->anodes[10] = n1;
08255     nd0->aedges[10] = eg;
08256     nd0->aelegs[10] = el;
08257
08258     nd1->anodes[11] = n0;
08259     nd1->aedges[11] = eg;
08260     nd1->aelegs[11] = eo;
08261
08262     ed->nodes[el] = n0;
08263     ed->nlegs[el] = 10;
08264
08265     ed->nodes[eo] = n1;
08266     ed->nlegs[eo] = 11;
08267 }
08268
08269 //-----
08270 Bool Assign::fillEGraph(int aid, BigInt nsym, BigInt esym, BigInt nsym1)
08271 {
08272     // Set variables of egraph with the result of particle assignment
08273     // Adjust the ordering of legs of vertices in a consistent way
08274     // with the interaction defined in the model.
08275
08276     ANode *an;
08277     ENode *en;
08278     int work[3][GRCC_MAXLEGS];
08279     int n, lr, e;
08280     int *elist;
08281     int lg, ed, eg;
08282 #ifdef CHECK
08283     int cl, ok = True;
08284 #endif
08285
08286     egraph->assigned = True;
08287
08288     egraph->aId = aid;
08289     egraph->nsym = nsym;
08290     egraph->esym = esym;
08291     egraph->bicount = -1;
08292     if (sproc != NULL) {
08293         egraph->extperm = sproc->extperm;
08294     } else {
08295         egraph->extperm = 1;
08296     }
08297     egraph->nsym1 = nsym1;
08298     egraph->multp = (egraph->extperm * egraph->nsym1) / egraph->nsym;
08299
08300 #ifdef CHECK
08301     checkAG("fillEGraph:0");
08302 #endif
08303     // external node
08304     for (n = 0; n < nNodes; n++) {
08305         // vertices
08306 #ifdef CHECK
08307         cl = pnclass->nd2cl[n];
08308         if (isATEXternal(pnclass->type[cl])) {
08309             ;
08310         } else if (nodes[n]->cand->st != AS_Assigned) {
08311             printf("*** fillEGraph : node %d is not assigned", n);
08312             prCand("fillEGraph: node");
08313             erEnd("fillEGraph : node is not assigned");
08314         }
08315 #endif
08316
08317         an = nodes[n];
08318         en = egraph->nodes[n];
08319
08320         en->initAss(egraph, n, an->deg);
08321
08322         en->intrct = an->cand->ilist[0];
08323         elist = reordLeg(n, work[0], work[1], work[2]);
08324         for (lr = 0; lr < nodes[n]->deg; lr++) {
08325             if (elist == NULL) {
08326                 lg = lr;
08327             } else {
08328                 lg = elist[lr];
08329             }
08330 #ifdef CHECK
08331             if (lg < 0 || lg >= nodes[n]->deg) {
08332                 printf("*** fillEGraph: n=%d, lr=%d: 0 <= lg=%d < %d\n",
08333                     n, lr, lg, nodes[n]->deg);
08334                 erEnd("fillEGrah: illegal reordering");
08335             }
08336 #endif
08337             ed = an->aedges[lg];

```

```

08338         eg = an->aelegs[lg];
08339
08340         en->edges[lr] = I2Vedge(ed, 2*eg-1);
08341
08342         egraph->edges[ed]->nodes[eg] = n;
08343         egraph->edges[ed]->nlegs[eg] = lr;
08344     }
08345 }
08346 // edges
08347 for (e = 0; e < nEdges; e++) {
08348 #ifdef CHECK
08349     if (edges[e]->cand->nplist != 1) {
08350         printf("*** fillEGraph : edge %d is not assigned", e);
08351         prCand("fillEGraph: edge");
08352         erEnd("fillEGraph : edge is not assigned");
08353     }
08354 #endif
08355     egraph->edges[e]->ptcl = edges[e]->cand->plist[0];
08356 }
08357 #ifdef EDGEORDER
08358 for (e = 0; e < nEdges; e++) {
08359     if (egraph->edges[e]->ptcl < 0) {
08360         egraph->edges[e]->ptcl = - edges[e]->cand->plist[0];
08361         n = egraph->edges[e]->nodes[0];
08362         lr = egraph->edges[e]->nlegs[0];
08363         egraph->edges[e]->nodes[0] = egraph->edges[e]->nodes[1];
08364         egraph->edges[e]->nlegs[0] = egraph->edges[e]->nlegs[1];
08365         egraph->edges[e]->nodes[1] = n;
08366         egraph->edges[e]->nlegs[1] = lr;
08367
08368         egraph->nodes[n]->edges[lr] = - egraph->nodes[n]->edges[lr];
08369         n = egraph->edges[e]->nodes[0];
08370         lr = egraph->edges[e]->nlegs[0];
08371         egraph->nodes[n]->edges[lr] = - egraph->nodes[n]->edges[lr];
08372     }
08373 }
08374 #endif
08375 #ifdef CHECK
08376 for (ed = 0; ed < nEdges; ed++) {
08377     n = egraph->edges[ed]->nodes[0];
08378     lr = egraph->edges[ed]->nlegs[0];
08379     if (egraph->nodes[n]->edges[lr] != -ed-1) {
08380         fprintf(GRCC_Stderr, "+++ node[%d][%d]=%d != - (edge[%d][0] + 1) = %d\n",
08381             n, lr, egraph->nodes[n]->edges[lr], e, -ed-1);
08382         ok = False;
08383     }
08384     n = egraph->edges[ed]->nodes[1];
08385     lr = egraph->edges[ed]->nlegs[1];
08386     if (egraph->nodes[n]->edges[lr] != ed+1) {
08387         fprintf(GRCC_Stderr, "+++ node[%d][%d]=%d != + (edge[%d][0] + 1) = %d\n",
08388             n, lr, egraph->nodes[n]->edges[lr], e, ed+1);
08389         ok = False;
08390     }
08391 }
08392 if (!ok) {
08393     egraph->print();
08394     erEnd("fillEGraph: illegal connection");
08395 }
08396 #endif
08397
08398 // analyse fermion line and determine Fermi statistical sign factor
08399 if (!model->skipFLine) {
08400     egraph->getFLines();
08401 } else {
08402     if (prlevel > 0) {
08403         fprintf(GRCC_Stderr, "+++ Sign factors related to "
08404             "Dirac/Majorana/Ghost particles are not calculated.\n");
08405     }
08406 }
08407 }
08408
08409 #ifdef CHECK
08410     checkAG("fillEGraph:0");
08411 #endif
08412     return True;
08413 }
08414 }
08415
08416 //-----
08417 int *Assign::reordLeg(int n, int *reord, int *plist, int *used)
08418 {
08419     // Reorder legs of node 'n' in accordance with the definition
08420     // of the interaction in the model
08421     //
08422     // plist[reord[j]] = model->interacts[ia]->plist[j]
08423
08424     int lg, ia, lr, deg, pt, ed;

```



```

08425     int *ilegs;
08426 #ifdef CHECK
08427     Bool found;
08428 #endif
08429
08430     // external node
08431     if (nodes[n]->cand->st == AS_AssExt) {
08432         reord[0] = 0;
08433         return 0;
08434     }
08435
08436     // degree of the node
08437     deg = nodes[n]->deg;
08438
08439     if (deg <= 1) {
08440         reord[0] = 0;
08441         return 0;
08442     }
08443
08444     // list of particles at the legs of the node 'n'
08445     for (lg = 0; lg < deg; lg++) {
08446         ed = nodes[n]->aedges[lg];
08447         pt = edges[ed]->cand->plist[0];
08448         plist[lg] = legEdgeParticle(n, lg, pt);
08449         used[lg] = 0;
08450     }
08451
08452     // list of legs in the interaction
08453     ia = nodes[n]->cand->ilist[0];
08454     ilegs = model->interacts[ia]->plist;
08455
08456     // reorder
08457 #ifdef CHECK
08458     found = False;
08459 #endif
08460     for (lr = 0; lr < deg; lr++) {
08461         for (lg = 0; lg < deg; lg++) {
08462             if (used[lg] == 0 && plist[lg] == ilegs[lr]) {
08463                 reord[lr] = lg;
08464                 used[lg] = 1;
08465 #ifdef CHECK
08466                 found = True;
08467 #endif
08468                 break;
08469             }
08470         }
08471     }
08472 #ifdef CHECK
08473     if (!found) {
08474         printf("*** reordLeg: illegal list of particles:"
08475                "interaction %d ", ia);
08476         prIntArray(deg, ilegs, " ");
08477         printf("vertex %d ", n);
08478         prIntArray(deg, plist, "\n");
08479         prCand("reordLeg");
08480         erEnd("reordLeg: illegal list of particles");
08481     }
08482 #endif
08483 }
08484
08485     return reord;
08486 }
08487
08488 //=====
08489 // Adjust the direction of a particle on an edge and on a leg of
08490 // node.
08491 //-----
08492 int Assign::getLegParticle(int n, int ln)
08493 {
08494     // Get particle code of (n, ln) when particle 'pt' runs
08495     // in the direction of the edge at (n, ln).
08496     //
08497     // Get particle code in the direction the edge at (n, ln)
08498     // when particle 'pt' is at leg (n, ln).
08499     //
08500     // These two cases are realized by the same function
08501
08502     ANode *nd;
08503     int elg, ed, pt;
08504
08505     nd = nodes[n];
08506     elg = nd->aelegs[ln];
08507     ed = nd->aedges[ln];
08508     pt = edges[ed]->cand->plist[0];
08509
08510     // normalization of sign for neutral particle
08511     if (elg == 0) {

```

```

08512         return model->normalParticle(-pt);
08513     } else {
08514         return model->normalParticle(pt);
08515     }
08516 }
08517
08518 //-----
08519 int Assign::legEdgeParticle(int n, int ln, int pt)
08520 {
08521     // Get particle code of (n, ln) when particle 'pt' runs
08522     // in the direction of the edge at (n, ln).
08523     //
08524     // Get particle code in the direction the edge at (n, ln)
08525     // when particle 'pt' is at leg (n, ln).
08526     //
08527     // These two cases are realized by the same function
08528
08529     // normalization of sign for neutral particle
08530
08531     if (nodes[n]->aelegs[ln] == 0) {
08532         return model->normalParticle(-pt);
08533     } else {
08534         return model->normalParticle(pt);
08535     }
08536 }
08537
08538 //-----
08539 int Assign::legPart(int v, int lg, int nplist, int *plist, int *rlist, const int size)
08540 {
08541     // Convert list of candidate particles in 'plist' at (v, lg) ==> edge
08542     // or edge ==> (v, lg)
08543
08544     int j, nrlist;
08545
08546     nrlist = 0;
08547     for (j = 0; j < nplist; j++) {
08548         nrlist = intSetAdd(nrlist, rlist,
08549                             legEdgeParticle(v, lg, plist[j]), size);
08550     }
08551     return nrlist;
08552 }
08553
08554 //-----
08555 int Assign::candPart(int v, int ln, int *plist, const int size)
08556 {
08557     // update candidate particles incoming to (v, ln)
08558
08559     int en;
08560     int ntplist;
08561     ECand *ec;
08562
08563     en = nodes[v]->aedges[ln];
08564
08565     #ifdef CHECK
08566     if (edges[en]->cand->det) {
08567         printf("*** candPart : particle of leg (%d, %d) "
08568                "is assigned to %d\n",
08569                v, ln, edges[en]->cand->plist[0]);
08570         checkCand("candPart");
08571         mgraph->printAdjMat(mgraph->curcl);
08572         prCand("candPart");
08573         erEnd("candPart : particle of leg is assigned");
08574     }
08575     #endif
08576
08577     ec = edges[en]->cand;
08578
08579     ntplist = legPart(v, ln, ec->nplist, ec->plist, plist, size);
08580
08581     return ntplist;
08582 }
08583
08584 //=====
08585 // tools for assignment
08586 //-----
08587
08588 int Assign::selUnAssVertexSimp(int lastv)
08589 {
08590     // Select a vertex for assignment to its leg.
08591     // We take one with minimum number of candidates
08592     //
08593     // There may be better method.
08594
08595     int v0, v;
08596
08597     // select vertex one by one in the sequential order of node number
08598     v0 = Max(lastv+1, nExtern);

```

```

08599     for (v = v0; v < nNodes; v++) {
08600         if (nodes[v]->cand->st == AS_UnAssLegs) {
08601             return v;
08602         }
08603     }
08604     return -1;
08605 }
08606
08607 //-----
08608 int Assign::selUnAssVertex(void)
08609 {
08610     // Select a vertex for assignment to its leg.
08611     // We take one with minimum number of candidates
08612     //
08613     // There may be better method.
08614
08615     int v0, v, nl;
08616
08617     // select vertex one by one in the sequential order of node number
08618     v0 = -1;
08619     nl = 0;
08620     for (v = 0; v < nNodes; v++) {
08621         if (nodes[v]->cand->st == AS_UnAssLegs) {
08622             if (nodes[v]->cand->nilist == 1) {
08623                 return v;
08624             } else if (nodes[v]->cand->nilist > 1) {
08625                 // select node with fewer candidates
08626                 if (v0 < 0 || nodes[v]->cand->nilist < nl) {
08627                     v0 = v;
08628                     nl = nodes[v]->cand->nilist;
08629                 }
08630             }
08631         }
08632     }
08633     return v0;
08634 }
08635
08636 //-----
08637 int Assign::selUnAssLeg(int v, int lastlg)
08638 {
08639     // Select a leg of vertex 'v' for the assignment.
08640     // The lastly selected leg was 'lastlg'.
08641
08642     int lg0, lg, e;
08643 #ifdef CHECK
08644     int n0, n1;
08645 #endif
08646
08647     // lg0 = Max(lastlg, lastlg + 1);
08648     lg0 = lastlg + 1;
08649
08650     for (lg = lg0; lg < nodes[v]->deg; lg++) {
08651
08652 #ifdef CHECK
08653         if (lastlg >= 0) {
08654             n0 = nodes[v]->anodes[lastlg];
08655         } else {
08656             n0 = -1;
08657         }
08658         n1 = nodes[v]->anodes[lg];
08659         if (n0 > n1) {
08660             printf("*** selUnAssLeg: n0=%d > n1=%d\n", n0, n1);
08661             printf("*** illegal connection\n");
08662             erEnd("selUnAssLeg: n0 > n1");
08663         }
08664 #endif
08665
08666         e = nodes[v]->aedges[lg];
08667         if (!edges[e]->cand->det) {
08668             return lg;
08669         }
08670     }
08671     return -1;
08672 }
08673
08674 //-----
08675 NCandSt Assign::assignIVertex(int v, int ia)
08676 {
08677     // Try to assign interaction 'ia' to 'v'
08678
08679     int lplist[GRCC_MAXLEGS], iplist[GRCC_MAXLEGS];
08680     int lg, e, pt, deg;
08681
08682     if (nodes[v]->cand->st == AS_Assigned || nodes[v]->cand->st == AS_AssExt) {
08683         return AS_Assigned0;
08684     }

```

```

08686     }
08687
08688     deg = nodes[v]->deg;
08689     if (deg != model->interacts[ia]->nlegs) {
08690         return AS_Impossible;
08691     }
08692
08693     // list of particles at the legs of the vertex
08694     for (lg = 0; lg < deg; lg++) {
08695         e = nodes[v]->aedges[lg];
08696         if (edges[e]->cand->nplist != 1) {
08697             return AS_UnAssLegs;
08698         } else {
08699             pt = edges[e]->cand->plist[0];
08700             lplist[lg] = legEdgeParticle(v, lg, pt);
08701         }
08702         iplist[lg] = model->interacts[ia]->plist[lg];
08703     }
08704
08705     sorti(deg, lplist);
08706     sorti(deg, iplist);
08707
08708     // from interaction
08709     if (cmpArray(deg, lplist, deg, iplist) == 0) {
08710         // orders of coupling constants should be checked ???
08711         astack->pushNode(v);
08712         nodes[v]->cand->st = AS_Assigned;
08713         nodes[v]->cand->nlist = 1;
08714         nodes[v]->cand->ilist[0] = ia;
08715         return AS_Assigned;
08716     }
08717     return AS_Impossible;
08718 }
08719
08720
08721 //-----
08722 Bool Assign::assignPLeg(int n, int ln, int pt)
08723 {
08724     // A particle 'pt' is assigned to leg 'ln' of node 'n'
08725
08726     ANode *nd;
08727     int e, ept;
08728     Bool ok;
08729
08730 #ifdef CHECK
08731     checkNode(n, "assignPLeg0");
08732     checkCand("assignPLeg");
08733 #endif
08734
08735     // nodes and the edge
08736     nd = nodes[n];
08737     e = nd->aedges[ln];
08738
08739     // already determined
08740     if (edges[e]->cand->det) {
08741         return True;
08742     }
08743
08744     if (!isOrdPLeg(n, ln, pt)) {
08745 #ifdef DEBUGM
08746         nopleg++;
08747 #endif
08748         return False;
08749     }
08750
08751     // particle on the edge
08752     ept = legEdgeParticle(n, ln, pt);
08753
08754 #ifdef CHECK
08755     if (!isIn(edges[e]->cand->nplist, edges[e]->cand->plist, ept)) {
08756         printf("*** assignPLeg: particle %d is not in the cand. of e=%d",
08757             ept, e);
08758         edges[e]->cand->prECand("\n");
08759         prCand("assignPLeg");
08760         erEnd("assignPLeg: particle is not in the cand.");
08761     }
08762 #endif
08763
08764     // save the current configuration
08765     astack->pushEdge(e);
08766
08767     // determine the particle on the edge
08768     edges[e]->cand->nplist = 1;
08769     edges[e]->cand->plist[0] = ept;
08770
08771     ok = detEdge(e);
08772

```

```

08773     return ok;
08774 }
08775
08776 //-----
08777 Bool Assign::detEdge(int e)
08778 {
08779     Bool ok0, ok1;
08780
08781     if (edges[e]->cand->nplist != 1) {
08782         return False;
08783     }
08784     edges[e]->cand->det = True;
08785
08786     // update candidate list
08787     ok0 = updateCandNode(edges[e]->nodes[0]);
08788     if (ok0) {
08789         ok1 = updateCandNode(edges[e]->nodes[1]);
08790     } else {
08791         ok1 = False;
08792     }
08793
08794     return (ok0 && ok1);
08795 }
08796
08797 //-----
08798 Bool Assign::isOrdPLeg(int n, int ln, int pt)
08799 {
08800     int e0, e1, ln0, ep0, pt0, nn, n0, n1, ln1, pt1, e1, ep1;
08801
08802     if (nodes[n]->deg < 2) {
08803         return True;
08804     }
08805     nn = nodes[n]->anodes[ln];
08806     if (n <= nn) {
08807         n0 = n;
08808         n1 = nn;
08809         ln0 = ln;
08810         pt0 = pt;
08811     } else {
08812         n0 = nn;
08813         n1 = n;
08814         e0 = nodes[n]->aedges[ln];
08815         e1 = nodes[n]->aelegs[ln];
08816         ln0 = edges[e0]->nlegs[1-e1];
08817         ep0 = legEdgeParticle(n, ln, pt);
08818         pt0 = legEdgeParticle(n0, ln0, ep0);
08819     }
08820
08821     for (ln1 = 0; ln1 < nodes[n0]->deg; ln1++) {
08822         if (ln1 == ln0 || n1 != nodes[n0]->anodes[ln1]) {
08823             continue;
08824         }
08825         e1 = nodes[n0]->aedges[ln1];
08826         if (!edges[e1]->cand->det) {
08827             continue;
08828         }
08829         ep1 = edges[e1]->cand->p1st[0];
08830         pt1 = legEdgeParticle(n0, ln1, ep1);
08831         if ((ln0 < ln1 && pt0 > pt1) ||
08832             (ln0 > ln1 && pt0 < pt1)) {
08833             return False;
08834         }
08835     }
08836     return True;
08837 }
08838
08839 //=====
08840 // Operations of candidate
08841 //-----
08842 Bool Assign::candPartClassify(int v, int *npdass, int *pdass, int *npuass, int *puass, const int size)
08843 {
08844     // Construct the classified lists of particles possible to assign to
08845     // the legs of node v
08846     //
08847     // pdass = (duplicated set of particles where edges has a unique
08848     // candidate)
08849     // puass = (duplicated set of other candidate particles)
08850
08851     int lg, e, npl, ass, jd, ju, j, pt, pts;
08852
08853     jd = 0;
08854     ju = 0;
08855     for (lg = 0; lg < nodes[v]->deg; lg++) {
08856         e = nodes[v]->aedges[lg];
08857         npl = edges[e]->cand->nplist;
08858         if (npl < 1) {
08859             return False;

```

```

08860     }
08861     ass = (npl == 1);
08862     for (j = 0; j < edges[e]->cand->nplist; j++) {
08863         pt = edges[e]->cand->plist[j];
08864         pts = legEdgeParticle(v, lg, pt);
08865         if (ass) {
08866             jd = intSListAdd(jd, pdass, pts, size);
08867         } else {
08868             ju = intSListAdd(ju, puass, pts, size);
08869         }
08870     }
08871 }
08872 *npdass = jd;
08873 *npuass = ju;
08874
08875 return True;
08876 }
08877
08878 //-----
08879 Bool Assign::updateCandNode(int v)
08880 {
08881     // Update the configuration
08882     // - nodes[v]->cand->ilist
08883     // - edges[e]->cand->plist for the adjacent edge
08884     // - nodes[n]->cand->unass for the adjacent node 'n' of 'e'
08885
08886     int npdass, pdass[GRCC_MAXPSLIST];
08887     int npuass, puass[GRCC_MAXPSLIST];
08888     int nsub0, sub0[GRCC_MAXPSLIST];
08889     int niplist, iplist[GRCC_MAXPSLIST];
08890     int nd0, d0[GRCC_MAXMPARTICLES2];
08891     int nd1, d1[GRCC_MAXMPARTICLES2];
08892     int ns0, s0[GRCC_MAXMPARTICLES2];
08893     int neplist, eplist[GRCC_MAXMPARTICLES2];
08894     int nitlist, itlist[GRCC_MAXMINTERACT];
08895     int e, it, j, k, i, lg;
08896
08897     // external node
08898     if (nodes[v]->cand->st == AS_AssExt) {
08899 #ifdef CHECK
08900         e = nodes[v]->aedges[0];
08901         if (e < 0 || edges[e]->cand->nplist != 1) {
08902             printf("*** illegal external node: v=%d e=%d :", v, e);
08903             prCand("updateCandNode");
08904             erEnd("illegal external node");
08905         }
08906 #endif
08907         return True;
08908     }
08909
08910     // impossible configuration.
08911     if (nodes[v]->cand->nlist < 1) {
08912         return False;
08913     }
08914
08915     // list of possible particles on the edges of the vertex.
08916     // pdass = (set of assigned particles)
08917     // puass = (list of particles of edges with two of more candidates)
08918
08919     niplist = 0;
08920     nitlist = 0;
08921     if (!candPartClassify(v, &npdass, pdass, &npuass, puass, GRCC_MAXPSLIST)) {
08922         return False;
08923     }
08924
08925     if (npdass < 1) {
08926         return False;
08927     }
08928
08929     // from interaction : slist
08930     // Conditions :
08931     // 1. pdass \subset slist
08932     // 2. slist-pdass \subset puass
08933     // from interaction : slist
08934     // conditions : pdass \subset slist \subset (pdass + puass)
08935     // sub0 = slist - pdass,
08936     nitlist = 0;
08937     for (j = 0; j < nodes[v]->cand->nlist; j++) {
08938         it = nodes[v]->cand->ilist[j];
08939         // conditions:
08940         // 1. pdass \subset slist <=> pdass-slist = \emptyset
08941         // 2. slist-pdass \subset puass <=> (slist-pdass)-puass = \emptyset
08942         if (isSubSList(npdass, pdass,
08943             model->interacts[it]->nslist,
08944             model->interacts[it]->slist) ) {
08945             nsub0 = subtrSet(model->interacts[it]->nslist,

```

```

08947             model->interacts[it]->slist,
08948             npdass, pdass, sub0, GRCC_MAXPSLIST);
08949         if (nsub0 < 1) {
08950             // slist == pdass : no additional candidate particle
08951             nitlist = intSetAdd(nitlist, itlist, it, GRCC_MAXMINTERACT);
08952         } else {
08953             if (isSubSList(nsub0, sub0, npuass, puass)) {
08954                 nitlist = intSetAdd(nitlist, itlist, it, GRCC_MAXMINTERACT);
08955                 for (k = 0; k < nsub0; k++) {
08956                     niplist = intSetAdd(niplist, iplist, sub0[k], GRCC_MAXPSLIST);
08957                 }
08958             }
08959         }
08960     }
08961 }
08962 }
08963
08964 if (nitlist < 1) {
08965     return False;
08966 }
08967
08968 if (nitlist != nodes[v]->cand->nilist) {
08969     astack->pushNode(v);
08970     nodes[v]->cand->nilist = nitlist;
08971     for (i = 0; i < nitlist; i++) {
08972         nodes[v]->cand->ilist[i] = itlist[i];
08973     }
08974 #ifdef CHECK
08975 } else if (nitlist > nodes[v]->cand->nilist) {
08976     erEnd("larger ncand");
08977 #endif
08978 }
08979
08980 // edge
08981 for (lg = 0; lg < nodes[v]->deg; lg++) {
08982     e = nodes[v]->aedges[lg];
08983     if (edges[e]->cand->nplist > 1) {
08984         // elist = {candidate particles in the direction of the edge}
08985         // s0 = plist \cap elist
08986         neplist = legPart(v, lg, niplist, iplist, eplist, GRCC_MAXMPARTICLES2);
08987         listDiff(edges[e]->cand->nplist, edges[e]->cand->plist,
08988                 neplist, eplist,
08989                 &nd0, d0, &ns0, s0, &nd1, d1);
08990
08991         if (ns0 < 1) {
08992             return False;
08993         }
08994         if (ns0 < edges[e]->cand->nplist) {
08995             astack->pushEdge(e);
08996             edges[e]->cand->nplist = ns0;
08997             for (i = 0; i < ns0; i++) {
08998                 edges[e]->cand->plist[i] = s0[i];
08999             }
09000         }
09001     }
09002 }
09003 for (lg = 0; lg < nodes[v]->deg; lg++) {
09004     e = nodes[v]->aedges[lg];
09005     if (edges[e]->cand->nplist == 1 && !edges[e]->cand->det) {
09006         if (!detEdge(e)) {
09007             return False;
09008         }
09009     }
09010 }
09011
09012 return True;
09013 }
09014
09015 //=====
09016 // Check order of coupling constants, duplication and isomorphism
09017 //-----
09018 Bool Assign::checkOrderCpl(void)
09019 {
09020     // Check order of coupling constants
09021
09022     int ord[GRCC_MAXNCPLG];
09023     int lcn, n, ia, j, cl;
09024
09025     // list of coupling constants
09026     lcn = model->ncouple;
09027
09028     if (lcn == 1) {
09029         return True;
09030     }
09031
09032     // sum up for each coupling constants
09033     for (j = 0; j < GRCC_MAXNCPLG; j++) {

```

```

09034     ord[j] = 0;
09035 }
09036 for (n = 0; n < nNodes; n++) {
09037     cl = pnclass->nd2cl[n];
09038     if (!isATEXternal(pnclass->type[cl])) {
09039         ia = nodes[n]->cand->ilist[0];
09040         for (j = 0; j < lcn; j++) {
09041             ord[j] += model->interacts[ia]->clist[j];
09042         }
09043     }
09044 }
09045
09046 // comparison
09047 for (j = 0; j < lcn; j++) {
09048     if (sproc->clist[j] != ord[j]) {
09049         return False;
09050     }
09051 }
09052 return True;
09053 }
09054
09055 //-----
09056 Bool Assign::isOrdLegs(void)
09057 {
09058     // Whether graph is duplicated because of the violation
09059     // of ordering in the case of multiple connections
09060     // v0 <-- v --> v1
09061
09062     int v, l0, v0, l1, v1, e0, e1, q0, q1, p0, p1, cl;
09063
09064     for (v = 0; v < nNodes; v++) {
09065         cl = pnclass->nd2cl[v];
09066         if (!isATEXternal(pnclass->type[cl])) {
09067             for (l0 = 0; l0 < nodes[v]->deg; l0++) {
09068                 v0 = nodes[v]->anodes[l0];
09069                 for (l1 = l0+1; l1 < nodes[v]->deg; l1++) {
09070                     v1 = nodes[v]->anodes[l1];
09071
09072                     if (v0 == v1 && v <= v0) {
09073                         // multiple connections (v, l0) and (v, l1)
09074                         e0 = nodes[v]->aedges[l0];
09075                         e1 = nodes[v]->aedges[l1];
09076                         q0 = edges[e0]->cand->plist[0];
09077                         q1 = edges[e1]->cand->plist[0];
09078                         p0 = legEdgeParticle(v, l0, q0);
09079                         p1 = legEdgeParticle(v, l1, q1);
09080                         if (p0 > p1) {
09081                             // violation of ordering
09082                             return False;
09083                         }
09084                     }
09085                 }
09086             }
09087         }
09088     }
09089     return True;
09090 }
09091 }
09092
09093 //-----
09094 Bool Assign::isIsomorphic(MNodeClass *cl, BigInt *nsym, BigInt *esym, BigInt *nsym1)
09095 {
09096     // Check whether the current graph is the representative of
09097     // a isomorphic class.
09098     // Returns (nsym, esym)
09099     // nsym = symmetry factor by the permutation of nodes.
09100     // esym = symmetry factor by the permutation of edge.
09101     // If this graph is not a representative, then returns (0,0).
09102
09103     int j, cmp, n, cln;
09104     BigInt ngelem;
09105     int *p;
09106     Bool invext;
09107
09108     ngelem = mgraph->group->nElem();
09109 #ifndef CHECK
09110     if (mgraph->nsym > 1 && ngelem <= 1) {
09111         printf("*** isIsomorphic: illegal group: "
09112             "ngelem=%ld, mgraph->sym=(%ld, %ld)\n",
09113             ngelem, mgraph->nsym, mgraph->esym);
09114         erEnd("Assign::isIsomorphic: illegal group");
09115     }
09116 #endif
09117     if (ngelem == 0) {
09118         *nsym = 1;
09119         *nsym1 = 1;
09120     } else {

```



```

09121     *nsym = 0;
09122     *nsym1 = 0;
09123     for (j = 0; j < ngelem; j++) {
09124         // check the graph is the representative and count 'nsym'.
09125
09126         //p = mgraph->group->nextElem();
09127         p = mgraph->group->elem[j];
09128
09129         cmp = cmpPermGraph(p, cl);
09130
09131         if (cmp < 0) {           // duplicated graph
09132 #ifdef DEBUGM
09133             niso1++;
09134 #endif
09135             return False;
09136         } else if (cmp == 0) { // not duplicated
09137             (*nsym)++;
09138
09139             if (opt->values[GRCC_OPT_SymmInitial] || opt->values[GRCC_OPT_SymmFinal]) {
09140                 invext = True;
09141                 for (n = 0; n < nNodes; n++) {
09142                     cln = pnclass->nd2cl[n];
09143                     if (!isATEXternal(pnclass->type[cln])) {
09144                         continue;
09145                     }
09146                     if (p[n] != n) {
09147                         invext = False;
09148                     }
09149                 }
09150                 if (invext) {
09151                     (*nsym1)++;
09152                 }
09153             } else {
09154                 (*nsym1)++;
09155             }
09156         }
09157     }
09158 }
09159 if (*nsym1 == 0) {
09160     *nsym1 = *nsym;
09161 }
09162
09163 // calculate permutations of edges
09164 *esym = edgeSym();
09165 if (*esym < 1) {
09166 #ifdef DEBUGM
09167     niso2++;
09168 #endif
09169     return False;
09170 }
09171
09172 return True;
09173 }
09174
09175 //-----
09176 int Assign::cmpPermGraph(int *p, MNodeClass *cl)
09177 {
09178     // compare the graph with one permutated by 'p'.
09179
09180     int n, cmp, n1, n2, p1, p2, j;
09181     ANode *nd1, *np1;
09182     // ANode *nd2, *np2;
09183     int njn, jn[GRCC_MAXLEGS];
09184     int njp, jp[GRCC_MAXLEGS];
09185
09186 #ifdef CHECK
09187     if (p == NULL) {
09188         erEnd("Assign::cmpPermGraph: p=NULL");
09189     }
09190 #endif
09191 #ifdef DEBUG1
09192     printf("cmpPermGraph:0: p=");
09193     prIntArray(nNodes, p, "\n");
09194 #endif
09195     for (n = 0; n < nNodes; n++) {
09196         if (!isATEXternal(pnclass->type[pnclass->nd2cl[n]])) {
09197             cmp = cmpNodes(n, p[n], cl);
09198             if (cmp != 0) {
09199 #ifdef DEBUGM
09200                 if (cmp < 0) { niso1++; }
09201 #endif
09202                 return cmp;
09203             }
09204         }
09205     }
09206
09207     njn = 0;

```

```

09208     njp = 0;
09209     for (n1 = 0; n1 < nNodes; n1++) {
09210         p1 = p[n1];
09211         nd1 = nodes[n1];
09212         np1 = nodes[p1];
09213         for (n2 = n1; n2 < nNodes; n2++) {
09214             if (mgraph->adjMat[n1][n2] == 0) {
09215                 continue;
09216             }
09217             p2 = p[n2];
09218             if (p1 == n1 && p2 == n2) {
09219                 continue;
09220             }
09221             cmp = mgraph->adjMat[n1][n2] - mgraph->adjMat[p1][p2];
09222             if (cmp != 0) {
09223                 #ifdef DEBUGM
09224                     if (cmp < 0) { nisol2++; }
09225                 #endif
09226                 return cmp;
09227             }
09228             njn = 0;
09229             njp = 0;
09230             for (j = 0; j < nd1->deg; j++) {
09231                 if (nd1->anodes[j] == n2) {
09232                     jn[njn++] = getLegParticle(n1, j);
09233                 }
09234             }
09235             for (j = 0; j < np1->deg; j++) {
09236                 if (np1->anodes[j] == p2) {
09237                     jp[njp++] = getLegParticle(p1, j);
09238                 }
09239             }
09240             cmp = njn - njp;
09241             if (cmp != 0) {
09242                 #ifdef DEBUGM
09243                     if (cmp < 0) { nisol3++; }
09244                 #endif
09245                 return cmp;
09246             }
09247             // ignore ordering for multiple connections
09248             if (mgraph->adjMat[n1][n2] >= 2) {
09249                 sorti(njn, jn);
09250                 sorti(njp, jp);
09251             }
09252             for (j = 0; j < njn; j++) {
09253                 cmp = jn[j] - jp[j];
09254                 if (cmp != 0) {
09255                     #ifdef DEBUGM
09256                         if (cmp < 0) { nisol4++; }
09257                     #endif
09258                     return cmp;
09259                 }
09260             }
09261         }
09262     }
09263     return 0;
09264 }
09265
09266 //-----
09267 int Assign::cmpNodes(int nd0, int nd1, MNodeClass *cn)
09268 {
09269     // Comparison of two nodes 'nd0' and 'nd1'
09270     // Ordering is lexicographical (class, connection configuration)
09271     //
09272     int cmp;
09273
09274     // Wether two nodes are in a same class or not.
09275     cmp = cn->ndcl[nd0] - cn->ndcl[nd1];
09276     if (cmp != 0) {
09277         #ifdef DEBUGM
09278             if (cmp < 0) { nisol11++; }
09279         #endif
09280         return cmp;
09281     }
09282
09283     // interaction
09284     cmp = nodes[nd0]->cand->ilist[0] - nodes[nd1]->cand->ilist[0];
09285     #ifdef DEBUGM
09286         if (cmp < 0) { nisol12++; }
09287     #endif
09288     return cmp;
09289 }
09290
09291 // interaction
09292 cmp = nodes[nd0]->cand->ilist[0] - nodes[nd1]->cand->ilist[0];
09293 #ifdef DEBUGM
09294     if (cmp < 0) { nisol12++; }
09295 #endif

```

```

09295 #endif
09296     return cmp;
09297 }
09298
09299 //-----
09300 BigInt Assign::edgeSym(void)
09301 {
09302     // calculate permutations of edges
09303
09304     int lg[GRCC_MAXLEGS], lt[GRCC_MAXLEGS], lc;
09305     ANode *nd1;
09306     int n1, n2, j, k, mult;
09307     BigInt esym;
09308
09309     esym = 1;
09310     for (n1 = 0; n1 < nNodes; n1++) {
09311         nd1 = nodes[n1];
09312         for (n2 = n1; n2 < nNodes; n2++) {
09313             if (mgraph->adjMat[n1][n2] > 1) {
09314                 lc = 0;
09315                 for (j = 0; j < nd1->deg; j++) {
09316                     if (nd1->anodes[j] == n2) {
09317                         lg[lc] = 1;
09318                         lt[lc] = getLegParticle(n1, j);
09319                         lc++;
09320                     }
09321                 }
09322                 for (j = 0; j < lc-1; j++) {
09323                     if (lg[j] > 0) {
09324                         for (k = lc-1; k > j; k--) {
09325                             if (lt[j] == lt[k]) {
09326                                 lg[j] += lg[k];
09327                                 lg[k] = 0;
09328                             }
09329                         }
09330                     }
09331                 }
09332
09333                 // calculate symmetry factor
09334                 for (j = 0; j < lc; j++) {
09335                     if (lg[j] < 1) {
09336                         continue;
09337                     }
09338                     mult = lg[j];
09339                     if (mult > 1) {
09340                         if (n1 == n2) {
09341                             // self-loop
09342                             esym *= ipow(2, mult/2) * factorial(mult/2);
09343                         } else {
09344                             esym *= factorial(mult);
09345                         }
09346                     }
09347                 }
09348             }
09349         }
09350     }
09351
09352     return esym;
09353 }
09354
09355 #ifdef CHECK
09356 //-----
09357 // check
09358 //-----
09359 Bool Assign::checkCand(const char *msg)
09360 {
09361     // Check the consistency of Candidate data
09362
09363     Bool ok;
09364     ANode *na;
09365     NCand *nc;
09366     ECand *ec;
09367     int n, lg, e;
09368
09369     // check 'nodes->cand'
09370     ok = True;
09371     for (n = 0; n < nNodes; n++) {
09372         na = nodes[n];
09373         nc = na->cand;
09374
09375         // check unassigned vertex
09376         if (nc->st == AS_UnAssLegs) {
09377
09378             // check assigned vertex
09379         } else if (nc->st == AS_Assigned) {
09380             if (nc->nlist < 1) {
09381                 printf("*** checkCand:7:%s:status (%d) of node %d says"

```

```

09382         " interaction is assigned to %d but ilist=",
09383         msg, nc->st, n, nc->st);
09384     prIntArray(nc->nilist, nc->ilist, "\n");
09385     ok = False;
09386 }
09387
09388     for (lg = 0; lg < nc->deg; lg++) {
09389         e = na->aedges[lg];
09390         ec = edges[e]->cand;
09391         if (ec->nplist != 1) {
09392             printf("*** checkCand:8:%s:status (%d) of node %d says"
09393                 " interaction is assigned "
09394                 " but unassigned edge %d is found\n",
09395                 msg, nc->st, n, e);
09396             ok = False;
09397         }
09398     }
09399
09400     // external particle
09401 } else if (nc->st == AS_AssExt) {
09402     // pt = nc->ilist[0];
09403     // *** pte = legEdgeParticle(n, 0, - pt);
09404 } else {
09405     printf("*** checkCand:10:%s:illegal status of node %d : %d",
09406         msg, n, nc->st);
09407     ok = False;
09408 }
09409 }
09410
09411 // check edges
09412
09413 for (e = 0; e < nEdges; e++) {
09414     ec = edges[e]->cand;
09415     if (ec != NULL && ec->nplist < 1) {
09416         printf("*** checkCand:12:%s:illegal edge %d\n", msg, e);
09417         ok = False;
09418     }
09419 }
09420
09421 if (!ok) {
09422     printf("*** checkCand:15:%s:illegal configuration\n", msg);
09423     prCand("checkCand");
09424     printf("*** checkCand:16:illegal configuration\n");
09425     erEnd("checkCand:16:illegal configuration");
09426 }
09427 return ok;
09428 }
09429
09430 //-----
09431 void Assign::checkNode(int n, const char *msg)
09432 {
09433     int j, it;
09434
09435     for (j = 0; j < nodes[n]->cand->nilist; j++) {
09436         it = nodes[n]->cand->ilist[j];
09437         if (Abs(it) >= GRCC_MAXMINTERACT) {
09438             printf("*** %s: n=%d, j=%d, it=%d\n", msg, n, j, it);
09439             nodes[n]->cand->prNCand(msg);
09440             printf("\n");
09441             erEnd("checkNode:illegal it");
09442         }
09443     }
09444 }
09445 #endif // CHECK
09446
09447 //*****
09448 // astack.cc
09449 //-----
09450 // Assign particles to the edges and interactions to nodes.
09451 // Completed graph data is saved in the form of EGraph.
09452
09453 //-----
09454 // stack operations
09455 //-----
09456 void NStack::print(const char *msg)
09457 {
09458     printf(" node=%d, deg=%d, st=%d, ilist=", noden, deg, st);
09459     prilist(nilist, ilist, msg);
09460 }
09461
09462 //-----
09463 void EStack::print(const char *msg)
09464 {
09465     printf(" edge=%d, det=%d, plist=", edgen, det);
09466     prilist(nplist, plist, msg);
09467 }
09468

```

```

09469 //-----
09470 AStack::AStack(int nsize, int esize)
09471 {
09472     int j;
09473
09474     agraph = NULL;
09475
09476     nSize = nsize;
09477     eSize = esize;
09478     if (nSize > 0) {
09479         nStack = new NStack*[nSize];
09480     } else {
09481         nStack = NULL;
09482     }
09483     if (eSize > 0) {
09484         eStack = new EStack*[eSize];
09485     } else {
09486         eStack = NULL;
09487     }
09488
09489     nStackP = 0;
09490     eStackP = 0;
09491
09492     for (j = 0; j < nSize; j++) {
09493         nStack[j] = new NStack();
09494     }
09495
09496     for (j = 0; j < eSize; j++) {
09497         eStack[j] = new EStack();
09498     }
09499 }
09500
09501 //-----
09502 AStack::~AStack(void)
09503 {
09504     int j;
09505
09506     if (eStack != NULL) {
09507         for (j = eSize-1; j >= 0; j--) {
09508             if (eStack[j] != NULL) {
09509                 delete eStack[j];
09510                 eStack[j] = NULL;
09511             }
09512         }
09513         delete[] eStack;
09514         eStack = NULL;
09515     }
09516
09517     if (nStack != NULL) {
09518         for (j = nSize-1; j >= 0; j--) {
09519             if (nStack[j] != NULL) {
09520                 delete nStack[j];
09521                 nStack[j] = NULL;
09522             }
09523         }
09524         delete[] nStack;
09525         nStack = NULL;
09526     }
09527 }
09528
09529 //-----
09530 void AStack::setAGraph(Assign *ag)
09531 {
09532     agraph = ag;
09533 }
09534
09535 //-----
09536 void AStack::pushNode(int n)
09537 {
09538     NCand *nc;
09539     NStack *ns;
09540     int j;
09541
09542     #ifndef CHECK
09543     if (agraph == NULL) {
09544         erEnd("pushNode: agraph is not defined");
09545     }
09546     #endif
09547     if (nStackP >= GRCC_MAXNSTACK) {
09548         erEnd("N-stack overflow (GRCC_MAXNSTACK)");
09549     }
09550     nc = agraph->nodelist[n]->cand;
09551     if (nc->nodelist >= GRCC_MAXMINTERACT) {
09552         erEnd("N-stack: too long list (GRCC_MAXMINTERACT)");
09553     }
09554     ns = nStack[nStackP];
09555     ns->noden = n;

```

```

09556     ns->deg      = nc->deg;
09557     ns->st       = nc->st;
09558     ns->nilist   = nc->nilist;
09559     for (j = 0; j < nc->nilist; j++) {
09560         ns->ilist[j] = nc->ilist[j];
09561     }
09562     nStackP++;
09563 }
09564
09565 //-----
09566 void AStack::pushEdge(int e)
09567 {
09568     ECand *ec;
09569     EStack *es;
09570     int j;
09571
09572     #ifdef CHECK
09573     if (agraph == NULL) {
09574         erEnd("pushEdge: agraph is not defined");
09575     }
09576     #endif
09577     if (eStackP >= GRCC_MAXESTACK) {
09578         erEnd("E-stack overflow (GRCC_MAXESTACK)");
09579     }
09580     ec = agraph->edges[e]->cand;
09581     if (ec->nplist >= GRCC_MAXMPARTICLES2) {
09582         erEnd("E-stack: Too long list (GRCC_MAXMPARTICLES)");
09583     }
09584     es = eStack[eStackP];
09585     es->edgen = e;
09586     es->det   = ec->det;
09587     es->nplist = ec->nplist;
09588     for (j = 0; j < ec->nplist; j++) {
09589         es->plist[j] = ec->plist[j];
09590     }
09591     eStackP++;
09592 }
09593
09594 //-----
09595 void AStack::checkPoint(CheckPt sav)
09596 {
09597     sav[0] = nStackP;
09598     sav[1] = eStackP;
09599 }
09600
09601 //-----
09602 void AStack::restoreNode(int spr)
09603 {
09604     NStack *stc;
09605     int     sp, n, j;
09606
09607     for (sp = nStackP-1; sp >= spr; sp--) {
09608         stc = nStack[sp];
09609         n = stc->noden;
09610         agraph->nodes[n]->cand->deg = stc->deg;
09611         agraph->nodes[n]->cand->st  = stc->st;
09612         agraph->nodes[n]->cand->nilist = stc->nilist;
09613         for (j = 0; j < stc->nilist; j++) {
09614             agraph->nodes[stc->noden]->cand->ilist[j] = stc->ilist[j];
09615         }
09616     }
09617     nStackP = spr;
09618 }
09619
09620 //-----
09621 void AStack::restoreEdge(int spr)
09622 {
09623     EStack *stc;
09624     int     sp, e, j;
09625
09626     for (sp = eStackP-1; sp >= spr; sp--) {
09627         stc = eStack[sp];
09628         e = eStack[sp]->edgen;
09629         agraph->edges[e]->cand->det = stc->det;
09630         agraph->edges[e]->cand->nplist = stc->nplist;
09631         for (j = 0; j < stc->nplist; j++) {
09632             agraph->edges[e]->cand->plist[j] = stc->plist[j];
09633         }
09634     }
09635     eStackP = spr;
09636 }
09637
09638 //-----
09639 void AStack::restore(CheckPt sav)
09640 {
09641     restoreNode(sav[0]);
09642     restoreEdge(sav[1]);

```

```

09643 }
09644
09645 #ifndef CHECK
09646 //-----
09647 void AStack::restoreMsg(CheckPt sav, const char *msg)
09648 {
09649     if (agraph == NULL) {
09650         erEnd("restore: agraph is not defined");
09651     }
09652     restore(sav);
09653
09654     if (!agraph->checkCand("restore")) {
09655         printf("restore is called from %s\n", msg);
09656     }
09657 }
09658 #endif
09659
09660 //-----
09661 void AStack::prStack(void)
09662 {
09663     int j;
09664
09665     printf("+++ prStack : (%d, %d)", nStackP, eStackP);
09666     for (j = 0; j < nStackP; j++) {
09667         printf("N:%4d ", j);
09668         nStack[j]->print("\n");
09669     }
09670     for (j = 0; j < eStackP; j++) {
09671         printf("E:%4d ", j);
09672         eStack[j]->print("\n");
09673     }
09674 }
09675 }
09676
09677 //*****
09678 // class Fraction
09679 //-----
09680 Fraction::Fraction(BigInt n, BigInt d)
09681 {
09682     BigInt g;
09683
09684     g = gcd(n, d);
09685     num = n/g;
09686     den = d/g;
09687     ratio = Real(n)/Real(d);
09688 }
09689
09690 //-----
09691 void Fraction::print(const char *msg)
09692 {
09693     double err = Abs(Real(num)/Real(den) - ratio);
09694
09695     if (err > GRCC_FRACERROR) {
09696         printf("%ld/%ld(%g) (overflow)%s", num, den, ratio, msg);
09697     } else {
09698         printf("%ld/%ld(%g)%s", num, den, ratio, msg);
09699     }
09700 }
09701
09702 //-----
09703 void Fraction::setValue(BigInt n, BigInt d)
09704 {
09705     num = n;
09706     den = d;
09707     ratio = Real(n)/Real(d);
09708 }
09709
09710 //-----
09711 void Fraction::setValue(Fraction &f)
09712 {
09713     num = f.num;
09714     den = f.den;
09715     ratio = f.ratio;
09716 }
09717
09718 //-----
09719 BigInt Fraction::gcd(BigInt n0, BigInt n1)
09720 {
09721     BigInt nn[2], r;
09722     int nc;
09723
09724     nn[0] = Abs(n0);
09725     nn[1] = Abs(n1);
09726     nc = 0;
09727
09728     while (1) {
09729         r = nn[nc] % nn[1-nc];

```

```

09730         if (r == 0) {
09731             return nn[l-nc];
09732         }
09733         nn[nc] = r;
09734         nc = 1 - nc;
09735     }
09736 }
09737
09738 //-----
09739 void Fraction::normal(void)
09740 {
09741     BigInt g;
09742
09743     if (den == 0) {
09744         erEnd("Fraction: den==0");
09745     }
09746     g = gcd(num, den);
09747     num /= g;
09748     den /= g;
09749     if (den < 0) {
09750         num = - num;
09751         den = - den;
09752     }
09753 }
09754
09755 //-----
09756 void Fraction::add(BigInt n, BigInt d)
09757 {
09758     BigInt g, dl;
09759
09760     if (d < 0) {
09761         n = -n;
09762         d = -d;
09763     }
09764     if (den < 0) {
09765         num = - num;
09766         den = - den;
09767     }
09768     g = gcd(den, d);
09769     dl = d/g;
09770     num = num * dl + n * (den/g);
09771     den = dl*den;
09772     g = gcd(num, den);
09773     num /= g;
09774     den /= g;
09775     ratio += Real(n)/Real(d);
09776 }
09777
09778 //-----
09779 void Fraction::add(Fraction f)
09780 {
09781     double r = ratio + f.ratio;
09782
09783     add(f.num, f.den);
09784     ratio = r;
09785 }
09786
09787 //-----
09788 void Fraction::sub(Fraction f)
09789 {
09790     double r = ratio - f.ratio;
09791
09792     add(-f.num, f.den);
09793     ratio = r;
09794 }
09795
09796 //-----
09797 Bool Fraction::isEq(Fraction f)
09798 {
09799     return (num == f.num && den == f.den);
09800 }
09801
09802 //*****
09803 // common.cc
09804 //=====
09805 static void erEnd(const char *msg)
09806 {
09807     if (erExit != NULL) {
09808         (*erExit)(msg, erExitArg);
09809     }
09810     fprintf(GRCC_Stderr, "*** Error : %s\n", msg);
09811     GRCC_ABORT();
09812 }
09813
09814 //-----
09815 static Bool nextPart(int nelelem, int nclist, int *clist, int *nl, int *r)
09816 {

```



```

09817     int rem, pn, j;
09818
09819     if (*r < 0) {
09820         *r = 0;
09821         rem = nelem;
09822         for (j = 0; j < nclist; j++) {
09823             nl[j] = rem/clist[j];
09824             rem -= nl[j]*clist[j];
09825         }
09826         if (rem == 0) {
09827             return True;
09828         }
09829     } else {
09830         rem = 0;
09831     }
09832     for (int c = 0; c < 100; c++) {
09833         rem += nl[nclist-1]*clist[nclist-1];
09834         for (pn = nclist-2; pn >= 0 && nl[pn] == 0; pn--) {
09835             ;
09836         }
09837         if (pn < 0) {
09838             return False;
09839         }
09840         rem += clist[pn];
09841         nl[pn]--;
09842         for (j = pn+1; j < nclist; j++) {
09843             nl[j] = rem/clist[j];
09844             rem -= nl[j]*clist[j];
09845         }
09846         if (rem == 0) {
09847             return True;
09848         }
09849     }
09850     printf("*** nextPart : too many repetition\n");
09851     return False;
09852 }
09853
09854 //-----
09855 static int *intdup(int n, int *a)
09856 {
09857     int *r, j;
09858
09859     r = new int[n];
09860     for (j = 0; j < n; j++) {
09861         r[j] = a[j];
09862     }
09863     return r;
09864 }
09865
09866 //-----
09867 static int *delintdup(int *a)
09868 {
09869     if (a != NULL) {
09870         delete[] a;
09871     }
09872     return NULL;
09873 }
09874
09875 //-----
09876 static void prilist(int n, const int *a, const char *msg)
09877 {
09878     printf("[");
09879     for (int j = 0; j < n; j++) {
09880         if (j!=0) printf(", ");
09881         printf("%d", a[j]);
09882     }
09883     printf("]%s", msg);
09884 }
09885
09886 //-----
09887 static int nextPerm(int nelem, int nclass, int *cl, int *r, int *q, int *p, int count)
09888 {
09889     // Sequential generation of all permutations.
09890
09891     int j, k, n, e, t;
09892     Bool b;
09893
09894     for (j = 0; j < nelem; j++) {
09895         p[j] = j;
09896     }
09897     if (count < 1) {
09898         for (j = 0; j < nelem; j++) {
09899             q[j] = 0;
09900             r[j] = 0;
09901         }
09902         j = 0;
09903         for (k = 0; k < nclass; k++) {

```

```

09904         n = cl[k];
09905         for (e = 0; e < n; e++) {
09906             r[j] = n - e - 1;
09907             j++;
09908         }
09909     }
09910 #ifdef CHECK
09911     if (j != nelem) {
09912         erEnd("inconsistent # elements");
09913     }
09914 #endif
09915     return 1;
09916 }
09917 b = False;
09918 for (j = nelem-1; j >= 0; j--) {
09919     if (q[j] < r[j]) {
09920         for (k = j+1; k < nelem; k++) {
09921             q[k] = 0;
09922         }
09923         q[j]++;
09924         b = True;
09925         break;
09926     }
09927 }
09928 if (!b) {
09929     return (-count);
09930 }
09931
09932 for (j = 0; j < nelem; j++) {
09933     k = j + q[j];
09934     t = p[j];
09935     p[j] = p[k];
09936     p[k] = t;
09937 }
09938 return count + 1;
09939 }
09940
09941 //-----
09942 static BigInt factorial(int n)
09943 {
09944     // returns 1 for n < 1
09945
09946     int r, j;
09947
09948     r = 1;
09949     for (j = 2; j <= n; j++) {
09950         r *= j;
09951     }
09952     return r;
09953 }
09954
09955 //-----
09956 static BigInt ipow(int n, int p)
09957 {
09958     int r, j;
09959
09960     r = 1;
09961     for (j = 0; j < p; j++) {
09962         r *= n;
09963     }
09964     return r;
09965 }
09966
09967 //-----
09968 static int *newArray(int size, int val)
09969 {
09970     // memory allocation of an array
09971
09972     int *a, j;
09973
09974     a = new int[size];
09975     for (j = 0; j < size; j++) {
09976         a[j] = val;
09977     }
09978     return a;
09979 }
09980
09981 //-----
09982 static int *deleteArray(int *a)
09983 {
09984     // memory allocation of an array
09985
09986     delete[] a;
09987     return NULL;
09988 }
09989
09990 //-----

```

```

09991 static int **newMat(int n0, int n1, int val)
09992 {
09993     int **m, j, k;
09994
09995     m = new int*[n0];
09996     for (j = 0; j < n0; j++) {
09997         m[j] = new int[n1];
09998         for (k = 0; k < n1; k++) {
09999             m[j][k] = val;
10000         }
10001     }
10002     return m;
10003 }
10004
10005 //-----
10006 static int **deleteMat(int **m, int n0)
10007 {
10008     int j;
10009
10010     for (j = n0-1; j >= 0; j--) {
10011         delete[] m[j];
10012     }
10013     delete[] m;
10014
10015     return NULL;
10016 }
10017
10018 //-----
10019 static void bsort(int n, int *ord, int *a)
10020 {
10021     int i, j, t;
10022
10023     for (j = n-1; j > 0; j--) {
10024         for (i = 0; i < j; i++) {
10025             if(a[ord[i+1]] < a[ord[i]]) {
10026                 t = ord[i];
10027                 ord[i] = ord[i+1];
10028                 ord[i+1] = t;
10029             }
10030         }
10031     }
10032 }
10033
10034 //-----
10035 static void prMomStr(int mom, const char *ms, int mn)
10036 {
10037     if (mom == 0) {
10038         return;
10039     } else if (mom == 1) {
10040         printf(" + %s%d", ms, mn);
10041     } else if (mom > 0) {
10042         printf(" + %d*%s%d", mom, ms, mn);
10043     } else if (mom == -1) {
10044         printf(" - %s%d", ms, mn);
10045     } else {
10046         printf(" - %d*%s%d", -mom, ms, mn);
10047     }
10048 }
10049
10050 //-----
10051 static void prIntArray(int n, int *p, const char *msg)
10052 {
10053
10054     int j;
10055
10056     printf("[");
10057     for (j = 0; j < n; j++) {
10058         if (j!=0) printf(", ");
10059         printf("%2d", p[j]);
10060     }
10061     printf("]%s", msg);
10062 }
10063
10064 //-----
10065 static void prIntArrayErr(int n, int *p, const char *msg)
10066 {
10067
10068     int j;
10069
10070     fprintf(GRCC_Stderr, "[");
10071     for (j = 0; j < n; j++) {
10072         if (j!=0) fprintf(GRCC_Stderr, ", ");
10073         fprintf(GRCC_Stderr, "%2d", p[j]);
10074     }
10075     fprintf(GRCC_Stderr, "]%s", msg);
10076 }
10077

```

```

10078 //-----
10079 static void sortrec(int l, int r, int *a)
10080 {
10081     // quick sort : taken from N. Wirth
10082     int i, j, x, w;
10083
10084     i = l;
10085     j = r;
10086     x = a[(l+r)/2];
10087     do {
10088         while(a[i] < x) i++;
10089         while(x < a[j]) j--;
10090         if (i <= j) {
10091             w = a[i]; a[i] = a[j]; a[j] = w;
10092             i++; j--;
10093         }
10094     } while (i < j);
10095     if (l < j) sortrec(l, j, a);
10096     if (i < r) sortrec(i, r, a);
10097 }
10098
10099 //-----
10100 static void sorti(int size, int *a)
10101 {
10102     sortrec(0, size-1, a);
10103 }
10104
10105 //-----
10106 static int sortb(int n, int *a)
10107 {
10108     int i, j, t, nswap;
10109
10110     nswap = 0;
10111     for (j = n-1; j > 0; j--) {
10112         for (i = 0; i < j; i++) {
10113             if(a[i+1] < a[i]) {
10114                 t = a[i];
10115                 a[i] = a[i+1];
10116                 a[i+1] = t;
10117                 nswap++;
10118             }
10119         }
10120     }
10121
10122     // the sign of the permutation is (-1)^{nswap}
10123     return nswap;
10124 }
10125
10126 #ifdef CHECK
10127 //-----
10128 static Bool isIn(int n, int *a, int v)
10129 {
10130     int j;
10131
10132     for (j = 0; j < n; j++) {
10133         if (a[j] == v) {
10134             return True;
10135         }
10136     }
10137     return False;
10138 }
10139 #endif
10140
10141 //-----
10142 static int intSetAdd(int n, int *a, int v, const int size)
10143 {
10144     // Add 'v' into 'a'.
10145     // 'n' is the current # of element.
10146     // 'a' is assumed sorted without duplicated elements.
10147     // Size of 'a' should be large enough.
10148
10149     int j, k;
10150
10151     if (n >= size) {
10152         fprintf(GRCC_Stderr, "*** intSetAdd : array out of range (>%d)\n", size);
10153         erEnd("intSetAdd : array out of range (GRCC_MAXPSLIST)");
10154     }
10155     for (j = 0; j < n; j++) {
10156         if (a[j] > v) {
10157             break;
10158         } else if (a[j] == v) {
10159             return n;
10160         }
10161     }
10162
10163     // 'j' can be equal to 'n'
10164     for (k = n-1; k >= j; k--) {

```

```

10165         a[k+1] = a[k];
10166     }
10167     a[j] = v;
10168     return n+1;
10169 }
10170
10171 //-----
10172 static int intSListAdd(int n, int *a, int v, const int size)
10173 {
10174     // Add 'v' into 'a'.
10175     // 'n' is the current # of element.
10176     // 'a' is assumed sorted without duplicated elements.
10177     // Size of 'a' should be large enough.
10178
10179     int j, k;
10180
10181     if (n >= size) {
10182         fprintf(GRCC_Stderr, "*** intSListAdd : array out of range (>%d)\n", size);
10183         erEnd("intSListAdd : array out of range");
10184     }
10185     for (j = 0; j < n; j++) {
10186         if (a[j] >= v) {
10187             break;
10188         }
10189     }
10190
10191     // 'j' can be equal to 'n'
10192     for (k = n-1; k >= j; k--) {
10193         a[k+1] = a[k];
10194     }
10195     a[j] = v;
10196     return n+1;
10197 }
10198
10199 //-----
10200 static int    cmpArray(int na, int *a, int nb, int *b)
10201 {
10202     int j, cmp;
10203
10204     cmp = na - nb;
10205     if (cmp != 0) {
10206         return cmp;
10207     }
10208     for (j = 0; j < na; j++) {
10209         cmp = a[j] - b[j];
10210         if (cmp != 0) {
10211             return cmp;
10212         }
10213     }
10214     return 0;
10215 }
10216
10217 //-----
10218 static Bool    leqArray(int n, int *a, int *b)
10219 {
10220     int j;
10221
10222     for (j = 0; j < n; j++) {
10223         if (a[j] > b[j]) {
10224             return False;
10225         }
10226     }
10227     return True;
10228 }
10229
10230 //-----
10231 // convert list to sorted list
10232 static int    toSList(int n, int *a)
10233 {
10234     sorti(n, a);
10235     return n;
10236 }
10237
10238 //-----
10239 // as set operation assuming a and b are sorted with duplication
10240 //     p := a \ b;  q := a \cap b;  r := b \ a;
10241 // p, q, r are sorted without duplication
10242 static void    listDiff(int na, int *a, int nb, int *b, int *np, int *p, int *nq, int *q, int *nr, int
10243 *r)
10244 {
10245     int ja, jb;
10246
10247     *np = *nq = *nr = 0;
10248     ja = jb = 0;
10249     while (ja < na && jb < nb) {
10250         while (ja < na && a[ja] < b[jb]) {

```

```

10251     }
10252     while (jb < nb && b[jb] < a[ja]) {
10253         r[(*nr)++] = b[jb++];
10254     }
10255     if (ja < na && jb < nb && a[ja] == b[jb]) {
10256         q[(*nq)++] = a[ja++];
10257         jb++;
10258     }
10259 }
10260 for (; ja < na; ja++) {
10261     p[(*np)++] = a[ja];
10262 }
10263 for (; jb < nb; jb++) {
10264     r[(*nr)++] = b[jb];
10265 }
10266 }
10267
10268 //-----
10269 static int isSubSList(int na, int *a, int nb, int *b)
10270 {
10271     int ja, jb;
10272
10273     ja = jb = 0;
10274     while (ja < na && jb < nb) {
10275         for (; jb < nb && b[jb] < a[ja]; jb++) {
10276             ;
10277         }
10278         if (jb >= nb || b[jb] > a[ja]) {
10279             return False;
10280         }
10281         for (; jb < nb && ja < na && b[jb] == a[ja]; jb++, ja++) {
10282             ;
10283         }
10284     }
10285     return (ja >= na);
10286 }
10287
10288 //-----
10289 static int subtrSet(int na, int *a, int nb, int *b, int *c, int size)
10290 {
10291     // set subtraction 'c' := 'a' - 'b'.
10292     // 'a' and 'b' are sorted lists (not always unique)
10293     // 'c' is the set (sorted and unique elements)
10294
10295     int ja, jb, jc;
10296
10297     jc = 0;
10298     ja = jb = 0;
10299     while (ja < na && jb < nb) {
10300         while (ja < na && a[ja] < b[jb]) {
10301             if (jc == 0 || c[jc-1] != a[ja]) {
10302                 if (jc >= size) {
10303                     erEnd("subsutSet: too small size of array (GRCC_MAXPSLIST)");
10304                 }
10305                 c[jc++] = a[ja];
10306             }
10307             ja++;
10308         }
10309         while (jb < nb && b[jb] < a[ja]) {
10310             ;
10311         }
10312         if (ja < na && jb < nb && a[ja] == b[jb]) {
10313             ja++;
10314             jb++;
10315         }
10316     }
10317     for (; ja < na; ja++) {
10318         if (jc == 0 || c[jc-1] != a[ja]) {
10319             c[jc++] = a[ja];
10320         }
10321     }
10322     return jc;
10323 }
10324
10325 // } } ]

```

## 4.58 grcc.h

```

00001 #pragma once
00002
00003 #include <stdio.h>
00004 #include <stdlib.h>
00005 #include <string.h>

```

```

00006 #include <time.h>
00007
00008 //=====
00009 extern "C" {
00010 #include "grccparam.h"
00011 }
00012
00013 //=====
00014 // Name space of this library
00015 // See 'grccparam.h' for #define
00016 #ifndef GRCC_NAMESPACE
00017 namespace Grcc {
00018 #endif
00019
00020 //=====
00021 // For debugging
00022 //
00023 // #define MONITOR      // in graph elimination
00024 // #define CHECK        // check consistency
00025 #define DEBUGABORT
00026 // #define DEBUG
00027 // #define DEBUG0
00028 // #define DEBUG1
00029 // #define DEBUGM
00030
00031 //=====
00032 // Common types
00033 //
00034 typedef int Bool;
00035 const int True  = 1;
00036 const int False = 0;
00037
00038
00039 //=====
00040 // Big integer (may be replaced by suitable one)
00041
00042 #define BigInt      long
00043 #define ToBigInt(x) ((BigInt) (x))
00044
00045 //=====
00046 // Macro functions
00047 #define Real(x)      ((double) (x))
00048 #define Abs(x)       ((x) >= 0 ? (x) : (-x))
00049 #define Sign(x)      ((x) >= 0 ? (1) : (-1))
00050 #define Max(x, y)    ((x) > (y) ? (x) : (y))
00051 #define Min(x, y)    ((x) < (y) ? (x) : (y))
00052 #define Second(t)    ((double) (*t)=(double) (clock() / ((double)CLOCKS_PER_SEC)))
00053 #define isATExternal(x) ((x)==GRCC_AT_Initial|| (x)==GRCC_AT_Final|| (x)==GRCC_AT_External)
00054
00055 //=====
00056 // types and classes
00057 typedef int  Edge2n[2]; // pair of nodes expressing an edge
00058 class Assign;
00059 class AStack;
00060 class EEdge;
00061 class EGraph;
00062 class ENode;
00063 class Interaction;
00064 class MGraph;
00065 class MNodeClass;
00066 class MCOpi;
00067 class MCBridge;
00068 class MCBlock;
00069 class MConn;
00070 class Model;
00071 class MOrbits;
00072 class Options;
00073 class Output;
00074 class Particle;
00075 class PNodeClass;
00076 class Process;
00077 class SGroup;
00078 class SProcess;
00079 class DCGraph;
00080
00081 //=====
00082 // type of output functions
00083 typedef Bool OutEGB(EGraph *, void *);
00084 typedef void OutEG(EGraph *, void *);
00085 typedef void ErExit(const char *msg, void *);
00086
00087 //=====
00088 class Options {
00089 public:
00090
00091     Model      *model;
00092     Process    *proc;

```

```

00093     SProcess    *sproc;
00094     Output      *out;           // for output
00095     OutEGB      *outmg;        // call back function of mgraph
00096     OutEGB      *endmg;        // call back function of end of mgraph
00097     OutEG       *outag;        // call back function of agraph
00098     void        *argmg;        // additional argument to outmg
00099     void        *argemg;       // additional argument to endmg
00100     void        *argag;        // additional argument to outag
00101
00102     //switches for graph generation
00103     int values[GRCC_OPT_Size];  // array of options
00104
00105     int          DUMMYPADDING;
00106
00107     // measuring time
00108     double time0;
00109     double time1;
00110
00111     //-----
00112     Options(void);
00113     ~Options(void);
00114     void setDefaultValue(void);
00115
00116     void setValue(int ind, int val);
00117     int  getValue(int ind);
00118
00119     void print(void);
00120
00121     void setOutputF(Bool outgrf, const char *fname);
00122     void setOutputP(Bool outgrp, const char *fname);
00123     void printLevel(int l);
00124
00125     void printModel(void);
00126     void setOutMG(OutEGB *omg, void *pt);
00127     void setOutAG(OutEG  *oag, void *pt);
00128     void setEndMG(OutEGB *omg, void *pt);
00129     void setErExit(ErExit *ere, void *pt);
00130     const OptDef *getDef(void);
00131
00132     void begin(Model *mdl);
00133     void end(void);
00134     void beginProc(Process *prc);
00135     void endProc(void);
00136     void beginSubProc(SProcess *sproc);
00137     void endSubProc(void);
00138     void newMGraph(MGraph *mgr);
00139     void newAGraph(EGraph *egr);
00140     void outModel(void);
00141 };
00142
00143 //=====
00144 class Output {
00145 public:
00146
00147     Options    *opt;
00148     Model      *model;
00149     Process     *proc;
00150     SProcess    *sproc;
00151     char        *outgrf;
00152     FILE        *outgrfp;
00153     char        *outgrp;
00154     FILE        *outgrpp;
00155     int         procId;
00156     Bool        outproc;
00157
00158     Output(Options *optn);
00159     ~Output(void);
00160
00161     void setOutgrf(const char *fname);
00162     void setOutgrp(const char *fname);
00163     Bool outBeginF(Model *mdl, Bool pr);
00164     Bool outBeginP(Model *mdl, Bool pr);
00165     void outEndF(void);
00166     void outEndP(void);
00167     void outProcBeginF(Process *prc);
00168     void outProcBeginP(Process *prc);
00169     void outProcBegin0(int next, int couple, int loop);
00170     void outSProcBeginF(SProcess *sproc);
00171     void outSProcBeginP(SProcess *sproc);
00172     void outProcEndF(void);
00173     void outProcEndP(void);
00174     void outEGraphF(EGraph *egraph);
00175     void outEGraphP(EGraph *egraph);
00176     void outModelF(void);
00177     void outModelP(void);
00178
00179 };

```



```

00180
00181 //=====
00182 class Fraction {
00183 public:
00184     BigInt num, den;
00185     double ratio;
00186
00187     Fraction() { num = 0; den = 1; };
00188     Fraction(BigInt n, BigInt d);
00189
00190     void print(const char *msg);
00191     void setValue(BigInt n, BigInt d);
00192     void setValue(Fraction &f);
00193     void add(BigInt n, BigInt d);
00194     void add(Fraction f);
00195     void sub(Fraction f);
00196     BigInt gcd(BigInt n0, BigInt n1);
00197     void normal(void);
00198     Bool isEq(Fraction f);
00199 };
00200
00201 //*****
00202 // Model
00203 //=====
00204 //-----
00205 class Particle {
00206 public:
00207     Model *mdl;           // the model
00208     char *name;           // the name of the particle
00209     char *aname;          // the name of the anti-particle
00210     int id;               // id of this particle
00211     int ptype;            // the type of particle : GRCC_PT_Scalar, etc.
00212     int neutral;          // True if particle==anti-particle
00213     int pcode;            // Particle code
00214     int acode;            // Anti-particle code
00215     int cmindeg;          // min(deg of connectable interactions)
00216     int cmaxdeg;          // max(deg of connectable interactions)
00217
00218     int extonly;          // can appear only as external particle
00219
00220     //-----
00221     Particle(Model *mdl, int pid, Pinput *pinp);
00222     ~Particle(void);
00223
00224     char *particleName(int p);
00225     int particleCode(int p);
00226     char *interactionName(int p);
00227     char *aparticle(void);
00228     void prParticle(void);
00229     int isNeutral(void);
00230     const char *typeName(void);
00231     const char *typeGName(void);
00232
00233 };
00234
00235 //-----
00236 class Interaction {
00237 public:
00238     Model *mdl;           // the model
00239     char *name;           // the name of the interaction
00240     int *plist;           // the list of particle codes
00241     int *clist;           // the orders of coupling constants
00242     int *slist;           // the sorted list of particle codes
00243
00244     int id;               // id of this interaction
00245     int csum;             // the total orders of coupling constants
00246     int nlegs;            // the number of legs
00247     int loop;             // the number of loops
00248     int icode;            // user defined interaction code
00249
00250     int nplist;           // the list of particle codes
00251     int nclist;           // the orders of coupling constants
00252     int nslist;           // the sorted list of particle codes
00253
00254     //-----
00255     Interaction(Model *mdl, int iid, const char* nam, int icode, int *cpl, int nlgs, int *plst, int
csm, int lp);
00256     ~Interaction(void);
00257
00258     void prInteraction(void);
00259 };
00260
00261 //-----
00262 class Model {
00263 public:
00264     char *name;           // the name of this model
00265     char **cnlist;         // the list of coupling constant names

```

```

00266     int          ncouple;      // the number of coupling consants.
00267
00268     int          nParticles;    // the number of particles
00269     Particle     **particles;    // the list of particles
00270     int          pdef;          // the def. of partcls is ended
00271
00272     // list of particles and anti-p. without Undef.
00273     int          nallPart;
00274     int          allPart[GRCC_MAXMPARTICLES2];
00275
00276     // list of particles and anti-p. without ones of extloop=True
00277     int          nintPart;
00278     int          intPart[GRCC_MAXMPARTICLES2];
00279
00280     int          nInteracts;    // the number of interactions.
00281     Interaction   **interacts;  // the list of interactions
00282     int          vdef;          // the definition of interaction is ended
00283
00284     int          maxnlegs;      // maximum number of legs of an interaction
00285     int          maxcpl;        // maximum number of coupling const.
00286     int          maxloop;       // maximum number of loops inside an intr.
00287
00288     // classification of interactions
00289     int          *cplgcp;       // coupling constants
00290     int          *cplglg;       // degree
00291     int          *cplgnvl;      // number of vertices
00292     int          **cplgv1;      // list of vertices
00293     int          ncplgcp;       // the number of classes
00294
00295     int          defpart;       // GRCC_DEFBYNAME or GRCC_DEFBYCODE
00296     int          skipFLine;     // skip analysis fermion line
00297
00298     int          DUMMYPPADDING;
00299
00300     // methods
00301     Model(MInput *minp);
00302     ~Model(void);
00303
00304     void prModel(void);
00305     void addParticle(PInput *pinp);
00306     void addParticleEnd(void);
00307     void addInteraction(IInput *iinp);
00308     void addInteractionEnd(void);
00309     int findParticleName(const char *name);
00310     int findParticleCode(int pcd);
00311     int findInteractionName(const char *name);
00312     int findInteractionCode(int icd);
00313     char *particleName(int p);
00314     int particleCode(int p);
00315     int normalParticle(int pt);
00316     int antiParticle(int pt);
00317     int *allParticles(int *len);
00318     int findMClass(const int cpl, const int dgr);
00319     void prParticleArray(int n, int *a, const char *msg);
00320 };
00321
00322 //*****
00323 // Process
00324 //=====
00325 class PNodeClass {
00326 public:
00327     SProcess *sproc;
00328     int *deg;      // The degree of each node
00329     int *type;     // The type of the class : GRCC_AT_XXX
00330     int *particle; // The particle code of external node.
00331     int *couple;   // The orders of coupling constants
00332     int *cmindeg;  // min(deg of connectable vertex)
00333     int *cmaxdeg;  // max(deg of connectable vertex)
00334     int *count;    // The number of nodes in the class
00335     int *cl2nd;    // cl2nd[class] <= nd < cl2nd[class+1] = set of nodes
00336     int *nd2cl;    // nd2cl[node] = class
00337     int *cl2mcl;   // cl2mcl[class] = class defined in the model.
00338     int nnodes;
00339     int nclass;
00340
00341     PNodeClass(SProcess *sprc, int nnods, int nclss, NCInput *cls);
00342     PNodeClass(SProcess *sprc, int nnods, int nclss, int *dgs, int *typ, int *ptcl, int *cpl, int
*cnt, int *cmincl, int *cmaxcl);
00343     ~PNodeClass(void);
00344
00345     void prPNodeClass(void);
00346     void prElem(int e);
00347 };
00348
00349 //=====
00350 class SProcess {
00351     // In sprocess the number of nodes and edges are fixed.

```

```

00352 // A node is considered as
00353 // (degree, total order of coupling constants),
00354 // which pair of data defines a class of nodes.
00355
00356 public:
00357     Model      *model;      // model
00358     Process    *proc;       // mother process
00359     Options    *opt;        // options
00360     PNodeClass *pnclass;    // list of classes (cpl and deg is determined)
00361     AStack     *astack;
00362
00363     MGraph     *mgraph;
00364     EGraph     *egraph;
00365     Assign     *agraph;
00366
00367     int        *cl2nd;      // (code of nodes in the class[c])
00368                        // = [cl2nd[c], ..., cl2nd[c+1]-1]
00369     int        *nd2cl;      // node nd is in the class pnclass[nd2cl[nd]]
00370     int        nclass;      // the number of classes
00371
00372     int        ninitl;      // the number of initial particles
00373     int        nfinal;      // the number of final particles
00374     int        nvert;       // the number of vertices
00375
00376     int        clist[GRCC_MAXNCPLG]; // coupling constans of the process
00377     int        id;          // sprocess id
00378     int        loop;        // the number of loops
00379     int        nNodes;      // the number of nodes
00380     int        nEdges;      // the number of edges
00381     int        nExtern;     // the number of external particles
00382     int        ncouple;     // the number of coupling constants
00383     int        tCouple;     // the total coupling constants
00384
00385     int        DUMMYPPADDING;
00386
00387     BigInt     mgrcount;     // count generated mgraph
00388     BigInt     agrcount;     // count generated agraph
00389     BigInt     extperm;      // count generated agraph
00390
00391     // the results of the graph generation
00392     BigInt     nMGraphs;     // the number of generated M-graphs
00393     BigInt     nMOPI;        // the number of 1PI M-graphs
00394     Fraction   wMGraphs;     // the weighted sum of M-graphs
00395     Fraction   wMOPI;        // the weighted sum of 1PI M-graphs
00396
00397     BigInt     nAGraphs;     // the number of generated A-graphs
00398     BigInt     nAOPI;        // the number of 1PI A-graphs
00399     Fraction   wAGraphs;     // the weighted sum of A-graphs
00400     Fraction   wAOPI;        // the weighted sum of 1PI A-graphs
00401
00402     // methods
00403     SProcess(Model *mdl, Process *prc, Options *opts, int sid, int *clst, int ncls, NCInput *cls);
00404     SProcess(Model *mdl, Process *prc, Options *opts, int sid, int *clst, int ncls, int *cdeg, int
*ctyp, int *ptcl, int *cpl, int *cnum, int *cmind, int *cmaxd);
00405
00406     ~SProcess(void);
00407
00408     void prSProcess(void);
00409     BigInt generate(void);
00410     void assign(MGraph *mgr);
00411
00412     int toMNodeClass(int *ctyp, int *cldeg, int *clnum, int *cmind, int *cmaxd);
00413     PNodeClass *match(MGraph *mgr);
00414
00415     void endMGraph(MGraph *mgr);
00416     void endAGraph(EGraph *egr);
00417
00418     void resultMGraph(BigInt nmgraphs, Fraction mwsum, BigInt nmopi, Fraction mwopi);
00419     void resultAGraph(BigInt nagraphs, Fraction awsum, BigInt naopi, Fraction awopi);
00420
00421 };
00422
00423 //=====
00424 class Process {
00425 public:
00426     Model      *model;      // model used for this process
00427     Options    *opt;        // options
00428     BigInt     mgrcount;     // count generated mgraph
00429     BigInt     agrcount;     // count generated agraph
00430     int        *initlPart;   // list of ids of initial particles
00431     int        *finalPart;   // list of ids of final particles
00432
00433     int        id;          // process id
00434     int        ninitl;      // the number of initial particles
00435     int        nfinal;      // the number of final particles
00436     int        ctotall;      // the total number of coupling constants
00437     int        nExtern;     // the number of external particles

```

```

00438     int          loop;                // the number of external particles
00439     int          maxnlegs;            // the maximum possible degree of node
00440     int          clist[GRCC_MAXNCPLG]; // coupling constants of the process
00441
00442
00443     // table of sprocesses
00444     int          nSubproc;            // the number of sprocesses
00445     SProcess     *sptbl[GRCC_MAXSUBPROCS]; // the table of sprocesses
00446     SProcess     *sproc;              // the current sprocess
00447
00448     // stack for the assignment;
00449     AStack       *astack;
00450
00451     // the number of graphs
00452     BigInt       ngraphs;
00453     BigInt       nopi;
00454     BigInt       wgraphs;
00455     BigInt       wopi;
00456
00457     // the results of the graph generation
00458     BigInt       nMGraphs;            // the number of generated M-graphs
00459     BigInt       nMOPI;              // the number of lPI M-graphs
00460     Fraction     wMGraphs;           // the weighted sum of M-graphs
00461     Fraction     wMOPI;              // the weighted sum of lPI M-graphs
00462
00463     BigInt       nAGraphs;            // the number of generated A-graphs
00464     BigInt       nAOPI;              // the number of lPI A-graphs
00465     Fraction     wAGraphs;           // the weighted sum of A-graphs
00466     Fraction     wAOPI;              // the weighted sum of lPI A-graphs
00467
00468     double       sec;
00469
00470     Process(int pid, Model *model, Options *opt, int nin, int *initlPart, int nfin, int *finalPart,
00471 int *coupling);
00472     Process(int pid, Model *model, Options *opt, FGInput *fgi);
00473     ~Process(void);
00474     void prProcess(void);
00475     void mkSProcess(void);
00476
00477 };
00478
00479 //*****
00480 // classes for EGraph
00481 //=====
00482 // Constants
00483
00484 #define GRCC_ED_Undef  0
00485 #define GRCC_ED_Deleted 1
00486 #define GRCC_ED_Extern 2
00487 #define GRCC_ED_Back 3
00488 #define GRCC_ED_Bridge 4
00489 #define GRCC_ED_Inloop 5
00490 #define GRCC_ED_Size 6
00491 #define GRCC_ED_NAMES {"Undef", "Deleted", "Extern", "Back_Edge", "Bridge", "In_Loop"}
00492
00493 #define GRCC_ND_Undef  0
00494 #define GRCC_ND_Deleted 1
00495 #define GRCC_ND_Initial 2
00496 #define GRCC_ND_Final 3
00497 #define GRCC_ND_CPoint 4
00498 #define GRCC_ND_VBlock 5
00499 #define GRCC_ND_Inblock 6
00500 #define GRCC_ND_Size 7
00501 #define GRCC_ND_NAMES {"Undef", "Deleted", "Init", "Final", "CPoint", "VBlock", "In_Block"}
00502
00503 //-----
00504 class ENode {
00505 public:
00506     EGraph *egraph;
00507     int *edges; // list of edges
00508                // the value is \pm [(edge index)+1]
00509
00510     int id; // id of the enode
00511     int maxdeg; // maximum degree
00512     int deg; // degree
00513     int extloop; // loop inside this node
00514     int ndtype; // type of the node
00515     int intrct; // assigned interaction/particle (ext.)
00516
00517     // used in EGraph::biconn()
00518     int *klow;
00519     int visited;
00520
00521     int DUMMYPADDING;
00522
00523     //-----

```

```

00524 // functions
00525 ENode(void);
00526 ENode(EGraph *egrph, int loops, int sdeg);
00527 ~ENode(void);
00528 void initAss(EGraph *egrph, int nid, int sdg);
00529 void setId(EGraph *egrph, const int nid);
00530 void copy(ENode *en);
00531 void setExtern(int typ, int pt);
00532 void setType(int typ);
00533 void print(void);
00534
00535 };
00536
00537 //-----
00538 class EEdge {
00539 // momentum is printed like: ("%s%d", (enode.ext)?"Q":"p", enode.momn)
00540 public:
00541     EGraph *egrph;           // egraph
00542
00543     int id;                  // id
00544     int ext;
00545     int ptcl;                // assigned particle (agraph)
00546     int deleted;             // deleted edge, if true
00547     int nodes[2];            // nodes of bothsides
00548     int nlegs[2];            // nodes of both side (agraph)
00549
00550     // for biconn
00551     int *emom;                // external momenta
00552     int *lmom;                // loop momenta
00553     int *extMom;              // set of external momenta.
00554
00555     Bool cut;
00556     int visited;
00557     int conid;                // connected component
00558     int edtype;               // type
00559     int opicomp;              // id of 1PI component
00560     int dir;                  // direction of momentum
00561
00562
00563 //-----
00564 // functions
00565 EEdge(void);
00566 EEdge(EGraph *egrph, int nedges, int nloops);
00567 ~EEdge(void);
00568 void copy(EEdge *ee);
00569 void setId(EGraph *egrph, const int eid);
00570 void print(void);
00571
00572 void setType(int typ);
00573 void setLMom(int k, int dir);
00574 void setEMom(int nedges, int *extn, int dir);
00575 };
00576
00577 //-----
00578 // type of fermion lines
00579
00580 typedef enum {FL_Open, FL_Closed} FLType;
00581
00582 //-----
00583 class EFLine {
00584 public:
00585     int elist[GRCC_MAXNODES]; // list of (\pm [(edge index)+1])
00586     FLType ftype;
00587     int fkind;
00588     int nlist;
00589
00590     EFLine(void);
00591     void print(const char *msg);
00592 };
00593
00594 //-----
00595 class EGraph {
00596 public:
00597     Options *opt;              // table of options
00598     Model *model;
00599     Process *proc;
00600     SProcess *sproc;
00601     MGraph *mgraph;
00602     MConn *econn;
00603
00604     ENode **nodes;
00605     EEdge **edges;            // edges[nEdges+1]: index starts from 1
00606
00607     BigInt mId;                // id of mgraph
00608     BigInt aId;                // id of agraph in the same mgraph
00609     BigInt sId;                // sequential no. of agraph
00610     BigInt gSubId;             // ???

```

```

00611     Bool    assigned;           // mgraph (False) or agraph (True)
00612
00613     int      fsign;
00614     BigInt   nsym, esym;        // symmetry factor with symm. ext.
00615     BigInt   nsym1;             // symmetry factor without symm. ext.
00616     BigInt   extperm;           // the order of group of symm. ext.
00617     BigInt   multp;             // multiplicity of graph in symm. ext.
00618
00619     int      pId;
00620
00621     int      sNodes;            // memory size
00622     int      sEdges;            // memory size
00623     int      sMaxdeg;           // memory size
00624     int      sLoops;            // memory size
00625
00626     int      nNodes;
00627     int      nEdges;
00628     int      nExtern;
00629     int      maxdeg;            // maximum value of degree of nodes
00630     int      nLoops;
00631     int      totalc;            // total order of coupling constants
00632
00633     // biconnect
00634     int      nopicomp;
00635     int      opi2plp;
00636     int      nopi2p;
00637     int      nadj2ptv;          // the no. of edges connecting 2point vertices
00638
00639     int      DUMMYPADDING;
00640
00641     int      *bidef;
00642     int      *bilow;
00643     int      *extMom;
00644     int      bconn;
00645     int      bicount;
00646     int      loopm;
00647     int      opiCount;
00648
00649     // Fermion lines
00650     EFLine *flines[GRCC_MAXFLINES];
00651     int      nflines;
00652
00653     int      DUMMYPADDING1;
00654
00655     //-----
00656     // functions
00657     EGraph(int nnodes, int nedges, int mxdeg);
00658     ~EGraph(void);
00659
00660     void copy(EGraph *eg);
00661     void print(void);
00662     void fromDGraph(DGraph *dg);
00663     void fromMGraph(MGraph *mgraph);
00664     Bool isOptE(void);
00665
00666     ENode *setExtern(int n0, int pt, int ndtp);
00667     Bool isExternal(int nd) { return (nodes[nd]->extloop < 0); };
00668     Bool isFermion(int nd);
00669
00670     void setExtLoop(int nd, int val);
00671     void endSetExtLoop(void);
00672
00673     int connComp(void);
00674     int connVisit(int nd, int ncc);
00675
00676     void biconnE(void);
00677     void biinitE(void);
00678     void bisearchE(int nd, int *extlst, int *intlst, int *opiext, int *opiloop);
00679
00680     int findRoot(void);
00681     int dirEdge(int n, int e);
00682     void extMomConsv(void);
00683     int cmpMom(int *lm0, int *em0, int *lm1, int *em1);
00684     int groupLMom(int *grp, int *ed2gr);
00685
00686     void chkMomConsv(void);
00687
00688     void prFLines(void);
00689     void getFLines(void);
00690     int fltrace(int fk, int nd0, int *fl);
00691     void addFLine(const FLType ft, int fk, int nfl, int *fl);
00692
00693     int legParticle(int ed, int lg);
00694
00695 };
00696
00697 //*****

```

```

00698 // Symmetry group of graphs
00699 //=====
00700 class SGroup {
00701 public:
00702     BigInt size;           // # of allocated elements
00703     BigInt nelelem;        // # of saved elements
00704     int **elem;            // saved elements
00705     int nnodes;            // # of nodes.
00706     int necclass;          // # of classes
00707     int eclass[GRCC_MAXNODES]; // table of classes
00708     int cgen;              // counter for the generation
00709     int csav;              // counter for the saved elements
00710     int permg[GRCC_MAXNODES]; // resulting permutation
00711     int perms[GRCC_MAXNODES]; // curr. elem of saved ones.
00712
00713     int pgr[GRCC_MAXNODES]; // work
00714     int pgq[GRCC_MAXNODES]; // work
00715     int psr[GRCC_MAXNODES]; // work
00716     int psq[GRCC_MAXNODES]; // work
00717
00718     //-----
00719     // functions
00720
00721     SGroup(void);
00722     ~SGroup(void);
00723
00724     void print(void);
00725     void newGroup(int nelm, int nclss, int *clss);
00726     void clearGroup(void);
00727     void delGroup(void);
00728     int *genNext(void);
00729     void addGroup(int *p);
00730     BigInt nElem(void);
00731     int *nextElem(void);
00732 };
00733
00734 //*****
00735 // MGraph
00736 //=====
00737 // class of nodes for MGraph
00738
00739 class MNode {
00740 public:
00741     int id;                // node id
00742     int deg;               // degree(node) = the number of legs
00743     int clss;              // initial class number in which the node belongs
00744     int extloop;
00745     int cmindeg;
00746     int cmaxdeg;
00747
00748     int freelg;           // the number of free legs
00749     int visited;
00750
00751     MNode(int id, int clss, NCInput *mgi);
00752     MNode(int id, int deg, int extloop, int clss, int cmin, int cmaxd);
00753 };
00754
00755 //=====
00756 // class of scalar graph expressed by matrix form
00757 //
00758 // Input : the classified set of nodes.
00759 // Output : control passed to 'EGraph(self)'
00760
00761 class MGraph {
00762 public:
00763
00764     // initial conditions
00765     Options *opt;          // options
00766
00767     // symmetry group
00768     SGroup *group;         // symmetry group
00769     MOrbits *orbits;       // Orbits of nodes with respect to symmetry group
00770
00771     MNode **nodes;         // table of MNode object
00772     BigInt mId;            // process/sprocess ID
00773     int *clist;            // list of initial classes
00774     int pId;               // process/sprocess ID
00775     int nNodes;            // the number of nodes
00776     int nEdges;            // the number of edges
00777     int nLoops;            // the number of loops
00778     int nExtern;           // the number of external nodes
00779     int nClasses;          // the number of initial classes
00780     int mindeg;            // minimum value of degree of nodes
00781     int maxdeg;            // maximum value of degree of nodes
00782
00783
00784     // the current graph

```

```

00785     int **adjMat;           // adjacency matrix
00786     MNodeClass *curcl;      // the current 'MNodeClass' object
00787     EGraph *egraph;
00788     BigInt nsym;            // symmetry factor from nodes
00789     BigInt esym;            // symmetry factor from edges
00790
00791     // generated set of graphs
00792     BigInt cDiag;           // the total number of generated graphs
00793     BigInt c1PI;            // the total number of 1PI graphs
00794     BigInt cNoTadpole;      // the total number of graphs without tadpoles
00795     BigInt cNoTadBlock;     // the total number of graphs without tad-blocks
00796     BigInt c1PINoTadBlock;  // the total number of 1PI graphs without tad-blocks
00797     Fraction wscon;         // weighted sum of graphs
00798     Fraction wsopi;         // weighted sum of 1PI graphs
00799
00800     // measures of efficiency
00801     BigInt ngen;            // generated graph before check
00802     BigInt ngconn;         // generated connected graph before check
00803
00804     BigInt nCallRefine;
00805     BigInt discardRefine;
00806     BigInt discardDisc;
00807     BigInt discardIso;
00808
00809     // for options
00810     Bool opi;
00811     Bool opiloop;
00812     Bool extself;
00813     Bool selfloop;
00814     Bool tadpole;
00815     Bool tadblock;
00816     Bool block;
00817
00818     int    DUMMYPADDING1;
00819
00820     // table of n edge-connected components
00821     MConn *mconn;
00822
00823     // work space for isomorphism
00824     int **modmat;          // permuted adjacency matrix
00825
00826     // work space for biconnected component
00827     int *bidef;
00828     int *bilow;
00829     int  bicount;
00830
00831     int    DUMMYPADDING2;
00832
00833     //-----
00834     // functions
00835     MGraph(int pid, int ncl, NCInput *mgi, Options *opt);
00836     MGraph(int pid, int ncl, int *cldeg, int *clnum, int *clexl, int *cmind, int *cmaxd, Options
*opt);
00837     ~MGraph(void);
00838
00839     void    init(void);
00840     BigInt  generate(void);
00841     Bool    isExternal(int nd) { return (nodes[nd]->extloop < 0); };
00842     void    printAdjMat(MNodeClass *cl);
00843     void    print(void);
00844
00845     Bool    isConnected(void);
00846     Bool    visit(int nd);
00847     Bool    isIsomorphic(MNodeClass *cl);
00848     void    permMat(int size, int *perm, int **mat0, int **mat1);
00849     int     compMat(int size, int **mat0, int **mat1);
00850     MNodeClass *refineClass(MNodeClass *cl);
00851     void    bisearchME(int nd, int pd, int ned, MCOpi *mopi, MCBlock *mblk, int *next, int *nart);
00852     void    biconnME(void);
00853     Bool    isOptM(void);
00854
00855     void    connectClass(MNodeClass *cl);
00856     void    connectNode(int sc, int ss, MNodeClass *cl);
00857     void    connectLeg(int sc, int sn, int tc, int ts, MNodeClass *cl);
00858     void    newGraph(MNodeClass *cl);
00859
00860 };
00861
00862 //=====
00863 // class of node-classes for MGraph
00864 //-----
00865 class MNodeClass {
00866 public:
00867     int    clmat[GRCC_MAXNODES][GRCC_MAXNODES]; // matrix used for classification
00868     int    clist[GRCC_MAXNODES]; // the number of nodes in each class
00869     int    ndcl[GRCC_MAXNODES]; // node --> class
00870     int    flist[GRCC_MAXNODES+1]; // the first node in each class

```



```

00871     int    clord[GRCC_MAXNODES];    // ordering of classes
00872     int    cmindeg[GRCC_MAXNODES];  // min(deg of connectable node)
00873     int    cmaxdeg[GRCC_MAXNODES];  // max(deg of connectable node)
00874     int    nNodes;                  // the number of nodes
00875     int    nClasses;                 // the number of classes
00876     int    maxdeg;                   // maximal value of degree(node)
00877
00878     int    flg0;
00879     int    flg1;
00880     int    flg2;
00881
00882     MNodeClass(int nnodes, int nclasses);
00883     ~MNodeClass(void);
00884
00885     void    init(int *cl, int mxdeg, int **adjmat);
00886     void    copy(MNodeClass* mnc);
00887     int     clCmp(int nd0, int nd1, int cn);
00888     void    printMat(void);
00889
00890     void    mkFlist(void);
00891     void    mkNdCl(void);
00892     void    mkClMat(int **adjmat);
00893     void    incMat(int nd, int td, int val);
00894     int     cmpMNCArray(int *a0, int *a1, int ma);
00895     void    reorder(MGraph *mg);
00896 };
00897
00898
00899 //=====
00900 //  class of 1PI component
00901 //-----
00902 class MCOp {
00903 public:
00904     int *nodes;    // array of nodes in
00905     int  nnodes;   // # nodes in the 1PI component
00906     int  nlegs;    // # leg of the 1PI component
00907     int  next;     // # external particles of the 1PI comp.
00908     int  nedges;   // # edges in the 1PI comp.
00909     int  loop;     // # loops in the 1PI comp.
00910     int  ctloop;   // # loops in the counter terms in the OP comp.
00911
00912     MCOp(void);
00913     ~MCOp(void);
00914
00915     void init(void);
00916 };
00917
00918 //=====
00919 //  class of bridge
00920 //-----
00921 class MCBridge {
00922 public:
00923     Edge2n nodes; // nodes at the both size of the bridge
00924     int  next;    // # momenta of ext. particles flowing on the bridge
00925
00926     MCBridge(void);
00927     ~MCBridge(void);
00928 };
00929
00930 //=====
00931 //  class of block
00932 //-----
00933 class MCBlock {
00934 public:
00935     Edge2n *edges; // array of edges in the block
00936     int  nedges;   // # edges in the block
00937     int  nartps;   // # articulation points of the block
00938     int  loop;     // # loop in the block
00939
00940     int  DUMMYPADDING;
00941
00942     MCBlock(void);
00943     ~MCBlock(void);
00944     void init(void);
00945 };
00946
00947 //=====
00948 //  class of table of MConn
00949 //-----
00950 class MConn {
00951 public:
00952     // 2-edge connected components
00953     MCOp *opics;    // table of n-edge-connected components
00954     MCBridge *bridges; // table of bridges
00955     MCBlock *blocks; // table of blocks
00956     int  *articuls; // buffer for nodes for articulation points
00957

```

```

00958     int      *opisp;          // buffer for nodes in lPI components
00959     int      *opistk;         // stack of nodes for lPI components.
00960     Edge2n   *blksp;          // buffer for edges in blocks
00961     Edge2n   *blkstk;         // stack of edges for blocks
00962     int      snodes;          // # nodes
00963     int      sedges;          // # edges
00964
00965     // opi components (edge-connected)
00966     int      nopic;           // # lPI components (nlPIComps)
00967     int      nlpopic;         // # looped lPI components (nlPIComps)
00968     int      nctopic;         // # lPI components of one counter term.
00969     int      nbridges;        // # bridges
00970     int      ne0bridges;      // # bridges whose next=0
00971     int      nelbridges;      // # bridges whose next=1
00972     int      nselfloops;
00973
00974     // blocks (node-connected)
00975     int      nblocks;         // # blocks
00976     int      nalblocks;       // # bridges whose next=0
00977     int      narticuls;       // # articulation points
00978     int      neblocks;        // # effective looped blocks
00979
00980     // indices to work spaces
00981     int      nopisp;          // # used in opisp
00982     int      opistkptra;      // # stack pointer of opistk
00983     int      nblksp;          // # used in blksp
00984     int      blkstkptra;      // # stack pointer of tlkstk
00985
00986     int      DUMMYPADDING;
00987
00988     MConn(int nnod, int nedg);
00989     ~MConn(void);
00990
00991     void init(void);
00992     void pushNode(int nd);
00993     void pushEdge(int n0, int n1);
00994     void addOPIC(MCOpi *mopi, int stp);
00995     void addBridge(int n0, int n1, int nex, int nextot);
00996     void addArtic(int nd, int mul);
00997     void addBlock(MCBlock *eblk, int stp);
00998     void addBlockSelf(int nd, int mul);
00999     void print(void);
01000 };
01001
01002 //=====
01003 // class of orbits of nodes
01004 //-----
01005 class MORbits {
01006     // Usage
01007     // 1. node ==> orbit
01008     // (class) = nd2or[(node)]
01009     //
01010     // 2. class ==> nodes
01011     // for (c = 0; c < nClass; c++) {
01012     //     for (j = flist[c]; j < flist[c+1]; j++) {
01013     //         (node) = or2nd[c];
01014     //     }
01015     // }
01016 public:
01017     int nOrbits;
01018     int nNodes;
01019     int nd2or[GRCC_MAXNODES];
01020     int or2nd[GRCC_MAXNODES];
01021     int flist[GRCC_MAXNODES+1];
01022
01023     MORbits(void);
01024     ~MORbits(void);
01025
01026     void print(void);
01027
01028     // construction of orbits
01029     void initPerm(int nnodes);
01030     void fromPerm(int *perm);
01031     void toOrbits(void);
01032 };
01033
01034 //*****
01035 // Particle assignment
01036 //=====
01037 typedef int CheckPt[2];
01038
01039 typedef enum {
01040     AS_UnAssLegs, AS_Assigned, AS_Assigned0, AS_AssExt, AS_Impossible
01041 } NCandSt;
01042
01043 //=====
01044 class NCand {

```

```

01045 // List of candidats for the assignment of interactions to a vertex
01046
01047 public:
01048     int     deg;                // degree of the node
01049     NCandSt st;                // status
01050     int     nilist;            // length of the list
01051     int     ilist[GRCC_MAXMINTERACT]; // list of candidates
01052
01053     //=====
01054     NCand(const NCandSt sta, const int dega, const int nilst, int *ilst);
01055     ~NCand(void);
01056
01057     // print
01058     void prNCand(const char* msg);
01059 };
01060
01061 //=====
01062 class ECand {
01063     // List of candidats for the assignment of particles to an edge
01064
01065     public:
01066         Bool det;                // determined or not
01067         int  nplist;            // size of the list of candidates
01068         int  plist[GRCC_MAXMPARTICLES2]; // list of candidates
01069
01070         ECand(int dt, int nplist, int *plist);
01071         ~ECand(void);
01072
01073         // print
01074         void prECand(const char *msg);
01075 };
01076
01077 //=====
01078 class ANode {
01079     // Connection information of a node to others
01080     //
01081     // (this node) -- (adjacent edge) -- (next node)
01082     //      (n0, j)           e           n1
01083     //
01084     // e = (aedges[j], aelegs[j])
01085     // n1 = anodes[j]
01086
01087     public:
01088         int     deg;                // degree of the node
01089         int     nlegs;             // the number of legs already assigned
01090         int     *anodes;           // anodes[j] = (next node of leg j)
01091         int     *aedges;           // aedges[j] = (next edge of leg j)
01092         int     *aelegs;           // aelegs[j] = (leg of the next edge)
01093         NCand   *cand;             // candidate list
01094
01095         ANode(int dg);
01096         ~ANode(void);
01097
01098         int newleg(void);          // get a new leg
01099 };
01100
01101 //=====
01102 class AEdge {
01103     // Connection information of an edge
01104     // (self) ==> nodes [(nodes[0], nlegs[0]), (nodes[1], nlegs[1])]
01105     // particle 'ptcl' flows from leg=0 to 1.
01106
01107     public:
01108
01109         ECand *cand;                // candidate
01110         int   nodes[2];            // nodes of both sides of the edge
01111         int   nlegs[2];            // nodes of both sides of the edge
01112         int   ptcl;                // particle defined in the model
01113
01114         int DUMMYPADDING;
01115
01116         AEdge(int n0, int l0, int n1, int l1);
01117         ~AEdge(void);
01118 };
01119
01120 //=====
01121 class Assign {
01122     // Class for particle/interaction assignment.
01123
01124     public:
01125
01126         EGraph *egraph;            // EGraph object
01127         MGraph *mgraph;           // MGraph object
01128         SProcess *sproc;          // Sub-process object
01129         Process *proc;            // Process object
01130         Model *model;             // Model object
01131         Options *opt;             // Option object

```

```

01132     AStack      *astack;          // Stack for saving candidates
01133     PNodeClass  *pnclass;         // class of nodes
01134     MOrbits     *orbits;          // orbits of nodes by symmetry group
01135
01136     int         nNodes;           // the number of nodes
01137     int         nEdges;           // the number of edges
01138     int         nExtern;          // the number of external particles.
01139     int         nETotal;          // the total number of edges
01140
01141     BigInt      nAGraphs;         // the number of assigned graphs
01142     Fraction    wAGraphs;         // the weighted sum of graphs
01143     BigInt      nAOPI;            // the number of assigned lPI
01144     Fraction    wAOPI;           // the weighted sum of lPI
01145
01146     ANode       **nodes;          // table of nodes
01147     AEdge       **edges;          // table of edges
01148
01149     CheckPt     checkpoint0;      // save stack pointers
01150
01151     int         cplleft[GRCC_MAXNCPLG]; // coupling constants left
01152
01153     //=====
01154     Assign(SProcess *sprc, MGraph *mgr, PNodeClass *pnc);
01155     ~Assign(void);
01156
01157     //=====
01158     // print lists of candidates
01159     void prCand(const char *msg);
01160
01161     // check
01162     void checkAG(const char *msg);
01163
01164     //=====
01165     // control of assignment
01166
01167     // entry point of assignment
01168     Bool assignAllVertices(void);
01169
01170     // select a source node for assignment
01171     Bool selectVertex(void);
01172     Bool selectVertexSimp(int lastv);
01173
01174     // select a source leg of the node for assignment
01175     Bool selectLeg(int v, int lastlg);
01176
01177     // assign a vertex a interaction
01178     Bool assignVertex(int v);
01179
01180     // assignment procedure is finished. Needs check.
01181     Bool allAssigned(void);
01182
01183     //=====
01184     // Input and output
01185
01186     // Input from mgraph
01187     Bool fromMGraph(void);
01188
01189     // add an edge
01190     void addEdge(int n0, int n1, int nplist, int *plist);
01191
01192     // connect nodes and edges.
01193     void connect(int n0, int l0, int eg, int el, int n1, int l1);
01194
01195     // Output to egraph
01196     Bool fillEGraph(int aid, BigInt nsym, BigInt esym, BigInt nsym1);
01197
01198     // reorder legs in accordance with the interaction definition
01199     int *reordLeg(int n, int *reord, int *plist, int *used);
01200
01201     //=====
01202     // direction of particle on an edge and at (node, leg)
01203
01204     // convert particle code
01205     // at node-leg ('n', 'ln') <==> edge at ('n', 'ln')
01206     int getLegParticle(int n, int ln);
01207
01208     // convert particle 'pt' on the edge to ('n', 'ln')
01209     int legEdgeParticle(int n, int ln, int pt);
01210
01211     // convert candidate list at ('v', 'lg') into in-coming direction
01212     int legPart(int v, int lg, int nplst, int *plst, int *rlist, const int size);
01213
01214     // convert candidate list at ('v', 'lg') into in-coming direction
01215     int candPart(int v, int ln, int *plist, const int size);
01216
01217     //=====
01218     // assignment

```

```

01219
01220 // find a vertex to be assigned
01221 int selUnAssVertex(void);
01222 int selUnAssVertexSimp(int lastv);
01223
01224 // find a leg of the vertex to be assigned
01225 int selUnAssLeg(int v, int lastlg);
01226
01227 // assign the interaction 'ia' to the vertex 'v'.
01228 NCandSt assignIVertex(int v, int ia);
01229
01230 // assign particle 'pt' the the leg 'ln' of the node 'n'.
01231 Bool assignPLeg(int n, int ln, int pt);
01232 Bool isOrdPLeg(int n, int ln, int pt);
01233 Bool detEdge(int e);
01234
01235 //=====
01236 // canndidates
01237
01238 // construct lists of assigned / unassigned particles at a node
01239 Bool candPartClassify(int v, int *npdass, int *pdass, int *npuass, int *puass, const int size);
01240
01241 // update candidate list for a vertex 'v'.
01242 Bool updateCandNode(int v);
01243
01244 //=====
01245 // filter
01246
01247 Bool checkOrderCpl(void);
01248 Bool isOrdLegs(void);
01249 Bool isIsomorphic(MNodeClass *cl, BigInt *nsym, BigInt *esym, BigInt *nsym1);
01250 int cmpPermGraph(int *p, MNodeClass *cl);
01251 int cmpNodes(int nd0, int nd1, MNodeClass *cn);
01252 BigInt edgeSym(void);
01253
01254 void saveCouple(int *sav);
01255 void restoreCouple(int *sav);
01256 Bool subCouple(int *cpl);
01257
01258 #ifdef CHECK
01259 //=====
01260 // check
01261 Bool checkCand(const char *msg);
01262
01263 void checkNode(int n, const char *msg);
01264 #endif
01265 };
01266
01267 //*****
01268 // Stack of candidate lists
01269 //=====
01270 class NStack {
01271 // Stack for backtracking method for a node.
01272
01273 public:
01274 int noden;
01275 int deg;
01276 NCandSt st;
01277 int nilist;
01278 int ilist[GRCC_MAXMINTERACT];
01279
01280 NStack() { };
01281 ~NStack() { };
01282
01283 void print(const char *msg);
01284
01285 };
01286
01287 //=====
01288 class EStack {
01289 // Stack for backtracking method for an edge.
01290
01291 public:
01292 int edgen;
01293 int det;
01294 int nplist;
01295 int plist[GRCC_MAXMPARTICLES2];
01296
01297 EStack() { };
01298 ~EStack() { };
01299
01300 void print(const char *msg);
01301 };
01302
01303
01304 //=====
01305 class AStack {

```

```

01306 // Stack for backtracking method for an edge.
01307
01308 public:
01309     Assign      *agraph;
01310
01311     NStack      **nStack;      // stack for nodes
01312     int          nStackP;      // stack pointer for nodes
01313     int          nSize;        // stack size
01314
01315     EStack      **eStack;      // stack for edges
01316     int          eStackP;      // stack pointer for edges
01317     int          eSize;        // stack size
01318
01319     AStack(int nSize, int eSize);
01320     ~AStack(void);
01321
01322     void setAGraph(Assign *ag);
01323
01324     void checkPoint(CheckPt sav);
01325     void restore(CheckPt sav);
01326     void restoreMsg(CheckPt sav, const char *msg);
01327     void prStack(void);
01328
01329     void pushNode(int n);
01330     void pushEdge(int e);
01331
01332 private:
01333     void restoreNode(int spr);
01334     void restoreEdge(int spr);
01335 };
01336
01337 //=====
01338 // end of namespace Grcc
01339 #ifndef GRCC_NAMESPACE
01340 }
01341 #endif

```

## 4.59 grccparam.h

```

00001 #ifndef GRCC_PARAM_H
00002 #else
00003 #define GRCC_PARAM_H
00004
00005 /*=====
00006  * Parameters
00007  */
00008 /*
00009 #define GRCC_NAMESPACE
00010 */
00011
00012 #define GRCC_MAXNCPLG      4
00013 #define GRCC_MAXLEGS      10
00014 #define GRCC_MAXMPARTICLES 50
00015 #define GRCC_MAXMININTERACT 200
00016 #define GRCC_MAXSUBPROCS  500
00017 #define GRCC_MAXNODES     20
00018 #define GRCC_MAXEDGES     100
00019 #define GRCC_MAXNSTACK    50
00020 #define GRCC_MAXESTACK    50
00021 #define GRCC_MAXGROUP     1000
00022 #define GRCC_MAXPSLIST    500
00023
00024 #define GRCC_MAXMPARTICLES2 (2*GRCC_MAXMPARTICLES-1)
00025 #define GRCC_MAXFLINES     GRCC_MAXEDGES
00026
00027 #define GRCC_FRACERROR     (1.0e-10)
00028
00029 /* error message */
00030 /*
00031 #define GRCC_Stderr      stderr
00032 */
00033 #define GRCC_Stderr      stdout
00034 /*
00035 #define GRCC_ABORT()      exit(1)
00036 */
00037 #define GRCC_ABORT()      abort()
00038
00039 /*-----
00040  * Constants
00041  */
00042
00043 /* model definition */
00044 #define GRCC_DEFBYNAME    1      /* define interactions by particle name */

```

```

00045 #define GRCC_DEFBYCODE 2 /* define interactions by particle code */
00046
00047 /* types of particles */
00048 #define GRCC_PT_Undef 0
00049 #define GRCC_PT_Scalar 1
00050 #define GRCC_PT_Dirac 2
00051 #define GRCC_PT_Majorana 3
00052 #define GRCC_PT_Vector 4
00053 #define GRCC_PT_Ghost 5
00054 #define GRCC_PT_Size 6
00055
00056 #define GRCC_PTS_Undef "undef"
00057 #define GRCC_PTS_Scalar "scalar"
00058 #define GRCC_PTS_Dirac "dirac"
00059 #define GRCC_PTS_Majorana "majorana"
00060 #define GRCC_PTS_Vector "vector"
00061 #define GRCC_PTS_Ghost "ghost"
00062
00063 #define GRCC_PTO_General 0
00064 #define GRCC_PTO_ExtOnly 1
00065
00066 #define GRCC_PT_Table { GRCC_PTS_Undef, GRCC_PTS_Scalar, GRCC_PTS_Dirac, GRCC_PTS_Majorana,
GRCC_PTS_Vector, GRCC_PTS_Ghost}
00067
00068 #define GRCC_PT_GTable { "Undef", "Scalar", "Fermion", "Majorana", "Vector", "Ghost"}
00069
00070 /* for mgraph generation */
00071 #define GRCC_AT_Vertex 0
00072 #define GRCC_AT_External -1
00073 #define GRCC_AT_Initial -2
00074 #define GRCC_AT_Final -3
00075
00076 #define GRCC_AT_NdStr(x) ((x)>=GRCC_AT_Vertex ?"Vertex": \
00077 ((x)==GRCC_AT_External ?"External": \
00078 ((x)==GRCC_AT_Final ?"Fianl": \
00079 ((x)==GRCC_AT_Initial ?"Initial":"Undef"))))
00080
00081 /* graph generation */
00082 #define GRCC_MGraph 0 /* mgraph (topology) generation */
00083 #define GRCC_AGraph 1 /* agraph generation */
00084
00085 /* options for graph greeneration */
00086 #define GRCC_OPT_Step 0
00087 #define GRCC_OPT_Outgrf 1
00088 #define GRCC_OPT_Outgrp 2
00089 #define GRCC_OPT_lPI 3
00090 #define GRCC_OPT_NoSelfLoop 4
00091 #define GRCC_OPT_NoTadpole 5
00092 #define GRCC_OPT_No1PtBlock 6
00093 #define GRCC_OPT_No2PtLlPI 7
00094 #define GRCC_OPT_NoExtSelf 8
00095 #define GRCC_OPT_NoAdj2PtV 9
00096 #define GRCC_OPT_Block 10
00097 #define GRCC_OPT_SymmInitial 11
00098 #define GRCC_OPT_SymmFinal 12
00099 #define GRCC_OPT_Size 13
00100
00101 typedef struct {
00102     const char *name;
00103     const char *mean;
00104     int defaultv;
00105     int DUMMYPADDING;
00106 } OptDef;
00107
00108 /*-----
00109 * Conversion of
00110 * eind : index (ed) of EGraph.edges[ed]
00111 * sind : value (v) of EGraph.nodes[nd]->edges[j])
00112 */
00113 #define I2Vedge(eind, sign) (((sign<=0)?(-1):(1))*((eind)+1))
00114 #define V2Iedge(sind) (abs(sind)-1)
00115 #define V2Ileg(sind) ((sind)<0 ? 0 : 1)
00116 /*-----
00117 * data types for model definition
00118 */
00119 typedef struct {
00120     const char *name; /* name of the model */
00121     int defpart; /* particle is defined by name or code */
00122     /* GRCC_DEFBYNAME or GRCC_DEFBYCODE */
00123     int ncouple; /* No. of coupling constants */
00124     const char *cnamlist[GRCC_MAXNCPLG]; /* Names of coupling constants */
00125 } MInput;
00126
00127 typedef struct {
00128     const char *name; /* name of particle */
00129     const char *aname; /* name of anti-particle */
00130     int pcode; /* code of particle */

```

```

00131         int    acode;                /* code of anti-particle */
00132     const char *ptypen;               /* type : 'GRCC PTS_Scalar' etc */
00133     int    ptypec;                   /* type : 'GRCC PT_Scalar' etc */
00134     int    extonly;                  /* only as external particle */
00135 } PInput;
00136
00137 typedef struct {
00138     const char *name;                /* name of interaction */
00139     int    icode;                   /* code of interaction */
00140     int    nplistn;                 /* the number of legs */
00141     const char *plistn[GRCC_MAXLEGS]; /* list of particle names */
00142     int    plistc[GRCC_MAXLEGS];    /* list of particle codes */
00143     int    cvallist[GRCC_MAXNCPLG]; /* coupling constants */
00144 } IInput;
00145
00146 /*-----
00147  * data type for node class input
00148  */
00149 typedef struct {
00150     /* used as input to MGraph() and SProcess() */
00151     int    cldeg;
00152     int    clnum;
00153     int    cltyp;
00154     int    cmind;
00155     int    cmaxd;
00156
00157     /* used as input to SProcess() */
00158     int    ptcl;
00159     int    cple;
00160 } NCInput;
00161
00162 /*-----
00163  * data types for process definition
00164  */
00165 typedef struct {
00166     long    ninitl;                /* the number of initial particles */
00167     const char *initln[GRCC_MAXNODES]; /* list of initial particles (name) */
00168     int    initlc[GRCC_MAXNODES];    /* list of final particles (code) */
00169     long    nfinal;                /* the number of final particles */
00170     const char *finaln[GRCC_MAXNODES]; /* list of final particles (name) */
00171     int    finalc[GRCC_MAXNODES];    /* list of final particles (code) */
00172     int    coupl[GRCC_MAXNCPLG];    /* list of orders of c. consts */
00173 } FGInput;
00174
00175 /* class of node : input for SProcess */
00176 typedef struct {
00177     int    cdeg;                   /* degree of each node */
00178     int    ctyp;                   /* typde : GRCC_AT_XXX */
00179     int    cnum;                   /* the number of nodes in the class */
00180     int    cple;                   /* total order of c.c. of each node */
00181     /* = 0 for external particle */
00182     const char *pname;             /* particle name defined in the model */
00183     /* = NULL for vertex */
00184     long    pcode;                 /* particle code defined in the model */
00185     /* = 0 for vertex */
00186     /* long = int + padding */
00187 } PGInput;
00188
00189 /*-----
00190  * minimum input data to EGraph
00191  */
00192 typedef struct {
00193     int    extloop;               /* external/loop */
00194     int    intrct;               /* particl/interaction */
00195 } DNode;
00196
00197 typedef int DEdge[2]; /* {node0, node1} */
00198
00199 typedef struct {
00200     int    nnodes;
00201     int    nedges;
00202     DNode *nodes;
00203     DEdge *edges;
00204 } DGraph;
00205
00206 /*-----
00207  */
00208 #endif

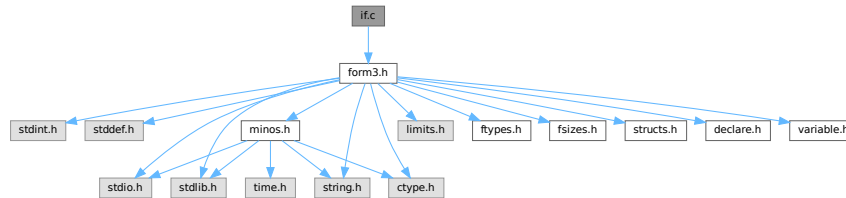
```



## 4.60 if.c File Reference

```
#include "form3.h"
```

Include dependency graph for if.c:



### Functions

- WORD [GetIfDollarNum](#) (WORD \*ifp, WORD \*ifstop)
- int [FindVar](#) (WORD \*v, WORD \*term)
- int [DolfStatement](#) (PHEAD WORD \*ifcode, WORD \*term)
- WORD [HowMany](#) (PHEAD WORD \*ifcode, WORD \*term)
- void [DoubleIfBuffers](#) (void)
- int [DoSwitch](#) (PHEAD WORD \*term, WORD \*lhs)
- int [DoEndSwitch](#) (PHEAD WORD \*term, WORD \*lhs)
- SWITCHTABLE \* [FindCase](#) (WORD nswitch, WORD ncase)
- int [DoubleSwitchBuffers](#) (void)
- void [SwitchSplitMergeRec](#) (SWITCHTABLE \*array, WORD num, SWITCHTABLE \*auxarray)
- void [SwitchSplitMerge](#) (SWITCHTABLE \*array, WORD num)

### 4.60.1 Detailed Description

Routines for the dealing with if statements.

Definition in file [if.c](#).

### 4.60.2 Function Documentation

#### 4.60.2.1 GetIfDollarNum()

```
WORD GetIfDollarNum (
    WORD * ifp,
    WORD * ifstop )
```

Definition at line [106](#) of file [if.c](#).

#### 4.60.2.2 FindVar()

```
int FindVar (
    WORD * v,
    WORD * term )
```

Definition at line [173](#) of file [if.c](#).

#### 4.60.2.3 DoIfStatement()

```
int DoIfStatement (
    PHEAD WORD * ifcode,
    WORD * term )
```

Definition at line [280](#) of file [if.c](#).

#### 4.60.2.4 HowMany()

```
WORD HowMany (
    PHEAD WORD * ifcode,
    WORD * term )
```

Definition at line [840](#) of file [if.c](#).

#### 4.60.2.5 DoubleIfBuffers()

```
void DoubleIfBuffers (
    void )
```

Definition at line [1033](#) of file [if.c](#).

#### 4.60.2.6 DoSwitch()

```
int DoSwitch (
    PHEAD WORD * term,
    WORD * lhs )
```

Definition at line [1070](#) of file [if.c](#).

#### 4.60.2.7 DoEndSwitch()

```
int DoEndSwitch (
    PHEAD WORD * term,
    WORD * lhs )
```

Definition at line [1086](#) of file [if.c](#).

#### 4.60.2.8 FindCase()

```
SWITCHTABLE * FindCase (
    WORD nswitch,
    WORD ncase )
```

Definition at line [1097](#) of file [if.c](#).

**4.60.2.9 DoubleSwitchBuffers()**

```
int DoubleSwitchBuffers (
    void )
```

Definition at line 1135 of file [if.c](#).

**4.60.2.10 SwitchSplitMergeRec()**

```
void SwitchSplitMergeRec (
    SWITCHTABLE * array,
    WORD num,
    SWITCHTABLE * auxarray )
```

Definition at line 1184 of file [if.c](#).

**4.60.2.11 SwitchSplitMerge()**

```
void SwitchSplitMerge (
    SWITCHTABLE * array,
    WORD num )
```

Definition at line 1213 of file [if.c](#).

**4.61 if.c**

[Go to the documentation of this file.](#)

```
00001
00005 /* #[] License : */
00006 /*
00007 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00008 * When using this file you are requested to refer to the publication
00009 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010 * This is considered a matter of courtesy as the development was paid
00011 * for by FOM the Dutch physics granting agency and we would like to
00012 * be able to track its scientific use to convince FOM of its value
00013 * for the community.
00014 *
00015 * This file is part of FORM.
00016 *
00017 * FORM is free software: you can redistribute it and/or modify it under the
00018 * terms of the GNU General Public License as published by the Free Software
00019 * Foundation, either version 3 of the License, or (at your option) any later
00020 * version.
00021 *
00022 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00023 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00024 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00025 * details.
00026 *
00027 * You should have received a copy of the GNU General Public License along
00028 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00029 */
00030 /* #[] License : */
00031 /*
00032 * #[] Includes : if.c
00033 */
00034
00035 #include "form3.h"
00036
00037 /*
00038 * #[] Includes :
00039 * #[] If statement :
00040 * #[] Syntax :
```

```

00041
00042     The `if` is a conglomerate of statements: if,else,endif
00043
00044     The if consists in principle of:
00045
00046     if ( number );
00047         statements
00048     else;
00049         statements
00050     endif;
00051
00052     The first set is taken when number != 0.
00053     The else is not mandatory.
00054     TRUE = 1 and FALSE = 0
00055
00056     The number can be built up via a logical expression:
00057
00058         expr1 condition expr2
00059
00060     each expression can be a subexpression again. It has to be
00061     enclosed in parentheses in that case.
00062     Conditions are:
00063         >, >=, <, <=, ==, !=, ||, &&
00064
00065     When Expressions are chained evaluation is from left to right,
00066     independent of whether this indicates nonsense.
00067     if ( a || b || c || d ); is a perfectly normal statement.
00068     if ( a >= b || c == d ); would be messed up. This should be:
00069     if ( ( a >= b ) || ( c == d ) );
00070
00071     The building blocks of the Expressions are:
00072
00073         Match(option,pattern)    The number of times pattern fits in term_
00074         Count(...)               The count value of term_
00075         Coeff[icient]            The coefficient of term_
00076         FindLoop(options)        Are there loops (as in ReplaceLoop).
00077
00078     Implementation for internal notation:
00079
00080     TYPEIF,length,gotolevel(if fail),EXPRTYPE,length,.....
00081
00082     EXPRTYPE can be:
00083         SHORTNUMBER              ->,4,sign,size
00084         LONGNUMBER               ->,|ncoef+2|,ncoef,number,denom
00085         MATCH                    ->,patternsiz+3,keyword,pattern
00086         MULTIPLEOF               ->,3,thenumber
00087         COUNT                    ->,countsiz+2,countinfo
00088         TYPEFINDLOOP             ->,7 (findloop info)
00089         COEFFICIENT              ->,2
00090         IFDOLLAR                 ->,3,dollarnumber
00091         SUBEXPR                  ->,size,dummy,size1,EXPRTYPE,length,...
00092                                 ,2,condition1,size2,...
00093                                 This is like functions.
00094
00095     Note that there must be a restriction to the number of nestings
00096     of parentheses in an if statement. It has been set to 10.
00097
00098     The syntax of match corresponds to the syntax of the left side
00099     of an id statement. The only difference is the keyword
00100     MATCH vs TYPEIDNEW.
00101
00102     #] Syntax :
00103     #[ GetIfDollarNum :
00104 */
00105
00106 WORD GetIfDollarNum(WORD *ifp, WORD *ifstop)
00107 {
00108     DOLLARS d;
00109     WORD num, *w;
00110     if ( ifp[2] < 0 ) { return(-ifp[2]-1); }
00111     d = Dollars+ifp[2];
00112     if ( ifp+3 < ifstop && ifp[3] == IFDOLLAREXTRA ) {
00113         if ( d->nfactors == 0 ) {
00114             MLOCK(ErrorMessageLock);
00115             MesPrint("Attempt to use a factor of an unfactored $-variable");
00116             MUNLOCK(ErrorMessageLock);
00117             Terminate(-1);
00118         }
00119         num = GetIfDollarNum(ifp+3,ifstop);
00120         if ( num > d->nfactors ) {
00121             MLOCK(ErrorMessageLock);
00122             MesPrint("Dollar factor number %s out of range",num);
00123             MUNLOCK(ErrorMessageLock);
00124             Terminate(-1);
00125         }
00126         if ( num == 0 ) {
00127             return(d->nfactors);

```

```

00128     }
00129     w = d->factors[num-1].where;
00130     if ( w == 0 ) return(d->factors[num].value);
00131 getnumber::
00132     if ( *w == 0 ) return(0);
00133     if ( *w == 4 && w[3] == 3 && w[2] == 1 && w[1] < MAXPOSITIVE && w[4] == 0 ) {
00134         return(w[1]);
00135     }
00136     if ( ( w[w[0]] != 0 ) || ( ABS(w[w[0]-1]) != w[0]-1 ) ) {
00137         MLOCK(ErrorMessageLock);
00138         MesPrint("Dollar factor number expected but found expression");
00139         MUNLOCK(ErrorMessageLock);
00140         Terminate(-1);
00141     }
00142     else {
00143         MLOCK(ErrorMessageLock);
00144         MesPrint("Dollar factor number out of range");
00145         MUNLOCK(ErrorMessageLock);
00146         Terminate(-1);
00147     }
00148     return(0);
00149 }
00150 /*
00151 Now we have just a dollar and should evaluate that into a short number
00152 */
00153 if ( d->type == DOLZERO ) {
00154     return(0);
00155 }
00156 else if ( d->type == DOLNUMBER || d->type == DOLTERMS ) {
00157     w = d->where; goto getnumber;
00158 }
00159 else {
00160     MLOCK(ErrorMessageLock);
00161     MesPrint("Dollar factor number is wrong type");
00162     MUNLOCK(ErrorMessageLock);
00163     Terminate(-1);
00164     return(0);
00165 }
00166 }
00167
00168 /*
00169 #] GetIfDollarNum :
00170 #[ FindVar :
00171 */
00172
00173 int FindVar(WORD *v, WORD *term)
00174 {
00175     WORD *t, *tstop, *m, *mstop, *f, *fstop, *a, *astop;
00176     GETSTOP(term,tstop);
00177     t = term+1;
00178     while ( t < tstop ) {
00179         if ( *v == *t && *v < FUNCTION ) { /* VECTOR, INDEX, SYMBOL, DOTPRODUCT */
00180             switch ( *v ) {
00181                 case SYMBOL:
00182                     m = t+2; mstop = t+t[1];
00183                     while ( m < mstop ) {
00184                         if ( *m == v[1] ) return(1);
00185                         m += 2;
00186                     }
00187                     break;
00188                 case INDEX:
00189                 case VECTOR:
00190 InVe:
00191                     m = t+2; mstop = t+t[1];
00192                     while ( m < mstop ) {
00193                         if ( *m == v[1] ) return(1);
00194                         m++;
00195                     }
00196                     break;
00197                 case DOTPRODUCT:
00198                     m = t+2; mstop = t+t[1];
00199                     while ( m < mstop ) {
00200                         if ( *m == v[1] && m[1] == v[2] ) return(1);
00201                         if ( *m == v[2] && m[1] == v[1] ) return(1);
00202                         m += 3;
00203                     }
00204                     break;
00205             }
00206         }
00207         else if ( *v == VECTOR && *t == INDEX ) goto InVe;
00208         else if ( *v == INDEX && *t == VECTOR ) goto InVe;
00209         else if ( ( *v == VECTOR || *v == INDEX ) && *t == DOTPRODUCT ) {
00210             m = t+2; mstop = t+t[1];
00211             while ( m < mstop ) {
00212                 if ( v[1] == m[0] || v[1] == m[1] ) return(1);
00213                 m += 3;
00214             }
00215         }
00216     }

```

```

00215     }
00216     else if ( *t >= FUNCTION ) {
00217         if ( *v == FUNCTION && v[1] == *t ) return(1);
00218         if ( functions[*t-FUNCTION].spec > 0 ) {
00219             if ( *v == VECTOR || *v == INDEX ) { /* we need to check arguments */
00220                 int i;
00221                 for ( i = FUNHEAD; i < t[1]; i++ ) {
00222                     if ( v[1] == t[i] ) return(1);
00223                 }
00224             }
00225         }
00226         else {
00227             fstop = t + t[1]; f = t + FUNHEAD;
00228             while ( f < fstop ) { /* Do the arguments one by one */
00229                 if ( *f <= 0 ) {
00230                     switch ( *f ) {
00231                         case -SYMBOL:
00232                             if ( *v == SYMBOL && v[1] == f[1] ) return(1);
00233                             f += 2;
00234                             break;
00235                         case -VECTOR:
00236                         case -MINVECTOR:
00237                         case -INDEX:
00238                             if ( ( *v == VECTOR || *v == INDEX )
00239                                 && ( v[1] == f[1] ) ) return(1);
00240                             f += 2;
00241                             break;
00242                         case -SNUMBER:
00243                             f += 2;
00244                             break;
00245                         default:
00246                             if ( *v == FUNCTION && v[1] == -*f && *f <= -FUNCTION ) return(1);
00247                             if ( *f <= -FUNCTION ) f++;
00248                             else f += 2;
00249                             break;
00250                     }
00251                 }
00252                 else {
00253                     a = f + ARGHEAD; astop = f + *f;
00254                     while ( a < astop ) {
00255                         if ( FindVar(v,a) == 1 ) return(1);
00256                         a += *a;
00257                     }
00258                     f = astop;
00259                 }
00260             }
00261         }
00262     }
00263     t += t[1];
00264 }
00265 return(0);
00266 }
00267
00268 /*
00269 #] FindVar :
00270 #[ DoIfStatement : WORD DoIfStatement(PHEAD ifcode,term)
00271
00272 The execution time part of the if-statement.
00273 The arguments are a pointer to the TYPEIF and a pointer to the term.
00274 The answer is either 1 (success) or 0 (fail).
00275 The calling routine can figure out where to go in case of failure
00276 by picking up gotolevel.
00277 Note that the whole setup asks for recursions.
00278 */
00279
00280 int DoIfStatement(PHEAD WORD *ifcode, WORD *term)
00281 {
00282     GETBIDENTITY
00283     WORD *ifstop, *ifp;
00284     UWORD *coef1 = 0, *coef2, *coef3, *cc;
00285     WORD ncoef1, ncoef2, ncoef3, i = 0, first, *r, acoef, ismul1, ismul2, j;
00286     UWORD *Spac1, *Spac2;
00287     ifstop = ifcode + ifcode[1];
00288     ifp = ifcode + 3;
00289     if ( ifp >= ifstop ) return(1);
00290     if ( ( ifp + ifp[1] ) >= ifstop ) {
00291         switch ( *ifp ) {
00292             case LONGNUMBER:
00293                 if ( ifp[2] ) return(1);
00294                 else return(0);
00295             case MATCH:
00296             case TYPEIF:
00297                 if ( HowMany(BHEAD ifp,term) ) return(1);
00298                 else return(0);
00299             case TYPEFINDLOOP:
00300                 if ( Lus(term,ifp[3],ifp[4],ifp[5],ifp[6],ifp[2]) ) return(1);
00301                 else return(0);

```

```

00302         case TYPECOUNT:
00303             if ( CountDo(term,ifp) ) return(1);
00304             else return(0);
00305         case COEFFI:
00306         case MULTIPLEOF:
00307             return(1);
00308         case IFDOLLAR:
00309             {
00310                 DOLLARS d = Dollars + ifp[2];
00311 #ifdef WITHPTHREADS
00312                 int nummodopt, dtype = -1;
00313                 if ( AS.MultiThreaded ) {
00314                     for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
00315                         if ( ifp[2] == ModOptdollars[nummodopt].number ) break;
00316                     }
00317                     if ( nummodopt < NumModOptdollars ) {
00318                         dtype = ModOptdollars[nummodopt].type;
00319                         if ( dtype == MODLOCAL ) {
00320                             d = ModOptdollars[nummodopt].dstruct+AT.identity;
00321                         }
00322                     }
00323                 }
00324                 dtype = d->type;
00325 #else
00326                 int dtype = d->type; /* We use dtype to make the operation atomic */
00327 #endif
00328                 if ( dtype == DOLZERO ) return(0);
00329                 if ( dtype == DOLUNDEFINED ) {
00330                     if ( AC.UnsureDollarMode == 0 ) {
00331                         MesPrint("$%s is undefined",AC.dollarnames->namebuffer+d->name);
00332                         Terminate(-1);
00333                     }
00334                 }
00335             }
00336             return(1);
00337         case IFEXPRESSION:
00338             r = ifp+2; j = ifp[1] - 2;
00339             while ( --j >= 0 ) {
00340                 if ( *r == AR.CurExpr ) return(1);
00341                 r++;
00342             }
00343             return(0);
00344         case IFISFACTORIZED:
00345             r = ifp+2; j = ifp[1] - 2;
00346             if ( j == 0 ) {
00347                 if ( ( Expressions[AR.CurExpr].vflags & ISFACTORIZED ) != 0 )
00348                     return(1);
00349                 else
00350                     return(0);
00351             }
00352             while ( --j >= 0 ) {
00353                 if ( ( Expressions[*r].vflags & ISFACTORIZED ) == 0 ) return(0);
00354                 r++;
00355             }
00356             return(1);
00357         case IFOCCURS:
00358             {
00359                 WORD *OccStop = ifp + ifp[1];
00360                 ifp += 2;
00361                 while ( ifp < OccStop ) {
00362                     if ( FindVar(ifp,term) == 1 ) return(1);
00363                     if ( *ifp == DOTPRODUCT ) ifp += 3;
00364                     else ifp += 2;
00365                 }
00366             }
00367             return(0);
00368         case IFUSERFLAG:
00369             if ( ( Expressions[AR.CurExpr].uflags & (1 << ifp[2]) ) != 0 )
00370                 return(1);
00371             return(0);
00372         default:
00373             /*
00374              Now we have a subexpression. Test first for one with a single item.
00375             */
00376             if ( ifp[3] == ( ifp[1] + 3 ) ) return(DoIfStatement(BHEAD ifp,term));
00377             ifstop = ifp + ifp[1];
00378             ifp += 3;
00379             break;
00380         }
00381     }
00382     /*
00383     Here is the composite condition.
00384     */
00385     coef3 = NumberMalloc("DoIfStatement");
00386     Spac1 = NumberMalloc("DoIfStatement");
00387     Spac2 = (UWORD *) (TermMalloc("DoIfStatement"));
00388     ncoef1 = 0; first = 1; isnull = 0;

```

```

00389     do {
00390         if ( !first ) {
00391             ifp += 2;
00392             if ( ifp[-2] == ORCOND && ncoef1 ) {
00393                 coef1 = Spac1;
00394                 ncoef1 = 1; coef1[0] = coef1[1] = 1;
00395                 goto SkipCond;
00396             }
00397             if ( ifp[-2] == ANDCOND && !ncoef1 ) goto SkipCond;
00398         }
00399         coef2 = Spac2;
00400         ncoef2 = 1;
00401         ismul2 = 0;
00402         switch ( *ifp ) {
00403             case LONGNUMBER:
00404                 ncoef2 = ifp[2];
00405                 j = 2*(ABS(ncoef2));
00406                 cc = (UWORD *) (ifp + 3);
00407                 for ( i = 0; i < j; i++ ) coef2[i] = cc[i];
00408                 break;
00409 #ifdef WITHFLOAT
00410             case IFFLOATNUMBER:
00411                 /*
00412                  * The sloppy solution is: Convert to rational.
00413                  * This way we can write it over coef2,ncoef2
00414                  */
00415                 ncoef2 = FloatFunToRat(BHEAD coef2,ifp);
00416                 break;
00417 #endif
00418             case MATCH:
00419             case TYPEIF:
00420                 coef2[0] = HowMany(BHEAD ifp,term);
00421                 coef2[1] = 1;
00422                 if ( coef2[0] == 0 ) ncoef2 = 0;
00423                 break;
00424             case TYPECOUNT:
00425                 acoef = CountDo(term,ifp);
00426                 coef2[0] = ABS(acoef);
00427                 coef2[1] = 1;
00428                 if ( acoef == 0 ) ncoef2 = 0;
00429                 else if ( acoef < 0 ) ncoef2 = -1;
00430                 break;
00431             case TYPEFINDLOOP:
00432                 acoef = Lus(term,ifp[3],ifp[4],ifp[5],ifp[6],ifp[2]);
00433                 coef2[0] = ABS(acoef);
00434                 coef2[1] = 1;
00435                 if ( acoef == 0 ) ncoef2 = 0;
00436                 else if ( acoef < 0 ) ncoef2 = -1;
00437                 break;
00438             case COEFFI:
00439                 r = term + *term;
00440                 ncoef2 = r[-1];
00441                 i = ABS(ncoef2);
00442                 cc = (UWORD *) (r - i);
00443                 if ( ncoef2 < 0 ) ncoef2 = (ncoef2+1)»1;
00444                 else ncoef2 = (ncoef2-1)»1;
00445                 i--; for ( j = 0; j < i; j++ ) coef2[j] = cc[j];
00446                 break;
00447             case SUBEXPR:
00448                 ncoef2 = coef2[0] = DoIfStatement(BHEAD ifp,term);
00449                 coef2[1] = 1;
00450                 break;
00451             case MULTIPLEOF:
00452                 ncoef2 = 1;
00453                 coef2[0] = ifp[2];
00454                 coef2[1] = 1;
00455                 ismul2 = 1;
00456                 break;
00457             case IFDOLLAREXTRA:
00458                 break;
00459             case IFDOLLAR:
00460                 {
00461                     /*
00462                      * We need to abstract a long rational in coef2
00463                      * with length ncoef2. What if that cannot be done?
00464                      */
00465                     DOLLARS d = Dollars + ifp[2];
00466 #ifdef WITHPTHREADS
00467                     int nummodopt, dtype = -1;
00468                     if ( AS.MultiThreaded ) {
00469                         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
00470                             if ( ifp[2] == ModOptdollars[nummodopt].number ) break;
00471                         }
00472                         if ( nummodopt < NumModOptdollars ) {
00473                             dtype = ModOptdollars[nummodopt].type;
00474                             if ( dtype == MODLOCAL ) {
00475                                 d = ModOptdollars[nummodopt].dstruct+AT.identity;

```



```

00476         }
00477     else {
00478         LOCK(d->pthreadlockread);
00479     }
00480 }
00481 }
00482 #endif
00483 /*
00484 We have to pick up the IFDOLLAREXTRA pieces for [1], [$y] etc.
00485 */
00486 if ( ifp+3 < ifstop && ifp[3] == IFDOLLAREXTRA ) {
00487     if ( d->nfactors == 0 ) {
00488         MLOCK(ErrorMessageLock);
00489         MesPrint("Attempt to use a factor of an unfactored $-variable");
00490         MUNLOCK(ErrorMessageLock);
00491         Terminate(-1);
00492     } {
00493         WORD num = GetIfDollarNum(ifp+3,ifstop);
00494         WORD *w;
00495         while ( ifp+3 < ifstop && ifp[3] == IFDOLLAREXTRA ) ifp += 3;
00496         if ( num > d->nfactors ) {
00497             MLOCK(ErrorMessageLock);
00498             MesPrint("Dollar factor number %s out of range",num);
00499             MUNLOCK(ErrorMessageLock);
00500             Terminate(-1);
00501         }
00502         if ( num == 0 ) {
00503             ncoef2 = 1; coef2[0] = d->nfactors; coef2[1] = 1;
00504             break;
00505         }
00506         w = d->factors[num-1].where;
00507         if ( w == 0 ) {
00508             if ( d->factors[num-1].value < 0 ) {
00509                 ncoef2 = -1; coef2[0] = -d->factors[num-1].value; coef2[1] = 1;
00510             }
00511             else {
00512                 ncoef2 = 1; coef2[0] = d->factors[num-1].value; coef2[1] = 1;
00513             }
00514             break;
00515         }
00516         if ( w[*w] == 0 ) {
00517             r = w + *w - 1;
00518             i = ABS(*r);
00519             if ( i == ( *w-1 ) ) {
00520                 ncoef2 = (i-1)/2;
00521                 if ( *r < 0 ) ncoef2 = -ncoef2;
00522                 i--; cc = coef2; r = w + 1;
00523                 while ( --i >= 0 ) *cc++ = (UWORD) (*r++);
00524                 break;
00525             }
00526         }
00527         goto generic;
00528     }
00529 }
00530 else {
00531     switch ( d->type ) {
00532     case DOLUNDEFINED:
00533         if ( AC.UnsureDollarMode == 0 ) {
00534             if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00535             MLOCK(ErrorMessageLock);
00536             MesPrint("%s is undefined",AC.dollarnames->namebuffer+d->name);
00537             MUNLOCK(ErrorMessageLock);
00538             Terminate(-1);
00539         }
00540         ncoef2 = 0; coef2[0] = 0; coef2[1] = 1;
00541         break;
00542     case DOLZERO:
00543         ncoef2 = coef2[0] = 0; coef2[1] = 1;
00544         break;
00545     case DOLSUBTERM:
00546         if ( d->where[0] != INDEX || d->where[1] != 3
00547             || d->where[2] < 0 || d->where[2] >= AM.OffsetIndex ) {
00548             if ( AC.UnsureDollarMode == 0 ) {
00549                 if ( dtype > 0 && dtype != MODLOCAL ) {
00550                     UNLOCK(d->pthreadlockread); }
00551                 MLOCK(ErrorMessageLock);
00552                 MesPrint("%s is of wrong
00553 type",AC.dollarnames->namebuffer+d->name);
00554                 MUNLOCK(ErrorMessageLock);
00555                 Terminate(-1);
00556             }
00557             ncoef2 = 0; coef2[0] = 0; coef2[1] = 1;
00558             break;
00559         }
00560     }

```

```

00561         }
00562         d->index = d->where[2];
00563         /* fall through */
00564     case DOLINDEX:
00565         if ( d->index == 0 ) {
00566             ncoef2 = coef2[0] = 0; coef2[1] = 1;
00567         }
00568         else if ( d->index > 0 && d->index < AM.OffsetIndex ) {
00569             ncoef2 = 1; coef2[0] = d->index; coef2[1] = 1;
00570         }
00571         else if ( AC.UnsureDollarMode == 0 ) {
00572             #ifndef WITHPTHREADS
00573                 if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00574             #endif
00575             MLOCK(ErrorMessageLock);
00576             MesPrint("%s is of wrong type",AC.dollarnames->namebuffer+d->name);
00577             MUNLOCK(ErrorMessageLock);
00578             Terminate(-1);
00579         }
00580         ncoef2 = coef2[0] = 0; coef2[1] = 1;
00581         break;
00582     case DOLWILDARGS:
00583         if ( d->where[0] <= -FUNCTION ||
00584             ( d->where[0] < 0 && d->where[2] != 0 )
00585             || ( d->where[0] > 0 && d->where[d->where[0]] != 0 )
00586         ) {
00587             if ( AC.UnsureDollarMode == 0 ) {
00588                 #ifndef WITHPTHREADS
00589                     if ( dtype > 0 && dtype != MODLOCAL ) {
00590                         UNLOCK(d->pthreadlockread); }
00591                 #endif
00592                 MLOCK(ErrorMessageLock);
00593                 MesPrint("%s is of wrong
00594 type",AC.dollarnames->namebuffer+d->name);
00595                 MUNLOCK(ErrorMessageLock);
00596                 Terminate(-1);
00597             }
00598             ncoef2 = coef2[0] = 0; coef2[1] = 1;
00599             break;
00600         }
00601         /* fall through */
00602     case DOLARGUMENT:
00603         if ( d->where[0] == -SNUMBER ) {
00604             if ( d->where[1] == 0 ) {
00605                 ncoef2 = coef2[0] = 0;
00606             }
00607             else if ( d->where[1] < 0 ) {
00608                 ncoef2 = -1;
00609                 coef2[0] = -d->where[1];
00610             }
00611             else {
00612                 ncoef2 = 1;
00613                 coef2[0] = d->where[1];
00614             }
00615             coef2[1] = 1;
00616         }
00617         else if ( d->where[0] == -INDEX
00618             && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
00619             if ( d->where[1] == 0 ) {
00620                 ncoef2 = coef2[0] = 0; coef2[1] = 1;
00621             }
00622             else {
00623                 ncoef2 = 1; coef2[0] = d->where[1];
00624                 coef2[1] = 1;
00625             }
00626         }
00627         else if ( d->where[0] > 0
00628             && d->where[ARGHEAD] == (d->where[0]-ARGHEAD)
00629             && ABS(d->where[d->where[0]-1]) ==
00630                 (d->where[0] - ARGHEAD-1) ) {
00631             i = d->where[d->where[0]-1];
00632             ncoef2 = (ABS(i)-1)/2;
00633             if ( i < 0 ) { ncoef2 = -ncoef2; i = -i; }
00634             i--; cc = coef2; r = d->where + ARGHEAD+1;
00635             while ( --i >= 0 ) *cc++ = (UWORD) (*r++);
00636         }
00637         else {
00638             if ( AC.UnsureDollarMode == 0 ) {
00639                 #ifndef WITHPTHREADS
00640                     if ( dtype > 0 && dtype != MODLOCAL ) {
00641                         UNLOCK(d->pthreadlockread); }
00642                     #endif
00643                     MLOCK(ErrorMessageLock);
00644                     MesPrint("%s is of wrong
00645 type",AC.dollarnames->namebuffer+d->name);
00646                     MUNLOCK(ErrorMessageLock);
00647                     Terminate(-1);

```

```

00644         }
00645         ncoef2 = 0; coef2[0] = 0; coef2[1] = 1;
00646     }
00647     break;
00648 case DOLNUMBER:
00649 case DOLTERMS:
00650     if ( d->where[d->where[0]] == 0 ) {
00651         r = d->where + d->where[0]-1;
00652         i = ABS(*r);
00653         if ( i == ( d->where[0]-1 ) ) {
00654             ncoef2 = (i-1)/2;
00655             if ( *r < 0 ) ncoef2 = -ncoef2;
00656             i--; cc = coef2; r = d->where + 1;
00657             while ( --i >= 0 ) *cc++ = (UWORD) (*r++);
00658             break;
00659         }
00660     }
00661 generic::
00662     if ( AC.UnsureDollarMode == 0 ) {
00663 #ifdef WITHPTHREADS
00664         if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00665 #endif
00666         MLOCK(ErrorMessageLock);
00667         MesPrint("%$s is of wrong type",AC.dollarnames->namebuffer+d->name);
00668         MUNLOCK(ErrorMessageLock);
00669         Terminate(-1);
00670     }
00671     ncoef2 = 0; coef2[0] = 0; coef2[1] = 1;
00672     break;
00673 }
00674 }
00675 #ifdef WITHPTHREADS
00676     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00677 #endif
00678     }
00679     break;
00680 case IFEXPRESSION:
00681     r = ifp+2; j = ifp[1] - 2; ncoef2 = 0;
00682     while ( --j >= 0 ) {
00683         if ( *r == AR.CurExpr ) { ncoef2 = 1; break; }
00684         r++;
00685     }
00686     coef2[0] = ncoef2;
00687     coef2[1] = 1;
00688     break;
00689 case IFISFACTORIZED:
00690     r = ifp+2; j = ifp[1] - 2;
00691     if ( j == 0 ) {
00692         ncoef2 = 0;
00693         if ( ( Expressions[AR.CurExpr].vflags & ISFACTORIZED ) != 0 ) {
00694             ncoef2 = 1;
00695         }
00696     }
00697     else {
00698         ncoef2 = 1;
00699         while ( --j >= 0 ) {
00700             if ( ( Expressions[*r].vflags & ISFACTORIZED ) == 0 ) {
00701                 ncoef2 = 0;
00702                 break;
00703             }
00704             r++;
00705         }
00706     }
00707     coef2[0] = ncoef2;
00708     coef2[1] = 1;
00709     break;
00710 case IFOCCURS:
00711     {
00712         WORD *OccStop = ifp + ifp[1], *ifpp = ifp+2;
00713         ncoef2 = 0;
00714         while ( ifpp < OccStop ) {
00715             if ( FindVar(ifpp,term) == 1 ) {
00716                 ncoef2 = 1; break;
00717             }
00718             if ( *ifpp == DOTPRODUCT ) ifpp += 3;
00719             else ifpp += 2;
00720         }
00721         coef2[0] = ncoef2;
00722         coef2[1] = 1;
00723     }
00724     break;
00725 case IFUSERFLAG:
00726     {
00727         ncoef2 = 0;
00728         if ( ( Expressions[AR.CurExpr].uflags & (1 << ifp[2]) ) != 0 )
00729             ncoef2 = 1;
00730         coef2[0] = ncoef2;

```

```

00731         coef2[1] = 1;
00732     }
00733     break;
00734 default:
00735     break;
00736 }
00737 if ( !first ) {
00738     if ( ifp[-2] != ORCOND && ifp[-2] != ANDCOND ) {
00739         if ( ( ifp[-2] == EQUAL || ifp[-2] == NOTEQUAL ) &&
00740             ( ismul2 || ismul1 ) ) {
00741             if ( ismul1 && ismul2 ) {
00742                 if ( coef1[0] == coef2[0] ) i = 1;
00743                 else i = 0;
00744             }
00745             else {
00746                 if ( ismul1 ) {
00747                     if ( ncoef2 )
00748                         Divvy(BHEAD coef2,&ncoef2,coef1,ncoef1);
00749                     cc = coef2; ncoef3 = ncoef2;
00750                 }
00751                 else {
00752                     if ( ncoef1 )
00753                         Divvy(BHEAD coef1,&ncoef1,coef2,ncoef2);
00754                     cc = coef1; ncoef3 = ncoef1;
00755                 }
00756                 if ( ncoef3 < 0 ) ncoef3 = -ncoef3;
00757                 if ( ncoef3 == 0 ) {
00758                     if ( ifp[-2] == EQUAL ) i = 1;
00759                     else i = 0;
00760                 }
00761                 else if ( cc[ncoef3] != 1 ) {
00762                     if ( ifp[-2] == EQUAL ) i = 0;
00763                     else i = 1;
00764                 }
00765                 else {
00766                     for ( j = 1; j < ncoef3; j++ ) {
00767                         if ( cc[ncoef3+j] != 0 ) break;
00768                     }
00769                     if ( j < ncoef3 ) {
00770                         if ( ifp[-2] == EQUAL ) i = 0;
00771                         else i = 1;
00772                     }
00773                     else if ( ifp[-2] == EQUAL ) i = 1;
00774                     else i = 0;
00775                 }
00776             }
00777             goto donemul;
00778         }
00779         else if ( AddRat(BHEAD coef1,ncoef1,coef2,-ncoef2,coef3,&ncoef3) ) {
00780             TermFree(Spac2,"DoIfStatement");
00781             MesCall("DoIfStatement"); return(-1);
00782         }
00783         switch ( ifp[-2] ) {
00784             case GREATER:
00785                 if ( ncoef3 > 0 ) i = 1;
00786                 else i = 0;
00787                 break;
00788             case GREATEREQUAL:
00789                 if ( ncoef3 >= 0 ) i = 1;
00790                 else i = 0;
00791                 break;
00792             case LESS:
00793                 if ( ncoef3 < 0 ) i = 1;
00794                 else i = 0;
00795                 break;
00796             case LESSEQUAL:
00797                 if ( ncoef3 <= 0 ) i = 1;
00798                 else i = 0;
00799                 break;
00800             case EQUAL:
00801                 if ( ncoef3 == 0 ) i = 1;
00802                 else i = 0;
00803                 break;
00804             case NOTEQUAL:
00805                 if ( ncoef3 != 0 ) i = 1;
00806                 else i = 0;
00807                 break;
00808         }
00809     donemul: if ( i ) { ncoef2 = 1; coef2 = Spac2; coef2[0] = coef2[1] = 1; }
00810     else ncoef2 = 0;
00811     ismul1 = ismul2 = 0;
00812 }
00813 }
00814 else {
00815     first = 0;
00816 }

```

```

00817     coef1 = Spac1;
00818     i = 2*ABS(ncoef2);
00819     for ( j = 0; j < i; j++ ) coef1[j] = coef2[j];
00820     ncoef1 = ncoef2;
00821 SkipCond:
00822     ifp += ifp[1];
00823     } while ( ifp < ifstop );
00824
00825     NumberFree(coef3,"DoIfStatement"); NumberFree(Spac1,"DoIfStatement");
TermFree(Spac2,"DoIfStatement");
00826     if ( ncoef1 ) return(1);
00827     else return(0);
00828 }
00829
00830 /*
00831     #] DoIfStatement :
00832     #[ HowMany :                WORD HowMany(ifcode,term)
00833
00834     Returns the number of times that the pattern in ifcode
00835     can be taken out from term. There is a subkey in ifcode[2];
00836     The notation is identical to the lhs of an id statement.
00837     Most of the code comes from TestMatch.
00838 */
00839
00840 WORD HowMany(PHEAD WORD *ifcode, WORD *term)
00841 {
00842     GETBIDENTITY
00843     WORD *m, *t, *r, *w, power, RetVal, i, topje, *newterm;
00844     WORD *OldWork, *ww, *mm;
00845     int *RepSto, RepVal;
00846     int numdollars = 0;
00847     m = ifcode + IDHEAD;
00848     AN.FullProto = m;
00849     AN.WildValue = w = m + SUBEXPSIZE;
00850     m += m[1];
00851     AN.WildStop = m;
00852     OldWork = AT.WorkPointer;
00853     if ( ( ifcode[4] & 1 ) != 0 ) { /* We have at least one dollar in the pattern */
00854         AR.Eside = LHSIDEX;
00855         ww = AT.WorkPointer; i = m[0]; mm = m;
00856         NCOPY(ww,mm,i);
00857         *OldWork += 3;
00858         *ww++ = 1; *ww++ = 1; *ww++ = 3;
00859         AT.WorkPointer = ww;
00860         RepSto = AN.RepPoint;
00861         RepVal = *RepSto;
00862         NewSort(BHEAD0);
00863         if ( Generator(BHEAD OldWork,AR.Cnumlhs) ) {
00864             LowerSortLevel();
00865             *RepSto = RepVal;
00866             AN.RepPoint = RepSto;
00867             AT.WorkPointer = OldWork;
00868             return(-1);
00869         }
00870         AT.WorkPointer = ww;
00871         if ( EndSort(BHEAD ww,0) < 0 ) {}
00872         *RepSto = RepVal;
00873         AN.RepPoint = RepSto;
00874         if ( *ww == 0 || *(ww+ww) != 0 ) {
00875             if ( AP.lhdollarerror == 0 ) {
00876                 MLOCK(ErrorMessageLock);
00877                 MesPrint("&LHS must be one term");
00878                 MUNLOCK(ErrorMessageLock);
00879                 AP.lhdollarerror = 1;
00880             }
00881             AT.WorkPointer = OldWork;
00882             return(-1);
00883         }
00884         m = ww; AT.WorkPointer = ww = m + *m;
00885         if ( m[*m-1] < 0 ) { m[*m-1] = -m[*m-1]; }
00886         *m -= m[*m-1];
00887         AR.Eside = RHSIDE;
00888     }
00889     else {
00890         ww = term + *term;
00891         if ( AT.WorkPointer < ww ) AT.WorkPointer = ww;
00892     }
00893     ClearWild(BHEAD0);
00894     while ( w < AN.WildStop ) {
00895         if ( *w == LOADDOLLAR ) numdollars++;
00896         w += w[1];
00897     }
00898     AN.RepFunNum = 0;
00899     AN.RepFunList = AT.WorkPointer;
00900     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
00901     topje = cbuf[AT.ebufnum].numrhs;
00902     if ( AT.WorkPointer >= AT.WorkTop ) {

```

```

00903         MLOCK(ErrorMessageLock);
00904         MesWork();
00905         MUNLOCK(ErrorMessageLock);
00906         return(-1);
00907     }
00908     AN.DisOrderFlag = ifcode[2] & SUBDISORDER;
00909     switch ( ifcode[2] & (~SUBDISORDER) ) {
00910         case SUBONLY :
00911             /* Must be an exact match */
00912             AN.UseFindOnly = 1; AN.ForFindOnly = 0;
00913             /*
00914             Copy the term first to scratchterm. This is needed
00915             because of the Substitute.
00916             */
00917             i = *term;
00918             t = term; newterm = r = AT.WorkPointer;
00919             NCOPY(r,t,i); AT.WorkPointer = r;
00920             RetVal = 0;
00921             if ( FindRest(BHEAD newterm,m) && ( AN.UsedOtherFind ||
00922             FindOnly(BHEAD newterm,m) ) ) {
00923                 Substitute(BHEAD newterm,m,1);
00924                 if ( numdollars ) {
00925                     WildDollars(BHEAD (WORD *)0);
00926                     numdollars = 0;
00927                 }
00928                 ClearWild(BHEAD0);
00929                 RetVal = 1;
00930             }
00931             else RetVal = 0;
00932             break;
00933         case SUBMANY :
00934             /*
00935             Copy the term first to scratchterm. This is needed
00936             because of the Substitute.
00937             */
00938             i = *term;
00939             t = term; newterm = r = AT.WorkPointer;
00940             NCOPY(r,t,i); AT.WorkPointer = r;
00941             RetVal = 0;
00942             AN.UseFindOnly = 0;
00943             if ( ( power = FindRest(BHEAD newterm,m) ) > 0 ) {
00944                 if ( ( power = FindOnce(BHEAD newterm,m) ) > 0 ) {
00945                     AN.UseFindOnly = 0;
00946                     do {
00947                         Substitute(BHEAD newterm,m,1);
00948                         if ( numdollars ) {
00949                             WildDollars(BHEAD (WORD *)0);
00950                             numdollars = 0;
00951                         }
00952                         ClearWild(BHEAD0);
00953                         RetVal++;
00954                     } while ( FindRest(BHEAD newterm,m) && (
00955                     AN.UsedOtherFind || FindOnce(BHEAD newterm,m) ) );
00956                 }
00957                 else if ( power < 0 ) {
00958                     do {
00959                         Substitute(BHEAD newterm,m,1);
00960                         if ( numdollars ) {
00961                             WildDollars(BHEAD (WORD *)0);
00962                             numdollars = 0;
00963                         }
00964                         ClearWild(BHEAD0);
00965                         RetVal++;
00966                     } while ( FindRest(BHEAD newterm,m) );
00967                 }
00968             }
00969             else if ( power < 0 ) {
00970                 if ( FindOnce(BHEAD newterm,m) ) {
00971                     do {
00972                         Substitute(BHEAD newterm,m,1);
00973                         if ( numdollars ) {
00974                             WildDollars(BHEAD (WORD *)0);
00975                             numdollars = 0;
00976                         }
00977                         ClearWild(BHEAD0);
00978                     } while ( FindOnce(BHEAD newterm,m) );
00979                     RetVal = 1;
00980                 }
00981             }
00982             break;
00983         case SUBONCE :
00984             /*
00985             Copy the term first to scratchterm. This is needed
00986             because of the Substitute.
00987             */
00988             i = *term;
00989             t = term; newterm = r = AT.WorkPointer;

```

```

00990         NCOPY(r,t,i); AT.WorkPointer = r;
00991         RetVal = 0;
00992         AN.UseFindOnly = 0;
00993         if ( FindRest(BHEAD newterm,m) && ( AN.UsedOtherFind || FindOnce(BHEAD newterm,m) ) ) {
00994             Substitute(BHEAD newterm,m,1);
00995             if ( numdollars ) {
00996                 WildDollars(BHEAD (WORD *)0);
00997                 numdollars = 0;
00998             }
00999             ClearWild(BHEAD0);
01000             RetVal = 1;
01001         }
01002         else RetVal = 0;
01003         break;
01004     case SUBMULTI :
01005         RetVal = FindMulti(BHEAD term,m);
01006         break;
01007     case SUBVECTOR :
01008         RetVal = 0;
01009         for ( i = 0; i < *term; i++ ) ww[i] = term[i];
01010         while ( ( power = FindAll(BHEAD ww,m,AR.Cnumlhs,ifcode) ) != 0 ) { RetVal += power; }
01011         break;
01012     case SUBSELECT :
01013         ifcode += IDHEAD; ifcode += ifcode[1]; ifcode += *ifcode;
01014         AN.UseFindOnly = 1; AN.ForFindOnly = ifcode;
01015         if ( FindRest(BHEAD term,m) && ( AN.UsedOtherFind ||
01016             FindOnly(BHEAD term,m) ) ) RetVal = 1;
01017         else RetVal = 0;
01018         break;
01019     default :
01020         RetVal = 0;
01021         break;
01022 }
01023 AT.WorkPointer = AN.RepFunList;
01024 cbuf[AT.ebufnum].numrhs = topje;
01025 return(RetVal);
01026 }
01027
01028 /*
01029     #] HowMany :
01030     #[ DoubleIfBuffers :
01031 */
01032
01033 void DoubleIfBuffers(void)
01034 {
01035     int newmax, i;
01036     WORD *newsumcheck;
01037     LONG *newheap, *newifcount;
01038     if ( AC.MaxIf == 0 ) newmax = 10;
01039     else newmax = 2*AC.MaxIf;
01040     newheap = (LONG *)Malloc1(sizeof(LONG)*(newmax+1),"IfHeap");
01041     newsumcheck = (WORD *)Malloc1(sizeof(WORD)*(newmax+1),"IfSumCheck");
01042     newifcount = (LONG *)Malloc1(sizeof(LONG)*(newmax+1),"IfCount");
01043     if ( AC.MaxIf ) {
01044         for ( i = 0; i < AC.MaxIf; i++ ) {
01045             newheap[i] = AC.IfHeap[i];
01046             newsumcheck[i] = AC.IfSumCheck[i];
01047             newifcount[i] = AC.IfCount[i];
01048         }
01049         AC.IfStack = (AC.IfStack-AC.IfHeap) + newheap;
01050         M_free(AC.IfHeap,"AC.IfHeap");
01051         M_free(AC.IfCount,"AC.IfCount");
01052         M_free(AC.IfSumCheck,"AC.IfSumCheck");
01053     }
01054     else {
01055         AC.IfStack = newheap;
01056     }
01057     AC.IfHeap = newheap;
01058     AC.IfSumCheck = newsumcheck;
01059     AC.IfCount = newifcount;
01060     AC.MaxIf = newmax;
01061 }
01062
01063 /*
01064     #] DoubleIfBuffers :
01065     #] If statement :
01066     #[ Switch statement :
01067     #[ DoSwitch :
01068 */
01069
01070 int DoSwitch(PHEAD WORD *term, WORD *lhs)
01071 {
01072     /*
01073     For the moment we ignore the compiler buffer problems.
01074     */
01075     WORD numdollar = lhs[2];
01076     WORD ncase = DoIToNumber(BHEAD numdollar);

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```

01077     SWITCHTABLE *swtab = FindCase(lhs[3],ncase);
01078     return(Generator(BHEAD term,swtab->value));
01079 }
01080
01081 /*
01082     #] DoSwitch :
01083     #[ DoEndSwitch :
01084 */
01085
01086 int DoEndSwitch(PHEAD WORD *term, WORD *lhs)
01087 {
01088     SWITCH *sw = AC.SwitchArray+lhs[2];
01089     return(Generator(BHEAD term,sw->endswitch.value+1));
01090 }
01091
01092 /*
01093     #] DoEndSwitch :
01094     #[ FindCase :
01095 */
01096
01097 SWITCHTABLE *FindCase(WORD nswitch, WORD ncase)
01098 {
01099     /*
01100     First find the switch table and determine how we have to search.
01101     */
01102     SWITCH *sw = AC.SwitchArray+nswitch;
01103     WORD hi, lo, med;
01104     if ( sw->typetable == DENSETABLE ) {
01105         med = ncase - sw->caseoffset;
01106         if ( med >= sw->numcases || med < 0 ) return(&sw->defaultcase);
01107     }
01108     else {
01109         /*
01110         We need a binary search in the table.
01111         */
01112         if ( ncase > sw->maxcase || ncase < sw->mincase ) return(&sw->defaultcase);
01113         hi = sw->numcases-1; lo = 0;
01114         for(;;) {
01115             med = (hi+lo)/2;
01116             if ( ncase == sw->table[med].ncase ) break;
01117             else if ( ncase > sw->table[med].ncase ) {
01118                 lo = med+1;
01119                 if ( lo > hi ) return(&sw->defaultcase);
01120             }
01121             else {
01122                 hi = med-1;
01123                 if ( hi < lo ) return(&sw->defaultcase);
01124             }
01125         }
01126     }
01127     return(&sw->table[med]);
01128 }
01129
01130 /*
01131     #] FindCase :
01132     #[ DoubleSwitchBuffers :
01133 */
01134
01135 int DoubleSwitchBuffers(void)
01136 {
01137     int newmax, i;
01138     SWITCH *newarray;
01139     WORD *newheap;
01140     if ( AC.MaxSwitch == 0 ) newmax = 10;
01141     else newmax = 2*AC.MaxSwitch;
01142     newarray = (SWITCH *)Malloca(sizeof(SWITCH)*(newmax+1),"SwitchArray");
01143     newheap = (WORD *)Malloca(sizeof(WORD)*(newmax+1),"SwitchHeap");
01144     if ( AC.MaxSwitch ) {
01145         for ( i = 0; i < AC.MaxSwitch; i++ ) {
01146             newarray[i] = AC.SwitchArray[i];
01147             newheap[i] = AC.SwitchHeap[i];
01148         }
01149         M_free(AC.SwitchHeap,"AC.SwitchHeap");
01150         M_free(AC.SwitchArray,"AC.SwitchArray");
01151     }
01152     for ( i = AC.MaxSwitch; i <= newmax; i++ ) {
01153         newarray[i].table = 0;
01154         newarray[i].tablesize = 0;
01155         newarray[i].defaultcase.ncase = 0;
01156         newarray[i].defaultcase.value = 0;
01157         newarray[i].defaultcase.compbuffer = 0;
01158         newarray[i].endswitch.ncase = 0;
01159         newarray[i].endswitch.value = 0;
01160         newarray[i].endswitch.compbuffer = 0;
01161         newarray[i].typetable = 0;
01162         newarray[i].mincase = 0;
01163         newarray[i].maxcase = 0;

```



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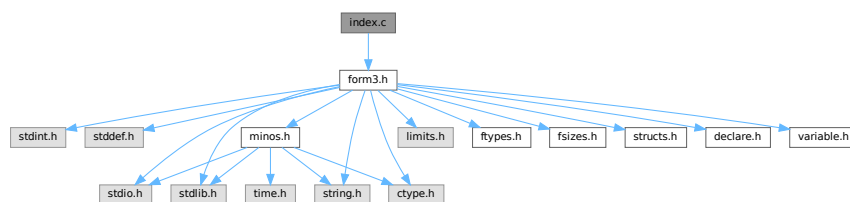
01164         newarray[i].numcases = 0;
01165         newarray[i].caseoffset = 0;
01166         newarray[i].iflevel = 0;
01167         newarray[i].whilelevel = 0;
01168         newarray[i].nestingsum = 0;
01169         newheap[i] = 0;
01170     }
01171     AC.SwitchArray = newarray;
01172     AC.SwitchHeap = newheap;
01173     AC.MaxSwitch = newmax;
01174     return(0);
01175 }
01176
01177 /*
01178     #] DoubleSwitchBuffers :
01179     #[ SwitchSplitMerge :
01180
01181     Sorts an array of WORDs. No adding of equal objects.
01182 */
01183
01184 void SwitchSplitMergeRec(SWITCHTABLE *array,WORD num,SWITCHTABLE *auxarray)
01185 {
01186     WORD n1,n2,i,j,k;
01187     SWITCHTABLE *t1,*t2, t;
01188     if ( num < 2 ) return;
01189     if ( num == 2 ) {
01190         if ( array[0].ncase > array[1].ncase ) {
01191             t = array[0]; array[0] = array[1]; array[1] = t;
01192         }
01193         return;
01194     }
01195     n1 = num/2;
01196     n2 = num - n1;
01197     SwitchSplitMergeRec(array,n1,auxarray);
01198     SwitchSplitMergeRec(array+n1,n2,auxarray);
01199     if ( array[n1-1].ncase <= array[n1].ncase ) return;
01200
01201     t1 = array; t2 = auxarray; i = n1; NCOPY(t2,t1,i);
01202     i = 0; j = n1; k = 0;
01203     while ( i < n1 && j < num ) {
01204         if ( auxarray[i].ncase <= array[j].ncase ) { array[k++] = auxarray[i++]; }
01205         else { array[k++] = array[j++]; }
01206     }
01207     while ( i < n1 ) array[k++] = auxarray[i++];
01208 /*
01209     Remember: remnants of j are still in place!
01210 */
01211 }
01212
01213 void SwitchSplitMerge(SWITCHTABLE *array,WORD num)
01214 {
01215     SWITCHTABLE *auxarray = (SWITCHTABLE *)Malloc1(sizeof(SWITCHTABLE)*num/2,"SwitchSplitMerge");
01216     SwitchSplitMergeRec(array,num,auxarray);
01217     M_free(auxarray,"SwitchSplitMerge");
01218 }
01219
01220 /*
01221     #] SwitchSplitMerge :
01222     #] Switch statement :
01223 */

```

## 4.62 index.c File Reference

```
#include "form3.h"
```

Include dependency graph for index.c:



## Functions

- [POSITION](#) \* [FindBracket](#) (WORD nexp, WORD \*bracket)
- void [PutBracketInIndex](#) (PHEAD WORD \*term, [POSITION](#) \*newpos)
- void [ClearBracketIndex](#) (WORD numexp)
- void [OpenBracketIndex](#) (WORD nexpr)
- int [PutInside](#) (PHEAD WORD \*term, WORD \*code)

### 4.62.1 Detailed Description

The routines that deal with bracket indexing It creates the bracket index and it can find the brackets using the index.

Definition in file [index.c](#).

### 4.62.2 Function Documentation

#### 4.62.2.1 FindBracket()

```
POSITION * FindBracket (
    WORD nexp,
    WORD * bracket )
```

Definition at line 65 of file [index.c](#).

#### 4.62.2.2 PutBracketInIndex()

```
void PutBracketInIndex (
    PHEAD WORD * term,
    POSITION * newPos )
```

Definition at line 331 of file [index.c](#).

#### 4.62.2.3 ClearBracketIndex()

```
void ClearBracketIndex (
    WORD numexp )
```

Definition at line 546 of file [index.c](#).

#### 4.62.2.4 OpenBracketIndex()

```
void OpenBracketIndex (
    WORD nexpr )
```

Definition at line 578 of file [index.c](#).

## 4.62.2.5 PutInside()

```
int PutInside (
    PHEAD WORD * term,
    WORD * code )
```

Definition at line 609 of file [index.c](#).

## 4.63 index.c

[Go to the documentation of this file.](#)

```
00001
00008 /* #[ License : */
00009 /*
00010 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00011 *   When using this file you are requested to refer to the publication
00012 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00013 *   This is considered a matter of courtesy as the development was paid
00014 *   for by FOM the Dutch physics granting agency and we would like to
00015 *   be able to track its scientific use to convince FOM of its value
00016 *   for the community.
00017 *
00018 *   This file is part of FORM.
00019 *
00020 *   FORM is free software: you can redistribute it and/or modify it under the
00021 *   terms of the GNU General Public License as published by the Free Software
00022 *   Foundation, either version 3 of the License, or (at your option) any later
00023 *   version.
00024 *
00025 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00026 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00027 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00028 *   details.
00029 *
00030 *   You should have received a copy of the GNU General Public License along
00031 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00032 */
00033 /* #] License : */
00034
00035 /*
00036 *   #[ Includes : index.c
00037 */
00038
00039 #include "form3.h"
00040
00041 /*
00042 *   #] Includes :
00043 *   #[ syntax and use :
00044
00045   The indexing of brackets is not automatic! It should only be used
00046   when one intends to use the contents of individual brackets.
00047   This is done with the addition of a + sign to the bracket statement:
00048   B+ a,b,c; or AB+ a,b,c;
00049   It does require resources! The index is kept in memory and is removed
00050   once the expression is treated and passed on to the output with different
00051   or no brackets.
00052   The index is limited to a given amount of space. Hence if there are too
00053   many brackets we will skip some in the index. Skipping goes by space
00054   occupied by the contents. We take the two adjacent bracket(s) with the
00055   least space together and represent them by the first one only. This gives
00056   a new spot.
00057   The expression struct has two pointers:
00058   bracketinfo      for using.
00059   newbracketinfo   for making new index.
00060
00061 *   #] syntax and use :
00062 *   #[ FindBracket :
00063 */
00064
00065 POSITION *FindBracket(WORD nexpt, WORD *bracket)
00066 {
00067     GETIDENTITY
00068     BRACKETINDEX *bi;
00069     BRACKETINFO *bracketinfo;
00070     EXPRESSIONS e = &(Expressions[nexpt]);
00071     LONG hi, low, med;
00072     int i;
```

```

00073     WORD oldsorttype = AR.SortType, *t1, *t2, j, bsize, *term, *p, *pstop, *pp;
00074     WORD *tstop, *cp, a[4], *bracketh;
00075     FILEHANDLE *fi;
00076     POSITION auxpos, toppos;
00077     switch ( e->status ) {
00078         case UNHIDELEXPRESSION:
00079         case UNHIDEGEXPRESSION:
00080         case DROPHLEXPRESSION:
00081         case DROPHGEXPRESSION:
00082         case HIDDENLEXPRESSION:
00083         case HIDDENGEXPRESSION:
00084         fi = AR.hidefile;
00085         break;
00086     default:
00087         fi = AR.infile;
00088         break;
00089     }
00090     if ( AT.bracketinfo ) bracketinfo = AT.bracketinfo;
00091     else bracketinfo = e->bracketinfo;
00092     hi = bracketinfo->indexfill; low = 0;
00093     if ( hi <= 0 ) return(0);
00094 /*
00095     The next code is needed for a problem with sorting when there are
00096     only functions outside the bracket. This gives ordinarily the wrong
00097     sorting. We solve that by taking HAAKJE along with the outside while
00098     running the Compare. But this means that we need to copy bracket.
00099 */
00100     bracketh = TermMalloc("FindBracket");
00101     i = *bracket; p = bracket; pp = bracketh; NCOPY(pp,p,i)
00102     pp -= 3; *pp++ = HAAKJE; *pp++ = 3; *pp++ = 0; *pp++ = 1; *pp++ = 1; *pp++ = 3;
00103     *bracketh += 3;
00104
00105     AT.fromindex = 1;
00106     AR.SortType = bracketinfo->SortType;
00107     bi = bracketinfo->indexbuffer + hi - 1;
00108     if ( *bracketh == 7 ) {
00109         if ( bracketinfo->bracketbuffer[bi->bracket] == 7 ) i = 0;
00110         else i = -1;
00111     }
00112     else if ( bracketinfo->bracketbuffer[bi->bracket] == 7 ) i = 1;
00113     else i = CompareTerms(BHEAD bracketh,bracketinfo->bracketbuffer+bi->bracket,0);
00114     if ( i < 0 ) {
00115         AR.SortType = oldsorttype;
00116         AT.fromindex = 0;
00117         TermFree(bracketh,"FindBracket");
00118         return(0);
00119     }
00120     else if ( i == 0 ) med = hi-1;
00121     else {
00122         for (;;) {
00123             med = (hi+low)/2;
00124             bi = bracketinfo->indexbuffer + med;
00125             if ( *bracketh == 7 ) {
00126                 if ( bracketinfo->bracketbuffer[bi->bracket] == 7 ) i = 0;
00127                 else i = -1;
00128             }
00129             else if ( bracketinfo->bracketbuffer[bi->bracket] == 7 ) i = 1;
00130             else i = CompareTerms(BHEAD bracketh,bracketinfo->bracketbuffer+bi->bracket,0);
00131             if ( i == 0 ) { break; }
00132             if ( i > 0 ) {
00133                 if ( low == med ) { /* no occurrence */
00134                     AR.SortType = oldsorttype;
00135                     AT.fromindex = 0;
00136                     TermFree(bracketh,"FindBracket");
00137                     return(0);
00138                 }
00139                 hi = med;
00140             }
00141             else if ( i < 0 ) {
00142                 if ( low == med ) break;
00143                 low = med;
00144             }
00145         }
00146 /*
00147     The bracket is now either bi itself or between bi and the next one
00148     or it is not present at all.
00149 */
00150     AN.theposition = AS.OldOnFile[nexp];
00151     ADD2POS(AN.theposition,bi->start);
00152 /*
00153     The seek will have to move closer to the actual read so that we
00154     can place a lock around the two.
00155     if ( fi->handle >= 0 ) SeekFile(fi->handle,&AN.theposition,SEEK_SET);
00156     else SetScratch(fi,&AN.theposition);
00157 */
00158 /*
00159     Put the bracket in the compress buffer as if it were the last term read.

```

```

00160      Have a look at AR.CompressPointer. (set it right)
00161  */
00162      term = AT.WorkPointer;
00163      t1 = bracketinfo->bracketbuffer+bi->bracket;
00164      j = *t1;
00165  /*
00166      Note that in the bracketbuffer, the bracket sits with HAAKJE.
00167
00168      The next is (hopefully) a bug fix. Originally the code read bsize = j
00169      but that overcounts one. We have the part outside the bracket and the
00170      coefficient which is 1,1,3. But we also have the length indicator.
00171      Where we use the variable bsize we do not include the length indicator,
00172      and we have the part outside plus 7,3,0 which is also three words.
00173  */
00174      bsize = j-1;
00175      t2 = AR.CompressPointer;
00176      NCOPY(t2,t1,j)
00177      if ( i == 0 ) { /* We found the proper bracket already */
00178          AR.SortType = oldsorttype;
00179          AT.fromindex = 0;
00180          TermFree(bracketh,"FindBracket");
00181          return(&AN.theposition);
00182      }
00183  /*
00184      Here we have to skip to the bracket if it exists (!)
00185      Let us first look whether the expression is in memory.
00186      If not we have to make a buffer to increase speed..
00187  */
00188      if ( fi->handle < 0 ) {
00189          p = (WORD *) ((UBYTE *) (fi->PObuffer)
00190              + BASEPOSITION(AS.OldOnFile[nexp])
00191              + BASEPOSITION(bi->start));
00192          pstop = (WORD *) ((UBYTE *) (fi->PObuffer)
00193              + BASEPOSITION(AS.OldOnFile[nexp])
00194              + BASEPOSITION(bi->next));
00195          while ( p < pstop ) {
00196  /*
00197              Check now: if size or part from previous term < size of bracket
00198              we have to setup the bracket again and test.
00199              Otherwise, skip immediately to the next term.
00200  */
00201              if ( *p <= -bsize ) { /* no change of bracket */
00202                  p++; p += *p + 1;
00203              }
00204              else if ( *p < 0 ) { /* changes bracket */
00205                  pp = p;
00206                  t2 = AR.CompressPointer;
00207                  t1 = t2 - *p++ + 1;
00208                  j = *p++;
00209                  NCOPY(t1,p,j)
00210                  t2++; while ( *t2 != HAAKJE ) t2 += t2[1];
00211                  t2 += t2[1];
00212                  a[1] = t2[0]; a[2] = t2[1]; a[3] = t2[2];
00213                  *t2++ = 1; *t2++ = 1; *t2++ = 3;
00214                  *AR.CompressPointer = t2 - AR.CompressPointer;
00215                  bsize = *AR.CompressPointer - 1;
00216                  if ( *bracketh == 7 ) {
00217                      if ( AR.CompressPointer[0] == 7 ) i = 0;
00218                      else i = -1;
00219                  }
00220                  else if ( AR.CompressPointer[0] == 7 ) i = 1;
00221                  else i = CompareTerms(BHEAD bracketh,AR.CompressPointer,0);
00222                  t2[-3] = a[1]; t2[-2] = a[2]; t2[-1] = a[3];
00223                  if ( i == 0 ) {
00224                      SETBASEPOSITION(AN.theposition,(pp-fi->PObuffer)*sizeof(WORD));
00225                      fi->POfill = pp;
00226                      goto found;
00227                  }
00228                  if ( i > 0 ) break; /* passed what was possible */
00229              }
00230              else { /* no compression. We have to check! */
00231                  WORD *oldworkpointer = AT.WorkPointer, *t3, *t4;
00232                  t2 = p + 1; while ( *t2 != HAAKJE ) t2 += t2[1];
00233                  t2 += t2[1];
00234  /*
00235                      Here we need to copy the term. Modifying has proven to
00236                      be NOT threadsafe.
00237  */
00238                  t3 = oldworkpointer; t4 = p;
00239                  while ( t4 < t2 ) *t3++ = *t4++;
00240                  *t3++ = 1; *t3++ = 1; *t3++ = 3;
00241                  *oldworkpointer = t3 - oldworkpointer;
00242                  bsize = *oldworkpointer - 1;
00243                  AT.WorkPointer = t3;
00244                  t3 = oldworkpointer;
00245                  if ( *bracketh == 7 ) {
00246                      if ( t3[0] == 7 ) i = 0;

```

```

00247         else i = -1;
00248     }
00249     else if ( t3[0] == 7 ) i = 1;
00250     else {
00251         i = CompareTerms(BHEAD bracketh,t3,0);
00252     }
00253     AT.WorkPointer = oldworkpointer;
00254     if ( i == 0 ) {
00255         SETBASEPOSITION(AN.theposition,(p-fi->PObuffer)*sizeof(WORD));
00256         fi->POfill = p;
00257         goto found;
00258     }
00259     if ( i > 0 ) break; /* passed what was possible */
00260     p += *p;
00261 }
00262 }
00263 AR.SortType = oldsorttype;
00264 AT.fromindex = 0;
00265 TermFree(bracketh,"FindBracket");
00266 return(0); /* Bracket does not exist */
00267 }
00268 else {
00269 /*
00270     In this case we can work with the old representation without HAAKJE.
00271     We stop searching when we reach toppos and we do not call Compare.
00272 */
00273     toppos = AS.OldOnFile[nexp];
00274     ADD2POS(toppos,bi->next);
00275     cp = AR.CompressPointer;
00276     for(;;) {
00277         auxpos = AN.theposition;
00278         GetOneTerm(BHEAD term,fi,&auxpos,0);
00279         if ( *term == 0 ) {
00280             AR.SortType = oldsorttype;
00281             AT.fromindex = 0;
00282             return(0); /* Bracket does not exist */
00283         }
00284         tstop = term + *term;
00285         tstop -= ABS(tstop[-1]);
00286         t1 = term + 1;
00287         while ( *t1 != HAAKJE && t1 < tstop ) t1 += t1[1];
00288         i = *bracket-4;
00289         if ( t1-term == *bracket-3 ) {
00290             t1 = term + 1; t2 = bracket+1;
00291             while ( i > 0 && *t1 == *t2 ) { t1++; t2++; i--; }
00292             if ( i <= 0 ) {
00293                 AR.CompressPointer = cp;
00294                 goto found;
00295             }
00296         }
00297         AR.CompressPointer = cp;
00298         AN.theposition = auxpos;
00299 /*
00300     Now check whether we passed the 'point'
00301 */
00302     if ( ISGEPOS(AN.theposition,toppos) ) {
00303         AR.SortType = oldsorttype;
00304         AR.CompressPointer = cp;
00305         AT.fromindex = 0;
00306         TermFree(bracketh,"FindBracket");
00307         return(0); /* Bracket does not exist */
00308     }
00309 }
00310 }
00311 found:
00312 AR.SortType = oldsorttype;
00313 AT.fromindex = 0;
00314 TermFree(bracketh,"FindBracket");
00315 return(&AN.theposition);
00316 }
00317
00318 /*
00319 #] FindBracket :
00320 #[ PutBracketInIndex :
00321
00322 Call via
00323 if ( AR.BracketOn ) PutBracketInIndex(BHEAD term);
00324
00325 This means that there should be a bracket somewhere
00326 Note that the brackets come in in proper order.
00327
00328 DON'T forget AR.SortType to be put into e->bracketinfo->SortType
00329 */
00330
00331 void PutBracketInIndex(PHEAD WORD *term, POSITION *newpos)
00332 {
00333     GETBIDENTITY

```

```

00334     BRACKETINDEX *bi, *b1, *b2, *b3;
00335     BRACKETINFO *b;
00336     POSITION thepos;
00337     EXPRESSIONS e = Expressions + AR.CurExpr;
00338     LONG hi, i, average;
00339     WORD *t, *tstop, *t1, *t2, *oldt, oldsize, a[4];
00340     if ( ( b = e->newbracketinfo ) == 0 ) return;
00341     DIFPOS(thepos,*newpos,e->onfile);
00342     tstop = term + *term;
00343     tstop -= ABS(tstop[-1]);
00344     t = term+1;
00345     while ( *t != HAAKJE && t < tstop ) t += t[1];
00346     if ( t >= tstop ) return; /* no ticket, no laundry */
00347     t += t[1]; /* include HAAKJE for the sorting */
00348     a[0] = t[0]; a[1] = t[1]; a[2] = t[2];
00349     oldt = t; oldsize = *term; *t++ = 1; *t++ = 1; *t++ = 3;
00350     *term = t - term;
00351     AT.fromindex = 1;
00352 /*
00353     Check now with the last bracket in the buffer.
00354     If it is the same we can abort.
00355 */
00356     hi = b->indexfill;
00357     if ( hi > 0 ) {
00358         bi = b->indexbuffer + hi - 1;
00359         bi->next = thepos;
00360         if ( *term == 7 ) {
00361             if ( b->bracketbuffer[bi->bracket] == 7 ) i = 0;
00362             else i = -1;
00363         }
00364         else if ( b->bracketbuffer[bi->bracket] == 7 ) i = 1;
00365         else i = CompareTerms(BHEAD term,b->bracketbuffer+bi->bracket,0);
00366         if ( i == 0 ) { /* still the same bracket */
00367             bi->termsinbracket++;
00368             goto bracketdone;
00369         }
00370         if ( i > 0 ) { /* We have a problem */
00371 /*
00372         There is a special case in which we have only functions and
00373         term is contained completely in the bracket
00374 */
00375 /*
00376         t = term + 1;
00377         tstop = term + *term - 3;
00378         while ( t < tstop && *t > HAAKJE ) t += t[1];
00379         if ( t < tstop ) goto problems;
00380 */
00381         for ( i = 1; i < *term - 3; i++ ) {
00382             if ( term[i] != b->bracketbuffer[bi->bracket+i] ) break;
00383         }
00384         if ( i < *term - 3 ) {
00385 /*
00386         problems::
00387 */
00388             *term = oldsize; oldt[0] = a[0]; oldt[1] = a[1]; oldt[2] = a[2];
00389             MLOCK(ErrorMessageLock);
00390             MesPrint("Error!!!! Illegal bracket sequence detected in PutBracketInIndex");
00391 #ifdef WITHPTHREADS
00392             MesPrint("Worker = %w");
00393 #endif
00394             PrintTerm(term,"term into index");
00395             PrintTerm(b->bracketbuffer+bi->bracket,"Last in index");
00396             MUNLOCK(ErrorMessageLock);
00397             AT.fromindex = 0;
00398             Terminate(-1);
00399         }
00400         i = -1;
00401     }
00402 }
00403 /*
00404     If there is room for more brackets, we add this one.
00405 */
00406     if ( b->bracketfill+*term >= b->bracketbuffersize
00407         && ( b->bracketbuffersize < AM.MaxBracketBufferSize
00408             || ( e->vflags & ISFACTORIZED ) != 0 ) ) {
00409 /*
00410         Enlarge bracket buffer
00411 */
00412         WORD *oldbracketbuffer = b->bracketbuffer;
00413         i = MAX(b->bracketbuffersize * 2, b->bracketfill+*term+1);
00414         if ( i > AM.MaxBracketBufferSize && ( e->vflags & ISFACTORIZED ) == 0 )
00415             i = AM.MaxBracketBufferSize;
00416         if ( i > b->bracketfill+*term ) {
00417             b->bracketbuffersize = i;
00418             b->bracketbuffer = (WORD *)Malloca1(b->bracketbuffersize*sizeof(WORD),
00419                 "new bracket buffer");
00420             t1 = b->bracketbuffer; t2 = oldbracketbuffer;

```

```

00421         i = b->bracketfill;
00422         NCOPY(t1,t2,i)
00423         if ( oldbracketbuffer ) M_free(oldbracketbuffer,"old bracket buffer");
00424     }
00425 }
00426 if ( b->bracketfill+*term < b->bracketbuffersize ) {
00427     if ( b->indexfill >= b->indexbuffersize ) {
00428 /*
00429         Enlarge index
00430 */
00431         BRACKETINDEX *oldindexbuffer = b->indexbuffer;
00432         b->indexbuffersize *= 2;
00433         b->indexbuffer = (BRACKETINDEX *)
00434             Malloc1(b->indexbuffersize*sizeof(BRACKETINDEX),"new bracket index");
00435         b1 = b->indexbuffer; b2 = oldindexbuffer;
00436         i = b->indexfill;
00437         NCOPY(b1,b2,i)
00438         if ( oldindexbuffer ) M_free(oldindexbuffer,"old bracket index");
00439     }
00440 }
00441 else {
00442 /*
00443     We have too many brackets in the buffer. Try to improve.
00444     This is the interesting algorithm. We try to eliminate about 1/4 to
00445     1/2 of the brackets from the index. This should be done by size of
00446     the bracket contents to make the searching as fast as possible.
00447     But! Do not touch the last bracket.
00448     Note that we are always filling from the back.
00449     Algorithm: Throw away every second bracket, unless b1+b2 is much longer
00450     than average. How much is something we can tune.
00451 */
00452     average = DIVPOS(thepos,b->indexfill+1);
00453     if ( average <= 0 ) {
00454         MLOCK(ErrorMessageLock);
00455         MesPrint("Problems with bracket buffer. Increase MaxBracketBufferSize in form.set");
00456         MesPrint("Current size is %1",AM.MaxBracketBufferSize*sizeof(WORD));
00457         MUNLOCK(ErrorMessageLock);
00458         Terminate(-1);
00459     }
00460     average *= 4; /* 2*2: one 2 for much longer, one 2 because we have pairs */
00461     t2 = b->bracketbuffer;
00462     b3 = b1 = b->indexbuffer;
00463     bi = b->indexbuffer + b->indexfill;
00464     b2 = b1+1;
00465     while ( b2+2 < bi ) {
00466         if ( DIFBASE(b2->next,b1->start) > average ) {
00467             t1 = b->bracketbuffer + b1->bracket;
00468             b1->bracket = t2 - b->bracketbuffer;
00469             i = *t1; NCOPY(t2,t1,i)
00470             *b3++ = *b1;
00471             t1 = b->bracketbuffer + b2->bracket;
00472             b2->bracket = t2 - b->bracketbuffer;
00473             i = *t1; NCOPY(t2,t1,i)
00474             *b3++ = *b2;
00475             if ( b3 <= b1 ) {
00476                 PUTZERO(b1->start);
00477                 PUTZERO(b1->next);
00478                 b1->bracket = 0;
00479                 b1->termsinbracket = 0;
00480             }
00481             if ( b3 <= b2 ) {
00482                 PUTZERO(b2->start);
00483                 PUTZERO(b2->next);
00484                 b2->bracket = 0;
00485                 b2->termsinbracket = 0;
00486             }
00487         }
00488         else {
00489             t1 = b->bracketbuffer + b1->bracket;
00490             b1->bracket = t2 - b->bracketbuffer;
00491             i = *t1; NCOPY(t2,t1,i)
00492             b1->next = b2->next;
00493             b1->termsinbracket += b2->termsinbracket;
00494             *b3++ = *b1;
00495             if ( b3 <= b1 ) {
00496                 PUTZERO(b1->start);
00497                 PUTZERO(b1->next);
00498                 b1->bracket = 0;
00499                 b1->termsinbracket = 0;
00500             }
00501             PUTZERO(b2->start);
00502             PUTZERO(b2->next);
00503             b2->bracket = 0;
00504             b2->termsinbracket = 0;
00505         }
00506         b1 += 2; b2 += 2;
00507     }

```



```

00508         while ( b1 < bi ) {
00509             t1 = b->bracketbuffer + b1->bracket;
00510             b1->bracket = t2 - b->bracketbuffer;
00511             i = *t1; NCOPY(t2,t1,i)
00512             *b3++ = *b1;
00513             if ( b3 <= b1 ) {
00514                 PUTZERO(b1->start);
00515                 PUTZERO(b1->next);
00516                 b1->bracket = 0;
00517                 b1->termsinbracket = 0;
00518             }
00519             b1++;
00520         }
00521         b->indexfill = b3 - b->indexbuffer;
00522         b->bracketfill = t2 - b->bracketbuffer;
00523     }
00524     bi = b->indexbuffer + b->indexfill;
00525     b->indexfill++;
00526     bi->bracket = b->bracketfill;
00527     bi->start = thepos;
00528     bi->next = thepos;
00529     bi->termsinbracket = 1;
00530 /*
00531     Copy the bracket into the buffer
00532 */
00533     t1 = term; t2 = b->bracketbuffer + bi->bracket; i = *t1;
00534     b->bracketfill += i;
00535     NCOPY(t2,t1,i)
00536 bracketdone:
00537     *term = oldsize; oldt[0] = a[0]; oldt[1] = a[1]; oldt[2] = a[2];
00538     AT.fromindex = 0;
00539 }
00540
00541 /*
00542     #] PutBracketInIndex :
00543     #[ ClearBracketIndex :
00544 */
00545
00546 void ClearBracketIndex(WORD numexp)
00547 {
00548     BRACKETINFO *b;
00549     if ( numexp >= 0 ) {
00550         b = Expressions[numexp].bracketinfo;
00551         Expressions[numexp].bracketinfo = 0;
00552     }
00553     else if ( numexp == -1 ) {
00554         GETIDENTITY
00555         b = AT.bracketinfo;
00556         AT.bracketinfo = 0;
00557     }
00558     else {
00559         numexp = -numexp-2;
00560         b = Expressions[numexp].newbracketinfo;
00561         Expressions[numexp].newbracketinfo = 0;
00562     }
00563     if ( b == 0 ) return;
00564     b->indexfill = b->indexbuffersize = 0;
00565     b->bracketfill = b->bracketbuffersize = 0;
00566     M_free(b->bracketbuffer,"ClearBracketBuffer");
00567     M_free(b->indexbuffer,"ClearIndexBuffer");
00568     M_free(b,"BracketInfo");
00569 }
00570
00571 /*
00572     #] ClearBracketIndex :
00573     #[ OpenBracketIndex :
00574
00575     Note: This routine is thread-safe
00576 */
00577
00578 void OpenBracketIndex(WORD nexpr)
00579 {
00580     EXPRESSIONS e = Expressions + nexpr;
00581     BRACKETINFO *bi;
00582     LONG i;
00583     bi = (BRACKETINFO *)Malloc1(sizeof(BRACKETINFO),"BracketInfo");
00584     e->newbracketinfo = bi;
00585 /*
00586     i = 20*AM.MaxTer/sizeof(WORD);
00587     if ( i < 1000 ) i = 1000;
00588 */
00589     i = 2000;
00590     bi->bracketbuffer = (WORD *)Malloc1(i*sizeof(WORD),"Bracket Buffer");
00591     bi->bracketbuffersize = i;
00592     bi->bracketfill = 0;
00593     i = 50;
00594     bi->indexbuffer = (BRACKETINDEX *)Malloc1(i*sizeof(BRACKETINDEX),"Bracket Index");

```

```

00595     bi->indexbuffersize = i;
00596     bi->indexfill = 0;
00597     bi->SortType = AC.SortType;
00598 }
00599
00600 /*
00601     #] OpenBracketIndex :
00602     #[ PutInside :
00603
00604     Puts a term, or a bracket determined part of a term inside a function.
00605
00606     AT.WorkPointer points at term+*term
00607 */
00608
00609 int PutInside(PHEAD WORD *term, WORD *code)
00610 {
00611     WORD *from, *to, *oldbuf, *tStop, *t, *tt, oldon, oldact, inc, argsize, *termout;
00612     int i, ii, error;
00613
00614     if ( code[1] == 4 && ( code[2] == 0 || code[2] == 1 ) ) {
00615 /*
00616         Put all inside. Move the term by 1+FUNHEAD+ARGHEAD
00617 */
00618         from = term+*term; to = from+1+ARGHEAD+FUNHEAD; i = ii = *term;
00619         to[0] = 1; to[1] = 1; to[2] = 3;
00620         while ( --i >= 0 ) *--to = *--from;
00621         to = term;
00622         *to++ = term[0]+4+ARGHEAD+FUNHEAD;
00623         *to++ = code[3];
00624         *to++ = ii+FUNHEAD+ARGHEAD;
00625         *to++ = 1; /* set dirty flags, because there could be a fast notation */
00626         FILLFUN3(to)
00627         *to++ = ii+ARGHEAD;
00628         *to++ = 1;
00629         FILLARG(to)
00630         return(0);
00631     }
00632 /*
00633     First we save the old bracket variables. Then we set variables to
00634     influence the PutBracket routine and call it.
00635     After that we set the values back and sort out the results by placing the
00636     inside of the bracket inside the function.
00637 */
00638     termout = AT.WorkPointer;
00639     oldbuf = AT.BrackBuf;
00640     oldon = AR.BracketOn;
00641     oldact = AT.PolyAct;
00642     AR.BracketOn = -code[2];
00643     AT.BrackBuf = code+4;
00644     AT.PolyAct = 0;
00645     error = PutBracket(BHEAD term);
00646     AT.PolyAct = oldact;
00647     AT.BrackBuf = oldbuf;
00648     AR.BracketOn = oldon;
00649     if ( error ) return(error);
00650     i = *termout; from = termout; to = term;
00651     NCOPY(to, from, i);
00652     tStop = term + *term; tStop -= tStop[-1];
00653     t = term+1;
00654     while ( t < tStop && *t != HAAKJE ) t += t[1];
00655     from = term + *term;
00656     inc = FUNHEAD+ARGHEAD-t[1]+1;
00657     tt = t + t[1];
00658     argsize = from-tt+1;
00659     to = from + inc;
00660     to[0] = 1;
00661     to[1] = 1;
00662     to[2] = 3;
00663     while ( from > tt ) *--to = *--from;
00664     *--to = argsize;
00665     *t++ = code[3];
00666     *t++ = argsize+FUNHEAD+ARGHEAD;
00667     *t++ = 1;
00668     FILLFUN3(t);
00669     *t++ = argsize+ARGHEAD;
00670     *t++ = 1;
00671     FILLARG(t);
00672     *term += inc+3;
00673     AT.WorkPointer = term+*term;
00674     if ( Normalize(BHEAD term) ) error = 1;
00675     return(error);
00676 }
00677
00678 /*
00679     #] PutInside :
00680 */
00681

```

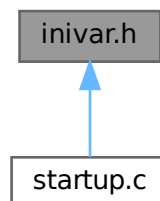
```

00682  /*
00683      The next routines are for indexing the local output files in a parallel
00684      sort. This indexing is needed to get a fast determination of the
00685      splitting terms needed to divide the terms evenly over the processors.
00686      Actually this method works well for ParFORM, but may not work well
00687      for TFORM.
00688
00689      #[ PutTermInIndex :
00690
00691      Puts a term in the term index.
00692      Action:
00693          if the index hasn't reached its full size
00694              if there is room, put the term
00695              if there is no room: extend the buffer, put the term
00696          else
00697              check if the last term has a number of the type skip*m+1
00698                  if no, overwrite the last term
00699                  if yes, check whether there is room for one more term
00700                      yes: add the term
00701                      no: drop all even terms, compress the list,
00702                          multiply skip by 2, and add this term.
00703
00704  int PutTermInIndex(WORD *term, POSITION *position)
00705  {
00706      return(0);
00707  }
00708
00709      #] PutTermInIndex :
00710  */

```

## 4.64 inivar.h File Reference

This graph shows which files directly or indirectly include this file:



### Data Structures

- struct **fixedfun**

### Variables

- [FIXEDGLOBALS FG](#)
- [ALLGLOBALS A](#)
- [FIXEDSET fixedsets \[\]](#)
- [UBYTE BufferForOutput \[MAXLINELENGTH+14\]](#)
- [char \\* setupfilename = "form.set"](#)

### 4.64.1 Detailed Description

Contains the initialization of a number of structs at compile time This file should only be included in the file [startup.c](#) !!!

Definition in file [inivar.h](#).

### 4.64.2 Variable Documentation

#### 4.64.2.1 FG

[FIXEDGLOBALS](#) FG

Definition at line 34 of file [inivar.h](#).

#### 4.64.2.2 A

[ALLGLOBALS](#) A

Definition at line 133 of file [inivar.h](#).

#### 4.64.2.3 fixedsets

[FIXEDSET](#) fixedsets[]

Initial value:

```
= {
    {"pos_", "integers > 0", CSYMBOL, 0}
    , {"pos0_", "integers >= 0", CSYMBOL, 0}
    , {"neg_", "integers < 0", CSYMBOL, 0}
    , {"neg0_", "integers <= 0", CSYMBOL, 0}
    , {"even_", "even integers", CSYMBOL, 0}
    , {"odd_", "odd integers", CSYMBOL, 0}
    , {"int_", "all integers", CSYMBOL, 0}
    , {"symbol_", "only symbols", CSYMBOL, MAXPOSITIVE}
    , {"fixed_", "fixed indices", CINDEX, 0}
    , {"index_", "all indices", CINDEX, 0}
    , {"number_", "all rationals", CSYMBOL, 0}
    , {"dummyindices_", "dummy indices", CINDEX, 0}
    , {"vector_", "only vectors", CVECTOR, 0}
}
```

Definition at line 258 of file [inivar.h](#).

#### 4.64.2.4 BufferForOutput

UBYTE BufferForOutput [MAXLINELENGTH+14]

Definition at line 274 of file [inivar.h](#).

#### 4.64.2.5 setupfilename

char\* setupfilename = "form.set"

Definition at line 276 of file [inivar.h](#).

## 4.65 inivar.h

[Go to the documentation of this file.](#)

```

00001
00007 /* #[ License : */
00008 /*
00009 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00010 * When using this file you are requested to refer to the publication
00011 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00012 * This is considered a matter of courtesy as the development was paid
00013 * for by FOM the Dutch physics granting agency and we would like to
00014 * be able to track its scientific use to convince FOM of its value
00015 * for the community.
00016 *
00017 * This file is part of FORM.
00018 *
00019 * FORM is free software: you can redistribute it and/or modify it under the
00020 * terms of the GNU General Public License as published by the Free Software
00021 * Foundation, either version 3 of the License, or (at your option) any later
00022 * version.
00023 *
00024 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00025 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00026 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00027 * details.
00028 *
00029 * You should have received a copy of the GNU General Public License along
00030 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00031 */
00032 /* #[ License : */
00033
00034 FIXEDGLOBALS FG = {
00035     {Traces /* Dummy as zeroeth argument */
00036     ,Traces
00037     ,EpFCon
00038     ,RatioGen
00039     ,SymGen
00040     ,TenVec
00041     ,DoSumF1
00042     ,DoSumF2}
00043
00044     ,{TraceFind /* Dummy as zeroeth argument */
00045     ,TraceFind
00046     ,EpFFind
00047     ,RatioFind
00048     ,SymFind
00049     ,TenVecFind}
00050
00051     ,{"symbol"
00052     , "index"
00053     , "vector"
00054     , "function"
00055     , "set"
00056     , "expression"
00057     , "dotproduct"
00058     , "number"
00059     , "subexp"
00060     , "delta"}
00061
00062     ,{"(local)"
00063     , "(skip/local)"
00064     , "(drop/local)"
00065     , "(dropped)"
00066     , "(global)"
00067     , "(skip/global)"
00068     , "(drop/global)"
00069     , "(dropped)"
00070     , "(stored)"
00071     , "(local-hidden)"
00072     , "(local-to be hidden)"
00073     , "(local-hidden-dropped)"
00074     , "(local-to be unhidden)"
00075     , "(global-hidden)"
00076     , "(global-to be hidden)"
00077     , "(global-hidden-dropped)"
00078     , "(global-to be unhidden)"
00079     , "(into-hide-local)"
00080     , "(into-hide-global)"
00081     , "(spectator)"
00082     , "(drop/spectator)"}
00083
00084     ,{" Functions"
00085     , " Commuting Functions"}
00086
00087     ,{"left "

```

```

00088     , "active "
00089     , "in output"}
00090
00091     , (char *)0
00092     , (char *)0
00093     , (UBYTE *) "1"
00094     , (WORD)0
00095     , (WORD)0
00096     , (WORD)0
00097     , (WORD)0
00098
00099 /* ASCII table of character types. Note that on some computers this
00100    table may be different from the ASCII table.
00101
00102    -1 Illegal character
00103     0 Alphabetic character
00104     1 Digit
00105     2 . $ _ ? # or '
00106     3 [,]
00107     4 ( ) = ; or ,
00108     5 + - * % / ^ :
00109     6 blank, tab, linefeed
00110     7 {,|,}
00111     8 ! & < >
00112     9 "
00113    10 The ultimate end.
00114 */
00115     , { 10,255,255,255,255,255,255,255,255, 6, 6,255,255, 6,255,255,
00116         255,255,255,255,255,255,255,255,255, 10,255,255,255,255,255,
00117         6, 8, 9, 2, 2, 5, 8, 2, 4, 4, 5, 5, 4, 5, 2, 5,
00118         1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 5, 4, 8, 4, 8, 2,
00119         255, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
00120         0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 3,255, 3, 5, 2,
00121         255, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
00122         0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 7, 7, 7,255,255,
00123         255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,
00124         255,255,255,255,255,255,255,255,255,255,255,255,255,255,
00125         255,255,255,255,255,255,255,255,255,255,255,255,255,255,
00126         255,255,255,255,255,255,255,255,255,255,255,255,255,255,
00127         255,255,255,255,255,255,255,255,255,255,255,255,255,255,
00128         255,255,255,255,255,255,255,255,255,255,255,255,255,255,
00129         255,255,255,255,255,255,255,255,255,255,255,255,255,255,
00130         255,255,255,255,255,255,255,255,255,255,255,255,255,255 }
00131 };
00132
00133 ALLGLOBALS A;
00134 #ifdef WITHPTHREADS
00135 ALLPRIVATES **AB;
00136 #endif
00137
00138 static struct fixedfun {
00139     char *name;
00140     int      commu , tensor      , complx      , symmetric;
00141 } fixedfunctions[] = {
00142     {"exp_"      , 1 , 0      , 0      , 0 } /* EXPONENT */
00143     , {"denom_"  , 1 , 0      , 0      , 0 } /* DENOMINATOR */
00144     , {"setfun_"  , 1 , 0      , 0      , 0 } /* SETFUNCTION */
00145     , {"g_"      , 1 , GAMMAFUNCTION , VARTYPEIMAGINARY, 0 } /* GAMMA */
00146     , {"gi_"     , 1 , GAMMAFUNCTION , VARTYPEIMAGINARY, 0 } /* GAMMAI */
00147     , {"g5_"     , 1 , GAMMAFUNCTION , VARTYPEIMAGINARY, 0 } /* GAMMAFIVE */
00148     , {"g6_"     , 1 , GAMMAFUNCTION , VARTYPEIMAGINARY, 0 } /* GAMMASIX */
00149     , {"g7_"     , 1 , GAMMAFUNCTION , VARTYPEIMAGINARY, 0 } /* GAMMASEVEN */
00150     , {"sum_"    , 1 , 0      , 0      , 0 } /* SUMF1 */
00151     , {"sump_"   , 1 , 0      , 0      , 0 } /* SUMF2 */
00152     , {"dum_"    , 1 , 0      , 0      , 0 } /* DUMFUN */
00153     , {"replace_" , 0 , 0      , 0      , 0 } /* REPLACEMENT */
00154     , {"reverse_" , 1 , 0      , 0      , 0 } /* REVERSEFUNCTION */
00155     , {"distrib_" , 1 , 0      , 0      , 0 } /* DISTRIBUTION */
00156     , {"dd_"     , 0 , TENSORFUNCTION , 0      , 0 } /* DELTA3 */
00157     , {"dummy_"  , 0 , 0      , 0      , 0 } /* DUMMYFUN */
00158     , {"dummyten_" , 0 , TENSORFUNCTION , 0      , 0 } /* DUMMYTEN */
00159     , {"e_"      , 0 , TENSORFUNCTION|1 , VARTYPEIMAGINARY, ANTISYMMETRIC }
00160     /* LEVICIVITA */
00161     , {"fac_"    , 0 , 0      , 0      , 0 } /* FACTORIAL */
00162     , {"invfac_" , 0 , 0      , 0      , 0 } /* INVERSEFACTORIAL */
00163     , {"binom_"  , 0 , 0      , 0      , 0 } /* BINOMIAL */
00164     , {"nargs_"  , 0 , 0      , 0      , 0 } /* NUMARGSFUN */
00165     , {"sign_"   , 0 , 0      , 0      , 0 } /* SIGNFUN */
00166     , {"mod_"    , 0 , 0      , 0      , 0 } /* MODFUNCTION */
00167     , {"mod2_"   , 0 , 0      , 0      , 0 } /* MOD2FUNCTION */
00168     , {"min_"    , 0 , 0      , 0      , 0 } /* MINFUNCTION */
00169     , {"max_"    , 0 , 0      , 0      , 0 } /* MAXFUNCTION */
00170     , {"abs_"    , 0 , 0      , 0      , 0 } /* ABSFUNCTION */
00171     , {"sig_"    , 0 , 0      , 0      , 0 } /* SIGFUNCTION */
00172     , {"integer_" , 0 , 0      , 0      , 0 } /* INTFUNCTION */
00173     , {"theta_"  , 0 , 0      , 0      , 0 } /* THETA */
00174     , {"thetap_" , 0 , 0      , 0      , 0 } /* THETA2 */

```

```

00175     {"delta_" ,0 ,0 ,0 ,0} /* DELTA2 */
00176     {"deltap_" ,0 ,0 ,0 ,0} /* DELTAP */
00177     {"bernoulli_" ,0 ,0 ,0 ,0} /* BERNOULLIFUNCTION */
00178     {"count_" ,0 ,0 ,0 ,0} /* COUNTFUNCTION */
00179     {"match_" ,0 ,0 ,0 ,0} /* MATCHFUNCTION */
00180     {"pattern_" ,0 ,0 ,VARTYPECOMPLEX ,0} /* PATTERNFUNCTION */
00181     {"term_" ,1 ,0 ,0 ,0} /* TERMFUNCTION */
00182     {"conjg_" ,1 ,0 ,VARTYPECOMPLEX ,0} /* CONJUGATEFUNCTION */
00183     {"root_" ,0 ,0 ,0 ,0} /* ROOTFUNCTION */
00184     {"table_" ,1 ,0 ,0 ,0} /* TABLEFUNCTION */
00185     {"firstbracket_" ,0 ,0 ,0 ,0} /* FIRSTBRACKET */
00186     {"termsin_" ,0 ,0 ,0 ,0} /* TERMSINEXPR */
00187     {"nterms_" ,0 ,0 ,0 ,0} /* NUMTERMSFUN */
00188     {"gcd_" ,0 ,0 ,0 ,0} /* GCDFUNCTION */
00189     {"div_" ,0 ,0 ,0 ,0} /* DIVFUNCTION */
00190     {"rem_" ,0 ,0 ,0 ,0} /* REMFUNCTION */
00191     {"maxpowerof_" ,0 ,0 ,0 ,0} /* MAXPOWEROF */
00192     {"minpowerof_" ,0 ,0 ,0 ,0} /* MINPOWEROF */
00193     {"tbl_" ,0 ,0 ,0 ,0} /* TABLESTUB */
00194     {"factorin_" ,0 ,0 ,0 ,0} /* FACTORIN */
00195     {"termsinbracket_" ,0 ,0 ,0 ,0} /* TERMSINBRACKET */
00196     {"farg_" ,0 ,0 ,0 ,0} /* WILDARGFUN */
00197 /*
00198     The following names are reserved for the floating point library.
00199     As long as we have no floating point numbers they do not do anything.
00200 */
00201     {"sqrt_" ,0 ,0 ,0 ,0} /* SQRTFUNCTION */
00202     {"ln_" ,0 ,0 ,0 ,0} /* LNFUNCTION */
00203     {"sin_" ,0 ,0 ,0 ,0} /* SINFUNCTION */
00204     {"cos_" ,0 ,0 ,0 ,0} /* COSFUNCTION */
00205     {"tan_" ,0 ,0 ,0 ,0} /* TANFUNCTION */
00206     {"asin_" ,0 ,0 ,0 ,0} /* ASINFUNCTION */
00207     {"acos_" ,0 ,0 ,0 ,0} /* ACOSFUNCTION */
00208     {"atan_" ,0 ,0 ,0 ,0} /* ATANFUNCTION */
00209     {"atan2_" ,0 ,0 ,0 ,0} /* ATAN2FUNCTION */
00210     {"sinh_" ,0 ,0 ,0 ,0} /* SINHFUNCTION */
00211     {"cosh_" ,0 ,0 ,0 ,0} /* COSHFUNCTION */
00212     {"tanh_" ,0 ,0 ,0 ,0} /* TANHFUNCTION */
00213     {"asinh_" ,0 ,0 ,0 ,0} /* ASINHFUNCTION */
00214     {"acosh_" ,0 ,0 ,0 ,0} /* ACOSHFUNCTION */
00215     {"atanh_" ,0 ,0 ,0 ,0} /* ATANHFUNCTION */
00216     {"li2_" ,0 ,0 ,0 ,0} /* LI2FUNCTION */
00217     {"lin_" ,0 ,0 ,0 ,0} /* LINFUNCTION */
00218 /*
00219     From here on we continue with new functions (26-sep-2010)
00220 */
00221     {"extrasymbol_" ,0 ,0 ,0 ,0} /* EXTRASYMFUN */
00222     {"random_" ,0 ,0 ,0 ,0} /* RANDOMFUNCTION */
00223     {"ranperm_" ,1 ,0 ,0 ,0} /* RANPERM */
00224     {"numfactors_" ,0 ,0 ,0 ,0} /* NUMFACTORS */
00225     {"firstterm_" ,0 ,0 ,0 ,0} /* FIRSTTERM */
00226     {"content_" ,0 ,0 ,0 ,0} /* CONTENTTERM */
00227     {"prime_" ,0 ,0 ,0 ,0} /* PRIMENUMBER */
00228     {"exteuclidean_" ,0 ,0 ,0 ,0} /* EXTEUCLIDEAN */
00229     {"makerational_" ,0 ,0 ,0 ,0} /* MAKERATIONAL */
00230     {"inverse_" ,0 ,0 ,0 ,0} /* INVERSEFUNCTION */
00231     {"id_" ,1 ,0 ,0 ,0} /* IDFUNCTION */
00232     {"putfirst_" ,1 ,0 ,0 ,0} /* PUTFIRST */
00233     {"perm_" ,1 ,0 ,0 ,0} /* PERMUTATIONS */
00234     {"partitions_" ,1 ,0 ,0 ,0} /* PARTITIONS */
00235     {"mul_" ,0 ,0 ,0 ,0} /* MULFUNCTION */
00236     {"moebius_" ,0 ,0 ,0 ,0} /* MOEBIUS */
00237     {"topologies_" ,0 ,0 ,0 ,0} /* TOPOLOGIES */
00238     {"diagrams_" ,0 ,0 ,0 ,0} /* DIAGRAMS */
00239     {"topo_" ,0 ,0 ,0 ,0} /* TOPO */
00240     {"node_" ,0 ,0 ,0 ,0} /* VERTEX */
00241     {"edge_" ,0 ,0 ,0 ,0} /* EDGE */
00242     {"sizeof_" ,0 ,0 ,0 ,0} /* SIZEOFFUNCTION */
00243     {"block_" ,0 ,0 ,0 ,0} /* BLOCK */
00244     {"onepi_" ,0 ,0 ,0 ,0} /* ONEPI */
00245     {"phi_" ,0 ,0 ,VERTEXFUNCTION ,0} /* PHI */
00246 #ifdef WITHFLOAT
00247     {"float_" ,0 ,0 ,0 ,0} /* FLOATFUN */
00248     {"tofloat_" ,0 ,0 ,0 ,0} /* TOFLOAT */
00249     {"torat_" ,0 ,0 ,0 ,0} /* TORAT */
00250     {"mzv_" ,0 ,0 ,0 ,0} /* MZV */
00251     {"euler_" ,0 ,0 ,0 ,0} /* EULER */
00252     {"mzvhalf_" ,0 ,0 ,0 ,0} /* MZVHALF */
00253     {"agm_" ,0 ,0 ,0 ,0} /* AGMFUNCTION */
00254     {"gamma_" ,0 ,0 ,0 ,0} /* GAMMAFUN */
00255 #endif
00256 };
00257
00258 FIXEDSET fixedsets[] = {
00259     {"pos_" , "integers > 0" , CSYMBOL , 0} /* POS_ 0 */
00260     {"pos0_" , "integers >= 0" , CSYMBOL , 0} /* POS0_ 1 */
00261     {"neg_" , "integers < 0" , CSYMBOL , 0} /* NEG_ 2 */

```

```

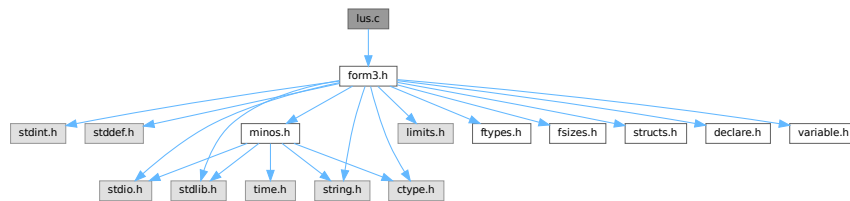
00262     ,{"neg0_", "integers <= 0", CSYMBOL, 0} /* NEG0_ 3 */
00263     ,{"even_", "even integers", CSYMBOL, 0} /* EVEN_ 4 */
00264     ,{"odd_", "odd integers", CSYMBOL, 0} /* ODD_ 5 */
00265     ,{"int_", "all integers", CSYMBOL, 0} /* Z_ 6 */
00266     ,{"symbol_", "only symbols", CSYMBOL, MAXPOSITIVE} /* SYMBOL_ 7 */
00267     ,{"fixed_", "fixed indices", CINDEX, 0} /* FIXED_ 8 */
00268     ,{"index_", "all indices", CINDEX, 0} /* INDEX_ 9 */
00269     ,{"number_", "all rationals", CSYMBOL, 0} /* Q_ 10 */
00270     ,{"dummyindices_", "dummy indices", CINDEX, 0} /* DUMMYINDEX_ 11 */
00271     ,{"vector_", "only vectors", CVECTOR, 0} /* VECTOR_ 12 */
00272 };
00273
00274 UBYTE BufferForOutput[MAXLINELENGTH+14];
00275
00276 char *setupfilename = "form.set";
00277
00278 #ifdef WITHPTHREADS
00279 INILOCK(ErrorMessageLock)
00280 INILOCK(FileReadLock)
00281 #endif

```

## 4.66 lus.c File Reference

#include "form3.h"

Include dependency graph for lus.c:



## Functions

- int [Lus](#) (WORD \*term, WORD funnum, WORD loopsize, WORD numargs, WORD outfun, WORD mode)
- int [FindLus](#) (int from, int level, int openindex)
- int [SortTheList](#) (int \*slist, int num)
- int [AllLoops](#) (PHEAD WORD \*term, WORD level)
- LONG [StartLoops](#) (PHEAD WORD \*term, WORD level, LONG vert, WORD nvert, WORD \*arglist, WORD nargs, WORD \*loop, WORD nloop)
- LONG [GenLoops](#) (PHEAD WORD \*term, WORD level, LONG vert, WORD nvert, WORD \*arglist, WORD nargs, WORD \*loop, WORD nloop)
- void [LoopOutput](#) (PHEAD WORD \*term, WORD level, WORD \*loop, WORD nloop)
- int [AllPaths](#) (PHEAD WORD \*term, WORD level)
- LONG [GenPaths](#) (PHEAD WORD \*term, WORD level, LONG vert, WORD nvert, WORD \*arglist, WORD nargs, WORD \*path, WORD npath)
- void [PathOutput](#) (PHEAD WORD \*term, WORD level, WORD \*path, WORD npath)
- WORD [AllOnePI](#) (WORD \*term, WORD level)
- int [RemoveBridges](#) (void)
- int [TakeOneLine](#) (WORD \*term, WORD level)
- int [OutputOnePI](#) (PHEAD WORD \*term, WORD level)



### 4.66.1 Detailed Description

Routines to find loops in index contractions. These routines allow for a category of topological statements. They were originally developed for the color library.

Definition in file [lus.c](#).

### 4.66.2 Function Documentation

#### 4.66.2.1 Lus()

```
int Lus (
    WORD * term,
    WORD funnum,
    WORD loopsize,
    WORD numargs,
    WORD outfun,
    WORD mode )
```

Definition at line [57](#) of file [lus.c](#).

#### 4.66.2.2 FindLus()

```
int FindLus (
    int from,
    int level,
    int openindex )
```

Definition at line [515](#) of file [lus.c](#).

#### 4.66.2.3 SortTheList()

```
int SortTheList (
    int * slist,
    int num )
```

Definition at line [578](#) of file [lus.c](#).

#### 4.66.2.4 AllLoops()

```
int AllLoops (
    PHEAD WORD * term,
    WORD level )
```

Definition at line [650](#) of file [lus.c](#).

#### 4.66.2.5 StartLoops()

```
LONG StartLoops (
    PHEAD WORD * term,
    WORD level,
    LONG vert,
    WORD nvert,
    WORD * arglist,
    WORD nargs,
    WORD * loop,
    WORD nloop )
```

Definition at line [894](#) of file [lus.c](#).

#### 4.66.2.6 GenLoops()

```
LONG GenLoops (
    PHEAD WORD * term,
    WORD level,
    LONG vert,
    WORD nvert,
    WORD * arglist,
    WORD nargs,
    WORD * loop,
    WORD nloop )
```

Definition at line [980](#) of file [lus.c](#).

#### 4.66.2.7 LoopOutput()

```
void LoopOutput (
    PHEAD WORD * term,
    WORD level,
    WORD * loop,
    WORD nloop )
```

Definition at line [1059](#) of file [lus.c](#).

#### 4.66.2.8 AllPaths()

```
int AllPaths (
    PHEAD WORD * term,
    WORD level )
```

Definition at line [1123](#) of file [lus.c](#).

#### 4.66.2.9 GenPaths()

```
LONG GenPaths (
    PHEAD WORD * term,
    WORD level,
    LONG vert,
    WORD nvert,
    WORD * arglist,
    WORD nargs,
    WORD * path,
    WORD npath )
```

Definition at line 1413 of file [lus.c](#).

#### 4.66.2.10 PathOutput()

```
void PathOutput (
    PHEAD WORD * term,
    WORD level,
    WORD * path,
    WORD npath )
```

Definition at line 1486 of file [lus.c](#).

#### 4.66.2.11 AllOnePI()

```
WORD AllOnePI (
    WORD * term,
    WORD level )
```

Definition at line 1566 of file [lus.c](#).

#### 4.66.2.12 RemoveBridges()

```
int RemoveBridges (
    void )
```

Definition at line 1586 of file [lus.c](#).

#### 4.66.2.13 TakeOneLine()

```
int TakeOneLine (
    WORD * term,
    WORD level )
```

Definition at line 1596 of file [lus.c](#).

#### 4.66.2.14 OutputOnePI()

```
int OutputOnePI (
    PHEAD WORD * term,
    WORD level )
```

Definition at line 1608 of file [lus.c](#).

### 4.67 lus.c

[Go to the documentation of this file.](#)

```
00001
00007 /* #[] License : */
00008 /*
00009  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00010  * When using this file you are requested to refer to the publication
00011  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00012  * This is considered a matter of courtesy as the development was paid
00013  * for by FOM the Dutch physics granting agency and we would like to
00014  * be able to track its scientific use to convince FOM of its value
00015  * for the community.
00016  *
00017  * This file is part of FORM.
00018  *
00019  * FORM is free software: you can redistribute it and/or modify it under the
00020  * terms of the GNU General Public License as published by the Free Software
00021  * Foundation, either version 3 of the License, or (at your option) any later
00022  * version.
00023  *
00024  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00025  * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00026  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00027  * details.
00028  *
00029  * You should have received a copy of the GNU General Public License along
00030  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00031  */
00032 /* #[] License : */
00033 /*
00034  * #[] Includes : lus.c
00035  */
00036
00037 #include "form3.h"
00038
00039 /*
00040  * #[] Includes :
00041  * #[] Lus :
00042  *
00043  * Routine to find loops.
00044  * Mode: 0: Just tell whether there is such a loop.
00045  * 1: Take out the functions and replace by outfun with the
00046  * remaining arguments of the loop function
00047  * > AM.OffsetIndex: This index must be included in the loop.
00048  * < -AM.OffsetIndex: This index must be included in the loop. Replace.
00049  * Return value: 0: no loop. 1: there is/was such a loop.
00050  * funnum: the function(s) in which we look for a loop.
00051  * numargs: the number of arguments admissible in the function.
00052  * outfun: the output function in case of a substitution.
00053  * loopsize: the size of the loop we are looking for.
00054  * if < 0 we look for all loops.
00055  */
00056
00057 int Lus(WORD *term, WORD funnum, WORD loopsize, WORD numargs, WORD outfun, WORD mode)
00058 {
00059     GETIDENTITY
00060     WORD *w, *t, *tt, *m, *r, **loc, *tstop, minloopsize;
00061     int nfun, i, j, jj, k, n, sign = 0, action = 0, L, ten, ten2, totnum,
00062     sign2, *alist, *wi, mini, maxi, medi = 0;
00063     if ( numargs <= 1 ) return(0);
00064     GETSTOP(term,tstop);
00065     /*
00066     * First count the number of functions with the proper number of arguments.
00067     */
00068     t = term+1; nfun = 0;
00069     if ( ( ten = functions[funnum-FUNCTION].spec ) >= TENSORFUNCTION ) {
00070         while ( t < tstop ) {
00071             if ( *t == funnum && t[1] == FUNHEAD+numargs ) { nfun++; }
```

```

00072         t += t[1];
00073     }
00074 }
00075 else {
00076     while ( t < tstop ) {
00077         if ( *t == funnum ) {
00078             i = 0; m = t+FUNHEAD; t += t[1];
00079             while ( m < t ) { i++; NEXTARG(m) }
00080             if ( i == numargs ) nfun++;
00081         }
00082         else t += t[1];
00083     }
00084 }
00085 if ( loopsize < 0 ) minloopsize = 2;
00086 else minloopsize = loopsize;
00087 if ( funnum < minloopsize ) return(0); /* quick abort */
00088 if ( ((functions[funnum-FUNCTION].symmetric) & ~REVERSEORDER) == ANTISYMMETRIC ) sign = 1;
00089 if ( mode == 1 || mode < 0 ) {
00090     ten2 = functions[outfun-FUNCTION].spec >= TENSORFUNCTION;
00091 }
00092 else ten2 = -1;
00093 /*
00094 Allocations:
00095 */
00096 if ( AN.numflocs < funnum ) {
00097     if ( AN.funlocs ) M_free(AN.funlocs,"Lus: AN.funlocs");
00098     AN.numflocs = funnum;
00099     AN.funlocs = (WORD **)Malloc1(sizeof(WORD *)*AN.numflocs,"Lus: AN.funlocs");
00100 }
00101 if ( AN.numfargs < funnum*numargs ) {
00102     if ( AN.funargs ) M_free(AN.funargs,"Lus: AN.funargs");
00103     AN.numfargs = funnum*numargs;
00104     AN.funargs = (int *)Malloc1(sizeof(int *)*AN.numfargs,"Lus: AN.funargs");
00105 }
00106 /*
00107 Make a list of relevant indices
00108 */
00109 alist = AN.funargs; loc = AN.funlocs;
00110 t = term+1;
00111 if ( ten >= TENSORFUNCTION ) {
00112     while ( t < tstop ) {
00113         if ( *t == funnum && t[1] == FUNHEAD+numargs ) {
00114             *loc++ = t;
00115             t += FUNHEAD;
00116             j = i = numargs; while ( --i >= 0 ) {
00117                 if ( *t >= AM.OffsetIndex &&
00118                     ( *t >= AM.OffsetIndex+WILDOFFSET ||
00119                     indices[*t-AM.OffsetIndex].dimension != 0 ) ) {
00120                     *alist++ = *t++; j--;
00121                 }
00122                 else t++;
00123             }
00124             while ( --j >= 0 ) *alist++ = -1;
00125         }
00126         else t += t[1];
00127     }
00128 }
00129 else {
00130     nfun = 0;
00131     while ( t < tstop ) {
00132         if ( *t == funnum ) {
00133             w = t;
00134             i = 0; m = t+FUNHEAD; t += t[1];
00135             while ( m < t ) { i++; NEXTARG(m) }
00136             if ( i == numargs ) {
00137                 m = w + FUNHEAD;
00138                 while ( m < t ) {
00139                     if ( *m == -INDEX && m[1] >= AM.OffsetIndex &&
00140                         ( m[1] >= AM.OffsetIndex+WILDOFFSET ||
00141                         indices[m[1]-AM.OffsetIndex].dimension != 0 ) ) {
00142                         *alist++ = m[1]; m += 2; i--;
00143                     }
00144                     else if ( ten2 >= TENSORFUNCTION && *m != -INDEX
00145                             && *m != -VECTOR && *m != -MINVECTOR &&
00146                             ( *m != -SNUMBER || *m < 0 || *m >= AM.OffsetIndex ) ) {
00147                         i = numargs; break;
00148                     }
00149                     else { NEXTARG(m) }
00150                 }
00151                 if ( i < numargs ) {
00152                     *loc++ = w;
00153                     nfun++;
00154                     while ( --i >= 0 ) *alist++ = -1;
00155                 }
00156             }
00157         }
00158         else t += t[1];

```

```

00159     }
00160     if ( nfun < minloopsize ) return(0);
00161 }
00162 /*
00163 We have now nfun objects. Not all indices may be usable though.
00164 If the list is not long, we use a quadratic algorithm to remove
00165 indices and vertices that cannot be used. If it becomes large we
00166 sort the list of available indices (and their multiplicity) and
00167 work with binary searches.
00168 */
00169 alist = AN.funargs; totnum = numargs*nfun;
00170 if ( nfun > 7 ) {
00171     if ( AN.funisize < totnum ) {
00172         if ( AN.funinds ) M_free(AN.funinds,"AN.funinds");
00173         AN.funisize = (totnum*3)/2;
00174         AN.funinds = (int *)Malloc1(AN.funisize*2*sizeof(int),"AN.funinds");
00175     }
00176     i = totnum; n = 0; wi = AN.funinds;
00177     while ( --i >= 0 ) {
00178         if ( *alist >= 0 ) { n++; *wi++ = *alist; *wi++ = 1; }
00179         alist++;
00180     }
00181     n = SortTheList(AN.funinds,n);
00182     do {
00183         action = 0;
00184         for ( i = 0; i < nfun; i++ ) {
00185             alist = AN.funargs + i*numargs;
00186             jj = numargs;
00187             for ( j = 0; j < jj; j++ ) {
00188                 if ( alist[j] < 0 ) break;
00189                 mini = 0; maxi = n-1;
00190                 while ( mini <= maxi ) {
00191                     medi = (mini + maxi) / 2; k = AN.funinds[2*medi];
00192                     if ( alist[j] > k ) mini = medi + 1;
00193                     else if ( alist[j] < k ) maxi = medi - 1;
00194                     else break;
00195                 }
00196                 if ( AN.funinds[2*medi+1] <= 1 ) {
00197                     (AN.funinds[2*medi+1])--;
00198                     jj--; k = j; while ( k < jj ) { alist[k] = alist[k+1]; k++; }
00199                     alist[jj] = -1; j--;
00200                 }
00201             }
00202             if ( jj < 2 ) {
00203                 if ( jj == 1 ) {
00204                     mini = 0; maxi = n-1;
00205                     while ( mini <= maxi ) {
00206                         medi = (mini + maxi) / 2; k = AN.funinds[2*medi];
00207                         if ( alist[0] > k ) mini = medi + 1;
00208                         else if ( alist[0] < k ) maxi = medi - 1;
00209                         else break;
00210                     }
00211                     (AN.funinds[2*medi+1])--;
00212                     if ( AN.funinds[2*medi+1] == 1 ) action++;
00213                 }
00214                 nfun--; totnum -= numargs; AN.funlocs[i] = AN.funlocs[nfun];
00215                 wi = AN.funargs + nfun*numargs;
00216                 for ( j = 0; j < numargs; j++ ) alist[j] = *wi++;
00217                 i--;
00218             }
00219         }
00220     } while ( action );
00221 }
00222 else {
00223     for ( i = 0; i < totnum; i++ ) {
00224         if ( alist[i] == -1 ) continue;
00225         for ( j = 0; j < totnum; j++ ) {
00226             if ( alist[j] == alist[i] && j != i ) break;
00227         }
00228         if ( j >= totnum ) alist[i] = -1;
00229     }
00230     do {
00231         action = 0;
00232         for ( i = 0; i < nfun; i++ ) {
00233             alist = AN.funargs + i*numargs;
00234             n = numargs;
00235             for ( k = 0; k < n; k++ ) {
00236                 if ( alist[k] < 0 ) { alist[k--] = alist[--n]; alist[n] = -1; }
00237             }
00238             if ( n <= 1 ) {
00239                 if ( n == 1 ) { j = alist[0]; }
00240                 else j = -1;
00241                 nfun--; totnum -= numargs; AN.funlocs[i] = AN.funlocs[nfun];
00242                 wi = AN.funargs + nfun * numargs;
00243                 for ( k = 0; k < numargs; k++ ) alist[k] = wi[k];
00244                 i--;
00245                 if ( j >= 0 ) {

```

```

00246         for ( k = 0, jj = 0, wi = AN.funargs; k < totnum; k++, wi++ ) {
00247             if ( *wi == j ) { jj++; if ( jj > 1 ) break; }
00248         }
00249         if ( jj <= 1 ) {
00250             for ( k = 0, wi = AN.funargs; k < totnum; k++, wi++ ) {
00251                 if ( *wi == j ) { *wi = -1; action = 1; }
00252             }
00253         }
00254     }
00255 }
00256 }
00257 } while ( action );
00258 }
00259 if ( nfun < minloopsize ) return(0);
00260 /*
00261 Now we have nfun objects, each with at least 2 indices, each of which
00262 occurs at least twice in our list. There will be a loop!
00263 */
00264 if ( mode != 0 && mode != 1 ) {
00265     if ( mode > 0 ) AN.tohunt = mode - 5;
00266     else AN.tohunt = -mode - 5;
00267     AN.nargs = numargs; AN.numoffuns = nfun;
00268     i = 0;
00269     if ( loopsize < 0 ) {
00270         if ( loopsize == -1 ) k = nfun;
00271         else { k = -loopsize-1; if ( k > nfun ) k = nfun; }
00272         for ( L = 2; L <= k; L++ ) {
00273             if ( FindLus(0,L,AN.tohunt) ) goto Success;
00274         }
00275     }
00276     else if ( FindLus(0,loopsize,AN.tohunt) ) { L = loopsize; goto Success; }
00277 }
00278 else {
00279     AN.nargs = numargs; AN.numoffuns = nfun;
00280     if ( loopsize < 0 ) {
00281         jj = 2; k = nfun;
00282         if ( loopsize < -1 ) { k = -loopsize-1; if ( k > nfun ) k = nfun; }
00283     }
00284     else { jj = k = loopsize; }
00285     for ( L = jj; L <= k; L++ ) {
00286         for ( i = 0; i <= nfun-L; i++ ) {
00287             alist = AN.funargs + i * numargs;
00288             for ( jj = 0; jj < numargs; jj++ ) {
00289                 if ( alist[jj] < 0 ) continue;
00290                 AN.tohunt = alist[jj];
00291                 for ( j = jj+1; j < numargs; j++ ) {
00292                     if ( alist[j] < 0 ) continue;
00293                     if ( FindLus(i+1,L-1,alist[j]) ) {
00294                         alist[0] = alist[jj];
00295                         alist[1] = alist[j];
00296                         goto Success;
00297                     }
00298                 }
00299             }
00300         }
00301     }
00302 }
00303 return(0);
00304 Success:;
00305 if ( mode == 0 || mode > 1 ) return(1);
00306 /*
00307 Now we have to make the replacement and fix the potential sign
00308 */
00309 sign2 = 1;
00310 wi = AN.funargs + i*numargs; loc = AN.funlocs + i;
00311 for ( k = 0; k < L; k++ ) *(loc[k]) = -1;
00312 if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
00313 w = AT.WorkPointer + 1;
00314 m = t = term + 1;
00315 while ( t < tstop ) {
00316     if ( *t == -1 ) break;
00317     t += t[1];
00318 }
00319 while ( m < t ) *w++ = *m++;
00320 r = w;
00321 *w++ = outfun;
00322 w++;
00323 *w++ = DIRTYFLAG;
00324 FILLFUN3(w)
00325 if ( functions[outfun-FUNCTION].spec >= TENSORFUNCTION ) {
00326     if ( ten >= TENSORFUNCTION ) {
00327         for ( i = 0; i < L; i++ ) {
00328             alist = wi + i*numargs;
00329             m = loc[i] + FUNHEAD;
00330             for ( k = 0; k < numargs; k++ ) {
00331                 if ( m[k] == alist[0] ) {
00332                     if ( k != 0 ) {

```

```

00333             jj = m[k]; m[k] = m[0]; m[0] = jj;
00334             sign = -sign;
00335         }
00336         break;
00337     }
00338 }
00339 for ( k = 1; k < numargs; k++ ) {
00340     if ( m[k] == alist[1] ) {
00341         if ( k != 1 ) {
00342             jj = m[k]; m[k] = m[1]; m[1] = jj;
00343             sign = -sign;
00344         }
00345         break;
00346     }
00347 }
00348 m += 2;
00349 for ( k = 2; k < numargs; k++ ) *w++ = *m++;
00350 }
00351 }
00352 else {
00353     WORD *t1, *t2, *t3;
00354     for ( i = 0; i < L; i++ ) {
00355         alist = wi + i*numargs;
00356         tt = loc[i];
00357         m = tt + FUNHEAD;
00358         for ( k = 0; k < numargs; k++ ) {
00359             if ( *m == -INDEX && m[1] == alist[0] ) {
00360                 if ( k != 0 ) {
00361                     if ( ( k & 1 ) != 0 ) sign = -sign;
00362                     /*
00363                     now move to position 0
00364                     */
00365                     t2 = m+2; t1 = m; t3 = tt+FUNHEAD;
00366                     while ( t1 > t3 ) { *--t2 = *--t1; }
00367                     t3[0] = -INDEX; t3[1] = alist[0];
00368                 }
00369                 break;
00370             }
00371             NEXTARG(m)
00372         }
00373         m = tt + FUNHEAD + 2;
00374         for ( k = 1; k < numargs; k++ ) {
00375             if ( *m == -INDEX && m[1] == alist[1] ) {
00376                 if ( k != 1 ) {
00377                     if ( ( k & 1 ) == 0 ) sign = -sign;
00378                     /*
00379                     now move to position 1
00380                     */
00381                     t2 = m+2; t1 = m; t3 = tt+FUNHEAD+2;
00382                     while ( t1 > t3 ) { *--t2 = *--t1; }
00383                     t3[0] = -INDEX; t3[1] = alist[1];
00384                 }
00385                 break;
00386             }
00387             NEXTARG(m)
00388         }
00389         /*
00390         now copy the remaining arguments to w
00391         keep in mind that the output function is a tensor!
00392         */
00393         t1 = tt + FUNHEAD + 4;
00394         t2 = tt + tt[1];
00395         while ( t1 < t2 ) {
00396             if ( *t1 == -INDEX || *t1 == -VECTOR ) {
00397                 *w++ = t1[1]; t1 += 2;
00398             }
00399             else if ( *t1 == -MINVECTOR ) {
00400                 *w++ = t1[1]; t1 += 2; sign2 = -sign2;
00401             }
00402             else if ( ( *t1 == -SNUMBER ) && ( t1[1] >= 0 ) && ( t1[1] < AM.OffsetIndex ) ) {
00403                 *w++ = t1[1]; t1 += 2; sign2 = -sign2;
00404             }
00405             else {
00406                 MLOCK(ErrorMessageLock);
00407                 MesPrint("Illegal attempt to use a non-index-like argument in a tensor in
ReplaceLoop statement");
00408                 MUNLOCK(ErrorMessageLock);
00409                 Terminate(-1);
00410             }
00411         }
00412     }
00413 }
00414 }
00415 else {
00416     if ( ten >= TENSORFUNCTION ) {
00417         for ( i = 0; i < L; i++ ) {
00418             alist = wi + i*numargs;

```



```

00419         m = loc[i] + FUNHEAD;
00420         for ( k = 0; k < numargs; k++ ) {
00421             if ( m[k] == alist[0] ) {
00422                 if ( k != 0 ) {
00423                     jj = m[k]; m[k] = m[0]; m[0] = jj;
00424                     sign = -sign;
00425                     break;
00426                 }
00427             }
00428         }
00429         for ( k = 1; k < numargs; k++ ) {
00430             if ( m[k] == alist[1] ) {
00431                 if ( k != 1 ) {
00432                     jj = m[k]; m[k] = m[1]; m[1] = jj;
00433                     sign = -sign;
00434                     break;
00435                 }
00436             }
00437         }
00438         m += 2;
00439         for ( k = 2; k < numargs; k++ ) {
00440             if ( *m >= AM.OffsetIndex ) { *w++ = -INDEX; }
00441             else if ( *m < 0 ) { *w++ = -VECTOR; }
00442             else { *w = -SNUMBER; }
00443             *w++ = *m++;
00444         }
00445     }
00446 }
00447 else {
00448     WORD *t1, *t2, *t3;
00449     for ( i = 0; i < L; i++ ) {
00450         alist = w1 + i*numargs;
00451         tt = loc[i];
00452         m = tt + FUNHEAD;
00453         for ( k = 0; k < numargs; k++ ) {
00454             if ( *m == -INDEX && m[1] == alist[0] ) {
00455                 if ( k != 0 ) {
00456                     if ( ( k & 1 ) != 0 ) sign = -sign;
00457                     /*
00458                        now move to position 0
00459                     */
00460                     t2 = m+2; t1 = m; t3 = tt+FUNHEAD;
00461                     while ( t1 > t3 ) { *--t2 = *--t1; }
00462                     t3[0] = -INDEX; t3[1] = alist[0];
00463                 }
00464                 break;
00465             }
00466             NEXTARG(m)
00467         }
00468         m = tt + FUNHEAD + 2;
00469         for ( k = 1; k < numargs; k++ ) {
00470             if ( *m == -INDEX && m[1] == alist[1] ) {
00471                 if ( k != 1 ) {
00472                     if ( ( k & 1 ) == 0 ) sign = -sign;
00473                     /*
00474                        now move to position 1
00475                     */
00476                     t2 = m+2; t1 = m; t3 = tt+FUNHEAD+2;
00477                     while ( t1 > t3 ) { *--t2 = *--t1; }
00478                     t3[0] = -INDEX; t3[1] = alist[1];
00479                 }
00480                 break;
00481             }
00482             NEXTARG(m)
00483         }
00484         /*
00485            now copy the remaining arguments to w
00486         */
00487         t1 = tt + FUNHEAD + 4;
00488         t2 = tt + tt[1];
00489         while ( t1 < t2 ) *w++ = *t1++;
00490     }
00491 }
00492 }
00493 r[1] = w-r;
00494 while ( t < tstop ) {
00495     if ( *t == -1 ) { t += t[1]; continue; }
00496     i = t[1];
00497     NCOPY(w,t,i)
00498 }
00499 tstop = term + *term;
00500 while ( t < tstop ) *w++ = *t++;
00501 if ( sign < 0 ) w[-1] = -w[-1];
00502 i = w - AT.WorkPointer;
00503 *AT.WorkPointer = i;
00504 t = term; w = AT.WorkPointer;
00505 NCOPY(t,w,i)

```

```

00506     *AN.RepPoint = 1;    /* For Repeat */
00507     return(1);
00508 }
00509
00510 /*
00511 #] Lus :
00512 #[ FindLus :
00513 */
00514
00515 int FindLus(int from, int level, int openindex)
00516 {
00517     GETIDENTITY
00518     int i, j, k, jj, *alist, *blist, *w, *m, partner;
00519     WORD **loc = AN.funlocs, *wor;
00520     if ( level == 1 ) {
00521         for ( i = from; i < AN.numoffuns; i++ ) {
00522             alist = AN.funargs + i*AN.nargs;
00523             for ( j = 0; j < AN.nargs; j++ ) {
00524                 if ( alist[j] == openindex ) {
00525                     for ( k = 0; k < AN.nargs; k++ ) {
00526                         if ( k == j ) continue;
00527                         if ( alist[k] == AN.tohunt ) {
00528                             loc[from] = loc[i];
00529                             alist = AN.funargs + from*AN.nargs;
00530                             alist[0] = openindex; alist[1] = AN.tohunt;
00531                             return(1);
00532                         }
00533                     }
00534                 }
00535             }
00536         }
00537     }
00538     else {
00539         for ( i = from; i < AN.numoffuns; i++ ) {
00540             alist = AN.funargs + i*AN.nargs;
00541             for ( j = 0; j < AN.nargs; j++ ) {
00542                 if ( alist[j] == openindex ) {
00543                     if ( from != i ) {
00544                         wor = loc[i]; loc[i] = loc[from]; loc[from] = wor;
00545                         blist = w = AN.funargs + from*AN.nargs;
00546                         m = alist;
00547                         k = AN.nargs;
00548                         while ( --k >= 0 ) { jj = *m; *m++ = *w; *w++ = jj; }
00549                     }
00550                     else blist = alist;
00551                     for ( k = 0; k < AN.nargs; k++ ) {
00552                         if ( k == j || blist[k] < 0 ) continue;
00553                         partner = blist[k];
00554                         if ( FindLus(from+1, level-1, partner) ) {
00555                             blist[0] = openindex; blist[1] = partner;
00556                             return(1);
00557                         }
00558                     }
00559                     if ( from != i ) {
00560                         wor = loc[i]; loc[i] = loc[from]; loc[from] = wor;
00561                         w = AN.funargs + from*AN.nargs;
00562                         m = alist;
00563                         k = AN.nargs;
00564                         while ( --k >= 0 ) { jj = *m; *m++ = *w; *w++ = jj; }
00565                     }
00566                 }
00567             }
00568         }
00569     }
00570     return(0);
00571 }
00572
00573 /*
00574 #] FindLus :
00575 #[ SortTheList :
00576 */
00577
00578 int SortTheList(int *slist, int num)
00579 {
00580     GETIDENTITY
00581     int i, nleft, nright, *t1, *t2, *t3, *rlist;
00582     if ( num <= 2 ) {
00583         if ( num <= 1 ) return(num);
00584         if ( slist[0] < slist[2] ) return(2);
00585         if ( slist[0] > slist[2] ) {
00586             i = slist[0]; slist[0] = slist[2]; slist[2] = i;
00587             i = slist[1]; slist[1] = slist[3]; slist[3] = i;
00588             return(2);
00589         }
00590         slist[1] += slist[3];
00591         return(1);
00592     }

```

```

00593     else {
00594         nleft = num/2; rlist = slist + 2*nleft;
00595         nright = SortTheList(rlist,num-nleft);
00596         nleft = SortTheList(slist,nleft);
00597         if ( AN.tlistsize < nleft ) {
00598             if ( AN.tlistbuf ) M_free(AN.tlistbuf,"AN.tlistbuf");
00599             AN.tlistsize = (nleft*3)/2;
00600             AN.tlistbuf = (int *)Malloc1(AN.tlistsize*2*sizeof(int),"AN.tlistbuf");
00601         }
00602         i = nleft; t1 = slist; t2 = AN.tlistbuf;
00603         while ( --i >= 0 ) { *t2++ = *t1++; *t2++ = *t1++; }
00604         i = nleft+nright; t1 = AN.tlistbuf; t2 = rlist; t3 = slist;
00605         while ( nleft > 0 && nright > 0 ) {
00606             if ( *t1 < *t2 ) {
00607                 *t3++ = *t1++; *t3++ = *t1++; nleft--;
00608             }
00609             else if ( *t1 > *t2 ) {
00610                 *t3++ = *t2++; *t3++ = *t2++; nright--;
00611             }
00612             else {
00613                 *t3++ = *t1++; t2++; *t3++ = (*t1++) + (*t2++); i--;
00614                 nleft--; nright--;
00615             }
00616         }
00617         while ( --nleft >= 0 ) { *t3++ = *t1++; *t3++ = *t1++; }
00618         while ( --nright >= 0 ) { *t3++ = *t2++; *t3++ = *t2++; }
00619         return(i);
00620     }
00621 }
00622
00623 /*
00624 #] SortTheList :
00625 #[ AllLoops :
00626
00627 Routine finds all possible loops that can be made in the
00628 arguments of the symmetric commuting vertex function v and creates
00629 for each loop a new term which is the original term times the loop.
00630 The occurrences of v can have different numbers of arguments.
00631 Each argument that occurs twice (and in different instances of v)
00632 in total will be considered.
00633
00634 The input parameters are in C->lhs[level] and are
00635 C->lhs[level][0] TYPEALLLOOPS
00636 C->lhs[level][1] 6
00637 C->lhs[level][2] Number of function v.
00638 C->lhs[level][3] Number of the output function loop.
00639 C->lhs[level][4] option1: the type of argument.
00640 C->lhs[level][5] option2: 0,1: what to do if no loop.
00641 Eligible arguments must be of the type SYMBOL, VECTOR, INDEX or SNUMBER.
00642 They must all be of the same type, indicated by option1.
00643 Extra restriction: In a loop, each v can be visited at most once.
00644 If there is no loop at all, option2 determines whether the term
00645 remains unmodified, or is replaced by zero.
00646 Function v can be either a regular function or a tensor.
00647 To facilitate this we copy the relevant arguments into the workspace.
00648 */
00649
00650 int AllLoops(PHEAD WORD *term,WORD level)
00651 {
00652     CBUF *C = cbuf+AM.rbufnum;
00653     WORD vcode = C->lhs[level][2]; /* The input function */
00654     WORD option1 = C->lhs[level][4]; /* type of argument */
00655     WORD option2 = C->lhs[level][5]; /* what to do when no loop */
00656     WORD *tstop, *t, *tend, *tstart, *tfrom;
00657     WORD i, j, jj;
00658     WORD *arglist, nargs, *loop, nloop;
00659     WORD *oldworkpointer = AT.WorkPointer;
00660     LONG oldpworkpointer = AT.pWorkPointer;
00661     LONG numgenerated = 0, vert;
00662     WORD *a, *a1, *a2, *a3, *v, *vv, nvert, *to, *from, *tos, action;
00663     /*
00664     Search for the first occurrence of vcode.
00665     */
00666     tstop = term+*term; tstop -= ABS(tstop[-1]);
00667     t = term + 1;
00668     while ( t < tstop && *t != vcode ) t += t[1];
00669     if ( t == tstop ) {
00670         if ( option2 == 0 ) return(0);
00671         else return(Generator(BHEAD term,level));
00672     }
00673     tstart = t;
00674     nvert = 0;
00675     do {
00676         t += t[1]; nvert++;
00677     } while ( t < tstop && *t == vcode );
00678     tend = t;
00679     /*

```

```

00680      Make room for 2*nvert pointers
00681  */
00682      WantAddPointers(2*nvert);
00683      vert = AT.pWorkPointer;
00684      AT.pWorkPointer += 2*nvert;
00685      nvert = 0;
00686  /*
00687      Next we copy these functions into the workspace, but only the arguments
00688      that agree with option1. Because of the difference between tensors and
00689      regular functions in the copy we strip the type of the argument.
00690  */
00691      to = AT.WorkPointer;
00692      from = tstart;
00693      if ( functions[vcode-FUNCTION].spec == TENSORFUNCTION ) {
00694          from = tstart;
00695          while ( from < tend ) {
00696              tos = to++;
00697              tfrom = from+from[1];
00698              from += FUNHEAD;
00699              while ( from < tfrom ) {
00700                  if ( option1 == -INDEX && *from >= 0 ) {
00701                      *to++ = *from++;
00702                  }
00703                  else if ( option1 == -VECTOR && *from < 0 ) {
00704                      *to++ = *from++ - AM.OffsetVector;
00705                  }
00706                  else {
00707                      from++;
00708                  }
00709              }
00710              *tos = to-from;
00711              if ( *tos < 3 ) to = tos;
00712              else AT.pWorkSpace[vert+nvert++] = tos;
00713          }
00714      }
00715      else if ( functions[vcode-FUNCTION].spec == 0 ) {
00716          from = tstart;
00717          while ( from < tend ) {
00718              tos = to++;
00719              tfrom = from + from[1];
00720              from += FUNHEAD;
00721              while ( from < tfrom ) {
00722                  if ( option1 == -VECTOR
00723                      && ( from[0] == -INDEX || from[0] == -VECTOR )
00724                      && from[1] < 0 ) {
00725                      from++;
00726                      *to++ = *from++ - AM.OffsetVector;
00727                  }
00728                  else if ( option1 == *from ) {
00729                      from++; *to++ = *from++;
00730                  }
00731                  else {
00732                      NEXTARG(from);
00733                  }
00734              }
00735              *tos = to-tos;
00736              if ( *tos < 3 ) to = tos;
00737              else AT.pWorkSpace[vert+nvert++] = tos;
00738          }
00739      }
00740      else {
00741          AT.WorkPointer = oldworkpointer;
00742          AT.pWorkPointer = oldpworkpointer;
00743          if ( option2 == 0 ) return(0);
00744          return(Generator(BHEAD term,level));
00745      }
00746      AT.WorkPointer = to;
00747  /*
00748      Now make a list of all relevant arguments.
00749  */
00750      a = arglist = AT.WorkPointer;
00751      for ( i = 0; i < nvert; i++ ) {
00752          v = AT.pWorkSpace[vert+i];
00753          j = *v-1; v++;
00754          NCOPY(a,v,j);
00755      }
00756      nargs = a-arglist;
00757      AT.WorkPointer = a;
00758  /*
00759      Now sort the list. Bubble type sort.
00760  */
00761      a1 = arglist; a2 = a1+1;
00762      while ( a2 < a ) {
00763          a3 = a2;
00764          while ( a3 > a1 && a3[-1] > a3[0] ) {
00765              EXCH(*a3,a3[-1]);
00766              a3--;

```

```

00767     }
00768     a2++;
00769 }
00770 /*
00771 Now remove the elements from the list that do not occur exactly twice.
00772 */
00773 a1 = arglist; a2 = a1; a3 = a1+nargs;
00774 while ( a2 < a3 ) {
00775     if ( a2+1 == a3 ) { break; }
00776     else if ( a2+2 == a3 ) {
00777         if ( a2[0] == a2[1] ) { *a1++ = a2[0]; }
00778         break;
00779     }
00780     else {
00781         if ( a2[0] != a2[1] ) { a2++; }
00782         else if ( a2[0] != a2[2] ) {
00783             *a1++ = a2[0]; a2 += 2;
00784         }
00785         else {
00786             a2++; while ( a2 < a3 && a2[-1] == a2[0] ) a2++;
00787         }
00788     }
00789 }
00790 nargs = a1-arglist;
00791 /*
00792 Now we need to redo the list of vertices and remove the elements
00793 that are not in our arglist.
00794 */
00795 do {
00796     action = 0;
00797     for ( i = 0; i < nvert; i++ ) {
00798         vv = v = AT.pWorkspace[vert+i];
00799         v++;
00800         for ( j = 1; j < vv[0]; j++ ) {
00801             for ( jj = 0; jj < nargs; jj++ ) {
00802                 if ( *v == arglist[jj] ) break;
00803             }
00804             if ( jj >= nargs ) { /* was not in the list */
00805                 vv[0] = vv[0]-1;
00806                 for ( jj = j; jj < vv[0]; jj++ ) vv[jj] = vv[jj+1];
00807             }
00808             v++;
00809         }
00810     }
00811 } /*
00812 Next we need to remove vertices that have only one object remaining.
00813 Also, the object can be removed from arglist. After that we go back
00814 to clean up the list.
00815 */
00816 for ( i = 0; i < nvert; i++ ) {
00817     vv = AT.pWorkspace[vert+i];
00818     if ( vv[0] == 1 ) {
00819         AT.pWorkspace[vert+i] = AT.pWorkspace[vert+nvert-1];
00820         nvert--; i--; continue;
00821     }
00822     else if ( vv[0] == 2 ) {
00823         for ( j = 0; j < nargs; j++ ) {
00824             if ( arglist[j] == vv[1] ) break;
00825         }
00826         while ( j < nargs-1 ) arglist[j] = arglist[j+1];
00827         nargs--;
00828         AT.pWorkspace[vert+i] = AT.pWorkspace[vert+nvert-1];
00829         nvert--; i--;
00830         action = 1;
00831     }
00832 }
00833 } while ( action );
00834 /*
00835 Because the user can remove tadpoles rather easily as in
00836 id v(?a,i?,?b,i?,?c) = v(?a,?b,?c)
00837 there is no need to avoid them as potential loops.
00838
00839 At this point we are ready to look for loops.
00840 We have
00841     arglist,nargs    list of eligible objects.
00842     vert,nvert       position in pWorkspace for vertices and their number.
00843                     The vertices themselves are in the Workspace.
00844     loop,nloop       The buildup of the loop.
00845 */
00846 loop = AT.WorkPointer;
00847 AT.WorkPointer += nargs;
00848 nloop = 0;
00849 numgenerated += StartLoops(BHEAD term,level,vert,nvert,arglist,nargs,loop,nloop);
00850 AT.WorkPointer = oldworkpointer;
00851 AT.pWorkPointer = oldpworkpointer;
00852
00853 if ( numgenerated == 0 && option2 != 0 ) return(Generator(BHEAD term,level));

```

```

00854     return(0);
00855 }
00856
00857 /*
00858 #] AllLoops :
00859 #[ StartLoops :
00860
00861 Algorithm.
00862 1: have a list of vertices and objects that can be part of the loops.
00863 2: starting with the first element of arglist, look for the two
00864    vertices that contain this object. The first is vstart.
00865 3: At the second, look at all possible objects to continue from here.
00866 4: For each, find the vertex with its partner.
00867 5: if the partner vertex is vstart, we have a loop. --> 9.
00868 6: The partner vertex is not allowed to be a vertex that we passed
00869    in the current build-up of the loop.
00870 7: Put the object in loop and increase nloop.
00871 8: Now sum over all allowed objects in the partner vertex. --> 4.
00872
00873 9: Create the function loop with the arguments from loop,nloop.
00874 10: If a reverse cyclic permutation gives a smaller result, drop this
00875     solution and continue with the last summing over objects at a vertex.
00876 11: If not, output the result by adding the function loop
00877     to the term and call Generator(term,level)
00878
00879 12: when all possibilities with the first element of arglist have been
00880     exhausted, raise arglist and lower nargs by one. --> 2
00881 13: when nargs == 1, we can stop.
00882
00883 The inclusion of a new object when we sum over the possibilities at a
00884 vertex can best be done in a recursion, because we have no idea how
00885 many nested loops we would otherwise need. Of course the recursion can
00886 be emulated, but that makes the algorithm rather messy.
00887
00888 We have two routines: StartLoops and GenLoops.
00889 StartLoops takes care of the first argument (and the summing over who
00890 is the first argument), while GenLoops treats an additional vertex until
00891 a loop is closed. GenLoops also sends completed loops off to Generator.
00892 */
00893
00894 LONG StartLoops(PHEAD WORD *term,WORD level,WORD vert,WORD nvert,
00895                WORD *arglist,WORD nargs,WORD *loop,WORD nloop)
00896 {
00897     LONG numgenerated = 0;
00898     WORD *v, *vv, *vstart, istart, *vpartner, ipartner, j;
00899     while ( nargs > 1 ) {
00900 /*
00901     Look for a vertex with arglist[0]. This is our starting vertex.
00902     Next we look for its partner and we put arglist[0] in loop.
00903 */
00904         nloop = 0;
00905         loop[nloop++] = arglist[0];
00906         for ( istart = 0; istart < nvert; istart++ ) {
00907             vstart = AT.pWorkSpace[vert+istart];
00908             v = vstart+1; vv = vstart + *vstart;
00909             do {
00910                 if ( *v == arglist[0] ) goto havestart;
00911                 v++;
00912             } while ( v < vv );
00913         }
00914 /*
00915         If we come here, we have a problem.
00916 */
00917         MesPrint("Internal error in StartLoops. Object not found.");
00918         Terminate(-1);
00919         return(-1);
00920 havestart:
00921         AT.pWorkSpace[vert+nvert] = vstart;
00922 /*
00923         Check for tadpole.
00924 */
00925         v++;
00926         while ( v < vv ) {
00927             if ( *v == arglist[0] ) { /* tadpole */
00928                 LoopOutput(BHEAD term,level,loop,nloop);
00929                 numgenerated++;
00930                 goto nextarg;
00931             }
00932             v++;
00933         }
00934 /*
00935         Now the partner vertex.
00936 */
00937         for ( ipartner = istart+1; ipartner < nvert; ipartner++ ) {
00938             vpartner = AT.pWorkSpace[vert+ipartner];
00939             vv = vpartner+*vpartner; v = vpartner+1;
00940             do {

```

```

00941         if ( *v == arglist[0] ) goto havepartner;
00942         v++;
00943     } while ( v < vv );
00944 }
00945 return(numgenerated);
00946 havepartner:
00947     AT.pWorkSpace[vert+nvert+1] = vpartner;
00948 /*
00949     Now we run through all other possibilities at vpartner.
00950     vv is still OK.
00951 */
00952     v = vpartner+1;
00953     while ( v < vv ) {
00954         if ( *v != arglist[0] ) {
00955             for ( j = 1; j < nargs; j++ ) {
00956                 if ( *v == arglist[j] ) {
00957                     loop[nloop++] = *v;
00958                     numgenerated += GenLoops(BHEAD term,level,vert,nvert,
00959                                             arglist,nargs,loop,nloop);
00960                     nloop--;
00961                     break;
00962                 }
00963             }
00964         }
00965         v++;
00966     }
00967 nextarg:
00968     arglist++; nargs--;
00969 }
00970 return(numgenerated);
00971 }
00972
00973 /*
00974 #] StartLoops :
00975 #[ GenLoops :
00976
00977     We enter with an open line in loop[nloop-1]
00978 */
00979
00980 LONG GenLoops(PHEAD WORD *term,WORD level,WORD vert,WORD nvert,
00981             WORD *arglist,WORD nargs,WORD *loop,WORD nloop)
00982 {
00983     LONG numgenerated = 0;
00984     WORD *vstart, *v, *vv, i, j, *vpartner;
00985 /*
00986     Start with checking whether the partner is in vstart (=vert[nvert])
00987 */
00988     vstart = AT.pWorkSpace[nvert];
00989     vv = vstart + *vstart; v = vstart+1;
00990     while ( v < vv ) {
00991         if ( *v == loop[nloop-1] ) {
00992 /*
00993             This closes the loop.
00994             Now we can output it.
00995 */
00996             LoopOutput(BHEAD term,level,loop,nloop);
00997             numgenerated++;
00998             return(numgenerated);
00999         }
01000         v++;
01001     }
01002 /*
01003     Start with finding the partner.
01004 */
01005     for ( i = 0; i < nvert; i++ ) {
01006         vpartner = AT.pWorkSpace[vert+i];
01007         if ( vpartner == vstart ) continue;
01008         for ( j = 0; j < nloop; j++ ) {
01009             if ( vpartner == AT.pWorkSpace[vert+nvert+j] ) break;
01010         }
01011         if ( j < nloop ) continue;
01012         v = vpartner+1; vv = vpartner + *vpartner;
01013         while ( v < vv ) {
01014             if ( *v == loop[nloop-1] ) {
01015 /*
01016                 Found the partner.
01017                 Now we can sum over the remaining permitted arguments of this vertex.
01018 */
01019                 v = vpartner + 1;
01020                 while ( v < vv ) {
01021 /*
01022                     *v should be in arglist.
01023 */
01024                     for ( j = 0; j < nargs; j++ ) {
01025                         if ( *v == arglist[j] ) break;
01026                     }
01027                     if ( j >= nargs ) { v++; continue; }

```

```

01028 /*
01029             *v should not be in loops.
01030 */
01031         for ( j = 0; j < nloop; j++ ) {
01032             if ( *v == loop[j] ) break;
01033         }
01034         if ( j >= nloop ) {
01035             AT.pWorkSpace[vert+nvert+nloop] = vpartner;
01036             loop[nloop++] = *v;
01037             numgenerated += GenLoops(BHEAD term, level, vert, nvert,
01038                                     arglist, nargs, loop, nloop);
01039             nloop--;
01040         }
01041         v++;
01042     }
01043     return(numgenerated);
01044 }
01045     v++;
01046 }
01047 }
01048 /*
01049 The partner is in a vertex we have finished before.
01050 */
01051     return(numgenerated);
01052 }
01053
01054 /*
01055 #] GenLoops :
01056 #[ LoopOutput :
01057 */
01058
01059 void LoopOutput(PHEAD WORD *term, WORD level, WORD *loop, WORD nloop)
01060 {
01061     CBUF *C = cbuf+AM.rbufnum;
01062     WORD loopfun = C->lhs[level][3]; /* The output function */
01063     WORD option1 = C->lhs[level][4]; /* type of argument */
01064     WORD *tstop, *tstop1, *t, *tt;
01065     WORD *outterm, *loop1;
01066     WORD i;
01067     tstop1 = term+*term; tstop = tstop1 - ABS(tstop1[-1]);
01068 /*
01069 Construct the rcycle symmetrized version of loop.
01070 */
01071     if ( nloop > 2 ) {
01072         loop1 = AT.WorkPointer;
01073         loop1[0] = loop[0];
01074         for ( i = 1; i < nloop; i++ ) { loop1[i] = loop[nloop-i]; }
01075         if ( loop1[1] < loop[1] ) {
01076             AT.WorkPointer += nloop;
01077             loop = loop1;
01078         }
01079     }
01080     outterm = AT.WorkPointer;
01081     tt = outterm; t = term;
01082     while ( t < tstop ) *tt++ = *t++;
01083     *tt++ = loopfun;
01084     if ( functions[loopfun-FUNCTION].spec == TENSORFUNCTION ) {
01085         *tt++ = FUNHEAD+nloop;
01086         FILLFUN(tt)
01087         if ( option1 == -VECTOR ) {
01088             for ( i = 0; i < nloop; i++ ) *tt++ = loop[i]+AM.OffsetVector;
01089         }
01090         else {
01091             for ( i = 0; i < nloop; i++ ) *tt++ = loop[i];
01092         }
01093     }
01094     else {
01095         *tt++ = FUNHEAD+nloop*2;
01096         FILLFUN(tt)
01097         for ( i = 0; i < nloop; i++ ) {
01098             *tt++ = option1;
01099             if ( option1 == -VECTOR ) *tt++ = loop[i] + AM.OffsetVector;
01100             else *tt++ = loop[i];
01101         }
01102     }
01103     while ( t < tstop1 ) *tt++ = *t++;
01104     *outterm = tt - outterm;
01105     AT.WorkPointer = tt;
01106     if ( Generator(BHEAD outterm, level) ) {
01107         MesCall("LoopOutput");
01108         Terminate(-1);
01109     }
01110     AT.WorkPointer = outterm;
01111 }
01112
01113 /*
01114 #] LoopOutput :

```



```

01115     #[ AllPaths :
01116
01117     This routine has many similarities with AllLoops.
01118     In AllLoops we have startingpoint and endpoint at the same vertex.
01119     In AllPaths the startingpoint and the endpoints are different and
01120     given in advance. This endfun can occur only twice.
01121 */
01122
01123 int AllPaths(PHEAD WORD *term,WORD level)
01124 {
01125     CBUF *C = cbuf+AM.rbufnum;
01126     WORD endcode = C->lhs[level][2]; /* The endpoint function */
01127     WORD vcode = C->lhs[level][3]; /* The intermediate function */
01128     WORD option1 = C->lhs[level][5]; /* type of argument */
01129     WORD option2 = C->lhs[level][6]; /* what to do when no loop */
01130     WORD *t, *tstop, *tendl, *tend2, *tstart, *tend, numend, nvert, npass;
01131     WORD *tfrom, *to, *tos, *from;
01132     WORD *arglist, nargs, *path, npath, *a, *a1, *a2, *a3;
01133     WORD i, j, jj, *v, *vv, action;
01134     LONG vert,vert1; /* ,vert2; */
01135     WORD numgenerated = 0;
01136     WORD *oldworkpointer = AT.WorkPointer;
01137     LONG oldpworkpointer = AT.pWorkPointer;
01138 /*
01139     Search for the two occurrences of endcode.
01140 */
01141     tstop = term+*term; tstop -= ABS(tstop[-1]);
01142     t = term + 1;
01143     while ( t < tstop && *t != endcode ) t += t[1];
01144     if ( t == tstop ) { /* no endpoints */
01145         if ( option2 == 0 ) return(0);
01146         else return(Generator(BHEAD term,level));
01147     }
01148     tendl = t;
01149     numend = 0;
01150     while ( t < tstop && *t == endcode ) { tend2 = t; t += t[1]; numend++; }
01151     if ( numend != 2 ) { /* not 2 endpoints -> no path */
01152         if ( option2 == 0 ) return(0);
01153         else return(Generator(BHEAD term,level));
01154     }
01155 /*
01156     Search for the intermediate functions.
01157 */
01158     t = term + 1;
01159     nvert = 0;
01160     while ( t < tstop && *t != vcode ) t += t[1];
01161     tstart = t;
01162     while ( t < tstop && *t == vcode ) {
01163         t += t[1]; nvert++;
01164     }
01165     tend = t;
01166 /*
01167     Next we copy the relevant content of the various functions into the
01168     Workspace. We keep pointers to the functions in pWorkspace.
01169     First the two endpoints and then the intermediate ones.
01170 */
01171     WantAddPointers((2*nvert+8));
01172     vert = AT.pWorkPointer+2;
01173     vert1 = AT.pWorkPointer;
01174 /* vert2 = AT.pWorkPointer+1; */
01175     AT.pWorkPointer += 2*nvert+8;
01176     nvert = 0;
01177 /*
01178     Copy the endpoints.
01179 */
01180     to = AT.WorkPointer;
01181
01182     if ( functions[endcode-FUNCTION].spec == TENSORFUNCTION ) {
01183         from = tendl; npass = 0;
01184     redol:
01185         tos = to++;
01186         tfrom = from+from[1];
01187         from += FUNHEAD;
01188         while ( from < tfrom ) {
01189             if ( option1 == -INDEX && *from >= 0 ) {
01190                 *to++ = *from++;
01191             }
01192             else if ( option1 == -VECTOR && *from < 0 ) {
01193                 *to++ = *from++ - AM.OffsetVector;
01194             }
01195             else {
01196                 from++;
01197             }
01198         }
01199         *tos = to-tos;
01200         if ( *tos < 2 ) to = tos;
01201         else AT.pWorkspace[vert1+npass] = tos;

```

```

01202         npass++;
01203         if ( from == tend2 ) goto redo1;
01204     }
01205     else if ( functions[endcode-FUNCTION].spec == 0 ) {
01206         from = tend1;
01207         npass = 0;
01208 redo2:
01209         tos = to++;
01210         tfrom = from + from[1];
01211         from += FUNHEAD;
01212         while ( from < tfrom ) {
01213             if ( option1 == -VECTOR
01214                 && ( from[0] == -INDEX || from[0] == -VECTOR )
01215                 && from[1] < 0 ) {
01216                 from++;
01217                 *to++ = *from++ - AM.OffsetVector;
01218             }
01219             else if ( option1 == *from ) {
01220                 from++; *to++ = *from++;
01221             }
01222             else {
01223                 NEXTARG(from);
01224             }
01225         }
01226         *tos = to-tos;
01227         if ( *tos < 2 ) to = tos;
01228         else AT.pWorkSpace[vert1+npass] = tos;
01229         npass++;
01230         if ( from == tend2 ) goto redo2;
01231     }
01232     else {
01233         AT.WorkPointer = oldworkpointer;
01234         AT.pWorkPointer = oldpworkpointer;
01235         if ( option2 == 0 ) return(0);
01236         return(Generator(BHEAD term,level));
01237     }
01238 /*
01239 And now the intermediate functions
01240 */
01241 from = tstart;
01242 if ( functions[vcode-FUNCTION].spec == TENSORFUNCTION ) {
01243     from = tstart;
01244     while ( from < tend ) {
01245         tos = to++;
01246         tfrom = from+from[1];
01247         from += FUNHEAD;
01248         while ( from < tfrom ) {
01249             if ( option1 == -INDEX && *from >= 0 ) {
01250                 *to++ = *from++;
01251             }
01252             else if ( option1 == -VECTOR && *from < 0 ) {
01253                 *to++ = *from++ - AM.OffsetVector;
01254             }
01255             else {
01256                 from++;
01257             }
01258         }
01259         *tos = to-tos;
01260         if ( *tos < 3 ) to = tos;
01261         else AT.pWorkSpace[vert+nvert++] = tos;
01262     }
01263 }
01264 else if ( functions[vcode-FUNCTION].spec == 0 ) {
01265     from = tstart;
01266     while ( from < tend ) {
01267         tos = to++;
01268         tfrom = from + from[1];
01269         from += FUNHEAD;
01270         while ( from < tfrom ) {
01271             if ( option1 == -VECTOR
01272                 && ( from[0] == -INDEX || from[0] == -VECTOR )
01273                 && from[1] < 0 ) {
01274                 from++;
01275                 *to++ = *from++ - AM.OffsetVector;
01276             }
01277             else if ( option1 == *from ) {
01278                 from++; *to++ = *from++;
01279             }
01280             else {
01281                 NEXTARG(from);
01282             }
01283         }
01284         *tos = to-tos;
01285         if ( *tos < 3 ) to = tos;
01286         else AT.pWorkSpace[vert+nvert++] = tos;
01287     }
01288 }

```

```

01289     else {
01290         AT.WorkPointer = oldworkpointer;
01291         AT.pWorkPointer = oldpworkpointer;
01292         if ( option2 == 0 ) return(0);
01293         return(Generator(BHEAD term,level));
01294     }
01295     AT.WorkPointer = to;
01296     /*
01297     Now make a list of all relevant arguments.
01298     */
01299     a = arglist = AT.WorkPointer;
01300     for ( i = -2; i < nvert; i++ ) {
01301         v = AT.pWorkSpace[vert+i];
01302         j = *v-1; v++;
01303         NCOPY(a,v,j);
01304     }
01305     nargs = a-arglist;
01306     AT.WorkPointer = a;
01307     /*
01308     Now sort the list. Bubble type sort.
01309     */
01310     a1 = arglist; a2 = a1+1;
01311     while ( a2 < a ) {
01312         a3 = a2;
01313         while ( a3 > a1 && a3[-1] > a3[0] ) {
01314             EXCH(*a3,a3[-1]);
01315             a3--;
01316         }
01317         a2++;
01318     }
01319     /*
01320     Now remove the elements from the list that do not occur exactly twice.
01321     */
01322     a1 = arglist; a2 = a1; a3 = a1+nargs;
01323     while ( a2 < a3 ) {
01324         if ( a2+1 == a3 ) { break; }
01325         else if ( a2+2 == a3 ) {
01326             if ( a2[0] == a2[1] ) { *a1++ = a2[0]; }
01327             break;
01328         }
01329         else {
01330             if ( a2[0] != a2[1] ) { a2++; }
01331             else if ( a2[0] != a2[2] ) {
01332                 *a1++ = a2[0]; a2 += 2;
01333             }
01334             else {
01335                 a2++; while ( a2 < a3 && a2[-1] == a2[0] ) a2++;
01336             }
01337         }
01338     }
01339     nargs = a1-arglist;
01340     /*
01341     Now we need to redo the list of vertices and remove the elements
01342     that are not in our arglist.
01343     */
01344     do {
01345         action = 0;
01346         for ( i = -2; i < nvert; i++ ) {
01347             vv = v = AT.pWorkSpace[vert+i];
01348             v++;
01349             for ( j = 1; j < vv[0]; j++ ) {
01350                 for ( jj = 0; jj < nargs; jj++ ) {
01351                     if ( *v == arglist[jj] ) break;
01352                 }
01353                 if ( jj >= nargs ) { /* was not in the list */
01354                     vv[0] = vv[0]-1;
01355                     for ( jj = j; jj < vv[0]; jj++ ) vv[jj] = vv[jj+1];
01356                 }
01357                 v++;
01358             }
01359         }
01360     } /*
01361     Next we need to remove vertices that have only one object remaining.
01362     Also, the object can be removed from arglist. After that we go back
01363     to clean up the list.
01364     */
01365     for ( i = -2; i < nvert; i++ ) {
01366         vv = AT.pWorkSpace[vert+i];
01367         if ( vv[0] == 1 ) {
01368             AT.pWorkSpace[vert+i] = AT.pWorkSpace[vert+nvert-1];
01369             nvert--; i--; continue;
01370         }
01371         else if ( vv[0] == 2 && i >= 0 ) {
01372             for ( j = 0; j < nargs; j++ ) {
01373                 if ( arglist[j] == vv[1] ) break;
01374             }
01375             while ( j < nargs-1 ) arglist[j] = arglist[j+1];

```

```

01376         nargs--;
01377         AT.pWorkSpace[vert+i] = AT.pWorkSpace[vert+nvert-1];
01378         nvert--; i--;
01379         action = 1;
01380     }
01381 }
01382 } while ( action );
01383 /*
01384 Now we have a clean list of connections and we can start building
01385 the path. We start with summing over all objects in vert1.
01386 */
01387 path = AT.WorkPointer; npath = 0;
01388 AT.WorkPointer += nvert+8;
01389
01390 t = AT.pWorkSpace[vert1];
01391 if ( *t >= 2 ) {
01392     for ( i = 1; i < *t; i++ ) {
01393         AT.pWorkSpace[vert+nvert] = t;
01394         path[npnath++] = t[i];
01395         numgenerated += GenPaths(BHEAD term,level,vert,nvert,arglist,nargs,path,npnath);
01396         npnath--;
01397     }
01398 }
01399 AT.WorkPointer = oldworkpointer;
01400 AT.pWorkPointer = oldpworkpointer;
01401 if ( numgenerated == 0 && option2 != 0 ) return(Generator(BHEAD term,level));
01402 return(0);
01403 }
01404
01405 /*
01406 #] AllPaths :
01407 #[ GenPaths :
01408
01409 We enter with an open line in path[npnath-1]
01410 We traverse though the vertices until we reach vert-1, the endpoint.
01411 */
01412
01413 LONG GenPaths(PHEAD WORD *term, WORD level, LONG vert, WORD nvert,
01414 WORD *arglist, WORD nargs, WORD *path, WORD npath)
01415 {
01416     LONG numgenerated = 0;
01417     WORD *t, *vpartner, *v, *vv;
01418     WORD i, j;
01419 /*
01420 Check whether path[npnath-1] is part of the endpoint.
01421 */
01422 t = AT.pWorkSpace[vert-1];
01423 for ( i = 1; i < *t; i++ ) {
01424     if ( t[i] == path[npnath-1] ) { /* Got a path! */
01425         PathOutput(BHEAD term,level,path,npnath);
01426         numgenerated++;
01427         return(numgenerated);
01428     }
01429 }
01430 /*
01431 Start with finding the partner.
01432 */
01433 for ( i = 0; i < nvert; i++ ) {
01434     vpartner = AT.pWorkSpace[vert+i];
01435     for ( j = 0; j < npath; j++ ) {
01436         if ( vpartner == AT.pWorkSpace[vert+nvert+j] ) break;
01437     }
01438     if ( j < npath ) continue;
01439     v = vpartner+1; vv = vpartner + *vpartner;
01440     while ( v < vv ) {
01441         if ( *v == path[npnath-1] ) {
01442 /*
01443 Found the partner.
01444 Now we can sum over the remaining permitted arguments of this vertex.
01445 */
01446 v = vpartner + 1;
01447 while ( v < vv ) {
01448 /*
01449 *v should be in arglist.
01450 */
01451 for ( j = 0; j < nargs; j++ ) {
01452     if ( *v == arglist[j] ) break;
01453 }
01454 if ( j >= nargs ) { v++; continue; }
01455 /*
01456 *v should not be in path.
01457 */
01458 for ( j = 0; j < npath; j++ ) {
01459     if ( *v == path[j] ) break;
01460 }
01461 if ( j >= npath ) {
01462     AT.pWorkSpace[vert+nvert+npnath] = vpartner;

```

```

01463             path[npath++] = *v;
01464             numgenerated += GenPaths(BHEAD term, level, vert, nvert,
01465                                     arglist, nargs, path, npath);
01466             npath--;
01467         }
01468         v++;
01469     }
01470     return(numgenerated);
01471 }
01472 v++;
01473 }
01474 }
01475 /*
01476 The partner is in a vertex we have finished before.
01477 */
01478 return(numgenerated);
01479 }
01480
01481 /*
01482 #] GenPaths :
01483 #[ PathOutput :
01484 */
01485
01486 void PathOutput(PHEAD WORD *term, WORD level, WORD *path, WORD npath)
01487 {
01488     CBUF *C = cbuf+AM.rbufnum;
01489     WORD pathfun = C->lhs[level][4]; /* The output function */
01490     WORD option1 = C->lhs[level][5]; /* type of argument */
01491     WORD *tstop, *tstop1, *t, *tt;
01492     WORD *outterm;
01493     WORD i;
01494     tstop1 = term+*term; tstop = tstop1 - ABS(tstop1[-1]);
01495     outterm = AT.WorkPointer;
01496     tt = outterm; t = term;
01497     while ( t < tstop ) *tt++ = *t++;
01498     *tt++ = pathfun;
01499     if ( functions[pathfun-FUNCTION].spec == TENSORFUNCTION ) {
01500         *tt++ = FUNHEAD+npath;
01501         FILLFUN(tt)
01502         if ( option1 == -VECTOR ) {
01503             for ( i = 0; i < npath; i++ ) *tt++ = path[i]+AM.OffsetVector;
01504         }
01505         else {
01506             for ( i = 0; i < npath; i++ ) *tt++ = path[i];
01507         }
01508     }
01509     else {
01510         *tt++ = FUNHEAD+npath*2;
01511         FILLFUN(tt)
01512         for ( i = 0; i < npath; i++ ) {
01513             *tt++ = option1;
01514             if ( option1 == -VECTOR ) *tt++ = path[i] + AM.OffsetVector;
01515             else *tt++ = path[i];
01516         }
01517     }
01518     while ( t < tstop1 ) *tt++ = *t++;
01519     *outterm = tt - outterm;
01520     AT.WorkPointer = tt;
01521     if ( Generator(BHEAD outterm, level) ) {
01522         MesCall("PathOutput");
01523         Terminate(-1);
01524     }
01525     AT.WorkPointer = outterm;
01526 }
01527
01528
01529
01530 /*
01531 #] PathOutput :
01532 #[ AllOnePI :
01533
01534 We assume a graph that has loops and is OnePI.
01535 This routine initializes the recursion that creates all onePI subgraphs.
01536
01537 Algorithm:
01538 a: have a routine that removes all bridges: RemoveBridges
01539 b: output an empty diagram.
01540 c: output the diagram itself
01541 d: cut a line, followed by removing all bridges.
01542 e: if the diagram still has lines, go to c, else go to the next line in d.
01543 In order to avoid factorial blowup, we need to prune the tree as fast as possible.
01544
01545 1: list of lines to be cut.
01546 2: once we have tried a line and removed its bridges, we do not have to
01547 try this line deeper in the tree, neither its bridges.
01548
01549

```

```

01550      1 2 3
01551      cut 1.                                x
01552      cut 2 -> bridge 3                    x
01553      cut 2.                                x
01554      cut 3 -> bridge 1      ????    mag niet
01555      cut 3.                                x
01556      Hence 1,2,3,4,5,6,7
01557      1 still to go 2,3,4,5,6,7
01558      2 still to go 3,4,5,6,7
01559      3 still to go 4,5,6,7
01560      etc.
01561      bridges: if x is bridge, we take x from the list_to_go.
01562      If we have a bridge that is a number lower than the list_to_go we skip this possibility.
01563      We reach the end when the list_to_go is empty.
01564  */
01565
01566 WORD AllOnePI(WORD *term,WORD level)
01567 {
01568     CBUF *C = cbuf+AM.rbufnum;
01569     WORD vcode = C->lhs[level][2];    /* The vertex function */
01570     WORD option1 = C->lhs[level][4];    /* type of argument */
01571  /*
01572     First we have to collect all relevant information about the diagram.
01573     We should start with removing bridges to make the real starting point onePI.
01574  */
01575     DUMMYUSE(term)
01576     DUMMYUSE(vcode)
01577     DUMMYUSE(option1)
01578     return(0);
01579 }
01580
01581  /*
01582     #] AllOnePI :
01583     #[ RemoveBridges :
01584  */
01585
01586 int RemoveBridges(void)
01587 {
01588     return(0);
01589 }
01590
01591  /*
01592     #] RemoveBridges :
01593     #[ TakeOneLine :
01594  */
01595
01596 int TakeOneLine(WORD*term,WORD level)
01597 {
01598     DUMMYUSE(term)
01599     DUMMYUSE(level)
01600     return(0);
01601 }
01602
01603  /*
01604     #] TakeOneLine :
01605     #[ OutputOnePI :
01606  */
01607
01608 int OutputOnePI(PHEAD WORD *term,WORD level)
01609 {
01610     return(Generator(BHEAD term,level));
01611 }
01612
01613  /*
01614     #] OutputOnePI :
01615  */
01616

```

## 4.68 mallocprotect.h

```

00001 #ifndef MALLOCPROTECT_H
00002 #define MALLOCPROTECT_H
00003
00010  /* #[ License : */
00011  /*
00012  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00013  *   When using this file you are requested to refer to the publication
00014  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00015  *   This is considered a matter of courtesy as the development was paid
00016  *   for by FOM the Dutch physics granting agency and we would like to
00017  *   be able to track its scientific use to convince FOM of its value
00018  *   for the community.
00019  *

```

```

00020 *   This file is part of FORM.
00021 *
00022 *   FORM is free software: you can redistribute it and/or modify it under the
00023 *   terms of the GNU General Public License as published by the Free Software
00024 *   Foundation, either version 3 of the License, or (at your option) any later
00025 *   version.
00026 *
00027 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00028 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00029 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00030 *   details.
00031 *
00032 *   You should have received a copy of the GNU General Public License along
00033 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00034 */
00035 /* #] License : */
00036
00037 /*
00038     #[ Documentation :
00039
00040 The enhanced malloc debugger. It intercepts access (both write AND
00041 read) to the memory outside the allocated chunk in the following
00042 manner: it prints out (to the stderr) the information about the event
00043 and goes to the spilock. The user is able to attach the debugger to
00044 the running process, investigate the problem and continue evaluation.
00045
00046 In case of error, the debugger will print to the stderr the following
00047 information:
00048 "***** PID: ###, Addr: ###, signal ### (code ###)"
00049 "Attach gdb -p ### and set var loopForever=0 in a corresponding frame to continue"
00050 or for TFORM:
00051 "Attach gdb -p ### and set var loopForever=0 in a corresponding frame
00052   of a thread with LPW id ### to continue"
00053
00054 and go to the spilock. The user may attach gdb by the command
00055 gdb -p ### and type "where" (in case of TFORM, the user should first
00056 switch to the proper thread). Then investigation of the corresponding
00057 frames should clarify the situation. In order to continue the program,
00058 the user might set (in the corresponding frame of the corresponding
00059 thread) the variable loopForever to 0. The debugger will try to remove
00060 local problem lead to the exception.
00061
00062 To activate the debugger, the macro MALLOCPROTECT should be
00063 defined. There are three possible values:
00064 #define MALLOCPROTECT -1 (any integer <0)
00065 #define MALLOCPROTECT 0
00066 #define MALLOCPROTECT 1 (any integer >0)
00067
00068 If MALLOCPROTECT < 0, the debugger will intercept any access to the
00069 memory before the allocated chunk, even one byte.
00070
00071 If MALLOCPROTECT == 0, the debugger will intercept any access to the
00072 memory before the allocated chunk, even one byte, and access to the
00073 memory page next to the allocated one.
00074
00075 If MALLOCPROTECT > 0, the debugger will intercept any access to the
00076 memory after the allocated chunk, even one byte, and access to the
00077 memory page before the allocated one.
00078
00079 The original FORM malloc debugger is able to catch errors when the
00080 allocated memory is freed improper, or if some small portion OUT of
00081 the allocated chunk is written. This debugger is a complementary
00082 one. It permits to catch situation when some small portion of memory
00083 before or after the allocated chunk is written or even just read.
00084
00085 The idea is to protect the beginning or the end of the allocated chunk
00086 for any kind of access and install the SIGSEGV handler in order to
00087 intercept the access to this memory. Moreover, the allocated memory is
00088 always immediately returned to the system so if the user tries to
00089 access the memory after it is freed then the handler is triggered,
00090 also.
00091
00092 The problem here is that we are able to allocate / protect only the
00093 whole page. So the user has to run the debugger twice: one time with
00094 the left alignment (#define MALLOCPROTECT <0), and then with the right
00095 alignment (#define MALLOCPROTECT >0). During the first run the possible
00096 errors like x[-1] will be caught, and the second run will manifest
00097 reading over the allocated ares.
00098
00099 The leftmost extra page is always allocated and protected. The size
00100 of the allocated chunk is stored in the beginning of this page.
00101
00102     #] Documentation :
00103     #[ Includes :
00104 */
00105
00106 #include <stdio.h>

```

```

00107 #include <stdlib.h>
00108 #include <unistd.h>
00109 #include <signal.h>
00110 #include <sys/mman.h>
00111
00112 #ifdef WITHPTHREADS
00113 #ifdef LINUX
00114 #include <sys/syscall.h>
00115 #endif
00116 #endif
00117
00118 /*
00119  #] Includes :
00120 */
00121
00122 static int pageSize=4096; /*the default value*/
00123
00124
00125 /*
00126  #[ segv_handler :
00127 */
00128
00129 /*The handler will be invoked on some signal (usually SIGSEGV). It
00130 will print some diagnostics to stderr and hands up in infinite loop
00131 waiting the user for debugging:*/
00132 static void segv_handler(int sig, siginfo_t *sip, void *xxx) {
00133     char *vadr;
00134     char *errStr;
00135     int actionBeforeExit=0; /* < 0 - unprotect, > 0 - map */
00136     switch(sip->si_signo){ /* All POSIX signals for which the field siginfo_t.si_addr is defined:*/
00137         case SIGILL:
00138             switch(sip->si_code){
00139                 case ILL_ILLOPC:
00140                     errStr="SIGILL: Illegal opcode";
00141                     break;
00142                 case ILL_ILLOPN:
00143                     errStr="SIGILL: Illegal operand";
00144                     break;
00145                 case ILL_ILLADR:
00146                     errStr="SIGILL: Illegal addressing mode";
00147                     break;
00148                 case ILL_ILLTRP:
00149                     errStr="SIGILL: Illegal trap";
00150                     break;
00151                 case ILL_PRVOPC:
00152                     errStr="SIGILL: Privileged opcode";
00153                     break;
00154                 case ILL_PRVREG:
00155                     errStr="SIGILL: Privileged register";
00156                     break;
00157                 case ILL_COPROC:
00158                     errStr="SIGILL: Coprocessor error";
00159                     break;
00160                 case ILL_BADSTK:
00161                     errStr="SIGILL: Internal stack error";
00162                     break;
00163                 default:
00164                     errStr="SIGILL: Unknown signal code";
00165             } /*case SIGILL:*/
00166             break;
00167         case SIGFPE:
00168             switch(sip->si_code){
00169                 case FPE_INTDIV:
00170                     errStr="SIGFPE: Integer divide-by-zero";
00171                     break;
00172                 case FPE_INTOVF:
00173                     errStr="SIGFPE: Integer overflow";
00174                     break;
00175                 case FPE_FLTDIV:
00176                     errStr="SIGFPE: Floating point divide-by-zero";
00177                     break;
00178                 case FPE_FLTOVF:
00179                     errStr="SIGFPE: Floating point overflow";
00180                     break;
00181                 case FPE_FLTUND:
00182                     errStr="SIGFPE: Floating point underflow";
00183                     break;
00184                 case FPE_FLTRES:
00185                     errStr="SIGFPE: Floating point inexact result";
00186                     break;
00187                 case FPE_FLTINV:
00188                     errStr="SIGFPE: Invalid floating point operation";
00189                     break;
00190                 case FPE_FLTSUB:
00191                     errStr="SIGFPE: Subscript out of range";
00192                     break;
00193                 default:

```



```

00194         errStr="SIGFPE: Unknown signal code";
00195     }/*switch(sip->si_code)*/
00196     break;
00197 case SIGSEGV:
00198     switch(sip->si_code){
00199         case SEGV_MAPERR:
00200             errStr="SIGSEGV: Address not mapped";
00201             actionBeforeExit = 1;
00202             break;
00203         case SEGV_ACCERR:
00204             errStr="SIGSEGV: Invalid permissions";
00205             actionBeforeExit = -1;
00206             break;
00207         default:
00208             errStr="SIGSEGV: Unknown signal code";
00209     }/*switch(sip->si_code)*/
00210     break;
00211 case SIGBUS:
00212     switch(sip->si_code){
00213         case BUS_ADRALN:
00214             errStr="SIGBUS: Invalid address alignment";
00215             break;
00216         case BUS_ADRERR:
00217             errStr="SIGBUS: Non-existent physical address";
00218             break;
00219         case BUS_OBJERR:
00220             errStr="SIGBUS: Object-specific hardware error";
00221             break;
00222         default:
00223             errStr="SIGBUS: Unknown signal code";
00224     }/*switch(sip->si_code)*/
00225     break;
00226 default:
00227     errStr="Unknown signal";
00228 }/*switch(sip->si_signo)*/
00229
00230 vadr = (caddr_t)sip->si_addr;
00231 fprintf(stderr, "\n***** PID: %ld, Addr: %p, signal %s (code %d)\n",
00232     (long) getpid(), vadr, errStr, sip->si_code);
00233
00234 /*Block*/
00235 /*The process hangs up at this block.
00236 Attach gdb to investigate and continue:
00237 */
00238 volatile int loopForever=1;
00239 size_t alignedAdr=((size_t)vadr) & (~ (pageSize-1));
00240 fprintf(stderr, " Attach gdb -p %ld and set var loopForever=0 in a corresponding frame",
00241     (long) getpid());
00242 #ifdef CORRECTCODE
00243 #ifdef WITHPTHREADS
00244 #ifdef LINUX
00245     fprintf(stderr, "\n of a thread with LPW id %ld", (long) syscall(SYS_gettid));
00246 #else
00247     /*If the compiler fails here, just remove the next line:*/
00248     fprintf(stderr, "\n of thread with LPW id %ld", (long) gettid());
00249 #endif
00250 #endif
00251 #endif
00252 fprintf(stderr, " to continue\n");
00253
00254 while(loopForever)
00255     sleep(1);
00256
00257 if(actionBeforeExit<0)/*After changing loopForever=0, unprotect the page to continue:*/
00258     mprotect((char*)alignedAdr, pageSize, PROT_READ | PROT_WRITE);
00259 if(actionBeforeExit>0)/*After changing loopForever=0, map the page to continue:*/
00260     mmap((void*)alignedAdr, pageSize, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, 0, 0); /*
00261 /* mmap((void*)alignedAdr, pageSize, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANON, 0, 0);
00262 */
00263 }/*Block*/
00264 }/*segv_handler*/
00265
00266 /*
00267 #] segv_handler :
00268 */
00269
00270 /*
00271 #[ mprotectInit :
00272 */
00273
00274 static FORM_INLINE int mprotectInit(void)
00275 {
00276     struct sigaction sa;
00277     pageSize = getpagesize();
00278     sa.sa_sigaction = &segv_handler;
00279
00280     sigemptyset(&sa.sa_mask);

```

```

00281     sigaddset(&sa.sa_mask, SIGIO);
00282     sigaddset(&sa.sa_mask, SIGALRM);
00283
00284     sa.sa_flags = SA_SIGINFO;
00285
00286     if (sigaction(SIGILL, &sa, NULL)) {
00287         fprintf(stderr, "Error on assigning %s.\n", "SIGILL");
00288         return SIGILL;
00289     }
00290     if (sigaction(SIGFPE, &sa, NULL)) {
00291         fprintf(stderr, "Error on assigning %s.\n", "SIGFPE");
00292         return SIGFPE;
00293     }
00294     if (sigaction(SIGSEGV, &sa, NULL)) {
00295         fprintf(stderr, "Error on assigning %s.\n", "SIGSEGV");
00296         return SIGSEGV;
00297     }
00298     if (sigaction(SIGBUS, &sa, NULL)) {
00299         fprintf(stderr, "Error on assigning %s.\n", "SIGBUS");
00300         return SIGBUS;
00301     }
00302     return 0;
00303 } /* mprotectInit */
00304
00305 /*
00306  #] mprotectInit :
00307  */
00308
00309 /*
00310  #[ mprotectMalloc :
00311  */
00312
00313 static void *mprotectMalloc(size_t theSize)
00314 {
00315     if (MALLOCPROTECT < 0
00316         /*Only one side is protected*/
00317         size_t nPages=1;
00318     #else
00319         /*Both sides are protected*/
00320         size_t nPages=2;
00321     #if MALLOCPROTECT > 0
00322         /*will need the original theSize*/
00323         size_t requestedSize=theSize;
00324     #endif
00325     #endif
00326
00327     char *ret=NULL;
00328     size_t *ptr;
00329
00330
00331
00332
00333     /*Align required size to the pagesize:*/
00334     if (theSize % pageSize)
00335         nPages++;
00336     theSize= (theSize/pageSize+nPages)*pageSize;
00337
00338     /* ret=(char*)mmap(0,theSize,PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, 0, 0); */
00339     ret=(char*)mmap(0,theSize,PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANON, 0, 0);
00340     if (ret == MAP_FAILED)
00341         return NULL;
00342     ptr=(size_t *)ret;
00343     *ptr=theSize;
00344     if (mprotect(ret, pageSize, PROT_NONE)) return NULL;
00345     #if MALLOCPROTECT < 0
00346         return ret+pageSize;
00347     #else
00348         if (mprotect(ret+(theSize-pageSize), pageSize, PROT_NONE)) return NULL;
00349     #if MALLOCPROTECT ==0
00350         return ret+pageSize;
00351     #endif
00352     #endif
00353     /*MALLOCPROTECT > 0, but we need conditional compilation
00354     since the variable requestedSize:*/
00355     #if MALLOCPROTECT > 0
00356
00357         /*Potential problems with alignment if the requested size is not
00358         a multiple of items. But no problems on x86-64:*/
00359         return ret+ (theSize-pageSize-requestedSize);
00360     #endif
00361 } /* mprotectMalloc */
00362 /*
00363  #] mprotectMalloc :
00364  */
00365
00366 /*
00367  #[ mprotectFree :

```

```

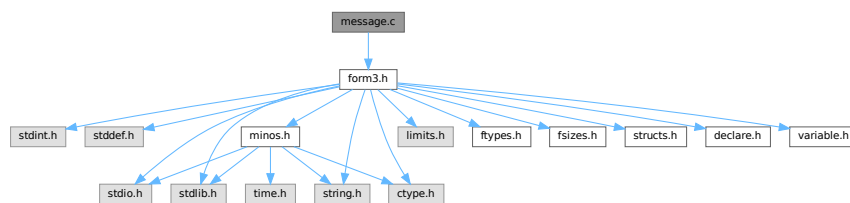
00368 */
00369
00370 static void mprotectFree(char *ptr)
00371 {
00372     size_t theSize;
00373     if(ptr==NULL) return;
00374 #if MALLOCPROTECT > 0
00375     /*The memory block was moved to the right, find the left page boundary:*/
00376     /*Block*/
00377     size_t alignedPtr=((size_t)ptr) & ~(pageSize-1);
00378     ptr=(char*)alignedPtr;
00379     /*Block*/
00380 #endif
00381
00382     ptr-=pageSize;
00383     mprotect(ptr, pageSize, PROT_READ);
00384     theSize=((size_t*)ptr);
00385     munmap(ptr,theSize);
00386 }/*mprotectFree*/
00387
00388 /*
00389 #] mprotectFree :
00390 */
00391
00392 #endif

```

## 4.69 message.c File Reference

#include "form3.h"

Include dependency graph for message.c:



### Functions

- void [Error0](#) (char \*s)
- void [Error1](#) (char \*s, UBYTE \*t)
- void [Error2](#) (char \*s1, char \*s2, UBYTE \*t)
- int [MesWork](#) (void)
- int [MesPrint](#) (va\_alist)
- void [Warning](#) (char \*s)
- void [HighWarning](#) (char \*s)
- int [MesCall](#) (char \*s)
- int [MesCerr](#) (char \*s, UBYTE \*t)
- int [MesComp](#) (char \*s, UBYTE \*p, UBYTE \*q)
- void [PrintTerm](#) (WORD \*term, char \*where)
- void [PrintTermC](#) (WORD \*term, char \*where)
- void [PrintSubTerm](#) (WORD \*term, char \*where)
- void [PrintWords](#) (WORD \*buffer, LONG number)
- void [PrintSeq](#) (WORD \*a, char \*text)

### 4.69.1 Detailed Description

Contains the routines that write messages. This includes the very important routine MesPrint which is the FORM equivalent of printf but then with escape sequences that are relevant for symbolic manipulation. The FORM statement Print "..." is passed almost literally to MesPrint.

Definition in file [message.c](#).

### 4.69.2 Function Documentation

#### 4.69.2.1 Error0()

```
void Error0 (
    char * s )
```

Definition at line 56 of file [message.c](#).

#### 4.69.2.2 Error1()

```
void Error1 (
    char * s,
    UBYTE * t )
```

Definition at line 67 of file [message.c](#).

#### 4.69.2.3 Error2()

```
void Error2 (
    char * s1,
    char * s2,
    UBYTE * t )
```

Definition at line 78 of file [message.c](#).

#### 4.69.2.4 MesWork()

```
int MesWork (
    void )
```

Definition at line 89 of file [message.c](#).

#### 4.69.2.5 MesPrint()

```
int MesPrint (
    va_alist )
```

Definition at line 140 of file [message.c](#).

#### 4.69.2.6 Warning()

```
void Warning (  
    char * s )
```

Definition at line 790 of file [message.c](#).

#### 4.69.2.7 HighWarning()

```
void HighWarning (  
    char * s )
```

Definition at line 802 of file [message.c](#).

#### 4.69.2.8 MesCall()

```
int MesCall (  
    char * s )
```

Definition at line 814 of file [message.c](#).

#### 4.69.2.9 MesCerr()

```
int MesCerr (  
    char * s,  
    UBYTE * t )
```

Definition at line 824 of file [message.c](#).

#### 4.69.2.10 MesComp()

```
int MesComp (  
    char * s,  
    UBYTE * p,  
    UBYTE * q )
```

Definition at line 843 of file [message.c](#).

#### 4.69.2.11 PrintTerm()

```
void PrintTerm (  
    WORD * term,  
    char * where )
```

Definition at line 857 of file [message.c](#).

#### 4.69.2.12 PrintTermC()

```
void PrintTermC (
    WORD * term,
    char * where )
```

Definition at line 887 of file [message.c](#).

#### 4.69.2.13 PrintSubTerm()

```
void PrintSubTerm (
    WORD * term,
    char * where )
```

Definition at line 921 of file [message.c](#).

#### 4.69.2.14 PrintWords()

```
void PrintWords (
    WORD * buffer,
    LONG number )
```

Definition at line 943 of file [message.c](#).

#### 4.69.2.15 PrintSeq()

```
void PrintSeq (
    WORD * a,
    char * text )
```

Definition at line 961 of file [message.c](#).

## 4.70 message.c

[Go to the documentation of this file.](#)

```
00001
00009 /* # [ License : */
00010 /*
00011 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00012 * When using this file you are requested to refer to the publication
00013 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00014 * This is considered a matter of courtesy as the development was paid
00015 * for by FOM the Dutch physics granting agency and we would like to
00016 * be able to track its scientific use to convince FOM of its value
00017 * for the community.
00018 *
00019 * This file is part of FORM.
00020 *
00021 * FORM is free software: you can redistribute it and/or modify it under the
00022 * terms of the GNU General Public License as published by the Free Software
00023 * Foundation, either version 3 of the License, or (at your option) any later
00024 * version.
00025 *
00026 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00027 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00028 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00029 * details.
00030 *
```

```

00031 *   You should have received a copy of the GNU General Public License along
00032 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
00033 */
00034 /* #] License : */
00035 /*
00036     #[ Includes :
00037
00038     The static variables for the messages can remain as such also for
00039     the parallel version as messages are to be locked to avoid problems
00040     with simultaneous messages.
00041 */
00042
00043 #include "form3.h"
00044
00045 static int iswarning = 0;
00046
00047 static char hex[] = {'0','1','2','3','4','5','6','7','8','9',
00048                     'A','B','C','D','E','F'};
00049
00050 /*
00051     #] Includes :
00052     #[ exit :
00053     #[ Error0 :
00054 */
00055
00056 void Error0(char *s)
00057 {
00058     MesPrint("=== %s",s);
00059     Terminate(-1);
00060 }
00061
00062 /*
00063     #] Error0 :
00064     #[ Error1 :
00065 */
00066
00067 void Error1(char *s, UBYTE *t)
00068 {
00069     MesPrint("@%s %s",s,t);
00070     Terminate(-1);
00071 }
00072
00073 /*
00074     #] Error1 :
00075     #[ Error2 :
00076 */
00077
00078 void Error2(char *s1, char *s2, UBYTE *t)
00079 {
00080     MesPrint("@%s%s %s",s1,s2,t);
00081     Terminate(-1);
00082 }
00083
00084 /*
00085     #] Error2 :
00086     #[ MesWork :
00087 */
00088
00089 int MesWork(void)
00090 {
00091     MesPrint("=== Workspace overflow. %l bytes is not enough.",AM.WorkSize);
00092     MesPrint("=== Change parameter WorkSpace in %s",setupfilename);
00093     Terminate(-1);
00094     return(-1);
00095 }
00096
00097 /*
00098     #] MesWork :
00099     #[ MesPrint :
00100
00101     Kind of a printf function for simple messages.
00102     The main concern is getting the arguments in a portable way.
00103     Note: many compilers have errors when sizeof(WORD) < sizeof(int)
00104     %a array of size n WORDs (two parameters, first is int, second WORD *)
00105     %b array of size n UBYTES (two parameters, first is int, second UBYTE *)
00106     %C array of size n chars (two parameters, first is int, second char *)
00107     %d word;
00108     %l long;
00109     %L long long *;
00110     %s string;
00111     %#i unsigned word filled
00112     %#d word positioned
00113     %#l long word positioned.
00114     %#L long long word * positioned.
00115     %#s string positioned.
00116     %#p position in file.
00117     %r The current term in raw format (internal representation)

```

```

00118      %t The current term (AN.currentTerm)
00119      %T The current term (AN.currentTerm) with its sign
00120      %w Number of the thread(worker)
00121      %$ The next $ in AN.listinprint
00122      %x hexadecimal. Takes 8 places. Mainly for debugging.
00123      %% %
00124      %# #
00125      # " ==> "
00126      @ " ==> " Preprocessor error
00127      & ' --> ' Regular compiler error
00128      Each call is terminated with a new line.
00129      Put a % at the end of the string to suppress the new line.
00130
00131      New feature (7-dec-2011): The & will only work when we do not block it
00132      from the execution of the print statement because we need the & also for
00133      the tabulator in the print "" statement.
00134  */
00135
00136  int
00137  #ifdef ANSI
00138  MesPrint(const char *fmt, ... )
00139  #else
00140  MesPrint(va_alist)
00141  va_dcl
00142  #endif
00143  {
00144      GETIDENTITY
00145      char Out[MAXLINELENGTH+14], *stopper, *t, *s, *u, c, *carray;
00146      UBYTE extrabuffer[MAXLINELENGTH+14];
00147      int w, x, i, specialerror = 0;
00148      LONG num, y;
00149      WORD *array;
00150      UBYTE *oldoutfill = AO.OutputLine, *barray;
00151      /*[19apr2004 mt]:*/
00152      LONG (*OldWrite)(int handle, UBYTE *buffer, LONG size) = WriteFile;
00153      /*:[19apr2004 mt]*/
00154      va_list ap;
00155      #ifdef ANSI
00156      va_start(ap,fmt);
00157      s = (char *)fmt;
00158      #else
00159      va_start(ap);
00160      s = va_arg(ap,char *);
00161      #endif
00162      #ifdef WITHMPI
00163      /*
00164      * On slaves, if AS.printflag is
00165      * = 0 : print nothing.
00166      * > 0 : synchronized output. All text will be sent to the master
00167      * in the next MUNLOCK().
00168      * < 0 : normal output.
00169      */
00170      if ( PF.me != MASTER && AS.printflag == 0 ) return(0);
00171      if ( PF.me == MASTER || AS.printflag < 0 )
00172      #endif
00173      FLUSHCONSOLE;
00174      /*
00175      * MesPrints() never prints a message to an external channel even if
00176      * WriteFile is set to &WriteToExternalChannel.
00177      */
00178      #ifdef WITHMPI
00179      WriteFile = PF.me == MASTER || AS.printflag > 0 ? &PF_WriteFileToFile : &WriteFileToFile;
00180      #else
00181      WriteFile = &WriteFileToFile;
00182      #endif
00183      AO.OutputLine = extrabuffer;
00184      t = Out;
00185      stopper = Out + AC.LineLength;
00186      while ( *s ) {
00187          if ( ( ( *s == '&' && AO.ErrorBlock == 0 ) || *s == '@' || *s == '#' ) && AC.CurrentStream !=
00188 0 ) {
00189              u = (char *)AC.CurrentStream->name;
00190              while ( *u ) {
00191                  *t++ = *u++;
00192                  if ( t >= stopper ) {
00193                      num = t - Out;
00194                      WriteString(ERROROUT, (UBYTE *)Out,num);
00195                      num = 0; t = Out;
00196                  }
00197              }
00198              *t++ = ' ';
00199              if ( t+20 >= stopper ) {
00200                  num = t - Out;
00201                  WriteString(ERROROUT, (UBYTE *)Out,num);
00202                  num = 0; t = Out;
00203              }
00204              *t++ = 'L'; *t++ = 'i'; *t++ = 'n'; *t++ = 'e'; *t++ = ' ';

```



```

00204         if ( *s == '&' ) y = AC.CurrentStream->prevline;
00205     else y = AC.CurrentStream->linenumber;
00206     t = LongCopy(y,t);
00207     if ( !iswarning && ( *s == '&' || *s == '@' ) ) {
00208         for ( i = 0; i < NumDoLoops; i++ ) DoLoops[i].errorsinloop = 1;
00209     }
00210 }
00211 if ( ( *s == '&' && AO.ErrorBlock == 0 ) ) {
00212     *t++ = ' '; *t++ = '-'; *t++ = '-'; *t++ = '>'; *t++ = ' '; s++;
00213 }
00214 else if ( *s == '@' || *s == '#' ) {
00215     *t++ = ' '; *t++ = '='; *t++ = '='; *t++ = '>'; *t++ = ' '; s++;
00216 }
00217 /*
00218 else if ( *s == '&' && AO.ErrorBlock == 1 ) {
00219 }
00220 */
00221 else if ( *s != '%' ) {
00222     *t++ = *s++;
00223     if ( t >= stopper ) {
00224         num = t - Out;
00225         WriteString(ERROROUT, (UBYTE *)Out,num);
00226         num = 0; t = Out;
00227     }
00228 }
00229 else {
00230     s++;
00231     if ( *s == 'd' ) {
00232         if ( ( w = va_arg(ap, int) ) < 0 ) { *t++ = '-'; w = -w; }
00233         t = (char *)NumCopy(w, (UBYTE *)t);
00234     }
00235     else if ( *s == 'l' ) {
00236         if ( ( y = va_arg(ap, LONG) ) < 0 ) { *t++ = '-'; y = -y; }
00237         t = LongCopy(y,t);
00238     }
00239     /* #ifdef __GLIBC_HAVE_LONG_LONG */
00240     else if ( *s == 'p' ) {
00241         POSITION *pp;
00242         off_t ly;
00243         pp = va_arg(ap, POSITION *);
00244         ly = BASEPOSITION(*pp);
00245         if ( ly < 0 ) { *t++ = '-'; ly = -ly; }
00246         /*-----change 10-feb-2003 did not have & */
00247         t = LongLongCopy(&(ly),t);
00248     }
00249     /* #endif */
00250     else if ( *s == 'c' ) {
00251         c = (char)(va_arg(ap, int));
00252         *t++ = c; *t = 0;
00253     }
00254     else if ( *s == 'a' ) {
00255         w = va_arg(ap, int);
00256         array = va_arg(ap,WORD *);
00257         while ( w > 0 ) {
00258             t = (char *)NumCopy(*array, (UBYTE *)t);
00259             if ( t >= stopper ) {
00260                 num = t - Out;
00261                 WriteString(ERROROUT, (UBYTE *)Out,num);
00262                 t = Out;
00263                 *t++ = ' ';
00264             }
00265             *t++ = ' ';
00266             w--; array++;
00267         }
00268     }
00269     else if ( *s == 'b' ) {
00270         w = va_arg(ap, int);
00271         barray = va_arg(ap,UBYTE *);
00272         while ( w > 0 ) {
00273             *t++ = hex[((*barray)>>4)&0xF];
00274             *t++ = hex[(*barray)&0xF];
00275             *t = 0;
00276             if ( t >= stopper ) {
00277                 num = t - Out;
00278                 WriteString(ERROROUT, (UBYTE *)Out,num);
00279                 t = Out;
00280                 *t++ = ' ';
00281             }
00282             *t++ = ' ';
00283             w--; barray++;
00284         }
00285     }
00286     else if ( *s == 'C' ) {
00287         w = va_arg(ap, int);
00288         carray = va_arg(ap,char *);
00289         while ( w > 0 ) {
00290

```

```

00291         if ( *carray < 32 ) *t++ = '^';
00292     else *t++ = *carray;
00293     *t = 0;
00294     if ( t >= stopper ) {
00295         num = t - Out;
00296         WriteString(ERROROUT, (UBYTE *)Out, num);
00297         t = Out;
00298         *t++ = ' ';
00299     }
00300     w--; carray++;
00301 }
00302 }
00303 else if ( *s == 'I' ) {
00304     int *iarray;
00305     w = va_arg(ap, int);
00306     iarray = va_arg(ap, int *);
00307     while ( w > 0 ) {
00308         t = (char *)LongCopy( (LONG) (*iarray), (char *)t);
00309         if ( t >= stopper ) {
00310             num = t - Out;
00311             WriteString(ERROROUT, (UBYTE *)Out, num);
00312             t = Out;
00313             *t++ = ' ';
00314         }
00315         *t++ = ' ';
00316         w--; array++;
00317     }
00318 }
00319 else if ( *s == 'E' ) {
00320     LONG *larray;
00321     w = va_arg(ap, int);
00322     larray = va_arg(ap, LONG *);
00323     while ( w > 0 ) {
00324         t = (char *)LongCopy(*larray, (char *)t);
00325         if ( t >= stopper ) {
00326             num = t - Out;
00327             WriteString(ERROROUT, (UBYTE *)Out, num);
00328             t = Out;
00329             *t++ = ' ';
00330         }
00331         *t++ = ' ';
00332         w--; array++;
00333     }
00334 }
00335 else if ( *s == 's' ) {
00336     u = va_arg(ap, char *);
00337     while ( *u ) {
00338         if ( t >= stopper ) {
00339             num = t - Out;
00340             WriteString(ERROROUT, (UBYTE *)Out, num);
00341             t = Out;
00342         }
00343         *t++ = *u++;
00344     }
00345     *t = 0;
00346 }
00347 else if ( *s == 't' || *s == 'T' ) {
00348     WORD oldskip = AO.OutSkip, noleadsign;
00349     WORD oldmode = AC.OutputMode;
00350     WORD oldbracket = AO.IsBracket;
00351     WORD oldlength = AC.LineLength;
00352     UBYTE *oldStop = AO.OutStop;
00353     if ( AN.currentTerm ) {
00354         if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
00355         AO.IsBracket = 0;
00356         AO.OutSkip = 1;
00357         AC.OutputMode = 0;
00358         AO.OutFill = AO.OutputLine;
00359         AO.OutStop = AO.OutputLine + AC.LineLength;
00360         *t = 0;
00361         AddToLine((UBYTE *)Out);
00362         if ( *s == 'T' ) noleadsign = 1;
00363         else noleadsign = 0;
00364         if ( WriteInnerTerm(AN.currentTerm, noleadsign) ) Terminate(-1);
00365         t = Out;
00366         u = (char *)AO.OutputLine;
00367         *(AO.OutFill) = 0;
00368         while ( u < (char *) (AO.OutFill) ) *t++ = *u++;
00369         *t = 0;
00370         AO.OutSkip = oldskip;
00371         AC.OutputMode = oldmode;
00372         AO.IsBracket = oldbracket;
00373         AC.LineLength = oldlength;
00374         AO.OutStop = oldStop;
00375     }
00376 }
00377 else if ( *s == 'r' ) {

```

```

00378         WORD oldskip = AO.OutSkip;
00379         WORD oldmode = AC.OutputMode;
00380         WORD oldbracket = AO.IsBracket;
00381         WORD oldlength = AC.LineLength;
00382         UBYTE *oldStop = AO.OutStop;
00383         if ( AN.currentTerm ) {
00384             WORD *tt = AN.currentTerm;
00385             if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
00386             AO.IsBracket = 0;
00387             AO.OutSkip = 1;
00388             AC.OutputMode = 0;
00389             AO.OutFill = AO.OutputLine;
00390             AO.OutStop = AO.OutputLine + AC.LineLength;
00391             *t = 0;
00392             i = *tt;
00393             while ( --i >= 0 ) {
00394                 t = (char *)NumCopy(*tt, (UBYTE *)t);
00395                 tt++;
00396                 if ( t >= stopper ) {
00397                     num = t - Out;
00398                     WriteString(ERROROUT, (UBYTE *)Out, num);
00399                     num = 0; t = Out;
00400                 }
00401                 *t++ = ' '; *t++ = ' ';
00402             }
00403             *t = 0;
00404             AO.OutSkip = oldskip;
00405             AC.OutputMode = oldmode;
00406             AO.IsBracket = oldbracket;
00407             AC.LineLength = oldlength;
00408             AO.OutStop = oldStop;
00409         }
00410     }
00411     else if ( *s == '$' ) {
00412         /*
00413         #[ dollars :
00414         */
00415         WORD oldskip = AO.OutSkip;
00416         WORD oldmode = AC.OutputMode;
00417         WORD oldbracket = AO.IsBracket;
00418         WORD oldlength = AC.LineLength;
00419         UBYTE *oldStop = AO.OutStop;
00420         WORD *term, indsubterm[3], *tt;
00421         WORD value[5], first, num;
00422         if ( *AN.listinprint != DOLLAREXPRESSION ) {
00423             specialerror = 1;
00424         }
00425         else {
00426             DOLLARS d = Dollars + AN.listinprint[1];
00427             #ifdef WITHPTHREADS
00428             int nummodopt, dtype;
00429             dtype = -1;
00430             if ( AS.MultiThreaded ) {
00431                 for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
00432                     if ( AN.listinprint[1] == ModOptdollars[nummodopt].number ) break;
00433                 }
00434                 if ( nummodopt < NumModOptdollars ) {
00435                     dtype = ModOptdollars[nummodopt].type;
00436                     if ( dtype == MODLOCAL ) {
00437                         d = ModOptdollars[nummodopt].dstruct+AT.identity;
00438                     }
00439                     else {
00440                         LOCK(d->pthreadlockread);
00441                     }
00442                 }
00443             }
00444             #endif
00445             AO.IsBracket = 0;
00446             AO.OutSkip = 0;
00447             AC.OutputMode = 0;
00448             AO.OutFill = AO.OutputLine;
00449             AO.OutStop = AO.OutputLine + AC.LineLength;
00450             *t = 0;
00451             AddToLine((UBYTE *)Out);
00452             if ( d->nfactors >= 1 && AN.listinprint[2] == DOLLAREXPR2 ) {
00453                 if ( d->type == 0 ||
00454                     ( d->factors == 0 && d->nfactors != 1 ) ) goto dollarzero;
00455                 num = EvalDoLoopArg(BHEAD AN.listinprint+2,-1);
00456                 if ( num == 0 ) {
00457                     value[0] = 4; value[1] = d->nfactors; value[2] = 1; value[3] = 3; value[4]
00458                     = 0;
00459                     term = value; goto printterms;
00460                 }
00461                 if ( num == 1 && d->nfactors == 1 ) {
00462                     term = d->where;
00463                     if ( *term == 0 ) goto dollarzero;
00464                     goto printterms;

```

```

00464         }
00465         if ( num > d->nfactors ) {
00466             MesPrint("\nFactor number for dollar is too large.");
00467             Terminate(-1);
00468         }
00469         term = d->factors[num-1].where;
00470         if ( term == 0 ) {
00471             if ( d->factors[num-1].value < 0 ) {
00472                 value[0] = 4; value[1] = -d->factors[num-1].value; value[2] = 1;
00473                 value[3] = -3; value[4] = 0;
00474             }
00475             else {
00476                 value[0] = 4; value[1] = d->factors[num-1].value; value[2] = 1;
00477                 value[3] = 3; value[4] = 0;
00478             }
00479             term = value;
00480         }
00481         goto printterms;
00482     }
00483     if ( d->type == DOLTERMS || d->type == DOLNUMBER ) {
00484         term = d->where;
00485         first = 1;
00486         do {
00487             if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
00488             AO.IsBracket = 0;
00489             AO.OutSkip = 1;
00490             AC.OutputMode = 0;
00491             AO.OutFill = AO.OutputLine;
00492             AO.OutStop = AO.OutputLine + AC.LineLength;
00493             *t = 0;
00494             AddToLine((UBYTE *)Out);
00495             if ( WriteInnerTerm(term,first) ) {
00496                 if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00497                 Terminate(-1);
00498             }
00499             first = 0;
00500             t = Out;
00501             u = (char *)AO.OutputLine;
00502             *(AO.OutFill) = 0;
00503             while ( u < (char *)AO.OutFill ) *t++ = *u++;
00504             *t = 0;
00505             AO.OutSkip = oldskip;
00506             AC.OutputMode = oldmode;
00507             AO.IsBracket = oldbracket;
00508             AC.LineLength = oldlength;
00509             AO.OutStop = oldStop;
00510             term += *term;
00511         } while ( *term );
00512         AO.OutSkip = oldskip;
00513     }
00514     else if ( d->type == DOLSUBTERM ) {
00515         tt = d->where;
00516         dosubterm:
00517         if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
00518         AO.IsBracket = 0;
00519         AO.OutSkip = 1;
00520         AC.OutputMode = 0;
00521         AO.OutFill = AO.OutputLine;
00522         AO.OutStop = AO.OutputLine + AC.LineLength;
00523         *t = 0;
00524         AddToLine((UBYTE *)Out);
00525         if ( WriteSubTerm(tt,1) ) {
00526             if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00527             Terminate(-1);
00528         }
00529         t = Out;
00530         u = (char *)AO.OutputLine;
00531         *(AO.OutFill) = 0;
00532         while ( u < (char *)AO.OutFill ) *t++ = *u++;
00533         *t = 0;
00534         AO.OutSkip = oldskip;
00535         AC.OutputMode = oldmode;
00536         AO.IsBracket = oldbracket;
00537         AC.LineLength = oldlength;
00538         AO.OutStop = oldStop;
00539     }
00540     else if ( d->type == DOLUNDEFINED ) {
00541         *t++ = '*'; *t++ = '*'; *t++ = '*'; *t = 0;
00542     }
00543     else if ( d->type == DOLZERO ) {
00544         *t++ = '0'; *t = 0;
00545     }
00546     dollarzero:
00547     else if ( d->type == DOLINDEX ) {
00548         tt = indsubterm; *tt = INDEX;

```

```

00549         tt[1] = 3; tt[2] = d->index;
00550         goto dosubterm;
00551     }
00552     else if ( d->type == DOLARGUMENT ) {
00553         if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
00554         AO.IsBracket = 0;
00555         AO.OutSkip = 1;
00556         AC.OutputMode = 0;
00557         AO.OutFill = AO.OutputLine;
00558         AO.OutStop = AO.OutputLine + AC.LineLength;
00559         *t = 0;
00560         AddToLine((UBYTE *)Out);
00561         WriteArgument(d->where);
00562         t = Out;
00563         u = (char *)AO.OutputLine;
00564         *(AO.OutFill) = 0;
00565         while ( u < (char *) (AO.OutFill) ) *t++ = *u++;
00566         *t = 0;
00567         AO.OutSkip = oldskip;
00568         AC.OutputMode = oldmode;
00569         AO.IsBracket = oldbracket;
00570         AC.LineLength = oldlength;
00571         AO.OutStop = oldStop;
00572     }
00573     else if ( d->type == DOLWILDARGS ) {
00574         tt = d->where;
00575         if ( *tt == 0 ) { tt++;
00576             while ( *tt ) {
00577                 if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
00578                 AO.IsBracket = 0;
00579                 AO.OutSkip = 1;
00580                 AC.OutputMode = 0;
00581                 AO.OutFill = AO.OutputLine;
00582                 AO.OutStop = AO.OutputLine + AC.LineLength;
00583                 *t = 0;
00584                 AddToLine((UBYTE *)Out);
00585                 WriteArgument(tt);
00586                 NEXTARG(tt);
00587                 if ( *tt ) TokenToLine((UBYTE *)",");
00588                 t = Out;
00589                 u = (char *)AO.OutputLine;
00590                 *(AO.OutFill) = 0;
00591                 while ( u < (char *) (AO.OutFill) ) *t++ = *u++;
00592                 *t = 0;
00593                 AO.OutSkip = oldskip;
00594                 AC.OutputMode = oldmode;
00595                 AO.IsBracket = oldbracket;
00596                 AC.LineLength = oldlength;
00597                 AO.OutStop = oldStop;
00598             }
00599         }
00600         else if ( *tt > 0 ) { /* Tensor arguments */
00601             i = *tt++;
00602             while ( --i >= 0 ) {
00603                 indsubterm[0] = INDEX;
00604                 indsubterm[1] = 3;
00605                 indsubterm[2] = *tt++;
00606                 if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
00607                 AO.IsBracket = 0;
00608                 AO.OutSkip = 1;
00609                 AC.OutputMode = 0;
00610                 AO.OutFill = AO.OutputLine;
00611                 AO.OutStop = AO.OutputLine + AC.LineLength;
00612                 *t = 0;
00613                 AddToLine((UBYTE *)Out);
00614                 if ( WriteSubTerm(indsubterm,1) ) Terminate(-1);
00615                 if ( i > 0 ) TokenToLine((UBYTE *)",");
00616                 t = Out;
00617                 u = (char *)AO.OutputLine;
00618                 *(AO.OutFill) = 0;
00619                 while ( u < (char *) (AO.OutFill) ) *t++ = *u++;
00620                 *t = 0;
00621                 AO.OutSkip = oldskip;
00622                 AC.OutputMode = oldmode;
00623                 AO.IsBracket = oldbracket;
00624                 AC.LineLength = oldlength;
00625                 AO.OutStop = oldStop;
00626             }
00627         }
00628     }
00629 #ifdef WITHPTHREADS
00630     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00631 #endif
00632     AN.listinprint += 2;
00633     while ( AN.listinprint[0] == DOLLAREXPR2 ) AN.listinprint += 2;
00634 }
00635 /*

```

```

00636         #] dollars :
00637     */
00638     }
00639 #ifdef WITHPTHREADS
00640     else if ( *s == 'W' ) { /* number of the thread with time */
00641         LONG millitime;
00642         WORD timepart;
00643         t = (char *)NumCopy(identity, (UBYTE *)t);
00644         millitime = TimeCPU(1);
00645         timepart = (WORD)(millitime%1000);
00646         millitime /= 1000;
00647         timepart /= 10;
00648         *t++ = '('; *t = 0;
00649         t = (char *)LongCopy(millitime, (char *)t);
00650         *t++ = '.'; *t = 0;
00651         t = (char *)NumCopy(timepart, (UBYTE *)t);
00652         *t++ = ')'; *t = 0;
00653         if ( t >= stopper ) {
00654             num = t - Out;
00655             WriteString(ERROROUT, (UBYTE *)Out, num);
00656             num = 0; t = Out;
00657         }
00658     }
00659     else if ( *s == 'w' ) { /* number of the thread */
00660         t = (char *)NumCopy(identity, (UBYTE *)t);
00661     }
00662 #elif defined(WITHMPI)
00663     else if ( *s == 'W' ) { /* number of the thread with time */
00664         LONG millitime;
00665         WORD timepart;
00666         t = (char *)NumCopy(PF.me, (UBYTE *)t);
00667         millitime = TimeCPU(1);
00668         timepart = (WORD)(millitime%1000);
00669         millitime /= 1000;
00670         timepart /= 10;
00671         *t++ = '('; *t = 0;
00672         t = (char *)LongCopy(millitime, (char *)t);
00673         *t++ = '.'; *t = 0;
00674         t = (char *)NumCopy(timepart, (UBYTE *)t);
00675         *t++ = ')'; *t = 0;
00676         if ( t >= stopper ) {
00677             num = t - Out;
00678             WriteString(ERROROUT, (UBYTE *)Out, num);
00679             num = 0; t = Out;
00680         }
00681     }
00682     else if ( *s == 'w' ) { /* number of the thread */
00683         t = (char *)NumCopy(PF.me, (UBYTE *)t);
00684     }
00685 #else
00686     else if ( *s == 'w' ) { }
00687     else if ( *s == 'W' ) { }
00688 #endif
00689     else if ( FG.cTable[(int)*s] == 1 ) {
00690         x = *s++ - '0';
00691         while ( FG.cTable[(int)*s] == 1 )
00692             x = 10 * x + *s++ - '0';
00693
00694         if ( *s == 'l' || *s == 'd' ) {
00695             if ( *s == 'l' ) { y = va_arg(ap, LONG); }
00696             else { y = va_arg(ap, int); }
00697             if ( y < 0 ) { y = -y; w = 1; }
00698             else w = 0;
00699             u = t + x;
00700             do { *--u = y%10+'0'; y /= 10; } while ( y && u > t );
00701             if ( w && u > t ) *--u = '-';
00702             while ( --u >= t ) *u = ' ';
00703             t += x;
00704         }
00705         else if ( *s == 's' ) {
00706             u = va_arg(ap, char *);
00707             i = 0;
00708             while ( *u ) { i++; u++; }
00709             if ( i > x ) i = x;
00710             while ( x > i ) { *t++ = ' '; x--; }
00711             t += x;
00712             while ( --i >= 0 ) { *--t = *--u; }
00713             t += x;
00714         }
00715         else if ( *s == 'p' ) {
00716             POSITION *pp;
00717             /*#ifdef __GLIBC_HAVE_LONG_LONG */
00718             off_t ly;
00719             /*
00720             #else
00721                 LONG ly;
00722             #endif

```

```

00723 */
00724         pp = va_arg(ap, POSITION *);
00725         ly = BASEPOSITION(*pp);
00726         u = t + x;
00727         do { *--u = ly%10+'0'; ly /= 10; } while ( ly && u > t );
00728         while ( --u >= t ) *u = ' ';
00729         t += x;
00730     }
00731     else if ( *s == 'i' ) {
00732         w = va_arg(ap, int);
00733         u = t + x;
00734         do { *--u = (char)(w%10+'0'); w /= 10; } while ( u > t );
00735         t += x;
00736     }
00737     else {
00738         w = va_arg(ap, int);
00739         u = t + x;
00740         do { *--u = (char )(w%10+'0'); w /= 10; } while ( w && u > t );
00741         while ( --u >= t ) *u = ' ';
00742         t += x;
00743     }
00744 }
00745 else if ( *s == 'x' ) {
00746     char ccc;
00747     y = va_arg(ap, LONG);
00748     i = 2*sizeof(LONG);
00749     while ( --i > 0 ) {
00750         ccc = ( y >> (i*4) ) & 0xF;
00751         if ( ccc ) break;
00752     }
00753     do {
00754         ccc = ( y >> (i*4) ) & 0xF;
00755         *t++ = hex[(int)ccc];
00756     } while ( --i >= 0 );
00757 }
00758 else if ( *s == '#' ) *t++ = *s;
00759 else if ( *s == '%' ) *t++ = *s;
00760 else if ( *s == 0 ) { *t++ = 0; break; }
00761 else if ( *s == '&' ) {
00762     *t++ = *s;
00763 }
00764 else {
00765     *t++ = '%';
00766     s--;
00767 }
00768 s++;
00769 }
00770 }
00771 num = t - Out;
00772 WriteString(ERROROUT, (UBYTE *)Out, num);
00773 va_end(ap);
00774 if ( specialerror == 1 ) {
00775     MesPrint("!!!Wrong object in Print statement!!!");
00776     MesPrint("!!!Object encountered is of a different type as in the format specifier");
00777 }
00778 AO.OutputLine = oldoutfill;
00779 /*[19apr2004 mt]:*/
00780 WriteFile=OldWrite;
00781 /*:[19apr2004 mt]*/
00782 return(-1);
00783 }
00784
00785 /*
00786     #] MesPrint :
00787     #[ Warning :
00788 */
00789
00790 void Warning(char *s)
00791 {
00792     iswarning = 1;
00793     if ( AC.WarnFlag ) MesPrint("&Warning: %s",s);
00794     iswarning = 0;
00795 }
00796
00797 /*
00798     #] Warning :
00799     #[ HighWarning :
00800 */
00801
00802 void HighWarning(char *s)
00803 {
00804     iswarning = 1;
00805     if ( AC.WarnFlag >= 2 ) MesPrint("&Warning: %s",s);
00806     iswarning = 0;
00807 }
00808
00809 /*

```

```

00810         #] HighWarning :
00811         #[ MesCall :
00812 */
00813
00814 int MesCall(char *s)
00815 {
00816     return(MesPrint((char *)"Called from %s",s));
00817 }
00818
00819 /*
00820         #] MesCall :
00821         #[ MesCerr :
00822 */
00823
00824 int MesCerr(char *s, UBYTE *t)
00825 {
00826     UBYTE *u, c;
00827     WORD i = 11;
00828     u = t;
00829     while ( *u && --i >= 0 ) u--;
00830     u++;
00831     c = *++t;
00832     *t = 0;
00833     MesPrint("&Illegal %s: %s",s,u);
00834     *t = c;
00835     return(-1);
00836 }
00837
00838 /*
00839         #] MesCerr :
00840         #[ MesComp :
00841 */
00842
00843 int MesComp(char *s, UBYTE *p, UBYTE *q)
00844 {
00845     UBYTE c;
00846     c = *++q; *q = 0;
00847     MesPrint("&%s: %s",s,p);
00848     *q = c;
00849     return(-1);
00850 }
00851
00852 /*
00853         #] MesComp :
00854         #[ PrintTerm :
00855 */
00856
00857 void PrintTerm(WORD *term, char *where)
00858 {
00859     UBYTE OutBuf[140];
00860     WORD *t, x;
00861     int i;
00862     AO.OutFill = AO.OutputLine = OutBuf;
00863     t = term;
00864     AO.OutSkip = 3;
00865     FiniLine();
00866     TokenToLine((UBYTE *)where);
00867     TokenToLine((UBYTE *)" ": "");
00868     i = *t;
00869     while ( --i >= 0 ) {
00870         x = *t++;
00871         if ( x < 0 ) {
00872             x = -x;
00873             TokenToLine((UBYTE *)" "-);
00874         }
00875         TalToLine((UWORD)(x));
00876         TokenToLine((UBYTE *)" " ");
00877     }
00878     AO.OutSkip = 0;
00879     FiniLine();
00880 }
00881
00882 /*
00883         #] PrintTerm :
00884         #[ PrintTermC :
00885 */
00886
00887 void PrintTermC(WORD *term, char *where)
00888 {
00889     UBYTE OutBuf[140];
00890     WORD *t, x;
00891     int i;
00892     if ( *term >= 0 ) {
00893         PrintTerm(term,where);
00894         return;
00895     }
00896     AO.OutFill = AO.OutputLine = OutBuf;

```



```

00897     t = term;
00898     AO.OutSkip = 3;
00899     FiniLine();
00900     TokenToLine((UBYTE *)where);
00901     TokenToLine((UBYTE *)": ");
00902     i = t[1]+2;
00903     while ( --i >= 0 ) {
00904         x = *t++;
00905         if ( x < 0 ) {
00906             x = -x;
00907             TokenToLine((UBYTE *)"-");
00908         }
00909         TalToLine((UWORD)(x));
00910         TokenToLine((UBYTE *)" ");
00911     }
00912     AO.OutSkip = 0;
00913     FiniLine();
00914 }
00915
00916 /*
00917     #] PrintTermC :
00918     #[ PrintSubTerm :
00919 */
00920
00921 void PrintSubTerm(WORD *term, char *where)
00922 {
00923     UBYTE OutBuf[140];
00924     WORD *t;
00925     int i;
00926     AO.OutFill = AO.OutputLine = OutBuf;
00927     t = term;
00928     AO.OutSkip = 3;
00929     FiniLine();
00930     TokenToLine((UBYTE *)where);
00931     TokenToLine((UBYTE *)": ");
00932     i = t[1];
00933     while ( --i >= 0 ) { TalToLine((UWORD)(*t++)); TokenToLine((UBYTE *)" "); }
00934     AO.OutSkip = 0;
00935     FiniLine();
00936 }
00937
00938 /*
00939     #] PrintSubTerm :
00940     #[ PrintWords :
00941 */
00942
00943 void PrintWords(WORD *buffer, LONG number)
00944 {
00945     UBYTE OutBuf[140];
00946     WORD *t;
00947     AO.OutFill = AO.OutputLine = OutBuf;
00948     t = buffer;
00949     AO.OutSkip = 3;
00950     FiniLine();
00951     while ( --number >= 0 ) { TalToLine((UWORD)(*t++)); TokenToLine((UBYTE *)" "); }
00952     AO.OutSkip = 0;
00953     FiniLine();
00954 }
00955
00956 /*
00957     #] PrintWords :
00958     #[ PrintSeq :
00959 */
00960
00961 void PrintSeq(WORD *a, char *text)
00962 {
00963     MesPrint(" %s:", text);
00964     while ( *a ) {
00965         MesPrint("      %a", a[0], a);
00966         a += *a;
00967     }
00968 }
00969
00970 /*
00971     #] PrintSeq :
00972     #[ exit :
00973 */

```

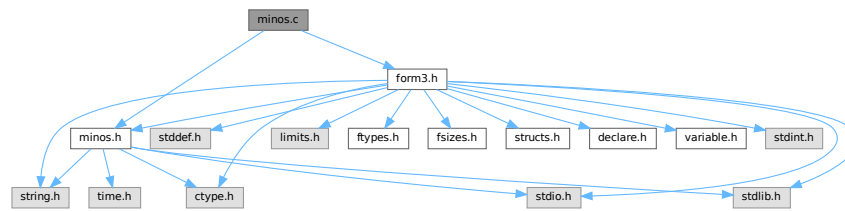
## 4.71 minos.c File Reference

```

#include "form3.h"
#include "minos.h"

```

Include dependency graph for minos.c:



## Macros

- `#define` [CFD](#)(y, s, type, x, j)
- `#define` [CTD](#)(y, s, type, x, j)

## Functions

- `int` [minosread](#) (`FILE *f`, `char *buffer`, `MLONG size`)
- `int` [minoswrite](#) (`FILE *f`, `char *buffer`, `MLONG size`)
- `char *` [str\\_dup](#) (`char *str`)
- `void` [convertblock](#) (`INDEXBLOCK *in`, `INDEXBLOCK *out`, `int mode`)
- `void` [convertnamesblock](#) (`NAMESBLOCK *in`, `NAMESBLOCK *out`, `int mode`)
- `void` [convertiniinfo](#) (`INIINFO *in`, `INIINFO *out`, `int mode`)
- `FILE *` [LocateBase](#) (`char **name`, `char **newname`, `char *iomode`)
- `int` [ReadIndex](#) (`DBASE *d`)
- `int` [WriteIndexBlock](#) (`DBASE *d`, `MLONG num`)
- `int` [WriteNamesBlock](#) (`DBASE *d`, `MLONG num`)
- `int` [WriteIndex](#) (`DBASE *d`)
- `int` [WriteNilInfo](#) (`DBASE *d`)
- `int` [ReadNilInfo](#) (`DBASE *d`)
- `DBASE *` [GetDbase](#) (`char *filename`, `MLONG rwmode`)
- `DBASE *` [NewDbase](#) (`char *name`, `MLONG number`)
- `void` [FreeTableBase](#) (`DBASE *d`)
- `int` [ComposeTableNames](#) (`DBASE *d`)
- `DBASE *` [OpenDbase](#) (`char *filename`)
- `MLONG` [AddTableName](#) (`DBASE *d`, `char *name`, `TABLES T`)
- `MLONG` [GetTableName](#) (`DBASE *d`, `char *name`)
- `int` [PutTableNames](#) (`DBASE *d`)
- `int` [AddToIndex](#) (`DBASE *d`, `MLONG number`)
- `MLONG` [AddObject](#) (`DBASE *d`, `MLONG tablename`, `char *arguments`, `char *rhs`)
- `MLONG` [FindTableNumber](#) (`DBASE *d`, `char *name`)
- `int` [WriteObject](#) (`DBASE *d`, `MLONG tablename`, `char *arguments`, `char *rhs`, `MLONG number`)
- `char *` [ReadObject](#) (`DBASE *d`, `MLONG tablename`, `char *arguments`)
- `char *` [ReadijObject](#) (`DBASE *d`, `MLONG i`, `MLONG j`, `char *arguments`)
- `int` [ExistsObject](#) (`DBASE *d`, `MLONG tablename`, `char *arguments`)
- `int` [DeleteObject](#) (`DBASE *d`, `MLONG tablename`, `char *arguments`)

## Variables

- `int` [withoutflush](#) = 0

### 4.71.1 Detailed Description

These are the low level functions for the database part of the tablebases. These routines have been copied (and then adapted) from the minos database program. This file goes together with [minos.h](#)

Definition in file [minos.c](#).

### 4.71.2 Macro Definition Documentation

#### 4.71.2.1 CFD

```
#define CFD(  
    y,  
    s,  
    type,  
    x,  
    j )
```

**Value:**

```
for(x=0, j=0; j<((int)sizeof(type)); j++) \  
{x=(x<<8)+((s++)&0x00FF); y=x;
```

Definition at line 55 of file [minos.c](#).

#### 4.71.2.2 CTD

```
#define CTD(  
    y,  
    s,  
    type,  
    x,  
    j )
```

**Value:**

```
x=y; for(j=sizeof(type)-1; j>=0; j--) {s[j]=x&0xFF; \  
x>=8;} s += sizeof(type);
```

Definition at line 57 of file [minos.c](#).

### 4.71.3 Function Documentation

#### 4.71.3.1 minosread()

```
int minosread (  
    FILE * f,  
    char * buffer,  
    MLONG size )
```

Definition at line 66 of file [minos.c](#).

#### 4.71.3.2 minoswrite()

```
int minoswrite (
    FILE * f,
    char * buffer,
    MLONG size )
```

Definition at line 83 of file [minos.c](#).

#### 4.71.3.3 str\_dup()

```
char * str_dup (
    char * str )
```

Definition at line 101 of file [minos.c](#).

#### 4.71.3.4 convertblock()

```
void convertblock (
    INDEXBLOCK * in,
    INDEXBLOCK * out,
    int mode )
```

Definition at line 120 of file [minos.c](#).

#### 4.71.3.5 convertnamesblock()

```
void convertnamesblock (
    NAMESBLOCK * in,
    NAMESBLOCK * out,
    int mode )
```

Definition at line 171 of file [minos.c](#).

#### 4.71.3.6 convertiniinfo()

```
void convertiniinfo (
    INIINFO * in,
    INIINFO * out,
    int mode )
```

Definition at line 197 of file [minos.c](#).

#### 4.71.3.7 LocateBase()

```
FILE * LocateBase (
    char ** name,
    char ** newname,
    char * iomode )
```

Definition at line 225 of file [minos.c](#).

#### 4.71.3.8 ReadIndex()

```
int ReadIndex (
    DBASE * d )
```

Definition at line 288 of file [minos.c](#).

#### 4.71.3.9 WriteIndexBlock()

```
int WriteIndexBlock (
    DBASE * d,
    MLONG num )
```

Definition at line 399 of file [minos.c](#).

#### 4.71.3.10 WriteNamesBlock()

```
int WriteNamesBlock (
    DBASE * d,
    MLONG num )
```

Definition at line 420 of file [minos.c](#).

#### 4.71.3.11 WriteIndex()

```
int WriteIndex (
    DBASE * d )
```

Definition at line 443 of file [minos.c](#).

#### 4.71.3.12 WriteIniInfo()

```
int WriteIniInfo (
    DBASE * d )
```

Definition at line 506 of file [minos.c](#).

#### 4.71.3.13 ReadIniInfo()

```
int ReadIniInfo (
    DBASE * d )
```

Definition at line 523 of file [minos.c](#).

#### 4.71.3.14 GetDbase()

```
DBASE * GetDbase (
    char * filename,
    MLONG rwmode )
```

Definition at line 546 of file [minos.c](#).

#### 4.71.3.15 NewDbase()

```
DBASE * NewDbase (
    char * name,
    MLONG number )
```

Definition at line 602 of file [minos.c](#).

#### 4.71.3.16 FreeTableBase()

```
void FreeTableBase (
    DBASE * d )
```

Definition at line 739 of file [minos.c](#).

#### 4.71.3.17 ComposeTableNames()

```
int ComposeTableNames (
    DBASE * d )
```

Definition at line 771 of file [minos.c](#).

#### 4.71.3.18 OpenDbase()

```
DBASE * OpenDbase (
    char * filename )
```

Definition at line 820 of file [minos.c](#).

#### 4.71.3.19 AddTableName()

```
MLONG AddTableName (
    DBASE * d,
    char * name,
    TABLES T )
```

Definition at line 854 of file [minos.c](#).

#### 4.71.3.20 GetTableName()

```
MLONG GetTableName (
    DBASE * d,
    char * name )
```

Definition at line 937 of file [minos.c](#).

#### 4.71.3.21 PutTableNames()

```
int PutTableNames (
    DBASE * d )
```

Definition at line 969 of file [minos.c](#).

#### 4.71.3.22 AddToIndex()

```
int AddToIndex (
    DBASE * d,
    MLONG number )
```

Definition at line 1065 of file [minos.c](#).

#### 4.71.3.23 AddObject()

```
MLONG AddObject (
    DBASE * d,
    MLONG tablename,
    char * arguments,
    char * rhs )
```

Definition at line 1152 of file [minos.c](#).

#### 4.71.3.24 FindTableNumber()

```
MLONG FindTableNumber (
    DBASE * d,
    char * name )
```

Definition at line 1166 of file [minos.c](#).

#### 4.71.3.25 WriteObject()

```
int WriteObject (
    DBASE * d,
    MLONG tablename,
    char * arguments,
    char * rhs,
    MLONG number )
```

Definition at line 1194 of file [minos.c](#).

#### 4.71.3.26 ReadObject()

```
char * ReadObject (
    DBASE * d,
    MLONG tablenumber,
    char * arguments )
```

Definition at line [1294](#) of file [minos.c](#).

#### 4.71.3.27 ReadijObject()

```
char * ReadijObject (
    DBASE * d,
    MLONG i,
    MLONG j,
    char * arguments )
```

Definition at line [1375](#) of file [minos.c](#).

#### 4.71.3.28 ExistsObject()

```
int ExistsObject (
    DBASE * d,
    MLONG tablenumber,
    char * arguments )
```

Definition at line [1427](#) of file [minos.c](#).

#### 4.71.3.29 DeleteObject()

```
int DeleteObject (
    DBASE * d,
    MLONG tablenumber,
    char * arguments )
```

Definition at line [1459](#) of file [minos.c](#).

### 4.71.4 Variable Documentation

#### 4.71.4.1 withoutflush

```
int withoutflush = 0
```

Definition at line [45](#) of file [minos.c](#).



## 4.72 minos.c

[Go to the documentation of this file.](#)

```

00001
00007 /* #[ License : */
00008 /*
00009  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00010  *   When using this file you are requested to refer to the publication
00011  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00012  *   This is considered a matter of courtesy as the development was paid
00013  *   for by FOM the Dutch physics granting agency and we would like to
00014  *   be able to track its scientific use to convince FOM of its value
00015  *   for the community.
00016  *
00017  *   This file is part of FORM.
00018  *
00019  *   FORM is free software: you can redistribute it and/or modify it under the
00020  *   terms of the GNU General Public License as published by the Free Software
00021  *   Foundation, either version 3 of the License, or (at your option) any later
00022  *   version.
00023  *
00024  *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00025  *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00026  *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00027  *   details.
00028  *
00029  *   You should have received a copy of the GNU General Public License along
00030  *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00031  */
00032 /* #[ License : */
00033 /*
00034  *   #[ Includes :
00035  *
00036  *   File contains the lowlevel routines for the management of primitive
00037  *   database-like files. The structures are explained in the file manage.h
00038  *   Original file for minos made by J.Vermaseren, april-1994.
00039  *
00040  *   Note: the minos primitives for writing are never invoked in parallel.
00041  */
00042
00043 #include "form3.h"
00044 #include "minos.h"
00045 int withoutflush = 0;
00046
00047 /*
00048  *   #[ Includes :
00049  *   #[ Variables :
00050  */
00051
00052 static INDEXBLOCK scratchblock;
00053 static NAMESBLOCK scratchnamesblock;
00054
00055 #define CFD(y,s,type,x,j) for(x=0,j=0;j<((int)sizeof(type));j++) \
00056     {x=(x<<8)+((*(s++)&0x00FF));} y=x;
00057 #define CTD(y,s,type,x,j) x=y;for(j=sizeof(type)-1;j>=0;j--){s[j]=x&0xFF; \
00058     x>>=8;} s += sizeof(type);
00059
00060 /*
00061  *   #[ Variables :
00062  *   #[ Utilities :
00063  *   #[ minosread :
00064  */
00065
00066 int minosread(FILE *f,char *buffer,MLONG size)
00067 {
00068     MLONG x;
00069     while ( size > 0 ) {
00070         x = fread(buffer,sizeof(char),size,f);
00071         if ( x <= 0 ) return(-1);
00072         buffer += x;
00073         size -= x;
00074     }
00075     return(0);
00076 }
00077
00078 /*
00079  *   #[ minosread :
00080  *   #[ minoswrite :
00081  */
00082
00083 int minoswrite(FILE *f,char *buffer,MLONG size)
00084 {
00085     MLONG x;
00086     while ( size > 0 ) {
00087         x = fwrite(buffer,sizeof(char),size,f);

```

```

00088         if ( x <= 0 ) return(-1);
00089         buffer += x;
00090         size -= x;
00091     }
00092     if ( withoutflush == 0 ) fflush(f);
00093     return(0);
00094 }
00095
00096 /*
00097     #] minoswrite :
00098     #[ str_dup :
00099 */
00100
00101 char *str_dup(char *str)
00102 {
00103     char *s, *t;
00104     int i;
00105     s = str;
00106     while ( *s ) s++;
00107     i = s - str + 1;
00108     if ( ( s = (char *)Malloc1((size_t)i,"a string copy") ) == 0 ) return(0);
00109     t = s;
00110     while ( *str ) *t++ = *str++;
00111     *t = 0;
00112     return(s);
00113 }
00114
00115 /*
00116     #] str_dup :
00117     #[ convertblock :
00118 */
00119
00120 void convertblock(INDEXBLOCK *in,INDEXBLOCK *out,int mode)
00121 {
00122     char *s,*t;
00123     MLONG i, x;
00124     int j;
00125     OBJECTS *obj;
00126     switch ( mode ) {
00127     case TODISK:
00128         s = (char *)out;
00129         CTD(in->flags,s,MLONG,x,j)
00130         CTD(in->previousblock,s,MLONG,x,j)
00131         CTD(in->position,s,MLONG,x,j)
00132         for ( i = 0, obj = in->objects; i < NUMOBJECTS; i++, obj++ ) {
00133             CTD(obj->position,s,MLONG,x,j)
00134             CTD(obj->size,s,MLONG,x,j)
00135             CTD(obj->date,s,MLONG,x,j)
00136             CTD(obj->tablenumber,s,MLONG,x,j)
00137             CTD(obj->uncompressed,s,MLONG,x,j)
00138             CTD(obj->spare1,s,MLONG,x,j)
00139             CTD(obj->spare2,s,MLONG,x,j)
00140             CTD(obj->spare3,s,MLONG,x,j)
00141             t = obj->element;
00142             for ( j = 0; j < ELEMENTSIZE; j++ ) *s++ = *t++;
00143         }
00144         break;
00145     case FROMDISK:
00146         s = (char *)in;
00147         CFD(out->flags,s,MLONG,x,j)
00148         CFD(out->previousblock,s,MLONG,x,j)
00149         CFD(out->position,s,MLONG,x,j)
00150         for ( i = 0, obj = out->objects; i < NUMOBJECTS; i++, obj++ ) {
00151             CFD(obj->position,s,MLONG,x,j)
00152             CFD(obj->size,s,MLONG,x,j)
00153             CFD(obj->date,s,MLONG,x,j)
00154             CFD(obj->tablenumber,s,MLONG,x,j)
00155             CFD(obj->uncompressed,s,MLONG,x,j)
00156             CFD(obj->spare1,s,MLONG,x,j)
00157             CFD(obj->spare2,s,MLONG,x,j)
00158             CFD(obj->spare3,s,MLONG,x,j)
00159             t = obj->element;
00160             for ( j = 0; j < ELEMENTSIZE; j++ ) *t++ = *s++;
00161         }
00162         break;
00163     }
00164 }
00165
00166 /*
00167     #] convertblock :
00168     #[ convertnamesblock :
00169 */
00170
00171 void convertnamesblock(NAMESBLOCK *in,NAMESBLOCK *out,int mode)
00172 {
00173     char *s;
00174     MLONG x;

```

```

00175     int j;
00176     switch ( mode ) {
00177     case TODISK:
00178         s = (char *)out;
00179         CTD(in->previousblock,s,MLONG,x,j)
00180         CTD(in->position,s,MLONG,x,j)
00181         for ( j = 0; j < NAMETABLESIZE; j++ ) out->names[j] = in->names[j];
00182         break;
00183     case FROMDISK:
00184         s = (char *)in;
00185         CFD(out->previousblock,s,MLONG,x,j)
00186         CFD(out->position,s,MLONG,x,j)
00187         for ( j = 0; j < NAMETABLESIZE; j++ ) out->names[j] = in->names[j];
00188         break;
00189     }
00190 }
00191
00192 /*
00193     #] convertnamesblock :
00194     #[ convertiniinfo :
00195 */
00196
00197 void convertiniinfo(INIINFO *in,INIINFO *out,int mode)
00198 {
00199     char *s;
00200     MLONG i, x, *y;
00201     int j;
00202     switch ( mode ) {
00203     case TODISK:
00204         s = (char *)out; y = (MLONG *)in;
00205         for ( i = sizeof(INIINFO)/sizeof(MLONG); i > 0; i-- ) {
00206             CTD(*y,s,MLONG,x,j)
00207             y++;
00208         }
00209         break;
00210     case FROMDISK:
00211         s = (char *)in; y = (MLONG *)out;
00212         for ( i = sizeof(INIINFO)/sizeof(MLONG); i > 0; i-- ) {
00213             CFD(*y,s,MLONG,x,j)
00214             y++;
00215         }
00216         break;
00217     }
00218 }
00219
00220 /*
00221     #] convertiniinfo :
00222     #[ LocateBase :
00223 */
00224
00225 FILE *LocateBase(char **name, char **newname, char *iomode)
00226 {
00227     FILE *handle;
00228     int namesize, i;
00229     UBYTE *s, *to, *u1, *u2, *indir;
00230
00231     if ( ( handle = fopen(*name,iomode) ) != 0 ) {
00232         *newname = (char *)strDup1((UBYTE *)(*name),"LocateBase");
00233         return(handle);
00234     }
00235     namesize = 2; s = (UBYTE *)(*name);
00236     while ( *s ) { s++; namesize++; }
00237     indir = AM.IncDir;
00238     if ( indir ) {
00239         s = indir; i = 0;
00240         while ( *s ) { s++; i++; }
00241         *newname = (char *)Malloca1(namesize+i,"LocateBase");
00242         s = indir; to = (UBYTE *)(*newname);
00243         while ( *s ) *to++ = *s++;
00244         if ( to > (UBYTE *)(*newname) && to[-1] != SEPARATOR ) *to++ = SEPARATOR;
00245         s = (UBYTE *)(*name);
00246         while ( *s ) *to++ = *s++;
00247         *to = 0;
00248         if ( ( handle = fopen(*newname,iomode) ) != 0 ) {
00249             return(handle);
00250         }
00251         M_free(*newname,"LocateBase, indir/file");
00252     }
00253     if ( AM.Path ) {
00254         u1 = AM.Path;
00255         while ( *u1 ) {
00256             u2 = u1; i = 0;
00257             while ( *u1 && *u1 != ':' ) {
00258                 if ( *u1 == '\\\\' ) u1++;
00259                 u1++; i++;
00260             }
00261             *newname = (char *)Malloca1(namesize+i,"LocateBase");

```

```

00262         s = u2; to = (UBYTE *) (*newname);
00263         while ( s < u1 ) {
00264             if ( *s == '\\\\' ) s++;
00265             *to++ = *s++;
00266         }
00267         if ( to > (UBYTE *) (*newname) && to[-1] != SEPARATOR ) *to++ = SEPARATOR;
00268         s = (UBYTE *) (*name);
00269         while ( *s ) *to++ = *s++;
00270         *to = 0;
00271         if ( ( handle = fopen(*newname,iomode) ) != 0 ) {
00272             return(handle);
00273         }
00274         M_free(*newname,"LocateBase Path/file");
00275         if ( *u1 ) u1++;
00276     }
00277 }
00278 /* Error1("LocateBase: Cannot find file",*name); */
00279 return(0);
00280 }
00281
00282 /*
00283     #] LocateBase :
00284     #] Utilities :
00285     #[ ReadIndex :
00286 */
00287
00288 int ReadIndex(DBASE *d)
00289 {
00290     MLONG i;
00291     INDEXBLOCK **ib;
00292     NAMESBLOCK **ina;
00293     MLONG position, size;
00294 /*
00295     Allocate the pieces one by one (makes it easier to give it back)
00296 */
00297     if ( d->info.numberofindexblocks <= 0 ) return(0);
00298     if ( sizeof(INDEXBLOCK)*d->info.numberofindexblocks > MAXINDEXSIZE ) {
00299         MesPrint("We need more than %ld bytes for the index.\n",MAXINDEXSIZE);
00300         MesPrint("The file %s may not be a proper database\n",d->name);
00301         return(-1);
00302     }
00303     size = sizeof(INDEXBLOCK *)*d->info.numberofindexblocks;
00304     if ( ( ib = (INDEXBLOCK **)Malloca1(size,"tb,index") ) == 0 ) return(-1);
00305     for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
00306         if ( ( ib[i] = (INDEXBLOCK *)Malloca1(sizeof(INDEXBLOCK),"index block") ) == 0 ) {
00307             for ( --i; i >= 0; i-- ) M_free(ib[i],"tb,indexblock");
00308             M_free(ib,"tb,index");
00309             return(-1);
00310         }
00311     }
00312     size = sizeof(NAMESBLOCK *)*d->info.numberofnamesblocks;
00313     if ( ( ina = (NAMESBLOCK **)Malloca1(size,"tb,indexnames") ) == 0 ) return(-1);
00314     for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
00315         if ( ( ina[i] = (NAMESBLOCK *)Malloca1(sizeof(NAMESBLOCK),"index names block") ) == 0 ) {
00316             for ( --i; i >= 0; i-- ) M_free(ina[i],"index names block");
00317             M_free(ina,"tb,indexnames");
00318             for ( i = 0; i < d->info.numberofindexblocks; i++ ) M_free(ib[i],"tb,indexblock");
00319             M_free(ib,"tb,index");
00320             return(-1);
00321         }
00322     }
00323 /*
00324     Read the index blocks, from the back to the front. The links are only
00325     reliable that way.
00326 */
00327     position = d->info.lastindexblock;
00328     for ( i = d->info.numberofindexblocks - 1; i >= 0; i-- ) {
00329         fseek(d->handle,position,SEEK_SET);
00330         if ( minosread(d->handle,(char *)&scratchblock,sizeof(INDEXBLOCK)) ) {
00331             MesPrint("Error while reading file %s\n",d->name);
00332             thisiswrong:
00333             for ( i = 0; i < d->info.numberofnamesblocks; i++ ) M_free(ina[i],"index names block");
00334             M_free(ina,"tb,indexnames");
00335             for ( i = 0; i < d->info.numberofindexblocks; i++ ) M_free(ib[i],"tb,indexblock");
00336             M_free(ib,"tb,index");
00337             return(-1);
00338         }
00339         convertblock(&scratchblock,ib[i],FROMDISK);
00340         if ( ib[i]->position != position ||
00341             ( ib[i]->previousblock <= 0 && i > 0 ) ) {
00342             MesPrint("File %s has inconsistent contents\n",d->name);
00343             goto thisiswrong;
00344         }
00345         position = ib[i]->previousblock;
00346     }
00347     d->info.firstindexblock = ib[0]->position;
00348     for ( i = 0; i < d->info.numberofindexblocks; i++ ) {

```

```

00349         ib[i]->flags &= MCLEANFLAG;
00350     }
00351     /*
00352     Read the names blocks, from the back to the front. The links are only
00353     reliable that way.
00354     */
00355     position = d->info.lastnameblock;
00356     for ( i = d->info.numberofnamesblocks - 1; i >= 0; i-- ) {
00357         fseek(d->handle,position,SEEK_SET);
00358         if ( minosread(d->handle,(char *)(&scratchnamesblock),sizeof(NAMESBLOCK)) ) {
00359             MesPrint("Error while reading file %s\n",d->name);
00360             goto thisiswrong;
00361         }
00362         convertnamesblock(&scratchnamesblock,ina[i],FROMDISK);
00363         if ( ina[i]->position != position ||
00364             ( ina[i]->previousblock <= 0 && i > 0 ) ) {
00365             MesPrint("File %s has inconsistent contents\n",d->name);
00366             goto thisiswrong;
00367         }
00368         position = ina[i]->previousblock;
00369     }
00370     d->info.firstnameblock = ina[0]->position;
00371     /*
00372     Give the old info back to the system.
00373     */
00374     if ( d->iblocks ) {
00375         for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
00376             if ( d->iblocks[i] ) M_free(d->iblocks[i],"d->iblocks[i]");
00377         }
00378         M_free(d->iblocks,"d->iblocks");
00379     }
00380     if ( d->nblocks ) {
00381         for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
00382             if ( d->nblocks[i] ) M_free(d->nblocks[i],"d->nblocks[i]");
00383         }
00384         M_free(d->nblocks,"d->nblocks");
00385     }
00386     /*
00387     And substitute the new blocks
00388     */
00389     d->iblocks = ib;
00390     d->nblocks = ina;
00391     return(0);
00392 }
00393
00394 /*
00395 #] ReadIndex :
00396 #[ WriteIndexBlock :
00397 */
00398
00399 int WriteIndexBlock(DBASE *d,MLONG num)
00400 {
00401     if ( num >= d->info.numberofindexblocks ) {
00402         MesPrint("Illegal number specified for number of index blocks\n");
00403         return(-1);
00404     }
00405     fseek(d->handle,d->iblocks[num]->position,SEEK_SET);
00406     convertblock(d->iblocks[num],&scratchblock,TODISK);
00407     if ( minoswrite(d->handle,(char *)(&scratchblock),sizeof(INDEXBLOCK)) ) {
00408         MesPrint("Error while writing an index block in file %s\n",d->name);
00409         MesPrint("File may be unreliable now\n");
00410         return(-1);
00411     }
00412     return(0);
00413 }
00414
00415 /*
00416 #] WriteIndexBlock :
00417 #[ WriteNamesBlock :
00418 */
00419
00420 int WriteNamesBlock(DBASE *d,MLONG num)
00421 {
00422     if ( num >= d->info.numberofnamesblocks ) {
00423         MesPrint("Illegal number specified for number of names blocks\n");
00424         return(-1);
00425     }
00426     fseek(d->handle,d->nblocks[num]->position,SEEK_SET);
00427     convertnamesblock(d->nblocks[num],&scratchnamesblock,TODISK);
00428     if ( minoswrite(d->handle,(char *)(&scratchnamesblock),sizeof(NAMESBLOCK)) ) {
00429         MesPrint("Error while writing a names block in file %s\n",d->name);
00430         MesPrint("File may be unreliable now\n");
00431         return(-1);
00432     }
00433     return(0);
00434 }
00435

```

```

00436 /*
00437     #] WriteNamesBlock :
00438     #[ WriteIndex :
00439
00440     Problem here is to get the links right.
00441 */
00442
00443 int WriteIndex(DBASE *d)
00444 {
00445     MLONG i, position;
00446     if ( d->iblocks == 0 ) return(0);
00447     if ( d->nblocks == 0 ) return(0);
00448     for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
00449         if ( d->iblocks[i] == 0 ) {
00450             MesPrint("Error: unassigned index blocks. Cannot write\n");
00451             return(-1);
00452         }
00453     }
00454     for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
00455         if ( d->nblocks[i] == 0 ) {
00456             MesPrint("Error: unassigned names blocks. Cannot write\n");
00457             return(-1);
00458         }
00459     }
00460     d->info.lastindexblock = -1;
00461     for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
00462         position = d->iblocks[i]->position;
00463         if ( position <= 0 ) {
00464             fseek(d->handle,0,SEEK_END);
00465             position = ftell(d->handle);
00466             d->iblocks[i]->position = position;
00467             if ( i <= 0 ) d->iblocks[i]->previousblock = -1;
00468             else d->iblocks[i]->previousblock = d->iblocks[i-1]->position;
00469         }
00470         else fseek(d->handle,position,SEEK_SET);
00471         convertblock(d->iblocks[i],&scratchblock,TODISK);
00472         if ( minoswrite(d->handle,(char *)(&scratchblock),sizeof(INDEXBLOCK)) ) {
00473             MesPrint("Error while writing index of file %s",d->name);
00474             d->iblocks[i]->position = -1;
00475             return(-1);
00476         }
00477         d->info.lastindexblock = position;
00478     }
00479     d->info.lastnameblock = -1;
00480     for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
00481         position = d->nblocks[i]->position;
00482         if ( position <= 0 ) {
00483             fseek(d->handle,0,SEEK_END);
00484             position = ftell(d->handle);
00485             d->nblocks[i]->position = position;
00486             if ( i <= 0 ) d->nblocks[i]->previousblock = -1;
00487             else d->nblocks[i]->previousblock = d->nblocks[i-1]->position;
00488         }
00489         else fseek(d->handle,position,SEEK_SET);
00490         convertnamesblock(d->nblocks[i],&scratchnamesblock,TODISK);
00491         if ( minoswrite(d->handle,(char *)(&scratchnamesblock),sizeof(NAMESBLOCK)) ) {
00492             MesPrint("Error while writing index of file %s",d->name);
00493             d->nblocks[i]->position = -1;
00494             return(-1);
00495         }
00496         d->info.lastnameblock = position;
00497     }
00498     return(0);
00499 }
00500
00501 /*
00502     #] WriteIndex :
00503     #[ WriteIniInfo :
00504 */
00505
00506 int WriteIniInfo(DBASE *d)
00507 {
00508     INIINFO inf;
00509     fseek(d->handle,0,SEEK_SET);
00510     convertiniinfo(&(d->info),&inf,TODISK);
00511     if ( minoswrite(d->handle,(char *)(&inf),sizeof(INIINFO)) ) {
00512         MesPrint("Error while writing masterindex of file %s",d->name);
00513         return(-1);
00514     }
00515     return(0);
00516 }
00517
00518 /*
00519     #] WriteIniInfo :
00520     #[ ReadIniInfo :
00521 */
00522

```

```

00523 int ReadIniInfo(DBASE *d)
00524 {
00525     INIINFO inf;
00526     fseek(d->handle,0,SEEK_SET);
00527     if ( minosread(d->handle,(char *)(&inf),sizeof(INIINFO)) ) {
00528         MesPrint("Error while reading masterindex of file %s",d->name);
00529         return(-1);
00530     }
00531     convertiniinfo(&inf,&(d->info),FROMDISK);
00532     if ( d->info.entriesinindex < 0
00533         || d->info.numberofindexblocks < 0
00534         || d->info.lastindexblock < 0 ) {
00535         MesPrint("The file %s is not a proper database\n",d->name);
00536         return(-1);
00537     }
00538     return(0);
00539 }
00540
00541 /*
00542 #] ReadIniInfo :
00543 #[ GetDbase :
00544 */
00545
00546 DBASE *GetDbase(char *filename, MLONG rwmode)
00547 {
00548     FILE *f;
00549     DBASE *d;
00550     char *newname;
00551     if ( rwmode == 0 ) {
00552         if ( ( f = LocateBase(&filename,&newname,"rb") ) == 0 ) {
00553
00554             MesPrint("&Trying to open non-existent TableBase in readonly mode: %s", filename);
00555             Terminate(-1);
00556         }
00557     } else {
00558         if ( ( f = LocateBase(&filename,&newname,"r+b") ) == 0 ) {
00559
00560             return(NewDbase(filename,0));
00561         }
00562     }
00563
00564     /* setbuf(f,0); */
00565     d = (DBASE *)From0List(&(AC.TableBaseList));
00566     d->mode = 0;
00567     d->tablenamessize = 0;
00568     d->topnumber = 0;
00569     d->tablenamefill = 0;
00570     d->iblocks = 0;
00571     d->nblocks = 0;
00572     d->tablenames = 0;
00573     d->rwmode = rwmode;
00574
00575     d->info.entriesinindex = 0;
00576     d->info.numberofindexblocks = 0;
00577     d->info.firstindexblock = 0;
00578     d->info.lastindexblock = 0;
00579     d->info.numberoftables = 0;
00580     d->info.numberofnamesblocks = 0;
00581     d->info.firstnameblock = 0;
00582     d->info.lastnameblock = 0;
00583
00584     d->name = str_dup(filename); /* For the moment just for the error messages */
00585     d->handle = f;
00586     if ( ReadIniInfo(d) || ReadIndex(d) ) { M_free(d,"index-d"); fclose(f); return(0); }
00587     if ( ComposeTableNames(d) < 0 ) { FreeTableBase(d); fclose(f); return(0); }
00588     // free allocation from previous str_dup
00589     M_free(d->name, "from str_dup");
00590     d->name = str_dup(filename);
00591     d->fullname = newname;
00592     return(d);
00593 }
00594
00595 /*
00596 #] GetDbase :
00597 #[ NewDbase :
00598
00599     Creates a new database with 'number' entries in the index.
00600 */
00601
00602 DBASE *NewDbase(char *name,MLONG number)
00603 {
00604     FILE *f;
00605     DBASE *d;
00606     MLONG numblocks, numnameblocks, i;
00607     char *s;
00608     /*-----change 10-feb-2003 */
00609     int j, jj;

```

```

00610     MLONG t = (MLONG) (time(0));
00611     /*-----*/
00612     if ( number < 0 ) number = 0;
00613     if ( ( f = fopen(name,"w+b") ) == 0 ) {
00614         MesPrint("Could not create a new file with name %s\n",name);
00615         return(0);
00616     }
00617     numblocks = (number+NUMOBJECTS-1)/NUMOBJECTS;
00618     numnameblocks = 1;
00619     if ( numblocks <= 0 ) numblocks = 1;
00620     if ( numnameblocks <= 0 ) numnameblocks = 1;
00621     d = (DBASE *)From0List(&(AC.TableBaseList));
00622     if ( ( d->iblocks = (INDEXBLOCK **)Malloca1(numblocks*sizeof(INDEXBLOCK *),
00623 "new database") ) == 0 ) {
00624         NumTableBases--;
00625         return(0);
00626     }
00627     d->tablenames = 0;
00628     d->tablenamessize = 0;
00629     d->topnumber = 0;
00630     d->tablenamefill = 0;
00631     d->rwmode = 1;
00632
00633     d->mode = 0;
00634     if ( ( d->nblocks = (NAMESBLOCK **)Malloca1(sizeof(NAMESBLOCK *)*numnameblocks,
00635 "new database") ) == 0 ) {
00636         M_free(d->iblocks,"new database");
00637         NumTableBases--;
00638         return(0);
00639     }
00640     if ( ( f = fopen(name,"w+b") ) == 0 ) {
00641         MesPrint("Could not create new file %s\n",name);
00642         NumTableBases--;
00643         return(0);
00644     }
00645     /* setbuf(f,0); */
00646     d->name = str_dup(name);
00647     d->fullname = str_dup(name);
00648     d->handle = f;
00649
00650     d->info.entriesinindex = number;
00651     d->info.numberofindexblocks = numblocks;
00652     d->info.numberofnamesblocks = numnameblocks;
00653     d->info.firstindexblock = 0;
00654     d->info.lastindexblock = 0;
00655     d->info.numberoftables = 0;
00656     d->info.firstnameblock = 0;
00657     d->info.lastnameblock = 0;
00658
00659     if ( WriteIniInfo(d) ) {
00660 getout:
00661         fclose(f);
00662         remove(d->fullname);
00663         if ( d->name ) { M_free(d->name,"name tablebase"); d->name = 0; }
00664         if ( d->fullname ) { M_free(d->fullname,"fullname tablebase"); d->fullname = 0; }
00665         M_free(d->nblocks,"new database");
00666         M_free(d->iblocks,"new database");
00667         NumTableBases--;
00668         return(0);
00669     }
00670     for ( i = 0; i < numblocks; i++ ) {
00671         if ( ( d->iblocks[i] = (INDEXBLOCK *)Malloca1(sizeof(INDEXBLOCK),
00672 "index blocks of new database") ) == 0 ) {
00673             while ( --i >= 0 ) M_free(d->iblocks[i],"index blocks of new database");
00674             goto getout;
00675         }
00676         if ( i > 0 ) d->iblocks[i]->previousblock = d->iblocks[i-1]->position;
00677         else d->iblocks[i]->previousblock = -1;
00678         d->iblocks[i]->position = ftell(f);
00679         // Initialise, to keep valgrind happy
00680         d->iblocks[i]->flags = -1;
00681         /*-----change 10-feb-2003 */
00682         /*
00683             Zero things properly. We don't want garbage in the file.
00684         */
00685         for ( j = 0; j < NUMOBJECTS; j++ ) {
00686             d->iblocks[i]->objects[j].date = t;
00687             d->iblocks[i]->objects[j].size = 0;
00688             d->iblocks[i]->objects[j].position = -1;
00689             d->iblocks[i]->objects[j].tablename = 0;
00690             d->iblocks[i]->objects[j].uncompressed = 0;
00691             d->iblocks[i]->objects[j].spare1 = 0;
00692             d->iblocks[i]->objects[j].spare2 = 0;
00693             d->iblocks[i]->objects[j].spare3 = 0;
00694             for ( jj = 0; jj < ELEMENTSIZE; jj++ ) d->iblocks[i]->objects[j].element[jj] = 0;
00695         }
00696         convertblock(d->iblocks[i],&scratchblock,TODISK);

```



```

00697         if ( minoswrite(d->handle, (char *)(&scratchblock), sizeof(INDEXBLOCK)) ) {
00698             MesPrint("Error while writing new index blocks\n");
00699             goto getout;
00700         }
00701     }
00702     for ( i = 0; i < numnameblocks; i++ ) {
00703         if ( ( d->nblocks[i] = (NAMESBLOCK *)Malloc1(sizeof(NAMESBLOCK),
00704             "names blocks of new database") ) == 0 ) {
00705             while ( --i >= 0 ) { M_free(d->nblocks[i], "names blocks of new database"); }
00706             for ( i = 0; i < numblocks; i++ ) M_free(d->iblocks[i], "index blocks of new database");
00707             goto getout;
00708         }
00709         if ( i > 0 ) d->nblocks[i]->previousblock = d->nblocks[i-1]->position;
00710         else d->nblocks[i]->previousblock = -1;
00711         d->nblocks[i]->position = ftell(f);
00712         s = d->nblocks[i]->names;
00713         for ( j = 0; j < NAMETABLESIZE; j++ ) *s++ = 0;
00714         convertnamesblock(d->nblocks[i], &scratchnamesblock, TODISK);
00715         if ( minoswrite(d->handle, (char *)(&scratchnamesblock), sizeof(NAMESBLOCK)) ) {
00716             MesPrint("Error while writing new names blocks\n");
00717             for ( i = 0; i < numnameblocks; i++ ) M_free(d->nblocks[i], "names blocks of new
database");
00718             for ( i = 0; i < numblocks; i++ ) M_free(d->iblocks[i], "index blocks of new database");
00719             goto getout;
00720         }
00721     }
00722     d->info.firstindexblock = d->iblocks[0]->position;
00723     d->info.lastindexblock = d->iblocks[numblocks-1]->position;
00724     d->info.firstnameblock = d->nblocks[0]->position;
00725     d->info.lastnameblock = d->nblocks[numnameblocks-1]->position;
00726     if ( WriteIniInfo(d) ) {
00727         for ( i = 0; i < numnameblocks; i++ ) M_free(d->nblocks[i], "names blocks of new database");
00728         for ( i = 0; i < numblocks; i++ ) M_free(d->iblocks[i], "index blocks of new database");
00729         goto getout;
00730     }
00731     return(d);
00732 }
00733
00734 /*
00735 #] NewDbase :
00736 #[ FreeTableBase :
00737 */
00738
00739 void FreeTableBase(DBASE *d)
00740 {
00741     int i, j, *old, *newL;
00742     LIST *L;
00743     for ( i = 0; i < d->info.numberofnamesblocks; i++ ) M_free(d->nblocks[i], "nblocks[i]");
00744     for ( i = 0; i < d->info.numberofindexblocks; i++ ) M_free(d->iblocks[i], "iblocks[i]");
00745     M_free(d->nblocks, "nblocks");
00746     M_free(d->iblocks, "iblocks");
00747     if ( d->tablenames ) M_free(d->tablenames, "d->tablenames");
00748     if ( d->name ) M_free(d->name, "d->name");
00749     if ( d->fullname ) M_free(d->fullname, "d->fullname");
00750     i = d - tablebases;
00751     if ( i < ( NumTableBases - 1 ) ) {
00752         L = &(AC.TableBaseList);
00753         j = ( ( NumTableBases - i - 1 ) * L->size ) / sizeof(int);
00754         old = (int *)d; newL = (int *) (d+1);
00755         while ( --j >= 0 ) *newL++ = *old++;
00756         j = L->size / sizeof(int);
00757         while ( --j >= 0 ) *newL++ = 0;
00758     }
00759     NumTableBases--;
00760     M_free(d, "tb,d");
00761 }
00762
00763 /*
00764 #] FreeTableBase :
00765 #[ ComposeTableNames :
00766
00767     The nameblocks are supposed to be in memory.
00768     Hence we have to go through them
00769 */
00770
00771 int ComposeTableNames(DBASE *d)
00772 {
00773     MLONG nsize = 0;
00774     int i, j, k;
00775     char *s, *t, *ss;
00776     d->topnumber = 0;
00777     i = 0; s = d->nblocks[i]->names; j = NAMETABLESIZE;
00778     while ( *s ) {
00779         if ( *s ) d->topnumber++;
00780         for ( k = 0; k < 2; k++ ) { /* name and argtail */
00781             while ( *s ) {
00782                 j--;

```

```

00783         if ( j <= 0 ) {
00784             i++; if ( i >= d->info.numberofnamesblocks ) goto gotall;
00785             s = d->nblocks[i]->names; j = NAMETABLESIZE;
00786         }
00787         else s++;
00788     }
00789     j--;
00790     if ( j <= 0 ) {
00791         i++; if ( i >= d->info.numberofnamesblocks ) goto gotall;
00792         s = d->nblocks[i]->names; j = NAMETABLESIZE;
00793     }
00794     else s++;
00795 }
00796 }
00797 gotall:;
00798 nsize = (d->info.numberofnamesblocks-1)*NAMETABLESIZE +
00799         (s-d->nblocks[i]->names)+1;
00800 if ( ( d->tablenames = (char *)Malloc1((2*nsize+30)*sizeof(char),"tablenames") )
00801     == 0 ) { return(-1); }
00802 t = d->tablenames;
00803 d->tablenamessize = 2*nsize+30;
00804 d->tablenamefill = nsize-1;
00805 for ( k = 0; k < i; k++ ) {
00806     ss = d->nblocks[k]->names;
00807     for ( j = 0; j < NAMETABLESIZE; j++ ) *t++ = *ss++;
00808 }
00809 ss = d->nblocks[i]->names;
00810 while ( ss < s ) *t++ = *ss++;
00811 *t = 0;
00812 return(0);
00813 }
00814
00815 /*
00816 #] ComposeTableNames :
00817 #[ OpenDbase :
00818 */
00819
00820 DBASE *OpenDbase(char *filename)
00821 {
00822     FILE *f;
00823     DBASE *d;
00824     char *newname;
00825     if ( ( f = LocateBase(&filename,&newname,"r+b") ) == 0 ) {
00826         MesPrint("Cannot open file %s\n",filename);
00827         return(0);
00828     }
00829     /* setbuf(f,0); */
00830     d = (DBASE *)From0List(&(AC.TableBaseList));
00831     d->name = filename; /* For the moment just for the error messages */
00832     d->handle = f;
00833     if ( ReadIniInfo(d) || ReadIndex(d) ) { M_free(d,"OpenDbase"); fclose(f); return(0); }
00834     if ( ComposeTableNames(d) ) {
00835         FreeTableBase(d);
00836         fclose(f);
00837         return(0);
00838     }
00839     d->name = str_dup(filename);
00840     d->fullname = newname;
00841     return(d);
00842 }
00843
00844 /*
00845 #] OpenDbase :
00846 #[ AddTableName :
00847
00848 Adds a name of a table. Writes the namelist to disk.
00849 Returns the number of this tablename in the database.
00850 If the name was already in the table we return its value in negative.
00851 Zero is an error!
00852 */
00853
00854 MLONG AddTableName(DBASE *d,char *name, TABLES T)
00855 {
00856     char *s, *t, *tt;
00857     int namesize, tailsize;
00858     MLONG newsize, i, num;
00859     /*
00860     First search for the name in what we have already
00861     */
00862     if ( d->tablenames ) {
00863         num = 0;
00864         s = d->tablenames;
00865         while ( *s ) {
00866             num++;
00867             t = name;
00868             while ( ( *s == *t ) && *t ) { s++; t++; }
00869             if ( *s == *t ) { return(-num); }

```

```

00870         while ( *s ) s++;
00871         s++;
00872         while ( *s ) s++;
00873         s++;
00874     }
00875 }
00876 /*
00877  This name has to be added
00878 */
00879 MesPrint("We add the name %s\n",name);
00880 t = name;
00881 while ( *t ) { t++; }
00882 namesize = t-name;
00883 if ( ( t = (char *) (T->argtail) ) != 0 ) {
00884     while ( *t ) { t++; }
00885     tailsizsize = t - (char *) (T->argtail);
00886 }
00887 else { tailsizsize = 0; }
00888 if ( d->tablenames == 0 ) {
00889     if ( ComposeTableNames(d) ) {
00890         FreeTableBase(d);
00891         M_free(d,"AddTableName");
00892         return(0);
00893     }
00894 }
00895 d->info.numberoftables++;
00896 while ( ( d->tablenameefill+namesize+tailsizsize+3 > d->tablenamessize )
00897     || ( d->tablenames == 0 ) ) {
00898     newsizsize = 2*d->tablenamessize + 2*namesize + 2*tailsizsize + 6;
00899     if ( ( t = (char *) Malloc1(newsizsize*sizeof(char),"AddTableName") ) == 0 )
00900         return(0);
00901     tt = t;
00902     if ( d->tablenames ) {
00903         s = d->tablenames;
00904         for ( i = 0; i < d->tablenameefill; i++ ) *t++ = *s++;
00905         *t = 0;
00906         M_free(d->tablenames,"d->tablenames");
00907     }
00908     d->tablenames = tt;
00909     d->tablenamessize = newsizsize;
00910 }
00911 s = d->tablenames + d->tablenameefill;
00912 t = name;
00913 while ( *t ) *s++ = *t++;
00914 *s++ = 0;
00915 t = (char *) (T->argtail);
00916 while ( *t ) *s++ = *t++;
00917 *s++ = 0;
00918 *s = 0;
00919 d->tablenameefill = s - d->tablenames;
00920 d->topnumber++;
00921 /*
00922  Now we have to synchronize
00923 */
00924 if ( PutTableNames(d) ) return(0);
00925 return(d->topnumber);
00926 }
00927
00928 /*
00929 #] AddTableName :
00930 #[ GetTableName :
00931
00932  Gets a name of a table.
00933  Returns the number of this tablename in the database.
00934  Zero -> error
00935 */
00936
00937 MLONG GetTableName(DBASE *d,char *name)
00938 {
00939     char *s, *t;
00940     MLONG num;
00941     /*
00942      search for the name in what we have
00943     */
00944     if ( d->tablenames ) {
00945         num = 0;
00946         s = d->tablenames;
00947         while ( *s ) {
00948             num++;
00949             t = name;
00950             while ( ( *s == *t ) && *t ) { s++; t++; }
00951             if ( *s == *t ) { return(num); }
00952             while ( *s ) s++;
00953             s++;
00954             while ( *s ) s++;
00955             s++;
00956         }

```

```

00957     }
00958     return(0);
00959 }
00960
00961 /*
00962 #] GetTableName :
00963 #[ PutTableNames :
00964
00965     Takes the names string in d->tablenames and puts it in the nblocks
00966     pieces. Writes what has been changed to disk.
00967 */
00968
00969 int PutTableNames(DBASE *d)
00970 {
00971     NAMESBLOCK **nnew;
00972     int i, j, firstdif;
00973     MLONG m;
00974     char *s, *t;
00975     /*
00976     Determine how many blocks are needed.
00977     */
00978     MLONG numblocks = d->tablenamefill/NAMETABLESIZE + 1;
00979     if ( d->info.numberofnamesblocks < numblocks ) {
00980     /*
00981         We need more blocks. First make sure of the space for nblocks.
00982     */
00983         if ( ( nnew = (NAMESBLOCK **)Malloc1(sizeof(NAMESBLOCK *)*numblocks,
00984             "new names block") ) == 0 ) {
00985             return(-1);
00986         }
00987         for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
00988             nnew[i] = d->nblocks[i];
00989         }
00990         free(d->nblocks);
00991         d->nblocks = nnew;
00992         for ( ; i < numblocks; i++ ) {
00993             if ( ( d->nblocks[i] = (NAMESBLOCK *)Malloc1(sizeof(NAMESBLOCK),
00994                 "additional names blocks ") ) == 0 ) {
00995                 FreeTableBase(d);
00996                 return(-1);
00997             }
00998             d->nblocks[i]->previousblock = -1;
00999             d->nblocks[i]->position = -1;
01000             s = d->nblocks[i]->names;
01001             for ( j = 0; j < NAMETABLESIZE; j++ ) *s++ = 0;
01002         }
01003         d->info.numberofnamesblocks = numblocks;
01004     }
01005     /*
01006     Now look till where the new contents agree with the old.
01007     */
01008     firstdif = 0;
01009     i = 0; t = d->nblocks[i]->names; j = 0; s = d->tablenames;
01010     for ( m = 0; m < d->tablenamefill; m++ ) {
01011         if ( *s == *t ) {
01012             s++; t++; j++;
01013             if ( j >= NAMETABLESIZE ) {
01014                 i++;
01015                 t = d->nblocks[i]->names;
01016                 j = 0;
01017             }
01018         }
01019         else {
01020             firstdif = i;
01021             for ( ; m < d->tablenamefill; m++ ) {
01022                 *t++ = *s++; j++;
01023                 if ( j >= NAMETABLESIZE ) {
01024                     i++;
01025                     t = d->nblocks[i]->names;
01026                     j = 0;
01027                 }
01028             }
01029             *t = 0;
01030             break;
01031         }
01032     }
01033     for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
01034         if ( i == firstdif ) break;
01035         if ( d->nblocks[i]->position < 0 ) { firstdif = i; break; }
01036     }
01037     /*
01038     Now we have to (re)write the blocks, starting at firstdif.
01039     */
01040     for ( i = firstdif; i < d->info.numberofnamesblocks; i++ ) {
01041         if ( i > 0 ) d->nblocks[i]->previousblock = d->nblocks[i-1]->position;
01042         else d->nblocks[i]->previousblock = -1;
01043         if ( d->nblocks[i]->position < 0 ) {

```

```

01044         fseek(d->handle,0,SEEK_END);
01045         d->nblocks[i]->position = ftell(d->handle);
01046     }
01047     else fseek(d->handle,d->nblocks[i]->position,SEEK_SET);
01048     convertnamesblock(d->nblocks[i],&scratchnamesblock,TODISK);
01049     if ( minoswrite(d->handle,(char *)(&scratchnamesblock),sizeof(NAMESBLOCK)) ) {
01050         MesPrint("Error while writing names blocks\n");
01051         FreeTableBase(d);
01052         return(-1);
01053     }
01054 }
01055 d->info.lastnameblock = d->nblocks[d->info.numberofnamesblocks-1]->position;
01056 d->info.firstnameblock = d->nblocks[0]->position;
01057 return(WriteIniInfo(d));
01058 }
01059
01060 /*
01061 #] PutTableNames :
01062 #[ AddToIndex :
01063 */
01064
01065 int AddToIndex(DBASE *d,MLONG number)
01066 {
01067     MLONG i, oldnumofindexblocks = d->info.numberofindexblocks;
01068     MLONG j, newnumofindexblocks, jj;
01069     INDEXBLOCK **ib;
01070     MLONG t = (MLONG)(time(0));
01071     if ( number == 0 ) return(0);
01072     else if ( number < 0 ) {
01073         if ( d->info.entriesinindex < -number ) {
01074             MesPrint("There are only %ld entries in the index of file %s\n",
01075                 d->info.entriesinindex,d->name);
01076             return(-1);
01077         }
01078         d->info.entriesinindex += number;
01079     dowrite:
01080         if ( WriteIniInfo(d) ) {
01081             d->info.entriesinindex -= number;
01082             MesPrint("File may be corrupted\n");
01083             return(-1);
01084         }
01085     }
01086     else if ( d->info.entriesinindex+number <=
01087         NUMOBJECTS*d->info.numberofindexblocks ) {
01088         d->info.entriesinindex += number;
01089         goto dowrite;
01090     }
01091     else {
01092         d->info.entriesinindex += number;
01093         newnumofindexblocks = d->info.numberofindexblocks + ((number -
01094             (NUMOBJECTS*d->info.numberofindexblocks - d->info.entriesinindex))
01095             +NUMOBJECTS-1)/NUMOBJECTS;
01096         if ( ( ib = (INDEXBLOCK **)Malloc1(sizeof(INDEXBLOCK *)*newnumofindexblocks,
01097             "index") ) == 0 ) return(-1);
01098         for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
01099             ib[i] = d->iblocks[i];
01100         }
01101         for ( i = d->info.numberofindexblocks; i < newnumofindexblocks; i++ ) {
01102             if ( ( ib[i] = (INDEXBLOCK *)Malloc1(sizeof(INDEXBLOCK),"index block") ) == 0 ) {
01103                 FreeTableBase(d);
01104                 return(-1);
01105             }
01106             if ( i > 0 ) ib[i]->previousblock = ib[i-1]->position;
01107             else ib[i]->previousblock = -1;
01108         }
01109     /*
01110     Zero things properly. We don't want garbage in the file.
01111     */
01112     for ( j = 0; j < NUMOBJECTS; j++ ) {
01113         ib[i]->objects[j].date = t;
01114         ib[i]->objects[j].size = 0;
01115         ib[i]->objects[j].position = -1;
01116         ib[i]->objects[j].tablennumber = 0;
01117         ib[i]->objects[j].uncompressed = 0;
01118         ib[i]->objects[j].spare1 = 0;
01119         ib[i]->objects[j].spare2 = 0;
01120         ib[i]->objects[j].spare3 = 0;
01121         for ( jj = 0; jj < ELEMENTSIZE; jj++ ) ib[i]->objects[j].element[jj] = 0;
01122     }
01123     fseek(d->handle,0,SEEK_END);
01124     ib[i]->position = ftell(d->handle);
01125     convertblock(ib[i],&scratchblock,TODISK);
01126     if ( minoswrite(d->handle,(char *)(&scratchblock),sizeof(INDEXBLOCK)) ) {
01127         MesPrint("Error while writing new index of file %s",d->name);
01128         FreeTableBase(d);
01129         return(-1);
01130     }
01131 }

```

```

01131         d->info.lastindexblock = ib[newnumofindexblocks-1]->position;
01132         d->info.firstindexblock = ib[0]->position;
01133         d->info.numberofindexblocks = newnumofindexblocks;
01134         if ( WriteIniInfo(d) ) {
01135             d->info.numberofindexblocks = oldnumofindexblocks;
01136             d->info.entriesinindex -= number;
01137             MesPrint("File may be corrupted\n");
01138             FreeTableBase(d);
01139             return(-1);
01140         }
01141         M_free(d->iblocks,"AddToIndex");
01142         d->iblocks = ib;
01143     }
01144     return(0);
01145 }
01146
01147 /*
01148 #] AddToIndex :
01149 #[ AddObject :
01150 */
01151
01152 MLONG AddObject(DBASE *d,MLONG tablenumber,char *arguments,char *rhs)
01153 {
01154     MLONG number;
01155     number = d->info.entriesinindex;
01156     if ( AddToIndex(d,1) ) return(-1);
01157     if ( WriteObject(d,tablenumber,arguments,rhs,number) ) return(-1);
01158     return(number);
01159 }
01160
01161 /*
01162 #] AddObject :
01163 #[ FindTableNumber :
01164 */
01165
01166 MLONG FindTableNumber(DBASE *d,char *name)
01167 {
01168     char *s = d->tablenames, *t, *ss;
01169     MLONG num = 0;
01170     ss = d->tablenames + d->tablenamefill;
01171     while ( s < ss ) {
01172         num++;
01173         t = name;
01174         while ( *s == *t && *t ) {
01175             s++; t++;
01176         }
01177         if ( *s == 0 && *t == 0 ) return(num);
01178         while ( *s ) s++;
01179         s++;
01180     }
01181     /* Skip also the argument tail */
01182     while ( *s ) s++;
01183     s++;
01184 }
01185 return(-1); /* Name not found */
01186 }
01187
01188 /*
01189 #] FindTableNumber :
01190 #[ WriteObject :
01191 */
01192
01193 int WriteObject(DBASE *d,MLONG tablenumber,char *arguments,char *rhs,MLONG number)
01194 {
01195     char *s, *a;
01196     #ifdef WITHZLIB
01197     char *buffer = 0;
01198     uLongf newsize = 0, oldsize = 0;
01199     uLong ssize;
01200     int error = 0;
01201     #endif
01202     MLONG i, j, position, size, n;
01203     OBJECTS *obj;
01204     if ( ( d->mode & INPUTONLY ) == INPUTONLY ) {
01205         MesPrint("Not allowed to write to input\n");
01206         return(-1);
01207     }
01208     if ( number >= d->info.entriesinindex ) {
01209         MesPrint("Reference to non-existing object number %ld\n",number+1);
01210         return(0);
01211     }
01212     j = number/NUMOBJECTS;
01213     i = number%NUMOBJECTS;
01214     obj = &(d->iblocks[j]->objects[i]);
01215     a = arguments;
01216     while ( *a ) a++;

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```

01218     a++; n = a - arguments;
01219     if ( n > ELEMENTSIZE ) {
01220         MesPrint("Table element %s has more than %ld characters.\n",arguments,
01221             (MLONG)ELEMENTSIZE);
01222         return(-1);
01223     }
01224     s = obj->element;
01225     a = arguments;
01226     while ( *a ) *s++ = *a++;
01227     *s++ = 0;
01228     while ( n < ELEMENTSIZE ) { *s++ = 0; n++; }
01229     obj->spare1 = obj->spare2 = obj->spare3 = 0;
01230
01231     fseek(d->handle,0,SEEK_END);
01232     position = ftell(d->handle);
01233     s = rhs;
01234     while ( *s ) s++;
01235     s++;
01236     size = s - rhs;
01237 #ifdef WITHZLIB
01238     if ( ( d->mode & NOCOMPRESS ) == 0 ) {
01239         newsize = size + size/1000 + 20;
01240         if ( ( buffer = (char *)Malloca1(newsize*sizeof(char),"compress buffer") )
01241             == 0 ) {
01242             MesPrint("No compress used for element %s in file %s\n",arguments,d->name);
01243         }
01244     }
01245     else buffer = 0;
01246     if ( buffer ) {
01247         ssize = size;
01248 #ifdef WITHZSTD
01249         // Force the use of zlib for compressed Tablebase entries, so that tablebases created
01250         // with zstd-supported FORM builds can be used by zstd-unsupported FORM builds.
01251         const int old_isUsingZSTDcompression = ZWRAP_isUsingZSTDcompression();
01252         ZWRAP_useZSTDcompression(0);
01253 #endif
01254         if ( ( error = compress((Bytef *)buffer,&newsize,(Bytef *)rhs,ssize) ) != Z_OK ) {
01255             MesPrint("Error = %d\n",error);
01256             MesPrint("Due to error no compress used for element %s in file %s\n",arguments,d->name);
01257             M_free(buffer,"tb,WriteObject");
01258             buffer = 0;
01259         }
01260 #ifdef WITHZSTD
01261         ZWRAP_useZSTDcompression(old_isUsingZSTDcompression);
01262 #endif
01263     }
01264     if ( buffer ) {
01265         rhs = buffer;
01266         oldsize = size;
01267         size = newsize;
01268     }
01269 #endif
01270     if ( minoswrite(d->handle,rhs,size) ) {
01271         MesPrint("Error while writing rhs\n");
01272         return(-1);
01273     }
01274     obj->position = position;
01275     obj->size = size;
01276     obj->date = (MLONG) (time(0));
01277     obj->tablenumber = tablenumber;
01278 #ifdef WITHZLIB
01279     obj->uncompressed = oldsize;
01280     if ( buffer ) M_free(buffer,"tb,WriteObject");
01281 #else
01282     obj->uncompressed = 0;
01283 #endif
01284     return(WriteIndexBlock(d,j));
01285 }
01286
01287 /*
01288  #] WriteObject :
01289  #[ ReadObject :
01290
01291  Returns a pointer to the proper rhs
01292 */
01293
01294 char *ReadObject(DBASE *d,MLONG tablenumber,char *arguments)
01295 {
01296     OBJECTS *obj;
01297     MLONG i, j;
01298     char *buffer1, *s, *t;
01299 #ifdef WITHZLIB
01300     char *buffer2 = 0;
01301     ulongf finallength = 0;
01302 #endif
01303     if ( tablenumber > d->topnumber ) {
01304         MesPrint("Reference to non-existing table number in tablebase %s: %ld\n",

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```

01305         d->name,tablename);
01306         return(0);
01307     }
01308     /*
01309     Start looking for the object
01310     */
01311     for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
01312         for ( j = 0; j < NUMOBJECTS; j++ ) {
01313             if ( d->iblocks[i]->objects[j].tablename != tablename ) continue;
01314             s = arguments; t = d->iblocks[i]->objects[j].element;
01315             while ( *s == *t && *s ) { s++; t++; }
01316             if ( *t == 0 && *s == 0 ) goto foundelement;
01317         }
01318     }
01319     s = d->tablenames; i = 1;
01320     while ( *s ) {
01321         if ( i == tablename ) break;
01322         while ( *s ) s++;
01323         s++;
01324         while ( *s ) s++;
01325         s++;
01326         i++;
01327     }
01328     MesPrint("%s(%s) not found in tablebase %s\n",s,arguments,d->name);
01329     return(0);
01330
01331 foundelement:;
01332     obj = &(d->iblocks[i]->objects[j]);
01333     fseek(d->handle,obj->position,SEEK_SET);
01334     if ( ( buffer1 = (char *)Malloc1(obj->size,"reading rhs buffer1") ) == 0 ) {
01335         return(0);
01336     }
01337     #ifdef WITHZLIB
01338     if ( obj->uncompressed > 0 ) {
01339         if ( ( buffer2 = (char *)Malloc1(obj->uncompressed,"reading rhs buffer2") ) == 0 ) {
01340             return(0);
01341         }
01342     }
01343     else buffer2 = 0;
01344     #endif
01345     if ( minosread(d->handle,buffer1,obj->size) ) {
01346         MesPrint("Could not read rhs %s in file %s\n",arguments,d->name);
01347         M_free(buffer1,"tb,ReadObject");
01348     }
01349     #ifdef WITHZLIB
01350     if ( buffer2 ) M_free(buffer2,"tb,ReadObject");
01351     #endif
01352     return(0);
01353 }
01354 #ifdef WITHZLIB
01355 if ( buffer2 == 0 ) return(buffer1);
01356 finallength = obj->uncompressed;
01357 if ( uncompress((Bytef *)buffer2,&finallength,(Bytef *)buffer1,obj->size) != Z_OK ) {
01358     MesPrint("Cannot uncompress element %s in file %s\n",arguments,d->name);
01359     M_free(buffer1,"tb,ReadObject"); M_free(buffer2,"tb,ReadObject");
01360     return(0);
01361 }
01362 M_free(buffer1,"tb,ReadObject");
01363 return(buffer2);
01364 #else
01365 return(buffer1);
01366 #endif
01367 }
01368 /*
01369 #] ReadObject :
01370 #[ ReadijObject :
01371
01372 Returns a pointer to the proper rhs
01373 */
01374
01375 char *ReadijObject(DBASE *d,MLONG i,MLONG j,char *arguments)
01376 {
01377     OBJECTS *obj;
01378     char *buffer1;
01379     #ifdef WITHZLIB
01380     char *buffer2 = 0;
01381     uLongf finallength = 0;
01382     #endif
01383     obj = &(d->iblocks[i]->objects[j]);
01384     fseek(d->handle,obj->position,SEEK_SET);
01385     if ( ( buffer1 = (char *)Malloc1(obj->size,"reading rhs buffer1") ) == 0 ) {
01386         return(0);
01387     }
01388     #ifdef WITHZLIB
01389     if ( obj->uncompressed > 0 ) {
01390         if ( ( buffer2 = (char *)Malloc1(obj->uncompressed,"reading rhs buffer2") ) == 0 ) {
01391             return(0);

```



```

01392     }
01393     }
01394     else buffer2 = 0;
01395 #endif
01396     if ( minosread(d->handle,buffer1,obj->size) ) {
01397         MesPrint("Could not read rhs %s in file %s\n",arguments,d->name);
01398         if ( buffer1 ) M_free(buffer1,"rhs buffer1");
01399 #ifdef WITHZLIB
01400         if ( buffer2 ) M_free(buffer2,"rhs buffer2");
01401 #endif
01402         return(0);
01403     }
01404 #ifdef WITHZLIB
01405     if ( buffer2 == 0 ) return(buffer1);
01406     finallength = obj->uncompressed;
01407     if ( uncompress((Bytef *)buffer2,&finallength,(Bytef *)buffer1,obj->size) != Z_OK ) {
01408         MesPrint("Cannot uncompress element %s in file %s\n",arguments,d->name);
01409         if ( buffer1 ) M_free(buffer1,"rhs buffer1");
01410         if ( buffer2 ) M_free(buffer2,"rhs buffer2");
01411         return(0);
01412     }
01413     M_free(buffer1,"rhs buffer1");
01414     return(buffer2);
01415 #else
01416     return(buffer1);
01417 #endif
01418 }
01419
01420 /*
01421  #] ReadijObject :
01422  #[ ExistsObject :
01423
01424  Returns 1 if Object exists
01425 */
01426
01427 int ExistsObject(DBASE *d,MLONG tablenumber,char *arguments)
01428 {
01429     MLONG i, j;
01430     char *s, *t;
01431     if ( tablenumber > d->topnumber ) {
01432         MesPrint("Reference to non-existing table number in tablebase %s: %ld\n",
01433             d->name,tablenumber);
01434         return(0);
01435     }
01436     /*
01437     Start looking for the object
01438     */
01439     for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
01440         for ( j = 0; j < NUMOBJECTS; j++ ) {
01441             if ( d->iblocks[i]->objects[j].tablenumber != tablenumber ) continue;
01442             s = arguments; t = d->iblocks[i]->objects[j].element;
01443             while ( *s == *t && *s ) { s++; t++; }
01444             if ( *t == 0 && *s == 0 ) return(1);
01445         }
01446     }
01447     return(0);
01448 }
01449
01450 /*
01451  #] ExistsObject :
01452  #[ DeleteObject :
01453
01454  Returns 1 if Object has been deleted.
01455  We leave a hole. Actually the object is still there but has been
01456  inactivated. It can be reactivated by calling this routine again.
01457 */
01458
01459 int DeleteObject(DBASE *d,MLONG tablenumber,char *arguments)
01460 {
01461     MLONG i, j;
01462     char *s, *t;
01463     if ( tablenumber > d->topnumber ) {
01464         MesPrint("Reference to non-existing table number in tablebase %s: %ld\n",
01465             d->name,tablenumber);
01466         return(0);
01467     }
01468     /*
01469     Start looking for the object
01470     */
01471     for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
01472         for ( j = 0; j < NUMOBJECTS; j++ ) {
01473             if ( d->iblocks[i]->objects[j].tablenumber != tablenumber ) continue;
01474             s = arguments; t = d->iblocks[i]->objects[j].element;
01475             while ( *s == *t && *s ) { s++; t++; }
01476             if ( *t == 0 && *s == 0 ) {
01477                 d->iblocks[i]->objects[j].tablenumber =
01478                     -d->iblocks[i]->objects[j].tablenumber - 1;

```

```

01479             return(1);
01480         }
01481     }
01482 }
01483     return(0);
01484 }
01485
01486 /*
01487  #] DeleteObject :
01488 */

```

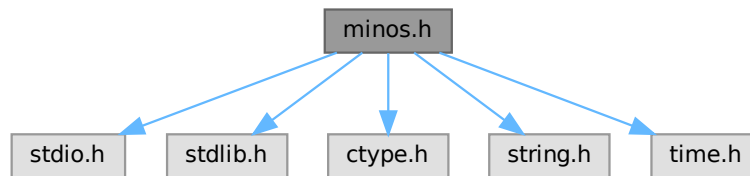
## 4.73 minos.h File Reference

```

#include <stdio.h>
#include <stdlib.h>
#include <ctype.h>
#include <string.h>
#include <time.h>

```

Include dependency graph for minos.h:



This graph shows which files directly or indirectly include this file:



### Data Structures

- struct [iniinfo](#)
- struct [objects](#)
- struct [indexblock](#)
- struct [nameblock](#)
- struct [dbase](#)

### Macros

- #define [TODISK](#) 0
- #define [FROMDISK](#) 1
- #define [MDIRTYFLAG](#) 1
- #define [MCLEANFLAG](#) (~MDIRTYFLAG)
- #define [INANDOUT](#) 0
- #define [INPUTONLY](#) 1
- #define [OUTPUTONLY](#) 2
- #define [NOCOMPRESS](#) 4

## Typedefs

- typedef struct [iniinfo](#) **INIINFO**
- typedef struct [objects](#) **OBJECTS**
- typedef struct [indexblock](#) **INDEXBLOCK**
- typedef struct [nameblock](#) **NAMESBLOCK**
- typedef struct [dbase](#) **DBASE**

## Functions

- int [minosread](#) (FILE \*f, char \*buffer, MLONG size)
- int [minoswrite](#) (FILE \*f, char \*buffer, MLONG size)

## Variables

- int [withoutflush](#)

### 4.73.1 Detailed Description

Contains all needed declarations and definitions for the tablebase low level file routines. These have been taken from the minos database system and modified somewhat.

!!!CAUTION!!! Changes in this file will most likely have consequences for the recovery mechanism (see [checkpoint.c](#)). You need to care for the code in [checkpoint.c](#) as well and modify the code there accordingly!

Definition in file [minos.h](#).

### 4.73.2 Macro Definition Documentation

#### 4.73.2.1 TODISK

```
#define TODISK 0
```

Definition at line [145](#) of file [minos.h](#).

#### 4.73.2.2 FROMDISK

```
#define FROMDISK 1
```

Definition at line [146](#) of file [minos.h](#).

#### 4.73.2.3 MDIRTYFLAG

```
#define MDIRTYFLAG 1
```

Definition at line [147](#) of file [minos.h](#).

#### 4.73.2.4 MCLEANFLAG

```
#define MCLEANFLAG (~MDIRTYFLAG)
```

Definition at line 148 of file [minos.h](#).

#### 4.73.2.5 INANDOUT

```
#define INANDOUT 0
```

Definition at line 149 of file [minos.h](#).

#### 4.73.2.6 INPUTONLY

```
#define INPUTONLY 1
```

Definition at line 150 of file [minos.h](#).

#### 4.73.2.7 OUTPUTONLY

```
#define OUTPUTONLY 2
```

Definition at line 151 of file [minos.h](#).

#### 4.73.2.8 NOCOMPRESS

```
#define NOCOMPRESS 4
```

Definition at line 152 of file [minos.h](#).

### 4.73.3 Function Documentation

#### 4.73.3.1 minosread()

```
int minosread (
    FILE * f,
    char * buffer,
    MLONG size )
```

Definition at line 66 of file [minos.c](#).

#### 4.73.3.2 minoswrite()

```
int minoswrite (
    FILE * f,
    char * buffer,
    MLONG size )
```

Definition at line 83 of file [minos.c](#).

## 4.73.4 Variable Documentation

### 4.73.4.1 withoutflush

int withoutflush [extern]

Definition at line 45 of file [minos.c](#).

## 4.74 minos.h

[Go to the documentation of this file.](#)

```

00001 #ifndef __MANAGE_H__
00002
00003 #define __MANAGE_H__
00004
00017 /* #[ License : */
00018 /*
00019  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00020  * When using this file you are requested to refer to the publication
00021  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00022  * This is considered a matter of courtesy as the development was paid
00023  * for by FOM the Dutch physics granting agency and we would like to
00024  * be able to track its scientific use to convince FOM of its value
00025  * for the community.
00026  *
00027  * This file is part of FORM.
00028  *
00029  * FORM is free software: you can redistribute it and/or modify it under the
00030  * terms of the GNU General Public License as published by the Free Software
00031  * Foundation, either version 3 of the License, or (at your option) any later
00032  * version.
00033  *
00034  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00035  * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00036  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00037  * details.
00038  *
00039  * You should have received a copy of the GNU General Public License along
00040  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00041 */
00042 /* #] License : */
00043
00044 #include <stdio.h>
00045 #include <stdlib.h>
00046 #include <ctype.h>
00047 #include <string.h>
00048 #include <time.h>
00049
00050 /*
00051  * The following typedef has been moved to form3.h where all the sizes
00052  * are defined for the various memory models.
00053  * We want MLONG to have a more or less fixed size.
00054  * In form3.h we try to fix it at 8 bytes.
00055  * This should make files exchangeable
00056  * between various 32-bits and 64-bits systems. At 4 bytes it might have
00057  * problems with files of more than 2 Gbytes.
00058
00059 typedef long MLONG;
00060 */
00061
00062 #if BITSINWORD == 32
00063     #define NUMOBJECTS 1024
00064     #define MAXINDEXSIZE 1000000000L
00065     #define NAMETABLESIZE 1008
00066     #define ELEMENTSIZE 256
00067 #elif BITSINWORD == 16
00068     #define NUMOBJECTS 100
00069     #define MAXINDEXSIZE 33000000L
00070     #define NAMETABLESIZE 1008
00071     #define ELEMENTSIZE 128
00072 #else
00073     #error Only 64-bit and 32-bit platforms are supported.
00074 #endif
00075
00076
00077 int minosread(FILE *f,char *buffer,MLONG size);
00078 int minoswrite(FILE *f,char *buffer,MLONG size);

```

```

00079
00080 /*
00081     ELEMENTSIZE should make a nice number of sizeof(OBJECTS)
00082     Usually this will be much too much, but there are cases.....
00083 */
00084
00085 typedef struct iniinfo {
00086     /* should contains only MLONG variables or convertiniinfo should be modified */
00087     MLONG     entriesinindex;
00088     MLONG     numberofindexblocks;
00089     MLONG     firstindexblock;
00090     MLONG     lastindexblock;
00091     MLONG     numberoftables;
00092     MLONG     numberofnamesblocks;
00093     MLONG     firstnameblock;
00094     MLONG     lastnameblock;
00095 } INIINFO;
00096
00097 typedef struct objects {
00098     /* if any changes, convertblock should be adapted too!!!! */
00099     MLONG position;           /* position of RHS= */
00100     MLONG size;              /* size on disk (could be compressed) */
00101     MLONG date;              /* Time stamp */
00102     MLONG tablenumber;        /* Number of table. Refers to name in special index */
00103     MLONG uncompressed;      /* uncompressed size if compressed. If not: 0 */
00104     MLONG spare1;
00105     MLONG spare2;
00106     MLONG spare3;
00107     char element[ELEMENTSIZE]; /* table element in character form */
00108 } OBJECTS;
00109
00110 typedef struct indexblock {
00111     MLONG flags;
00112     MLONG previousblock;
00113     MLONG position;
00114     OBJECTS objects[NUMOBJECTS];
00115 } INDEXBLOCK;
00116
00117 typedef struct nameblock {
00118     MLONG previousblock;
00119     MLONG position;
00120     char names[NAMETABLESIZE];
00121 } NAMESBLOCK;
00122
00123 typedef struct dbase {
00124     INIINFO info;
00125     MLONG mode;
00126     MLONG tablenamessize;
00127     MLONG topnumber;
00128     MLONG tablenamefill;
00129     MLONG rwmode;
00130     INDEXBLOCK **iblocks;
00131     NAMESBLOCK **nblocks;
00132     FILE *handle;
00133     char *name;
00134     char *fullname;
00135     char *tablenames;
00136 } DBASE;
00137
00138 /*
00139 typedef int (*SFUN)(char *);
00140 typedef struct compile {
00141     char *keyword;
00142     SFUN func;
00143 } MCFUNCTION;
00144 */
00145 #define TODISK 0
00146 #define FROMDISK 1
00147 #define MDIRTYFLAG 1
00148 #define MCLEANFLAG (~MDIRTYFLAG)
00149 #define INANDOUT 0
00150 #define INPUTONLY 1
00151 #define OUTPUTONLY 2
00152 #define NOCOMPRESS 4
00153
00154 extern int withoutflush;
00155
00156 #endif

```

## 4.75 model.c

```

00001 /*
00002     #[ Explanations :

```

```

00003
00004     This is a second generation much simplified version of model.c
00005     Syntax:
00006         Model QED;
00007             Particle electron,positron,-2;
00008             Particle photon,+3;
00009             Vertex electron,positron,photon: g;
00010         EndModel;
00011     or:
00012         Model QCD;
00013             Particle quark,antiquark,-2;
00014             Particle ghost,antighost,-1;
00015             Particle gluon,+3;
00016             Vertex quark,antiquark,gluon: g;
00017             Vertex qghost,antighost,gluon: g;
00018             Vertex gluon,gluon,gluon: g;
00019             Vertex gluon,gluon,gluon,gluon: g^2;
00020         EndModel;
00021     Remarks:
00022     1: if no second particle is given, a particle and its antiparticle are the same
00023     2: Spin statistics is by dimension of SU(2) representation with a sign.
00024     3: In a vertex we need to mention the coupling constants.
00025     4: Internally a particle gives a vertex with only two lines.
00026     5: All particles must have been declared before any vertex.
00027
00028     The diagrams should be provided as a collection of nodes and edges
00029     with momenta with a direction substituted. The other parameters (like
00030     Lorentz indices, color indices etc.) can then be provided in a procedure
00031     that should go with this mode definition.
00032     In principle the edges are not needed if there are no 'insertions', but
00033     because we would like to have the insertions as well we need the edges.
00034
00035     #] Explanations :
00036     #[ Includes : model.c
00037 */
00038
00039 #include "form3.h"
00040
00041 /*
00042     #] Includes :
00043     #[ CreateVertex :
00044 */
00045
00046 VERTEX *CreateVertex(MODEL *m)
00047 {
00048     VERTEX *v, **new;
00049     WORD newsz;
00050     int i;
00051     if ( m->invertices >= m->sizevertices ) {
00052         if ( m->sizevertices == 0 ) newsz = 20;
00053         else newsz = 2*m->sizevertices;
00054         new = (VERTEX **)Malloc1(newsz*sizeof(VERTEX *),"m->vertices");
00055         for ( i = 0; i < m->sizevertices; i++ ) new[i] = m->vertices[i];
00056         if ( m->sizevertices > 0 ) M_free(m->vertices,"m->vertices");
00057         m->vertices = new;
00058         m->sizevertices = newsz;
00059     }
00060     v = (VERTEX *)Malloc1(sizeof(VERTEX),"VERTEX");
00061     m->vertices[m->invertices++] = v;
00062     v->nparticles = 0;
00063     v->ncouplings = 0;
00064     v->type = 0;
00065     v->error = 0;
00066     v->externonly = 0;
00067     v->spare = 0;
00068     return(v);
00069 }
00070
00071 /*
00072     #] CreateVertex :
00073     #[ ReadParticle :
00074 */
00075
00076 UBYTE *ReadParticle(UBYTE *s, VERTEX *v, MODEL *m, int par)
00077 {
00078     UBYTE *name, c;
00079     PARTICLE *p = v->particles + v->nparticles++;
00080     WORD type, funnum;
00081     int i, j;
00082     SKIPBLANKS(s)
00083     name = s; s = SkipName(s); c = *s; *s = 0;
00084     if ( GetVar(name,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND ) {
00085         p->number = AddFunction(name,0,VERTEXFUNCTION,0,0,0,-1,-1) + FUNCTION;
00086     }
00087     else if ( par == 1 && type == CFUNCTION && functions[funnum].spec == VERTEXFUNCTION ) {
00088         p->number = funnum+FUNCTION;
00089     }

```

```

00090     else if ( par == 0 && type == CFUNCTION && functions[funnum].spec == VERTEXFUNCTION ) {
00091 /*
00092     We should check whether this particle exists already in this model.
00093     If so, this will be an error.
00094 */
00095     for ( i = 0; i < m->nparticles-1; i++ ) {
00096         for ( j = 0; j < m->vertices[i]->nparticles; j++ ) {
00097             if ( m->vertices[i]->particles[j].number == funnum ) {
00098                 MesPrint("&Illegal attempt to redefine a particle in the same model: %s",name);
00099                 v->error = 1;
00100             }
00101         }
00102     }
00103     p->number = funnum+FUNCTION;
00104 }
00105 else {
00106     MesPrint("&Name of particle previously declared as another variable: %s",name);
00107     v->error = 1;
00108 }
00109 *s = c;
00110 while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
00111 return(s);
00112 }
00113
00114 /*
00115 #] ReadParticle :
00116 #[ CoModel :
00117 */
00118
00119 int CoModel(UBYTE *s)
00120 {
00121     UBYTE *name, c;
00122     MODEL *m = (MODEL *)Malloc1(sizeof(MODEL),"Model");
00123     WORD numberofset, *e;
00124     SETS set;
00125     SKIPBLANKS(s)
00126     AC.ModelLevel++;
00127     int i;
00128 /*
00129     We now have to add the model to the list of models.
00130 */
00131     if ( AC.modelspace == 0 || AC.nummodels >= AC.modelspace ) {
00132         int newspace = 2*AC.modelspace+2, i;
00133         MODEL **models = (MODEL **)Malloc1((sizeof(MODEL *))*newspace,"AC.models");
00134         for ( i = 0; i < AC.modelspace; i++ ) models[i] = AC.models[i];
00135         if ( AC.models ) M_free(AC.models,"AC.models");
00136         AC.modelspace = newspace;
00137         AC.models = models;
00138     }
00139
00140     AC.models[AC.nummodels++] = m;
00141
00142     m->vertices = 0;
00143     m->couplings = 0;
00144     m->nparticles = m->nvertices = m->sizevertices = m->invertices = 0;
00145     m->sizecouplings = 0;
00146     m->error = 0;
00147     m->grccmodel = NULL;
00148
00149     name = s;
00150     if ( FG.cTable[*s] == 0 ) {
00151         while ( FG.cTable[*s] <= 1 ) s++;
00152         c = *s; *s = 0;
00153         m->name = strDup1(name,"Model name");
00154         *s = c;
00155         SKIPBLANKS(s)
00156         if ( *s != 0 ) {
00157             MesPrint("&Illegal option in model statement: %s",s);
00158             return(1);
00159         }
00160     }
00161     else {
00162         m->name = strDup1((UBYTE *) "---", "Model name");
00163         m->error = 1;
00164         MesPrint("&Illegal name for model: %s",name);
00165         return(1);
00166     }
00167 /*
00168     Now make it a set with one element. The type of the set is rather special
00169 */
00170     if ( GetName(AC.varnames,m->name,&numberofset,NOAUTO) != NAMENOTFOUND ) {
00171         MesPrint("&Name conflict with name %s of model",m->name);
00172         return(1);
00173     }
00174     numberofset = AddSet(name,0);
00175     set = Sets + numberofset;
00176     set->type = CMODEL;

```



```

00177     e = (WORD *)FromVarList (&AC.SetElementList);
00178     *e = AC.nummodels-1;
00179     set->last++;
00180     AC.SetList.numtemp = AC.SetList.num;
00181     AC.SetElementList.numtemp = AC.SetElementList.num;
00182
00183     for ( i = 0; i <= MAXLEGS; i++ ) m->legcouple[i] = 0;
00184     m->legcouple[2] = 1;
00185
00186     return(0);
00187 }
00188
00189 /*
00190  #] CoModel :
00191  #[ CoParticle :
00192 */
00193
00194 int CoParticle(UBYTE *s)
00195 {
00196     MODEL *m;
00197     VERTEX *v;
00198     int i;
00199     if ( AC.nummodels <= 0 ) {
00200         MesPrint("&No open model for particle statement");
00201         return(1);
00202     }
00203     m = AC.models[AC.nummodels-1];
00204     if ( m->nvertices > 0 ) {
00205         MesPrint("&In a model description Particle statements should be before Vertex statements.");
00206         m->error = 1;
00207         return(1);
00208     }
00209     v = CreateVertex(m);
00210     m->nparticles++;
00211     s = ReadParticle(s,v,m,0);
00212     v->particles[0].type = 1;
00213     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
00214     if ( v->error == 0 ) {
00215         if ( FG.cTable[*s] == 0 ) {
00216             s = ReadParticle(s,v,m,0);
00217             v->particles[1].type = -1;
00218             if ( v->particles[0].number == v->particles[1].number ) {
00219                 v->particles[0].type = 0;
00220                 v->particles[1].type = 0;
00221             }
00222         }
00223         else { /* Particle = antiparticle */
00224             v->particles[1] = v->particles[0];
00225             v->nparticles++;
00226             v->particles[0].type = 0;
00227             v->particles[1].type = 0;
00228         }
00229     }
00230     if ( v->error == 0 && ( *s == '+' || *s == '-' ) ) {
00231         int x = 0, sign = ( *s == '-' ) ? -1: 1;
00232         s++;
00233         while ( FG.cTable[*s] == 1 ) { x = 10*x + *s++ - '0'; }
00234         if ( x == 0 ) {
00235             v->error = 1;
00236             MesPrint("&Spin goes by dimension of SU(2) representation. Zero is not allowed.");
00237         }
00238         else { v->particles[0].spin = v->particles[1].spin = sign*x; }
00239         SKIPBLANKS(s)
00240     }
00241     else if ( v->error == 0 ) {
00242         v->particles[0].spin = v->particles[1].spin = 1;
00243         v->particles[0].type = 0;
00244         v->particles[1].type = 0;
00245     }
00246 /*
00247 Here will come future options
00248 */
00249 while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
00250 while ( v->error == 0 && *s != 0 && FG.cTable[*s] == 0 ) {
00251     UBYTE *opt = s, c;
00252     while ( FG.cTable[*s] == 0 ) s++;
00253     c = *s; *s = 0;
00254     if ( StrICont(opt,(UBYTE *)"external") == 0 ) { v->externonly = 1; }
00255     else {
00256         MesPrint("&Unrecognized option %s in particle statement.",opt);
00257         v->error = 1;
00258     }
00259     *s = c;
00260     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
00261 }
00262 /*
00263 The couplings array will be used for registering in what kind of vertices the

```

```

00264     particle is involved. (example: in QCD the quarks only in 2 and 3-point vertices)
00265 */
00266     for ( i = 0; i < MAXPARTICLES; i++ ) v->couplings[i] = 0;
00267     v->couplings[2] = 1;
00268     return(v->error);
00269 }
00270
00271 /*
00272 #] CoParticle :
00273 #[ CoVertex :
00274 */
00275
00276 int CoVertex(UBYTE *s)
00277 {
00278     UBYTE *ss, *name, c;
00279     WORD type;
00280     MODEL *m;
00281     VERTEX *v;
00282     if ( AC.ModelLevel <= 0 ) {
00283         MesPrint("&No open model for vertex statement");
00284         return(1);
00285     }
00286     m = AC.models[AC.nummodels-1];
00287 /*
00288     Get an object of type VERTEX
00289 */
00290     v = CreateVertex(m);
00291     m->nvertices++;
00292     SKIPBLANKS(s);
00293     while ( *s && *s != ':' ) {
00294         if ( v->error == 0 ) {
00295             s = ReadParticle(s,v,m,1);
00296             if ( v->error ) m->error = 1;
00297         }
00298         else {
00299             while ( *s && *s != ':' ) {
00300                 if ( *s == '[' ) { SKIPBRA1(s) }
00301                 else if ( *s == '(' ) { SKIPBRA3(s) }
00302                 else if ( *s == 0 ) {
00303                     v->error = 1;
00304                     MesPrint("&No coupling constant in vertex statement.");
00305                     break;
00306                 }
00307             }
00308             break;
00309         }
00310     }
00311     if ( v->error == 0 ) {
00312         if ( *s == ':' ) { /* read symbols and powers */
00313             s++; SKIPBLANKS(s);
00314             while ( *s ) {
00315                 if ( v->ncouplings >= 2*MAXCOUPLINGS ) {
00316                     MesPrint("&More than %d coupling constants in vertex.",(WORD)MAXCOUPLINGS);
00317                     MesPrint("      Recompile with a larger value for MAXCOUPLINGS.");
00318                     return(1);
00319                 }
00320                 ss = s; s = SkipName(s); c = *s; *s = 0; name = ConstructName(ss,0);
00321                 if ( GetVar(name,&type,&v->couplings[v->ncouplings],CSYMBOL,WITHAUTO)
00322                     == NAMENOTFOUND ) {
00323                     WORD minpow = -MAXPOWER;
00324                     WORD maxpow = MAXPOWER;
00325                     WORD cplx = 0, dim = 0;
00326                     v->couplings[v->ncouplings] = AddSymbol(name,minpow,maxpow,cplx,dim);
00327                 }
00328                 *s = c;
00329                 v->ncouplings++;
00330             }
00331             /*
00332             Now possibly a power
00333             */
00334             if ( *s == '^' ) { /* we need an integer */
00335                 WORD x = 0; WORD sign = 1;
00336                 s++;
00337                 while ( *s == '-' || *s == '+' ) {
00338                     if ( *s == '-' ) sign = -sign;
00339                     s++;
00340                 }
00341                 while ( FG.cTable[*s] == 1 ) x = 10*x + *s++ - '0';
00342                 v->couplings[v->ncouplings++] = x;
00343             }
00344             else {
00345                 v->couplings[v->ncouplings++] = 1;
00346             }
00347             SKIPBLANKS(s);
00348             if ( *s == '*' ) { s++; continue; }
00349             if ( *s != ',' ) break;
00350             s++;
00351             SKIPBLANKS(s);

```

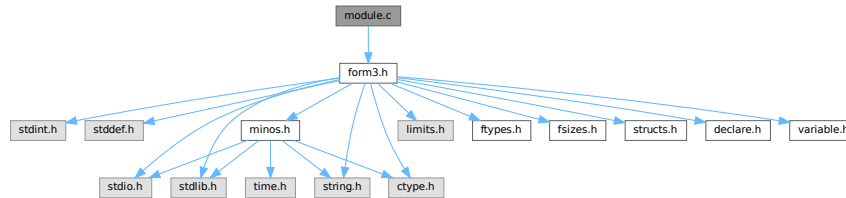
```

00351     }
00352   }
00353   else {
00354     v->error = 1;
00355     MesPrint("%A vertex statement needs at least one coupling constant.");
00356   }
00357 }
00358 if ( v->error == 0 ) {
00359 /*
00360     Register which types of vertices are possible for each particle.
00361 */
00362     int i, j;
00363     for ( i = 0; i < v->nparticles; i++ ) {
00364         for ( j = 0; j < m->nparticles; j++ ) {
00365             if ( m->vertices[j]->particles[0].number == v->particles[i].number ) break;
00366             if ( m->vertices[j]->particles[1].number == v->particles[i].number ) break;
00367         }
00368         m->vertices[j]->couplings[v->nparticles] = 1;
00369     }
00370 }
00371 return(v->error);
00372 }
00373
00374 /*
00375 #] CoVertex :
00376 #[ CoEndModel :
00377 */
00378
00379 int CoEndModel(UBYTE *s)
00380 {
00381     int i, j, k, kk;
00382     WORD csize = 0, *newcouplings, newsize;
00383     MODEL *m;
00384     VERTEX *v;
00385     if ( AC.ModelLevel <= 0 ) {
00386         MesPrint("&EndModel statement without matching Model statement");
00387         return(1);
00388     }
00389     m = AC.models[AC.nummodels-1];
00390 /*
00391     Now create a list of all coupling constants.
00392     Note that we do not expect an astronomical number of them and hence
00393     we use a simple insertion algorithm.
00394 */
00395     m->ncouplings = 0;
00396     for ( i = 0; i < m->nvertices; i++ ) {
00397         v = m->vertices[i+m->nparticles];
00398         m->legcouple[v->nparticles] = 1;
00399         if ( m->ncouplings + v->ncouplings > csize ) {
00400             if ( csize == 0 ) newsize = v->ncouplings + 10;
00401             else newsize = 2*csize;
00402             newcouplings = (WORD *)Mallco1(newsize*sizeof(WORD), "m->couplings");
00403             for ( k = 0; k < m->ncouplings; k++ ) newcouplings[k] = m->couplings[k];
00404             if ( csize > 0 ) M_free(m->couplings, "m->couplings");
00405             m->couplings = newcouplings;
00406             csize = newsize;
00407         }
00408         for ( j = 0; j < v->ncouplings; j += 2 ) {
00409             WORD sym = v->couplings[j];
00410             for ( k = 0; k < m->ncouplings; k++ ) {
00411                 if ( sym == m->couplings[k] ) break;
00412                 if ( sym < m->couplings[k] ) {
00413                     for ( kk = m->ncouplings; kk > k; k-- )
00414                         m->couplings[kk] = m->couplings[kk-1];
00415                     m->couplings[k] = sym;
00416                     m->ncouplings++;
00417                     break;
00418                 }
00419             }
00420             if ( k >= m->ncouplings ) m->couplings[m->ncouplings++] = sym;
00421         }
00422     }
00423     AC.ModelLevel--;
00424     DUMMYUSE(s)
00425     return(LoadModel(m));
00426 }
00427
00428 /*
00429 #] CoEndModel :
00430 */

```

## 4.76 module.c File Reference

#include "form3.h"  
 Include dependency graph for module.c:



### Functions

- int [ModuleInstruction](#) (int \*moduletype, int \*specialtype)
- int [CoModuleOption](#) (UBYTE \*s)
- int [CoModOption](#) (UBYTE \*s)
- void [SetSpecialMode](#) (int moduletype, int specialtype)
- int [ExecModule](#) (int moduletype)
- int [ExecStore](#) (void)
- void [FullCleanUp](#) (void)
- int [DoPolyfun](#) (UBYTE \*s)
- int [DoPolyratfun](#) (UBYTE \*s)
- int [DoNoParallel](#) (UBYTE \*s)
- int [DoParallel](#) (UBYTE \*s)
- int [DoModSum](#) (UBYTE \*s)
- int [DoModMax](#) (UBYTE \*s)
- int [DoModMin](#) (UBYTE \*s)
- int [DoModLocal](#) (UBYTE \*s)
- int [DoProcessBucket](#) (UBYTE \*s)
- UBYTE \* [DoModDollar](#) (UBYTE \*s, int type)
- int [DoinParallel](#) (UBYTE \*s)
- int [DonotinParallel](#) (UBYTE \*s)
- int [DoExecStatement](#) (void)
- int [DoPipeStatement](#) (void)

### 4.76.1 Detailed Description

A number of routines that deal with the moduleoption statement and the execution of modules. Additionally there are the execution of the exec and pipe instructions.

Definition in file [module.c](#).

### 4.76.2 Function Documentation

#### 4.76.2.1 ModuleInstruction()

```

int ModuleInstruction (
    int * moduletype,
    int * specialtype )
  
```

Definition at line 76 of file [module.c](#).

#### 4.76.2.2 CoModuleOption()

```
int CoModuleOption (
    UBYTE * s )
```

Definition at line 143 of file [module.c](#).

#### 4.76.2.3 CoModOption()

```
int CoModOption (
    UBYTE * s )
```

Definition at line 212 of file [module.c](#).

#### 4.76.2.4 SetSpecialMode()

```
void SetSpecialMode (
    int moduletype,
    int specialtype )
```

Definition at line 257 of file [module.c](#).

#### 4.76.2.5 ExecModule()

```
int ExecModule (
    int moduletype )
```

Definition at line 274 of file [module.c](#).

#### 4.76.2.6 ExecStore()

```
int ExecStore (
    void )
```

Definition at line 299 of file [module.c](#).

#### 4.76.2.7 FullCleanUp()

```
void FullCleanUp (
    void )
```

Definition at line 314 of file [module.c](#).

#### 4.76.2.8 DoPolyfun()

```
int DoPolyfun (
    UBYTE * s )
```

Definition at line 395 of file [module.c](#).

#### 4.76.2.9 DoPolyratfun()

```
int DoPolyratfun (
    UBYTE * s )
```

Definition at line 458 of file [module.c](#).

#### 4.76.2.10 DoNoParallel()

```
int DoNoParallel (
    UBYTE * s )
```

Definition at line 529 of file [module.c](#).

#### 4.76.2.11 DoParallel()

```
int DoParallel (
    UBYTE * s )
```

Definition at line 544 of file [module.c](#).

#### 4.76.2.12 DoModSum()

```
int DoModSum (
    UBYTE * s )
```

Definition at line 559 of file [module.c](#).

#### 4.76.2.13 DoModMax()

```
int DoModMax (
    UBYTE * s )
```

Definition at line 579 of file [module.c](#).

#### 4.76.2.14 DoModMin()

```
int DoModMin (
    UBYTE * s )
```

Definition at line 599 of file [module.c](#).

#### 4.76.2.15 DoModLocal()

```
int DoModLocal (
    UBYTE * s )
```

Definition at line 619 of file [module.c](#).

#### 4.76.2.16 DoProcessBucket()

```
int DoProcessBucket (
    UBYTE * s )
```

Definition at line 639 of file [module.c](#).

#### 4.76.2.17 DoModDollar()

```
UBYTE * DoModDollar (
    UBYTE * s,
    int type )
```

Definition at line 657 of file [module.c](#).

#### 4.76.2.18 DoinParallel()

```
int DoinParallel (
    UBYTE * s )
```

Definition at line 758 of file [module.c](#).

#### 4.76.2.19 DonotinParallel()

```
int DonotinParallel (
    UBYTE * s )
```

Definition at line 768 of file [module.c](#).

#### 4.76.2.20 DoExecStatement()

```
int DoExecStatement (
    void )
```

Definition at line 780 of file [module.c](#).

#### 4.76.2.21 DoPipeStatement()

```
int DoPipeStatement (
    void )
```

Definition at line 797 of file [module.c](#).

## 4.77 module.c

[Go to the documentation of this file.](#)

```

00001
00007 /* #[] License : */
00008 /*
00009 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00010 *   When using this file you are requested to refer to the publication
00011 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00012 *   This is considered a matter of courtesy as the development was paid
00013 *   for by FOM the Dutch physics granting agency and we would like to
00014 *   be able to track its scientific use to convince FOM of its value
00015 *   for the community.
00016 *
00017 *   This file is part of FORM.
00018 *
00019 *   FORM is free software: you can redistribute it and/or modify it under the
00020 *   terms of the GNU General Public License as published by the Free Software
00021 *   Foundation, either version 3 of the License, or (at your option) any later
00022 *   version.
00023 *
00024 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00025 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00026 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00027 *   details.
00028 *
00029 *   You should have received a copy of the GNU General Public License along
00030 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00031 */
00032 /* #[] License : */
00033 /*
00034 *   #[] Includes :
00035 */
00036
00037 #include "form3.h"
00038
00039 /*
00040 *   #[] Includes :
00041 *   #[] Modules :
00042 *   #[] ModuleInstruction :
00043
00044 *       Enters after the . of .sort etc
00045 *       We have word[[ (options) ][:commentary];]
00046 *       Word is one of 'clear end global sort store'
00047 *       Options are 'polyfun, endloopifunchanged, endloopifzero'
00048 *       Additions for solving equations can be added to 'word'
00049
00050 *       The flag in the moduleoptions is for telling whether the current
00051 *       option can be followed by others. If it is > 0 it cannot.
00052 */
00053
00054 static KEYWORD ModuleWords[] = {
00055     {"clear", (TFUN)0, CLEARMODULE, 0}
00056     , {"end", (TFUN)0, ENDMODULE, 0}
00057     , {"global", (TFUN)0, GLOBALMODULE, 0}
00058     , {"sort", (TFUN)0, SORTMODULE, 0}
00059     , {"store", (TFUN)0, STOREMODULE, 0}
00060 };
00061
00062 static KEYWORD ModuleOptions[] = {
00063     {"inparallel", DoinParallel, 1, 1}
00064     , {"local", DoModLocal, MODLOCAL, 0}
00065     , {"maximum", DoModMax, MODMAX, 0}
00066     , {"minimum", DoModMin, MODMIN, 0}
00067     , {"noparallel", DoNoParallel, NOPARALLEL_USER, 0}
00068     , {"notinparallel", DonotinParallel, 0, 1}
00069     , {"parallel", DoParallel, PARALLELFLAG, 0}
00070     , {"polyfun", DoPolyfun, POLYFUN, 0}
00071     , {"polyratfun", DoPolyratfun, POLYFUN, 0}
00072     , {"processbucketsize", DoProcessBucket, 0, 0}
00073     , {"sum", DoModSum, MODSUM, 0}
00074 };
00075
00076 int ModuleInstruction(int *moduletype, int *specialtype)
00077 {
00078     UBYTE *t, *s, *u, c;
00079     KEYWORD *key;
00080     int addit = 0, error = 0, i, j;
00081     DUMMYUSE(specialtype);
00082     LoadInstruction(0);
00083     AC.firstctypemessage = 0;
00084     s = AP.preStart; SKIPBLANKS(s)
00085     t = EndOfToken(s); c = *t; *t = 0;
00086     AC.origin = FROMPOINTINSTRUCTION;
00087     key = FindKeyWord(AP.preStart, ModuleWords, sizeof(ModuleWords)/sizeof(KEYWORD));

```



```

00088     if ( key == 0 ) {
00089         MesPrint("@Unrecognized module terminator: %s",s);
00090         error = 1;
00091         key = ModuleWords;
00092         while ( StrCmp((UBYTE *)key->name,(UBYTE *)"end") ) key++;
00093     }
00094     *t = c;
00095     *moduletype = key->type;
00096     SKIPBLANKS(t);
00097     while ( *t == '(' ) { /* There are options */
00098         s = t+1; SKIPBLANKS(t);
00099         if ( *t == 0 ) {
00100             MesPrint("@Improper options field in . instruction");
00101             error = 1;
00102         }
00103         else {
00104             *t = 0;
00105             if ( CoModOption(s) ) error = 1;
00106             *t++ = ')';
00107         }
00108     }
00109     if ( *t == ':' ) { /* There is an 'advertisement' */
00110         t++;
00111         SKIPBLANKS(t);
00112         s = t; i = 0;
00113         while ( *t && *t != ';' ) {
00114             if ( *t == '\\\\' ) t++;
00115             t++; i++;
00116         }
00117         u = t;
00118         while ( u > s && u[-1] == ' ' ) { u--; i--; }
00119         if ( *u == '\\\\' ) { u++; i++; }
00120         for ( j = COMMERCIALSIZE-1; j >= 0; j-- ) {
00121             if ( i <= 0 ) break;
00122             AC.Commercial[j] = *--u; i--;
00123             if ( u > s && u[-1] == '\\\\' ) u--;
00124         }
00125         for ( ; j >= 0; j-- ) AC.Commercial[j] = ' ';
00126         AC.Commercial[COMMERCIALSIZE] = 0;
00127         addit += 2;
00128     }
00129     if ( addit && *t != ';' ) {
00130         MesPrint("@Improper ending of . instruction");
00131         error = -1;
00132     }
00133     return(error);
00134 }
00135
00136 /*
00137     #] ModuleInstruction :
00138     #[ CoModuleOption :
00139
00140     ModuleOption, options;
00141 */
00142
00143 int CoModuleOption(UBYTE *s)
00144 {
00145     UBYTE *t,*tt,c;
00146     KEYWORD *option;
00147     int error = 0, polyflag = 0;
00148     AC.origin = FROMMODULEOPTION;
00149     if ( *s ) do {
00150         s = ToToken(s);
00151         t = EndOfToken(s);
00152         c = *t; *t = 0;
00153         option = FindKeyWord(s,ModuleOptions,
00154             sizeof(ModuleOptions)/sizeof(KEYWORD));
00155         if ( option == 0 ) {
00156             if ( polyflag ) {
00157                 *t = c; t++; s = SkipAName(t);
00158                 polyflag = 0;
00159                 continue;
00160             }
00161             else {
00162                 MesPrint("@Unrecognized module option: %s",s);
00163                 error = 1;
00164                 polyflag = 0;
00165                 *t = c;
00166             }
00167         }
00168         else {
00169             *t = c;
00170             SKIPBLANKS(t);
00171             if ( (option->func)(t) ) error = 1;
00172         }
00173         if ( StrCmp((UBYTE *) (option->name), (UBYTE *) ("polyfun")) == 0
00174             || StrCmp((UBYTE *) (option->name), (UBYTE *) ("polyratfun")) == 0 ) {

```

```

00175         polyflag = 1;
00176         t++;
00177         t = SkipAName(t);
00178     }
00179     else polyflag = 0;
00180     if ( option->flags > 0 ) return(error);
00181     while ( *t ) {
00182         if ( *t == ',' ) {
00183             tt = t+1;
00184             while ( *tt == ',' ) tt++;
00185             if ( *tt != '$' ) break;
00186             t = tt+1;
00187         }
00188         if ( *t == ')' ) break;
00189         if ( *t == '(' ) SKIPBRA3(t)
00190         else if ( *t == '{' ) SKIPBRA2(t)
00191         else if ( *t == '[' ) SKIPBRA1(t)
00192         t++;
00193     }
00194     s = t;
00195 } while ( *s == ',' );
00196 if ( *s ) {
00197     MesPrint("@Unrecognized module option: %s",s);
00198     error = 1;
00199 }
00200 return(error);
00201 }
00202
00203 /*
00204     #] CoModuleOption :
00205     #[ CoModOption :
00206
00207     To be called from a .instruction.
00208     Only recognizes polyfun. The newer ones should be via the
00209     ModuleOption statement.
00210 */
00211
00212 int CoModOption(UBYTE *s)
00213 {
00214     UBYTE *t,c;
00215     int error = 0;
00216     AC.origin = FROMPOINTINSTRUCTION;
00217     if ( *s ) do {
00218         s = ToToken(s);
00219         t = EndOfToken(s);
00220         c = *t; *t = 0;
00221         if ( StrICmp(s,(UBYTE *)"polyfun") == 0 ) {
00222             *t = c;
00223             SKIPBLANKS(t)
00224             if ( DoPolyfun(t) ) error = 1;
00225         }
00226         else if ( StrICmp(s,(UBYTE *)"polyratfun") == 0 ) {
00227             *t = c;
00228             SKIPBLANKS(t)
00229             if ( DoPolyratfun(t) ) error = 1;
00230         }
00231         else {
00232             MesPrint("@Unrecognized module option in .instruction: %s",s);
00233             error = 1;
00234             *t = c;
00235         }
00236         while ( *t ) {
00237             if ( *t == ',' || *t == ')' ) break;
00238             if ( *t == '(' ) SKIPBRA3(t)
00239             else if ( *t == '{' ) SKIPBRA2(t)
00240             else if ( *t == '[' ) SKIPBRA1(t)
00241             t++;
00242         }
00243         s = t;
00244     } while ( *s == ',' );
00245     if ( *s ) {
00246         MesPrint("@Unrecognized module option in .instruction: %s",s);
00247         error = 1;
00248     }
00249     return(error);
00250 }
00251
00252 /*
00253     #] CoModOption :
00254     #[ SetSpecialMode :
00255 */
00256
00257 void SetSpecialMode(int moduletype, int specialtype)
00258 {
00259     DUMMYUSE(moduletype); DUMMYUSE(specialtype);
00260 }
00261

```

```

00262 /*
00263     #] SetSpecialMode :
00264     #[ MakeGlobal :
00265
00266 void MakeGlobal()
00267 {
00268 }
00269
00270     #] MakeGlobal :
00271     #[ ExecModule :
00272 */
00273
00274 int ExecModule(int moduletype)
00275 {
00276 /*
00277     Here we check module options and other things that should
00278     be done at the start of a module.
00279 */
00280 #ifdef WITHFLOAT
00281     if ( ( AC.tMaxWeight != 0 && AC.tMaxWeight != AC.MaxWeight )
00282         || ( AC.tDefaultPrecision != 0 && AC.tDefaultPrecision != AC.DefaultPrecision ) ) {
00283         AC.DefaultPrecision = AC.tDefaultPrecision;
00284         AC.tDefaultPrecision = 0;
00285         AC.MaxWeight = AC.tMaxWeight;
00286         AC.tMaxWeight = 0;
00287         ClearMZVTables();
00288         SetupMZVTables();
00289     }
00290 #endif
00291     return (DoExecute(moduletype,0));
00292 }
00293
00294 /*
00295     #] ExecModule :
00296     #[ ExecStore :
00297 */
00298
00299 int ExecStore(void)
00300 {
00301     return(0);
00302 }
00303
00304 /*
00305     #] ExecStore :
00306     #[ FullCleanup :
00307
00308     Remark 27-oct-2005 by JV
00309     This routine (and Cleanup in startup.c) may still need some work:
00310         What to do with preprocessor variables
00311         What to do with files we write to
00312 */
00313
00314 void FullCleanup(void)
00315 {
00316     int j;
00317
00318     while ( AC.CurrentStream->previous >= 0 )
00319         AC.CurrentStream = CloseStream(AC.CurrentStream);
00320     AP.PreSwitchLevel = AP.PreIfLevel = 0;
00321
00322     for ( j = NumProcedures-1; j >= 0; j-- ) {
00323         if ( Procedures[j].name ) M_free(Procedures[j].name,"name of procedure");
00324         if ( Procedures[j].p.buffer ) M_free(Procedures[j].p.buffer,"buffer of procedure");
00325     }
00326     NumProcedures = 0;
00327
00328     while ( NumPre > AP.gNumPre ) {
00329         NumPre--;
00330         M_free(PreVar[NumPre].name,"PreVar[NumPre].name");
00331         PreVar[NumPre].name = PreVar[NumPre].value = 0;
00332     }
00333
00334     AC.DidClean = 0;
00335     for ( j = 0; j < NumExpressions; j++ ) {
00336         AC.exprnames->namenode[Expressions[j].node].type = CDELETE;
00337         AC.DidClean = 1;
00338     }
00339
00340     CompactifyTree(AC.exprnames,EXPRNAMES);
00341
00342     for ( j = AO.NumDictionaries-1; j >= 0; j-- ) {
00343         RemoveDictionary(AO.Dictionaries[j]);
00344         M_free(AO.Dictionaries[j],"Dictionary");
00345     }
00346     AO.NumDictionaries = AO.gNumDictionaries = 0;
00347     M_free(AO.Dictionaries,"Dictionaries");
00348     AO.Dictionaries = 0;

```

```

00349     AO.SizeDictionaries = 0;
00350     AP.OpenDictionary = 0;
00351     AO.CurrentDictionary = 0;
00352
00353     AP.ComChar = AP.cComChar;
00354     if ( AP.procedureExtension ) M_free(AP.procedureExtension,"procedureextension");
00355     AP.procedureExtension = strDupl(AP.cprocedureExtension,"procedureextension");
00356
00357     AC.StatsFlag = AM.gStatsFlag = AM.ggStatsFlag;
00358     AC.extrasymbols = AM.gextrasymbols = AM.ggextrasymbols;
00359     AC.extrasym[0] = AM.gextrasym[0] = AM.ggextrasym[0] = 'Z';
00360     AC.extrasym[1] = AM.gextrasym[1] = AM.ggextrasym[1] = 0;
00361     AO.NoSpacesInNumbers = AM.gNoSpacesInNumbers = AM.ggNoSpacesInNumbers;
00362     AO.IndentSpace = AM.gIndentSpace = AM.ggIndentSpace;
00363     AC.ThreadStats = AM.gThreadStats = AM.ggThreadStats;
00364     AC.OldFactArgFlag = AM.gOldFactArgFlag = AM.ggOldFactArgFlag;
00365     AC.FinalStats = AM.gFinalStats = AM.ggFinalStats;
00366     AC.OldGCDflag = AM.gOldGCDflag = AM.ggOldGCDflag;
00367     AC.ThreadsFlag = AM.gThreadsFlag = AM.ggThreadsFlag;
00368     if ( AC.ThreadsFlag && AM.totalnumberofthreads > 1 ) AS.MultiThreaded = 1;
00369     AC.ThreadBucketSize = AM.gThreadBucketSize = AM.ggThreadBucketSize;
00370     AC.ThreadBalancing = AM.gThreadBalancing = AM.ggThreadBalancing;
00371     AC.ThreadSortFileSynch = AM.gThreadSortFileSynch = AM.ggThreadSortFileSynch;
00372     AC.ShortStatsMax = AM.gShortStatsMax = AM.ggShortStatsMax;
00373     AC.SizeCommuteInSet = AM.gSizeCommuteInSet = 0;
00374     #ifdef WITHFLOAT
00375     AC.MaxWeight = AM.gMaxWeight = AM.ggMaxWeight;
00376     AC.DefaultPrecision = AM.gDefaultPrecision = AM.ggDefaultPrecision;
00377     #endif
00378
00379     NumExpressions = 0;
00380     if ( DeleteStore(0) < 0 ) {
00381         MesPrint("@Cannot restart the storage file");
00382         Terminate(-1);
00383     }
00384     RemoveDollars();
00385     CleanUp(1);
00386     ResetVariables(2);
00387     IniVars();
00388 }
00389
00390 /*
00391     #] FullCleanUp :
00392     #[ DoPolyfun :
00393 */
00394
00395 int DoPolyfun(UBYTE *s)
00396 {
00397     GETIDENTITY
00398     UBYTE *t, c;
00399     WORD funnum, eqsign = 0;
00400     if ( AC.origin == FROMPOINTINSTRUCTION ) {
00401         if ( *s == 0 || *s == ',' || *s == ')' ) {
00402             AR.PolyFun = 0; AR.PolyFunType = 0;
00403             return(0);
00404         }
00405         if ( *s != '=' ) {
00406             MesPrint("@Proper use in point instructions is: PolyFun[=functionname]");
00407             return(-1);
00408         }
00409         eqsign = 1;
00410     }
00411     else {
00412         if ( *s == 0 ) {
00413             AR.PolyFun = 0; AR.PolyFunType = 0;
00414             return(0);
00415         }
00416         if ( *s != '=' && *s != ',' ) {
00417             MesPrint("@Proper use is: PolyFun[{ ,= }functionname]");
00418             return(-1);
00419         }
00420         if ( *s == '=' ) eqsign = 1;
00421     }
00422     s++;
00423     SKIPBLANKS(s)
00424     t = EndOfToken(s);
00425     c = *t; *t = 0;
00426
00427     if ( GetName(AC.varnames,s,&funnum,WITHAUTO) != CFUNCTION ) {
00428         if ( AC.origin != FROMPOINTINSTRUCTION && eqsign == 0 ) {
00429             AR.PolyFun = 0; AR.PolyFunType = 0;
00430             return(0);
00431         }
00432         MesPrint("@ %s is not a properly declared function",s);
00433         *t = c;
00434         return(-1);
00435     }

```

```

00436     if ( functions[funnum].spec > 0 || functions[funnum].commute != 0 ) {
00437         MesPrint("@The PolyFun must be a regular commuting function!");
00438         *t = c;
00439         return(-1);
00440     }
00441     AR.PolyFun = funnum+FUNCTION; AR.PolyFunType = 1;
00442     *t = c;
00443     SKIPBLANKS(t)
00444     if ( *t && *t != ',' && *t != ')' ) {
00445         t++; c = *t; *t = 0;
00446         MesPrint("@Improper ending of end-of-module instruction: %s",s);
00447         *t = c;
00448         return(-1);
00449     }
00450     return(0);
00451 }
00452
00453 /*
00454     #] DoPolyfun :
00455     #[ DoPolyratfun :
00456 */
00457
00458 int DoPolyratfun(UBYTE *s)
00459 {
00460     GETIDENTITY
00461     UBYTE *t, c;
00462     WORD funnum;
00463     if ( AC.origin == FROMPOINTINSTRUCTION ) {
00464         if ( *s == 0 || *s == ',' || *s == ')' ) {
00465             AR.PolyFun = 0; AR.PolyFunType = 0; AR.PolyFunInv = 0; AR.PolyFunExp = 0;
00466             return(0);
00467         }
00468         if ( *s != '=' ) {
00469             MesPrint("@Proper use in point instructions is:
PolyRatFun[=functionname[+functionname]]");
00470             return(-1);
00471         }
00472     }
00473     else {
00474         if ( *s == 0 ) {
00475             AR.PolyFun = 0; AR.PolyFunType = 0; AR.PolyFunInv = 0; AR.PolyFunExp = 0;
00476             return(0);
00477         }
00478         if ( *s != '=' && *s != ',' ) {
00479             MesPrint("@Proper use is: PolyRatFun[{ ,= }functionname[+functionname]]");
00480             return(-1);
00481         }
00482     }
00483     s++;
00484     SKIPBLANKS(s)
00485     t = EndOfToken(s);
00486     c = *t; *t = 0;
00487
00488     if ( GetName(AC.varnames,s,&funnum,WITHAUTO) != CFUNCTION ) {
00489 Error1:;
00490         MesPrint("@ %s is not a properly declared function",s);
00491         *t = c;
00492         return(-1);
00493     }
00494     if ( functions[funnum].spec > 0 || functions[funnum].commute != 0 ) {
00495 Error2:;
00496         MesPrint("@The PolyRatFun must be a regular commuting function!");
00497         *t = c;
00498         return(-1);
00499     }
00500     AR.PolyFun = funnum+FUNCTION; AR.PolyFunType = 2;
00501     AR.PolyFunInv = 0;
00502     AR.PolyFunExp = 0;
00503     AC.PolyRatFunChanged = 1;
00504     *t = c;
00505     if ( *t == '+' ) {
00506         t++; s = t;
00507         t = EndOfToken(s);
00508         c = *t; *t = 0;
00509         if ( GetName(AC.varnames,s,&funnum,WITHAUTO) != CFUNCTION ) goto Error1;
00510         if ( functions[funnum].spec > 0 || functions[funnum].commute != 0 ) goto Error2;
00511         AR.PolyFunInv = funnum+FUNCTION;
00512         *t = c;
00513     }
00514     SKIPBLANKS(t)
00515     if ( *t && *t != ',' && *t != ')' ) {
00516         t++; c = *t; *t = 0;
00517         MesPrint("@Improper ending of end-of-module instruction: %s",s);
00518         *t = c;
00519         return(-1);
00520     }
00521     return(0);

```

```

00522 }
00523
00524 /*
00525     #] DoPolyratfun :
00526     #[ DoNoParallel :
00527 */
00528
00529 int DoNoParallel(UBYTE *s)
00530 {
00531     if ( *s == 0 || *s == ',' || *s == ')' ) {
00532         AC.mparallelflag |= NOPARALLEL_USER;
00533         return(0);
00534     }
00535     MesPrint("@NoParallel should not have extra parameters");
00536     return(-1);
00537 }
00538
00539 /*
00540     #] DoNoParallel :
00541     #[ DoParallel :
00542 */
00543
00544 int DoParallel(UBYTE *s)
00545 {
00546     if ( *s == 0 || *s == ',' || *s == ')' ) {
00547         AC.mparallelflag &= ~NOPARALLEL_USER;
00548         return(0);
00549     }
00550     MesPrint("@Parallel should not have extra parameters");
00551     return(-1);
00552 }
00553
00554 /*
00555     #] DoParallel :
00556     #[ DoModSum :
00557 */
00558
00559 int DoModSum(UBYTE *s)
00560 {
00561     while ( *s == ',' ) s++;
00562     if ( *s != '$' ) {
00563         MesPrint("@Module Sum should mention which $-variables");
00564         return(-1);
00565     }
00566     s = DoModDollar(s,MODSUM);
00567     if ( s && *s != 0 && *s != ')' ) {
00568         MesPrint("@Irregular end of Sum option of Module statement");
00569         return(-1);
00570     }
00571     return(0);
00572 }
00573
00574 /*
00575     #] DoModSum :
00576     #[ DoModMax :
00577 */
00578
00579 int DoModMax(UBYTE *s)
00580 {
00581     while ( *s == ',' ) s++;
00582     if ( *s != '$' ) {
00583         MesPrint("@Module Maximum should mention which $-variables");
00584         return(-1);
00585     }
00586     s = DoModDollar(s,MODMAX);
00587     if ( s && *s != 0 ) {
00588         MesPrint("@Irregular end of Maximum option of Module statement");
00589         return(-1);
00590     }
00591     return(0);
00592 }
00593
00594 /*
00595     #] DoModMax :
00596     #[ DoModMin :
00597 */
00598
00599 int DoModMin(UBYTE *s)
00600 {
00601     while ( *s == ',' ) s++;
00602     if ( *s != '$' ) {
00603         MesPrint("@Module Minimum should mention which $-variables");
00604         return(-1);
00605     }
00606     s = DoModDollar(s,MODMIN);
00607     if ( s && *s != 0 ) {
00608         MesPrint("@Irregular end of Minimum option of Module statement");

```

```

00609         return(-1);
00610     }
00611     return(0);
00612 }
00613
00614 /*
00615     #] DoModMin :
00616     #[ DoModLocal :
00617 */
00618
00619 int DoModLocal(UBYTE *s)
00620 {
00621     while ( *s == ',' ) s++;
00622     if ( *s != '$' ) {
00623         MesPrint("@ModuleOption Local should mention which $-variables");
00624         return(-1);
00625     }
00626     s = DoModDollar(s,MODLOCAL);
00627     if ( s && *s != 0 ) {
00628         MesPrint("@Irregular end of Local option of ModuleOption statement");
00629         return(-1);
00630     }
00631     return(0);
00632 }
00633
00634 /*
00635     #] DoModLocal :
00636     #[ DoProcessBucket :
00637 */
00638
00639 int DoProcessBucket(UBYTE *s)
00640 {
00641     LONG x;
00642     while ( *s == ',' || *s == '=' ) s++;
00643     ParseNumber(x,s);
00644     if ( *s && *s != ' ' && *s != '\t' ) {
00645         MesPrint("&Numerical value expected for ProcessBucketSize");
00646         return(1);
00647     }
00648     AC.mProcessBucketSize = x;
00649     return(0);
00650 }
00651
00652 /*
00653     #] DoProcessBucket :
00654     #[ DoModDollar :
00655 */
00656
00657 UBYTE * DoModDollar(UBYTE *s, int type)
00658 {
00659     UBYTE *name, c;
00660     WORD number;
00661     MODOPTDOLLAR *md;
00662     while ( *s == '$' ) {
00663         /*
00664             Read the name of the dollar
00665             Mark the type
00666         */
00667         s++;
00668         name = s;
00669         if ( FG.cTable[*s] == 0 ) {
00670             while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
00671             c = *s; *s = 0;
00672             number = GetDollar(name);
00673             if ( number < 0 ) {
00674                 number = AddDollar(s,0,0,0);
00675                 Warning("&Undefined $-variable in module statement");
00676             }
00677             md = (MODOPTDOLLAR *)FromList(&AC.ModOptDolList);
00678             md->number = number;
00679             md->type = type;
00680 #ifdef WITHPTHREADS
00681             if ( type == MODLOCAL ) {
00682                 int j, i;
00683                 DOLLARS dglobal, dlocal;
00684                 md->dstruct = (DOLLARS)Malloc1(
00685                     sizeof(struct DOLLARS)*AM.totalnumberofthreads,"Local DOLLARS");
00686                 /*
00687                     Now copy the global dollar into the local copies.
00688                     This can be nontrivial if the value needs an allocation.
00689                     We don't really need the locks.
00690                 */
00691                 dglobal = Dollars + number;
00692                 for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
00693                     dlocal = md->dstruct + j;
00694                     dlocal->index = dglobal->index;
00695                     dlocal->node = dglobal->node;

```

```

00696         dlocal->type = dglobal->type;
00697         dlocal->name = dglobal->name;
00698         dlocal->size = dglobal->size;
00699         dlocal->where = dglobal->where;
00700         if ( dlocal->size > 0 ) {
00701             dlocal->where = (WORD *)Malloc1((dlocal->size+1)*sizeof(WORD),"Local dollar
value");
00702             for ( i = 0; i < dlocal->size; i++ )
00703                 dlocal->where[i] = dglobal->where[i];
00704             dlocal->where[dlocal->size] = 0;
00705         }
00706         dlocal->pthreadlockread = dummylock;
00707         dlocal->pthreadlockwrite = dummylock;
00708         dlocal->nfactors = dglobal->nfactors;
00709         if ( dglobal->nfactors > 1 ) {
00710             int nsize;
00711             WORD *t, *m;
00712             dlocal->factors = (FACDOLLAR
*)Malloc1(dglobal->nfactors*sizeof(FACDOLLAR),"Dollar factors");
00713             for ( i = 0; i < dglobal->nfactors; i++ ) {
00714                 nsize = dglobal->factors[i].size;
00715                 dlocal->factors[i].type = DOLUNDEFINED;
00716                 dlocal->factors[i].value = dglobal->factors[i].value;
00717                 if ( ( dlocal->factors[i].size = nsize ) > 0 ) {
00718                     dlocal->factors[i].where = t = (WORD
*)Malloc1(sizeof(WORD)*(nsize+1),"DollarCopyFactor");
00719                     m = dglobal->factors[i].where;
00720                     NCOPY(t,m,nsize);
00721                     *t = 0;
00722                 }
00723                 else {
00724                     dlocal->factors[i].where = 0;
00725                 }
00726             }
00727             else { dlocal->factors = 0; }
00728         }
00729     }
00730     else {
00731         md->dstruct = 0;
00732     }
00733 }
00734 #endif
00735 *s = c;
00736 }
00737 else {
00738     MesPrint("&Illegal name for $-variable in module option");
00739     while ( *s != ',' && *s != 0 && *s != ')' ) s++;
00740 }
00741 while ( *s == ',' ) s++;
00742 }
00743 return(s);
00744 }
00745
00746 /*
00747 #] DoModDollar :
00748 #[ DoinParallel :
00749
00750 The idea is that we should have the commands
00751 ModuleOption,InParallel;
00752 ModuleOption,InParallel,name1,name2,...,namen;
00753 ModuleOption,NotInParallel,name1,name2,...,namen;
00754 The advantage over the InParallel statement is that this statement
00755 comes after the definition of the expressions.
00756 */
00757
00758 int DoinParallel(UBYTE *s)
00759 {
00760     return(DoInParallel(s,1));
00761 }
00762
00763 /*
00764 #] DoinParallel :
00765 #[ DonotinParallel :
00766 */
00767
00768 int DonotinParallel(UBYTE *s)
00769 {
00770     return(DoInParallel(s,0));
00771 }
00772
00773 /*
00774 #] DonotinParallel :
00775 #] Modules :
00776 #[ External :
00777 #[ DoExecStatement :
00778 */
00779

```



```

00780 int DoExecStatement(void)
00781 {
00782     #ifdef WITHSYSTEM
00783         FLUSHCONSOLE;
00784         if ( system((char *) (AP.preStart)) ) return(-1);
00785         return(0);
00786     #else
00787         Error0("External programs not implemented on this computer/system");
00788         return(-1);
00789     #endif
00790 }
00791
00792 /*
00793     #] DoExecStatement :
00794     #[ DoPipeStatement :
00795 */
00796
00797 int DoPipeStatement(void)
00798 {
00799     #ifdef WITHPIPE
00800         FLUSHCONSOLE;
00801         if ( OpenStream(AP.preStart,PIPESTREAM,0,PREENOACTION) == 0 ) return(-1);
00802         return(0);
00803     #else
00804         Error0("Pipes not implemented on this computer/system");
00805         return(-1);
00806     #endif
00807 }
00808
00809 /*
00810     #] DoPipeStatement :
00811     #] External :
00812 */

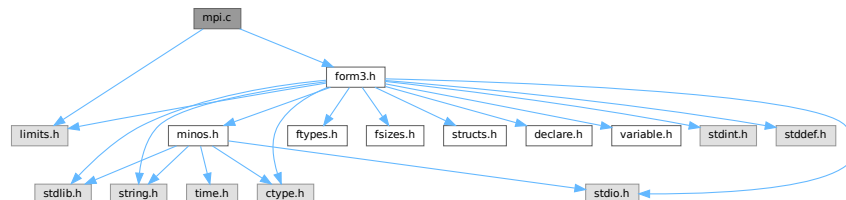
```

## 4.78 mpi.c File Reference

```
#include <limits.h>
```

```
#include "form3.h"
```

Include dependency graph for mpi.c:



### Data Structures

- struct [longMultiStruct](#)

### Macros

- #define [PF\\_PACKSIZE](#) 1600
- #define [MPI\\_ERRCODE\\_CHECK](#)(err)

### Typedefs

- typedef struct [longMultiStruct](#) [PF\\_LONGMULTI](#)

## Functions

- LONG [PF\\_RealTime](#) (int i)
- int [PF\\_LibInit](#) (int \*argcp, char \*\*\*argvp)
- int [PF\\_LibTerminate](#) (int error)
- int [PF\\_Probe](#) (int \*src)
- int [PF\\_ISendSbuf](#) (int to, int tag)
- int [PF\\_RecvWbuf](#) (WORD \*b, LONG \*s, int \*src)
- int [PF\\_IRecvRbuf](#) ([PF\\_BUFFER](#) \*r, int bn, int from)
- int [PF\\_WaitRbuf](#) ([PF\\_BUFFER](#) \*r, int bn, LONG \*size)
- int [PF\\_Bcast](#) (void \*buffer, int count)
- int [PF\\_RawSend](#) (int dest, void \*buf, LONG l, int tag)
- LONG [PF\\_RawRecv](#) (int \*src, void \*buf, LONG thesize, int \*tag)
- int [PF\\_RawProbe](#) (int \*src, int \*tag, int \*bytesize)
- int [PF\\_PrintPackBuf](#) (char \*s, int size)
- int [PF\\_PreparePack](#) (void)
- int [PF\\_Pack](#) (const void \*buffer, size\_t count, MPI\_Datatype type)
- int [PF\\_Unpack](#) (void \*buffer, size\_t count, MPI\_Datatype type)
- int [PF\\_PackString](#) (const UBYTE \*str)
- int [PF\\_UnpackString](#) (UBYTE \*str)
- int [PF\\_Send](#) (int to, int tag)
- int [PF\\_Receive](#) (int src, int tag, int \*psrc, int \*ptag)
- int [PF\\_Broadcast](#) (void)
- int [PF\\_PrepareLongSinglePack](#) (void)
- int [PF\\_LongSinglePack](#) (const void \*buffer, size\_t count, MPI\_Datatype type)
- int [PF\\_LongSingleUnpack](#) (void \*buffer, size\_t count, MPI\_Datatype type)
- int [PF\\_LongSingleSend](#) (int to, int tag)
- int [PF\\_LongSingleReceive](#) (int src, int tag, int \*psrc, int \*ptag)
- int [PF\\_PrepareLongMultiPack](#) (void)
- int [PF\\_LongMultiPackImpl](#) (const void \*buffer, size\_t count, size\_t eSize, MPI\_Datatype type)
- int [PF\\_LongMultiUnpackImpl](#) (void \*buffer, size\_t count, size\_t eSize, MPI\_Datatype type)
- int [PF\\_LongMultiBroadcast](#) (void)

## Variables

- LONG [PF\\_maxDollarChunkSize](#) = 0

### 4.78.1 Detailed Description

MPI dependent functions of parform

This file contains all the functions for the parallel version of form3 that explicitly need to call mpi routines. This is the only file that really needs to be linked to the mpi-library!

Definition in file [mpi.c](#).

### 4.78.2 Macro Definition Documentation

#### 4.78.2.1 PF\_PACKSIZE

```
#define PF_PACKSIZE 1600
```

Definition at line 58 of file [mpi.c](#).

### 4.78.2.2 MPI\_ERRCODE\_CHECK

```
#define MPI_ERRCODE_CHECK(  
    err )
```

#### Value:

```
do { \
    int _tmp_err = (err); \
    if ( _tmp_err != MPI_SUCCESS ) return _tmp_err != 0 ? _tmp_err : -1; \
} while (0)
```

A macro which exits the caller with a non-zero return value if *err* is not MPI\_SUCCESS.

#### Parameters

|            |                                                   |
|------------|---------------------------------------------------|
| <i>err</i> | The return value of a MPI function to be checked. |
|------------|---------------------------------------------------|

#### Remarks

The MPI standard defines MPI\_SUCCESS == 0. Then ( \_tmp\_err == 0 ) appears twice and we can expect the second evaluation will be eliminated by the compiler optimization.

Definition at line 84 of file [mpi.c](#).

## 4.78.3 Function Documentation

### 4.78.3.1 PF\_RealTime()

```
LONG PF_RealTime (  
    int i )
```

Returns the realtime in 1/100 sec. as a LONG.

#### Parameters

|          |                                    |
|----------|------------------------------------|
| <i>i</i> | the timer will be reset if i == 0. |
|----------|------------------------------------|

#### Returns

the real elapsed time in 1/100 second.

Definition at line 101 of file [mpi.c](#).

Referenced by [PF\\_Init\(\)](#).

### 4.78.3.2 PF\_LibInit()

```
int PF_LibInit (  
    int * argcp,  
    char *** argvp )
```

Performs all library dependent initializations.

**Parameters**

|              |                                     |
|--------------|-------------------------------------|
| <i>argcp</i> | pointer to the number of arguments. |
| <i>argvp</i> | pointer to the arguments.           |

**Returns**

0 if OK, nonzero on error.

Definition at line 123 of file [mpi.c](#).

Referenced by [PF\\_Init\(\)](#).

**4.78.3.3 PF\_LibTerminate()**

```
int PF_LibTerminate (
    int error )
```

Exits mpi, when there is an error either indicated or happening, returnvalue is 1, else returnvalue is 0.

**Parameters**

|              |                |
|--------------|----------------|
| <i>error</i> | an error code. |
|--------------|----------------|

**Returns**

0 if OK, nonzero on error.

Definition at line 209 of file [mpi.c](#).

Referenced by [PF\\_Terminate\(\)](#).

**4.78.3.4 PF\_Probe()**

```
int PF_Probe (
    int * src )
```

Probes the next incoming message. If src == PF\_ANY\_SOURCE this operation is blocking, otherwise nonblocking.

**Parameters**

|                |            |                                                                                  |
|----------------|------------|----------------------------------------------------------------------------------|
| <i>in, out</i> | <i>src</i> | the source process number. In output, the process number of actual found source. |
|----------------|------------|----------------------------------------------------------------------------------|

**Returns**

the tag value of the next incoming message if found, 0 if a nonblocking probe (input src != PF\_ANY\_SOURCE) did not find any messages. A negative returned value indicates an error.

Definition at line 230 of file [mpi.c](#).

#### 4.78.3.5 PF\_ISendSbuf()

```
int PF_ISendSbuf (
    int to,
    int tag )
```

Nonblocking send operation. It sends everything from `buff` to `fill` of the active buffer. Depending on `tag` it also can do waiting for other sends to finish or set the active buffer to the next one.

##### Parameters

|            |                                 |
|------------|---------------------------------|
| <i>to</i>  | the destination process number. |
| <i>tag</i> | the message tag.                |

##### Returns

0 if OK, nonzero on error.

Definition at line 261 of file [mpi.c](#).

Referenced by [FlushOut\(\)](#), [PF\\_Processor\(\)](#), and [PutOut\(\)](#).

#### 4.78.3.6 PF\_RecvWbuf()

```
int PF_RecvWbuf (
    WORD * b,
    LONG * s,
    int * src )
```

Blocking receive of a `WORD` buffer.

##### Parameters

|                |            |                                                                                     |
|----------------|------------|-------------------------------------------------------------------------------------|
| <i>out</i>     | <i>b</i>   | the buffer to store the received data.                                              |
| <i>in, out</i> | <i>s</i>   | the size of the buffer. The output value is the actual size of the received data.   |
| <i>in, out</i> | <i>src</i> | the source process number. The output value is the process number of actual source. |

##### Returns

the received message tag. A negative value indicates an error.

Definition at line 337 of file [mpi.c](#).

#### 4.78.3.7 PF\_IRecvRbuf()

```
int PF_IRecvRbuf (
    PF_BUFFER * r,
    int bn,
    int from )
```

Posts nonblocking receive for the active receive buffer. The buffer is filled from `full` to `stop`.

## Parameters

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| <i>r</i>    | the <a href="#">PF_BUFFER</a> struct for the nonblocking receive. |
| <i>bn</i>   | the index of the cyclic buffer.                                   |
| <i>from</i> | the source process number.                                        |

## Returns

0 if OK, nonzero on error.

Definition at line [366](#) of file [mpi.c](#).

**4.78.3.8 PF\_WaitRbuf()**

```
int PF_WaitRbuf (
    PF\_BUFFER * r,
    int bn,
    LONG * size )
```

Waits for the buffer *bn* to finish a pending nonblocking receive. It returns the received tag and in *\*size* the number of field received. If the receive is already finished, just return the flag and size, else wait for it to finish, but also check for other pending receives.

## Parameters

|     |             |                                                                           |
|-----|-------------|---------------------------------------------------------------------------|
|     | <i>r</i>    | the <a href="#">PF_BUFFER</a> struct for the pending nonblocking receive. |
|     | <i>bn</i>   | the index of the cyclic buffer.                                           |
| out | <i>size</i> | the actual size of received data.                                         |

## Returns

the received message tag. A negative value indicates an error.

Definition at line [400](#) of file [mpi.c](#).

**4.78.3.9 PF\_Bcast()**

```
int PF_Bcast (
    void * buffer,
    int count )
```

Broadcasts a message from the master to slaves.

## Parameters

|         |               |                                                                                                                       |
|---------|---------------|-----------------------------------------------------------------------------------------------------------------------|
| in, out | <i>buffer</i> | the starting address of buffer. The contents in this buffer on the master will be transferred to those on the slaves. |
|         | <i>count</i>  | the length of the buffer in bytes.                                                                                    |

**Returns**

0 if OK, nonzero on error.

Definition at line 440 of file [mpi.c](#).

Referenced by [PF\\_BroadcastBuffer\(\)](#), and [PF\\_BroadcastNumber\(\)](#).

**4.78.3.10 PF\_RawSend()**

```
int PF_RawSend (
    int dest,
    void * buf,
    LONG l,
    int tag )
```

Sends *l* bytes from *buf* to *dest*. Returns 0 on success, or -1.

**Parameters**

|             |                                               |
|-------------|-----------------------------------------------|
| <i>dest</i> | the destination process number.               |
| <i>buf</i>  | the send buffer.                              |
| <i>l</i>    | the size of data in the send buffer in bytes. |
| <i>tag</i>  | the message tag.                              |

**Returns**

0 if OK, nonzero on error.

Definition at line 463 of file [mpi.c](#).

Referenced by [PF\\_MUnlock\(\)](#), and [PF\\_SendFile\(\)](#).

**4.78.3.11 PF\_RawRecv()**

```
LONG PF_RawRecv (
    int * src,
    void * buf,
    LONG thesize,
    int * tag )
```

Receives not more than *thesize* bytes from *src*, returns the actual number of received bytes, or -1 on failure.

**Parameters**

|         |                |                                                                            |
|---------|----------------|----------------------------------------------------------------------------|
| in, out | <i>src</i>     | the source process number. In output, that of the actual received message. |
| out     | <i>buf</i>     | the receive buffer.                                                        |
|         | <i>thesize</i> | the size of the receive buffer in bytes.                                   |
| out     | <i>tag</i>     | the message tag of the actual received message.                            |

**Returns**

the actual sizeof received data in bytes, or -1 on failure.

Definition at line 484 of file [mpi.c](#).

Referenced by [PF\\_RecvFile\(\)](#).

**4.78.3.12 PF\_RawProbe()**

```
int PF_RawProbe (
    int * src,
    int * tag,
    int * bytesize )
```

Probes an incoming message.

**Parameters**

|         |                 |                                                                            |
|---------|-----------------|----------------------------------------------------------------------------|
| in, out | <i>src</i>      | the source process number. In output, that of the actual received message. |
| in, out | <i>tag</i>      | the message tag. In output, that of the actual received message.           |
| out     | <i>bytesize</i> | the size of incoming data in bytes.                                        |

**Returns**

0 if OK, nonzero on error.

Definition at line 508 of file [mpi.c](#).

**4.78.3.13 PF\_PrintPackBuf()**

```
int PF_PrintPackBuf (
    char * s,
    int size )
```

Prints the contents in the pack buffer.

**Parameters**

|             |                                       |
|-------------|---------------------------------------|
| <i>s</i>    | a message to be shown.                |
| <i>size</i> | the length of the buffer to be shown. |

**Returns**

0 if OK, nonzero on error.

Definition at line 594 of file [mpi.c](#).



#### 4.78.3.14 PF\_PreparePack()

```
int PF_PreparePack (
    void )
```

Prepares for the next pack operations on the sender.

##### Returns

0 if OK, nonzero on error.

Definition at line 624 of file [mpi.c](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastPreDollar\(\)](#), [PF\\_BroadcastString\(\)](#), and [PF\\_Init\(\)](#).

#### 4.78.3.15 PF\_Pack()

```
int PF_Pack (
    const void * buffer,
    size_t count,
    MPI_Datatype type )
```

Adds data into the pack buffer.

##### Parameters

|               |                                                          |
|---------------|----------------------------------------------------------|
| <i>buffer</i> | the pointer to the buffer storing the data to be packed. |
| <i>count</i>  | the number of elements in the buffer.                    |
| <i>type</i>   | the data type of elements in the buffer.                 |

##### Returns

0 if OK, nonzero on error.

Definition at line 642 of file [mpi.c](#).

References [MPI\\_ERRCODE\\_CHECK](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastPreDollar\(\)](#), and [PF\\_Init\(\)](#).

#### 4.78.3.16 PF\_Unpack()

```
int PF_Unpack (
    void * buffer,
    size_t count,
    MPI_Datatype type )
```

Retrieves the next data in the pack buffer.

**Parameters**

|                  |               |                                                       |
|------------------|---------------|-------------------------------------------------------|
| <code>out</code> | <i>buffer</i> | the pointer to the buffer to store the unpacked data. |
|                  | <i>count</i>  | the number of elements of data to be received.        |
|                  | <i>type</i>   | the data type of elements of data to be received.     |

**Returns**

0 if OK, nonzero on error.

Definition at line 671 of file [mpi.c](#).

References [MPI\\_ERRCODE\\_CHECK](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastPreDollar\(\)](#), and [PF\\_Init\(\)](#).

**4.78.3.17 PF\_PackString()**

```
int PF_PackString (
    const UBYTE * str )
```

Packs a string *str* into the packed buffer `PF_packbuf`, including the trailing zero.

The first element (`PF_INT`) is the length of the packed portion of the string. If the string does not fit to the buffer `PF_packbuf`, the function packs only the initial portion. It returns the number of packed bytes, so if (`str[length-1]=='\0'`) then the whole string fits to the buffer, if not, then the rest (`str+length`) must be packed and send again. On error, the function returns the negative error code.

One exception: the string `"\0\0"` is used as an image of the NULL, so all 3 characters will be packed.

**Parameters**

|            |                        |
|------------|------------------------|
| <i>str</i> | a string to be packed. |
|------------|------------------------|

**Returns**

the number of packed bytes, or a negative value on failure.

Definition at line 706 of file [mpi.c](#).

Referenced by [PF\\_BroadcastString\(\)](#).

**4.78.3.18 PF\_UnpackString()**

```
int PF_UnpackString (
    UBYTE * str )
```

Unpacks a string to *str* from the packed buffer `PF_packbuf`, including the trailing zero.

It returns the number of unpacked bytes, so if (`str[length-1]=='\0'`) then the whole string was unpacked, if not, then the rest must be appended to (`str+length`). On error, the function returns the negative error code.

## Parameters

|            |            |                                         |
|------------|------------|-----------------------------------------|
| <i>out</i> | <i>str</i> | the buffer to store the unpacked string |
|------------|------------|-----------------------------------------|

## Returns

the number of unpacked bytes, or a negative value on failure.

Definition at line 774 of file [mpi.c](#).

Referenced by [PF\\_BroadcastString\(\)](#).

#### 4.78.3.19 PF\_Send()

```
int PF_Send (
    int to,
    int tag )
```

Sends the contents in the pack buffer to the process specified by *to*.

## Example:

```
if ( PF.me == SRC ) {
    PF_PreparePack();
    // Packing operations here...
    PF_Send(DEST, TAG);
}
else if ( PF.me == DEST ) {
    PF_Receive(SRC, TAG, &actual_src, &actual_tag);
    // Unpacking operations here...
}
```

## Parameters

|            |                                 |
|------------|---------------------------------|
| <i>to</i>  | the destination process number. |
| <i>tag</i> | the message tag.                |

## Returns

0 if OK, nonzero on error.

Definition at line 822 of file [mpi.c](#).

References [MPI\\_ERRCODE\\_CHECK](#).

#### 4.78.3.20 PF\_Receive()

```
int PF_Receive (
    int src,
    int tag,
    int * psrc,
    int * ptag )
```

Receives data into the pack buffer from the process specified by *src*. This function allows *&src == psrc* or *&tag == ptag*. Either *psrc* or *ptag* can be NULL.

See the example of [PF\\_Send\(\)](#).

**Parameters**

|     |             |                                                       |
|-----|-------------|-------------------------------------------------------|
|     | <i>src</i>  | the source process number (can be PF_ANY_SOURCE).     |
|     | <i>tag</i>  | the source message tag (can be PF_ANY_TAG).           |
| out | <i>psrc</i> | the actual source process number of received message. |
| out | <i>ptag</i> | the received message tag.                             |

**Returns**

0 if OK, nonzero on error.

Definition at line 848 of file [mpi.c](#).

References [MPI\\_ERRCODE\\_CHECK](#).

**4.78.3.21 PF\_Broadcast()**

```
int PF_Broadcast (
    void )
```

Broadcasts the contents in the pack buffer on the master to those on the slaves.

**Example:**

```
if ( PF.me == MASTER ) {
    PF_PreparePack();
    // Packing operations here...
}
PF_Broadcast();
if ( PF.me != MASTER ) {
    // Unpacking operations here...
}
```

**Returns**

0 if OK, nonzero on error.

Definition at line 883 of file [mpi.c](#).

References [MPI\\_ERRCODE\\_CHECK](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastPreDollar\(\)](#), [PF\\_BroadcastString\(\)](#), and [PF\\_Init\(\)](#).

**4.78.3.22 PF\_PrepareLongSinglePack()**

```
int PF_PrepareLongSinglePack (
    void )
```

Prepares for the next long-single-pack operations on the sender.

**Returns**

0 if OK, nonzero on error.

Definition at line 1451 of file [mpi.c](#).

Referenced by [PF\\_CollectModifiedDollars\(\)](#), and [PF\\_Processor\(\)](#).

**4.78.3.23 PF\_LongSinglePack()**

```
int PF_LongSinglePack (
    const void * buffer,
    size_t count,
    MPI_Datatype type )
```

Adds data into the "long single" pack buffer.

## Parameters

|               |                                                          |
|---------------|----------------------------------------------------------|
| <i>buffer</i> | the pointer to the buffer storing the data to be packed. |
| <i>count</i>  | the number of elements in the buffer.                    |
| <i>type</i>   | the data type of elements in the buffer.                 |

## Returns

0 if OK, nonzero on error.

Definition at line 1469 of file [mpi.c](#).

Referenced by [PF\\_CollectModifiedDollars\(\)](#), and [PF\\_Processor\(\)](#).

#### 4.78.3.24 PF\_LongSingleUnpack()

```
int PF_LongSingleUnpack (
    void * buffer,
    size_t count,
    MPI_Datatype type )
```

Retrieves the next data in the "long single" pack buffer.

## Parameters

|            |               |                                                       |
|------------|---------------|-------------------------------------------------------|
| <i>out</i> | <i>buffer</i> | the pointer to the buffer to store the unpacked data. |
|            | <i>count</i>  | the number of elements of data to be received.        |
|            | <i>type</i>   | the data type of elements of data to be received.     |

## Returns

0 if OK, nonzero on error.

Definition at line 1503 of file [mpi.c](#).

Referenced by [PF\\_CollectModifiedDollars\(\)](#), and [PF\\_Processor\(\)](#).

#### 4.78.3.25 PF\_LongSingleSend()

```
int PF_LongSingleSend (
    int to,
    int tag )
```

Sends the contents in the "long single" pack buffer to the process specified by *to*.

## Example:

```
if ( PF.me == SRC ) {
    PF_PrepareLongSinglePack();
    // Packing operations here...
    PF_LongSingleSend(DEST, TAG);
}
else if ( PF.me == DEST ) {
    PF_LongSingleReceive(SRC, TAG, &actual_src, &actual_tag);
    // Unpacking operations here...
}
```

## Parameters

|            |                                 |
|------------|---------------------------------|
| <i>to</i>  | the destination process number. |
| <i>tag</i> | the message tag.                |

## Returns

0 if OK, nonzero on error.

Definition at line 1540 of file [mpi.c](#).

Referenced by [PF\\_CollectModifiedDollars\(\)](#), and [PF\\_Processor\(\)](#).

**4.78.3.26 PF\_LongSingleReceive()**

```
int PF_LongSingleReceive (
    int src,
    int tag,
    int * psrc,
    int * ptag )
```

Receives data into the "long single" pack buffer from the process specified by *src*. This function allows &*src* == *psrc* or &*tag* == *ptag*. Either *psrc* or *ptag* can be NULL.

See the example of [PF\\_LongSingleSend\(\)](#).

## Parameters

|     |             |                                                       |
|-----|-------------|-------------------------------------------------------|
|     | <i>src</i>  | the source process number (can be PF_ANY_SOURCE).     |
|     | <i>tag</i>  | the source message tag (can be PF_ANY_TAG).           |
| out | <i>psrc</i> | the actual source process number of received message. |
| out | <i>ptag</i> | the received message tag.                             |

## Returns

0 if OK, nonzero on error.

Definition at line 1583 of file [mpi.c](#).

Referenced by [PF\\_CollectModifiedDollars\(\)](#), and [PF\\_Processor\(\)](#).

**4.78.3.27 PF\_PrepareLongMultiPack()**

```
int PF_PrepareLongMultiPack (
    void )
```

Prepares for the next long-multi-pack operations on the sender.

## Returns

0 if OK, nonzero on error.

Definition at line 1643 of file [mpi.c](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastCBuf\(\)](#), [PF\\_BroadcastExpFlags\(\)](#), [PF\\_BroadcastModifiedDollars\(\)](#), and [PF\\_BroadcastRedefinedPreVars\(\)](#).

### 4.78.3.28 PF\_LongMultiPackImpl()

```
int PF_LongMultiPackImpl (
    const void * buffer,
    size_t count,
    size_t eSize,
    MPI_Datatype type )
```

Adds data into the "long multi" pack buffer.

#### Parameters

|               |                                                          |
|---------------|----------------------------------------------------------|
| <i>buffer</i> | the pointer to the buffer storing the data to be packed. |
| <i>count</i>  | the number of elements in the buffer.                    |
| <i>eSize</i>  | the byte size of each element of data.                   |
| <i>type</i>   | the data type of elements in the buffer.                 |

#### Returns

0 if OK, nonzero on error.

Definition at line 1662 of file [mpi.c](#).

References [PF\\_LongMultiPackImpl\(\)](#).

Referenced by [PF\\_LongMultiPackImpl\(\)](#).

Here is the call graph for this function:



### 4.78.3.29 PF\_LongMultiUnpackImpl()

```
int PF_LongMultiUnpackImpl (
    void * buffer,
    size_t count,
    size_t eSize,
    MPI_Datatype type )
```

Retrieves the next data in the "long multi" pack buffer.

**Parameters**

|     |               |                                                       |
|-----|---------------|-------------------------------------------------------|
| out | <i>buffer</i> | the pointer to the buffer to store the unpacked data. |
|     | <i>count</i>  | the number of elements of data to be received.        |
|     | <i>eSize</i>  | the byte size of each element of data.                |
|     | <i>type</i>   | the data type of elements of data to be received.     |

**Returns**

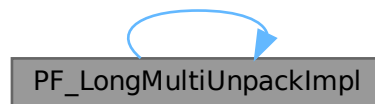
0 if OK, nonzero on error.

Definition at line 1721 of file [mpi.c](#).

References [PF\\_LongMultiUnpackImpl\(\)](#).

Referenced by [PF\\_LongMultiUnpackImpl\(\)](#).

Here is the call graph for this function:

**4.78.3.30 PF\_LongMultiBroadcast()**

```
int PF_LongMultiBroadcast (
    void )
```

Broadcasts the contents in the "long multi" pack buffer on the master to those on the slaves.

**Example:**

```
if ( PF.me == MASTER ) {
    PF_PrepareLongMultiPack();
    // Packing operations here...
}
PF_LongMultiBroadcast();
if ( PF.me != MASTER ) {
    // Unpacking operations here...
}
```

**Returns**

0 if OK, nonzero on error.

Definition at line 1807 of file [mpi.c](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastCBuf\(\)](#), [PF\\_BroadcastExpFlags\(\)](#), [PF\\_BroadcastModifiedDollars\(\)](#), and [PF\\_BroadcastRedefinedPreVars\(\)](#).



## 4.78.4 Variable Documentation

### 4.78.4.1 PF\_maxDollarChunkSize

LONG PF\_maxDollarChunkSize = 0

Definition at line 69 of file [mpi.c](#).

## 4.79 mpi.c

[Go to the documentation of this file.](#)

```

00001
00009 /* #[ License : */
00010 /*
00011 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00012 *   When using this file you are requested to refer to the publication
00013 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00014 *   This is considered a matter of courtesy as the development was paid
00015 *   for by FOM the Dutch physics granting agency and we would like to
00016 *   be able to track its scientific use to convince FOM of its value
00017 *   for the community.
00018 *
00019 *   This file is part of FORM.
00020 *
00021 *   FORM is free software: you can redistribute it and/or modify it under the
00022 *   terms of the GNU General Public License as published by the Free Software
00023 *   Foundation, either version 3 of the License, or (at your option) any later
00024 *   version.
00025 *
00026 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00027 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00028 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00029 *   details.
00030 *
00031 *   You should have received a copy of the GNU General Public License along
00032 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00033 */
00034 /* #[ License : */
00035 /*
00036 *   #[ Includes and variables :
00037 */
00038
00039 #include <limits.h>
00040 #include "form3.h"
00041
00042 #ifdef MPICH_PROFILING
00043 # include "mpe.h"
00044 #endif
00045
00046 #ifdef MPIDEBUGGING
00047 #include "mpidbg.h"
00048 #endif
00049
00050 /*[12oct2005 mt]:*/
00051 /*
00052 *   Today there was some cleanup, some stuff is moved into another place
00053 *   in this file, and PF.packsize is removed and PF_packsize is used
00054 *   instead. It is rather difficult to proper comment it, so not all these
00055 *   changing are marked by "[12oct2005 mt]"
00056 */
00057
00058 #define PF_PACKSIZE 1600
00059
00060 /*
00061 *   Size in bytes, will be initialized soon as
00062 *   PF_packsize=PF_PACKSIZE/sizeof(int)*sizeof(int); for possible
00063 *   future developing we prefer to do this initialization not here,
00064 *   but in PF_LibInit:
00065 */
00066
00067 static int PF_packsize = 0;
00068 static MPI_Status PF_status;
00069 LONG PF_maxDollarChunkSize = 0; /*:[04oct2005 mt]*/
00070
00071 static int PF_ShortPackInit(void);
00072 static int PF_longPackInit(void); /*:[12oct2005 mt]*/
00073

```

```

00084 #define MPI_ERRCODE_CHECK(err) \
00085     do { \
00086         int _tmp_err = (err); \
00087         if ( _tmp_err != MPI_SUCCESS ) return _tmp_err != 0 ? _tmp_err : -1; \
00088     } while (0)
00089
00090 /*
00091     #] Includes and variables :
00092     #[ PF_RealTime :
00093 */
00094
00101 LONG PF_RealTime(int i)
00102 {
00103     static double starttime;
00104     if ( i == PF_RESET ) {
00105         starttime = MPI_Wtime();
00106         return((LONG)0);
00107     }
00108     return((LONG)( 100. * (MPI_Wtime() - starttime) ));
00109 }
00110
00111 /*
00112     #] PF_RealTime :
00113     #[ PF_LibInit :
00114 */
00115
00123 int PF_LibInit(int *argcp, char ***argvp)
00124 {
00125     int ret;
00126     ret = MPI_Init(argcp,argvp);
00127     if ( ret != MPI_SUCCESS ) return(ret);
00128     ret = MPI_Comm_rank(PF_COMM,&PF.me);
00129     if ( ret != MPI_SUCCESS ) return(ret);
00130     ret = MPI_Comm_size(PF_COMM,&PF.numtasks);
00131     if ( ret != MPI_SUCCESS ) return(ret);
00132
00133     /* Initialization of packed communications. */
00134     PF_packsize = PF_PACKSIZE/sizeof(int)*sizeof(int);
00135     if ( PF_ShortPackInit() ) return -1;
00136     if ( PF_longPackInit() ) return -1;
00137
00138     /*Block*/
00139     int bytes, totalbytes=0;
00140
00141     /*
00142         There is one problem with maximal possible packing: there is no API to
00143         convert bytes to the record number. So, here we calculate the buffer
00144         size needed for storing dollarvars:
00145
00146         LONG PF_maxDollarChunkSize is the size for the portion of the dollar
00147         variable buffer suitable for broadcasting. This variable should be
00148         visible from parallel.c
00149
00150         Evaluate PF_Pack(numterms,1,PF_INT):
00151     */
00152     if ( ( ret = MPI_Pack_size(1,PF_INT,PF_COMM,&bytes) )!=MPI_SUCCESS )
00153         return(ret);
00154     totalbytes+=bytes;
00155
00156     /*
00157         Evaluate PF_Pack( newsize,1,PF_LONG):
00158     */
00159     if ( ( ret = MPI_Pack_size(1,PF_LONG,PF_COMM,&bytes) )!=MPI_SUCCESS )
00160         return(ret);
00161     totalbytes += bytes;
00162
00163     /*
00164         Now available room is PF_packsize-totalbytes
00165     */
00166     totalbytes = PF_packsize-totalbytes;
00167
00168     /*
00169         Now totalbytes is the size of chunk in bytes.
00170         Evaluate this size in number of records:
00171
00172         Rough estimate:
00173     */
00174     PF_maxDollarChunkSize=totalbytes/sizeof(WORD);
00175
00176     /*
00177         Go to the up limit:
00178     */
00179     do {
00180         if ( ( ret = MPI_Pack_size(
00181             ++PF_maxDollarChunkSize,PF_WORD,PF_COMM,&bytes) )!=MPI_SUCCESS )
00182             return(ret);
00183     } while ( bytes<totalbytes );
00184
00185     /*
00186         Now the chunk size is too large
00187         And now evaluate the exact value:
00188     */

```

```

00184 */
00185     do {
00186         if ( ( ret = MPI_Pack_size(
00187             --PF_maxDollarChunkSize,PF_WORD,PF_COMM,&bytes) )!=MPI_SUCCESS )
00188             return(ret);
00189     } while ( bytes>totalbytes );
00190 /*
00191     Now PF_maxDollarChunkSize is the size of chunk of PF_WORD fitting the
00192     buffer <= (PF_packsize-PF_INT-PF_LONG)
00193 */
00194 }/*Block*/
00195 return(0);
00196 }
00197 /*
00198 #] PF_LibInit :
00199 #[ PF_LibTerminate :
00200 */
00201
00209 int PF_LibTerminate(int error)
00210 {
00211     DUMMYUSE(error);
00212     return(MPI_Finalize());
00213 }
00214
00215 /*
00216 #] PF_LibTerminate :
00217 #[ PF_Probe :
00218 */
00219
00230 int PF_Probe(int *src)
00231 {
00232     int ret, flag;
00233     if ( *src == PF_ANY_SOURCE ) { /*Blocking call*/
00234         ret = MPI_Probe(*src,MPI_ANY_TAG,PF_COMM,&PF_status);
00235         flag = 1;
00236     }
00237     else { /*Non-blocking call*/
00238         ret = MPI_Iprobe(*src,MPI_ANY_TAG,PF_COMM,&flag,&PF_status);
00239     }
00240     *src = PF_status.MPI_SOURCE;
00241     if ( ret != MPI_SUCCESS ) { if ( ret > 0 ) ret *= -1; return(ret); }
00242     if ( !flag ) return(0);
00243     return(PF_status.MPI_TAG);
00244 }
00245
00246 /*
00247 #] PF_Probe :
00248 #[ PF_ISendSbuf :
00249 */
00250
00261 int PF_ISendSbuf(int to, int tag)
00262 {
00263     PF_BUFFER *s = PF.sbuf;
00264     int a = s->active;
00265     int size = s->fill[a] - s->buff[a];
00266     int r = 0;
00267
00268     static int finished;
00269
00270     s->fill[a] = s->buff[a];
00271     if ( s->numbufs == 1 ) {
00272         r = MPI_Ssend(s->buff[a],size,PF_WORD,MASTER,tag,PF_COMM);
00273         if ( r != MPI_SUCCESS ) {
00274             fprintf(stderr,"%d%d] PF_ISendSbuf: MPI_Ssend returns: %d \n",
00275                 PF.me,(int)AC.CModule,r);
00276             fflush(stderr);
00277             return(r);
00278         }
00279         return(0);
00280     }
00281
00282     switch ( tag ) { /* things to do before sending */
00283     case PF_TERM_MSGTAG:
00284         if ( PF.sbuf->request[to] != MPI_REQUEST_NULL)
00285             r = MPI_Wait(&PF.sbuf->request[to],&PF.sbuf->retstat[to]);
00286         if ( r != MPI_SUCCESS ) return(r);
00287         break;
00288     default:
00289         break;
00290     }
00291
00292     r = MPI_Isend(s->buff[a],size,PF_WORD,to,tag,PF_COMM,&s->request[a]);
00293
00294     if ( r != MPI_SUCCESS ) return(r);
00295
00296     switch ( tag ) { /* things to do after initialising sending */
00297     case PF_TERM_MSGTAG:

```

```

00298         finished = 0;
00299         break;
00300     case PF_ENDSORT_MSGTAG:
00301         if ( ++finished == PF.numtasks - 1 )
00302             r = MPI_Waitall(s->numbufs,s->request,s->status);
00303         if ( r != MPI_SUCCESS ) return(r);
00304         break;
00305     case PF_BUFFER_MSGTAG:
00306         if ( ++s->active >= s->numbufs ) s->active = 0;
00307         while ( s->request[s->active] != MPI_REQUEST_NULL ) {
00308             r = MPI_Waitsome(s->numbufs,s->request,&size,s->index,s->retstat);
00309             if ( r != MPI_SUCCESS ) return(r);
00310         }
00311         break;
00312     case PF_ENDBUFFER_MSGTAG:
00313         if ( ++s->active >= s->numbufs ) s->active = 0;
00314         r = MPI_Waitall(s->numbufs,s->request,s->status);
00315         if ( r != MPI_SUCCESS ) return(r);
00316         break;
00317     default:
00318         return(-99);
00319         break;
00320 }
00321 return(0);
00322 }
00323
00324 /*
00325  #] PF_ISendSbuf :
00326  #[ PF_RecvWbuf :
00327  */
00328
00337 int PF_RecvWbuf(WORD *b, LONG *s, int *src)
00338 {
00339     int i, r = 0;
00340
00341     r = MPI_Recv(b, (int)*s, PF_WORD, *src, PF_ANY_MSGTAG, PF_COMM, &PF_status);
00342     if ( r != MPI_SUCCESS ) { if ( r > 0 ) r *= -1; return(r); }
00343
00344     r = MPI_Get_count(&PF_status, PF_WORD, &i);
00345     if ( r != MPI_SUCCESS ) { if ( r > 0 ) r *= -1; return(r); }
00346
00347     *s = (LONG)i;
00348     *src = PF_status.MPI_SOURCE;
00349     return(PF_status.MPI_TAG);
00350 }
00351
00352 /*
00353  #] PF_RecvWbuf :
00354  #[ PF_IRecvRbuf :
00355  */
00356
00366 int PF_IRecvRbuf(PF_BUFFER *r, int bn, int from)
00367 {
00368     int ret;
00369     r->type[bn] = PF_WORD;
00370
00371     if ( r->numbufs == 1 ) {
00372         r->tag[bn] = MPI_ANY_TAG;
00373         r->from[bn] = from;
00374     }
00375     else {
00376         ret = MPI_Irecv(r->full[bn], (int)(r->stop[bn] - r->full[bn]), PF_WORD, from,
00377             MPI_ANY_TAG, PF_COMM, &r->request[bn]);
00378         if (ret != MPI_SUCCESS) { if(ret > 0) ret *= -1; return(ret); }
00379     }
00380     return(0);
00381 }
00382
00383 /*
00384  #] PF_IRecvRbuf :
00385  #[ PF_WaitRbuf :
00386  */
00387
00400 int PF_WaitRbuf(PF_BUFFER *r, int bn, LONG *size)
00401 {
00402     int ret, rsize;
00403
00404     if ( r->numbufs == 1 ) {
00405         *size = r->stop[bn] - r->full[bn];
00406         ret = MPI_Recv(r->full[bn], (int)*size, r->type[bn], r->from[bn], r->tag[bn],
00407             PF_COMM, &(r->status[bn]));
00408         if ( ret != MPI_SUCCESS ) { if ( ret > 0 ) ret *= -1; return(ret); }
00409         ret = MPI_Get_count(&(r->status[bn]), r->type[bn], &rsize);
00410         if ( ret != MPI_SUCCESS ) { if ( ret > 0 ) ret *= -1; return(ret); }
00411         if ( rsize > *size ) return(-99);
00412         *size = (LONG)rsize;
00413     }

```

```

00414     else {
00415         while ( r->request[bn] != MPI_REQUEST_NULL ) {
00416             ret = MPI_Waitsome(r->numbufrs, r->request, &rsiz, r->index, r->retstat);
00417             if ( ret != MPI_SUCCESS ) { if ( ret > 0 ) ret *= -1; return(ret); }
00418             while ( --rsiz >= 0 ) r->status[r->index[rsiz]] = r->retstat[rsiz];
00419         }
00420         ret = MPI_Get_count(&(r->status[bn]), r->type[bn], &rsiz);
00421         if ( ret != MPI_SUCCESS ) { if ( ret > 0 ) ret *= -1; return(ret); }
00422         *size = (LONG)rsiz;
00423     }
00424     return(r->status[bn].MPI_TAG);
00425 }
00426
00427 /*
00428 #] PF_WaitRbuf :
00429 #[ PF_Bcast :
00430 */
00431
00440 int PF_Bcast(void *buffer, int count)
00441 {
00442     if ( MPI_Bcast(buffer, count, MPI_BYTE, MASTER, PF_COMM) != MPI_SUCCESS )
00443         return(-1);
00444     return(0);
00445 }
00446
00447 /*
00448 #] PF_Bcast :
00449 #[ PF_RawSend :
00450 */
00451
00463 int PF_RawSend(int dest, void *buf, LONG l, int tag)
00464 {
00465     int ret=MPI_Ssend(buf, (int)l, MPI_BYTE, dest, tag, PF_COMM);
00466     if ( ret != MPI_SUCCESS ) return(-1);
00467     return(0);
00468 }
00469 /*
00470 #] PF_RawSend :
00471 #[ PF_RawRecv :
00472 */
00473
00484 LONG PF_RawRecv(int *src, void *buf, LONG thesize, int *tag)
00485 {
00486     MPI_Status stat;
00487     int ret=MPI_Recv(buf, (int)thesize, MPI_BYTE, *src, MPI_ANY_TAG, PF_COMM, &stat);
00488     if ( ret != MPI_SUCCESS ) return(-1);
00489     if ( MPI_Get_count(&stat, MPI_BYTE, &ret) != MPI_SUCCESS ) return(-1);
00490     *tag = stat.MPI_TAG;
00491     *src = stat.MPI_SOURCE;
00492     return(ret);
00493 }
00494
00495 /*
00496 #] PF_RawRecv :
00497 #[ PF_RawProbe :
00498 */
00499
00508 int PF_RawProbe(int *src, int *tag, int *bytesize)
00509 {
00510     MPI_Status stat;
00511     int srcval = src != NULL ? *src : PF_ANY_SOURCE;
00512     int tagval = tag != NULL ? *tag : PF_ANY_MSTAG;
00513     int ret = MPI_Probe(srcval, tagval, PF_COMM, &stat);
00514     if ( ret != MPI_SUCCESS ) return -1;
00515     if ( src != NULL ) *src = stat.MPI_SOURCE;
00516     if ( tag != NULL ) *tag = stat.MPI_TAG;
00517     if ( bytesize != NULL ) {
00518         ret = MPI_Get_count(&stat, MPI_BYTE, bytesize);
00519         if ( ret != MPI_SUCCESS ) return -1;
00520     }
00521     return 0;
00522 }
00523
00524 /*
00525 #] PF_RawProbe :
00526 #[ The pack buffer :
00527 #[ Variables :
00528 */
00529
00530 /*
00531 * The pack buffer with the fixed size (= PF_packsize).
00532 */
00533 static UBYTE *PF_packbuf = NULL;
00534 static UBYTE *PF_packstop = NULL;
00535 static int PF_packpos = 0;
00536
00537 /*

```

```

00538         #] Variables :
00539         #[ PF_ShortPackInit :
00540     */
00541
00542 static int PF_ShortPackInit(void)
00543 {
00544     PF_packbuf = (UBYTE *)Malloc1(sizeof(UBYTE) * PF_packsize, "PF_ShortPackInit");
00545     if ( PF_packbuf == NULL ) return -1;
00546     PF_packstop = PF_packbuf + PF_packsize;
00547     return 0;
00548 }
00549
00550 /*
00551         #] PF_ShortPackInit :
00552         #[ PF_InitPackBuf :
00553     */
00554
00555 static inline int PF_InitPackBuf(void)
00556 {
00557     /*
00558         This is definitely not the best place for allocating the
00559         buffer! Moved to PF_LibInit():
00560     */
00561     if ( PF_packbuf == 0 ) {
00562         PF_packbuf = (UBYTE *)Malloc1(sizeof(UBYTE)*PF_packsize,"PF_InitPackBuf");
00563         if ( PF_packbuf == 0 ) return(-1);
00564         PF_packstop = PF_packbuf + PF_packsize;
00565     }
00566     /*
00567         PF_packpos = 0;
00568         return(0);
00569     */
00570 }
00571
00572 /*
00573         #] PF_InitPackBuf :
00574         #[ PF_PrintPackBuf :
00575     */
00576
00577 int PF_PrintPackBuf(char *s, int size)
00578 {
00579     #ifdef NOMESPRINTYET
00580     /*
00581         The use of printf should be discouraged. The results are flushed to
00582         the output at unpredictable moments. We should use printf only
00583         during startup when MesPrint doesn't have its buffers and output
00584         channels initialized.
00585     */
00586     int i;
00587     printf("[%d] %s: ",PF.me,s);
00588     for(i=0;i<size;i++) printf("%d ",PF_packbuf[i]);
00589     printf("\n");
00590     #else
00591     MesPrint("[%d] %s: %a",PF.me,s,size,(WORD *) (PF_packbuf));
00592     #endif
00593     return(0);
00594 }
00595
00596 /*
00597         #] PF_PrintPackBuf :
00598         #[ PF_PreparePack :
00599     */
00600
00601 int PF_PreparePack(void)
00602 {
00603     return PF_InitPackBuf();
00604 }
00605
00606 /*
00607         #] PF_PreparePack :
00608         #[ PF_Pack :
00609     */
00610
00611 int PF_Pack(const void *buffer, size_t count, MPI_Datatype type)
00612 {
00613     int err, bytes;
00614
00615     if ( count > INT_MAX ) return -99;
00616
00617     err = MPI_Pack_size((int)count, type, PF_COMM, &bytes);
00618     MPI_ERRCODE_CHECK(err);
00619     if ( PF_packpos + bytes > PF_packstop - PF_packbuf ) return -99;
00620
00621     err = MPI_Pack((void *)buffer, (int)count, type, PF_packbuf, PF_packsize, &PF_packpos, PF_COMM);
00622     MPI_ERRCODE_CHECK(err);
00623
00624     return 0;
00625 }

```

```

00657
00658 /*
00659     #] PF_Pack :
00660     #[ PF_Unpack :
00661 */
00662
00671 int PF_Unpack(void *buffer, size_t count, MPI_Datatype type)
00672 {
00673     int err;
00674
00675     if ( count > INT_MAX ) return -99;
00676
00677     err = MPI_Unpack(PF_packbuf, PF_packsize, &PF_packpos, buffer, (int)count, type, PF_COMM);
00678     MPI_ERRCODE_CHECK(err);
00679
00680     return 0;
00681 }
00682
00683 /*
00684     #] PF_Unpack :
00685     #[ PF_PackString :
00686 */
00687
00706 int PF_PackString(const UBYTE *str)
00707 {
00708     int ret, buflength, bytes, length;
00709     /*
00710         length will be packed in the beginning.
00711         Decrement buffer size by the length of the field "length":
00712     */
00713     if ( ( ret = MPI_Pack_size(1, PF_INT, PF_COMM, &bytes) ) != MPI_SUCCESS )
00714         return(ret);
00715     buflength = PF_packsize - bytes;
00716     /*
00717         Calculate the string length (INCLUDING the trailing zero!):
00718     */
00719     for ( length = 0; length < buflength; length++ ) {
00720         if ( str[length] == '\0' ) {
00721             length++; /* since the trailing zero must be accounted */
00722             break;
00723         }
00724     }
00725     /*
00726         The string "\0!\0" is used as an image of the NULL.
00727     */
00728     if ( ( str[0] == '\0' ) /* empty string */
00729         && ( str[1] == '!' ) /* Special case? */
00730         && ( str[2] == '\0' ) /* Yes, pass 3 initial symbols */
00731         ) length += 2; /* all 3 characters will be packed */
00732     length++; /* Will be decremented in the following loop */
00733     /*
00734         The problem: packed size of byte may be not equal 1! So first, suppose
00735         it is 1, and if this is not the case decrease the length of the string
00736         until it fits the buffer:
00737     */
00738     do {
00739         if ( ( ret = MPI_Pack_size(--length, PF_BYTE, PF_COMM, &bytes) )
00740             != MPI_SUCCESS ) return(ret);
00741     } while ( bytes > buflength );
00742     /*
00743         Note, now if str[length-1] == '\0' then the string fits to the buffer
00744         (INCLUDING the trailing zero!); if not, the rest must be packed further!
00745
00746         Pack the length to PF_packbuf:
00747     */
00748     if ( ( ret = MPI_Pack(&length, 1, PF_INT, PF_packbuf, PF_packsize,
00749         &PF_packpos, PF_COMM) ) != MPI_SUCCESS ) return(ret);
00750     /*
00751         Pack the string to PF_packbuf:
00752     */
00753     if ( ( ret = MPI_Pack((UBYTE *)str, length, PF_BYTE, PF_packbuf, PF_packsize,
00754         &PF_packpos, PF_COMM) ) != MPI_SUCCESS ) return(ret);
00755     return(length);
00756 }
00757
00758 /*
00759     #] PF_PackString :
00760     #[ PF_UnpackString :
00761 */
00762
00774 int PF_UnpackString(UBYTE *str)
00775 {
00776     int ret, length;
00777     /*
00778         Unpack the length:
00779     */
00780     if( (ret = MPI_Unpack(PF_packbuf, PF_packsize, &PF_packpos,

```

```

00781         &length,1,PF_INT,PF_COMM))!= MPI_SUCCESS )
00782             return(ret);
00783  /*
00784     Unpack the string:
00785  */
00786     if ( ( ret = MPI_Unpack(PF_packbuf,PF_packsize,&PF_packpos,
00787         str,length,PF_BYTE,PF_COMM) ) != MPI_SUCCESS ) return(ret);
00788  /*
00789     Now if str[length-1]=='\0' then the whole string
00790     (INCLUDING the trailing zero!) was unpacked ;if not, the rest
00791     must be unpacked to str+length.
00792  */
00793     return(length);
00794 }
00795
00796  /*
00797     #] PF_UnpackString :
00798     #[ PF_Send :
00799  */
00800
00822 int PF_Send(int to, int tag)
00823 {
00824     int err;
00825     err = MPI_Ssend(PF_packbuf, PF_packpos, MPI_PACKED, to, tag, PF_COMM);
00826     MPI_ERRCODE_CHECK(err);
00827     return 0;
00828 }
00829
00830  /*
00831     #] PF_Send :
00832     #[ PF_Receive :
00833  */
00834
00848 int PF_Receive(int src, int tag, int *psrc, int *ptag)
00849 {
00850     int err;
00851     MPI_Status status;
00852     PF_InitPackBuf();
00853     err = MPI_Recv(PF_packbuf, PF_packsize, MPI_PACKED, src, tag, PF_COMM, &status);
00854     MPI_ERRCODE_CHECK(err);
00855     if ( psrc ) *psrc = status.MPI_SOURCE;
00856     if ( ptag ) *ptag = status.MPI_TAG;
00857     return 0;
00858 }
00859
00860  /*
00861     #] PF_Receive :
00862     #[ PF_Broadcast :
00863  */
00864
00883 int PF_Broadcast(void)
00884 {
00885     int err;
00886  /*
00887   * If PF_SHORTBROADCAST is defined, then the broadcasting will be performed in
00888   * 2 steps. First, the size of the buffer will be broadcast, then the buffer of
00889   * exactly used size. This should be faster with slow connections, but slower on
00890   * SMP shmem MPI because of the latency.
00891   */
00892  #ifdef PF_SHORTBROADCAST
00893     int pos = PF_packpos;
00894  #endif
00895     if ( PF.me != MASTER ) {
00896         err = PF_InitPackBuf();
00897         if ( err ) return err;
00898     }
00899  #ifdef PF_SHORTBROADCAST
00900     err = MPI_Bcast(&pos, 1, MPI_INT, MASTER, PF_COMM);
00901     MPI_ERRCODE_CHECK(err);
00902     err = MPI_Bcast(PF_packbuf, pos, MPI_PACKED, MASTER, PF_COMM);
00903  #else
00904     err = MPI_Bcast(PF_packbuf, PF_packsize, MPI_PACKED, MASTER, PF_COMM);
00905  #endif
00906     MPI_ERRCODE_CHECK(err);
00907     return 0;
00908 }
00909
00910  /*
00911     #] PF_Broadcast :
00912     #] The pack buffer :
00913     #[ Long pack stuff :
00914     #[ Explanations :
00915
00916     The problems here are:
00917     1. We need to send/receive long dollar variables. For
00918     preprocessor-defined dollarvars we used multiply
00919     packing/broadcasting (see parallel.c:PF_BroadcastPreDollar())

```



```

00920     since each variable must be broadcast immediately. For run-time
00921     the changed dollar variables, collecting and broadcasting are
00922     performed at the end of the module and all modified dollarvars
00923     are transferred "at once", that is why the size of packed and
00924     transferred buffers may be really very large.
00925     2. There is some strange feature of MPI_Bcast() on Altix MPI
00926     implementation, namely, sometimes it silently fails with big
00927     buffers. For better performance, it would be useful to send one
00928     big buffer instead of several small ones (since the latency is more
00929     important than the bandwidth). That is why we need two different
00930     sets of routines: for long point-to-point communication we collect
00931     big re-allocatable buffer, the corresponding routines have the
00932     prefix PF_longSingle, and for broadcasting we pack data into
00933     several smaller buffers, the corresponding routines have the
00934     prefix PF_longMulti.
00935     Note, from portability reasons we cannot split large packed
00936     buffer into small chunks, send them and collect back on the other
00937     side, see "Advice to users" on page 180 MPI--The Complete Reference
00938     Volume1, second edition.
00939     OPTIMIZING:
00940     We assume, for most communications, the single buffer of size
00941     PF_packsize is enough.
00942
00943     How does it work:
00944     For point-to-point, there is one big re-allocatable
00945     buffer PF_longPackBuf with two integer positions: PF_longPackPos
00946     and PF_longPackTop (due to re-allocatable character of the buffer,
00947     it is better to use integers rather than pointers).
00948     Each time of re-allocation, the size of the buffer
00949     PF_longPackBuf is incremented by the same size of a "standard" chunk
00950     PF_packsize.
00951     For broadcasting there is one linked list (PF_longMultiRoot),
00952     which contains either positions of a chunk of PF_longPackBuf, or
00953     it's own buffer. This is done for better memory utilisation:
00954     longSingle and longMulti are never used simultaneously.
00955     When a new cell is needed for longMulti packing, we increment
00956     the counter PF_longPackN and just follow the list. If it is not
00957     possible, we allocate the cell's own buffer and link it to the end
00958     of the list PF_longMultiRoot.
00959     When PF_longPackPos is reallocated, we link new chunks into
00960     existing PF_longMultiRoot list before the first longMulti allocated
00961     cell's own buffer. The pointer PF_longMultiLastChunk points to the last
00962     cell of PF_longMultiRoot containing the pointer to the chunk of
00963     PF_longPackBuf.
00964     Initialization PF_longPackBuf is made by the function
00965     PF_longSingleReset(). In the begin of the PF_longPackBuf it packs
00966     the size of the last sent buffer. Upon sending, the program checks,
00967     whether there was at list one re-allocation (PF_longPackN>1) .
00968     If so, the sender first packs and sends small buffer
00969     (PF_longPackSmallBuf) containing one integer number -- the
00970     _negative_ new size of the send buffer. Getting the buffer, a
00971     receiver unpacks one integer and checks whether it is <0 . If so,
00972     the receiver will repeat receiving, but first it checks whether
00973     it has enough buffer and increase it, if necessary.
00974     Initialization PF_longMultiRoot is made by the function
00975     PF_longMultiReset(). In the begin of the first chunk it packs
00976     one integer -- the number 1. Upon sending, the program checks,
00977     how many cells were packed (PF_longPackN). If more than 1, the
00978     sender packs to the next cell the integer PF_longPackN, than
00979     packs PF_longPackN pairs of integers -- the information about how many
00980     times chunk on each cell was accessed by the packing procedure,
00981     this information is contained by the nPacks field of the cell
00982     structure, and how many non-complete items was at the end of this
00983     chunk the structure field lastLen. Then the sender sends first
00984     this auxiliary chunk.
00985     The receiver unpacks the integer from obtained chunk and, if this
00986     integer is more than 1, it gets more chunks, unpacking information
00987     from the first auxiliary chunk into the corresponding nPacks
00988     fields. Unpacking information from multiple chunks, the receiver
00989     knows, when the chunk is expired and it must switch to the next cell,
00990     successively decrementing corresponding nPacks field.
00991
00992     XXX: There are still some flaws:
00993     PF_LongSingleSend/PF_LongSingleReceive may fail, for example, for data
00994     transfers from the master to many slaves. Suppose that the master sends big
00995     data to slaves, which needs an increase of the buffer of the receivers. For
00996     the first data transfer, the master sends the new buffer size as the first
00997     message, and then sends the data as the second message, because
00998     PF_LongSinglePack records the increase of the buffer size on the master. For
00999     the next time, however, the master sends the data without sending the new
01000     buffer size, and then MPI_Recv fails due to the data overflow.
01001     In parallel.c, they are used for the communication from slaves to the
01002     master. In this case, this problem does not occur because the master always
01003     has enough buffer.
01004     The maximum size that PF_LongMultiBroadcast can broadcast is limited to
01005     around 320kB because the current implementation tries to pack all
01006     information of chained buffers into one buffer, whose size is PF_packsize

```

```

01007     = 1600B.
01008
01009     #] Explanations :
01010     #[ Variables :
01011 */
01012
01013 typedef struct longMultiStruct {
01014     UBYTE *buffer; /* NULL if */
01015     int bufpos; /* if >=0, PF_longPackBuf+bufpos is the chunk start */
01016     int packpos; /* the current position */
01017     int nPacks; /* How many times PF_longPack operates on this cell */
01018     int lastLen; /* if > 0, the last packing didn't fit completely to this
01019                  chunk, only lastLen items was packed, the rest is in
01020                  the next cell. */
01021     struct longMultiStruct *next; /* next linked cell, or NULL */
01022 } PF_LONGMULTI;
01023
01024 static UBYTE *PF_longPackBuf = NULL;
01025 static void *PF_longPackSmallBuf = NULL;
01026 static int PF_longPackPos = 0;
01027 static int PF_longPackTop = 0;
01028 static PF_LONGMULTI *PF_longMultiRoot = NULL;
01029 static PF_LONGMULTI *PF_longMultiTop = NULL;
01030 static PF_LONGMULTI *PF_longMultiLastChunk = NULL;
01031 static int PF_longPackN = 0;
01032
01033 /*
01034     #] Variables :
01035     #[ Long pack private functions :
01036     #[ PF_longMultiNewCell :
01037 */
01038
01039 static inline int PF_longMultiNewCell(void)
01040 {
01041     /*
01042         Allocate a new cell:
01043     */
01044     PF_longMultiTop->next = (PF_LONGMULTI *)
01045         Malloc1(sizeof(PF_LONGMULTI), "PF_longMultiCell");
01046     if ( PF_longMultiTop->next == NULL ) return(-1);
01047     /*
01048         Allocate a private buffer:
01049     */
01050     PF_longMultiTop->next->buffer=(UBYTE*)
01051         Malloc1(sizeof(UBYTE)*PF_packsize, "PF_longMultiChunk");
01052     if ( PF_longMultiTop->next->buffer == NULL ) return(-1);
01053     /*
01054         For the private buffer position is -1:
01055     */
01056     PF_longMultiTop->next->bufpos = -1;
01057     /*
01058         This is the last cell in the chain:
01059     */
01060     PF_longMultiTop->next->next = NULL;
01061     /*
01062         packpos and nPacks are not initialized!
01063     */
01064     return(0);
01065 }
01066
01067 /*
01068     #] PF_longMultiNewCell :
01069     #[ PF_longMultiPack2NextCell :
01070 */
01071 static inline int PF_longMultiPack2NextCell(void)
01072 {
01073     /*
01074         Is there a free cell in the chain?
01075     */
01076     if ( PF_longMultiTop->next == NULL ) {
01077         /*
01078             No, allocate the new cell with a private buffer:
01079         */
01080         if ( PF_longMultiNewCell() ) return(-1);
01081     }
01082     /*
01083         Move to the next cell in the chain:
01084     */
01085     PF_longMultiTop = PF_longMultiTop->next;
01086     /*
01087         if >=0, the cell buffer is the chunk of PF_longPackBuf, initialize it:
01088     */
01089     if ( PF_longMultiTop->bufpos > -1 )
01090         PF_longMultiTop->buffer = PF_longPackBuf+PF_longMultiTop->bufpos;
01091     /*
01092         else -- the cell has it's own private buffer.
01093         Initialize the cell fields:

```

```

01094 */
01095     PF_longMultiTop->nPacks = 0;
01096     PF_longMultiTop->lastLen = 0;
01097     PF_longMultiTop->packpos = 0;
01098     return(0);
01099 }
01100
01101 /*
01102     #] PF_longMultiPack2NextCell :
01103     #[ PF_longMultiNewChunkAdded :
01104 */
01105
01106 static inline int PF_longMultiNewChunkAdded(int n)
01107 {
01108     /*
01109         Store the list tail:
01110     */
01111     PF_LONGMULTI *MemCell = PF_longMultiLastChunk->next;
01112     int pos = PF_longPackTop;
01113
01114     while ( n-- > 0 ) {
01115         /*
01116             Allocate a new cell:
01117         */
01118         PF_longMultiLastChunk->next = (PF_LONGMULTI *)
01119             Malloc1(sizeof(PF_LONGMULTI), "PF_longMultiCell");
01120         if ( PF_longMultiLastChunk->next == NULL ) return(-1);
01121         /*
01122             Update the Last Chunk Pointer:
01123         */
01124         PF_longMultiLastChunk = PF_longMultiLastChunk->next;
01125         /*
01126             Initialize the new cell:
01127         */
01128         PF_longMultiLastChunk->bufpos = pos;
01129         pos += PF_packsize;
01130         PF_longMultiLastChunk->buffer = NULL;
01131         PF_longMultiLastChunk->packpos = 0;
01132         PF_longMultiLastChunk->nPacks = 0;
01133         PF_longMultiLastChunk->lastLen = 0;
01134     }
01135     /*
01136         Hitch the tail:
01137     */
01138     PF_longMultiLastChunk->next = MemCell;
01139     return(0);
01140 }
01141
01142 /*
01143     #] PF_longMultiNewChunkAdded :
01144     #[ PF_longCopyChunk :
01145 */
01146
01147 static inline void PF_longCopyChunk(int *to, int *from, int n)
01148 {
01149     NCOPYI(to, from, n)
01150     /* for ( ; n > 0; n-- ) *to++ = *from++; */
01151 }
01152
01153 /*
01154     #] PF_longCopyChunk :
01155     #[ PF_longAddChunk :
01156
01157     The chunk must be increased by n*PF_packsize.
01158 */
01159
01160 static int PF_longAddChunk(int n, int mustRealloc)
01161 {
01162     UBYTE *newbuf;
01163     if ( ( newbuf = (UBYTE*)Malloc1(sizeof(UBYTE)*(PF_longPackTop+n*PF_packsize),
01164         "PF_longPackBuf" ) ) == NULL ) return(-1);
01165     /*
01166         Allocate and chain a new cell for longMulti:
01167     */
01168     if ( PF_longMultiNewChunkAdded(n) ) return(-1);
01169     /*
01170         Copy the content to the new buffer:
01171     */
01172     if ( mustRealloc ) {
01173         PF_longCopyChunk( (int*)newbuf, (int*)PF_longPackBuf, PF_longPackTop/sizeof(int) );
01174     }
01175     /*
01176         Note, PF_packsize is multiple by sizeof(int) by construction!
01177     */
01178     PF_longPackTop += (n*PF_packsize);
01179     /*
01180         Free the old buffer and store the new one:

```

```

01181 */
01182     M_free(PF_longPackBuf,"PF_longPackBuf");
01183     PF_longPackBuf = newbuf;
01184 /*
01185         Count number of re-allocs:
01186 */
01187     PF_longPackN += n;
01188     return(0);
01189 }
01190
01191 /*
01192     #] PF_longAddChunk :
01193     #[ PF_longMultiHowSplit :
01194
01195     "count" of "type" elements in an input buffer occupy "bytes" bytes.
01196     We know from the algorithm, that it is too many. How to split
01197     the buffer so that the head fits to rest of a storage buffer?*/
01198 static inline int PF_longMultiHowSplit(int count, MPI_Datatype type, int bytes)
01199 {
01200     int ret, items, totalbytes;
01201
01202     if ( count < 2 ) return(0); /* Nothing to split */
01203 /*
01204         A rest of a storage buffer:
01205 */
01206     totalbytes = PF_packsize - PF_longMultiTop->packpos;
01207 /*
01208     Rough estimate:
01209 */
01210     items = (int)((double)totalbytes*count/bytes);
01211 /*
01212         Go to the up limit:
01213 */
01214     do {
01215         if ( ( ret = MPI_Pack_size(++items,type,PF_COMM,&bytes) )
01216             !=MPI_SUCCESS ) return(ret);
01217     } while ( bytes < totalbytes );
01218 /*
01219         Now the value of "items" is too large
01220         And now evaluate the exact value:
01221 */
01222     do {
01223         if ( ( ret = MPI_Pack_size(--items,type,PF_COMM,&bytes) )
01224             !=MPI_SUCCESS ) return(ret);
01225         if ( items == 0 ) /* Nothing about MPI_Pack_size(0) == 0 in standards! */
01226             return(0);
01227     } while ( bytes > totalbytes );
01228     return(items);
01229 }
01230 /*
01231     #] PF_longMultiHowSplit :
01232     #[ PF_longPackInit :
01233 */
01234
01235 static int PF_longPackInit(void)
01236 {
01237     int ret;
01238     PF_longPackBuf = (UBYTE*)Malloc1(sizeof(UBYTE)*PF_packsize,"PF_longPackBuf");
01239     if ( PF_longPackBuf == NULL ) return(-1);
01240 /*
01241         PF_longPackTop is not initialized yet, use in as a return value:
01242 */
01243     ret = MPI_Pack_size(1,MPI_INT,PF_COMM,&PF_longPackTop);
01244     if ( ret != MPI_SUCCESS ) return(ret);
01245
01246     PF_longPackSmallBuf =
01247         (void*)Malloc1(sizeof(UBYTE)*PF_longPackTop,"PF_longPackSmallBuf");
01248
01249     PF_longPackTop = PF_packsize;
01250     PF_longMultiRoot =
01251         (PF_LONGMULTI *)Malloc1(sizeof(PF_LONGMULTI),"PF_longMultiRoot");
01252     if ( PF_longMultiRoot == NULL ) return(-1);
01253     PF_longMultiRoot->bufpos = 0;
01254     PF_longMultiRoot->buffer = NULL;
01255     PF_longMultiRoot->next = NULL;
01256     PF_longMultiLastChunk = PF_longMultiRoot;
01257
01258     PF_longPackPos = 0;
01259     PF_longMultiRoot->packpos = 0;
01260     PF_longMultiTop = PF_longMultiRoot;
01261     PF_longPackN = 1;
01262     return(0);
01263 }
01264
01265 /*
01266     #] PF_longPackInit :
01267     #[ PF_longMultiPreparePrefix :

```

```

01268 */
01269
01270 static inline int PF_longMultiPreparePrefix(void)
01271 {
01272     int ret;
01273     PF_LONGMULTI *thePrefix;
01274     int i = PF_longPackN;
01275     /*
01276         Here we have PF_longPackN>1!
01277         New cell (at the list end) to create the auxiliary chunk:
01278     */
01279     if ( PF_longMultiPack2NextCell() ) return(-1);
01280     /*
01281         Store the pointer to the chunk we will proceed:
01282     */
01283     thePrefix = PF_longMultiTop;
01284     /*
01285         Pack PF_longPackN:
01286     */
01287     ret = MPI_Pack(&(PF_longPackN),
01288                   1,
01289                   MPI_INT,
01290                   thePrefix->buffer,
01291                   PF_packsize,
01292                   &(thePrefix->packpos),
01293                   PF_COMM);
01294     if ( ret != MPI_SUCCESS ) return(ret);
01295     /*
01296         And start from the beginning:
01297     */
01298     for ( PF_longMultiTop = PF_longMultiRoot; i > 0; i-- ) {
01299         /*
01300             Pack number of Pack hits:
01301         */
01302         ret = MPI_Pack(&(PF_longMultiTop->nPacks),
01303                       1,
01304                       MPI_INT,
01305                       thePrefix->buffer,
01306                       PF_packsize,
01307                       &(thePrefix->packpos),
01308                       PF_COMM);
01309         /*
01310             Pack the length of the last fit portion:
01311         */
01312         ret |= MPI_Pack(&(PF_longMultiTop->lastLen),
01313                       1,
01314                       MPI_INT,
01315                       thePrefix->buffer,
01316                       PF_packsize,
01317                       &(thePrefix->packpos),
01318                       PF_COMM);
01319         /*
01320             Check the size -- not necessary, MPI_Pack did it.
01321         */
01322         if ( ret != MPI_SUCCESS ) return(ret);
01323         /*
01324             Go to the next cell:
01325         */
01326         PF_longMultiTop = PF_longMultiTop->next;
01327     }
01328     PF_longMultiTop = thePrefix;
01329     /*
01330         PF_longMultiTop is ready!
01331     */
01332     return(0);
01333 }
01334
01335
01336 /*
01337     #] PF_longMultiPreparePrefix :
01338     #[ PF_longMultiProcessPrefix :
01339 */
01340
01341 static inline int PF_longMultiProcessPrefix(void)
01342 {
01343     int ret,i;
01344     /*
01345         We have PF_longPackN records packed in PF_longMultiRoot->buffer,
01346         pairs nPacks and lastLen. Loop through PF_longPackN cells,
01347         unpacking these integers into proper fields:
01348     */
01349     for ( PF_longMultiTop = PF_longMultiRoot, i = 0; i < PF_longPackN; i++ ) {
01350         /*
01351             Go to th next cell, allocating, when necessary:
01352         */
01353         if ( PF_longMultiPack2NextCell() ) return(-1);
01354         /*

```

```

01355         Unpack the number of Pack hits:
01356  */
01357     ret = MPI_Unpack(PF_longMultiRoot->buffer,
01358                     PF_packsize,
01359                     &( PF_longMultiRoot->packpos),
01360                     &(PF_longMultiTop->nPacks),
01361                     1,
01362                     MPI_INT,
01363                     PF_COMM);
01364     if ( ret != MPI_SUCCESS ) return(ret);
01365  */
01366         Unpack the length of the last fit portion:
01367  */
01368     ret = MPI_Unpack(PF_longMultiRoot->buffer,
01369                     PF_packsize,
01370                     &( PF_longMultiRoot->packpos),
01371                     &(PF_longMultiTop->lastLen),
01372                     1,
01373                     MPI_INT,
01374                     PF_COMM);
01375     if ( ret != MPI_SUCCESS ) return(ret);
01376 }
01377 return(0);
01378 }
01379
01380  */
01381     #] PF_longMultiProcessPrefix :
01382     #[ PF_longSingleReset :
01383  */
01384
01392 static inline int PF_longSingleReset(int is_sender)
01393 {
01394     int ret;
01395     PF_longPackPos=0;
01396     if ( is_sender ) {
01397         ret = MPI_Pack(&PF_longPackTop,1,MPI_INT,
01398                     PF_longPackBuf,PF_longPackTop,&PF_longPackPos,PF_COMM);
01399         if ( ret != MPI_SUCCESS ) return(ret);
01400         PF_longPackN = 1;
01401     }
01402     else {
01403         PF_longPackN=0;
01404     }
01405     return(0);
01406 }
01407
01408  */
01409     #] PF_longSingleReset :
01410     #[ PF_longMultiReset :
01411  */
01412
01420 static inline int PF_longMultiReset(int is_sender)
01421 {
01422     int ret = 0, theone = 1;
01423     PF_longMultiRoot->packpos = 0;
01424     if ( is_sender ) {
01425         ret = MPI_Pack(&theone,1,MPI_INT,
01426                     PF_longPackBuf,PF_longPackTop,&(PF_longMultiRoot->packpos),PF_COMM);
01427         PF_longPackN = 1;
01428     }
01429     else {
01430         PF_longPackN = 0;
01431     }
01432     PF_longMultiRoot->nPacks = 0; /* The auxiliary field is not counted */
01433     PF_longMultiRoot->lastLen = 0;
01434     PF_longMultiTop = PF_longMultiRoot;
01435     PF_longMultiRoot->buffer = PF_longPackBuf;
01436     return ret;
01437 }
01438
01439  */
01440     #] PF_longMultiReset :
01441     #] Long pack private functions :
01442     #[ PF_PrepareLongSinglePack :
01443  */
01444
01451 int PF_PrepareLongSinglePack(void)
01452 {
01453     return PF_longSingleReset(1);
01454 }
01455
01456  */
01457     #] PF_PrepareLongSinglePack :
01458     #[ PF_LongSinglePack :
01459  */
01460
01469 int PF_LongSinglePack(const void *buffer, size_t count, MPI_Datatype type)

```

```

01470 {
01471     int ret, bytes;
01472     /* XXX: Limited by int size. */
01473     if ( count > INT_MAX ) return -99;
01474     ret = MPI_Pack_size((int)count, type, PF_COMM, &bytes);
01475     if ( ret != MPI_SUCCESS ) return(ret);
01476
01477     while ( PF_longPackPos+bytes > PF_longPackTop ) {
01478         if ( PF_longAddChunk(1, 1) ) return(-1);
01479     }
01480     /*
01481         PF_longAddChunk(1, 1) means, the chunk must
01482         be increased by 1 and re-allocated
01483     */
01484     ret = MPI_Pack((void *)buffer, (int)count, type,
01485                   PF_longPackBuf, PF_longPackTop, &PF_longPackPos, PF_COMM);
01486     if ( ret != MPI_SUCCESS ) return(ret);
01487     return(0);
01488 }
01489
01490 /*
01491     #] PF_LongSinglePack :
01492     #[ PF_LongSingleUnpack :
01493 */
01494
01503 int PF_LongSingleUnpack(void *buffer, size_t count, MPI_Datatype type)
01504 {
01505     int ret;
01506     /* XXX: Limited by int size. */
01507     if ( count > INT_MAX ) return -99;
01508     ret = MPI_Unpack(PF_longPackBuf, PF_longPackTop, &PF_longPackPos,
01509                     buffer, (int)count, type, PF_COMM);
01510     if ( ret != MPI_SUCCESS ) return(ret);
01511     return(0);
01512 }
01513
01514 /*
01515     #] PF_LongSingleUnpack :
01516     #[ PF_LongSingleSend :
01517 */
01518
01540 int PF_LongSingleSend(int to, int tag)
01541 {
01542     int ret, pos = 0;
01543     /*
01544         Note, here we assume that this function couldn't be used
01545         with to == PF_ANY_SOURCE!
01546     */
01547     if ( PF_longPackN > 1 ) {
01548         /* The buffer was incremented, pack send the new size first: */
01549         int tmp = -PF_longPackTop;
01550         /*
01551             Negative value means there will be the second buffer
01552         */
01553         ret = MPI_Pack(&tmp, 1, PF_INT,
01554                       PF_longPackSmallBuf, PF_longPackTop, &pos, PF_COMM);
01555         if ( ret != MPI_SUCCESS ) return(ret);
01556         ret = MPI_Ssend(PF_longPackSmallBuf, pos, MPI_PACKED, to, tag, PF_COMM);
01557         if ( ret != MPI_SUCCESS ) return(ret);
01558     }
01559     ret = MPI_Ssend(PF_longPackBuf, PF_longPackPos, MPI_PACKED, to, tag, PF_COMM);
01560     if ( ret != MPI_SUCCESS ) return(ret);
01561     return(0);
01562 }
01563
01564 /*
01565     #] PF_LongSingleSend :
01566     #[ PF_LongSingleReceive :
01567 */
01568
01583 int PF_LongSingleReceive(int src, int tag, int *psrc, int *ptag)
01584 {
01585     int ret, missed, oncemore;
01586     MPI_Status status;
01587     PF_longSingleReset(0);
01588     do {
01589         ret = MPI_Recv(PF_longPackBuf, PF_longPackTop, MPI_PACKED, src, tag,
01590                       PF_COMM, &status);
01591         if ( ret != MPI_SUCCESS ) return(ret);
01592     } /*
01593         The source and tag must be specified here for the case if
01594         MPI_Recv is performed more than once:
01595     */
01596     src = status.MPI_SOURCE;
01597     tag = status.MPI_TAG;
01598     if ( psrc ) *psrc = status.MPI_SOURCE;
01599     if ( ptag ) *ptag = status.MPI_TAG;

```

```

01600 /*
01601     Now we got either small buffer with the new PF_longPackTop,
01602     or just a regular chunk.
01603 */
01604     ret = MPI_Unpack(PF_longPackBuf, PF_longPackTop, &PF_longPackPos,
01605                     &missed, 1, MPI_INT, PF_COMM);
01606     if ( ret != MPI_SUCCESS ) return(ret);
01607
01608     if ( missed < 0 ) { /* The small buffer was received. */
01609         oncemore = 1; /* repeat receiving afterwards */
01610         /* Reallocate the buffer and get the data */
01611         missed = -missed;
01612     /*
01613         restore after unpacking small from buffer:
01614     */
01615         PF_longPackPos = 0;
01616     }
01617     else {
01618         oncemore = 0; /* That's all, no repetition */
01619     }
01620     if ( missed > PF_longPackTop ) {
01621         /*
01622          * The room must be increased. We need a re-allocation for the
01623          * case that there is no repetition.
01624          */
01625         if ( PF_longAddChunk( (missed-PF_longPackTop)/PF_packsize, !oncemore ) )
01626             return(-1);
01627     }
01628     while ( oncemore );
01629     return(0);
01630 }
01631
01632 /*
01633     #] PF_LongSingleReceive :
01634     #[ PF_PrepareLongMultiPack :
01635 */
01636
01643 int PF_PrepareLongMultiPack(void)
01644 {
01645     return PF_longMultiReset(1);
01646 }
01647
01648 /*
01649     #] PF_PrepareLongMultiPack :
01650     #[ PF_LongMultiPackImpl :
01651 */
01652
01662 int PF_LongMultiPackImpl(const void*buffer, size_t count, size_t eSize, MPI_Datatype type)
01663 {
01664     int ret, items;
01665
01666     /* XXX: Limited by int size. */
01667     if ( count > INT_MAX ) return -99;
01668
01669     ret = MPI_Pack_size((int)count, type, PF_COMM, &items);
01670     if ( ret != MPI_SUCCESS ) return(ret);
01671
01672     if ( PF_longMultiTop->packpos + items <= PF_packsize ) {
01673         ret = MPI_Pack((void *)buffer, (int)count, type, PF_longMultiTop->buffer,
01674                       PF_packsize, &(PF_longMultiTop->packpos), PF_COMM);
01675         if ( ret != MPI_SUCCESS ) return(ret);
01676         PF_longMultiTop->nPacks++;
01677         return(0);
01678     }
01679     /*
01680         The data do not fit to the rest of the buffer.
01681         There are two possibilities here: go to the next cell
01682         immediately, or first try to pack some portion. The function
01683         PF_longMultiHowSplit() returns the number of items could be
01684         packed in the end of the current cell:
01685     */
01686     if ( ( items = PF_longMultiHowSplit((int)count, type, items) ) < 0 ) return(items);
01687
01688     if ( items > 0 ) { /* store the head */
01689         ret = MPI_Pack((void *)buffer, items, type, PF_longMultiTop->buffer,
01690                       PF_packsize, &(PF_longMultiTop->packpos), PF_COMM);
01691         if ( ret != MPI_SUCCESS ) return(ret);
01692         PF_longMultiTop->nPacks++;
01693         PF_longMultiTop->lastLen = items;
01694     }
01695     /*
01696         Now the rest should be packed to the new cell.
01697         Slide to the new cell:
01698     */
01699     if ( PF_longMultiPack2NextCell() ) return(-1);
01700     PF_longPackN++;
01701     /*

```



```

01702         Pack the rest to the next cell:
01703     */
01704     return(PF_LongMultiPackImpl((char *)buffer+items*eSize,count-items,eSize,type));
01705 }
01706
01707 /*
01708     #] PF_LongMultiPackImpl :
01709     #[ PF_LongMultiUnpackImpl :
01710 */
01711
01721 int PF_LongMultiUnpackImpl(void *buffer, size_t count, size_t eSize, MPI_Datatype type)
01722 {
01723     int ret;
01724
01725     /* XXX: Limited by int size. */
01726     if ( count > INT_MAX ) return -99;
01727
01728     if ( PF_longPackN < 2 ) { /* Just unpack the buffer from the single cell */
01729         ret = MPI_Unpack(
01730             PF_longMultiTop->buffer,
01731             PF_packsize,
01732             &(PF_longMultiTop->packpos),
01733             buffer,
01734             count,type,PF_COMM);
01735         if ( ret != MPI_SUCCESS ) return(ret);
01736         return(0);
01737     }
01738     /*
01739         More than one cell is in use.
01740     */
01741     if ( ( PF_longMultiTop->nPacks > 1 ) /* the cell is not expired */
01742         || /* The last cell contains exactly required portion: */
01743         ( ( PF_longMultiTop->nPacks == 1 ) && ( PF_longMultiTop->lastLen == 0 ) ) )
01744     { /* Just unpack the buffer from the current cell */
01745         ret = MPI_Unpack(
01746             PF_longMultiTop->buffer,
01747             PF_packsize,
01748             &(PF_longMultiTop->packpos),
01749             buffer,
01750             count,type,PF_COMM);
01751         if ( ret != MPI_SUCCESS ) return(ret);
01752         (PF_longMultiTop->nPacks)--;
01753         return(0);
01754     }
01755     if ( ( PF_longMultiTop->nPacks == 1 ) && ( PF_longMultiTop->lastLen != 0 ) ) {
01756     /*
01757         Unpack the head:
01758     */
01759         ret = MPI_Unpack(
01760             PF_longMultiTop->buffer,
01761             PF_packsize,
01762             &(PF_longMultiTop->packpos),
01763             buffer,
01764             PF_longMultiTop->lastLen,type,PF_COMM);
01765         if ( ret != MPI_SUCCESS ) return(ret);
01766     /*
01767         Decrement the counter by read items:
01768     */
01769         count -= PF_longMultiTop->lastLen;
01770         if ( count <= 0 ) return(-1); /*Something is wrong! */
01771     /*
01772         Shift the output buffer position:
01773     */
01774         buffer = (char *)buffer + PF_longMultiTop->lastLen * eSize;
01775         (PF_longMultiTop->nPacks)--;
01776     }
01777     /*
01778         Here PF_longMultiTop->nPacks == 0
01779     */
01780     if ( ( PF_longMultiTop = PF_longMultiTop->next ) == NULL ) return(-1);
01781     return(PF_LongMultiUnpackImpl(buffer,count,eSize,type));
01782 }
01783
01784 /*
01785     #] PF_LongMultiUnpackImpl :
01786     #[ PF_LongMultiBroadcast :
01787 */
01788
01807 int PF_LongMultiBroadcast(void)
01808 {
01809     int ret, i;
01810
01811     if ( PF.me == MASTER ) {
01812     /*
01813         PF_longPackN is the number of packed chunks. If it is more
01814         than 1, we have to pack a new one and send it first
01815     */

```

```

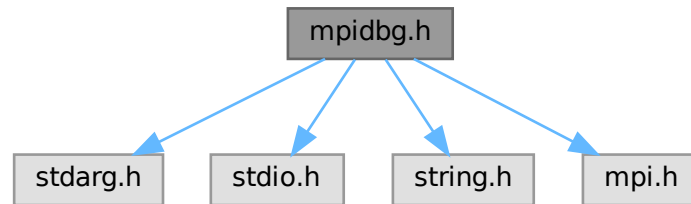
01816         if ( PF_longPackN > 1 ) {
01817             if ( PF_longMultiPreparePrefix() ) return(-1);
01818             ret = MPI_Bcast((void*)PF_longMultiTop->buffer,
01819                             PF_packsize,MPI_PACKED,MASTER,PF_COMM);
01820             if ( ret != MPI_SUCCESS ) return(ret);
01821         /*
01822             PF_longPackN was not incremented by PF_longMultiPreparePrefix()!
01823         */
01824     }
01825     /*
01826         Now we start from the beginning:
01827     */
01828     PF_longMultiTop = PF_longMultiRoot;
01829     /*
01830         Just broadcast all the chunks:
01831     */
01832     for ( i = 0; i < PF_longPackN; i++ ) {
01833         ret = MPI_Bcast((void*)PF_longMultiTop->buffer,
01834                         PF_packsize,MPI_PACKED,MASTER,PF_COMM);
01835         if ( ret != MPI_SUCCESS ) return(ret);
01836         PF_longMultiTop = PF_longMultiTop->next;
01837     }
01838     return(0);
01839 }
01840 /*
01841     else - the slave
01842 */
01843 PF_longMultiReset(0);
01844 /*
01845     Get the first chunk; it can be either the only data chunk, or
01846     an auxiliary chunk, if the data do not fit the single chunk:
01847 */
01848 ret = MPI_Bcast((void*)PF_longMultiRoot->buffer,
01849                 PF_packsize,MPI_PACKED,MASTER,PF_COMM);
01850 if ( ret != MPI_SUCCESS ) return(ret);
01851
01852 ret = MPI_Unpack((void*)PF_longMultiRoot->buffer,
01853                 PF_packsize,
01854                 &(PF_longMultiRoot->packpos),
01855                 &PF_longPackN,1,MPI_INT,PF_COMM);
01856 if ( ret != MPI_SUCCESS ) return(ret);
01857 /*
01858     Now in PF_longPackN we have the number of cells used
01859     for broadcasting. If it is >1, then we have to allocate
01860     enough cells, initialize them and receive all the chunks.
01861 */
01862 if ( PF_longPackN < 2 ) /* That's all, the single chunk is received. */
01863     return(0);
01864 /*
01865     Here we have to get PF_longPackN chunks. But, first,
01866     initialize cells by info from the received auxiliary chunk.
01867 */
01868 if ( PF_longMultiProcessPrefix() ) return(-1);
01869 /*
01870     Now we have free PF_longPackN cells, starting
01871     from PF_longMultiRoot->next, with properly initialized
01872     nPacks and lastLen fields. Get chunks:
01873 */
01874 for ( PF_longMultiTop = PF_longMultiRoot->next, i = 0; i < PF_longPackN; i++ ) {
01875     ret = MPI_Bcast((void*)PF_longMultiTop->buffer,
01876                     PF_packsize,MPI_PACKED,MASTER,PF_COMM);
01877     if ( ret != MPI_SUCCESS ) return(ret);
01878     if ( i == 0 ) { /* The first chunk, it contains extra "1". */
01879         int tmp;
01880     /*
01881         Extract this 1 into tmp and forget about it.
01882     */
01883         ret = MPI_Unpack((void*)PF_longMultiTop->buffer,
01884                         PF_packsize,
01885                         &(PF_longMultiTop->packpos),
01886                         &tmp,1,MPI_INT,PF_COMM);
01887         if ( ret != MPI_SUCCESS ) return(ret);
01888     }
01889     PF_longMultiTop = PF_longMultiTop->next;
01890 }
01891 /*
01892     multiUnPack starts with PF_longMultiTop, skip auxiliary chunk in
01893     PF_longMultiRoot:
01894 */
01895 PF_longMultiTop = PF_longMultiRoot->next;
01896 return(0);
01897 }
01898 /*
01899     #] PF_LongMultiBroadcast :
01900     #] Long pack stuff :
01901 */

```

## 4.80 mpidbg.h File Reference

```
#include <stdarg.h>
#include <stdio.h>
#include <string.h>
#include <mpi.h>
```

Include dependency graph for mpidbg.h:



### Macros

- #define [MPIDBG\\_RANK](#) MPIDBG\_Get\_rank()
- #define [MPIDBG\\_Out](#)(...) MPIDBG\_Out(file, line, func, \_\_VA\_ARGS\_\_)
- #define [MPIDBG\\_EXTARG](#) const char \*file, int line, const char \*func
- #define [MPI\\_Send](#)(...) MPIDBG\_Send(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Recv](#)(...) MPIDBG\_Recv(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Bsend](#)(...) MPIDBG\_Bsend(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Ssend](#)(...) MPIDBG\_Ssend(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Rsend](#)(...) MPIDBG\_Rsend(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Isend](#)(...) MPIDBG\_Isend(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Ibsend](#)(...) MPIDBG\_Ibsend(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Issend](#)(...) MPIDBG\_Issend(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Irsend](#)(...) MPIDBG\_Irsend(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Irecv](#)(...) MPIDBG\_Irecv(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Wait](#)(...) MPIDBG\_Wait(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Test](#)(...) MPIDBG\_Test(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Waitany](#)(...) MPIDBG\_Waitany(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Testany](#)(...) MPIDBG\_Testany(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Waitall](#)(...) MPIDBG\_Waitall(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Testall](#)(...) MPIDBG\_Testall(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Waitsome](#)(...) MPIDBG\_Waitsome(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Testsome](#)(...) MPIDBG\_Testsome(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Iprobe](#)(...) MPIDBG\_Iprobe(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Probe](#)(...) MPIDBG\_Probe(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Cancel](#)(...) MPIDBG\_Cancel(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Test\\_cancelled](#)(...) MPIDBG\_Test\_cancelled(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_,  
\_\_)
- #define [MPI\\_Barrier](#)(...) MPIDBG\_Barrier(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)
- #define [MPI\\_Bcast](#)(...) MPIDBG\_Bcast(\_\_VA\_ARGS\_\_, \_\_FILE\_\_, \_\_LINE\_\_, \_\_func\_\_)

### 4.80.1 Detailed Description

MPI APIs with the logging feature. NOTE: This file needs C99.

Definition in file [mpidbg.h](#).

### 4.80.2 Macro Definition Documentation

#### 4.80.2.1 MPIDBG\_RANK

```
#define MPIDBG_RANK MPIDBG_Get_rank()
```

Definition at line 55 of file [mpidbg.h](#).

#### 4.80.2.2 MPIDBG\_Out

```
#define MPIDBG_Out(  
    ... ) MPIDBG_Out(file, line, func, __VA_ARGS__)
```

Definition at line 72 of file [mpidbg.h](#).

#### 4.80.2.3 MPIDBG\_EXTARG

```
#define MPIDBG_EXTARG const char *file, int line, const char *func
```

Definition at line 139 of file [mpidbg.h](#).

#### 4.80.2.4 MPI\_Send

```
#define MPI_Send(  
    ... ) MPIDBG_Send(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 157 of file [mpidbg.h](#).

#### 4.80.2.5 MPI\_Recv

```
#define MPI_Recv(  
    ... ) MPIDBG_Recv(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 178 of file [mpidbg.h](#).

#### 4.80.2.6 MPI\_Bsend

```
#define MPI_Bsend(  
    ... ) MPIDBG_Bsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 197 of file [mpidbg.h](#).

#### 4.80.2.7 MPI\_Ssend

```
#define MPI_Ssend(  
    ... ) MPIDBG_Ssend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 216 of file [mpidbg.h](#).

#### 4.80.2.8 MPI\_Rsend

```
#define MPI_Rsend(  
    ... ) MPIDBG_Rsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 235 of file [mpidbg.h](#).

#### 4.80.2.9 MPI\_Isend

```
#define MPI_Isend(  
    ... ) MPIDBG_Isend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 254 of file [mpidbg.h](#).

#### 4.80.2.10 MPI\_Ibsend

```
#define MPI_Ibsend(  
    ... ) MPIDBG_Ibsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 273 of file [mpidbg.h](#).

#### 4.80.2.11 MPI\_Issend

```
#define MPI_Issend(  
    ... ) MPIDBG_Issend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 292 of file [mpidbg.h](#).

#### 4.80.2.12 MPI\_Irsend

```
#define MPI_Irsend(  
    ... ) MPIDBG_Irsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 311 of file [mpidbg.h](#).

#### 4.80.2.13 MPI\_Irecv

```
#define MPI_Irecv(  
    ... ) MPIDBG_Irecv(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 330 of file [mpidbg.h](#).

#### 4.80.2.14 MPI\_Wait

```
#define MPI_Wait(  
    ... ) MPIDBG_Wait(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 353 of file [mpidbg.h](#).

#### 4.80.2.15 MPI\_Test

```
#define MPI_Test(  
    ... ) MPIDBG_Test(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 381 of file [mpidbg.h](#).

#### 4.80.2.16 MPI\_Waitany

```
#define MPI_Waitany(  
    ... ) MPIDBG_Waitany(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 407 of file [mpidbg.h](#).

#### 4.80.2.17 MPI\_Testany

```
#define MPI_Testany(  
    ... ) MPIDBG_Testany(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 438 of file [mpidbg.h](#).

#### 4.80.2.18 MPI\_Waitall

```
#define MPI_Waitall(  
    ... ) MPIDBG_Waitall(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 462 of file [mpidbg.h](#).

#### 4.80.2.19 MPI\_Testall

```
#define MPI_Testall(  
    ... ) MPIDBG_Testall(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 491 of file [mpidbg.h](#).

#### 4.80.2.20 MPI\_Waitsome

```
#define MPI_Waitsome(  
    ... ) MPIDBG_Waitsome(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 515 of file [mpidbg.h](#).

#### 4.80.2.21 MPI\_Testsome

```
#define MPI_Testsome(  
    ... ) MPIDBG_Testsome(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 539 of file [mpidbg.h](#).

#### 4.80.2.22 MPI\_Iprobe

```
#define MPI_Iprobe(  
    ... ) MPIDBG_Iprobe(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 565 of file [mpidbg.h](#).

#### 4.80.2.23 MPI\_Probe

```
#define MPI_Probe(  
    ... ) MPIDBG_Probe(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 586 of file [mpidbg.h](#).

#### 4.80.2.24 MPI\_Cancel

```
#define MPI_Cancel(  
    ... ) MPIDBG_Cancel(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 605 of file [mpidbg.h](#).

#### 4.80.2.25 MPI\_Test\_cancelled

```
#define MPI_Test_cancelled(  
    ... ) MPIDBG_Test_cancelled(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 629 of file [mpidbg.h](#).

#### 4.80.2.26 MPI\_Barrier

```
#define MPI_Barrier(  
    ... ) MPIDBG_Barrier(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 648 of file [mpidbg.h](#).

#### 4.80.2.27 MPI\_Bcast

```
#define MPI_Bcast(  
    ... ) MPIDBG_Bcast(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 667 of file [mpidbg.h](#).

## 4.81 mpidbg.h

[Go to the documentation of this file.](#)

```

00001 #ifndef MPIDBG_H
00002 #define MPIDBG_H
00003
00010 /* #[] License : */
00011 /*
00012  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00013  * When using this file you are requested to refer to the publication
00014  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00015  * This is considered a matter of courtesy as the development was paid
00016  * for by FOM the Dutch physics granting agency and we would like to
00017  * be able to track its scientific use to convince FOM of its value
00018  * for the community.
00019  *
00020  * This file is part of FORM.
00021  *
00022  * FORM is free software: you can redistribute it and/or modify it under the
00023  * terms of the GNU General Public License as published by the Free Software
00024  * Foundation, either version 3 of the License, or (at your option) any later
00025  * version.
00026  *
00027  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00028  * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00029  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00030  * details.
00031  *
00032  * You should have received a copy of the GNU General Public License along
00033  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00034  */
00035 /* #[] License : */
00036
00037 /*
00038  #[] Includes :
00039 */
00040
00041 #include <stdarg.h>
00042 #include <stdio.h>
00043 #include <string.h>
00044 #include <mpi.h>
00045
00046 /*
00047  #[] Includes :
00048  #[] Utilities :
00049  #[] MPIDBG_RANK :
00050 */
00051
00052 static inline int MPIDBG_Get_rank(void) {
00053     return PF.me; /* Assume we are working with ParFORM. */
00054 }
00055 #define MPIDBG_RANK MPIDBG_Get_rank()
00056
00057 /*
00058  #[] MPIDBG_RANK :
00059  #[] MPIDBG_Out :
00060 */
00061
00062 static inline void MPIDBG_Out(const char *file, int line, const char *func, const char *fmt, ...) {
00063     char buf[1024]; /* Enough. */
00064     va_list ap;
00065     va_start(ap, fmt);
00066     snprintf(buf, 1024, "*** [%d] %10s %4d @ %-16s: ", MPIDBG_RANK, file, line, func);
00067     vsnprintf(buf + strlen(buf), 1024 - strlen(buf), fmt, ap);
00068     va_end(ap);
00069     /* Assume fprintf with a line will work well even in multi-processes. */
00070     fprintf(stderr, "%s\n", buf);
00071 }
00072 #define MPIDBG_Out(...) MPIDBG_Out(file, line, func, __VA_ARGS__)
00073
00074 /*
00075  #[] MPIDBG_Out :
00076  #[] MPIDBG_sprint_requests :
00077 */
00078
00079 static inline void MPIDBG_sprint_requests(char *buf, int count, const MPI_Request *requests)
00080 {
00081     /* Assume sprintf never fail and returns >= 0... */
00082     buf += sprintf(buf, "(");
00083     int i, first = 1;
00084     for (i = 0; i < count; i++) {
00085         if (first) {
00086             first = 0;
00087         }
00088         else {

```



```

00089         buf += sprintf(buf, ",");
00090     }
00091     if ( requests[i] != MPI_REQUEST_NULL ) {
00092         buf += sprintf(buf, "%d", i);
00093     }
00094     else {
00095         buf += sprintf(buf, "*");
00096     }
00097 }
00098 buf += sprintf(buf, " ");
00099 }
00100
00101 /*
00102     #] MPIDBG_sprint_requests :
00103     #[ MPIDBG_sprint_statuses :
00104 */
00105
00106 static inline void MPIDBG_sprint_statuses(char *buf, int count, const MPI_Request *old_requests, const
MPI_Request *new_requests, const MPI_Status *statuses)
00107 {
00108     /* Assume sprintf never fail and returns >= 0... */
00109     buf += sprintf(buf, "(");
00110     int i, first = 1;
00111     for ( i = 0; i < count; i++ ) {
00112         if ( first ) {
00113             first = 0;
00114         }
00115         else {
00116             buf += sprintf(buf, ",");
00117         }
00118         if ( old_requests[i] != MPI_REQUEST_NULL && new_requests[i] == MPI_REQUEST_NULL ) {
00119             int ret_size = 0;
00120             MPI_Get_count((MPI_Status *)&statuses[i], MPI_BYTE, &ret_size);
00121             buf += sprintf(buf, "(source=%d,tag=%d,size=%d)", statuses[i].MPI_SOURCE,
statuses[i].MPI_TAG, ret_size);
00122         } else {
00123             buf += sprintf(buf, "*");
00124         }
00125     }
00126     buf += sprintf(buf, " ");
00127 }
00128
00129 /*
00130     #] MPIDBG_sprint_statuses :
00131     #] Utilities :
00132     #[ MPI APIs :
00133 */
00134
00135 /*
00136 * The followings are the MPI APIs with the logging.
00137 */
00138
00139 #define MPIDBG_EXTARG const char *file, int line, const char *func
00140
00141 /*
00142     #[ MPI_Send :
00143 */
00144
00145 static inline int MPIDBG_Send(void *buf, int count, MPI_Datatype datatype, int dest, int tag, MPI_Comm
comm, MPIDBG_EXTARG)
00146 {
00147     MPIDBG_Out("MPI_Send: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
00148     int ret = MPI_Send(buf, count, datatype, dest, tag, comm);
00149     if ( ret == MPI_SUCCESS ) {
00150         MPIDBG_Out("MPI_Send: OK");
00151     }
00152     else {
00153         MPIDBG_Out("MPI_Send: Failed");
00154     }
00155     return ret;
00156 }
00157 #define MPI_Send(...) MPIDBG_Send(__VA_ARGS__, __FILE__, __LINE__, __func__)
00158
00159 /*
00160     #] MPI_Send :
00161     #[ MPI_Recv :
00162 */
00163
00164 static inline int MPIDBG_Recv(void* buf, int count, MPI_Datatype datatype, int source, int tag,
MPI_Comm comm, MPI_Status *status, MPIDBG_EXTARG)
00165 {
00166     MPIDBG_Out("MPI_Recv: src=%d dest=%d tag=%d", source, MPIDBG_RANK, tag);
00167     int ret = MPI_Recv(buf, count, datatype, source, tag, comm, status);
00168     if ( ret == MPI_SUCCESS ) {
00169         int ret_count = 0;
00170         MPI_Get_count(status, datatype, &ret_count);
00171         MPIDBG_Out("MPI_Recv: OK src=%d dest=%d tag=%d count=%d", status->MPI_SOURCE, MPIDBG_RANK,

```

```

        status->MPI_TAG, ret_count);
00172     }
00173     else {
00174         MPIDBG_Out("MPI_Recv: Failed");
00175     }
00176     return ret;
00177 }
00178 #define MPI_Recv(...) MPIDBG_Recv(__VA_ARGS__, __FILE__, __LINE__, __func__)
00179
00180 /*
00181     #] MPI_Recv :
00182     #[ MPI_Bsend :
00183 */
00184
00185 static inline int MPIDBG_Bsend(void* buf, int count, MPI_Datatype datatype, int dest, int tag,
MPI_Comm comm, MPIDBG_EXTARG)
00186 {
00187     MPIDBG_Out("MPI_Bsend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
00188     int ret = MPI_Bsend(buf, count, datatype, dest, tag, comm);
00189     if ( ret == MPI_SUCCESS ) {
00190         MPIDBG_Out("MPI_Bsend: OK");
00191     }
00192     else {
00193         MPIDBG_Out("MPI_Bsend: Failed");
00194     }
00195     return ret;
00196 }
00197 #define MPI_Bsend(...) MPIDBG_Bsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
00198
00199 /*
00200     #] MPI_Bsend :
00201     #[ MPI_Ssend :
00202 */
00203
00204 static inline int MPIDBG_Ssend(void* buf, int count, MPI_Datatype datatype, int dest, int tag,
MPI_Comm comm, MPIDBG_EXTARG)
00205 {
00206     MPIDBG_Out("MPI_Ssend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
00207     int ret = MPI_Ssend(buf, count, datatype, dest, tag, comm);
00208     if ( ret == MPI_SUCCESS ) {
00209         MPIDBG_Out("MPI_Ssend: OK");
00210     }
00211     else {
00212         MPIDBG_Out("MPI_Ssend: Failed");
00213     }
00214     return ret;
00215 }
00216 #define MPI_Ssend(...) MPIDBG_Ssend(__VA_ARGS__, __FILE__, __LINE__, __func__)
00217
00218 /*
00219     #] MPI_Ssend :
00220     #[ MPI_Rsend :
00221 */
00222
00223 static inline int MPIDBG_Rsend(void* buf, int count, MPI_Datatype datatype, int dest, int tag,
MPI_Comm comm, MPIDBG_EXTARG)
00224 {
00225     MPIDBG_Out("MPI_Rsend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
00226     int ret = MPI_Rsend(buf, count, datatype, dest, tag, comm);
00227     if ( ret == MPI_SUCCESS ) {
00228         MPIDBG_Out("MPI_Rsend: OK");
00229     }
00230     else {
00231         MPIDBG_Out("MPI_Rsend: Failed");
00232     }
00233     return ret;
00234 }
00235 #define MPI_Rsend(...) MPIDBG_Rsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
00236
00237 /*
00238     #] MPI_Rsend :
00239     #[ MPI_Isend :
00240 */
00241
00242 static inline int MPIDBG_Isend(void* buf, int count, MPI_Datatype datatype, int dest, int tag,
MPI_Comm comm, MPI_Request *request, MPIDBG_EXTARG)
00243 {
00244     MPIDBG_Out("MPI_Isend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
00245     int ret = MPI_Isend(buf, count, datatype, dest, tag, comm, request);
00246     if ( ret == MPI_SUCCESS ) {
00247         MPIDBG_Out("MPI_Isend: OK");
00248     }
00249     else {
00250         MPIDBG_Out("MPI_Isend: Failed");
00251     }
00252     return ret;
00253 }

```

```

00254 #define MPI_Isend(...) MPIDBG_Isend(__VA_ARGS__, __FILE__, __LINE__, __func__)
00255
00256 /*
00257     #] MPI_Isend :
00258     #[ MPI_Ibsend :
00259 */
00260
00261 static inline int MPIDBG_Ibsend(void* buf, int count, MPI_Datatype datatype, int dest, int tag,
    MPI_Comm comm, MPI_Request *request, MPIDBG_EXTARG)
00262 {
00263     MPIDBG_Out("MPI_Ibsend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
00264     int ret = MPI_Ibsend(buf, count, datatype, dest, tag, comm, request);
00265     if ( ret == MPI_SUCCESS ) {
00266         MPIDBG_Out("MPI_Ibsend: OK");
00267     }
00268     else {
00269         MPIDBG_Out("MPI_Ibsend: Failed");
00270     }
00271     return ret;
00272 }
00273 #define MPI_Ibsend(...) MPIDBG_Ibsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
00274
00275 /*
00276     #] MPI_Ibsend :
00277     #[ MPI_Issend :
00278 */
00279
00280 static inline int MPIDBG_Issend(void* buf, int count, MPI_Datatype datatype, int dest, int tag,
    MPI_Comm comm, MPI_Request *request, MPIDBG_EXTARG)
00281 {
00282     MPIDBG_Out("MPI_Issend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
00283     int ret = MPI_Issend(buf, count, datatype, dest, tag, comm, request);
00284     if ( ret == MPI_SUCCESS ) {
00285         MPIDBG_Out("MPI_Issend: OK");
00286     }
00287     else {
00288         MPIDBG_Out("MPI_Issend: Failed");
00289     }
00290     return ret;
00291 }
00292 #define MPI_Issend(...) MPIDBG_Issend(__VA_ARGS__, __FILE__, __LINE__, __func__)
00293
00294 /*
00295     #] MPI_Issend :
00296     #[ MPI_Irsend :
00297 */
00298
00299 static inline int MPIDBG_Irsend(void* buf, int count, MPI_Datatype datatype, int dest, int tag,
    MPI_Comm comm, MPI_Request *request, MPIDBG_EXTARG)
00300 {
00301     MPIDBG_Out("MPI_Irsend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
00302     int ret = MPI_Irsend(buf, count, datatype, dest, tag, comm, request);
00303     if ( ret == MPI_SUCCESS ) {
00304         MPIDBG_Out("MPI_Irsend: OK");
00305     }
00306     else {
00307         MPIDBG_Out("MPI_Irsend: Failed");
00308     }
00309     return ret;
00310 }
00311 #define MPI_Irsend(...) MPIDBG_Irsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
00312
00313 /*
00314     #] MPI_Irsend :
00315     #[ MPI_Irecv :
00316 */
00317
00318 static inline int MPIDBG_Irecv(void* buf, int count, MPI_Datatype datatype, int source, int tag,
    MPI_Comm comm, MPI_Request *request, MPIDBG_EXTARG)
00319 {
00320     MPIDBG_Out("MPI_Irecv: src=%d dest=%d tag=%d", source, MPIDBG_RANK, tag);
00321     int ret = MPI_Irecv(buf, count, datatype, source, tag, comm, request);
00322     if ( ret == MPI_SUCCESS ) {
00323         MPIDBG_Out("MPI_Irecv: OK dest=%d", MPIDBG_RANK);
00324     }
00325     else {
00326         MPIDBG_Out("MPI_Irecv: Failed");
00327     }
00328     return ret;
00329 }
00330 #define MPI_Irecv(...) MPIDBG_Irecv(__VA_ARGS__, __FILE__, __LINE__, __func__)
00331
00332 /*
00333     #] MPI_Irecv :
00334     #[ MPI_Wait :
00335 */
00336

```

```

00337 static inline int MPIDBG_Wait(MPI_Request *request, MPI_Status *status, MPIDBG_EXTARG)
00338 {
00339     char buf[256 * 1]; /* Enough. */
00340     MPI_Request old_request = *request;
00341     MPIDBG_sprint_requests(buf, 1, request);
00342     MPIDBG_Out("MPI_Wait: rank=%d request=%s", MPIDBG_RANK, buf);
00343     int ret = MPI_Wait(request, status);
00344     if ( ret == MPI_SUCCESS ) {
00345         MPIDBG_sprint_statuses(buf, 1, request, &old_request, status);
00346         MPIDBG_Out("MPI_Wait: OK rank=%d result=%s", MPIDBG_RANK, buf);
00347     }
00348     else {
00349         MPIDBG_Out("MPI_Wait: Failed");
00350     }
00351     return ret;
00352 }
00353 #define MPI_Wait(...) MPIDBG_Wait(__VA_ARGS__, __FILE__, __LINE__, __func__)
00354
00355 /*
00356     #] MPI_Wait :
00357     #[ MPI_Test :
00358 */
00359
00360 static inline int MPIDBG_Test(MPI_Request *request, int *flag, MPI_Status *status, MPIDBG_EXTARG)
00361 {
00362     char buf[256 * 1]; /* Enough. */
00363     MPI_Request old_request = *request;
00364     MPIDBG_sprint_requests(buf, 1, request);
00365     MPIDBG_Out("MPI_Test: rank=%d request=%s", MPIDBG_RANK, buf);
00366     int ret = MPI_Test(request, flag, status);
00367     if ( ret == MPI_SUCCESS ) {
00368         if ( *flag ) {
00369             MPIDBG_sprint_statuses(buf, 1, request, &old_request, status);
00370             MPIDBG_Out("MPI_Test: OK rank=%d result=%s", MPIDBG_RANK, buf);
00371         }
00372         else {
00373             MPIDBG_Out("MPI_Test: OK flag=false");
00374         }
00375     }
00376     else {
00377         MPIDBG_Out("MPI_Test: Failed");
00378     }
00379     return ret;
00380 }
00381 #define MPI_Test(...) MPIDBG_Test(__VA_ARGS__, __FILE__, __LINE__, __func__)
00382
00383 /*
00384     #] MPI_Test :
00385     #[ MPI_Waitany :
00386 */
00387
00388 static inline int MPIDBG_Waitany(int count, MPI_Request *array_of_requests, int *index, MPI_Status
*status, MPIDBG_EXTARG)
00389 {
00390     char buf[256]; /* Enough. */
00391     MPI_Request old_requests[count];
00392     memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * count);
00393     MPIDBG_sprint_requests(buf, count, array_of_requests);
00394     MPIDBG_Out("MPI_Waitany: rank=%d request=%s", MPIDBG_RANK, buf);
00395     int ret = MPI_Waitany(count, array_of_requests, index, status);
00396     if ( ret == MPI_SUCCESS ) {
00397         MPI_Status statuses[count];
00398         statuses[*index] = *status;
00399         MPIDBG_sprint_statuses(buf, count, old_requests, array_of_requests, statuses);
00400         MPIDBG_Out("MPI_Waitany: OK rank=%d result=%s", MPIDBG_RANK, buf);
00401     }
00402     else {
00403         MPIDBG_Out("MPI_Waitany: Failed");
00404     }
00405     return ret;
00406 }
00407 #define MPI_Waitany(...) MPIDBG_Waitany(__VA_ARGS__, __FILE__, __LINE__, __func__)
00408
00409 /*
00410     #] MPI_Waitany :
00411     #[ MPI_Testany :
00412 */
00413
00414 static inline int MPIDBG_Testany(int count, MPI_Request *array_of_requests, int *index, int *flag,
MPI_Status *status, MPIDBG_EXTARG)
00415 {
00416     char buf[256]; /* Enough. */
00417     MPI_Request old_requests[count];
00418     memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * count);
00419     MPIDBG_sprint_requests(buf, count, array_of_requests);
00420     MPIDBG_Out("MPI_Testany: rank=%d request=%s", MPIDBG_RANK, buf);
00421     int ret = MPI_Testany(count, array_of_requests, index, flag, status);

```

```

00422     if ( ret == MPI_SUCCESS ) {
00423         if ( *flag ) {
00424             MPI_Status statuses[count];
00425             statuses[*index] = *status;
00426             MPIDBG_sprint_statuses(buf, count, old_requests, array_of_requests, statuses);
00427             MPIDBG_Out("MPI_Testany: OK rank=%d result=%s", MPIDBG_RANK, buf);
00428         }
00429         else {
00430             MPIDBG_Out("MPI_Testany: OK flag=false");
00431         }
00432     }
00433     else {
00434         MPIDBG_Out("MPI_Testany: Failed");
00435     }
00436     return ret;
00437 }
00438 #define MPI_Testany(...) MPIDBG_Testany(__VA_ARGS__, __FILE__, __LINE__, __func__)
00439
00440 /*
00441     #] MPI_Testany :
00442     #[ MPI_Waitall :
00443 */
00444
00445 static inline int MPIDBG_Waitall(int count, MPI_Request *array_of_requests, MPI_Status
*array_of_statuses, MPIDBG_EXTARG)
00446 {
00447     char buf[256 * count]; /* Enough. */
00448     MPI_Request old_requests[count];
00449     memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * count);
00450     MPIDBG_sprint_requests(buf, count, array_of_requests);
00451     MPIDBG_Out("MPI_Waitall: rank=%d request=%s", MPIDBG_RANK, buf);
00452     int ret = MPI_Waitall(count, array_of_requests, array_of_statuses);
00453     if ( ret == MPI_SUCCESS ) {
00454         MPIDBG_sprint_statuses(buf, count, old_requests, array_of_requests, array_of_statuses);
00455         MPIDBG_Out("MPI_Waitall: OK rank=%d result=%s", MPIDBG_RANK, buf);
00456     }
00457     else {
00458         MPIDBG_Out("MPI_Waitall: Failed");
00459     }
00460     return ret;
00461 }
00462 #define MPI_Waitall(...) MPIDBG_Waitall(__VA_ARGS__, __FILE__, __LINE__, __func__)
00463
00464 /*
00465     #] MPI_Waitall :
00466     #[ MPI_Testall :
00467 */
00468
00469 static inline int MPIDBG_Testall(int count, MPI_Request *array_of_requests, int *flag, MPI_Status
*array_of_statuses, MPIDBG_EXTARG)
00470 {
00471     char buf[256 * count]; /* Enough. */
00472     MPI_Request old_requests[count];
00473     memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * count);
00474     MPIDBG_sprint_requests(buf, count, array_of_requests);
00475     MPIDBG_Out("MPI_Testall: rank=%d request=%s", MPIDBG_RANK, buf);
00476     int ret = MPI_Testall(count, array_of_requests, flag, array_of_statuses);
00477     if ( ret == MPI_SUCCESS ) {
00478         if ( *flag ) {
00479             MPIDBG_sprint_statuses(buf, count, old_requests, array_of_requests, array_of_statuses);
00480             MPIDBG_Out("MPI_Testall: OK rank=%d result=%s", MPIDBG_RANK, buf);
00481         }
00482         else {
00483             MPIDBG_Out("MPI_Testall: OK flag=false");
00484         }
00485     }
00486     else {
00487         MPIDBG_Out("MPI_Testall: Failed");
00488     }
00489     return ret;
00490 }
00491 #define MPI_Testall(...) MPIDBG_Testall(__VA_ARGS__, __FILE__, __LINE__, __func__)
00492
00493 /*
00494     #] MPI_Testall :
00495     #[ MPI_Waitsome :
00496 */
00497
00498 static inline int MPIDBG_Waitsome(int incount, MPI_Request *array_of_requests, int *outcount, int
*array_of_indices, MPI_Status *array_of_statuses, MPIDBG_EXTARG)
00499 {
00500     char buf[256 * incount]; /* Enough. */
00501     MPI_Request old_requests[incount];
00502     memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * incount);
00503     MPIDBG_sprint_requests(buf, incount, array_of_requests);
00504     MPIDBG_Out("MPI_Waitsome: rank=%d request=%s", MPIDBG_RANK, buf);
00505     int ret = MPI_Waitsome(incount, array_of_requests, outcount, array_of_indices, array_of_statuses);

```

```

00506     if ( ret == MPI_SUCCESS ) {
00507         MPIDBG_sprint_statuses(buf, incount, old_requests, array_of_requests, array_of_statuses);
00508         MPIDBG_Out("MPI_Waitosome: OK rank=%d result=%s", MPIDBG_RANK, buf);
00509     }
00510     else {
00511         MPIDBG_Out("MPI_Waitosome: Failed");
00512     }
00513     return ret;
00514 }
00515 #define MPI_Waitosome(...) MPIDBG_Waitosome(__VA_ARGS__, __FILE__, __LINE__, __func__)
00516
00517 /*
00518     #] MPI_Waitosome :
00519     #[ MPI_Testsome :
00520 */
00521
00522 static inline int MPIDBG_Testsome(int incount, MPI_Request *array_of_requests, int *outcount, int
    *array_of_indices, MPI_Status *array_of_statuses, MPIDBG_EXTARG)
00523 {
00524     char buf[256 * incount]; /* Enough. */
00525     MPI_Request old_requests[incount];
00526     memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * incount);
00527     MPIDBG_sprint_requests(buf, incount, array_of_requests);
00528     MPIDBG_Out("MPI_Testsome: rank=%d request=%s", MPIDBG_RANK, buf);
00529     int ret = MPI_Testsome(incount, array_of_requests, outcount, array_of_indices, array_of_statuses);
00530     if ( ret == MPI_SUCCESS ) {
00531         MPIDBG_sprint_statuses(buf, incount, old_requests, array_of_requests, array_of_statuses);
00532         MPIDBG_Out("MPI_Testsome: OK rank=%d result=%s", MPIDBG_RANK, buf);
00533     }
00534     else {
00535         MPIDBG_Out("MPI_Testsome: Failed");
00536     }
00537     return ret;
00538 }
00539 #define MPI_Testsome(...) MPIDBG_Testsome(__VA_ARGS__, __FILE__, __LINE__, __func__)
00540
00541 /*
00542     #] MPI_Testsome :
00543     #[ MPI_Iprobe :
00544 */
00545
00546 static inline int MPIDBG_Iprobe(int source, int tag, MPI_Comm comm, int *flag, MPI_Status *status,
    MPIDBG_EXTARG)
00547 {
00548     MPIDBG_Out("MPI_Iprobe: src=%d dest=%d tag=%d", source, MPIDBG_RANK, tag);
00549     int ret = MPI_Iprobe(source, tag, comm, flag, status);
00550     if ( ret == MPI_SUCCESS ) {
00551         if ( *flag ) {
00552             int ret_size = 0;
00553             MPI_Get_count(status, MPI_BYTE, &ret_size);
00554             MPIDBG_Out("MPI_Iprobe: OK src=%d dest=%d tag=%d size=%d", status->MPI_SOURCE,
    MPIDBG_RANK, status->MPI_TAG, ret_size);
00555         }
00556         else {
00557             MPIDBG_Out("MPI_Iprobe: OK flag=false");
00558         }
00559     }
00560     else {
00561         MPIDBG_Out("MPI_Iprobe: Failed");
00562     }
00563     return ret;
00564 }
00565 #define MPI_Iprobe(...) MPIDBG_Iprobe(__VA_ARGS__, __FILE__, __LINE__, __func__)
00566
00567 /*
00568     #] MPI_Iprobe :
00569     #[ MPI_Probe :
00570 */
00571
00572 static inline int MPIDBG_Probe(int source, int tag, MPI_Comm comm, MPI_Status *status, MPIDBG_EXTARG)
00573 {
00574     MPIDBG_Out("MPI_Probe: src=%d dest=%d tag=%d", source, MPIDBG_RANK, tag);
00575     int ret = MPI_Probe(source, tag, comm, status);
00576     if ( ret == MPI_SUCCESS ) {
00577         int ret_size = 0;
00578         MPI_Get_count(status, MPI_BYTE, &ret_size);
00579         MPIDBG_Out("MPI_Probe: OK src=%d dest=%d tag=%d size=%d", status->MPI_SOURCE, MPIDBG_RANK,
    status->MPI_TAG, ret_size);
00580     }
00581     else {
00582         MPIDBG_Out("MPI_Probe: Failed");
00583     }
00584     return ret;
00585 }
00586 #define MPI_Probe(...) MPIDBG_Probe(__VA_ARGS__, __FILE__, __LINE__, __func__)
00587
00588 /*

```

```

00589         #] MPI_Probe :
00590         #[ MPI_Cancel :
00591     */
00592
00593     static inline int MPIDBG_Cancel(MPI_Request *request, MPIDBG_EXTARG)
00594     {
00595         MPIDBG_Out("MPI_Cancel: rank=%d", MPIDBG_RANK);
00596         int ret = MPI_Cancel(request);
00597         if ( ret == MPI_SUCCESS ) {
00598             MPIDBG_Out("MPI_Cancel: OK");
00599         }
00600         else {
00601             MPIDBG_Out("MPI_Cancel: Failed");
00602         }
00603         return ret;
00604     }
00605     #define MPI_Cancel(...) MPIDBG_Cancel(__VA_ARGS__, __FILE__, __LINE__, __func__)
00606
00607     /*
00608         #] MPI_Cancel :
00609         #[ MPI_Test_cancelled :
00610     */
00611
00612     static inline int MPIDBG_Test_cancelled(MPI_Status *status, int *flag, MPIDBG_EXTARG)
00613     {
00614         MPIDBG_Out("MPI_Test_cancelled: rank=%d", MPIDBG_RANK);
00615         int ret = MPI_Test_cancelled(status, flag);
00616         if ( ret == MPI_SUCCESS ) {
00617             if ( *flag ) {
00618                 MPIDBG_Out("MPI_Test_cancelled: OK flag=true");
00619             }
00620             else {
00621                 MPIDBG_Out("MPI_Test_cancelled: OK flag=false");
00622             }
00623         }
00624         else {
00625             MPIDBG_Out("MPI_Test_cancelled: Failed");
00626         }
00627         return ret;
00628     }
00629     #define MPI_Test_cancelled(...) MPIDBG_Test_cancelled(__VA_ARGS__, __FILE__, __LINE__, __func__)
00630
00631     /*
00632         #] MPI_Test_cancelled :
00633         #[ MPI_Barrier :
00634     */
00635
00636     static inline int MPIDBG_Barrier(MPI_Comm comm, MPIDBG_EXTARG)
00637     {
00638         MPIDBG_Out("MPI_Barrier: rank=%d", MPIDBG_RANK);
00639         int ret = MPI_Barrier(comm);
00640         if ( ret == MPI_SUCCESS ) {
00641             MPIDBG_Out("MPI_Barrier: OK");
00642         }
00643         else {
00644             MPIDBG_Out("MPI_Barrier: Failed");
00645         }
00646         return ret;
00647     }
00648     #define MPI_Barrier(...) MPIDBG_Barrier(__VA_ARGS__, __FILE__, __LINE__, __func__)
00649
00650     /*
00651         #] MPI_Barrier :
00652         #[ MPI_Bcast :
00653     */
00654
00655     static inline int MPIDBG_Bcast(void* buffer, int count, MPI_Datatype datatype, int root, MPI_Comm
comm, MPIDBG_EXTARG)
00656     {
00657         MPIDBG_Out("MPI_Bcast: root=%d count=%d", root, count);
00658         int ret = MPI_Bcast(buffer, count, datatype, root, comm);
00659         if ( ret == MPI_SUCCESS ) {
00660             MPIDBG_Out("MPI_Bcast: OK");
00661         }
00662         else {
00663             MPIDBG_Out("MPI_Bcast: Failed");
00664         }
00665         return ret;
00666     }
00667     #define MPI_Bcast(...) MPIDBG_Bcast(__VA_ARGS__, __FILE__, __LINE__, __func__)
00668
00669     /*
00670         #] MPI_Bcast :
00671         #] MPI APIs :
00672     */
00673
00674     #endif /* MPIDBG_H */

```

## 4.82 mytime.cc

```

00001 #ifdef HAVE_CONFIG_H
00002 #include <config.h>
00003 #endif
00004
00005 // A timing routine for debugging. Only on Unix (where sys/time.h is available).
00006 #ifdef UNIX
00007
00008 #include <sys/time.h>
00009 #include <cstdlib>
00010 #include <stdio>
00011 #include <string>
00012
00013 #ifndef timersub
00014 /* timersub is not in POSIX, but presents on most BSD derivatives.
00015    This implementation is borrowed from glibc. (TU 23 Oct 2011) */
00016 #define timersub(a, b, result) \
00017     do { \
00018         (result)->tv_sec = (a)->tv_sec - (b)->tv_sec; \
00019         (result)->tv_usec = (a)->tv_usec - (b)->tv_usec; \
00020         if ((result)->tv_usec < 0) { \
00021             --(result)->tv_sec; \
00022             (result)->tv_usec += 1000000; \
00023         } \
00024     } while (0)
00025 #endif
00026
00027 bool starttime_set = false;
00028 timeval starttime;
00029
00030 double thetime () {
00031     if (!starttime_set) {
00032         gettimeofday(&starttime, NULL);
00033         starttime_set=true;
00034     }
00035
00036     timeval now,diff;
00037     gettimeofday(&now, NULL);
00038     timersub(&now,&starttime,&diff);
00039     return diff.tv_sec+diff.tv_usec/1000000.0;
00040 }
00041
00042 std::string thetime_str() {
00043     char res[10];
00044     snprintf (res,10,"%4.1f", thetime());
00045     return res;
00046 }
00047
00048 #endif // UNIX

```

## 4.83 mytime.h

```

00001 #pragma once
00002 /* #[] License : */
00003 /*
00004  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00005  * When using this file you are requested to refer to the publication
00006  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00007  * This is considered a matter of courtesy as the development was paid
00008  * for by FOM the Dutch physics granting agency and we would like to
00009  * be able to track its scientific use to convince FOM of its value
00010  * for the community.
00011  *
00012  * This file is part of FORM.
00013  *
00014  * FORM is free software: you can redistribute it and/or modify it under the
00015  * terms of the GNU General Public License as published by the Free Software
00016  * Foundation, either version 3 of the License, or (at your option) any later
00017  * version.
00018  *
00019  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00020  * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00021  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00022  * details.
00023  *
00024  * You should have received a copy of the GNU General Public License along
00025  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00026  */
00027 /* #[] License : */
00028
00029 double thetime();
00030 std::string thetime_str();

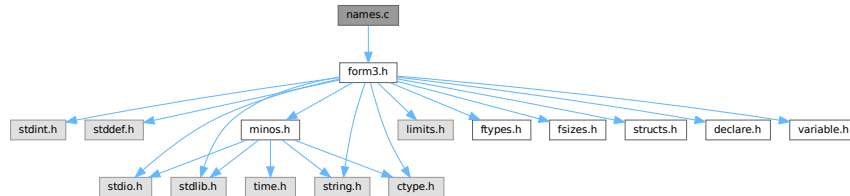
```



## 4.84 names.c File Reference

```
#include "form3.h"
```

Include dependency graph for names.c:



### Functions

- [NAMENODE](#) \* [GetNode](#) ([NAMETREE](#) \*nametree, [UBYTE](#) \*name)
- int [AddName](#) ([NAMETREE](#) \*nametree, [UBYTE](#) \*name, [WORD](#) type, [WORD](#) number, int \*nodenum)
- int [GetName](#) ([NAMETREE](#) \*nametree, [UBYTE](#) \*namein, [WORD](#) \*number, int par)
- [UBYTE](#) \* [GetFunction](#) ([UBYTE](#) \*s, [WORD](#) \*funnum)
- [UBYTE](#) \* [GetNumber](#) ([UBYTE](#) \*s, [WORD](#) \*num)
- int [GetLastExprName](#) ([UBYTE](#) \*name, [WORD](#) \*number)
- int [GetOName](#) ([NAMETREE](#) \*nametree, [UBYTE](#) \*name, [WORD](#) \*number, int par)
- int [GetAutoName](#) ([UBYTE](#) \*name, [WORD](#) \*number)
- int [GetVar](#) ([UBYTE](#) \*name, [WORD](#) \*type, [WORD](#) \*number, int wantedtype, int par)
- int [EntVar](#) ([WORD](#) type, [UBYTE](#) \*name, [WORD](#) x, [WORD](#) y, [WORD](#) z, [WORD](#) d)
- int [GetDollar](#) ([UBYTE](#) \*name)
- void [DumpTree](#) ([NAMETREE](#) \*nametree)
- void [DumpNode](#) ([NAMETREE](#) \*nametree, [WORD](#) node, [WORD](#) depth)
- int [CompactifyTree](#) ([NAMETREE](#) \*nametree, [WORD](#) par)
- void [CopyTree](#) ([NAMETREE](#) \*newtree, [NAMETREE](#) \*oldtree, [WORD](#) node, [WORD](#) par)
- void [LinkTree](#) ([NAMETREE](#) \*tree, [WORD](#) offset, [WORD](#) numnodes)
- [NAMETREE](#) \* [MakeNameTree](#) (void)
- void [FreeNameTree](#) ([NAMETREE](#) \*n)
- void [ClearWildcardNames](#) (void)
- int [AddWildcardName](#) ([UBYTE](#) \*name)
- int [GetWildcardName](#) ([UBYTE](#) \*name)
- int [AddSymbol](#) ([UBYTE](#) \*name, int minpow, int maxpow, int cplx, int dim)
- int [CoSymbol](#) ([UBYTE](#) \*s)
- int [AddIndex](#) ([UBYTE](#) \*name, int dim, int dim4)
- int [CoIndex](#) ([UBYTE](#) \*s)
- [UBYTE](#) \* [DoDimension](#) ([UBYTE](#) \*s, int \*dim, int \*dim4)
- int [CoDimension](#) ([UBYTE](#) \*s)
- int [AddVector](#) ([UBYTE](#) \*name, int cplx, int dim)
- int [CoVector](#) ([UBYTE](#) \*s)
- int [AddFunction](#) ([UBYTE](#) \*name, int comm, int istensor, int cplx, int symprop, int dim, int argmax, int argmin)
- int [CoCommuteInSet](#) ([UBYTE](#) \*s)
- int [CoFunction](#) ([UBYTE](#) \*s, int comm, int istensor)
- int [CoNFunction](#) ([UBYTE](#) \*s)
- int [CoCFunction](#) ([UBYTE](#) \*s)
- int [CoNTensor](#) ([UBYTE](#) \*s)
- int [CoCTensor](#) ([UBYTE](#) \*s)

- int [DoTable](#) (UBYTE \*s, int par)
- int [CoTable](#) (UBYTE \*s)
- int [CoNTable](#) (UBYTE \*s)
- int [CoCTable](#) (UBYTE \*s)
- void [EmptyTable](#) ([TABLES](#) T)
- int [AddSet](#) (UBYTE \*name, WORD dim)
- int [DoElements](#) (UBYTE \*s, [SETS](#) set, UBYTE \*name)
- int [CoSet](#) (UBYTE \*s)
- int [DoTempSet](#) (UBYTE \*from, UBYTE \*to)
- int [CoAuto](#) (UBYTE \*inp)
- int [AddDollar](#) (UBYTE \*name, WORD type, WORD \*start, LONG size)
- int [ReplaceDollar](#) (WORD number, WORD newtype, WORD \*newstart, LONG newsize)
- int [AddDubious](#) (UBYTE \*name)
- int [MakeDubious](#) ([NAMETREE](#) \*nametree, UBYTE \*name, WORD \*number)
- int [NameConflict](#) (int type, UBYTE \*name)
- int [AddExpression](#) (UBYTE \*name, int x, int y)
- int [GetLabel](#) (UBYTE \*name)
- void [ResetVariables](#) (int par)
- void [RemoveDollars](#) (void)
- void [Globalize](#) (int par)
- int [TestName](#) (UBYTE \*name)

#### 4.84.1 Detailed Description

The complete names administration. All variables with a name have to pass here to be properly registered, have structs of the proper type assigned to them etc. The file also contains the utility routines for maintaining the balanced trees that make searching for names rather fast.

Definition in file [names.c](#).

#### 4.84.2 Function Documentation

##### 4.84.2.1 GetNode()

```
NAMENODE * GetNode (
    NAMETREE * nametree,
    UBYTE * name )
```

Definition at line [49](#) of file [names.c](#).

##### 4.84.2.2 AddName()

```
int AddName (
    NAMETREE * nametree,
    UBYTE * name,
    WORD type,
    WORD number,
    int * nodenum )
```

Definition at line [71](#) of file [names.c](#).

#### 4.84.2.3 GetName()

```
int GetName (
    NAMETREE * nametree,
    UBYTE * namein,
    WORD * number,
    int par )
```

Definition at line 252 of file [names.c](#).

#### 4.84.2.4 GetFunction()

```
UBYTE * GetFunction (
    UBYTE * s,
    WORD * funnum )
```

Definition at line 342 of file [names.c](#).

#### 4.84.2.5 GetNumber()

```
UBYTE * GetNumber (
    UBYTE * s,
    WORD * num )
```

Definition at line 388 of file [names.c](#).

#### 4.84.2.6 GetLastExprName()

```
int GetLastExprName (
    UBYTE * name,
    WORD * number )
```

Definition at line 438 of file [names.c](#).

#### 4.84.2.7 GetOName()

```
int GetOName (
    NAMETREE * nametree,
    UBYTE * name,
    WORD * number,
    int par )
```

Definition at line 460 of file [names.c](#).

#### 4.84.2.8 GetAutoName()

```
int GetAutoName (
    UBYTE * name,
    WORD * number )
```

Definition at line 479 of file [names.c](#).

#### 4.84.2.9 GetVar()

```
int GetVar (
    UBYTE * name,
    WORD * type,
    WORD * number,
    int wantedtype,
    int par )
```

Definition at line [524](#) of file [names.c](#).

#### 4.84.2.10 EntVar()

```
int EntVar (
    WORD type,
    UBYTE * name,
    WORD x,
    WORD y,
    WORD z,
    WORD d )
```

Definition at line [557](#) of file [names.c](#).

#### 4.84.2.11 GetDollar()

```
int GetDollar (
    UBYTE * name )
```

Definition at line [590](#) of file [names.c](#).

#### 4.84.2.12 DumpTree()

```
void DumpTree (
    NAMETREE * nametree )
```

Definition at line [602](#) of file [names.c](#).

#### 4.84.2.13 DumpNode()

```
void DumpNode (
    NAMETREE * nametree,
    WORD node,
    WORD depth )
```

Definition at line [615](#) of file [names.c](#).

#### 4.84.2.14 CompactifyTree()

```
int CompactifyTree (
    NAMETREE * nametree,
    WORD par )
```

Definition at line 634 of file [names.c](#).

#### 4.84.2.15 CopyTree()

```
void CopyTree (
    NAMETREE * newtree,
    NAMETREE * oldtree,
    WORD node,
    WORD par )
```

Definition at line 696 of file [names.c](#).

#### 4.84.2.16 LinkTree()

```
void LinkTree (
    NAMETREE * tree,
    WORD offset,
    WORD numnodes )
```

Definition at line 784 of file [names.c](#).

#### 4.84.2.17 MakeNameTree()

```
NAMETREE * MakeNameTree (
    void )
```

Definition at line 815 of file [names.c](#).

#### 4.84.2.18 FreeNameTree()

```
void FreeNameTree (
    NAMETREE * n )
```

Definition at line 834 of file [names.c](#).

#### 4.84.2.19 ClearWildcardNames()

```
void ClearWildcardNames (
    void )
```

Definition at line 849 of file [names.c](#).

#### 4.84.2.20 AddWildcardName()

```
int AddWildcardName (
    UBYTE * name )
```

Definition at line [854](#) of file [names.c](#).

#### 4.84.2.21 GetWildcardName()

```
int GetWildcardName (
    UBYTE * name )
```

Definition at line [898](#) of file [names.c](#).

#### 4.84.2.22 AddSymbol()

```
int AddSymbol (
    UBYTE * name,
    int minpow,
    int maxpow,
    int cplx,
    int dim )
```

Definition at line [920](#) of file [names.c](#).

#### 4.84.2.23 CoSymbol()

```
int CoSymbol (
    UBYTE * s )
```

Definition at line [946](#) of file [names.c](#).

#### 4.84.2.24 AddIndex()

```
int AddIndex (
    UBYTE * name,
    int dim,
    int dim4 )
```

Definition at line [1088](#) of file [names.c](#).

#### 4.84.2.25 CoIndex()

```
int CoIndex (
    UBYTE * s )
```

Definition at line [1112](#) of file [names.c](#).

#### 4.84.2.26 DoDimension()

```
UBYTE * DoDimension (
    UBYTE * s,
    int * dim,
    int * dim4 )
```

Definition at line 1160 of file [names.c](#).

#### 4.84.2.27 CoDimension()

```
int CoDimension (
    UBYTE * s )
```

Definition at line 1226 of file [names.c](#).

#### 4.84.2.28 AddVector()

```
int AddVector (
    UBYTE * name,
    int cplx,
    int dim )
```

Definition at line 1244 of file [names.c](#).

#### 4.84.2.29 CoVector()

```
int CoVector (
    UBYTE * s )
```

Definition at line 1267 of file [names.c](#).

#### 4.84.2.30 AddFunction()

```
int AddFunction (
    UBYTE * name,
    int comm,
    int istensor,
    int cplx,
    int symprop,
    int dim,
    int argmax,
    int argmin )
```

Definition at line 1326 of file [names.c](#).

#### 4.84.2.31 CoCommuteInSet()

```
int CoCommuteInSet (
    UBYTE * s )
```

Definition at line 1356 of file [names.c](#).

#### 4.84.2.32 CoFunction()

```
int CoFunction (
    UBYTE * s,
    int comm,
    int istensor )
```

Definition at line 1446 of file [names.c](#).

#### 4.84.2.33 CoNFunction()

```
int CoNFunction (
    UBYTE * s )
```

Definition at line 1624 of file [names.c](#).

#### 4.84.2.34 CoCFunction()

```
int CoCFunction (
    UBYTE * s )
```

Definition at line 1625 of file [names.c](#).

#### 4.84.2.35 CoNTensor()

```
int CoNTensor (
    UBYTE * s )
```

Definition at line 1626 of file [names.c](#).

#### 4.84.2.36 CoCTensor()

```
int CoCTensor (
    UBYTE * s )
```

Definition at line 1627 of file [names.c](#).

#### 4.84.2.37 DoTable()

```
int DoTable (
    UBYTE * s,
    int par )
```

Definition at line 1666 of file [names.c](#).



#### 4.84.2.38 CoTable()

```
int CoTable (
    UBYTE * s )
```

Definition at line 2048 of file [names.c](#).

#### 4.84.2.39 CoNTable()

```
int CoNTable (
    UBYTE * s )
```

Definition at line 2058 of file [names.c](#).

#### 4.84.2.40 CoCTable()

```
int CoCTable (
    UBYTE * s )
```

Definition at line 2068 of file [names.c](#).

#### 4.84.2.41 EmptyTable()

```
void EmptyTable (
    TABLES T )
```

Definition at line 2078 of file [names.c](#).

#### 4.84.2.42 AddSet()

```
int AddSet (
    UBYTE * name,
    WORD dim )
```

Definition at line 2122 of file [names.c](#).

#### 4.84.2.43 DoElements()

```
int DoElements (
    UBYTE * s,
    SETS set,
    UBYTE * name )
```

Definition at line 2155 of file [names.c](#).

#### 4.84.2.44 CoSet()

```
int CoSet (
    UBYTE * s )
```

Definition at line [2355](#) of file [names.c](#).

#### 4.84.2.45 DoTempSet()

```
int DoTempSet (
    UBYTE * from,
    UBYTE * to )
```

Definition at line [2480](#) of file [names.c](#).

#### 4.84.2.46 CoAuto()

```
int CoAuto (
    UBYTE * inp )
```

Definition at line [2572](#) of file [names.c](#).

#### 4.84.2.47 AddDollar()

```
int AddDollar (
    UBYTE * name,
    WORD type,
    WORD * start,
    LONG size )
```

Definition at line [2602](#) of file [names.c](#).

#### 4.84.2.48 ReplaceDollar()

```
int ReplaceDollar (
    WORD number,
    WORD newtype,
    WORD * newstart,
    LONG newsize )
```

Definition at line [2646](#) of file [names.c](#).

#### 4.84.2.49 AddDubious()

```
int AddDubious (
    UBYTE * name )
```

Definition at line [2679](#) of file [names.c](#).

#### 4.84.2.50 MakeDubious()

```
int MakeDubious (
    NAMETREE * nametree,
    UBYTE * name,
    WORD * number )
```

Definition at line 2693 of file [names.c](#).

#### 4.84.2.51 NameConflict()

```
int NameConflict (
    int type,
    UBYTE * name )
```

Definition at line 2728 of file [names.c](#).

#### 4.84.2.52 AddExpression()

```
int AddExpression (
    UBYTE * name,
    int x,
    int y )
```

Definition at line 2744 of file [names.c](#).

#### 4.84.2.53 GetLabel()

```
int GetLabel (
    UBYTE * name )
```

Definition at line 2787 of file [names.c](#).

#### 4.84.2.54 ResetVariables()

```
void ResetVariables (
    int par )
```

Definition at line 2832 of file [names.c](#).

#### 4.84.2.55 RemoveDollars()

```
void RemoveDollars (
    void )
```

Definition at line 3128 of file [names.c](#).

#### 4.84.2.56 Globalize()

```
void Globalize (
    int par )
```

Definition at line 3155 of file [names.c](#).

#### 4.84.2.57 TestName()

```
int TestName (
    UBYTE * name )
```

Definition at line 3254 of file [names.c](#).

### 4.85 names.c

[Go to the documentation of this file.](#)

```
00001
00009 /* #[] License : */
00010 /*
00011 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00012 *   When using this file you are requested to refer to the publication
00013 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00014 *   This is considered a matter of courtesy as the development was paid
00015 *   for by FOM the Dutch physics granting agency and we would like to
00016 *   be able to track its scientific use to convince FOM of its value
00017 *   for the community.
00018 *
00019 *   This file is part of FORM.
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00024 *   version.
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00026 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00027 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00028 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00029 *   details.
00030 *
00031 *   You should have received a copy of the GNU General Public License along
00032 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00033 */
00034 /* #[] License : */
00035 /*
00036 *   #[] Includes :
00037 */
00038 #include "form3.h"
00039
00040 /* EXTERNLOCK(dummylock) */
00041
00042 /*
00043 *   #[] Includes :
00044 *
00045 *   #[] GetNode :
00046 */
00047
00048 NAMENODE *GetNode(NAMETREE *nametree, UBYTE *name)
00049 {
00050     NAMENODE *n;
00051     int node, newnode, i;
00052     if ( nametree->namenode == 0 ) return(0);
00053     newnode = nametree->headnode;
00054     do {
00055         node = newnode;
00056         n = nametree->namenode+node;
00057         if ( ( i = StrCmp(name,nametree->namebuffer+n->name) ) < 0 )
00058             newnode = n->left;
00059         else if ( i > 0 ) newnode = n->right;
00060         else { return(n); }
00061     } while ( newnode >= 0 );
```

```

00063     return(0);
00064 }
00065
00066 /*
00067  #] GetNode :
00068  #[ AddName :
00069  */
00070
00071 int AddName(NAMETREE *nametree, UBYTE *name, WORD type, WORD number, int *nodenum)
00072 {
00073     NAMENODE *n, *nn, *nnn;
00074     UBYTE *s, *ss, *sss;
00075     LONG *c1, *c2, j, newsize;
00076     int node, newnode, node3, r, rr = 0, i, retval = 0;
00077     if ( nametree->namenode == 0 ) {
00078         s = name; i = 1; while ( *s ) { i++; s++; }
00079         j = INITNAMESIZE;
00080         if ( i > j ) j = i;
00081         nametree->namenode = (NAMENODE *)Malloc1(INITNODESIZE*sizeof(NAMENODE),
00082             "new nametree in AddName");
00083         nametree->namebuffer = (UBYTE *)Malloc1(j,
00084             "new namebuffer in AddName");
00085         nametree->nodesize = INITNODESIZE;
00086         nametree->namesize = j;
00087         nametree->namefill = i;
00088         nametree->nodefill = 1;
00089         nametree->headnode = 0;
00090         n = nametree->namenode;
00091         n->parent = n->left = n->right = -1;
00092         n->balance = 0;
00093         n->type = type;
00094         n->number = number;
00095         n->name = 0;
00096         s = name;
00097         ss = nametree->namebuffer;
00098         while ( *s ) *ss++ = *s++;
00099         *ss = 0;
00100         *nodenum = 0;
00101         return(retval);
00102     }
00103     newnode = nametree->headnode;
00104     do {
00105         node = newnode;
00106         n = nametree->namenode+node;
00107         if ( StrCmp(name, nametree->namebuffer+n->name) < 0 ) {
00108             newnode = n->left; r = -1;
00109         }
00110         else {
00111             newnode = n->right; r = 1;
00112         }
00113     } while ( newnode >= 0 );
00114     /*
00115     We are at the insertion point. Add the node.
00116     */
00117     if ( nametree->nodefill >= nametree->nodesize ) { /* Double allocation */
00118         newsize = nametree->nodesize * 2;
00119         if ( newsize > MAXINNAMETREE ) newsize = MAXINNAMETREE;
00120         if ( nametree->nodefill >= MAXINNAMETREE ) {
00121             MesPrint("!!!More than %l names in one object", (LONG)MAXINNAMETREE);
00122             Terminate(-1);
00123         }
00124         nnn = (NAMENODE *)Malloc1(2*((LONG)newsize*sizeof(NAMENODE)),
00125             "extra names in AddName");
00126         c1 = (LONG *)nnn; c2 = (LONG *)nametree->namenode;
00127         i = (nametree->nodefill * sizeof(NAMENODE))/sizeof(LONG);
00128         while ( --i >= 0 ) *c1++ = *c2++;
00129         M_free(nametree->namenode, "nametree->namenode");
00130         nametree->namenode = nnn;
00131         nametree->nodesize = newsize;
00132         n = nametree->namenode+node;
00133     }
00134     *nodenum = newnode = nametree->nodefill++;
00135     nn = nametree->namenode+newnode;
00136     nn->parent = node;
00137     if ( r < 0 ) n->left = newnode; else n->right = newnode;
00138     nn->left = nn->right = -1;
00139     nn->type = type;
00140     nn->number = number;
00141     nn->balance = 0;
00142     i = 1; s = name; while ( *s ) { i++; s++; }
00143     while ( nametree->namefill + i >= nametree->namesize ) { /* Double alloc */
00144         sss = (UBYTE *)Malloc1(2*nametree->namesize,
00145             "extra names in AddName");
00146         s = sss; ss = nametree->namebuffer; j = nametree->namefill;
00147         while ( --j >= 0 ) *s++ = *ss++;
00148         M_free(nametree->namebuffer, "nametree->namebuffer");
00149         nametree->namebuffer = sss;

```

```

00150         nametree->namesize *= 2;
00151     }
00152     s = nametree->namebuffer+nametree->namefill;
00153     nn->name = nametree->namefill;
00154     retval = nametree->namefill;
00155     nametree->namefill += i;
00156     while ( *name ) *s++ = *name++;
00157     *s = 0;
00158 /*
00159 Adjust the balance factors
00160 */
00161 while ( node >= 0 ) {
00162     n = nametree->namenode + node;
00163     if ( newnode == n->left ) rr = -1;
00164     else rr = 1;
00165     if ( n->balance == -rr ) { n->balance = 0; return(retval); }
00166     else if ( n->balance == rr ) break;
00167     n->balance = rr;
00168     newnode = node;
00169     node = n->parent;
00170 }
00171 if ( node < 0 ) return(retval);
00172 /*
00173 We have to rebalance the tree. There are two basic operations.
00174 n/node is the unbalanced node. newnode is its child.
00175 rr is the old balance of n/node.
00176 */
00177 nn = nametree->namenode + newnode;
00178 if ( nn->balance == -rr ) { /* The difficult case */
00179     if ( rr > 0 ) {
00180         node3 = nn->left;
00181         nnn = nametree->namenode + node3;
00182         nnn->parent = n->parent;
00183         n->parent = nn->parent = node3;
00184         if ( nnn->right >= 0 ) nametree->namenode[nnn->right].parent = newnode;
00185         if ( nnn->left >= 0 ) nametree->namenode[nnn->left].parent = node;
00186         n->right = nnn->left; nnn->left = node;
00187         nn->left = nnn->right; nnn->right = newnode;
00188         if ( nnn->balance > 0 ) { n->balance = -1; nn->balance = 0; }
00189         else if ( nnn->balance == 0 ) { n->balance = nn->balance = 0; }
00190         else { nn->balance = 1; n->balance = 0; }
00191     }
00192     else {
00193         node3 = nn->right;
00194         nnn = nametree->namenode + node3;
00195         nnn->parent = n->parent;
00196         n->parent = nn->parent = node3;
00197         if ( nnn->right >= 0 ) nametree->namenode[nnn->right].parent = node;
00198         if ( nnn->left >= 0 ) nametree->namenode[nnn->left].parent = newnode;
00199         n->left = nnn->right; nnn->right = node;
00200         nn->right = nnn->left; nnn->left = newnode;
00201         if ( nnn->balance < 0 ) { n->balance = 1; nn->balance = 0; }
00202         else if ( nnn->balance == 0 ) { n->balance = nn->balance = 0; }
00203         else { nn->balance = -1; n->balance = 0; }
00204     }
00205     nnn->balance = 0;
00206     if ( nnn->parent >= 0 ) {
00207         nn = nametree->namenode + nnn->parent;
00208         if ( node == nn->left ) nn->left = node3;
00209         else nn->right = node3;
00210     }
00211     if ( node == nametree->headnode ) nametree->headnode = node3;
00212 }
00213 else if ( nn->balance == rr ) { /* The easy case */
00214     nn->parent = n->parent; n->parent = newnode;
00215     if ( rr > 0 ) {
00216         if ( nn->left >= 0 ) nametree->namenode[nn->left].parent = node;
00217         n->right = nn->left; nn->left = node;
00218     }
00219     else {
00220         if ( nn->right >= 0 ) nametree->namenode[nn->right].parent = node;
00221         n->left = nn->right; nn->right = node;
00222     }
00223     if ( nn->parent >= 0 ) {
00224         nnn = nametree->namenode + nn->parent;
00225         if ( node == nnn->left ) nnn->left = newnode;
00226         else nnn->right = newnode;
00227     }
00228     nn->balance = n->balance = 0;
00229     if ( node == nametree->headnode ) nametree->headnode = newnode;
00230 }
00231 #ifdef DEBUGON
00232 else { /* Cannot be. Code here for debugging only */
00233     MesPrint("We ran into an impossible case in AddName\n");
00234     DumpTree(nametree);
00235     Terminate(-1);
00236 }

```

```

00237 #endif
00238     return(retval);
00239 }
00240
00241 /*
00242  #] AddName :
00243  #[ GetName :
00244
00245  When AutoDeclare is an active statement.
00246  If par == WITHAUTO and the variable is not found we have to check:
00247  1: that nametree != AC.exprnames && nametree != AC.dollarnames
00248  2: check that the variable is not in AC.exprnames after all.
00249  3: call GetAutoName and return its values.
00250 */
00251
00252 int GetName(NAMETREE *nametree, UBYTE *namein, WORD *number, int par)
00253 {
00254     NAMENODE *n;
00255     int node, newnode, i;
00256     UBYTE *s, *t, *u, *name;
00257     /* name = ConstructName(namein,0); */
00258     name = namein;
00259     if ( nametree->namenode == 0 || nametree->namefill == 0 ) goto NotFound;
00260     newnode = nametree->headnode;
00261     do {
00262         node = newnode;
00263         n = nametree->namenode+node;
00264         if ( ( i = StrCmp(name,nametree->namebuffer+n->name) ) < 0 )
00265             newnode = n->left;
00266         else if ( i > 0 ) newnode = n->right;
00267         else {
00268             *number = n->number;
00269             return(n->type);
00270         }
00271     } while ( newnode >= 0 );
00272     s = name;
00273     while ( *s ) s++;
00274     if ( s > name && s[-1] == '_' && nametree == AC.varnames ) {
00275     /*
00276         The Kronecker delta d_ is very special. It is not really a function.
00277     */
00278         if ( s == name+2 && ( *name == 'd' || *name == 'D' ) ) {
00279             *number = DELTA-FUNCTION;
00280             return(CDELTA);
00281         }
00282     /*
00283         Test for N#_? type variables (summed indices)
00284     */
00285         if ( s > name+2 && *name == 'N' ) {
00286             t = name+1; i = 0;
00287             while ( FG.cTable[*t] == 1 ) i = 10*i + *t++ - '0';
00288             if ( s == t+1 ) {
00289                 *number = i + AM.IndDum - AM.OffsetIndex;
00290                 return(CINDEX);
00291             }
00292         }
00293     /*
00294         Now test for any built in object
00295     */
00296         newnode = nametree->headnode;
00297         do {
00298             node = newnode;
00299             n = nametree->namenode+node;
00300             if ( ( i = StrHICmp(name,nametree->namebuffer+n->name) ) < 0 )
00301                 newnode = n->left;
00302             else if ( i > 0 ) newnode = n->right;
00303             else {
00304                 *number = n->number; return(n->type);
00305             }
00306         } while ( newnode >= 0 );
00307     /*
00308         Now we test for the extra symbols of the type STR###_
00309         The string sits in AC.extrasym and is followed by digits.
00310         The name is only legal if the number is in the
00311         range 1,...,cbuf[AM.sbufnum].numrhs
00312     */
00313         t = name; u = AC.extrasym;
00314         while ( *t == *u ) { t++; u++; }
00315         if ( *u == 0 && *t != 0 ) { /* potential hit */
00316             WORD x = 0;
00317             while ( FG.cTable[*t] == 1 ) {
00318                 x = 10*x + (*t++ - '0');
00319             }
00320             if ( *t == '_' && x > 0 && x <= cbuf[AM.sbufnum].numrhs ) { /* Hit */
00321                 *number = MAXVARIABLES-x;
00322                 return(CSYMBOL);
00323             }

```

```

00324     }
00325 }
00326 NotFound;;
00327 if ( par != WITHAUTO || nametree == AC.autonames ) return(NAMENOTFOUND);
00328 return(GetAutoName(name,number));
00329 }
00330
00331 /*
00332 #] GetName :
00333 #[ GetFunction :
00334
00335 Gets either a function or a $ that should expand into a function
00336 during runtime. In the case of the $ the value in funnum is -dolnum-1.
00337 The return value is the position after the name of the function or the $.
00338 */
00339
00340 static WORD one = 1;
00341
00342 UBYTE *GetFunction(UBYTE *s,WORD *funnum)
00343 {
00344     int type;
00345     WORD numfun;
00346     UBYTE *t1, c;
00347     if ( *s == '$' ) {
00348         t1 = s+1; while ( FG.cTable[*t1] < 2 ) t1++;
00349         c = *t1; *t1 = 0;
00350         if ( ( type = GetName(AC.dollarnames,s+1,&numfun,NOAUTO) ) == CDOLLAR ) {
00351             *funnum = -numfun-2;
00352         }
00353         else {
00354             MesPrint("%s is undefined",s);
00355             numfun = AddDollar(s+1,DOLINDEX,&one,1);
00356             *funnum = 0;
00357         }
00358     }
00359     else {
00360         t1 = SkipAName(s);
00361         c = *t1; *t1 = 0;
00362         if ( ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
00363             || ( functions[numfun].spec > 0 ) ) {
00364             MesPrint("%s should be a regular function",s);
00365             *funnum = 0;
00366             if ( type < 0 ) {
00367                 if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
00368                     AddFunction(s,0,0,0,0,0,-1,-1);
00369             }
00370             *t1 = c;
00371             return(t1);
00372         }
00373         *funnum = numfun+FUNCTION;
00374     }
00375     *t1 = c;
00376     return(t1);
00377 }
00378
00379 /*
00380 #] GetFunction :
00381 #[ GetNumber :
00382
00383 Gets either a number or a $ that should expand into a number
00384 during runtime. In the case of the $ the value in num is -dolnum-2.
00385 The return value is the position after the number or the $.
00386 */
00387
00388 UBYTE *GetNumber(UBYTE *s,WORD *num)
00389 {
00390     int type;
00391     WORD numfun;
00392     UBYTE *t1, c;
00393     while ( *s == '+' ) s++;
00394     if ( *s == '$' ) {
00395         t1 = s+1; while ( FG.cTable[*t1] < 2 ) t1++;
00396         c = *t1; *t1 = 0;
00397         if ( ( type = GetName(AC.dollarnames,s+1,&numfun,NOAUTO) ) == CDOLLAR ) {
00398             *num = -numfun-2;
00399         }
00400         else {
00401             MesPrint("%s is undefined",s);
00402             numfun = AddDollar(s+1,DOLINDEX,&one,1);
00403             *num = -1;
00404         }
00405     }
00406     else if ( *s >= '0' && *s <= '9' ) {
00407         ULONG x = *s++ - '0';
00408         while ( *s >= '0' && *s <= '9' ) { x = 10*x + (*s++-'0'); }
00409         t1 = s;
00410         if ( x >= MAXPOSITIVE ) goto illegal;

```



```

00411         *num = (WORD)x;
00412         return(t1);
00413     }
00414     else {
00415         if ( *s == '-' ) { s++; }
00416         if ( *s >= '0' && *s <= '9' ) { while ( *s >= '0' && *s <= '9' ) s++; t1 = s; }
00417         else { t1 = SkipAName(s); }
00418     illegal:
00419         *num = -1;
00420         MesPrint("%Illegal option in Canonicalize statement. Should be a nonnegative number or $
variable.");
00421         return(t1);
00422     }
00423     *t1 = c;
00424     return(t1);
00425 }
00426
00427 /*
00428 #] GetNumber :
00429 #[ GetLastExprName :
00430
00431 When AutoDeclare is an active statement.
00432 If par == WITHAUTO and the variable is not found we have to check:
00433 1: that nametree != AC.exprnames && nametree != AC.dollarnames
00434 2: check that the variable is not in AC.exprnames after all.
00435 3: call GetAutoName and return its values.
00436 */
00437
00438 int GetLastExprName(UBYTE *name, WORD *number)
00439 {
00440     int i;
00441     EXPRESSIONS e;
00442     for ( i = NumExpressions; i > 0; i-- ) {
00443         e = Expressions+i-1;
00444         if ( StrCmp(AC.exprnames->namebuffer+e->name,name) == 0 ) {
00445             *number = i-1;
00446             return(1);
00447         }
00448     }
00449     return(0);
00450 }
00451
00452 /*
00453 #] GetLastExprName :
00454 #[ GetOName :
00455
00456 Adds the proper offsets, so we do not have to do that in the calling
00457 routine.
00458 */
00459
00460 int GetOName(NAMETREE *nametree, UBYTE *name, WORD *number, int par)
00461 {
00462     int retval = GetName(nametree,name,number,par);
00463     switch ( retval ) {
00464         case CVECTOR: *number += AM.OffsetVector; break;
00465         case CINDEX: *number += AM.OffsetIndex; break;
00466         case CFUNCTION: *number += FUNCTION; break;
00467         default: break;
00468     }
00469     return(retval);
00470 }
00471
00472 /*
00473 #] GetOName :
00474 #[ GetAutoName :
00475
00476 This routine gets the automatic declarations
00477 */
00478
00479 int GetAutoName(UBYTE *name, WORD *number)
00480 {
00481     UBYTE *s, c;
00482     int type;
00483     if ( GetName(AC.exprnames,name,number,NOAUTO) != NAMENOTFOUND )
00484         return(NAMENOTFOUND);
00485     s = name;
00486     while ( *s ) { s++; }
00487     if ( s[-1] == '-' ) {
00488         return(NAMENOTFOUND);
00489     }
00490     while ( s > name ) {
00491         c = *s; *s = 0;
00492         type = GetName(AC.autonames,name,number,NOAUTO);
00493         *s = c;
00494         switch(type) {
00495             case CSYMBOL: {
00496                 SYMBOLS sym = ((SYMBOLS) (AC.AutoSymbolList.lijst)) + *number;

```

```

00497         *number = AddSymbol(name,sym->minpower,sym->maxpower,sym->complex,sym->dimension);
00498         return(type); }
00499     case CVECTOR: {
00500         VECTORS vec = ((VECTORS) (AC.AutoVectorList.lijst)) + *number;
00501         *number = AddVector(name,vec->complex,vec->dimension);
00502         return(type); }
00503     case CINDEX: {
00504         INDICES ind = ((INDICES) (AC.AutoIndexList.lijst)) + *number;
00505         *number = AddIndex(name,ind->dimension,ind->nmin4);
00506         return(type); }
00507     case CFUNCTION: {
00508         FUNCTIONS fun = ((FUNCTIONS) (AC.AutoFunctionList.lijst)) + *number;
00509         *number =
AddFunction(name,fun->commute,fun->spec,fun->complex,fun->symmetric,fun->dimension,fun->maxnumargs,fun->minnumargs);
00510         return(type); }
00511     default:
00512         break;
00513 }
00514     s--;
00515 }
00516     return(NAMENOTFOUND);
00517 }
00518
00519 /*
00520 #] GetAutoName :
00521 #[ GetVar :
00522 */
00523
00524 int GetVar(UBYTE *name, WORD *type, WORD *number, int wantedtype, int par)
00525 {
00526     WORD funnum;
00527     int typ;
00528     if ( ( typ = GetName(AC.varnames,name,number,par) ) != wantedtype ) {
00529         if ( typ != NAMENOTFOUND ) {
00530             if ( wantedtype == -1 ) {
00531                 *type = typ;
00532                 return(1);
00533             }
00534             NameConflict(typ,name);
00535             MakeDubious(AC.varnames,name,&funnum);
00536             return(-1);
00537         }
00538         if ( ( typ = GetName(AC.exprnames,name,&funnum,par) ) != NAMENOTFOUND ) {
00539             if ( typ == wantedtype || wantedtype == -1 ) {
00540                 *number = funnum; *type = typ; return(1);
00541             }
00542             NameConflict(typ,name);
00543             return(-1);
00544         }
00545         return(NAMENOTFOUND);
00546     }
00547     if ( typ == -1 ) { return(0); }
00548     *type = typ;
00549     return(1);
00550 }
00551
00552 /*
00553 #] GetVar :
00554 #[ EntVar :
00555 */
00556
00557 int EntVar(WORD type, UBYTE *name, WORD x, WORD y, WORD z, WORD d)
00558 {
00559     switch ( type ) {
00560     case CSYMBOL:
00561         return(AddSymbol(name,y,z,x,d));
00562         break;
00563     case CINDEX:
00564         return(AddIndex(name,x,z));
00565         break;
00566     case CVECTOR:
00567         return(AddVector(name,x,d));
00568         break;
00569     case CFUNCTION:
00570         return(AddFunction(name,y,z,x,0,d,-1,-1));
00571         break;
00572     case CSET:
00573         AC.SetList.numtemp++;
00574         return(AddSet(name,d));
00575         break;
00576     case CEXPRESSION:
00577         return(AddExpression(name,x,y));
00578         break;
00579     default:
00580         break;
00581     }
00582     return(-1);

```

```

00583 }
00584
00585 /*
00586  #] EntVar :
00587  #[ GetDollar :
00588 */
00589
00590 int GetDollar(UBYTE *name)
00591 {
00592     WORD number;
00593     if ( GetName(AC.dollarnames,name,&number,NOAUTO) == NAMENOTFOUND ) return(-1);
00594     return((int)number);
00595 }
00596
00597 /*
00598  #] GetDollar :
00599  #[ DumpTree :
00600 */
00601
00602 void DumpTree(NAMETREE *nametree)
00603 {
00604     if ( nametree->headnode >= 0
00605         && nametree->namebuffer && nametree->namenode ) {
00606         DumpNode(nametree,nametree->headnode,0);
00607     }
00608 }
00609
00610 /*
00611  #] DumpTree :
00612  #[ DumpNode :
00613 */
00614
00615 void DumpNode(NAMETREE *nametree, WORD node, WORD depth)
00616 {
00617     NAMENODE *n;
00618     int i;
00619     char *name;
00620     n = nametree->namenode + node;
00621     if ( n->left >= 0 ) DumpNode(nametree,n->left,depth+1);
00622     for ( i = 0; i < depth; i++ ) printf(" ");
00623     name = (char *) (nametree->namebuffer+n->name);
00624     printf("%s(%d): { %d } ( %d ) ( %d ) [ %d ] \n",
00625         name,node,n->parent,n->left,n->right,n->balance);
00626     if ( n->right >= 0 ) DumpNode(nametree,n->right,depth+1);
00627 }
00628
00629 /*
00630  #] DumpNode :
00631  #[ CompactifyTree :
00632 */
00633
00634 int CompactifyTree(NAMETREE *nametree,WORD par)
00635 {
00636     NAMETREE newtree;
00637     NAMENODE *n;
00638     LONG i, j, ns, k;
00639     UBYTE *s;
00640
00641     for ( i = 0, j = 0, k = 0, n = nametree->namenode, ns = 0;
00642           i < nametree->nodefill; i++, n++ ) {
00643         if ( n->type != CDELETE ) {
00644             s = nametree->namebuffer+n->name;
00645             while ( *s ) { s++; ns++; }
00646             j++;
00647         }
00648         else k++;
00649     }
00650     if ( k == 0 ) return(0);
00651     if ( j == 0 ) {
00652         if ( nametree->namebuffer ) M_free(nametree->namebuffer,"nametree->namebuffer");
00653         if ( nametree->namenode ) M_free(nametree->namenode,"nametree->namenode");
00654         nametree->namebuffer = 0;
00655         nametree->namenode = 0;
00656         nametree->namesize = nametree->namefill =
00657         nametree->nodesize = nametree->nodefill =
00658         nametree->oldnamefill = nametree->oldnodefill = 0;
00659         nametree->globalnamefill = nametree->globalnodefill =
00660         nametree->clearnamefill = nametree->clearnodefill = 0;
00661         nametree->headnode = -1;
00662         return(0);
00663     }
00664     ns += j;
00665     if ( j < 10 ) j = 10;
00666     if ( ns < 100 ) ns = 100;
00667     newtree.namenode = (NAMENODE *) Malloc1(2*j*sizeof(NAMENODE),"compactify nametree");
00668     newtree.nodefill = 0; newtree.nodesize = 2*j;
00669     newtree.namebuffer = (UBYTE *) Malloc1(2*ns,"compactify nametree");

```

```

00670     newtree.namefill = 0; newtree.namesize = 2*ns;
00671     CopyTree(&newtree,nametree,nametree->headnode,par);
00672     newtree.namenode[newtree.nodefill»1].parent = -1;
00673     LinkTree(&newtree,(WORD)0,newtree.nodefill);
00674     newtree.headnode = newtree.nodefill » 1;
00675     M_free(nametree->namebuffer,"nametree->namebuffer");
00676     M_free(nametree->namenode,"nametree->namenode");
00677     nametree->namebuffer = newtree.namebuffer;
00678     nametree->namenode = newtree.namenode;
00679     nametree->namesize = newtree.namesize;
00680     nametree->namefill = newtree.namefill;
00681     nametree->nodesize = newtree.nodesize;
00682     nametree->nodefill = newtree.nodefill;
00683     nametree->oldnamefill = newtree.namefill;
00684     nametree->oldnodefill = newtree.nodefill;
00685     nametree->headnode = newtree.headnode;
00686
00687     /* DumpTree(nametree); */
00688     return(0);
00689 }
00690
00691 /*
00692  #] CompactifyTree :
00693  #[ CopyTree :
00694  */
00695
00696 void CopyTree(NAMETREE *newtree, NAMETREE *oldtree, WORD node, WORD par)
00697 {
00698     NAMENODE *n, *m;
00699     UBYTE *s, *t;
00700     n = oldtree->namenode+node;
00701     if ( n->left >= 0 ) CopyTree(newtree,oldtree,n->left,par);
00702     if ( n->type != CDELETE ) {
00703         m = newtree->namenode+newtree->nodefill;
00704         m->type = n->type;
00705         m->number = n->number;
00706         m->name = newtree->namefill;
00707         m->left = m->right = -1;
00708         m->balance = 0;
00709         switch ( n->type ) {
00710             case CSYMBOL:
00711                 if ( par == AUTONAMES ) {
00712                     autosymbols[n->number].name = newtree->namefill;
00713                     autosymbols[n->number].node = newtree->nodefill;
00714                 }
00715                 else {
00716                     symbols[n->number].name = newtree->namefill;
00717                     symbols[n->number].node = newtree->nodefill;
00718                 }
00719                 break;
00720             case CINDEX :
00721                 if ( par == AUTONAMES ) {
00722                     autoindices[n->number].name = newtree->namefill;
00723                     autoindices[n->number].node = newtree->nodefill;
00724                 }
00725                 else {
00726                     indices[n->number].name = newtree->namefill;
00727                     indices[n->number].node = newtree->nodefill;
00728                 }
00729                 break;
00730             case CVECTOR:
00731                 if ( par == AUTONAMES ) {
00732                     autovectors[n->number].name = newtree->namefill;
00733                     autovectors[n->number].node = newtree->nodefill;
00734                 }
00735                 else {
00736                     vectors[n->number].name = newtree->namefill;
00737                     vectors[n->number].node = newtree->nodefill;
00738                 }
00739                 break;
00740             case CFUNCTION:
00741                 if ( par == AUTONAMES ) {
00742                     autofunctions[n->number].name = newtree->namefill;
00743                     autofunctions[n->number].node = newtree->nodefill;
00744                 }
00745                 else {
00746                     functions[n->number].name = newtree->namefill;
00747                     functions[n->number].node = newtree->nodefill;
00748                 }
00749                 break;
00750             case CSET:
00751                 Sets[n->number].name = newtree->namefill;
00752                 Sets[n->number].node = newtree->nodefill;
00753                 break;
00754             case CEXPRESSION:
00755                 Expressions[n->number].name = newtree->namefill;
00756                 Expressions[n->number].node = newtree->nodefill;

```

```

00757         break;
00758     case CDUBIOUS:
00759         Dubious[n->number].name = newtree->namefill;
00760         Dubious[n->number].node = newtree->nodefill;
00761         break;
00762     case CDOLLAR:
00763         Dollars[n->number].name = newtree->namefill;
00764         Dollars[n->number].node = newtree->nodefill;
00765         break;
00766     default:
00767         MesPrint("Illegal variable type in CopyTree: %d",n->type);
00768         break;
00769     }
00770     newtree->nodefill++;
00771     s = newtree->namebuffer + newtree->namefill;
00772     t = oldtree->namebuffer + n->name;
00773     while ( *t ) { *s++ = *t++; newtree->namefill++; }
00774     *s = 0; newtree->namefill++;
00775 }
00776 if ( n->right >= 0 ) CopyTree(newtree,oldtree,n->right,par);
00777 }
00778
00779 /*
00780 #] CopyTree :
00781 #[ LinkTree :
00782 */
00783
00784 void LinkTree(NAMETREE *tree, WORD offset, WORD numnodes)
00785 {
00786     /*
00787      Makes the tree into a binary tree
00788     */
00789     int med,numleft,numright,medleft,medright;
00790     med = numnodes » 1;
00791     numleft = med;
00792     numright = numnodes - med - 1;
00793     medleft = numleft » 1;
00794     medright = ( numright » 1 ) + med + 1;
00795     if ( numleft > 0 ) {
00796         tree->namenode[offset+med].left = offset+medleft;
00797         tree->namenode[offset+medleft].parent = offset+med;
00798     }
00799     if ( numright > 0 ) {
00800         tree->namenode[offset+med].right = offset+medright;
00801         tree->namenode[offset+medright].parent = offset+med;
00802     }
00803     if ( numleft > 0 ) LinkTree(tree,offset,numleft);
00804     if ( numright > 0 ) LinkTree(tree,offset+med+1,numright);
00805     while ( numleft && numright ) { numleft »= 1; numright »= 1; }
00806     if ( numleft ) tree->namenode[offset+med].balance = -1;
00807     else if ( numright ) tree->namenode[offset+med].balance = 1;
00808 }
00809
00810 /*
00811 #] LinkTree :
00812 #[ MakeNameTree :
00813 */
00814
00815 NAMETREE *MakeNameTree(void)
00816 {
00817     NAMETREE *n;
00818     n = (NAMETREE *)Malloc1(sizeof(NAMETREE),"new nametree");
00819     n->namebuffer = 0;
00820     n->namenode = 0;
00821     n->namesize = n->namefill = n->nodesize = n->nodefill =
00822     n->oldnamefill = n->oldnodefill = 0;
00823     n->globalnamefill = n->globalnodefill =
00824     n->clearnamefill = n->clearnodefill = 0;
00825     n->headnode = -1;
00826     return(n);
00827 }
00828
00829 /*
00830 #] MakeNameTree :
00831 #[ FreeNameTree :
00832 */
00833
00834 void FreeNameTree(NAMETREE *n)
00835 {
00836     if ( n ) {
00837         if ( n->namebuffer ) M_free(n->namebuffer,"nametree->namebuffer");
00838         if ( n->namenode ) M_free(n->namenode,"nametree->namenode");
00839         M_free(n,"nametree");
00840     }
00841 }
00842
00843 /*

```

```

00844     #] FreeNameTree :
00845
00846     #[ WildcardNames :
00847     /*
00848
00849 void ClearWildcardNames(void)
00850 {
00851     AC.NumWildcardNames = 0;
00852 }
00853
00854 int AddWildcardName(UBYTE *name)
00855 {
00856     GETIDENTITY
00857     int size = 0, tocopy, i;
00858     UBYTE *s = name, *t, *newbuffer;
00859     while ( *s ) { s++; size++; }
00860     for ( i = 0, t = AC.WildcardNames; i < AC.NumWildcardNames; i++ ) {
00861         s = name;
00862         while ( ( *s == *t ) && *s ) { s++; t++; }
00863         if ( *s == 0 && *t == 0 ) return(i+1);
00864         while ( *t ) t++;
00865         t++;
00866     }
00867     tocopy = t - AC.WildcardNames;
00868     if ( tocopy + size + 1 > AC.WildcardBufferSize ) {
00869         if ( AC.WildcardBufferSize == 0 ) {
00870             AC.WildcardBufferSize = size+1;
00871             if ( AC.WildcardBufferSize < 100 ) AC.WildcardBufferSize = 100;
00872         }
00873         else if ( size+1 >= AC.WildcardBufferSize ) {
00874             AC.WildcardBufferSize += size+1;
00875         }
00876         else {
00877             AC.WildcardBufferSize *= 2;
00878         }
00879         newbuffer = (UBYTE *)Malloc1((LONG)AC.WildcardBufferSize,"argument list names");
00880         t = newbuffer;
00881         if ( AC.WildcardNames ) {
00882             s = AC.WildcardNames;
00883             while ( tocopy > 0 ) { *t++ = *s++; tocopy--; }
00884             M_free(AC.WildcardNames,"AC.WildcardNames");
00885         }
00886         AC.WildcardNames = newbuffer;
00887         M_free(AT.WildArgTaken,"AT.WildArgTaken");
00888         AT.WildArgTaken = (WORD *)Malloc1((LONG)AC.WildcardBufferSize*sizeof(WORD)/2
00889             ,"argument list names");
00890     }
00891     s = name;
00892     while ( *s ) *t++ = *s++;
00893     *t = 0;
00894     AC.NumWildcardNames++;
00895     return(AC.NumWildcardNames);
00896 }
00897
00898 int GetWildcardName(UBYTE *name)
00899 {
00900     UBYTE *s, *t;
00901     int i;
00902     for ( i = 0, t = AC.WildcardNames; i < AC.NumWildcardNames; i++ ) {
00903         s = name;
00904         while ( ( *s == *t ) && *s ) { s++; t++; }
00905         if ( *s == 0 && *t == 0 ) return(i+1);
00906         while ( *t ) t++;
00907         t++;
00908     }
00909     return(0);
00910 }
00911
00912 /*
00913     #] WildcardNames :
00914
00915     #[ AddSymbol :
00916
00917     The actual addition. Special routine for additions 'on the fly'
00918     /*
00919
00920 int AddSymbol(UBYTE *name, int minpow, int maxpow, int cplx, int dim)
00921 {
00922     int nodenum, numsymbol = AC.Symbols->num;
00923     UBYTE *s = name;
00924     SYMBOLS sym = (SYMBOLS)FromVarList(AC.Symbols);
00925     bzero(sym,sizeof(struct SyMb01));
00926     sym->name = AddName(*AC.activenames,name,CSYMBOL,numsymbol,&nodenum);
00927     sym->minpower = minpow;
00928     sym->maxpower = maxpow;
00929     sym->complex = cplx;
00930     sym->flags = 0;

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00931     sym->node      = nodenum;
00932     sym->dimension= dim;
00933     while ( *s ) s++;
00934     sym->namesize = (s-name)+1;
00935     return(numsymbol);
00936 }
00937
00938 /*
00939 #] AddSymbol :
00940 #[ CoSymbol :
00941
00942     Symbol declarations.  name[#{R|I|C}][{[min]:[max]}]
00943     Note that we know already that the parentheses match properly
00944 */
00945
00946 int CoSymbol(UBYTE *s)
00947 {
00948     int type, error = 0, minpow, maxpow, cplx, sgn, dim;
00949     WORD numsymbol;
00950     UBYTE *name, *oldc, c, cc;
00951     do {
00952         minpow = -MAXPOWER;
00953         maxpow =  MAXPOWER;
00954         cplx = 0;
00955         dim = 0;
00956         name = s;
00957         if ( ( s = SkipAName(s) ) == 0 ) {
00958             IllForm:    MesPrint("&Illegally formed name in symbol statement");
00959             error = 1;
00960             s = SkipField(name,0);
00961             goto eol;
00962         }
00963         oldc = s; cc = c = *s; *s = 0;
00964         if ( TestName(name) ) { *s = c; goto IllForm; }
00965         if ( cc == '#' ) {
00966             s++;
00967             if ( tolower(*s) == 'r' ) cplx = VARTYPENONE;
00968             else if ( tolower(*s) == 'c' ) cplx = VARTYPECOMPLEX;
00969             else if ( tolower(*s) == 'i' ) cplx = VARTYPEIMAGINARY;
00970             else if ( ( ( *s == '-' || *s == '+' || *s == '=' )
00971                 && ( s[1] >= '0' && s[1] <= '9' ) )
00972                 || ( *s >= '0' && *s <= '9' ) ) {
00973                 LONG x;
00974                 sgn = 0;
00975                 if ( *s == '-' ) { sgn = VARTYPEMINUS; s++; }
00976                 else if ( *s == '+' || *s == '=' ) { sgn = 0; s++; }
00977                 x = *s - '0';
00978                 while ( s[1] >= '0' && s[1] <= '9' ) {
00979                     x = 10*x + (s[1] - '0'); s++;
00980                 }
00981                 if ( x >= MAXPOWER || x <= 1 ) {
00982                     MesPrint("&Illegal value for root of unity %s",name);
00983                     error = 1;
00984                 }
00985                 else {
00986                     maxpow = x;
00987                 }
00988                 cplx = VARTYPEROOTOFUNITY | sgn;
00989             }
00990             else {
00991                 MesPrint("&Illegal specification for complexity of symbol %s",name);
00992                 *oldc = c;
00993                 error = 1;
00994                 s = SkipField(s,0);
00995                 goto eol;
00996             }
00997             s++; cc = *s;
00998         }
00999         if ( cc == '{' ) {
01000             s++;
01001             if ( ( *s == 'd' || *s == 'D' ) && s[1] == '=' ) {
01002                 s += 2;
01003                 if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
01004                     ParseSignedNumber(dim,s)
01005                     if ( dim < -HALFMAX || dim > HALFMAX ) {
01006                         MesPrint("&Warning: dimension of %s (%d) out of range"
01007                             ,name,dim);
01008                     }
01009                 }
01010                 if ( *s != '}' ) goto IllDim;
01011                 else s++;
01012             }
01013             else {
01014                 IllDim:    MesPrint("&Error: Illegal dimension field for variable %s",name);
01015                 error = 1;
01016                 s = SkipField(s,0);
01017                 goto eol;

```

```

01018         }
01019         cc = *s;
01020     }
01021     if ( cc == '(' ) {
01022         if ( ( cplx & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY ) {
01023             MesPrint("&Root of unity property for %s cannot be combined with power
restrictions",name);
01024             error = 1;
01025         }
01026         s++;
01027         if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
01028             ParseSignedNumber(minpow,s)
01029             if ( minpow < -MAXPOWER ) {
01030                 minpow = -MAXPOWER;
01031                 if ( AC.WarnFlag )
01032                     MesPrint("&Warning: minimum power of %s corrected to %d"
,name,-MAXPOWER);
01033             }
01034         }
01035     }
01036     if ( *s != ':' ) {
01037 skippar: error = 1;
01038         s = SkipField(s,1);
01039         goto eol;
01040     }
01041     else s++;
01042     if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
01043         ParseSignedNumber(maxpow,s)
01044         if ( maxpow > MAXPOWER ) {
01045             maxpow = MAXPOWER;
01046             if ( AC.WarnFlag )
01047                 MesPrint("&Warning: maximum power of %s corrected to %d"
,name,MAXPOWER);
01048         }
01049     }
01050 }
01051 if ( *s != ')' ) goto skippar;
01052 s++;
01053 }
01054 if ( ( AC.AutoDeclareFlag == 0 &&
01055     ( ( type = GetName(AC.exprnames,name,&numsymbol,NOAUTO) )
01056       != NAMENOTFOUND ) )
01057 || ( ( type = GetName(* (AC.activenames),name,&numsymbol,NOAUTO) ) != NAMENOTFOUND ) ) {
01058     if ( type != CSYMBOL ) error = NameConflict(type,name);
01059     else {
01060         SYMBOLS sym = (SYMBOLS)(AC.Symbols->lijst) + numsymbol;
01061         if ( ( numsymbol == AC.lPolyFunVar ) && ( AC.lPolyFunType > 0 )
01062             && ( AC.lPolyFun != 0 ) && ( minpow > -MAXPOWER || maxpow < MAXPOWER ) ) {
01063             MesPrint("&The symbol %s is used by power expansions in the PolyRatFun!",name);
01064             error = 1;
01065         }
01066         sym->complex = cplx;
01067         sym->minpower = minpow;
01068         sym->maxpower = maxpow;
01069         sym->dimension= dim;
01070     }
01071 }
01072 else {
01073     AddSymbol(name,minpow,maxpow,cplx,dim);
01074 }
01075 *oldc = c;
01076 eol: while ( *s == ',' ) s++;
01077 } while ( *s );
01078 return(error);
01079 }
01080
01081 /*
01082 #] CoSymbol :
01083 #[ AddIndex :
01084
01085 The actual addition. Special routine for additions 'on the fly'
01086 */
01087
01088 int AddIndex(UBYTE *name, int dim, int dim4)
01089 {
01090     int nodenum, numindex = AC.Indices->num;
01091     INDICES ind = (INDICES)FromVarList(AC.Indices);
01092     UBYTE *s = name;
01093     bzero(ind,sizeof(struct InDeX));
01094     ind->name = AddName(*AC.activenames,name,CINDEX,numindex,&nodenum);
01095     ind->type = 0;
01096     ind->dimension = dim;
01097     ind->flags = 0;
01098     ind->nmin4 = dim4;
01099     ind->node = nodenum;
01100     while ( *s ) s++;
01101     ind->namesize = (s-name)+1;
01102     return(numindex);
01103 }

```



```

01104
01105 /*
01106 #] AddIndex :
01107 #[ CoIndex :
01108
01109 Index declarations. name[]={number|symbol[:othersymbol]}}
01110 */
01111
01112 int CoIndex(UBYTE *s)
01113 {
01114     int type, error = 0, dim, dim4;
01115     WORD numindex;
01116     UBYTE *name, *oldc, c;
01117     do {
01118         dim = AC.lDefDim;
01119         dim4 = AC.lDefDim4;
01120         name = s;
01121         if ( ( s = SkipAName(s) ) == 0 ) {
01122 IllForm: MesPrint("&Illegally formed name in index statement");
01123             error = 1;
01124             s = SkipField(name,0);
01125             goto eol;
01126         }
01127         oldc = s; c = *s; *s = 0;
01128         if ( TestName(name) ) { *s = c; goto IllForm; }
01129         if ( c == '=' ) {
01130             s++;
01131             if ( ( s = DoDimension(s,&dim,&dim4) ) == 0 ) {
01132                 *oldc = c;
01133                 error = 1;
01134                 s = SkipField(name,0);
01135                 goto eol;
01136             }
01137         }
01138         if ( ( AC.AutoDeclareFlag == 0 &&
01139             ( ( type = GetName(AC.exprnames,name,&numindex,NOAUTO) )
01140             != NAMENOTFOUND ) )
01141             || ( ( type = GetName(*AC.activenames),name,&numindex,NOAUTO) ) != NAMENOTFOUND ) ) {
01142             if ( type != CINDEX ) error = NameConflict(type,name);
01143             else { /* reset the dimensions */
01144                 indices[numindex].dimension = dim;
01145                 indices[numindex].nmin4 = dim4;
01146             }
01147         }
01148         else AddIndex(name,dim,dim4);
01149         *oldc = c;
01150 eol: while ( *s == ',' ) s++;
01151     } while ( *s );
01152     return(error);
01153 }
01154
01155 /*
01156 #] CoIndex :
01157 #[ DoDimension :
01158 */
01159
01160 UBYTE *DoDimension(UBYTE *s, int *dim, int *dim4)
01161 {
01162     UBYTE c, *t = s;
01163     int type, error = 0;
01164     WORD numsymbol;
01165     NAMETREE **oldtree = AC.activenames;
01166     LIST* oldsymbols = AC.Symbols;
01167     *dim4 = -NMIN4SHIFT;
01168     if ( FG.cTable[*s] == 1 ) {
01169 retry: ParseNumber(*dim,s)
01170         if ( BITSINWORD/8 < 4 )
01171             if ( *dim >= (1 << (BITSINWORD-1)) ) goto illeg;
01172     #endif
01173         *dim4 = *dim - 4;
01174         return(s);
01175     }
01176     else if ( ( ( FG.cTable[*s] == 0 ) || ( *s == '[' ) )
01177         && ( s = SkipAName(s) ) != 0 ) {
01178         AC.activenames = &(AC.varnames);
01179         AC.Symbols = &(AC.SymbolList);
01180         c = *s; *s = 0;
01181         if ( ( ( type = GetName(AC.exprnames,t,&numsymbol,NOAUTO) ) != NAMENOTFOUND )
01182             || ( ( type = GetName(AC.varnames,t,&numsymbol,WITHAUTO) ) != NAMENOTFOUND ) ) {
01183             if ( type != CSYMBOL ) error = NameConflict(type,t);
01184         }
01185         else {
01186             numsymbol = AddSymbol(t,-MAXPOWER,MAXPOWER,0,0);
01187             if ( AC.WarnFlag )
01188                 MesPrint("&Warning: Implicit declaration of %s as a symbol",t);
01189         }
01190     }

```

```

01191         *dim = -numsymbol;
01192         if ( ( *s = c ) == ':' ) {
01193             s++;
01194             t = s;
01195             if ( ( s = SkipAName(s) ) == 0 ) goto illeg;
01196             if ( ( ( type = GetName(AC.exprnames,t,&numsymbol,NOAUTO) ) != NAMENOTFOUND )
01197             || ( ( type = GetName(AC.varnames,t,&numsymbol,WITHAUTO) ) != NAMENOTFOUND ) ) {
01198                 if ( type != CSYMBOL ) error = NameConflict(type,t);
01199             }
01200             else {
01201                 numsymbol = AddSymbol(t,-MAXPOWER,MAXPOWER,0,0);
01202                 if ( AC.WarnFlag )
01203                     MesPrint("&Warning: Implicit declaration of %s as a symbol",t);
01204             }
01205             *dim4 = -numsymbol-NMIN4SHIFT;
01206         }
01207     }
01208     else if ( *s == '+' && FG.cTable[s[1]] == 1 ) {
01209         s++; goto retry;
01210     }
01211     else {
01212 illeg: MesPrint("&Illegal dimension specification. Should be number >= 0, symbol or symbol:symbol");
01213         return(0);
01214     }
01215     AC.Symbols = oldsymbols;
01216     AC.activenames = oldtree;
01217     if ( error ) return(0);
01218     return(s);
01219 }
01220
01221 /*
01222 #] DoDimension :
01223 #[ CoDimension :
01224 */
01225
01226 int CoDimension(UBYTE *s)
01227 {
01228     s = DoDimension(s,&AC.lDefDim,&AC.lDefDim4);
01229     if ( s == 0 ) return(1);
01230     if ( *s != 0 ) {
01231         MesPrint("&Argument of dimension statement should be number >= 0, symbol or symbol:symbol");
01232         return(1);
01233     }
01234     return(0);
01235 }
01236
01237 /*
01238 #] CoDimension :
01239 #[ AddVector :
01240
01241 The actual addition. Special routine for additions 'on the fly'
01242 */
01243
01244 int AddVector(UBYTE *name, int cplx, int dim)
01245 {
01246     int nodenum, numvector = AC.Vectors->num;
01247     VECTORS v = (VECTORS)FromVarList(AC.Vectors);
01248     UBYTE *s = name;
01249     bzero(v,sizeof(struct VeCtOr));
01250     v->name = AddName(*AC.activenames,name,CVECTOR,numvector,&nodenum);
01251     v->complex = cplx;
01252     v->node = nodenum;
01253     v->dimension = dim;
01254     v->flags = 0;
01255     while ( *s ) s++;
01256     v->namesize = (s-name)+1;
01257     return(numvector);
01258 }
01259
01260 /*
01261 #] AddVector :
01262 #[ CoVector :
01263
01264 Vector declarations. The descriptor string is "(,%n)"
01265 */
01266
01267 int CoVector(UBYTE *s)
01268 {
01269     int type, error = 0, dim;
01270     WORD numvector;
01271     UBYTE *name, c, *endname;
01272     do {
01273         name = s;
01274         dim = 0;
01275         if ( ( s = SkipAName(s) ) == 0 ) {
01276 IllForm: MesPrint("&Illegally formed name in vector statement");
01277             error = 1;

```

```

01278         s = SkipField(s,0);
01279     }
01280     else {
01281         c = *s; *s = 0, endname = s;
01282         if ( TestName(name) ) { *s = c; goto IllForm; }
01283         if ( c == '{' ) {
01284             s++;
01285             if ( ( *s == 'd' || *s == 'D' ) && s[1] == '=' ) {
01286                 s += 2;
01287                 if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
01288                     ParseSignedNumber(dim,s)
01289                     if ( dim < -HALFMAX || dim > HALFMAX ) {
01290                         MesPrint("&Warning: dimension of %s (%d) out of range"
01291                             ,name,dim);
01292                     }
01293                 }
01294                 if ( *s != '}' ) goto IllDim;
01295                 else s++;
01296             }
01297             else {
01298 IllDim: MesPrint("&Error: Illegal dimension field for variable %s",name);
01299                 error = 1;
01300                 s = SkipField(s,0);
01301                 while ( *s == ',' ) s++;
01302                 continue;
01303             }
01304         }
01305         if ( ( AC.AutoDeclareFlag == 0 &&
01306             ( ( type = GetName(AC.exprnames,name,&numvector,NOAUTO) )
01307               != NAMENOTFOUND ) )
01308             || ( ( type = GetName(*AC.activenames,name,&numvector,NOAUTO) ) != NAMENOTFOUND ) ) {
01309             if ( type != CVECTOR ) error = NameConflict(type,name);
01310         }
01311         else AddVector(name,0,dim);
01312         *endname = c;
01313     }
01314     while ( *s == ',' ) s++;
01315 } while ( *s );
01316 return(error);
01317 }
01318
01319 /*
01320 #] CoVector :
01321 #[ AddFunction :
01322
01323 The actual addition. Special routine for additions 'on the fly'
01324 */
01325
01326 int AddFunction(UBYTE *name, int comm, int istensor, int cplx, int symprop, int dim, int argmax, int
    argmin)
01327 {
01328     int nodenum, numfunction = AC.Functions->num;
01329     FUNCTIONS fun = (FUNCTIONS)FromVarList(AC.Functions);
01330     UBYTE *s = name;
01331     bzero(fun,sizeof(struct FuNcTiOn));
01332     fun->name = AddName(*AC.activenames,name,CFUNCTION,numfunction,&nodenum);
01333     fun->commute = comm;
01334     fun->spec = istensor;
01335     fun->complex = cplx;
01336     fun->tabl = 0;
01337     fun->flags = 0;
01338     fun->node = nodenum;
01339     fun->symminfo = 0;
01340     fun->symmetric = symprop;
01341     fun->dimension = dim;
01342     fun->maxnumargs = argmax;
01343     fun->minnumargs = argmin;
01344     while ( *s ) s++;
01345     fun->namesize = (s-name)+1;
01346     return(numfunction);
01347 }
01348
01349 /*
01350 #] AddFunction :
01351 #[ CoCommuteInSet :
01352
01353 Commuting, f1, ..., fn;
01354 */
01355
01356 int CoCommuteInSet(UBYTE *s)
01357 {
01358     UBYTE *name, *ss, c, *start = s;
01359     WORD number, type, *g, *gg;
01360     int error = 0, i, len = StrLen(s), len2 = 0;
01361     if ( AC.CommuteInSet != 0 ) {
01362         g = AC.CommuteInSet;
01363         while ( *g ) g += *g;

```

```

01364     len2 = g - AC.CommuteInSet;
01365     if ( len2+len+3 > AC.SizeCommuteInSet ) {
01366         gg = (WORD *)Malloc1((len2+len+3)*sizeof(WORD),"CommuteInSet");
01367         for ( i = 0; i < len2; i++ ) gg[i] = AC.CommuteInSet[i];
01368         gg[len2] = 0;
01369         M_free(AC.CommuteInSet,"CommuteInSet");
01370         AC.CommuteInSet = gg;
01371         AC.SizeCommuteInSet = len+len2+3;
01372         g = AC.CommuteInSet+len2;
01373     }
01374 }
01375 else {
01376     AC.SizeCommuteInSet = len+2;
01377     g = AC.CommuteInSet = (WORD *)Malloc1((len+3)*sizeof(WORD),"CommuteInSet");
01378     *g = 0;
01379 }
01380 gg = g++;
01381 ss = s-1;
01382 for(;;) {
01383     while ( *s == ' ' || *s == '\t' ) s++;
01384     if ( *s == 0 ) {
01385         if ( s - start >= len ) break;
01386         *s = ' '; s++;
01387         *g = 0;
01388         *gg = g-gg;
01389         if ( *gg < 2 ) {
01390             MesPrint("&There should be at least two noncommuting functions or tensors in a
commuting statement.");
01391             error = 1;
01392         }
01393         else if ( *gg == 2 ) {
01394             gg[2] = gg[1]; gg[3] = 0; gg[0] = 3;
01395         }
01396         gg = g++;
01397         continue;
01398     }
01399     if ( s > ss ) {
01400         if ( *s != '{' ) {
01401             MesPrint("&The CommuteInSet statement should have sets enclosed in {}.");
01402             error = 1;
01403             break;
01404         }
01405         ss = s;
01406         SKIPBRA2(ss) /* Note that parentheses were tested before */
01407         *ss = 0;
01408         s++;
01409     }
01410     name = s;
01411     s = SkipAName(s);
01412     c = *s; *s = 0;
01413     if ( ( type = GetName(AC.varnames,name,&number,NOAUTO) ) != CFUNCTION ) {
01414         MesPrint("&%s is not a function or tensor",name);
01415         error = 1;
01416     }
01417     else if ( functions[number].commute == 0 ){
01418         MesPrint("&%s is not a noncommuting function or tensor",name);
01419         error = 1;
01420     }
01421     else {
01422         *g++ = number+FUNCTION;
01423         functions[number].flags |= COULDCOMMUTE;
01424         if ( number+FUNCTION >= GAMMA && number+FUNCTION <= GAMMASEVEN ) {
01425             functions[GAMMA-FUNCTION].flags |= COULDCOMMUTE;
01426             functions[GAMMAI-FUNCTION].flags |= COULDCOMMUTE;
01427             functions[GAMMAFIVE-FUNCTION].flags |= COULDCOMMUTE;
01428             functions[GAMMASIX-FUNCTION].flags |= COULDCOMMUTE;
01429             functions[GAMMASEVEN-FUNCTION].flags |= COULDCOMMUTE;
01430         }
01431     }
01432     *s = c;
01433 }
01434 return(error);
01435 }
01436
01437 /*
01438 #] CoCommuteInSet :
01439 #[ CoFunction + ...:
01440
01441 Function declarations.
01442 The second parameter indicates commutation properties.
01443 The third parameter tells whether we have a tensor.
01444 */
01445
01446 int CoFunction(UBYTE *s, int comm, int istensor)
01447 {
01448     int type, error = 0, cplx, symtype, dim, argmax, argmin;
01449     WORD numfunction, reverseorder = 0, addone;

```

```

01450     UBYTE *name, *oldc, *par, c, cc;
01451     do {
01452         symtype = cplx = 0, argmin = argmax = -1;
01453         dim = 0;
01454         name = s;
01455         if ( ( s = SkipAName(s) ) == 0 ) {
01456 IllForm:     MesPrint("&Illegally formed function/tensor name");
01457                 error = 1;
01458                 s = SkipField(name,0);
01459                 goto eol;
01460         }
01461         oldc = s; cc = c = *s; *s = 0;
01462         if ( TestName(name) ) { *s = c; goto IllForm; }
01463         if ( c == '#' ) {
01464             s++;
01465             if ( tolower(*s) == 'r' ) cplx = VARTYPENONE;
01466             else if ( tolower(*s) == 'c' ) cplx = VARTYPECOMPLEX;
01467             else if ( tolower(*s) == 'i' ) cplx = VARTYPEIMAGINARY;
01468             else {
01469                 MesPrint("&Illegal specification for complexity of %s",name);
01470                 *oldc = c;
01471                 error = 1;
01472                 s = SkipField(s,0);
01473                 goto eol;
01474             }
01475             s++; cc = *s;
01476         }
01477         if ( cc == '{' ) {
01478             s++;
01479             if ( ( *s == 'd' || *s == 'D' ) && s[1] == '=' ) {
01480                 s += 2;
01481                 if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
01482                     ParseSignedNumber(dim,s)
01483                     if ( dim < -HALFMAX || dim > HALFMAX ) {
01484                         MesPrint("&Warning: dimension of %s (%d) out of range",
01485                             name,dim);
01486                     }
01487                 }
01488                 if ( *s != '}' ) goto IllDim;
01489                 else s++;
01490             }
01491             else {
01492 IllDim:     MesPrint("&Error: Illegal dimension field for variable %s",name);
01493                 error = 1;
01494                 s = SkipField(s,0);
01495                 goto eol;
01496             }
01497             cc = *s;
01498         }
01499         if ( cc == '(' ) {
01500             s++;
01501             if ( *s == '-' ) {
01502                 reverseorder = REVERSEORDER;
01503                 s++;
01504             }
01505             else {
01506                 reverseorder = 0;
01507             }
01508             par = s;
01509             while ( FG.cTable[*s] == 0 ) s++;
01510             cc = *s; *s = 0;
01511             if ( s <= par ) {
01512 illegsym:   *s = cc;
01513                 MesPrint("&Illegal specification for symmetry of %s",name);
01514                 *oldc = c;
01515                 error = 1;
01516                 s = SkipField(s,1);
01517                 goto eol;
01518             }
01519             if ( StrICont(par,(UBYTE *)"symmetric") == 0 ) symtype = SYMMETRIC;
01520             else if ( StrICont(par,(UBYTE *)"antisymmetric") == 0 ) symtype = ANTISYMMETRIC;
01521             else if ( ( StrICont(par,(UBYTE *)"cyclesymmetric") == 0 )
01522                 || ( StrICont(par,(UBYTE *)"cyclic") == 0 ) ) symtype = CYCLESYMMETRIC;
01523             else if ( ( StrICont(par,(UBYTE *)"rcyclesymmetric") == 0 )
01524                 || ( StrICont(par,(UBYTE *)"rcyclic") == 0 )
01525                 || ( StrICont(par,(UBYTE *)"reversecyclic") == 0 ) ) symtype = RCYCLESYMMETRIC;
01526             else goto illegsym;
01527             *s = cc;
01528             if ( *s != ')' || ( s[1] && s[1] != ',' && s[1] != '<' ) ) {
01529                 Warning("&Excess information in symmetric properties currently ignored");
01530                 s = SkipField(s,1);
01531             }
01532             else s++;
01533             symtype |= reverseorder;
01534             cc = *s;
01535         }
01536 retry;;

```

```

01537         if ( cc == '<' ) {
01538             s++; addone = 0;
01539             if ( *s == '=' ) { addone++; s++; }
01540             argmax = 0;
01541             while ( FG.cTable[*s] == 1 ) { argmax = 10*argmax + *s++ - '0'; }
01542             argmax += addone;
01543             par = s;
01544             while ( FG.cTable[*s] == 0 ) s++;
01545             if ( s > par ) {
01546                 cc = *s; *s = 0;
01547                 if ( ( StrICont(par,(UBYTE *)"arguments") == 0 )
01548                     || ( StrICont(par,(UBYTE *)"args") == 0 ) ) {}
01549                 else {
01550                     Warning("&Illegal information in number of arguments properties currently
01551 ignored");
01552                     error = 1;
01553                     }
01554                 *s = cc;
01555             }
01556             if ( argmax <= 0 ) {
01557                 MesPrint("&Error: Cannot have fewer than 0 arguments for variable %s",name);
01558                 error = 1;
01559             }
01560             cc = *s;
01561         }
01562         if ( cc == '>' ) {
01563             s++; addone = 1;
01564             if ( *s == '=' ) { addone = 0; s++; }
01565             argmin = 0;
01566             while ( FG.cTable[*s] == 1 ) { argmin = 10*argmin + *s++ - '0'; }
01567             argmin += addone;
01568             par = s;
01569             while ( FG.cTable[*s] == 0 ) s++;
01570             if ( s > par ) {
01571                 cc = *s; *s = 0;
01572                 if ( ( StrICont(par,(UBYTE *)"arguments") == 0 )
01573                     || ( StrICont(par,(UBYTE *)"args") == 0 ) ) {}
01574                 else {
01575                     Warning("&Illegal information in number of arguments properties currently
01576 ignored");
01577                     error = 1;
01578                     }
01579                 *s = cc;
01580             }
01581             cc = *s;
01582             if ( cc == '<' ) goto retry;
01583             if ( ( AC.AutoDeclareFlag == 0 &&
01584                 ( ( type = GetName(AC.exprnames,name,&numfunction,NOAUTO) )
01585                   != NAMENOTFOUND ) )
01586               || ( ( type = GetName(*AC.activenames),name,&numfunction,NOAUTO) ) != NAMENOTFOUND ) ) {
01587                 if ( type != CFUNCTION ) error = NameConflict(type,name);
01588                 else {
01589                     /* FUNCTIONS fun = (FUNCTIONS) (AC.Functions->lijst) + numfunction-FUNCTION; */
01590                     FUNCTIONS fun = (FUNCTIONS) (AC.Functions->lijst) + numfunction;
01591                     if ( fun->tabl != 0 ) {
01592                         MesPrint("&Illegal attempt to change table into function");
01593                         error = 1;
01594                     }
01595                     fun->complex = cplx;
01596                     fun->commute = comm;
01597                     if ( istensor && fun->spec == 0 ) {
01598                         MesPrint("&Function %s changed to tensor",name);
01599                         error = 1;
01600                     }
01601                     else if ( istensor == 0 && fun->spec > 0 ) {
01602                         MesPrint("&Tensor %s changed to function",name);
01603                         error = 1;
01604                     }
01605                     fun->spec = istensor;
01606                     if ( fun->symmetric != symtype ) {
01607                         fun->symmetric = symtype;
01608                         AC.SymChangeFlag = 1;
01609                     }
01610                     fun->maxnumargs = argmax;
01611                     fun->minnumargs = argmin;
01612                 }
01613             }
01614             else {
01615                 AddFunction(name,comm,istensor,cplx,symtype,dim,argmax,argmin);
01616             }
01617             *oldc = c;
01618             eol: while ( *s == ',' ) s++;
01619         } while ( *s );
01620         return(error);

```

```

01622 }
01623
01624 int CoNFunction(UBYTE *s) { return(CoFunction(s,1,0)); }
01625 int CoCFunction(UBYTE *s) { return(CoFunction(s,0,0)); }
01626 int CoNTensor(UBYTE *s) { return(CoFunction(s,1,2)); }
01627 int CoCTensor(UBYTE *s) { return(CoFunction(s,0,2)); }
01628
01629 /*
01630 #] CoFunction + ...:
01631 #[ DoTable :
01632
01633     Syntax:
01634     Table [check] [strict|relax] [zerofill] name(:1:2,...,regular arguments);
01635     name must be the name of a regular function.
01636     the table indices must be the first arguments.
01637     The parenthesis indicates 'name' as opposed to the options.
01638
01639     We leave behind:
01640     a struct tabl in the FUNCTION struct
01641     Regular table:
01642         an array tablepointers for the pointers to elements of rhs
01643         in the compiler struct cbuf[T->bufnum]
01644         an array MINMAX T->mm with the minima and maxima
01645         a prototype array
01646         an offset in the compiler buffer for the pattern to be matched
01647     Sparse table:
01648         Just the number of dimensions
01649         We will keep track of the number of defined elements in totind
01650         and in tablepointers we will have numind+1 positions for each
01651         element. The first numind elements for the indices and the
01652         last one for the element in cbuf[T->bufnum].rhs
01653
01654         If the number of dimensions is *<number>, there is not a fixed
01655         number of dimensions. Just a maximum. In that case the first
01656         index should be the number of other dimensions. This first index
01657         does not count in <number>.
01658
01659     Complication: to preserve speed we need a prototype and a pattern
01660     for each thread when we use WITHPTHREADS. This is because we write
01661     into those when looking for the pattern.
01662 */
01663
01664 static int nwarntab = 1;
01665
01666 int DoTable(UBYTE *s, int par)
01667 {
01668     GETIDENTITY
01669     UBYTE *name, *p, *inp, c;
01670     int i, j, k, sparseflag = 0, rflag = 0, checkflag = 0;
01671     int error = 0, ret, oldcbufnum, oldEside;
01672     WORD funnum, type, *OldWork, *w, *ww, *t, *tt, *flags1, oldnumrhs,oldnumlhs;
01673     LONG oldcpointer;
01674     MINMAX *mm, *mml;
01675     LONG x, y;
01676     TABLES T;
01677     CBUF *C;
01678
01679     while ( *s == ',' ) s++;
01680     do {
01681         name = s;
01682         if ( ( s = SkipAName(s) ) == 0 ) {
01683             IllForm: MesPrint("&Illegal name or option in table declaration");
01684             return(1);
01685         }
01686         c = *s; *s = 0;
01687         if ( TestName(name) ) { *s = c; goto IllForm; }
01688         *s = c;
01689         if ( *s == '(' ) break;
01690         if ( *s != ',' ) {
01691             MesPrint("&Illegal definition of table");
01692             return(1);
01693         }
01694         *s = 0;
01695     } while ( *s++ );
01696
01697     Secondary options
01698     if ( StrICmp(name, (UBYTE *)("check" )) == 0 ) checkflag = 1;
01699     else if ( StrICmp(name, (UBYTE *)("zero" )) == 0 ) checkflag = 2;
01700     else if ( StrICmp(name, (UBYTE *)("one" )) == 0 ) checkflag = 3;
01701     else if ( StrICmp(name, (UBYTE *)("strict")) == 0 ) rflag = 1;
01702     else if ( StrICmp(name, (UBYTE *)("relax" )) == 0 ) rflag = -1;
01703     else if ( StrICmp(name, (UBYTE *)("zerofill" )) == 0 ) { rflag = -2; checkflag = 2; }
01704     else if ( StrICmp(name, (UBYTE *)("onefill" )) == 0 ) { rflag = -3; checkflag = 3; }
01705     else if ( StrICmp(name, (UBYTE *)("sparse")) == 0 ) sparseflag |= 1;
01706     else if ( StrICmp(name, (UBYTE *)("base")) == 0 ) sparseflag |= 3;
01707     else if ( StrICmp(name, (UBYTE *)("tablebase")) == 0 ) sparseflag |= 3;
01708     else {

```

```

01709         MesPrint("&Illegal option in table definition: '%s'",name);
01710         error = 1;
01711     }
01712     *s++ = ',';
01713     while ( *s == ',' ) s++;
01714 } while ( *s );
01715 if ( name == s || *s == 0 ) {
01716     MesPrint("&Illegal name or option in table declaration");
01717     return(1);
01718 }
01719 *s = 0;          /* *s could only have been a parenthesis */
01720 if ( sparseflag ) {
01721     if ( checkflag == 1 ) rflag = 0;
01722     else if ( checkflag == 2 ) rflag = -2;
01723     else if ( checkflag == 3 ) rflag = -3;
01724     else rflag = -1;
01725 }
01726 if ( ( ret = GetVar(name,&type,&funnum,CFUNCTION,NOAUTO) ) ==
01727     NAMENOTFOUND ) {
01728     if ( par == 0 ) {
01729         funnum = EntVar(CFUNCTION,name,0,1,0,0);
01730     }
01731     else if ( par == 1 || par == 2 ) {
01732         funnum = EntVar(CFUNCTION,name,0,0,0,0);
01733     }
01734 }
01735 else if ( ret <= 0 ) {
01736     funnum = EntVar(CFUNCTION,name,0,0,0,0);
01737     error = 1;
01738 }
01739 else {
01740     if ( par == 2 ) {
01741         if ( nwarntab ) {
01742             Warning("Table now declares its (commuting) function.");
01743             Warning("Earlier definition in Function statement obsolete. Please remove.");
01744             nwarntab = 0;
01745         }
01746     }
01747     else {
01748         error = 1;
01749         MesPrint("&(N)(C)Tables should not be declared previously");
01750     }
01751 }
01752 if ( functions[funnum].spec > 0 ) {
01753     MesPrint("&Tensors cannot become tables");
01754     return(1);
01755 }
01756 if ( functions[funnum].symmetric > 0 ) {
01757     MesPrint("&Functions with nontrivial symmetrization properties cannot become tables");
01758     return(1);
01759 }
01760 if ( functions[funnum].tabl ) {
01761     MesPrint("&Redefinition of an existing table is not allowed.");
01762     return(1);
01763 }
01764 functions[funnum].tabl = T = (TABLES)Malloc1(sizeof(struct TableS),"table");
01765 /*
01766 Next we find the size of the table (if it is not sparse)
01767 */
01768 T->defined = T->mdefined = 0; T->sparse = sparseflag; T->mm = 0; T->flags = 0;
01769 T->numtree = 0; T->rootnum = 0; T->MaxTreeSize = 0;
01770 T->boomlijst = 0;
01771 T->strict = rflag;
01772 T->bounds = checkflag;
01773 T->bufnum = inicbufs();
01774 T->argtail = 0;
01775 T->sparse = 0;
01776 T->buffersize = 8;
01777 T->buffers = (WORD *)Malloc1(sizeof(WORD)*T->buffersize,"Table buffers");
01778 T->buffersfill = 0;
01779 T->buffers[T->buffersfill++] = T->bufnum;
01780 T->mode = 0;
01781 T->numdummies = 0;
01782 mm = T->mm;
01783 T->numind = 0;
01784 if ( rflag > 0 ) AC.MustTestTable++;
01785 T->totind = 0;          /* Table hasn't been checked */
01786
01787 p = s; *s = '(';
01788 if ( sparseflag ) {
01789 /*
01790     First copy the tail, just in case we will construct a tablebase
01791     Note that we keep the ( to indicate a tail
01792     The actual arguments can be found after the comma. Before we have
01793     the dimension which the tablebase will need for consistency checking.
01794 */
01795     inp = p+1;

```



```

01796     SKIPBRA3(inp)
01797     c = *inp; *inp = 0;
01798     T->argtail = strDupl(p,"argtail");
01799     *inp = c;
01800     /*
01801     Now the regular compilation
01802     */
01803     inp = p++;
01804     if ( *p == '<' ) {
01805         WORD inc = 1;
01806         p++;
01807         if ( *p == '=' ) { inc++; p++; }
01808     /*
01809         We will use one extra number for telling how many there really are.
01810     */
01811     x = 0;
01812     while ( *p <= '9' && *p >= '0' ) x = 10*x + (*p++-'0');
01813     x = -x-inc;
01814     if ( x == -1 ) {
01815         MesPrint("&Maximum number of dimensions in *-table should be at least one.");
01816         error = 1;
01817         goto FinishUp2;
01818     }
01819 }
01820 else {
01821     ParseNumber(x,p)
01822     if ( FG.cTable[p[-1]] != 1 || ( *p != ',' && *p != ')' ) ) {
01823         p = inp;
01824         MesPrint("&First argument in a sparse table must be a number of dimensions");
01825         error = 1;
01826         x = 1;
01827     }
01828 }
01829 T->numind = x;
01830 T->mm = (MINMAX *)Malloc1(ABS(x)*sizeof(MINMAX),"table dimensions");
01831 T->flags = (WORD *)Malloc1(ABS(x)*sizeof(WORD),"table flags");
01832 mm = T->mm;
01833 inp = p;
01834 if ( *inp != ')' ) inp++;
01835 T->totind = 0; /* At the moment there are this many */
01836 T->tablepointers = 0;
01837 T->reserved = 0;
01838 }
01839 else {
01840     T->numind = 0;
01841     T->totind = 1;
01842     for(;;) { /* Read the dimensions as far as they can be recognized */
01843         inp = ++p;
01844         if ( FG.cTable[*p] != 1 && *p != '+' && *p != '-' ) break;
01845         ParseSignedNumber(x,p)
01846         if ( FG.cTable[p[-1]] != 1 || *p != ':' ) break;
01847         p++;
01848         ParseSignedNumber(y,p)
01849         if ( FG.cTable[p[-1]] != 1 || ( *p != ',' && *p != ')' ) ) {
01850             MesPrint("&Illegal dimension field in table declaration");
01851             return(1);
01852         }
01853         mml = (MINMAX *)Malloc1((T->numind+1)*sizeof(MINMAX),"table dimensions");
01854         flags1 = (WORD *)Malloc1((T->numind+1)*sizeof(WORD),"table flags");
01855         for ( i = 0; i < T->numind; i++ ) { mml[i] = T->mm[i]; flags1[i] = T->flags[i]; }
01856         if ( T->mm ) M_free(T->mm,"table dimensions");
01857         if ( T->flags ) M_free(T->flags,"table flags");
01858         T->mm = mml;
01859         T->flags = flags1;
01860         mm = T->mm + T->numind;
01861         mm->mini = x; mm->maxi = y;
01862         T->totind += mm->maxi-mm->mini+1;
01863         T->numind++;
01864         if ( *p == ')' ) { inp = p; break; }
01865     }
01866     w = T->tablepointers
01867     = (WORD *)Malloc1(TABLEEXTENSION*sizeof(WORD)*(T->totind),"table pointers");
01868     i = T->totind;
01869     for ( i = TABLEEXTENSION*T->totind; i > 0; i-- ) *w++ = -1; /* means: undefined */
01870     for ( i = T->numind-1, x = 1; i >= 0; i-- ) {
01871         T->mm[i].size = x; /* Defines increment in this dimension */
01872         x *= T->mm[i].maxi - T->mm[i].mini + 1;
01873     }
01874 }
01875 /*
01876 Now we redo the 'function part' and send it to the compiler.
01877 The prototype has to be picked up properly.
01878 */
01879 AT.WorkPointer++; /* We need one extra word later */
01880 OldWork = AT.WorkPointer;
01881 oldcbufnum = AC.cbufnum;
01882 AC.cbufnum = T->bufnum;

```

```

01883     C = cbuf+AC.cbufnum;
01884     oldcpointer = C->Pointer - C->Buffer;
01885     oldnumlhs = C->numlhs;
01886     oldnumrhs = C->numrhs;
01887     AddLHS(AC.cbufnum);
01888     while ( s >= name ) *--inp = *s--;
01889     w = AT.WorkPointer;
01890     AC.ProtoType = w;
01891     *w++ = SUBEXPRESSION;
01892     *w++ = SUBEXPSIZE;
01893     *w++ = 0;
01894     *w++ = 1;
01895     *w++ = AC.cbufnum;
01896     FILLSUB(w);
01897     AC.WildC = w;
01898     AC.NwildC = 0;
01899     AT.WorkPointer = w + 4*AM.MaxWildcards;
01900     if ( ( ret = CompileAlgebra(inp,LHSIDE,AC.ProtoType) ) < 0 ) {
01901         error = 1; goto FinishUp;
01902     }
01903     if ( AC.NwildC && SortWild(w,AC.NwildC) ) error = 1;
01904     w += AC.NwildC;
01905     i = w-OldWork;
01906     OldWork[1] = i;
01907 /*
01908     Basically we have to pull this pattern through Generator in case
01909     there are functions inside functions, or parentheses.
01910     We have to temporarily disable the .tabl to avoid problems with
01911     TestSub.
01912     Essential: we need to start NewSort twice to avoid the PutOut routines.
01913     The ground pattern is sitting in C->numrhs, but it could be that it
01914     has subexpressions in it. Hence it has to be worked out as the lhs in
01915     id statements (in comexpr.c).
01916 */
01917     OldWork[2] = C->numrhs;
01918     *w++ = 1; *w++ = 1; *w++ = 3;
01919     OldWork[-1] = w-OldWork+1;
01920     AT.WorkPointer = w;
01921     ww = C->rhs[C->numrhs];
01922     for ( j = 0; j < *ww; j++ ) w[j] = ww[j];
01923     AT.WorkPointer = w+*w;
01924     if ( *ww == 0 || ww[*ww] != 0 ) {
01925         MesPrint("&Illegal table pattern definition");
01926         AC.lhdollarflag = 0;
01927         error = 1;
01928     }
01929     if ( error ) goto FinishUp;
01930
01931     if ( NewSort(BHEAD0) || NewSort(BHEAD0) ) { error = 1; goto FinishUp; }
01932     AN.RepPoint = AT.RepCount + 1;
01933     AC.lhdollarflag = 0; oldEside = AR.Eside; AR.Eside = LHSIDE;
01934     AR.Cnumlhs = C->numlhs;
01935     functions[funnum].tabl = 0;
01936     if ( Generator(BHEAD w,C->numlhs) ) {
01937         functions[funnum].tabl = T;
01938         AR.Eside = oldEside;
01939         LowerSortLevel(); LowerSortLevel(); goto FinishUp;
01940     }
01941     functions[funnum].tabl = T;
01942     AR.Eside = oldEside;
01943     AT.WorkPointer = w;
01944     if ( EndSort(BHEAD w,0) < 0 ) { LowerSortLevel(); goto FinishUp; }
01945     if ( *w == 0 || *(w+*w) != 0 ) {
01946         MesPrint("&Irregular pattern in table definition");
01947         error = 1;
01948         goto FinishUp;
01949     }
01950     LowerSortLevel();
01951     if ( AC.lhdollarflag ) {
01952         MesPrint("&Unexpanded dollar variables are not allowed in table definition");
01953         error = 1;
01954         goto FinishUp;
01955     }
01956     AT.WorkPointer = ww = w + *w;
01957     if ( ww[-1] != 3 || ww[-2] != 1 || ww[-3] != 1 ) {
01958         MesPrint("&Coefficient of pattern in table definition should be 1.");
01959         error = 1;
01960         goto FinishUp;
01961     }
01962     AC.DumNum = 0;
01963 /*
01964     Now we have to allocate space for prototype+pattern
01965     In the case of TFORM we need extra pointers, because each worker has its own
01966 */
01967     j = *w + ABS(T->numind)*2-3;
01968 #ifdef WITHPTHREADS
01969     { int n;

```

```

01970     T->prototypeSize = ((i+j)*sizeof(WORD)+2*sizeof(WORD *)) * AM.totalnumberofthreads;
01971     T->prototype = (WORD **)Malloc1(T->prototypeSize, "table prototype");
01972     T->pattern = T->prototype + AM.totalnumberofthreads;
01973     t = (WORD *) (T->pattern + AM.totalnumberofthreads);
01974     for ( n = 0; n < AM.totalnumberofthreads; n++ ) {
01975         T->prototype[n] = t;
01976         for ( k = 0; k < i; k++ ) *t++ = OldWork[k];
01977     }
01978     T->pattern[0] = t;
01979     j--; w++;
01980     w[1] += ABS(T->numind)*2;
01981     for ( k = 0; k < FUNHEAD; k++ ) *t++ = *w++;
01982     j -= FUNHEAD;
01983     for ( k = 0; k < ABS(T->numind); k++ ) { *t++ = -SNUMBER; *t++ = 0; j -= 2; }
01984     for ( k = 0; k < j; k++ ) *t++ = *w++;
01985     if ( sparseflag ) T->pattern[0][1] = t - T->pattern[0];
01986     k = t - T->pattern[0];
01987     for ( n = 1; n < AM.totalnumberofthreads; n++ ) {
01988         T->pattern[n] = t; tt = T->pattern[0];
01989         for ( i = 0; i < k; i++ ) *t++ = *tt++;
01990     }
01991 }
01992 #else
01993     T->prototypeSize = (i+j)*sizeof(WORD);
01994     T->prototype = (WORD *)Malloc1(T->prototypeSize, "table prototype");
01995     T->pattern = T->prototype + i;
01996     for ( k = 0; k < i; k++ ) T->prototype[k] = OldWork[k];
01997     t = T->pattern;
01998     j--; w++;
01999     w[1] += ABS(T->numind)*2;
02000     for ( k = 0; k < FUNHEAD; k++ ) *t++ = *w++;
02001     j -= FUNHEAD;
02002     for ( k = 0; k < ABS(T->numind); k++ ) { *t++ = -SNUMBER; *t++ = 0; j -= 2; }
02003     for ( k = 0; k < j; k++ ) *t++ = *w++;
02004     if ( sparseflag ) T->pattern[1] = t - T->pattern;
02005 #endif
02006 /*
02007     At this point we can pop the compilerbuffer.
02008 */
02009     C->Pointer = C->Buffer + oldcpointer;
02010     C->numrhs = oldnumrhs;
02011     C->numlhs = oldnumlhs;
02012 /*
02013     Now check whether wildcards get converted to dollars (for PARALLEL)
02014     We give a warning!
02015 */
02016 #ifdef WITHPTHREADS
02017     t = T->prototype[0];
02018 #else
02019     t = T->prototype;
02020 #endif
02021     tt = t + t[1]; t += SUBEXPSIZE;
02022     while ( t < tt ) {
02023         if ( *t == LOADDOLLAR ) {
02024             Warning("The use of $-variable assignments in tables disables parallel\
02025 execution for the whole program.");
02026             AM.hparallelflag |= NOPARALLEL_TBLDOLLAR;
02027             AC.mparallelflag |= NOPARALLEL_TBLDOLLAR;
02028             AddPotModdollar(t[2]);
02029         }
02030         t += t[1];
02031     }
02032 FinishUp:;
02033     AT.WorkPointer = OldWork - 1;
02034     AC.cbufnum = oldcbufnum;
02035 FinishUp2:;
02036     if ( T->sparse ) ClearTableTree(T);
02037     if ( ( sparseflag & 2 ) != 0 ) {
02038         if ( T->sparse == 0 ) { SpareTable(T); }
02039     }
02040     return(error);
02041 }
02042
02043 /*
02044 #] DoTable :
02045 #[ CoTable :
02046 */
02047
02048 int CoTable(UBYTE *s)
02049 {
02050     return(DoTable(s,2));
02051 }
02052
02053 /*
02054 #] CoTable :
02055 #[ CoNTable :
02056 */

```

```

02057
02058 int CoNTable(UBYTE *s)
02059 {
02060     return(DoTable(s,0));
02061 }
02062
02063 /*
02064     #] CoNTable :
02065     #[ CoCTable :
02066 */
02067
02068 int CoCTable(UBYTE *s)
02069 {
02070     return(DoTable(s,1));
02071 }
02072
02073 /*
02074     #] CoCTable :
02075     #[ EmptyTable :
02076 */
02077
02078 void EmptyTable(TABLES T)
02079 {
02080     int j;
02081     if ( T->sparse ) ClearTableTree(T);
02082     if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
02083     T->boomlijst = 0;
02084     for ( j = 0; j < T->buffersfill; j++ ) { /* was <= */
02085         finishcbuf(T->buffers[j]);
02086     }
02087     if ( T->buffers ) M_free(T->buffers,"Table buffers");
02088     finishcbuf(T->bufnum);
02089     T->bufnum = inicbufs();
02090     T->bufferssize = 8;
02091     T->buffers = (WORD *)Malloc1(sizeof(WORD)*T->bufferssize,"Table buffers");
02092     T->buffersfill = 0;
02093     T->buffers[T->buffersfill++] = T->bufnum;
02094     T->defined = T->mdefined = 0; T->flags = 0;
02095     T->numtree = 0; T->rootnum = 0; T->MaxTreeSize = 0;
02096     T->sparse = 0; T->reserved = 0;
02097     if ( T->spare ) {
02098         TABLES TT = T->spare;
02099         if ( TT->mm ) M_free(TT->mm,"tableminmax");
02100         if ( TT->flags ) M_free(TT->flags,"tableflags");
02101         if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
02102         for ( j = 0; j < TT->buffersfill; j++ ) {
02103             finishcbuf(TT->buffers[j]);
02104         }
02105         if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
02106         if ( TT->buffers ) M_free(TT->buffers,"Table buffers");
02107         M_free(TT,"table");
02108         SpareTable(T);
02109     }
02110     else {
02111         WORD *w = T->tablepointers;
02112         j = T->totind;
02113         for ( j = TABLEEXTENSION*T->totind; j > 0; j-- ) *w++ = -1; /* means: undefined */
02114     }
02115 }
02116
02117 /*
02118     #] EmptyTable :
02119     #[ AddSet :
02120 */
02121
02122 int AddSet(UBYTE *name, WORD dim)
02123 {
02124     int nodenum, numset = AC.SetList.num;
02125     SETS set = (SETS)FromVarList(&AC.SetList);
02126     UBYTE *s;
02127     if ( name ) {
02128         set->name = AddName(AC.varnames,name,CSET,numset,&nodenum);
02129         s = name;
02130         while ( *s ) s++;
02131         set->namesize = (s-name)+1;
02132         set->node = nodenum;
02133     }
02134     else {
02135         set->name = 0;
02136         set->namesize = 0;
02137         set->node = -1;
02138     }
02139     set->first =
02140     set->last = AC.SetElementList.num; /* set has no elements yet */
02141     set->type = -1; /* undefined as of yet */
02142     set->dimension = dim;
02143     set->flags = 0;

```

```

02144     return(numset);
02145 }
02146
02147 /*
02148 #] AddSet :
02149 #[ DoElements :
02150
02151 Remark (25-mar-2011): If the dimension has been set (dim != MAXPOSITIVE)
02152 we want to test dimensions. Numbers count as dimension zero?
02153 */
02154
02155 int DoElements(UBYTE *s, SETS set, UBYTE *cname)
02156 {
02157     int type, error = 0, x, sgn, i;
02158     WORD numset, *e;
02159     UBYTE c, *cname;
02160     while ( *s ) {
02161         if ( *s == ',' ) { s++; continue; }
02162         sgn = 0;
02163         while ( *s == '-' || *s == '+' ) {
02164             if ( *s == '-' ) sgn ^= 1;
02165             s++;
02166         }
02167         cname = s;
02168         if ( FG.cTable[*s] == 0 || *s == '_' || *s == '[' ) {
02169             if ( ( s = SkipAName(s) ) == 0 ) {
02170                 MesPrint("&Illegal name in set definition");
02171                 return(1);
02172             }
02173             c = *s; *s = 0;
02174             if ( ( ( type = GetName(AC.exprnames,cname,&numset,NOAUTO) ) == NAMENOTFOUND )
02175                 && ( ( type = GetOName(AC.varnames,cname,&numset,WITHAUTO) ) == NAMENOTFOUND ) ) {
02176                 DUBIOUSV dv;
02177                 int nodenum;
02178                 MesPrint("&%s has not been declared",cname);
02179             /*
02180              We enter a 'dubious' declaration to cut down on errors
02181             */
02182             numset = AC.DubiousList.num;
02183             dv = (DUBIOUSV)FromVarList(&AC.DubiousList);
02184             dv->name = AddName(AC.varnames,cname,CDUBIOUS,numset,&nodenum);
02185             dv->node = nodenum;
02186             set->type = type = CDUBIOUS;
02187             set->dimension = 0;
02188             error = 1;
02189         }
02190         if ( set->type == -1 ) {
02191             if ( type == CSYMBOL ) {
02192                 for ( i = set->first; i < set->last; i++ ) {
02193                     SetElements[i] += 2*MAXPOWER;
02194                 }
02195             }
02196             set->type = type;
02197         }
02198         if ( set->type != type && set->type != CDUBIOUS
02199             && type != CDUBIOUS ) {
02200             if ( set->type != CNUMBER || ( type != CSYMBOL
02201                 && type != CINDEXT ) ) {
02202                 MesPrint(
02203                     "&%s has not the same type as the other members of the set"
02204                     ,cname);
02205                 error = 1;
02206                 set->type = CDUBIOUS;
02207             }
02208             else {
02209                 if ( type == CSYMBOL ) {
02210                     for ( i = set->first; i < set->last; i++ ) {
02211                         SetElements[i] += 2*MAXPOWER;
02212                     }
02213                 }
02214                 set->type = type;
02215             }
02216         }
02217         if ( set->dimension != MAXPOSITIVE ) { /* Dimension check */
02218             switch ( set->type ) {
02219                 case CSYMBOL:
02220                     if ( symbols[numset].dimension != set->dimension ) {
02221                         MesPrint("&Dimension check failed in set %s, symbol %s",
02222                             VARNAME(Sets,(set-Sets)),
02223                             VARNAME(symbols,numset));
02224                         error = 1;
02225                         set->dimension = MAXPOSITIVE;
02226                     }
02227                     break;
02228                 case CVECTOR:
02229                     if ( vectors[numset-AM.OffsetVector].dimension != set->dimension ) {
02230                         MesPrint("&Dimension check failed in set %s, vector %s",

```

```

02231             VARNAME(Sets, (set-Sets)),
02232             VARNAME(vectors, (numset-AM.OffsetVector)));
02233         error = 1;
02234         set->dimension = MAXPOSITIVE;
02235     }
02236     break;
02237     case CFUNCTION:
02238         if ( functions[numset-FUNCTION].dimension != set->dimension ) {
02239             MesPrint("&Dimension check failed in set %s, function %s",
02240                     VARNAME(Sets, (set-Sets)),
02241                     VARNAME(functions, (numset-FUNCTION)));
02242             error = 1;
02243         }
02244         break;
02245         set->dimension = MAXPOSITIVE;
02246     }
02247 }
02248 if ( sgn ) {
02249     if ( type != CVECTOR ) {
02250         MesPrint("&Illegal use of - sign in set. Can use only with vector or number");
02251         error = 1;
02252     }
02253 }
02254 /*
02255     numset = AM.OffsetVector - numset;
02256     numset |= SPECMASK;
02257     numset = AM.OffsetVector - numset;
02258 */
02259     numset -= WILDMASK;
02260 }
02261 *s = c;
02262 if ( name == 0 && *s == '?' ) {
02263     s++;
02264     switch ( set->type ) {
02265         case CSYMBOL:
02266             numset = -numset; break;
02267         case CVECTOR:
02268             numset += WILD OFFSET; break;
02269         case CINDEX:
02270             numset |= WILDMASK; break;
02271         case CFUNCTION:
02272             numset |= WILDMASK; break;
02273     }
02274     AC.wildflag = 1;
02275 }
02276 /*
02277     Now add the element to the set.
02278 */
02279     e = (WORD *)FromVarList(&AC.SetElementList);
02280     *e = numset;
02281     (set->last)++;
02282 }
02283 else if ( FG.cTable[*s] == 1 ) {
02284     ParseNumber(x,s)
02285     if ( sgn ) x = -x;
02286     if ( x >= MAXPOWER || x <= -MAXPOWER ||
02287         ( set->type == CINDEX && ( x < 0 || x >= AM.OffsetIndex ) ) ) {
02288         MesPrint("&Illegal value for set element: %d",x);
02289         if ( AC.firstconstindex ) {
02290             MesPrint("&0 <= Fixed indices < ConstIndex(which is %d)",
02291                     AM.OffsetIndex-1);
02292             MesPrint("&For setting ConstIndex, read the chapter on the setup file");
02293             AC.firstconstindex = 0;
02294         }
02295         error = 1;
02296         x = 0;
02297     }
02298 }
02299 /*
02300     Check what is allowed with the type.
02301 */
02302 if ( set->type == -1 ) {
02303     if ( x < 0 || x >= AM.OffsetIndex ) {
02304         for ( i = set->first; i < set->last; i++ ) {
02305             SetElements[i] += 2*MAXPOWER;
02306         }
02307         set->type = CSYMBOL;
02308     }
02309     else set->type = CNUMBER;
02310 }
02311 else if ( set->type == CDUBIOUS ) {}
02312 else if ( set->type == CNUMBER && x < 0 ) {
02313     for ( i = set->first; i < set->last; i++ ) {
02314         SetElements[i] += 2*MAXPOWER;
02315     }
02316     set->type = CSYMBOL;
02317 }
02318 else if ( set->type != CSYMBOL && ( x < 0 ||
02319     ( set->type != CINDEX && set->type != CNUMBER ) ) ) {

```

```

02318         MesPrint("&Illegal mixture of element types in set");
02319         error = 1;
02320         set->type = CDUBIOUS;
02321     }
02322     /*
02323         Allocate an element
02324     */
02325     e = (WORD *)FromVarList(&AC.SetElementList);
02326     (set->last)++;
02327     if ( set->type == CSYMBOL ) *e = x + 2*MAXPOWER;
02328     /* else if ( set->type == CINDEX ) *e = x; */
02329     else *e = x;
02330 }
02331 else {
02332     MesPrint("&Illegal object in list of set elements");
02333     return(1);
02334 }
02335 }
02336 if ( error == 0 && ( ( set->flags & ORDEREDSET ) == ORDEREDSET ) ) {
02337     /*
02338         The set->last-set->first list of numbers must be sorted.
02339         Because we plan here potentially thousands of elements we use
02340         a simple version of splitmerge. In ordered sets we can search
02341         later with a binary search.
02342     */
02343     SimpleSplitMerge(SetElements+set->first,set->last-set->first);
02344 }
02345 return(error);
02346 }
02347
02348 /*
02349     #] DoElements :
02350     #[ CoSet :
02351
02352     Set declarations.
02353 */
02354
02355 int CoSet(UBYTE *s)
02356 {
02357     int type, error = 0, ordered = 0;
02358     UBYTE *name = s, c, *ss;
02359     SETS set;
02360     WORD numberofset, dim = MAXPOSITIVE;
02361     #ifdef WITHFLOAT
02362     /*-----*/
02363     {
02364         WORD numeq = 0;
02365         LONG x;
02366         ss = s;
02367         while ( *ss && *ss != ':' ) { if ( *ss == '=' ) numeq++; ss++; }
02368         if ( *ss == 0 && numeq == 1 ) { /* We have the Set var = value; variety */
02369             while ( FG.cTable[*s] == 0 ) s++;
02370             ss = s; c = *s; *s = 0;
02371             if ( c != '=' ) {
02372 Proper:
02373                 MesPrint("&Proper syntax for value-set is `Set name = value'");
02374                 return(1);
02375             }
02376             x = 0; s++;
02377             while ( *s >= '0' && *s <= '9' ) x = 10*x + (*s++-'0');
02378             if ( *s ) goto Proper;
02379             if ( StrICmp(name,(UBYTE *)"maxweight") == 0 ) {
02380                 AC.tMaxWeight = x; /* Temporary. Made permanent later */
02381             }
02382             else if ( StrICmp(name,(UBYTE *)"defaultprecision") == 0 ) {
02383                 AC.tDefaultPrecision = x; /* Temporary. Made permanent later */
02384             }
02385             else {
02386                 MesPrint("&Illegal subkey in value set: %s",name);
02387                 return(1);
02388             }
02389         }
02390     }
02391     /*-----*/
02392     #endif
02393     if ( ( s = SkipAName(s) ) == 0 ) {
02394 IllForm:MesPrint("&Illegal name for set");
02395         return(1);
02396     }
02397     c = *s; *s = 0;
02398     if ( TestName(name) ) goto IllForm;
02399     if ( ( ( type = GetName(AC.exprnames,name,&numberofset,NOAUTO) ) != NAMENOTFOUND )
02400 || ( ( type = GetName(AC.varnames,name,&numberofset,NOAUTO) ) != NAMENOTFOUND ) ) {
02401         if ( type != CSET ) NameConflict(type,name);
02402     }
02403     else {
02404         MesPrint("&There is already a set with the name %s",name);
02405     }
02406 }

```

```

02405         return(1);
02406     }
02407     if ( c == 0 ) {
02408         numberofset = AddSet(name,0);
02409         set = Sets + numberofset;
02410         return(0); /* empty set */
02411     }
02412     *s = c; ss = s; /* ss marks the end of the name */
02413     if ( *s == '(' ) {
02414         UBYTE *sss, cc;
02415         s++; sss = s; /* Beginning of option */
02416         while ( *s != ',' && *s != ')' && *s ) s++;
02417         cc = *s; *s = 0;
02418         if ( StrICont(sss,(UBYTE *)"ordered") == 0 ) {
02419             ordered = ORDEREDSET;
02420         }
02421         else {
02422             MesPrint("&Error: Illegal option in set definition: %s",sss);
02423             error = 1;
02424         }
02425         *s = cc;
02426         if ( *s != ')' ) {
02427             MesPrint("&Error: Currently only one option allowed in set definition.");
02428             error = 1;
02429             while ( *s && *s != ')' ) s++;
02430         }
02431         s++;
02432     }
02433     if ( *s == '(' ) {
02434         s++;
02435         if ( ( *s == 'd' || *s == 'D' ) && s[1] == '=' ) {
02436             s += 2;
02437             if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
02438                 ParseSignedNumber(dim,s)
02439                 if ( dim < -HALFMAX || dim > HALFMAX ) {
02440                     MesPrint("&Warning: dimension of %s (%d) out of range",
02441                         name,dim);
02442                 }
02443             }
02444             if ( *s != ')' ) goto IllDim;
02445             else s++;
02446         }
02447         else {
02448             IllDim: MesPrint("&Error: Illegal dimension field for set %s",name);
02449             error = 1;
02450             s = SkipField(s,0);
02451         }
02452         while ( *s == ',' ) s++;
02453     }
02454     c = *ss; *ss = 0;
02455     numberofset = AddSet(name,dim);
02456     *ss = c;
02457     set = Sets + numberofset;
02458     set->flags |= ordered;
02459     if ( *s != ':' ) {
02460         MesPrint("&Proper syntax is `Set name:elements'");
02461         return(1);
02462     }
02463     s++;
02464     error = DoElements(s,set,name);
02465     AC.SetList.numtemp = AC.SetList.num;
02466     AC.SetElementList.numtemp = AC.SetElementList.num;
02467     return(error);
02468 }
02469
02470 /*
02471 #] CoSet :
02472 #[ DoTempSet :
02473
02474     Gets a {} set definition and returns a set number if the set is
02475     properly structured. This number refers either to an already
02476     existing set, or to a set that is defined here.
02477     From and to refer to the contents. They exclude the {}.
02478 */
02479
02480 int DoTempSet(UBYTE *from, UBYTE *to)
02481 {
02482     int i, num, j, sgn;
02483     WORD *e, *ep;
02484     UBYTE c;
02485     int setnum = AddSet(0,MAXPOSITIVE);
02486     SETS set = Sets + setnum, setp;
02487     set->name = -1;
02488     set->type = -1;
02489     c = *to; *to = 0;
02490     AC.wildflag = 0;
02491     while ( *from == ',' ) from++;

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02492     if ( *from == '<' || *from == '>' ) {
02493         set->type = CRANGE;
02494         set->first = 3*MAXPOWER;
02495         set->last = -3*MAXPOWER;
02496         while ( *from == '<' || *from == '>' ) {
02497             if ( *from == '<' ) {
02498                 j = 1; from++;
02499                 if ( *from == '=' ) { from++; j++; }
02500             }
02501             else {
02502                 j = -1; from++;
02503                 if ( *from == '=' ) { from++; j--; }
02504             }
02505             sgn = 1;
02506             while ( *from == '-' || *from == '+' ) {
02507                 if ( *from == '-' ) sgn = -sgn;
02508                 from++;
02509             }
02510             ParseNumber(num,from)
02511             if ( *from && *from != ',' ) {
02512                 MesPrint("&Illegal number in ranged set definition");
02513                 return(-1);
02514             }
02515             if ( sgn < 0 ) num = -num;
02516             if ( num >= MAXPOWER || num <= -MAXPOWER ) {
02517                 Warning("Value in ranged set too big. Adjusted to infinity.");
02518                 if ( num > 0 ) num = 3*MAXPOWER;
02519                 else num = -3*MAXPOWER;
02520             }
02521             else if ( j == 2 ) num += 2*MAXPOWER;
02522             else if ( j == -2 ) num -= 2*MAXPOWER;
02523             if ( j > 0 ) set->first = num;
02524             else set->last = num;
02525             while ( *from == ',' ) from++;
02526         }
02527         if ( *from ) {
02528             MesPrint("&Definition of ranged set contains illegal objects");
02529             return(-1);
02530         }
02531     }
02532     else if ( DoElements(from,set,(UBYTE *)0) != 0 ) {
02533         AC.SetElementList.num = set->first;
02534         AC.SetList.num--; *to = c;
02535         return(-1);
02536     }
02537     *to = c;
02538     /*
02539     Now we have to test whether this set exists already.
02540     */
02541     num = set->last - set->first;
02542     for ( setp = Sets, i = 0; i < AC.SetList.num-1; i++, setp++ ) {
02543         if ( num != setp->last - setp->first ) continue;
02544         if ( set->type != setp->type ) continue;
02545         if ( set->type == CRANGE ) {
02546             if ( set->first == setp->first ) return(setp-Sets);
02547         }
02548         else {
02549             e = SetElements + set->first;
02550             ep = SetElements + setp->first;
02551             j = num;
02552             while ( --j >= 0 ) if ( *e++ != *ep++ ) break;
02553             if ( j < 0 ) {
02554                 AC.SetElementList.num = set->first;
02555                 AC.SetList.num--;
02556                 return(setp - Sets);
02557             }
02558         }
02559     }
02560     return(setnum);
02561 }
02562
02563 /*
02564 #] DoTempSet :
02565 #[ CoAuto :
02566
02567 To prepare first:
02568     Use of the proper pointers in the various declaration routines
02569     Proper action in .store and .clear
02570 */
02571
02572 int CoAuto(UBYTE *inp)
02573 {
02574     int retval;
02575
02576     AC.Symbols = &(AC.AutoSymbolList);
02577     AC.Vectors = &(AC.AutoVectorList);
02578     AC.Indices = &(AC.AutoIndexList);

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```

02579     AC.Functions = &(AC.AutoFunctionList);
02580     AC.activenames = &(AC.autonames);
02581     AC.AutoDeclareFlag = WITHAUTO;
02582
02583     while ( *inp == ',' ) inp++;
02584     retval = CompileStatement(inp);
02585
02586     AC.AutoDeclareFlag = 0;
02587     AC.Symbols = &(AC.SymbolList);
02588     AC.Vectors = &(AC.VectorList);
02589     AC.Indices = &(AC.IndexList);
02590     AC.Functions = &(AC.FunctionList);
02591     AC.activenames = &(AC.varnames);
02592     return(retval);
02593 }
02594
02595 /*
02596 #] CoAuto :
02597 #[ AddDollar :
02598
02599     The actual addition. Special routine for additions 'on the fly'
02600 */
02601
02602 int AddDollar(UBYTE *name, WORD type, WORD *start, LONG size)
02603 {
02604     int nodenum, numdollar = AP.DollarList.num;
02605     WORD *s, *t;
02606     DOLLARS dol = (DOLLARS)FromVarList(&AP.DollarList);
02607     dol->name = AddName(AC.dollarnames, name, CDOLLAR, numdollar, &nodenum);
02608     dol->type = type;
02609     dol->node = nodenum;
02610     dol->zero = 0;
02611     dol->numdummies = 0;
02612 #ifdef WITHPTHREADS
02613     dol->pthreadlockread = dummylock;
02614     dol->pthreadlockwrite = dummylock;
02615 #endif
02616     dol->nfactors = 0;
02617     dol->factors = 0;
02618     AddRHS(AM.dbufnum, 1);
02619     AddLHS(AM.dbufnum);
02620     if ( start && size > 0 ) {
02621         dol->size = size;
02622         dol->where =
02623             s = (WORD *)Malloc1((size+1)*sizeof(WORD), "$-variable contents");
02624         t = start;
02625         while ( --size >= 0 ) *s++ = *t++;
02626         *s = 0;
02627     }
02628     else { dol->where = &(AM.dollarzero); dol->size = 0; }
02629     cbuf[AM.dbufnum].rhs[numdollar] = dol->where;
02630     cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
02631     cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
02632
02633     return(numdollar);
02634 }
02635
02636 /*
02637 #] AddDollar :
02638 #[ ReplaceDollar :
02639
02640     Replacements of dollar variables can happen at any time.
02641     For debugging purposes we should have a tracing facility.
02642
02643     Not in use????
02644 */
02645
02646 int ReplaceDollar(WORD number, WORD newtype, WORD *newstart, LONG newsize)
02647 {
02648     int error = 0;
02649     DOLLARS dol = Dollars + number;
02650     WORD *s, *t;
02651     LONG i;
02652     dol->type = newtype;
02653     if ( dol->size == newsize && newsize > 0 && newstart ) {
02654         s = dol->where; t = newstart; i = newsize;
02655         while ( --i >= 0 ) { if ( *s++ != *t++ ) break; }
02656         if ( i < 0 ) return(0);
02657     }
02658     if ( dol->where && dol->where != &(dol->zero) ) {
02659         M_free(dol->where, "dollar->where"); dol->where = &(dol->zero); dol->size = 0;
02660     }
02661     if ( newstart && newsize > 0 ) {
02662         dol->size = newsize;
02663         dol->where =
02664             s = (WORD *)Malloc1((newsize+1)*sizeof(WORD), "$-variable contents");
02665         t = newstart; i = newsize;

```

```

02666         while ( --i >= 0 ) *s++ = *t++;
02667         *s = 0;
02668     }
02669     return(error);
02670 }
02671
02672 /*
02673  #] ReplaceDollar :
02674  #[ AddDubious :
02675
02676     This adds a variable of which we do not know the proper type.
02677 */
02678
02679 int AddDubious(UBYTE *name)
02680 {
02681     int nodenum, numdubious = AC.DubiousList.num;
02682     DUBIOUSV dub = (DUBIOUSV)FromVarList (&AC.DubiousList);
02683     dub->name = AddName(AC.varnames,name,CDUBIOUS,numdubious,&nodenum);
02684     dub->node = nodenum;
02685     return(numdubious);
02686 }
02687
02688 /*
02689  #] AddDubious :
02690  #[ MakeDubious :
02691 */
02692
02693 int MakeDubious(NAMETREE *nametree, UBYTE *name, WORD *number)
02694 {
02695     NAMENODE *n;
02696     int node, newnode, i;
02697     if ( nametree->namenode == 0 ) return(-1);
02698     newnode = nametree->headnode;
02699     do {
02700         node = newnode;
02701         n = nametree->namenode+node;
02702         if ( ( i = StrCmp(name,nametree->namebuffer+n->name) ) < 0 )
02703             newnode = n->left;
02704         else if ( i > 0 ) newnode = n->right;
02705         else {
02706             if ( n->type != CDUBIOUS ) {
02707                 int numdubious = AC.DubiousList.num;
02708                 FUNCTIONS dub = (FUNCTIONS)FromVarList (&AC.DubiousList);
02709                 dub->name = n->name;
02710                 n->number = numdubious;
02711             }
02712             *number = n->number;
02713             return(CDUBIOUS);
02714         }
02715     } while ( newnode >= 0 );
02716     return(-1);
02717 }
02718
02719 /*
02720  #] MakeDubious :
02721  #[ NameConflict :
02722 */
02723
02724 static char *nametype[] = { "symbol", "index", "vector", "function",
02725     "set", "expression" };
02726 static char *plural[] = { "", "n", "", "", "", "n" };
02727
02728 int NameConflict(int type, UBYTE *name)
02729 {
02730     if ( type == NAMENOTFOUND ) {
02731         MesPrint("%s has not been declared",name);
02732     }
02733     else if ( type != CDUBIOUS )
02734         MesPrint("%s has been declared as a%s %s already",
02735             name,plural[type],nametype[type]);
02736     return(1);
02737 }
02738
02739 /*
02740  #] NameConflict :
02741  #[ AddExpression :
02742 */
02743
02744 int AddExpression(UBYTE *name, int x, int y)
02745 {
02746     int nodenum, numexpr = AC.ExpressionList.num;
02747     EXPRESSIONS expr = (EXPRESSIONS)FromVarList (&AC.ExpressionList);
02748     UBYTE *s;
02749     expr->status = x;
02750     expr->printflag = y;
02751     PUTZERO(expr->onfile);
02752     PUTZERO(expr->size);

```

```

02753     expr->renum = 0;
02754     expr->renumlists = 0;
02755     expr->hidelevel = 0;
02756     expr->inmem = 0;
02757     expr->bracketinfo = expr->newbracketinfo = 0;
02758     if ( name ) {
02759         expr->name = AddName(AC.exprnames, name, CEXPRESSION, numexpr, &nodenum);
02760         expr->node = nodenum;
02761         expr->replace = NEWLYDEFINEDEXPRESSION ;
02762         s = name;
02763         while ( *s ) s++;
02764         expr->namesize = (s-name)+1;
02765     }
02766     else {
02767         expr->replace = REDEFINEDEXPRESSION;
02768         expr->name = AC.TransEname;
02769         expr->node = -1;
02770         expr->namesize = 0;
02771     }
02772     expr->vflags = 0;
02773     expr->numdummies = 0;
02774     expr->numfactors = 0;
02775     #ifdef PARALLELCODE
02776     expr->partodo = 0;
02777 #endif
02778     expr->uflags = 0;
02779     return(numexpr);
02780 }
02781
02782 /*
02783  #] AddExpression :
02784  #[ GetLabel :
02785  */
02786
02787 int GetLabel(UBYTE *name)
02788 {
02789     int i;
02790     LONG newnum;
02791     UBYTE **NewLabelNames;
02792     int *NewLabel;
02793     for ( i = 0; i < AC.NumLabels; i++ ) {
02794         if ( StrCmp(name, AC.LabelNames[i]) == 0 ) return(i);
02795     }
02796     if ( AC.NumLabels >= AC.MaxLabels ) {
02797         newnum = 2*AC.MaxLabels;
02798         if ( newnum == 0 ) newnum = 10;
02799         if ( newnum > 32765 ) newnum = 32765;
02800         if ( newnum == AC.MaxLabels ) {
02801             MesPrint("&More than 32765 labels in one module. Please simplify.");
02802             Terminate(-1);
02803         }
02804         NewLabelNames = (UBYTE **)Malloc1((sizeof(UBYTE *)+sizeof(int))
02805             *newnum, "Labels");
02806         NewLabel = (int *) (NewLabelNames+newnum);
02807         for ( i = 0; i < AC.MaxLabels; i++ ) {
02808             NewLabelNames[i] = AC.LabelNames[i];
02809             NewLabel[i] = AC.Labels[i];
02810         }
02811         if ( AC.LabelNames ) M_free(AC.LabelNames, "Labels");
02812         AC.LabelNames = NewLabelNames;
02813         AC.Labels = NewLabel;
02814         AC.MaxLabels = newnum;
02815     }
02816     i = AC.NumLabels++;
02817     AC.LabelNames[i] = strDup1(name, "Labels");
02818     AC.Labels[i] = -1;
02819     return(i);
02820 }
02821
02822 /*
02823  #] GetLabel :
02824  #[ ResetVariables :
02825
02826  Resets the variables.
02827  par = 0  The list of temporary sets (after each .sort)
02828  par = 1  The list of local variables (after each .store)
02829  par = 2  All variables (after each .clear)
02830  */
02831
02832 void ResetVariables(int par)
02833 {
02834     int i, j;
02835     TABLES T;
02836     switch ( par ) {
02837     case 0 : /* Only the sets without a name */
02838         AC.SetList.num = AC.SetList.numtemp;
02839         AC.SetElementList.num = AC.SetElementList.numtemp;

```

```

02840         break;
02841     case 2 :
02842         for ( i = AC.SymbolList.numclear; i < AC.SymbolList.num; i++ )
02843             AC.varnames->namenode[symbols[i].node].type = CDELETE;
02844         AC.SymbolList.num = AC.SymbolList.numglobal = AC.SymbolList.numclear;
02845         for ( i = AC.VectorList.numclear; i < AC.VectorList.num; i++ )
02846             AC.varnames->namenode[vectors[i].node].type = CDELETE;
02847         AC.VectorList.num = AC.VectorList.numglobal = AC.VectorList.numclear;
02848         for ( i = AC.IndexList.numclear; i < AC.IndexList.num; i++ )
02849             AC.varnames->namenode[indices[i].node].type = CDELETE;
02850         AC.IndexList.num = AC.IndexList.numglobal = AC.IndexList.numclear;
02851         for ( i = AC.FunctionList.numclear; i < AC.FunctionList.num; i++ ) {
02852             AC.varnames->namenode[functions[i].node].type = CDELETE;
02853             if ( ( T = functions[i].tabl ) != 0 ) {
02854                 if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
02855                 if ( T->prototype ) M_free(T->prototype,"tableprototype");
02856                 if ( T->mm ) M_free(T->mm,"tableminmax");
02857                 if ( T->flags ) M_free(T->flags,"tableflags");
02858                 if ( T->argtail ) M_free(T->argtail,"table arguments");
02859                 if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
02860                 for ( j = 0; j < T->buffersfill; j++ ) { /* was <= */
02861                     finishcbuf(T->buffers[j]);
02862                 }
02863                 /*[07apr2004 mt]:*/ /*memory leak*/
02864                 if ( T->buffers ) M_free(T->buffers,"Table buffers");
02865                 /*:[07apr2004 mt]*/
02866                 finishcbuf(T->bufnum);
02867                 if ( T->spare ) {
02868                     TABLES TT = T->spare;
02869                     if ( TT->mm ) M_free(TT->mm,"tableminmax");
02870                     if ( TT->flags ) M_free(TT->flags,"tableflags");
02871                     if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
02872                     for ( j = 0; j < TT->buffersfill; j++ ) { /* was <= */
02873                         finishcbuf(TT->buffers[j]);
02874                     }
02875                     if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
02876                     /*[07apr2004 mt]:*/ /*memory leak*/
02877                     if ( TT->buffers ) M_free(TT->buffers,"Table buffers");
02878                     /*:[07apr2004 mt]*/
02879                     M_free(TT,"table");
02880                 }
02881                 M_free(T,"table");
02882             }
02883         }
02884         AC.FunctionList.num = AC.FunctionList.numglobal = AC.FunctionList.numclear;
02885         for ( i = AC.SetList.numclear; i < AC.SetList.num; i++ ) {
02886             if ( Sets[i].node >= 0 )
02887                 AC.varnames->namenode[Sets[i].node].type = CDELETE;
02888         }
02889         AC.SetList.numtemp = AC.SetList.num = AC.SetList.numglobal = AC.SetList.numclear;
02890         for ( i = AC.DubiousList.numclear; i < AC.DubiousList.num; i++ )
02891             AC.varnames->namenode[Dubious[i].node].type = CDELETE;
02892         AC.DubiousList.num = AC.DubiousList.numglobal = AC.DubiousList.numclear;
02893         AC.SetElementList.numtemp = AC.SetElementList.num =
02894             AC.SetElementList.numglobal = AC.SetElementList.numclear;
02895         CompactifyTree(AC.varnames,VARNAMES);
02896         AC.varnames->namefill = AC.varnames->globalnamefill = AC.varnames->clearnamefill;
02897         AC.varnames->nodefill = AC.varnames->globalnodefill = AC.varnames->clearnodefill;
02898
02899         for ( i = AC.AutoSymbolList.numclear; i < AC.AutoSymbolList.num; i++ )
02900             AC.autonames->namenode[
02901                 ((SYMBOLS)(AC.AutoSymbolList.lijt))[i].node].type = CDELETE;
02902         AC.AutoSymbolList.num = AC.AutoSymbolList.numglobal
02903             = AC.AutoSymbolList.numclear;
02904         for ( i = AC.AutoVectorList.numclear; i < AC.AutoVectorList.num; i++ )
02905             AC.autonames->namenode[
02906                 ((VECTORS)(AC.AutoVectorList.lijt))[i].node].type = CDELETE;
02907         AC.AutoVectorList.num = AC.AutoVectorList.numglobal
02908             = AC.AutoVectorList.numclear;
02909         for ( i = AC.AutoIndexList.numclear; i < AC.AutoIndexList.num; i++ )
02910             AC.autonames->namenode[
02911                 ((INDICES)(AC.AutoIndexList.lijt))[i].node].type = CDELETE;
02912         AC.AutoIndexList.num = AC.AutoIndexList.numglobal
02913             = AC.AutoIndexList.numclear;
02914         for ( i = AC.AutoFunctionList.numclear; i < AC.AutoFunctionList.num; i++ ) {
02915             AC.autonames->namenode[
02916                 ((FUNCTIONS)(AC.AutoFunctionList.lijt))[i].node].type = CDELETE;
02917             if ( ( T = ((FUNCTIONS)(AC.AutoFunctionList.lijt))[i].tabl ) != 0 ) {
02918                 if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
02919                 if ( T->prototype ) M_free(T->prototype,"tableprototype");
02920                 if ( T->mm ) M_free(T->mm,"tableminmax");
02921                 if ( T->flags ) M_free(T->flags,"tableflags");
02922                 if ( T->argtail ) M_free(T->argtail,"table arguments");
02923                 if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
02924                 for ( j = 0; j < T->buffersfill; j++ ) { /* was <= */
02925                     finishcbuf(T->buffers[j]);
02926                 }
02927             }
02928         }

```

```

02927         if ( T->spare ) {
02928             TABLES TT = T->spare;
02929             if ( TT->mm ) M_free(TT->mm,"tableminmax");
02930             if ( TT->flags ) M_free(TT->flags,"tableflags");
02931             if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
02932             for ( j = 0; j < TT->buffersfill; j++ ) { /* was <= */
02933                 finishcbuf(TT->buffers[j]);
02934             }
02935             if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
02936             M_free(TT,"table");
02937         }
02938         M_free(T,"table");
02939     }
02940 }
02941 AC.AutoFunctionList.num = AC.AutoFunctionList.numglobal
02942                         = AC.AutoFunctionList.numclear;
02943 CompactifyTree(AC.autonames,AUTONAMES);
02944 AC.autonames->namefill = AC.autonames->globalnamefill
02945                       = AC.autonames->clearnamefill;
02946 AC.autonames->nodefill = AC.autonames->globalnodefill
02947                       = AC.autonames->clearnodefill;
02948 ReleaseTB();
02949 break;
02950 case 1 :
02951     for ( i = AC.SymbolList.numglobal; i < AC.SymbolList.num; i++ )
02952         AC.varnames->namenode[symbols[i].node].type = CDELETE;
02953     AC.SymbolList.num = AC.SymbolList.numglobal;
02954     for ( i = AC.VectorList.numglobal; i < AC.VectorList.num; i++ )
02955         AC.varnames->namenode[vectors[i].node].type = CDELETE;
02956     AC.VectorList.num = AC.VectorList.numglobal;
02957     for ( i = AC.IndexList.numglobal; i < AC.IndexList.num; i++ )
02958         AC.varnames->namenode[indices[i].node].type = CDELETE;
02959     AC.IndexList.num = AC.IndexList.numglobal;
02960     for ( i = AC.FunctionList.numglobal; i < AC.FunctionList.num; i++ ) {
02961         AC.varnames->namenode[functions[i].node].type = CDELETE;
02962         if ( ( T = functions[i].tabl ) != 0 ) {
02963             if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
02964             if ( T->prototype ) M_free(T->prototype,"tableprototype");
02965             if ( T->mm ) M_free(T->mm,"tableminmax");
02966             if ( T->flags ) M_free(T->flags,"tableflags");
02967             if ( T->argtail ) M_free(T->argtail,"table arguments");
02968             if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
02969             for ( j = 0; j < T->buffersfill; j++ ) { /* was <= */
02970                 finishcbuf(T->buffers[j]);
02971             }
02972             /*[07apr2004 mt]:*/ /*memory leak*/
02973             if ( T->buffers ) M_free(T->buffers,"Table buffers");
02974             /*:[07apr2004 mt]*/
02975             finishcbuf(T->bufnum);
02976             if ( T->spare ) {
02977                 TABLES TT = T->spare;
02978                 if ( TT->mm ) M_free(TT->mm,"tableminmax");
02979                 if ( TT->flags ) M_free(TT->flags,"tableflags");
02980                 if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
02981                 for ( j = 0; j < TT->buffersfill; j++ ) { /* was <= */
02982                     finishcbuf(TT->buffers[j]);
02983                 }
02984                 if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
02985                 /*[07apr2004 mt]:*/ /*memory leak*/
02986                 if ( TT->buffers ) M_free(TT->buffers,"Table buffers");
02987                 /*:[07apr2004 mt]*/
02988                 M_free(TT,"table");
02989             }
02990             M_free(T,"table");
02991         }
02992     }
02993 #ifdef TABLECLEANUP
02994 {
02995     int j;
02996     WORD *tp;
02997     for ( i = 0; i < AC.FunctionList.numglobal; i++ ) {
02998         /*
02999         Now, if the table definition is from after the .global
03000         while the function is from before, there is a problem.
03001         This could be resolved by defining CTable (=Table), Ntable
03002         and do away with the previous function definition.
03003         */
03004         if ( ( T = functions[i].tabl ) != 0 ) {
03005             /*
03006             First restore overwritten definitions.
03007             */
03008             if ( T->sparse ) {
03009                 T->totind = T->mdefined;
03010                 for ( j = 0, tp = T->tablepointers; j < T->totind; j++ ) {
03011                     tp += ABS(T->numind);
03012 #if TABLEEXTENSION == 2
03013                     tp[0] = tp[1];

```

```

03014 #else
03015         tp[0] = tp[2];
03016         tp[1] = tp[3];
03017         tp[4] = tp[5];
03018 #endif
03019         tp += TABLEEXTENSION;
03020     }
03021     RedoTableTree(T,T->totind);
03022     if ( T->spare ) {
03023         TABLES TT = T->spare;
03024         TT->totind = TT->mdefined;
03025         for ( j = 0, tp = TT->tablepointers; j < TT->totind; j++ ) {
03026             tp += ABS(TT->numind);
03027             #if TABLEEXTENSION == 2
03028                 tp[0] = tp[1];
03029             #else
03030                 tp[0] = tp[2];
03031                 tp[1] = tp[3];
03032                 tp[4] = tp[5];
03033             #endif
03034             tp += TABLEEXTENSION;
03035         }
03036         RedoTableTree(TT,TT->totind);
03037         cbuf[TT->bufnum].numlhs = cbuf[TT->bufnum].mnumlhs;
03038         cbuf[TT->bufnum].numrhs = cbuf[TT->bufnum].mnumrhs;
03039     }
03040 }
03041 else {
03042     for ( j = 0, tp = T->tablepointers; j < T->totind; j++ ) {
03043         #if TABLEEXTENSION == 2
03044             tp[0] = tp[1];
03045         #else
03046             tp[0] = tp[2];
03047             tp[1] = tp[3];
03048             tp[4] = tp[5];
03049         #endif
03050     }
03051     T->defined = T->mdefined;
03052 }
03053 cbuf[T->bufnum].numlhs = cbuf[T->bufnum].mnumlhs;
03054 cbuf[T->bufnum].numrhs = cbuf[T->bufnum].mnumrhs;
03055 }
03056 }
03057 }
03058 #endif
03059 AC.FunctionList.num = AC.FunctionList.numglobal;
03060 for ( i = AC.SetList.numglobal; i < AC.SetList.num; i++ ) {
03061     if ( Sets[i].node >= 0 )
03062         AC.varnames->namenode[Sets[i].node].type = CDELETE;
03063 }
03064 AC.SetList.numtemp = AC.SetList.num = AC.SetList.numglobal;
03065 for ( i = AC.DubiousList.numglobal; i < AC.DubiousList.num; i++ )
03066     AC.varnames->namenode[Dubious[i].node].type = CDELETE;
03067 AC.DubiousList.num = AC.DubiousList.numglobal;
03068 AC.SetElementList.numtemp = AC.SetElementList.num =
03069     AC.SetElementList.numglobal;
03070 CompactifyTree(AC.varnames,VARNAMES);
03071 AC.varnames->namefill = AC.varnames->globalnamefill;
03072 AC.varnames->nodefill = AC.varnames->globalnodefill;
03073
03074 for ( i = AC.AutoSymbolList.numglobal; i < AC.AutoSymbolList.num; i++ )
03075     AC.autonames->namenode[
03076         ((SYMBOLS)(AC.AutoSymbolList.lijst))[i].node].type = CDELETE;
03077 AC.AutoSymbolList.num = AC.AutoSymbolList.numglobal;
03078 for ( i = AC.AutoVectorList.numglobal; i < AC.AutoVectorList.num; i++ )
03079     AC.autonames->namenode[
03080         ((VECTORS)(AC.AutoVectorList.lijst))[i].node].type = CDELETE;
03081 AC.AutoVectorList.num = AC.AutoVectorList.numglobal;
03082 for ( i = AC.AutoIndexList.numglobal; i < AC.AutoIndexList.num; i++ )
03083     AC.autonames->namenode[
03084         ((INDICES)(AC.AutoIndexList.lijst))[i].node].type = CDELETE;
03085 AC.AutoIndexList.num = AC.AutoIndexList.numglobal;
03086 for ( i = AC.AutoFunctionList.numglobal; i < AC.AutoFunctionList.num; i++ ) {
03087     AC.autonames->namenode[
03088         ((FUNCTIONS)(AC.AutoFunctionList.lijst))[i].node].type = CDELETE;
03089     if ( ( T = ((FUNCTIONS)(AC.AutoFunctionList.lijst))[i].tabl ) != 0 ) {
03090         if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
03091         if ( T->prototype ) M_free(T->prototype,"tableprototype");
03092         if ( T->mm ) M_free(T->mm,"tableminmax");
03093         if ( T->flags ) M_free(T->flags,"tableflags");
03094         if ( T->argtail ) M_free(T->argtail,"table arguments");
03095         if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
03096         for ( j = 0; j < T->buffersfill; j++ ) { /* was <= */
03097             finishcbuf(T->buffers[j]);
03098         }
03099         if ( T->spare ) {
03100             TABLES TT = T->spare;

```

```

03101         if ( TT->mm ) M_free(TT->mm,"tableminmax");
03102         if ( TT->flags ) M_free(TT->flags,"tableflags");
03103         if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
03104         for ( j = 0; j < TT->buffersfill; j++ ) { /* was <= */
03105             finishcbuf(TT->buffers[j]);
03106         }
03107         if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
03108         M_free(TT,"table");
03109     }
03110     M_free(T,"table");
03111 }
03112 }
03113 AC.AutoFunctionList.num = AC.AutoFunctionList.numglobal;
03114
03115 CompactifyTree(AC.autonames,AUTONAMES);
03116
03117 AC.autonames->namefill = AC.autonames->globalnamefill;
03118 AC.autonames->nodefill = AC.autonames->globalnodefill;
03119 break;
03120 }
03121 }
03122
03123 /*
03124 #] ResetVariables :
03125 #[ RemoveDollars :
03126 */
03127
03128 void RemoveDollars(void)
03129 {
03130     DOLLARS d;
03131     CBUF *C = cbuf + AM.dbufnum;
03132     int numdollar = AP.DollarList.num;
03133     if ( numdollar > 0 ) {
03134         while ( numdollar > AM.gcNumDollars ) {
03135             numdollar--;
03136             d = Dollars + numdollar;
03137             if ( d->where && d->where != &(d->zero) && d->where != &(AM.dollarzero) ) {
03138                 M_free(d->where,"dollar->where"); d->where = &(d->zero); d->size = 0;
03139             }
03140             AC.dollarnames->namenode[d->node].type = CDELETE;
03141         }
03142         AP.DollarList.num = AM.gcNumDollars;
03143         CompactifyTree(AC.dollarnames,DOLLARNAMES);
03144
03145         C->numrhs = C->mnumrhs;
03146         C->numlhs = C->mnumlhs;
03147     }
03148 }
03149
03150 /*
03151 #] RemoveDollars :
03152 #[ Globalize :
03153 */
03154
03155 void Globalize(int par)
03156 {
03157     int i, j;
03158     WORD *tp;
03159     if ( par == 1 ) {
03160         AC.SymbolList.numclear = AC.SymbolList.num;
03161         AC.VectorList.numclear = AC.VectorList.num;
03162         AC.IndexList.numclear = AC.IndexList.num;
03163         AC.FunctionList.numclear = AC.FunctionList.num;
03164         AC.SetList.numclear = AC.SetList.num;
03165         AC.DubiousList.numclear = AC.DubiousList.num;
03166         AC.SetElementList.numclear = AC.SetElementList.num;
03167         AC.varnames->clearnamefill = AC.varnames->namefill;
03168         AC.varnames->clearnodefill = AC.varnames->nodefill;
03169
03170         AC.AutoSymbolList.numclear = AC.AutoSymbolList.num;
03171         AC.AutoVectorList.numclear = AC.AutoVectorList.num;
03172         AC.AutoIndexList.numclear = AC.AutoIndexList.num;
03173         AC.AutoFunctionList.numclear = AC.AutoFunctionList.num;
03174         AC.autonames->clearnamefill = AC.autonames->namefill;
03175         AC.autonames->clearnodefill = AC.autonames->nodefill;
03176     }
03177     /* for ( i = AC.FunctionList.numglobal; i < AC.FunctionList.num; i++ ) { */
03178     for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
03179         /*
03180             We need here not only the not-yet-global functions. The already
03181             global ones may have obtained extra elements.
03182         */
03183         if ( functions[i].tabl ) {
03184             TABLES T = functions[i].tabl;
03185             if ( T->sparse ) {
03186                 T->mdefined = T->totind;
03187                 for ( j = 0, tp = T->tablepointers; j < T->totind; j++ ) {

```



```

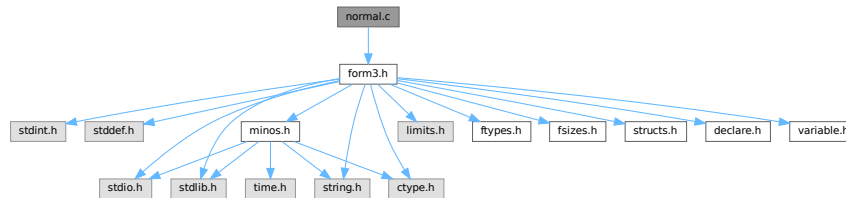
03188         tp += ABS(T->numind);
03189 #if TABLEEXTENSION == 2
03190         tp[1] = tp[0];
03191 #else
03192         tp[2] = tp[0]; tp[3] = tp[1]; tp[5] = tp[4] & (~ELEMENTUSED);
03193 #endif
03194         tp += TABLEEXTENSION;
03195     }
03196     if ( T->spare ) {
03197         TABLES TT = T->spare;
03198         TT->mdefined = TT->totind;
03199         for ( j = 0, tp = TT->tablepointers; j < TT->totind; j++ ) {
03200             tp += ABS(TT->numind);
03201 #if TABLEEXTENSION == 2
03202             tp[1] = tp[0];
03203 #else
03204             tp[2] = tp[0]; tp[3] = tp[1]; tp[5] = tp[4] & (~ELEMENTUSED);
03205 #endif
03206             tp += TABLEEXTENSION;
03207         }
03208         cbuf[TT->bufnum].mnumlhs = cbuf[TT->bufnum].numlhs;
03209         cbuf[TT->bufnum].mnumrhs = cbuf[TT->bufnum].numrhs;
03210     }
03211 }
03212 else {
03213     T->mdefined = T->defined;
03214     for ( j = 0, tp = T->tablepointers; j < T->totind; j++ ) {
03215 #if TABLEEXTENSION == 2
03216         tp[1] = tp[0];
03217 #else
03218         tp[2] = tp[0]; tp[3] = tp[1]; tp[5] = tp[4] & (~ELEMENTUSED);
03219 #endif
03220     }
03221 }
03222 cbuf[T->bufnum].mnumlhs = cbuf[T->bufnum].numlhs;
03223 cbuf[T->bufnum].mnumrhs = cbuf[T->bufnum].numrhs;
03224 }
03225 }
03226 for ( i = AC.AutoFunctionList.numglobal; i < AC.AutoFunctionList.num; i++ ) {
03227     if ( ((FUNCTIONS) (AC.AutoFunctionList.lijst))[i].tabl )
03228         ((FUNCTIONS) (AC.AutoFunctionList.lijst))[i].tabl->mdefined =
03229             ((FUNCTIONS) (AC.AutoFunctionList.lijst))[i].tabl->defined;
03230 }
03231 AC.SymbolList.numglobal = AC.SymbolList.num;
03232 AC.VectorList.numglobal = AC.VectorList.num;
03233 AC.IndexList.numglobal = AC.IndexList.num;
03234 AC.FunctionList.numglobal = AC.FunctionList.num;
03235 AC.SetList.numglobal = AC.SetList.num;
03236 AC.DubiousList.numglobal = AC.DubiousList.num;
03237 AC.SetElementList.numglobal = AC.SetElementList.num;
03238 AC.varnames->globalnamefill = AC.varnames->namefill;
03239 AC.varnames->globalnodefill = AC.varnames->nodefill;
03240
03241 AC.AutoSymbolList.numglobal = AC.AutoSymbolList.num;
03242 AC.AutoVectorList.numglobal = AC.AutoVectorList.num;
03243 AC.AutoIndexList.numglobal = AC.AutoIndexList.num;
03244 AC.AutoFunctionList.numglobal = AC.AutoFunctionList.num;
03245 AC.autonames->globalnamefill = AC.autonames->namefill;
03246 AC.autonames->globalnodefill = AC.autonames->nodefill;
03247 }
03248
03249 /*
03250 #] Globalize :
03251 #[ TestName :
03252 */
03253
03254 int TestName(UBYTE *name)
03255 {
03256     if ( *name == '[' ) {
03257         while ( *name ) name++;
03258         if ( name[-1] == ']' ) return(0);
03259         return(-1);
03260     }
03261     while ( *name ) {
03262         if ( *name == '_' ) return(-1);
03263         name++;
03264     }
03265     return(0);
03266 }
03267
03268 /*
03269 #] TestName :
03270 */

```

## 4.86 normal.c File Reference

```
#include "form3.h"
```

Include dependency graph for normal.c:



### Macros

- `#define` [MAXNUMBEROFNONCOMTERMS](#) 2

### Functions

- int [CompareFunctions](#) (WORD \*left, WORD \*fright)
- int [Commute](#) (WORD \*left, WORD \*fright)
- int [Normalize](#) (PHEAD WORD \*term)
- int [ExtraSymbol](#) (WORD sym, WORD pow, WORD nsym, WORD \*ppsym, WORD \*ncoef)
- int [DoTheta](#) (PHEAD WORD \*t)
- int [DoDelta](#) (WORD \*t)
- void [DoRevert](#) (WORD \*fun, WORD \*tmp)
- WORD [DetCommu](#) (WORD \*terms)
- WORD [DoesCommu](#) (WORD \*term)
- int [TreatPolyRatFun](#) (PHEAD WORD \*prf)
- void [DropCoefficient](#) (PHEAD WORD \*term)
- void [DropSymbols](#) (PHEAD WORD \*term)
- int [SymbolNormalize](#) (WORD \*term)
- int [TestFunFlag](#) (PHEAD WORD \*tfun)
- int [BracketNormalize](#) (PHEAD WORD \*term)

### 4.86.1 Detailed Description

Mainly the routine `Normalize`. This routine brings terms to standard FORM. Currently it has one serious drawback. Its buffers are all in the stack. This means these buffers have a fixed size (`NORMSIZE`). In the past this has caused problems and `NORMSIZE` had to be increased.

It is not clear whether `Normalize` can be called recursively.

Definition in file [normal.c](#).

## 4.86.2 Macro Definition Documentation

### 4.86.2.1 MAXNUMBEROFNONCOMTERMS

```
#define MAXNUMBEROFNONCOMTERMS 2
```

Definition at line [4509](#) of file [normal.c](#).

## 4.86.3 Function Documentation

### 4.86.3.1 CompareFunctions()

```
int CompareFunctions (  
    WORD * fleft,  
    WORD * fright )
```

Definition at line [54](#) of file [normal.c](#).

### 4.86.3.2 Commute()

```
int Commute (  
    WORD * fleft,  
    WORD * fright )
```

Definition at line [118](#) of file [normal.c](#).

### 4.86.3.3 Normalize()

```
int Normalize (  
    PHEAD WORD * term )
```

Definition at line [193](#) of file [normal.c](#).

### 4.86.3.4 ExtraSymbol()

```
int ExtraSymbol (  
    WORD sym,  
    WORD pow,  
    WORD nsym,  
    WORD * ppsym,  
    WORD * ncoef )
```

Definition at line [4197](#) of file [normal.c](#).

### 4.86.3.5 DoTheta()

```
int DoTheta (  
    PHEAD WORD * t )
```

Definition at line [4259](#) of file [normal.c](#).

#### 4.86.3.6 DoDelta()

```
int DoDelta (
    WORD * t )
```

Definition at line [4356](#) of file [normal.c](#).

#### 4.86.3.7 DoRevert()

```
void DoRevert (
    WORD * fun,
    WORD * tmp )
```

Definition at line [4423](#) of file [normal.c](#).

#### 4.86.3.8 DetCommu()

```
WORD DetCommu (
    WORD * terms )
```

Definition at line [4511](#) of file [normal.c](#).

#### 4.86.3.9 DoesCommu()

```
WORD DoesCommu (
    WORD * term )
```

Definition at line [4570](#) of file [normal.c](#).

#### 4.86.3.10 TreatPolyRatFun()

```
int TreatPolyRatFun (
    PHEAD WORD * prf )
```

Definition at line [4961](#) of file [normal.c](#).

#### 4.86.3.11 DropCoefficient()

```
void DropCoefficient (
    PHEAD WORD * term )
```

Definition at line [5097](#) of file [normal.c](#).

#### 4.86.3.12 DropSymbols()

```
void DropSymbols (
    PHEAD WORD * term )
```

Definition at line [5115](#) of file [normal.c](#).

**4.86.3.13 SymbolNormalize()**

```
int SymbolNormalize (
    WORD * term )
```

Routine normalizes terms that contain only symbols. Regular minimum and maximum properties are ignored.

We check whether there are negative powers in the output. This is not allowed.

Definition at line 5145 of file [normal.c](#).

Referenced by [InFunction\(\)](#), and [poly\\_sort\(\)](#).

**4.86.3.14 TestFunFlag()**

```
int TestFunFlag (
    PHEAD WORD * tfun )
```

Definition at line 5241 of file [normal.c](#).

**4.86.3.15 BracketNormalize()**

```
int BracketNormalize (
    PHEAD WORD * term )
```

Definition at line 5272 of file [normal.c](#).

**4.87 normal.c**

[Go to the documentation of this file.](#)

```
00001
00010 /* #[] License : */
00011 /*
00012 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00013 * When using this file you are requested to refer to the publication
00014 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00015 * This is considered a matter of courtesy as the development was paid
00016 * for by FOM the Dutch physics granting agency and we would like to
00017 * be able to track its scientific use to convince FOM of its value
00018 * for the community.
00019 *
00020 * This file is part of FORM.
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00026 *
00027 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
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00029 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00030 * details.
00031 *
00032 * You should have received a copy of the GNU General Public License along
00033 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00034 */
00035 /* #[] License : */
00036 /*
00037 #[] Includes : normal.c
00038 */
00039
00040 #include "form3.h"
00041 #ifdef WITHFLOAT
```

```

00042 #include <gmp.h>
00043
00044 int PackFloat(WORD *,mpf_t);
00045 int UnpackFloat(mpf_t, WORD *);
00046 void RatToFloat(mpf_t result, UWORD *format, int ratsize);
00047 #endif
00048 /*
00049     #] Includes :
00050     #[ Normalize :
00051         #[ CompareFunctions :
00052 */
00053
00054 int CompareFunctions(WORD *fleft,WORD *fright)
00055 {
00056     WORD k, kk;
00057     if ( AC.properorderflag ) {
00058         if ( ( *fleft >= (FUNCTION+WILDOFFSET)
00059             && functions[*fleft-FUNCTION-WILDOFFSET].spec >= TENSORFUNCTION )
00060             || ( *fleft >= FUNCTION && *fleft < (FUNCTION + WILDOFFSET)
00061             && functions[*fleft-FUNCTION].spec >= TENSORFUNCTION ) ) {}
00062         else {
00063             WORD *s1, *s2, *ss1, *ss2;
00064             s1 = fleft+FUNHEAD; s2 = fright+FUNHEAD;
00065             ss1 = fleft + fleft[1]; ss2 = fright + fright[1];
00066             while ( s1 < ss1 && s2 < ss2 ) {
00067                 k = CompArg(s1,s2);
00068                 if ( k > 0 ) return(1);
00069                 if ( k < 0 ) return(0);
00070                 NEXTARG(s1);
00071                 NEXTARG(s2);
00072             }
00073             if ( s1 < ss1 ) return(1);
00074             return(0);
00075         }
00076         k = fleft[1] - FUNHEAD;
00077         kk = fright[1] - FUNHEAD;
00078         fleft += FUNHEAD;
00079         fright += FUNHEAD;
00080         while ( k > 0 && kk > 0 ) {
00081             if ( *fleft < *fright ) return(0);
00082             else if ( *fleft++ > *fright++ ) return(1);
00083             k--; kk--;
00084         }
00085         if ( k > 0 ) return(1);
00086         return(0);
00087     }
00088     else {
00089         k = fleft[1] - FUNHEAD;
00090         kk = fright[1] - FUNHEAD;
00091         fleft += FUNHEAD;
00092         fright += FUNHEAD;
00093         while ( k > 0 && kk > 0 ) {
00094             if ( *fleft < *fright ) return(0);
00095             else if ( *fleft++ > *fright++ ) return(1);
00096             k--; kk--;
00097         }
00098         if ( k > 0 ) return(1);
00099         return(0);
00100     }
00101 }
00102
00103 /*
00104     #] CompareFunctions :
00105     #[ Commute :
00106
00107     This function gets two adjacent function pointers and decides
00108     whether these two functions should be exchanged to obtain a
00109     natural ordering.
00110
00111     Currently there is only an ordering of gamma matrices belonging
00112     to different spin lines.
00113
00114     Note that we skip for now the cases of (F)^(3/2) or 1/F and a few more
00115     of such funny functions.
00116 */
00117
00118 int Commute(WORD *fleft, WORD *fright)
00119 {
00120     WORD fun1, fun2;
00121     if ( *fleft == DOLLAREXPRESSION || *fright == DOLLAREXPRESSION ) return(0);
00122     fun1 = ABS(*fleft); fun2 = ABS(*fright);
00123     if ( *fleft >= GAMMA && *fleft <= GAMMASEVEN
00124         && *fright >= GAMMA && *fright <= GAMMASEVEN ) {
00125         if ( fleft[FUNHEAD] < AM.OffsetIndex && fleft[FUNHEAD] > fright[FUNHEAD] )
00126             return(1);
00127         return(0);
00128     }

```

```

00129     if ( fun1 >= WILDOFFSET ) fun1 -= WILDOFFSET;
00130     if ( fun2 >= WILDOFFSET ) fun2 -= WILDOFFSET;
00131     if ( ( ( functions[fun1-FUNCTION].flags & COULDCOMMUTE ) == 0 )
00132         || ( ( functions[fun2-FUNCTION].flags & COULDCOMMUTE ) == 0 ) ) return(0);
00133 /*
00134     if other conditions will come here, keep in mind that if *fleft < 0
00135     or *fright < 0 they are arguments in the exponent function as in f^(3/2)
00136 */
00137     if ( AC.CommuteInSet == 0 ) return(0);
00138 /*
00139     The code for CompareFunctions can be stolen from the commuting case.
00140
00141     We need the syntax:
00142         Commute Fun1,Fun2,...,Fun'n';
00143     For this Fun1,...,Fun'n' need to be noncommuting functions.
00144     These functions will commute with all members of the group.
00145     In the AC.paircommute buffer the representation is
00146         'n'+1,element1,...,element'n','m'+1,element1,...,element'm',0
00147     A function can belong to more than one group.
00148     If a function commutes with itself, it is most efficient to make a separate
00149     group of two elements for it as in
00150         Commute T,T;    -> 3,T,T
00151 */
00152     if ( fun1 >= fun2 ) {
00153         WORD *group = AC.CommuteInSet, *g1, *g2, *g3;
00154         while ( *group > 0 ) {
00155             g3 = group + *group;
00156             g1 = group+1;
00157             while ( g1 < g3 ) {
00158                 if ( *g1 == fun1 || ( fun1 <= GAMMASEVEN && fun1 >= GAMMA
00159                     && *g1 <= GAMMASEVEN && *g1 >= GAMMA ) ) {
00160                     g2 = group+1;
00161                     while ( g2 < g3 ) {
00162                         if ( g1 != g2 && ( *g2 == fun2 ||
00163                             ( fun2 <= GAMMASEVEN && fun2 >= GAMMA
00164                             && *g2 <= GAMMASEVEN && *g2 >= GAMMA ) ) ) {
00165                             if ( fun1 != fun2 ) return(1);
00166                             if ( *fleft < 0 ) return(0);
00167                             if ( *fright < 0 ) return(1);
00168                             return(CompareFunctions(fleft,fright));
00169                         }
00170                         g2++;
00171                     }
00172                     break;
00173                 }
00174                 g1++;
00175             }
00176             group = g3;
00177         }
00178     }
00179     return(0);
00180 }
00181
00182 /*
00183     #] Commute :
00184     #[ Normalize :
00185
00186     This is the big normalization routine. It has a great need
00187     to be economical.
00188     There is a fixed limit to the number of objects coming in.
00189     Something should be done about it.
00190
00191 */
00192
00193 int Normalize(PHEAD WORD *term)
00194 {
00195 /*
00196     #[ Declarations :
00197 */
00198     GETBIDENTITY
00199     WORD *t, *m, *r, i, j, k, l, nsym, *ss, *tt, *u;
00200     WORD shortnum, stype;
00201     WORD *stop, *to = 0, *from = 0;
00202 /*
00203     The next variables could be better off in the AT.WorkSpace (?)
00204     Now they make stackallocations rather bothersome.
00205 */
00206     WORD psym[7*NORMSIZE],*ppsym;
00207     WORD pvec[NORMSIZE],*ppvec,nvec;
00208     WORD pdot[3*NORMSIZE],*ppdot,ndot;
00209     WORD pdel[2*NORMSIZE],*ppdel,ndel;
00210     WORD pind[NORMSIZE],nind;
00211     WORD *peps[NORMSIZE/3],neps;
00212     WORD *pden[NORMSIZE/3],nden;
00213     WORD *pcom[NORMSIZE],ncom;
00214     WORD *pnco[NORMSIZE],nnco;
00215     WORD *pcon[2*NORMSIZE],ncon;          /* Pointer to contractable indices */

```

```

00216     WORD *n_coef, ncoef;                      /* Accumulator for the coefficient */
00217     WORD *n_llnum, *lnum, nnum;
00218     WORD *termout, oldtoprhs = 0, subtype;
00219     WORD ReplaceType, ReplaceVeto = 0, didcontr;
00220     int regval = 0;
00221     WORD *ReplaceSub;
00222     WORD *fillsetexp;
00223     CBUF *C = cbuf+AT.ebufnum;
00224     WORD *ANsc = 0, *ANsm = 0, *ANsr = 0, PolyFunMode;
00225 #ifdef WITHFLOAT
00226     WORD withfloat = 0;
00227     WORD *firstfloat = 0;
00228 #endif
00229     LONG oldcpointer = 0, x;
00230     n_coef = TermMalloc("NormCoef");
00231     n_llnum = TermMalloc("n_llnum");
00232     lnum = n_llnum+1;
00233 /*
00234     int termflag;
00235 */
00236 /*
00237     #] Declarations :
00238     #[ Setup :
00239 PrintTerm(term, "Normalize");
00240 */
00241
00242 Restart:
00243     didcontr = 0;
00244     ReplaceType = -1;
00245     t = term;
00246     if ( !*t ) { TermFree(n_coef, "NormCoef"); TermFree(n_llnum, "n_llnum"); return(regval); }
00247     r = t + *t;
00248     ncoef = r[-1];
00249     i = ABS(ncoef);
00250     r -= i;
00251     m = r;
00252     t = n_coef;
00253     NCOPY(t, r, i);
00254     termout = AT.WorkPointer;
00255     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
00256     fillsetexp = termout+1;
00257     AN.PolyNormFlag = 0; PolyFunMode = AN.PolyFunTodo;
00258 /*
00259     termflag = 0;
00260 */
00261 /*
00262     #] Setup :
00263     #[ First scan :
00264 */
00265     nsym = nvec = ndot = ndel = neps = nden =
00266     nind = ncom = nnco = ncon = 0;
00267     ppsym = psym;
00268     ppvec = pvec;
00269     ppdot = pdot;
00270     ppdel = pdel;
00271     t = term + 1;
00272 conscan:;
00273     if ( t < m ) do {
00274         r = t + t[1];
00275         switch ( *t ) {
00276             case SYMBOL :
00277                 t += 2;
00278                 from = m;
00279                 do {
00280                     if ( t[1] == 0 ) {
00281 /*
00282                     if ( *t == 0 || *t == MAXPOWER ) goto NormZZ; */
00283                     t += 2;
00284                     goto NextSymbol;
00285                 }
00286 /*
00287                 if ( *t <= DENOMINATORSYMBOL && *t >= COEFFSYMBOL ) {
00288                     if ( AN.NoScrat2 == 0 ) {
00289                         AN.NoScrat2 = (UWORD *)Malloc1((AM.MaxTal+2)*sizeof(UWORD), "Normalize");
00290                     }
00291                     if ( AN.cTerm ) m = AN.cTerm;
00292                     else m = term;
00293                     m += *m;
00294                     ncoef = REDLENG(ncoef);
00295                     if ( *t == COEFFSYMBOL ) {
00296                         i = t[1];
00297                         nnum = REDLENG(m[-1]);
00298                         m -= ABS(m[-1]);
00299                         if ( i > 0 ) {
00300                             while ( i > 0 ) {
00301                                 if ( MulRat(BHEAD (UWORD *)n_coef, ncoef, (UWORD *)m, nnum,
00302                                     (UWORD *)n_coef, &ncoef) ) goto FromNorm;

```



```

00303         i--;
00304     }
00305 }
00306 else if ( i < 0 ) {
00307     while ( i < 0 ) {
00308         if ( DivRat (BHEAD (UWORD *)n_coef,ncoef, (UWORD *)m,nnum,
00309             (UWORD *)n_coef,&ncoef) ) goto FromNorm;
00310         i++;
00311     }
00312 }
00313 }
00314 else {
00315     i = m[-1];
00316     nnum = (ABS(i)-1)/2;
00317     if ( *t == NUMERATORSYMBOL ) { m -= nnum + 1; }
00318     else { m--; }
00319     while ( *m == 0 && nnum > 1 ) { m--; nnum--; }
00320     m -= nnum;
00321     if ( i < 0 && *t == NUMERATORSYMBOL ) nnum = -nnum;
00322     i = t[1];
00323     if ( i > 0 ) {
00324         while ( i > 0 ) {
00325             if ( Mully(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)m,nnum) )
00326                 goto FromNorm;
00327             i--;
00328         }
00329     }
00330     else if ( i < 0 ) {
00331         while ( i < 0 ) {
00332             if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)m,nnum) )
00333                 goto FromNorm;
00334             i++;
00335         }
00336     }
00337 }
00338 ncoef = INCLENG(ncoef);
00339 t += 2;
00340 goto NextSymbol;
00341 }
00342 else if ( *t == DIMENSIONSYMBOL ) {
00343     if ( AN.cTerm ) m = AN.cTerm;
00344     else m = term;
00345     k = DimensionTerm(m);
00346     if ( k == 0 ) goto NormZero;
00347     if ( k == MAXPOSITIVE ) {
00348         MLOCK(ErrorMessageLock);
00349         MesPrint("Dimension_ is undefined in term %t");
00350         MUNLOCK(ErrorMessageLock);
00351         goto NormMin;
00352     }
00353     if ( k == -MAXPOSITIVE ) {
00354         MLOCK(ErrorMessageLock);
00355         MesPrint("Dimension_ out of range in term %t");
00356         MUNLOCK(ErrorMessageLock);
00357         goto NormMin;
00358     }
00359     if ( k > 0 ) { *((UWORD *)lnum) = k; nnum = 1; }
00360     else { *((UWORD *)lnum) = -k; nnum = -1; }
00361     ncoef = REDLENG(ncoef);
00362     if ( Mully(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)lnum,nnum) ) goto FromNorm;
00363     ncoef = INCLENG(ncoef);
00364     t += 2;
00365     goto NextSymbol;
00366 }
00367 if ( ( *t >= MAXPOWER && *t < 2*MAXPOWER )
00368     || ( *t < -MAXPOWER && *t > -2*MAXPOWER ) ) {
00369 /*
00370 #[ TO SNUMBER :
00371 */
00372     if ( *t < 0 ) {
00373         *t += MAXPOWER;
00374         *t = -*t;
00375         if ( t[1] & 1 ) ncoef = -ncoef;
00376     }
00377     else if ( *t == MAXPOWER ) {
00378         if ( t[1] > 0 ) goto NormZero;
00379         goto NormInf;
00380     }
00381     else {
00382         *t -= MAXPOWER;
00383     }
00384     lnum[0] = *t;
00385     nnum = 1;
00386     if ( t[1] && RaisPow(BHEAD (UWORD *)lnum,&nnum, (UWORD) (ABS(t[1]))) )
00387         goto FromNorm;
00388     ncoef = REDLENG(ncoef);
00389     if ( t[1] < 0 ) {

```

```

00390         if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)lnum,nnum) )
00391             goto FromNorm;
00392     }
00393     else if ( t[1] > 0 ) {
00394         if ( Mully(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)lnum,nnum) )
00395             goto FromNorm;
00396     }
00397     ncoef = INCLENG(ncoef);
00398 /*
00399 #] TO SNUMBER :
00400 */
00401     t += 2;
00402     goto NextSymbol;
00403 }
00404 if ( ( *t <= NumSymbols && *t > -MAXPOWER )
00405     && ( symbols[*t].complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY ) {
00406     if ( t[1] <= 2*MAXPOWER && t[1] >= -2*MAXPOWER ) {
00407         t[1] %= symbols[*t].maxpower;
00408         if ( t[1] < 0 ) t[1] += symbols[*t].maxpower;
00409         if ( ( symbols[*t].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
00410             if ( ( ( symbols[*t].maxpower & 1 ) == 0 ) &&
00411                 ( t[1] >= symbols[*t].maxpower/2 ) ) {
00412                 t[1] -= symbols[*t].maxpower/2; ncoef = -ncoef;
00413             }
00414         }
00415         if ( t[1] == 0 ) { t += 2; goto NextSymbol; }
00416     }
00417 }
00418 i = nsym;
00419 m = ppsym;
00420 if ( i > 0 ) do {
00421     m -= 2;
00422     if ( *t == *m ) {
00423         t++; m++;
00424         if ( *t > 2*MAXPOWER || *t < -2*MAXPOWER
00425             || *m > 2*MAXPOWER || *m < -2*MAXPOWER ) {
00426             MLOCK(ErrorMessageLock);
00427             MesPrint("Illegal wildcard power combination.");
00428             MUNLOCK(ErrorMessageLock);
00429             goto NormMin;
00430         }
00431         *m += *t;
00432         if ( ( t[-1] <= NumSymbols && t[-1] > -MAXPOWER )
00433             && ( symbols[t[-1]].complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY
00434         ) {
00435             *m %= symbols[t[-1]].maxpower;
00436             if ( *m < 0 ) *m += symbols[t[-1]].maxpower;
00437             if ( ( symbols[t[-1]].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
00438                 if ( ( ( symbols[t[-1]].maxpower & 1 ) == 0 ) &&
00439                     ( *m >= symbols[t[-1]].maxpower/2 ) ) {
00440                     *m -= symbols[t[-1]].maxpower/2; ncoef = -ncoef;
00441                 }
00442             }
00443             if ( *m >= 2*MAXPOWER || *m <= -2*MAXPOWER ) {
00444                 MLOCK(ErrorMessageLock);
00445                 MesPrint("Power overflow during normalization");
00446                 MUNLOCK(ErrorMessageLock);
00447                 goto NormMin;
00448             }
00449             if ( !*m ) {
00450                 m--;
00451                 while ( i < nsym )
00452                     { *m = m[2]; m++; *m = m[2]; m++; i++; }
00453                 ppsym -= 2;
00454                 nsym--;
00455             }
00456             t++;
00457             goto NextSymbol;
00458         }
00459     } while ( *t < *m && --i > 0 );
00460     m = ppsym;
00461     while ( i < nsym )
00462         { m--; m[2] = *m; m--; m[2] = *m; i++; }
00463     *m++ = *t++;
00464     *m = *t++;
00465     ppsym += 2;
00466     nsym++;
00467 NextSymbol;;
00468     } while ( t < r );
00469     m = from;
00470     break;
00471 case VECTOR :
00472     t += 2;
00473     do {
00474         if ( t[0] == AM.vectorzero ) goto NormZero;
00475         if ( t[1] == FUNNYVEC ) {

```

```

00476         pind[nind++] = *t;
00477         t += 2;
00478     }
00479     else if ( t[1] < 0 ) {
00480         if ( *t == NOINDEX && t[1] == NOINDEX ) t += 2;
00481         else {
00482             if ( t[1] == AM.vectorzero ) goto NormZero;
00483             *ppdot++ = *t++; *ppdot++ = *t++; *ppdot++ = 1; ndot++;
00484         }
00485     }
00486     else { *ppvec++ = *t++; *ppvec++ = *t++; nvec += 2; }
00487 } while ( t < r );
00488 break;
00489 case DOTPRODUCT :
00490     t += 2;
00491     do {
00492         if ( t[2] == 0 ) t += 3;
00493         else if ( ndot > 0 && t[0] == ppdot[-3]
00494             && t[1] == ppdot[-2] ) {
00495             ppdot[-1] += t[2];
00496             t += 3;
00497             if ( ppdot[-1] == 0 ) { ppdot -= 3; ndot--; }
00498         }
00499         else if ( t[0] == AM.vectorzero || t[1] == AM.vectorzero ) {
00500             if ( t[2] > 0 ) goto NormZero;
00501             goto NormInf;
00502         }
00503         else {
00504             *ppdot++ = *t++; *ppdot++ = *t++;
00505             *ppdot++ = *t++; ndot++;
00506         }
00507     } while ( t < r );
00508     break;
00509 case HAAKJE :
00510     break;
00511 case SETSET:
00512     if ( WildFill(BHEAD termout,term,AT.dummysubexp) < 0 ) goto FromNorm;
00513     i = *termout;
00514     t = termout; m = term;
00515     NCOPY(m,t,i);
00516     goto Restart;
00517 case DOLLAREXPRESSION :
00518 /*
00519     We have DOLLAREXPRESSION,4,number,power
00520     Replace by SUBEXPRESSION and exit elegantly to let
00521     TestSub pick it up. Of course look for special cases first.
00522     Note that we have a special compiler buffer for the values.
00523 */
00524     if ( AR.Eside != LHSIDE ) {
00525         DOLLARS d = Dollars + t[2];
00526 #ifdef WITHPTHREADS
00527         int nummodopt, ptype = -1;
00528         if ( AS.MultiThreaded ) {
00529             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
00530                 if ( t[2] == ModOptdollars[nummodopt].number ) break;
00531             }
00532             if ( nummodopt < NumModOptdollars ) {
00533                 ptype = ModOptdollars[nummodopt].type;
00534                 if ( ptype == MODLOCAL ) {
00535                     d = ModOptdollars[nummodopt].dstruct+AT.identity;
00536                 }
00537                 else {
00538                     LOCK(d->pthreadlockread);
00539                 }
00540             }
00541         }
00542 #endif
00543         if ( d->type == DOLZERO ) {
00544 #ifdef WITHPTHREADS
00545             if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00546 #endif
00547             if ( t[3] == 0 ) goto NormZZ;
00548             if ( t[3] < 0 ) goto NormInf;
00549             goto NormZero;
00550         }
00551         else if ( d->type == DOLNUMBER ) {
00552             nnum = d->where[0];
00553             if ( nnum > 0 ) {
00554                 nnum = d->where[nnum-1];
00555                 if ( nnum < 0 ) { ncoef = -ncoef; nnum = -nnum; }
00556                 nnum = (nnum-1)/2;
00557                 for ( i = 1; i <= nnum; i++ ) lnum[i-1] = d->where[i];
00558             }
00559             if ( nnum == 0 || ( nnum == 1 && lnum[0] == 0 ) ) {
00560 #ifdef WITHPTHREADS
00561                 if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00562 #endif

```

```

00563         if ( t[3] < 0 ) goto NormInf;
00564         else if ( t[3] == 0 ) goto NormZZ;
00565         goto NormZero;
00566     }
00567     if ( t[3] && RaisPow(BHEAD (UWORD *)lnum,&nnum, (UWORD) (ABS(t[3]))) ) goto
FromNorm;
00568     ncoef = REDLENG(ncoef);
00569     if ( t[3] < 0 ) {
00570         if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)lnum,nnum) ) {
00571 #ifdef WITHPTHREADS
00572             if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00573 #endif
00574             goto FromNorm;
00575         }
00576     }
00577     else if ( t[3] > 0 ) {
00578         if ( Mully(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)lnum,nnum) ) {
00579 #ifdef WITHPTHREADS
00580             if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00581 #endif
00582             goto FromNorm;
00583         }
00584     }
00585     ncoef = INCLENG(ncoef);
00586 }
00587     else if ( d->type == DOLINDEX ) {
00588         if ( d->index == 0 ) {
00589 #ifdef WITHPTHREADS
00590             if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00591 #endif
00592             goto NormZero;
00593         }
00594         if ( d->index != NOINDEX ) pind[nind++] = d->index;
00595     }
00596     else if ( d->type == DOLTERMS ) {
00597         if ( t[3] >= MAXPOWER || t[3] <= -MAXPOWER ) {
00598             if ( d->where[0] == 0 ) goto NormZero;
00599             if ( d->where[d->where[0]] != 0 ) {
00600 IllDollarExp:
00601                 MLOCK(ErrorMessageLock);
00602                 MesPrint("!!!Illegal $ expansion with wildcard power!!!");
00603                 MUNLOCK(ErrorMessageLock);
00604                 goto FromNorm;
00605             }
00606 /*
00607 At this point we should only admit symbols and dotproducts
00608 We expand the dollar directly and do not send it back.
00609 */
00610 { WORD *td, *tdstop, dj;
00611     td = d->where+1;
00612     tdstop = d->where+d->where[0];
00613     if ( tdstop[-1] != 3 || tdstop[-2] != 1
00614         || tdstop[-3] != 1 ) goto IllDollarExp;
00615     tdstop -= 3;
00616     if ( td >= tdstop ) goto IllDollarExp;
00617     while ( td < tdstop ) {
00618         if ( *td == SYMBOL ) {
00619             for ( dj = 2; dj < td[1]; dj += 2 ) {
00620                 if ( td[dj+1] == 1 ) {
00621                     *ppsym++ = td[dj];
00622                     *ppsym++ = t[3];
00623                     nsym++;
00624                 }
00625                 else if ( td[dj+1] == -1 ) {
00626                     *ppsym++ = td[dj];
00627                     *ppsym++ = -t[3];
00628                     nsym++;
00629                 }
00630                 else goto IllDollarExp;
00631             }
00632         }
00633         else if ( *td == DOTPRODUCT ) {
00634             for ( dj = 2; dj < td[1]; dj += 3 ) {
00635                 if ( td[dj+2] == 1 ) {
00636                     *ppdot++ = td[dj];
00637                     *ppdot++ = td[dj+1];
00638                     *ppdot++ = t[3];
00639                     ndot++;
00640                 }
00641                 else if ( td[dj+2] == -1 ) {
00642                     *ppdot++ = td[dj];
00643                     *ppdot++ = td[dj+1];
00644                     *ppdot++ = -t[3];
00645                     ndot++;
00646                 }
00647                 else goto IllDollarExp;
00648             }
00649         }
00650     }
00651 }
00652     }
00653 }

```

```

00649             }
00650             else goto IllDollarExp;
00651             td += td[1];
00652         }
00653         regval = 2;
00654         break;
00655     }
00656 }
00657 t[0] = SUBEXPRESSION;
00658 t[4] = AM.dbufnum;
00659 if ( t[3] == 0 ) {
00660 #ifdef WITHPTHREADS
00661     if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00662 #endif
00663     break;
00664 }
00665 regval = 2;
00666 t = r;
00667 while ( t < m ) {
00668     if ( *t == DOLLAREXPRESSION ) {
00669 #ifdef WITHPTHREADS
00670         if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00671 #endif
00672         d = Dollars + t[2];
00673 #ifdef WITHPTHREADS
00674         if ( AS.MultiThreaded ) {
00675             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
00676                 if ( t[2] == ModOptdollars[nummodopt].number ) break;
00677             }
00678             if ( nummodopt < NumModOptdollars ) {
00679                 ptype = ModOptdollars[nummodopt].type;
00680                 if ( ptype == MODLOCAL ) {
00681                     d = ModOptdollars[nummodopt].dstruct+AT.identity;
00682                 }
00683                 else {
00684                     LOCK(d->pthreadlockread);
00685                 }
00686             }
00687         }
00688 #endif
00689         if ( d->type == DOLTERMS ) {
00690             t[0] = SUBEXPRESSION;
00691             t[4] = AM.dbufnum;
00692         }
00693     }
00694     t += t[1];
00695 }
00696 #ifdef WITHPTHREADS
00697 if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00698 #endif
00699 goto RegEnd;
00700 }
00701 else {
00702 #ifdef WITHPTHREADS
00703     if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00704 #endif
00705     MLOCK(ErrorMessageLock);
00706     MesPrint("!!!This $ variation has not been implemented yet!!!");
00707     MUNLOCK(ErrorMessageLock);
00708     goto NormMin;
00709 }
00710 #ifdef WITHPTHREADS
00711 if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
00712 #endif
00713 }
00714 else {
00715     pnco[ncco++] = t;
00716 /*
00717     The next statement should be safe as the value is used
00718     only by the compiler (ie the master).
00719 */
00720     AC.lhdollarflag = 1;
00721 }
00722 break;
00723 case DELTA :
00724     t += 2;
00725     do {
00726         if ( *t < 0 ) {
00727             if ( *t == SUMMEDIND ) {
00728                 if ( t[1] < -NMIN4SHIFT ) {
00729                     k = -t[1]-NMIN4SHIFT;
00730                     k = ExtraSymbol(k,1,nsym,ppsym,&ncoef);
00731                     nsym += k;
00732                     ppsym += (k * 2);
00733                 }
00734                 else if ( t[1] == 0 ) goto NormZero;
00735                 else {

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00736             if ( t[1] < 0 ) {
00737                 lnum[0] = -t[1];
00738                 nnum = -1;
00739             }
00740             else {
00741                 lnum[0] = t[1];
00742                 nnum = 1;
00743             }
00744             ncoef = REDLENG(ncoef);
00745             if ( Mully(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)lnum,nnum) )
00746                 goto FromNorm;
00747             ncoef = INCLENG(ncoef);
00748         }
00749         t += 2;
00750     }
00751     else if ( *t == NOINDEX && t[1] == NOINDEX ) t += 2;
00752     else if ( *t == EMPTYINDEX && t[1] == EMPTYINDEX ) {
00753         *ppdel++ = *t++; *ppdel++ = *t++; ndel += 2;
00754     }
00755     else
00756         if ( t[1] < 0 ) {
00757             *ppdot++ = *t++; *ppdot++ = *t++; *ppdot++ = 1; ndot++;
00758         }
00759         else {
00760             *ppvec++ = *t++; *ppvec++ = *t++; nvec += 2;
00761         }
00762     }
00763     else {
00764         if ( t[1] < 0 ) {
00765             *ppvec++ = t[1]; *ppvec++ = *t; t+=2; nvec += 2;
00766         }
00767         else { *ppdel++ = *t++; *ppdel++ = *t++; ndel += 2; }
00768     }
00769     } while ( t < r );
00770     break;
00771     case FACTORIAL :
00772     /*
00773         (FACTORIAL,FUNHEAD+2,...,-SNUMBER,number)
00774     */
00775     if ( t[FUNHEAD] == -SNUMBER && t[1] == FUNHEAD+2
00776         && t[FUNHEAD+1] >= 0 ) {
00777         if ( Factorial(BHEAD t[FUNHEAD+1],(UWORD *)lnum,&nnum) )
00778             goto FromNorm;
00779     MulIn:
00780         ncoef = REDLENG(ncoef);
00781         if ( Mully(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)lnum,nnum) ) goto FromNorm;
00782         ncoef = INCLENG(ncoef);
00783     }
00784     else pcom[ncom++] = t;
00785     break;
00786     case BERNOULLIFUNCTION :
00787     /*
00788         (BERNOULLIFUNCTION,FUNHEAD+2,...,-SNUMBER,number)
00789     */
00790     if ( ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] >= 0 )
00791         && ( t[1] == FUNHEAD+2 || ( t[1] == FUNHEAD+4 &&
00792         t[FUNHEAD+2] == -SNUMBER && ABS(t[FUNHEAD+3]) == 1 ) ) ) {
00793         WORD inum, mnum;
00794         if ( Bernoulli(t[FUNHEAD+1],(UWORD *)lnum,&nnum) )
00795             goto FromNorm;
00796         if ( nnum == 0 ) goto NormZero;
00797         inum = nnum; if ( inum < 0 ) inum = -inum;
00798         inum--; inum /= 2;
00799         mnum = inum;
00800         while ( lnum[mnum-1] == 0 ) mnum--;
00801         if ( nnum < 0 ) mnum = -mnum;
00802         ncoef = REDLENG(ncoef);
00803         if ( Mully(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)lnum,mnum) ) goto FromNorm;
00804         mnum = inum;
00805         while ( lnum[inum+mnum-1] == 0 ) mnum--;
00806         if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *) (lnum+inum),mnum) ) goto
FromNorm;
00807         ncoef = INCLENG(ncoef);
00808         if ( t[1] == FUNHEAD+4 && t[FUNHEAD+1] == 1
00809             && t[FUNHEAD+3] == -1 ) ncoef = -ncoef;
00810     }
00811     else pcom[ncom++] = t;
00812     break;
00813     case NUMARGSFUN:
00814     /*
00815         Numerical function giving the number of arguments.
00816     */
00817     k = 0;
00818     t += FUNHEAD;
00819     while ( t < r ) {
00820         k++;
00821         NEXTARG(t);

```

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00822     }
00823     if ( k == 0 ) goto NormZero;
00824     *((UWORD *)lnum) = k;
00825     nnum = 1;
00826     goto MulIn;
00827 case NUMFACTORS:
00828 /*
00829     Numerical function giving the number of factors in an expression.
00830 */
00831     t += FUNHEAD;
00832     if ( *t == -EXPRESSION ) {
00833         k = AS.OldNumFactors[t[1]];
00834     }
00835     else if ( *t == -DOLLAREXPRESSION ) {
00836         k = Dollars[t[1]].nfactors;
00837     }
00838     else {
00839         pcom[ncom++] = t;
00840         break;
00841     }
00842     if ( k == 0 ) goto NormZero;
00843     *((UWORD *)lnum) = k;
00844     nnum = 1;
00845     goto MulIn;
00846 case NUMTERMSFUN:
00847 /*
00848     Numerical function giving the number of terms in the single argument.
00849 */
00850     if ( t[FUNHEAD] < 0 ) {
00851         if ( t[FUNHEAD] <= -FUNCTION && t[1] == FUNHEAD+1 ) break;
00852         if ( t[FUNHEAD] > -FUNCTION && t[1] == FUNHEAD+2 ) {
00853             if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] == 0 ) goto NormZero;
00854             break;
00855         }
00856         pcom[ncom++] = t;
00857         break;
00858     }
00859     if ( t[FUNHEAD] > 0 && t[FUNHEAD] == t[1]-FUNHEAD ) {
00860         k = 0;
00861         t += FUNHEAD+ARGHEAD;
00862         while ( t < r ) {
00863             k++;
00864             t += *t;
00865         }
00866         if ( k == 0 ) goto NormZero;
00867         *((UWORD *)lnum) = k;
00868         nnum = 1;
00869         goto MulIn;
00870     }
00871     else pcom[ncom++] = t;
00872     break;
00873 case COUNTFUNCTION:
00874     if ( AN.cTerm ) {
00875         k = CountFun(AN.cTerm,t);
00876     }
00877     else {
00878         k = CountFun(term,t);
00879     }
00880     if ( k == 0 ) goto NormZero;
00881     if ( k > 0 ) { *((UWORD *)lnum) = k; nnum = 1; }
00882     else { *((UWORD *)lnum) = -k; nnum = -1; }
00883     goto MulIn;
00884     break;
00885 case MAKERATIONAL:
00886     if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+2] == -SNUMBER
00887         && t[1] == FUNHEAD+4 && t[FUNHEAD+3] > 1 ) {
00888         WORD x1[2], sgn;
00889         if ( t[FUNHEAD+1] == 0 ) goto NormZero;
00890         if ( t[FUNHEAD+1] < 0 ) { t[FUNHEAD+1] = -t[FUNHEAD+1]; sgn = -1; }
00891         else sgn = 1;
00892         if ( MakeRational(t[FUNHEAD+1],t[FUNHEAD+3],x1,x1+1) ) {
00893             static int warnflag = 1;
00894             if ( warnflag ) {
00895                 MesPrint("%w Warning: fraction could not be reconstructed in
MakeRational_");
00896                 warnflag = 0;
00897             }
00898             x1[0] = t[FUNHEAD+1]; x1[1] = 1;
00899         }
00900         if ( sgn < 0 ) { t[FUNHEAD+1] = -t[FUNHEAD+1]; x1[0] = -x1[0]; }
00901         if ( x1[0] < 0 ) { sgn = -1; x1[0] = -x1[0]; }
00902         else sgn = 1;
00903         ncoef = REDLENG(ncoef);
00904         if ( MulRat(BHEAD (UWORD *)n_coef,ncoef, (UWORD *)x1,sgn,
00905             (UWORD *)n_coef,&ncoef) ) goto FromNorm;
00906         ncoef = INCLENG(ncoef);
00907     }

```

```

00908     else {
00909         WORD narg = 0, *tt, *ttstop, *arg1 = 0, *arg2 = 0;
00910         UWORD *x1, *x2, *xx;
00911         WORD nx1, nx2, nxx;
00912         ttstop = t + t[1]; tt = t + FUNHEAD;
00913         while ( tt < ttstop ) {
00914             narg++;
00915             if ( narg == 1 ) arg1 = tt;
00916             else arg2 = tt;
00917             NEXTARG(tt);
00918         }
00919         if ( narg != 2 ) goto defaultcase;
00920         if ( *arg2 == -SNUMBER && arg2[1] <= 1 ) goto defaultcase;
00921         else if ( *arg2 > 0 && ttstop[-1] < 0 ) goto defaultcase;
00922         x1 = NumberMalloc("Norm-MakeRational");
00923         if ( *arg1 == -SNUMBER ) {
00924             if ( arg1[1] == 0 ) {
00925                 NumberFree(x1, "Norm-MakeRational");
00926                 goto NormZero;
00927             }
00928             if ( arg1[1] < 0 ) { x1[0] = -arg1[1]; nx1 = -1; }
00929             else { x1[0] = arg1[1]; nx1 = 1; }
00930         }
00931         else if ( *arg1 > 0 ) {
00932             WORD *tc;
00933             nx1 = (ABS(arg2[-1]) - 1) / 2;
00934             tc = arg1 + ARGHEAD + 1 + nx1;
00935             if ( tc[0] != 1 ) {
00936                 NumberFree(x1, "Norm-MakeRational");
00937                 goto defaultcase;
00938             }
00939             for ( i = 1; i < nx1; i++ ) if ( tc[i] != 0 ) {
00940                 NumberFree(x1, "Norm-MakeRational");
00941                 goto defaultcase;
00942             }
00943             tc = arg1 + ARGHEAD + 1;
00944             for ( i = 0; i < nx1; i++ ) x1[i] = tc[i];
00945             if ( arg2[-1] < 0 ) nx1 = -nx1;
00946         }
00947         else {
00948             NumberFree(x1, "Norm-MakeRational");
00949             goto defaultcase;
00950         }
00951         x2 = NumberMalloc("Norm-MakeRational");
00952         if ( *arg2 == -SNUMBER ) {
00953             if ( arg2[1] <= 1 ) {
00954                 NumberFree(x2, "Norm-MakeRational");
00955                 NumberFree(x1, "Norm-MakeRational");
00956                 goto defaultcase;
00957             }
00958             else { x2[0] = arg2[1]; nx2 = 1; }
00959         }
00960         else if ( *arg2 > 0 ) {
00961             WORD *tc;
00962             nx2 = (ttstop[-1] - 1) / 2;
00963             tc = arg2 + ARGHEAD + 1 + nx2;
00964             if ( tc[0] != 1 ) {
00965                 NumberFree(x2, "Norm-MakeRational");
00966                 NumberFree(x1, "Norm-MakeRational");
00967                 goto defaultcase;
00968             }
00969             for ( i = 1; i < nx2; i++ ) if ( tc[i] != 0 ) {
00970                 NumberFree(x2, "Norm-MakeRational");
00971                 NumberFree(x1, "Norm-MakeRational");
00972                 goto defaultcase;
00973             }
00974             tc = arg2 + ARGHEAD + 1;
00975             for ( i = 0; i < nx2; i++ ) x2[i] = tc[i];
00976         }
00977         else {
00978             NumberFree(x2, "Norm-MakeRational");
00979             NumberFree(x1, "Norm-MakeRational");
00980             goto defaultcase;
00981         }
00982         if ( BigLong(x1, ABS(nx1), x2, nx2) >= 0 ) {
00983             UWORD *x3 = NumberMalloc("Norm-MakeRational");
00984             UWORD *x4 = NumberMalloc("Norm-MakeRational");
00985             WORD nx3, nx4;
00986             DivLong(x1, nx1, x2, nx2, x3, &nx3, x4, &nx4);
00987             for ( i = 0; i < ABS(nx4); i++ ) x1[i] = x4[i];
00988             nx1 = nx4;
00989             NumberFree(x4, "Norm-MakeRational");
00990             NumberFree(x3, "Norm-MakeRational");
00991         }
00992         xx = (UWORD *) (TermMalloc("Norm-MakeRational"));
00993         if ( MakeLongRational(BHEAD x1, nx1, x2, nx2, xx, &nxx) ) {
00994             static int warnflag = 1;

```



```

00995             if ( warnflag ) {
00996                 MesPrint("%w Warning: fraction could not be reconstructed in
MakeRational_");
00997                 warnflag = 0;
00998             }
00999             ncoef = REDLENG(ncoef);
01000             if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,x1,nx1) )
01001                 goto FromNorm;
01002         }
01003     else {
01004         ncoef = REDLENG(ncoef);
01005         if ( MulRat(BHEAD (UWORD *)n_coef,ncoef,xx,nxx,
(UWORD *)n_coef,&ncoef) ) goto FromNorm;
01006     }
01007     ncoef = INCLENG(ncoef);
01008     TermFree(xx,"Norm-MakeRational");
01009     NumberFree(x2,"Norm-MakeRational");
01010     NumberFree(x1,"Norm-MakeRational");
01011 }
01012 break;
01013 case TERMFUNCTION:
01014     if ( t[1] == FUNHEAD && AN.cTerm ) {
01015         ANsr = r; ANsm = m; ANsc = AN.cTerm;
01016         AN.cTerm = 0;
01017         t = ANsc + 1;
01018         m = ANsc + *ANsc;
01019         ncoef = REDLENG(ncoef);
01020         nnum = REDLENG(m[-1]);
01021         m -= ABS(m[-1]);
01022         if ( MulRat(BHEAD (UWORD *)n_coef,ncoef,(UWORD *)m,nnum,
(UWORD *)n_coef,&ncoef) ) goto FromNorm;
01023         ncoef = INCLENG(ncoef);
01024         r = t;
01025     }
01026     break;
01027 case FIRSTBRACKET:
01028     if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -EXPRESSION ) {
01029         if ( GetFirstBracket(termout,t[FUNHEAD+1]) < 0 ) goto FromNorm;
01030         if ( *termout == 0 ) goto NormZero;
01031         if ( *termout > 4 ) {
01032             WORD *r1, *r2, *r3;
01033             while ( r < m ) *t++ = *r++;
01034             r1 = term + *term;
01035             r2 = termout + *termout; r2 -= ABS(r2[-1]);
01036             while ( r < r1 ) *r2++ = *r++;
01037             r3 = termout + 1;
01038             while ( r3 < r2 ) *t++ = *r3++;
01039             *term = t - term;
01040             if ( AT.WorkPointer > term && AT.WorkPointer < t )
01041                 AT.WorkPointer = t;
01042             goto Restart;
01043         }
01044     }
01045     break;
01046 case FIRSTTERM:
01047 case CONTENTTERM:
01048     if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -EXPRESSION ) {
01049         EXPRESSIONS e = Expressions+t[FUNHEAD+1];
01050         POSITION oldondisk = AS.OldOnFile[t[FUNHEAD+1]];
01051         if ( e->replace == NEWLYDEFINEDEXPRESSION ) {
01052             AS.OldOnFile[t[FUNHEAD+1]] = e->onfile;
01053         }
01054         if ( *t == FIRSTTERM ) {
01055             if ( GetFirstTerm(termout,t[FUNHEAD+1],0) < 0 ) goto FromNorm;
01056         }
01057         else if ( *t == CONTENTTERM ) {
01058             if ( GetContent(termout,t[FUNHEAD+1]) < 0 ) goto FromNorm;
01059         }
01060         AS.OldOnFile[t[FUNHEAD+1]] = oldondisk;
01061         if ( *termout == 0 ) goto NormZero;
01062     }
01063     PasteIn::
01064     {
01065         WORD *r1, *r2, *r3, *r4, *r5, nrl, *rterm;
01066         r2 = termout + *termout; lnum = r2 - ABS(r2[-1]);
01067         nnum = REDLENG(r2[-1]);
01068         r1 = term + *term; r3 = r1 - ABS(r1[-1]);
01069         nrl = REDLENG(r1[-1]);
01070         if ( Mully(BHEAD (UWORD *)lnum,&nnum,(UWORD *)r3,nrl) ) goto FromNorm;
01071         nnum = INCLENG(nnum); nrl = ABS(nnum); lnum[nrl-1] = nnum;
01072         rterm = TermMalloc("FirstTerm/ContentTerm");
01073         r4 = rterm+1; r5 = term+1; while ( r5 < t ) *r4++ = *r5++;
01074         r5 = termout+1; while ( r5 < lnum ) *r4++ = *r5++;
01075         r5 = r; while ( r5 < r3 ) *r4++ = *r5++;
01076         r5 = lnum; NCOPY(r4,r5,nrl);
01077     }
01078 }
01079 }
01080

```

```

01081         *rterm = r4-rterm;
01082         nr1 = *rterm; r1 = term; r2 = rterm; NCOPY(r1,r2,nr1);
01083         TermFree(rterm,"FirstTerm/ContentTerm");
01084         if ( AT.WorkPointer > term && AT.WorkPointer < r1 )
01085             AT.WorkPointer = r1;
01086         goto Restart;
01087     }
01088 }
01089 else if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -DOLLAREXPRESSION ) {
01090     DOLLARS d = Dollars + t[FUNHEAD+1], newd = 0;
01091     int idol, ido;
01092 #ifdef WITHPTHREADS
01093     int nummodopt, dtype = -1;
01094     if ( AS.MultiThreaded && ( AC.mparallelfld == PARALLELFLAG ) ) {
01095         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01096             if ( t[FUNHEAD+1] == ModOptdollars[nummodopt].number ) break;
01097         }
01098         if ( nummodopt < NumModOptdollars ) {
01099             dtype = ModOptdollars[nummodopt].type;
01100             if ( dtype == MODLOCAL ) {
01101                 d = ModOptdollars[nummodopt].dstruct+AT.identity;
01102             }
01103         }
01104     }
01105 #endif
01106     if ( d->where && ( d->type == DOLTERMS || d->type == DOLNUMBER ) ) {
01107         newd = d;
01108     }
01109     else {
01110         if ( ( newd = DolToTerms(BHEAD t[FUNHEAD+1]) ) == 0 )
01111             goto NormZero;
01112     }
01113     if ( newd->where[0] == 0 ) {
01114         M_free(newd,"Copy of dollar variable");
01115         goto NormZero;
01116     }
01117     if ( *t == FIRSTTERM ) {
01118         idol = newd->where[0];
01119         for ( ido = 0; ido < idol; ido++ ) termout[idol] = newd->where[idol];
01120     }
01121     else if ( *t == CONTENTTERM ) {
01122         WORD *tterm;
01123         tterm = newd->where;
01124         idol = tterm[0];
01125         for ( ido = 0; ido < idol; ido++ ) termout[idol] = tterm[idol];
01126         tterm += *tterm;
01127         while ( *tterm ) {
01128             if ( ContentMerge(BHEAD termout,tterm) < 0 ) goto FromNorm;
01129             tterm += *tterm;
01130         }
01131     }
01132     if ( newd != d ) {
01133         if ( newd->factors ) M_free(newd->factors,"Dollar factors");
01134         M_free(newd,"Copy of dollar variable");
01135         newd = 0;
01136     }
01137     goto PasteIn;
01138 }
01139 break;
01140 case TERMSINEXPR:
01141 {
01142     if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -EXPRESSION ) {
01143         x = TermsInExpression(t[FUNHEAD+1]);
01144         if ( x == 0 ) goto NormZero;
01145         if ( x < 0 ) {
01146             x = -x;
01147             if ( x > (LONG)WORDMASK ) { lnum[0] = x & WORDMASK;
01148                 lnum[1] = x >> BITSINWORD; nnum = -2;
01149             }
01150             else { lnum[0] = x; nnum = -1; }
01151         }
01152         else if ( x > (LONG)WORDMASK ) {
01153             lnum[0] = x & WORDMASK;
01154             lnum[1] = x >> BITSINWORD;
01155             nnum = 2;
01156         }
01157         else { lnum[0] = x; nnum = 1; }
01158         ncoef = REDLENG(ncoef);
01159         if ( Mully(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)lnum,nnum) )
01160             goto FromNorm;
01161         ncoef = INCLENG(ncoef);
01162     }
01163     else if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -DOLLAREXPRESSION ) {
01164         x = TermsInDollar(t[FUNHEAD+1]);
01165         goto multermnum;
01166     }
01167     else { pcom[ncom++] = t; }

```

```

01168         }
01169         break;
01170     case SIZEOFFUNCTION:
01171     {
01172         if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -EXPRESSION ) {
01173             x = SizeOfExpression(t[FUNHEAD+1]);
01174             goto multermnum;
01175         }
01176         else if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -DOLLAREXPRESSION ) {
01177             x = SizeOfDollar(t[FUNHEAD+1]);
01178             goto multermnum;
01179         }
01180         else { pcom[ncom++] = t; }
01181     }
01182     break;
01183 case MATCHFUNCTION:
01184 case PATTERNFUNCTION:
01185     break;
01186 case BINOMIAL:
01187     /*
01188         Binomial function for internal use for the moment.
01189         The routine in reken.c should be more efficient.
01190     */
01191     if ( t[1] == FUNHEAD+4 && t[FUNHEAD] == -SNUMBER
01192         && t[FUNHEAD+1] >= 0 && t[FUNHEAD+2] == -SNUMBER
01193         && t[FUNHEAD+3] >= 0 && t[FUNHEAD+1] >= t[FUNHEAD+3] ) {
01194         if ( t[FUNHEAD+1] > t[FUNHEAD+3] ) {
01195             if ( GetBinom((UWORD *)lnum,&nnum,
01196                 t[FUNHEAD+1],t[FUNHEAD+3]) ) goto FromNorm;
01197             if ( nnum == 0 ) goto NormZero;
01198             goto MulIn;
01199         }
01200     }
01201     else pcom[ncom++] = t;
01202     break;
01203 case SIGNFUN:
01204     /*
01205         Numerical function giving (-1)^arg
01206     */
01207     if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER ) {
01208         if ( ( t[FUNHEAD+1] & 1 ) != 0 ) ncoef = -ncoef;
01209     }
01210     else if ( ( t[FUNHEAD] > 0 ) && ( t[1] == FUNHEAD+t[FUNHEAD] )
01211         && ( t[FUNHEAD] == ARGHEAD+1+abs(t[t[1]-1]) ) ) {
01212         UWORD *numer1,*denom1;
01213         WORD nsize = abs(t[t[1]-1]), nnsiz, isize;
01214         nnsiz = (nsize-1)/2;
01215         numer1 = (UWORD *) (t + FUNHEAD+ARGHEAD+1);
01216         denom1 = numer1 + nnsiz;
01217         for ( isize = 1; isize < nnsiz; isize++ ) {
01218             if ( denom1[isize] ) break;
01219         }
01220         if ( ( denom1[0] != 1 ) || isize < nnsiz ) {
01221             pcom[ncom++] = t;
01222         }
01223         else {
01224             if ( ( numer1[0] & 1 ) != 0 ) ncoef = -ncoef;
01225         }
01226     }
01227     else {
01228         goto doflags;
01229         pcom[ncom++] = t; /*
01230     */
01231     }
01232     break;
01233 case SIGFUNCTION:
01234     /*
01235         Numerical function giving the sign of the numerical argument
01236         The sign of zero is 1.
01237         If there are roots of unity they are part of the sign.
01238     */
01239     if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER ) {
01240         if ( t[FUNHEAD+1] < 0 ) ncoef = -ncoef;
01241     }
01242     else if ( ( t[1] == FUNHEAD+2 ) && ( t[FUNHEAD] == -SYMBOL )
01243         && ( ( t[FUNHEAD+1] <= NumSymbols && t[FUNHEAD+1] > -MAXPOWER )
01244         && ( symbols[t[FUNHEAD+1]].complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY )
01245     ) {
01246         k = t[FUNHEAD+1];
01247         from = m;
01248         i = nsym;
01249         m = ppsym;
01250         if ( i > 0 ) do {
01251             m -= 2;
01252             if ( k == *m ) {
01253                 m++;
01254                 *m = *m + 1;
01255                 *m %= symbols[k].maxpower;

```

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01254         if ( ( symbols[k].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
01255             if ( ( ( symbols[k].maxpower & 1 ) == 0 ) &&
01256                 ( *m >= symbols[k].maxpower/2 ) ) {
01257                 *m -= symbols[k].maxpower/2; ncoef = -ncoef;
01258             }
01259         }
01260         if ( !*m ) {
01261             m--;
01262             while ( i < nsym )
01263                 { *m = m[2]; m++; *m = m[2]; m++; i++; }
01264             ppsym -= 2;
01265             nsym--;
01266         }
01267         goto sigDoneSymbol;
01268     }
01269     } while ( k < *m && --i > 0 );
01270     m = ppsym;
01271     while ( i < nsym )
01272         { m--; m[2] = *m; m--; m[2] = *m; i++; }
01273     *m++ = k;
01274     *m = 1;
01275     ppsym += 2;
01276     nsym++;
01277 sigDoneSymbol:
01278     m = from;
01279 }
01280 else if ( ( t[FUNHEAD] > 0 ) && ( t[1] == FUNHEAD+t[FUNHEAD] ) ) {
01281     if ( t[FUNHEAD] == ARGHEAD+1+abs(t[t[1]-1]) ) {
01282         if ( t[t[1]-1] < 0 ) ncoef = -ncoef;
01283     }
01284     /*
01285     Now we should fish out the roots of unity
01286     */
01287     else if ( ( t[FUNHEAD+ARGHEAD]+FUNHEAD+ARGHEAD == t[1] )
01288         && ( t[FUNHEAD+ARGHEAD+1] == SYMBOL ) ) {
01289         WORD *ts = t + FUNHEAD+ARGHEAD+3;
01290         WORD its = ts[-1]-2;
01291         while ( its > 0 ) {
01292             if ( ( *ts != 0 ) && ( ( *ts > NumSymbols || *ts <= -MAXPOWER )
01293                 || ( symbols[*ts].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
01294             ) {
01295                 goto signogood;
01296             }
01297             ts += 2; its -= 2;
01298         }
01299         /*
01300         Now we have only roots of unity which should be
01301         registered in the list of symbols.
01302         */
01303         if ( t[t[1]-1] < 0 ) ncoef = -ncoef;
01304         ts = t + FUNHEAD+ARGHEAD+3;
01305         its = ts[-1]-2;
01306         from = m;
01307         while ( its > 0 ) {
01308             i = nsym;
01309             m = ppsym;
01310             if ( i > 0 ) do {
01311                 m -= 2;
01312                 if ( *ts == *m ) {
01313                     ts++; m++;
01314                     *m += *ts;
01315                     if ( ( ts[-1] <= NumSymbols && ts[-1] > -MAXPOWER ) &&
01316                         ( symbols[ts[-1]].complex & VARTYPEROOTOFUNITY ) ==
01317                         VARTYPEROOTOFUNITY ) {
01318                         *m %= symbols[ts[-1]].maxpower;
01319                         if ( *m < 0 ) *m += symbols[ts[-1]].maxpower;
01320                         if ( ( symbols[ts[-1]].complex & VARTYPEMINUS ) ==
01321                             VARTYPEMINUS ) {
01322                             if ( ( ( symbols[ts[-1]].maxpower & 1 ) == 0 ) &&
01323                                 ( *m >= symbols[ts[-1]].maxpower/2 ) ) {
01324                                 *m -= symbols[ts[-1]].maxpower/2; ncoef = -ncoef;
01325                             }
01326                         }
01327                     }
01328                     if ( !*m ) {
01329                         m--;
01330                         while ( i < nsym )
01331                             { *m = m[2]; m++; *m = m[2]; m++; i++; }
01332                         ppsym -= 2;
01333                         nsym--;
01334                     }
01335                     ts++; its -= 2;
01336                     goto sigNextSymbol;
01337                 }
01338             } while ( *ts < *m && --i > 0 );
01339             m = ppsym;
01340             while ( i < nsym )

```

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01338             { m--; m[2] = *m; m--; m[2] = *m; i++; }
01339             *m++ = *ts++;
01340             *m = *ts++;
01341             ppsym += 2;
01342             nsym++; its -= 2;
01343 sigNextSymbol:;
01344             }
01345             m = from;
01346             }
01347             else {
01348 signogood:             pcom[ncom++] = t;
01349             }
01350             }
01351             else pcom[ncom++] = t;
01352             break;
01353 case ABSFUNCTION:
01354 /*
01355      Numerical function giving the absolute value of the
01356      numerical argument. Or roots of unity.
01357 */
01358     if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER ) {
01359         k = t[FUNHEAD+1];
01360         if ( k < 0 ) k = -k;
01361         if ( k == 0 ) goto NormZero;
01362         *((UWORD *)lnum) = k; nnum = 1;
01363         goto MulIn;
01364     }
01365     }
01366     else if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SYMBOL ) {
01367         k = t[FUNHEAD+1];
01368         if ( ( k > NumSymbols || k <= -MAXPOWER )
01369             || ( symbols[k].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
01370             goto absnogood;
01371     }
01372     else if ( ( t[FUNHEAD] > 0 ) && ( t[1] == FUNHEAD+t[FUNHEAD] )
01373     && ( t[1] == FUNHEAD+ARGHEAD+t[FUNHEAD+ARGHEAD] ) ) {
01374         if ( t[FUNHEAD] == ARGHEAD+1+abs(t[t[1]-1]) ) {
01375             WORD *ts;
01376             ts = t + t[1] - 1;
01377             ncoef = REDLENG(ncoef);
01378             nnum = REDLENG(*ts);
01379             if ( nnum < 0 ) nnum = -nnum;
01380             if ( MulRat(BHEAD (UWORD *)n_coef,ncoef,
01381             (UWORD *) (ts-ABS(*ts)+1),nnum,
01382             (UWORD *)n_coef,&ncoef) ) goto FromNorm;
01383             ncoef = INCLENG(ncoef);
01384         }
01385     /*
01386      Now get rid of the roots of unity. This includes i_
01387 */
01388     else if ( t[FUNHEAD+ARGHEAD+1] == SYMBOL ) {
01389         WORD *ts = t+FUNHEAD+ARGHEAD+1;
01390         WORD its = ts[1] - 2;
01391         ts += 2;
01392         while ( its > 0 ) {
01393             if ( *ts == 0 ) { }
01394             else if ( ( *ts > NumSymbols || *ts <= -MAXPOWER )
01395                 || ( symbols[*ts].complex & VARTYPEROOTOFUNITY )
01396                 != VARTYPEROOTOFUNITY ) goto absnogood;
01397             its -= 2; ts += 2;
01398         }
01399         goto absnosymbols;
01400     }
01401     else {
01402 absnogood:             pcom[ncom++] = t;
01403     }
01404     }
01405     else pcom[ncom++] = t;
01406     break;
01407 case MODFUNCTION:
01408 case MOD2FUNCTION:
01409 /*
01410      Mod function. Does work if two arguments and the
01411      second argument is a positive short number
01412 */
01413     if ( t[1] == FUNHEAD+4 && t[FUNHEAD] == -SNUMBER
01414     && t[FUNHEAD+2] == -SNUMBER && t[FUNHEAD+3] > 1 ) {
01415         WORD tmod;
01416         tmod = (t[FUNHEAD+1]%t[FUNHEAD+3]);
01417         if ( tmod < 0 ) tmod += t[FUNHEAD+3];
01418         if ( *t == MOD2FUNCTION && tmod > t[FUNHEAD+3]/2 )
01419             tmod -= t[FUNHEAD+3];
01420         if ( tmod < 0 ) {
01421             *((UWORD *)lnum) = -tmod;
01422             nnum = -1;
01423         }
01424         else if ( tmod > 0 ) {

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01425         *((UWORD *)lnum) = tmod;
01426         nnum = 1;
01427     }
01428     else goto NormZero;
01429     goto MulIn;
01430 }
01431 else if ( t[1] > t[FUNHEAD+2] && t[FUNHEAD] > 0
01432 && t[FUNHEAD+t[FUNHEAD]] == -SNUMBER
01433 && t[FUNHEAD+t[FUNHEAD]+1] > 1
01434 && t[1] == FUNHEAD+2+t[FUNHEAD] ) {
01435     WORD *ttt = t+FUNHEAD, iii;
01436     iii = ttt[ttt-1];
01437     if ( *ttt == ttt[ARGHEAD]+ARGHEAD &&
01438         ttt[ARGHEAD] == ABS(iii)+1 ) {
01439         WORD ncm = 1;
01440         WORD cm = ttt[*ttt+1];
01441         iii = REDLENG(iii);
01442         if ( *t == MODFUNCTION ) {
01443             if ( TakeModulus((UWORD *) (ttt+ARGHEAD+1)
01444                 , &iii, (UWORD *) (&cm), ncm, UNPACK|NOINVERSES) )
01445                 goto FromNorm;
01446         }
01447         else {
01448             if ( TakeModulus((UWORD *) (ttt+ARGHEAD+1)
01449                 , &iii, (UWORD *) (&cm), ncm, UNPACK|POSNEG|NOINVERSES) )
01450                 goto FromNorm;
01451         }
01452         if ( *t == MOD2FUNCTION && ttt[ARGHEAD+1] > cm/2 && iii > 0 ) {
01453             ttt[ARGHEAD+1] -= cm;
01454         }
01455         if ( ttt[ARGHEAD+1] < 0 ) {
01456             *((UWORD *)lnum) = -ttt[ARGHEAD+1];
01457             nnum = -1;
01458         }
01459         else if ( ttt[ARGHEAD+1] > 0 ) {
01460             *((UWORD *)lnum) = ttt[ARGHEAD+1];
01461             nnum = 1;
01462         }
01463         else goto NormZero;
01464         goto MulIn;
01465     }
01466 }
01467 else if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER ) {
01468     *((UWORD *)lnum) = t[FUNHEAD+1];
01469     if ( *lnum == 0 ) goto NormZero;
01470     nnum = 1;
01471     goto MulIn;
01472 }
01473 else if ( ( t[FUNHEAD] < 0 ) && ( t[FUNHEAD] == -SNUMBER )
01474 && ( t[1] >= ( FUNHEAD+6+ARGHEAD ) )
01475 && ( t[FUNHEAD+2] >= 4+ARGHEAD )
01476 && ( t[t[1]-1] == t[FUNHEAD+2+ARGHEAD]-1 ) ) ||
01477 ( t[FUNHEAD] > 0 )
01478 && ( t[FUNHEAD]-ARGHEAD-1 == ABS(t[FUNHEAD+t[FUNHEAD]-1]) )
01479 && ( t[FUNHEAD+t[FUNHEAD]]-ARGHEAD-1 == t[t[1]-1] ) ) {
01480 /*
01481     Check that the last (long) number is integer
01482 */
01483     WORD *ttt = t + t[1], iii, iiil;
01484     UWORD coefbuf[2], *coef2, ncoef2;
01485     iii = (ttt[-1]-1)/2;
01486     ttt -= iii;
01487     if ( ttt[-1] != 1 ) {
01488 exitfromhere:
01489         pcom[ncom++] = t;
01490         break;
01491     }
01492     iii--;
01493     for ( iiil = 0; iiil < iii; iiil++ ) {
01494         if ( ttt[iiil] != 0 ) goto exitfromhere;
01495     }
01496 /*
01497     Now we have a hit!
01498     The first argument will be put in lnum.
01499     It will be a rational.
01500     The second argument will be a long integer in coef2.
01501 */
01502     ttt = t + FUNHEAD;
01503     if ( *ttt < 0 ) {
01504         if ( ttt[1] < 0 ) {
01505             nnum = -1; lnum[0] = -ttt[1]; lnum[1] = 1;
01506         }
01507         else {
01508             nnum = 1; lnum[0] = ttt[1]; lnum[1] = 1;
01509         }
01510     }
01511     else {

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01512         nnum = ABS(ttt[ttt[0]-1] - 1);
01513         for ( iii = 0; iii < nnum; iii++ ) {
01514             lnum[iii] = ttt[ARGHEAD+1+iii];
01515         }
01516         nnum = nnum/2;
01517         if ( ttt[ttt[0]-1] < 0 ) nnum = -nnum;
01518     }
01519     NEXTARG(ttt);
01520     if ( *ttt < 0 ) {
01521         coef2 = coefbuf;
01522         ncoef2 = 3; *coef2 = (UWORD) (ttt[1]);
01523         coef2[1] = 1;
01524     }
01525     else {
01526         coef2 = (UWORD *) (ttt+ARGHEAD+1);
01527         ncoef2 = (ttt[ttt[0]-1]-1)/2;
01528     }
01529     if ( TakeModulus((UWORD *)lnum,&nnum,(UWORD *)coef2,ncoef2,
01530         UNPACK|NOINVERSES|FROMFUNCTION) ) {
01531         goto FromNorm;
01532     }
01533     if ( *t == MOD2FUNCTION && nnum > 0 ) {
01534         UWORD *coef3 = NumberMalloc("Mod2Function"), two = 2;
01535         WORD ncoef3;
01536         if ( MulLong((UWORD *)lnum,nnum,&two,1,coef3,&ncoef3) )
01537             goto FromNorm;
01538         if ( BigLong(coef3,ncoef3,(UWORD *)coef2,ncoef2) > 0 ) {
01539             nnum = -nnum;
01540             AddLong((UWORD *)lnum,nnum,(UWORD *)coef2,ncoef2
01541                 , (UWORD *)lnum,&nnum);
01542             nnum = -nnum;
01543         }
01544         NumberFree(coef3,"Mod2Function");
01545     }
01546     /*
01547     Do we have to pack? No, because the answer is not a fraction
01548     */
01549     ncoef = REDLENG(ncoef);
01550     if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) )
01551         goto FromNorm;
01552     ncoef = INCLENG(ncoef);
01553 }
01554 else pcom[ncom++] = t;
01555 break;
01556 case EXTEUCLIDEAN:
01557 {
01558     WORD argcount = 0, *tc, *ts, xc, xs, *tcc;
01559     UWORD *Num1, *Num2, *Num3, *Num4;
01560     WORD size1, size2, size3, size4, space;
01561     tc = t+FUNHEAD; ts = t + t[1];
01562     while ( argcount < 3 && tc < ts ) { NEXTARG(tc); argcount++; }
01563     if ( argcount != 2 ) goto defaultcase;
01564     if ( t[FUNHEAD] == -SNUMBER ) {
01565         if ( t[FUNHEAD+1] <= 1 ) goto defaultcase;
01566         if ( t[FUNHEAD+2] == -SNUMBER ) {
01567             if ( t[FUNHEAD+3] <= 1 ) goto defaultcase;
01568             Num2 = NumberMalloc("modinverses");
01569             *Num2 = t[FUNHEAD+3]; size2 = 1;
01570         }
01571         else {
01572             if ( ts[-1] < 0 ) goto defaultcase;
01573             if ( ts[-1] != t[FUNHEAD+2]-ARGHEAD-1 ) goto defaultcase;
01574             xs = (ts[-1]-1)/2;
01575             tcc = ts-xs-1;
01576             if ( *tcc != 1 ) goto defaultcase;
01577             for ( i = 1; i < xs; i++ ) {
01578                 if ( tcc[i] != 0 ) goto defaultcase;
01579             }
01580             Num2 = NumberMalloc("modinverses");
01581             size2 = xs;
01582             for ( i = 0; i < xs; i++ ) Num2[i] = t[FUNHEAD+ARGHEAD+3+i];
01583         }
01584         Num1 = NumberMalloc("modinverses");
01585         *Num1 = t[FUNHEAD+1]; size1 = 1;
01586     }
01587     else {
01588         tc = t + FUNHEAD + t[FUNHEAD];
01589         if ( tc[-1] < 0 ) goto defaultcase;
01590         if ( tc[-1] != t[FUNHEAD]-ARGHEAD-1 ) goto defaultcase;
01591         xc = (tc[-1]-1)/2;
01592         tcc = tc-xc-1;
01593         if ( *tcc != 1 ) goto defaultcase;
01594         for ( i = 1; i < xc; i++ ) {
01595             if ( tcc[i] != 0 ) goto defaultcase;
01596         }
01597         if ( *tc == -SNUMBER ) {
01598             if ( tc[1] <= 1 ) goto defaultcase;

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01599         Num2 = NumberMalloc("modinverses");
01600         *Num2 = tc[1]; size2 = 1;
01601     }
01602     else {
01603         if ( ts[-1] < 0 ) goto defaultcase;
01604         if ( ts[-1] != t[FUNHEAD+2]-ARGHEAD-1 ) goto defaultcase;
01605         xs = (ts[-1]-1)/2;
01606         tcc = ts-xs-1;
01607         if ( *tcc != 1 ) goto defaultcase;
01608         for ( i = 1; i < xs; i++ ) {
01609             if ( tcc[i] != 0 ) goto defaultcase;
01610         }
01611         Num2 = NumberMalloc("modinverses");
01612         size2 = xs;
01613         for ( i = 0; i < xs; i++ ) Num2[i] = tc[ARGHEAD+1+i];
01614     }
01615     Num1 = NumberMalloc("modinverses");
01616     size1 = xc;
01617     for ( i = 0; i < xc; i++ ) Num1[i] = t[FUNHEAD+ARGHEAD+1+i];
01618 }
01619 Num3 = NumberMalloc("modinverses");
01620 Num4 = NumberMalloc("modinverses");
01621 GetLongModInverses(BHEAD Num1, size1, Num2, size2
01622                   , Num3, &size3, Num4, &size4);
01623 /*
01624     Now we have to compose the answer. This needs more space
01625     and hence we have to put this inside the term.
01626     Compute first how much extra space we need.
01627     Then move the trailing part of the term upwards.
01628     Do not forget relevant pointers!!! (r, m, termout, AT.WorkPointer)
01629 */
01630 space = 0;
01631 if ( ( size3 == 1 || size3 == -1 ) && (*Num3&TOPBITONLY) == 0 ) space += 2;
01632 else space += ARGHEAD + 2*ABS(size3) + 2;
01633 if ( ( size4 == 1 || size4 == -1 ) && (*Num4&TOPBITONLY) == 0 ) space += 2;
01634 else space += ARGHEAD + 2*ABS(size4) + 2;
01635 tt = term + *term; u = tt + space;
01636 while ( tt >= ts ) *--u = *--tt;
01637 m += space; r += space;
01638 *term += space;
01639 t[1] += space;
01640 if ( ( size3 == 1 || size3 == -1 ) && (*Num3&TOPBITONLY) == 0 ) {
01641     *ts++ = -SNUMBER; *ts = (WORD) (*Num3);
01642     if ( size3 < 0 ) *ts = -*ts;
01643     ts++;
01644 }
01645 else {
01646     *ts++ = 2*ABS(size3)+ARGHEAD+2;
01647     *ts++ = 0; FILLARG(ts)
01648     *ts++ = 2*ABS(size3)+1;
01649     for ( i = 0; i < ABS(size3); i++ ) *ts++ = Num3[i];
01650     *ts++ = 1;
01651     for ( i = 1; i < ABS(size3); i++ ) *ts++ = 0;
01652     if ( size3 < 0 ) *ts++ = 2*size3-1;
01653     else *ts++ = 2*size3+1;
01654 }
01655 if ( ( size4 == 1 || size4 == -1 ) && (*Num4&TOPBITONLY) == 0 ) {
01656     *ts++ = -SNUMBER; *ts = *Num4;
01657     if ( size4 < 0 ) *ts = -*ts;
01658     ts++;
01659 }
01660 else {
01661     *ts++ = 2*ABS(size4)+ARGHEAD+2;
01662     *ts++ = 0; FILLARG(ts)
01663     *ts++ = 2*ABS(size4)+2;
01664     for ( i = 0; i < ABS(size4); i++ ) *ts++ = Num4[i];
01665     *ts++ = 1;
01666     for ( i = 1; i < ABS(size4); i++ ) *ts++ = 0;
01667     if ( size4 < 0 ) *ts++ = 2*size4-1;
01668     else *ts++ = 2*size4+1;
01669 }
01670 NumberFree(Num4, "modinverses");
01671 NumberFree(Num3, "modinverses");
01672 NumberFree(Num1, "modinverses");
01673 NumberFree(Num2, "modinverses");
01674 t[2] = 0; /* mark function as clean. */
01675 goto Restart;
01676 }
01677 break;
01678 case GCDFUNCTION:
01679 #ifdef EVALUATEGCD
01680 #ifdef NEWGCDFUNCTION
01681 {
01682 /*
01683     Has two integer arguments
01684     Four cases: S,S, S,L, L,S, L,L
01685 */

```



```

01686 WORD *num1, *num2, size1, size2, stor1, stor2, *ttt, ti;
01687 if ( t[1] == FUNHEAD+4 && t[FUNHEAD] == -SNUMBER
01688 && t[FUNHEAD+2] == -SNUMBER && t[FUNHEAD+1] != 0
01689 && t[FUNHEAD+3] != 0 ) { /* Short,Short */
01690     stor1 = t[FUNHEAD+1];
01691     stor2 = t[FUNHEAD+3];
01692     if ( stor1 < 0 ) stor1 = -stor1;
01693     if ( stor2 < 0 ) stor2 = -stor2;
01694     num1 = &stor1; num2 = &stor2;
01695     size1 = size2 = 1;
01696     goto gcdcalc;
01697 }
01698 else if ( t[1] > FUNHEAD+4 ) {
01699     if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] != 0
01700 && t[FUNHEAD+2] == t[1]-FUNHEAD-2 &&
01701 ABS(t[t[1]-1]) == t[FUNHEAD+2]-1-ARGHEAD ) { /* Short,Long */
01702     num2 = t + t[1];
01703     size2 = ABS(num2[-1]);
01704     ttt = num2-1;
01705     num2 -= size2;
01706     size2 = (size2-1)/2;
01707     ti = size2;
01708     while ( ti > 1 && ttt[-1] == 0 ) { ttt--; ti--; }
01709     if ( ti == 1 && ttt[-1] == 1 ) {
01710         stor1 = t[FUNHEAD+1];
01711         if ( stor1 < 0 ) stor1 = -stor1;
01712         num1 = &stor1;
01713         size1 = 1;
01714         goto gcdcalc;
01715     }
01716     else pcom[ncom++] = t;
01717 }
01718 else if ( t[FUNHEAD] > 0 &&
01719 t[FUNHEAD]-1-ARGHEAD == ABS(t[t[FUNHEAD]+FUNHEAD-1]) ) {
01720     num1 = t + FUNHEAD + t[FUNHEAD];
01721     size1 = ABS(num1[-1]);
01722     ttt = num1-1;
01723     num1 -= size1;
01724     size1 = (size1-1)/2;
01725     ti = size1;
01726     while ( ti > 1 && ttt[-1] == 0 ) { ttt--; ti--; }
01727     if ( ti == 1 && ttt[-1] == 1 ) {
01728         if ( t[1]-FUNHEAD == t[FUNHEAD]+2 && t[t[1]-2] == -SNUMBER
01729 && t[t[1]-1] != 0 ) { /* Long,Short */
01730             stor2 = t[t[1]-1];
01731             if ( stor2 < 0 ) stor2 = -stor2;
01732             num2 = &stor2;
01733             size2 = 1;
01734             goto gcdcalc;
01735         }
01736         else if ( t[1]-FUNHEAD == t[FUNHEAD]+t[FUNHEAD+t[FUNHEAD]]
01737 && ABS(t[t[1]-1]) == t[FUNHEAD+t[FUNHEAD]] - ARGHEAD-1 ) {
01738             num2 = t + t[1];
01739             size2 = ABS(num2[-1]);
01740             ttt = num2-1;
01741             num2 -= size2;
01742             size2 = (size2-1)/2;
01743             ti = size2;
01744             while ( ti > 1 && ttt[-1] == 0 ) { ttt--; ti--; }
01745             if ( ti == 1 && ttt[-1] == 1 ) {
01746 gcdcalc:
01747                 if ( GcdLong(BHEAD (UWORD *)num1,size1,(UWORD *)num2,size2
01748 , (UWORD *)lnum,&nnum) ) goto FromNorm;
01749                 goto MulIn;
01750             }
01751             else pcom[ncom++] = t;
01752         }
01753         else pcom[ncom++] = t;
01754     }
01755     else pcom[ncom++] = t;
01756 }
01757 }
01758 else pcom[ncom++] = t;
01759 }
01760 #else
01761 {
01762     WORD *gcd = AT.WorkPointer;
01763     if ( ( gcd = EvaluateGcd(BHEAD t) ) == 0 ) goto FromNorm;
01764     if ( *gcd == 4 && gcd[1] == 1 && gcd[2] == 1 && gcd[4] == 0 ) {
01765         AT.WorkPointer = gcd;
01766     }
01767     else if ( gcd[*gcd] == 0 ) {
01768         WORD *t1, iii, change, *num, *den, numsize, densize;
01769         if ( gcd[*gcd-1] < *gcd-1 ) {
01770             t1 = gcd+1;
01771             for ( iii = 2; iii < t1[1]; iii += 2 ) {
01772                 change = ExtraSymbol(t1[iii],t1[iii+1],nsym,ppsym,&ncoef);

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```

01773             nsym += change;
01774             ppsym += change * 2;
01775         }
01776     }
01777     t1 = gcd + *gcd;
01778     iii = t1[-1]; num = t1-iii; numsize = (iii-1)/2;
01779     den = num + numsize; densize = numsize;
01780     while ( numsize > 1 && num[numsize-1] == 0 ) numsize--;
01781     while ( densize > 1 && den[densize-1] == 0 ) densize--;
01782     if ( numsize > 1 || num[0] != 1 ) {
01783         ncoef = REDLENG(ncoef);
01784         if ( Mullly(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)num,numsize) ) goto
FromNorm;
01785         ncoef = INCLENG(ncoef);
01786     }
01787     if ( densize > 1 || den[0] != 1 ) {
01788         ncoef = REDLENG(ncoef);
01789         if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)den,densize) ) goto
FromNorm;
01790         ncoef = INCLENG(ncoef);
01791     }
01792     AT.WorkPointer = gcd;
01793 }
01794 else { /* a whole expression */
01795 /*
01796     Action: Put the expression in a compiler buffer.
01797     Insert a SUBEXPRESSION subterm
01798     Set the return value of the routine such that in
01799     Generator the term gets sent again to TestSub.
01800
01801     1: put in C (ebufnum)
01802     2: after that the WorkSpace is free again.
01803     3: insert the SUBEXPRESSION
01804     4: copy the top part of the term down
01805 */
01806     LONG size = AT.WorkPointer - gcd;
01807
01808     ss = AddrRHS(AT.ebufnum,1);
01809     while ( (ss + size + 10) > C->Top ) ss = DoubleCbuffer(AT.ebufnum,ss,13);
01810     tt = gcd;
01811     NCOPY(ss,tt,size);
01812     C->rhs[C->numrhs+1] = ss;
01813     C->Pointer = ss;
01814
01815     t[0] = SUBEXPRESSION;
01816     t[1] = SUBEXPSIZE;
01817     t[2] = C->numrhs;
01818     t[3] = 1;
01819     t[4] = AT.ebufnum;
01820     t += 5;
01821     tt = term + *term;
01822     while ( r < tt ) *t++ = *r++;
01823     *term = t - term;
01824
01825     regval = 1;
01826     goto RegEnd;
01827 }
01828 }
01829 #endif
01830 #else
01831     MesPrint(" Unexpected call to EvaluateGCD");
01832     Terminate(-1);
01833 #endif
01834 break;
01835 case MINFUNCTION:
01836 case MAXFUNCTION:
01837     if ( t[1] == FUNHEAD ) break;
01838     {
01839         WORD *ttt = t + FUNHEAD;
01840         WORD *tttstop = t + t[1];
01841         WORD tterm[4], iii;
01842         while ( ttt < tttstop ) {
01843             if ( *ttt > 0 ) {
01844                 if ( ttt[ARGHEAD]-1 > ABS(ttt[*ttt-1]) ) goto nospec;
01845                 ttt += *ttt;
01846             }
01847             else {
01848                 if ( *ttt != -SNUMBER ) goto nospec;
01849                 ttt += 2;
01850             }
01851         }
01852     }
01853 /*
01854     Function has only numerical arguments
01855     Pick up the first argument.
01856 */
01857     ttt = t + FUNHEAD;
01858     if ( *ttt > 0 ) {

```

```

01858 loadnew1:
01859     for ( iii = 0; iii < ttt[ARGHEAD]; iii++ )
01860         n_llnum[iii] = ttt[ARGHEAD+iii];
01861     ttt += *ttt;
01862 }
01863 else {
01864 loadnew2:
01865     if ( ttt[1] == 0 ) {
01866         n_llnum[0] = n_llnum[1] = n_llnum[2] = n_llnum[3] = 0;
01867     }
01868     else {
01869         n_llnum[0] = 4;
01870         if ( ttt[1] > 0 ) { n_llnum[1] = ttt[1]; n_llnum[3] = 3; }
01871         else { n_llnum[1] = -ttt[1]; n_llnum[3] = -3; }
01872         n_llnum[2] = 1;
01873     }
01874     ttt += 2;
01875 }
01876 /*
01877 Now loop over the other arguments
01878 */
01879 while ( ttt < tttstop ) {
01880     if ( *ttt > 0 ) {
01881         if ( n_llnum[0] == 0 ) {
01882             if ( ( *t == MINFUNCTION && ttt[*ttt-1] < 0 )
01883                 || ( *t == MAXFUNCTION && ttt[*ttt-1] > 0 ) )
01884                 goto loadnew1;
01885         }
01886         else {
01887             ttt += ARGHEAD;
01888             iii = CompCoef(n_llnum,ttt);
01889             if ( ( iii > 0 && *t == MINFUNCTION )
01890                 || ( iii < 0 && *t == MAXFUNCTION ) ) {
01891                 for ( iii = 0; iii < ttt[0]; iii++ )
01892                     n_llnum[iii] = ttt[iii];
01893             }
01894         }
01895         ttt += *ttt;
01896     }
01897     else {
01898         if ( n_llnum[0] == 0 ) {
01899             if ( ( *t == MINFUNCTION && ttt[1] < 0 )
01900                 || ( *t == MAXFUNCTION && ttt[1] > 0 ) )
01901                 goto loadnew2;
01902         }
01903         else if ( ttt[1] == 0 ) {
01904             if ( ( *t == MINFUNCTION && n_llnum[*n_llnum-1] > 0 )
01905                 || ( *t == MAXFUNCTION && n_llnum[*n_llnum-1] < 0 ) ) {
01906                 n_llnum[0] = 0;
01907             }
01908         }
01909         else {
01910             tterm[0] = 4; tterm[2] = 1;
01911             if ( ttt[1] < 0 ) { tterm[1] = -ttt[1]; tterm[3] = -3; }
01912             else { tterm[1] = ttt[1]; tterm[3] = 3; }
01913             iii = CompCoef(n_llnum,tterm);
01914             if ( ( iii > 0 && *t == MINFUNCTION )
01915                 || ( iii < 0 && *t == MAXFUNCTION ) ) {
01916                 for ( iii = 0; iii < 4; iii++ )
01917                     n_llnum[iii] = tterm[iii];
01918             }
01919             ttt += 2;
01920         }
01921     }
01922 }
01923 if ( n_llnum[0] == 0 ) goto NormZero;
01924 ncoef = REDLENG(ncoef);
01925 nnum = REDLENG(n_llnum[*n_llnum-1]);
01926 if ( MulRat(BHEAD (UWORD *)n_coef,ncoef, (UWORD *)lnum,nnum,
01927 (UWORD *)n_coef,&ncoef) ) goto FromNorm;
01928 ncoef = INCLENG(ncoef);
01929 }
01930 break;
01931 case INVERSEFACTORIAL:
01932     if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] >= 0 ) {
01933         if ( Factorial(BHEAD t[FUNHEAD+1], (UWORD *)lnum,&nnum) )
01934             goto FromNorm;
01935         ncoef = REDLENG(ncoef);
01936         if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)lnum,nnum) ) goto FromNorm;
01937         ncoef = INCLENG(ncoef);
01938     }
01939     else {
01940 nospec:
01941         pcom[ncom++] = t;
01942     }
01943     break;
01944 case MAXPOWEROF:
01945     if ( ( t[FUNHEAD] == -SYMBOL )

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```

01945         && ( t[FUNHEAD+1] > 0 ) && ( t[1] == FUNHEAD+2 ) ) {
01946         *((UWORD *)lnum) = symbols[t[FUNHEAD+1]].maxpower;
01947         nnum = 1;
01948         goto MulIn;
01949     }
01950     else { pcom[ncom++] = t; }
01951     break;
01952 case MINPOWEROF:
01953     if ( ( t[FUNHEAD] == -SYMBOL )
01954         && ( t[FUNHEAD] > 0 ) && ( t[1] == FUNHEAD+2 ) ) {
01955         *((UWORD *)lnum) = symbols[t[FUNHEAD+1]].minpower;
01956         nnum = 1;
01957         goto MulIn;
01958     }
01959     else { pcom[ncom++] = t; }
01960     break;
01961 case PRIMENUMBER :
01962     if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER
01963         && t[FUNHEAD+1] > 0 ) {
01964         UWORD xp = (UWORD) (NextPrime (BHEAD t[FUNHEAD+1]));
01965         ncoef = REDLENG(ncoef);
01966         if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,&xp,1) ) goto FromNorm;
01967         ncoef = INCLENG(ncoef);
01968     }
01969     else goto defaultcase;
01970     break;
01971 case MOEBIUS:
01972 /*
01973     Numerical function giving -1,0,1 or no evaluation
01974 */
01975     if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER
01976         && t[FUNHEAD+1] > 0 ) {
01977         WORD val = Moebius(BHEAD t[FUNHEAD+1]);
01978         if ( val == 0 ) goto NormZero;
01979         if ( val < 0 ) ncoef = -ncoef;
01980     }
01981     else {
01982         pcom[ncom++] = t;
01983     }
01984     break;
01985 case LNUMBER :
01986     ncoef = REDLENG(ncoef);
01987     if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *) (t+3),t[2]) ) goto FromNorm;
01988     ncoef = INCLENG(ncoef);
01989     break;
01990 case SNUMBER :
01991     if ( t[2] < 0 ) {
01992         t[2] = -t[2];
01993         if ( t[3] & 1 ) ncoef = -ncoef;
01994     }
01995     else if ( t[2] == 0 ) {
01996         if ( t[3] < 0 ) goto NormInf;
01997         goto NormZero;
01998     }
01999     lnum[0] = t[2];
02000     nnum = 1;
02001     if ( t[3] && RaisPow(BHEAD (UWORD *)lnum,&nnum,(UWORD) (ABS(t[3]))) ) goto FromNorm;
02002     ncoef = REDLENG(ncoef);
02003     if ( t[3] < 0 ) {
02004         if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) )
02005             goto FromNorm;
02006     }
02007     else if ( t[3] > 0 ) {
02008         if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) )
02009             goto FromNorm;
02010     }
02011     ncoef = INCLENG(ncoef);
02012     break;
02013 case GAMMA :
02014 case GAMMAI :
02015 case GAMMAFIVE :
02016 case GAMMASIX :
02017 case GAMMASEVEN :
02018     if ( t[1] == FUNHEAD ) {
02019         MLOCK(ErrorMessageLock);
02020         MesPrint("Gamma matrix without spin line encountered.");
02021         MUNLOCK(ErrorMessageLock);
02022         goto NormMin;
02023     }
02024     pnco[nco++] = t;
02025     t += FUNHEAD+1;
02026     goto ScanCont;
02027 case LEVICIVITA :
02028     peps[neps++] = t;
02029     if ( ( t[2] & DIRTYFLAG ) == DIRTYFLAG ) {
02030         t[2] &= ~DIRTYFLAG;
02031         t[2] |= DIRTYSYMLFLAG;

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```

02032     }
02033     t += FUNHEAD;
02034 ScanCont: while ( t < r ) {
02035     if ( *t >= AM.OffsetIndex &&
02036         ( *t >= AM.DumInd || ( *t < AM.WilInd &&
02037             indices[*t-AM.OffsetIndex].dimension ) ) )
02038         pcon[ncon++] = t;
02039         t++;
02040     }
02041     break;
02042 case EXPONENT :
02043     {
02044         WORD *rr;
02045         k = 1;
02046         rr = t + FUNHEAD;
02047         if ( *rr == ARGHEAD || ( *rr == -SNUMBER && rr[1] == 0 ) )
02048             k = 0;
02049         if ( *rr == -SNUMBER && rr[1] == 1 ) break;
02050         if ( *rr <= -FUNCTION ) k = *rr;
02051         NEXTARG(rr)
02052         if ( *rr == ARGHEAD || ( *rr == -SNUMBER && rr[1] == 0 ) ) {
02053             if ( k == 0 ) goto NormZZ;
02054             break;
02055         }
02056         if ( *rr == -SNUMBER && rr[1] > 0 && rr[1] < MAXPOWER && k < 0 ) {
02057             k = -k;
02058             if ( functions[k-FUNCTION].commute ) {
02059                 for ( i = 0; i < rr[1]; i++ ) pnco[nnco++] = rr-1;
02060             }
02061             else {
02062                 for ( i = 0; i < rr[1]; i++ ) pcom[ncom++] = rr-1;
02063             }
02064             break;
02065         }
02066         if ( k == 0 ) goto NormZero;
02067         if ( t[FUNHEAD] == -SYMBOL && *rr == -SNUMBER && t[1] == FUNHEAD+4 ) {
02068             if ( rr[1] < MAXPOWER ) {
02069                 t[FUNHEAD+2] = t[FUNHEAD+1]; t += FUNHEAD+2;
02070                 from = m;
02071                 goto NextSymbol;
02072             }
02073         }
02074 /*
02075         if ( ( t[FUNHEAD] > 0 && t[FUNHEAD+1] != 0 )
02076             || ( *rr > 0 && rr[1] != 0 ) ) {}
02077         else
02078             t[2] &= ~DIRTYSYMFLAG;
02079 */
02080         pnco[nnco++] = t;
02081     }
02082     break;
02083 case DENOMINATOR :
02084     t[2] &= ~DIRTYSYMFLAG;
02085     pden[nden++] = t;
02086     pnco[nnco++] = t;
02087     break;
02088 case INDEX :
02089     t += 2;
02090     do {
02091         if ( *t == 0 || *t == AM.vectorzero ) goto NormZero;
02092         if ( *t > 0 && *t < AM.OffsetIndex ) {
02093             lnum[0] = *t++;
02094             nnum = 1;
02095             ncoef = REDLENG(ncoef);
02096             if ( Mully(BHEAD (UWORD *)n_coef,&ncoef, (UWORD *)lnum,nnum) )
02097                 goto FromNorm;
02098             ncoef = INCLENG(ncoef);
02099         }
02100         else if ( *t == NOINDEX ) t++;
02101         else pind[nind++] = *t++;
02102     } while ( t < r );
02103     break;
02104 case SUBEXPRESSION :
02105     if ( t[3] == 0 ) break;
02106 case EXPRESSION :
02107     goto RegEnd;
02108 case ROOTFUNCTION :
02109 /*
02110     Tries to take the n-th root inside the rationals
02111     If this is not possible, it clears all flags and
02112     hence tries no more.
02113     Notation:
02114     root_(power(=integer),(rational)number)
02115 */
02116     { WORD nc;
02117         if ( t[2] == 0 ) goto defaultcase;

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02119         if ( t[FUNHEAD] != -SNUMBER || t[FUNHEAD+1] < 0 ) goto defaultcase;
02120         if ( t[FUNHEAD+2] == -SNUMBER ) {
02121             if ( t[FUNHEAD+1] == 0 && t[FUNHEAD+3] == 0 ) goto NormZZ;
02122             if ( t[FUNHEAD+1] == 0 ) break;
02123             if ( t[FUNHEAD+3] < 0 ) {
02124                 AT.WorkPointer[0] = -t[FUNHEAD+3];
02125                 nc = -1;
02126             }
02127             else {
02128                 AT.WorkPointer[0] = t[FUNHEAD+3];
02129                 nc = 1;
02130             }
02131             AT.WorkPointer[1] = 1;
02132         }
02133         else if ( t[FUNHEAD+2] == t[1]-FUNHEAD-2
02134             && t[FUNHEAD+2] == t[FUNHEAD+2+ARGHEAD]+ARGHEAD
02135             && ABS(t[t[1]-1]) == t[FUNHEAD+2+ARGHEAD] - 1 ) {
02136             WORD *r1, *r2;
02137             if ( t[FUNHEAD+1] == 0 ) break;
02138             i = t[t[1]-1]; r1 = t + FUNHEAD+ARGHEAD+3;
02139             nc = REDLENG(i);
02140             i = ABS(i) - 1;
02141             r2 = AT.WorkPointer;
02142             while ( --i >= 0 ) *r2++ = *r1++;
02143         }
02144         else goto defaultcase;
02145         if ( TakeRatRoot((UWORD *)AT.WorkPointer,&nc,t[FUNHEAD+1]) ) {
02146             t[2] = 0;
02147             goto defaultcase;
02148         }
02149         ncoef = REDLENG(ncoef);
02150         if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)AT.WorkPointer,nc) )
02151             goto FromNorm;
02152         if ( nc < 0 ) nc = -nc;
02153         if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *) (AT.WorkPointer+nc),nc) )
02154             goto FromNorm;
02155         ncoef = INCLENG(ncoef);
02156     }
02157     break;
02158 case RANDOMFUNCTION :
02159     {
02160         WORD nnc, nc, nca, nr;
02161         UWORD xx;
02162     /*
02163         Needs one positive integer argument.
02164         returns (wranf()%argument)+1.
02165         We may call wranf several times to paste UWORDS together
02166         when we need long numbers.
02167         We make little errors when taking the % operator
02168         (not 100% uniform). We correct for that by redoing the
02169         calculation in the (unlikely) case that we are in leftover area
02170     */
02171         if ( t[1] == FUNHEAD ) goto defaultcase;
02172         if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER &&
02173             t[FUNHEAD+1] > 0 ) {
02174             if ( t[FUNHEAD+1] == 1 ) break;
02175         redoshort:
02176             ((UWORD *)AT.WorkPointer)[0] = wranf(BHEAD0);
02177             ((UWORD *)AT.WorkPointer)[1] = wranf(BHEAD0);
02178             nr = 2;
02179             if ( ((UWORD *)AT.WorkPointer)[1] == 0 ) {
02180                 nr = 1;
02181                 if ( ((UWORD *)AT.WorkPointer)[0] == 0 ) {
02182                     nr = 0;
02183                 }
02184             }
02185             xx = (UWORD) (t[FUNHEAD+1]);
02186             if ( nr ) {
02187                 DivLong((UWORD *)AT.WorkPointer,nr
02188                     ,&xx,1
02189                     ,((UWORD *)AT.WorkPointer)+4,&nnc
02190                     ,((UWORD *)AT.WorkPointer)+2,&nc);
02191                 ((UWORD *)AT.WorkPointer)[4] = 0;
02192                 ((UWORD *)AT.WorkPointer)[5] = 0;
02193                 ((UWORD *)AT.WorkPointer)[6] = 1;
02194                 DivLong((UWORD *)AT.WorkPointer+4,3
02195                     ,&xx,1
02196                     ,((UWORD *)AT.WorkPointer)+9,&nnc
02197                     ,((UWORD *)AT.WorkPointer)+7,&nca);
02198                 AddLong((UWORD *)AT.WorkPointer+4,3
02199                     ,((UWORD *)AT.WorkPointer)+7,-nca
02200                     ,((UWORD *)AT.WorkPointer)+9,&nnc);
02201                 if ( BigLong((UWORD *)AT.WorkPointer,nr
02202                     ,((UWORD *)AT.WorkPointer)+9,nnc) >= 0 ) goto redoshort;
02203             }
02204             else nc = 0;
02205             if ( nc == 0 ) {

```

```

02206         AT.WorkPointer[2] = (WORD)xx;
02207         nc = 1;
02208     }
02209     ncoef = REDLENG(ncoef);
02210     if ( Mully(BHEAD (UWORD *)n_coef,&ncoef, ((UWORD *) (AT.WorkPointer))+2,nc) )
02211         goto FromNorm;
02212     ncoef = INCLENG(ncoef);
02213 }
02214 else if ( t[FUNHEAD] > 0 && t[1] == t[FUNHEAD]+FUNHEAD
02215 && ABS(t[t[1]-1]) == t[FUNHEAD]-1-ARGHEAD && t[t[1]-1] > 0 ) {
02216     WORD nna, nnb, nni, nnb2, nnb2a;
02217     UWORD *nnt;
02218     nna = t[t[1]-1];
02219     nnb2 = nna-1;
02220     nnb = nnb2/2;
02221     nnt = (UWORD *) (t+t[1]-1-nnb); /* start of denominator */
02222     if ( *nnt != 1 ) goto defaultcase;
02223     for ( nni = 1; nni < nnb; nni++ ) {
02224         if ( nnt[nni] != 0 ) goto defaultcase;
02225     }
02226     nnt = (UWORD *) (t + FUNHEAD + ARGHEAD + 1);
02227
02228     for ( nni = 0; nni < nnb2; nni++ ) {
02229         ((UWORD *)AT.WorkPointer)[nni] = wranf(BHEAD0);
02230     }
02231     nnb2a = nnb2;
02232     while ( nnb2a > 0 && ((UWORD *)AT.WorkPointer)[nnb2a-1] == 0 ) nnb2a--;
02233     if ( nnb2a > 0 ) {
02234         DivLong((UWORD *)AT.WorkPointer,nnb2a
02235             ,nnt,nnb
02236             ,((UWORD *)AT.WorkPointer)+2*nnb2,&nnc
02237             ,((UWORD *)AT.WorkPointer)+nnb2,&nc);
02238         for ( nni = 0; nni < nnb2; nni++ ) {
02239             ((UWORD *)AT.WorkPointer)[nni+2*nnb2] = 0;
02240         }
02241         ((UWORD *)AT.WorkPointer)[3*nnb2] = 1;
02242         DivLong((UWORD *)AT.WorkPointer+2*nnb2,nnb2+1
02243             ,nnt,nnb
02244             ,((UWORD *)AT.WorkPointer)+4*nnb2+1,&nnc
02245             ,((UWORD *)AT.WorkPointer)+3*nnb2+1,&nca);
02246         AddLong((UWORD *)AT.WorkPointer+2*nnb2,nnb2+1
02247             ,((UWORD *)AT.WorkPointer)+3*nnb2+1,-nca
02248             ,((UWORD *)AT.WorkPointer)+4*nnb2+1,&nnc);
02249         if ( BigLong((UWORD *)AT.WorkPointer,nnb2a
02250             ,((UWORD *)AT.WorkPointer)+4*nnb2+1,nnc) >= 0 ) goto redoshort;
02251     }
02252     else nc = 0;
02253     if ( nc == 0 ) {
02254         for ( nni = 0; nni < nnb; nni++ ) {
02255             ((UWORD *)AT.WorkPointer)[nnb2+nni] = nnt[nni];
02256         }
02257         nc = nnb;
02258     }
02259     ncoef = REDLENG(ncoef);
02260     if ( Mully(BHEAD (UWORD *)n_coef,&ncoef, ((UWORD *) (AT.WorkPointer))+nnb2,nc) )
02261         goto FromNorm;
02262     ncoef = INCLENG(ncoef);
02263 }
02264 else goto defaultcase;
02265 }
02266 break;
02267 case RANPERM :
02268     if ( *t == RANPERM && t[1] > FUNHEAD && t[FUNHEAD] <= -FUNCTION ) {
02269         WORD **pwork;
02270         WORD *mm, *ww, *ow = AT.WorkPointer;
02271         WORD *Array, *targ, *argstop, narg = 0, itot;
02272         int ie;
02273         argstop = t+t[1];
02274         targ = t+FUNHEAD+1;
02275         while ( targ < argstop ) {
02276             narg++; NEXTARG(targ);
02277         }
02278         WantAddPointers(narg);
02279         pwork = AT.pWorkSpace + AT.pWorkPointer;
02280         targ = t+FUNHEAD+1; narg = 0;
02281         while ( targ < argstop ) {
02282             pwork[narg++] = targ;
02283             NEXTARG(targ);
02284         }
02285         /*
02286         Make a random permutation of the numbers 0,...,narg-1
02287         The following code works also for narg == 0 and narg == 1
02288         */
02289         ow = AT.WorkPointer;
02290         Array = AT.WorkPointer;
02291         AT.WorkPointer += narg;
02292         for ( i = 0; i < narg; i++ ) Array[i] = i;

```

```

02293         for ( i = 2; i <= nargs; i++ ) {
02294             itot = (WORD) (iranf(BHEAD i));
02295             for ( j = 0; j < itot; j++ ) CYCLE1(WORD,Array,i)
02296         }
02297         mm = AT.WorkPointer;
02298         *mm++ = -t[FUNHEAD];
02299         *mm++ = t[1] - 1;
02300         for ( ie = 2; ie < FUNHEAD; ie++ ) *mm++ = t[ie];
02301         for ( i = 0; i < nargs; i++ ) {
02302             ww = pwork[Array[i]];
02303             CopyArg(mm,ww);
02304         }
02305         mm = AT.WorkPointer; t++; ww = t;
02306         i = mm[1]; NCOPY(ww,mm,i)
02307         AT.WorkPointer = ow;
02308         goto TryAgain;
02309     }
02310     pnco[nnco++] = t;
02311     break;
02312 case PUTFIRST : /* First argument should be a function, second a number */
02313     if ( ( t[2] & DIRTYFLAG ) != 0 && t[FUNHEAD] <= -FUNCTION
02314         && t[FUNHEAD+1] == -SNUMBER && t[FUNHEAD+2] > 0 ) {
02315         WORD *rr = t+t[1], *mm = t+FUNHEAD+3, *tt, *ttl, *tt2, num = 0;
02316     /*
02317
02318     */
02319         while ( mm < rr ) { num++; NEXTARG(mm); }
02320         if ( num < t[FUNHEAD+2] ) { pnco[nnco++] = t; break; }
02321         /* Replace "putfirst_" with the resulting function code. mm goes to arg start. */
02322         *t = -t[FUNHEAD]; mm = t+FUNHEAD+3;
02323         /* Set i to the arg number we are putting first, then move mm to its start. */
02324         i = t[FUNHEAD+2];
02325         while ( --i > 0 ) { NEXTARG(mm); }
02326         /* Keep a pointer to the arguments trailing the selected: */
02327         WORD *argTail = mm;
02328         NEXTARG(argTail);
02329         tt = TermMalloc("Select_"); /* Move selected out of the way into tmp space */
02330         ttl = tt;
02331         /* Normal argument: */
02332         if ( *mm > 0 ) {
02333             for ( i = 0; i < *mm; i++ ) *ttl++ = mm[i];
02334         }
02335         /* Fast-notation function: single word */
02336         else if ( *mm <= -FUNCTION ) { *ttl++ = *mm; }
02337         /* Fast-notation symbol, number etc: two words */
02338         else { *ttl++ = mm[0]; *ttl++ = mm[1]; }
02339         /* Put tt2 at the start of the original arguments */
02340         tt2 = t+FUNHEAD+3;
02341         /* Copy leading original arguments after the new first, in the tmp space. */
02342         while ( tt2 < mm ) *ttl++ = *tt2++;
02343         /* i contains the size so far. ttl goes to the start of the tmp space.
02344            tt2 to the final argument location (we overwrite "putfirst_") */
02345         i = ttl-tt; ttl = tt; tt2 = t+FUNHEAD;
02346         /* Copy everything so far to its final place */
02347         NCOPY(tt2,ttl,i);
02348         /* We are finished with the tmp space */
02349         TermFree(tt,"Select_");
02350         /* Now copy the trailing args. Use the stored pointer, *mm has been edited
02351            during the copy to tt2 above! NEXTARG(mm) would produce nonsense. */
02352         while ( argTail < rr ) *tt2++ = *argTail++;
02353         /* Set the size of all function args */
02354         t[1] = tt2 - t;
02355         /* Now copy the rest of the term, and set the final term size */
02356         rr = term + *term;
02357         while ( argTail < rr ) *tt2++ = *argTail++;
02358         *term = tt2-term;
02359         goto Restart;
02360     }
02361     else pnco[nnco++] = t;
02362     break;
02363 #ifndef WITHFLOAT
02364 case FLOATFUN :
02365     /*
02366     If it is a proper float_ we give it special treatment.
02367     If it is not proper, we treat it as a regular commuting function.
02368     */
02369     if ( withfloat == 0 ) {
02370         if ( TestFloat(t) == 0 ) goto defaultcase;
02371         firstfloat = t;
02372         withfloat = 1;
02373     }
02374     else { /* There are now at least two float_'s. Time for action. */
02375         if ( TestFloat(t) == 0 ) goto defaultcase;
02376         if ( withfloat == 1 ) UnpackFloat(aux4,firstfloat);
02377         withfloat++;
02378         UnpackFloat(aux5,t);
02379         mpf_mul(aux4,aux4,aux5);

```



```

02380     }
02381     break;
02382 #endif
02383 case INTFUNCTION :
02384 /*
02385     Can be resolved if the first argument is a number
02386     and the second argument either doesn't exist or has
02387     the value +1, 0, -1
02388     +1 : rounding up
02389     0  : rounding towards zero
02390     -1 : rounding down (same as no argument)
02391 */
02392     if ( t[1] <= FUNHEAD ) break;
02393     {
02394         WORD *rr, den, num;
02395         to = t + FUNHEAD;
02396         if ( *to > 0 ) {
02397             if ( *to == ARGHEAD ) goto NormZero;
02398             rr = to + *to;
02399             i = rr[-1];
02400             j = ABS(i);
02401             if ( to[ARGHEAD] != j+1 ) goto NoInteg;
02402             if ( rr >= r ) k = -1;
02403             else if ( *rr == ARGHEAD ) { k = 0; rr += ARGHEAD; }
02404             else if ( *rr == -SNUMBER ) { k = rr[1]; rr += 2; }
02405             else goto NoInteg;
02406             if ( rr != r ) goto NoInteg;
02407             if ( k > 1 || k < -1 ) goto NoInteg;
02408             to += ARGHEAD+1;
02409             j = (j-1) >> 1;
02410             i = ( i < 0 ) ? -j: j;
02411             UnPack((UWORD *)to,i,&den,&num);
02412 /*
02413             Potentially the use of NoScrat2 is unsafe.
02414             It makes the routine not reentrant, but because it is
02415             used only locally and because we only call the
02416             low level routines DivLong and AddLong which never
02417             make calls involving Normalize, things are OK after all
02418 */
02419             if ( AN.NoScrat2 == 0 ) {
02420                 AN.NoScrat2 = (UWORD *)Malloc1((AM.MaxTal+2)*sizeof(UWORD),"Normalize");
02421             }
02422             if ( DivLong((UWORD *)to,num,(UWORD *) (to+j),den
02423 , (UWORD *)AT.WorkPointer,&num,AN.NoScrat2,&den) ) goto FromNorm;
02424             if ( k < 0 && den < 0 ) {
02425                 *AN.NoScrat2 = 1;
02426                 den = -1;
02427                 if ( AddLong((UWORD *)AT.WorkPointer,num
02428 ,AN.NoScrat2,den,(UWORD *)AT.WorkPointer,&num) )
02429                     goto FromNorm;
02430             }
02431             else if ( k > 0 && den > 0 ) {
02432                 *AN.NoScrat2 = 1;
02433                 den = 1;
02434                 if ( AddLong((UWORD *)AT.WorkPointer,num,
02435 AN.NoScrat2,den,(UWORD *)AT.WorkPointer,&num) )
02436                     goto FromNorm;
02437             }
02438         }
02439     }
02440     else if ( *to == -SNUMBER ) { /* No rounding needed */
02441         if ( to[1] < 0 ) { *AT.WorkPointer = -to[1]; num = -1; }
02442         else if ( to[1] == 0 ) goto NormZero;
02443         else { *AT.WorkPointer = to[1]; num = 1; }
02444     }
02445     else goto NoInteg;
02446     if ( num == 0 ) goto NormZero;
02447     ncoef = REDLENG(ncoef);
02448     if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)AT.WorkPointer,num) )
02449         goto FromNorm;
02450     ncoef = INCLENG(ncoef);
02451     break;
02452 }
02453 NoInteg;;
02454 /*
02455     Fall through if it cannot be resolved
02456 */
02457     default :
02458     defaultcase:;
02459     if ( *t < FUNCTION ) {
02460         MLOCK(ErrorMessageLock);
02461         MesPrint("Illegal code in Norm");
02462 #ifdef DEBUGON
02463     {
02464         UBYTE OutBuf[140];
02465         AO.OutFill = AO.OutputLine = OutBuf;
02466         t = term;

```

```

02467         AO.OutSkip = 3;
02468         FiniLine();
02469         i = *t;
02470         while ( --i >= 0 ) {
02471             TalToLine((UWORD) (*t++));
02472             TokenToLine((UBYTE *) " ");
02473         }
02474         AO.OutSkip = 0;
02475         FiniLine();
02476     }
02477 #endif
02478     MUNLOCK(ErrorMessageLock);
02479     goto NormMin;
02480 }
02481 if ( *t == REPLACEMENT ) {
02482     if ( AR.Eside != LHSIDE ) ReplaceVeto--;
02483     pcom[ncom++] = t;
02484     break;
02485 }
02486 /*
02487 if ( *t == AM.termfunnum && t[1] == FUNHEAD+2
02488 && t[FUNHEAD] == -DOLLAREXPRESSION ) termflag++;
02489 */
02490 if ( *t == DUMMYFUN || *t == DUMMYTEN ) {}
02491 else {
02492     if ( *t < (FUNCTION + WILDOFFSET) ) {
02493         if ( ( functions[*t-FUNCTION].maxnumargs > 0 )
02494             || ( functions[*t-FUNCTION].minnumargs > 0 ) )
02495             && ( t[2] & DIRTYFLAG ) != 0 ) {
02496             /*
02497                 Number of arguments is bounded. And we have not checked.
02498             */
02499             WORD *ta = t + FUNHEAD, *tb = t + t[1];
02500             int numarg = 0;
02501             while ( ta < tb ) { numarg++; NEXTARG(ta) }
02502             if ( ( functions[*t-FUNCTION].maxnumargs > 0 )
02503                 && ( numarg >= functions[*t-FUNCTION].maxnumargs ) )
02504                 goto NormZero;
02505             if ( ( functions[*t-FUNCTION].minnumargs > 0 )
02506                 && ( numarg < functions[*t-FUNCTION].minnumargs ) )
02507                 goto NormZero;
02508         }
02509         doflags:
02510         if ( ( ( t[2] & DIRTYFLAG ) != 0 ) && ( functions[*t-FUNCTION].tabl == 0 ) ) {
02511             t[2] &= ~DIRTYFLAG;
02512             t[2] |= DIRTYSYMFLAG;
02513         }
02514         if ( functions[*t-FUNCTION].commute ) { pnco[nnco++] = t; }
02515         else { pcom[ncom++] = t; }
02516     }
02517     else {
02518         if ( ( ( t[2] & DIRTYFLAG ) != 0 ) && ( functions[*t-FUNCTION-WILDOFFSET].tabl
02519 == 0 ) ) {
02519             t[2] &= ~DIRTYFLAG;
02520             t[2] |= DIRTYSYMFLAG;
02521         }
02522         if ( functions[*t-FUNCTION-WILDOFFSET].commute ) {
02523             pnco[nnco++] = t;
02524         }
02525         else { pcom[ncom++] = t; }
02526     }
02527 }
02528
02529 /* Now hunt for contractible indices */
02530
02531 if ( ( *t < (FUNCTION + WILDOFFSET)
02532 && functions[*t-FUNCTION].spec >= TENSORFUNCTION ) || (
02533 *t >= (FUNCTION + WILDOFFSET)
02534 && functions[*t-FUNCTION-WILDOFFSET].spec >= TENSORFUNCTION ) ) {
02535     if ( *t >= GAMMA && *t <= GAMMASEVEN ) t++;
02536     t += FUNHEAD;
02537     while ( t < r ) {
02538         if ( *t == AM.vectorzero ) goto NormZero;
02539         if ( *t >= AM.OffsetIndex && ( *t >= AM.DumInd
02540 || ( *t < AM.WilInd && indices[*t-AM.OffsetIndex].dimension ) ) ) {
02541             pcon[ncon++] = t;
02542         }
02543         else if ( *t == FUNNYWILD ) { t++; }
02544         t++;
02545     }
02546 }
02547 else {
02548     t += FUNHEAD;
02549     while ( t < r ) {
02550         if ( *t > 0 ) {
02551             /*
02552                 Here we should worry about a recursion

```

```

02553             A problem is the possibility of a construct
02554             like f(mu+nu)
02555 */
02556         t += *t;
02557     }
02558     else if ( *t <= -FUNCTION ) t++;
02559     else if ( *t == -INDEX ) {
02560         if ( t[1] >= AM.OffsetIndex &&
02561             ( t[1] >= AM.DumInd || ( t[1] < AM.WilInd
02562                 && indices[t[1]-AM.OffsetIndex].dimension ) ) )
02563             pcon[ncon++] = t+1;
02564         t += 2;
02565     }
02566     else if ( *t == -SYMBOL ) {
02567         if ( t[1] >= MAXPOWER && t[1] < 2*MAXPOWER ) {
02568             *t = -SNUMBER;
02569             t[1] -= MAXPOWER;
02570         }
02571         else if ( t[1] < -MAXPOWER && t[1] > -2*MAXPOWER ) {
02572             *t = -SNUMBER;
02573             t[1] += MAXPOWER;
02574         }
02575         else t += 2;
02576     }
02577     else t += 2;
02578 }
02579 }
02580 break;
02581 }
02582 t = r;
02583 TryAgain;;
02584 } while ( t < m );
02585 if ( ANsc ) {
02586     AN.cTerm = ANsc;
02587     r = t = ANsr; m = ANsm;
02588     ANsc = ANsm = ANsr = 0;
02589     goto conscan;
02590 }
02591 /*
02592 #] First scan :
02593 #[ Easy denominators :
02594
02595 Easy denominators are denominators that can be replaced by
02596 negative powers of individual subterms. This may add to all
02597 our sublists.
02598
02599 */
02600 if ( nden ) {
02601     for ( k = 0, i = 0; i < nden; i++ ) {
02602         t = pden[i];
02603         if ( ( t[2] & DIRTYFLAG ) == 0 ) continue;
02604         r = t + t[1]; m = t + FUNHEAD;
02605         if ( m >= r ) {
02606             for ( j = i+1; j < nden; j++ ) pden[j-1] = pden[j];
02607             nden--;
02608             for ( j = 0; j < nnco; j++ ) if ( pnco[j] == t ) break;
02609             for ( j++; j < nnco; j++ ) pnco[j-1] = pnco[j];
02610             nnco--;
02611             i--;
02612         }
02613         else {
02614             NEXTARG(m);
02615             if ( m >= r ) continue;
02616         }
02617         /*
02618         We have more than one argument. Split the function.
02619         */
02619         if ( k == 0 ) {
02620             k = 1; to = termout; from = term;
02621         }
02622         while ( from < t ) *to++ = *from++;
02623         m = t + FUNHEAD;
02624         while ( m < r ) {
02625             stop = to;
02626             *to++ = DENOMINATOR;
02627             for ( j = 1; j < FUNHEAD; j++ ) *to++ = 0;
02628             if ( *m < -FUNCTION ) *to++ = *m++;
02629             else if ( *m < 0 ) { *to++ = *m++; *to++ = *m++; }
02630             else {
02631                 j = *m; while ( --j >= 0 ) *to++ = *m++;
02632             }
02633             stop[1] = WORDDIF(to,stop);
02634         }
02635         from = r;
02636         if ( i == nden - 1 ) {
02637             stop = term + *term;
02638             while ( from < stop ) *to++ = *from++;
02639             i = *termout = WORDDIF(to,termout);

```

```

02640         to = term; from = termout;
02641         while ( --i >= 0 ) *to++ = *from++;
02642         goto Restart;
02643     }
02644 }
02645 }
02646 for ( i = 0; i < nden; i++ ) {
02647     t = pden[i];
02648     if ( ( t[2] & DIRTYFLAG ) == 0 ) continue;
02649     t[2] = 0;
02650     if ( t[FUNHEAD] == -SYMBOL ) {
02651         WORD change;
02652         t += FUNHEAD+1;
02653         change = ExtraSymbol(*t,-1,nsym,ppsym,&ncoef);
02654         nsym += change;
02655         ppsym += change * 2;
02656         goto DropDen;
02657     }
02658     else if ( t[FUNHEAD] == -SNUMBER ) {
02659         t += FUNHEAD+1;
02660         if ( *t == 0 ) goto NormInf;
02661         if ( *t < 0 ) { *AT.WorkPointer = -*t; j = -1; }
02662         else { *AT.WorkPointer = *t; j = 1; }
02663         ncoef = REDLENG(ncoef);
02664         if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)AT.WorkPointer,j) )
02665             goto FromNorm;
02666         ncoef = INCLENG(ncoef);
02667         goto DropDen;
02668     }
02669     else if ( t[FUNHEAD] == ARGHEAD ) goto NormInf;
02670     else if ( t[FUNHEAD] > 0 && t[FUNHEAD+ARGHEAD] ==
02671 t[FUNHEAD]-ARGHEAD ) {
02672         /* Only one term */
02673         r = t + t[1] - 1;
02674         t += FUNHEAD + ARGHEAD + 1;
02675         j = *r;
02676         m = r - ABS(*r) + 1;
02677         if ( j != 3 || ( ( *m != 1 ) || ( m[1] != 1 ) ) ) {
02678             ncoef = REDLENG(ncoef);
02679             if ( DivRat(BHEAD (UWORD *)n_coef,ncoef,(UWORD *)m,REDLENG(j),(UWORD
*)n_coef,&ncoef) ) goto FromNorm;
02680             ncoef = INCLENG(ncoef);
02681             j = ABS(j) - 3;
02682             t[-FUNHEAD-ARGHEAD] -= j;
02683             t[-ARGHEAD-1] -= j;
02684             t[-1] -= j;
02685             m[0] = m[1] = 1;
02686             m[2] = 3;
02687         }
02688         while ( t < m ) {
02689             r = t + t[1];
02690             if ( *t == SYMBOL || *t == DOTPRODUCT ) {
02691                 k = t[1];
02692                 pden[i][1] -= k;
02693                 pden[i][FUNHEAD] -= k;
02694                 pden[i][FUNHEAD+ARGHEAD] -= k;
02695                 m -= k;
02696                 stop = m + 3;
02697                 tt = to = t;
02698                 from = r;
02699                 if ( *t == SYMBOL ) {
02700                     t += 2;
02701                     while ( t < r ) {
02702                         WORD change;
02703                         change = ExtraSymbol(*t,-t[1],nsym,ppsym,&ncoef);
02704                         nsym += change;
02705                         ppsym += change * 2;
02706                         t += 2;
02707                     }
02708                 }
02709                 else {
02710                     t += 2;
02711                     while ( t < r ) {
02712                         *ppdot++ = *t++;
02713                         *ppdot++ = *t++;
02714                         *ppdot++ = -*t++;
02715                         ndot++;
02716                     }
02717                 }
02718                 while ( to < stop ) *to++ = *from++;
02719                 r = tt;
02720             }
02721 #ifdef WITHFLOAT
02722             else if ( *t == FLOATFUN && TestFloat(t) ) {
02723                 k = t[1];
02724                 pden[i][1] -= k;
02725                 pden[i][FUNHEAD] -= k;

```

```

02726         pden[i][FUNHEAD+ARGHEAD] -= k;
02727         UnpackFloat(aux5,t);
02728         if ( withfloat == 0 ) {
02729             mpf_ui_div(aux4,1,aux5);
02730             withfloat = 2;
02731         }
02732         else {
02733             if ( withfloat == 1 ) UnpackFloat(aux4,firstfloat);
02734             mpf_div(aux4,aux4,aux5);
02735             withfloat++;
02736         }
02737         tt = to = t;
02738         while ( r < m ) *to++ = *r++;
02739         *to++ = 1; *to++ = 1; *to++ = 3;
02740         if ( ncoef < 0 ) t[-1] = -t[-1];
02741         m -= k;
02742         stop = m + 3;
02743         r = tt;
02744     }
02745 #endif
02746     t = r;
02747 }
02748 if ( pden[i][1] == 4+FUNHEAD+ARGHEAD ) {
02749 DropDen:
02750     for ( j = 0; j < nnco; j++ ) {
02751         if ( pden[i] == pnco[j] ) {
02752             --nnco;
02753             while ( j < nnco ) {
02754                 pnco[j] = pnco[j+1];
02755                 j++;
02756             }
02757             break;
02758         }
02759     }
02760     pden[i--] = pden[--nden];
02761 }
02762 }
02763 }
02764 }
02765 /*
02766 #] Easy denominators :
02767 #[ Index Contractions :
02768 */
02769 if ( ndel ) {
02770     t = pdel;
02771     for ( i = 0; i < ndel; i += 2 ) {
02772         if ( t[0] == t[1] ) {
02773             if ( t[0] == EMPTYINDEX ) {}
02774             else if ( *t < AM.OffsetIndex ) {
02775                 k = AC.FixIndices[*t];
02776                 if ( k < 0 ) { j = -1; k = -k; }
02777                 else if ( k > 0 ) j = 1;
02778                 goto NormZero;
02779                 goto WithFix;
02780             }
02781             else if ( *t >= AM.DumInd ) {
02782                 k = AC.lDefDim;
02783                 if ( k ) goto docontract;
02784             }
02785             else if ( *t >= AM.WilInd ) {
02786                 k = indices[*t-AM.OffsetIndex-WILDOFFSET].dimension;
02787                 if ( k ) goto docontract;
02788             }
02789             else if ( ( k = indices[*t-AM.OffsetIndex].dimension ) != 0 ) {
02790 docontract:
02791                 if ( k > 0 ) {
02792                     j = 1;
02793                     shortnum = k;
02794                     ncoef = REDLENG(ncoef);
02795                     if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)(&shortnum),j) )
02796                         goto FromNorm;
02797                     ncoef = INCLENG(ncoef);
02798                 }
02799                 else {
02800                     WORD change;
02801                     change = ExtraSymbol((WORD)(-k),(WORD)1,nsym,ppsym,&ncoef);
02802                     nsym += change;
02803                     ppsym += change * 2;
02804                 }
02805                 t[1] = pdel[ndel-1];
02806                 t[0] = pdel[ndel-2];
02807 HaveCon:
02808                 ndel -= 2;
02809                 i -= 2;
02810             }
02811         }
02812     }
    else {

```

```

02813         if ( *t < AM.OffsetIndex && t[1] < AM.OffsetIndex ) goto NormZero;
02814         j = *t - AM.OffsetIndex;
02815         if ( j >= 0 && ( ( *t >= AM.DumInd && AC.lDefDim )
02816         || ( *t < AM.WilInd && indices[j].dimension ) ) ) {
02817             for ( j = i + 2, m = pdel+j; j < ndel; j += 2, m += 2 ) {
02818                 if ( *t == *m ) {
02819                     *t = m[1];
02820                     *m++ = pdel[ndel-2];
02821                     *m = pdel[ndel-1];
02822                     goto HaveCon;
02823                 }
02824                 else if ( *t == m[1] ) {
02825                     *t = *m;
02826                     *m++ = pdel[ndel-2];
02827                     *m = pdel[ndel-1];
02828                     goto HaveCon;
02829                 }
02830             }
02831         }
02832         j = t[1]-AM.OffsetIndex;
02833         if ( j >= 0 && ( ( t[1] >= AM.DumInd && AC.lDefDim )
02834         || ( t[1] < AM.WilInd && indices[j].dimension ) ) ) {
02835             for ( j = i + 2, m = pdel+j; j < ndel; j += 2, m += 2 ) {
02836                 if ( t[1] == *m ) {
02837                     t[1] = m[1];
02838                     *m++ = pdel[ndel-2];
02839                     *m = pdel[ndel-1];
02840                     goto HaveCon;
02841                 }
02842                 else if ( t[1] == m[1] ) {
02843                     t[1] = *m;
02844                     *m++ = pdel[ndel-2];
02845                     *m = pdel[ndel-1];
02846                     goto HaveCon;
02847                 }
02848             }
02849         }
02850         t += 2;
02851     }
02852 }
02853 if ( ndel > 0 ) {
02854     if ( nvec ) {
02855         t = pdel;
02856         for ( i = 0; i < ndel; i++ ) {
02857             if ( *t >= AM.OffsetIndex && ( ( *t >= AM.DumInd && AC.lDefDim ) ||
02858             ( *t < AM.WilInd && indices[*t-AM.OffsetIndex].dimension ) ) ) {
02859                 r = pvec + 1;
02860                 for ( j = 1; j < nvec; j += 2 ) {
02861                     if ( *r == *t ) {
02862                         if ( i & 1 ) {
02863                             *r = t[-1];
02864                             *t-- = pdel[--ndel];
02865                             i -= 2;
02866                         }
02867                         else {
02868                             *r = t[1];
02869                             t[1] = pdel[--ndel];
02870                             i--;
02871                         }
02872                         *t-- = pdel[--ndel];
02873                         break;
02874                     }
02875                     r += 2;
02876                 }
02877             }
02878             t++;
02879         }
02880     }
02881     if ( ndel > 0 && ncon ) {
02882         t = pdel;
02883         for ( i = 0; i < ndel; i++ ) {
02884             if ( *t >= AM.OffsetIndex && ( ( *t >= AM.DumInd && AC.lDefDim ) ||
02885             ( *t < AM.WilInd && indices[*t-AM.OffsetIndex].dimension ) ) ) {
02886                 for ( j = 0; j < ncon; j++ ) {
02887                     if ( *pcon[j] == *t ) {
02888                         if ( i & 1 ) {
02889                             *pcon[j] = t[-1];
02890                             *t-- = pdel[--ndel];
02891                             i -= 2;
02892                         }
02893                         else {
02894                             *pcon[j] = t[1];
02895                             t[1] = pdel[--ndel];
02896                             i--;
02897                         }
02898                         *t-- = pdel[--ndel];
02899                         didcontr++;

```

```

02900             r = pcon[j];
02901         for ( j = 0; j < nnco; j++ ) {
02902             m = pnco[j];
02903             if ( r > m && r < m+m[1] ) {
02904                 m[2] |= DIRTYSYMFLAG;
02905                 break;
02906             }
02907         }
02908         for ( j = 0; j < ncom; j++ ) {
02909             m = pcom[j];
02910             if ( r > m && r < m+m[1] ) {
02911                 m[2] |= DIRTYSYMFLAG;
02912                 break;
02913             }
02914         }
02915         for ( j = 0; j < neps; j++ ) {
02916             m = peps[j];
02917             if ( r > m && r < m+m[1] ) {
02918                 m[2] |= DIRTYSYMFLAG;
02919                 break;
02920             }
02921         }
02922         break;
02923     }
02924 }
02925 }
02926     t++;
02927 }
02928 }
02929 }
02930 }
02931 if ( nvec ) {
02932     t = pvec + 1;
02933     for ( i = 3; i < nvec; i += 2 ) {
02934         k = *t - AM.OffsetIndex;
02935         if ( k >= 0 && ( (*t > AM.DumInd && AC.lDefDim )
02936             || ( *t < AM.WilInd && indices[k].dimension ) ) ) {
02937             r = t + 2;
02938             for ( j = i; j < nvec; j += 2 ) {
02939                 if ( *r == *t ) { /* Another dotproduct */
02940                     *ppdot++ = t[-1];
02941                     *ppdot++ = r[-1];
02942                     *ppdot++ = 1;
02943                     ndot++;
02944                     *r-- = pvec[--nvec];
02945                     *r = pvec[--nvec];
02946                     *t-- = pvec[--nvec];
02947                     *t-- = pvec[--nvec];
02948                     i -= 2;
02949                     break;
02950                 }
02951                 r += 2;
02952             }
02953         }
02954         t += 2;
02955     }
02956     if ( nvec > 0 && ncon ) {
02957         t = pvec + 1;
02958         for ( i = 1; i < nvec; i += 2 ) {
02959             k = *t - AM.OffsetIndex;
02960             if ( k >= 0 && ( (*t >= AM.DumInd && AC.lDefDim )
02961                 || ( *t < AM.WilInd && indices[k].dimension ) ) ) {
02962                 for ( j = 0; j < ncon; j++ ) {
02963                     if ( *pcon[j] == *t ) {
02964                         *pcon[j] = t[-1];
02965                         *t-- = pvec[--nvec];
02966                         *t-- = pvec[--nvec];
02967                         r = pcon[j];
02968                         pcon[j] = pcon[--ncon];
02969                         i -= 2;
02970                         for ( j = 0; j < nnco; j++ ) {
02971                             m = pnco[j];
02972                             if ( r > m && r < m+m[1] ) {
02973                                 m[2] |= DIRTYSYMFLAG;
02974                                 break;
02975                             }
02976                         }
02977                         for ( j = 0; j < ncom; j++ ) {
02978                             m = pcom[j];
02979                             if ( r > m && r < m+m[1] ) {
02980                                 m[2] |= DIRTYSYMFLAG;
02981                                 break;
02982                             }
02983                         }
02984                         for ( j = 0; j < neps; j++ ) {
02985                             m = peps[j];
02986                             if ( r > m && r < m+m[1] ) {

```

```

02987                                     m[2] |= DIRTYSYMFLAG;
02988                                     break;
02989                                 }
02990                             }
02991                             break;
02992                         }
02993                     }
02994                 }
02995                 t += 2;
02996             }
02997         }
02998     }
02999 /*
03000 #] Index Contractions :
03001 #[ NonCommuting Functions :
03002 */
03003 m = fillsetexp;
03004 if ( nnco ) {
03005     for ( i = 0; i < nnco; i++ ) {
03006         t = pnco[i];
03007         if ( ( *t >= (FUNCTION+WILDOFFSET)
03008             && functions[*t-FUNCTION-WILDOFFSET].spec <= 0 )
03009             || ( *t >= FUNCTION && *t < (FUNCTION + WILDOFFSET)
03010             && functions[*t-FUNCTION].spec <= 0 ) ) {
03011             DoRevert(t,m);
03012             if ( didcontr ) {
03013                 r = t + FUNHEAD;
03014                 t += t[1];
03015                 while ( r < t ) {
03016                     if ( *r == -INDEX && r[1] >= 0 && r[1] < AM.OffsetIndex ) {
03017                         *r = -SNUMBER;
03018                         didcontr--;
03019                         pnco[i][2] |= DIRTYSYMFLAG;
03020                     }
03021                     NEXTARG(r)
03022                 }
03023             }
03024         }
03025     }
03026
03027     /* First should come the code for function properties. */
03028
03029     /* First we test for symmetric properties and the DIRTYSYMFLAG */
03030
03031     for ( i = 0; i < nnco; i++ ) {
03032         t = pnco[i];
03033         if ( *t > 0 && ( t[2] & DIRTYSYMFLAG ) && *t != DOLLAREXPRESSION ) {
03034             l = 0; /* to make the compiler happy */
03035             if ( ( *t >= (FUNCTION+WILDOFFSET)
03036                 && ( l = functions[*t-FUNCTION-WILDOFFSET].symmetric ) > 0 )
03037                 || ( *t >= FUNCTION && *t < (FUNCTION + WILDOFFSET)
03038                 && ( l = functions[*t-FUNCTION].symmetric ) > 0 ) ) {
03039                 if ( *t >= (FUNCTION+WILDOFFSET) ) {
03040                     *t -= WILDOFFSET;
03041                     j = FullSymmetrize(BHEAD t,l);
03042                     *t += WILDOFFSET;
03043                 }
03044                 else j = FullSymmetrize(BHEAD t,l);
03045                 if ( (l & ~REVERSEORDER) == ANTISYMMETRIC ) {
03046                     if ( ( j & 2 ) != 0 ) goto NormZero;
03047                     if ( ( j & 1 ) != 0 ) ncoef = -ncoef;
03048                 }
03049             }
03050             else t[2] &= ~DIRTYSYMFLAG;
03051         }
03052     }
03053
03054     /* Non commuting functions are then tested for commutation
03055        rules. If needed their order is exchanged. */
03056
03057     k = nnco - 1;
03058     for ( i = 0; i < k; i++ ) {
03059         j = i;
03060         while ( Commute(pnco[j],pnco[j+1]) ) {
03061             t = pnco[j]; pnco[j] = pnco[j+1]; pnco[j+1] = t;
03062             l = j-1;
03063             while ( l >= 0 && Commute(pnco[l],pnco[l+1]) ) {
03064                 t = pnco[l]; pnco[l] = pnco[l+1]; pnco[l+1] = t;
03065                 l--;
03066             }
03067             if ( ++j >= k ) break;
03068         }
03069     }
03070
03071     /* Finally they are written to output. gamma matrices
03072        are bundled if possible */
03073

```



```

03074     for ( i = 0; i < nnco; i++ ) {
03075         t = pnco[i];
03076         if ( *t == IDFUNCTION ) AN.idfunctionflag = 1;
03077         if ( *t >= GAMMA && *t <= GAMMASEVEN ) {
03078             WORD gtype;
03079             to = m;
03080             *m++ = GAMMA;
03081             m++;
03082             FILLFUN(m)
03083             *m++ = stype = t[FUNHEAD];      /* type of string */
03084             j = 0;
03085             nnum = 0;
03086             do {
03087                 r = t + t[1];
03088                 if ( *t == GAMMAFIVE ) {
03089                     gtype = GAMMA5; t += FUNHEAD; goto onegammamatrix; }
03090                 else if ( *t == GAMMASIX ) {
03091                     gtype = GAMMA6; t += FUNHEAD; goto onegammamatrix; }
03092                 else if ( *t == GAMMASEVEN ) {
03093                     gtype = GAMMA7; t += FUNHEAD; goto onegammamatrix; }
03094                 t += FUNHEAD+1;
03095                 while ( t < r ) {
03096                     gtype = *t;
03097 onegammamatrix:
03098                     if ( gtype == GAMMA5 ) {
03099                         if ( j == GAMMA1 ) j = GAMMA5;
03100                         else if ( j == GAMMA5 ) j = GAMMA1;
03101                         else if ( j == GAMMA7 ) ncoef = -ncoef;
03102                         if ( nnum & 1 ) ncoef = -ncoef;
03103                     }
03104                     else if ( gtype == GAMMA6 || gtype == GAMMA7 ) {
03105                         if ( nnum & 1 ) {
03106                             if ( gtype == GAMMA6 ) gtype = GAMMA7;
03107                             else gtype = GAMMA6;
03108                         }
03109                         if ( j == GAMMA1 ) j = gtype;
03110                         else if ( j == GAMMA5 ) {
03111                             j = gtype;
03112                             if ( j == GAMMA7 ) ncoef = -ncoef;
03113                         }
03114                         else if ( j != gtype ) goto NormZero;
03115                         else {
03116                             shortnum = 2;
03117                             ncoef = REDLENG(ncoef);
03118                             if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)(&shortnum),1) ) goto
FromNorm;
03119                             ncoef = INCLENG(ncoef);
03120                         }
03121                     }
03122                     else {
03123                         *m++ = gtype; nnum++;
03124                     }
03125                     t++;
03126                 }
03127             } while ( ( ++i < nnco ) && ( *(t = pnco[i]) >= GAMMA
&& *t <= GAMMASEVEN ) && ( t[FUNHEAD] == stype ) );
03128             i--;
03129             if ( j ) {
03130                 k = WORDDIF(m,to) - FUNHEAD-1;
03131                 r = m;
03132                 from = m++;
03133                 while ( --k >= 0 ) *from-- = *--r;
03134                 *from = j;
03135             }
03136             to[1] = WORDDIF(m,to);
03137         }
03138     }
03139     else if ( *t < 0 ) {
03140         *m++ = -*t; *m++ = FUNHEAD; *m++ = 0;
03141         FILLFUN3(m)
03142     }
03143     else {
03144         if ( ( t[2] & DIRTYFLAG ) == DIRTYFLAG
&& *t != REPLACEMENT && *t != DOLLAREXPRESSION
&& TestFunFlag(BHEAD t) ) ReplaceVeto = 1;
03145         k = t[1];
03146         NCOPY(m,t,k);
03147     }
03148 }
03149 }
03150 }
03151 }
03152 }
03153 }
03154 /*
03155 #] NonCommuting Functions :
03156 #[ Commuting Functions :
03157 */
03158 if ( ncom ) {
03159     for ( i = 0; i < ncom; i++ ) {

```

```

03160         t = pcom[i];
03161         if ( ( *t >= (FUNCTION+WILDOFFSET)
03162             && functions[*t-FUNCTION-WILDOFFSET].spec <= 0 )
03163             || ( *t >= FUNCTION && *t < (FUNCTION + WILDOFFSET)
03164             && functions[*t-FUNCTION].spec <= 0 ) ) {
03165             DoRevert(t,m);
03166             if ( didcontr ) {
03167                 r = t + FUNHEAD;
03168                 t += t[1];
03169                 while ( r < t ) {
03170                     if ( *r == -INDEX && r[1] >= 0 && r[1] < AM.OffsetIndex ) {
03171                         *r = -SNUMBER;
03172                         didcontr--;
03173                         pcom[i][2] |= DIRTYSYMFLAG;
03174                     }
03175                     NEXTARG(r)
03176                 }
03177             }
03178         }
03179     }
03180
03181     /* Now we test for symmetric properties and the DIRTYSYMFLAG */
03182
03183     for ( i = 0; i < ncom; i++ ) {
03184         t = pcom[i];
03185         if ( *t > 0 && ( t[2] & DIRTYSYMFLAG ) ) {
03186             l = 0; /* to make the compiler happy */
03187             if ( ( *t >= (FUNCTION+WILDOFFSET)
03188                 && ( l = functions[*t-FUNCTION-WILDOFFSET].symmetric ) > 0 )
03189                 || ( *t >= FUNCTION && *t < (FUNCTION + WILDOFFSET)
03190                 && ( l = functions[*t-FUNCTION].symmetric ) > 0 ) ) {
03191                 if ( *t >= (FUNCTION+WILDOFFSET) ) {
03192                     *t -= WILDOFFSET;
03193                     j = FullSymmetrize(BHEAD t,l);
03194                     *t += WILDOFFSET;
03195                 }
03196                 else j = FullSymmetrize(BHEAD t,l);
03197                 if ( ( l & ~REVERSEORDER ) == ANTISYMMETRIC ) {
03198                     if ( ( j & 2 ) != 0 ) goto NormZero;
03199                     if ( ( j & 1 ) != 0 ) ncoef = -ncoef;
03200                 }
03201             }
03202             else t[2] &= ~DIRTYSYMFLAG;
03203         }
03204     }
03205     /*
03206     Sort the functions
03207     From a purists point of view this can be improved.
03208     There are slow and fast arguments and no conversions are
03209     taken into account here.
03210     */
03211     for ( i = 1; i < ncom; i++ ) {
03212         for ( j = i; j > 0; j-- ) {
03213             WORD jj, kk;
03214             jj = j-1;
03215             t = pcom[jj];
03216             r = pcom[j];
03217             if ( *t < 0 ) {
03218                 if ( *r < 0 ) { if ( *t >= *r ) goto NextI; }
03219                 else { if ( -*t <= *r ) goto NextI; }
03220                 goto jexch;
03221             }
03222             else if ( *r < 0 ) {
03223                 if ( *t < -*r ) goto NextI;
03224                 goto jexch;
03225             }
03226             else if ( *t != *r ) {
03227                 if ( *t < *r ) goto NextI;
03228                 t = pcom[j]; pcom[j] = pcom[jj]; pcom[jj] = t;
03229                 continue;
03230             }
03231             if ( AC.properorderflag ) {
03232                 if ( ( *t >= (FUNCTION+WILDOFFSET)
03233                     && functions[*t-FUNCTION-WILDOFFSET].spec >= TENSORFUNCTION )
03234                     || ( *t >= FUNCTION && *t < (FUNCTION + WILDOFFSET)
03235                     && functions[*t-FUNCTION].spec >= TENSORFUNCTION ) ) {}
03236                 else {
03237                     WORD *s1, *s2, *ss1, *ss2;
03238                     s1 = t+FUNHEAD; s2 = r+FUNHEAD;
03239                     ss1 = t + t[1]; ss2 = r + r[1];
03240                     while ( s1 < ss1 && s2 < ss2 ) {
03241                         k = CompArg(s1,s2);
03242                         if ( k > 0 ) goto jexch;
03243                         if ( k < 0 ) goto NextI;
03244                         NEXTARG(s1)
03245                         NEXTARG(s2)
03246                     }

```

```

03247         if ( s1 < ss1 ) goto jexch;
03248         goto NextI;
03249     }
03250     k = t[1] - FUNHEAD;
03251     kk = r[1] - FUNHEAD;
03252     t += FUNHEAD;
03253     r += FUNHEAD;
03254     while ( k > 0 && kk > 0 ) {
03255         if ( *t < *r ) goto NextI;
03256         else if ( *t++ > *r++ ) goto jexch;
03257         k--; kk--;
03258     }
03259     if ( k > 0 ) goto jexch;
03260     goto NextI;
03261 }
03262 else
03263 {
03264     k = t[1] - FUNHEAD;
03265     kk = r[1] - FUNHEAD;
03266     t += FUNHEAD;
03267     r += FUNHEAD;
03268     while ( k > 0 && kk > 0 ) {
03269         if ( *t < *r ) goto NextI;
03270         else if ( *t++ > *r++ ) goto jexch;
03271         k--; kk--;
03272     }
03273     if ( k > 0 ) goto jexch;
03274     goto NextI;
03275 }
03276 }
03277 NextI;;
03278 }
03279 for ( i = 0; i < ncom; i++ ) {
03280     t = pcom[i];
03281     if ( *t == THETA || *t == THETA2 ) {
03282         if ( ( k = DoTheta(BHEAD t) ) == 0 ) goto NormZero;
03283         else if ( k < 0 ) {
03284             k = t[1];
03285             NCOPY(m,t,k);
03286         }
03287     }
03288     else if ( *t == DELTA2 || *t == DELTAP ) {
03289         if ( ( k = DoDelta(t) ) == 0 ) goto NormZero;
03290         else if ( k < 0 ) {
03291             k = t[1];
03292             NCOPY(m,t,k);
03293         }
03294     }
03295     else if ( *t == AR.PolyFunInv && AR.PolyFunType == 2 ) {
03296         /*
03297         If there are two arguments, exchange them, change the
03298         name of the function and go to dealing with PolyRatFun.
03299         */
03300         WORD *mm, *tt = t, numt = 0;
03301         tt += FUNHEAD;
03302         while ( tt < t+t[1] ) { numt++; NEXTARG(tt) }
03303         if ( numt == 2 ) {
03304             tt = t; mm = m; k = t[1];
03305             NCOPY(mm,tt,k)
03306             mm = m+FUNHEAD;
03307             NEXTARG(mm);
03308             tt = t+FUNHEAD;
03309             if ( *mm < 0 ) {
03310                 if ( *mm <= -FUNCTION ) { *tt++ = *mm++; }
03311                 else { *tt++ = *mm++; *tt++ = *mm++; }
03312             }
03313             else {
03314                 k = *mm; NCOPY(tt,mm,k)
03315             }
03316             mm = m+FUNHEAD;
03317             if ( *mm < 0 ) {
03318                 if ( *mm <= -FUNCTION ) { *tt++ = *mm++; }
03319                 else { *tt++ = *mm++; *tt++ = *mm++; }
03320             }
03321             else {
03322                 k = *mm; NCOPY(tt,mm,k)
03323             }
03324             *t = AR.PolyFun;
03325             t[2] |= MUSTCLEANPRF;
03326             goto regularratfun;
03327         }
03328     }
03329     else if ( *t == AR.PolyFun ) {
03330         if ( AR.PolyFunType == 1 ) { /* Regular PolyFun with one argument */
03331             if ( t[FUNHEAD+1] == 0 && AR.Eside != LHSIDE &&
03332                 t[1] == FUNHEAD + 2 && t[FUNHEAD] == -SNUMBER ) goto NormZero;
03333             if ( i > 0 && pcom[i-1][0] == AR.PolyFun ) {

```

```

03334         if ( AN.PolyNormFlag == 0 ) {
03335             AN.PolyNormFlag = 1;
03336             AN.PolyFunTodo = 0;
03337         }
03338     }
03339     k = t[1];
03340     NCOPY(m,t,k);
03341 }
03342 else if ( AR.PolyFunType == 2 ) {
03343     /*
03344     PolyRatFun.
03345     Regular type: Two arguments
03346     Power expanded: One argument. Here to be treated as
03347     AR.PolyFunType == 1, but with power cutoff.
03348     */
03349     regulararratfun;;
03350     /*
03351     First check for zeroes.
03352     */
03353     if ( t[FUNHEAD+1] == 0 && AR.Eside != LHSIDE &&
03354         t[1] > FUNHEAD + 2 && t[FUNHEAD] == -SNUMBER ) {
03355         u = t + FUNHEAD + 2;
03356         if ( *u < 0 ) {
03357             if ( *u <= -FUNCTION ) {}
03358             else if ( t[1] == FUNHEAD+4 && t[FUNHEAD+2] == -SNUMBER
03359                 && t[FUNHEAD+3] == 0 ) goto NormPRE;
03360             else if ( t[1] == FUNHEAD+4 ) goto NormZero;
03361         }
03362         else if ( t[1] == *u+FUNHEAD+2 ) goto NormZero;
03363     }
03364     else {
03365         u = t+FUNHEAD; NEXTARG(u);
03366         if ( *u == -SNUMBER && u[1] == 0 ) goto NormInf;
03367     }
03368     if ( i > 0 && pcom[i-1][0] == AR.PolyFun ) AN.PolyNormFlag = 1;
03369     else if ( i < ncom-1 && pcom[i+1][0] == AR.PolyFun ) AN.PolyNormFlag = 1;
03370     k = t[1];
03371     if ( AN.PolyNormFlag ) {
03372         if ( AR.PolyFunExp == 0 ) {
03373             AN.PolyFunTodo = 0;
03374             NCOPY(m,t,k);
03375         }
03376         else if ( AR.PolyFunExp == 1 ) { /* get highest divergence */
03377             if ( PolyFunMode == 0 ) {
03378                 NCOPY(m,t,k);
03379                 AN.PolyFunTodo = 1;
03380             }
03381             else {
03382                 WORD *mmm = m;
03383                 NCOPY(m,t,k);
03384                 if ( TreatPolyRatFun(BHEAD mmm) != 0 )
03385                     goto FromNorm;
03386                 m = mmm+mmm[1];
03387             }
03388         }
03389         else {
03390             if ( PolyFunMode == 0 ) {
03391                 NCOPY(m,t,k);
03392                 AN.PolyFunTodo = 1;
03393             }
03394             else {
03395                 WORD *mmm = m;
03396                 NCOPY(m,t,k);
03397                 if ( ExpandRat(BHEAD mmm) != 0 )
03398                     goto FromNorm;
03399                 m = mmm+mmm[1];
03400             }
03401         }
03402     }
03403     else {
03404         if ( AR.PolyFunExp == 0 ) {
03405             AN.PolyFunTodo = 0;
03406             NCOPY(m,t,k);
03407         }
03408         else if ( AR.PolyFunExp == 1 ) { /* get highest divergence */
03409             WORD *mmm = m;
03410             NCOPY(m,t,k);
03411             if ( TreatPolyRatFun(BHEAD mmm) != 0 )
03412                 goto FromNorm;
03413             m = mmm+mmm[1];
03414         }
03415         else {
03416             WORD *mmm = m;
03417             NCOPY(m,t,k);
03418             if ( ExpandRat(BHEAD mmm) != 0 )
03419                 goto FromNorm;
03420             m = mmm+mmm[1];

```

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03421         }
03422     }
03423 }
03424 }
03425     else if ( *t > 0 ) {
03426         if ( ( t[2] & DIRTYFLAG ) == DIRTYFLAG
03427             && *t != REPLACEMENT && TestFunFlag(BHEAD t) ) ReplaceVeto = 1;
03428         k = t[1];
03429         NCOPY(m,t,k);
03430     }
03431     else {
03432         *m++ = -*t; *m++ = FUNHEAD; *m++ = 0;
03433         FILLFUN3(m)
03434     }
03435 }
03436 }
03437 /*
03438 #] Commuting Functions :
03439 #[ Track Replace_ :
03440 */
03441 if ( ReplaceVeto < 0 ) {
03442 /*
03443     We found one (or more) replace_ functions and all other
03444     functions are 'clean' (no dirty flag).
03445     Now we check whether one of these functions can be used.
03446     Thus far the functions go from fillsetexp to m.
03447     Somewhere in there there are -ReplaceVeto occurrences of REPLACEMENT.
03448     Hunt for the first one that fits the bill.
03449     Note that replace_ is a commuting function.
03450 */
03451     WORD *ma = fillsetexp, *mb, *mc;
03452     while ( ma < m ) {
03453         mb = ma + ma[1];
03454         if ( *ma != REPLACEMENT ) {
03455             ma = mb;
03456             continue;
03457         }
03458         if ( *ma == REPLACEMENT && ReplaceType == -1 ) {
03459             mc = ma;
03460             ReplaceType = 0;
03461             if ( AN.RSsize < 2*ma[1]+SUBEXPSIZE ) {
03462                 if ( AN.ReplaceSrat ) M_free(AN.ReplaceSrat,"AN.ReplaceSrat");
03463                 AN.RSsize = 2*ma[1]+SUBEXPSIZE+40;
03464                 AN.ReplaceSrat = (WORD *)Malloc1((AN.RSsize+1)*sizeof(WORD),"AN.ReplaceSrat");
03465             }
03466             ma += FUNHEAD;
03467             ReplaceSub = AN.ReplaceSrat;
03468             ReplaceSub += SUBEXPSIZE;
03469             while ( ma < mb ) {
03470                 if ( *ma > 0 ) goto NoRep;
03471                 if ( *ma <= -FUNCTION ) {
03472                     *ReplaceSub++ = FUNTOFUN;
03473                     *ReplaceSub++ = 4;
03474                     *ReplaceSub++ = -*ma++;
03475                     if ( *ma > -FUNCTION ) goto NoRep;
03476                     *ReplaceSub++ = -*ma++;
03477                 }
03478                 else if ( ma+4 > mb ) goto NoRep;
03479                 else {
03480                     if ( *ma == -SYMBOL ) {
03481                         if ( ma[2] == -SYMBOL && ma+4 <= mb )
03482                             *ReplaceSub++ = SYMTOSYM;
03483                     }
03484                     else if ( ma[2] == -SNUMBER && ma+4 <= mb ) {
03485                         *ReplaceSub++ = SYMTONUM;
03486                         if ( ReplaceType == 0 ) {
03487                             oldtoprhs = C->numrhs;
03488                             oldcpointer = C->Pointer - C->Buffer;
03489                         }
03490                         ReplaceType = 1;
03491                     }
03492                     else if ( ma[2] == ARGHEAD && ma+2+ARGHEAD <= mb ) {
03493                         *ReplaceSub++ = SYMTONUM;
03494                         *ReplaceSub++ = 4;
03495                         *ReplaceSub++ = ma[1];
03496                         *ReplaceSub++ = 0;
03497                         ma += 2+ARGHEAD;
03498                         continue;
03499                     }
03500                 }
03501             }
03502             /*
03503             Next is the subexpression. We have to test that
03504             it isn't vector-like or index-like
03505             */
03506             else if ( ma[2] > 0 ) {
03507                 WORD *sstop, *ttstop, n;
03508                 ss = ma+2;
03509                 sstop = ss + *ss;
03510                 ss += ARGHEAD;

```

```

03508         while ( ss < sstop ) {
03509             tt = ss + *ss;
03510             ttstop = tt - ABS(tt[-1]);
03511             ss++;
03512             while ( ss < ttstop ) {
03513                 if ( *ss == INDEX ) goto NoRep;
03514                 ss += ss[1];
03515             }
03516             ss = tt;
03517         }
03518         subtype = SYMTOSUB;
03519         if ( ReplaceType == 0 ) {
03520             oldtoprhs = C->numrhs;
03521             oldcpointer = C->Pointer - C->Buffer;
03522         }
03523         ReplaceType = 1;
03524         ss = AddrHS(AT.ebufnum,1);
03525         tt = ma+2;
03526         n = *tt - ARGHEAD;
03527         tt += ARGHEAD;
03528         while ( (ss + n + 10) > C->Top ) ss = DoubleCbuffer(AT.ebufnum,ss,14);
03529         while ( --n >= 0 ) *ss++ = *tt++;
03530         *ss++ = 0;
03531         C->rhs[C->numrhs+1] = ss;
03532         C->Pointer = ss;
03533         *ReplaceSub++ = subtype;
03534         *ReplaceSub++ = 4;
03535         *ReplaceSub++ = ma[1];
03536         *ReplaceSub++ = C->numrhs;
03537         ma += 2 + ma[2];
03538         continue;
03539     }
03540     else goto NoRep;
03541 }
03542 else if ( ( *ma == -VECTOR || *ma == -MINVECTOR ) && ma+4 <= mb ) {
03543     if ( ma[2] == -VECTOR ) {
03544         if ( *ma == -VECTOR ) *ReplaceSub++ = VECTOVEC;
03545         else *ReplaceSub++ = VECTOMIN;
03546     }
03547     else if ( ma[2] == -MINVECTOR ) {
03548         if ( *ma == -VECTOR ) *ReplaceSub++ = VECTOMIN;
03549         else *ReplaceSub++ = VECTOVEC;
03550     }
03551     /*
03552     Next is a vector-like subexpression
03553     Search for vector nature first
03554     */
03555     else if ( ma[2] > 0 ) {
03556         WORD *sstop, *ttstop, *w, *mm, n, count;
03557         WORD *v1, *v2 = 0;
03558         if ( *ma == -MINVECTOR ) {
03559             ss = ma+2;
03560             sstop = ss + *ss;
03561             ss += ARGHEAD;
03562             while ( ss < sstop ) {
03563                 ss += *ss;
03564                 ss[-1] = -ss[-1];
03565             }
03566             *ma = -VECTOR;
03567         }
03568         ss = ma+2;
03569         sstop = ss + *ss;
03570         ss += ARGHEAD;
03571         while ( ss < sstop ) {
03572             tt = ss + *ss;
03573             ttstop = tt - ABS(tt[-1]);
03574             ss++;
03575             count = 0;
03576             while ( ss < ttstop ) {
03577                 if ( *ss == INDEX ) {
03578                     n = ss[1] - 2; ss += 2;
03579                     while ( --n >= 0 ) {
03580                         if ( *ss < MINSPEC ) count++;
03581                         ss++;
03582                     }
03583                 }
03584                 else ss += ss[1];
03585             }
03586             if ( count != 1 ) goto NoRep;
03587             ss = tt;
03588         }
03589         subtype = VECTOSUB;
03590         if ( ReplaceType == 0 ) {
03591             oldtoprhs = C->numrhs;
03592             oldcpointer = C->Pointer - C->Buffer;
03593         }
03594         ReplaceType = 1;

```

```

03595 mm = AddRHS(AT.ebufnum,1);
03596 *ReplaceSub++ = subtype;
03597 *ReplaceSub++ = 4;
03598 *ReplaceSub++ = ma[1];
03599 *ReplaceSub++ = C->numrhs;
03600 w = ma+2;
03601 n = *w - ARGHEAD;
03602 w += ARGHEAD;
03603 while ( (mm + n + 10) > C->Top )
03604     mm = DoubleCbuffer(AT.ebufnum,mm,15);
03605 while ( --n >= 0 ) *mm++ = *w++;
03606 *mm++ = 0;
03607 C->rhs[C->numrhs+1] = mm;
03608 C->Pointer = mm;
03609 mm = AddRHS(AT.ebufnum,1);
03610 w = ma+2;
03611 n = *w - ARGHEAD;
03612 w += ARGHEAD;
03613 while ( (mm + n + 13) > C->Top )
03614     mm = DoubleCbuffer(AT.ebufnum,mm,16);
03615 sstop = w + n;
03616 while ( w < sstop ) {
03617     tt = w + *w; ttstop = tt - ABS(tt[-1]);
03618     ss = mm; mm++; w++;
03619     while ( w < ttstop ) {          /* Subterms */
03620         if ( *w != INDEX ) {
03621             n = w[1];
03622             NCOPY(mm,w,n);
03623         }
03624         else {
03625             v1 = mm;
03626             *mm++ = *w++;
03627             *mm++ = n = *w++;
03628             n -= 2;
03629             while ( --n >= 0 ) {
03630                 if ( *w >= MINSPEC ) *mm++ = *w++;
03631                 else v2 = w++;
03632             }
03633             n = WORDDIF(mm,v1);
03634             if ( n != v1[1] ) {
03635                 if ( n <= 2 ) mm -= 2;
03636                 else v1[1] = n;
03637                 *mm++ = VECTOR;
03638                 *mm++ = 4;
03639                 *mm++ = *v2;
03640                 *mm++ = FUNNYVEC;
03641             }
03642         }
03643     }
03644     while ( w < tt ) *mm++ = *w++;
03645     *ss = WORDDIF(mm,ss);
03646 }
03647 *mm++ = 0;
03648 C->rhs[C->numrhs+1] = mm;
03649 C->Pointer = mm;
03650 if ( mm > C->Top ) {
03651     MLOCK(ErrorMessageLock);
03652     MesPrint("Internal error in Normalize with extra compiler
buffer");
03653     MUNLOCK(ErrorMessageLock);
03654     Terminate(-1);
03655 }
03656 ma += 2 + ma[2];
03657 continue;
03658 }
03659 else goto NoRep;
03660 }
03661 else if ( *ma == -INDEX ) {
03662     if ( ( ma[2] == -INDEX || ma[2] == -VECTOR )
03663         && ma+4 <= mb )
03664         *ReplaceSub++ = INDTOIND;
03665     else if ( ma[1] >= AM.OffsetIndex ) {
03666         if ( ma[2] == -SNUMBER && ma+4 <= mb
03667             && ma[3] >= 0 && ma[3] < AM.OffsetIndex )
03668             *ReplaceSub++ = INDTOIND;
03669         else if ( ma[2] == ARGHEAD && ma+2+ARGHEAD <= mb ) {
03670             *ReplaceSub++ = INDTOIND;
03671             *ReplaceSub++ = 4;
03672             *ReplaceSub++ = ma[1];
03673             *ReplaceSub++ = 0;
03674             ma += 2+ARGHEAD;
03675             continue;
03676         }
03677         else goto NoRep;
03678     }
03679     else goto NoRep;
03680 }

```

```

03681         else goto NoRep;
03682         *ReplaceSub++ = 4;
03683         *ReplaceSub++ = ma[1];
03684         *ReplaceSub++ = ma[3];
03685         ma += 4;
03686     }
03687 }
03688 }
03689 AN.ReplaceScrat[1] = ReplaceSub-AN.ReplaceScrat;
03690 /*
03691     Success. This means that we have to remove the replace_
03692     from the functions. It starts at mc and end at mb.
03693 */
03694     while ( mb < m ) *mc++ = *mb++;
03695     m = mc;
03696     break;
03697 NoRep:
03698     if ( ReplaceType > 0 ) {
03699         C->numrhs = oldtoprhs;
03700         C->Pointer = C->Buffer + oldcpointer;
03701     }
03702     ReplaceType = -1;
03703     if ( ++ReplaceVeto >= 0 ) break;
03704 }
03705     ma = mb;
03706 }
03707 }
03708 /*
03709     #] Track Replace_ :
03710     #[ LeviCivita tensors :
03711 */
03712     if ( neps ) {
03713         to = m;
03714         for ( i = 0; i < neps; i++ ) { /* Put the indices in order */
03715             t = peps[i];
03716             if ( ( t[2] & DIRTYSYMFLAG ) != DIRTYSYMFLAG ) continue;
03717             t[2] &= ~DIRTYSYMFLAG;
03718             if ( AR.Eside == LHSIDE || AR.Eside == LHSIDEX ) {
03719                 /* Potential problems with FUNNYWILD */
03720             }
03721             /*
03722                 First make sure all FUNNIES are at the end.
03723                 Then sort separately
03724             */
03725             r = t + FUNHEAD;
03726             m = tt = t + t[1];
03727             while ( r < m ) {
03728                 if ( *r != FUNNYWILD ) { r++; continue; }
03729                 k = r[1]; u = r + 2;
03730                 while ( u < tt ) {
03731                     u[-2] = *u;
03732                     if ( *u != FUNNYWILD ) ncoef = -ncoef;
03733                     u++;
03734                 }
03735                 tt[-2] = FUNNYWILD; tt[-1] = k; m -= 2;
03736             }
03737             t += FUNHEAD;
03738             do {
03739                 for ( r = t + 1; r < m; r++ ) {
03740                     if ( *r < *t ) { k = *r; *r = *t; *t = k; ncoef = -ncoef; }
03741                     else if ( *r == *t ) goto NormZero;
03742                 }
03743                 t++;
03744             } while ( t < m );
03745             do {
03746                 for ( r = t + 2; r < tt; r += 2 ) {
03747                     if ( r[1] < t[1] ) {
03748                         k = r[1]; r[1] = t[1]; t[1] = k; ncoef = -ncoef; }
03749                     else if ( r[1] == t[1] ) goto NormZero;
03750                 }
03751                 t += 2;
03752             } while ( t < tt );
03753         }
03754     }
03755     else {
03756         m = t + t[1];
03757         t += FUNHEAD;
03758         do {
03759             for ( r = t + 1; r < m; r++ ) {
03760                 if ( *r < *t ) { k = *r; *r = *t; *t = k; ncoef = -ncoef; }
03761                 else if ( *r == *t ) goto NormZero;
03762             }
03763             t++;
03764         } while ( t < m );
03765     }
03766 }
03767 /* Sort the tensors */

```



```

03768     for ( i = 0; i < (neps-1); i++ ) {
03769         t = peps[i];
03770         for ( j = i+1; j < neps; j++ ) {
03771             r = peps[j];
03772             if ( t[1] > r[1] ) {
03773                 peps[i] = m = r; peps[j] = r = t; t = m;
03774             }
03775             else if ( t[1] == r[1] ) {
03776                 k = t[1] - FUNHEAD;
03777                 m = t + FUNHEAD;
03778                 r += FUNHEAD;
03779                 do {
03780                     if ( *r < *m ) {
03781                         m = peps[j]; peps[j] = t; peps[i] = t = m;
03782                         break;
03783                     }
03784                     else if ( *r++ > *m++ ) break;
03785                 } while ( --k > 0 );
03786             }
03787         }
03788     }
03789     m = to;
03790     for ( i = 0; i < neps; i++ ) {
03791         t = peps[i];
03792         k = t[1];
03793         NCOPY(m,t,k);
03794     }
03795 }
03796 /*
03797 #] LeviCivita tensors :
03798 #[ Delta :
03799 */
03800 if ( ndel ) {
03801     r = t = pdel;
03802     for ( i = 0; i < ndel; i += 2, r += 2 ) {
03803         if ( r[1] < r[0] ) { k = *r; *r = r[1]; r[1] = k; }
03804     }
03805     for ( i = 2; i < ndel; i += 2, t += 2 ) {
03806         r = t + 2;
03807         for ( j = i; j < ndel; j += 2 ) {
03808             if ( *r > *t ) { r += 2; }
03809             else if ( *r < *t ) {
03810                 k = *r; *r++ = *t; *t++ = k;
03811                 k = *r; *r++ = *t; *t-- = k;
03812             }
03813             else {
03814                 if ( **r < t[1] ) {
03815                     k = *r; *r = t[1]; t[1] = k;
03816                 }
03817                 r++;
03818             }
03819         }
03820     }
03821     t = pdel;
03822     *m++ = DELTA;
03823     *m++ = ndel + 2;
03824     i = ndel;
03825     NCOPY(m,t,i);
03826 }
03827 /*
03828 #] Delta :
03829 #[ Loose Vectors/Indices :
03830 */
03831 if ( nind ) {
03832     t = pind;
03833     for ( i = 0; i < nind; i++ ) {
03834         r = t + 1;
03835         for ( j = i+1; j < nind; j++ ) {
03836             if ( *r < *t ) {
03837                 k = *r; *r = *t; *t = k;
03838             }
03839             r++;
03840         }
03841         t++;
03842     }
03843     t = pind;
03844     *m++ = INDEX;
03845     *m++ = nind + 2;
03846     i = nind;
03847     NCOPY(m,t,i);
03848 }
03849 /*
03850 #] Loose Vectors/Indices :
03851 #[ Vectors :
03852 */
03853 if ( nvec ) {
03854     t = pvec;

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```

03855     for ( i = 2; i < nvec; i += 2 ) {
03856         r = t + 2;
03857         for ( j = i; j < nvec; j += 2 ) {
03858             if ( *r == *t ) {
03859                 if ( *++r < t[1] ) {
03860                     k = *r; *r = t[1]; t[1] = k;
03861                 }
03862                 r++;
03863             }
03864             else if ( *r < *t ) {
03865                 k = *r; *r++ = *t; *t++ = k;
03866                 k = *r; *r++ = *t; *t-- = k;
03867             }
03868             else { r += 2; }
03869         }
03870         t += 2;
03871     }
03872     t = pvec;
03873     *m++ = VECTOR;
03874     *m++ = nvec + 2;
03875     i = nvec;
03876     NCOPY(m,t,i);
03877 }
03878 /*
03879 #] Vectors :
03880 #[ Dotproducts :
03881 */
03882 if ( ndot ) {
03883     to = m;
03884     m = t = pdot;
03885     i = ndot;
03886     while ( --i >= 0 ) {
03887         if ( *t > t[1] ) { j = *t; *t = t[1]; t[1] = j; }
03888         t += 3;
03889     }
03890     t = m;
03891     ndot *= 3;
03892     m += ndot;
03893     while ( t < (m-3) ) {
03894         r = t + 3;
03895         do {
03896             if ( *r == *t ) {
03897                 if ( *++r == *++t ) {
03898                     r++;
03899                     if ( ( *r < MAXPOWER && t[1] < MAXPOWER )
03900                         || ( *r > -MAXPOWER && t[1] > -MAXPOWER ) ) {
03901                         t++;
03902                         *t += *r;
03903                         if ( *t > MAXPOWER || *t < -MAXPOWER ) {
03904                             MLOCK(ErrorMessageLock);
03905                             MesPrint("Exponent of dotproduct out of range: %d",*t);
03906                             MUNLOCK(ErrorMessageLock);
03907                             goto NormMin;
03908                         }
03909                         ndot -= 3;
03910                         *r-- = *--m;
03911                         *r-- = *--m;
03912                         *r = *--m;
03913                         if ( !*t ) {
03914                             ndot -= 3;
03915                             *t-- = *--m;
03916                             *t-- = *--m;
03917                             *t = *--m;
03918                             t -= 3;
03919                             break;
03920                         }
03921                     }
03922                     else if ( *r < *++t ) {
03923                         k = *r; *r++ = *t; *t = k;
03924                     }
03925                     else r++;
03926                     t -= 2;
03927                 }
03928                 else if ( *r < *t ) {
03929                     k = *r; *r++ = *t; *t++ = k;
03930                     k = *r; *r++ = *t; *t = k;
03931                     t -= 2;
03932                 }
03933                 else { r += 2; t--; }
03934             }
03935             else if ( *r < *t ) {
03936                 k = *r; *r++ = *t; *t++ = k;
03937                 k = *r; *r++ = *t; *t++ = k;
03938                 k = *r; *r++ = *t; *t = k;
03939                 t -= 2;
03940             }
03941             else { r += 3; }

```

```

03942         } while ( r < m );
03943         t += 3;
03944     }
03945     m = to;
03946     t = pdot;
03947     if ( ( i = ndot ) > 0 ) {
03948         *m++ = DOTPRODUCT;
03949         *m++ = i + 2;
03950         NCOPY(m,t,i);
03951     }
03952 }
03953 /*
03954 #] Dotproducts :
03955 #[ Symbols :
03956 */
03957 if ( nsym ) {
03958     nsym <= 1;
03959     t = psym;
03960     *m++ = SYMBOL;
03961     r = m;
03962     *m++ = ( i = nsym ) + 2;
03963     if ( i ) { do {
03964         if ( !*t ) {
03965             if ( t[1] < (2*MAXPOWER) ) { /* powers of i */
03966                 if ( t[1] & 1 ) { *m++ = 0; *m++ = 1; }
03967                 else *r -= 2;
03968                 if ( *++t & 2 ) ncoef = -ncoef;
03969                 t++;
03970             }
03971         }
03972         else if ( *t <= NumSymbols && *t > -2*MAXPOWER ) { /* Put powers in range */
03973             if ( ( ( t[1] > symbols[*t].maxpower ) && ( symbols[*t].maxpower < MAXPOWER ) ) ||
03974                 ( t[1] < symbols[*t].minpower ) && ( symbols[*t].minpower > -MAXPOWER ) ) )
03975                 &&
03976                 ( t[1] < 2*MAXPOWER ) && ( t[1] > -2*MAXPOWER ) ) {
03977                 if ( i <= 2 || t[2] != *t ) goto NormZero;
03978             }
03979             if ( AN.ncmod == 1 && ( AC.modmode & ALSOPOWERS ) != 0 ) {
03980                 if ( AC.cmod[0] == 1 ) t[1] = 0;
03981                 else if ( t[1] >= 0 ) t[1] = 1 + (t[1]-1)%AC.cmod[0]-1;
03982                 else {
03983                     t[1] = -1 - (-t[1]-1)%AC.cmod[0]-1;
03984                     if ( t[1] < 0 ) t[1] += AC.cmod[0]-1;
03985                 }
03986             }
03987             if ( ( t[1] < (2*MAXPOWER) && t[1] >= MAXPOWER )
03988                 || ( t[1] > -(2*MAXPOWER) && t[1] <= -MAXPOWER ) ) {
03989                 MLOCK(ErrorMessageLock);
03990                 MesPrint("Exponent out of range: %d",t[1]);
03991                 MUNLOCK(ErrorMessageLock);
03992                 goto NormMin;
03993             }
03994             if ( AT.TrimPower && AR.PolyFunVar == *t && t[1] > AR.PolyFunPow ) {
03995                 goto NormZero;
03996             }
03997             else if ( t[1] ) {
03998                 *m++ = *t++;
03999                 *m++ = *t++;
04000             }
04001             else { *r -= 2; t += 2; }
04002         }
04003         else {
04004             *m++ = *t++; *m++ = *t++;
04005         }
04006     } while ( (i-=2) > 0 ); }
04007     if ( *r <= 2 ) m = r-1;
04008 }
04009 /*
04010 #] Symbols :
04011 #[ float_ :
04012
04013 Here we treat float_ functions and combined them with the regular
04014 coefficient.
04015 */
04016 #ifdef WITHFLOAT
04017     if ( withfloat ) {
04018         WORD floatsign = 3;
04019     /*
04020         First check whether the coefficient is already 1/1
04021     */
04022     if ( ABS(ncoef) == 3 && n_coef[0] == 1 && n_coef[1] == 1 ) {
04023         if ( withfloat == 1 ) {
04024             t = firstfloat;
04025         /*
04026             We only interfere by making the sign positive.
04027             The float_ function has already been tested.

```

```

04028 */
04029         if ( t[FUNHEAD+3] < 0 ) {
04030             t[FUNHEAD+3] = -t[FUNHEAD+3];
04031             floatsign = -floatsign;
04032         }
04033         if ( ncoef < 0 ) floatsign = -floatsign;
04034 /*
04035         Copy to output;
04036 */
04037         AT.FloatPos = m-termout;
04038         i = t[1]; NCOPY(m,t,i)
04039     }
04040     else {
04041         PackFloat(m,aux4);
04042         if ( m[FUNHEAD+3] < 0 ) {
04043             m[FUNHEAD+3] = -m[FUNHEAD+3];
04044             floatsign = -floatsign;
04045         }
04046         if ( ncoef < 0 ) floatsign = -floatsign;
04047         AT.FloatPos = m-termout;
04048         m += m[1];
04049     }
04050 }
04051 else {
04052     if ( withfloat == 1 ) UnpackFloat(aux4,firstfloat);
04053     RatToFloat(aux5,(UWORD *)n_coef,ncoef);
04054     mpf_mul(aux4,aux4,aux5);
04055     PackFloat(m,aux4);
04056     AT.FloatPos = m-termout;
04057     if ( m[FUNHEAD+3] < 0 ) {
04058         m[FUNHEAD+3] = -m[FUNHEAD+3];
04059         floatsign = -floatsign;
04060     }
04061     m += m[1];
04062 }
04063 n_coef[0] = 1; n_coef[1] = 1; ncoef = floatsign;
04064 }
04065 else AT.FloatPos = 0;
04066 #endif
04067 /*
04068 #] float_ :
04069 #[ Do Replace_ :
04071 */
04072 stop = (WORD *)(((UBYTE *) (termout)) + AM.MaxTer);
04073 i = ABS(ncoef);
04074 if ( ( m + i ) > stop ) {
04075     MLOCK(ErrorMessageLock);
04076     MesPrint("Term too complex during normalization");
04077     MUNLOCK(ErrorMessageLock);
04078     goto NormMin;
04079 }
04080 if ( ReplaceType >= 0 ) {
04081     t = n_coef;
04082     i--;
04083     NCOPY(m,t,i);
04084     *m++ = ncoef;
04085     t = termout;
04086     *t = WORDDIFF(m,t);
04087     if ( ReplaceType == 0 ) {
04088         AT.WorkPointer = termout+*termout;
04089         WildFill(BHEAD term,termout,AN.ReplaceScrat);
04090         termout = term + *term;
04091     }
04092     else {
04093         AT.WorkPointer = r = termout + *termout;
04094         WildFill(BHEAD r,termout,AN.ReplaceScrat);
04095         i = *r; m = term;
04096         NCOPY(m,r,i);
04097         termout = m;
04098
04099         r = m = term;
04100         r += *term; r -= ABS(r[-1]);
04101         m++;
04102         while ( m < r ) {
04103             if ( *m >= FUNCTION && m[1] > FUNHEAD &&
04104                 functions[*m-FUNCTION].spec != TENSORFUNCTION )
04105                 m[2] |= DIRTYFLAG;
04106             m += m[1];
04107         }
04108     }
04109 }
04110 /*
04111 The next 'reset' cannot be done. We still need the expression
04112 in the buffer. Note though that this may cause a runaway pointer
04113 if we are not very careful.
04114

```

```

04115         C->numrhs = oldtoprhs;
04116         C->Pointer = C->Buffer + oldcpointer;
04117     */
04118     TermFree(n_llnum, "n_llnum");
04119     TermFree(n_coef, "NormCoef");
04120     return(1);
04121 }
04122 else {
04123     t = termout;
04124     k = WORDDIF(m,t);
04125     *t = k + i;
04126     m = term;
04127     NCOPY(m,t,k);
04128     i--;
04129     t = n_coef;
04130     NCOPY(m,t,i);
04131     *m++ = ncoef;
04132 }
04133 /*
04134     #] Do Replace_ :
04135     #[ Errors and Finish :
04136 */
04137 RegEnd:
04138     if ( termout < term + *term && termout >= term ) AT.WorkPointer = term + *term;
04139     else AT.WorkPointer = termout;
04140 /*
04141     if ( termflag ) { We have to assign the term to $variable(s)
04142         TermAssign(term);
04143     }
04144 */
04145     TermFree(n_llnum, "n_llnum");
04146     TermFree(n_coef, "NormCoef");
04147     return(regval);
04148
04149 NormInf:
04150     MLOCK(ErrorMessageLock);
04151     MesPrint("Division by zero during normalization");
04152     MUNLOCK(ErrorMessageLock);
04153     Terminate(-1);
04154
04155 NormZZ:
04156     MLOCK(ErrorMessageLock);
04157     MesPrint("0^0 during normalization of term");
04158     MUNLOCK(ErrorMessageLock);
04159     Terminate(-1);
04160
04161 NormPRF:
04162     MLOCK(ErrorMessageLock);
04163     MesPrint("0/0 in polyratfun during normalization of term");
04164     MUNLOCK(ErrorMessageLock);
04165     Terminate(-1);
04166
04167 NormZero:
04168     *term = 0;
04169     AT.WorkPointer = termout;
04170     TermFree(n_llnum, "n_llnum");
04171     TermFree(n_coef, "NormCoef");
04172     return(regval);
04173
04174 NormMin:
04175     TermFree(n_llnum, "n_llnum");
04176     TermFree(n_coef, "NormCoef");
04177     return(-1);
04178
04179 FromNorm:
04180     MLOCK(ErrorMessageLock);
04181     MesCall("Norm");
04182     MUNLOCK(ErrorMessageLock);
04183     TermFree(n_llnum, "n_llnum");
04184     TermFree(n_coef, "NormCoef");
04185     return(-1);
04186
04187 /*
04188     #] Errors and Finish :
04189 */
04190 }
04191
04192 /*
04193     #] Normalize :
04194     #[ ExtraSymbol :
04195 */
04196
04197 int ExtraSymbol(WORD sym, WORD pow, WORD nsym, WORD *ppsym, WORD *ncoef)
04198 {
04199     WORD *m, i;
04200     i = nsym;
04201     m = ppsym - 2;

```

```

04202     while ( i > 0 ) {
04203         if ( sym == *m ) {
04204             m++;
04205             if ( pow > 2*MAXPOWER || pow < -2*MAXPOWER
04206                 || *m > 2*MAXPOWER || *m < -2*MAXPOWER ) {
04207                 MLOCK(ErrorMessageLock);
04208                 MesPrint("Illegal wildcard power combination.");
04209                 MUNLOCK(ErrorMessageLock);
04210                 Terminate(-1);
04211             }
04212             *m += pow;
04213
04214             if ( ( sym <= NumSymbols && sym > -MAXPOWER )
04215                 && ( symbols[sym].complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY ) {
04216                 *m %= symbols[sym].maxpower;
04217                 if ( *m < 0 ) *m += symbols[sym].maxpower;
04218                 if ( ( symbols[sym].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
04219                     if ( ( ( symbols[sym].maxpower & 1 ) == 0 ) &&
04220                         ( *m >= symbols[sym].maxpower/2 ) ) {
04221                         *m -= symbols[sym].maxpower/2; *ncoef = -*ncoef;
04222                     }
04223                 }
04224             }
04225
04226             if ( *m >= 2*MAXPOWER || *m <= -2*MAXPOWER ) {
04227                 MLOCK(ErrorMessageLock);
04228                 MesPrint("Power overflow during normalization");
04229                 MUNLOCK(ErrorMessageLock);
04230                 return(-1);
04231             }
04232             if ( !*m ) {
04233                 m--;
04234                 while ( i < nsym )
04235                     { *m = m[2]; m++; *m = m[2]; m++; i++; }
04236                 return(-1);
04237             }
04238             return(0);
04239         }
04240         else if ( sym < *m ) {
04241             m -= 2;
04242             i--;
04243         }
04244         else break;
04245     }
04246     m = ppsym;
04247     while ( i < nsym )
04248         { m--; m[2] = *m; m--; m[2] = *m; i++; }
04249     *m++ = sym;
04250     *m = pow;
04251     return(1);
04252 }
04253
04254 /*
04255     #] ExtraSymbol :
04256     #[ DoTheta :
04257 */
04258
04259 int DoTheta(PHEAD WORD *t)
04260 {
04261     GETBIDENTITY
04262     WORD k, *r1, *r2, *tstop, type;
04263     WORD ia, *ta, *tb, *stopa, *stopb;
04264     if ( AC.BraceNormalize ) return(-1);
04265     type = *t;
04266     k = t[1];
04267     tstop = t + k;
04268     t += FUNHEAD;
04269     if ( k <= FUNHEAD ) return(1);
04270     r1 = t;
04271     NEXTARG(r1)
04272     if ( r1 == tstop ) {
04273 /*
04274         One argument
04275 */
04276         if ( *t == ARGHEAD ) {
04277             if ( type == THETA ) return(1);
04278             else return(0); /* THETA2 */
04279         }
04280         if ( *t < 0 ) {
04281             if ( *t == -SNUMBER ) {
04282                 if ( t[1] < 0 ) return(0);
04283                 else {
04284                     if ( type == THETA2 && t[1] == 0 ) return(0);
04285                     else return(1);
04286                 }
04287             }
04288             return(-1);

```

```

04289     }
04290     k = t[*t-1];
04291     if ( *t == ABS(k)+1+ARGHEAD ) {
04292         if ( k > 0 ) return(1);
04293         else return(0);
04294     }
04295     return(-1);
04296 }
04297 /*
04298 At least two arguments
04299 */
04300 r2 = r1;
04301 NEXTARG(r2)
04302 if ( r2 < tstop ) return(-1); /* More than 2 arguments */
04303 /*
04304 Note now that zero has to be treated specially
04305 We take the criteria from the symmetrize routine
04306 */
04307 if ( *t == -SNUMBER && *r1 == -SNUMBER ) {
04308     if ( t[1] > r1[1] ) return(0);
04309     else if ( t[1] < r1[1] ) {
04310         return(1);
04311     }
04312     else if ( type == THETA ) return(1);
04313     else return(0); /* THETA2 */
04314 }
04315 else if ( t[1] == 0 && *t == -SNUMBER ) {
04316     if ( *r1 > 0 ) { }
04317     else if ( *t < *r1 ) return(1);
04318     else if ( *t > *r1 ) return(0);
04319 }
04320 else if ( r1[1] == 0 && *r1 == -SNUMBER ) {
04321     if ( *t > 0 ) { }
04322     else if ( *t < *r1 ) return(1);
04323     else if ( *t > *r1 ) return(0);
04324 }
04325 r2 = AT.WorkPointer;
04326 if ( *t < 0 ) {
04327     ta = r2;
04328     ToGeneral(t,ta,0);
04329     r2 += *r2;
04330 }
04331 else ta = t;
04332 if ( *r1 < 0 ) {
04333     tb = r2;
04334     ToGeneral(r1,tb,0);
04335 }
04336 else tb = r1;
04337 stopa = ta + *ta;
04338 stopb = tb + *tb;
04339 ta += ARGHEAD; tb += ARGHEAD;
04340 while ( ta < stopa ) {
04341     if ( tb >= stopb ) return(0);
04342     if ( ( ia = CompareTerms(BHEAD ta,tb,(WORD)1) ) < 0 ) return(0);
04343     if ( ia > 0 ) return(1);
04344     ta += *ta;
04345     tb += *tb;
04346 }
04347 if ( type == THETA ) return(1);
04348 else return(0); /* THETA2 */
04349 }
04350
04351 /*
04352 #] DoTheta :
04353 #[ DoDelta :
04354 */
04355
04356 int DoDelta(WORD *t)
04357 {
04358     WORD k, *r1, *r2, *tstop, isnum, isnum2, type = *t;
04359     if ( AC.BracketNormalize ) return(-1);
04360     k = t[1];
04361     if ( k <= FUNHEAD ) goto argzero;
04362     if ( k == FUNHEAD+ARGHEAD && t[FUNHEAD] == ARGHEAD ) goto argzero;
04363     tstop = t + k;
04364     t += FUNHEAD;
04365     r1 = t;
04366     NEXTARG(r1)
04367     if ( *t < 0 ) {
04368         k = 1;
04369         if ( *t == -SNUMBER ) { isnum = 1; k = t[1]; }
04370         else isnum = 0;
04371     }
04372     else {
04373         k = t[*t-1];
04374         k = ABS(k);
04375         if ( k == *t-ARGHEAD-1 ) isnum = 1;

```

```

04376         else isnum = 0;
04377         k = 1;
04378     }
04379     if ( r1 >= tstop ) {          /* Single argument */
04380         if ( !isnum ) return(-1);
04381         if ( k == 0 ) goto argzero;
04382         goto argnonzero;
04383     }
04384     r2 = r1;
04385     NEXTARG(r2)
04386     if ( r2 < tstop ) return(-1);
04387     if ( *r1 < 0 ) {
04388         if ( *r1 == -SNUMBER ) { isnum2 = 1; }
04389         else isnum2 = 0;
04390     }
04391     else {
04392         k = r1[*r1-1];
04393         k = ABS(k);
04394         if ( k == *r1-ARGHEAD-1 ) isnum2 = 1;
04395         else isnum2 = 0;
04396     }
04397     if ( isnum != isnum2 ) return(-1);
04398     tstop = r1;
04399     while ( t < tstop && r1 < r2 ) {
04400         if ( *t != *r1 ) {
04401             if ( !isnum ) return(-1);
04402             goto argnonzero;
04403         }
04404         t++; r1++;
04405     }
04406     if ( t != tstop || r1 != r2 ) {
04407         if ( !isnum ) return(-1);
04408         goto argnonzero;
04409     }
04410 argzero:
04411     if ( type == DELTA2 ) return(1);
04412     else return(0);
04413 argnonzero:
04414     if ( type == DELTA2 ) return(0);
04415     else return(1);
04416 }
04417
04418 /*
04419     #] DoDelta :
04420     #[ DoRevert :
04421 */
04422
04423 void DoRevert(WORD *fun, WORD *tmp)
04424 {
04425     WORD *t, *r, *m, *to, *tt, *mm, i, j;
04426     to = fun + fun[1];
04427     r = fun + FUNHEAD;
04428     while ( r < to ) {
04429         if ( *r <= 0 ) {
04430             if ( *r == -REVERSEFUNCTION ) {
04431                 m = r; mm = m+1;
04432                 while ( mm < to ) *m++ = *mm++;
04433                 to--;
04434                 (fun[1])--;
04435                 fun[2] |= DIRTSYMLFLAG;
04436             }
04437             else if ( *r <= -FUNCTION ) r++;
04438             else {
04439                 if ( *r == -INDEX && r[1] < MINSPEC ) *r = -VECTOR;
04440                 r += 2;
04441             }
04442         }
04443         else {
04444             if ( ( *r > ARGHEAD )
04445                 && ( r[ARGHEAD+1] == REVERSEFUNCTION )
04446                 && ( *r == (r[ARGHEAD]+ARGHEAD) )
04447                 && ( r[ARGHEAD] == (r[ARGHEAD+2]+4) )
04448                 && ( *(r+r-3) == 1 )
04449                 && ( *(r+r-2) == 1 )
04450                 && ( *(r+r-1) == 3 ) ) {
04451                 mm = r;
04452                 r += ARGHEAD + 1;
04453                 tt = r + r[1];
04454                 r += FUNHEAD;
04455                 m = tmp;
04456                 t = r;
04457                 j = 0;
04458                 while ( t < tt ) {
04459                     NEXTARG(t)
04460                     j++;
04461                 }
04462                 while ( --j >= 0 ) {

```



```

04463         i = j;
04464         t = r;
04465         while ( --i >= 0 ) {
04466             NEXTARG(t)
04467         }
04468         if ( *t > 0 ) {
04469             i = *t;
04470             NCOPY(m,t,i);
04471         }
04472         else if ( *t <= -FUNCTION ) *m++ = *t++;
04473         else { *m++ = *t++; *m++ = *t++; }
04474     }
04475     i = WORDDIF(m,tmp);
04476     m = tmp;
04477     t = mm;
04478     r = t + *t;
04479     NCOPY(t,m,i);
04480     m = r;
04481     r = t;
04482     i = WORDDIF(to,m);
04483     NCOPY(t,m,i);
04484     fun[1] = WORDDIF(t,fun);
04485     to = t;
04486     fun[2] |= DIRTYSYMFLAG;
04487 }
04488     else r += *r;
04489 }
04490 }
04491 }
04492
04493 /*
04494     #] DoRevert :
04495     #] Normalize :
04496     #[ DetCommu :
04497
04498     Determines the number of terms in an expression that contain
04499     noncommuting objects. This can be used to see whether products of
04500     this expression can be evaluated with binomial coefficients.
04501
04502     We don't try to be fancy. If a term contains noncommuting objects
04503     we are not looking whether they can commute with complete other
04504     terms.
04505
04506     If the number gets too large we cut it off.
04507 */
04508
04509 #define MAXNUMBEROFNONCOMTERMS 2
04510
04511 WORD DetCommu(WORD *terms)
04512 {
04513     WORD *t, *tnext, *tstop;
04514     WORD num = 0;
04515     if ( *terms == 0 ) return(0);
04516     if ( terms[*terms] == 0 ) return(0);
04517     t = terms;
04518     while ( *t ) {
04519         tnext = t + *t;
04520         tstop = tnext - ABS(tnext[-1]);
04521         t++;
04522         while ( t < tstop ) {
04523             if ( *t >= FUNCTION ) {
04524                 if ( functions[*t-FUNCTION].commute ) {
04525                     num++;
04526                     if ( num >= MAXNUMBEROFNONCOMTERMS ) return(num);
04527                     break;
04528                 }
04529             }
04530             else if ( *t == SUBEXPRESSION ) {
04531                 if ( cbuf[t[4]].CanCommu[t[2]] ) {
04532                     num++;
04533                     if ( num >= MAXNUMBEROFNONCOMTERMS ) return(num);
04534                     break;
04535                 }
04536             }
04537             else if ( *t == EXPRESSION ) {
04538                 num++;
04539                 if ( num >= MAXNUMBEROFNONCOMTERMS ) return(num);
04540                 break;
04541             }
04542             else if ( *t == DOLLAREXPRESSION ) {
04543
04544                 /*
04545                 Technically this is not correct. We have to test first
04546                 whether this is MODLOCAL (in TFORM) and if so, use the
04547                 local version. Anyway, this should be rare to never
04548                 occurring because dollars should be replaced.
04549                 */
04549                 if ( cbuf[AM.dbufnum].CanCommu[t[2]] ) {

```

```

04550             num++;
04551             if ( num >= MAXNUMBEROFNONCOMTERMS ) return(num);
04552             break;
04553         }
04554     }
04555     t += t[1];
04556 }
04557 t = tnext;
04558 }
04559 return(num);
04560 }
04561
04562 /*
04563 #] DetCommu :
04564 #[ DoesCommu :
04565
04566     Determines the number of noncommuting objects in a term.
04567     If the number gets too large we cut it off.
04568 */
04569
04570 WORD DoesCommu(WORD *term)
04571 {
04572     WORD *tstop;
04573     WORD num = 0;
04574     if ( *term == 0 ) return(0);
04575     tstop = term + *term;
04576     tstop = tstop - ABS(tstop[-1]);
04577     term++;
04578     while ( term < tstop ) {
04579         if ( ( *term >= FUNCTION ) && ( functions[*term-FUNCTION].commute ) ) {
04580             num++;
04581             if ( num >= MAXNUMBEROFNONCOMTERMS ) return(num);
04582         }
04583         term += term[1];
04584     }
04585     return(num);
04586 }
04587
04588 /*
04589 #] DoesCommu :
04590 #[ PolyNormPoly :
04591
04592     Normalizes a polynomial
04593 */
04594
04595 #ifdef EVALUATEGCD
04596 WORD *PolyNormPoly (PHEAD WORD *Poly) {
04597
04598     GETBIDENTITY;
04599     WORD *buffer = AT.WorkPointer;
04600     WORD *p;
04601     if ( NewSort(BHEAD0) ) { Terminate(-1); }
04602     AR.CompareRoutine = (COMPAREDUMMY) (&CompareSymbols);
04603     while ( *Poly ) {
04604         p = Poly + *Poly;
04605         if ( SymbolNormalize(Poly) < 0 ) return(0);
04606         if ( StoreTerm(BHEAD Poly) ) {
04607             AR.CompareRoutine = (COMPAREDUMMY) (&Compare1);
04608             LowerSortLevel();
04609             Terminate(-1);
04610         }
04611         Poly = p;
04612     }
04613     if ( EndSort(BHEAD buffer,1) < 0 ) {
04614         AR.CompareRoutine = (COMPAREDUMMY) (&Compare1);
04615         Terminate(-1);
04616     }
04617     p = buffer;
04618     while ( *p ) p += *p;
04619     AR.CompareRoutine = (COMPAREDUMMY) (&Compare1);
04620     AT.WorkPointer = p + 1;
04621     return(buffer);
04622 }
04623 #endif
04624
04625 /*
04626 #] PolyNormPoly :
04627 #[ EvaluateGcd :
04628
04629     Try to evaluate the GCDFUNCTION gcd_.
04630     This function can have a number of arguments which can be numbers
04631     and/or polynomials. If there are objects that aren't SYMBOLS or numbers
04632     it cannot work currently.
04633
04634     To make this work properly we have to intervene in proces.c
04635     proces.c:             if ( Normalize(BHEAD m) ) {
04636 1060 proces.c:             if ( Normalize(BHEAD r) ) {

```

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04637 1126?proces.c:                                if ( Normalize(BHEAD term) ) {
04638     proces.c:                                if ( Normalize(BHEAD AT.WorkPointer) ) goto PasErr;
04639 2308!proces.c:    if ( ( retnorm = Normalize(BHEAD term) ) != 0 ) {
04640     proces.c:                                ReNumber(BHEAD term); Normalize(BHEAD term);
04641     proces.c:                                if ( Normalize(BHEAD v) ) Terminate(-1);
04642     proces.c:    if ( Normalize(BHEAD w) ) { LowerSortLevel(); goto PolyCall; }
04643     proces.c:    if ( Normalize(BHEAD term) ) goto PolyCall;
04644 */
04645 #ifdef EVALUATEGCD
04646
04647 WORD *EvaluateGcd(PHEAD WORD *subterm)
04648 {
04649     GETBIDENTITY
04650     WORD *oldworkpointer = AT.WorkPointer, *work1, *work2, *work3;
04651     WORD *t, *ttt, *ttt, *t1, *t2, *t3, *t4, *tstop;
04652     WORD ct, nnum;
04653     UWORD gcdnum, stor;
04654     WORD *lnum=n_llnum+1;
04655     WORD *num1, *num2, *num3, *den1, *den2, *den3;
04656     WORD sizenum1, sizenum2, sizenum3, sizeden1, sizeden2, sizeden3;
04657     int i, isnumeric = 0, numarg = 0 /*, sizearg */;
04658     LONG size;
04659 /*
04660     Step 1: Look for -SNUMBER or -SYMBOL arguments.
04661     If encountered, treat everybody with it.
04662 */
04663     t = subterm + subterm[1]; t = subterm + FUNHEAD;
04664
04665     while ( t < tt ) {
04666         numarg++;
04667         if ( *t == -SNUMBER ) {
04668             if ( t[1] == 0 ) {
04669 gcdzero:;
04670                 MLOCK(ErrorMessageLock);
04671                 MesPrint("Trying to take the GCD involving a zero term.");
04672                 MUNLOCK(ErrorMessageLock);
04673                 return(0);
04674             }
04675             gcdnum = ABS(t[1]);
04676             t1 = subterm + FUNHEAD;
04677             while ( gcdnum > 1 && t1 < tt ) {
04678                 if ( *t1 == -SNUMBER ) {
04679                     stor = ABS(t1[1]);
04680                     if ( stor == 0 ) goto gcdzero;
04681                     if ( GcdLong(BHEAD (UWORD *)&stor,1,(UWORD *)&gcdnum,1,
04682                             (UWORD *)lnum,&nnum) ) goto FromGCD;
04683                     gcdnum = lnum[0];
04684                     t1 += 2;
04685                     continue;
04686                 }
04687                 else if ( *t1 == -SYMBOL ) goto gcdisone;
04688                 else if ( *t1 < 0 ) goto gcdillegal;
04689 /*
04690                 Now we have to go through all the terms in the argument.
04691                 This includes long numbers.
04692 */
04693                 ttt = t1 + *t1;
04694                 ct = *ttt; *ttt = 0;
04695                 if ( t1[1] != 0 ) { /* First normalize the argument */
04696                     t1 = PolyNormPoly(BHEAD t1+ARGHEAD);
04697                 }
04698                 else t1 += ARGHEAD;
04699                 while ( *t1 ) {
04700                     t1 += *t1;
04701                     i = ABS(t1[-1]);
04702                     t2 = t1 - i;
04703                     i = (i-1)/2;
04704                     t3 = t2+i-1;
04705                     while ( t3 > t2 && *t3 == 0 ) { t3--; i--; }
04706                     if ( GcdLong(BHEAD (UWORD *)t2,(WORD)i,(UWORD *)&gcdnum,1,
04707                             (UWORD *)lnum,&nnum) ) {
04708                         *ttt = ct;
04709                         goto FromGCD;
04710                     }
04711                     gcdnum = lnum[0];
04712                     if ( gcdnum == 1 ) {
04713                         *ttt = ct;
04714                         goto gcdisone;
04715                     }
04716                 }
04717                 *ttt = ct;
04718                 t1 = ttt;
04719                 AT.WorkPointer = oldworkpointer;
04720             }
04721             if ( gcdnum == 1 ) goto gcdisone;
04722             oldworkpointer[0] = 4;
04723             oldworkpointer[1] = gcdnum;

```

```

04724         oldworkpointer[2] = 1;
04725         oldworkpointer[3] = 3;
04726         oldworkpointer[4] = 0;
04727         AT.WorkPointer = oldworkpointer + 5;
04728         return(oldworkpointer);
04729     }
04730     else if ( *t == -SYMBOL ) {
04731         t1 = subterm + FUNHEAD;
04732         i = t[1];
04733         while ( t1 < tt ) {
04734             if ( *t1 == -SNUMBER ) goto gcdisone;
04735             if ( *t1 == -SYMBOL ) {
04736                 if ( t1[1] != i ) goto gcdisone;
04737                 t1 += 2; continue;
04738             }
04739             if ( *t1 < 0 ) goto gcdillegal;
04740             ttt = t1 + *t1;
04741             ct = *ttt; *ttt = 0;
04742             if ( t1[1] != 0 ) { /* First normalize the argument */
04743                 t2 = PolyNormPoly(BHEAD t1+ARGHEAD);
04744             }
04745             else t2 = t1 + ARGHEAD;
04746             while ( *t2 ) {
04747                 t3 = t2+1;
04748                 t2 = t2 + *t2;
04749                 tstop = t2 - ABS(t2[-1]);
04750                 while ( t3 < tstop ) {
04751                     if ( *t3 != SYMBOL ) {
04752                         *ttt = ct;
04753                         goto gcdillegal;
04754                     }
04755                     t4 = t3 + 2;
04756                     t3 += t3[1];
04757                     while ( t4 < t3 ) {
04758                         if ( *t4 == i && t4[1] > 0 ) goto nextterminarg;
04759                         t4 += 2;
04760                     }
04761                 }
04762                 *ttt = ct;
04763                 goto gcdisone;
04764 nextterminarg:;
04765             }
04766             *ttt = ct;
04767             t1 = ttt;
04768             AT.WorkPointer = oldworkpointer;
04769         }
04770         oldworkpointer[0] = 8;
04771         oldworkpointer[1] = SYMBOL;
04772         oldworkpointer[2] = 4;
04773         oldworkpointer[3] = t[1];
04774         oldworkpointer[4] = 1;
04775         oldworkpointer[5] = 1;
04776         oldworkpointer[6] = 1;
04777         oldworkpointer[7] = 3;
04778         oldworkpointer[8] = 0;
04779         AT.WorkPointer = oldworkpointer+9;
04780         return(oldworkpointer);
04781     }
04782     else if ( *t < 0 ) {
04783 gcdillegal:;
04784         MLOCK(ErrorMessageLock);
04785         MesPrint("Illegal object in gcd_ function. Object not a number or a symbol.");
04786         MUNLOCK(ErrorMessageLock);
04787         goto FromGCD;
04788     }
04789     else if ( ABS(t[*t-1]) == *t-ARGHEAD-1 ) isnumeric = numarg;
04790     else if ( t[1] != 0 ) {
04791         ttt = t + *t; ct = *ttt; *ttt = 0;
04792         t = PolyNormPoly(BHEAD t+ARGHEAD);
04793         *ttt = ct;
04794         if ( t[*t] == 0 && ABS(t[*t-1]) == *t-ARGHEAD-1 ) isnumeric = numarg;
04795         AT.WorkPointer = oldworkpointer;
04796         t = ttt;
04797     }
04798     t += *t;
04799 }
04800 /*
04801 At this point there are only generic arguments.
04802 There are however still two cases:
04803 1: There is an argument that is purely numerical
04804    In that case we have to take the gcd of all coefficients
04805 2: All arguments are nontrivial polynomials.
04806    Here we don't worry so much about the factor. (???)
04807    We know whether case 1 occurs when isnumeric > 0.
04808    We can look up numarg to get a good starting value.
04809 */
04810 AT.WorkPointer = oldworkpointer;

```

```

04811     if ( isnumeric ) {
04812         t = subterm + FUNHEAD;
04813         for ( i = 1; i < isnumeric; i++ ) {
04814             NEXTARG(t);
04815         }
04816         if ( t[1] != 0 ) { /* First normalize the argument */
04817             ttt = t + *t; ct = *ttt; *ttt = 0;
04818             t = PolyNormPoly(BHEAD t+ARGHEAD);
04819             *ttt = ct;
04820         }
04821         t += *t;
04822         i = (ABS(t[-1])-1)/2;
04823         den1 = t - 1 - i;
04824         num1 = den1 - i;
04825         sizenum1 = sizeden1 = i;
04826         while ( sizenum1 > 1 && num1[sizenum1-1] == 0 ) sizenum1--;
04827         while ( sizeden1 > 1 && den1[sizeden1-1] == 0 ) sizeden1--;
04828         work1 = AT.WorkPointer+1; work2 = work1+sizenum1;
04829         for ( i = 0; i < sizenum1; i++ ) work1[i] = num1[i];
04830         for ( i = 0; i < sizeden1; i++ ) work2[i] = den1[i];
04831         num1 = work1; den1 = work2;
04832         AT.WorkPointer = work2 = work2 + sizeden1;
04833         t = subterm + FUNHEAD;
04834         while ( t < tt ) {
04835             ttt = t + *t; ct = *ttt; *ttt = 0;
04836             if ( t[1] != 0 ) {
04837                 t = PolyNormPoly(BHEAD t+ARGHEAD);
04838             }
04839             else t += ARGHEAD;
04840             while ( *t ) {
04841                 t += *t;
04842                 i = (ABS(t[-1])-1)/2;
04843                 den2 = t - 1 - i;
04844                 num2 = den2 - i;
04845                 sizenum2 = sizeden2 = i;
04846                 while ( sizenum2 > 1 && num2[sizenum2-1] == 0 ) sizenum2--;
04847                 while ( sizeden2 > 1 && den2[sizeden2-1] == 0 ) sizeden2--;
04848                 num3 = AT.WorkPointer;
04849                 if ( GcdLong(BHEAD (UWORD *)num2,sizenum2,(UWORD *)num1,sizenum1,
04850                     (UWORD *)num3,&sizenum3) ) goto FromGCD;
04851                 sizenum1 = sizenum3;
04852                 for ( i = 0; i < sizenum1; i++ ) num1[i] = num3[i];
04853                 den3 = AT.WorkPointer;
04854                 if ( GcdLong(BHEAD (UWORD *)den2,sizeden2,(UWORD *)den1,sizeden1,
04855                     (UWORD *)den3,&sizeden3) ) goto FromGCD;
04856                 sizeden1 = sizeden3;
04857                 for ( i = 0; i < sizeden1; i++ ) den1[i] = den3[i];
04858                 if ( sizenum1 == 1 && num1[0] == 1 && sizeden1 == 1 && den1[1] == 1 )
04859                     goto gcdisone;
04860             }
04861             *ttt = ct;
04862             t = ttt;
04863             AT.WorkPointer = work2;
04864         }
04865         AT.WorkPointer = oldworkpointer;
04866     /*
04867     Now copy the GCD to the 'output'
04868     */
04869     if ( sizenum1 > sizeden1 ) {
04870         while ( sizenum1 > sizeden1 ) den1[sizeden1++] = 0;
04871     }
04872     else if ( sizenum1 < sizeden1 ) {
04873         while ( sizenum1 < sizeden1 ) num1[sizenum1++] = 0;
04874     }
04875     t = oldworkpointer;
04876     i = 2*sizenum1+1;
04877     *t++ = i+1;
04878     if ( num1 != t ) { NCOPY(t,num1,sizenum1); }
04879     else t += sizenum1;
04880     if ( den1 != t ) { NCOPY(t,den1,sizeden1); }
04881     else t += sizeden1;
04882     *t++ = i;
04883     *t++ = 0;
04884     AT.WorkPointer = t;
04885     return(oldworkpointer);
04886 }
04887 /*
04888 Now the real stuff with only polynomials.
04889 Pick up the shortest term to start.
04890 We are a bit brutish about this.
04891 */
04892 t = subterm + FUNHEAD;
04893 AT.WorkPointer += AM.MaxTer/sizeof(WORD);
04894 work2 = AT.WorkPointer;
04895 /*
04896 sizearg = subterm[1];
04897 */

```

```

04898     i = 0; work3 = 0;
04899     while ( t < tt ) {
04900         i++;
04901         work1 = AT.WorkPointer;
04902         ttt = t + *t; ct = *ttt; *ttt = 0;
04903         t = PolyNormPoly(BHEAD t+ARGHEAD);
04904         if ( *work1 < AT.WorkPointer-work1 ) {
04905             /*
04906                 sizearg = AT.WorkPointer-work1;
04907             */
04908                 numarg = i;
04909                 work3 = work1;
04910             }
04911             *ttt = ct; t = ttt;
04912         }
04913         *AT.WorkPointer++ = 0;
04914     /*
04915         We have properly normalized arguments and the shortest is indicated in work3
04916     */
04917     work1 = work3;
04918     while ( *work2 ) {
04919         if ( work2 != work3 ) {
04920             work1 = PolyGCD2(BHEAD work1,work2);
04921         }
04922         while ( *work2 ) work2 += *work2;
04923         work2++;
04924     }
04925     work2 = work1;
04926     while ( *work2 ) work2 += *work2;
04927     size = work2 - work1 + 1;
04928     t = oldworkpointer;
04929     NCOPY(t,work1,size);
04930     AT.WorkPointer = t;
04931     return(oldworkpointer);
04932
04933 gcdisone;;
04934     oldworkpointer[0] = 4;
04935     oldworkpointer[1] = 1;
04936     oldworkpointer[2] = 1;
04937     oldworkpointer[3] = 3;
04938     oldworkpointer[4] = 0;
04939     AT.WorkPointer = oldworkpointer+5;
04940     return(oldworkpointer);
04941 FromGCD:
04942     MLOCK(ErrorMessageLock);
04943     MesCall("EvaluateGcd");
04944     MUNLOCK(ErrorMessageLock);
04945     return(0);
04946 }
04947
04948 #endif
04949
04950 /*
04951     #] EvaluateGcd :
04952     #[ TreatPolyRatFun :
04953
04954     if ( AR.PolyFunExp == 1 ) we have to trim the contents of the polyratfun
04955     down to its most divergent term and give it coefficient +1. This is done
04956     by taking the terms with the least power in the variable in the numerator
04957     and in the denominator and then combine them.
04958     Answer is either PolyRatFun(ep^n,1) or PolyRatFun(1,1) or PolyRatFun(1,ep^n)
04959 */
04960
04961 int TreatPolyRatFun(PHEAD WORD *prf)
04962 {
04963     WORD *t, *tstop, *r, *rstop, *m, *mstop;
04964     WORD expl = MAXPOWER, exp2 = MAXPOWER;
04965     t = prf+FUNHEAD;
04966     if ( *t < 0 ) {
04967         if ( *t == -SYMBOL && t[1] == AR.PolyFunVar ) {
04968             if ( expl > 1 ) expl = 1;
04969             t += 2;
04970         }
04971         else {
04972             if ( expl > 0 ) expl = 0;
04973             NEXTARG(t)
04974         }
04975     }
04976     else {
04977         tstop = t + *t;
04978         t += ARGHEAD;
04979         while ( t < tstop ) {
04980             /*
04981                 Now look for the minimum power of AR.PolyFunVar
04982             */
04983                 r = t+1;
04984                 t += *t;

```

```

04985         rstop = t - ABS(t[-1]);
04986         while ( r < rstop ) {
04987             if ( *r != SYMBOL ) { r += r[1]; continue; }
04988             m = r;
04989             mstop = m + m[1];
04990             m += 2;
04991             while ( m < mstop ) {
04992                 if ( *m == AR.PolyFunVar ) {
04993                     if ( m[1] < expl ) expl = m[1];
04994                     break;
04995                 }
04996                 m += 2;
04997             }
04998             if ( m == mstop ) {
04999                 if ( expl > 0 ) expl = 0;
05000             }
05001             break;
05002         }
05003         if ( r == rstop ) {
05004             if ( expl > 0 ) expl = 0;
05005         }
05006     }
05007     t = tstop;
05008 }
05009 if ( *t < 0 ) {
05010     if ( *t == -SYMBOL && t[1] == AR.PolyFunVar ) {
05011         if ( exp2 > 1 ) exp2 = 1;
05012     }
05013     else {
05014         if ( exp2 > 0 ) exp2 = 0;
05015     }
05016 }
05017 else {
05018     tstop = t + *t;
05019     t += ARGHEAD;
05020     while ( t < tstop ) {
05021 /*
05022         Now look for the minimum power of AR.PolyFunVar
05023 */
05024         r = t+1;
05025         t += *t;
05026         rstop = t - ABS(t[-1]);
05027         while ( r < rstop ) {
05028             if ( *r != SYMBOL ) { r += r[1]; continue; }
05029             m = r;
05030             mstop = m + m[1];
05031             m += 2;
05032             while ( m < mstop ) {
05033                 if ( *m == AR.PolyFunVar ) {
05034                     if ( m[1] < exp2 ) exp2 = m[1];
05035                     break;
05036                 }
05037                 m += 2;
05038             }
05039             if ( m == mstop ) {
05040                 if ( exp2 > 0 ) exp2 = 0;
05041             }
05042             break;
05043         }
05044         if ( r == rstop ) {
05045             if ( exp2 > 0 ) exp2 = 0;
05046         }
05047     }
05048 }
05049 /*
05050     Now we can compose the output.
05051     Notice that the output can never be longer than the input provided
05052     we never can have arguments that consist of just a function.
05053 */
05054 expl = expl-exp2;
05055 /* if ( expl > 0 ) expl = 0; */
05056 t = prf+FUNHEAD;
05057 if ( expl == 0 ) {
05058     *t++ = -SNUMBER; *t++ = 1;
05059     *t++ = -SNUMBER; *t++ = 1;
05060 }
05061 else if ( expl > 0 ) {
05062     if ( expl == 1 ) {
05063         *t++ = -SYMBOL; *t++ = AR.PolyFunVar;
05064     }
05065     else {
05066         *t++ = 8+ARGHEAD;
05067         *t++ = 0;
05068         FILLARG(t);
05069         *t++ = 8; *t++ = SYMBOL; *t++ = 4; *t++ = AR.PolyFunVar;
05070         *t++ = expl; *t++ = 1; *t++ = 1; *t++ = 3;
05071     }

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```

05072     *t++ = -SNUMBER; *t++ = 1;
05073 }
05074 else {
05075     *t++ = -SNUMBER; *t++ = 1;
05076     if ( expl == -1 ) {
05077         *t++ = -SYMBOL; *t++ = AR.PolyFunVar;
05078     }
05079     else {
05080         *t++ = 8+ARGHEAD;
05081         *t++ = 0;
05082         FILLARG(t);
05083         *t++ = 8; *t++ = SYMBOL; *t++ = 4; *t++ = AR.PolyFunVar;
05084         *t++ = -expl; *t++ = 1; *t++ = 1; *t++ = 3;
05085     }
05086 }
05087 prf[2] = 0; /* Clean */
05088 prf[1] = t - prf;
05089 return(0);
05090 }
05091
05092 /*
05093 #] TreatPolyRatFun :
05094 #[ DropCoefficient :
05095 */
05096
05097 void DropCoefficient(PHEAD WORD *term)
05098 {
05099     GETBIDENTITY
05100     WORD *t = term + *term;
05101     WORD n, na;
05102     n = t[-1]; na = ABS(n);
05103     t -= na;
05104     if ( n == 3 && t[0] == 1 && t[1] == 1 ) return;
05105     *AN.RepPoint = 1;
05106     t[0] = 1; t[1] = 1; t[2] = 3;
05107     *term -= (na-3);
05108 }
05109
05110 /*
05111 #] DropCoefficient :
05112 #[ DropSymbols :
05113 */
05114
05115 void DropSymbols(PHEAD WORD *term)
05116 {
05117     GETBIDENTITY
05118     WORD *tend = term + *term, *t1, *t2, *tstop;
05119     tstop = tend - ABS(tend[-1]);
05120     t1 = term+1;
05121     while ( t1 < tstop ) {
05122         if ( *t1 == SYMBOL ) {
05123             *AN.RepPoint = 1;
05124             t2 = t1+t1[1];
05125             while ( t2 < tend ) *t1++ = *t2++;
05126             *term = t1 - term;
05127             break;
05128         }
05129         t1 += t1[1];
05130     }
05131 }
05132
05133 /*
05134 #] DropSymbols :
05135 #[ SymbolNormalize :
05136 */
05137
05138 int SymbolNormalize(WORD *term)
05139 {
05140     GETIDENTITY
05141     WORD buffer[7*NORMSIZE], *t, *b, *bb, *tt, *m, *tstop;
05142     int i;
05143     b = buffer;
05144     *b++ = SYMBOL; *b++ = 2;
05145     t = term + *term;
05146     tstop = t - ABS(t[-1]);
05147     t = term + 1;
05148     while ( t < tstop ) { /* Step 1: collect symbols */
05149         if ( *t == SYMBOL && t < tstop ) {
05150             for ( i = 2; i < t[1]; i += 2 ) {
05151                 bb = buffer+2;
05152                 while ( bb < b ) {
05153                     if ( bb[0] == t[i] ) { /* add powers */
05154                         bb[1] += t[i+1];
05155                         if ( bb[1] > MAXPOWER || bb[1] < -MAXPOWER ) {
05156                             MLOCK(ErrorMessageLock);
05157                             MesPrint("Power in SymbolNormalize out of range");
05158                             MUNLOCK(ErrorMessageLock);
05159                             return(-1);
05160                         }
05161                     }
05162                 }
05163             }
05164         }
05165     }
05166 }

```



```

05167         }
05168         if ( bb[1] == 0 ) {
05169             b -= 2;
05170             while ( bb < b ) {
05171                 bb[0] = bb[2]; bb[1] = bb[3]; bb += 2;
05172             }
05173         }
05174         goto Nexti;
05175     }
05176     else if ( bb[0] > t[i] ) { /* insert it */
05177         m = b;
05178         while ( m > bb ) { m[1] = m[-1]; m[0] = m[-2]; m -= 2; }
05179         b += 2;
05180         bb[0] = t[i];
05181         bb[1] = t[i+1];
05182         goto Nexti;
05183     }
05184     bb += 2;
05185 }
05186 if ( bb >= b ) { /* add it to the end */
05187     *b++ = t[i]; *b++ = t[i+1];
05188 }
05189 Nexti;;
05190 }
05191 }
05192 else {
05193     MLOCK(ErrorMessageLock);
05194     MesPrint("Illegal term in SymbolNormalize");
05195     MUNLOCK(ErrorMessageLock);
05196     return(-1);
05197 }
05198 t += t[1];
05199 }
05200 buffer[1] = b - buffer;
05201 /*
05202 Veto negative powers
05203 */
05204 if ( AT.LeaveNegative == 0 ) {
05205     b = buffer; bb = b + b[1]; b += 3;
05206     while ( b < bb ) {
05207         if ( *b < 0 ) {
05208             MLOCK(ErrorMessageLock);
05209             MesPrint("Negative power in SymbolNormalize");
05210             MUNLOCK(ErrorMessageLock);
05211             return(-1);
05212         }
05213         b += 2;
05214     }
05215 }
05216 /*
05217 Now we use the fact that the new term will not be longer than the old one
05218 Actually it should be shorter when there is more than one subterm!
05219 Copy back.
05220 */
05221 i = buffer[1];
05222 b = buffer; tt = term + 1;
05223 if ( i > 2 ) { NCOPY(tt,b,i) }
05224 if ( tt < tstop ) {
05225     i = term[*term-1];
05226     if ( i < 0 ) i = -i;
05227     *term -= (tstop-tt);
05228     NCOPY(tt,tstop,i)
05229 }
05230 return(0);
05231 }
05232
05233 /*
05234 #] SymbolNormalize :
05235 #[ TestFunFlag :
05236
05237 Tests whether a function still has unsubstituted subexpressions
05238 This function has its dirtyflag on!
05239 */
05240
05241 int TestFunFlag(PHEAD WORD *tfun)
05242 {
05243     WORD *t, *tstop, *r, *rstop, *m, *mstop;
05244     if ( functions[*tfun-FUNCTION].spec <= 0 ) return(0);
05245     tstop = tfun + tfun[1];
05246     t = tfun + FUNHEAD;
05247     while ( t < tstop ) {
05248         if ( *t < 0 ) { NEXTARG(t); continue; }
05249         rstop = t + *t;
05250         if ( t[1] == 0 ) { t = rstop; continue; }
05251         r = t + ARGHEAD;
05252         while ( r < rstop ) { /* Here we loop over terms */
05253             m = r+1; mstop = r+r; mstop -= ABS(mstop[-1]);

```

```

05254         while ( m < mstop ) { /* Loop over the subterms */
05255             if ( *m == SUBEXPRESSION || *m == EXPRESSION || *m == DOLLAREXPRESSION ) return(1);
05256             if ( ( *m >= FUNCTION ) && ( ( m[2] & DIRTYFLAG ) == DIRTYFLAG )
05257                 && ( *m != REPLACEMENT ) && TestFunFlag(BHEAD m) ) return(1);
05258             m += m[1];
05259         }
05260         r += *r;
05261     }
05262     t += *t;
05263 }
05264 return(0);
05265 }
05266 /*
05267 #] TestFunFlag :
05268 #[ BracketNormalize :
05270 */
05271
05272 int BracketNormalize(PHEAD WORD *term)
05273 {
05274     WORD *stop = term+*term-3, *t, *tt, *tstart, *r;
05275     WORD *oldwork = AT.WorkPointer;
05276     WORD *termout;
05277     WORD i, ii, j;
05278     termout = AT.WorkPointer = term+*term;
05279     /*
05280     First collect all functions and sort them
05281     */
05282     tt = termout+1; t = term+1;
05283     while ( t < stop ) {
05284         if ( *t >= FUNCTION ) { i = t[1]; NCOPY(tt,t,i); }
05285         else t += t[1];
05286     }
05287     if ( tt > termout+1 && tt-termout-1 > termout[2] ) { /* sorting */
05288         r = termout+1; ii = tt-r;
05289         for ( i = 0; i < ii-FUNHEAD; i += FUNHEAD ) { /* Bubble sort */
05290             for ( j = i+FUNHEAD; j > 0; j -= FUNHEAD ) {
05291                 if ( functions[r[j-FUNHEAD]-FUNCTION].commute
05292                     && functions[r[j]-FUNCTION].commute == 0 ) break;
05293                 if ( r[j-FUNHEAD] > r[j] ) EXCH(r[j-FUNHEAD],r[j])
05294                 else break;
05295             }
05296         }
05297     }
05298
05299     tstart = tt; t = term + 1; *tt++ = DELTA; *tt++ = 2;
05300     while ( t < stop ) {
05301         if ( *t == DELTA ) { i = t[1]-2; t += 2; tstart[1] += i; NCOPY(tt,t,i); }
05302         else t += t[1];
05303     }
05304     if ( tstart[1] > 2 ) {
05305         for ( r = tstart+2; r < tstart+tstart[1]; r += 2 ) {
05306             if ( r[0] > r[1] ) EXCH(r[0],r[1])
05307         }
05308     }
05309     if ( tstart[1] > 4 ) { /* sorting */
05310         r = tstart+2; ii = tstart[1]-2;
05311         for ( i = 0; i < ii-2; i += 2 ) { /* Bubble sort */
05312             for ( j = i+2; j > 0; j -= 2 ) {
05313                 if ( r[j-2] > r[j] ) {
05314                     EXCH(r[j-2],r[j])
05315                     EXCH(r[j-1],r[j+1])
05316                 }
05317                 else if ( r[j-2] < r[j] ) break;
05318                 else {
05319                     if ( r[j-1] > r[j+1] ) EXCH(r[j-1],r[j+1])
05320                     else break;
05321                 }
05322             }
05323         }
05324         tt = tstart+tstart[1];
05325     }
05326     else if ( tstart[1] == 2 ) { tt = tstart; }
05327     else tt = tstart+4;
05328
05329     tstart = tt; t = term + 1; *tt++ = INDEX; *tt++ = 2;
05330     while ( t < stop ) {
05331         if ( *t == INDEX ) { i = t[1]-2; t += 2; tstart[1] += i; NCOPY(tt,t,i); }
05332         else t += t[1];
05333     }
05334     if ( tstart[1] >= 4 ) { /* sorting */
05335         r = tstart+2; ii = tstart[1]-2;
05336         for ( i = 0; i < ii-1; i += 1 ) { /* Bubble sort */
05337             for ( j = i+1; j > 0; j -= 1 ) {
05338                 if ( r[j-1] > r[j] ) EXCH(r[j-1],r[j])
05339                 else break;
05340             }

```

```

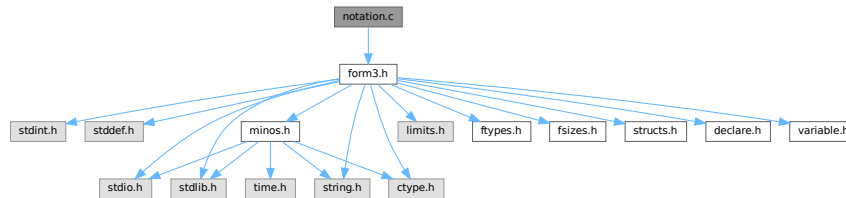
05341     }
05342     tt = tstart+tstart[1];
05343 }
05344 else if ( tstart[1] == 2 ) { tt = tstart; }
05345 else tt = tstart+3;
05346
05347 tstart = tt; t = term + 1; *tt++ = DOTPRODUCT; *tt++ = 2;
05348 while ( t < stop ) {
05349     if ( *t == DOTPRODUCT ) { i = t[1]-2; t += 2; tstart[1] += i; NCOPY(tt,t,i); }
05350     else t += t[1];
05351 }
05352 if ( tstart[1] > 5 ) { /* sorting */
05353     r = tstart+2; ii = tstart[1]-2;
05354     for ( i = 0; i < ii; i += 3 ) {
05355         if ( r[i] > r[i+1] ) EXCH(r[i],r[i+1])
05356     }
05357     for ( i = 0; i < ii-3; i += 3 ) { /* Bubble sort */
05358         for ( j = i+3; j > 0; j -= 3 ) {
05359             if ( r[j-3] < r[j] ) break;
05360             if ( r[j-3] > r[j] ) {
05361                 EXCH(r[j-3],r[j])
05362                 EXCH(r[j-2],r[j+1])
05363             }
05364             else {
05365                 if ( r[j-2] > r[j+1] ) EXCH(r[j-2],r[j+1])
05366                 else break;
05367             }
05368         }
05369     }
05370     tt = tstart+tstart[1];
05371 }
05372 else if ( tstart[1] == 2 ) { tt = tstart; }
05373 else {
05374     if ( tstart[2] > tstart[3] ) EXCH(tstart[2],tstart[3])
05375     tt = tstart+5;
05376 }
05377
05378 tstart = tt; t = term + 1; *tt++ = SYMBOL; *tt++ = 2;
05379 while ( t < stop ) {
05380     if ( *t == SYMBOL ) { i = t[1]-2; t += 2; tstart[1] += i; NCOPY(tt,t,i); }
05381     else t += t[1];
05382 }
05383 if ( tstart[1] > 4 ) { /* sorting */
05384     r = tstart+2; ii = tstart[1]-2;
05385     for ( i = 0; i < ii-2; i += 2 ) { /* Bubble sort */
05386         for ( j = i+2; j > 0; j -= 2 ) {
05387             if ( r[j-2] > r[j] ) EXCH(r[j-2],r[j])
05388             else break;
05389         }
05390     }
05391     tt = tstart+tstart[1];
05392 }
05393 else if ( tstart[1] == 2 ) { tt = tstart; }
05394 else tt = tstart+4;
05395
05396 tstart = tt; t = term + 1; *tt++ = SETSET; *tt++ = 2;
05397 while ( t < stop ) {
05398     if ( *t == SETSET ) { i = t[1]-2; t += 2; tstart[1] += i; NCOPY(tt,t,i); }
05399     else t += t[1];
05400 }
05401 if ( tstart[1] > 4 ) { /* sorting */
05402     r = tstart+2; ii = tstart[1]-2;
05403     for ( i = 0; i < ii-2; i += 2 ) { /* Bubble sort */
05404         for ( j = i+2; j > 0; j -= 2 ) {
05405             if ( r[j-2] > r[j] ) {
05406                 EXCH(r[j-2],r[j])
05407                 EXCH(r[j-1],r[j+1])
05408             }
05409             else break;
05410         }
05411     }
05412     tt = tstart+tstart[1];
05413 }
05414 else if ( tstart[1] == 2 ) { tt = tstart; }
05415 else tt = tstart+4;
05416 *tt++ = 1; *tt++ = 1; *tt++ = 3;
05417 t = term; i = *termout = tt - termout; tt = termout;
05418 NCOPY(t,tt,i);
05419 AT.WorkPointer = oldwork;
05420 return(0);
05421 }
05422
05423 /*
05424 #] BracketNormalize :
05425 */

```

## 4.88 notation.c File Reference

```
#include "form3.h"
```

Include dependency graph for notation.c:



### Functions

- int [NormPolyTerm](#) (PHEAD WORD \*term)
- int [ConvertToPoly](#) (PHEAD WORD \*term, WORD \*outterm, WORD \*comlist, WORD par)
- int [LocalConvertToPoly](#) (PHEAD WORD \*term, WORD \*outterm, WORD startbuf, WORD par)
- int [ConvertFromPoly](#) (PHEAD WORD \*term, WORD \*outterm, WORD from, WORD to, WORD offset, WORD par)
- int [FindSubterm](#) (WORD \*subterm)
- int [FindLocalSubterm](#) (PHEAD WORD \*subterm, WORD startbuf)
- void [PrintSubtermList](#) (int from, int to)
- void [PrintExtraSymbol](#) (int num, WORD \*terms, int par)
- int [FindSubexpression](#) (WORD \*subexpr)
- int [ExtraSymFun](#) (PHEAD WORD \*term, WORD level)
- int [PruneExtraSymbols](#) (WORD downto)

### 4.88.1 Detailed Description

Contains the functions that deal with the rewriting and manipulation of expressions/terms in polynomial representation.

Definition in file [notation.c](#).

### 4.88.2 Function Documentation

#### 4.88.2.1 NormPolyTerm()

```
int NormPolyTerm (
    PHEAD WORD * term )
```

Definition at line 53 of file [notation.c](#).

#### 4.88.2.2 ConvertToPoly()

```
int ConvertToPoly (
    PHEAD WORD * term,
    WORD * outterm,
    WORD * comlist,
    WORD par )
```

Definition at line 307 of file [notation.c](#).

#### 4.88.2.3 LocalConvertToPoly()

```
int LocalConvertToPoly (
    PHEAD WORD * term,
    WORD * outterm,
    WORD startebuf,
    WORD par )
```

Converts a generic term to polynomial notation in which there are only symbols and brackets. During conversion there will be only symbols. Brackets are stripped. Objects that need 'translation' are put inside a special compiler buffer and represented by a symbol. The numbering of the extra symbols is down from the maximum. In principle there can be a problem when running into the already assigned ones. This uses the FindTree for searching in the global tree and then looks further in the AT.ebufnum. This allows fully parallel processing. Hence we need no locks. Cannot be used in the same module as ConvertToPoly.

Definition at line 510 of file [notation.c](#).

Referenced by [poly\\_factorize\\_expression\(\)](#).

#### 4.88.2.4 ConvertFromPoly()

```
int ConvertFromPoly (
    PHEAD WORD * term,
    WORD * outterm,
    WORD from,
    WORD to,
    WORD offset,
    WORD par )
```

Definition at line 660 of file [notation.c](#).

#### 4.88.2.5 FindSubterm()

```
int FindSubterm (
    WORD * subterm )
```

Definition at line 773 of file [notation.c](#).

#### 4.88.2.6 FindLocalSubterm()

```
int FindLocalSubterm (
    PHEAD WORD * subterm,
    WORD startbuf )
```

Definition at line [843](#) of file [notation.c](#).

#### 4.88.2.7 PrintSubtermList()

```
void PrintSubtermList (
    int from,
    int to )
```

Definition at line [897](#) of file [notation.c](#).

#### 4.88.2.8 PrintExtraSymbol()

```
void PrintExtraSymbol (
    int num,
    WORD * terms,
    int par )
```

Definition at line [1009](#) of file [notation.c](#).

#### 4.88.2.9 FindSubexpression()

```
int FindSubexpression (
    WORD * subexpr )
```

Definition at line [1096](#) of file [notation.c](#).

#### 4.88.2.10 ExtraSymFun()

```
int ExtraSymFun (
    PHEAD WORD * term,
    WORD level )
```

Definition at line [1149](#) of file [notation.c](#).

#### 4.88.2.11 PruneExtraSymbols()

```
int PruneExtraSymbols (
    WORD downto )
```

Definition at line [1217](#) of file [notation.c](#).

## 4.89 notation.c

[Go to the documentation of this file.](#)

```

00001
00006 /* #[ License : */
00007 /*
00008 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 *   When using this file you are requested to refer to the publication
00010 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 *   This is considered a matter of courtesy as the development was paid
00012 *   for by FOM the Dutch physics granting agency and we would like to
00013 *   be able to track its scientific use to convince FOM of its value
00014 *   for the community.
00015 *
00016 *   This file is part of FORM.
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00026 *   details.
00027 *
00028 *   You should have received a copy of the GNU General Public License along
00029 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #[ License : */
00032 /*
00033     #[ Includes :
00034 */
00035
00036 #include "form3.h"
00037
00038 /*
00039     #[ Includes :
00040         #[ NormPolyTerm :
00041
00042         Brings a term to normal form.
00043
00044         This routine knows objects of the following types:
00045             SYMBOL
00046             HAAKJE
00047             SNUMBER
00048             LNUMBER
00049         The SNUMBER and LNUMBER are worked into the coefficient.
00050         One of the essences here is that everything can be done in place.
00051 */
00052
00053 int NormPolyTerm(PHEAD WORD *term)
00054 {
00055     WORD *tcoef, ncoef, *tstop, *tfill, *t, *tt;
00056     int equal, i;
00057     WORD *r1, *r2, *r3, *r4, *r5, *rfirst, rv;
00058     WORD *lnum, nnum; /* Scratch, originally for factorials */
00059 /*
00060     One: find the coefficient
00061 */
00062     tcoef = term+*term;
00063     ncoef = tcoef[-1];
00064     tstop = tcoef - ABS(tcoef[-1]);
00065     tfill = t = term + 1;
00066     rfirst = 0;
00067     if ( t >= tstop ) { return(*term); }
00068     while ( t < tstop ) {
00069         switch ( *t ) {
00070             case SYMBOL:
00071                 if ( rfirst == 0 ) {
00072 /*
00073                     Here we only need to sort
00074                     1: assume no equals. Bubble.
00075 */
00076                     rfirst = t;
00077                     r2 = rfirst+4; tt = r3 = t + t[1]; equal = 0;
00078                     while ( r2 < r3 ) {
00079                         r1 = r2 - 2;
00080                         if ( *r2 > *r1 ) { r2 += 2; continue; }
00081                         if ( *r2 == *r1 ) { r2 += 2; equal = 1; continue; }
00082                         rv = *r1; *r1 = *r2; *r2 = rv;
00083                         r1 -= 2; r2 -= 2; r4 = r2 + 2;
00084                         while ( r1 > t ) {
00085                             if ( *r2 >= *r1 ) { r2 = r4; break; }
00086                             rv = *r1; *r1 = *r2; *r2 = rv;

```

```

00087         r1 -= 2; r2 -= 2;
00088     }
00089 }
00090 /*
00091     2: hunt down the equal objects
00092     postpone eliminating zero powers.
00093 */
00094     if ( equal ) {
00095         r1 = t+2; r2 = r1+2;
00096         while ( r2 < r3 ) {
00097             if ( *r1 == *r2 ) {
00098                 r1[1] += r2[1];
00099                 r4 = r2+2;
00100                 while ( r4 < r3 ) *r2++ = *r4++;
00101                 t[1] -= 2;
00102                 r2 = r1 + 2; r3 -= 2;
00103             }
00104         }
00105     }
00106 }
00107 else {
00108     /*
00109         Here we only need to insert
00110     */
00111     r1 = t + 2; tt = r3 = t + t[1];
00112     while ( r1 < r3 ) {
00113         r2 = rfirst+2; r4 = rfirst + rfirst[1];
00114         while ( r2 < r4 ) {
00115             if ( *r1 == *r2 ) {
00116                 r2[1] += r1[1];
00117                 break;
00118             }
00119             else if ( *r2 > *r1 ) {
00120                 r5 = r4;
00121                 while ( r5 > r2 ) { r5[1] = r5[-1]; r5[0] = r5[-2]; r5 -= 2; }
00122                 rfirst[1] += 2;
00123                 *r2 = *r1; r2[1] = r1[1];
00124                 break;
00125             }
00126             r2 += 2;
00127         }
00128         if ( r2 == r4 ) {
00129             rfirst[1] += 2;
00130             *r2++ = *r1++; *r2++ = *r1++;
00131         }
00132         else r1 += 2;
00133     }
00134 }
00135 t = tt;
00136 break;
00137 case HAAKJE: /* Here we skip brackets */
00138     t += t[1];
00139     break;
00140 case SNUMBER:
00141     if ( t[2] < 0 ) {
00142         t[2] = -t[2];
00143         if ( t[3] & 1 ) ncoef = -ncoef;
00144     }
00145     else if ( t[2] == 0 ) {
00146         if ( t[3] < 0 ) goto NormInf;
00147         goto NormZero;
00148     }
00149     lnum = TermMalloc("lnum");
00150     lnum[0] = t[2];
00151     nnum = 1;
00152     if ( t[3] && RaisPow(BHEAD (UWORD *)lnum,&nnum, (UWORD) (ABS(t[3])) ) ) goto FromNorm;
00153     ncoef = REDLENG(ncoef);
00154     if ( t[3] < 0 ) {
00155         if ( Divvy(BHEAD (UWORD *)tstop,&ncoef, (UWORD *)lnum,nnum) )
00156             goto FromNorm;
00157     }
00158     else if ( t[3] > 0 ) {
00159         if ( Mully(BHEAD (UWORD *)tstop,&ncoef, (UWORD *)lnum,nnum) )
00160             goto FromNorm;
00161     }
00162     ncoef = INCLENG(ncoef);
00163     t += t[1];
00164     TermFree(lnum,"lnum");
00165     break;
00166 case LNUMBER:
00167     ncoef = REDLENG(ncoef);
00168     if ( Mully(BHEAD (UWORD *)tstop,&ncoef, (UWORD *) (t+3),t[2]) ) goto FromNorm;
00169     ncoef = INCLENG(ncoef);
00170     t += t[1];
00171     break;
00172 default:
00173     MLOCK(ErrorMessageLock);

```



```

00174         MesPrint("Illegal code in NormPolyTerm");
00175         MUNLOCK(ErrorMessageLock);
00176         Terminate(-1);
00177         break;
00178     }
00179 }
00180 /*
00181 Now we try to eliminate objects to the power zero.
00182 */
00183 if ( rfirst ) {
00184     r2 = rfirst+2;
00185     r3 = rfirst + rfirst[1];
00186     while ( r2 < r3 ) {
00187         if ( r2[1] == 0 ) {
00188             r1 = r2 + 2;
00189             while ( r1 < r3 ) { r1[-2] = r1[0]; r1[-1] = r1[1]; r1 += 2; }
00190             r3 -= 2;
00191             rfirst[1] -= 2;
00192         }
00193         else { r2 += 2; }
00194     }
00195     if ( rfirst[1] < 4 ) rfirst = 0;
00196 }
00197 /*
00198 Finally we put the term together
00199 */
00200 if ( rfirst ) {
00201     i = rfirst[1];
00202     NCOPY(tfill,rfirst,i)
00203 }
00204 i = ABS(ncoef)-1;
00205 NCOPY(tfill,tstop,i)
00206 *tfill++ = ncoef;
00207 *term = tfill - term;
00208 return(*term);
00209 NormZero:
00210 *term = 0;
00211 return(0);
00212 NormInf:
00213 MLOCK(ErrorMessageLock);
00214 MesPrint("0^0 in NormPolyTerm");
00215 MUNLOCK(ErrorMessageLock);
00216 Terminate(-1);
00217 return(-1);
00218 FromNorm:
00219 MLOCK(ErrorMessageLock);
00220 MesCall("NormPolyTerm");
00221 MUNLOCK(ErrorMessageLock);
00222 Terminate(-1);
00223 return(-1);
00224 }
00225
00226 /*
00227 #] NormPolyTerm :
00228 #[ ComparePoly :
00229 */
00251 #ifdef WITHCOMPAREPOLY
00252
00253 WORD ComparePoly(WORD *term1, WORD *term2, WORD level)
00254 {
00255     WORD *t1, *t2, *t3, *t4, *tstop1, *tstop2;
00256     tstop1 = term1 + *term1;
00257     tstop1 -= ABS(tstop1[-1]);
00258     tstop2 = term2 + *term2;
00259     tstop2 -= ABS(tstop2[-1]);
00260     t1 = term1+1;
00261     t2 = term2+1;
00262     while ( t1 < tstop1 && t2 < tstop2 ) {
00263         if ( *t1 == *t2 ) {
00264             if ( *t1 == HAAKJE ) {
00265                 if ( t1[2] != t2[2] ) return(t2[2]-t1[2]);
00266                 t1 += t1[1]; t2 += t2[1];
00267             }
00268             else { /* must be type SYMBOL */
00269                 t3 = t1 + t1[1]; t4 = t2 + t2[1];
00270                 t1 += 2; t2 += 2;
00271                 while ( t1 < t3 && t2 < t4 ) {
00272                     if ( *t1 != *t2 ) return(*t2-*t1);
00273                     if ( t1[1] != t2[1] ) return(t2[1]-t1[1]);
00274                     t1 += 2; t2 += 2;
00275                 }
00276                 if ( t1 < t3 ) return(-1);
00277                 if ( t2 < t4 ) return(1);
00278             }
00279         }
00280         else return(*t2-*t1);
00281     }

```

```

00282     if ( t1 < tstop1 ) return(-1);
00283     if ( t2 < tstop2 ) return(1);
00284     return(0);
00285 }
00286
00287 #endif
00288
00289 /*
00290     #] ComparePoly :
00291     #[ ConvertToPoly :
00292 */
00305 static int FirstWarnConvertToPoly = 1;
00306
00307 int ConvertToPoly(PHEAD WORD *term, WORD *outterm, WORD *comlist, WORD par)
00308 {
00309     WORD *tout, *tstop, ncoef, *t, *r, *tt, *ttwo = 0;
00310     int i, action = 0;
00311     tt = term + *term;
00312     ncoef = ABS(tt[-1]);
00313     tstop = tt - ncoef;
00314     tout = outterm+1;
00315     t = term + 1;
00316     if ( comlist[2] == DOALL ) {
00317         while ( t < tstop ) {
00318             if ( *t == SYMBOL ) {
00319                 r = t+2;
00320                 t += t[1];
00321                 while ( r < t ) {
00322                     if ( r[1] > 0 ) {
00323                         *tout++ = SYMBOL;
00324                         *tout++ = 4;
00325                         *tout++ = r[0];
00326                         *tout++ = r[1];
00327                     }
00328                     else {
00329                         tout[1] = SYMBOL;
00330                         tout[2] = 4;
00331                         tout[3] = r[0];
00332                         tout[4] = -1;
00333                         i = FindSubterm(tout+1);
00334                         *tout++ = SYMBOL;
00335                         *tout++ = 4;
00336                         *tout++ = MAXVARIABLES-i;
00337                         *tout++ = -r[1];
00338                         action = 1;
00339                     }
00340                     r += 2;
00341                 }
00342             }
00343             else if ( *t == DOTPRODUCT ) {
00344                 r = t + 2;
00345                 t += t[1];
00346                 while ( r < t ) {
00347                     tout[1] = DOTPRODUCT;
00348                     tout[2] = 5;
00349                     tout[3] = r[0];
00350                     tout[4] = r[1];
00351                     if ( r[2] < 0 ) {
00352                         tout[5] = -1;
00353                     }
00354                     else {
00355                         tout[5] = 1;
00356                     }
00357                     i = FindSubterm(tout+1);
00358                     *tout++ = SYMBOL;
00359                     *tout++ = 4;
00360                     *tout++ = MAXVARIABLES-i;
00361                     *tout++ = ABS(r[2]);
00362                     r += 3;
00363                     action = 1;
00364                 }
00365             }
00366             else if ( *t == VECTOR ) {
00367                 r = t + 2;
00368                 t += t[1];
00369                 while ( r < t ) {
00370                     tout[1] = VECTOR;
00371                     tout[2] = 4;
00372                     tout[3] = r[0];
00373                     tout[4] = r[1];
00374                     i = FindSubterm(tout+1);
00375                     *tout++ = SYMBOL;
00376                     *tout++ = 4;
00377                     *tout++ = MAXVARIABLES-i;
00378                     *tout++ = 1;
00379                     r += 2;
00380                     action = 1;

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```

00381     }
00382 }
00383 else if ( *t == INDEX ) {
00384     r = t + 2;
00385     t += t[1];
00386     while ( r < t ) {
00387         tout[1] = INDEX;
00388         tout[2] = 3;
00389         tout[3] = r[0];
00390         i = FindSubterm(tout+1);
00391         *tout++ = SYMBOL;
00392         *tout++ = 4;
00393         *tout++ = MAXVARIABLES-i;
00394         *tout++ = 1;
00395         r++;
00396         action = 1;
00397     }
00398 }
00399 else if ( *t == HAAKJE ) {
00400     if ( par ) {
00401         tout[0] = 1; tout[1] = 1; tout[2] = 3;
00402         *outterm = (tout+3)-outterm;
00403         if ( NormPolyTerm(BHEAD outterm) < 0 ) return(-1);
00404         tout = outterm + *outterm;
00405         tout -= 3;
00406         i = t[1]; NCOPY(tout,t,i);
00407         ttwo = tout-1;
00408     }
00409     else { t += t[1]; }
00410 }
00411 else if ( *t >= FUNCTION ) {
00412     i = FindSubterm(t);
00413     t += t[1];
00414     *tout++ = SYMBOL;
00415     *tout++ = 4;
00416     *tout++ = MAXVARIABLES-i;
00417     *tout++ = 1;
00418     action = 1;
00419 }
00420 else {
00421     if ( FirstWarnConvertToPoly ) {
00422         MLOCK(ErrorMessageLock);
00423         MesPrint("Illegal object in conversion to polynomial notation");
00424         MUNLOCK(ErrorMessageLock);
00425         FirstWarnConvertToPoly = 0;
00426     }
00427     return(-1);
00428 }
00429 }
00430 NCOPY(tout,tstop,ncoef)
00431 if ( ttwo ) {
00432     WORD hh = *ttwo;
00433     *ttwo = tout-ttwo;
00434     if ( ( i = NormPolyTerm(BHEAD ttwo) ) >= 0 ) i = action;
00435     tout = ttwo + *ttwo;
00436     *ttwo = hh;
00437     *outterm = tout - outterm;
00438 }
00439 else {
00440     *outterm = tout-outterm;
00441     if ( ( i = NormPolyTerm(BHEAD outterm) ) >= 0 ) i = action;
00442 }
00443 }
00444 else if ( comlist[2] == ONLYFUNCTIONS ) {
00445     while ( t < tstop ) {
00446         if ( *t >= FUNCTION ) {
00447             if ( comlist[1] == 3 ) {
00448                 i = FindSubterm(t);
00449                 t += t[1];
00450                 *tout++ = SYMBOL;
00451                 *tout++ = 4;
00452                 *tout++ = MAXVARIABLES-i;
00453                 *tout++ = 1;
00454                 action = 1;
00455             }
00456             else {
00457                 for ( i = 3; i < comlist[1]; i++ ) {
00458                     if ( *t == comlist[i] ) break;
00459                 }
00460                 if ( i < comlist[1] ) {
00461                     i = FindSubterm(t);
00462                     t += t[1];
00463                     *tout++ = SYMBOL;
00464                     *tout++ = 4;
00465                     *tout++ = MAXVARIABLES-i;
00466                     *tout++ = 1;
00467                     action = 1;

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```

00468         }
00469         else {
00470             i = t[1]; NCOPY(tout,t,i);
00471         }
00472     }
00473 }
00474 else {
00475     i = t[1]; NCOPY(tout,t,i);
00476 }
00477 }
00478 NCOPY(tout,tstop,ncoef)
00479 *outterm = tout-outterm;
00480 Normalize(BHEAD outterm);
00481 i = action;
00482 }
00483 else {
00484     MLOCK(ErrorMessageLock);
00485     MesPrint("Illegal internal code in conversion to polynomial notation");
00486     MUNLOCK(ErrorMessageLock);
00487     i = -1;
00488 }
00489 return(i);
00490 }
00491
00492 /*
00493     #] ConvertToPoly :
00494     #[ LocalConvertToPoly :
00495 */
00510 int LocalConvertToPoly(PHEAD WORD *term, WORD *outterm, WORD startebuf, WORD par)
00511 {
00512     WORD *tout, *tstop, ncoef, *t, *r, *tt, *ttwo = 0;
00513     int i, action = 0;
00514     tt = term + *term;
00515     ncoef = ABS(tt[-1]);
00516     tstop = tt - ncoef;
00517     tout = outterm+1;
00518     t = term + 1;
00519     while ( t < tstop ) {
00520         if ( *t == SYMBOL ) {
00521             r = t+2;
00522             t += t[1];
00523             while ( r < t ) {
00524                 if ( r[1] > 0 ) {
00525                     *tout++ = SYMBOL;
00526                     *tout++ = 4;
00527                     *tout++ = r[0];
00528                     *tout++ = r[1];
00529                 }
00530                 else {
00531                     tout[1] = SYMBOL;
00532                     tout[2] = 4;
00533                     tout[3] = r[0];
00534                     tout[4] = -1;
00535                     i = FindLocalSubterm(BHEAD tout+1,startebuf);
00536                     *tout++ = SYMBOL;
00537                     *tout++ = 4;
00538                     *tout++ = MAXVARIABLES-i;
00539                     *tout++ = -r[1];
00540                     action = 1;
00541                 }
00542                 r += 2;
00543             }
00544         }
00545         else if ( *t == DOTPRODUCT ) {
00546             r = t + 2;
00547             t += t[1];
00548             while ( r < t ) {
00549                 tout[1] = DOTPRODUCT;
00550                 tout[2] = 5;
00551                 tout[3] = r[0];
00552                 tout[4] = r[1];
00553                 if ( r[2] < 0 ) {
00554                     tout[5] = -1;
00555                 }
00556                 else {
00557                     tout[5] = 1;
00558                 }
00559                 i = FindLocalSubterm(BHEAD tout+1,startebuf);
00560                 *tout++ = SYMBOL;
00561                 *tout++ = 4;
00562                 *tout++ = MAXVARIABLES-i;
00563                 *tout++ = ABS(r[2]);
00564                 r += 3;
00565                 action = 1;
00566             }
00567         }
00568         else if ( *t == VECTOR ) {

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```

00569         r = t + 2;
00570         t += t[1];
00571         while ( r < t ) {
00572             tout[1] = VECTOR;
00573             tout[2] = 4;
00574             tout[3] = r[0];
00575             tout[4] = r[1];
00576             i = FindLocalSubterm(BHEAD tout+1,startebuf);
00577             *tout++ = SYMBOL;
00578             *tout++ = 4;
00579             *tout++ = MAXVARIABLES-i;
00580             *tout++ = 1;
00581             r += 2;
00582             action = 1;
00583         }
00584     }
00585     else if ( *t == INDEX ) {
00586         r = t + 2;
00587         t += t[1];
00588         while ( r < t ) {
00589             tout[1] = INDEX;
00590             tout[2] = 3;
00591             tout[3] = r[0];
00592             i = FindLocalSubterm(BHEAD tout+1,startebuf);
00593             *tout++ = SYMBOL;
00594             *tout++ = 4;
00595             *tout++ = MAXVARIABLES-i;
00596             *tout++ = 1;
00597             r++;
00598             action = 1;
00599         }
00600     }
00601     else if ( *t == HAAKJE ) {
00602         if ( par ) {
00603             tout[0] = 1; tout[1] = 1; tout[2] = 3;
00604             *outterm = (tout+3)-outterm;
00605             if ( NormPolyTerm(BHEAD outterm) < 0 ) return(-1);
00606             tout = outterm + *outterm;
00607             tout -= 3;
00608             i = t[1]; NCOPY(tout,t,i);
00609             ttwo = tout-1;
00610         }
00611         else { t += t[1]; }
00612     }
00613     else if ( *t >= FUNCTION ) {
00614         i = FindLocalSubterm(BHEAD t,startebuf);
00615         t += t[1];
00616         *tout++ = SYMBOL;
00617         *tout++ = 4;
00618         *tout++ = MAXVARIABLES-i;
00619         *tout++ = 1;
00620         action = 1;
00621     }
00622     else {
00623         if ( FirstWarnConvertToPoly ) {
00624             MLOCK(ErrorMessageLock);
00625             MesPrint("Illegal object in conversion to polynomial notation");
00626             MUNLOCK(ErrorMessageLock);
00627             FirstWarnConvertToPoly = 0;
00628         }
00629         return(-1);
00630     }
00631 }
00632 NCOPY(tout,tstop,ncoef)
00633 if ( ttwo ) {
00634     WORD hh = *ttwo;
00635     *ttwo = tout-ttwo;
00636     if ( ( i = NormPolyTerm(BHEAD ttwo) ) >= 0 ) i = action;
00637     tout = ttwo + *ttwo;
00638     *ttwo = hh;
00639     *outterm = tout - outterm;
00640 }
00641 else {
00642     *outterm = tout-outterm;
00643     if ( ( i = NormPolyTerm(BHEAD outterm) ) >= 0 ) i = action;
00644 }
00645 return(i);
00646 }
00647
00648 /*
00649 #] LocalConvertToPoly :
00650 #[ ConvertFromPoly :
00651
00652 Converts a generic term from polynomial notation to the original
00653 in which the extra symbols have been replaced by their values.
00654 The output is in outterm.
00655 We only deal with the extra symbols in the range from < i <= to

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00656         The output has to be sent to TestSub because it may contain
00657         subexpressions when extra symbols have been replaced.
00658     */
00659
00660 int ConvertFromPoly(PHEAD WORD *term, WORD *outterm, WORD from, WORD to, WORD offset, WORD par)
00661 {
00662     WORD *tout, *tstop, *tstop1, ncoef, *t, *r, *tt;
00663     int i;
00664     /* first = 1; */
00665     tt = term + *term;
00666     tout = outterm+1;
00667     ncoef = ABS(tt[-1]);
00668     tstop = tt - ncoef;
00669     /*
00670     r = t = term + 1;
00671     while ( t < tstop ) {
00672         if ( *t == SYMBOL ) {
00673             tstop1 = t + t[1];
00674             tt = t + 2;
00675             while ( tt < tstop1 ) {
00676                 if ( ( *tt < MAXVARIABLES - to )
00677                     || ( *tt >= MAXVARIABLES - from ) ) {
00678                     tt += 2;
00679                 }
00680                 else break;
00681             }
00682             if ( tt >= tstop1 ) { t = tstop1; continue; }
00683             while ( r < t ) *tout++ = *r++;
00684             t += 2;
00685             first = 0;
00686             while ( t < tstop1 ) {
00687                 if ( ( *t < MAXVARIABLES - to )
00688                     || ( *t >= MAXVARIABLES - from ) ) {
00689                     *tout++ = SYMBOL;
00690                     *tout++ = 4;
00691                     *tout++ = *t++;
00692                     *tout++ = *t++;
00693                 }
00694                 else {
00695                     *tout++ = SUBEXPRESSION;
00696                     *tout++ = SUBEXPSIZE;
00697                     *tout++ = MAXVARIABLES - *t++ + offset;
00698                     *tout++ = *t++;
00699                     if ( par ) *tout++ = AT.ebufnum;
00700                     else *tout++ = AM.sbufnum;
00701                     FILLSUB(tout)
00702                 }
00703             }
00704             r = t;
00705         }
00706         else {
00707             t += t[1];
00708         }
00709     }
00710     if ( first ) {
00711         i = *term; t = term;
00712         NCOPY(outterm,t,i);
00713         return(*term);
00714     }
00715     while ( r < t ) *tout++ = *r++;
00716     NCOPY(tout,tstop,ncoef)
00717     *outterm = tout-outterm;
00718     */
00719     t = term + 1;
00720     while ( t < tstop ) {
00721         if ( *t == SYMBOL ) {
00722             tstop1 = t + t[1];
00723             tt = t + 2;
00724             while ( tt < tstop1 ) {
00725                 if ( ( *tt < MAXVARIABLES - to )
00726                     || ( *tt >= MAXVARIABLES - from ) ) {
00727                     tt += 2;
00728                 }
00729                 else {
00730                     *tout++ = SUBEXPRESSION;
00731                     *tout++ = SUBEXPSIZE;
00732                     *tout++ = MAXVARIABLES - *tt++ + offset;
00733                     *tout++ = *tt++;
00734                     if ( par ) *tout++ = AT.ebufnum;
00735                     else *tout++ = AM.sbufnum;
00736                     FILLSUB(tout)
00737                 }
00738             }
00739             r = tout; t += 2;
00740             *tout++ = SYMBOL; *tout++ = 0;
00741             while ( t < tstop1 ) {
00742                 if ( ( *t < MAXVARIABLES - to )

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```

00743         || ( *t >= MAXVARIABLES - from ) ) {
00744             *tout++ = *t++;
00745             *tout++ = *t++;
00746         }
00747         else { t += 2; }
00748     }
00749     r[1] = tout - r;
00750     if ( r[1] <= 2 ) tout = r;
00751 }
00752 else {
00753     i = t[1]; NCOPY(tout,t,i)
00754 }
00755 }
00756 NCOPY(tout,tstop,ncoef)
00757 *outterm = tout-outterm;
00758 return(*outterm);
00759 }
00760
00761 /*
00762     #] ConvertFromPoly :
00763     #[ FindSubterm :
00764
00765     In this routine we look up a variable.
00766     If we don't find it we will enter it in the subterm compiler buffer
00767     Searching is by tree structure.
00768     Adding changes the tree.
00769
00770     Notice that in TFORM we should be in sequential mode.
00771 */
00772
00773 int FindSubterm(WORD *subterm)
00774 {
00775     WORD old[5], *ss, *term;
00776     int number;
00777     CBUF *C = cbuf + AM.sbufnum;
00778     LONG oldCpointer;
00779     term = subterm-1;
00780     ss = subterm+subterm[1];
00781 /*
00782     Convert to proper term
00783 */
00784     old[0] = *term; old[1] = ss[0]; old[2] = ss[1]; old[3] = ss[2]; old[4] = ss[3];
00785     ss[0] = 1; ss[1] = 1; ss[2] = 3; ss[3] = 0; *term = subterm[1]+4;
00786 /*
00787     We may have to add the term to the compiler
00788     buffer and then to the tree. This cannot be done in parallel and
00789     hence we have to set a lock.
00790 */
00791     LOCK(AM.sbuflock);
00792
00793     oldCpointer = C->Pointer-C->Buffer; /* Offset of course !!!!!*/
00794     AddRHS(AM.sbufnum,1);
00795     AddNtoC(AM.sbufnum,*term,term,8);
00796     AddToCB(C,0)
00797 /*
00798     See whether we have this one already. If not, insert it in the tree.
00799 */
00800     number = InsTree(AM.sbufnum,C->numrhs);
00801 /*
00802     Restore old values and return what is needed.
00803 */
00804     if ( number < (C->numrhs) ) { /* It existed already */
00805         C->Pointer = oldCpointer + C->Buffer;
00806         C->numrhs--;
00807     }
00808     else {
00809         GETIDENTITY
00810         WORD dim = DimensionSubterm(subterm);
00811
00812         if ( dim == -MAXPOSITIVE ) { /* Give error message but continue */
00813             WORD *old = AN.currentTerm;
00814             AN.currentTerm = term;
00815             MLOCK(ErrorMessageLock);
00816             MesPrint("Dimension out of range in %t");
00817             MUNLOCK(ErrorMessageLock);
00818             AN.currentTerm = old;
00819         }
00820 /*
00821         Store the dimension
00822 */
00823         C->dimension[number] = dim;
00824     }
00825     UNLOCK(AM.sbuflock);
00826
00827     *term = old[0]; ss[0] = old[1]; ss[1] = old[2]; ss[2] = old[3]; ss[3] = old[4];
00828     return(number);
00829 }

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```

00830
00831 /*
00832     #] FindSubterm :
00833     #[ FindLocalSubterm :
00834
00835     In this routine we look up a variable.
00836     If we don't find it we will enter it in the subterm compiler buffer
00837     Searching is by tree structure.
00838     Adding changes the tree.
00839
00840     Notice that in TFORM we should be in sequential mode.
00841 */
00842
00843 int FindLocalSubterm(PHEAD WORD *subterm, WORD startebuf)
00844 {
00845     WORD old[5], *ss, *term, i, j, *t1, *t2;
00846     int number;
00847     CBUF *C = cbuf + AT.ebufnum;
00848     term = subterm-1;
00849     ss = subterm+subterm[1];
00850 /*
00851     Convert to proper term
00852 */
00853     old[0] = *term; old[1] = ss[0]; old[2] = ss[1]; old[3] = ss[2]; old[4] = ss[3];
00854     ss[0] = 1; ss[1] = 1; ss[2] = 3; ss[3] = 0; *term = subterm[1]+4;
00855 /*
00856     First see whether we have this one already in the global buffer.
00857 */
00858     number = FindTree(AM.sbufnum,term);
00859     if ( number > 0 ) goto wearehappy;
00860 /*
00861     Now look whether it is in the ebufnum between startebuf and numrhs
00862     Note however that we need an offset of (numxsymbol-startebuf)
00863 */
00864     for ( i = startebuf+1; i <= C->numrhs; i++ ) {
00865         t1 = C->rhs[i]; t2 = term;
00866         if ( *t1 == *t2 ) {
00867             j = *t1;
00868             while ( *t1 == *t2 && j > 0 ) { t1++; t2++; j--; }
00869             if ( j <= 0 ) {
00870                 number = i-startebuf+numxsymbol;
00871                 goto wearehappy;
00872             }
00873         }
00874     }
00875 /*
00876     Now we have to add it to cbuf[AT.ebufnum]
00877 */
00878     AddRHS(AT.ebufnum,1);
00879     AddNtoC(AT.ebufnum,*term,term,9);
00880     AddToCB(C,0)
00881     number = C->numrhs-startebuf+numxsymbol;
00882 wearehappy:
00883     *term = old[0]; ss[0] = old[1]; ss[1] = old[2]; ss[2] = old[3]; ss[3] = old[4];
00884     return(number);
00885 }
00886
00887 /*
00888     #] FindLocalSubterm :
00889     #[ PrintSubtermList :
00890
00891     Prints all the expressions in the subterm compiler buffer.
00892     The format is such that they give definitions of the temporary
00893     variables of which the contents are stored in this buffer.
00894     These variables have the names Z_123 etc.
00895 */
00896
00897 void PrintSubtermList(int from,int to)
00898 {
00899     UBYTE buffer[80], *out, outbuffer[300];
00900     int first, i, ii, inc = 1;
00901     WORD *term;
00902     CBUF *C = cbuf + AM.sbufnum;
00903 /*
00904     if ( to < from ) inc = -1;
00905     if ( to == from ) inc = 0;
00906 */
00907     if ( from <= to ) {
00908         inc = 1; to += inc;
00909     }
00910     else {
00911         inc = -1; to += inc;
00912     }
00913     AO.OutFill = AO.OutputLine = outbuffer;
00914     AO.OutStop = AO.OutputLine+AC.LineLength;
00915     AO.IsBracket = 0;
00916     AO.OutSkip = 3;

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00917
00918     if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
00919         TokenToLine((UBYTE *)"      ");
00920         AO.OutSkip = 7;
00921     }
00922     else if ( ( AO.Optimize.debugflags & 1 ) == 1 ) {}
00923     else if ( AO.OutSkip > 0 ) {
00924         for ( i = 0; i < AO.OutSkip; i++ ) TokenToLine((UBYTE *)" ");
00925     }
00926     i = from;
00927     do {
00928         if ( ( AO.Optimize.debugflags & 1 ) == 1 ) {
00929             TokenToLine((UBYTE *)"id ");
00930             for ( ii = 3; ii < AO.OutSkip; ii++ ) TokenToLine((UBYTE *)" ");
00931         }
00932         /*
00933         if ( AC.OutputMode == NORMALFORMAT ) {
00934             TokenToLine((UBYTE *)"id ");
00935         }
00936         */
00937         else if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {}
00938         else { TokenToLine((UBYTE *)" "); }
00939
00940         out = StrCopy((UBYTE *)AC.extrasym,buffer);
00941         if ( AC.extrasymbols == 0 ) {
00942             out = NumCopy(i,out);
00943             out = StrCopy((UBYTE *)"_",out);
00944         }
00945         else if ( AC.extrasymbols == 1 ) {
00946             out = AddArrayIndex(i,out);
00947         }
00948         out = StrCopy((UBYTE *)"=",out);
00949         TokenToLine(buffer);
00950         term = C->rhs[i];
00951         first = 1;
00952         if ( *term == 0 ) {
00953             out = StrCopy((UBYTE *)"0",buffer);
00954             if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE ) {
00955                 out = StrCopy((UBYTE *)";",out);
00956             }
00957             TokenToLine(buffer);
00958         }
00959         else {
00960             while ( *term ) {
00961                 if ( WriteInnerTerm(term,first) ) Terminate(-1);
00962                 term += *term;
00963                 first = 0;
00964             }
00965             if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE ) {
00966                 out = StrCopy((UBYTE *)";",buffer);
00967                 TokenToLine(buffer);
00968             }
00969         }
00970         /*
00971         There is a problem with FiniLine because it prepares for a
00972         continuation line in fortran mode.
00973         But the next statement should start on a blank line.
00974         */
00975         /*
00976         FiniLine();
00977         if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
00978             AO.OutFill = AO.OutputLine;
00979             TokenToLine((UBYTE *)"      ");
00980             AO.OutSkip = 7;
00981         }
00982         */
00983         if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
00984             AO.OutSkip = 6;
00985             FiniLine();
00986             AO.OutSkip = 7;
00987         }
00988         else {
00989             FiniLine();
00990         }
00991         i += inc;
00992     } while ( i != to );
00993 }
00994
00995 /*
00996     #] PrintSubtermList :
00997     #[ PrintExtraSymbol :
00998
00999     Prints the definition of extra symbol num as the contents
01000     of the expression in terms.
01001     The parameter par has three options:
01002         EXTRASYMBOL    num is interpreted as the number of an extra symbol
01003         REGULARSYMBOL  num is interpreted as the number of a symbol.

```

```

01004                                     It could still be an extra symbol.
01005     EXPRESSIONNUMBER num is the number of an expression.
01006     terms contains the rhs expression.
01007 */
01008
01009 void PrintExtraSymbol(int num, WORD *terms,int par)
01010 {
01011     UBYTE buffer[80], *out, outbuffer[300];
01012     int first, i;
01013     WORD *term;
01014
01015     AO.OutFill = AO.OutputLine = outbuffer;
01016     AO.OutStop = AO.OutputLine+AC.LineLength;
01017     AO.IsBracket = 0;
01018
01019     if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
01020         TokenToLine((UBYTE *) "      ");
01021         AO.OutSkip = 7;
01022     }
01023     else if ( ( AO.Optimize.debugflags & 1 ) == 1 ) {
01024         TokenToLine((UBYTE *) "id ");
01025         for ( i = 3; i < AO.OutSkip; i++ ) TokenToLine((UBYTE *) " ");
01026     }
01027     else if ( AO.OutSkip > 0 ) {
01028         for ( i = 0; i < AO.OutSkip; i++ ) TokenToLine((UBYTE *) " ");
01029     }
01030     out = buffer;
01031     switch ( par ) {
01032     case REGULARSYMBOL:
01033         if ( num >= MAXVARIABLES-cbuf[AM.sbufnum].numrhs ) {
01034             num = MAXVARIABLES-num;
01035         }
01036         else {
01037             out = StrCopy(FindSymbol(num),out);
01038             /* out = StrCopy(VARNAME(symbols,num),out); */
01039             break;
01040         }
01041         /* fall through */
01042     case EXTRASYMBOL:
01043         out = StrCopy(FindExtraSymbol(num),out);
01044         /*
01045         out = StrCopy((UBYTE *)AC.extrasym,out);
01046         if ( AC.extrasymbols == 0 ) {
01047             out = NumCopy(num,out);
01048             out = StrCopy((UBYTE *) "_",out);
01049         }
01050         else if ( AC.extrasymbols == 1 ) {
01051             out = AddArrayIndex(num,out);
01052         }
01053         */
01054         break;
01055     case EXPRESSIONNUMBER:
01056         out = StrCopy(EXPRNAME(num),out);
01057         break;
01058     default:
01059         MesPrint("Illegal option in PrintExtraSymbol");
01060         Terminate(-1);
01061     }
01062     out = StrCopy((UBYTE *) "=",out);
01063     TokenToLine(buffer);
01064     term = terms;
01065     first = 1;
01066     if ( *term == 0 ) {
01067         out = StrCopy((UBYTE *) "0",buffer);
01068         TokenToLine(buffer);
01069     }
01070     else {
01071         while ( *term ) {
01072             if ( WriteInnerTerm(term,first) ) Terminate(-1);
01073             term += *term;
01074             first = 0;
01075         }
01076     }
01077     if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE ) {
01078         out = StrCopy((UBYTE *) ";",buffer);
01079         TokenToLine(buffer);
01080     }
01081     FiniLine();
01082 }
01083
01084 /*
01085     #] PrintExtraSymbol :
01086     #[ FindSubexpression :
01087
01088     In this routine we look up a subexpression.
01089     If we don't find it we will enter it in the subterm compiler buffer
01090     Searching is by tree structure.

```

```

01091         Adding changes the tree.
01092
01093         Notice that in TFORM we should be in sequential mode.
01094 */
01095
01096 int FindSubexpression(WORD *subexpr)
01097 {
01098     WORD *term;
01099     int number;
01100     CBUF *C = cbuf + AM.sbufnum;
01101     LONG oldCpointer;
01102
01103     term = subexpr;
01104     while ( *term ) term += *term;
01105     number = term - subexpr;
01106 /*
01107         We may have to add the subexpression to the tree.
01108         This requires a lock.
01109 */
01110     LOCK(AM.sbuflock);
01111
01112     oldCpointer = C->Pointer-C->Buffer; /* Offset of course !!!!!*/
01113     AddRHS(AM.sbufnum,1);
01114 /*
01115         Add the terms to the compiler buffer. Paste on a zero.
01116 */
01117     AddNtoC(AM.sbufnum,number,subexpr,10);
01118     AddToCB(C,0)
01119 /*
01120         See whether we have this one already. If not, insert it in the tree.
01121 */
01122     number = InsTree(AM.sbufnum,C->numrhs);
01123 /*
01124         Restore old values and return what is needed.
01125 */
01126     if ( number < (C->numrhs) ) { /* It existed already */
01127         C->Pointer = oldCpointer + C->Buffer;
01128         C->numrhs--;
01129     }
01130     else {
01131         GETIDENTITY
01132         WORD dim = DimensionExpression(BHEAD subexpr);
01133 /*
01134         Store the dimension
01135 */
01136         C->dimension[number] = dim;
01137     }
01138
01139     UNLOCK(AM.sbuflock);
01140
01141     return(number);
01142 }
01143
01144 /*
01145     #] FindSubexpression :
01146     #[ ExtraSymFun :
01147 */
01148
01149 int ExtraSymFun(PHEAD WORD *term,WORD level)
01150 {
01151     WORD *oldworkpointer = AT.WorkPointer;
01152     WORD *termout, *t1, *t2, *t3, *tstop, *tend, i;
01153     int retval = 0;
01154     tend = termout = term + *term;
01155     tstop = tend - ABS(tend[-1]);
01156     t3 = t1 = term+1; t2 = termout+1;
01157 /*
01158     First refind the function(s). There is at least one.
01159 */
01160     while ( t1 < tstop ) {
01161         if ( *t1 == EXTRASYMFUN && t1[1] == FUNHEAD+2 ) {
01162             if ( t1[FUNHEAD] == -SNUMBER && t1[FUNHEAD+1] <= numxsymbol
01163                 && t1[FUNHEAD+1] > 0 ) {
01164                 i = t1[FUNHEAD+1];
01165             }
01166             else if ( t1[FUNHEAD] == -SYMBOL && t1[FUNHEAD+1] < MAXVARIABLES
01167                 && t1[FUNHEAD+1] >= MAXVARIABLES-numxsymbol ) {
01168                 i = MAXVARIABLES - t1[FUNHEAD+1];
01169             }
01170             else goto nocase;
01171             while ( t3 < t1 ) *t2++ = *t3++;
01172 /*
01173                 Now inset the rhs pointer
01174 */
01175                 *t2++ = SUBEXPRESSION;
01176                 *t2++ = SUBEXPSIZE;
01177                 *t2++ = i;

```

```

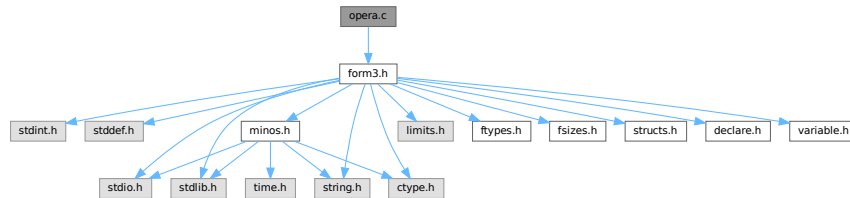
01178         *t2++ = 1;
01179         *t2++ = AM.sbufnum;
01180         FILLSUB(t2)
01181         t3 = t1 = t1 + t1[1];
01182     }
01183     else if ( *t1 == EXTRASYMFUN && t1[1] == FUNHEAD ) {
01184         while ( t3 < t1 ) *t2++ = *t3++;
01185         t3 = t1 = t1 + t1[1];
01186     }
01187     else {
01188 nocase:;
01189         t1 = t1 + t1[1];
01190     }
01191 }
01192 while ( t3 < tend ) *t2++ = *t3++;
01193 *termout = t2 - termout;
01194 AT.WorkPointer = t2;
01195 if ( AT.WorkPointer >= AT.WorkTop ) {
01196     MLOCK(ErrorMessageLock);
01197     MesWork();
01198     MUNLOCK(ErrorMessageLock);
01199     AT.WorkPointer = oldworkpointer;
01200     return(-1);
01201 }
01202 retval = Generator(BHEAD termout,level);
01203 AT.WorkPointer = oldworkpointer;
01204 if ( retval < 0 ) {
01205     MLOCK(ErrorMessageLock);
01206     MesCall("ExtraSymFun");
01207     MUNLOCK(ErrorMessageLock);
01208 }
01209 return(retval);
01210 }
01211
01212 /*
01213     #] ExtraSymFun :
01214     #[ PruneExtraSymbols :
01215 */
01216
01217 int PruneExtraSymbols(WORD downto)
01218 {
01219     CBUF *C = cbuf + AM.sbufnum;
01220     if ( downto < C->numrhs && downto >= 0 ) { /* !!!!! */
01221         ClearTree(AM.sbufnum);
01222         C->numrhs = downto;
01223         if ( downto == 0 ) {
01224             C->Pointer = C->Buffer;
01225         }
01226         else {
01227             WORD *w = C->rhs[downto], i;
01228             while ( *w ) w += *w;
01229             C->Pointer = w+1;
01230             for ( i = 1; i <= downto; i++ ) {
01231                 InsTree(AM.sbufnum,i);
01232             }
01233         }
01234     }
01235     return(0);
01236 }
01237
01238 /*
01239     #] PruneExtraSymbols :
01240 */

```

## 4.90 opera.c File Reference

```
#include "form3.h"
```

Include dependency graph for opera.c:



### Functions

- int [EpfFind](#) (PHEAD WORD \*term, WORD \*params)
- int [EpfCon](#) (PHEAD WORD \*term, WORD \*params, WORD num, WORD level)
- WORD [EpfGen](#) (WORD number, WORD \*inlist, WORD \*kron, WORD \*perm, WORD sgn)
- WORD [Trick](#) (WORD \*in, [TRACES](#) \*t)
- int [Trace4no](#) (WORD number, WORD \*kron, [TRACES](#) \*t)
- int [Trace4](#) (PHEAD WORD \*term, WORD \*params, WORD num, WORD level)
- int [Trace4Gen](#) (PHEAD [TRACES](#) \*t, WORD number)
- WORD [TraceNno](#) (WORD number, WORD \*kron, [TRACES](#) \*t)
- int [TraceN](#) (PHEAD WORD \*term, WORD \*params, WORD num, WORD level)
- int [TraceNgen](#) (PHEAD [TRACES](#) \*t, WORD number)
- int [Traces](#) (PHEAD WORD \*term, WORD \*params, WORD num, WORD level)
- int [TraceFind](#) (PHEAD WORD \*term, WORD \*params)
- int [Chisholm](#) (PHEAD WORD \*term, WORD level)
- int [TenVecFind](#) (PHEAD WORD \*term, WORD \*params)
- int [TenVec](#) (PHEAD WORD \*term, WORD \*params, WORD num, WORD level)

### 4.90.1 Detailed Description

Contains the 'operations' These are the trace routines, the contractions of the Levi-Civita tensors and the tensor to vector/vector to tensor routines. The trace and contraction routines are done in a special way (see commentary with the FIXEDGLOBALS struct)

Definition in file [opera.c](#).

### 4.90.2 Function Documentation

#### 4.90.2.1 EpfFind()

```
int EpfFind (
    PHEAD WORD * term,
    WORD * params )
```

Definition at line 58 of file [opera.c](#).

#### 4.90.2.2 EpfCon()

```
int EpfCon (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level )
```

Definition at line [212](#) of file [opera.c](#).

#### 4.90.2.3 EpfGen()

```
WORD EpfGen (
    WORD number,
    WORD * inlist,
    WORD * kron,
    WORD * perm,
    WORD sgn )
```

Definition at line [282](#) of file [opera.c](#).

#### 4.90.2.4 Trick()

```
WORD Trick (
    WORD * in,
    TRACES * t )
```

Definition at line [331](#) of file [opera.c](#).

#### 4.90.2.5 Trace4no()

```
int Trace4no (
    WORD number,
    WORD * kron,
    TRACES * t )
```

Definition at line [418](#) of file [opera.c](#).

#### 4.90.2.6 Trace4()

```
int Trace4 (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level )
```

Definition at line [695](#) of file [opera.c](#).

#### 4.90.2.7 Trace4Gen()

```
int Trace4Gen (
    PHEAD TRACES * t,
    WORD number )
```

Definition at line [858](#) of file [opera.c](#).

#### 4.90.2.8 TraceNno()

```
WORD TraceNno (
    WORD number,
    WORD * kron,
    TRACES * t )
```

Definition at line [1343](#) of file [opera.c](#).

#### 4.90.2.9 TraceN()

```
int TraceN (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level )
```

Definition at line [1384](#) of file [opera.c](#).

#### 4.90.2.10 TraceNgen()

```
int TraceNgen (
    PHEAD TRACES * t,
    WORD number )
```

Definition at line [1449](#) of file [opera.c](#).

#### 4.90.2.11 Traces()

```
int Traces (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level )
```

Definition at line [1834](#) of file [opera.c](#).

#### 4.90.2.12 TraceFind()

```
int TraceFind (
    PHEAD WORD * term,
    WORD * params )
```

Definition at line [1856](#) of file [opera.c](#).

#### 4.90.2.13 Chisholm()

```
int Chisholm (
    PHEAD WORD * term,
    WORD level )
```

Definition at line 1966 of file [opera.c](#).

#### 4.90.2.14 TenVecFind()

```
int TenVecFind (
    PHEAD WORD * term,
    WORD * params )
```

Definition at line 2181 of file [opera.c](#).

#### 4.90.2.15 TenVec()

```
int TenVec (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level )
```

Definition at line 2315 of file [opera.c](#).

## 4.91 opera.c

[Go to the documentation of this file.](#)

```
00001
00009 /* #[] License : */
00010 /*
00011 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00012 * When using this file you are requested to refer to the publication
00013 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00014 * This is considered a matter of courtesy as the development was paid
00015 * for by FOM the Dutch physics granting agency and we would like to
00016 * be able to track its scientific use to convince FOM of its value
00017 * for the community.
00018 *
00019 * This file is part of FORM.
00020 *
00021 * FORM is free software: you can redistribute it and/or modify it under the
00022 * terms of the GNU General Public License as published by the Free Software
00023 * Foundation, either version 3 of the License, or (at your option) any later
00024 * version.
00025 *
00026 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00027 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00028 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00029 * details.
00030 *
00031 * You should have received a copy of the GNU General Public License along
00032 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00033 */
00034 /* #[] License : */
00035 /*
00036 * #[] Includes : opera.c
00037 */
00038
00039 #include "form3.h"
00040 /*
00041 int hulp;
```



```

00042 */
00043
00044 /*
00045  #] Includes :
00046  #[ Operations :
00047      #[ EpfFind :          WORD EpfFind(term,params)
00048
00049      Searches for a pair of Levi-Civita tensors that should be
00050      contracted.
00051      If a match is found its settings are recorded in AT.TMout.
00052      type indicates the number of indices that is searched for,
00053      unless all are searched for (type = 0).
00054      number is the number of tensors that should survive contraction.
00055
00056 */
00057
00058 int EpfFind(PHEAD WORD *term, WORD *params)
00059 {
00060     GETBIDENTITY
00061     WORD *t, *m, *r, n1 = 0, n2, min = -1, count, fac;
00062     WORD *c1 = 0, *c2 = 0, sgn = 1;
00063     WORD *tstop, *mstop;
00064     UWORD *facto = (UWORD *)AT.WorkPointer;
00065     WORD ncoef, nfac;
00066     WORD number, type;
00067     if ( ( AT.WorkPointer = (WORD *) (facto + AM.MaxTal) ) > AT.WorkTop ) {
00068         MLOCK(ErrorMessageLock);
00069         MesWork();
00070         MUNLOCK(ErrorMessageLock);
00071         return(-1);
00072     }
00073     number = params[3];
00074     type = params[4];
00075     t = term;
00076     GETSTOP(t,tstop);
00077     t++;
00078     if ( !type ) {
00079         while ( *t != LEVICIVITA && t < tstop ) t += t[1];
00080         if ( t >= tstop ) return(0);
00081         m = t;
00082         while ( *m == LEVICIVITA && m < tstop ) { n1++; m += m[1]; }
00083     AllLev:
00084         if ( n1 <= (number+1) || n1 <= 1 ) return(0);
00085         mstop = m;
00086         m = t + t[1];
00087         do {
00088             while ( m[1] == t[1] ) {
00089                 m += FUNHEAD;
00090                 r = t+FUNHEAD;
00091                 count = fac = n1 = n2 = t[1] - FUNHEAD;
00092                 while ( n1 && n2 ) {
00093                     if ( *m > *r ) {
00094                         r++; n2--;
00095                     }
00096                     else if ( *m < *r ) {
00097                         m++; n1--;
00098                     }
00099                     else {
00100                         if ( *m >= AM.OffsetIndex &&
00101                             ( ( *m >= AM.IndDum && AC.lDefDim == fac ) ||
00102                               ( *m < AM.IndDum &&
00103                                 indices[*m-AM.OffsetIndex].dimension == fac ) ) ) {
00104                             count--;
00105                         }
00106                         r++; m++; n1--; n2--;
00107                     }
00108                 }
00109                 m += n1;
00110                 if ( min < 0 || count < min ) {
00111                     c1 = t;
00112                     c2 = m - fac - FUNHEAD;
00113                     min = count;
00114                 }
00115                 if ( m >= mstop ) break;
00116             }
00117             t += t[1];
00118         } while ( ( m = t + t[1] ) < mstop );
00119     }
00120     else {
00121         fac = type + FUNHEAD;
00122         while ( *t != LEVICIVITA && t < tstop ) t += t[1];
00123         while ( *t == LEVICIVITA && t < tstop && t[1] != fac ) t += t[1];
00124         if ( t >= tstop ) return(0);
00125         m = t;
00126         while ( *m == LEVICIVITA && m < tstop && m[1] == fac ) { n1++; m += m[1]; }
00127         goto AllLev;
00128     }
}

```

```

00129 /*
00130 We have now the two tensors that give the minimum contraction
00131 in c1 and c2.
00132 Prepare the AT.TMout array;
00133 */
00134 if ( min < 0 ) return(0); /* No matching pair! */
00135 t = c1;
00136 mstop = c2;
00137 fac = t[1] - FUNHEAD;
00138 m = AT.TMout;
00139 *m++ = 3 + (min*2); /* The full length */
00140 *m++ = CONTRACT;
00141 if ( number < 0 ) *m++ = 1;
00142 else *m++ = 0;
00143 n1 = n2 = t[1] - FUNHEAD;
00144 r = c1 + FUNHEAD;
00145 c1 = m;
00146 m = c2 + FUNHEAD;
00147 c2 = c1 + min;
00148 while ( n1 && n2 ) {
00149     if ( *m > *r ) { *c1++ = *r++; n2--; }
00150     else if ( *m < *r ) { *c2++ = *m++; n1--; }
00151     else {
00152         if ( *m < AM.OffsetIndex || ( *m < AM.IndDum &&
00153             ( indices[*m-AM.OffsetIndex].dimension != fac ) ) ||
00154             ( *m >= AM.IndDum && AC.lDefDim != fac ) ) {
00155             *c1++ = *r++; *c2++ = *m++;
00156         }
00157         else { if ( ( n1 ^ n2 ) & 1 ) sgn = -sgn; r++; m++; }
00158         n1--; n2--;
00159     }
00160 }
00161 if ( n1 ) { NCOPY(c2,m,n1); }
00162 else if ( n2 ) { NCOPY(c1,r,n2); }
00163 fac -= min;
00164 m = t + t[1];
00165 while ( m < mstop ) *t++ = *m++;
00166 m += m[1];
00167 while ( m < tstop ) *t++ = *m++;
00168 *t++ = SUBEXPRESSION;
00169 *t++ = SUBEXPSIZE;
00170 *t++ = -1;
00171 *t++ = 1;
00172 *t++ = DUMMYBUFFER;
00173 FILLSUB(t)
00174 r = term;
00175 r += *r - 1;
00176 mstop = r;
00177 ncoef = REDLENG(*r);
00178 tstop = t;
00179 while ( m < mstop ) *t++ = *m++;
00180 if ( Factorial(BHEAD fac,facto,&nfac) || Mully(BHEAD (UWORD *)tstop,&ncoef,facto,nfac) ) {
00181     MLOCK(ErrorMessageLock);
00182     MesCall("EpFFind");
00183     MUNLOCK(ErrorMessageLock);
00184     SETERROR(-1)
00185 }
00186 tstop += (ABS(ncoef))*2;
00187 if ( sgn < 0 ) ncoef = -ncoef;
00188 ncoef *= 2;
00189 *tstop++ = (ncoef<0)?(ncoef-1):(ncoef+1);
00190 *term = WORDDIF(tstop,term);
00191 return(1);
00192 }
00193
00194 /*
00195 #] EpFFind :
00196 #[ EpFCon : WORD EpFCon(term,params,num,level)
00197
00198 Contraction of two strings of indices/vectors. They come
00199 from LeviCivita tensors that are being contracted.
00200 term is the term with the subterm to be replaced.
00201 params is the full indicator:
00202     Length, number, raise, parameters.
00203 Length is the length of the parameter field.
00204 number is the number of the operation.
00205 raise tells whether level should be raised afterwards.
00206 the parameters are the two strings.
00207 level is the id level.
00208 The factorial has been multiplied in already.
00209
00210 */
00211
00212 int EpFCon(PHEAD WORD *term, WORD *params, WORD num, WORD level)
00213 {
00214     GETBIDENTITY
00215     WORD *kron, *perm, *termout, *tstop, size2;

```

```

00216     WORD *m, *t, sizes, sgn = 0, i;
00217     sizes = *params - 3;
00218     kron = AT.WorkPointer;
00219     perm = (AT.WorkPointer += sizes);
00220     termout = (AT.WorkPointer += sizes);
00221     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
00222     if ( AT.WorkPointer > AT.WorkTop ) {
00223         MLOCK(ErrorMessageLock);
00224         MesWork();
00225         MUNLOCK(ErrorMessageLock);
00226         return(-1);
00227     }
00228     params += 2;
00229     if ( !(*params++) ) level--;
00230     size2 = sizes>>1;
00231     if ( !size2 ) goto DoOnce;
00232     while ( ( sgn = EpfGen(size2,params,kron,perm,sgn) ) != 0 ) {
00233 DoOnce:
00234         t = term;
00235         GETSTOP(t,tstop);
00236         m = termout;
00237         tstop -= SUBEXPSIZE;
00238         while ( t < tstop ) *m++ = *t++;
00239         if ( t[2] != num || *t != SUBEXPRESSION ) {
00240             MLOCK(ErrorMessageLock);
00241             MesPrint("Serious error in EpfCon");
00242             MUNLOCK(ErrorMessageLock);
00243             return(-1);
00244         }
00245         tstop += SUBEXPSIZE;
00246         if ( sizes ) {
00247             *m++ = DELTA;
00248             *m++ = sizes + 2;
00249             t = kron;
00250             i = sizes;
00251             if ( i ) { NCOPY(m,t,i); }
00252         }
00253         t = tstop;
00254         tstop = term + *term;
00255         while ( t < tstop ) *m++ = *t++;
00256         *termout = WORDDIF(m,termout);
00257         m--;
00258         if ( sgn < 0 ) *m = - *m;
00259         if ( *termout ) {
00260             *AN.RepPoint = 1;
00261             AR.expchanged = 1;
00262             AT.WorkPointer = termout + *termout;
00263             if ( Generator(BHEAD termout,level) < 0 ) goto EpfCall;
00264         }
00265     }
00266     AT.WorkPointer = kron;
00267     return(0);
00268 EpfCall:
00269     if ( AM.tracebackflag ) {
00270         MLOCK(ErrorMessageLock);
00271         MesCall("EpfCon");
00272         MUNLOCK(ErrorMessageLock);
00273     }
00274     return(-1);
00275 }
00276
00277 /*
00278     #] EpfCon :
00279     #[ EpfGen :                WORD EpfGen(number,inlist,kron,perm,sgn)
00280 */
00281
00282 WORD EpfGen(WORD number, WORD *inlist, WORD *kron, WORD *perm, WORD sgn)
00283 {
00284     WORD i, *in2, k, a;
00285     if ( !sgn ) {
00286         in2 = inlist + number;
00287         number *= 2;
00288         for ( i = 1; i < number; i += 2 ) {
00289             *perm++ = i;
00290             *perm++ = i;
00291             *kron++ = *inlist++;
00292             *kron++ = *in2++;
00293         }
00294         if ( number <= 0 ) return(0);
00295         else return(1);
00296     }
00297     number *= 2;
00298     i = number - 1;
00299     while ( ( i -= 2 ) >= 0 ) {
00300         if ( ( k = perm[i] ) != i ) {
00301             sgn = -sgn;
00302             a = kron[i];

```

```

00303         kron[i] = kron[k];
00304         kron[k] = a;
00305     }
00306     if ( ( k = ( perm[i] += 2 ) ) < number ) {
00307         a = kron[i];
00308         kron[i] = kron[k];
00309         kron[k] = a;
00310         sgn = - sgn;
00311         for ( k = i + 2; k < number; k += 2 ) perm[k] = k;
00312         return(sgn);
00313     }
00314 }
00315 return(0);
00316 }
00317
00318 /*
00319     #] EpfGen :
00320     #[ Trick :          WORD Trick(in,t)
00321
00322     This routine implements the identity:
00323     g_(j,mu)*g_(j,nu)*g_(j,ro)=e_(mu,nu,ro,si)*g5_(j)*g_(j,si)
00324     +d_(mu,nu)*g_(j,ro)-d_(mu,ro)*g_(j,nu)+d_(nu,ro)*g_(j,mu)
00325     which is for 4 dimensions only!
00326
00327     Note that z->gamm = 1 if there is no g5 present.
00328
00329 */
00330
00331 WORD Trick(WORD *in, TRACES *t)
00332 {
00333     WORD n, n1;
00334     n = t->stap;
00335     n1 = t->step1;
00336     switch ( t->eers[n] ) {
00337     case 0: {
00338         WORD *p;
00339         p = t->pepf + t->mepf[n];
00340         *p++ = *in++;
00341         *p++ = *in++;
00342         *p++ = *in;
00343         *p = ++(t->mdum);
00344         (t->mepf[n1]) += 4;
00345         *in = t->mdum;
00346         t->gamm = - t->gamm;
00347         t->eers[n] = 5;
00348         break;
00349     }
00350     case 4: {
00351         WORD *p;
00352         p = t->pdel + t->mdel[n];
00353         (t->mepf[n1]) -= 4;
00354         (t->mdum)--;
00355         *p++ = *in++;
00356         *p = *in++;
00357         *in = *(t->pepf + t->mepf[n] + 2);
00358         (t->mdel[n1]) += 2;
00359         t->gamm = - t->gamm;
00360         break;
00361     }
00362     case 3: {
00363         t->sign1 = - t->sign1;
00364         *(t->pdel + t->mdel[n] + 1) = in[2];
00365         in[2] = in[1];
00366         break;
00367     }
00368     case 2: {
00369         t->sign1 = - t->sign1;
00370         in[2] = in[0];
00371         *(t->pdel + t->mdel[n]) = in[1];
00372         break;
00373     }
00374     case 1: {
00375         in[2] = *(t->pdel + t->mdel[n] + 1);
00376         (t->mdel[n1]) -= 2;
00377         break;
00378     }
00379     default: {
00380         return(0);
00381     }
00382 }
00383 return(--(t->eers[n]));
00384 }
00385
00386 /*
00387     #] Trick :
00388     #[ Trace4no :          WORD Trace4no(number,kron,t)
00389

```

```

00390     Takes the trace of a string of gamma matrices in 4 dimensions.
00391     There is no test for indices or vectors that are the same.
00392     The four dimensions refer to the contraction in the algebra:
00393     g_(i,a,b,c) = e_(a,b,c,d)*g_(i,5_,d) + g_(i,a)*d_(b,c)
00394                 - g_(i,b)*d_(a,c) + g_(i,c)*d_(a,b)
00395     This simplifies life very much and leads to shorter expressions
00396     than in the n dimensional case.
00397
00398     Parameters:
00399     number: the number of vectors/indices in inlist.
00400     inlist: the indices/vectors in the string.
00401     kron:   the output delta's.
00402     gamma5: the potential gamma5 in front.
00403     t:      the struct for scratch manipulations.
00404     stack:  the space to put all scratch arrays in.
00405
00406     The return value is zero if there are no more terms, 1 if a
00407     term was generated with a positive sign and -1 if a term was
00408     generated with a negative sign.
00409     The value of one is increased to two if the first 4 values
00410     in kron should be interpreted as a Levi-Civita tensor.
00411
00412     Note that kron should have more places reserved than the number
00413     of indices in inlist, because it will contain dummy indices
00414     temporarily. In principle there can be 'number*1/4' extra dummies.
00415
00416 */
00417
00418 int Trace4no(WORD number, WORD *kron, TRACES *t)
00419 {
00420     WORD i;
00421     WORD *p, *m;
00422     WORD *stop, oldsign;
00423     int retval;
00424     if ( !t->sgn ) { /* Startup */
00425         if ( ( number < 0 ) || ( number & 1 ) ) return(0);
00426         if ( number <= 2 ) {
00427             if ( t->gamma5 == GAMMA5 ) return(0);
00428             if ( number == 2 ) {
00429                 *kron++ = *t->inlist;
00430                 *kron++ = t->inlist[1];
00431             }
00432             return(1);
00433         }
00434         t->sgn = 1;
00435         {
00436             WORD nhalf = number >> 1;
00437             WORD ndouble = number * 2;
00438             p = t->eers;
00439             t->eers = p;    p += nhalf;
00440             t->mepf = p;    p += nhalf;
00441             t->mdel = p;    p += nhalf;
00442             t->pdel = p;    p += number + nhalf;
00443             t->pepf = p;    p += ndouble;
00444             t->e4 = p;      p += number;
00445             t->e3 = p;      p += ndouble;
00446             t->nt3 = p;     p += nhalf;
00447             t->nt4 = p;     p += nhalf;
00448             t->j3 = p;      p += ndouble;
00449             t->j4 = p;
00450         }
00451         t->mepf[0] = 0;
00452         t->mdel[0] = 0;
00453         t->mdum = AM.mTraceDum;
00454         t->kstep = -2;
00455         t->step1 = 0;
00456         t->sign1 = 1;
00457         t->lc3 = -1;
00458         t->lc4 = -1;
00459         t->gamm = 1;
00460
00461         do {
00462             t->stap = (t->step1)++;
00463             t->kstep += 2;
00464             t->eers[t->stap] = 0;
00465             t->mepf[t->step1] = t->mepf[t->stap];
00466             t->mdel[t->step1] = t->mdel[t->stap];
00467             CallTrick: while ( !Trick(t->inlist+t->kstep,t) ) {
00468                 t->kstep -= 2;
00469                 t->step1 = (t->stap)--;
00470                 if ( t->stap < 0 ) {
00471                     return(0);
00472                 }
00473             }
00474         } while ( t->kstep < (number-4) );
00475     /*
00476     Take now the trace of the leftover matrices.

```

```

00477      If gamma5 causes the term to vanish there will be a
00478      renewed call to Trick for its next term.
00479  */
00480      t->sign2 = t->sign1;
00481      if ( ( t->gamma5 == GAMMA7 ) && ( t->gamm == -1 ) ) {
00482          t->sign2 = - t->sign2;
00483      }
00484      else if ( ( t->gamma5 == GAMMA5 ) && ( t->gamm == 1 ) ) {
00485          goto CallTrick;
00486      }
00487      else if ( ( t->gamma5 == GAMMA1 ) && ( t->gamm == -1 ) ) {
00488          goto CallTrick;
00489      }
00490      p = t->pdel + t->mdel[t->step1];
00491      *p++ = t->inlist[t->kstep+2];
00492      *p++ = t->inlist[t->kstep+3];
00493  /*
00494      Now the trace has been expressed in terms of Levi-Civita tensors
00495      and Kronecker delta's.
00496      The Levi-Civita tensors are in t->pepf
00497      and there are t->mepf[step1] elements in this array.
00498      The Kronecker delta's are in t->pdel
00499      and there are t->mdel[step1] elements in this array.
00500
00501      Next we rake the Levi-Civita tensors together such that there
00502      is an optimal use of the contractions.
00503  */
00504      {
00505          WORD ae;
00506          ae = t->mepf[t->step1];
00507          t->ad = t->mdel[t->step1]+2;
00508          t->a4 = 0;
00509          t->a3 = 0;
00510          while ( ( ae -= 4 ) >= 0 ) {
00511              if ( t->pepf[ae] > AM.mTraceDum && t->pepf[ae] <= t->mdum ) {
00512                  p = t->e3 + t->a3;
00513                  m = t->pepf + ae;
00514                  for ( i = 0; i < 3; i++ ) {
00515                      p[3] = m[3-i];
00516                      *p++ = m[i-4];
00517                  }
00518                  t->a3 += 6;
00519                  ae -= 4;
00520              }
00521              else {
00522                  p = t->e4 + t->a4;
00523                  m = t->pepf + ae;
00524                  for ( i = 0; i < 4; i++ ) *p++ = *m++;
00525                  t->a4 += 4;
00526              }
00527          }
00528      }
00529  /*
00530      Now e3 contains pairs of LeviCivita tensors that have
00531      three indices each and a3 is the total number of indices.
00532      e4 contains individual tensors with 4 indices.
00533      Some indices may be contracted with Kronecker delta's.
00534
00535      Contract the e3 tensors first.
00536  */
00537      while ( t->a3 > 0 ) {
00538          t->nt3[++(t->lc3)] = 0;
00539          while ( ( t->nt3[t->lc3] = EpfGen(3,t->e3+t->a3-6,
00540              t->pdel+t->ad,t->j3+6*t->lc3,oldsign = t->nt3[t->lc3]) ) == 0 ) {
00541              if ( oldsign < 0 ) t->sign2 = - t->sign2;
00542              (t->lc3)--;
00543              NextE3: if ( t->lc3 < 0 ) goto CallTrick;
00544                  t->ad -= 6;
00545                  t->a3 += 6;
00546              }
00547          if ( oldsign ) {
00548              if ( oldsign != t->nt3[t->lc3] ) t->sign2 = - t->sign2;
00549          }
00550          else if ( t->nt3[t->lc3] < 0 ) t->sign2 = - t->sign2;
00551          t->a3 -= 6;
00552          t->ad += 6;
00553      }
00554  }
00555  /*
00556      Contract the e4 tensors.
00557  */
00558      while ( t->a4 > 4 ) {
00559          t->nt4[++(t->lc4)] = 0;
00560          while ( ( t->nt4[t->lc4] = EpfGen(4,t->e4+t->a4-8,
00561              t->pdel+t->ad,t->j4+8*t->lc4,oldsign = t->nt4[t->lc4]) ) == 0 ) {
00562              if ( oldsign < 0 ) t->sign2 = - t->sign2;
00563              (t->lc4)--;

```

```

00564 NextE4:      if ( t->lc4 < 0 ) goto NextE3;
00565                t->ad -= 8;
00566                t->a4 += 8;
00567            }
00568            if ( oldsign ) {
00569                if ( oldsign != t->nt4[t->lc4] ) t->sign2 = - t->sign2;
00570            }
00571            else if ( t->nt4[t->lc4] < 0 ) t->sign2 = - t->sign2;
00572            t->a4 -= 8;
00573            t->ad += 8;
00574        }
00575    /*
00576        Finally the extra dummy indices can be eliminated.
00577        Note that there are currently t->ad words in t->pdel forming
00578        t->ad / 2 Kronecker delta's. We are however not allowed to
00579        alter anything in these arrays, so the results should be
00580        copied to kron.
00581    */
00582    m = kron;
00583    if ( t->a4 == 4 ) {
00584        p = t->e4;
00585        *m++ = *p++; *m++ = *p++; *m++ = *p++; *m++ = *p++;
00586        retval = 2;
00587    }
00588    else retval = 1;
00589    if ( t->sign2 < 0 ) retval = - retval;
00590    p = t->pdel;
00591    for ( i = 0; i < t->ad; i++ ) *m++ = *p++;
00592    p = kron;
00593    if ( t->a4 == 4 ) {
00594    /*
00595        Test for dummies in the last position of the e_.
00596    */
00597        stop = p + t->ad + 4;
00598        p += 3;
00599        while ( *p >= AM.mTraceDum && *p <= t->mdum ) {
00600            m = p + 1;
00601            do {
00602                if ( *m == *p ) {
00603                    *p = m[1];
00604                    *m = *--stop;
00605                    m[1] = *--stop;
00606                    break;
00607                }
00608                else if ( m[1] == *p ) {
00609                    *p = *m;
00610                    *m = *--stop;
00611                    m[1] = *--stop;
00612                    break;
00613                }
00614                else m += 2;
00615            } while ( m < stop );
00616        }
00617        p++;
00618    }
00619    else stop = p + t->ad;
00620    while ( p < (stop-2) ) {
00621        while ( *p >= AM.mTraceDum && *p <= t->mdum ) {
00622            m = p + 2;
00623            do {
00624                if ( *m == *p ) {
00625                    *p = m[1];
00626                    *m = *--stop;
00627                    m[1] = *--stop;
00628                    break;
00629                }
00630                else if ( m[1] == *p ) {
00631                    *p = *m;
00632                    *m = *--stop;
00633                    m[1] = *--stop;
00634                    break;
00635                }
00636                else m += 2;
00637            } while ( m < stop );
00638        }
00639        p++;
00640        while ( *p >= AM.mTraceDum && *p <= t->mdum ) {
00641            m = p + 1;
00642            do {
00643                if ( *m == *p ) {
00644                    *p = m[1];
00645                    *m = *--stop;
00646                    m[1] = *--stop;
00647                    break;
00648                }
00649                else if ( m[1] == *p ) {
00650                    *p = *m;

```

```

00651             *m = *--stop;
00652             m[1] = *--stop;
00653             break;
00654         }
00655         else m += 2;
00656     } while ( m < stop );
00657 }
00658 p++;
00659 }
00660 return(retval);
00661 }
00662 if ( number <= 2 ) return(0);
00663 else { goto NextE4; }
00664 }
00665
00666 /*
00667     #] Trace4no :
00668     #[ Trace4 :          WORD Trace4(term,params,num,level)
00669
00670     Generates traces of the string of gamma matrices in 'instring'.
00671
00672     The difference with the routine tracen ( for n dimensions )
00673     lies in the treatment of gamma 5 and the specific form
00674     of the Chisholm identities. The identities used here are
00675     g(mu)*g(al)*...*g(an)*g(mu)=
00676     n=odd: -2*g(an)*...*g(al)      ( reversed order )
00677     n=even: 2*g(an)*g(al)*...*g(a(n-1))
00678             +2*g(a(n-1))*...*g(al)*g(an)
00679     There is a special case for n=2 : 4*d(al,a2)*gi
00680
00681     The main difference with the old fortran version lies in
00682     the recursion that is used here. That cleans up the variables
00683     very much.
00684
00685     The contents of the AT.TMout array are:
00686     length,type,gamma5,gamma's
00687
00688     The space for the vectors in t is at most 14 * number.
00689
00690     The condition params[5] == 0 corresponds to finding gamma6*gamma7
00691     during the pick up of the matrices.
00692 */
00693
00694 int Trace4(PHEAD WORD *term, WORD *params, WORD num, WORD level)
00695 {
00696     GETBIDENTITY
00697     TRACES *t;
00698     WORD *p, *m, number, i;
00699     WORD *OldW;
00700     WORD j, minimum, minimum2, *min, *stopper;
00701     int ret;
00702     OldW = AT.WorkPointer;
00703     if ( AN.numtracesctack >= AN.intracestack ) {
00704         number = AN.intracestack + 2;
00705         t = (TRACES *)Malloca(number*sizeof(TRACES),"TRACES-struct");
00706         if ( AN.tracestack ) {
00707             for ( i = 0; i < AN.intracestack; i++ ) { t[i] = AN.tracestack[i]; }
00708             M_free(AN.tracestack,"TRACES-struct");
00709         }
00710         AN.tracestack = t;
00711         AN.intracestack = number;
00712     }
00713
00714     number = *params - 6;
00715     if ( number < 0 || ( number & 1 ) || !params[5] ) return(0);
00716
00717     t = AN.tracestack + AN.numtracesctack;
00718     AN.numtracesctack++;
00719
00720     t->finalstep = ( params[2] & 16 ) ? 1 : 0;
00721     t->gamma5 = params[3];
00722     if ( t->finalstep && t->gamma5 != GAMMA1 ) {
00723         MLOCK(ErrorMessageLock);
00724         MesPrint("Gamma5 not allowed in this option of the trace command");
00725         MUNLOCK(ErrorMessageLock);
00726         AN.numtracesctack--;
00727         SETERROR(-1)
00728     }
00729     t->inlist = AT.WorkPointer;
00730     t->accup = t->accu = t->inlist + number;
00731     t->perm = t->accu + (number*2);
00732     t->eers = t->perm + number;
00733     if ( ( AT.WorkPointer += 19 * number ) >= AT.WorkTop ) {
00734         MLOCK(ErrorMessageLock);
00735         MesWork();
00736         MUNLOCK(ErrorMessageLock);
00737     }

```



```

00738         return(-1);
00739     }
00740     t->num = num;
00741     t->level = level;
00742     p = t->inlist;
00743     m = params+6;
00744     for ( i = 0; i < number; i++ ) *p++ = *m++;
00745     t->termp = term;
00746     t->factor = params[4];
00747     t->allsign = params[5];
00748     if ( number >= 10 || ( t->gamma5 != GAMMA1 && number > 4 ) ) {
00749 /*
00750         The next code should `normal order' the string
00751         We need the lexicographic smallest string, taking also the
00752         reverse strings into account.
00753 */
00754         minimum = 0; min = t->inlist;
00755         stopper = min + number;
00756         for ( i = 1; i < number; i++ ) {
00757             p = min;
00758             m = t->inlist + i;
00759             for ( j = 0; j < number; j++ ) {
00760                 if ( *p < *m ) break;
00761                 if ( *p > *m ) {
00762                     min = t->inlist+i;
00763                     minimum = i;
00764                     break;
00765                 }
00766                 p++; m++;
00767                 if ( m >= stopper ) m = t->inlist;
00768                 if ( p >= stopper ) p = t->inlist;
00769             }
00770         }
00771         p = min;
00772         min = m = AT.WorkPointer;
00773         i = number;
00774         while ( --i >= 0 ) {
00775             *m++ = *p++;
00776             if ( p >= stopper ) p = t->inlist;
00777         }
00778         p = t->inlist;
00779         m = min;
00780         i = number;
00781         while ( --i >= 0 ) *p++ = *m++;
00782         p = t->inlist;
00783         m = stopper;
00784         while ( p < m ) { /* reverse string */
00785             i = *p; *p++ = *--m; *m = i;
00786         }
00787         minimum2 = 0;
00788         for ( i = 0; i < number; i++ ) {
00789             p = min;
00790             m = t->inlist + i;
00791             for ( j = 0; j < number; j++ ) {
00792                 if ( *p < *m ) break;
00793                 if ( *p > *m ) {
00794                     m = t->inlist + i;
00795                     p = min;
00796                     j = number;
00797                     while ( --j >= 0 ) {
00798                         *p++ = *m++;
00799                         if ( m >= stopper ) m = t->inlist;
00800                     }
00801                     minimum2 = i;
00802                     break;
00803                 }
00804                 p++; m++;
00805                 if ( m >= stopper ) m = t->inlist;
00806             }
00807         }
00808         minimum ^= minimum2;
00809         if ( ( minimum & 1 ) != 0 ) {
00810             if ( t->gamma5 == GAMMA5 ) t->allsign = - t->allsign;
00811             else if ( t->gamma5 != GAMMA1 )
00812                 t->gamma5 = GAMMA6 + GAMMA7 - t->gamma5;
00813         }
00814         p = min; m = t->inlist; i = number;
00815         while ( --i >= 0 ) *m++ = *p++;
00816 /*
00817         Now the trace is in normal order
00818 */
00819     }
00820     ret = Trace4Gen(BHEAD t,number);
00821     AT.WorkPointer = OldW;
00822     AN.numtracesctack--;
00823     return(ret);
00824 }

```

```

00825
00826 /*
00827 #] Trace4 :
00828 #[ Trace4Gen :          WORD Trace4Gen(t,number)
00829
00830 The recursive breakdown of a trace in 4 dimensions.
00831 We test first whether the trace has zero or two gamma's left.
00832 This case can be done quickly.
00833 Next we test whether we can eliminate adjacent objects that are
00834 the same.
00835 Then we test for Chisholm identities (I). First for identities
00836 with an odd number of gamma matrices (II), then for those with an
00837 even number of matrices (III). The special thing here is the demand
00838 that the contraction be between indices with 4 dimensions only.
00839 Then there is a scan for objects that are the same, not regarding
00840 their type (IV). This is exactly the same as in n dimensions.
00841 Finally we have a string left in which all objects are different (V).
00842 This case is treated by the routine Trace4no (no stands for no
00843 objects are the same).
00844
00845 In case I we have one d_ of which the result of the contraction
00846 has not yet been fixed.
00847 Case II gives just a reordering of the matrices and a factor -2.
00848 Case III gives two terms: one for the anti commutation, such that
00849 the number of intermediate matrices becomes odd and the other
00850 from the Chisholm rule for an odd number of matrices. Both have
00851 a factor 2.
00852 Case IV gives m+1 terms when m is the number of matrices inbetween.
00853 We take the shortest path. The sign alternates and all terms have
00854 a factor two, except for the last one.
00855
00856 */
00857
00858 int Trace4Gen(PHEAD TRACES *t, WORD number)
00859 {
00860     GETBIDENTITY
00861     WORD *termout, *stop;
00862     WORD *p, *m, oldval;
00863     WORD *pold, *mold, diff, *oldstring, cp;
00864 /*
00865     #] Special cases :
00866 */
00867     if ( number <= 2 ) { /* Special cases */
00868         if ( t->gamma5 == GAMMA5 ) return(0);
00869         termout = AT.WorkPointer;
00870         p = t->termp;
00871         stop = p + *p;
00872         m = termout;
00873         p++;
00874         if ( p < stop ) do {
00875             if ( *p == SUBEXPRESSION && p[2] == t->num ) {
00876                 oldstring = p;
00877                 p = t->termp;
00878                 do { *m++ = *p++; } while ( p < oldstring );
00879                 p += p[1];
00880                 *m++ = AC.lUniTrace[0];
00881                 *m++ = AC.lUniTrace[1];
00882                 *m++ = AC.lUniTrace[2];
00883                 *m++ = AC.lUniTrace[3];
00884                 if ( number == 2 || t->accup > t->accu ) {
00885                     oldstring = m;
00886                     *m++ = DELTA;
00887                     *m++ = 4;
00888                     if ( number == 2 ) {
00889                         *m++ = t->inlist[0];
00890                         *m++ = t->inlist[1];
00891                     }
00892                     if ( t->accup > t->accu ) {
00893                         pold = p;
00894                         p = t->accu;
00895                         while ( p < t->accup ) *m++ = *p++;
00896                         oldstring[1] = WORDDIFF(m,oldstring);
00897                         p = pold;
00898                     }
00899                 }
00900                 if ( t->factor ) {
00901                     *m++ = SNUMBER;
00902                     *m++ = 4;
00903                     *m++ = 2;
00904                     *m++ = t->factor;
00905                 }
00906                 do { *m++ = *p++; } while ( p < stop );
00907                 *termout = WORDDIFF(m,termout);
00908                 if ( t->allsign < 0 ) m[-1] = -m[-1];
00909                 if ( ( AT.WorkPointer = m ) > AT.WorkTop ) {
00910                     MLOCK(ErrorMessageLock);
00911                     MesWork();

```

```

00912             MUNLOCK(ErrorMessageLock);
00913             return(-1);
00914         }
00915         *AN.RepPoint = 1;
00916         AR.expchanged = 1;
00917         if ( *termout ) {
00918             *AN.RepPoint = 1;
00919             AR.expchanged = 1;
00920             if ( Generator(BHEAD termout,t->level) ) goto TracCall;
00921         }
00922         AT.WorkPointer= termout;
00923         return(0);
00924     }
00925     p += p[1];
00926 } while ( p < stop );
00927 return(0);
00928 }
00929 /*
00930     #] Special cases :
00931     #[ Adjacent objects :
00932 */
00933 p = t->inlist;
00934 stop = p + number - 1;
00935 if ( *p == *stop ) {          /* First and last of string */
00936     oldval = *p;
00937     *(t->accup)++ = *p;
00938     *(t->accup)++ = *p;
00939     m = p+1;
00940     while ( m < stop ) *p++ = *m++;
00941     if ( t->gamma5 != GAMMA1 ) {
00942         if ( t->gamma5 == GAMMA5 ) t->allsign = - t->allsign;
00943         else if ( t->gamma5 == GAMMA6 ) t->gamma5 = GAMMA7;
00944         else if ( t->gamma5 == GAMMA7 ) t->gamma5 = GAMMA6;
00945     }
00946     if ( Trace4Gen(BHEAD t,number-2) ) goto TracCall;
00947     t = AN.tracestack + AN.numtracesctack - 1;
00948     if ( t->gamma5 != GAMMA1 ) {
00949         if ( t->gamma5 == GAMMA5 ) t->allsign = - t->allsign;
00950         else if ( t->gamma5 == GAMMA6 ) t->gamma5 = GAMMA7;
00951         else if ( t->gamma5 == GAMMA7 ) t->gamma5 = GAMMA6;
00952     }
00953     while ( p > t->inlist ) *--m = *--p;
00954     *p = *stop = oldval;
00955     t->accup -= 2;
00956     return(0);
00957 }
00958 do {
00959     if ( *p == p[1] ) {          /* Adjacent in string */
00960         oldval = *p;
00961         pold = p;
00962         m = p+2;
00963         *(t->accup)++ = *p;
00964         *(t->accup)++ = *p;
00965         while ( m <= stop ) *p++ = *m++;
00966         if ( Trace4Gen(BHEAD t,number-2) ) goto TracCall;
00967         t = AN.tracestack + AN.numtracesctack - 1;
00968         while ( p > pold ) *--m = *--p;
00969         *p++ = oldval;
00970         *p++ = oldval;
00971         t->accup -= 2;
00972         return(0);
00973     }
00974     p++;
00975 } while ( p < stop );
00976 /*
00977     #] Adjacent objects :
00978     #[ Odd Contraction :
00979 */
00980 p = t->inlist;
00981 stop = p + number;
00982 do {
00983     if ( *p >= AM.OffsetIndex && (
00984         ( *p < WILDOFFSET + AM.OffsetIndex &&
00985         indices[*p-AM.OffsetIndex].dimension == 4 )
00986         || ( *p >= WILDOFFSET + AM.OffsetIndex && AC.lDefDim == 4 ) ) ) {
00987         m = p+2;
00988         while ( m < stop ) {
00989             if ( *p == *m ) {
00990                 pold = p;
00991                 mold = m;
00992                 oldval = *p;
00993                 (t->factor)++;
00994                 t->allsign = - t->allsign;
00995                 *p++ = *--m;
00996                 m--;
00997                 while ( m > p ) { diff = *p; *p++ = *m; *m-- = diff; }
00998                 p = mold - 1;

```

```

00999          m = mold + 1;
01000          while ( m < stop ) *p++ = *m++;
01001          if ( Trace4Gen(BHEAD t,number-2) ) goto TracCall;
01002          t = AN.tracestack + AN.numtracesetack - 1;
01003          m--;
01004          while ( m > mold ) *m-- = *--p;
01005          p = pold;
01006          *m-- = oldval;
01007          *m-- = *p;
01008          *p++ = oldval;
01009          while ( m > p ) { diff = *p; *p++ = *m; *m-- = diff; }
01010          t->allsign = - t->allsign;
01011          (t->factor)--;
01012          return(0);
01013      }
01014      m += 2;
01015  }
01016  }
01017  p++;
01018  } while ( p < stop );
01019  /*
01020      #] Odd Contraction :
01021      #[ Even Contraction :
01022      First the case with two matrices inbetween.
01023  */
01024  p = t->inlist;
01025  stop = p + number;
01026  do {
01027      if ( *p >= AM.OffsetIndex && (
01028          ( *p < WILDOFFSET + AM.OffsetIndex &&
01029              indices[*p-AM.OffsetIndex].dimension == 4 )
01030          || ( *p >= WILDOFFSET + AM.OffsetIndex && AC.lDefDim == 4 ) ) ) {
01031          m = p+3;
01032          if ( m >= stop ) m -= number;
01033          if ( *p == *m ) {
01034              WORD oldfactor, old5;
01035              oldstring = AT.WorkPointer;
01036              AT.WorkPointer += number;
01037              oldfactor = t->allsign;
01038              old5 = t->gamma5;
01039              if ( m < p ) cp = (WORDDIF(m,t->inlist) + 1) & 1;
01040              else cp = 0;
01041              if ( cp && ( t->gamma5 != GAMMA1 ) ) {
01042                  if ( t->gamma5 == GAMMA5 ) t->allsign = -t->allsign;
01043                  else if ( t->gamma5 == GAMMA6 ) t->gamma5 = GAMMA7;
01044                  else if ( t->gamma5 == GAMMA7 ) t->gamma5 = GAMMA6;
01045              }
01046              mold = m;
01047              p = oldstring;
01048              m = t->inlist;
01049              while ( m < stop ) *p++ = *m++;
01050  /*
01051      Rotate the string
01052  */
01053          m = mold + 1;
01054          p = t->inlist;
01055          while ( m < stop ) *p++ = *m++;
01056          m = oldstring;
01057          if ( !cp && ((WORDDIF(stop,p))&1) != 0 && ( t->gamma5 != GAMMA1 ) ) {
01058              if ( t->gamma5 == GAMMA5 ) t->allsign = -t->allsign;
01059              else if ( t->gamma5 == GAMMA6 ) t->gamma5 = GAMMA7;
01060              else if ( t->gamma5 == GAMMA7 ) t->gamma5 = GAMMA6;
01061          }
01062          while ( p < stop ) *p++ = *m++;
01063          t->factor += 2;
01064          m = p - 3;
01065          p = t->inlist;
01066          oldval = number - 4;
01067          while ( oldval > 0 ) {
01068              if ( *p >= AM.OffsetIndex && (
01069                  ( *p < WILDOFFSET + AM.OffsetIndex &&
01070                      indices[*p-AM.OffsetIndex].dimension )
01071                  || ( *p >= WILDOFFSET + AM.OffsetIndex && AC.lDefDim ) ) ) {
01072                  if ( *p == *m ) {
01073                      *p = m[1];
01074                      break;
01075                  }
01076                  else if ( *p == m[1] ) {
01077                      *p = *m;
01078                      break;
01079                  }
01080              }
01081              p++; oldval--;
01082          }
01083          if ( oldval <= 0 ) {
01084              *(t->accup)++ = *m++;
01085              *(t->accup)++ = *m++;

```

```

01086         }
01087         if ( Trace4Gen(BHEAD t,number-4) ) goto TracCall;
01088         t = AN.tracestack + AN.numtracesctack - 1;
01089         t->factor -= 2;
01090         if ( oldval <= 0 ) t->accup -= 2;
01091         t->gamma5 = old5;
01092         t->allsign = oldfactor;
01093         AT.WorkPointer = p = oldstring;
01094         m = t->inlist;
01095         while ( m < stop ) *m++ = *p++;
01096         return(0);
01097     }
01098 }
01099 p++;
01100 } while ( p < stop );
01101 /*
01102     The case with at least 4 matrices inbetween.
01103 */
01104 p = t->inlist;
01105 stop = p + number;
01106 do {
01107     if ( *p >= AM.OffsetIndex && (
01108         ( *p < WILDOFFSET + AM.OffsetIndex &&
01109         indices[*p-AM.OffsetIndex].dimension == 4 )
01110         || ( *p >= WILDOFFSET + AM.OffsetIndex && AC.lDefDim == 4 ) ) ) {
01111         m = p+5;
01112         while ( m < stop ) {
01113             if ( *p == *m ) {
01114                 WORD *pex, *mex;
01115                 pold = p;
01116                 mold = m;
01117                 oldval = *p;
01118             /*
01119                 g_(1,mu)*g_(1,a1)*...*g_(1,aj)*g_(1,an)*g_(1,mu) ->
01120                 first:
01121                 2*g_(1,an)*g_(1,a1)*...*g_(1,aj)
01122             */
01123                 (t->factor)++;
01124             /*
01125                 The variable hulp seems unnecessary
01126                 *p = hulp = m[-1];
01127             */
01128                 *p = m[-1];
01129                 p = m - 1;
01130                 m++;
01131                 while ( m < stop ) *p++ = *m++;
01132                 if ( Trace4Gen(BHEAD t,number-2) ) goto TracCall;
01133                 t = AN.tracestack + AN.numtracesctack - 1;
01134                 pex = p; mex = m;
01135                 p = pold;
01136                 m = mold - 2;
01137                 while ( m > p ) { diff = *p; *p++ = *m; *m-- = diff; }
01138             /*
01139                 and then:
01140                 2*g_(1,aj)*...*g_(1,a1)*g_(1,an)
01141             */
01142                 if ( Trace4Gen(BHEAD t,number-2) ) goto TracCall;
01143                 t = AN.tracestack + AN.numtracesctack - 1;
01144                 p = pold;
01145                 m = mold - 2;
01146                 while ( m > p ) { diff = *p; *p++ = *m; *m-- = diff; }
01147                 m = mex;
01148                 p = pex;
01149                 m--;
01150                 while ( m > mold ) *m-- = *--p;
01151                 m = mold;
01152                 *m-- = oldval;
01153                 p = pold;
01154                 *m = *p;
01155                 *p = oldval;
01156                 (t->factor)--;
01157                 return(0);
01158             }
01159             m += 2;
01160         }
01161     }
01162     p++;
01163 } while ( p < stop );
01164 /*
01165     #] Even Contraction :
01166     #[ Same Objects :
01167 */
01168 p = t->inlist;
01169 stop = p + number - 1;
01170 diff = 2;
01171 do {
01172     p = t->inlist;

```

```

01173     while ( p <= stop ) {
01174         m = p + diff;
01175         if ( m > stop ) m -= number;
01176         if ( *p == *m ) {
01177             WORD oldfactor, c, old5;
01178             oldfactor = t->allsign;
01179             old5 = t->gamma5;
01180             cp = (WORDDIFF(m,t->inlist)) & 1;
01181             if ( !cp && ( t->gamma5 != GAMMA1 ) ) {
01182                 if ( t->gamma5 == GAMMA5 ) t->allsign = -t->allsign;
01183                 else if ( t->gamma5 == GAMMA6 ) t->gamma5 = GAMMA7;
01184                 else if ( t->gamma5 == GAMMA7 ) t->gamma5 = GAMMA6;
01185             }
01186             oldstring = AT.WorkPointer;
01187             AT.WorkPointer += number;
01188             mold = m;
01189             oldval = *p;
01190             p = oldstring;
01191             m = t->inlist;
01192             while ( m <= stop ) *p++ = *m++;
01193         /*
01194         Rotate the string
01195         */
01196         m = mold + 1;
01197         p = t->inlist;
01198         while ( m <= stop ) *p++ = *m++;
01199         m = oldstring;
01200         while ( p <= stop ) *p++ = *m++;
01201         (t->factor)++;
01202         p -= diff + 1;
01203         m = stop;
01204         *(t->accup) = oldval;
01205         t->accup += 2;
01206         m--;
01207         while ( m > p ) {
01208             c = t->accup[-1];
01209             t->accup[-1] = *m;
01210             *m = c;
01211             if ( Trace4Gen(BHEAD t,number-2) ) goto Trac4Call;
01212             t = AN.tracestack + AN.numtracesctack - 1;
01213             m--;
01214             t->allsign = - t->allsign;
01215         }
01216         c = t->accup[-1];
01217         t->accup[-1] = *m;
01218         *m = c;
01219         (t->factor)--;
01220         if ( Trace4Gen(BHEAD t,number-2) ) goto Trac4Call;
01221         t = AN.tracestack + AN.numtracesctack - 1;
01222         t->accup -= 2;
01223         t->allsign = oldfactor;
01224         AT.WorkPointer = p = oldstring;
01225         m = t->inlist;
01226         while ( m <= stop ) *m++ = *p++;
01227         t->gamma5 = old5;
01228         return(0);
01229     }
01230     p++;
01231 }
01232 } while ( ++diff <= (number>>1) );
01233 /*
01234     #] Same Objects :
01235     #[ All Different :
01236
01237     Here we have a string with all different objects.
01238
01239 */
01240 t->sgn = 0;
01241 termout = AT.WorkPointer;
01242 for(;;) {
01243     if ( t->finalstep == 0 ) diff = Trace4no(number,t->accup,t);
01244     else diff = TraceNno(number,t->accup,t);
01245     /* while ( ( diff = Trace4no(number,t->accup,t) ) != 0 ) */
01246     if ( diff == 0 ) break;
01247     p = t->term;
01248     stop = p + *p;
01249     m = termout;
01250     p++;
01251     if ( p < stop ) do {
01252         if ( *p == SUBEXPRESSION && p[2] == t->num ) {
01253             oldstring = p;
01254             p = t->term;
01255             do { *m++ = *p++; } while ( p < oldstring );
01256             p += p[1];
01257             pold = p;
01258             *m++ = AC.lUniTrace[0];
01259             *m++ = AC.lUniTrace[1];

```

```

01260      *m++ = AC.lUniTrace[2];
01261      *m++ = AC.lUniTrace[3];
01262      *m++ = SNUMBER;
01263      *m++ = 4;
01264      *m++ = 2;
01265      *m++ = t->factor;
01266      p = t->accup;
01267      oldval = number;
01268      if ( diff == 2 || diff == -2 ) {
01269          *m++ = LEVICIVITA;
01270          *m++ = 4+FUNHEAD;
01271          *m++ = DIRTYFLAG;
01272          FILLFUN3(m)
01273          *m++ = *p++; *m++ = *p++; *m++ = *p++; *m++ = *p++;
01274          oldval -= 4;
01275      }
01276      if ( oldval > 0 || t->accup > t->accu ) {
01277          oldstring = m;
01278          *m++ = DELTA;
01279          *m++ = oldval + 2;
01280          if ( oldval > 0 ) NCOPY(m,p,oldval);
01281          if ( t->accup > t->accu ) {
01282              p = t->accu;
01283              while ( p < t->accup ) *m++ = *p++;
01284              oldstring[1] = WORDDIF(m,oldstring);
01285          }
01286      }
01287      p = pold;
01288      do { *m++ = *p++; } while ( p < stop );
01289      *termout = WORDDIF(m,termout);
01290      if ( ( diff ^ t->allsign ) < 0 ) m[-1] = - m[-1];
01291      if ( ( AT.WorkPointer = m ) > AT.WorkTop ) {
01292          MLOCK(ErrorMessageLock);
01293          MesWork();
01294          MUNLOCK(ErrorMessageLock);
01295          return(-1);
01296      }
01297      if ( *termout ) {
01298          *AN.RepPoint = 1;
01299          AR.expchanged = 1;
01300          if ( Generator(BHEAD termout,t->level) ) {
01301              AT.WorkPointer = termout;
01302              goto TracCall;
01303          }
01304          t = AN.tracestack + AN.numtracesctack - 1;
01305      }
01306      break;
01307  }
01308  p += p[1];
01309  } while ( p < stop );
01310  }
01311  AT.WorkPointer = termout;
01312  return(0);
01313  */
01314  /*
01315      #] All Different :
01316  */
01317  Trac4Call:
01318      AT.WorkPointer = oldstring;
01319  TracCall:
01320      if ( AM.tracebackflag ) {
01321          MLOCK(ErrorMessageLock);
01322          MesCall("Trace4Gen");
01323          MUNLOCK(ErrorMessageLock);
01324      }
01325      return(-1);
01326  }
01327  */
01328  /*
01329      #] Trace4Gen :
01330      #[ TraceNno :          WORD TraceNno(number,kron,t)
01331
01332      Routine takes the trace in N dimensions of a string
01333      of gamma matrices. It is assumed that there are no
01334      contractions and no vectors that are the same. For
01335      the treatment of those cases there are special routines,
01336      that call this routine as a final stage.
01337      The calling routine must reserve 'number' WORDs for perm
01338      and kron each.
01339      kron and perm may not be altered during the generation!
01340
01341  */
01342
01343  WORD TraceNno(WORD number, WORD *kron, TRACES *t)
01344  {
01345      WORD i, j, a, *p;
01346      if ( !t->sgn ) {

```

```

01347         if ( !number || ( number & 1 ) ) return(0);
01348         p = t->inlist;
01349         for ( i = 0; i < number; i++ ) {
01350             t->perm[i] = i;
01351             *kron++ = *p++;
01352         }
01353         t->sgn = 1;
01354         return(1);
01355     }
01356     else {
01357         i = number - 3;
01358         while ( i > 0 ) {
01359             a = kron[i];
01360             p = t->perm + i;
01361             for ( j = i + 1; j <= *p; j++ ) kron[j-1] = kron[j];
01362             kron[( *p )++] = a;
01363             if ( *p < number ) {
01364                 a = kron[*p];
01365                 j = *p;
01366                 while ( j >= (i+1) ) { kron[j] = kron[j-1]; j--; }
01367                 kron[i] = a;
01368                 number -= 2;
01369                 for ( j = i+2; j < number; j += 2 ) t->perm[j] = j;
01370                 t->sgn = - t->sgn;
01371                 return(t->sgn);
01372             }
01373             i -= 2;
01374         }
01375     }
01376     return(0);
01377 }
01378
01379 /*
01380     #] TraceNno :
01381     #[ TraceN :          WORD TraceN(term,params,num,level)
01382 */
01383
01384 int TraceN(PHEAD WORD *term, WORD *params, WORD num, WORD level)
01385 {
01386     GETBIDENTITY
01387     TRACES *t;
01388     WORD *p, *m, number, i;
01389     WORD *OldW;
01390     int ret;
01391     if ( params[3] != GAMMA1 ) {
01392         MLOCK(ErrorMessageLock);
01393         MesPrint("Gamma5 not allowed in n-trace");
01394         MUNLOCK(ErrorMessageLock);
01395         SETERROR(-1)
01396     }
01397     OldW = AT.WorkPointer;
01398     if ( AN.numtracesctack >= AN.intracesctack ) {
01399         number = AN.intracesctack + 2;
01400         t = (TRACES *)Malloc1(number*sizeof(TRACES),"TRACES-struct");
01401         if ( AN.tracesctack ) {
01402             for ( i = 0; i < AN.intracesctack; i++ ) { t[i] = AN.tracesctack[i]; }
01403             M_free(AN.tracesctack,"TRACES-struct");
01404         }
01405         AN.tracesctack = t;
01406         AN.intracesctack = number;
01407     }
01408     number = *params - 6;
01409     if ( number < 0 || ( number & 1 ) || !params[5] ) return(0);
01410
01411     t = AN.tracesctack + AN.numtracesctack;
01412     AN.numtracesctack++;
01413
01414     t->inlist = AT.WorkPointer;
01415     t->accup = t->accu = t->inlist + number;
01416     t->perm = t->accu + number;
01417     if ( ( AT.WorkPointer += 3 * number ) >= AT.WorkTop ) {
01418         AN.numtracesctack--;
01419         MLOCK(ErrorMessageLock);
01420         MesWork();
01421         MUNLOCK(ErrorMessageLock);
01422         return(-1);
01423     }
01424     t->num = num;
01425     t->level = level;
01426     p = t->inlist;
01427     m = params+6;
01428     for ( i = 0; i < number; i++ ) *p++ = *m++;
01429     t->termp = term;
01430     t->factor = params[4];
01431     t->allsign = params[5];
01432     ret = TraceNgen(BHEAD t,number);
01433     AT.WorkPointer = OldW;

```



```

01434     AN.numtracesctack--;
01435     return(ret);
01436 }
01437
01438 /*
01439     #] TraceN :
01440     #[ TraceNgen :      WORD TraceNgen(t,number)
01441
01442     This routine is a simplified version of Trace4Gen. We know here
01443     only three cases: Adjacent objects, same objects and all different.
01444     The other difference lies of course in the struct which is now
01445     not of type TRACES, but of type TRACES.
01446
01447 */
01448
01449 int TraceNgen(PHEAD TRACES *t, WORD number)
01450 {
01451     GETBIDENTITY
01452     WORD *termout, *stop;
01453     WORD *p, *m, oldval;
01454     WORD *pold, *mold, diff, *oldstring;
01455 /*
01456     #[ Special cases :
01457 */
01458     if ( number <= 2 ) { /* Special cases */
01459         termout = AT.WorkPointer;
01460         p = t->term;
01461         stop = p + *p;
01462         m = termout;
01463         p++;
01464         if ( p < stop ) do {
01465             if ( *p == SUBEXPRESSION && p[2] == t->num ) {
01466                 oldstring = p;
01467                 p = t->term;
01468                 do { *m++ = *p++; } while ( p < oldstring );
01469                 p += p[1];
01470                 *m++ = AC.lUniTrace[0];
01471                 *m++ = AC.lUniTrace[1];
01472                 *m++ = AC.lUniTrace[2];
01473                 *m++ = AC.lUniTrace[3];
01474                 if ( number == 2 || t->accup > t->accu ) {
01475                     oldstring = m;
01476                     *m++ = DELTA;
01477                     *m++ = 4;
01478                     if ( number == 2 ) {
01479                         *m++ = t->inlist[0];
01480                         *m++ = t->inlist[1];
01481                     }
01482                     if ( t->accup > t->accu ) {
01483                         pold = p;
01484                         p = t->accu;
01485                         while ( p < t->accup ) *m++ = *p++;
01486                         oldstring[1] = WORDDIFF(m,oldstring);
01487                         p = pold;
01488                     }
01489                 }
01490                 if ( t->factor ) {
01491                     *m++ = SNUMBER;
01492                     *m++ = 4;
01493                     *m++ = 2;
01494                     *m++ = t->factor;
01495                 }
01496                 do { *m++ = *p++; } while ( p < stop );
01497                 *termout = WORDDIFF(m,termout);
01498                 if ( t->allsign < 0 ) m[-1] = -m[-1];
01499                 if ( ( AT.WorkPointer = m ) > AT.WorkTop ) {
01500                     MLOCK(ErrorMessageLock);
01501                     MesWork();
01502                     MUNLOCK(ErrorMessageLock);
01503                     return(-1);
01504                 }
01505                 if ( *termout ) {
01506                     *AN.RepPoint = 1;
01507                     AR.expchanged = 1;
01508                     if ( Generator(BHEAD termout,t->level) ) goto TracCall;
01509                 }
01510                 AT.WorkPointer= termout;
01511                 return(0);
01512             }
01513             p += p[1];
01514         } while ( p < stop );
01515         return(0);
01516     }
01517 /*
01518     #[ Special cases :
01519     #[ Adjacent objects :
01520 */

```

```

01521     p = t->inlist;
01522     stop = p + number - 1;
01523     if ( *p == *stop ) {          /* First and last of string */
01524         oldval = *p;
01525         *(t->accup)++ = *p;
01526         *(t->accup)++ = *p;
01527         m = p+1;
01528         while ( m < stop ) *p++ = *m++;
01529         if ( TraceNgen(BHEAD t,number-2) ) goto TracCall;
01530         t = AN.tracestack + AN.numtracesctack - 1;
01531         while ( p > t->inlist ) *--m = *--p;
01532         *p = *stop = oldval;
01533         t->accup -= 2;
01534         return(0);
01535     }
01536     do {
01537         if ( *p == p[1] ) {        /* Adjacent in string */
01538             oldval = *p;
01539             pold = p;
01540             m = p+2;
01541             *(t->accup)++ = *p;
01542             *(t->accup)++ = *p;
01543             while ( m <= stop ) *p++ = *m++;
01544             if ( TraceNgen(BHEAD t,number-2) ) goto TracCall;
01545             t = AN.tracestack + AN.numtracesctack - 1;
01546             while ( p > pold ) *--m = *--p;
01547             *p++ = oldval;
01548             *p++ = oldval;
01549             t->accup -= 2;
01550             return(0);
01551         }
01552         p++;
01553     } while ( p < stop );
01554 /*
01555     #] Adjacent objects :
01556     #[ Same Objects :
01557 */
01558     p = t->inlist;
01559     stop = p + number - 1;
01560     diff = 2;
01561     do {
01562         p = t->inlist;
01563         while ( p <= stop ) {
01564             m = p + diff;
01565             if ( m > stop ) m -= number;
01566             if ( *p == *m ) {
01567                 WORD oldfactor, c;
01568                 oldstring = AT.WorkPointer;
01569                 AT.WorkPointer += number;
01570                 mold = m;
01571                 oldval = *p;
01572                 p = oldstring;
01573                 m = t->inlist;
01574                 while ( m <= stop ) *p++ = *m++;
01575 /*
01576                 Rotate the string
01577 */
01578                 {
01579                     m = mold + 1;
01580                     p = t->inlist;
01581                     while ( m <= stop ) *p++ = *m++;
01582                     m = oldstring;
01583                     while ( p <= stop ) *p++ = *m++;
01584                 }
01585                 oldfactor = t->allsign;
01586                 (t->factor)++;
01587                 p -= diff + 1;
01588                 m = stop;
01589                 if ( oldval >= ( AM.OffsetIndex + WILDOFFSET ) ||
01590                     ( oldval >= AM.OffsetIndex
01591                       && indices[oldval-AM.OffsetIndex].dimension ) ) {
01592                     m--;
01593 /*
01594                     We distinguish 4 cases:
01595                     m-p=1   Use g_(1,mu,a,mu) = (2-d_(mu,mu))*g_(1,a)
01596                     m-p=2   Use g_(1,mu,a,b,mu) = 4*d_(a,b)+(d_(mu,mu)-4)*g_(1,a,b)
01597                     m-p=3   Use g_(1,mu,a,b,c,mu) = -2*g_(1,c,b,a)
01598                             -(d_(mu,mu)-4)*g_(1,a,b,c)
01599                     m-p>3   Reduce down to m-p=3 with the old technique
01600 */
01601                     while ( m > (p+3) ) {
01602                         c = *p;
01603                         *p = *m;
01604                         *m = c;
01605                         if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01606                         t = AN.tracestack + AN.numtracesctack - 1;
01607                         m--;

```

```

01608         t->allsign = - t->allsign;
01609     }
01610     switch ( WORDDIF(m,p) ) {
01611     case 1:
01612         c = *p;
01613         *p = *m;
01614         *m = c;
01615         if ( oldval < ( AM.OffsetIndex + WILDOFFSET )
01616             && indices[oldval-AM.OffsetIndex].nmin4
01617             != -NMIN4SHIFT ) {
01618             t->allsign = - t->allsign;
01619             if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01620             t = AN.tracestack + AN.numtracesctack - 1;
01621             (t->factor)--;
01622             * (t->accup)++ = SUMMEDIND;
01623             * (t->accup)++ =
01624                 indices[oldval-AM.OffsetIndex].nmin4;
01625         }
01626     else
01627     {
01628         if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01629         t = AN.tracestack + AN.numtracesctack - 1;
01630         t->allsign = - t->allsign;
01631         (t->factor)--;
01632         * (t->accup)++ = oldval;
01633         * (t->accup)++ = oldval;
01634     }
01635     if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01636     t = AN.tracestack + AN.numtracesctack - 1;
01637     t->accup -= 2;
01638     break;
01639 case 2:
01640     { WORD one, two;
01641     one = *p = p[1];
01642     two = p[1] = *m;
01643     (t->factor)++;
01644     * (t->accup)++ = *p; /* 4 */
01645     * (t->accup)++ = *m; /* d_(a,b) */
01646     if ( TraceNgen(BHEAD t,number-4) ) goto TracnCall;
01647     t = AN.tracestack + AN.numtracesctack - 1;
01648     *p = one; p[1] = two;
01649     t->accup -= 2;
01650     if ( oldval < ( AM.OffsetIndex + WILDOFFSET )
01651         && indices[oldval-AM.OffsetIndex].nmin4
01652         != -NMIN4SHIFT ) {
01653         t->factor -= 2;
01654         * (t->accup)++ = SUMMEDIND;
01655         * (t->accup)++ =
01656             indices[oldval-AM.OffsetIndex].nmin4;
01657     }
01658     else {
01659         t->allsign = - t->allsign;
01660         if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01661         t = AN.tracestack + AN.numtracesctack - 1;
01662         t->allsign = - t->allsign;
01663         t->factor -= 2;
01664         * (t->accup)++ = oldval;
01665         * (t->accup)++ = oldval;
01666     }
01667     if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01668     t = AN.tracestack + AN.numtracesctack - 1;
01669     t->accup -= 2;
01670     }
01671     break;
01672 default:
01673     c = *p;
01674     *p = *m;
01675     *m = c;
01676     c = m[-1]; m[-1] = m[-2]; m[-2] = c;
01677     t->allsign = - t->allsign;
01678     if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01679     t = AN.tracestack + AN.numtracesctack - 1;
01680     m--;
01681     c = *p;
01682     *p = *m;
01683     *m = c;
01684     (t->factor)--;
01685     if ( oldval < ( AM.OffsetIndex + WILDOFFSET )
01686         && indices[oldval-AM.OffsetIndex].nmin4
01687         != -NMIN4SHIFT ) {
01688         * (t->accup)++ = SUMMEDIND;
01689         * (t->accup)++ =
01690             indices[oldval-AM.OffsetIndex].nmin4;
01691         if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01692         t = AN.tracestack + AN.numtracesctack - 1;
01693         t->accup -= 2;
01694         t->allsign = - t->allsign;

```

```

01695         }
01696         else
01697         {
01698             *(t->accup)++ = oldval;
01699             *(t->accup)++ = oldval;
01700             if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01701             t = AN.tracestack + AN.numtracesctack - 1;
01702             t->accup -= 2;
01703             t->allsign = - t->allsign;
01704             t->factor += 2;
01705             if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01706             t = AN.tracestack + AN.numtracesctack - 1;
01707             t->factor -= 2;
01708         }
01709         break;
01710     }
01711 }
01712 else {
01713     *(t->accup) = oldval;
01714     t->accup += 2;
01715     m--;
01716     while ( m > p ) {
01717         c = t->accup[-1];
01718         t->accup[-1] = *m;
01719         *m = c;
01720         if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01721         t = AN.tracestack + AN.numtracesctack - 1;
01722         m--;
01723         t->allsign = - t->allsign;
01724     }
01725     c = t->accup[-1];
01726     t->accup[-1] = *m;
01727     *m = c;
01728     (t->factor)--;
01729     if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
01730     t = AN.tracestack + AN.numtracesctack - 1;
01731     t->accup -= 2;
01732 }
01733 t->allsign = oldfactor;
01734 p = oldstring;
01735 m = t->inlist;
01736 while ( m <= stop ) *m++ = *p++;
01737 AT.WorkPointer = oldstring;
01738 return(0);
01739 }
01740 p++;
01741 }
01742 diff++;
01743 } while ( diff <= (number>>1) );
01744 /*
01745     #] Same Objects :
01746     #[ All Different :
01747
01748     Here we have a string with all different objects.
01749 */
01750 /*
01751     t->sgn = 0;
01752     termout = AT.WorkPointer;
01753     while ( ( diff = TraceNno(number,t->accup,t) ) != 0 ) {
01754         p = t->termp;
01755         stop = p + *p;
01756         m = termout;
01757         p++;
01758         if ( p < stop ) do {
01759             if ( *p == SUBEXPRESSION && p[2] == t->num ) {
01760                 oldstring = p;
01761                 p = t->termp;
01762                 do { *m++ = *p++; } while ( p < oldstring );
01763                 p += p[1];
01764                 pold = p;
01765                 *m++ = AC.lUniTrace[0];
01766                 *m++ = AC.lUniTrace[1];
01767                 *m++ = AC.lUniTrace[2];
01768                 *m++ = AC.lUniTrace[3];
01769                 *m++ = SNUMBER;
01770                 *m++ = 4;
01771                 *m++ = 2;
01772                 *m++ = t->factor;
01773                 p = t->accup;
01774                 oldval = number;
01775                 oldstring = m;
01776                 *m++ = DELTA;
01777                 *m++ = oldval + 2;
01778                 NCOPY(m,p,oldval);
01779                 if ( t->accup > t->accu ) {
01780                     p = t->accu;
01781                     while ( p < t->accup ) *m++ = *p++;

```

```

01782         oldstring[1] = WORDDIFF(m,oldstring);
01783     }
01784     p = pold;
01785     do { *m++ = *p++; } while ( p < stop );
01786     *termout = WORDDIFF(m,termout);
01787     if ( ( diff ^ t->allsign ) < 0 ) m[-1] = - m[-1];
01788     if ( ( AT.WorkPointer = m ) > AT.WorkTop ) {
01789         MLOCK(ErrorMessageLock);
01790         MesWork();
01791         MUNLOCK(ErrorMessageLock);
01792         return(-1);
01793     }
01794     if ( *termout ) {
01795         *AN.RepPoint = 1;
01796         AR.expchanged = 1;
01797         if ( Generator(BHEAD termout,t->level) ) {
01798             AT.WorkPointer = termout;
01799             goto TracCall;
01800         }
01801         t = AN.tracestack + AN.numtracesctack - 1;
01802     }
01803     break;
01804 }
01805 p += p[1];
01806 } while ( p < stop );
01807 }
01808 AT.WorkPointer = termout;
01809 return(0);
01810
01811 /*
01812     #] All Different :
01813 */
01814 TracnCall:
01815     AT.WorkPointer = oldstring;
01816 TracCall:
01817     if ( AM.tracebackflag ) {
01818         MLOCK(ErrorMessageLock);
01819         MesCall("TraceNgen");
01820         MUNLOCK(ErrorMessageLock);
01821     }
01822     return(-1);
01823 }
01824
01825 /*
01826     #] TraceNgen :
01827     #[ Traces :          WORD Traces(term,params,num,level)
01828
01829     The contents of the AT.TMout array are:
01830     length,type,subtype,gamma5,factor,sign,gamma's
01831 */
01832
01833 int Traces(PHEAD WORD *term, WORD *params, WORD num, WORD level)
01834 {
01835     GETBIDENTITY
01836     switch ( AT.TMout[2] ) { /* Subtype gives dimension */
01837     case 0:
01838         return(TraceN(BHEAD term,params,num,level));
01839     case 4:
01840         return(Trace4(BHEAD term,params,num,level));
01841     case 12:
01842         return(Trace4(BHEAD term,params,num,level));
01843     case 20:
01844         return(Trace4(BHEAD term,params,num,level));
01845     default:
01846         return(0);
01847     }
01848 }
01849
01850
01851 /*
01852     #] Traces :
01853     #[ TraceFind :      WORD TraceFind(term,params)
01854 */
01855
01856 int TraceFind(PHEAD WORD *term, WORD *params)
01857 {
01858     GETBIDENTITY
01859     WORD *p, *m, *to;
01860     WORD *termout, *stop, *stop2, number = 0;
01861     WORD first = 1;
01862     WORD type, spinline, sp;
01863     type = params[3];
01864     spinline = params[4];
01865     if ( spinline < 0 ) { /* $ variable. Evaluate */
01866         sp = DolToIndex(BHEAD -spinline);
01867         if ( AN.ErrorInDollar || sp < 0 ) {
01868             MLOCK(ErrorMessageLock);

```

```

01869         MesPrint("$%s does not have an index value in trace statement in module %1",
01870                 DOLLARNAME(Dollars,-spline),AC.CModule);
01871         MUNLOCK(ErrorMessageLock);
01872         return(0);
01873     }
01874     spline = sp;
01875 }
01876 to = AT.TMout;
01877 to++;
01878 *to++ = TAKETRACE;
01879 *to++ = type;
01880 *to++ = GAMMA1;
01881 *to++ = 0;           /* Powers of two */
01882 *to++ = 1;           /* sign */
01883 p = term;
01884 m = p + *p - 1;
01885 stop = m - ABS(*m);
01886 termout = m = AT.WorkPointer;
01887 m++;
01888 p++;
01889 while ( p < stop ) {
01890     stop2 = p + p[1];
01891     if ( *p == GAMMA && p[FUNHEAD] == spline ) {
01892         if ( first ) {
01893             *m++ = SUBEXPRESSION;
01894             *m++ = SUBEXPSIZE;
01895             *m++ = -1;
01896             *m++ = 1;
01897             *m++ = DUMMYBUFFER;
01898             FILLSUB(m)
01899             first = 0;
01900         }
01901         p += FUNHEAD+1;
01902         while ( p < stop2 ) {
01903             if ( *p == GAMMA5 ) {
01904                 if ( AT.TMout[3] == GAMMA5 ) AT.TMout[3] = GAMMA1;
01905                 else if ( AT.TMout[3] == GAMMA1 ) AT.TMout[3] = GAMMA5;
01906                 else if ( AT.TMout[3] == GAMMA7 ) AT.TMout[5] = -AT.TMout[5];
01907                 if ( number & 1 ) AT.TMout[5] = - AT.TMout[5];
01908                 p++;
01909             }
01910             else if ( *p == GAMMA6 ) {
01911                 if ( number & 1 ) goto F7;
01912                 if ( AT.TMout[3] == GAMMA6 ) (AT.TMout[4])++;
01913                 else if ( AT.TMout[3] == GAMMA1 ) AT.TMout[3] = GAMMA6;
01914                 else if ( AT.TMout[3] == GAMMA5 ) AT.TMout[3] = GAMMA6;
01915                 else if ( AT.TMout[3] == GAMMA7 ) AT.TMout[5] = 0;
01916                 p++;
01917             }
01918             else if ( *p == GAMMA7 ) {
01919                 if ( number & 1 ) goto F6;
01920                 if ( AT.TMout[3] == GAMMA7 ) (AT.TMout[4])++;
01921                 else if ( AT.TMout[3] == GAMMA1 ) AT.TMout[3] = GAMMA7;
01922                 else if ( AT.TMout[3] == GAMMA5 ) {
01923                     AT.TMout[3] = GAMMA7;
01924                     AT.TMout[5] = -AT.TMout[5];
01925                 }
01926                 else if ( AT.TMout[3] == GAMMA6 ) AT.TMout[5] = 0;
01927                 p++;
01928             }
01929             else {
01930                 *to++ = *p++;
01931                 number++;
01932             }
01933         }
01934     }
01935     else {
01936         while ( p < stop2 ) *m++ = *p++;
01937     }
01938 }
01939 if ( first ) return(0);
01940 AT.TMout[0] = WORDDIF(to,AT.TMout);
01941 to = term;
01942 to += *to;
01943 while ( p < to ) *m++ = *p++;
01944 *termout = WORDDIF(m,termout);
01945 to = term;
01946 p = termout;
01947 do { *to++ = *p++; } while ( p < m );
01948 AT.WorkPointer = term + *term;
01949 return(1);
01950 }
01951
01952 /*
01953     #] TraceFind :
01954     #[ Chisholm :           WORD Chisholm(term,level,num)
01955

```

```

01956      Routines for reorganizing traces.
01957      The command
01958          Chisholm,1;
01959      will collect the gamma matrices in spinline 1 and see whether
01960      they have an index in common with another gamma matrix. If this
01961      is the case the identity
01962           $g_{(2,\mu)} \text{Tr}[g_{(1,\mu)} S(2)] = S(2) + SR(2)$ 
01963      is applied (SR is the reversed string).
01964  */
01965
01966  int Chisholm(PHEAD WORD *term, WORD level)
01967  {
01968      GETBIDENTITY
01969      WORD *t, *r, *m, *s, *tt, *rr;
01970      WORD *mat, *matpoint, *termout, *rdo;
01971      CBUF *C = cbuf+AM.rbufnum;
01972      WORD i, j, num = C->lhs[level][2], gam5;
01973      WORD norm = 0, k, *matp;
01974  /*
01975      #[ Find : Find possible matrices
01976  */
01977      mat = matpoint = AT.WorkPointer;
01978      t = term;
01979      r = t + *t - 1; r -= ABS(*r);
01980      t++;
01981      i = 0;
01982      gam5 = GAMMA1;
01983      while ( t < r ) {
01984          if ( *t == GAMMA && t[FUNHEAD] == num ) {
01985              m = t + t[1];
01986              t += FUNHEAD+1;
01987              while ( t < m ) {
01988                  if ( *t >= 0 || *t < MINSPEC ) i++;
01989                  else {
01990                      if ( gam5 == GAMMA1 ) gam5 = *t;
01991                      else if ( gam5 == GAMMA5 ) {
01992                          if ( *t == GAMMA5 ) gam5 = GAMMA1;
01993                          else if ( *t != GAMMA1 ) gam5 = *t;
01994                      }
01995                  }
01996                  *matpoint++ = *t++;
01997              }
01998          }
01999          else t += t[1];
02000      }
02001      if ( ( i & 1 ) != 0 ) return(0); /* odd trace */
02002  /*
02003      #] Find :
02004      #[ Test : Test for contracted index
02005
02006      This code should be modified.
02007
02008      We have to check for all possible matches if C->lhs[level][3] == 1
02009      and the trace contains a gamma5, gamma6 or gamma7.
02010      Then we normalize by the number of possible contractions (norm) and
02011      do all of them. This way the Levi-Civita tensors have a maximum
02012      chance of cancelling each other. This option is activated with
02013      'contract' and 'symmetrize'. Defaults are that they are on, but
02014      they can be switched off with nocontract and nosymmetrize.
02015  */
02016      s = mat;
02017      while ( s < matpoint ) {
02018  /*
02019          if ( *s < AM.OffsetIndex || ( *s < ( AM.OffsetIndex + WILDOFFSET ) &&
02020              indices[*s-AM.OffsetIndex].dimension == 0 ) ) {
02021  /*
02022              if ( *s < AM.OffsetIndex || ( *s < ( AM.OffsetIndex + WILDOFFSET ) &&
02023                  indices[*s-AM.OffsetIndex].dimension != 4 )
02024              || ( ( AC.lDefDim != 4 ) && ( *s >= ( AM.OffsetIndex + WILDOFFSET ) ) ) ) {
02025                  s++; continue;
02026              }
02027              t = term+1;
02028              while ( t < r ) {
02029                  if ( *t == GAMMA && t[FUNHEAD] != num ) {
02030                      m = t + t[1];
02031                      t += FUNHEAD+1;
02032                      while ( t < m ) {
02033                          if ( *t == *s ) {
02034                              norm++;
02035                          }
02036                          t++;
02037                      }
02038                  }
02039                  else t += t[1];
02040              }
02041              s++;
02042          }

```

```

02043     if ( norm == 0 ) return(Generator(BHEAD term,level)); /* No Action */
02044 /*
02045 #] Test :
02046 #[ Do : Process the string
02047
02048 tt: The subterm
02049 t: The matrix
02050 s: The matrix in the relevant string
02051
02052 Cycle the string in mat so that s is at the end.
02053 Copy the part till the critical GAMMA.
02054 Copy inside the critical string, copy S, copy tail inside string.
02055 Important to remember where S is so that we can reverse it later.
02056 Add term UnitTrace/2/norm.
02057 Copy rest of term.
02058 Continue execution with S.
02059 Reverse S.
02060 Continue execution with SR.
02061 */
02062
02063 if ( C->lhs[level][3] == 0 /* || gam5 == GAMMA1 */ ) norm = 1;
02064
02065 matp = matpoint;
02066 for ( k = 0; k < norm; k++ ) {
02067     matpoint = matp;
02068     s = mat;
02069     while ( s < matpoint ) {
02070 /*
02071         if ( *s < AM.OffsetIndex || ( *s < ( AM.OffsetIndex + WILDOFFSET ) &&
02072         indices[*s-AM.OffsetIndex].dimension == 0 ) ) {
02073 */
02074         if ( *s < AM.OffsetIndex || ( *s < ( AM.OffsetIndex + WILDOFFSET ) &&
02075         indices[*s-AM.OffsetIndex].dimension != 4 ) ) {
02076             s++; continue;
02077         }
02078         t = term+1;
02079         while ( t < r ) {
02080             if ( *t == GAMMA && t[FUNHEAD] != num ) {
02081                 tt = t;
02082                 m = t + t[1];
02083                 t += FUNHEAD+1;
02084                 while ( t < m ) {
02085                     if ( *t == *s ) {
02086                         i = WORDDIF(t,tt);
02087                         m = mat;
02088                         while ( m <= s ) *matpoint++ = *m++;
02089                         t = mat;
02090                         while ( m < matpoint ) *t++ = *m++;
02091                         termout = t;
02092                         m = termout + 1;
02093                         t = term + 1;
02094                         while ( t < tt ) {
02095                             if ( *t != GAMMA || t[FUNHEAD] != num ) {
02096                                 j = t[1];
02097                                 NCOPY(m,t,j);
02098                             }
02099                             else t += t[1];
02100                         }
02101                     }
02102                     tt += tt[1];
02103                     rdo = m;
02104                     j = i;
02105                     while ( --j >= 0 ) *m++ = *t++;
02106                     matpoint = m;
02107                     s = mat;
02108                     while ( s < termout ) *m++ = *s++;
02109                     m--;
02110                     t++;
02111                     while ( t < tt ) *m++ = *t++;
02112                     rdo[1] = WORDDIF(m,rdo);
02113
02114                     *m++ = AC.lUniTrace[0];
02115                     *m++ = AC.lUniTrace[1];
02116                     *m++ = AC.lUniTrace[2];
02117                     *m++ = AC.lUniTrace[3];
02118                     *m++ = SNUMBER;
02119                     *m++ = 4;
02120                     *m++ = 2*norm;
02121                     *m++ = -1;
02122
02123                     while ( t < r ) {
02124                         if ( *t != GAMMA || t[FUNHEAD] != num ) {
02125                             j = t[1];
02126                             NCOPY(m,t,j);
02127                         }
02128                         else t += t[1];
02129                     }

```



```

02130             rr = term + *term;
02131             while ( t < rr ) *m++ = *t++;
02132
02133             *termout = WORDDIFF(m,termout);
02134             rr = m;
02135             t = termout;
02136             j = *termout;
02137             NCOPY(m,t,j);
02138             AT.WorkPointer = m;
02139             if ( Generator(BHEAD t,level) ) goto ChisCall;
02140
02141             j = WORDDIFF(termout,mat)-1;
02142             t = matpoint;
02143             m = t + j;
02144             AT.WorkPointer = rr;
02145             while ( m > t ) {
02146                 i = *--m; *m = *t; *t++ = i;
02147             }
02148
02149             if ( Generator(BHEAD termout,level) ) goto ChisCall;
02150             AT.WorkPointer = mat;
02151
02152             goto NextK;
02153         }
02154         t++;
02155     }
02156 }
02157 else t += t[1];
02158 }
02159 s++;
02160 }
02161 NextK;
02162 }
02163 return(0);
02164 /*
02165 #] Do :
02166 */
02167 ChisCall:
02168     if ( AM.tracebackflag ) {
02169         MLOCK(ErrorMessageLock);
02170         MesCall("Chisholm");
02171         MUNLOCK(ErrorMessageLock);
02172     }
02173     return(-1);
02174 }
02175
02176 /*
02177 #] Chisholm :
02178 #[ TenVecFind :      WORD TenVecFind(term,params)
02179 */
02180
02181 int TenVecFind(PHEAD WORD *term, WORD *params)
02182 {
02183     GETBIDENTITY
02184     WORD *t, *w, *m, *tstop;
02185     WORD i, mode, thevector, thetensor, spectator;
02186     thetensor = params[3];
02187     thevector = params[4];
02188     mode = params[5];
02189     if ( thetensor < 0 ) { /* $-expression */
02190         thetensor = DolToTensor(BHEAD -thetensor);
02191         if ( thetensor < FUNCTION ) {
02192             if ( thevector > 0 ) {
02193                 thetensor = DolToTensor(BHEAD thevector);
02194                 if ( thetensor < FUNCTION ) {
02195                     MLOCK(ErrorMessageLock);
02196                     MesPrint("%s should have been a tensor in module %1"
02197                             ,DOLLARNAME(Dollars,params[4]),AC.CModule);
02198                     MUNLOCK(ErrorMessageLock);
02199                     return(-1);
02200                 }
02201                 thevector = DolToVector(BHEAD -params[3]);
02202                 if ( thevector >= 0 ) {
02203                     MLOCK(ErrorMessageLock);
02204                     MesPrint("%s should have been a vector in module %1"
02205                             ,DOLLARNAME(Dollars,-params[3]),AC.CModule);
02206                     MUNLOCK(ErrorMessageLock);
02207                     return(-1);
02208                 }
02209             }
02210             else {
02211                 MLOCK(ErrorMessageLock);
02212                 MesPrint("%s should have been a tensor in module %1"
02213                         ,DOLLARNAME(Dollars,-params[3]),AC.CModule);
02214                 MUNLOCK(ErrorMessageLock);
02215                 return(-1);
02216             }
02217         }
02218     }

```

```

02217     }
02218 }
02219 if ( thevector > 0 ) { /* $-expression */
02220     thevector = DolToVector(BHEAD thevector);
02221     if ( thevector >= 0 ) {
02222         MLOCK(ErrorMessageLock);
02223         MesPrint("$%s should have been a vector in module %1"
02224             , DOLLARNAME(Dollars,params[4]),AC.CModule);
02225         MUNLOCK(ErrorMessageLock);
02226         return(-1);
02227     }
02228 }
02229 if ( ( mode & 1 ) != 0 ) { /* Vector to tensor */
02230     GETSTOP(term,tstop);
02231     t = term + 1;
02232     while ( t < tstop ) {
02233         if ( *t == DOTPRODUCT ) {
02234             i = t[1] - 2; t += 2;
02235             while ( i > 0 ) {
02236                 spectator = 0;
02237                 if ( t[2] < 0 ) {}
02238                 else if ( *t == thevector && t[1] == thevector ) {
02239                     if ( ( mode & 2 ) == 0 ) spectator = thevector;
02240                 }
02241                 else if ( *t == thevector ) spectator = t[1];
02242                 else if ( t[1] == thevector ) spectator = *t;
02243                 if ( spectator ) {
02244                     if ( ( mode & 8 ) == 0 ) goto match;
02245                     w = SetElements + Sets[params[6]].first;
02246                     m = SetElements + Sets[params[6]].last;
02247                     while ( w < m ) {
02248                         if ( *w == spectator ) break;
02249                         w++;
02250                     }
02251                     if ( w >= m ) goto match;
02252                 }
02253                 t += 3;
02254                 i -= 3;
02255             }
02256         }
02257         else if ( *t == VECTOR ) {
02258             i = t[1] - 2; t += 2;
02259             while ( i > 0 ) {
02260                 if ( *t == thevector ) goto match;
02261                 t += 2;
02262                 i -= 2;
02263             }
02264         }
02265         else if ( *t == thetensor ) t += t[1];
02266         else if ( *t >= FUNCTION ) {
02267             if ( functions[*t-FUNCTION].spec > 0 ) {
02268                 w = t + t[1];
02269                 t += FUNHEAD;
02270                 while ( t < w ) {
02271                     if ( *t == thevector ) goto match;
02272                     t++;
02273                 }
02274             }
02275             else if ( ( mode & 4 ) != 0 ) {
02276                 w = t + t[1];
02277                 t += FUNHEAD;
02278                 while ( t < w ) {
02279                     if ( *t == -VECTOR && t[1] == thevector ) goto match;
02280                     else if ( *t > 0 ) t += *t;
02281                     else if ( *t <= -FUNCTION ) t++;
02282                     else t += 2;
02283                 }
02284             }
02285             else t += t[1];
02286         }
02287         else t += t[1];
02288     }
02289 }
02290 else { /* Tensor to Vector */
02291     GETSTOP(term,tstop);
02292     t = term+1;
02293     while ( t < tstop ) {
02294         if ( *t == thetensor ) goto match;
02295         t += t[1];
02296     }
02297 }
02298 return(0);
02299 match:
02300     AT.TMout[0] = 5;
02301     AT.TMout[1] = TENVEC;
02302     AT.TMout[2] = thetensor;
02303     AT.TMout[3] = thevector;

```

```

02304     AT.TMout[4] = mode;
02305     if ( ( mode & 8 ) != 0 ) { AT.TMout[0] = 6; AT.TMout[5] = params[6]; }
02306     return(1);
02307 }
02308 }
02309
02310 /*
02311     #] TenVecFind :
02312     #[ TenVec :           WORD TenVec(term,params,num,level)
02313 */
02314
02315 int TenVec(PHEAD WORD *term, WORD *params, WORD num, WORD level)
02316 {
02317     GETBIDENTITY
02318     WORD *t, *m, *w, *termout, *tstop, *outlist, *ou, *ww, *mm;
02319     WORD i, j, k, x, mode, thevector, thetensor, DumNow, spectator;
02320     DUMMYUSE(num);
02321     thetensor = params[2];
02322     thevector = params[3];
02323     mode = params[4];
02324     termout = AT.WorkPointer;
02325     DumNow = AR.CurDum = DetCurDum(BHEAD term);
02326     if ( ( mode & 1 ) != 0 ) { /* Vector to tensor */
02327         AT.WorkPointer += *term;
02328         ou = outlist = AT.WorkPointer;
02329         GETSTOP(term,tstop);
02330         t = term + 1;
02331         m = termout + 1;
02332         while ( t < tstop ) {
02333             if ( *t == DOTPRODUCT ) {
02334                 i = t[1] - 2;
02335                 w = m;
02336                 *m++ = *t++; *m++ = *t++;
02337                 while ( i > 0 ) {
02338                     spectator = 0;
02339                     if ( t[2] < 0 ) {
02340                         *m++ = *t++; *m++ = *t++; *m++ = *t++;
02341                     }
02342                     else if ( *t == thevector && t[1] == thevector ) {
02343                         if ( ( mode & 2 ) == 0 ) spectator = thevector;
02344                         else {
02345                             *m++ = *t++; *m++ = *t++; *m++ = *t++;
02346                         }
02347                     }
02348                     else if ( *t == thevector ) spectator = t[1];
02349                     else if ( t[1] == thevector ) spectator = *t;
02350                     else {
02351                         *m++ = *t++; *m++ = *t++; *m++ = *t++;
02352                     }
02353                     if ( spectator ) {
02354                         if ( ( mode & 8 ) == 0 ) goto noveto;
02355                         ww = SetElements + Sets[params[5]].first;
02356                         mm = SetElements + Sets[params[5]].last;
02357                         while ( ww < mm ) {
02358                             if ( *ww == spectator ) break;
02359                             ww++;
02360                         }
02361                         if ( ww < mm ) {
02362                             *m++ = *t++; *m++ = *t++; *m++ = *t++;
02363                         }
02364                         else {
02365                             if ( spectator == thevector ) {
02366                                 for ( j = 0; j < t[2]; j++ ) {
02367                                     *ou++ = ++AR.CurDum;
02368                                     *ou++ = AR.CurDum;
02369                                 }
02370                                 t += 3;
02371                             }
02372                             else {
02373                                 for ( j = 0; j < t[2]; j++ ) *ou++ = spectator;
02374                                 t += 3;
02375                             }
02376                         }
02377                         i -= 3;
02378                     }
02379                     w[1] = WORDDIF(m,w);
02380                     if ( w[1] == 2 ) m = w;
02381                 }
02382             else if ( *t == VECTOR ) {
02383                 i = t[1] - 2; w = m;
02384                 *m++ = *t++; *m++ = *t++;
02385                 while ( i > 0 ) {
02386                     if ( *t == thevector ) {
02387                         *ou++ = t[1];
02388                         t += 2;
02389                     }
02390                     else { *m++ = *t++; *m++ = *t++; }

```

```

02391         i -= 2;
02392     }
02393     w[1] = WORDDIF(m,w);
02394     if ( w[1] == 2 ) m = w;
02395 }
02396 else if ( *t == thetensor ) {
02397     i = t[1] - FUNHEAD;
02398     t += FUNHEAD;
02399     NCOPY(ou,t,i);
02400 }
02401 else if ( *t >= FUNCTION ) {
02402     if ( functions[*t-FUNCTION].spec > 0 ) {
02403         w = t + t[1];
02404         i = FUNHEAD;
02405         NCOPY(m,t,i);
02406         while ( t < w ) {
02407             if ( *t == thevector ) {
02408                 *m++ = ++AR.CurDum;
02409                 *ou++ = AR.CurDum;
02410                 t++;
02411             }
02412             else *m++ = *t++;
02413         }
02414     }
02415     else if ( ( mode & 4 ) != 0 ) {
02416         w = t + t[1];
02417         i = FUNHEAD;
02418         NCOPY(m,t,i);
02419         while ( t < w ) {
02420             if ( *t == -VECTOR && t[1] == thevector ) {
02421                 *m++ = -INDEX;
02422                 *m++ = ++AR.CurDum;
02423                 *ou++ = AR.CurDum;
02424                 t += 2;
02425             }
02426             else if ( *t > 0 ) {
02427                 i = *t;
02428                 NCOPY(m,t,i);
02429             }
02430             else if ( *t <= -FUNCTION ) *m++ = *t++;
02431             else { *m++ = *t++; *m++ = *t++; }
02432         }
02433     }
02434     else goto docopy;
02435 }
02436 else {
02437 docopy:
02438     i = t[1];
02439     NCOPY(m,t,i);
02440 }
02441 }
02442 i = WORDDIF(ou,outlist);
02443 if ( i > 0 ) {
02444     for ( j = 1; j < i; j++ ) {
02445         if ( outlist[j-1] > outlist[j] ) {
02446             x = outlist[j-1]; outlist[j-1] = outlist[j]; outlist[j] = x;
02447             for ( k = j-1; k > 0; k-- ) {
02448                 if ( outlist[k-1] <= outlist[k] ) break;
02449                 x = outlist[k-1]; outlist[k-1] = outlist[k]; outlist[k] = x;
02450             }
02451         }
02452     }
02453 }
02454 *m++ = thetensor;
02455 *m++ = FUNHEAD + i;
02456 *m++ = DIRTYSYMFLAG;
02457 FILLFUN3(m)
02458 ou = outlist;
02459 NCOPY(m,ou,i);
02460 }
02461 w = term + *term;
02462 while ( t < w ) *m++ = *t++;
02463 }
02464 else { /* Tensor to Vector */
02465     GETSTOP(term,tstop);
02466     t = term+1;
02467     m = termout+1;
02468     while ( t < tstop ) {
02469         if ( *t != thetensor ) {
02470             i = t[1];
02471             NCOPY(m,t,i);
02472         }
02473         else {
02474             i = t[1] - FUNHEAD;
02475             t += FUNHEAD;
02476             if ( i > 0 ) {
02477                 w = m; m += 2;

```

```

02478             while ( --i >= 0 ) {
02479                 *m++ = thevector;
02480                 *m++ = *t++;
02481             }
02482             *w = DELTA;
02483             w[1] = WORDDIF(m,w);
02484         }
02485     }
02486 }
02487 w = term + *term;
02488 while ( t < w ) *m++ = *t++;
02489 }
02490 *termout = WORDDIF(m,termout);
02491 AT.WorkPointer = m;
02492 *AT.TMout = 0;
02493 if ( Generator(BHEAD termout,level) ) goto fromTenVec;
02494 AR.CurDum = DumNow;
02495 AT.WorkPointer = termout;
02496 return(0);
02497 fromTenVec:
02498 if ( AM.tracebackflag ) {
02499     MLOCK(ErrorMessageLock);
02500     MesCall("TenVec");
02501     MUNLOCK(ErrorMessageLock);
02502 }
02503 return(-1);
02504 }
02505
02506 /*
02507     #] TenVec :
02508     #] Operations :
02509 */

```

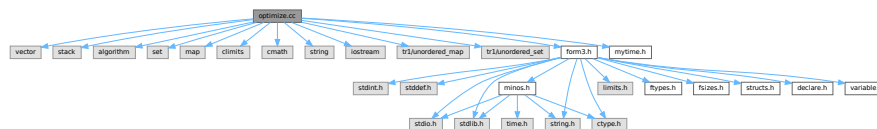
## 4.92 optimize.cc File Reference

```

#include <vector>
#include <stack>
#include <algorithm>
#include <set>
#include <map>
#include <climits>
#include <cmath>
#include <string>
#include <iostream>
#include <tr1/unordered_map>
#include <tr1/unordered_set>
#include "form3.h"
#include "mytime.h"

```

Include dependency graph for optimize.cc:



### Data Structures

- class [tree\\_node](#)
- struct [CSEHash](#)
- struct [CSEEq](#)
- struct [node](#)
- struct [NodeHash](#)
- struct [NodeEq](#)
- class [optimization](#)

## Typedefs

- typedef struct [node](#) **NODE**

## Functions

- void [print\\_instr](#) (const vector< WORD > &instr, WORD num)
- template<class RandomAccessIterator >  
void [my\\_random\\_shuffle](#) (PHEAD RandomAccessIterator fr, RandomAccessIterator to)
- LONG [get\\_expression](#) (int exprnr)
- vector< vector< WORD > > [get\\_brackets](#) ()
- int [count\\_operators](#) (const WORD \*expr, bool print=false)
- int [count\\_operators](#) (const vector< WORD > &instr, bool print=false)
- vector< WORD > [occurrence\\_order](#) (const WORD \*expr, bool rev)
- WORD [getpower](#) (const WORD \*term, int var, int pos)
- void [fixarg](#) (UWORD \*t, WORD &n)
- void [GcdLong\\_fix\\_args](#) (PHEAD UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc)
- void [DivLong\\_fix\\_args](#) (UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc, UWORD \*d, WORD \*nd)
- void [build\\_Horner\\_tree](#) (const WORD \*\*terms, int numterms, int var, int maxvar, int pos, vector< WORD > \*res)
- bool [term\\_compare](#) (const WORD \*a, const WORD \*b)
- vector< WORD > [Horner\\_tree](#) (const WORD \*expr, const vector< WORD > &order)
- void [print\\_tree](#) (const vector< WORD > &tree)
- template<typename T >  
size\_t [hash\\_range](#) (T \*array, int size)
- vector< WORD > [generate\\_instructions](#) (const vector< WORD > &tree, bool do\_CSE)
- int [count\\_operators\\_cse](#) (const vector< WORD > &tree)
- **NODE** \* [buildTree](#) (vector< WORD > &tree)
- int [count\\_operators\\_cse\\_topdown](#) (vector< WORD > &tree)
- vector< WORD > [simulated\\_annealing](#) ()
- void [find\\_Horner\\_MCTS\\_expand\\_tree](#) ()
- void [find\\_Horner\\_MCTS](#) ()
- vector< WORD > [merge\\_operators](#) (const vector< WORD > &all\_instr, bool move\_coeff)
- vector< [optimization](#) > [find\\_optimizations](#) (const vector< WORD > &instr)
- bool [do\\_optimization](#) (const [optimization](#) optim, vector< WORD > &instr, int newid)
- int [partial\\_factorize](#) (vector< WORD > &instr, int n, int improve)
- vector< WORD > [optimize\\_greedy](#) (vector< WORD > instr, LONG time\_limit)
- vector< WORD > [recycle\\_variables](#) (const vector< WORD > &all\_instr)
- void [optimize\\_expression\\_given\\_Horner](#) ()
- void [generate\\_output](#) (const vector< WORD > &instr, int exprnr, int extraoffset, const vector< vector< WORD > > &brackets)
- WORD [generate\\_expression](#) (WORD exprnr)
- void [optimize\\_print\\_code](#) (int print\_expr)
- int [Optimize](#) (WORD exprnr, int do\_print)
- int [ClearOptimize](#) ()

## Variables

- const WORD [OPER\\_ADD](#) = -1
- const WORD [OPER\\_MUL](#) = -2
- const WORD [OPER\\_COMMA](#) = -3
- WORD \* [optimize\\_expr](#)
- vector< vector< WORD > > [optimize\\_best\\_Horner\\_schemes](#)
- int [optimize\\_num\\_vars](#)
- int [optimize\\_best\\_num\\_oper](#)
- vector< WORD > [optimize\\_best\\_instr](#)
- vector< WORD > [optimize\\_best\\_vars](#)
- bool [mcts\\_factorized](#)
- bool [mcts\\_separated](#)
- vector< WORD > [mcts\\_vars](#)
- [tree\\_node](#) [mcts\\_root](#)
- int [mcts\\_expr\\_score](#)
- set< pair< int, vector< WORD > > > [mcts\\_best\\_schemes](#)

### 4.92.1 Detailed Description

experimental routines for the optimization of FORTRAN or C output.

Definition in file [optimize.cc](#).

### 4.92.2 Function Documentation

#### 4.92.2.1 [print\\_instr\(\)](#)

```
void print_instr (
    const vector< WORD > & instr,
    WORD num )
```

Definition at line [180](#) of file [optimize.cc](#).

#### 4.92.2.2 [my\\_random\\_shuffle\(\)](#)

```
template<class RandomAccessIterator >
void my_random_shuffle (
    PHEAD RandomAccessIterator fr,
    RandomAccessIterator to )
```

Random shuffle

### 4.92.3 Description

Randomly permutes elements in the range [fr,to). Functionality is the same as C++'s "random\_shuffle", but it uses Form's "wranf".

Definition at line [202](#) of file [optimize.cc](#).

Referenced by [find\\_Horner\\_MCTS\(\)](#).

#### 4.92.3.1 `get_expression()`

```
LONG get_expression (
    int exprnr )
```

Get expression

#### 4.92.4 Description

Reads an expression from the input file into a buffer (called `optimize_expr`). This buffer is used during the optimization process. Non-symbols are removed by `ConvertToPoly` and are put in temporary symbols.

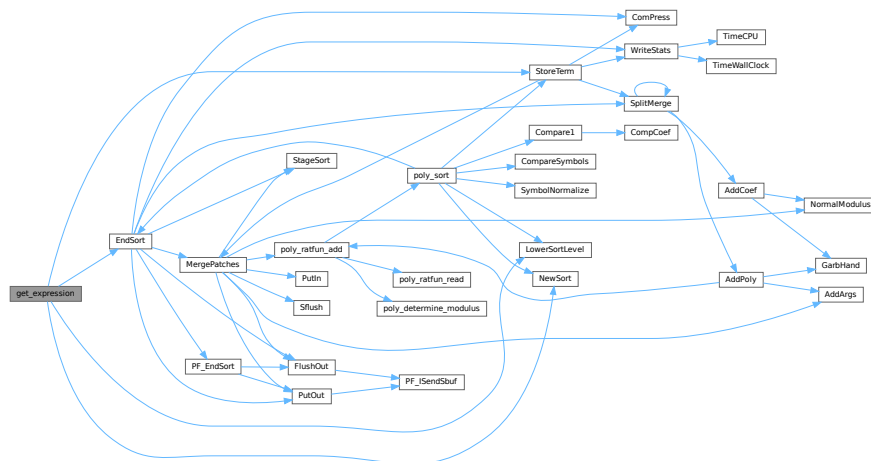
The return value is the length of the expression in WORDs, or a negative number if it failed.

Definition at line 226 of file [optimize.cc](#).

References [EndSort\(\)](#), [LowerSortLevel\(\)](#), [NewSort\(\)](#), and [StoreTerm\(\)](#).

Referenced by [Optimize\(\)](#).

Here is the call graph for this function:



#### 4.92.4.1 `get_brackets()`

```
vector< vector< WORD > > get_brackets ( )
```

Get brackets

#### 4.92.5 Description

Checks whether the input expression (stored in `optimize_expr`) contains brackets. If so, this method replaces terms outside brackets by powers of `SEPERATESYMBOL` (equal brackets have equal powers) and the brackets are returned. If not, the result is empty.

Brackets are used for simultaneous optimization. The symbol `SEPERATESYMBOL` is always the first one used in a Horner scheme.

Definition at line 303 of file [optimize.cc](#).

Referenced by [Optimize\(\)](#).



#### 4.92.5.1 `count_operators()` [1/2]

```
int count_operators (
    const WORD * expr,
    bool print = false )
```

Count operators

### 4.92.6 Description

Counts the number of operators in a Form-style expression.

Definition at line 423 of file [optimize.cc](#).

Referenced by [find\\_Horner\\_MCTS\(\)](#), [generate\\_instructions\(\)](#), [Optimize\(\)](#), [optimize\\_expression\\_given\\_Horner\(\)](#), and [optimize\\_greedy\(\)](#).

#### 4.92.6.1 `count_operators()` [2/2]

```
int count_operators (
    const vector< WORD > & instr,
    bool print = false )
```

Count operators

### 4.92.7 Description

Counts the number of operators in a vector of instructions

Definition at line 477 of file [optimize.cc](#).

#### 4.92.7.1 `occurrence_order()`

```
vector< WORD > occurrence_order (
    const WORD * expr,
    bool rev )
```

Occurrence order

### 4.92.8 Description

Extracts all variables from an expression and sorts them with most occurring first (or last, with `rev=true`)

Definition at line 520 of file [optimize.cc](#).

Referenced by [Optimize\(\)](#).

#### 4.92.8.1 `getpower()`

```
WORD getpower (
    const WORD * term,
    int var,
    int pos )
```

Horner tree building

#### 4.92.9 Description

Given a Form-style expression (in a buffer in memory), this builds an expression tree. The tree is determined by a multivariate Horner scheme, i.e., something of the form:

$$1+y+x*(2+y*(1+y)+x*(3-y*(...)))$$

The order of the variables is given to the method "Horner\_tree", which rennumbers ad reorders the terms of the expression. Next, the recursive method "build\_Horner\_tree" does the actual tree construction.

The tree is represented in postfix notation. Tokens are of the following forms:

- SNUMBER tokenlength num den coefflength
- SYMBOL tokenlength variable power
- OPER\_ADD or OPER\_MUL

#### 4.92.10 Note

Sets AN.poly\_num\_vars and allocates AN.poly\_vars. The latter should be freed later. Get power of variable (helper function for build\_Horner\_tree)

#### 4.92.11 Description

Returns the power of the variable "var", which is at position "pos" in this term, if it is present.

Definition at line 601 of file [optimize.cc](#).

Referenced by [build\\_Horner\\_tree\(\)](#).

#### 4.92.11.1 `fixarg()`

```
void fixarg (
    UWORD * t,
    WORD & n )
```

Call GcdLong/DivLong with leading zeroes

## 4.92.12 Description

These method remove leading zeroes, so that GcdLong and DivLong can safely be called.

Definition at line 615 of file [optimize.cc](#).

### 4.92.12.1 GcdLong\_fix\_args()

```
void GcdLong_fix_args (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc )
```

Definition at line 621 of file [optimize.cc](#).

### 4.92.12.2 DivLong\_fix\_args()

```
void DivLong_fix_args (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc,
    UWORD * d,
    WORD * nd )
```

Definition at line 627 of file [optimize.cc](#).

### 4.92.12.3 build\_Horner\_tree()

```
void build_Horner_tree (
    const WORD ** terms,
    int numterms,
    int var,
    int maxvar,
    int pos,
    vector< WORD > * res )
```

Build the Horner tree

### 4.92.13 Description

Constructs the Horner tree. The method processes one variable and continues recursively until the Horner scheme is completed.

"terms" is a pointer to the starts of the terms. "numterms" is the number of terms to be processed. "var" is the next variable to be processed (index between 0 and #maxvar) and "maxvar" is the last variable to be processed, so that partial Horner trees can also be constructed. "pos" is the position that the power of "var" should be in (one level further in the recursion, "pos" has increased by 0 or 1 depending on whether the previous power was 0 or not). The result is written at the pointer "res".

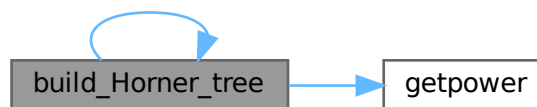
This method also factors out gcds of the coefficients. The result should end with "gcd OPER\_MUL" at all times, so that one level higher gcds can be extracted again.

Definition at line 653 of file [optimize.cc](#).

References [build\\_Horner\\_tree\(\)](#), and [getpower\(\)](#).

Referenced by [build\\_Horner\\_tree\(\)](#), and [Horner\\_tree\(\)](#).

Here is the call graph for this function:



#### 4.92.13.1 term\_compare()

```

bool term_compare (
    const WORD * a,
    const WORD * b )
  
```

Term compare (helper function for Horner\_tree)

### 4.92.14 Description

Compares two terms of the form "L SYM 4 x n coeff" or "L coeff". Lower powers of lower-indexed symbols come first. This is used to sort the terms in correct order.

Definition at line 853 of file [optimize.cc](#).

Referenced by [Horner\\_tree\(\)](#).

#### 4.92.14.1 Horner\_tree()

```
vector< WORD > Horner_tree (
    const WORD * expr,
    const vector< WORD > & order )
```

Prepare Horner tree building

#### 4.92.15 Description

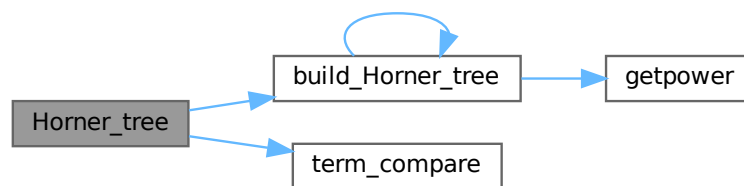
This method rennumbers the variables to 0...#vars-1 according to the specified order. Next, it stored pointer to individual terms and sorts the terms with higher power first. Then the sorted lists of power is used for the construction of the Horner tree.

Definition at line [874](#) of file [optimize.cc](#).

References [build\\_Horner\\_tree\(\)](#), and [term\\_compare\(\)](#).

Referenced by [optimize\\_expression\\_given\\_Horner\(\)](#).

Here is the call graph for this function:



#### 4.92.15.1 print\_tree()

```
void print_tree (
    const vector< WORD > & tree )
```

Definition at line [1002](#) of file [optimize.cc](#).

#### 4.92.15.2 hash\_range()

```
template<typename T >
size_t hash_range (
    T * array,
    int size )
```

Definition at line [1071](#) of file [optimize.cc](#).

#### 4.92.15.3 generate\_instructions()

```
vector< WORD > generate_instructions (
    const vector< WORD > & tree,
    bool do_CSE )
```

Generate instructions

#### 4.92.16 Description

Converts the expression tree to a list of instructions that directly translate to code. Instructions are of the form:

expr.nr operator length [operands]+ trailing.zero

The operands are of the form:

length [(EXTRA)SYMBOL length variable power] coeff

This method only generates binary operators. Merging is done later. The method also checks for common subexpressions and eliminates them and the flag "do\_CSE" is set.

#### 4.92.17 Implementation details

The map "ID" keeps track of which subexpressions already exist. The key is formatted as one of the following:

SYMBOL x n SNUMBER coeff OPERATOR LHS RHS

with LHS/RHS formatted as one of the following:

SNUMBER idx 0 (EXTRA)SYMBOL x n

ID[symbol] or ID[operator] equals a subexpression number. ID[coeff] equals the position of the number in the input.

The stack s is used to process the postfix expression tree. Three-word tokens of the form:

SNUMBER idx.of.coeff 0 SYMBOL x n EXTRASYMBOL x 1

are pushed onto it. Operators pop two operands and push the resulting expression.

(Extra)symbols are 1-indexed, because -X is also needed to represent -1 times this term.

There is currently a bug. The notation cannot tell if there is a single bracket and then ignores the bracket.

TODO: check if this method performs properly if do\_CSE=false

Definition at line 1133 of file [optimize.cc](#).

References [count\\_operators\(\)](#).

Referenced by [optimize\\_expression\\_given\\_Horner\(\)](#).

Here is the call graph for this function:



#### 4.92.17.1 count\_operators\_cse()

```
int count_operators_cse (
    const vector< WORD > & tree )
```

Count number of operators in a binary tree, while removing CSEs on the fly. The instruction set is not created, which makes this method slightly faster.

A hash is created on the fly and is passed through the stack. TODO: find better hash functions

Definition at line 1342 of file [optimize.cc](#).

#### 4.92.17.2 buildTree()

```
NODE * buildTree (
    vector< WORD > & tree )
```

Definition at line 1564 of file [optimize.cc](#).

#### 4.92.17.3 count\_operators\_cse\_topdown()

```
int count_operators_cse_topdown (
    vector< WORD > & tree )
```

Definition at line 1626 of file [optimize.cc](#).

#### 4.92.17.4 simulated\_annealing()

```
vector< WORD > simulated_annealing ( )
```

Definition at line 1681 of file [optimize.cc](#).

#### 4.92.17.5 find\_Horner\_MCTS\_expand\_tree()

```
void find_Horner_MCTS_expand_tree ( )
```

Definition at line 2054 of file [optimize.cc](#).

#### 4.92.17.6 find\_Horner\_MCTS()

```
void find_Horner_MCTS ( )
```

Find best Horner schemes using MCTS

### 4.92.18 Description

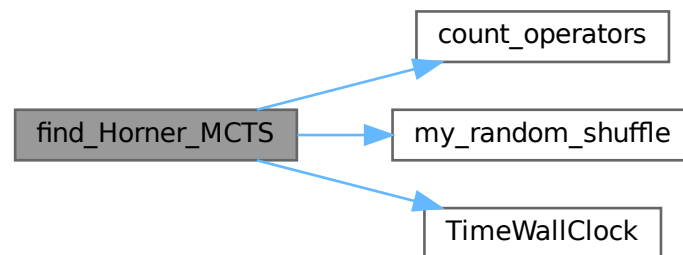
The method governs the MCTS for the best Horner schemes. It does some pre-processing, calls "find\_Horner\_MCTS\_expand\_tree" a number of times and does some post-processing.

Definition at line 2255 of file [optimize.cc](#).

References [count\\_operators\(\)](#), [my\\_random\\_shuffle\(\)](#), and [TimeWallClock\(\)](#).

Referenced by [Optimize\(\)](#).

Here is the call graph for this function:



#### 4.92.18.1 merge\_operators()

```
vector< WORD > merge_operators (
    const vector< WORD > & all_instr,
    bool move_coeff )
```

Merge operators

### 4.92.19 Description

The input instructions form a binary DAG. This method merges expressions like

$Z1 = a+b; Z2 = Z1+c;$

into

$Z2 = a+b+c;$

An instruction is merged iff it only has one parent and the operator equals its parent's operator.

This still leaves some freedom: where should the coefficients end up in cases as:

$Z1 = Z2 + x \Leftrightarrow Z1 = 2*Z2 + x \quad Z2 = 2*x*y \quad Z2 = x*y$

Both are relevant, e.g. for CSE of the form "2\*x" and "2\*Z2". The flag "move\_coeff" moves coefficients from LHS-like expressions to RHS-like expressions.

Furthermore, this method removes empty equation ( $Z1=0$ ), that are introduced by some "optimize\_greedy" substitutions.



### 4.92.20 Implementation details

Expressions are mostly traversed via a stack, so that parents are evaluated before their children.

With "move\_coeff" set coefficients are moved, but this leads to some tricky cases, e.g.

$$Z1 = Z2 + x \quad Z2 = 2*y$$

Here Z2 reduces to the trivial equation  $Z2=y$ , which should be eliminated. Here the array skip[i] comes in.

Furthermore in the case

$$Z1 = Z2 + x \quad Z2 = 2*Z3 \quad Z3 = x*Z4 \quad Z4 = y*z$$

after substituting  $Z1 = 2*Z3 + x$ , the parent expression for Z4 becomes Z3 instead of Z2. This is where renum\_par[i] comes in.

Finally, once a coefficient has been moved, skip\_coeff[i] is set and this coefficient is copied into the new expression anymore.

Definition at line 2403 of file [optimize.cc](#).

Referenced by [optimize\\_expression\\_given\\_Horner\(\)](#).

#### 4.92.20.1 find\_optimizations()

```
vector< optimization > find_optimizations (
    const vector< WORD > & instr )
```

Find optimizations

### 4.92.21 Description

This method find all optimization of the form described in "class Optimization". It process every equation, looking for possible optimizations and stores them in a fast-access data structure to count the total improvement of an optimization.

Definition at line 2698 of file [optimize.cc](#).

Referenced by [optimize\\_greedy\(\)](#).

#### 4.92.21.1 do\_optimization()

```
bool do_optimization (
    const optimization optim,
    vector< WORD > & instr,
    int newid )
```

Do optimization

### 4.92.22 Description

This method performs an optimization. It scans through the equations of "optim.eqnidxs" and looks in which this optimization can still be performed (due to other performed optimizations this isn't always the case). If possible, it substitutes the common subexpression by a new extra symbol numbered "newid". Finally, the new extrasymbol is defined accordingly.

Substitutions may lead to trivial equations of the form " $Z_i=Z_j$ ", but these are removed in the end of the method. The method returns whether the substitution has been done once or more (or not).

Definition at line 2931 of file [optimize.cc](#).

Referenced by [optimize\\_greedy\(\)](#).

#### 4.92.22.1 partial\_factorize()

```
int partial_factorize (
    vector< WORD > & instr,
    int n,
    int improve )
```

Partial factorization of instructions

### 4.92.23 Description

This method performs partial factorization of instructions. In particular the following instructions

$Z_1 = x*a*b$   $Z_2 = x*c*d*e$   $Z_3 = 2*x + Z_1 + Z_2 + \text{more}$

are replaced by

$Z_1 = a*b$   $Z_2 = c*d*e$   $Z_3 = Z_j + \text{more}$   $Z_i = 2 + Z_1 + Z_2$   $Z_j = x*Z_i$

Here it is necessary that no other equations refer to  $Z_1$  and  $Z_2$ . The generation of trivial instructions ( $Z_i=Z_j$  or  $Z_i=x$ ) is prevented.

Definition at line 3480 of file [optimize.cc](#).

Referenced by [optimize\\_greedy\(\)](#).

#### 4.92.23.1 optimize\_greedy()

```
vector< WORD > optimize_greedy (
    vector< WORD > instr,
    LONG time_limit )
```

Optimize instructions greedily

### 4.92.24 Description

This method optimizes an expression greedily. It calls "find\_optimizations" to obtain candidates and performs the best one(s) by calling "do\_optimization".

How many different optimization are done, before "find\_optimization" is called again, is determined by the settings "greedyminnum" and "greedymaxperc".

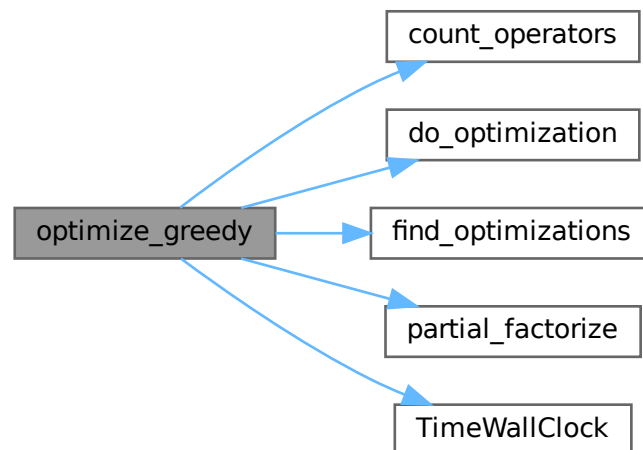
During the optimization process, sequences of zeroes are introduced in the instructions, since moving all instructions when one gets optimized, is very costly. Therefore, in the end, the instructions are "compressed" again to remove these extra zeroes.

Definition at line 3774 of file [optimize.cc](#).

References [count\\_operators\(\)](#), [do\\_optimization\(\)](#), [find\\_optimizations\(\)](#), [partial\\_factorize\(\)](#), and [TimeWallClock\(\)](#).

Referenced by [optimize\\_expression\\_given\\_Horner\(\)](#).

Here is the call graph for this function:



#### 4.92.24.1 recycle\_variables()

```
vector< WORD > recycle_variables (
    const vector< WORD > & all_instr )
```

Recycle variables

### 4.92.25 Description

The current input uses many temporary variables. Many of them become obsolete at some point during the evaluation of the code, so can be recycled. This method rennumbers the temporary variables, so that they are recycled. Furthermore, the input is order in depth-first order, so that the instructions can be performed consecutively.

### 4.92.26 Implementation details

First, for each subDAG, an estimate for the number of variables needed is made. This is done by the following recursive formula:

$$\#vars(x) = \max(\#vars(ch\_i(x)) + i),$$

with  $ch\_i(x)$  the  $i$ -th child of  $x$ , where the childs are ordered w.r.t.  $\#vars(ch\_i)$ . This formula is exact if the input forms a tree, and otherwise gives a reasonable estimate.

Then, the instructions are reordered in a depth-first order with childs ordered w.r.t.  $\#vars$ . Next, the times that variables become obsolete are found. Each LHS of an instruction is renumbered to the lowest-numbered temporary variable that is available at that time.

Definition at line 3914 of file [optimize.cc](#).

Referenced by [optimize\\_expression\\_given\\_Horner\(\)](#).

#### 4.92.26.1 optimize\_expression\_given\_Horner()

```
void optimize_expression_given_Horner ( )
```

Optimize expression given a Horner scheme

### 4.92.27 Description

This method picks one Horner scheme from the list of best Horner schemes, applies this scheme to the expression and then, depending on `optimize.settings`, does a common subexpression elimination (CSE) or performs greedy optimizations.

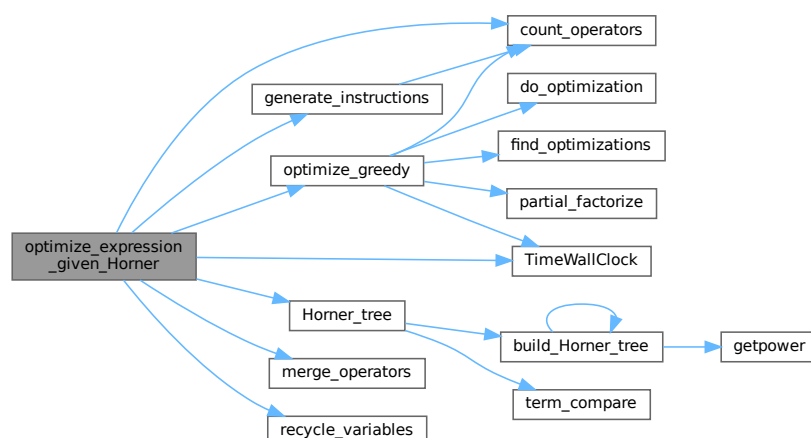
CSE is fast, while greedy might be slow. CSE followed by greedy is faster than greedy alone, but typically results in slightly worse code (not proven; just observed). eventually do greedy optimizations

Definition at line 4061 of file [optimize.cc](#).

References [count\\_operators\(\)](#), [generate\\_instructions\(\)](#), [Horner\\_tree\(\)](#), [merge\\_operators\(\)](#), [optimize\\_greedy\(\)](#), [recycle\\_variables\(\)](#), and [TimeWallClock\(\)](#).

Referenced by [Optimize\(\)](#).

Here is the call graph for this function:



#### 4.92.27.1 generate\_output()

```
void generate_output (
    const vector< WORD > & instr,
    int exprnr,
    int extraoffset,
    const vector< vector< WORD > > & brackets )
```

Generate output

#### 4.92.28 Description

This method prepares the instructions for printing. They are stored in Form format, so that they can be printed by "PrintExtraSymbol". The results are stored in the buffer AO.OptimizeResult.

Definition at line [4292](#) of file [optimize.cc](#).

Referenced by [Optimize\(\)](#).

#### 4.92.28.1 generate\_expression()

```
WORD generate_expression (
    WORD exprnr )
```

Generate expression

#### 4.92.29 Description

This method modifies the original optimized expression by an expression with extra symbols. This is used for "#Optimize".

Definition at line [4441](#) of file [optimize.cc](#).

References [EndSort\(\)](#), [Generator\(\)](#), [LowerSortLevel\(\)](#), [NewSort\(\)](#), and [PutOut\(\)](#).

Referenced by [Optimize\(\)](#).



### 4.92.31 Description

This method takes an input expression and generates optimized code to calculate its value. The following methods are called to do so:

- (1) `get_expression` : to read to expression
- (2) `get_brackets` : find brackets for simultaneous optimization
- (3) `occurrence_order` or `find_Horner_MCTS` : to determine (the) Horner scheme(s) to use; this depends on `AO.optimize.horner`
- (4) `optimize_expression_given_Horner` : to do the optimizations for each Horner scheme; this method does either CSE or greedy optimizations dependings on `AO.optimize.method`
- (5) `generate_output` : to format the output in Form notation and store it in a buffer
- (6a) `optimize_print_code` : to print the expression (for "Print") or (6b) `generate_expression` : to modify the expression (for "#Optimize")

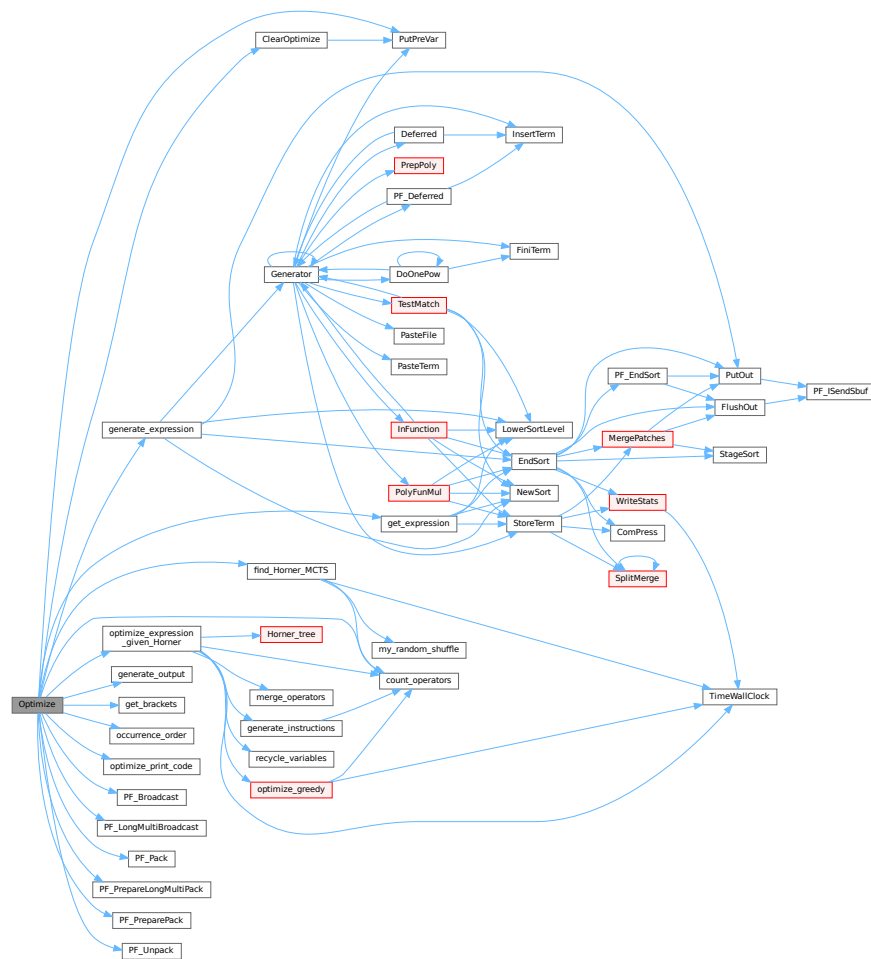
On ParFORM, all the processes must call this function at the same time. Then

- (1) Because only the master can access to the expression to be optimized, the master broadcast the expression to all the slaves after reading the expression (`PF_get_expression`).
- (2) `get_brackets` reads `optimize_expr` as the input and it works also on the slaves. We leave it although the bracket information is not needed on the slaves (used in (5) on the master).
- (3) and (4) `find_Horner_MCTS` and `optimize_expression_given_Horner` are parallelized.
- (5), (6a) and (6b) are needed only on the master.

Definition at line 4638 of file [optimize.cc](#).

References [ClearOptimize\(\)](#), [count\\_operators\(\)](#), [find\\_Horner\\_MCTS\(\)](#), [generate\\_expression\(\)](#), [generate\\_output\(\)](#), [get\\_brackets\(\)](#), [get\\_expression\(\)](#), [occurrence\\_order\(\)](#), [optimize\\_expression\\_given\\_Horner\(\)](#), [optimize\\_print\\_code\(\)](#), [PF\\_Broadcast\(\)](#), [PF\\_LongMultiBroadcast\(\)](#), [PF\\_Pack\(\)](#), [PF\\_PrepareLongMultiPack\(\)](#), [PF\\_PreparePack\(\)](#), [PF\\_Unpack\(\)](#), and [PutPreVar\(\)](#).

Here is the call graph for this function:



#### 4.92.31.1 ClearOptimize()

```
int ClearOptimize (
    void )
```

Optimization of expression

#### 4.92.32 Description

Clears the buffers that were used for optimization output. Clears the expression from the buffers (marks it to be dropped). Note: we need to use the expression by its name, because numbers may change if we drop other expressions between when we do the optimizations and clear the results (in [execute.c](#)). Also this is not 100% safe, because we could overwrite the optimized expression. But that can be done only in a Local or Global statement and hence we only have to test there that we might have to call ClearOptimize first. (in file [comexpr.c](#))

Definition at line 4975 of file [optimize.cc](#).

References [PutPreVar\(\)](#).



Referenced by [Optimize\(\)](#).

Here is the call graph for this function:



## 4.92.33 Variable Documentation

### 4.92.33.1 OPER\_ADD

```
const WORD OPER_ADD = -1
```

Definition at line [120](#) of file [optimize.cc](#).

### 4.92.33.2 OPER\_MUL

```
const WORD OPER_MUL = -2
```

Definition at line [121](#) of file [optimize.cc](#).

### 4.92.33.3 OPER\_COMMA

```
const WORD OPER_COMMA = -3
```

Definition at line [122](#) of file [optimize.cc](#).

### 4.92.33.4 optimize\_expr

```
WORD* optimize_expr
```

Definition at line [157](#) of file [optimize.cc](#).

### 4.92.33.5 optimize\_best\_Horner\_schemes

```
vector<vector<WORD>> > optimize_best_Horner_schemes
```

Definition at line [158](#) of file [optimize.cc](#).

#### 4.92.33.6 optimize\_num\_vars

```
int optimize_num_vars
```

Definition at line 159 of file [optimize.cc](#).

#### 4.92.33.7 optimize\_best\_num\_oper

```
int optimize_best_num_oper
```

Definition at line 160 of file [optimize.cc](#).

#### 4.92.33.8 optimize\_best\_instr

```
vector<WORD> optimize_best_instr
```

Definition at line 161 of file [optimize.cc](#).

#### 4.92.33.9 optimize\_best\_vars

```
vector<WORD> optimize_best_vars
```

Definition at line 162 of file [optimize.cc](#).

#### 4.92.33.10 mcts\_factorized

```
bool mcts_factorized
```

Definition at line 165 of file [optimize.cc](#).

#### 4.92.33.11 mcts\_separated

```
bool mcts_separated
```

Definition at line 165 of file [optimize.cc](#).

#### 4.92.33.12 mcts\_vars

```
vector<WORD> mcts_vars
```

Definition at line 166 of file [optimize.cc](#).

#### 4.92.33.13 mcts\_root

```
tree\_node mcts_root
```

Definition at line 167 of file [optimize.cc](#).

**4.92.33.14 mcts\_expr\_score**

```
int mcts_expr_score
```

Definition at line 168 of file [optimize.cc](#).

**4.92.33.15 mcts\_best\_schemes**

```
set<pair<int,vector<WORD>> > > mcts_best_schemes
```

Definition at line 169 of file [optimize.cc](#).

**4.93 optimize.cc**

[Go to the documentation of this file.](#)

```
00001
00006 /*  #[ License : */
00007 /*
00008  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009  *   When using this file you are requested to refer to the publication
00010  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011  *   This is considered a matter of courtesy as the development was paid
00012  *   for by FOM the Dutch physics granting agency and we would like to
00013  *   be able to track its scientific use to convince FOM of its value
00014  *   for the community.
00015  *
00016  *   This file is part of FORM.
00017  *
00018  *   FORM is free software: you can redistribute it and/or modify it under the
00019  *   terms of the GNU General Public License as published by the Free Software
00020  *   Foundation, either version 3 of the License, or (at your option) any later
00021  *   version.
00022  *
00023  *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024  *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025  *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026  *   details.
00027  *
00028  *   You should have received a copy of the GNU General Public License along
00029  *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00030  */
00031 /*
00032  *   #[ License :
00033  *   #[ includes :
00034  */
00035
00036 // #define DEBUG
00037 // #define DEBUG_MORE
00038 // #define DEBUG_MCTS
00039 // #define DEBUG_GREEDY
00040
00041 #ifdef HAVE_CONFIG_H
00042 #ifndef CONFIG_H_INCLUDED
00043 #define CONFIG_H_INCLUDED
00044 #include <config.h>
00045 #endif
00046 #endif
00047
00048 #include <vector>
00049 #include <stack>
00050 #include <algorithm>
00051 #include <set>
00052 #include <map>
00053 #include <climits>
00054 #include <cmath>
00055 #include <string>
00056 #include <algorithm>
00057 #include <iostream>
00058
00059 #ifdef APPLE64
00060 #ifndef HAVE_UNORDERED_MAP
00061 #define HAVE_UNORDERED_MAP
00062 #endif
```

```

00063 #ifndef HAVE_UNORDERED_SET
00064 #define HAVE_UNORDERED_SET
00065 #endif
00066 #endif
00067
00068 #ifdef HAVE_UNORDERED_MAP
00069     #include <unordered_map>
00070     using std::unordered_map;
00071 #elif !defined(HAVE_TR1_UNORDERED_MAP) && defined(HAVE_BOOST_UNORDERED_MAP_HPP)
00072     #include <boost/unordered_map.hpp>
00073     using boost::unordered_map;
00074 #else
00075     #include <tr1/unordered_map>
00076     using std::tr1::unordered_map;
00077 #endif
00078 #ifdef HAVE_UNORDERED_SET
00079     #include <unordered_set>
00080     using std::unordered_set;
00081 #elif !defined(HAVE_TR1_UNORDERED_SET) && defined(HAVE_BOOST_UNORDERED_SET_HPP)
00082     #include <boost/unordered_set.hpp>
00083     using boost::unordered_set;
00084 #else
00085     #include <tr1/unordered_set>
00086     using std::tr1::unordered_set;
00087 #endif
00088
00089 #if defined(HAVE_BUILTIN_POPCOUNT)
00090     static inline int popcount(unsigned int x) {
00091         return __builtin_popcount(x);
00092     }
00093 #elif defined(HAVE_POPCNT)
00094     #include <intrin.h>
00095     static inline int popcount(unsigned int x) {
00096         return __popcnt(x);
00097     }
00098 #else
00099     static inline int popcount(unsigned int x) {
00100         int count = 0;
00101         while (x > 0) {
00102             if ((x & 1) == 1) count++;
00103             x >>= 1;
00104         }
00105         return count;
00106     }
00107 #endif
00108
00109 extern "C" {
00110     #include "form3.h"
00111 }
00112
00113 // #ifdef DEBUG
00114 #include "mytime.h"
00115 // #endif
00116
00117 using namespace std;
00118
00119 // operators
00120 const WORD OPER_ADD = -1;
00121 const WORD OPER_MUL = -2;
00122 const WORD OPER_COMMA = -3;
00123
00124 // class for a node of the MCTS tree
00125 class tree_node {
00126 public:
00127     vector<tree_node> childs;
00128     double sum_results;
00129     int num_visits;
00130     WORD var;
00131     bool finished;
00132     PADPOINTER(1,1,1,1);
00133
00134     tree_node(int _var=0):
00135         sum_results(0), num_visits(0), var(_var), finished(false) {}
00136
00137     tree_node(const tree_node& other):
00138         childs(other.childs),
00139         sum_results(other.sum_results),
00140         num_visits(other.num_visits),
00141         var(other.var),
00142         finished(other.finished) {}
00143
00144     tree_node& operator=(const tree_node& other) {
00145         if (this != &other) {
00146             childs = other.childs;
00147             sum_results = other.sum_results;
00148             num_visits = other.num_visits;
00149             var = other.var;

```

```

00150         finished = other.finished;
00151     }
00152     return *this;
00153 }
00154 };
00155
00156 // global variables for multithreading
00157 WORD *optimize_expr;
00158 vector<vector<WORD> > optimize_best_Horner_schemes;
00159 int optimize_num_vars;
00160 int optimize_best_num_oper;
00161 vector<WORD> optimize_best_instr;
00162 vector<WORD> optimize_best_vars;
00163
00164 // global variables for MCTS
00165 bool mcts_factorized, mcts_separated;
00166 vector<WORD> mcts_vars;
00167 tree_node mcts_root;
00168 int mcts_expr_score;
00169 set<pair<int, vector<WORD> > > mcts_best_schemes;
00170
00171 #ifdef WITHPTHREADS
00172 pthread_mutex_t optimize_lock;
00173 #endif
00174
00175 /*
00176     #] includes :
00177     #[ print_instr :
00178 */
00179
00180 void print_instr (const vector<WORD> &instr, WORD num)
00181 {
00182     const WORD *tbegin = &instr.begin();
00183     const WORD *tend = tbegin+instr.size();
00184     for (const WORD *t=tbegin; t!=tend; t+=*(t+2)) {
00185         MesPrint("< %d> %a", num, t[2], t);
00186     }
00187 }
00188
00189 /*
00190     #] print_instr :
00191     #[ my_random_shuffle :
00192 */
00193
00194 template <class RandomAccessIterator>
00195 void my_random_shuffle (PHEAD RandomAccessIterator fr, RandomAccessIterator to) {
00196     for (int i=to-fr-1; i>0; --i)
00197         swap (fr[i], fr[wrnf(BHEAD0) % (i+1)]);
00198 }
00199
00200 /*
00201     #] my_random_shuffle :
00202     #[ get_expression :
00203 */
00204
00205 static WORD comlist[3] = {TYPETOPOLYNOMIAL, 3, DOALL};
00206
00207 LONG get_expression (int exprnr) {
00208     GETIDENTITY;
00209
00210     AR.NoCompress = 1;
00211
00212     NewSort(BHEAD0);
00213     EXPRESSIONS e = Expressions+exprnr;
00214     SetScratch(AR.infile, &(e->onfile));
00215
00216     // get header term
00217     WORD *term = AT.WorkPointer;
00218     GetTerm(BHEAD term);
00219
00220     NewSort(BHEAD0);
00221
00222     // get terms
00223     while (GetTerm(BHEAD term) > 0) {
00224         AT.WorkPointer = term + *term;
00225         WORD *t1 = term;
00226         WORD *t2 = term + *term;
00227         if (ConvertToPoly(BHEAD t1, t2, comlist, 1) < 0) return -1;
00228         int n = *t2;
00229         NCOPY(t1, t2, n);
00230         AT.WorkPointer = term + *term;
00231         if (StoreTerm(BHEAD term)) return -1;
00232     }
00233
00234     // sort and store in buffer
00235     LONG len = EndSort(BHEAD (WORD *) ((void *) (&optimize_expr)), 2);

```

```

00256     LowerSortLevel();
00257     AT.WorkPointer = term;
00258
00259     return len;
00260 }
00261
00262 /*
00263     #] get_expression :
00264     #[ PF_get_expression :
00265 */
00266 #ifdef WITHMPI
00267
00268 // get_expression for ParFORM
00269 LONG PF_get_expression (int exprnr) {
00270     LONG len;
00271     if (PF.me == MASTER) {
00272         len = get_expression(exprnr);
00273     }
00274     if (PF.numtasks > 1) {
00275         PF_BroadcastBuffer(&optimize_expr, &len);
00276     }
00277     return len;
00278 }
00279
00280 // replace get_expression called in Optimize
00281 #define get_expression PF_get_expression
00282
00283 #endif
00284 /*
00285     #] PF_get_expression :
00286     #[ get_brackets :
00287 */
00288
00303 vector<vector<WORD> > get_brackets () {
00304
00305     // check for brackets in expression
00306     bool has_brackets = false;
00307     for (WORD *t=optimize_expr; *t!=0; t+=*t) {
00308         WORD *tend=t+*t; tend-=ABS(*tend-1));
00309         for (WORD *t1=t+1; t1<tend; t1+=*(t1+1))
00310             if (*t1 == HAAKJE)
00311                 has_brackets=true;
00312     }
00313
00314     // replace brackets by SEPARATESYMBOL
00315     vector<vector<WORD> > brackets;
00316
00317     if (has_brackets) {
00318         int exprlen=10; // we need potential space for an empty bracket
00319         for (WORD *t=optimize_expr; *t!=0; t+=*t)
00320             exprlen += *t;
00321         WORD *newexpr = (WORD *)Malloc1(exprlen*sizeof(WORD), "optimize newexpr");
00322
00323         int i=0;
00324         int sep_power = 0;
00325
00326         for (WORD *t=optimize_expr; *t!=0; t+=*t) {
00327             WORD *t1 = t+1;
00328
00329             // scan for bracket
00330             vector<WORD> bracket;
00331             for (; *t1!=HAAKJE; t1+=*(t1+1))
00332                 bracket.insert(bracket.end(), t1, t1+*(t1+1));
00333
00334             if (brackets.size()==0 || bracket!=brackets.back()) {
00335                 sep_power++;
00336                 brackets.push_back(bracket);
00337             }
00338             t1+=*(t1+1);
00339
00340             WORD left = t + *t - t1;
00341             bool more_symbols = (left != ABS(*t+*t-1));
00342
00343             // add power of SEPARATESYMBOL
00344             newexpr[i++] = 1 + left + (more_symbols ? 2 : 4);
00345             newexpr[i++] = SYMBOL;
00346             newexpr[i++] = (more_symbols ? *(t1+1) + 2 : 4);
00347             newexpr[i++] = SEPARATESYMBOL;
00348             newexpr[i++] = sep_power;
00349
00350             // add remaining terms
00351             if (more_symbols) {
00352                 t1+=2;
00353                 left-=2;
00354             }
00355             while (left-->0)
00356                 newexpr[i++] = *(t1++);

```

```

00357     }
00358     /*
00359     We insert here an empty bracket that is zero.
00360     It is used for the case that there is only a single bracket which is
00361     outside the notation for trees at a later stage.
00362
00363     There may be a problem now in that in the case of sep_power==1
00364     newexpr is bigger than optimize_expr. We have to check that.
00365     */
00366     if ( sep_power == 1 )
00367     {
00368         WORD *t;
00369         vector<WORD> bracket;
00370         bracket.push_back(0);
00371         bracket.push_back(0);
00372         bracket.push_back(0);
00373         bracket.push_back(0);
00374         sep_power++;
00375         brackets.push_back(bracket);
00376         newexpr[i++] = 8;
00377         newexpr[i++] = SYMBOL;
00378         newexpr[i++] = 4;
00379         newexpr[i++] = SEPARATESYMBOL;
00380         newexpr[i++] = sep_power;
00381         newexpr[i++] = 1;
00382         newexpr[i++] = 1;
00383         newexpr[i++] = 3;
00384         newexpr[i++] = 0;
00385         for (t=optimize_expr; *t!=0; t+=*t) {}
00386         if ( t-optimize_expr+1 < i ) { // We have to redo this
00387             M_free(optimize_expr,"$-sort space");
00388             optimize_expr = (WORD *)Malloca(1*i*sizeof(WORD),"$-sort space");
00389         }
00390     }
00391     else {
00392         newexpr[i++] = 0;
00393     }
00394     memcpy(optimize_expr, newexpr, i*sizeof(WORD));
00395     M_free(newexpr,"optimize newexpr");
00396
00397     // if factorized, replace SEP by FAC and remove brackets
00398     if (brackets[0].size()>0 && brackets[0][2]==FACTORSYMBOL) {
00399         for (WORD *t=optimize_expr; *t!=0; t+=*t) {
00400             if (*t == ABS(*(t+*t-1))+1) continue;
00401             if (t[1]==SYMBOL)
00402                 for (int i=3; i<t[2]; i+=2)
00403                     if (t[i]==SEPARATESYMBOL) t[i]=FACTORSYMBOL;
00404         }
00405         return vector<vector<WORD> >();
00406     }
00407 }
00408
00409 return brackets;
00410 }
00411
00412 /*
00413 #] get_brackets :
00414 #[ count_operators :
00415 */
00416
00423 int count_operators (const WORD *expr, bool print=false) {
00424
00425     int n=0;
00426     while (*(expr+n)!=0) n+=*(expr+n);
00427
00428     int cntpow=0, cntmul=0, cntadd=0, sumpow=0;
00429     WORD maxpowfac=1, maxpowsep=1;
00430
00431     for (const WORD *t=expr; *t!=0; t+=*t) {
00432         if (t!=expr) cntadd++; // new term
00433         if (*t==ABS(*(t+*t-1))+1) continue; // only coefficient
00434
00435         int cntsym=0;
00436
00437         if (t[1]==SYMBOL)
00438             for (int i=3; i<t[2]; i+=2) {
00439                 if (t[i]==FACTORSYMBOL) {
00440                     maxpowfac = max(maxpowfac, t[i+1]);
00441                     continue;
00442                 }
00443                 if (t[i]==SEPARATESYMBOL) {
00444                     maxpowsep = max(maxpowsep, t[i+1]);
00445                     continue;
00446                 }
00447                 if (t[i+1]>2) { // (extra)symbol power>2
00448                     cntpow++;
00449                     sumpow += (int)floor(log(t[i+1])/log(2.0)) + popcount(t[i+1]) - 1;

```

```

00450         }
00451         if (t[i+1]==2) cntmul++; // (extra)symbol squared
00452         cntsym++;
00453     }
00454
00455     if (ABS(*(t++t-1))!=3 || *(t++t-2)!=1 || *(t++t-3)!=1) cntsym++; // non +/-1 coefficient
00456
00457     if (cntsym > 0) cntmul+=cntsym-1;
00458 }
00459
00460 cntadd -= maxpowfac-1;
00461 cntmul += maxpowfac-1;
00462
00463 cntadd -= maxpowsep-1;
00464
00465 if (print)
00466     MesPrint ("*** STATS: original %1P %1M %1A : %1", cntpow,cntmul,cntadd,sumpow+cntmul+cntadd);
00467
00468 return sumpow+cntmul+cntadd;
00469 }
00470
00471 int count_operators (const vector<WORD> &instr, bool print=false) {
00472
00473     int cntpow=0, cntmul=0, cntadd=0, sumpow=0;
00474
00475     const WORD *ebegin = &instr.begin();
00476     const WORD *eend = ebegin+instr.size();
00477
00478     for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {
00479         for (const WORD *t=e+3; *t!=0; t+=*t) {
00480             if (t!=e+3) {
00481                 if (*(e+1)==OPER_ADD) cntadd++; // new term
00482                 if (*(e+1)==OPER_MUL) cntmul++; // new term
00483             }
00484             if (*t == ABS(*(t++t-1))+1) continue; // only coefficient
00485             if (*(t+1)==SYMBOL || *(t+1)==EXTRASYMBOL) {
00486                 if (*(t+4)==2) cntmul++; // (extra)symbol squared
00487                 if (*(t+4)>2) { // (extra)symbol power>2
00488                     cntpow++;
00489                     sumpow += (int)floor(log(*(t+4))/log(2.0)) + popcount(*(t+4)) - 1;
00490                 }
00491             }
00492             if (ABS(*(t++t-1))!=3 || *(t++t-2)!=1 || *(t++t-3)!=1) cntmul++; // non +/-1 coefficient
00493         }
00494     }
00495
00496     if (print)
00497         MesPrint ("*** STATS: optimized %1P %1M %1A : %1", cntpow,cntmul,cntadd,sumpow+cntmul+cntadd);
00498
00499     return sumpow+cntmul+cntadd;
00500 }
00501
00502 if (print)
00503     MesPrint ("*** STATS: optimized %1P %1M %1A : %1", cntpow,cntmul,cntadd,sumpow+cntmul+cntadd);
00504
00505 return sumpow+cntmul+cntadd;
00506 }
00507
00508 /*
00509 #] count_operators :
00510 #[ occurrence_order :
00511 */
00512
00520 vector<WORD> occurrence_order (const WORD *expr, bool rev) {
00521
00522     // count the number of occurrences of variables
00523     map<WORD,int> cnt;
00524     for (const WORD *t=expr; *t!=0; t+=*t) {
00525         if (*t == ABS(*(t++t-1))+1) continue; // skip constant term
00526         if (t[1] == SYMBOL)
00527             for (int i=3; i<t[2]; i+=2)
00528                 cnt[t[i]]++;
00529     }
00530
00531     bool is_fac=false, is_sep=false;
00532     if (cnt.count(FACTORSYMBOL)) {
00533         is_fac=true;
00534         cnt.erase(FACTORSYMBOL);
00535     }
00536     if (cnt.count(SEPARATESYMBOL)) {
00537         is_sep=true;
00538         cnt.erase(SEPARATESYMBOL);
00539     }
00540
00541     // determine the order of the variables
00542     vector<pair<int,WORD> > cnt_order;
00543     for (map<WORD,int>::iterator i=cnt.begin(); i!=cnt.end(); i++)
00544         cnt_order.push_back(make_pair(i->second, i->first));
00545     sort(cnt_order.rbegin(), cnt_order.rend());
00546
00547     // create resulting order
00548     vector<WORD> order;
00549     for (int i=0; i<(int)cnt_order.size(); i++)

```



```

00550         order.push_back(cnt_order[i].second);
00551
00552         if (rev) reverse(order.begin(),order.end());
00553
00554         // add FACTORSYMBOL/SEPARATESYMBOL
00555         if (is_fac) order.insert(order.begin(), FACTORSYMBOL);
00556         if (is_sep) order.insert(order.begin(), SEPARATESYMBOL);
00557
00558         return order;
00559     }
00560
00561     /*
00562     #] occurrence_order :
00563     #[ Horner_tree :
00564     */
00565
00601 WORD getpower (const WORD *term, int var, int pos) {
00602     if (*term == ABS>(*term+*term-1))+1) return 0; // constant term
00603     if (2*pos+2 >= term[2]) return 0; // too few symbols
00604     if (term[2*pos+3] != var) return 0; // incorrect symbol
00605     return term[2*pos+4]; // return power
00606 }
00607
00615 void fixarg (UWORD *t, WORD &n) {
00616     int an=ABS(n), sn=SGN(n);
00617     while (*(t+an-1)==0) an--;
00618     n=an*sn;
00619 }
00620
00621 void GcdLong_fix_args (PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc) {
00622     fixarg(a,na);
00623     fixarg(b,nb);
00624     GcdLong(BHEAD a,na,b,nb,c,nc);
00625 }
00626
00627 void DivLong_fix_args(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc, UWORD *d, WORD *nd)
00628 {
00629     fixarg(a,na);
00629     fixarg(b,nb);
00630     DivLong(a,na,b,nb,c,nc,d,nd);
00631 }
00632
00653 void build_Horner_tree (const WORD **terms, int numterms, int var, int maxvar, int pos, vector<WORD>
*res) {
00654
00655     GETIDENTITY;
00656
00657     if (var == maxvar) {
00658         // Horner tree is finished, so add remaining terms unfactorized
00659         // (note: since only complete Horner schemes seem to be useful, numterms=1 here
00660
00661         for (int fr=0; fr<numterms; fr++) {
00662
00663             bool empty = true;
00664             const WORD *t = terms[fr];
00665
00666             // add symbols
00667             if (*t != ABS(*t+*t-1))+1)
00668                 for (int i=3+2*pos; i<t[2]; i+=2) {
00669                     res->push_back(SYMBOL);
00670                     res->push_back(4);
00671                     res->push_back(t[i]);
00672                     res->push_back(t[i+1]);
00673                     if (!empty) res->push_back(OPER_MUL);
00674                     empty = false;
00675                 }
00676
00677             // if empty, add a 1, since the result should look like "... coeff *"
00678             if (empty) {
00679                 res->push_back(SNUMBER);
00680                 res->push_back(5);
00681                 res->push_back(1);
00682                 res->push_back(1);
00683                 res->push_back(3);
00684             }
00685
00686             // add coefficient
00687             res->push_back(SNUMBER);
00688             WORD n = ABS(*t+*t-1);
00689             res->push_back(n+2);
00690             for (int i=0; i<n; i++)
00691                 res->push_back(*t+*t-n+i);
00692             res->push_back(OPER_MUL);
00693
00694             if (fr>0) res->push_back(OPER_ADD);
00695         }
00696

```

```

00697 // result should end with gcd of the terms; right now this never
00698 // triggers, but if one would allow for incomplete Horner schemes,
00699 // one should extract the gcd here
00700 if (numterms > 1) {
00701     res->push_back(SNUMBER);
00702     res->push_back(5);
00703     res->push_back(1);
00704     res->push_back(1);
00705     res->push_back(3);
00706     res->push_back(OPER_MUL);
00707 }
00708 }
00709 else {
00710     // extract variable "var" and the gcd and proceed recursively
00711
00712     WORD nnum = 0, nden = 0, ntmp = 0, ndum = 0;
00713     UWORD *num = NumberMalloc("build_Horner_tree");
00714     UWORD *den = NumberMalloc("build_Horner_tree");
00715     UWORD *tmp = NumberMalloc("build_Horner_tree");
00716     UWORD *dum = NumberMalloc("build_Horner_tree");
00717
00718     // previous coefficient for gcd extraction or coefficient multiplication
00719     int prev_coeff_idx = -1;
00720
00721     for (int fr=0; fr<numterms; ) {
00722
00723         // find power of current term
00724         WORD pow = getpower(terms[fr], var, pos);
00725
00726         // find all terms with that power
00727         int to=fr+1;
00728         while (to<numterms && getpower(terms[to], var, pos) == pow) to++;
00729
00730         // recursively build Horner tree of all terms proportional to var^pow
00731         build_Horner_tree(terms+fr, to-fr, var+1, maxvar, pow==0?pos:pos+1, res);
00732
00733         if (AN.poly_vars[var] != FACTORSYMBOL && AN.poly_vars[var] != SEPARATESYMBOL) {
00734             // if normal symbol, find gcd(numerators) and gcd(denominators)
00735             WORD n1 = res->at(res->size()-2) / 2;
00736             WORD *t1 = &res->at(res->size()-2-2*ABS(n1));
00737
00738             WORD *t2 = fr==0 ? t1 : &res->at(prev_coeff_idx);
00739             WORD n2 = fr==0 ? n1 : *(t2+*(t2+1)-1) / 2;
00740             if (fr>0) t2+=2;
00741
00742             GcdLong_fix_args(BHEAD (UWORD *)t1,n1,(UWORD *)t2,n2,num,&nnum);
00743             GcdLong_fix_args(BHEAD (UWORD *)t1+ABS(n1),ABS(n1),(UWORD
00744 *)t2+ABS(n2),ABS(n2),den,&nden);
00745
00746             // divide out gcds; note: leading zeroes can be added here
00747             for (int i=0; i<2; i++) {
00748                 if (i==1 && fr==0) break;
00749
00750                 WORD *t = i==0 ? t1 : t2;
00751                 WORD n = i==0 ? n1 : n2;
00752
00753                 DivLong_fix_args((UWORD *)t, n, num, nnum, tmp, &ntmp, dum, &ndum);
00754                 for (int j=0; j<ABS(ntmp); j++) *t++ = tmp[j];
00755                 for (int j=0; j<ABS(n)-ABS(ntmp); j++) *t++ = 0;
00756
00757                 if (SGN(ntmp) != SGN(n)) n=-n;
00758
00759                 DivLong_fix_args((UWORD *)t, n, den, nden, tmp, &ntmp, dum, &nden);
00760                 for (int j=0; j<ABS(ntmp); j++) *t++ = tmp[j];
00761                 for (int j=0; j<ABS(n)-ABS(ntmp); j++) *t++ = 0;
00762
00763                 *t++ = SGN(n) * (2*ABS(n)+1);
00764             }
00765
00766             // add the addition operator
00767             if (fr>0) res->push_back(OPER_ADD);
00768
00769             // add the power of "var"
00770             WORD nextpow = (to==numterms ? 0 : getpower(terms[to], var, pos));
00771
00772             if (pow-nextpow > 0) {
00773                 res->push_back(SYMBOL);
00774                 res->push_back(4);
00775                 res->push_back(var);
00776                 res->push_back(pow-nextpow);
00777                 res->push_back(OPER_MUL);
00778             }
00779
00780             // add the extracted gcd
00781             res->push_back(SNUMBER);
00782             WORD n = MaX(ABS(nnum),nden);
00783             res->push_back(n*2+3);

```

```

00783         for (int i=0; i<ABS(nnum); i++) res->push_back(num[i]);
00784         for (int i=0; i<n-ABS(nnum); i++) res->push_back(0);
00785         for (int i=0; i<nden; i++) res->push_back(den[i]);
00786         for (int i=0; i<n-ABS(nnden); i++) res->push_back(0);
00787         res->push_back(SGN(nnum)*(2*n+1));
00788         res->push_back(OPER_MUL);
00789
00790         prev_coeff_idx = res->size() - ABS(res->at(res->size()-2)) - 3;
00791     }
00792     else if (AN.poly_vars[var]==FACTORSYMBOL) {
00793
00794         // if factorsymbol, multiply overall integer contents
00795
00796         if (fr>0) {
00797             WORD n1 = res->at(res->size()-2) / 2;
00798             WORD *t1 = &res->at(res->size()-2-2*ABS(n1));
00799             WORD *t2 = &res->at(prev_coeff_idx);
00800             WORD n2 = *(t2+(t2+1)-1) / 2;
00801             t2+=2;
00802
00803             MulRat(BHEAD (UWORD *)t1,n1, (UWORD *)t2,n2,tmp,&ntmp);
00804
00805             // replace previous coefficient with 1
00806             n2=ABS(n2);
00807             for (int i=0; i<ABS(n2); i++)
00808                 t2[i] = t2[n2+i] = i==0 ? 1 : 0;
00809             t2[2*n2] = 2*n2+1;
00810
00811             // remove this coefficient
00812             for (int i=0; i<2*ABS(n1)+2; i++)
00813                 res->pop_back();
00814
00815             // add product
00816             res->back() = 2*ABS(ntmp)+3; // adjust size of term
00817             res->insert(res->end(), tmp, tmp+2*ABS(ntmp)); // num/den coefficient
00818             res->push_back(SGN(ntmp)*(2*ABS(ntmp)+1)); // size of coefficient
00819             res->push_back(OPER_MUL); // operator
00820         }
00821
00822         prev_coeff_idx = res->size() - ABS(res->at(res->size()-2)) - 3;
00823
00824         // multiply previous factors with this factor
00825         if (fr>0)
00826             res->push_back(OPER_MUL);
00827     }
00828     else { // AN.poly_vars[var]==SEPARATESYMBOL
00829         if (fr>0)
00830             res->push_back(OPER_COMMA);
00831         prev_coeff_idx = -1;
00832     }
00833
00834     fr=to;
00835 }
00836
00837 // cleanup
00838 NumberFree(dum,"build_Horner_tree");
00839 NumberFree(tmp,"build_Horner_tree");
00840 NumberFree(den,"build_Horner_tree");
00841 NumberFree(num,"build_Horner_tree");
00842 }
00843 }
00844
00853 bool term_compare (const WORD *a, const WORD *b) {
00854     if (*a == ABS(*(a++a-1))+1) return true; // coefficient comes first
00855     if (*b == ABS(*(b++b-1))+1) return false;
00856     if (a[1]!=SYMBOL) return true;
00857     if (b[1]!=SYMBOL) return false;
00858     for (int i=3; i<a[2] && i<b[2]; i+=2) {
00859         if (a[i] !=b[i] ) return a[i] >b[i];
00860         if (a[i+1]!=b[i+1]) return a[i+1]<b[i+1];
00861     }
00862     return a[2]<b[2];
00863 }
00864
00874 vector<WORD> Horner_tree (const WORD *expr, const vector<WORD> &order) {
00875     #ifdef DEBUG
00876         MesPrint ("*** [%s, w=%w] CALL: Horner_tree(%a)", thetime_str().c_str(), order.size(), &order[0]);
00877     #endif
00878
00879     GETIDENTITY;
00880
00881     // find the renumbering scheme (new numbers are 0,1,...,#vars-1)
00882     map<WORD,WORD> renum;
00883     AN.poly_num_vars = order.size();
00884     AN.poly_vars = (WORD *)Mallloc1(AN.poly_num_vars*sizeof(WORD), "AN.poly_vars");
00885     for (int i=0; i<AN.poly_num_vars; i++) {
00886         AN.poly_vars[i] = order[i];

```

```

00887         renum[order[i]] = i;
00888     }
00889
00890     // sort variables in individual terms using bubble sort
00891     WORD* sorted;
00892     WORD* dynamicAlloc = 0;
00893     LONG sumsize = 0;
00894
00895     for (const WORD *t=expr; *t!=0; t+=*t) {
00896         sumsize += *t;
00897     }
00898
00899     // We actually need sumsize + 1 WORDS available, due to the "*sorted = 0;"
00900     // at the end of the sort loop.
00901     if ( AT.WorkPointer + sumsize + 1 > AT.WorkTop ) {
00902         dynamicAlloc = (WORD*)Malloc1(sizeof(*dynamicAlloc)*(sumsize+1), "Horner_tree alloc");
00903         sorted = dynamicAlloc;
00904     }
00905     else {
00906         sorted = AT.WorkPointer;
00907     }
00908
00909     for (const WORD *t=expr; *t!=0; t+=*t) {
00910         memcpy(sorted, t, *t*sizeof(WORD));
00911
00912         if (*t != ABS(*(t+*t-1))+1) {
00913             // non-constant term
00914
00915             // renumber variables, adding new variables if necessary
00916             for (int i=3; i<sorted[2]; i+=2) {
00917                 if (!renum.count(sorted[i])) {
00918                     renum[sorted[i]] = AN.poly_num_vars;
00919
00920                     WORD *new_poly_vars = (WORD *)Malloc1((AN.poly_num_vars+1)*sizeof(WORD),
00921 "AN.poly_vars");
00922                     memcpy(new_poly_vars, AN.poly_vars, AN.poly_num_vars*sizeof(WORD));
00923                     M_free(AN.poly_vars, "poly_vars");
00924                     AN.poly_vars = new_poly_vars;
00925                     AN.poly_vars[AN.poly_num_vars] = sorted[i];
00926                     AN.poly_num_vars++;
00927                 }
00928                 sorted[i] = renum[sorted[i]];
00929             }
00930             // order variables
00931             for (int i=0; i<sorted[2]/2; i++)
00932                 for (int j=3; j+2<sorted[2]; j+=2)
00933                     if (sorted[j] > sorted[j+2]) {
00934                         swap(sorted[j], sorted[j+2]);
00935                         swap(sorted[j+1], sorted[j+3]);
00936                     }
00937
00938             sorted += *sorted;
00939         }
00940
00941         *sorted = 0;
00942         if ( dynamicAlloc ) {
00943             sorted = dynamicAlloc;
00944         }
00945         else {
00946             sorted = AT.WorkPointer;
00947         }
00948
00949         // find pointers to all terms and sort them efficiently
00950         vector<const WORD *> terms;
00951         for (const WORD *t=sorted; *t!=0; t+=*t)
00952             terms.push_back(t);
00953         sort(terms.rbegin(), terms.rend(), term_compare);
00954
00955         // construct the Horner tree
00956         vector<WORD> res;
00957         build_Horner_tree(&terms[0], terms.size(), 0, AN.poly_num_vars, 0, &res);
00958
00959         // remove leading zeroes in coefficients
00960         int j=0;
00961         for (int i=0; i<(int)res.size(); ) {
00962             if (res[i]==OPER_ADD || res[i]==OPER_MUL || res[i]==OPER_COMMA)
00963                 res[j++] = res[i++];
00964             else if (res[i]==SYMBOL) {
00965                 memmove(&res[j], &res[i], res[i+1]*sizeof(WORD));
00966                 i+=res[j+1];
00967                 j+=res[j+1];
00968             }
00969             else if (res[i]==SNUMBER) {
00970                 int n = (res[i+1]-2)/2;
00971                 int dn = 0;
00972                 while (res[i+1+n-dn]==0 && res[i+1+2*n-dn]==0) dn++;

```

```

00973         res[j++] = SNUMBER;
00974         res[j++] = 2*(n-dn) + 3;
00975         memmove(&res[j], &res[i+2], (n-dn)*sizeof(WORD));
00976         j+=n-dn;
00977         memmove(&res[j], &res[i+n+2], (n-dn)*sizeof(WORD));
00978         j+=n-dn;
00979         res[j++] = SGN(res[i+2*n+2])*(2*(n-dn)+1);
00980         i+=2*n+3;
00981     }
00982 }
00983 res.resize(j);
00984
00985 if ( dynamicAlloc ) {
00986     M_free(dynamicAlloc, "Horner_tree alloc");
00987 }
00988
00989 #ifdef DEBUG
00990     MesPrint ("*** [%s, w=%w] DONE: Horner_tree(%a)", thetime_str().c_str(), order.size(), &order[0]);
00991 #endif
00992
00993     return res;
00994 }
00995
00996 /*
00997  #] Horner_tree :
00998  #[ print_tree :
00999  */
01000
01001 // print Horner tree (for debugging)
01002 void print_tree (const vector<WORD> &tree) {
01003
01004     GETIDENTITY;
01005
01006     for (int i=0; i<(int)tree.size();) {
01007         if (tree[i]==OPER_ADD) {
01008             MesPrint ("%");
01009             i++;
01010         }
01011         else if (tree[i]==OPER_MUL) {
01012             MesPrint ("%");
01013             i++;
01014         }
01015         else if (tree[i]==OPER_COMMA) {
01016             MesPrint (",");
01017             i++;
01018         }
01019         else if (tree[i]==SNUMBER) {
01020             UBYTE buf[100];
01021             int n = tree[i+tree[i+1]-1]/2;
01022             PrtLong((UWORD *)&tree[i+2], n, buf);
01023             int l = strlen((char *)buf);
01024             buf[l]='/';
01025             n=ABS(n);
01026             PrtLong((UWORD *)&tree[i+2+n], n, buf+l+1);
01027             MesPrint ("%s",buf);
01028             i+=tree[i+1];
01029         }
01030         else if (tree[i]==SYMBOL) {
01031             if (AN.poly_vars[tree[i+2]] < 10000)
01032                 MesPrint ("%s^%d", VARNAME(symbols,AN.poly_vars[tree[i+2]]), tree[i+3]);
01033             else
01034                 MesPrint ("Z%d^%d", MAXVARIABLES-AN.poly_vars[tree[i+2]], tree[i+3]);
01035             i+=tree[i+1];
01036         }
01037         else {
01038             MesPrint ("error");
01039             exit(1);
01040         }
01041
01042         MesPrint (" %");
01043     }
01044
01045     MesPrint ("");
01046 }
01047
01048 /*
01049  #] print_tree :
01050  #[ generate_instructions :
01051  */
01052
01053
01054 struct CSEHash {
01055     size_t operator()(const vector<WORD>& n) const {
01056         return n[0];
01057     }
01058 };
01059

```

```

01060 struct CSEEq {
01061     bool operator()(const vector<WORD>& lhs, const vector<WORD>& rhs) const {
01062         if (lhs.size() != rhs.size()) return false;
01063         for (unsigned int i = 1; i < lhs.size(); i++) {
01064             if (lhs[i] != rhs[i]) return false;
01065         }
01066         return true;
01067     }
01068 };
01069
01070
01071 template<typename T> size_t hash_range(T* array, int size) {
01072     size_t hash = 0;
01073
01074     for (int i = 0; i < size; i++) {
01075         hash ^= array[i] + 0x9e3779b9 + (hash << 6) + (hash >> 2);
01076     }
01077
01078     return hash;
01079 }
01080
01133 vector<WORD> generate_instructions (const vector<WORD> &tree, bool do_CSE) {
01134     #ifdef DEBUG
01135         MesPrint ("*** [%s, w=%w] CALL: generate_instructions(cse=%d)",
01136                 thetime_str().c_str(), do_CSE?1:0);
01137     #endif
01138
01139     typedef unordered_map<vector<WORD>, int, CSEHash, CSEEq> csemap;
01140     csemap ID;
01141
01142     // reserve lots of space, to prevent later rehashes
01143     // TODO: what if this is too large? make a parameter?
01144     if (do_CSE) {
01145         ID.rehash(mcts_expr_score * 2);
01146     }
01147
01148     // s is a stack of operands to process when you encounter operators
01149     // in the postfix expression tree. Operands consist of three WORDs,
01150     // formatted as the LHS/RHS of the keys in ID.
01151     stack<int> s;
01152     vector<WORD> instr;
01153     WORD numinstr = 0;
01154     vector<WORD> x;
01155
01156     // process the expression tree
01157     for (int i=0; i<(int)tree.size(); ) {
01158         x.clear();
01159
01160         if (tree[i]==SNUMBER) {
01161             WORD hash = hash_range(&tree[i], tree[i + 1]);
01162             x.push_back(hash);
01163             x.push_back(SNUMBER);
01164             x.insert(x.end(), &tree[i], &tree[i]+tree[i+1]);
01165             int sign = SGN(x.back());
01166             x.back() *= sign;
01167
01168             std::pair<csemap::iterator, bool> suc = ID.insert(csemap::value_type(x, i + 1));
01169             s.push(0);
01170             s.push(sign * suc.first->second);
01171             s.push(SNUMBER);
01172             s.push(hash);
01173             i+=tree[i+1];
01174         }
01175         else if (tree[i]==SYMBOL) {
01176             WORD hash = hash_range(&tree[i], tree[i + 1]);
01177             if (tree[i+3]>1) {
01178                 x.push_back(hash);
01179                 x.push_back(SYMBOL);
01180                 x.push_back(tree[i+2]+1); // variable (1-indexed)
01181                 x.push_back(tree[i+3]); // power
01182
01183                 csemap::iterator it = ID.find(x);
01184                 if (do_CSE && it != ID.end()) {
01185                     // already-seen power of a symbol
01186                     s.push(1);
01187                     s.push(it->second);
01188                     s.push(EXTRASYMBOL);
01189                 } else {
01190                     //MesPrint("strange");
01191                     if (numinstr == MAXPOSITIVE) {
01192                         MesPrint((char *) "ERROR: too many temporary variables needed in
optimization");
01193                         Terminate(-1);
01194                     }
01195
01196                     // new power greater than 1 of a symbol
01197                     instr.push_back(numinstr); // expr.nr

```

```

01198         instr.push_back(OPER_MUL); // operator
01199         instr.push_back(9+OPTHEAD); // length total
01200         instr.push_back(8); // length operand
01201         instr.push_back(SYMBOL); // SYMBOL
01202         instr.push_back(4); // length symbol
01203         instr.push_back(tree[i+2]); // variable
01204         instr.push_back(tree[i+3]); // power
01205         instr.push_back(1); // numerator
01206         instr.push_back(1); // denominator
01207         instr.push_back(3); // length coeff
01208         instr.push_back(0); // trailing 0
01209
01210         ID[x] = ++numinstr;
01211         s.push(1);
01212         s.push(numinstr);
01213         s.push(EXTRASymbol);
01214     }
01215 }
01216 else {
01217     // power of 1
01218     s.push(tree[i+3]); // power
01219     s.push(tree[i+2]+1); // variable (1-indexed)
01220     s.push(SYMBOL);
01221 }
01222
01223 s.push(hash); // push hash
01224 i+=tree[i+1];
01225 }
01226 else { // tree[i]==OPERATOR
01227     int oper = tree[i];
01228     i++;
01229
01230     x.push_back(0); // placeholder for hash
01231     x.push_back(oper);
01232     vector<WORD> hash;
01233     hash.push_back(oper);
01234
01235     // get two operands from the stack
01236     for (int operand=0; operand<2; operand++) {
01237         hash.push_back(s.top()); s.pop();
01238         x.push_back(s.top()); s.pop();
01239         x.push_back(s.top()); s.pop();
01240         x.push_back(s.top()); s.pop();
01241     }
01242
01243     x[0] = hash_range(&hash[0], 3);
01244
01245     // get rid of multiplications by +/-1
01246     if (oper==OPER_MUL) {
01247         bool do_continue = false;
01248
01249         for (int operand=0; operand<2; operand++) {
01250             int idx_oper1 = operand==0 ? 2 : 5;
01251             int idx_oper2 = operand==0 ? 5 : 2;
01252
01253             // check whether operand 1 equals +/-1
01254             if (x[idx_oper1]==SNUMBER) {
01255
01256                 int idx = ABS(x[idx_oper1+1])-1;
01257                 if (tree[idx+2]==1 && tree[idx+3]==1 && ABS(tree[idx+4])==3) {
01258                     // push +/- other operand and continue
01259                     s.push(x[idx_oper2+2]);
01260                     s.push(x[idx_oper2+1]*SGN(x[idx_oper1+1]));
01261                     s.push(x[idx_oper2]);
01262                     s.push(hash[1 + (operand + 1) % 2]);
01263                     do_continue = true;
01264                     break;
01265                 }
01266             }
01267         }
01268
01269         if (do_continue) continue;
01270     }
01271
01272     // check whether this subexpression has been seen before
01273     // if not, generate instruction to define it
01274     csemap::iterator it = ID.find(x);
01275     if (!do_CSE || it == ID.end()) {
01276         if (numinstr == MAXPOSITIVE) {
01277             MesPrint((char *) "ERROR: too many temporary variables needed in optimization");
01278             Terminate(-1);
01279         }
01280
01281         instr.push_back(numinstr); // expr.nr.
01282         instr.push_back(x[1]); // operator
01283         instr.push_back(3); // length
01284     }

```

```

01285         ID[x] = ++numinstr;
01286
01287         int lenidx = instr.size()-1;
01288
01289         for (int j=0; j<2; j++)
01290             if (x[3*j+2]==SYMBOL || x[3*j+2]==EXTRASymbol) {
01291                 instr.push_back(8); // length total
01292                 instr.push_back(x[3*j+2]); // (extra)symbol
01293                 instr.push_back(4); // length (extra)symbol
01294                 instr.push_back(ABS(x[3*j+3])-1); // variable (0-indexed)
01295                 instr.push_back(x[3*j+4]); // power
01296                 instr.push_back(1); // numerator
01297                 instr.push_back(1); // denominator
01298                 instr.push_back(3*SGN(x[3*j+3])); // length coeff
01299                 instr[lenidx] += 8;
01300             }
01301             else { // x[3*j+1]==SNUMBER
01302                 int t = ABS(x[3*j+3])-1;
01303                 instr.push_back(tree[t+1]-1); // length number
01304                 instr.insert(instr.end(), &tree[t+2], &tree[t+tree[t+1]]); // digits
01305                 instr.back() *= SGN(instr.back()) * SGN(x[3*j+3]);
01306                 instr[lenidx] += tree[t+1]-1;
01307             }
01308
01309         instr.push_back(0); // trailing 0
01310         instr[lenidx]++;
01311     }
01312
01313     // push new expression on the stack
01314     s.push(1);
01315     s.push(ID[x]);
01316     s.push(EXTRASymbol);
01317     s.push(x[0]); // push hash
01318 }
01319 }
01320
01321 #ifdef DEBUG
01322     MesPrint ("*** [%s, w=%w] DONE: generate_instructions(cse=%d) : numoper=%d",
01323             thetime_str().c_str(), do_CSE?1:0, count_operators(instr));
01324 #endif
01325
01326     return instr;
01327 }
01328
01329 /*
01330 #] generate_instructions :
01331 #[ count_operators_cse :
01332 */
01333
01334
01342 int count_operators_cse (const vector<WORD> &tree) {
01343     //MesPrint ("*** [%s] Starting CSEE", thetime_str().c_str());
01344
01345     typedef unordered_map<vector<WORD>, int, CSEHash, CSEEq> csemap;
01346     csemap ID;
01347
01348     // reserve lots of space, to prevent later rehashes
01349     // TODO: what if this is too large? make a parameter?
01350     ID.rehash(mcts_expr_score * 2);
01351
01352     // s is a stack of operands to process when you encounter operators
01353     // in the postfix expression tree. Operands consist of three WORDs,
01354     // formatted as the LHS/RHS of the keys in ID.
01355     stack<int> s;
01356     int numinstr = 0, numcommas = 0;
01357     vector<WORD> x;
01358
01359     // process the expression tree
01360     for (int i=0; i<(int)tree.size(); i) {
01361         x.clear();
01362
01363         if (tree[i]==SNUMBER) {
01364             WORD hash = hash_range(&tree[i], tree[i + 1]);
01365             x.push_back(hash);
01366             x.push_back(SNUMBER);
01367             x.insert(x.end(), &tree[i], &tree[i+tree[i+1]]);
01368             int sign = SGN(x.back());
01369             x.back() *= sign;
01370
01371             std::pair<csemap::iterator, bool> suc = ID.insert(csemap::value_type(x, i + 1));
01372             s.push(0);
01373             s.push(sign * suc.first->second);
01374             s.push(SNUMBER);
01375             s.push(hash);
01376             i+=tree[i+1];
01377         }
01378         else if (tree[i]==SYMBOL) {

```



```

01379         WORD hash = hash_range(&tree[i], tree[i + 1]);
01380         if (tree[i+3]>1) {
01381             x.push_back(hash);
01382             x.push_back(SYMBOL);
01383             x.push_back(tree[i+2]+1); // variable (1-indexed)
01384             x.push_back(tree[i+3]); // power
01385
01386             csemaph::iterator it = ID.find(x);
01387             if (it != ID.end()) {
01388                 // already-seen power of a symbol
01389                 s.push(1);
01390                 s.push(it->second);
01391                 s.push(EXTRASymbol);
01392             } else {
01393                 if (tree[i + 3] == 2)
01394                     numinstr++;
01395                 else
01396                     numinstr += (int)floor(log(tree[i+3])/log(2.0)) + popcount(tree[i+3]) - 1;
01397
01398                 ID[x] = numinstr;
01399                 s.push(1);
01400                 s.push(numinstr);
01401                 s.push(EXTRASymbol);
01402             }
01403         }
01404         else {
01405             // power of 1
01406             s.push(tree[i+3]); // power
01407             s.push(tree[i+2]+1); // variable (1-indexed)
01408             s.push(SYMBOL);
01409         }
01410
01411         s.push(hash); // push hash
01412
01413         i+=tree[i+1];
01414     }
01415     else { // tree[i]==OPERATOR
01416         int oper = tree[i];
01417         i++;
01418
01419         x.push_back(0); // placeholder for hash
01420         x.push_back(oper);
01421         vector<WORD> hash;
01422         hash.push_back(oper);
01423
01424         // get two operands from the stack
01425         for (int operand=0; operand<2; operand++) {
01426             hash.push_back(s.top()); s.pop();
01427             x.push_back(s.top()); s.pop();
01428             x.push_back(s.top()); s.pop();
01429             x.push_back(s.top()); s.pop();
01430         }
01431
01432
01433         x[0] = hash_range(&hash[0], 3);
01434
01435         // get rid of multiplications by +/-1
01436         if (oper==OPER_MUL) {
01437             bool do_continue = false;
01438
01439             for (int operand=0; operand<2; operand++) {
01440                 int idx_oper1 = operand==0 ? 2 : 5;
01441                 int idx_oper2 = operand==0 ? 5 : 2;
01442
01443                 // check whether operand 1 equals +/-1
01444                 if (x[idx_oper1]==SNUMBER) {
01445                     int idx = ABS(x[idx_oper1+1])-1;
01446                     if (tree[idx+2]==1 && tree[idx+3]==1 && ABS(tree[idx+4])==3) {
01447                         // push +/- other operand and continue
01448                         s.push(x[idx_oper2+2]);
01449                         s.push(x[idx_oper2+1]*SGN(x[idx_oper1+1]));
01450                         s.push(x[idx_oper2]);
01451                         s.push(hash[1 + (operand + 1) % 2]);
01452                         do_continue = true;
01453                         break;
01454                     }
01455                 }
01456             }
01457         }
01458
01459         if (do_continue) continue;
01460     }
01461
01462     // check whether this subexpression has been seen before
01463     // if not, generate instruction to define it
01464     csemaph::iterator it = ID.find(x);
01465     if (it == ID.end()) {

```

```

01466         if (numinstr == MAXPOSITIVE) {
01467             MesPrint((char *) "ERROR: too many temporary variables needed in optimization");
01468             Terminate(-1);
01469         }
01470
01471         if (oper == OPER_COMMA) numcommas++;
01472         ID[x] = ++numinstr;
01473         s.push(1);
01474         s.push(numinstr);
01475         s.push(EXTRASymbol);
01476     } else {
01477         // push new expression on the stack
01478         s.push(1);
01479         s.push(it->second);
01480         s.push(EXTRASymbol);
01481     }
01482
01483     s.push(x[0]); // push hash
01484 }
01485 }
01486
01487 //MesPrint ("*** [%s] Stopping CSEE", thetime_str().c_str());
01488 return numinstr - numcommas;
01489 }
01490
01491 /*
01492 #] count_operators_cse :
01493 #[ count_operators_cse_topdown :
01494 */
01495
01496 typedef struct node {
01497     const WORD* data;
01498     struct node* l;
01499     struct node* r; // TODO: add l,r to data?
01500     WORD sign; // TODO: use data for this?
01501     UWORD hash;
01502
01503     node() : l(NULL), r(NULL), sign(1), hash(0) {};
01504     node(const WORD* data) : data(data), l(NULL), r(NULL), sign(1), hash(0) {};
01505
01506     // a minus sign in the tree should only count as a different entry if
01507     // it is a compound expression: a = -a, but T+-V != T+V
01508     int cmp(const struct node* rhs) const {
01509         if (this == rhs) return 0;
01510
01511         if (data[0] != rhs->data[0]) return data[0] < rhs->data[0] ? -1 : 1;
01512         int mod = data[0] == SNUMBER ? -1 : 0; // don't check sign, for numbers
01513         if (data[0] == SYMBOL || data[0] == SNUMBER) {
01514             for (int i = 0; i < data[1] + mod; i++) {
01515                 if (data[i] != rhs->data[i]) return data[i] < rhs->data[i] ? -1 : 1;
01516             }
01517         } else {
01518             int lv = l->cmp(rhs->l);
01519             if (lv != 0) return lv;
01520             int rv = r->cmp(rhs->r);
01521             if (rv != 0) return rv;
01522
01523             // TODO: only for ADD operation
01524             if (l->sign != rhs->l->sign) return l->sign < rhs->l->sign ? -1 : 1;
01525             if (r->sign != rhs->r->sign) return r->sign < rhs->r->sign ? -1 : 1;
01526         }
01527
01528         return 0;
01529     }
01530
01531     // less than operator
01532     bool operator() (const struct node* lhs, const struct node* rhs) const
01533     {
01534         return lhs->cmp(rhs) < 0;
01535     }
01536
01537     void calcHash() {
01538         int mod = data[0] == SNUMBER ? -1 : 0; // don't check sign, for numbers
01539         if (data[0] == SYMBOL || data[0] == SNUMBER) {
01540             hash = hash_range(data, data[1] + mod);
01541         } else {
01542             if (l->hash == 0) l->calcHash();
01543             if (r->hash == 0) r->calcHash();
01544
01545             // signs only matter for compound expressions
01546             size_t newr[] = {(size_t)data[0], l->hash, (size_t)l->sign, r->hash, (size_t)r->sign};
01547             hash = hash_range(newr, 5);
01548         }
01549     }
01550 } NODE;
01551
01552 struct NodeHash {

```

```

01553     size_t operator()(const NODE* n) const {
01554         return n->hash; // already computed
01555     }
01556 };
01557
01558 struct NodeEq {
01559     bool operator()(const NODE* lhs, const NODE* rhs) const {
01560         return lhs->cmp(rhs) == 0;
01561     }
01562 };
01563
01564 NODE* buildTree(vector<WORD> &tree) {
01565     //MesPrint ("*** [%s] Starting CSEE topdown", thetime_str().c_str());
01566
01567     // allocate spaces for the tree, cannot be more nodes than tree size
01568     NODE* ar = (NODE*)Mallocl(tree.size() * sizeof(NODE), "CSE tree");
01569     NODE* c = 0;
01570     unsigned int curIndex = 0;
01571
01572     stack<NODE*> st;
01573     for (int i=0; i<(int)tree.size(); ) {
01574         c = ar + curIndex;
01575         new (c) NODE(&tree[i]); // placement new
01576         curIndex++;
01577
01578         if (tree[i]==SYMBOL || tree[i] == SNUMBER) {
01579             // extract the sign to a new class member
01580             if (tree[i] == SNUMBER) {
01581                 c->sign = SGN(tree[i + tree[i + 1] -1]);
01582             }
01583
01584             c->calcHash();
01585             st.push(c);
01586             i+=tree[i+1];
01587         } else {
01588             c->r = st.top(); st.pop();
01589             c->l = st.top(); st.pop();
01590
01591             // filter *1 and *-1
01592             // TODO: also multiply if there are two numbers?
01593             if (c->data[0] == OPER_MUL) {
01594                 NODE* ch[] = {c->r, c->l};
01595                 for (int j = 0; j < 2; j++)
01596                     if (ch[j]->data[0] == SNUMBER && ch[j]->data[1] == 5 && ch[j]->data[2]==1 &&
ch[j]->data[3]==1) {
01597                         ch[(j+1)%2]->sign *= ch[j]->sign; // transfer sign
01598                         c = ch[(j+1)%2];
01599                         break;
01600                     }
01601             }
01602
01603             c->calcHash();
01604             st.push(c);
01605             i++;
01606         }
01607     }
01608
01609     // TODO: reallocate to smaller size? Could save memory
01610     //MesPrint("Memory difference: %d vs %d", curIndex, tree.size());
01611
01612     // we want to make the root of the tree the first element
01613     // so that we can easily free the array.
01614     // we swap the first element with the root
01615     // we need to change the pointer in the operator node that has this element as a child
01616     // TODO: check performance
01617     for (unsigned int i = 0; i < curIndex; i++) {
01618         if (ar[i].l == ar) ar[i].l = st.top();
01619         if (ar[i].r == ar) ar[i].r = st.top();
01620     }
01621
01622     swap(ar[0], *st.top());
01623     return ar;
01624 }
01625
01626 int count_operators_cse_topdown (vector<WORD> &tree) {
01627     typedef unordered_set<NODE*, NodeHash, NodeEq> nodeset;
01628     nodeset ID;
01629
01630     // reserve lots of space, to prevent later rehashes
01631     // TODO: what if this is too large? make a parameter?
01632     ID.rehash(mcts_expr_score * 2);
01633
01634     int numinstr = 0;
01635
01636     NODE* root = buildTree(tree);
01637
01638     stack<NODE*> stack;

```

```

01639     stack.push(root);
01640     while (!stack.empty())
01641     {
01642         NODE* c = stack.top();
01643         stack.pop();
01644
01645         if (c->data[0] == SYMBOL) {
01646             if (c->data[3] > 1) {
01647                 std::pair<nodeset::iterator, bool> suc = ID.insert(c);
01648                 if (suc.second) { // new
01649                     if (c->data[3] == 2)
01650                         numinstr++;
01651                     else
01652                         numinstr += (int)floor(log(c->data[3])/log(2.0)) + popcount(c->data[3]) - 1;
01653                 }
01654             }
01655         } else {
01656             if (c->data[0] != SNUMBER) {
01657                 // operator
01658                 std::pair<nodeset::iterator, bool> suc = ID.insert(c);
01659                 if (suc.second) {
01660                     stack.push(c->r);
01661                     stack.push(c->l);
01662
01663                     // ignore OPER_COMMA
01664                     if (c->data[0] == OPER_MUL || c->data[0] == OPER_ADD)
01665                         numinstr++;
01666                 }
01667             }
01668         }
01669     }
01670
01671     //MesPrint ("*** [%s] Stopping CSEE", thetime_str().c_str());
01672     M_free(root, "CSE tree");
01673
01674     return numinstr;
01675 }
01676
01677 /*
01678 #] count_operators_cse_topdown :
01679 #[ simulated_annealing :
01680 */
01681 vector<WORD> simulated_annealing() {
01682     float minT = AO.Optimize.saMinT.fval;
01683     float maxT = AO.Optimize.saMaxT.fval;
01684     float T = maxT;
01685     float coolrate = pow(minT / maxT, 1 / (float)AO.Optimize.saIter);
01686
01687     GETIDENTITY;
01688
01689     // create a valid state where FACTORSYMBOL/SEPARATESYMBOL remains first
01690     vector<WORD> state = occurrence_order(optimize_expr, false);
01691     int startindex = 0;
01692     if ((state.size() > 0 && state[0] == SEPARATESYMBOL) || (state.size() > 1 && state[1] ==
FACTORSYMBOL)) startindex++;
01693     if (state.size() > 1 && state[1] == FACTORSYMBOL) startindex++;
01694
01695     my_random_shuffle(BHEAD state.begin() + startindex, state.end()); // start from random scheme
01696
01697     vector<WORD> tree = Horner_tree(optimize_expr, state);
01698     int curscore = count_operators_cse_topdown(tree);
01699
01700     // clean poly_vars, that are allocated by Horner_tree
01701     AN.poly_num_vars = 0;
01702     M_free(AN.poly_vars, "poly_vars");
01703
01704     std::vector<WORD> best = state; // best state
01705     int bestscore = curscore;
01706
01707     if (startindex == (int)state.size() || (int)state.size() - startindex < 2) {
01708         return best;
01709     }
01710
01711     for (int o = 0; o < AO.Optimize.saIter; o++) {
01712         int inda = iranf(BHEAD state.size() - startindex) + startindex;
01713         int indb = iranf(BHEAD state.size() - startindex) + startindex;
01714
01715         if (inda == indb) {
01716             continue;
01717         }
01718
01719         swap(state[inda], state[indb]); // swap works best for Horner
01720
01721         vector<WORD> tree = Horner_tree(optimize_expr, state);
01722         int newscore = count_operators_cse_topdown(tree);
01723
01724         // clean poly_vars, that are allocated by Horner_tree

```

```

01725     AN.poly_num_vars = 0;
01726     M_free(AN.poly_vars, "poly_vars");
01727
01728     if (newscore <= curscore || 2.0 * wranf(BHEAD0) / (float)(UWORD) (-1) < exp((curscore -
newscore) / T)) {
01729         curscore = newscore;
01730
01731         if (curscore < bestscore) {
01732             bestscore = curscore;
01733             best = state;
01734         }
01735     } else {
01736         swap(state[inda], state[indb]);
01737     }
01738
01739 #ifdef DEBUG_SA
01740     MesPrint("Score at step %d: %d", o, curscore);
01741 #endif
01742     T *= coolrate;
01743 }
01744
01745 #ifdef DEBUG_SA
01746     MesPrint("Simulated annealing score: %d", bestscore);
01747 #endif
01748
01749     return best;
01750 }
01751
01752 /*
01753     #] simulated_annealing :
01754     #[ printpstree :
01755 */
01756
01757 /*
01758 // print MCTS tree with LaTeX/pstricks (for analysis)
01759 void printpstree_rec (tree_node x, string pre="") {
01760
01761     if (x.num_visits==1) {
01762         MesPrint("%s\\TR{%d}", pre.c_str(), x.var);
01763     }
01764     else {
01765         MesPrint("%s\\pstree%s{\\TR{%d}}{", pre.c_str(),
01766             pre==" "? "[nodesep=0, levels=40]": "",
01767             x.var);
01768         for (int i=0; i<(int)x.chlds.size(); i++)
01769             if (x.chlds[i].num_visits>0)
01770                 printpstree_rec(x.chlds[i], pre+" ");
01771         MesPrint("%s", pre.c_str());
01772     }
01773 }
01774
01775 void printpstree () {
01776     // draw tree with pstricks
01777     MesPrint ("\\documentclass{article}");
01778     MesPrint ("\\usepackage{pstricks,pst-node,pst-tree,graphicx}");
01779     MesPrint ("\\begin{document}");
01780     MesPrint ("\\scalebox{0.02}{");
01781     printpstree_rec(mcts_root, " ");
01782     MesPrint ("}");
01783     MesPrint ("\\end{document}");
01784 }
01785 */
01786
01787 /*
01788     #] printpstree :
01789     #[ find_Horner_MCTS_expand_tree :
01790 */
01791
01792 /*
01793     #[ next_MCTS_scheme :
01794 */
01795
01796 // find a Horner scheme to be used for the next simulation
01797 inline static void next_MCTS_scheme (PHEAD vector<WORD> *porder, vector<WORD> *pscheme,
vector<tree_node> *ppath) {
01798
01799     vector<WORD> &order = *porder;
01800     vector<WORD> &schemev = *pscheme;
01801     vector<tree_node> &path = *ppath;
01802     int depth = 0, nchild0;
01803     float slide_down_factor = 1.0;
01804
01805     order.clear();
01806     path.clear();
01807
01808     // MCTS step I: select
01809     tree_node *select = &mcts_root;

```

```

01841     path.push_back(select);
01842     nchild0 = select->childs.size();
01843     while (select->childs.size() > 0) {
01844         // add virtual loss
01845         select->num_visits++;
01846         select->sum_results+=mcts_expr_score;
01847
01848         //-----
01849         switch ( AO.Optimize.mctsdecaymode ) {
01850             case 1: // Based on http://arxiv.org/abs/arXiv:1312.0841
01851                 slide_down_factor = 1.0-(1.0*AT.optimitimes)/(1.0*AO.Optimize.mctsnumexpand);
01852                 break;
01853             case 2: // This gives a bit more cleanup time at the end.
01854                 if ( 2*AT.optimitimes < AO.Optimize.mctsnumexpand ) {
01855                     slide_down_factor = 1.0*(AO.Optimize.mctsnumexpand-2*AT.optimitimes);
01856                     slide_down_factor /= 1.0*AO.Optimize.mctsnumexpand;
01857                 }
01858                 else {
01859                     slide_down_factor = 0.0001;
01860                 }
01861                 break;
01862             case 3: // depth dependent factor combined with case 1
01863                 float dd = 1.0-(1.0*depth)/(1.0*nchild0);
01864                 slide_down_factor = 1.0-(1.0*AT.optimitimes)/(1.0*AO.Optimize.mctsnumexpand);
01865                 if ( dd <= 0.000001 ) slide_down_factor = 1.0;
01866                 else slide_down_factor /= dd;
01867                 if ( slide_down_factor > 1.0 ) slide_down_factor = 1.0;
01868                 break;
01869         }
01870         //-----
01871
01872         #ifdef DEBUG_MCTS
01873             MesPrint("select %d",select->var);
01874         #endif
01875
01876         // find most-promising node
01877         double best=0;
01878         tree_node *next=NULL;
01879         for (vector<tree_node>::iterator p=select->childs.begin(); p<select->childs.end(); p++) {
01880             double score;
01881             if (p->num_visits >= 1) {
01882
01883                 // there are results calculated, so select with the UCT formula
01884                 score = mcts_expr_score / (p->sum_results/p->num_visits) +
01885                 //-----
01886                     slide_down_factor *
01887                 //-----
01888                     2 * AO.Optimize.mctsconstant.fval * sqrt(2*log(select->num_visits) /
01889                     p->num_visits);
01890
01891                 #ifdef DEBUG_MCTS
01892                     printf("%d: %.2lf [x=%.2lf n=%d fin=%i]\n",p->var,score,mcts_expr_score /
01893                     (p->sum_results/p->num_visits),
01894                     p->num_visits,p->finished?1:0);
01895                     fflush(stdout);
01896                 #endif
01897             }
01898             else {
01899                 // no results yet, so select this node by setting score=infinite
01900                 score = 1e100;
01901             }
01902             #ifdef DEBUG_MCTS
01903                 printf("%d: inf\n",p->var); fflush(stdout);
01904             #endif
01905
01906             // update best candidate
01907             if (!p->finished && score>best) {
01908                 best=score;
01909                 next=*p;
01910             }
01911
01912             // if no node is found, this node is finished
01913             if (next==NULL) {
01914                 select->finished=true;
01915                 break;
01916             }
01917
01918             // traverse down the tree
01919             select = next;
01920             path.push_back(select);
01921             order.push_back(select->var);
01922             depth++;
01923         }
01924
01925         // MCTS step II: expand

```

```

01926
01927 #ifdef DEBUG_MCTS
01928     MesPrint("expand %d",select->var);
01929 #endif
01930
01931     // variables used so far
01932     set<WORD> var_used;
01933
01934     for (int i=0; i<(int)order.size(); i++)
01935         var_used.insert(ABS(order[i])-1);
01936
01937     // if this a new node, create node and add children
01938     if (!select->finished && select->childs.size()==0) {
01939         tree_node new_node(select->var);
01940         int sign = (order.size() == 0) ? 1 : SGN(order.back());
01941         for (int i=0; i<(int)mcts_vars.size(); i++)
01942             if (!var_used.count(mcts_vars[i])) {
01943                 new_node.childs.push_back(tree_node(sign*(mcts_vars[i]+1)));
01944                 if (AO.Optimize.hornerdirection==O_FORWARDANDBACKWARD)
01945                     new_node.childs.push_back(tree_node(-sign*(mcts_vars[i]+1)));
01946             }
01947         my_random_shuffle(BHEAD new_node.childs.begin(), new_node.childs.end());
01948
01949         // here locking is necessary, since operator=(tree_node) is a
01950         // non-atomic operation (using pointers makes this lock obsolete)
01951         LOCK(optimize_lock);
01952         *select = new_node;
01953         UNLOCK(optimize_lock);
01954     }
01955     // set finished if necessary
01956     if (select->childs.size()==0)
01957         select->finished = true;
01958
01959     // add virtual loss of number of operators in original expression
01960     select->num_visits++;
01961     select->sum_results+=mcts_expr_score;
01962
01963     // MCTS step III: simulation
01964
01965     // create complete Horner scheme
01966     deque<WORD> scheme;
01967
01968     for (int i=0; i<(int)mcts_vars.size(); i++)
01969         if (!var_used.count(mcts_vars[i]))
01970             scheme.push_back(mcts_vars[i]);
01971     my_random_shuffle(BHEAD scheme.begin(), scheme.end());
01972
01973     for (int i=(int)order.size()-1; i>=0; i--) {
01974         if (order[i] > 0)
01975             scheme.push_front(order[i]-1);
01976         else
01977             scheme.push_back(-order[i]-1);
01978     }
01979
01980     // add FACTORSYMBOL/SEPARATESYMBOL is necessary
01981     if (mcts_factorized)
01982         scheme.push_front(FACTORSYMBOL);
01983     if (mcts_separated)
01984         scheme.push_front(SEPARATESYMBOL);
01985
01986     // Horner scheme as a vector
01987     schemev = vector<WORD>(scheme.begin(), scheme.end());
01988
01989 }
01990
01991 /*
01992     #] next_MCTS_scheme :
01993     #[ try_MCTS_scheme :
01994 */
01995
01996 // count the number of operators in the given Horner scheme
01997 inline static void try_MCTS_scheme (PHEAD const vector<WORD> &scheme, int *pnum_oper) {
01998
01999     // do Horner, CSE and count the number of operators
02000     vector<WORD> tree = Horner_tree(optimize_expr, scheme);
02001     //vector<WORD> instr = generate_instructions(tree, true);
02002     //int num_oper = count_operators(instr);
02003     //int num_oper2 = count_operators_cse(tree);
02004     //int num_oper2 = count_operators_cse_topdown(tree);
02005     //MesPrint("%d %d", num_oper, num_oper2);
02006
02007     int num_oper = count_operators_cse_topdown(tree);
02008
02009     // clean poly_vars, that is allocated by Horner_tree
02010     AN.poly_num_vars = 0;
02011     M_free(AN.poly_vars,"poly_vars");
02012

```

```

02013     *pnnum_oper = num_oper;
02014
02015 }
02016
02017 /*
02018     #] try_MCTS_scheme :
02019     #[ update_MCTS_scheme :
02020 */
02021
02022 // update the best score and the search tree
02023 inline static void update_MCTS_scheme (int num_oper, const vector<WORD> &scheme, vector<tree_node *>
    *ppath) {
02024
02025     vector<tree_node *> &path = *ppath;
02026
02027     // update the (global) list of best Horner scheme
02028     if ((int)mcts_best_schemes.size() < AO.Optimize.mctsnumkeep ||
02029         (--mcts_best_schemes.end()->first > num_oper) {
02030         // here locking is necessary, for otherwise best_schemes may
02031         // become corrupted; lock can be prevented if each thread keeps
02032         // track of it's own list and those lists are merged in the end,
02033         // but this seems not useful to implement
02034         LOCK(optimize_lock);
02035         mcts_best_schemes.insert(make_pair(num_oper,scheme));
02036         if ((int)mcts_best_schemes.size() > AO.Optimize.mctsnumkeep)
02037             mcts_best_schemes.erase(--mcts_best_schemes.end());
02038         UNLOCK(optimize_lock);
02039     }
02040
02041     // MCTS step IV: backpropagate
02042
02043     // add number of operator and subtract mcts_expr_score, which
02044     // behaves as a "virtual loss"
02045     for (vector<tree_node *>::iterator p=path.begin(); p<path.end(); p++)
02046         (*p)->sum_results += num_oper - mcts_expr_score;
02047
02048 }
02049
02050 /*
02051     #] update_MCTS_scheme :
02052 */
02053
02054 void find_Horner_MCTS_expand_tree () {
02055
02056     #ifdef DEBUG
02057         MesPrint ("*** [%s, w=%w] CALL: find_Horner_MCTS_expand_tree", thetime_str().c_str());
02058     #endif
02059
02060     GETIDENTITY;
02061
02062     // the order for the Horner scheme up to the selected node, with signs
02063     // indicating forward or backward.
02064     vector<WORD> order;
02065
02066     // complete Horner scheme
02067     vector<WORD> scheme;
02068
02069     // path to the selected node
02070     vector<tree_node *> path;
02071
02072     // the number of operations obtained by the simulation
02073     int num_oper;
02074
02075     next_MCTS_scheme(BHEAD &order, &scheme, &path);
02076     try_MCTS_scheme(BHEAD scheme, &num_oper);
02077     #ifdef DEBUG_MCTS
02078         // Actually "order" is needed only for this debug output.
02079         MesPrint ("[%a] -> [%a] -> %d", order.size(), &order[0], scheme.size(), &scheme[0], num_oper);
02080     #endif
02081     update_MCTS_scheme(num_oper, scheme, &path);
02082
02083     #ifdef DEBUG
02084         MesPrint ("*** [%s, w=%w] DONE: find_Horner_MCTS_expand_tree(%a-> %d)",
02085             thetime_str().c_str(), scheme.size(), &scheme[0], num_oper);
02086     #endif
02087
02088 }
02089
02090 /*
02091     #] find_Horner_MCTS_expand_tree :
02092     #[ PF_find_Horner_MCTS_expand_tree :
02093 */
02094 #ifdef WITHMPI
02095
02096 // To remember which task is assigned to each slave. A task is represented by
02097 // a pair of a Horner scheme and the selected path for the scheme.
02098 // The index range is from 1 to PF.numtasks-1.

```



```

02099 vector<pair<vector<WORD>, vector<tree_node *> > > PF_opt_MCTS_tasks;
02100
02101 // Initialization.
02102 void PF_find_Horner_MCTS_expand_tree_master_init () {
02103     PF_opt_MCTS_tasks.resize(PF.numtasks);
02104     for (int i = 1; i < PF.numtasks; i++) {
02105         pair<vector<WORD>, vector<tree_node *> > &p = PF_opt_MCTS_tasks[i];
02106         p.first.clear();
02107         p.second.clear();
02108     }
02109 }
02110
02111 }
02112
02113 // Wait for an idle slave and return the process number.
02114 int PF_find_Horner_MCTS_expand_tree_master_next () {
02115     // Find an idle slave.
02116     int next;
02117     PF_Receive(PF_ANY_SOURCE, PF_OPT_MCTS_MSGTAG, &next, NULL);
02118
02119     // Check if the slave had a task.
02120     pair<vector<WORD>, vector<tree_node *> > &p = PF_opt_MCTS_tasks[next];
02121     if (!p.first.empty()) {
02122         // If so, update the result.
02123         int num_oper;
02124         PF_Unpack(&num_oper, 1, PF_INT);
02125         update_MCTS_scheme(num_oper, p.first, &p.second);
02126
02127         // Clear the task.
02128         p.first.clear();
02129         p.second.clear();
02130     }
02131     return next;
02132 }
02133
02134 }
02135
02136 // The main function on the master.
02137 void PF_find_Horner_MCTS_expand_tree_master () {
02138     #ifdef DEBUG
02139         MesPrint ("*** [%s, w=%w] CALL: PF_find_Horner_MCTS_expand_tree_master", thetime_str().c_str());
02140     #endif
02141     vector<WORD> order;
02142     vector<WORD> scheme;
02143     vector<tree_node *> path;
02144
02145     next_MCTS_scheme(BHEAD &order, &scheme, &path);
02146
02147     // Find an idle slave.
02148     int next = PF_find_Horner_MCTS_expand_tree_master_next();
02149
02150     // Send a new task to the slave.
02151     PF_PrepareLongSinglePack();
02152     int len = scheme.size();
02153     PF_LongSinglePack(&len, 1, PF_INT);
02154     PF_LongSinglePack(&scheme[0], len, PF_WORD);
02155     PF_LongSingleSend(next, PF_OPT_MCTS_MSGTAG);
02156
02157     // Remember the task.
02158     pair<vector<WORD>, vector<tree_node *> > &p = PF_opt_MCTS_tasks[next];
02159     p.first = scheme;
02160     p.second = path;
02161
02162     #ifdef DEBUG
02163         MesPrint ("*** [%s, w=%w] DONE: PF_find_Horner_MCTS_expand_tree_master", thetime_str().c_str());
02164     #endif
02165 }
02166
02167 // Wait for all the slaves to finish their tasks.
02168 void PF_find_Horner_MCTS_expand_tree_master_wait () {
02169     #ifdef DEBUG
02170         MesPrint ("*** [%s, w=%w] CALL: PF_find_Horner_MCTS_expand_tree_master_wait",
02171             thetime_str().c_str());
02172     #endif
02173
02174     // Wait for all the slaves.
02175     for (int i = 1; i < PF.numtasks; i++) {
02176         int next = PF_find_Horner_MCTS_expand_tree_master_next();
02177         // Send a null task.
02178         PF_PrepareLongSinglePack();
02179         int len = 0;
02180         PF_LongSinglePack(&len, 1, PF_INT);
02181     }
02182 }

```

```

02185         PF_LongSingleSend(next, PF_OPT_MCTS_MSGTAG);
02186     }
02187
02188 #ifdef DEBUG
02189     MesPrint ("*** [%s, w=%w] DONE: PF_find_Horner_MCTS_expand_tree_master_wait",
02190         thetime_str().c_str());
02191 #endif
02192 }
02193
02194 // The main function on the slaves.
02195 void PF_find_Horner_MCTS_expand_tree_slave () {
02196
02197 #ifdef DEBUG
02198     MesPrint ("*** [%s, w=%w] CALL: PF_find_Horner_MCTS_expand_tree_slave", thetime_str().c_str());
02199 #endif
02200
02201     vector<WORD> scheme;
02202
02203     {
02204         // Send the first message to the master, which indicates I am idle.
02205         PF_PreparePack();
02206         int dummy = 0;
02207         PF_Pack(&dummy, 1, PF_INT);
02208         PF_Send(MASTER, PF_OPT_MCTS_MSGTAG);
02209     }
02210
02211     for (;;) {
02212         // Get a task from the master.
02213         PF_LongSingleReceive(MASTER, PF_OPT_MCTS_MSGTAG, NULL, NULL);
02214
02215         // Length of the task.
02216         int len;
02217         PF_LongSingleUnpack(&len, 1, PF_INT);
02218
02219         // No task remains.
02220         if (len == 0) break;
02221
02222         // Perform the given task.
02223         scheme.resize(len);
02224         PF_LongSingleUnpack(&scheme[0], len, PF_WORD);
02225         int num_oper;
02226         try_MCTS_scheme(scheme, &num_oper);
02227
02228         // Send the result to the master.
02229         PF_PreparePack();
02230         PF_Pack(&num_oper, 1, PF_INT);
02231         PF_Send(MASTER, PF_OPT_MCTS_MSGTAG);
02232     }
02233
02234 #ifdef DEBUG
02235     MesPrint ("*** [%s, w=%w] DONE: PF_find_Horner_MCTS_expand_tree_slave", thetime_str().c_str());
02236 #endif
02237 }
02238
02239 #endif
02240
02241 /*
02242 #] PF_find_Horner_MCTS_expand_tree :
02243 #[ find_Horner_MCTS :
02244 */
02245
02246 //vector<vector<WORD>> find_Horner_MCTS () {
02247 void find_Horner_MCTS () {
02248
02249 #ifdef WITHMPI
02250     if (PF.me != MASTER) {
02251         if (PF.numtasks <= 1) return;
02252         PF_find_Horner_MCTS_expand_tree_slave();
02253         return;
02254     }
02255 #endif
02256
02257 #ifdef DEBUG
02258     MesPrint ("*** [%s, w=%w] CALL: find_Horner_MCTS", thetime_str().c_str());
02259 #endif
02260
02261     GETIDENTITY;
02262
02263     LONG start_time = TimeWallClock(1);
02264
02265     // initialize the used global variables
02266     mcts_expr_score = count_operators(optimize_expr);
02267     mcts_root = tree_node();
02268
02269     // extract all symbols from the expression
02270     set<WORD> var_set;

```

```

02279     for (WORD *t=optimize_expr; *t!=0; t+=*t)
02280         if (t[1] == SYMBOL)
02281             for (int i=3; i<t[2]; i+=2)
02282                 var_set.insert(t[i]);
02283
02284     // check for factorized/separated expression and make sure that
02285     // FACTORSYMBOL/SEPARATESYMBOL isn't included in the MCTS
02286     mcts_factorized = var_set.count(FACTORSYMBOL);
02287     if (mcts_factorized) var_set.erase(FACTORSYMBOL);
02288     mcts_separated = var_set.count(SEPARATESYMBOL);
02289     if (mcts_separated) var_set.erase(SEPARATESYMBOL);
02290
02291     mcts_vars = vector<WORD>(var_set.begin(), var_set.end());
02292     optimize_num_vars = (int)mcts_vars.size();
02293     // initialize MCTS tree root
02294     for (int i=0; i<(int)mcts_vars.size(); i++) {
02295         if (AO.Optimize.hornerdirection != O_BACKWARD)
02296             mcts_root.childs.push_back(tree_node(+ (mcts_vars[i]+1)));
02297         if (AO.Optimize.hornerdirection != O_FORWARD)
02298             mcts_root.childs.push_back(tree_node(- (mcts_vars[i]+1)));
02299     }
02300     my_random_shuffle(BHEAD mcts_root.childs.begin(), mcts_root.childs.end());
02301
02302     #if defined(WITHMPI)
02303         PF_find_Horner_MCTS_expand_tree_master_init();
02304     #endif
02305
02306     // initialize a potential variable mctsconstant scheme.
02307     AT.optentimes = 0;
02308
02309     // call expand_tree until it is called "mctsnunexpand" times, the
02310     // time limit is reached or the tree is fully finished
02311     for (int times=0; times<AO.Optimize.mctsnunexpand && !mcts_root.finished &&
02312          (AO.Optimize.mctsttimelimit==0 || (TimeWallClock(1)-start_time)/100 <
02313          AO.Optimize.mctsttimelimit);
02314          times++) {
02315         AT.optentimes = times;
02316         // call expand_tree routine depending on threading mode
02317         #if defined(WITHPTHREADS)
02318             if (AM.totalnumberofthreads > 1)
02319                 find_Horner_MCTS_expand_tree_threaded();
02320             else
02321                 #elif defined(WITHMPI)
02322                     if (PF.numtasks > 1)
02323                         PF_find_Horner_MCTS_expand_tree_master();
02324                     else
02325                         #endif
02326                             find_Horner_MCTS_expand_tree();
02327
02328         // if TForm or ParForm, wait for everyone to finish
02329         #ifdef WITHPTHREADS
02330             MasterWaitAll();
02331         #endif
02332         #ifdef WITHMPI
02333             PF_find_Horner_MCTS_expand_tree_master_wait();
02334         #endif
02335         #ifdef DEBUG
02336             MesPrint ("*** [%s, w=%w] DONE: find_Horner_MCTS", thetime_str().c_str());
02337         #endif
02338     }
02339 }
02340
02341 /*
02342     #] find_Horner_MCTS :
02343     #[ merge_operators :
02344 */
02345
02403 vector<WORD> merge_operators (const vector<WORD> &all_instr, bool move_coeff) {
02404
02405     #ifdef DEBUG_MORE
02406         MesPrint ("*** [%s, w=%w] CALL: merge_operators", thetime_str().c_str());
02407     #endif
02408
02409     // get starting positions of instructions
02410     vector<const WORD *> instr;
02411     const WORD *tbegin = &*all_instr.begin();
02412     const WORD *tend = tbegin+all_instr.size();
02413     // copy all instructions to temp space. There will be n of them in instr.
02414     for (const WORD *t=tbegin; t!=tend; t+=*(t+2)) {
02415         instr.push_back(t);
02416     }
02417     int n = instr.size();
02418
02419     // find parents and number of parents of instructions
02420     vector<int> par(n), numpar(n,0);
02421     for (int i=0; i<n; i++) par[i]=i;

```

```

02422
02423     for (int i=0; i<n; i++) {
02424         for (const WORD *t=instr[i]+OPTHEAD; *t!=0; t+=*t) {
02425             if ( (*t+1)==EXTRASymbol && *t!=1+ABS(*t+*t-1)) ) {
02426                 // extra symbol t[3] is referred to in instr i.
02427                 par[*t+3]=i;
02428                 numpar[*t+3]++;
02429
02430                 // if coefficient isn't +/-1 or power > 1, increase numpar,
02431                 // so that this is not merged
02432                 if (*t+*t-3)!=1 || *t+*t-2)!=1 || ABS(*t+*t-1)!=3) numpar[*t+3]++;
02433                 if (*t+4)>1) numpar[*t+3]++;
02434             }
02435         }
02436     }
02437     // determine which instructions to merge
02438     stack<int> s;
02439     s.push(n-1);
02440     vector<bool> vis(n,false);
02441
02442     while (!s.empty()) {
02443         int i=s.top(); s.pop();
02444         if (vis[i]) continue;
02445         vis[i]=true;
02446
02447         for (const WORD *t=instr[i]+OPTHEAD; *t!=0; t+=*t)
02448             if ( (*t+1)==EXTRASymbol && *t!=1+ABS(*t+*t-1)) )
02449                 s.push(*t+3);
02450
02451         // condition: one parent and equal operator as parent
02452         if (numpar[i]==1 && *(instr[i+1])==*(instr[par[i]+1]))
02453             par[i] = par[par[i]]; // The expr into which we subst par[i] to get i
02454         else
02455             par[i] = i;
02456     }
02457
02458     // merge instructions into new instructions
02459     vector<WORD> newinstr;
02460
02461     // stack of new expressions, all 0-indexed
02462     stack<WORD> new_expr;
02463     new_expr.push(n-1);
02464     vis = vector<bool>(n,false);
02465
02466     // skip empty equations (might be introduced by greedy optimizations)
02467     vector<bool> skip(n,false), skipcoeff(n,false);
02468     for (int i=0; i<n; i++)
02469         if (*(instr[i]+OPTHEAD) == 0) skip[i]=skipcoeff[i]=true;
02470
02471     // for renumbering merged parents
02472     vector<int> renum_par(n);
02473     for (int i=0; i<n; i++)
02474         renum_par[i]=i;
02475
02476     while (!new_expr.empty()) {
02477         int x = new_expr.top(); new_expr.pop();
02478         if (vis[x]) continue;
02479         vis[x] = true;
02480
02481         // find all instructions with parent=x and copy their arguments
02482         // into a new expression
02483         bool first_copy=true;
02484         int lenidx = newinstr.size()+2;
02485
02486         // 1-indexed, since signs may occur
02487         stack<WORD> this_expr;
02488         this_expr.push(x+1);
02489
02490         while (!this_expr.empty()) {
02491             // pop from stack, determine expr.nr and sign
02492             int i = this_expr.top(); this_expr.pop();
02493             int sign = SGN(i);
02494             i = ABS(i)-1;
02495             for (const WORD *t=instr[i]+OPTHEAD; *t!=0; t+=*t) { // terms in i
02496                 // don't copy a term if:
02497                 // (1) skip=true, since then it's merged into the parent
02498                 // (2) extrasymbol with parent=x, because its children should be copied
02499                 // (3) coefficient with skipcoeff=true, since it's already copied
02500                 bool copy = !skip[i];
02501                 if (*t!=1+ABS(*t+*t-1)) && *t+1==EXTRASymbol) {
02502                     if (par[*t+3] == x) {
02503                         // parent of term refers to x. we push it with its sign if no skip is true
02504                         // and the sign of the expr.
02505                         this_expr.push(sign * (skip[i]||skipcoeff[i] ? 1 : SGN(*t+*t-1))) *
02506                             (1+*t+3));
02507                     if (*(instr[i+1]) == OPER_MUL) sign=1;

```

```

02508         copy=false;
02509     }
02510     else {
02511         new_expr.push(*(t+3));
02512     }
02513 }
02514
02515 if (*t == 1+ABS(*(t+t-1)) && skipcoeff[i]) {
02516     copy=false;
02517 }
02518
02519 if (copy) {
02520     // first term, so add header
02521     if (first_copy) {
02522         newinstr.push_back(renum_par[x]); // expr.nr.
02523         newinstr.push_back(*(instr[x]+1)); // operator
02524         newinstr.push_back(3); // length OPTHEAD?
02525         first_copy=false;
02526     }
02527
02528     // copy term and adjust sign
02529     int thislenidx = newinstr.size();
02530     newinstr.insert(newinstr.end(), t, t+t); // Put the whole term in newinstr
02531     newinstr.back() *= sign;
02532     if (*(instr[i]+1) == OPER_MUL) sign=1;
02533     newinstr[thislenidx] += *t;
02534
02535     // check for moving coefficients up
02536     // necessary condition: MUL-expression with 1 parent
02537     if (move_coeff && *t!=1+ABS(*(t+t-1)) && *(instr[i]+1)!=OPER_COMMA &&
02538         *(t+1)==EXTRASymbol && numpar[*(t+3)]=1 && *(instr[*(t+3)+1]==OPER_MUL)
02539 {
02540
02541     // coefficient is always the first term (that's how Horner+generate works)
02542     const WORD *t1 = instr[*(t+3)]+OPTHEAD;
02543     const WORD *t2 = t1+t1;
02544
02545     if (*t1 == 1+ABS(*(t1+t1-1))) {
02546         // t1 pointer to a coefficient, so move it
02547
02548         // remove old coefficient of 1
02549         WORD *t3 = &newinstr.end(); //
02550         int sign2 = SGN(t3[-1]); //
02551         newinstr.erase(newinstr.end()-3, newinstr.end());
02552         // count number of arguments; if it is 2 move the (extra)symbol too
02553         int numargs=0;
02554         for (const WORD *tt=t1; *tt!=0; tt+=*tt) {
02555             numargs++;
02556         }
02557         if (numargs==2 && *(t2+4)==1) {
02558             // replace (extra)symbol
02559             newinstr[newinstr.size()-4] = *(t2+1);
02560             newinstr[newinstr.size()-3] = *(t2+2);
02561             newinstr[newinstr.size()-2] = *(t2+3);
02562             newinstr[newinstr.size()-1] = *(t2+4);
02563             sign2 *= SGN(*(t2+t2-1)); // was t2[7]
02564
02565             // ignore this expression from now on
02566             skip[*(t+3)]=true;
02567             if (*(t2+1)==EXTRASymbol)
02568                 renum_par[*(t+3)] = *(t2+3);
02569         }
02570         else {
02571             // otherwise, ignore coefficient from now on
02572             // we need to collect the signs of the terms
02573             // first and set them to one. This was forgotten
02574             // before. Gave occasional errors.
02575             if ( numargs > 2 || ( numargs == 2 && t2[4] > 1 ) ) {
02576                 for (WORD *tt=(WORD *)t2; *tt!=0; tt+=*tt) {
02577                     if ( tt[*tt-1] < 0 ) {
02578                         tt[*tt-1] = -tt[*tt-1];
02579                         sign2 = -sign2;
02580                     }
02581                 }
02582             }
02583             skipcoeff[*(t+3)]=true;
02584         }
02585
02586         // add new coefficient
02587         newinstr.insert(newinstr.end(), t1+1, t1+t1);
02588         newinstr.back() *= sign2;
02589         newinstr[thislenidx] += ABS(newinstr.back()) - 3;
02590         newinstr[thislenidx] += ABS(newinstr.back()) - 3;
02591     }
02592 }
02593 }

```

```

02594     }
02595
02596     // if something has been copied, add trailing zero
02597     if (!first_copy) {
02598         newinstr.push_back(0);
02599         newinstr[lenidx]++;
02600     }
02601 }
02602
02603 // renumber the expressions to 0,1,2,...; only keep expressions with
02604 // skip=false which are their own parent after a renumbering in case
02605 // of moved coefficients
02606
02607 // find renumber scheme
02608 vector<int> renum(n,-1);
02609 int next=0;
02610 for (int i=0; i<n; i++)
02611     if (!skip[i] && renum_par[par[i]]==i) renum[renum_par[i]]=next++;
02612
02613 // find new instruction index
02614 tbegin = &*newinstr.begin();
02615 tend = tbegin+newinstr.size();
02616 for (const WORD *t=tbegin; t!=tend; t+=*(t+2))
02617     instr[renum[*t]] = t;
02618
02619 // renumbering expressions and sort them lexicographically (in
02620 // new_instr they are in the preorder of the expression tree)
02621 vector<WORD> sortinstr;
02622 for (int i=0; i<next; i++) {
02623     int idx = sortinstr.size();
02624     sortinstr.insert(sortinstr.end(), instr[i], instr[i]+(instr[i]+2));
02625
02626     sortinstr[idx] = renum[sortinstr[idx]];
02627
02628     // renumber content of an expression
02629     for (WORD *t2=&sortinstr[idx]+3; *t2!=0; t2+=*t2)
02630         if (*t2!=1+ABS(*t2+*t2-1)) && *(t2+1)==EXTRASYMBOL
02631             *(t2+3) = renum[*t2+3];
02632 }
02633
02634 #ifdef DEBUG_MORE
02635     MesPrint ("*** [%s, w=%w] DONE: merge_operators", thetime_str().c_str());
02636 #endif
02637
02638     return sortinstr;
02639 }
02640
02641 /*
02642 #] merge_operators :
02643 #[ class Optimization :
02644 */
02645
02646 class optimization {
02647 public:
02648     int type, arg1, arg2, improve;
02649     vector<WORD> coeff;
02650     vector<int> eqnidxs;
02651
02652     bool operator< (const optimization &a) const {
02653         if (arg1 != a.arg1) return arg1 < a.arg1;
02654         if (arg2 != a.arg2) return arg2 < a.arg2;
02655         if (type != a.type) return type < a.type;
02656         return coeff < a.coeff;
02657     }
02658 };
02659
02660 /*
02661 #] class Optimization :
02662 #[ find_optimizations :
02663 */
02664
02665 vector<optimization> find_optimizations (const vector<WORD> &instr) {
02666
02667 #ifdef DEBUG_GREEDY
02668     MesPrint ("*** [%s, w=%w] CALL: find_optimizations", thetime_str().c_str());
02669 #endif
02670
02671 // #[ Startup :
02672 // the resulting vector of optimizations
02673 vector<optimization> res;
02674
02675 // a map to count the improvement of an optimization; the
02676 // improvement is stored as a vector<int> with equation numbers
02677 map<optimization, vector<int> > cnt;
02678
02679 // a map to identify coefficients
02680 map<vector<int>,int> idx_coeff;

```

```

02714
02715     const WORD *ebegin = &*instr.begin();
02716     const WORD *eend = ebegin+instr.size();
02717
02718     for (int optim_type=0; optim_type<=4; optim_type++) {
02719
02720         cnt.clear();
02721         idx_coeff.clear();
02722
02723         optimization optim;
02724         optim.type = optim_type;
02725         // #] Startup :
02726
02727         // #[ type 0 : find optimizations of the form z=x^n (optim.type==0)
02728         if (optim_type == 0) {
02729             for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {
02730                 if (*(e+1) != OPER_MUL) continue;
02731                 for (const WORD *t=e+3; *t!=0; t+=*t) {
02732                     if (*t == ABS(*(t+*t-1))+1) continue;
02733                     if (*(t+4) > 1) {
02734                         optim.arg1 = (*(t+1)==SYMBOL ? 1 : -1) * (*(t+3) + 1);
02735                         optim.arg2 = *(t+4);
02736                         cnt[optim].push_back(e-ebegin);
02737                     }
02738                 }
02739             }
02740         }
02741         // #] type 0 :
02742         // #[ type 1 : find optimizations of the form z=x*y (optim.type==1)
02743         if (optim_type == 1) {
02744             for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {
02745                 if (*(e+1) != OPER_MUL) continue;
02746
02747                 for (const WORD *t1=e+3; *t1!=0; t1+=*t1) {
02748                     if (*t1 == ABS(*(t1+*t1-1))+1) continue;
02749                     int x1 = (*(t1+1)==SYMBOL ? 1 : -1) * (*(t1+3) + 1);
02750
02751                     for (const WORD *t2=t1+*t1; *t2!=0; t2+=*t2) {
02752                         if (*t2 == ABS(*(t2+*t2-1))+1) continue;
02753                         int x2 = (*(t2+1)==SYMBOL ? 1 : -1) * (*(t2+3) + 1);
02754
02755                         if (*(t1+4) == *(t2+4)) {
02756                             optim.arg1 = x1;
02757                             optim.arg2 = x2;
02758                             if (optim.arg1 > optim.arg2)
02759                                 swap(optim.arg1, optim.arg2);
02760
02761                             if (*(t1+4) == 1)
02762                                 cnt[optim].push_back(e-ebegin);
02763                             else {
02764                                 // E=x^n*y^n -> z=x*y; E=z^n is double improvement
02765                                 cnt[optim].push_back(e-ebegin);
02766                                 cnt[optim].push_back(e-ebegin);
02767                             }
02768                         }
02769                     }
02770                 }
02771             }
02772         }
02773         // #] type 1 :
02774         // #[ type 2 : find optimizations of the form z=c*x (optim.type==2)
02775         if (optim_type == 2) {
02776             for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {
02777
02778                 if (*(e+1) == OPER_ADD) {
02779                     // in ADD-equation
02780
02781                     for (const WORD *t=e+3; *t!=0; t+=*t) {
02782                         if (*t == ABS(*(t+*t-1))+1) continue;
02783
02784                         if (*(t+4)==1) {
02785                             if (ABS(*(t+*t-1))==3 && *(t+*t-2)==1 && *(t+*t-3)==1) continue;
02786
02787                             optim.coeff = vector<WORD>(t+*t-ABS(*(t+*t-1)), t+*t);
02788                             optim.coeff.back() = ABS(optim.coeff.back());
02789
02790                             optim.arg1 = (*(t+1)==SYMBOL ? 1 : -1) * (*(t+3) + 1);
02791                             optim.arg2 = 0;
02792
02793                             cnt[optim].push_back(e-ebegin);
02794                         }
02795                     }
02796                 }
02797                 else if (*(e+1) == OPER_MUL) {
02798                     // in MUL-equation
02799                     optim.coeff.clear();
02800

```

```

02801         for (const WORD *t=e+3; *t!=0; t+=*t)
02802             if (*t == ABS(* (t+*t-1))+1) {
02803                 if (ABS(* (t+*t-1))==3 && * (t+*t-2)==1 && * (t+*t-3)==1) continue;
02804                 optim.coeff = vector<WORD>(t+*t-ABS(* (t+*t-1)), t+*t);
02805                 optim.coeff.back() = ABS(optim.coeff.back());
02806             }
02807
02808         if (!optim.coeff.empty())
02809             for (const WORD *t=e+3; *t!=0; t+=*t) {
02810                 if (*t == ABS(* (t+*t-1))+1) continue;
02811                 if (* (t+4) != 1) continue;
02812                 optim.arg1 = (* (t+1)==SYMBOL ? 1 : -1) * (* (t+3) + 1);
02813                 optim.arg2 = 0;
02814                 cnt[optim].push_back(e-ebegin);
02815             }
02816     }
02817 }
02818 }
02819 // #] type 2 :
02820 // #[ type 3 : find optimizations of the form z=x+c (optim.type==3)
02821     if (optim_type == 3) {
02822         for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {
02823
02824             if (* (e+1) != OPER_ADD) continue;
02825
02826             optim.coeff.clear();
02827
02828             for (const WORD *t=e+3; *t!=0; t+=*t)
02829                 if (*t == ABS(* (t+*t-1))+1) {
02830                     optim.coeff = vector<WORD>(t+*t-ABS(* (t+*t-1)), t+*t);
02831                 }
02832
02833             if (!optim.coeff.empty()) {
02834                 for (const WORD *t=e+3; *t!=0; t+=*t) {
02835                     if (*t == ABS(* (t+*t-1))+1) continue;
02836                     if (ABS(* (t+*t-1))!=3 || * (t+*t-2)!=1 || * (t+*t-3)!=1) continue;
02837                     if (* (t+*t-1)==-3) optim.coeff.back() *= -1;
02838
02839                     optim.arg1 = (* (t+1)==SYMBOL ? 1 : -1) * (* (t+3) + 1);
02840                     optim.arg2 = 0;
02841                     cnt[optim].push_back(e-ebegin);
02842                     if (* (t+*t-1)==-3) optim.coeff.back() *= -1;
02843                 }
02844             }
02845         }
02846     }
02847 // #] type 3 :
02848 // #[ type 4,5 : find optimizations of the form z=x+y or z=x-y (optim.type==4 or 5)
02849     if (optim_type == 4) {
02850         for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {
02851             if (* (e+1) != OPER_ADD) continue;
02852
02853             for (const WORD *t1=e+3; *t1!=0; t1+=*t1) {
02854                 if (*t1 == ABS(* (t1+*t1-1))+1) continue;
02855                 int x1 = (* (t1+1)==SYMBOL ? 1 : -1) * (* (t1+3) + 1);
02856
02857                 for (const WORD *t2=t1+*t1; *t2!=0; t2+=*t2) {
02858                     if (*t2 == ABS(* (t2+*t2-1))+1) continue;
02859                     int x2 = (* (t2+1)==SYMBOL ? 1 : -1) * (* (t2+3) + 1);
02860
02861                     int sign1 = SGN(* (t1+*t1-1));
02862                     int sign2 = SGN(* (t2+*t2-1));
02863
02864                     if (BigLong((UWORD *)t1+5, ABS(* (t1+*t1-1))-1, (UWORD *)t2+5,
02865 ABS(* (t2+*t2-1))-1) == 0) {
02866
02867                         optim.type = (sign1 * sign2 == 1 ? 4 : 5); // optimization type
02868                         optim.arg1 = x1;
02869                         optim.arg2 = x2;
02870                         if (optim.arg1 > optim.arg2) {
02871                             swap(optim.arg1, optim.arg2);
02872                         }
02873
02874                         if (ABS(* (t1+*t1-1))==3 && * (t1+*t1-2)==1 && * (t1+*t1-3)==1)
02875                             cnt[optim].push_back(e-ebegin);
02876                         else {
02877                             // E=2x+2y -> z=x+y; E=2z is improvement bby itself
02878                             cnt[optim].push_back(e-ebegin);
02879                             cnt[optim].push_back(e-ebegin);
02880                         }
02881                     }
02882                 }
02883             }
02884         }
02885 // #] type 4,5 :
02886 // #[ add :

```



```

02887
02888 // add optimizations with positive improvement to the result
02889 for (map<optimization, vector<int> >::iterator i=cnt.begin(); i!=cnt.end(); i++) {
02890     int improve = i->second.size() - 1;
02891     if (improve > 0) {
02892         res.push_back(i->first);
02893         res.back().improve = improve;
02894         res.back().eqnidxs = i->second;
02895
02896         // remove duplicates, that were add to get the correct improvement
02897         res.back().eqnidxs.erase(unique(res.back().eqnidxs.begin(), res.back().eqnidxs.end()),
res.back().eqnidxs.end());
02898     }
02899 }
02900 }
02901 // #] add :
02902
02903 #ifdef DEBUG_GREEDY
02904     MesPrint ("*** [%s, w=%w] DONE: find_optimizations", thetime_str().c_str());
02905 #endif
02906
02907     return res;
02908 }
02909
02910 /*
02911     #] find_optimizations :
02912     #[ do_optimization :
02913 */
02914
02931 bool do_optimization (const optimization optim, vector<WORD> &instr, int newid) {
02932
02933 // #] Debug code :
02934 #ifdef DEBUG_GREEDY
02935     if (optim.type==0)
02936         MesPrint ("*** [%s, w=%w] CALL: do_optimization(improve=%d, %c%d^%d)",
02937 thetime_str().c_str(), optim.improve,
02938 optim.arg1>0?'x':'z', ABS(optim.arg1)-1, optim.arg2);
02939     else if (optim.type==1 || optim.type==4)
02940         MesPrint ("*** [%s, w=%w] CALL: do_optimization(improve=%d, %c%d%c%c%d)",
02941 thetime_str().c_str(), optim.improve,
02942 optim.arg1>0?'x':'z', ABS(optim.arg1)-1,
02943 optim.type==1 ? '*' : optim.type==4 ? '+' : '-',
02944 optim.arg2>0?'x':'z', ABS(optim.arg2)-1);
02945     else {
02946         WORD n = optim.coeff.back()/2;
02947         UBYTE num[BITSinWORD*ABS(n)], den[BITSinWORD*ABS(n)];
02948         PrtLong((UWORD *)&optim.coeff[0], n, num);
02949         PrtLong((UWORD *)&optim.coeff[ABS(n)], ABS(n), den);
02950         MesPrint ("*** [%s, w=%w] CALL: do_optimization(improve=%d, %c%d%c%s/%s)",
02951 thetime_str().c_str(), optim.improve,
02952 optim.arg1>0?'x':'z', ABS(optim.arg1)-1,
02953 optim.type==2 ? '*' : '+', num, den);
02954     }
02955 #endif
02956 // #] Debug code :
02957
02958     bool substituted = false;
02959     WORD *ebegin = &instr.begin();
02960
02961 // #] type 0 : substitution of the form z=x^n (optim.type==0)
02962     if (optim.type == 0) {
02963
02964         int vartypex = optim.arg1>0 ? SYMBOL : EXTRASYMBOL;
02965         int varnumx = ABS(optim.arg1) - 1;
02966         int n = optim.arg2;
02967
02968         for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
02969             WORD *e = ebegin + optim.eqnidxs[i];
02970             if (*(e+1) != OPER_MUL) continue;
02971
02972             // scan through equation
02973             for (WORD *t=e+3; *t!=0; t+=*t) {
02974                 if (*t == ABS(*(t+*t-1))+1) continue;
02975                 if (*(t+1)==vartypex &&
02976                     *(t+3)==varnumx &&
02977                     *(t+4) % n == 0) {
02978
02979                     // substitute
02980                     *(t+1) = EXTRASYMBOL;
02981                     *(t+3) = newid;
02982                     *(t+4) /= n;
02983
02984                     substituted = true;
02985                 }
02986             }
02987         }
02988

```

```

02989         if (!substituted) {
02990 #ifdef DEBUG_GREEDY
02991     MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=false", thetime_str().c_str(),
optim.improve);
02992 #endif
02993         return false;
02994     }
02995
02996     // add extra equation (Tnew = x^n)
02997     instr.push_back(newid); // eqn.nr
02998     instr.push_back(OPER_MUL); // operator
02999     instr.push_back(12); // total length
03000     instr.push_back(8); // term length
03001     instr.push_back(vartypex); // (extra)symbol
03002     instr.push_back(4); // symbol length
03003     instr.push_back(varnumx); // symbol id
03004     instr.push_back(n); // power
03005     instr.push_back(1); // coefficient 1
03006     instr.push_back(1); // coefficient 1
03007     instr.push_back(3); // trailing 0
03008     instr.push_back(0); // trailing 0
03009 }
03010 // #] type 0 :
03011 // #[ type 1 : substitution of the form z=x*y (optim.type==1)
03012 if (optim.type == 1) {
03013
03014     int vartypex = optim.arg1>0 ? SYMBOL : EXTRASYMBOL;
03015     int varnumx = ABS(optim.arg1) - 1;
03016     int vartypey = optim.arg2>0 ? SYMBOL : EXTRASYMBOL;
03017     int varnumy = ABS(optim.arg2) - 1;
03018
03019     for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
03020         WORD *e = ebegin + optim.eqnidxs[i];
03021         if (*(e+1) != OPER_MUL) continue;
03022
03023         // scan through equation
03024         int powx=0, powy=0;
03025         for (WORD *t=e+3; *t!=0; t+=*t) {
03026             if (*t == ABS(*(t+*t-1))+1) continue;
03027             if (*(t+1)==vartypex && *(t+3)==varnumx) powx = *(t+4);
03028             if (*(t+1)==vartypey && *(t+3)==varnumy) powy = *(t+4);
03029         }
03030
03031         // substitute if found
03032         if (powx>0 && powy>0 && powx==powy) {
03033
03034             WORD sign = 1;
03035             WORD *newt = e+3;
03036
03037             for (WORD *t=e+3; *t!=0; t+=*t) {
03038                 int dt=*t;
03039
03040                 if (*t == ABS(*(t+*t-1))+1 ||
03041                     (!(*(t+1)==vartypex && *(t+3)==varnumx) &&
03042                      !(*(t+1)==vartypey && *(t+3)==varnumy))) {
03043                     memmove(newt, t, *t*sizeof(WORD));
03044                     newt += dt;
03045                 }
03046                 else {
03047                     sign *= SGN(*(t+*t-1));
03048                 }
03049
03050                 t+=dt;
03051             }
03052
03053             *newt++ = 8; // term length
03054             *newt++ = EXTRASYMBOL; // extrasymbol
03055             *newt++ = 4; // symbol length
03056             *newt++ = newid; // symbol id
03057             *newt++ = powx; // power
03058             *newt++ = 1;
03059             *newt++ = 1; // coefficient +/-1
03060             *newt++ = 3*sign;
03061             *newt++ = 0; // trailing 0
03062
03063             substituted = true;
03064         }
03065     }
03066
03067     if (!substituted) {
03068 #ifdef DEBUG_GREEDY
03069     MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=false", thetime_str().c_str(),
optim.improve);
03070 #endif
03071         return false;
03072     }
03073 }

```

```

03074         // add extra equation (Tnew = x*y)
03075         instr.push_back(newid); // eqn.nr
03076         instr.push_back(OPER_MUL); // operator
03077         instr.push_back(20); // total length
03078         instr.push_back(8); // LHS length
03079         instr.push_back(vartypex); // (extra)symbol
03080         instr.push_back(4); // symbol length
03081         instr.push_back(varnumx); // symbol id
03082         instr.push_back(1); // power 1
03083         instr.push_back(1);
03084         instr.push_back(1); // coefficient 1
03085         instr.push_back(3);
03086         instr.push_back(8); // RHS length
03087         instr.push_back(vartypey); // (extra)symbol
03088         instr.push_back(4); // symbol length
03089         instr.push_back(varnumy); // symbol id
03090         instr.push_back(1); // power 1
03091         instr.push_back(1);
03092         instr.push_back(1); // coefficient 1
03093         instr.push_back(3);
03094         instr.push_back(0); // trailing 0
03095     }
03096     // #] type 1 :
03097     // #[ type 2 : substitution of the form z=c*x (optim.type==2)
03098
03099     if (optim.type == 2) {
03100         int vartype = optim.arg1>0 ? SYMBOL : EXTRASYMBOL;
03101         int varnum = ABS(optim.arg1) - 1;
03102
03103         WORD ncoeff = optim.coeff.back();
03104
03105         // scan through equations
03106         for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
03107             WORD *e = ebegin + optim.eqnidxs[i];
03108
03109             if (*(e+1) == OPER_ADD) {
03110                 // scan through ADD-equation
03111                 for (WORD *t=e+3; *t!=0; t+=*t) {
03112
03113                     if (*t == ABS(*(t+*t-1))+1) continue;
03114
03115                     if (*(t+1)==vartype && ABS(*(t+3))==varnum && *(t+4)==1 &&
03116                         BigLong((UWORD *) &optim.coeff[0], ncoeff-1,
03117                             (UWORD *) t+*t-ABS(*(t+*t-1)), ABS(*(t+*t-1))-1) == 0) {
03118                         // substitute
03119
03120                         int sign = SGN(*(t+*t-1));
03121
03122                         WORD *tend = t;
03123                         while (*tend!=0) tend+=*tend;
03124                         WORD nmove = tend - t - *t;
03125                         memmove(t, t+*t, nmove*sizeof(WORD));
03126                         t += nmove;
03127
03128                         *t++ = 8; // term length
03129                         *t++ = EXTRASYMBOL; // (extra)symbol
03130                         *t++ = 4; // symbol length
03131                         *t++ = newid; // symbol id
03132                         *t++ = 1; // power of 1
03133                         *t++ = 1;
03134                         *t++ = 1; // coefficient of +/-1
03135                         *t++ = 3 * sign;
03136                         *t++ = 0; // trailing 0
03137
03138                         substituted = true;
03139                         break;
03140                     }
03141                 }
03142             }
03143             else if (*(e+1) == OPER_MUL) {
03144
03145                 bool coeff_match=false, var_match=false;
03146                 int sign = 1;
03147
03148                 // scan through MUL-equation
03149                 for (WORD *t=e+3; *t!=0; t+=*t) {
03150                     if (*t == ABS(*(t+*t-1))+1 && BigLong((UWORD *) &optim.coeff[0], ncoeff-1,
03151                         (*t+*t-ABS(*(t+*t-1)), ABS(*(t+*t-1))-1) == 0) {
03152                         coeff_match = true;
03153                         sign *= SGN(*(t+*t-1));
03154                     }
03155                     else if (*(t+1)==vartype && ABS(*(t+3))==varnum && *(t+4)==1) {
03156                         var_match = true;
03157                         sign *= SGN(*(t+*t-1));
03158                     }
03159                 }

```

```

03160
03161         // substitute if found
03162         if (coeff_match && var_match) {
03163
03164             WORD *newt = e+3;
03165
03166             for (WORD *t=e+3; *t!=0;) {
03167
03168                 int dt=*t;
03169
03170                 if (*t!=ABS(*(t+*t-1))+1 && !(*t+1)==vartype && ABS(*(t+3))==varnum &&
03171                     *(t+4)==1) {
03172                     memmove(newt, t, dt*sizeof(WORD));
03173                     newt += dt;
03174                 }
03175                 t+=dt;
03176             }
03177
03178             *newt++ = 8;           // term length
03179             *newt++ = EXTRASYMBOL; // extrasymbol
03180             *newt++ = 4;           // symbol length
03181             *newt++ = newid;       // symbol id
03182             *newt++ = 1;           // power of 1
03183             *newt++ = 1;
03184             *newt++ = 1;           // coefficient of +/-1
03185             *newt++ = 3 * sign;
03186             *newt++ = 0;           // trailing 0
03187
03188             substituted = true;
03189         }
03190     }
03191 }
03192
03193     if (!substituted) {
03194 #ifdef DEBUG_GREEDY
03195         MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=false", thetime_str().c_str(),
03196             optim.improve);
03197 #endif
03198         return false;
03199     }
03200
03201     // add extra equation (Tnew = c*y)
03202     instr.push_back(newid);           // eqn.nr
03203     instr.push_back(OPER_ADD);        // operator
03204     instr.push_back(9+ABS(ncoeff));   // total length
03205     instr.push_back(5+ABS(ncoeff));   // term length
03206     instr.push_back(vartype);         // (extra)symbol
03207     instr.push_back(4);               // symbol length
03208     instr.push_back(varnum);          // symbol id
03209     instr.push_back(1);               // power of 1
03210     for (int i=0; i<ABS(ncoeff); i++) // coefficient
03211         instr.push_back(optim.coeff[i]);
03212     instr.push_back(0);               // trailing 0
03213 }
03214 // #] type 2 :
03215 // #[ type 3 : substitution of the form z=x+c (optim.type==3)
03216 if (optim.type == 3) {
03217     int vartype = optim.arg1>0 ? SYMBOL : EXTRASYMBOL;
03218     int varnum = ABS(optim.arg1) - 1;
03219
03220     WORD ncoeff = optim.coeff.back();
03221
03222     // scan through equation
03223     for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
03224         WORD *e = ebegin + optim.eqnidxs[i];
03225
03226         if (*(e+1) != OPER_ADD) continue;
03227
03228         int coeff_match=0, var_match=0;
03229
03230         for (WORD *t=e+3; *t!=0; t+=*t) {
03231             if (*t == ABS(*(t+*t-1))+1 && BigLong((UWORD *) &optim.coeff[0], ABS(ncoeff)-1,
03232                 *(t+*t-ABS(*(t+*t-1)), ABS(*(t+*t-1))-1) == 0)
03233                 coeff_match = SGN(ncoeff) * SGN(*(t+*t-1));
03234             else if (*t+1==vartype && ABS(*(t+3))==varnum && *(t+4)==1)
03235                 var_match = SGN(*(t+7));
03236         }
03237
03238         // substitute if found (x+c and -x-c and matches)
03239         if (coeff_match * var_match == 1) {
03240             WORD *newt = e+3;
03241
03242             for (WORD *t=e+3; *t!=0;) {
03243                 int dt=*t;

```

```

03244         if (*t!=ABS(*t+*t-1))+1 && !(*t+1)==vartype && ABS(*t+3)==varnum &&
    *(t+4)==1) {
03245             memmove(newt, t, dt*sizeof(WORD));
03246             newt += dt;
03247         }
03248         t+=dt;
03249     }
03250
03251     *newt++ = 8;           // term length
03252     *newt++ = EXTRASYMBOL; // extrasymbol
03253     *newt++ = 4;           // symbol length
03254     *newt++ = newid;       // symbol id
03255     *newt++ = 1;           // power of 1
03256     *newt++ = 1;
03257     *newt++ = 1;           // coefficient of +/-1
03258     *newt++ = 3*coeff_match;
03259     *newt++ = 0;           // trailing zero
03260
03261     substituted = true;
03262 }
03263 }
03264
03265     if (!substituted) {
03266 #ifdef DEBUG_GREEDY
03267         MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=false", thetime_str().c_str(),
    optim.improve);
03268 #endif
03269         return false;
03270     }
03271
03272     // add extra equation (Tnew = x+c)
03273     instr.push_back(newid);           // eqn.nr
03274     instr.push_back(OPER_ADD);        // operator
03275     instr.push_back(13+ABS(ncoeff)); // total length
03276     instr.push_back(8);               // x-term length
03277     instr.push_back(vartype);         // (extra)symbol
03278     instr.push_back(4);               // symbol length
03279     instr.push_back(varnum);          // symbol id
03280     instr.push_back(1);               // power of 1
03281     instr.push_back(1);
03282     instr.push_back(1);               // coefficient of 1
03283     instr.push_back(3);
03284     instr.push_back(ABS(ncoeff)+1); // c-term length
03285     for (int i=0; i<ABS(ncoeff); i++) // coefficient
03286         instr.push_back(optim.coeff[i]);
03287     instr.push_back(0);               // trailing zero
03288 }
03289 // #] type 3 :
03290 // #[ type 4,5 : substitution of the form z=x+y or z=x-y (optim.type=4 or 5)
03291 if (optim.type >= 4) {
03292
03293     int vartypex = optim.arg1>0 ? SYMBOL : EXTRASYMBOL;
03294     int varnumx  = ABS(optim.arg1) - 1;
03295     int vartypey = optim.arg2>0 ? SYMBOL : EXTRASYMBOL;
03296     int varnumy  = ABS(optim.arg2) - 1;
03297
03298     // scan through equations
03299     for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
03300         WORD *e = ebegin + optim.eqnidxs[i];
03301
03302         if (*(e+1) != OPER_ADD) continue;
03303
03304         const WORD *coeffx=NULL, *coeffy=NULL;
03305         WORD ncoeffx=0,ncoeffy=0;
03306
03307         // looks for terms
03308         for (WORD *t=e+3; *t!=0; t+=*t) {
03309             if (*t == ABS(*t+*t-1))+1 continue; // constant
03310             if (*t+1)==vartypex && *t+3==varnumx && *t+4==1) {
03311                 coeffx = t+5;
03312                 ncoeffx = *(t+*t-1);
03313             }
03314             if (*t+1)==vartypey && *t+3==varnumy && *t+4==1) {
03315                 coeffy = t+5;
03316                 ncoeffy = *(t+*t-1);
03317             }
03318         }
03319
03320         // check signs (type=4: x+y and -x-y, type=5: x-y and -x+y) ??????
03321         // check signs (type=4: x+y, type=5: x-y) !!!!!!!!!!!
03322         if (SGN(ncoeffx) * SGN(ncoeffy) * (optim.type==4 ? 1 : -1) == 1) {
03323             // check absolute value of coefficients
03324             if (BigLong((UWORD *)coeffx, ABS(ncoeffx)-1, (UWORD *)coeffy, ABS(ncoeffy)-1) == 0) {
03325                 // substitute
03326                 vector<WORD> coeff(coeffx, coeffx+ABS(ncoeffx));
03327
03328                 WORD *newt = e+3;

```

```

03329 /*
03330 if ( optim.type == 5 ) {
03331     while ( *newt ) newt+=*newt;
03332     int i = (newt - e) - 3;
03333     MesPrint(" < %a",i,e+3);
03334     newt = e+3;
03335 }
03336 */
03337         for (WORD *t=e+3; *t!=0;) {
03338             int dt=*t;
03339             if (*t == ABS(*(t+*t-1))+1 ||
03340                 (!(*t+1)==vartypex && *(t+3)==varnumx) &&
03341                 !(*t+1)==vartypey && *(t+3)==varnumy)) {
03342                 memmove(newt, t, dt*sizeof(WORD));
03343                 newt += dt;
03344             }
03345             t+=dt;
03346         }
03347
03348         *newt++ = 5 + ABS(ncoeffx);           // term length
03349         *newt++ = EXTRASYMBOL;               // extrasymbol
03350         *newt++ = 4;                         // symbol length
03351         *newt++ = newid;                     // symbol id
03352         *newt++ = 1;                         // power of 1
03353         for (int j=0; j<ABS(ncoeffx); j++) // coefficient
03354             *newt++ = coeff[j];
03355         *newt++ = 0;                         // trailing 0
03356         substituted = true;
03357 /*
03358 if ( optim.type == 5 ) {
03359     int i = (newt - e) - 4;
03360     MesPrint(" > %a",i,e+3);
03361 }
03362 */
03363     }
03364 }
03365 }
03366
03367     if (!substituted) {
03368 #ifdef DEBUG_GREEDY
03369         MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=false", thetime_str().c_str(),
03370             optim.improve);
03371 #endif
03372         return false;
03373 }
03374 /*
03375 if ( optim.type == 5 )
03376     MesPrint ("improve=%d, %c%d%c%c%d", optim.improve,
03377         optim.arg1>0?'x':'Z', ABS(optim.arg1)-1,
03378         optim.type==1?'*': optim.type==4?'+' : '- ',
03379         optim.arg2>0?'x':'Z', ABS(optim.arg2)-1);
03380 */
03381         // add extra equation (Tnew = x+/-y)
03382         instr.push_back(newid);           // eqn.nr
03383         instr.push_back(OPER_ADD);        // operator
03384         instr.push_back(20);              // total length
03385         instr.push_back(8);              // term length
03386         instr.push_back(vartypex);        // (extra)symbol
03387         instr.push_back(4);              // symbol length
03388         instr.push_back(varnumx);         // symbol id
03389         instr.push_back(1);              // power of 1
03390         instr.push_back(1);
03391         instr.push_back(1);              // coefficient of 1
03392         instr.push_back(8);              // term length
03393         instr.push_back(vartypey);        // (extra)symbol
03394         instr.push_back(4);              // symbol length
03395         instr.push_back(varnumy);         // symbol id
03396         instr.push_back(1);              // power of 1
03397         instr.push_back(1);
03398         instr.push_back(1);              // coefficient of +/-1
03399         instr.push_back(3*(optim.type==4?1:-1));
03400         instr.push_back(0);              // trailing 0
03401     }
03402 // #] type 4,5 :
03403 // #[ trivial : remove trivial equations of the form Zi = +/-Zj
03404 vector<int> renum(newid+1, 0);
03405 bool do_renum=false;
03406
03407 // vector may be moved when it is extended
03408 ebegin = &*instr.begin();
03409
03410 for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
03411     WORD *e = ebegin + optim.eqnidxs[i];
03412     WORD *t = e+3;
03413     if (*t==0) continue;           // empty (removed) equation
03414     if (*(t+*t)!=0) continue;      // more than 1 term

```

```

03415         if (*t!=8) continue;          // term not of correct form
03416         if (*t!=1) continue;          // power != 1
03417         if (*t!=1 || *(t+6)!=1) continue; // coeff != +/-1
03418
03419         // trivial term, so renumber this one
03420         renum[*e] = SGN(*(t+7)) * (*(t+3) + 1);
03421         do_renum = true;
03422
03423         // remove equation
03424         *t=0;
03425     }
03426 // #] trivial :
03427 // #[ renumbering :
03428
03429 // there are renumberings to be done, so loop through all equations
03430 if (do_renum) {
03431     WORD *eend = ebegin+instr.size();
03432
03433     for (WORD *e=ebegin; e!=eend; e+=*(e+2)) {
03434         for (WORD *t=e+3; *t!=0; t+=*t) {
03435             if (*t == ABS(*(t+*t-1))+1) continue;
03436             if (*(t+1)==EXTRASymbol && renum[*t+3]!=0) {
03437                 *(t+*t-1) *= SGN(renum[*t+3]);
03438                 *(t+3) = ABS(renum[*t+3]) - 1;
03439             }
03440         }
03441     }
03442 }
03443 // #] renumbering :
03444
03445 #ifdef DEBUG_GREEDY
03446     MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=true", thetime_str().c_str(),
03447             optim.improve);
03447 #endif
03448
03449     return true;
03450 }
03451
03452 /*
03453 #] do_optimization :
03454 #[ partial_factorize :
03455 */
03456
03480 int partial_factorize (vector<WORD> &instr, int n, int improve) {
03481
03482 #ifdef DEBUG_GREEDY
03483     MesPrint ("*** [%s, w=%w] CALL: partial_factorize (n=%d)", thetime_str().c_str(), n);
03484 #endif
03485
03486     GETIDENTITY;
03487
03488     // get starting positions of instructions
03489     vector<int> instr_idx(n);
03490     WORD *ebegin = &*instr.begin();
03491     WORD *eend = ebegin+instr.size();
03492     for (WORD *e=ebegin; e!=eend; e+=*(e+2)) {
03493         instr_idx[*e] = e - ebegin;
03494     }
03495
03496     // get reference counts
03497 /*
03498 * The next construction replaces the vector construction which is
03499 * rather costly for valgrind (and maybe also in normal running)
03500 */
03501     int nmax = 2*n;
03502     WORD *numpar = (WORD *)Malloc1(nmax*sizeof(WORD), "numpar");
03503     for (int i = 0; i < nmax; i++) numpar[i] = 0;
03504 // vector<WORD> numpar(n);
03505     for (WORD *e=ebegin; e!=eend; e+=*(e+2))
03506         for (WORD *t=e+3; *t!=0; t+=*t) {
03507             if (*t == ABS(*(t+*t-1))+1) continue;
03508             if (*(t+1) == EXTRASymbol) numpar[*t+3]++;
03509         }
03510
03511     // find factorizable expressions
03512     for (int i=0; i<n; i++) {
03513         WORD *e = &*instr.begin() + instr_idx[i];
03514         if (*(e+1) != OPER_ADD) continue;
03515
03516         // count symbol occurrences
03517         map<WORD,WORD> cnt; // 1-indexed, <0:EXTRASymbol, >0:Symbol
03518
03519         for (WORD *t=e+3; *t!=0; t+=*t) {
03520             if (*t==ABS(*(t+*t-1))+1) continue;
03521
03522             // count symbols in t
03523             if (*(t+4)==1)

```

```

03524         cnt[(*(t+1)==SYMBOL ? 1 : -1) * (*(t+3)+1)]++;
03525
03526     // count symbols in extrasymbols of t
03527     if (*(t+1)==EXTRASYMBOL && *(t+4)==1 && numpar[*(t+3)]==1) {
03528         WORD *t2 = &*instr.begin() + instr_idx[*(t+3)];
03529         if (*(t2+1) != OPER_MUL) continue;
03530         for (t2+=3; *t2!=0; t2+=*t2) {
03531             if (*t2 == ABS(*(t2+*t2-1))+1) continue;
03532             if (*(t2+4)==1)
03533                 cnt[(*(t2+1)==SYMBOL ? 1 : -1) * (*(t2+3)+1)]++;
03534         }
03535     }
03536 }
03537
03538 // find most-occurring symbol
03539 WORD x=0, best=0;
03540 for (map<WORD,WORD>::iterator it=cnt.begin(); it!=cnt.end(); it++)
03541     if (it->second > best) { x=it->first; best=it->second; }
03542
03543 // occurrence>=2 and occurrence>improve, so factorize
03544
03545 if (best>=2 && best>improve) {
03546     // initialize new equation (Zi from example above)
03547     vector<WORD> new_eqn;
03548     new_eqn.push_back(n);
03549     new_eqn.push_back(OPER_ADD);
03550     new_eqn.push_back(0); // length
03551
03552     WORD dt;
03553     WORD *newt=e+3;
03554     for (WORD *t=e+3; *t!=0; t+=dt) {
03555         dt = *t;
03556         bool keep=true;
03557
03558         if (*(t!=ABS(*(t+*t-1))+1) {
03559
03560             // factorized symbol is in t itself
03561             if (*(t+4)==1) {
03562                 WORD y = (*(t+1)==SYMBOL ? 1 : -1) * (*(t+3)+1);
03563                 if (y==x) {
03564                     new_eqn.push_back(*t-4);
03565                     new_eqn.insert(new_eqn.end(), t+5, t+dt);
03566                     keep=false;
03567                 }
03568             }
03569
03570             // look in extrasymbol of t with ref.count=1
03571             if (*(t+1)==EXTRASYMBOL && *(t+4)==1 && numpar[*(t+3)]==1) {
03572                 WORD *t2 = &*instr.begin() + instr_idx[*(t+3)];
03573                 if (*(t2+1) == OPER_MUL) {
03574                     bool has_x=false;
03575                     for (t2+=3; *t2!=0; t2+=*t2) {
03576                         if (*t2 == ABS(*(t2+*t2-1))+1) continue;
03577                         WORD y = (*(t2+1)==SYMBOL ? 1 : -1) * (*(t2+3)+1);
03578                         // extrasymbol has factorized symbol
03579                         if (y==x && *(t2+4)==1) {
03580                             has_x=true;
03581                             // copy remaining part
03582                             WORD *tend=t2+*t2;
03583                             WORD sign = SGN(*(tend-1));
03584                             while (*(tend!=0) tend+=*tend;
03585                             int dt2 = tend - (t2+*t2);
03586                             memmove(t2, t2+*t2, (dt2+1)*sizeof(WORD));
03587                             t2 += dt2;
03588                             *(t2-1) *= sign;
03589                             break;
03590                         }
03591                     }
03592                     if (has_x) {
03593                         // extrasymbol has x, so add it to new equation
03594                         keep=false;
03595                         int thisidx=new_eqn.size();
03596                         new_eqn.insert(new_eqn.end(), t, t+dt);
03597                         t2 = &*instr.begin() + instr_idx[*(t+3)] + 3;
03598                         // if becomes trivial, substitute the term
03599                         if (*(t2+*t2)==0) {
03600                             // it's a number
03601                             if (*t2 == ABS(*(t2+*t2-1))+1) {
03602                                 if (ABS(new_eqn[new_eqn.size()-1])==3 &&
new_eqn[new_eqn.size()-2]==1 && new_eqn[new_eqn.size()-3]==1) {
03603                                     // original equation has coefficient of +/-1, so replace
it
03604                                     WORD sign = SGN(new_eqn.back());
03605                                     new_eqn.erase(new_eqn.begin()+thisidx, new_eqn.end());
03606                                     new_eqn.insert(new_eqn.end(), t2, t2+*t2);
03607                                     new_eqn.back() *= sign;
03608                                     *t2 = 0;

```



```

03609     }
03610     else {
03611         // two non-trivial coefficients, so multiply them
03612         // note: untested code (found no way to trigger it)
03613         UWORD *tmp = NumberMalloc("partial_factorize");
03614         WORD ntmp=0;
03615         MulRat(BHEAD (UWORD *)t2+*t2-ABS(* (t2+*t2-1)),
03616 * (t2+*t2-1),
03617 (UWORD
*) &* (new_eqn.end()-ABS(new_eqn.back())) , new_eqn.back(),
tmp, &ntmp);
03618 new_eqn.erase(new_eqn.begin()+thisidx, new_eqn.end());
03619 new_eqn.push_back(ABS(ntmp)+1);
03620 new_eqn.insert(new_eqn.end(), tmp, tmp+ABS(ntmp));
03621 NumberFree(tmp, "partial_factorize");
03622 *t2 = 0;
03623     }
03624 }
03625 else if (* (t2+4)==1) {
03626     // it's a variable
03627     new_eqn.back() *= SGN(* (t2+*t2-1));
03628     new_eqn[thisidx+1] = * (t2+1);
03629     new_eqn[thisidx+2] = * (t2+2);
03630     new_eqn[thisidx+3] = * (t2+3);
03631     new_eqn[thisidx+4] = * (t2+4);
03632     *t2 = 0;
03633 }
03634 }
03635 }
03636 }
03637 }
03638 }
03639 }
03640 // no x, so copy it
03641 if (keep) {
03642     memmove(newt, t, dt*sizeof(WORD));
03643     newt += dt;
03644 }
03645 }
03646 // finalize new equation
03647 new_eqn.push_back(0);
03648 new_eqn[2] = new_eqn.size();
03649 bool empty = newt == e+3;
03650 if ( n+1 >= nmax ) {
03651     int i, newnmax = nmax*2;
03652     WORD *newnumpar = (WORD *) Malloc1(newnmax*sizeof(WORD), "newnumpar");
03653     for ( i = 0; i < n; i++ ) newnumpar[i] = numpar[i];
03654     for ( ; i < newnmax; i++ ) newnumpar[i] = 0;
03655     M_free(numpar, "numpar");
03656     numpar = newnumpar;
03657     nmax = newnmax;
03658 }
03659 // numpar.push_back(0);
03660 n++;
03661 // if original is not empty, add new equation (Zj) to it
03662 // otherwise replace it later
03663 if (!empty) {
03664     *newt++ = 8;
03665     *newt++ = EXTRASYMBOL;
03666     *newt++ = 4;
03667     *newt++ = n;
03668     *newt++ = 1;
03669     *newt++ = 1;
03670     *newt++ = 1;
03671     *newt++ = 3;
03672     *newt++ = 0;
03673 }
03674 // add new equation to instructions
03675 instr_idx.push_back(instr.size());
03676 instr.insert(instr.end(), new_eqn.begin(), new_eqn.end());
03677 // generate another new equation (Zj=x*Zi)
03678 new_eqn.clear();
03679 new_eqn.push_back(n);
03680 new_eqn.push_back(OPER_MUL);
03681 new_eqn.push_back(20);
03682 new_eqn.push_back(8);
03683 // add factorized symbol
03684 if (x>0) {
03685     new_eqn.push_back(SYMBOL);
03686     new_eqn.push_back(4);
03687     new_eqn.push_back(x-1);
03688 }
03689 }
03690 }
03691 }
03692 }
03693 }

```

```

03694         new_eqn.push_back(1);
03695     }
03696     else {
03697         new_eqn.push_back(EXTRASymbol);
03698         new_eqn.push_back(4);
03699         new_eqn.push_back(-x-1);
03700         new_eqn.push_back(1);
03701     }
03702     new_eqn.push_back(1);
03703     new_eqn.push_back(1);
03704     new_eqn.push_back(3);
03705     new_eqn.push_back(8);
03706     // add new equation (Zi)
03707     new_eqn.push_back(EXTRASymbol);
03708     new_eqn.push_back(4);
03709     new_eqn.push_back(n-1);
03710     new_eqn.push_back(1);
03711     new_eqn.push_back(1);
03712     new_eqn.push_back(1);
03713     new_eqn.push_back(3);
03714     new_eqn.push_back(0);
03715
03716     if (!empty) {
03717         // add new equation (Zj) to instructions
03718         instr_idx.push_back(instr.size());
03719         instr.insert(instr.end(), new_eqn.begin(), new_eqn.end());
03720         if (n+1 >= nmax) {
03721             int i, newnmax = nmax*2;
03722             WORD *newnumpar = (WORD *)Malloca1(newnmax*sizeof(WORD), "newnumpar");
03723             for (i = 0; i < n; i++) newnumpar[i] = numpar[i];
03724             for (; i < newnmax; i++) newnumpar[i] = 0;
03725             M_free(numpar, "numpar");
03726             numpar = newnumpar;
03727             nmax = newnmax;
03728         }
03729         // numpar.push_back(0);
03730         n++;
03731     }
03732     else {
03733         // replace e with Zj
03734         e = &*instr.begin() + instr_idx[i];
03735         e[1] = OPER_MUL;
03736         memcpy(e+3, &new_eqn[3], (new_eqn.size()-3)*sizeof(WORD));
03737     }
03738
03739     // decrease i, so this expression is factorized again if possible
03740     i--;
03741 }
03742 }
03743
03744 #ifdef DEBUG_GREEDY
03745     MesPrint ("*** [%s, w=%w] DONE: partial_factorize (n=%d)", thetime_str().c_str(), n);
03746 #endif
03747     M_free(numpar, "numpar");
03748     return n;
03749 }
03750
03751 /*
03752 #] partial_factorize :
03753 #[ optimize_greedy :
03754 */
03755
03774 vector<WORD> optimize_greedy (vector<WORD> instr, LONG time_limit) {
03775
03776 #ifdef DEBUG
03777     int old_num_oper = count_operators(instr);
03778     MesPrint ("*** [%s, w=%w] CALL: optimize_greedy(numoper=%d)",
03779             thetime_str().c_str(), old_num_oper);
03780 #endif
03781
03782     LONG start_time = TimeWallClock(1);
03783
03784     WORD *ebegin = &*instr.begin();
03785     WORD *eend = ebegin+instr.size();
03786
03787     // store final equation, since it must be the last equation later
03788     int final_eqn_idx = 0;
03789     int next_eqn = 0;
03790
03791     for (WORD *e=ebegin; e!=eend; e+=*(e+2)) {
03792         next_eqn = *e + 1;
03793         final_eqn_idx = e-ebegin;
03794     }
03795     // optimize instructions
03796     while (TimeWallClock(1)-start_time < time_limit) {
03797         int old_next_eqn = next_eqn;
03798

```

```

03799         // find optimizations
03800         vector<optimization> optim = find_optimizations(instr);
03801
03802         // add_eqnidxs contains modified equations, which might have to be updated later again
03803         vector<int> add_eqnidxs;
03804
03805         // number of optimizations to do
03806         int num_do_optims = max(AO.Optimize.greedyminnum,
(int)optim.size()*AO.Optimize.greedymaxperc/100);
03807
03808         // if best improvement is one, do all optimizations
03809         int best_improve=0;
03810         for (int i=0; i<(int)optim.size(); i++)
03811             best_improve = max(best_improve, optim[i].improve);
03812         if (best_improve <= 1)
03813             num_do_optims = MAXPOSITIVE;
03814         // do a number of optimizations
03815         while (optim.size() > 0 && num_do_optims-- > 0) {
03816
03817             // find best optimization
03818             int best=0;
03819             best_improve=0;
03820             for (int i=0; i<(int)optim.size(); i++)
03821                 if (optim[i].improve > best_improve) {
03822                     best=i;
03823                     best_improve=optim[i].improve;
03824                 }
03825
03826             // add extra equations
03827             for (int i=0; i<(int)add_eqnidxs.size(); i++)
03828                 optim[best].eqnidxs.push_back(add_eqnidxs[i]);
03829
03830             // do optimization, update next_eqn if successful
03831             int next_idx = instr.size();
03832             if (do_optimization(optim[best], instr, next_eqn)) {
03833                 next_eqn++;
03834                 add_eqnidxs.push_back(next_idx);
03835             }
03836
03837             optim.erase(optim.begin()+best);
03838         }
03839
03840         // partially factorize with improve >= best_improve
03841         next_eqn = partial_factorize(instr, next_eqn, best_improve);
03842
03843         // check whether nothing has changed
03844         if (next_eqn == old_next_eqn) break;
03845     }
03846
03847     // add final equation to the back (must be by definition)
03848     instr.push_back(next_eqn);
03849     instr.insert(instr.end(), instr.begin()+final_eqn_idx+1,
instr.begin()+final_eqn_idx+instr[final_eqn_idx+2]);
03850
03851     // removed original final equation
03852     instr[final_eqn_idx+3] = 0;
03853
03854     // remove extra zeroes
03855     WORD *t = &instr[0];
03856
03857     ebegin = &*instr.begin();
03858     eend = ebegin+instr.size();
03859     int de=0;
03860
03861     for (WORD *e=ebegin; e!=eend; e+=de) {
03862         de = *(e+2);
03863         int n=3;
03864         while (*(e+n) != 0) n+=*(e+n);
03865         n++;
03866         memmove (t, e, n*sizeof(WORD));
03867         *(t+2) = n;
03868         t += n;
03869     }
03870
03871     instr.resize(t - &instr[0]);
03872
03873 #ifdef DEBUG
03874     MesPrint ("*** [%s, w=%w] DONE: optimize_greedy(numoper=%d) : numoper=%d",
03875             thetime_str().c_str(), old_num_oper, count_operators(instr));
03876 #endif
03877
03878     return instr;
03879 }
03880
03881 /*
03882 #] optimize_greedy :
03883 #[ recycle_variables :

```

```

03884 */
03885
03914 vector<WORD> recycle_variables (const vector<WORD> &all_instr) {
03915
03916 #ifdef DEBUG_MORE
03917     MesPrint ("*** [%s, w=%w] CALL: recycle_variables", thetime_str().c_str());
03918 #endif
03919
03920     // get starting positions of instructions
03921     vector<const WORD *> instr;
03922     const WORD *tbegin = &*all_instr.begin();
03923     const WORD *tend = tbegin+all_instr.size();
03924     for (const WORD *t=tbegin; t!=tend; t+=*(t+2))
03925         instr.push_back(t);
03926     int n = instr.size();
03927
03928     // determine with expressions are connected, how many intermediate
03929     // are needed (assuming it's a expression tree instead of a DAG) and
03930     // sort the leaves such that you need a minimal number of variables
03931     vector<int> vars_needed(n);
03932     vector<bool> vis(n,false);
03933     vector<vector<int> > conn(n);
03934
03935     stack<int> s;
03936     s.push(n);
03937
03938     while (!s.empty()) {
03939         int i=s.top(); s.pop();
03940         if (i>0) {
03941             i--;
03942             if (vis[i]) continue;
03943             vis[i]=true;
03944             s.push(-(i+1));
03945
03946             // find all connections
03947             for (const WORD *t=instr[i]+3; *t!=0; t+=*t)
03948                 if (*t!=1+ABS(*(t+*t-1)) && *(t+1)==EXTRASYMBOL) {
03949                     int k = *(t+3);
03950                     conn[i].push_back(k);
03951                     s.push(k+1);
03952                 }
03953         }
03954         else {
03955             i=-i-1;
03956
03957             // sort the childs w.r.t. needed variables
03958             vector<pair<int,int> > need;
03959             for (int j=0; j<(int)conn[i].size(); j++)
03960                 need.push_back(make_pair(vars_needed[conn[i][j]], conn[i][j]));
03961
03962             // keep the comma expression in proper order
03963             if (*(instr[i]+1) != OPER_COMMA)
03964                 sort(need.rbegin(), need.rend());
03965
03966             vars_needed[i] = 1;
03967             for (int j=0; j<(int)need.size(); j++) {
03968                 vars_needed[i] = max(vars_needed[i], need[j].first+j);
03969                 conn[i][j] = need[j].second;
03970             }
03971         }
03972     }
03973
03974     // order the instructions in depth-first order and determine the first
03975     // and last occurrences of variables
03976     vector<int> order, first(n,0), last(n,0);
03977     vis = vector<bool>(n,false);
03978     s.push(n);
03979
03980     while (!s.empty()) {
03981
03982         int i=s.top(); s.pop();
03983
03984         if (i>0) {
03985             i--;
03986             if (vis[i]) continue;
03987             vis[i]=true;
03988             s.push(-(i+1));
03989             for (int j=(int)conn[i].size()-1; j>=0; j--)
03990                 s.push(conn[i][j]+1);
03991         }
03992         else {
03993             i=-i-1;
03994             first[i] = last[i] = order.size();
03995             order.push_back(i);
03996             for (int j=0; j<(int)conn[i].size(); j++) {
03997                 int k = conn[i][j];
03998                 last[k] = max(last[k], first[i]);
03999             }
04000         }
04001     }
04002 }

```

```

03999     }
04000     }
04001 }
04002
04003 // find the renumbering to recycled variables, where at any time the
04004 // lowest-indexed variable that can be used is chosen
04005 int numvar=0;
04006 set<int> var;
04007 vector<int> renum(n);
04008
04009 for (int i=0; i<(int)order.size(); i++) {
04010     for (int j=0; j<(int)conn[order[i]].size(); j++) {
04011         int k = conn[order[i]][j];
04012         if (last[k] == i) var.insert(renum[k]);
04013     }
04014
04015     if (var.empty()) var.insert(numvar++);
04016     renum[order[i]] = *var.begin(); var.erase(var.begin());
04017 }
04018
04019 // put the number of variables used in a preprocessor variable
04020
04021 // generate new instructions with the renumbering
04022 vector<WORD> newinstr;
04023
04024 for (int i=0; i<(int)order.size(); i++) {
04025     int x = order[i];
04026     int j = newinstr.size();
04027     newinstr.insert(newinstr.end(), instr[x], instr[x]+(instr[x]+2));
04028
04029     newinstr[j] = renum[newinstr[j]];
04030
04031     for (WORD *t=&newinstr[j+3]; *t!=0; t+=*t)
04032         if (*t!=1+ABS(*t+*t-1)) && *(t+1)==EXTRASymbol
04033             *(t+3) = renum[*t+3];
04034 }
04035
04036 #ifdef DEBUG_MORE
04037     MesPrint ("*** [%s, w=%w] DONE: recycle_variables", thetime_str().c_str());
04038 #endif
04039
04040     return newinstr;
04041 }
04042
04043 /*
04044 #] recycle_variables :
04045 #[ optimize_expression_given_Horner :
04046 */
04047
04061 void optimize_expression_given_Horner () {
04062
04063     #ifdef DEBUG
04064         MesPrint ("*** [%s, w=%w] CALL: optimize_expression_given_Horner", thetime_str().c_str());
04065     #endif
04066
04067     GETIDENTITY;
04068
04069     // initialize timer
04070     LONG start_time = TimeWallClock(1);
04071     LONG time_limit = 100 * AO.Optimize.greedytimelimit / (AO.Optimize.horner == O_MCTS ?
AO.Optimize.mctsnumkeep : 1);
04072     if (time_limit == 0) time_limit=MAXPOSITIVE;
04073
04074     // pick a Horner scheme from the list
04075     LOCK(optimize_lock);
04076     vector<WORD> Horner_scheme = optimize_best_Horner_schemes.back();
04077     optimize_best_Horner_schemes.pop_back();
04078     UNLOCK(optimize_lock);
04079
04080     // if ( ( AO.Optimize.debugflags&2 ) == 2 ) {
04081     //     MesPrint ("Scheme: %a", Horner_scheme.size(), &(Horner_scheme[0]));
04082     // }
04083
04084     // apply Horner scheme
04085     vector<WORD> tree = Horner_tree(optimize_expr, Horner_scheme);
04086
04087     // generate instructions, eventually with CSE
04088     vector<WORD> instr;
04089
04090     if (AO.Optimize.method == O_CSE || AO.Optimize.method == O_CSEGREEDY)
04091         instr = generate_instructions(tree, true);
04092     else
04093         instr = generate_instructions(tree, false);
04094     if (AO.Optimize.method == O_CSEGREEDY || AO.Optimize.method == O_GREEDY) {
04095         instr = merge_operators(instr, false);
04096         instr = optimize_greedy(instr, time_limit-(TimeWallClock(1)-start_time));
04097         instr = merge_operators(instr, true);
04098     }

```

```

04099     instr = optimize_greedy(instr, time_limit-(TimeWallClock(1)-start_time));
04100 }
04101 instr = merge_operators(instr, true);
04102
04103 // recycle the temporary variables
04104 instr = recycle_variables(instr);
04105
04106 // determine the quality of the code and possibly update the best code
04107 int num_oper = count_operators(instr);
04108
04109 LOCK(optimize_lock);
04110 if (num_oper < optimize_best_num_oper) {
04111     optimize_num_vars = Horner_scheme.size();
04112     optimize_best_num_oper = num_oper;
04113     optimize_best_instr = instr;
04114     optimize_best_vars = vector<WORD>(AN.poly_vars, AN.poly_vars+AN.poly_num_vars);
04115 }
04116 UNLOCK(optimize_lock);
04117
04118 // clean poly_vars, that are allocated by Horner_tree
04119 AN.poly_num_vars = 0;
04120 M_free(AN.poly_vars, "poly_vars");
04121
04122 #ifdef DEBUG
04123     MesPrint ("*** [%s, w=%w] DONE: optimize_expression_given_Horner", thetime_str().c_str());
04124 #endif
04125 }
04126
04127 /*
04128     #] optimize_expression_given_Horner :
04129     #[ PF_optimize_expression_given_Horner :
04130 */
04131 #ifdef WITHMPI
04132
04133 // Initialization.
04134 void PF_optimize_expression_given_Horner_master_init () {
04135     // Nothing to do for now.
04136 }
04137
04138 // Wait for an idle slave and return the process number.
04139 int PF_optimize_expression_given_Horner_master_next () {
04140
04141     // Find an idle slave.
04142     int next;
04143     PF_LongSingleReceive(PF_ANY_SOURCE, PF_OPT_HORNER_MSGTAG, &next, NULL);
04144     return next;
04145 }
04146
04147 // The main function on the master.
04148 void PF_optimize_expression_given_Horner_master () {
04149
04150     #ifdef DEBUG
04151         MesPrint ("*** [%s, w=%w] CALL: PF_optimize_expression_given_Horner_master",
04152             thetime_str().c_str());
04153     #endif
04154
04155     // pick a Horner scheme from the list
04156     vector<WORD> Horner_scheme = optimize_best_Horner_schemes.back();
04157     optimize_best_Horner_schemes.pop_back();
04158
04159     // Find an idle slave.
04160     int next = PF_optimize_expression_given_Horner_master_next();
04161
04162     // Send a new task to the slave.
04163     PF_PrepareLongSinglePack();
04164     int len = Horner_scheme.size();
04165     PF_LongSinglePack(&len, 1, PF_INT);
04166     PF_LongSinglePack(&Horner_scheme[0], len, PF_WORD);
04167     PF_LongSingleSend(next, PF_OPT_HORNER_MSGTAG);
04168
04169     #ifdef DEBUG
04170         MesPrint ("*** [%s, w=%w] DONE: PF_optimize_expression_given_Horner_master",
04171             thetime_str().c_str());
04172     #endif
04173 }
04174
04175 // Wait for all the slaves to finish their tasks.
04176 void PF_optimize_expression_given_Horner_master_wait () {
04177
04178     #ifdef DEBUG
04179         MesPrint ("*** [%s, w=%w] CALL: PF_optimize_expression_given_Horner_master_wait",
04180             thetime_str().c_str());
04181     #endif
04182
04183     // Wait for all the slaves.

```

```

04183     for (int i = 1; i < PF.numtasks; i++) {
04184         int next = PF_optimize_expression_given_Horner_master_next();
04185         // Send a null task.
04186         PF_PrepareLongSinglePack();
04187         int len = 0;
04188         PF_LongSinglePack(&len, 1, PF_INT);
04189         PF_LongSingleSend(next, PF_OPT_HORNER_MSGTAG);
04190     }
04191
04192     // Combine the result from all the slaves.
04193     optimize_best_num_oper = INT_MAX;
04194     for (int i = 1; i < PF.numtasks; i++) {
04195         PF_LongSingleReceive(PF_ANY_SOURCE, PF_OPT_COLLECT_MSGTAG, NULL, NULL);
04196
04197         int len;
04198
04199         // The first integer is the number of operations.
04200         PF_LongSingleUnpack(&len, 1, PF_INT);
04201
04202         if (len >= optimize_best_num_oper) continue;
04203
04204         // Update the best result.
04205         optimize_best_num_oper = len;
04206         PF_LongSingleUnpack(&len, 1, PF_INT);
04207         optimize_best_instr.resize(len);
04208         PF_LongSingleUnpack(&optimize_best_instr[0], len, PF_WORD);
04209         PF_LongSingleUnpack(&len, 1, PF_INT);
04210         optimize_best_vars.resize(len);
04211         PF_LongSingleUnpack(&optimize_best_vars[0], len, PF_WORD);
04212
04213         optimize_num_vars = optimize_best_vars.size(); // TODO
04214     }
04215
04216 #ifdef DEBUG
04217     MesPrint ("*** [%s, w=%w] DONE: PF_optimize_expression_given_Horner_master_wait",
04218             thetime_str().c_str());
04219 #endif
04220 }
04221
04222 // The main function on the slaves.
04223 void PF_optimize_expression_given_Horner_slave () {
04224
04225 #ifdef DEBUG
04226     MesPrint ("*** [%s, w=%w] CALL: PF_optimize_expression_given_Horner_slave",
04227             thetime_str().c_str());
04228 #endif
04229
04230     optimize_best_Horner_schemes.clear();
04231     optimize_best_num_oper = INT_MAX;
04232
04233     int dummy = 0;
04234     int len;
04235
04236     for (;;) {
04237         // Ask the master for the next task.
04238         PF_PrepareLongSinglePack();
04239         PF_LongSinglePack(&dummy, 1, PF_INT);
04240         PF_LongSingleSend(MASTER, PF_OPT_HORNER_MSGTAG);
04241
04242         // Get a task from the master.
04243         PF_LongSingleReceive(MASTER, PF_OPT_HORNER_MSGTAG, NULL, NULL);
04244
04245         // Length of the task.
04246         PF_LongSingleUnpack(&len, 1, PF_INT);
04247
04248         // No task remains.
04249         if (len == 0) break;
04250
04251         // Perform the given task.
04252         optimize_best_Horner_schemes.push_back(vector<WORD>());
04253         vector<WORD> &Horner_scheme = optimize_best_Horner_schemes.back();
04254         Horner_scheme.resize(len);
04255         PF_LongSingleUnpack(&Horner_scheme[0], len, PF_WORD);
04256         optimize_expression_given_Horner ();
04257     }
04258
04259     // Send the result to the master.
04260     PF_PrepareLongSinglePack();
04261     PF_LongSinglePack(&optimize_best_num_oper, 1, PF_INT);
04262     if (optimize_best_num_oper != INT_MAX) {
04263         len = optimize_best_instr.size();
04264         PF_LongSinglePack(&len, 1, PF_INT);
04265         PF_LongSinglePack(&optimize_best_instr[0], len, PF_WORD);
04266         len = optimize_best_vars.size();
04267         PF_LongSinglePack(&len, 1, PF_INT);
04268         PF_LongSinglePack(&optimize_best_vars[0], len, PF_WORD);

```

```

04268     }
04269     PF_LongSingleSend(MASTER, PF_OPT_COLLECT_MSGTAG);
04270
04271 #ifdef DEBUG
04272     MesPrint ("*** [%s, w=%w] DONE: PF_optimize_expression_given_Horner_slave",
04273             thetime_str().c_str());
04274 #endif
04275 }
04276
04277 #endif
04278 /*
04279 #] PF_optimize_expression_given_Horner :
04280 #[ generate_output :
04281 */
04282
04292 void generate_output (const vector<WORD> &instr, int exprnr, int extraoffset, const
vector<vector<WORD> > &brackets) {
04293
04294 #ifdef DEBUG
04295     MesPrint ("*** [%s, w=%w] CALL: generate_output", thetime_str().c_str());
04296 #endif
04297
04298     GETIDENTITY;
04299     vector<WORD> output;
04300
04301     // one-indexed instead of zero-indexed
04302     extraoffset++;
04303     int num = 0;
04304     int maxsize = (int)instr.size();
04305     for (int i=0; i<maxsize; i+=instr[i+2]) num++;
04306     int *tstart = (int *)Malloca(num*sizeof(int), "nplaces");
04307     num = 0;
04308     for (int i=0; i<maxsize; i+=instr[i+2]) tstart[num++] = i;
04309     for (int j=0; j<num; j++) {
04310         int i = tstart[j];
04311
04312         // copy arguments
04313         WCOPY(AT.WorkPointer, &instr[i+3], (instr[i+2]-3));
04314
04315         // update maximal variable number
04316         AO.OptimizeResult.maxvar = MaX(AO.OptimizeResult.maxvar, instr[i]+extraoffset);
04317
04318         // renumber symbols and extrasymbols to correct values
04319         for (WORD *t=AT.WorkPointer; *t!=0; t+=*t) {
04320             if (*t == ABS(*(t+*t-1))+1) continue;
04321             if (*(t+1)==SYMBOL) *(t+3) = optimize_best_vars[*(t+3)];
04322             if (*(t+1)==EXTRASymbol) {
04323                 *(t+1) = SYMBOL;
04324                 *(t+3) = MAXVARIABLES-*(t+3)-extraoffset;
04325             }
04326         }
04327
04328         // reformat multiplication instructions, since their current
04329         // format is "expr.nr OPER_MUL length arguments", which differs
04330         // from Form's format for a product of symbols
04331         if (instr[i+1]==OPER_MUL) {
04332
04333             WORD *now=AT.WorkPointer+1;
04334             int dt;
04335             bool coeff=false;
04336             int coeff_sign=1;
04337
04338             for (WORD *t=AT.WorkPointer; *t!=0; t+=dt) {
04339                 dt = *t;
04340                 if (*t == ABS(*(t+*t-1))+1) {
04341                     // copy coefficient
04342                     memmove(AT.WorkPointer+instr[i+2], t, *t*sizeof(WORD));
04343                     coeff = true;
04344                 }
04345                 else {
04346                     // move symbol
04347                     int n = *(t+2);
04348                     memmove(now, t+1, n*sizeof(WORD));
04349                     now += n;
04350                     coeff_sign *= SGN(*(t+dt-1));
04351                 }
04352             }
04353
04354             if (coeff) {
04355                 // add existing coefficient
04356                 int n = *(AT.WorkPointer + instr[i+2]) - 1;
04357                 memmove(now, AT.WorkPointer+instr[i+2]+1, n*sizeof(WORD));
04358                 now += n;
04359             }
04360             else {
04361                 // add coefficient of one

```



```

04362         *now+=1;
04363         *now+=1;
04364         *now+=3;
04365     }
04366
04367     *(now-1) *= coeff_sign;
04368
04369     *AT.WorkPointer = now - AT.WorkPointer;
04370     *now++ = 0;
04371 }
04372
04373 // in the case of simultaneous optimization of expressions, add the
04374 // brackets to the final expression
04375 if (instr[i+1]==OPER_COMMA) {
04376     WORD *start = AT.WorkPointer + instr[i+2];
04377     WORD *now = start;
04378     int b=0;
04379     for (const WORD *t=AT.WorkPointer; *t!=0; t+=*t) {
04380         if ( ( brackets[b].size() != 0 ) && ( brackets[b][0] == 0 ) ) break;
04381         *now++ = *t + brackets[b].size();
04382         if (brackets[b].size() != 0) {
04383             memcpy(now, &brackets[b][0], brackets[b].size()*sizeof(WORD));
04384             now += brackets[b].size();
04385         }
04386         memcpy(now, t+1, (*t-1)*sizeof(WORD));
04387         now += *t-1;
04388         b++;
04389     }
04390     *now++ = 0;
04391     memmove(AT.WorkPointer, start, (now-start)*sizeof(WORD));
04392 }
04393
04394 // add the number of the extra symbol; if it is the last one,
04395 // replace the extra symbol number with the expression number
04396 if (i+instr[i+2]<(int)instr.size())
04397     output.push_back(instr[i]+extraoffset);
04398 else {
04399     output.push_back(-(exprnr+1));
04400 }
04401
04402 // add code for this symbol
04403 int n=0;
04404 while (*(AT.WorkPointer+n)!=0)
04405     n += *(AT.WorkPointer+n);
04406 n++;
04407 output.insert(output.end(), AT.WorkPointer, AT.WorkPointer+n);
04408 }
04409
04410 // trailing zero
04411 output.push_back(0);
04412
04413 // clear buffer
04414 if (AO.OptimizeResult.code != NULL)
04415     M_free(AO.OptimizeResult.code, "optimize output");
04416
04417 M_free(tstart, "nplaces");
04418
04419 // copy buffer
04420 AO.OptimizeResult.codesize = output.size();
04421 AO.OptimizeResult.code = (WORD *)Malloc1(output.size()*sizeof(WORD), "optimize output");
04422 memcpy(AO.OptimizeResult.code, &output[0], output.size()*sizeof(WORD));
04423
04424 #ifdef DEBUG
04425     MesPrint ("*** [%s, w=%w] DONE: generate_output", thetime_str().c_str());
04426 #endif
04427 }
04428
04429 /*
04430 #] generate_output :
04431 #[ generate_expression :
04432 */
04433
04441 WORD generate_expression (WORD exprnr) {
04442
04443 #ifdef DEBUG
04444     MesPrint ("*** [%s, w=%w] CALL: generate_expression", thetime_str().c_str());
04445 #endif
04446
04447     GETIDENTITY;
04448
04449     WORD *oldWorkPointer = AT.WorkPointer;
04450
04451     CBUF *C = cbuf+AC.cbunum;
04452     WORD *term = AT.WorkPointer, oldcurexpr = AR.CurExpr;
04453     POSITION position;
04454     EXPRESSIONS e = Expressions+exprnr;
04455     SetScratch(AR.infile, &(e->onfile));

```

```

04456
04457     if ( GetTerm(BHEAD term) <= 0 ) {
04458         MesPrint("Expression %d has problems in scratchfile",exprnr);
04459         Terminate(-1);
04460     }
04461     SeekScratch(AR.outfile,&position);
04462     e->onfile = position;
04463     if ( PutOut(BHEAD term,&position,AR.outfile,0) < 0 ) {
04464         MesPrint("Expression %d has problems in output scratchfile",exprnr);
04465         Terminate(-1);
04466     }
04467
04468     AR.CurExpr = exprnr;
04469     NewSort(BHEAD0);
04470
04471     // scan for the original expression (marked by *t<0) and give the
04472     // terms to Generator
04473     WORD *t = AO.OptimizeResult.code;
04474     {
04475         WORD old = AR.Cnumlhs; AR.Cnumlhs = 0;
04476         // We can use the remaining part of the WorkSpace for every term that
04477         // goes through Generator. We have WorkSpace problems here if we use
04478         // the (modified) AT.WorkPointer every time in the loop.
04479         WORD* currentWorkPointer = AT.WorkPointer;
04480         while (*t!=0) {
04481             bool is_expr = *t < 0;
04482             t++;
04483             while (*t!=0) {
04484                 if (is_expr) {
04485                     memcpy(currentWorkPointer, t, *t*sizeof(WORD));
04486                     Generator(BHEAD currentWorkPointer, C->numlhs);
04487                 }
04488                 t+=*t;
04489             }
04490             t++;
04491         }
04492         AR.Cnumlhs = old;
04493     }
04494
04495     // final sorting
04496     if (EndSort(BHEAD NULL,0) < 0) {
04497         LowerSortLevel();
04498         Terminate(-1);
04499     }
04500
04501     AT.WorkPointer = oldWorkPointer;
04502     AR.CurExpr = oldcurexpr;
04503
04504 #ifdef DEBUG
04505     MesPrint ("*** [%s, w=%w] DONE: generate_expression", thetime_str().c_str());
04506 #endif
04507     return 0;
04508 }
04509
04510 /*
04511 #] generate_expression :
04512 #[ optimize_print_code :
04513 */
04514
04525 void optimize_print_code (int print_expr) {
04526
04527 #ifdef DEBUG
04528     MesPrint ("*** [%s, w=%w] CALL: optimize_print_code", thetime_str().c_str());
04529 #endif
04530     if ( ( AO.Optimize.debugflags & 1 ) != 0 ) {
04531         /*
04532          * The next code is for debugging purposes. We may want the statements
04533          * in reverse order to substitute them all back.
04534          * Jan used a Mathematica program to do this. Here we make that
04535          * Format Ox,debugflag=1;
04536          * Creates reverse order during printing.
04537          * All we have to do is put id in front of the statements. This is done
04538          * in PrintExtraSymbol.
04539          */
04540         WORD *t = AO.OptimizeResult.code;
04541         WORD num = 0; // First we count the number of objects.
04542         while (*t!=0) {
04543             num++;
04544             t++; while (*t!=0) t+=*t; t++;
04545         }
04546         WORD **tstart = (WORD **)Malloca1(num*sizeof(WORD *),"print objects");
04547         t = AO.OptimizeResult.code; num = 0; // Now we get the addresses
04548         while (*t!=0) {
04549             tstart[num++] = t;
04550             t++; while (*t!=0) t+=*t; t++;
04551         }

```

```

04552         // Flip the addresses
04553         int halfnum = num/2;
04554         for (int i=0; i<halfnum; i++) { swap(tstart[i],tstart[num-1-i]); }
04555         for ( int i = 0; i < num; i++ ) {
04556             t = tstart[i];
04557             if (*t > 0)
04558                 PrintExtraSymbol(*t, t+1, EXTRASYMBOL);
04559             else if (print_expr)
04560                 PrintExtraSymbol(-*t-1, t+1, EXPRESSIONNUMBER);
04561         }
04562         CBUF *C = cbuf + AM.sbufnum;
04563         if (C->numrhs >= AO.OptimizeResult.minvar)
04564             PrintSubtermList(AO.OptimizeResult.minvar, C->numrhs);
04565     }
04566     else {
04567         // print extra symbols from ConvertToPoly in optimization
04568         CBUF *C = cbuf + AM.sbufnum;
04569         if (C->numrhs >= AO.OptimizeResult.minvar)
04570             PrintSubtermList(AO.OptimizeResult.minvar, C->numrhs);
04571
04572         WORD *t = AO.OptimizeResult.code;
04573         while (*t!=0) {
04574             if (*t > 0) {
04575                 PrintExtraSymbol(*t, t+1, EXTRASYMBOL);
04576             }
04577             else if (print_expr)
04578                 PrintExtraSymbol(-*t-1, t+1, EXPRESSIONNUMBER);
04579             t++;
04580             while (*t!=0) t+=*t;
04581             t++;
04582         }
04583     }
04584
04585 #ifdef DEBUG
04586     MesPrint ("*** [%s, w=%w] DONE: optimize_print_code", thetime_str().c_str());
04587 #endif
04588 }
04589
04590 /*
04591 #] optimize_print_code :
04592 #[ Optimize :
04593 */
04594
04638 int Optimize (WORD exprnr, int do_print) {
04639
04640 #if defined(WITHMPI) && (defined(DEBUG) || defined(DEBUG_MORE) || defined(DEBUG_MCTS) ||
    defined(DEBUG_GREEDY))
04641     // set AS.printflag negative temporary.
04642     struct save_printflag {
04643         save_printflag() {
04644             oldprintflag = AS.printflag;
04645             AS.printflag = -1;
04646         }
04647         ~save_printflag() {
04648             AS.printflag = oldprintflag;
04649         }
04650         int oldprintflag;
04651     } save_printflag_;
04652 #endif
04653
04654 #ifdef DEBUG
04655     MesPrint ("*** [%s, w=%w] CALL: Optimize", thetime_str().c_str());
04656     MesPrint ("*** %"); PrintRunningTime();
04657 #endif
04658
04659 #ifdef WITHPTHREADS
04660     optimize_lock = dummylock;
04661 #endif
04662
04663     AO.OptimizeResult.minvar = (cbuf + AM.sbufnum)->numrhs + 1;
04664
04665     if (get_expression(exprnr) < 0) return -1;
04666     vector<vector<WORD>> > brackets = get_brackets();
04667
04668 #ifdef DEBUG
04669 #ifdef WITHMPI
04670     if (PF.me == MASTER)
04671 #endif
04672     MesPrint ("*** runtime after preparing the expression: %"); PrintRunningTime();
04673 #endif
04674
04675     if (optimize_expr[0]==0 ||
04676         (optimize_expr[optimize_expr[0]]==0 &&
04677          optimize_expr[0]==ABS(optimize_expr[optimize_expr[0]-1])+1) ||
04677         (optimize_expr[optimize_expr[0]]==0 && optimize_expr[0]==8 &&
04678          optimize_expr[5]==1 && optimize_expr[6]==1 && ABS(optimize_expr[7])==3)) {
04679         if (AO.OptimizeResult.code != NULL)

```

```

04680         M_free(AO.OptimizeResult.code, "optimize output");
04681
04682         // zero terms or one trivial term (number or +/-variable), so no
04683         // optimization, so copy expression; special case because without
04684         // operators the optimization crashes
04685         AO.OptimizeResult.code = (WORD *)Malloc1((optimize_expr[0]+3)*sizeof(WORD), "optimize
output");
04686         AO.OptimizeResult.code[0] = -(exprnr+1);
04687         memcpy(AO.OptimizeResult.code+1, optimize_expr, (optimize_expr[0]+1)*sizeof(WORD));
04688         AO.OptimizeResult.code[optimize_expr[0]+2] = 0;
04689     }
04690     else {
04691         // find Horner scheme(s)
04692         optimize_best_Horner_schemes.clear();
04693         if (AO.Optimize.horner == O_OCCURRENCE) {
04694             if (AO.Optimize.hornerdirection != O_BACKWARD)
04695                 optimize_best_Horner_schemes.push_back(occurrence_order(optimize_expr, false));
04696             if (AO.Optimize.hornerdirection != O_FORWARD)
04697                 optimize_best_Horner_schemes.push_back(occurrence_order(optimize_expr, true));
04698         }
04699         else {
04700             if (AO.Optimize.horner == O_SIMULATED_ANNEALING) {
04701                 optimize_best_Horner_schemes.push_back(simulated_annealing());
04702             }
04703             else {
04704                 mcts_best_schemes.clear();
04705                 if (AO.inscheme) {
04706                     optimize_best_Horner_schemes.push_back(vector<WORD>(AO.schemenum));
04707                     for (int i=0; i < AO.schemenum; i++)
04708                         optimize_best_Horner_schemes[0][i] = AO.inscheme[i];
04709                 }
04710                 else {
04711                     for (int i = 0; i < AO.Optimize.mctsnumrepeat; i++)
04712                         find_Horner_MCTS();
04713                     // generate results
04714                     for (set<pair<int, vector<WORD>> >>::iterator i=mcts_best_schemes.begin();
i!=mcts_best_schemes.end(); i++) {
04715                         optimize_best_Horner_schemes.push_back(i->second);
04716                     #ifdef DEBUG_MCTS
04717                         MesPrint ("%a -> %d", i->second.size(), &i->second[0], i->first);
04718                     #endif
04719                 }
04720             }
04721             // clear the tree by making a new empty one.
04722             mcts_root = tree_node();
04723         }
04724     }
04725     #ifdef DEBUG
04726     #ifdef WITHMPI
04727         if (PF.me == MASTER)
04728     #endif
04729     MesPrint ("*** runtime after Horner: %"); PrintRunningTime();
04730     #endif
04731
04732     #ifdef WITHMPI
04733         if (PF.me == MASTER) {
04734             PF_optimize_expression_given_Horner_master_init();
04735         }
04736     #endif
04737     // find best Horner scheme and results
04738     optimize_best_num_oper = INT_MAX;
04739
04740     int imax = (int)optimize_best_Horner_schemes.size();
04741
04742     for (int i=0; i<imax; i++) {
04743         #if defined(WITHPTHREADS)
04744             if (AM.totalnumberofthreads > 1)
04745                 optimize_expression_given_Horner_threaded();
04746             else
04747         #elif defined(WITHMPI)
04748             if (PF.numtasks > 1)
04749                 PF_optimize_expression_given_Horner_master();
04750             else
04751         #endif
04752                 optimize_expression_given_Horner();
04753     }
04754
04755     #ifdef WITHMPI
04756         PF_optimize_expression_given_Horner_master_wait();
04757     }
04758     else {
04759         if (PF.numtasks > 1)
04760             PF_optimize_expression_given_Horner_slave();
04761     }
04762     #endif
04763
04764     #ifdef WITHPTHREADS

```

```

04765     MasterWaitAll();
04766 #endif
04767     // format results, then print it (for "Print") or modify
04768     // expression (for "#Optimize")
04769 #ifdef WITHMPI
04770     if (PF.me == MASTER)
04771 #endif
04772     generate_output(optimize_best_instr, exprnr, cbuf[AM.sbufnum].numrhs, brackets);
04773 #ifdef WITHMPI
04774     else {
04775         // non-null dummy code for slaves
04776         if (AO.OptimizeResult.code == NULL) {
04777             AO.OptimizeResult.code = (WORD *)Malloca(sizeof(WORD), "optimize output");
04778         }
04779     }
04780 #endif
04781 }
04782
04783 #ifdef WITHMPI
04784 if (PF.me == MASTER) {
04785     PF_PrepPack();
04786     PF_Pack(&AO.OptimizeResult.minvar, 1, PF_WORD);
04787     PF_Pack(&AO.OptimizeResult.maxvar, 1, PF_WORD);
04788 }
04789 PF_Broadcast();
04790 if (PF.me != MASTER) {
04791     PF_Unpack(&AO.OptimizeResult.minvar, 1, PF_WORD);
04792     PF_Unpack(&AO.OptimizeResult.maxvar, 1, PF_WORD);
04793 }
04794 #endif
04795
04796 // set preprocessor variables
04797 char str[100];
04798 snprintf(str, 100, "%d", AO.OptimizeResult.minvar);
04799 PutPreVar((UBYTE *) "optimminvar_", (UBYTE *) str, 0, 1);
04800 snprintf(str, 100, "%d", AO.OptimizeResult.maxvar);
04801 PutPreVar((UBYTE *) "optimmaxvar_", (UBYTE *) str, 0, 1);
04802
04803 if (do_print) {
04804 #ifdef WITHMPI
04805     if (PF.me == MASTER)
04806 #endif
04807     optimize_print_code(1);
04808     ClearOptimize();
04809 }
04810 else {
04811 #ifdef WITHMPI
04812     if (PF.me == MASTER)
04813 #endif
04814     generate_expression(exprnr);
04815 }
04816
04817 #ifdef WITHMPI
04818 if (PF.me == MASTER) {
04819 #endif
04820
04821     if (AO.Optimize.printstats > 0) {
04822         char str[20];
04823         MesPrint("");
04824         count_operators(optimize_expr, true);
04825         int numop = count_operators(optimize_best_instr, true);
04826         snprintf(str, 20, "%d", numop);
04827         PutPreVar((UBYTE *) "optimvalue_", (UBYTE *) str, 0, 1);
04828     }
04829     else {
04830         char str[20];
04831         int numop = count_operators(optimize_best_instr, false);
04832         snprintf(str, 20, "%d", numop);
04833         PutPreVar((UBYTE *) "optimvalue_", (UBYTE *) str, 0, 1);
04834     }
04835
04836     if ( ( AO.Optimize.schemeflags&1 ) == 1 ) {
04837         GETIDENTITY
04838         UBYTE *OutScr, *Out, *old1 = AO.OutputLine, *old2 = AO.OutFill;
04839         int i, sym;
04840         AO.OutputLine = AO.OutFill = (UBYTE *) AT.WorkPointer;
04841         FiniLine();
04842         OutScr = (UBYTE *) AT.WorkPointer + ( TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer) ) / 2;
04843         TokenToLine((UBYTE *) " Scheme selected: ");
04844         for ( i = 0; i < optimize_num_vars; i++ ) {
04845             Out = OutScr;
04846             sym = optimize_best_vars[i];
04847             if ( i > 0 ) TokenToLine((UBYTE *) ",");
04848             if ( sym < NumSymbols ) {
04849                 StrCopy(FindSymbol(sym), OutScr);
04850                 StrCopy(VARNAME(symbols, sym), OutScr);
04851             }

```

```

04852         else {
04853             Out = StrCopy((UBYTE *)AC.extrasym,Out);
04854             if ( AC.extrasymbols == 0 ) {
04855                 Out = NumCopy((MAXVARIABLES-sym),Out);
04856                 Out = StrCopy((UBYTE *)"_",Out);
04857             }
04858             else if ( AC.extrasymbols == 1 ) {
04859                 Out = AddArrayIndex((MAXVARIABLES-sym),Out);
04860             }
04861         }
04862         TokenToLine(OutScr);
04863     }
04864     TokenToLine((UBYTE *)";");
04865     FiniLine();
04866     AO.OutFill = old2;
04867     AO.OutputLine = old1;
04868 }
04869
04870 {
04871     GETIDENTITY
04872     UBYTE *OutScr, *Out, *outstring = 0;
04873     int i, sym;
04874     AO.OutputLine = AO.OutFill = (UBYTE *)AT.WorkPointer;
04875     OutScr = (UBYTE *)AT.WorkPointer + ( TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer) ) /2;
04876     for ( i = 0; i < optimize_num_vars; i++ ) {
04877         Out = OutScr;
04878         sym = optimize_best_vars[i];
04879         if ( sym < NumSymbols ) {
04880             StrCopy(FindSymbol(sym),OutScr);
04881             /* StrCopy(VARNAME(symbols,sym),OutScr); */
04882         }
04883         else {
04884             Out = StrCopy((UBYTE *)AC.extrasym,Out);
04885             if ( AC.extrasymbols == 0 ) {
04886                 Out = NumCopy((MAXVARIABLES-sym),Out);
04887                 Out = StrCopy((UBYTE *)"_",Out);
04888             }
04889             else if ( AC.extrasymbols == 1 ) {
04890                 Out = AddArrayIndex((MAXVARIABLES-sym),Out);
04891             }
04892         }
04893         outstring = AddToString(outstring,OutScr,1);
04894     }
04895     if ( outstring == 0 ) {
04896         PutPreVar((UBYTE *)"optimscheme_",(UBYTE *)"",0,1);
04897     }
04898     else {
04899         PutPreVar((UBYTE *)"optimscheme_",(UBYTE *)outstring,0,1);
04900         M_free(outstring,"AddToString");
04901     }
04902 }
04903
04904 #ifdef WITHMPI
04905 {
04906     // synchronize optimvalue_ and optimscheme_
04907     if ( PF.me == MASTER ) {
04908         UBYTE *value;
04909         int bytes;
04910
04911         PF_PrepLongMultiPack();
04912
04913         value = GetPreVar((UBYTE *)"optimvalue_", WITHERROR);
04914         bytes = strlen((char *)value);
04915         PF_LongMultiPack(&bytes, 1, PF_INT);
04916         PF_LongMultiPack(value, bytes, PF_BYTE);
04917
04918         value = GetPreVar((UBYTE *)"optimscheme_", WITHERROR);
04919         bytes = strlen((char *)value);
04920         PF_LongMultiPack(&bytes, 1, PF_INT);
04921         PF_LongMultiPack(value, bytes, PF_BYTE);
04922     }
04923     PF_LongMultiBroadcast();
04924     if ( PF.me != MASTER ) {
04925         static vector<UBYTE> prevarbuf;
04926         UBYTE *value;
04927         int bytes;
04928
04929         PF_LongMultiUnpack(&bytes, 1, PF_INT);
04930         prevarbuf.reserve(bytes + 1);
04931         value = &prevarbuf[0];
04932         PF_LongMultiUnpack(value, bytes, PF_BYTE);
04933         value[bytes] = '\0'; // null terminator
04934         PutPreVar((UBYTE *)"optimvalue_", value, NULL, 1);
04935
04936         PF_LongMultiUnpack(&bytes, 1, PF_INT);
04937         prevarbuf.reserve(bytes + 1);

```

```

04939     value = &prevarbuf[0];
04940     PF_LongMultiUnpack(value, bytes, PF_BYTE);
04941     value[bytes] = '\0'; // null terminator
04942     PutPreVar((UBYTE *) "optimscheme_", value, NULL, 1);
04943 }
04944 #endif
04945
04946 // cleanup
04947 M_free(optimize_expr, "LoadOptim");
04948
04949 #ifdef DEBUG
04950     MesPrint ("*** [%s, w=%w] DONE: Optimize", thetime_str().c_str());
04951 #endif
04952     return 0;
04953 }
04954
04955 /*
04956 #] Optimize :
04957 #[ ClearOptimize :
04958 */
04959
04960 int ClearOptimize()
04961 {
04962     char str[20];
04963     WORD numexpr, *w;
04964     int error = 0;
04965     if ( AO.OptimizeResult.code != NULL ) {
04966         M_free(AO.OptimizeResult.code, "optimize output");
04967         AO.OptimizeResult.code = NULL;
04968         AO.OptimizeResult.codesize = 0;
04969         PutPreVar((UBYTE *) "optimminvar_", (UBYTE *) ("0"), 0, 1);
04970         PutPreVar((UBYTE *) "optimmaxvar_", (UBYTE *) ("0"), 0, 1);
04971         PruneExtraSymbols(AO.OptimizeResult.minvar-1);
04972         cbuf[AM.sbufnum].numrhs = AO.OptimizeResult.minvar-1;
04973         AO.OptimizeResult.minvar = AO.OptimizeResult.maxvar = 0;
04974     }
04975     if ( AO.OptimizeResult.nameofexpr != NULL ) {
04976         /*
04977         We have to pick up the expression by its name. Numbers may change.
04978         Note that this requires that when we overwrite an expression, we
04979         check that it is not an optimized expression. See execute.c and
04980         comexpr.c
04981         */
04982         if ( GetName(AC.exprnames, AO.OptimizeResult.nameofexpr, &numexpr, NOAUTO) != CEXPRESSION ) {
04983             MesPrint("@Internal error while clearing optimized expression %s", AO.OptimizeResult.nameofexpr);
04984             Terminate(-1);
04985         }
04986         M_free(AO.OptimizeResult.nameofexpr, "optimize expression name");
04987         AO.OptimizeResult.nameofexpr = NULL;
04988         w = &(Expressions[numexpr].status);
04989         *w = SetExprCases(DROP, 1, *w);
04990         if ( *w < 0 ) error = 1;
04991     }
04992     snprintf (str, 20, "%d", cbuf[AM.sbufnum].numrhs);
04993     PutPreVar (AM.oldnumextrasyms, (UBYTE *) str, 0, 1);
04994     PutPreVar ((UBYTE *) "optimvalue_", (UBYTE *) ("0"), 0, 1);
04995     PutPreVar ((UBYTE *) "optimscheme_", (UBYTE *) ("0"), 0, 1);
04996     return(error);
04997 }
04998
04999 /*
05000 #] ClearOptimize :
05001 */

```

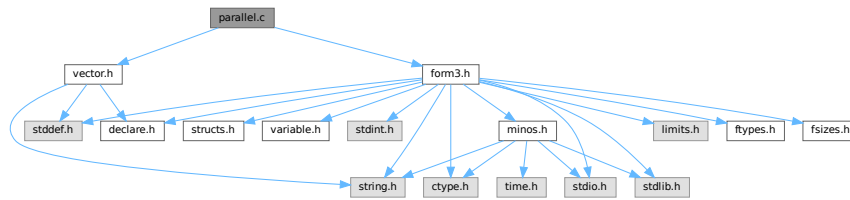
## 4.94 parallel.c File Reference

```

#include "form3.h"
#include "vector.h"

```

Include dependency graph for parallel.c:



## Data Structures

- struct [NoDe](#)
- struct [dollar\\_buf](#)
- struct [bufIPstruct](#)

## Macros

- `#define PRINTFBUF(TEXT, TERM, SIZE) {}`
- `#define SWAP(x, y)`
- `#define PACK\_LONG(p, n)`
- `#define UNPACK\_LONG(p, n)`
- `#define CHECK(condition) __CHECK(condition, __FILE__, __LINE__)`
- `#define \_\_CHECK(condition, file, line) __CHECK(condition, file, line)`
- `#define \_\_CHECK(condition, file, line)`
- `#define DBGOUT(lv1, lv2, a) do { if ( lv1 >= lv2 ) { printf a; fflush(stdout); } } while (0)`
- `#define DBGOUT\_NINTERMS(lv, a)`
- `#define PF\_STATS\_SIZE 5`
- `#define PF\_SNDFILEBUFSIZE 4096`
- `#define recvBuffer logBuffer /* (master) The buffer for receiving messages. */`

## Typedefs

- typedef struct [NoDe](#) [NODE](#)
- typedef struct [bufIPstruct](#) [bufIPstruct\\_t](#)

## Functions

- LONG [PF\\_RealTime](#) (int)
- int [PF\\_LibInit](#) (int \*, char \*\*\*)
- int [PF\\_LibTerminate](#) (int)
- int [PF\\_Probe](#) (int \*)
- int [PF\\_RecvWbuf](#) (WORD \*, LONG \*, int \*)
- int [PF\\_IRecvRbuf](#) ([PF\\_BUFFER](#) \*, int, int)
- int [PF\\_WaitRbuf](#) ([PF\\_BUFFER](#) \*, int, LONG \*)
- int [PF\\_RawSend](#) (int dest, void \*buf, LONG l, int tag)
- LONG [PF\\_RawRecv](#) (int \*src, void \*buf, LONG thesize, int \*tag)
- int [PF\\_RawProbe](#) (int \*src, int \*tag, int \*bytesize)
- int [PF\\_EndSort](#) (void)



- WORD [PF\\_Deferred](#) (WORD \*term, WORD level)
- int [PF\\_Processor](#) (EXPRESSIONS e, WORD i, WORD LastExpression)
- int [PF\\_Init](#) (int \*argc, char \*\*\*argv)
- int [PF\\_Terminate](#) (int errorcode)
- LONG [PF\\_GetSlaveTimes](#) (void)
- LONG [PF\\_BroadcastNumber](#) (LONG x)
- void [PF\\_BroadcastBuffer](#) (WORD \*\*buffer, LONG \*length)
- int [PF\\_BroadcastString](#) (UBYTE \*str)
- int [PF\\_BroadcastPreDollar](#) (WORD \*\*dbuffer, LONG \*newsize, int \*numterms)
- int [PF\\_CollectModifiedDollars](#) (void)
- int [PF\\_BroadcastModifiedDollars](#) (void)
- int [PF\\_BroadcastRedefinedPreVars](#) (void)
- int [PF\\_StoreInsideInfo](#) (void)
- int [PF\\_RestoreInsideInfo](#) (void)
- int [PF\\_BroadcastCBuf](#) (int bufnum)
- int [PF\\_BroadcastExpFlags](#) (void)
- int [PF\\_BroadcastExpr](#) (EXPRESSIONS e, FILEHANDLE \*file)
- int [PF\\_BroadcastRHS](#) (void)
- int [PF\\_InParallelProcessor](#) (void)
- int [PF\\_SendFile](#) (int to, FILE \*fd)
- int [PF\\_RecvFile](#) (int from, FILE \*fd)
- void [PF\\_MLock](#) (void)
- void [PF\\_MUnlock](#) (void)
- LONG [PF\\_WriteFileToFile](#) (int handle, UBYTE \*buffer, LONG size)
- void [PF\\_FlushStdOutBuffer](#) (void)
- void [PF\\_FreeErrorMessageBuffers](#) (void)

## Variables

- [PARALLELVAR](#) PF

### 4.94.1 Detailed Description

Message passing library independent functions of parform

This file contains functions needed for the parallel version of form3 these functions need no real link to the message passing libraries, they only need some interface dependent preprocessor definitions (check [parallel.h](#)). So there still need two different objectfiles to be compiled for mpi and pvm!

Definition in file [parallel.c](#).

### 4.94.2 Macro Definition Documentation

#### 4.94.2.1 PRINTFBUF

```
#define PRINTFBUF(
    TEXT,
    TERM,
    SIZE ) {}
```

Definition at line 117 of file [parallel.c](#).

#### 4.94.2.2 SWAP

```
#define SWAP(
    x,
    y )
```

##### Value:

```
do { \
    char swap_tmp__[sizeof(x) == sizeof(y) ? (int)sizeof(x) : -1]; \
    memcpy(swap_tmp__, &y, sizeof(x)); \
    memcpy(&y, &x, sizeof(x)); \
    memcpy(&x, swap_tmp__, sizeof(x)); \
} while (0)
```

Swaps the variables *x* and *y*. If `sizeof(x) != sizeof(y)` then a compilation error will occur. A set of `memcpy` calls with constant sizes is expected to be inlined by the optimisation.

Definition at line 124 of file [parallel.c](#).

#### 4.94.2.3 PACK\_LONG

```
#define PACK_LONG(
    p,
    n )
```

##### Value:

```
do { \
    *p)++ = (UWORD)((ULONG)(n) & (ULONG)WORDMASK); \
    *p)++ = (UWORD)((ULONG)(n) >> BITSINWORD) & (ULONG)WORDMASK); \
} while (0)
```

Packs a LONG value *n* to a WORD buffer *p*.

Definition at line 135 of file [parallel.c](#).

#### 4.94.2.4 UNPACK\_LONG

```
#define UNPACK_LONG(
    p,
    n )
```

##### Value:

```
do { \
    (n) = (LONG)((((ULONG)(p)[1] & (ULONG)WORDMASK) << BITSINWORD) | ((ULONG)(p)[0] & (ULONG)WORDMASK)); \
    (p) += 2; \
} while (0)
```

Unpacks a LONG value *n* from a WORD buffer *p*.

Definition at line 144 of file [parallel.c](#).

#### 4.94.2.5 CHECK

```
#define CHECK(
    condition ) _CHECK(condition, __FILE__, __LINE__)
```

A simple check for unrecoverable errors.

Definition at line 153 of file [parallel.c](#).

#### 4.94.2.6 \_CHECK

```
#define _CHECK(  
    condition,  
    file,  
    line ) __CHECK(condition, file, line)
```

Definition at line 154 of file [parallel.c](#).

#### 4.94.2.7 \_\_CHECK

```
#define __CHECK(  
    condition,  
    file,  
    line )
```

##### Value:

```
do { \  
    if ( !(condition) ) { \  
        Error0("Fatal error at " file ":" #line); \  
        Terminate(-1); \  
    } \  
} while (0)
```

Definition at line 155 of file [parallel.c](#).

#### 4.94.2.8 DBGOUT

```
#define DBGOUT(  
    lv1,  
    lv2,  
    a ) do { if ( lv1 >= lv2 ) { printf a; fflush(stdout); } } while (0)
```

Definition at line 166 of file [parallel.c](#).

#### 4.94.2.9 DBGOUT\_NINTERMS

```
#define DBGOUT_NINTERMS(  
    lv,  
    a )
```

Definition at line 169 of file [parallel.c](#).

#### 4.94.2.10 PF\_STATS\_SIZE

```
#define PF_STATS_SIZE 5
```

Definition at line 178 of file [parallel.c](#).

#### 4.94.2.11 PF\_SNDFILEBUFSIZE

```
#define PF_SNDFILEBUFSIZE 4096
```

Definition at line 4212 of file [parallel.c](#).

#### 4.94.2.12 recvBuffer

```
#define recvBuffer logBuffer /* (master) The buffer for receiving messages. */
```

Definition at line 4320 of file [parallel.c](#).

### 4.94.3 Typedef Documentation

#### 4.94.3.1 NODE

```
typedef struct NoDe NODE
```

A node for the tree of losers in the final sorting on the master.

### 4.94.4 Function Documentation

#### 4.94.4.1 PF\_RealTime()

```
LONG PF_RealTime (  
    int i )
```

Returns the realtime in 1/100 sec. as a LONG.

##### Parameters

|          |                                       |
|----------|---------------------------------------|
| <i>i</i> | the timer will be reset if $i == 0$ . |
|----------|---------------------------------------|

##### Returns

the real elapsed time in 1/100 second.

Definition at line 101 of file [mpi.c](#).

Referenced by [PF\\_Init\(\)](#).

#### 4.94.4.2 PF\_LibInit()

```
int PF_LibInit (  
    int * argcp,  
    char *** argvp )
```

Performs all library dependent initializations.

## Parameters

|              |                                     |
|--------------|-------------------------------------|
| <i>argcp</i> | pointer to the number of arguments. |
| <i>argvp</i> | pointer to the arguments.           |

## Returns

0 if OK, nonzero on error.

Definition at line 123 of file [mpi.c](#).

Referenced by [PF\\_Init\(\)](#).

**4.94.4.3 PF\_LibTerminate()**

```
int PF_LibTerminate (
    int error )
```

Exits mpi, when there is an error either indicated or happening, returnvalue is 1, else returnvalue is 0.

## Parameters

|              |                |
|--------------|----------------|
| <i>error</i> | an error code. |
|--------------|----------------|

## Returns

0 if OK, nonzero on error.

Definition at line 209 of file [mpi.c](#).

Referenced by [PF\\_Terminate\(\)](#).

**4.94.4.4 PF\_Probe()**

```
int PF_Probe (
    int * src )
```

Probes the next incoming message. If `src == PF_ANY_SOURCE` this operation is blocking, otherwise nonblocking.

## Parameters

|                |            |                                                                                  |
|----------------|------------|----------------------------------------------------------------------------------|
| <i>in, out</i> | <i>src</i> | the source process number. In output, the process number of actual found source. |
|----------------|------------|----------------------------------------------------------------------------------|

## Returns

the tag value of the next incoming message if found, 0 if a nonblocking probe (input `src != PF_ANY_SOURCE`) did not find any messages. A negative returned value indicates an error.

Definition at line 230 of file [mpi.c](#).

#### 4.94.4.5 PF\_RecvWbuf()

```
int PF_RecvWbuf (
    WORD * b,
    LONG * s,
    int * src )
```

Blocking receive of a WORD buffer.

##### Parameters

|         |            |                                                                                     |
|---------|------------|-------------------------------------------------------------------------------------|
| out     | <i>b</i>   | the buffer to store the received data.                                              |
| in, out | <i>s</i>   | the size of the buffer. The output value is the actual size of the received data.   |
| in, out | <i>src</i> | the source process number. The output value is the process number of actual source. |

##### Returns

the received message tag. A negative value indicates an error.

Definition at line 337 of file [mpi.c](#).

#### 4.94.4.6 PF\_IRecvRbuf()

```
int PF_IRecvRbuf (
    PF_BUFFER * r,
    int bn,
    int from )
```

Posts nonblocking receive for the active receive buffer. The buffer is filled from `full` to `stop`.

##### Parameters

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| <i>r</i>    | the <a href="#">PF_BUFFER</a> struct for the nonblocking receive. |
| <i>bn</i>   | the index of the cyclic buffer.                                   |
| <i>from</i> | the source process number.                                        |

##### Returns

0 if OK, nonzero on error.

Definition at line 366 of file [mpi.c](#).

#### 4.94.4.7 PF\_WaitRbuf()

```
int PF_WaitRbuf (
    PF_BUFFER * r,
    int bn,
    LONG * size )
```

Waits for the buffer *bn* to finish a pending nonblocking receive. It returns the received tag and in *\*size* the number of field received. If the receive is already finished, just return the flag and size, else wait for it to finish, but also check for other pending receives.

## Parameters

|     |             |                                                                           |
|-----|-------------|---------------------------------------------------------------------------|
|     | <i>r</i>    | the <a href="#">PF_BUFFER</a> struct for the pending nonblocking receive. |
|     | <i>bn</i>   | the index of the cyclic buffer.                                           |
| out | <i>size</i> | the actual size of received data.                                         |

## Returns

the received message tag. A negative value indicates an error.

Definition at line [400](#) of file [mpi.c](#).

**4.94.4.8 PF\_RawSend()**

```
int PF_RawSend (
    int dest,
    void * buf,
    LONG l,
    int tag )
```

Sends *l* bytes from *buf* to *dest*. Returns 0 on success, or -1.

## Parameters

|             |                                               |
|-------------|-----------------------------------------------|
| <i>dest</i> | the destination process number.               |
| <i>buf</i>  | the send buffer.                              |
| <i>l</i>    | the size of data in the send buffer in bytes. |
| <i>tag</i>  | the message tag.                              |

## Returns

0 if OK, nonzero on error.

Definition at line [463](#) of file [mpi.c](#).

Referenced by [PF\\_MUnlock\(\)](#), and [PF\\_SendFile\(\)](#).

**4.94.4.9 PF\_RawRecv()**

```
LONG PF_RawRecv (
    int * src,
    void * buf,
    LONG thesize,
    int * tag )
```

Receives not more than *thesize* bytes from *src*, returns the actual number of received bytes, or -1 on failure.

**Parameters**

|         |                |                                                                            |
|---------|----------------|----------------------------------------------------------------------------|
| in, out | <i>src</i>     | the source process number. In output, that of the actual received message. |
| out     | <i>buf</i>     | the receive buffer.                                                        |
|         | <i>thesize</i> | the size of the receive buffer in bytes.                                   |
| out     | <i>tag</i>     | the message tag of the actual received message.                            |

**Returns**

the actual sizeof received data in bytes, or -1 on failure.

Definition at line 484 of file [mpi.c](#).

Referenced by [PF\\_RecvFile\(\)](#).

**4.94.4.10 PF\_RawProbe()**

```
int PF_RawProbe (
    int * src,
    int * tag,
    int * bytesize )
```

Probes an incoming message.

**Parameters**

|         |                 |                                                                            |
|---------|-----------------|----------------------------------------------------------------------------|
| in, out | <i>src</i>      | the source process number. In output, that of the actual received message. |
| in, out | <i>tag</i>      | the message tag. In output, that of the actual received message.           |
| out     | <i>bytesize</i> | the size of incoming data in bytes.                                        |

**Returns**

0 if OK, nonzero on error.

Definition at line 508 of file [mpi.c](#).

**4.94.4.11 PF\_EndSort()**

```
int PF_EndSort (
    void )
```

Finishes a master sorting with collecting terms from slaves. Called by [EndSort\(\)](#).

If this is not the masterprocess, just initialize the sendbuffers and return 0, else [PF\\_EndSort\(\)](#) sends the rest of the terms in the sendbuffer to the next slave and a dummy message to all slaves with tag PF\_ENDSORT\_MSGTAG. Then it receives the sorted terms, sorts them using a recursive 'tree of losers' (PF\_GetLoser()) and writes them to the outputfile.



**Returns**

1 if the sorting on the master was done. 0 if [EndSort\(\)](#) still must perform a regular sorting because it is not at the ground level or not on the master or in the sequential mode or in the InParallel mode. -1 if an error occurred.

**Remarks**

The slaves will send the sorted terms back to the master in the regular sorting (after the initialization of the send buffer in [PF\\_EndSort\(\)](#)). See [PutOut\(\)](#) and [FlushOut\(\)](#).

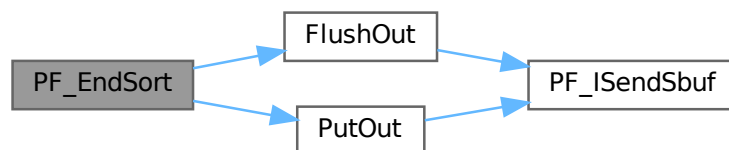
This function has been changed such that when it returns 1, AM.S0->TermsLeft is set correctly. But AM.S0->GenTerms is not set: it will be set after collecting the statistics from the slaves at the end of [PF\\_Processor\(\)](#). (TU 30 Jun 2011)

Definition at line 864 of file [parallel.c](#).

References [FlushOut\(\)](#), and [PutOut\(\)](#).

Referenced by [EndSort\(\)](#).

Here is the call graph for this function:

**4.94.4.12 PF\_Deferred()**

```
WORD PF_Deferred (
    WORD * term,
    WORD level )
```

Replaces [Deferred\(\)](#) on the slaves.

**Parameters**

|              |                                                                          |
|--------------|--------------------------------------------------------------------------|
| <i>term</i>  | the term that must be multiplied by the contents of the current bracket. |
| <i>level</i> | the compiler level.                                                      |

**Returns**

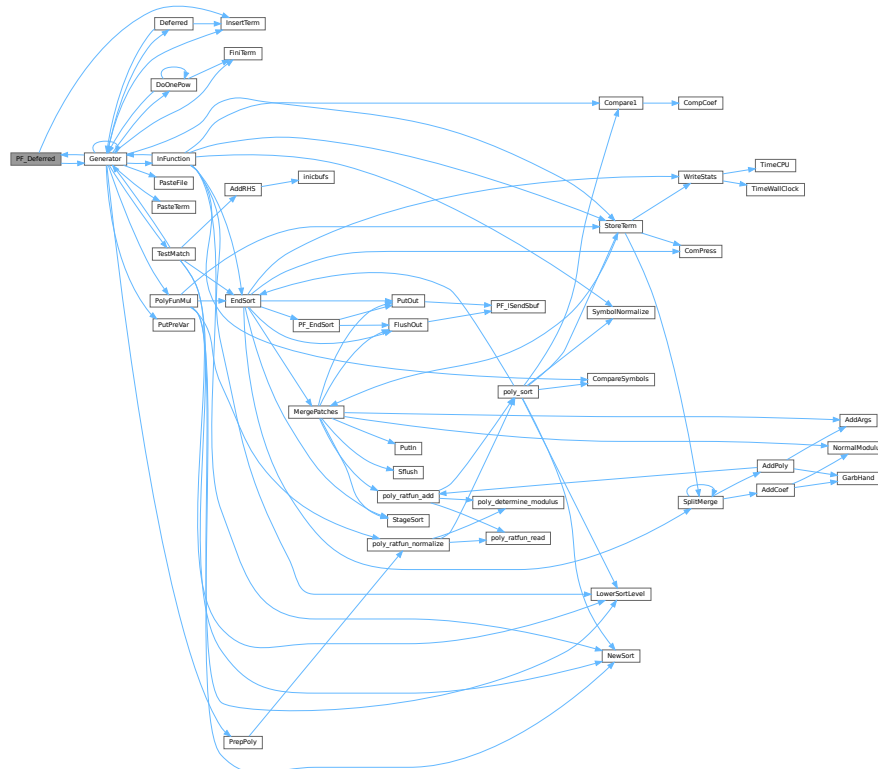
0 if OK, nonzero on error.

Definition at line 1208 of file [parallel.c](#).

References [Generator\(\)](#), and [InsertTerm\(\)](#).

Referenced by [Generator\(\)](#).

Here is the call graph for this function:



#### 4.94.4.13 PF\_Processor()

```
int PF_Processor (
    EXPRESSIONS e,
    WORD i,
    WORD LastExpression )
```

Replaces parts of [Processor\(\)](#) on the masters and slaves. On the master [PF\\_Processor\(\)](#) is responsible for proper distribution of terms from the input file to the slaves. On the slaves it calls [Generator\(\)](#) for all the terms that this process gets, but [PF\\_GetTerm\(\)](#) gets terms from the master (not directly from infile).

##### Parameters

|                       |                                                        |
|-----------------------|--------------------------------------------------------|
| <i>e</i>              | The pointer to the current expression.                 |
| <i>i</i>              | The index for the current expression.                  |
| <i>LastExpression</i> | The flag indicating whether it is the last expression. |

##### Returns

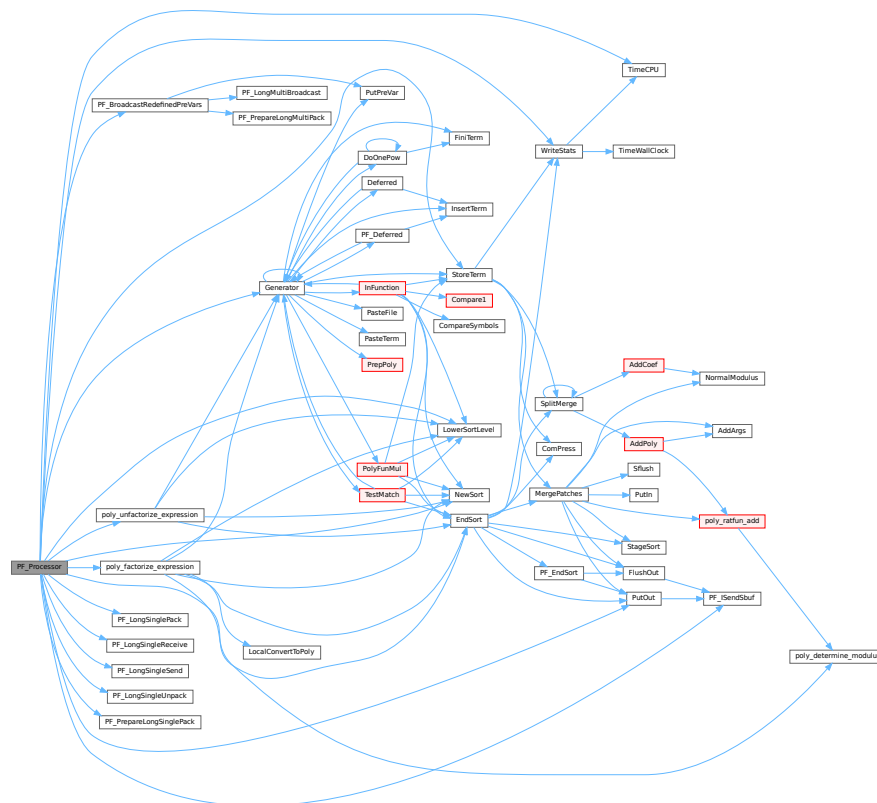
0 if OK, nonzero on error.

Definition at line 1540 of file [parallel.c](#).

References [EndSort\(\)](#), [Generator\(\)](#), [LowerSortLevel\(\)](#), [NewSort\(\)](#), [PACK\\_LONG](#), [PF\\_BroadcastRedefinedPreVars\(\)](#), [PF\\_ISendSbuf\(\)](#), [PF\\_LongSinglePack\(\)](#), [PF\\_LongSingleReceive\(\)](#), [PF\\_LongSingleSend\(\)](#), [PF\\_LongSingleUnpack\(\)](#), [PF\\_PrepareLongSinglePack\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [PutOut\(\)](#), [StoreTerm\(\)](#), [SWAP](#), [TimeCPU\(\)](#), and [WriteStats\(\)](#).

Referenced by [Processor\(\)](#).

Here is the call graph for this function:



#### 4.94.4.14 PF\_Init()

```
int PF_Init (
    int * argc,
    char *** argv )
```

All the library independent stuff. [PF\\_LibInit\(\)](#) should do all library dependent initializations.

##### Parameters

|             |                                     |
|-------------|-------------------------------------|
| <i>argc</i> | pointer to the number of arguments. |
| <i>argv</i> | pointer to the arguments.           |

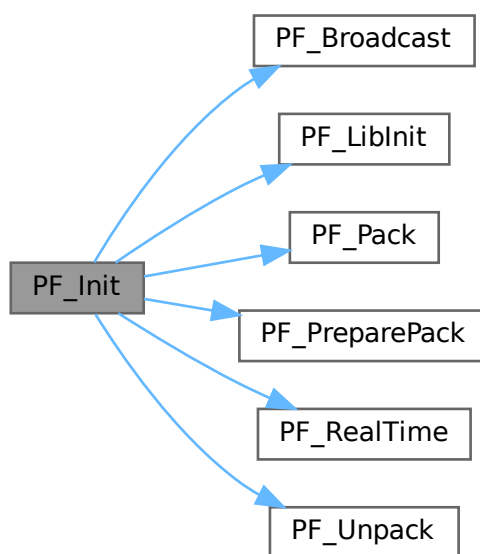
**Returns**

0 if OK, nonzero on error.

Definition at line 1963 of file [parallel.c](#).

References [PF\\_Broadcast\(\)](#), [PF\\_LibInit\(\)](#), [PF\\_Pack\(\)](#), [PF\\_PreparePack\(\)](#), [PF\\_RealTime\(\)](#), and [PF\\_Unpack\(\)](#).

Here is the call graph for this function:

**4.94.4.15 PF\_Terminate()**

```
int PF_Terminate (
    int errorcode )
```

Performs the finalization of ParFORM. To be called by Terminate().

**Parameters**

|              |                |
|--------------|----------------|
| <i>error</i> | an error code. |
|--------------|----------------|

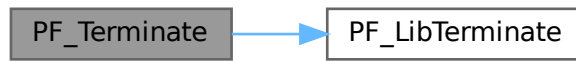
**Returns**

0 if OK, nonzero on error.

Definition at line 2058 of file [parallel.c](#).

References [PF\\_LibTerminate\(\)](#).

Here is the call graph for this function:



#### 4.94.4.16 PF\_GetSlaveTimes()

```
LONG PF_GetSlaveTimes (
    void )
```

Returns the total CPU time of all slaves together. This function must be called on the master and all slaves.

##### Returns

on the master, the sum of CPU times on all slaves.

Definition at line 2074 of file [parallel.c](#).

References [TimeCPU\(\)](#).

Here is the call graph for this function:



#### 4.94.4.17 PF\_BroadcastNumber()

```
LONG PF_BroadcastNumber (
    LONG x )
```

Broadcasts a LONG value from the master to the all slaves.

##### Parameters

|   |                                                 |
|---|-------------------------------------------------|
| x | the number to be broadcast (set on the master). |
|---|-------------------------------------------------|

**Returns**

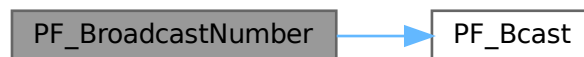
the synchronised result.

Definition at line 2094 of file [parallel.c](#).

References [PF\\_Bcast\(\)](#).

Referenced by [DoCheckpoint\(\)](#), [PF\\_BroadcastBuffer\(\)](#), and [StartVariables\(\)](#).

Here is the call graph for this function:

**4.94.4.18 PF\_BroadcastBuffer()**

```

void PF_BroadcastBuffer (
    WORD ** buffer,
    LONG * length )
  
```

Broadcasts a buffer from the master to all the slaves.

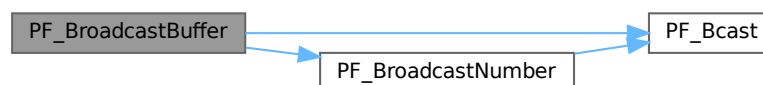
**Parameters**

|                |               |                                                                                                                                                                                        |
|----------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>in, out</i> | <i>buffer</i> | on the master, the buffer to be broadcast. On the slaves, the buffer will be allocated if the length is greater than 0. The caller must free it.                                       |
| <i>in, out</i> | <i>length</i> | on the master, the length of the buffer to be broadcast. On the slaves, it receives the length of transferred buffer. The actual transfer occurs only if the length is greater than 0. |

Definition at line 2121 of file [parallel.c](#).

References [PF\\_Bcast\(\)](#), and [PF\\_BroadcastNumber\(\)](#).

Here is the call graph for this function:



**4.94.4.19 PF\_BroadcastString()**

```
int PF_BroadcastString (
    UBYTE * str )
```

Broadcasts a string from the master to all slaves.

**Parameters**

|                |            |                                          |
|----------------|------------|------------------------------------------|
| <i>in, out</i> | <i>str</i> | The pointer to a null-terminated string. |
|----------------|------------|------------------------------------------|

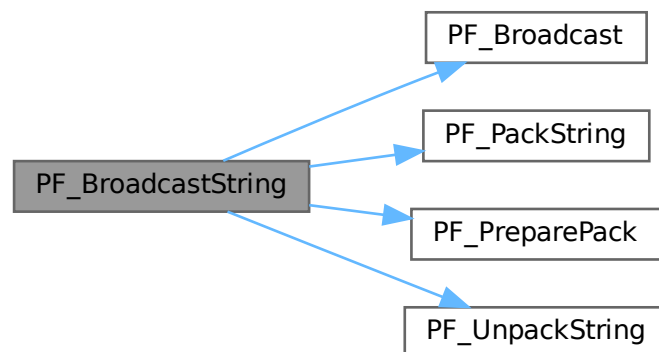
**Returns**

0 if OK, nonzero on error.

Definition at line 2163 of file [parallel.c](#).

References [PF\\_Broadcast\(\)](#), [PF\\_PackString\(\)](#), [PF\\_PreparePack\(\)](#), and [PF\\_UnpackString\(\)](#).

Here is the call graph for this function:

**4.94.4.20 PF\_BroadcastPreDollar()**

```
int PF_BroadcastPreDollar (
    WORD ** dbuffer,
    LONG * newsize,
    int * numterms )
```

Broadcasts dollar variables set as a preprocessor variables. Only the master is able to make an assignment like `#$a=g;` where `g` is an expression: only the master has an access to the expression. So, the master broadcasts the result to slaves.

The result is in `*dbuffer` of the size is `*newsize` (in number of WORDs), +1 for trailing zero. For slave `newsize` and `numterms` are output parameters.

**Parameters**

|                |                 |                                             |
|----------------|-----------------|---------------------------------------------|
| <i>in, out</i> | <i>dbuffer</i>  | the buffer for a dollar variable.           |
| <i>in, out</i> | <i>newsize</i>  | the size of the dollar variable in WORDs.   |
| <i>in, out</i> | <i>numterms</i> | the number of terms in the dollar variable. |

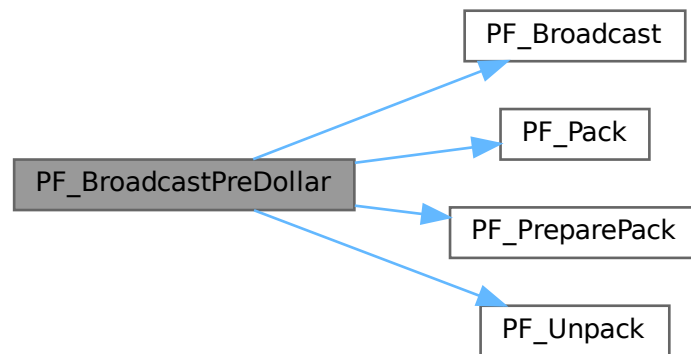
**Returns**

0 if OK, nonzero on error.

Definition at line 2218 of file [parallel.c](#).

References [PF\\_Broadcast\(\)](#), [PF\\_Pack\(\)](#), [PF\\_PreparePack\(\)](#), and [PF\\_Unpack\(\)](#).

Here is the call graph for this function:

**4.94.4.21 PF\_CollectModifiedDollars()**

```
int PF_CollectModifiedDollars (
    void )
```

Combines modified dollar variables on the all slaves, and store them into those on the master.

The potentially modified dollar variables are given in `PotModdollars`, and the number of them is given by `NumPotModdollars`.

The current module could be executed in parallel only if all potentially modified variables are listed in `ModOptdollars`, otherwise the module was switched to the sequential mode.



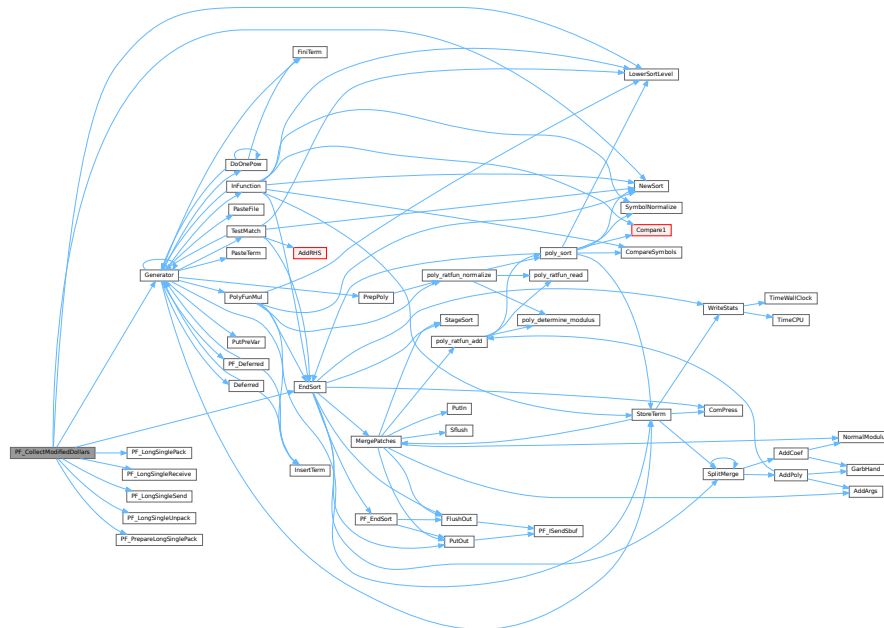
## Returns

0 if OK, nonzero on error.

Definition at line 2506 of file parallel.c.

References [EndSort\(\)](#), [Generator\(\)](#), [LowerSortLevel\(\)](#), [NewSort\(\)](#), [PF\\_LongSinglePack\(\)](#), [PF\\_LongSingleReceive\(\)](#), [PF\\_LongSingleSend\(\)](#), [PF\\_LongSingleUnpack\(\)](#), [PF\\_PrepareLongSinglePack\(\)](#), [VectorInit](#), [VectorPtr](#), [VectorReserve](#), and [VectorSize](#).

Here is the call graph for this function:



#### 4.94.4.22 PF\_BroadcastModifiedDollars()

```
int PF_BroadcastModifiedDollars (
    void )
```

Broadcasts modified dollar variables on the master to the all slaves.

The potentially modified dollar variables are given in PotModdollars, and the number of them is given by NumPotModdollars.

The current module could be executed in parallel only if all potentially modified variables are listed in `ModOptdollars`, otherwise the module was switched to the sequential mode. In either cases, we need to broadcast them.

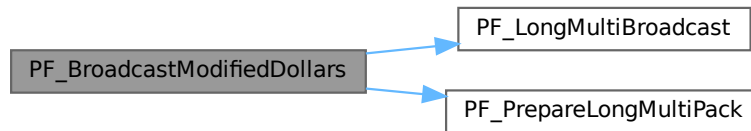
**Returns**

0 if OK, nonzero on error.

Definition at line 2785 of file [parallel.c](#).

References [PF\\_LongMultiBroadcast\(\)](#), and [PF\\_PrepareLongMultiPack\(\)](#).

Here is the call graph for this function:

**4.94.4.23 PF\_BroadcastRedefinedPreVars()**

```
int PF_BroadcastRedefinedPreVars (  
    void )
```

Broadcasts preprocessor variables, which were changed by the Redefine statements in the current module, from the master to the all slaves.

The potentially redefined preprocessor variables are given in `AC.pfirstnum`, and the number of them is given by `AC.numpfirstnum`. For an actually redefined variable, the corresponding value in `AC.inputnumbers` is non-negative.

**Returns**

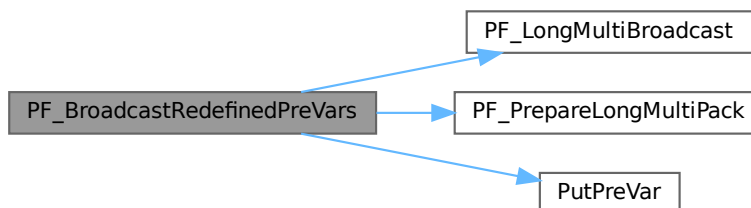
0 if OK, nonzero on error.

Definition at line 3002 of file [parallel.c](#).

References [PF\\_LongMultiBroadcast\(\)](#), [PF\\_PrepareLongMultiPack\(\)](#), [PutPreVar\(\)](#), [VectorPtr](#), and [VectorReserve](#).

Referenced by [PF\\_Processor\(\)](#).

Here is the call graph for this function:



**4.94.4.24 PF\_StoreInsideInfo()**

```
int PF_StoreInsideInfo (
    void )
```

Definition at line 3092 of file [parallel.c](#).

**4.94.4.25 PF\_RestoreInsideInfo()**

```
int PF_RestoreInsideInfo (
    void )
```

Definition at line 3118 of file [parallel.c](#).

**4.94.4.26 PF\_BroadcastCBuf()**

```
int PF_BroadcastCBuf (
    int bufnum )
```

Broadcasts a compiler buffer specified by *bufnum* from the master to the all slaves.

**Parameters**

|               |                                                   |
|---------------|---------------------------------------------------|
| <i>bufnum</i> | The index of the compiler buffer to be broadcast. |
|---------------|---------------------------------------------------|

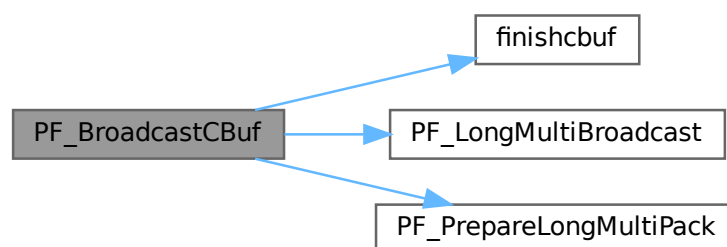
**Returns**

0 if OK, nonzero on error.

Definition at line 3144 of file [parallel.c](#).

References [CbUf::boomlijst](#), [CbUf::Buffer](#), [CbUf::BufferSize](#), [CbUf::CanCommu](#), [CbUf::dimension](#), [finishcbuf\(\)](#), [CbUf::lhs](#), [CbUf::numdum](#), [CbUf::NumTerms](#), [PF\\_LongMultiBroadcast\(\)](#), [PF\\_PrepareLongMultiPack\(\)](#), [CbUf::Pointer](#), [CbUf::rhs](#), and [CbUf::Top](#).

Here is the call graph for this function:



#### 4.94.4.27 PF\_BroadcastExpFlags()

```
int PF_BroadcastExpFlags (
    void )
```

Broadcasts AR.expflags and several properties of each expression, e.g., e->vflags, from the master to all slaves.

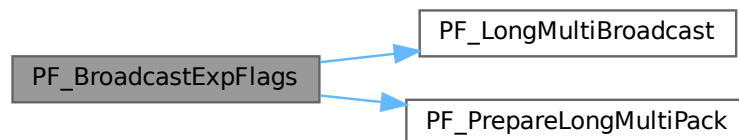
##### Returns

0 if OK, nonzero on error.

Definition at line 3255 of file [parallel.c](#).

References [PF\\_LongMultiBroadcast\(\)](#), and [PF\\_PrepareLongMultiPack\(\)](#).

Here is the call graph for this function:



#### 4.94.4.28 PF\_BroadcastExpr()

```
int PF_BroadcastExpr (
    EXPRESSIONS e,
    FILEHANDLE * file )
```

Broadcasts an expression from the master to the all slaves.

##### Parameters

|             |                                              |
|-------------|----------------------------------------------|
| <i>e</i>    | The expression to be broadcast.              |
| <i>file</i> | The file in which the expression is sitting. |

##### Returns

0 if OK, nonzero on error.

Definition at line 3549 of file [parallel.c](#).

Referenced by [PF\\_BroadcastRHS\(\)](#).

#### 4.94.4.29 PF\_BroadcastRHS()

```
int PF_BroadcastRHS (
    void )
```

Broadcasts expressions appearing in the right-hand side from the master to the all slaves.

##### Returns

0 if OK, nonzero on error.

Definition at line 3577 of file [parallel.c](#).

References [PF\\_BroadcastExpr\(\)](#).

Referenced by [Processor\(\)](#).

Here is the call graph for this function:



#### 4.94.4.30 PF\_InParallelProcessor()

```
int PF_InParallelProcessor (
    void )
```

Processes expressions in the InParallel mode, i.e., dividing expressions marked by partodo over the slaves.

##### Returns

0 if OK, nonzero on error.

Definition at line 3624 of file [parallel.c](#).

Referenced by [Processor\(\)](#).

#### 4.94.4.31 PF\_SendFile()

```
int PF_SendFile (
    int to,
    FILE * fd )
```

Sends a file to the process specified by *to*.

**Parameters**

|           |                                 |
|-----------|---------------------------------|
| <i>to</i> | the destination process number. |
| <i>fd</i> | the file to be sent.            |

**Returns**

the size of sent data in bytes, or -1 on error.

Definition at line 4221 of file [parallel.c](#).

References [PF\\_RawSend\(\)](#).

Referenced by [DoCheckpoint\(\)](#).

Here is the call graph for this function:

**4.94.4.32 PF\_RecvFile()**

```
int PF_RecvFile (
    int from,
    FILE * fd )
```

Receives a file from the process specified by *from*.

**Parameters**

|             |                                     |
|-------------|-------------------------------------|
| <i>from</i> | the source process number.          |
| <i>fd</i>   | the file to save the received data. |

**Returns**

the size of received data in bytes, or -1 on error.

Definition at line 4259 of file [parallel.c](#).

References [PF\\_RawRecv\(\)](#).

Referenced by [DoCheckpoint\(\)](#).

Here is the call graph for this function:



#### 4.94.4.33 PF\_MLock()

```
void PF_MLock (
    void )
```

A function called by MLOCK(ErrorMessageLock) for slaves.

Definition at line 4340 of file [parallel.c](#).

References [VectorClear](#).

#### 4.94.4.34 PF\_MUnlock()

```
void PF_MUnlock (
    void )
```

A function called by MUNLOCK(ErrorMessageLock) for slaves.

Definition at line 4356 of file [parallel.c](#).

References [PF\\_RawSend\(\)](#), [VectorEmpty](#), [VectorPtr](#), and [VectorSize](#).

Here is the call graph for this function:



#### 4.94.4.35 PF\_WriteFileToFile()

```
LONG PF_WriteFileToFile (
    int handle,
    UBYTE * buffer,
    LONG size )
```

Replaces WriteFileToFile() on the master and slaves.

It copies the given buffer into internal buffers if called between MLOCK(ErrorMessageLock) and MUNLOCK(ErrorMessageLock) for slaves and handle is StdOut or LogHandle, otherwise calls WriteFileToFile().

**Parameters**

|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| <i>handle</i> | a file handle that specifies the output.                          |
| <i>buffer</i> | a pointer to the source buffer containing the data to be written. |
| <i>size</i>   | the size of data to be written in bytes.                          |

**Returns**

the actual size of data written to the output in bytes.

Definition at line [4385](#) of file [parallel.c](#).

References [VectorClear](#), [VectorPtr](#), [VectorPushBacks](#), and [VectorSize](#).

**4.94.4.36 PF\_FlushStdOutBuffer()**

```
void PF_FlushStdOutBuffer (
    void )
```

Explicitly Flushes the buffer for the standard output on the master, which is used if PF\_ENABLE\_STDOUT\_↵  
BUFFERING is defined.

Definition at line [4479](#) of file [parallel.c](#).

References [VectorClear](#), [VectorPtr](#), and [VectorSize](#).

**4.94.4.37 PF\_FreeErrorMessageBuffers()**

```
void PF_FreeErrorMessageBuffers (
    void )
```

Frees the buffers allocated for the synchronized output.

Currently, not used anywhere, but could be used in [PF\\_Terminate\(\)](#).

Definition at line [4615](#) of file [parallel.c](#).

References [VectorFree](#).

**4.94.5 Variable Documentation****4.94.5.1 PF**

[PARALLELVARS](#) PF

Definition at line [90](#) of file [parallel.c](#).



## 4.95 parallel.c

[Go to the documentation of this file.](#)

```

00001
00011 /* #[ License : */
00012 /*
00013 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00014 *   When using this file you are requested to refer to the publication
00015 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00016 *   This is considered a matter of courtesy as the development was paid
00017 *   for by FOM the Dutch physics granting agency and we would like to
00018 *   be able to track its scientific use to convince FOM of its value
00019 *   for the community.
00020 *
00021 *   This file is part of FORM.
00022 *
00023 *   FORM is free software: you can redistribute it and/or modify it under the
00024 *   terms of the GNU General Public License as published by the Free Software
00025 *   Foundation, either version 3 of the License, or (at your option) any later
00026 *   version.
00027 *
00028 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00029 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00030 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00031 *   details.
00032 *
00033 *   You should have received a copy of the GNU General Public License along
00034 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00035 */
00036 /* #] License : */
00037 /*
00038 *   #[ includes :
00039 */
00040 #include "form3.h"
00041 #include "vector.h"
00042
00043 /*
00044 #define PF_DEBUG_BCAST_LONG
00045 #define PF_DEBUG_BCAST_BUF
00046 #define PF_DEBUG_BCAST_PREDOLLAR
00047 #define PF_DEBUG_BCAST_RHSEXP
00048 #define PF_DEBUG_BCAST_DOLLAR
00049 #define PF_DEBUG_BCAST_PREVAR
00050 #define PF_DEBUG_BCAST_CBUF
00051 #define PF_DEBUG_BCAST_EXPRFLAGS
00052 #define PF_DEBUG_REDUCE_DOLLAR
00053 */
00054
00055 /* mpi.c */
00056 LONG PF_RealTime(int);
00057 int PF_LibInit(int*, char**);
00058 int PF_LibTerminate(int);
00059 int PF_Probe(int*);
00060 int PF_RecvWbuf(WORD*, LONG*, int*);
00061 int PF_IRecvRbuf(PF_BUFFER*, int, int);
00062 int PF_WaitRbuf(PF_BUFFER *, int, LONG *);
00063 int PF_RawSend(int dest, void *buf, LONG l, int tag);
00064 LONG PF_RawRecv(int *src, void *buf, LONG thesize, int *tag);
00065 int PF_RawProbe(int *src, int *tag, int *bytesize);
00066
00067 /* Private functions */
00068
00069 static int PF_WaitAllSlaves(void);
00070
00071 static void PF_PackRedefinedPreVars(void);
00072 static void PF_UnpackRedefinedPreVars(void);
00073
00074 static int PF_Wait4MasterIP(int tag);
00075 static int PF_DoOneExpr(void);
00076 static int PF_ReadMaster(void); /* reads directly to its scratch! */
00077 static int PF_Slave2MasterIP(int src); /* both master and slave */
00078 static int PF_Master2SlaveIP(int dest, EXPRESSIONS e);
00079 static int PF_WalkThrough(WORD *t, LONG l, LONG chunk, LONG *count);
00080 static int PF_SendChunkIP(FILEHANDLE *curfile, POSITION *position, int to, LONG thesize);
00081 static int PF_RecvChunkIP(FILEHANDLE *curfile, int from, LONG thesize);
00082
00083 static void PF_ReceiveErrorMessage(int src, int tag);
00084 static void PF_CatchErrorMessages(int *src, int *tag);
00085 static void PF_CatchErrorMessagesForAll(void);
00086 static int PF_ProbeWithCatchingErrorMessages(int *src);
00087
00088 /* Variables */
00089
00090 PARALLELVARS PF;
00091 #ifdef MPI2

```

```

00092 WORD *PF_shared_buff;
00093 #endif
00094
00095 static LONG PF_goutterms; /* (master) Total out terms at PF_EndSort(), used in PF_Statistics().
    */
00096 static POSITION PF_exprsize; /* (master) The size of the expression at PF_EndSort(), used in
    PF_Processor(). */
00097
00098 /*
00099 This will work well only under Linux, see
00100 #ifdef PF_WITH_SCHED_YIELD
00101 below in PF_WaitAllSlaves().
00102 */
00103 #ifdef PF_WITH_SCHED_YIELD
00104 #include <sched.h>
00105 #endif
00106
00107 #ifdef PF_WITHLOG
00108 #define PRINTFBUF(TEXT,TERM,SIZE) { UBYTE lbuf[24]; if(PF.log){ WORD iii;\
00109 NumToStr(lbuf,AC.CModule); \
00110 fprintf(stderr,"%d|s] %s : ",PF.me,lbuf,(char*)TEXT);\
00111 if(TERM){ fprintf(stderr,"%d] ",(int)(*TERM));\
00112 if((SIZE)<500 && (SIZE)>0) for(iii=1;iii<(SIZE);iii++)\
00113 fprintf(stderr,"%d ",TERM[iii]); }\
00114 fprintf(stderr,"\n");\
00115 fflush(stderr); } }
00116 #else
00117 #define PRINTFBUF(TEXT,TERM,SIZE) {}
00118 #endif
00119
00124 #define SWAP(x, y) \
00125 do { \
00126 char swap_tmp__[sizeof(x) == sizeof(y) ? (int)sizeof(x) : -1]; \
00127 memcpy(swap_tmp__, &y, sizeof(x)); \
00128 memcpy(&y, &x, sizeof(x)); \
00129 memcpy(&x, swap_tmp__, sizeof(x)); \
00130 } while (0)
00131
00135 #define PACK_LONG(p, n) \
00136 do { \
00137 * (p)++ = (UWORD)((ULONG)(n) & (ULONG)WORDMASK); \
00138 * (p)++ = (UWORD)((ULONG)(n) » BITSINWORD) & (ULONG)WORDMASK); \
00139 } while (0)
00140
00144 #define UNPACK_LONG(p, n) \
00145 do { \
00146 (n) = (LONG)((((ULONG)(p)[1] & (ULONG)WORDMASK) « BITSINWORD) | ((ULONG)(p)[0] &
(ULONG)WORDMASK)); \
00147 (p) += 2; \
00148 } while (0)
00149
00153 #define CHECK(condition) _CHECK(condition, __FILE__, __LINE__)
00154 #define _CHECK(condition, file, line) __CHECK(condition, file, line)
00155 #define __CHECK(condition, file, line) \
00156 do { \
00157 if ( !(condition) ) { \
00158 Error0("Fatal error at " file ":" #line); \
00159 Terminate(-1); \
00160 } \
00161 } while (0)
00162
00163 /*
00164 * For debugging.
00165 */
00166 #define DBGOUT(lv1, lv2, a) do { if ( lv1 >= lv2 ) { printf a; fflush(stdout); } } while (0)
00167
00168 /* (AN.ninterms of master) == max(AN.ninterms of slaves) == sum(PF_linterms of slaves) at EndSort().
    */
00169 #define DBGOUT_NINTERMS(lv, a)
00170 /* #define DBGOUT_NINTERMS(lv, a) DBGOUT(1, lv, a) */
00171
00172 /*
00173 #] includes :
00174 #[ statistics :
00175 #] variables : (should be part of a struct?)
00176 */
00177 static LONG PF_linterms; /* local interms on this proces: PF_Proces */
00178 #define PF_STATS_SIZE 5
00179 static LONG **PF_stats = NULL; /* space for collecting statistics of all procs */
00180 static LONG PF_laststat; /* last realtime when statistics were printed */
00181 static LONG PF_statsinterval; /* timeinterval for printing statistics */
00182 /*
00183 #] variables :
00184 #[ PF_Statistics :
00185 */
00186
00195 static int PF_Statistics(LONG **stats, int proc)

```

```

00196 {
00197     GETIDENTITY
00198     LONG real, cpu;
00199     WORD rpart, cpart;
00200     int i, j;
00201
00202     if ( AT.SS == AM.S0 && PF.me == MASTER ) {
00203         real = PF_RealTime(PF_TIME); rpart = (WORD)(real%100); real /= 100;
00204
00205         if ( PF_stats == NULL ) {
00206             PF_stats = (LONG**)Malloc1(PF.numtasks*sizeof(LONG*), "PF_stats 1");
00207             for ( i = 0; i < PF.numtasks; i++ ) {
00208                 PF_stats[i] = (LONG*)Malloc1(PF_STATS_SIZE*sizeof(LONG), "PF_stats 2");
00209                 for ( j = 0; j < PF_STATS_SIZE; j++ ) PF_stats[i][j] = 0;
00210             }
00211         }
00212         if ( proc > 0 ) for ( i = 0; i < PF_STATS_SIZE; i++ ) PF_stats[proc][i] = stats[0][i];
00213
00214         if ( real >= PF_laststat + PF_statsinterval || proc == 0 ) {
00215             LONG sum[PF_STATS_SIZE];
00216
00217             for ( i = 0; i < PF_STATS_SIZE; i++ ) sum[i] = 0;
00218             sum[0] = cpu = TimeCPU(1);
00219             cpart = (WORD)(cpu%1000);
00220             cpu /= 1000;
00221             cpart /= 10;
00222             if ( AC.OldParallelStats ) MesPrint("");
00223             if ( proc > 0 && AC.StatsFlag && AC.OldParallelStats ) {
00224                 MesPrint("proc      CPU      in      gen      left      byte");
00225                 MesPrint("%3d : %7l.%2i %10l", 0, cpu, cpart, AN.ninterms);
00226             }
00227             else if ( AC.StatsFlag && AC.OldParallelStats ) {
00228                 MesPrint("proc      CPU      in      gen      out      byte");
00229                 MesPrint("%3d : %7l.%2i %10l %10l %10l", 0, cpu, cpart, AN.ninterms, 0, PF_goutterms);
00230             }
00231
00232             for ( i = 1; i < PF.numtasks; i++ ) {
00233                 cpart = (WORD)(PF_stats[i][0]%1000);
00234                 cpu = PF_stats[i][0] / 1000;
00235                 cpart /= 10;
00236                 if ( AC.StatsFlag && AC.OldParallelStats )
00237                     MesPrint("%3d : %7l.%2i %10l %10l %10l", i, cpu, cpart,
00238                             PF_stats[i][2], PF_stats[i][3], PF_stats[i][4]);
00239                 for ( j = 0; j < PF_STATS_SIZE; j++ ) sum[j] += PF_stats[i][j];
00240             }
00241             cpart = (WORD)(sum[0]%1000);
00242             cpu = sum[0] / 1000;
00243             cpart /= 10;
00244             if ( AC.StatsFlag && AC.OldParallelStats ) {
00245                 MesPrint("Sum = %7l.%2i %10l %10l %10l", cpu, cpart, sum[2], sum[3], sum[4]);
00246                 MesPrint("Real = %7l.%2i %20s (%1) %16s",
00247                         real, rpart, AC.Commercial, AC.CModule, EXPRNAME(AR.CurExpr));
00248                 MesPrint("");
00249             }
00250             PF_laststat = real;
00251         }
00252     }
00253     return(0);
00254 }
00255 /*
00256     #] PF_Statistics :
00257     #] statistics :
00258     #[ sort.c :
00259     #[ sort variables :
00260 */
00261
00265 typedef struct NoDe {
00266     struct NoDe *left;
00267     struct NoDe *rghet;
00268     int lloser;
00269     int rloser;
00270     int lsrc;
00271     int rsrc;
00272 } NODE;
00273
00274 /*
00275     should/could be put in one struct
00276 */
00277 static NODE *PF_root;          /* root of tree of losers */
00278 static WORD PF_loser;          /* this is the last loser */
00279 static WORD **PF_term;         /* these point to the active terms */
00280 static WORD **PF_newcpos;      /* new coeffs of merged terms */
00281 static WORD *PF_newcclen;      /* length of new coefficients */
00282
00283 /*
00284     preliminary: could also write somewhere else?
00285 */

```

```

00286
00287 static WORD *PF_WorkSpace; /* used in PF_EndSort() */
00288 static UWORD *PF_ScratchSpace; /* used in PF_GetLoser() */
00289
00290 /*
00291     #] sort variables :
00292     #[ PF_AllocBuf :
00293 */
00294
00311 static PF_BUFFER *PF_AllocBuf(int nbufs, LONG bsize, WORD free)
00312 {
00313     PF_BUFFER *buf;
00314     UBYTE *p, *stop;
00315     LONG allocsize;
00316     int i;
00317
00318     allocsize =
00319         (LONG)(sizeof(PF_BUFFER) + 4*nbufs*sizeof(WORD*) + (nbufs-free)*bsize);
00320
00321     allocsize +=
00322         (LONG)( nbufs * ( 2 * sizeof(MPI_Status)
00323                         + sizeof(MPI_Request)
00324                         + sizeof(MPI_Datatype)
00325                     ) );
00326     allocsize += (LONG)( nbufs * 3 * sizeof(int) );
00327
00328     if ( ( buf = (PF_BUFFER*)Malloc1(allocsize,"PF_AllocBuf") ) == NULL ) return(NULL);
00329
00330     p = ((UBYTE *)buf) + sizeof(PF_BUFFER);
00331     stop = ((UBYTE *)buf) + allocsize;
00332
00333     buf->nbufs = nbufs;
00334     buf->active = 0;
00335
00336     buf->buff = (WORD**)p; p += buf->nbufs*sizeof(WORD*);
00337     buf->fill = (WORD**)p; p += buf->nbufs*sizeof(WORD*);
00338     buf->full = (WORD**)p; p += buf->nbufs*sizeof(WORD*);
00339     buf->stop = (WORD**)p; p += buf->nbufs*sizeof(WORD*);
00340     buf->status = (MPI_Status *)p; p += buf->nbufs*sizeof(MPI_Status);
00341     buf->retstat = (MPI_Status *)p; p += buf->nbufs*sizeof(MPI_Status);
00342     buf->request = (MPI_Request *)p; p += buf->nbufs*sizeof(MPI_Request);
00343     buf->type = (MPI_Datatype *)p; p += buf->nbufs*sizeof(MPI_Datatype);
00344     buf->index = (int *)p; p += buf->nbufs*sizeof(int);
00345
00346     for ( i = 0; i < buf->nbufs; i++ ) buf->request[i] = MPI_REQUEST_NULL;
00347     buf->tag = (int *)p; p += buf->nbufs*sizeof(int);
00348     buf->from = (int *)p; p += buf->nbufs*sizeof(int);
00349 /*
00350     and finally the real bufferspace
00351 */
00352     for ( i = free; i < buf->nbufs; i++ ) {
00353         buf->buff[i] = (WORD*)p; p += bsize;
00354         buf->stop[i] = (WORD*)p;
00355         buf->fill[i] = buf->full[i] = buf->buff[i];
00356     }
00357     if ( p != stop ) {
00358         MesPrint("Error in PF_AllocBuf p = %x stop = %x\n",p,stop);
00359         return(NULL);
00360     }
00361     return(buf);
00362 }
00363
00364 /*
00365     #] PF_AllocBuf :
00366     #[ PF_InitTree :
00367 */
00368
00380 static int PF_InitTree(void)
00381 {
00382     GETIDENTITY
00383     PF_BUFFER **rbuf = PF.rbufs;
00384     UBYTE *p, *stop;
00385     int numrbufs,numtasks = PF.numtasks;
00386     int i, j, src, numnodes;
00387     int numslaves = numtasks - 1;
00388     LONG size;
00389 /*
00390     #] the buffers : for the new coefficients and the terms
00391     we need one for each slave
00392 */
00393     if ( PF_term == NULL ) {
00394         size = 2*numtasks*sizeof(WORD*) + sizeof(WORD)*
00395             ( numtasks*(1 + AM.MaxTal) + (AM.MaxTer/sizeof(WORD)+1) + 2*(AM.MaxTal+2) );
00396
00397         PF_term = (WORD **)Malloc1(size,"PF_term");
00398         stop = ((UBYTE*)PF_term) + size;
00399         p = ((UBYTE*)PF_term) + numtasks*sizeof(WORD*);

```

```

00400
00401     PF_newcpos = (WORD **)p; p += sizeof(WORD*) * numtasks;
00402     PF_newclen = (WORD *)p; p += sizeof(WORD) * numtasks;
00403     for ( i = 0; i < numtasks; i++ ) {
00404         PF_newcpos[i] = (WORD *)p; p += sizeof(WORD)*AM.MaxTal;
00405         PF_newclen[i] = 0;
00406     }
00407     PF_WorkSpace = (WORD *)p; p += AM.MaxTer+sizeof(WORD);
00408     PF_ScratchSpace = (UWORD*)p; p += 2*(AM.MaxTal+2)*sizeof(UWORD);
00409
00410     if ( p != stop ) { MesPrint("error in PF_InitTree"); return(-1); }
00411 }
00412 /*
00413     #] the buffers :
00414     #[ the receive buffers :
00415 */
00416     numrbufs = PF.numrbufs;
00417 /*
00418     this is the size we have in the combined sortbufs for one slave
00419 */
00420     size = (AT.SS->sTop2 - AT.SS->lBuffer - 1)/(PF.numtasks - 1);
00421
00422     if ( rbuf == NULL ) {
00423         if ( ( rbuf = (PF_BUFFER**)Malloca1(numtasks*sizeof(PF_BUFFER*), "Master: rbufs") ) == NULL )
00424             return(-1);
00425         if ( ( rbuf[0] = PF_AllocBuf(1,0,1) ) == NULL ) return(-1);
00426         for ( i = 1; i < numtasks; i++ ) {
00427             if (!(rbuf[i] = PF_AllocBuf(numrbufs,sizeof(WORD)*size,1))) return(-1);
00428         }
00429         rbuf[0]->buff[0] = AT.SS->lBuffer;
00430         rbuf[0]->full[0] = rbuf[0]->fill[0] = rbuf[0]->buff[0];
00431         rbuf[0]->stop[0] = rbuf[1]->buff[0] = rbuf[0]->buff[0] + 1;
00432         rbuf[1]->full[0] = rbuf[1]->fill[0] = rbuf[1]->buff[0];
00433         for ( i = 2; i < numtasks; i++ ) {
00434             rbuf[i-1]->stop[0] = rbuf[i]->buff[0] = rbuf[i-1]->buff[0] + size;
00435             rbuf[i]->full[0] = rbuf[i]->fill[0] = rbuf[i]->buff[0];
00436         }
00437         rbuf[numtasks-1]->stop[0] = rbuf[numtasks-1]->buff[0] + size;
00438
00439         for ( i = 1; i < numtasks; i++ ) {
00440             for ( j = 0; j < rbuf[i]->numbufs; j++ ) {
00441                 rbuf[i]->full[j] = rbuf[i]->fill[j] = rbuf[i]->buff[j] + AM.MaxTer/sizeof(WORD) + 2;
00442             }
00443             PF_term[i] = rbuf[i]->fill[rbuf[i]->active];
00444             *PF_term[i] = 0;
00445             PF_IRecvRbuf(rbuf[i],rbuf[i]->active,i);
00446         }
00447         rbuf[0]->active = 0;
00448         PF_term[0] = rbuf[0]->buff[0];
00449         PF_term[0][0] = 0; /* PF_term[0] is used for a zero term. */
00450         PF.rbufs = rbuf;
00451 /*
00452         #] the receive buffers :
00453         #[ the actual tree :
00454
00455         calculate number of nodes in mergetree and allocate space for them
00456 */
00457         if ( numslaves < 3 ) numnodes = 1;
00458         else {
00459             numnodes = 2;
00460             while ( numnodes < numslaves ) numnodes *= 2;
00461             numnodes -= 1;
00462         }
00463
00464         if ( PF_root == NULL )
00465             if ( ( PF_root = (NODE*)Malloca1(sizeof(NODE)*numnodes,"nodes in mergetree") ) == NULL )
00466                 return(-1);
00467 /*
00468         then initialize all the nodes
00469 */
00470         src = 1;
00471         for ( i = 0; i < numnodes; i++ ) {
00472             if ( 2*(i+1) <= numnodes ) {
00473                 PF_root[i].left = &(PF_root[2*(i+1)-1]);
00474                 PF_root[i].lsrc = 0;
00475             }
00476             else {
00477                 PF_root[i].left = 0;
00478                 if ( src < numtasks ) PF_root[i].lsrc = src++;
00479                 else PF_root[i].lsrc = 0;
00480             }
00481             PF_root[i].lloser = 0;
00482         }
00483         for ( i = 0; i < numnodes; i++ ) {
00484             if ( 2*(i+1)+1 <= numnodes ) {
00485                 PF_root[i].rght = &(PF_root[2*(i+1)]);

```

```

00486         PF_root[i].rsrc = 0;
00487     }
00488     else {
00489         PF_root[i].rghet = 0;
00490         if (src<numtasks) PF_root[i].rsrc = src++;
00491         else PF_root[i].rsrc = 0;
00492     }
00493     PF_root[i].rloser = 0;
00494 }
00495 /*
00496     #] the actual tree :
00497 */
00498     return(numnodes);
00499 }
00500
00501 /*
00502     #] PF_InitTree :
00503     #[ PF_PutIn :
00504 */
00505
00524 static WORD *PF_PutIn(int src)
00525 {
00526     int tag;
00527     WORD im, r;
00528     WORD *m1, *m2;
00529     LONG size;
00530     PF_BUFFER *rbuf = PF.rbufs[src];
00531     int a = rbuf->active;
00532     int next = a+1 >= rbuf->numbufs ? 0 : a+1 ;
00533     WORD *lastterm = PF_term[src];
00534     WORD *term = rbuf->fill[a];
00535
00536     if ( src <= 0 ) return(PF_term[0]);
00537
00538     if ( rbuf->full[a] == rbuf->buff[a] + AM.MaxTer/sizeof(WORD) + 2 ) {
00539 /*
00540         very first term from this src
00541 */
00542         tag = PF_WaitRbuf(rbuf,a,&size);
00543         rbuf->full[a] += size;
00544         if ( tag == PF_ENDBUFFER_MSGTAG ) *rbuf->full[a]++ = 0;
00545         else if ( rbuf->numbufs > 1 ) {
00546 /*
00547             post a nonblock. recv. for the next buffer
00548 */
00549             rbuf->full[next] = rbuf->buff[next] + AM.MaxTer/sizeof(WORD) + 2;
00550             size = (LONG)(rbuf->stop[next] - rbuf->full[next]);
00551             PF_IRecvRbuf(rbuf,next,src);
00552         }
00553     }
00554     if ( *term == 0 && term != rbuf->full[a] ) return(PF_term[0]);
00555 /*
00556     exception is for rare cases when the terms fitted exactly into buffer
00557 */
00558     if ( term + *term > rbuf->full[a] || term + 1 >= rbuf->full[a] ) {
00559 newterms:
00560         m1 = rbuf->buff[next] + AM.MaxTer/sizeof(WORD) + 1;
00561         if ( *term < 0 || term == rbuf->full[a] ) {
00562 /*
00563             copy term and lastterm to the new buffer, so that they end at m1
00564 */
00565             m2 = rbuf->full[a] - 1;
00566             while ( m2 >= term ) *m1-- = *m2--;
00567             rbuf->fill[next] = term = m1 + 1;
00568             m2 = lastterm + *lastterm - 1;
00569             while ( m2 >= lastterm ) *m1-- = *m2--;
00570             lastterm = m1 + 1;
00571         }
00572         else {
00573 /*
00574             copy beginning of term to the next buffer so that it ends at m1
00575 */
00576             m2 = rbuf->full[a] - 1;
00577             while ( m2 >= term ) *m1-- = *m2--;
00578             rbuf->fill[next] = term = m1 + 1;
00579         }
00580         if ( rbuf->numbufs == 1 ) {
00581             rbuf->full[a] = rbuf->buff[a] + AM.MaxTer/sizeof(WORD) + 2;
00582             size = (LONG)(rbuf->stop[a] - rbuf->full[a]);
00583             PF_IRecvRbuf(rbuf,a,src);
00584         }
00585 /*
00586         wait for new terms in the next buffer
00587 */
00588         rbuf->full[next] = rbuf->buff[next] + AM.MaxTer/sizeof(WORD) + 2;
00589         tag = PF_WaitRbuf(rbuf,next,&size);
00590         rbuf->full[next] += size;

```

```

00591         if ( tag == PF_ENDBUFFER_MSGTAG ) {
00592             *rbuf->full[next]++ = 0;
00593         }
00594         else if ( rbuf->numbufs > 1 ) {
00595             /*
00596                 post a nonblock. recv. for active buffer, it is not needed anymore
00597             */
00598             rbuf->full[a] = rbuf->buff[a] + AM.MaxTer/sizeof(WORD) + 2;
00599             size = (LONG)(rbuf->stop[a] - rbuf->full[a]);
00600             PF_IRecvRbuf(rbuf,a,src);
00601         }
00602         /*
00603             now safely make next buffer active
00604         */
00605         a = rbuf->active = next;
00606     }
00607     if ( *term < 0 ) {
00608         /*
00609             We need to decompress the term
00610         */
00611         im = *term;
00612         r = term[1] - im + 1;
00613         m1 = term + 2;
00614         m2 = lastterm - im + 1;
00615         while ( ++im <= 0 ) *--m1 = *--m2;
00616         *--m1 = r;
00617         rbuf->fill[a] = term = m1;
00618         if ( term + *term > rbuf->full[a] ) goto newterms;
00619     }
00620     rbuf->fill[a] += *term;
00621     return(term);
00622 }
00623
00624
00625 /*
00626     #] PF_PutIn :
00627     #[ PF_GetLoser :
00628 */
00629
00648 static int PF_GetLoser(NODE *n)
00649 {
00650     GETIDENTITY
00651     WORD comp;
00652
00653     if ( PF_loser == 0 ) {
00654         /*
00655             this is for the right initialization of the tree only
00656         */
00657         if ( n->left ) n->lloser = PF_GetLoser(n->left);
00658         else {
00659             n->lloser = n->lsrc;
00660             if ( *(PF_term[n->lsrc] = PF_PutIn(n->lsrc)) == 0 ) n->lloser = 0;
00661         }
00662         PF_loser = 0;
00663         if ( n->right ) n->rloser = PF_GetLoser(n->right);
00664         else {
00665             n->rloser = n->rsrc;
00666             if ( *(PF_term[n->rsrc] = PF_PutIn(n->rsrc)) == 0 ) n->rloser = 0;
00667         }
00668         PF_loser = 0;
00669     }
00670     else if ( PF_loser == n->lloser ) {
00671         if ( n->left ) n->lloser = PF_GetLoser(n->left);
00672         else {
00673             n->lloser = n->lsrc;
00674             if ( *(PF_term[n->lsrc] = PF_PutIn(n->lsrc)) == 0 ) n->lloser = 0;
00675         }
00676     }
00677     else if ( PF_loser == n->rloser ) {
00678 newright:
00679         if ( n->right ) n->rloser = PF_GetLoser(n->right);
00680         else {
00681             n->rloser = n->rsrc;
00682             if ( *(PF_term[n->rsrc] = PF_PutIn(n->rsrc)) == 0 ) n->rloser = 0;
00683         }
00684     }
00685     if ( n->lloser > 0 && n->rloser > 0 ) {
00686         comp = CompareTerms(BHEAD PF_term[n->lloser],PF_term[n->rloser],(WORD)0);
00687         if ( comp > 0 ) return(n->lloser);
00688         else if (comp < 0 ) return(n->rloser);
00689         else {
00690             /*
00691                 #[ terms are equal :
00692             */
00693             WORD *lcpos, *rcpos;
00694             UWORD *newcpos;
00695             WORD lclen, rclen, newclen, newnlen;

```

```

00696         SORTING *S = AT.SS;
00697
00698         if ( S->PolyWise ) {
00699             /*
00700             #[ Here we work with PolyFun :
00701             */
00702                 WORD *ttl, *w;
00703                 WORD r1,r2;
00704                 WORD *ml = PF_term[n->lloser];
00705                 WORD *mr = PF_term[n->rloser];
00706
00707                 if ( ( r1 = (int)*PF_term[n->lloser] ) <= 0 ) r1 = 20;
00708                 if ( ( r2 = (int)*PF_term[n->rloser] ) <= 0 ) r2 = 20;
00709                 ttl = ml;
00710                 ml += S->PolyWise;
00711                 mr += S->PolyWise;
00712                 if ( S->PolyFlag == 2 ) {
00713                     w = poly_ratfun_add(BHEAD ml,mr);
00714                     if ( *ttl + w[1] - ml[1] > AM.MaxTer/((LONG)sizeof(WORD)) ) {
00715                         MesPrint("Term too complex in PolyRatFun addition. MaxTermSize of %10l is too
small",AM.MaxTer);
00716                         Terminate(-1);
00717                     }
00718                     AT.WorkPointer = w;
00719                 }
00720                 else {
00721                     w = AT.WorkPointer;
00722                     if ( w + ml[1] + mr[1] > AT.WorkTop ) {
00723                         MesPrint("A WorkSpace of %10l is too small",AM.WorkSize);
00724                         Terminate(-1);
00725                     }
00726                     AddArgs(BHEAD ml,mr,w);
00727                 }
00728                 r1 = w[1];
00729                 if ( r1 <= FUNHEAD || ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) ) {
00730                     goto cancelled;
00731                 }
00732                 if ( r1 == ml[1] ) {
00733                     NCOPY(ml,w,r1);
00734                 }
00735                 else if ( r1 < ml[1] ) {
00736                     r2 = ml[1] - r1;
00737                     mr = w + r1;
00738                     ml += ml[1];
00739                     while ( --r1 >= 0 ) *--ml = *--mr;
00740                     mr = ml - r2;
00741                     r1 = S->PolyWise;
00742                     while ( --r1 >= 0 ) *--ml = *--mr;
00743                     *ml -= r2;
00744                     PF_term[n->lloser] = ml;
00745                 }
00746                 else {
00747                     r2 = r1 - ml[1];
00748                     if ( r2 > 2*AM.MaxTal )
00749                         MesPrint("warning: new term in polyfun is large");
00750                     mr = ttl - r2;
00751                     r1 = S->PolyWise;
00752                     ml = ttl;
00753                     *ml += r2;
00754                     PF_term[n->lloser] = mr;
00755                     NCOPY(mr,ml,r1);
00756                     r1 = w[1];
00757                     NCOPY(mr,w,r1);
00758                 }
00759                 PF_newclen[n->rloser] = 0;
00760                 PF_loser = n->rloser;
00761                 goto newright;
00762             /*
00763             #[ Here we work with PolyFun :
00764             */
00765                 }
00766                 if ( ( lclen = PF_newclen[n->lloser] ) != 0 ) lcpos = PF_newcpos[n->lloser];
00767                 else {
00768                     lcpos = PF_term[n->lloser];
00769                     lclen = *(lcpos += *lcpos - 1);
00770                     lcpos -= ABS(lclen) - 1;
00771                 }
00772                 if ( ( rclen = PF_newclen[n->rloser] ) != 0 ) rcpos = PF_newcpos[n->rloser];
00773                 else {
00774                     rcpos = PF_term[n->rloser];
00775                     rclen = *(rcpos += *rcpos - 1);
00776                     rcpos -= ABS(rclen) - 1;
00777                 }
00778                 lclen = ( (lclen > 0) ? (lclen-1) : (lclen+1) ) >> 1;
00779                 rclen = ( (rclen > 0) ? (rclen-1) : (rclen+1) ) >> 1;
00780                 newcpos = PF_ScratchSpace;
00781                 if ( AddRat(BHEAD (UWORD *)lcpos,lclen,(UWORD *)rcpos,rclen,newcpos,&newnlen) )

```



```

    return(-1);
00782     if ( AN.ncmod != 0 ) {
00783         if ( ( AC.modmode & POSNEG ) != 0 ) {
00784             NormalModulus(newcpos,&newnlen);
00785         }
00786         if ( BigLong(newcpos,newnlen,(UWORD *)AC.cmod,ABS(AN.ncmod)) >=0 ) {
00787             WORD ii;
00788             SubPLon(newcpos,newnlen,(UWORD *)AC.cmod,ABS(AN.ncmod),newcpos,&newnlen);
00789             newcpos[newnlen] = 1;
00790             for ( ii = 1; ii < newnlen; ii++ ) newcpos[newnlen+ii] = 0;
00791         }
00792     }
00793     if ( newnlen == 0 ) {
00794         /*
00795             terms cancel, get loser of left subtree and then of right subtree
00796         */
00797         cancelled:
00798             PF_loser = n->lloser;
00799             PF_newcpos[n->lloser] = 0;
00800             if ( n->left ) n->lloser = PF_GetLoser(n->left);
00801             else {
00802                 n->lloser = n->lsrc;
00803                 if ( *(PF_term[n->lsrc] = PF_PutIn(n->lsrc)) == 0 ) n->lloser = 0;
00804             }
00805             PF_loser = n->rloser;
00806             PF_newcpos[n->rloser] = 0;
00807             goto newright;
00808         }
00809         else {
00810             /*
00811                 keep the left term and get the loser of right subtree
00812             */
00813             newnlen *= 2;
00814             newcpos = ( newnlen > 0 ) ? ( newnlen + 1 ) : ( newnlen - 1 );
00815             if ( newnlen < 0 ) newnlen = -newnlen;
00816             PF_newcpos[n->lloser] = newcpos;
00817             lcpos = PF_newcpos[n->lloser];
00818             if ( newcpos < 0 ) newcpos = -newcpos;
00819             while ( newcpos-- ) *lcpos++ = *newcpos++;
00820             PF_loser = n->rloser;
00821             PF_newcpos[n->rloser] = 0;
00822             goto newright;
00823         }
00824         /*
00825             #] terms are equal :
00826         */
00827     }
00828 }
00829 if ( n->lloser > 0 ) return(n->lloser);
00830 if ( n->rloser > 0 ) return(n->rloser);
00831 return(0);
00832 }
00833 /*
00834     #] PF_GetLoser :
00835     #[ PF_EndSort :
00836 */
00837
00864 int PF_EndSort(void)
00865 {
00866     GETIDENTITY
00867     FILEHANDLE *fout = AR.outfile;
00868     PF_BUFFER *sbuf=PF.sbuf;
00869     SORTING *S = AT.SS;
00870     WORD *outterm,*pp;
00871     LONG size, noutterms;
00872     POSITION position, oldposition;
00873     WORD i,cc;
00874     int oldgzipCompress;
00875
00876     if ( AT.SS != AT.S0 || !PF.parallel ) return 0;
00877
00878     if ( PF.me != MASTER ) {
00879         /*
00880             #[ the slaves have to initialize their sendbuffer :
00881
00882             this is a slave and it's PObuffer should be the minimum of the
00883             sortiosize on the master and the PObuffer of our file.
00884             First save the original PObuffer and PObuffer of the outfile
00885         */
00886         size = (S->sTop2 - S->lBuffer - 1)/(PF.numtasks - 1);
00887         size -= (AM.MaxTer/sizeof(WORD) + 2);
00888         if ( fout->POsize < (LONG)(size*sizeof(WORD)) ) size = fout->POsize/sizeof(WORD);
00889         if ( sbuf == NULL ) {
00890             if ( (sbuf = PF_AllocBuf(PF.numsbuffers, size*sizeof(WORD), 1)) == NULL ) return -1;
00891             sbuf->active = 0;
00892             PF.sbuf = sbuf;
00893         }

```

```

00894     sbuf->buff[0] = fout->PObuffer;
00895     sbuf->stop[0] = fout->PObuffer+size;
00896     if ( sbuf->stop[0] > fout->POstop ) return -1;
00897     for ( i = 0; i < PF.numsbuffers; i++ )
00898         sbuf->fill[i] = sbuf->full[i] = sbuf->buff[i];
00899
00900     fout->PObuffer = sbuf->buff[sbuf->active];
00901     fout->POstop = sbuf->stop[sbuf->active];
00902     fout->POsize = size*sizeof(WORD);
00903     fout->POfill = fout->POfull = fout->PObuffer;
00904 /*
00905     #] the slaves have to initialize their sendbuffer :
00906 */
00907     return(0);
00908 }
00909 /*
00910     this waits for all slaves to be ready to send terms back
00911 */
00912 PF_WaitAllSlaves(); /* Note, the returned value should be 0 on success. */
00913 /*
00914     Now collect the terms of all slaves and merge them.
00915     PF_GetLoser gives the position of the smallest term, which is the real
00916     work. The smallest term needs to be copied to the outbuf: use PutOut.
00917 */
00918 PF_InitTree();
00919 if ( AR.PolyFun == 0 ) { S->PolyFlag = 0; }
00920 else if ( AR.PolyFunType == 1 ) { S->PolyFlag = 1; }
00921 else if ( AR.PolyFunType == 2 ) {
00922     if ( AR.PolyFunExp == 2
00923         || AR.PolyFunExp == 3 ) S->PolyFlag = 1;
00924     else S->PolyFlag = 2;
00925 }
00926 *AR.CompressPointer = 0;
00927 SeekScratch(fout, &position);
00928 oldposition = position;
00929 oldgzipCompress = AR.gzipCompress;
00930 AR.gzipCompress = 0;
00931
00932 noutterms = 0;
00933
00934 while ( PF_loser >= 0 ) {
00935     if ( (PF_loser = PF_GetLoser(PF_root)) == 0 ) break;
00936     outterm = PF_term[PF_loser];
00937     noutterms++;
00938
00939     if ( PF_newclen[PF_loser] != 0 ) {
00940 /*
00941         #[ this is only when new coeff was too long :
00942 */
00943         outterm = PF_WorkSpace;
00944         pp = PF_term[PF_loser];
00945         cc = *pp;
00946         while ( cc-- ) *outterm++ = *pp++;
00947         outterm = (outterm[-1] > 0) ? outterm-outterm[-1] : outterm+outterm[-1];
00948         if ( PF_newclen[PF_loser] > 0 ) cc = (WORD)PF_newclen[PF_loser] - 1;
00949         else cc = -(WORD)PF_newclen[PF_loser] - 1;
00950         pp = PF_newcpos[PF_loser];
00951         while ( cc-- ) *outterm++ = *pp++;
00952         *outterm++ = PF_newclen[PF_loser];
00953         *PF_WorkSpace = outterm - PF_WorkSpace;
00954         outterm = PF_WorkSpace;
00955         *PF_newcpos[PF_loser] = 0;
00956         PF_newclen[PF_loser] = 0;
00957 /*
00958         #] this is only when new coeff was too long :
00959 */
00960     }
00961     PRINTFBUF("PF_EndSort to PutOut: ",outterm,*outterm);
00962     PutOut(BHEAD outterm,&position,fout,1);
00963 }
00964 if ( FlushOut(&position,fout,0) ) {
00965     AR.gzipCompress = oldgzipCompress;
00966     return(-1);
00967 }
00968 S->TermsLeft = PF_goutterms = noutterms;
00969 DIFPOS(PF_exprsize, position, oldposition);
00970 AR.gzipCompress = oldgzipCompress;
00971 return(1);
00972 }
00973
00974 /*
00975     #] PF_EndSort :
00976     #] sort.c :
00977     #[ proces.c :
00978         #[ variables :
00979 */
00980

```

```

00981 static WORD *PF_CurrentBracket;
00982
00983 /*
00984     #] variables :
00985     #[ PF_GetTerm :
00986 */
00987
01006 static WORD PF_GetTerm(WORD *term)
01007 {
01008     GETIDENTITY
01009     FILEHANDLE *fi = AC.RhsExprInModuleFlag && PF.rhsInParallel ? &PF.slavebuf : AR.infile;
01010     WORD i;
01011     WORD *next, *np, *last, *lp = 0, *nextstop, *tp=term;
01012
01013     /* Only on the slaves. */
01014
01015     AN.deferskipped = 0;
01016     if ( fi->POfill >= fi->POfull || fi->POfull == fi->PObuffer ) {
01017 ReceiveNew:
01018     {
01019     /*
01020         #[ receive new terms from master :
01021     */
01022         int src = MASTER, tag;
01023         int follow = 0;
01024         LONG size,cpu,space = 0;
01025
01026         if ( PF.log ) {
01027             fprintf(stderr,"%d] Starting to send to Master\n",PF.me);
01028             fflush(stderr);
01029         }
01030
01031         cpu = TimeCPU(1);
01032         PF_PrepPack();
01033         PF_Pack(&cpu, 1,PF_LONG);
01034         PF_Pack(&space, 1,PF_LONG);
01035         PF_Pack(&PF_linterms, 1,PF_LONG);
01036         PF_Pack(&(AM.S0->GenTerms), 1,PF_LONG);
01037         PF_Pack(&(AM.S0->TermsLeft), 1,PF_LONG);
01038         PF_Pack(&follow, 1,PF_INT );
01039
01040         if ( PF.log ) {
01041             fprintf(stderr,"%d] Now sending with tag = %d\n",PF.me,PF_READY_MSGTAG);
01042             fflush(stderr);
01043         }
01044
01045         PF_Send(MASTER, PF_READY_MSGTAG);
01046
01047         if ( PF.log ) {
01048             fprintf(stderr,"%d] returning from send\n",PF.me);
01049             fflush(stderr);
01050         }
01051
01052         size = fi->POstop - fi->PObuffer - 1;
01053 #ifdef AbsolutelyExtra
01054         PF_Receive(MASTER,PF_ANY_MSGTAG,&src,&tag);
01055 #ifdef MPI2
01056         if ( tag == PF_TERM_MSGTAG ) {
01057             PF_Unpack(&size, 1, PF_LONG);
01058             if ( PF_Put_target(src) == 0 ) {
01059                 printf("PF_Put_target error ...\n");
01060             }
01061         }
01062         else {
01063             PF_RecvWbuf(fi->PObuffer,&size,&src);
01064         }
01065 #else
01066         PF_RecvWbuf(fi->PObuffer,&size,&src);
01067 #endif
01068 #endif
01069         tag=PF_RecvWbuf(fi->PObuffer,&size,&src);
01070
01071         fi->POfill = fi->PObuffer;
01072         /* Get AN.ninterms which sits in the first 2 WORDs. */
01073         {
01074             LONG ninterms;
01075             UNPACK_LONG(fi->POfill, ninterms);
01076             if ( fi->POfill < fi->POfull ) {
01077                 DBGOUT_NINTERMS(2, ("PF.me=%d AN.ninterms=%d PF_linterms=%d ninterms=%d GET\n",
(int)PF.me, (int)AN.ninterms, (int)PF_linterms, (int)ninterms));
01078                 AN.ninterms = ninterms - 1;
01079             } else {
01080                 DBGOUT_NINTERMS(2, ("PF.me=%d AN.ninterms=%d PF_linterms=%d ninterms=%d GETEND\n",
(int)PF.me, (int)AN.ninterms, (int)PF_linterms, (int)ninterms));
01081             }
01082         }
01083         fi->POfull = fi->PObuffer + size;

```

```

01084         if ( tag == PF_ENDSORT_MSGTAG ) *fi->POfull++ = 0;
01085 /*
01086      #] receive new terms from master :
01087 */
01088     }
01089     if ( PF_CurrentBracket ) *PF_CurrentBracket = 0;
01090 }
01091 if ( *fi->POfill == 0 ) {
01092     fi->POfill = fi->POfull = fi->PObuffer;
01093     *term = 0;
01094     goto RegRet;
01095 }
01096 if ( AR.DeferFlag ) {
01097     if ( !PF_CurrentBracket ) {
01098 /*
01099      #[ alloc space :
01100 */
01101         PF_CurrentBracket =
01102             (WORD*)Malloc1(AM.MaxTer,"PF_CurrentBracket");
01103         *PF_CurrentBracket = 0;
01104 /*
01105      #] alloc space :
01106 */
01107     }
01108     while ( *PF_CurrentBracket ) { /* "for each term in the buffer" */
01109 /*
01110      #[ test : bracket & skip if it's equal to the last in PF_CurrentBracket
01111 */
01112         next = fi->POfill;
01113         nextstop = next + *next; nextstop -= ABS(nextstop[-1]);
01114         next++;
01115         last = PF_CurrentBracket+1;
01116         while ( next < nextstop ) {
01117 /*
01118             scan the next term and PF_CurrentBracket
01119 */
01120             if ( *last == HAAKJE && *next == HAAKJE ) {
01121 /*
01122                 the part outside brackets is equal => skip this term
01123 */
01124                 PRINTFBUF("PF_GetTerm skips",fi->POfill,*fi->POfill);
01125                 break;
01126             }
01127 /*
01128             check if the current subterms are equal
01129 */
01130             np = next; next += next[1];
01131             lp = last; last += last[1];
01132             while ( np < next ) if ( *lp++ != *np++ ) goto strip;
01133         }
01134 /*
01135         go on to next term
01136 */
01137         fi->POfill += *fi->POfill;
01138         AN.deferskipped++;
01139 /*
01140         the usual checks
01141 */
01142         if ( fi->POfill >= fi->POfull || fi->POfull == fi->PObuffer )
01143             goto ReceiveNew;
01144         if ( *fi->POfill == 0 ) {
01145             fi->POfill = fi->POfull = fi->PObuffer;
01146             *term = 0;
01147             goto RegRet;
01148         }
01149 /*
01150      #] test :
01151 */
01152     }
01153 /*
01154      #[ copy :
01155 */
01156     this term to CurrentBracket and the part outside of bracket
01157     to Workspace at term
01158 */
01159 strip:
01160     next = fi->POfill;
01161     nextstop = next + *next; nextstop -= ABS(nextstop[-1]);
01162     next++;
01163     tp++;
01164     lp = PF_CurrentBracket + 1;
01165     while ( next < nextstop ) {
01166         if ( *next == HAAKJE ) {
01167             fi->POfill += *fi->POfill;
01168             while ( next < fi->POfill ) *lp++ = *next++;
01169             *PF_CurrentBracket = lp - PF_CurrentBracket;
01170             *lp = 0;

```

```

01171             *tp++ = 1;
01172             *tp++ = 1;
01173             *tp++ = 3;
01174             *term = WORDDIFF(tp,term);
01175             PRINTFBUF("PF_GetTerm new brack",PF_CurrentBracket,*PF_CurrentBracket);
01176             PRINTFBUF("PF_GetTerm POfill",fi->POfill,*fi->POfill);
01177             goto RegRet;
01178         }
01179         np = next; next += next[1];
01180         while ( np < next ) *tp++ = *lp++ = *np++;
01181     }
01182     tp = term;
01183     /*
01184     #] copy :
01185     */
01186 }
01187
01188     i = *fi->POfill;
01189     while ( i-- ) *tp++ = *fi->POfill++;
01190 RegRet:
01191     PRINTFBUF("PF_GetTerm returns",term,*term);
01192     return(*term);
01193 }
01194
01195 /*
01196     #] PF_GetTerm :
01197     #[ PF_Deferred :
01198 */
01199
01200 WORD PF_Deferred(WORD *term, WORD level)
01201 {
01210     GETIDENTITY
01211     WORD *bra, *bstop;
01212     WORD *tstart;
01213     FILEHANDLE *fi = AC.RhsExprInModuleFlag && PF.rhsInParallel ? &PF.slavebuf : AR.infile;
01214     WORD *next = fi->POfill;
01215     WORD *termout = AT.WorkPointer;
01216     WORD *oldwork = AT.WorkPointer;
01217
01218     AT.WorkPointer = (WORD *) ((UBYTE *) (AT.WorkPointer) + AM.MaxTer);
01219     AR.DeferFlag = 0;
01220
01221     PRINTFBUF("PF_Deferred (Term) ",term,*term);
01222     PRINTFBUF("PF_Deferred (Bracket)",PF_CurrentBracket,*PF_CurrentBracket);
01223
01224     bra = bstop = PF_CurrentBracket;
01225     if ( *bstop > 0 ) {
01226         bstop += *bstop;
01227         bstop -= ABS(bstop[-1]);
01228     }
01229     bra++;
01230     while ( *bra != HAAKJE && bra < bstop ) bra += bra[1];
01231     if ( bra >= bstop ) { /* No deferred action! */
01232         AT.WorkPointer = term + *term;
01233         if ( Generator(BHEAD term,level) ) goto DefCall;
01234         AR.DeferFlag = 1;
01235         AT.WorkPointer = oldwork;
01236         return(0);
01237     }
01238     bstop = bra;
01239     tstart = bra + bra[1];
01240     bra = PF_CurrentBracket;
01241     tstart--;
01242     *tstart = bra + *bra - tstart;
01243     bra++;
01244     /*
01245         Status of affairs:
01246         First bracket content starts at tstart.
01247         Next term starts at next.
01248         The outside of the bracket runs from bra = PF_CurrentBracket to bstop.
01249     */
01250     for(;;) {
01251         if ( InsertTerm(BHEAD term,0,AM.rbufnum,tstart,termout,0) < 0 ) {
01252             goto DefCall;
01253         }
01254         /*
01255             call Generator with new composed term
01256         */
01257         AT.WorkPointer = termout + *termout;
01258         if ( Generator(BHEAD termout,level) ) goto DefCall;
01259         AT.WorkPointer = termout;
01260         tstart = next + 1;
01261         if ( tstart >= fi->POfull ) goto ThatsIt;
01262         next += *next;
01263         /*
01264             compare with current bracket
01265         */

```

```

01266         while ( bra <= bstop ) {
01267             if ( *bra != *tstart ) goto ThatsIt;
01268             bra++; tstart++;
01269         }
01270 /*
01271         now bra and tstart should both be a HAAKJE
01272 */
01273     bra--; tstart--;
01274     if ( *bra != HAAKJE || *tstart != HAAKJE ) goto ThatsIt;
01275     tstart += tstart[1];
01276     tstart--;
01277     *tstart = next - tstart;
01278     bra = PF_CurrentBracket + 1;
01279 }
01280
01281 ThatsIt:
01282 /*
01283     AT.WorkPointer = oldwork;
01284 */
01285     AR.DeferFlag = 1;
01286     return(0);
01287 DefCall:
01288     MesCall("PF_Deferred");
01289     SETERROR(-1);
01290 }
01291
01292 /*
01293     #] PF_Deferred :
01294     #[ PF_Wait4Slave :
01295 */
01296
01297 static LONG **PF_W4Sstats = 0;
01298
01305 static int PF_Wait4Slave(int src)
01306 {
01307     int j, tag, next;
01308
01309     tag = PF_ANY_MSGTAG;
01310     PF_CatchErrorMessages(&src, &tag);
01311     PF_Receive(src, tag, &next, &tag);
01312
01313     if ( tag != PF_READY_MSGTAG ) {
01314         MesPrint("[%d] PF_Wait4Slave: received MSGTAG %d", (WORD)PF.me, (WORD)tag);
01315         return(-1);
01316     }
01317     if ( PF_W4Sstats == 0 ) {
01318         PF_W4Sstats = (LONG**)Malloca(sizeof(LONG*), "");
01319         PF_W4Sstats[0] = (LONG*)Malloca(PF_STATS_SIZE*sizeof(LONG), "");
01320     }
01321     PF_Unpack(PF_W4Sstats[0], PF_STATS_SIZE, PF_LONG);
01322     PF_Statistics(PF_W4Sstats, next);
01323
01324     PF_Unpack(&j, 1, PF_INT);
01325
01326     if ( j ) {
01327 /*
01328         actions depending on rest of information in last message
01329 */
01330     }
01331     return(next);
01332 }
01333
01334 /*
01335     #] PF_Wait4Slave :
01336     #[ PF_Wait4SlaveIP :
01337 */
01338 /*
01339     array of expression numbers for PF_InParallel processor.
01340     Each time the master sends expression "i" to the slave
01341     "next" it sets partodoexr[next]=i:
01342 */
01343 static WORD *partodoexr=NULL;
01344
01352 static int PF_Wait4SlaveIP(int *src)
01353 {
01354     int j, tag, next;
01355
01356     tag = PF_ANY_MSGTAG;
01357     PF_CatchErrorMessages(src, &tag);
01358     PF_Receive(*src, tag, &next, &tag);
01359     *src=tag;
01360     if ( PF_W4Sstats == 0 ) {
01361         PF_W4Sstats = (LONG**)Malloca(sizeof(LONG*), "");
01362         PF_W4Sstats[0] = (LONG*)Malloca(PF_STATS_SIZE*sizeof(LONG), "");
01363     }
01364
01365     PF_Unpack(PF_W4Sstats[0], PF_STATS_SIZE, PF_LONG);

```

```

01366     if ( tag == PF_DATA_MSGTAG )
01367         AR.CurExpr = partodoexr[next];
01368     PF_Statistics(PF_W4Sstats,next);
01369
01370     PF_Unpack(&j,1,PF_INT);
01371
01372     if ( j ) {
01373         /* actions depending on rest of information in last message */
01374     }
01375
01376     return(next);
01377 }
01378 /*
01379     #] PF_Wait4SlaveIP :
01380     #[ PF_WaitAllSlaves :
01381 */
01382
01391 static int PF_WaitAllSlaves(void)
01392 {
01393     int i, readySlaves, tag, next = PF_ANY_SOURCE;
01394     UBYTE *has_sent = 0;
01395
01396     has_sent = (UBYTE*)Malloc1(sizeof(UBYTE)*(PF.numtasks + 1),"PF_WaitAllSlaves");
01397     for ( i = 0; i < PF.numtasks; i++ ) has_sent[i] = 0;
01398
01399     for ( readySlaves = 1; readySlaves < PF.numtasks; ) {
01400         if ( next != PF_ANY_SOURCE ) { /*Go to the next slave:*/
01401             do{ /*Note, here readySlaves<PF.numtasks, so this loop can't be infinite*/
01402                 if ( ++next >= PF.numtasks ) next = 1;
01403             } while ( has_sent[next] == 1 );
01404         }
01405         /*
01406             Here PF_ProbeWithCatchingErrorMessages() is BLOCKING function if next = PF_ANY_SOURCE:
01407         */
01408         tag = PF_ProbeWithCatchingErrorMessages(&next);
01409         /*
01410             Here next != PF_ANY_SOURCE
01411         */
01412         switch ( tag ) {
01413             case PF_BUFFER_MSGTAG:
01414             case PF_ENDBUFFER_MSGTAG:
01415                 /*
01416                     Slaves are ready to send their results back
01417                 */
01418                 if ( has_sent[next] == 0 ) {
01419                     has_sent[next] = 1;
01420                     readySlaves++;
01421                 }
01422                 else { /*error?*/
01423                     fprintf(stderr,"ERROR next=%d tag=%d\n",next,tag);
01424                 }
01425                 /*
01426                     Note, we do NOT read results here! Messages from these slaves will be read
01427                     only after all slaves are ready, further in caller function
01428                 */
01429                 break;
01430             case 0:
01431                 /*
01432                     The slave is not ready. Just go to the next slave.
01433                     It may appear that there are no more ready slaves, and the master
01434                     will wait them in infinite loop. Stupid situation - the master can
01435                     receive buffers from ready slaves!
01436                 */
01437                 #ifdef PF_WITH_SCHED_YIELD
01438                 /*
01439                     Relinquish the processor:
01440                 */
01441                 sched_yield();
01442                 #endif
01443                 break;
01444             case PF_DATA_MSGTAG:
01445                 tag=next;
01446                 next=PF_Wait4SlaveIP(&tag);
01447             /*
01448                 tag must be == PF_DATA_MSGTAG!
01449             */
01450             PF_Statistics(PF_stats,0);
01451             PF_Slave2MasterIP(next);
01452             PF_Master2SlaveIP(next,NULL);
01453             if ( has_sent[next] == 0 ) {
01454                 has_sent[next]=1;
01455                 readySlaves++;
01456             }else{
01457                 /*error?*/
01458                 fprintf(stderr,"ERROR next=%d tag=%d\n",next,tag);
01459             }/*if ( has_sent[next] == 0 )*/
01460             break;

```

```

01461         case PF_EMPTY_MSGTAG:
01462             tag=next;
01463             next=PF_Wait4SlaveIP(&tag);
01464     /*
01465     tag must be == PF_EMPTY_MSGTAG!
01466     */
01467         PF_Master2SlaveIP(next,NULL);
01468         if ( has_sent[next] == 0 ) {
01469             has_sent[next]=1;
01470             readySlaves++;
01471         }else{
01472             /*error?*/
01473             fprintf(stderr,"ERROR next=%d tag=%d\n",next,tag);
01474             /*if ( has_sent[next] == 0 )*/
01475             break;
01476         case PF_READY_MSGTAG:
01477     /*
01478         idle slave
01479         May be only PF_READY_MSGTAG:
01480     */
01481         next = PF_Wait4Slave(next);
01482         if ( next == -1 ) return(next); /*Cannot be!*/
01483         if ( has_sent[0] == 0 ) { /*Send the last chunk to the slave*/
01484             PF.sbuf->active = 0;
01485             has_sent[0] = 1;
01486         }
01487         else {
01488     /*
01489         Last chunk was sent, so just send to slave ENDSORT
01490         AN.ninterms must be sent because the slave expects it:
01491     */
01492             PACK_LONG(PF.sbuf->fill[next], AN.ninterms);
01493     /*
01494         This will tell to the slave that there are no more terms:
01495     */
01496             *(PF.sbuf->fill[next])++ = 0;
01497             PF.sbuf->active = next;
01498         }
01499     /*
01500         Send ENDSORT
01501     */
01502         PF_ISendSbuf(next,PF_ENDSORT_MSGTAG);
01503         break;
01504     default:
01505     /*
01506         Error?
01507         Indicates the error. This will force exit from the main loop:
01508     */
01509         MesPrint("!!!Unexpected MPI message src=%d tag=%d.", next, tag);
01510         readySlaves = PF.numtasks+1;
01511         break;
01512     }
01513 }
01514
01515 if ( has_sent ) M_free(has_sent,"PF_WaitAllSlaves");
01516 /*
01517 0 on success (exit from the main loop by loop condition), or -1 if fails
01518 (exit from the main loop since readySlaves=PF.numtasks+1):
01519 */
01520 return(PF.numtasks-readySlaves);
01521 }
01522
01523 /*
01524 #] PF_WaitAllSlaves :
01525 #[ PF_Processor :
01526 */
01527
01540 int PF_Processor(EXPRESSIONS e, WORD i, WORD LastExpression)
01541 {
01542     GETIDENTITY
01543     WORD *term = AT.WorkPointer;
01544     LONG dd = 0;
01545     PF_BUFFER *sb = PF.sbuf;
01546     WORD j, *s, next;
01547     LONG size, cpu;
01548     POSITION position;
01549     int k, src, tag;
01550     FILEHANDLE *oldoutfile = AR.outfile;
01551
01552 #ifndef MPI2
01553     if ( PF_shared_buff == NULL ) {
01554         if ( PF_SMWin_Init() == 0 ) {
01555             MesPrint("PF_SMWin_Init error");
01556             exit(-1);
01557         }
01558     }
01559 #endif

```



```

01560
01561     if ( ( (WORD *) ((UBYTE *) (AT.WorkPointer)) + AM.MaxTer ) > AT.WorkTop ) return(MesWork());
01562
01563     /* For redefine statements. */
01564     if ( AC.numpfirstnum > 0 ) {
01565         for ( j = 0; j < AC.numpfirstnum; j++ ) {
01566             AC.inputnumbers[j] = -1;
01567         }
01568     }
01569
01570     if ( AC.mparalleelflag != PARALLELEFLAG ) return(0);
01571
01572     if ( PF.me == MASTER ) {
01573         /*
01574             #[ Master:
01575             #[ write prototype to outfile:
01576         */
01577         WORD oldBracketOn = AR.BraceOn;
01578         WORD *oldBrackBuf = AT.BrackBuf;
01579         WORD oldbracketindexflag = AT.bracketindexflag;
01580
01581         LONG maxinterms; /* the maximum number of terms in the bucket */
01582         int cmaxinterms; /* a variable controlling the transition of maxinterms */
01583         LONG termsinbucket; /* the number of filled terms in the bucket */
01584         LONG ProcessBucketSize = AC.mProcessBucketSize;
01585
01586         if ( PF.log && AC.CModule >= PF.log )
01587             MesPrint("[%d] working on expression %s in module %l", PF.me, EXPRNAME(i), AC.CModule);
01588         if ( GetTerm(BHEAD term) <= 0 ) {
01589             MesPrint("[%d] Expression %d has problems in scratchfile", PF.me, i);
01590             return(-1);
01591         }
01592         term[3] = i;
01593         if ( AR.outtohide ) {
01594             SeekScratch(AR.hidefile, &position);
01595             e->onfile = position;
01596             if ( PutOut(BHEAD term, &position, AR.hidefile, 0) < 0 ) return(-1);
01597         }
01598         else {
01599             SeekScratch(AR.outfile, &position);
01600             e->onfile = position;
01601             if ( PutOut(BHEAD term, &position, AR.outfile, 0) < 0 ) return(-1);
01602         }
01603         AR.DeferFlag = 0; /* The master leave the brackets!!! */
01604         AR.Eside = RHSIDE;
01605         if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
01606             AR.BraceOn = 1;
01607             AT.BrackBuf = AM.BracketFactors;
01608             AT.bracketindexflag = 1;
01609         }
01610         if ( AT.bracketindexflag > 0 ) OpenBracketIndex(i);
01611         /*
01612             #[ write prototype to outfile:
01613             #[ initialize sendbuffer if necessary:
01614
01615             the size of the sendbufs is:
01616             MIN(1/PF.numtasks*(AT.SS->sBufsize+AT.SS->lBufsize), AR.infile->Psize)
01617             No allocation for extra buffers necessary, just make sb->buf... point
01618             to the right places in the sortbuffers.
01619         */
01620         NewSort(BHEAD0); /* we need AT.SS to be set for this!!! */
01621         if ( sb == 0 || sb->buff[0] != AT.SS->lBuffer ) {
01622             size = (LONG) ((AT.SS->sTop2 - AT.SS->lBuffer) / (PF.numtasks));
01623             if ( size > (LONG) (AR.infile->Psize/sizeof(WORD) - 1) )
01624                 size = AR.infile->Psize/sizeof(WORD) - 1;
01625             if ( sb == 0 ) {
01626                 if ( ( sb = PF_AllocBuf(PF.numtasks, size*sizeof(WORD), PF.numtasks) ) == NULL )
01627                     return(-1);
01628             }
01629             sb->buff[0] = AT.SS->lBuffer;
01630             sb->full[0] = sb->fill[0] = sb->buff[0];
01631             for ( j = 1; j < PF.numtasks; j++ ) {
01632                 sb->stop[j-1] = sb->buff[j] = sb->buff[j-1] + size;
01633             }
01634             sb->stop[PF.numtasks-1] = sb->buff[PF.numtasks-1] + size;
01635             PF.sbuf = sb;
01636         }
01637         for ( j = 0; j < PF.numtasks; j++ ) {
01638             sb->full[j] = sb->fill[j] = sb->buff[j];
01639         }
01640         /*
01641             #[ initialize sendbuffer if necessary:
01642             #[ loop for all terms in infile:
01643         */
01644         /*
01645             * The initial value of maxinterms is determined by the user given
01646             * ProcessBucketSize and the number of terms in the current expression.

```

```

01647      * We make the initial maxinterms smaller, so that we get the all
01648      * workers busy as soon as possible.
01649      */
01650      maxinterms = ProcessBucketSize / 100;
01651      if ( maxinterms > e->counter / (PF.numtasks - 1) / 4 )
01652          maxinterms = e->counter / (PF.numtasks - 1) / 4;
01653      if ( maxinterms < 1 ) maxinterms = 1;
01654      cmaxinterms = 0;
01655      /*
01656      * Copy them always to sb->buff[0]. When that is full, wait for
01657      * the next slave to accept terms, exchange sb->buff[0] and
01658      * sb->buff[next], send sb->buff[next] to next slave and go on
01659      * filling the now empty sb->buff[0].
01660      */
01661      AN.ninterms = 0;
01662      termsinbucket = 0;
01663      PACK_LONG(sb->fill[0], 1);
01664      while ( GetTerm(BHEAD term) ) {
01665          AN.ninterms++; dd = AN.deferskipped;
01666          if ( AC.CollectFun && *term <= (LONG)(AM.MaxTer/(2*sizeof(WORD))) ) {
01667              if ( GetMoreTerms(term) < 0 ) {
01668                  LowerSortLevel(); return(-1);
01669              }
01670          }
01671          PRINTFBUF("PF_Processor gets",term,*term);
01672          if ( termsinbucket >= maxinterms || sb->fill[0] + *term >= sb->stop[0] ) {
01673              next = PF_Wait4Slave(PF_ANY_SOURCE);
01674
01675              sb->fill[next] = sb->fill[0];
01676              sb->full[next] = sb->full[0];
01677              SWAP(sb->stop[next], sb->stop[0]);
01678              SWAP(sb->buff[next], sb->buff[0]);
01679              sb->fill[0] = sb->full[0] = sb->buff[0];
01680              sb->active = next;
01681
01682          #ifdef MPI2
01683              if ( PF_Put_origin(next) == 0 ) {
01684                  printf("PF_Put_origin error...\n");
01685              }
01686          #else
01687              PF_ISendSbuf(next,PF_TERM_MSGTAG);
01688          #endif
01689
01690          /* Initialize the next bucket. */
01691          termsinbucket = 0;
01692          PACK_LONG(sb->fill[0], AN.ninterms);
01693          /*
01694          * For the "slow startup". We double maxinterms up to ProcessBucketSize
01695          * after (hopefully) the all workers got some terms.
01696          */
01697          if ( cmaxinterms >= PF.numtasks - 2 ) {
01698              maxinterms *= 2;
01699              if ( maxinterms >= ProcessBucketSize ) {
01700                  cmaxinterms = -1;
01701                  maxinterms = ProcessBucketSize;
01702              }
01703          }
01704          else if ( cmaxinterms >= 0 ) {
01705              cmaxinterms++;
01706          }
01707          j = *(s = term);
01708          NCOPY(sb->fill[0], s, j);
01709          termsinbucket++;
01710      }
01711      /* NOTE: The last chunk will be sent to a slave at EndSort() => PF_EndSort()
01712      *      => PF_WaitAllSlaves(). */
01713      AN.ninterms += dd;
01714      /*
01715      #] loop for all terms in infile:
01716      #[ Clean up & EndSort:
01717      */
01718      if ( LastExpression ) {
01719          UpdateMaxSize();
01720          if ( AR.infile->handle >= 0 ) {
01721              CloseFile(AR.infile->handle);
01722              AR.infile->handle = -1;
01723              remove(AR.infile->name);
01724              PUTZERO(AR.infile->POposition);
01725          }
01726          AR.infile->POfill = AR.infile->POfull = AR.infile->PObuffer;
01727      }
01728      if ( AR.outtohide ) AR.outfile = AR.hidefile;
01729      PF.parallel = 1;
01730      if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) return(-1);
01731      PF.parallel = 0;
01732      if ( AR.outtohide ) {
01733          AR.outfile = oldoutfile;

```

```

01734         AR.hidefile->POfull = AR.hidefile->POfill;
01735     }
01736     UpdateMaxSize();
01737     AR.BraceOn = oldBracketOn;
01738     AT.BrackBuf = oldBrackBuf;
01739     if ( ( e->vflags & TOBEFACTORED ) != 0 )
01740         poly_factorize_expression(e);
01741     else if ( ( ( e->vflags & TOBEUNFACTORED ) != 0 )
01742             && ( ( e->vflags & ISFACTORIZED ) != 0 ) )
01743         poly_unfactorize_expression(e);
01744     AT.bracketindexflag = oldbracketindexflag;
01745     AR.GetFile = 0;
01746     AR.outtohide = 0;
01747     /*
01748     * NOTE: e->numdummies, e->vflags and AR.exprflags will be updated
01749     *       after gathering the information from all slaves.
01750     */
01751 /*
01752     #] Clean up & EndSort:
01753     #[ Collect (stats,prepro,...):
01754 */
01755     DBGOUT_NINTERMS(1, ("PF.me=%d AN.ninterms=%d ENDSORT\n", (int)PF.me, (int)AN.ninterms));
01756     PF_CatchErrorMessageForAll();
01757     e->numdummies = 0;
01758     for ( k = 1; k < PF.numtasks; k++ ) {
01759         PF_LongSingleReceive(PF_ANY_SOURCE, PF_ENDSORT_MSGTAG, &src, &tag);
01760         PF_LongSingleUnpack(PF_stats[src], PF_STATS_SIZE, PF_LONG);
01761         {
01762             WORD numdummies, expchanged;
01763             PF_LongSingleUnpack(&numdummies, 1, PF_WORD);
01764             PF_LongSingleUnpack(&expchanged, 1, PF_WORD);
01765             if ( e->numdummies < numdummies ) e->numdummies = numdummies;
01766             AR.expchanged |= expchanged;
01767         }
01768         /* Now handle redefined preprocessor variables. */
01769         if ( AC.numpfirstnum > 0 ) PF_UnpackRedefinedPreVars();
01770     }
01771     /* Broadcast redefined preprocessor variables. */
01772     if ( AC.numpfirstnum > 0 ) {
01773         int RetCode = PF_BroadcastRedefinedPreVars();
01774         if ( RetCode ) return RetCode;
01775     }
01776     if ( ! AC.OldParallelStats ) {
01777         /* Now we can calculate AT.SS->GenTerms from the statistics of the slaves. */
01778         LONG genterms = 0;
01779         for ( k = 1; k < PF.numtasks; k++ ) {
01780             genterms += PF_stats[k][3];
01781         }
01782         AT.SS->GenTerms = genterms;
01783         WriteStats(&PF_exprsize, STATSPOSTSORT, NOCHECKLOGTYPE);
01784         Expressions[AR.CurExpr].size = PF_exprsize;
01785     }
01786     PF_Statistics(PF_stats,0);
01787 /*
01788     #] Collect (stats,prepro,...):
01789     #[ Update flags :
01790 */
01791     if ( AM.S0->TermsLeft ) e->vflags &= ~ISZERO;
01792     else e->vflags |= ISZERO;
01793     if ( AR.expchanged == 0 ) e->vflags |= ISUNMODIFIED;
01794     if ( AM.S0->TermsLeft ) AR.expflags |= ISZERO;
01795     if ( AR.expchanged ) AR.expflags |= ISUNMODIFIED;
01796 /*
01797     #] Update flags :
01798     #] Master:
01799 */
01800 }
01801 else {
01802 /*
01803     #[ Slave :
01804 */
01805 /*
01806     #[ Generator Loop & EndSort :
01807
01808     loop for all terms to get from master, call Generator for each of them
01809     then call EndSort and do cleanup (to be implemented)
01810 */
01811     WORD oldBracketOn = AR.BraceOn;
01812     WORD *oldBrackBuf = AT.BrackBuf;
01813     WORD oldbracketindexflag = AT.bracketindexflag;
01814
01815     /* For redefine statements. */
01816     if ( AC.numpfirstnum > 0 ) {
01817         for ( j = 0; j < AC.numpfirstnum; j++ ) {
01818             AC.inputnumbers[j] = -1;
01819         }
01820     }

```

```

01821
01822     SeekScratch(AR.outfile,&position);
01823     e->onfile = position;
01824     AR.DeferFlag = AC.ComDefer;
01825     AR.Eside = RHSIDE;
01826     if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
01827         AR.BracketOn = 1;
01828         AT.BrackBuf = AM.BracketFactors;
01829         AT.bracketindexflag = 1;
01830     }
01831     NewSort(BHEAD0);
01832     AR.MaxDum = AM.IndDum;
01833     AN.ninterms = 0;
01834     PF_linterms = 0;
01835     PF.parallel = 1;
01836 #ifdef MPI2
01837     AR.infile->POfull = AR.infile->POfill = AR.infile->PObuffer = PF_shared_buff;
01838 #endif
01839     {
01840         FILEHANDLE *fi = AC.RhsExprInModuleFlag && PF.rhsInParallel ? &PF.slavebuf : AR.infile;
01841         fi->POfull = fi->POfill = fi->PObuffer;
01842     }
01843     /* FIXME: AN.ninterms is still broken when AN.deferskipped is non-zero.
01844      *      It still needs some work, also in PF_GetTerm(). (TU 30 Aug 2011) */
01845     while ( PF_GetTerm(term) ) {
01846         PF_linterms++; AN.ninterms++; dd = AN.deferskipped;
01847         AT.WorkPointer = term + *term;
01848         AN.RepPoint = AT.RepCount + 1;
01849         if ( ( e->vflags & ISFACTORIZED ) != 0 && term[1] == HAAKJE ) {
01850             StoreTerm(BHEAD term);
01851             continue;
01852         }
01853         if ( AR.DeferFlag ) {
01854             AR.CurDum = AN.IndDum = Expressions[AR.CurExpr].numdummies + AM.IndDum;
01855         }
01856         else {
01857             AN.IndDum = AM.IndDum;
01858             AR.CurDum = ReNumber(BHEAD term);
01859         }
01860         if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMLAG);
01861         if ( AN.ncmod ) {
01862             if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
01863             else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
01864         }
01865         else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
01866         if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
01867             && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
01868             PolyFunClean(BHEAD term);
01869         }
01870         if ( Generator(BHEAD term,0) ) {
01871             MesPrint("[%d] PF_Processor: Error in Generator",PF.me);
01872             LowerSortLevel(); return(-1);
01873         }
01874         PF_linterms += dd; AN.ninterms += dd;
01875     }
01876     PF_linterms += dd; AN.ninterms += dd;
01877     {
01878         /*
01879          * EndSort() overrides AR.outfile->PObuffer etc. (See also PF_EndSort()),
01880          * but it causes a problem because
01881          * (1) PF_EndSort() sets AR.outfile->PObuffer to a send-buffer.
01882          * (2) RevertScratch() clears AR.infile, but then swaps buffers of AR.infile
01883          *     and AR.outfile.
01884          * (3) RHS expressions are stored to AR.infile->PObuffer.
01885          * (4) Again, PF_EndSort() sets AR.outfile->PObuffer, but now AR.outfile->PObuffer
01886          *     == AR.infile->PObuffer because of (1) and (2).
01887          * (5) The result goes to AR.outfile. This breaks the RHS expressions,
01888          *     which may be needed for the next expression.
01889          * Solution: backup & restore AR.outfile->PObuffer etc. (TU 14 Sep 2011)
01890          */
01891         FILEHANDLE *fout = AR.outfile;
01892         WORD *oldbuff = fout->PObuffer;
01893         WORD *oldstop = fout->POstop;
01894         LONG oldsize = fout->POsize;
01895         if ( EndSort(BHEAD AM.S0->sBuffer, 0) < 0 ) return -1;
01896         fout->PObuffer = oldbuff;
01897         fout->POstop = oldstop;
01898         fout->POsize = oldsize;
01899         fout->POfill = fout->POfull = fout->PObuffer;
01900     }
01901     AR.BracketOn = oldBracketOn;
01902     AT.BrackBuf = oldBrackBuf;
01903     AT.bracketindexflag = oldbracketindexflag;
01904     /*
01905      *#] Generator Loop & EndSort :
01906      *#[ Collect (stats,prepro...) :
01907      */

```

```

01908     DBGOUT_NINTERMS(1, ("PF.me=%d AN.ninterms=%d PF_linterms=%d ENDSORT\n", (int)PF.me,
(int)AN.ninterms, (int)PF_linterms));
01909     PF_PrepereLongSinglePack();
01910     cpu = TimeCPU(1);
01911     size = 0;
01912     PF_LongSinglePack(&cpu, 1, PF_LONG);
01913     PF_LongSinglePack(&size, 1, PF_LONG);
01914     PF_LongSinglePack(&PF_linterms, 1, PF_LONG);
01915     PF_LongSinglePack(&AM.S0->GenTerms, 1, PF_LONG);
01916     PF_LongSinglePack(&AM.S0->TermsLeft, 1, PF_LONG);
01917     {
01918         WORD numdummies = AR.MaxDum - AM.IndDum;
01919         PF_LongSinglePack(&numdummies, 1, PF_WORD);
01920         PF_LongSinglePack(&AR.expchanged, 1, PF_WORD);
01921     }
01922     /* Now handle redefined preprocessor variables. */
01923     if ( AC.numpfirstnum > 0 ) PF_PackRedefinedPreVars();
01924     PF_LongSingleSend(MASTER, PF_ENDSORT_MSGTAG);
01925     /* Broadcast redefined preprocessor variables. */
01926     if ( AC.numpfirstnum > 0 ) {
01927         int RetCode = PF_BroadcastRedefinedPreVars();
01928         if ( RetCode ) return RetCode;
01929     }
01930 /*
01931     #] Collect (stats,prepro...) :
01932
01933     This operation is moved to the beginning of each block, see PreProcessor
01934     in pre.c.
01935
01936     #] Slave :
01937 */
01938     if ( PF.log ) {
01939         UBYTE lbuf[24];
01940         NumToStr(lbuf, AC.CModule);
01941         fprintf(stderr, "[%d] [%s] Endsort,Collect,Broadcast done\n", PF.me, lbuf);
01942         fflush(stderr);
01943     }
01944 }
01945 return(0);
01946 }
01947
01948 /*
01949     #] PF_Processor :
01950     #] proces.c :
01951     #[ startup :, prepro & compile
01952     #[ PF_Init :
01953 */
01954
01963 int PF_Init(int *argc, char ***argv)
01964 {
01965     /*
01966         this should definitely be somewhere else ...
01967     */
01968     PF_CurrentBracket = 0;
01969
01970     PF.numtasks = 0; /* number of tasks, is determined in PF_LibInit ! */
01971     PF.numbufs = 2; /* might be changed by the environment variable on the master ! */
01972     PF.numrbufs = 2; /* might be changed by the environment variable on the master ! */
01973
01974     int ret = PF_LibInit(argc, argv);
01975     if (ret) { return ret; }
01976     PF_RealTime(PF_RESET);
01977
01978     PF.log = 0;
01979     PF.parallel = 0;
01980     PF_statsinterval = 10;
01981     PF.rhsInParallel=1;
01982     PF.exprbufsize=4096; /*in WORDs*/
01983
01984 #ifdef PF_WITHGETENV
01985     if ( PF.me == MASTER ) {
01986         char *c;
01987     /*
01988         get these from the environment at the moment should be in setfile/tail
01989     */
01990     if ( ( c = getenv("PF_LOG") ) != 0 ) {
01991         if ( *c ) PF.log = (int)atoi(c);
01992         else PF.log = 1;
01993         fprintf(stderr, "[%d] changing PF.log to %d\n", PF.me, PF.log);
01994         fflush(stderr);
01995     }
01996     if ( ( c = (char*)getenv("PF_RBUFS") ) != 0 ) {
01997         PF.numrbufs = (int)atoi(c);
01998         fprintf(stderr, "[%d] changing numrbufs to: %d\n", PF.me, PF.numrbufs);
01999         fflush(stderr);
02000     }
02001     if ( ( c = (char*)getenv("PF_SBUFS") ) != 0 ) {

```

```

02002         PF.numsbufs = (int)atoi(c);
02003         fprintf(stderr, "[%d] changing numsbufs to: %d\n", PF.me, PF.numsbufs);
02004         fflush(stderr);
02005     }
02006     if ( PF.numsbufs > 10 ) PF.numsbufs = 10;
02007     if ( PF.numsbufs < 1 ) PF.numsbufs = 1;
02008     if ( PF.numrbufs > 2 ) PF.numrbufs = 2;
02009     if ( PF.numrbufs < 1 ) PF.numrbufs = 1;
02010
02011     if ( ( c = getenv("PF_STATS") ) ) {
02012         UBYTE lbuf[24];
02013         PF_statsinterval = (int)atoi(c);
02014         NumToStr(lbuf, PF_statsinterval);
02015         fprintf(stderr, "[%d] changing PF_statsinterval to %s\n", PF.me, lbuf);
02016         fflush(stderr);
02017         if ( PF_statsinterval < 1 ) PF_statsinterval = 10;
02018     }
02019 }
02020 #endif
02021 /*
02022  #[ Broadcast settings from getenv: could also be done in PF_DoSetup
02023 */
02024 if ( PF.me == MASTER ) {
02025     PF_PreparePack();
02026     PF_Pack(&PF.log, 1, PF_INT);
02027     PF_Pack(&PF.numrbufs, 1, PF_WORD);
02028     PF_Pack(&PF.numsbufs, 1, PF_WORD);
02029 }
02030 PF_Broadcast();
02031 if ( PF.me != MASTER ) {
02032     PF_Unpack(&PF.log, 1, PF_INT);
02033     PF_Unpack(&PF.numrbufs, 1, PF_WORD);
02034     PF_Unpack(&PF.numsbufs, 1, PF_WORD);
02035     if ( PF.log ) {
02036         fprintf(stderr, "[%d] log=%d rbufs=%d sbufs=%d\n",
02037             PF.me, PF.log, PF.numrbufs, PF.numsbufs);
02038         fflush(stderr);
02039     }
02040 }
02041 /*
02042  #] Broadcast settings from getenv:
02043 */
02044 return(0);
02045 }
02046 /*
02047     #] PF_Init :
02048     #[ PF_Terminate :
02049 */
02050
02058 int PF_Terminate(int errorcode)
02059 {
02060     return PF_LibTerminate(errorcode);
02061 }
02062
02063 /*
02064     #] PF_Terminate :
02065     #[ PF_GetSlaveTimes :
02066 */
02067
02074 LONG PF_GetSlaveTimes(void)
02075 {
02076     LONG slavetimes = 0;
02077     LONG t = PF.me == MASTER ? 0 : AM.SumTime + TimeCPU(1);
02078     MPI_Reduce(&t, &slavetimes, 1, PF_LONG, MPI_SUM, MASTER, PF_COMM);
02079     return slavetimes;
02080 }
02081
02082 /*
02083     #] PF_GetSlaveTimes :
02084     #] startup :
02085     #[ PF_BroadcastNumber :
02086 */
02087
02094 LONG PF_BroadcastNumber(LONG x)
02095 {
02096     #ifdef PF_DEBUG_BCAST_LONG
02097         if ( PF.me == MASTER ) {
02098             MesPrint(">> Broadcast LONG: %l", x);
02099         }
02100     #endif
02101     PF_Bcast(&x, sizeof(LONG));
02102     return x;
02103 }
02104
02105 /*
02106     #] PF_BroadcastNumber :
02107     #[ PF_BroadcastBuffer :

```

```

02108 */
02109
02121 void PF_BroadcastBuffer(WORD **buffer, LONG *length)
02122 {
02123     WORD *p;
02124     LONG rest;
02125 #ifdef PF_DEBUG_BCAST_BUF
02126     if ( PF.me == MASTER ) {
02127         MesPrint("» Broadcast Buffer: length=%l", *length);
02128     }
02129 #endif
02130     /* Initialize the buffer on the slaves. */
02131     if ( PF.me != MASTER ) {
02132         *buffer = NULL;
02133     }
02134     /* Broadcast the length of the buffer. */
02135     *length = PF_BroadcastNumber(*length);
02136     if ( *length <= 0 ) return;
02137     /* Allocate the buffer on the slaves. */
02138     if ( PF.me != MASTER ) {
02139         *buffer = (WORD *)Malloc1(*length * sizeof(WORD), "PF_BroadcastBuffer");
02140     }
02141     /* Broadcast the data in the buffer. */
02142     p = *buffer;
02143     rest = *length;
02144     while ( rest > 0 ) {
02145         int l = rest < (LONG)PF.exprbufsize ? (int)rest : PF.exprbufsize;
02146         PF_Bcast(p, l * sizeof(WORD));
02147         p += l;
02148         rest -= l;
02149     }
02150 }
02151
02152 /*
02153 #] PF_BroadcastBuffer :
02154 #[ PF_BroadcastString :
02155 */
02156
02163 int PF_BroadcastString(UBYTE *str)
02164 {
02165     int clength = 0;
02166     /*
02167         If string does not fit to the PF_buffer, it
02168         will be split into chunks. Next chunk is started at str+clength
02169     */
02170     UBYTE *cstr=str;
02171     /*
02172         Note, compilation is performed INDEPENDENTLY on AC.mparallelfalg!
02173         No if ( AC.mparallelfalg == PARALLELFALG ) !!
02174     */
02175     do {
02176         cstr += clength; /*at each step for all slaves and master */
02177         if ( MASTER == PF.me ) { /*Pack str*/
02178             /*
02179             initialize buffers
02180             */
02181             if ( PF_PreparePack() != 0 ) Terminate(-1);
02182             if ( ( clength = PF_PackString(cstr) ) < 0 ) Terminate(-1);
02183             PF_Broadcast();
02184             if ( MASTER != PF.me ) {
02185                 /*
02186                 Slave - unpack received string
02187                 For slaves buffers are initialised automatically.
02188                 */
02189                 if ( ( clength = PF_UnpackString(cstr) ) < 0 ) Terminate(-1);
02190             } while ( cstr[clength-1] != '\0' );
02191             return (0);
02192         }
02193     } while ( cstr[clength-1] != '\0' );
02194     return (0);
02195 }
02196
02197
02198 /*
02199 #] PF_BroadcastString :
02200 #[ PF_BroadcastPreDollar :
02201 */
02202
02218 int PF_BroadcastPreDollar(WORD **dbuffer, LONG *newsize, int *numterms)
02219 {
02220     int err = 0;
02221     LONG i;
02222     /*
02223         Note, compilation is performed INDEPENDENTLY on AC.mparallelfalg!
02224         No if(AC.mparallelfalg==PARALLELFALG) !!
02225     */
02226     if ( MASTER == PF.me ) {

```

```

02227 /*
02228     The problem is that sometimes dollar variables are longer
02229     than PF_packbuf! So we split long expression into chunks.
02230     There are n filled chunks and one partially filled chunk:
02231 */
02232 LONG n = ((*newsize)+1)/PF_maxDollarChunkSize;
02233 /*
02234     ...and one more chunk for the rest; if the expression fits to
02235     the buffer without splitting, the latter will be the only one.
02236
02237     PF_maxDollarChunkSize is the maximal number of items fitted to
02238     the buffer. It is calculated in PF_LibInit() in mpi.c.
02239     PF_maxDollarChunkSize is calculated for the first step, when
02240     two fields (numterms and newsize, see below) are already packed.
02241     For simplicity, this value is used also for all steps, in
02242     despite of it is a bit less than maximally available space.
02243 */
02244 WORD *thechunk = *dbuffer;
02245
02246 err = PF_PreparePack();
02247 err |= PF_Pack(numterms,1,PF_INT);
02248 err |= PF_Pack(newsize,1,PF_LONG); /* pack the size */
02249 /*
02250     Pack and broadcast completely filled chunks.
02251     It may happen, this loop is not entered at all:
02252 */
02253 for ( i = 0; i < n; i++ ) {
02254     err |= PF_Pack(thechunk,PF_maxDollarChunkSize,PF_WORD);
02255     err |= PF_Broadcast();
02256     thechunk += PF_maxDollarChunkSize;
02257     PF_PreparePack();
02258 }
02259 /*
02260     Pack and broadcast the rest:
02261 */
02262 if ( ( n = ( (*newsize)+1)%PF_maxDollarChunkSize ) != 0 ) {
02263     err |= PF_Pack(thechunk,n,PF_WORD);
02264     err |= PF_Broadcast();
02265 }
02266 #ifdef PF_DEBUG_BCAST_PREDOLLAR
02267     MesPrint("> Broadcast PreDollar: newsize=%d numterms=%d", (int)*newsize, *numterms);
02268 #endif
02269 }
02270 if ( MASTER != PF.me ) { /* Slave - unpack received buffer */
02271     WORD *thechunk;
02272     LONG n, therest, thesize;
02273     err |= PF_Broadcast();
02274     err |= PF_Unpack(numterms,1,PF_INT);
02275     err |= PF_Unpack(newsize,1,PF_LONG);
02276 /*
02277     Now we know the buffer size.
02278 */
02279 thesize = (*newsize)+1;
02280 /*
02281     Evaluate the number of completely filled chunks. The last step must be
02282     treated separately, so -1:
02283 */
02284 n = (thesize/PF_maxDollarChunkSize) - 1;
02285 /*
02286     Note, here n can be <0, this is ok.
02287 */
02288 therest = thesize % PF_maxDollarChunkSize;
02289 thechunk = *dbuffer =
02290     (WORD*)Malloc1( thesize * sizeof(WORD), "$-buffer slave");
02291 if ( thechunk == NULL ) return(err|4);
02292 /*
02293     Unpack completely filled chunks and receive the next portion.
02294     It may happen, this loop is not entered at all:
02295 */
02296 for ( i = 0; i < n; i++ ) {
02297     err |= PF_Unpack(thechunk,PF_maxDollarChunkSize,PF_WORD);
02298     thechunk += PF_maxDollarChunkSize;
02299     err |= PF_Broadcast();
02300 }
02301 /*
02302     Now the last completely filled chunk:
02303 */
02304 if ( n >= 0 ) {
02305     err |= PF_Unpack(thechunk,PF_maxDollarChunkSize,PF_WORD);
02306     thechunk += PF_maxDollarChunkSize;
02307     if ( therest != 0 ) err |= PF_Broadcast();
02308 }
02309 /*
02310     Unpack the rest (it is already received!):
02311 */
02312 if ( therest != 0 ) err |= PF_Unpack(thechunk,therest,PF_WORD);
02313 }

```



```

02314     return (err);
02315 }
02316
02317 /*
02318 #] PF_BroadcastPreDollar :
02319 #[ Synchronization of modified dollar variables :
02320     #[ Helper functions :
02321         #[ dollarlen :
02322 */
02323
02327 static inline LONG dollarlen(const WORD *terms)
02328 {
02329     const WORD *p = terms;
02330     while ( *p ) p += *p;
02331     return p - terms; /* Not including the null terminator. */
02332 }
02333
02334 /*
02335     #] dollarlen :
02336     #[ dollar_mod_type :
02337 */
02338
02343 static inline WORD dollar_mod_type(WORD index)
02344 {
02345     int i;
02346     for ( i = 0; i < NumModOptdollars; i++ )
02347         if ( ModOptdollars[i].number == index ) break;
02348     if ( i >= NumModOptdollars ) return -1;
02349     return ModOptdollars[i].type;
02350 }
02351
02352
02353 /*
02354     #] dollar_mod_type :
02355     #] Helper functions :
02356     #[ PF_CollectModifiedDollars :
02357 */
02358
02359 /*
02360     #[ dollar_to_be_collected :
02361 */
02362
02367 static inline int dollar_to_be_collected(WORD index)
02368 {
02369     switch ( dollar_mod_type(index) ) {
02370     case MODSUM:
02371     case MODMAX:
02372     case MODMIN:
02373         return 1;
02374     default:
02375         return 0;
02376     }
02377 }
02378
02379 /*
02380     #] dollar_to_be_collected :
02381     #[ copy_dollar :
02382 */
02383
02388 static inline void copy_dollar(WORD index, WORD type, const WORD *where, LONG size)
02389 {
02390     DOLLARS d = Dollars + index;
02391
02392     CleanDollarFactors(d);
02393
02394     if ( type != DOLZERO && where != NULL && where != &AM.dollarzero && where[0] != 0 && size > 0 ) {
02395         if ( size > d->size || size < d->size / 4 ) { /* Reallocate if not enough or too much. */
02396             if ( d->where && d->where != &AM.dollarzero )
02397                 M_free(d->where, "old content of dollar");
02398             d->where = Malloc1(sizeof(WORD) * size, "copy buffer to dollar");
02399             d->size = size;
02400         }
02401         d->type = type;
02402         WCOPY(d->where, where, size);
02403     }
02404     else {
02405         if ( d->where && d->where != &AM.dollarzero )
02406             M_free(d->where, "old content of dollar");
02407         d->type = DOLZERO;
02408         d->where = &AM.dollarzero;
02409         d->size = 0;
02410     }
02411 }
02412
02413 /*
02414     #] copy_dollar :
02415     #[ compare_two_expressions :

```

```

02416 */
02417
02422 static inline int compare_two_expressions(const WORD *e1, const WORD *e2)
02423 {
02424     GETIDENTITY
02425     /*
02426      * We consider the cases that
02427      *   (1) the expression has no term,
02428      *   (2) the expression has only one term and it is a number,
02429      *   (3) otherwise.
02430      * Assume that the expressions are sorted and all terms are normalized.
02431      * The numerators of the coefficients must never be zero.
02432      *
02433      * Note that TwoExprCompare() is not adequate for our purpose
02434      * (as of 6 Aug. 2013), e.g., TwoExprCompare({0}, {4, 1, 1, -1}, LESS)
02435      * returns TRUE.
02436      */
02437     if ( e1[0] == 0 ) {
02438         if ( e2[0] == 0 ) {
02439             return(0);
02440         }
02441         else if ( e2[e2[0]] == 0 && e2[0] == ABS(e2[e2[0] - 1]) + 1 ) {
02442             if ( e2[e2[0] - 1] > 0 )
02443                 return(-1);
02444             else
02445                 return(+1);
02446         }
02447     }
02448     else if ( e1[e1[0]] == 0 && e1[0] == ABS(e1[e1[0] - 1]) + 1 ) {
02449         if ( e2[0] == 0 ) {
02450             if ( e1[e1[0] - 1] > 0 )
02451                 return(+1);
02452             else
02453                 return(-1);
02454         }
02455         else if ( e2[e2[0]] == 0 && e2[0] == ABS(e2[e2[0] - 1]) + 1 ) {
02456             return(CompCoef((WORD *)e1, (WORD *)e2));
02457         }
02458     }
02459     /* The expressions are not so simple. Define the order by each term. */
02460     while ( e1[0] && e2[0] ) {
02461         int c = CompareTerms(BHEAD (WORD *)e1, (WORD *)e2, 1);
02462         if ( c < 0 )
02463             return(-1);
02464         else if ( c > 0 )
02465             return(+1);
02466         e1 += e1[0];
02467         e2 += e2[0];
02468     }
02469     if ( e1[0] ) return(+1);
02470     if ( e2[0] ) return(-1);
02471     return(0);
02472 }
02473
02474 /*
02475     #] compare_two_expressions :
02476     #[ Variables :
02477 */
02478
02479 typedef struct {
02480     VectorStruct(WORD) buf;
02481     LONG size;
02482     WORD type;
02483     PADPOINTER(1,0,1,0);
02484 } dollar_buf;
02485
02486 /* Buffers used to store data for each variable from each slave. */
02487 static Vector(dollar_buf, dollar_slave_bufs);
02488
02489 /*
02490     #] Variables :
02491 */
02492
02506 int PF_CollectModifiedDollars(void)
02507 {
02508     int i, j, ndollars;
02509     /*
02510      * If the current module was executed in the sequential mode,
02511      * there are no modified module on the slaves.
02512      */
02513     if ( AC.mparallelfalg != PARALLELFLAG && !AC.partodoflag ) return 0;
02514     /*
02515      * Count the number of (potentially) modified dollar variables, which we need to collect.
02516      * Here we need to collect all max/min/sum variables.
02517      */
02518     ndollars = 0;
02519     for ( i = 0; i < NumPotModdollars; i++ ) {

```

```

02520     WORD index = PotModdollars[i];
02521     if ( dollar_to_be_collected(index) ) ndollars++;
02522 }
02523 if ( ndollars == 0 ) return 0; /* No dollars to be collected. */
02524
02525 if ( PF.me == MASTER ) {
02526 /*
02527     #[ Master :
02528 */
02529     int nslaves, nvars;
02530     /* Prepare receive buffers. We need ndollars*(PF.numtasks-1) buffers. */
02531     int nbufs = ndollars * (PF.numtasks - 1);
02532     VectorReserve(dollar_slave_bufs, nbufs);
02533     for ( i = VectorSize(dollar_slave_bufs); i < nbufs; i++ ) {
02534         VectorInit(VectorPtr(dollar_slave_bufs)[i].buf);
02535     }
02536     VectorSize(dollar_slave_bufs) = nbufs;
02537     /* Receive data from each slave. */
02538     for ( nslaves = 1; nslaves < PF.numtasks; nslaves++ ) {
02539         int src;
02540         PF_LongSingleReceive(PF_ANY_SOURCE, PF_DOLLAR_MSGTAG, &src, NULL);
02541         nvars = 0;
02542         for ( i = 0; i < NumPotModdollars; i++ ) {
02543             WORD index = PotModdollars[i];
02544             dollar_buf *b;
02545             if ( !dollar_to_be_collected(index) ) continue;
02546             b = &VectorPtr(dollar_slave_bufs)[(PF.numtasks - 1) * nvars + (src - 1)];
02547             PF_LongSingleUnpack(&b->type, 1, PF_WORD);
02548             if ( b->type != DOLZERO ) {
02549                 LONG size;
02550                 WORD *where;
02551                 PF_LongSingleUnpack(&size, 1, PF_LONG);
02552                 VectorReserve(b->buf, size + 1);
02553                 where = VectorPtr(b->buf);
02554                 PF_LongSingleUnpack(where, size, PF_WORD);
02555                 where[size] = 0; /* The null terminator is needed. */
02556                 b->size = size + 1; /* Including the null terminator. */
02557                 /* Note that we don't collect factored stuff for max/min/sum variables. */
02558             }
02559             else {
02560                 VectorReserve(b->buf, 1);
02561                 VectorPtr(b->buf)[0] = 0;
02562                 b->size = 0;
02563             }
02564             nvars++;
02565         }
02566     }
02567     /*
02568     * Combine received dollars. The FORM reference manual says maximum/minimum/sum
02569     * $-variables must have a numerical value, however, this routine should work also
02570     * for non-numerical cases, although the maximum/minimum value for non-numerical
02571     * terms has ambiguity.
02572     */
02573     nvars = 0;
02574     for ( i = 0; i < NumPotModdollars; i++ ) {
02575         WORD index = PotModdollars[i];
02576         WORD dtype;
02577         DOLLARS d;
02578         dollar_buf *b;
02579         if ( !dollar_to_be_collected(index) ) continue;
02580         d = Dollars + index;
02581         b = &VectorPtr(dollar_slave_bufs)[(PF.numtasks - 1) * nvars];
02582         dtype = dollar_mod_type(index);
02583         switch ( dtype ) {
02584             case MODMAX:
02585             case MODMIN: {
02586 /*
02587             #[ MODMAX & MODMIN :
02588 */
02589                 int selected = 0;
02590                 for ( j = 1; j < PF.numtasks - 1; j++ ) {
02591                     int c = compare_two_expressions(VectorPtr(b[j].buf),
02592 VectorPtr(b[selected].buf));
02593                     if ( (dtype == MODMAX && c > 0) || (dtype == MODMIN && c < 0) )
02594                         selected = j;
02595                 }
02596                 b = b + selected;
02597                 copy_dollar(index, b->type, VectorPtr(b->buf), b->size);
02598             }
02599             case MODSUM: {
02600 /*
02601             #[ MODSUM :
02602 */
02603                 break;
02604             }
02605         }
02606     }
02607 }

```

```

02606         GETIDENTITY
02607         int err = 0;
02608
02609         CBUF *C = cbuf + AM.rbufnum;
02610         WORD *oldwork = AT.WorkPointer, *oldcterm = AN.cTerm;
02611         WORD olddefer = AR.DeferFlag, oldnumlhs = AR.Cnumlhs, oldnumrhs = C->numrhs;
02612
02613         LONG size;
02614         WORD type, *dbuf;
02615
02616         AN.cTerm = 0;
02617         AR.DeferFlag = 0;
02618
02619         if ( ((WORD *) ((UBYTE *) AT.WorkPointer + AM.MaxTer)) > AT.WorkTop ) {
02620             err = -1;
02621             goto cleanup;
02622             MesWork();
02623         }
02624
02625         if ( NewSort(BHEAD0) ) {
02626             err = -1;
02627             goto cleanup;
02628         }
02629         if ( NewSort(BHEAD0) ) {
02630             LowerSortLevel();
02631             err = -1;
02632             goto cleanup;
02633         }
02634
02635         /*
02636          * Sum up the original $-variable in the master and $-variables on all slaves.
02637          * Note that $-variables on the slaves are set to zero at the beginning of
02638          * the module (See also DoExecute()).
02639          */
02640         for ( j = 0; j < PF.numtasks; j++ ) {
02641             const WORD *r;
02642             for ( r = j == 0 ? Dollars[index].where : VectorPtr(b[j - 1].buf); *r; r += *r
02643         ) {
02644             WCOPY(AT.WorkPointer, r, *r);
02645             AT.WorkPointer += *r;
02646             AR.Cnumlhs = 0;
02647             if ( Generator(BHEAD oldwork, 0) ) {
02648                 LowerSortLevel(); LowerSortLevel();
02649                 err = -1;
02650                 goto cleanup;
02651             }
02652             AT.WorkPointer = oldwork;
02653         }
02654
02655         size = EndSort(BHEAD (WORD *)&dbuf, 2);
02656         if ( size < 0 ) {
02657             LowerSortLevel();
02658             err = -1;
02659             goto cleanup;
02660         }
02661         LowerSortLevel();
02662
02663         /* Find special cases. */
02664         type = DOLTERMS;
02665         if ( dbuf[0] == 0 ) {
02666             type = DOLZERO;
02667         }
02668         else if ( dbuf[dbuf[0]] == 0 ) {
02669             const WORD *t = dbuf, *w;
02670             WORD n, nsize;
02671             n = *t;
02672             nsize = t[n - 1];
02673             if ( nsize < 0 ) nsize = -nsize;
02674             if ( nsize == n - 1 ) {
02675                 nsize = (nsize - 1) / 2;
02676                 w = t + 1 + nsize;
02677                 if ( *w == 1 ) {
02678                     w++; while ( w < t + n - 1 ) { if ( *w ) break; w++; }
02679                     if ( w >= t + n - 1 ) type = DOLNUMBER;
02680                 }
02681                 else if ( n == 7 && t[6] == 3 && t[5] == 1 && t[4] == 1 && t[1] == INDEX
02682                     && t[2] == 3 ) {
02683                     type = DOLINDEX;
02684                     d->index = t[3];
02685                 }
02686             }
02687         }
02688         copy_dollar(index, type, dbuf, dollarlen(dbuf) + 1);
02689         M_free(dbuf, "temporary dollar buffer");
02690         cleanup:
02691         AR.Cnumlhs = oldnumlhs;

```

```

02691             C->numrhs = oldnumrhs;
02692             AR.DeferFlag = olddefer;
02693             AN.cTerm = oldcterm;
02694             AT.WorkPointer = oldwork;
02695
02696             if ( err ) return err;
02697 /*
02698             #] MODSUM :
02699 */
02700             break;
02701         }
02702     }
02703     if ( d->type == DOLTERMS )
02704         cbuf[AM.dbufnum].CanCommu[index] = numcommute(d->where,
02705 &cbuf[AM.dbufnum].NumTerms[index]);
02706         cbuf[AM.dbufnum].rhs[index] = d->where;
02707         nvars++;
02708 #ifdef PF_DEBUG_REDUCE_DOLLAR
02709     MesPrint("« Reduce $-var: %s", AC.dollarnames->namebuffer + d->name);
02710 #endif
02711 }
02712 /*
02713 #] Master :
02714 */
02715 }
02716 else {
02717 /*
02718 #[ Slave :
02719 */
02720     PF_PreparesLongSinglePack();
02721     /* Pack each variable. */
02722     for ( i = 0; i < NumPotModdollars; i++ ) {
02723         WORD index = PotModdollars[i];
02724         DOLLARS d;
02725         if ( !dollar_to_be_collected(index) ) continue;
02726         d = Dollars + index;
02727         PF_LongSinglePack(&d->type, 1, PF_WORD);
02728         if ( d->type != DOLZERO ) {
02729             /*
02730              * NOTE: d->size is the allocated buffer size for d->where in WORDs.
02731              * So dollarlen(d->where) can be < d->size-1. (TU 15 Dec 2011)
02732              */
02733             LONG size = dollarlen(d->where);
02734             PF_LongSinglePack(&size, 1, PF_LONG);
02735             PF_LongSinglePack(d->where, size, PF_WORD);
02736             /* Note that we don't collect factored stuff for max/min/sum variables. */
02737         }
02738     }
02739     PF_LongSingleSend(MASTER, PF_DOLLAR_MSGTAG);
02740 /*
02741 #] Slave :
02742 */
02743 }
02744 return 0;
02745 }
02746 /*
02747 #] PF_CollectModifiedDollars :
02748 #[ PF_BroadcastModifiedDollars :
02749 */
02750 }
02751 /*
02752 #[ dollar_to_be_broadcast :
02753 */
02754 }
02755 static inline int dollar_to_be_broadcast(WORD index)
02756 {
02757     switch ( dollar_mod_type(index) ) {
02758     case MODLOCAL:
02759         return 0;
02760     default:
02761         return 1;
02762     }
02763 }
02764 }
02765 /*
02766 #] dollar_to_be_broadcast :
02767 */
02768 }
02769 /*
02770 #] PF_BroadcastModifiedDollars(void)
02771 */
02772 int PF_BroadcastModifiedDollars(void)
02773 {
02774     int i, j, ndollars;
02775     /*
02776      * Count the number of (potentially) modified dollar variables, which we need to broadcast.
02777      * Here we need to broadcast all non-local variables.
02778      */
02779     ndollars = 0;

```

```

02793     for ( i = 0; i < NumPotModdollars; i++ ) {
02794         WORD index = PotModdollars[i];
02795         if ( dollar_to_be_broadcast(index) ) ndollars++;
02796     }
02797     if ( ndollars == 0 ) return 0; /* No dollars to be broadcast. */
02798
02799     if ( PF.me == MASTER ) {
02800 /*
02801         #[ Master :
02802 */
02803         PF_PrepareLongMultiPack();
02804         /* Pack each variable. */
02805         for ( i = 0; i < NumPotModdollars; i++ ) {
02806             WORD index = PotModdollars[i];
02807             DOLLARS d;
02808             if ( !dollar_to_be_broadcast(index) ) continue;
02809             d = Dollars + index;
02810             PF_LongMultiPack(&d->type, 1, PF_WORD);
02811             if ( d->type != DOLZERO ) {
02812                 /*
02813                  * NOTE: d->size is the allocated buffer size for d->where in WORDs.
02814                  *       So dollarlen(d->where) can be < d->size-1. (TU 15 Dec 2011)
02815                  */
02816                 LONG size = dollarlen(d->where);
02817                 PF_LongMultiPack(&size, 1, PF_LONG);
02818                 PF_LongMultiPack(d->where, size, PF_WORD);
02819                 /* ...and the factored stuff. */
02820                 PF_LongMultiPack(&d->nfactors, 1, PF_WORD);
02821                 if ( d->nfactors > 1 ) {
02822                     for ( j = 0; j < d->nfactors; j++ ) {
02823                         FACDOLLAR *f = &d->factors[j];
02824                         PF_LongMultiPack(&f->type, 1, PF_WORD);
02825                         PF_LongMultiPack(&f->size, 1, PF_LONG);
02826                         if ( f->size > 0 )
02827                             PF_LongMultiPack(f->where, f->size, PF_WORD);
02828                         else
02829                             PF_LongMultiPack(&f->value, 1, PF_WORD);
02830                     }
02831                 }
02832             }
02833 #ifdef PF_DEBUG_BCAST_DOLLAR
02834             MesPrint("» Broadcast $-var: %s", AC.dollarnames->namebuffer + d->name);
02835 #endif
02836         }
02837 /*
02838         #] Master :
02839 */
02840     }
02841     if ( PF_LongMultiBroadcast() ) return -1;
02842     if ( PF.me != MASTER ) {
02843 /*
02844         #[ Slave :
02845 */
02846         for ( i = 0; i < NumPotModdollars; i++ ) {
02847             WORD index = PotModdollars[i];
02848             DOLLARS d;
02849             if ( !dollar_to_be_broadcast(index) ) continue;
02850             d = Dollars + index;
02851             /* Clear the contents of the dollar variable. */
02852             if ( d->where && d->where != &AM.dollarzero )
02853                 M_free(d->where, "old content of dollar");
02854             d->where = &AM.dollarzero;
02855             d->size = 0;
02856             CleanDollarFactors(d);
02857             /* Unpack and store the contents. */
02858             PF_LongMultiUnpack(&d->type, 1, PF_WORD);
02859             if ( d->type != DOLZERO ) {
02860                 LONG size;
02861                 PF_LongMultiUnpack(&size, 1, PF_LONG);
02862                 d->size = size + 1;
02863                 d->where = (WORD *)Malloca(sizeof(WORD) * d->size, "dollar content");
02864                 PF_LongMultiUnpack(d->where, size, PF_WORD);
02865                 d->where[size] = 0; /* The null terminator is needed. */
02866                 /* ...and the factored stuff. */
02867                 PF_LongMultiUnpack(&d->nfactors, 1, PF_WORD);
02868                 if ( d->nfactors > 1 ) {
02869                     d->factors = (FACDOLLAR *)Malloca(sizeof(FACDOLLAR) * d->nfactors, "dollar
factored stuff");
02870                     for ( j = 0; j < d->nfactors; j++ ) {
02871                         FACDOLLAR *f = &d->factors[j];
02872                         PF_LongMultiUnpack(&f->type, 1, PF_WORD);
02873                         PF_LongMultiUnpack(&f->size, 1, PF_LONG);
02874                         if ( f->size > 0 ) {
02875                             f->where = (WORD *)Malloca(sizeof(WORD) * (f->size + 1), "dollar factor
content");
02876                             PF_LongMultiUnpack(f->where, f->size, PF_WORD);
02877                             f->where[f->size] = 0; /* The null terminator is needed. */

```

```

02878             f->value = 0;
02879         }
02880         else {
02881             f->where = NULL;
02882             PF_LongMultiUnpack(&f->value, 1, PF_WORD);
02883         }
02884     }
02885 }
02886 }
02887 if ( d->type == DOLTERMS )
02888     cbuf[AM.dbufnum].CanCommu[index] = numcommute(d->where,
02889 &cbuf[AM.dbufnum].NumTerms[index]);
02889     cbuf[AM.dbufnum].rhs[index] = d->where;
02890 }
02891 /*
02892     #] Slave :
02893 */
02894 }
02895 return 0;
02896 }
02897
02898 /*
02899     #] PF_BroadcastModifiedDollars :
02900     #] Synchronization of modified dollar variables :
02901     #[ Synchronization of redefined preprocessor variables :
02902     #[ Variables :
02903 */
02904
02905 /* A buffer used in receivers. */
02906 static Vector(UBYTE, prevarbuf);
02907
02908 /*
02909     #] Variables :
02910     #[ PF_PackRedefinedPreVars :
02911 */
02912
02913 static void PF_PackRedefinedPreVars(void)
02914 {
02915     int i;
02916     /* First, pack the number of redefined preprocessor variables. */
02917     int nredefs = 0;
02918     for ( i = 0; i < AC.numpfirstnum; i++ )
02919         if ( AC.inputnumbers[i] >= 0 ) nredefs++;
02920     PF_LongSinglePack(&nredefs, 1, PF_INT);
02921     /* Then, pack each variable. */
02922     for ( i = 0; i < AC.numpfirstnum; i++ )
02923         if ( AC.inputnumbers[i] >= 0 ) {
02924             WORD index = AC.pfirstnum[i];
02925             UBYTE *value = PreVar[index].value;
02926             int bytes = strlen((char *)value);
02927             PF_LongSinglePack(&index, 1, PF_WORD);
02928             PF_LongSinglePack(&bytes, 1, PF_INT);
02929             PF_LongSinglePack(value, bytes, PF_BYTE);
02930             PF_LongSinglePack(&AC.inputnumbers[i], 1, PF_LONG);
02931         }
02932 }
02933
02934 /*
02935     #] PF_PackRedefinedPreVars :
02936     #[ PF_UnpackRedefinedPreVars :
02937 */
02938
02939 static void PF_UnpackRedefinedPreVars(void)
02940 {
02941     int i, j;
02942     /* Unpack the number of redefined preprocessor variables. */
02943     int nredefs;
02944     PF_LongSingleUnpack(&nredefs, 1, PF_INT);
02945     if ( nredefs > 0 ) {
02946         /* Then unpack each variable. */
02947         for ( i = 0; i < nredefs; i++ ) {
02948             WORD index;
02949             int bytes;
02950             UBYTE *value;
02951             LONG inputnumber;
02952             PF_LongSingleUnpack(&index, 1, PF_WORD);
02953             PF_LongSingleUnpack(&bytes, 1, PF_INT);
02954             VectorReserve(prevarbuf, bytes + 1);
02955             value = VectorPtr(prevarbuf);
02956             PF_LongSingleUnpack(value, bytes, PF_BYTE);
02957             value[bytes] = '\0'; /* The null terminator is needed. */
02958             PF_LongSingleUnpack(&inputnumber, 1, PF_LONG);
02959             /* Put this variable if it must be updated. */
02960             for ( j = 0; j < AC.numpfirstnum; j++ )
02961                 if ( AC.pfirstnum[j] == index ) break;
02962             if ( AC.inputnumbers[j] < inputnumber ) {
02963                 AC.inputnumbers[j] = inputnumber;

```

```

02981         PutPreVar(PreVar[index].name, value, NULL, 1);
02982     }
02983 }
02984 }
02985 }
02986
02987 /*
02988     #] PF_UnpackRedefinedPreVars :
02989     #[ PF_BroadcastRedefinedPreVars :
02990 */
02991
03002 int PF_BroadcastRedefinedPreVars(void)
03003 {
03004     /*
03005      * NOTE: Because the compilation is performed on the all processes
03006      * independently on AC.mparallelfalg, we always have to broadcast redefined
03007      * preprocessor variables from the master to the all slaves.
03008      */
03009     if ( PF.me == MASTER ) {
03010         /*
03011             #[ Master :
03012         */
03013         int i, nredefs;
03014         PF_PrepareLongMultiPack();
03015         /* First, pack the number of redefined preprocessor variables. */
03016         nredefs = 0;
03017         for ( i = 0; i < AC.numpfirstnum; i++ )
03018             if ( AC.inputnumbers[i] >= 0 ) nredefs++;
03019         PF_LongMultiPack(&nredefs, 1, PF_INT);
03020         /* Then, pack each variable. */
03021         for ( i = 0; i < AC.numpfirstnum; i++ )
03022             if ( AC.inputnumbers[i] >= 0 ) {
03023                 WORD index = AC.pfirstnum[i];
03024                 UBYTE *value = PreVar[index].value;
03025                 int bytes = strlen((char *)value);
03026                 PF_LongMultiPack(&index, 1, PF_WORD);
03027                 PF_LongMultiPack(&bytes, 1, PF_INT);
03028                 PF_LongMultiPack(value, bytes, PF_BYTE);
03029 #ifdef PF_DEBUG_BCAST_PREVAR
03030                 MesPrint("» Broadcast PreVar: %s = \"%s\"", PreVar[index].name, value);
03031 #endif
03032             }
03033         /*
03034             #[ Master :
03035         */
03036     }
03037     if ( PF_LongMultiBroadcast() ) return -1;
03038     if ( PF.me != MASTER ) {
03039         /*
03040             #[ Slave :
03041         */
03042         int i, nredefs;
03043         /* Unpack the number of redefined preprocessor variables. */
03044         PF_LongMultiUnpack(&nredefs, 1, PF_INT);
03045         if ( nredefs > 0 ) {
03046             /* Then unpack each variable and put it. */
03047             for ( i = 0; i < nredefs; i++ ) {
03048                 WORD index;
03049                 int bytes;
03050                 UBYTE *value;
03051                 PF_LongMultiUnpack(&index, 1, PF_WORD);
03052                 PF_LongMultiUnpack(&bytes, 1, PF_INT);
03053                 VectorReserve(prevarbuf, bytes + 1);
03054                 value = VectorPtr(prevarbuf);
03055                 PF_LongMultiUnpack(value, bytes, PF_BYTE);
03056                 value[bytes] = '\0'; /* The null terminator is needed. */
03057                 PutPreVar(PreVar[index].name, value, NULL, 1);
03058             }
03059         }
03060         /*
03061             #[ Slave :
03062         */
03063     }
03064     return 0;
03065 }
03066
03067 /*
03068     #] PF_BroadcastRedefinedPreVars :
03069     #] Synchronization of redefined preprocessor variables :
03070     #[ Preprocessor Inside instruction :
03071     #[ Variables :
03072 */
03073
03074 /* Saved values of AC.RhsExprInModuleFlag, PotModdollars and AC.pfirstnum. */
03075 static WORD oldRhsExprInModuleFlag;
03076 static Vector (WORD, oldPotModdollars);
03077 static Vector (WORD, oldpfirstnum);

```



```

03078
03079 /*
03080     #] Variables :
03081     #[ PF_StoreInsideInfo :
03082 */
03083
03084 /*
03085  * Saves the current values of AC.RhsExprInModuleFlag, PotModdollars
03086  * and AC.pfirstnum.
03087  *
03088  * Called by DoInside().
03089  *
03090  * @return 0 if OK, nonzero on error.
03091  */
03092 int PF_StoreInsideInfo(void)
03093 {
03094     int i;
03095     oldRhsExprInModuleFlag = AC.RhsExprInModuleFlag;
03096     VectorClear(oldPotModdollars);
03097     for ( i = 0; i < NumPotModdollars; i++ )
03098         VectorPushBack(oldPotModdollars, PotModdollars[i]);
03099     VectorClear(oldpfirstnum);
03100     for ( i = 0; i < AC.numpfirstnum; i++ )
03101         VectorPushBack(oldpfirstnum, AC.pfirstnum[i]);
03102     return 0;
03103 }
03104
03105 /*
03106     #] PF_StoreInsideInfo :
03107     #[ PF_RestoreInsideInfo :
03108 */
03109
03110 /*
03111  * Restores the saved values of AC.RhsExprInModuleFlag, PotModdollars
03112  * and AC.pfirstnum.
03113  *
03114  * Called by DoEndInside().
03115  *
03116  * @return 0 if OK, nonzero on error.
03117  */
03118 int PF_RestoreInsideInfo(void)
03119 {
03120     int i;
03121     AC.RhsExprInModuleFlag = oldRhsExprInModuleFlag;
03122     NumPotModdollars = VectorSize(oldPotModdollars);
03123     for ( i = 0; i < NumPotModdollars; i++ )
03124         PotModdollars[i] = VectorPtr(oldPotModdollars)[i];
03125     AC.numpfirstnum = VectorSize(oldpfirstnum);
03126     for ( i = 0; i < AC.numpfirstnum; i++ )
03127         AC.pfirstnum[i] = VectorPtr(oldpfirstnum)[i];
03128     return 0;
03129 }
03130
03131 /*
03132     #] PF_RestoreInsideInfo :
03133     #] Preprocessor Inside instruction :
03134     #[ PF_BroadcastCBuf :
03135 */
03136
03144 int PF_BroadcastCBuf(int bufnum)
03145 {
03146     CBUF *C = cbuf + bufnum;
03147     int i;
03148     LONG l;
03149     if ( PF.me == MASTER ) {
03150 /*
03151     #[ Master :
03152 */
03153         PF_PrepareLongMultiPack();
03154         /* Pack CBUF struct except pointers. */
03155         PF_LongMultiPack(&C->BufferSize, 1, PF_LONG);
03156         PF_LongMultiPack(&C->numlhs, 1, PF_INT);
03157         PF_LongMultiPack(&C->numrhs, 1, PF_INT);
03158         PF_LongMultiPack(&C->maxlhs, 1, PF_INT);
03159         PF_LongMultiPack(&C->maxrhs, 1, PF_INT);
03160         PF_LongMultiPack(&C->mnumlhs, 1, PF_INT);
03161         PF_LongMultiPack(&C->mnumrhs, 1, PF_INT);
03162         PF_LongMultiPack(&C->numtree, 1, PF_INT);
03163         PF_LongMultiPack(&C->rootnum, 1, PF_INT);
03164         PF_LongMultiPack(&C->MaxTreeSize, 1, PF_INT);
03165         /* Now pointers. Pointer, lhs and rhs are packed as offsets. We don't pack Top. */
03166         l = C->Pointer - C->Buffer;
03167         PF_LongMultiPack(&l, 1, PF_LONG);
03168         PF_LongMultiPack(C->Buffer, 1, PF_WORD);
03169         for ( i = 0; i < C->numlhs + 1; i++ ) {
03170             l = C->lhs[i] - C->Buffer;
03171             PF_LongMultiPack(&l, 1, PF_LONG);

```

```

03172     }
03173     for ( i = 0; i < C->numrhs + 1; i++ ) {
03174         l = C->rhs[i] - C->Buffer;
03175         PF_LongMultiPack(&l, 1, PF_LONG);
03176     }
03177     PF_LongMultiPack(C->CanCommu, C->numrhs + 1, PF_LONG);
03178     PF_LongMultiPack(C->NumTerms, C->numrhs + 1, PF_LONG);
03179     PF_LongMultiPack(C->numdum, C->numrhs + 1, PF_WORD);
03180     PF_LongMultiPack(C->dimension, C->numrhs + 1, PF_WORD);
03181     if ( C->MaxTreeSize > 0 )
03182         PF_LongMultiPack(C->boomlijst, (C->numtree + 1) * (sizeof(COMPTREE) / sizeof(int)),
PF_INT);
03183 #ifdef PF_DEBUG_BCAST_CBUF
03184     MesPrint("» Broadcast CBuf %d", bufnum);
03185 #endif
03186 /*
03187     #] Master :
03188 */
03189 }
03190 if ( PF_LongMultiBroadcast() ) return -1;
03191 if ( PF.me != MASTER ) {
03192 /*
03193     #[ Slave :
03194 */
03195     /* First, free already allocated buffers. */
03196     finishcbuf(bufnum);
03197     /* Unpack CBUF struct except pointers. */
03198     PF_LongMultiUnpack(&C->BufferSize, 1, PF_LONG);
03199     PF_LongMultiUnpack(&C->numlhs, 1, PF_INT);
03200     PF_LongMultiUnpack(&C->numrhs, 1, PF_INT);
03201     PF_LongMultiUnpack(&C->maxlhs, 1, PF_INT);
03202     PF_LongMultiUnpack(&C->maxrhs, 1, PF_INT);
03203     PF_LongMultiUnpack(&C->mnumlhs, 1, PF_INT);
03204     PF_LongMultiUnpack(&C->mnumrhs, 1, PF_INT);
03205     PF_LongMultiUnpack(&C->numtree, 1, PF_INT);
03206     PF_LongMultiUnpack(&C->rootnum, 1, PF_INT);
03207     PF_LongMultiUnpack(&C->MaxTreeSize, 1, PF_INT);
03208     /* Allocate new buffers. */
03209     C->Buffer = (WORD *)Malloc1(C->BufferSize * sizeof(WORD), "compiler buffer");
03210     C->Top = C->Buffer + C->BufferSize;
03211     C->lhs = (WORD **)Malloc1(C->maxlhs * sizeof(WORD *), "compiler buffer");
03212     C->rhs = (WORD **)Malloc1(C->maxrhs * (sizeof(WORD *) + 2 * sizeof(LONG) + 2 * sizeof(WORD)),
"compiler buffer");
03213     C->CanCommu = (LONG *) (C->rhs + C->maxrhs);
03214     C->NumTerms = C->CanCommu + C->maxrhs;
03215     C->numdum = (WORD *) (C->NumTerms + C->maxrhs);
03216     C->dimension = C->numdum + C->maxrhs;
03217     if ( C->MaxTreeSize > 0 )
03218         C->boomlijst = (COMPTREE *)Malloc1(C->MaxTreeSize * sizeof(COMPTREE), "compiler buffer");
03219     /* Unpack buffers. */
03220     PF_LongMultiUnpack(&l, 1, PF_LONG);
03221     PF_LongMultiUnpack(C->Buffer, 1, PF_WORD);
03222     C->Pointer = C->Buffer + 1;
03223     for ( i = 0; i < C->numlhs + 1; i++ ) {
03224         PF_LongMultiUnpack(&l, 1, PF_LONG);
03225         C->lhs[i] = C->Buffer + 1;
03226     }
03227     for ( i = 0; i < C->numrhs + 1; i++ ) {
03228         PF_LongMultiUnpack(&l, 1, PF_LONG);
03229         C->rhs[i] = C->Buffer + 1;
03230     }
03231     PF_LongMultiUnpack(C->CanCommu, C->numrhs + 1, PF_LONG);
03232     PF_LongMultiUnpack(C->NumTerms, C->numrhs + 1, PF_LONG);
03233     PF_LongMultiUnpack(C->numdum, C->numrhs + 1, PF_WORD);
03234     PF_LongMultiUnpack(C->dimension, C->numrhs + 1, PF_WORD);
03235     if ( C->MaxTreeSize > 0 )
03236         PF_LongMultiUnpack(C->boomlijst, (C->numtree + 1) * (sizeof(COMPTREE) / sizeof(int)),
PF_INT);
03237 /*
03238     #[ Slave :
03239 */
03240 }
03241 return 0;
03242 }
03243 /*
03244     #] PF_BroadcastCBuf :
03245     #[ PF_BroadcastExpFlags :
03246 */
03247 int PF_BroadcastExpFlags(void)
03248 {
03249     WORD i;
03250     EXPRESSIONS e;
03251     if ( PF.me == MASTER ) {
03252 /*
03253     #[ Master :

```

```

03262 */
03263     PF_PrepareLongMultiPack();
03264     PF_LongMultiPack(&AR.expflags, 1, PF_WORD);
03265     for ( i = 0; i < NumExpressions; i++ ) {
03266         e = &Expressions[i];
03267         PF_LongMultiPack(&e->counter, 1, PF_WORD);
03268         PF_LongMultiPack(&e->vflags, 1, PF_WORD);
03269         PF_LongMultiPack(&e->uflags, 1, PF_WORD);
03270         PF_LongMultiPack(&e->numdummies, 1, PF_WORD);
03271         PF_LongMultiPack(&e->numfactors, 1, PF_WORD);
03272     #ifdef PF_DEBUG_BCAST_EXPRFLAGS
03273         MesPrint("» Broadcast ExprFlags: %s", AC.exprnames->namebuffer + e->name);
03274     #endif
03275     }
03276     /*
03277     #] Master :
03278     */
03279     }
03280     if ( PF_LongMultiBroadcast() ) return -1;
03281     if ( PF.me != MASTER ) {
03282     /*
03283     #[ Slave :
03284     */
03285         PF_LongMultiUnpack(&AR.expflags, 1, PF_WORD);
03286         for ( i = 0; i < NumExpressions; i++ ) {
03287             e = &Expressions[i];
03288             PF_LongMultiUnpack(&e->counter, 1, PF_WORD);
03289             PF_LongMultiUnpack(&e->vflags, 1, PF_WORD);
03290             PF_LongMultiUnpack(&e->uflags, 1, PF_WORD);
03291             PF_LongMultiUnpack(&e->numdummies, 1, PF_WORD);
03292             PF_LongMultiUnpack(&e->numfactors, 1, PF_WORD);
03293         }
03294     /*
03295     #] Slave :
03296     */
03297     }
03298     return 0;
03299 }
03300
03301 /*
03302 #] PF_BroadcastExpFlags :
03303 #[ PF_SetScratch :
03304 */
03305
03312 static void PF_SetScratch(FILEHANDLE *f, POSITION *position)
03313 {
03314     if(
03315         ( f->handle >= 0 ) && ISGEPOS(*position, f->POposition) &&
03316         ( ISGEPOSINC(*position, f->POposition, (f->POfull-f->PObuffer)*sizeof(WORD)) ==0 )
03317     ) /*position is inside the buffer! SetScratch() will do nothing.*/
03318         f->POfull=f->PObuffer; /*force SetScratch() to re-read the position from the beginning:*/
03319     SetScratch(f, position);
03320 }
03321
03322 /*
03323 #] PF_SetScratch :
03324 #[ PF_pushScratch :
03325 */
03326
03333 static int PF_pushScratch(FILEHANDLE *f)
03334 {
03335     LONG size, RetCode;
03336     if ( f->handle < 0 ){
03337         /*Create the file*/
03338         if ( ( RetCode = CreateFile(f->name) ) >= 0 ) {
03339             f->handle = (WORD)RetCode;
03340             PUTZERO(f->filesize);
03341             PUTZERO(f->POposition);
03342         }
03343         else{
03344             MesPrint("Cannot create scratch file %s", f->name);
03345             return(-1);
03346         }
03347     } /*if ( f->handle < 0 )*/
03348     size = (f->POfill-f->PObuffer)*sizeof(WORD);
03349     if( size > 0 ){
03350         SeekFile(f->handle, &(f->POposition), SEEK_SET);
03351         if ( WriteFile(f->handle, (UBYTE *) (f->PObuffer), size) != size ){
03352             MesPrint("Error while writing to disk. Disk full?");
03353             return(-1);
03354         }
03355         ADDPOS(f->filesize, size);
03356         ADDPOS(f->POposition, size);
03357         f->POfill = f->POfull=f->PObuffer;
03358     } /*if( size > 0 )*/
03359     return(0);
03360 }

```

```

03361
03362 /*
03363 #] PF_pushScratch :
03364 #[ Broadcasting RHS expressions :
03365 #] PF_WalkThroughExprMaster :
03366 Returns <=0 if the expression is ready, or dl+1;
03367 */
03368
03369 static int PF_WalkThroughExprMaster(FILEHANDLE *curfile, int dl)
03370 {
03371     LONG l=0;
03372     for(;;){
03373         if(curfile->POfull-curfile->POfill < dl){
03374             POSITION pos;
03375             SeekScratch(curfile,&pos);
03376             PF_SetScratch(curfile,&pos);
03377         }/*if(curfile->POfull-curfile->POfill < dl)*/
03378         curfile->POfill+=dl;
03379         l+=dl;
03380         if( l >= PF.exprbufsize){
03381             if( l == PF.exprbufsize){
03382                 if( *(curfile->POfill) == 0)/*expression is ready*/
03383                     return(0);
03384             }
03385             l-=PF.exprbufsize;
03386             curfile->POfill-=l;
03387             return l+1;
03388         }
03389
03390         dl=*(curfile->POfill);
03391         if(dl == 0)
03392             return l-PF.exprbufsize;
03393
03394         if(dl<0){/*compressed term*/
03395             if(curfile->POfull-curfile->POfill < 1){
03396                 POSITION pos;
03397                 SeekScratch(curfile,&pos);
03398                 PF_SetScratch(curfile,&pos);
03399             }/*if(curfile->POfull-curfile->POfill < 1)*/
03400             dl=*(curfile->POfill+1)+2;
03401         }/*if(*(curfile->POfill)<0)*/
03402     }/*for(;;)*/
03403 }
03404
03405 /*
03406 #] PF_WalkThroughExprMaster :
03407 #] PF_WalkThroughExprSlave :
03408 Returns <=0 if the expression is ready, or dl+1;
03409 */
03410
03411 static int PF_WalkThroughExprSlave(FILEHANDLE *curfile, LONG *counter, int dl)
03412 {
03413     LONG l=0;
03414     for(;;){
03415         if(curfile->POstop-curfile->POfill < dl){
03416             if(PF_pushScratch(curfile))
03417                 return(-PF.exprbufsize-1);
03418         }
03419         curfile->POfill+=dl;
03420         curfile->POfull=curfile->POfill;
03421         l+=dl;
03422         if( l >= PF.exprbufsize){
03423             if( l == PF.exprbufsize){
03424                 /*
03425                  * This access is valid because PF.exprbufsize+1 WORDs are
03426                  * broadcasted, this shortcut is not mandatory though. (TU 15 Sep 2011)
03427                  */
03428                 if( *(curfile->POfill) == 0)/*expression is ready*/
03429                     return(0);
03430             }
03431             l-=PF.exprbufsize;
03432             curfile->POfill-=l;
03433             curfile->POfull=curfile->POfill;
03434             return l+1;
03435         }
03436
03437         dl=*(curfile->POfill);
03438         if(dl == 0)
03439             return l-PF.exprbufsize;
03440         (*counter)++;
03441         if(dl<0){/*compressed term*/
03442             if(curfile->POstop-curfile->POfill < 1){
03443                 if(PF_pushScratch(curfile))
03444                     return(-PF.exprbufsize-1);
03445             }
03446             /*
03447              * This access is always valid because PF.exprbufsize+1 WORDs are

```

```

03448         * broadcasted. (TU 15 Sep 2011)
03449         */
03450         dl=*(curfile->POfill+1)+2;
03451     }/*if*(curfile->POfill)<0)*/
03452 }/*for(;;)*/
03453 }
03454
03455 /*
03456     #] PF_WalkThroughExprSlave :
03457     #[ PF_rhsBCastMaster :
03458 */
03459
03460 static int PF_rhsBCastMaster(FILEHANDLE *curfile, EXPRESSIONS e)
03461 {
03462     LONG l=1; /*PF_WalkThroughExpr returns length + 1*/
03463     SetScratch(curfile, &(e->onfile));
03464     do{
03465         /*
03466          * We need to broadcast PF.exprbufsize+1 WORDs because PF_WalkThroughExprSlave
03467          * may access to an additional 1 WORD. It is better to rewrite the routines
03468          * in such a way as to broadcast only PF.exprbufsize WORDs. (TU 15 Sep 2011)
03469          */
03470         if ( curfile->POfull - curfile->POfill < PF.exprbufsize + 1 ) {
03471             POSITION pos;
03472             SeekScratch(curfile, &pos);
03473             PF_SetScratch(curfile, &pos);
03474         }
03475         if ( PF_Bcast(curfile->POfill, (PF.exprbufsize + 1) * sizeof(WORD)) )
03476             return -1;
03477         l=PF_WalkThroughExprMaster(curfile, l-1);
03478     }while(l>0);
03479     if(l<0)/*The tail is extra, decrease POfill*/
03480         curfile->POfill-=l;
03481     return(0);
03482 }
03483
03484 /*
03485     #] PF_rhsBCastMaster :
03486     #[ PF_rhsBCastSlave :
03487 */
03488
03489 static int PF_rhsBCastSlave(FILEHANDLE *curfile, EXPRESSIONS e)
03490 {
03491     LONG l=1; /*PF_WalkThroughExpr returns length + 1*/
03492     LONG counter = 0;
03493     do{
03494         /*
03495          * We need to broadcast PF.exprbufsize+1 WORDs because PF_WalkThroughExprSlave
03496          * may access to an additional 1 WORD. It is better to rewrite the routines
03497          * in such a way as to broadcast only PF.exprbufsize WORDs. (TU 15 Sep 2011)
03498          */
03499         if ( curfile->POstop - curfile->POfill < PF.exprbufsize + 1 ) {
03500             if(PF_pushScratch(curfile))
03501                 return(-1);
03502         }
03503         if ( PF_Bcast(curfile->POfill, (PF.exprbufsize + 1) * sizeof(WORD)) )
03504             return(-1);
03505         l = PF_WalkThroughExprSlave(curfile, &counter, l - 1);
03506     }while(l>0);
03507     if(l<0){/*The tail is extra, decrease POfill*/
03508         if(l<-PF.exprbufsize)/*error due to a PF_pushScratch() failure */
03509             return(-1);
03510         curfile->POfill-=l;
03511     }
03512     if ( curfile->handle >= 0 ) {
03513         if ( PF_pushScratch(curfile) ) return -1;
03514     }
03515     curfile->POfull=curfile->POfill;
03516     if ( curfile != AR.hidefile ) AR.InInBuf = curfile->POfull-curfile->PObuffer;
03517     else AR.InHiBuf = curfile->POfull-curfile->PObuffer;
03518     CHECK(counter == e->counter + 1); /* The first term is the prototype. */
03519     return(0);
03520 }
03521
03522 /*
03523     #] PF_rhsBCastSlave :
03524     #[ PF_BroadcastExpr :
03525 */
03526
03527 int PF_BroadcastExpr(EXPRESSIONS e, FILEHANDLE *file)
03528 {
03529     if ( PF.me == MASTER ) {
03530         if ( PF_rhsBCastMaster(file, e) ) return -1;
03531     }
03532     #ifdef PF_DEBUG_BCAST_RHSEXP
03533         MesPrint(">> Broadcast RhsExpr: %s", AC.exprnames->namebuffer + e->name);
03534     #endif
03535 }

```

```

03557     else {
03558         POSITION pos;
03559         SetEndHScratch(file, &pos);
03560         e->onfile = pos;
03561         if ( PF_rhsBCastSlave(file, e) ) return -1;
03562     }
03563     return 0;
03564 }
03565
03566 /*
03567     #] PF_BroadcastExpr :
03568     #[ PF_BroadcastRHS :
03569 */
03570
03571 int PF_BroadcastRHS(void)
03572 {
03573     int i;
03574     for ( i = 0; i < NumExpressions; i++ ) {
03575         EXPRESSIONS e = &Expressions[i];
03576         if ( !(e->vflags & ISINRHS) ) continue;
03577         switch ( e->status ) {
03578             case LOCALEXPRESSION:
03579             case SKIPLEXPRESSION:
03580             case DROPLEXPRESSION:
03581             case GLOBALEXPRESSION:
03582             case SKIPGEXPRESSION:
03583             case DROPGEXPRESSION:
03584             case HIDELEXPRESSION:
03585             case HIDEGEXPRESSION:
03586             case INTOHIDELEXPRESSION:
03587             case INTOHIDEGEXPRESSION:
03588                 if ( PF_BroadcastExpr(e, AR.infile) ) return -1;
03589                 break;
03590             case HIDDENLEXPRESSION:
03591             case HIDDENGEXPRESSION:
03592             case DROPHLEXPRESSION:
03593             case DROPHGEXPRESSION:
03594             case UNHIDELEXPRESSION:
03595             case UNHIDEGEXPRESSION:
03596                 if ( PF_BroadcastExpr(e, AR.hidefile) ) return -1;
03597                 break;
03598         }
03599     }
03600     if ( PF.me != MASTER )
03601         UpdatePositions();
03602     return 0;
03603 }
03604
03605 /*
03606     #] PF_BroadcastRHS :
03607     #] Broadcasting RHS expressions :
03608     #[ InParallel mode :
03609     #[ PF_InParallelProcessor :
03610 */
03611
03612 int PF_InParallelProcessor(void)
03613 {
03614     GETIDENTITY
03615     int i, next, tag;
03616     EXPRESSIONS e;
03617     /*
03618      * Skip expressions with zero terms. All the master and slaves need to
03619      * change the "partodo" flag.
03620      */
03621     if ( PF.numtasks >= 3 ) {
03622         for ( i = 0; i < NumExpressions; i++ ) {
03623             e = Expressions + i;
03624             if ( e->partodo > 0 && e->counter == 0 ) {
03625                 e->partodo = 0;
03626             }
03627         }
03628     }
03629     if (PF.me == MASTER){
03630         if ( PF.numtasks >= 3 ) {
03631             partodoexpr = (WORD*)Malloc1(sizeof(WORD)*(PF.numtasks+1), "PF_InParallelProcessor");
03632             for ( i = 0; i < NumExpressions; i++ ) {
03633                 e = Expressions+i;
03634                 if ( e->partodo <= 0 ) continue;
03635                 switch(e->status){
03636                     case LOCALEXPRESSION:
03637                     case GLOBALEXPRESSION:
03638                     case UNHIDELEXPRESSION:
03639                     case UNHIDEGEXPRESSION:
03640                     case INTOHIDELEXPRESSION:
03641                     case INTOHIDEGEXPRESSION:
03642                         tag=PF_ANY_SOURCE;
03643                         next=PF_Wait4SlaveIP(&tag);

```

```

03656         if(next<0)
03657             return(-1);
03658         if(tag == PF_DATA_MSGTAG){
03659             PF_Statistics(PF_stats,0);
03660             if(PF_Slave2MasterIP(next))
03661                 return(-1);
03662         }
03663         if(PF_Master2SlaveIP(next,e))
03664             return(-1);
03665         partodoexr[next]=i;
03666         break;
03667     default:
03668         e->partodo = 0;
03669         continue;
03670     }/*switch(e->status)*/
03671     }/*for ( i = 0; i < NumExpressions; i++ )*/
03672     /*Here some slaves are working, other are waiting on PF_Send.
03673     Wait all of them.*/
03674     /*At this point no new slaves may be launched so PF_WaitAllSlaves()
03675     does not modify partodoexr[].*
03676     if(PF_WaitAllSlaves())
03677         return(-1);
03678
03679     if ( AC.CollectFun ) AR.DeferFlag = 0;
03680     if(partodoexr){
03681         M_free(partodoexr,"PF_InParallelProcessor");
03682         partodoexr=NULL;
03683     }/*if(partodoexr)*/
03684     }/*if ( PF.numtasks >= 3 ) */
03685     else {
03686         for ( i = 0; i < NumExpressions; i++ ) {
03687             Expressions[i].partodo = 0;
03688         }
03689     }
03690     return(0);
03691 }/*if(PF.me == MASTER)*/
03692 /*Slave:*/
03693 if(PF_Wait4MasterIP(PF_EMPTY_MSGTAG))
03694     return(-1);
03695 /*master is ready to listen to me*/
03696 do{
03697     WORD *oldwork= AT.WorkPointer;
03698     tag=PF_ReadMaster();/*reads directly to its scratch!*/
03699     if(tag<0)
03700         return(-1);
03701     if(tag == PF_DATA_MSGTAG){
03702         oldwork = AT.WorkPointer;
03703
03704         /* For redefine statements. */
03705         if ( AC.numpfirstnum > 0 ) {
03706             int j;
03707             for ( j = 0; j < AC.numpfirstnum; j++ ) {
03708                 AC.inputnumbers[j] = -1;
03709             }
03710         }
03711
03712         if(PF_DoOneExpr())/*the processor*/
03713             return(-1);
03714         if(PF_Wait4MasterIP(PF_DATA_MSGTAG))
03715             return(-1);
03716         if(PF_Slave2MasterIP(PF.me))/*both master and slave*/
03717             return(-1);
03718         AT.WorkPointer=oldwork;
03719     }/*if(tag == PF_DATA_MSGTAG)*/
03720     }while(tag!=PF_EMPTY_MSGTAG);
03721     PF.exprtado=-1;
03722     return(0);
03723 }/*PF_InParallelProcessor*/
03724
03725 /*
03726     #] PF_InParallelProcessor :
03727     #[ PF_Wait4MasterIP :
03728 */
03729
03730 static int PF_Wait4MasterIP(int tag)
03731 {
03732     int follow = 0;
03733     LONG cpu,space = 0;
03734
03735     if(PF.log){
03736         fprintf(stderr,"%d] Starting to send to Master\n",PF.me);
03737         fflush(stderr);
03738     }
03739
03740     PF_PreparePack();
03741     cpu = TimeCPU(1);
03742     PF_Pack(&cpu,1,PF_LONG);

```

```

03743     PF_Pack(&space, 1, PF_LONG);
03744     PF_Pack(&PF_linterms, 1, PF_LONG);
03745     PF_Pack(&(AM.S0->GenTerms), 1, PF_LONG);
03746     PF_Pack(&(AM.S0->TermsLeft), 1, PF_LONG);
03747     PF_Pack(&follow, 1, PF_INT );
03748
03749     if(PF.log){
03750         fprintf(stderr, "[%d] Now sending with tag = %d\n", PF.me, tag);
03751         fflush(stderr);
03752     }
03753
03754     PF_Send(MASTER, tag);
03755
03756     if(PF.log){
03757         fprintf(stderr, "[%d] returning from send\n", PF.me);
03758         fflush(stderr);
03759     }
03760     return(0);
03761 }
03762 /*
03763     #] PF_Wait4MasterIP :
03764     #[ PF_DoOneExpr :
03765 */
03766
03774 static int PF_DoOneExpr(void) /*the processor*/
03775 {
03776     GETIDENTITY
03777     EXPRESSIONS e;
03778     int i;
03779     WORD *term;
03780     POSITION position, outposition;
03781     FILEHANDLE *fi, *fout;
03782     LONG dd = 0;
03783     WORD oldBracketOn = AR.BracketOn;
03784     WORD *oldBrackBuf = AT.BrackBuf;
03785     WORD oldbracketindexflag = AT.bracketindexflag;
03786     e = Expressions + PF.exprtodo;
03787     i = PF.exprtodo;
03788     AR.CurExpr = i;
03789     AR.SortType = AC.SortType;
03790     AR.expchanged = 0;
03791     if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
03792         AR.BracketOn = 1;
03793         AT.BrackBuf = AM.BracketFactors;
03794         AT.bracketindexflag = 1;
03795     }
03796
03797     position = AS.OldOnFile[i];
03798     if ( e->status == HIDDENLEXPRESSION || e->status == HIDDENGEXPRESSION ) {
03799         AR.GetFile = 2; fi = AR.hidefile;
03800     }
03801     else {
03802         AR.GetFile = 0; fi = AR.infile;
03803     }
03804 /*
03805     PUTZERO(fi->PPosition);
03806     if ( fi->handle >= 0 ) {
03807         fi->POfill = fi->POfull = fi->PObuffer;
03808     }
03809 */
03810     SetScratch(fi, &position);
03811     term = AT.WorkPointer;
03812     AR.CompressPointer = AR.CompressBuffer;
03813     AR.CompressPointer[0] = 0;
03814     AR.KeptInHold = 0;
03815     if ( GetTerm(BHEAD term) <= 0 ) {
03816         MesPrint("Expression %d has problems in scratchfile", i);
03817         Terminate(-1);
03818     }
03819     if ( AT.bracketindexflag > 0 ) OpenBracketIndex(i);
03820     term[3] = i;
03821     PUTZERO(outposition);
03822     fout = AR.outfile;
03823     fout->POfill = fout->POfull = fout->PObuffer;
03824     fout->PPosition = outposition;
03825     if ( fout->handle >= 0 ) {
03826         fout->PPosition = outposition;
03827     }
03828 /*
03829     The next statement is needed because we need the system
03830     to believe that the expression is at position zero for
03831     the moment. In this worker, with no memory of other expressions,
03832     it is. This is needed for when a bracket index is made
03833     because there e->onfile is an offset. Afterwards, when the
03834     expression is written to its final location in the masters
03835     output e->onfile will get its real value.
03836 */

```



```

03837         PUTZERO(e->onfile);
03838         if ( PutOut(BHEAD term,&outposition,fout,0) < 0 ) return -1;
03839
03840         AR.DeferFlag = AC.ComDefer;
03841
03842  /*      AR.sLevel = AB[0]->R.sLevel;*/
03843         term = AT.WorkPointer;
03844         NewSort(BHEAD0);
03845         AR.MaxDum = AM.IndDum;
03846         AN.ninterms = 0;
03847         while ( GetTerm(BHEAD term) ) {
03848             SeekScratch(fi,&position);
03849             AN.ninterms++; dd = AN.deferskipped;
03850             if ( ( e->vflags & ISFACTORIZED ) != 0 && term[1] == HAAKJE ) {
03851                 StoreTerm(BHEAD term);
03852             }
03853             else {
03854                 if ( AC.CollectFun && *term <= (AM.MaxTer/(2*(LONG)sizeof(WORD))) ) {
03855                     if ( GetMoreTerms(term) < 0 ) {
03856                         LowerSortLevel(); return(-1);
03857                     }
03858                     SeekScratch(fi,&position);
03859                 }
03860                 AT.WorkPointer = term + *term;
03861                 AN.RepPoint = AT.RepCount + 1;
03862                 if ( AR.DeferFlag ) {
03863                     AR.CurDum = AN.IndDum = Expressions[PF.exprtodo].numdummies;
03864                 }
03865                 else {
03866                     AN.IndDum = AM.IndDum;
03867                     AR.CurDum = ReNumber(BHEAD term);
03868                 }
03869                 if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMFLAG);
03870                 if ( AN.ncmod ) {
03871                     if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
03872                     else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
03873                 }
03874                 else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
03875                 if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
03876                     && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
03877                     PolyFunClean(BHEAD term);
03878                 }
03879                 if ( Generator(BHEAD term,0) ) {
03880                     LowerSortLevel(); return(-1);
03881                 }
03882                 AN.ninterms += dd;
03883             }
03884             SetScratch(fi,&position);
03885             if ( fi == AR.hidefile ) {
03886                 AR.InHiBuf = (fi->POfull-fi->PObuffer)
03887                     -DIFBASE(position,fi->POposition)/sizeof(WORD);
03888             }
03889             else {
03890                 AR.InInBuf = (fi->POfull-fi->PObuffer)
03891                     -DIFBASE(position,fi->POposition)/sizeof(WORD);
03892             }
03893         }
03894         AN.ninterms += dd;
03895         if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) return(-1);
03896         e->numdummies = AR.MaxDum - AM.IndDum;
03897         AR.BraceOn = oldBracketOn;
03898         AT.BrackBuf = oldBrackBuf;
03899         if ( ( e->vflags & TOBEFACTORED ) != 0 )
03900             poly_factorize_expression(e);
03901         else if ( ( ( e->vflags & TOBEUNFACTORED ) != 0 )
03902             && ( ( e->vflags & ISFACTORIZED ) != 0 ) )
03903             poly_unfactorize_expression(e);
03904         if ( AM.S0->TermsLeft ) e->vflags &= ~ISZERO;
03905         else e->vflags |= ISZERO;
03906         if ( AR.expchanged == 0 ) e->vflags |= ISUNMODIFIED;
03907  /*      if ( AM.S0->TermsLeft ) AR.expflags |= ISZERO;
03908         if ( AR.expchanged ) AR.expflags |= ISUNMODIFIED;*/
03909         AR.GetFile = 0;
03910         AT.bracketindexflag = oldbracketindexflag;
03911         fout->POfull = fout->POfill;
03912         return(0);
03913     }
03914 }
03915
03916  /*
03917     #] PF_DoOneExpr :
03918     #[ PF_Slave2MasterIP :
03919  */
03920
03921 typedef struct bufIPstruct {
03922     LONG i;
03923     struct ExprEssiOn e;

```

```

03924 } bufIPstruct_t;
03925
03926 static int PF_Slave2MasterIP(int src)/*both master and slave*/
03927 {
03928     EXPRESSIONS e;
03929     bufIPstruct_t exprData;
03930     int i,l;
03931     FILEHANDLE *fout=AR.outfile;
03932     POSITION pos;
03933     /*Here we know the length of data to send in advance:
03934     slave has the only one expression in its scratch file, and it sends
03935     this information to the master.*/
03936     if (PF.me != MASTER){/*slave*/
03937         e = Expressions + PF.exprtodo;
03938         /*Fill in the expression data:*/
03939         memcpy(&(exprData.e), e, sizeof(struct ExprEsSiOn));
03940         SeekScratch(fout,&pos);
03941         exprData.i=BASEPOSITION(pos);
03942         /*Send the metadata:*/
03943         if (PF_RawSend(MASTER,&exprData,sizeof(bufIPstruct_t),0))
03944             return(-1);
03945         i=exprData.i;
03946         SETBASEPOSITION(pos,0);
03947         do{
03948             int blen=PF.exprbufsize*sizeof(WORD);
03949             if(i<blen)
03950                 blen=i;
03951             l=PF_SendChunkIP(fout,&pos, MASTER, blen);
03952             /*Here always l == blen!*/
03953             if(l<0)
03954                 return(-1);
03955             ADDPOS(pos,l);
03956             i-=l;
03957         }while(i>0);
03958         if ( fout->handle >= 0 ) { /* Now get rid of the file */
03959             CloseFile(fout->handle);
03960             fout->handle = -1;
03961             remove(fout->name);
03962             PUTZERO(fout->POposition);
03963             PUTZERO(fout->filesize);
03964             fout->POfill = fout->POfull = fout->PObuffer;
03965         }
03966         /* Now handle redefined preprocessor variables. */
03967         if ( AC.numpfirstnum > 0 ) {
03968             PF_PrepareLongSinglePack();
03969             PF_PackRedefinedPreVars();
03970             PF_LongSingleSend(MASTER, PF_MISC_MSGTAG);
03971         }
03972         return(0);
03973     }/*if(PF.me != MASTER)*/
03974     /*Master*/
03975     /*partodoexpr[src] is the number of expression.*/
03976     e = Expressions +partodoexpr[src];
03977     /*Get metadata:*/
03978     if (PF_RawRecv(&src, &exprData,sizeof(bufIPstruct_t),&i)!= sizeof(bufIPstruct_t))
03979         return(-1);
03980     /*Fill in the expression data:*/
03981     /* memcpy(e, &(exprData.e), sizeof(struct ExprEsSiOn)); */
03982     e->counter = exprData.e.counter;
03983     e->vflags = exprData.e.vflags;
03984     e->uflags = exprData.e.uflags;
03985     e->numdummies = exprData.e.numdummies;
03986     e->numfactors = exprData.e.numfactors;
03987     if ( !(e->vflags & ISZERO) ) AR.expflags |= ISZERO;
03988     if ( !(e->vflags & ISUNMODIFIED) ) AR.expflags |= ISUNMODIFIED;
03989     SeekScratch(fout,&pos);
03990     e->onfile = pos;
03991     i=exprData.i;
03992     while(i>0){
03993         int blen=PF.exprbufsize*sizeof(WORD);
03994         if(i<blen)
03995             blen=i;
03996         l=PF_RecvChunkIP(fout,src,blen);
03997         /*Here always l == blen!*/
03998         if(l<0)
03999             return(-1);
04000         i-=l;
04001     }
04002     /* Now handle redefined preprocessor variables. */
04003     if ( AC.numpfirstnum > 0 ) {
04004         PF_LongSingleReceive(src, PF_MISC_MSGTAG, NULL, NULL);
04005         PF_UnpackRedefinedPreVars();
04006     }
04007     return(0);
04008 }
04009
04010 /*

```

```

04011         #] PF_Slave2MasterIP :
04012         #[ PF_Master2SlaveIP :
04013     */
04014
04015     static int PF_Master2SlaveIP(int dest, EXPRESSIONS e)
04016     {
04017         bufIPstruct_t exprData;
04018         FILEHANDLE *fi;
04019         POSITION pos;
04020         int l;
04021         LONG ll=0, count=0;
04022         WORD *t;
04023         if(e==NULL){/*Say to the slave that no more job:*/
04024             if(PF_RawSend(dest, &exprData, sizeof(bufIPstruct_t), PF_EMPTY_MSGTAG))
04025                 return(-1);
04026             return(0);
04027         }
04028         memcpy(&(exprData.e), e, sizeof(struct ExprEsSiOn));
04029         exprData.i=e-Expressions;
04030         if ( AC.StatsFlag && AC.OldParallelStats ) {
04031             MesPrint("");
04032             MesPrint(" Sending expression %s to slave %d", EXPRNAME(exprData.i), dest);
04033         }
04034         if(PF_RawSend(dest, &exprData, sizeof(bufIPstruct_t), PF_DATA_MSGTAG))
04035             return(-1);
04036         if ( e->status == HIDDENLEXPRESSION || e->status == HIDDENGEXPRESSION )
04037             fi = AR.hidefile;
04038         else
04039             fi = AR.infile;
04040         pos=e->onfile;
04041         SetScratch(fi, &pos);
04042         do{
04043             l=PF_SendChunkIP(fi, &pos, dest, PF.exprbufsize*sizeof(WORD));
04044             if(l<0)
04045                 return(-1);
04046             t=fi->PObuffer+ (DIFBASE(pos, fi->POposition))/sizeof(WORD);
04047             ll=PF_WalkThrough(t, ll, l/sizeof(WORD), &count);
04048             ADDPOS(pos, l);
04049         }while(ll>=2);
04050         return(0);
04051     }
04052
04053     /*
04054         #] PF_Master2SlaveIP :
04055         #[ PF_ReadMaster :
04056     */
04057
04058     static int PF_ReadMaster(void)/*reads directly to its scratch!*/
04059     {
04060         bufIPstruct_t exprData;
04061         int tag, m=MASTER;
04062         EXPRESSIONS e;
04063         FILEHANDLE *fi;
04064         POSITION pos;
04065         LONG count=0;
04066         WORD *t;
04067         LONG ll=0;
04068         int l;
04069         /*Get metadata:*/
04070         if (PF_RawRecv(&m, &exprData, sizeof(bufIPstruct_t), &tag) != sizeof(bufIPstruct_t))
04071             return(-1);
04072
04073         if(tag == PF_EMPTY_MSGTAG)/*No data, no job*/
04074             return(tag);
04075
04076         /*data expected, tag must be == PF_DATA_MSTAG!*/
04077         PF.exprtodo=exprData.i;
04078         e=Expressions + PF.exprtodo;
04079         /*Fill in the expression data:*/
04080         /* memcpy(e, &(exprData.e), sizeof(struct ExprEsSiOn)); */
04081         if ( e->status == HIDDENLEXPRESSION || e->status == HIDDENGEXPRESSION )
04082             fi = AR.hidefile;
04083         else
04084             fi = AR.infile;
04085         SetEndHScratch(fi, &pos);
04086         e->onfile=AS.OldOnFile[PF.exprtodo]=pos;
04087
04088         do{
04089             l=PF_RecvChunkIP(fi, MASTER, PF.exprbufsize*sizeof(WORD));
04090             if(l<0)
04091                 return(-1);
04092             t=fi->POfull-l/sizeof(WORD);
04093             ll=PF_WalkThrough(t, ll, l/sizeof(WORD), &count);
04094         }while(ll>=2);
04095         /*Now -ll-2 is the number of "extra" elements transferred from the master.*/
04096         fi->POfull=-ll-2;
04097         fi->POfill=fi->POfull;

```

```

04098     return(PF_DATA_MSGTAG);
04099 }
04100
04101 /*
04102     #] PF_ReadMaster :
04103     #[ PF_SendChunkIP :
04104     thesize is in bytes. Returns the number of sent bytes or <0 on error:
04105 */
04106
04107 static int PF_SendChunkIP(FILEHANDLE *curfile, POSITION *position, int to, LONG thesize)
04108 {
04109     LONG l=thesize;
04110     if(
04111         ISLESSPOS(*position,curfile->POposition) ||
04112         ISGEPOSINC(*position,curfile->POposition,
04113             ((curfile->POfull-curfile->PObuffer)*sizeof(WORD)-thesize) )
04114     ){
04115         if(curfile->handle< 0)
04116             l=(curfile->POfull-curfile->PObuffer)*sizeof(WORD) - (LONG)(position->p1);
04117         else{
04118             PF_SetScratch(curfile,position);
04119             if(
04120                 ISGEPOSINC(*position,curfile->POposition,
04121                     ((curfile->POfull-curfile->PObuffer)*sizeof(WORD)-thesize) )
04122             )
04123                 l=(curfile->POfull-curfile->PObuffer)*sizeof(WORD) - (LONG)position->p1;
04124         }
04125     }
04126     /*Now we are able to sent l bytes from the
04127     curfile->PObuffer[position-curfile->POposition]*/
04128     if(PF_RawSend(to,curfile->PObuffer+ (DIFBASE(*position,curfile->POposition))/sizeof(WORD),l,0))
04129         return(-1);
04130     return(l);
04131 }
04132
04133 /*
04134     #] PF_SendChunkIP :
04135     #[ PF_RecvChunkIP :
04136     thesize is in bytes. Returns the number of sent bytes or <0 on error:
04137 */
04138
04139 static int PF_RecvChunkIP(FILEHANDLE *curfile, int from, LONG thesize)
04140 {
04141     LONG receivedBytes;
04142
04143     if( (LONG)((curfile->POstop - curfile->POfull)*sizeof(WORD)) < thesize )
04144         if(PF_pushScratch(curfile))
04145             return(-1);
04146     /*Now there is enough space from curfile->POfill to curfile->POstop*/
04147     /*Block:*/
04148     int tag=0;
04149     receivedBytes=PF_RawRecv(&from,curfile->POfull,thesize,&tag);
04150     /*:Block*/
04151     if(receivedBytes >= 0 ){
04152         curfile->POfull+=receivedBytes/sizeof(WORD);
04153         curfile->POfill=curfile->POfull;
04154     }/*if(receivedBytes >= 0 )*/
04155     return(receivedBytes);
04156 }
04157
04158 /*
04159     #] PF_RecvChunkIP :
04160     #[ PF_WalkThrough :
04161     Returns:
04162     >= 0 -- initial offset,
04163     -1 -- the first element of t contains the length of the tail of compressed term,
04164     <= -2 -- -(d+2), where d is the number of extra transferred elements.
04165     Expects:
04166     l -- initial offset or -1,
04167     chunk -- number of transferred elements (not bytes!)
04168     *count -- incremented each time a new term is found
04169 */
04170
04171 static int PF_WalkThrough(WORD *t, LONG l, LONG chunk, LONG *count)
04172 {
04173     if(l<0) /*== -1!*/
04174         l=(*t)+1; /*the first element of t contains the length of
04175             the tail of compressed term*/
04176     else{
04177         if(l>=chunk) /*next term is out of the chunk*/
04178             return(l-chunk);
04179         t+=l;
04180         chunk-=l; /*note, l was less than chunk so chunk >0!*/
04181         l=*t;
04182     }
04183     /*Main loop:*/
04184     while(l!=0){

```

```

04185         if(l>0){/*an offset to the next term*/
04186             if(l<chunk){
04187                 t+=1;
04188                 chunk-=1;/*note, l was less than chunk so chunk >0!*/
04189                 l=*t;
04190                 (*count)++;
04191             }/*if(l<chunk)*/
04192             else
04193                 return(l-chunk);
04194         }/*if(l>0)*/
04195         else{ /* l<0 */
04196             if(chunk < 2){/*i.e., chunk == 1*/
04197                 return(-1);/*the first WORD in the next chunk is length of the tail of the compressed
term*/
04198                 l=(t+1)+2; /*+2 since
04199                     1. t points to the length field -1,
04200                     2. the size of a tail of compressed term is equal to the number of WORDs in this
tail*/
04201             }
04202         }/*while(l!=0)*/
04203         return(-l-chunk);/* -(2+(chunk-1)), chunk>0 ! */
04204     }
04205     /*
04206     #] PF_WalkThrough :
04207     #] InParallel mode :
04208     #[ PF_SendFile :
04209     */
04210     /*
04211     #define PF_SNDFILEBUFSIZE 4096
04212     */
04213     int PF_SendFile(int to, FILE *fd)
04214     {
04215         size_t len=0;
04216         if(fd == NULL){
04217             if(PF_RawSend(to,&to,sizeof(int),PF_EMPTY_MSGTAG))
04218                 return(-1);
04219             return(0);
04220         }
04221         for(;;){
04222             char buf[PF_SNDFILEBUFSIZE];
04223             size_t l;
04224             l=fread(buf, 1, PF_SNDFILEBUFSIZE, fd);
04225             len+=l;
04226             if(l==PF_SNDFILEBUFSIZE){
04227                 if(PF_RawSend(to,buf,PF_SNDFILEBUFSIZE,PF_BUFFER_MSGTAG))
04228                     return(-1);
04229             }
04230             else{
04231                 if(PF_RawSend(to,buf,l,PF_ENDBUFFER_MSGTAG))
04232                     return(-1);
04233                 break;
04234             }
04235         }/*for(;;)*/
04236         return(len);
04237     }
04238     /*
04239     #] PF_SendFile :
04240     #[ PF_RecvFile :
04241     */
04242     int PF_RecvFile(int from, FILE *fd)
04243     {
04244         size_t len=0;
04245         int tag;
04246         do{
04247             char buf[PF_SNDFILEBUFSIZE];
04248             int l;
04249             l=PF_RawRecv(&from,buf,PF_SNDFILEBUFSIZE,&tag);
04250             if(l<0)
04251                 return(-1);
04252             if(tag == PF_EMPTY_MSGTAG)
04253                 return(-1);
04254             if( fwrite(buf,l,1,fd) !=1 )
04255                 return(-1);
04256             len+=l;
04257         }while(tag!=PF_ENDBUFFER_MSGTAG);
04258         return(len);
04259     }
04260     /*
04261     #] PF_RecvFile :
04262     #[ Synchronised output :
04263     #[ Explanations :
04264     */

```

```

04284
04285 /*
04286  * If the master and slaves output statistics or error messages to the same stream
04287  * or file (e.g., the standard output or the log file) simultaneously, then
04288  * a mixing of their outputs can occur. To avoid this, TFORM uses a lock of
04289  * ErrorMessageLock, but there is no locking functionality in the original MPI
04290  * specification. We need to synchronise the output from the master and slaves.
04291  *
04292  * The idea of the synchronised output (by, e.g., MesPrint()) implemented here is
04293  * Slaves:
04294  *   1. Save the output by WriteFile() (set to PF_WriteFileToFile())
04295  *      into some buffers between MLOCK(ErrorMessageLock) and
04296  *      MUNLOCK(ErrorMessageLock), which call PF_MLock() and PF_MUnlock(),
04297  *      respectively. The output for AM.Stdout and AC.LogHandle are saved to
04298  *      the buffers.
04299  *   2. At MUNLOCK(ErrorMessageLock), send the output in the buffer to the master,
04300  *      with PF_STDOUT_MSGTAG or PF_LOG_MSGTAG.
04301  * Master:
04302  *   1. Receive the buffered output from slaves, and write them by
04303  *      WriteFileToFile().
04304  * The main problem is how and where the master receives messages from
04305  * the slaves (PF_ReceiveErrorMessage()). For this purpose there are three
04306  * helper functions: PF_CatchErrorMessage() and PF_CatchErrorMessagesForAll()
04307  * which remove messages with PF_STDOUT_MSGTAG or PF_LOG_MSGTAG from the top
04308  * of the message queue, and PF_ProbeWithCatchingErrorMessages() which is same as
04309  * PF_Probe() except removing these messages.
04310 */
04311
04312 /*
04313     #] Explanations :
04314     #[ Variables :
04315 */
04316
04317 static int errorMessageLock = 0; /* (slaves) The lock count. See PF_MLock() and PF_MUnlock(). */
04318 static Vector(UBYTE, stdoutBuffer); /* (slaves) The buffer for AM.Stdout. */
04319 static Vector(UBYTE, logBuffer); /* (slaves) The buffer for AC.LogHandle. */
04320 #define recvBuffer logBuffer /* (master) The buffer for receiving messages. */
04321
04322 /*
04323  * If PF_ENABLE_STDOUT_BUFFERING is defined, the master performs the line buffering
04324  * (using stdoutBuffer) at PF_WriteFileToFile().
04325  */
04326 #ifndef PF_ENABLE_STDOUT_BUFFERING
04327 #ifndef UNIX
04328 #define PF_ENABLE_STDOUT_BUFFERING
04329 #endif
04330 #endif
04331
04332 /*
04333     #] Variables :
04334     #[ PF_MLock :
04335 */
04336
04337 void PF_MLock(void)
04338 {
04339     /* Only on slaves. */
04340     if ( errorMessageLock++ > 0 ) return;
04341     VectorClear(stdoutBuffer);
04342     VectorClear(logBuffer);
04343 }
04344
04345 /*
04346     #] PF_MLock :
04347     #[ PF_MUnlock :
04348 */
04349
04350 void PF_MUnlock(void)
04351 {
04352     /* Only on slaves. */
04353     if ( --errorMessageLock > 0 ) return;
04354     if ( !VectorEmpty(stdoutBuffer) ) {
04355         PF_RawSend(MASTER, VectorPtr(stdoutBuffer), VectorSize(stdoutBuffer), PF_STDOUT_MSGTAG);
04356     }
04357     if ( !VectorEmpty(logBuffer) ) {
04358         PF_RawSend(MASTER, VectorPtr(logBuffer), VectorSize(logBuffer), PF_LOG_MSGTAG);
04359     }
04360 }
04361
04362 /*
04363     #] PF_MUnlock :
04364     #[ PF_WriteFileToFile :
04365 */
04366
04367 LONG PF_WriteFileToFile(int handle, UBYTE *buffer, LONG size)
04368 {
04369     if ( PF.me != MASTER && errorMessageLock > 0 ) {
04370         if ( handle == AM.Stdout ) {

```

```

04389         VectorPushBacks(stdoutBuffer, buffer, size);
04390         return size;
04391     }
04392     else if ( handle == AC.LogHandle ) {
04393         VectorPushBacks(logBuffer, buffer, size);
04394         return size;
04395     }
04396 }
04397 #ifndef PF_ENABLE_STDOUT_BUFFERING
04398 /*
04399  * On my computer, sometimes a single linefeed "\n" sent to the standard
04400  * output is ignored on the execution of mpiexec. A typical example is:
04401  * $ cat foo.c
04402  * #include <unistd.h>
04403  * int main() {
04404  *     write(1, "    ", 4);
04405  *     write(1, "\n", 1);
04406  *     write(1, "    ", 4);
04407  *     write(1, "123\n", 4);
04408  *     return 0;
04409  * }
04410  * or even as a shell script:
04411  * $ cat foo.sh
04412  * #! bin/sh
04413  * printf "    "
04414  * printf "\n"
04415  * printf "    "
04416  * printf "123\n"
04417  * When I ran it on mpiexec
04418  * $ while :; do mpiexec -np 1 ./foo.sh; done
04419  * I observed the single linefeed (printf "\n") was sometimes ignored. Even
04420  * though this phenomenon might be specific to my environment, I added this
04421  * code because someone may encounter a similar phenomenon and feel it
04422  * frustrating. (TU 16 Jun 2011)
04423  *
04424  * Phenomenon:
04425  * A single linefeed sent to the standard output occasionally ignored
04426  * on mpiexec.
04427  *
04428  * Environment:
04429  * openSUSE 11.4 (x86_64)
04430  * kernel: 2.6.37.6-0.5-desktop
04431  * gcc: 4.5.1 20101208
04432  * mpich2-1.3.2pl configured with '--enable-shared --with-pm=smpd'
04433  *
04434  * Solution:
04435  * In Unix (in which Uwrite() calls write() system call without any buffering),
04436  * we perform the line buffering here. A single linefeed is also buffered.
04437  *
04438  * XXX:
04439  * At the end of the program the buffered output (text without LF) will not be flushed,
04440  * i.e., will not be written to the standard output. This is not problematic at a normal run.
04441  * The buffer can be explicitly flushed by PF_FlushStdOutBuffer().
04442  */
04443 if ( PF.me == MASTER && handle == AM.StdOut ) {
04444     size_t oldsize;
04445     /* Assume the newline character is LF (when UNIX is defined). */
04446     if ( (size > 0 && buffer[size - 1] != LINEFEED) || (size == 1 && buffer[0] == LINEFEED) ) {
04447         VectorPushBacks(stdoutBuffer, buffer, size);
04448         return size;
04449     }
04450     if ( (oldsize = VectorSize(stdoutBuffer)) > 0 ) {
04451         LONG ret;
04452         VectorPushBacks(stdoutBuffer, buffer, size);
04453         ret = WriteFileToFile(handle, VectorPtr(stdoutBuffer), VectorSize(stdoutBuffer));
04454         VectorClear(stdoutBuffer);
04455         if ( ret < 0 ) {
04456             return ret;
04457         }
04458         else if ( ret < (LONG)oldsize ) {
04459             return 0; /* This means the buffered output in previous calls is lost. */
04460         }
04461         else {
04462             return ret - (LONG)oldsize;
04463         }
04464     }
04465 }
04466 #endif
04467 return WriteFileToFile(handle, buffer, size);
04468 }
04469 /*
04470 #] PF_WriteFileToFile :
04471 #[ PF_FlushStdOutBuffer :
04472 */
04473 void PF_FlushStdOutBuffer(void)

```

```

04480 {
04481 #ifdef PF_ENABLE_STDOUT_BUFFERING
04482     if ( PF.me == MASTER && VectorSize(stdoutBuffer) > 0 ) {
04483         WriteFileToFile(AM.Stdout, VectorPtr(stdoutBuffer), VectorSize(stdoutBuffer));
04484         VectorClear(stdoutBuffer);
04485     }
04486 #endif
04487 }
04488
04489 /*
04490     #] PF_FlushStdOutBuffer :
04491     #[ PF_ReceiveErrorMessage :
04492 */
04493
04502 static void PF_ReceiveErrorMessage(int src, int tag)
04503 {
04504     /* Only on the master. */
04505     int size;
04506     int ret = PF_RawProbe(&src, &tag, &size);
04507     CHECK(ret == 0);
04508     switch ( tag ) {
04509         case PF_STDOUT_MSGTAG:
04510         case PF_LOG_MSGTAG:
04511             VectorReserve(recvBuffer, size);
04512             ret = PF_RawRecv(&src, VectorPtr(recvBuffer), size, &tag);
04513             CHECK(ret == size);
04514             if ( size > 0 ) {
04515                 int handle = (tag == PF_STDOUT_MSGTAG) ? AM.Stdout : AC.LogHandle;
04516 #ifdef PF_ENABLE_STDOUT_BUFFERING
04517                 if ( handle == AM.Stdout ) PF_WriteFileToFile(handle, VectorPtr(recvBuffer), size);
04518                 else
04519 #endif
04520                 WriteFileToFile(handle, VectorPtr(recvBuffer), size);
04521             }
04522             break;
04523     }
04524 }
04525
04526 /*
04527     #] PF_ReceiveErrorMessage :
04528     #[ PF_CatchErrorMessages :
04529 */
04530
04539 static void PF_CatchErrorMessages(int *src, int *tag)
04540 {
04541     /* Only on the master. */
04542     for (;;) {
04543         int asrc = *src;
04544         int atag = *tag;
04545         int ret = PF_RawProbe(&asrc, &atag, NULL);
04546         CHECK(ret == 0);
04547         if ( atag == PF_STDOUT_MSGTAG || atag == PF_LOG_MSGTAG ) {
04548             PF_ReceiveErrorMessage(asrc, atag);
04549             continue;
04550         }
04551         *src = asrc;
04552         *tag = atag;
04553         break;
04554     }
04555 }
04556
04557 /*
04558     #] PF_CatchErrorMessages :
04559     #[ PF_CatchErrorMessagesForAll :
04560 */
04561
04566 static void PF_CatchErrorMessagesForAll(void)
04567 {
04568     /* Only on the master. */
04569     int i;
04570     for ( i = 1; i < PF.numtasks; i++ ) {
04571         int src = i;
04572         int tag = PF_ANY_MSGTAG;
04573         PF_CatchErrorMessages(&src, &tag);
04574     }
04575 }
04576
04577 /*
04578     #] PF_CatchErrorMessagesForAll :
04579     #[ PF_ProbeWithCatchingErrorMessages :
04580 */
04581
04591 static int PF_ProbeWithCatchingErrorMessages(int *src)
04592 {
04593     for (;;) {
04594         int newsrc = *src;
04595         int tag = PF_Probe(&newsrc);

```



```

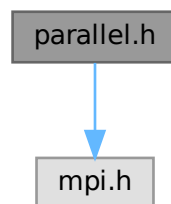
04596         if ( tag == PF_STDOUT_MSGTAG || tag == PF_LOG_MSGTAG ) {
04597             PF_ReceiveErrorMessage(newsrc, tag);
04598             continue;
04599         }
04600         if ( tag > 0 ) *src = newsrc;
04601         return tag;
04602     }
04603 }
04604
04605 /*
04606     #] PF_ProbeWithCatchingErrorMessages :
04607     #[ PF_FreeErrorMessageBuffers :
04608 */
04609
04615 void PF_FreeErrorMessageBuffers(void)
04616 {
04617     VectorFree(stdoutBuffer);
04618     VectorFree(logBuffer);
04619 }
04620
04621 /*
04622     #] PF_FreeErrorMessageBuffers :
04623     #] Synchronised output :
04624 */

```

## 4.96 parallel.h File Reference

#include <mpi.h>

Include dependency graph for parallel.h:



### Data Structures

- struct [PF\\_BUFFER](#)
- struct [ParallelVars](#)

### Macros

- #define [MASTER](#) 0
- #define [PF\\_RESET](#) 0
- #define [PF\\_TIME](#) 1
- #define [PF\\_TERM\\_MSGTAG](#) 10 /\* master -> slave: sending terms \*/
- #define [PF\\_ENDSORT\\_MSGTAG](#) 11 /\* master -> slave: no more terms to be distributed, slave -> master: after [EndSort\(\)](#) \*/
- #define [PF\\_DOLLAR\\_MSGTAG](#) 12 /\* slave -> master: sending \$-variables \*/
- #define [PF\\_BUFFER\\_MSGTAG](#) 20 /\* slave -> master: sending sorted terms, or in [PF\\_SendFile\(\)](#)/[PF\\_RecvFile\(\)](#) \*/

- `#define PF_ENDBUFFER_MSGTAG 21 /* same as PF_BUFFER_MSGTAG, but indicates the end of operation */`
- `#define PF_READY_MSGTAG 30 /* slave -> master: slave is idle and can accept terms */`
- `#define PF_DATA_MSGTAG 50 /* InParallel, DoCheckpoint() */`
- `#define PF_EMPTY_MSGTAG 52 /* InParallel, DoCheckpoint(), PF_SendFile(), PF_RecvFile() */`
- `#define PF_STDOUT_MSGTAG 60 /* slave -> master: sending text to the stdout */`
- `#define PF_LOG_MSGTAG 61 /* slave -> master: sending text to the log file */`
- `#define PF_OPT_MCTS_MSGTAG 70 /* master <-> slave: optimization */`
- `#define PF_OPT_HORNER_MSGTAG 71 /* master <-> slave: optimization */`
- `#define PF_OPT_COLLECT_MSGTAG 72 /* slave -> master: optimization */`
- `#define PF_MISC_MSGTAG 100`
- `#define GNUC_PREREQ(major, minor, patchlevel) 0`
- `#define OMPI_SKIP_MPICXX 1`
- `#define indices ((INDICES))(AC.IndexList.lijst)`
- `#define PF_ANY_SOURCE MPI_ANY_SOURCE`
- `#define PF_ANY_MSGTAG MPI_ANY_TAG`
- `#define PF_COMM MPI_COMM_WORLD`
- `#define PF_BYTE MPI_BYTE`
- `#define PF_INT MPI_INT`
- `#define PF_LongMultiPack(buffer, count, type) PF_LongMultiPackImpl(buffer, count, sizeof_datatype(type), type)`
- `#define PF_LongMultiUnpack(buffer, count, type) PF_LongMultiUnpackImpl(buffer, count, sizeof_↵ datatype(type), type)`

## Typedefs

- `typedef struct ParallelVars PARALLELVARS`

## Functions

- `int PF_ISendSbuf (int, int)`
- `int PF_Bcast (void *buffer, int count)`
- `int PF_RawSend (int, void *, LONG, int)`
- `LONG PF_RawRecv (int *, void *, LONG, int *)`
- `int PF_PreparePack (void)`
- `int PF_Pack (const void *buffer, size_t count, MPI_Datatype type)`
- `int PF_Unpack (void *buffer, size_t count, MPI_Datatype type)`
- `int PF_PackString (const UBYTE *str)`
- `int PF_UnpackString (UBYTE *str)`
- `int PF_Send (int to, int tag)`
- `int PF_Receive (int src, int tag, int *psrc, int *ptag)`
- `int PF_Broadcast (void)`
- `int PF_PrepareLongSinglePack (void)`
- `int PF_LongSinglePack (const void *buffer, size_t count, MPI_Datatype type)`
- `int PF_LongSingleUnpack (void *buffer, size_t count, MPI_Datatype type)`
- `int PF_LongSingleSend (int to, int tag)`
- `int PF_LongSingleReceive (int src, int tag, int *psrc, int *ptag)`
- `int PF_PrepareLongMultiPack (void)`
- `int PF_LongMultiPackImpl (const void *buffer, size_t count, size_t eSize, MPI_Datatype type)`
- `int PF_LongMultiUnpackImpl (void *buffer, size_t count, size_t eSize, MPI_Datatype type)`
- `int PF_LongMultiBroadcast (void)`
- `int PF_EndSort (void)`
- `WORD PF_Deferred (WORD *, WORD)`

- int [PF\\_Processor](#) ([EXPRESSIONS](#), WORD, WORD)
- int [PF\\_Init](#) (int \*, char \*\*\*)
- int [PF\\_Terminate](#) (int)
- LONG [PF\\_GetSlaveTimes](#) (void)
- LONG [PF\\_BroadcastNumber](#) (LONG)
- void [PF\\_BroadcastBuffer](#) (WORD \*\*buffer, LONG \*length)
- int [PF\\_BroadcastString](#) (UBYTE \*)
- int [PF\\_BroadcastPreDollar](#) (WORD \*\*, LONG \*, int \*)
- int [PF\\_CollectModifiedDollars](#) (void)
- int [PF\\_BroadcastModifiedDollars](#) (void)
- int [PF\\_BroadcastRedefinedPreVars](#) (void)
- int [PF\\_BroadcastCBuf](#) (int bufnum)
- int [PF\\_BroadcastExpFlags](#) (void)
- int [PF\\_StoreInsideInfo](#) (void)
- int [PF\\_RestoreInsideInfo](#) (void)
- int [PF\\_BroadcastExpr](#) ([EXPRESSIONS](#) e, [FILEHANDLE](#) \*file)
- int [PF\\_BroadcastRHS](#) (void)
- int [PF\\_InParallelProcessor](#) (void)
- int [PF\\_SendFile](#) (int to, FILE \*fd)
- int [PF\\_RecvFile](#) (int from, FILE \*fd)
- void [PF\\_MLock](#) (void)
- void [PF\\_MUnlock](#) (void)
- LONG [PF\\_WriteFileToFile](#) (int, UBYTE \*, LONG)
- void [PF\\_FlushStdOutBuffer](#) (void)

## Variables

- [PARALLELVARS](#) PF
- LONG [PF\\_maxDollarChunkSize](#)

## 4.96.1 Detailed Description

Header file with things relevant to ParForm.

Definition in file [parallel.h](#).

## 4.96.2 Macro Definition Documentation

### 4.96.2.1 MASTER

```
#define MASTER 0
```

Definition at line 39 of file [parallel.h](#).

### 4.96.2.2 PF\_RESET

```
#define PF_RESET 0
```

Definition at line 41 of file [parallel.h](#).

#### 4.96.2.3 PF\_TIME

```
#define PF_TIME 1
```

Definition at line 42 of file [parallel.h](#).

#### 4.96.2.4 PF\_TERM\_MSGTAG

```
#define PF_TERM_MSGTAG 10 /* master -> slave:  sending terms */
```

Definition at line 44 of file [parallel.h](#).

#### 4.96.2.5 PF\_ENDSORT\_MSGTAG

```
#define PF_ENDSORT_MSGTAG 11 /* master -> slave:  no more terms to be distributed, slave ->
master:  after EndSort() */
```

Definition at line 45 of file [parallel.h](#).

#### 4.96.2.6 PF\_DOLLAR\_MSGTAG

```
#define PF_DOLLAR_MSGTAG 12 /* slave -> master:  sending $-variables */
```

Definition at line 46 of file [parallel.h](#).

#### 4.96.2.7 PF\_BUFFER\_MSGTAG

```
#define PF_BUFFER_MSGTAG 20 /* slave -> master:  sending sorted terms, or in PF_SendFile()/PF_RecvFile()
*/
```

Definition at line 47 of file [parallel.h](#).

#### 4.96.2.8 PF\_ENDBUFFER\_MSGTAG

```
#define PF_ENDBUFFER_MSGTAG 21 /* same as PF_BUFFER_MSGTAG, but indicates the end of operation
*/
```

Definition at line 48 of file [parallel.h](#).

#### 4.96.2.9 PF\_READY\_MSGTAG

```
#define PF_READY_MSGTAG 30 /* slave -> master:  slave is idle and can accept terms */
```

Definition at line 49 of file [parallel.h](#).

#### 4.96.2.10 PF\_DATA\_MSGTAG

```
#define PF_DATA_MSGTAG 50 /* InParallel, DoCheckpoint() */
```

Definition at line 50 of file [parallel.h](#).

#### 4.96.2.11 PF\_EMPTY\_MSGTAG

```
#define PF_EMPTY_MSGTAG 52 /* InParallel, DoCheckpoint(), PF_SendFile(), PF_RecvFile() */
```

Definition at line 51 of file [parallel.h](#).

#### 4.96.2.12 PF\_STDOUT\_MSGTAG

```
#define PF_STDOUT_MSGTAG 60 /* slave -> master: sending text to the stdout */
```

Definition at line 52 of file [parallel.h](#).

#### 4.96.2.13 PF\_LOG\_MSGTAG

```
#define PF_LOG_MSGTAG 61 /* slave -> master: sending text to the log file */
```

Definition at line 53 of file [parallel.h](#).

#### 4.96.2.14 PF\_OPT\_MCTS\_MSGTAG

```
#define PF_OPT_MCTS_MSGTAG 70 /* master <-> slave: optimization */
```

Definition at line 54 of file [parallel.h](#).

#### 4.96.2.15 PF\_OPT\_HORNER\_MSGTAG

```
#define PF_OPT_HORNER_MSGTAG 71 /* master <-> slave: optimization */
```

Definition at line 55 of file [parallel.h](#).

#### 4.96.2.16 PF\_OPT\_COLLECT\_MSGTAG

```
#define PF_OPT_COLLECT_MSGTAG 72 /* slave -> master: optimization */
```

Definition at line 56 of file [parallel.h](#).

#### 4.96.2.17 PF\_MISC\_MSGTAG

```
#define PF_MISC_MSGTAG 100
```

Definition at line 57 of file [parallel.h](#).

#### 4.96.2.18 GNUC\_PREREQ

```
#define GNUC_PREREQ(  
    major,  
    minor,  
    patchlevel ) 0
```

Definition at line 67 of file [parallel.h](#).

#### 4.96.2.19 OMPI\_SKIP\_MPICXX

```
#define OMPI_SKIP_MPICXX 1
```

Definition at line 108 of file [parallel.h](#).

#### 4.96.2.20 indices

```
#define indices ((INDICES) (AC.IndexList.lijst))
```

Definition at line 113 of file [parallel.h](#).

#### 4.96.2.21 PF\_ANY\_SOURCE

```
#define PF_ANY_SOURCE MPI_ANY_SOURCE
```

Definition at line 123 of file [parallel.h](#).

#### 4.96.2.22 PF\_ANY\_MSGTAG

```
#define PF_ANY_MSGTAG MPI_ANY_TAG
```

Definition at line 124 of file [parallel.h](#).

#### 4.96.2.23 PF\_COMM

```
#define PF_COMM MPI_COMM_WORLD
```

Definition at line 125 of file [parallel.h](#).

#### 4.96.2.24 PF\_BYTE

```
#define PF_BYTE MPI_BYTE
```

Definition at line 126 of file [parallel.h](#).

#### 4.96.2.25 PF\_INT

```
#define PF_INT MPI_INT
```

Definition at line 127 of file [parallel.h](#).

#### 4.96.2.26 PF\_LongMultiPack

```
#define PF_LongMultiPack(  
    buffer,  
    count,  
    type ) PF_LongMultiPackImpl(buffer, count, sizeof_datatype(type), type)
```

Definition at line 234 of file [parallel.h](#).

#### 4.96.2.27 PF\_LongMultiUnpack

```
#define PF_LongMultiUnpack(  
    buffer,  
    count,  
    type ) PF_LongMultiUnpackImpl(buffer, count, sizeof_datatype(type), type)
```

Definition at line 235 of file [parallel.h](#).

### 4.96.3 Function Documentation

#### 4.96.3.1 PF\_IsendSbuf()

```
int PF_IsendSbuf (  
    int to,  
    int tag ) [extern]
```

Nonblocking send operation. It sends everything from *buff* to *fill* of the active buffer. Depending on *tag* it also can do waiting for other sends to finish or set the active buffer to the next one.

##### Parameters

|            |                                 |
|------------|---------------------------------|
| <i>to</i>  | the destination process number. |
| <i>tag</i> | the message tag.                |

##### Returns

0 if OK, nonzero on error.

Definition at line 261 of file [mpi.c](#).

Referenced by [FlushOut\(\)](#), [PF\\_Processor\(\)](#), and [PutOut\(\)](#).

### 4.96.3.2 PF\_Bcast()

```
int PF_Bcast (
    void * buffer,
    int count ) [extern]
```

Broadcasts a message from the master to slaves.

#### Parameters

|                |               |                                                                                                                       |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------|
| <i>in, out</i> | <i>buffer</i> | the starting address of buffer. The contents in this buffer on the master will be transferred to those on the slaves. |
|                | <i>count</i>  | the length of the buffer in bytes.                                                                                    |

#### Returns

0 if OK, nonzero on error.

Definition at line 440 of file [mpi.c](#).

Referenced by [PF\\_BroadcastBuffer\(\)](#), and [PF\\_BroadcastNumber\(\)](#).

### 4.96.3.3 PF\_RawSend()

```
int PF_RawSend (
    int dest,
    void * buf,
    LONG l,
    int tag ) [extern]
```

Sends *l* bytes from *buf* to *dest*. Returns 0 on success, or -1.

#### Parameters

|             |                                               |
|-------------|-----------------------------------------------|
| <i>dest</i> | the destination process number.               |
| <i>buf</i>  | the send buffer.                              |
| <i>l</i>    | the size of data in the send buffer in bytes. |
| <i>tag</i>  | the message tag.                              |

#### Returns

0 if OK, nonzero on error.

Definition at line 463 of file [mpi.c](#).

### 4.96.3.4 PF\_RawRecv()

```
LONG PF_RawRecv (
    int * src,
```



```
void * buf,
LONG thesize,
int * tag ) [extern]
```

Receives not more than *thesize* bytes from *src*, returns the actual number of received bytes, or -1 on failure.

#### Parameters

|         |                |                                                                            |
|---------|----------------|----------------------------------------------------------------------------|
| in, out | <i>src</i>     | the source process number. In output, that of the actual received message. |
| out     | <i>buf</i>     | the receive buffer.                                                        |
|         | <i>thesize</i> | the size of the receive buffer in bytes.                                   |
| out     | <i>tag</i>     | the message tag of the actual received message.                            |

#### Returns

the actual sizeof received data in bytes, or -1 on failure.

Definition at line 484 of file [mpi.c](#).

#### 4.96.3.5 PF\_PreparePack()

```
int PF_PreparePack (
    void ) [extern]
```

Prepares for the next pack operations on the sender.

#### Returns

0 if OK, nonzero on error.

Definition at line 624 of file [mpi.c](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastPreDollar\(\)](#), [PF\\_BroadcastString\(\)](#), and [PF\\_Init\(\)](#).

#### 4.96.3.6 PF\_Pack()

```
int PF_Pack (
    const void * buffer,
    size_t count,
    MPI_Datatype type ) [extern]
```

Adds data into the pack buffer.

#### Parameters

|               |                                                          |
|---------------|----------------------------------------------------------|
| <i>buffer</i> | the pointer to the buffer storing the data to be packed. |
| <i>count</i>  | the number of elements in the buffer.                    |
| <i>type</i>   | the data type of elements in the buffer.                 |

**Returns**

0 if OK, nonzero on error.

Definition at line 642 of file [mpi.c](#).

References [MPI\\_ERRCODE\\_CHECK](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastPreDollar\(\)](#), and [PF\\_Init\(\)](#).

**4.96.3.7 PF\_Unpack()**

```
int PF_Unpack (
    void * buffer,
    size_t count,
    MPI_Datatype type ) [extern]
```

Retrieves the next data in the pack buffer.

**Parameters**

|     |               |                                                       |
|-----|---------------|-------------------------------------------------------|
| out | <i>buffer</i> | the pointer to the buffer to store the unpacked data. |
|     | <i>count</i>  | the number of elements of data to be received.        |
|     | <i>type</i>   | the data type of elements of data to be received.     |

**Returns**

0 if OK, nonzero on error.

Definition at line 671 of file [mpi.c](#).

References [MPI\\_ERRCODE\\_CHECK](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastPreDollar\(\)](#), and [PF\\_Init\(\)](#).

**4.96.3.8 PF\_PackString()**

```
int PF_PackString (
    const UBYTE * str ) [extern]
```

Packs a string *str* into the packed buffer `PF_packbuf`, including the trailing zero.

The first element (`PF_INT`) is the length of the packed portion of the string. If the string does not fit to the buffer `PF_packbuf`, the function packs only the initial portion. It returns the number of packed bytes, so if `(str[length-1]=='\0')` then the whole string fits to the buffer, if not, then the rest (`str+length`) must be packed and send again. On error, the function returns the negative error code.

One exception: the string `"\0\0"` is used as an image of the NULL, so all 3 characters will be packed.

**Parameters**

|            |                        |
|------------|------------------------|
| <i>str</i> | a string to be packed. |
|------------|------------------------|

#### Returns

the number of packed bytes, or a negative value on failure.

Definition at line 706 of file [mpi.c](#).

Referenced by [PF\\_BroadcastString\(\)](#).

#### 4.96.3.9 PF\_UnpackString()

```
int PF_UnpackString (
    UBYTE * str ) [extern]
```

Unpacks a string to *str* from the packed buffer PF\_packbuf, including the trailing zero.

It returns the number of unpacked bytes, so if (str[length-1]=='\0') then the whole string was unpacked, if not, then the rest must be appended to (str+length). On error, the function returns the negative error code.

#### Parameters

|     |            |                                         |
|-----|------------|-----------------------------------------|
| out | <i>str</i> | the buffer to store the unpacked string |
|-----|------------|-----------------------------------------|

#### Returns

the number of unpacked bytes, or a negative value on failure.

Definition at line 774 of file [mpi.c](#).

Referenced by [PF\\_BroadcastString\(\)](#).

#### 4.96.3.10 PF\_Send()

```
int PF_Send (
    int to,
    int tag ) [extern]
```

Sends the contents in the pack buffer to the process specified by *to*.

#### Example:

```
if ( PF.me == SRC ) {
    PF_PreparePack();
    // Packing operations here...
    PF_Send(DEST, TAG);
}
else if ( PF.me == DEST ) {
    PF_Receive(SRC, TAG, &actual_src, &actual_tag);
    // Unpacking operations here...
}
```

#### Parameters

|            |                                 |
|------------|---------------------------------|
| <i>to</i>  | the destination process number. |
| <i>tag</i> | the message tag.                |

**Returns**

0 if OK, nonzero on error.

Definition at line 822 of file [mpi.c](#).

References [MPI\\_ERRCODE\\_CHECK](#).

**4.96.3.11 PF\_Receive()**

```
int PF_Receive (
    int src,
    int tag,
    int * psrc,
    int * ptag ) [extern]
```

Receives data into the pack buffer from the process specified by *src*. This function allows &*src* == *psrc* or &*tag* == *ptag*. Either *psrc* or *ptag* can be NULL.

See the example of [PF\\_Send\(\)](#).

**Parameters**

|     |             |                                                       |
|-----|-------------|-------------------------------------------------------|
|     | <i>src</i>  | the source process number (can be PF_ANY_SOURCE).     |
|     | <i>tag</i>  | the source message tag (can be PF_ANY_TAG).           |
| out | <i>psrc</i> | the actual source process number of received message. |
| out | <i>ptag</i> | the received message tag.                             |

**Returns**

0 if OK, nonzero on error.

Definition at line 848 of file [mpi.c](#).

References [MPI\\_ERRCODE\\_CHECK](#).

**4.96.3.12 PF\_Broadcast()**

```
int PF_Broadcast (
    void ) [extern]
```

Broadcasts the contents in the pack buffer on the master to those on the slaves.

**Example:**

```
if ( PF.me == MASTER ) {
    PF_PreparePack();
    // Packing operations here...
}
PF_Broadcast();
if ( PF.me != MASTER ) {
    // Unpacking operations here...
}
```

**Returns**

0 if OK, nonzero on error.

Definition at line 883 of file [mpi.c](#).

References [MPI\\_ERRCODE\\_CHECK](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastPreDollar\(\)](#), [PF\\_BroadcastString\(\)](#), and [PF\\_Init\(\)](#).

**4.96.3.13 PF\_PrepareLongSinglePack()**

```
int PF_PrepareLongSinglePack (
    void ) [extern]
```

Prepares for the next long-single-pack operations on the sender.

**Returns**

0 if OK, nonzero on error.

Definition at line 1451 of file [mpi.c](#).

Referenced by [PF\\_CollectModifiedDollars\(\)](#), and [PF\\_Processor\(\)](#).

**4.96.3.14 PF\_LongSinglePack()**

```
int PF_LongSinglePack (
    const void * buffer,
    size_t count,
    MPI_Datatype type ) [extern]
```

Adds data into the "long single" pack buffer.

**Parameters**

|               |                                                          |
|---------------|----------------------------------------------------------|
| <i>buffer</i> | the pointer to the buffer storing the data to be packed. |
| <i>count</i>  | the number of elements in the buffer.                    |
| <i>type</i>   | the data type of elements in the buffer.                 |

**Returns**

0 if OK, nonzero on error.

Definition at line 1469 of file [mpi.c](#).

Referenced by [PF\\_CollectModifiedDollars\(\)](#), and [PF\\_Processor\(\)](#).

**4.96.3.15 PF\_LongSingleUnpack()**

```
int PF_LongSingleUnpack (
    void * buffer,
    size_t count,
    MPI_Datatype type ) [extern]
```

Retrieves the next data in the "long single" pack buffer.

**Parameters**

|     |               |                                                       |
|-----|---------------|-------------------------------------------------------|
| out | <i>buffer</i> | the pointer to the buffer to store the unpacked data. |
|     | <i>count</i>  | the number of elements of data to be received.        |
|     | <i>type</i>   | the data type of elements of data to be received.     |

**Returns**

0 if OK, nonzero on error.

Definition at line 1503 of file [mpi.c](#).

Referenced by [PF\\_CollectModifiedDollars\(\)](#), and [PF\\_Processor\(\)](#).

**4.96.3.16 PF\_LongSingleSend()**

```
int PF_LongSingleSend (
    int to,
    int tag ) [extern]
```

Sends the contents in the "long single" pack buffer to the process specified by *to*.

**Example:**

```
if ( PF.me == SRC ) {
    PF_PrepareLongSinglePack();
    // Packing operations here...
    PF_LongSingleSend(DEST, TAG);
}
else if ( PF.me == DEST ) {
    PF_LongSingleReceive(SRC, TAG, &actual_src, &actual_tag);
    // Unpacking operations here...
}
```

**Parameters**

|            |                                 |
|------------|---------------------------------|
| <i>to</i>  | the destination process number. |
| <i>tag</i> | the message tag.                |

**Returns**

0 if OK, nonzero on error.

Definition at line 1540 of file [mpi.c](#).

Referenced by [PF\\_CollectModifiedDollars\(\)](#), and [PF\\_Processor\(\)](#).

**4.96.3.17 PF\_LongSingleReceive()**

```
int PF_LongSingleReceive (
    int src,
    int tag,
    int * psrc,
    int * ptag ) [extern]
```

Receives data into the "long single" pack buffer from the process specified by *src*. This function allows *&src == psrc* or *&tag == ptag*. Either *psrc* or *ptag* can be NULL.

See the example of [PF\\_LongSingleSend\(\)](#).

## Parameters

|     |             |                                                       |
|-----|-------------|-------------------------------------------------------|
|     | <i>src</i>  | the source process number (can be PF_ANY_SOURCE).     |
|     | <i>tag</i>  | the source message tag (can be PF_ANY_TAG).           |
| out | <i>psrc</i> | the actual source process number of received message. |
| out | <i>ptag</i> | the received message tag.                             |

## Returns

0 if OK, nonzero on error.

Definition at line 1583 of file [mpi.c](#).

Referenced by [PF\\_CollectModifiedDollars\(\)](#), and [PF\\_Processor\(\)](#).

**4.96.3.18 PF\_PrepareLongMultiPack()**

```
int PF_PrepareLongMultiPack (
    void ) [extern]
```

Prepares for the next long-multi-pack operations on the sender.

## Returns

0 if OK, nonzero on error.

Definition at line 1643 of file [mpi.c](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastCBuf\(\)](#), [PF\\_BroadcastExpFlags\(\)](#), [PF\\_BroadcastModifiedDollars\(\)](#), and [PF\\_BroadcastRedefinedPreVars\(\)](#).

**4.96.3.19 PF\_LongMultiPackImpl()**

```
int PF_LongMultiPackImpl (
    const void * buffer,
    size_t count,
    size_t eSize,
    MPI_Datatype type ) [extern]
```

Adds data into the "long multi" pack buffer.

## Parameters

|               |                                                          |
|---------------|----------------------------------------------------------|
| <i>buffer</i> | the pointer to the buffer storing the data to be packed. |
| <i>count</i>  | the number of elements in the buffer.                    |
| <i>eSize</i>  | the byte size of each element of data.                   |
| <i>type</i>   | the data type of elements in the buffer.                 |

**Returns**

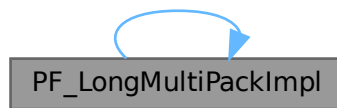
0 if OK, nonzero on error.

Definition at line 1662 of file [mpi.c](#).

References [PF\\_LongMultiPackImpl\(\)](#).

Referenced by [PF\\_LongMultiPackImpl\(\)](#).

Here is the call graph for this function:

**4.96.3.20 PF\_LongMultiUnpackImpl()**

```

int PF_LongMultiUnpackImpl (
    void * buffer,
    size_t count,
    size_t eSize,
    MPI_Datatype type ) [extern]
  
```

Retrieves the next data in the "long multi" pack buffer.

**Parameters**

|     |               |                                                       |
|-----|---------------|-------------------------------------------------------|
| out | <i>buffer</i> | the pointer to the buffer to store the unpacked data. |
|     | <i>count</i>  | the number of elements of data to be received.        |
|     | <i>eSize</i>  | the byte size of each element of data.                |
|     | <i>type</i>   | the data type of elements of data to be received.     |

**Returns**

0 if OK, nonzero on error.

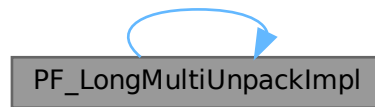
Definition at line 1721 of file [mpi.c](#).

References [PF\\_LongMultiUnpackImpl\(\)](#).

Referenced by [PF\\_LongMultiUnpackImpl\(\)](#).



Here is the call graph for this function:



#### 4.96.3.21 PF\_LongMultiBroadcast()

```
int PF_LongMultiBroadcast (
    void ) [extern]
```

Broadcasts the contents in the "long multi" pack buffer on the master to those on the slaves.

Example:

```
if ( PF.me == MASTER ) {
    PF_PrepareLongMultiPack();
    // Packing operations here...
}
PF_LongMultiBroadcast();
if ( PF.me != MASTER ) {
    // Unpacking operations here...
}
```

##### Returns

0 if OK, nonzero on error.

Definition at line 1807 of file [mpi.c](#).

Referenced by [Optimize\(\)](#), [PF\\_BroadcastCBuf\(\)](#), [PF\\_BroadcastExpFlags\(\)](#), [PF\\_BroadcastModifiedDollars\(\)](#), and [PF\\_BroadcastRedefinedPreVars\(\)](#).

#### 4.96.3.22 PF\_EndSort()

```
int PF_EndSort (
    void ) [extern]
```

Finishes a master sorting with collecting terms from slaves. Called by [EndSort\(\)](#).

If this is not the masterprocess, just initialize the sendbuffers and return 0, else [PF\\_EndSort\(\)](#) sends the rest of the terms in the sendbuffer to the next slave and a dummy message to all slaves with tag PF\_ENDSORT\_MSGTAG. Then it receives the sorted terms, sorts them using a recursive 'tree of losers' ([PF\\_GetLoser\(\)](#)) and writes them to the outputfile.

##### Returns

1 if the sorting on the master was done. 0 if [EndSort\(\)](#) still must perform a regular sorting because it is not at the ground level or not on the master or in the sequential mode or in the InParallel mode. -1 if an error occurred.

**Remarks**

The slaves will send the sorted terms back to the master in the regular sorting (after the initialization of the send buffer in [PF\\_EndSort\(\)](#)). See [PutOut\(\)](#) and [FlushOut\(\)](#).

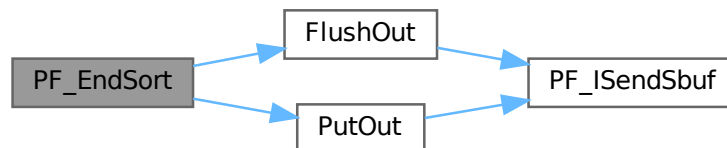
This function has been changed such that when it returns 1, AM.S0->TermsLeft is set correctly. But AM.S0->GenTerms is not set: it will be set after collecting the statistics from the slaves at the end of [PF\\_Processor\(\)](#). (TU 30 Jun 2011)

Definition at line 864 of file [parallel.c](#).

References [FlushOut\(\)](#), and [PutOut\(\)](#).

Referenced by [EndSort\(\)](#).

Here is the call graph for this function:

**4.96.3.23 PF\_Deferred()**

```
WORD PF_Deferred (
    WORD * term,
    WORD level ) [extern]
```

Replaces [Deferred\(\)](#) on the slaves.

**Parameters**

|              |                                                                          |
|--------------|--------------------------------------------------------------------------|
| <i>term</i>  | the term that must be multiplied by the contents of the current bracket. |
| <i>level</i> | the compiler level.                                                      |

**Returns**

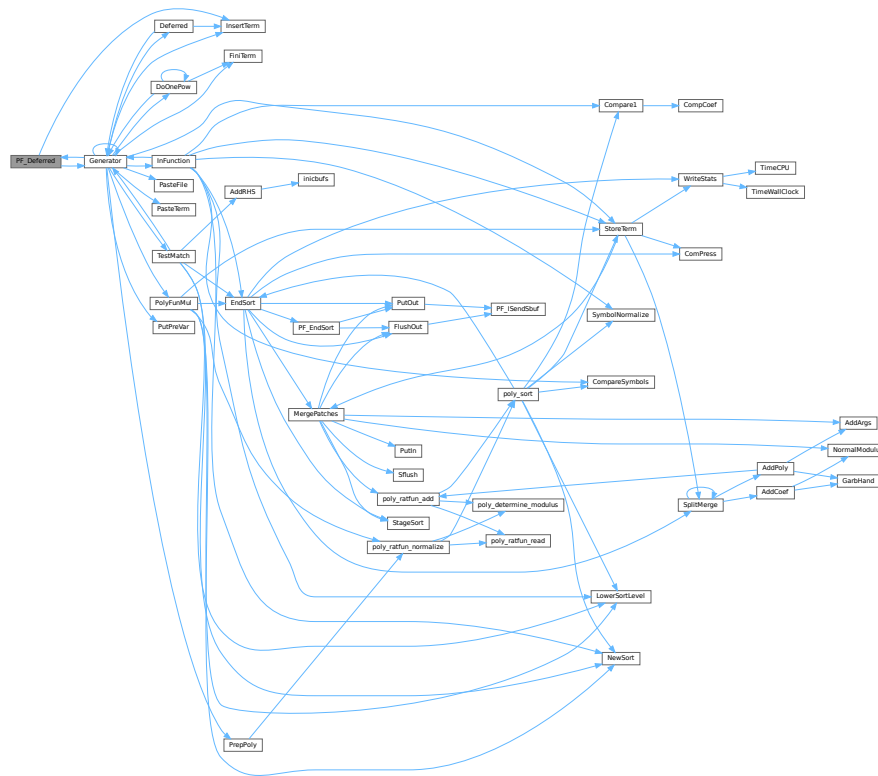
0 if OK, nonzero on error.

Definition at line 1208 of file [parallel.c](#).

References [Generator\(\)](#), and [InsertTerm\(\)](#).

Referenced by [Generator\(\)](#).

Here is the call graph for this function:



#### 4.96.3.24 PF\_Processor()

```
int PF_Processor (
    EXPRESSIONS e,
    WORD i,
    WORD LastExpression ) [extern]
```

Replaces parts of [Processor\(\)](#) on the masters and slaves. On the master [PF\\_Processor\(\)](#) is responsible for proper distribution of terms from the input file to the slaves. On the slaves it calls [Generator\(\)](#) for all the terms that this process gets, but [PF\\_GetTerm\(\)](#) gets terms from the master (not directly from infile).

##### Parameters

|                       |                                                        |
|-----------------------|--------------------------------------------------------|
| <i>e</i>              | The pointer to the current expression.                 |
| <i>i</i>              | The index for the current expression.                  |
| <i>LastExpression</i> | The flag indicating whether it is the last expression. |

##### Returns

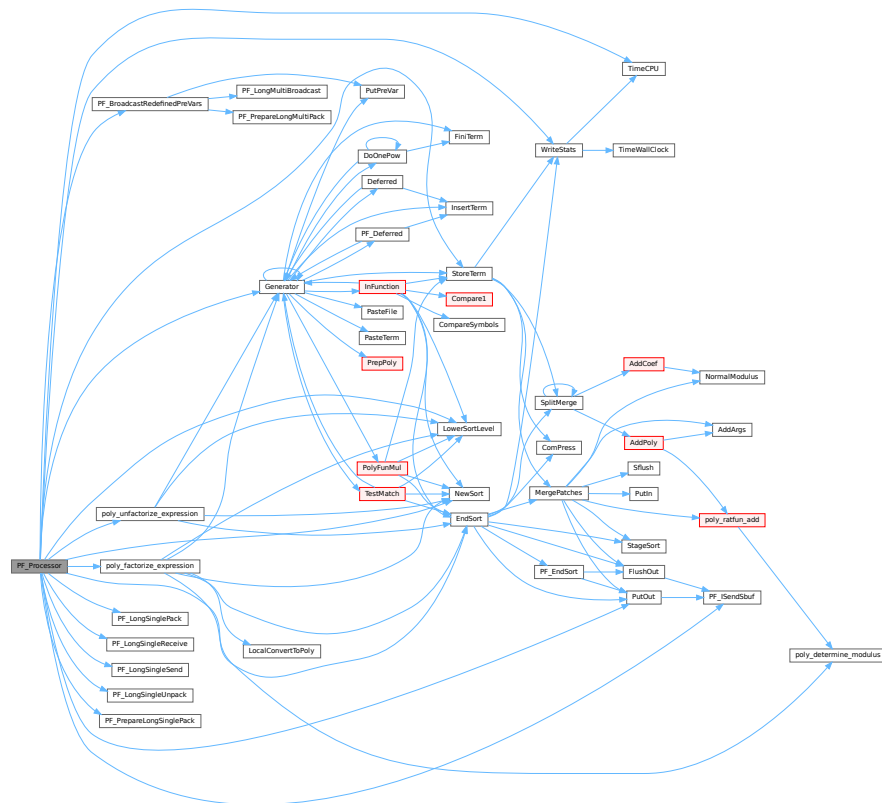
0 if OK, nonzero on error.

Definition at line 1540 of file [parallel.c](#).

References [EndSort\(\)](#), [Generator\(\)](#), [LowerSortLevel\(\)](#), [NewSort\(\)](#), [PACK\\_LONG](#), [PF\\_BroadcastRedefinePreVars\(\)](#), [PF\\_ISendSbuf\(\)](#), [PF\\_LongSinglePack\(\)](#), [PF\\_LongSingleReceive\(\)](#), [PF\\_LongSingleSend\(\)](#), [PF\\_LongSingleUnpack\(\)](#), [PF\\_PrepareLongSinglePack\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [PutOut\(\)](#), [StoreTerm\(\)](#), [SWAP](#), [TimeCPU\(\)](#), and [WriteStats\(\)](#).

Referenced by [Processor\(\)](#).

Here is the call graph for this function:



#### 4.96.3.25 PF\_Init()

```
int PF_Init (
    int * argc,
    char *** argv ) [extern]
```

All the library independent stuff. [PF\\_LibInit\(\)](#) should do all library dependent initializations.

##### Parameters

|             |                                     |
|-------------|-------------------------------------|
| <i>argc</i> | pointer to the number of arguments. |
| <i>argv</i> | pointer to the arguments.           |

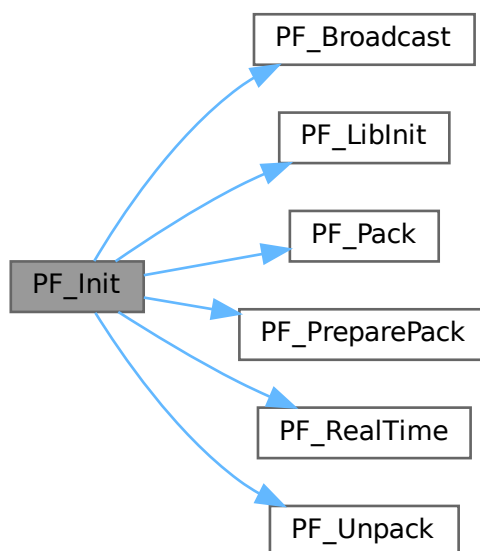
##### Returns

0 if OK, nonzero on error.

Definition at line 1963 of file [parallel.c](#).

References [PF\\_Broadcast\(\)](#), [PF\\_LibInit\(\)](#), [PF\\_Pack\(\)](#), [PF\\_PreparePack\(\)](#), [PF\\_RealTime\(\)](#), and [PF\\_Unpack\(\)](#).

Here is the call graph for this function:



#### 4.96.3.26 PF\_Terminate()

```
int PF_Terminate (
    int errorcode ) [extern]
```

Performs the finalization of ParFORM. To be called by `Terminate()`.

##### Parameters

|              |                |
|--------------|----------------|
| <i>error</i> | an error code. |
|--------------|----------------|

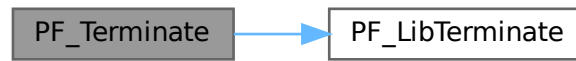
##### Returns

0 if OK, nonzero on error.

Definition at line 2058 of file [parallel.c](#).

References [PF\\_LibTerminate\(\)](#).

Here is the call graph for this function:



#### 4.96.3.27 PF\_GetSlaveTimes()

```

LONG PF_GetSlaveTimes (
    void ) [extern]
  
```

Returns the total CPU time of all slaves together. This function must be called on the master and all slaves.

##### Returns

on the master, the sum of CPU times on all slaves.

Definition at line [2074](#) of file [parallel.c](#).

References [TimeCPU\(\)](#).

Here is the call graph for this function:



#### 4.96.3.28 PF\_BroadcastNumber()

```

LONG PF_BroadcastNumber (
    LONG x ) [extern]
  
```

Broadcasts a LONG value from the master to the all slaves.

##### Parameters

|   |                                                 |
|---|-------------------------------------------------|
| x | the number to be broadcast (set on the master). |
|---|-------------------------------------------------|

**Returns**

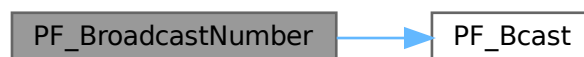
the synchronised result.

Definition at line 2094 of file [parallel.c](#).

References [PF\\_Bcast\(\)](#).

Referenced by [DoCheckpoint\(\)](#), [PF\\_BroadcastBuffer\(\)](#), and [StartVariables\(\)](#).

Here is the call graph for this function:

**4.96.3.29 PF\_BroadcastBuffer()**

```

void PF_BroadcastBuffer (
    WORD ** buffer,
    LONG * length ) [extern]
  
```

Broadcasts a buffer from the master to all the slaves.

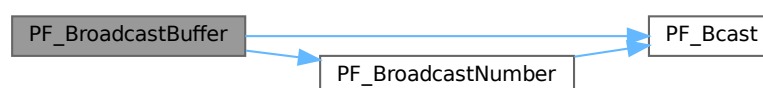
**Parameters**

|         |               |                                                                                                                                                                                        |
|---------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| in, out | <i>buffer</i> | on the master, the buffer to be broadcast. On the slaves, the buffer will be allocated if the length is greater than 0. The caller must free it.                                       |
| in, out | <i>length</i> | on the master, the length of the buffer to be broadcast. On the slaves, it receives the length of transferred buffer. The actual transfer occurs only if the length is greater than 0. |

Definition at line 2121 of file [parallel.c](#).

References [PF\\_Bcast\(\)](#), and [PF\\_BroadcastNumber\(\)](#).

Here is the call graph for this function:



#### 4.96.3.30 PF\_BroadcastString()

```
int PF_BroadcastString (
    UBYTE * str ) [extern]
```

Broadcasts a string from the master to all slaves.

##### Parameters

|                |            |                                          |
|----------------|------------|------------------------------------------|
| <i>in, out</i> | <i>str</i> | The pointer to a null-terminated string. |
|----------------|------------|------------------------------------------|

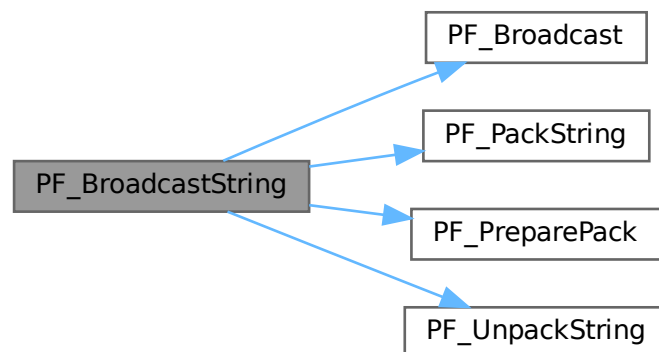
##### Returns

0 if OK, nonzero on error.

Definition at line 2163 of file [parallel.c](#).

References [PF\\_Broadcast\(\)](#), [PF\\_PackString\(\)](#), [PF\\_PreparePack\(\)](#), and [PF\\_UnpackString\(\)](#).

Here is the call graph for this function:



#### 4.96.3.31 PF\_BroadcastPreDollar()

```
int PF_BroadcastPreDollar (
    WORD ** dbuffer,
    LONG * newsize,
    int * numterms ) [extern]
```

Broadcasts dollar variables set as a preprocessor variables. Only the master is able to make an assignment like `#$a=g;` where `g` is an expression: only the master has an access to the expression. So, the master broadcasts the result to slaves.

The result is in `*dbuffer` of the size is `*newsize` (in number of WORDs), +1 for trailing zero. For slave `newsize` and `numterms` are output parameters.



## Parameters

|                |                 |                                             |
|----------------|-----------------|---------------------------------------------|
| <i>in, out</i> | <i>dbuffer</i>  | the buffer for a dollar variable.           |
| <i>in, out</i> | <i>newsize</i>  | the size of the dollar variable in WORDs.   |
| <i>in, out</i> | <i>numterms</i> | the number of terms in the dollar variable. |

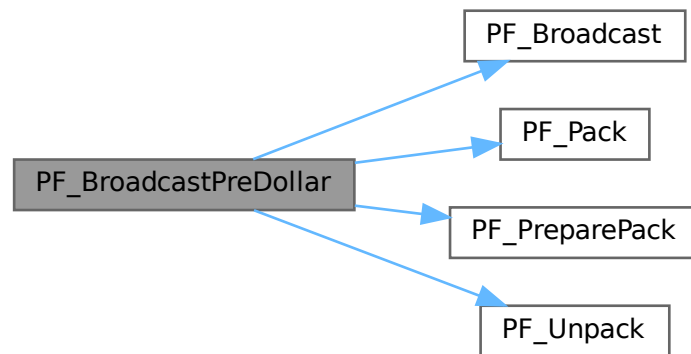
## Returns

0 if OK, nonzero on error.

Definition at line 2218 of file [parallel.c](#).

References [PF\\_Broadcast\(\)](#), [PF\\_Pack\(\)](#), [PF\\_PreparePack\(\)](#), and [PF\\_Unpack\(\)](#).

Here is the call graph for this function:



## 4.96.3.32 PF\_CollectModifiedDollars()

```
int PF_CollectModifiedDollars (
    void ) [extern]
```

Combines modified dollar variables on the all slaves, and store them into those on the master.

The potentially modified dollar variables are given in `PotModdollars`, and the number of them is given by `NumPotModdollars`.

The current module could be executed in parallel only if all potentially modified variables are listed in `ModOptdollars`, otherwise the module was switched to the sequential mode.



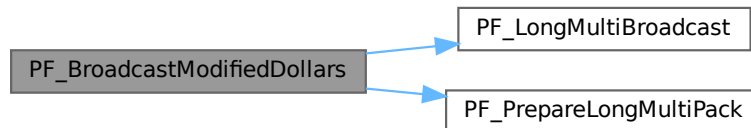
**Returns**

0 if OK, nonzero on error.

Definition at line 2785 of file [parallel.c](#).

References [PF\\_LongMultiBroadcast\(\)](#), and [PF\\_PrepareLongMultiPack\(\)](#).

Here is the call graph for this function:

**4.96.3.34 PF\_BroadcastRedefinedPreVars()**

```
int PF_BroadcastRedefinedPreVars (
    void ) [extern]
```

Broadcasts preprocessor variables, which were changed by the Redefine statements in the current module, from the master to the all slaves.

The potentially redefined preprocessor variables are given in `AC.pfirstnum`, and the number of them is given by `AC.numpfirstnum`. For an actually redefined variable, the corresponding value in `AC.inputnumbers` is non-negative.

**Returns**

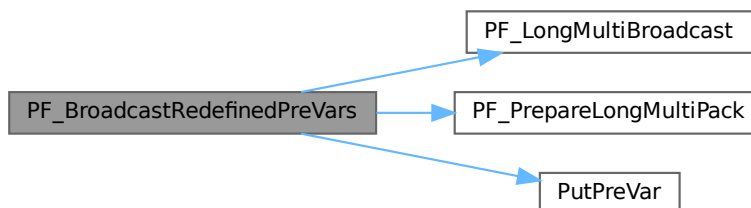
0 if OK, nonzero on error.

Definition at line 3002 of file [parallel.c](#).

References [PF\\_LongMultiBroadcast\(\)](#), [PF\\_PrepareLongMultiPack\(\)](#), [PutPreVar\(\)](#), [VectorPtr](#), and [VectorReserve](#).

Referenced by [PF\\_Processor\(\)](#).

Here is the call graph for this function:



#### 4.96.3.35 PF\_BroadcastCBuf()

```
int PF_BroadcastCBuf (
    int bufnum ) [extern]
```

Broadcasts a compiler buffer specified by *bufnum* from the master to the all slaves.

##### Parameters

|               |                                                   |
|---------------|---------------------------------------------------|
| <i>bufnum</i> | The index of the compiler buffer to be broadcast. |
|---------------|---------------------------------------------------|

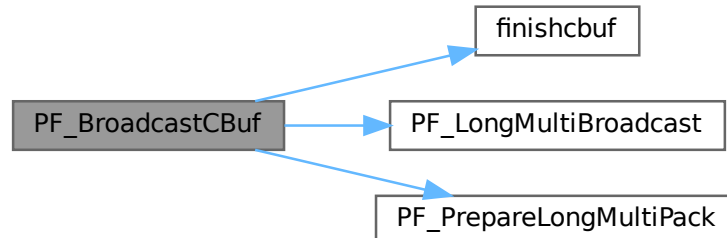
##### Returns

0 if OK, nonzero on error.

Definition at line 3144 of file [parallel.c](#).

References [CbUf::boomlijst](#), [CbUf::Buffer](#), [CbUf::BufferSize](#), [CbUf::CanCommu](#), [CbUf::dimension](#), [finishcbuf\(\)](#), [CbUf::lhs](#), [CbUf::numdum](#), [CbUf::NumTerms](#), [PF\\_LongMultiBroadcast\(\)](#), [PF\\_PrepareLongMultiPack\(\)](#), [CbUf::Pointer](#), [CbUf::rhs](#), and [CbUf::Top](#).

Here is the call graph for this function:



#### 4.96.3.36 PF\_BroadcastExpFlags()

```
int PF_BroadcastExpFlags (
    void ) [extern]
```

Broadcasts `AR.expflags` and several properties of each expression, e.g., `e->vflags`, from the master to all slaves.

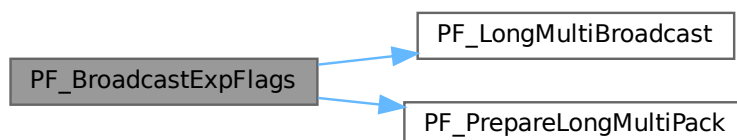
**Returns**

0 if OK, nonzero on error.

Definition at line 3255 of file [parallel.c](#).

References [PF\\_LongMultiBroadcast\(\)](#), and [PF\\_PrepareLongMultiPack\(\)](#).

Here is the call graph for this function:

**4.96.3.37 PF\_StoreInsideInfo()**

```
int PF_StoreInsideInfo (
    void ) [extern]
```

Definition at line 3092 of file [parallel.c](#).

**4.96.3.38 PF\_RestoreInsideInfo()**

```
int PF_RestoreInsideInfo (
    void ) [extern]
```

Definition at line 3118 of file [parallel.c](#).

**4.96.3.39 PF\_BroadcastExpr()**

```
int PF_BroadcastExpr (
    EXPRESSIONS e,
    FILEHANDLE * file ) [extern]
```

Broadcasts an expression from the master to the all slaves.

**Parameters**

|             |                                              |
|-------------|----------------------------------------------|
| <i>e</i>    | The expression to be broadcast.              |
| <i>file</i> | The file in which the expression is sitting. |

**Returns**

0 if OK, nonzero on error.

Definition at line [3549](#) of file [parallel.c](#).

Referenced by [PF\\_BroadcastRHS\(\)](#).

**4.96.3.40 PF\_BroadcastRHS()**

```
int PF_BroadcastRHS (
    void ) [extern]
```

Broadcasts expressions appearing in the right-hand side from the master to the all slaves.

**Returns**

0 if OK, nonzero on error.

Definition at line [3577](#) of file [parallel.c](#).

References [PF\\_BroadcastExpr\(\)](#).

Referenced by [Processor\(\)](#).

Here is the call graph for this function:

**4.96.3.41 PF\_InParallelProcessor()**

```
int PF_InParallelProcessor (
    void ) [extern]
```

Processes expressions in the InParallel mode, i.e., dividing expressions marked by `partodo` over the slaves.

**Returns**

0 if OK, nonzero on error.

Definition at line [3624](#) of file [parallel.c](#).

Referenced by [Processor\(\)](#).

**4.96.3.42 PF\_SendFile()**

```
int PF_SendFile (
    int to,
    FILE * fd ) [extern]
```

Sends a file to the process specified by `to`.

## Parameters

|           |                                 |
|-----------|---------------------------------|
| <i>to</i> | the destination process number. |
| <i>fd</i> | the file to be sent.            |

## Returns

the size of sent data in bytes, or -1 on error.

Definition at line 4221 of file [parallel.c](#).

References [PF\\_RawSend\(\)](#).

Referenced by [DoCheckpoint\(\)](#).

Here is the call graph for this function:



#### 4.96.3.43 PF\_RecvFile()

```
int PF_RecvFile (
    int from,
    FILE * fd ) [extern]
```

Receives a file from the process specified by *from*.

## Parameters

|             |                                     |
|-------------|-------------------------------------|
| <i>from</i> | the source process number.          |
| <i>fd</i>   | the file to save the received data. |

## Returns

the size of received data in bytes, or -1 on error.

Definition at line 4259 of file [parallel.c](#).

References [PF\\_RawRecv\(\)](#).

Referenced by [DoCheckpoint\(\)](#).

Here is the call graph for this function:



#### 4.96.3.44 PF\_MLock()

```
void PF_MLock (
    void ) [extern]
```

A function called by MLOCK(ErrorMessageLock) for slaves.

Definition at line 4340 of file [parallel.c](#).

References [VectorClear](#).

#### 4.96.3.45 PF\_MUnlock()

```
void PF_MUnlock (
    void ) [extern]
```

A function called by MUNLOCK(ErrorMessageLock) for slaves.

Definition at line 4356 of file [parallel.c](#).

References [PF\\_RawSend\(\)](#), [VectorEmpty](#), [VectorPtr](#), and [VectorSize](#).

Here is the call graph for this function:



#### 4.96.3.46 PF\_WriteFileToFile()

```
LONG PF_WriteFileToFile (
    int handle,
    UBYTE * buffer,
    LONG size ) [extern]
```

Replaces WriteFileToFile() on the master and slaves.

It copies the given buffer into internal buffers if called between MLOCK(ErrorMessageLock) and MUNLOCK(ErrorMessageLock) for slaves and handle is StdOut or LogHandle, otherwise calls WriteFileToFile().



## Parameters

|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| <i>handle</i> | a file handle that specifies the output.                          |
| <i>buffer</i> | a pointer to the source buffer containing the data to be written. |
| <i>size</i>   | the size of data to be written in bytes.                          |

## Returns

the actual size of data written to the output in bytes.

Definition at line 4385 of file [parallel.c](#).

References [VectorClear](#), [VectorPtr](#), [VectorPushBacks](#), and [VectorSize](#).

#### 4.96.3.47 PF\_FlushStdOutBuffer()

```
void PF_FlushStdOutBuffer (
    void ) [extern]
```

Explicitly Flushes the buffer for the standard output on the master, which is used if PF\_ENABLE\_STDOUT\_↵  
BUFFERING is defined.

Definition at line 4479 of file [parallel.c](#).

References [VectorClear](#), [VectorPtr](#), and [VectorSize](#).

### 4.96.4 Variable Documentation

#### 4.96.4.1 PF

```
PARALLELVARS PF [extern]
```

Definition at line 90 of file [parallel.c](#).

#### 4.96.4.2 PF\_maxDollarChunkSize

```
LONG PF_maxDollarChunkSize [extern]
```

Definition at line 69 of file [mpi.c](#).

## 4.97 parallel.h

[Go to the documentation of this file.](#)

```

00001 #ifndef __PARALLEL__
00002 #define __PARALLEL__
00003
00009 /* #[ License : */
00010 /*
00011 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00012 * When using this file you are requested to refer to the publication
00013 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00014 * This is considered a matter of courtesy as the development was paid
00015 * for by FOM the Dutch physics granting agency and we would like to
00016 * be able to track its scientific use to convince FOM of its value
00017 * for the community.
00018 *
00019 * This file is part of FORM.
00020 *
00021 * FORM is free software: you can redistribute it and/or modify it under the
00022 * terms of the GNU General Public License as published by the Free Software
00023 * Foundation, either version 3 of the License, or (at your option) any later
00024 * version.
00025 *
00026 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00027 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00028 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00029 * details.
00030 *
00031 * You should have received a copy of the GNU General Public License along
00032 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00033 */
00034 /* #[ License : */
00035
00036 /*
00037      #[ macros & definitions :
00038 */
00039 #define MASTER 0
00040
00041 #define PF_RESET 0
00042 #define PF_TIME 1
00043
00044 #define PF_TERM_MSGTAG 10 /* master -> slave: sending terms */
00045 #define PF_ENDSORT_MSGTAG 11 /* master -> slave: no more terms to be distributed, slave ->
master: after EndSort() */
00046 #define PF_DOLLAR_MSGTAG 12 /* slave -> master: sending $-variables */
00047 #define PF_BUFFER_MSGTAG 20 /* slave -> master: sending sorted terms, or in
PF_SendFile()/PF_RecvFile() */
00048 #define PF_ENDBUFFER_MSGTAG 21 /* same as PF_BUFFER_MSGTAG, but indicates the end of operation */
00049 #define PF_READY_MSGTAG 30 /* slave -> master: slave is idle and can accept terms */
00050 #define PF_DATA_MSGTAG 50 /* InParallel, DoCheckpoint() */
00051 #define PF_EMPTY_MSGTAG 52 /* InParallel, DoCheckpoint(), PF_SendFile(), PF_RecvFile() */
00052 #define PF_STDOUT_MSGTAG 60 /* slave -> master: sending text to the stdout */
00053 #define PF_LOG_MSGTAG 61 /* slave -> master: sending text to the log file */
00054 #define PF_OPT_MCTS_MSGTAG 70 /* master <-> slave: optimization */
00055 #define PF_OPT_HORNER_MSGTAG 71 /* master <-> slave: optimization */
00056 #define PF_OPT_COLLECT_MSGTAG 72 /* slave -> master: optimization */
00057 #define PF_MISC_MSGTAG 100
00058
00059 /*
00060 * A macro for checking the version of gcc.
00061 */
00062 #if defined(__GNUC__) && defined(__GNUC_MINOR__) && defined(__GNUC_PATCHLEVEL__)
00063 # define GNUC_PREREQ(major, minor, patchlevel) \
00064     ((__GNUC__ < 16) + ((__GNUC_MINOR__ < 8) + __GNUC_PATCHLEVEL__ >= \
00065     ((major) < 16) + ((minor) < 8) + (patchlevel)))
00066 #else
00067 # define GNUC_PREREQ(major, minor, patchlevel) 0
00068 #endif
00069
00070 /*
00071 * The macro "indices" defined in variable.h collides with some function
00072 * argument names in the MPI-3.0 standard.
00073 */
00074 #undef indices
00075
00076 /* Avoid messy padding warnings which may appear in mpi.h. */
00077 #if GNUC_PREREQ(4, 6, 0)
00078 # pragma GCC diagnostic push
00079 # pragma GCC diagnostic ignored "-Wpadded"
00080 # pragma GCC diagnostic ignored "-Wunused-parameter"
00081 #endif
00082 #if defined(__clang__) && defined(__has_warning)
00083 # pragma clang diagnostic push
00084 # if __has_warning("-Wpadded")
00085 # pragma clang diagnostic ignored "-Wpadded"

```

```

00086 # endif
00087 # if __has_warning("-Wunused-parameter")
00088 # pragma clang diagnostic ignored "-Wunused-parameter"
00089 # endif
00090 #endif
00091
00092 # ifdef __cplusplus
00093 /*
00094  * form3.h (which includes parallel.h) is included from newpoly.h as
00095  * extern "C" {
00096  * #include "form3.h"
00097  * }
00098  * On the other hand, C++ interfaces to MPI are defined in mpi.h if it is
00099  * included from C++ sources. We first leave from the C-linkage, include
00100  * mpi.h, and then go back to the C-linkage.
00101  * (TU 7 Jun 2011)
00102  */
00103 }
00104 # define OMPI_SKIP_MPICXX 1
00105 # include <mpi.h>
00106 extern "C" {
00107 # else
00108 # define OMPI_SKIP_MPICXX 1
00109 # include <mpi.h>
00110 # endif
00111
00112 /* Now redefine "indices" in the same way as in variable.h. */
00113 #define indices ((INDICES) (AC.IndexList.lijst))
00114
00115 /* Restore the warning settings. */
00116 #if GNUC_PREREQ(4, 6, 0)
00117 # pragma GCC diagnostic pop
00118 #endif
00119 #if defined(__clang__) && defined(__has_warning)
00120 # pragma clang diagnostic pop
00121 #endif
00122
00123 # define PF_ANY_SOURCE MPI_ANY_SOURCE
00124 # define PF_ANY_MSGTAG MPI_ANY_TAG
00125 # define PF_COMM MPI_COMM_WORLD
00126 # define PF_BYTE MPI_BYTE
00127 # define PF_INT MPI_INT
00128 #if BITSINWORD == 32
00129 # define PF_WORD MPI_INT32_T
00130 # define PF_LONG MPI_INT64_T
00131 #elif BITSINWORD == 16
00132 # define PF_WORD MPI_INT16_T
00133 # define PF_LONG MPI_INT32_T
00134 #else
00135 # error Can not detect if this is a 32-bit or 64-bit platform.
00136 #endif
00137
00138 /*
00139  #] macros & definitions :
00140  #[ s/r-bufs :
00141  */
00142
00147 typedef struct {
00148     WORD **buff;
00149     WORD **fill;
00150     WORD **full;
00151     WORD **stop;
00152     MPI_Status *status;
00153     MPI_Status *retstat;
00154     MPI_Request *request;
00155     MPI_Datatype *type; /* this is needed in PF_Wait for Get_count */
00156     int *index; /* dummies for returnvalues */
00157     int *tag; /* for the version with blocking send/receives */
00158     int *from;
00159     int numbufs; /* number of cyclic buffers */
00160     int active; /* flag telling which buffer is active */
00161     PADPOINTER(0,2,0,0);
00162 } PF_BUFFER;
00163
00164 /*
00165  #] s/r-bufs :
00166  #[ global variables used by the PF_functions : need to be known everywhere
00167  */
00168
00169 typedef struct ParallelVars {
00170     FILEHANDLE slavebuf; /* (slave) allocated if there are RHS expressions */
00171     /* special buffers for nonblocking, unbuffered send/receives */
00172     PF_BUFFER *sbuf; /* set of cyclic send buffers for master_and_slave */
00173     PF_BUFFER **rbufs; /* array of sets of cyclic receive buffers for master */
00174     int me; /* Internal number of task: master is 0 */
00175     int numtasks; /* total number of tasks */
00176     int parallel; /* flags telling the master and slaves to do the sorting parallel */

```

```

00177                                     /* [05nov2003 mt] This flag must be set to 0 in iniModule! */
00178     int      rhsInParallel; /* flag for parallel executing even if there are RHS expressions */
00179     int      mkSlaveInfile; /* flag tells that slavebuf is used on the slaves */
00180     int      exprbufsize; /* buffer size in WORDs to be used for transferring expressions */
00181     int      exprtodo; /* >= 0: the expression to do in InParallel, -1: otherwise */
00182     int      log; /* flag for logging mode */
00183     WORD     numsbufs; /* number of cyclic send buffers (PF.sbuf->numsbufs) */
00184     WORD     numrbufs; /* number of cyclic receive buffers (PF.rbufs[i]->numsbufs,
                                i=1,...,numtasks-1) */
00185     PADPOSITION(2,0,8,2,0);
00186 } PARALLELVARS;
00187
00188 extern PARALLELVARS PF;
00189 /*[04oct2005 mt]:*/
00190 /*for broadcasting dollarvars, see parallel.c:PF_BroadcastPreDollar():*/
00191 extern LONG PF_maxDollarChunkSize;
00192 /*:[04oct2005 mt]:*/
00193
00194 /*
00195     #] global variables used by the PF_functions :
00196     #[ Function prototypes :
00197 */
00198
00199 /* mpi.c */
00200 extern int PF_ISendSbuf(int,int);
00201 extern int PF_Bcast(void *buffer, int count);
00202 extern int PF_RawSend(int,void *,LONG,int);
00203 extern LONG PF_RawRecv(int *,void *,LONG,int *);
00204
00205 extern int PF_PreparePack(void);
00206 extern int PF_Pack(const void *buffer, size_t count, MPI_Datatype type);
00207 extern int PF_Unpack(void *buffer, size_t count, MPI_Datatype type);
00208 extern int PF_PackString(const UBYTE *str);
00209 extern int PF_UnpackString(UBYTE *str);
00210 extern int PF_Send(int to, int tag);
00211 extern int PF_Receive(int src, int tag, int *psrc, int *ptag);
00212 extern int PF_Broadcast(void);
00213
00214 extern int PF_PrepareLongSinglePack(void);
00215 extern int PF_LongSinglePack(const void *buffer, size_t count, MPI_Datatype type);
00216 extern int PF_LongSingleUnpack(void *buffer, size_t count, MPI_Datatype type);
00217 extern int PF_LongSingleSend(int to, int tag);
00218 extern int PF_LongSingleReceive(int src, int tag, int *psrc, int *ptag);
00219
00220 extern int PF_PrepareLongMultiPack(void);
00221 extern int PF_LongMultiPackImpl(const void *buffer, size_t count, size_t eSize, MPI_Datatype type);
00222 extern int PF_LongMultiUnpackImpl(void *buffer, size_t count, size_t eSize, MPI_Datatype type);
00223 extern int PF_LongMultiBroadcast(void);
00224
00225 static inline size_t sizeof_datatype(MPI_Datatype type)
00226 {
00227     if ( type == PF_BYTE ) return sizeof(char);
00228     if ( type == PF_INT ) return sizeof(int);
00229     if ( type == PF_WORD ) return sizeof(WORD);
00230     if ( type == PF_LONG ) return sizeof(LONG);
00231     return(0);
00232 }
00233
00234 #define PF_LongMultiPack(buffer, count, type) PF_LongMultiPackImpl(buffer, count,
                                sizeof_datatype(type), type)
00235 #define PF_LongMultiUnpack(buffer, count, type) PF_LongMultiUnpackImpl(buffer, count,
                                sizeof_datatype(type), type)
00236
00237 /* parallel.c */
00238 extern int PF_EndSort(void);
00239 extern WORD PF_Deferred(WORD *,WORD);
00240 extern int PF_Processor(EXPRESSIONS,WORD,WORD);
00241 extern int PF_Init(int*,char ***);
00242 extern int PF_Terminate(int);
00243 extern LONG PF_GetSlaveTimes(void);
00244 extern LONG PF_BroadcastNumber(LONG);
00245 extern void PF_BroadcastBuffer(WORD **buffer, LONG *length);
00246 extern int PF_BroadcastString(UBYTE *);
00247 extern int PF_BroadcastPreDollar(WORD **, LONG *,int *);
00248 extern int PF_CollectModifiedDollars(void);
00249 extern int PF_BroadcastModifiedDollars(void);
00250 extern int PF_BroadcastRedefinedPreVars(void);
00251 extern int PF_BroadcastCBuf(int bufnum);
00252 extern int PF_BroadcastExpFlags(void);
00253 extern int PF_StoreInsideInfo(void);
00254 extern int PF_RestoreInsideInfo(void);
00255 extern int PF_BroadcastExpr(EXPRESSIONS e, FILEHANDLE *file);
00256 extern int PF_BroadcastRHS(void);
00257 extern int PF_InParallelProcessor(void);
00258 extern int PF_SendFile(int to, FILE *fd);
00259 extern int PF_RecvFile(int from, FILE *fd);
00260 extern void PF_MLock(void);

```

```

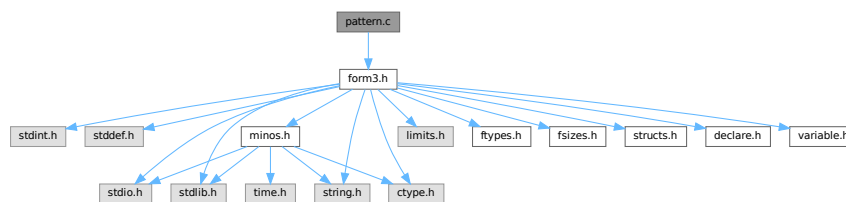
00261 extern void    PF_MUnlock(void);
00262 extern LONG     PF_WriteFileToFile(int, UBYTE *, LONG);
00263 extern void     PF_FlushStdOutBuffer(void);
00264
00265 /*
00266     #] Function prototypes :
00267 */
00268
00269 #endif

```

## 4.98 pattern.c File Reference

```
#include "form3.h"
```

Include dependency graph for pattern.c:



### Macros

- `#define PutInBuffers(pow)`

### Functions

- `int TestMatch (PHEAD WORD *term, WORD *level)`
- `void Substitute (PHEAD WORD *term, WORD *pattern, WORD power)`
- `int FindAll (PHEAD WORD *term, WORD *pattern, WORD level, WORD *par)`
- `int TestSelect (WORD *term, WORD *setp)`
- `void SubsInAll (PHEAD0)`
- `void TransferBuffer (int from, int to, int spectator)`
- `int TakeIDfunction (PHEAD WORD *term)`

### 4.98.1 Detailed Description

Top level pattern matching routines. More pattern matching is found in [findpat.c](#), [function.c](#), [symmetr.c](#) and [smart.c](#). The last three files contain the matching inside functions. The file [pattern.c](#) contains also the very important routine `Substitute`. All regular pattern matching is just the finding of the pattern and indicating what are the wildcards etc. The routine `Substitute` does the actual removal of the pattern and replaces it by a subterm of the type SUBEXPRES-

Definition in file [pattern.c](#).

## 4.98.2 Macro Definition Documentation

### 4.98.2.1 PutInBuffers

```
#define PutInBuffers(
    pow )
```

**Value:**

```
AddRHS(AT.ebufnum,1); \
*out++ = SUBEXPRESSION; \
*out++ = SUBEXPSIZE; \
*out++ = C->numrhs; \
*out++ = pow; \
*out++ = AT.ebufnum; \
FILLSUB(out) \
r = AT.pWorkSpace[rhs+i]; \
if ( *r > 0 ) { \
    oldinr = r[*r]; r[*r] = 0; \
    AddNtoC(AT.ebufnum, (*r+1-ARGHEAD), (r+ARGHEAD), 14); \
    r[*r] = oldinr; \
} \
else { \
    ToGeneral(r,buffer,1); \
    buffer[buffer[0]] = 0; \
    AddNtoC(AT.ebufnum,buffer[0]+1,buffer,15); \
}
```

Definition at line 2193 of file [pattern.c](#).

## 4.98.3 Function Documentation

### 4.98.3.1 TestMatch()

```
int TestMatch (
    PHEAD WORD * term,
    WORD * level )
```

This routine governs the pattern matching. If it decides that a substitution should be made, this can be either the insertion of a right hand side (C->rhs) or the automatic generation of terms as a result of an operation (like trace). The object to be replaced is removed from term and a subexpression pointer is inserted. If the substitution is made more than once there can be more subexpression pointers. Its number is positive as it corresponds to the level at which the C->rhs can be found in the compiler output. The subexpression pointer contains the wildcard substitution information. The power is found in \*AT.TMout. For operations the subexpression pointer is negative and corresponds to an address in the array AT.TMout. In this array are then the instructions for the routine to be called and its number in the array 'Operations' The format is here: length,functionnumber,length-2 parameters

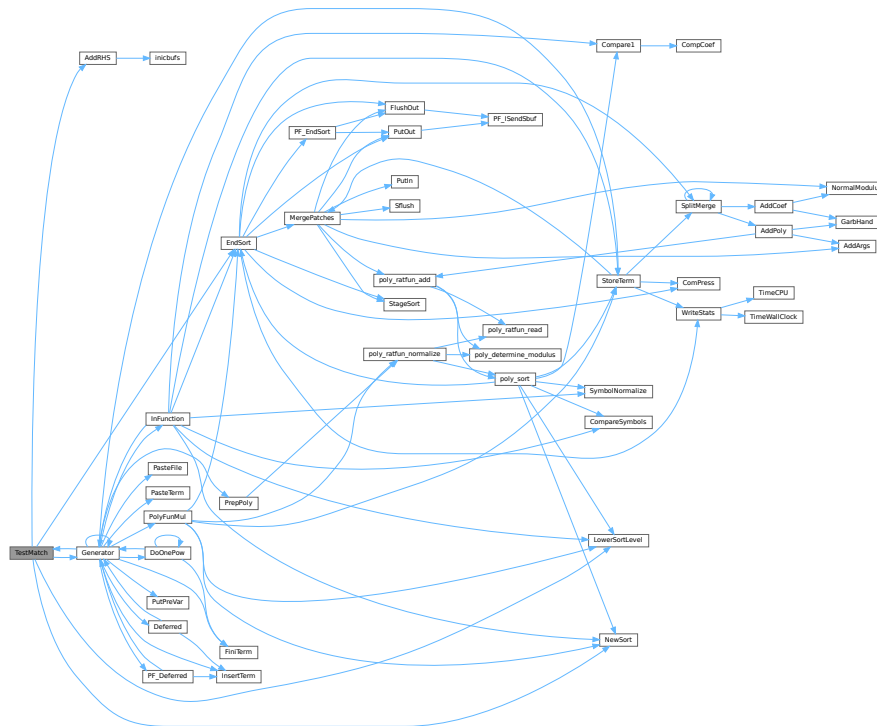
There is a certain complexity wrt repeat levels. Another complication is the poking of the wildcard values in the subexpression prototype in the compiler buffer. This was how things were done in the past with sequential FORM, but with the advent of TFORM this cannot be maintained. Now, for TFORM we make a copy of it. 7-may-2008 (JV): We cannot yet guarantee that this has been done 100% correctly. There are errors that occur in TFORM only and that may indicate problems.

Definition at line 97 of file [pattern.c](#).

References [AddRHS\(\)](#), [EndSort\(\)](#), [Generator\(\)](#), [CbUf::lhs](#), [LowerSortLevel\(\)](#), and [NewSort\(\)](#).

Referenced by [Generator\(\)](#).

Here is the call graph for this function:



#### 4.98.3.2 Substitute()

```
void Substitute (
    PHEAD WORD * term,
    WORD * pattern,
    WORD power )
```

Definition at line 641 of file [pattern.c](#).

#### 4.98.3.3 FindAll()

```
int FindAll (
    PHEAD WORD * term,
    WORD * pattern,
    WORD level,
    WORD * par )
```

Definition at line 1197 of file [pattern.c](#).

#### 4.98.3.4 TestSelect()

```
int TestSelect (
    WORD * term,
    WORD * setp )
```

Definition at line 1748 of file [pattern.c](#).

#### 4.98.3.5 SubsInAll()

```
void SubsInAll (
    PHEAD0 )
```

Definition at line 1982 of file [pattern.c](#).

#### 4.98.3.6 TransferBuffer()

```
void TransferBuffer (
    int from,
    int to,
    int spectator )
```

Definition at line 2143 of file [pattern.c](#).

#### 4.98.3.7 TakeIDfunction()

```
int TakeIDfunction (
    PHEAD WORD * term )
```

Definition at line 2213 of file [pattern.c](#).

## 4.99 pattern.c

[Go to the documentation of this file.](#)

```
00001
00012 /* #[] License : */
00013 /*
00014 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00015 * When using this file you are requested to refer to the publication
00016 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00017 * This is considered a matter of courtesy as the development was paid
00018 * for by FOM the Dutch physics granting agency and we would like to
00019 * be able to track its scientific use to convince FOM of its value
00020 * for the community.
00021 *
00022 * This file is part of FORM.
00023 *
00024 * FORM is free software: you can redistribute it and/or modify it under the
00025 * terms of the GNU General Public License as published by the Free Software
00026 * Foundation, either version 3 of the License, or (at your option) any later
00027 * version.
00028 *
00029 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00030 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00031 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00032 * details.
00033 *
00034 * You should have received a copy of the GNU General Public License along
00035 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00036 */
00037 /* #[] License : */
00038 /*
00039 !!! Notice the change in OnePV in FindAll (7-may-2008 JV).
00040
00041 #[] Includes : pattern.c
00042 */
00043
00044 #include "form3.h"
00045
00046 /*
00047 #[] Includes :
00048 #[] Patterns :
00049 #[] Rules :
```



```

00050
00051     There are several rules governing the allowable replacements.
00052     1: Multi with anything but symbols or dotproducts reverts
00053         to many.
00054     2: Each symbol can have only one (wildcard) power, so
00055         x^2*x^n? is illegal.
00056     3: when a single vector is used it replaces all occurrences
00057         of the vector. Therefore q*q(mu) or q*q(mu) cannot occur.
00058         Also q*q cannot be done.
00059     4: Loose vector elements are replaced with p(mu), dotproducts
00060         with p?.q.
00061     5: p?.q? is allowed.
00062     6: x^n? can revert to n = 0 if there is no power of x.
00063     7: x?^n? must match some x. There could be an ambiguity otherwise.
00064
00065     #] Rules :
00066     #[ TestMatch :          WORD TestMatch(term,level)
00067 */
00068
00097 int TestMatch(PHEAD WORD *term, WORD *level)
00098 {
00099     GETBIDENTITY
00100     WORD *ll, *m, *w, *llf, *OldWork, *StartWork, *ww, *mm, *t, *OldTermBuffer = 0;
00101     WORD power = 0, i, msign = 0, ll2;
00102     int match = 0;
00103     int numdollars = 0, protosize, oldallnumrhs;
00104     CBUF *C = cbuf+AM.rbufnum, *CC;
00105     AT.idallflag = 0;
00106     do {
00107 /*
00108         #] Preliminaries :
00109 */
00110         ll = C->lhs[*level];
00111         if ( *ll == TYPEEXPRESSION ) {
00112 /*
00113             Expressions are not subject to anything.
00114 */
00115             return(0);
00116         }
00117         else if ( *ll == TYPEREPEAT ) {
00118             ***AN.RepPoint = 0;
00119             return(0); /* Will force the next level */
00120         }
00121         else if ( *ll == TYPEENDREPEAT ) {
00122             if ( *AN.RepPoint ) {
00123                 AN.RepPoint[-1] = 1; /* Mark the higher level as dirty */
00124                 *AN.RepPoint = 0;
00125                 *level = ll[2]; /* Level to jump back to */
00126             }
00127             else {
00128                 AN.RepPoint--;
00129                 if ( AN.RepPoint < AT.RepCount ) {
00130                     MLOCK(ErrorMessageLock);
00131                     MesPrint("Internal problems with REPEAT count");
00132                     MUNLOCK(ErrorMessageLock);
00133                     Terminate(-1);
00134                 }
00135             }
00136             return(0); /* Force the next level */
00137         }
00138         else if ( *ll == TYPEOPERATION ) {
00139 /*
00140             Operations have always their own level.
00141 */
00142             if ( (*(FG.OperaFind[ll[2]]))(BHEAD term,ll) ) return(-1);
00143             else return(0);
00144         }
00145 /*
00146         #] Preliminaries :
00147 */
00148         OldWork = AT.WorkPointer;
00149         if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
00150         ww = AT.WorkPointer;
00151 /*
00152         Here we need to make a copy of the subexpression object because we
00153         will be writing the values of the wildcards in it.
00154         Originally we copied it into the private version of the compiler buffer
00155         that is used for scratch space (ebufnum). This caused errors in the
00156         routines like ScanFunctions when the ebufnum Buffer was expanded
00157         and inpat was still pointing at the old Buffer. This expansion
00158         could be done in AddWild and hence cannot be fixed at > 100 places.
00159         The solution is to use AN.patternbuffer (JV 16-mar-2009).
00160 */
00161         {
00162             WORD *ta = ll, *ma;
00163             int ja = ta[1];
00164 /*

```

```

00165         New code (16-mar-2009) JV
00166     */
00167     if ( ( ja + 2 ) > AN.patternbuffersize ) {
00168         if ( AN.patternbuffer ) M_free(AN.patternbuffer,"AN.patternbuffer");
00169         AN.patternbuffersize = 2 * ja + 2;
00170         AN.patternbuffer = (WORD *)Malloc(AN.patternbuffersize * sizeof(WORD),
00171             "AN.patternbuffer");
00172     }
00173     ma = AN.patternbuffer;
00174     m = ma + IDHEAD;
00175     NCOPY(ma,ta,ja);
00176     *ma = 0;
00177 }
00178 AN.FullProto = m;
00179 AN.WildValue = w = m + SUBEXPSIZE;
00180 protosize = IDHEAD + m[1];
00181 m += m[1];
00182 AN.WildStop = m;
00183 StartWork = ww;
00184 ll2 = ll[2];
00185 /*
00186     #! Expand dollars :
00187 */
00188 if ( ( ll[4] & DOLLARFLAG ) != 0 ) { /* We have at least one dollar in the pattern */
00189     WORD oldRepPoint = *AN.RepPoint, olddefer = AR.DeferFlag;
00190     AR.Eside = LHSIDEX;
00191 /*
00192     Copy into WorkSpace. This means that AN.patternbuffer will be free.
00193 */
00194     ww = AT.WorkPointer; i = m[0]; mm = m;
00195     NCOPY(ww,mm,i);
00196     *StartWork += 3;
00197     *ww++ = 1; *ww++ = 1; *ww++ = 3;
00198     AT.WorkPointer = ww;
00199     AR.DeferFlag = 0;
00200     NewSort(BHEAD0);
00201     if ( Generator(BHEAD StartWork,AR.Cnumlhs) ) {
00202         LowerSortLevel();
00203         AT.WorkPointer = OldWork;
00204         AR.DeferFlag = olddefer;
00205         return(-1);
00206     }
00207     AT.WorkPointer = ww;
00208     if ( EndSort(BHEAD ww,0) < 0 ) {}
00209     AR.DeferFlag = olddefer;
00210     if ( *ww == 0 || *(ww+ww) != 0 ) {
00211         if ( AP.lhdollarerror == 0 ) {
00212 /*
00213             If race condition we just get more error messages
00214 */
00215             MLOCK(ErrorMessageLock);
00216             MesPrint("&LHS must be one term");
00217             MUNLOCK(ErrorMessageLock);
00218             AP.lhdollarerror = 1;
00219         }
00220         AT.WorkPointer = OldWork;
00221         return(-1);
00222     }
00223     m = ww; ww = m + *m;
00224     if ( m[*m-1] < 0 ) { msign = 1; m[*m-1] = -m[*m-1]; }
00225     if ( *ww || m[*m-1] != 3 || m[*m-2] != 1 || m[*m-3] != 1 ) {
00226         MLOCK(ErrorMessageLock);
00227         MesPrint("Dollar variable develops into an illegal pattern in id-statement");
00228         MUNLOCK(ErrorMessageLock);
00229         return(-1);
00230     }
00231     *m -= m[*m-1];
00232     if ( ( *m + 1 + protosize ) > AN.patternbuffersize ) {
00233         if ( AN.patternbuffer ) M_free(AN.patternbuffer,"AN.patternbuffer");
00234         AN.patternbuffersize = 2 * (*m) + 2 + protosize;
00235         AN.patternbuffer = (WORD *)Malloc(AN.patternbuffersize * sizeof(WORD),
00236             "AN.patternbuffer");
00237         mm = ll; ww = AN.patternbuffer; i = protosize;
00238         NCOPY(ww,mm,i);
00239         AN.FullProto = AN.patternbuffer + IDHEAD;
00240         AN.WildValue = w = AN.FullProto + SUBEXPSIZE;
00241         AN.WildStop = AN.patternbuffer + protosize;
00242     }
00243     mm = AN.patternbuffer + protosize;
00244     i = *m;
00245     NCOPY(mm,m,i);
00246     m = AN.patternbuffer + protosize;
00247     AR.Eside = RHSIDE;
00248     *mm = 0;
00249 /*
00250     Test the pattern. If only wildcard powers -> SUBONCE
00251 */

```

```

00252     {
00253         WORD *mmm = m + *m, *m1 = m+1, jm, noveto = 0;
00254         while ( m1 < mmm ) {
00255             if ( *m1 == SYMBOL ) {
00256                 for ( jm = 2; jm < m1[1]; jm+=2 ) {
00257                     if ( m1[jm+1] < MAXPOWER && m1[jm+1] > -MAXPOWER ) break;
00258                 }
00259                 if ( jm < m1[1] ) { noveto = 1; break; }
00260             }
00261             else if ( *m1 == DOTPRODUCT ) {
00262                 for ( jm = 2; jm < m1[1]; jm+=3 ) {
00263                     if ( m1[jm+2] < MAXPOWER && m1[jm+2] > -MAXPOWER ) break;
00264                 }
00265                 if ( jm < m1[1] ) { noveto = 1; break; }
00266             }
00267             else { noveto = 1; break; }
00268             m1 += m1[1];
00269         }
00270         if ( noveto == 0 ) {
00271             ll2 = ll2 & ~SUBMASK;
00272             ll2 |= SUBONCE;
00273         }
00274     }
00275     AT.WorkPointer = ww = StartWork;
00276     *AN.RepPoint = oldRepPoint;
00277 }
00278 /*
00279     #] Expand dollars :
00280
00281     In case of id,all we have to check at this point that there are only
00282     functions in the pattern.
00283 */
00284 if ( ( ll2 & SUBMASK ) == SUBALL ) {
00285     WORD *t = AN.patternbuffer+IDHEAD, *tt;
00286     WORD *tstop, *ttstop, ii;
00287     t += t[1]; tstop = t + *t; t++;
00288     while ( t < tstop ) {
00289         if ( *t < FUNCTION ) break;
00290         t += t[1];
00291     }
00292     if ( t < tstop ) {
00293         MLOCK(ErrorMessageLock);
00294         MesPrint("Error: id,all can only be used with (products of) functions and/or tensors.");
00295         MUNLOCK(ErrorMessageLock);
00296         return(-1);
00297     }
00298     OldTermBuffer = AN.termbuffer;
00299     AN.termbuffer = TermMalloc("id,all");
00300 /*
00301     Now make sure that only regular functions and tensors can take part.
00302 */
00303     tt = term; ttstop = tt+*tt; ttstop -= ABS(ttstop[-1]); tt++;
00304     t = AN.termbuffer+1;
00305     while ( tt < ttstop ) {
00306         if ( *tt >= FUNCTION && *tt != AR.PolyFun && *tt != AR.PolyFunInv ) {
00307             ii = tt[1]; NCOPY(t,tt,ii);
00308         }
00309         else tt += tt[1];
00310     }
00311     *t++ = 1; *t++ = 1; *t++ = 3; AN.termbuffer[0] = t-AN.termbuffer;
00312 }
00313 /*
00314     To be puristic, we need to check that all wildcards in the prototype
00315     are actually present. If the LHS contained a replace_ this may not be
00316     the case.
00317 */
00318 ClearWild(BHEAD0);
00319 while ( w < AN.WildStop ) {
00320     if ( *w == LOADDOLLAR ) numdollars++;
00321     w += w[1];
00322 }
00323 AN.RepFunNum = 0;
00324 /* rep = */ AN.RepFunList = AT.WorkPointer;
00325 AT.WorkPointer = (WORD *) ((UBYTE *) (AT.WorkPointer)) + AM.MaxTer/2;
00326 if ( AT.WorkPointer >= AT.WorkTop ) {
00327     MLOCK(ErrorMessageLock);
00328     MesWork();
00329     MUNLOCK(ErrorMessageLock);
00330     return(-1);
00331 }
00332 AN.DisOrderFlag = ll2 & SUBDISORDER;
00333 AN.nogroundlevel = 0;
00334 switch ( ll2 & SUBMASK ) {
00335     case SUBONLY :
00336         /* Must be an exact match */
00337         AN.UseFindOnly = 1; AN.ForFindOnly = 0;
00338         if ( FindRest(BHEAD term,m) && ( AN.UsedOtherFind ||

```

```

00339         FindOnly(BHEAD term,m) ) ) {
00340             power = 1;
00341             if ( msign ) term[term[0]-1] = -term[term[0]-1];
00342         }
00343         else power = 0;
00344         break;
00345     case SUBMANY :
00346         AN.UseFindOnly = -1;
00347         if ( ( power = FindRest(BHEAD term,m) ) > 0 ) {
00348             if ( ( power = FindOnce(BHEAD term,m) ) > 0 ) {
00349                 AN.UseFindOnly = 0;
00350                 do {
00351                     if ( msign ) term[term[0]-1] = -term[term[0]-1];
00352                     Substitute(BHEAD term,m,1);
00353                     if ( numdollars ) {
00354                         WildDollars(BHEAD (WORD *)0);
00355                         numdollars = 0;
00356                     }
00357                     if ( ww < term+term[0] ) ww = term+term[0];
00358                     ClearWild(BHEAD0);
00359                     AT.WorkPointer = ww;
00360                     /* if ( rep < ww ) { */
00361                         AN.RepFunNum = 0;
00362                         /* rep = */ AN.RepFunList = ww;
00363                         AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer/2);
00364                         if ( AT.WorkPointer >= AT.WorkTop ) {
00365                             MLOCK(ErrorMessageLock);
00366                             MesWork();
00367                             MUNLOCK(ErrorMessageLock);
00368                             return(-1);
00369                         }
00370                     /* }
00371                 }
00372                 else {
00373                     AN.RepFunList = rep;
00374                     AN.RepFunNum = 0;
00375                 }
00376             */
00377                 AN.nogroundlevel = 0;
00378             } while ( FindRest(BHEAD term,m) && ( AN.UsedOtherFind ||
00379                 FindOnce(BHEAD term,m) ) );
00380             match = 1;
00381         }
00382         else if ( power < 0 ) {
00383             do {
00384                 if ( msign ) term[term[0]-1] = -term[term[0]-1];
00385                 Substitute(BHEAD term,m,1);
00386                 if ( numdollars ) {
00387                     WildDollars(BHEAD (WORD *)0);
00388                     numdollars = 0;
00389                 }
00390                 if ( ww < term+term[0] ) ww = term+term[0];
00391                 ClearWild(BHEAD0);
00392                 AT.WorkPointer = ww;
00393                 /* if ( rep < ww ) { */
00394                     AN.RepFunNum = 0;
00395                     /* rep = */ AN.RepFunList = ww;
00396                     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer/2);
00397                     if ( AT.WorkPointer >= AT.WorkTop ) {
00398                         MLOCK(ErrorMessageLock);
00399                         MesWork();
00400                         MUNLOCK(ErrorMessageLock);
00401                         return(-1);
00402                     }
00403                 /* }
00404             }
00405             else {
00406                 AN.RepFunList = rep;
00407                 AN.RepFunNum = 0;
00408             }
00409         */
00410             } while ( FindRest(BHEAD term,m) );
00411             match = 1;
00412         }
00413     }
00414     else if ( power < 0 ) {
00415         if ( FindOnce(BHEAD term,m) ) {
00416             do {
00417                 if ( msign ) term[term[0]-1] = -term[term[0]-1];
00418                 Substitute(BHEAD term,m,1);
00419                 if ( numdollars ) {
00420                     WildDollars(BHEAD (WORD *)0);
00421                     numdollars = 0;
00422                 }
00423                 if ( ww < term+term[0] ) ww = term+term[0];
00424                 ClearWild(BHEAD0);
00425                 AT.WorkPointer = ww;

```

```

00426 /*                      if ( rep < ww ) { */
00427                          AN.RepFunNum = 0;
00428                          /* rep = */ AN.RepFunList = ww;
00429                          AT.WorkPointer = (WORD *) ((UBYTE *) (AT.WorkPointer)) + AM.MaxTer/2;
00430                          if ( AT.WorkPointer >= AT.WorkTop ) {
00431                              MLOCK(ErrorMessageLock);
00432                              MesWork();
00433                              MUNLOCK(ErrorMessageLock);
00434                              return(-1);
00435                          }
00436 /*                      }
00437                      }
00438                      else {
00439                          AN.RepFunList = rep;
00440                          AN.RepFunNum = 0;
00441                      }
00442 */
00443                      } while ( FindOnce(BHEAD term,m) );
00444                      match = 1;
00445                      }
00446                  }
00447                  if ( match ) {
00448                      if ( ( ll2 & SUBAFTER ) != 0 ) *level = AC.Labels[ll[3]];
00449                  }
00450                  else {
00451                      if ( ( ll2 & SUBAFTERNOT ) != 0 ) *level = AC.Labels[ll[3]];
00452                  }
00453                  goto nextlevel;
00454              case SUBONCE :
00455                  AN.UseFindOnly = 0;
00456                  if ( FindRest(BHEAD term,m) && ( AN.UsedOtherFind || FindOnce(BHEAD term,m) ) ) {
00457                      power = 1;
00458                      if ( msign ) term[term[0]-1] = -term[term[0]-1];
00459                  }
00460                  else power = 0;
00461                  break;
00462              case SUBMULTI :
00463                  power = FindMulti(BHEAD term,m);
00464                  if ( ( power & 1 ) != 0 && msign ) term[term[0]-1] = -term[term[0]-1];
00465                  break;
00466              case SUBVECTOR :
00467                  while ( ( power = FindAll(BHEAD term,m,*level,(WORD *)0) ) != 0 ) {
00468                      if ( ( power & 1 ) != 0 && msign ) term[term[0]-1] = -term[term[0]-1];
00469                      match = 1;
00470                  }
00471                  break;
00472              case SUBSELECT :
00473                  llf = ll + IDHEAD; llf += llf[1]; llf += *llf;
00474                  AN.UseFindOnly = 1; AN.ForFindOnly = llf;
00475                  if ( FindRest(BHEAD term,m) && ( AN.UsedOtherFind || FindOnly(BHEAD term,m) ) ) {
00476                      if ( msign ) term[term[0]-1] = -term[term[0]-1];
00477 /*
00478                      The following code needs to be hacked a bit to allow for
00479                      all types of sets and for occurrence anywhere in the term
00480                      The code at the end of FindOnly is a bit mysterious.
00481 */
00482                      if ( llf[1] > 2 ) {
00483                          WORD *t1, *t2;
00484                          if ( *term > AN.sizeselecttermundo ) {
00485                              if ( AN.selecttermundo ) M_free(AN.selecttermundo,"AN.selecttermundo");
00486                              AN.sizeselecttermundo = *term + 10;
00487                              AN.selecttermundo = (WORD *)Malloc1(
00488                                  AN.sizeselecttermundo*sizeof(WORD),"AN.selecttermundo");
00489                          }
00490                          t1 = term; t2 = AN.selecttermundo; i = *term;
00491                          NCOPY(t2,t1,i);
00492                      }
00493                      power = 1;
00494                      Substitute(BHEAD term,m,power);
00495                      if ( llf[1] > 2 ) {
00496                          if ( TestSelect(term,llf) ) {
00497                              WORD *t1, *t2;
00498                              power = 0;
00499                              t1 = term; t2 = AN.selecttermundo; i = *t2;
00500                              NCOPY(t1,t2,i);
00501 #if IDHEAD > 3
00502                              if ( ( ll2 & SUBAFTERNOT ) != 0 ) {
00503                                  *level = AC.Labels[ll[3]];
00504                              }
00505 #endif
00506                              goto nextlevel;
00507                          }
00508                      }
00509                      if ( numdollars ) {
00510                          WildDollars(BHEAD (WORD *)0);
00511                          numdollars = 0;
00512                      }

```

```

00513         match = 1;
00514         if ( ( ll2 & SUBAFTER ) != 0 ) {
00515             *level = AC.Labels[ll[3]];
00516         }
00517     }
00518     else {
00519         if ( ( ll2 & SUBAFTERNOT ) != 0 ) {
00520             *level = AC.Labels[ll[3]];
00521         }
00522         power = 0;
00523     }
00524     goto nextlevel;
00525 case SUBALL:
00526     AN.UseFindOnly = 0;
00527     CC = cbuf+AT.allbufnum;
00528     oldallnumrhs = CC->numrhs;
00529     t = AddRHS(AT.allbufnum,1);
00530     *t = 0;
00531     AT.idallflag = 1;
00532     AT.idallmaxnum = ll[5];
00533     AT.idallnum = 0;
00534     if ( FindRest(BHEAD AN.termbuffer,m) || AT.idallflag > 1 ) {
00535         WORD *t, *tstop, *tt, first = 1, ii;
00536         power = 1;
00537         *CC->Pointer++ = 0;
00538         if ( msign ) term[term[0]-1] = -term[term[0]-1];
00539     /*
00540
00541         If we come here the matches are all already in the
00542         compiler buffer. All we need to do is take out all
00543         functions and replace them by a SUBEXPRESSION that
00544         points to this buffer.
00545         Note: the PolyFun/PolyRatFun should be excluded from this.
00546         This works because each match writes incrementally to
00547         the buffer using the routine SubsInAll.
00548
00549         The call to WildDollars should be made in Generator.....
00550     */
00551     t = term; tstop = t + *t; ii = ABS(tstop[-1]); tstop -= ii;
00552     tt = AT.WorkPointer+1;
00553     t++;
00554     while ( t < tstop ) {
00555         if ( *t >= FUNCTION && *t != AR.PolyFun && *t != AR.PolyFunInv ) {
00556             if ( first ) { /* SUBEXPRESSION */
00557                 *tt++ = SUBEXPRESSION;
00558                 *tt++ = SUBEXPSIZE;
00559                 *tt++ = CC->numrhs;
00560                 *tt++ = 1;
00561                 *tt++ = AT.allbufnum;
00562                 FILLSUB(tt)
00563                 first = 0;
00564             }
00565             t += t[1];
00566         }
00567         else {
00568             i = t[1]; NCOPY(tt,t,i);
00569         }
00570     }
00571     if ( ( ll[4] & NORMALIZEFLAG ) != 0 ) {
00572     /*
00573
00574         In case of the normalization option, we have to divide
00575         by AT.idallnum;
00576
00577         WORD na = t[ii-1];
00578         na = REDLENG(na);
00579         for ( i = 0; i < ii; i++ ) tt[i] = t[i];
00580         Divvy(BHEAD (UWORD *)tt,&na,(UWORD *)&(AT.idallnum),1);
00581         na = INCLENG(na);
00582         ii = ABS(na);
00583         tt[ii-1] = na;
00584         tt += ii;
00585     */
00586     }
00587     else {
00588         NCOPY(tt,t,ii);
00589     }
00590     ii = tt-AT.WorkPointer;
00591     *(AT.WorkPointer) = ii;
00592     tt = AT.WorkPointer; t = term;
00593     NCOPY(t,tt,ii);
00594
00595     if ( ( ll2 & SUBAFTER ) != 0 ) { /* ifmatch -> */
00596         *level = AC.Labels[ll[3]];
00597     }
00598     TermFree(AN.termbuffer,"id,all");
00599     AN.termbuffer = OldTermBuffer;
00600     AT.WorkPointer = AN.RepFunList;
00601     AT.idallflag = 0;
00602     CC->Pointer[0] = 0;

```

```

00600         TransferBuffer(AT.aebufnum,AT.ebufnum,AT.allbufnum);
00601         return(1);
00602     }
00603     AT.idallflag = 0;
00604     power = 0;
00605     CC->numrhs = oldallnumrhs;
00606     TermFree(AN.termbuffer,"id,all");
00607     AN.termbuffer = OldTermBuffer;
00608     break;
00609     default :
00610         break;
00611 }
00612 if ( power ) {
00613     Substitute(BHEAD term,m,power);
00614     if ( numdollars ) {
00615         WildDollars(BHEAD (WORD *)0);
00616         numdollars = 0;
00617     }
00618     match = 1;
00619     if ( ( ll2 & SUBAFTER ) != 0 ) { /* ifmatch -> */
00620         *level = AC.Labels[ll[3]];
00621     }
00622 }
00623 else {
00624     AT.WorkPointer = AN.RepFunList;
00625     if ( ( ll2 & SUBAFTERNOT ) != 0 ) { /* ifnomatch -> */
00626         *level = AC.Labels[ll[3]];
00627     }
00628 }
00629 nextlevel:;
00630 } while ( (*level)++ < AR.Cnumlhs && C->lhs[*level][0] == TYPEIDOLD );
00631 (*level)--;
00632 AT.WorkPointer = AN.RepFunList;
00633 return(match);
00634 }
00635
00636 /*
00637     #] TestMatch :
00638     #[ Substitute :          void Substitute(term,pattern,power)
00639 */
00640
00641 void Substitute(PHEAD WORD *term, WORD *pattern, WORD power)
00642 {
00643     GETBIDENTITY
00644     WORD *TemTerm;
00645     WORD *t, *m;
00646     WORD *tstop, *mstop;
00647     WORD *xstop, *ystop;
00648     WORD nt, *fill, nq, mt;
00649     WORD *q, *subterm, *tcoef, oldval1 = 0, newval3, i = 0;
00650     WORD PutExpr = 0, sign = 0;
00651     TemTerm = AT.WorkPointer;
00652     if ( ( (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer*2) ) > AT.WorkTop ) {
00653         MLOCK(ErrorMessageLock);
00654         MesWork();
00655         MUNLOCK(ErrorMessageLock);
00656         Terminate(-1);
00657     }
00658     m = pattern;
00659     mstop = m + *m;
00660     m++;
00661     t = term;
00662     t += *term - 1;
00663     tcoef = t;
00664     tstop = t - ABS(*t) + 1;
00665     t = term;
00666     t++;
00667     fill = TemTerm;
00668     fill++;
00669     if ( m < mstop ) { do {
00670 /*
00671         #[ SYMBOLS :
00672 */
00673         if ( *m == SYMBOL ) {
00674             ystop = m + m[1];
00675             m += 2;
00676             while ( *t != SYMBOL && t < tstop ) {
00677                 nq = t[1];
00678                 NCOPY(fill,t,nq);
00679             }
00680             if ( t >= tstop ) goto SubCoeef;
00681             *fill++ = SYMBOL;
00682             fill++;
00683             subterm = fill;
00684             xstop = t + t[1];
00685             t += 2;
00686             do {

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00687         if ( *m == *t && t < xstop ) {
00688             nt = t[1];
00689             mt = m[1];
00690             if ( mt >= 2*MAXPOWER ) {
00691                 if ( CheckWild(BHEAD mt-2*MAXPOWER, SYMTONUM, -MAXPOWER, &newval3) ) {
00692                     nt -= AN.oldvalue;
00693                     goto SubsL1;
00694                 }
00695             }
00696             else if ( mt <= -2*MAXPOWER ) {
00697                 if ( CheckWild(BHEAD -mt-2*MAXPOWER, SYMTONUM, -MAXPOWER, &newval3) ) {
00698                     nt += AN.oldvalue;
00699                     goto SubsL1;
00700                 }
00701             }
00702             else {
00703                 nt -= mt * power;
00704             SubsL1:
00705                 if ( nt ) {
00706                     *fill++ = *t;
00707                     *fill++ = nt;
00708                 }
00709                 m += 2; t += 2;
00710             }
00711             else if ( *m >= 2*MAXPOWER ) {
00712                 while ( t < xstop ) { *fill++ = *t++; *fill++ = *t++; }
00713                 nq = WORDDIF(fill, subterm);
00714                 fill = subterm;
00715                 while ( nq > 0 ) {
00716                     if ( !CheckWild(BHEAD *m-2*MAXPOWER, SYMTOSYM, *fill, &newval3) ) {
00717                         mt = m[1];
00718                         if ( mt >= 2*MAXPOWER ) {
00719                             if ( CheckWild(BHEAD mt-2*MAXPOWER, SYMTONUM, -MAXPOWER, &newval3) ) {
00720                                 if ( fill[1] -= AN.oldvalue ) goto SubsL2;
00721                             }
00722                         }
00723                         else if ( mt <= -2*MAXPOWER ) {
00724                             if ( CheckWild(BHEAD -mt-2*MAXPOWER, SYMTONUM, -MAXPOWER, &newval3) ) {
00725                                 if ( fill[1] += AN.oldvalue ) goto SubsL2;
00726                             }
00727                         }
00728                         else {
00729                             if ( fill[1] -= mt * power ) {
00730             SubsL2:
00731                                 fill += nq;
00732                                 nq = 0;
00733                             }
00734                             break;
00735                         }
00736                     }
00737                     nq -= 2;
00738                     fill += 2;
00739                 }
00740                 if ( nq ) {
00741                     nq -= 2;
00742                     q = fill + 2;
00743                     while ( --nq >= 0 ) *fill++ = *q++;
00744                 }
00745                 m += 2;
00746             }
00747             else if ( *m < *t || t >= xstop ) { m += 2; }
00748             else { *fill++ = *t++; *fill++ = *t++; }
00749         } while ( m < ystop );
00750         while ( t < xstop ) *fill++ = *t++;
00751         nq = WORDDIF(fill, subterm);
00752         if ( nq > 0 ) {
00753             nq += 2;
00754             subterm[-1] = nq;
00755         }
00756         else { fill = subterm; fill -= 2; }
00757     }
00758     /*
00759     #] SYMBOLS :
00760     #[ DOTPRODUCTS :
00761     */
00762     else if ( *m == DOTPRODUCT ) {
00763         ystop = m + m[1];
00764         m += 2;
00765         while ( *t > DOTPRODUCT && t < tstop ) {
00766             nq = t[1];
00767             NCOPY(fill, t, nq);
00768         }
00769         if ( t >= tstop ) goto SubCoef;
00770         if ( *t != DOTPRODUCT ) {
00771             m = ystop;
00772             goto EndLoop;
00773         }
00774         *fill++ = DOTPRODUCT;

```



```

00774         fill++;
00775         subterm = fill;
00776         xstop = t + t[1];
00777         t += 2;
00778         do {
00779             if ( *m == *t && m[1] == t[1] && t < xstop ) {
00780                 nt = t[2];
00781                 mt = m[2];
00782                 if ( mt >= 2*MAXPOWER ) {
00783                     if ( CheckWild(BHEAD mt-2*MAXPOWER, SYMTONUM, -MAXPOWER, &newval3) ) {
00784                         nt -= AN.oldvalue;
00785                         goto SubsL3;
00786                     }
00787                 }
00788                 else if ( mt <= -2*MAXPOWER ) {
00789                     if ( CheckWild(BHEAD -mt-2*MAXPOWER, SYMTONUM, -MAXPOWER, &newval3) ) {
00790                         nt += AN.oldvalue;
00791                         goto SubsL3;
00792                     }
00793                 }
00794                 else {
00795                     nt -= mt * power;
00796 SubsL3:
00797                     if ( nt ) {
00798                         *fill++ = *t++;
00799                         *fill++ = *t;
00800                         *fill++ = nt;
00801                         t += 2;
00802                     }
00803                     else t += 3;
00804                 }
00805                 m += 3;
00806             }
00807             else if ( *m >= (AM.OffsetVector+WILDOFFSET) ) {
00808                 while ( t < xstop ) {
00809                     *fill++ = *t++; *fill++ = *t++; *fill++ = *t++;
00810                 }
00811                 oldvall = 1;
00812                 goto SubsL4;
00813             }
00814             else if ( m[1] >= (AM.OffsetVector+WILDOFFSET) ) {
00815                 while ( *m >= *t && t < xstop ) {
00816                     *fill++ = *t++; *fill++ = *t++; *fill++ = *t++;
00817                 }
00818                 oldvall = 0;
00819 SubsL4:
00820                 nq = WORDDIF(fill, subterm);
00821                 fill = subterm;
00822                 while ( nq > 0 ) {
00823                     if ( ( oldvall && ( (
00824                         !CheckWild(BHEAD *m-WILDOFFSET, VECTOVEC, *fill, &newval3)
00825                         && !CheckWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, fill[1], &newval3)
00826                     ) || (
00827                         !CheckWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, *fill, &newval3)
00828                         && !CheckWild(BHEAD *m-WILDOFFSET, VECTOVEC, fill[1], &newval3)
00829                     ) ) || ( !oldvall && ( (
00830                         *m == *fill
00831                         && !CheckWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, fill[1], &newval3)
00832                     ) || (
00833                         !CheckWild(BHEAD m[1]-WILDOFFSET, VECTOVEC, *fill, &newval3)
00834                         && *m == fill[1] ) ) ) ) {
00835                         mt = m[2];
00836                         if ( mt >= 2*MAXPOWER ) {
00837                             if ( CheckWild(BHEAD mt-2*MAXPOWER, SYMTONUM, -MAXPOWER, &newval3) ) {
00838                                 if ( fill[2] -= AN.oldvalue )
00839                                     goto SubsL5;
00840                             }
00841                             else if ( mt <= -2*MAXPOWER ) {
00842                                 if ( CheckWild(BHEAD -mt-2*MAXPOWER, SYMTONUM, -MAXPOWER, &newval3) ) {
00843                                     if ( fill[2] += AN.oldvalue )
00844                                         goto SubsL5;
00845                                 }
00846                             }
00847                             else {
00848 SubsL5:
00849                                 if ( fill[2] -= mt * power ) {
00850                                     fill += nq;
00851                                     nq = 0;
00852                                 }
00853                                 }
00854                                 m += 3;
00855                                 break;
00856                             }
00857                             fill += 3; nq -= 3;
00858                         }
00859                     }
00860                     if ( nq ) {
00861                         nq -= 3;
00862                         q = fill + 3;
00863                         while ( --nq >= 0 ) *fill++ = *q++;

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```

00861         }
00862     }
00863     else if ( t >= xstop || *m < *t || ( *m == *t && m[1] < t[1] ) )
00864     { m += 3; }
00865     else {
00866         *fill++ = *t++; *fill++ = *t++; *fill++ = *t++;
00867     }
00868     while ( m < ystop );
00869     while ( t < xstop ) *fill++ = *t++;
00870     nq = WORDDIF(fill,subterm);
00871     if ( nq > 0 ) {
00872         nq += 2;
00873         subterm[-1] = nq;
00874     }
00875     else { fill = subterm; fill -= 2; }
00876 }
00877 /*
00878     #] DOTPRODUCTS :
00879     #[ FUNCTIONS :
00880 */
00881 else if ( *m >= FUNCTION ) {
00882     while ( *t >= FUNCTION || *t == SUBEXPRESSION ) {
00883         nt = WORDDIF(t,term);
00884         for ( mt = 0; mt < AN.RepFunNum; mt += 2 ) {
00885             if ( nt == AN.RepFunList[mt] ) break;
00886         }
00887         if ( mt >= AN.RepFunNum ) {
00888             nq = t[1];
00889             NCOPY(fill,t,nq);
00890         }
00891         else {
00892             WORD *oldt = 0;
00893             if ( *m == GAMMA && m[1] != FUNHEAD+1 ) {
00894                 oldt = t;
00895                 if ( ( i = AN.RepFunList[mt+1] ) > 0 ) {
00896                     *fill++ = GAMMA;
00897                     *fill++ = i + FUNHEAD+1;
00898                     FILLFUN(fill)
00899                     nq = i + 1;
00900                     t += FUNHEAD;
00901                     NCOPY(fill,t,nq);
00902                 }
00903                 t = oldt;
00904             }
00905             else if ( ( *t == LEVICIVITA ) || ( *t >= FUNCTION
00906                 && (functions[*t-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC )
00907                 ) sign += AN.RepFunList[mt+1];
00908             else if ( *m >= FUNCTION+WILDOFFSET
00909                 && (functions[*m-FUNCTION-WILDOFFSET].symmetric & ~REVERSEORDER) == ANTISYMMETRIC
00910                 ) sign += AN.RepFunList[mt+1];
00911             if ( !PutExpr ) {
00912                 xstop = t + t[1];
00913                 t = AN.FullProto;
00914                 nq = t[1];
00915                 t[3] = power;
00916                 NCOPY(fill,t,nq);
00917                 t = xstop;
00918                 PutExpr = 1;
00919             }
00920             else t += t[1];
00921             if ( *m == GAMMA && m[1] != FUNHEAD+1 ) {
00922                 i = oldt[1] - m[1] - i;
00923                 if ( i > 0 ) {
00924                     *fill++ = GAMMA;
00925                     *fill++ = i + FUNHEAD+1;
00926                     FILLFUN(fill)
00927                     *fill++ = oldt[FUNHEAD];
00928                     t = t - i;
00929                     NCOPY(fill,t,i);
00930                 }
00931             }
00932             break;
00933         }
00934     }
00935     m += m[1];
00936 }
00937 /*
00938     #] FUNCTIONS :
00939     #[ VECTORS :
00940 */
00941 else if ( *m == VECTOR ) {
00942     while ( *t > VECTOR ) {
00943         nq = t[1];
00944         NCOPY(fill,t,nq);
00945     }
00946     xstop = t + t[1];
00947     ystop = m + m[1];

```

```

00948         t += 2;
00949         m += 2;
00950         *fill++ = VECTOR;
00951         fill++;
00952         subterm = fill;
00953         do {
00954             if ( *m == *t && m[1] == t[1] ) {
00955                 m += 2; t += 2;
00956             }
00957             else if ( *m >= (AM.OffsetVector+WILDOFFSET) ) {
00958                 while ( t < xstop ) *fill++ = *t++;
00959                 nq = WORDDIF(fill,subterm);
00960                 fill = subterm;
00961                 if ( m[1] < (AM.OffsetIndex+WILDOFFSET) ) {
00962                     do {
00963                         if ( m[1] == fill[1] &&
00964                             !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*fill,&newval3) )
00965                             break;
00966                         fill += 2;
00967                         nq -= 2;
00968                     } while ( nq > 0 );
00969                 }
00970                 else { /* Double wildcard */
00971                     do {
00972                         if ( !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,fill[1],&newval3)
00973                             && !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*fill,&newval3) )
00974                             break;
00975                         if ( *fill == oldvall && fill[1] == AN.oldvalue ) break;
00976                         fill += 2;
00977                         nq -= 2;
00978                     } while ( nq > 0 );
00979                 }
00980                 nq -= 2;
00981                 q = fill + 2;
00982                 if ( nq > 0 ) { NCOPY(fill,q,nq); }
00983                 m += 2;
00984             }
00985             else if ( *m <= *t &&
00986                 m[1] >= (AM.OffsetIndex + WILDOFFSET) ) {
00987                 while ( *m == *t && t < xstop )
00988                     { *fill++ = *t++; *fill++ = *t++; }
00989                 nq = WORDDIF(fill,subterm);
00990                 fill = subterm;
00991                 do {
00992                     if ( *m == *fill &&
00993                         !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,fill[1],&newval3) )
00994                         break;
00995                     nq -= 2;
00996                     fill += 2;
00997                 } while ( nq > 0 );
00998                 nq -= 2;
00999                 q = fill + 2;
01000                 if ( nq > 0 ) { NCOPY(fill,q,nq); }
01001                 m += 2;
01002             }
01003             else { *fill++ = *t++; *fill++ = *t++; }
01004         } while ( m < ystop );
01005         while ( t < xstop ) *fill++ = *t++;
01006         nq = WORDDIF(fill,subterm);
01007         if ( nq > 0 ) {
01008             nq += 2;
01009             subterm[-1] = nq;
01010         }
01011         else { fill = subterm; fill -= 2; }
01012     }
01013     /*
01014         #] VECTORS :
01015         #[ INDICES :
01016
01017         Currently without wildcards
01018     */
01019     else if ( *m == INDEX ) {
01020         while ( *t > INDEX ) {
01021             nq = t[1];
01022             NCOPY(fill,t,nq);
01023         }
01024         xstop = t + t[1];
01025         ystop = m + m[1];
01026         t += 2;
01027         m += 2;
01028         *fill++ = INDEX;
01029         fill++;
01030         subterm = fill;
01031         do {
01032             if ( *m == *t ) {
01033                 m += 1; t += 1;
01034             }

```

```

01035         else if ( *m >= (AM.OffsetIndex+WILDOFFSET) ) {
01036             while ( t < xstop ) *fill++ = *t++;
01037             nq = WORDDIF(fill, subterm);
01038             fill = subterm;
01039             do {
01040                 if ( !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*fill,&newval3) ) {
01041                     break;
01042                 }
01043                 fill += 1;
01044                 nq -= 1;
01045             } while ( nq > 0 );
01046             nq -= 1;
01047             if ( nq > 0 ) {
01048                 q = fill + 1;
01049                 NCOPY(fill,q,nq);
01050             }
01051             m += 1;
01052         }
01053         else {
01054             *fill++ = *t++;
01055         }
01056     } while ( m < ystop );
01057     while ( t < xstop ) *fill++ = *t++;
01058     nq = WORDDIF(fill,subterm);
01059     if ( nq > 0 ) {
01060         nq += 2;
01061         subterm[-1] = nq;
01062     }
01063     else { fill = subterm; fill -= 2; }
01064 }
01065 /*
01066     #] INDICES :
01067     #[ DELTAS :
01068 */
01069     else if ( *m == DELTA ) {
01070         while ( *t > DELTA ) {
01071             nq = t[1];
01072             NCOPY(fill,t,nq);
01073         }
01074         xstop = t + t[1];
01075         ystop = m + m[1];
01076         t += 2;
01077         m += 2;
01078         *fill++ = DELTA;
01079         fill++;
01080         subterm = fill;
01081         do {
01082             if ( *t == *m && t[1] == m[1] ) { m += 2; t += 2; }
01083             else if ( *m >= (AM.OffsetIndex+WILDOFFSET) ) { /* Two dummies */
01084                 while ( t < xstop ) *fill++ = *t++;
01085                 fill = subterm; /*
01086                 oldvall = 1;
01087                 goto SubsL6;
01088             }
01089             else if ( m[1] >= (AM.OffsetIndex+WILDOFFSET) ) {
01090                 while ( (*m == *t || *m == t[1]) && ( t < xstop ) ) {
01091                     *fill++ = *t++; *fill++ = *t++;
01092                 }
01093                 oldvall = 0;
01094                 SubsL6:
01095                 nq = WORDDIF(fill,subterm);
01096                 fill = subterm;
01097                 do {
01098                     if ( ( oldvall && ( (
01099                         !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*fill,&newval3)
01100                         && !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,fill[1],&newval3)
01101                     ) || (
01102                         !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,*fill,&newval3)
01103                         && !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,fill[1],&newval3)
01104                     ) ) ) || ( !oldvall && ( (
01105                         *m == *fill
01106                         && !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,fill[1],&newval3)
01107                     ) || (
01108                         *m == fill[1]
01109                         && !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,*fill,&newval3)
01110                     ) ) ) ) break;
01111                     fill += 2;
01112                     nq -= 2;
01113                 } while ( nq > 0 );
01114                 nq -= 2;
01115                 if ( nq > 0 ) {
01116                     q = fill + 2;
01117                     NCOPY(fill,q,nq);
01118                 }
01119                 m += 2;
01120             }
01121             else {
01122                 *fill++ = *t++; *fill++ = *t++;

```

```

01122         }
01123         } while ( m < ystop );
01124         while ( t < xstop ) *fill++ = *t++;
01125         nq = WORDDIF(fill,subterm);
01126         if ( nq > 0 ) {
01127             nq += 2;
01128             subterm[-1] = nq;
01129         }
01130         else { fill = subterm; fill -= 2; }
01131     }
01132     /*
01133         #] DELTAS :
01134     */
01135 EndLoop;;
01136     } while ( m < mstop ); }
01137     while ( t < tstop ) *fill++ = *t++;
01138 SubCoef:
01139     if ( !PutExpr ) {
01140         t = AN.FullProto;
01141         nq = t[1];
01142         t[3] = power;
01143         NCOPY(fill,t,nq);
01144     }
01145     t = tcoef;
01146     nq = ABS(*t);
01147     t = tstop;
01148     NCOPY(fill,t,nq);
01149     nq = WORDDIF(fill,TemTerm);
01150     fill = term;
01151     t = TemTerm;
01152     *fill++ = nq--;
01153     t++;
01154     NCOPY(fill,t,nq);
01155     if ( sign ) {
01156         if ( ( sign & 1 ) != 0 ) fill[-1] = -fill[-1];
01157     }
01158     if ( AT.WorkPointer < fill ) AT.WorkPointer = fill;
01159     AN.RepFunNum = 0;
01160 }
01161
01162 /*
01163     #] Substitute :
01164     #[ FindSpecial :          WORD FindSpecial(term)
01165
01166     Routine to detect simplifications regarding the special functions
01167     exponent, denominator.
01168
01169 void FindSpecial(WORD *term)
01170 {
01171     WORD *t;
01172     WORD *tstop;
01173     t = term; t += *t - 1; tstop = t - ABS(*t) + 1; t = term;
01174     t++;
01175     if ( t < tstop ) { do {
01176         if ( *t == EXPONENT ) {
01177             Exponents can become simpler when:
01178             a: the exponent of an expression becomes an integer.
01179             b: The expression becomes zero.
01180         }
01181         else if ( *t == DENOMINATOR ) {
01182             Denominators can become simpler when:
01183             a: The denominator is a single term without functions.
01184             b: An overall coefficient can be removed.
01185             c: An overall object can be removed.
01186             The task is here to bring the denominator in an unique form.
01187         }
01188         t += *t;
01189     } while ( t < tstop ); }
01190 }
01191
01192     #] FindSpecial :
01193     #[ FindAll :              WORD FindAll(term,pattern,level,par)
01194
01195 */
01196
01197 int FindAll(PHEAD WORD *term, WORD *pattern, WORD level, WORD *par)
01198 {
01199     GETBIDENTITY
01200     WORD *t, *m, *r, *mm, rnum;
01201     WORD *tstop, *mstop, *TwoProto, *vwhere = 0, oldv, oldvv, vv, level2;
01202     WORD v, nq, OffNum = AM.OffsetVector + WILD OFFSET, i, ii = 0, jj;
01203     WORD fromindex, *intens, notflag1 = 0, notflag2 = 0;
01204     CBUF *C;
01205     C = cbuf+AM.rbufnum;
01206     v = pattern[3]; /* The vector to be found */
01207     m = t = term;
01208     m += *m;

```

```

01209     m -= ABS(m[-1]);
01210     t++;
01211     if ( t < m ) do {
01212         tstop = t + t[1];
01213         fromindex = 2;
01214     /*
01215         #[ VECTOR :
01216     */
01217     if ( *t == VECTOR ) {
01218         r = t;
01219         r += 2;
01220 InVect:
01221         while ( r < tstop ) {
01222             oldv = *r;
01223             if ( v >= OffNum ) {
01224                 vwhere = AN.FullProto + 3 + SUBEXPSIZE;
01225                 if ( vwhere[1] == FROMSET || vwhere[1] == SETTONUM ) {
01226                     WORD *afirst, *alast, j;
01227                     j = vwhere[3];
01228                     if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag1 = 1; }
01229                     else { notflag1 = 0; }
01230                     afirst = SetElements + Sets[j].first;
01231                     alast = SetElements + Sets[j].last;
01232                     ii = 1;
01233                     if ( notflag1 == 0 ) {
01234                         do {
01235                             if ( *afirst == *r ) {
01236                                 if ( vwhere[1] == SETTONUM ) {
01237                                     AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
01238                                     AN.FullProto[11+SUBEXPSIZE] = ii;
01239                                 }
01240                                 else if ( vwhere[4] >= 0 ) {
01241                                     oldv = *(afirst - Sets[j].first
01242                                         + Sets[vwhere[4]].first);
01243                                 }
01244                                 goto DoVect;
01245                             }
01246                             ii++;
01247                         } while ( ++afirst < alast );
01248                     }
01249                     else {
01250                         do {
01251                             if ( *afirst == *r ) break;
01252                         } while ( ++afirst < alast );
01253                         if ( afirst >= alast ) goto DoVect;
01254                     }
01255                 }
01256                 else goto DoVect;
01257             }
01258             else if ( v == *r ) {
01259 DoVect:
01260                 m = AT.WorkPointer;
01261                 tstop = t;
01262                 t = term;
01263                 mstop = t + *t;
01264                 do { *m++ = *t++; } while ( t < tstop );
01265                 vwhere = m;
01266                 t = AN.FullProto;
01267                 nq = t[1];
01268                 t[3] = 1;
01269                 NCOPY(m,t,nq);
01270                 t = tstop;
01271                 if ( fromindex == 1 ) m[-1] = FUNNYVEC;
01272                 else m[-1] = r[1]; /* The index is always here! */
01273                 if ( v >= OffNum ) vwhere[3+SUBEXPSIZE] = oldv;
01274                 if ( vwhere[1] > 12+SUBEXPSIZE ) {
01275                     vwhere[11+SUBEXPSIZE] = ii;
01276                     vwhere[8+SUBEXPSIZE] = SYMTONUM;
01277                 }
01278                 if ( t[1] > fromindex+2 ) {
01279                     *m++ = *t++;
01280                     *m++ = *t++ - fromindex;
01281                     while ( t < r ) *m++ = *t++;
01282                     t += fromindex;
01283                 }
01284                 else t += t[1];
01285                 do { *m++ = *t++; } while ( t < mstop );
01286                 *AT.WorkPointer = nq = WORDDIF(m,AT.WorkPointer);
01287                 m = AT.WorkPointer;
01288                 t = term;
01289                 NCOPY(t,m,nq);
01290                 AT.WorkPointer = t;
01291                 return(1);
01292             }
01293             r += fromindex;
01294         }
01295     /*

```

```

01296         #] VECTOR :
01297         #[ DOTPRODUCT :
01298     */
01299     else if ( *t == DOTPRODUCT ) {
01300         r = t;
01301         r += 2;
01302         do {
01303             if ( ( i = r[2] ) < 0 ) goto NextDot;
01304             if ( *r == r[1] ) { /* p.p */
01305                 oldv = *r;
01306                 if ( v == *r ) { /* v.v */
01307                     TwoVec:
01308                         m = AT.WorkPointer;
01309                         tstop = t;
01310                         t = term;
01311                         mstop = t + *t;
01312                         do { *m++ = *t++; } while ( t < tstop );
01313                         do {
01314                             vwhere = m;
01315                             t = AN.FullProto;
01316                             nq = t[1];
01317                             t[3] = 2;
01318                             NCOPY(m,t,nq);
01319                             m[-1] = ++AR.CurDum;
01320                             if ( v >= OffNum ) vwhere[3+SUBEXPSIZE] = oldv;
01321                         } while ( --i > 0 );
01322                         t = tstop;
01323                         if ( t[1] > 5 ) {
01324                             *m++ = *t++;
01325                             *m++ = *t++ - 3;
01326                             while ( t < r ) *m++ = *t++;
01327                             t += 3;
01328                         }
01329                         else t += t[1];
01330                         do { *m++ = *t++; } while ( t < mstop );
01331                         *AT.WorkPointer = nq = WORDDIF(m,AT.WorkPointer);
01332                         m = AT.WorkPointer;
01333                         t = term;
01334                         NCOPY(t,m,nq);
01335                         AT.WorkPointer = t;
01336                         return(1);
01337                     }
01338                 else if ( v >= OffNum ) { /* v?.v? */
01339                     vwhere = AN.FullProto + 3+SUBEXPSIZE;
01340                     if ( vwhere[1] == FROMSET || vwhere[1] == SETTONUM ) {
01341                         WORD *afirst, *alast, j;
01342                         j = vwhere[3];
01343                         if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag1 = 1; }
01344                         else { notflag1 = 0; }
01345                         afirst = SetElements + Sets[j].first;
01346                         alast = SetElements + Sets[j].last;
01347                         ii = 1;
01348                         if ( notflag1 == 0 ) {
01349                             do {
01350                                 if ( *afirst == *r ) {
01351                                     if ( vwhere[1] == SETTONUM ) {
01352                                         AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
01353                                         AN.FullProto[11+SUBEXPSIZE] = ii;
01354                                     }
01355                                     else if ( vwhere[4] >= 0 ) {
01356                                         oldv = *(afirst - Sets[j].first
01357                                             + Sets[vwhere[4]].first);
01358                                         goto TwoVec;
01359                                     }
01360                                 }
01361                                 ii++;
01362                             } while ( ++afirst < alast );
01363                         }
01364                         else {
01365                             do {
01366                                 if ( *afirst == *r ) break;
01367                             } while ( ++afirst < alast );
01368                             if ( afirst >= alast ) goto TwoVec;
01369                         }
01370                     }
01371                     else goto TwoVec;
01372                 }
01373             }
01374         }
01375         else {
01376             if ( v == r[1] ) { r[1] = *r; *r = v; }
01377             oldv = *r;
01378             oldvv = r[1];
01379             if ( v == *r ) {
01380                 if ( !par ) { while ( ++level <= AR.Cnumlhs
01381                     && C->lhs[level][0] == TYPEIDOLD ) {
01382                     m = C->lhs[level];
01383                     m += IDHEAD;
01384                     if ( m[-IDHEAD+2] == SUBVECTOR ) {

```

```

01383         if ( ( vv = m[m[1]+3] ) == r[1] ) {
01384 OnePV:         TwoProto = AN.FullProto;
01385 TwoPV:         m = AT.WorkPointer;
01386                 tstop = t;
01387                 t = term;
01388                 mstop = t + *t;
01389                 do { *m++ = *t++; } while ( t < tstop );
01390                 do {
01391                     t = AN.FullProto;
01392                     vwhere = m + 3 + SUBEXPSIZE;
01393                     nq = t[1];
01394                     t[3] = 1;
01395                     NCOPY(m,t,nq);
01396                     m[-1] = ++AR.CurDum;
01397                     if ( v >= OffNum ) *vwhere = oldv;
01398                     if ( vwhere[-2-SUBEXPSIZE] > 12+SUBEXPSIZE ) {
01399                         vwhere[8] = ii;
01400                         vwhere[5] = SYMTONUM;
01401                     }
01402                     t = TwoProto;
01403                     vwhere = m + 3+SUBEXPSIZE;
01404                     mm = m;
01405                     nq = t[1];
01406                     t[3] = 1;
01407                     NCOPY(m,t,nq);
01408 /*
01409 The next two lines repair a bug. without them it takes twice
01410 the rhs of the first vector.
01411 */
01412                     mm[2] = C->lhs[level][IDHEAD+2];
01413                     mm[4] = C->lhs[level][IDHEAD+4];
01414                     m[-1] = AR.CurDum;
01415                     if ( vv >= OffNum ) *vwhere = oldvv;
01416                 } while ( --i > 0 );
01417                 goto CopRest;
01418             }
01419             else if ( vv > OffNum ) {
01420                 vwhere = AN.FullProto + 3+SUBEXPSIZE;
01421                 if ( vwhere[1] == FROMSET || vwhere[1] == SETTONUM ) {
01422                     WORD *afirst, *alast, j;
01423                     j = vwhere[3];
01424                     if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag1 = 1; }
01425                     else { notflag1 = 0; }
01426                     afirst = SetElements + Sets[j].first;
01427                     alast = SetElements + Sets[j].last;
01428                     if ( notflag1 == 0 ) {
01429                         ii = 1;
01430                         do {
01431                             if ( *afirst == r[1] ) {
01432                                 if ( vwhere[1] == SETTONUM ) {
01433                                     AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
01434                                     AN.FullProto[11+SUBEXPSIZE] = ii;
01435                                 }
01436                                 else if ( vwhere[4] >= 0 ) {
01437                                     oldvv = *(afirst - Sets[j].first
01438                                         + Sets[vwhere[4]].first);
01439                                 }
01440                                 goto OnePV;
01441                             }
01442                             ii++;
01443                             } while ( ++afirst < alast );
01444                         }
01445                     else {
01446                         do {
01447                             if ( *afirst == *r ) break;
01448                             } while ( ++afirst < alast );
01449                             if ( afirst >= alast ) goto OnePV;
01450                         }
01451                     }
01452                 else goto OnePV;
01453             }
01454         }
01455     }}
01456 /*
01457 v.q with v matching and no match for the q, also
01458 not in following idold statements.
01459 Notice that a following q.p? cannot match.
01460 */
01461     rnum = r[1];
01462 OneOnly:     m = AT.WorkPointer;
01463             tstop = t;
01464             t = term;
01465             mstop = t + *t;
01466             do { *m++ = *t++; } while ( t < tstop );
01467             vwhere = m;
01468             t = AN.FullProto;
01469             nq = t[1];

```



```

01470         t[3] = i;
01471         NCOPY(m,t,nq);
01472         m[-4] = INDTOIND;
01473         m[-1] = rnum;
01474         if ( v >= OffNum ) vwhere[3+SUBEXPSIZE] = oldv;
01475         goto CopRest;
01476     }
01477     else if ( v >= OffNum ) {
01478         vwhere = AN.FullProto + 3+SUBEXPSIZE;
01479         if ( vwhere[1] == FROMSET || vwhere[1] == SETTONUM ) {
01480             WORD *afirst, *alast, *bfirst, *blast, j;
01481             j = vwhere[3];
01482             if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag1 = 1; }
01483             else { notflag1 = 0; }
01484             afirst = SetElements + Sets[j].first;
01485             alast = SetElements + Sets[j].last;
01486             ii = 1;
01487             if ( notflag1 == 0 ) {
01488                 do {
01489                     if ( *afirst == *r ) {
01490                         if ( vwhere[1] == SETTONUM ) {
01491                             AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
01492                             AN.FullProto[11+SUBEXPSIZE] = ii;
01493                         }
01494                         else if ( vwhere[4] >= 0 ) {
01495                             oldv = *(afirst - Sets[j].first
01496                                 + Sets[vwhere[4]].first);
01497                         }
01498                         level2 = level;
01499                         do {
01500                             if ( !par ) m = C->lhs[level2];
01501                             else m = par;
01502                             m += IDHEAD;
01503                             if ( m[-IDHEAD+2] == SUBVECTOR ) {
01504                                 if ( ( vv = m[m[1]+3] ) == r[1] )
01505                                     goto OnePV;
01506                                 else if ( vv >= OffNum ) {
01507                                     if ( m[SUBEXPSIZE+4] != FROMSET &&
01508                                         m[SUBEXPSIZE+4] != SETTONUM ) goto OnePV;
01509                                     j = m[SUBEXPSIZE+6];
01510                                     if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag2 = 1; }
01511                                     else { notflag2 = 0; }
01512                                     bfirst = SetElements + Sets[j].first;
01513                                     blast = SetElements + Sets[j].last;
01514                                     jj = 1;
01515                                     if ( notflag2 == 0 ) {
01516                                         do {
01517                                             if ( *bfirst == r[1] ) {
01518                                                 if ( m[SUBEXPSIZE+4] == SETTONUM ) {
01519                                                     m[SUBEXPSIZE+8] = SYMTONUM;
01520                                                     m[SUBEXPSIZE+11] = jj;
01521                                                 }
01522                                                 else if ( m[SUBEXPSIZE+7] >= 0 ) {
01523                                                     oldvv = *(bfirst - Sets[j].first
01524                                                         + Sets[m[SUBEXPSIZE+7]].first);
01525                                                 }
01526                                                 goto OnePV;
01527                                             }
01528                                             jj++;
01529                                         } while ( ++bfirst < blast );
01530                                     }
01531                                     else {
01532                                         do {
01533                                             if ( *bfirst == r[1] ) break;
01534                                         } while ( ++bfirst < blast );
01535                                         if ( bfirst >= blast ) goto OnePV;
01536                                     }
01537                                 }
01538                             } while ( ++level2 < AR.Cnumlhs &&
01539                                 C->lhs[level2][0] == TYPEIDOLD );
01540                             rnum = r[1];
01541                             goto OneOnly;
01542                         }
01543                     }
01544                     else if ( *afirst == r[1] ) {
01545                         if ( vwhere[1] == SETTONUM ) {
01546                             AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
01547                             AN.FullProto[11+SUBEXPSIZE] = ii;
01548                         }
01549                         else if ( vwhere[4] >= 0 ) {
01550                             oldv = *(afirst - Sets[j].first
01551                                 + Sets[vwhere[4]].first);
01552                         }
01553                         level2 = level;
01554                         while ( ++level2 < AR.Cnumlhs &&
01555                             C->lhs[level2][0] == TYPEIDOLD ) {
01556                             if ( !par ) m = C->lhs[level2];

```

```

01557         else m = par;
01558         m += IDHEAD;
01559         if ( m[-IDHEAD+2] == SUBVECTOR ) {
01560             if ( ( vv = m[6] ) == *r )
01561                 goto OnePV;
01562             else if ( vv >= OffNum ) {
01563                 if ( m[SUBEXPSIZE+4] != FROMSET && m[SUBEXPSIZE+4]
01564                     != SETTONUM ) {
01565                     j = *r;
01566                     *r = r[1];
01567                     r[1] = j;
01568                     goto OnePV;
01569                 }
01570                 j = m[SUBEXPSIZE+6];
01571                 bfirst = SetElements + Sets[j].first;
01572                 blast = SetElements + Sets[j].last;
01573                 jj = 1;
01574                 do {
01575                     if ( *bfirst == *r ) {
01576                         if ( m[SUBEXPSIZE+4] == SETTONUM ) {
01577                             m[SUBEXPSIZE+8] = SYMTONUM;
01578                             m[SUBEXPSIZE+11] = jj;
01579                         }
01580                         else if ( m[SUBEXPSIZE+7] >= 0 ) {
01581                             oldvv = *(bfirst + Sets[j].first
01582                                 + Sets[m[SUBEXPSIZE+7]].first);
01583                         }
01584                         j = *r;
01585                         *r = r[1];
01586                         r[1] = j;
01587                         j = oldv; oldv = oldvv; oldvv = j;
01588                         goto OnePV;
01589                     }
01590                     jj++;
01591                 } while ( ++bfirst < blast );
01592             }
01593         }
01594     }
01595     jj = *r; *r = r[1]; r[1] = jj;
01596     jj = oldv; oldv = oldvv; oldvv = j;
01597     rnum = r[1];
01598     goto OneOnly;
01599 }
01600 ii++;
01601 } while ( ++afirst < alast );
01602 }
01603 else {
01604     do {
01605         if ( *afirst == *r ) break;
01606     } while ( ++afirst < alast );
01607     if ( afirst >= alast ) goto Hitlevel1;
01608     do {
01609         if ( *afirst == r[1] ) break;
01610     } while ( ++afirst < alast );
01611     if ( afirst >= alast ) goto Hitlevel2;
01612 }
01613 }
01614 else { /* Matches twice */
01615     vv = v;
01616     TwoProto = AN.FullProto;
01617     goto TwoPV;
01618 }
01619 }
01620 }
01621 NextDot:    r += 3;
01622 } while ( r < tstop );
01623 }
01624 /*
01625     #] DOTPRODUCT :
01626     #[ LEVICIVITA :
01627 */
01628 else if ( *t == LEVICIVITA ) {
01629     intens = 0;
01630     r = t;
01631     r += FUNHEAD;
01632 OneVect:;
01633     while ( r < tstop ) {
01634         oldv = *r;
01635         if ( v >= OffNum && *r < -10 ) {
01636             vwhere = AN.FullProto + 3+SUBEXPSIZE;
01637             if ( vwhere[1] == FROMSET || vwhere[1] == SETTONUM ) {
01638                 WORD *afirst, *alast, j;
01639                 j = vwhere[3];
01640                 if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag1 = 1; }
01641                 else { notflag1 = 0; }
01642                 afirst = SetElements + Sets[j].first;
01643                 alast = SetElements + Sets[j].last;

```

```

01644             ii = 1;
01645             if ( notflag1 == 0 ) {
01646                 do {
01647                     if ( *afirst == *r ) {
01648                         if ( vwhere[1] == SETTONUM ) {
01649                             AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
01650                             AN.FullProto[11+SUBEXPSIZE] = ii;
01651                         }
01652                         else if ( vwhere[4] >= 0 ) {
01653                             oldv = *(afirst - Sets[j].first
01654                                 + Sets[vwhere[4]].first);
01655                         }
01656                         goto DoVect;
01657                     }
01658                     ii++;
01659                 } while ( ++afirst < alast );
01660             }
01661             else {
01662                 do {
01663                     if ( *afirst == *r ) break;
01664                 } while ( ++afirst < alast );
01665                 if ( afirst >= alast ) goto DoVect;
01666             }
01667         }
01668         else goto LeVect;
01669     }
01670     else if ( v == *r ) {
01671 LeVect:
01672         m = AT.WorkPointer;
01673         mstop = term + *term;
01674         t = term;
01675         *r = ++AR.CurDum;
01676         if ( intens ) *intens = DIRTYSYMFLAG;
01677         do { *m++ = *t++; } while ( t < tstop );
01678         t = AN.FullProto;
01679         nq = t[1];
01680         t[3] = 1;
01681         if ( v >= OffNum ) *vwhere = oldv;
01682         NCOPY(m,t,nq);
01683         m[-1] = AR.CurDum;
01684         t = tstop;
01685         do { *m++ = *t++; } while ( t < mstop );
01686         *AT.WorkPointer = nq = WORDDIF(m,AT.WorkPointer);
01687         m = AT.WorkPointer;
01688         t = term;
01689         NCOPY(t,m,nq);
01690         AT.WorkPointer = t;
01691         return(1);
01692     }
01693     r++;
01694 }
01695 /*
01696     #] LEVICIVITA :
01697     #[ GAMMA :
01698 */
01699     else if ( *t == GAMMA ) {
01700         intens = 0;
01701         r = t;
01702         r += FUNHEAD+1;
01703         if ( r < tstop ) goto OneVect;
01704     }
01705 /*
01706     #] GAMMA :
01707     #[ INDEX :
01708 */
01709     else if ( *t == INDEX ) { /* The 'forgotten' part */
01710         r = t;
01711         r += 2;
01712         fromindex = 1;
01713         goto InVect;
01714     }
01715 /*
01716     #] INDEX :
01717     #[ FUNCTION :
01718 */
01719     else if ( *t >= FUNCTION ) {
01720         if ( *t >= FUNCTION
01721             && functions[*t-FUNCTION].spec >= TENSORFUNCTION
01722             && t[1] > FUNHEAD ) {
01723 /*
01724             Tensors are linear in their vectors!
01725 */
01726             r = t;
01727             r += FUNHEAD;
01728             intens = t+2;
01729             goto OneVect;
01730         }

```

```

01731     }
01732  /*
01733     #] FUNCTION :
01734  */
01735     t += t[1];
01736   } while ( t < m );
01737   return(0);
01738 }
01739
01740 /*
01741     #] FindAll :
01742     #[ TestSelect :
01743
01744     Returns 1 if any of the objects in any of the sets in setp
01745     occur anywhere in the term
01746  */
01747
01748 int TestSelect(WORD *term, WORD *setp)
01749 {
01750     WORD *tstop, *t, *s, *el, *elstop, *termstop, *tt, n, ns;
01751     GETSTOP(term,tstop);
01752     term += 1;
01753     while ( term < tstop ) {
01754         switch ( *term ) {
01755             case SYMBOL:
01756                 n = term[1] - 2;
01757                 t = term + 2;
01758                 while ( n > 0 ) {
01759                     ns = setp[1] - 2;
01760                     s = setp + 2;
01761                     while ( --ns >= 0 ) {
01762                         if ( Sets[*s].type != CSYMBOL ) { s++; continue; }
01763                         el = SetElements + Sets[*s].first;
01764                         elstop = SetElements + Sets[*s].last;
01765                         while ( el < elstop ) {
01766                             if ( *el++ == *t ) return(1);
01767                         }
01768                         s++;
01769                     }
01770                     n -= 2;
01771                     t += 2;
01772                 }
01773                 break;
01774             case VECTOR:
01775                 n = term[1] - 2;
01776                 t = term + 2;
01777                 while ( n > 0 ) {
01778                     ns = setp[1] - 2;
01779                     s = setp + 2;
01780                     while ( --ns >= 0 ) {
01781                         if ( Sets[*s].type != CVECTOR ) { s++; continue; }
01782                         el = SetElements + Sets[*s].first;
01783                         elstop = SetElements + Sets[*s].last;
01784                         while ( el < elstop ) {
01785                             if ( *el++ == *t ) return(1);
01786                         }
01787                         s++;
01788                     }
01789                     t++;
01790                     ns = setp[1] - 2;
01791                     s = setp + 2;
01792                     while ( --ns >= 0 ) {
01793                         if ( Sets[*s].type != CINDEX
01794                             && Sets[*s].type != CNUMBER ) { s++; continue; }
01795                         el = SetElements + Sets[*s].first;
01796                         elstop = SetElements + Sets[*s].last;
01797                         while ( el < elstop ) {
01798                             if ( *el++ == *t ) return(1);
01799                         }
01800                         s++;
01801                     }
01802                     n -= 2;
01803                     t++;
01804                 }
01805                 break;
01806             case INDEX:
01807                 n = term[1] - 2;
01808                 t = term + 2;
01809                 goto dotensor;
01810             case DOTPRODUCT:
01811                 n = term[1] - 2;
01812                 t = term + 2;
01813                 while ( n > 0 ) {
01814                     ns = setp[1] - 2;
01815                     s = setp + 2;
01816                     while ( --ns >= 0 ) {
01817                         if ( Sets[*s].type != CVECTOR ) { s++; continue; }

```

```

01818         el = SetElements + Sets[*s].first;
01819         elstop = SetElements + Sets[*s].last;
01820         while ( el < elstop ) {
01821             if ( *el++ == *t ) return(1);
01822         }
01823         s++;
01824     }
01825     t++;
01826     ns = setp[1] - 2;
01827     s = setp + 2;
01828     while ( --ns >= 0 ) {
01829         if ( Sets[*s].type != CVECTOR ) { s++; continue; }
01830         el = SetElements + Sets[*s].first;
01831         elstop = SetElements + Sets[*s].last;
01832         while ( el < elstop ) {
01833             if ( *el++ == *t ) return(1);
01834         }
01835         s++;
01836     }
01837     n -= 3;
01838     t += 2;
01839 }
01840 break;
01841 case DELTA:
01842     n = term[1] - 2;
01843     t = term + 2;
01844     goto dotensor;
01845 default:
01846     if ( *term < FUNCTION ) break;
01847     ns = setp[1] - 2;
01848     s = setp + 2;
01849     while ( --ns >= 0 ) {
01850         if ( Sets[*s].type != CFUNCTION ) { s++; continue; }
01851         el = SetElements + Sets[*s].first;
01852         elstop = SetElements + Sets[*s].last;
01853         while ( el < elstop ) {
01854             if ( *el++ == *term ) return(1);
01855         }
01856         s++;
01857     }
01858     if ( functions[*term-FUNCTION].spec > 0 ) {
01859         n = term[1] - FUNHEAD;
01860         t = term + FUNHEAD;
01861 dotensor:
01862         while ( n > 0 ) {
01863             ns = setp[1] - 2;
01864             s = setp + 2;
01865             while ( --ns >= 0 ) {
01866                 if ( *t < MINSPEC ) {
01867                     if ( Sets[*s].type != CVECTOR ) { s++; continue; }
01868                 }
01869                 else if ( *t >= 0 ) {
01870                     if ( Sets[*s].type != CINDEX
01871                         && Sets[*s].type != CNUMBER ) { s++; continue; }
01872                 }
01873                 else { s++; continue; }
01874                 el = SetElements + Sets[*s].first;
01875                 elstop = SetElements + Sets[*s].last;
01876                 while ( el < elstop ) {
01877                     if ( *el++ == *t ) return(1);
01878                 }
01879                 s++;
01880             }
01881             t++;
01882             n--;
01883         }
01884     }
01885     else {
01886         termstop = term + term[1];
01887         tt = term + FUNHEAD;
01888         while ( tt < termstop ) {
01889             if ( *tt < 0 ) {
01890                 if ( *tt == -SYMBOL ) {
01891                     ns = setp[1] - 2;
01892                     s = setp + 2;
01893                     while ( --ns >= 0 ) {
01894                         if ( Sets[*s].type != CSYMBOL ) { s++; continue; }
01895                         el = SetElements + Sets[*s].first;
01896                         elstop = SetElements + Sets[*s].last;
01897                         while ( el < elstop ) {
01898                             if ( *el++ == tt[1] ) return(1);
01899                         }
01900                         s++;
01901                     }
01902                     tt += 2;
01903                 }
01904                 else if ( *tt == -VECTOR || *tt == -MINVECTOR ) {

```

```

01905         ns = setp[1] - 2;
01906         s = setp + 2;
01907         while ( --ns >= 0 ) {
01908             if ( Sets[*s].type != CVECTOR ) { s++; continue; }
01909             el = SetElements + Sets[*s].first;
01910             elstop = SetElements + Sets[*s].last;
01911             while ( el < elstop ) {
01912                 if ( *el++ == tt[1] ) return(1);
01913             }
01914             s++;
01915         }
01916         tt += 2;
01917     }
01918     else if ( *tt == -INDEX ) {
01919         ns = setp[1] - 2;
01920         s = setp + 2;
01921         while ( --ns >= 0 ) {
01922             if ( Sets[*s].type != CINDEX
01923                 && Sets[*s].type != CNUMBER ) { s++; continue; }
01924             el = SetElements + Sets[*s].first;
01925             elstop = SetElements + Sets[*s].last;
01926             while ( el < elstop ) {
01927                 if ( *el++ == tt[1] ) return(1);
01928             }
01929             s++;
01930         }
01931         tt += 2;
01932     }
01933     else if ( *tt <= -FUNCTION ) {
01934         ns = setp[1] - 2;
01935         s = setp + 2;
01936         while ( --ns >= 0 ) {
01937             if ( Sets[*s].type != CFUNCTION ) { s++; continue; }
01938             el = SetElements + Sets[*s].first;
01939             elstop = SetElements + Sets[*s].last;
01940             while ( el < elstop ) {
01941                 if ( *el++ == -( *tt ) ) return(1);
01942             }
01943             s++;
01944         }
01945         tt++;
01946     }
01947     else tt += 2;
01948 }
01949 else {
01950     t = tt + ARGHEAD;
01951     tt += *tt;
01952     while ( t < tt ) {
01953         if ( TestSelect(t,setp) ) return(1);
01954         t += *t;
01955     }
01956 }
01957 }
01958 }
01959 break;
01960 }
01961 term += term[1];
01962 }
01963 return(0);
01964 }
01965
01966 /*
01967     #] TestSelect :
01968     #[ SubsInAll :      void SubsInAll()
01969
01970     This routine takes a match in id,all and stores it away in
01971     the AT.allbufnum 'compiler' buffer, after taking out the pattern.
01972     The main problem here is that id,all usually has (lots of) wildcards
01973     and their assignments are on stack and the difficult ones are in
01974     AT.ebufnum. Popping the stack while looking for more matches would
01975     loose those. Hence we have to copy them into yet another compiler
01976     buffer: AT.aebufnum. Because this may involve many matches and
01977     because the original term has only a limited number of arguments,
01978     it will pay to look for already existing ones in this buffer.
01979     (to be done later).
01980 */
01981
01982 void SubsInAll(PHEAD0)
01983 {
01984     GETBIDENTITY
01985     WORD *TemTerm;
01986     WORD *t, *m, *term;
01987     WORD *tstop, *mstop, *xstop;
01988     WORD nt, *fill, nq, mt;
01989     WORD *tcoef, i = 0;
01990     WORD PutExpr = 0, sign = 0;
01991 }

```

```

01992     We start with building the term in the Workspace.
01993     Afterwards we will transfer it to AT.allbufnum.
01994     We have to make sure there is room in the Workspace.
01995 */
01996     AT.idallflag = 2;
01997     TemTerm = AT.WorkPointer;
01998     if ( ( (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer*2) ) > AT.WorkTop ) {
01999         MLOCK(ErrorMessageLock);
02000         MesWork();
02001         MUNLOCK(ErrorMessageLock);
02002         Terminate(-1);
02003     }
02004     m = AN.patternbuffer + IDHEAD; m += m[1];
02005     mstop = m + *m;
02006     m++;
02007     term = AN.termbuffer;
02008     tstop = term + *term; tcoef = tstop-1; tstop -= ABS(tstop[-1]);
02009     t = term;
02010     t++;
02011     fill = TemTerm;
02012     fill++;
02013     while ( m < mstop ) {
02014         while ( t < tstop ) {
02015             nt = WORDDIF(t,term);
02016             for ( mt = 0; mt < AN.RepFunNum; mt += 2 ) {
02017                 if ( nt == AN.RepFunList[mt] ) break;
02018             }
02019             if ( mt >= AN.RepFunNum ) {
02020                 nq = t[1];
02021                 NCOPY(fill,t,nq);
02022             }
02023             else {
02024                 WORD *oldt = 0;
02025                 if ( *m == GAMMA && m[1] != FUNHEAD+1 ) {
02026                     oldt = t;
02027                     if ( ( i = AN.RepFunList[mt+1] ) > 0 ) {
02028                         *fill++ = GAMMA;
02029                         *fill++ = i + FUNHEAD+1;
02030                         FILLFUN(fill)
02031                         nq = i + 1;
02032                         t += FUNHEAD;
02033                         NCOPY(fill,t,nq);
02034                     }
02035                     t = oldt;
02036                 }
02037                 else if ( ( *t == LEVICIVITA ) || ( *t >= FUNCTION
02038                     && (functions[*t-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC )
02039                     ) sign += AN.RepFunList[mt+1];
02040                 else if ( *m >= FUNCTION+WILDOFFSET
02041                     && (functions[*m-FUNCTION-WILDOFFSET].symmetric & ~REVERSEORDER) == ANTISYMMETRIC
02042                     ) sign += AN.RepFunList[mt+1];
02043                 if ( !PutExpr ) {
02044                     WORD *pstart = fill, *p, *w, *ww;
02045                     xstop = t + t[1];
02046                     t = AN.FullProto;
02047                     nq = t[1];
02048                     t[3] = 1;
02049                     NCOPY(fill,t,nq);
02050                     t = xstop;
02051                     PutExpr = 1;
02052 */
02053                     Here we need provisions for keeping wildcard matches
02054                     that reside in AT.ebufnum. We will move them to
02055                     AT.aebufnum.
02056                     Problem: the SUBEXPRESSION assumes automatically
02057                     that the compiler buffer is AT.ebufnum. We have to
02058                     correct that in TransferBuffer.
02059 */
02060                     p = pstart + SUBEXPSIZE;
02061                     while ( p < fill ) {
02062                         switch ( *p ) {
02063                             case SYMTOSUB:
02064                             case VECTOSUB:
02065                             case INDTOSUB:
02066                             case ARGTOARG:
02067                             case ARLTOARL:
02068                                 w = cbuf[AT.ebufnum].rhs[p[3]];
02069                                 ww = cbuf[AT.ebufnum].rhs[p[3]+1];
02070 */
02071                                 Here we could search for whether this
02072                                 object sits in the buffer already.
02073                                 To be done later.
02074                                 By the way: ww-w fits inside a WORD.
02075 */
02076                                 AddRHS(AT.aebufnum,1);
02077                                 AddNtoC(AT.aebufnum,ww-w,w,11);
02078                                 p[3] = cbuf[AT.aebufnum].numrhs;

```

```

02079             cbuf[AT.aebufnum].rhs[p[3]+1] = cbuf[AT.aebufnum].Pointer;
02080             p += p[1];
02081             break;
02082             case FROMSET:
02083             case SETTONUM:
02084             case LOADDOLLAR:
02085                 p += p[1];
02086                 break;
02087             default:
02088                 p += p[1];
02089                 break;
02090         }
02091     }
02092 }
02093 }
02094 else t += t[1];
02095 if ( *m == GAMMA && m[1] != FUNHEAD+1 ) {
02096     i = oldt[1] - m[1] - i;
02097     if ( i > 0 ) {
02098         *fill++ = GAMMA;
02099         *fill++ = i + FUNHEAD+1;
02100         FILLFUN(fill)
02101         *fill++ = oldt[FUNHEAD];
02102         t = t - i;
02103         NCOPY(fill,t,i);
02104     }
02105 }
02106 break;
02107 }
02108 }
02109 m += m[1];
02110 }
02111 while ( t < tstop ) *fill++ = *t++;
02112 if ( !PutExpr ) {
02113     t = AN.FullProto;
02114     nq = t[1];
02115     t[3] = 1;
02116     NCOPY(fill,t,nq);
02117 }
02118 t = tcoef;
02119 nq = ABS(*t);
02120 t = tstop;
02121 NCOPY(fill,t,nq);
02122 if ( sign ) {
02123     if ( ( sign & 1 ) != 0 ) fill[-1] = -fill[-1];
02124 }
02125 *TemTerm = fill-TemTerm;
02126 /*
02127 And now we copy this to AT.allbufnum
02128 */
02129 AddNtoC(AT.allbufnum,TemTerm[0],TemTerm,12);
02130 cbuf[AT.allbufnum].Pointer[0] = 0;
02131 AN.RepFunNum = 0;
02132 }
02133
02134 /*
02135 #] SubsInAll :
02136 #[ TransferBuffer :
02137
02138 Adds the whole content of a (compiler)buffer to another buffer.
02139 In spectator we have an expression in the RHS that needs the
02140 wildcard resolutions adapted by an offset.
02141 */
02142
02143 void TransferBuffer(int from,int to,int spectator)
02144 {
02145     CBUF *C = cbuf + spectator;
02146     CBUF *Cf = cbuf + from;
02147     CBUF *Ct = cbuf + to;
02148     int offset = Ct->numrhs;
02149     LONG i;
02150     WORD *t, *tt, *ttt, *tstop, size;
02151     for ( i = 1; i <= Cf->numrhs; i++ ) {
02152         size = Cf->rhs[i+1]-Cf->rhs[i];
02153         AddRHS(to,1);
02154         AddNtoC(to,size,Cf->rhs[i],13);
02155     }
02156     Ct->rhs[Ct->numrhs+1] = Ct->Pointer;
02157     Cf->numrhs = 0;
02158 /*
02159 Now we have to update the 'pointers' in the spectator.
02160 */
02161 t = C->rhs[C->numrhs];
02162 while ( *t ) {
02163     tt = t+1; t += *t;
02164     tstop = t-ABS(t[-1]);
02165     while ( tt < tstop ) {

```



```

02166         if ( *tt == SUBEXPRESSION ) {
02167             ttt = tt+SUBEXPSIZE; tt += tt[1];
02168             while ( ttt < tt ) {
02169                 switch ( *ttt ) {
02170                     case SYMTOSUB:
02171                     case VECTOSUB:
02172                     case INDIOSUB:
02173                     case ARGTOARG:
02174                     case ARLTOARL:
02175                         ttt[3] += offset;
02176                         break;
02177                     default:
02178                         break;
02179                 }
02180                 ttt += 4;
02181             }
02182         }
02183         else tt += tt[1];
02184     }
02185 }
02186 }
02187
02188 /*
02189     #] TransferBuffer :
02190     #[ TakeIDfunction :
02191 */
02192
02193 #define PutInBuffers(pow) \
02194     AddRHS(AT.ebufnum,1); \
02195     *out++ = SUBEXPRESSION; \
02196     *out++ = SUBEXPSIZE; \
02197     *out++ = C->numrhs; \
02198     *out++ = pow; \
02199     *out++ = AT.ebufnum; \
02200     FILLSUB(out) \
02201     r = AT.pWorkSpace[rhs+i]; \
02202     if ( *r > 0 ) { \
02203         oldinr = r[*r]; r[*r] = 0; \
02204         AddNtoC(AT.ebufnum, (*r+1-ARGHEAD), (r+ARGHEAD), 14); \
02205         r[*r] = oldinr; \
02206     } \
02207     else { \
02208         ToGeneral(r,buffer,1); \
02209         buffer[buffer[0]] = 0; \
02210         AddNtoC(AT.ebufnum,buffer[0]+1,buffer,15); \
02211     }
02212
02213 int TakeIDfunction(PHEAD WORD *term)
02214 {
02215     WORD *tstop, *t, *r, *m, *f, *nextf, *funstop, *left, *l, *newterm;
02216     WORD *out, oldinr, pow;
02217     WORD buffer[20];
02218     int i, ii, j, numsub, numfound = 0, first;
02219     LONG lhs,rhs;
02220     CBUF *C;
02221     GETSTOP(term,tstop);
02222     for ( t = term+1; t < tstop; t += t[1] ) { if ( *t == IDFUNCTION ) break; }
02223     if ( t >= tstop ) return(0);
02224 /*
02225     Step 1: test validity
02226 */
02227     funstop = t + t[1]; f = t + FUNHEAD;
02228     left = term + *term;
02229     l = left+1; numsub = 0;
02230     while ( f < funstop ) {
02231         nextf = f; NEXTARG(nextf)
02232         if ( nextf >= funstop ) { return(0); } /* odd number of arguments */
02233         if ( *f == -SYMBOL ) { *l++ = SYMBOL; *l++ = 4; *l++ = f[1]; *l++ = 1; }
02234         else if ( *f < -FUNCTION ) { *l++ = *f; *l++ = FUNHEAD; FILLFUN(1) }
02235         else if ( *f > 0 ) {
02236             if ( *f != f[ARGHEAD]+ARGHEAD ) goto noaction;
02237             if ( nextf[-1] != 3 || nextf[-2] != 1 || nextf[-3] != 1 ) goto noaction;
02238             if ( f[ARGHEAD] <= 4 ) goto noaction;
02239             if ( f[ARGHEAD] != f[ARGHEAD+2]+4 ) goto noaction;
02240             if ( f[ARGHEAD] == 8 && f[ARGHEAD+1] == SYMBOL ) {
02241                 for ( i = 0; i < 4; i++ ) *l++ = f[ARGHEAD+1+i];
02242             }
02243             else if ( f[ARGHEAD] == 9 && f[ARGHEAD+1] == DOTPRODUCT ) {
02244                 for ( i = 0; i < 5; i++ ) *l++ = f[ARGHEAD+1+i];
02245             }
02246             else if ( f[ARGHEAD+1] >= FUNCTION ) {
02247                 for ( i = 0; i < f[ARGHEAD+1]-4; i++ ) *l++ = f[ARGHEAD+1+i];
02248             }
02249             else goto noaction;
02250         }
02251         else goto noaction;
02252         numsub++;

```

```

02253         f = nextf;
02254         NEXTARG(f)
02255     }
02256     C = cbuf+AT.ebufnum;
02257     AT.WorkPointer = l;
02258     *left = l-left;
02259 /*
02260 Put the pointers to the lhs and the rhs in the pointer workspace
02261 */
02262 WantAddPointers(2*numsub);
02263 lhs = AT.pWorkPointer;
02264 rhs = lhs+numsub;
02265 AT.pWorkPointer = rhs+numsub;
02266 f = t + FUNHEAD; l = left+1;
02267 for ( i = 0; i < numsub; i++ ) {
02268     AT.pWorkSpace[lhs+i] = l; l += l[1];
02269     NEXTARG(f);
02270     AT.pWorkSpace[rhs+i] = f;
02271     NEXTARG(f);
02272 }
02273 /*
02274 Take out the patterns and replace them by SUBEXPRESSIONs pointing at
02275 the e buffer. We put the resulting term above the left sides.
02276 Note that we take out only the first id_ if there is more than one!
02277 */
02278 first = 1;
02279 t = term+1; newterm = AT.WorkPointer; out = newterm+1;
02280 while ( t < tstop ) {
02281     if ( *t == IDFUNCTION && first ) { first = 0; t += t[1]; continue; }
02282     if ( *t >= FUNCTION ) {
02283         for ( i = 0; i < numsub; i++ ) {
02284             m = AT.pWorkSpace[lhs+i];
02285             if ( *m != *t ) continue;
02286             for ( j = 1; j < t[1]; j++ ) {
02287                 if ( m[j] != t[j] ) break;
02288             }
02289             if ( j != t[1] ) continue;
02290             numfound++;
02291 /*
02292 We have a match! Set up a SUBEXPRESSION subterm and put the
02293 corresponding rhs in the eBuffer.
02294 */
02295             PutInBuffers(1)
02296             t += t[1];
02297         }
02298         if ( i == numsub ) { /* no match. Just copy to output. */
02299             j = t[1]; NCOPY(out,t,j)
02300         }
02301     }
02302     else if ( *t == SYMBOL ) {
02303         for ( i = 0; i < numsub; i++ ) {
02304             m = AT.pWorkSpace[lhs+i];
02305             if ( *m != SYMBOL ) continue;
02306             for ( ii = 2; ii < t[1]; ii += 2 ) {
02307                 if ( m[2] != t[ii] ) continue;
02308                 pow = t[ii+1]/m[3];
02309                 if ( pow <= 0 ) continue;
02310                 t[ii+1] = t[ii+1]*m[3];
02311                 numfound++;
02312 /*
02313 Create the proper rhs in the eBuffer and set up a
02314 SUBEXPRESSION subterm.
02315 */
02316                 PutInBuffers(pow)
02317             }
02318         }
02319 /*
02320 Now we copy whatever remains of the SYMBOL subterm to the output
02321 */
02322         m = out; *out++ = t[0]; *out++ = t[1];
02323         for ( ii = 2; ii < t[1]; ii += 2 ) {
02324             if ( t[ii+1] ) { *out++ = t[ii]; *out++ = t[ii+1]; }
02325         }
02326         m[1] = out-m;
02327         if ( m[1] == 2 ) out = m;
02328         t += t[1];
02329     }
02330     else if ( *t == DOTPRODUCT ) {
02331         for ( i = 0; i < numsub; i++ ) {
02332             m = AT.pWorkSpace[lhs+i];
02333             if ( *m != DOTPRODUCT ) continue;
02334             for ( ii = 2; ii < t[1]; ii += 3 ) {
02335                 if ( m[2] != t[ii] || m[3] != t[ii+1] ) continue;
02336                 pow = t[ii+2]/m[4];
02337                 if ( pow <= 0 ) continue;
02338                 t[ii+2] = t[ii+2]*m[4];
02339                 numfound++;

```

```

02340 /*
02341             Create the proper rhs in the eBuffer and set up a
02342             SUBEXPRESSION subterm.
02343 */
02344             PutInBuffers(pow)
02345         }
02346     }
02347 /*
02348             Now we copy whatever remains of the DOTPRODUCT subterm to the output
02349 */
02350     m = out; *out++ = t[0]; *out++ = t[1];
02351     for ( ii = 2; ii < t[1]; ii += 3 ) {
02352         if ( t[ii+2] ) { *out++ = t[ii]; *out++ = t[ii+1]; *out++ = t[ii+2]; }
02353     }
02354     m[1] = out-m;
02355     if ( m[1] == 2 ) out = m;
02356     t += t[1];
02357 }
02358 else {
02359     j = t[1]; NCOPY(out,t,j)
02360 }
02361 }
02362 /*
02363 Copy the coefficient and set the size.
02364 */
02365 t = tstop; r = term+term; while ( t < r ) *out++ = *t++;
02366 *newterm = out-newterm;
02367 /*
02368 Finally we move the new term over the original term.
02369 */
02370 i = *newterm;
02371 t = term; r = newterm; NCOPY(t,r,i)
02372 /*
02373 At this point we can return and if the calling Generator jumps back to
02374 its start, TestSub can take care of the expansions of SUBEXPRESSIONs.
02375 */
02376 AT.pWorkPointer = lhs;
02377 AT.WorkPointer = t;
02378 return(numfound);
02379 noaction:
02380 return(0);
02381 }
02382
02383 /*
02384             #] TakeIDfunction :
02385             #] Patterns :
02386 */
02387

```

## 4.100 poly.cc

```

00001 /* @file poly.cc
00002 *
00003 * Contains the class for representing sparse multivariate
00004 * polynomials with integer coefficients
00005 */
00006 /* #] License : */
00007 /*
00008 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 * When using this file you are requested to refer to the publication
00010 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 * This is considered a matter of courtesy as the development was paid
00012 * for by FOM the Dutch physics granting agency and we would like to
00013 * be able to track its scientific use to convince FOM of its value
00014 * for the community.
00015 *
00016 * This file is part of FORM.
00017 *
00018 * FORM is free software: you can redistribute it and/or modify it under the
00019 * terms of the GNU General Public License as published by the Free Software
00020 * Foundation, either version 3 of the License, or (at your option) any later
00021 * version.
00022 *
00023 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 * details.
00027 *
00028 * You should have received a copy of the GNU General Public License along
00029 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #] License : */
00032 /*

```

```

00033     #[ includes :
00034 */
00035
00036 #include "poly.h"
00037 #include "polygcd.h"
00038
00039 #include <cassert>
00040 #include <cctype>
00041 #include <cmath>
00042 #include <algorithm>
00043 #include <map>
00044 #include <iostream>
00045
00046 using namespace std;
00047
00048 /*
00049     #] includes :
00050     #[ constructor (small constant polynomial) :
00051 */
00052
00053 // constructor for a small constant polynomial
00054 poly::poly (PHEAD int a, WORD modp, WORD modn):
00055     size_of_terms(AM.MaxTer/(LONG) sizeof(WORD)),
00056     modp(0),
00057     modn(1)
00058 {
00059     POLY_STOREIDENTITY;
00060
00061     terms = TermMalloc("poly::poly(int)");
00062
00063     if (a == 0) {
00064         terms[0] = 1; // length
00065     }
00066     else {
00067         terms[0] = 4 + AN.poly_num_vars; // length
00068         terms[1] = 3 + AN.poly_num_vars; // length
00069         for (int i=0; i<AN.poly_num_vars; i++) terms[2+i] = 0; // powers
00070         terms[2+AN.poly_num_vars] = ABS(a); // coefficient
00071         terms[3+AN.poly_num_vars] = a>0 ? 1 : -1; // length coefficient
00072     }
00073
00074     if (modp!=-1) setmod(modp,modn);
00075 }
00076
00077 /*
00078     #] constructor (small constant polynomial) :
00079     #[ constructor (large constant polynomial) :
00080 */
00081
00082 // constructor for a large constant polynomial
00083 poly::poly (PHEAD const UWORD *a, WORD na, WORD modp, WORD modn):
00084     size_of_terms(AM.MaxTer/(LONG) sizeof(WORD)),
00085     modp(0),
00086     modn(1)
00087 {
00088     POLY_STOREIDENTITY;
00089
00090     terms = TermMalloc("poly::poly(UWORD*,WORD)");
00091
00092     // remove leading zeros
00093     while (*(a+ABS(na)-1)==0)
00094         na -= SGN(na);
00095
00096     terms[0] = 3 + AN.poly_num_vars + ABS(na); // length
00097     terms[1] = terms[0] - 1; // length
00098     for (int i=0; i<AN.poly_num_vars; i++) terms[2+i] = 0; // powers
00099     termscopy((WORD *)a, 2+AN.poly_num_vars, ABS(na)); // coefficient
00100     terms[2+AN.poly_num_vars+ABS(na)] = na; // length coefficient
00101
00102     if (modp!=-1) setmod(modp,modn);
00103 }
00104
00105 /*
00106     #] constructor (large constant polynomial) :
00107     #[ constructor (deep copy polynomial) :
00108 */
00109
00110 // copy constructor: makes a deep copy of a polynomial
00111 poly::poly (const poly &a, WORD modp, WORD modn):
00112     size_of_terms(AM.MaxTer/(LONG) sizeof(WORD))
00113 {
00114     #ifdef POLY_MOVE_DEBUG
00115         std::cout << "poly copy ctor" << std::endl;
00116     #endif
00117     POLY_GETIDENTITY(a);
00118     POLY_STOREIDENTITY;
00119 }

```

```

00120     terms = TermMalloc("poly::poly(poly&)");
00121
00122     *this = a;
00123
00124     if (modp!= -1) setmod(modp,modn);
00125 }
00126
00127 /*
00128  #] constructor (deep copy polynomial) :
00129  #[ destructor :
00130  */
00131
00132 // destructor
00133 poly::~poly () {
00134
00135     POLY_GETIDENTITY(*this);
00136
00137     if (size_of_terms == AM.MaxTer/(LONG)sizeof(WORD))
00138         TermFree(terms, "poly::~poly");
00139     else
00140         M_free(terms, "poly::~poly");
00141 }
00142
00143 /*
00144  #] destructor :
00145  #[ expand_memory :
00146  */
00147
00148 // expands the memory allocated for terms to at least twice its size
00149 void poly::expand_memory (int i) {
00150
00151     POLY_GETIDENTITY(*this);
00152
00153     LONG new_size_of_terms = MaX(2*size_of_terms, i);
00154     if (new_size_of_terms > MAXPOSITIVE) {
00155         MLOCK(ErrorMessageLock);
00156         MesPrint ((char*)"ERROR: polynomials too large (> WORDSIZE)");
00157         MUNLOCK(ErrorMessageLock);
00158         Terminate(1);
00159     }
00160
00161     WORD *newterms = (WORD *)Malloca(new_size_of_terms * sizeof(WORD), "poly::expand_memory");
00162     WCOPY(newterms, terms, size_of_terms);
00163
00164     if (size_of_terms == AM.MaxTer/(LONG)sizeof(WORD))
00165         TermFree(terms, "poly::expand_memory");
00166     else
00167         M_free(terms, "poly::expand_memory");
00168
00169     terms = newterms;
00170     size_of_terms = new_size_of_terms;
00171 }
00172
00173 /*
00174  #] expand_memory :
00175  #[ setmod :
00176  */
00177
00178 // sets the coefficient space to ZZ/p^n
00179 void poly::setmod(WORD _modp, WORD _modn) {
00180
00181     POLY_GETIDENTITY(*this);
00182
00183     if (_modp>0 && (_modp!=modp || _modn<modn)) {
00184         modp = _modp;
00185         modn = _modn;
00186
00187         WORD nmodq=0;
00188         UWORD *modq=NULL;
00189         bool smallq;
00190
00191         if (modn == 1) {
00192             modq = (UWORD *)&modp;
00193             nmodq = 1;
00194             smallq = true;
00195         }
00196         else {
00197             RaisPowCached(BHEAD modp,modn,&modq,&nmodq);
00198             smallq = false;
00199         }
00200
00201         coefficients_modulo(modq,nmodq,smallq);
00202     }
00203     else {
00204         modp = _modp;
00205         modn = _modn;
00206     }

```

```

00207 }
00208
00209 /*
00210  #] setmod :
00211  #[ coefficients_modulo :
00212 */
00213
00214 // reduces all coefficients of the polynomial modulo a
00215 void poly::coefficients_modulo (UWORD *a, WORD na, bool small) {
00216     POLY_GETIDENTITY(*this);
00217
00218     int j=1;
00219     for (int i=1, di; i<terms[0]; i+=di) {
00220         di = terms[i];
00221
00222         if (i!=j)
00223             for (int k=0; k<di; k++)
00224                 terms[j+k] = terms[i+k];
00225
00226         WORD n = terms[j+terms[j]-1];
00227
00228         if (ABS(n)==1 && small) {
00229             // small number modulo small number
00230             terms[j+AN.poly_num_vars] = n*((UWORD)terms[j+1+AN.poly_num_vars] % a[0]);
00231             if (terms[j+1+AN.poly_num_vars] == 0)
00232                 n=0;
00233             else {
00234                 if (terms[j+1+AN.poly_num_vars] > +(WORD)a[0]/2) terms[j+1+AN.poly_num_vars]-=a[0];
00235                 if (terms[j+1+AN.poly_num_vars] < -(WORD)a[0]/2) terms[j+1+AN.poly_num_vars]+=a[0];
00236                 n = SGN(terms[j+1+AN.poly_num_vars]);
00237                 terms[j+1+AN.poly_num_vars] = ABS(terms[j+1+AN.poly_num_vars]);
00238             }
00239         }
00240         else
00241             TakeNormalModulus((UWORD *)&terms[j+1+AN.poly_num_vars], &n, a, na, NOUNPACK);
00242
00243         if (n!=0) {
00244             terms[j] = 2+AN.poly_num_vars+ABS(n);
00245             terms[j+terms[j]-1] = n;
00246             j += terms[j];
00247         }
00248     }
00249     terms[0] = j;
00250 }
00251
00252 }
00253
00254 /*
00255  #] coefficients_modulo :
00256  #[ to_string :
00257 */
00258
00259 // converts an integer to a string
00260 const string int_to_string (WORD x) {
00261     char res[41];
00262     snprintf (res, 41, "%i",x);
00263     return res;
00264 }
00265
00266 // converts a polynomial to a string
00267 const string poly::to_string() const {
00268     POLY_GETIDENTITY(*this);
00269
00270     string res;
00271
00272     int printtimes;
00273     UBYTE *scoeff = (UBYTE *)NumberMalloc("poly::to_string");
00274
00275     if (terms[0]==1)
00276         // zero
00277         res = "0";
00278     else {
00279         for (int i=1; i<terms[0]; i+=terms[i]) {
00280             // sign
00281             WORD ncoeff = terms[i+terms[i]-1];
00282             if (ncoeff < 0) {
00283                 ncoeff*=-1;
00284                 res += "-";
00285             }
00286             else {
00287                 if (i>1) res += "+";
00288             }
00289
00290             if (ncoeff==1 && terms[i+terms[i]-1-ncoeff]==1) {
00291                 // coeff=1, so don't print coefficient and '*'

```

```

00294         printtimes = 0;
00295     }
00296     else {
00297         // print coefficient
00298         PRTLong((UWORD*)&terms[i+terms[i]-1-ncoeff], ncoeff, scoeff);
00299         res += string((char *)scoeff);
00300         printtimes=1;
00301     }
00302
00303     // print variables
00304     for (int j=0; j<AN.poly_num_vars; j++) {
00305         if (terms[i+1+j] > 0) {
00306             if (printtimes) res += "*";
00307             res += string(1,'a'+j);
00308             if (terms[i+1+j] > 1) res += "^" + int_to_string(terms[i+1+j]);
00309             printtimes = 1;
00310         }
00311     }
00312
00313     // iff coeff=1 and all power=0, print '1' after all
00314     if (!printtimes) res += "1";
00315 }
00316 }
00317
00318 // eventual modulo
00319 if (modp>0) {
00320     res += " (mod ";
00321     res += int_to_string(modp);
00322     if (modn>1) {
00323         res += "^";
00324         res += int_to_string(modn);
00325     }
00326     res += ")";
00327 }
00328
00329 NumberFree(scoeff,"poly::to_string");
00330
00331 return res;
00332 }
00333
00334 /*
00335 #] to_string :
00336 #[ ostream operator :
00337 */
00338
00339 // output stream operator
00340 ostream& operator<< (ostream &out, const poly &a) {
00341     return out << a.to_string();
00342 }
00343
00344 /*
00345 #] ostream operator :
00346 #[ monomial_compare :
00347 */
00348
00349 // compares two monomials (0:equal, <0:a smaller, >0:b smaller)
00350 int poly::monomial_compare (PHEAD const WORD *a, const WORD *b) {
00351     for (int i=0; i<AN.poly_num_vars; i++)
00352         if (a[i+1]!=b[i+1]) return a[i+1]-b[i+1];
00353     return 0;
00354 }
00355
00356 /*
00357 #] monomial_compare :
00358 #[ normalize :
00359 */
00360
00361 // brings a polynomial to normal form
00362 const poly & poly::normalize() {
00363
00364     POLY_GETIDENTITY(*this);
00365
00366     WORD nmodq=0;
00367     UWORD *modq=NULL;
00368
00369     if (modp!=0) {
00370         if (modn == 1) {
00371             modq = (UWORD *)&modp;
00372             nmodq = 1;
00373         }
00374         else {
00375             RaisPowCached(BHEAD modp,modn,&modq,&nmodq);
00376         }
00377     }
00378
00379     // find and sort all monomials
00380     // terms[0]/num_vars+3 is an upper bound for number of terms in a

```

```

00381
00382     LONG poffset = AT.pWorkPointer;
00383     WantAddPointers((terms[0]/(AN.poly_num_vars+3)));
00384     AT.pWorkPointer += terms[0]/(AN.poly_num_vars+3);
00385     #define p (&(AT.pWorkspace[poffset]))
00386
00387 // WORD **p = (WORD **)Malloc1(terms[0]/(AN.poly_num_vars+3) * sizeof(WORD*), "poly::normalize");
00388
00389     int nterms = 0;
00390     for (int i=1; i<terms[0]; i+=terms[i])
00391         p[nterms++] = terms + i;
00392
00393     sort(p, p + nterms, monomial_larger(BHEAD0));
00394
00395     WORD *tmp;
00396     if (size_of_terms == AM.MaxTer/(LONG)sizeof(WORD)) {
00397         tmp = (WORD *)TermMalloc("poly::normalize");
00398     }
00399     else {
00400         tmp = (WORD *)Malloc1(size_of_terms * sizeof(WORD), "poly::normalize");
00401     }
00402     int j=1;
00403     int prevj=0;
00404     tmp[0]=0;
00405     tmp[1]=0;
00406
00407     for (int i=0; i<nterms; i++) {
00408         if (i>0 && monomial_compare(BHEAD &tmp[j], p[i])==0) {
00409             // duplicate term, so add coefficients
00410             WORD ncoeff = tmp[j+tmp[j]-1];
00411             AddLong((UWORD *)&tmp[j+1+AN.poly_num_vars], ncoeff,
00412                 (UWORD *)&p[i][1+AN.poly_num_vars], p[i][p[i][0]-1],
00413                 (UWORD *)&tmp[j+1+AN.poly_num_vars], &ncoeff);
00414
00415             tmp[j+1+AN.poly_num_vars+ABS(ncoeff)] = ncoeff;
00416             tmp[j] = 2+AN.poly_num_vars+ABS(ncoeff);
00417         }
00418         else {
00419             // new term
00420             prevj = j;
00421             j += tmp[j];
00422             WCOPY(&tmp[j],p[i],p[i][0]);
00423         }
00424
00425         if (modp!=0) {
00426             // bring coefficient to normal form mod p^n
00427             WORD ntmp = tmp[j+tmp[j]-1];
00428             TakeNormalModulus((UWORD *)&tmp[j+1+AN.poly_num_vars], &ntmp,
00429                 modq,nmodq, NOUNPACK);
00430
00431             tmp[j] = 2+AN.poly_num_vars+ABS(ntmp);
00432             tmp[j+tmp[j]-1] = ntmp;
00433         }
00434
00435         // add terms to polynomial
00436         if (tmp[j+tmp[j]-1]==0) {
00437             tmp[j]=0;
00438             j=prevj;
00439         }
00440
00441         j+=tmp[j];
00442
00443         tmp[0] = j;
00444         WCOPY(terms,tmp,tmp[0]);
00445
00446 // M_free(p, "poly::normalize");
00447
00448         if (size_of_terms == AM.MaxTer/(LONG)sizeof(WORD))
00449             TermFree(tmp, "poly::normalize");
00450         else
00451             M_free(tmp, "poly::normalize");
00452
00453         AT.pWorkPointer = poffset;
00454         #undef p
00455
00456         return *this;
00457     }
00458
00459 /*
00460     #] normalize :
00461     #[ last_monomial_index :
00462 */
00463
00464 // index of the last monomial, i.e., the constant term
00465 WORD poly::last_monomial_index () const {
00466     POLY_GETIDENTITY(*this);
00467     return terms[0] - ABS(terms[terms[0]-1]) - AN.poly_num_vars - 2;

```



```

00468 }
00469
00470
00471 // pointer to the last monomial (the constant term)
00472 WORD * poly::last_monomial () const {
00473     return &terms[last_monomial_index()];
00474 }
00475
00476 /*
00477     #] last_monomial_index :
00478     #[ compare_degree_vector :
00479 */
00480
00481 int poly::compare_degree_vector(const poly & b) const {
00482     POLY_GETIDENTITY(*this);
00483
00484     // special cases if one or both are 0
00485     if (terms[0] == 1 && b[0] == 1) return 0;
00486     if (terms[0] == 1) return -1;
00487     if (b[0] == 1) return 1;
00488
00489     for (int i = 0; i < AN.poly_num_vars; i++) {
00490         int d1 = degree(i);
00491         int d2 = b.degree(i);
00492         if (d1 != d2) return d1 - d2;
00493     }
00494
00495     return 0;
00496 }
00497
00498 /*
00499     #] compare_degree_vector :
00500     #[ degree_vector :
00501 */
00502
00503 std::vector<int> poly::degree_vector() const {
00504     POLY_GETIDENTITY(*this);
00505
00506     std::vector<int> degrees(AN.poly_num_vars);
00507     for (int i = 0; i < AN.poly_num_vars; i++) {
00508         degrees[i] = degree(i);
00509     }
00510
00511     return degrees;
00512 }
00513
00514 /*
00515     #] degree_vector :
00516     #[ compare_degree_vector :
00517 */
00518
00519 int poly::compare_degree_vector(const vector<int> & b) const {
00520     POLY_GETIDENTITY(*this);
00521
00522     if (terms[0] == 1) return -1;
00523
00524     for (int i = 0; i < AN.poly_num_vars; i++) {
00525         int d1 = degree(i);
00526         if (d1 != b[i]) return d1 - b[i];
00527     }
00528
00529     return 0;
00530 }
00531
00532 /*
00533     #] compare_degree_vector :
00534     #[ add :
00535 */
00536
00537 // addition of polynomials by merging
00538 void poly::add (const poly &a, const poly &b, poly &c) {
00539     POLY_GETIDENTITY(a);
00540
00541     c.modp = a.modp;
00542     c.modn = a.modn;
00543
00544     WORD nmodq=0;
00545     UWORD *modq=NULL;
00546
00547     bool both_mod_small=false;
00548
00549     if (c.modp!=0) {
00550         if (c.modn == 1) {
00551             modq = (UWORD *)&c.modp;
00552             nmodq = 1;
00553             if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)

```

```

00555         both_mod_small=true;
00556     }
00557     else {
00558         RaisPowCached(BHEAD c.modp, c.modn, &modq, &nmodq);
00559     }
00560 }
00561
00562 int ai=1, bi=1, ci=1;
00563
00564 while (ai<a[0] || bi<b[0]) {
00565
00566     c.check_memory(ci);
00567
00568     int cmp = ai<a[0] && bi<b[0] ? monomial_compare(BHEAD &a[ai], &b[bi]) : 0;
00569
00570     if (bi==b[0] || cmp>0) {
00571         // insert term from a
00572         c.termscopy(&a[ai], ci, a[ai]);
00573         ci+=a[ai];
00574         ai+=a[ai];
00575     }
00576     else if (ai==a[0] || cmp<0) {
00577         // insert term from b
00578         c.termscopy(&b[bi], ci, b[bi]);
00579         ci+=b[bi];
00580         bi+=b[bi];
00581     }
00582     else {
00583         // insert term from a+b
00584         c.termscopy(&a[ai], ci, MaX(a[ai], b[bi]));
00585         WORD nc=0;
00586         if (both_mod_small) {
00587             c[ci+1+AN.poly_num_vars] = ((LONG)a[ai+1+AN.poly_num_vars]*a[ai+a[ai]-1]+
00588 (LONG)b[bi+1+AN.poly_num_vars]*b[bi+b[bi]-1]) % c.modp;
00589             if ((WORD)c[ci+1+AN.poly_num_vars] > +c.modp/2) c[ci+1+AN.poly_num_vars] -= c.modp;
00590             if ((WORD)c[ci+1+AN.poly_num_vars] < -c.modp/2) c[ci+1+AN.poly_num_vars] += c.modp;
00591             nc = (c[ci+1+AN.poly_num_vars]==0 ? 0 : SGN((WORD)c[ci+1+AN.poly_num_vars]));
00592             c[ci+1+AN.poly_num_vars] = ABS((WORD)c[ci+1+AN.poly_num_vars]);
00593         }
00594         else {
00595             AddLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
00596 (UWORD *)&b[bi+1+AN.poly_num_vars], b[bi+b[bi]-1],
00597 (UWORD *)&c[ci+1+AN.poly_num_vars], &nc);
00598             if (c.modp!=0) TakeNormalModulus((UWORD *)&c[ci+1+AN.poly_num_vars], &nc,
00599 modq, nmodq,
00600 NOUNPACK);
00601         }
00602
00603         if (nc!=0) {
00604             c[ci] = 2+AN.poly_num_vars+ABS(nc);
00605             c[ci+c[ci]-1] = nc;
00606             ci += c[ci];
00607         }
00608
00609         ai+=a[ai];
00610         bi+=b[bi];
00611     }
00612 }
00613 c[0]=ci;
00614 }
00615
00616 /*
00617 #] add :
00618 #[ sub :
00619 */
00620
00621 // subtraction of polynomials by merging
00622 void poly::sub (const poly &a, const poly &b, poly &c) {
00623
00624     POLY_GETIDENTITY(a);
00625
00626     c.modp = a.modp;
00627     c.modn = a.modn;
00628
00629     WORD nmodq=0;
00630     UWORD *modq=NULL;
00631
00632     bool both_mod_small=false;
00633
00634     if (c.modp!=0) {
00635         if (c.modn == 1) {
00636             modq = (UWORD *)&c.modp;
00637             nmodq = 1;
00638             if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)

```

```

00639         both_mod_small=true;
00640     }
00641     else {
00642         RaisPowCached(BHEAD c.modp, c.modn, &modq, &nmodq);
00643     }
00644 }
00645
00646 int ai=1, bi=1, ci=1;
00647
00648 while (ai<a[0] || bi<b[0]) {
00649     c.check_memory(ci);
00650
00651     int cmp = ai<a[0] && bi<b[0] ? monomial_compare(BHEAD &a[ai], &b[bi]) : 0;
00652
00653     if (bi==b[0] || cmp>0) {
00654         // insert term from a
00655         c.termscopy(&a[ai], ci, a[ai]);
00656         ci+=a[ai];
00657         ai+=a[ai];
00658     }
00659     else if (ai==a[0] || cmp<0) {
00660         // insert term from b
00661         c.termscopy(&b[bi], ci, b[bi]);
00662         ci+=b[bi];
00663         bi+=b[bi];
00664         c[ci-1]*=-1;
00665     }
00666     else {
00667         // insert term from a+b
00668         c.termscopy(&a[ai], ci, MaX(a[ai], b[bi]));
00669         WORD nc=0;
00670         if (both_mod_small) {
00671             c[ci+1+AN.poly_num_vars] = ((LONG)a[ai+1+AN.poly_num_vars]*a[ai+a[ai]-1]-
00672 (LONG)b[bi+1+AN.poly_num_vars]*b[bi+b[bi]-1]) % c.modp;
00673             if ((WORD)c[ci+1+AN.poly_num_vars] > +c.modp/2) c[ci+1+AN.poly_num_vars] -= c.modp;
00674             if ((WORD)c[ci+1+AN.poly_num_vars] < -c.modp/2) c[ci+1+AN.poly_num_vars] += c.modp;
00675             nc = (c[ci+1+AN.poly_num_vars]==0 ? 0 : SGN((WORD)c[ci+1+AN.poly_num_vars]));
00676             c[ci+1+AN.poly_num_vars] = ABS((WORD)c[ci+1+AN.poly_num_vars]);
00677         }
00678         else {
00679             AddLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
00680 (UWORD *)&b[bi+1+AN.poly_num_vars], -b[bi+b[bi]-1], // -b[...] causes
00681 subtraction
00682 (UWORD *)&c[ci+1+AN.poly_num_vars], &nc);
00683             if (c.modp!=0) TakeNormalModulus((UWORD *)&c[ci+1+AN.poly_num_vars], &nc,
00684 modq, nmodq,
00685 NOUNPACK);
00686         }
00687         if (nc!=0) {
00688             c[ci] = 2+AN.poly_num_vars+ABS(nc);
00689             c[ci+c[ci]-1] = nc;
00690             ci += c[ci];
00691         }
00692         ai+=a[ai];
00693         bi+=b[bi];
00694     }
00695 }
00696 }
00697
00698 c[0]=ci;
00699 }
00700
00701 /*
00702 #] sub :
00703 #[ pop_heap :
00704 */
00705
00706 // pops the largest monomial from the heap and stores it in heap[n]
00707 void poly::pop_heap (PHEAD WORD **heap, int n) {
00708     WORD *old = heap[0];
00709     heap[0] = heap[--n];
00710
00711     int i=0;
00712     while (2*i+2<n && (monomial_compare(BHEAD heap[2*i+1]+3, heap[i]+3)>0 ||
00713 monomial_compare(BHEAD heap[2*i+2]+3, heap[i]+3)>0)) {
00714         if (monomial_compare(BHEAD heap[2*i+1]+3, heap[2*i+2]+3)>0) {
00715             swap(heap[i], heap[2*i+1]);
00716             i=2*i+1;
00717         }
00718         else {
00719             swap(heap[i], heap[2*i+2]);
00720             i=2*i+2;
00721         }
00722     }

```

```

00722         swap(heap[i], heap[2*i+2]);
00723         i=2*i+2;
00724     }
00725 }
00726
00727 if (2*i+1<n && monomial_compare(BHEAD heap[2*i+1]+3, heap[i]+3)>0)
00728     swap(heap[i], heap[2*i+1]);
00729
00730 heap[n] = old;
00731 }
00732
00733 /*
00734 #] pop_heap :
00735 #[ push_heap :
00736 */
00737
00738 // pushes the monomial in heap[n] onto the heap
00739 void poly::push_heap (PHEAD WORD **heap, int n) {
00740
00741     int i=n-1;
00742
00743     while (i>0 && monomial_compare(BHEAD heap[i]+3, heap[(i-1)/2]+3) > 0) {
00744         swap(heap[(i-1)/2], heap[i]);
00745         i=(i-1)/2;
00746     }
00747 }
00748
00749 /*
00750 #] push_heap :
00751 #[ mul_one_term :
00752 */
00753
00754 // multiplies a polynomial with a monomial
00755 void poly::mul_one_term (const poly &a, const poly &b, poly &c) {
00756
00757     POLY_GETIDENTITY(a);
00758
00759     int ci=1;
00760
00761     WORD nmodq=0;
00762     UWORD *modq=NULL;
00763
00764     bool both_mod_small=false;
00765
00766     if (c.modp!=0) {
00767         if (c.modn == 1) {
00768             modq = (UWORD *)&c.modp;
00769             nmodq = 1;
00770             if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
00771                 both_mod_small=true;
00772         }
00773         else {
00774             RaisPowCached(BHEAD c.modp, c.modn, &modq, &nmodq);
00775         }
00776     }
00777
00778     for (int ai=1; ai<a[0]; ai+=a[ai])
00779         for (int bi=1; bi<b[0]; bi+=b[bi]) {
00780
00781             c.check_memory(ci);
00782
00783             for (int i=0; i<AN.poly_num_vars; i++)
00784                 c[ci+1+i] = a[ai+1+i] + b[bi+1+i];
00785             WORD nc;
00786
00787             // if both polynomials are modulo p^1, use integer calculus
00788             if (both_mod_small) {
00789                 c[ci+1+AN.poly_num_vars] =
00790                     ((LONG)a[ai+1+AN.poly_num_vars] * a[ai+2+AN.poly_num_vars] *
00791                      b[bi+1+AN.poly_num_vars] * b[bi+2+AN.poly_num_vars]) % c.modp;
00792                 nc = (c[ci+1+AN.poly_num_vars]==0 ? 0 : 1);
00793             }
00794             else {
00795                 // otherwise, use form long calculus
00796                 MulLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
00797                        (UWORD *)&b[bi+1+AN.poly_num_vars], b[bi+b[bi]-1],
00798                        (UWORD *)&c[ci+1+AN.poly_num_vars], &nc);
00799                 if (c.modp!=0) TakeNormalModulus((UWORD *)&c[ci+1+AN.poly_num_vars], &nc,
00800                                                  modq, nmodq,
00801
00802 NOUNPACK);
00803             }
00804
00805             if (nc!=0) {
00806                 c[ci] = 2+AN.poly_num_vars+ABS(nc);
00807                 ci += c[ci];
00808
00809                 if (both_mod_small) {

```

```

00808             if (c[ci-2] > +c.modp/2) c[ci-2] -= c.modp;
00809             if (c[ci-2] < -c.modp/2) c[ci-2] += c.modp;
00810             c[ci-1] = SGN(c[ci-2]);
00811             c[ci-2] = ABS(c[ci-2]);
00812         }
00813         else {
00814             c[ci-1] = nc;
00815         }
00816     }
00817 }
00818
00819 c[0]=ci;
00820 }
00821
00822 /*
00823 #] mul_one_term :
00824 #[ mul_univar :
00825 */
00826
00827 // dense univariate multiplication, i.e., for each power find all
00828 // pairs of monomials that result in that power
00829 void poly::mul_univar (const poly &a, const poly &b, poly &c, int var) {
00830
00831     POLY_GETIDENTITY(a);
00832
00833     WORD nmodq=0;
00834     UWORD *modq=NULL;
00835
00836     bool both_mod_small=false;
00837
00838     if (c.modp!=0) {
00839         if (c.modn == 1) {
00840             modq = (UWORD *)&c.modp;
00841             nmodq = 1;
00842             if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
00843                 both_mod_small=true;
00844         }
00845         else {
00846             RaisPowCached(BHEAD c.modp, c.modn, &modq, &nmodq);
00847         }
00848     }
00849
00850     poly t(BHEAD 0);
00851     WORD nt;
00852
00853     int ci=1;
00854
00855     // bounds on the powers in a*b
00856     int minpow = AN.poly_num_vars==0 ? 0 : a.last_monomial()[1+var] + b.last_monomial()[1+var];
00857     int maxpow = AN.poly_num_vars==0 ? 0 : a[2+var]+b[2+var];
00858     int afirst=1, blast=1;
00859
00860     for (int pow=maxpow; pow>=minpow; pow--) {
00861
00862         c.check_memory(ci);
00863
00864         WORD nc=0;
00865
00866         // adjust range in a or b
00867         if (a[afirst+1+var] + b[blast+1+var] > pow) {
00868             if (blast+b[blast] < b[0])
00869                 blast+=b[blast];
00870             else
00871                 afirst+=a[afirst];
00872         }
00873
00874         // find terms that result in the correct power
00875         for (int ai=afirst, bi=blast; ai<a[0] && bi>=1;) {
00876
00877             int thispow = AN.poly_num_vars==0 ? 0 : a[ai+1+var] + b[bi+1+var];
00878
00879             if (thispow == pow) {
00880
00881                 // if both polynomials are modulo p^1, use integer calculus
00882                 if (both_mod_small) {
00883                     c[ci+1+AN.poly_num_vars] =
00884                         ((nc==0 ? 0 : (LONG)c[ci+1+AN.poly_num_vars] * nc) +
00885                         (LONG)a[ai+1+AN.poly_num_vars] * a[ai+2+AN.poly_num_vars] *
00886                         b[bi+1+AN.poly_num_vars] * b[bi+2+AN.poly_num_vars]) % c.modp;
00887                     nc = (c[ci+1+AN.poly_num_vars]==0 ? 0 : 1);
00888                 }
00889                 else {
00890                     // otherwise, use form long calculus
00891
00892                     MulLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
00893                             (UWORD *)&b[bi+1+AN.poly_num_vars], b[bi+b[bi]-1],
00894                             (UWORD *)&t[0], &nt);

```

```

00895
00896         AddLong ((UWORD *)t[0], nt,
00897                 (UWORD *)&c[ci+1+AN.poly_num_vars], nc,
00898                 (UWORD *)&c[ci+1+AN.poly_num_vars], &nc);
00899
00900         if (c.modp!=0) TakeNormalModulus((UWORD *)&c[ci+1+AN.poly_num_vars], &nc,
00901                                         modq, nmodq,
NOUNPACK);
00902     }
00903
00904     ai += a[ai];
00905     bi -= ABS(b[bi-1]) + 2 + AN.poly_num_vars;
00906 }
00907 else if (thispow > pow)
00908     ai += a[ai];
00909 else
00910     bi -= ABS(b[bi-1]) + 2 + AN.poly_num_vars;
00911 }
00912
00913 // add term to result
00914 if (nc != 0) {
00915     for (int j=0; j<AN.poly_num_vars; j++)
00916         c[ci+1+j] = 0;
00917     if (AN.poly_num_vars > 0)
00918         c[ci+1+var] = pow;
00919
00920     c[ci] = 2+AN.poly_num_vars+ABS(nc);
00921     ci += c[ci];
00922
00923     // if necessary, adjust to range [-p/2,p/2]
00924     if (both_mod_small) {
00925         if (c[ci-2] > +c.modp/2) c[ci-2] -= c.modp;
00926         if (c[ci-2] < -c.modp/2) c[ci-2] += c.modp;
00927         c[ci-1] = SGN(c[ci-2]);
00928         c[ci-2] = ABS(c[ci-2]);
00929     }
00930     else {
00931         c[ci-1] = nc;
00932     }
00933 }
00934 }
00935
00936 c[0] = ci;
00937 }
00938
00939 /*
00940 #] mul_univar :
00941 #[ mul_heap :
00942 */
00943
00961 void poly::mul_heap (const poly &a, const poly &b, poly &c) {
00962
00963     POLY_GETIDENTITY(a);
00964
00965     WORD nmodq=0;
00966     UWORD *modq=NULL;
00967
00968     bool both_mod_small=false;
00969
00970     if (c.modp!=0) {
00971         if (c.modn == 1) {
00972             modq = (UWORD *)&c.modp;
00973             nmodq = 1;
00974             if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
00975                 both_mod_small=true;
00976         }
00977         else {
00978             RaisPowCached(BHEAD c.modp, c.modn, &modq, &nmodq);
00979         }
00980     }
00981
00982     // find maximum powers in different variables
00983     WORD *maxpower = AT.WorkPointer;
00984     AT.WorkPointer += AN.poly_num_vars;
00985     WORD *maxpowera = AT.WorkPointer;
00986     AT.WorkPointer += AN.poly_num_vars;
00987     WORD *maxpowerb = AT.WorkPointer;
00988     AT.WorkPointer += AN.poly_num_vars;
00989
00990     for (int i=0; i<AN.poly_num_vars; i++)
00991         maxpowera[i] = maxpowerb[i] = 0;
00992
00993     for (int ai=1; ai<a[0]; ai+=a[ai])
00994         for (int j=0; j<AN.poly_num_vars; j++)
00995             maxpowera[j] = MaX(maxpowera[j], a[ai+1+j]);
00996
00997     for (int bi=1; bi<b[0]; bi+=b[bi])

```

```

00998         for (int j=0; j<AN.poly_num_vars; j++)
00999             maxpowerb[j] = MaX(maxpowerb[j], b[bi+1+j]);
01000
01001     for (int i=0; i<AN.poly_num_vars; i++)
01002         maxpower[i] = maxpowera[i] + maxpowerb[i];
01003
01004     // if PROD(maxpower) small, use hash array
01005     bool use_hash = true;
01006     int nhash = 1;
01007
01008     for (int i=0; i<AN.poly_num_vars; i++) {
01009         if (nhash > POLY_MAX_HASH_SIZE / (maxpower[i]+1)) {
01010             nhash = 1;
01011             use_hash = false;
01012             break;
01013         }
01014         nhash *= maxpower[i]+1;
01015     }
01016
01017     // allocate heap and hash
01018     int nheap=a.number_of_terms();
01019
01020     WantAddPointers(nheap+nhash);
01021     WORD **heap = AT.pWorkSpace + AT.pWorkPointer;
01022
01023     for (int ai=1, i=0; ai<a[0]; ai+=a[ai], i++) {
01024         heap[i] = (WORD *) NumberMalloc("poly::mul_heap");
01025         heap[i][0] = ai;
01026         heap[i][1] = 1;
01027         heap[i][2] = -1;
01028         heap[i][3] = 0;
01029         for (int j=0; j<AN.poly_num_vars; j++)
01030             heap[i][4+j] = MAXPOSITIVE;
01031     }
01032
01033     WORD **hash = AT.pWorkSpace + AT.pWorkPointer + nheap;
01034     for (int i=0; i<nhash; i++)
01035         hash[i] = NULL;
01036
01037     int ci = 1;
01038
01039     // multiply
01040     while (nheap > 0) {
01041
01042         c.check_memory(ci);
01043
01044         pop_heap(BHEAD heap, nheap--);
01045         WORD *p = heap[nheap];
01046
01047         // if non-zero
01048         if (p[3] != 0) {
01049             if (use_hash) hash[p[2]] = NULL;
01050
01051             c[0] = ci;
01052
01053             // append this term to the result
01054             if (use_hash || ci==1 || monomial_compare(BHEAD p+3, c.last_monomial())!=0) {
01055                 p[4 + AN.poly_num_vars + ABS(p[3])] = p[3];
01056                 p[3] = 2 + AN.poly_num_vars + ABS(p[3]);
01057                 c.termscopy(&p[3],ci,p[3]);
01058                 ci += c[ci];
01059             }
01060             else {
01061                 // add this term to the last term of the result
01062                 ci = c.last_monomial_index();
01063                 WORD nc = c[ci+c[ci]-1];
01064
01065                 // if both polynomials are modulo p^1, use integer calculus
01066                 if (both_mod_small) {
01067                     c[ci+AN.poly_num_vars+1] = ((LONG)c[ci+AN.poly_num_vars+1]*nc +
01068 p[4+AN.poly_num_vars]*p[3]) % c.modp;
01069                     if (c[ci+1+AN.poly_num_vars]==0)
01070                         nc = 0;
01071                     else {
01072                         if (c[ci+1+AN.poly_num_vars] > +c.modp/2) c[ci+1+AN.poly_num_vars] -= c.modp;
01073                         if (c[ci+1+AN.poly_num_vars] < -c.modp/2) c[ci+1+AN.poly_num_vars] += c.modp;
01074                         nc = SGN(c[ci+1+AN.poly_num_vars]);
01075                         c[ci+1+AN.poly_num_vars] = ABS(c[ci+1+AN.poly_num_vars]);
01076                     }
01077                 }
01078                 else {
01079                     // otherwise, use form long calculus
01080                     AddLong ((UWORD *)&p[4+AN.poly_num_vars], p[3],
01081                             (UWORD *)&c[ci+AN.poly_num_vars+1], nc,
01082                             (UWORD *)&c[ci+AN.poly_num_vars+1],&nc);
01083
01084                     if (c.modp!=0) TakeNormalModulus((UWORD *)&c[ci+1+AN.poly_num_vars], &nc,

```

```

01084                                     modq, nmodq,
NOUNPACK);
01085     }
01086
01087     if (nc!=0) {
01088         c[ci] = 2 + AN.poly_num_vars + ABS(nc);
01089         ci += c[ci];
01090         c[ci-1] = nc;
01091     }
01092 }
01093 }
01094
01095 // add new term to the heap (ai, bi+1)
01096 while (p[1] < b[0]) {
01097
01098     for (int j=0; j<AN.poly_num_vars; j++)
01099         p[4+j] = a[p[0]+1+j] + b[p[1]+1+j];
01100
01101     // if both polynomials are modulo p^1, use integer calculus
01102     if (both_mod_small) {
01103         p[4+AN.poly_num_vars] = ((LONG)a[p[0]+1+AN.poly_num_vars]*a[p[0]+a[p[0]]-1]*
01104 b[p[1]+1+AN.poly_num_vars]*b[p[1]+b[p[1]]-1]) % c.modp;
01105         if (p[4+AN.poly_num_vars]==0)
01106             p[3]=0;
01107         else {
01108             if (p[4+AN.poly_num_vars] > +c.modp/2) p[4+AN.poly_num_vars] -= c.modp;
01109             if (p[4+AN.poly_num_vars] < -c.modp/2) p[4+AN.poly_num_vars] += c.modp;
01110             p[3] = SGN(p[4+AN.poly_num_vars]);
01111             p[4+AN.poly_num_vars] = ABS(p[4+AN.poly_num_vars]);
01112         }
01113     }
01114     else {
01115         // otherwise, use form long calculus
01116         MulLong ((UWORD *)&a[p[0]+1+AN.poly_num_vars], a[p[0]+a[p[0]]-1],
01117 (UWORD *)&b[p[1]+1+AN.poly_num_vars], b[p[1]+b[p[1]]-1],
01118 (UWORD *)&p[4+AN.poly_num_vars], &p[3]);
01119
01120         if (c.modp!=0) TakeNormalModulus((UWORD *)&p[4+AN.poly_num_vars], &p[3],
01121                                     modq, nmodq,
NOUNPACK);
01122     }
01123
01124     p[1] += b[p[1]];
01125
01126     if (use_hash) {
01127         int ID = 0;
01128         for (int i=0; i<AN.poly_num_vars; i++)
01129             ID = (maxpower[i]+1)*ID + p[4+i];
01130
01131         // if hash and unused, push onto heap
01132         if (hash[ID] == NULL) {
01133             p[2] = ID;
01134             hash[ID] = p;
01135             push_heap(BHEAD heap, ++nheap);
01136             break;
01137         }
01138         else {
01139             // if hash and used, add to heap element
01140             WORD *h = hash[ID];
01141             // if both polynomials are modulo p^1, use integer calculus
01142             if (both_mod_small) {
01143                 h[4+AN.poly_num_vars] = ((LONG)p[4+AN.poly_num_vars]*p[3] +
h[4+AN.poly_num_vars]*h[3]) % c.modp;
01144                 if (h[4+AN.poly_num_vars]==0)
01145                     h[3]=0;
01146                 else {
01147                     if (h[4+AN.poly_num_vars] > +c.modp/2) h[4+AN.poly_num_vars] -= c.modp;
01148                     if (h[4+AN.poly_num_vars] < -c.modp/2) h[4+AN.poly_num_vars] += c.modp;
01149                     h[3] = SGN(h[4+AN.poly_num_vars]);
01150                     h[4+AN.poly_num_vars] = ABS(h[4+AN.poly_num_vars]);
01151                 }
01152             }
01153             else {
01154                 // otherwise, use form long calculus
01155                 AddLong ((UWORD *)&p[4+AN.poly_num_vars], p[3],
01156 (UWORD *)&h[4+AN.poly_num_vars], h[3],
01157 (UWORD *)&h[4+AN.poly_num_vars], &h[3]);
01158
01159                 if (c.modp!=0) TakeNormalModulus((UWORD *)&h[4+AN.poly_num_vars], &h[3],
01160                                     modq, nmodq,
NOUNPACK);
01161             }
01162         }
01163     }
01164     else {
01165         // if no hash, push onto heap

```



```

01166         p[2] = -1;
01167         push_heap(BHEAD heap, ++nheap);
01168         break;
01169     }
01170 }
01171 }
01172
01173 c[0] = ci;
01174
01175 for (int ai=1, i=0; ai<a[0]; ai+=a[ai], i++)
01176     NumberFree(heap[i], "poly::mul_heap");
01177 AT.WorkPointer -= 3 * AN.poly_num_vars;
01178 }
01179
01180 /*
01181  #] mul_heap :
01182  #[ mul :
01183  */
01184
01195 void poly::mul (const poly &a, const poly &b, poly &c) {
01196
01197     c.modp = a.modp;
01198     c.modn = a.modn;
01199
01200     if (a.is_zero() || b.is_zero()) { c[0]=1; return; }
01201     if (a.is_one()) {
01202         c.check_memory(b[0]);
01203         c.termscopy(b.terms,0,b[0]);
01204         // normalize if necessary
01205         if (a.modp!=b.modp || a.modn!=b.modn) {
01206             c.modp=0;
01207             c.setmod(a.modp,a.modn);
01208         }
01209         return;
01210     }
01211     if (b.is_one()) {
01212         c.check_memory(a[0]);
01213         c.termscopy(a.terms,0,a[0]);
01214         return;
01215     }
01216
01217     int na=a.number_of_terms();
01218     int nb=b.number_of_terms();
01219
01220     if (na==1 || nb==1) {
01221         mul_one_term(a,b,c);
01222         return;
01223     }
01224
01225     int vara = a.is_dense_univariate();
01226     int varb = b.is_dense_univariate();
01227
01228     if (vara!=-2 && varb!=-2 && vara==varb) {
01229         mul_univar(a,b,c,vara);
01230         return;
01231     }
01232
01233     if (na < nb)
01234         mul_heap(a,b,c);
01235     else
01236         mul_heap(b,a,c);
01237 }
01238
01239 /*
01240  #] mul :
01241  #[ divmod_one_term : :
01242  */
01243
01244 // divides a polynomial by a monomial
01245 void poly::divmod_one_term (const poly &a, const poly &b, poly &q, poly &r, bool only_divides) {
01246
01247     POLY_GETIDENTITY(a);
01248
01249     int qi=1, ri=1;
01250
01251     WORD nmodq=0;
01252     UWORD *modq=NULL;
01253
01254     WORD nltbinv=0;
01255     UWORD *ltbinv=NULL;
01256
01257     bool both_mod_small=false;
01258
01259     if (q.modp!=0) {
01260         if (q.modn == 1) {
01261             modq = (UWORD *) &q.modp;
01262             nmodq = 1;

```

```

01263         if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
01264             both_mod_small=true;
01265     }
01266     else {
01267         RaisPowCached(BHEAD q.modp,q.modn,&modq,&nmodq);
01268     }
01269     ltbinv = NumberMalloc("poly::div_one_term");
01270
01271     if (both_mod_small) {
01272         WORD ltb = b[b[1]]*b[2+AN.poly_num_vars];
01273         GetModInverses(ltb + (ltb<0?q.modp:0), q.modp, (WORD*)ltbinv, NULL);
01274         nltbinv = 1;
01275     }
01276     else
01277         GetLongModInverses(BHEAD (UWORD *)&b[2+AN.poly_num_vars], b[b[1]], modq, nmodq, ltbinv,
01278 &nltbinv, NULL, NULL);
01279 }
01280
01281 for (int ai=1; ai<a[0]; ai+=a[ai]) {
01282
01283     q.check_memory(qi);
01284     r.check_memory(ri);
01285
01286     // check divisibility of powers
01287     bool div=true;
01288     for (int j=0; j<AN.poly_num_vars; j++) {
01289         q[qi+1+j] = a[ai+1+j]-b[2+j];
01290         r[ri+1+j] = a[ai+1+j];
01291         if (q[qi+1+j] < 0) div=false;
01292     }
01293
01294     WORD nq,nr;
01295
01296     if (div) {
01297         // if variables are divisible, divide coefficient
01298         if (q.modp==0) {
01299             DivLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
01300 (UWORD *)&b[2+AN.poly_num_vars], b[b[1]],
01301 (UWORD *)&q[qi+1+AN.poly_num_vars], &nq,
01302 (UWORD *)&r[ri+1+AN.poly_num_vars], &nr);
01303         }
01304         else {
01305             // if both polynomials are modulo p^1, use integer calculus
01306             if (both_mod_small) {
01307                 q[qi+1+AN.poly_num_vars] =
01308 (LONG)a[ai+1+AN.poly_num_vars] * a[ai+a[ai]-1] * ltbinv[0] * nltbinv) %
01309 q.modp;
01310                 nq = (q[qi+1+AN.poly_num_vars]==0 ? 0 : 1);
01311             }
01312             else {
01313                 // otherwise, use form long calculus
01314                 MulLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
01315 ltbinv, nltbinv,
01316 (UWORD *)&q[qi+1+AN.poly_num_vars], &nq);
01317                 TakeNormalModulus((UWORD *)&q[qi+1+AN.poly_num_vars], &nq,
01318 modq,nmodq, NOUNPACK);
01319             }
01320             nr=0;
01321         }
01322     }
01323     else {
01324         // if not, term becomes part of the remainder
01325         nq=0;
01326         nr=a[ai+a[ai]-1];
01327         r.termscopy(&a[ai+1+AN.poly_num_vars], ri+1+AN.poly_num_vars, ABS(nr));
01328     }
01329
01330     // add terms to quotient/remainder
01331     if (nq!=0) {
01332         q[qi] = 2+AN.poly_num_vars+ABS(nq);
01333         qi += q[qi];
01334
01335         // if necessary, adjust to range [-p/2,p/2]
01336         if (both_mod_small) {
01337             if (q[qi-2] > +q.modp/2) q[qi-2] -= q.modp;
01338             if (q[qi-2] < -q.modp/2) q[qi-2] += q.modp;
01339             q[qi-1] = SGN(q[qi-2]);
01340             q[qi-2] = ABS(q[qi-2]);
01341         }
01342         else {
01343             q[qi-1] = nq;
01344         }
01345     }
01346
01347     if (nr != 0) {

```

```

01348         if (only_divides) { r = poly(BHEAD 1); ri=r[0]; break; }
01349         r[ri] = 2+AN.poly_num_vars+ABS(nr);
01350         ri += r[ri];
01351         r[ri-1] = nr;
01352     }
01353 }
01354
01355 q[0]=qi;
01356 r[0]=ri;
01357
01358 if (q.modp!=0 || ltbinv != NULL) NumberFree(ltbinv,"poly::div_one_term");
01359 }
01360
01361 /*
01362 #] divmod_one_term :
01363 #[ divmod_univar : :
01364 */
01365
01376 void poly::divmod_univar (const poly &a, const poly &b, poly &q, poly &r, int var, bool only_divides)
01377 {
01378     POLY_GETIDENTITY(a);
01379
01380     WORD nmodq=0;
01381     UWORD *modq=NULL;
01382
01383     WORD nltbinv=0;
01384     UWORD *ltbinv=NULL;
01385
01386     bool both_mod_small=false;
01387
01388     if (q.modp!=0) {
01389         if (q.modn == 1) {
01390             modq = (UWORD *) &q.modp;
01391             nmodq = 1;
01392             if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
01393                 both_mod_small=true;
01394         }
01395         else {
01396             RaisPowCached(BHEAD q.modp,q.modn,&modq,&nmodq);
01397         }
01398         ltbinv = NumberMalloc("poly::div_univar");
01399
01400         if (both_mod_small) {
01401             WORD ltb = b[b[1]]*b[2+AN.poly_num_vars];
01402             GetModInverses(ltb + (ltb<0?q.modp:0), q.modp, (WORD*)ltbinv, NULL);
01403             nltbinv = 1;
01404         }
01405         else
01406             GetLongModInverses(BHEAD (UWORD *) &b[2+AN.poly_num_vars], b[b[1]], modq, nmodq, ltbinv,
01407 &nltbinv, NULL, NULL);
01408     }
01409
01410     WORD ns=0;
01411     WORD nt;
01412     UWORD *s = NumberMalloc("poly::div_univar");
01413     UWORD *t = NumberMalloc("poly::div_univar");
01414     s[0]=0;
01415
01416     int bpow = b[2+var];
01417
01418     int ai=1, qi=1, ri=1;
01419
01420     for (int pow=a[2+var]; pow>=0; pow--) {
01421         q.check_memory(qi);
01422         r.check_memory(ri);
01423
01424         // look for the correct power in a
01425         while (ai<a[0] && a[ai+1+var] > pow)
01426             ai+=a[ai];
01427
01428         // first term of the r.h.s. of the above equation
01429         if (ai<a[0] && a[ai+1+var] == pow) {
01430             ns = a[ai+a[ai]-1];
01431             WCOPY(s, &a[ai+1+AN.poly_num_vars], ABS(ns));
01432         }
01433         else {
01434             ns = 0;
01435         }
01436
01437         int bi=1, qj=qi;
01438
01439         // second term(s) of the r.h.s. of the above equation
01440         while (qj>1 && bi<b[0]) {
01441
01442             qj -= 2 + AN.poly_num_vars + ABS(q[qj-1]);

```

```

01443
01444 while (bi<b[0] && b[bi+1+var]+q[qj+1+var] > pow)
01445     bi += b[bi];
01446
01447 if (bi<b[0] && b[bi+1+var]+q[qj+1+var] == pow) {
01448     // if both polynomials are modulo p^1, use integer calculus
01449
01450     if (both_mod_small) {
01451         s[0] = ((WORD)s[0]*ns - (LONG)b[bi+1+AN.poly_num_vars] * b[bi+b[bi]-1] *
01452             q[qj+1+AN.poly_num_vars] * q[qj+q[qj]-1]) % q.modp;
01453         ns = (s[0]==0 ? 0 : 1);
01454     }
01455     else {
01456         // otherwise, use form long calculus
01457         MulLong((UWORD *)&b[bi+1+AN.poly_num_vars], b[bi+b[bi]-1],
01458             (UWORD *)&q[qj+1+AN.poly_num_vars], q[qj+q[qj]-1],
01459             t, &nt);
01460         nt *= -1;
01461         AddLong(t,nt,s,ns,s,&ns);
01462         if (q.modp!=0) TakeNormalModulus((UWORD *)s,&ns,modq, nmodq, NOUNPACK);
01463     }
01464 }
01465 }
01466
01467 if (ns != 0) {
01468     // if necessary, adjust to range [-p/2,p/2]
01469     if (both_mod_small) {
01470         s[0]*=ns;
01471         if ((WORD)s[0] > +q.modp/2) s[0] -= q.modp;
01472         if ((WORD)s[0] < -q.modp/2) s[0] += q.modp;
01473         ns = SGN((WORD)s[0]);
01474         s[0] = ABS((WORD)s[0]);
01475     }
01476
01477     if (pow >= bpow) {
01478         // large power, so divide by b
01479         if (q.modp == 0) {
01480             DivLong(s, ns,
01481                 (UWORD *)&b[2+AN.poly_num_vars], b[b[1]],
01482                 (UWORD *)&q[qi+1+AN.poly_num_vars], &ns, t, &nt);
01483         }
01484         else {
01485             if (both_mod_small) {
01486                 q[qi+1+AN.poly_num_vars] = ((LONG)s[0]*ns*ltbinv[0]*nltbinv) % q.modp;
01487                 if ((WORD)q[qi+1+AN.poly_num_vars] > +q.modp/2) q[qi+1+AN.poly_num_vars] -=
q.modp;
01488                 if ((WORD)q[qi+1+AN.poly_num_vars] < -q.modp/2) q[qi+1+AN.poly_num_vars] +=
q.modp;
01489                 ns = (q[qi+1+AN.poly_num_vars]==0 ? 0 : SGN((WORD)q[qi+1+AN.poly_num_vars]));
01490                 q[qi+1+AN.poly_num_vars] = ABS((WORD)q[qi+1+AN.poly_num_vars]);
01491             }
01492             else {
01493                 MulLong(s, ns, ltbinv, nltbinv, (UWORD *)&q[qi+1+AN.poly_num_vars], &ns);
01494                 TakeNormalModulus((UWORD *)&q[qi+1+AN.poly_num_vars], &ns,
01495                     modq,nmodq, NOUNPACK);
01496             }
01497             nt=0;
01498         }
01499     }
01500     else {
01501         // small power, so remainder
01502         WCOPY(t,s,ABS(ns));
01503         nt = ns;
01504         ns = 0;
01505     }
01506
01507     // add terms to quotient/remainder
01508     if (ns!=0) {
01509         for (int i=0; i<AN.poly_num_vars; i++)
01510             q[qi+1+i] = 0;
01511         q[qi+1+var] = pow-bpow;
01512
01513         q[qi] = 2+AN.poly_num_vars+ABS(ns);
01514         qi += q[qi];
01515         q[qi-1] = ns;
01516     }
01517
01518     if (nt != 0) {
01519         if (only_divides) { r=poly(BHEAD 1); ri=r[0]; break; }
01520         for (int i=0; i<AN.poly_num_vars; i++)
01521             r[ri+1+i] = 0;
01522         r[ri+1+var] = pow;
01523
01524         for (int i=0; i<ABS(nt); i++)
01525             r[ri+1+AN.poly_num_vars+i] = t[i];
01526

```

```

01527             r[ri] = 2+AN.poly_num_vars+ABS(nt);
01528             ri += r[ri];
01529             r[ri-1] = nt;
01530         }
01531     }
01532 }
01533
01534 q[0] = qi;
01535 r[0] = ri;
01536
01537 NumberFree(s,"poly::div_univar");
01538 NumberFree(t,"poly::div_univar");
01539
01540 if (q.modp!=0 || ltbinv!=NULL) NumberFree(ltbinv,"poly::div_univar");
01541 }
01542
01543 /*
01544 #] divmod_univar :
01545 #[ divmod_heap :
01546 */
01547
01579 void poly::divmod_heap (const poly &a, const poly &b, poly &q, poly &r, bool only_divides, bool
check_div, bool &div_fail) {
01580
01581     POLY_GETIDENTITY(a);
01582
01583     div_fail = false;
01584     q[0] = r[0] = 1;
01585
01586     WORD nmodq=0;
01587     UWORD *modq=NULL;
01588
01589     WORD nltbinv=0;
01590     UWORD *ltbinv=NULL;
01591     LONG oldpWorkPointer = AT.pWorkPointer;
01592
01593     bool both_mod_small=false;
01594
01595     if (q.modp!=0) {
01596         if (q.modn == 1) {
01597             modq = (UWORD *) &q.modp;
01598             nmodq = 1;
01599             if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
01600                 both_mod_small=true;
01601         }
01602         else {
01603             RaisPowCached(BHEAD q.modp,q.modn,&modq,&nmodq);
01604         }
01605         ltbinv = NumberMalloc("poly::div_heap-a");
01606
01607         if (both_mod_small) {
01608             WORD ltb = b[b[1]]*b[2+AN.poly_num_vars];
01609             GetModInverses(ltb + (ltb<0?q.modp:0), q.modp, (WORD*)ltbinv, NULL);
01610             nltbinv = 1;
01611         }
01612         else
01613             GetLongModInverses(BHEAD (UWORD *) &b[2+AN.poly_num_vars], b[b[1]], modq, nmodq, ltbinv,
&nltbinv, NULL, NULL);
01614     }
01615
01616     // allocate heap
01617     int nb=b.number_of_terms();
01618     WantAddPointers(nb);
01619     WORD **heap = AT.pWorkSpace + AT.pWorkPointer;
01620     AT.pWorkPointer += nb;
01621
01622     for (int i=0; i<nb; i++)
01623         heap[i] = (WORD *) NumberMalloc("poly::div_heap-b");
01624     int nheap = 1;
01625     heap[0][0] = 1;
01626     heap[0][1] = 0;
01627     heap[0][2] = -1;
01628     WCOPY(&heap[0][3], &a[1], a[1]);
01629     heap[0][3] = a[a[1]];
01630
01631     int qi=1, ri=1;
01632
01633     int s = nb;
01634     WORD *t = (WORD *) NumberMalloc("poly::div_heap-c");
01635
01636     // insert contains element that still have to be inserted to the heap
01637     // (exists to avoid code duplication).
01638     vector<pair<int,int> > insert;
01639
01640     while (insert.size()>0 || nheap>0) {
01641
01642         q.check_memory(qi);

```

```

01643     r.check_memory(ri);
01644
01645     // collect a term t for the quotient/remainder
01646     t[0] = -1;
01647
01648     do {
01649
01650         WORD *p = heap[nheap];
01651         bool this_insert;
01652
01653         if (insert.empty()) {
01654             // extract element from the heap and prepare adding new ones
01655             this_insert = false;
01656
01657             pop_heap(BHEAD heap, nheap--);
01658             p = heap[nheap];
01659
01660             if (t[0] == -1) {
01661                 WCOPY(t, p, (5+ABS(p[3])+AN.poly_num_vars));
01662             }
01663             else {
01664                 // if both polynomials are modulo p^1, use integer calculus
01665                 if (both_mod_small) {
01666                     t[4+AN.poly_num_vars] = ((LONG)t[4+AN.poly_num_vars]*t[3] +
p[4+AN.poly_num_vars]*p[3]) % q.modp;
01667                     if (t[4+AN.poly_num_vars]==0)
01668                         t[3]=0;
01669                     else {
01670                         if (t[4+AN.poly_num_vars] > +q.modp/2) t[4+AN.poly_num_vars] -= q.modp;
01671                         if (t[4+AN.poly_num_vars] < -q.modp/2) t[4+AN.poly_num_vars] += q.modp;
01672                         t[3] = SGN(t[4+AN.poly_num_vars]);
01673                         t[4+AN.poly_num_vars] = ABS(t[4+AN.poly_num_vars]);
01674                     }
01675                 }
01676                 else {
01677                     // otherwise, use form long calculus
01678                     AddLong ((UWORD *)&p[4+AN.poly_num_vars], p[3],
                                (UWORD *)&t[4+AN.poly_num_vars], t[3],
                                (UWORD *)&t[4+AN.poly_num_vars], &t[3]);
01679                     if (q.modp!=0) TakeNormalModulus((UWORD *)&t[4+AN.poly_num_vars], &t[3],
                                modq, nmodq,
01680 NOUNPACK);
01681                 }
01682             }
01683         }
01684     }
01685 }
01686 else {
01687     // prepare adding an element of insert to the heap
01688     this_insert = true;
01689
01690     p[0] = insert.back().first;
01691     p[1] = insert.back().second;
01692     insert.pop_back();
01693 }
01694
01695 // add elements to the heap
01696 while (true) {
01697     // prepare the element
01698     if (p[1]==0) {
01699         p[0] += a[p[0]];
01700         if (p[0]==a[0]) break;
01701         WCOPY(&p[3], &a[p[0]], a[p[0]]);
01702         p[3] = p[2+p[3]];
01703     }
01704     else {
01705         if (!this_insert)
01706             p[1] += q[p[1]];
01707         this_insert = false;
01708
01709         if (p[1]==qi) { s++; break; }
01710
01711         for (int i=0; i<AN.poly_num_vars; i++)
01712             p[4+i] = b[p[0]+1+i] + q[p[1]+1+i];
01713
01714         // if both polynomials are modulo p^1, use integer calculus
01715         if (both_mod_small) {
01716             p[4+AN.poly_num_vars] = ((LONG)b[p[0]+1+AN.poly_num_vars]*b[p[0]+b[p[0]]-1]*
q[p[1]+1+AN.poly_num_vars]*q[p[1]+q[p[1]]-1]) % q.modp;
01717             if (p[4+AN.poly_num_vars]==0)
01718                 p[3]=0;
01719             else {
01720                 if (p[4+AN.poly_num_vars] > +q.modp/2) p[4+AN.poly_num_vars] -= q.modp;
01721                 if (p[4+AN.poly_num_vars] < -q.modp/2) p[4+AN.poly_num_vars] += q.modp;
01722                 p[3] = SGN(p[4+AN.poly_num_vars]);
01723                 p[4+AN.poly_num_vars] = ABS(p[4+AN.poly_num_vars]);
01724             }
01725         }
01726     }
}

```

```

01727         else {
01728             // otherwise, use form long calculus
01729             MulLong((UWORD *) &b[p[0]+1+AN.poly_num_vars], b[p[0]+b[p[0]]-1],
01730                     (UWORD *) &q[p[1]+1+AN.poly_num_vars], q[p[1]+q[p[1]]-1],
01731                     (UWORD *) &p[4+AN.poly_num_vars], &p[3]);
01732             if (q.modp!=0) TakeNormalModulus((UWORD *) &p[4+AN.poly_num_vars], &p[3],
01733                                             modq, nmodq,
NOUNPACK);
01734         }
01735
01736         p[3] *= -1;
01737     }
01738
01739     // no hashing
01740     p[2] = -1;
01741
01742     // add it to a heap element
01743     swap (heap[nheap],p);
01744     push_heap(BHEAD heap, ++nheap);
01745     break;
01746 }
01747 }
01748 while (t[0]==-1 || (nheap>0 && monomial_compare(BHEAD heap[0]+3, t+3)==0));
01749
01750 if (t[3] == 0) continue;
01751
01752 // check divisibility
01753 bool div = true;
01754 for (int i=0; i<AN.poly_num_vars; i++)
01755     if (t[4+i] < b[2+i]) div=false;
01756
01757 if (!div) {
01758     // not divisible, so append it to the remainder
01759     if (only_divides) { r=poly(BHEAD 1); ri=r[0]; break;}
01760     t[4 + AN.poly_num_vars + ABS(t[3])] = t[3];
01761     t[3] = 2 + AN.poly_num_vars + ABS(t[3]);
01762     r.termscopy(&t[3], ri, t[3]);
01763     ri += t[3];
01764 }
01765 else {
01766     // divisible, so divide coefficient as well
01767     WORD nq, nr;
01768
01769     if (q.modp==0) {
01770         DivLong((UWORD *) &t[4+AN.poly_num_vars], t[3],
01771                 (UWORD *) &b[2+AN.poly_num_vars], b[b[1]],
01772                 (UWORD *) &q[qi+1+AN.poly_num_vars], &nq,
01773                 (UWORD *) &r[ri+1+AN.poly_num_vars], &nr);
01774         if(check_div && nr != 0)
01775         {
01776             // non-zero remainder from coefficient division => result not expressible in
integers
01777             div_fail = true;
01778             break;
01779         }
01780     }
01781     else {
01782         // if both polynomials are modulo p^1, use integer calculus
01783         if (both_mod_small) {
01784             q[qi+1+AN.poly_num_vars] = ((LONG)t[4+AN.poly_num_vars]*t[3]*ltbinv[0]*nltbinv) %
q.modp;
01785             if (q[qi+1+AN.poly_num_vars]==0)
01786                 nq=0;
01787             else {
01788                 if (q[qi+1+AN.poly_num_vars] > +q.modp/2) q[qi+1+AN.poly_num_vars] -= q.modp;
01789                 if (q[qi+1+AN.poly_num_vars] < -q.modp/2) q[qi+1+AN.poly_num_vars] += q.modp;
01790                 nq = SGN(q[qi+1+AN.poly_num_vars]);
01791                 q[qi+1+AN.poly_num_vars] = ABS(q[qi+1+AN.poly_num_vars]);
01792             }
01793         }
01794         else {
01795             // otherwise, use form long calculus
01796             MulLong((UWORD *) &t[4+AN.poly_num_vars], t[3], ltbinv, nltbinv, (UWORD
*) &q[qi+1+AN.poly_num_vars], &nq);
01797             TakeNormalModulus((UWORD *) &q[qi+1+AN.poly_num_vars], &nq, modq, nmodq, NOUNPACK);
01798         }
01799
01800         nr=0;
01801     }
01802
01803     // add terms to quotient and remainder
01804     if (nq != 0) {
01805         int bi = 1;
01806         for (int j=1; j<s; j++) {
01807             bi += b[bi];
01808             insert.push_back(make_pair(bi,qi));
01809         }

```

```

01810             s=1;
01811
01812             q[qi] = 2+AN.poly_num_vars+ABS(nq);
01813             for (int i=0; i<AN.poly_num_vars; i++)
01814                 q[qi+1+i] = t[4+i] - b[2+i];
01815             qi += q[qi];
01816             q[qi-1] = nq;
01817         }
01818
01819         if (nr != 0) {
01820             r[ri] = 2+AN.poly_num_vars+ABS(nr);
01821             for (int i=0; i<AN.poly_num_vars; i++)
01822                 r[ri+1+i] = t[4+i];
01823             ri += r[ri];
01824             r[ri-1] = nr;
01825         }
01826     }
01827 }
01828
01829 q[0] = qi;
01830 r[0] = ri;
01831
01832 for (int i=0; i<nb; i++)
01833     NumberFree(heap[i], "poly::div_heap-b");
01834
01835 NumberFree(t, "poly::div_heap-c");
01836
01837 if (q.modp!=0 || ltbinv!=NULL) NumberFree(ltbinv, "poly::div_heap-a");
01838 AT.pWorkPointer = oldpWorkPointer;
01839 }
01840
01841 /*
01842 #] divmod_heap :
01843 #[ divmod :
01844 */
01845
01856 void poly::divmod (const poly &a, const poly &b, poly &q, poly &r, bool only_divides) {
01857     q.modp = r.modp = a.modp;
01858     q.modn = r.modn = a.modn;
01859
01860     if (a.is_zero()) {
01861         q[0]=1;
01862         r[0]=1;
01863         return;
01864     }
01865
01866     if (b.is_zero()) {
01867         MLOCK(ErrorMessageLock);
01868         MesPrint ((char*)"ERROR: division by zero polynomial");
01869         MUNLOCK(ErrorMessageLock);
01870         Terminate(1);
01871     }
01872
01873     if (b.is_one()) {
01874         q.check_memory(a[0]);
01875         q.termscopy(a.terms,0,a[0]);
01876         r[0]=1;
01877         return;
01878     }
01879
01880     if (b.is_monomial()) {
01881         divmod_one_term(a,b,q,r,only_divides);
01882         return;
01883     }
01884
01885     int vara = a.is_dense_univariate();
01886     int varb = b.is_dense_univariate();
01887
01888     if (vara!=-2 && varb!=-2 && (vara==--1 || varb==--1 || vara==varb)) {
01889         divmod_univar(a,b,q,r,Max(vara,varb),only_divides);
01890     }
01891     else {
01892         bool div_fail;
01893         divmod_heap(a,b,q,r,only_divides,false,div_fail);
01894     }
01895 }
01896
01897 /*
01898 #] divmod :
01899 #[ divides :
01900 */
01901 // returns whether a exactly divides b
01902 bool poly::divides (const poly &a, const poly &b) {
01903     POLY_GETIDENTITY(a);
01904     poly q(BHEAD 0), r(BHEAD 0);
01905     divmod(b,a,q,r,true);
01906     return r.is_zero();

```



```

01907 }
01908
01909 /*
01910     #] divides :
01911     #[ div :
01912 */
01913
01914 // the quotient of two polynomials
01915 void poly::div (const poly &a, const poly &b, poly &c) {
01916     POLY_GETIDENTITY(a);
01917     poly d(BHEAD 0);
01918     divmod(a,b,c,d,false);
01919 }
01920
01921 /*
01922     #] div :
01923     #[ mod :
01924 */
01925
01926 // the remainder of division of two polynomials
01927 void poly::mod (const poly &a, const poly &b, poly &c) {
01928     POLY_GETIDENTITY(a);
01929     poly d(BHEAD 0);
01930     divmod(a,b,c,d,false);
01931 }
01932
01933 /*
01934     #] mod :
01935     #[ copy operator :
01936 */
01937
01938 // copy operator
01939 poly & poly::operator= (const poly &a) {
01940     #ifdef POLY_MOVE_DEBUG
01941         std::cout << "poly copy assign" << std::endl;
01942     #endif
01943     if (&a != this) {
01944         modp = a.modp;
01945         modn = a.modn;
01946         check_memory(a[0]);
01947         termscopy(a.terms,0,a[0]);
01948     }
01949     return *this;
01950 }
01951
01952
01953 /*
01954     #] copy operator :
01955     #[ operator overloads :
01956 */
01957
01958 // binary operators for polynomial arithmetic
01959 const poly poly::operator+ (const poly &a) const { POLY_GETIDENTITY(a); poly b(BHEAD 0);
    add(*this,a,b); return b; }
01960 const poly poly::operator- (const poly &a) const { POLY_GETIDENTITY(a); poly b(BHEAD 0);
    sub(*this,a,b); return b; }
01961 const poly poly::operator* (const poly &a) const { POLY_GETIDENTITY(a); poly b(BHEAD 0);
    mul(*this,a,b); return b; }
01962 const poly poly::operator/ (const poly &a) const { POLY_GETIDENTITY(a); poly b(BHEAD 0);
    div(*this,a,b); return b; }
01963 const poly poly::operator% (const poly &a) const { POLY_GETIDENTITY(a); poly b(BHEAD 0);
    mod(*this,a,b); return b; }
01964
01965 // assignment operators for polynomial arithmetic
01966 poly& poly::operator+= (const poly &a) { return *this = *this + a; }
01967 poly& poly::operator-= (const poly &a) { return *this = *this - a; }
01968 poly& poly::operator*= (const poly &a) { return *this = *this * a; }
01969 poly& poly::operator/= (const poly &a) { return *this = *this / a; }
01970 poly& poly::operator%= (const poly &a) { return *this = *this % a; }
01971
01972 // comparison operators
01973 bool poly::operator== (const poly &a) const {
01974     for (int i=0; i<terms[0]; i++)
01975         if (terms[i] != a[i]) return 0;
01976     return 1;
01977 }
01978
01979 bool poly::operator!= (const poly &a) const { return !(*this == a); }
01980
01981 /*
01982     #] operator overloads :
01983     #[ num_terms :
01984 */
01985
01986 int poly::number_of_terms () const {
01987
01988     int res=0;

```

```

01989     for (int i=1; i<terms[0]; i+=terms[i])
01990         res++;
01991     return res;
01992 }
01993
01994 /*
01995     #] num_terms :
01996     #[ first_variable :
01997 */
01998
01999 // returns the lexicographically first variable of a polynomial
02000 int poly::first_variable () const {
02001
02002     POLY_GETIDENTITY(*this);
02003
02004     int var = AN.poly_num_vars;
02005     for (int j=0; j<var; j++)
02006         if (terms[2+j]>0) return j;
02007     return var;
02008 }
02009
02010 /*
02011     #] first_variable :
02012     #[ all_variables :
02013 */
02014
02015 // returns a list of all variables of a polynomial
02016 const vector<int> poly::all_variables () const {
02017
02018     POLY_GETIDENTITY(*this);
02019
02020     vector<bool> used(AN.poly_num_vars, false);
02021
02022     for (int i=1; i<terms[0]; i+=terms[i])
02023         for (int j=0; j<AN.poly_num_vars; j++)
02024             if (terms[i+1+j]>0) used[j] = true;
02025
02026     vector<int> vars;
02027     for (int i=0; i<AN.poly_num_vars; i++)
02028         if (used[i]) vars.push_back(i);
02029
02030     return vars;
02031 }
02032
02033 /*
02034     #] all_variables :
02035     #[ sign :
02036 */
02037
02038 // returns the sign of the leading coefficient
02039 int poly::sign () const {
02040     if (terms[0]==1) return 0;
02041     return terms[terms[1]] > 0 ? 1 : -1;
02042 }
02043
02044 /*
02045     #] sign :
02046     #[ degree :
02047 */
02048
02049 // returns the degree of x of a polynomial (deg=-1 iff a=0)
02050 int poly::degree (int x) const {
02051     int deg = -1;
02052     for (int i=1; i<terms[0]; i+=terms[i])
02053         deg = MaX(deg, terms[i+1+x]);
02054     return deg;
02055 }
02056
02057 /*
02058     #] degree :
02059     #[ total_degree :
02060 */
02061
02062 // returns the total degree of a polynomial (deg=-1 iff a=0)
02063 int poly::total_degree () const {
02064
02065     POLY_GETIDENTITY(*this);
02066
02067     int tot_deg = -1;
02068     for (int i=1; i<terms[0]; i+=terms[i]) {
02069         int deg=0;
02070         for (int j=0; j<AN.poly_num_vars; j++)
02071             deg += terms[i+1+j];
02072         tot_deg = MaX(deg, tot_deg);
02073     }
02074     return tot_deg;
02075 }

```

```

02076
02077 /*
02078     #] total_degree :
02079     #[ integer_lcoeff :
02080 */
02081
02082 // returns the integer coefficient of the leading monomial
02083 const poly poly::integer_lcoeff () const {
02084
02085     POLY_GETIDENTITY(*this);
02086
02087     poly res(BHEAD 0);
02088     res.modp = modp;
02089     res.modn = modn;
02090
02091     res.termscopy(&terms[1],1,terms[1]);
02092     res[0] = res[1] + 1; // length
02093     for (int i=0; i<AN.poly_num_vars; i++)
02094         res[2+i] = 0; // powers
02095
02096     return res;
02097 }
02098
02099 /*
02100     #] integer_lcoeff :
02101     #[ coefficient :
02102 */
02103
02104 // returns the polynomial coefficient of x^n
02105 const poly poly::coefficient (int x, int n) const {
02106
02107     POLY_GETIDENTITY(*this);
02108
02109     poly res(BHEAD 0);
02110     res.modp = modp;
02111     res.modn = modn;
02112     res[0] = 1;
02113
02114     for (int i=1; i<terms[0]; i+=terms[i])
02115         if (terms[i+1+x] == n) {
02116             res.check_memory(res[0]+terms[i]);
02117             res.termscopy(&terms[i], res[0], terms[i]);
02118             res[res[0]+1+x] -= n; // power of x
02119             res[0] += res[res[0]]; // length
02120         }
02121
02122     return res;
02123 }
02124
02125 /*
02126     #] coefficient :
02127     #[ lcoeff_multivar :
02128 */
02129
02130 // returns the leading coefficient with the polynomial viewed as a
02131 // polynomial in x, so that the result is a polynomial in the
02132 // remaining variables
02133 const poly poly::lcoeff_univar (int x) const {
02134     return coefficient(x, degree(x));
02135 }
02136
02137 // returns the leading coefficient with the polynomial viewed as a
02138 // polynomial with coefficients in ZZ[x], so that the result
02139 // polynomial is in ZZ[x] too
02140 const poly poly::lcoeff_multivar (int x) const {
02141
02142     POLY_GETIDENTITY(*this);
02143
02144     poly res(BHEAD 0,modp,modn);
02145
02146     for (int i=1; i<terms[0]; i+=terms[i]) {
02147         bool same_powers = true;
02148         for (int j=0; j<AN.poly_num_vars; j++)
02149             if (j!=x && terms[i+1+j]!=terms[2+j]) {
02150                 same_powers = false;
02151                 break;
02152             }
02153         if (!same_powers) break;
02154
02155         res.check_memory(res[0]+terms[i]);
02156         res.termscopy(&terms[i],res[0],terms[i]);
02157         for (int k=0; k<AN.poly_num_vars; k++)
02158             if (k!=x) res[res[0]+1+k]=0;
02159
02160         res[0] += terms[i];
02161     }
02162

```

```

02163     return res;
02164 }
02165
02166 /*
02167  #] lcoeff_multivar :
02168  #[ derivative :
02169 */
02170
02171 // returns the derivative with respect to x
02172 const poly poly::derivative (int x) const {
02173
02174     POLY_GETIDENTITY(*this);
02175
02176     poly b(BHEAD 0);
02177     int bi=1;
02178
02179     for (int i=1; i<terms[0]; i+=terms[i]) {
02180
02181         int power = terms[i+1+x];
02182
02183         if (power > 0) {
02184             b.check_memory(bi);
02185             b.termscopy(&terms[i], bi, terms[i]);
02186             b[bi+1+x]--;
02187
02188             WORD nb = b[bi+b[bi]-1];
02189             Product((UWORD *)&b[bi+1+AN.poly_num_vars], &nb, power);
02190
02191             b[bi] = 2 + AN.poly_num_vars + ABS(nb);
02192             b[bi+b[bi]-1] = nb;
02193
02194             bi += b[bi];
02195         }
02196     }
02197
02198     b[0] = bi;
02199     b.setmod(modp, modn);
02200     return b;
02201 }
02202
02203 /*
02204  #] derivative :
02205  #[ is_zero :
02206 */
02207
02208 // returns whether the polynomial is zero
02209 bool poly::is_zero () const {
02210     return terms[0] == 1;
02211 }
02212
02213 /*
02214  #] is_zero :
02215  #[ is_one :
02216 */
02217
02218 // returns whether the polynomial is one
02219 bool poly::is_one () const {
02220
02221     POLY_GETIDENTITY(*this);
02222
02223     if (terms[0] != 4+AN.poly_num_vars) return false;
02224     if (terms[1] != 3+AN.poly_num_vars) return false;
02225     for (int i=0; i<AN.poly_num_vars; i++)
02226         if (terms[2+i] != 0) return false;
02227     if (terms[2+AN.poly_num_vars] != 1) return false;
02228     if (terms[3+AN.poly_num_vars] != 1) return false;
02229
02230     return true;
02231 }
02232
02233 /*
02234  #] is_one :
02235  #[ is_integer :
02236 */
02237
02238 // returns whether the polynomial is an integer
02239 bool poly::is_integer () const {
02240
02241     POLY_GETIDENTITY(*this);
02242
02243     if (terms[0] == 1) return true;
02244     if (terms[0] != terms[1]+1) return false;
02245
02246     for (int j=0; j<AN.poly_num_vars; j++)
02247         if (terms[2+j] != 0)
02248             return false;
02249

```

```

02250     return true;
02251 }
02252
02253 /*
02254 #] is_integer :
02255 #[ is_monomial :
02256 */
02257
02258 // returns whether the polynomial consist of one term
02259 bool poly::is_monomial () const {
02260     return terms[0]>1 && terms[0]==terms[1]+1;
02261 }
02262
02263 /*
02264 #] is_monomial :
02265 #[ is_dense_univariate :
02266 */
02267
02284 int poly::is_dense_univariate () const {
02285     POLY_GETIDENTITY(*this);
02286
02287     int num_terms=0, res=-1;
02288
02289     // test univariate
02290     for (int i=1; i<terms[0]; i+=terms[i]) {
02291         for (int j=0; j<AN.poly_num_vars; j++)
02292             if (terms[i+1+j] > 0) {
02293                 if (res == -1) res = j;
02294                 if (res != j) return -2;
02295             }
02296         num_terms++;
02297     }
02298
02301     // constant polynomial
02302     if (res == -1) return -1;
02303
02304     // test density
02305     int deg = terms[2+res];
02306     if (2*num_terms < deg+1) return -2;
02307
02308     return res;
02309 }
02310
02311 /*
02312 #] is_dense_univariate :
02313 #[ simple_poly (small) :
02314 */
02315
02316 // returns the polynomial (x-a)^b mod p^n with a small
02317 const poly poly::simple_poly (PHEAD int x, int a, int b, int p, int n) {
02318     poly tmp(BHEAD 0,p,n);
02319
02320     int idx=1;
02321     tmp[idx++] = 3 + AN.poly_num_vars; // length
02322     for (int i=0; i<AN.poly_num_vars; i++)
02323         tmp[idx++] = i==x ? 1 : 0; // powers
02324     tmp[idx++] = 1; // coefficient
02325     tmp[idx++] = 1; // length coefficient
02326
02327     if (a != 0) {
02328         tmp[idx++] = 3 + AN.poly_num_vars; // length
02329         for (int i=0; i<AN.poly_num_vars; i++) tmp[idx++] = 0; // powers
02330         tmp[idx++] = ABS(a); // coefficient
02331         tmp[idx++] = -SGN(a); // length coefficient
02332     }
02333
02334     tmp[0] = idx; // length
02335
02336     if (b == 1) return tmp;
02337
02338     poly res(BHEAD 1,p,n);
02339
02340     while (b!=0) {
02341         if (b&1) res*=tmp;
02342         tmp*=tmp;
02343         b>>=1;
02344     }
02345
02346     return res;
02347 }
02348
02349 /*
02350 #] simple_poly (small) :
02351 #[ simple_poly (large) :

```

```

02353 */
02354
02355 // returns the polynomial (x-a)^b mod p^n with a large
02356 const poly poly::simple_poly (PHEAD int x, const poly &a, int b, int p, int n) {
02357
02358     poly res(BHEAD 1,p,n);
02359     poly tmp(BHEAD 0,p,n);
02360
02361     int idx=1;
02362
02363     tmp[idx++] = 3 + AN.poly_num_vars; // length
02364     for (int i=0; i<AN.poly_num_vars; i++)
02365         tmp[idx++] = i==x ? 1 : 0; // powers
02366     tmp[idx++] = 1; // coefficient
02367     tmp[idx++] = 1; // length coefficient
02368
02369     if (!a.is_zero()) {
02370         tmp[idx++] = 2 + AN.poly_num_vars + ABS(a[a[0]-1]); // length
02371         for (int i=0; i<AN.poly_num_vars; i++) tmp[idx++] = 0; // powers
02372         for (int i=0; i<ABS(a[a[0]-1]); i++)
02373             tmp[idx++] = a[2 + AN.poly_num_vars + i]; // coefficient
02374         tmp[idx++] = -a[a[0]-1]; // length coefficient
02375     }
02376
02377     tmp[0] = idx; // length
02378
02379     while (b!=0) {
02380         if (b&1) res*=tmp;
02381         tmp*=tmp;
02382         b>>=1;
02383     }
02384
02385     return res;
02386 }
02387
02388 /*
02389 #] simple_poly (large) :
02390 #[ get_variables :
02391 */
02392
02393 // gets all variables in the expressions and stores them in AN.poly_vars
02394 // (it is assumed that AN.poly_vars=NULL)
02395 void poly::get_variables (PHEAD const vector<WORD *> &es, bool with_arghead, bool sort_vars) {
02396     assert(AN.poly_vars == NULL);
02397
02398     AN.poly_num_vars = 0;
02399
02400     vector<int> vars;
02401     vector<int> degrees;
02402     map<int,int> var_to_idx;
02403
02404     // extract all variables
02405     for (int ei=0; ei<(int)es.size(); ei++) {
02406         const WORD *e = es[ei];
02407
02408         // fast notation
02409         if (*e == -SNUMBER) {
02410         }
02411         else if (*e == -SYMBOL) {
02412             if (!var_to_idx.count(e[1])) {
02413                 vars.push_back(e[1]);
02414                 var_to_idx[e[1]] = AN.poly_num_vars++;
02415                 degrees.push_back(1);
02416             }
02417         }
02418         // JD: Here we need to check for non-symbol/number terms in fast notation.
02419         else if (*e < 0) {
02420             MLOCK(ErrorMessageLock);
02421             MesPrint ((char*)"ERROR: polynomials and polyratfuns must contain symbols only");
02422             MUNLOCK(ErrorMessageLock);
02423             Terminate(1);
02424         }
02425         else {
02426             for (int i=with_arghead ? ARGHEAD : 0; with_arghead ? i<e[0] : e[i]!=0; i+=e[i]) {
02427                 if (i+1<i+e[i]-ABS(e[i+e[i]-1]) && e[i+1]!=SYMBOL) {
02428                     MLOCK(ErrorMessageLock);
02429                     MesPrint ((char*)"ERROR: polynomials and polyratfuns must contain symbols only");
02430                     MUNLOCK(ErrorMessageLock);
02431                     Terminate(1);
02432                 }
02433
02434                 for (int j=i+3; j<i+e[i]-ABS(e[i+e[i]-1]); j+=2) {
02435                     if (!var_to_idx.count(e[j])) {
02436                         vars.push_back(e[j]);
02437                         var_to_idx[e[j]] = AN.poly_num_vars++;
02438                         degrees.push_back(e[j+1]);
02439                     }

```

```

02440         else {
02441             degrees[var_to_idx[e[j]]] = MaX(degrees[var_to_idx[e[j]]], e[j+1]);
02442         }
02443     }
02444 }
02445 }
02446 }
02447
02448 // make sure an eventual FACTORSYMBOL appear as last
02449 if (var_to_idx.count(FACTORSYMBOL)) {
02450     int i = var_to_idx[FACTORSYMBOL];
02451     while (i+1<AN.poly_num_vars) {
02452         swap(vars[i], vars[i+1]);
02453         swap(degrees[i], degrees[i+1]);
02454         i++;
02455     }
02456     degrees[i] = -1; // makes sure it stays last
02457 }
02458
02459 // AN.poly_vars will be deleted in calling functions from polywrap.c
02460 // For that we use the function poly_free_poly_vars
02461 // We add one to the allocation to avoid copying in poly_divmod
02462
02463 if ( (AN.poly_num_vars+1)*sizeof(WORD) > (size_t)(AM.MaxTer) ) {
02464     // This can happen only in expressions with excessively many variables.
02465     AN.poly_vars = (WORD *)Malloc1((AN.poly_num_vars+1)*sizeof(WORD), "AN.poly_vars");
02466     AN.poly_vars_type = 1;
02467 }
02468 else {
02469     AN.poly_vars = TermMalloc("AN.poly_vars");
02470     AN.poly_vars_type = 0;
02471 }
02472
02473 for (int i=0; i<AN.poly_num_vars; i++)
02474     AN.poly_vars[i] = vars[i];
02475
02476 if (sort_vars) {
02477     // bubble sort variables in decreasing order of degree
02478     // (this seems better for factorization)
02479     for (int i=0; i<AN.poly_num_vars; i++)
02480         for (int j=0; j+1<AN.poly_num_vars; j++)
02481             if (degrees[j] < degrees[j+1]) {
02482                 swap(degrees[j], degrees[j+1]);
02483                 swap(AN.poly_vars[j], AN.poly_vars[j+1]);
02484             }
02485 }
02486 else {
02487     // sort lexicographically
02488     sort(AN.poly_vars, AN.poly_vars+AN.poly_num_vars - var_to_idx.count(FACTORSYMBOL));
02489 }
02490 }
02491
02492 /*
02493 #] get_variables :
02494 #[ argument_to_poly :
02495 */
02496
02497 // converts a form expression to a polynomial class "poly"
02498 const poly poly::argument_to_poly (PHEAD WORD *e, bool with_arghead, bool sort_univar, poly *denpoly)
02499 {
02500     map<int,int> var_to_idx;
02501     for (int i=0; i<AN.poly_num_vars; i++)
02502         var_to_idx[AN.poly_vars[i]] = i;
02503
02504     poly res(BHEAD 0);
02505
02506     // fast notation
02507     if (*e == -SNUMBER) {
02508         if (denpoly!=NULL) *denpoly = poly(BHEAD 1);
02509
02510         if (e[1] == 0) {
02511             res[0] = 1;
02512             return res;
02513         }
02514         else {
02515             res[0] = 4 + AN.poly_num_vars;
02516             res[1] = 3 + AN.poly_num_vars;
02517             for (int i=0; i<AN.poly_num_vars; i++)
02518                 res[2+i] = 0;
02519             res[2+AN.poly_num_vars] = ABS(e[1]);
02520             res[3+AN.poly_num_vars] = SGN(e[1]);
02521
02522             return res;
02523         }
02524     }
02525 }

```

```

02526     if (*e == -SYMBOL) {
02527         if (denpoly!=NULL) *denpoly = poly(BHEAD 1);
02528
02529         res[0] = 4 + AN.poly_num_vars;
02530         res[1] = 3 + AN.poly_num_vars;
02531         for (int i=0; i<AN.poly_num_vars; i++)
02532             res[2+i] = 0;
02533         res[2+var_to_idx.find(e[1])>second] = 1;
02534         res[2+AN.poly_num_vars] = 1;
02535         res[3+AN.poly_num_vars] = 1;
02536         return res;
02537     }
02538
02539     // find LCM of denominators of all terms
02540     WORD nden=1, npro=0, ngcd=0, ndum=0;
02541     UWORD *den = NumberMalloc("poly::argument_to_poly");
02542     UWORD *pro = NumberMalloc("poly::argument_to_poly");
02543     UWORD *gcd = NumberMalloc("poly::argument_to_poly");
02544     UWORD *dum = NumberMalloc("poly::argument_to_poly");
02545     den[0]=1;
02546
02547     for (int i=with_arghead ? ARGHEAD : 0; with_arghead ? i<e[0] : e[i]!=0; i+=e[i]) {
02548         int ncoe = ABS(e[i+e[i]-1]/2);
02549         UWORD *coe = (UWORD *)&e[i+e[i]-ncoe-1];
02550         while (ncoe>0 && coe[ncoe-1]==0) ncoe--;
02551         Mullong(den, nden, coe, ncoe, pro, &npro);
02552         GcdLong(BHEAD den, nden, coe, ncoe, gcd, &ngcd);
02553         DivLong(pro, npro, gcd, ngcd, den, &nden, dum, &ndum);
02554     }
02555
02556     if (denpoly!=NULL) *denpoly = poly(BHEAD den, nden);
02557
02558     int ri=1;
02559
02560     // ordinary notation
02561     for (int i=with_arghead ? ARGHEAD : 0; with_arghead ? i<e[0] : e[i]!=0; i+=e[i]) {
02562         res.check_memory(ri);
02563         WORD nc = e[i+e[i]-1]; // length coefficient
02564         for (int j=0; j<AN.poly_num_vars; j++) // powers=0
02565             res[ri+1+j]=0; // coefficient to dummy
02566         WCOPY(dum, &e[i+e[i]-ABS(nc)], ABS(nc)); // remove denominator
02567         nc /= 2; // multiply with
02568         Mully(BHEAD dum, &nc, den, nden);
02569         overall den
02570             res.termscopy((WORD *)dum, ri+1+AN.poly_num_vars, ABS(nc)); // coefficient to res
02571         res[ri] = ABS(nc) + AN.poly_num_vars + 2; // length
02572         res[ri+res[ri]-1] = nc; // length coefficient
02573         for (int j=i+3; j<i+e[i]-ABS(e[i+e[i]-1]); j+=2)
02574             res[ri+1+var_to_idx.find(e[j])>second] = e[j+1]; // powers
02575         ri += res[ri]; // length
02576     }
02577
02578     res[0] = ri;
02579
02580     // JD: For univariate cases, check whether the ordering is correct or not.
02581     // There are various ways to arrive here, but with an incorrect ordering, such
02582     // as interactions with collect, moving a polyratfun out of another function
02583     // argument, etc.
02584     // If the ordering is not correct, make sure we normalize. In multivariate cases,
02585     // always normalize as before.
02586     if (sort_univar == false && AN.poly_num_vars == 1) {
02587         if (res.number_of_terms() >= 2) {
02588             if (res[2] < res.last_monomial()[2]) {
02589                 sort_univar = true;
02590             }
02591         }
02592     }
02593
02594     if (sort_univar || AN.poly_num_vars>1) {
02595         res.normalize();
02596     }
02597
02598     NumberFree(den, "poly::argument_to_poly");
02599     NumberFree(pro, "poly::argument_to_poly");
02600     NumberFree(gcd, "poly::argument_to_poly");
02601     NumberFree(dum, "poly::argument_to_poly");
02602
02603     return res;
02604 }
02605
02606 /*
02607 #] argument_to_poly :
02608 #[ poly_to_argument :
02609 */
02610 // converts a polynomial class "poly" to a form expression
02611 void poly::poly_to_argument (const poly &a, WORD *res, bool with_arghead) {

```



```

02612
02613     POLY_GETIDENTITY(a);
02614
02615     // special case: a=0
02616     if (a[0]==1) {
02617         if (with_arghead) {
02618             res[0] = -SNUMBER;
02619             res[1] = 0;
02620         }
02621         else {
02622             res[0] = 0;
02623         }
02624         return;
02625     }
02626
02627     if (with_arghead) {
02628         res[1] = AN.poly_num_vars>1 ? DIRTYFLAG : 0; // dirty flag
02629         for (int i=2; i<ARGHEAD; i++)
02630             res[i] = 0; // remainder of arghead
02631     }
02632
02633     int L = with_arghead ? ARGHEAD : 0;
02634
02635     for (int i=1; i!=a[0]; i+=a[i]) {
02636
02637         res[L]=1; // length
02638
02639         bool first=true;
02640
02641         for (int j=0; j<AN.poly_num_vars; j++)
02642             if (a[i+1+j] > 0) {
02643                 if (first) {
02644                     first=false;
02645                     res[L+1] = 1; // symbols
02646                     res[L+2] = 2; // length
02647                 }
02648                 res[L+1+res[L+2]++] = AN.poly_vars[j]; // symbol
02649                 res[L+1+res[L+2]++] = a[i+1+j]; // power
02650             }
02651
02652         if (!first) res[L] += res[L+2]; // fix length
02653
02654         WORD nc = a[i+a[i]-1];
02655         WCOPY(&res[L+res[L]], &a[i+a[i]-1-ABS(nc)], ABS(nc)); // numerator
02656
02657         res[L] += ABS(nc); // fix length
02658         memset(&res[L+res[L]], 0, ABS(nc)*sizeof(WORD)); // denominator one
02659         res[L+res[L]] = 1; // denominator one
02660         res[L] += ABS(nc); // fix length
02661         res[L+res[L]] = SGN(nc) * (2*ABS(nc)+1); // length of coefficient
02662         res[L]++; // fix length
02663         L += res[L]; // fix length
02664     }
02665
02666     if (with_arghead) {
02667         res[0] = L;
02668         // convert to fast notation if possible
02669         ToFast(res,res);
02670     }
02671     else {
02672         res[L] = 0;
02673     }
02674 }
02675
02676 /*
02677 #] poly_to_argument :
02678 #[ poly_to_argument_with_den :
02679 */
02680
02681 // converts a polynomial class "poly" divided by a number (nden, den) to a form expression
02682 // cf. poly::poly_to_argument()
02683 void poly::poly_to_argument_with_den (const poly &a, WORD nden, const UWORD *den, WORD *res, bool
    with_arghead) {
02684
02685     POLY_GETIDENTITY(a);
02686
02687     // special case: a=0
02688     if (a[0]==1) {
02689         if (with_arghead) {
02690             res[0] = -SNUMBER;
02691             res[1] = 0;
02692         }
02693         else {
02694             res[0] = 0;
02695         }
02696         return;
02697     }

```

```

02698
02699     if (with_arghead) {
02700         res[1] = AN.poly_num_vars>1 ? DIRTYFLAG : 0; // dirty flag
02701         for (int i=2; i<ARGHEAD; i++)
02702             res[i] = 0; // remainder of arghead
02703     }
02704
02705     WORD nden1;
02706     UWORD *den1 = (UWORD *)NumberMalloc("poly_to_argument_with_den");
02707
02708     int L = with_arghead ? ARGHEAD : 0;
02709
02710     for (int i=1; i!=a[0]; i+=a[i]) {
02711
02712         res[L]=1; // length
02713
02714         bool first=true;
02715
02716         for (int j=0; j<AN.poly_num_vars; j++)
02717             if (a[i+1+j] > 0) {
02718                 if (first) {
02719                     first=false;
02720                     res[L+1] = 1; // symbols
02721                     res[L+2] = 2; // length
02722                 }
02723                 res[L+1+res[L+2]++] = AN.poly_vars[j]; // symbol
02724                 res[L+1+res[L+2]++] = a[i+1+j]; // power
02725             }
02726
02727         if (!first) res[L] += res[L+2]; // fix length
02728
02729         // numerator
02730         WORD nc = a[i+a[i]-1];
02731         WCOPY(&res[L+res[L]], &a[i+a[i]-1-ABS(nc)], ABS(nc));
02732
02733         // denominator
02734         nden1 = nden;
02735         WCOPY(den1, den, ABS(nden));
02736
02737         if (nden != 1 || den[0] != 1) {
02738             // remove gcd(num,den)
02739             Simplify(BHEAD (UWORD *)&res[L+res[L]], &nc, den1, &nden1);
02740         }
02741
02742         Pack((UWORD *)&res[L+res[L]], &nc, den1, nden1); // format
02743         res[L] += 2*ABS(nc)+1; // fix length
02744         res[L+res[L]-1] = SGN(nc)*(2*ABS(nc)+1); // length of coefficient
02745         L += res[L]; // fix length
02746     }
02747
02748     NumberFree(den1, "poly_to_argument_with_den");
02749
02750     if (with_arghead) {
02751         res[0] = L;
02752         // convert to fast notation if possible
02753         ToFast(res,res);
02754     }
02755     else {
02756         res[L] = 0;
02757     }
02758 }
02759
02760 /*
02761 #] poly_to_argument_with_den :
02762 #[ size_of_form_notation :
02763 */
02764
02765 // the size of the polynomial in form notation (without argheads and fast notation)
02766 int poly::size_of_form_notation () const {
02767
02768     POLY_GETIDENTITY(*this);
02769
02770     // special case: a=0
02771     if (terms[0]==1) return 0;
02772
02773     int len = 0;
02774
02775     for (int i=1; i!=terms[0]; i+=terms[i]) {
02776         len++;
02777         int npow = 0;
02778         for (int j=0; j<AN.poly_num_vars; j++)
02779             if (terms[i+1+j] > 0) npow++;
02780         if (npow > 0) len += 2*npow + 2;
02781         len += 2 * ABS(terms[i+terms[i]-1]) + 1;
02782     }
02783
02784     return len;

```

```

02785 }
02786
02787 /*
02788 #] size_of_form_notation :
02789 #[ size_of_form_notation_with_den :
02790 */
02791
02792 // the size of the polynomial divided by a number (its size is given by nden)
02793 // in form notation (without argheads and fast notation)
02794 // cf. poly::size_of_form_notation()
02795 int poly::size_of_form_notation_with_den (WORD nden) const {
02796
02797     POLY_GETIDENTITY(*this);
02798
02799     // special case: a=0
02800     if (terms[0]==1) return 0;
02801
02802     nden = ABS(nden);
02803     int len = 0;
02804
02805     for (int i=1; i!=terms[0]; i+=terms[i]) {
02806         len++;
02807         int npow = 0;
02808         for (int j=0; j<AN.poly_num_vars; j++)
02809             if (terms[i+1+j] > 0) npow++;
02810         if (npow > 0) len += 2*npow + 2;
02811         len += 2 * MaX(ABS(terms[i+terms[i]-1]), nden) + 1;
02812     }
02813
02814     return len;
02815 }
02816
02817 /*
02818 #] size_of_form_notation_with_den :
02819 #[ to_coefficient_list :
02820 */
02821
02822 // returns the coefficient list of a univariate polynomial
02823 const vector<WORD> poly::to_coefficient_list (const poly &a) {
02824
02825     POLY_GETIDENTITY(a);
02826
02827     if (a.is_zero()) return vector<WORD>();
02828
02829     int x = a.first_variable();
02830     if (x == AN.poly_num_vars) x=0;
02831
02832     vector<WORD> res(1+a[2+x],0);
02833
02834     for (int i=1; i<a[0]; i+=a[i])
02835         res[a[i+1+x]] = (a[i+a[i]-1] * a[i+a[i]-2]) % a.modp;
02836
02837     return res;
02838 }
02839
02840 /*
02841 #] to_coefficient_list :
02842 #[ coefficient_list_divmod :
02843 */
02844
02845 // divides two polynomials represented by coefficient lists
02846 const vector<WORD> poly::coefficient_list_divmod (const vector<WORD> &a, const vector<WORD> &b, WORD
p, int divmod) {
02847
02848     int bsize = (int)b.size();
02849     while (b[bsize-1]==0) bsize--;
02850
02851     WORD inv;
02852     GetModInverses(b[bsize-1] + (b[bsize-1]<0?p:0), p, &inv, NULL);
02853
02854     vector<WORD> q(a.size(),0);
02855     vector<WORD> r(a);
02856
02857     while ((int)r.size() >= bsize) {
02858         LONG mul = ((LONG)inv * r.back()) % p;
02859         int off = r.size()-bsize;
02860         q[off] = mul;
02861         for (int i=0; i<bsize; i++)
02862             r[off+i] = (r[off+i] - mul*b[i]) % p;
02863         while (r.size()>0 && r.back()==0)
02864             r.pop_back();
02865     }
02866
02867     if (divmod==0) {
02868         while (q.size()>0 && q.back()==0)
02869             q.pop_back();
02870         return q;

```

```

02871     }
02872     else {
02873         while (r.size()>0 && r.back()==0)
02874             r.pop_back();
02875         return r;
02876     }
02877 }
02878
02879 /*
02880  #] coefficient_list_divmod :
02881  #[ from_coefficient_list :
02882 */
02883
02884 // converts a coefficient list to a "poly"
02885 const poly poly::from_coefficient_list (PHEAD const vector<WORD> &a, int x, WORD p) {
02886     poly res(BHEAD 0);
02887     int ri=1;
02888
02889     for (int i=(int)a.size()-1; i>=0; i--)
02890         if (a[i] != 0) {
02891             res.check_memory(ri);
02892             res[ri] = AN.poly_num_vars+3; // length
02893             for (int j=0; j<AN.poly_num_vars; j++)
02894                 res[ri+1+j]=0; // powers
02895             res[ri+1+x] = i; // power of x
02896             res[ri+1+AN.poly_num_vars] = ABS(a[i]); // coefficient
02897             res[ri+1+AN.poly_num_vars+1] = SGN(a[i]); // sign
02898             ri += res[ri];
02899         }
02900
02901
02902     res[0]=ri; // total length
02903     res.setmod(p,1);
02904
02905     return res;
02906 }
02907
02908 /*
02909  #] from_coefficient_list :
02910 */

```

## 4.101 poly.h

```

00001 #pragma once
00002 /* #] License : */
00003 /*
00004  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00005  * When using this file you are requested to refer to the publication
00006  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00007  * This is considered a matter of courtesy as the development was paid
00008  * for by FOM the Dutch physics granting agency and we would like to
00009  * be able to track its scientific use to convince FOM of its value
00010  * for the community.
00011  *
00012  * This file is part of FORM.
00013  *
00014  * FORM is free software: you can redistribute it and/or modify it under the
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00016  * Foundation, either version 3 of the License, or (at your option) any later
00017  * version.
00018  *
00019  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00020  * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00021  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00022  * details.
00023  *
00024  * You should have received a copy of the GNU General Public License along
00025  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00026 */
00027 /* #] License : */
00028
00029 extern "C" {
00030 #include "form3.h"
00031 }
00032
00033 #include <iostream>
00034 #include <string>
00035 #include <vector>
00036
00037 // for debugging
00038 // #define POLY_MOVE_DEBUG
00039
00040 // macros for tform

```

```

00041 #ifndef WITHPTHREADS
00042 #define POLY_GETIDENTITY(X)
00043 #define POLY_STOREIDENTITY
00044 #else
00045 #define POLY_GETIDENTITY(X) ALLPRIVATES *B = (X).Bpointer
00046 #define POLY_STOREIDENTITY Bpointer = B
00047 #endif
00048
00049 // maximum size of the hash table used for multiplication and division
00050
00051 const int POLY_MAX_HASH_SIZE = Min(1<<20, MAXPOSITIVE);
00052
00053 class poly {
00054 public:
00055     // variables
00056     #ifdef WITHPTHREADS
00057         ALLPRIVATES *Bpointer;
00058     #endif
00059     WORD *terms;
00060     LONG size_of_terms;
00061     WORD modp, modn;
00062
00063     // constructors/destructor
00064     poly (PHEAD int, WORD=-1, WORD=1);
00065     poly (PHEAD const UWORD *, WORD, WORD=-1, WORD=1);
00066     poly (const poly &, WORD=-1, WORD=1);
00067     ~poly ();
00068
00069     // operators
00070     poly& operator+= (const poly&);
00071     poly& operator-= (const poly&);
00072     poly& operator*= (const poly&);
00073     poly& operator/= (const poly&);
00074     poly& operator%= (const poly&);
00075
00076     const poly operator+ (const poly&) const;
00077     const poly operator- (const poly&) const;
00078     const poly operator* (const poly&) const;
00079     const poly operator/ (const poly&) const;
00080     const poly operator% (const poly&) const;
00081
00082     bool operator== (const poly&) const;
00083     bool operator!= (const poly&) const;
00084     poly& operator= (const poly &);
00085     WORD& operator[] (int);
00086     const WORD& operator[] (int) const;
00087
00088     #if __cplusplus >= 201103L
00089         // support move semantics
00090         poly (poly &&, WORD=-1, WORD=1) noexcept;
00091         poly& operator= (poly &&) noexcept;
00092     #endif
00093
00094     // memory management
00095     void termscopy (const WORD *, int, int);
00096     void check_memory(int);
00097     void expand_memory(int);
00098
00099     // check type of polynomial
00100     bool is_zero () const;
00101     bool is_one () const;
00102     bool is_integer () const;
00103     bool is_monomial () const;
00104     int is_dense_univariate () const;
00105
00106     // properties
00107     int sign () const;
00108     int degree (int) const;
00109     int total_degree () const;
00110     int first_variable () const;
00111     int number_of_terms () const;
00112     const std::vector<int> all_variables () const;
00113     const poly integer_lcoeff () const;
00114     const poly lcoeff_univar (int) const;
00115     const poly lcoeff_multivar (int) const;
00116     const poly coefficient (int, int) const;
00117     const poly derivative (int) const;
00118
00119     // modulo calculus
00120     void setmod(WORD, WORD=1);
00121     void coefficients_modulo (UWORD *, WORD, bool);
00122
00123     // simple polynomials
00124     static const poly simple_poly (PHEAD int, int=0, int=1, int=0, int=1);

```

```

00128     static const poly simple_poly (PHEAD int, const poly&, int=1, int=0, int=1);
00129
00130     // conversion from/to form notation
00131     static void get_variables (PHEAD const std::vector<WORD *> &, bool, bool);
00132     static const poly argument_to_poly (PHEAD WORD *, bool, bool, poly *den=NULL);
00133     static void poly_to_argument (const poly &, WORD *, bool);
00134     static void poly_to_argument_with_den (const poly &, WORD, const UWORD *, WORD *, bool);
00135     int size_of_form_notation () const;
00136     int size_of_form_notation_with_den (WORD) const;
00137     const poly & normalize ();
00138
00139     // operations for coefficient lists
00140     static const poly from_coefficient_list (PHEAD const std::vector<WORD> &, int, WORD);
00141     static const std::vector<WORD> to_coefficient_list (const poly &);
00142     static const std::vector<WORD> coefficient_list_divmod (const std::vector<WORD> &, const
std::vector<WORD> &, WORD, int);
00143
00144     // string output for debugging
00145     const std::string to_string() const;
00146
00147     // monomials
00148     static int monomial_compare (PHEAD const WORD *, const WORD *);
00149     WORD last_monomial_index () const;
00150     WORD* last_monomial () const;
00151     int compare_degree_vector(const poly &) const;
00152     std::vector<int> degree_vector() const;
00153     int compare_degree_vector(const std::vector<int> &) const;
00154
00155     // (internal) mathematical operations
00156     static void add (const poly &, const poly &, poly &);
00157     static void sub (const poly &, const poly &, poly &);
00158     static void mul (const poly &, const poly &, poly &);
00159     static void div (const poly &, const poly &, poly &);
00160     static void mod (const poly &, const poly &, poly &);
00161     static void divmod (const poly &, const poly &, poly &, poly &, bool only_divides);
00162     static bool divides (const poly &, const poly &);
00163
00164     static void mul_one_term (const poly &, const poly &, poly &);
00165     static void mul_univar (const poly &, const poly &, poly &, int);
00166     static void mul_heap (const poly &, const poly &, poly &);
00167
00168     static void divmod_one_term (const poly &, const poly &, poly &, poly &, bool);
00169     static void divmod_univar (const poly &, const poly &, poly &, poly &, int, bool);
00170     static void divmod_heap (const poly &, const poly &, poly &, poly &, bool, bool, bool&);
00171
00172     static void push_heap (PHEAD WORD **, int);
00173     static void pop_heap (PHEAD WORD **, int);
00174
00175     PADPOINTER(1,0,2,0);
00176 };
00177
00178 // comparison class for monomials (for std::sort)
00179 class monomial_larger {
00180
00181 public:
00182
00183     #ifndef WITHPTHREADS
00184         monomial_larger() {}
00185     #else
00186         ALLPRIVATES *B;
00187         monomial_larger(ALLPRIVATES *b): B(b) {}
00188     #endif
00189
00190     bool operator()(const WORD *a, const WORD *b) {
00191         return poly::monomial_compare(BHEAD a, b) > 0;
00192     }
00193 };
00194
00195 // stream operator
00196 std::ostream& operator<< (std::ostream &, const poly &);
00197
00198 // inline function definitions
00199
00200 #if __cplusplus >= 201103L
00201
00202 inline poly::poly (poly &&a, WORD modp, WORD modn) noexcept {
00203     #ifdef POLY_MOVE_DEBUG
00204         std::cout << "poly move ctor" << std::endl;
00205     #endif
00206     POLY_GETIDENTITY(a);
00207     POLY_STOREIDENTITY;
00208     terms = a.terms;
00209     size_of_terms = a.size_of_terms;
00210     this->modp = a.modp;
00211     this->modn = a.modn;
00212     if (modp != -1) {
00213         setmod(modp,modn);

```

```

00214     }
00215     a.terms = nullptr;
00216     a.size_of_terms = -1;
00217 }
00218
00219 inline poly& poly::operator= (poly &a) noexcept {
00220 #ifdef POLY_MOVE_DEBUG
00221     std::cout << "poly move assign" << std::endl;
00222 #endif
00223     if (this != &a) {
00224         WORD *old_terms = terms;
00225         LONG old_size_of_terms = size_of_terms;
00226         terms = a.terms;
00227         size_of_terms = a.size_of_terms;
00228         modp = a.modp;
00229         modn = a.modn;
00230         a.terms = old_terms;
00231         a.size_of_terms = old_size_of_terms;
00232     }
00233     return *this;
00234 }
00235
00236 #endif
00237
00238 /* Checks whether the terms array is large enough to add another
00239 * term (of size AM.MaxTal) to the polynomials. In case not, it is
00240 * expanded.
00241 */
00242 inline void poly::check_memory (int i) {
00243     POLY_GETIDENTITY(*this);
00244     if (i + 3 + AN.poly_num_vars + AM.MaxTal >= size_of_terms) expand_memory(i + AM.MaxTal);
00245 // Used to be i+2 but there should also be space for a trailing zero
00246 }
00247
00248 // indexing operators
00249 inline WORD& poly::operator[] (int i) {
00250     return terms[i];
00251 }
00252
00253 inline const WORD& poly::operator[] (int i) const {
00254     return terms[i];
00255 }
00256
00257 /* Copies "num" WORD-sized terms from the pointer "source" to the
00258 * current polynomial at index "dest"
00259 */
00260 inline void poly::termscopy (const WORD *source, int dest, int num) {
00261 #ifdef POLY_MOVE_DEBUG
00262     std::cout << "poly termscopy: num=" << num << std::endl;
00263 #endif
00264     memcpy (terms+dest, source, num*sizeof(WORD));
00265 }

```

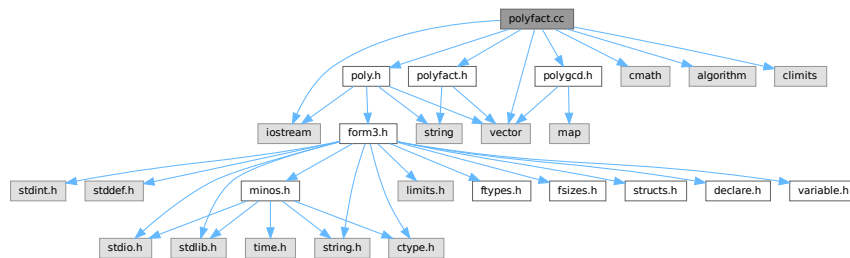
## 4.102 polyfact.cc File Reference

```

#include "poly.h"
#include "polygcd.h"
#include "polyfact.h"
#include <cmath>
#include <vector>
#include <iostream>
#include <algorithm>
#include <climits>

```

Include dependency graph for polyfact.cc:



## Functions

- ostream & [operator<<](#) (ostream &out, const [factorized\\_poly](#) &a)
- [template<class T >](#)  
ostream & [operator<<](#) (ostream &out, const vector< T > &v)

### 4.102.1 Detailed Description

Contains the routines for factorizing multivariate polynomials

Definition in file [polyfact.cc](#).

### 4.102.2 Function Documentation

#### 4.102.2.1 [operator<<\(\)](#) [1/2]

```
ostream & operator<< (
    ostream & out,
    const factorized\_poly & a )
```

Definition at line [104](#) of file [polyfact.cc](#).

#### 4.102.2.2 [operator<<\(\)](#) [2/2]

```
template<class T >
ostream & operator<< (
    ostream & out,
    const vector< T > & v )
```

Definition at line [109](#) of file [polyfact.cc](#).



## 4.103 polyfact.cc

[Go to the documentation of this file.](#)

```

00001
00005 /* #[ License : */
00006 /*
00007 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00008 *   When using this file you are requested to refer to the publication
00009 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010 *   This is considered a matter of courtesy as the development was paid
00011 *   for by FOM the Dutch physics granting agency and we would like to
00012 *   be able to track its scientific use to convince FOM of its value
00013 *   for the community.
00014 *
00015 *   This file is part of FORM.
00016 *
00017 *   FORM is free software: you can redistribute it and/or modify it under the
00018 *   terms of the GNU General Public License as published by the Free Software
00019 *   Foundation, either version 3 of the License, or (at your option) any later
00020 *   version.
00021 *
00022 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00023 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00024 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00025 *   details.
00026 *
00027 *   You should have received a copy of the GNU General Public License along
00028 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00029 */
00030 /* #[ License : */
00031 /*
00032     #[ include :
00033 */
00034
00035 #include "poly.h"
00036 #include "polygcd.h"
00037 #include "polyfact.h"
00038
00039 #include <cmath>
00040 #include <vector>
00041 #include <iostream>
00042 #include <algorithm>
00043 #include <climits>
00044
00045 // #define DEBUG
00046
00047 #ifdef DEBUG
00048 #include "mytime.h"
00049 #endif
00050
00051 using namespace std;
00052
00053 /*
00054     #[ include :
00055     #[ tostring :
00056 */
00057
00058 // Turns a factorized_poly into a readable string
00059 const string factorized_poly::tostring () const {
00060
00061     // empty
00062     if (factor.size()==0)
00063         return "no_factors";
00064
00065     string res;
00066
00067     // polynomial
00068     for (int i=0; i<(int)factor.size(); i++) {
00069         if (i>0) res += "*";
00070         res += "(";
00071         res += poly(factor[i],0,1).to_string();
00072         res += ")";
00073         if (power[i]>1) {
00074             res += "^";
00075             char tmp[100];
00076             snprintf (tmp,100,"%i",power[i]);
00077             res += tmp;
00078         }
00079     }
00080
00081     // modulo p^n
00082     if (factor[0].modp>0) {
00083         res += " (mod ";
00084         char tmp[12];
00085         snprintf (tmp,12,"%i",factor[0].modp);

```

```

00086         res += tmp;
00087         if (factor[0].modn > 1) {
00088             snprintf(tmp,12,"%i",factor[0].modn);
00089             res += "^";
00090             res += tmp;
00091         }
00092         res += ")";
00093     }
00094
00095     return res;
00096 }
00097
00098 /*
00099  #] toString :
00100  #[ ostream operator :
00101  */
00102
00103 // ostream operator for outputting a factorized_poly
00104 ostream& operator<< (ostream &out, const factorized_poly &a) {
00105     return out << a.toString();
00106 }
00107
00108 // ostream operator for outputting a vector<T>
00109 template<class T> ostream& operator<< (ostream &out, const vector<T> &v) {
00110     out<<"{";
00111     for (int i=0; i<(int)v.size(); i++) {
00112         if (i>0) out<<",";
00113         out<<v[i];
00114     }
00115     out<<"}";
00116     return out;
00117 }
00118
00119 /*
00120  #] ostream operator :
00121  #[ add_factor :
00122  */
00123
00124 // adds a factor f^p to a factorization
00125 void factorized_poly::add_factor(const poly &f, int p) {
00126     factor.push_back(f);
00127     power.push_back(p);
00128 }
00129
00130 /*
00131  #] add_factor :
00132  #[ extended_gcd_Euclidean_lifted :
00133  */
00134
00152 const vector<poly> polyfact::extended_gcd_Euclidean_lifted (const poly &a, const poly &b) {
00153
00154 #ifdef DEBUGALL
00155     cout << "*** [" << thetime() << " ] CALL: extended_Euclidean_lifted(" <<a<<","<<b<<")<<endl;
00156 #endif
00157
00158     POLY_GETIDENTITY(a);
00159
00160     // Calculate s,t,gcd (mod p) with the Extended Euclidean Algorithm.
00161     poly s(a,a.modp,1);
00162     poly t(b,b.modp,1);
00163     poly sa(BHEAD 1,a.modp,1);
00164     poly sb(BHEAD 0,b.modp,1);
00165     poly ta(BHEAD 0,a.modp,1);
00166     poly tb(BHEAD 1,b.modp,1);
00167
00168     while (!t.is_zero()) {
00169         poly x(s/t);
00170         swap(s -=x*t , t);
00171         swap(sa-=x*ta, ta);
00172         swap(sb-=x*tb, tb);
00173     }
00174
00175     // Normalize the result.
00176     sa /= s.integer_lcoeff();
00177     sb /= s.integer_lcoeff();
00178
00179     // Lift the result to modulo p^n with p-adic Newton's iteration.
00180     poly samodp(sa);
00181     poly sbmodp(sb);
00182     poly term(BHEAD 1);
00183
00184     sa.setmod(0,1);
00185     sb.setmod(0,1);
00186
00187     poly amodp(a,a.modp,1);
00188     poly bmodp(b,a.modp,1);
00189     poly error(poly(BHEAD 1) - sa*a - sb*b);

```

```

00190     poly p(BHEAD a.modp);
00191
00192     for (int n=1; n<a.modn && !error.is_zero(); n++) {
00193         error /= p;
00194         term *= p;
00195         poly errormodp(error,a.modp,1);
00196         poly dsa((samodp * errormodp) % bmodp);
00197         // equivalent, but faster than the symmetric
00198         // poly dsb((sbmodp * errormodp) % amodp);
00199         poly dsb((errormodp - dsa*amodp) / bmodp);
00200         sa += term * dsa;
00201         sb += term * dsb;
00202         error -= a*dsa + b*dsb;
00203     }
00204
00205     sa.setmod(a.modp,a.modn);
00206     sb.setmod(a.modp,a.modn);
00207
00208     // Output the result
00209     vector<poly> res;
00210     res.push_back(sa);
00211     res.push_back(sb);
00212
00213 #ifdef DEBUGALL
00214     cout << "*** [" << thetime() << "]" RES : extended_Euclidean_lifted("«a«","«b«") = "«res«endl;
00215 #endif
00216
00217     return res;
00218 }
00219
00220 /*
00221 #] extended_gcd_Euclidean_lifted :
00222 #[ solve_Diophantine_univariate :
00223 */
00224
00225 const vector<poly> polyfact::solve_Diophantine_univariate (const vector<poly> &a, const poly &b) {
00226
00227 #ifdef DEBUGALL
00228     cout << "*** [" << thetime() << "]" CALL: solve_Diophantine_univariate(" «a«","«b«")«endl;
00229 #endif
00230
00231     POLY_GETIDENTITY(b);
00232
00233     vector<poly> s(1,b);
00234
00235     for (int i=0; i+1<(int)a.size(); i++) {
00236         poly A(BHEAD 1,b.modp,b.modn);
00237         for (int j=i+1; j<(int)a.size(); j++) A *= a[j];
00238
00239         vector<poly> t(extended_gcd_Euclidean_lifted(a[i],A));
00240         poly prev(s.back());
00241         s.back() = t[1] * prev % a[i];
00242         s.push_back(t[0] * prev % A);
00243     }
00244
00245 #ifdef DEBUGALL
00246     cout << "*** [" << thetime() << "]" RES : solve_Diophantine_univariate(" «a«","«b«") = "«s«endl;
00247 #endif
00248
00249     return s;
00250 }
00251
00252 /*
00253 #] solve_Diophantine_univariate :
00254 #[ solve_Diophantine_multivariate :
00255 */
00256
00257 const vector<poly> polyfact::solve_Diophantine_multivariate (const vector<poly> &a, const poly &b,
00258     const vector<int> &x, const vector<int> &c, int d) {
00259
00260 #ifdef DEBUGALL
00261     cout << "*** [" << thetime() << "]" CALL: solve_Diophantine_multivariate("
00262 «a«","«b«","«x«","«c«","«d«")«endl;
00263 #endif
00264
00265     POLY_GETIDENTITY(b);
00266
00267     if (b.is_zero()) return vector<poly>(a.size(),poly(BHEAD 0));
00268
00269     if (x.size() == 1) return solve_Diophantine_univariate(a,b);
00270
00271     // Reduce the polynomial with the homomorphism <xm-c{m-1}>
00272     poly simple(poly::simple_poly(BHEAD x.back(),c.back()));
00273
00274     vector<poly> ared(a);
00275     for (int i=0; i<(int)ared.size(); i++)
00276         ared[i] %= simple;

```

```

00337
00338     poly bred(b % simple);
00339     vector<int> xred(x.begin(), x.end()-1);
00340     vector<int> cred(c.begin(), c.end()-1);
00341
00342     // Solve the equation in one less variable
00343     vector<poly> s(solve_Diophantine_multivariate(ared, bred, xred, cred, d));
00344     if (s == vector<poly>()) return vector<poly>();
00345
00346     // Cache the Ai = product(aj | j!=i).
00347     vector<poly> A(a.size(), poly(BHEAD 1, b.modp, b.modn));
00348     for (int i=0; i<(int)a.size(); i++)
00349         for (int j=0; j<(int)a.size(); j++)
00350             if (i!=j) A[i] *= a[j];
00351
00352     // Add the powers (xm-c{m-1})^k with ideal-adic Newton iteration.
00353     poly term(BHEAD 1, b.modp, b.modn);
00354
00355     poly error(b);
00356     for (int i=0; i<(int)A.size(); i++)
00357         error -= s[i] * A[i];
00358
00359     for (int deg=1; deg<=d; deg++) {
00360
00361         if (error.is_zero()) break;
00362
00363         error /= simple;
00364         term *= simple;
00365
00366         vector<poly> ds(solve_Diophantine_multivariate(ared, error%simple, xred, cred, d));
00367         if (ds == vector<poly>()) return vector<poly>();
00368
00369         for (int i=0; i<(int)s.size(); i++) {
00370             s[i] += ds[i] * term;
00371             error -= ds[i] * A[i];
00372         }
00373     }
00374
00375     if (!error.is_zero()) return vector<poly>();
00376
00377 #ifdef DEBUGALL
00378     cout << "*** [" << thetime() << "] RES : solve_Diophantine_multivariate("
00379         << a<< ", " << b<< ", " << x<< ", " << c<< ", " << d<< ") = " << s<< endl;
00380 #endif
00381     return s;
00382 }
00383
00384 /*
00385 #] solve_Diophantine_multivariate :
00386 #[ lift_coefficients :
00387 */
00388
00421 const vector<poly> polyfact::lift_coefficients (const poly &_A, const vector<poly> &_a) {
00422
00423 #ifdef DEBUG
00424     cout << "*** [" << thetime() << "] CALL: lift_coefficients(" << _A<< ", " << _a<< ")" << endl;
00425 #endif
00426
00427     POLY_GETIDENTITY(_A);
00428
00429     poly A(_A);
00430     vector<poly> a(_a);
00431     poly term(BHEAD 1);
00432
00433     int x = A.first_variable();
00434
00435     // Replace the leading term of all factors with lterm(A) mod p
00436     poly lead(A.integer_lcoeff());
00437     for (int i=0; i<(int)a.size(); i++) {
00438         a[i] *= lead / a[i].integer_lcoeff();
00439         if (i>0) A*=lead;
00440     }
00441
00442     // Solve Diophantine equation
00443     vector<poly> s(solve_Diophantine_univariate(a, poly(BHEAD 1, A.modp, 1)));
00444
00445     // Replace the leading term of all factors with lterm(A) mod p^n
00446     for (int i=0; i<(int)a.size(); i++) {
00447         a[i].setmod(A.modp, A.modn);
00448         a[i] += (lead - a[i].integer_lcoeff()) * poly::simple_poly(BHEAD x, 0, a[i].degree(x));
00449     }
00450
00451     // Calculate the error, express it in terms of ai and add corrections.
00452     for (int k=2; k<=A.modn; k++) {
00453         term *= poly(BHEAD A.modp);

```

```

00454
00455     poly error(BHEAD -1);
00456     for (int i=0; i<(int)a.size(); i++) error *= a[i];
00457     error += A;
00458     if (error.is_zero()) break;
00459
00460     error /= term;
00461     error.setmod(A.modp,1);
00462
00463     for (int i=0; i<(int)a.size(); i++)
00464         a[i] += term * (error * s[i] % a[i]);
00465 }
00466
00467 // Fix leading coefficients by dividing out integer contents.
00468 for (int i=0; i<(int)a.size(); i++)
00469     a[i] /= polygcd::integer_content(poly(a[i],0,1));
00470
00471 #ifdef DEBUG
00472     cout << "*** [" << thetime() << "]" RES : lift_coefficients("<_A<","<_a<") = "<a<<endl;
00473 #endif
00474
00475     return a;
00476 }
00477
00478 /*
00479     #] lift_coefficients :
00480     #[ predetermine :
00481 */
00482
00497 void polyfact::predetermine (int dep, const vector<vector<int> > &state, vector<vector<vector<int> > >
    &terms, vector<int> &term, int sumdeg) {
00498     // store the term
00499     if (dep == (int)state.size()) {
00500         terms[sumdeg].push_back(term);
00501         return;
00502     }
00503
00504     // recursively create new terms
00505     term.push_back(0);
00506
00507     for (int deg=0; sumdeg+deg<(int)state[dep].size(); deg++)
00508         if (state[dep][deg] > 0) {
00509             term.back() = deg;
00510             predetermine(dep+1, state, terms, term, sumdeg+deg);
00511         }
00512
00513     term.pop_back();
00514 }
00515
00516 /*
00517     #] predetermine :
00518     #[ lift_variables :
00519 */
00520
00558 const vector<poly> polyfact::lift_variables (const poly &A, const vector<poly> &_a, const vector<int>
    &x, const vector<int> &c, const vector<poly> &lc) {
00559
00560 #ifdef DEBUG
00561     cout << "*** [" << thetime() << "]" CALL: lift_variables("<A<","<_a<","<x<","<c<","<lc<")\n";
00562 #endif
00563
00564     // If univariate, don't lift
00565     if (x.size()<=1) return _a;
00566
00567     POLY_GETIDENTITY(A);
00568
00569     vector<poly> a(_a);
00570
00571     // First method: predetermine coefficients
00572
00573     // check feasibility, otherwise it tries too many possibilities
00574     int cnt = POLYFACT_MAX_PREDETERMINATION;
00575     for (int i=0; i<(int)a.size(); i++) {
00576         if (a[i].number_of_terms() == 0) return vector<poly>();
00577         cnt /= a[i].number_of_terms();
00578     }
00579
00580     if (cnt>0) {
00581         // state[n][d]: coefficient of x^d in a[n] is
00582         // 0: non-existent, 1: undetermined, 2: determined
00583         int D = A.degree(x[0]);
00584         vector<vector<int> > state(a.size(), vector<int>(D+1, 0));
00585
00586         for (int i=0; i<(int)a.size(); i++)
00587             for (int j=1; j<a[i][0]; j+=a[i][j])
00588                 state[i][a[i][j+1+x[0]]] = j==1 ? 2 : 1;
00589

```

```

00590         // collect all products of terms
00591         vector<vector<vector<int> > > terms(D+1);
00592         vector<int> term;
00593         predetermine(0, state, terms, term);
00594
00595         // count the number of undetermined coefficients
00596         vector<int> num_undet (terms.size(), 0);
00597         for (int i=0; i<(int)terms.size(); i++)
00598             for (int j=0; j<(int)terms[i].size(); j++)
00599                 for (int k=0; k<(int)terms[i][j].size(); k++)
00600                     if (state[k][terms[i][j][k]] == 1) num_undet[i]++;
00601
00602         // replace the current leading coefficients by the correct ones
00603         for (int i=0; i<(int)a.size(); i++)
00604             a[i] += (lc[i] - a[i].lcoeff_univar(x[0])) * poly::simple_poly(BHEAD
x[0], 0, a[i].degree(x[0]));
00605
00606         bool changed;
00607         do {
00608             changed = false;
00609
00610             for (int i=0; i<(int)terms.size(); i++) {
00611                 // is only one coefficient in a equation is undetermined, solve
00612                 // the equation to determine this coefficient
00613                 if (num_undet[i] == 1) {
00614                     // generate equation
00615                     poly lhs(BHEAD 0), rhs(A.coefficient(x[0], i), A.modp, A.modn);
00616                     int which_idx=-1, which_deg=-1;
00617                     for (int j=0; j<(int)terms[i].size(); j++) {
00618                         poly coeff(BHEAD 1, A.modp, A.modn);
00619                         bool undet=false;
00620                         for (int k=0; k<(int)terms[i][j].size(); k++) {
00621                             if (state[k][terms[i][j][k]] == 1) {
00622                                 undet = true;
00623                                 which_idx=k;
00624                                 which_deg=terms[i][j][k];
00625                             }
00626                             else
00627                                 coeff *= a[k].coefficient(x[0], terms[i][j][k]);
00628                         }
00629                         if (undet)
00630                             lhs = coeff;
00631                         else
00632                             rhs -= coeff;
00633                     }
00634                     // solve equation
00635                     if (A.modn > 1) rhs.setmod(0, 1);
00636                     if (lhs.is_zero() || !(rhs%lhs).is_zero()) return vector<poly>();
00637                     a[which_idx] += (rhs / lhs - a[which_idx].coefficient(x[0], which_deg)) *
poly::simple_poly(BHEAD x[0], 0, which_deg);
00638                     state[which_idx][which_deg] = 2;
00639
00640                     // update number of undetermined coefficients
00641                     for (int j=0; j<(int)terms.size(); j++)
00642                         for (int k=0; k<(int)terms[j].size(); k++)
00643                             if (terms[j][k][which_idx] == which_deg)
00644                                 num_undet[j]--;
00645
00646                     changed = true;
00647                 }
00648             }
00649         } while (changed);
00650
00651         // if this is the complete result, skip lifting
00652         poly check(BHEAD 1, A.modn>1?A.modp, 1);
00653         for (int i=0; i<(int)a.size(); i++)
00654             check *= a[i];
00655
00656         if (check == A) return a;
00657     }
00658
00659     // Second method: Hensel lifting
00660
00661     // Calculate A and lc's modulo Ii = <xi-c[i-1], ..., xm-c[m-1]> (for i=2, ..., m)
00662     vector<poly> simple(x.size(), poly(BHEAD 0));
00663     for (int i=(int)x.size()-2; i>=0; i--)
00664         simple[i] = poly::simple_poly(BHEAD x[i+1], c[i], 1);
00665
00666     // Calculate the maximum degree of A in x2, ..., xm
00667     int maxdegA=0;
00668     for (int i=1; i<(int)x.size(); i++)
00669         maxdegA = MaX(maxdegA, A.degree(x[i]));
00670
00671     // Iteratively add the variables x2, ..., xm
00672     for (int xi=1; xi<(int)x.size(); xi++) {
00673         // replace the current leading coefficients by the correct ones

```

```

00675         for (int i=0; i<(int)a.size(); i++)
00676             a[i] += (lc[i] - a[i].lcoeff_univar(x[0])) * poly::simple_poly(BHEAD
x[0],0,a[i].degree(x[0]));
00677
00678         vector<poly> anew(a);
00679         for (int i=0; i<(int)anew.size(); i++)
00680             for (int j=xi-1; j<(int)c.size(); j++)
00681                 anew[i] %= simple[j];
00682
00683         vector<int> xnew(x.begin(), x.begin()+xi);
00684         vector<int> cnew(c.begin(), c.begin()+xi-1);
00685         poly term(BHEAD 1,A.modp,A.modn);
00686
00687         // Iteratively add the powers xi^k
00688         for (int deg=1, maxdeg=A.degree(x[xi]); deg<=maxdeg; deg++) {
00689
00690             term *= simple[xi-1];
00691
00692             // Calculate the error, express it in terms of ai and add corrections.
00693             poly error(BHEAD -1,A.modp,A.modn);
00694             for (int i=0; i<(int)a.size(); i++) error += a[i];
00695             error += A;
00696             for (int i=xi; i<(int)c.size(); i++) error %= simple[i];
00697
00698             if (error.is_zero()) break;
00699
00700             error /= term;
00701             error %= simple[xi-1];
00702
00703             vector<poly> s(solve_Diophantine_multivariate(anew,error,xnew,cnew,maxdegA));
00704             if (s == vector<poly>()) return vector<poly>();
00705
00706             for (int i=0; i<(int)a.size(); i++)
00707                 a[i] += s[i] * term;
00708         }
00709
00710         // check whether PRODUCT(a[i]) = A mod <xi-c{i-1},...,xm-c{m-1}> over the integers or ZZ/p
00711         poly check(BHEAD -1, A.modn>1?0:A.modp, 1);
00712         for (int i=0; i<(int)a.size(); i++) check *= a[i];
00713         check += A;
00714         for (int i=xi; i<(int)c.size(); i++) check %= simple[i];
00715         if (!check.is_zero()) return vector<poly>();
00716     }
00717
00718 #ifdef DEBUG
00719     cout << "*** [" << thetime() << "]" RES : lift_variables("«A«", "«_a«", "«x«", "«c«", "«lc«") = " << a <<
endl;
00720 #endif
00721
00722     return a;
00723 }
00724
00725 /*
00726 #] lift_variables :
00727 #[ choose_prime :
00728 */
00729
00741 WORD polyfact::choose_prime (const poly &a, const vector<int> &x, WORD p) {
00742
00743 #ifdef DEBUG
00744     cout << "*** [" << thetime() << "]" CALL: choose_prime("«a«", "«x«", "«p«") << endl;
00745 #endif
00746
00747     POLY_GETIDENTITY(a);
00748
00749     poly icont_lcoeff(polygcd::integer_content(a.lcoeff_univar(x[0])));
00750
00751     if (p==0) p = POLYFACT_FIRST_PRIME;
00752
00753     poly icont_lcoeff_modp(BHEAD 0);
00754
00755     do {
00756
00757         bool is_prime;
00758
00759         do {
00760             p += 2;
00761             is_prime = true;
00762             for (int d=2; d*d<=p; d++)
00763                 if (p%d==0) { is_prime=false; break; }
00764         }
00765         while (!is_prime);
00766
00767         icont_lcoeff_modp = icont_lcoeff;
00768         icont_lcoeff_modp.setmod(p,1);
00769     }
00770     while (icont_lcoeff_modp.is_zero());

```

```

00771
00772 #ifdef DEBUG
00773     cout << "*** [" << thetime() << "]" RES : choose_prime("«a«", "«x«", ?) = "«p«endl;
00774 #endif
00775
00776     return p;
00777 }
00778
00779 /*
00780 #] choose_prime :
00781 #[ choose_prime_power :
00782 */
00783
00798 WORD polyfact::choose_prime_power (const poly &a, WORD p) {
00799
00800 #ifdef DEBUG
00801     cout << "*** [" << thetime() << "]" CALL: choose_prime_power("«a«", "«p«") <<endl;
00802 #endif
00803
00804     POLY_GETIDENTITY(a);
00805
00806     // analyse the polynomial for calculating the bound
00807     double maxdegree=0, maxlogcoeff=0, numterms=0;
00808
00809     for (int i=1; i<a[0]; i+=a[i]) {
00810         for (int j=0; j<AN.poly_num_vars; j++)
00811             maxdegree = MaX(maxdegree, a[i+1+j]);
00812
00813         maxlogcoeff = MaX(maxlogcoeff,
00814             log(1.0+(UWORD)a[i+a[i]-2]) + // most
00815             significant digit + 1
00816             BITSINWORD*log(2.0)*(ABS(a[i+a[i]-1])-1)); // number of
00817             digits
00818             numterms++;
00819     }
00820
00821     WORD res = (WORD)ceil((log((sqrt(5.0)+1)/2)*maxdegree + maxlogcoeff + 0.5*log(numterms)) /
00822         log((double)p));
00823
00824 #ifdef DEBUG
00825     cout << "*** [" << thetime() << "]" CALL: choose_prime_power("«a«", "«p«") = "«res«endl;
00826 #endif
00827
00828     return res;
00829 }
00830
00831 /*
00832 #] choose_prime_power :
00833 #[ choose_ideal :
00834 */
00835
00860 const vector<int> polyfact::choose_ideal (const poly &a, int p, const factorized_poly &lc, const
vector<int> &x) {
00861
00862 #ifdef DEBUG
00863     cout << "*** [" << thetime() << "]" CALL: polyfact::choose_ideal("
00864         «a«", "«p«", "«lc«", "«x«") <<endl;
00865 #endif
00866
00867     if (x.size()==1) return vector<int>();
00868
00869     POLY_GETIDENTITY(a);
00870
00871     vector<int> c(x.size()-1);
00872
00873     int dega = a.degree(x[0]);
00874     poly amodI(a);
00875
00876     // choose random c
00877     for (int i=0; i<(int)c.size(); i++) {
00878         c[i] = 1 + wranf(BHEAD0) % ((p-1) / POLYFACT_IDEAL_FRACTION);
00879         amodI %= poly::simple_poly(BHEAD x[i+1], c[i], 1);
00880     }
00881
00882     poly amodIp(amodI);
00883     amodIp.setmod(p, 1);
00884
00885     // check if leading coefficient is non-zero [equivalent to degree=old_degree]
00886     if (amodIp.degree(x[0]) != dega)
00887         return c = vector<int>();
00888
00889     // check if leading coefficient is squarefree [equivalent to gcd(a,a')==const]
00890     if (!polygcd::gcd_Euclidean(amodIp, amodIp.derivative(x[0])).is_integer())
00891         return c = vector<int>();
00892
00893     if (a.modp>0 && a.modn==1) return c;
00894

```



```

00895 // check for unique prime factors in each factor lc[i] of the leading coefficient
00896 vector<poly> d(1, polygcd::integer_content(amodI));
00897
00898 for (int i=0; i<(int)lc.factor.size(); i++) {
00899     // constant factor
00900     if (i==0 && lc.factor[i].is_integer()) {
00901         d[0] *= lc.factor[i];
00902         continue;
00903     }
00904
00905     // factor modulo I
00906     poly q(lc.factor[i]);
00907     for (int j=0; j<(int)c.size(); j++)
00908         q %= poly::simple_poly(BHEAD x[j+1],c[j]);
00909     if (q.sign() == -1) q *= poly(BHEAD -1);
00910
00911     // divide out common factors
00912     for (int j=(int)d.size()-1; j>=0; j--) {
00913         poly r(d[j]);
00914         while (!r.is_one()) {
00915             r = polygcd::integer_gcd(r,q);
00916             q /= r;
00917         }
00918     }
00919
00920     // check whether there is some factor left
00921     if (q.is_one()) return vector<int>();
00922     d.push_back(q);
00923 }
00924
00925 #ifdef DEBUG
00926     cout << "*** [" << thetime() << "] RES : polyfact::choose_ideal("
00927         <<a<<","<p<<","<lc<<","<xx<<") = "<c<<endl;
00928 #endif
00929     return c;
00930 }
00931
00932 /*
00933 #] choose_ideal :
00934 #[ squarefree_factors_Yun :
00935 #] squarefree_factors_modp :
00936 */
00937
00944 const factorized_poly polyfact::squarefree_factors_Yun (const poly &a) {
00945
00946     factorized_poly res;
00947     poly a(_a);
00948
00949     int pow = 1;
00950     int x = a.first_variable();
00951
00952     poly b(a.derivative(x));
00953     poly c(polygcd::gcd(a,b));
00954
00955     while (true) {
00956         a /= c;
00957         b /= c;
00958         b -= a.derivative(x);
00959         if (b.is_zero()) break;
00960         c = polygcd::gcd(a,b);
00961         if (!c.is_one()) res.add_factor(c,pow);
00962         pow++;
00963     }
00964
00965     if (!a.is_one()) res.add_factor(a,pow);
00966     return res;
00967 }
00968
00969 /*
00970 #] squarefree_factors_Yun :
00971 #[ squarefree_factors_modp :
00972 */
00973
00980 const factorized_poly polyfact::squarefree_factors_modp (const poly &a) {
00981
00982     factorized_poly res;
00983     poly a(_a);
00984
00985     int pow = 1;
00986     int x = a.first_variable();
00987     poly b(a.derivative(x));
00988
00989     // poly contains terms of the form c(x)^n (n!=c*p)
00990     if (!b.is_zero()) {
00991         poly c(polygcd::gcd(a,b));
00992         a /= c;
00993     }

```

```

00994         while (!a.is_one()) {
00995             b = polygcd::gcd(a,c);
00996             a /= b;
00997             if (!a.is_one()) res.add_factor(a,pow);
00998             pow++;
00999             a = b;
01000             c /= a;
01001         }
01002
01003         a = c;
01004     }
01005
01006     // polynomial contains terms of the form c(x)^p
01007     if (!a.is_one()) {
01008         for (int i=1; i<a[1]; i+=a[i])
01009             a[i+1+x] /= a.modp;
01010         factorized_poly res2(squarefree_factors(a));
01011         for (int i=0; i<(int)res2.factor.size(); i++) {
01012             res.factor.push_back(res2.factor[i]);
01013             res.power.push_back(a.modp*res2.power[i]);
01014         }
01015     }
01016
01017     return res;
01018 }
01019
01020 /*
01021  #] squarefree_factors_modp :
01022  #[ squarefree_factors :
01023  */
01024
01045 const factorized_poly polyfact::squarefree_factors (const poly &a) {
01046
01047 #ifdef DEBUG
01048     cout << "*** [" << thetime() << " ] CALL: squarefree_factors("«a«")\n";
01049 #endif
01050
01051     if (a.is_one()) return factorized_poly();
01052
01053     factorized_poly res;
01054
01055     if (a.modp==0)
01056         res = squarefree_factors_Yun(a);
01057     else
01058         res = squarefree_factors_modp(a);
01059
01060 #ifdef DEBUG
01061     cout << "*** [" << thetime() << " ] RES : squarefree_factors("«a«") = "«res«"\n";
01062 #endif
01063
01064     return res;
01065 }
01066
01067 /*
01068  #] squarefree_factors :
01069  #[ Berlekamp_Qmatrix :
01070  */
01071
01094 const vector<vector<WORD> > polyfact::Berlekamp_Qmatrix (const poly &a) {
01095
01096 #ifdef DEBUG
01097     cout << "*** [" << thetime() << " ] CALL: Berlekamp_Qmatrix("«_a«")\n";
01098 #endif
01099
01100     if (_a.all_variables() == vector<int>())
01101         return vector<vector<WORD> >(0);
01102
01103     POLY_GETIDENTITY(_a);
01104
01105     poly a(_a);
01106     int x = a.first_variable();
01107     int n = a.degree(x);
01108     int p = a.modp;
01109
01110     poly lc(a.integer_lcoeff());
01111     a /= lc;
01112
01113     vector<vector<WORD> > Q(n, vector<WORD>(n));
01114
01115     // c is the vector of coefficients of the polynomial a
01116     vector<WORD> c(n+1,0);
01117     for (int j=1; j<a[0]; j+=a[j])
01118         c[a[j+1+x]] = a[j+1+AN.poly_num_vars] * a[j+2+AN.poly_num_vars];
01119
01120     // d is the vector of coefficients of x^i mod a, starting with i=0
01121     vector<WORD> d(n,0);
01122     d[0]=1;

```

```

01123
01124     for (int i=0; i<=(n-1)*p; i++) {
01125         // store the coefficients of x^(i*p) mod a
01126         if (i%p==0) Q[i/p] = d;
01127
01128         // transform d=x^i mod a into d=x^(i+1) mod a
01129         vector<WORD> e(n);
01130         for (int j=0; j<n; j++) {
01131             e[j] = (- (LONG)d[n-1]*c[j] + (j>0?d[j-1]:0)) % p;
01132             if (e[j]<0) e[j]+=p;
01133         }
01134
01135         d=e;
01136     }
01137
01138     // Q = Q - I
01139     for (int i=0; i<n; i++)
01140         Q[i][i] = (Q[i][i] - 1 + p) % p;
01141
01142     // Gaussian elimination
01143     for (int i=0; i<n; i++) {
01144         // Find pivot
01145         int ii=i; while (ii<n && Q[ii][ii]==0) ii++;
01146         if (ii==n) continue;
01147
01148         for (int k=0; k<n; k++)
01149             swap(Q[k][ii],Q[k][i]);
01150
01151         // normalize row i, which becomes the pivot
01152         WORD mul;
01153         GetModInverses (Q[i][i], p, &mul, NULL);
01154         vector<int> idx;
01155
01156         for (int k=0; k<n; k++) if (Q[k][i] != 0) {
01157             // store indices of non-zero elements for sparse matrices
01158             idx.push_back(k);
01159             Q[k][i] = ((LONG)Q[k][i] * mul) % p;
01160         }
01161
01162         // reduce
01163         for (int j=0; j<n; j++)
01164             if (j!=i && Q[i][j]!=0) {
01165                 LONG mul = Q[i][j];
01166                 for (int k=0; k<(int)idx.size(); k++) {
01167                     Q[idx[k]][j] = (Q[idx[k]][j] - mul*Q[idx[k]][i]) % p;
01168                     if (Q[idx[k]][j] < 0) Q[idx[k]][j]+=p;
01169                 }
01170             }
01171     }
01172
01173     for (int i=0; i<n; i++) {
01174         // Q = Q - I
01175         Q[i][i] = Q[i][i]-1;
01176
01177         // reduce all coefficients in the range 0,1,...,p-1
01178         for (int j=0; j<n; j++) {
01179             Q[i][j] = -Q[i][j]%p;
01180             if (Q[i][j]<0) Q[i][j]+=p;
01181         }
01182     }
01183
01184 #ifndef DEBUG
01185     cout << "*** [" << thetime() << "]" RES : Berlekamp_Qmatrix("«_a«") = "«Q«"\n";
01186 #endif
01187
01188     return Q;
01189 }
01190
01191 /*
01192 #] Berlekamp_Qmatrix :
01193 #[ Berlekamp_find_factors :
01194 */
01195
01219 const vector<poly> polyfact::Berlekamp_find_factors (const poly &a, const vector<vector<WORD> > &q)
01220 {
01221 #ifndef DEBUG
01222     cout << "*** [" << thetime() << "]" CALL: Berlekamp_find_factors("«_a«","«_q«")\n";
01223 #endif
01224
01225     if (_a.all_variables() == vector<int>()) return vector<poly>(1,_a);
01226
01227     POLY_GETIDENTITY(_a);
01228
01229     vector<vector<WORD> > q=_q;
01230
01231     int rank=0;

```

```

01232     for (int i=0; i<(int)q.size(); i++)
01233         if (q[i]!=vector<WORD>(q[i].size(),0)) rank++;
01234
01235     poly a(_a);
01236     int x = a.first_variable();
01237     int n = a.degree(x);
01238     int p = a.modp;
01239
01240     a /= a.integer_lcoeff();
01241
01242     // Vector of factors, represented as dense polynomials mod p
01243     vector<vector<WORD> > fac(1, vector<WORD>(n+1,0));
01244
01245     fac[0] = poly::to_coefficient_list(a);
01246     bool finished=false;
01247
01248     // Loop over the columns of q + constant, i.e., an exhaustive list of possible factors
01249     for (int i=1; i<n && !finished; i++) {
01250         if (q[i] == vector<WORD>(n,0)) continue;
01251
01252         for (int s=0; s<p && !finished; s++) {
01253             for (int j=0; j<(int)fac.size() && !finished; j++) {
01254                 vector<WORD> c = polygcd::coefficient_list_gcd(fac[j], q[i], p);
01255
01256                 // If a non-trivial factor is found, add it to the list
01257                 if (c.size()!=1 && c.size()!=fac[j].size()) {
01258                     fac.push_back(c);
01259                     fac[j] = poly::coefficient_list_divmod(fac[j], c, p, 0);
01260                     if ((int)fac.size() == rank) finished=true;
01261                 }
01262             }
01263
01264             // Increase the constant term by one
01265             q[i][0] = (q[i][0]+1) % p;
01266         }
01267     }
01268
01269     // Convert the densely represented polynomials to sparse ones
01270     vector<poly> res(fac.size(),poly(BHEAD 0, p));
01271     for (int i=0; i<(int)fac.size(); i++)
01272         res[i] = poly::from_coefficient_list(BHEAD fac[i],x,p);
01273
01274 #ifdef DEBUG
01275     cout << "*** [" << thetime() << "]" RES : Berlekamp_find_factors("<a_a","<a_q") = "<res"<n";
01276 #endif
01277
01278     return res;
01279 }
01280
01281 /*
01282 #] Berlekamp_find_factors :
01283 #[ combine_factors :
01284 */
01285
01307 const vector<poly> polyfact::combine_factors (const poly &a, const vector<poly> &f) {
01308
01309 #ifdef DEBUG
01310     cout << "*** [" << thetime() << "]" CALL: combine_factors("<a_a","<a_f")<n";
01311 #endif
01312
01313     POLY_GETIDENTITY(a);
01314
01315     poly a0(a,0,1);
01316     vector<poly> res;
01317
01318     int num_used = 0;
01319     vector<bool> used(f.size(), false);
01320
01321     // Loop over all bitmasks with num=1,2,...,size(factors)/2 bits
01322     // set, that contain only unused factors
01323     for (int num=1; num<=(int)(f.size() - num_used)/2; num++) {
01324         vector<int> next(f.size() - num_used, 0);
01325         for (int i=0; i<num; i++) next[next.size()-1-i] = 1;
01326
01327         do {
01328             poly fac(BHEAD 1,a.modp,a.modn);
01329             for (int i=0, j=0; i<(int)f.size(); i++)
01330                 if (!used[i] && next[j++]) fac *= f[i];
01331             fac /= fac.integer_lcoeff();
01332             fac *= a.integer_lcoeff();
01333             fac /= polygcd::integer_content(poly(fac,0,1));
01334
01335             if ((a0 % fac).is_zero()) {
01336                 res.push_back(fac);
01337                 for (int i=0, j=0; i<(int)f.size(); i++)
01338                     if (!used[i]) used[i] = next[j++];
01339                 num_used += num;

```

```

01340         num--;
01341         break;
01342     }
01343 }
01344 while (next_permutation(next.begin(), next.end()));
01345 }
01346
01347 // All unused factors together form one more factor
01348 if (num_used != (int)f.size()) {
01349     poly fac(BHEAD 1, a.modp, a.modn);
01350     for (int i=0; i<(int)f.size(); i++)
01351         if (!used[i]) fac *= f[i];
01352     fac /= fac.integer_lcoeff();
01353     fac *= a.integer_lcoeff();
01354     fac /= polygcd::integer_content(poly(fac, 0, 1));
01355     res.push_back(fac);
01356 }
01357
01358 #ifdef DEBUG
01359     cout << "*** [" << thetime() << " ] RES : combine_factors(" << a << ", " << f << ") = " << res << "\n";
01360 #endif
01361 return res;
01362 }
01363
01364 /*
01365 #] combine_factors :
01366 #[ factorize_squarefree :
01367 */
01368
01369 const vector<poly> polyfact::factorize_squarefree (const poly &a, const vector<int> &x) {
01370
01371     #ifdef DEBUG
01372         cout << "*** [" << thetime() << " ] CALL: factorize_squarefree(" << a << ") \n";
01373     #endif
01374
01375     POLY_GETIDENTITY(a);
01376
01377     WORD p=a.modp, n=a.modn;
01378
01379     try_again:
01380
01381     int bestp=0;
01382     int min_factors = INT_MAX;
01383     poly amodI(BHEAD 0), bestamodI(BHEAD 0);
01384     vector<int> c,d,bestc,bestd;
01385     vector<vector<WORD>> > q,bestq;
01386
01387     // Factorize leading coefficient
01388     factorized_poly lc(factorize(a.lcoeff_univar(x[0])));
01389
01390     // Try a number of primes
01391     int prime_tries = 0;
01392
01393     while (prime_tries<POLYFACT_NUM_CONFIRMATIONS && min_factors>1) {
01394         if (a.modp == 0) {
01395             p = choose_prime(a,x,p);
01396             n = 0;
01397             if (a.degree(x[0]) % p == 0) continue;
01398
01399             // Univariate case: check whether the polynomial mod p is squarefree
01400             // Multivariate case: this check is done after choosing I (for efficiency)
01401             if (x.size()==1) {
01402                 poly amodp(a,p,1);
01403                 if (polygcd::gcd_Euclidean(amodp, amodp.derivative(x[0])).degree(x[0]) != 0)
01404                     continue;
01405             }
01406
01407             // Try a number of ideals
01408             if (x.size()>1)
01409                 for (int ideal_tries=0; ideal_tries<POLYFACT_MAX_IDEAL_TRIES; ideal_tries++) {
01410                     c = choose_ideal(a,p,lc,x);
01411                     if (c.size()>0) break;
01412                 }
01413
01414             if (x.size()==1 || c.size()>0) {
01415                 amodI = a;
01416                 for (int i=0; i<(int)c.size(); i++)
01417                     amodI %= poly::simple_poly(BHEAD x[i+1],c[i]);
01418
01419                 // Determine Q-matrix and its rank. Smaller rank is better.
01420                 q = Berlekamp_Qmatrix(poly(amodI,p,1));
01421                 int rank=0;
01422                 for (int i=0; i<(int)q.size(); i++)
01423                     if (q[i]!=vector<WORD>(q[i].size(),0)) rank++;
01424
01425                 bestp=p;
01426                 min_factors = rank;
01427                 bestamodI = amodI;
01428                 bestc=c;
01429                 bestd=d;
01430                 bestq=q;
01431             }
01432             prime_tries++;
01433         }
01434     }
01435
01436     if (bestp==0)
01437         return vector<poly>();
01438
01439     vector<poly> res;
01440     for (int i=0; i<(int)bestc.size(); i++)
01441         res.push_back(polyfact::factorize_squarefree(bestamodI, bestc[i]));
01442
01443     return res;
01444 }

```

```

01445         if (rank<min_factors) {
01446             bestp=p;
01447             bestc=c;
01448             bestq=q;
01449             bestamodI=amodI;
01450             min_factors = rank;
01451             prime_tries = 0;
01452         }
01453
01454         if (rank==min_factors)
01455             prime_tries++;
01456
01457 #ifdef DEBUG
01458     cout << "*** [" << thetime() << "] (A) : factorize_squarefree("a«
01459         " try p=" << p << " #factors=" << rank << " (min="«min_factors«"x"«prime_tries«")" << endl;
01460 #endif
01461     }
01462 }
01463
01464 p=bestp;
01465 c=bestc;
01466 q=bestq;
01467 amodI=bestamodI;
01468
01469 // Determine to which power of p to lift
01470 if (n==0) {
01471     n = choose_prime_power(amodI,p);
01472     n = MaX(n, choose_prime_power(a,p));
01473 }
01474
01475 amodI.setmod(p,n);
01476
01477 #ifdef DEBUG
01478     cout << "*** [" << thetime() << "] (B) : factorize_squarefree("a«
01479         " chosen c = " << c << ", p^n = "«p«"^"«n«endl;
01480     cout << "*** [" << thetime() << "] (C) : factorize_squarefree("a«
01481         " #factors = " << min_factors << endl;
01482 #endif
01483
01484 // Find factors
01485 vector<poly> f(Berlekamp_find_factors(poly(amodI,p,1),q));
01486
01487 // Lift coefficients
01488 if (f.size() > 1) {
01489     f = lift_coefficients(amodI,f);
01490     if (f==vector<poly>()) {
01491 #ifdef DEBUG
01492         cout << "factor_squarefree failed (lift_coeff step) : " << endl;
01493 #endif
01494         goto try_again;
01495     }
01496
01497 // Combine factors
01498 if (a.modp==0) f = combine_factors(amodI,f);
01499 }
01500
01501 // Lift variables
01502 if (f.size() == 1)
01503     f = vector<poly>(1, a);
01504
01505 if (x.size() > 1 && f.size() > 1) {
01506
01507     // The correct leading coefficients of the factors can be
01508     // reconstructed from prime number factors of the leading
01509     // coefficients modulo I. This is possible since all factors of
01510     // the leading coefficient have unique prime factors for the ideal
01511     // I is chosen as such.
01512
01513     poly amodI(a);
01514     for (int i=0; i<(int)c.size(); i++)
01515         amodI %= poly::simple_poly(BHEAD x[i+1],c[i]);
01516     poly delta(polygcd::integer_content(amodI));
01517
01518     vector<poly> lcmmodI(lc.factor.size(), poly(BHEAD 0));
01519     for (int i=0; i<(int)lc.factor.size(); i++) {
01520         lcmmodI[i] = lc.factor[i];
01521         for (int j=0; j<(int)c.size(); j++)
01522             lcmmodI[i] %= poly::simple_poly(BHEAD x[j+1],c[j]);
01523     }
01524
01525     vector<poly> correct_lc(f.size(), poly(BHEAD 1,p,n));
01526
01527     for (int j=0; j<(int)f.size(); j++) {
01528         poly lc_f(f[j].integer_lcoeff() * delta);
01529         WORD nlc_f = lc_f[lc_f[1]];
01530         poly quo(BHEAD 0,rem(BHEAD 0);
01531         WORD nquo,nrem;

```

```

01532
01533     for (int i=(int)lcmmodI.size()-1; i>=0; i--) {
01534
01535         if (i==0 && lc.factor[i].is_integer()) continue;
01536
01537         do {
01538             DivLong((UWORD *)&lc_f[2+AN.poly_num_vars], nlc_f,
01539                     (UWORD *)&lcmmodI[i][2+AN.poly_num_vars], lcmmodI[i][lcmmodI[i][1]],
01540                     (UWORD *)&quo[0], &nquo,
01541                     (UWORD *)&rem[0], &nrem);
01542
01543             if (nrem == 0) {
01544                 correct_lc[j] *= lc.factor[i];
01545                 lc_f.termscopy(&quo[0], 2+AN.poly_num_vars, ABS(nquo));
01546                 nlc_f = nquo;
01547             }
01548         } while (nrem == 0);
01549     }
01550 }
01551
01552
01553 for (int i=0; i<(int)correct_lc.size(); i++) {
01554     poly correct_modI(correct_lc[i]);
01555     for (int j=0; j<(int)c.size(); j++)
01556         correct_modI %= poly::simple_poly(BHEAD x[j+1],c[j]);
01557
01558     poly d(polygcd::integer_gcd(correct_modI, f[i].integer_lcoeff()));
01559     correct_lc[i] *= f[i].integer_lcoeff() / d;
01560     delta /= correct_modI / d;
01561     f[i] *= correct_modI / d;
01562 }
01563
01564 // increase n, because of multiplying with delta
01565 if (!delta.is_one()) {
01566     poly deltapow(BHEAD 1);
01567     for (int i=1; i<(int)correct_lc.size(); i++)
01568         deltapow *= delta;
01569     while (!deltapow.is_zero()) {
01570         deltapow /= poly(BHEAD p);
01571         n++;
01572     }
01573
01574     for (int i=0; i<(int)f.size(); i++) {
01575         f[i].modn = n;
01576         correct_lc[i].modn = n;
01577     }
01578 }
01579
01580 poly aa(a,p,n);
01581
01582 for (int i=0; i<(int)correct_lc.size(); i++) {
01583     correct_lc[i] *= delta;
01584     f[i] *= delta;
01585     if (i>0) aa *= delta;
01586 }
01587
01588 f = lift_variables(aa,f,x,c,correct_lc);
01589
01590 for (int i=0; i<(int)f.size(); i++)
01591     if (a.modp == 0)
01592         f[i] /= polygcd::integer_content(poly(f[i],0,1));
01593     else
01594         f[i] /= polygcd::content_univar(f[i], x[0]);
01595
01596 if (f==vector<poly>()) {
01597 #ifdef DEBUG
01598     cout << "factor_squarefree failed (lift_var step)" << endl;
01599 #endif
01600     goto try_again;
01601 }
01602
01603 // set modulus of the factors correctly
01604 if (a.modp==0)
01605     for (int i=0; i<(int)f.size(); i++)
01606         f[i].setmod(0,1);
01607
01608 // Final check (not sure if this is necessary, but it doesn't hurt)
01609 poly check(BHEAD 1,a.modp,a.modn);
01610 for (int i=0; i<(int)f.size(); i++)
01611     check *= f[i];
01612
01613 if (check != a) {
01614 #ifdef DEBUG
01615     cout << "factor_squarefree failed (final check) : " << f << endl;
01616 #endif
01617     goto try_again;
01618 }

```

```

01619     }
01620
01621 #ifdef DEBUG
01622     cout << "*** [" << thetime() << "]" RES : factorize_squarefree("«a«", "«x«", "«c«") = "«f«"\n";
01623 #endif
01624
01625     return f;
01626 }
01627
01628 /*
01629     #] factorize_squarefree :
01630     #[ factorize :
01631 */
01632
01641 const factorized_poly polyfact::factorize (const poly &a) {
01642
01643 #ifdef DEBUG
01644     cout << "*** [" << thetime() << "]" CALL: factorize("«a«")\n";
01645 #endif
01646     vector<int> x = a.all_variables();
01647
01648     // No variables, so just one factor
01649     if (x.size() == 0) {
01650         factorized_poly res;
01651         if (a.is_one()) return res;
01652         res.add_factor(a,1);
01653         return res;
01654     }
01655
01656     // Remove content
01657     poly conta(polygcd::content_univar(a,x[0]));
01658
01659     factorized_poly faca(factorize(conta));
01660
01661     poly ppa(a / conta);
01662
01663     // Find a squarefree factorization
01664     factorized_poly b(squarefree_factors(ppa));
01665
01666 #ifdef DEBUG
01667     cout << "*** [" << thetime() << "]" ... : factorize("«a«") : SFF = "«b«"\n";
01668 #endif
01669
01670     // Factorize each squarefree factor and build the "factorized_poly"
01671     for (int i=0; i<(int)b.factor.size(); i++) {
01672
01673         vector<poly> c(factorize_squarefree(b.factor[i], x));
01674
01675         for (int j=0; j<(int)c.size(); j++) {
01676             faca.factor.push_back(c[j]);
01677             faca.power.push_back(b.power[i]);
01678         }
01679     }
01680
01681 #ifdef DEBUG
01682     cout << "*** [" << thetime() << "]" RES : factorize("«a«") = "«faca«"\n";
01683 #endif
01684
01685     return faca;
01686 }
01687
01688 /*
01689     #] factorize :
01690 */

```

## 4.104 polyfact.h

```

00001 #pragma once
00002 /* #[ License : */
00003 /*
00004 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00005 *   When using this file you are requested to refer to the publication
00006 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00007 *   This is considered a matter of courtesy as the development was paid
00008 *   for by FOM the Dutch physics granting agency and we would like to
00009 *   be able to track its scientific use to convince FOM of its value
00010 *   for the community.
00011 *
00012 *   This file is part of FORM.
00013 *
00014 *   FORM is free software: you can redistribute it and/or modify it under the
00015 *   terms of the GNU General Public License as published by the Free Software
00016 *   Foundation, either version 3 of the License, or (at your option) any later

```



```

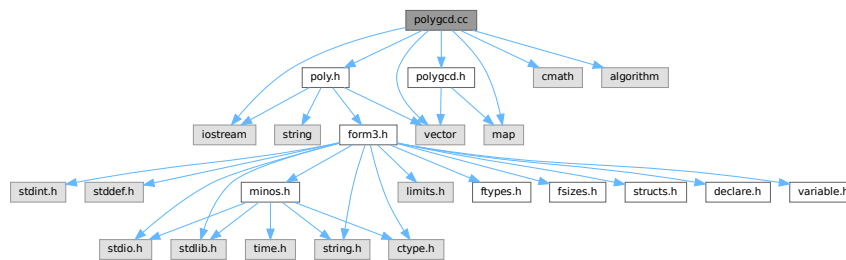
00017 *   version.
00018 *
00019 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00020 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00021 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00022 *   details.
00023 *
00024 *   You should have received a copy of the GNU General Public License along
00025 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00026 */
00027 /* #] License : */
00028
00029 #include <vector>
00030 #include <string>
00031
00032 // First prime modulo which factorization is tried. Too small results
00033 // in more unsuccessful attempts; too large is slower.
00034 const int POLYFACT_FIRST_PRIME = 17;
00035
00036 // Fraction of [1,p) that is used for substitutions of variables. Too
00037 // small results in more unsuccessful attempts; too large is slower.
00038 const int POLYFACT_IDEAL_FRACTION = 5;
00039
00040 // Number of ideals that are tried before failure due to unlucky
00041 // choices is accepted.
00042 const int POLYFACT_MAX_IDEAL_TRIES = 3;
00043
00044 // Number of confirmations for the minimal number of factors before
00045 // Hensel lifting is started.
00046 const int POLYFACT_NUM_CONFIRMATIONS = 3;
00047
00048 // Maximum number of equations for predetermination of coefficients
00049 // for multivariate Hensel lifting
00050 const int POLYFACT_MAX_PREDETERMINATION = 10000;
00051
00052 class poly;
00053
00054 class factorized_poly {
00055     /* Class for representing a factorized polynomial
00056      * The polynomial is given by: PRODUCT(factor[i] ^ power[i])
00057      */
00058 public:
00059     std::vector<poly> factor;
00060     std::vector<int> power;
00061
00062     void add_factor(const poly &f, int p=1);
00063     const std::string toString() const;
00064     friend std::ostream& operator<< (std::ostream &out, const poly &p);
00065 };
00066
00067 std::ostream& operator<< (std::ostream &out, const factorized_poly &a);
00068
00069 namespace polyfact {
00070
00071     // factorization routine
00072     const factorized_poly factorize (const poly &a);
00073
00074     // methods for squarefree factorization
00075     const factorized_poly squarefree_factors (const poly &a);
00076     const factorized_poly squarefree_factors_Yun (const poly &a);
00077     const factorized_poly squarefree_factors_modp (const poly &a);
00078
00079     // methods for choosing suitable reductions
00080     const std::vector<poly> factorize_squarefree (const poly &a, const std::vector<int> &x);
00081     WORD choose_prime (const poly &a, const std::vector<int> &x, WORD p=0);
00082     WORD choose_prime_power (const poly &a, WORD p);
00083     const std::vector<int> choose_ideal (const poly &a, int p, const factorized_poly &lc, const
std::vector<int> &x);
00084
00085     // methods for univariate factorization
00086     const std::vector<std::vector<WORD> > Berlekamp_Qmatrix (const poly &a);
00087     const std::vector<poly> Berlekamp_find_factors (const poly &a, const std::vector<std::vector<WORD>
> &Q);
00088     const std::vector<poly> combine_factors (const poly &a, const std::vector<poly> &f);
00089
00090     // methods for Hensel lifting
00091     const std::vector<poly> extended_gcd_Euclidean_lifted (const poly &a, const poly &b);
00092     const std::vector<poly> solve_Diophantine_univariate (const std::vector<poly> &a, const poly &b);
00093     const std::vector<poly> solve_Diophantine_multivariate (const std::vector<poly> &a, const poly &b,
const std::vector<int> &x, const std::vector<int> &c, int d);
00094     const std::vector<poly> lift_coefficients (const poly &a, const std::vector<poly> &f);
00095     const std::vector<poly> lift_variables (const poly &a, const std::vector<poly> &f, const
std::vector<int> &x, const std::vector<int> &c, const std::vector<poly> &lc);
00096     void predetermine (int dep, const std::vector<std::vector<int> > &state,
std::vector<std::vector<std::vector<int> > > &terms, std::vector<int> &term, int sumdeg=0);
00097
00098 }

```

## 4.105 polygcd.cc File Reference

```
#include "poly.h"
#include "polygcd.h"
#include <iostream>
#include <vector>
#include <cmath>
#include <map>
#include <algorithm>
```

Include dependency graph for polygcd.cc:



### Data Structures

- struct [BracketInfo](#)

### Functions

- bool [gcd\\_heuristic\\_possible](#) (const [poly](#) &a)
- const [poly](#) [gcd\\_linear\\_helper](#) (const [poly](#) &a, const [poly](#) &b)

### 4.105.1 Detailed Description

Contains the routines for calculating greatest commons divisors of multivariate polynomials

Definition in file [polygcd.cc](#).

### 4.105.2 Function Documentation

#### 4.105.2.1 gcd\_heuristic\_possible()

```
bool gcd_heuristic_possible (
    const poly & a )
```

Heuristic greatest common divisor of multivariate polynomials

### 4.105.3 Description

Checks whether the heuristic seems possible by estimating

$\text{MAX\_terms} \{ \text{coeff} \wedge \text{PROD}_{\{i=1..\#\text{vars}\}} (\text{pow}_i+1) \}$

and comparing this with GCD\_HEURISTIC\_MAX\_DIGITS.

### 4.105.4 Notes

- For small polynomials, this consumes time and never triggers.

Definition at line 1142 of file [polygcd.cc](#).

#### 4.105.4.1 gcd\_linear\_helper()

```
const poly gcd_linear_helper (
    const poly & a,
    const poly & b )
```

Definition at line 1403 of file [polygcd.cc](#).

## 4.106 polygcd.cc

[Go to the documentation of this file.](#)

```
00001
00006 /* #[] License : */
00007 /*
00008 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 * When using this file you are requested to refer to the publication
00010 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 * This is considered a matter of courtesy as the development was paid
00012 * for by FOM the Dutch physics granting agency and we would like to
00013 * be able to track its scientific use to convince FOM of its value
00014 * for the community.
00015 *
00016 * This file is part of FORM.
00017 *
00018 * FORM is free software: you can redistribute it and/or modify it under the
00019 * terms of the GNU General Public License as published by the Free Software
00020 * Foundation, either version 3 of the License, or (at your option) any later
00021 * version.
00022 *
00023 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 * details.
00027 *
00028 * You should have received a copy of the GNU General Public License along
00029 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #[] License : */
00032 /*
00033 #[] include :
00034 */
00035
00036 #include "poly.h"
00037 #include "polygcd.h"
00038
00039 #include <iostream>
00040 #include <vector>
00041 #include <cmath>
00042 #include <map>
00043 #include <algorithm>
```

```

00044
00045 //define DEBUG
00046 //define DEBUGALL
00047
00048 #ifdef DEBUG
00049 #include "mytime.h"
00050 #endif
00051
00052 using namespace std;
00053
00054 /*
00055     #] include :
00056     #[ ostream operator :
00057 */
00058
00059 #ifdef DEBUG
00060 // ostream operator for outputting vector<T>s for debugging purposes
00061 template<class T> ostream& operator<< (ostream &out, const vector<T> &x) {
00062     out<<"{";
00063     for (int i=0; i<(int)x.size(); i++) {
00064         if (i>0) out<<",";
00065         out<<x[i];
00066     }
00067     out<<"}";
00068     return out;
00069 }
00070 #endif
00071
00072 /*
00073     #] ostream operator :
00074     #[ integer_gcd :
00075 */
00076
00091 const poly polygcd::integer_gcd (const poly &a, const poly &b) {
00092
00093     #ifdef DEBUGALL
00094         cout << "*** [" << thetime() << " ] CALL: integer_gcd(" << a << "," << b << ")" << endl;
00095     #endif
00096
00097     POLY_GETIDENTITY(a);
00098
00099     if (a.is_zero()) return b;
00100     if (b.is_zero()) return a;
00101
00102     poly c(BHEAD 0, a.modp, a.modn);
00103     WORD nc;
00104
00105     GcdLong(BHEAD
00106             (UWORD *)&a[AN.poly_num_vars+2],a[a[0]-1],
00107             (UWORD *)&b[AN.poly_num_vars+2],b[b[0]-1],
00108             (UWORD *)&c[AN.poly_num_vars+2],&nc);
00109
00110     WORD x = 2 + AN.poly_num_vars + ABS(nc);
00111     c[1] = x; // term length
00112     c[0] = x+1; // total length
00113     c[x] = nc; // coefficient length
00114
00115     for (int i=0; i<AN.poly_num_vars; i++)
00116         c[2+i] = 0; // powers
00117
00118     #ifdef DEBUGALL
00119         cout << "*** [" << thetime() << " ] RES : integer_gcd(" << a << "," << b << ") = " << c << endl;
00120     #endif
00121
00122     return c;
00123 }
00124
00125 /*
00126     #] integer_gcd :
00127     #[ integer_content :
00128 */
00129
00143 const poly polygcd::integer_content (const poly &a) {
00144
00145     #ifdef DEBUGALL
00146         cout << "*** [" << thetime() << " ] CALL: integer_content(" << a << ")" << endl;
00147     #endif
00148
00149     POLY_GETIDENTITY(a);
00150
00151     if (a.modp>0) return a.integer_lcoeff();
00152
00153     poly c(BHEAD 0, 0, 1);
00154     WORD *d = (WORD *)NumberMalloc("polygcd::integer_content");
00155     WORD nc=0;
00156
00157     for (int i=0; i<AN.poly_num_vars; i++)

```

```

00158         c[2+i] = 0;
00159
00160     for (int i=1; i<a[0]; i+=a[i]) {
00161
00162         WCOPY(d,&c[2+AN.poly_num_vars],nc);
00163
00164         GcdLong(BHEAD (UWORD *)d, nc,
00165                 (UWORD *)&a[i+1+AN.poly_num_vars], a[i+a[i]-1],
00166                 (UWORD *)&c[2+AN.poly_num_vars], &nc);
00167
00168         WORD x = 2 + AN.poly_num_vars + ABS(nc);
00169         c[1] = x; // term length
00170         c[0] = x+1; // total length
00171         c[x] = nc; // coefficient length
00172     }
00173
00174     if (a.sign() != c.sign()) c *= poly(BHEAD -1);
00175
00176     NumberFree(d,"polygcd::integer_content");
00177
00178 #ifdef DEBUGALL
00179     cout << "*** [" << thetime() << "]" RES : integer_content(" << a << ") = " << c << endl;
00180 #endif
00181
00182     return c;
00183 }
00184
00185 /*
00186 #] integer_content :
00187 #[ content_univar :
00188 */
00189
00205 const poly polygcd::content_univar (const poly &a, int x) {
00206
00207 #ifdef DEBUG
00208     cout << "*** [" << thetime() << "]" CALL: content_univar(" << a << "," << string(1,'a'+x) << ") << endl;
00209 #endif
00210
00211     POLY_GETIDENTITY(a);
00212
00213     poly res(BHEAD 0, a.modp, a.modn);
00214
00215     for (int i=1; i<a[0];) {
00216         poly b(BHEAD 0, a.modp, a.modn);
00217         int deg = a[i+1+x];
00218
00219         for (; i<a[0] && a[i+1+x]==deg; i+=a[i]) {
00220             b.check_memory(b[0]+a[i]);
00221             b.termscopy(&a[i],b[0],a[i]);
00222             b[b[0]+1+x] = 0;
00223             b[0] += a[i];
00224         }
00225
00226         res = gcd(res, b);
00227
00228         if (res.is_integer()) {
00229             res = integer_content(a);
00230             break;
00231         }
00232     }
00233
00234     if (a.sign() != res.sign()) res *= poly(BHEAD -1);
00235
00236 #ifdef DEBUG
00237     cout << "*** [" << thetime() << "]" RES : content_univar(" << a << "," << string(1,'a'+x) << ") = " << res
00238     << endl;
00239 #endif
00240     return res;
00241 }
00242
00243 /*
00244 #] content_univar :
00245 #[ content_multivar :
00246 */
00247
00263 const poly polygcd::content_multivar (const poly &a, int x) {
00264
00265 #ifdef DEBUGALL
00266     cout << "*** [" << thetime() << "]" CALL: content_multivar(" << a << "," << string(1,'a'+x) << ") <<
00267     endl;
00268 #endif
00269
00270     POLY_GETIDENTITY(a);
00271
00272     poly res(BHEAD 0, a.modp, a.modn);
00273

```

```

00273     for (int i=1,j; i<a[0]; i=j) {
00274         poly b(BHEAD 0, a.modp, a.modn);
00275
00276         for (j=i; j<a[0]; j+=a[j]) {
00277             bool same_powers = true;
00278             for (int k=0; k<AN.poly_num_vars; k++)
00279                 if (k!=x && a[i+1+k]!=a[j+1+k]) {
00280                     same_powers = false;
00281                     break;
00282                 }
00283             if (!same_powers) break;
00284
00285             b.check_memory(b[0]+a[j]);
00286             b.termscopy(&a[j],b[0],a[j]);
00287             for (int k=0; k<AN.poly_num_vars; k++)
00288                 if (k!=x) b[b[0]+1+k]=0;
00289
00290             b[0] += a[j];
00291         }
00292
00293         res = gcd_Euclidean(res, b);
00294
00295         if (res.is_integer()) {
00296             res = poly(BHEAD a.sign(),a.modp,a.modn);
00297             break;
00298         }
00299     }
00300
00301 #ifdef DEBUGALL
00302     cout << "*** [" << thetime() << "] RES : content_multivar(" << a << ", " << string(1,'a'+x) << ") = " <<
00303     res << endl;
00304 #endif
00305     return res;
00306 }
00307
00308 /*
00309     #] content_multivar :
00310         #[ coefficient_list_gcd :
00311 */
00312
00325 const vector<WORD> polygcd::coefficient_list_gcd (const vector<WORD> &a, const vector<WORD> &b, WORD
p) {
00326
00327 #ifdef DEBUGALL
00328     cout << "*** [" << thetime() << "] CALL: coefficient_list_gcd("<<_a<<","<<_b<<","<<p<<")<<endl;
00329 #endif
00330
00331     vector<WORD> a(_a), b(_b);
00332
00333     while (b.size() != 0) {
00334         a = poly::coefficient_list_divmod(a,b,p,1);
00335         swap(a,b);
00336     }
00337
00338     while (a.back()==0) a.pop_back();
00339
00340     WORD inv;
00341     GetModInverses(a.back() + (a.back()<0?p:0), p, &inv, NULL);
00342
00343     for (int i=0; i<(int)a.size(); i++)
00344         a[i] = (LONG)inv*a[i] % p;
00345
00346 #ifdef DEBUGALL
00347     cout << "*** [" << thetime() << "] RES : coefficient_list_gcd("<<_a<<","<<_b<<","<<p<<") = "<<a<<endl;
00348 #endif
00349
00350     return a;
00351 }
00352
00353 /*
00354     #] coefficient_list_gcd :
00355     #[ gcd_Euclidean :
00356 */
00357
00374 const poly polygcd::gcd_Euclidean (const poly &a, const poly &b) {
00375
00376 #ifdef DEBUG
00377     cout << "*** [" << thetime() << "] CALL: gcd_Euclidean("<<a<<","<<b<<")<<endl;
00378 #endif
00379
00380     POLY_GETIDENTITY(a);
00381
00382     if (a.is_zero()) return b;
00383     if (b.is_zero()) return a;
00384     if (a.is_integer() || b.is_integer()) return integer_gcd(a,b);
00385

```

```

00386     poly res(BHEAD 0);
00387
00388     if (a.is_dense_univariate()>=-1 && b.is_dense_univariate()>=-1) {
00389         vector<WORD> coeff = coefficient_list_gcd(poly::to_coefficient_list(a),
00390 poly::to_coefficient_list(b), a.modp);
00391         res = poly::from_coefficient_list(BHEAD coeff, a.first_variable(), a.modp);
00392     }
00393     else {
00394         res = a;
00395         poly rem(b);
00396         while (!rem.is_zero())
00397             swap(res%=rem, rem);
00398         res /= res.integer_lcoeff();
00399     }
00400
00401 #ifdef DEBUG
00402     cout << "*** [" << thetime() << "] RES : gcd_Euclidean(" << a << ", " << b << ") = " << res << endl;
00403 #endif
00404
00405     return res;
00406 }
00407
00408 /*
00409 #] gcd_Euclidean :
00410 #[ chinese_remainder :
00411 */
00412
00426 const poly polygcd::chinese_remainder (const poly &a1, const poly &m1, const poly &a2, const poly &m2)
00427 {
00428 #ifdef DEBUGALL
00429     cout << "*** [" << thetime() << "] CALL: chinese_remainder(" << a1 << ", " << m1 << ", " << a2 << ", " << m2 <<
00430 " << endl;
00431 #endif
00432
00433     POLY_GETIDENTITY(a1);
00434
00435     WORD nx,ny,nz;
00436     UWORD *x = (UWORD *)NumberMalloc("polygcd::chinese_remainder");
00437     UWORD *y = (UWORD *)NumberMalloc("polygcd::chinese_remainder");
00438     UWORD *z = (UWORD *)NumberMalloc("polygcd::chinese_remainder");
00439
00440     GetLongModInverses(BHEAD (UWORD *)&m1[2+AN.poly_num_vars], m1[m1[1]],
00441 (UWORD *)&m2[2+AN.poly_num_vars], m2[m2[1]],
00442 (UWORD *)x, &nx, NULL, NULL);
00443
00444     AddLong((UWORD *)&a2[2+AN.poly_num_vars], a2.is_zero() ? 0 : a2[a2[1]],
00445 (UWORD *)&a1[2+AN.poly_num_vars], a1.is_zero() ? 0 : -a1[a1[1]],
00446 y, &ny);
00447
00448     MulLong (x,nx,y,ny,z,&nz);
00449     MulLong (z,nz,(UWORD *)&m1[2+AN.poly_num_vars],m1[m1[1]],x,&nx);
00450
00451     AddLong (x,nx,(UWORD *)&a1[2+AN.poly_num_vars], a1.is_zero() ? 0 : a1[a1[1]],y,&ny);
00452
00453     MulLong ((UWORD *)&m1[2+AN.poly_num_vars], m1[m1[1]],
00454 (UWORD *)&m2[2+AN.poly_num_vars], m2[m2[1]],
00455 (UWORD *)z,&nz);
00456
00457     TakeNormalModulus (y,&ny,(UWORD *)z,nz,NOUNPACK);
00458
00459     poly res(BHEAD y,ny);
00460
00461     NumberFree(x,"polygcd::chinese_remainder");
00462     NumberFree(y,"polygcd::chinese_remainder");
00463     NumberFree(z,"polygcd::chinese_remainder");
00464
00465 #ifdef DEBUGALL
00466     cout << "*** [" << thetime() << "] RES : chinese_remainder(" << a1 << ", " << m1 << ", " << a2 << ", " << m2 <<
00467 " << res << endl;
00468 #endif
00469
00470     return res;
00471 }
00472
00473 /*
00474 #] chinese_remainder :
00475 #[ substitute :
00476 */
00477
00488 const poly polygcd::substitute(const poly &a, int x, int c) {
00489
00490     POLY_GETIDENTITY(a);
00491
00492     poly b(BHEAD 0);
00493

```

```

00494     if (a.is_zero()) {
00495         return b;
00496     }
00497
00498     bool zero=true;
00499     int bi=1;
00500
00501     // cache size is bounded by the degree in x, twice the number of terms of
00502     // the polynomial and a constant
00503     vector<WORD> cache(min(a.degree(x)+1,min(2*a.number_of_terms(),
00504 POLYGCD_RAISPOWMOD_CACHE_SIZE)), 0);
00505
00506     for (int ai=1; ai<=a[0]; ai+=a[ai]) {
00507         // last term or different power, then add term to b iff non-zero
00508         if (!zero) {
00509             bool add=false;
00510             if (ai==a[0])
00511                 add=true;
00512             else {
00513                 for (int i=0; i<AN.poly_num_vars; i++)
00514                     if (i!=x && a[ai+1+i]!=b[bi+1+i]) {
00515                         zero=true;
00516                         add=true;
00517                         break;
00518                     }
00519             }
00520
00521             if (add) {
00522                 if (b[bi+AN.poly_num_vars+1] < 0)
00523                     b[bi+AN.poly_num_vars+1] += a.modp;
00524                 bi+=b[bi];
00525             }
00526
00527             if (ai==a[0]) break;
00528         }
00529
00530         b.check_memory(bi);
00531
00532         // create new term in b
00533         if (zero) {
00534             b[bi] = 3+AN.poly_num_vars;
00535             for (int i=0; i<AN.poly_num_vars; i++)
00536                 b[bi+1+i] = a[ai+1+i];
00537             b[bi+1+x] = 0;
00538             b[bi+AN.poly_num_vars+1] = 0;
00539             b[bi+AN.poly_num_vars+2] = 1;
00540         }
00541
00542         // add term of a to the current term in b
00543         LONG coeff = a[ai+1+AN.poly_num_vars] * a[ai+2+AN.poly_num_vars];
00544         int pow = a[ai+1+x];
00545
00546         if (pow<(int)cache.size()) {
00547             if (cache[pow]==0)
00548                 cache[pow] = RaisPowMod(c, pow, a.modp);
00549             coeff = (coeff * cache[pow]) % a.modp;
00550         }
00551         else {
00552             coeff = (coeff * RaisPowMod(c, pow, a.modp)) % a.modp;
00553         }
00554
00555         b[bi+AN.poly_num_vars+1] = (coeff + b[bi+AN.poly_num_vars+1]) % a.modp;
00556         if (b[bi+AN.poly_num_vars+1] != 0) zero=false;
00557     }
00558
00559     b[0]=bi;
00560     b.setmod(a.modp);
00561
00562     return b;
00563 }
00564
00565 /*
00566     #] substitute :
00567     #[ sparse_interpolation helper functions :
00568 */
00569
00570 // Returns a list of size #terms(a) with entries PROD(ci^powi, i=2..n)
00571 const vector<int> polygcd::sparse_interpolation_get_mul_list (const poly &a, const vector<int> &x,
const vector<int> &c) {
00572     // cache size for variable x is bounded by the degree in x, twice
00573     // the number of terms of the polynomial and a constant
00574     vector<vector<WORD>> > cache(c.size());
00575     int max_cache_size = min(2*a.number_of_terms(),POLYGCD_RAISPOWMOD_CACHE_SIZE);
00576     for (int i=0; i<(int)c.size(); i++)
00577         cache[i] = vector<WORD>(min(a.degree(x[i+1])+1,max_cache_size), 0);
00578

```



```

00579     vector<int> res;
00580     for (int i=1; i<a[0]; i+=a[i]) {
00581         LONG coeff=1;
00582         for (int j=0; j<(int)c.size(); j++) {
00583             int pow = a[i+1+x[j+1]];
00584             if (pow<(int)cache[j].size()) {
00585                 if (cache[j][pow]==0)
00586                     cache[j][pow] = RaisPowMod(c[j], pow, a.modp);
00587                 coeff = (coeff * cache[j][pow]) % a.modp;
00588             }
00589             else {
00590                 coeff = (coeff * RaisPowMod(c[j], pow, a.modp)) % a.modp;
00591             }
00592         }
00593         res.push_back(coeff);
00594     }
00595     return res;
00596 }
00597
00598 // Multiplies the coefficients of a with the entries of mul
00599 void polygcd::sparse_interpolation_mul_poly (poly &a, const vector<int> &mul) {
00600     for (int i=1, j=0; i<a[0]; i+=a[i], j++)
00601         a[i+a[i]-2] = ((LONG)a[i+a[i]-2]*mul[j]) % a.modp;
00602 }
00603
00604 // Sets all coefficients to the range 0..modp-1 and the powers of x2...xn to 0
00605 const poly polygcd::sparse_interpolation_reduce_poly (const poly &a, const vector<int> &x) {
00606     poly res(a);
00607     for (int i=1; i<a[0]; i+=a[i]) {
00608         for (int j=1; j<(int)x.size(); j++)
00609             res[i+1+x[j]]=0;
00610         if (res[i+a[i]-1]==-1) {
00611             res[i+a[i]-1]=1;
00612             res[i+a[i]-2]=a.modp-res[i+a[i]-2];
00613         }
00614     }
00615     return res;
00616 }
00617
00618 // Collects entries with equal powers, so that the result is a proper polynomial
00619 const poly polygcd::sparse_interpolation_fix_poly (const poly &a, int x) {
00620
00621     POLY_GETIDENTITY(a);
00622     poly res(BHEAD 0, a.modp, 1);
00623
00624     int j=1;
00625     bool newterm=true;
00626
00627     for (int i=1; i<a[0]; i+=a[i]) {
00628         if (newterm)
00629             res.termscopy(&a[i], j, a[i]);
00630         else
00631             res[j+res[j]-2] = ((LONG)res[j+res[j]-2] + a[i+a[i]-2]) % a.modp;
00632
00633         newterm = i+a[i] == a[0] || res[j+1+x] != a[i+a[i]+1+x];
00634         if (newterm && res[j+res[j]-2]!=0) j += res[j];
00635     }
00636
00637     res[0]=j;
00638     return res;
00639 }
00640
00641 /*
00642     #] sparse_interpolation helper functions :
00643     #[ gcd_modular_sparse_interpolation :
00644 */
00645
00677 const poly polygcd::gcd_modular_sparse_interpolation (const poly &origa, const poly &origb, const
vector<int> &x, const poly &s) {
00678
00679 #ifdef DEBUG
00680     cout << "*** [" << thetime() << " ] CALL: gcd_modular_sparse_interpolation("
00681         << origa << ", " << origb << ", " << x << ", " << " << s << ")" << endl;
00682 #endif
00683
00684     POLY_GETIDENTITY(origa);
00685
00686     // strip multivariate content
00687     poly conta(content_multivar(origa, x.back()));
00688     poly contb(content_multivar(origb, x.back()));
00689     poly gcdconts(gcd_Euclidean(conta, contb));
00690     const poly& a = conta.is_one() ? origa : origa/conta;
00691     const poly& b = contb.is_one() ? origb : origb/contb;
00692
00693     // for non-monic cases, we need to normalize with the gcd of the lcoeffs of a poly in x[0]
00694     // or else the shape fitting does not work.
00695     // FIXME: the current implementation still rejects some valid shapes.

```

```

00696     poly lcgcd(BHEAD 1, a.modp);
00697     if (!s.lcoeff_univar(x[0]).is_integer()) {
00698         lcgcd = gcd_modular_dense_interpolation(a.lcoeff_univar(x[0]), b.lcoeff_univar(x[0]), x,
poly(BHEAD 0));
00699     }
00700
00701     // reduce polynomials
00702     poly ared(sparse_interpolation_reduce_poly(a,x));
00703     poly bred(sparse_interpolation_reduce_poly(b,x));
00704     poly sred(sparse_interpolation_reduce_poly(s,x));
00705     poly lred(sparse_interpolation_reduce_poly(lcgcd,x));
00706
00707     // set all coefficients to 1
00708     for (int i=1; i<sred[0]; i+=sred[i]) {
00709         sred[i+sred[i]-2] = sred[i+sred[i]-1] = 1;
00710     }
00711
00712     // generate random numbers and check there this set doesn't result
00713     // in a singular matrix
00714     vector<int> c(x.size()-1);
00715     vector<int> smul;
00716
00717     bool duplicates;
00718     do {
00719         for (int i=0; i<(int)c.size(); i++)
00720             c[i] = 1 + wranf(BHEAD0) % (a.modp-1);
00721         smul = sparse_interpolation_get_mul_list(s,x,c);
00722
00723         duplicates = false;
00724
00725         int fr=0,to=0;
00726         for (int i=1; i<s[0];) {
00727             int pow = s[i+1+x[0]];
00728             while (i<s[0] && s[i+1+x[0]]==pow) i+=s[i], to++;
00729             for (int j=fr; j<to; j++)
00730                 for (int k=fr; k<j; k++)
00731                     if (smul[j] == smul[k])
00732                         duplicates = true;
00733             fr=to;
00734         }
00735     } while (duplicates);
00736
00737     // get the lists to multiply the polynomials with every iteration
00738     vector<int> amul(sparse_interpolation_get_mul_list(a,x,c));
00739     vector<int> bmul(sparse_interpolation_get_mul_list(b,x,c));
00740     vector<int> lmul(sparse_interpolation_get_mul_list(lcgcd,x,c));
00741
00742     vector<vector<vector<LONG> > > M;
00743     vector<vector<LONG> > V;
00744
00745     int maxMsize=0;
00746
00747     // create (empty) matrices
00748     for (int i=1; i<s[0]; i+=s[i]) {
00749         if (i==1 || s[i+1+x[0]]!=s[i+1+x[0]-s[i]]) {
00750             M.push_back(vector<vector<LONG> >());
00751             V.push_back(vector<LONG>());
00752         }
00753         M.back().push_back(vector<LONG>());
00754         V.back().push_back(0);
00755         maxMsize = max(maxMsize, (int)M.back().size());
00756     }
00757
00758     // generate linear equations
00759     for (int numg=0; numg<maxMsize; numg++) {
00760         poly amodI(sparse_interpolation_fix_poly(ared,x[0]));
00761         poly bmodI(sparse_interpolation_fix_poly(bred,x[0]));
00762         poly lmodI(sparse_interpolation_fix_poly(lred,x[0]));
00763
00764         // A fix for non-monic gcds. This could be slow if lmodI has many terms,
00765         // since it overfits the gcd now. Another gcd has to be run to remove the
00766         // extra terms.
00767         poly gcd(lmodI * gcd_Euclidean(amodI,bmodI));
00768
00769         // if correct gcd
00770         if (!gcd.is_zero() && gcd[2+x[0]]==sred[2+x[0]]) {
00771             // for each power in the gcd, generate an equation if needed
00772             int gi=1, midx=0;
00773
00774             for (int si=1; si<s[0];) {
00775                 // if the term exists, set Vi=coeff, otherwise Vi remains 0
00776                 if (gi<gcd[0] && gcd[gi+1+x[0]]==sred[si+1+x[0]]) {
00777                     if (numg < (int)V[midx].size())
00778                         V[midx][numg] = gcd[gi+gcd[gi]-1]*gcd[gi+gcd[gi]-2];
00779                 }
00780                 gi+=s[si];
00781             }

```

```

00782         gi += gcd[gi];
00783     }
00784
00785     // add the coefficients of s to the matrix M
00786     for (int i=0; i<(int)M[midx].size(); i++) {
00787         if (numg < (int)M[midx].size())
00788             M[midx][numg].push_back(sred[si+1+AN.poly_num_vars]);
00789         si += s[si];
00790     }
00791
00792     midx++;
00793 }
00794 }
00795 else {
00796     // incorrect gcd
00797     if (!gcd.is_zero() && gcd[2+x[0]]<sred[2+x[0]])
00798         return poly(BHEAD 0);
00799     numg--;
00800 }
00801
00802 // multiply polynomials by the lists to obtain new ones
00803 sparse_interpolation_mul_poly(ared, amul);
00804 sparse_interpolation_mul_poly(bred, bmul);
00805 sparse_interpolation_mul_poly(sred, smul);
00806 sparse_interpolation_mul_poly(lred, lmul);
00807 }
00808
00809 // solve the linear equations
00810 for (int i=0; i<(int)M.size(); i++) {
00811     int n = M[i].size();
00812
00813     // Gaussian elimination
00814     for (int j=0; j<n; j++) {
00815         for (int k=0; k<j; k++) {
00816             LONG x = M[i][j][k];
00817             for (int l=k; l<n; l++)
00818                 M[i][j][l] = (M[i][j][l] - M[i][k][l]*x) % a.modp;
00819             V[i][j] = (V[i][j] - V[i][k]*x) % a.modp;
00820         }
00821
00822         // normalize row
00823         WORD x = M[i][j][j]; // WORD for GetModInverses
00824         GetModInverses(x + (x<0?a.modp:0), a.modp, &x, NULL);
00825         for (int k=0; k<n; k++)
00826             M[i][j][k] = (M[i][j][k]*x) % a.modp;
00827         V[i][j] = (V[i][j]*x) % a.modp;
00828     }
00829
00830     // solve
00831     for (int j=n-1; j>=0; j--)
00832         for (int k=j+1; k<n; k++)
00833             V[i][j] = (V[i][j] - M[i][j][k]*V[i][k]) % a.modp;
00834 }
00835
00836 // create coefficient list
00837 vector<LONG> coeff;
00838 for (int i=0; i<(int)V.size(); i++)
00839     for (int j=0; j<(int)V[i].size(); j++)
00840         coeff.push_back(V[i][j]);
00841
00842 // create resulting polynomial
00843 poly res(BHEAD 0);
00844 int ri=1, i=0;
00845 for (int si=1; si<s[0]; si+=s[si]) {
00846     res.check_memory(ri);
00847     res[ri] = 3 + AN.poly_num_vars; // term length
00848     for (int j=0; j<AN.poly_num_vars; j++)
00849         res[ri+1+j] = s[si+1+j]; // powers
00850     res[ri+1+AN.poly_num_vars] = ABS(coeff[i]); // coefficient
00851     res[ri+2+AN.poly_num_vars] = SGN(coeff[i]); // coefficient length
00852     i++;
00853     ri += res[ri];
00854 }
00855 res[0]=ri; // total length
00856 res.setmod(a.modp,1);
00857
00858 if (!poly::divides(res, lcgcd * a) || !poly::divides(res, lcgcd * b)) {
00859     return poly(BHEAD 0); // bad shape
00860 } else {
00861     // refine gcd
00862     if (!poly::divides(res, a))
00863         res = gcd_modular_dense_interpolation(res, a, x, poly(BHEAD 0));
00864     if (!poly::divides(res, b))
00865         res = gcd_modular_dense_interpolation(res, b, x, poly(BHEAD 0));
00866 }
00867
00868 #ifdef DEBUG

```

```

00869     cout << "*** [" << thetime() << "]" RES : gcd_modular_sparse_interpolation("
00870         << a << ", " << b << ", " << x << ", " << ", " << s << ") = " << res << endl;
00871 #endif
00872
00873     return gcdconts * res;
00874 }
00875
00876 /*
00877 #] gcd_modular_sparse_interpolation :
00878 #[ gcd_modular_dense_interpolation :
00879 */
00880
00899 const poly polygcd::gcd_modular_dense_interpolation (const poly &a, const poly &b, const vector<int>
&x, const poly &s) {
00900
00901 #ifdef DEBUG
00902     cout << "*** [" << thetime() << "]" CALL: gcd_modular_dense_interpolation(" << a << ", " << b << ", " << x <<
", " << s << ") << endl;
00903 #endif
00904
00905     POLY_GETIDENTITY(a);
00906
00907     // if univariate, then use Euclidean algorithm
00908     if (x.size() == 1) {
00909         return gcd_Euclidean(a,b);
00910     }
00911
00912     // if shape is known, use sparse interpolation
00913     if (!s.is_zero()) {
00914         poly res = gcd_modular_sparse_interpolation (a,b,x,s);
00915         if (!res.is_zero()) return res;
00916         // apparently the shape was not correct. continue.
00917     }
00918
00919     // divide out multivariate content in last variable
00920     int X = x.back();
00921
00922     poly conta(content_multivar(a,X));
00923     poly contb(content_multivar(b,X));
00924     poly gcdconts(gcd_Euclidean(conta,contb));
00925     const poly& ppa = conta.is_one() ? a : poly(a/conta);
00926     const poly& ppb = contb.is_one() ? b : poly(b/contb);
00927
00928     // gcd of leading coefficients
00929     poly lcoeffa(ppa.lcoeff_multivar(X));
00930     poly lcoeffb(ppb.lcoeff_multivar(X));
00931     poly gcdlcoeffs(gcd_Euclidean(lcoeffa,lcoeffb));
00932
00933     // calculate the degree bound for each variable
00934     int m = Min(ppa.degree(x[x.size() - 2]),ppb.degree(x[x.size() - 2]));
00935
00936     poly res(BHEAD 0);
00937     poly oldres(BHEAD 0);
00938     poly newshape(BHEAD 0);
00939     poly modpoly(BHEAD 1,a.modp);
00940
00941     while (true) {
00942         // generate random constants and substitute it
00943         int c = 1 + wranf(BHEAD0) % (a.modp-1);
00944         if (substitute(gcdlcoeffs,X,c).is_zero()) continue;
00945         if (substitute(modpoly,X,c).is_zero()) continue;
00946
00947         poly amodc(substitute(ppa,X,c));
00948         poly bmodc(substitute(ppb,X,c));
00949
00950         // calculate gcd recursively
00951         poly gcdmodc(gcd_modular_dense_interpolation(amodc,bmodc,vector<int>(x.begin(),x.end()-1),
newshape));
00952         int n = gcdmodc.degree(x[x.size() - 2]);
00953
00954         // normalize
00955         gcdmodc = (gcdmodc * substitute(gcdlcoeffs,X,c)) / gcdmodc.integer_lcoeff();
00956         poly simple(poly::simple_poly(BHEAD X,c,1,a.modp)); // (X-c) mod p
00957
00958         // if power is smaller, the old one was wrong
00959         if ((res.is_zero() && n == m) || n < m) {
00960             m = n;
00961             res = gcdmodc;
00962             newshape = gcdmodc; // set a new shape (interpolation does not change it)
00963             modpoly = simple;
00964         }
00965         else if (n == m) {
00966             oldres = res;
00967             // equal powers, so interpolate results
00968             poly coeff_poly(substitute(modpoly,X,c));
00969             WORD coeff_word = coeff_poly[2+AN.poly_num_vars] * coeff_poly[3+AN.poly_num_vars];
00970             if (coeff_word < 0) coeff_word += a.modp;

```

```

00971
00972         GetModInverses(coeff_word, a.modp, &coeff_word, NULL);
00973
00974         res.setmod(a.modp); // make sure the mod is set before substituting
00975         res += poly(BHEAD coeff_word, a.modp, 1) * modpoly * (gcdmodc - substitute(res,X,c));
00976         modpoly *= simple;
00977     }
00978
00979     // check whether this is the complete gcd
00980     if (!res.is_zero() && res == oldres) {
00981         poly nres = res / content_multivar(res, X);
00982         if (poly::divides(nres,ppa) && poly::divides(nres,ppb)) {
00983 #ifdef DEBUG
00984             cout << "*** [" << thetime() << "] RES : gcd_modular_dense_interpolation(" << a << ", " << b
00985             << ", "
00986             << x << ", " << ", " << s << ") = " << gcdconts * nres << endl;
00987 #endif
00988             return gcdconts * nres;
00989         }
00990         // At this point, the gcd may be too large due to bad luck
00991         // TODO: create an efficient fail state that tries to find a smaller
00992         // polynomial without interpolating bad ones?
00993         newshape = poly(BHEAD 0); // reset the shape, important!
00994     }
00995 }
00996 }
00997
00998 /*
00999 #] gcd_modular_dense_interpolation : :
01000 #[ gcd_modular :
01001 */
01002
01019 const poly polygcd::gcd_modular (const poly &origa, const poly &origb, const vector<int> &x) {
01020
01021 #ifdef DEBUG
01022     cout << "*** [" << thetime() << "] CALL: gcd_modular(" << origa << ", " << origb << ", " << x << ")" << endl;
01023 #endif
01024
01025     POLY_GETIDENTITY(origa);
01026
01027     if (origa.is_zero()) return origa.modp==0 ? origb : origb / origb.integer_lcoeff();
01028     if (origb.is_zero()) return origa.modp==0 ? origa : origa / origa.integer_lcoeff();
01029     if (origa==origb) return origa.modp==0 ? origa : origa / origa.integer_lcoeff();
01030
01031     poly ac = integer_content(origa);
01032     poly bc = integer_content(origb);
01033     const poly& a = ac.is_one() ? origa : poly(origa/ac);
01034     const poly& b = bc.is_one() ? origb : poly(origb/bc);
01035     poly ic = integer_gcd(ac, bc);
01036     poly g = integer_gcd(a.integer_lcoeff(), b.integer_lcoeff());
01037
01038     int pnun=0;
01039
01040     poly d(BHEAD 0);
01041     poly m1(BHEAD 1);
01042     int mindeg=MAXPOSITIVE;
01043
01044     while (true) {
01045         // choose a prime and solve modulo the prime
01046         WORD p = NextPrime(BHEAD pnun++);
01047         if (poly(a.integer_lcoeff(),p).is_zero()) continue;
01048         if (poly(b.integer_lcoeff(),p).is_zero()) continue;
01049
01050         poly c(gcd_modular_dense_interpolation(poly(a,p),poly(b,p),x,poly(d,p)));
01051         c = (c * poly(g,p)) / c.integer_lcoeff(); // normalize so that lcoeff(c) = g mod p
01052
01053         if (c.is_zero()) {
01054             // unlucky choices somewhere, so start all over again
01055             d = poly(BHEAD 0);
01056             m1 = poly(BHEAD 1);
01057             mindeg = MAXPOSITIVE;
01058             continue;
01059         }
01060
01061         if (!(poly(a,p)%c).is_zero()) continue;
01062         if (!(poly(b,p)%c).is_zero()) continue;
01063
01064         int deg = c.degree(x[0]);
01065
01066         if (deg < mindeg) {
01067             // small degree, so the old one is wrong
01068             d=c;
01069             d.modp=a.modp;
01070             d.modn=a.modn;
01071             m1 = poly(BHEAD p);
01072             mindeg=deg;

```

```

01073     }
01074     else if (deg == mindeg) {
01075         // same degree, so use Chinese Remainder Algorithm
01076         poly newd(BHEAD 0);
01077
01078         for (int ci=1,di=1; ci<c[0]||di<d[0]; ) {
01079             int comp = ci==c[0] ? -1 : di==d[0] ? +1 : poly::monomial_compare(BHEAD
&c[ci],&d[di]);
01080             poly a1(BHEAD 0),a2(BHEAD 0);
01081
01082             newd.check_memory(newd[0]);
01083
01084             if (comp <= 0) {
01085                 newd.termscopy(&d[di],newd[0],1+AN.poly_num_vars);
01086                 a1 = poly(BHEAD (UWORD *)&d[di+1+AN.poly_num_vars],d[di+d[di]-1]);
01087                 di+=d[di];
01088             }
01089             if (comp >= 0) {
01090                 newd.termscopy(&c[ci],newd[0],1+AN.poly_num_vars);
01091                 a2 = poly(BHEAD (UWORD *)&c[ci+1+AN.poly_num_vars],c[ci+c[ci]-1]);
01092                 ci+=c[ci];
01093             }
01094
01095             poly e(chinese_remainder(a1,m1,a2,poly(BHEAD p)));
01096             newd.termscopy(&e[2+AN.poly_num_vars], newd[0]+1+AN.poly_num_vars, ABS(e[e[1]])+1);
01097             newd[newd[0]] = 2 + AN.poly_num_vars + ABS(e[e[1]]);
01098             newd[0] += newd[newd[0]];
01099         }
01100
01101         m1 *= poly(BHEAD p);
01102         d=newd;
01103     }
01104
01105     // divide out spurious integer content
01106     poly ppd(d / integer_content(d));
01107
01108     // check whether this is the complete gcd
01109     if (poly::divides(ppd,a) && poly::divides(ppd,b)) {
01110 #ifdef DEBUG
01111         cout << "*** [" << thetime() << "]" RES : gcd_modular(" << origa << "," << origb << "," << x << ")
= "
01112             << ic * ppd << endl;
01113 #endif
01114         return ic * ppd;
01115     }
01116 #ifdef DEBUG
01117     cout << "*** [" << thetime() << "]" Retrying modular_gcd with new prime" << endl;
01118 #endif
01119 }
01120 }
01121
01122 /*
01123 #] gcd_modular :
01124 #[ gcd_heuristic_possible :
01125 */
01126
01142 bool gcd_heuristic_possible (const poly &a) {
01143     POLY_GETIDENTITY(a);
01144
01145     double prod_deg = 1;
01146     for (int j=0; j<AN.poly_num_vars; j++)
01147         prod_deg *= a[2+j]+1;
01148
01149     double digits = ABS(a[1+a[1]-1]);
01150     double lead = a[1+1+AN.poly_num_vars];
01151
01152     return prod_deg*(digits-1+log(2*ABS(lead))/log(2.0)/(BITSINWORD/2)) <
POLYGCD_HEURISTIC_MAX_DIGITS;
01154 }
01155
01156 /*
01157 #] gcd_heuristic_possible :
01158 #[ gcd_heuristic :
01159 */
01160
01161
01188 const poly polygcd::gcd_heuristic (const poly &a, const poly &b, const vector<int> &x, int max_tries)
{
01189
01190 #ifdef DEBUG
01191     cout << "*** [" << thetime() << "]" CALL: gcd_heuristic("<a<","<b<","<x<")\n";
01192 #endif
01193
01194     if (a.is_integer()) return integer_gcd(a,integer_content(b));
01195     if (b.is_integer()) return integer_gcd(integer_content(a),b);
01196

```

```

01197     POLY_GETIDENTITY(a);
01198
01199     // Calculate xi = 2*min(max(coefficients a),max(coefficients b))+2
01200
01201     UWORD *pxi=NULL;
01202     WORD nxi=0;
01203
01204     for (int i=1; i<a[0]; i+=a[i]) {
01205         WORD na = ABS(a[i+a[i]-1]);
01206         if (BigLong((UWORD *)&a[i+a[i]-1-na], na, pxi, nxi) > 0) {
01207             pxi = (UWORD *)&a[i+a[i]-1-na];
01208             nxi = na;
01209         }
01210     }
01211
01212     for (int i=1; i<b[0]; i+=b[i]) {
01213         WORD nb = ABS(b[i+b[i]-1]);
01214         if (BigLong((UWORD *)&b[i+b[i]-1-nb], nb, pxi, nxi) > 0) {
01215             pxi = (UWORD *)&b[i+b[i]-1-nb];
01216             nxi = nb;
01217         }
01218     }
01219
01220     poly xi(BHEAD pxi,nxi);
01221
01222     // Addition of another random factor gives better performance
01223     xi = xi*poly(BHEAD 2) + poly(BHEAD 2 + wranf(BHEAD0)%POLYgcd_HEURISTIC_MAX_ADD_RANDOM);
01224
01225     // If degree*digits(xi) is too large, throw exception
01226     if (max(a.degree(x[0]),b.degree(x[0])) * xi[xi[1]] >= Min(AM.MaxTal,
POLYgcd_HEURISTIC_MAX_DIGITS)) {
01227 #ifdef DEBUG
01228     cout << "*** [" << thetime() << " ] RES : gcd_heuristic(" << a << ", " << b << ", " << xi << ") = overflow\n";
01229 #endif
01230     throw(gcd_heuristic_failed());
01231 }
01232
01233     for (int times=0; times<max_tries; times++) {
01234
01235         // Recursively calculate the gcd
01236
01237         poly gamma(gcd_heuristic(a % poly::simple_poly(BHEAD x[0],xi),
01238                             b % poly::simple_poly(BHEAD x[0],xi),
01239                             vector<int>(x.begin()+1,x.end()),1));
01240
01241         // If a gcd is found, reconstruct the powers of x
01242         if (!gamma.is_zero()) {
01243             // res is construct is reverse order. idx/len are for reversing
01244             // it in the correct order
01245             poly res(BHEAD 0), c(BHEAD 0);
01246             vector<int> idx, len;
01247
01248             for (int power=0; !gamma.is_zero(); power++) {
01249
01250                 // calculate c = gamma % xi (c and gamma are polynomials, xi is integer)
01251                 c = gamma;
01252                 c.coefficients_modulo((UWORD *)&xi[2+AN.poly_num_vars], xi[xi[0]-1], false);
01253
01254                 // Add the terms c * x^power to res
01255                 res.check_memory(res[0]+c[0]);
01256                 res.termscopy(&c[1],res[0],c[0]-1);
01257                 for (int i=1; i<c[0]; i+=c[i])
01258                     res[res[0]-1+i+1+x[0]] = power;
01259
01260                 // Store idx/len for reversing
01261                 if (!c.is_zero()) {
01262                     idx.push_back(res[0]);
01263                     len.push_back(c[0]-1);
01264                 }
01265
01266                 res[0] += c[0]-1;
01267
01268                 // Divide gamma by xi
01269                 gamma = (gamma - c) / xi;
01270             }
01271
01272             // Reverse the resulting polynomial
01273             poly rev(BHEAD 0);
01274             rev.check_memory(res[0]);
01275
01276             rev[0] = 1;
01277             for (int i=idx.size()-1; i>=0; i--) {
01278                 rev.termscopy(&res[idx[i]], rev[0], len[i]);
01279                 rev[0] += len[i];
01280             }
01281             res = rev;
01282

```

```

01283         poly ppres = res / integer_content(res);
01284
01285         if ((a%ppres).is_zero() && (b%ppres).is_zero()) {
01286 #ifdef DEBUG
01287             cout << "*** [" << thetime() << "] RES : gcd_heuristic(" << a << ", " << b << ", " << x << ") = " << res << "\n";
01288 #endif
01289             return res;
01290         }
01291     }
01292
01293     // Next xi by multiplying with the golden ratio to avoid correlated errors
01294     xi = xi * poly(BHEAD 28657) / poly(BHEAD 17711) + poly(BHEAD wranf(BHEAD0) %
POLYGCD_HEURISTIC_MAX_ADD_RANDOM);
01295 }
01296
01297 #ifdef DEBUG
01298     cout << "*** [" << thetime() << "] RES : gcd_heuristic(" << a << ", " << b << ", " << x << ") = failed\n";
01299 #endif
01300
01301     return poly(BHEAD 0);
01302 }
01303
01304 /*
01305 #] gcd_heuristic :
01306 #[ bracket :
01307 */
01308
01309 const map<vector<int>,poly> polygcd::full_bracket(const poly &a, const vector<int>& filter) {
01310     POLY_GETIDENTITY(a);
01311
01312     map<vector<int>,poly> bracket;
01313     for (int ai=1; ai<a[0]; ai+=a[ai]) {
01314         vector<int> varpattern(AN.poly_num_vars);
01315         for (int i=0; i<AN.poly_num_vars; i++)
01316             if (filter[i] == 1 && a[ai + i + 1] > 0)
01317                 varpattern[i] = a[ai + i + 1];
01318
01319         // create monomial
01320         poly mon(BHEAD 1);
01321         mon.setmod(a.modp);
01322         mon[0] = a[ai] + 1;
01323         for (int i=0; i<a[ai]; i++)
01324             if (i > 0 && i <= AN.poly_num_vars && varpattern[i - 1])
01325                 mon[1+i] = 0;
01326             else
01327                 mon[1+i] = a[ai+i];
01328
01329         map<vector<int>,poly>::iterator i = bracket.find(varpattern);
01330         if (i == bracket.end()) {
01331             bracket.insert(std::make_pair(varpattern, mon));
01332         } else {
01333             i->second += mon;
01334         }
01335     }
01336
01337     return bracket;
01338 }
01339
01340 const poly polygcd::bracket(const poly &a, const vector<int>& pattern, const vector<int>& filter) {
01341     POLY_GETIDENTITY(a);
01342
01343     poly bracket(BHEAD 0);
01344     for (int ai=1; ai<a[0]; ai+=a[ai]) {
01345         bool ispat = true;
01346         for (int i=0; i<AN.poly_num_vars; i++)
01347             if (filter[i] == 1 && pattern[i] != a[ai + i + 1]) {
01348                 ispat = false;
01349                 break;
01350             }
01351
01352         if (ispat) {
01353             poly mon(BHEAD 1);
01354             mon.setmod(a.modp);
01355             mon[0] = a[ai] + 1;
01356             for (int i=0; i<a[ai]; i++)
01357                 if (i > 0 && i <= AN.poly_num_vars && pattern[i - 1])
01358                     mon[1+i] = 0;
01359                 else
01360                     mon[1+i] = a[ai+i];
01361             bracket += mon;
01362         }
01363     }
01364
01365     return bracket;
01366 }
01367
01368 const map<vector<int>,int> polygcd::bracket_count(const poly &a, const vector<int>& filter) {

```



```

01369     POLY_GETIDENTITY(a);
01370
01371     map<vector<int>,int> bracket;
01372     for (int ai=1; ai<a[0]; ai+=a[ai]) {
01373         vector<int> varpattern(AN.poly_num_vars);
01374         for (int i=0; i<AN.poly_num_vars; i++)
01375             if (filter[i] == 1 && a[ai + i + 1] > 0)
01376                 varpattern[i] = a[ai + i + 1];
01377
01378         map<vector<int>,int>::iterator i = bracket.find(varpattern);
01379         if (i == bracket.end()) {
01380             bracket.insert(std::make_pair(varpattern, 0));
01381         } else {
01382             i->second++;
01383         }
01384     }
01385
01386     return bracket;
01387 }
01388
01389 struct BracketInfo {
01390     std::vector<int> pattern;
01391     int num_terms, dummy = 0;
01392     const poly* p;
01393
01394     BracketInfo(const std::vector<int>& pattern, int num_terms, const poly* p) : pattern(pattern),
01395     num_terms(num_terms), p(p) {}
01396
01397     bool operator<(const BracketInfo& rhs) const { return num_terms > rhs.num_terms; } // biggest
01398     // should be first!
01399 };
01400
01401 /*
01402 #] bracket :
01403 #[ gcd_linear:
01404 */
01405
01406 const poly gcd_linear_helper (const poly &a, const poly &b) {
01407     POLY_GETIDENTITY(a);
01408
01409     for (int i = 0; i < AN.poly_num_vars; i++)
01410         if (a.degree(i) == 1) {
01411             vector<int> filter(AN.poly_num_vars);
01412             filter[i] = 1;
01413
01414             // bracket the linear variable
01415             map<vector<int>,poly> ba = polygcd::full_bracket(a, filter);
01416
01417             poly subgcd(BHEAD 1);
01418             if (ba.size() == 2)
01419                 subgcd = polygcd::gcd_linear(ba.begin()->second, (++ba.begin()->second);
01420             else
01421                 subgcd = ba.begin()->second;
01422
01423             poly linfac = a / subgcd;
01424             if (poly::divides(linfac,b))
01425                 return linfac * polygcd::gcd_linear(subgcd, b / linfac);
01426
01427             return polygcd::gcd_linear(subgcd, b);
01428         }
01429
01430     return poly(BHEAD 0);
01431 }
01432
01433 const poly polygcd::gcd_linear (const poly &a, const poly &b) {
01434     #ifdef DEBUG
01435         cout << "*** [" << thetime() << " ] CALL: gcd_linear(\"a<<\", \"b<<\")\n";
01436     #endif
01437
01438     POLY_GETIDENTITY(a);
01439
01440     if (a.is_zero()) return a.modp==0 ? b : b / b.integer_lcoeff();
01441     if (b.is_zero()) return a.modp==0 ? a : a / a.integer_lcoeff();
01442     if (a==b) return a.modp==0 ? a : a / a.integer_lcoeff();
01443
01444     if (a.is_integer() || b.is_integer()) {
01445         if (a.modp > 0) return poly(BHEAD 1,a.modp,a.modn);
01446         return poly(integer_gcd(integer_content(a),integer_content(b)),0,1);
01447     }
01448
01449     poly h = gcd_linear_helper(a, b);
01450     if (!h.is_zero()) return h;
01451
01452     h = gcd_linear_helper(b, a);
01453     if (!h.is_zero()) return h;
01454
01455     vector<int> xa = a.all_variables();
01456     vector<int> xb = b.all_variables();

```

```

01458
01459     vector<int> used(AN.poly_num_vars,0);
01460     for (int i=0; i<(int)xa.size(); i++) used[xa[i]]++;
01461     for (int i=0; i<(int)xb.size(); i++) used[xb[i]]++;
01462     vector<int> x;
01463     for (int i=0; i<AN.poly_num_vars; i++)
01464         if (used[i]) x.push_back(i);
01465
01466     return gcd_modular(a,b,x);
01467 }
01468
01469 /*
01470 #] gcd_linear:
01471 #[ gcd :
01472 */
01473
01494 const poly polygcd::gcd (const poly &a, const poly &b) {
01495
01496 #ifdef DEBUG
01497     cout << "*** [" << thetime() << " ] CALL: gcd(" << a << ", " << b << ") \n";
01498 #endif
01499
01500     POLY_GETIDENTITY(a);
01501
01502     if (a.is_zero()) return a.modp==0 ? b : b / b.integer_lcoeff();
01503     if (b.is_zero()) return a.modp==0 ? a : a / a.integer_lcoeff();
01504     if (a==b) return a.modp==0 ? a : a / a.integer_lcoeff();
01505
01506     if (a.is_integer() || b.is_integer()) {
01507         if (a.modp > 0) return poly(BHEAD 1,a.modp,a.modn);
01508         return poly(integer_gcd(integer_content(a),integer_content(b)),0,1);
01509     }
01510
01511     // Generate a list of variables of a and b
01512     vector<int> xa = a.all_variables();
01513     vector<int> xb = b.all_variables();
01514
01515     vector<int> used(AN.poly_num_vars,0);
01516     for (int i=0; i<(int)xa.size(); i++) used[xa[i]]++;
01517     for (int i=0; i<(int)xb.size(); i++) used[xb[i]]++;
01518     vector<int> x;
01519     for (int i=0; i<AN.poly_num_vars; i++)
01520         if (used[i]) x.push_back(i);
01521
01522     if (a.is_monomial() || b.is_monomial()) {
01523
01524         poly res(BHEAD 1,a.modp,a.modn);
01525         if (a.modp == 0) res = integer_gcd(integer_content(a),integer_content(b));
01526
01527         for (int i=0; i<(int)x.size(); i++)
01528             res[2+x[i]] = 1<<(BITSINWORD-2);
01529
01530         for (int i=1; i<a[0]; i+=a[i])
01531             for (int j=0; j<(int)x.size(); j++)
01532                 res[2+x[j]] = MiN(res[2+x[j]], a[i+1+x[j]]);
01533
01534         for (int i=1; i<b[0]; i+=b[i])
01535             for (int j=0; j<(int)x.size(); j++)
01536                 res[2+x[j]] = MiN(res[2+x[j]], b[i+1+x[j]]);
01537
01538         return res;
01539     }
01540
01541     // Calculate the contents, their gcd and the primitive parts
01542     poly conta(x.size()==1 ? integer_content(a) : content_univar(a,x[0]));
01543     poly contb(x.size()==1 ? integer_content(b) : content_univar(b,x[0]));
01544     poly gcdconts(x.size()==1 ? integer_gcd(conta,contb) : gcd(conta,contb));
01545     const poly& ppa = conta.is_one() ? a : poly(a/conta);
01546     const poly& ppb = contb.is_one() ? b : poly(b/contb);
01547
01548     if (ppa == ppb)
01549         return ppa * gcdconts;
01550
01551     poly res(BHEAD 0);
01552
01553 #ifdef POLY_GCD_USE_HEURISTIC
01554     // Try the heuristic gcd algorithm
01555     if (a.modp==0 && gcd_heuristic_possible(a) && gcd_heuristic_possible(b)) {
01556         try {
01557             res = gcd_heuristic(ppa,ppb,x);
01558             if (!res.is_zero()) res /= integer_content(res);
01559         }
01560         catch (gcd_heuristic_failed) {}
01561     }
01562 #endif
01563
01564     // If gcd==0, the heuristic algorithm failed, so we do more extensive checks.

```

```

01565 // First, we filter out variables that appear in only one of the expressions.
01566 if (res.is_zero()) {
01567     bool unusedVars = false;
01568     for (unsigned int i = 0; i < used.size(); i++) {
01569         if (used[i] == 1) {
01570             unusedVars = true;
01571             break;
01572         }
01573     }
01574
01575     // if there are no unused variables, go to the linear routine directly
01576     if (!unusedVars) {
01577         res = gcd_linear(ppa,ppb);
01578 #ifdef DEBUG
01579         cout << "New GCD attempt (unused vars): " << res << endl;
01580 #endif
01581     }
01582
01583     // if res is not the gcd, it is 0 or larger than the gcd.
01584     // we bracket the expression in all the variables that appear only in one expr.
01585     // and we refine the gcd.
01586     bool diva = !res.is_zero() && poly::divides(res,ppa);
01587     bool divb = !res.is_zero() && poly::divides(res,ppb);
01588     if (!diva || !divb) {
01589         vector<BracketInfo> bracketinfo;
01590
01591         if (!diva) {
01592             map<vector<int>,int> ba = bracket_count(ppa, used);
01593             for (map<vector<int>,int>::iterator it = ba.begin(); it != ba.end(); it++)
01594                 bracketinfo.push_back(BracketInfo(it->first, it->second, &ppa));
01595         }
01596
01597         if (!divb) {
01598             map<vector<int>,int> bb = bracket_count(ppb, used);
01599             for (map<vector<int>,int>::iterator it = bb.begin(); it != bb.end(); it++)
01600                 bracketinfo.push_back(BracketInfo(it->first, it->second, &ppb));
01601         }
01602
01603         // sort so that the smallest bracket will be last
01604         sort(bracketinfo.begin(), bracketinfo.end());
01605
01606         if (res.is_zero()) {
01607             res = bracket(*bracketinfo.back().p, bracketinfo.back().pattern, used);
01608             bracketinfo.pop_back();
01609         }
01610
01611         while (bracketinfo.size() > 0) {
01612             poly subpoly(bracket(*bracketinfo.back().p, bracketinfo.back().pattern, used));
01613             if (!poly::divides(res,subpoly)) {
01614                 // if we can filter out more variables, call gcd again
01615                 if (res.all_variables() != subpoly.all_variables())
01616                     res = gcd(subpoly,res);
01617             }
01618             else
01619                 res = gcd_linear(subpoly,res);
01620
01621             bracketinfo.pop_back();
01622         }
01623     }
01624
01625     if (res.is_zero() || !poly::divides(res,ppa) || !poly::divides(res,ppb)) {
01626         MesPrint("Bad gcd found.");
01627         std::cout << "Bad gcd:" << res << " for " << ppa << " " << ppb << std::endl;
01628         Terminate(1);
01629     }
01630 }
01631
01632 res *= gcdconts * poly(BHEAD res.sign());
01633
01634 #ifdef DEBUG
01635     cout << "*** [" << thetime() << "] RES : gcd(" << a << ", " << b << ") = " << res << endl;
01636 #endif
01637
01638 return res;
01639 }
01640
01641 /*
01642 #] gcd :
01643 */

```

## 4.107 polygcd.h

```
00001 #pragma once
```

```

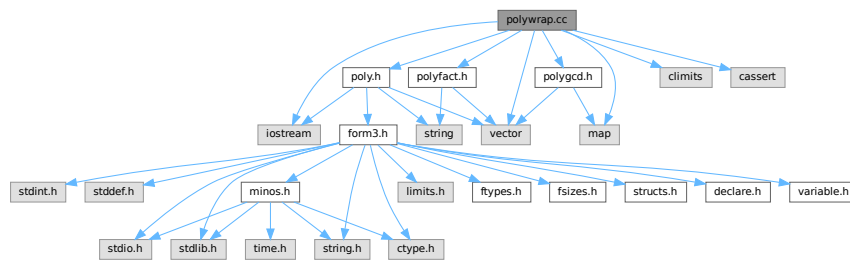
00002 /* #[ License : */
00003 /*
00004 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00005 *   When using this file you are requested to refer to the publication
00006 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00007 *   This is considered a matter of courtesy as the development was paid
00008 *   for by FOM the Dutch physics granting agency and we would like to
00009 *   be able to track its scientific use to convince FOM of its value
00010 *   for the community.
00011 *
00012 *   This file is part of FORM.
00013 *
00014 *   FORM is free software: you can redistribute it and/or modify it under the
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00016 *   Foundation, either version 3 of the License, or (at your option) any later
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00018 *
00019 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00020 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00021 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00022 *   details.
00023 *
00024 *   You should have received a copy of the GNU General Public License along
00025 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00026 */
00027 /* #[ License : */
00028
00029 #include <vector>
00030 #include <map>
00031
00032 class poly; // polynomial class
00033 class gcd_heuristic_failed {}; // class for throwing exceptions
00034
00035 // whether or not to use the heuristic before Zippel's algorithm
00036 #define POLYGCD_USE_HEURISTIC
00037
00038 // maximum number of words in a coefficient for gcd_heuristic to continue
00039 const int POLYGCD_HEURISTIC_MAX_DIGITS = 1000;
00040
00041 // maximum number of retries after the heuristic has failed
00042 const int POLYGCD_HEURISTIC_MAX_TRIES = 10;
00043
00044 // a fudge factor, which improves efficiency
00045 const int POLYGCD_HEURISTIC_MAX_ADD_RANDOM = 10;
00046
00047 // maximum cached power in substitute_last and sparse_interpolation_get_mul_list
00048 const int POLYGCD_RAISPOWMOD_CACHE_SIZE = 1000;
00049
00050 namespace polygcd {
00051
00052     // functions to call the gcd routines
00053     const poly integer_gcd (const poly &a, const poly &b);
00054     const poly integer_content (const poly &a);
00055
00056     const poly gcd (const poly &a, const poly &b);
00057     const poly content_univar (const poly &a, int x);
00058     const poly content_multivar (const poly &a, int x);
00059
00060     const std::vector<WORD> coefficient_list_gcd (const std::vector<WORD> &a, const std::vector<WORD>
&b, WORD p);
00061
00062     // internal functions
00063     const poly gcd_heuristic (const poly &a, const poly &b, const std::vector<int> &x, int
max_tries=POLYGCD_HEURISTIC_MAX_TRIES);
00064     const poly gcd_Euclidean (const poly &a, const poly &b);
00065     const poly gcd_modular (const poly &a, const poly &b, const std::vector<int> &x);
00066     const poly gcd_modular_dense_interpolation (const poly &a, const poly &b, const std::vector<int>
&x, const poly &s);
00067     const poly gcd_modular_sparse_interpolation (const poly &a, const poly &b, const std::vector<int>
&x, const poly &s);
00068
00069     const std::vector<int> sparse_interpolation_get_mul_list (const poly &a, const std::vector<int>
&x, const std::vector<int> &c);
00070     void sparse_interpolation_mul_poly (poly &a, const std::vector<int> &m);
00071     const poly sparse_interpolation_reduce_poly (const poly &a, const std::vector<int> &x);
00072     const poly sparse_interpolation_fix_poly (const poly &a, int x);
00073
00074     const poly chinese_remainder (const poly &a1, const poly &m1, const poly &a2, const poly &m2);
00075     const poly substitute(const poly &a, int x, int c);
00076     const std::map<std::vector<int>,poly> full_bracket(const poly &a, const std::vector<int>& filter);
00077     const poly bracket(const poly &a, const std::vector<int>& pattern, const std::vector<int>&
filter);
00078     const std::map<std::vector<int>,int> bracket_count(const poly &a, const std::vector<int>& filter);
00079     const poly gcd_linear (const poly &a, const poly &b);
00080 }

```

## 4.108 polywrap.cc File Reference

```
#include "poly.h"
#include "polygcd.h"
#include "polyfact.h"
#include <iostream>
#include <vector>
#include <map>
#include <climits>
#include <cassert>
```

Include dependency graph for polywrap.cc:



### Functions

- WORD [poly\\_determine\\_modulus](#) (PHEAD bool multi\_error, bool is\_fun\_arg, string message)
- WORD \* [poly\\_gcd](#) (PHEAD WORD \*a, WORD \*b, WORD fit)
- WORD \* [poly\\_divmod](#) (PHEAD WORD \*a, WORD \*b, int divmod, WORD fit)
- WORD \* [poly\\_div](#) (PHEAD WORD \*a, WORD \*b, WORD fit)
- WORD \* [poly\\_rem](#) (PHEAD WORD \*a, WORD \*b, WORD fit)
- void [poly\\_ratfun\\_read](#) (WORD \*a, poly &num, poly &den)
- void [poly\\_sort](#) (PHEAD WORD \*a)
- WORD \* [poly\\_ratfun\\_add](#) (PHEAD WORD \*t1, WORD \*t2)
- int [poly\\_ratfun\\_normalize](#) (PHEAD WORD \*term)
- void [poly\\_fix\\_minus\\_signs](#) (factorized\_poly &a)
- WORD \* [poly\\_factorize](#) (PHEAD WORD \*argin, WORD \*argout, bool with\_arghead, bool is\_fun\_arg)
- int [poly\\_factorize\\_argument](#) (PHEAD WORD \*argin, WORD \*argout)
- WORD \* [poly\\_factorize\\_dollar](#) (PHEAD WORD \*argin)
- int [poly\\_factorize\\_expression](#) (EXPRESSIONS expr)
- int [poly\\_unfactorize\\_expression](#) (EXPRESSIONS expr)
- WORD \* [poly\\_inverse](#) (PHEAD WORD \*arga, WORD \*argb)
- WORD \* [poly\\_mul](#) (PHEAD WORD \*a, WORD \*b)
- void [poly\\_free\\_poly\\_vars](#) (PHEAD const char \*text)

### Variables

- const int [POLYWRAP\\_DENOMPOWER\\_INCREASE\\_FACTOR](#) = 2

#### 4.108.1 Detailed Description

Contains methods to call the polynomial methods (written in C++) from the rest of Form (written in C). These include polynomial gcd computation, factorization and polyratfuns.

Definition in file [polywrap.cc](#).

## 4.108.2 Function Documentation

### 4.108.2.1 `poly_determine_modulus()`

```
WORD poly_determine_modulus (
    PHEAD bool multi_error,
    bool is_fun_arg,
    string message )
```

Modulus for polynomial algebra

### 4.108.3 Description

This method determines whether polynomial algebra is done with a modulus or not. This depends on AC.ncmod. If only\_funargs is set it also depends on (AC.modmode & ALSOFUNARGS).

The program terminates if the feature is not implemented. Polynomial algebra modulo  $M > \text{WORDSIZE}$  is not implemented. If multi\_error is set, multivariate algebra mod  $M$  is not implemented.

### 4.108.4 Notes

- If AC.ncmod>0 and only\_funargs=true and AC.modmode&ALSOFUNARGS=false, AN.ncmod is set to zero, for otherwise RaisPow calculates mod  $M$ .

Definition at line 79 of file [polywrap.cc](#).

Referenced by [poly\\_factorize\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_gcd\(\)](#), [poly\\_ratfun\\_add\(\)](#), and [poly\\_ratfun\\_normalize\(\)](#).

#### 4.108.4.1 `poly_gcd()`

```
WORD * poly_gcd (
    PHEAD WORD * a,
    WORD * b,
    WORD fit )
```

Polynomial gcd

### 4.108.5 Description

This method calculates the greatest common divisor of two polynomials, given by two zero-terminated Form-style term lists.

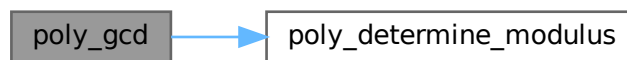
### 4.108.6 Notes

- The result is written at newly allocated memory
- Called from [ratio.c](#)
- Calls `polygcd::gcd`

Definition at line [124](#) of file [polywrap.cc](#).

References [poly\\_determine\\_modulus\(\)](#).

Here is the call graph for this function:



#### 4.108.6.1 poly\_divmod()

```
WORD * poly_divmod (  
    PHEAD WORD * a,  
    WORD * b,  
    int divmod,  
    WORD fit )
```

Definition at line [203](#) of file [polywrap.cc](#).

#### 4.108.6.2 poly\_div()

```
WORD * poly_div (  
    PHEAD WORD * a,  
    WORD * b,  
    WORD fit )
```

Definition at line [435](#) of file [polywrap.cc](#).

#### 4.108.6.3 poly\_rem()

```
WORD * poly_rem (  
    PHEAD WORD * a,  
    WORD * b,  
    WORD fit )
```

Definition at line [463](#) of file [polywrap.cc](#).

#### 4.108.6.4 poly\_ratfun\_read()

```
void poly_ratfun_read (
    WORD * a,
    poly & num,
    poly & den )
```

Read a PolyRatFun

#### 4.108.7 Description

This method reads a polyratfun starting at the pointer a. The resulting numerator and denominator are written in num and den. If MUSTCLEANPRF, the result is normalized.

#### 4.108.8 Notes

- Calls polygcd::gcd

Definition at line 496 of file [polywrap.cc](#).

Referenced by [poly\\_ratfun\\_add\(\)](#), and [poly\\_ratfun\\_normalize\(\)](#).

#### 4.108.8.1 poly\_sort()

```
void poly_sort (
    PHEAD WORD * a )
```

Sort the polynomial terms

#### 4.108.9 Description

Sorts the terms of a polynomial in Form [poly\(rat\)](#)fun order, i.e. lexicographical order with highest degree first.



### 4.108.10 Notes

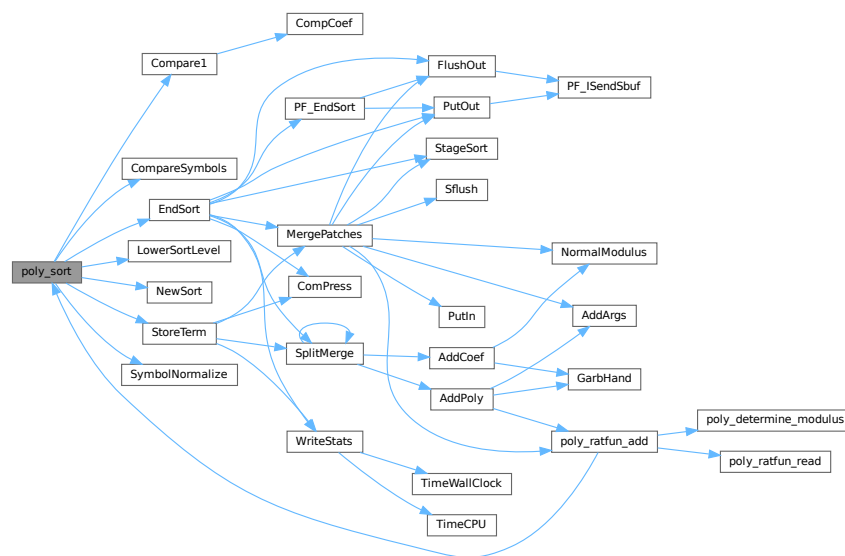
- Uses Form sort routines with custom compare

Definition at line 590 of file [polywrap.cc](#).

References [Compare1\(\)](#), [CompareSymbols\(\)](#), [EndSort\(\)](#), [LowerSortLevel\(\)](#), [NewSort\(\)](#), [StoreTerm\(\)](#), and [SymbolNormalize\(\)](#).

Referenced by [poly\\_ratfun\\_add\(\)](#), and [poly\\_ratfun\\_normalize\(\)](#).

Here is the call graph for this function:



#### 4.108.10.1 poly\_ratfun\_add()

```
WORD * poly_ratfun_add (
    PHEAD WORD * t1,
    WORD * t2 )
```

Addition of PolyRatFuns

### 4.108.11 Description

This method gets two pointers to polyratfuns with up to two arguments each and calculates the sum.



### 4.108.14 Notes

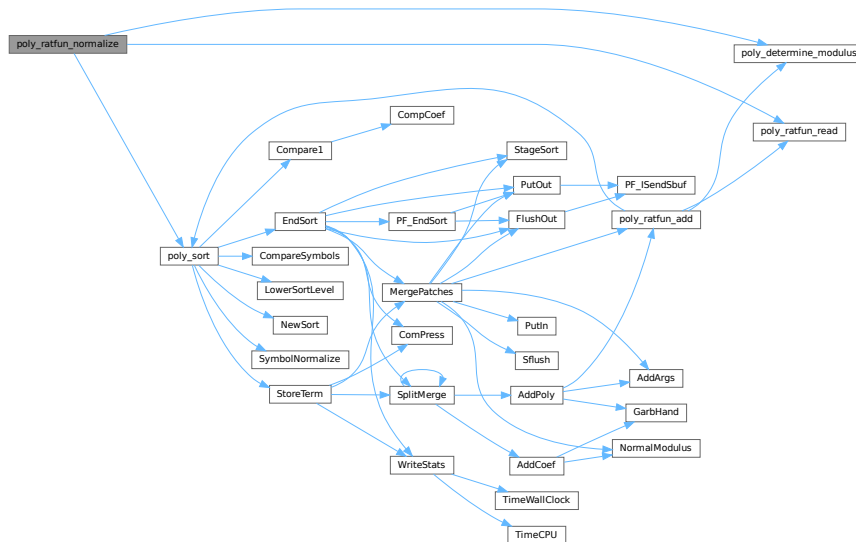
- The result overwrites the original term
- Called from [proces.c](#)
- Calls `poly::operators` and `polygcd::gcd`

Definition at line 769 of file [polywrap.cc](#).

References [poly\\_determine\\_modulus\(\)](#), [poly\\_ratfun\\_read\(\)](#), and [poly\\_sort\(\)](#).

Referenced by [PolyFunMul\(\)](#), and [PrepPoly\(\)](#).

Here is the call graph for this function:



#### 4.108.14.1 poly\_fix\_minus\_signs()

```
void poly_fix_minus_signs (
    factorized_poly & a )
```

Definition at line 921 of file [polywrap.cc](#).

#### 4.108.14.2 poly\_factorize()

```
WORD * poly_factorize (
    PHEAD WORD * argin,
    WORD * argout,
    bool with_arghead,
    bool is_fun_arg )
```

Factorization of function arguments / dollars

### 4.108.15 Description

This method factorizes a Form style argument or zero-terminated term list.

### 4.108.16 Notes

- Called from `poly_factorize_{argument,dollar}`
- Calls `polyfact::factorize`

Definition at line 986 of file `polywrap.cc`.

References `poly_determine_modulus()`.

Referenced by `poly_factorize_argument()`, and `poly_factorize_dollar()`.

Here is the call graph for this function:



#### 4.108.16.1 poly\_factorize\_argument()

```
int poly_factorize_argument (
    PHEAD WORD * argin,
    WORD * argout )
```

Factorization of function arguments

### 4.108.17 Description

This method factorizes the Form-style argument `argin`.

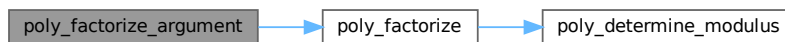
### 4.108.18 Notes

- The result is written at argout
- Called from [argument.c](#)
- Calls `poly_factorize`

Definition at line 1112 of file [polywrap.cc](#).

References [poly\\_factorize\(\)](#).

Here is the call graph for this function:



### 4.108.18.1 poly\_factorize\_dollar()

```
WORD * poly_factorize_dollar (
    PHEAD WORD * argin )
```

Factorization of dollar variables

### 4.108.19 Description

This method factorizes a dollar variable.

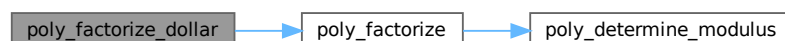
### 4.108.20 Notes

- The result is written at newly allocated memory.
- Called from [dollar.c](#)
- Calls `poly_factorize`

Definition at line 1146 of file [polywrap.cc](#).

References [poly\\_factorize\(\)](#).

Here is the call graph for this function:



```
int poly_factorize_expression (
    EXPRESSIONS expr )
```

#### 4.108.21 Description

#### 4.108.22 Notes

- Definition at line 1178 of file polywrap.cc.

Referenced by [PF\\_Processor\(\)](#), and [Processor\(\)](#).

#### 4.108.22.1 poly\_unfactorize\_expression()

```
int poly_unfactorize_expression (
    EXPRESSIONS expr )
```

## Unfactorization of expressions

#### 4.108.23 Description

This method expands a factorized expression.

#### 4.108.24 Notes

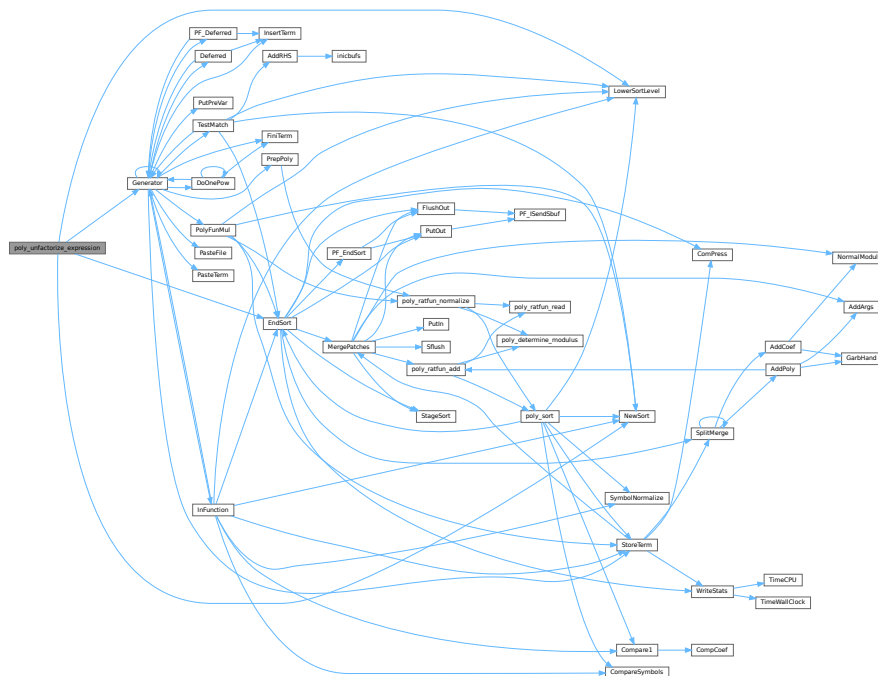
- The result overwrites the input expression
- Called from `proces.c`

Definition at line 1535 of file polywrap.cc.

References [EndSort\(\)](#), [Generator\(\)](#), [FiLe::handle](#), [LowerSortLevel\(\)](#), and [NewSort\(\)](#).

Referenced by [PF\\_Processor\(\)](#), and [Processor\(\)](#).

Here is the call graph for this function:



**4.108.24.1 poly\_inverse()**

```
WORD * poly_inverse (
    PHEAD WORD * arga,
    WORD * argb )
```

Definition at line 1743 of file [polywrap.cc](#).

**4.108.24.2 poly\_mul()**

```
WORD * poly_mul (
    PHEAD WORD * a,
    WORD * b )
```

Definition at line 1948 of file [polywrap.cc](#).

**4.108.24.3 poly\_free\_poly\_vars()**

```
void poly_free_poly_vars (
    PHEAD const char * text )
```

Definition at line 2014 of file [polywrap.cc](#).

**4.108.25 Variable Documentation****4.108.25.1 POLYWRAP\_DENOMPOWER\_INCREASE\_FACTOR**

```
const int POLYWRAP_DENOMPOWER_INCREASE_FACTOR = 2
```

Definition at line 52 of file [polywrap.cc](#).

**4.109 polywrap.cc**

[Go to the documentation of this file.](#)

```
00001
00008 /* # [ License : */
00009 /*
00010 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00011 * When using this file you are requested to refer to the publication
00012 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00013 * This is considered a matter of courtesy as the development was paid
00014 * for by FOM the Dutch physics granting agency and we would like to
00015 * be able to track its scientific use to convince FOM of its value
00016 * for the community.
00017 *
00018 * This file is part of FORM.
00019 *
00020 * FORM is free software: you can redistribute it and/or modify it under the
00021 * terms of the GNU General Public License as published by the Free Software
00022 * Foundation, either version 3 of the License, or (at your option) any later
00023 * version.
00024 *
00025 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00026 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00027 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00028 * details.
00029 *
```



```

00030 *   You should have received a copy of the GNU General Public License along
00031 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
00032 */
00033 /* #] License : */
00034
00035 #include "poly.h"
00036 #include "polygcd.h"
00037 #include "polyfact.h"
00038
00039 #include <iostream>
00040 #include <vector>
00041 #include <map>
00042 #include <climits>
00043 #include <cassert>
00044
00045 // #define DEBUG
00046
00047 #ifdef DEBUG
00048 #include "mytime.h"
00049 #endif
00050
00051 // denompower is increased by this factor when divmod_heap fails
00052 const int POLYWRAP_DENOMPOWER_INCREASE_FACTOR = 2;
00053
00054 using namespace std;
00055
00056 /*
00057  #[ poly_determine_modulus :
00058 */
00059
00079 WORD poly_determine_modulus (PHEAD bool multi_error, bool is_fun_arg, string message) {
00080
00081     if (AC.ncmod==0) return 0;
00082
00083     if (!is_fun_arg || (AC.modmode & ALSOFUNARGS)) {
00084
00085         if (ABS(AC.ncmod)>1) {
00086             MLOCK(ErrorMessageLock);
00087             MesPrint ((char*)"ERROR: %s with modulus > WORDSIZE not implemented",message.c_str());
00088             MUNLOCK(ErrorMessageLock);
00089             Terminate(-1);
00090         }
00091
00092         if (multi_error && AN.poly_num_vars>1) {
00093             MLOCK(ErrorMessageLock);
00094             MesPrint ((char*)"ERROR: multivariate %s with modulus not implemented",message.c_str());
00095             MUNLOCK(ErrorMessageLock);
00096             Terminate(-1);
00097         }
00098
00099         return *AC.cmod;
00100     }
00101
00102     AN.ncmod = 0;
00103     return 0;
00104 }
00105
00106 /*
00107  #] poly_determine_modulus :
00108  #[ poly_gcd :
00109 */
00110
00124 WORD *poly_gcd(PHEAD WORD *a, WORD *b, WORD fit) {
00125
00126 #ifdef WITHFLINT
00127     if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
00128         WORD *ret = flint_gcd(BHEAD a, b, fit);
00129         return ret;
00130     }
00131 #endif
00132
00133 #ifdef DEBUG
00134     cout << "*** [" << thetime() << "]" << " CALL : poly_gcd" << endl;
00135 #endif
00136
00137 //
00138 //MesPrint("Calling poly_gcd with:");
00139 //{
00140 // WORD *at = a;
00141 // MesPrint("      a:");
00142 // while ( *at ) {
00143 //     MesPrint("      %a",*at,at);
00144 //     at += *at;
00145 // }
00146 // MesPrint("      b:");
00147 // at = b;
00148 // while ( *at ) {

```

```

00149 //      MesPrint("    %a",*at,at);
00150 //      at += *at;
00151 //  }
00152 //}
00153
00154 // Extract variables
00155 vector<WORD *> e;
00156 e.reserve(2);
00157 e.push_back(a);
00158 e.push_back(b);
00159 poly::get_variables(BHEAD e, false, true);
00160
00161 // Check for modulus calculus
00162 WORD modp=poly_determine_modulus(BHEAD true, true, "polynomial GCD");
00163
00164 // Convert to polynomials
00165 poly pa(poly::argument_to_poly(BHEAD a, false, true), modp, 1);
00166 poly pb(poly::argument_to_poly(BHEAD b, false, true), modp, 1);
00167
00168 // Calculate gcd
00169 poly gcd(polygcd::gcd(pa,pb));
00170
00171 // Allocate new memory and convert to Form notation
00172 int newsize = (gcd.size_of_form_notation()+1);
00173 WORD *res;
00174 if ( fit ) {
00175     if ( newsize*sizeof(WORD) >= (size_t)(AM.MaxTer) ) {
00176         MLOCK(ErrorMessageLock);
00177         MesPrint("poly_gcd: Term too complex (%d words). Maybe increasing MaxTermSize (%d words)
can help",
00178                 newsize, AM.MaxTer/sizeof(WORD));
00179         MUNLOCK(ErrorMessageLock);
00180         Terminate(-1);
00181     }
00182     res = TermMalloc("poly_gcd");
00183 }
00184 else {
00185     res = (WORD *)Malloca1(newsize*sizeof(WORD), "poly_gcd");
00186 }
00187 poly::poly_to_argument(gcd, res, false);
00188
00189 poly_free_poly_vars(BHEAD "AN.poly_vars_gcd");
00190
00191 // reset modulo calculation
00192 AN.ncmod = AC.ncmod;
00193 return res;
00194 }
00195
00196 /*
00197 #] poly_gcd :
00198 #[ poly_divmod :
00199
00200 if fit == 1 the answer must fit inside a term.
00201 */
00202
00203 WORD *poly_divmod(PHEAD WORD *a, WORD *b, int divmod, WORD fit) {
00204
00205 #ifdef DEBUG
00206     cout << "*** [" << thetime() << "]" CALL : poly_divmod" << endl;
00207 #endif
00208
00209 // check for modulus calculus
00210 WORD modp=poly_determine_modulus(BHEAD false, true, "polynomial division");
00211
00212 // get variables
00213 vector<WORD *> e;
00214 e.reserve(2);
00215 e.push_back(a);
00216 e.push_back(b);
00217 poly::get_variables(BHEAD e, false, false);
00218
00219 // add extra variables to keep track of denominators
00220 const int DENOMSYMBOL = MAXPOSITIVE;
00221
00222 // WORD *new_poly_vars = (WORD *)Malloca1((AN.poly_num_vars+1)*sizeof(WORD), "AN.poly_vars");
00223 // WCOPY(new_poly_vars, AN.poly_vars, AN.poly_num_vars);
00224 // new_poly_vars[AN.poly_num_vars] = DENOMSYMBOL;
00225 // if (AN.poly_num_vars > 0)
00226 //     M_free(AN.poly_vars, "AN.poly_vars");
00227 // AN.poly_num_vars++;
00228 // AN.poly_vars = new_poly_vars;
00229 // AN.poly_vars[AN.poly_num_vars++] = DENOMSYMBOL;
00230
00231 // convert to polynomials
00232 poly dena(BHEAD 0);
00233 poly denb(BHEAD 0);
00234 poly pa(poly::argument_to_poly(BHEAD a, false, true, &dena), modp, 1);

```

```

00235     poly pb(poly::argument_to_poly(BHEAD b, false, true, &denb), modp, 1);
00236
00237     // remove contents
00238     poly numres(polygcd::integer_content(pa));
00239     poly denres(polygcd::integer_content(pb));
00240     pa /= numres;
00241     pb /= denres;
00242
00243     if (divmod==0) {
00244         numres *= denb;
00245         denres *= dena;
00246     }
00247     else {
00248         denres = dena;
00249     }
00250
00251     poly gcdres(polygcd::integer_gcd(numres,denres));
00252     numres /= gcdres;
00253     denres /= gcdres;
00254
00255     // determine lcoeff(b)
00256     poly lcoeffb(pb.integer_lcoeff());
00257
00258     poly pres(BHEAD 0);
00259
00260     if (!lcoeffb.is_one()) {
00261
00262         if (AN.poly_num_vars > 2) {
00263             // the original polynomial is multivariate (one dummy variable has
00264             // been added), so it is not trivial to determine which power of
00265             // lcoeff(b) can be in the answer
00266
00267             int denompower = 0, prevdenompower = 0;
00268             poly pq(BHEAD 0);
00269             poly pr(BHEAD 0);
00270
00271             // try denompower = 0, if this fails increase denompower until division succeeds
00272             bool div_fail = true;
00273             do
00274             {
00275                 if(denompower < prevdenompower)
00276                 {
00277                     // denompower increased beyond INT_MAX
00278                     MLOCK(ErrorMessageLock);
00279                     MesPrint ((char*)"ERROR: pseudo-division failed in poly_divmod (denompower >
INT_MAX)");
00280                     MUNLOCK(ErrorMessageLock);
00281                     Terminate(1);
00282                 }
00283
00284                 if(denompower != 0)
00285                 {
00286                     // multiply a by lcoeffb^(denompower-prevdenompower)
00287                     WORD n = lcoeffb[lcoeffb[1]];
00288                     RaisPow(BHEAD (UWORD *)&lcoeffb[2+AN.poly_num_vars], &n,
denompower-prevdenompower);
00289                     lcoeffb[1] = 2 + AN.poly_num_vars + ABS(n);
00290                     lcoeffb[0] = 1 + lcoeffb[1];
00291                     lcoeffb[lcoeffb[1]] = n;
00292
00293                     pa *= lcoeffb;
00294                     denres *= lcoeffb;
00295                 }
00296
00297                 // try division
00298                 poly ppow(BHEAD 0);
00299                 poly::divmod_heap(pa,pb,pq,pr,false,true,div_fail); // sets div_fail
00300
00301                 // increase denompower for next iteration
00302                 prevdenompower = denompower;
00303                 denompower = (denompower==0 ? 1 : denompower*POLYWRAP_DENOMPPOWER_INCREASE_FACTOR+1 );
// generates 2^n-1 when POLYWRAP_DENOMPPOWER_INCREASE_FACTOR = 2
00304             }
00305             while(div_fail);
00306
00307             pres = (divmod==0 ? pq : pr);
00308
00309         }
00310         else {
00311             // one variable, so the power is the difference of the degrees
00312
00313             int denompower = MaX(0, pa.degree(0) - pb.degree(0) + 1);
00314
00315             // multiply a by that power
00316             WORD n = lcoeffb[lcoeffb[1]];
00317             RaisPow(BHEAD (UWORD *)&lcoeffb[2+AN.poly_num_vars], &n, denompower);
00318             lcoeffb[1] = 2 + AN.poly_num_vars + ABS(n);

```

```

00319         lcoeffb[0] = 1 + lcoeffb[1];
00320         lcoeffb[lcoeffb[1]] = n;
00321
00322         pa *= lcoeffb;
00323         denres *= lcoeffb;
00324
00325         pres = (divmod==0 ? pa/pb : pa%pb);
00326
00327     }
00328 }
00329 }
00330 else {
00331
00332     pres = (divmod==0 ? pa/pb : pa%pb);
00333
00334 }
00335
00336 // convert to Form notation
00337 // NOTE: this part can be rewritten with poly::size_of_form_notation_with_den()
00338 // and poly::poly_to_argument_with_den().
00339 WORD *res;
00340
00341 // special case: a=0
00342 if (pres[0]==1) {
00343     if ( fit ) {
00344         res = TermMalloc("poly_divmod");
00345     }
00346     else {
00347         res = (WORD *)Malloca(sizeof(WORD), "poly_divmod");
00348     }
00349     res[0] = 0;
00350 }
00351 else {
00352     pres *= numres;
00353
00354     WORD nden;
00355     UWORD *den = (UWORD *)NumberMalloc("poly_divmod");
00356
00357     // allocate the memory; note that this overestimates the size,
00358     // since the estimated denominators are too large
00359     int ressize = pres.size_of_form_notation() + pres.number_of_terms()*2*ABS(denres[denres[1]]) +
1;
00360
00361     if ( fit ) {
00362         if ( ressize*sizeof(WORD) > (size_t)(AM.MaxTer) ) {
00363             MLOCK(ErrorMessageLock);
00364             MesPrint("poly_divmod: Term too complex (%d words). Maybe increasing MaxTermSize (%d
words) can help",
00365                     ressize, AM.MaxTer/sizeof(WORD));
00366             MUNLOCK(ErrorMessageLock);
00367             Terminate(-1);
00368         }
00369         res = TermMalloc("poly_divmod");
00370     }
00371     else {
00372         res = (WORD *)Malloca(ressize*sizeof(WORD), "poly_divmod");
00373     }
00374     int L=0;
00375
00376     for (int i=1; i!=pres[0]; i+=pres[i]) {
00377
00378         res[L]=1; // length
00379         bool first = true;
00380
00381         for (int j=0; j<AN.poly_num_vars; j++)
00382             if (pres[i+1+j] > 0) {
00383                 if (first) {
00384                     first = false;
00385                     res[L+1] = 1; // symbols
00386                     res[L+2] = 2; // length
00387                 }
00388                 res[L+1+res[L+2]++] = AN.poly_vars[j]; // symbol
00389                 res[L+1+res[L+2]++] = pres[i+1+j]; // power
00390             }
00391
00392         if (!first) res[L] += res[L+2]; // fix length
00393
00394         // numerator
00395         WORD nnum = pres[i+pres[i]-1];
00396         WCOPY(&res[L+res[L]], &pres[i+pres[i]-1-ABS(nnum)], ABS(nnum));
00397
00398         // calculate denominator
00399         nden = denres[denres[1]];
00400         WCOPY(den, &denres[2+AN.poly_num_vars], ABS(nden));
00401
00402         if (nden!=1 || den[0]!=1)
00403             Simplify(BHEAD (UWORD *)&res[L+res[L]], &nnum, den, &nden); // gcd(num,den)

```

```

00404         Pack((UWORD *)&res[L+res[L]], &nnum, den, nden);           // format
00405         res[L] += 2*ABS(nnum)+1;                                     // fix length
00406         res[L+res[L]-1] = SGN(nnum)*(2*ABS(nnum)+1);               // length of coefficient
00407         L += res[L];                                               // fix length
00408     }
00409
00410     res[L] = 0;
00411
00412     NumberFree(den, "poly_divmod");
00413 }
00414
00415 // clean up
00416 poly_free_poly_vars(BHEAD "AN.poly_vars_divmod");
00417
00418 // reset modulo calculation
00419 AN.ncmod = AC.ncmod;
00420
00421 return res;
00422 }
00423
00424 /*
00425 #] poly_divmod :
00426 #[ poly_div :
00427
00428 Routine divides the expression in arg1 by the expression in arg2.
00429 We did not take out special cases.
00430 The arguments are zero terminated sequences of term(s).
00431 The action is to divide arg1 by arg2: [arg1/arg2].
00432 The answer should be a buffer (allocated by Malloc1) with a zero
00433 terminated sequence of terms (or just zero).
00434 */
00435 WORD *poly_div(PHEAD WORD *a, WORD *b, WORD fit) {
00436
00437 #ifdef WITHFLINT
00438     if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
00439         WORD *ret = flint_div(BHEAD a, b, fit);
00440         return ret;
00441     }
00442 #endif
00443
00444 #ifdef DEBUG
00445     cout << "*** [" << thetime() << "]" CALL : poly_div" << endl;
00446 #endif
00447
00448     return poly_divmod(BHEAD a, b, 0, fit);
00449 }
00450
00451 /*
00452 #] poly_div :
00453 #[ poly_rem :
00454
00455 Routine divides the expression in arg1 by the expression in arg2
00456 and takes the remainder.
00457 We did not take out special cases.
00458 The arguments are zero terminated sequences of term(s).
00459 The action is to divide arg1 by arg2 and take the remainder: [arg1%arg2].
00460 The answer should be a buffer (allocated by Malloc1) with a zero
00461 terminated sequence of terms (or just zero).
00462 */
00463 WORD *poly_rem(PHEAD WORD *a, WORD *b, WORD fit) {
00464
00465 #ifdef WITHFLINT
00466     if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
00467         WORD *ret = flint_rem(BHEAD a, b, fit);
00468         return ret;
00469     }
00470 #endif
00471
00472 #ifdef DEBUG
00473     cout << "*** [" << thetime() << "]" CALL : poly_rem" << endl;
00474 #endif
00475
00476     return poly_divmod(BHEAD a, b, 1, fit);
00477 }
00478
00479 /*
00480 #] poly_rem :
00481 #[ poly_ratfun_read :
00482 */
00483
00496 void poly_ratfun_read (WORD *a, poly &nnum, poly &den) {
00497
00498 #ifdef DEBUG
00499     cout << "*** [" << thetime() << "]" CALL : poly_ratfun_read" << endl;
00500 #endif
00501
00502     POLY_GETIDENTITY(nnum);

```

```

00503
00504     int modp = num.modp;
00505
00506     WORD *astop = a+a[1];
00507
00508     bool clean = (a[2] & MUSTCLEANPRF) == 0;
00509
00510     a += FUNHEAD;
00511     if (a >= astop) {
00512         MLOCK(ErrorMessageLock);
00513         MesPrint ((char*)"ERROR: PolyRatFun cannot have zero arguments");
00514         MUNLOCK(ErrorMessageLock);
00515         Terminate(-1);
00516     }
00517
00518     poly den_num(BHEAD 1), den_den(BHEAD 1);
00519
00520     num = poly::argument_to_poly(BHEAD a, true, !clean, &den_num);
00521     num.setmod(modp,1);
00522     NEXTARG(a);
00523
00524     if (a < astop) {
00525         den = poly::argument_to_poly(BHEAD a, true, !clean, &den_den);
00526         den.setmod(modp,1);
00527         NEXTARG(a);
00528     }
00529     else {
00530         den = poly(BHEAD 1, modp, 1);
00531     }
00532
00533     if (a < astop) {
00534         MLOCK(ErrorMessageLock);
00535         MesPrint ((char*)"ERROR: PolyRatFun cannot have more than two arguments");
00536         MUNLOCK(ErrorMessageLock);
00537         Terminate(-1);
00538     }
00539
00540     // JD: At this point, num and den are certainly sorted into the correct order by
00541     // poly::argument_to_poly, but we can't rely on the clean flag to know if there
00542     // are any negative powers. Check for them, and set clean = false if there are any.
00543     vector<WORD> minpower(AN.poly_num_vars, MAXPOSITIVE);
00544
00545     for (int i=1; i<num[0]; i+=num[i]) {
00546         for (int j=0; j<AN.poly_num_vars; j++) {
00547             minpower[j] = Min(minpower[j], num[i+1+j]);
00548         }
00549     }
00550     for (int i=1; i<den[0]; i+=den[i]) {
00551         for (int j=0; j<AN.poly_num_vars; j++) {
00552             minpower[j] = Min(minpower[j], den[i+1+j]);
00553             if ( minpower[j] < 0 ) clean = false;
00554         }
00555     }
00556
00557     if (!clean) {
00558         for (int i=1; i<num[0]; i+=num[i])
00559             for (int j=0; j<AN.poly_num_vars; j++)
00560                 num[i+1+j] -= minpower[j];
00561         for (int i=1; i<den[0]; i+=den[i])
00562             for (int j=0; j<AN.poly_num_vars; j++)
00563                 den[i+1+j] -= minpower[j];
00564
00565         num *= den_den;
00566         den *= den_num;
00567
00568         poly gcd = polygcd::gcd(num,den);
00569         num /= gcd;
00570         den /= gcd;
00571     }
00572 }
00573
00574 /*
00575 #] poly_ratfun_read :
00576 #[ poly_sort :
00577 */
00578
00590 void poly_sort(PHEAD WORD *a) {
00591
00592 #ifdef DEBUG
00593     cout << "*** [" << thetime() << "]" CALL : poly_sort" << endl;
00594 #endif
00595     if (NewSort(BHEAD0)) { Terminate(-1); }
00596     AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);
00597
00598     for (int i=ARGHEAD; i<a[0]; i+=a[i]) {
00599         if (SymbolNormalize(a+i)<0 || StoreTerm(BHEAD a+i)) {
00600             AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);

```

```

00601         LowerSortLevel();
00602         Terminate(-1);
00603     }
00604 }
00605
00606 if (EndSort(BHEAD a+ARGHEAD,1) < 0) {
00607     AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
00608     Terminate(-1);
00609 }
00610
00611 AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
00612 a[1] = 0; // set dirty flag to zero
00613 }
00614
00615 /*
00616 #] poly_sort :
00617 #[ poly_ratfun_add :
00618 */
00619
00633 WORD *poly_ratfun_add (PHEAD WORD *t1, WORD *t2) {
00634
00635 #ifdef WITHFLINT
00636     if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
00637         WORD *ret = flint_ratfun_add(BHEAD t1, t2);
00638         return ret;
00639     }
00640 #endif
00641
00642     if ( AR.PolyFunExp == 1 ) return PolyRatFunSpecial(BHEAD t1, t2);
00643
00644 #ifdef DEBUG
00645     cout << "*** [" << thetime() << "]" CALL : poly_ratfun_add" << endl;
00646 #endif
00647
00648     WORD *oldworkpointer = AT.WorkPointer;
00649
00650     // Extract variables
00651     vector<WORD *> e;
00652     e.reserve(4);
00653
00654     for (WORD *t=t1+FUNHEAD; t<t1+t1[1];) {
00655         e.push_back(t);
00656         NEXTARG(t);
00657     }
00658     for (WORD *t=t2+FUNHEAD; t<t2+t2[1];) {
00659         e.push_back(t);
00660         NEXTARG(t);
00661     }
00662
00663     assert(e.size() == 4);
00664
00665     poly::get_variables(BHEAD e, true, true);
00666
00667     // Check for modulus calculus
00668     WORD modp=poly_determine_modulus(BHEAD true, true, "PolyRatFun");
00669
00670     // Find numerators / denominators
00671     poly num1(BHEAD 0,modp,1), den1(BHEAD 0,modp,1), num2(BHEAD 0,modp,1), den2(BHEAD 0,modp,1);
00672
00673     poly_ratfun_read(t1, num1, den1);
00674     poly_ratfun_read(t2, num2, den2);
00675
00676     poly num(BHEAD 0),den(BHEAD 0),gcd(BHEAD 0);
00677
00678     // Calculate result
00679     if (den1 != den2) {
00680         gcd = polygcd::gcd(den1,den2);
00681 #ifdef OLDADDITION
00682         num = num1*(den2/gcd) + num2*(den1/gcd);
00683         den = (den1/gcd)*den2;
00684         gcd = polygcd::gcd(num,den);
00685 #else
00686         den = den1/gcd;
00687         num = num1*(den2/gcd) + num2*den;
00688         den = den*den2;
00689         gcd = polygcd::gcd(num,gcd);
00690 #endif
00691     }
00692     else {
00693         num = num1 + num2;
00694         den = den1;
00695         gcd = polygcd::gcd(num,den);
00696     }
00697
00698     num /= gcd;
00699     den /= gcd;
00700

```

```

00701 // Fix sign
00702 if (den.sign() == -1) { num*=poly(BHEAD -1); den*=poly(BHEAD -1); }
00703
00704 // Check size: include FUNHEAD for the prf itself, an ARGHEAD each for num and den,
00705 // and 3 for the final coeff "1/1". We don't know here what the rest of the term looks like,
00706 // but it certainly has at least its total size (so +1):
00707 if ((num.size_of_form_notation() + den.size_of_form_notation() + FUNHEAD + 2*ARGHEAD + 3 + 1)
00708 > AM.MaxTer/(int)sizeof(WORD)) {
00709
00710     MLOCK(ErrorMessageLock);
00711     MesPrint ("ERROR: PolyRatFun doesn't fit in a term");
00712     MesPrint ("(1) num size = %d, den size = %d, rest = %d, MaxTermSize = %d words",
00713             num.size_of_form_notation()+ARGHEAD,
00714             den.size_of_form_notation()+ARGHEAD,
00715             FUNHEAD + 3 + 1,
00716             AM.MaxTer/sizeof(WORD));
00717     MUNLOCK(ErrorMessageLock);
00718     Terminate(-1);
00719 }
00720
00721 // Format result in Form notation
00722 WORD *t = oldworkpointer;
00723
00724 *t++ = AR.PolyFun; // function
00725 *t++ = 0; // length (to be determined)
00726 // *t++ &= ~MUSTCLEANPRF; // clean polyratfun
00727 *t++ = 0;
00728 FILLFUN3(t); // header
00729 poly::poly_to_argument(num,t, true); // argument 1 (numerator)
00730 if (*t>0 && t[1]==DIRTYFLAG) // to Form order
00731     poly_sort(BHEAD t);
00732 t += (*t>0 ? *t : 2);
00733 poly::poly_to_argument(den,t, true); // argument 2 (denominator)
00734 if (*t>0 && t[1]==DIRTYFLAG) // to Form order
00735     poly_sort(BHEAD t);
00736 t += (*t>0 ? *t : 2);
00737
00738 oldworkpointer[1] = t - oldworkpointer; // length
00739 AT.WorkPointer = t;
00740
00741 poly_free_poly_vars(BHEAD "AN.poly_vars_ratfun_add");
00742
00743 // reset modulo calculation
00744 AN.ncmod = AC.ncmod;
00745
00746 return oldworkpointer;
00747 }
00748
00749 /*
00750 #] poly_ratfun_add :
00751 #[ poly_ratfun_normalize :
00752 */
00753
00769 int poly_ratfun_normalize (PHEAD WORD *term) {
00770
00771 #ifdef WITHFLINT
00772     if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
00773         flint_ratfun_normalize(BHEAD term);
00774         return 0;
00775     }
00776 #endif
00777
00778 #ifdef DEBUG
00779     cout << "*** [" << thetime() << "]" CALL : poly_ratfun_normalize" << endl;
00780 #endif
00781
00782 // Strip coefficient
00783 WORD *tstop = term + *term;
00784 int ncoeff = tstop[-1];
00785 tstop -= ABS(ncoeff);
00786
00787 // if only one clean polyratfun, return immediately
00788 int num_polyratfun = 0;
00789
00790 for (WORD *t=term+1; t<tstop; t+=t[1])
00791     if (*t == AR.PolyFun) {
00792         num_polyratfun++;
00793         if ((t[2] & MUSTCLEANPRF) != 0)
00794             num_polyratfun = INT_MAX;
00795         if (num_polyratfun > 1) break;
00796     }
00797
00798 if (num_polyratfun <= 1) return 0;
00799
00800 WORD oldsorttype = AR.SortType;
00801 AR.SortType = SORTHIGFIRST;
00802

```



```

00803 /*
00804     When there are polyratfun's with only one variable: rename them
00805     temporarily to TMPPOLYFUN.
00806 */
00807 for (WORD *t=term+1; t<tstop; t+=t[1]) {
00808     if (*t == AR.PolyFun && (t[1] == FUNHEAD+t[FUNHEAD]
00809         || t[1] == FUNHEAD+2 ) ) { *t = TMPPOLYFUN; }
00810 }
00811
00812 // Extract all variables in the polyfuns
00813 vector<WORD *> e;
00814
00815 for (WORD *t=term+1; t<tstop; t+=t[1]) {
00816     if (*t == AR.PolyFun)
00817         for (WORD *t2 = t+FUNHEAD; t2<t+t[1];) {
00818             e.push_back(t2);
00819             NEXTARG(t2);
00820         }
00821 }
00822
00823 poly::get_variables(BHEAD e, true, true);
00824
00825 // Check for modulus calculus
00826 WORD modp=poly_determine_modulus(BHEAD true, true, "PolyRatFun");
00827
00828 // Accumulate total denominator/numerator and copy the remaining terms
00829 // We start with 'trivial' polynomials
00830 poly num1(BHEAD (UWORD *)tstop, ncoeff/2, modp, 1);
00831 poly den1(BHEAD (UWORD *)tstop+ABS(ncoeff/2), ABS(ncoeff)/2, modp, 1);
00832
00833 WORD *s = term+1;
00834
00835 for (WORD *t=term+1; t<tstop;)
00836     if (*t == AR.PolyFun) {
00837
00838         poly num2(BHEAD 0, modp, 1);
00839         poly den2(BHEAD 0, modp, 1);
00840         poly_ratfun_read(t, num2, den2);
00841
00842         if ((t[2] & MUSTCLEANPRF) != 0) { // first normalize
00843             poly gcd1(polygcd::gcd(num2, den2));
00844             num2 = num2/gcd1;
00845             den2 = den2/gcd1;
00846         }
00847         t += t[1];
00848         poly gcd1(polygcd::gcd(num1, den2));
00849         poly gcd2(polygcd::gcd(num2, den1));
00850
00851         num1 = (num1 / gcd1) * (num2 / gcd2);
00852         den1 = (den1 / gcd2) * (den2 / gcd1);
00853     }
00854     else {
00855         int i = t[1];
00856         if (s != t) { NCOPY(s, t, i) }
00857         else { t += i; s += i; }
00858     }
00859
00860 // Fix sign
00861 if (den1.sign() == -1) { num1*=poly(BHEAD -1); den1*=poly(BHEAD -1); }
00862
00863 // Check size: include FUNHEAD for the prf itself, an ARGHEAD each for num and den,
00864 // s-term for the copied term so far, and 3 for final coeff "1/1"
00865 if ((num1.size_of_form_notation() + den1.size_of_form_notation() + FUNHEAD + 2*ARGHEAD
00866     + s-term + 3) > AM.MaxTer/(int)sizeof(WORD)) {
00867
00868     MLOCK(ErrorMessageLock);
00869     MesPrint ("ERROR: PolyRatFun doesn't fit in a term");
00870     MesPrint ("(2) num size = %d, den size = %d, rest = %d, MaxTermSize = %d words",
00871         num1.size_of_form_notation()+ARGHEAD,
00872         den1.size_of_form_notation()+ARGHEAD,
00873         FUNHEAD + s-term + 3,
00874         AM.MaxTer/sizeof(WORD));
00875     MUNLOCK(ErrorMessageLock);
00876     Terminate(-1);
00877 }
00878
00879 // Format result in Form notation
00880 WORD *t = s;
00881 *t++ = AR.PolyFun; // function
00882 *t++ = 0; // size (to be determined)
00883 *t++ &= ~MUSTCLEANPRF; // clean polyratfun
00884 FILLFUN3(t); // header
00885 poly::poly_to_argument(num1, t, true); // argument 1 (numerator)
00886 if (*t>0 && t[1]==DIRTYFLAG) // to Form order
00887     poly_sort(BHEAD t);
00888 t += (*t>0 ? *t : 2);
00889 poly::poly_to_argument(den1, t, true); // argument 2 (denominator)

```

```

00890     if (*t>0 && t[1]==DIRTYFLAG)           // to Form order
00891         poly_sort(BHEAD t);
00892     t += (*t>0 ? *t : 2);
00893
00894     s[1] = t - s;                           // function length
00895
00896     *t++ = 1;                               // term coefficient
00897     *t++ = 1;
00898     *t++ = 3;
00899
00900     term[0] = t-term;                       // term length
00901
00902     poly_free_poly_vars(BHEAD "AN.poly_vars_ratfun_normalize");
00903
00904     // reset modulo calculation
00905     AN.ncmod = AC.ncmod;
00906
00907     tstop = term + *term; tstop -= ABS(tstop[-1]);
00908     for (WORD *t=term+1; t<tstop; t+=t[1]) {
00909         if (*t == TMPPOLYFUN) *t = AR.PolyFun;
00910     }
00911
00912     AR.SortType = oldsorttype;
00913     return 0;
00914 }
00915
00916 /*
00917 #] poly_ratfun_normalize :
00918 #[ poly_fix_minus_signs :
00919 */
00920
00921 void poly_fix_minus_signs (factorized_poly &a) {
00922
00923     if ( a.factor.empty() ) return;
00924
00925     POLY_GETIDENTITY(a.factor[0]);
00926
00927     int overall_sign = +1;
00928
00929     // find term with maximum power of highest symbol
00930     for (int i=0; i<(int)a.factor.size(); i++) {
00931
00932         int maxvar = -1;
00933         int maxpow = -1;
00934         int sign = +1;
00935
00936         WORD *tstop = a.factor[i].terms; tstop += *tstop;
00937         for (WORD *t=a.factor[i].terms+1; t<tstop; t+=*t)
00938             for (int j=0; j<AN.poly_num_vars; j++) {
00939                 int var = AN.poly_vars[j];
00940                 int pow = t[1+j];
00941                 if (pow>0 && (var>maxvar || (var==maxvar && pow>maxpow))) {
00942                     maxvar = var;
00943                     maxpow = pow;
00944                     sign = SGN(*(t+*t-1));
00945                 }
00946             }
00947
00948         // if negative coefficient, multiply by -1
00949         if (sign== -1) {
00950             a.factor[i] *= poly(BHEAD sign);
00951             if (a.power[i] % 2 == 1) overall_sign*=-1;
00952         }
00953     }
00954
00955     // if overall minus sign
00956     if (overall_sign == -1) {
00957         // look at constant factor and multiply by -1
00958         for (int i=0; i<(int)a.factor.size(); i++)
00959             if (a.factor[i].is_integer()) {
00960                 a.factor[i] *= poly(BHEAD -1);
00961                 return;
00962             }
00963
00964         // otherwise, add a factor of -1
00965         a.add_factor(poly(BHEAD -1), 1);
00966     }
00967 }
00968
00969 /*
00970 #] poly_fix_minus_signs :
00971 #[ poly_factorize :
00972 */
00973
00986 WORD *poly_factorize (PHEAD WORD *argin, WORD *argout, bool with_arghead, bool is_fun_arg) {
00987
00988 #ifdef DEBUG

```

```

00989     cout << "*** [" << thetime() << "]" CALL : poly_factorize" << endl;
00990 #endif
00991
00992     poly::get_variables(BHEAD vector<WORD*>(1,argin), with_arghead, true);
00993     poly den(BHEAD 0);
00994     poly a(poly::argument_to_poly(BHEAD argin, with_arghead, true, &den));
00995
00996     // check for modulus calculus
00997     WORD modp=poly_determine_modulus(BHEAD true, is_fun_arg, "polynomial factorization");
00998     a.setmod(modp,1);
00999
01000     // factorize
01001     factorized_poly f(polyfact::factorize(a));
01002     poly_fix_minus_signs(f);
01003
01004     poly num(BHEAD 1);
01005     for (int i=0; i<(int)f.factor.size(); i++)
01006         if (f.factor[i].is_integer())
01007             num = f.factor[i];
01008
01009     // determine size
01010     int len = with_arghead ? ARGHEAD : 0;
01011
01012     if (!num.is_one() || !den.is_one()) {
01013         len++;
01014         len += MAX(ABS(num[num[1]]), den[den[1]])*2+1;
01015         len += with_arghead ? ARGHEAD : 1;
01016     }
01017
01018     for (int i=0; i<(int)f.factor.size(); i++) {
01019         if (!f.factor[i].is_integer()) {
01020             len += f.power[i] * f.factor[i].size_of_form_notation();
01021             len += f.power[i] * (with_arghead ? ARGHEAD : 1);
01022         }
01023     }
01024
01025     len++;
01026
01027     if (argout != NULL) {
01028         // check size
01029         if (len >= AM.MaxTer) {
01030             MLOCK(ErrorMessageLock);
01031             MesPrint ("ERROR: factorization doesn't fit in a term (len = %d, MaxTermSize = %d words)",
01032                 len/sizeof(WORD), AM.MaxTer/sizeof(WORD));
01033             MUNLOCK(ErrorMessageLock);
01034             Terminate(-1);
01035         }
01036     }
01037     else {
01038         // allocate size
01039         argout = (WORD*) Malloc1(len*sizeof(WORD), "poly_factorize");
01040     }
01041
01042     WORD *old_argout = argout;
01043
01044     // constant factor
01045     if (!num.is_one() || !den.is_one()) {
01046         int n = max(ABS(num[num[1]]), ABS(den[den[1]]));
01047
01048         if (with_arghead) {
01049             *argout++ = ARGHEAD + 2 + 2*n;
01050             for (int i=1; i<ARGHEAD; i++)
01051                 *argout++ = 0;
01052         }
01053
01054         *argout++ = 2 + 2*n;
01055
01056         for (int i=0; i<n; i++)
01057             *argout++ = i<ABS(num[num[1]]) ? num[2+AN.poly_num_vars+i] : 0;
01058         for (int i=0; i<n; i++)
01059             *argout++ = i<ABS(den[den[1]]) ? den[2+AN.poly_num_vars+i] : 0;
01060
01061         *argout++ = SGN(num[num[1]]) * (2*n+1);
01062
01063         if (!with_arghead)
01064             *argout++ = 0;
01065         else {
01066             if (ToFast(old_argout, old_argout))
01067                 argout = old_argout+2;
01068         }
01069     }
01070
01071     // non-constant factors
01072     for (int i=0; i<(int)f.factor.size(); i++)
01073         if (!f.factor[i].is_integer())
01074             for (int j=0; j<f.power[i]; j++) {
01075                 poly::poly_to_argument(f.factor[i], argout, with_arghead);

```

```

01076
01077         if (with_arghead)
01078             argout += *argout > 0 ? *argout : 2;
01079         else {
01080             while (*argout!=0) argout+=*argout;
01081             argout++;
01082         }
01083     }
01084
01085     *argout=0;
01086
01087     poly_free_poly_vars(BHEAD "AN.poly_vars_factorize");
01088
01089     // reset modulo calculation
01090     AN.ncmod = AC.ncmod;
01091
01092     return old_argout;
01093 }
01094
01095 /*
01096 #] poly_factorize :
01097 #[ poly_factorize_argument :
01098 */
01099
01100 int poly_factorize_argument(PHEAD WORD *argin, WORD *argout) {
01101
01102     #ifdef DEBUG
01103         cout << "*** [" << thetime() << " ] CALL : poly_factorize_argument" << endl;
01104     #endif
01105
01106     #ifdef WITHFLINT
01107         if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
01108             flint_factorize_argument(BHEAD argin, argout);
01109             return 0;
01110         }
01111     #endif
01112
01113     poly_factorize(BHEAD argin,argout,true,true);
01114     return 0;
01115 }
01116
01117 /*
01118 #] poly_factorize_argument :
01119 #[ poly_factorize_dollar :
01120 */
01121
01122 WORD *poly_factorize_dollar (PHEAD WORD *argin) {
01123
01124     #ifdef DEBUG
01125         cout << "*** [" << thetime() << " ] CALL : poly_factorize_dollar" << endl;
01126     #endif
01127
01128     #ifdef WITHFLINT
01129         if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
01130             return flint_factorize_dollar(BHEAD argin);
01131         }
01132     #endif
01133
01134     return poly_factorize(BHEAD argin,NULL,false,false);
01135 }
01136
01137 /*
01138 #] poly_factorize_dollar :
01139 #[ poly_factorize_expression :
01140 */
01141
01142 int poly_factorize_expression(EXPRESSIONS expr) {
01143
01144     #ifdef DEBUG
01145         cout << "*** [" << thetime() << " ] CALL : poly_factorize_expression" << endl;
01146     #endif
01147
01148     GETIDENTITY;
01149
01150     if (AT.WorkPointer + AM.MaxTer > AT.WorkTop ) {
01151         MLOCK(ErrorMessageLock);
01152         MesWork();
01153         MUNLOCK(ErrorMessageLock);
01154         Terminate(-1);
01155     }
01156
01157     WORD *term = AT.WorkPointer;
01158     WORD startebuf = cbuf[AT.ebufnum].numrhs;
01159     FILEHANDLE *file;
01160     POSITION pos;
01161
01162     FILEHANDLE *oldinfile = AR.infile;

```

```

01199 FILEHANDLE *oldoutfile = AR.outfile;
01200 WORD oldBracketOn = AR.BraceOn;
01201 WORD *oldBrackBuf = AT.BrackBuf;
01202 WORD oldbracketindexflag = AT.bracketindexflag;
01203 char oldCommercial[COMMERCIALSIZE+2];
01204
01205 strcpy(oldCommercial, (char*)AC.Commercial);
01206 strcpy((char*)AC.Commercial, "factorize");
01207
01208 // locate is the input
01209 if (expr->status == HIDDENEXPRESSION || expr->status == HIDDENLEXPRESSION ||
01210     expr->status == INTOHIDEEXPRESSION || expr->status == INTOHIDELEXPRESSION) {
01211     AR.InHiBuf = 0; file = AR.hidefile; AR.GetFile = 2;
01212 }
01213 else {
01214     AR.InInBuf = 0; file = AR.outfile; AR.GetFile = 0;
01215 }
01216
01217 // read and write to expression file
01218 AR.infile = AR.outfile = file;
01219
01220 // dummy indices are not allowed
01221 if (expr->numdummies > 0) {
01222     MesPrint("ERROR: factorization with dummy indices not implemented");
01223     Terminate(-1);
01224 }
01225
01226 // determine whether the expression is on file or in memory
01227 if (file->handle >= 0) {
01228     pos = expr->onfile;
01229     SeekFile(file->handle, &pos, SEEK_SET);
01230     if (ISNOTEQUALPOS(pos, expr->onfile)) {
01231         MesPrint("ERROR: something wrong in scratch file [poly_factorize_expression]");
01232         Terminate(-1);
01233     }
01234     file->POposition = expr->onfile;
01235     file->POfull = file->PObuffer;
01236     if (expr->status == HIDDENEXPRESSION)
01237         AR.InHiBuf = 0;
01238     else
01239         AR.InInBuf = 0;
01240 }
01241 else {
01242     file->POfill = (WORD *) ((UBYTE *) (file->PObuffer) + BASEPOSITION(expr->onfile));
01243 }
01244
01245 SetScratch(AR.infile, &(expr->onfile));
01246
01247 // read the first header term
01248 WORD size = GetTerm(BHEAD term);
01249 if (size <= 0) {
01250     MesPrint("ERROR: something wrong with expression [poly_factorize_expression]");
01251     Terminate(-1);
01252 }
01253
01254 // store position: this is where the output will go
01255 pos = expr->onfile;
01256 ADDPOS(pos, size*sizeof(WORD));
01257
01258 // use polynomial as buffer, because it is easy to extend
01259 poly buffer(BHEAD 0);
01260 int bufpos = 0;
01261 int sumcommu = 0;
01262
01263 // read all terms
01264 while (GetTerm(BHEAD term)) {
01265     // substitute non-symbols by extra symbols
01266     sumcommu += DoesCommu(term);
01267     if (sumcommu > 1) {
01268         MesPrint("ERROR: Cannot factorize an expression with more than one noncommuting object");
01269         Terminate(-1);
01270     }
01271     buffer.check_memory(bufpos);
01272     if (LocalConvertToPoly(BHEAD term, buffer.terms + bufpos, startebuf, 0) < 0) {
01273         MesPrint("ERROR: in LocalConvertToPoly [factorize_expression]");
01274         Terminate(-1);
01275     }
01276     bufpos += *(buffer.terms + bufpos);
01277 }
01278 buffer[bufpos] = 0;
01279
01280 // parse the polynomial
01281
01282 AN.poly_num_vars = 0;
01283 poly::get_variables(BHEAD vector<WORD*>(1, buffer.terms), false, true);
01284 poly den(BHEAD 0);
01285 poly a(poly::argument_to_poly(BHEAD buffer.terms, false, true, &den));

```

```

01286
01287 // check for modulus calculus
01288 WORD modp=poly_determine_modulus(BHEAD true, false, "polynomial factorization");
01289 a.setmod(modp,1);
01290
01291 // create output
01292 SetScratch(file, &pos);
01293 NewSort(BHEAD0);
01294
01295 CBUF *C = cbuf+AC.cbufnum;
01296 CBUF *CC = cbuf+AT.ebufnum;
01297
01298 // turn brackets on. We force the existence of a bracket index.
01299 WORD nexpr = expr - Expressions;
01300 AR.BracketOn = 1;
01301 AT.BrackBuf = AM.BracketFactors;
01302 AT.bracketindexflag = 1;
01303 ClearBracketIndex(~nexpr-2); // Clears the index made during primary generation
01304 OpenBracketIndex(nexpr); // Set up a new index
01305
01306 if (a.is_zero()) {
01307     expr->numfactors = 1;
01308 }
01309 else if (a.is_one() && den.is_one()) {
01310     expr->numfactors = 1;
01311
01312     term[0] = 8;
01313     term[1] = SYMBOL;
01314     term[2] = 4;
01315     term[3] = FACTORSYMBOL;
01316     term[4] = 1;
01317     term[5] = 1;
01318     term[6] = 1;
01319     term[7] = 3;
01320
01321     AT.WorkPointer += *term;
01322     Generator(BHEAD term, C->numlhs);
01323     AT.WorkPointer = term;
01324 }
01325 else {
01326     factorized_poly fac;
01327     bool iszero = false;
01328
01329     if (!(expr->vflags & ISFACTORIZED)) {
01330         // factorize the polynomial
01331         fac = polyfact::factorize(a);
01332         poly_fix_minus_signs(fac);
01333     }
01334     else {
01335         int factorsymbol=-1;
01336         for (int i=0; i<AN.poly_num_vars; i++)
01337             if (AN.poly_vars[i] == FACTORSYMBOL)
01338                 factorsymbol = i;
01339
01340         poly denpow(BHEAD 1);
01341
01342         // already factorized, so factorize the factors
01343         for (int i=1; i<=expr->numfactors; i++) {
01344             poly origfac(a.coefficient(factorsymbol, i));
01345             factorized_poly fac2;
01346             if (origfac.is_zero())
01347                 iszero=true;
01348             else {
01349                 fac2 = polyfact::factorize(origfac);
01350                 poly_fix_minus_signs(fac2);
01351                 denpow *= den;
01352             }
01353             for (int j=0; j<(int)fac2.power.size(); j++)
01354                 fac.add_factor(fac2.factor[j], fac2.power[j]);
01355         }
01356
01357         // update denominator, since each factor was scaled
01358         den=denpow;
01359     }
01360
01361     expr->numfactors = 0;
01362
01363     // coefficient
01364     poly num(BHEAD 1);
01365     for (int i=0; i<(int)fac.factor.size(); i++)
01366         if (fac.factor[i].is_integer())
01367             num *= fac.factor[i];
01368
01369     poly gcd(polygcd::integer_gcd(num,den));
01370     den/=gcd;
01371     num/=gcd;
01372

```

```

01373         if (iszero)
01374             expr->numfactors++;
01375
01376         if (!iszero || (expr->vflags & KEEPZERO)) {
01377             if (!num.is_one() || !den.is_one()) {
01378                 expr->numfactors++;
01379
01380                 int n = max(ABS(num[num[1]]), ABS(den[den[1]]));
01381
01382                 term[0] = 6 + 2*n;
01383                 term[1] = SYMBOL;
01384                 term[2] = 4;
01385                 term[3] = FACTORSYMBOL;
01386                 term[4] = expr->numfactors;
01387                 for (int i=0; i<n; i++) {
01388                     term[5+i] = i<ABS(num[num[1]]) ? num[2+AN.poly_num_vars+i] : 0;
01389                     term[5+n+i] = i<ABS(den[den[1]]) ? den[2+AN.poly_num_vars+i] : 0;
01390                 }
01391                 term[5+2*n] = SGN(num[num[1]]) * (2*n+1);
01392                 AT.WorkPointer += *term;
01393                 Generator(BHEAD term, C->numlhs);
01394                 AT.WorkPointer = term;
01395             }
01396
01397             vector<poly> fac_arg(fac.factor.size(), poly(BHEAD 0));
01398
01399             // convert the non-constant factors to Form-style arguments
01400             for (int i=0; i<(int)fac.factor.size(); i++)
01401                 if (!fac.factor[i].is_integer()) {
01402                     buffer.check_memory(fac.factor[i].size_of_form_notation()+1);
01403                     poly::poly_to_argument(fac.factor[i], buffer.terms, false);
01404                     NewSort(BHEAD0);
01405
01406                     for (WORD *t=buffer.terms; *t!=0; t+=*t) {
01407                         // substitute extra symbols
01408                         if (ConvertFromPoly(BHEAD t, term, numxsymbol,
01409                             CC->numrhs-startebuf+numxsymbol,
01410                             startebuf-numxsymbol, 1) <= 0 ) {
01411                             MesPrint("ERROR: in ConvertFromPoly [factorize_expression]");
01412                             Terminate(-1);
01413                             return(-1);
01414                         }
01415
01416                         // store term
01417                         AT.WorkPointer += *term;
01418                         Generator(BHEAD term, C->numlhs);
01419                         AT.WorkPointer = term;
01420                     }
01421
01422                     // sort and store in buffer
01423                     WORD *buffer;
01424                     if (EndSort(BHEAD (WORD *)((void *)(&buffer)),2) < 0) return -1;
01425
01426                     LONG bufsize=0;
01427                     for (WORD *t=buffer; *t!=0; t+=*t)
01428                         bufsize+=*t;
01429
01430                     fac_arg[i].check_memory(bufsize+ARGHEAD+1);
01431
01432                     for (int j=0; j<ARGHEAD; j++)
01433                         fac_arg[i].terms[j] = 0;
01434                     fac_arg[i].terms[0] = ARGHEAD + bufsize;
01435                     WCOPY(fac_arg[i].terms+ARGHEAD, buffer, bufsize);
01436                     M_free(buffer, "polynomial factorization");
01437                 }
01438
01439             // compare and sort the factors in Form notation
01440             vector<int> order;
01441             vector<vector<int>> > comp(fac.factor.size(), vector<int>(fac.factor.size(), 0));
01442
01443             for (int i=0; i<(int)fac.factor.size(); i++)
01444                 if (!fac.factor[i].is_integer()) {
01445                     order.push_back(i);
01446
01447                     for (int j=i+1; j<(int)fac.factor.size(); j++)
01448                         if (!fac.factor[j].is_integer()) {
01449                             comp[i][j] = CompArg(fac_arg[j].terms, fac_arg[i].terms);
01450                             comp[j][i] = -comp[i][j];
01451                         }
01452                 }
01453
01454             for (int i=0; i<(int)order.size(); i++)
01455                 for (int j=0; j+1<(int)order.size(); j++)
01456                     if (comp[order[i]][order[j]] == 1)
01457                         swap(order[i], order[j]);
01458
01459             // create the final expression

```

```

01459         for (int i=0; i<(int)order.size(); i++)
01460             for (int j=0; j<fac.power[order[i]]; j++) {
01461
01462                 expr->numfactors++;
01463
01464                 WORD *tstop = fac_arg[order[i]].terms + *fac_arg[order[i]].terms;
01465                 for (WORD *t=fac_arg[order[i]].terms+ARGHEAD; t<tstop; t+=*t) {
01466
01467                     WCOPY(term+4, t, *t);
01468
01469                     // add special symbol "factor_"
01470                     *term = *(term+4) + 4;
01471                     *(term+1) = SYMBOL;
01472                     *(term+2) = 4;
01473                     *(term+3) = FACTORSYMBOL;
01474                     *(term+4) = expr->numfactors;
01475
01476                     // store term
01477                     AT.WorkPointer += *term;
01478                     Generator(BHEAD term, C->numlhs);
01479                     AT.WorkPointer = term;
01480                 }
01481             }
01482     }
01483 }
01484
01485 // final sorting
01486 if (EndSort(BHEAD NULL,0) < 0) {
01487     LowerSortLevel();
01488     Terminate(-1);
01489 }
01490
01491 // set factorized flag
01492 if (expr->numfactors > 0)
01493     expr->vflags |= ISFACTORIZED;
01494
01495 // clean up
01496 AR.infile = oldinfile;
01497 AR.outfile = oldoutfile;
01498 AR.BracketOn = oldBracketOn;
01499 AT.BrackBuf = oldBrackBuf;
01500 AT.bracketindexflag = oldbracketindexflag;
01501 strcpy((char*)AC.Commercial, oldCommercial);
01502
01503 poly_free_poly_vars(BHEAD "AN.poly_vars_factorize_expression");
01504
01505 return 0;
01506 }
01507
01508 /*
01509 #] poly_factorize_expression :
01510 #[ poly_unfactorize_expression :
01511 */
01512
01513 #if ( SUBEXPSIZE == 5 )
01514 static WORD genericterm[] = {38,1,4,FACTORSYMBOL,0
01515 ,EXPRESSION,15,0,1,0,13,10,8,1,4,FACTORSYMBOL,0,1,1,3
01516 ,EXPRESSION,15,0,1,0,13,10,8,1,4,FACTORSYMBOL,0,1,1,3
01517 ,1,1,3,0};
01518 static WORD genericterm2[] = {23,1,4,FACTORSYMBOL,0
01519 ,EXPRESSION,15,0,1,0,13,10,8,1,4,FACTORSYMBOL,0,1,1,3
01520 ,1,1,3,0};
01521 #endif
01522
01523 int poly_unfactorize_expression(EXPRESSIONS expr)
01524 {
01525     GETIDENTITY;
01526     int i, j, nfac = expr->numfactors, nfaccp, nexpr = expr - Expressions;
01527     int expriszero = 0;
01528
01529     FILEHANDLE *oldinfile = AR.infile;
01530     FILEHANDLE *oldoutfile = AR.outfile;
01531     char oldCommercial[COMMERCIALSIZE+2];
01532
01533     WORD *oldworkpointer = AT.WorkPointer;
01534     WORD *term = AT.WorkPointer, *t, *w, size;
01535
01536     FILEHANDLE *file;
01537     POSITION pos, oldpos;
01538
01539     WORD oldBracketOn = AR.BracketOn;
01540     WORD *oldBrackBuf = AT.BrackBuf;
01541     CBUF *C = cbuf+AC.cbunum;
01542
01543     if ( ( expr->vflags & ISFACTORIZED ) == 0 ) return(0);
01544
01545     if ( AT.WorkPointer + AM.MaxTer > AT.WorkTop ) {

```



```

01558         MLOCK(ErrorMessageLock);
01559         MesWork();
01560         MUNLOCK(ErrorMessageLock);
01561         Terminate(-1);
01562     }
01563
01564     oldpos = AS.OldOnFile[nexpr];
01565     AS.OldOnFile[nexpr] = expr->onfile;
01566
01567     strcpy(oldCommercial, (char*)AC.Commercial);
01568     strcpy((char*)AC.Commercial, "unfactorize");
01569 /*
01570     locate the input
01571 */
01572     if ( expr->status == HIDDENEXPRESSION || expr->status == HIDDENLEXPRESSION ||
01573         expr->status == INTOHIDEGEXPRESSION || expr->status == INTOHIDELEXPRESSION ) {
01574         AR.InHiBuf = 0; file = AR.hidefile; AR.GetFile = 2;
01575     }
01576     else {
01577         AR.InInBuf = 0; file = AR.outfile; AR.GetFile = 0;
01578     }
01579 /*
01580     read and write to expression file
01581 */
01582     AR.infile = AR.outfile = file;
01583 /*
01584     set the input file to the correct position
01585 */
01586     if ( file->handle >= 0 ) {
01587         pos = expr->onfile;
01588         SeekFile(file->handle, &pos, SEEK_SET);
01589         if (ISNOTEQUALPOS(pos, expr->onfile)) {
01590             MesPrint("ERROR: something wrong in scratch file unfactorize_expression");
01591             Terminate(-1);
01592         }
01593         file->POposition = expr->onfile;
01594         file->POfull = file->PObuffer;
01595         if ( expr->status == HIDDENEXPRESSION )
01596             AR.InHiBuf = 0;
01597         else
01598             AR.InInBuf = 0;
01599     }
01600     else {
01601         file->POfill = (WORD *) ((UBYTE *) (file->PObuffer) + BASEPOSITION(expr->onfile));
01602     }
01603     SetScratch(AR.infile, &(expr->onfile));
01604 /*
01605     Test for whether the first factor is zero.
01606 */
01607     if ( GetFirstBracket(term, nexpr) < 0 ) Terminate(-1);
01608     if ( term[4] != 1 || *term != 8 || term[1] != SYMBOL || term[3] != FACTORSYMBOL || term[4] != 1 )
01609     {
01610         expriszero = 1;
01611     }
01612     SetScratch(AR.infile, &(expr->onfile));
01613 /*
01614     Read the prototype. After this we have the file ready for the output at pos.
01615 */
01616     size = GetTerm(BHEAD term);
01617     if ( size <= 0 ) {
01618         MesPrint ("ERROR: something wrong with expression unfactorize_expression");
01619         Terminate(-1);
01620     }
01621     pos = expr->onfile;
01622     ADDPOS(pos, size*sizeof(WORD));
01623 /*
01624     Set the brackets straight
01625 */
01626     AR.BracketOn = 1;
01627     AT.BrackBuf = AM.BracketFactors;
01628     AT.bracketinfo = 0;
01629     while ( nfac > 2 ) {
01630         nfacp = nfac - nfac%2;
01631 /*
01632     Prepare the bracket index. We have:
01633         e->bracketinfo: the old input bracket index
01634         e->newbracketinfo: the bracket index made for our current input
01635     We need to keep e->bracketinfo in case other workers need it (InParallel)
01636     Hence we work with AT.bracketinfo which takes priority.
01637     Note that in Processor we forced a newbracketinfo to be made.
01638 */
01639     if ( AT.bracketinfo != 0 ) ClearBracketIndex(-1);
01640     AT.bracketinfo = expr->newbracketinfo;
01641     OpenBracketIndex(nexpr);
01642 /*
01643     Now emulate the terms:
01644         sum_(i,0,nfacp,2,factor_^(i/2+1)*F[factor_^(i+1)]*F[factor_^(i+2)])

```

```

01644         +factor_^(nfacp/2+1)*F[factor_^nfac]
01645     */
01646     NewSort(BHEAD0);
01647     if ( expriszero == 0 ) {
01648         for ( i = 0; i < nfacp; i += 2 ) {
01649             t = genericterm; w = term = oldworkpointer;
01650             j = *t; NCOPY(w,t,j);
01651             term[4] = i/2+1;
01652             term[7] = nexpr;
01653             term[16] = i+1;
01654             term[22] = nexpr;
01655             term[31] = i+2;
01656             AT.WorkPointer = term + *term;
01657             Generator(BHEAD term, C->numlhs);
01658         }
01659         if ( nfac > nfacp ) {
01660             t = genericterm2; w = term = oldworkpointer;
01661             j = *t; NCOPY(w,t,j);
01662             term[4] = i/2+1;
01663             term[7] = nexpr;
01664             term[16] = nfac;
01665             AT.WorkPointer = term + *term;
01666             Generator(BHEAD term, C->numlhs);
01667         }
01668     }
01669     if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) {
01670         LowerSortLevel();
01671         Terminate(-1);
01672     }
01673     /*
01674     Set the file back into reading position
01675     */
01676     SetScratch(file, &pos);
01677     nfac = (nfac+1)/2;
01678     if ( expriszero ) { nfac = 1; }
01679 }
01680 if ( AT.bracketinfo != 0 ) ClearBracketIndex(-1);
01681 AT.bracketinfo = expr->newbracketinfo;
01682 expr->newbracketinfo = 0;
01683 /*
01684 Reset the brackets to make them ready for the final pass
01685 */
01686 AR.BracketOn = oldBracketOn;
01687 AT.BrackBuf = oldBrackBuf;
01688 if ( AR.BracketOn ) OpenBracketIndex(nexpr);
01689 /*
01690 We distinguish two cases: nfac == 2 and nfac == 1
01691 After preparing the term we skip the factor_ part.
01692 */
01693 NewSort(BHEAD0);
01694 if ( expriszero == 0 ) {
01695     if ( nfac == 1 ) {
01696         t = genericterm2; w = term = oldworkpointer;
01697         j = *t; NCOPY(w,t,j);
01698         term[7] = nexpr;
01699         term[16] = nfac;
01700     }
01701     else if ( nfac == 2 ) {
01702         t = genericterm; w = term = oldworkpointer;
01703         j = *t; NCOPY(w,t,j);
01704         term[7] = nexpr;
01705         term[16] = 1;
01706         term[22] = nexpr;
01707         term[31] = 2;
01708     }
01709     else {
01710         AS.OldOnFile[nexpr] = oldpos;
01711         return(-1);
01712     }
01713     term[4] = term[0]-4;
01714     term += 4;
01715     AT.WorkPointer = term + *term;
01716     Generator(BHEAD term, C->numlhs);
01717 }
01718 if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) {
01719     LowerSortLevel();
01720     Terminate(-1);
01721 }
01722 /*
01723 Final Cleanup
01724 */
01725 expr->numfactors = 0;
01726 expr->vflags &= ~ISFACTORIZED;
01727 if ( AT.bracketinfo != 0 ) ClearBracketIndex(-1);
01728
01729 AR.infile = oldinfile;
01730 AR.outfile = oldoutfile;

```

```

01731     strcpy((char*)AC.Commercial, oldCommercial);
01732     AT.WorkPointer = oldworkpointer;
01733     AS.OldOnFile[nexpr] = oldpos;
01734
01735     return(0);
01736 }
01737
01738 /*
01739 #] poly_unfactorize_expression :
01740 #[ poly_inverse :
01741 */
01742
01743 WORD *poly_inverse(PHEAD WORD *arga, WORD *argb) {
01744
01745 #ifdef WITHFLINT
01746     if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
01747         return flint_inverse(BHEAD arga, argb);
01748     }
01749 #endif
01750
01751 #ifdef DEBUG
01752     cout << "*** [" << thetime() << "]" << " CALL : poly_inverse" << endl;
01753 #endif
01754
01755     // Extract variables
01756     vector<WORD *> e;
01757     e.reserve(2);
01758     e.push_back(arga);
01759     e.push_back(argb);
01760     poly::get_variables(BHEAD e, false, true);
01761
01762     if (AN.poly_num_vars > 1) {
01763         MLOCK(ErrorMessageLock);
01764         MesPrint ((char*)"ERROR: multivariate polynomial inverse is generally impossible");
01765         MUNLOCK(ErrorMessageLock);
01766         Terminate(-1);
01767     }
01768
01769     poly finalden(BHEAD 1), finalres(BHEAD 1);
01770     int ressize = 0;
01771     WORD *res = NULL;
01772
01773     // Convert to polynomials
01774     poly dena(BHEAD 0); // We need to keep the overall denominator of arga, to multiply the result
01775     poly a(poly::argument_to_poly(BHEAD arga, false, true, &dena));
01776     poly b(poly::argument_to_poly(BHEAD argb, false, true));
01777
01778     // Divide out the integer content, FORM has not already done this.
01779     poly content_a(BHEAD 0), content_b(BHEAD 0);
01780     content_a = polygcd::integer_content(a);
01781     content_b = polygcd::integer_content(b);
01782     a /= content_a;
01783     b /= content_b;
01784
01785     poly invamodp(BHEAD 0), invbmodp(BHEAD 0);
01786     WORD modp = 0;
01787
01788     // Special cases:
01789     // Possibly strange that we give 1 for inverse_(x1,1) but here we take MMA's convention.
01790     if ((a.is_one() && b.is_one()) || a.is_one()) {
01791         finalres = poly(BHEAD 1);
01792     }
01793     else if (b.is_one()) {
01794         finalres = poly(BHEAD 0);
01795     }
01796     else {
01797         // Check for modulus calculus
01798         modp=poly_determine_modulus(BHEAD true, true, "polynomial inverse");
01799         const bool mod_calc = modp==0 ? false : true;
01800         a.setmod(modp,1);
01801         b.setmod(modp,1);
01802
01803         // Check the gcd of a,b: if it is != 1, the inverse does not exist.
01804         poly gcd(polygcd::gcd(a,b));
01805         if (!gcd.is_one()) {
01806             MLOCK(ErrorMessageLock);
01807             if (mod_calc) {
01808                 MesPrint ((char*)"ERROR: polynomial inverse does not exist (mod %d)", modp);
01809             }
01810             else {
01811                 MesPrint ((char*)"ERROR: polynomial inverse does not exist");
01812             }
01813             MUNLOCK(ErrorMessageLock);
01814             Terminate(-1);
01815         }
01816
01817         bool inv_exists = true;

```

```

01818     do {
01819         // If we are not using modulus calculus, find a suitable prime for xgcd:
01820         if (!mod_calc) {
01821             vector<int> x(1,0);
01822             modp = polyfact::choose_prime(a.integer_lcoeff()*b.integer_lcoeff(), x, modp);
01823         }
01824
01825         poly amodp(a,modp,1);
01826         poly bmodp(b,modp,1);
01827
01828         // Calculate gcd
01829         vector<poly> xgcd(polyfact::extended_gcd_Euclidean_lifted(amodp,bmodp));
01830         invamodp = poly(xgcd[0]);
01831         invbmodp = poly(xgcd[1]);
01832
01833         inv_exists = ((invamodp * amodp) % bmodp).is_one();
01834         if (!inv_exists && mod_calc) {
01835             // Control should not reach here!
01836             MLOCK(ErrorMessageLock);
01837             MesPrint ((char*)"ERROR: polynomial inverse does not exist (mod %d) B", modp);
01838             MUNLOCK(ErrorMessageLock);
01839             Terminate(-1);
01840         }
01841
01842         // If the inverse does not exist and we are not working in modulus calculus,
01843         // choose a new prime and try again. This loop should always terminate, as
01844         // we have already checked that gcd(a,b) == 1.
01845     } while (!inv_exists);
01846
01847     // estimate of the size of the Form notation; might be extended later
01848     ressize = invamodp.size_of_form_notation()+1;
01849     res = (WORD *)Malloca(ressize*sizeof(WORD), "poly_inverse");
01850
01851     // initialize polynomials to store the result
01852     poly primepower(BHEAD modp);
01853     poly inva(invamodp,modp,1);
01854     poly invb(invbmop,modp,1);
01855
01856     while (true) {
01857         // convert to Form notation
01858         int j=0;
01859         WORD n=0;
01860         for (int i=1; i<inva[0]; i+=inva[i]) {
01861
01862             // check whether res should be extended
01863             while (ressize < j + 2*ABS(inva[i+inva[i]-1]) + (inva[i+1]>0?4:0) + 3) {
01864                 int newressize = 2*ressize;
01865
01866                 WORD *newres = (WORD *)Malloca(newressize*sizeof(WORD), "poly_inverse");
01867                 WCOPY(newres, res, ressize);
01868                 M_free(res, "poly_inverse");
01869                 res = newres;
01870                 ressize = newressize;
01871             }
01872
01873             res[j] = 1;
01874             if (inva[i+1]>0) {
01875                 res[j+res[j]++] = SYMBOL;
01876                 res[j+res[j]++] = 4;
01877                 res[j+res[j]++] = AN.poly_vars[0];
01878                 res[j+res[j]++] = inva[i+1];
01879             }
01880             MakeLongRational(BHEAD (UWORD *)&inva[i+2], inva[i+inva[i]-1],
01881                             (UWORD*)&primepower.terms[3],
01882                             primepower.terms[primepower.terms[1]],
01883                             (UWORD *)&res[j+res[j]], &n);
01884             res[j] += 2*ABS(n);
01885             res[j+res[j]++] = SGN(n)*(2*ABS(n)+1);
01886             j += res[j];
01887         }
01888         res[j]=0;
01889
01890         // if modulus calculus is set, this is the answer
01891         if (a.modp != 0) break;
01892
01893         // otherwise check over integers
01894         poly den(BHEAD 0);
01895         poly check(poly::argument_to_poly(BHEAD res, false, true, &den));
01896         // Shortcut: if b's lcoeff doesn't divide the lcoeff of check*a,
01897         // b certainly doesn't divide check*a:
01898         if (poly::divides(b.integer_lcoeff(), check.integer_lcoeff()*a.integer_lcoeff())) {
01899             check = check*a - den;
01900             if (poly::divides(b, check)) break;
01901         }
01902
01903         // if incorrect, lift with quadratic p-adic Newton's iteration.
01904         poly error((poly(BHEAD 1) - a*inva - b*invb) / primepower);

```

```

01904         poly errormodpp(error, modp, inva.modn);
01905
01906         inva.modn *= 2;
01907         invb.modn *= 2;
01908
01909         poly dinva((inva * errormodpp) % b);
01910         poly dinvb((invb * errormodpp) % a);
01911
01912         inva += dinva * primepower;
01913         invb += dinvb * primepower;
01914
01915         primepower *= primepower;
01916     }
01917
01918     // One more round trip from form -> poly -> form, to multiply by dena in an easy way
01919     finalres = poly(poly::argument_to_poly(BHEAD res, false, true, &finalden));
01920 }
01921
01922 finalres *= dena;
01923 // The overall denominator additionally needs to be multiplied by content_a:
01924 finalden *= content_a;
01925 const WORD finalden_size = finalden.terms[finalden.terms[1]];
01926 const int finalsize = finalres.size_of_form_notation_with_den(finalden_size)+1;
01927 if (ressize < finalsize) {
01928     if (res != NULL) {
01929         M_free(res, "poly_inverse");
01930     }
01931     res = (WORD *)Malloca(finalsize*sizeof(WORD), "poly_inverse");
01932 }
01933 poly::poly_to_argument_with_den(finalres, finalden_size,
01934     (UWORD*)&(finalden.terms[finalden.terms[1] - ABS(finalden_size)]), res, false);
01935
01936 // clean up and reset modulo calculation
01937 poly_free_poly_vars(BHEAD "AN.poly_vars_inverse");
01938
01939 AN.ncmod = AC.ncmod;
01940 return res;
01941 }
01942
01943 /*
01944 #] poly_inverse :
01945 #[ poly_mul :
01946 */
01947
01948 WORD *poly_mul(PHEAD WORD *a, WORD *b) {
01949
01950 #ifdef DEBUG
01951     cout << "*** [" << thetime() << "]" CALL : poly_mul" << endl;
01952 #endif
01953
01954 #ifdef WITHFLINT
01955     if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
01956         return flint_mul(BHEAD a, b);
01957     }
01958 #endif
01959
01960     // Extract variables
01961     vector<WORD *> e;
01962     e.reserve(2);
01963     e.push_back(a);
01964     e.push_back(b);
01965     poly::get_variables(BHEAD e, false, false); // TODO: any performance effect by sort_vars=true?
01966
01967     // Convert to polynomials
01968     poly dena(BHEAD 0);
01969     poly denb(BHEAD 0);
01970     poly pa(poly::argument_to_poly(BHEAD a, false, true, &dena));
01971     poly pb(poly::argument_to_poly(BHEAD b, false, true, &denb));
01972
01973     // Check for modulus calculus
01974     WORD modp = poly_determine_modulus(BHEAD true, true, "polynomial multiplication");
01975     pa.setmod(modp, 1);
01976
01977     // NOTE: mul_ is currently implemented by translating negative powers of
01978     // symbols to extra symbols. For future improvement, it may be better to
01979     // compute
01980     // (content(a) * content(b)) * (a/content(a)) * (b/content(b))
01981     // to avoid introducing extra symbols for "mixed" cases, e.g.,
01982     // (1+x) * (1/x) -> (1+x) * (1+Z1_).
01983     assert(dena.is_integer());
01984     assert(denb.is_integer());
01985     assert(modp == 0 || dena.is_one());
01986     assert(modp == 0 || denb.is_one());
01987
01988     // multiplication
01989     pa *= pb;
01990

```

```

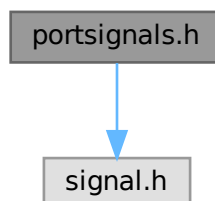
01991 // convert to Form notation
01992 WORD *res;
01993 if (dena.is_one() && denb.is_one()) {
01994     res = (WORD *)Malloca1((pa.size_of_form_notation() + 1) * sizeof(WORD), "poly_mul");
01995     poly::poly_to_argument(pa, res, false);
01996 } else {
01997     dena *= denb;
01998     res = (WORD *)Malloca1((pa.size_of_form_notation_with_den(dena[dena[1]]) + 1) * sizeof(WORD),
01999 "poly_mul");
01999     poly::poly_to_argument_with_den(pa, dena[dena[1]], (const UWORD *)&dena[2+AN.poly_num_vars],
02000 res, false);
02001 }
02002 // clean up and reset modulo calculation
02003 poly_free_poly_vars(BHEAD "AN.poly_vars_mul");
02004 AN.ncmod = AC.ncmod;
02005 return res;
02006 }
02007 }
02008
02009 /*
02010 #] poly_mul :
02011 #[ poly_free_poly_vars :
02012 */
02013
02014 void poly_free_poly_vars(PHEAD const char *text)
02015 {
02016     if ( AN.poly_vars_type == 0 ) {
02017         TermFree(AN.poly_vars, text);
02018     }
02019     else {
02020         M_free(AN.poly_vars, text);
02021     }
02022     AN.poly_num_vars = 0;
02023     AN.poly_vars = 0;
02024 }
02025
02026 /*
02027 #] poly_free_poly_vars :
02028 */

```

## 4.110 portsignals.h File Reference

#include <signal.h>

Include dependency graph for portsignals.h:



### Macros

- #define [FATAL\\_SIG\\_ERROR](#) 4
- #define [NSIG](#) (1024)
- #define [SIGSEGV](#) (NSIG+1)
- #define [SIGFPE](#) (NSIG+2)

- `#define SIGILL (NSIG+3)`
- `#define SIGEMT (NSIG+4)`
- `#define SIGSYS (NSIG+5)`
- `#define SIGPIPE (NSIG+6)`
- `#define SIGLOST (NSIG+7)`
- `#define SIGXCPU (NSIG+8)`
- `#define SIGXFSZ (NSIG+9)`
- `#define SIGTERM (NSIG+10)`
- `#define SIGINT (NSIG+11)`
- `#define SIGQUIT (NSIG+12)`
- `#define SIGHUP (NSIG+13)`
- `#define SIGALRM (NSIG+14)`
- `#define SIGVTALRM (NSIG+15)`
- `#define SIGPROF (NSIG+16)`

### 4.110.1 Detailed Description

Contains definitions for signals used/intercepted in FORM.

Some systems (especially LINUX) have not enough signals available so some of the (!documented!) signals are not defined. This file contains the definition of all signals used in the program. If the signal is not defined we define it as unused ( $>NSIG$ ).

The include of signal.h must be first, before we try to define undefined signals.

Definition in file [portsignals.h](#).

### 4.110.2 Macro Definition Documentation

#### 4.110.2.1 FATAL\_SIG\_ERROR

```
#define FATAL_SIG_ERROR 4
```

Definition at line 47 of file [portsignals.h](#).

#### 4.110.2.2 NSIG

```
#define NSIG (1024)
```

Definition at line 53 of file [portsignals.h](#).

#### 4.110.2.3 SIGSEGV

```
#define SIGSEGV (NSIG+1)
```

Definition at line 57 of file [portsignals.h](#).

#### 4.110.2.4 SIGFPE

```
#define SIGFPE (NSIG+2)
```

Definition at line 60 of file [portsignals.h](#).

#### 4.110.2.5 SIGILL

```
#define SIGILL (NSIG+3)
```

Definition at line 63 of file [portsignals.h](#).

#### 4.110.2.6 SIGEMT

```
#define SIGEMT (NSIG+4)
```

Definition at line 66 of file [portsignals.h](#).

#### 4.110.2.7 SIGSYS

```
#define SIGSYS (NSIG+5)
```

Definition at line 69 of file [portsignals.h](#).

#### 4.110.2.8 SIGPIPE

```
#define SIGPIPE (NSIG+6)
```

Definition at line 72 of file [portsignals.h](#).

#### 4.110.2.9 SIGLOST

```
#define SIGLOST (NSIG+7)
```

Definition at line 75 of file [portsignals.h](#).

#### 4.110.2.10 SIGXCPU

```
#define SIGXCPU (NSIG+8)
```

Definition at line 78 of file [portsignals.h](#).

#### 4.110.2.11 SIGXFSZ

```
#define SIGXFSZ (NSIG+9)
```

Definition at line 81 of file [portsignals.h](#).



#### 4.110.2.12 SIGTERM

```
#define SIGTERM (NSIG+10)
```

Definition at line 84 of file [portsignals.h](#).

#### 4.110.2.13 SIGINT

```
#define SIGINT (NSIG+11)
```

Definition at line 87 of file [portsignals.h](#).

#### 4.110.2.14 SIGQUIT

```
#define SIGQUIT (NSIG+12)
```

Definition at line 90 of file [portsignals.h](#).

#### 4.110.2.15 SIGHUP

```
#define SIGHUP (NSIG+13)
```

Definition at line 93 of file [portsignals.h](#).

#### 4.110.2.16 SIGALRM

```
#define SIGALRM (NSIG+14)
```

Definition at line 96 of file [portsignals.h](#).

#### 4.110.2.17 SIGVTALRM

```
#define SIGVTALRM (NSIG+15)
```

Definition at line 99 of file [portsignals.h](#).

#### 4.110.2.18 SIGPROF

```
#define SIGPROF (NSIG+16)
```

Definition at line 102 of file [portsignals.h](#).

## 4.111 portsignals.h

[Go to the documentation of this file.](#)

```

00001 #ifndef PORTSIGNAL_H
00002 #define PORTSIGNAL_H
00003
00018 /* #[] License : */
00019 /*
00020  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00021  * When using this file you are requested to refer to the publication
00022  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00023  * This is considered a matter of courtesy as the development was paid
00024  * for by FOM the Dutch physics granting agency and we would like to
00025  * be able to track its scientific use to convince FOM of its value
00026  * for the community.
00027  *
00028  * This file is part of FORM.
00029  *
00030  * FORM is free software: you can redistribute it and/or modify it under the
00031  * terms of the GNU General Public License as published by the Free Software
00032  * Foundation, either version 3 of the License, or (at your option) any later
00033  * version.
00034  *
00035  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00036  * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00037  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00038  * details.
00039  *
00040  * You should have received a copy of the GNU General Public License along
00041  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00042  */
00043 /* #[] License : */
00044
00045 #include <signal.h>
00046
00047 #define FATAL_SIG_ERROR 4
00048
00049 #ifndef NSIG
00050 /*
00051  * The value of NSIG must be enough to fall outside the range of defined signals
00052  */
00053 #define NSIG (1024)
00054 #endif
00055
00056 #ifndef SIGSEGV
00057 #define SIGSEGV (NSIG+1)
00058 #endif
00059 #ifndef SIGFPE
00060 #define SIGFPE (NSIG+2)
00061 #endif
00062 #ifndef SIGILL
00063 #define SIGILL (NSIG+3)
00064 #endif
00065 #ifndef SIGEMT
00066 #define SIGEMT (NSIG+4)
00067 #endif
00068 #ifndef SIGSYS
00069 #define SIGSYS (NSIG+5)
00070 #endif
00071 #ifndef SIGPIPE
00072 #define SIGPIPE (NSIG+6)
00073 #endif
00074 #ifndef SIGLOST
00075 #define SIGLOST (NSIG+7)
00076 #endif
00077 #ifndef SIGXCPU
00078 #define SIGXCPU (NSIG+8)
00079 #endif
00080 #ifndef SIGXFSZ
00081 #define SIGXFSZ (NSIG+9)
00082 #endif
00083 #ifndef SIGTERM
00084 #define SIGTERM (NSIG+10)
00085 #endif
00086 #ifndef SIGINT
00087 #define SIGINT (NSIG+11)
00088 #endif
00089 #ifndef SIGQUIT
00090 #define SIGQUIT (NSIG+12)
00091 #endif
00092 #ifndef SIGHUP
00093 #define SIGHUP (NSIG+13)
00094 #endif
00095 #ifndef SIGALRM
00096 #define SIGALRM (NSIG+14)

```

```

00097 #endif
00098 #ifndef SIGVTALRM
00099 #define SIGVTALRM (NSIG+15)
00100 #endif
00101 #ifndef SIGPROF
00102 #define SIGPROF (NSIG+16)
00103 #endif
00104
00105 #endif

```

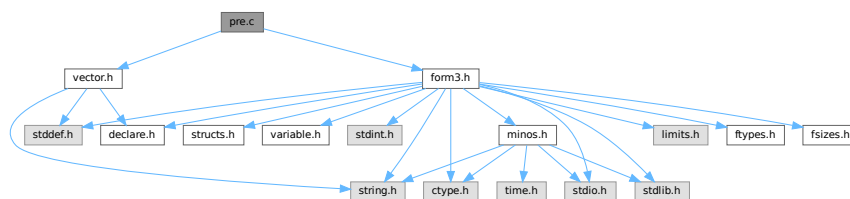
## 4.112 pre.c File Reference

```

#include "form3.h"
#include "vector.h"

```

Include dependency graph for pre.c:



### Macros

- #define [STRINGIFY](#)(x) [STRINGIFY\\_\\_](#)(x)
- #define [STRINGIFY\\_\\_](#)(x) #x
- #define [SKIPBUFSIZE](#) 20
- #define [KILL](#) "kill"
- #define [KILLALL](#) "killall"
- #define [DAEMON](#) "daemon"
- #define [SHELL](#) "shell"
- #define [STDERR](#) "stderr"
- #define [TRUE\\_EXPR](#) "true"
- #define [FALSE\\_EXPR](#) "false"
- #define [NOSHELL](#) "noshell"
- #define [TERMINAL](#) "terminal"

### Functions

- UBYTE [GetInput](#) (void)
- void [ClearPushback](#) (void)
- UBYTE [GetChar](#) (int level)
- void [CharOut](#) (UBYTE c)
- void [UnsetAllowDelay](#) (void)
- UBYTE \* [GetPreVar](#) (UBYTE \*name, int flag)
- int [PutPreVar](#) (UBYTE \*name, UBYTE \*value, UBYTE \*args, int mode)
- void [PopPreVars](#) (int tonumber)
- void [IniModule](#) (int type)
- void [IniSpecialModule](#) (int type)
- void [PreProcessor](#) (void)

- int [PreProInstruction](#) (void)
- int [LoadInstruction](#) (int mode)
- int [LoadStatement](#) (int type)
- int [ExpandTripleDots](#) (int par)
- [KEYWORD](#) \* [FindKeyWord](#) (UBYTE \*theword, [KEYWORD](#) \*table, int size)
- [KEYWORD](#) \* [FindInKeyWord](#) (UBYTE \*theword, [KEYWORD](#) \*table, int size)
- int [TheDefine](#) (UBYTE \*s, int mode)
- int [DoCommentChar](#) (UBYTE \*s)
- int [DoPreAssign](#) (UBYTE \*s)
- int [DoDefine](#) (UBYTE \*s)
- int [DoRedefine](#) (UBYTE \*s)
- int [ClearMacro](#) (UBYTE \*name)
- int [TheUndefine](#) (UBYTE \*name)
- int [DoUndefine](#) (UBYTE \*s)
- int [DoInclude](#) (UBYTE \*s)
- int [DoReverseInclude](#) (UBYTE \*s)
- int [Include](#) (UBYTE \*s, int type)
- int [DoPreExchange](#) (UBYTE \*s)
- int [DoCall](#) (UBYTE \*s)
- int [DoDebug](#) (UBYTE \*s)
- int [DoTerminate](#) (UBYTE \*s)
- int [DoContinueDo](#) (UBYTE \*s)
- int [DoDo](#) (UBYTE \*s)
- int [DoBreakDo](#) (UBYTE \*s)
- int [DoElse](#) (UBYTE \*s)
- int [DoElseif](#) (UBYTE \*s)
- int [DoEnddo](#) (UBYTE \*s)
- int [DoEndif](#) (UBYTE \*s)
- int [DoEndprocedure](#) (UBYTE \*s)
- int [Dolf](#) (UBYTE \*s)
- int [Dolfdef](#) (UBYTE \*s, int par)
- int [Dolfydef](#) (UBYTE \*s)
- int [Dolfndef](#) (UBYTE \*s)
- int [DoInside](#) (UBYTE \*s)
- int [DoEndInside](#) (UBYTE \*s)
- int [DoMessage](#) (UBYTE \*s)
- int [DoPipe](#) (UBYTE \*s)
- int [DoProcExtension](#) (UBYTE \*s)
- int [DoPreOut](#) (UBYTE \*s)
- int [DoPrePrintTimes](#) (UBYTE \*s)
- int [DoPreSortReallocate](#) (UBYTE \*s)
- int [DoPreAppend](#) (UBYTE \*s)
- int [DoPreCreate](#) (UBYTE \*s)
- int [DoPreRemove](#) (UBYTE \*s)
- int [DoPreClose](#) (UBYTE \*s)
- int [DoPreWrite](#) (UBYTE \*s)
- int [DoProcedure](#) (UBYTE \*s)
- int [DoPreBreak](#) (UBYTE \*s)
- int [DoPreCase](#) (UBYTE \*s)
- int [DoPreDefault](#) (UBYTE \*s)
- int [DoPreEndSwitch](#) (UBYTE \*s)
- int [DoPreSwitch](#) (UBYTE \*s)
- int [DoPreShow](#) (UBYTE \*s)
- int [DoSystem](#) (UBYTE \*s)
- int [PreLoad](#) ([PRELOAD](#) \*p, UBYTE \*start, UBYTE \*stop, int mode, char \*message)

- int [PreSkip](#) (UBYTE \*start, UBYTE \*stop, int mode)
- void [StartPrepro](#) (void)
- int [EvalPrelf](#) (UBYTE \*s)
- UBYTE \* [PrelfEval](#) (UBYTE \*s, int \*value)
- int [PreCmp](#) (int type, int val, UBYTE \*t, int type2, int val2, UBYTE \*t2, int cmpop)
- int [PreEq](#) (int type, int val, UBYTE \*t, int type2, int val2, UBYTE \*t2, int eqop)
- UBYTE \* [pParseObject](#) (UBYTE \*s, int \*type, LONG \*val2)
- UBYTE \* [PreCalc](#) (void)
- UBYTE \* [PreEval](#) (UBYTE \*s, LONG \*x)
- void [AddToPreTypes](#) (int type)
- void [MessPreNesting](#) (int par)
- int [DoPreAddSeparator](#) (UBYTE \*s)
- int [DoPreRmSeparator](#) (UBYTE \*s)
- int [DoExternal](#) (UBYTE \*s)
- int [DoPrompt](#) (UBYTE \*s)
- int [DoSetExternal](#) (UBYTE \*s)
- int [DoSetExternalAttr](#) (UBYTE \*s)
- int [DoRmExternal](#) (UBYTE \*s)
- int [DoFromExternal](#) (UBYTE \*s)
- int [DoToExternal](#) (UBYTE \*s)
- UBYTE \* [defineChannel](#) (UBYTE \*s, [HANDLERS](#) \*h)
- int [writeToChannel](#) (int wtype, UBYTE \*s, [HANDLERS](#) \*h)
- int [DoFactDollar](#) (UBYTE \*s)
- WORD [GetDollarNumber](#) (UBYTE \*\*inp, [DOLLARS](#) d)
- int [DoSetRandom](#) (UBYTE \*s)
- int [DoOptimize](#) (UBYTE \*s)
- int [DoClearOptimize](#) (UBYTE \*s)
- int [DoSkipExtraSymbols](#) (UBYTE \*s)
- int [DoPreReset](#) (UBYTE \*s)
- int [DoPreAppendPath](#) (UBYTE \*s)
- int [DoPrePrependPath](#) (UBYTE \*s)
- int [DoTimeOutAfter](#) (UBYTE \*s)
- int [DoNamespace](#) (UBYTE \*s)
- int [DoEndNamespace](#) (UBYTE \*s)
- UBYTE \* [SkipName](#) (UBYTE \*s)
- UBYTE \* [ConstructName](#) (UBYTE \*s, UBYTE type)
- int [DoUse](#) (UBYTE \*s)
- int [UserFlags](#) (UBYTE \*s, int par)
- int [DoClearUserFlag](#) (UBYTE \*s)
- int [DoSetUserFlag](#) (UBYTE \*s)

### 4.112.1 Detailed Description

This is the preprocessor and all its routines.

Definition in file [pre.c](#).

### 4.112.2 Macro Definition Documentation

#### 4.112.2.1 STRINGIFY

```
#define STRINGIFY(  
    x ) STRINGIFY__(x)
```

Definition at line [4251](#) of file [pre.c](#).

#### 4.112.2.2 STRINGIFY\_\_

```
#define STRINGIFY__(  
    x ) #x
```

Definition at line [4252](#) of file [pre.c](#).

#### 4.112.2.3 SKIPBUFSIZE

```
#define SKIPBUFSIZE 20
```

Definition at line [4375](#) of file [pre.c](#).

#### 4.112.2.4 KILL

```
#define KILL "kill"
```

Definition at line [5595](#) of file [pre.c](#).

#### 4.112.2.5 KILLALL

```
#define KILLALL "killall"
```

Definition at line [5596](#) of file [pre.c](#).

#### 4.112.2.6 DAEMON

```
#define DAEMON "daemon"
```

Definition at line [5597](#) of file [pre.c](#).

#### 4.112.2.7 SHELL

```
#define SHELL "shell"
```

Definition at line [5598](#) of file [pre.c](#).

#### 4.112.2.8 STDERR

```
#define STDERR "stderr"
```

Definition at line [5599](#) of file [pre.c](#).

#### 4.112.2.9 TRUE\_EXPR

```
#define TRUE_EXPR "true"
```

Definition at line 5601 of file [pre.c](#).

#### 4.112.2.10 FALSE\_EXPR

```
#define FALSE_EXPR "false"
```

Definition at line 5602 of file [pre.c](#).

#### 4.112.2.11 NOSHELL

```
#define NOSHELL "noshell"
```

Definition at line 5603 of file [pre.c](#).

#### 4.112.2.12 TERMINAL

```
#define TERMINAL "terminal"
```

Definition at line 5604 of file [pre.c](#).

### 4.112.3 Function Documentation

#### 4.112.3.1 GetInput()

```
UBYTE GetInput (  
    void )
```

Definition at line 132 of file [pre.c](#).

#### 4.112.3.2 ClearPushback()

```
void ClearPushback (  
    void )
```

Definition at line 161 of file [pre.c](#).

#### 4.112.3.3 GetChar()

```
UBYTE GetChar (  
    int level )
```

Definition at line 179 of file [pre.c](#).

#### 4.112.3.4 CharOut()

```
void CharOut (
    UBYTE c )
```

Definition at line 489 of file [pre.c](#).

#### 4.112.3.5 UnsetAllowDelay()

```
void UnsetAllowDelay (
    void )
```

Definition at line 510 of file [pre.c](#).

#### 4.112.3.6 GetPreVar()

```
UBYTE * GetPreVar (
    UBYTE * name,
    int flag )
```

Definition at line 536 of file [pre.c](#).

#### 4.112.3.7 PutPreVar()

```
int PutPreVar (
    UBYTE * name,
    UBYTE * value,
    UBYTE * args,
    int mode )
```

Inserts/Updates a preprocessor variable in the name administration.

##### Parameters

|              |                                                                                                                                                                                                                    |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>name</i>  | Character string with the variable name.                                                                                                                                                                           |
| <i>value</i> | Character string with a possible value. Special case: if this argument is zero, then we have no value. Note: This is different from having an empty argument! This should only occur when the name starts with a ? |
| <i>args</i>  | Character string with possible arguments.                                                                                                                                                                          |
| <i>mode</i>  | =0: always create a new name entry, =1: try to do a redefinition if possible.                                                                                                                                      |

##### Returns

Index of used entry in name list.

Definition at line 718 of file [pre.c](#).

Referenced by [ClearOptimize\(\)](#), [Generator\(\)](#), [Optimize\(\)](#), [PF\\_BroadcastRedefinedPreVars\(\)](#), [StartVariables\(\)](#), and [TheDefine\(\)](#).



#### 4.112.3.8 PopPreVars()

```
void PopPreVars (
    int tonumber )
```

Definition at line 820 of file [pre.c](#).

#### 4.112.3.9 IniModule()

```
void IniModule (
    int type )
```

Definition at line 835 of file [pre.c](#).

#### 4.112.3.10 IniSpecialModule()

```
void IniSpecialModule (
    int type )
```

Definition at line 937 of file [pre.c](#).

#### 4.112.3.11 PreProcessor()

```
void PreProcessor (
    void )
```

Definition at line 947 of file [pre.c](#).

#### 4.112.3.12 PreProInstruction()

```
int PreProInstruction (
    void )
```

Definition at line 1166 of file [pre.c](#).

#### 4.112.3.13 LoadInstruction()

```
int LoadInstruction (
    int mode )
```

Definition at line 1254 of file [pre.c](#).

#### 4.112.3.14 LoadStatement()

```
int LoadStatement (
    int type )
```

Definition at line 1457 of file [pre.c](#).

#### 4.112.3.15 ExpandTripleDots()

```
int ExpandTripleDots (
    int par )
```

Definition at line 1595 of file [pre.c](#).

#### 4.112.3.16 FindKeyWord()

```
KEYWORD * FindKeyWord (
    UBYTE * theword,
    KEYWORD * table,
    int size )
```

Definition at line 1963 of file [pre.c](#).

#### 4.112.3.17 FindInKeyWord()

```
KEYWORD * FindInKeyWord (
    UBYTE * theword,
    KEYWORD * table,
    int size )
```

Definition at line 1994 of file [pre.c](#).

#### 4.112.3.18 TheDefine()

```
int TheDefine (
    UBYTE * s,
    int mode )
```

Preprocessor assignment. Possible arguments and values are treated and the new preprocessor variable is put into the name administration.

##### Parameters

|             |                                                                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>s</i>    | Pointer to the character string following the preprocessor command.                                                                                 |
| <i>mode</i> | Bitmask. 0-bit clear: always create a new name entry, 0-bit set: try to redefine an existing name, 1-bit set: ignore preprocessor if/switch status. |

##### Returns

zero: no errors, negative number: errors.

Definition at line 2024 of file [pre.c](#).

References [PutPreVar\(\)](#).

Here is the call graph for this function:



#### 4.112.3.19 DoCommentChar()

```
int DoCommentChar (  
    UBYTE * s )
```

Definition at line 2097 of file [pre.c](#).

#### 4.112.3.20 DoPreAssign()

```
int DoPreAssign (  
    UBYTE * s )
```

Definition at line 2128 of file [pre.c](#).

#### 4.112.3.21 DoDefine()

```
int DoDefine (  
    UBYTE * s )
```

Definition at line 2159 of file [pre.c](#).

#### 4.112.3.22 DoRedefine()

```
int DoRedefine (  
    UBYTE * s )
```

Definition at line 2169 of file [pre.c](#).

#### 4.112.3.23 ClearMacro()

```
int ClearMacro (  
    UBYTE * name )
```

Definition at line 2181 of file [pre.c](#).

#### 4.112.3.24 TheUndefine()

```
int TheUndefine (
    UBYTE * name )
```

Definition at line 2209 of file [pre.c](#).

#### 4.112.3.25 DoUndefine()

```
int DoUndefine (
    UBYTE * s )
```

Definition at line 2263 of file [pre.c](#).

#### 4.112.3.26 DoInclude()

```
int DoInclude (
    UBYTE * s )
```

Definition at line 2315 of file [pre.c](#).

#### 4.112.3.27 DoReverseInclude()

```
int DoReverseInclude (
    UBYTE * s )
```

Definition at line 2322 of file [pre.c](#).

#### 4.112.3.28 Include()

```
int Include (
    UBYTE * s,
    int type )
```

Definition at line 2329 of file [pre.c](#).

#### 4.112.3.29 DoPreExchange()

```
int DoPreExchange (
    UBYTE * s )
```

Definition at line 2490 of file [pre.c](#).

#### 4.112.3.30 DoCall()

```
int DoCall (
    UBYTE * s )
```

Definition at line 2551 of file [pre.c](#).

#### 4.112.3.31 DoDebug()

```
int DoDebug (
    UBYTE * s )
```

Definition at line 2720 of file [pre.c](#).

#### 4.112.3.32 DoTerminate()

```
int DoTerminate (
    UBYTE * s )
```

Definition at line 2757 of file [pre.c](#).

#### 4.112.3.33 DoContinueDo()

```
int DoContinueDo (
    UBYTE * s )
```

Definition at line 2780 of file [pre.c](#).

#### 4.112.3.34 DoDo()

```
int DoDo (
    UBYTE * s )
```

Definition at line 2811 of file [pre.c](#).

#### 4.112.3.35 DoBreakDo()

```
int DoBreakDo (
    UBYTE * s )
```

Definition at line 3026 of file [pre.c](#).

#### 4.112.3.36 DoElse()

```
int DoElse (
    UBYTE * s )
```

Definition at line 3095 of file [pre.c](#).

#### 4.112.3.37 DoElseif()

```
int DoElseif (
    UBYTE * s )
```

Definition at line 3132 of file [pre.c](#).

#### 4.112.3.38 DoEnddo()

```
int DoEnddo (
    UBYTE * s )
```

Definition at line 3167 of file [pre.c](#).

#### 4.112.3.39 DoEndif()

```
int DoEndif (
    UBYTE * s )
```

Definition at line 3313 of file [pre.c](#).

#### 4.112.3.40 DoEndprocedure()

```
int DoEndprocedure (
    UBYTE * s )
```

Definition at line 3341 of file [pre.c](#).

#### 4.112.3.41 Dolf()

```
int DoIf (
    UBYTE * s )
```

Definition at line 3367 of file [pre.c](#).

#### 4.112.3.42 Dolfdef()

```
int DoIfdef (
    UBYTE * s,
    int par )
```

Definition at line 3391 of file [pre.c](#).

#### 4.112.3.43 Dolfydef()

```
int DoIfydef (
    UBYTE * s )
```

Definition at line 3416 of file [pre.c](#).

#### 4.112.3.44 Dolfndef()

```
int DoIfndef (
    UBYTE * s )
```

Definition at line 3426 of file [pre.c](#).

**4.112.3.45 DoInside()**

```
int DoInside (
    UBYTE * s )
```

Definition at line [3451](#) of file [pre.c](#).

**4.112.3.46 DoEndInside()**

```
int DoEndInside (
    UBYTE * s )
```

Definition at line [3544](#) of file [pre.c](#).

**4.112.3.47 DoMessage()**

```
int DoMessage (
    UBYTE * s )
```

Definition at line [3676](#) of file [pre.c](#).

**4.112.3.48 DoPipe()**

```
int DoPipe (
    UBYTE * s )
```

Definition at line [3690](#) of file [pre.c](#).

**4.112.3.49 DoPrcExtension()**

```
int DoPrcExtension (
    UBYTE * s )
```

Definition at line [3713](#) of file [pre.c](#).

**4.112.3.50 DoPreOut()**

```
int DoPreOut (
    UBYTE * s )
```

Definition at line [3747](#) of file [pre.c](#).

**4.112.3.51 DoPrePrintTimes()**

```
int DoPrePrintTimes (
    UBYTE * s )
```

Definition at line [3770](#) of file [pre.c](#).

#### 4.112.3.52 DoPreSortReallocate()

```
int DoPreSortReallocate (
    UBYTE * s )
```

Definition at line [3784](#) of file [pre.c](#).

#### 4.112.3.53 DoPreAppend()

```
int DoPreAppend (
    UBYTE * s )
```

Definition at line [3804](#) of file [pre.c](#).

#### 4.112.3.54 DoPreCreate()

```
int DoPreCreate (
    UBYTE * s )
```

Definition at line [3847](#) of file [pre.c](#).

#### 4.112.3.55 DoPreRemove()

```
int DoPreRemove (
    UBYTE * s )
```

Definition at line [3887](#) of file [pre.c](#).

#### 4.112.3.56 DoPreClose()

```
int DoPreClose (
    UBYTE * s )
```

Definition at line [3920](#) of file [pre.c](#).

#### 4.112.3.57 DoPreWrite()

```
int DoPreWrite (
    UBYTE * s )
```

Definition at line [3964](#) of file [pre.c](#).

#### 4.112.3.58 DoProcedure()

```
int DoProcedure (
    UBYTE * s )
```

Definition at line [4005](#) of file [pre.c](#).



**4.112.3.59 DoPreBreak()**

```
int DoPreBreak (
    UBYTE * s )
```

Definition at line [4047](#) of file [pre.c](#).

**4.112.3.60 DoPreCase()**

```
int DoPreCase (
    UBYTE * s )
```

Definition at line [4071](#) of file [pre.c](#).

**4.112.3.61 DoPreDefault()**

```
int DoPreDefault (
    UBYTE * s )
```

Definition at line [4110](#) of file [pre.c](#).

**4.112.3.62 DoPreEndSwitch()**

```
int DoPreEndSwitch (
    UBYTE * s )
```

Definition at line [4134](#) of file [pre.c](#).

**4.112.3.63 DoPreSwitch()**

```
int DoPreSwitch (
    UBYTE * s )
```

Definition at line [4161](#) of file [pre.c](#).

**4.112.3.64 DoPreShow()**

```
int DoPreShow (
    UBYTE * s )
```

Definition at line [4215](#) of file [pre.c](#).

**4.112.3.65 DoSystem()**

```
int DoSystem (
    UBYTE * s )
```

Definition at line [4254](#) of file [pre.c](#).

#### 4.112.3.66 PreLoad()

```
int PreLoad (
    PRELOAD * p,
    UBYTE * start,
    UBYTE * stop,
    int mode,
    char * message )
```

Definition at line 4294 of file [pre.c](#).

#### 4.112.3.67 PreSkip()

```
int PreSkip (
    UBYTE * start,
    UBYTE * stop,
    int mode )
```

Definition at line 4377 of file [pre.c](#).

#### 4.112.3.68 StartPrepro()

```
void StartPrepro (
    void )
```

Definition at line 4432 of file [pre.c](#).

#### 4.112.3.69 EvalPrelf()

```
int EvalPreIf (
    UBYTE * s )
```

Definition at line 4460 of file [pre.c](#).

#### 4.112.3.70 PrelfEval()

```
UBYTE * PreIfEval (
    UBYTE * s,
    int * value )
```

Definition at line 4495 of file [pre.c](#).

#### 4.112.3.71 PreCmp()

```
int PreCmp (
    int type,
    int val,
    UBYTE * t,
    int type2,
    int val2,
    UBYTE * t2,
    int cmpop )
```

Definition at line 4624 of file [pre.c](#).

#### 4.112.3.72 PreEq()

```
int PreEq (
    int type,
    int val,
    UBYTE * t,
    int type2,
    int val2,
    UBYTE * t2,
    int eqop )
```

Definition at line 4648 of file [pre.c](#).

#### 4.112.3.73 pParseObject()

```
UBYTE * pParseObject (
    UBYTE * s,
    int * type,
    LONG * val2 )
```

Definition at line 4677 of file [pre.c](#).

#### 4.112.3.74 PreCalc()

```
UBYTE * PreCalc (
    void )
```

Definition at line 5113 of file [pre.c](#).

#### 4.112.3.75 PreEval()

```
UBYTE * PreEval (
    UBYTE * s,
    LONG * x )
```

Definition at line 5191 of file [pre.c](#).

#### 4.112.3.76 AddToPreTypes()

```
void AddToPreTypes (
    int type )
```

Definition at line 5341 of file [pre.c](#).

#### 4.112.3.77 MessPreNesting()

```
void MessPreNesting (
    int par )
```

Definition at line 5359 of file [pre.c](#).

#### 4.112.3.78 DoPreAddSeparator()

```
int DoPreAddSeparator (
    UBYTE * s )
```

Definition at line [5382](#) of file [pre.c](#).

#### 4.112.3.79 DoPreRmSeparator()

```
int DoPreRmSeparator (
    UBYTE * s )
```

Definition at line [5407](#) of file [pre.c](#).

#### 4.112.3.80 DoExternal()

```
int DoExternal (
    UBYTE * s )
```

Definition at line [5424](#) of file [pre.c](#).

#### 4.112.3.81 DoPrompt()

```
int DoPrompt (
    UBYTE * s )
```

Definition at line [5506](#) of file [pre.c](#).

#### 4.112.3.82 DoSetExternal()

```
int DoSetExternal (
    UBYTE * s )
```

Definition at line [5539](#) of file [pre.c](#).

#### 4.112.3.83 DoSetExternalAttr()

```
int DoSetExternalAttr (
    UBYTE * s )
```

Definition at line [5609](#) of file [pre.c](#).

#### 4.112.3.84 DoRmExternal()

```
int DoRmExternal (
    UBYTE * s )
```

Definition at line [5726](#) of file [pre.c](#).

**4.112.3.85 DoFromExternal()**

```
int DoFromExternal (
    UBYTE * s )
```

Definition at line [5791](#) of file [pre.c](#).

**4.112.3.86 DoToExternal()**

```
int DoToExternal (
    UBYTE * s )
```

Definition at line [5986](#) of file [pre.c](#).

**4.112.3.87 defineChannel()**

```
UBYTE * defineChannel (
    UBYTE * s,
    HANDLERS * h )
```

Definition at line [6033](#) of file [pre.c](#).

**4.112.3.88 writeToChannel()**

```
int writeToChannel (
    int wtype,
    UBYTE * s,
    HANDLERS * h )
```

Definition at line [6068](#) of file [pre.c](#).

**4.112.3.89 DoFactDollar()**

```
int DoFactDollar (
    UBYTE * s )
```

Definition at line [6633](#) of file [pre.c](#).

**4.112.3.90 GetDollarNumber()**

```
WORD GetDollarNumber (
    UBYTE ** inp,
    DOLLARS d )
```

Definition at line [6670](#) of file [pre.c](#).

#### 4.112.3.91 DoSetRandom()

```
int DoSetRandom (
    UBYTE * s )
```

Definition at line 6760 of file [pre.c](#).

#### 4.112.3.92 DoOptimize()

```
int DoOptimize (
    UBYTE * s )
```

Definition at line 6803 of file [pre.c](#).

#### 4.112.3.93 DoClearOptimize()

```
int DoClearOptimize (
    UBYTE * s )
```

Definition at line 6936 of file [pre.c](#).

#### 4.112.3.94 DoSkipExtraSymbols()

```
int DoSkipExtraSymbols (
    UBYTE * s )
```

Definition at line 6956 of file [pre.c](#).

#### 4.112.3.95 DoPreReset()

```
int DoPreReset (
    UBYTE * s )
```

Definition at line 6995 of file [pre.c](#).

#### 4.112.3.96 DoPreAppendPath()

```
int DoPreAppendPath (
    UBYTE * s )
```

Appends the given path (absolute or relative to the current file directory) to the FORM path.

Syntax: #appendpath <path>

Definition at line 7153 of file [pre.c](#).

**4.112.3.97 DoPrePrependPath()**

```
int DoPrePrependPath (
    UBYTE * s )
```

Prepends the given path (absolute or relative to the current file directory) to the FORM path.

Syntax: #prependpath <path>

Definition at line 7170 of file [pre.c](#).

**4.112.3.98 DoTimeOutAfter()**

```
int DoTimeOutAfter (
    UBYTE * s )
```

Definition at line 7182 of file [pre.c](#).

**4.112.3.99 DoNamespace()**

```
int DoNamespace (
    UBYTE * s )
```

Definition at line 7234 of file [pre.c](#).

**4.112.3.100 DoEndNamespace()**

```
int DoEndNamespace (
    UBYTE * s )
```

Definition at line 7282 of file [pre.c](#).

**4.112.3.101 SkipName()**

```
UBYTE * SkipName (
    UBYTE * s )
```

Definition at line 7305 of file [pre.c](#).

**4.112.3.102 ConstructName()**

```
UBYTE * ConstructName (
    UBYTE * s,
    UBYTE type )
```

Definition at line 7405 of file [pre.c](#).

#### 4.112.3.103 DoUse()

```
int DoUse (
    UBYTE * s )
```

Definition at line 7497 of file [pre.c](#).

#### 4.112.3.104 UserFlags()

```
int UserFlags (
    UBYTE * s,
    int par )
```

Definition at line 7545 of file [pre.c](#).

#### 4.112.3.105 DoClearUserFlag()

```
int DoClearUserFlag (
    UBYTE * s )
```

Definition at line 7652 of file [pre.c](#).

#### 4.112.3.106 DoSetUserFlag()

```
int DoSetUserFlag (
    UBYTE * s )
```

Definition at line 7662 of file [pre.c](#).

### 4.113 pre.c

[Go to the documentation of this file.](#)

```
00001
00005 /* #[] License : */
00006 /*
00007  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00008  * When using this file you are requested to refer to the publication
00009  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010  * This is considered a matter of courtesy as the development was paid
00011  * for by FOM the Dutch physics granting agency and we would like to
00012  * be able to track its scientific use to convince FOM of its value
00013  * for the community.
00014  *
00015  * This file is part of FORM.
00016  *
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00025  * details.
00026  *
00027  * You should have received a copy of the GNU General Public License along
00028  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00029 */
00030 /* #[] License : */
```



```

00031 /*
00032     #[ Includes :
00033 */
00034 #include "form3.h"
00035
00036 static UBYTE pushbackchar = 0;
00037 static int oldmode = 0;
00038 static int stopdelay = 0;
00039 static STREAM *oldstream = 0;
00040 static UBYTE underscore[2] = {'_', 0};
00041 static PREVAR *ThePreVar = 0;
00042
00043 static KEYWORD precommands[] = {
00044     {"add" , DoPreAdd , 0, 0}
00045     , {"addseparator" , DoPreAddSeparator, 0, 0}
00046     , {"append" , DoPreAppend , 0, 0}
00047     , {"appendpath" , DoPreAppendPath, 0, 0}
00048     , {"assign" , DoPreAssign , 0, 0}
00049     , {"break" , DoPreBreak , 0, 0}
00050     , {"breakdo" , DoBreakDo , 0, 0}
00051     , {"call" , DoCall , 0, 0}
00052     , {"case" , DoPreCase , 0, 0}
00053     , {"clearflag" , DoClearUserFlag, 0, 0}
00054     , {"clearoptimize" , DoClearOptimize, 0, 0}
00055     , {"close" , DoPreClose , 0, 0}
00056     , {"closedictionary" , DoPreCloseDictionary, 0, 0}
00057     , {"commentchar" , DoCommentChar , 0, 0}
00058     , {"continuedo" , DoContinueDo , 0, 0}
00059     , {"create" , DoPreCreate , 0, 0}
00060     , {"debug" , DoDebug , 0, 0}
00061     , {"default" , DoPreDefault , 0, 0}
00062     , {"define" , DoDefine , 0, 0}
00063     , {"do" , DoDo , 0, 0}
00064     , {"else" , DoElse , 0, 0}
00065     , {"elseif" , DoElseif , 0, 0}
00066     , {"enddo" , DoEnddo , 0, 0}
00067 #ifdef WITHFLOAT
00068     , {"endfloat" , DoEndFloat , 0, 0}
00069 #endif
00070     , {"endif" , DoEndif , 0, 0}
00071     , {"endinside" , DoEndInside , 0, 0}
00072     , {"endnamespace" , DoEndNamespace , 0, 0}
00073     , {"endprocedure" , DoEndprocedure , 0, 0}
00074     , {"endswitch" , DoPreEndSwitch , 0, 0}
00075     , {"exchange" , DoPreExchange , 0, 0}
00076     , {"external" , DoExternal , 0, 0}
00077     , {"factdollar" , DoFactDollar , 0, 0}
00078     , {"fromexternal" , DoFromExternal , 0, 0}
00079     , {"if" , DoIf , 0, 0}
00080     , {"ifdef" , DoIfydef , 0, 0}
00081     , {"ifndef" , DoIfndef , 0, 0}
00082     , {"include" , DoInclude , 0, 0}
00083     , {"inside" , DoInside , 0, 0}
00084     , {"message" , DoMessage , 0, 0}
00085     , {"namespace" , DoNamespace , 0, 0}
00086     , {"opendictionary" , DoPreOpenDictionary, 0, 0}
00087     , {"optimize" , DoOptimize , 0, 0}
00088     , {"pipe" , DoPipe , 0, 0}
00089     , {"preout" , DoPreOut , 0, 0}
00090     , {"prependpath" , DoPrePrependPath, 0, 0}
00091     , {"printtimes" , DoPrePrintTimes, 0, 0}
00092     , {"procedure" , DoProcedure , 0, 0}
00093     , {"procedureextension" , DoPrcExtension , 0, 0}
00094     , {"prompt" , DoPrompt , 0, 0}
00095     , {"redefine" , DoRedefine , 0, 0}
00096     , {"remove" , DoPreRemove , 0, 0}
00097     , {"reset" , DoPreReset , 0, 0}
00098     , {"reverseinclude" , DoReverseInclude , 0, 0}
00099     , {"rmexternal" , DoRmExternal , 0, 0}
00100     , {"rmseparator" , DoPreRmSeparator, 0, 0}
00101     , {"setexternal" , DoSetExternal , 0, 0}
00102     , {"setexternalattr" , DoSetExternalAttr , 0, 0}
00103     , {"setflag" , DoSetUserFlag , 0, 0}
00104     , {"setrandom" , DoSetRandom , 0, 0}
00105     , {"show" , DoPreShow , 0, 0}
00106     , {"skipextrasymbols" , DoSkipExtraSymbols , 0, 0}
00107     , {"sortreallocate" , DoPreSortReallocate , 0, 0}
00108 #ifdef WITHFLOAT
00109     , {"startfloat" , DoStartFloat , 0, 0}
00110 #endif
00111     , {"switch" , DoPreSwitch , 0, 0}
00112     , {"system" , DoSystem , 0, 0}
00113     , {"terminate" , DoTerminate , 0, 0}
00114     , {"timeoutafter" , DoTimeOutAfter , 0, 0}
00115     , {"toexternal" , DoToExternal , 0, 0}
00116     , {"undefine" , DoUndefine , 0, 0}
00117     , {"use" , DoUse , 0, 0}

```

```

00118     ,{"usedictionary", DoPreUseDictionary,0,0}
00119     ,{"write"         , DoPreWrite      , 0, 0}
00120 };
00121
00122 /*
00123  #] Includes :
00124  # [ PreProcessor :
00125  # [ GetInput :
00126
00127      Gets one input character. If we reach the end of a stream
00128      we pop to the previous stream and try again.
00129      If there are no more streams we let this be known.
00130 */
00131
00132 UBYTE GetInput(void)
00133 {
00134     UBYTE c;
00135     while ( AC.CurrentStream ) {
00136         c = GetFromStream(AC.CurrentStream);
00137         if ( c != ENDOFSTREAM ) {
00138 #ifdef WITHMPI
00139             if ( PF.me == MASTER
00140                 && AC.NoShowInput <= 0
00141                 && AC.CurrentStream->type != PREVARSTREAM )
00142 #else
00143             if ( AC.NoShowInput <= 0 && AC.CurrentStream->type != PREVARSTREAM )
00144 #endif
00145                 CharOut(c);
00146             return(c);
00147         }
00148         AC.CurrentStream = CloseStream(AC.CurrentStream);
00149         if ( stopdelay && AC.CurrentStream == oldstream ) {
00150             stopdelay = 0; AP.AllowDelay = 1;
00151         }
00152     }
00153     return(ENDOFINPUT);
00154 }
00155
00156 /*
00157  #] GetInput :
00158  # [ ClearPushback :
00159 */
00160
00161 void ClearPushback(void)
00162 {
00163     pushbackchar = 0;
00164 }
00165
00166 /*
00167  #] ClearPushback :
00168  # [ GetChar :
00169
00170      Reads one character. If it encounters a quote it immediately
00171      takes the whole preprocessor variable and opens a stream
00172      for it and starts reading the stream.
00173      Note that we have to take special precautions for escaped quotes.
00174      That is why we remember the previous character. We allow the
00175      (dubious?) construction of ending a stream with a backslash and
00176      then using it to escape an object in the parent stream.
00177 */
00178
00179 UBYTE GetChar(int level)
00180 {
00181     UBYTE namebuf[MAXPRENAMESIZE+2], c, *s, *t;
00182     static UBYTE lastchar, charinbuf = 0;
00183     int i, j, raiselow, olddelay;
00184     STREAM *stream;
00185     if ( level > 0 ) {
00186         lastchar = '';
00187         goto higherlevel;
00188     }
00189     if ( pushbackchar ) { c = pushbackchar; pushbackchar = 0; return(c); }
00190     if ( charinbuf ) { c = charinbuf; charinbuf = 0; return(c); }
00191     c = GetInput();
00192     for(;;) {
00193         if ( c == '\\\\' ) {
00194             charinbuf = GetInput();
00195             if ( charinbuf != LINEFEED ) {
00196                 pushbackchar = charinbuf;
00197                 charinbuf = 0;
00198                 break;
00199             }
00200             charinbuf = 0; /* Escaped linefeed -> skip leading blanks */
00201             while ( ( c = GetInput() ) == ' ' || c == '\t' ) {}
00202         }
00203         else if ( c == '\"' || c == '\'' ) {
00204             if ( AP.DelayPrevar == 1 && c == '\"' ) {

```

```

00205         AP.DelayPrevar = 0;
00206         break;
00207     }
00208     lastchar = c;
00209 higherlevel:
00210     c = GetInput();
00211     if ( c == '!' && lastchar == ' ' ) {
00212         if ( stopdelay == 0 ) oldstream = AC.CurrentStream;
00213         AP.AllowDelay = 0;
00214         stopdelay = 1;
00215         c = GetInput();
00216     }
00217     if ( c == '~' && lastchar == ' ' ) {
00218         if ( AP.AllowDelay ) {
00219             pushbackchar = c;
00220             c = lastchar;
00221             AP.DelayPrevar = 1;
00222             break;
00223         }
00224     }
00225     else {
00226         pushbackchar = c;
00227     }
00228     olddelay = AP.DelayPrevar;
00229     AP.DelayPrevar = 0;
00230     i = 0; lastchar = 0;
00231     for (;;) {
00232         if ( pushbackchar ) { c = pushbackchar; pushbackchar = 0; }
00233         else { c = GetInput(); }
00234         if ( c == ENDOFINPUT || ( ( c == '\"' || c == LINEFEED )
00235             && lastchar != '\\') ) {
00236             break;
00237         }
00238         if ( c == '{' ) { /* Try the preprocessor calculator */
00239             if ( PreCalc() == 0 ) Terminate(-1);
00240             c = GetInput(); /* This is either a { or a number */
00241             if ( c == '{' ) {
00242                 MesPrint("@Illegal set inside preprocessor variable name");
00243                 Terminate(-1);
00244             }
00245         }
00246         if ( c == ' ' && lastchar != '\\') {
00247             c = GetChar(1);
00248             if ( c == ENDOFINPUT || ( ( c == '\"' || c == LINEFEED )
00249                 && lastchar != '\\') ) {
00250                 break;
00251             }
00252         }
00253         if ( lastchar == '\\') { i--; lastchar = 0; }
00254         else lastchar = c;
00255         namebuf[i++] = c;
00256         if ( i > MAXPRENAME_SIZE ) {
00257             namebuf[i] = 0;
00258             Error1("Preprocessor variable name too long: ",namebuf);
00259         }
00260     }
00261     namebuf[i++] = 0;
00262     if ( c != '\"' ) {
00263         Error1("Unmatched quotes for preprocessor variable",namebuf);
00264     }
00265     AP.DelayPrevar = olddelay;
00266     if ( namebuf[0] == '$' ) {
00267         raiselow = PRENOACTION;
00268         if ( AP.PreproFlag && *AP.preStart ) {
00269             s = EndOfToken(AP.preStart);
00270             c = *s; *s = 0;
00271             if ( ( StrICmp(AP.preStart,(UBYTE *) "ifdef") == 0
00272                 || StrICmp(AP.preStart,(UBYTE *) "ifndef") == 0 )
00273                 && GetDollar(namebuf+1) < 0 ) {
00274                 *s = c; c = ' ';
00275                 break;
00276             }
00277             *s = c;
00278         }
00279         else {
00280             s = EndOfToken(namebuf+1);
00281             if ( *s == '[' ) { while ( *s ) s++; }
00282         }
00283         if ( *s == '-' && s[1] == '-' && s[2] == 0 )
00284             raiselow = PRELOWERAFTER;
00285         else if ( *s == '+' && s[1] == '+' && s[2] == 0 )
00286             raiselow = PRERAISEAFTER;
00287         c = *s; *s = 0;
00288         if ( OpenStream(namebuf+1,DOLLARSTREAM,0,raiselow) == 0 ) {
00289             *s = c;
00290             MesPrint("@Undefined variable %s used as preprocessor variable",
00291                 namebuf);

```

```

00292         Terminate(-1);
00293     }
00294     *s = c;
00295 }
00296 else {
00297     raiselow = PRENOACTION;
00298     if ( AP.PreproFlag && *AP.preStart) {
00299         s = EndOfToken(AP.preStart);
00300         c = *s; *s = 0;
00301         if ( ( StrICmp(AP.preStart, (UBYTE *) "ifndef") == 0
00302             || StrICmp(AP.preStart, (UBYTE *) "ifndef") == 0 )
00303             && GetPreVar(namebuf, WITHOUTERROR) == 0 ) {
00304             *s = c; c = ' ';
00305             break;
00306         }
00307         *s = c;
00308     }
00309     s = EndOfToken(namebuf);
00310     if ( *s == '_' ) s++;
00311     if ( *s == '-' && s[1] == '-' && s[2] == 0 )
00312         raiselow = PRELOWERAFTER;
00313     else if ( *s == '+' && s[1] == '+' && s[2] == 0 )
00314         raiselow = PRERAISEAFTER;
00315     else if ( *s == '(' && namebuf[i-2] == ')' ) {
00316 /*
00317         Now count the arguments and separate them by zeroes
00318         Check on the ?var construction and if present, reset
00319         some comma's.
00320         Make the assignments of the variables
00321         Run the macro.
00322         Undefine the variables
00323 */
00324         int nargs = 1;
00325         PREVAR *p;
00326         size_t p_offset;
00327         *s++ = 0; namebuf[i-2] = 0;
00328         if ( StrICmp(namebuf, (UBYTE *) "random_") == 0 ) {
00329             UBYTE *ranvalue;
00330             ranvalue = PreRandom(s);
00331             PutPreVar(namebuf, ranvalue, (UBYTE *) "?a", 1);
00332             M_free(ranvalue, "PreRandom");
00333             goto dostream;
00334         }
00335         else if ( StrICmp(namebuf, (UBYTE *) "tolower_") == 0 ) {
00336             UBYTE *ss = s;
00337             while ( *ss ) { *ss = (UBYTE) (tolower(*ss)); ss++; }
00338             PutPreVar(namebuf, s, (UBYTE *) "?a", 1);
00339             goto dostream;
00340         }
00341         else if ( StrICmp(namebuf, (UBYTE *) "toupper_") == 0 ) {
00342             UBYTE *ss = s;
00343             while ( *ss ) { *ss = (UBYTE) (toupper(*ss)); ss++; }
00344             PutPreVar(namebuf, s, (UBYTE *) "?a", 1);
00345             goto dostream;
00346         }
00347         else if ( StrICmp(namebuf, (UBYTE *) "takeleft_") == 0 ) {
00348             UBYTE *ss = s;
00349             int x = 0, nsize;
00350             while ( *ss != ',' && *ss ) ss++;
00351             nsize = ss-s;
00352             if ( *ss ) {
00353                 *ss++ = 0;
00354                 while ( FG.cTable[*ss] == 1 ) x = 10*x + (*ss++ - '0');
00355                 if ( x > nsize ) x = nsize;
00356             }
00357             else x = 0;
00358             PutPreVar(namebuf, s+x, (UBYTE *) "?a", 1);
00359             goto dostream;
00360         }
00361         else if ( StrICmp(namebuf, (UBYTE *) "takeright_") == 0 ) {
00362             UBYTE *ss = s;
00363             int x = 0, nsize;
00364             while ( *ss != ',' && *ss ) ss++;
00365             nsize = ss-s;
00366             if ( *ss ) {
00367                 *ss++ = 0;
00368                 while ( FG.cTable[*ss] == 1 ) x = 10*x + (*ss++ - '0');
00369                 if ( x > nsize ) x = nsize;
00370             }
00371             else x = 0;
00372             x = nsize - x;
00373             s[x] = 0;
00374             PutPreVar(namebuf, s, (UBYTE *) "?a", 1);
00375             goto dostream;
00376         }
00377         else if ( StrICmp(namebuf, (UBYTE *) "keepleft_") == 0 ) {
00378             UBYTE *ss = s;

```

```

00379         int x = 0, nsize;
00380         while ( *ss != ',' && *ss ) ss++;
00381         nsize = ss-s;
00382         if ( *ss ) {
00383             *ss++ = 0;
00384             while ( FG.cTable[*ss] == 1 ) x = 10*x + (*ss++ - '0');
00385             if ( x > nsize ) x = nsize;
00386         }
00387         else x = nsize;
00388         s[x] = 0;
00389         PutPreVar(namebuf,s,(UBYTE *)"?a",1);
00390         goto dostream;
00391     }
00392     else if ( StrICmp(namebuf,(UBYTE *)"keepright_") == 0 ) {
00393         UBYTE *ss = s;
00394         int x = 0, nsize;
00395         while ( *ss != ',' && *ss ) ss++;
00396         nsize = ss-s;
00397         if ( *ss ) {
00398             *ss++ = 0;
00399             while ( FG.cTable[*ss] == 1 ) x = 10*x + (*ss++ - '0');
00400             if ( x > nsize ) x = nsize;
00401         }
00402         else x = nsize;
00403         x = nsize-x;
00404         PutPreVar(namebuf,s+x,(UBYTE *)"?a",1);
00405         goto dostream;
00406     }
00407     while ( *s ) {
00408         if ( *s == '\\\\' ) s++;
00409         if ( *s == ',' ) { *s = 0; nargs++; }
00410         s++;
00411     }
00412     GetPreVar(namebuf,WITHERROR);
00413     p = ThePreVar;
00414     if ( p == 0 ) {
00415         MesPrint("@Illegal use of arguments in preprocessor variable %s",namebuf);
00416         Terminate(-1);
00417     }
00418     if ( p->nargs <= 0 || ( p->wildarg == 0 && nargs != p->nargs )
00419         || ( p->wildarg > 0 && nargs < p->nargs-1 ) ) {
00420         MesPrint("@Arguments of macro %s do not match",namebuf);
00421         Terminate(-1);
00422     }
00423     if ( p->wildarg > 0 ) {
00424         /*
00425          * Change some zeroes into commas
00426          */
00427         s = namebuf;
00428         for ( j = 0; j < p->wildarg; j++ ) {
00429             while ( *s ) s++;
00430             s++;
00431         }
00432         for ( j = 0; j < nargs-p->nargs; j++ ) {
00433             while ( *s ) s++;
00434             *s++ = ',';
00435         }
00436     }
00437     /*
00438      * Now we can make the assignments
00439      */
00440     s = namebuf;
00441     while ( *s ) s++;
00442     s++;
00443     t = p->argnames;
00444     p_offset = p - PreVar;
00445     for ( j = 0; j < p->nargs; j++ ) {
00446         if ( ( nargs == p->nargs-1 ) && ( *t == '?' ) ) {
00447             PutPreVar(t,0,0,0);
00448         }
00449         else {
00450             PutPreVar(t,s,0,0);
00451             while ( *s ) s++;
00452             s++;
00453         }
00454         p = PreVar + p_offset;
00455         while ( *t ) t++;
00456         t++;
00457     }
00458 }
00459 dostream:;
00460 if ( ( stream = OpenStream(namebuf,PREVARSTREAM,0,raiselow) ) == 0 ) {
00461     /*
00462      * Eat comma before or after. This is `no value'
00463      */
00464 }
00465 else if ( stream->inbuffer == 0 ) {

```

```

00466             c = GetInput();
00467             if ( level > 0 && c == '\\' ) return(c);
00468             goto endofloop;
00469         }
00470     }
00471     c = GetInput();
00472 }
00473 else if ( c == '{' ) { /* Try the preprocessor calculator */
00474     if ( PreCalc() == 0 ) Terminate(-1);
00475     c = GetInput(); /* This is either a { or a number */
00476     break;
00477 }
00478 else break;
00479 endofloop;
00480 }
00481 return(c);
00482 }
00483
00484 /*
00485     #] GetChar :
00486     #[ CharOut :
00487 */
00488
00489 void CharOut(UBYTE c)
00490 {
00491     if ( c == LINEFEED ) {
00492         AM.OutBuffer[AP.InOutBuf++] = c;
00493         WriteString(INPUTOUT,AM.OutBuffer,AP.InOutBuf);
00494         AP.InOutBuf = 0;
00495     }
00496     else {
00497         if ( AP.InOutBuf >= AM.OutBufSize || c == LINEFEED ) {
00498             WriteString(INPUTOUT,AM.OutBuffer,AP.InOutBuf);
00499             AP.InOutBuf = 0;
00500         }
00501         AM.OutBuffer[AP.InOutBuf++] = c;
00502     }
00503 }
00504
00505 /*
00506     #] CharOut :
00507     #[ UnsetAllowDelay :
00508 */
00509
00510 void UnsetAllowDelay(void)
00511 {
00512     if ( ThePreVar != 0 ) {
00513         if ( ThePreVar->nargs > 0 ) AP.AllowDelay = 0;
00514     }
00515 }
00516
00517 /*
00518     #] UnsetAllowDelay :
00519     #[ GetPreVar :
00520
00521     We use the model of a heap. If the same name has been used more
00522     than once the last definition is used. This gives the impression
00523     of local variables.
00524
00525     There are two types: The regular ones and the expression variables.
00526     The last ones are like UNCHANGED_exprname and ZERO_exprname or
00527     UNCHANGED_ and ZERO_.
00528 */
00529
00530 static UBYTE *yes = (UBYTE *)"1";
00531 static UBYTE *no = (UBYTE *)"0";
00532 static UBYTE numintopolynomial[12];
00533 #include "vector.h"
00534 static Vector(UBYTE, exprstr); /* Used for numactiveexprs_ and activeexprnames_. */
00535
00536 UBYTE *GetPreVar(UBYTE *name, int flag)
00537 {
00538     GETIDENTITY
00539     int i, mode;
00540     WORD number;
00541     UBYTE *t, c = 0, *tt = 0;
00542     t = name; while ( *t ) t++;
00543     if ( t[-1] == '-' && t[-2] == '-' && t-2 > name && t[-3] != '_' ) {
00544         t -= 2; c = *t; *t = 0; tt = t;
00545     }
00546     else if ( t[-1] == '+' && t[-2] == '+' && t-2 > name && t[-3] != '_' ) {
00547         t -= 2; c = *t; *t = 0; tt = t;
00548     }
00549     else if ( StrICmp(name,(UBYTE *)"time_") == 0 ) {
00550         UBYTE millibuf[24];
00551         LONG millitime, timepart;
00552         int timepart1, timepart2;

```

```

00553     static char timestring[40];
00554 /*     millitime = TimeCPU(1); */
00555     millitime = GetRunningTime();
00556     timepart = millitime%1000;
00557     millitime /= 1000;
00558     timepart /= 10;
00559     timepart1 = timepart / 10;
00560     timepart2 = timepart % 10;
00561     NumToStr(millibuf,millitime);
00562     snprintf(timestring,40,"%s.%ld%ld",millibuf,timepart1,timepart2);
00563     return((UBYTE *)timestring);
00564 }
00565 else if ( ( StrICmp(name,(UBYTE *)"timer_") == 0 )
00566 || ( StrICmp(name,(UBYTE *)"stopwatch_") == 0 ) ) {
00567     static char timestring[40];
00568     snprintf(timestring,40,"%ld", (GetRunningTime() - AP.StopWatchZero));
00569     return((UBYTE *)timestring);
00570 }
00571 else if ( StrICmp(name, (UBYTE *)"numactiveexprs_") == 0 ) {
00572     /* the number of active expressions */
00573     int n = 0;
00574     for ( i = 0; i < NumExpressions; i++ ) {
00575         EXPRESSIONS e = Expressions + i;
00576         switch ( e->status ) {
00577             case LOCALEXPRESSION:
00578             case GLOBALEXPRESSION:
00579             case UNHIDELEXPRESSION:
00580             case UNHIDEGEXPRESSION:
00581             case INTOHIDELEXPRESSION:
00582             case INTOHIDEGEXPRESSION:
00583                 n++;
00584                 break;
00585         }
00586     }
00587     VectorReserve(exprstr, 41); /* up to 128-bit */
00588     LongCopy(n, (char *)VectorPtr(exprstr));
00589     return VectorPtr(exprstr);
00590 }
00591 else if ( StrICmp(name, (UBYTE *)"activeexprnames_") == 0 ) {
00592     /* the list of active expressions separated by commas */
00593     int j = 0;
00594     VectorReserve(exprstr, 16); /* at least 1 character for '\0' */
00595     for ( i = 0; i < NumExpressions; i++ ) {
00596         UBYTE *p, *s;
00597         int len, k;
00598         EXPRESSIONS e = Expressions + i;
00599         switch ( e->status ) {
00600             case LOCALEXPRESSION:
00601             case GLOBALEXPRESSION:
00602             case UNHIDELEXPRESSION:
00603             case UNHIDEGEXPRESSION:
00604             case INTOHIDELEXPRESSION:
00605             case INTOHIDEGEXPRESSION:
00606                 s = AC.exprnames->namebuffer + e->name;
00607                 len = StrLen(s);
00608                 VectorSize(exprstr) = j; /* j bytes must be copied in extending the buffer. */
00609                 VectorReserve(exprstr, j + len * 2 + 1);
00610                 p = VectorPtr(exprstr);
00611                 if ( j > 0 ) p[j++] = ',';
00612                 for ( k = 0; k < len; k++ ) {
00613                     if ( s[k] == ',' || s[k] == '|' ) p[j++] = '\\';
00614                     p[j++] = s[k];
00615                 }
00616                 break;
00617             }
00618         }
00619         VectorPtr(exprstr)[j] = '\0';
00620         return VectorPtr(exprstr);
00621     }
00622 else if ( StrICmp(name, (UBYTE *)"path_") == 0 ) {
00623     /* the current FORM path (for debugging both in .c and .frm) */
00624     if ( AM.Path ) {
00625         return(AM.Path);
00626     }
00627     else {
00628         return((UBYTE *) "");
00629     }
00630 }
00631 t = name;
00632 while ( *t && *t != '_' ) t++;
00633 for ( i = NumPre-1; i >= 0; i-- ) {
00634     if ( *t == '_' && ( StrICmp(name,PreVar[i].name) == 0 ) ) {
00635         if ( c ) *tt = c;
00636         ThePreVar = PreVar+i;
00637         return(PreVar[i].value);
00638     }
00639     else if ( StrCmp(name,PreVar[i].name) == 0 ) {

```

```

00640         if ( c ) *tt = c;
00641         ThePreVar = PreVar+i;
00642         return(PreVar[i].value);
00643     }
00644 }
00645 if ( *t == '_' ) {
00646     if ( StrICmp(name, (UBYTE *)"EXTRASYNBOLS_") == 0 ) goto extrashort;
00647     *t = 0;
00648     if ( StrICmp(name, (UBYTE *)"UNCHANGED") == 0 ) mode = 1;
00649     else if ( StrICmp(name, (UBYTE *)"ZERO") == 0 ) mode = 0;
00650     else if ( StrICmp(name, (UBYTE *)"SHOWINPUT") == 0 ) {
00651         *t++ = '_';
00652         if ( c ) *tt = c;
00653         if ( AC.NoShowInput > 0 ) return(no);
00654         else return(yes);
00655     }
00656     else if ( StrICmp(name, (UBYTE *)"EXTRASYNBOLS") == 0 ) {
00657         *t++ = '_';
00658 extrashort:;
00659         number = cbuf[AM.sbufnum].numrhs;
00660         t = numintopolynomial;
00661         NumCopy(number,t);
00662         return(numintopolynomial);
00663     }
00664     else mode = -1;
00665     *t++ = '_';
00666     if ( mode >= 0 ) {
00667         ThePreVar = 0;
00668         if ( *t ) {
00669             if ( GetName(AC.exprnames,t,&number,NOAUTO) == CEXPRESSION ) {
00670                 if ( c ) *tt = c;
00671                 if ( ( Expressions[number].vflags & ( 1 << mode ) ) != 0 )
00672                     return(yes);
00673                 else return(no);
00674             }
00675         }
00676         else {
00677 /*
00678             Here we have to test all active results.
00679             These are in 'negative' so the flags have to be zero.
00680 */
00681             if ( c ) *tt = c;
00682             if ( ( AR.expflags & ( 1 << mode ) ) == 0 ) return(yes);
00683             else return(no);
00684         }
00685     }
00686 }
00687 if ( ( t = (UBYTE *) (getenv((char *) (name))) ) != 0 ) {
00688     if ( c ) *tt = c;
00689     ThePreVar = 0;
00690     return(t);
00691 }
00692 if ( c ) *tt = c;
00693 if ( flag == WITHERROR ) {
00694     Error1("Undefined preprocessor variable",name);
00695 }
00696 return(0);
00697 }
00698
00699 /*
00700     #] GetPreVar :
00701     #[ PutPreVar :
00702 */
00703
00718 int PutPreVar(UBYTE *name, UBYTE *value, UBYTE *args, int mode)
00719 {
00720     int i, ii, num = 2, nnum = 2, numargs = 0;
00721     UBYTE *s, *t, *u = 0;
00722     PREVAR *p;
00723     if ( value == 0 && name[0] != '?' ) {
00724         MesPrint("@Illegal empty value for preprocessor variable %s",name);
00725         Terminate(-1);
00726     }
00727     if ( args ) {
00728         s = args; num++;
00729         while ( *s ) {
00730             if ( *s != ' ' && *s != '\t' ) num++;
00731             s++;
00732         }
00733     }
00734     if ( mode == 1 ) {
00735         i = NumPre;
00736         while ( --i >= 0 ) {
00737             if ( StrCmp(name,PreVar[i].name) == 0 ) {
00738                 u = PreVar[i].name;
00739                 break;
00740             }

```



```

00741     }
00742 }
00743 else i = -1;
00744 if ( i < 0 ) { p = (PREVAR *)FromList(&AP.PreVarList); ii = p - PreVar; }
00745 else { p = &(PreVar[i]); ii = i; }
00746 if ( value ) {
00747     s = value; while ( *s ) { s++; num++; }
00748 }
00749 else num = 1;
00750 if ( i >= 0 ) {
00751     if ( p->value ) {
00752         s = p->value;
00753         while ( *s ) { s++; nnum++; }
00754     }
00755     else nnum = 1;
00756     if ( nnum >= num ) {
00757 /*
00758         We can keep this in place
00759 */
00760         if ( value && p->value ) {
00761             s = value;
00762             t = p->value;
00763             while ( *s ) *t++ = *s++;
00764             *t = 0;
00765         }
00766         else p->value = 0;
00767         return(ii);
00768     }
00769 }
00770 s = name; while ( *s ) { s++; num++; }
00771 t = (UBYTE *)Malloc1(num,"PreVariable");
00772 p->name = t;
00773 s = name; while ( *s ) *t++ = *s++; *t++ = 0;
00774 if ( value ) {
00775     p->value = t;
00776     s = value; while ( *s ) *t++ = *s++; *t = 0;
00777     if ( AM.atstartup && t[-1] == '\n' ) t[-1] = 0;
00778 }
00779 else p->value = 0;
00780 p->wildarg = 0;
00781 if ( args ) {
00782     int first = 1;
00783     t++; p->argnames = t;
00784     s = args;
00785     while ( *s ) {
00786         if ( *s == ' ' || *s == '\t' ) { s++; continue; }
00787         if ( *s == ',' ) {
00788             s++; *t++ = 0; numargs++;
00789             while ( *s == ' ' || *s == '\t' ) s++;
00790             if ( *s == '?' ) {
00791                 if ( p->wildarg > 0 ) {
00792                     Error0("More than one ?var in #define");
00793                 }
00794                 p->wildarg = numargs;
00795             }
00796         }
00797         else if ( *s == '?' && first ) {
00798             p->wildarg = 1; *t++ = *s++;
00799         }
00800         else { *t++ = *s++; }
00801         first = 0;
00802     }
00803     *t = 0;
00804     numargs++;
00805     p->nargs = numargs;
00806 }
00807 else {
00808     p->nargs = 0;
00809     p->argnames = 0;
00810 }
00811 if ( u ) M_free(u,"replace PreVar value");
00812 return(ii);
00813 }
00814
00815 /*
00816     #] PutPreVar :
00817     #[ PopPreVars :
00818 */
00819
00820 void PopPreVars(int tonumber)
00821 {
00822     PREVAR *p = &(PreVar[NumPre]);
00823     while ( NumPre > tonumber ) {
00824         NumPre--; p--;
00825         M_free(p->name,"popping PreVar");
00826         p->name = p->value = 0;
00827     }

```

```

00828 }
00829
00830 /*
00831     #] PopPreVars :
00832     #[ IniModule :
00833 */
00834
00835 void IniModule(int type)
00836 {
00837     GETIDENTITY
00838     WORD **w, i;
00839     CBUF *C = cbuf+AC.cbunum;
00840     /*[05nov2003 mt]:*/
00841     #ifdef WITHMPI
00842     /* To prevent
00843      * (1) FlushOut() and PutOut() on the slaves to send a mess to the master
00844      * compiling a module,
00845      * (2) EndSort() called from poly_factorize_expression() on the master
00846      * waits for the slaves.
00847      */
00848     PF.parallel=0;
00849     /*BTW, this was the bug preventing usage of more than 1 expression!*/
00850     #endif
00851
00852     AR.BracketOn = 0;
00853     AR.StoreData.dirtyflag = 0;
00854     AC.bracketindexflag = 0;
00855     AT.bracketindexflag = 0;
00856
00857     /*[06nov2003 mt]:*/
00858     #ifdef WITHMPI
00859     /* This flag may be set in the procedure tokenize(). */
00860     AC.RhsExprInModuleFlag = 0;
00861     /*[20oct2009 mt]:*/
00862     PF.mkSlaveInfile=0;
00863     PF.slavebuf.PObuffer=NULL;
00864     for(i=0; i<NumExpressions; i++)
00865         Expressions[i].vflags &= ~ISINRHS;
00866     /*:[20oct2009 mt]:*/
00867     #endif
00868     /*:[06nov2003 mt]:*/
00869
00870     /*[19nov2003 mt]:*/
00871     /*The module counter:*/
00872     (AC.CModule)++;
00873     /*:[19nov2003 mt]:*/
00874
00875     if ( !type ) {
00876         if ( C->rhs ) {
00877             w = C->rhs; i = C->maxrhs;
00878             do { *w++ = 0; } while ( --i > 0 );
00879         }
00880         if ( C->lhs ) {
00881             w = C->lhs; i = C->maxlhs;
00882             do { *w++ = 0; } while ( --i > 0 );
00883         }
00884     }
00885     C->numlhs = C->numrhs = 0;
00886     ClearTree(AC.cbunum);
00887     while ( AC.NumLabels > 0 ) {
00888         AC.NumLabels--;
00889         if ( AC.LabelNames[AC.NumLabels] ) M_free(AC.LabelNames[AC.NumLabels], "LabelName");
00890     }
00891
00892     C->Pointer = C->Buffer;
00893
00894     AC.Commercial[0] = 0;
00895
00896     AC.IfStack = AC.IfHeap;
00897     AC.arglevel = 0;
00898     AC.termlevel = 0;
00899     AC.IfLevel = 0;
00900     AC.WhileLevel = 0;
00901     AC.RepLevel = 0;
00902     AC.insidelevel = 0;
00903     AC.dolooplevel = 0;
00904     AC.MustTestTable = 0;
00905     AO.PrintType = 0; /* Otherwise statistics can get spoiled */
00906     AC.ComDefer = 0;
00907     AC.CollectFun = 0;
00908     AM.S0->PolyWise = 0;
00909     AC.SymChangeFlag = 0;
00910     AP.lhdollarerror = 0;
00911     AR.PolyFun = AC.lPolyFun;
00912     AR.PolyFunInv = AC.lPolyFunInv;
00913     AR.PolyFunType = AC.lPolyFunType;
00914     AR.PolyFunExp = AC.lPolyFunExp;

```

```

00915     AR.PolyFunVar = AC.lPolyFunVar;
00916     AR.PolyFunPow = AC.lPolyFunPow;
00917     AC.mparallelfldflag = AC.pparallelfldflag | AM.hparallelfldflag;
00918     AC.inparallelfldflag = 0;
00919     AC.mProcessBucketSize = AC.ProcessBucketSize;
00920     NumPotModdollars = 0;
00921     AC.topolynomialflag = 0;
00922     #ifdef WITHPTHREADS
00923         if ( AM.totalnumberofthreads > 1 ) AS.MultiThreaded = 1;
00924     else AS.MultiThreaded = 0;
00925     for ( i = 1; i < AM.totalnumberofthreads; i++ ) {
00926         AB[i]->T.S0->PolyWise = 0;
00927     }
00928     #endif
00929     OpenTemp();
00930 }
00931
00932 /*
00933     #] IniModule :
00934     #[ IniSpecialModule :
00935 */
00936
00937 void IniSpecialModule(int type)
00938 {
00939     DUMMYUSE(type);
00940 }
00941
00942 /*
00943     #] IniSpecialModule :
00944     #[ PreProcessor :
00945 */
00946
00947 void PreProcessor(void)
00948 {
00949     int moduletype = FIRSTMODULE;
00950     int specialtype = 0;
00951     int error1 = 0, error2 = 0, retcode, retval;
00952     UBYTE c, *t, *s;
00953     AP.StopWatchZero = GetRunningTime();
00954     AC.compiletype = 0;
00955     AP.PreContinuation = 0;
00956     AP.PreAssignLevel = 0;
00957     AP.gNumPre = NumPre;
00958     AC.iPointer = AC.iBuffer;
00959     AC.iPointer[0] = 0;
00960
00961     if ( AC.CheckpointFlag == -1 ) DoRecovery(&moduletype);
00962     AC.CheckpointStamp = Timer(0);
00963
00964     for(;;) {
00965         /* if ( A.StatisticsFlag ) CharOut(LINEFEED); */
00966
00967         IniModule(moduletype);
00968
00969         /*Re-define preprocessor variable CMODULE_ as a current module number, starting from 1*/
00970         /*The module counter is AC.CModule, it is incremented in IniModule*/
00971         {
00972             UBYTE buf[24];/*64/Log_2[10] = 19.3, this is enough for any integer*/
00973             NumToStr(buf,AC.CModule);
00974             PutPreVar((UBYTE *) "CMODULE_",buf,0,1);
00975         }
00976
00977         if ( specialtype ) IniSpecialModule(specialtype);
00978
00979         for(;;) { /* Read a single line/statement */
00980             c = GetChar(0);
00981             if ( c == AP.ComChar ) { /* This line is commentary */
00982                 LoadInstruction(5);
00983                 if ( AC.CurrentStream->FoldName ) {
00984                     t = AP.preStart;
00985                     if ( *t && t[1] && t[2] == '#' && t[3] == ']' ) {
00986                         t += 4;
00987                         while ( *t == ' ' || *t == '\t' ) t++;
00988                         s = AC.CurrentStream->FoldName;
00989                         while ( *s == *t ) { s++; t++; }
00990                         if ( *s == 0 && ( *t == ' ' || *t == '\t'
00991                             || *t == ':' ) ) {
00992                             while ( *t == ' ' || *t == '\t' ) t++;
00993                             if ( *t == ':' ) {
00994                                 AC.CurrentStream = CloseStream(AC.CurrentStream);
00995                             }
00996                         }
00997                     }
00998                 }
00999                 *AP.preStart = 0;
01000                 continue;
01001             }

```

```

01002         while ( c == ' ' || c == '\t' ) c = GetChar(0);
01003         if ( c == LINEFEED ) continue;
01004         if ( c == ENDOFINPUT ) {
01005             /* CharOut(LINEFEED); */
01006             Warning("end instruction generated");
01007             moduletype = ENDMODULE; specialtype = 0;
01008             goto endmodule; /* Fake one */
01009         }
01010         if ( c == '#' ) {
01011             if ( PreProInstruction() ) { error1++; error2++; AP.preError++; }
01012             *AP.preStart = 0;
01013         }
01014         else if ( c == '.' ) {
01015             if ( ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) ||
01016                 ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) ) {
01017                 LoadInstruction(1);
01018                 continue;
01019             }
01020             if ( ModuleInstruction(&moduletype,&specialtype) ) { error2++; AP.preError++; }
01021             if ( specialtype ) SetSpecialMode(moduletype,specialtype);
01022             if ( AP.PreInsideLevel != 0 ) {
01023                 MesPrint("@end of module instructions may not be used inside");
01024                 MesPrint("@the scope of a %#inside %#endinside construction.");
01025                 Terminate(-1);
01026             }
01027             if ( AC.RepLevel > 0 ) {
01028                 MesPrint("&EndRepeat statement(s) missing");
01029                 error2++; AP.preError++;
01030             }
01031             if ( AC.tablecheck == 0 ) {
01032                 AC.tablecheck = 1;
01033                 if ( TestTables() ) { error2++; AP.preError++; }
01034             }
01035             if ( AP.PreContinuation ) {
01036                 error1++; error2++;
01037                 MesPrint("&Unfinished statement. Missing ;?");
01038             }
01039             if ( moduletype == GLOBALMODULE ) MakeGlobal();
01040             else {
01041 endmodule:
01042                 if ( error2 == 0 && AM.qError == 0 ) {
01043                     retcode = ExecModule(moduletype);
01044                     #ifdef WITHMPI
01045                         if(PF.slavebuf.PObuffer!=NULL){
01046                             M_free(PF.slavebuf.PObuffer,"PF inbuf");
01047                             PF.slavebuf.PObuffer=NULL;
01048                         }
01049                     #endif
01050                     UpdatePositions();
01051                     if ( retcode < 0 ) error1++;
01052                     if ( retcode ) { error2++; AP.preError++; }
01053                 }
01054                 else {
01055                     EXPRESSIONS e;
01056                     WORD j;
01057                     for ( j = 0, e = Expressions; j < NumExpressions; j++, e++ ) {
01058                         if ( e->replace == NEWLYDEFINEDEXPRESSION ) e->replace =
01059                             REGULAREXPRESSION;
01060                     }
01061                 }
01062                 switch ( moduletype ) {
01063                     case STOREMODULE:
01064                         if ( ExecStore() ) error1++;
01065                         break;
01066                     case CLEARMODULE:
01067                         FullCleanUp();
01068                         error1 = error2 = AP.preError = 0;
01069                         AM.atstartup = 1;
01070                         PutPreVar((UBYTE *) "DATE_", (UBYTE *) MakeDate(), 0, 1);
01071                         AM.atstartup = 0;
01072                         if ( AM.resetTimeOnClear ) {
01073                             #ifdef WITHPTHREADS
01074                                 ClearAllThreads();
01075                             #endif
01076                             AM.SumTime += TimeCPU(1);
01077                             TimeCPU(0);
01078                         }
01079                         AP.StopWatchZero = GetRunningTime();
01080                         break;
01081                     case ENDMODULE:
01082                         Terminate( -( error1 | error2 ) );
01083                 }
01084             }
01085             AC.tablecheck = 0;
01086             AC.compiletype = 0;
01087             if ( AC.exprfillwarning > 0 ) {
01088                 AC.exprfillwarning = 0;
01089             }

```

```

01088         if ( AC.CheckpointFlag && error1 == 0 && error2 == 0 ) DoCheckpoint(moduletype);
01089         break; /* start a new module */
01090     }
01091     else {
01092         if ( ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) ||
01093             ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) ) {
01094             pushbackchar = c;
01095             LoadInstruction(5);
01096             continue;
01097         }
01098         UngetChar(c);
01099         if ( AP.PreContinuation ) {
01100             retval = LoadStatement(OLDSTATEMENT);
01101         }
01102         else {
01103             AC.CurrentStream->prevline = AC.CurrentStream->linenumber;
01104             retval = LoadStatement(NEWSTATEMENT);
01105         }
01106         if ( retval < 0 ) {
01107             error1++;
01108             if ( retval == -1 ) AP.PreContinuation = 0;
01109             else AP.PreContinuation = 1;
01110             TryRecover(0);
01111         }
01112         else if ( retval > 0 ) AP.PreContinuation = 0;
01113         else AP.PreContinuation = 1;
01114         if ( error1 == 0 && !AP.PreContinuation ) {
01115             if ( ( AP.PreDebug & PREPROONLY ) == 0 ) {
01116                 int onpmd = NumPotModdollars;
01117                 #ifdef WITHMPI
01118                 WORD oldRhsExprInModuleFlag = AC.RhsExprInModuleFlag;
01119                 if ( AP.PreAssignFlag ) AC.RhsExprInModuleFlag = 0;
01120                 #endif
01121                 if ( AP.PreOut || ( AP.PreDebug & DUMPTOCOMPILER )
01122                     == DUMPTOCOMPILER )
01123                     MesPrint( " %s", AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]);
01124                 retcode = CompileStatement(AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]);
01125                 if ( retcode < 0 ) error1++;
01126                 if ( retcode ) { error2++; AP.preError++; }
01127                 if ( AP.PreAssignFlag ) {
01128                     if ( retcode == 0 ) {
01129                         if ( ( retcode = CatchDollar(0) ) < 0 ) error1++;
01130                         else if ( retcode > 0 ) { error2++; AP.preError++; }
01131                     }
01132                     else CatchDollar(-1);
01133                     POPPREASSIGNLEVEL;
01134                     if ( AP.PreAssignLevel <= 0 )
01135                         AP.PreAssignFlag = 0;
01136                     NumPotModdollars = onpmd;
01137                     #ifdef WITHMPI
01138                     AC.RhsExprInModuleFlag = oldRhsExprInModuleFlag;
01139                     #endif
01140                 }
01141             }
01142             else {
01143                 MesPrint( " %s", AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]);
01144             }
01145         }
01146         else if ( !AP.PreContinuation ) {
01147             if ( AP.PreAssignLevel > 0 ) {
01148                 POPPREASSIGNLEVEL;
01149                 if ( AP.PreAssignLevel <= 0 )
01150                     AP.PreAssignFlag = 0;
01151             }
01152         }
01153     }
01154     if ( !AP.PreContinuation ) AP.PreAssignFlag = 0;
01155 }
01156 }
01157 }
01158 }
01159 }
01160
01161 /*
01162     #] PreProcessor :
01163     #[ PreProInstruction :
01164 */
01165
01166 int PreProInstruction(void)
01167 {
01168     UBYTE *s, *t;
01169     KEYWORD *key;
01170     AP.PreproFlag = 1;
01171     AP.preFill = 0;
01172     AP.AllowDelay = 0;
01173     AP.DelayPrevar = 0;
01174 }

```

```

01175     oldmode = 0;
01176     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) {
01177         LoadInstruction(3);
01178         if ( ( StrICmp(AP.preStart,(UBYTE *)"case") == 0
01179             || StrICmp(AP.preStart,(UBYTE *)"default") == 0 )
01180             && AP.PreSwitchModes[AP.PreSwitchLevel] == SEARCHINGPRECASE ) {
01181             LoadInstruction(0);
01182         }
01183         else if ( StrICmp(AP.preStart,(UBYTE *)"assign ") == 0 ) {}
01184         else { LoadInstruction(1); }
01185     }
01186     else if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) {
01187         LoadInstruction(3);
01188         if ( ( StrICmp(AP.preStart,(UBYTE *)"else") == 0
01189             || StrICmp(AP.preStart,(UBYTE *)"elseif") == 0 )
01190             && AP.PreIfStack[AP.PreIfLevel] == LOOKINGFORELSE ) {
01191             LoadInstruction(0);
01192         }
01193         else if ( StrICmp(AP.preStart,(UBYTE *)"assign ") == 0 ) {}
01194         else {
01195             LoadInstruction(1);
01196         }
01197     }
01198     else {
01199         LoadInstruction(0);
01200     }
01201     AP.PreproFlag = 0;
01202     t = AP.preStart;
01203     if ( *t == '-' ) {
01204         if ( AP.PreSwitchModes[AP.PreSwitchLevel] == EXECUTINGPRESWITCH
01205             && AP.PreIfStack[AP.PreIfLevel] == EXECUTINGIF )
01206             AC.NoShowInput = 1;
01207     }
01208     else if ( *t == '+' ) {
01209         if ( AP.PreSwitchModes[AP.PreSwitchLevel] == EXECUTINGPRESWITCH
01210             && AP.PreIfStack[AP.PreIfLevel] == EXECUTINGIF )
01211             AC.NoShowInput = 0;
01212     }
01213     else if ( *t == ':' ) {}
01214     else {
01215     retry:;
01216         key = FindKeyWord(t,precommands,sizeof(precommands)/sizeof(KEYWORD));
01217         s = EndOfToken(t);
01218         if ( key == 0 ) {
01219             if ( *s == ';' ) {
01220                 *s = 0; goto retry;
01221             }
01222             else {
01223                 *s = 0;
01224                 MesPrint("@Unrecognized preprocessor instruction: %s",t);
01225                 return(-1);
01226             }
01227         }
01228         while ( *s == ' ' || *s == '\t' || *s == ',' ) s++;
01229         t = s;
01230         while ( *t ) t++;
01231         while ( ( t[-1] == ';' ) && ( t[-2] != '\\\' ) ) {
01232             t--; *t = 0;
01233         }
01234         return((key->func)(s));
01235     }
01236     return(0);
01237 }
01238
01239 /*
01240     #] PreProInstruction :
01241     #[ LoadInstruction :
01242
01243     0: preprocessor instruction that may involve matching of brackets
01244     1: runs straight to end-of-line
01245     2: runs to ;
01246     3: only gets one word without ` interpretation.
01247     5: with pushbackchar, but inside commentary. -> 1
01248
01249 To be added:
01250     In define, redefine, call and listed do we may have delayed substitution
01251     of preprocessor variables.
01252 */
01253
01254 int LoadInstruction(int mode)
01255 {
01256     UBYTE *s, *sstart, *t, c, cp;
01257     LONG position, fillpos = 0;
01258     int bralevel = 0, parlevel = 0, first = 1;
01259     int quotelevel = 0;
01260     if ( AP.preFill ) {
01261         s = AP.preFill;

```

```

01262     AP.preFill = 0;
01263     if ( s[1] != LINEFEED && s[1] != ENDOFINPUT ) {
01264         s[0] = s[1]; s++;
01265     }
01266     else { oldmode = mode; return(0); }
01267 }
01268 else { s = AP.preStart; }
01269 sstart = s; *s = 0;
01270 for(;;) {
01271     if ( ( mode & 1 ) == 1 ) {
01272         if ( pushbackchar && ( mode == 3 || mode == 5 ) ) {
01273             c = pushbackchar; pushbackchar = 0;
01274         }
01275         else c = GetInput();
01276     }
01277     else {
01278         c = GetChar(0);
01279     }
01280
01281     if ( mode == 2 && c == ';' ) break;
01282     if ( ( mode == 1 || mode == 5 ) && c == LINEFEED ) break;
01283     if ( mode == 3 && FG.cTable[c] != 0 ) {
01284         if ( c == '$' ) {
01285             pushbackchar = '$';
01286             *s++ = 'a'; *s++ = 's'; *s++ = 's'; *s++ = 'i';
01287             *s++ = 'g'; *s++ = 'n'; *s++ = ' '; *s = 0;
01288         }
01289         if ( c == '\\' || c == '"' ) { /* we do not expand preprocessor variables */
01290             mode = 1;
01291         }
01292         else {
01293             AP.preFill = s; *s++ = 0; *s = c;
01294             oldmode = mode;
01295             return(0);
01296         }
01297     }
01298     if ( mode == 0 && first ) {
01299         if ( c == '$' ) {
01300             dodollar:
01301                 s = sstart;
01302                 *s++ = 'a'; *s++ = 's'; *s++ = 's'; *s++ = 'i';
01303                 *s++ = 'g'; *s++ = 'n'; *s = 0;
01304                 pushbackchar = c;
01305                 oldmode = mode;
01306                 return(0);
01307             }
01308             if ( c == ' ' || c == '\t' || c == ',' ) {}
01309             else first = 0;
01310         }
01311         else if ( mode == 1 && first && c == '$' && oldmode == 3 ) goto dodollar;
01312         if ( c == ENDOFINPUT || ( c == LINEFEED
01313             /* && bralevel == 0 */
01314             && quotelevel == 0 ) ) {
01315             if ( mode == 2 && c == ENDOFINPUT ) {
01316                 MesPrint("@Unexpected end of instruction");
01317                 oldmode = mode;
01318                 return(-1);
01319             }
01320             /*
01321             if ( mode == 0 && bralevel ) {
01322                 MesPrint("@Unmatched brackets");
01323                 oldmode = mode;
01324                 return(-1);
01325             }
01326             */
01327             if ( mode != 2 ) break;
01328         }
01329         if ( quotelevel ) {
01330             if ( c == '\\' ) {
01331                 if ( ( mode == 1 ) || ( mode == 5 ) ) c = GetInput();
01332                 else {
01333                     c = GetChar(0);
01334                 }
01335             }
01336             if ( c == ENDOFINPUT ) {
01337                 MesPrint("@Unmatched \"");
01338                 if ( mode == 2 && c == ENDOFINPUT ) {
01339                     MesPrint("@Unexpected end of instruction");
01340                 }
01341             }
01342             /*
01343             if ( mode == 0 && bralevel ) {
01344                 MesPrint("@Unmatched brackets");
01345             }
01346             */
01347             oldmode = mode;
01348             return(-1);
01349         }
01350         else if ( c == LINEFEED ) {}
01351         else if ( c == '"' ) { *s++ = '\\'; }

```

```

01349         else {
01350             *s++ = '\\';
01351         }
01352     }
01353     else if ( c == '"' ) {
01354         quotelevel = 0;
01355         AP.AllowDelay = 0;
01356     }
01357 }
01358 else if ( c == '\\' ) {
01359     if ( ( mode == 1 ) || ( mode == 5 ) ) cp = GetInput();
01360     else {
01361         cp = GetChar(0);
01362     }
01363     if ( cp == LINEFEED ) continue;
01364     if ( mode != 2 || cp != ';' ) *s++ = c;
01365     c = cp;
01366 }
01367 else if ( c == '"' ) {
01368     /*
01369         Now look back in the buffer and determine what the keyword is.
01370         If it is define or redefine, put AllowDelay to 1.
01371     */
01372     t = AP.preStart;
01373     while ( FG.cTable[*t] <= 1 ) t++;
01374     cp = *t; *t = 0;
01375     if ( ( StrICmp(AP.preStart, (UBYTE *) "define") == 0 )
01376         || ( StrICmp(AP.preStart, (UBYTE *) "redefine") == 0 ) ) {
01377         AP.AllowDelay = 1;
01378         oldstream = AC.CurrentStream;
01379     }
01380     *t = cp;
01381     quotelevel = 1;
01382 }
01383 else if ( quotelevel == 0 && bralevel == 0 && c == '(' ) {
01384     t = AP.preStart;
01385     while ( FG.cTable[*t] <= 1 ) t++;
01386     cp = *t; *t = 0;
01387     if ( ( parlevel == 0 )
01388         && ( StrICmp(AP.preStart, (UBYTE *) "call") == 0 ) ) {
01389         AP.AllowDelay = 1;
01390         oldstream = AC.CurrentStream;
01391     }
01392     *t = cp;
01393     parlevel++;
01394 }
01395 else if ( quotelevel == 0 && bralevel == 0 && c == ')' ) {
01396     parlevel--;
01397 }
01398 else if ( quotelevel == 0 && parlevel == 0 && c == '{' ) {
01399     t = AP.preStart;
01400     while ( FG.cTable[*t] <= 1 ) t++;
01401     cp = *t; *t = 0;
01402     if ( ( bralevel == 0 )
01403         && ( ( StrICmp(AP.preStart, (UBYTE *) "call") == 0 )
01404             || ( StrICmp(AP.preStart, (UBYTE *) "do") == 0 ) ) ) {
01405         AP.AllowDelay = 1;
01406         oldstream = AC.CurrentStream;
01407     }
01408     *t = cp;
01409     bralevel++;
01410 }
01411 else if ( quotelevel == 0 && parlevel == 0 && c == '}' ) {
01412     bralevel--;
01413     if ( bralevel < 0 ) {
01414         if ( mode != 5 ) {
01415             MesPrint("@Unmatched brackets");
01416             oldmode = mode;
01417             return(-1);
01418         }
01419         bralevel = 0;
01420     }
01421 }
01422 if ( s >= (AP.preStop-1) ) {
01423     UBYTE **ppp;
01424     position = s - AP.preStart;
01425     if ( AP.preFill ) fillpos = AP.preFill - AP.preStart;
01426     ppp = &(AP.preStart); /* to avoid a compiler warning */
01427     if ( DoubleList((void ***) ppp, &AP.pSize, sizeof(UBYTE),
01428         "instruction buffer") ) { *s = 0; oldmode = mode; return(-1); }
01429     AP.preStop = AP.preStart + AP.pSize-3;
01430     s = AP.preStart + position;
01431     if ( AP.preFill ) AP.preFill = fillpos + AP.preStart;
01432 }
01433 *s++ = c;
01434 }
01435 *s = 0;

```



```

01436     oldmode = mode;
01437     if ( mode == 0 ) {
01438         if ( ExpandTripleDots(1) < 0 ) return(-1);
01439     }
01440     return(0);
01441 }
01442
01443 /*
01444     #] LoadInstruction :
01445     #[ LoadStatement :
01446
01447     Puts the current string together in the input buffer.
01448     Does things like placing comma's where needed and expand ...
01449     We force a comma after the keyword. Before 8-sep-2009 the program might
01450     not put a comma if a + or - followed. And then the compiler ate
01451     the + or - and we needed repair code in the routines that used the
01452     + or - (Print, modulus, multiply and (a)bracket). This worked but
01453     the problem was with statements like Dimension -4; which then would
01454     be processed as Dimension 4; (JV)
01455 */
01456
01457 int LoadStatement(int type)
01458 {
01459     UBYTE *s, c, cp;
01460     int retval = 0, stringlevel = 0, newstatement = 0;
01461     if ( type == NEWSTATEMENT ) { AP.eat = 1; newstatement = 1;
01462         s = AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]; }
01463     else { s = AC.iPointer; *s = 0; c = ' '; goto blank; }
01464     *s = 0;
01465     for(;;) {
01466         c = GetChar(0);
01467         if ( c == ENDOFINPUT ) { retval = -1; break; }
01468         if ( stringlevel == 0 ) {
01469             if ( c == LINEFEED ) {
01470                 if ( AP.eat < 0 ) { s--; AP.eat = 0; }
01471                 retval = 0; break;
01472             }
01473             if ( c == ';' ) {
01474                 if ( AP.eat < 0 ) { s--; AP.eat = 0; }
01475                 while ( ( c = GetChar(0) ) == ' ' || c == '\t' ) {}
01476                 if ( c != LINEFEED ) UngetChar(c);
01477                 retval = 1;
01478                 break;
01479             }
01480         }
01481         if ( c == '\\\\' ) {
01482             cp = GetChar(0);
01483             if ( cp == LINEFEED ) continue;
01484             *s++ = c;
01485             c = cp;
01486         }
01487         if ( c == '"' ) {
01488             if ( stringlevel == 0 ) stringlevel = 1;
01489             else stringlevel = 0;
01490             AP.eat = 0;
01491         }
01492         else if ( stringlevel == 0 ) {
01493             if ( c == '\t' ) c = ' ';
01494             if ( c == ' ' ) {
01495                 if ( newstatement < 0 ) newstatement = 0;
01496                 if ( AP.eat && ( newstatement == 0 ) ) continue;
01497                 c = ',';
01498                 AP.eat = -2;
01499                 if ( newstatement > 0 ) newstatement = -1;
01500             }
01501             else if ( chartype[c] <= 3 ) {
01502                 AP.eat = 0;
01503                 if ( newstatement < 0 ) newstatement = 0;
01504             }
01505             else if ( c == ',' ) {
01506                 if ( newstatement > 0 ) {
01507                     newstatement = -1;
01508                     AP.eat = -2;
01509                 }
01510             }
01511             /* else if ( AP.eat == -2 ) { s--; } */
01512             else if ( AP.eat == -2 ) { AP.eat = 1; continue; }
01513             else { goto doall; }
01514         }
01515         else {
01516             if ( AP.eat < 0 ) {
01517                 if ( newstatement == 0 ) s--;
01518                 else { newstatement = 0; }
01519             }
01520             else if ( newstatement == 1 ) newstatement = 0;
01521             AP.eat = 1;
01522             if ( c == '*' && s > AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel] && s[-1] == '*' )

```

```

01522         s[-1] = '^';
01523         continue;
01524     }
01525 }
01526 }
01527 if ( s >= AC.iStop ) {
01528     if ( !AP.iBufError ) {
01529         LONG position = s - AC.iBuffer;
01530         LONG position2 = AC.iPointer - AC.iBuffer;
01531         UBYTE *ppp = &(AC.iBuffer); /* to avoid a compiler warning */
01532         if ( DoubleLList((void ***)ppp,&AC.iBufferSize
01533             ,sizeof(UBYTE),"statement buffer") ) {
01534             *s = 0; retval = -1; AP.iBufError = 1;
01535         }
01536         AC.iPointer = AC.iBuffer + position2;
01537         AC.iStop = AC.iBuffer + AC.iBufferSize-2;
01538         s = AC.iBuffer + position;
01539     }
01540     if ( AP.iBufError ) {
01541         for(;;){
01542             c = GetChar(0);
01543             if ( c == ENDOFINPUT ) { retval = -1; break; }
01544             if ( c == '"' ) {
01545                 if ( stringlevel > 0 ) stringlevel = 0;
01546                 else stringlevel = 1;
01547             }
01548             else if ( c == LINEFEED && !stringlevel ) { retval = -2; break; }
01549             else if ( c == ';' && !stringlevel ) {
01550                 while ( ( c = GetChar(0) ) == ' ' || c == '\t' ) {}
01551                 if ( c != LINEFEED ) UngetChar(c);
01552                 retval = -1;
01553                 break;
01554             }
01555             else if ( c == '\\\' ) c = GetChar(0);
01556         }
01557         break;
01558     }
01559 }
01560 *s++ = c;
01561 }
01562 AC.iPointer = s;
01563 *s = 0;
01564 if ( stringlevel > 0 ) {
01565     MesPrint("@Unbalanced \". Runaway string");
01566     retval = -1;
01567 }
01568 if ( retval == 1 ) {
01569     if ( ExpandTripleDots(0) < 0 ) retval = -1;
01570 }
01571 return(retval);
01572 }
01573
01574 /*
01575     #] LoadStatement :
01576     #[ ExpandTripleDots :
01577 */
01578
01579 static inline int IsSignChar(UBYTE c)
01580 {
01581     return c == '+' || c == '-';
01582 }
01583
01584 static inline int IsAlphanumericChar(UBYTE c)
01585 {
01586     return FG.cTable[c] == 0 || FG.cTable[c] == 1;
01587 }
01588
01589 static inline int CanParseSignedNumber(const UBYTE *s)
01590 {
01591     while ( IsSignChar(*s) ) s++;
01592     return FG.cTable[*s] == 1;
01593 }
01594
01595 int ExpandTripleDots(int par)
01596 {
01597     UBYTE *s, *s1, *s2, *n1, *n2, *t1, *t2, *startp, operator1, operator2, c, cc;
01598     UBYTE *nBuffer, *strngs, *Buffer, *Stop;
01599     LONG withquestion, x1, x2, y1, y2, number, inc, newsize, pow, fullsize;
01600     int i, error = 0, i1, i2, ii, *nums = 0;
01601
01602     if ( par == 0 ) {
01603         Buffer = AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]; Stop = AC.iStop;
01604     }
01605     else {
01606         Buffer = AP.preStart; Stop = AP.preStop;
01607     }
01608     s = Buffer; while ( *s ) s++;

```

```

01609     fullsize = s - Buffer;
01610     if ( fullsize < 7 ) return(error);
01611
01612     s = Buffer+2;
01613     while ( *s ) {
01614         if ( *s != '.' || ( s[-1] != ',' && FG.cTable[s[-1]] != 5 ) )
01615             { s++; continue; }
01616         if ( s[-1] == '%' || s[-1] == '^' || s[1] != '.' || s[2] != '.' )
01617             { s++; continue; }
01618         s1 = s - 2;
01619         s += 3;
01620         if ( *s != s[-4] && ( *s != '+' || s[-4] != '-' )
01621             && ( *s != '-' || s[-4] != '+' ) ) {
01622             MesPrint("&Improper operators for ...");
01623             error = -1;
01624         }
01625         operator1 = s[-4];
01626         operator2 = *s++;
01627         if ( operator1 == ':' ) operator1 = '.';
01628         if ( operator2 == ':' ) operator2 = '.';
01629     /*
01630         We have now O1...O2 (O stands for operator)
01631         Full syntax is
01632         [str]#1[?]O1...O2[str]#2[?]    (Special case)
01633         in which both strings are identical and if one ? then also the other.
01634         <pattern1>O1...O2<pattern2>    (General case)
01635         in which the difference in the patterns is just numerical.
01636     */
01637     s2 = s;          /* the beginning of the second string */
01638     if ( *s2 != '<' || *s1 != '>' ) { /* Special case */
01639         startp = s1+1;
01640         withquestion = ( *s1 == '?' ); s1--;
01641         while ( FG.cTable[*s1] == 1 && s1 >= Buffer ) s1--;
01642         n1 = s1+1;    /* Beginning of first number */
01643         if ( FG.cTable[*n1] != 1 ) {
01644             MesPrint("&No first number in ... operator");
01645             error = -1;
01646         }
01647         while ( FG.cTable[*s1] <= 1 && s1 >= Buffer ) s1--;
01648         s1++;
01649     /*
01650         We have now the first string from s1 to n1, number from n1
01651     */
01652     t1 = s1; t2 = s2;
01653     while ( t1 < n1 && *t1 == *t2 ) { t1++; t2++; }
01654     n2 = t2;
01655     if ( FG.cTable[*t2] != 1 ) {
01656         MesPrint("&No second number in ... operator");
01657         error = -1;
01658     }
01659     x2 = 0;
01660     while ( FG.cTable[*t2] == 1 ) x2 = 10*x2 + *t2++ - '0';
01661     x1 = 0;
01662     while ( FG.cTable[*t1] == 1 ) x1 = 10*x1 + *t1++ - '0';
01663     if ( withquestion != ( *t2 == '?' ) ) {
01664         MesPrint("&Improper use of ? in ... operator");
01665         if ( *t2 == '?' ) t2++;
01666         error = -1;
01667     }
01668     else if ( withquestion ) t2++;
01669     if ( FG.cTable[*t2] <= 2 ) {
01670         MesPrint("&Illegal object after ... construction");
01671         error = -1;
01672     }
01673     c = *n1; *n1 = 0; s = t2;
01674     if ( error ) continue;
01675 /*
01676     At this point the syntax has been fulfilled. We have
01677     str in s1.
01678     x1,x2 are #1,#2
01679     operator1,operator2 are the two operators.
01680     s points at whatever comes after.
01681     Expansion will have to be computed.
01682 */
01683     if ( x2 < x1 ) { number = x1-x2; inc = -1; y1 = x2; y2 = x1; }
01684     else { number = x2-x1; inc = 1; y1 = x1; y2 = x2; }
01685     newsize = (number+1)*(n1-s1) /* the strings */
01686         + number /* the operators */
01687         +(number+1)*(withquestion?1:0) /* questionmarks */
01688         +(number+1); /* last digits */
01689     pow = 10;
01690     for ( i = 1; i < 10; i++, pow *= 10 ) {
01691         if ( y1 >= pow ) newsize += number+1;
01692         else if ( y2 >= pow ) newsize += y2-pow+1;
01693         else break;
01694     }
01695     while ( Buffer+(fullsize+newsize-(s-s1)) >= Stop ) {

```

```

01696         LONG strpos = s1-Buffer;
01697         LONG endstr = n1-Buffer;
01698         LONG startq = startp - Buffer;
01699         LONG position = s - Buffer;
01700         UBYTE *ppp;
01701         if ( par == 0 ) {
01702             LONG position2 = AC.iPointer - AC.iBuffer;
01703             ppp = &(AC.iBuffer); /* to avoid a compiler warning */
01704             if ( DoubleList((void ***)ppp,&AC.iBufferSize
01705                 ,sizeof(UBYTE),"statement buffer") ) {
01706                 Terminate(-1);
01707             }
01708             AC.iPointer = AC.iBuffer + position2;
01709             AC.iStop = AC.iBuffer + AC.iBufferSize-2;
01710             Buffer = AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]; Stop = AC.iStop;
01711         }
01712         else {
01713             LONG fillpos = 0;
01714             if ( AP.preFill ) fillpos = AP.preFill - AP.preStart;
01715             ppp = &(AP.preStart); /* to avoid a compiler warning */
01716             if ( DoubleList((void ***)ppp,&AP.pSize,sizeof(UBYTE),
01717                 "instruction buffer") ) {
01718                 Terminate(-1);
01719             }
01720             AP.preStop = AP.preStart + AP.pSize-3;
01721             if ( AP.preFill ) AP.preFill = fillpos + AP.preStart;
01722             Buffer = AP.preStart; Stop = AP.preStop;
01723         }
01724         s = Buffer + position;
01725         n1 = Buffer + endstr;
01726         s1 = Buffer + strpos;
01727         startp = Buffer + startq;
01728     }
01729     /*
01730     We have space for the expansion in the buffer.
01731     There are two cases: new size > old size
01732                        old size >= new size
01733     Note that wherever we move things, it will be at least startp.
01734     */
01735     if ( newsize > (s-s1) ) {
01736         t2 = Buffer + fullsize;
01737         t1 = t2 + (newsize - (s-s1));
01738         *t1 = 0;
01739         while ( t2 > s ) { *--t1 = *--t2; }
01740     }
01741     else if ( newsize < (s-s1) ) {
01742         t1 = s1 + newsize; t2 = s; s = t1;
01743         while ( *t2 ) *t1++ = *t2++;
01744         *t1 = 0;
01745     }
01746     for ( x1 += inc, t1 = startp; number > 0; number--, x1 += inc ) {
01747         *t1++ = operator1;
01748         cc = operator1; operator1 = operator2; operator2 = cc;
01749         t2 = s1; while ( *t2 ) *t1++ = *t2++;
01750         x2 = x1; n2 = t1;
01751         do {
01752             *t1++ = '0' + x2 % 10;
01753             x2 /= 10;
01754         } while ( x2 );
01755         s2 = t1 - 1;
01756         while ( s2 > n2 ) { cc = *s2; *s2 = *n2; *n2++ = cc; s2--; }
01757         if ( withquestion ) *t1++ = '?';
01758     }
01759     fullsize += newsize - ( s - s1 );
01760     *n1 = c;
01761 }
01762 else { /* General case. Find the patterns first */
01763     t1 = s1; s1--;
01764     while ( s1 > Buffer ) {
01765         if ( *s1 == '<' ) break;
01766         s1--;
01767     }
01768     t2 = s2;
01769     while ( *t2 ) {
01770         if ( *t2 == '>' ) break;
01771         t2++;
01772     }
01773     if ( *s1 != '<' || *t2 != '>' ) {
01774         MesPrint("&Illegal attempt to use ... operator");
01775         return(-1);
01776     }
01777     s1++; s2++; /* Pointers to the patterns */
01778     nums = (int *)Malloc1((t1-s1)*2*(sizeof(int)+sizeof(UBYTE))
01779         ,"Expand ...");
01780     strngs = (UBYTE *) (nums + 2*(t1-s1));
01781     n1 = s1; n2 = s2; ii = -1; i = 0;
01782     s = strngs;

```

```

01783 while ( n1 < t1 || n2 < t2 ) {
01784     /* Check the next characters can be parsed as numbers including signs. */
01785     if ( CanParseSignedNumber(n1) && CanParseSignedNumber(n2) ) {
01786         /*
01787          * Don't allow the cases that one has the sign and the other doesn't,
01788          * and the meaning changes without the sign. For example,
01789          * <f(1)>+...+<f(3)>      Allowed
01790          * <f(-2)>+...+<f(2)>     Allowed
01791          * <f(x-2)>+...+<f(x+2)>   Allowed
01792          * <f(x-2)>+...+<f(x2)>    Not allowed
01793          */
01794         int sign1 = IsSignChar(*n1);
01795         int sign2 = IsSignChar(*n2);
01796         int inword1 = s1 < n1 && IsAlphanumericChar(n1[-1]);
01797         int inword2 = s2 < n2 && IsAlphanumericChar(n2[-1]);
01798         if ( ( sign1 ^ sign2 ) && ( inword1 || inword2 ) ) break; /* Not allowed. */
01799         if ( sign1 || sign2 ) {
01800             *s++ = '+'; /* Marker indicating we need the sign. */
01801         }
01802     } else {
01803         /* If they are not numbers, they should be same. */
01804         if ( *n1 == *n2 ) { *s++ = *n1++; n2++; continue; }
01805         else break;
01806     }
01807     ParseSignedNumber(x1,n1)
01808     ParseSignedNumber(x2,n2)
01809     if ( x1 == x2 ) {
01810         if ( s != strngs && ( s[-1] == '+' || s[-1] == '-' ) ) {
01811             /* We need the sign. */
01812             s--;
01813             if ( x1 >= 0 ) {
01814                 *s++ = '+';
01815             }
01816         }
01817         s = NumCopy(x1, s);
01818     }
01819     else {
01820         nums[2*i] = x1; nums[2*i+1] = x2;
01821         i++; *s++ = 0;
01822     }
01823 }
01824 if ( n1 < t1 || n2 < t2 ) {
01825     MesPrint("&Improper use of ... operator.");
01826 theend: M_free(nums,"Expand ...");
01827     return(-1);
01828 }
01829 *s = 0;
01830 if ( i == 0 ) ii = 0;
01831 else {
01832     ii = nums[0] - nums[1];
01833     if ( ii < 0 ) ii = -ii;
01834     for ( x1 = 1; x1 < i; x1++ ) {
01835         x2 = nums[2*x1]-nums[2*x1+1];
01836         if ( x2 < 0 ) x2 = -x2;
01837         if ( x2 != ii ) {
01838             MesPrint("&Improper synchronization of numbers in ... operator");
01839             goto theend;
01840         }
01841     }
01842 }
01843 ii++;
01844 /*
01845 We have now proper syntax.
01846 There are i+1 strings in strngs and i pairs of numbers
01847 in nums. Each time a start value and a finish value.
01848 We have ii steps. If ii <= 2, it will fit in the existing
01849 allocation. But this is hardly useful.
01850 We make a new allocation and copy from the old.
01851 Compute space.
01852 */
01853 x2 = s - strngs - i; /* -1 for end-of-string and +1 for the operator*/
01854 for ( i1 = 0; i1 < i; i1++ ) {
01855     i2 = nums[2*i1];
01856     x1 = nums[2*i1+1];
01857     if ( i2 < 0 ) i2 = -i2;
01858     if ( x1 < 0 ) x1 = -x1;
01859     if ( x1 > i2 ) i2 = x1;
01860     x1 = 2;
01861     while ( i2 > 0 ) { i2 /= 10; x1++; }
01862     x2 += x1;
01863 }
01864 x2 *= ii; /* Space for the expanded string (a bit more) */
01865 x2 += fullsize;
01866 x2 += 5; /* This will definitely hold everything */
01867 x2 += sizeof(UBYTE *);
01868 x2 = x2 - (x2 & (sizeof(UBYTE *)-1));
01869

```

```

01870         nBuffer = (UBYTE *)Malloca1(x2,"input buffer");
01871         n1 = nBuffer; s = Buffer; s1--;
01872         while ( s < s1 ) *n1++ = *s++;
01873     /*
01874     Solution of the special case that no comma was generated
01875     due to the presence of < to start the pattern.
01876     We get a comma when the word before ends in an alphanumeric
01877     character, a _ or a ] and the word inside starts with an
01878     alphanumeric character, a [ (or an _ (for future considerations))
01879     */
01880     if ( ( ( n1 > nBuffer ) && ( ( FG.cTable[n1[-1]] <= 1 )
01881     || ( n1[-1] == '_' ) || ( n1[-1] == ']' ) ) ) &&
01882     ( ( FG.cTable[strngs[0]] <= 1 ) || ( strngs[0] == '[' )
01883     || ( strngs[0] == '_' ) ) ) *n1++ = ',';
01884
01885     for ( i1 = 0; i1 < ii; i1++ ) {
01886         s = strngs; while ( *s ) *n1++ = *s++;
01887         for ( i2 = 0; i2 < i; i2++ ) {
01888             if ( n1 > nBuffer && IsSignChar(n1[-1]) ) {
01889                 /* We need the sign of counters. */
01890                 n1--;
01891                 if ( nums[2*i2] >= 0 ) {
01892                     *n1++ = '+';
01893                 }
01894             }
01895             n1 = NumCopy((WORD)(nums[2*i2]),n1);
01896             if ( nums[2*i2] > nums[2*i2+1] ) nums[2*i2]--;
01897             else nums[2*i2]++;
01898             s++; while ( *s ) *n1++ = *s++;
01899         }
01900         if ( ( i1 & 1 ) == 0 ) *n1++ = operator1;
01901         else *n1++ = operator2;
01902     }
01903     n1--; /* drop the trailing operator */
01904     s = t2 + 1; n2 = n1;
01905 /*
01906 Similar extra comma
01907 */
01908     if ( ( ( ( FG.cTable[n1[-1]] <= 1 )
01909     || ( n1[-1] == '_' ) || ( n1[-1] == ']' ) ) ) &&
01910     ( ( FG.cTable[s[0]] <= 1 ) || ( s[0] == '[' )
01911     || ( s[0] == '_' ) ) ) *n1++ = ',';
01912
01913     while ( *s ) *n1++ = *s++;
01914     *n1 = 0;
01915     if ( par == 0 ) {
01916         LONG nnn1 = n1-nBuffer;
01917         LONG nnn2 = n2-nBuffer;
01918         LONG nnn3;
01919         while ( AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel] + x2 >= AC.iStop ) {
01920             LONG position = s-Buffer;
01921             LONG position2 = AC.iPointer - AC.iBuffer;
01922             UBYTE **ppp;
01923             ppp = &(AC.iBuffer); /* to avoid a compiler warning */
01924             if ( DoubleLList((void ***)ppp,&AC.iBufferSize
01925             ,sizeof(UBYTE),"statement buffer") ) {
01926                 Terminate(-1);
01927             }
01928             AC.iPointer = AC.iBuffer + position2;
01929             AC.iStop = AC.iBuffer + AC.iBufferSize-2;
01930             Buffer = AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]; Stop = AC.iStop;
01931             s = Buffer + position;
01932         }
01933     /*
01934     This can be improved. We only have to start from the first term.
01935     */
01936     for ( nnn3 = 0; nnn3 < nnn1; nnn3++ ) Buffer[nnn3] = nBuffer[nnn3];
01937     Buffer[nnn3] = 0;
01938     n1 = Buffer + nnn1;
01939     n2 = Buffer + nnn2;
01940     M_free(nBuffer,"input buffer");
01941     M_free(nums,"Expand ...");
01942 }
01943 else { /* Comes here only inside a real preprocessor instruction */
01944     AP.preStop = nBuffer + x2 - 2;
01945     AP.pSize = x2;
01946     M_free(AP.preStart,"input buffer");
01947     M_free(nums,"Expand ...");
01948     AP.preStart = nBuffer;
01949     Buffer = AP.preStart; Stop = AP.preStop;
01950 }
01951 fullsize = n1 - Buffer;
01952 s = n2;
01953 }
01954 }
01955 return(error);
01956 }

```

```

01957
01958 /*
01959     #] ExpandTripleDots :
01960     #[ FindKeyWord :
01961 */
01962
01963 KEYWORD *FindKeyWord(UBYTE *theword, KEYWORD *table, int size)
01964 {
01965     int low,med,hi;
01966     UBYTE *s1, *s2;
01967     low = 0;
01968     hi = size-1;
01969     while ( hi >= low ) {
01970         med = (hi+low)/2;
01971         s1 = (UBYTE *) (table[med].name);
01972         s2 = theword;
01973         while ( *s1 && tolower(*s1) == tolower(*s2) ) { s1++; s2++; }
01974         if ( *s1 == 0 &&
01975 /*[30apr2004 mt]:*/
01976 /* The bug!:
01977         FG.cTable[*s2] != 1 && FG.cTable[*s2] != 2
01978 */
01979         FG.cTable[*s2] != 0 && FG.cTable[*s2] != 1
01980 /*         ( *s2 == ' ' || *s2 == '\t' || *s2 == 0 || *s2 == ',' || *s2 == '(' ) */
01981         )
01982             return(table+med);
01983         if ( tolower(*s2) > tolower(*s1) ) low = med+1;
01984         else hi = med - 1;
01985     }
01986     return(0);
01987 }
01988
01989 /*
01990     #] FindKeyWord :
01991     #[ FindInKeyWord :
01992 */
01993
01994 KEYWORD *FindInKeyWord(UBYTE *theword, KEYWORD *table, int size)
01995 {
01996     int i;
01997     UBYTE *s1, *s2;
01998     for ( i = 0; i < size; i++ ) {
01999         s1 = (UBYTE *) (table[i].name);
02000         s2 = theword;
02001         while ( *s1 && tolower(*s1) == tolower(*s2) ) { s1++; s2++; }
02002         if ( *s2 == 0 || *s2 == ' ' || *s2 == ',' || *s2 == '\t' )
02003             return(table+i);
02004     }
02005     return(0);
02006 }
02007
02008 /*
02009     #] FindInKeyWord :
02010     #[ TheDefine :
02011 */
02012
02024 int TheDefine(UBYTE *s, int mode)
02025 {
02026     UBYTE *name, *value, *valpoin, *args = 0, c;
02027     if ( ( mode & 2 ) == 0 ) {
02028         if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
02029         if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
02030     }
02031     else { mode &= ~2; }
02032     name = s;
02033     if ( chartype[*s] != 0 ) goto illname;
02034     s++;
02035     while ( chartype[*s] <= 1 ) s++;
02036     value = s;
02037     while ( *s == ' ' || *s == '\t' ) s++;
02038     c = *s; *value = 0;
02039     if ( c == 0 ) {
02040         if ( PutPreVar(name, (UBYTE *) "1", 0, mode) < 0 ) return(-1);
02041         return(0);
02042     }
02043     if ( c == '(' ) { /* arguments. scan for correctness */
02044         s++; args = s;
02045         for (;;) {
02046             if ( chartype[*s] != 0 ) goto illarg;
02047             s++;
02048             while ( chartype[*s] <= 1 ) s++;
02049             while ( *s == ' ' || *s == '\t' ) s++;
02050             if ( *s == ')' ) break;
02051             if ( *s != ',' ) goto illargs;
02052             s++;
02053             while ( *s == ' ' || *s == '\t' ) s++;
02054         }

```

```

02055     *s++ = 0;
02056     while ( *s == ' ' || *s == '\t' ) s++;
02057     c = *s;
02058 }
02059 if ( c == '"' ) {
02060     s++; valpoin = value = s;
02061     while ( *s != '"' ) {
02062         if ( *s == '\\' ) {
02063             if ( s[1] == 'n' ) { *valpoin++ = LINEFEED; s += 2; }
02064             else if ( s[1] == '"' ) { *valpoin++ = '"'; s += 2; }
02065             else if ( s[1] == 0 ) goto illval;
02066             else { *valpoin++ = *s++; *valpoin++ = *s++; }
02067         }
02068         else *valpoin++ = *s++;
02069     }
02070     *valpoin = 0;
02071     if ( PutPreVar(name,value,args,mode) < 0 ) return(-1);
02072 }
02073 else {
02074     MesPrint("@Illegal string for preprocessor variable %s. Forgotten double quotes (\") ?",name);
02075     return(-1);
02076 }
02077 return(0);
02078 illname:;
02079 MesPrint("@Illegally formed name of preprocessor variable");
02080 return(-1);
02081 illarg:;
02082 MesPrint("@Illegally formed name of argument of preprocessor definition");
02083 return(-1);
02084 illargs:;
02085 MesPrint("@Illegally formed arguments of preprocessor definition");
02086 return(-1);
02087 illval:;
02088 MesPrint("@Illegal valpoin for preprocessor variable %s",name);
02089 return(-1);
02090 }
02091
02092 /*
02093     #] TheDefine :
02094     #[ DoCommentChar :
02095 */
02096
02097 int DoCommentChar(UBYTE *s)
02098 {
02099     UBYTE c;
02100     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
02101     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
02102     while ( *s == ' ' || *s == '\t' ) s++;
02103     if ( *s == 0 || *s == '\n' ) {
02104         MesPrint("@No valid comment character specified");
02105         return(-1);
02106     }
02107     c = *s++;
02108     while ( *s == ' ' || *s == '\t' ) s++;
02109     if ( *s != 0 && *s != '\n' ) {
02110         MesPrint("@Comment character should be a single valid character");
02111         return(-1);
02112     }
02113     AP.ComChar = c;
02114     return(0);
02115 }
02116
02117 /*
02118     #] DoCommentChar :
02119     #[ DoPreAssign :
02120
02121     Routine assigns a 'value' to a $variable.
02122     Syntax: #assign
02123           next line(s) a statement of the type
02124           $name = expression;
02125     Note: at the moment of the assign there cannot be an 'open' statement.
02126 */
02127
02128 int DoPreAssign(UBYTE *s)
02129 {
02130     int error = 0;
02131     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) {
02132         return(0);
02133     }
02134     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) {
02135         return(0);
02136     }
02137     if ( *s ) {
02138         MesPrint("@Illegal characters in #assign instruction");
02139         error = 1;
02140     }
02141     PUSHPREASSIGNLEVEL;

```



```

02142     AP.PreAssignFlag = 1;
02143 /*
02144     if ( AP.PreContinuation ) {
02145         MesPrint("@Assign instructions cannot occur inside statements");
02146         MesPrint("@Missing ; ?");
02147         AP.PreContinuation = 0;
02148         error = 1;
02149     }
02150 */
02151     return(error);
02152 }
02153
02154 /*
02155     #] DoPreAssign :
02156     #[ DoDefine :
02157 */
02158
02159 int DoDefine(UBYTE *s)
02160 {
02161     return(TheDefine(s,0));
02162 }
02163
02164 /*
02165     #] DoDefine :
02166     #[ DoRedefine :
02167 */
02168
02169 int DoRedefine(UBYTE *s)
02170 {
02171     return(TheDefine(s,1));
02172 }
02173
02174 /*
02175     #] DoRedefine :
02176     #[ ClearMacro :
02177
02178     Undefines the arguments of a macro after its use.
02179 */
02180
02181 int ClearMacro(UBYTE *name)
02182 {
02183     int i;
02184     PREVAR *p;
02185     UBYTE *s;
02186     for ( i = NumPre-1, p = &(PreVar[NumPre-1]); i >= 0; i--, p-- ) {
02187         if ( StrCmp(name,p->name) == 0 ) break;
02188     }
02189     if ( i < 0 ) return(-1);
02190     if ( p->nargs <= 0 ) return(0);
02191     s = p->argnames;
02192     for ( i = 0; i < p->nargs; i++ ) {
02193         TheUndefine(s);
02194         while ( *s ) s++;
02195         s++;
02196     }
02197     return(0);
02198 }
02199
02200 /*
02201     #] ClearMacro :
02202     #[ TheUndefine :
02203
02204     There is a complication here. If there are redefine statements
02205     they will be pointing at the wrong variable if their number is
02206     greater than the number of the variable we pop.
02207 */
02208
02209 int TheUndefine(UBYTE *name)
02210 {
02211     int i, inum, error = 0;
02212     PREVAR *p;
02213     for ( i = NumPre-1, p = &(PreVar[NumPre-1]); i >= 0; i--, p-- ) {
02214         if ( StrCmp(name,p->name) == 0 ) {
02215             M_free(p->name,"undefining PreVar");
02216             NumPre--;
02217             inum = i;
02218             while ( i < NumPre ) {
02219                 p->name = p[1].name;
02220                 p->value = p[1].value;
02221                 p++; i++;
02222             }
02223             p->name = 0; p->value = 0;
02224             {
02225                 CBUF *CC = cbuf + AC.cbufnum;
02226                 int j, k;
02227                 for ( j = 1; j <= CC->numlhs; j++ ) {
02228                     if ( CC->lhs[j][0] == TYPEREDEFPRE ) {

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```

02229             if ( CC->lhs[j][2] > inum ) CC->lhs[j][2]--;
02230             else if ( CC->lhs[j][2] == inum ) {
02231                 for ( k = inum - 1; k >= 0; k-- )
02232                     if ( StrCmp(name, PreVar[k].name) == 0 ) break;
02233                 if ( k >= 0 ) CC->lhs[j][2] = k;
02234             }
02235             MesPrint("@Conflict between undefining a preprocessor variable and a
redefine statement");
02236             error = 1;
02237         }
02238     }
02239 }
02240 }
02241 #ifdef PARALLELCODE
02242     for ( j = 0; j < AC.numpfirstnum; j++ ) {
02243         if ( AC.pfirstnum[j] > inum ) AC.pfirstnum[j]--;
02244         else if ( AC.pfirstnum[j] == inum ) {
02245             for ( k = inum - 1; k >= 0; k-- )
02246                 if ( StrCmp(name, PreVar[k].name) == 0 ) break;
02247             if ( k >= 0 ) AC.pfirstnum[j] = k;
02248         }
02249     }
02250 #endif
02251 }
02252 break;
02253 }
02254 }
02255 return(error);
02256 }
02257
02258 /*
02259     #] TheUndefine :
02260     #[ DoUndefine :
02261 */
02262
02263 int DoUndefine(UBYTE *s)
02264 {
02265     UBYTE *name, *t;
02266     int error = 0, retval;
02267 /*
02268     int i;
02269     PREVAR *p;
02270 */
02271     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
02272     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
02273     name = s;
02274     if ( chartype[*s] != 0 ) goto illname;
02275     s++;
02276     while ( chartype[*s] <= 1 ) s++;
02277     t = s;
02278     if ( *s && *s != ' ' && *s != '\t' ) goto illname;
02279     while ( *s == ' ' || *s == '\t' ) s++;
02280     if ( *s ) {
02281         MesPrint("@Undefine should just have a variable name");
02282         error = -1;
02283     }
02284     *t = 0;
02285     if ( ( retval = TheUndefine(name) ) != 0 ) {
02286         if ( error == 0 ) return(retval);
02287         if ( error > 0 ) error = retval;
02288     }
02289 /*
02290     for ( i = NumPre-1, p = &(PreVar[NumPre-1]); i >= 0; i--, p-- ) {
02291         if ( StrCmp(name,p->name) == 0 ) {
02292             M_free(p->name,"undefining PreVar");
02293             NumPre--;
02294             while ( i < NumPre ) {
02295                 p->name = p[1].name;
02296                 p->value = p[1].value;
02297                 p++; i++;
02298             }
02299             p->name = 0; p->value = 0;
02300             break;
02301         }
02302     }
02303 */
02304     return(error);
02305 illname:;
02306     MesPrint("@Illegally formed name of preprocessor variable");
02307     return(-1);
02308 }
02309
02310 /*
02311     #] DoUndefine :
02312     #[ DoInclude :
02313 */
02314

```

```

02315 int DoInclude(UBYTE *s) { return(Include(s,FILESTREAM)); }
02316
02317 /*
02318     #] DoInclude :
02319     #[ DoReverseInclude :
02320 */
02321
02322 int DoReverseInclude(UBYTE *s) { return(Include(s,REVERSEFILESTREAM)); }
02323
02324 /*
02325     #] DoReverseInclude :
02326     #[ Include :
02327 */
02328
02329 int Include(UBYTE *s, int type)
02330 {
02331     UBYTE *name = s, *fold, *t, c, c1 = 0, c2 = 0, c3 = 0;
02332     int strloffset, withnolist = AC.NoShowInput;
02333     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
02334     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
02335     if ( *s == '-' || *s == '+' ) {
02336         if ( *s == '-' ) withnolist = 1;
02337         else withnolist = 0;
02338         s++;
02339         while ( *s == ' ' || *s == '\t' ) s++;
02340         name = s;
02341     }
02342     if ( *s == '"' ) {
02343         while ( *s && *s != '"' ) {
02344             if ( *s == '\\' ) s++;
02345             s++;
02346         }
02347         t = s++;
02348     }
02349     else {
02350         while ( *s && *s != ' ' && *s != '\t' ) {
02351             if ( *s == '\\' ) s++;
02352             s++;
02353         }
02354         t = s;
02355     }
02356     while ( *s == ' ' || *s == '\t' ) s++;
02357     if ( *s == '#' ) {
02358         *t = 0;
02359         s++;
02360         while ( *s == ' ' || *s == '\t' ) s++;
02361         fold = s;
02362         if ( *s == 0 ) {
02363             MesPrint("@Empty fold name");
02364             return(-1);
02365         }
02366     continue_fold:
02367         while ( *s && *s != ' ' && *s != '\t' ) {
02368             if ( *s == '\\' ) s++;
02369             s++;
02370         }
02371         t = s;
02372         while ( *s == ' ' || *s == '\t' ) s++;
02373         if ( *s ) {
02374             /*
02375              * A non-whitespace character is found. Continue parsing the fold.
02376              */
02377             goto continue_fold;
02378         }
02379     }
02380     else if ( *s == 0 ) {
02381         fold = 0;
02382     }
02383     else {
02384         MesPrint("@Improper syntax for file name");
02385         return(-1);
02386     }
02387     *t = 0;
02388     if ( fold ) {
02389         fold = strDup1(fold,"foldname");
02390     }
02391     /*
02392     We have the name of the file in 'name' and the fold in 'fold' (or NULL)
02393     */
02394     if ( OpenStream(name,type,0,PREENOACTION) == 0 ) {
02395         if ( fold ) { M_free(fold,"foldname"); fold = 0; }
02396         return(-1);
02397     }
02398     if ( fold ) {
02399         LONG position = -1;
02400         int foldopen = 0;
02401         LONG linenum = 0, prevline = 0;

```

```

02402     name = strDup1(name,"name of include file");
02403     AC.CurrentStream->FoldName = strDup1(fold,"name of fold");
02404     AC.NoShowInput++;
02405     for(;;) {
02406         c = GetFromStream(AC.CurrentStream);
02407         if ( c == ENDOFSTREAM ) {
02408             AC.CurrentStream = CloseStream(AC.CurrentStream);
02409             goto nofold;
02410         }
02411         if ( c == AP.ComChar ) {
02412             strloffset = AC.CurrentStream-AC.Streams;
02413             LoadInstruction(1);
02414             if ( AC.CurrentStream != strloffset+AC.Streams ) {
02415                 c = ENDOFSTREAM;
02416             }
02417             else {
02418                 t = AP.preStart;
02419                 if ( t[2] == '#' && ( t[3] == '[' && !foldopen )
02420                     || ( t[3] == ']' && foldopen ) ) {
02421                     t += 4;
02422                     while ( *t == ' ' || *t == '\t' ) t++;
02423                     s = AC.CurrentStream->FoldName;
02424                     while ( *s == *t ) { s++; t++; }
02425                     if ( *s == 0 && ( *t == ' ' || *t == '\t'
02426                         || *t == ':' ) ) {
02427                         while ( *t == ' ' || *t == '\t' ) t++;
02428                         if ( *t == ':' ) {
02429                             if ( foldopen == 0 ) {
02430                                 foldopen = 1;
02431                                 position = GetStreamPosition(AC.CurrentStream);
02432                                 linenum = AC.CurrentStream->linenum;
02433                                 prevline = AC.CurrentStream->prevline;
02434                                 c3 = AC.CurrentStream->isnextchar;
02435                                 c1 = AC.CurrentStream->nextchar[0];
02436                                 c2 = AC.CurrentStream->nextchar[1];
02437                             }
02438                             else {
02439                                 foldopen = 0;
02440                                 PositionStream(AC.CurrentStream,position);
02441                                 AC.CurrentStream->linenum = linenum;
02442                                 AC.CurrentStream->prevline = prevline;
02443                                 AC.CurrentStream->eqnum = 1;
02444                                 AC.NoShowInput--;
02445                                 AC.CurrentStream->isnextchar = c3;
02446                                 AC.CurrentStream->nextchar[0] = c1;
02447                                 AC.CurrentStream->nextchar[1] = c2;
02448                                 break;
02449                             }
02450                         }
02451                     }
02452                 }
02453             }
02454         }
02455         else {
02456             while ( c != LINEFEED && c != ENDOFSTREAM ) {
02457                 c = GetFromStream(AC.CurrentStream);
02458                 if ( c == ENDOFSTREAM ) {
02459                     AC.CurrentStream = CloseStream(AC.CurrentStream);
02460                     break;
02461                 }
02462             }
02463         }
02464         if ( c == ENDOFSTREAM ) {
02465 nofold:
02466             MesPrint("@Cannot find fold %s in file %s",fold,name);
02467             UngetChar(c);
02468             AC.NoShowInput--;
02469             M_free(name,"name of include file");
02470             Terminate(-1);
02471         }
02472     }
02473     M_free(name,"name of include file");
02474 }
02475 AC.NoShowInput = withnolist;
02476 if ( fold ) { M_free(fold,"foldname"); fold = 0; }
02477 return(0);
02478 }
02479
02480 /*
02481 #] Include :
02482 #[ DoPreExchange :
02483
02484 Exchanges the names of expressions or the contents of dollars
02485 Syntax:
02486     #exchange expr1,expr2
02487     #exchange $var1,$var2
02488 */

```

```

02489
02490 int DoPreExchange(UBYTE *s)
02491 {
02492     int error = 0;
02493     UBYTE *s1, *s2;
02494     WORD num1, num2;
02495     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
02496     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
02497     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
02498     if ( *s == '$' ) {
02499         s++; s1 = s; while ( FG.cTable[*s] <= 1 ) s++;
02500         if ( *s != ',' && *s != ' ' && *s != '\t' ) goto syntax;
02501         *s++ = 0;
02502         while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
02503         if ( *s != '$' ) goto syntax;
02504         s++; s2 = s; while ( FG.cTable[*s] <= 1 ) s++;
02505         if ( *s != 0 && *s != ';' ) goto syntax;
02506         *s = 0;
02507         if ( ( num1 = GetDollar(s1) ) <= 0 ) {
02508             MesPrint("@$%s has not been defined (yet)",s1);
02509             error = 1;
02510         }
02511         if ( ( num2 = GetDollar(s2) ) <= 0 ) {
02512             MesPrint("@$%s has not been defined (yet)",s2);
02513             error = 1;
02514         }
02515         if ( error == 0 ) {
02516             ExchangeDollars((int)num1,(int)num2);
02517         }
02518     }
02519     else {
02520         s1 = s; s = SkipAName(s);
02521         if ( *s != ',' && *s != ' ' && *s != '\t' ) goto syntax;
02522         *s++ = 0;
02523         while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
02524         if ( FG.cTable[*s] != 0 && *s != '[' ) goto syntax;
02525         s2 = s; s = SkipAName(s);
02526         if ( *s != 0 && *s != ';' ) goto syntax;
02527         *s = 0;
02528         if ( GetName(AC.exprnames,s1,&num1,NOAUTO) != CEXPRESSION ) {
02529             MesPrint("@%s is not an expression",s1);
02530             error = 1;
02531         }
02532         if ( GetName(AC.exprnames,s2,&num2,NOAUTO) != CEXPRESSION ) {
02533             MesPrint("@%s is not an expression",s2);
02534             error = 1;
02535         }
02536         if ( error == 0 ) {
02537             ExchangeExpressions((int)num1,(int)num2);
02538         }
02539     }
02540     return(error);
02541 syntax:
02542     MesPrint("@Proper syntax: %#exchange expr1,expr2 or %#exchange $var1,$var2");
02543     return(1);
02544 }
02545
02546 /*
02547     #] DoPreExchange :
02548     #[ DoCall :
02549 */
02550
02551 int DoCall(UBYTE *s)
02552 {
02553     UBYTE *t, *u, *v, *name, c, cp, *args1, *args2, *t1, *t2, *wild = 0;
02554     int bratype = 0, wildargs = 0, inwildargs = 0, nwildargs = 0;
02555     PROCEDURE *p;
02556     int streamoffset;
02557     int i, namesize, narg1, narg2, bralevel, numpre;
02558     LONG i1, i2;
02559     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
02560     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
02561     /*
02562     1: Get the name of the procedure.
02563     2: Locate the procedure.
02564     */
02565     name = s; s = EndOfToken(s); c = *s; *s = 0;
02566     for ( i = NumProcedures-1; i >= 0; i-- ) {
02567         if ( StrCmp(Procedures[i].name,name) == 0 ) break;
02568     }
02569     p = (PROCEDURE *)FromList(&AP.ProcList);
02570     if ( i < 0 ) { /* Try to find a file */
02571         namesize = 0;
02572         t = name;
02573         while ( *t ) { t++; namesize++; }
02574         t = AP.procedureExtension;
02575         while ( *t ) { t++; namesize++; }

```

```

02576     t = p->name = (UBYTE *)Malloca1(namesize+2,"procedure");
02577     u = name;
02578     while ( *u ) *t++ = *u++;
02579     *t++ = '.';
02580     v = AP.procedureExtension;
02581     while ( *v ) *t++ = *v++;
02582     *t = 0;
02583     p->loadmode = 0; /* buffer should be freed at end */
02584     p->p.buffer = LoadInputFile(p->name,PROCEDUREFILE);
02585     if ( p->p.buffer == 0 ) return(-1);
02586     t[-4] = 0;
02587 }
02588 else {
02589     p->p.buffer = Procedures[i].p.buffer;
02590     p->name = Procedures[i].name;
02591     p->loadmode = 1;
02592 }
02593 t = p->p.buffer;
02594 SKIPBLANKS(t)
02595 if ( *t++ != '#' ) goto wrongfile;
02596 SKIPBLANKS(t)
02597 t += 9;
02598 SKIPBLANKS(t)
02599 u = EndOfToken(t);
02600 cp = *u; *u = 0;
02601 if ( StrCmp(t,name) != 0 ) goto wrongfile;
02602 *u = cp;
02603 *s = c;
02604 /*
02605 The pointer p points to the contents of the procedure (in memory)
02606 Now we have to match the arguments. u points to after the name
02607 in the 'file', s to after the name in the call statement.
02608 */
02609 bralevel = nargs1 = nargs2 = 0; args2 = u;
02610 SKIPBLANKS(u)
02611 if ( *u == '(' ) {
02612     u++; SKIPBLANKS(u)
02613     args2 = u;
02614     while ( *u != ')' ) {
02615         if ( *u == '?' ) { wildargs++; u++; nwildargs = nargs2+1; }
02616         nargs2++; u = EndOfToken(u); SKIPBLANKS(u)
02617         if ( *u == ',' ) { u++; SKIPBLANKS(u) }
02618         else if ( *u != ')' || ( wildargs > 1 ) ) {
02619             MesPrint("@Illegal argument field in procedure %s",p->name);
02620             return(-1);
02621         }
02622     }
02623 }
02624 while ( *u != LINEFEED ) u++;
02625 SKIPBLANKS(s)
02626 args1 = s+1;
02627 if ( *s == '(' ) bratype = 1;
02628 do {
02629     if ( *s == '{' && bratype == 0 ) bralevel++;
02630     else if ( *s == '(' && bratype == 1 ) bralevel++;
02631     else if ( *s == ')' && bratype == 0 ) {
02632         bralevel--;
02633         if ( bralevel == 0 ) {
02634             *s = 0; nargs1++;
02635             if ( wildargs && nargs1 == nwildargs ) wild = s;
02636         }
02637     }
02638     else if ( *s == ')' && bratype == 1 ) {
02639         bralevel--;
02640         if ( bralevel == 0 ) {
02641             *s = 0; nargs1++;
02642             if ( wildargs && nargs1 == nwildargs ) wild = s;
02643         }
02644     }
02645     /*[12dec2003 mt]:*/
02646     /*else if ( *s == ',' || *s == '|' ) {*/
02647     else if (set_in(*s,AC.separators)) {/*Function set_in see in
02648                                         file tools.c*/
02649         /*:[12dec2003 mt]*/
02650         *s = 0; nargs1++;
02651         if ( wildargs && nargs1 == nwildargs ) wild = s;
02652     }
02653     else if ( *s == '\\\\' ) s++;
02654     s++;
02655 } while ( bralevel > 0 );
02656 if ( wildargs && nargs1 >= nargs2-1 ) {
02657     inwildargs = nargs1-nargs2+1;
02658     if ( inwildargs == 0 ) nwildargs = 0;
02659     else {
02660         while ( inwildargs > 1 ) {
02661             *wild = ',';
02662             while ( *wild ) wild++;

```

```

02663         inwildargs--;
02664     }
02665 }
02666 }
02667 else if ( narg1 != narg2 && ( narg2 != 0 || narg1 != 1 || *args1 != 0 ) ) {
02668     MesPrint("@Arguments of procedure %s are not matching",p->name);
02669     return(-1);
02670 }
02671 numpre = -NumPre-1; /* For the stream */
02672 for ( i = 0; i < narg2; i++ ) {
02673     t = args2;
02674     if ( *t == '?' ) {
02675         args2++;
02676     }
02677     if ( *t == '?' && inwildargs == 0 ) {
02678         args2 = EndOfToken(args2); c = *args2; *args2 = 0;
02679         if ( PutPreVar(t, (UBYTE *)"", 0, 0) < 0 ) return(-1);
02680     }
02681     else {
02682         args2 = EndOfToken(args2); c = *args2; *args2 = 0;
02683         t1 = t2 = args1;
02684         while ( *t1 ) {
02685             if ( *t1 == '\\\\' ) t1++;
02686             if ( t1 != t2 ) *t2 = *t1;
02687             t2++; t1++;
02688         }
02689         *t2 = 0;
02690         if ( PutPreVar(t, args1, 0, 0) < 0 ) return(-1);
02691         args1 = t1+1; /* Next argument */
02692     }
02693     *args2 = c; SKIPBLANKS(args2) /* skip to next name */
02694     args2++; SKIPBLANKS(args2)
02695 }
02696 streamoffset = AC.CurrentStream - AC.Streams;
02697 args1 = AC.CurrentStream->name;
02698 AC.CurrentStream->name = p->name;
02699 i1 = AC.CurrentStream->linenumber;
02700 i2 = AC.CurrentStream->prevline;
02701 AC.CurrentStream->prevline =
02702 AC.CurrentStream->linenumber = 2;
02703 OpenStream(u+1, PREREADSTREAM3, numpre, PRENOACTION);
02704 AC.Streams[streamoffset].name = args1;
02705 AC.Streams[streamoffset].linenumber = i1;
02706 AC.Streams[streamoffset].prevline = i2;
02707 AddToPreTypes(PRETYPEPROCEDURE);
02708 return(0);
02709 wrongfile;;
02710 if ( i < 0 ) MesPrint("@File %s is not a proper procedure",p->name);
02711 else MesPrint("!!!Internal error with procedure names: %s",name);
02712 return(-1);
02713 }
02714
02715 /*
02716     #] DoCall :
02717     #[ DoDebug :
02718 */
02719
02720 int DoDebug(UBYTE *s)
02721 {
02722     int x;
02723     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
02724     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
02725     NeedNumber(x,s,nonumber)
02726     if ( x < 0 || x > (PREPROONLY
02727         | DUMPTOCOMPILER
02728         | DUMPOUTTERMS
02729         | DUMPINTERMS
02730         | DUMPTOSORT
02731         | DUMPTOPARALLEL
02732 #ifdef WITHPTHREADS
02733         | THREADSDEBUG
02734 #endif
02735     ) ) goto nonumber;
02736     AP.PreDebug = 0;
02737     if ( ( x & PREPROONLY ) != 0 ) AP.PreDebug |= PREPROONLY; /* 1 */
02738     if ( ( x & DUMPTOCOMPILER ) != 0 ) AP.PreDebug |= DUMPTOCOMPILER; /* 2 */
02739     if ( ( x & DUMPOUTTERMS ) != 0 ) AP.PreDebug |= DUMPOUTTERMS; /* 4 */
02740     if ( ( x & DUMPINTERMS ) != 0 ) AP.PreDebug |= DUMPINTERMS; /* 8 */
02741     if ( ( x & DUMPTOSORT ) != 0 ) AP.PreDebug |= DUMPTOSORT; /* 16 */
02742     if ( ( x & DUMPTOPARALLEL ) != 0 ) AP.PreDebug |= DUMPTOPARALLEL; /* 32 */
02743 #ifdef WITHPTHREADS
02744     if ( ( x & THREADSDEBUG ) != 0 ) AP.PreDebug |= THREADSDEBUG; /* 64 */
02745 #endif
02746     return(0);
02747 nonumber:
02748     MesPrint("@Illegal argument for debug instruction");
02749     return(1);

```

```

02750 }
02751
02752 /*
02753     #] DoDebug :
02754     #[ DoTerminate :
02755 */
02756
02757 int DoTerminate(UBYTE *s)
02758 {
02759     int x;
02760     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
02761     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
02762     if ( *s ) {
02763         NeedNumber(x,s,nonumber)
02764         Terminate(x);
02765     }
02766     else {
02767         Terminate(-1);
02768     }
02769     return(0);
02770 nonumber:
02771     MesPrint("@Illegal argument for terminate instruction");
02772     return(1);
02773 }
02774
02775 /*
02776     #] DoTerminate :
02777     #[ DoContinueDo :
02778 */
02779
02780 int DoContinueDo(UBYTE *s)
02781 {
02782     DOLOOP *loop;
02783
02784     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
02785     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
02786
02787     if ( NumDoLoops <= 0 ) {
02788         MesPrint("@%#continuedo without %#do");
02789         return(1);
02790     }
02791
02792     loop = &(DoLoops[NumDoLoops-1]);
02793     AP.NumPreTypes = loop->NumPreTypes+1;
02794     AP.PreIfLevel = loop->PreIfLevel;
02795     AP.PreSwitchLevel = loop->PreSwitchLevel;
02796
02797     return(DoEnddo(s));
02798 }
02799
02800 /*
02801     #] DoContinueDo :
02802     #[ DoDo :
02803
02804     The do loop has three varieties:
02805     #do i = num1,num2 [,num3]
02806     #do i = {string1,string2,...,stringn}
02807         The | as separator is also allowed for backwards compatibility
02808     #do i = expression      One by one all terms of the expression
02809 */
02810
02811 int DoDo(UBYTE *s)
02812 {
02813     GETIDENTITY
02814     UBYTE *t, c, *u, *uu;
02815     DOLOOP *loop;
02816     WORD expnum;
02817     LONG linenum = AC.CurrentStream->linenumber;
02818     int oldNoShowInput = AC.NoShowInput, i, oldpreassignflag;
02819
02820     if ( ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH )
02821         || ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) ) {
02822         if ( PreSkip((UBYTE *) "do", (UBYTE *) "enddo",1) ) return(-1);
02823         return(0);
02824     }
02825
02826 /*
02827     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
02828     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
02829 */
02830     AddToPreTypes(PRETYPEPO);
02831
02832     loop = (DOLOOP *)FromList(&AP.LoopList);
02833     loop->firstdollar = loop->lastdollar = loop->incdollar = -1;
02834     loop->NumPreTypes = AP.NumPreTypes-1;
02835     loop->PreIfLevel = AP.PreIfLevel;
02836     loop->PreSwitchLevel = AP.PreSwitchLevel;

```



```

02837     AC.NoShowInput = 1;
02838     if ( PreLoad(&(loop->p), (UBYTE *) "do", (UBYTE *) "enddo", 1, "doloop") ) return(-1);
02839     AC.NoShowInput = oldNoShowInput;
02840     loop->NoShowInput = AC.NoShowInput;
02841     /*
02842     Get now the name. We have to take great care when the name is terminated!
02843     */
02844     s = loop->p.buffer + (s - AP.preStart);
02845     SKIPBLANKS(s);
02846     loop->name = s;
02847     if ( chartye[*s] != 0 ) goto illname;
02848     s++;
02849     while ( chartye[*s] <= 1 ) s++;
02850     t = s;
02851     while ( *s == ' ' || *s == '\t' ) s++;
02852     if ( *s != '=' ) goto illdo;
02853     s++;
02854     while ( *s == ' ' || *s == '\t' ) s++;
02855     *t = 0;
02856
02857     if ( *s == '{' ) {
02858         loop->type = LISTEDLOOP;
02859         s++; loop->vars = s;
02860         loop->lastnum = 0;
02861         while ( *s != '}' && *s != 0 ) {
02862             if ( set_in(*s, AC.separators) ) { *s = 0; loop->lastnum++; }
02863             else if ( *s == '\\' ) s++;
02864             s++;
02865         }
02866         if ( *s == 0 ) goto illdo;
02867         *s++ = 0;
02868         loop->lastnum++;
02869         loop->firstnum = 0;
02870         loop->contents = s;
02871     }
02872     else if ( *s == '-' || *s == '+' || chartye[*s] == 1 || *s == '$' ) {
02873         loop->type = NUMERICLOOP;
02874         t = s;
02875         while ( *s && *s != ',' ) s++;
02876         if ( *s == 0 ) goto illdo;
02877         if ( *t == '$' ) {
02878             c = *s; *s = 0;
02879             if ( GetName(AC.dollarnames, t+1, &loop->firstdollar, NOAUTO) != CDOLLAR ) {
02880                 MesPrint("@%s is undefined in first parameter in %sdo instruction", t);
02881                 return(-1);
02882             }
02883             loop->firstnum = DolToLong(BHEAD loop->firstdollar);
02884             if ( AN.ErrorInDollar ) {
02885                 MesPrint("@%s does not evaluate into a valid loop parameter", t);
02886                 return(-1);
02887             }
02888             *s++ = c;
02889         }
02890         else {
02891             *s = ',';
02892             if ( PreEval(t, &loop->firstnum) == 0 ) goto illdo;
02893             *s++ = ',';
02894         }
02895         t = s;
02896         while ( *s && *s != ',' && *s != ';' && *s != LINEFEED ) s++;
02897         c = *s;
02898         if ( *t == '$' ) {
02899             *s = 0;
02900             if ( GetName(AC.dollarnames, t+1, &loop->lastdollar, NOAUTO) != CDOLLAR ) {
02901                 MesPrint("@%s is undefined in second parameter in %sdo instruction", t);
02902                 return(-1);
02903             }
02904             loop->lastnum = DolToLong(BHEAD loop->lastdollar);
02905             if ( AN.ErrorInDollar ) {
02906                 MesPrint("@%s does not evaluate into a valid loop parameter", t);
02907                 return(-1);
02908             }
02909             *s++ = c;
02910         }
02911         else {
02912             *s = ',';
02913             if ( PreEval(t, &loop->lastnum) == 0 ) goto illdo;
02914             *s++ = c;
02915         }
02916         if ( c == ',' ) {
02917             t = s;
02918             while ( *s && *s != ';' && *s != LINEFEED ) s++;
02919             if ( *t == '$' ) {
02920                 c = *s; *s = 0;
02921                 if ( GetName(AC.dollarnames, t+1, &loop->incdollar, NOAUTO) != CDOLLAR ) {
02922                     MesPrint("@%s is undefined in third parameter in %sdo instruction", t);
02923                     return(-1);

```

```

02924         }
02925         loop->incnum = DolToLong(BHEAD loop->incdollar);
02926         if ( AN.ErrorInDollar ) {
02927             MesPrint("@%s does not evaluate into a valid loop parameter",t);
02928             return(-1);
02929         }
02930         *s++ = c;
02931     }
02932     else {
02933         c = *s; *s = ' ';
02934         if ( PreEval(t,&loop->incnum) == 0 ) goto illdo;
02935         *s++ = c;
02936     }
02937 }
02938 else loop->incnum = 1;
02939 loop->contents = s;
02940 }
02941 else if ( ( chartype[*s] == 0 ) || ( *s == '[' ) ) {
02942     int oldNumPotModdollars = NumPotModdollars;
02943 #ifdef WITHMPI
02944     WORD oldRhsExprInModuleFlag = AC.RhsExprInModuleFlag;
02945     AC.RhsExprInModuleFlag = 0;
02946 #endif
02947     t = s;
02948     if ( ( s = SkipAName(s) ) == 0 ) goto illdo;
02949     c = *s; *s = 0;
02950     if ( GetName(AC.exprnames,t,&expnum,NOAUTO) == CEXPRESSION ) {
02951         loop->type = ONEEXPRESSION;
02952         /*
02953          * We should remember the expression by name for when it gets
02954          * renumbered!!! If it gets deleted there will be a crash or at
02955          * least the loop terminates.
02956          */
02957         loop->vars = t;
02958     }
02959     else goto illdo;
02960     if ( c == ',' || c == '\t' || c == ';' ) { s++; }
02961     else if ( c != 0 && c != '\n' ) goto illdo;
02962     while ( *s == ',' || *s == '\t' || *s == ';' ) s++;
02963     if ( *s != 0 && *s != '\n' ) goto illdo;
02964     loop->firstnum = 0;
02965     s++;
02966     loop->contents = s;
02967     loop->incnum = 0;
02968     /*
02969      * Next determine size of statement and allocate space
02970      */
02971     while ( *t ) t++;
02972     i = t - loop->vars;
02973     t = loop->name;
02974     while ( *t ) { t++; i++; }
02975     i += 4;
02976     loop->dollarname = Malloc1((LONG)i,"do-loop instruction");
02977     /*
02978      * Construct the statement
02979      */
02980     u = loop->dollarname;
02981     *u++ = '$'; t = loop->name; while ( *t ) *u++ = *t++;
02982     *u++ = '_'; uu = u; *u++ = '='; t = loop->vars;
02983     while ( *t ) *u++ = *t++;
02984     *t = 0; *u = 0;
02985     /*
02986      * Compile and put in dollar variable.
02987      * Note that we remember the dollar by name and that this name ends in _
02988      */
02989     oldpreassignflag = AP.PreAssignFlag;
02990     AP.PreAssignFlag = 2;
02991     CompileStatement(loop->dollarname);
02992     if ( CatchDollar(0) ) {
02993         MesPrint("@Cannot load expression in do loop");
02994         return(-1);
02995     }
02996     AP.PreAssignFlag = oldpreassignflag;
02997     NumPotModdollars = oldNumPotModdollars;
02998 #ifdef WITHMPI
02999     AC.RhsExprInModuleFlag = oldRhsExprInModuleFlag;
03000 #endif
03001     *uu = 0;
03002 }
03003 else goto illdo; /* Syntax problems */
03004 loop->errorsinloop = 0;
03005 /* loop->startlinenumber = linenum+1; 5-oct-2000 One too much? */
03006 loop->startlinenumber = linenum;
03007 PutPreVar(loop->name,(UBYTE *)"0",0,0);
03008 loop->firstloopcall = 1;
03009 return(DoEnddo(s));
03010 illname:;

```

```

03011     MesPrint("@Improper name for do loop variable");
03012     return(-1);
03013 illdo;
03014     MesPrint("@Improper syntax in do loop instruction");
03015     return(-1);
03016 }
03017
03018 /*
03019     #] DoDo :
03020     #[ DoBreakDo :
03021
03022     #breakdo [num]
03023     jumps out of num #do-loops (if there are that many) (default is 1)
03024 */
03025
03026 int DoBreakDo(UBYTE *s)
03027 {
03028     DOLOOP *loop;
03029     WORD levels;
03030
03031     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03032     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03033
03034     if ( NumDoLoops <= 0 ) {
03035         MesPrint("@%#breakdo without %#do");
03036         return(1);
03037     }
03038 /*
03039     if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEDO ) { MessPreNesting(4); return(-1); }
03040 */
03041     while ( *s && ( *s == ',' || *s == ' ' || *s == '\t' ) ) s++;
03042     if ( *s == 0 ) {
03043         levels = 1;
03044     }
03045     else if ( FG.cTable[*s] == 1 ) {
03046         levels = 0;
03047         while ( *s >= '0' && *s <= '9' ) { levels = 10*levels + *s++ - '0'; }
03048         if ( *s != 0 ) goto improper;
03049     }
03050     else {
03051 improper:
03052         MesPrint("@Improper syntax of %#breakdo instruction");
03053         return(1);
03054     }
03055     if ( levels > NumDoLoops ) {
03056         MesPrint("@Too many loop levels requested in %#breakdo instruction");
03057         Terminate(-1);
03058     }
03059     while ( levels > 0 ) {
03060         while ( AC.CurrentStream->type != PREREADSTREAM
03061                && AC.CurrentStream->type != PREREADSTREAM2
03062                && AC.CurrentStream->type != PREREADSTREAM3 ) {
03063             AC.CurrentStream = CloseStream(AC.CurrentStream);
03064         }
03065         while ( AP.PreTypes[AP.NumPreTypes] != PRETYPEDO
03066                && AP.PreTypes[AP.NumPreTypes] != PRETYPEPROCEDURE ) AP.NumPreTypes--;
03067         if ( AC.CurrentStream->type == PREREADSTREAM3
03068             || AP.PreTypes[AP.NumPreTypes] == PRETYPEPROCEDURE ) {
03069             MesPrint("@Trying to jump out of a procedure with a %#breakdo instruction");
03070             Terminate(-1);
03071         }
03072         loop = &(DoLoops[NumDoLoops-1]);
03073         AP.NumPreTypes = loop->NumPreTypes;
03074         AP.PreIfLevel = loop->PreIfLevel;
03075         AP.PreSwitchLevel = loop->PreSwitchLevel;
03076 /*
03077         AP.NumPreTypes--;
03078 */
03079         NumDoLoops--;
03080         DoUndefine(loop->name);
03081         M_free(loop->p.buffer, "loop->p.buffer");
03082         loop->firstloopcall = 0;
03083
03084         AC.CurrentStream = CloseStream(AC.CurrentStream);
03085         levels--;
03086     }
03087     return(0);
03088 }
03089
03090 /*
03091     #] DoBreakDo :
03092     #[ DoElse :
03093 */
03094
03095 int DoElse(UBYTE *s)
03096 {
03097     if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEIF ) {

```

```

03098         if ( AP.PreIfLevel <= 0 ) MesPrint("@%#else without corresponding %#if");
03099     else MessPreNesting(1);
03100     return(-1);
03101 }
03102 if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03103 while ( *s == ' ' ) s++;
03104 if ( tolower(*s) == 'i' && tolower(s[1]) == 'f' && s[2]
03105     && FG.cTable[s[2]] > 1 && s[2] != '_' ) {
03106     s += 2;
03107     while ( *s == ' ' ) s++;
03108     return(DoElseif(s));
03109 }
03110 if ( AP.PreIfLevel <= 0 ) {
03111     MesPrint("@%#else without corresponding %#if");
03112     return(-1);
03113 }
03114 switch ( AP.PreIfStack[AP.PreIfLevel] ) {
03115     case EXECUTINGIF:
03116         AP.PreIfStack[AP.PreIfLevel] = LOOKINGFORENDIF;
03117         break;
03118     case LOOKINGFORELSE:
03119         AP.PreIfStack[AP.PreIfLevel] = EXECUTINGIF;
03120         break;
03121     case LOOKINGFORENDIF:
03122         break;
03123 }
03124 return(0);
03125 }
03126
03127 /*
03128     #] DoElse :
03129     #[ DoElseif :
03130 */
03131
03132 int DoElseif(UBYTE *s)
03133 {
03134     int condition;
03135     if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEIF ) {
03136         if ( AP.PreIfLevel <= 0 ) MesPrint("@%#elseif without corresponding %#if");
03137         else MessPreNesting(2);
03138         return(-1);
03139     }
03140     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03141     if ( AP.PreIfLevel <= 0 ) {
03142         MesPrint("@%#elseif without corresponding %#if");
03143         return(-1);
03144     }
03145     switch ( AP.PreIfStack[AP.PreIfLevel] ) {
03146         case EXECUTINGIF:
03147             AP.PreIfStack[AP.PreIfLevel] = LOOKINGFORENDIF;
03148             break;
03149         case LOOKINGFORELSE:
03150             if ( ( condition = EvalPreIf(s) ) < 0 ) return(-1);
03151             AP.PreIfStack[AP.PreIfLevel] = condition;
03152             break;
03153         case LOOKINGFORENDIF:
03154             break;
03155     }
03156     return(0);
03157 }
03158
03159 /*
03160     #] DoElseif :
03161     #[ DoEnddo :
03162
03163     At the first call there is no stream yet.
03164     After that we have to close the stream and start a new one.
03165 */
03166
03167 int DoEnddo(UBYTE *s)
03168 {
03169     GETIDENTITY
03170     DOLOOP *loop;
03171     UBYTE *t, *tt, *value, numstr[16];
03172     LONG xval;
03173     int xsign, retval;
03174     DUMMYUSE(s);
03175     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03176     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03177     /*
03178     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ||
03179         AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) {
03180         if ( AP.PreTypes[AP.NumPreTypes] == PRETYPEDO ) AP.NumPreTypes--;
03181         else { MessPreNesting(3); return(-1); }
03182         return(0);
03183     }
03184     */

```

```

03185     if ( NumDoLoops <= 0 ) {
03186         MesPrint("@%#enddo without %#do");
03187         return(1);
03188     }
03189     if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEDO ) { MessPreNesting(4); return(-1); }
03190     loop = &(DoLoops[NumDoLoops-1]);
03191     if ( !loop->firstloopcall ) AC.CurrentStream = CloseStream(AC.CurrentStream);
03192
03193     if ( loop->errorsinloop ) {
03194         MesPrint("+++Errors in Loop");
03195         goto finish;
03196     }
03197     if ( loop->type == LISTEDLOOP ) {
03198         if ( loop->firstnum >= loop->lastnum ) goto finish;
03199         loop->firstnum++;
03200         t = value = loop->vars;
03201         while ( *value ) value++;
03202         value++;
03203         loop->vars = value;
03204         value = tt = t;
03205         while ( *value ) {
03206             if ( *value == '\\\\' ) value++;
03207             *tt++ = *value++;
03208         }
03209         *tt = 0;
03210         PutPreVar(loop->name,t,0,1); /* We overwrite the definition */
03211     }
03212     else if ( loop->type == NUMERICALLOOP ) {
03213
03214         if ( !loop->firstloopcall ) {
03215             /*
03216              Test whether the variable was changed inside the loop into
03217              a different numerical value. If so, adjust.
03218             */
03219             t = GetPreVar(loop->name,WITHOUTERROR);
03220             if ( t ) {
03221                 value = t;
03222                 xsign = 1;
03223                 while ( *value && ( *value == ' ' ||
03224                     *value == '-' || *value == '+' ) ) {
03225                     if ( *value == '-' ) xsign = -xsign;
03226                     value++;
03227                 }
03228                 t = value; xval = 0;
03229                 while ( *value >= '0' && *value <= '9' ) xval = 10*xval + *value++ - '0';
03230                 while ( *value && *value == ' ' ) value++;
03231                 if ( *value == 0 ) {
03232                     /*
03233                      Now we may substitute the loopvalue.
03234                     */
03235                     if ( xsign < 0 ) xval = -xval;
03236                     if ( loop->incdollar >= 0 ) {
03237                         loop->incnum = DolToLong(BHEAD loop->incdollar);
03238                         if ( AN.ErrorInDollar ) {
03239                             MesPrint("@%s does not evaluate into a valid third loop
03240 parameter",DOLLARNAME(Dollars,loop->incdollar));
03241                             return(-1);
03242                         }
03243                         loop->firstnum = xval + loop->incnum;
03244                     }
03245                     if ( loop->lastdollar >= 0 ) {
03246                         loop->lastnum = DolToLong(BHEAD loop->lastdollar);
03247                         if ( AN.ErrorInDollar ) {
03248                             MesPrint("@%s does not evaluate into a valid second loop
03249 parameter",DOLLARNAME(Dollars,loop->lastdollar));
03250                             return(-1);
03251                         }
03252                     }
03253                 }
03254                 if ( ( loop->incnum > 0 && loop->firstnum > loop->lastnum )
03255                     || ( loop->incnum < 0 && loop->firstnum < loop->lastnum ) ) goto finish;
03256                 NumToStr(numstr,loop->firstnum);
03257                 t = numstr;
03258                 loop->firstnum += loop->incnum;
03259                 PutPreVar(loop->name,t,0,1); /* We overwrite the definition */
03260             }
03261         }
03262         else if ( loop->type == ONEEXPRESSION ) {
03263             /*
03264              Find the dollar expression
03265             */
03266             WORD numdollar = GetDollar(loop->dollarname+1);
03267             DOLLARS d = Dollars + numdollar;
03268             WORD *w, *dw, v, *ww;
03269             if ( (d->where) == 0 ) {
03270                 d->type = DOLUNDEFINED;

```

```

03270         M_free(loop->dollarname,"do-loop instruction");
03271         goto finish;
03272     }
03273     w = d->where + loop->incnum;
03274     if ( *w == 0 ) {
03275         M_free(d->where,"dollar");
03276         d->where = 0;
03277         d->type = DOLUNDEFINED;
03278         M_free(loop->dollarname,"do-loop instruction");
03279         goto finish;
03280     }
03281     loop->incnum += *w;
03282     /*
03283     Now the term has to be converted to text.
03284     */
03285     ww = w + *w; v = *ww; *ww = 0;
03286     dw = d->where; d->where = w;
03287     t = WriteDollarToBuffer(numdollar,1);
03288     d->where = dw; *ww = v;
03289     PutPreVar(loop->name,t,0,1); /* We overwrite the definition */
03290     M_free(t,"dollar");
03291 }
03292 if ( loop->firstloopcall ) OpenStream(loop->contents,PREREADSTREAM2,0,PRENOACTION);
03293 else OpenStream(loop->contents,PREREADSTREAM,0,PRENOACTION);
03294 AC.CurrentStream->prevline =
03295 AC.CurrentStream->linenumber = loop->startlinenumber;
03296 AC.CurrentStream->eqnum = 0;
03297 loop->firstloopcall = 0;
03298 return(0);
03299 finish:;
03300 NumDoLoops--;
03301 retval = DoUndefine(loop->name);
03302 M_free(loop->p.buffer,"loop->p.buffer");
03303 loop->firstloopcall = 0;
03304 AP.NumPreTypes--;
03305 return(retval);
03306 }
03307
03308 /*
03309     #] DoEnddo :
03310     #[ DoEndif :
03311 */
03312
03313 int DoEndif(UBYTE *s)
03314 {
03315     DUMMYUSE(s);
03316     if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEIF ) {
03317         if ( AP.PreIfLevel <= 0 ) MesPrint("@%#endif without corresponding %#if");
03318         else MessPreNesting(5);
03319         return(-1);
03320     }
03321     AP.NumPreTypes--;
03322     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03323     if ( AP.PreIfLevel <= 0 ) {
03324         MesPrint("@%#endif without corresponding %#if");
03325         return(-1);
03326     }
03327     AP.PreIfLevel--;
03328     return(0);
03329 }
03330
03331 /*
03332     #] DoEndif :
03333     #[ DoEndprocedure :
03334
03335     Action is simple: close the current stream if it is still
03336     the stream from which the statement came.
03337     Then pop the current procedure and all its local derivatives.
03338     if loadmode > 1 the procedure was defined locally.
03339 */
03340
03341 int DoEndprocedure(UBYTE *s)
03342 {
03343     DUMMYUSE(s);
03344     if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEPROCEDURE ) {
03345         MessPreNesting(6);
03346         return(-1);
03347     }
03348     AP.NumPreTypes--;
03349     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03350     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03351     AC.CurrentStream = CloseStream(AC.CurrentStream);
03352     do {
03353         NumProcedures--;
03354         if ( Procedures[NumProcedures].loadmode == 0 ) {
03355             M_free(Procedures[NumProcedures].p.buffer,"procedures buffer");
03356             M_free(Procedures[NumProcedures].name,"procedures name");

```

```

03357     }
03358     } while ( Procedures[NumProcedures].loadmode > 1 );
03359     return(0);
03360 }
03361
03362 /*
03363     #] DoEndprocedure :
03364     #[ DoIf :
03365 */
03366
03367 int DoIf(UBYTE *s)
03368 {
03369     int condition;
03370     AddToPreTypes(PRETYPEIF);
03371     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03372     if ( AP.PreIfStack[AP.PreIfLevel] == EXECUTINGIF ) {
03373         condition = EvalPreIf(s);
03374         if ( condition < 0 ) return(-1);
03375     }
03376     else condition = LOOKINGFORENDIF;
03377     if ( AP.PreIfLevel+1 >= AP.MaxPreIfLevel ) {
03378         int **ppp = &AP.PreIfStack; /* To avoid a compiler warning */
03379         if ( DoubleList((void ***)ppp,&AP.MaxPreIfLevel,sizeof(int),
03380             "PreIfLevels" ) ) return(-1);
03381     }
03382     AP.PreIfStack[++AP.PreIfLevel] = condition;
03383     return(0);
03384 }
03385
03386 /*
03387     #] DoIf :
03388     #[ DoIfdef :
03389 */
03390
03391 int DoIfdef(UBYTE *s, int par)
03392 {
03393     int condition;
03394     AddToPreTypes(PRETYPEIF);
03395     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03396     if ( AP.PreIfStack[AP.PreIfLevel] == EXECUTINGIF ) {
03397         while ( *s == ' ' || *s == '\t' ) s++;
03398         if ( ( *s == 0 ) == ( par == 1 ) ) condition = LOOKINGFORELSE;
03399         else condition = EXECUTINGIF;
03400     }
03401     else condition = LOOKINGFORENDIF;
03402     if ( AP.PreIfLevel+1 >= AP.MaxPreIfLevel ) {
03403         int **ppp = &AP.PreIfStack; /* to avoid a compiler warning */
03404         if ( DoubleList((void ***)ppp,&AP.MaxPreIfLevel,sizeof(int),
03405             "PreIfLevels" ) ) return(-1);
03406     }
03407     AP.PreIfStack[++AP.PreIfLevel] = condition;
03408     return(0);
03409 }
03410
03411 /*
03412     #] DoIfdef :
03413     #[ DoIfydef :
03414 */
03415
03416 int DoIfydef(UBYTE *s)
03417 {
03418     return DoIfdef(s,1);
03419 }
03420
03421 /*
03422     #] DoIfydef :
03423     #[ DoIfndef :
03424 */
03425
03426 int DoIfndef(UBYTE *s)
03427 {
03428     return DoIfdef(s,2);
03429 }
03430
03431 /*
03432     #] DoIfndef :
03433     #[ DoInside :
03434
03435     #inside $var1,...,$varn
03436     statements without .sort
03437     #endinside
03438
03439     executes the statements on the contents of the $ variables as if they
03440     are a module. The results are put back in the dollar variables.
03441     To do this right we need a struct with
03442         old compiler buffer
03443         list of numbers of dollars

```

```

03444     length of the list
03445     length of the array containing the list
03446     Because we need to compose statements, the statement buffer must be
03447     empty. This means that we have to test for that. Same at the end. We
03448     must have a completed statement.
03449 */
03450
03451 int DoInside(UBYTE *s)
03452 {
03453     GETIDENTITY
03454     int numdol, error = 0;
03455     WORD *nb, newsize, i;
03456     UBYTE *name, c;
03457     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03458     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03459     if ( AP.PreInsideLevel != 0 ) {
03460         MesPrint("@Illegal nesting of %inside/%endinside instructions");
03461         return(-1);
03462     }
03463 /*
03464     if ( AP.PreContinuation ) {
03465         error = -1;
03466         MesPrint("@%inside cannot be inside a regular statement");
03467     }
03468 */
03469     PUSHPREASSIGNLEVEL
03470 /*
03471     Now the dollars to do
03472 */
03473     AP.inside.numdollars = 0;
03474     for(;;) {
03475         while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
03476         if ( *s == 0 ) break;
03477         if ( *s != '$' ) {
03478             MesPrint("@%inside instruction can have only $ variables for parameters");
03479             return(-1);
03480         }
03481         s++;
03482         name = s;
03483         while (chartype[*s] <= 1 ) s++;
03484         c = *s; *s = 0;
03485         if ( ( numdol = GetDollar(name) ) < 0 ) {
03486             MesPrint("@%inside: %s has not (yet) been defined",name);
03487             *s = c;
03488             error = -1;
03489         }
03490         else {
03491             *s = c;
03492             if ( AP.inside.numdollars >= AP.inside.size ) {
03493                 if ( AP.inside.buffer == 0 ) newsize = 20;
03494                 else newsize = 2*AP.inside.size;
03495                 nb = (WORD *)Malloc1(newsize*sizeof(WORD),"insidebuffer");
03496                 if ( AP.inside.buffer ) {
03497                     for ( i = 0; i < AP.inside.size; i++ ) nb[i] = AP.inside.buffer[i];
03498                     M_free(AP.inside.buffer,"insidebuffer");
03499                 }
03500                 AP.inside.buffer = nb;
03501                 AP.inside.size = newsize;
03502             }
03503             AP.inside.buffer[AP.inside.numdollars++] = numdol;
03504         }
03505     }
03506 /*
03507     We have to store the configuration of the compiler buffer, so that
03508     we know where to start executing and how to reset the buffer.
03509 */
03510     AP.inside.oldcompletetype = AC.completetype;
03511     AP.inside.oldparallelflag = AC.mparallelflag;
03512     AP.inside.oldnumpotmoddollars = NumPotModdollars;
03513     AP.inside.olddcbuf = AC.cbufnum;
03514     AP.inside.olddrbuf = AM.rbufnum;
03515     AP.inside.olddcnmlhs = AR.Cnumlhs,
03516     AddToPreTypes(PRETYPEINSIDE);
03517     AP.PreInsideLevel = 1;
03518     AC.cbufnum = AP.inside.inscbuf;
03519     AM.rbufnum = AP.inside.inscbuf;
03520     clearcbuf(AC.cbufnum);
03521     AC.completetype = 0;
03522     AC.mparallelflag = PARALLELEFLAG;
03523 #ifdef WITHMPI
03524 /*
03525     * We use AC.RhsExprInModuleFlag, PotModdollars, and AC.pfirstnum
03526     * in order to check (1) whether there are expression names in RHS,
03527     * (2) which dollar variables can be modified, and (3) which
03528     * preprocessor variables can be redefined, in #inside.
03529     * We store the current values of them, and then reset them.
03530     */

```



```

03531     PF_StoreInsideInfo();
03532     AC.RhsExprInModuleFlag = 0;
03533     NumPotModdollars = 0;
03534     AC.numpfirstnum = 0;
03535 #endif
03536     return(error);
03537 }
03538
03539 /*
03540     #] DoInside :
03541     #[ DoEndInside :
03542 */
03543
03544 int DoEndInside(UBYTE *s)
03545 {
03546     GETIDENTITY
03547     WORD numdol, *oldworkpointer = AT.WorkPointer, *term, *t, j, i;
03548     DOLLARS d, nd;
03549     WORD oldbracketon = AR.BraceOn;
03550     WORD *oldcompresspointer = AR.CompressPointer;
03551     int oldmultithreaded = AS.MultiThreaded;
03552     /* int oldmparallelflag = AC.mparallelflag; */
03553     FILEHANDLE *f;
03554 #ifdef WITHMPI
03555     int error = 0;
03556 #endif
03557     DUMMYUSE(s);
03558     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03559     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03560     if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEINSIDE ) {
03561         if ( AP.PreInsideLevel != 1 ) MesPrint("@%#endinside without corresponding %#inside");
03562         else MessPreNesting(11);
03563         return(-1);
03564     }
03565     AP.NumPreTypes--;
03566     if ( AP.PreInsideLevel != 1 ) {
03567         MesPrint("@%#endinside without corresponding %#inside");
03568         return(-1);
03569     }
03570     if ( AP.PreContinuation ) {
03571         MesPrint("@%#endinside: previous statement not terminated.");
03572         Terminate(-1);
03573     }
03574     AC.compiletype = AP.inside.oldcompiletype;
03575     AR.Cnumlhs = cbuf[AM.rbufnum].numlhs;
03576 #ifdef WITHMPI
03577     /*
03578      * If the #inside...#endinside contains expressions in RHS, only the master executes it
03579      * and then broadcasts the result to the all slaves. If not, the all processes execute
03580      * it and in this case no MPI interactions are needed.
03581      */
03582     if ( PF.me == MASTER || !AC.RhsExprInModuleFlag ) {
03583 #endif
03584         AR.BraceOn = 0;
03585         AS.MultiThreaded = 0;
03586         /* AC.mparallelflag = PARALLELFLAG; */
03587         if ( AR.CompressPointer == 0 ) AR.CompressPointer = AR.CompressBuffer;
03588         f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
03589     /*
03590      Now we have to execute the statements on the proper dollars.
03591     */
03592     for ( i = 0; i < AP.inside.numdollars; i++ ) {
03593         numdol = AP.inside.buffer[i];
03594         nd = d = Dollars + numdol;
03595         if ( d->type != DOLZERO ) {
03596             if ( d->type != DOLTERMS ) nd = DolToTerms(BHEAD numdol);
03597             term = nd->where;
03598             NewSort(BHEAD0);
03599             NewSort(BHEAD0);
03600             AR.MaxDum = AM.IndDum;
03601             while ( *term ) {
03602                 t = oldworkpointer; j = *term;
03603                 NCOPY(t,term,j);
03604                 AT.WorkPointer = t;
03605                 AN.IndDum = AM.IndDum;
03606                 AR.CurDum = ReNumber(BHEAD term);
03607                 if ( Generator(BHEAD oldworkpointer,0) ) {
03608                     MesPrint("@Called from %#endinside");
03609                     MesPrint("@Evaluating variable %s",DOLLARNAME(Dollars,numdol));
03610                     Terminate(-1);
03611                 }
03612             }
03613             AT.WorkPointer = oldworkpointer;
03614             CleanDollarFactors(d);
03615             if ( d->where ) { M_free(d->where,"dollar contents"); d->where = 0; }
03616             EndSort(BHEAD (WORD *) ((void *) (&(d->where))),2);
03617             LowerSortLevel();

```

```

03618         term = d->where; while ( *term ) term += *term;
03619         d->size = term - d->where;
03620         if ( nd != d ) M_free(nd,"Copy of dollar variable");
03621         if ( d->where[0] == 0 ) {
03622             M_free(d->where,"dollar contents"); d->where = 0;
03623             d->type = DOLZERO;
03624         }
03625     }
03626 }
03627 #ifdef WITHMPI
03628 {
03629     if ( AC.RhsExprInModuleFlag ) {
03630         /*
03631          * The only master executed the statements in #inside.
03632          * We need to broadcast the result to the all slaves.
03633          */
03634         for ( i = 0; i < AP.inside.numdollars; i++ ) {
03635             /*
03636              * Mark $-variables specified in the #inside instruction as modified
03637              * such that they will be broadcast.
03638              */
03639             AddPotModdollar(AP.inside.buffer[i]);
03640         }
03641         /* Now actual broadcast of modified variables. */
03642         if ( NumPotModdollars > 0 ) {
03643             error = PF_BroadcastModifiedDollars();
03644             if ( error ) goto cleanup;
03645         }
03646         if ( AC.numpfirstnum > 0 ) {
03647             error = PF_BroadcastRedefinedPreVars();
03648             if ( error ) goto cleanup;
03649         }
03650     }
03651 cleanup:
03652 #endif
03653     f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
03654     AC.cbufnum = AP.inside.oldcbuf;
03655     AM.rbufnum = AP.inside.olldrbuf;
03656     AR.Cnumlhs = AP.inside.oldcnumlhs;
03657     AR.BracketOn = oldbracketon;
03658     AP.PreInsideLevel = 0;
03659     AR.CompressPointer = oldcompresspointer;
03660     AS.MultiThreaded = oldmultithreaded;
03661     AC.mparallelflag = AP.inside.oldparallelflag;
03662     NumPotModdollars = AP.inside.oldnumpotmoddollars;
03663     POPPREASSIGNLEVEL
03664 #ifdef WITHMPI
03665     PF_RestoreInsideInfo();
03666     if ( error ) return error;
03667 #endif
03668     return(0);
03669 }
03670
03671 /*
03672     #] DoEndInside :
03673     #[ DoMessage :
03674 */
03675
03676 int DoMessage(UBYTE *s)
03677 {
03678     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03679     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03680     while ( *s == ' ' || *s == '\t' ) s++;
03681     MesPrint("~~~%s",s);
03682     return(0);
03683 }
03684
03685 /*
03686     #] DoMessage :
03687     #[ DoPipe :
03688 */
03689
03690 int DoPipe(UBYTE *s)
03691 {
03692 #ifndef WITHPIPE
03693     DUMMYUSE(s);
03694 #endif
03695     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03696     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03697 #ifdef WITHPIPE
03698     FLUSHCONSOLE;
03699     while ( *s == ' ' || *s == '\t' ) s++;
03700     if ( OpenStream(s,PIPESTREAM,0,PREENOACTION) == 0 ) return(-1);
03701     return(0);
03702 #else
03703     Error0("Pipes not implemented on this computer/system");
03704     return(-1);
03705 #endif

```

```

03705 #endif
03706 }
03707
03708 /*
03709     #] DoPipe :
03710     #[ DoProcExtension :
03711 */
03712
03713 int DoProcExtension(UBYTE *s)
03714 {
03715     UBYTE *t, *u, c;
03716     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03717     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03718     while ( *s == ' ' || *s == '\t' ) s++;
03719     if ( *s == 0 || *s == '\n' ) {
03720         MesPrint("@No valid procedure extension specified");
03721         return(-1);
03722     }
03723     if ( FG.cTable[*s] != 0 ) {
03724         MesPrint("@Procedure extension should be a string starting with an alphabetic character. No
03725         whitespace.");
03726         return(-1);
03727     }
03728     t = s;
03729     while ( *s && *s != '\n' && *s != ' ' && *s != '\t' ) s++;
03730     u = s;
03731     while ( *s == ' ' || *s == '\t' ) s++;
03732     if ( *s != 0 && *s != '\n' ) {
03733         MesPrint("@Too many parameters in ProcedureExtension instruction");
03734         return(-1);
03735     }
03736     c = *u; *u = 0;
03737     if ( AP.procedureExtension ) M_free(AP.procedureExtension, "ProcedureExtension");
03738     AP.procedureExtension = strDupl(t, "ProcedureExtension");
03739     *u = c;
03740     return(0);
03741 }
03742 /*
03743     #] DoProcExtension :
03744     #[ DoPreOut :
03745 */
03746
03747 int DoPreOut(UBYTE *s)
03748 {
03749     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03750     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03751     if ( tolower(*s) == 'o' ) {
03752         if ( tolower(s[1]) == 'n' && s[2] == 0 ) {
03753             AP.PreOut = 1;
03754             return(0);
03755         }
03756         if ( tolower(s[1]) == 'f' && tolower(s[2]) == 'f' && s[3] == 0 ) {
03757             AP.PreOut = 0;
03758             return(0);
03759         }
03760     }
03761     MesPrint("@Illegal option in PreOut instruction");
03762     return(-1);
03763 }
03764
03765 /*
03766     #] DoPreOut :
03767     #[ DoPrePrintTimes :
03768 */
03769
03770 int DoPrePrintTimes(UBYTE *s)
03771 {
03772     DUMMYUSE(s);
03773     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03774     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03775     PrintRunningTime();
03776     return(0);
03777 }
03778
03779 /*
03780     #] DoPrePrintTimes :
03781     #[ DoPreSortReallocate :
03782 */
03783
03784 int DoPreSortReallocate(UBYTE *s)
03785 {
03786     DUMMYUSE(s);
03787     if ( AC.SortReallocateFlag == 0 ) {
03788         /* Currently off, so set to 2. Then the reallocation code knows the flag was
03789         set here, since "On sortreallocate;" sets it to 1. */
03790         AC.SortReallocateFlag = 2;

```

```

03791     }
03792     /* If the flag is already on, do nothing. */
03793     return(0);
03794 }
03795
03796 /*
03797     #] DoPreSortReallocate :
03798     #[ DoPreAppend :
03799
03800     Syntax:
03801     #append <filename>
03802 */
03803
03804 int DoPreAppend(UBYTE *s)
03805 {
03806     UBYTE *name, *to;
03807
03808     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03809     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03810     if ( AP.preError ) return(0);
03811     while ( *s == ' ' || *s == '\t' ) s++;
03812 /*
03813     Determine where to write
03814 */
03815     if ( *s == '<' ) {
03816         s++;
03817         name = to = s;
03818         while ( *s && *s != '>' ) {
03819             if ( *s == '\\\\' ) s++;
03820             *to++ = *s++;
03821         }
03822         if ( *s == 0 ) {
03823             MesPrint("@Improper termination of filename");
03824             return(-1);
03825         }
03826         s++;
03827         *to = 0;
03828         if ( *name ) { GetAppendChannel((char *)name); }
03829         else goto improper;
03830     }
03831     else {
03832 improper:
03833         MesPrint("@Proper syntax is: %#append <filename>");
03834         return(-1);
03835     }
03836     return(0);
03837 }
03838
03839 /*
03840     #] DoPreAppend :
03841     #[ DoPreCreate :
03842
03843     Syntax:
03844     #create <filename>
03845 */
03846
03847 int DoPreCreate(UBYTE *s)
03848 {
03849     UBYTE *name, *to;
03850
03851     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03852     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03853     if ( AP.preError ) return(0);
03854     while ( *s == ' ' || *s == '\t' ) s++;
03855 /*
03856     Determine where to write
03857 */
03858     if ( *s == '<' ) {
03859         s++;
03860         name = to = s;
03861         while ( *s && *s != '>' ) {
03862             if ( *s == '\\\\' ) s++;
03863             *to++ = *s++;
03864         }
03865         if ( *s == 0 ) {
03866             MesPrint("@Improper termination of filename");
03867             return(-1);
03868         }
03869         s++;
03870         *to = 0;
03871         if ( *name ) { GetChannel((char *)name,0); }
03872         else goto improper;
03873     }
03874     else {
03875 improper:
03876         MesPrint("@Proper syntax is: %#create <filename>");
03877         return(-1);

```

```

03878     }
03879     return(0);
03880 }
03881
03882 /*
03883     #] DoPreCreate :
03884     #[ DoPreRemove :
03885 */
03886
03887 int DoPreRemove(UBYTE *s)
03888 {
03889     UBYTE *name, *to;
03890     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03891     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03892     if ( AP.preError ) return(0);
03893     while ( *s == ' ' || *s == '\t' ) s++;
03894     if ( *s == '<' ) { s++; }
03895     else {
03896         MesPrint("@Proper syntax is: %#remove <filename>");
03897         return(-1);
03898     }
03899     name = to = s;
03900     while ( *s && *s != '>' ) {
03901         if ( *s == '\\\\' ) s++;
03902         *to++ = *s++;
03903     }
03904     if ( *s == 0 ) {
03905         MesPrint("@Improper filename");
03906         return(-1);
03907     }
03908     s++;
03909     *to = 0;
03910     CloseChannel((char *)name);
03911     remove((char *)name);
03912     return(0);
03913 }
03914
03915 /*
03916     #] DoPreRemove :
03917     #[ DoPreClose :
03918 */
03919
03920 int DoPreClose(UBYTE *s)
03921 {
03922     UBYTE *name, *to;
03923     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03924     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03925     if ( AP.preError ) return(0);
03926     while ( *s == ' ' || *s == '\t' ) s++;
03927     if ( *s == '<' ) { s++; }
03928     else {
03929         MesPrint("@Proper syntax is: %#close <filename>");
03930         return(-1);
03931     }
03932     name = to = s;
03933     while ( *s && *s != '>' ) {
03934         if ( *s == '\\\\' ) s++;
03935         *to++ = *s++;
03936     }
03937     if ( *s == 0 ) {
03938         MesPrint("@Improper filename");
03939         return(-1);
03940     }
03941     s++;
03942     *to = 0;
03943     return(CloseChannel((char *)name));
03944 }
03945
03946 /*
03947     #] DoPreClose :
03948     #[ DoPreWrite :
03949
03950     Syntax:
03951     #write [<filename>] "formatstring" [,objects]
03952     The format string can contain the following special objects/codes
03953     \n newline
03954     \t tab
03955     \! if last entry in string: no linefeed at end
03956     \b put \ in output
03957     $$ $-variable (to be found among the objects)
03958     %e expression (name to be found among the objects)
03959     %E expression without ; (name to be found among the objects)
03960     %s string (to be found among the objects) (with or without "")
03961     %S subterms (see PrintSubtermList)
03962 */
03963
03964 int DoPreWrite(UBYTE *s)

```

```

03965 {
03966     HANDLERS h;
03967
03968     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
03969     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
03970     if ( AP.preError ) return(0);
03971
03972 #ifdef WITHMPI
03973     if ( PF.me != MASTER ) return 0;
03974 #endif
03975
03976     h.oldsilent      = AM.silent;
03977     h.newlogonly     = h.oldlogonly = AM.FileOnlyFlag;
03978     h.newhandle      = h.oldhandle  = AC.LogHandle;
03979     h.oldprintttype = AO.PrintType;
03980
03981     while ( *s == ' ' || *s == '\t' ) s++;
03982 /*
03983     Determine where to write
03984 */
03985     if( (s=defineChannel(s,&h))==0 ) return(-1);
03986
03987     return(writeToChannel(WRITEOUT,s,&h));
03988 }
03989
03990 /*
03991     #] DoPreWrite :
03992     #[ DoProcedure :
03993
03994     We have to read this procedure into a buffer.
03995     The only complications are:
03996     1: we have to seek through the file to do this efficiently
03997        the file operations under VMS cannot do this properly
03998        (unless we use the proper ANSI structs?)
03999        This is the reason why we read whole input files under VMS.
04000     2: what to do when the same name is used twice.
04001     Note that we have to do the reading without substitution of
04002     preprocessor variables.
04003 */
04004
04005 int DoProcedure(UBYTE *s)
04006 {
04007     UBYTE c;
04008     PROCEDURE *p;
04009     LONG i;
04010     if ( ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH )
04011         || ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) ) {
04012         if ( PreSkip((UBYTE *)"procedure", (UBYTE *)"endprocedure",1) ) return(-1);
04013         return(0);
04014     }
04015     p = (PROCEDURE *)FromList(&AP.ProcList);
04016     if ( PreLoad(&(p->p), (UBYTE *)"procedure", (UBYTE *)"endprocedure"
04017         ,1,(char *)"procedure") ) return(-1);
04018
04019     p->loadmode = 2;
04020     s = p->p.buffer + 10;
04021     while ( *s == ' ' || *s == LINEFEED ) s++;
04022     if ( chartype[*s] ) {
04023         MesPrint("@Illegal name for procedure");
04024         return(-1);
04025     }
04026     p->name = s++;
04027     while ( chartype[*s] == 0 || chartype[*s] == 1 ) s++;
04028     c = *s; *s = 0;
04029     p->name = strDup1(p->name,"procedure");
04030     *s = c;
04031 /*
04032     Check for double names
04033 */
04034     for ( i = NumProcedures-2; i >= 0; i-- ) {
04035         if ( StrCmp(Procedures[i].name,p->name) == 0 ) {
04036             Error1("Multiple occurrence of procedure name ",p->name);
04037         }
04038     }
04039     return(0);
04040 }
04041
04042 /*
04043     #] DoProcedure :
04044     #[ DoPreBreak :
04045 */
04046
04047 int DoPreBreak(UBYTE *s)
04048 {
04049     DUMMYUSE(s);
04050     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
04051     if ( AP.PreTypes[AP.NumPreTypes] != PRETYPESWITCH ) {

```

```

04052         if ( AP.PreSwitchLevel <= 0 )
04053             MesPrint("@Break without corresponding Switch");
04054         else MessPreNesting(7);
04055         return(-1);
04056     }
04057     if ( AP.PreSwitchLevel <= 0 ) {
04058         MesPrint("@Break without corresponding Switch");
04059         return(-1);
04060     }
04061     if ( AP.PreSwitchModes[AP.PreSwitchLevel] == EXECUTINGPRESWITCH )
04062         AP.PreSwitchModes[AP.PreSwitchLevel] = SEARCHINGPREENDSWITCH;
04063     return(0);
04064 }
04065
04066 /*
04067     #] DoPreBreak :
04068     #[ DoPreCase :
04069 */
04070
04071 int DoPreCase(UBYTE *s)
04072 {
04073     UBYTE *t;
04074     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
04075     if ( AP.PreTypes[AP.NumPreTypes] != PRETYPESWITCH ) {
04076         if ( AP.PreSwitchLevel <= 0 )
04077             MesPrint("@Case without corresponding Switch");
04078         else MessPreNesting(8);
04079         return(-1);
04080     }
04081     if ( AP.PreSwitchLevel <= 0 ) {
04082         MesPrint("@Case without corresponding Switch");
04083         return(-1);
04084     }
04085     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != SEARCHINGPRECASE ) return(0);
04086
04087     SKIPBLANKS(s)
04088     t = s;
04089     while ( *s ) { if ( *s == '\\\\' ) s++; s++; }
04090     while ( s > t && ( s[-1] == ' ' || s[-1] == '\\t' ) && s[-2] != '\\\\' ) {
04091         if ( s[-2] == '\\\\' ) s--;
04092         s--;
04093     }
04094     if ( *t == '"' && s > t+1 && s[-1] == '"' && s[-2] != '\\\\' ) {
04095         t++; s--; *s = 0;
04096     }
04097     else *s = 0;
04098     s = AP.PreSwitchStrings[AP.PreSwitchLevel];
04099     while ( *t == *s && *t ) { s++; t++; }
04100     if ( *t || *s ) return(0); /* case did not match */
04101     AP.PreSwitchModes[AP.PreSwitchLevel] = EXECUTINGPRESWITCH;
04102     return(0);
04103 }
04104
04105 /*
04106     #] DoPreCase :
04107     #[ DoPreDefault :
04108 */
04109
04110 int DoPreDefault(UBYTE *s)
04111 {
04112     DUMMYUSE(s);
04113     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
04114     if ( AP.PreTypes[AP.NumPreTypes] != PRETYPESWITCH ) {
04115         if ( AP.PreSwitchLevel <= 0 )
04116             MesPrint("@Default without corresponding Switch");
04117         else MessPreNesting(9);
04118         return(-1);
04119     }
04120     if ( AP.PreSwitchLevel <= 0 ) {
04121         MesPrint("@Default without corresponding Switch");
04122         return(-1);
04123     }
04124     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != SEARCHINGPRECASE ) return(0);
04125     AP.PreSwitchModes[AP.PreSwitchLevel] = EXECUTINGPRESWITCH;
04126     return(0);
04127 }
04128
04129 /*
04130     #] DoPreDefault :
04131     #[ DoPreEndSwitch :
04132 */
04133
04134 int DoPreEndSwitch(UBYTE *s)
04135 {
04136     DUMMYUSE(s);
04137     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
04138     if ( AP.PreTypes[AP.NumPreTypes] != PRETYPESWITCH ) {

```

```

04139         if ( AP.PreSwitchLevel <= 0 )
04140             MesPrint("@EndSwitch without corresponding Switch");
04141         else MessPreNesting(10);
04142         return(-1);
04143     }
04144     AP.NumPreTypes--;
04145     if ( AP.PreSwitchLevel <= 0 ) {
04146         MesPrint("@EndSwitch without corresponding Switch");
04147         return(-1);
04148     }
04149     M_free(AP.PreSwitchStrings[AP.PreSwitchLevel--],"pre switch string");
04150     return(0);
04151 }
04152
04153 /*
04154     #] DoPreEndSwitch :
04155     #[ DoPreSwitch :
04156
04157     There should be a string after this.
04158     We have to store it somewhere.
04159 */
04160
04161 int DoPreSwitch(UBYTE *s)
04162 {
04163     UBYTE *t, *switchstring, **newstrings;
04164     int newnum, i, *newmodes;
04165     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
04166     SKIPBLANKS(s)
04167     t = s;
04168     while ( *s ) { if ( *s == '\\\\' ) s++; s++; }
04169     while ( s > t && ( s[-1] == ' ' || s[-1] == '\\t' ) && s[-2] != '\\\\' ) {
04170         if ( s[-2] == '\\\\' ) s--;
04171         s--;
04172     }
04173     if ( *t == '"' && s > t+1 && s[-1] == '"' && s[-2] != '\\\\' ) {
04174         t++; s--; *s = 0;
04175     }
04176     else *s = 0;
04177     switchstring = (UBYTE *)Malloc1((s-t)+1,"case string");
04178     s = switchstring;
04179     while ( *t ) {
04180         if ( *t == '\\\\' ) t++;
04181         *s++ = *t++;
04182     }
04183     *s = 0;
04184     if ( AP.PreSwitchLevel >= AP.NumPreSwitchStrings ) {
04185         newnum = 2*AP.NumPreSwitchStrings;
04186         newstrings = (UBYTE **)Malloc1(sizeof(UBYTE *)*(newnum+1),"case strings");
04187         newmodes = (int *)Malloc1(sizeof(int)*(newnum+1),"case strings");
04188         for ( i = 0; i < AP.NumPreSwitchStrings; i++ )
04189             newstrings[i] = AP.PreSwitchStrings[i];
04190         M_free(AP.PreSwitchStrings,"AP.PreSwitchStrings");
04191         for ( i = 0; i <= AP.NumPreSwitchStrings; i++ )
04192             newmodes[i] = AP.PreSwitchModes[i];
04193         M_free(AP.PreSwitchModes,"AP.PreSwitchModes");
04194         AP.PreSwitchStrings = newstrings;
04195         AP.PreSwitchModes = newmodes;
04196         AP.NumPreSwitchStrings = newnum;
04197     }
04198     AP.PreSwitchStrings[++AP.PreSwitchLevel] = switchstring;
04199     if ( ( AP.PreSwitchLevel > 1 )
04200         && ( AP.PreSwitchModes[AP.PreSwitchLevel-1] != EXECUTINGPRESWITCH ) )
04201         AP.PreSwitchModes[AP.PreSwitchLevel] = SEARCHINGPREENDSWITCH;
04202     else
04203         AP.PreSwitchModes[AP.PreSwitchLevel] = SEARCHINGPRECASE;
04204     AddToPreTypes(PRETYPEPSWITCH);
04205     return(0);
04206 }
04207
04208 /*
04209     #] DoPreSwitch :
04210     #[ DoPreShow :
04211
04212     Print the contents of the preprocessor variables
04213 */
04214
04215 int DoPreShow(UBYTE *s)
04216 {
04217     int i;
04218     UBYTE *name, c;
04219     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
04220     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
04221     while ( *s == ' ' || *s == '\\t' ) s++;
04222     if ( *s == 0 ) {
04223         MesPrint("%#The preprocessor variables:");
04224         for ( i = 0; i < NumPre; i++ ) {
04225             MesPrint("%d: %s = \"%s\"",i,PreVar[i].name,PreVar[i].value);

```



```

04226     }
04227 }
04228 else {
04229     while ( *s ) {
04230         name = s; while ( *s && *s != ' ' && *s != '\t' && *s != ',' ) s++;
04231         c = *s; *s = 0;
04232         for ( i = 0; i < NumPre; i++ ) {
04233             if ( StrCmp(PreVar[i].name,name) == 0 )
04234                 MesPrint("%d: %s = \"%s\"",i,PreVar[i].name,PreVar[i].value);
04235         }
04236         *s = c;
04237         while ( *s == ' ' || *s == '\t' ) s++;
04238     }
04239 }
04240 return(0);
04241 }
04242
04243 /*
04244     #] DoPreShow :
04245     #[ DoSystem :
04246 */
04247
04248 /*
04249  * A macro for translating the contents of `x' into a string after expanding.
04250 */
04251 #define STRINGIFY(x) STRINGIFY__(x)
04252 #define STRINGIFY__(x) #x
04253
04254 int DoSystem(UBYTE *s)
04255 {
04256     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
04257     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
04258     if ( AP.preError ) return(0);
04259 #ifdef WITHSYSTEM
04260     FLUSHCONSOLE;
04261     while ( *s == ' ' || *s == '\t' ) s++;
04262     if ( *s == '-' && s[1] == 'e' ) {
04263         LONG err;
04264         UBYTE str[24];
04265         s += 2;
04266         if ( *s != ' ' ) {
04267             MesPrint("@Syntax error in #system command.");
04268             return(-1);
04269         }
04270         while ( *s == ' ' || *s == '\t' ) s++;
04271         err = system((char *)s);
04272         NumToStr(str,err);
04273         PutPreVar((UBYTE *) "SYSTEMERROR_",str,0,1);
04274     }
04275     else if ( system((char *)s) ) {
04276         MesPrint("@System call returned with error condition");
04277         Terminate(-1);
04278     }
04279     return(0);
04280 #else
04281     Error0("External programs not implemented on this computer/system");
04282     return(-1);
04283 #endif
04284 }
04285
04286 /*
04287     #] DoSystem :
04288     #[ PreLoad :
04289
04290     Loads a loop or procedure into a special buffer.
04291     Note: The current instruction is already in the preStart buffer
04292 */
04293
04294 int PreLoad(PRELOAD *p, UBYTE *start, UBYTE *stop, int mode, char *message)
04295 {
04296     UBYTE *s, *t, *top, *newbuffer, c;
04297     LONG i, ppsize, linenum = AC.CurrentStream->linenum;
04298     int size1, size2, level, com=0, last=1, strng = 0;
04299     p->size = AP.pSize;
04300     p->buffer = (UBYTE *)Malloc1(p->size+1,message);
04301     top = p->buffer + p->size - 2;
04302     t = p->buffer; *t++ = '#';
04303     s = start; size1 = size2 = 0;
04304     while ( *s ) { s++; size1++; }
04305     s = stop; while ( *s ) { s++; size2++; }
04306     s = AP.preStart; while ( *s ) *t++ = *s++; *t++ = LINEFEED;
04307     level = 1;
04308     i = 100;
04309     for (;;) {
04310         c = GetInput();
04311         if ( c == ENDOFINPUT ) {
04312             MesPrint("@Missing %s, Should match line %d",stop,linenum);

```

```

04313         return(-1);
04314     }
04315     if ( c == AP.ComChar && last == 1 ) com = 1;
04316     if ( c == LINEFEED ) { last = 1; com = 0; }
04317     else last = 0;
04318
04319     if ( ( c == '"' ) && ( com == 0 ) ) { strng ^= 1; }
04320
04321     if ( ( c == '#' ) && ( com == 0 ) ) i = 0;
04322     else i++;
04323
04324     if ( t >= top ) {
04325         ppsize = t - p->buffer;
04326         p->size *= 2;
04327         newbuffer = (UBYTE *)Malloc1(p->size,message);
04328         t = newbuffer; s = p->buffer;
04329         while ( --ppsize >= 0 ) *t++ = *s++;
04330         M_free(p->buffer,"loading do loop");
04331         p->buffer = newbuffer;
04332         top = p->buffer + p->size - 2;
04333     }
04334     *t++ = c;
04335     if ( strng == 0 ) {
04336         if ( ( i == size2 ) && ( com == 0 ) ) {
04337             *t = 0;
04338             if ( StrICmp(t-size2,(UBYTE *) (stop)) == 0 ) {
04339                 while ( ( c = GetInput() ) != LINEFEED && c != ENDOFINPUT ) {}
04340                 level--;
04341                 if ( level <= 0 ) break;
04342                 if ( c == ENDOFINPUT ) Error1("Missing #",stop);
04343                 *t++ = LINEFEED; *t = 0; last = 1;
04344             }
04345         }
04346         if ( ( i == size1 ) && mode && ( com == 0 ) ) {
04347             *t = 0;
04348             if ( StrICmp(t-size1,(UBYTE *) (start)) == 0 ) {
04349 /*
04350                 while ( ( c = GetInput() ) != LINEFEED && c != ENDOFINPUT ) {}
04351                 if ( c == ENDOFINPUT ) Error1("Missing #",stop);
04352 */
04353                 level++;
04354             }
04355         }
04356         if ( i == 1 && t[-2] == LINEFEED ) {
04357             if ( c == '-' ) AC.NoShowInput = 1;
04358             else if ( c == '+' ) AC.NoShowInput = 0;
04359         }
04360     }
04361 }
04362 *t++ = LINEFEED;
04363 *t = 0;
04364 return(0);
04365 }
04366
04367 /*
04368     #] PreLoad :
04369     #[ PreSkip :
04370
04371     Skips a loop or procedure.
04372     Note: The current instruction is already in the preStart buffer
04373 */
04374
04375 #define SKIPBUFSIZE 20
04376
04377 int PreSkip(UBYTE *start, UBYTE *stop, int mode)
04378 {
04379     UBYTE *s, *t, buffer[SKIPBUFSIZE+2], c;
04380     LONG i, linenum = AC.CurrentStream->linenum;
04381     int size1, size2, level, com=0, last=1;
04382
04383     t = buffer; *t++ = '#';
04384     s = start; size1 = size2 = 0;
04385     while ( *s ) { s++; size1++; }
04386     s = stop; while ( *s ) { s++; size2++; }
04387     level = 1;
04388     i = 0;
04389     for (;;) {
04390         c = GetInput();
04391         if ( c == ENDOFINPUT ) {
04392             MesPrint("@Missing %s, Should match line %1",stop,linenum);
04393             return(-1);
04394         }
04395         if ( c == AP.ComChar && last == 1 ) com = 1;
04396         if ( c == LINEFEED ) { last = 1; com = 0; i = 0; t = buffer; }
04397         else last = 0;
04398         if ( ( c == '#' ) && ( com == 0 ) ) { i = 0; t = buffer; }
04399         else i++;

```

```

04400
04401     if ( i < SKIPBUFSIZE ) *t++ = c;
04402     if ( ( i == size2 ) && ( com == 0 ) ) {
04403         *t = 0;
04404         if ( StrICmp(t-size2,(UBYTE *) (stop)) == 0 ) {
04405             while ( ( c = GetInput() ) != LINEFEED && c != ENDOFINPUT ) {}
04406             level--;
04407             if ( level <= 0 ) {
04408                 pushbackchar = LINEFEED;
04409                 break;
04410             }
04411             if ( c == ENDOFINPUT ) Error1("Missing #",stop);
04412             i = 0; t = buffer;
04413         }
04414     }
04415     if ( ( i == size1 ) && mode && ( com == 0 ) ) {
04416         *t = 0;
04417         if ( StrICmp(t-size1,(UBYTE *) (start)) == 0 ) {
04418             while ( ( c = GetInput() ) != LINEFEED && c != ENDOFINPUT ) {}
04419             level++;
04420             i = 0; t = buffer;
04421         }
04422     }
04423 }
04424 return(0);
04425 }
04426
04427 /*
04428     #] PreSkip :
04429     #[ StartPrepro :
04430 */
04431
04432 void StartPrepro(void)
04433 {
04434     int **ppp;
04435     AP.MaxPreIfLevel = 2;
04436     ppp = &AP.PreIfStack;
04437     if ( DoubleList((void ***)ppp,&AP.MaxPreIfLevel,sizeof(int),
04438         "PreIfLevels" ) Terminate(-1);
04439     AP.PreIfLevel = 0; AP.PreIfStack[0] = EXECUTINGIF;
04440
04441     AP.NumPreSwitchStrings = 10;
04442     AP.PreSwitchStrings = (UBYTE **)Malloc1(sizeof(UBYTE *)*
04443         (AP.NumPreSwitchStrings+1),"case strings");
04444     AP.PreSwitchModes = (int *)Malloc1(sizeof(int)*
04445         (AP.NumPreSwitchStrings+1),"case strings");
04446     AP.PreSwitchModes[0] = EXECUTINGPRESWITCH;
04447     AP.PreSwitchLevel = 0;
04448 }
04449
04450 /*
04451     #] StartPrepro :
04452     #[ EvalPreIf :
04453
04454     Evaluates the condition in an if instruction.
04455     The return value is EXECUTINGIF if the condition is true.
04456     If it is false the returnvalue is LOOKINGFORELSE.
04457     An error gives a return value of -1
04458 */
04459
04460 int EvalPreIf(UBYTE *s)
04461 {
04462     UBYTE *t, *u;
04463     int val;
04464     t = s;
04465     while ( *t ) t++;
04466     *t++ = '\0';
04467     *t = 0;
04468     if ( ( u = PreIfEval(s,&val) ) == 0 ) return(-1);
04469     if ( u < t ) {
04470         MesPrint("@Unmatched parentheses in condition");
04471         return(-1);
04472     }
04473     if ( val ) return(EXECUTINGIF);
04474     else      return(LOOKINGFORELSE);
04475 }
04476
04477 /*
04478     #] EvalPreIf :
04479     #[ PreIfEval :
04480
04481     Used for recursions in the evaluation of a preprocessor if-condition.
04482     It determines whether the contents of ( ) is true or false
04483     (or in error).
04484     The return value is the address of the first character after the
04485     closing parenthesis or null if there is an error.
04486     In value we find true(1) or false(0)

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04487         We enter after the opening parenthesis.
04488         There are levels:
04489             0: orlevel:  a || b
04490             1: andlevel: a && b
04491             2: eqlevel:  a == b or a != b or a = b
04492             3: cmplevel: a > b or a >= b or a < b or a <= b or a >~ b etc
04493 */
04494
04495 UBYTE *PreIfEval(UBYTE *s, int *value)
04496 {
04497     int orlevel = 0, andlevel = 0, eqlevel = 0, cmplevel = 0;
04498     int type, val;
04499     LONG val2;
04500     int orte, orval, cmptype, cmpval, eqtype, eqval, andtype, andval;
04501     UBYTE *t, *eqt, *cmpt, c;
04502     int eqop, cmpop;
04503     orte = orval = cmptype = cmpval = eqtype = eqval = andtype = andval = 0;
04504     eqop = cmpop = 0;
04505     eqt = cmpt = 0;
04506     *value = 0;
04507     while ( *s != ')' ) {
04508         while ( *s == ' ' || *s == '\t' || *s == '\n' || *s == '\r' ) s++;
04509         t = s;
04510         s = pParseObject(s,&type,&val2);
04511         if ( s == 0 ) return(0);
04512         val = val2;
04513         c = *s;
04514         *s++ = 0; /* in case the object is a string without " */
04515         while ( c == ' ' || c == '\t' || c == '\n' || c == '\r' ) {
04516             c = *s; *s++ = 0;
04517         }
04518         if ( *t == '"' ) t++;
04519         switch(c) {
04520             case '|':
04521                 if ( *s != '|' ) goto illoper;
04522                 s++;
04523                 /* fall through */
04524             case ')':
04525                 if ( cmplevel ) {
04526                     if ( type == 0 || cmptype == 0 ) goto illobject;
04527                     val = PreCmp(type,val,t,cmptype,cmpval,cmpt,cmpop);
04528                     type = 0;
04529                     cmplevel = 0;
04530                 }
04531                 if ( eqlevel ) {
04532                     val = PreEq(type,val,t,eqtype,eqval,eqt,eqop);
04533                     type = 0;
04534                     eqlevel = 0;
04535                 }
04536                 if ( andlevel ) {
04537                     if ( andtype != 0 || type != 0 ) goto illobject;
04538                     val &= andval;
04539                     andlevel = 0;
04540                 }
04541                 if ( orlevel ) {
04542                     if ( orte != 0 || type != 0 ) goto illobject;
04543                     val |= orval;
04544                 }
04545                 if ( c == ')' ) {
04546                     *value = val;
04547                     return(s);
04548                 }
04549                 orlevel = 1;
04550                 orval = val;
04551                 orte = type;
04552                 break;
04553             case '&':
04554                 if ( *s != '&' ) goto illoper;
04555                 s++;
04556                 if ( cmplevel ) {
04557                     if ( type == 0 || cmptype == 0 ) goto illobject;
04558                     val = PreCmp(type,val,t,cmptype,cmpval,cmpt,cmpop);
04559                     type = 0;
04560                     cmplevel = 0;
04561                 }
04562                 if ( eqlevel ) {
04563                     val = PreEq(type,val,t,eqtype,eqval,eqt,eqop);
04564                     type = 0;
04565                     eqlevel = 0;
04566                 }
04567                 if ( andlevel ) {
04568                     if ( andtype != 0 || type != 0 ) goto illobject;
04569                     val &= andval;
04570                 }
04571                 andlevel = 1;
04572                 andval = val;
04573                 andtype = type;

```

```

04574         break;
04575     case '!':
04576     case '=':
04577         if ( eqlevel ) goto illorder;
04578         if ( cmplevel ) {
04579             if ( type == 0 || cmptype == 0 ) goto illobject;
04580             val = PreCmp(type,val,t,cmptype,cmpval,cmpt,cmpop);
04581             type = 0;
04582             cmplevel = 0;
04583         }
04584         if ( c == '!' && *s != '=' ) goto illoper;
04585         if ( *s == '=' ) s++;
04586         if ( c == '!' ) eqop = 1;
04587         else eqop = 0;
04588         eqlevel = 1; eqt = t; eqval = val; eqtype = type;
04589         break;
04590     case '>':
04591     case '<':
04592         if ( cmplevel ) goto illorder;
04593         if ( c == '<' ) cmpop = -1;
04594         else cmpop = 1;
04595         cmplevel = 1; cmpt = t; cmpval = val; cmptype = type;
04596         if ( *s == '=' ) {
04597             s++;
04598             if ( *s == '~' ) { s++; cmpop *= 4; }
04599             else cmpop *= 2;
04600         }
04601         else if ( *s == '~' ) { s++; cmpop *= 3; }
04602         break;
04603     default:
04604         goto illoper;
04605     }
04606 }
04607 return(s);
04608 illorder:
04609     MesPrint("@illegal order of operators");
04610     return(0);
04611 illobject:
04612     MesPrint("@illegal object for this operator");
04613     return(0);
04614 illoper:
04615     MesPrint("@illegal operator");
04616     return(0);
04617 }
04618
04619 /*
04620     #] PreIfEval :
04621     #[ PreCmp :
04622 */
04623
04624 int PreCmp(int type, int val, UBYTE *t, int type2, int val2, UBYTE *t2, int cmpop)
04625 {
04626     if ( type == 2 || type2 == 2 || cmpop < -2 || cmpop > 2 ) {
04627         if ( cmpop < 0 && cmpop > -3 ) cmpop -= 2;
04628         if ( cmpop > 0 && cmpop < 3 ) cmpop += 2;
04629         if ( cmpop == 3 ) val = StrCmp(t2,t) > 0;
04630         else if ( cmpop == 4 ) val = StrCmp(t2,t) >= 0;
04631         else if ( cmpop == -3 ) val = StrCmp(t2,t) < 0;
04632         else if ( cmpop == -4 ) val = StrCmp(t2,t) <= 0;
04633     }
04634     else {
04635         if ( cmpop == 1 ) val = ( val2 > val );
04636         else if ( cmpop == 2 ) val = ( val2 >= val );
04637         else if ( cmpop == -1 ) val = ( val2 < val );
04638         else if ( cmpop == -2 ) val = ( val2 <= val );
04639     }
04640     return(val);
04641 }
04642
04643 /*
04644     #] PreCmp :
04645     #[ PreEq :
04646 */
04647
04648 int PreEq(int type, int val, UBYTE *t, int type2, int val2, UBYTE *t2, int eqop)
04649 {
04650     UBYTE str[20];
04651     if ( type == 2 || type2 == 2 ) {
04652         if ( type != 2 ) { NumToStr(str,val); t = str; }
04653         if ( type2 != 2 ) { NumToStr(str,val2); t2 = str; }
04654         if ( eqop == 1 ) val = StrCmp(t,t2) != 0;
04655         else val = StrCmp(t,t2) == 0;
04656     }
04657     else {
04658         if ( eqop ) val = val != val2;
04659         else val = val == val2;
04660     }

```



```

04748     }
04749     else {
04750         tt = SkipAName(t);
04751         c = *tt; *tt = 0;
04752         if ( GetName(AC.exprnames,t,&numexp,NOAUTO) == NAMENOTFOUND ) {
04753             MesPrint("@%s has not (yet) been defined",t);
04754             *tt = c;
04755             Terminate(-1);
04756         }
04757         else {
04758             *tt = c;
04759             if ( ttype == 1 ) {
04760                 x = SizeOfExpression(numexp);
04761             }
04762             else {
04763                 x = TermsInExpression(numexp);
04764             }
04765         }
04766     }
04767     while ( *tt == ' ' || *tt == '\t'
04768             || *tt == '\n' || *tt == '\r' ) tt++;
04769     if ( *tt != ' ' ) {
04770         MesPrint("@Improper use of terms($var) or terms(expr)");
04771         Terminate(-1);
04772     }
04773     *type = 3;
04774     s = tt+1;
04775     *val2 = x;
04776     return(s);
04777 }
04778 else if ( StrICmp(s,(UBYTE *)"sizeof") == 0 ) {
04779     ttype = 1;
04780     goto together;
04781 }
04782 else if ( StrICmp(s,(UBYTE *)"exists") == 0 ) {
04783     UBYTE *tt;
04784     WORD numdol, numexp;
04785     *t++ = c;
04786     while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
04787     if ( *t == '$' ) {
04788         t++; tt = t; while ( chartype[*tt] <= 1 ) tt++;
04789         c = *tt; *tt = 0;
04790         if ( ( numdol = GetDollar(t) ) >= 0 ) { x = 1; }
04791         else { x = 0; }
04792         *tt = c;
04793     }
04794     else if ( *t == '"' ) { /* see whether a file exists */
04795         /* UBYTE *name, *oldname; */
04796         t++; tt = t;
04797         for (;;) {
04798             if ( *tt == '\\ ' ) tt++;
04799             else if ( *tt == '"' ) break;
04800             tt++;
04801         }
04802         c = *tt; *tt = 0;
04803         /*
04804             Try to open the file. If possible, return 1.
04805             Afterwards close it.
04806             We do have to run through the FORMPATH. Hence we use LocateFile.
04807             This routine may change the name to the full name.
04808
04809             oldname = name = strDupl(t,"name in exists");
04810             x = LocateFile(&name,-1);
04811         */
04812         x = OpenFile((char *)t);
04813         if ( x >= 0 ) {
04814             CloseFile(x);
04815             x = 1;
04816             /*
04817                 if ( name != oldname ) M_free(name,"name from LocateFile");
04818             */
04819         }
04820         else x = 0;
04821         /*
04822             M_free(oldname,"name in exists");
04823         */
04824         *tt++ = c;
04825     }
04826     else {
04827         tt = SkipAName(t);
04828         c = *tt; *tt = 0;
04829         if ( GetName(AC.exprnames,t,&numexp,NOAUTO) == NAMENOTFOUND ) { x = 0; }
04830         else { x = 1; }
04831         *tt = c;
04832     }
04833     while ( *tt == ' ' || *tt == '\t'
04834             || *tt == '\n' || *tt == '\r' ) tt++;

```

```

04835         if ( *tt != ' ' ) {
04836             MesPrint("@Improper use of exists($var) or exists(expr)");
04837             Terminate(-1);
04838         }
04839         *type = 3;
04840         s = tt+1;
04841         *val2 = x;
04842         return(s);
04843     }
04844     else if ( StrICmp(s, (UBYTE *) "isnumerical") == 0 ) {
04845         GETIDENTITY
04846         UBYTE *tt;
04847         WORD numdol, numexp;
04848         *tt++ = c;
04849         while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
04850         if ( *t == '$' ) {
04851             t++; tt = t; while ( chartype[*tt] <= 1 ) tt++;
04852             c = *tt; *tt = 0;
04853             if ( ( numdol = GetDollar(t) ) < 0 ) {
04854                 MesPrint("@$ variable in isnumerical(%s) does not exist",t);
04855                 Terminate(-1);
04856             }
04857             x = DolToLong(BHEAD numdol);
04858             if ( AN.ErrorInDollar ) {
04859                 DOLLARS d = Dollars + numdol;
04860                 x = 0;
04861                 if ( d->type == DOLNUMBER || d->type == DOLTERMS ) {
04862                     if ( d->where[0] == 0 ) x = 1;
04863                     else if ( d->where[d->where[0]] == 0 ) {
04864                         if ( ABS(d->where[d->where[0]-1]) == d->where[0]-1 )
04865                             x = 1;
04866                     }
04867                 }
04868             }
04869             else x = 1;
04870             *tt = c;
04871         }
04872         else {
04873             tt = SkipAName(t);
04874             c = *tt; *tt = 0;
04875             if ( GetName(AC.exprnames,t,&numexp,NOAUTO) == NAMENOTFOUND ) {
04876                 MesPrint("@expression in isnumerical(%s) does not exist",t);
04877                 Terminate(-1);
04878             }
04879             x = TermsInExpression(numexp);
04880             if ( x != 1 ) x = 0;
04881             else {
04882                 WORD *term = AT.WorkPointer;
04883                 if ( GetFirstTerm(term,numexp,1) < 0 ) {
04884                     MesPrint("@error reading expression in isnumerical(%s)",t);
04885                     Terminate(-1);
04886                 }
04887                 if ( *term == ABS(term[*term-1])+1 ) x = 1;
04888                 else x = 0;
04889             }
04890             *tt = c;
04891         }
04892         while ( *tt == ' ' || *tt == '\t'
04893             || *tt == '\n' || *tt == '\r' ) tt++;
04894         if ( *tt != ' ' ) {
04895             MesPrint("@Improper use of isnumerical($var) or numerical(expr)");
04896             Terminate(-1);
04897         }
04898         *type = 3;
04899         s = tt+1;
04900         *val2 = x;
04901         return(s);
04902     }
04903     else if ( StrICmp(s, (UBYTE *) ("maxpowerof")) == 0 ) {
04904         UBYTE *tt;
04905         WORD numsym;
04906         int stype;
04907         *tt++ = c;
04908         while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
04909         tt = SkipAName(t);
04910         c = *tt; *tt = 0;
04911         if ( ( stype = GetName(AC.varnames,t,&numsym,NOAUTO) ) == NAMENOTFOUND ) {
04912             MesPrint("@%s has not (yet) been defined",t);
04913             *tt = c;
04914             Terminate(-1);
04915         }
04916         else if ( stype != CSYMBOL ) {
04917             MesPrint("@%s should be a symbol",t);
04918             *tt = c;
04919             Terminate(-1);
04920         }
04921         else {

```



```

04922         *tt = c;
04923         x = symbols[numsym].maxpower;
04924     }
04925     while ( *tt == ' ' || *tt == '\t'
04926            || *tt == '\n' || *tt == '\r' ) tt++;
04927     if ( *tt != ')' ) {
04928         MesPrint("@Improper use of maxpowerof(symbol)");
04929         Terminate(-1);
04930     }
04931     *type = 3;
04932     s = tt+1;
04933     *val2 = x;
04934     return(s);
04935 }
04936 else if ( StrICmp(s, (UBYTE *)("minpowerof")) == 0 ) {
04937     UBYTE *tt;
04938     WORD numsym;
04939     int stype;
04940     *tt++ = c;
04941     while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
04942     tt = SkipAName(t);
04943     c = *tt; *tt = 0;
04944     if ( ( stype = GetName(AC.varnames,t,&numsym,NOAUTO) ) == NAMENOTFOUND ) {
04945         MesPrint("@%s has not (yet) been defined",t);
04946         *tt = c;
04947         Terminate(-1);
04948     }
04949     else if ( stype != CSYMBOL ) {
04950         MesPrint("@%s should be a symbol",t);
04951         *tt = c;
04952         Terminate(-1);
04953     }
04954     else {
04955         *tt = c;
04956         x = symbols[numsym].minpower;
04957     }
04958     while ( *tt == ' ' || *tt == '\t'
04959            || *tt == '\n' || *tt == '\r' ) tt++;
04960     if ( *tt != ')' ) {
04961         MesPrint("@Improper use of minpowerof(symbol)");
04962         Terminate(-1);
04963     }
04964     *type = 3;
04965     s = tt+1;
04966     *val2 = x;
04967     return(s);
04968 }
04969 else if ( StrICmp(s, (UBYTE *)("isfactorized")) == 0 ) {
04970     UBYTE *tt;
04971     WORD numdol, numexp;
04972     *tt++ = c;
04973     while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
04974     if ( *t == '$' ) {
04975         t++; tt = t; while ( chartype[*tt] <= 1 ) tt++;
04976         c = *tt; *tt = 0;
04977         if ( ( numdol = GetDollar(t) ) > 0 ) {
04978             if ( Dollars[numdol].factors != 0 ) x = 1;
04979             else x = 0;
04980         }
04981         else {
04982             MesPrint("@ %s should be the name of an expression or a $ variable",t-1);
04983             Terminate(-1);
04984         }
04985         *tt = c;
04986     }
04987     else {
04988         tt = SkipAName(t);
04989         c = *tt; *tt = 0;
04990         if ( GetName(AC.exprnames,t,&numexp,NOAUTO) == NAMENOTFOUND ) {
04991             MesPrint("@ %s should be the name of an expression or a $ variable",t);
04992             Terminate(-1);
04993         }
04994         else {
04995             if ( ( Expressions[numexp].vflags & ISFACTORIZED ) != 0 ) x = 1;
04996             else x = 0;
04997         }
04998         *tt = c;
04999     }
05000     while ( *tt == ' ' || *tt == '\t'
05001            || *tt == '\n' || *tt == '\r' ) tt++;
05002     if ( *tt != ')' ) {
05003         MesPrint("@Improper use of isfactorized($var) or isfactorized(expr)");
05004         Terminate(-1);
05005     }
05006     *type = 3;
05007     s = tt+1;
05008     *val2 = x;

```

```

05009         return(s);
05010     }
05011     else if ( StrICmp(s, (UBYTE *) "isdefined") == 0 ) {
05012         UBYTE *tt;
05013         *t++ = c;
05014         while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
05015         tt = SkipAName(t);
05016         c = *tt; *tt = 0;
05017         if ( GetPreVar(t, WITHOUTERROR) != 0 ) x = 1;
05018         else x = 0;
05019         *tt = c;
05020         while ( *tt == ' ' || *tt == '\t'
05021             || *tt == '\n' || *tt == '\r' ) tt++;
05022         if ( *tt != ')' ) {
05023             MesPrint("@Improper use of isdefined(var)");
05024             Terminate(-1);
05025         }
05026         *type = 3;
05027         s = tt+1;
05028         *val2 = x;
05029         return(s);
05030     }
05031     else if ( StrICmp(s, (UBYTE *) "flag") == 0 ) {
05032         UBYTE *tt;
05033         WORD x = 0, numexp;
05034         {
05035             *t++ = c;
05036             while ( *t == ' ' || *t == '\t' ) t++;
05037             if ( FG.cTable[*t] != 1 ) goto flagerror;
05038             while ( FG.cTable[*t] == 1 ) x = 10*x + (*t++ - '0');
05039             if ( x < 1 || x > BITSINWORD ) {
05040                 MesPrint("@Illegal number %d for flag in flag condition", x);
05041                 goto flagerror;
05042             }
05043             while ( *t == ' ' || *t == '\t' ) t++;
05044             if ( *t != ',' ) goto flagerror;
05045             t++;
05046             while ( *t == ' ' || *t == '\t' ) t++;
05047             tt = SkipAName(t);
05048             c = *tt; *tt = 0;
05049             if ( GetName(AC.exprnames, t, &numexp, NOAUTO) == NAMENOTFOUND ) {
05050                 MesPrint("@ %s should be the name of an expression", t);
05051                 goto flagerror;
05052             }
05053             *tt = c;
05054             while ( *t == ' ' || *t == '\t' ) t++;
05055             if ( *tt != ')' ) {
05056 flagerror:
05057                 MesPrint("@Improper use of flag(num,expr)");
05058                 Terminate(-1);
05059                 return(0);
05060             }
05061             if ( ( Expressions[numexp].uflags & ( 1 << (x-1) ) ) != 0 )
05062                 *val2 = 1;
05063             else *val2 = 0;
05064             *type = 3;
05065             s = tt+1;
05066             return(s);
05067         }
05068     }
05069     else *t = c;
05070 }
05071 else if ( *t == '=' || *t == '<' || *t == '>' || *t == '!'
05072 || *t == ')' || *t == ' ' || *t == '\t' || *t == 0 || *t == '\n' ) {
05073     *val2 = 0;
05074     *type = 2;
05075     return(t);
05076 }
05077 else {
05078     MesPrint("@Illegal use of string in preprocessor condition: %s", s);
05079     Terminate(-1);
05080 }
05081 }
05082 while ( *t == '-' || *t == '+' || *t == ' ' || *t == '\t' ) {
05083     if ( *t == '-' ) sign = -sign;
05084     t++;
05085 }
05086 while ( chartype[*t] == 1 ) { x = 10*x + *t++ - '0'; }
05087 while ( *t == ' ' || *t == '\t' ) t++;
05088 if ( chartype[*t] == 8 || *t == ')' || *t == '=' || *t == 0 ) {
05089     *val2 = sign > 0 ? x: -x;
05090     *type = 1;
05091     return(t);
05092 }
05093 while ( chartype[*t] != 8 && *t != ')' && *t != '=' && *t ) t++;
05094 while ( ( t > s ) && ( t[-1] == ' ' || t[-1] == '\t' ) ) t--;
05095 *type = 2;

```

```

05096     *val2 = val;
05097     return(t);
05098 }
05099
05100 /*
05101     #] pParseObject :
05102     #[ PreCalc :
05103
05104     To be called when a { is encountered.
05105     Action: read first till matching }. This is to be stored.
05106     Next we look whether this is a set or whether it can be
05107     evaluated. If it is a set we consider it as a new stream.
05108     The stream will have to be deallocated when read completely.
05109     If it is to be evaluated we do that and put the result in
05110     a stream.
05111 */
05112
05113 UBYTE *PreCalc(void)
05114 {
05115     UBYTE *buff, *s = 0, *t, *newb, c;
05116     int size, i, n, parlevel = 0, bralevel = 0;
05117     LONG answer;
05118     ULONG uanswer;
05119     size = n = 0;
05120     buff = 0; c = '{';
05121     for (;;) {
05122         if ( n >= size ) {
05123             if ( size == 0 ) size = 72;
05124             else size *= 2;
05125             if ( ( newb = (UBYTE *)Malloc1(size+2,"{ }") ) == 0 ) return(0);
05126             s = newb;
05127             if ( buff ) {
05128                 i = n;
05129                 t = buff;
05130                 NCOPYB(s,t,i);
05131                 M_free(buff,"pre calc buffer");
05132             }
05133             else s = newb;
05134             buff = newb;
05135         }
05136         *s++ = c; n++;
05137         c = GetChar(0);
05138         if ( c == 0 ) {
05139             Error0("Unmatched {}");
05140             M_free(buff,"precalc buffer");
05141             return(0);
05142         }
05143         else if ( c == '{' ) { bralevel++; }
05144         else if ( c == '}' ) {
05145             if ( --bralevel < 0 ) { *s++ = c; *s = 0; break; }
05146         }
05147         else if ( c == '(' ) { parlevel++; }
05148         else if ( c == ')' ) {
05149             if ( --parlevel < 0 ) { *s++ = c; *s = 0; goto setstring; }
05150         }
05151         else if ( chartype[c] != 1 && chartype[c] != 5
05152             && chartype[c] != 6 && c != '!' && c != '&'
05153             && c != '|' && c != '\\') { *s++ = c; *s = 0; goto setstring; }
05154     }
05155     if ( parlevel > 0 ) goto setstring;
05156 /*
05157     Try now to evaluate the string.
05158     If it works, copy the resulting value back into buff as a string.
05159 */
05160     answer = 0;
05161     if ( PreEval(buff+1,&answer) == 0 ) goto setstring;
05162     t = buff + size;
05163     s = buff;
05164     if ( answer < 0 ) { *s++ = '-'; }
05165     uanswer = LongAbs(answer);
05166     n = 0;
05167     do {
05168         *--t = ( uanswer % 10 ) + '0';
05169         uanswer /= 10;
05170         n++;
05171     } while ( uanswer > 0 );
05172     NCOPYB(s,t,n);
05173     *s = 0;
05174     setstring;
05175 /*
05176     Open a stream that contains the current string.
05177     Mark it to be removed after termination.
05178 */
05179     if ( OpenStream(buff,PRECALCSTREAM,0,PRENOACTION) == 0 ) return(0);
05180     return(buff);
05181 }
05182

```

```

05183 /*
05184     #] PreCalc :
05185     #[ PreEval :
05186
05187     Operations are:
05188     +, -, *, /, %, &, |, ^, !,  ^% (postfix 2log), ^/ (postfix sqrt)
05189 */
05190
05191 UBYTE *PreEval(UBYTE *s, LONG *x)
05192 {
05193     LONG y, z, a;
05194     int tobemultiplied, tobeadded = 1, expsign, i;
05195     UBYTE *t;
05196     *x = 0; a = 1;
05197     while ( *s == ' ' || *s == '\t' ) s++;
05198     for(;;){
05199         if ( *s == '+' || *s == '-' ) {
05200             if ( *s == '-' ) tobeadded = -1;
05201             else tobeadded = 1;
05202             s++;
05203             while ( *s == '-' || *s == '+' || *s == ' ' || *s == '\t' ) {
05204                 if ( *s == '-' ) tobeadded = -tobeadded;
05205                 s++;
05206             }
05207         }
05208         tobemultiplied = 0;
05209         for(;;){
05210             while ( *s == ' ' || *s == '\t' ) s++;
05211             if ( *s <= '9' && *s >= '0' ) {
05212                 ULONG uy;
05213                 ParseNumber(uy,s)
05214                 y = uy; /* may cause an implementation-defined behaviour */
05215             }
05216             else if ( *s == '(' || *s == '{' ) {
05217                 if ( ( t = PreEval(s+1,&y) ) == 0 ) return(0);
05218                 s = t;
05219             }
05220             else return(0);
05221             while ( *s == ' ' || *s == '\t' ) s++;
05222             expsign = 1;
05223             while ( *s == '^' || *s == '!' ) {
05224                 s++;
05225                 if ( s[-1] == '!' ) { /* factorial of course */
05226                     while ( *s == ' ' || *s == '\t' ) s++;
05227                     if ( y < 0 ) {
05228                         MesPrint("@Negative value in preprocessor factorial: %1",y);
05229                         return(0);
05230                     }
05231                     else if ( y == 0 ) y = 1;
05232                     else if ( y > 1 ) {
05233                         z = y-1;
05234                         while ( z > 0 ) { y = y*z; z--; }
05235                     }
05236                     continue;
05237                 }
05238                 else if ( *s == '%' ) { /* ^% is postfix 2log */
05239                     s++;
05240                     while ( *s == ' ' || *s == '\t' ) s++;
05241                     z = y;
05242                     if ( z <= 0 ) {
05243                         MesPrint("@Illegal value in preprocessor logarithm: %1",z);
05244                         return(0);
05245                     }
05246                     y = 0; z >= 1;
05247                     while ( z ) { y++; z >= 1; }
05248                     continue;
05249                 }
05250                 else if ( *s == '/' ) { /* ^/ is postfix sqrt */
05251                     LONG yy, zz;
05252                     s++;
05253                     while ( *s == ' ' || *s == '\t' ) s++;
05254                     z = y;
05255                     if ( z <= 0 ) {
05256                         MesPrint("@Illegal value in preprocessor square root: %1",z);
05257                         return(0);
05258                     }
05259                     if ( z > 8 ) { /* Very crude integer square root */
05260                         zz = z;
05261                         yy = 0; zz >= 1;
05262                         while ( zz ) { yy++; zz >= 1; }
05263                         zz = z >> (yy/2); i = 10; y = 0;
05264                         do {
05265                             yy = zz/2 + z/(2*zz); i--;
05266                             if ( y == yy ) break;
05267                             y = zz; zz = yy;
05268                         } while ( y != yy && i > 0 );
05269                         while ( y*y < z ) y++;

```

```

05270         while ( y*y > z ) y--;
05271     }
05272     else if ( z >= 4 ) y = 2;
05273     else if ( z == 0 ) y = 0;
05274     else y = 1;
05275     continue;
05276 }
05277 while ( *s == ' ' || *s == '\t' ) s++;
05278 while ( *s == '-' || *s == '+' || *s == ' ' || *s == '\t' ) {
05279     if ( *s == '-' ) expsign = -expsign;
05280 }
05281 if ( *s <= '9' && *s >= '0' ) {
05282     ParseNumber(z,s)
05283 }
05284 else if ( *s == '(' || *s == '{' ) {
05285     if ( ( t = PreEval(s+1,&z) ) == 0 ) return(0);
05286     s = t;
05287 }
05288 else return(0);
05289 while ( *s == ' ' || *s == '\t' ) s++;
05290 y = iexp(y,(int)z);
05291 }
05292 if ( tobemultiplied == 0 ) {
05293     if ( expsign < 0 ) a = 1/y;
05294     else a = y;
05295 }
05296 else {
05297     if ( tobemultiplied > 2 && expsign != 1 ) {
05298         MesPrint("&Incorrect use of ^ with & or |. Use brackets!");
05299         Terminate(-1);
05300     }
05301     tobemultiplied *= expsign;
05302     if ( tobemultiplied == 1 ) a *= y;
05303     else if ( tobemultiplied == 3 ) a &= y;
05304     else if ( tobemultiplied == 4 ) a |= y;
05305     else {
05306         if ( y == 0 || tobemultiplied == -2 ) {
05307             MesPrint("@Division by zero in preprocessor calculator");
05308             Terminate(-1);
05309         }
05310         if ( tobemultiplied == 2 ) a %= y;
05311         else a /= y;
05312     }
05313 }
05314 if ( *s == '%' ) tobemultiplied = 2;
05315 else if ( *s == '*' ) tobemultiplied = 1;
05316 else if ( *s == '/' ) tobemultiplied = -1;
05317 else if ( *s == '&' ) tobemultiplied = 3;
05318 else if ( *s == '|' ) tobemultiplied = 4;
05319 else {
05320     ULONG ux, ua;
05321     ux = *x;
05322     ua = a;
05323     if ( tobeadded >= 0 ) ux += ua;
05324     else ux -= ua;
05325     *x = ULONGToLong(ux);
05326     if ( *s == ')' || *s == '}' ) return(s+1);
05327     else if ( *s == '-' || *s == '+' ) { tobeadded = 1; break; }
05328     else return(0);
05329 }
05330 s++;
05331 }
05332 }
05333 /* return(0); */
05334 }
05335
05336 /*
05337     #] PreEval :
05338     #[ AddToPreTypes :
05339 */
05340
05341 void AddToPreTypes(int type)
05342 {
05343     if ( AP.NumPreTypes >= AP.MaxPreTypes ) {
05344         int i, *newlist = (int *)Malloca(sizeof(int)*(2*AP.MaxPreTypes+1)
05345             , "preprocessor type lists");
05346         for ( i = 0; i <= AP.MaxPreTypes; i++ ) newlist[i] = AP.PreTypes[i];
05347         M_free(AP.PreTypes, "preprocessor type lists");
05348         AP.PreTypes = newlist;
05349         AP.MaxPreTypes = 2*AP.MaxPreTypes;
05350     }
05351     AP.PreTypes[AP.NumPreTypes] = type;
05352 }
05353
05354 /*
05355     #] AddToPreTypes :
05356     #[ MessPreNesting :

```

```

05357 */
05358
05359 void MessPreNesting(int par)
05360 {
05361     MesPrint("@(%d)Illegal nesting of %sif, %sdo, %sprocedure and/or %sswitch",par);
05362 }
05363
05364 /*
05365     #] MessPreNesting :
05366     #[ DoPreAddSeparator :
05367
05368     Preprocessor directives "addseparator" and "rmseparator" add/remove
05369     separator characters used to separate function arguments.
05370     Example:
05371
05372         #define QQ "a|g|a"
05373         #addseparator %
05374         *Comma must be quoted!:
05375         #rmseparator " ,"
05376         #rmseparator |
05377         #call H(a,a%`QQ')
05378
05379     Characters ' ', '\t' and '"' are ignored!
05380 */
05381
05382 int DoPreAddSeparator(UBYTE *s)
05383 {
05384     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
05385     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
05386     for(;*s != '\0';s++){
05387         while ( *s == ' ' || *s == '\t' || *s == '"' ) s++;
05388         /* Todo:
05389         if ( set_in(*s,invalidseparators) ) {
05390             MesPrint("@Invalid separator specified");
05391             return(-1);
05392         }
05393         */
05394         set_set(*s,AC.separators);
05395     }
05396     return(0);
05397 }
05398
05399 /*
05400     #] DoPreAddSeparator :
05401     #[ DoPreRmSeparator :
05402
05403     See commentary with DoPreAddSeparator
05404
05405     Characters ' ', '\t' and '"' are ignored!
05406 */
05407 int DoPreRmSeparator(UBYTE *s)
05408 {
05409     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
05410     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
05411     for(;*s != '\0';s++){
05412         while ( *s == ' ' || *s == '\t' || *s == '"' ) s++;
05413         set_del(*s,AC.separators);
05414     }
05415     return(0);
05416 }
05417
05418 /*
05419     #] DoPreRmSeparator :
05420     #[ DoExternal:
05421
05422     #external ["prevar"] command
05423 */
05424 int DoExternal(UBYTE *s)
05425 {
05426     #ifdef WITHEXTERNALCHANNEL
05427         UBYTE *prevar=0;
05428         int externalD= 0;
05429     #else
05430         DUMMYUSE(s);
05431     #endif
05432     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
05433     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
05434     if ( AP.preError ) return(0);
05435
05436     #ifdef WITHEXTERNALCHANNEL
05437         while ( *s == ' ' || *s == '\t' ) s++;
05438         if(*s == '"'){/*prevar to store the descriptor is defined*/
05439             prevar=++s;
05440
05441             if ( chartype[*s] == 0 )for(;*s != '"'; s++)switch(chartype[*s]){
05442                 case 10:/*'\0' fits here*/
05443                     MesPrint("@Can't finde closing \"");

```

```

05444         Terminate(-1);
05445         case 0:case 1: continue;
05446         default:
05447             break;
05448     }
05449     if(*s != '"'){
05450         MesPrint("@Illegal name of preprocessor variable to store external channel");
05451         return(-1);
05452     }
05453     *s='\0';
05454     for(s++; *s == ' ' || *s == '\t'; s++);
05455 }
05456
05457 if(*s == '\0'){
05458     MesPrint("@Illegal external command");
05459     return(-1);
05460 }
05461 /*here s is a command*/
05462 /*See the file extcmd.c*/
05463 /*[08may2006 mt]:*/
05464 externalD=openExternalChannel(
05465     s,
05466     AX.daemonize,
05467     AX.shellname,
05468     AX.stderrname);
05469 /*:[08may2006 mt]*/
05470 if(externalD<1){/*error?*/
05471     /*Not quite correct - terminate the program on error:*/
05472     Error1("Can't start external program",s);
05473     return(-1);
05474 }
05475 /*Now external command runs.*/
05476
05477 if(prevar){/*Store the external channel descriptor in the provided variable:*/
05478     UBYTE buf[21];/* 64/Log_2[10] = 19.3, so this is enough forever...*/
05479     NumToStr(buf,externalD);
05480     if ( PutPreVar(prevar,buf,0,1) < 0 ) return(-1);
05481 }
05482
05483 AX.currentExternalChannel=externalD;
05484 /*[08may2006 mt]:*/
05485 if(AX.currentPrompt!=0){/*Change default terminator*/
05486     if(setTerminatorForExternalChannel( (char *)AX.currentPrompt)){
05487         MesPrint("@Prompt is too long");
05488         return(-1);
05489     }
05490 }
05491 setKillModeForExternalChannel(AX.killSignal,AX.killWholeGroup);
05492 /*:[08may2006 mt]*/
05493 return(0);
05494 #else /*ifdef WITHEXTERNALCHANNEL*/
05495 Error0("External channel: not implemented on this computer/system");
05496 return(-1);
05497 #endif /*ifdef WITHEXTERNALCHANNEL ... else*/
05498 }
05499
05500 /*
05501     #] DoExternal:
05502     #[ DoPrompt:
05503         #prompt string
05504 */
05505
05506 int DoPrompt(UBYTE *s)
05507 {
05508     #ifndef WITHEXTERNALCHANNEL
05509         DUMMYUSE(s);
05510     #endif
05511     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
05512     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
05513
05514     #ifndef WITHEXTERNALCHANNEL
05515         while ( *s == ' ' || *s == '\t' ) s++;
05516         if ( AX.currentPrompt )
05517             M_free(AX.currentPrompt,"external channel prompt");
05518         if ( *s == '\0' )
05519             AX.currentPrompt = (UBYTE *)strDup1((UBYTE *)"", "external channel prompt");
05520         else
05521             AX.currentPrompt = strDup1(s, "external channel prompt");
05522         if( setTerminatorForExternalChannel( (char *)AX.currentPrompt) > 0 ){
05523             MesPrint("@Prompt is too long");
05524             return(-1);
05525         }
05526         /*else: if 0, ok; if -1, there is no current channel-ok, just prompt is stored.*/
05527         return(0);
05528     #else /*ifdef WITHEXTERNALCHANNEL*/
05529         Error0("External channel: not implemented on this computer/system");
05530         return(-1);
05531     #endif
05532 }

```

```

05531 #endif /*ifdef WITHEXTERNALCHANNEL ... else*/
05532 }
05533 /*
05534     #] DoPrompt:
05535     #[ DoSetExternal:
05536         #setexternal n
05537 */
05538
05539 int DoSetExternal(UBYTE *s)
05540 {
05541     #ifdef WITHEXTERNALCHANNEL
05542         int n=0;
05543     #else
05544         DUMMYUSE(s);
05545     #endif
05546     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
05547     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
05548     if ( AP.preError ) return(0);
05549
05550     #ifdef WITHEXTERNALCHANNEL
05551         while ( *s == ' ' || *s == '\t' ) s++;
05552         while ( chartype[*s] == 1 ) { n = 10*n + *s++ - '0'; }
05553         while ( *s == ' ' || *s == '\t' ) s++;
05554         if (*s!='\0'){
05555             MesPrint("@setexternal: number expected");
05556             return(-1);
05557         }
05558         if(selectExternalChannel(n)<0){
05559             MesPrint("@setexternal: invalid number");
05560             return(-1);
05561         }
05562         AX.currentExternalChannel=n;
05563         return(0);
05564     #else /*ifdef WITHEXTERNALCHANNEL*/
05565         Error0("External channel: not implemented on this computer/system");
05566         return(-1);
05567     #endif /*ifdef WITHEXTERNALCHANNEL ... else*/
05568 }
05569 /*
05570     #] DoSetExternal:
05571     #[ DoSetExternalAttr:
05572 */
05573
05574 static FORM_INLINE UBYTE *pickupword(UBYTE *s)
05575 {
05576     for(;*s>' ';s++)switch(*s){
05577         case '=':
05578         case ',':
05579         case ';':
05580             return(s);
05581     }/*for(;*s>' ';s++)switch(*s)*/
05582     return(s);
05583 }
05584
05585 /*Returns 0 if the first string (case insensitively) equal to
05586 the beginning of the second string (of length n):
05587 */
05588 static inline int strINCmp(UBYTE *a, UBYTE *b, int n)
05589 {
05590     for(;n>0;n--)if(tolower(*a++)!=tolower(*b++))
05591         return(1);
05592     return(*a != '\0');
05593 }
05594
05595 #define KILL "kill"
05596 #define KILLALL "killall"
05597 #define DAEMON "daemon"
05598 #define SHELL "shell"
05599 #define STDERR "stderr"
05600
05601 #define TRUE_EXPR "true"
05602 #define FALSE_EXPR "false"
05603 #define NOSHELL "noshell"
05604 #define TERMINAL "terminal"
05605
05606 /*
05607     Expects comma-separated list of pairs name=value
05608 */
05609 int DoSetExternalAttr(UBYTE *s)
05610 {
05611     #ifdef WITHEXTERNALCHANNEL
05612         int lnam,lval;
05613         UBYTE *nam,*val;
05614     #else
05615         DUMMYUSE(s);
05616     #endif
05617     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);

```



```

05618     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
05619     if ( AP.preError ) return(0);
05620
05621 #ifdef WITHEXTERNALCHANNEL
05622     do{
05623         /*Read the name:*/
05624         while ( *s == ' ' || *s == '\t' ) s++;
05625         s=pickupword(nam=s);
05626         lnam=s-nam;
05627         while ( *s == ' ' || *s == '\t' ) s++;
05628         if(*s++!='='){
05629             MesPrint("@External channel:'=' expected instead of %s",s-1);
05630             return(-1);
05631         }
05632         /*Read the value:*/
05633         while ( *s == ' ' || *s == '\t' ) s++;
05634         val=s;
05635
05636         for(;;){
05637             UBYTE *m;
05638             s=pickupword(s);
05639             m=s;
05640             while ( *s == ' ' || *s == '\t' ) s++;
05641             if( (*s == ',') || (*s == '\n') || (*s == ';') || (*s == '\0') ){
05642                 s=m;
05643                 break;
05644             }
05645         }/*for(;;)*/
05646
05647         lval=s-val;
05648         while ( *s == ' ' || *s == '\t' ) s++;
05649
05650         if(strINComp((UBYTE *)SHELL,nam,lnam)==0){
05651             if(AX.shellname!=NULL)
05652                 M_free(AX.shellname,"external channel shellname");
05653             if(strINComp((UBYTE *)NOSHELL,val,lval)==0)
05654                 AX.shellname=NULL;
05655             else{
05656                 UBYTE *ch,*b;
05657                 b=ch=AX.shellname=Malloc1(lval+1,"external channel shellname");
05658                 while(ch-b<lval)
05659                     *ch++=*val++;
05660                 *ch='\0';
05661             }
05662         }else if(strINComp((UBYTE *)DAEMON,nam,lnam)==0){
05663             if(strINComp((UBYTE *)TRUE_EXPR,val,lval)==0)
05664                 AX.daemonize = 1;
05665             else if(strINComp((UBYTE *)FALSE_EXPR,val,lval)==0)
05666                 AX.daemonize = 0;
05667             else{
05668                 MesPrint("@External channel:true or false expected for %s",DAEMON);
05669                 return(-1);
05670             }
05671         }else if(strINComp((UBYTE *)KILLALL,nam,lnam)==0){
05672             if(strINComp((UBYTE *)TRUE_EXPR,val,lval)==0)
05673                 AX.killWholeGroup = 1;
05674             else if(strINComp((UBYTE *)FALSE_EXPR,val,lval)==0)
05675                 AX.killWholeGroup = 0;
05676             else{
05677                 MesPrint("@External channel: true or false expected for %s",KILLALL);
05678                 return(-1);
05679             }
05680         }else if(strINComp((UBYTE *)KILL,nam,lnam)==0){
05681             int i,n=0;
05682             for(i=0;i<lval;i++){
05683                 if( *val>='0' && *val<= '9' )
05684                     n = 10*n + *val++ - '0';
05685                 else{
05686                     MesPrint("@External channel: number expected for %s",KILL);
05687                     return(-1);
05688                 }
05689             }
05690             AX.killSignal=n;
05691         }else if(strINComp((UBYTE *)STDERR,nam,lnam)==0){
05692             if( AX.stdername != NULL ) {
05693                 M_free(AX.stdername,"external channel stderrname");
05694             }
05695             if(strINComp((UBYTE *)TERMINAL,val,lval)==0)
05696                 AX.stdername = NULL;
05697             else{
05698                 UBYTE *ch,*b;
05699                 b=ch=AX.stdername=Malloc1(lval+1,"external channel stderrname");
05700                 while(ch-b<lval)
05701                     *ch++=*val++;
05702                 *ch='\0';
05703             }
05704         }else{

```

```

05705         nam[lname+1]='\0';
05706         MesPrint("@External channel: unrecognized attribute",nam);
05707         return(-1);
05708     }
05709 }while(*s++ == ' ');
05710 if( (*s-1)>' ')&&(*s-1)!=';' ){
05711     MesPrint("@External channel: syntax error: %s",s-1);
05712     return(-1);
05713 }
05714 return(0);
05715 #else /*ifdef WITHEXTERNALCHANNEL*/
05716     Error0("External channel: not implemented on this computer/system");
05717     return(-1);
05718 #endif /*ifdef WITHEXTERNALCHANNEL ... else*/
05719 }
05720 /*
05721     #] DoSetExternalAttr:
05722     #[ DoRmExternal:
05723         #rmexternal [n] (if 0, close all)
05724 */
05725
05726 int DoRmExternal(UBYTE *s)
05727 {
05728     #ifdef WITHEXTERNALCHANNEL
05729         int n = -1;
05730     #else
05731         DUMMYUSE(s);
05732     #endif
05733     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
05734     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
05735     if ( AP.preError ) return(0);
05736
05737     #ifdef WITHEXTERNALCHANNEL
05738         while ( *s == ' ' || *s == '\t' ) s++;
05739         if( chartype[*s] == 1 ){
05740             for(n=0; chartype[*s] == 1 ; s++) { n = 10*n + *s - '0'; }
05741             while ( *s == ' ' || *s == '\t' ) s++;
05742         }
05743         if(*s!='\0'){
05744             MesPrint("@rmexternal: invalid number");
05745             return(-1);
05746         }
05747         switch(n){
05748             case 0:/*Close all opened channels*/
05749                 closeAllExternalChannels();
05750                 AX.currentExternalChannel=0;
05751                 /*Do not clean AX.currentPrompt!*/
05752                 return(0);
05753             case -1:/*number is not specified - try current*/
05754                 n=AX.currentExternalChannel;
05755                 /* fall through */
05756             default:
05757                 closeExternalChannel(n);/*No reaction for possible error*/
05758         }
05759         if (n == AX.currentExternalChannel)/*cleaned up by closeExternalChannel()*/
05760             AX.currentExternalChannel=0;
05761         return(0);
05762     #else /*ifdef WITHEXTERNALCHANNEL*/
05763         Error0("External channel: not implemented on this computer/system");
05764         return(-1);
05765     #endif /*ifdef WITHEXTERNALCHANNEL ... else*/
05766 }
05767 }
05768 /*
05769     #] DoRmExternal:
05770     #[ DoFromExternal :
05771         #fromexternal
05772             is used to read the text from the running external
05773             program, the syntax is similar to the #include
05774             directive.
05775         #fromexternal "varname"
05776             is used to read the text from the running external
05777             program into the preprocessor variable varname.
05778             directive.
05779         #fromexternal "varname" maxlength
05780             is used to read the text from the running external
05781             program into the preprocessor variable varname.
05782             directive. Only first maxlength characters are
05783             stored.
05784
05785             FORM continues to read the running external
05786             program output until the external program outputs a
05787             prompt.
05788 */
05789 */
05790
05791 int DoFromExternal(UBYTE *s)

```

```

05792 {
05793 #ifdef WITHEXTERNALCHANNEL
05794     UBYTE *prevar=0;
05795     int lbuf=-1;
05796     int withNoList=AC.NoShowInput;
05797     int oldpreassignflag;
05798 #else
05799     DUMMYUSE(s);
05800 #endif
05801     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
05802     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
05803     if ( AP.preError ) return(0);
05804 #ifdef WITHEXTERNALCHANNEL
05805     FLUSHCONSOLE;
05806
05807     while ( *s == ' ' || *s == '\t' ) s++;
05808     /*[17may2006 mt]:*/
05809     if ( *s == '-' || *s == '+' ) {
05810         if ( *s == '-' )
05811             withNoList = 1;
05812         else
05813             withNoList = 0;
05814         s++;
05815         while ( *s == ' ' || *s == '\t' ) s++;
05816     }/*if ( *s == '-' || *s == '+' )*/
05817     /*:[17may2006 mt]*/
05818     /*[02feb2006 mt]:*/
05819     if(*s == '"'){/*prevar to store the output is defined*/
05820         prevar=++s;
05821
05822         if ( *s=='$' || chartype[*s] == 0 )for(;*s != '"'; s++)switch(chartype[*s]){
05823             case 10:/*'\0' fits here*/
05824                 MesPrint("@Can't finde closing \"");
05825                 Terminate(-1);
05826             case 0:case 1: continue;
05827             default:
05828                 break;
05829         }
05830         if(*s != '"'){
05831             MesPrint("@Illegal name to store output of external channel");
05832             return(-1);
05833         }
05834         *s='\0';
05835         for(s++; *s == ' ' || *s == '\t'; s++);
05836     }/*if(*s == '"')*/
05837
05838     if(*s != '\0'){
05839         if( chartype[*s] == 1 ){
05840             for(lbuf=0; chartype[*s] == 1 ; s++) { lbuf = 10*lbuf + *s - '0'; }
05841             while ( *s == ' ' || *s == '\t' ) s++;
05842         }
05843         if( (*s!='\0')||(lbuf<0) ){
05844             MesPrint("@Illegal buffer length in fromexternal");
05845             return(-1);
05846         }
05847     }
05848     }/*if(*s != '\0')*/
05849     /*:[02feb2006 mt]*/
05850     if(getCurrentExternalChannel()!=AX.currentExternalChannel)
05851         /*[08may20006 mt]:*/
05852         /*selectExternalChannel(AX.currentExternalChannel);*/
05853         if(selectExternalChannel(AX.currentExternalChannel)){
05854             MesPrint("@No current external channel");
05855             return(-1);
05856         }
05857         /*:[08may20006 mt]*/
05858
05859     /*[02feb2006 mt]:*/
05860     if(prevar!=0){/*The result must be stored into preprovar*/
05861         UBYTE *buf;
05862         int cc = 0;
05863         if(lbuf == -1){/*Unlimited buffer, everything must be stored*/
05864             int i;
05865             buf=Malloc1( (lbuf=255)+1,"Fromexternal");
05866             /*[18may20006 mt]:*/
05867             /*for(i=0; (cc=getcFromExtChannel())!=EOF;i++){*/
05868             /* May 2006: now getcFromExtChannelOk returns EOF while
05869             getcFromExtChannelFailure returns -2 (see comments in
05870             exctcmd.c):*/
05871             for(i=0; (cc=getcFromExtChannel())>0;i++){
05872                 /*:[18may20006 mt]*/
05873                 if(i==lbuf){
05874                     int j;
05875                     UBYTE *tmp=Malloc1( (lbuf*=2)+1,"Fromexternal");
05876                     for (j=0; j<i; j++) tmp[j]=buf[j];
05877                     M_free(buf,"Fromexternal");
05878                     buf=tmp;

```

```

05879         }
05880         buf[i]=(UBYTE)(cc);
05881     }/*for(i=0; (cc=getcFromExtChannel())>0;i++)*/
05882     /*[18may20006 mt]:*/
05883     if(cc == -2){
05884         MesPrint("@No current external channel");
05885         return(-1);
05886     }
05887     lbuf=i;
05888     /*:[18may20006 mt]*/
05889     buf[i]='\0';
05890 }else{/*Fixed buffer, only lbuf chars must be stored*/
05891     int i;
05892     buf=Malloc1(lbuf+1,"Fromexternal");
05893     for(i=0; i<lbuf;i++){
05894         /*[18may20006 mt]:*/
05895         /*if( (cc=getcFromExtChannel())==EOF )*/
05896         /* May 2006: now getcFromExtChannelOk returns EOF while
05897            getcFromExtChannelFailure returns -2 (see comments in
05898            exctcmd.c):*/
05899         if( (cc=getcFromExtChannel())<1 )
05900         /*:[18may20006 mt]*/
05901             break;
05902         buf[i]=(UBYTE)(cc);
05903     }
05904     buf[i]='\0';
05905     /*[18may20006 mt]:*/
05906     /*if(cc!=EOF)
05907        while(getcFromExtChannel()!=EOF);**Eat the rest*/
05908     /* May 2006: now getcFromExtChannelOk returns EOF while
05909        getcFromExtChannelFailure returns -2 (see comments in
05910        exctcmd.c):*/
05911     if(cc>0)
05912         while(getcFromExtChannel())>0; /*Eat the rest*/
05913     else if(cc == -2){
05914         MesPrint("@No current external channel");
05915         return(-1);
05916     }
05917     /*:[18may20006 mt]*/
05918 }
05919 /*[18may20006 mt]:*/
05920 if(*prevar == '$'){/*Put the answer to the dollar variable*/
05921     int oldNumPotModdollars = NumPotModdollars;
05922 #ifdef WITHMPI
05923     WORD oldRhsExprInModuleFlag = AC.RhsExprInModuleFlag;
05924     AC.RhsExprInModuleFlag = 0;
05925 #endif
05926     /*Here lbuf is the actual length of buf!*/
05927     /*"prevar=buf'\0'":*/
05928     UBYTE *pbuf=Malloc1(StrLen(prevar)+1+lbuf+1,"Fromexternal to dollar");
05929     UBYTE *c=pbuf;
05930     UBYTE *b=prevar;
05931     while(*b!='\0'){*c++ = *b++;}
05932     *c++='=';
05933     b=buf;
05934     while( (*c++*b++)!='\0' );
05935     oldpreassignflag = AP.PreAssignFlag;
05936     AP.PreAssignFlag = 1;
05937     if ( ( cc = CompileStatement(pbuf) ) || ( cc = CatchDollar(0) ) ) {
05938         Error1("External channel: can't assign output to dollar variable ",prevar);
05939     }
05940     AP.PreAssignFlag = oldpreassignflag;
05941     NumPotModdollars = oldNumPotModdollars;
05942 #ifdef WITHMPI
05943     AC.RhsExprInModuleFlag = oldRhsExprInModuleFlag;
05944 #endif
05945     M_free(pbuf,"Fromexternal to dollar");
05946 }else{
05947     cc = PutPreVar(prevar, buf, 0, 1) < 0;
05948 }
05949 /*:[18may20006 mt]*/
05950 M_free(buf,"Fromexternal");
05951 if ( cc ) return(-1);
05952 return(0);
05953 }
05954 /*:[02feb2006 mt]*/
05955 if ( OpenStream(s,EXTERNALCHANNELSTREAM,0,PRENOACTION) == 0 ) return(-1);
05956 /*[17may2006 mt]:*/
05957 AC.NoShowInput = withNoList;
05958 /*:[17may2006 mt]*/
05959 return(0);
05960 #else
05961 Error0("External channel: not implemented on this computer/system");
05962 return(-1);
05963 #endif
05964 }
05965

```

```

05966 /*
05967     #] DoFromExternal :
05968     #[ DoToExternal :
05969         #toexeternal
05970 */
05971
05972 #ifdef WITHEXTERNALCHANNEL
05973
05974 /*A wrapper to writeBufToExtChannel, see the file extcmd.c:*/
05975 LONG WriteToExternalChannel(int handle, UBYTE *buffer, LONG size)
05976 {
05977     /*ATT! handle is not used! Actual output is performed to
05978         the current external channel, see extcmd.c!*/
05979     DUMMYUSE(handle);
05980     if(writeBufToExtChannel((char*)buffer,size))
05981         return(-1);
05982     return(size);
05983 }
05984 #endif /*ifdef WITHEXTERNALCHANNEL*/
05985
05986 int DoToExternal(UBYTE *s)
05987 {
05988     #ifdef WITHEXTERNALCHANNEL
05989         HANDLERS h;
05990         LONG (*OldWrite)(int handle, UBYTE *buffer, LONG size) = WriteFile;
05991         int ret=-1;
05992     #else
05993         DUMMYUSE(s);
05994     #endif
05995     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
05996     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
05997     if ( AP.preError ) return(0);
05998     #ifdef WITHEXTERNALCHANNEL
05999
06000         h.oldsilent=AM.silent;
06001         h.newlogonly = h.oldlogonly = AM.FileOnlyFlag;
06002         h.newhandle = h.oldhandle = AC.LogHandle;
06003         h.oldprinttype = AO.PrintType;
06004
06005         WriteFile=&WriteToExternalChannel;
06006
06007         while ( *s == ' ' || *s == '\t' ) s++;
06008
06009         if(AX.currentExternalChannel==0){
06010             MesPrint("@No current external channel");
06011             goto DoToExternalReady;
06012         }
06013
06014         if(getCurrentExternalChannel()!=AX.currentExternalChannel)
06015             selectExternalChannel(AX.currentExternalChannel);
06016
06017         ret=writeToChannel(EXTERNALCHANNELOUT,s,&h);
06018         DoToExternalReady:
06019             WriteFile=OldWrite;
06020             return(ret);
06021     #else /*ifdef WITHEXTERNALCHANNEL*/
06022         Error0("External channel: not implemented on this computer/system");
06023         return(-1);
06024     #endif /*ifdef WITHEXTERNALCHANNEL ... else*/
06025 }
06026
06027 /*
06028     #] DoToExternal :
06029     #[ defineChannel :
06030 */
06031
06032 UBYTE *defineChannel(UBYTE *s, HANDLERS *h)
06033 {
06034     UBYTE *name,*to;
06035
06036     if ( *s != '<' )
06037         return(s);
06038
06039     s++;
06040     name = to = s;
06041     while ( *s && *s != '>' ) {
06042         if ( *s == '\\\\' ) s++;
06043         *to++ = *s++;
06044     }
06045     if ( *s == 0 ) {
06046         MesPrint("@Improper termination of filename");
06047         return(0);
06048     }
06049     s++;
06050     *to = 0;
06051     if ( *name ) {

```

```

06053         h->newhandle = GetChannel((char *)name,0);
06054         h->newlogonly = 1;
06055     }
06056     else if ( AC.LogHandle >= 0 ) {
06057         h->newhandle = AC.LogHandle;
06058         h->newlogonly = 1;
06059     }
06060     return(s);
06061 }
06062
06063 /*
06064     #] defineChannel :
06065     #[ writeToChannel :
06066 */
06067
06068 int writeToChannel(int wtype, UBYTE *s, HANDLERS *h)
06069 {
06070     UBYTE *to, *fstring, *ss, *sss, *sl, c, cl;
06071     WORD num, number, nfac;
06072     WORD oldOptimizationLevel;
06073     UBYTE Out[MAXLINELENGTH+14], *stopper;
06074     int nosemi, i;
06075     int plus = 0;
06076
06077     /*
06078     Now determine the format string
06079     */
06080     while ( *s == ',' || *s == ' ' ) s++;
06081     if ( *s != '"' ) {
06082         MesPrint("@No format string present");
06083         return(-1);
06084     }
06085     s++; fstring = to = s;
06086     while ( *s ) {
06087         if ( *s == '\\' ) {
06088             s++;
06089             if ( *s == '\\' ) {
06090                 *to++ = *s++;
06091                 if ( *s == '\\' ) *to++ = *s++;
06092             }
06093             else if ( *s == '"' ) *to++ = *s++;
06094             else { *to++ = '\\'; *to++ = *s++; }
06095         }
06096         else if ( *s == '"' ) break;
06097         else *to++ = *s++;
06098     }
06099     if ( *s != '"' ) {
06100         MesPrint("@No closing \" in format string");
06101         return(-1);
06102     }
06103     *to = 0; s++;
06104     if ( AC.LineLength > 20 && AC.LineLength <= MAXLINELENGTH ) stopper = Out + AC.LineLength;
06105     else stopper = Out + MAXLINELENGTH;
06106     to = Out;
06107     /*
06108     s points now at the list of objects (if any)
06109     we can start executing the format string.
06110     */
06111     AM.silent = 0;
06112     AC.LogHandle = h->newhandle;
06113     AM.FileOnlyFlag = h->newlogonly;
06114     if ( h->newhandle >= 0 ) {
06115         AO.PrintType |= PRINTLFILE;
06116     }
06117     while ( *fstring ) {
06118         if ( to >= stopper ) {
06119             if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
06120                 *to++ = '&';
06121             }
06122             num = to - Out;
06123             WriteString(wtype,Out,num);
06124             to = Out;
06125             if ( AC.OutputMode == FORTRANMODE
06126                 || AC.OutputMode == PFORTRANMODE ) {
06127                 number = 7;
06128                 for ( i = 0; i < number; i++ ) *to++ = ' ';
06129                 to[-2] = '&';
06130             }
06131         }
06132         if ( *fstring == '\\' ) {
06133             fstring++;
06134             if ( *fstring == 'n' ) {
06135                 num = to - Out;
06136                 WriteString(wtype,Out,num);
06137                 to = Out;
06138                 fstring++;
06139             }

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```

06140         else if ( *fstring == 't' ) { *to++ = '\t'; fstring++; }
06141         else if ( *fstring == 'b' ) { *to++ = '\\'; fstring++; }
06142         else *to++ = *fstring++;
06143     }
06144     else if ( *fstring == '%' ) {
06145         plus = 0;
06146     retry:
06147         fstring++;
06148         if ( *fstring == 'd' ) {
06149             int sign,dig;
06150             number = -1;
06151         donumber:
06152             while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
06153             sign = 1;
06154             while ( *s == '+' || *s == '-' ) {
06155                 if ( *s == '-' ) sign = -sign;
06156                 s++;
06157             }
06158             dig = 0; ss = s; if ( sign < 0 ) { ss--; *ss = '-'; dig++; }
06159             while ( *s >= '0' && *s <= '9' ) { s++; dig++; }
06160             if ( number < 0 ) {
06161                 while ( ss < s ) {
06162                     if ( to >= stopper ) {
06163                         num = to - Out;
06164                         WriteString(wtype,Out,num);
06165                         to = Out;
06166                     }
06167                     if ( *ss == '\\ ' ) ss++;
06168                     *to++ = *ss++;
06169                 }
06170             }
06171             else {
06172                 if ( number < dig ) { dig = number; ss = s - dig; }
06173                 while ( number > dig ) {
06174                     if ( to >= stopper ) {
06175                         num = to - Out;
06176                         WriteString(wtype,Out,num);
06177                         to = Out;
06178                     }
06179                     *to++ = ' '; number--;
06180                 }
06181                 while ( ss < s ) {
06182                     if ( to >= stopper ) {
06183                         num = to - Out;
06184                         WriteString(wtype,Out,num);
06185                         to = Out;
06186                     }
06187                     if ( *ss == '\\ ' ) ss++;
06188                     *to++ = *ss++;
06189                 }
06190             }
06191             fstring++;
06192         }
06193     else if ( *fstring == '$' ) {
06194         UBYTE *dolalloc;
06195         number = AO.OutSkip;
06196     dodollar:
06197         while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
06198         if ( AC.OutputMode == FORTRANMODE
06199             || AC.OutputMode == PFORTRANMODE ) {
06200             number = 7;
06201         }
06202         if ( *s != '$' ) {
06203     nodollar:
06204             MesPrint("@$-variable expected in #write instruction");
06205             AM.FileOnlyFlag = h->oldlogonly;
06206             AC.LogHandle = h->oldhandle;
06207             AO.PrintType = h->oldprinttype;
06208             AM.silent = h->oldsilent;
06209             return(-1);
06210         }
06211         s++; ss = s;
06212         while ( chartype[*s] <= 1 ) s++;
06213         if ( s == ss ) goto nodollar;
06214         c = *s; *s = 0;
06215         num = GetDollar(ss);
06216         if ( num < 0 ) {
06217             MesPrint("@#write instruction: $%s has not been defined",ss);
06218             AM.FileOnlyFlag = h->oldlogonly;
06219             AC.LogHandle = h->oldhandle;
06220             AO.PrintType = h->oldprinttype;
06221             AM.silent = h->oldsilent;
06222             return(-1);
06223         }
06224         *s = c;
06225         if ( *s == '[' ) {
06226             if ( Dollars[num].nfactors <= 0 ) {
06227                 *s = 0;

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```

06227         MesPrint("@#write instruction: $%s has not been factorized",ss);
06228         AM.FileOnlyFlag = h->oldlogonly;
06229         AC.LogHandle = h->oldhandle;
06230         AO.PrintType = h->oldprinttype;
06231         AM.silent = h->oldsilent;
06232         return(-1);
06233     }
06234     /*
06235     Now get the number between the []
06236     */
06237     nfac = GetDollarNumber(&s,Dollars+num);
06238
06239     if ( Dollars[num].nfactors == 1 && nfac == 1 ) goto writewhole;
06240
06241     if ( ( dolalloc = WriteDollarFactorToBuffer(num,nfac,0) ) == 0 ) {
06242         AM.FileOnlyFlag = h->oldlogonly;
06243         AC.LogHandle = h->oldhandle;
06244         AO.PrintType = h->oldprinttype;
06245         AM.silent = h->oldsilent;
06246         return(-1);
06247     }
06248     goto writealloc;
06249 }
06250 else if ( *s && *s != ' ' && *s != ',' && *s != '\t' ) {
06251     MesPrint("@#write instruction: illegal characters after $-variable");
06252     AM.FileOnlyFlag = h->oldlogonly;
06253     AC.LogHandle = h->oldhandle;
06254     AO.PrintType = h->oldprinttype;
06255     AM.silent = h->oldsilent;
06256     return(-1);
06257 }
06258 else {
06259 writewhole:
06260     if ( ( dolalloc = WriteDollarToBuffer(num,0) ) == 0 ) {
06261         AM.FileOnlyFlag = h->oldlogonly;
06262         AC.LogHandle = h->oldhandle;
06263         AO.PrintType = h->oldprinttype;
06264         AM.silent = h->oldsilent;
06265         return(-1);
06266     }
06267     else {
06268 writealloc:
06269         ss = dolalloc;
06270         while ( *ss ) {
06271             if ( to >= stopper ) {
06272                 if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
06273                     *to++ = '&';
06274                 }
06275                 num = to - Out;
06276                 WriteString(wtype,Out,num);
06277                 to = Out;
06278                 for ( i = 0; i < number; i++ ) *to++ = ' ';
06279                 if ( AC.OutputMode == FORTRANMODE
06280                     || AC.OutputMode == PFORTRANMODE ) to[-2] = '&';
06281             }
06282             if ( chartye[*ss] > 3 ) { *to++ = *ss++; }
06283             else {
06284                 sss = ss; while ( chartye[*ss] <= 3 ) ss++;
06285                 if ( ( to + (ss-sss) ) >= stopper ) {
06286                     if ( (ss-sss) >= (stopper-Out) ) {
06287                         if ( ( to - stopper ) < 10 ) {
06288                             if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 ==
06289                                 ISFORTRAN90 ) {
06290                                 *to++ = '&';
06291                             }
06292                             num = to - Out;
06293                             WriteString(wtype,Out,num);
06294                             to = Out;
06295                             for ( i = 0; i < number; i++ ) *to++ = ' ';
06296                             if ( AC.OutputMode == FORTRANMODE
06297                                 || AC.OutputMode == PFORTRANMODE ) to[-2] = '&';
06298                         }
06299                         while ( (ss-sss) >= (stopper-Out) ) {
06300                             while ( to < stopper-1 ) {
06301                                 *to++ = *sss++;
06302                             }
06303                             if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 ==
06304                                 ISFORTRAN90 ) {
06305                                 *to++ = '&';
06306                             }
06307                             else {
06308                                 *to++ = '\\';
06309                             }
06310                             num = to - Out;
06311                             WriteString(wtype,Out,num);
06312                             to = Out;
06313                             if ( AC.OutputMode == FORTRANMODE

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06312         || AC.OutputMode == PFORTRANMODE ) {
06313             for ( i = 0; i < number; i++ ) *to++ = ' ';
06314             to[-2] = '&';
06315         }
06316     }
06317 }
06318 else {
06319     if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90
) {
06320         *to++ = '&';
06321     }
06322     num = to - Out;
06323     WriteString(wtype,Out,num);
06324     to = Out;
06325     for ( i = 0; i < number; i++ ) *to++ = ' ';
06326     if ( AC.OutputMode == FORTRANMODE
06327         || AC.OutputMode == PFORTRANMODE ) to[-2] = '&';
06328     }
06329 }
06330 while ( sss < ss ) *to++ = *sss++;
06331 }
06332 }
06333 }
06334 M_free(dolalloc,"written dollar");
06335 fstring++;
06336 }
06337 }
06338 else if ( *fstring == 's' ) {
06339     fstring++;
06340     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
06341     if ( *s == '"' ) {
06342         s++; ss = s;
06343         while ( *s ) {
06344             if ( *s == '\\' ) s++;
06345             else if ( *s == '"' ) break;
06346             s++;
06347         }
06348         if ( *s == 0 ) {
06349             MesPrint("@#write instruction: Missing \" in string");
06350             AM.FileOnlyFlag = h->oldlogonly;
06351             AC.LogHandle = h->oldhandle;
06352             AO.PrintType = h->oldprinttype;
06353             AM.silent = h->oldsilent;
06354             return(-1);
06355         }
06356         while ( ss < s ) {
06357             if ( to >= stopper ) {
06358                 num = to - Out;
06359                 WriteString(wtype,Out,num);
06360                 to = Out;
06361             }
06362             if ( *ss == '\\' ) ss++;
06363             *to++ = *ss++;
06364         }
06365         s++;
06366     }
06367     else {
06368         sss = ss = s;
06369         while ( *s && *s != ',' ) {
06370             if ( *s == '\\' ) { s++; sss = s+1; }
06371             s++;
06372         }
06373         while ( s > sss+1 && ( s[-1] == ' ' || s[-1] == '\t' ) ) s--;
06374         while ( ss < s ) {
06375             if ( to >= stopper ) {
06376                 num = to - Out;
06377                 WriteString(wtype,Out,num);
06378                 to = Out;
06379             }
06380             if ( *ss == '\\' ) ss++;
06381             *to++ = *ss++;
06382         }
06383     }
06384 }
06385 else if ( *fstring == 'X' ) {
06386     fstring++;
06387     if ( cbuf[AM.sbufnum].numrhs > 0 ) {
06388         /*
06389             This should be only to the value of AM.oldnumextrasyms
06390             */
06391         UBYTE *s = GetPreVar(AM.oldnumextrasyms,0);
06392         WORD x = 0;
06393         while ( *s >= '0' && *s <= '9' ) x = 10*x + *s++ - '0';
06394         if ( x > 0 )
06395             PrintSubtermList(1,x);
06396         else
06397             PrintSubtermList(1,cbuf[AM.sbufnum].numrhs);

```

```

06398     }
06399 }
06400 else if ( *fstring == 'O' ) {
06401     number = AO.OutSkip;
06402 dooptim:
06403     fstring++;
06404 /*
06405     First test whether there is an optimization buffer
06406 */
06407     if ( AO.OptimizeResult.code == NULL && AO.OptimizationLevel != 0 ) {
06408         MesPrint("@In #write instruction: no optimization results available!");
06409         return(-1);
06410     }
06411     num = to - Out;
06412     WriteString(wtype,Out,num);
06413     to = Out;
06414     if ( AO.OptimizationLevel != 0 ) {
06415         WORD oldoutskip = AO.OutSkip;
06416         AO.OutSkip = number;
06417         optimize_print_code(0);
06418         AO.OutSkip = oldoutskip;
06419     }
06420 }
06421 else if ( *fstring == 'e' || *fstring == 'E' ) {
06422     if ( *fstring == 'E'
06423         || AC.OutputMode == FORTRANMODE
06424         || AC.OutputMode == PFORTRANMODE ) nosemi = 1;
06425     else nosemi = 0;
06426     fstring++;
06427     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
06428     if ( chartype[*s] != 0 && *s != '[' ) {
06429 noexpr:
06430         MesPrint("@expression name expected in #write instruction");
06431         AM.FileOnlyFlag = h->oldlogonly;
06432         AC.LogHandle = h->oldhandle;
06433         AO.PrintType = h->oldprinttype;
06434         AM.silent = h->oldsilent;
06435         return(-1);
06436     }
06437     ss = s;
06438     if ( ( s = SkipAName(ss) ) == 0 || s[-1] == '_' ) goto noexpr;
06439     s1 = s; c = c1 = *s1;
06440     if ( c1 == '(' ) {
06441         SKIPBRA3(s)
06442         if ( *s == ')' ) {
06443             AO.CurBufWrt = s1+1;
06444             c = *s; *s = 0;
06445         }
06446         else {
06447             MesPrint("@Illegal () specifier in expression name in #write");
06448             AM.FileOnlyFlag = h->oldlogonly;
06449             AC.LogHandle = h->oldhandle;
06450             AO.PrintType = h->oldprinttype;
06451             AM.silent = h->oldsilent;
06452             return(-1);
06453         }
06454     }
06455     else AO.CurBufWrt = (UBYTE *)underscore;
06456     *s1 = 0;
06457     num = to - Out;
06458     if ( num > 0 ) WriteUnfinString(wtype,Out,num);
06459     to = Out;
06460     oldOptimizationLevel = AO.OptimizationLevel;
06461     AO.OptimizationLevel = 0;
06462     if ( WriteOne(ss,(int)num,nosemi,plus) < 0 ) {
06463         AM.FileOnlyFlag = h->oldlogonly;
06464         AC.LogHandle = h->oldhandle;
06465         AO.PrintType = h->oldprinttype;
06466         AM.silent = h->oldsilent;
06467         return(-1);
06468     }
06469     AO.OptimizationLevel = oldOptimizationLevel;
06470     *s1 = c1;
06471     if ( s > s1 ) *s++ = c;
06472 }
06473 /*
06474 File content
06475 */
06476 else if ( ( *fstring == 'f' ) || ( *fstring == 'F' ) ) {
06477     LONG n;
06478     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
06479     ss = s;
06480     while ( *s && *s != ',' ) {
06481         if ( *s == '\\\ ' ) s++;
06482         s++;
06483     }
06484     c = *s; *s = 0;
06485     s1 = LoadInputFile(ss,HEADERFILE);

```

```

06485         *s = c;
06486     /*
06487         There should have been a way to pass the file size.
06488         Also there should be conversions for \r\n etc.
06489     */
06490     if ( sl ) {
06491         ss = sl; while ( *ss ) ss++;
06492         n = ss-sl;
06493         WriteString(wtype,sl,n);
06494         M_free(sl,"copy file");
06495     }
06496     else if ( *fstring == 'F' ) {
06497         *s = 0;
06498         MesPrint("@Error in #write: could not open file %s",ss);
06499         *s = c;
06500         goto ReturnWithError;
06501     }
06502     fstring++;
06503 }
06504 else if ( *fstring == '%' ) {
06505     *to++ = *fstring++;
06506 }
06507 else if ( FG.cTable[*fstring] == 1 ) { /* %#S */
06508     number = 0;
06509     while ( FG.cTable[*fstring] == 1 ) {
06510         number = 10*number + *fstring++ - '0';
06511     }
06512     if ( *fstring == 'O' ) goto dooptim;
06513     else if ( *fstring == 'd' ) goto donumber;
06514     else if ( *fstring == '$' ) goto dodollar;
06515     else if ( *fstring == 't' ) { /* 'tab' position */
06516         if ( number < (WORD)(stopper-Out) ) {
06517             while ( (WORD)(to-Out) < number ) *to++ = ' ';
06518         }
06519         fstring++;
06520     }
06521     else if ( *fstring == 'X' || *fstring == 'x' ) {
06522         if ( number > 0 && number <= cbuf[AM.sbufnum].numrhs ) {
06523             UBYTE buffer[80], *out, *old1, *old2, *old3;
06524             WORD *term, first;
06525             if ( *fstring == 'X' ) {
06526                 out = StrCopy((UBYTE *)AC.extrasym,buffer);
06527                 if ( AC.extrasymbols == 0 ) {
06528                     out = NumCopy(number,out);
06529                     out = StrCopy((UBYTE *)"_",out);
06530                 }
06531                 else if ( AC.extrasymbols == 1 ) {
06532                     if ( AC.OutputMode == CMODE ) {
06533                         out = StrCopy((UBYTE *)"[",out);
06534                         out = NumCopy(number,out);
06535                         out = StrCopy((UBYTE *)"]",out);
06536                     }
06537                     else {
06538                         out = StrCopy((UBYTE *)"(",out);
06539                         out = NumCopy(number,out);
06540                         out = StrCopy((UBYTE *)")",out);
06541                     }
06542                 }
06543                 out = StrCopy((UBYTE *)"=",out);
06544                 ss = buffer;
06545                 while ( ss < out ) {
06546                     if ( to >= stopper ) {
06547                         num = to - Out;
06548                         WriteString(wtype,Out,num);
06549                         to = Out;
06550                     }
06551                     *to++ = *ss++;
06552                 }
06553             }
06554             term = cbuf[AM.sbufnum].rhs[number];
06555             first = 1;
06556             if ( *term == 0 ) {
06557                 *to++ = '0';
06558             }
06559             else {
06560                 old1 = AO.OutFill;
06561                 old2 = AO.OutputLine;
06562                 old3 = AO.OutStop;
06563                 AO.OutFill = to;
06564                 AO.OutputLine = Out;
06565                 AO.OutStop = Out + AC.LineLength;
06566                 while ( *term ) {
06567                     if ( WriteInnerTerm(term,first) ) Terminate(-1);
06568                     term += *term;
06569                     first = 0;
06570                 }
06571                 to = Out + (AO.OutFill-AO.OutputLine);

```

```

06572             AO.OutFill = old1;
06573             AO.OutputLine = old2;
06574             AO.OutStop = old3;
06575         }
06576     }
06577     fstring++;
06578 }
06579     else {
06580         goto IllegControlSequence;
06581     }
06582 }
06583     else if ( *fstring == '+' ) {
06584         plus = 1; goto retry;
06585     }
06586     else if ( *fstring == 0 ) {
06587         *to++ = 0;
06588     }
06589     else {
06590 IllegControlSequence:
06591         MesPrint("@Illegal control sequence in format string in #write instruction");
06592 ReturnWithError:
06593         AM.FileOnlyFlag = h->oldlogonly;
06594         AC.LogHandle = h->oldhandle;
06595         AO.PrintType = h->oldprinttype;
06596         AM.silent = h->oldsilent;
06597         return(-1);
06598     }
06599 }
06600     else {
06601         *to++ = *fstring++;
06602     }
06603 }
06604 /*
06605 Now flush the output
06606 */
06607 num = to - Out;
06608 /*[15apr2004 mt]:*/
06609 if (wtype==EXTERNALCHANNELOUT) {
06610     if (num!=0)
06611         WriteUnfinString(wtype,Out,num);
06612 }else
06613 /*[15apr2004 mt]*/
06614 WriteString(wtype,Out,num);
06615 /*
06616 and restore original parameters
06617 */
06618 AM.FileOnlyFlag = h->oldlogonly;
06619 AC.LogHandle = h->oldhandle;
06620 AO.PrintType = h->oldprinttype;
06621 AM.silent = h->oldsilent;
06622 return(0);
06623 }
06624
06625 /*
06626 #] writeToChannel :
06627 #[ DoFactDollar :
06628
06629 Executes the #factdollar $var
06630 instruction
06631 */
06632
06633 int DoFactDollar(UBYTE *s)
06634 {
06635     GETIDENTITY
06636     WORD numdollar, *oldworkpointer;
06637
06638     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
06639     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
06640     while ( *s == ' ' || *s == '\t' ) s++;
06641     if ( *s == '$' ) {
06642         if ( GetName(AC.dollarnames,s+1,&numdollar,NOAUTO) != CDOLLAR ) {
06643             MesPrint("@%s is undefined",s);
06644             return(-1);
06645         }
06646         s = SkipAName(s+1);
06647         if ( *s != 0 ) {
06648             MesPrint("@#FactDollar should have a single $variable for its argument");
06649             return(-1);
06650         }
06651         NewSort(BHEAD0);
06652         oldworkpointer = AT.WorkPointer;
06653         if ( DollarFactorize(BHEAD numdollar) ) return(-1);
06654         AT.WorkPointer = oldworkpointer;
06655         LowerSortLevel();
06656         return(0);
06657     }
06658     else if ( ParenthesesTest(s) ) return(-1);

```

```

06659     else {
06660         MesPrint("@#FactDollar should have a single $variable for its argument");
06661         return -1;
06662     }
06663 }
06664
06665 /*
06666     #] DoFactDollar :
06667     #[ GetDollarNumber :
06668 */
06669
06670 WORD GetDollarNumber(UBYTE **inp, DOLLARS d)
06671 {
06672     UBYTE *s = *inp, c, *name;
06673     WORD number, nfac, *w;
06674     DOLLARS dd;
06675     s++;
06676     if ( *s == '$' ) {
06677         s++; name = s;
06678         while ( FG.cTable[*s] < 2 ) s++;
06679         c = *s; *s = 0;
06680         if ( GetName(AC.dollarnames,name,&number,NOAUTO) == NAMENOTFOUND ) {
06681             MesPrint("@dollar in #write should have been defined previously");
06682             Terminate(-1);
06683         }
06684         *s = c;
06685         dd = Dollars + number;
06686         if ( c == '[' ) {
06687             *inp = s;
06688             nfac = GetDollarNumber(inp,dd);
06689             s = *inp;
06690             if ( *s != ']' ) {
06691                 MesPrint("@Illegal factor for dollar variable");
06692                 Terminate(-1);
06693             }
06694             *inp = s+1;
06695             if ( nfac == 0 ) {
06696                 if ( dd->nfactors > d->nfactors ) {
06697 TooBig:
06698                     MesPrint("@Factor number for dollar variable too large");
06699                     Terminate(-1);
06700                 }
06701                 return(dd->nfactors);
06702             }
06703             w = dd->factors[nfac-1].where;
06704             if ( w == 0 ) {
06705                 if ( dd->factors[nfac-1].value > d->nfactors ||
06706                     dd->factors[nfac-1].value < 0 ) goto TooBig;
06707                 return(dd->factors[nfac-1].value);
06708             }
06709             if ( *w == 4 && w[4] == 0 && w[3] == 3 && w[2] == 1
06710                 && w[1] <= d->nfactors ) return(w[1]);
06711             if ( w[*w] == 0 && w[*w-1] == *w-1 ) goto TooBig;
06712 IllNum:
06713             MesPrint("@Illegal factor number for dollar variable");
06714             Terminate(-1);
06715         }
06716     } else { /* The dollar should be a number */
06717         if ( dd->type == DOLZERO ) {
06718             return(0);
06719         }
06720         else if ( dd->type == DOLTERMS || dd->type == DOLNUMBER ) {
06721             w = dd->where;
06722             if ( *w == 4 && w[4] == 0 && w[3] == 3 && w[2] == 1
06723                 && w[1] <= d->nfactors ) return(w[1]);
06724             if ( w[*w] == 0 && w[*w-1] == *w-1 ) goto TooBig;
06725             goto IllNum;
06726         }
06727         else goto IllNum;
06728     }
06729 }
06730 else if ( FG.cTable[*s] == 1 ) {
06731     WORD x = *s++ - '0';
06732     while ( FG.cTable[*s] == 1 ) {
06733         x = 10*x + *s++ - '0';
06734         if ( x > d->nfactors ) {
06735             MesPrint("@Factor number %d for dollar variable too large",x);
06736             Terminate(-1);
06737         }
06738     }
06739     if ( *s != ']' ) {
06740         MesPrint("@Illegal factor number for dollar variable");
06741         Terminate(-1);
06742     }
06743     s++; *inp = s;
06744     return(x);
06745 }

```

```

06746     else {
06747         MesPrint("@Illegal factor indicator for dollar variable");
06748         Terminate(-1);
06749     }
06750     return(-1);
06751 }
06752
06753 /*
06754     #] GetDollarNumber :
06755     #[ DoSetRandom :
06756
06757     Executes the #SetRandom number
06758 */
06759 int DoSetRandom(UBYTE *s)
06760 {
06761     ULONG x;
06762     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
06763     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
06764     while ( *s == ' ' || *s == '\t' ) s++;
06765     x = 0;
06766     while ( FG.cTable[*s] == 1 ) {
06767         x = 10*x + (*s++-'0');
06768     }
06769     while ( *s == ' ' || *s == '\t' ) s++;
06770     if ( *s == 0 ) {
06771         #ifdef WITHPTHREADS
06772         #ifdef WITHSORTBOTS
06773             int id, totnum = MaX(2*AM.totalnumberofthreads-3,AM.totalnumberofthreads);
06774         #else
06775             int id, totnum = AM.totalnumberofthreads;
06776         #endif
06777         for ( id = 0; id < totnum; id++ ) {
06778             AB[id]->R.wranfseed = x;
06779             if ( AB[id]->R.wranfia ) M_free(AB[id]->R.wranfia,"wranf");
06780             AB[id]->R.wranfia = 0;
06781         }
06782         #else
06783             AR.wranfseed = x;
06784             if ( AR.wranfia ) M_free(AR.wranfia,"wranf");
06785             AR.wranfia = 0;
06786         #endif
06787         return(0);
06788     }
06789     else {
06790         MesPrint("@proper syntax is #SetRandom number");
06791         return(-1);
06792     }
06793 }
06794
06795 /*
06796     #] DoSetRandom :
06797     #[ DoOptimize :
06798
06799     Executes the #Optimize(expr) instruction.
06800 */
06801 int DoOptimize(UBYTE *s)
06802 {
06803     GETIDENTITY
06804     UBYTE *exprname;
06805     WORD numexpr;
06806     int error = 0, i;
06807     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
06808     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
06809     DUMMYUSE(*s)
06810     exprname = s; s = SkipAName(s);
06811     if ( *s != 0 && *s != ';' ) {
06812         MesPrint("@proper syntax is #Optimize,expression");
06813         return(-1);
06814     }
06815     *s = 0;
06816     if ( GetName(AC.exprnames,exprname,&numexpr,NOAUTO) != CEXPRESSION ) {
06817         MesPrint("@%s is not an expression",exprname);
06818         error = 1;
06819     }
06820     else if ( AP.preError == 0 ) {
06821         EXPRESSIONS e = Expressions + numexpr;
06822         POSITION position;
06823         int firstterm;
06824         WORD *term = AT.WorkPointer;
06825         ClearOptimize();
06826         if ( AO.OptimizationLevel == 0 ) return(0);
06827         switch ( e->status ) {
06828             case LOCALEXPRESSION:
06829             case GLOBALEXPRESSION:
06830                 break;

```

```

06833         default:
06834             MesPrint("@Expression %s is not an active unhidden local or global
expression.", exprname);
06835             Terminate(-1);
06836             break;
06837     }
06838 #ifdef WITHMPI
06839     if ( PF.me == MASTER )
06840 #endif
06841     RevertScratch();
06842     for ( i = NumExpressions-1; i >= 0; i-- ) {
06843         AS.OldOnFile[i] = Expressions[i].onfile;
06844         AS.OldNumFactors[i] = Expressions[i].numfactors;
06845         AS.Oldvflags[i] = Expressions[i].vflags;
06846         Expressions[i].vflags &= ~(ISUNMODIFIED|ISZERO);
06847     }
06848     for ( i = 0; i < NumExpressions; i++ ) {
06849         if ( i == numexpr ) {
06850             PutPreVar(AM.oldnumextrasymbols,
06851                 GetPreVar((UBYTE *) "EXTRASYMBOLS_", 0), 0, 1);
06852             Optimize(numexpr, 0);
06853             AO.OptimizeResult.nameofexpr = strDupl(exprname, "optimize expression name");
06854             continue;
06855         }
06856 #ifdef WITHMPI
06857         if ( PF.me == MASTER ) {
06858 #endif
06859             e = Expressions + i;
06860             switch ( e->status ) {
06861                 case LOCALEXPRESSION:
06862                 case SKIPLEXPRESSION:
06863                 case DROPLEXPRESSION:
06864                 case DROPEDEXPRESSION:
06865                 case GLOBALEXPRESSION:
06866                 case SKIPGEXPRESSION:
06867                 case DROPGEXPRESSION:
06868                 case HIDELEXPRESSION:
06869                 case HIDEGEXPRESSION:
06870                 case DROPHLEXPRESSION:
06871                 case DROPHGEXPRESSION:
06872                 case INTOHIDELEXPRESSION:
06873                 case INTOHIDEGEXPRESSION:
06874                     break;
06875                 default:
06876                     continue;
06877             }
06878             AR.GetFile = 0;
06879             SetScratch(AR.infile, &(e->onfile));
06880             if ( GetTerm(BHEAD term) <= 0 ) {
06881                 MesPrint("@Expression %d has problems reading from scratchfile", i);
06882                 Terminate(-1);
06883             }
06884             term[3] = i;
06885             AR.DeferFlag = 0;
06886             SeekScratch(AR.outfile, &position);
06887             e->onfile = position;
06888             *AM.S0->sBuffer = 0; firstterm = -1;
06889             do {
06890                 WORD *oldipointer = AR.CompressPointer;
06891                 WORD *comprtop = AR.ComprTop;
06892                 AR.ComprTop = AM.S0->sTop;
06893                 AR.CompressPointer = AM.S0->sBuffer;
06894                 if ( firstterm > 0 ) {
06895                     if ( PutOut(BHEAD term, &position, AR.outfile, 1) < 0 ) goto DoSerr;
06896                 }
06897                 else if ( firstterm < 0 ) {
06898                     if ( PutOut(BHEAD term, &position, AR.outfile, 0) < 0 ) goto DoSerr;
06899                     firstterm++;
06900                 }
06901                 else {
06902                     if ( PutOut(BHEAD term, &position, AR.outfile, -1) < 0 ) goto DoSerr;
06903                     firstterm++;
06904                 }
06905                 AR.CompressPointer = oldipointer;
06906                 AR.ComprTop = comprtop;
06907             } while ( GetTerm(BHEAD term) );
06908             if ( FlushOut(&position, AR.outfile, 1) ) {
06909 DoSerr:
06910                 MesPrint("@Expression %d has problems writing to scratchfile", i);
06911                 Terminate(-1);
06912             }
06913 #ifdef WITHMPI
06914         }
06915 #endif
06916     }
06917     /*
06918     Now some administration and we are done

```

```

06919 */
06920     UpdateMaxSize();
06921 }
06922 else {
06923     ClearOptimize();
06924 }
06925 return(error);
06926
06927 }
06928
06929 /*
06930     #] DoOptimize :
06931     #[ DoClearOptimize :
06932
06933     Clears all relevant buffers of the output optimization
06934 */
06935
06936 int DoClearOptimize(UBYTE *s)
06937 {
06938     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
06939     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
06940     DUMMYUSE(*s);
06941     return(ClearOptimize());
06942 }
06943
06944 /*
06945     #] DoClearOptimize :
06946     #[ DoSkipExtraSymbols :
06947
06948     Adds the intermediate variables of the previous optimization
06949     to the list of extra symbols, provided it has not yet been erased
06950     by a #clearoptimize
06951     To remove them again one needs to use the 'delete extrasymbols;'
06952     or the 'delete extrasymbols>num;' statement in which num is the
06953     old number of extra symbols.
06954 */
06955
06956 int DoSkipExtraSymbols(UBYTE *s)
06957 {
06958     CBUF *C = cbuf + AM.sbufnum;
06959     WORD tt = 0, j = 0, oldval = AO.OptimizeResult.minvar;
06960     if ( AO.OptimizeResult.code == NULL ) return(0);
06961     if ( AO.OptimizationLevel == 0 ) return(0);
06962     while ( *s == ' ' ) s++;
06963     if ( *s == 0 ) {
06964         AO.OptimizeResult.minvar = AO.OptimizeResult.maxvar+1;
06965     }
06966     else {
06967         while ( *s <= '9' && *s >= '0' ) j = 10*j + *s++ - '0';
06968         if ( *s ) {
06969             MesPrint("@Illegal use of #SkipExtraSymbols instruction");
06970             Terminate(-1);
06971         }
06972         AO.OptimizeResult.minvar += j;
06973         if ( AO.OptimizeResult.minvar > AO.OptimizeResult.maxvar )
06974             AO.OptimizeResult.minvar = AO.OptimizeResult.maxvar+1;
06975     }
06976     j = AO.OptimizeResult.minvar - oldval;
06977     while ( j > 0 ) {
06978         AddRHS(AM.sbufnum,1);
06979         AddNtoC(AM.sbufnum,1,&tt,16);
06980         AddToCB(C,0)
06981         InsTree(AM.sbufnum,C->numrhs);
06982         j--;
06983     }
06984     return(0);
06985 }
06986
06987 /*
06988     #] DoSkipExtraSymbols :
06989     #[ DoPreReset :
06990
06991     Does a reset of variables.
06992     Currently only the timer (stopwatch) of 'timer_'
06993 */
06994
06995 int DoPreReset(UBYTE *s)
06996 {
06997     UBYTE *ss, c;
06998     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
06999     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
07000     while ( *s == ' ' || *s == '\t' ) s++;
07001     if ( *s == 0 ) {
07002         MesPrint("@proper syntax is #Reset variable");
07003         return(-1);
07004     }
07005     ss = s;

```



```

07006     while ( FG.cTable[*s] == 0 ) s++;
07007     c = *s; *s = 0;
07008     if ( ( StrICmp(ss, (UBYTE *) "timer") == 0 )
07009         || ( StrICmp(ss, (UBYTE *) "stopwatch") == 0 ) ) {
07010         *s = c;
07011         AP.StopWatchZero = GetRunningTime();
07012         return(0);
07013     }
07014     else {
07015         *s = c;
07016         MesPrint("@proper syntax is #Reset variable");
07017         return(-1);
07018     }
07019 }
07020
07021 /*
07022     #] DoPreReset :
07023     #[ DoPreAppendPath :
07024 */
07025
07026 static int DoAddPath(UBYTE *s, int bPrepend)
07027 {
07028     /* NOTE: this doesn't support some file systems, e.g., 0x5c with CP932. */
07029
07030     UBYTE *path, *path_end, *current_dir, *current_dir_end, *NewPath, *t;
07031     int bRelative, n;
07032
07033     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
07034     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
07035
07036     /* Parse the path in the input. */
07037     while ( *s == ' ' || *s == '\\t' ) s++; /* skip spaces */
07038     if ( *s == '"' ) { /* the path is given by "..." */
07039         path = ++s;
07040         while ( *s && *s != '"' ) {
07041             if ( SEPARATOR != '\\\\' && *s == '\\\\' ) { /* escape character, e.g., "\\\" */
07042                 if ( !s[1] ) goto ImproperPath;
07043                 s++;
07044             }
07045             s++;
07046         }
07047         if ( *s != '"' ) goto ImproperPath;
07048         path_end = s++;
07049     }
07050     else {
07051         path = s;
07052         while ( *s && *s != ' ' && *s != '\\t' ) {
07053             if ( SEPARATOR != '\\\\' && *s == '\\\\' ) { /* escape character, e.g., "\\ " */
07054                 if ( !s[1] ) goto ImproperPath;
07055                 s++;
07056             }
07057             s++;
07058         }
07059         path_end = s;
07060     }
07061     if ( path == path_end ) goto ImproperPath; /* empty path */
07062     while ( *s == ' ' || *s == '\\t' ) s++; /* skip spaces */
07063     if ( *s ) goto ImproperPath; /* extra tokens found */
07064
07065     /* Check if the path is an absolute path. */
07066     bRelative = 1;
07067     if ( path[0] == SEPARATOR ) { /* starts with the directory separator */
07068         bRelative = 0;
07069     }
07070 #ifdef WINDOWS
07071     else if ( chartype[path[0]] == 0 && path[1] == ':' ) { /* starts with (drive letter): */
07072         bRelative = 0;
07073     }
07074 #endif
07075
07076     /* Get the current file directory when a relative path is given. */
07077     if ( bRelative ) {
07078         if ( !AC.CurrentStream ) goto FileNameUnavailable;
07079         if ( AC.CurrentStream->type != FILESTREAM && AC.CurrentStream->type != REVERSEFILESTREAM )
07080             goto FileNameUnavailable;
07081         if ( !AC.CurrentStream->name ) goto FileNameUnavailable;
07082         s = current_dir = current_dir_end = AC.CurrentStream->name;
07083         while ( *s ) {
07084             if ( SEPARATOR != '\\\\' && *s == '\\\\' && s[1] ) { /* escape character, e.g., "\\\" */
07085                 s += 2;
07086                 continue;
07087             }
07088             if ( *s == SEPARATOR ) {
07089                 current_dir_end = s;
07090             }
07091             s++;
07092         }

```

```

07092     }
07093     else {
07094         current_dir = current_dir_end = NULL;
07095     }
07096
07097     /* Allocate a buffer for new AM.Path. */
07098     n = path_end - path;
07099     if ( AM.Path ) n += StrLen(AM.Path) + 1;
07100     if ( current_dir != current_dir_end ) n+= current_dir_end - current_dir + 1;
07101     s = NewPath = (UBYTE *)Malloca(n + 1,"add path");
07102
07103     /* Construct new FORM path. */
07104     if ( bPrepend ) {
07105         if ( current_dir != current_dir_end ) {
07106             t = current_dir;
07107             while ( t != current_dir_end ) *s++ = *t++;
07108             *s++ = SEPARATOR;
07109         }
07110         t = path;
07111         while ( t != path_end ) *s++ = *t++;
07112         if ( AM.Path ) *s++ = PATHSEPARATOR;
07113     }
07114     if ( AM.Path ) {
07115         t = AM.Path;
07116         while ( *t ) *s++ = *t++;
07117     }
07118     if ( !bPrepend ) {
07119         if ( AM.Path ) *s++ = PATHSEPARATOR;
07120         if ( current_dir != current_dir_end ) {
07121             t = current_dir;
07122             while ( t != current_dir_end ) *s++ = *t++;
07123             *s++ = SEPARATOR;
07124         }
07125         t = path;
07126         while ( t != path_end ) *s++ = *t++;
07127     }
07128     *s = '\0';
07129
07130     /* Update AM.Path. */
07131     if ( AM.Path ) M_free(AM.Path,"add path");
07132     AM.Path = NewPath;
07133
07134     return(0);
07135
07136 ImproperPath:
07137     MesPrint("@Improper syntax for %%%sPath", bPrepend ? "Prepend" : "Append");
07138     return(-1);
07139
07140 FileNameUnavailable:
07141     /* This may be improved in future. */
07142     MesPrint("@Sorry, %%%sPath can't resolve the current file name from here", bPrepend ? "Prepend" :
"Append");
07143     return(-1);
07144 }
07145
07153 int DoPreAppendPath(UBYTE *s)
07154 {
07155     return DoAddPath(s, 0);
07156 }
07157
07158 /*
07159     #] DoPreAppendPath :
07160     #[ DoPrePrependPath :
07161 */
07162
07170 int DoPrePrependPath(UBYTE *s)
07171 {
07172     return DoAddPath(s, 1);
07173 }
07174
07175 /*
07176     #] DoPrePrependPath :
07177     #[ DoTimeOutAfter :
07178
07179     Executes the #timeoutafter number
07180 */
07181
07182 int DoTimeOutAfter(UBYTE *s)
07183 {
07184     #ifdef WITH_ALARM
07185         ULONG x;
07186     #else
07187         DUMMYUSE(s);
07188     #endif
07189     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
07190     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
07191     #ifdef WITH_ALARM

```

```

07192     while ( *s == ' ' || *s == '\t' ) s++;
07193     x = 0;
07194     while ( FG.cTable[*s] == 1 ) {
07195         x = 10*x + (*s++-'0');
07196     }
07197     while ( *s == ' ' || *s == '\t' ) s++;
07198     if ( *s == 0 ) {
07199         alarm(x);
07200         return(0);
07201     }
07202     else {
07203         MesPrint("@proper syntax is #TimeoutAfter number");
07204         return(-1);
07205     }
07206 #else
07207     Error0("#timeoutafter not implemented on this computer/system");
07208     return(-1);
07209 #endif
07210 }
07211
07212 /*
07213     #] DoTimeOutAfter :
07214     #[ DoNamespace :
07215
07216     Syntax:
07217         #Namespace name
07218         .....
07219         #use variables
07220         .....
07221         #EndNamespace
07222     Effect:
07223         All variables/expressions defined inside the range of the
07224         namespace get name_ prepended.
07225         This holds also for $-variables, names of procedures and
07226         names of files.
07227     Namespaces can be used in a nested way. cf this_is_deep_x
07228     A leading _ takes the role of what is super:: in some other languages.
07229     Remarks:
07230         Names of preprocessor variables are excluded!
07231         Names of built in objects are excluded! (like sum_, d_ etc.)
07232 */
07233
07234 int DoNamespace (UBYTE *s)
07235 {
07236     UBYTE *s1, *s2, c;
07237     NAMESPACE *namespace;
07238     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
07239     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
07240     while ( *s == ' ' || *s == '\t' || *s == ',' || *s == '\n' ) s++;
07241     if ( FG.cTable[*s] != 0 ) {
07242         MesPrint("@Illegal name in #namespace instruction: %s",s);
07243         return(-1);
07244     }
07245     s1 = s;
07246     while ( FG.cTable[*s1] <= 1 ) s1++;
07247     s2 = s1;
07248     while ( *s2 == ' ' || *s2 == '\t' || *s2 == ',' || *s2 == '\n' ) s2++;
07249     if ( *s2 != 0 ) {
07250         MesPrint("@A #namespace instruction can only have one name with only alphanumeric
characters.");
07251         return(-1);
07252     }
07253     c = *s1; *s1 = 0;
07254 /*
07255     Now we have the name and the statement is legal.
07256     We can proceed creating the namespace and its use tree.
07257 */
07258     namespace = (NAMESPACE *)Malloc1(sizeof(NAMESPACE),"namespace");
07259     namespace->name = strDup1(s,"namespace_name");
07260     namespace->usenames = MakeNameTree();
07261     if ( AP.firstnamespace == 0 ) {
07262         namespace->previous = 0;
07263         namespace->next = 0;
07264         AP.firstnamespace = namespace;
07265         AP.lastnamespace = namespace;
07266     }
07267     else {
07268         AP.lastnamespace->next = namespace;
07269         namespace->next = 0;
07270         namespace->previous = AP.lastnamespace;
07271         AP.lastnamespace = namespace;
07272     }
07273     *s1 = c;
07274     return(0);
07275 }
07276
07277 /*

```

```

07278         #] DoNamespace :
07279         #[ DoEndNamespace :
07280     */
07281
07282     int DoEndNamespace(UBYTE *s)
07283     {
07284         NAMESPACE *namespace;
07285         if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
07286         if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
07287         while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
07288         if ( *s != 0 ) {
07289             MesPrint("@Illegal #endnamespace instruction");
07290             return(-1);
07291         }
07292         namespace = AP.lastnamespace;
07293         AP.lastnamespace = namespace->previous;
07294         M_free(namespace->name, "namespace_name");
07295         FreeNameTree(namespace->usenames);
07296         M_free(namespace, "namespace");
07297         return(0);
07298     }
07299
07300     /*
07301         #] DoEndNamespace :
07302         #[ SkipName :
07303     */
07304
07305     UBYTE *SkipName(UBYTE *s)
07306     {
07307         UBYTE *t = s, *s1, c;
07308         int num = 0, block = 0;
07309         if ( *s == '[' ) {
07310             straight:
07311             SKIPBRAL(s)
07312             if ( *s == 0 ) {
07313                 MesPrint("&Illegal name: '%s'",t);
07314                 return(0);
07315             }
07316             s++; s1 = s;
07317             while ( FG.cTable[*s] <= 1 || *s == '_' ) s++;
07318             if ( s1 != s ) goto witherror;
07319         }
07320         else if ( *s == '$' ) {
07321             s++;
07322             while ( *s == '_' ) { s++; num++; }
07323             block = 1;
07324             while ( FG.cTable[*s] <= 1 || *s == '_' ) {
07325                 if ( FG.cTable[*s] != 0 && block == 1 ) {
07326                     blocked:
07327                         MesPrint("&Illegally formed name: %s",t);
07328                         return(0);
07329                     }
07330                     if ( *s == '_' ) { num++; block = 1; }
07331                     else block = 0;
07332                     s++;
07333                 }
07334                 if ( s[-1] == '_' && num > 1 ) goto built;
07335             }
07336             else if ( FG.cTable[*s] == 0 ) {
07337                 regular:
07338                 while ( FG.cTable[*s] <= 1 || *s == '_' ) {
07339                     if ( FG.cTable[*s] != 0 && block == 1 ) goto blocked;
07340                     if ( *s == '_' ) { block = 1; num++; }
07341                     else block = 0;
07342                     s++;
07343                 }
07344                 if ( *s == '[' ) goto straight;
07345                 if ( s[-1] == '_' && num > 1 ) {
07346                     built:
07347                         c = *s; *s = 0;
07348                         MesPrint("&Built in objects cannot be part of namespaces: %s",t);
07349                         *s = c;
07350                         return(0);
07351                     }
07352                 }
07353                 else if ( *s == '_' ) {
07354                     while ( *s == '_' ) { s++; num++; }
07355                     if ( FG.cTable[*s] == 0 ) { block = 0; goto regular; }
07356                     goto witherror;
07357                 }
07358                 else if ( *s == '@' ) {
07359                     s++; block = 1;
07360                     if ( *s == '_' || FG.cTable[*s] == 1 ) {
07361                         MesPrint("@Illegally formed name: %s",s-1);
07362                     }
07363                     goto regular;
07364                 }

```

```

07365     else if ( *s == '#' ) { /* name of a procedure */
07366         s++;
07367         while ( *s == '_' ) { s++; num++; }
07368         block = 1;
07369         while ( FG.cTable[*s] <= 1 || *s == '_' ) {
07370             if ( FG.cTable[*s] != 0 && block == 1 ) goto blocked;
07371             if ( *s == '_' ) { num++; block = 1; }
07372             else block = 0;
07373             s++;
07374         }
07375         if ( s[-1] == '_' ) {
07376 witherror:
07377             c = *s; *s = 0;
07378             MesPrint("%Illegally formed name: %s",t);
07379             *s = c;
07380             return(0);
07381         }
07382     }
07383     else if ( *s == '<' ) { /* name of a file. Can be anything (more or less) */
07384         s++;
07385         while ( *s && *s != '>' ) s++;
07386         if ( *s != '>' ) goto witherror;
07387         s++;
07388     }
07389     return(s);
07390 }
07391
07392 /*
07393     #] SkipName :
07394     #[ ConstructName :
07395
07396     Routine gets a 'raw' name and modifies it if the namespace
07397     settings ask for it. It puts the new name in a buffer that
07398     may be expanded if the names become rather long.
07399     Note that eventually that name needs to be copied, because
07400     we do not allocate new buffers for each name.
07401
07402     type tells what kind of name we look for
07403 */
07404
07405 UBYTE *ConstructName(UBYTE *s,UBYTE type)
07406 {
07407     int len;
07408     UBYTE *t, *u;
07409     WORD number;
07410     NAMESPACE *namespace;
07411     if ( AP.lastnamespace == 0 ) return(s);
07412     if ( *s == '@' ) return(s+1);
07413     if ( GetName(AP.lastnamespace->usenames,s,&number,NOAUTO) !=
07414         NAMENOTFOUND ) return(s);
07415 /*
07416     Now the real stuff
07417     First we have to compute the size of the new name.
07418 */
07419     len = StrLen(s) + 1;
07420     namespace = AP.firstnamespace;
07421     while ( namespace ) {
07422         len += StrLen(namespace->name)+1;
07423         namespace = namespace->previous;
07424     }
07425     if ( len > AP.fullnamesize ) {
07426         while ( len > AP.fullnamesize ) AP.fullnamesize *= 2;
07427         M_free(AP.fullname,"AP.fullname");
07428         AP.fullname = (UBYTE *)Malloc1(AP.fullnamesize*sizeof(UBYTE *),"AP.fullname");
07429     }
07430     namespace = AP.firstnamespace;
07431     t = AP.fullname;
07432     switch ( type ) {
07433     case ' ':
07434     case 0:
07435         while ( namespace ) {
07436             u = namespace->name;
07437             while ( *u ) *t++ = *u++;
07438             *t++ = '_';
07439             namespace = namespace->previous;
07440         }
07441         while ( *s ) *t++ = *s++;
07442         *t = 0;
07443         break;
07444     case '$':
07445     case '#':
07446         *t++ = type;
07447         while ( namespace ) {
07448             u = namespace->name;
07449             while ( *u ) *t++ = *u++;
07450             *t++ = '_';
07451             namespace = namespace->previous;

```

```

07452     }
07453     if ( type == '$' ) s++;
07454     while ( *s ) *t++ = *s++;
07455     *t = 0;
07456     break;
07457 case '<':
07458     while ( namespace ) {
07459         u = namespace->name;
07460         while ( *u ) *t++ = *u++;
07461         *t++ = '_';
07462         namespace = namespace->previous;
07463     }
07464     s++;
07465     while ( *s ) *t++ = *s++;
07466     t--; /* strip the '>' */
07467     *t = 0;
07468     break;
07469 default:
07470     MesPrint("Unrecognized datatype in ConstructName");
07471     *t = 0;
07472     break;
07473 }
07474 return(AP.fullname);
07475 }
07476
07477 /*
07478 #] ConstructName :
07479 #[ DoUse :
07480
07481 Routine makes (inside the confines of the current namespace)
07482 a list of variables that are excluded from the namespace.
07483 Once the namespace is ended, the list is removed.
07484 The list can include names of variables, dollars and procedures.
07485 Preprocessor variables are excluded from the namespace. Their
07486 inclusion would be too complicated for the input streams.
07487 Note that the preprocessor variables that are arguments in a #do
07488 or in a procedure are on a stack and should not cause problems.
07489 Names of variables can be just that.
07490 Names of $-variables are also straightforward.
07491 Names of procedures should be preceded by a # character.
07492 Names of files (like in #include) should be enclosed by <>.
07493 The names are stored in a balanced tree. Each namespace may have
07494 its own tree. The toplevel (no namespace) does not allow a #use.
07495 */
07496
07497 int DoUse(UBYTE *s)
07498 {
07499     NAMESPACE *namespace;
07500     UBYTE *t, c;
07501     int number;
07502     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
07503     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
07504     if ( AP.lastnamespace == 0 ) {
07505         MesPrint("@It is not allowed to use #use outside the scope of a namespace.");
07506         return(-1);
07507     }
07508     namespace = AP.lastnamespace;
07509     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
07510     while ( *s ) {
07511         t = s;
07512         if ( ( s = SkipName(t) ) == 0 ) return(-1);
07513         if ( s == t ) {
07514             MesPrint("@Unrecognized object in #use instruction: %s",t);
07515             return(-1);
07516         }
07517         c = *s; *s = 0;
07518     }
07519     /*
07520     In usernames we only need the names to know whether they are 'protected'.
07521     We need to keep the $, # and <> to avoid potential double names.
07522 */
07523     AddName(namespace->usenames,t,0,0,&number);
07524     *s = c;
07525     while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
07526 }
07527
07528
07529 /*
07530 #] DoUse :
07531 #[ UserFlags :
07532
07533 Syntax:
07534 #ClearFlag number(s),expression(s)
07535 #ClearFlag number(s)
07536 #ClearFlag expression(s)
07537 #ClearFlag
07538 #SetFlag number(s),expression(s)

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07539         #SetFlag number(s)
07540         #SetFlag expression(s)
07541         #SetFlag
07542         par == 0: Clear, par == 1: Set.
07543     */
07544
07545 int UserFlags(UBYTE *s,int par)
07546 {
07547     int mask = 0, error = 0, i;
07548     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
07549     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
07550     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
07551     if ( *s == 0 ) { /* Treat all flags in all active expressions */
07552     allexp:
07553         for ( i = 0; i < NumExpressions; i++ ) {
07554             switch ( Expressions[i].status ) {
07555                 case UNHIDELEXPRESSSION:
07556                 case UNHIDEGEXPRESSSION:
07557                 case INTOHIDELEXPRESSSION:
07558                 case INTOHIDEGEXPRESSSION:
07559                 case LOCALEXPRESSSION:
07560                 case GLOBALEXPRESSSION:
07561                 case SKIPLEXPRESSSION:
07562                 case SKIPGEXPRESSSION:
07563                 case HIDELEXPRESSSION:
07564                 case HIDEGEXPRESSSION:
07565                     if ( par == 1 ) Expressions[i].uflags |= ~mask;
07566                     else Expressions[i].uflags &= mask;
07567                     break;
07568                 case DROPEXPRESSSION:
07569                 case DROPLEXPRESSSION:
07570                 case DROPHLEXPRESSSION:
07571                 case DROPHGEXPRESSSION:
07572                 case STOREXPRESSSION:
07573                 case HIDDENLEXPRESSSION:
07574                 case HIDDENGEXPRESSSION:
07575                 case SPECTATOREXPRESSSION:
07576                 default:
07577                     break;
07578             }
07579         }
07580     }
07581 }
07582 else if ( FG.cTable[*s] == 1 ) {
07583     mask = (int)WORDMASK;
07584     while ( FG.cTable[*s] == 1 ) {
07585         int x = 0;
07586         while ( FG.cTable[*s] == 1 ) { x = 10*x + (*s++-'0'); }
07587         if ( x < 1 || x > BITSINWORD ) {
07588             MesPrint("@Illegal number %d for flag in #...Flag instruction",x);
07589             return(1);
07590         }
07591         mask ^= (1<<(x-1));
07592         while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
07593     }
07594     if ( *s == 0 ) goto allexp;
07595 }
07596 else { /* Clear all flags in all expressions that are specified */
07597     mask = (int)WORDMASK;
07598 }
07599 while ( *s ) { /* now read the expressions */
07600     UBYTE *s1, c;
07601     WORD num1;
07602     if ( FG.cTable[*s] != 0 && *s != '[' ) goto syntax;
07603     s1 = s; s = SkipAName(s);
07604     c = *s; *s = 0;
07605     if ( GetName(AC.exprnames,s1,&num1,NOAUTO) != CEXPRESSION ) {
07606         MesPrint("@%s is not an active expression",s1);
07607         error = 1;
07608         return(error);
07609     }
07610     switch ( Expressions[num1].status ) {
07611         case UNHIDELEXPRESSSION:
07612         case UNHIDEGEXPRESSSION:
07613         case INTOHIDELEXPRESSSION:
07614         case INTOHIDEGEXPRESSSION:
07615         case LOCALEXPRESSSION:
07616         case GLOBALEXPRESSSION:
07617         case SKIPLEXPRESSSION:
07618         case SKIPGEXPRESSSION:
07619         case HIDELEXPRESSSION:
07620         case HIDEGEXPRESSSION:
07621             if ( par == 1 ) Expressions[num1].uflags |= ~mask;
07622             else Expressions[num1].uflags &= mask;
07623             break;
07624         case DROPEXPRESSSION:
07625         case DROPLEXPRESSSION:

```

```

07626         case DROPGEXPRESSION:
07627         case DROPHLEXPRESSION:
07628         case DROPHGEXPRESSION:
07629         case STOREDEXPRESSION:
07630         case HIDDENLEXPRESSION:
07631         case HIDDENGEXPRESSION:
07632         case SPECTATOREXPRESSION:
07633         default:
07634             MesPrint("@%s is not an active expression",s1);
07635             error = 1;
07636             break;
07637     }
07638     *s = c;
07639     while ( *s == ',' || *s == ' ' || *s == '\\t' ) s++;
07640 }
07641 return(error);
07642 syntax:
07643 MesPrint("@Illegal name in #...Flag instruction.");
07644 return(1);
07645 }
07646
07647 /*
07648     #] UserFlags :
07649     #[ DoClearUserFlag :
07650 */
07651 int DoClearUserFlag(UBYTE *s)
07652 {
07653     return(UserFlags(s,0));
07654 }
07655
07656 /*
07657     #] DoClearUserFlag :
07658     #[ DoSetUserFlag :
07659 */
07660 int DoSetUserFlag(UBYTE *s)
07661 {
07662     return(UserFlags(s,1));
07663 }
07664
07665 /*
07666     #] DoSetUserFlag :
07667     #[ DoStartFloat :
07668
07669     If there is a number follwing, it will be the new default precision.
07670     If float has been started before, the old one will be removed first.
07671     If there are two numbers, the second one is the maximum weight for
07672     MZV's.
07673 */
07674 #ifdef WITHFLOAT
07675 int DoStartFloat(UBYTE *s)
07676 {
07677     GETIDENTITY
07678     int error = 0;
07679     LONG x;
07680     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
07681     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
07682     if ( AR.PolyFun != 0 ) {
07683         MesPrint("@Simultaneous use of Poly(Rat)Fun and float_ is not allowed.");
07684         error = 1;
07685     }
07686     if ( AC.ncmod != 0 ) {
07687         MesPrint("@Simultaneous use of floating point and modulus arithmetic makes no sense.");
07688         error = 1;
07689     }
07690     while ( *s == ',' || *s == ' ' || *s == '\\t' ) s++;
07691     if ( *s >= '0' && *s <= '9' ) {
07692         x = 0;
07693         do {
07694             x = 10*x + (*s++ - '0');
07695         } while ( *s >= '0' && *s <= '9' );
07696         AC.tDefaultPrecision = x;
07697         while ( *s == ',' || *s == ' ' || *s == '\\t' ) s++;
07698         if ( *s >= '0' && *s <= '9' ) {
07699             x = 0;
07700             do {
07701                 x = 10*x + (*s++ - '0');
07702             } while ( *s >= '0' && *s <= '9' );
07703             AC.tMaxWeight = x;
07704             while ( *s == ',' || *s == ' ' || *s == '\\t' ) s++;
07705         }
07706         else {
07707             AC.tMaxWeight = 0;
07708         }
07709         if ( *s ) goto IllPar;
07710     }
07711 }

```



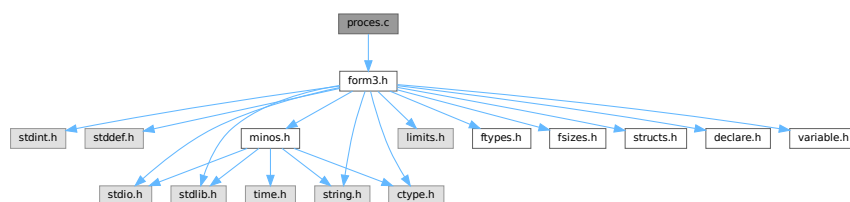
```

07713     }
07714     else if ( *s != 0 ) {
07715     IllPar:
07716         MesPrint("@Illegal parameter in %#StartFloat instruction: %s ",s);
07717         error = 1;
07718     }
07719     if ( error == 0 ) {
07720         if ( AC.tDefaultPrecision && ( AC.tDefaultPrecision != AC.DefaultPrecision
07721         || AT.aux_ == 0 ) ) {
07722             AC.DefaultPrecision = AC.tDefaultPrecision;
07723             AC.tDefaultPrecision = 0;
07724         }
07725         if ( AC.tMaxWeight && ( AC.tMaxWeight != AC.MaxWeight
07726         || AT.aux_ == 0 ) ) {
07727             AC.MaxWeight = AC.tMaxWeight;
07728             AC.tMaxWeight = 0;
07729         }
07730         SetFloatPrecision(AC.DefaultPrecision+AC.MaxWeight+1);
07731         SetupMPFTTables();
07732         if ( AC.MaxWeight > 0 ) SetupMZVTables();
07733         SetFloatPrecision(AC.DefaultPrecision);
07734     }
07735     else {
07736         AC.tDefaultPrecision = 0;
07737         AC.tMaxWeight = 0;
07738     }
07739     return(error);
07740 }
07741
07742 #endif
07743 /*
07744     #] DoStartFloat :
07745     #[ DoEndFloat :
07746 */
07747 #ifdef WITHFLOAT
07748
07749 int DoEndFloat(UBYTE *s)
07750 {
07751     int error = 0;
07752     if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
07753     if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
07754     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
07755     if ( *s != 0 ) {
07756         MesPrint("@Illegal parameter in %#EndFloat instruction: %s ",s);
07757         error = 1;
07758     }
07759     if ( error == 0 ) {
07760         ClearFloat();
07761         ClearMZVTables();
07762     }
07763     return(error);
07764 }
07765
07766 #endif
07767 /*
07768     #] DoEndFloat :
07769     # ] PreProcessor :
07770 */

```

## 4.114 proces.c File Reference

#include "form3.h"  
 Include dependency graph for proces.c:



## Macros

- `#define DONE(x) { retvalue = x; goto Done; }`

## Functions

- `int Processor (void)`
- `WORD TestSub (PHEAD WORD *term, WORD level)`
- `int InFunction (PHEAD WORD *term, WORD *termout)`
- `int InsertTerm (PHEAD WORD *term, WORD replac, WORD extractbuff, WORD *position, WORD *termout, WORD tepos)`
- `LONG PasteFile (PHEAD WORD number, WORD *accum, POSITION *position, WORD **accfill, RENUMBER renumber, WORD *freeze, WORD nexpr)`
- `WORD * PasteTerm (PHEAD WORD number, WORD *accum, WORD *position, WORD times, WORD divby)`
- `int FiniTerm (PHEAD WORD *term, WORD *accum, WORD *termout, WORD number, WORD tepos)`
- `int Generator (PHEAD WORD *term, WORD level)`
- `int DoOnePow (PHEAD WORD *term, WORD power, WORD nexpr, WORD *accum, WORD *aa, WORD level, WORD *freeze)`
- `int Deferred (PHEAD WORD *term, WORD level)`
- `int PrepPoly (PHEAD WORD *term, WORD par)`
- `int PolyFunMul (PHEAD WORD *term)`

## Variables

- `WORD printscratch [2]`

### 4.114.1 Detailed Description

Contains the central terms processor routines. This is the core of the virtual machine. All other files are to help these routines.

Definition in file [proces.c](#).

### 4.114.2 Macro Definition Documentation

#### 4.114.2.1 DONE

```
#define DONE(
    x ) { retvalue = x; goto Done; }
```

TestSub hunts for subexpression pointers. If one is found its power is given in AN.TeSuOut. and the returnvalue is 'expressionnumber'. If the expression number is negative it is an expression on disk.

In addition this routine tries to locate subexpression pointers in functions. It also notices that action must be taken with any of the special functions.

#### Parameters

|              |                                                      |
|--------------|------------------------------------------------------|
| <i>term</i>  | The term in which TestSub hunts for potential action |
| <i>level</i> | The number of the 'level' in the compiler buffer.    |

### Returns

The number of the (sub)expression that was encountered.

Other values that are returned are in AN.TeSuOut, AR.TePos, AT.TMbuff, AN.TeInFun, AN.Frozen, AT.TMaddr

The level in the compiler buffer is more or less the number of the statement in the module. Hence it refers to the element in the lhs array.

This routine is one of the most important routines in FORM.

Definition at line 683 of file [proces.c](#).

## 4.114.3 Function Documentation

### 4.114.3.1 Processor()

```
int Processor (
    void )
```

This is the central processor. It accepts a stream of Expressions which is accessed by calls to GetTerm. The expressions reside either in AR.infile or AR.hidefile The definitions of an expression are seen as an id-statement, so the primary Expressions should be written to the system of scratch files as single terms with an expression pointer. Each expression is terminated with a zero and the whole is terminated by two zeroes.

The routine DoExecute should determine whether results are to be printed, should revert the scratch I/O directions etc. In principle it is DoExecute that calls Processor.

### Returns

if everything OK: 0. Otherwise error. The preprocessor may continue with compilation though. Really fatal errors should return on the spot by calling Terminate.

Definition at line 64 of file [proces.c](#).

References [CbUf::Buffer](#), [EndSort\(\)](#), [FlushOut\(\)](#), [Generator\(\)](#), [CbUf::lhs](#), [LowerSortLevel\(\)](#), [NewSort\(\)](#), [PF\\_BroadcastRHS\(\)](#), [PF\\_InParallelProcessor\(\)](#), [PF\\_Processor\(\)](#), [CbUf::Pointer](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [PutOut\(\)](#), [CbUf::rhs](#), and [StoreTerm\(\)](#).



## Returns

0: everything is fine, Negative: fatal, Positive: error.

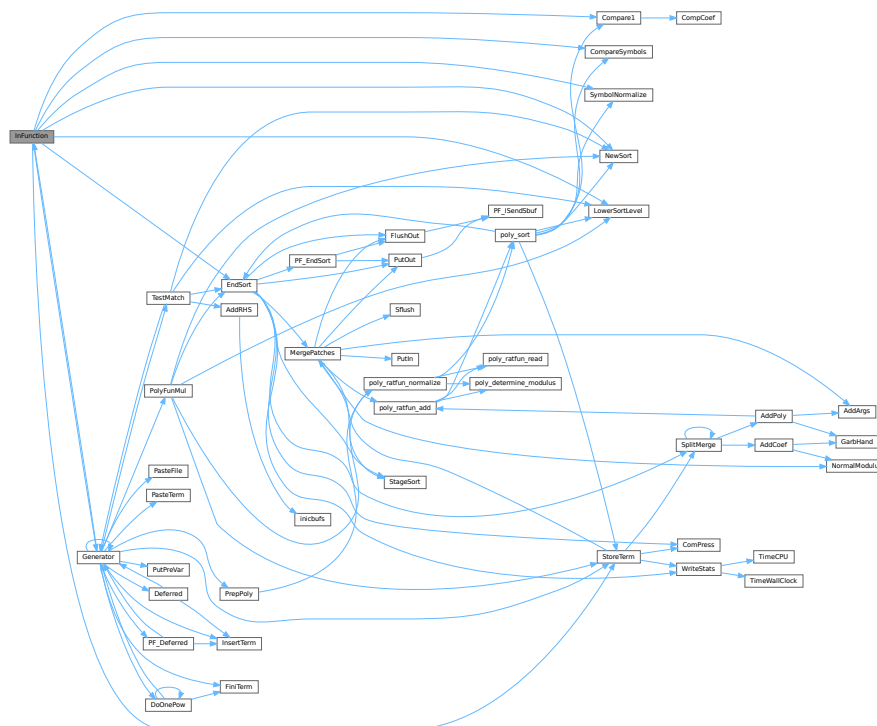
Special attention should be given to nested functions!

Definition at line 2175 of file [proces.c](#).

References [Compare1\(\)](#), [CompareSymbols\(\)](#), [EndSort\(\)](#), [Generator\(\)](#), [LowerSortLevel\(\)](#), [NewSort\(\)](#), [StoreTerm\(\)](#), and [SymbolNormalize\(\)](#).

Referenced by [Generator\(\)](#).

Here is the call graph for this function:



## 4.114.3.4 InsertTerm()

```
int InsertTerm (
    PHEAD WORD * term,
    WORD replac,
    WORD extractbuff,
    WORD * position,
    WORD * termout,
    WORD tepos )
```

Puts the terms 'term' and 'position' together into a single legal term in termout. replac is the number of the subexpression that should be replaced. It must be a positive term. When action is needed in the argument of a function all terms in that argument are dealt with recursively. The subexpression is sorted. Only one subexpression is done at a time this way.

**Parameters**

|                    |                                                             |
|--------------------|-------------------------------------------------------------|
| <i>term</i>        | the input term                                              |
| <i>replac</i>      | number of the subexpression pointer to replace              |
| <i>extractbuff</i> | number of the compiler buffer <i>replac</i> refers to       |
| <i>position</i>    | position from where to take the term in the compiler buffer |
| <i>termout</i>     | the output term                                             |
| <i>tepos</i>       | offset in term where the subexpression is.                  |

**Returns**

Normal conventions (OK = 0).

Definition at line 2745 of file [proces.c](#).

Referenced by [Deferred\(\)](#), [Generator\(\)](#), and [PF\\_Deferred\(\)](#).

**4.114.3.5 PasteFile()**

```
LONG PasteFile (
    PHEAD WORD number,
    WORD * accum,
    POSITION * position,
    WORD ** accfill,
    RENUMBER renumber,
    WORD * freeze,
    WORD nexpr )
```

Gets a term from stored expression *expr* and puts it in the accumulator at position *number*. It returns the length of the term that came from file.

**Parameters**

|                 |                                                                |
|-----------------|----------------------------------------------------------------|
| <i>number</i>   | number of partial terms to skip in <i>accum</i>                |
| <i>accum</i>    | the accumulator                                                |
| <i>position</i> | file position from where to get the stored term                |
| <i>accfill</i>  | returns tail position in <i>accum</i>                          |
| <i>renumber</i> | the renumber struct for the variables in the stored expression |
| <i>freeze</i>   | information about if we need only the contents of a bracket    |
| <i>nexpr</i>    | the number of the stored expression                            |

**Returns**

Normal conventions (OK = 0).

Definition at line 2881 of file [proces.c](#).

Referenced by [Generator\(\)](#).

### 4.114.3.6 PasteTerm()

```
WORD * PasteTerm (
    PHEAD WORD number,
    WORD * accum,
    WORD * position,
    WORD times,
    WORD divby )
```

Puts the term at position in the accumulator *accum* at position '*number*+1'. if *times* > 0 the coefficient of this term is multiplied by *times*/*divby*.

#### Parameters

|                 |                                                                                   |
|-----------------|-----------------------------------------------------------------------------------|
| <i>number</i>   | The number of term fragments in <i>accum</i> that should be skipped               |
| <i>accum</i>    | The accumulator of term fragments                                                 |
| <i>position</i> | A position in (typically) a compiler buffer from where a (piece of a) term comes. |
| <i>times</i>    | Multiply the result by this                                                       |
| <i>divby</i>    | Divide the result by this.                                                        |

This routine is typically used when we have to replace a (sub)expression pointer by a power of a (sub)expression. This uses mostly a binomial expansion and the new term is the old term multiplied one by one by terms of the new expression. The factors *times* and *divby* keep track of the binomial coefficient. Once this is complete, the routine *FiniTerm* will make the contents of the accumulator into a proper term that still needs to be normalized.

Definition at line 3003 of file [proces.c](#).

Referenced by [Generator\(\)](#).

### 4.114.3.7 FiniTerm()

```
int FiniTerm (
    PHEAD WORD * term,
    WORD * accum,
    WORD * termout,
    WORD number,
    WORD tepos )
```

Concatenates the contents of the accumulator into a single legal term, which replaces the subexpression pointer

#### Parameters

|                |                                                           |
|----------------|-----------------------------------------------------------|
| <i>term</i>    | the input term with the (sub)expression subterm           |
| <i>accum</i>   | the accumulator with the term fragments                   |
| <i>termout</i> | the location where the output should be written           |
| <i>number</i>  | the number of term fragments in the accumulator           |
| <i>tepos</i>   | the position of the subterm in <i>term</i> to be replaced |

Definition at line 3068 of file [proces.c](#).

Referenced by [DoOnePow\(\)](#), and [Generator\(\)](#).

#### 4.114.3.8 Generator()

```
int Generator (
    PHEAD WORD * term,
    WORD level )
```

The heart of the program. Here the expansion tree is set up in one giant recursion

##### Parameters

|              |                                                        |
|--------------|--------------------------------------------------------|
| <i>term</i>  | the input term. may be overwritten                     |
| <i>level</i> | the level in the compiler buffer (number of statement) |

##### Returns

Normal conventions (OK = 0).

The routine looks first whether there are unsubstituted (sub)expressions. If so, one of them gets inserted term by term and the new term is used in a renewed call to Generator. If there are no (sub)expressions, the term is normalized, the compiler level is raised (next statement) and the program looks what type of statement this is. If this is a special statement it is either treated on the spot or the appropriate routine is called. If it is a substitution, the pattern matcher is called (TestMatch) which tells whether there was a match. If so we need to call TestSub again to test for (sub)expressions. If we run out of levels, the term receives a final treatment for modulus calculus and/or brackets and is then sent off to the sorting routines.

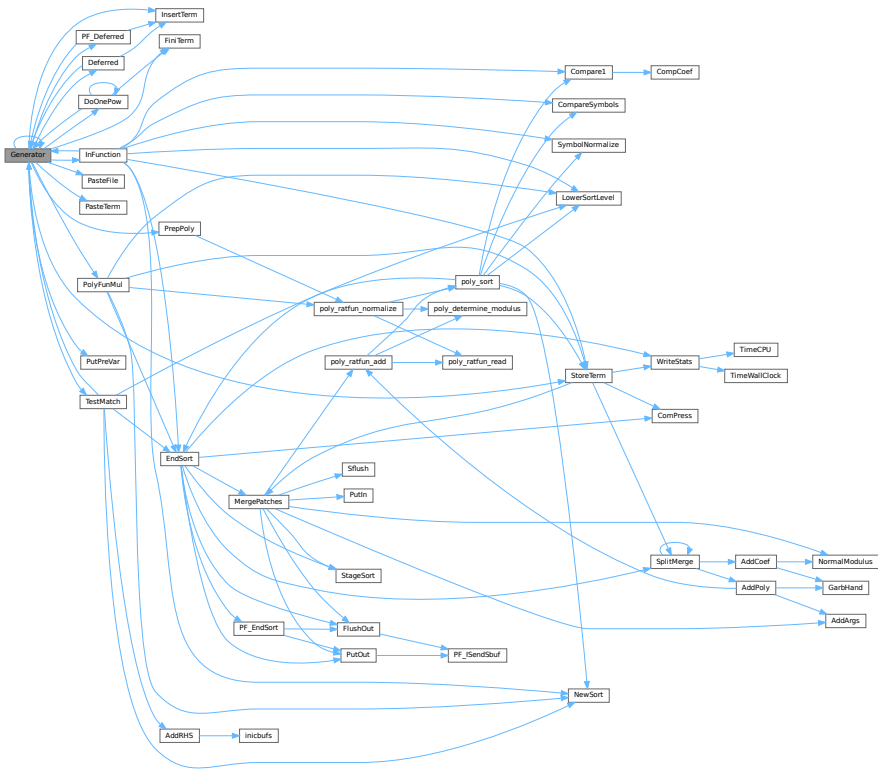
Definition at line 3267 of file [proces.c](#).

References [Deferred\(\)](#), [DoOnePow\(\)](#), [FiniTerm\(\)](#), [Generator\(\)](#), [InFunction\(\)](#), [InsertTerm\(\)](#), [CbUf::lhs](#), [VaRrEnUm::lo](#), [PasteFile\(\)](#), [PasteTerm\(\)](#), [PF\\_Deferred\(\)](#), [CbUf::Pointer](#), [PolyFunMul\(\)](#), [PrepPoly\(\)](#), [PutPreVar\(\)](#), [CbUf::rhs](#), [StoreTerm\(\)](#), [ReNuMbEr::symb](#), and [TestMatch\(\)](#).

Referenced by [Deferred\(\)](#), [DoOnePow\(\)](#), [generate\\_expression\(\)](#), [Generator\(\)](#), [InFunction\(\)](#), [MakeDollarInteger\(\)](#), [MakeDollarMod\(\)](#), [PF\\_CollectModifiedDollars\(\)](#), [PF\\_Deferred\(\)](#), [PF\\_Processor\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [Processor\(\)](#), and [TestMatch\(\)](#).



Here is the call graph for this function:



4.114.3.9 DoOnePow()

```
int DoOnePow (
    PHEAD WORD * term,
    WORD power,
    WORD nex,
    WORD * accum,
    WORD * aa,
    WORD level,
    WORD * freeze )
```

Routine gets one power of an expression in the scratch system. If there are more powers needed there will be a recursion.

No attempt is made to use binomials because we have no information about commuting properties.

There is a searching for the contents of brackets if needed. This searching may be rather slow because of the single links.

Parameters

|              |                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------|
| <i>term</i>  | is the term we are adding to.                                                                          |
| <i>power</i> | is the power of the expression that we need.                                                           |
| <i>nex</i>   | is the number of the expression.                                                                       |
| <i>accum</i> | is the accumulator of terms. It accepts the termfragments that are made into a proper term in FiniTerm |

## Parameters

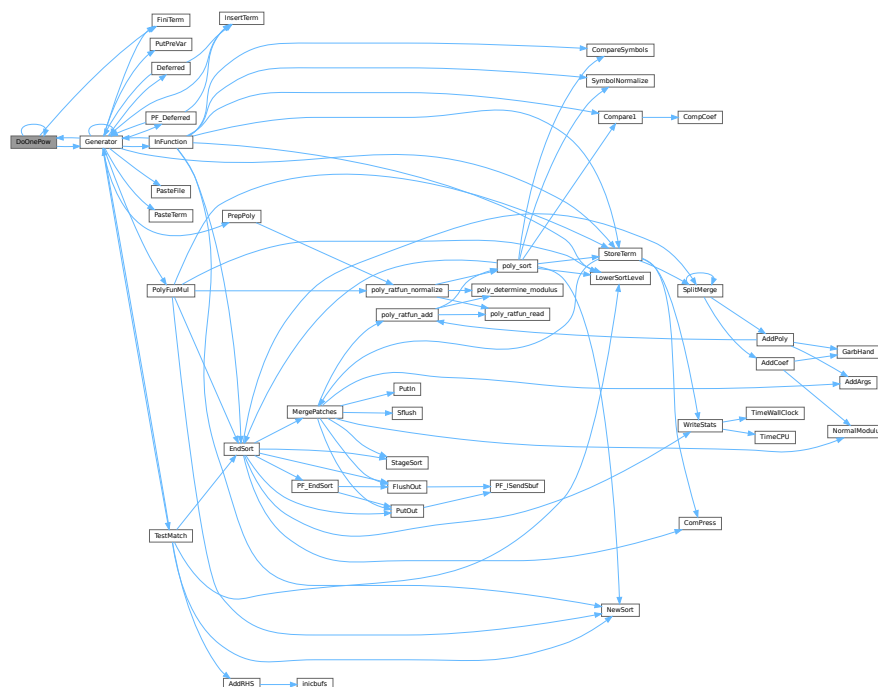
|               |                                                                                                                                      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <i>aa</i>     | points to the start of the entire accumulator. In *aa we store the number of term fragments that are in the accumulator.             |
| <i>level</i>  | is the current depth in the tree of statements. It is needed to continue to the next operation/substitution with each generated term |
| <i>freeze</i> | is the pointer to the bracket information that should be matched.                                                                    |

Definition at line 4625 of file [proces.c](#).

References [DoOnePow\(\)](#), [FiniTerm\(\)](#), [Generator\(\)](#), and [FiLe::handle](#).

Referenced by [DoOnePow\(\)](#), and [Generator\(\)](#).

Here is the call graph for this function:



#### 4.114.3.10 Deferred()

```
int Deferred (
    PHEAD WORD * term,
    WORD level )
```

Picks up the deferred brackets. These are the bracket contents of which we postpone the reading when we use the 'Keep Brackets' statement. These contents are multiplying the terms just before they are sent to the sorting system. Special attention goes to having it thread-safe We have to lock positioning the file and reading it in a thread specific buffer.

## Parameters

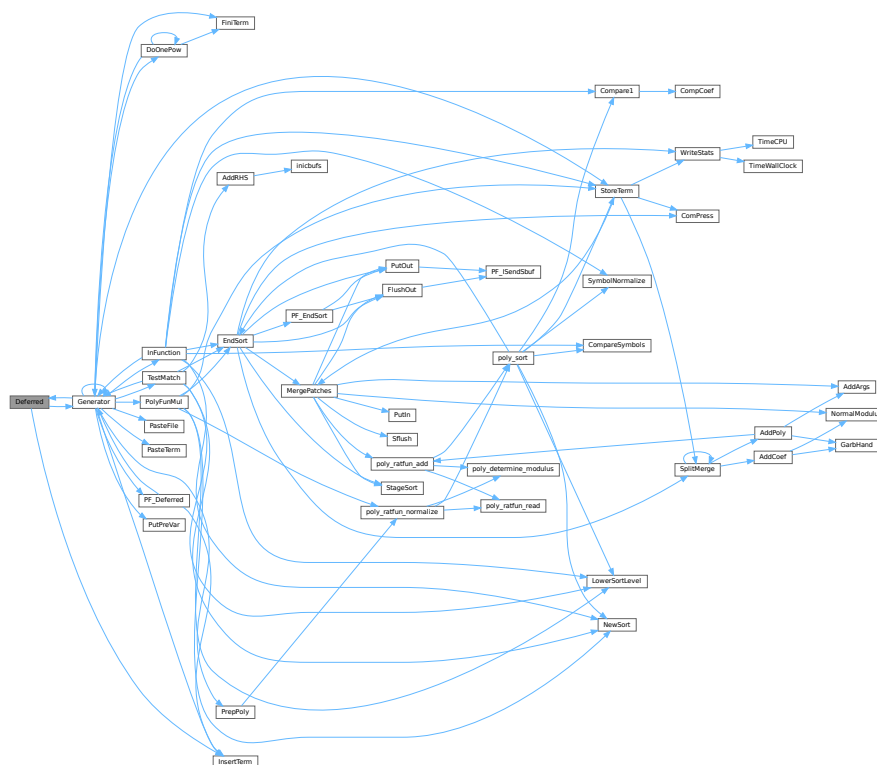
|              |                                                                                                    |
|--------------|----------------------------------------------------------------------------------------------------|
| <i>term</i>  | The term that must be multiplied by the contents of the current bracket                            |
| <i>level</i> | The compiler level. This is needed because after multiplying term by term we call Generator again. |

Definition at line 4846 of file proces.c.

References [Generator\(\)](#), and [InsertTerm\(\)](#).

Referenced by [Generator\(\)](#).

Here is the call graph for this function:



#### 4.114.3.11 PrepPoly()

```
int PrepPoly (
    PHEAD WORD * term,
    WORD par )
```

Routine checks whether the count of function AR.PolyFun is zero or one. If it is one and it has one scalarlike argument the coefficient of the term is pulled inside the argument. If the count is zero a new function is made with the coefficient as its only argument. The function should be placed at its proper position.

When this function is active it places the PolyFun as last object before the coefficient. This is needed because otherwise the compress algorithm has problems in MergePatches.

The bracket routine should also place the PolyFun at a comparable spot. The compression should then stop at the PolyFun. It doesn't really have to stop when writing the final result but this may be too complicated.

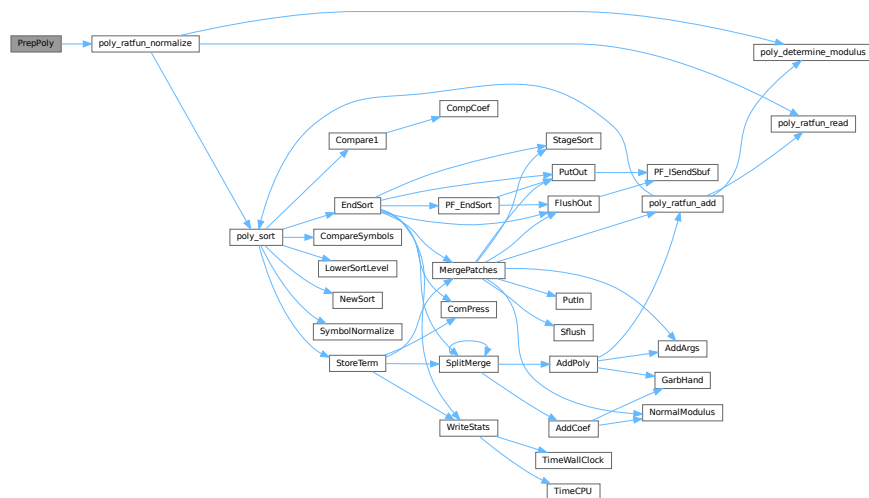
The parameter `par` tells whether we are at groundlevel or inside a function or dollar variable.

Definition at line 4974 of file [proces.c](#).

References [poly\\_ratfun\\_normalize\(\)](#).

Referenced by [Generator\(\)](#).

Here is the call graph for this function:



#### 4.114.3.12 PolyFunMul()

```
int PolyFunMul (
    PHEAD WORD * term )
```

Multiplies the arguments of multiple occurrences of the polyfun. In this routine we do the original PolyFun with one argument only. The PolyRatFun (PolyFunType = 2) is done in a dedicated routine in the file [polywrap.cc](#). The new result is written over the old result.

##### Parameters

|             |                                                  |
|-------------|--------------------------------------------------|
| <i>term</i> | It contains the input term and later the output. |
|-------------|--------------------------------------------------|

##### Returns

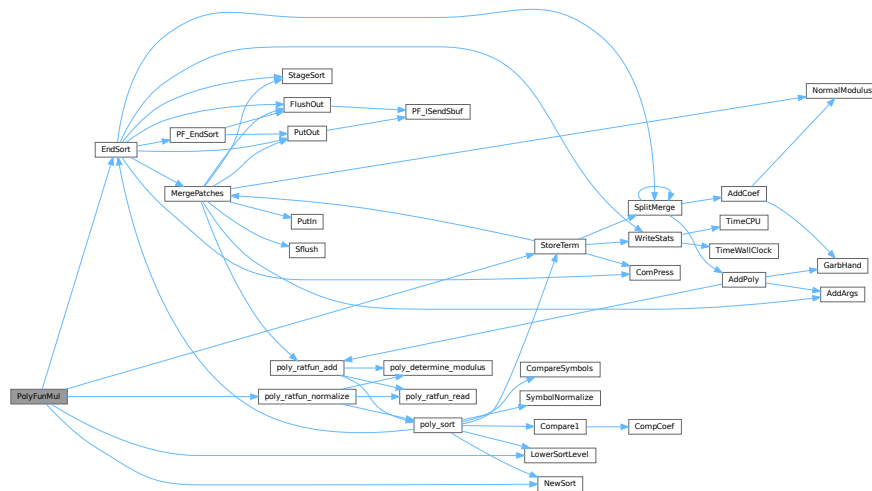
Normal conventions (OK = 0).

Definition at line 5362 of file [proces.c](#).

References [EndSort\(\)](#), [LowerSortLevel\(\)](#), [NewSort\(\)](#), [poly\\_ratfun\\_normalize\(\)](#), and [StoreTerm\(\)](#).

Referenced by [Generator\(\)](#).

Here is the call graph for this function:



## 4.114.4 Variable Documentation

### 4.114.4.1 printscratch

WORD printscratch[2]

Definition at line 39 of file [proces.c](#).

## 4.115 proces.c

[Go to the documentation of this file.](#)

```

00001
00006 /* # [ License : */
00007 /*
00008 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 * When using this file you are requested to refer to the publication
00010 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 * This is considered a matter of courtesy as the development was paid
00012 * for by FOM the Dutch physics granting agency and we would like to
00013 * be able to track its scientific use to convince FOM of its value
00014 * for the community.
00015 *
00016 * This file is part of FORM.
00017 *
00018 * FORM is free software: you can redistribute it and/or modify it under the
00019 * terms of the GNU General Public License as published by the Free Software
00020 * Foundation, either version 3 of the License, or (at your option) any later
00021 * version.
00022 *
00023 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 * details.
00027 *
00028 * You should have received a copy of the GNU General Public License along
00029 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* # [ License : */
00032 /*
00033 #define HIDEDEBUG

```

```

00034     # [ Includes : proces.c
00035     */
00036
00037     #include "form3.h"
00038
00039     WORD printscratch[2];
00040
00041     /*
00042     #] Includes :
00043     # [ Processor :
00044     # [ Processor :          WORD Processor()
00045     */
00064     int Processor(void)
00065     {
00066         GETIDENTITY
00067         WORD *term, *t, i, size;
00068         int retval = 0;
00069         EXPRESSIONS e;
00070         POSITION position;
00071         WORD last, LastExpression, fromspectator;
00072         LONG dd = 0;
00073         CBUF *C = cbuf+AC.cbunum;
00074         int firstterm;
00075         CBUF *CC = cbuf+AT.ebunum;
00076         WORD **w, *cpo, *cbo;
00077         FILEHANDLE *curfile, *oldoutfile = AR.outfile;
00078         WORD oldBracketOn = AR.BracketOn;
00079         WORD *oldBrackBuf = AT.BrackBuf;
00080         WORD oldbracketindexflag = AT.bracketindexflag;
00081     #ifdef WITHPTHREADS
00082         int OldMultiThreaded = AS.MultiThreaded, Oldmparallelflag = AC.mparallelflag;
00083     #endif
00084         if ( CC->numrhs > 0 || CC->numlhs > 0 ) {
00085             if ( CC->rhs ) {
00086                 w = CC->rhs; i = CC->numrhs;
00087                 do { *w++ = 0; } while ( --i > 0 );
00088             }
00089             if ( CC->lhs ) {
00090                 w = CC->lhs; i = CC->numlhs;
00091                 do { *w++ = 0; } while ( --i > 0 );
00092             }
00093             CC->numlhs = CC->numrhs = 0;
00094             ClearTree(AT.ebunum);
00095             CC->Pointer = CC->Buffer;
00096         }
00097
00098         if ( NumExpressions == 0 ) return(0);
00099         AR.expflags = 0;
00100         AR.CompressPointer = AR.CompressBuffer;
00101         AR.NoCompress = AC.NoCompress;
00102         term = AT.WorkPointer;
00103         if ( ( (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer) ) > AT.WorkTop ) return(MesWork());
00104         UpdatePositions();
00105         C->rhs[C->numrhs+1] = C->Pointer;
00106         AR.KeptInHold = 0;
00107         if ( AC.CollectFun ) AR.DeferFlag = 0;
00108         AR.outtohide = 0;
00109         AN.PolyFunTodo = 0;
00110     #ifdef HIIDEBUG
00111         MesPrint("Status at the start of Processor (HideLevel = %d)", AC.HideLevel);
00112         for ( i = 0; i < NumExpressions; i++ ) {
00113             e = Expressions+i;
00114             ExprStatus(e);
00115         }
00116     #endif
00117     /*
00118         Next determine the last expression. This is used for removing the
00119         input file when the final stage of the sort of this expression is
00120         reached. That can save up to 1/3 in disk space.
00121     */
00122         for ( i = NumExpressions-1; i >= 0; i-- ) {
00123             e = Expressions+i;
00124             if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
00125                 || e->status == HIDELEXPRESSION || e->status == HIDEGEXPRESSION
00126                 || e->status == SKIPLEXPRESSION || e->status == SKIPGEXPRESSION
00127                 || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
00128                 || e->status == INTOHIDELEXPRESSION || e->status == INTOHIDEGEXPRESSION
00129             ) break;
00130         }
00131         last = i;
00132         for ( i = NumExpressions-1; i >= 0; i-- ) {
00133             AS.OldOnFile[i] = Expressions[i].onfile;
00134             AS.OldNumFactors[i] = Expressions[i].numfactors;
00135             /* AS.Oldvflags[i] = e[i].vflags; */
00136             AS.Oldvflags[i] = Expressions[i].vflags;
00137             AS.Olduflags[i] = Expressions[i].uflags;
00138             Expressions[i].vflags &= ~(ISUNMODIFIED|ISZERO);

```

```

00139     }
00140 #ifndef WITHPTHREADS
00141 /*
00142     When we run with threads we have to make sure that all local input
00143     buffers are pointed correctly. Of course this isn't needed if we
00144     run on a single thread only.
00145 */
00146     if ( AC.partodoflag && AM.totalnumberofthreads > 1 ) {
00147         AS.MultiThreaded = 1; AC.mparallelflag = PARALLELFLAG;
00148     }
00149     if ( AS.MultiThreaded && AC.mparallelflag == PARALLELFLAG ) {
00150         SetWorkerFiles();
00151     }
00152 /*
00153     We start with running the expressions with expr->partodo in parallel.
00154     The current model is: give each worker an expression. Wait for
00155     workers to finish and tell them where to write.
00156     Then give them a new expression. Workers may have to wait for each
00157     other. This is also the case with the last one.
00158 */
00159     if ( AS.MultiThreaded && AC.mparallelflag == PARALLELFLAG ) {
00160         if ( InParallelProcessor() ) {
00161             retval = 1;
00162         }
00163         AS.MultiThreaded = OldMultiThreaded;
00164         AC.mparallelflag = Oldmparallelflag;
00165     }
00166 #endif
00167 #ifdef WITHMPI
00168     if ( AC.RhsExprInModuleFlag && PF.rhsInParallel && (AC.mparallelflag == PARALLELFLAG ||
00169 AC.partodoflag) ) {
00170         if ( PF_BroadcastRHS() ) {
00171             retval = -1;
00172         }
00173         PF.exprtodo = -1; /* This means, the slave does not perform inparallel */
00174         if ( AC.partodoflag > 0 ) {
00175             if ( PF_InParallelProcessor() ) {
00176                 retval = -1;
00177             }
00178         }
00179     #endif
00180     for ( i = 0; i < NumExpressions; i++ ) {
00181         #ifdef INNERTEST
00182             if ( AC.InnerTest ) {
00183                 if ( StrCmp(AC.TestValue,(UBYTE *)INNERTEST) == 0 ) {
00184                     MesPrint("Testing(Processor): value = %s",AC.TestValue);
00185                 }
00186             }
00187         #endif
00188         e = Expressions+i;
00189         #ifdef WITHPTHREADS
00190         if ( AC.partodoflag > 0 && e->partodo > 0 && AM.totalnumberofthreads > 2 ) {
00191             e->partodo = 0;
00192             continue;
00193         }
00194         #endif
00195         #ifdef WITHMPI
00196         if ( AC.partodoflag > 0 && e->partodo > 0 && PF.numtasks > 2 ) {
00197             e->partodo = 0;
00198             continue;
00199         }
00200         #endif
00201         AS.CollectOverFlag = 0;
00202         AR.exchanged = 0;
00203         if ( i == last ) LastExpression = 1;
00204         else LastExpression = 0;
00205         if ( e->inmem ) {
00206             /*
00207                 #! in memory : Memory allocated by poly.c only thusfar.
00208                 Here GetTerm cannot work.
00209                 For the moment we ignore this for parallelization.
00210             */
00211             WORD j;
00212
00213             AR.GetFile = 0;
00214             SetScratch(AR.infile,&(e->onfile));
00215             if ( GetTerm(BHEAD term) <= 0 ) {
00216                 MesPrint("(1) Expression %d has problems in scratchfile",i);
00217                 retval = -1;
00218                 break;
00219             }
00220             term[3] = i;
00221             AR.CurExpr = i;
00222             SeekScratch(AR.outfile,&position);
00223             e->onfile = position;
00224             if ( PutOut (BHEAD term,&position,AR.outfile,0) < 0 ) goto ProcErr;

```

```

00225     AR.DeferFlag = AC.ComDefer;
00226     NewSort(BHEAD0);
00227     AN.ninterms = 0;
00228     t = e->inmem;
00229     while ( *t ) {
00230         for ( j = 0; j < *t; j++ ) term[j] = t[j];
00231         t += *t;
00232         AN.ninterms++; dd = AN.deferskipped;
00233         if ( AC.CollectFun && *term <= (AM.MaxTer/(2*(LONG)(sizeof(WORD)))) ) {
00234             if ( GetMoreFromMem(term,&t) ) {
00235                 LowerSortLevel(); goto ProcErr;
00236             }
00237         }
00238         AT.WorkPointer = term + *term;
00239         AN.RepPoint = AT.RepCount + 1;
00240         AN.IndDum = AM.IndDum;
00241         AR.CurDum = ReNumber(BHEAD term);
00242         if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMLAG);
00243         if ( AN.ncmod ) {
00244             if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
00245             else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
00246         }
00247         else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
00248         if ( Generator(BHEAD term,0) ) {
00249             LowerSortLevel(); goto ProcErr;
00250         }
00251         AN.ninterms += dd;
00252     }
00253     AN.ninterms += dd;
00254     if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) goto ProcErr;
00255     if ( AM.S0->TermsLeft ) e->vflags &= ~ISZERO;
00256     else e->vflags |= ISZERO;
00257     if ( AR.expchanged == 0 ) e->vflags |= ISUNMODIFIED;
00258     if ( AM.S0->TermsLeft ) AR.expflags |= ISZERO;
00259     if ( AR.expchanged ) AR.expflags |= ISUNMODIFIED;
00260     AR.GetFile = 0;
00261     /*
00262         #] in memory :
00263     */
00264     }
00265     else {
00266         AR.CurExpr = i;
00267         switch ( e->status ) {
00268             case UNHIDELEXPRESSION:
00269             case UNHIDEGEXPRESSION:
00270                 AR.GetFile = 2;
00271             #ifdef WITHMPI
00272                 if ( PF.me == MASTER ) SetScratch(AR.hidefile,&(e->onfile));
00273             #else
00274                 SetScratch(AR.hidefile,&(e->onfile));
00275                 AR.InHiBuf = AR.hidefile->POfull-AR.hidefile->POfill;
00276             #ifdef HIIDEDEBUG
00277                 MesPrint("Hidefile: onfile: %15p, POposition: %15p, filesize: %15p",&(e->onfile)
00278                     ,&(AR.hidefile->POposition),&(AR.hidefile->filesize));
00279                 MesPrint("Set hidefile to buffer position %1/%1; AR.InHiBuf = %1"
00280                     ,(AR.hidefile->POfill-AR.hidefile->PObuffer)*sizeof(WORD)
00281                     ,(AR.hidefile->POfull-AR.hidefile->PObuffer)*sizeof(WORD)
00282                     ,AR.InHiBuf
00283                 );
00284             #endif
00285             #endif
00286                 curfile = AR.hidefile;
00287                 goto commonread;
00288             case INTOHIDELEXPRESSION:
00289             case INTOHIDEGEXPRESSION:
00290                 AR.outtohide = 1;
00291             /*
00292                 BugFix 12-feb-2016
00293                 This may not work when the file is open and we move around.
00294                 AR.hidefile->POfill = AR.hidefile->POfull;
00295             */
00296                 SetEndHScratch(AR.hidefile,&position);
00297                 /* fall through */
00298             case LOCALEXPRESSION:
00299             case GLOBALEXPRESSION:
00300                 AR.GetFile = 0;
00301             /*[20oct2009 mt]:*/
00302             #ifdef WITHMPI
00303                 if ( ( PF.me == MASTER ) || (PF.mkSlaveInfile) )
00304             #endif
00305                 SetScratch(AR.infile,&(e->onfile));
00306             /*:[20oct2009 mt]*/
00307                 curfile = AR.infile;
00308             commonread:;
00309             #ifdef WITHMPI
00310                 if ( PF_Processor(e,i,LastExpression) ) {
00311                     MesPrint("Error in PF_Processor");

```



```

00312             goto ProcErr;
00313         }
00314         /*[20oct2009 mt]:*/
00315         if ( AC.mparallelflag != PARALLELFLAG ){
00316             if (PF.me != MASTER)
00317                 break;
00318         #endif
00319         /*[20oct2009 mt]:*/
00320         if ( GetTerm(BHEAD term) <= 0 ) {
00321             #ifdef HIDEDEBUG
00322                 MesPrint("Error condition 1a");
00323                 ExprStatus(e);
00324             #endif
00325             MesPrint("(2) Expression %d has problems in scratchfile(process)",i);
00326             retval = -1;
00327             break;
00328         }
00329         term[3] = i;
00330         if ( term[5] < 0 ) { /* Fill with spectator */
00331             fromspectator = -term[5];
00332             PUTZERO (AM.SpectatorFiles[fromspectator-1].readpos);
00333             term[5] = AC.cbufnum;
00334         }
00335         else fromspectator = 0;
00336         if ( AR.outtohide ) {
00337             SeekScratch(AR.hidefile,&position);
00338             e->onfile = position;
00339             if ( PutOut (BHEAD term,&position,AR.hidefile,0) < 0 ) goto ProcErr;
00340         }
00341         else {
00342             SeekScratch(AR.outfile,&position);
00343             e->onfile = position;
00344             if ( PutOut (BHEAD term,&position,AR.outfile,0) < 0 ) goto ProcErr;
00345         }
00346         AR.DeferFlag = AC.ComDefer;
00347         AR.Eside = RHSIDE;
00348         if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
00349             AR.BracketOn = 1;
00350             AT.BrackBuf = AM.BracketFactors;
00351             AT.bracketindexflag = 1;
00352         }
00353         if ( AT.bracketindexflag > 0 ) OpenBracketIndex(i);
00354         #ifdef WITHPTHREADS
00355         if ( AS.MultiThreaded && AC.mparallelflag == PARALLELFLAG ) {
00356             if ( ThreadsProcessor(e,LastExpression,fromspectator) ) {
00357                 MesPrint("Error in ThreadsProcessor");
00358                 goto ProcErr;
00359             }
00360             if ( AR.outtohide ) {
00361                 AR.outfile = oldoutfile;
00362                 AR.hidefile->POfull = AR.hidefile->POfill;
00363             }
00364         }
00365         else
00366         #endif
00367         {
00368             NewSort (BHEAD0);
00369             AR.MaxDum = AM.IndDum;
00370             AN.ninterms = 0;
00371             for (;;) {
00372                 if ( fromspectator ) size = GetFromSpectator(term,fromspectator-1);
00373                 else size = GetTerm(BHEAD term);
00374                 if ( size <= 0 ) break;
00375                 SeekScratch(curfile,&position);
00376                 if ( ( e->vflags & ISFACTORIZED ) != 0 && term[1] == HAAKJE ) {
00377                     StoreTerm(BHEAD term);
00378                 }
00379                 else {
00380                     AN.ninterms++; dd = AN.deferskipped;
00381                     if ( AC.CollectFun && *term <= (AM.MaxTer/(2*(LONG)(sizeof(WORD)))) ) {
00382                         if ( GetMoreTerms(term) < 0 ) {
00383                             LowerSortLevel(); goto ProcErr;
00384                         }
00385                         SeekScratch(curfile,&position);
00386                     }
00387                     AT.WorkPointer = term + *term;
00388                     AN.RepPoint = AT.RepCount + 1;
00389                     if ( AR.DeferFlag ) {
00390                         AN.IndDum = Expressions[AR.CurExpr].numdummies + AM.IndDum;
00391                         AR.CurDum = AN.IndDum;
00392                     }
00393                     else {
00394                         AN.IndDum = AM.IndDum;
00395                         AR.CurDum = ReNumber (BHEAD term);
00396                     }
00397                     if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMLAG);
00398                     if ( AN.ncmod ) {

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00399         if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
00400     else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
00401     }
00402     else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
00403     if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
00404         && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
00405         PolyFunClean(BHEAD term);
00406     }
00407     if ( Generator(BHEAD term,0) ) {
00408         LowerSortLevel(); goto ProcErr;
00409     }
00410     AN.ninterms += dd;
00411 }
00412 SetScratch(curfile,&position);
00413 if ( AR.GetFile == 2 ) {
00414     AR.InHiBuf = (curfile->POfull-curfile->PObuffer)
00415         -DIFBASE(position,curfile->POposition)/sizeof(WORD);
00416 }
00417 else {
00418     AR.InInBuf = (curfile->POfull-curfile->PObuffer)
00419         -DIFBASE(position,curfile->POposition)/sizeof(WORD);
00420 }
00421 }
00422 AN.ninterms += dd;
00423 if ( LastExpression ) {
00424     UpdateMaxSize();
00425     if ( AR.infile->handle >= 0 ) {
00426         CloseFile(AR.infile->handle);
00427         AR.infile->handle = -1;
00428         remove(AR.infile->name);
00429         PUTZERO(AR.infile->POposition);
00430     }
00431     AR.infile->POfill = AR.infile->POfull = AR.infile->PObuffer;
00432 }
00433 if ( AR.outtohide ) AR.outfile = AR.hidefile;
00434 if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) goto ProcErr;
00435 if ( AR.outtohide ) {
00436     AR.outfile = oldoutfile;
00437     AR.hidefile->POfull = AR.hidefile->POfill;
00438 }
00439 e->numdummies = AR.MaxDum - AM.IndDum;
00440 UpdateMaxSize();
00441 }
00442 AR.BracketOn = oldBracketOn;
00443 AT.BrackBuf = oldBrackBuf;
00444 if ( ( e->vflags & TOBEFACTORED ) != 0 ) {
00445     poly_factorize_expression(e);
00446 }
00447 else if ( ( ( e->vflags & TOBEUNFACTORED ) != 0 )
00448         && ( ( e->vflags & ISFACTORIZED ) != 0 ) ) {
00449     poly_unfactorize_expression(e);
00450 }
00451 AT.bracketindexflag = oldbracketindexflag;
00452 if ( AM.S0->TermsLeft ) e->vflags &= ~ISZERO;
00453 else e->vflags |= ISZERO;
00454 if ( AR.expchanged == 0 ) e->vflags |= ISUNMODIFIED;
00455 if ( AM.S0->TermsLeft ) AR.expflags |= ISZERO;
00456 if ( AR.expchanged ) AR.expflags |= ISUNMODIFIED;
00457 AR.GetFile = 0;
00458 AR.outtohide = 0;
00459 /*[20oct2009 mt]:*/
00460 #ifdef WITHMPI
00461     }
00462 #endif
00463 #ifdef WITHPTHREADS
00464     if ( e->status == INTOHIDELEXPRESSION ||
00465         e->status == INTOHIDEGEXPRESSSION ) {
00466         SetHideFiles();
00467     }
00468 #endif
00469     break;
00470 case SKIPLEXPRESSION:
00471 case SKIPGEXPRESSSION:
00472     /*
00473         This can be greatly improved of course by file-to-file copy.
00474     */
00475     #ifdef WITHMPI
00476         if ( PF.me != MASTER ) break;
00477     #endif
00478     AR.GetFile = 0;
00479     SetScratch(AR.infile,&(e->onfile));
00480     if ( GetTerm(BHEAD term) <= 0 ) {
00481     #ifdef HIDEDEBUG
00482         MesPrint("Error condition 1b");
00483         ExprStatus(e);
00484     #endif
00485         MesPrint("(3) Expression %d has problems in scratchfile",i);

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00486         retval = -1;
00487         break;
00488     }
00489     term[3] = i;
00490     AR.DeferFlag = 0;
00491     SeekScratch(AR.outfile,&position);
00492     e->onfile = position;
00493     *AM.S0->sBuffer = 0; firstterm = -1;
00494     do {
00495         WORD *oldipointer = AR.CompressPointer;
00496         WORD *comprtop = AR.ComprTop;
00497         AR.ComprTop = AM.S0->sTop;
00498         AR.CompressPointer = AM.S0->sBuffer;
00499         if ( firstterm > 0 ) {
00500             if ( PutOut(BHEAD term,&position,AR.outfile,1) < 0 ) goto ProcErr;
00501         }
00502         else if ( firstterm < 0 ) {
00503             if ( PutOut(BHEAD term,&position,AR.outfile,0) < 0 ) goto ProcErr;
00504             firstterm++;
00505         }
00506         else {
00507             if ( PutOut(BHEAD term,&position,AR.outfile,-1) < 0 ) goto ProcErr;
00508             firstterm++;
00509         }
00510         AR.CompressPointer = oldipointer;
00511         AR.ComprTop = comprtop;
00512     } while ( GetTerm(BHEAD term) );
00513     if ( FlushOut(&position,AR.outfile,1) ) goto ProcErr;
00514     UpdateMaxSize();
00515     break;
00516     case HIDELEXPRESSION:
00517     case HIDEGEXPRESSION:
00518     #ifdef WITHMPI
00519         if ( PF.me != MASTER ) break;
00520     #endif
00521     AR.GetFile = 0;
00522     SetScratch(AR.infile,&(e->onfile));
00523     if ( GetTerm(BHEAD term) <= 0 ) {
00524     #ifdef HIDEDEBUG
00525         MesPrint("Error condition 1c");
00526         ExprStatus(e);
00527     #endif
00528         MesPrint("(4) Expression %d has problems in scratchfile",i);
00529         retval = -1;
00530         break;
00531     }
00532     term[3] = i;
00533     AR.DeferFlag = 0;
00534     SetEndHScratch(AR.hidefile,&position);
00535     e->onfile = position;
00536     #ifdef HIDEDEBUG
00537     if ( AR.hidefile->handle >= 0 ) {
00538         POSITION possize,pos;
00539         PUTZERO(possize);
00540         PUTZERO(pos);
00541         SeekFile(AR.hidefile->handle,&pos,SEEK_CUR);
00542         SeekFile(AR.hidefile->handle,&possize,SEEK_END);
00543         SeekFile(AR.hidefile->handle,&pos,SEEK_SET);
00544         MesPrint("Processor Hidel: filesize(th) = %12p, filesize(ex) = %12p",&(position),
00545                 &(possize));
00546         MesPrint("      in buffer:
00547 %1", (AR.hidefile->POfill-AR.hidefile->PObuffer)*sizeof(WORD));
00548     #endif
00549     *AM.S0->sBuffer = 0; firstterm = -1;
00550     cbo = cpo = AM.S0->sBuffer;
00551     do {
00552         WORD *oldipointer = AR.CompressPointer;
00553         WORD *oldibuffer = AR.CompressBuffer;
00554         WORD *comprtop = AR.ComprTop;
00555         AR.ComprTop = AM.S0->sTop;
00556         AR.CompressPointer = cpo;
00557         AR.CompressBuffer = cbo;
00558         if ( firstterm > 0 ) {
00559             if ( PutOut(BHEAD term,&position,AR.hidefile,1) < 0 ) goto ProcErr;
00560         }
00561         else if ( firstterm < 0 ) {
00562             if ( PutOut(BHEAD term,&position,AR.hidefile,0) < 0 ) goto ProcErr;
00563             firstterm++;
00564         }
00565         else {
00566             if ( PutOut(BHEAD term,&position,AR.hidefile,-1) < 0 ) goto ProcErr;
00567             firstterm++;
00568         }
00569         cpo = AR.CompressPointer;
00570         cbo = AR.CompressBuffer;
00571         AR.CompressPointer = oldipointer;

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00572         AR.CompressBuffer = oldibuffer;
00573         AR.ComprTop = comprtop;
00574     } while ( GetTerm(BHEAD term) );
00575 #ifdef HIIDEDEBUG
00576     if ( AR.hidefile->handle >= 0 ) {
00577         POSITION posize,pos;
00578         PUTZERO(posize);
00579         PUTZERO(pos);
00580         SeekFile(AR.hidefile->handle,&pos,SEEK_CUR);
00581         SeekFile(AR.hidefile->handle,&posize,SEEK_END);
00582         SeekFile(AR.hidefile->handle,&pos,SEEK_SET);
00583         MesPrint("Processor Hide2: filesize(th) = %12p, filesize(ex) = %12p",&(position),
00584                 &(posize));
00585         MesPrint("    in buffer:
%1", (AR.hidefile->POfill-AR.hidefile->PObuffer)*sizeof(WORD));
00586     }
00587 #endif
00588     if ( FlushOut(&position,AR.hidefile,1) ) goto ProcErr;
00589     AR.hidefile->POfull = AR.hidefile->POfill;
00590 #ifdef HIIDEDEBUG
00591     if ( AR.hidefile->handle >= 0 ) {
00592         POSITION posize,pos;
00593         PUTZERO(posize);
00594         PUTZERO(pos);
00595         SeekFile(AR.hidefile->handle,&pos,SEEK_CUR);
00596         SeekFile(AR.hidefile->handle,&posize,SEEK_END);
00597         SeekFile(AR.hidefile->handle,&pos,SEEK_SET);
00598         MesPrint("Processor Hide3: filesize(th) = %12p, filesize(ex) = %12p",&(position),
00599                 &(posize));
00600         MesPrint("    in buffer:
%1", (AR.hidefile->POfill-AR.hidefile->PObuffer)*sizeof(WORD));
00601     }
00602 #endif
00603 /*
00604     Because we direct the e->onfile already to the hide file, we
00605     need to change the status of the expression. Otherwise the use
00606     of parts (or the whole) of the expression looks in the infile
00607     while the position is that of the hide file.
00608     We choose to get everything from the hide file. On average that
00609     should give least file activity.
00610 */
00611     if ( e->status == HIDELEXPRESSION ) {
00612         e->status = HIDDENLEXPRESSION;
00613         AS.OldOnFile[i] = e->onfile;
00614         AS.OldNumFactors[i] = Expressions[i].numfactors;
00615     }
00616     if ( e->status == HIDEGEXPRESSION ) {
00617         e->status = HIDDENGEXPRESSION;
00618         AS.OldOnFile[i] = e->onfile;
00619         AS.OldNumFactors[i] = Expressions[i].numfactors;
00620     }
00621 #ifdef WITHPTHREADS
00622     SetHideFiles();
00623 #endif
00624     UpdateMaxSize();
00625     break;
00626 case DROPEDEXPRESSION:
00627 case DROPLEXPRESSION:
00628 case DROPGEXPRESSION:
00629 case DROPHLEXPRESSION:
00630 case DROPHGEXPRESSION:
00631 case STOREDEXPRESSION:
00632 case HIDDENLEXPRESSION:
00633 case HIDDENGEXPRESSION:
00634 case SPECTATOREXPRESSION:
00635     default:
00636         break;
00637 }
00638 }
00639     AR.KeptInHold = 0;
00640 }
00641     AR.DeferFlag = 0;
00642     AT.WorkPointer = term;
00643 #ifdef HIIDEDEBUG
00644     MesPrint("Status at the end of Processor (HideLevel = %d)",AC.HideLevel);
00645     for ( i = 0; i < NumExpressions; i++ ) {
00646         e = Expressions+i;
00647         ExprStatus(e);
00648     }
00649 #endif
00650     return(retval);
00651 ProcErr:
00652     AT.WorkPointer = term;
00653     if ( AM.tracebackflag ) MesCall("Processor");
00654     return(-1);
00655 }
00656 /*

```

```

00657         #] Processor :
00658         #[ TestSub :             WORD TestSub(term,level)
00659     */
00683 #define DONE(x) { retvalue = x; goto Done; }
00684
00685 WORD TestSub(PHEAD WORD *term, WORD level)
00686 {
00687     GETBIDENTITY
00688     WORD *m, *t, *r, retvalue = 0, funflag, j, oldncmod, nexpr, *Tpattern = 0;
00689     WORD *stop, *t1, *t2, funnum, wilds, tbufnum, stilldirty = 0;
00690     NESTING n;
00691     CBUF *C = cbuf+AT.ebufnum;
00692     LONG isp, i;
00693     TABLES T;
00694     COMPARE oldcompareroutine = (COMPARE)(AR.CompareRoutine);
00695     WORD oldsorttype = AR.SortType;
00696     ReStart:
00697     tbufnum = 0; i = 0; retvalue = 0;
00698     AT.TMbuff = AM.rbufnum;
00699     funflag = 0;
00700     t = term;
00701     r = t + *t - 1;
00702     m = r - ABS(*r) + 1;
00703     t++;
00704     if ( t < m ) do {
00705         if ( *t == SUBEXPRESSION ) {
00706             /*
00707                 Subexpression encountered
00708                 There may be more than one.
00709                 The old strategy was to take the last.
00710                 A newer strategy was to take the lowest power first.
00711                 The current strategy is that we compute the number of terms
00712                 generated by this subexpression and take the minimum of that.
00713             */
00714
00715 #ifdef WHICHSUBEXPRESSION
00716
00717             WORD *tmin = t, AN.nbino;
00718             /* LONG minval = MAXLONG; */
00719             LONG minval = -1;
00720             LONG mm, mnum1 = 1;
00721             if ( AN.BinoScrat == 0 ) {
00722                 AN.BinoScrat = (UWORD *)Malloc1((AM.MaxTal+2)*sizeof(UWORD),"GetBinoScrat");
00723             }
00724 #endif
00725             if ( t[3] ) {
00726                 r = t + t[1];
00727                 while ( AN.subsubveto == 0 &&
00728                     *r == SUBEXPRESSION && r < m && r[3] ) {
00729 #ifdef WHICHSUBEXPRESSION
00730                 mnum1++;
00731 #endif
00732                 if ( r[1] == t[1] && r[2] == t[2] && r[4] == t[4] ) {
00733                     j = t[1] - SUBEXPSIZE;
00734                     t1 = t + SUBEXPSIZE;
00735                     t2 = r + SUBEXPSIZE;
00736                     while ( j > 0 && *t1++ == *t2++ ) j--;
00737                     if ( j <= 0 ) {
00738                         t[3] += r[3];
00739                         if ( t[3] == 0 ) {
00740                             t1 = r + r[1];
00741                             t2 = term + *term;
00742                             *term -= r[1]+t[1];
00743                             r = t;
00744                             while ( t1 < t2 ) *r++ = *t1++;
00745                             goto ReStart;
00746                         }
00747                         else {
00748                             t1 = r + r[1];
00749                             t2 = term + *term;
00750                             *term -= r[1];
00751                             m -= r[1];
00752                             while ( t1 < t2 ) *r++ = *t1++;
00753                             r = t;
00754                         }
00755                     }
00756                 }
00757 #ifdef WHICHSUBEXPRESSION
00758
00759                 else if ( t[2] >= 0 ) {
00760                     /*
00761                         Compute Binom(numterms+power-1,power-1)
00762                         We need potentially long arithmetic.
00763                         That is why we had to allocate AN.BinoScrat
00764                     */
00765                     if ( AN.last1 == t[3] && AN.last2 == cbuf[t[4]].NumTerms[t[2]] + t[3] - 1 ) {
00766                         if ( AN.last3 > minval ) {

```

```

00767             minval = AN.last3; tmin = t;
00768         }
00769     }
00770     else {
00771         AN.last1 = t[3]; mm = AN.last2 = cbuf[t[4]].NumTerms[t[2]] + t[3] - 1;
00772         if ( t[3] == 1 ) {
00773             if ( mm > minval ) {
00774                 minval = mm; tmin = t;
00775             }
00776         }
00777         else if ( t[3] > 0 ) {
00778             if ( mm > MAXPOSITIVE ) goto TooMuch;
00779             GetBinom(AN.BinoScrat,&AN.nbino,(WORD)mm,t[3]);
00780             if ( AN.nbino > 2 ) goto TooMuch;
00781             if ( AN.nbino == 2 ) {
00782                 mm = AN.BinoScrat[1];
00783                 mm = ( mm << BITSINWORD ) + AN.BinoScrat[0];
00784             }
00785             else if ( AN.nbino == 1 ) mm = AN.BinoScrat[0];
00786             else mm = 0;
00787             if ( mm > minval ) {
00788                 minval = mm; tmin = t;
00789             }
00790         }
00791         AN.last3 = mm;
00792     }
00793 }
00794 #endif
00795 t = r;
00796 r += r[1];
00797 }
00798 #ifdef WHICHSUBEXPRESSION
00799     if ( mnum1 > 1 && t[2] >= 0 ) {
00800 /*
00801     To keep the flowcontrol simple we duplicate some code here
00802 */
00803     if ( AN.last1 == t[3] && AN.last2 == cbuf[t[4]].NumTerms[t[2]] + t[3] - 1 ) {
00804         if ( AN.last3 > minval ) {
00805             minval = AN.last3; tmin = t;
00806         }
00807     }
00808     else {
00809         AN.last1 = t[3]; mm = AN.last2 = cbuf[t[4]].NumTerms[t[2]] + t[3] - 1;
00810         if ( t[3] == 1 ) {
00811             if ( mm > minval ) {
00812                 minval = mm; tmin = t;
00813             }
00814         }
00815         else if ( t[3] > 0 ) {
00816             if ( mm > MAXPOSITIVE ) {
00817 /*
00818                 We will generate more terms than we can count
00819 */
00820 TooMuch;;
00821
00822                 MLOCK(ErrorMessageLock);
00823                 MesPrint("Attempt to generate more terms than FORM can count");
00824                 MUNLOCK(ErrorMessageLock);
00825                 Terminate(-1);
00826             }
00827             GetBinom(AN.BinoScrat,&AN.nbino,(WORD)mm,t[3]);
00828             if ( AN.nbino > 2 ) goto TooMuch;
00829             if ( AN.nbino == 2 ) {
00830                 mm = AN.BinoScrat[1];
00831                 mm = ( mm << BITSINWORD ) + AN.BinoScrat[0];
00832             }
00833             else if ( AN.nbino == 1 ) mm = AN.BinoScrat[0];
00834             else mm = 0;
00835             if ( mm > minval ) {
00836                 minval = mm; tmin = t;
00837             }
00838         }
00839         AN.last3 = mm;
00840     }
00841     t = tmin;
00842 #endif
00843 /*
00844     AR.TePos = 0;
00845     AR.TePos = WORDDIF(t,term);
00846     AT.TMbuff = t[4];
00847     if ( t[4] == AM.dbufnum && (t+t[1]) < m && t[t[1]] == DOLLAREXP2 ) {
00848         if ( t[t[1]+2] < 0 ) AT.TMdolfac = -t[t[1]+2];
00849         else { /* resolve the element number */
00850             AT.TMdolfac = GetDolNum(BHEAD t+t[1],m)+1;
00851         }
00852     }
00853     else AT.TMdolfac = 0;
00854     if ( t[3] < 0 ) {

```

```

00854         AN.TeInFun = 1;
00855         AR.TePos = WORDDIF(t,term);
00856         DONE(t[2])
00857     }
00858     else {
00859         AN.TeInFun = 0;
00860         AN.TeSuOut = t[3];
00861     }
00862     if ( t[2] < 0 ) {
00863         AN.TeSuOut = -t[3];
00864         DONE(-t[2])
00865     }
00866     DONE(t[2])
00867 }
00868 }
00869 else if ( *t == EXPRESSION ) {
00870     WORD *toTMaddr;
00871     i = -t[2] - 1;
00872     if ( t[3] < 0 ) {
00873         AN.TeInFun = 1;
00874         AR.TePos = WORDDIF(t,term);
00875         DONE(i)
00876     }
00877     nexpr = t[3];
00878     toTMaddr = m = AT.WorkPointer;
00879     AN.Frozen = 0;
00880 /*
00881     We have to be very careful with respect to setting variables
00882     like AN.TeInFun, because we may still call Generator and that
00883     may change those variables. That is why we set them at the
00884     last moment only.
00885 */
00886     j = t[1];
00887     AT.WorkPointer += j;
00888     r = t;
00889     NCOPY(m,r,j);
00890     r = t + t[1];
00891     t += SUBEXPSIZE;
00892     while ( t < r ) {
00893         if ( *t == FROMBRAC ) {
00894             WORD *tttstop,*tttstop;
00895 /*
00896             Note: Convention is that wildcards are done
00897             after the expression has been picked up. So
00898             no wildcard substitutions are needed here.
00899 */
00900             t += 2;
00901             AN.Frozen = m = AT.WorkPointer;
00902 /*
00903             We should check now for subexpressions and if necessary
00904             we substitute them. Keep in mind: only one term allowed!
00905
00906             In retrospect (26-jan-2010): take also functions that
00907             have a dirty flag on
00908 */
00909             j = *t; tttstop = t + j;
00910             GETSTOP(t,tttstop);
00911             *m++ = j; t++;
00912             while ( t < tttstop ) {
00913                 if ( *t == SUBEXPRESSION ) break;
00914                 if ( *t >= FUNCTION && ( ( t[2] & DIRTYFLAG ) == DIRTYFLAG ) ) break;
00915                 j = t[1]; NCOPY(m,t,j);
00916             }
00917             if ( t < tttstop ) {
00918 /*
00919                 We ran into a subexpression or a function with a
00920                 'dirty' argument. It could also be a $ or
00921                 just e[(a^2)*b]. In all cases we should evaluate
00922 */
00923                 while ( t < tttstop ) *m++ = *t++;
00924                 *AT.WorkPointer = m-AT.WorkPointer;
00925                 m = AT.WorkPointer;
00926                 AT.WorkPointer = m + *m;
00927                 NewSort(BHEAD0);
00928                 if ( Generator(BHEAD m,AR.Cnumlhs) ) {
00929                     LowerSortLevel(); goto EndTest;
00930                 }
00931                 if ( EndSort(BHEAD m,0) < 0 ) goto EndTest;
00932                 AN.Frozen = m;
00933                 if ( *m == 0 ) {
00934                     *m++ = 4; *m++ = 1; *m++ = 1; *m++ = 3;
00935                 }
00936                 else if ( m[*m] != 0 ) {
00937                     MLOCK(ErrorMessageLock);
00938                     MesPrint("Bracket specification in expression should be one single term");
00939                     MUNLOCK(ErrorMessageLock);
00940                     Terminate(-1);

```

```

00941         }
00942     else {
00943         m += *m;
00944         m -= ABS(m[-1]);
00945         *m++ = 1; *m++ = 1; *m++ = 3;
00946         *AN.Frozen = m - AN.Frozen;
00947     }
00948 }
00949 else {
00950     while ( t < tttstop ) *m++ = *t++;
00951     *AT.WorkPointer = m-AT.WorkPointer;
00952     m = AT.WorkPointer;
00953     AT.WorkPointer = m + *m;
00954     if ( Normalize(BHEAD m) ) {
00955         MLOCK(ErrorMessageLock);
00956         MesPrint("Error while picking up contents of bracket");
00957         MUNLOCK(ErrorMessageLock);
00958         Terminate(-1);
00959     }
00960     if ( !*m ) {
00961         *m++ = 4; *m++ = 1; *m++ = 1; *m++ = 3;
00962     }
00963     else m += *m;
00964 }
00965 AT.WorkPointer = m;
00966 break;
00967 }
00968 t += t[1];
00969 }
00970 AN.TeInFun = 0;
00971 AR.TePos = 0;
00972 AN.TeSuOut = nexpr;
00973 AT.TMaddr = toTMaddr;
00974 DONE(i)
00975 }
00976 else if ( *t >= FUNCTION ) {
00977     if ( t[0] == EXPONENT ) {
00978         if ( t[1] == FUNHEAD+4 && t[FUNHEAD] == -SYMBOL &&
00979             t[FUNHEAD+2] == -SNUMBER && t[FUNHEAD+3] < MAXPOWER
00980             && t[FUNHEAD+3] > -MAXPOWER ) {
00981             t[0] = SYMBOL;
00982             t[1] = 4;
00983             t[2] = t[FUNHEAD+1];
00984             t[3] = t[FUNHEAD+3];
00985             r = term + *term;
00986             m = t + FUNHEAD+4;
00987             t += 4;
00988             while ( m < r ) *t++ = *m++;
00989             *term = WORDDIF(t,term);
00990             goto ReStart;
00991         }
00992     else if ( t[1] == FUNHEAD+ARGHEAD+11 && t[FUNHEAD] == ARGHEAD+9
00993         && t[FUNHEAD+ARGHEAD] == 9 && t[FUNHEAD+ARGHEAD+1] == DOTPRODUCT
00994         && t[FUNHEAD+ARGHEAD+8] == 3
00995         && t[FUNHEAD+ARGHEAD+7] == 1
00996         && t[FUNHEAD+ARGHEAD+6] == 1
00997         && t[FUNHEAD+ARGHEAD+5] == 1
00998         && t[FUNHEAD+ARGHEAD+9] == -SNUMBER
00999         && t[FUNHEAD+ARGHEAD+10] < MAXPOWER
01000         && t[FUNHEAD+ARGHEAD+10] > -MAXPOWER ) {
01001         t[0] = DOTPRODUCT;
01002         t[1] = 5;
01003         t[2] = t[FUNHEAD+ARGHEAD+3];
01004         t[3] = t[FUNHEAD+ARGHEAD+4];
01005         t[4] = t[FUNHEAD+ARGHEAD+10];
01006         r = term + *term;
01007         m = t + FUNHEAD+ARGHEAD+11;
01008         t += 5;
01009         while ( m < r ) *t++ = *m++;
01010         *term = WORDDIF(t,term);
01011         goto ReStart;
01012     }
01013 }
01014 funnum = *t;
01015 if ( *t >= FUNCTION + WILDOFFSET ) funnum -= WILDOFFSET;
01016 if ( *t == EXPONENT ) {
01017     /*
01018     Test whether the second argument is an integer
01019     */
01020     r = t+FUNHEAD;
01021     NEXTARG(r)
01022     if ( *r == -SNUMBER && r[1] < MAXPOWER && r+2 == t+t[1] &&
01023         t[FUNHEAD] > -FUNCTION && ( t[FUNHEAD] != -SNUMBER
01024         || t[FUNHEAD+1] != 0 ) && t[FUNHEAD] != ARGHEAD ) {
01025         if ( r[1] == 0 ) {
01026             if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] == 0 ) {
01027                 MLOCK(ErrorMessageLock);

```



```

01028         MesPrint("Encountered 0^0. Fatal error.");
01029         MUNLOCK(ErrorMessageLock);
01030         SETERROR(-1);
01031     }
01032     *t = DUMMYFUN;
01033 /*
01034         Now mark it clean to avoid further interference.
01035         Normalize will remove this object.
01036 */
01037     t[2] = 0;
01038 }
01039 else {
01040     /* Note that the case 0^ is treated in Normalize */
01041
01042     t1 = AddRHS(AT.ebufnum,1);
01043     m = t + FUNHEAD;
01044     if ( *m > 0 ) {
01045         m += ARGHEAD;
01046         i = t[FUNHEAD] - ARGHEAD;
01047         while ( (t1 + i + 10) > C->Top )
01048             t1 = DoubleCbuffer(AT.ebufnum,t1,9);
01049         while ( --i >= 0 ) *t1++ = *m++;
01050     }
01051     else {
01052         if ( (t1 + 20) > C->Top )
01053             t1 = DoubleCbuffer(AT.ebufnum,t1,10);
01054         ToGeneral(m,t1,1);
01055         t1 += *t1;
01056     }
01057     *t1++ = 0;
01058     C->rhs[C->numrhs+1] = t1;
01059     C->Pointer = t1;
01060
01061     /* No provisions yet for commuting objects */
01062
01063     C->CanCommu[C->numrhs] = 1;
01064     *t++ = SUBEXPRESSION;
01065     *t++ = SUBEXPSIZE;
01066     *t++ = C->numrhs;
01067     *t++ = r[1];
01068     *t++ = AT.ebufnum;
01069 #if SUBEXPSIZE > 5
01070 Important: we may not have enough spots here
01071 #endif
01072     FILLSUB(t) /* Important: We have maybe only 5 spots! */
01073     r += 2;
01074     m = term + *term;
01075     do { *t++ = *r++; } while ( r < m );
01076     *term -= WORDDIF(r,t);
01077     goto ReStart;
01078 }
01079 }
01080 }
01081 else if ( *t == SUMF1 || *t == SUMF2 ) {
01082 /*
01083     What we are looking for is:
01084     1-st argument: Single symbol or index.
01085     2-nd argument: Number.
01086     3-rd argument: Number.
01087     (4-th argument):Number.
01088     One more argument.
01089     This would activate the summation procedure.
01090     Note that the initiated recursion here can be done
01091     without upsetting the regular procedures.
01092 */
01093     WORD *tstop, lcounter, lcmin, lcmax, lcinc;
01094     tstop = t + t[1];
01095     r = t+FUNHEAD;
01096     if ( r+6 < tstop && r[2] == -SNUMBER && r[4] == -SNUMBER
01097         && ( r[0] == -SYMBOL )
01098         || ( r[0] == -INDEX && r[1] >= AM.OffsetIndex
01099             && r[3] >= 0 && r[3] < AM.OffsetIndex
01100             && r[5] >= 0 && r[5] < AM.OffsetIndex ) ) ) {
01101         lcounter = r[0] == -INDEX ? -r[1]: r[1]; /* The loop counter */
01102         lcmin = r[3];
01103         lcmax = r[5];
01104         r += 6;
01105         if ( *r == -SNUMBER && r+2 < tstop ) {
01106             lcinc = r[1];
01107             r += 2;
01108         }
01109         else lcinc = 1;
01110         if ( r < tstop && ( (*r > 0 && (r+r) == tstop )
01111             || (*r <= -FUNCTION && r+1 == tstop )
01112             || (*r > -FUNCTION && *r < 0 && r+2 == tstop ) ) ) {
01113             m = AddRHS(AT.ebufnum,1);
01114             if ( *r > 0 ) {

```

```

01115         i = *r - ARGHEAD;
01116         r += ARGHEAD;
01117         while ( ( m + i + 10 ) > C->Top )
01118             m = DoubleCbuffer(AT.ebufnum,m,11);
01119         while ( --i >= 0 ) *m++ = *r++;
01120     }
01121     else {
01122         while ( ( m + 20 ) > C->Top )
01123             m = DoubleCbuffer(AT.ebufnum,m,12);
01124         ToGeneral(r,m,1);
01125         m += *m;
01126     }
01127     *m++ = 0;
01128     C->rhs[C->numrhs+1] = m;
01129     C->Pointer = m;
01130     m = AT.TMout;
01131     *m++ = 6;
01132     if ( *t == SUMF1 ) *m++ = SUMNUM1;
01133     else *m++ = SUMNUM2;
01134     *m++ = lcounter;
01135     *m++ = lcmin;
01136     *m++ = lcmax;
01137     *m++ = lcinc;
01138     m = t + t[1];
01139     r = C->rhs[C->numrhs];
01140     /*
01141
01142     Test now if the argument was already evaluated.
01143     In that case it needs a new subexpression prototype.
01144     In either case we replace the function now by a
01145     subexpression prototype.
01146
01147     if ( *r >= (SUBEXPSIZE+4)
01148         && ABS(*(r++r-1)) < (*r - 1)
01149         && r[1] == SUBEXPRESSION ) {
01150         r++;
01151         i = r[1] - 5;
01152         *t++ = *r++; *t++ = *r++; *t++ = C->numrhs;
01153         r++; *t++ = *r++; *t++ = AT.ebufnum; r++;
01154         while ( --i >= 0 ) *t++ = *r++;
01155     }
01156     else {
01157         *t++ = SUBEXPRESSION;
01158         *t++ = 4+SUBEXPSIZE;
01159         *t++ = C->numrhs;
01160         *t++ = 1;
01161         *t++ = AT.ebufnum;
01162         FILLSUB(t)
01163         if ( lcounter < 0 ) {
01164             *t++ = INDTOIND;
01165             *t++ = 4;
01166             *t++ = -lcounter;
01167         }
01168         else {
01169             *t++ = SYMTONUM;
01170             *t++ = 4;
01171             *t++ = lcounter;
01172         }
01173         *t++ = lcmin;
01174     }
01175     t2 = term + *term;
01176     while ( m < t2 ) *t++ = *m++;
01177     *term = WORDDIF(t,term);
01178     AN.TeInFun = -C->numrhs;
01179     AR.TePos = 0;
01180     AN.TeSuOut = 0;
01181     AT.TMbuff = AT.ebufnum;
01182     DONE(C->numrhs)
01183 }
01184 }
01185 else if ( *t == TOPOLOGIES ) {
01186     /*
01187     Syntax:
01188     topologies_(nloops,nlegs,setvertexsizes,setext,setint[,options])
01189     */
01190     t1 = t+FUNHEAD; t2 = t+t[1];
01191     if ( *t1 == -SNUMBER && t1[1] >= 0 &&
01192         t1[2] == -SNUMBER && ( t1[3] >= 0 || t1[3] == -2 ) &&
01193         t1[4] == -SETSET && Sets[t1[5]].type == CNUMBER &&
01194         t1[6] == -SETSET && Sets[t1[7]].type == CVECTOR &&
01195         t1[8] == -SETSET && Sets[t1[9]].type == CVECTOR &&
01196         t1+10 <= t2 ) {
01197         if ( t1+10 == t2 || ( t1+12 <= t2 && ( t1[10] == -SNUMBER ||
01198             ( t1[10] == -SETSET &&
01199                 Sets[t1[5]].last-Sets[t1[5]].first ==
01200                 Sets[t1[11]].last-Sets[t1[11]].first ) ) ) ) {
01201             AN.TeInFun = -15;

```

```

01202             AN.TeSuOut = 0;
01203             AR.TePos = -1;
01204             AR.funoffset = t - term;
01205             DONE(1)
01206         }
01207     }
01208 }
01209 else if ( *t == DIAGRAMS ) {
01210 /*
01211     Syntax:
01212     diagrams_(model,setinparticles,setoutparticles,
01213         setextmomenta,setintmomenta,couplings or loops)
01214 */
01215     if ( AC.nummodels > 0 ) { /* No model, no diagrams */
01216         t1 = t+FUNHEAD; t2 = t+t[1];
01217         if (
01218             t1[0] == -SETSET && Sets[t1[1]].type == CMODEL &&
01219             t1[2] == -SETSET && ( Sets[t1[3]].type == CFUNCTION
01220                 || ( Sets[t1[3]].type == ANYTYPE && ( Sets[t1[3]].first == Sets[t1[3]].last ) ) )
01221             &&
01222             t1[4] == -SETSET && ( Sets[t1[5]].type == CFUNCTION
01223                 || ( Sets[t1[5]].type == ANYTYPE && ( Sets[t1[5]].first == Sets[t1[5]].last ) ) )
01224             &&
01225             t1[6] == -SETSET && Sets[t1[7]].type == CVECTOR &&
01226             t1[8] == -SETSET && Sets[t1[9]].type == CVECTOR &&
01227             t1+12 <= t2 ) {
01228             /*
01229             Test that the sets of particles correspond to particles
01230             of the set model.
01231             */
01232             MODEL *m = AC.models[SetElements[Sets[t1[1]].first]];
01233             int nn0,nn1,nn2;
01234             for ( nn0 = 3; nn0 <= 5; nn0 += 2 ) {
01235                 for ( nn1 = Sets[t1[nn0]].first; nn1 < Sets[t1[nn0]].last; nn1++ ) {
01236                     for ( nn2 = 0; nn2 < m->nparticles; nn2++ ) {
01237                         if ( m->vertices[nn2]->particles[0].number == SetElements[nn1]
01238                             || m->vertices[nn2]->particles[1].number == SetElements[nn1] ) break;
01239                     }
01240                     if ( nn2 >= m->nparticles ) goto doesnotwork;
01241                 }
01242             }
01243             /*
01244             Now test for a single argument indicating the order
01245             in perturbation theory.
01246             */
01247             if ( ( t1[10] == -SNUMBER && t1[11] >= 0 && t1+12 == t2 )
01248                 || ( t1[10] == -SYMBOL && t1+12 == t2 )
01249                 || ( t1+10+t1[10] == t2 && t1+10+ARGHEAD+t1[10+ARGHEAD] == t2
01250                     && t1[11+ARGHEAD] == SYMBOL ) ) {
01251             /*
01252             Now test that all symbols are valid coupling constants.
01253             */
01254             if ( t1+12 > t2 ) {
01255                 WORD *ttl = t1+13+ARGHEAD, im;
01256                 t2 -= ABS(t2[-1]);
01257                 while ( ttl < t2 ) {
01258                     for ( im = 0; im < m->ncouplings; im++ ) {
01259                         if ( *ttl == m->couplings[im] ) break;
01260                     }
01261                     if ( im >= m->ncouplings ) goto doesnotwork;
01262                     ttl += 2;
01263                 }
01264                 AN.TeInFun = -16;
01265                 AN.TeSuOut = 0;
01266                 AR.TePos = -1;
01267                 AR.funoffset = t - term;
01268                 DONE(1)
01269             }
01270             else if ( ( ( t1[10] == -SNUMBER && t1[11] >= 0 && t1+12 == t2-2 )
01271                 || ( t1[10] == -SYMBOL && t1+12 == t2-2 )
01272                 || ( t1+10+t1[10] == t2-2 && t1+10+ARGHEAD+t1[10+ARGHEAD] == t2-2
01273                     && t1[11+ARGHEAD] == SYMBOL ) ) && t2[-2] == -SNUMBER ) {
01274             /*
01275             With options at t2[-2],t2[-1]
01276             Now test that all symbols are valid coupling constants.
01277             */
01278             t2 -= 2;
01279             if ( t1+12 > t2 ) {
01280                 WORD *ttl = t1+13+ARGHEAD, im;
01281                 t2 -= ABS(t2[-1]);
01282                 while ( ttl < t2 ) {
01283                     for ( im = 0; im < m->ncouplings; im++ ) {
01284                         if ( *ttl == m->couplings[im] ) break;
01285                     }
01286                     if ( im >= m->ncouplings ) goto doesnotwork;
01287                     ttl += 2;
01288                 }

```

```

01287         }
01288     }
01289     AN.TeInFun = -16;
01290     AN.TeSuOut = 0;
01291     AR.TePos = -1;
01292     AR.funoffset = t - term;
01293     DONE(1)
01294 }
01295 doesnotwork;;
01296     }
01297 }
01298 }
01299 if ( functions[funnum-FUNCTION].spec <= 0
01300     || ( t[2] & (DIRTYFLAG|MUSTCLEANPRF) ) != 0 ) {
01301     funflag = 1;
01302 }
01303 if ( *t <= MAXBUILTINFUNCTION ) {
01304     if ( *t <= DELTAP && *t >= THETA ) { /* Speeds up by 2 or 3 compares */
01305         if ( *t == THETA || *t == THETA2 ) {
01306             WORD *tstop, *tt2, kk;
01307             tstop = t + t[1];
01308             tt2 = t + FUNHEAD;
01309             while ( tt2 < tstop ) {
01310                 if ( *tt2 > 0 && tt2[1] != 0 ) goto DoSpec;
01311                 NEXTARG(tt2)
01312             }
01313             if ( !AT.RecFlag ) {
01314                 if ( ( kk = DoTheta(BHEAD t) ) == 0 ) {
01315                     *term = 0;
01316                     DONE(0)
01317                 }
01318                 else if ( kk > 0 ) {
01319                     m = t + t[1];
01320                     r = term + *term;
01321                     while ( m < r ) *t++ = *m++;
01322                     *term = WORDDIF(t,term);
01323                     goto ReStart;
01324                 }
01325             }
01326         }
01327         else if ( *t == DELTA2 || *t == DELTAP ) {
01328             WORD *tstop, *tt2, kk;
01329             tstop = t + t[1];
01330             tt2 = t + FUNHEAD;
01331             while ( tt2 < tstop ) {
01332                 if ( *tt2 > 0 && tt2[1] != 0 ) goto DoSpec;
01333                 NEXTARG(tt2)
01334             }
01335             if ( !AT.RecFlag ) {
01336                 if ( ( kk = DoDelta(t) ) == 0 ) {
01337                     *term = 0;
01338                     DONE(0)
01339                 }
01340                 else if ( kk > 0 ) {
01341                     m = t + t[1];
01342                     r = term + *term;
01343                     while ( m < r ) *t++ = *m++;
01344                     *term = WORDDIF(t,term);
01345                     goto ReStart;
01346                 }
01347             }
01348         }
01349         else if ( *t == DISTRIBUTION && t[FUNHEAD] == -SNUMBER
01350             && t[FUNHEAD+1] >= -2 && t[FUNHEAD+1] <= 2
01351             && t[FUNHEAD+2] == -SNUMBER
01352             && t[FUNHEAD+4] <= -FUNCTION
01353             && t[FUNHEAD+5] <= -FUNCTION ) {
01354             WORD *ttt = t+FUNHEAD+6, *tttstop = t+t[1];
01355             while ( ttt < tttstop ) {
01356                 if ( *ttt == -DOLLAREXPRESSION ) break;
01357                 NEXTARG(ttt);
01358             }
01359             if ( ttt >= tttstop ) {
01360                 AN.TeInFun = -1;
01361                 AN.TeSuOut = 0;
01362                 AR.TePos = -1;
01363                 DONE(1)
01364             }
01365         }
01366         else if ( *t == DELTA3 && ((t[1]-FUNHEAD) & 1) == 0 ) {
01367             AN.TeInFun = -2;
01368             AN.TeSuOut = 0;
01369             AR.TePos = -1;
01370             DONE(1)
01371         }
01372         else if ( ( *t == TABLEFUNCTION ) && ( t[FUNHEAD] <= -FUNCTION )
01373             && ( T = functions[-t[FUNHEAD]-FUNCTION].tabl ) != 0

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```

01374      && ( t[1] >= FUNHEAD+1+2*ABS(T->numind) )
01375      && ( t[FUNHEAD+1] == -SYMBOL ) ) {
01376  /*
01377      The case of table_(tab,sym1,...,symn)
01378  */
01379      for ( isp = 0; isp < ABS(T->numind); isp++ ) {
01380          if ( t[FUNHEAD+1+2*isp] != -SYMBOL ) break;
01381      }
01382      if ( isp >= ABS(T->numind) ) {
01383          AN.TeInFun = -3;
01384          AN.TeSuOut = 0;
01385          AR.TePos = -1;
01386          DONE(1)
01387      }
01388  }
01389  else if ( *t == TABLEFUNCTION && t[FUNHEAD] <= -FUNCTION
01390  && ( T = functions[-t[FUNHEAD]-FUNCTION].tabl ) != 0
01391  && ( t[1] == FUNHEAD+2 )
01392  && ( t[FUNHEAD+1] <= -FUNCTION ) ) {
01393  /*
01394      The case of table_(tab,fun)
01395  */
01396      AN.TeInFun = -3;
01397      AN.TeSuOut = 0;
01398      AR.TePos = -1;
01399      DONE(1)
01400  }
01401  else if ( *t == FACTORIN ) {
01402      if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -DOLLAREXPRESSION ) {
01403          AN.TeInFun = -4;
01404          AN.TeSuOut = 0;
01405          AR.TePos = -1;
01406          DONE(1)
01407      }
01408      else if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -EXPRESSION ) {
01409          AN.TeInFun = -5;
01410          AN.TeSuOut = 0;
01411          AR.TePos = -1;
01412          DONE(1)
01413      }
01414  }
01415  else if ( *t == TERMSINBRACKET ) {
01416      if ( t[1] == FUNHEAD || (
01417          t[1] == FUNHEAD+2
01418          && t[FUNHEAD] == -SNUMBER
01419          && t[FUNHEAD+1] == 0
01420      ) ) {
01421          AN.TeInFun = -6;
01422          AN.TeSuOut = 0;
01423          AR.TePos = -1;
01424          DONE(1)
01425      }
01426  /*
01427      The other cases have not yet been implemented
01428      We still have to add the case of short arguments
01429      First the different bracket in same expression
01430
01431      else if ( t[1] > FUNHEAD+ARGHEAD
01432          && t[FUNHEAD] == t[1]-FUNHEAD
01433          && t[FUNHEAD+ARGHEAD] == t[1]-FUNHEAD-ARGHEAD
01434          && t[t[1]-1] == 3
01435          && t[t[1]-2] == 1
01436          && t[t[1]-3] == 1 ) {
01437          AN.TeInFun = -6;
01438          AN.TeSuOut = 0;
01439          AR.TePos = -1;
01440          DONE(1)
01441      }
01442
01443      Next the bracket in an other expression
01444
01445      else if ( t[1] > FUNHEAD+ARGHEAD+2
01446          && t[FUNHEAD] == -EXPRESSION
01447          && t[FUNHEAD+2] == t[1]-FUNHEAD-2
01448          && t[FUNHEAD+ARGHEAD+2] == t[1]-FUNHEAD-ARGHEAD-2
01449          && t[t[1]-1] == 3
01450          && t[t[1]-2] == 1
01451          && t[t[1]-3] == 1 ) {
01452          AN.TeInFun = -6;
01453          AN.TeSuOut = 0;
01454          AR.TePos = -1;
01455          DONE(1)
01456      }
01457  */
01458  }
01459  else if ( *t == EXTRASYMFUN ) {
01460      if ( t[1] == FUNHEAD+2 && (

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01461         ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] <= cbuf[AM.sbufnum].numrhs
01462             && t[FUNHEAD+1] > 0 ) ||
01463         ( t[FUNHEAD] == -SYMBOL && t[FUNHEAD+1] < MAXVARIABLES
01464             && t[FUNHEAD+1] >= MAXVARIABLES-cbuf[AM.sbufnum].numrhs ) ) {
01465     AN.TeInFun = -7;
01466     AN.TeSuOut = 0;
01467     AR.TePos = -1;
01468     DONE(1)
01469 }
01470 else if ( t[1] == FUNHEAD ) {
01471     AN.TeInFun = -7;
01472     AN.TeSuOut = 0;
01473     AR.TePos = -1;
01474     DONE(1)
01475 }
01476 }
01477 else if ( *t == DIVFUNCTION || *t == REMFUNCTION
01478     || *t == INVERSEFUNCTION || *t == MULFUNCTION
01479     || *t == GCDFUNCTION ) {
01480     WORD *tf;
01481     int todo = 1, numargs = 0;
01482     tf = t + FUNHEAD;
01483     while ( tf < t + t[1] ) {
01484         DOLLARS d;
01485         if ( *tf == -DOLLAREXPRESSION ) {
01486             d = Dollars + tf[1];
01487             if ( d->type == DOLWILDARGS ) {
01488                 WORD *tterm = AT.WorkPointer, *tw;
01489                 WORD *ta = term, *tb = tterm, *tc, *td = term + *term;
01490                 while ( ta < t ) *tb++ = *ta++;
01491                 tc = tb;
01492                 while ( ta < tf ) *tb++ = *ta++;
01493                 tw = d->where+1;
01494                 while ( *tw ) {
01495                     if ( *tw < 0 ) {
01496                         if ( *tw > -FUNCTION ) *tb++ = *tw++;
01497                         *tb++ = *tw++;
01498                     }
01499                     else {
01500                         int ia;
01501                         for ( ia = 0; ia < *tw; ia++ ) *tb++ = *tw++;
01502                     }
01503                 }
01504                 NEXTARG(ta)
01505                 while ( ta < t+t[1] ) *tb++ = *ta++;
01506                 tc[1] = tb-tc;
01507                 while ( ta < td ) *tb++ = *ta++;
01508                 *tterm = tb - tterm;
01509                 {
01510                     int ia, na = *tterm;
01511                     ta = tterm; tb = term;
01512                     for ( ia = 0; ia < na; ia++ ) *tb++ = *ta++;
01513                 }
01514                 if ( tb > AT.WorkTop ) {
01515                     MLOCK(ErrorMessageLock);
01516                     MesWork();
01517                     goto EndTest2;
01518                 }
01519                 AT.WorkPointer = tb;
01520                 goto Restart;
01521             }
01522         }
01523         NEXTARG(tf);
01524     }
01525     tf = t + FUNHEAD;
01526     while ( tf < t + t[1] ) {
01527         numargs++;
01528         if ( *tf > 0 && tf[1] != 0 ) todo = 0;
01529         NEXTARG(tf);
01530     }
01531     if ( todo && numargs == 2 ) {
01532         if ( *t == DIVFUNCTION ) AN.TeInFun = -9;
01533         else if ( *t == REMFUNCTION ) AN.TeInFun = -10;
01534         else if ( *t == INVERSEFUNCTION ) AN.TeInFun = -11;
01535         else if ( *t == MULFUNCTION ) AN.TeInFun = -14;
01536         else if ( *t == GCDFUNCTION ) AN.TeInFun = -8;
01537         AN.TeSuOut = 0;
01538         AR.TePos = -1;
01539         DONE(1)
01540     }
01541     else if ( todo && numargs == 3 ) {
01542         if ( *t == DIVFUNCTION ) AN.TeInFun = -9;
01543         else if ( *t == REMFUNCTION ) AN.TeInFun = -10;
01544         else if ( *t == GCDFUNCTION ) AN.TeInFun = -8;
01545         AN.TeSuOut = 0;
01546         AR.TePos = -1;
01547         DONE(1)

```

```

01548         }
01549         else if ( todo && *t == GCDFUNCTION ) {
01550             AN.TeInFun = -8;
01551             AN.TeSuOut = 0;
01552             AR.TePos = -1;
01553             DONE(1)
01554         }
01555     }
01556     else if ( *t == PERMUTATIONS && ( ( t[1] >= FUNHEAD+1
01557         && t[FUNHEAD] <= -FUNCTION ) || ( t[1] >= FUNHEAD+3
01558         && t[FUNHEAD] == -SNUMBER && t[FUNHEAD+2] <= -FUNCTION ) ) ) {
01559         AN.TeInFun = -12;
01560         AN.TeSuOut = 0;
01561         AR.TePos = -1;
01562         DONE(1)
01563     }
01564     else if ( *t == PARTITIONS ) {
01565         if ( TestPartitions(t,&(AT.partitions)) ) {
01566             AT.partitions.where = t-term;
01567             AN.TeInFun = -13;
01568             AN.TeSuOut = 0;
01569             AR.TePos = -1;
01570             DONE(1)
01571         }
01572     }
01573 }
01574 }
01575 t += t[1];
01576 } while ( t < m );
01577 if ( funflag ) { /* Search in functions */
01578 DoSpec:
01579     t = term;
01580     AT.NestPoin->termsize = t;
01581     if ( AT.NestPoin == AT.Nest ) AN.EndNest = t + *t;
01582     t++;
01583     oldncmod = AN.ncmod;
01584     if ( t < m ) do {
01585         if ( *t < FUNCTION ) {
01586             t += t[1]; continue;
01587         }
01588         if ( AN.ncmod && ( ( AC.modmode & ALSOFUNARGS ) == 0 ) ) {
01589             if ( *t != AR.PolyFun ) AN.ncmod = 0;
01590             else AN.ncmod = oldncmod;
01591         }
01592         r = t + t[1];
01593         funnum = *t;
01594         if ( *t >= FUNCTION + WILDOFFSET ) funnum -= WILDOFFSET;
01595         if ( ( *t == NUMFACTORS || *t == FIRSTTERM || *t == CONTENTTERM )
01596             && t[1] == FUNHEAD+2 &&
01597             ( t[FUNHEAD] == -EXPRESSION || t[FUNHEAD] == -DOLLAREXPRESSION ) ) {
01598             /*
01599                 if ( *t == NUMFACTORS ) {
01600                     This we leave for Normalize
01601                 }
01602             */
01603         }
01604         else if ( functions[funnum-FUNCTION].spec <= 0 ) {
01605             AT.NestPoin->funsizesize = t + 1;
01606             t1 = t;
01607             t += FUNHEAD;
01608             while ( t < r ) { /* Sum over arguments */
01609                 if ( *t > 0 && t[1] ) { /* Argument is dirty */
01610                     AT.NestPoin->argsize = t;
01611                     AT.NestPoin++;
01612                     stop = t + *t; /*
01613                         t2 = t;
01614                         t += ARGHEAD;
01615                         while ( t < AT.NestPoin[-1].argsize+(AT.NestPoin[-1].argsize) ) {
01616                             /* Sum over terms */
01617                             AT.RecFlag++;
01618                             i = *t;
01619                             AN.subsubveto = 1;
01620                             /*
01621                                 AN.subsubveto repairs a bug that became apparent
01622                                 in an example by York Schroeder:
01623                                 f(k1.k1)*replace_(k1,2*k2)
01624                                 Is it possible to repair the counting of the various
01625                                 length indicators? (JV 1-jun-2010)
01626                             */
01627                                 if ( ( retvalue = TestSub(BHEAD t,level) ) != 0 ) {
01628                                     /*
01629                                         Possible size changes:
01630                                         Note defs at 471,467,460,400,425,328
01631                                     */
01632                                 }
01633                                 redosize:
01634                                 if ( i > *t ) {

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```

01635         i -= *t;
01636         *t2 -= i;
01637         t1[1] -= i;
01638         t += *t;
01639         r = t + i;
01640         m = term + *term;
01641         while ( r < m ) *t++ = *r++;
01642         *term -= i;
01643     */
01644
01645         i -= *t;
01646         t += *t;
01647         r = t + i;
01648         m = AN.EndNest;
01649         while ( r < m ) *t++ = *r++;
01650         t = AT.NestPoin[-1].argsize + ARGHEAD;
01651         n = AT.Nest;
01652         while ( n < AT.NestPoin ) {
01653             *(n->argsize) -= i;
01654             *(n->funsize) -= i;
01655             *(n->termsize) -= i;
01656             n++;
01657         }
01658         AN.EndNest -= i;
01659     }
01660     AN.subsubveto = 0;
01661     t1[2] = 1;
01662     if ( *t1 == AR.PolyFun && AR.PolyFunType == 2 )
01663         t1[2] |= MUSTCLEANPRF;
01664     AT.RecFlag--;
01665     AT.NestPoin--;
01666     AN.TeInFun++;
01667     AR.TePos = 0;
01668     AN.ncmod = oldncmod;
01669     DONE(retvalue)
01670 }
01671 else {
01672     /*
01673     * Somehow the next line fixes Issue #106.
01674     */
01675     i = *t;
01676     Normalize(BHEAD t);
01677     if ( i > *t ) { retvalue = 1; goto redosize; } */
01678     /*
01679     * Experimentally, the next line fixes Issue #105.
01680     */
01681     if ( *t == 0 ) { retvalue = 1; goto redosize; }
01682     {
01683         WORD *tend = t + *t, *tt = t+1;
01684         stilldirty = 0;
01685         tend -= ABS(tend[-1]);
01686         while ( tt < tend ) {
01687             if ( *tt == SUBEXPRESSION || *tt == EXPRESSION ) {
01688                 stilldirty = 1; break;
01689             }
01690             tt += tt[1];
01691         }
01692     }
01693     if ( i > *t ) {
01694         We should not forget to correct the Nest
01695         stack. That caused trouble in the past.
01696     */
01697         retvalue = 1;
01698         i -= *t;
01699         t += *t;
01700         r = t + i;
01701         m = AN.EndNest;
01702         while ( r < m ) *t++ = *r++;
01703         t = AT.NestPoin[-1].argsize + ARGHEAD;
01704         n = AT.Nest;
01705         while ( n < AT.NestPoin ) {
01706             *(n->argsize) -= i;
01707             *(n->funsize) -= i;
01708             *(n->termsize) -= i;
01709             n++;
01710         }
01711         AN.EndNest -= i;
01712     }
01713 }
01714 AN.subsubveto = 0;
01715 AT.RecFlag--;
01716 t += *t;
01717 }
01718 AT.NestPoin--;
01719 /*
01720 Argument contains no subexpressions.
01721 It should be normalized and sorted.

```



```

01722                                     The main problem is the storage.
01723 */
01724 t = AT.NestPoin->argsize;
01725 j = *t;
01726 t += ARGHEAD;
01727 NewSort(BHEAD0);
01728 if ( *t1 == AR.PolyFun && AR.PolyFunType == 2 ) {
01729     AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);
01730     AR.SortType = SORTHIGHFIRST;
01731 }
01732 if ( AT.WorkPointer < term + *term )
01733     AT.WorkPointer = term + *term;
01734
01735 while ( t < AT.NestPoin->argsize+(AT.NestPoin->argsize) ) {
01736     m = AT.WorkPointer;
01737     r = t + *t;
01738     do { *m++ = *t++; } while ( t < r );
01739     r = AT.WorkPointer;
01740     AT.WorkPointer = r + *r;
01741     if ( Normalize(BHEAD r) ) {
01742         if ( *t1 == AR.PolyFun && AR.PolyFunType == 2 ) {
01743             AR.SortType = oldsorttype;
01744             AR.CompareRoutine = (COMPAREDUMMY)oldcompareroutine;
01745             t1[2] |= MUSTCLEANPRF;
01746         }
01747         LowerSortLevel(); goto EndTest;
01748     }
01749     if ( AN.ncmod != 0 ) {
01750         if ( *r ) {
01751             if ( Modulus(r) ) {
01752                 LowerSortLevel();
01753                 AT.WorkPointer = r;
01754                 if ( *t1 == AR.PolyFun && AR.PolyFunType == 2 ) {
01755                     AR.SortType = oldsorttype;
01756                     AR.CompareRoutine = (COMPAREDUMMY)oldcompareroutine;
01757                     t1[2] |= MUSTCLEANPRF;
01758                 }
01759                 goto EndTest;
01760             }
01761         }
01762     }
01763     if ( AR.PolyFun > 0 ) {
01764         if ( PrepPoly(BHEAD r,1) != 0 ) goto EndTest;
01765     }
01766     if ( *r ) StoreTerm(BHEAD r);
01767     AT.WorkPointer = r;
01768 }
01769 if ( EndSort(BHEAD AT.WorkPointer+ARGHEAD,1) < 0 ) goto EndTest;
01770 m = AT.WorkPointer+ARGHEAD;
01771 if ( *t1 == AR.PolyFun && AR.PolyFunType == 2 ) {
01772     AR.SortType = oldsorttype;
01773     AR.CompareRoutine = (COMPAREDUMMY)oldcompareroutine;
01774     t1[2] |= MUSTCLEANPRF;
01775 }
01776 while ( *m ) m += *m;
01777 i = WORDDIF(m,AT.WorkPointer);
01778 *AT.WorkPointer = i;
01779 AT.WorkPointer[1] = stilldirty;
01780 if ( ToFast(AT.WorkPointer,AT.WorkPointer) ) {
01781     m = AT.WorkPointer;
01782     if ( *m <= -FUNCTION ) { m++; i = 1; }
01783     else { m += 2; i = 2; }
01784 }
01785 j = i - j;
01786 if ( j > 0 ) {
01787     r = m + j;
01788     if ( r > AT.WorkTop ) {
01789         MLOCK(ErrorMessageLock);
01790         MesWork();
01791         goto EndTest2;
01792     }
01793     do { *--r = *--m; } while ( m > AT.WorkPointer );
01794     AT.WorkPointer = r;
01795     m = AN.EndNest;
01796     r = m + j;
01797     stop = AT.NestPoin->argsize+(AT.NestPoin->argsize);
01798     do { *--r = *--m; } while ( m >= stop );
01799 }
01800 else if ( j < 0 ) {
01801     m = AT.NestPoin->argsize+(AT.NestPoin->argsize);
01802     r = m + j;
01803     do { *r++ = *m++; } while ( m < AN.EndNest );
01804 }
01805 m = AT.NestPoin->argsize;
01806 r = AT.WorkPointer;
01807 while ( --i >= 0 ) *m++ = *r++;
01808 n = AT.Nest;

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```

01809         while ( n <= AT.NestPoin ) {
01810             if ( *(n->argsize) > 0 && n != AT.NestPoin )
01811                 *(n->argsize) += j;
01812             *(n->funsize) += j;
01813             *(n->termsize) += j;
01814             n++;
01815         }
01816         AN.EndNest += j;
01817         (AT.NestPoin->argsize)[1] = 0; /*
01818         if ( funnum == DENOMINATOR || funnum == EXPONENT ) {
01819             if ( Normalize(BHEAD term) ) {
01820                 /*
01821                     In this case something has been substituted
01822                     Either a $ or a replace_????
01823                     Originally we had here:
01824
01825                     goto EndTest;
01826
01827                     It seems better to restart.
01828                 */
01829                 AN.ncmod = oldncmod;
01830                 goto ReStart;
01831             }
01832             /*
01833                 And size changes here????
01834             */
01835         }
01836         AN.ncmod = oldncmod;
01837         goto ReStart;
01838     }
01839     else if ( *t == -DOLLAREXPRESSION ) {
01840         if ( ( *t1 == TERMSINEXPR || *t1 == SIZEOFFUNCTION )
01841             && t1[1] == FUNHEAD+2 ) {}
01842         else {
01843             if ( AR.Eside != LHSIDE ) {
01844                 AN.TeInFun = 1; AR.TePos = 0;
01845                 AT.TMbuff = AM.dbufnum; t1[2] |= DIRTYFLAG;
01846                 AN.ncmod = oldncmod;
01847                 DONE(1)
01848             }
01849             AC.lhdollarflag = 1;
01850         }
01851     }
01852     else if ( *t == -TERMSINBRACKET ) {
01853         if ( AR.Eside != LHSIDE ) {
01854             AN.TeInFun = 1; AR.TePos = 0;
01855             t1[2] |= DIRTYFLAG;
01856             AN.ncmod = oldncmod;
01857             DONE(1)
01858         }
01859     }
01860     else if ( AN.ncmod != 0 && *t == -SNUMBER ) {
01861         if ( AN.ncmod == 1 || AN.ncmod == -1 ) {
01862             isp = (UWORD)(AC.cmod[0]);
01863             isp = t[1] % isp;
01864             if ( ( AC.modmode & POSNEG ) != 0 ) {
01865                 if ( isp > (UWORD)(AC.cmod[0])/2 ) isp = isp - (UWORD)(AC.cmod[0]);
01866                 else if ( -isp > (UWORD)(AC.cmod[0])/2 ) isp = isp +
(UWORD)(AC.cmod[0]);
01867             }
01868             else {
01869                 if ( isp < 0 ) isp += (UWORD)(AC.cmod[0]);
01870             }
01871             if ( isp <= MAXPOSITIVE && isp >= -MAXPOSITIVE ) {
01872                 t[1] = isp;
01873             }
01874         }
01875     }
01876     NEXTARG(t)
01877 }
01878 if ( funnum >= FUNCTION && functions[funnum-FUNCTION].tabl ) {
01879     /*
01880         Test whether the table catches
01881         Test 1: index arguments and range. i will be the number
01882         of the element in the table.
01883     */
01884     WORD rhsnumber, *oldwork = AT.WorkPointer;
01885     WORD ii, *p, *pp, *ppstop;
01886     MINMAX *mm;
01887     T = functions[funnum-FUNCTION].tabl;
01888     /*
01889         Because of tables with a variable number of indices
01890         we need to make a copy of the pattern.
01891         If we do this in the Workspace we get problems with EndNest.
01892         This is why we use TermMalloc.
01893         Now Tpattern is a copy that can be modified.
01894     */

```

```

01895 #ifdef WITHPTHREADS
01896     pp = T->pattern[AT.identity];
01897 #else
01898     pp = T->pattern;
01899 #endif
01900     if ( Tpattern == 0 ) Tpattern = TermMalloc("Tpattern");
01901     p = Tpattern;
01902     ii = pp[1];
01903     for ( i = 0; i < ii; i++ ) *p++ = *pp++;
01904
01905     p = Tpattern + FUNHEAD+1;
01906     mm = T->mm;
01907     if ( T->sparse ) {
01908         WORD xx;
01909         t = t1+FUNHEAD;
01910         if ( T->numind == 0 ) { isp = 0; xx = 0; }
01911         else {
01912             if ( T->numind < 0 ) {
01913                 if ( *t != -SNUMBER && t[2] != -SNUMBER ) {
01914                     xx = ABS(T->numind);
01915                     goto teststrict;
01916                 }
01917                 xx = t[1]+1;
01918                 if ( xx < 2 || xx > -T->numind ) goto teststrict;
01919             }
01920             else xx = T->numind;
01921             for ( i = 0; i < xx; i++, t += 2 ) {
01922                 if ( *t != -SNUMBER ) break;
01923             }
01924             if ( i < xx ) goto teststrict;
01925             isp = FindTableTree(T,t1+FUNHEAD,2);
01926         }
01927         if ( isp < 0 ) {
01928 teststrict:
01929             if ( T->strict == -2 ) {
01930                 rhsnumber = AM.zerorhs;
01931                 tbufnum = AM.zbufnum;
01932             }
01933             else if ( T->strict == -3 ) {
01934                 rhsnumber = AM.onerhs;
01935                 tbufnum = AM.zbufnum;
01936             }
01937             else if ( T->strict < 0 ) goto NextFun;
01938             else {
01939                 MLOCK(ErrorMessageLock);
01940                 MesPrint("Element in table is undefined");
01941                 if ( Tpattern ) {
01942                     TermFree(Tpattern,"Tpattern");
01943                     Tpattern = 0;
01944                 }
01945                 goto showtable;
01946             }
01947             Copy the indices;
01948             t = t1+FUNHEAD+1;
01949             for ( i = 0; i < xx; i++ ) {
01950                 *p = *t; p+=2; t+=2;
01951             }
01952         }
01953         else {
01954             rhsnumber = T->tablepointers[isp+ABS(T->numind)];
01955             tbufnum = T->bbufnum;
01956             if ( TABLEEXTENSION == 2 )
01957                 tbufnum = T->bbufnum;
01958             else
01959                 tbufnum = T->tablepointers[isp+ABS(T->numind)+1];
01960         }
01961         t = t1+FUNHEAD+1;
01962         ii = xx;
01963         while ( --ii >= 0 ) {
01964             *p = *t; t += 2; p += 2;
01965         }
01966     }
01967     if ( xx < ABS(T->numind) ) {
01968         p--; ppstop = Tpattern+Tpattern[1];
01969         pp = p+2*(-T->numind-xx);
01970         while ( pp < ppstop ) *p++ = *pp++;
01971         Tpattern[1] = p - Tpattern;
01972     }
01973     goto caughttable;
01974 }
01975 else {
01976     i = 0;
01977     t = t1 + FUNHEAD;
01978     j = T->numind;
01979     while ( --j >= 0 ) {
01980         if ( *t != -SNUMBER ) goto NextFun;
01981         t++;

```

```

01982         if ( *t < mm->mini || *t > mm->maxi ) {
01983             if ( T->bounds ) {
01984                 MLOCK(ErrorMessageLock);
01985                 MesPrint("Table boundary check. Argument %d",
01986                     T->numind-j);
01987 showtable:
01988                 AO.OutFill = AO.OutputLine = (UBYTE *)m;
01989                 AO.OutSkip = 8;
01990                 IniLine(0);
01991                 WriteSubTerm(tl,1);
01992                 FiniLine();
01993                 MUNLOCK(ErrorMessageLock);
01994                 if ( Tpattern ) {
01995                     TermFree(Tpattern,"Tpattern");
01996                     Tpattern = 0;
01997                 }
01998                 SETERROR(-1)
01999             }
02000             if ( Tpattern ) {
02001                 TermFree(Tpattern,"Tpattern");
02002                 Tpattern = 0;
02003             }
02004             goto NextFun;
02005         }
02006         i += ( *t - mm->mini ) * (LONG)(mm->size);
02007         *p = *t++;
02008         p += 2;
02009         mm++;
02010     }
02011
02012     /*
02013     Test now whether the entry exists.
02014     */
02015     i *= TABLEEXTENSION;
02016     if ( T->tablepointers[i] == -1 ) {
02017         if ( T->strict == -2 ) {
02018             rhsnumber = AM.zerorhs;
02019             tbufnum = AM.zbufnum;
02020         }
02021         else if ( T->strict == -3 ) {
02022             rhsnumber = AM.onerhs;
02023             tbufnum = AM.zbufnum;
02024         }
02025         else if ( T->strict < 0 ) {
02026             if ( Tpattern ) {
02027                 TermFree(Tpattern,"Tpattern");
02028                 Tpattern = 0;
02029             }
02030             goto NextFun;
02031         }
02032         else {
02033             MLOCK(ErrorMessageLock);
02034             MesPrint("Element in table is undefined");
02035             if ( Tpattern ) {
02036                 TermFree(Tpattern,"Tpattern");
02037                 Tpattern = 0;
02038             }
02039             goto showtable;
02040         }
02041     }
02042     else {
02043         rhsnumber = T->tablepointers[i];
02044         tbufnum = T->tbufnum;
02045         tbufnum = T->tablepointers[i+1];
02046     }
02047 }
02048
02049 /*
02050 If there are more arguments we have to do some
02051 pattern matching. This should be easy. We adapted the
02052 pattern, so that the array indices match already.
02053 Note that if there is no match the program will become
02054 very slow.
02055 */
02056 caughttable:
02057 #ifdef WITHPTHREADS
02058     AN.FullProto = T->prototype[AT.identity];
02059 #else
02060     AN.FullProto = T->prototype;
02061 #endif
02062     AN.WildValue = AN.FullProto + SUBEXPSIZE;
02063     AN.WildStop = AN.FullProto+AN.FullProto[1];
02064     ClearWild(BHEAD0);
02065     AN.RepFunNum = 0;
02066     AN.RepFunList = AN.EndNest;
02067     AT.WorkPointer = (WORD *)(((UBYTE *) (AN.EndNest)) + AM.MaxTer/2);
02068     if ( AT.WorkPointer >= AT.WorkTop ) {

```

```

02069         MLOCK(ErrorMessageLock);
02070         MesWork();
02071         MUNLOCK(ErrorMessageLock);
02072     }
02073     wilds = 0;
02074     if ( MatchFunction(BHEAD Tpattern,t1,&wilds) > 0 ) {
02075         AT.WorkPointer = oldwork;
02076         if ( AT.NestPoin != AT.Nest ) {
02077             AN.ncmod = oldncmod;
02078             if ( Tpattern ) {
02079                 TermFree(Tpattern,"Tpattern");
02080                 Tpattern = 0;
02081             }
02082             DONE(1)
02083         }
02084
02085         m = AN.FullProto;
02086         retvalue = m[2] = rhsnumber;
02087         m[4] = tbufnum;
02088         t = t1;
02089         j = t[1];
02090         i = m[1];
02091         if ( j > i ) {
02092             j = i - j;
02093             NCOPY(t,m,i);
02094             m = term + *term;
02095             while ( r < m ) *t++ = *r++;
02096             *term += j;
02097         }
02098         else if ( j < i ) {
02099             j = i-j;
02100             t = term + *term;
02101             while ( t >= r ) { t[j] = *t; t--; }
02102             t = t1;
02103             NCOPY(t,m,i);
02104             *term += j;
02105         }
02106         else {
02107             NCOPY(t,m,j);
02108         }
02109         AN.TeInFun = 0;
02110         AR.TePos = 0;
02111         AN.TeSuOut = -1;
02112         if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
02113         AT.TMbuff = tbufnum;
02114         AN.ncmod = oldncmod;
02115         DONE(retvalue);
02116     }
02117     AT.WorkPointer = oldwork;
02118     if ( Tpattern ) {
02119         TermFree(Tpattern,"Tpattern");
02120         Tpattern = 0;
02121     }
02122 }
02123 NextFun::
02124     }
02125     else if ( ( t[2] & DIRTYFLAG ) != 0 ) {
02126         t += FUNHEAD;
02127         while ( t < r ) {
02128             if ( *t == FUNNYDOLLAR ) {
02129                 if ( AR.Eside != LHSIDE ) {
02130                     AN.TeInFun = 1;
02131                     AR.TePos = 0;
02132                     AT.TMbuff = AM.dbufnum;
02133                     AN.ncmod = oldncmod;
02134                     DONE(1)
02135                 }
02136                 AC.lhdollarflag = 1;
02137             }
02138             t++;
02139         }
02140     }
02141     t = r;
02142     AN.ncmod = oldncmod;
02143     } while ( t < m );
02144 }
02145 Done:
02146     if ( Tpattern ) {
02147         TermFree(Tpattern,"Tpattern");
02148         Tpattern = 0;
02149     }
02150     return(retvalue);
02151 EndTest::
02152     MLOCK(ErrorMessageLock);
02153 EndTest2:
02154     MesCall("TestSub");
02155     MUNLOCK(ErrorMessageLock);

```

```

02156     SETERROR(-1)
02157 }
02158
02159 /*
02160     #] TestSub :
02161     #[ InFunction :          WORD InFunction(term,termout)
02162 */
02175 int InFunction(PHEAD WORD *term, WORD *termout)
02176 {
02177     GETBIDENTITY
02178     WORD *m, *t, *r, *rr, sign = 1, oldncmod;
02179     WORD *u, *v, *w, *from, *to,
02180     ipp, olddefer = AR.DeferFlag, oldPolyFun = AR.PolyFun, i, j;
02181     LONG numterms;
02182     from = t = term;
02183     r = t + *t - 1;
02184     m = r - ABS(*r) + 1;
02185     t++;
02186     while ( t < m ) {
02187         if ( *t >= FUNCTION+WILDOFFSET ) ipp = *t - WILDOFFSET;
02188         else ipp = *t;
02189         if ( AR.TePos ) {
02190             if ( ( term + AR.TePos ) == t ) {
02191                 m = termout;
02192                 while ( from < t ) *m++ = *from++;
02193                 *m++ = DENOMINATOR;
02194                 *m++ = t[1] + 4 + FUNHEAD + ARGHEAD;
02195                 *m++ = DIRTYFLAG;
02196                 FILLFUN3(m)
02197                 *m++ = t[1] + 4 + ARGHEAD;
02198                 *m++ = 1;
02199                 FILLARG(m)
02200                 *m++ = t[1] + 4;
02201                 t[3] = -t[3];
02202                 v = t + t[1];
02203                 while ( t < v ) *m++ = *t++;
02204                 from[3] = -from[3];
02205                 *m++ = 1;
02206                 *m++ = 1;
02207                 *m++ = 3;
02208                 r = term + *term;
02209                 while ( t < r ) *m++ = *t++;
02210                 if ( (m-termout) > (LONG)(AM.MaxTer/sizeof(WORD)) ) {
02211                     MLOCK(ErrorMessageLock);
02212                     MesPrint("Output term too large (%d words) (MaxTermSize: %d words)", m-termout,
02213                     AM.MaxTer/sizeof(WORD));
02214                     MUNLOCK(ErrorMessageLock);
02215                     goto TooLarge;
02216                 }
02217                 *termout = WORDDIF(m,termout);
02218                 return(0);
02219             }
02220             else if ( ( *t >= FUNCTION && functions[ipp-FUNCTION].spec <= 0 )
02221             && ( t[2] & DIRTYFLAG ) == DIRTYFLAG ) {
02222                 m = termout;
02223                 r = t + t[1];
02224                 u = t;
02225                 t += FUNHEAD;
02226                 oldncmod = AN.ncmod;
02227                 while ( t < r ) { /* t points at an argument */
02228                     if ( *t > 0 && t[1] ) { /* Argument has been modified */
02229                         WORD oldsorttype = AR.SortType;
02230                         /* This whole argument must be redone */
02231
02232                         if ( ( AN.ncmod != 0 )
02233                             && ( ( AC.modmode & ALSOFUNARGS ) == 0 )
02234                             && ( *u != AR.PolyFun ) ) { AN.ncmod = 0; }
02235                         AR.DeferFlag = 0;
02236                         v = t + *t;
02237                         t += ARGHEAD; /* First term */
02238                         w = 0; /* to appease the compilers warning devices */
02239                         while ( from < t ) {
02240                             if ( from == u ) w = m;
02241                             *m++ = *from++;
02242                         }
02243                         to = m;
02244                         NewSort(BHEAD0);
02245                         if ( *u == AR.PolyFun && AR.PolyFunType == 2 ) {
02246                             AR.CompareRoutine = (COMPAREDDUMMY)(&CompareSymbols);
02247                             AR.SortType = SORTHIGHFIRST;
02248                         }
02249                     }
02250                     AR.PolyFun = 0;
02251                 }
02252                 while ( t < v ) {
02253                     i = *t;

```

```

02254         NCOPY(m,t,i);
02255         m = to;
02256         if ( AT.WorkPointer < m+m ) AT.WorkPointer = m + *m;
02257         if ( Generator(BHEAD m,AR.Cnumlhs) ) {
02258             AN.ncmod = oldncmod;
02259             LowerSortLevel(); goto InFunc;
02260         }
02261     }
02262     /* w = the function */
02263     /* v = the next argument */
02264     /* u = the function */
02265     /* to is new argument */
02266
02267     to -= ARGHEAD;
02268     if ( EndSort(BHEAD m,1) < 0 ) {
02269         AN.ncmod = oldncmod;
02270         goto InFunc;
02271     }
02272     AR.PolyFun = oldPolyFun;
02273     if ( *u == AR.PolyFun && AR.PolyFunType == 2 ) {
02274         AR.CompareRoutine = (COMPAREDDUMMY)(&Compare1);
02275         AR.SortType = oldsortttype;
02276     }
02277     while ( *m ) m += *m;
02278     *to = WORDDIF(m,to);
02279     to[1] = 1; /* ??????? or rather 0?. 24-mar-2006 JV */
02280     if ( ToFast(to,to) ) {
02281         if ( *to <= -FUNCTION ) m = to+1;
02282         else m = to+2;
02283     }
02284     w[1] = WORDDIF(m,w) + WORDDIF(r,v);
02285     r = term + *term;
02286     t = v;
02287     while ( t < r ) *m++ = *t++;
02288     if ( (m-termout) > (LONG)(AM.MaxTer/sizeof(WORD)) ) {
02289         MLOCK(ErrorMessageLock);
02290         MesPrint("Output term too large (%d words) (MaxTermSize: %d words)",
m-termout, AM.MaxTer/sizeof(WORD));
02291         MUNLOCK(ErrorMessageLock);
02292         goto TooLarge;
02293     }
02294     *termout = WORDDIF(m,termout);
02295     AR.DeferFlag = olddefer;
02296     AN.ncmod = oldncmod;
02297     return(0);
02298 }
02299 else if ( *t == -DOLLAREXPRESSION ) {
02300     if ( AR.Eside == LHSIDE ) {
02301         NEXTARG(t)
02302         AC.lhdollarflag = 1;
02303     }
02304     else {
02305         /*
02306             This whole argument must be redone
02307         */
02308         DOLLARS d = Dollars + t[1];
02309 #ifdef WITHPTHREADS
02310         int nummodopt, dtype = -1;
02311         if ( AS.MultiThreaded ) {
02312             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
02313                 if ( t[1] == ModOptdollars[nummodopt].number ) break;
02314             }
02315             if ( nummodopt < NumModOptdollars ) {
02316                 dtype = ModOptdollars[nummodopt].type;
02317                 if ( dtype == MODLOCAL ) {
02318                     d = ModOptdollars[nummodopt].dstruct+AT.identity;
02319                 }
02320                 else {
02321                     LOCK(d->pthreadlockread);
02322                 }
02323             }
02324         }
02325 #endif
02326         oldncmod = AN.ncmod;
02327         if ( ( AN.ncmod != 0 )
02328             && ( ( AC.modmode & ALSOFUNARGS ) == 0 )
02329             && ( *u != AR.PolyFun ) ) { AN.ncmod = 0; }
02330         AR.DeferFlag = 0;
02331         v = t + 2;
02332         w = 0; /* to appease the compilers warning devices */
02333         while ( from < t ) {
02334             if ( from == u ) w = m;
02335             *m++ = *from++;
02336         }
02337         to = m;
02338         switch ( d->type ) {
02339             case DOLINDEX:

```

```

02340         if ( d->index >= 0 && d->index < AM.OffsetIndex ) {
02341             *m++ = -SNUMBER; *m++ = d->index;
02342         }
02343         else { *m++ = -INDEX; *m++ = d->index; }
02344         break;
02345     case DOLZERO:
02346         *m++ = -SNUMBER; *m++ = 0; break;
02347     case DOLNUMBER:
02348         if ( d->where[0] == 4 &&
02349             ( d->where[1] & MAXPOSITIVE ) == d->where[1] ) {
02350             *m++ = -SNUMBER;
02351             if ( d->where[3] >= 0 ) *m++ = d->where[1];
02352             else *m++ = -d->where[1];
02353             break;
02354         }
02355         /* fall through */
02356     case DOLTERMS:
02357         /*
02358             Here we have the special case of the PolyRatFun
02359             That function may have a different sort of the
02360             terms in the argument.
02361         */
02362         to = m; r = d->where;
02363         *m++ = 0; *m++ = 1;
02364         FILLARG(m)
02365         while ( *r ) {
02366             i = *r; NCOPY(m,r,i)
02367         }
02368         *to = m-to;
02369         if ( ToFast(to,to) ) {
02370             if ( *to <= -FUNCTION ) m = to+1;
02371             else m = to+2;
02372         }
02373         else if ( *u == AR.PolyFun && AR.PolyFunType == 2 ) {
02374             AR.PolyFun = 0;
02375             NewSort(BHEAD0);
02376             AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);
02377             r = to + ARGHEAD;
02378             while ( r < m ) {
02379                 rr = r; r += *r;
02380                 if ( SymbolNormalize(rr) ) goto InFunc;
02381                 if ( StoreTerm(BHEAD rr) ) {
02382                     AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
02383                     LowerSortLevel();
02384                     Terminate(-1);
02385                 }
02386             }
02387             if ( EndSort(BHEAD to+ARGHEAD,1) < 0 ) goto InFunc;
02388             AR.PolyFun = oldPolyFun;
02389             AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
02390             m = to+ARGHEAD;
02391             if ( *m == 0 ) {
02392                 *to = -SNUMBER;
02393                 to[1] = 0;
02394                 m = to + 2;
02395             }
02396             else {
02397                 while ( *m ) m += *m;
02398                 *t = m - to;
02399                 if ( ToFast(to,to) ) {
02400                     if ( *to <= -FUNCTION ) m = to+1;
02401                     else m = to+2;
02402                 }
02403             }
02404         }
02405         w[1] = w[1] - 2 + (m-to);
02406         break;
02407     case DOLSUBTERM:
02408         to = m; r = d->where;
02409         i = r[1];
02410         *m++ = i+4+ARGHEAD; *m++ = 1;
02411         FILLARG(m)
02412         *m++ = i+4;
02413         while ( --i >= 0 ) *m++ = *r++;
02414         *m++ = 1; *m++ = 1; *m++ = 3;
02415         if ( ToFast(to,to) ) {
02416             if ( *to <= -FUNCTION ) m = to+1;
02417             else m = to+2;
02418         }
02419         w[1] = w[1] - 2 + (m-to);
02420         break;
02421     case DOLARGUMENT:
02422         to = m; r = d->where;
02423         if ( *r > 0 ) {
02424             i = *r - 2;
02425             *m++ = *r++; *m++ = 1; r++;
02426             while ( --i >= 0 ) *m++ = *r++;

```



```

02427     }
02428     else if ( *r <= -FUNCTION ) *m++ = *r++;
02429     else { *m++ = *r++; *m++ = *r++; }
02430     w[1] = w[1] - 2 + (m-to);
02431     break;
02432 case DOLWILDARGS:
02433     to = m; r = d->where;
02434     if ( *r > 0 ) { /* Tensor arguments */
02435         i = *r++;
02436         while ( --i >= 0 ) {
02437             if ( *r < 0 ) {
02438                 *m++ = -VECTOR; *m++ = *r++;
02439             }
02440             else if ( *r >= AM.OffsetIndex ) {
02441                 *m++ = -INDEX; *m++ = *r++;
02442             }
02443             else { *m++ = -SNUMBER; *m++ = *r++; }
02444         }
02445     }
02446     else { /* Regular arguments */
02447         r++;
02448         while ( *r ) {
02449             if ( *r > 0 ) {
02450                 i = *r - 2;
02451                 *m++ = *r++; *m++ = 1; r++;
02452                 while ( --i >= 0 ) *m++ = *r++;
02453             }
02454             else if ( *r <= -FUNCTION ) *m++ = *r++;
02455             else { *m++ = *r++; *m++ = *r++; }
02456         }
02457     }
02458     w[1] = w[1] - 2 + (m-to);
02459     break;
02460 case DOLUNDEFINED:
02461 default:
02462     MLOCK(ErrorMessageLock);
02463     MesPrint("!!!Undefined $-variable: %s!!!",
02464             AC.dollarnames->namebuffer+d->name);
02465     MUNLOCK(ErrorMessageLock);
02466 #ifdef WITHPTHREADS
02467     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
02468 #endif
02469     Terminate(-1);
02470 }
02471 #ifdef WITHPTHREADS
02472     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
02473 #endif
02474     r = term + *term;
02475     t = v;
02476     while ( t < r ) *m++ = *t++;
02477     if ( (m-termout) > (LONG)(AM.MaxTer/sizeof(WORD)) ) {
02478         MLOCK(ErrorMessageLock);
02479         MesPrint("Output term too large (%d words) (MaxTermSize: %d words)",
02480                 m-termout, AM.MaxTer/sizeof(WORD));
02481         MUNLOCK(ErrorMessageLock);
02482         goto TooLarge;
02483     }
02484     *termout = WORDDIFF(m,termout);
02485     AR.DeferFlag = olddefer;
02486     AN.ncmod = oldncmod;
02487     return(0);
02488 }
02489 else if ( *t == -TERMSINBRACKET ) {
02490     if ( AC.ComDefer ) numterms = CountTerms1(BHEAD0);
02491     else numterms = 1;
02492 /*
02493     Compose the output term
02494     First copy the part till this function argument
02495     m points at the output term space
02496     u points at the start of the function
02497     t points at the start of the argument
02498 */
02499     w = 0;
02500     while ( from < t ) {
02501         if ( from == u ) w = m;
02502         *m++ = *from++;
02503     }
02504     if ( ( numterms & MAXPOSITIVE ) == numterms ) {
02505         *m++ = -SNUMBER; *m++ = numterms & MAXPOSITIVE;
02506         w[1] += 1;
02507     }
02508     else if ( ( i = numterms » BITSINWORD ) == 0 ) {
02509         *m++ = ARGHEAD+4;
02510         for ( j = 1; j < ARGHEAD; j++ ) *m++ = 0;
02511         *m++ = 4; *m++ = numterms & WORDMASK; *m++ = 1; *m++ = 3;
02512         w[1] += ARGHEAD+3;

```

```

02513     }
02514     else {
02515         *m++ = ARGHEAD+6;
02516         for ( j = 1; j < ARGHEAD; j++ ) *m++ = 0;
02517         *m++ = 6; *m++ = numterms & WORDMASK;
02518         *m++ = i; *m++ = 1; *m++ = 0; *m++ = 5;
02519         w[1] += ARGHEAD+5;
02520     }
02521     from++; /* Skip our function */
02522     r = term + *term;
02523     while ( from < r ) *m++ = *from++;
02524     if ( (m-termout) > (LONG)(AM.MaxTer/sizeof(WORD)) ) {
02525         MLOCK(ErrorMessageLock);
02526         MesPrint("Output term too large (%d words) (MaxTermSize: %d words)",
m-termout, AM.MaxTer/sizeof(WORD));
02527         MUNLOCK(ErrorMessageLock);
02528         goto TooLarge;
02529     }
02530     *termout = WORDDIF(m,termout);
02531     return(0);
02532 }
02533 else { NEXTARG(t) }
02534 }
02535 t = u;
02536 }
02537 else if ( ( *t >= FUNCTION && functions[ipp-FUNCTION].spec > 0 )
&& ( t[2] & DIRTYFLAG ) == DIRTYFLAG ) { /* Could be FUNNYDOLLAR */
02538     u = t; v = t + t[1];
02539     t += FUNHEAD;
02540     while ( t < v ) {
02541         if ( *t == FUNNYDOLLAR ) {
02542             if ( AR.Eside != LHSIDE ) {
02543                 DOLLARS d = Dollars + t[1];
02544 #ifdef WITHPTHREADS
02545                 int nummodopt, dtype = -1;
02546                 if ( AS.MultiThreaded ) {
02547                     for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
02548                         if ( t[1] == ModOptdollars[nummodopt].number ) break;
02549                     }
02550                     if ( nummodopt < NumModOptdollars ) {
02551                         dtype = ModOptdollars[nummodopt].type;
02552                         if ( dtype == MODLOCAL ) {
02553                             d = ModOptdollars[nummodopt].dstruct+AT.identity;
02554                         }
02555                         else {
02556                             LOCK(d->pthreadslackread);
02557                         }
02558                     }
02559                 }
02560             }
02561 #endif
02562                 oldncmod = AN.ncmod;
02563                 if ( ( AN.ncmod != 0 )
&& ( ( AC.modmode & ALSOFUNARGS ) == 0 )
&& ( *u != AR.PolyFun ) ) { AN.ncmod = 0; }
02564                 m = termout; w = 0;
02565                 while ( from < t ) {
02566                     if ( from == u ) w = m;
02567                     *m++ = *from++;
02568                 }
02569                 to = m;
02570                 switch ( d->type ) {
02571                     case DOLINDEX:
02572                         *m++ = d->index; break;
02573                     case DOLZERO:
02574                         *m++ = 0; break;
02575                     case DOLNUMBER:
02576                     case DOLTERMS:
02577                         if ( d->where[0] == 4 && d->where[4] == 0
&& d->where[3] == 3 && d->where[2] == 1
&& d->where[1] < AM.OffsetIndex ) {
02578                             *m++ = d->where[1];
02579                         }
02580                         else {
02581                             wrongtype;;
02582                             #ifdef WITHPTHREADS
02583                                 if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslackread); }
02584                             #endif
02585                             MLOCK(ErrorMessageLock);
02586                             MesPrint("$%s has wrong type for tensor substitution",
AC.dollarnames->namebuffer+d->name);
02587                             MUNLOCK(ErrorMessageLock);
02588                             AN.ncmod = oldncmod;
02589                             return(-1);
02590                         }
02591                     break;
02592                     case DOLARGUMENT:
02593                         if ( d->where[0] == -INDEX ) {

```

```

02599         *m++ = d->where[1]; break;
02600     }
02601     else if ( d->where[0] == -VECTOR ) {
02602         *m++ = d->where[1]; break;
02603     }
02604     else if ( d->where[0] == -MINVECTOR ) {
02605         *m++ = d->where[1];
02606         sign = -sign;
02607         break;
02608     }
02609     else if ( d->where[0] == -SNUMBER ) {
02610         if ( d->where[1] >= 0
02611             && d->where[1] < AM.OffsetIndex ) {
02612             *m++ = d->where[1]; break;
02613         }
02614     }
02615     goto wrongtype;
02616 case DOLWILDARGS:
02617     if ( d->where[0] > 0 ) {
02618         r = d->where; i = *r++;
02619         while ( --i >= 0 ) *m++ = *r++;
02620     }
02621     else {
02622         r = d->where + 1;
02623         while ( *r ) {
02624             if ( *r == -INDEX ) {
02625                 *m++ = r[1]; r += 2; continue;
02626             }
02627             else if ( *r == -VECTOR ) {
02628                 *m++ = r[1]; r += 2; continue;
02629             }
02630             else if ( *r == -MINVECTOR ) {
02631                 *m++ = r[1]; r += 2;
02632                 sign = -sign; continue;
02633             }
02634             else if ( *r == -SNUMBER ) {
02635                 if ( r[1] >= 0
02636                     && r[1] < AM.OffsetIndex ) {
02637                     *m++ = r[1]; r += 2; continue;
02638                 }
02639             }
02640             goto wrongtype;
02641         }
02642     }
02643     break;
02644 case DOLSUBTERM:
02645     r = d->where;
02646     if ( *r == INDEX && r[1] == 3 ) {
02647         *m++ = r[2];
02648     }
02649     else goto wrongtype;
02650     break;
02651 case DOLUNDEFINED:
02652     #ifdef WITHPTHREADS
02653         if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
02654     #endif
02655     MLOCK(ErrorMessageLock);
02656     MesPrint("%s is undefined in tensor substitution",
02657             AC.dollarnames->namebuffer+d->name);
02658     MUNLOCK(ErrorMessageLock);
02659     AN.ncmod = oldncmod;
02660     return(-1);
02661 }
02662 #ifdef WITHPTHREADS
02663     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
02664 #endif
02665     w[1] = w[1] - 2 + (m-to);
02666     from += 2;
02667     term += *term;
02668     while ( from < term ) *m++ = *from++;
02669     if ( sign < 0 ) m[-1] = -m[-1];
02670     if ( (m-termout) > (LONG)(AM.MaxTer/sizeof(WORD)) ) {
02671         MLOCK(ErrorMessageLock);
02672         MesPrint("Output term too large (%d words) (MaxTermSize: %d words)",
02673             m-termout, AM.MaxTer/sizeof(WORD));
02674         MUNLOCK(ErrorMessageLock);
02675         goto TooLarge;
02676     }
02677     *termout = m - termout;
02678     AN.ncmod = oldncmod;
02679     return(0);
02680 }
02681 else {
02682     AC.lhdollarflag = 1;
02683 }
02684 t++;

```

```

02685         }
02686         t = u;
02687     }
02688     t += t[1];
02689 }
02690 MLOCK(ErrorMessageLock);
02691 MesPrint("Internal error in InFunction: Function not encountered.");
02692 if ( AM.tracebackflag ) {
02693     MesPrint("%w: AR.TePos = %d",AR.TePos);
02694     MesPrint("%w: AN.TeInFun = %d",AN.TeInFun);
02695     termout = term;
02696     AO.OutFill = AO.OutputLine = (UBYTE *)AT.WorkPointer + AM.MaxTer;
02697     AO.OutSkip = 3;
02698     FiniLine();
02699     i = *termout;
02700     while ( --i >= 0 ) {
02701         TalToLine((UWORD) (*termout++));
02702         TokenToLine((UBYTE *) " ");
02703     }
02704     AO.OutSkip = 0;
02705     FiniLine();
02706     MesCall("InFunction");
02707 }
02708 MUNLOCK(ErrorMessageLock);
02709 return(1);
02710
02711 InFunc:
02712     MLOCK(ErrorMessageLock);
02713     MesCall("InFunction");
02714     MUNLOCK(ErrorMessageLock);
02715     SETERROR(-1)
02716
02717 TooLarge:
02718     MLOCK(ErrorMessageLock);
02719     MesCall("InFunction");
02720     MUNLOCK(ErrorMessageLock);
02721     SETERROR(-1)
02722 }
02723
02724 /*
02725     #] InFunction :
02726     #[ InsertTerm :          WORD InsertTerm(term,replac,extractbuff,position,termout)
02727 */
02745 int InsertTerm(PHEAD WORD *term, WORD replac, WORD extractbuff, WORD *position, WORD *termout,
02746               WORD tepos)
02747 {
02748     GETBIDENTITY
02749     WORD *m, *t, *r, i, l2, j;
02750     WORD *u, *v, ll, *coef;
02751     coef = AT.WorkPointer;
02752     if ( ( AT.WorkPointer = coef + 2*AM.MaxTal ) > AT.WorkTop ) {
02753         MLOCK(ErrorMessageLock);
02754         MesWork();
02755         MUNLOCK(ErrorMessageLock);
02756         return(-1);
02757     }
02758     t = term;
02759     r = t + *t;
02760     ll = l2 = r[-1];
02761     m = r - ABS(l2);
02762     if ( tepos > 0 ) {
02763         t = term + tepos;
02764         goto foundit;
02765     }
02766     t++;
02767     while ( t < m ) {
02768         if ( *t == SUBEXPRESSION && t[2] == replac && t[3] && t[4] == extractbuff ) {
02769             r = t + t[1];
02770             while ( *r == SUBEXPRESSION && r[2] == replac && r[3] && r < m && r[4] == extractbuff ) {
02771                 t = r; r += r[1];
02772             }
02773             foundit;;
02774             u = m;
02775             r = term;
02776             m = termout;
02777             do { *m++ = *r++; } while ( r < t );
02778             if ( t[1] > SUBEXPSIZE ) {
02779                 /*
02780                     if this is a dollar expression there are no wildcards
02781                 */
02782                 i = *--m;
02783                 if ( ( l2 = WildFill(BHEAD m,position,t) ) < 0 ) goto InsCall;
02784                 *m = i;
02785                 m += l2-1;
02786                 l2 = *m;
02787                 i = ( j = ABS(l2) ) - 1;
02788                 r = coef + i;

```

```

02789         do { *--r = *--m; } while ( --i > 0 );
02790     }
02791     else {
02792         v = t;
02793         t = position;
02794         r = t + *t;
02795         l2 = r[-1];
02796         r -= ( j = ABS(l2) );
02797         t++;
02798         if ( t < r ) do { *m++ = *t++; } while ( t < r );
02799         t = v;
02800     }
02801     t += t[1];
02802     while ( t < u && *t == DOLLAREXPR2 ) t += t[1];
02803 ComAct: if ( t < u ) do { *m++ = *t++; } while ( t < u );
02804 if ( *r == 1 && r[1] == 1 && j == 3 ) {
02805     if ( l2 < 0 ) l1 = -l1;
02806     i = ABS(l1)-1;
02807     NCOPY(m,t,i);
02808     *m++ = l1;
02809 }
02810 else {
02811     if ( MulRat(BHEAD (UWORD *)u,REDLENG(l1), (UWORD *)r,REDLENG(l2),
02812         (UWORD *)m,&l1) ) goto InsCall;
02813     l2 = l1;
02814     l2 *= 2;
02815     if ( l2 < 0 ) {
02816         m -= l2;
02817         *m++ = l2-1;
02818     }
02819     else {
02820         m += l2;
02821         *m++ = l2+1;
02822     }
02823 }
02824 *termout = WORDDIF(m,termout);
02825 if ( (*termout)*((LONG)sizeof(WORD)) > AM.MaxTer ) {
02826     MLOCK(ErrorMessageLock);
02827     MesPrint("Term too complex during substitution (%d words). MaxTermSize (%l words) is
too small.", *termout, AM.MaxTer/(LONG)sizeof(WORD) );
02828     goto InsCall2;
02829 }
02830 AT.WorkPointer = coef;
02831 return(0);
02832 }
02833 t += t[1];
02834 }
02835 /*
02836 The next action is for when there is no subexpression pointer.
02837 We append the extra term. Effectively the routine becomes now a
02838 merge routine for two terms.
02839 */
02840 v = t;
02841 u = m;
02842 r = term;
02843 m = termout;
02844 do { *m++ = *r++; } while ( r < t );
02845 t = position;
02846 r = t + *t;
02847 l2 = r[-1];
02848 r -= ( j = ABS(l2) );
02849 t++;
02850 if ( t < r ) do { *m++ = *t++; } while ( t < r );
02851 t = v;
02852 goto ComAct;
02853
02854 InsCall:
02855     MLOCK(ErrorMessageLock);
02856 InsCall2:
02857     MesCall("InsertTerm");
02858     MUNLOCK(ErrorMessageLock);
02859     SETERROR(-1)
02860 }
02861
02862 /*
02863 #] InsertTerm :
02864 #[ PasteFile :          WORD PasteFile(num,acc,pos,accf,renum,freeze,nexpr)
02865 */
02881 LONG PasteFile(PHEAD WORD number, WORD *accum, POSITION *position, WORD **accfill,
02882     RENUMBER renumber, WORD *freeze, WORD nexpr)
02883 {
02884     GETBIDENTITY
02885     WORD *r, l, *m, i;
02886     WORD *stop, *s1, *s2;
02887 /* POSITION AccPos; bug 12-apr-2008 JV */
02888     WORD InCompState;
02889     WORD *oldipointer;

```

```

02890     LONG retlength;
02891     stop = (WORD *) ((UBYTE *) (accum)) + 2*AM.MaxTer;
02892     *accum++ = number;
02893     while ( --number >= 0 ) accum += *accum;
02894     if ( freeze ) {
02895 /*       AccPos = *position; bug 12-apr-2008 JV */
02896         oldipointer = AR.CompressPointer;
02897         do {
02898             AR.CompressPointer = oldipointer;
02899 /*       if ( ( l = GetFromStore(accum,&AccPos,renumber,&InCompState,nexpr) ) < 0 ) bug 12-apr-2008
JV */
02900             if ( ( l = GetFromStore(accum,position,renumber,&InCompState,nexpr) ) < 0 )
02901                 goto PasErr;
02902             if ( !l ) { *accum = 0; return(0); }
02903             r = accum;
02904             m = r + *r;
02905             m -= ABS(m[-1]);
02906             r++;
02907             while ( r < m && *r != HAAKJE ) r += r[1];
02908             if ( r >= m ) {
02909                 if ( *freeze != 4 ) l = -1;
02910             }
02911             else {
02912 /*
02913                 The algorithm for accepting terms with a given (freeze)
02914                 representation outside brackets is rather crude. A refinement
02915                 would be to store the part outside the bracket and skip the
02916                 term when this part doesn't alter (and is unacceptable).
02917                 Once accepting one can keep accepting till the bracket alters
02918                 and then one may stop the generation. It is necessary to
02919                 set up a struct to remember the bracket and the progress
02920                 status.
02921 */
02922                 m = AT.WorkPointer;
02923                 s2 = r;
02924                 r = accum;
02925                 *m++ = WORDDIF(s2,r) + 3;
02926                 r++;
02927                 while ( r < s2 ) *m++ = *r++;
02928                 *m++ = 1; *m++ = 1; *m++ = 3;
02929                 m = AT.WorkPointer;
02930                 if ( Normalize(BHEAD AT.WorkPointer) ) goto PasErr;
02931                 r = freeze;
02932                 i = *m;
02933                 while ( --i >= 0 && *m++ == *r++ ) {}
02934                 if ( i > 0 ) {
02935                     l = -1;
02936                 }
02937                 else { /* Term to be accepted */
02938                     r = accum;
02939                     s1 = r + *r;
02940                     r++;
02941                     m = s2;
02942                     m += m[1];
02943                     do { *r++ = *m++; } while ( m < s1 );
02944                     *accum = 1 = WORDDIF(r,accum);
02945                 }
02946             }
02947         } while ( l < 0 );
02948         retlength = InCompState;
02949 /*       retlength = DIFBASE(AccPos,*position) / sizeof(WORD); bug 12-apr-2008 JV */
02950     }
02951     else {
02952         if ( ( l = GetFromStore(accum,position,renumber,&InCompState,nexpr) ) < 0 ) {
02953             MLOCK(ErrorMessageLock);
02954             MesCall("PasteFile");
02955             MUNLOCK(ErrorMessageLock);
02956             SETERROR(-1)
02957         }
02958         if ( l == 0 ) { *accum = 0; return(0); }
02959         retlength = InCompState;
02960     }
02961     accum += 1;
02962     if ( accum > stop ) {
02963         MLOCK(ErrorMessageLock);
02964         MesPrint("Buffer too small in PasteFile");
02965         MUNLOCK(ErrorMessageLock);
02966         SETERROR(-1)
02967     }
02968     *accum = 0;
02969     *accfill = accum;
02970     return(retlength);
02971 PasErr:
02972     MLOCK(ErrorMessageLock);
02973     MesCall("PasteFile");
02974     MUNLOCK(ErrorMessageLock);
02975     SETERROR(-1)

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```

02976 }
02977
02978 /*
02979      #] PasteFile :
02980      #[ PasteTerm :          WORD PasteTerm(number,accum,position,times,divby)
02981 */
03003 WORD *PasteTerm(PHEAD WORD number, WORD *accum, WORD *position, WORD times, WORD divby)
03004 {
03005     GETBIDENTITY
03006     WORD *t, *r, x, y, z;
03007     WORD *m, *u, ll, a[2];
03008     m = (WORD *)(((UBYTE *) (accum)) + AM.MaxTer);
03009     /* m = (WORD *)(((UBYTE *) (accum)) + 2*AM.MaxTer); */
03010     *accum++ = number;
03011     while ( --number >= 0 ) accum += *accum;
03012     if ( times == divby ) {
03013         t = position;
03014         r = t + *t;
03015         if ( t < r ) do { *accum++ = *t++; } while ( t < r );
03016     }
03017     else {
03018         u = accum;
03019         t = position;
03020         r = t + *t - 1;
03021         ll = *r;
03022         r -= ABS(*r) - 1;
03023         if ( t < r ) do { *accum++ = *t++; } while ( t < r );
03024         if ( divby > times ) { x = divby; y = times; }
03025         else { x = times; y = divby; }
03026         z = x*y;
03027         while ( z ) { x = y; y = z; z = x*y; }
03028         if ( y != 1 ) { divby /= y; times /= y; }
03029         a[1] = divby;
03030         a[0] = times;
03031         if ( MulRat(BHEAD (UWORD *)t, REDLENG(ll), (UWORD *)a, 1, (UWORD *)accum, &ll) ) {
03032             MLOCK(ErrorMessageLock);
03033             MesCall("PasteTerm");
03034             MUNLOCK(ErrorMessageLock);
03035             return(0);
03036         }
03037         x = ll;
03038         x *= 2;
03039         if ( x < 0 ) { accum -= x; *accum++ = x - 1; }
03040         else { accum += x; *accum++ = x + 1; }
03041         *u = WORDDIFF(accum,u);
03042     }
03043     if ( accum >= m ) {
03044         MLOCK(ErrorMessageLock);
03045         MesPrint("Buffer too small in PasteTerm");
03046         MUNLOCK(ErrorMessageLock);
03047         return(0);
03048     }
03049     *accum = 0;
03050     return(accum);
03051 }
03052
03053 /*
03054      #] PasteTerm :
03055      #[ FiniTerm :          WORD FiniTerm(term,accum,termout,number)
03056 */
03068 int FiniTerm(PHEAD WORD *term, WORD *accum, WORD *termout, WORD number, WORD tepos)
03069 {
03070     GETBIDENTITY
03071     WORD *m, *t, *r, i, numacc, l2, ipp;
03072     WORD *u, *v, ll, *coef = AT.WorkPointer, *oldaccum;
03073     if ( ( AT.WorkPointer = coef + 2*AM.MaxTal ) > AT.WorkTop ) {
03074         MLOCK(ErrorMessageLock);
03075         MesWork();
03076         MUNLOCK(ErrorMessageLock);
03077         return(-1);
03078     }
03079     oldaccum = accum;
03080     t = term;
03081     m = t + *t - 1;
03082     ll = REDLENG(*m);
03083     i = ABS(*m) - 1;
03084     r = coef + i;
03085     do { *--r = *--m; } while ( --i > 0 ); /* Copies coefficient */
03086     if ( tepos > 0 ) {
03087         t = term + tepos;
03088         goto foundit;
03089     }
03090     t++;
03091     if ( t < m ) do {
03092         if ( ( ( *t == SUBEXPRESSION && ( *(r=t+t[1]) != SUBEXPRESSION
03093             || r >= m || !r[3] ) ) || *t == EXPRESSION ) && t[2] == number && t[3] ) {
03094             foundit++;

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```

03095         u = m;
03096         r = term;
03097         m = termout;
03098         if ( r < t ) do { *m++ = *r++; } while ( r < t );
03099         numacc = *accum++;
03100         if ( numacc >= 0 ) do {
03101             if ( *t == EXPRESSION ) {
03102                 v = t + t[1];
03103                 r = t + SUBEXPSIZE;
03104                 while ( r < v ) {
03105                     if ( *r == WILDCARDS ) {
03106                         r += 2;
03107                         i = *--m;
03108                         if ( ( l2 = WildFill(BHEAD m,accum,r) ) < 0 ) goto FiniCall;
03109                         goto AllWild;
03110                     }
03111                     r += r[1];
03112                 }
03113                 goto NoWild;
03114             }
03115             else if ( t[1] > SUBEXPSIZE && t[SUBEXPSIZE] != FROMBRAC ) {
03116                 i = *--m;
03117                 if ( ( l2 = WildFill(BHEAD m,accum,t) ) < 0 ) goto FiniCall;
03118 AllWild:
03119                 *m = i;
03120                 m += l2-1;
03121                 l2 = *m;
03122                 m -= ABS(l2) - 1;
03123                 r = m;
03124             }
03125 NoWild:
03126             r = accum;
03127             v = r + *r - 1;
03128             l2 = *v;
03129             v -= ABS(l2) - 1;
03130             r++;
03131             if ( r < v ) do { *m++ = *r++; } while ( r < v );
03132             if ( *r == 1 && r[1] == 1 && ABS(l2) == 3 ) {
03133                 if ( l2 < 0 ) l1 = -l1;
03134             }
03135             else {
03136                 l2 = REDLENG(l2);
03137                 if ( l2 == 0 ) {
03138                     t = oldaccum;
03139                     numacc = *t++;
03140                     AO.OutSkip = 3;
03141                     FiniLine();
03142                     while ( --numacc >= 0 ) {
03143                         i = *t;
03144                         while ( --i >= 0 ) {
03145                             TalToLine((UWORD) (*t++));
03146                             TokenToLine((UBYTE *) " ");
03147                         }
03148                     }
03149                     AO.OutSkip = 0;
03150                     FiniLine();
03151                     goto FiniCall;
03152                 }
03153                 if ( MulRat(BHEAD (UWORD *)coef,l1,(UWORD *)r,l2,(UWORD *)coef,&l1) ) goto
03154 FiniCall;
03155                 if ( AN.ncmod != 0 && TakeModulus((UWORD
03156 *)coef,&l1,AC.cmod,AN.ncmod,UNPACK|AC.modmode) ) goto FiniCall;
03157             }
03158             accum += *accum;
03159             while ( --numacc >= 0 );
03160             if ( *t == SUBEXPRESSION ) {
03161                 while ( t+t[1] < u && t[t[1]] == DOLLAREXP2 ) t += t[1];
03162             }
03163             t += t[1];
03164             if ( t < u ) do { *m++ = *t++; } while ( t < u );
03165             l2 = l1;
03166 /*
03167 Code to economize when taking x = (a+b)/2
03168 */
03169         r = termout+1;
03170         while ( r < m ) {
03171             if ( *r == SUBEXPRESSION ) {
03172                 t = r + r[1];
03173                 l1 = (WORD) (cbuf[r[4]].CanCommu[r[2]]);
03174                 while ( t < m ) {
03175                     if ( *t == SUBEXPRESSION &&
03176                         t[1] == r[1] && t[2] == r[2] && t[4] == r[4] ) {
03177                         i = t[1] - SUBEXPSIZE;
03178                         u = r + SUBEXPSIZE; v = t + SUBEXPSIZE;
03179                         while ( i > 0 ) {
03180                             if ( *v++ != *u++ ) break;
03181                             i--;

```



```

03180                                     }
03181                                     if ( i <= 0 ) {
03182                                         u = r;
03183                                         r[3] += t[3];
03184                                         r = t + t[1];
03185                                         while ( r < m ) *t++ = *r++;
03186                                         m = t;
03187                                         r = u;
03188                                         goto Nexttr;
03189                                     }
03190                                     if ( l1 && cbuf[t[4]].CanCommu[t[2]] ) break;
03191                                     while ( t+t[1] < m && t[t[1]] == DOLLAREXP2 ) t += t[1];
03192                                 }
03193                                 else if ( l1 ) {
03194                                     if ( *t == SUBEXPRESSION && cbuf[t[4]].CanCommu[t[2]] )
03195                                         break;
03196                                     if ( *t >= FUNCTION+WILDOFFSET )
03197                                         ipp = *t - WILDOFFSET;
03198                                     else ipp = *t;
03199                                     if ( *t >= FUNCTION
03200                                         && functions[ipp-FUNCTION].commute && l1 ) break;
03201                                     if ( *t == EXPRESSION ) break;
03202                                 }
03203                                 t += t[1];
03204                             }
03205                             r += r[1];
03206                         }
03207                         else r += r[1];
03208 Nexttr;;
03209                     }
03210
03211                     i = ABS(l2);
03212                     i *= 2;
03213                     i++;
03214                     l2 = ( l2 >= 0 ) ? i: -i;
03215                     r = coef;
03216                     while ( --i > 0 ) *m++ = *r++;
03217                     *m++ = l2;
03218                     *termout = WORDDIF(m,termout);
03219                     AT.WorkPointer = coef;
03220                     return(0);
03221                 }
03222                 t += t[1];
03223             } while ( t < m );
03224             AT.WorkPointer = coef;
03225             return(1);
03226
03227 FiniCall:
03228     MLOCK(ErrorMessageLock);
03229     MesCall("FiniTerm");
03230     MUNLOCK(ErrorMessageLock);
03231     SETERROR(-1)
03232 }
03233
03234 /*
03235     #] FiniTerm :
03236     #[ Generator :          WORD Generator(BHEAD term,level)
03237 */
03238
03239 static WORD zeroDollar[] = { 0, 0 };
03240 /*
03241 static LONG debugcounter = 0;
03242 */
03243
03267 int Generator(PHEAD WORD *term, WORD level)
03268 {
03269     GETBIDENTITY
03270     WORD replac, *accum, *termout, *t, i, j, tepos, applyflag = 0, *StartBuf;
03271     WORD *a, power, powerl, DumNow = AR.CurDum, oldtoprhs, oldatoprhs, extractbuff;
03272     int ret;
03273     int *RepSto = AN.RepPoint, iscopy = 0;
03274     CBUF *C = cbuf+AM.rbufnum, *CC = cbuf + AT.ebufnum, *CCC = cbuf + AT.aebufnum;
03275     LONG posisub, oldcpointer, oldacpointer;
03276     DOLLARS d = 0;
03277     WORD numfac[5], idfunctionflag;
03278 #ifdef WITHPTHREADS
03279     int nummodopt, dtype = -1, id;
03280 #endif
03281     oldtoprhs = CC->numrhs;
03282     oldcpointer = CC->Pointer - CC->Buffer;
03283     oldatoprhs = CCC->numrhs;
03284     oldacpointer = CCC->Pointer - CCC->Buffer;
03285 ReStart:
03286     if ( ( replac = TestSub(BHEAD term,level) ) == 0 ) {
03287         if ( applyflag ) { TableReset(); applyflag = 0; }
03288     /*
03289         if ( AN.PolyNormFlag > 1 ) {

```

```

03290         if ( PolyFunMul(BHEAD term) < 0 ) goto GenCall;
03291         AN.PolyNormFlag = 0;
03292         if ( !*term ) goto Return0;
03293     }
03294     */
03295 Renormalize:
03296     AN.PolyNormFlag = 0;
03297     AN.idfunctionflag = 0;
03298     if ( ( ret = Normalize(BHEAD term) ) != 0 ) {
03299         if ( ret > 0 ) {
03300             if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
03301             goto ReStart;
03302         }
03303         goto GenCall;
03304     }
03305     idfunctionflag = AN.idfunctionflag;
03306     if ( !*term ) { AN.PolyNormFlag = 0; goto Return0; }
03307
03308     if ( AN.PolyNormFlag ) {
03309         if ( AN.PolyFunTodo == 0 ) {
03310             if ( PolyFunMul(BHEAD term) < 0 ) goto GenCall;
03311             if ( !*term ) { AN.PolyNormFlag = 0; goto Return0; }
03312         }
03313         else {
03314             WORD oldPolyFunExp = AR.PolyFunExp;
03315             AR.PolyFunExp = 0;
03316             if ( PolyFunMul(BHEAD term) < 0 ) goto GenCall;
03317             AT.WorkPointer = term+*term;
03318             AR.PolyFunExp = oldPolyFunExp;
03319             if ( !*term ) { AN.PolyNormFlag = 0; goto Return0; }
03320             if ( Normalize(BHEAD term) < 0 ) goto GenCall;
03321             if ( !*term ) { AN.PolyNormFlag = 0; goto Return0; }
03322             AT.WorkPointer = term+*term;
03323             if ( AN.PolyNormFlag ) {
03324                 if ( PolyFunMul(BHEAD term) < 0 ) goto GenCall;
03325                 if ( !*term ) { AN.PolyNormFlag = 0; goto Return0; }
03326                 AT.WorkPointer = term+*term;
03327             }
03328             AN.PolyFunTodo = 0;
03329         }
03330     }
03331     if ( idfunctionflag > 0 ) {
03332         if ( TakeIDfunction(BHEAD term) ) {
03333             AT.WorkPointer = term + *term;
03334             goto ReStart;
03335         }
03336     }
03337     if ( AT.WorkPointer < (WORD *)(((UBYTE *) (term)) + AM.MaxTer) )
03338         AT.WorkPointer = (WORD *)(((UBYTE *) (term)) + AM.MaxTer);
03339     do {
03340 SkipCount: level++;
03341         if ( level > AR.Cnumlhs ) {
03342             if ( AR.DeferFlag && AR.sLevel <= 0 ) {
03343 #ifdef WITHMPI
03344                 if ( PF.me != MASTER && AC.mparallelflag == PARALLELFLAG && PF.exprtodo < 0 ) {
03345                     if ( PF_Deferred(term,level) ) goto GenCall;
03346                 }
03347             else
03348 #endif
03349                 {
03350                     if ( Deferred(BHEAD term,level) ) goto GenCall;
03351                 }
03352             goto Return0;
03353         }
03354         if ( AN.ncmod != 0 ) {
03355             if ( Modulus(term) ) goto GenCall;
03356             if ( !*term ) goto Return0;
03357         }
03358         if ( AR.CurDum > AM.IndDum && AR.sLevel <= 0 ) {
03359             WORD olddummies = AN.IndDum;
03360             AN.IndDum = AM.IndDum;
03361             ReNumber(BHEAD term);
03362             Normalize(BHEAD term);
03363             AN.IndDum = olddummies;
03364             if ( !*term ) goto Return0;
03365             olddummies = DetCurDum(BHEAD term);
03366             if ( olddummies > AR.MaxDum ) AR.MaxDum = olddummies;
03367         }
03368         if ( AR.PolyFun > 0 && ( AR.sLevel <= 0 || AN.FunSorts[AR.sLevel]->PolyFlag > 0 ) ) {
03369             if ( PrepPoly(BHEAD term,0) != 0 ) goto Return0;
03370         }
03371         else if ( AR.PolyFun > 0 ) {
03372             if ( PrepPoly(BHEAD term,1) != 0 ) goto Return0;
03373         }
03374         if ( AR.sLevel <= 0 && AR.BracketOn ) {
03375             if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
03376             termout = AT.WorkPointer;

```

```

03377         if ( AT.WorkPointer + *term + 3 > AT.WorkTop ) goto OverWork;
03378         if ( PutBracket(BHEAD term) ) return(-1);
03379         AN.RepPoint = RepSto;
03380         *AT.WorkPointer = 0;
03381         ret = StoreTerm(BHEAD termout);
03382         AT.WorkPointer = termout;
03383         CC->numrhs = oldtoprhs;
03384         CC->Pointer = CC->Buffer + oldcpointer;
03385         CCC->numrhs = oldatoprhs;
03386         CCC->Pointer = CCC->Buffer + oldacpointer;
03387         return(ret);
03388     }
03389     else {
03390         if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
03391         if ( AT.WorkPointer >= AT.WorkTop ) goto OverWork;
03392         *AT.WorkPointer = 0;
03393         AN.RepPoint = RepSto;
03394         ret = StoreTerm(BHEAD term);
03395         CC->numrhs = oldtoprhs;
03396         CC->Pointer = CC->Buffer + oldcpointer;
03397         CCC->numrhs = oldatoprhs;
03398         CCC->Pointer = CCC->Buffer + oldacpointer;
03399         return(ret);
03400     }
03401 }
03402 i = C->lhs[level][0];
03403 if ( i >= TYPECOUNT ) {
03404     /*
03405     #[ Special action :
03406     */
03407     switch ( i ) {
03408     case TYPECOUNT:
03409         if ( CountDo(term,C->lhs[level]) < C->lhs[level][2] ) {
03410             AT.WorkPointer = term + *term;
03411             goto Return0;
03412         }
03413         break;
03414     case TYPEMULT:
03415         if ( MultDo(BHEAD term,C->lhs[level]) ) goto GenCall;
03416         goto ReStart;
03417     case TYPEGOTO:
03418         level = AC.Labels[C->lhs[level][2]];
03419         break;
03420     case TYPEDISCARD:
03421         AT.WorkPointer = term + *term;
03422         goto Return0;
03423     case TYPEIF:
03424         #ifdef WITHPTHREADS
03425         {
03426             /*
03427             We may be writing in the space here when wildcards
03428             are involved in a match(). Hence we have to make
03429             a private copy here!!!!
03430             */
03431             WORD ic, jc, *ifcode, *jfcodes;
03432             jfcodes = C->lhs[level]; jc = jfcodes[1];
03433             ifcode = AT.WorkPointer; AT.WorkPointer += jc;
03434             for ( ic = 0; ic < jc; ic++ ) ifcode[ic] = jfcodes[ic];
03435             while ( !DoIfStatement(BHEAD ifcode,term) ) {
03436                 level = C->lhs[level][2];
03437                 if ( C->lhs[level][0] != TYPEELIF ) break;
03438             }
03439             AT.WorkPointer = ifcode;
03440         }
03441         #else
03442         while ( !DoIfStatement(BHEAD C->lhs[level],term) ) {
03443             level = C->lhs[level][2];
03444             if ( C->lhs[level][0] != TYPEELIF ) break;
03445         }
03446         #endif
03447         break;
03448     case TYPEELIF:
03449         do {
03450             level = C->lhs[level][2];
03451         } while ( C->lhs[level][0] == TYPEELIF );
03452         break;
03453     case TYPEELSE:
03454     case TYPEENDIF:
03455         level = C->lhs[level][2];
03456         break;
03457     case TYPESUMFIX:
03458     {
03459         WORD *cp = AR.CompressPointer, *op = AR.CompressPointer;
03460         WORD *tlhs = C->lhs[level] + 3, *m, *lhs;
03461         WORD theindex = C->lhs[level][2];
03462         if ( theindex < 0 ) { /* $-variable */
03463             #ifdef WITHPTHREADS

```

```

03464         int ddtype = -1;
03465         theindex = -theindex;
03466         d = Dollars + theindex;
03467         if ( AS.MultiThreaded ) {
03468             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
03469                 if ( theindex == ModOptdollars[nummodopt].number ) break;
03470             }
03471             if ( nummodopt < NumModOptdollars ) {
03472                 ddtype = ModOptdollars[nummodopt].type;
03473                 if ( ddtype == MODLOCAL ) {
03474                     d = ModOptdollars[nummodopt].dstruct+AT.identity;
03475                 }
03476                 else {
03477                     LOCK(d->pthreadlockread);
03478                 }
03479             }
03480         }
03481     #else
03482         theindex = -theindex;
03483         d = Dollars + theindex;
03484     #endif
03485
03486     if ( d->type != DOLINDEX
03487         || d->index < AM.OffsetIndex
03488         || d->index >= AM.OffsetIndex + WILDOffset ) {
03489         MLOCK(ErrorMessageLock);
03490         MesPrint("%s should have been an index"
03491             ,AC.dollarnames->namebuffer+d->name);
03492         AN.currentTerm = term;
03493         MesPrint("Current term: %t");
03494         AN.listinprint = printscratch;
03495         printscratch[0] = DOLLAREXPRESSION;
03496         printscratch[1] = theindex;
03497         MesPrint("$s = %s"
03498             ,AC.dollarnames->namebuffer+d->name);
03499         MUNLOCK(ErrorMessageLock);
03500     #ifdef WITHPTHREADS
03501         if ( ddtype > 0 && ddtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
03502     #endif
03503         goto GenCall;
03504     }
03505     theindex = d->index;
03506     #ifdef WITHPTHREADS
03507         if ( ddtype > 0 && ddtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
03508     #endif
03509 }
03510 cp[1] = SUBEXPSIZE+4;
03511 cp += SUBEXPSIZE;
03512 *cp++ = INDTOIND;
03513 *cp++ = 4;
03514 *cp++ = theindex;
03515 i = C->lhs[level][1] - 3;
03516 cp++;
03517 AR.CompressPointer = cp;
03518 while ( --i >= 0 ) {
03519     cp[-1] = *tlhs++;
03520     termout = AT.WorkPointer;
03521     if ( ( jlhs = WildFill(BHEAD termout,term,op)) < 0 )
03522         goto GenCall;
03523     m = term;
03524     jlhs = *m;
03525     while ( --jlhs >= 0 ) {
03526         if ( *m++ != *termout++ ) break;
03527     }
03528     if ( jlhs >= 0 ) {
03529         termout = AT.WorkPointer;
03530         AT.WorkPointer = termout + *termout;
03531         if ( Generator(BHEAD termout,level) ) goto GenCall;
03532         AT.WorkPointer = termout;
03533     }
03534     else {
03535         AR.CompressPointer = op;
03536         goto SkipCount;
03537     }
03538 }
03539 AR.CompressPointer = op;
03540 goto CommonEnd;
03541 }
03542 case TYPESUM:
03543 {
03544     WORD *wp, *cp = AR.CompressPointer, *op = AR.CompressPointer;
03545     WORD theindex;
03546     WORD *ow;
03547     /*
03548     At this point it is safest to determine CurDum
03549     */
03550     AR.CurDum = DetCurDum(BHEAD term);

```

```

03551         i = C->lhs[level][1]-2;
03552         wp = C->lhs[level] + 2;
03553         cp[1] = SUBEXPSIZE+4*i;
03554         cp += SUBEXPSIZE;
03555         while ( --i >= 0 ) {
03556             theindex = *wp++;
03557             if ( theindex < 0 ) { /* $-variable */
03558 #ifdef WITHPTHREADS
03559                 int ddtype = -1;
03560                 theindex = -theindex;
03561                 d = Dollars + theindex;
03562                 if ( AS.MultiThreaded ) {
03563                     for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
03564                         if ( theindex == ModOptdollars[nummodopt].number ) break;
03565                     }
03566                     if ( nummodopt < NumModOptdollars ) {
03567                         ddtype = ModOptdollars[nummodopt].type;
03568                         if ( ddtype == MODLOCAL ) {
03569                             d = ModOptdollars[nummodopt].dstruct+AT.identity;
03570                         }
03571                         else {
03572                             LOCK(d->pthreadlockread);
03573                         }
03574                     }
03575                 }
03576 #else
03577                 theindex = -theindex;
03578                 d = Dollars + theindex;
03579 #endif
03580                 if ( d->type != DOLINDEX
03581                     || d->index < AM.OffsetIndex
03582                     || d->index >= AM.OffsetIndex + WILD OFFSET ) {
03583                     MLOCK(ErrorMessageLock);
03584                     MesPrint("$s should have been an index"
03585                             ,AC.dollarnames->namebuffer+d->name);
03586                     AN.currentTerm = term;
03587                     MesPrint("Current term: %t");
03588                     AN.listinprint = printscratch;
03589                     printscratch[0] = DOLLAREXPRESSION;
03590                     printscratch[1] = theindex;
03591                     MesPrint("$s = %s"
03592                             ,AC.dollarnames->namebuffer+d->name);
03593                     MUNLOCK(ErrorMessageLock);
03594 #ifdef WITHPTHREADS
03595                     if ( ddtype > 0 && ddtype != MODLOCAL ) {
03596                         UNLOCK(d->pthreadlockread); }
03597 #endif
03598                     goto GenCall;
03599                 }
03600                 theindex = d->index;
03601 #ifdef WITHPTHREADS
03602                 if ( ddtype > 0 && ddtype != MODLOCAL ) { UNLOCK(d->pthreadlockread); }
03603 #endif
03604             }
03605             *cp++ = INDTOIND;
03606             *cp++ = 4;
03607             *cp++ = theindex;
03608             *cp++ = ++AR.CurDum;
03609         }
03610         ow = AT.WorkPointer;
03611         AR.CompressPointer = cp;
03612         if ( WildFill(BHEAD ow,term,op) < 0 ) goto GenCall;
03613         AR.CompressPointer = op;
03614         i = ow[0];
03615         WORD term_changed = 0;
03616         for ( j = 0; j < i; j++ ) {
03617             if ( term[j] != ow[j] ) term_changed = 1;
03618             term[j] = ow[j];
03619         }
03620         // If the term was modified by WildFill, set RepCount.
03621         if ( term_changed ) *AN.RepPoint = 1;
03622         AT.WorkPointer = ow;
03623         ReNumber(BHEAD term);
03624         goto Renormalize;
03625     }
03626 case TYPECHISHOLM:
03627     if ( Chisholm(BHEAD term,level) ) goto GenCall;
03628 CommonEnd:
03629     AT.WorkPointer = term + *term;
03630     goto Return0;
03631 case TYPEARG:
03632     if ( ( i = execarg(BHEAD term,level) ) < 0 ) goto GenCall;
03633     level = C->lhs[level][2];
03634     if ( i > 0 ) goto ReStart;
03635     break;
03636 case TYPENORM:

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```

03636         case TYPENORM2:
03637         case TYPENORM3:
03638         case TYPENORM4:
03639         case TYPESPLITARG:
03640         case TYPESPLITARG2:
03641         case TYPESPLITFIRSTARG:
03642         case TYPESPLITLASTARG:
03643         case TYPEARGTOEXTRASymbol:
03644             if ( execarg(BHEAD term,level) < 0 ) goto GenCall;
03645             level = C->lhs[level][2];
03646             break;
03647         case TYPEFACTARG:
03648         case TYPEFACTARG2:
03649             { WORD jjj;
03650             if ( ( jjj = execarg(BHEAD term,level) ) < 0 ) goto GenCall;
03651             if ( jjj > 0 ) goto ReStart;
03652             level = C->lhs[level][2];
03653             break; }
03654         case TYPEEXIT:
03655             if ( C->lhs[level][2] > 0 ) {
03656                 MLOCK(ErrorMessageLock);
03657                 MesPrint("%s",C->lhs[level]+3);
03658                 MUNLOCK(ErrorMessageLock);
03659             }
03660             Terminate(-1);
03661             goto GenCall;
03662         case TYPESETEXIT:
03663             AM.exitflag = 1; /* no danger of race conditions */
03664             break;
03665         case TYPEPRINT:
03666             AN.currentTerm = term;
03667             AN.numlistinprint = (C->lhs[level][1] - C->lhs[level][4] - 5)/2;
03668             AN.listinprint = C->lhs[level]+5+C->lhs[level][4];
03669             MLOCK(ErrorMessageLock);
03670             AO.ErrorBlock = 1;
03671             MesPrint((char *) (C->lhs[level]+5));
03672             AO.ErrorBlock = 0;
03673             MUNLOCK(ErrorMessageLock);
03674             break;
03675         case TYPEFPRINT:
03676             {
03677                 int oldFOflag;
03678                 WORD oldPrintType, oldLogHandle = AC.LogHandle;
03679                 AC.LogHandle = C->lhs[level][2];
03680                 MLOCK(ErrorMessageLock);
03681                 oldFOflag = AM.FileOnlyFlag;
03682                 oldPrintType = AO.PrintType;
03683                 if ( AC.LogHandle >= 0 ) {
03684                     AM.FileOnlyFlag = 1;
03685                     AO.PrintType |= PRINTLFILE;
03686                 }
03687                 AO.PrintType |= C->lhs[level][3];
03688                 AN.currentTerm = term;
03689                 AN.numlistinprint = (C->lhs[level][1] - C->lhs[level][4] - 5)/2;
03690                 AN.listinprint = C->lhs[level]+5+C->lhs[level][4];
03691                 MesPrint((char *) (C->lhs[level]+5));
03692                 AO.PrintType = oldPrintType;
03693                 AM.FileOnlyFlag = oldFOflag;
03694                 MUNLOCK(ErrorMessageLock);
03695                 AC.LogHandle = oldLogHandle;
03696             }
03697             break;
03698         case TYPEREDEFPRE:
03699             j = C->lhs[level][2];
03700 #ifdef WITHMPI
03701         {
03702             /*
03703              * Regardless of parallel/nonparallel switch, we need to set
03704              * AC.inputnumbers[ii], which indicates that the corresponding
03705              * preprocessor variable is redefined and so we need to
03706              * send/broadcast it.
03707              */
03708             int ii;
03709             for ( ii = 0; ii < AC.numpfirstnum; ii++ ) {
03710                 if ( AC.pfirstnum[ii] == j ) break;
03711             }
03712             AC.inputnumbers[ii] = AN.ninterms;
03713         }
03714 #endif
03715 #ifdef WITHPTHREADS
03716         if ( AS.MultiThreaded ) {
03717             int ii;
03718             for ( ii = 0; ii < AC.numpfirstnum; ii++ ) {
03719                 if ( AC.pfirstnum[ii] == j ) break;
03720             }
03721             if ( AN.inputnumber < AC.inputnumbers[ii] ) break;
03722             LOCK(AP.PreVarLock);

```

```

03723         if ( AN.inputnumber >= AC.inputnumbers[ii] ) {
03724             a = C->lhs[level]+4;
03725             if ( a[a[-1]] == 0 )
03726                 PutPreVar(PreVar[j].name, (UBYTE *) (a), 0, 1);
03727             else
03728                 PutPreVar(PreVar[j].name, (UBYTE *) (a)
03729                     , (UBYTE *) (a+a[-1]+1), 1);
03730 /*
03731             PutPreVar(PreVar[j].name, (UBYTE *) (C->lhs[level]+4), 0, 1);
03732 */
03733             AC.inputnumbers[ii] = AN.inputnumber;
03734         }
03735         UNLOCK(AP.PreVarLock);
03736     }
03737     else
03738 #endif
03739     {
03740         a = C->lhs[level]+4;
03741         LOCK(AP.PreVarLock);
03742         if ( a[a[-1]] == 0 )
03743             PutPreVar(PreVar[j].name, (UBYTE *) (a), 0, 1);
03744         else
03745             PutPreVar(PreVar[j].name, (UBYTE *) (a)
03746                 , (UBYTE *) (a+a[-1]+1), 1);
03747         UNLOCK(AP.PreVarLock);
03748     }
03749     break;
03750 case TYPERENUMBER:
03751     AT.WorkPointer = term + *term;
03752     if ( FullRenummer(BHEAD term, C->lhs[level][2]) ) goto GenCall;
03753     AT.WorkPointer = term + *term;
03754     if ( *term == 0 ) goto Return0;
03755     break;
03756 case TYPETRY:
03757     if ( TryDo(BHEAD term, C->lhs[level], level) ) goto GenCall;
03758     AT.WorkPointer = term + *term;
03759     goto Return0;
03760 case TYPEASSIGN:
03761     { WORD onc = AR.NoCompress, oldEside = AR.Eside;
03762       WORD oldrepeat = *AN.RepPoint;
03763 /*
03764       Here we have to assign an expression to a $ variable.
03765 */
03766       AR.Eside = RHSIDE;
03767       AR.NoCompress = 1;
03768       AN.cTerm = AN.currentTerm = term;
03769       AT.WorkPointer = term + *term;
03770       *AT.WorkPointer++ = 0;
03771       if ( AssignDollar(BHEAD term, level) ) goto GenCall;
03772       AT.WorkPointer = term + *term;
03773       AN.cTerm = 0;
03774       *AN.RepPoint = oldrepeat;
03775       AR.NoCompress = onc;
03776       AR.Eside = oldEside;
03777       break;
03778     }
03779 case TYPEFINDLOOP:
03780     if ( Lus(term, C->lhs[level][3], C->lhs[level][4],
03781         C->lhs[level][5], C->lhs[level][6], C->lhs[level][2]) ) {
03782         AT.WorkPointer = term + *term;
03783         goto Renormalize;
03784     }
03785     break;
03786 case TYPEINSIDE:
03787     if ( InsideDollar(BHEAD C->lhs[level], level) < 0 ) goto GenCall;
03788     level = C->lhs[level][2];
03789     break;
03790 case TYPETERM:
03791     ret = execterm(BHEAD term, level);
03792     AN.RepPoint = RepSto;
03793     AR.CurDum = DumNow;
03794     CC->numrhs = oldtoprhs;
03795     CC->Pointer = CC->Buffer + oldcpointer;
03796     CCC->numrhs = oldatoprhs;
03797     CCC->Pointer = CCC->Buffer + oldacpointer;
03798     return(ret);
03799 case TYPEDETCURDUM:
03800     AT.WorkPointer = term + *term;
03801     AR.CurDum = DetCurDum(BHEAD term);
03802     break;
03803 case TYPEINEXPRESSION:
03804     {WORD *ll = C->lhs[level];
03805       int numexprs = (int)(ll[1]-3);
03806       ll += 3;
03807       while ( numexprs-- >= 0 ) {
03808           if ( *ll == AR.CurExpr ) break;
03809           ll++;

```

```

03810         }
03811         if ( numexprs < 0 ) level = C->lhs[level][2];
03812     }
03813     break;
03814 case TYPEMERGE:
03815     AT.WorkPointer = term + *term;
03816     if ( DoShuffle(term,level,C->lhs[level][2],C->lhs[level][3]) )
03817         goto GenCall;
03818     AT.WorkPointer = term + *term;
03819     goto Return0;
03820 case TYPESTUFFLE:
03821     AT.WorkPointer = term + *term;
03822     if ( DoStuffle(term,level,C->lhs[level][2],C->lhs[level][3]) )
03823         goto GenCall;
03824     AT.WorkPointer = term + *term;
03825     goto Return0;
03826 case TYPETESTUSE:
03827     AT.WorkPointer = term + *term;
03828     if ( TestUse(term,level) ) goto GenCall;
03829     AT.WorkPointer = term + *term;
03830     break;
03831 case TYPEAPPLY:
03832     AT.WorkPointer = term + *term;
03833     if ( ApplyExec(term,C->lhs[level][2],level) < C->lhs[level][2] ) {
03834         AT.WorkPointer = term + *term;
03835         *AN.RepPoint = 1;
03836         goto ReStart;
03837     }
03838     AT.WorkPointer = term + *term;
03839     break;
03840 /*
03841 case TYPEAPPLYRESET:
03842     AT.WorkPointer = term + *term;
03843     if ( ApplyReset(level) ) goto GenCall;
03844     AT.WorkPointer = term + *term;
03845     break;
03846 */
03847 case TYPECHAININ:
03848     { int lter = *term;
03849     AT.WorkPointer = term + *term;
03850     if ( ChainIn(BHEAD term,C->lhs[level][2]) ) goto GenCall;
03851     AT.WorkPointer = term + *term;
03852     /* Symmetry properties might mean the term has vanished */
03853     if ( *term == 0 ) goto Return0;
03854     if ( *term != lter ) *AN.RepPoint = 1;
03855     }
03856     break;
03857 case TYPECHAINOUT:
03858     { int lter = *term;
03859     AT.WorkPointer = term + *term;
03860     if ( ChainOut(BHEAD term,C->lhs[level][2]) ) goto GenCall;
03861     AT.WorkPointer = term + *term;
03862     if ( *term != lter ) *AN.RepPoint = 1;
03863     }
03864     break;
03865 case TYPEFACTOR:
03866     AT.WorkPointer = term + *term;
03867     if ( DollarFactorize(BHEAD C->lhs[level][2]) ) goto GenCall;
03868     AT.WorkPointer = term + *term;
03869     break;
03870 case TYPEARGIMPLode:
03871     AT.WorkPointer = term + *term;
03872     if ( ArgumentImplode(BHEAD term,C->lhs[level]) ) goto GenCall;
03873     AT.WorkPointer = term + *term;
03874     break;
03875 case TYPEARGEXPLODE:
03876     AT.WorkPointer = term + *term;
03877     if ( ArgumentExplode(BHEAD term,C->lhs[level]) ) goto GenCall;
03878     AT.WorkPointer = term + *term;
03879     break;
03880 case TYPEDENOMINATORS:
03881     if ( DenToFunction(term,C->lhs[level][2]) ) goto ReStart;
03882     break;
03883 case TYPEDROPCOEFFICIENT:
03884     DropCoefficient(BHEAD term);
03885     break;
03886 case TYPETRANSFORM:
03887     AT.WorkPointer = term + *term;
03888     if ( RunTransform(BHEAD term,C->lhs[level]+2) ) goto GenCall;
03889     AT.WorkPointer = term + *term;
03890     if ( *term == 0 ) goto Return0;
03891     goto ReStart;
03892 case TYPETOPOLYNOMIAL:
03893     AT.WorkPointer = term + *term;
03894     termout = AT.WorkPointer;
03895     if ( ConvertToPoly(BHEAD term,termout,C->lhs[level],0) < 0 ) goto GenCall;
03896     if ( *termout == 0 ) goto Return0;

```



```

03897         i = termout[0]; t = term; NCOPY(t,termout,i);
03898         AT.WorkPointer = term + *term;
03899         break;
03900     case TYPEFROMPOLYNOMIAL:
03901         AT.WorkPointer = term + *term;
03902         termout = AT.WorkPointer;
03903         if ( ConvertFromPoly(BHEAD term,termout,0,numxsymbol,0,0) < 0 ) goto GenCall;
03904         if ( *term == 0 ) goto Return0;
03905         i = termout[0]; t = term; NCOPY(t,termout,i);
03906         AT.WorkPointer = term + *term;
03907         goto ReStart;
03908     case TYPEDOLOOP:
03909         level = TestDoLoop(BHEAD C->lhs[level],level);
03910         if ( level < 0 ) goto GenCall;
03911         break;
03912     case TYPEENDDOLOOP:
03913         level = TestEndDoLoop(BHEAD C->lhs[C->lhs[level][2]],C->lhs[level][2]);
03914         if ( level < 0 ) goto GenCall;
03915         break;
03916     case TYPEDROPSYMBOLS:
03917         DropSymbols(BHEAD term);
03918         break;
03919     case TYPEPUTINSIDE:
03920         AT.WorkPointer = term + *term;
03921         if ( PutInside(BHEAD term,C->lhs[level]) < 0 ) goto GenCall;
03922         AT.WorkPointer = term + *term;
03923         /*
03924          * We need to call Generator() to convert slow notation to
03925          * fast notation, which fixes Issue #30.
03926          */
03927         if ( Generator(BHEAD term,level) < 0 ) goto GenCall;
03928         goto Return0;
03929     case TYPEOTOSPECTATOR:
03930         if ( PutInSpectator(term,C->lhs[level][2]) < 0 ) goto GenCall;
03931         goto Return0;
03932     case TYPECANONICALIZE:
03933         AT.WorkPointer = term + *term;
03934         if ( DoCanonicalize(BHEAD term,C->lhs[level]) ) goto GenCall;
03935         AT.WorkPointer = term + *term;
03936         if ( *term == 0 ) goto Return0;
03937         break;
03938     case TYPESWITCH:
03939         AT.WorkPointer = term + *term;
03940         if ( DoSwitch(BHEAD term,C->lhs[level]) ) goto GenCall;
03941         goto Return0;
03942     case TYPEENDSWITCH:
03943         AT.WorkPointer = term + *term;
03944         if ( DoEndSwitch(BHEAD term,C->lhs[level]) ) goto GenCall;
03945         goto Return0;
03946     case TYPESETUSERFLAG:
03947         Expressions[AR.CurExpr].uflags |= 1 << (C->lhs[level][2]);
03948         break;
03949     case TYPECLEARUSERFLAG:
03950         Expressions[AR.CurExpr].uflags &= ~(1 << (C->lhs[level][2]));
03951         break;
03952     case TYPEALLLOOPS:
03953         AT.WorkPointer = term + *term;
03954         if ( AllLoops(BHEAD term,level) ) goto GenCall;
03955         goto Return0;
03956     case TYPEALLPATHS:
03957         AT.WorkPointer = term + *term;
03958         if ( AllPaths(BHEAD term,level) ) goto GenCall;
03959         goto Return0;
03960 #ifdef WITHFLOAT
03961     case TYPEEVALUATE:
03962         AT.WorkPointer = term + *term;
03963         if ( C->lhs[level][2] == MZV
03964             || C->lhs[level][2] == EULER
03965             || C->lhs[level][2] == MZVHALF
03966             || C->lhs[level][2] == ALLMZVFUNCTIONS
03967         ) {
03968             if ( EvaluateEuler(BHEAD term,level,C->lhs[level][2]) ) goto GenCall;
03969         }
03970         else {
03971             if ( EvaluateFun(BHEAD term,level,C->lhs[level]) ) goto GenCall;
03972         }
03973         /*
03974         else if ( C->lhs[level][2] == SQRTFUNCTION ) {
03975             if ( EvaluateSqrt(BHEAD term,level,C->lhs[level][2]) ) goto GenCall;
03976         }
03977         else {
03978             MLOCK(ErrorMessageLock);
03979             MesPrint("Illegal function %d in evaluate statement.",C->lhs[level][2]);
03980             MUNLOCK(ErrorMessageLock);
03981             goto GenCall;
03982         }
03983         */

```

```

03984         goto Return0;
03985     case TYPETOFLOAT:
03986         AT.WorkPointer = term + *term;
03987         if ( ToFloat(BHEAD term,level) ) goto GenCall;
03988         goto Return0;
03989     case TYPETORAT:
03990         AT.WorkPointer = term + *term;
03991         if ( ToRat(BHEAD term,level) ) goto GenCall;
03992         goto Return0;
03993 #endif
03994     }
03995     goto SkipCount;
03996 /*
03997     #] Special action :
03998 */
03999     }
04000     } while ( ( i = TestMatch(BHEAD term,&level) ) == 0 );
04001     if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
04002     if ( i > 0 ) replac = TestSub(BHEAD term,level);
04003     else replac = i;
04004     if ( replac >= 0 || AT.TMout[1] != SYMMETRIZE ) {
04005         *AN.RepPoint = 1;
04006         AR.expchanged = 1;
04007     }
04008     if ( replac < 0 ) { /* Terms come from automatic generation */
04009 AutoGen:    i = *AT.TMout;
04010             t = termout = AT.WorkPointer;
04011             if ( ( AT.WorkPointer += i ) > AT.WorkTop ) goto OverWork;
04012             accum = AT.TMout;
04013             while ( --i >= 0 ) *t++ = *accum++;
04014             if ( (*FG.Operation[termout[1]])(BHEAD term,termout,replac,level) ) goto GenCall;
04015             AT.WorkPointer = termout;
04016             goto Return0;
04017     }
04018 }
04019 if ( applyflag ) { TableReset(); applyflag = 0; }
04020
04021 if ( AN.TeInFun ) { /* Match in function argument */
04022     if ( AN.TeInFun < 0 && !AN.TeSuOut ) {
04023
04024         if ( AR.TePos >= 0 ) goto AutoGen;
04025         switch ( AN.TeInFun ) {
04026             case -1:
04027                 if ( DoDistrib(BHEAD term,level) ) goto GenCall;
04028                 break;
04029             case -2:
04030                 if ( DoDelta3(BHEAD term,level) ) goto GenCall;
04031                 break;
04032             case -3:
04033                 if ( DoTableExpansion(term,level) ) goto GenCall;
04034                 break;
04035             case -4:
04036                 if ( FactorIn(BHEAD term,level) ) goto GenCall;
04037                 break;
04038             case -5:
04039                 if ( FactorInExpr(BHEAD term,level) ) goto GenCall;
04040                 break;
04041             case -6:
04042                 if ( TermsInBracket(BHEAD term,level) < 0 ) goto GenCall;
04043                 break;
04044             case -7:
04045                 if ( ExtraSymFun(BHEAD term,level) < 0 ) goto GenCall;
04046                 break;
04047             case -8:
04048                 if ( GCDfunction(BHEAD term,level) < 0 ) goto GenCall;
04049                 break;
04050             case -9:
04051                 if ( DIVfunction(BHEAD term,level,0) < 0 ) goto GenCall;
04052                 break;
04053             case -10:
04054                 if ( DIVfunction(BHEAD term,level,1) < 0 ) goto GenCall;
04055                 break;
04056             case -11:
04057                 if ( DIVfunction(BHEAD term,level,2) < 0 ) goto GenCall;
04058                 break;
04059             case -12:
04060                 if ( DoPermutations(BHEAD term,level) ) goto GenCall;
04061                 break;
04062             case -13:
04063                 if ( DoPartitions(BHEAD term,level) ) goto GenCall;
04064                 break;
04065             case -14:
04066                 if ( DIVfunction(BHEAD term,level,3) < 0 ) goto GenCall;
04067                 break;
04068             case -15:
04069                 if ( GenTopologies(BHEAD term,level) < 0 ) goto GenCall;
04070                 break;

```

```

04071         case -16:
04072             if ( GenDiagrams(BHEAD term,level) < 0 ) goto GenCall;
04073             break;
04074         }
04075     }
04076     else {
04077         termout = AT.WorkPointer;
04078         AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
04079         if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
04080         if ( InFunction(BHEAD term,termout) ) goto GenCall;
04081         AT.WorkPointer = termout + *termout;
04082         *AN.RepPoint = 1;
04083         AR.expchanged = 1;
04084         if ( *termout && Generator(BHEAD termout,level) < 0 ) goto GenCall;
04085         AT.WorkPointer = termout;
04086     }
04087 }
04088 else if ( replac > 0 ) {
04089     power = AN.TeSuOut;
04090     tepos = AR.TePos;
04091     if ( power < 0 ) { /* Table expansion */
04092         power = -power; tepos = 0;
04093     }
04094     extractbuff = AT.TMbuff;
04095     if ( extractbuff == AM.dbufnum ) {
04096         d = DolToTerms(BHEAD replac);
04097         if ( d && d->where != 0 ) {
04098             iscopy = 1;
04099             if ( AT.TMdolfac > 0 ) { /* We need a factor */
04100                 if ( AT.TMdolfac == 1 ) {
04101                     if ( d->nfactors ) {
04102                         numfac[0] = 4;
04103                         numfac[1] = d->nfactors;
04104                         numfac[2] = 1;
04105                         numfac[3] = 3;
04106                         numfac[4] = 0;
04107                     }
04108                     else {
04109                         numfac[0] = 0;
04110                     }
04111                     StartBuf = numfac;
04112                 }
04113                 else {
04114                     if ( (AT.TMdolfac-1) > d->nfactors && d->nfactors > 0 ) {
04115                         MLOCK(ErrorMessageLock);
04116                         MesPrint("Attempt to use an nonexistent factor %d of a
$-variable", (WORD) (AT.TMdolfac-1));
04117                     }
04118                     if ( d->nfactors == 1 )
04119                         MesPrint("There is only one factor");
04120                     else
04121                         MesPrint("There are only %d factors", (WORD) (d->nfactors));
04122                     MUNLOCK(ErrorMessageLock);
04123                     goto GenCall;
04124                 }
04125                 if ( d->nfactors > 1 ) {
04126                     DOLLARS dd;
04127                     LONG dsize;
04128                     WORD *td1, *td2;
04129                     dd = Dollars + replac;
04130                     #ifdef WITHPTHREADS
04131                     {
04132                         int nummodopt, dtype = -1;
04133                         if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
04134                             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
04135                                 if ( replac == ModOptdollars[nummodopt].number ) break;
04136                             }
04137                             if ( nummodopt < NumModOptdollars ) {
04138                                 dtype = ModOptdollars[nummodopt].type;
04139                                 if ( dtype == MODLOCAL ) {
04140                                     dd = ModOptdollars[nummodopt].dstruct+AT.identity;
04141                                 }
04142                             }
04143                         }
04144                     }
04145                     #endif
04146                     dsize = dd->factors[AT.TMdolfac-2].size;
04147                     /*
04148                     We copy only the factor we need
04149                     */
04150                     if ( dsize == 0 ) {
04151                         numfac[0] = 4;
04152                         numfac[1] = d->factors[AT.TMdolfac-2].value;
04153                         numfac[2] = 1;
04154                         numfac[3] = 3;
04155                         numfac[4] = 0;
04156                         StartBuf = numfac;
04157                         if ( numfac[1] < 0 ) {

```

```

04157             numfac[1] = -numfac[1];
04158             numfac[3] = -numfac[3];
04159         }
04160     }
04161     else {
04162         d->factors[AT.TMdolfac-2].where = td2 = (WORD *)Malloc1(
04163             (dsize+1)*sizeof(WORD), "Copy of factor");
04164         td1 = dd->factors[AT.TMdolfac-2].where;
04165         StartBuf = td2;
04166         d->size = dsize; d->type = DOLTERMS;
04167         NCOPY(td2, td1, dsize);
04168         *td2 = 0;
04169     }
04170 }
04171 else if ( d->nfactors == 1 ) {
04172     StartBuf = d->where;
04173 }
04174 else {
04175     MLOCK(ErrorMessageLock);
04176     if ( d->nfactors == 0 ) {
04177         MesPrint("Attempt to use factor %d of an unfactored
$-variable", (WORD) (AT.TMdolfac-1));
04178     }
04179     else {
04180         MesPrint("Internal error. Illegal number of factors for $-variable");
04181     }
04182     MUNLOCK(ErrorMessageLock);
04183     goto GenCall;
04184 }
04185 }
04186 }
04187     else StartBuf = d->where;
04188 }
04189 else {
04190     d = Dollars + replac;
04191     StartBuf = zeroDollar;
04192 }
04193 posisub = 0;
04194 i = DetCommu(d->where);
04195 #ifdef WITHPTHREADS
04196     if ( AS.MultiThreaded ) {
04197         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
04198             if ( replac == ModOptdollars[nummodopt].number ) break;
04199         }
04200         if ( nummodopt < NumModOptdollars ) {
04201             dtype = ModOptdollars[nummodopt].type;
04202             if ( dtype != MODLOCAL && dtype != MODSUM ) {
04203                 if ( StartBuf[0] && StartBuf[StartBuf[0]] ) {
04204                     MLOCK(ErrorMessageLock);
04205                     MesPrint("A dollar variable with modoption max or min can have only one
term");
04206                     MUNLOCK(ErrorMessageLock);
04207                     goto GenCall;
04208                 }
04209                 LOCK(d->pthreadlockread);
04210             }
04211         }
04212     }
04213 #endif
04214 }
04215 else {
04216     StartBuf = cbuf[extractbuff].Buffer;
04217     posisub = cbuf[extractbuff].rhs[replac] - StartBuf;
04218     i = (WORD) cbuf[extractbuff].CanCommu[replac];
04219 }
04220 if ( power == 1 ) { /* Just a single power */
04221     termout = AT.WorkPointer;
04222     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
04223     if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
04224     while ( StartBuf[posisub] ) {
04225         if ( extractbuff == AT.allbufnum ) WildDollars(BHEAD &(StartBuf[posisub]));
04226         AT.WorkPointer = (WORD *)(((UBYTE *) (termout)) + AM.MaxTer);
04227         if ( InsertTerm(BHEAD term, replac, extractbuff,
&(StartBuf[posisub]), termout, tepos) < 0 ) goto GenCall;
04228         AT.WorkPointer = termout + *termout;
04229         *AN.RepPoint = 1;
04230         AR.expchanged = 1;
04231         posisub += StartBuf[posisub];
04232     }
04233     /*
04234         For multiple table substitutions it may be better to
04235         do modulus arithmetic right here
04236         Turns out to be not very effective.
04237     */
04238     if ( AN.ncmod != 0 ) {
04239         if ( Modulus(termout) ) goto GenCall;
04240         if ( !*termout ) goto Return0;
04241     }

```

```

04242 */
04243 #ifndef WITHPTHREADS
04244     if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) {
        UNLOCK(d->pthreadlockread); }
04245     if ( ( AS.Balancing && CC->numrhs == 0 ) && StartBuf[posisub] ) {
04246         if ( ( id = ConditionalGetAvailableThread() ) >= 0 ) {
04247             if ( BalanceRunThread(BHEAD id,termout,level) < 0 ) goto GenCall;
04248         }
04249     }
04250     else
04251 #endif
04252     if ( Generator(BHEAD termout,level) < 0 ) goto GenCall;
04253 #ifndef WITHPTHREADS
04254     if ( dtype > 0 && dtype != MODLOCAL ) { dtype = 0; break; }
04255 #endif
04256     if ( iscopy == 0 && ( extractbuff != AM.dbufnum ) ) {
04257 /*
04258     There are cases in which a bigger buffer is created
04259     on the fly, like with wildcard buffers.
04260     We play it safe here. Maybe we can be more selective
04261     in some distant future?
04262 */
04263     StartBuf = cbuf[extractbuff].Buffer;
04264 }
04265 }
04266 if ( extractbuff == AT.allbufnum ) {
04267     CBUF *Ce = cbuf + extractbuff;
04268     Ce->Pointer = Ce->rhs[Ce->numrhs--];
04269 }
04270 #ifndef WITHPTHREADS
04271     if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) { UNLOCK(d->pthreadlockread);
        dtype = 0; }
04272 #endif
04273     if ( iscopy ) {
04274         if ( d->nfactors > 1 ) {
04275             int j;
04276             for ( j = 0; j < d->nfactors; j++ ) {
04277                 if ( d->factors[j].where ) M_free(d->factors[j].where,"Copy of factor");
04278             }
04279             M_free(d->factors,"Dollar factors");
04280         }
04281         M_free(d,"Copy of dollar variable");
04282         d = 0; iscopy = 0;
04283     }
04284     AT.WorkPointer = termout;
04285 }
04286 else if ( i <= 1 ) { /* Use binomials */
04287     LONG posit, olw;
04288     WORD *same, *ow = AT.WorkPointer;
04289     LONG olpw = AT.posWorkPointer;
04290     power1 = power+1;
04291     WantAddLongs(power1);
04292     olw = posit = AT.lWorkPointer; AT.lWorkPointer += power1;
04293     same = ++AT.WorkPointer;
04294     a = accum = ( AT.WorkPointer += power1+1 );
04295     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + 2*AM.MaxTer);
04296     if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
04297     AT.lWorkspace[posit] = posisub;
04298     same[-1] = 0;
04299     *same = 1;
04300     *accum = 0;
04301     tepos = AR.TePos;
04302     i = 1;
04303     do {
04304         if ( StartBuf[AT.lWorkspace[posit]] ) {
04305             if ( ( a = PasteTerm(BHEAD i-1,accum,
04306                 &(StartBuf[AT.lWorkspace[posit]]),i,*same) ) == 0 )
04307                 goto GenCall;
04308             AT.lWorkspace[posit+1] = AT.lWorkspace[posit];
04309             same[1] = *same + 1;
04310             if ( i > 1 && AT.lWorkspace[posit] < AT.lWorkspace[posit-1] ) *same = 1;
04311             AT.lWorkspace[posit] += StartBuf[AT.lWorkspace[posit]];
04312             i++;
04313             posit++;
04314             same++;
04315         }
04316         else {
04317             i--; posit--; same--;
04318         }
04319         if ( i > power ) {
04320             termout = AT.WorkPointer = a;
04321             AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + 2*AM.MaxTer);
04322             if ( AT.WorkPointer > AT.WorkTop )
04323                 goto OverWork;
04324             if ( FiniTerm(BHEAD term,accum,termout,replac,tepos) ) goto GenCall;
04325             AT.WorkPointer = termout + *termout;
04326             *AN.RepPoint = 1;

```

```

04327         AR.expchanged = 1;
04328 #ifndef WITHPTHREADS
04329     if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) {
        UNLOCK(d->pthreadlockread); }
04330     if ( ( AS.Balancing && CC->numrhs == 0 ) && ( i > 0 )
04331         && ( id = ConditionalGetAvailableThread() ) >= 0 ) {
04332         if ( BalanceRunThread(BHEAD id,termout,level) < 0 ) goto GenCall;
04333     }
04334     else
04335 #endif
04336     if ( Generator(BHEAD termout,level) ) goto GenCall;
04337 #ifndef WITHPTHREADS
04338     if ( dtype > 0 && dtype != MODLOCAL ) { dtype = 0; break; }
04339 #endif
04340     if ( iscopy == 0 && ( extractbuff != AM.dbufnum ) )
04341         StartBuf = cbuf[extractbuff].Buffer;
04342     i--; posit--; same--;
04343 }
04344 } while ( i > 0 );
04345 #ifndef WITHPTHREADS
04346     if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) { UNLOCK(d->pthreadlockread);
        dtype = 0; }
04347 #endif
04348     if ( iscopy ) {
04349         if ( d->nfactors > 1 ) {
04350             int j;
04351             for ( j = 0; j < d->nfactors; j++ ) {
04352                 if ( d->factors[j].where ) M_free(d->factors[j].where,"Copy of factor");
04353             }
04354             M_free(d->factors,"Dollar factors");
04355         }
04356         M_free(d,"Copy of dollar variable");
04357         d = 0; iscopy = 0;
04358     }
04359     AT.WorkPointer = ow; AT.lWorkPointer = olw; AT.posWorkPointer = olpw;
04360 }
04361 else { /* No binomials */
04362     LONG posit, olw, olpw = AT.posWorkPointer;
04363     WantAddLongs(power);
04364     posit = olw = AT.lWorkPointer; AT.lWorkPointer += power;
04365     a = accum = AT.WorkPointer;
04366     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + 2*AM.MaxTer);
04367     if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
04368     for ( i = 0; i < power; i++ ) AT.lWorkspace[posit++] = posisub;
04369     posit = olw;
04370     *accum = 0;
04371     tepos = AR.TePos;
04372     i = 0;
04373     while ( i >= 0 ) {
04374         if ( StartBuf[AT.lWorkspace[posit]] ) {
04375             if ( ( a = PasteTerm(BHEAD i,accum,
04376                 &(StartBuf[AT.lWorkspace[posit]]),1,1) ) == 0 ) goto GenCall;
04377             AT.lWorkspace[posit] += StartBuf[AT.lWorkspace[posit]];
04378             i++; posit++;
04379         }
04380         else {
04381             AT.lWorkspace[posit--] = posisub;
04382             i--;
04383         }
04384         if ( i >= power ) {
04385             termout = AT.WorkPointer = a;
04386             AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + 2*AM.MaxTer);
04387             if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
04388             if ( FiniTerm(BHEAD term,accum,termout,replac,tepos) ) goto GenCall;
04389             AT.WorkPointer = termout + *termout;
04390             *AN.RepPoint = 1;
04391             AR.expchanged = 1;
04392 #ifndef WITHPTHREADS
04393             if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) {
        UNLOCK(d->pthreadlockread); }
04394             if ( ( AS.Balancing && CC->numrhs == 0 ) && ( i > 0 ) && ( id =
        ConditionalGetAvailableThread() ) >= 0 ) {
04395                 if ( BalanceRunThread(BHEAD id,termout,level) < 0 ) goto GenCall;
04396             }
04397             else
04398 #endif
04399             if ( Generator(BHEAD termout,level) ) goto GenCall;
04400 #ifndef WITHPTHREADS
04401             if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) { dtype = 0; break; }
04402 #endif
04403             if ( iscopy == 0 && ( extractbuff != AM.dbufnum ) )
04404                 StartBuf = cbuf[extractbuff].Buffer;
04405             i--; posit--;
04406         }
04407     }
04408 #ifndef WITHPTHREADS
04409     if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) { UNLOCK(d->pthreadlockread);

```

```

dtype = 0; }
04410 #endif
04411         if ( iscopy ) {
04412             if ( d->nfactors > 1 ) {
04413                 int j;
04414                 for ( j = 0; j < d->nfactors; j++ ) {
04415                     if ( d->factors[j].where ) M_free(d->factors[j].where,"Copy of factor");
04416                 }
04417                 M_free(d->factors,"Dollar factors");
04418             }
04419             M_free(d,"Copy of dollar variable");
04420             d = 0; iscopy = 0;
04421         }
04422         AT.WorkPointer = accum;
04423         AT.lWorkPointer = olw;
04424         AT.posWorkPointer = olpw;
04425     }
04426 }
04427 else {                                     /* Expression from disk */
04428     POSITION StartPos;
04429     LONG position, olpw, opw, comprev, extra;
04430     RENUMBER renumber;
04431     WORD *Freeze, *aa, *dummies;
04432     replac = -replac-1;
04433     power = AN.TeSuOut;
04434     Freeze = AN.Frozen;
04435     if ( Expressions[replac].status == STOREDEXPRESSION ) {
04436         POSITION firstpos;
04437         SETSTARTPOS(firstpos);
04438
04439         /* Note that AT.TMaddr is needed for GetTable just once! */
04440         /*
04441          We need space for the previous term in the compression
04442          This is made available in AR.CompressBuffer, although we may get
04443          problems with this sooner or later. Hence we need to keep
04444          a set of pointers in AR.CompressBuffer
04445          Note that after the last call there has been no use made
04446          of AR.CompressPointer, so it points automatically at its original
04447          position!
04448          */
04449         WantAddPointers(power+1);
04450         comprev = opw = AT.pWorkPointer;
04451         AT.pWorkPointer += power+1;
04452         WantAddPositions(power+1);
04453         position = olpw = AT.posWorkPointer;
04454         AT.posWorkPointer += power + 1;
04455
04456         AT.pWorkSpace[comprev++] = AR.CompressPointer;
04457
04458         for ( i = 0; i < power; i++ ) {
04459             PUTZERO(AT.posWorkSpace[position]); position++;
04460         }
04461         position = olpw;
04462         if ( ( renumber = GetTable(replac,&(AT.posWorkSpace[position]),1) ) == 0 ) goto GenCall;
04463         dummies = AT.WorkPointer;
04464         *dummies++ = AR.CurDum;
04465         AT.WorkPointer += power+2;
04466         accum = AT.WorkPointer;
04467         AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + 2*AM.MaxTer);
04468         if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
04469         aa = AT.WorkPointer;
04470         *accum = 0;
04471         i = 0; StartPos = AT.posWorkSpace[position];
04472         dummies[i] = AR.CurDum;
04473         while ( i >= 0 ) {
04474             skippedfirst:
04475                 AR.CompressPointer = AT.pWorkSpace[comprev-1];
04476                 if ( ( extra = PasteFile(BHEAD i,accum,&(AT.posWorkSpace[position])
04477                     ,&a,renumber,Freeze,replac) ) < 0 ) goto GenCall;
04478                 if ( Expressions[replac].numdummies > 0 ) {
04479                     AR.CurDum = dummies[i] + Expressions[replac].numdummies;
04480                 }
04481                 if ( NOTSTARTPOS(firstpos) ) {
04482                     if ( ISMINPOS(firstpos) || ISEQUALPOS(firstpos,AT.posWorkSpace[position]) ) {
04483                         firstpos = AT.posWorkSpace[position];
04484                     /*
04485                      ADDPOS(AT.posWorkSpace[position],extra * sizeof(WORD));
04486                      */
04487                     goto skippedfirst;
04488                 }
04489             }
04490             if ( extra ) {
04491                 ADDPOS(AT.posWorkSpace[position],extra * sizeof(WORD));
04492             /*
04493              i++; AT.posWorkSpace[++position] = StartPos;
04494              AT.pWorkSpace[comprev++] = AR.CompressPointer;
04495              */

```

```

04496         dummies[i] = AR.CurDum;
04497     }
04498     else {
04499         PUTZERO(AT.posWorkSpace[position]); position--; i--;
04500         AR.CurDum = dummies[i];
04501         comprev--;
04502     }
04503     if ( i >= power ) {
04504         termout = AT.WorkPointer = a;
04505         AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + 2*AM.MaxTer);
04506         if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
04507         if ( FiniTerm(BHEAD term,accum,termout,replac,0) ) goto GenCall;
04508         if ( *termout ) {
04509             AT.WorkPointer = termout + *termout;
04510             *AN.RepPoint = 1;
04511             AR.expchanged = 1;
04512 #ifndef WITHPTHREADS
04513             if ( ( AS.Balancing && CC->numrhs == 0 ) && ( i > 0 ) && ( id =
ConditionalGetAvailableThread() ) >= 0 ) {
04514                 if ( BalanceRunThread(BHEAD id,termout,level) < 0 ) goto GenCall;
04515             }
04516             else
04517 #endif
04518                 if ( Generator(BHEAD termout,level) ) goto GenCall;
04519             }
04520             i--; position--;
04521             AR.CurDum = dummies[i];
04522             comprev--;
04523         }
04524         AT.WorkPointer = aa;
04525     }
04526     AT.WorkPointer = accum;
04527     AT.posWorkPointer = olpw;
04528     AT.pWorkPointer = opw;
04529 /*
04530 Bug fix. See also GetTable
04531 #ifndef WITHPTHREADS
04532     M_free(renumber->symb.lo,"VarSpace");
04533     M_free(renumber,"Renummer");
04534 #endif
04535 */
04536     if ( renumber->symb.lo != AN.dummyrenumlist )
04537         M_free(renumber->symb.lo,"VarSpace");
04538     M_free(renumber,"Renummer");
04539 }
04540 }
04541 else { /* Active expression */
04542     aa = accum = AT.WorkPointer;
04543     if ( ( (WORD *)(((UBYTE *) (AT.WorkPointer)) + 2 * AM.MaxTer + sizeof(WORD)) ) > AT.WorkTop
04544 )
04545         goto OverWork;
04546     *accum++ = -1; AT.WorkPointer++;
04547     if ( DoOnePow(BHEAD term,power,replac,accum,aa,level,Freeze) ) goto GenCall;
04548     AT.WorkPointer = aa;
04549 }
04550 }
04551 Return0:
04552     AR.CurDum = DumNow;
04553     AN.RepPoint = RepSto;
04554     CC->numrhs = oldtoprhs;
04555     CC->Pointer = CC->Buffer + oldcpointer;
04556     CCC->numrhs = oldatoprhs;
04557     CCC->Pointer = CCC->Buffer + oldacpointer;
04558     return(0);
04559
04560 GenCall:
04561     if ( AM.tracebackflag ) {
04562         termout = term;
04563         MLOCK(ErrorMessageLock);
04564         AO.OutFill = AO.OutputLine = (UBYTE *)AT.WorkPointer;
04565         AO.OutSkip = 3;
04566         FiniLine();
04567         i = *termout;
04568         while ( --i >= 0 ) {
04569             TalToLine((UWORD) (*termout++));
04570             TokenToLine((UBYTE *) " ");
04571         }
04572         AO.OutSkip = 0;
04573         FiniLine();
04574         MesCall("Generator");
04575         MUNLOCK(ErrorMessageLock);
04576     }
04577     CC->numrhs = oldtoprhs;
04578     CC->Pointer = CC->Buffer + oldcpointer;
04579     CCC->numrhs = oldatoprhs;
04580     CCC->Pointer = CCC->Buffer + oldacpointer;

```



```

04581     return(-1);
04582 OverWork:
04583     CC->numrhs = oldtoprhs;
04584     CC->Pointer = CC->Buffer + oldcpointer;
04585     CCC->numrhs = oldatoprhs;
04586     CCC->Pointer = CCC->Buffer + oldacpointer;
04587     MLOCK(ErrorMessageLock);
04588     MesWork();
04589     MUNLOCK(ErrorMessageLock);
04590     return(-1);
04591 }
04592
04593 /*
04594     #] Generator :
04595     #[ DoOnePow :          WORD DoOnePow(term,power,nexp,accum,aa,level,freeze)
04596 */
04621 #ifdef WITHPTHREADS
04622 char freezestring[] = "freeze<-xxxx";
04623 #endif
04624
04625 int DoOnePow(PHEAD WORD *term, WORD power, WORD nexp, WORD * accum,
04626             WORD *aa, WORD level, WORD *freeze)
04627 {
04628     GETBIDENTITY
04629     POSITION oldposition, startposition;
04630     WORD *acc, *termout, fromfreeze = 0;
04631     WORD *oldipointer = AR.CompressPointer;
04632     FILEHANDLE *fi;
04633     WORD type, retval;
04634     WORD oldGetOneFile = AR.GetOneFile;
04635     WORD olddummies = AR.CurDum;
04636     WORD extradummies = Expressions[nexp].numdummies;
04637 /*
04638     The next code is for some tricky debugging. (5-jan-2010 JV)
04639     Normally it should be disabled.
04640 */
04641 /*
04642 #ifdef WITHPTHREADS
04643     if ( freeze ) {
04644         MLOCK(ErrorMessageLock);
04645         if ( AT.identity < 10 ) {
04646             freezestring[8] = '0'+AT.identity;
04647             freezestring[9] = '>';
04648             freezestring[10] = 0;
04649         }
04650         else if ( AT.identity < 100 ) {
04651             freezestring[8] = '0'+AT.identity/10;
04652             freezestring[9] = '0'+AT.identity%10;
04653             freezestring[10] = '>';
04654             freezestring[11] = 0;
04655         }
04656         else {
04657             freezestring[8] = 0;
04658         }
04659         PrintTerm(freeze,freezestring);
04660         MUNLOCK(ErrorMessageLock);
04661     }
04662 #else
04663     if ( freeze ) PrintTerm(freeze,"freeze");
04664 #endif
04665 */
04666     type = Expressions[nexp].status;
04667     if ( type == HIDDENLEXPRESSION || type == HIDDENEXPRESSION
04668         || type == DROPHLEXPRESSION || type == DROPHGEXPRESSION
04669         || type == UNHIDELEXPRESSION || type == UNHIDEGEXPRESSION ) {
04670         AR.GetOneFile = 2; fi = AR.hidefile;
04671     }
04672     else {
04673         AR.GetOneFile = 0; fi = AR.infile;
04674     }
04675     if ( fi->handle >= 0 ) {
04676         PUTZERO(oldposition);
04677 #ifdef WITHSEEK
04678         LOCK(AS.inputslock);
04679         SeekFile(fi->handle,&oldposition,SEEK_CUR);
04680         UNLOCK(AS.inputslock);
04681 #endif
04682     }
04683     else {
04684         SETBASEPOSITION(oldposition,fi->POfill-fi->PObuffer);
04685     }
04686     if ( freeze && ( Expressions[nexp].bracketinfo != 0 ) ) {
04687         POSITION *brapos;
04688 /*
04689         There is a bracket index
04690         AR.CompressPointer = oldipointer;
04691 */

```

```

04692         (*aa)++;
04693         power--;
04694         if ( ( brapos = FindBracket(nexp,freeze) ) == 0 )
04695             goto EndExpr;
04696         startposition = *brapos;
04697         goto doterms;
04698     }
04699     startposition = AS.OldOnFile[nexp];
04700     retval = GetOneTerm(BHEAD accum,fi,&startposition,0);
04701     if ( retval > 0 ) {          /* Skip prototype */
04702         (*aa)++;
04703         power--;
04704     doterms:
04705         AR.CompressPointer = oldipointer;
04706         for (;;) {
04707             retval = GetOneTerm(BHEAD accum,fi,&startposition,0);
04708             if ( retval <= 0 ) break;
04709         /*
04710             Here should come the code to test for [].
04711         */
04712         if ( freeze ) {
04713             WORD *t, *m, *r, *mstop;
04714             WORD *tset;
04715             t = accum;
04716             m = freeze;
04717             m += *m;
04718             m -= ABS(m[-1]);
04719             mstop = m;
04720             m = freeze + 1;
04721             r = t;
04722             r += *t;
04723             r -= ABS(r[-1]);
04724             t++;
04725             tset = t;
04726             while ( t < r && *t != HAAKJE ) t += t[1];
04727             if ( t >= r ) {
04728                 if ( m < mstop ) {
04729                     if ( fromfreeze ) goto EndExpr;
04730                     goto NextTerm;
04731                 }
04732                 t = tset;
04733             }
04734             else {
04735                 r = tset;
04736                 while ( r < t && m < mstop ) {
04737                     if ( *r == *m ) { m++; r++; }
04738                     else {
04739                         if ( fromfreeze ) goto EndExpr;
04740                         goto NextTerm;
04741                     }
04742                 }
04743                 if ( r < t || m < mstop ) {
04744                     if ( fromfreeze ) goto EndExpr;
04745                     goto NextTerm;
04746                 }
04747             }
04748             fromfreeze = 1;
04749             r = tset;
04750             m = accum;
04751             m += *m;
04752             while ( t < m ) *r++ = *t++;
04753             *accum = WORDDIF(r,accum);
04754         }
04755         if ( extradummies > 0 ) {
04756             if ( olddummies > AM.IndDum ) {
04757                 MoveDummies(BHEAD accum,olddummies-AM.IndDum);
04758             }
04759             AR.CurDum = olddummies+extradummies;
04760         }
04761         acc = accum;
04762         acc += *acc;
04763         if ( power <= 0 ) {
04764             termout = acc;
04765             AT.WorkPointer = (WORD *)(((UBYTE *) (acc)) + 2*AM.MaxTer);
04766             if ( AT.WorkPointer > AT.WorkTop ) {
04767                 MLOCK(ErrorMessageLock);
04768                 MesWork();
04769                 MUNLOCK(ErrorMessageLock);
04770                 return(-1);
04771             }
04772             if ( FiniTerm(BHEAD term,aa,termout,nexp,0) ) goto PowCall;
04773             if ( *termout ) {
04774                 MarkPolyRatFunDirty(termout)
04775                 PolyFunDirty(BHEAD termout); /*
04776                 AT.WorkPointer = termout + *termout;
04777                 *AN.RepPoint = 1;
04778                 AR.expchanged = 1;

```

```

04779             if ( Generator(BHEAD termout,level) ) goto PowCall;
04780         }
04781     }
04782     else {
04783         if ( acc > AT.WorkTop ) {
04784             MLOCK(ErrorMessageLock);
04785             MesWork();
04786             MUNLOCK(ErrorMessageLock);
04787             return(-1);
04788         }
04789         if ( DoOnePow(BHEAD term,power,nexp,acc,aa,level,freeze) ) goto PowCall;
04790     }
04791 NextTerm;;
04792     AR.CompressPointer = olddipointer;
04793 }
04794 EndExpr:
04795     (*aa)--;
04796 }
04797     AR.CompressPointer = olddipointer;
04798     if ( fi->handle >= 0 ) {
04799 #ifdef WITHSEEK
04800         LOCK(AS.inputslock);
04801         SeekFile(fi->handle,&oldposition,SEEK_SET);
04802         UNLOCK(AS.inputslock);
04803         if ( ISNEGPOS(oldposition) ) {
04804             MLOCK(ErrorMessageLock);
04805             MesPrint("File error");
04806             goto PowCall2;
04807         }
04808 #endif
04809     }
04810     else {
04811         fi->POfill = fi->PObuffer + BASEPOSITION(oldposition);
04812     }
04813     AR.GetOneFile = oldGetOneFile;
04814     AR.CurDum = olddummies;
04815     return(0);
04816 PowCall:;
04817     MLOCK(ErrorMessageLock);
04818 #ifdef WITHSEEK
04819 PowCall2:;
04820 #endif
04821     MesCall("DoOnePow");
04822     MUNLOCK(ErrorMessageLock);
04823     SETERROR(-1)
04824 }
04825
04826 /*
04827     #] DoOnePow :
04828     #[ Deferred :      WORD Deferred(term,level)
04829 */
04846 int Deferred(PHEAD WORD *term, WORD level)
04847 {
04848     GETBIDENTITY
04849     POSITION startposition;
04850     WORD *t, *m, *mstop, *tstart, decr, oldb, *termout, i, *oldwork, retval;
04851     WORD *olddipointer = AR.CompressPointer, *oldPOfill = AR.infile->POfill;
04852     WORD oldGetOneFile = AR.GetOneFile;
04853     AR.GetOneFile = 1;
04854     oldwork = AT.WorkPointer;
04855     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
04856     termout = AT.WorkPointer;
04857     AR.DeferFlag = 0;
04858     startposition = AR.DefPosition;
04859 /*
04860     Store old position
04861 */
04862     if ( AR.infile->handle >= 0 ) {
04863 /*
04864         PUTZERO(oldposition);
04865         SeekFile(AR.infile->handle,&oldposition,SEEK_CUR);
04866 */
04867     }
04868     else {
04869 /*
04870         SETBASEPOSITION(oldposition,AR.infile->POfill-AR.infile->PObuffer);
04871 */
04872         AR.infile->POfill = (WORD *)(((UBYTE *) (AR.infile->PObuffer)
04873             +BASEPOSITION(startposition)));
04874     }
04875 /*
04876     Look in the CompressBuffer where the bracket contents start
04877 */
04878     t = m = AR.CompressBuffer;
04879     t += *t;
04880     mstop = t - ABS(t[-1]);
04881     m++;

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04882     while ( *m != HAAKJE && m < mstop ) m += m[1];
04883     if ( m >= mstop ) { /* No deferred action! */
04884         AT.WorkPointer = term + *term;
04885         if ( Generator(BHEAD term,level) ) goto DefCall;
04886         AR.DeferFlag = 1;
04887         AT.WorkPointer = oldwork;
04888         AR.GetOneFile = oldGetOneFile;
04889         return(0);
04890     }
04891     mstop = m + m[1];
04892     decr = WORDDIFF(mstop,AR.CompressBuffer)-1;
04893     tstart = AR.CompressPointer + decr;
04894
04895     m = AR.CompressBuffer;
04896     t = AR.CompressPointer;
04897     i = *m;
04898     NCOPY(t,m,i);
04899     oldb = *tstart;
04900     AR.TePos = 0;
04901     AN.TeSuOut = 0;
04902 /*
04903     Status:
04904     First bracket content starts at mstop.
04905     Next term starts at startposition.
04906     Decompression information is in AR.CompressPointer.
04907     The outside of the bracket runs from AR.CompressBuffer+1 to mstop.
04908 */
04909     for(;;) {
04910         *tstart = *(AR.CompressPointer)-decr;
04911         AR.CompressPointer = AR.CompressPointer+AR.CompressPointer[0];
04912         if ( InsertTerm(BHEAD term,0,AM.rbufnum,tstart,termout,0) < 0 ) {
04913             goto DefCall;
04914         }
04915         *tstart = oldb;
04916         AT.WorkPointer = termout + *termout;
04917         if ( Generator(BHEAD termout,level) ) goto DefCall;
04918         AR.CompressPointer = oldipointer;
04919         AT.WorkPointer = termout;
04920         retval = GetOneTerm(BHEAD AT.WorkPointer,AR.infile,&startposition,0);
04921         if ( retval >= 0 ) AR.CompressPointer = oldipointer;
04922         if ( retval <= 0 ) break;
04923         t = AR.CompressPointer;
04924         if ( *t < (1 + decr + ABS(*(t+*t-1))) ) break;
04925         t++;
04926         m = AR.CompressBuffer+1;
04927         while ( m < mstop ) {
04928             if ( *m != *t ) goto Thatsit;
04929             m++; t++;
04930         }
04931     }
04932 Thatsit:;
04933 /*
04934     Finished. Reposition the file, restore information and return.
04935 */
04936     if ( AR.infile->handle < 0 ) AR.infile->POfill = oldPOfill;
04937     AR.DeferFlag = 1;
04938     AR.GetOneFile = oldGetOneFile;
04939     AT.WorkPointer = oldwork;
04940     return(0);
04941 DefCall:;
04942     MLOCK(ErrorMessageLock);
04943     MesCall("Deferred");
04944     MUNLOCK(ErrorMessageLock);
04945     SETERROR(-1)
04946 }
04947 /*
04948     #] Deferred :
04949     #[ PrepPoly :          WORD PrepPoly(term,par)
04950 */
04974 int PrepPoly(PHEAD WORD *term,WORD par)
04975 {
04976     GETBIDENTITY
04977     WORD count = 0, i, jcoef, ncoef;
04978     WORD *t, *m, *r, *tstop, *poly = 0, *v, *w, *vv, *ww;
04979     WORD *oldworkpointer = AT.WorkPointer;
04980 /*
04981     The problem here is that the function will be forced into 'long'
04982     notation. After this -SNUMBER,1 becomes 6,0,4,1,1,3 and the
04983     pattern matcher cannot match a short 1 with a long 1.
04984     But because this is an undocumented feature for very special
04985     purposes, we don't do anything about it. (30-aug-2011)
04986 */
04987     if ( AR.PolyFunType == 2 && AR.PolyFunExp != 2 ) {
04988         WORD oldtype = AR.SortType;
04989         AR.SortType = SORTHIGHFIRST;
04990         if ( poly_ratfun_normalize(BHEAD term) != 0 ) Terminate(-1);

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04991 /*      if ( ReadPolyRatFun(BHEAD term) != 0 ) Terminate(-1); */
04992      oldworkpointer = AT.WorkPointer;
04993      AR.SortType = oldtype;
04994  }
04995  AT.PolyAct = 0;
04996  t = term;
04997  GETSTOP(t,tstop);
04998  t++;
04999  while ( t < tstop ) {
05000      if ( *t == AR.PolyFun ) {
05001          if ( count > 0 ) return(0);
05002          poly = t;
05003          count++;
05004      }
05005      t += t[1];
05006  }
05007  r = m = term + *term;
05008  i = ABS(m[-1]);
05009  if ( par > 0 ) {
05010      if ( count == 0 ) return(0);
05011      else if ( AR.PolyFunType == 1 || (AR.PolyFunType == 2 && AR.PolyFunExp == 2) )
05012          goto DoOne;
05013      else if ( AR.PolyFunType == 2 )
05014          goto DoTwo;
05015      else
05016          goto DoError;
05017  }
05018  else if ( count == 0 ) {
05019      /*      #[ Create a PolyFun :
05020      05021      */
05022      poly = t = tstop;
05023      if ( i == 3 && m[-2] == 1 && (m[-3]&MAXPOSITIVE) == m[-3] ) {
05024          *m++ = AR.PolyFun;
05025          if ( AR.PolyFunType == 1 || (AR.PolyFunType == 2 && AR.PolyFunExp == 2) ) {
05026              *m++ = FUNHEAD+2;
05027              FILLFUN(m)
05028              *m++ = -SNUMBER;
05029              *m = m[-2-FUNHEAD] < 0 ? -m[-4-FUNHEAD]: m[-4-FUNHEAD];
05030              m++;
05031          }
05032          else if ( AR.PolyFunType == 2 ) {
05033              *m++ = FUNHEAD+4;
05034              FILLFUN(m)
05035              *m++ = -SNUMBER;
05036              *m = m[-2-FUNHEAD] < 0 ? -m[-4-FUNHEAD]: m[-4-FUNHEAD];
05037              m++;
05038              *m++ = -SNUMBER;
05039              *m++ = 1;
05040          }
05041      }
05042      else {
05043          WORD *vm;
05044          r = tstop;
05045          if ( AR.PolyFunType == 1 || (AR.PolyFunType == 2 && AR.PolyFunExp == 2) ) {
05046              *m++ = AR.PolyFun;
05047              *m++ = FUNHEAD+ARGHEAD+i+1;
05048              FILLFUN(m)
05049              *m++ = ARGHEAD+i+1;
05050              *m++ = 0;
05051              FILLARG(m)
05052              *m++ = i+1;
05053              NCOPY(m,r,i);
05054          }
05055          else if ( AR.PolyFunType == 2 ) {
05056              WORD *num, *den, size, sign, sizenum, sizeden;
05057              if ( m[-1] < 0 ) { sign = -1; size = -m[-1]; }
05058              else { sign = 1; size = m[-1]; }
05059              num = m - size; size = (size-1)/2; den = num + size;
05060              sizenum = size; while ( num[sizenum-1] == 0 ) sizenum--;
05061              sizeden = size; while ( den[sizeden-1] == 0 ) sizeden--;
05062              v = m;
05063              AT.PolyAct = WORDDIF(v,term);
05064              *v++ = AR.PolyFun;
05065              v++;
05066              FILLFUN(v);
05067              vm = v;
05068              *v++ = ARGHEAD+2*sizenum+2;
05069              *v++ = 0;
05070              FILLARG(v);
05071              *v++ = 2*sizenum+2;
05072              for ( i = 0; i < sizenum; i++ ) *v++ = num[i];
05073              *v++ = 1;
05074              for ( i = 1; i < sizenum; i++ ) *v++ = 0;
05075              *v++ = sign*(2*sizenum+1);
05076              if ( ToFast(vm,vm) ) v = vm+2;
05077              vm = v;

```

```

05078         *v++ = ARGHEAD+2*sizeden+2;
05079         *v++ = 0;
05080         FILLARG(v);
05081         *v++ = 2*sizeden+2;
05082         for ( i = 0; i < sizeden; i++ ) *v++ = den[i];
05083         *v++ = 1;
05084         for ( i = 1; i < sizeden; i++ ) *v++ = 0;
05085         *v++ = 2*sizeden+1;
05086         if ( ToFast(vm,vm) ) v = vm+2;
05087         i = v-m;
05088         m[1] = i;
05089         w = num;
05090         NCOPY(w,m,i);
05091         *w++ = 1; *w++ = 1; *w++ = 3; *term = w - term;
05092         return(0);
05093     }
05094 }
05095 /*
05096 #] Create a PolyFun :
05097 */
05098 }
05099 else if ( AR.PolyFunType == 1 || (AR.PolyFunType == 2 && AR.PolyFunExp == 2) ) {
05100     DoOne;;
05101 /*
05102 #[ One argument :
05103 */
05104 m = term + *term;
05105 r = poly + poly[1];
05106 if ( ( poly[1] == FUNHEAD+2 && poly[FUNHEAD+1] == 0
05107 && poly[FUNHEAD] == -SNUMBER ) || poly[1] == FUNHEAD ) return(1);
05108 t = poly + FUNHEAD;
05109 if ( t >= r ) return(0);
05110 if ( m[-1] == 3 && *tstop == 1 && tstop[1] == 1 ) {
05111     i = poly[1];
05112     t = poly;
05113     NCOPY(m,t,i);
05114 }
05115 else if ( *t <= -FUNCTION ) {
05116     if ( t+1 < r ) return(0); /* More than one argument */
05117     r = tstop;
05118     *m++ = AR.PolyFun;
05119     *m++ = FUNHEAD*2+ARGHEAD+i+1;
05120     FILLFUN(m)
05121     *m++ = FUNHEAD+ARGHEAD+i+1;
05122     *m++ = 0;
05123     FILLARG(m)
05124     *m++ = FUNHEAD+i+1;
05125     *m++ = -*t++;
05126     *m++ = FUNHEAD;
05127     FILLFUN(m)
05128     NCOPY(m,r,i);
05129 }
05130 else if ( *t < 0 ) {
05131     if ( t+2 < r ) return(0); /* More than one argument */
05132     r = tstop;
05133     if ( *t == -SNUMBER ) {
05134         if ( t[1] == 0 ) return(1); /* Term should be zero now */
05135         *m = AR.PolyFun;
05136         w = m+1;
05137         m += FUNHEAD+ARGHEAD;
05138         v = m;
05139         *m++ = 5+i;
05140         *m++ = SNUMBER;
05141         *m++ = 4;
05142         *m++ = t[1];
05143         *m++ = 1;
05144         NCOPY(m,r,i);
05145         if ( m >= AT.WorkSpace && m < AT.WorkTop )
05146             AT.WorkPointer = m;
05147         if ( Normalize(BHEAD v) ) Terminate(-1);
05148         AT.WorkPointer = oldworkpointer;
05149         m = w;
05150         if ( *v == 4 && v[2] == 1 && (v[1]&MAXPOSITIVE) == v[1] ) {
05151             *m++ = FUNHEAD+2;
05152             FILLFUN(m)
05153             *m++ = -SNUMBER;
05154             *m++ = v[3] < 0 ? -v[1] : v[1];
05155         }
05156         else if ( *v == 0 ) return(1);
05157         else {
05158             *m++ = FUNHEAD+ARGHEAD+*v;
05159             FILLFUN(m)
05160             *m++ = ARGHEAD+*v;
05161             *m++ = 0;
05162             FILLARG(m)
05163             m = v + *v;
05164         }
05165     }

```

```

05165     }
05166     else if ( *t == -SYMBOL ) {
05167         *m++ = AR.PolyFun;
05168         *m++ = FUNHEAD+ARGHEAD+5+i;
05169         FILLFUN(m)
05170         *m++ = ARGHEAD+5+i;
05171         *m++ = 0;
05172         FILLARG(m)
05173         *m++ = 5+i;
05174         *m++ = SYMBOL;
05175         *m++ = 4;
05176         *m++ = t[1];
05177         *m++ = 1;
05178         NCOPY(m,r,i);
05179     }
05180     else return(0);          /* Not symbol-like */
05181 }
05182 else {
05183     if ( t + *t < r ) return(0); /* More than one argument */
05184     i = m[-1];
05185     *m++ = AR.PolyFun;
05186     w = m;
05187     m += ARGHEAD+FUNHEAD-1;
05188     t += ARGHEAD;
05189     jcoef = i < 0 ? (i+1)>>1:(i-1)>>1;
05190     v = t;
05191 /*
05192     Test now the scalar nature of the argument.
05193     No indices allowed.
05194 */
05195     while ( t < r ) {
05196         WORD *vstop;
05197         vv = t + *t;
05198         vstop = vv - ABS(vv[-1]);
05199         t++;
05200         while( t < vstop ) {
05201             if ( *t == INDEX ) return(0);
05202             t += t[1];
05203         }
05204         t = vv;
05205     }
05206 /*
05207     Now multiply each term by the coefficient.
05208 */
05209     t = v;
05210     while ( t < r ) {
05211         ww = m;
05212         v = t + *t;
05213         ncoef = v[-1];
05214         vv = v - ABS(ncoef);
05215         if ( ncoef < 0 ) ncoef++;
05216         else ncoef--;
05217         ncoef >= 1;
05218         while ( t < vv ) *m++ = *t++;
05219         if ( MulRat(BHEAD (UWORD *)vv,ncoef, (UWORD *)tstop,jcoef,
05220             (UWORD *)m,&ncoef) ) Terminate(-1);
05221         ncoef *= 2;
05222         m += ABS(ncoef);
05223         if ( ncoef < 0 ) ncoef--;
05224         else ncoef++;
05225         *m++ = ncoef;
05226         *ww = WORDDIF(m,ww);
05227         if ( AN.ncmod != 0 ) {
05228             if ( Modulus(ww) ) Terminate(-1);
05229             if ( *ww == 0 ) return(1);
05230             m = ww + *ww;
05231         }
05232         t = vv;
05233     }
05234     *w = (WORDDIF(m,w))+1;
05235     w[FUNHEAD-1] = w[0] - FUNHEAD;
05236     w[FUNHEAD] = 0;
05237     w[1] = 0; /* omission survived for years. 23-mar-2006 JV */
05238     w += FUNHEAD-1;
05239     if ( ToFast(w,w) ) {
05240         if ( *w <= -FUNCTION ) { w[-FUNHEAD+1] = FUNHEAD+1; m = w+1; }
05241         else { w[-FUNHEAD+1] = FUNHEAD+2; m = w+2; }
05242     }
05243 }
05244 }
05245 t = poly + poly[1];
05246 while ( t < tstop ) *poly++ = *t++;
05247 /*
05248     #] One argument :
05249 */
05250 }
05251 else if ( AR.PolyFunType == 2 ) {

```

```

05252         DoTwo;;
05253     /*
05254         #] Two arguments :
05255     */
05256     WORD *num, *den, size, sign, sizenum, sizeden;
05257     /*
05258         First make sure that the PolyFun is last
05259     */
05260     m = term + *term;
05261     if ( poly + poly[1] < tstop ) {
05262         for ( i = 0; i < poly[1]; i++ ) m[i] = poly[i];
05263         t = poly; v = poly + poly[1];
05264         while ( v < tstop ) *t++ = *v++;
05265         poly = t;
05266         for ( i = 0; i < m[1]; i++ ) t[i] = m[i];
05267         t += m[1];
05268     }
05269     AT.PolyAct = WORDDIF(poly,term);
05270     /*
05271         If needed we convert the coefficient into a PolyRatFun and then
05272         we call poly_ratfun_normalize
05273     */
05274     if ( m[-1] == 3 && m[-2] == 1 && m[-3] == 1 ) return(0);
05275     if ( AR.PolyFunExp != 1 ) {
05276         if ( m[-1] < 0 ) { sign = -1; size = -m[-1]; } else { sign = 1; size = m[-1]; }
05277         num = m - size; size = (size-1)/2; den = num + size;
05278         sizenum = size; while ( num[sizenum-1] == 0 ) sizenum--;
05279         sizeden = size; while ( den[sizeden-1] == 0 ) sizeden--;
05280         v = m;
05281         *v++ = AR.PolyFun;
05282         *v++ = FUNHEAD + 2*(ARGHEAD+sizenum+sizeden+2);
05283     /*
05284         *v++ = MUSTCLEANPRF; */
05285         *v++ = 0;
05286         FILLFUN3(v);
05287         *v++ = ARGHEAD+2*sizenum+2;
05288         *v++ = 0;
05289         FILLARG(v);
05290         *v++ = 2*sizenum+2;
05291         for ( i = 0; i < sizenum; i++ ) *v++ = num[i];
05292         *v++ = 1;
05293         for ( i = 1; i < sizenum; i++ ) *v++ = 0;
05294         *v++ = sign*(2*sizenum+1);
05295         *v++ = ARGHEAD+2*sizeden+2;
05296         *v++ = 0;
05297         FILLARG(v);
05298         *v++ = 2*sizeden+2;
05299         for ( i = 0; i < sizeden; i++ ) *v++ = den[i];
05300         *v++ = 1;
05301         for ( i = 1; i < sizeden; i++ ) *v++ = 0;
05302         *v++ = 2*sizeden+1;
05303         w = num;
05304         i = v - m;
05305         NCOPY(w,m,i);
05306     }
05307     else {
05308         w = m-ABS(m[-1]);
05309     }
05310     *w++ = 1; *w++ = 1; *w++ = 3; *term = w - term;
05311     {
05312         WORD oldtype = AR.SortType;
05313         AR.SortType = SORTHIGHFIRST;
05314     /*
05315         if ( count > 0 )
05316             poly_ratfun_normalize(BHEAD term);
05317         else
05318             ReadPolyRatFun(BHEAD term);
05319     */
05320     /*
05321         poly_ratfun_normalize(BHEAD term);
05322     */
05323     /*
05324         oldworkpointer = AT.WorkPointer; */
05325         AR.SortType = oldtype;
05326     }
05327     goto endofit;
05328     /*
05329         #] Two arguments :
05330     */
05331     }
05332     else {
05333         DoError;;
05334         MLOCK(ErrorMessageLock);
05335         MesPrint("Illegal value for PolyFunType in PrepPoly");
05336         MUNLOCK(ErrorMessageLock);
05337         Terminate(-1);
05338     }
05339     r = term + *term;
05340     AT.PolyAct = WORDDIF(poly,term);
05341     while ( r < m ) *poly++ = *r++;

```



```

05339     *poly++ = 1;
05340     *poly++ = 1;
05341     *poly++ = 3;
05342     *term = WORDDIFF(poly,term);
05343 endofit;;
05344     return(0);
05345 }
05346
05347 /*
05348     #] PrepPoly :
05349     #[ PolyFunMul :      WORD PolyFunMul(term)
05350 */
05362 int PolyFunMul(PHEAD WORD *term)
05363 {
05364     GETBIDENTITY
05365     WORD *t, *fun1, *fun2, *t1, *t2, *m, *w, *ww, *tt1, *tt2, *tt4, *arg1, *arg2;
05366     WORD *tstop, i, dirty = 0, OldPolyFunPow = AR.PolyFunPow, minp1, minp2;
05367     WORD n1, n2, i1, i2, l1, l2, l3, l4, action = 0, noac = 0;
05368     int retval = 0;
05369     if ( AR.PolyFunType == 2 && AR.PolyFunExp == 1 ) {
05370         WORD pow = 0, pow1;
05371         t = term + 1; t1 = term + *term; t1 -= ABS(t1[-1]);
05372         w = t;
05373         while ( t < t1 ) {
05374             if ( *t != AR.PolyFun ) {
05375 SkipFun:
05376                 if ( t == w ) { t += t[1]; w = t; }
05377                 else { i = t[1]; NCOPY(w,t,i) }
05378                 continue;
05379             }
05380             pow1 = 0;
05381             t2 = t + t[1]; t += FUNHEAD;
05382             if ( *t < 0 ) {
05383                 if ( *t == -SYMBOL && t[1] == AR.PolyFunVar ) pow1++;
05384                 else if ( *t != -SNUMBER ) goto NoLegal;
05385                 t += 2;
05386             }
05387             else if ( t[0] == ARGHEAD+8 && t[ARGHEAD] == 8
05388                 && t[ARGHEAD+1] == SYMBOL && t[ARGHEAD+3] == AR.PolyFunVar
05389                 && t[ARGHEAD+5] == 1 && t[ARGHEAD+6] == 1 && t[ARGHEAD+7] == 3 ) {
05390                 pow1 += t[ARGHEAD+4];
05391                 t += *t;
05392             }
05393             else {
05394 NoLegal:
05395                 MLOCK(ErrorMessageLock);
05396                 MesPrint("Illegal term with divergence in PolyRatFun");
05397                 MesCall("PolyFunMul");
05398                 MUNLOCK(ErrorMessageLock);
05399                 Terminate(-1);
05400             }
05401             if ( *t < 0 ) {
05402                 if ( *t == -SYMBOL && t[1] == AR.PolyFunVar ) pow1--;
05403                 else if ( *t != -SNUMBER ) goto NoLegal;
05404                 t += 2;
05405             }
05406             else if ( t[0] == ARGHEAD+8 && t[ARGHEAD] == 8
05407                 && t[ARGHEAD+1] == SYMBOL && t[ARGHEAD+3] == AR.PolyFunVar
05408                 && t[ARGHEAD+5] == 1 && t[ARGHEAD+6] == 1 && t[ARGHEAD+7] == 3 ) {
05409                 pow1 -= t[ARGHEAD+4];
05410                 t += *t;
05411             }
05412             else goto NoLegal;
05413             if ( t == t2 ) pow += pow1;
05414             else goto SkipFun;
05415         }
05416         m = w;
05417         *w++ = AR.PolyFun; *w++ = 0; FILLFUN(w);
05418         if ( pow > 1 ) {
05419             *w++ = 8+ARGHEAD; *w++ = 0; FILLARG(w);
05420             *w++ = 8; *w++ = SYMBOL; *w++ = 4; *w++ = AR.PolyFunVar; *w++ = pow;
05421             *w++ = 1; *w++ = 1; *w++ = 3; *w++ = -SNUMBER; *w++ = 1;
05422         }
05423         else if ( pow == 1 ) {
05424             *w++ = -SYMBOL; *w++ = AR.PolyFunVar; *w++ = -SNUMBER; *w++ = 1;
05425         }
05426         else if ( pow < -1 ) {
05427             *w++ = -SNUMBER; *w++ = 1; *w++ = 8+ARGHEAD; *w++ = 0; FILLARG(w);
05428             *w++ = 8; *w++ = SYMBOL; *w++ = 4; *w++ = AR.PolyFunVar; *w++ = -pow;
05429             *w++ = 1; *w++ = 1; *w++ = 3;
05430         }
05431         else if ( pow == -1 ) {
05432             *w++ = -SNUMBER; *w++ = 1; *w++ = -SYMBOL; *w++ = AR.PolyFunVar;
05433         }
05434         else {
05435             *w++ = -SNUMBER; *w++ = 1; *w++ = -SNUMBER; *w++ = 1;
05436         }

```

```

05437         m[1] = w - m;
05438         *w++ = 1; *w++ = 1; *w++ = 3;
05439         *term = w - term;
05440         if ( w > AT.WorkSpace && w < AT.WorkTop ) AT.WorkPointer = w;
05441         return(0);
05442     }
05443 ReStart:
05444     if ( AR.PolyFunType == 2 && ( ( AR.PolyFunExp != 2 )
05445         || ( AR.PolyFunExp == 2 && AN.PolyNormFlag > 1 ) ) ) {
05446         WORD count1 = 0, count2 = 0, count3;
05447         WORD oldtype = AR.SortType;
05448         t = term + 1; t1 = term + *term; t1 -= ABS(t1[-1]);
05449         while ( t < t1 ) {
05450             if ( *t == AR.PolyFun ) {
05451                 if ( t[2] && dirty == 0 ) { /* Any dirty flag on? */
05452                     dirty = 1;
05453                     /* ReadPolyRatFun(BHEAD term); */
05454                     /* ToPolyFunGeneral(BHEAD term); */
05455                     poly_ratfun_normalize(BHEAD term);
05456                     if ( term[0] == 0 ) return(0);
05457                     count1 = 0;
05458                     action++;
05459                     goto ReStart;
05460                 }
05461                 t2 = t + t[1]; tt2 = t+FUNHEAD; count3 = 0;
05462                 while ( tt2 < t2 ) { count3++; NEXTARG(tt2); }
05463                 if ( count3 == 2 ) {
05464                     count1++;
05465                     if ( ( t[2] & MUSTCLEANPRF ) != 0 ) { /* Better civilize this guy */
05466                         action++;
05467                         w = AT.WorkPointer;
05468                         AR.SortType = SORTHIGHFIRST;
05469                         t2 = t + t[1]; tt2 = t+FUNHEAD;
05470                         while ( tt2 < t2 ) {
05471                             if ( *tt2 > 0 ) {
05472                                 tt4 = tt2; tt1 = tt2 + ARGHEAD; tt2 += *tt2;
05473                                 NewSort(BHEAD0);
05474                                 while ( tt1 < tt2 ) {
05475                                     i = *tt1; ww = w; NCOPY(ww,tt1,i);
05476                                     AT.WorkPointer = ww;
05477                                     Normalize(BHEAD w);
05478                                     StoreTerm(BHEAD w);
05479                                 }
05480                                 EndSort(BHEAD w,1);
05481                                 ww = w; while ( *ww ) ww += *ww;
05482                                 if ( ww-w != *tt4-ARGHEAD ) { /* Little problem */
05483                                     /*
05484                                     Solution: brute force copy
05485                                     Maybe it will never come here????
05486                                     */
05487                                     WORD *r1 = TermMalloc("PolyFunMul");
05488                                     WORD ii = (ww-w)-(*tt4-ARGHEAD); /* increment */
05489                                     WORD *r2 = tt4+ARGHEAD, *r3, *r4 = r1;
05490                                     i = r2 - term; r3 = term; NCOPY(r4,r3,i);
05491                                     i = ww-w; ww = w; NCOPY(r4,ww,i);
05492                                     r3 = tt2; i = term+*term-tt2; NCOPY(r4,r3,i);
05493                                     *r1 = i = r4-r1; r4 = term; r3 = r1;
05494                                     NCOPY(r4,r3,i);
05495                                     t[1] += ii; t1 += ii; *tt4 += ii;
05496                                     tt2 = tt4 + *tt4;
05497                                     TermFree(r1,"PolyFunMul");
05498                                 }
05499                                 else {
05500                                     i = ww-w; ww = w; tt1 = tt4+ARGHEAD;
05501                                     NCOPY(tt1,ww,i);
05502                                     AT.WorkPointer = w;
05503                                 }
05504                             }
05505                             else if ( *tt2 <= -FUNCTION ) tt2++;
05506                             else tt2 += 2;
05507                         }
05508                         AR.SortType = oldtype;
05509                     }
05510                 }
05511             }
05512             t += t[1];
05513         }
05514         if ( count1 <= 1 ) { goto checkaction; }
05515         if ( AR.PolyFunExp == 1 ) {
05516             t = term + *term; t -= ABS(t[-1]);
05517             *t++ = 1; *t++ = 1; *t++ = 3; *term = t - term;
05518         }
05519         {
05520             AR.SortType = SORTHIGHFIRST;
05521             /* retval = ReadPolyRatFun(BHEAD term); */
05522             /* ToPolyFunGeneral(BHEAD term); */
05523             retval = poly_ratfun_normalize(BHEAD term);

```

```

05524         if ( *term == 0 ) return(retval);
05525         AR.SortType = oldtype;
05526     }
05527
05528     t = term + 1; t1 = term + *term; t1 -= ABS(t1[-1]);
05529     while ( t < t1 ) {
05530         if ( *t == AR.PolyFun ) {
05531             t2 = t + t[1]; tt2 = t+FUNHEAD; count3 = 0;
05532             while ( tt2 < t2 ) { count3++; NEXTARG(tt2); }
05533             if ( count3 == 2 ) {
05534                 count2++;
05535             }
05536         }
05537         t += t[1];
05538     }
05539     if ( count1 >= count2 ) {
05540         t = term + 1;
05541         while ( t < t1 ) {
05542             if ( *t == AR.PolyFun ) {
05543                 t2 = t;
05544                 t = t + t[1];
05545                 t2[2] |= (DIRTYFLAG|MUSTCLEANPRF);
05546                 t2 += FUNHEAD;
05547                 while ( t2 < t ) {
05548                     if ( *t2 > 0 ) t2[1] = DIRTYFLAG;
05549                     NEXTARG(t2);
05550                 }
05551             }
05552             else t += t[1];
05553         }
05554     }
05555
05556     w = term + *term;
05557     if ( w > AT.WorkSpace && w < AT.WorkTop ) AT.WorkPointer = w;
05558     checkaction:
05559     if ( action ) retval = action;
05560     return(retval);
05561 }
05562 retry:
05563     if ( term >= AT.WorkSpace && term+*term < AT.WorkTop )
05564         AT.WorkPointer = term + *term;
05565     GETSTOP(term,tstop);
05566     t = term+1;
05567     while ( *t != AR.PolyFun && t < tstop ) t += t[1];
05568     while ( t < tstop && *t == AR.PolyFun ) {
05569         if ( t[1] > FUNHEAD ) {
05570             if ( t[FUNHEAD] < 0 ) {
05571                 if ( t[FUNHEAD] <= -FUNCTION && t[1] == FUNHEAD+1 ) break;
05572                 if ( t[FUNHEAD] > -FUNCTION && t[1] == FUNHEAD+2 ) {
05573                     if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] == 0 ) {
05574                         *term = 0;
05575                         return(0);
05576                     }
05577                     break;
05578                 }
05579             }
05580             else if ( t[FUNHEAD] == t[1] - FUNHEAD ) break;
05581         }
05582         noac = 1;
05583         t += t[1];
05584     }
05585     if ( *t != AR.PolyFun || t >= tstop ) goto done;
05586     fun1 = t;
05587     t += t[1];
05588     while ( t < tstop && *t == AR.PolyFun ) {
05589         if ( t[1] > FUNHEAD ) {
05590             if ( t[FUNHEAD] < 0 ) {
05591                 if ( t[FUNHEAD] <= -FUNCTION && t[1] == FUNHEAD+1 ) break;
05592                 if ( t[FUNHEAD] > -FUNCTION && t[1] == FUNHEAD+2 ) {
05593                     if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] == 0 ) {
05594                         *term = 0;
05595                         return(0);
05596                     }
05597                     break;
05598                 }
05599             }
05600             else if ( t[FUNHEAD] == t[1] - FUNHEAD ) break;
05601         }
05602         noac = 1;
05603         t += t[1];
05604     }
05605     if ( *t != AR.PolyFun || t >= tstop ) goto done;
05606     fun2 = t;
05607 /*
05608     We have two functions of the proper type.
05609     Count terms (needed for the specials)
05610 */

```

```

05611     t = fun1 + FUNHEAD;
05612     if ( *t < 0 ) {
05613         n1 = 1; arg1 = AT.WorkPointer;
05614         ToGeneral(t,arg1,1);
05615         AT.WorkPointer = arg1 + *arg1;
05616     }
05617     else {
05618         t += ARGHEAD;
05619         n1 = 0; t1 = fun1 + fun1[1]; arg1 = t;
05620         while ( t < t1 ) { n1++; t += *t; }
05621     }
05622     t = fun2 + FUNHEAD;
05623     if ( *t < 0 ) {
05624         n2 = 1; arg2 = AT.WorkPointer;
05625         ToGeneral(t,arg2,1);
05626         AT.WorkPointer = arg2 + *arg2;
05627     }
05628     else {
05629         t += ARGHEAD;
05630         n2 = 0; t2 = fun2 + fun2[1]; arg2 = t;
05631         while ( t < t2 ) { n2++; t += *t; }
05632     }
05633     /*
05634     Now we can start the multiplications. We first multiply the terms
05635     without coefficients, then normalize, and finally put the coefficients
05636     in place. This is because one has often truncated series and the
05637     high powers may get killed, while their coefficients are the most
05638     expensive ones.
05639     Note: We may run into fun(-SNUMBER,value)
05640     */
05641     w = AT.WorkPointer;
05642     NewSort(BHEAD0);
05643     if ( AR.PolyFunType == 2 && AR.PolyFunExp == 2 ) {
05644         AT.TrimPower = 1;
05645     /*
05646     We have to find the lowest power in both polynomials.
05647     This will be needed to temporarily correct the AR.PolyFunPow
05648     */
05649     minp1 = MAXPOWER;
05650     for ( t1 = arg1, i1 = 0; i1 < n1; i1++, t1 += *t1 ) {
05651         if ( *t1 == 4 ) {
05652             if ( minp1 > 0 ) minp1 = 0;
05653         }
05654         else if ( ABS(t1[*t1-1]) == (*t1-1) ) {
05655             if ( minp1 > 0 ) minp1 = 0;
05656         }
05657         else {
05658             if ( t1[1] == SYMBOL && t1[2] == 4 && t1[3] == AR.PolyFunVar ) {
05659                 if ( t1[4] < minp1 ) minp1 = t1[4];
05660             }
05661             else {
05662                 MesPrint("Illegal term in expanded polyratfun.");
05663                 goto PolyCall;
05664             }
05665         }
05666     }
05667     minp2 = MAXPOWER;
05668     for ( t2 = arg2, i2 = 0; i2 < n2; i2++, t2 += *t2 ) {
05669         if ( *t2 == 4 ) {
05670             if ( minp2 > 0 ) minp2 = 0;
05671         }
05672         else if ( ABS(t2[*t2-1]) == (*t2-1) ) {
05673             if ( minp2 > 0 ) minp2 = 0;
05674         }
05675         else {
05676             if ( t2[1] == SYMBOL && t2[2] == 4 && t2[3] == AR.PolyFunVar ) {
05677                 if ( t2[4] < minp2 ) minp2 = t2[4];
05678             }
05679             else {
05680                 MesPrint("Illegal term in expanded polyratfun.");
05681                 goto PolyCall;
05682             }
05683         }
05684     }
05685     AR.PolyFunPow += minp1+minp2;
05686 }
05687 for ( t1 = arg1, i1 = 0; i1 < n1; i1++, t1 += *t1 ) {
05688 for ( t2 = arg2, i2 = 0; i2 < n2; i2++, t2 += *t2 ) {
05689     m = w;
05690     m++;
05691     GETSTOP(t1,tt1);
05692     t = t1 + 1;
05693     while ( t < tt1 ) *m++ = *t++;
05694     GETSTOP(t2,tt2);
05695     t = t2+1;
05696     while ( t < tt2 ) *m++ = *t++;
05697     *m++ = 1; *m++ = 1; *m++ = 3; *w = WORDDIF(m,w);

```

```

05698     AT.WorkPointer = m;
05699     if ( Normalize(BHEAD w) ) { LowerSortLevel(); goto PolyCall; }
05700     if ( *w ) {
05701         m = w + *w;
05702         if ( m[-1] != 3 || m[-2] != 1 || m[-3] != 1 ) {
05703             l3 = REDLENG(m[-1]);
05704             m -= ABS(m[-1]);
05705             t = t1 + *t1 - 1;
05706             l1 = REDLENG(*t);
05707             if ( MulRat(BHEAD (UWORD *)m,l3,(UWORD *)t1,l1,(UWORD *)m,&l4) ) {
05708                 LowerSortLevel(); goto PolyCall; }
05709             if ( AN.ncmod != 0 && TakeModulus((UWORD *)m,&l4,AC.cmod,AN.ncmod,UNPACK|AC.modmode) )
{
05710                 LowerSortLevel(); goto PolyCall; }
05711                 if ( l4 == 0 ) continue;
05712                 t = t2 + *t2 - 1;
05713                 l2 = REDLENG(*t);
05714                 if ( MulRat(BHEAD (UWORD *)m,l4,(UWORD *)t2,l2,(UWORD *)m,&l3) ) {
05715                     LowerSortLevel(); goto PolyCall; }
05716                     if ( AN.ncmod != 0 && TakeModulus((UWORD *)m,&l3,AC.cmod,AN.ncmod,UNPACK|AC.modmode) )
{
05717                         LowerSortLevel(); goto PolyCall; }
05718                         }
05719                         else {
05720                             m -= 3;
05721                             t = t1 + *t1 - 1;
05722                             l1 = REDLENG(*t);
05723                             t = t2 + *t2 - 1;
05724                             l2 = REDLENG(*t);
05725                             if ( MulRat(BHEAD (UWORD *)t1,l1,(UWORD *)t2,l2,(UWORD *)m,&l3) ) {
05726                                 LowerSortLevel(); goto PolyCall; }
05727                                 if ( AN.ncmod != 0 && TakeModulus((UWORD *)m,&l3,AC.cmod,AN.ncmod,UNPACK|AC.modmode) )
{
05728                                     LowerSortLevel(); goto PolyCall; }
05729                                     }
05730                                     if ( l3 == 0 ) continue;
05731                                     l3 = INCLENG(l3);
05732                                     m += ABS(l3);
05733                                     m[-1] = l3;
05734                                     *w = WORDDIF(m,w);
05735                                     AT.WorkPointer = m;
05736                                     if ( StoreTerm(BHEAD w) ) { LowerSortLevel(); goto PolyCall; }
05737                                     }
05738                                     }
05739                                     }
05740                                     if ( EndSort(BHEAD w,0) < 0 ) goto PolyCall;
05741                                     AR.PolyFunPow = OldPolyFunPow;
05742                                     AT.TrimPower = 0;
05743                                     if ( *w == 0 ) {
05744                                         *term = 0;
05745                                         return(0);
05746                                     }
05747                                     t = w;
05748                                     while ( *t ) t += *t;
05749                                     AT.WorkPointer = t;
05750                                     n1 = WORDDIF(t,w);
05751                                     t1 = term;
05752                                     while ( t1 < fun1 ) *t++ = *t1++;
05753                                     t2 = t;
05754                                     *t++ = AR.PolyFun;
05755                                     *t++ = FUNHEAD+ARGHEAD+n1;
05756                                     *t++ = 0;
05757                                     FILLFUN3(t)
05758                                     *t++ = ARGHEAD+n1;
05759                                     *t++ = 0;
05760                                     FILLARG(t)
05761                                     NCOPY(t,w,n1);
05762                                     if ( ToFast(t2+FUNHEAD,t2+FUNHEAD) ) {
05763                                         if ( t2[FUNHEAD] > -FUNCTION ) t2[1] = FUNHEAD+2;
05764                                         else t2[FUNHEAD] = FUNHEAD+1;
05765                                         t = t2 + t2[1];
05766                                     }
05767                                     t1 = fun1 + fun1[1];
05768                                     while ( t1 < fun2 ) *t++ = *t1++;
05769                                     t1 = fun2 + fun2[1];
05770                                     t2 = term + *term;
05771                                     while ( t1 < t2 ) *t++ = *t1++;
05772                                     *AT.WorkPointer = n1 = WORDDIF(t,AT.WorkPointer);
05773                                     if ( n1*(LONG)sizeof(WORD) > AM.MaxTer ) {
05774                                         MLOCK(ErrorMessageLock);
05775                                         MesPrint("Term too complex (%d words). MaxTermSize (%l words) is too small.", n1,
AM.MaxTer/(LONG)sizeof(WORD) );
05776                                         goto PolyCall2;
05777                                     }
05778                                     m = term; t = AT.WorkPointer;
05779                                     NCOPY(m,t,n1);
05780                                     action++;

```

```

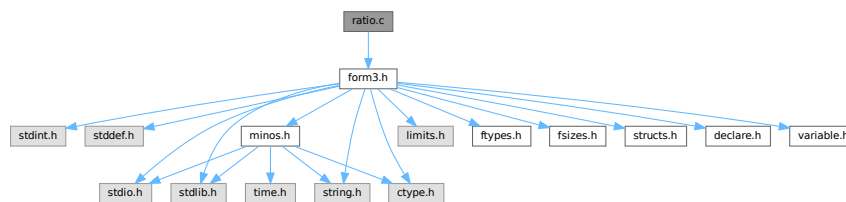
05781     goto retry;
05782 done:
05783     AT.WorkPointer = term + *term;
05784     if ( action && noac ) {
05785         if ( Normalize(BHEAD term) ) goto PolyCall;
05786         AT.WorkPointer = term + *term;
05787     }
05788     return(0);
05789 PolyCall:;
05790     MLOCK(ErrorMessageLock);
05791 PolyCall2:;
05792     AR.PolyFunPow = OldPolyFunPow;
05793     MesCall("PolyFunMul");
05794     MUNLOCK(ErrorMessageLock);
05795     SETERROR(-1)
05796 }
05797
05798 /*
05799     #] PolyFunMul :
05800     #] Processor :
05801 */

```

## 4.116 ratio.c File Reference

```
#include "form3.h"
```

Include dependency graph for ratio.c:



### Data Structures

- struct [ARGBUFFER](#)

### Functions

- int [RatioFind](#) (PHEAD WORD \*term, WORD \*params)
- int [RatioGen](#) (PHEAD WORD \*term, WORD \*params, WORD num, WORD level)
- int [BinomGen](#) (PHEAD WORD \*term, WORD level, WORD \*\*tstops, WORD x1, WORD x2, WORD pow1, WORD pow2, WORD sign, UWORD \*coef, WORD ncoef)
- int [DoSumF1](#) (PHEAD WORD \*term, WORD \*params, WORD replac, WORD level)
- int [Glue](#) (PHEAD WORD \*term1, WORD \*term2, WORD \*sub, WORD insert)
- int [DoSumF2](#) (PHEAD WORD \*term, WORD \*params, WORD replac, WORD level)
- int [GCDfunction](#) (PHEAD WORD \*term, WORD level)
- WORD \* [GCDfunction3](#) (PHEAD WORD \*in1, WORD \*in2)
- WORD \* [PutExtraSymbols](#) (PHEAD WORD \*in, WORD startbuf, int \*actionflag)
- WORD \* [TakeExtraSymbols](#) (PHEAD WORD \*in, WORD startbuf)
- WORD \* [MultiplyWithTerm](#) (PHEAD WORD \*in, WORD \*term, WORD par)
- WORD \* [TakeContent](#) (PHEAD WORD \*in, WORD \*term)
- int [MergeSymbolLists](#) (PHEAD WORD \*old, WORD \*extra, int par)
- int [MergeDotproductLists](#) (PHEAD WORD \*old, WORD \*extra, int par)

- WORD \* [CreateExpression](#) (PHEAD WORD nexp)
- int [GCDterms](#) (PHEAD WORD \*term1, WORD \*term2, WORD \*termout)
- int [ReadPolyRatFun](#) (PHEAD WORD \*term)
- int [FromPolyRatFun](#) (PHEAD WORD \*fun, WORD \*\*numout, WORD \*\*denout)
- WORD \* [TakeSymbolContent](#) (PHEAD WORD \*in, WORD \*term)
- void [GCDclean](#) (PHEAD WORD \*num, WORD \*den)
- WORD \* [PolyDiv](#) (PHEAD WORD \*a, WORD \*b, char \*text)
- int [DIVfunction](#) (PHEAD WORD \*term, WORD level, int par)
- WORD \* [MULfunc](#) (PHEAD WORD \*p1, WORD \*p2)
- WORD \* [ConvertArgument](#) (PHEAD WORD \*arg, int \*type)
- int [ExpandRat](#) (PHEAD WORD \*fun)
- int [InvPoly](#) (PHEAD WORD \*inpoly, WORD maxpow, WORD sym)

## Variables

- WORD [divrem](#) [4] = { DIVFUNCTION, REMFUNCTION, INVERSEFUNCTION, MULFUNCTION }
- char \* [TheErrorMessage](#) []

## 4.116.1 Detailed Description

A variety of routines: The ratio command for partial fractioning (rather old. Schoonschip inheritance) The sum routines.

Definition in file [ratio.c](#).

## 4.116.2 Function Documentation

### 4.116.2.1 RatioFind()

```
int RatioFind (
    PHEAD WORD * term,
    WORD * params )
```

Definition at line 62 of file [ratio.c](#).

### 4.116.2.2 RatioGen()

```
int RatioGen (
    PHEAD WORD * term,
    WORD * params,
    WORD num,
    WORD level )
```

Definition at line 170 of file [ratio.c](#).

#### 4.116.2.3 BinomGen()

```
int BinomGen (
    PHEAD WORD * term,
    WORD level,
    WORD ** tstops,
    WORD x1,
    WORD x2,
    WORD pow1,
    WORD pow2,
    WORD sign,
    UWORD * coef,
    WORD ncoef )
```

Definition at line 320 of file [ratio.c](#).

#### 4.116.2.4 DoSumF1()

```
int DoSumF1 (
    PHEAD WORD * term,
    WORD * params,
    WORD replac,
    WORD level )
```

Definition at line 395 of file [ratio.c](#).

#### 4.116.2.5 Glue()

```
int Glue (
    PHEAD WORD * term1,
    WORD * term2,
    WORD * sub,
    WORD insert )
```

Definition at line 461 of file [ratio.c](#).

#### 4.116.2.6 DoSumF2()

```
int DoSumF2 (
    PHEAD WORD * term,
    WORD * params,
    WORD replac,
    WORD level )
```

Definition at line 520 of file [ratio.c](#).

#### 4.116.2.7 GCDfunction()

```
int GCDfunction (
    PHEAD WORD * term,
    WORD level )
```

Definition at line 609 of file [ratio.c](#).



#### 4.116.2.8 GCDfunction3()

```
WORD * GCDfunction3 (
    PHEAD WORD * in1,
    WORD * in2 )
```

Definition at line 1141 of file [ratio.c](#).

#### 4.116.2.9 PutExtraSymbols()

```
WORD * PutExtraSymbols (
    PHEAD WORD * in,
    WORD startebuf,
    int * actionflag )
```

Definition at line 1244 of file [ratio.c](#).

#### 4.116.2.10 TakeExtraSymbols()

```
WORD * TakeExtraSymbols (
    PHEAD WORD * in,
    WORD startebuf )
```

Definition at line 1273 of file [ratio.c](#).

#### 4.116.2.11 MultiplyWithTerm()

```
WORD * MultiplyWithTerm (
    PHEAD WORD * in,
    WORD * term,
    WORD par )
```

Definition at line 1312 of file [ratio.c](#).

#### 4.116.2.12 TakeContent()

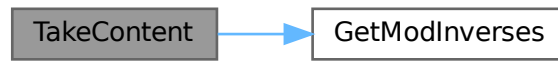
```
WORD * TakeContent (
    PHEAD WORD * in,
    WORD * term )
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. Here the input is a sequence of terms in 'in' and the answer is a content-free sequence of terms. This sequence has been allocated by the Malloc1 routine in a call to EndSort, unless the expression was already content-free. In that case the input pointer is returned. The content is returned in term. This is supposed to be a separate allocation, made by TermMalloc in the calling routine.

Definition at line 1376 of file [ratio.c](#).

References [GetModInverses\(\)](#).

Here is the call graph for this function:



#### 4.116.2.13 MergeSymbolLists()

```
int MergeSymbolLists (
    PHEAD WORD * old,
    WORD * extra,
    int par )
```

Definition at line 1808 of file [ratio.c](#).

#### 4.116.2.14 MergeDotproductLists()

```
int MergeDotproductLists (
    PHEAD WORD * old,
    WORD * extra,
    int par )
```

Definition at line 1927 of file [ratio.c](#).

#### 4.116.2.15 CreateExpression()

```
WORD * CreateExpression (
    PHEAD WORD nexp )
```

Definition at line 2020 of file [ratio.c](#).

#### 4.116.2.16 GCDterms()

```
int GCDterms (
    PHEAD WORD * term1,
    WORD * term2,
    WORD * termout )
```

Definition at line 2076 of file [ratio.c](#).

#### 4.116.2.17 ReadPolyRatFun()

```
int ReadPolyRatFun (
    PHEAD WORD * term )
```

Definition at line 2264 of file [ratio.c](#).

#### 4.116.2.18 FromPolyRatFun()

```
int FromPolyRatFun (
    PHEAD WORD * fun,
    WORD ** numout,
    WORD ** denout )
```

Definition at line 2383 of file [ratio.c](#).

#### 4.116.2.19 TakeSymbolContent()

```
WORD * TakeSymbolContent (
    PHEAD WORD * in,
    WORD * term )
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. We allow only symbols as this code is used for the polyratfun only. We have a special routine, because the generic TakeContent does too much work and speed is at a premium here. Input: *in* is the input expression as a sequence of terms. **Output:** *term*: the content return value: the contentfree expression. it is in new allocation, made by TermMalloc. (should be in a TermMalloc space?)

Definition at line 2434 of file [ratio.c](#).

References [GetModInverses\(\)](#).

Here is the call graph for this function:



#### 4.116.2.20 GCDclean()

```
void GCDclean (
    PHEAD WORD * num,
    WORD * den )
```

Definition at line 2644 of file [ratio.c](#).

#### 4.116.2.21 PolyDiv()

```
WORD * PolyDiv (
    PHEAD WORD * a,
    WORD * b,
    char * text )
```

Definition at line [2741](#) of file [ratio.c](#).

#### 4.116.2.22 DIVfunction()

```
int DIVfunction (
    PHEAD WORD * term,
    WORD level,
    int par )
```

Definition at line [2788](#) of file [ratio.c](#).

#### 4.116.2.23 MULfunc()

```
WORD * MULfunc (
    PHEAD WORD * p1,
    WORD * p2 )
```

Definition at line [2957](#) of file [ratio.c](#).

#### 4.116.2.24 ConvertArgument()

```
WORD * ConvertArgument (
    PHEAD WORD * arg,
    int * type )
```

Definition at line [3018](#) of file [ratio.c](#).

#### 4.116.2.25 ExpandRat()

```
int ExpandRat (
    PHEAD WORD * fun )
```

Definition at line [3113](#) of file [ratio.c](#).

#### 4.116.2.26 InvPoly()

```
int InvPoly (
    PHEAD WORD * inpoly,
    WORD maxpow,
    WORD sym )
```

Definition at line [3592](#) of file [ratio.c](#).

## 4.116.3 Variable Documentation

### 4.116.3.1 divrem

WORD divrem[4] = { DIVFUNCTION, REMFUNCTION, INVERSEFUNCTION, MULFUNCTION }

Definition at line 2786 of file [ratio.c](#).

### 4.116.3.2 TheErrorMessage

char\* TheErrorMessage[ ]

#### Initial value:

```
= {
    "PolyRatFun not of a type that FORM will expand: incorrect variable inside."
    , "Division by zero in PolyRatFun encountered in ExpandRat."
    , "Irregular code in PolyRatFun encountered in ExpandRat."
    , "Called from ExpandRat."
    , "WorkSpace overflow. Change parameter WorkSpace in setup file?"
}
```

Definition at line 3105 of file [ratio.c](#).

## 4.117 ratio.c

[Go to the documentation of this file.](#)

```
00001
00008 /* # [ License : */
00009 /*
00010 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00011 * When using this file you are requested to refer to the publication
00012 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00013 * This is considered a matter of courtesy as the development was paid
00014 * for by FOM the Dutch physics granting agency and we would like to
00015 * be able to track its scientific use to convince FOM of its value
00016 * for the community.
00017 *
00018 * This file is part of FORM.
00019 *
00020 * FORM is free software: you can redistribute it and/or modify it under the
00021 * terms of the GNU General Public License as published by the Free Software
00022 * Foundation, either version 3 of the License, or (at your option) any later
00023 * version.
00024 *
00025 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00026 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00027 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00028 * details.
00029 *
00030 * You should have received a copy of the GNU General Public License along
00031 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00032 */
00033 /* # [ License : */
00034 /*
00035 * # [ Includes : ratio.c
00036 */
00037
00038 #include "form3.h"
00039
00040 /*
00041 * # [ Includes :
00042 * # [ Ratio :
00043
00044 * These are the special operations regarding simple polynomials.
00045 * The first and most needed is the partial fractioning expansion.
00046 * Ratio,x1,x2,x3
00047
00048 * The files belonging to the ratio command serve also as a good example
00049 * of how to implement a new operation.
00050
```

```

00051      # [ RatioFind :
00052
00053      The routine that should locate the need for a ratio command.
00054      If located the corresponding symbols are removed and the
00055      operational parameters are loaded. A subexpression pointer
00056      is inserted and the code for success is returned.
00057
00058      params points at the compiler output block defined in RatioComp.
00059
00060  */
00061
00062  int RatioFind(PHEAD WORD *term, WORD *params)
00063  {
00064      GETBIDENTITY
00065      WORD *t, *m, *r;
00066      WORD x1, x2, i;
00067      WORD *y1, *y2, n1 = 0, n2 = 0;
00068      x1 = params[3];
00069      x2 = params[4];
00070      m = t = term;
00071      m += *m;
00072      m -= ABS(m[-1]);
00073      t++;
00074      if ( t < m ) do {
00075          if ( *t == SYMBOL ) {
00076              y1 = 0;
00077              y2 = 0;
00078              r = t + t[1];
00079              m = t + 2;
00080              do {
00081                  if ( *m == x1 ) { y1 = m; n1 = m[1]; }
00082                  else if ( *m == x2 ) { y2 = m; n2 = m[1]; }
00083                  m += 2;
00084              } while ( m < r );
00085              if ( !y1 || !y2 || ( n1 > 0 && n2 > 0 ) ) return(0);
00086              m -= 2;
00087              if ( y1 > y2 ) { r = y1; y1 = y2; y2 = r; }
00088              *y2 = *m; y2[1] = m[1];
00089              m -= 2;
00090              *y1 = *m; y1[1] = m[1];
00091              i = WORDDIFF(m,t);
00092              #if SUBEXPSIZE > 6
00093              We have to revise the code for the second case.
00094              #endif
00095              if ( i > 2 ) {          /* Subexpression fits exactly */
00096                  t[1] = i;
00097                  y1 = term+*term;
00098                  y2 = y1+SUBEXPSIZE-4;
00099                  r = m+4;
00100                  while ( y1 > r ) --y2 = --y1;
00101                  *m++ = SUBEXPRESSION;
00102                  *m++ = SUBEXPSIZE;
00103                  *m++ = -1;
00104                  *m++ = 1;
00105                  *m++ = DUMMYBUFFER;
00106                  FILLSUB(m)
00107                  *term += SUBEXPSIZE-4;
00108              }
00109              else {                  /* All symbols are gone. Rest has to be moved */
00110                  m -= 2;
00111                  *m++ = SUBEXPRESSION;
00112                  *m++ = SUBEXPSIZE;
00113                  *m++ = -1;
00114                  *m++ = 1;
00115                  *m++ = DUMMYBUFFER;
00116                  FILLSUB(m)
00117                  t = term;
00118                  t += *t;
00119                  *term += SUBEXPSIZE-6;
00120                  r = m + 6-SUBEXPSIZE;
00121                  do { *m++ = *r++; } while ( r < t );
00122              }
00123              t = AT.TMout;          /* Load up the TM out array for the generator */
00124              *t++ = 7;
00125              *t++ = RATIO;
00126              *t++ = x1;
00127              *t++ = x2;
00128              *t++ = params[5];
00129              *t++ = n1;
00130              *t++ = n2;
00131              return(1);
00132          }
00133          t += t[1];
00134      } while ( t < m );
00135      return(0);
00136  }
00137

```

```

00138 /*
00139     #] RatioFind :
00140     #[ RatioGen :
00141
00142     The algorithm:
00143     x1^-n1*x2^n2 ==> x2 --> x1 + x3
00144     x1^n1*x2^-n2 ==> x1 --> x2 - x3
00145     x1^-n1*x2^-n2 ==>
00146
00147         +sum(i=0,n1-1){(-1)^i*binom(n2-1+i,n2-1)
00148             *x3^-(n2+i)*x1^-(n1-i)}
00149         +sum(i=0,n2-1){(-1)^(n1)*binom(n1-1+i,n1-1)
00150             *x3^-(n1+i)*x2^-(n2-i)}
00151
00152     Actually there is an amount of arbitrariness in the first two
00153     formulae and the replacement x2 -> x1 + x3 could be made 'by hand'.
00154     It is better to use the nontrivial 'minimal change' formula:
00155
00156     x1^-n1*x2^n2:   if ( n1 >= n2 ) {
00157                     +sum(i=0,n2){x3^i*x1^-(n1-n2+i)*binom(n2,i)}
00158                     }
00159                     else {
00160                         sum(i=0,n2-n1){x2^(n2-n1-i)*x3^i*binom(n1-1+i,n1-1)}
00161                         +sum(i=0,n1-1){x3^(n2-i)*x1^-(n1-i)*binom(n2,i)}
00162                     }
00163     x1^n1*x2^-n2:   Same but x3 -> -x3.
00164
00165     The contents of the AT.TMout/params array are:
00166     length,type,x1,x2,x3,n1,n2
00167
00168 */
00169
00170 int RatioGen(PHEAD WORD *term, WORD *params, WORD num, WORD level)
00171 {
00172     GETBIDENTITY
00173     WORD *t, *m;
00174     WORD *tstops[3];
00175     WORD n1, n2, i, j;
00176     WORD x1,x2,x3;
00177     UWORD *coef;
00178     WORD ncoef, sign = 0;
00179     coef = (UWORD *)AT.WorkPointer;
00180     t = term;
00181     tstops[2] = m = t + *t;
00182     m -= ABS(m[-1]);
00183     t++;
00184     do {
00185         if ( *t == SUBEXPRESSION && t[2] == num ) break;
00186         t += t[1];
00187     } while ( t < m );
00188     tstops[0] = t;
00189     tstops[1] = t + t[1];
00190
00191     /* Copying to termout will be from term to tstop1, then the induced part
00192     and finally from tstop2 to tstop3
00193
00194     Now separate the various cases:
00195
00196 */
00197     t = params + 2;
00198     x1 = *t++;
00199     x2 = *t++;
00200     x3 = *t++;
00201     n1 = *t++;
00202     n2 = *t++;
00203     if ( n1 > 0 ) { /* Flip the variables and indicate -x3 */
00204         n2 = -n2;
00205         sign = 1;
00206         i = n1; n1 = n2; n2 = i;
00207         i = x1; x1 = x2; x2 = i;
00208         goto PosNeg;
00209     }
00210     else if ( n2 > 0 ) {
00211         n1 = -n1;
00212 PosNeg:
00213         if ( n2 <= n1 ) { /* x1 -> x2 + x3 */
00214             *coef = 1;
00215             ncoef = 1;
00216             AT.WorkPointer = (WORD *) (coef + 1);
00217             j = n2;
00218             for ( i = 0; i <= n2; i++ ) {
00219                 if ( BinomGen(BHEAD term,level,tstops,x1,x3,n2-n1-i,i,sign&i
00220                     ,coef,ncoef) ) goto RatioCall;
00221                 if ( i < n2 ) {
00222                     if ( Product(coef,&ncoef,j) ) goto RatioCall;
00223                     if ( Quotient(coef,&ncoef,i+1) ) goto RatioCall;
00224                     j--;

```

```

00225         AT.WorkPointer = (WORD *) (coef + ABS(ncoef));
00226     }
00227 }
00228 AT.WorkPointer = (WORD *) (coef);
00229 return(0);
00230 }
00231 else {
00232 /*
00233     sum(i=0,n2-n1){x2^(n2-n1-i)*x3^i*binom(n1-1+i,n1-1)}
00234 +sum(i=0,n1-1){x3^(n2-i)*x1^(n1-i)*binom(n2,i)}
00235 */
00236     *coef = 1;
00237     ncoef = 1;
00238     AT.WorkPointer = (WORD *) (coef + 1);
00239     j = n2 - n1;
00240     for ( i = 0; i <= j; i++ ) {
00241         if ( BinomGen(BHEAD term,level,tstops,x2,x3,n2-n1-i,i,sign&i
00242             ,coef,ncoef) ) goto RatioCall;
00243         if ( i < j ) {
00244             if ( Product(coef,&ncoef,n1+i) ) goto RatioCall;
00245             if ( Quotient(coef,&ncoef,i+1) ) goto RatioCall;
00246             AT.WorkPointer = (WORD *) (coef + ABS(ncoef));
00247         }
00248     }
00249     *coef = 1;
00250     ncoef = 1;
00251     AT.WorkPointer = (WORD *) (coef + 1);
00252     j = n1-1;
00253     for ( i = 0; i <= j; i++ ) {
00254         if ( BinomGen(BHEAD term,level,tstops,x1,x3,i-n1,n2-i,sign&(n2-i)
00255             ,coef,ncoef) ) goto RatioCall;
00256         if ( i < j ) {
00257             if ( Product(coef,&ncoef,n2-i) ) goto RatioCall;
00258             if ( Quotient(coef,&ncoef,i+1) ) goto RatioCall;
00259             AT.WorkPointer = (WORD *) (coef + ABS(ncoef));
00260         }
00261     }
00262     AT.WorkPointer = (WORD *) (coef);
00263     return(0);
00264 }
00265 }
00266 else {
00267     n2 = -n2;
00268     n1 = -n1;
00269 /*
00270     +sum(i=0,n1-1){(-1)^i*binom(n2-1+i,n2-1)
00271     *x3^(n2+i)*x1^(n1-i)}
00272 +sum(i=0,n2-1){(-1)^(n1)*binom(n1-1+i,n1-1)
00273     *x3^(n1+i)*x2^(n2-i)}
00274 */
00275     *coef = 1;
00276     ncoef = 1;
00277     AT.WorkPointer = (WORD *) (coef + 1);
00278     j = n1-1;
00279     for ( i = 0; i <= j; i++ ) {
00280         if ( BinomGen(BHEAD term,level,tstops,x1,x3,i-n1,-n2-i,i&1
00281             ,coef,ncoef) ) goto RatioCall;
00282         if ( i < j ) {
00283             if ( Product(coef,&ncoef,n2+i) ) goto RatioCall;
00284             if ( Quotient(coef,&ncoef,i+1) ) goto RatioCall;
00285             AT.WorkPointer = (WORD *) (coef + ABS(ncoef));
00286         }
00287     }
00288     *coef = 1;
00289     ncoef = 1;
00290     AT.WorkPointer = (WORD *) (coef + 1);
00291     j = n2-1;
00292     for ( i = 0; i <= j; i++ ) {
00293         if ( BinomGen(BHEAD term,level,tstops,x2,x3,i-n2,-n1-i,n1&1
00294             ,coef,ncoef) ) goto RatioCall;
00295         if ( i < j ) {
00296             if ( Product(coef,&ncoef,n1+i) ) goto RatioCall;
00297             if ( Quotient(coef,&ncoef,i+1) ) goto RatioCall;
00298             AT.WorkPointer = (WORD *) (coef + ABS(ncoef));
00299         }
00300     }
00301     AT.WorkPointer = (WORD *) (coef);
00302     return(0);
00303 }
00304 }
00305 RatioCall:
00306     MLOCK(ErrorMessageLock);
00307     MesCall("RatioGen");
00308     MUNLOCK(ErrorMessageLock);
00309     SETERROR(-1)
00310 }
00311

```



```

00312 /*
00313     #] RatioGen :
00314     #[ BinomGen :
00315
00316     Routine for the generation of terms in a binomialtype expansion.
00317
00318 */
00319
00320 int BinomGen(PHEAD WORD *term, WORD level, WORD **tstops, WORD x1, WORD x2,
00321             WORD pow1, WORD pow2, WORD sign, UWORD *coef, WORD ncoef)
00322 {
00323     GETBIDENTITY
00324     WORD *t, *r;
00325     WORD *termout;
00326     WORD k;
00327     termout = AT.WorkPointer;
00328     t = termout;
00329     r = term;
00330     do { *t++ = *r++; } while ( r < tstops[0] );
00331     *t++ = SYMBOL;
00332     if ( pow2 == 0 ) {
00333         if ( pow1 == 0 ) t--;
00334         else { *t++ = 4; *t++ = x1; *t++ = pow1; }
00335     }
00336     else if ( pow1 == 0 ) {
00337         *t++ = 4; *t++ = x2; *t++ = pow2;
00338     }
00339     else {
00340         *t++ = 6; *t++ = x1; *t++ = pow1; *t++ = x2; *t++ = pow2;
00341     }
00342     *t++ = LNUMBER;
00343     *t++ = ABS(ncoef) + 3;
00344     *t = ncoef;
00345     if ( sign ) *t = -*t;
00346     t++;
00347     ncoef = ABS(ncoef);
00348     for ( k = 0; k < ncoef; k++ ) *t++ = coef[k];
00349     r = tstops[1];
00350     do { *t++ = *r++; } while ( r < tstops[2] );
00351     *termout = WORDDIF(t,termout);
00352     AT.WorkPointer = t;
00353     if ( AT.WorkPointer > AT.WorkTop ) {
00354         MLOCK(ErrorMessageLock);
00355         MesWork();
00356         MUNLOCK(ErrorMessageLock);
00357         return(-1);
00358     }
00359     *AN.RepPoint = 1;
00360     AR.expchanged = 1;
00361     if ( Generator(BHEAD termout,level) ) {
00362         MLOCK(ErrorMessageLock);
00363         MesCall("BinomGen");
00364         MUNLOCK(ErrorMessageLock);
00365         SETERROR(-1)
00366     }
00367     AT.WorkPointer = termout;
00368     return(0);
00369 }
00370
00371 /*
00372     #] BinomGen :
00373     #] Ratio :
00374     #[ Sum :
00375     #[ DoSumF1 :
00376
00377     Routine expands a sum_ function.
00378     Its arguments are:
00379     The term in which the function occurs.
00380     The parameter list:
00381         length of parameter field
00382         function number (SUMNUM1)
00383         number of the symbol that is loop parameter
00384         min value
00385         max value
00386         increment
00387     the number of the subexpression to be removed
00388     the level in the generation tree.
00389
00390     Note that the insertion of the loop parameter in the argument
00391     is done via the regular wildcard substitution mechanism.
00392
00393 */
00394
00395 int DoSumF1(PHEAD WORD *term, WORD *params, WORD replac, WORD level)
00396 {
00397     GETBIDENTITY
00398     WORD *termout, *t, extractbuff = AT.TMbuff;

```

```

00399     WORD isum, ival, iinc;
00400     LONG from;
00401     CBUF *C;
00402     ival = params[3];
00403     iinc = params[5];
00404     if ( ( iinc > 0 && params[4] >= ival )
00405         || ( iinc < 0 && params[4] <= ival ) ) {
00406         isum = (params[4] - ival)/iinc + 1;
00407     }
00408     else return(0);
00409     termout = AT.WorkPointer;
00410     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
00411     if ( AT.WorkPointer > AT.WorkTop ) {
00412         MLOCK(ErrorMessageLock);
00413         MesWork();
00414         MUNLOCK(ErrorMessageLock);
00415         return(-1);
00416     }
00417     t = term + 1;
00418     while ( *t != SUBEXPRESSION || t[2] != replac || t[4] != extractbuff )
00419         t += t[1];
00420     C = cbuf+t[4];
00421     t += SUBEXPSIZE;
00422     if ( params[2] < 0 ) {
00423         while ( *t != INDTOIND || t[2] != -params[2] ) t += t[1];
00424         *t = INDTOIND;
00425     }
00426     else {
00427         while ( *t > SYMTOSUB || t[2] != params[2] ) t += t[1];
00428         *t = SYMTONUM;
00429     }
00430     do {
00431         t[3] = ival;
00432         from = C->rhs[replac] - C->Buffer;
00433         while ( C->Buffer[from] ) {
00434             if ( InsertTerm(BHEAD term,replac,extractbuff,C->Buffer+from,termout,0) < 0 ) goto
SumFlCall;
00435             AT.WorkPointer = termout + *termout;
00436             if ( Generator(BHEAD termout,level) < 0 ) goto SumFlCall;
00437             from += C->Buffer[from];
00438         }
00439         ival += iinc;
00440     } while ( --isum > 0 );
00441     AT.WorkPointer = termout;
00442     return(0);
00443 SumFlCall:
00444     MLOCK(ErrorMessageLock);
00445     MesCall("DoSumFl");
00446     MUNLOCK(ErrorMessageLock);
00447     SETERROR(-1)
00448 }
00449
00450 /*
00451     #] DoSumFl :
00452     #[ Glue :
00453
00454     Routine multiplies two terms. The second term is subject
00455     to the wildcard substitutions in sub.
00456     Output in the first term. This routine is a variation on
00457     the routine InsertTerm.
00458
00459 */
00460
00461 int Glue(PHEAD WORD *term1, WORD *term2, WORD *sub, WORD insert)
00462 {
00463     GETBIDENTITY
00464     UWORD *coef;
00465     WORD ncoef, *t, *t1, *t2, i, nc2, nc3, old, newer;
00466     coef = (UWORD *) (TermMalloc("Glue"));
00467     t = term1;
00468     t += *t;
00469     i = t[-1];
00470     t -= ABS(i);
00471     old = WORDDIF(t,term1);
00472     ncoef = REDLENG(i);
00473     if ( i < 0 ) i = -i;
00474     i--;
00475     t1 = t;
00476     t2 = (WORD *)coef;
00477     while ( --i >= 0 ) *t2++ = *t1++;
00478     i = *--t;
00479     nc2 = WildFill(BHEAD t,term2,sub);
00480     *t = i;
00481     t += nc2;
00482     nc2 = t[-1];
00483     t -= ABS(nc2);
00484     newer = WORDDIF(t,term1);

```

```

00485     if ( MulRat (BHEAD (UWORD *)t, REDLENG(nc2), coef, ncoef, (UWORD *)t, &nc3) ) {
00486         MLOCK(ErrorMessageLock);
00487         MesCall("Glue");
00488         MUNLOCK(ErrorMessageLock);
00489         TermFree(coef, "Glue");
00490         SETERROR(-1)
00491     }
00492     i = (ABS(nc3))*2;
00493     t += i++;
00494     *t++ = (nc3 >= 0)?i:-i;
00495     *term1 = WORDDIF(t, term1);
00496 /*
00497     Switch the new piece with the old tail, so that noncommuting
00498     variables get into their proper spot.
00499 */
00500     i = old - insert;
00501     t1 = t;
00502     t2 = term1+insert;
00503     NCOPY(t1, t2, i);
00504     i = newer - old;
00505     t1 = term1+insert;
00506     t2 = term1+old;
00507     NCOPY(t1, t2, i);
00508     t2 = t;
00509     i = old - insert;
00510     NCOPY(t1, t2, i);
00511     TermFree(coef, "Glue");
00512     return(0);
00513 }
00514
00515 /*
00516     #] Glue :
00517     #[ DoSumF2 :
00518 */
00519
00520 int DoSumF2(PHEAD WORD *term, WORD *params, WORD replac, WORD level)
00521 {
00522     GETBIDENTITY
00523     WORD *termout, *t, *from, *sub, *to, extractbuff = AT.TMbuff;
00524     WORD isum, ival, iinc, insert, i;
00525     CBUF *C;
00526     ival = params[3];
00527     iinc = params[5];
00528     if ( ( iinc > 0 && params[4] >= ival )
00529         || ( iinc < 0 && params[4] <= ival ) ) {
00530         isum = (params[4] - ival)/iinc + 1;
00531     }
00532     else return(0);
00533     termout = AT.WorkPointer;
00534     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
00535     if ( AT.WorkPointer > AT.WorkTop ) {
00536         MLOCK(ErrorMessageLock);
00537         MesWork();
00538         MUNLOCK(ErrorMessageLock);
00539         return(-1);
00540     }
00541     t = term + 1;
00542     while ( *t != SUBEXPRESSION || t[2] != replac || t[4] != extractbuff ) t += t[1];
00543     insert = WORDDIF(t, term);
00544
00545     from = term;
00546     to = termout;
00547     while ( from < t ) *to++ = *from++;
00548     from += t[1];
00549     sub = term + *term;
00550     while ( from < sub ) *to++ = *from++;
00551     *termout -= t[1];
00552
00553     sub = t;
00554     C = cbuf+t[4];
00555     t += SUBEXPSIZE;
00556     if ( params[2] < 0 ) {
00557         while ( *t != INDTOIND || t[2] != -params[2] ) t += t[1];
00558         *t = INDTOIND;
00559     }
00560     else {
00561         while ( *t > SYMTOSUB || t[2] != params[2] ) t += t[1];
00562         *t = SYMTONUM;
00563     }
00564     t[3] = ival;
00565     for(;;) {
00566         AT.WorkPointer = termout + *termout;
00567         to = AT.WorkPointer;
00568         if ( ( to + *termout ) > AT.WorkTop ) {
00569             MLOCK(ErrorMessageLock);
00570             MesWork();
00571             MUNLOCK(ErrorMessageLock);

```

```

00572         return(-1);
00573     }
00574     from = termout;
00575     i = *termout;
00576     NCOPY(to, from, i);
00577     from = AT.WorkPointer;
00578     AT.WorkPointer = to;
00579     if ( Generator(BHEAD from, level) < 0 ) goto SumF2Call;
00580     if ( --isum <= 0 ) break;
00581     ival += iinc;
00582     t[3] = ival;
00583     if ( Glue(BHEAD termout, C->rhs[replac], sub, insert) < 0 ) goto SumF2Call;
00584 }
00585 AT.WorkPointer = termout;
00586 return(0);
00587 SumF2Call:
00588 MLOCK(ErrorMessageLock);
00589 MesCall("DoSumF2");
00590 MUNLOCK(ErrorMessageLock);
00591 SETERROR(-1)
00592 }
00593
00594 /*
00595     #] DoSumF2 :
00596     #] Sum :
00597     #[ GCDfunction :
00598     #[ GCDfunction :
00599 */
00600
00601 typedef struct {
00602     WORD *buffer;
00603     DOLLARS dollar;
00604     LONG size;
00605     int type;
00606     int dummy;
00607 } ARGBUFFER;
00608
00609 int GCDfunction(PHEAD WORD *term, WORD level)
00610 {
00611     GETBIDENTITY
00612     WORD *t, *tstop, *tf, *termout, *tin, *tout, *m, *mnext, *mstop, *mm;
00613     int todo, i, ii, j, istart, sign = 1, action = 0;
00614     WORD firstshort = 0, firstvalue = 0, gcdisone = 0, mlength, tlength, newlength;
00615     WORD totargs = 0, numargs, argsdone = 0, *mh, oldvall, *g, *gcdout = 0;
00616     WORD *arg1, *arg2;
00617     UWORD x1, x2, x3;
00618     LONG args;
00619     #if ( FUNHEAD > 4 )
00620     WORD sh[FUNHEAD+5];
00621     #else
00622     WORD sh[9];
00623     #endif
00624     DOLLARS d;
00625     ARGBUFFER *abuf = 0, ab;
00626 /*
00627     #[ Find Function. Count arguments :
00628
00629     First find the proper function
00630 */
00631     t = term + *term; tlength = t[-1];
00632     tstop = t - ABS(tlength);
00633     t = term + 1;
00634     while ( t < tstop ) {
00635         if ( *t != GCDFUNCTION ) { t += t[1]; continue; }
00636         todo = 1; totargs = 0;
00637         tf = t + FUNHEAD;
00638         while ( tf < t + t[1] ) {
00639             totargs++;
00640             if ( *tf > 0 && tf[1] != 0 ) todo = 0;
00641             NEXTARG(tf);
00642         }
00643         if ( todo ) break;
00644         t += t[1];
00645     }
00646     if ( t >= tstop ) {
00647         MLOCK(ErrorMessageLock);
00648         MesPrint("Internal error. Indicated gcd_ function not encountered.");
00649         MUNLOCK(ErrorMessageLock);
00650         Terminate(-1);
00651     }
00652     WantAddPointers(totargs);
00653     args = AT.pWorkPointer; AT.pWorkPointer += totargs;
00654 /*
00655     #] Find Function. Count arguments :
00656     #[ Do short arguments :
00657
00658     The function we need, in agreement with TestSub, is now in t

```

```

00659     Make first a compilation of the short arguments (except $-s and expressions)
00660     to see whether we need to do much work.
00661     This means that after this scan we can ignore all short arguments with
00662     the exception of unevaluated $-s and expressions.
00663  */
00664     numargs = 0;
00665     firstshort = 0;
00666     tf = t + FUNHEAD;
00667     while ( tf < t + t[1] ) {
00668         if ( *tf == -SNUMBER && tf[1] == 0 ) { NEXTARG(tf); continue; }
00669         if ( *tf > 0 || *tf == -DOLLAREXPRESSION || *tf == -EXPRESSION ) {
00670             AT.pWorkspace[args+numargs++] = tf;
00671             NEXTARG(tf); continue;
00672         }
00673         if ( firstshort == 0 ) {
00674             firstshort = *tf;
00675             if ( *tf <= -FUNCTION ) { firstvalue = -( *tf ); }
00676             else { firstvalue = tf[1]; }
00677             NEXTARG(tf);
00678             argsdone++;
00679             continue;
00680         }
00681         else if ( *tf != firstshort ) {
00682             if ( *tf != -INDEX && *tf != -VECTOR && *tf != -MINVECTOR ) {
00683                 argsdone++; gcdisone = 1; break;
00684             }
00685             if ( firstshort != -INDEX && firstshort != -VECTOR && firstshort != -MINVECTOR ) {
00686                 argsdone++; gcdisone = 1; break;
00687             }
00688             if ( tf[1] != firstvalue ) {
00689                 argsdone++; gcdisone = 1; break;
00690             }
00691             if ( *t == -MINVECTOR ) { firstshort = -VECTOR; }
00692             if ( firstshort == -MINVECTOR ) { firstshort = -VECTOR; }
00693         }
00694         else if ( *tf > -FUNCTION && *tf != -SNUMBER && tf[1] != firstvalue ) {
00695             argsdone++; gcdisone = 1; break;
00696         }
00697         if ( *tf == -SNUMBER && firstvalue != tf[1] ) {
00698             /*
00699                 make a new firstvalue which is gcd_(firstvalue,tf[1])
00700             */
00701             if ( firstvalue == 1 || tf[1] == 1 ) { gcdisone = 1; break; }
00702             if ( firstvalue < 0 && tf[1] < 0 ) {
00703                 x1 = -firstvalue; x2 = -tf[1]; sign = -1;
00704             }
00705             else {
00706                 x1 = ABS(firstvalue); x2 = ABS(tf[1]); sign = 1;
00707             }
00708             while ( ( x3 = x1%x2 ) != 0 ) { x1 = x2; x2 = x3; }
00709             firstvalue = ((WORD)x2)*sign;
00710             argsdone++;
00711             if ( firstvalue == 1 ) { gcdisone = 1; break; }
00712         }
00713         NEXTARG(tf);
00714     }
00715     termout = AT.WorkPointer;
00716     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
00717     if ( AT.WorkPointer > AT.WorkTop ) {
00718         MLOCK(ErrorMessageLock);
00719         MesWork();
00720         MUNLOCK(ErrorMessageLock);
00721         return(-1);
00722     }
00723  */
00724     #] Do short arguments :
00725     #[ Do trivial GCD :
00726
00727     Copy head
00728  */
00729     i = t - term; tin = term; tout = termout;
00730     NCOPY(tout,tin,i);
00731     if ( gcdisone || ( firstshort == -SNUMBER && firstvalue == 1 ) ) {
00732         sign = 1;
00733     gcdone:
00734         tin += t[1]; tstop = term + *term;
00735         while ( tin < tstop ) *tout++ = *tin++;
00736         *termout = tout - termout;
00737         if ( sign < 0 ) tout[-1] = -tout[-1];
00738         AT.WorkPointer = tout;
00739         if ( argsdone && Generator(BHEAD termout,level) < 0 ) goto CalledFrom;
00740         AT.WorkPointer = termout;
00741         AT.pWorkPointer = args;
00742         return(0);
00743     }
00744  */
00745     #] Do trivial GCD :

```

```

00746     #[ Do short argument GCD :
00747     /*
00748     if ( numargs == 0 ) {      /* basically we are done */
00749 doshort:
00750     sign = 1;
00751     if ( firstshort == 0 ) goto gcdone;
00752     if ( firstshort == -SNUMBER ) {
00753         *tout++ = SNUMBER; *tout++ = 4; *tout++ = firstvalue; *tout++ = 1;
00754         goto gcdone;
00755     }
00756     else if ( firstshort == -SYMBOL ) {
00757         *tout++ = SYMBOL; *tout++ = 4; *tout++ = firstvalue; *tout++ = 1;
00758         goto gcdone;
00759     }
00760     else if ( firstshort == -VECTOR || firstshort == -INDEX ) {
00761         *tout++ = INDEX; *tout++ = 3; *tout++ = firstvalue; goto gcdone;
00762     }
00763     else if ( firstshort == -MINVECTOR ) {
00764         sign = -1;
00765         *tout++ = INDEX; *tout++ = 3; *tout++ = firstvalue; goto gcdone;
00766     }
00767     else if ( firstshort <= -FUNCTION ) {
00768         *tout++ = firstvalue; *tout++ = FUNHEAD; FILLFUN(tout);
00769         goto gcdone;
00770     }
00771     else {
00772         MLOCK(ErrorMessageLock);
00773         MesPrint("Internal error. Illegal short argument in GCDfunction.");
00774         MUNLOCK(ErrorMessageLock);
00775         Terminate(-1);
00776     }
00777 }
00778 /*
00779 #] Do short argument GCD :
00780 #[ Convert short argument :
00781
00782 Now we allocate space for the arguments in general notation.
00783 First the special one if there were short arguments
00784 */
00785 if ( firstshort ) {
00786     switch ( firstshort ) {
00787     case -SNUMBER:
00788         sh[0] = 4; sh[1] = ABS(firstvalue); sh[2] = 1;
00789         if ( firstvalue < 0 ) sh[3] = -3;
00790         else sh[3] = 3;
00791         sh[4] = 0;
00792         break;
00793     case -MINVECTOR:
00794     case -VECTOR:
00795     case -INDEX:
00796         sh[0] = 8; sh[1] = INDEX; sh[2] = 3; sh[3] = firstvalue;
00797         sh[4] = 1; sh[5] = 1;
00798         if ( firstshort == -MINVECTOR ) sh[6] = -3;
00799         else sh[6] = 3;
00800         sh[7] = 0;
00801         break;
00802     case -SYMBOL:
00803         sh[0] = 8; sh[1] = SYMBOL; sh[2] = 4; sh[3] = firstvalue; sh[4] = 1;
00804         sh[5] = 1; sh[6] = 1; sh[7] = 3; sh[8] = 0;
00805         break;
00806     default:
00807         sh[0] = FUNHEAD+4; sh[1] = firstshort; sh[2] = FUNHEAD;
00808         for ( i = 2; i < FUNHEAD; i++ ) sh[i+1] = 0;
00809         sh[FUNHEAD+1] = 1; sh[FUNHEAD+2] = 1; sh[FUNHEAD+3] = 3; sh[FUNHEAD+4] = 0;
00810         break;
00811     }
00812 }
00813 /*
00814 #] Convert short argument :
00815 #[ Sort arguments :
00816
00817 Now we should sort the arguments in a way that the dollars and the
00818 expressions come last. That way we may never need them.
00819 */
00820 for ( i = 1; i < numargs; i++ ) {
00821     for ( ii = i; ii > 0; ii-- ) {
00822         arg1 = AT.pWorkSpace[args+ii];
00823         arg2 = AT.pWorkSpace[args+ii-1];
00824         if ( *arg1 < 0 ) {
00825             if ( *arg2 < 0 ) {
00826                 if ( *arg1 == -EXPRESSION ) break;
00827                 if ( *arg2 == -DOLLAREXPRESSION ) break;
00828                 AT.pWorkSpace[args+ii] = arg2;
00829                 AT.pWorkSpace[args+ii-1] = arg1;
00830             }
00831             else break;
00832         }
00833     }
}

```

```

00833         else if ( *arg2 < 0 ) {
00834             AT.pWorkspace[args+ii] = arg2;
00835             AT.pWorkspace[args+ii-1] = arg1;
00836         }
00837         else {
00838             if ( *arg1 > *arg2 ) {
00839                 AT.pWorkspace[args+ii] = arg2;
00840                 AT.pWorkspace[args+ii-1] = arg1;
00841             }
00842             else break;
00843         }
00844     }
00845 }
00846 /*
00847 #] Sort arguments :
00848 #[ There is a single term argument :
00849 */
00850 if ( firstshort ) {
00851     mh = sh; istart = 0;
00852 oneterm:;
00853     for ( i = istart; i < numargs; i++ ) {
00854         arg1 = AT.pWorkspace[args+i];
00855         if ( *arg1 > 0 ) {
00856             oldvall = arg1[*arg1]; arg1[*arg1] = 0;
00857             m = arg1+ARGHEAD;
00858             while ( *m ) {
00859                 GCDterms(BHEAD mh,m,mh); m += *m;
00860                 if ( mh[0] == 4 && mh[1] == 1 && mh[2] == 1 && mh[3] == 3 ) {
00861                     gcdisone = 1; sign = 1; arg1[*arg1] = oldvall; goto gcdone;
00862                 }
00863             }
00864             arg1[*arg1] = oldvall;
00865         }
00866         else if ( *arg1 == -DOLLAREXPRESSION ) {
00867             if ( ( d = DolToTerms(BHEAD arg1[1]) ) != 0 ) {
00868                 m = d->where;
00869                 while ( *m ) {
00870                     GCDterms(BHEAD mh,m,mh); m += *m;
00871                     argsdone++;
00872                     if ( mh[0] == 4 && mh[1] == 1 && mh[2] == 1 && mh[3] == 3 ) {
00873                         gcdisone = 1; sign = 1;
00874                         if ( d->factors ) M_free(d->factors,"Dollar factors");
00875                         M_free(d,"Copy of dollar variable"); goto gcdone;
00876                     }
00877                 }
00878                 if ( d->factors ) M_free(d->factors,"Dollar factors");
00879                 M_free(d,"Copy of dollar variable");
00880             }
00881         }
00882         else {
00883             mm = CreateExpression(BHEAD arg1[1]);
00884             m = mm;
00885             while ( *m ) {
00886                 GCDterms(BHEAD mh,m,mh); m += *m;
00887                 argsdone++;
00888                 if ( mh[0] == 4 && mh[1] == 1 && mh[2] == 1 && mh[3] == 3 ) {
00889                     gcdisone = 1; sign = 1; M_free(mm,"CreateExpression"); goto gcdone;
00890                 }
00891             }
00892             M_free(mm,"CreateExpression");
00893         }
00894     }
00895     if ( firstshort ) {
00896         if ( mh[0] == 4 ) {
00897             firstshort = -SNUMBER; firstvalue = mh[1] * (mh[3]/3);
00898         }
00899         else if ( mh[1] == SYMBOL ) {
00900             firstshort = -SYMBOL; firstvalue = mh[3];
00901         }
00902         else if ( mh[1] == INDEX ) {
00903             firstshort = -INDEX; firstvalue = mh[3];
00904             if ( mh[6] == -3 ) firstshort = -MINVECTOR;
00905         }
00906         else if ( mh[1] >= FUNCTION ) {
00907             firstshort = -mh[1]; firstvalue = mh[1];
00908         }
00909         goto doshort;
00910     }
00911     else {
00912 /*
00913         We have a GCD that is only a single term.
00914         Paste it in and combine the coefficients.
00915 */
00916         mh[mh[0]] = 0;
00917         mm = mh;
00918         ii = 0;
00919         goto multiterms;

```

```

00920     }
00921 }
00922 /*
00923 Now we have only regular arguments.
00924 But some have not yet been expanded.
00925 Check whether there are proper long arguments and if so if there is
00926 one with just a single term
00927 */
00928 for ( i = 0; i < numargs; i++ ) {
00929     arg1 = AT.pWorkSpace[args+i];
00930     if ( *arg1 > 0 && arg1[ARGHEAD]+ARGHEAD == *arg1 ) {
00931 /*
00932         We have an argument with a single term
00933 */
00934         if ( i != 0 ) {
00935             arg2 = AT.pWorkSpace[args];
00936             AT.pWorkSpace[args] = arg1;
00937             AT.pWorkSpace[args+1] = arg2;
00938         }
00939         m = mh = AT.WorkPointer;
00940         mm = arg1+ARGHEAD; i = *mm;
00941         NCOPY(m,mm,i);
00942         AT.WorkPointer = m;
00943         istart = 1;
00944         argsdone++;
00945         goto oneterm;
00946     }
00947 }
00948 /*
00949 #] There is a single term argument :
00950 #[ Expand $ and expr :
00951
00952 We have: 1: regular multiterm arguments
00953          2: dollars
00954          3: expressions.
00955 The sum of them is numargs. Their addresses are in args. The problem is
00956 that expansion will lead to allocations that we have to return and all
00957 these allocations are different in nature.
00958 */
00959 action = 1;
00960 abuf = (ARGBUFFER *)Malloc1(numargs*sizeof(ARGBUFFER),"argbuffer");
00961 for ( i = 0; i < numargs; i++ ) {
00962     arg1 = AT.pWorkSpace[args+i];
00963     if ( *arg1 > 0 ) {
00964         m = (WORD *)Malloc1(*arg1*sizeof(WORD),"argbuffer type 0");
00965         abuf[i].buffer = m;
00966         abuf[i].type = 0;
00967         mm = arg1+ARGHEAD;
00968         j = *arg1-ARGHEAD;
00969         abuf[i].size = j;
00970         if ( j ) argsdone++;
00971         NCOPY(m,mm,j);
00972         *m = 0;
00973     }
00974     else if ( *arg1 == -DOLLAREXPRESSION ) {
00975         d = DolToTerms(BHEAD arg1[1]);
00976         abuf[i].buffer = d->where;
00977         abuf[i].type = 1;
00978         abuf[i].dollar = d;
00979         m = abuf[i].buffer;
00980         if ( *m ) argsdone++;
00981         while ( *m ) m+= *m;
00982         abuf[i].size = m-abuf[i].buffer;
00983     }
00984     else if ( *arg1 == -EXPRESSION ) {
00985         abuf[i].buffer = CreateExpression(BHEAD arg1[1]);
00986         abuf[i].type = 2;
00987         m = abuf[i].buffer;
00988         if ( *m ) argsdone++;
00989         while ( *m ) m+= *m;
00990         abuf[i].size = m-abuf[i].buffer;
00991     }
00992     else {
00993         MLOCK(ErrorMessageLock);
00994         MesPrint("What argument is this?");
00995         MUNLOCK(ErrorMessageLock);
00996         goto CalledFrom;
00997     }
00998 }
00999 for ( i = 0; i < numargs; i++ ) {
01000     arg1 = abuf[i].buffer;
01001     if ( *arg1 == 0 ) {}
01002     else if ( arg1[*arg1] == 0 ) {
01003 /*
01004         After expansion there is an argument with a single term
01005 */
01006         ab = abuf[i]; abuf[i] = abuf[0]; abuf[0] = ab;

```



```

01007         mh = abuf[0].buffer;
01008         for ( j = 1; j < numargs; j++ ) {
01009             m = abuf[j].buffer;
01010             while ( *m ) {
01011                 GCDterms(BHEAD mh,m,mh); m += *m;
01012                 argsdone++;
01013                 if ( mh[0] == 4 && mh[1] == 1 && mh[2] == 1 && mh[3] == 3 ) {
01014                     gcdisone = 1; sign = 1; break;
01015                 }
01016             }
01017             if ( *m ) break;
01018         }
01019         mm = mh + *mh; if ( mm[-1] < 0 ) { sign = -1; mm[-1] = -mm[-1]; }
01020         mstop = mm - mm[-1]; m = mh+1; mlength = mm[-1];
01021         while ( tin < t ) *tout++ = *tin++;
01022         while ( m < mstop ) *tout++ = *m++;
01023         tin += tin[1];
01024         while ( tin < tstop ) *tout++ = *tin++;
01025         tlength = REDLENG(tlength);
01026         mlength = REDLENG(mlength);
01027         if ( MulRat(BHEAD (UWORD *)tstop,tlength,(UWORD *)mstop,mlength,
01028             (UWORD *)tout,&newlength) < 0 ) goto CalledFrom;
01029         mlength = INCLENG(newlength);
01030         tout += ABS(mlength);
01031         tout[-1] = mlength*sign;
01032         *termout = tout - termout;
01033         AT.WorkPointer = tout;
01034         if ( argsdone && Generator(BHEAD termout,level) < 0 ) goto CalledFrom;
01035         goto cleanup;
01036     }
01037 }
01038 /*
01039  There are only arguments with more than one term.
01040  We order them by size to make the computations as easy as possible.
01041 */
01042 for ( i = 1; i < numargs; i++ ) {
01043     for ( ii = i; ii > 0; ii-- ) {
01044         if ( abuf[ii-1].size <= abuf[ii].size ) break;
01045         ab = abuf[ii-1]; abuf[ii-1] = abuf[ii]; abuf[ii] = ab;
01046     }
01047 }
01048 /*
01049 #] Expand $ and expr :
01050 #[ Multiterm subexpressions :
01051 */
01052 ii = 0;
01053 gcdout = abuf[ii].buffer;
01054 for ( i = 0; i < numargs; i++ ) {
01055     if ( abuf[i].buffer[0] ) { gcdout = abuf[i].buffer; ii = i; i++; argsdone++; break; }
01056 }
01057 for ( ; i < numargs; i++ ) {
01058     if ( abuf[i].buffer[0] ) {
01059         g = GCDfunction3(BHEAD gcdout,abuf[i].buffer);
01060         argsdone++;
01061         if ( gcdout != abuf[ii].buffer ) M_free(gcdout,"gcdout");
01062         gcdout = g;
01063         if ( gcdout[*gcdout] == 0 && gcdout[0] == 4 && gcdout[1] == 1
01064             && gcdout[2] == 1 && gcdout[3] == 3 ) break;
01065     }
01066 }
01067 mm = gcdout;
01068 multiterms:;
01069 tlength = REDLENG(tlength);
01070 while ( *mm ) {
01071     tin = term; tout = termout; while ( tin < t ) *tout++ = *tin++;
01072     tin += t[1];
01073     mnext = mm + *mm; mlength = mnext[-1]; mstop = mnext - ABS(mlength);
01074     mm++;
01075     while ( mm < mstop ) *tout++ = *mm++;
01076     while ( tin < tstop ) *tout++ = *tin++;
01077     mlength = REDLENG(mlength);
01078     if ( MulRat(BHEAD (UWORD *)tstop,tlength,(UWORD *)mm,mlength,
01079         (UWORD *)tout,&newlength) < 0 ) goto CalledFrom;
01080     mlength = INCLENG(newlength);
01081     tout += ABS(mlength);
01082     tout[-1] = mlength;
01083     *termout = tout - termout;
01084     AT.WorkPointer = tout;
01085     if ( argsdone && Generator(BHEAD termout,level) < 0 ) goto CalledFrom;
01086     mm = mnext; /* next term */
01087 }
01088 if ( action && ( gcdout != abuf[ii].buffer ) ) M_free(gcdout,"gcdout");
01089 /*
01090 #] Multiterm subexpressions :
01091 #[ Cleanup :
01092 */
01093 cleanup:;

```

```

01094     if ( action ) {
01095         for ( i = 0; i < numargs; i++ ) {
01096             if ( abuf[i].type == 0 ) { M_free(abuf[i].buffer,"argbuffer type 0"); }
01097             else if ( abuf[i].type == 1 ) {
01098                 d = abuf[i].dollar;
01099                 if ( d->factors ) M_free(d->factors,"Dollar factors");
01100                 M_free(d,"Copy of dollar variable");
01101             }
01102             else if ( abuf[i].type == 2 ) { M_free(abuf[i].buffer,"CreateExpression"); }
01103         }
01104         M_free(abuf,"argbuffer");
01105     }
01106 /*
01107     #] Cleanup :
01108 */
01109     AT.pWorkPointer = args;
01110     AT.WorkPointer = termout;
01111     return(0);
01112
01113 CalledFrom:
01114     MLOCK(ErrorMessageLock);
01115     MesCall("GCDfunction");
01116     MUNLOCK(ErrorMessageLock);
01117     SETERROR(-1)
01118     return(-1);
01119 }
01120
01121 /*
01122     #] GCDfunction :
01123     #[ GCDfunction3 :
01124
01125     Finds the GCD of the two arguments which are buffers with terms.
01126     In principle the first buffer can have only one term.
01127
01128     If both buffers have more than one term, we need to replace all
01129     non-symbolic objects by generated symbols and substitute that back
01130     afterwards. The rest we leave to the powerful routines.
01131     Philosophical problem: What do we do with GCD_(x/z+y,x+y*z) ?
01132
01133     Method:
01134     If we have only negative powers of z and no positive powers we let
01135     the EXTRASYMBOLS do their job. When mixed, multiply the arguments with
01136     the negative powers with enough powers of z to eliminate the negative powers.
01137     The DENOMINATOR function is always eliminated with the mechanism as we
01138     cannot tell whether there are positive powers of its contents.
01139 */
01140 WORD *GCDfunction3(PHEAD WORD *in1, WORD *in2)
01141 {
01142     GETBIDENTITY
01143     WORD oldsorttype = AR.SortType, *ow = AT.WorkPointer;;
01144     WORD *t, *tt, *gcdout, *term1, *term2, *confree1, *confree2, *gcdout1, *proper1, *proper2;
01145     int i, actionflag1, actionflag2;
01146     WORD startebuf = cbuf[AT.ebufnum].numrhs;
01147     WORD tryterm1, tryterm2;
01148     if ( in2[*in2] == 0 ) { t = in1; in1 = in2; in2 = t; }
01149     if ( in1[*in1] == 0 ) { /* First input with only one term */
01150         gcdout = (WORD *)Malloc1((*in1+1)*sizeof(WORD),"gcdout");
01151         i = *in1; t = gcdout; tt = in1; NCOPY(t,tt,i); *t = 0;
01152         t = in2;
01153         while ( *t ) {
01154             GCDterms(BHEAD gcdout,t,gcdout);
01155             if ( gcdout[0] == 4 && gcdout[1] == 1
01156                 && gcdout[2] == 1 && gcdout[3] == 3 ) break;
01157             t += *t;
01158         }
01159         AT.WorkPointer = ow;
01160         return(gcdout);
01161     }
01162 }
01163 /*
01164     We need to take out the content from the two expressions
01165     and determine their GCD. This plays with the negative powers!
01166 */
01167     AR.SortType = SORTHIGHFIRST;
01168     term1 = TermMalloc("GCDfunction3-a");
01169     term2 = TermMalloc("GCDfunction3-b");
01170
01171     confree1 = TakeContent(BHEAD in1,term1);
01172     tryterm1 = AN.tryterm; AN.tryterm = 0;
01173     confree2 = TakeContent(BHEAD in2,term2);
01174     tryterm2 = AN.tryterm; AN.tryterm = 0;
01175 /*
01176     confree1 = TakeSymbolContent(BHEAD in1,term1);
01177     confree2 = TakeSymbolContent(BHEAD in2,term2);
01178 */
01179     GCDterms(BHEAD term1,term2,term1);
01180     TermFree(term2,"GCDfunction3-b");

```

```

01181 /*
01182     Now we have to replace all non-symbols and symbols to a negative power
01183     by extra symbols.
01184 */
01185 if ( ( proper1 = PutExtraSymbols(BHEAD confree1,startebuf,&actionflag1) ) == 0 ) goto CalledFrom;
01186 if ( confree1 != in1 ) {
01187     if ( tryterm1 ) { TermFree(confree1,"TakeContent"); }
01188     else { M_free(confree1,"TakeContent"); }
01189 }
01190 /*
01191     TermFree(confree1,"TakeSymbolContent");
01192 */
01193 if ( ( proper2 = PutExtraSymbols(BHEAD confree2,startebuf,&actionflag2) ) == 0 ) goto CalledFrom;
01194 if ( confree2 != in2 ) {
01195     if ( tryterm2 ) { TermFree(confree2,"TakeContent"); }
01196     else { M_free(confree2,"TakeContent"); }
01197 }
01198 /*
01199     TermFree(confree2,"TakeSymbolContent");
01200 */
01201 /*
01202     And now the real work:
01203 */
01204 gcdout1 = poly_gcd(BHEAD proper1,proper2,0);
01205 M_free(proper1,"PutExtraSymbols");
01206 M_free(proper2,"PutExtraSymbols");
01207
01208 AR.SortType = oldsorttype;
01209 if ( actionflag1 || actionflag2 ) {
01210     if ( ( gcdout = TakeExtraSymbols(BHEAD gcdout1,startebuf) ) == 0 ) goto CalledFrom;
01211     M_free(gcdout1,"gcdout");
01212 }
01213 else {
01214     gcdout = gcdout1;
01215 }
01216
01217 cbuf[AT.ebufnum].numrhs = startebuf;
01218 /*
01219     Now multiply gcdout by term1
01220 */
01221 if ( term1[0] != 4 || term1[3] != 3 || term1[1] != 1 || term1[2] != 1 ) {
01222     AN.tryterm = -1;
01223     if ( ( gcdout1 = MultiplyWithTerm(BHEAD gcdout,term1,2) ) == 0 ) goto CalledFrom;
01224     AN.tryterm = 0;
01225     M_free(gcdout,"gcdout");
01226     gcdout = gcdout1;
01227 }
01228 TermFree(term1,"GCDfunction3-a");
01229 AT.WorkPointer = ow;
01230 return(gcdout);
01231 CalledFrom:
01232 AN.tryterm = 0;
01233 MLOCK(ErrorMessageLock);
01234 MesCall("GCDfunction3");
01235 MUNLOCK(ErrorMessageLock);
01236 return(0);
01237 }
01238
01239 /*
01240     #] GCDfunction3 :
01241     #[ PutExtraSymbols :
01242 */
01243
01244 WORD *PutExtraSymbols(PHEAD WORD *in,WORD startebuf,int *actionflag)
01245 {
01246     WORD *termout = AT.WorkPointer;
01247     int action;
01248     *actionflag = 0;
01249     NewSort(BHEAD0);
01250     while ( *in ) {
01251         if ( ( action = LocalConvertToPoly(BHEAD in,termout,startebuf,0) ) < 0 ) {
01252             LowerSortLevel();
01253             goto CalledFrom;
01254         }
01255         if ( action > 0 ) *actionflag = 1;
01256         StoreTerm(BHEAD termout);
01257         in += *in;
01258     }
01259     if ( EndSort(BHEAD (WORD *)((void *)(&termout)),2) < 0 ) goto CalledFrom;
01260     return(termout);
01261 CalledFrom:
01262     MLOCK(ErrorMessageLock);
01263     MesCall("PutExtraSymbols");
01264     MUNLOCK(ErrorMessageLock);
01265     return(0);
01266 }
01267

```

```

01268 /*
01269      #] PutExtraSymbols :
01270      #[ TakeExtraSymbols :
01271 */
01272
01273 WORD *TakeExtraSymbols(PHEAD WORD *in,WORD startebuf)
01274 {
01275     CBUF *C = cbuf+AC.cbunum;
01276     CBUF *CC = cbuf+AT.ebunum;
01277     WORD *oldworkpointer = AT.WorkPointer, *termout;
01278
01279     termout = AT.WorkPointer;
01280     NewSort(BHEAD0);
01281     while ( *in ) {
01282         if ( ConvertFromPoly(BHEAD
in,termout,numxsymbol,CC->numrhs-startebuf+numxsymbol,startebuf-numxsymbol,1) <= 0 ) {
01283             LowerSortLevel();
01284             goto CalledFrom;
01285         }
01286         in += *in;
01287         AT.WorkPointer = termout + *termout;
01288     /*
01289         ConvertFromPoly leaves terms with subexpressions. Hence:
01290     */
01291         if ( Generator(BHEAD termout,C->numlhs) ) {
01292             LowerSortLevel();
01293             goto CalledFrom;
01294         }
01295     }
01296     AT.WorkPointer = oldworkpointer;
01297     if ( EndSort(BHEAD (WORD *)((void *)(&termout)),2) < 0 ) goto CalledFrom;
01298     return(termout);
01299
01300 CalledFrom:
01301     MLOCK(ErrorMessageLock);
01302     MesCall("TakeExtraSymbols");
01303     MUNLOCK(ErrorMessageLock);
01304     return(0);
01305 }
01306
01307 /*
01308      #] TakeExtraSymbols :
01309      #[ MultiplyWithTerm :
01310 */
01311
01312 WORD *MultiplyWithTerm(PHEAD WORD *in, WORD *term, WORD par)
01313 {
01314     WORD *termout, *t, *tt, *tstop, *ttstop;
01315     WORD length, length1, length2;
01316     WORD oldsorttype = AR.SortType;
01317     COMPARE oldcompareroutine = (COMPARE) (AR.CompareRoutine);
01318     AR.CompareRoutine = (COMPAREDUMMY) (&CompareSymbols);
01319
01320     if ( par == 0 || par == 2 ) AR.SortType = SORTHIGHFIRST;
01321     else AR.SortType = SORTLOWFIRST;
01322     termout = AT.WorkPointer;
01323     NewSort(BHEAD0);
01324     while ( *in ) {
01325         tt = termout + 1;
01326         tstop = in + *in; tstop -= ABS(tstop[-1]); t = in + 1;
01327         while ( t < tstop ) *t++ = *t++;
01328         ttstop = term + *term; ttstop -= ABS(ttstop[-1]); t = term + 1;
01329         while ( t < ttstop ) *t++ = *t++;
01330         length1 = REDLENG(in[*in-1]); length2 = REDLENG(term[*term-1]);
01331         if ( MulRat(BHEAD (UWORD *)tstop,length1,
01332             (UWORD *)ttstop,length2,(UWORD *)tt,&length) ) goto CalledFrom;
01333         length = INCLENG(length);
01334         tt += ABS(length); tt[-1] = length;
01335         *termout = tt - termout;
01336         SymbolNormalize(termout);
01337         StoreTerm(BHEAD termout);
01338         in += *in;
01339     }
01340     if ( par == 2 ) {
01341     /*
01342         if ( AN.tryterm == 0 ) AN.tryterm = 1; */
01343         AN.tryterm = 0; /* For now */
01344         if ( EndSort(BHEAD (WORD *)((void *)(&termout)),2) < 0 ) goto CalledFrom;
01345     }
01346     else {
01347         if ( EndSort(BHEAD termout,1) < 0 ) goto CalledFrom;
01348     }
01349     AR.CompareRoutine = (COMPAREDUMMY)oldcompareroutine;
01350
01351     AR.SortType = oldsorttype;
01352     return(termout);
01353 }

```

```

01354 CalledFrom:
01355     MLOCK(ErrorMessageLock);
01356     MesCall("MultiplyWithTerm");
01357     MUNLOCK(ErrorMessageLock);
01358     return(0);
01359 }
01360
01361 /*
01362     #] MultiplyWithTerm :
01363     #[ TakeContent :
01364 */
01376 WORD *TakeContent(PHEAD WORD *in, WORD *term)
01377 {
01378     GETBIDENTITY
01379     WORD *t, *tstop, *tcom, *tout, *tstore, *r, *rstop, *m, *mm, *w, *ww, *wterm;
01380     WORD *tnext, *tt, *tterm, code[2];
01381     WORD *inp, a, *den;
01382     int i, j, k, action = 0, sign;
01383     UWORD *GCDbuffer, *GCDbuffer2, *LCMbuffer, *LCMbuffer2, *ap;
01384     WORD GCDlen, GCDlen2, LCMlen, LCMlen2, length, redlength, len1, len2;
01385     tout = tstore = term+1;
01386 /*
01387     #[ INDEX :
01388 */
01389     t = in;
01390     tnext = t + *t;
01391     tstop = tnext-ABS(tnext[-1]);
01392     t++;
01393     while ( t < tstop ) {
01394         if ( *t == INDEX ) {
01395             i = t[1]; NCOPY(tout,t,i); break;
01396         }
01397         else t += t[1];
01398     }
01399     if ( tout > tstore ) { /* There are indices in the first term */
01400         t = tnext;
01401         while ( *t ) {
01402             tnext = t + *t;
01403             rstop = tnext - ABS(tnext[-1]);
01404             r = t+1;
01405             if ( r == rstop ) goto noindices;
01406             while ( r < rstop ) {
01407                 if ( *r != INDEX ) { r += r[1]; continue; }
01408                 m = tstore+2;
01409                 while ( m < tout ) {
01410                     for ( i = 2; i < r[1]; i++ ) {
01411                         if ( *m == r[i] ) break;
01412                         if ( *m > r[i] ) continue;
01413                         mm = m+1;
01414                         while ( mm < tout ) { mm[-1] = mm[0]; mm++; }
01415                         tout--; tstore[1]--; m--;
01416                         break;
01417                     }
01418                     m++;
01419                 }
01420             }
01421             if ( r >= rstop || tout <= tstore+2 ) {
01422                 tout = tstore; break;
01423             }
01424         }
01425         if ( tout > tstore+2 ) { /* Now we have to take out what is in tstore */
01426             t = in; w = in;
01427             while ( *t ) {
01428                 wterm = w;
01429                 tnext = t + *t; t++; w++;
01430                 while ( *t != INDEX ) { i = t[1]; NCOPY(w,t,i); }
01431                 tt = t + t[1]; t += 2; r = tstore+2; ww = w; *w++ = INDEX; w++;
01432                 while ( r < tout && t < tt ) {
01433                     if ( *r > *t ) { *w++ = *t++; }
01434                     else if ( *r == *t ) { r++; t++; }
01435                     else goto CalledFrom;
01436                 }
01437                 if ( r < tout ) goto CalledFrom;
01438                 while ( t < tt ) *w++ = *t++;
01439                 ww[1] = w - ww;
01440                 if ( ww[1] == 2 ) w = ww;
01441                 while ( t < tnext ) *w++ = *t++;
01442                 *wterm = w - wterm;
01443             }
01444             *w = 0;
01445         }
01446     noindices:
01447         tstore = tout;
01448     }
01449 /*
01450     #] INDEX :
01451     #[ VECTOR/DELTA :

```

```

01452 */
01453 code[0] = VECTOR; code[1] = DELTA;
01454 for ( k = 0; k < 2; k++ ) {
01455     t = in;
01456     tnext = t + *t;
01457     tstop = tnext-ABS(tnext[-1]);
01458     t++;
01459     while ( t < tstop ) {
01460         if ( *t == code[k] ) {
01461             i = t[1]; NCOPY(tout,t,i); break;
01462         }
01463         else t += t[1];
01464     }
01465     if ( tout > tstore ) { /* There are vectors in the first term */
01466         t = tnext;
01467         while ( *t ) {
01468             tnext = t + *t;
01469             rstop = tnext - ABS(tnext[-1]);
01470             r = t+1;
01471             if ( r == rstop ) { tstore = tout; goto novectors; }
01472             while ( r < rstop ) {
01473                 if ( *r != code[k] ) { r += r[1]; continue; }
01474                 m = tstore+2;
01475                 while ( m < tout ) {
01476                     for ( i = 2; i < r[1]; i += 2 ) {
01477                         if ( *m == r[i] && m[1] == r[i+1] ) break;
01478                         if ( *m > r[i] || ( *m == r[i] && m[1] > r[i+1] ) ) continue;
01479                         mm = m+2;
01480                         while ( mm < tout ) { mm[-2] = mm[0]; mm[-1] = mm[1]; mm += 2; }
01481                         tout -= 2; tstore[1] -= 2; m -= 2;
01482                         break;
01483                     }
01484                     m += 2;
01485                 }
01486             }
01487             if ( r >= rstop || tout <= tstore+2 ) {
01488                 tout = tstore; break;
01489             }
01490         }
01491         if ( tout > tstore+2 ) { /* Now we have to take out what is in tstore */
01492             t = in; w = in;
01493             while ( *t ) {
01494                 wterm = w;
01495                 tnext = t + *t; t++; w++;
01496                 while ( *t != code[k] ) { i = t[1]; NCOPY(w,t,i); }
01497                 tt = t + t[1]; t += 2; r = tstore+2; ww = w; *w++ = code[k]; w++;
01498                 while ( r < tout && t < tt ) {
01499                     if ( ( *r > *t ) || ( *r == *t && r[1] > t[1] ) )
01500                         { *w++ = *t++; *w++ = *t++; }
01501                     else if ( *r == *t && r[1] == t[1] ) { r += 2; t += 2; }
01502                     else goto CalledFrom;
01503                 }
01504                 if ( r < tout ) goto CalledFrom;
01505                 while ( t < tt ) *w++ = *t++;
01506                 ww[1] = w - ww;
01507                 if ( ww[1] == 2 ) w = ww;
01508                 while ( t < tnext ) *w++ = *t++;
01509                 *wterm = w - wterm;
01510             }
01511             *w = 0;
01512         }
01513         tstore = tout;
01514     }
01515 }
01516 novectors;;
01517 /*
01518 #] VECTOR/DELTA :
01519 #[ FUNCTIONS :
01520 */
01521 t = in;
01522 tnext = t + *t;
01523 tstop = tnext-ABS(tnext[-1]);
01524 t++;
01525 tcom = 0;
01526 while ( t < tstop ) {
01527     if ( *t >= FUNCTION ) {
01528         if ( functions[*t-FUNCTION].commute ) {
01529             if ( tcom == 0 ) { tcom = tstore; }
01530             else {
01531                 for ( i = 0; i < t[1]; i++ ) {
01532                     if ( t[i] != tcom[i] ) {
01533                         MLOCK(ErrorMessageLock);
01534                         MesPrint("GCD or factorization of more than one noncommuting object not
allowed");
01535                         MUNLOCK(ErrorMessageLock);
01536                         goto CalledFrom;
01537                     }

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```

01538         }
01539     }
01540 }
01541     i = t[1]; NCOPY(tstore,t,i);
01542 }
01543     else t += t[1];
01544 }
01545 if ( tout > tstore ) {
01546     t = tnext;
01547     while ( *t ) {
01548         tnext = t + *t; tstop = tnext - ABS(tnext[-1]); t++;
01549         if ( t == tstop ) goto nofunctions;
01550         r = tstore;
01551         while ( r < tout ) {
01552             tt = t;
01553             while ( tt < tstop ) {
01554                 for ( i = 0; i < r[1]; i++ ) {
01555                     if ( r[i] != tt[i] ) break;
01556                 }
01557                 if ( i == r[1] ) { r += r[1]; goto nextrl; }
01558             }
01559             /*
01560              Not encountered in this term. Scratch from list
01561             */
01562             m = r; mm = r + r[1];
01563             while ( mm < tout ) *m++ = *mm++;
01564             tout = m;
01565 nextrl:;
01566         }
01567         if ( tout <= tstore ) break;
01568         t += *t;
01569     }
01570 }
01571 if ( tout > tstore ) {
01572     /*
01573      Now we have one or more functions left that are common in all terms.
01574      Take them out. We do this one by one.
01575     */
01576     r = tstore;
01577     while ( r < tout ) {
01578         t = in; ww = in; w = ww+1;
01579         while ( *t ) {
01580             tnext = t + *t;
01581             t++;
01582             for(;;) {
01583                 for ( i = 0; i < r[1]; i++ ) {
01584                     if ( t[i] != r[i] ) {
01585                         j = t[1]; NCOPY(w,t,j);
01586                         break;
01587                     }
01588                 }
01589                 if ( i == r[1] ) {
01590                     t += t[1];
01591                     while ( t < tnext ) *w++ = *t++;
01592                     *ww = w - ww;
01593                     break;
01594                 }
01595             }
01596         }
01597         r += r[1];
01598         *w = 0;
01599     }
01600 nofunctions:
01601     tstore = tout;
01602 }
01603 /*
01604  #] FUNCTIONS :
01605  #[ SYMBOL :
01606
01607  We make a list of symbols and their minimal powers.
01608  This includes negative powers. In the end we have to multiply by the
01609  inverse of this list. That takes out all negative powers and leaves
01610  things ready for further processing.
01611  */
01612 tterm = AT.WorkPointer; tt = tterm+1;
01613 tout[0] = SYMBOL; tout[1] = 2;
01614 t = in;
01615 tnext = t + *t; tstop = tnext - ABS(tnext[-1]); t++;
01616 while ( t < tstop ) {
01617     if ( *t == SYMBOL ) {
01618         for ( i = 0; i < t[1]; i++ ) tout[i] = t[i];
01619         break;
01620     }
01621     t += t[1];
01622 }
01623 t = tnext;
01624 while ( *t ) {

```

```

01625         tnext = t + *t; tstop = tnext - ABS(tnext[-1]); t++;
01626         if ( t == tstop ) {
01627             tout[1] = 2;
01628             break;
01629         }
01630         else {
01631             while ( t < tstop ) {
01632                 if ( *t == SYMBOL ) {
01633                     MergeSymbolLists(BHEAD tout,t,-1);
01634                     break;
01635                 }
01636                 t += t[1];
01637             }
01638             t = tnext;
01639         }
01640     }
01641     if ( tout[1] > 2 ) {
01642         t = tout;
01643         tt[0] = t[0]; tt[1] = t[1];
01644         for ( i = 2; i < t[1]; i += 2 ) {
01645             tt[i] = t[i]; tt[i+1] = -t[i+1];
01646         }
01647         tt += tt[1];
01648         tout += tout[1];
01649         action++;
01650     }
01651     /*
01652     #] SYMBOL :
01653     #[ DOTPRODUCT :
01654
01655     We make a list of dotproducts and their minimal powers.
01656     This includes negative powers. In the end we have to multiply by the
01657     inverse of this list. That takes out all negative powers and leaves
01658     things ready for further processing.
01659     */
01660     tout[0] = DOTPRODUCT; tout[1] = 2;
01661     t = in;
01662     while ( *t ) {
01663         tnext = t + *t; tstop = tnext - ABS(tnext[-1]); t++;
01664         if ( t == tstop ) {
01665             tout[1] = 2;
01666             break;
01667         }
01668         while ( t < tstop ) {
01669             if ( *t == DOTPRODUCT ) {
01670                 MergeDotproductLists(BHEAD tout,t,-1);
01671                 break;
01672             }
01673             t += t[1];
01674         }
01675         t = tnext;
01676     }
01677     if ( tout[1] > 2 ) {
01678         t = tout;
01679         tt[0] = t[0]; tt[1] = t[1];
01680         for ( i = 2; i < t[1]; i += 3 ) {
01681             tt[i] = t[i]; tt[i+1] = t[i+1]; tt[i+2] = -t[i+2];
01682         }
01683         tt += tt[1];
01684         tout += tout[1];
01685         action++;
01686     }
01687     /*
01688     #] DOTPRODUCT :
01689     #[ Coefficient :
01690
01691     Now we have to collect the GCD of the numerators
01692     and the LCM of the denominators.
01693     */
01694     AT.WorkPointer = tt;
01695     if ( AN.cmod != 0 ) {
01696         WORD x, ix, ip;
01697         t = in; tnext = t + *t; tstop = tnext - ABS(tnext[-1]);
01698         x = tstop[0];
01699         if ( tnext[-1] < 0 ) x += AC.cmod[0];
01700         if ( GetModInverses(x, (WORD) (AN.cmod[0]), &ix, &ip) ) goto CalledFrom;
01701         *tout++ = x; *tout++ = 1; *tout++ = 3;
01702         *tt++ = ix; *tt++ = 1; *tt++ = 3;
01703     }
01704     else {
01705         GCDBuffer = NumberMalloc("MakeInteger");
01706         GCDBuffer2 = NumberMalloc("MakeInteger");
01707         LCMbuffer = NumberMalloc("MakeInteger");
01708         LCMbuffer2 = NumberMalloc("MakeInteger");
01709         t = in;
01710         tnext = t + *t; length = tnext[-1];
01711         if ( length < 0 ) { sign = -1; length = -length; }

```



```

01712     else { sign = 1; }
01713     tstop = tnext - length;
01714     redlength = (length-1)/2;
01715     for ( i = 0; i < redlength; i++ ) {
01716         GCDbuffer[i] = (UWORD) (tstop[i]);
01717         LCMBuffer[i] = (UWORD) (tstop[redlength+i]);
01718     }
01719     GCDlen = LCMLen = redlength;
01720     while ( GCDbuffer[GCDlen-1] == 0 ) GCDlen--;
01721     while ( LCMBuffer[LCMLen-1] == 0 ) LCMLen--;
01722     t = tnext;
01723     while ( *t ) {
01724         tnext = t + *t; length = ABS(tnext[-1]);
01725         tstop = tnext - length; redlength = (length-1)/2;
01726         len1 = len2 = redlength;
01727         den = tstop + redlength;
01728         while ( tstop[len1-1] == 0 ) len1--;
01729         while ( den[len2-1] == 0 ) len2--;
01730         if ( GCDlen == 1 && GCDbuffer[0] == 1 ) {}
01731         else {
01732             GcdLong(BHEAD (UWORD *)tstop, len1, GCDbuffer, GCDlen, GCDbuffer2, &GCDlen2);
01733             ap = GCDbuffer; GCDbuffer = GCDbuffer2; GCDbuffer2 = ap;
01734             a = GCDlen; GCDlen = GCDlen2; GCDlen2 = a;
01735         }
01736         if ( len2 == 1 && den[0] == 1 ) {}
01737         else {
01738             GcdLong(BHEAD LCMBuffer, LCMLen, (UWORD *)den, len2, LCMBuffer2, &LCMLen2);
01739             DivLong((UWORD *)den, len2, LCMBuffer2, LCMLen2,
01740                 GCDbuffer2, &GCDlen2, (UWORD *)AT.WorkPointer, &a);
01741             Mullong(LCMBuffer, LCMLen, GCDbuffer2, GCDlen2, LCMBuffer2, &LCMLen2);
01742             ap = LCMBuffer; LCMBuffer = LCMBuffer2; LCMBuffer2 = ap;
01743             a = LCMLen; LCMLen = LCMLen2; LCMLen2 = a;
01744         }
01745         t = tnext;
01746     }
01747     if ( GCDlen != 1 || GCDbuffer[0] != 1 || LCMLen != 1 || LCMBuffer[0] != 1 ) {
01748         redlength = GCDlen; if ( LCMLen > GCDlen ) redlength = LCMLen;
01749         for ( i = 0; i < GCDlen; i++ ) *tout++ = (WORD) (GCDbuffer[i]);
01750         for ( ; i < redlength; i++ ) *tout++ = 0;
01751         for ( i = 0; i < LCMLen; i++ ) *tout++ = (WORD) (LCMBuffer[i]);
01752         for ( ; i < redlength; i++ ) *tout++ = 0;
01753         *tout++ = (2*redlength+1)*sign;
01754         for ( i = 0; i < LCMLen; i++ ) *ttt++ = (WORD) (LCMBuffer[i]);
01755         for ( ; i < redlength; i++ ) *ttt++ = 0;
01756         for ( i = 0; i < GCDlen; i++ ) *ttt++ = (WORD) (GCDbuffer[i]);
01757         for ( ; i < redlength; i++ ) *ttt++ = 0;
01758         *ttt++ = (2*redlength+1)*sign;
01759         action++;
01760     }
01761     else {
01762         *tout++ = 1; *tout++ = 1; *tout++ = 3*sign;
01763         *ttt++ = 1; *ttt++ = 1; *ttt++ = 3*sign;
01764         if ( sign != 1 ) action++;
01765     }
01766     *tout = 0;
01767     NumberFree(LCMBuffer2, "MakeInteger");
01768     NumberFree(LCMBuffer, "MakeInteger");
01769     NumberFree(GCDbuffer2, "MakeInteger");
01770     NumberFree(GCDbuffer, "MakeInteger");
01771 }
01772 /*
01773 #] Coefficient :
01774 #[ Multiply by the inverse content :
01775 */
01776 if ( action ) {
01777     *tterm = tt - tterm;
01778     AT.WorkPointer = tt;
01779     inp = MultiplyWithTerm(BHEAD in, tterm, 2);
01780     AT.WorkPointer = tterm;
01781     in = inp;
01782 }
01783 /*
01784 #] Multiply by the inverse content :
01785 */
01786 *term = tout - term;
01787 AT.WorkPointer = tterm;
01788 return(in);
01789 CalledFrom:
01790 MLOCK(ErrorMessageLock);
01791 MesCall("TakeContent");
01792 MUNLOCK(ErrorMessageLock);
01793 return(0);
01794 }
01795
01796 /*
01797 #] TakeContent :
01798 #[ MergeSymbolLists :

```

```

01799
01800     Merges the extra list into the old.
01801     If par == -1 we take minimum powers
01802     If par == 1 we take maximum powers
01803     If par == 0 we take minimum of the absolute value of the powers
01804             if one is positive and the other negative we get zero.
01805     We assume that the symbols are in order in both lists
01806 */
01807
01808 int MergeSymbolLists(PHEAD WORD *old, WORD *extra, int par)
01809 {
01810     GETBIDENTITY
01811     WORD *new = TermMalloc("MergeSymbolLists");
01812     WORD *t1, *t2, *fill;
01813     int i1, i2;
01814     fill = new + 2; *new = SYMBOL;
01815     i1 = old[1] - 2; i2 = extra[1] - 2;
01816     t1 = old + 2; t2 = extra + 2;
01817     switch ( par ) {
01818     case -1:
01819         while ( i1 > 0 && i2 > 0 ) {
01820             if ( *t1 > *t2 ) {
01821                 if ( t2[1] < 0 ) { *fill++ = *t2++; *fill++ = *t2++; }
01822                 else t2 += 2;
01823                 i2 -= 2;
01824             }
01825             else if ( *t1 < *t2 ) {
01826                 if ( t1[1] < 0 ) { *fill++ = *t1++; *fill++ = *t1++; }
01827                 else t1 += 2;
01828                 i1 -= 2;
01829             }
01830             else if ( t1[1] < t2[1] ) {
01831                 *fill++ = *t1++; *fill++ = *t1++; t2 += 2;
01832                 i1 -= 2; i2 -= 2;
01833             }
01834             else {
01835                 *fill++ = *t2++; *fill++ = *t2++; t1 += 2;
01836                 i1 -= 2; i2 -= 2;
01837             }
01838         }
01839         for ( ; i1 > 0; i1 -= 2 ) {
01840             if ( t1[1] < 0 ) { *fill++ = *t1++; *fill++ = *t1++; }
01841             else t1 += 2;
01842         }
01843         for ( ; i2 > 0; i2 -= 2 ) {
01844             if ( t2[1] < 0 ) { *fill++ = *t2++; *fill++ = *t2++; }
01845             else t2 += 2;
01846         }
01847         break;
01848     case 1:
01849         while ( i1 > 0 && i2 > 0 ) {
01850             if ( *t1 > *t2 ) {
01851                 if ( t2[1] > 0 ) { *fill++ = *t2++; *fill++ = *t2++; }
01852                 else t2 += 2;
01853                 i2 -= 2;
01854             }
01855             else if ( *t1 < *t2 ) {
01856                 if ( t1[1] > 0 ) { *fill++ = *t1++; *fill++ = *t1++; }
01857                 else t1 += 2;
01858                 i1 -= 2;
01859             }
01860             else if ( t1[1] > t2[1] ) {
01861                 *fill++ = *t1++; *fill++ = *t1++; t2 += 2;
01862                 i1 -= 2; i2 -= 2;
01863             }
01864             else {
01865                 *fill++ = *t2++; *fill++ = *t2++; t1 += 2;
01866                 i1 -= 2; i2 -= 2;
01867             }
01868         }
01869         for ( ; i1 > 0; i1 -= 2 ) {
01870             if ( t1[1] > 0 ) { *fill++ = *t1++; *fill++ = *t1++; }
01871             else t1 += 2;
01872         }
01873         for ( ; i2 > 0; i2 -= 2 ) {
01874             if ( t2[1] > 0 ) { *fill++ = *t2++; *fill++ = *t2++; }
01875             else t2 += 2;
01876         }
01877         break;
01878     case 0:
01879         while ( i1 > 0 && i2 > 0 ) {
01880             if ( *t1 > *t2 ) {
01881                 t2 += 2; i2 -= 2;
01882             }
01883             else if ( *t1 < *t2 ) {
01884                 t1 += 2; i1 -= 2;
01885             }

```

```

01886         else if ( ( t1[1] > 0 ) && ( t2[1] < 0 ) ) { t1 += 2; t2 += 2; i1 -= 2; i2 -= 2; }
01887         else if ( ( t1[1] < 0 ) && ( t2[1] > 0 ) ) { t1 += 2; t2 += 2; i1 -= 2; i2 -= 2; }
01888         else if ( t1[1] > 0 ) {
01889             if ( t1[1] < t2[1] ) {
01890                 *fill++ = *t1++; *fill++ = *t1++; t2 += 2; i2 -= 2;
01891             }
01892             else {
01893                 *fill++ = *t2++; *fill++ = *t2++; t1 += 2; i1 -= 2;
01894             }
01895         }
01896         else {
01897             if ( t2[1] < t1[1] ) {
01898                 *fill++ = *t2++; *fill++ = *t2++; t1 += 2; i1 -= 2; i2 -= 2;
01899             }
01900             else {
01901                 *fill++ = *t1++; *fill++ = *t1++; t2 += 2; i1 -= 2; i2 -= 2;
01902             }
01903         }
01904     }
01905     for ( ; i1 > 0; i1-- ) *fill++ = *t1++;
01906     for ( ; i2 > 0; i2-- ) *fill++ = *t2++;
01907     break;
01908 }
01909 i1 = new[1] = fill - new;
01910 t2 = new; t1 = old; NCOPY(t1,t2,i1);
01911 TermFree(new,"MergeSymbolLists");
01912 return(0);
01913 }
01914
01915 /*
01916     #] MergeSymbolLists :
01917     #[ MergeDotproductLists :
01918
01919     Merges the extra list into the old.
01920     If par == -1 we take minimum powers
01921     If par == 1 we take maximum powers
01922     If par == 0 we take minimum of the absolute value of the powers
01923         if one is positive and the other negative we get zero.
01924     We assume that the dotproducts are in order in both lists
01925 */
01926
01927 int MergeDotproductLists(PHEAD WORD *old, WORD *extra, int par)
01928 {
01929     GETBIDENTITY
01930     WORD *new = TermMalloc("MergeDotproductLists");
01931     WORD *t1, *t2, *fill;
01932     int i1,i2;
01933     fill = new + 2;
01934     i1 = old[1] - 2; i2 = extra[1] - 2;
01935     t1 = old + 2; t2 = extra + 2;
01936     switch ( par ) {
01937         case -1:
01938             while ( i1 > 0 && i2 > 0 ) {
01939                 if ( ( *t1 > *t2 ) || ( *t1 == *t2 && t1[1] > t2[1] ) ) {
01940                     if ( t2[2] < 0 ) { *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; }
01941                     else t2 += 3;
01942                 }
01943                 else if ( ( *t1 < *t2 ) || ( *t1 == *t2 && t1[1] < t2[1] ) ) {
01944                     if ( t1[2] < 0 ) { *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; }
01945                     else t1 += 3;
01946                 }
01947                 else if ( t1[2] < t2[2] ) {
01948                     *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; t2 += 3;
01949                 }
01950                 else {
01951                     *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; t1 += 3;
01952                 }
01953             }
01954             break;
01955         case 1:
01956             while ( i1 > 0 && i2 > 0 ) {
01957                 if ( ( *t1 > *t2 ) || ( *t1 == *t2 && t1[1] > t2[1] ) ) {
01958                     if ( t2[2] > 0 ) { *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; }
01959                     else t2 += 3;
01960                 }
01961                 else if ( ( *t1 < *t2 ) || ( *t1 == *t2 && t1[1] < t2[1] ) ) {
01962                     if ( t1[2] > 0 ) { *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; }
01963                     else t1 += 3;
01964                 }
01965                 else if ( t1[2] > t2[2] ) {
01966                     *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; t2 += 3;
01967                 }
01968                 else {
01969                     *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; t1 += 3;
01970                 }
01971             }
01972             break;

```

```

01973         case 0:
01974             while ( i1 > 0 && i2 > 0 ) {
01975                 if ( ( *t1 > *t2 ) || ( *t1 == *t2 && t1[1] > t2[1] ) ) {
01976                     t2 += 3;
01977                 }
01978                 else if ( ( *t1 < *t2 ) || ( *t1 == *t2 && t1[1] < t2[1] ) ) {
01979                     t1 += 3;
01980                 }
01981                 else if ( ( t1[2] > 0 ) && ( t2[2] < 0 ) ) { t1 += 3; t2 += 3; }
01982                 else if ( ( t1[2] < 0 ) && ( t2[2] > 0 ) ) { t1 += 3; t2 += 3; }
01983                 else if ( t1[2] > 0 ) {
01984                     if ( t1[2] < t2[2] ) {
01985                         *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; t2 += 3;
01986                     }
01987                     else {
01988                         *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; t1 += 3;
01989                     }
01990                 }
01991                 else {
01992                     if ( t2[2] < t1[2] ) {
01993                         *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; t1 += 3;
01994                     }
01995                     else {
01996                         *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; t2 += 3;
01997                     }
01998                 }
01999             }
02000             break;
02001         }
02002         i1 = new[1] = fill - new;
02003         t2 = new; t1 = old; NCOPY(t1,t2,i1);
02004         TermFree(new, "MergeDotproductLists");
02005         return(0);
02006     }
02007
02008     /*
02009         #] MergeDotproductLists :
02010         #[ CreateExpression :
02011
02012         Looks for the expression in the argument, reads it and puts it
02013         in a buffer. Returns the address of the buffer.
02014         We send the expression through the Generator system, because there
02015         may be unsubstituted (sub)expressions as in
02016         Local F = (a+b);
02017         Local G = gcd_(F,...);
02018     */
02019
02020     WORD *CreateExpression(PHEAD WORD nexp)
02021     {
02022         GETBIDENTITY
02023         CBUF *C = cbuf+AC.cbunum;
02024         POSITION startposition, oldposition;
02025         FILEHANDLE *fi;
02026         WORD *term, *oldipointer = AR.CompressPointer;
02027     ;
02028         switch ( Expressions[nexp].status ) {
02029             case HIDDENLEXPRESSION:
02030             case HIDDENGEXPRESSION:
02031             case DROPHLEXPRESSION:
02032             case DROPHGEXPRESSION:
02033             case UNHIDELEXPRESSION:
02034             case UNHIDEGEXPRESSION:
02035                 AR.GetOneFile = 2; fi = AR.hidefile;
02036                 break;
02037             default:
02038                 AR.GetOneFile = 0; fi = AR.infile;
02039                 break;
02040         }
02041         SeekScratch(fi,&oldposition);
02042         startposition = AS.OldOnFile[nexp];
02043         term = AT.WorkPointer;
02044         if ( GetOneTerm(BHEAD term,fi,&startposition,0) <= 0 ) goto CalledFrom;
02045         NewSort (BHEAD0);
02046         AR.CompressPointer = oldipointer;
02047         while ( GetOneTerm(BHEAD term,fi,&startposition,0) > 0 ) {
02048             AT.WorkPointer = term + *term;
02049             if ( Generator(BHEAD term,C->numlhs) ) {
02050                 LowerSortLevel();
02051                 goto CalledFrom;
02052             }
02053             AR.CompressPointer = oldipointer;
02054         }
02055         AT.WorkPointer = term;
02056         if ( EndSort (BHEAD (WORD *) ((void *) (&term)),2) < 0 ) goto CalledFrom;
02057         SetScratch(fi,&oldposition);
02058         return (term);
02059     CalledFrom:

```

```

02060     MLOCK(ErrorMessageLock);
02061     MesCall("CreateExpression");
02062     MUNLOCK(ErrorMessageLock);
02063     Terminate(-1);
02064     return(0);
02065 }
02066
02067 /*
02068     #] CreateExpression :
02069     #[ GCDterms :      GCD of two terms
02070
02071     Computes the GCD of two terms.
02072     Output in termout.
02073     termout may overlap with term1.
02074 */
02075
02076 int GCDterms(PHEAD WORD *term1, WORD *term2, WORD *termout)
02077 {
02078     GETBIDENTITY
02079     WORD *t1, *t1stop, *t1next, *t2, *t2stop, *t2next, *tout, *tt1, *tt2;
02080     int count1, count2, i, ii, x1, sign;
02081     WORD length1, length2;
02082     t1 = term1 + *term1; t1stop = t1 - ABS(t1[-1]); t1 = term1+1;
02083     t2 = term2 + *term2; t2stop = t2 - ABS(t2[-1]); t2 = term2+1;
02084     tout = termout+1;
02085     while ( t1 < t1stop ) {
02086         t1next = t1 + t1[1];
02087         t2 = term2+1;
02088         if ( *t1 == SYMBOL ) {
02089             while ( t2 < t2stop && *t2 != SYMBOL ) t2 += t2[1];
02090             if ( t2 < t2stop && *t2 == SYMBOL ) {
02091                 t2next = t2+t2[1];
02092                 tt1 = t1+2; tt2 = t2+2; count1 = 0;
02093                 while ( tt1 < t1next && tt2 < t2next ) {
02094                     if ( *tt1 < *tt2 ) tt1 += 2;
02095                     else if ( *tt1 > *tt2 ) tt2 += 2;
02096                     else if ( ( tt1[1] > 0 && tt2[1] < 0 ) ||
02097                             ( tt2[1] > 0 && tt1[1] < 0 ) ) {
02098                         tt1 += 2; tt2 += 2;
02099                     }
02100                     else {
02101                         x1 = tt1[1];
02102                         if ( tt1[1] < 0 ) { if ( tt2[1] > x1 ) x1 = tt2[1]; }
02103                         else { if ( tt2[1] < x1 ) x1 = tt2[1]; }
02104                         tout[count1+2] = *tt1;
02105                         tout[count1+3] = x1;
02106                         tt1 += 2; tt2 += 2;
02107                         count1 += 2;
02108                     }
02109                 }
02110                 if ( count1 > 0 ) {
02111                     *tout = SYMBOL; tout[1] = count1+2; tout += tout[1];
02112                 }
02113             }
02114         }
02115         else if ( *t1 == DOTPRODUCT ) {
02116             while ( t2 < t2stop && *t2 != DOTPRODUCT ) t2 += t2[1];
02117             if ( t2 < t2stop && *t2 == DOTPRODUCT ) {
02118                 t2next = t2+t2[1];
02119                 tt1 = t1+2; tt2 = t2+2; count1 = 0;
02120                 while ( tt1 < t1next && tt2 < t2next ) {
02121                     if ( *tt1 < *tt2 || ( *tt1 == *tt2 && tt1[1] < tt2[1] ) ) tt1 += 3;
02122                     else if ( *tt1 > *tt2 || ( *tt1 == *tt2 && tt1[1] > tt2[1] ) ) tt2 += 3;
02123                     else if ( ( tt1[2] > 0 && tt2[2] < 0 ) ||
02124                             ( tt2[2] > 0 && tt1[2] < 0 ) ) {
02125                         tt1 += 3; tt2 += 3;
02126                     }
02127                     else {
02128                         x1 = tt1[2];
02129                         if ( tt1[2] < 0 ) { if ( tt2[2] > x1 ) x1 = tt2[2]; }
02130                         else { if ( tt2[2] < x1 ) x1 = tt2[2]; }
02131                         tout[count1+2] = *tt1;
02132                         tout[count1+3] = tt1[1];
02133                         tout[count1+4] = x1;
02134                         tt1 += 3; tt2 += 3;
02135                         count1 += 3;
02136                     }
02137                 }
02138                 if ( count1 > 0 ) {
02139                     *tout = DOTPRODUCT; tout[1] = count1+2; tout += tout[1];
02140                 }
02141             }
02142         }
02143         else if ( *t1 == VECTOR ) {
02144             while ( t2 < t2stop && *t2 != VECTOR ) t2 += t2[1];
02145             if ( t2 < t2stop && *t2 == VECTOR ) {
02146                 t2next = t2+t2[1];

```

```

02147         tt1 = t1+2; tt2 = t2+2; count1 = 0;
02148         while ( tt1 < tlnext && tt2 < t2next ) {
02149             if ( *tt1 < *tt2 || ( *tt1 == *tt2 && tt1[1] < tt2[1] ) ) tt1 += 2;
02150             else if ( *tt1 > *tt2 || ( *tt1 == *tt2 && tt1[1] > tt2[1] ) ) tt2 += 2;
02151             else {
02152                 tout[count1+2] = *tt1;
02153                 tout[count1+3] = tt1[1];
02154                 tt1 += 2; tt2 += 2;
02155                 count1 += 2;
02156             }
02157         }
02158         if ( count1 > 0 ) {
02159             *tout = VECTOR; tout[1] = count1+2; tout += tout[1];
02160         }
02161     }
02162 }
02163 else if ( *t1 == INDEX ) {
02164     while ( t2 < t2stop && *t2 != INDEX ) t2 += t2[1];
02165     if ( t2 < t2stop && *t2 == INDEX ) {
02166         t2next = t2+t2[1];
02167         tt1 = t1+2; tt2 = t2+2; count1 = 0;
02168         while ( tt1 < tlnext && tt2 < t2next ) {
02169             if ( *tt1 < *tt2 ) tt1 += 1;
02170             else if ( *tt1 > *tt2 ) tt2 += 1;
02171             else {
02172                 tout[count1+2] = *tt1;
02173                 tt1 += 1; tt2 += 1;
02174                 count1 += 1;
02175             }
02176         }
02177         if ( count1 > 0 ) {
02178             *tout = INDEX; tout[1] = count1+2; tout += tout[1];
02179         }
02180     }
02181 }
02182 else if ( *t1 == DELTA ) {
02183     while ( t2 < t2stop && *t2 != DELTA ) t2 += t2[1];
02184     if ( t2 < t2stop && *t2 == DELTA ) {
02185         t2next = t2+t2[1];
02186         tt1 = t1+2; tt2 = t2+2; count1 = 0;
02187         while ( tt1 < tlnext && tt2 < t2next ) {
02188             if ( *tt1 < *tt2 || ( *tt1 == *tt2 && tt1[1] < tt2[1] ) ) tt1 += 2;
02189             else if ( *tt1 > *tt2 || ( *tt1 == *tt2 && tt1[1] > tt2[1] ) ) tt2 += 2;
02190             else {
02191                 tout[count1+2] = *tt1;
02192                 tout[count1+3] = tt1[1];
02193                 tt1 += 2; tt2 += 2;
02194                 count1 += 2;
02195             }
02196         }
02197         if ( count1 > 0 ) {
02198             *tout = DELTA; tout[1] = count1+2; tout += tout[1];
02199         }
02200     }
02201 }
02202 else if ( *t1 >= FUNCTION ) { /* noncommuting functions? Forbidden! */
02203     /*
02204         Count how many times this function occurs.
02205         Then count how many times it is in term2.
02206     */
02207     count1 = 1;
02208     while ( tlnext < t1stop && *t1 == *tlnext && t1[1] == tlnext[1] ) {
02209         for ( i = 2; i < t1[1]; i++ ) {
02210             if ( t1[i] != tlnext[i] ) break;
02211         }
02212         if ( i < t1[1] ) break;
02213         count1++;
02214         tlnext += tlnext[1];
02215     }
02216     count2 = 0;
02217     while ( t2 < t2stop ) {
02218         if ( *t2 == *t1 && t2[1] == t1[1] ) {
02219             for ( i = 2; i < t1[1]; i++ ) {
02220                 if ( t2[i] != t1[i] ) break;
02221             }
02222             if ( i >= t1[1] ) count2++;
02223         }
02224         t2 += t2[1];
02225     }
02226     if ( count1 < count2 ) count2 = count1; /* number of common occurrences */
02227     if ( count2 > 0 ) {
02228         if ( tout == t1 ) {
02229             while ( count2 > 0 ) { tout += tout[1]; count2--; }
02230         }
02231         else {
02232             i = t1[1]*count2;
02233             NCOPY(tout,t1,i);

```

```

02234     }
02235     }
02236     }
02237     t1 = t1next;
02238 }
02239 /*
02240     Now the coefficients. They are in t1stop and t2stop. Should go to tout.
02241 */
02242     sign = 1;
02243     length1 = term1[*term1-1]; ii = i = ABS(length1); t1 = t1stop;
02244     if ( t1 != tout ) { NCOPY(tout,t1,i); tout -= ii; }
02245     length2 = term2[*term2-1];
02246     if ( length1 < 0 && length2 < 0 ) sign = -1;
02247     if ( AccumGCD(BHEAD (UWORD *)tout,&length1,(UWORD *)t2stop,length2) ) {
02248         MLOCK(ErrorMessageLock);
02249         MesCall("GCDterms");
02250         MUNLOCK(ErrorMessageLock);
02251         SETERROR(-1)
02252     }
02253     if ( sign < 0 && length1 > 0 ) length1 = -length1;
02254     tout += ABS(length1); tout[-1] = length1;
02255     *termout = tout - termout; *tout = 0;
02256     return(0);
02257 }
02258 /*
02259     #] GCDterms :
02260     #[ ReadPolyRatFun :
02261     #[ ReadPolyRatFun :
02262 */
02263
02264 int ReadPolyRatFun(PHEAD WORD *term)
02265 {
02266     WORD *oldworkpointer = AT.WorkPointer;
02267     int flag, i;
02268     WORD *t, *fun, *nextt, *num, *den, *t1, *t2, size, numsize, densize;
02269     WORD *term1, *term2, *confree1, *confree2, *gcd, *num1, *den1, move, *newnum, *newden;
02270     WORD *tstop, *m1, *m2;
02271     WORD oldsorttype = AR.SortType;
02272     COMPARE oldcompareroutine = (COMPARE) (AR.CompareRoutine);
02273     AR.SortType = SORTHIGFIRST;
02274     AR.CompareRoutine = (COMPAREDUMMY) (&CompareSymbols);
02275
02276     tstop = term + *term; tstop -= ABS(tstop[-1]);
02277     if ( term + *term == AT.WorkPointer ) flag = 1;
02278     else flag = 0;
02279     t = term+1;
02280     while ( t < tstop ) {
02281         if ( *t != AR.PolyFun ) { t += t[1]; continue; }
02282         if ( ( t[2] & MUSTCLEANPRF ) == 0 ) { t += t[1]; continue; }
02283         fun = t;
02284         nextt = t + t[1];
02285         if ( fun[1] > FUNHEAD && fun[FUNHEAD] == -SNUMBER && fun[FUNHEAD+1] == 0 )
02286             { *term = 0; break; }
02287         if ( FromPolyRatFun(BHEAD fun, &num, &den) > 0 ) { t = nextt; continue; }
02288         if ( *num == ARGHEAD ) { *term = 0; break; }
02289     /*
02290         Now we have num and den. Both are in general argument notation,
02291         but can also be used as expressions as in num+ARGHEAD, den+ARGHEAD.
02292         We need the gcd. For this we have to take out the contents
02293         because PreGCD does not like contents.
02294     */
02295         term1 = TermMalloc("ReadPolyRatFun");
02296         term2 = TermMalloc("ReadPolyRatFun");
02297         confree1 = TakeSymbolContent(BHEAD num+ARGHEAD,term1);
02298         confree2 = TakeSymbolContent(BHEAD den+ARGHEAD,term2);
02299         GCDclean(BHEAD term1,term2);
02300     /*
02301         gcd = PreGCD(BHEAD confree1,confree2,1); */
02302         gcd = poly_gcd(BHEAD confree1,confree2,1);
02303         newnum = PolyDiv(BHEAD confree1,gcd,"ReadPolyRatFun");
02304         newden = PolyDiv(BHEAD confree2,gcd,"ReadPolyRatFun");
02305         TermFree(confree2,"ReadPolyRatFun");
02306         TermFree(confree1,"ReadPolyRatFun");
02307         num1 = MULfunc(BHEAD term1,newnum);
02308         den1 = MULfunc(BHEAD term2,newden);
02309         TermFree(newnum,"ReadPolyRatFun");
02310         TermFree(newden,"ReadPolyRatFun");
02311     /*
02312         M_free(gcd,"poly_gcd"); */
02313         TermFree(gcd,"poly_gcd");
02314         TermFree(term1,"ReadPolyRatFun");
02315         TermFree(term2,"ReadPolyRatFun");
02316     /*
02317         Now we can put the function back together.
02318         Notice that we cannot use ToFast, because there is no reservation
02319         for the header of the argument. Fortunately there are only two
02320         types of fast arguments.
02321     */
02322         if ( num1[0] == 4 && num1[4] == 0 && num1[2] == 1 && num1[1] > 0 ) {

```

```

02321         numsize = 2; numl[0] = -SNUMBER;
02322         if ( numl[3] < 0 ) numl[1] = -numl[1];
02323     }
02324     else if ( numl[0] == 8 && numl[8] == 0 && numl[7] == 3 && numl[6] == 1
02325         && numl[5] == 1 && numl[1] == SYMBOL && numl[4] == 1 ) {
02326         numsize = 2; numl[0] = -SYMBOL; numl[1] = numl[3];
02327     }
02328     else { m1 = numl; while ( *m1 ) m1 += *m1; numsize = (m1-numl)+ARGHEAD; }
02329     if ( denl[0] == 4 && denl[4] == 0 && denl[2] == 1 && denl[1] > 0 ) {
02330         densize = 2; denl[0] = -SNUMBER;
02331         if ( denl[3] < 0 ) denl[1] = -denl[1];
02332     }
02333     else if ( denl[0] == 8 && denl[8] == 0 && denl[7] == 3 && denl[6] == 1
02334         && denl[5] == 1 && denl[1] == SYMBOL && denl[4] == 1 ) {
02335         densize = 2; denl[0] = -SYMBOL; denl[1] = denl[3];
02336     }
02337     else { m2 = denl; while ( *m2 ) m2 += *m2; densize = (m2-denl)+ARGHEAD; }
02338     size = FUNHEAD+numsize+densize;
02339
02340     if ( size > fun[1] ) {
02341         move = size - fun[1];
02342         t1 = term+*term; t2 = t1+move;
02343         while ( t1 > nexttt ) *--t2 = *--t1;
02344         tstop += move; nexttt += move;
02345         *term += move;
02346     }
02347     else if ( size < fun[1] ) {
02348         move = fun[1]-size;
02349         t2 = fun+size; t1 = nexttt;
02350         tstop -= move; nexttt -= move;
02351         t = term+*term;
02352         while ( t1 < t ) *t2++ = *t1++;
02353         *term -= move;
02354     }
02355     else { /* no need to move anything */ }
02356     fun[1] = size; fun[2] = 0;
02357     t2 = fun+FUNHEAD; t1 = numl;
02358     if ( *numl < 0 ) { *t2++ = numl[0]; *t2++ = numl[1]; }
02359     else { *t2++ = numsize; *t2++ = 0; FILLARG(t2);
02360         i = numsize-ARGHEAD; NCOPY(t2,t1,i) }
02361     t1 = denl;
02362     if ( *denl < 0 ) { *t2++ = denl[0]; *t2++ = denl[1]; }
02363     else { *t2++ = densize; *t2++ = 0; FILLARG(t2);
02364         i = densize-ARGHEAD; NCOPY(t2,t1,i) }
02365
02366     TermFree(numl,"MULfunc");
02367     TermFree(denl,"MULfunc");
02368     t = nexttt;
02369 }
02370
02371 if ( flag ) AT.WorkPointer = term + *term;
02372 else AT.WorkPointer = oldworkpointer;
02373 AR.CompareRoutine = (COMPAREDUMMY)oldcompareroutine;
02374 AR.SortType = oldsorttype;
02375 return(0);
02376 }
02377
02378 /*
02379     #] ReadPolyRatFun :
02380     #[ FromPolyRatFun :
02381 */
02382
02383 int FromPolyRatFun(PHEAD WORD *fun, WORD **numout, WORD **denout)
02384 {
02385     WORD *nextfun, *tt, *num, *den;
02386     int i;
02387     nextfun = fun + fun[1];
02388     fun += FUNHEAD;
02389     num = AT.WorkPointer;
02390     if ( *fun < 0 ) {
02391         if ( *fun != -SNUMBER && *fun != -SYMBOL ) goto Improper;
02392         ToGeneral(fun,num,0);
02393         tt = num + *num; *tt++ = 0;
02394         fun += 2;
02395     }
02396     else { i = *fun; tt = num; NCOPY(tt,fun,i); *tt++ = 0; }
02397     den = tt;
02398     if ( *fun < 0 ) {
02399         if ( *fun != -SNUMBER && *fun != -SYMBOL ) goto Improper;
02400         ToGeneral(fun,den,0);
02401         tt = den + *den; *tt++ = 0;
02402         fun += 2;
02403     }
02404     else { i = *fun; tt = den; NCOPY(tt,fun,i); *tt++ = 0; }
02405     *numout = num; *denout = den;
02406     if ( fun != nextfun ) { return(1); }
02407     AT.WorkPointer = tt;

```



```

02408     return(0);
02409 Improper:
02410     MLOCK(ErrorMessageLock);
02411     MesPrint("Improper use of PolyRatFun");
02412     MesCall("FromPolyRatFun");
02413     MUNLOCK(ErrorMessageLock);
02414     SETERROR(-1);
02415 }
02416
02417 /*
02418     #] FromPolyRatFun :
02419     #[ TakeSymbolContent :
02420 */
02434 WORD *TakeSymbolContent (PHEAD WORD *in, WORD *term)
02435 {
02436     GETBIDENTITY
02437     WORD *t, *tstop, *tout, *tstore;
02438     WORD *tnext, *tt, *tterm;
02439     WORD *inp, a, *den, *oldworkpointer = AT.WorkPointer;
02440     int i, action = 0, sign, first;
02441     UWORD *GCDBuffer, *GCDBuffer2, *LCMbuffer, *LCMbuffer2, *ap;
02442     WORD GCDlen, GCDlen2, LCMlen, LCMlen2, length, redlength, len1, len2;
02443     LONG j;
02444     tout = tstore = term+1;
02445 /*
02446     #[ SYMBOL :
02447
02448     We make a list of symbols and their minimal powers.
02449     This includes negative powers. In the end we have to multiply by the
02450     inverse of this list. That takes out all negative powers and leaves
02451     things ready for further processing.
02452 */
02453     tterm = AT.WorkPointer; tt = tterm+1;
02454     tout[0] = SYMBOL; tout[1] = 2;
02455     t = in; first = 1;
02456     while ( *t ) {
02457         tnext = t + *t; tstop = tnext - ABS(tnext[-1]); t++;
02458         while ( t < tstop ) {
02459             if ( first ) {
02460                 if ( *t == SYMBOL ) {
02461                     for ( i = 0; i < t[1]; i++ ) tout[i] = t[i];
02462                     goto didwork;
02463                 }
02464                 else {
02465                     MLOCK(ErrorMessageLock);
02466                     MesPrint ((char*)"ERROR: polynomials and polyratfuns must contain symbols only");
02467                     MUNLOCK(ErrorMessageLock);
02468                     Terminate(1);
02469                 }
02470             }
02471             else if ( *t == SYMBOL ) {
02472                 MergeSymbolLists(BHEAD tout,t,-1);
02473                 goto didwork;
02474             }
02475             else {
02476                 t += t[1];
02477             }
02478         }
02479 /*
02480         Here we come when there were no symbols. Only keep the negative ones.
02481 */
02482         if ( first == 0 ) {
02483             int j = 2;
02484             for ( i = 2; i < tout[1]; i += 2 ) {
02485                 if ( tout[i+1] < 0 ) {
02486                     if ( i == j ) { j += 2; }
02487                     else { tout[j] = tout[i]; tout[j+1] = tout[i+1]; j += 2; }
02488                 }
02489             }
02490             tout[1] = j;
02491         }
02492     didwork:;
02493     first = 0;
02494     t = tnext;
02495 }
02496 if ( tout[1] > 2 ) {
02497     t = tout;
02498     tt[0] = t[0]; tt[1] = t[1];
02499     for ( i = 2; i < t[1]; i += 2 ) {
02500         tt[i] = t[i]; tt[i+1] = -t[i+1];
02501     }
02502     tt += tt[1];
02503     tout += tout[1];
02504     action++;
02505 }
02506 /*
02507     #] SYMBOL :

```

```

02508     #[ Coefficient :
02509
02510     Now we have to collect the GCD of the numerators
02511     and the LCM of the denominators.
02512 */
02513     AT.WorkPointer = tt;
02514     if ( AN.cmod != 0 ) {
02515         WORD x, ix, ip;
02516         t = in; tnext = t + *t; tstop = tnext - ABS(tnext[-1]);
02517         x = tstop[0];
02518         if ( tnext[-1] < 0 ) x += AC.cmod[0];
02519         if ( GetModInverses(x, (WORD)(AN.cmod[0]), &ix, &ip) ) goto CalledFrom;
02520         *tout++ = x; *tout++ = 1; *tout++ = 3;
02521         *tt++ = ix; *tt++ = 1; *tt++ = 3;
02522     }
02523     else {
02524         GCDbuffer = NumberMalloc("MakeInteger");
02525         GCDbuffer2 = NumberMalloc("MakeInteger");
02526         LCMbuffer = NumberMalloc("MakeInteger");
02527         LCMbuffer2 = NumberMalloc("MakeInteger");
02528         t = in;
02529         tnext = t + *t; length = tnext[-1];
02530         if ( length < 0 ) { sign = -1; length = -length; }
02531         else { sign = 1; }
02532         tstop = tnext - length;
02533         redlength = (length-1)/2;
02534         for ( i = 0; i < redlength; i++ ) {
02535             GCDbuffer[i] = (UWORD)(tstop[i]);
02536             LCMbuffer[i] = (UWORD)(tstop[redlength+i]);
02537         }
02538         GCDlen = LCMlen = redlength;
02539         while ( GCDbuffer[GCDlen-1] == 0 ) GCDlen--;
02540         while ( LCMbuffer[LCMlen-1] == 0 ) LCMlen--;
02541         t = tnext;
02542         while ( *t ) {
02543             tnext = t + *t; length = ABS(tnext[-1]);
02544             tstop = tnext - length; redlength = (length-1)/2;
02545             len1 = len2 = redlength;
02546             den = tstop + redlength;
02547             while ( tstop[len1-1] == 0 ) len1--;
02548             while ( den[len2-1] == 0 ) len2--;
02549             if ( GCDlen == 1 && GCDbuffer[0] == 1 ) {}
02550             else {
02551                 GcdLong(BHEAD (UWORD *)tstop, len1, GCDbuffer, GCDlen, GCDbuffer2, &GCDlen2);
02552                 ap = GCDbuffer; GCDbuffer = GCDbuffer2; GCDbuffer2 = ap;
02553                 a = GCDlen; GCDlen = GCDlen2; GCDlen2 = a;
02554             }
02555             if ( len2 == 1 && den[0] == 1 ) {}
02556             else {
02557                 GcdLong(BHEAD LCMbuffer, LCMlen, (UWORD *)den, len2, LCMbuffer2, &LCMlen2);
02558                 DivLong((UWORD *)den, len2, LCMbuffer2, LCMlen2,
02559                     GCDbuffer2, &GCDlen2, (UWORD *)AT.WorkPointer, &a);
02560                 MullLong(LCMbuffer, LCMlen, GCDbuffer2, GCDlen2, LCMbuffer2, &LCMlen2);
02561                 ap = LCMbuffer; LCMbuffer = LCMbuffer2; LCMbuffer2 = ap;
02562                 a = LCMlen; LCMlen = LCMlen2; LCMlen2 = a;
02563             }
02564             t = tnext;
02565         }
02566         if ( GCDlen != 1 || GCDbuffer[0] != 1 || LCMlen != 1 || LCMbuffer[0] != 1 ) {
02567             redlength = GCDlen; if ( LCMlen > GCDlen ) redlength = LCMlen;
02568             for ( i = 0; i < GCDlen; i++ ) *tout++ = (WORD)(GCDbuffer[i]);
02569             for ( ; i < redlength; i++ ) *tout++ = 0;
02570             for ( i = 0; i < LCMlen; i++ ) *tout++ = (WORD)(LCMbuffer[i]);
02571             for ( ; i < redlength; i++ ) *tout++ = 0;
02572             *tout++ = (2*redlength+1)*sign;
02573             for ( i = 0; i < LCMlen; i++ ) *tt++ = (WORD)(LCMbuffer[i]);
02574             for ( ; i < redlength; i++ ) *tt++ = 0;
02575             for ( i = 0; i < GCDlen; i++ ) *tt++ = (WORD)(GCDbuffer[i]);
02576             for ( ; i < redlength; i++ ) *tt++ = 0;
02577             *tt++ = (2*redlength+1)*sign;
02578             action++;
02579         }
02580         else {
02581             *tout++ = 1; *tout++ = 1; *tout++ = 3*sign;
02582             *tt++ = 1; *tt++ = 1; *tt++ = 3*sign;
02583             if ( sign != 1 ) action++;
02584         }
02585         NumberFree(LCMbuffer2, "MakeInteger");
02586         NumberFree(LCMbuffer, "MakeInteger");
02587         NumberFree(GCDbuffer2, "MakeInteger");
02588         NumberFree(GCDbuffer, "MakeInteger");
02589     }
02590 */
02591     #[ Coefficient :
02592     #[ Multiply by the inverse content :
02593 */
02594     if ( action ) {

```

```

02595     *term = tout - term; *tout = 0;
02596     *tterm = tt - tterm; *tt = 0;
02597     AT.WorkPointer = tt;
02598     inp = MultiplyWithTerm(BHEAD in,tterm,2);
02599     AT.WorkPointer = tterm;
02600     t = inp; while ( *t ) t += *t;
02601     j = (t-inp); t = inp;
02602     if ( j*sizeof(WORD) > (size_t)(AM.MaxTer) ) goto OverWork;
02603     in = tout = TermMalloc("TakeSymbolContent");
02604     NCOPY(tout,t,j); *tout = 0;
02605     if ( AN.tryterm > 0 ) { TermFree(inp,"MultiplyWithTerm"); AN.tryterm = 0; }
02606     else { M_free(inp,"MultiplyWithTerm"); }
02607 }
02608 else {
02609     t = in; while ( *t ) t += *t;
02610     j = (t-in); t = in;
02611     if ( j*sizeof(WORD) > (size_t)(AM.MaxTer) ) goto OverWork;
02612     in = tout = TermMalloc("TakeSymbolContent");
02613     NCOPY(tout,t,j); *tout = 0;
02614     term[0] = 4; term[1] = 1; term[2] = 1; term[3] = 3; term[4] = 0;
02615 }
02616 /*
02617 #] Multiply by the inverse content :
02618 AT.WorkPointer = tterm + *tterm;
02619 */
02620 AT.WorkPointer = oldworkpointer;
02621
02622 return(in);
02623 OverWork:
02624 MLOCK(ErrorMessageLock);
02625 MesPrint("Term too complex. Maybe increasing MaxTermSize can help");
02626 MUNLOCK(ErrorMessageLock);
02627 CalledFrom:
02628 MLOCK(ErrorMessageLock);
02629 MesCall("TakeSymbolContent");
02630 MUNLOCK(ErrorMessageLock);
02631 Terminate(-1);
02632 return(0);
02633 }
02634
02635 /*
02636 #] TakeSymbolContent :
02637 #[ GCDclean :
02638
02639     Takes a numerator and a denominator that each consist of a
02640     single term with only a coefficient and symbols and makes them
02641     into a proper fraction. Output overwrites input.
02642 */
02643
02644 void GCDclean(PHEAD WORD *num, WORD *den)
02645 {
02646     WORD *out1 = TermMalloc("GCDclean");
02647     WORD *out2 = TermMalloc("GCDclean");
02648     WORD *t1, *t2, *r1, *r2, *t1stop, *t2stop, csizel, csize2, csize3, pow, sign;
02649     int i;
02650     t1stop = num+*num; sign = ( t1stop[-1] < 0 ) ? -1 : 1;
02651     csizel = ABS(t1stop[-1]); t1stop -= csizel;
02652     t2stop = den+*den; if ( t2stop[-1] < 0 ) sign = -sign;
02653     csize2 = ABS(t2stop[-1]); t2stop -= csize2;
02654     t1 = num+1; t2 = den+1;
02655     r1 = out1+3; r2 = out2+3;
02656     if ( t1 == t1stop ) {
02657         if ( t2 < t2stop ) {
02658             for ( i = 2; i < t2[1]; i += 2 ) {
02659                 if ( t2[i+1] < 0 ) { *r1++ = t2[i]; *r1++ = -t2[i+1]; }
02660                 else { *r2++ = t2[i]; *r2++ = t2[i+1]; }
02661             }
02662         }
02663     }
02664     else if ( t2 == t2stop ) {
02665         for ( i = 2; i < t1[1]; i += 2 ) {
02666             if ( t1[i+1] < 0 ) { *r2++ = t1[i]; *r2++ = -t1[i+1]; }
02667             else { *r1++ = t1[i]; *r1++ = t1[i+1]; }
02668         }
02669     }
02670     else {
02671         t1 += 2; t2 += 2;
02672         while ( t1 < t1stop && t2 < t2stop ) {
02673             if ( *t1 < *t2 ) {
02674                 if ( t1[1] > 0 ) { *r1++ = *t1; *r1++ = t1[1]; t1 += 2; }
02675                 else if ( t1[1] < 0 ) { *r2++ = *t1; *r2++ = -t1[1]; t1 += 2; }
02676             }
02677             else if ( *t1 > *t2 ) {
02678                 if ( t2[1] > 0 ) { *r2++ = *t2; *r2++ = t2[1]; t2 += 2; }
02679                 else if ( t2[1] < 0 ) { *r1++ = *t2; *r1++ = -t2[1]; t2 += 2; }
02680             }
02681             else {

```

```

02682         pow = t1[1]-t2[1];
02683         if ( pow > 0 ) { *r1++ = *t1; *r1++ = pow; }
02684         else if ( pow < 0 ) { *r2++ = *t1; *r2++ = -pow; }
02685         t1 += 2; t2 += 2;
02686     }
02687 }
02688 while ( t1 < t1stop ) {
02689     if ( t1[1] < 0 ) { *r2++ = *t1; *r2++ = -t1[1]; }
02690     else { *r1++ = *t1; *r1++ = t1[1]; }
02691     t1 += 2;
02692 }
02693 while ( t2 < t2stop ) {
02694     if ( t2[1] < 0 ) { *r1++ = *t2; *r1++ = -t2[1]; }
02695     else { *r2++ = *t2; *r2++ = t2[1]; }
02696     t2 += 2;
02697 }
02698 }
02699 if ( r1 > out1+3 ) { out1[1] = SYMBOL; out1[2] = r1 - out1 - 1; }
02700 else r1 = out1+1;
02701 if ( r2 > out2+3 ) { out2[1] = SYMBOL; out2[2] = r2 - out2 - 1; }
02702 else r2 = out2+1;
02703 /*
02704     Now the coefficients.
02705 */
02706 csize1 = REDLENG(csize1);
02707 csize2 = REDLENG(csize2);
02708 if ( DivRat(BHEAD (UWORD *)t1stop,csize1,(UWORD *)t2stop,csize2,(UWORD *)r1,&csize3) ) {
02709     MLOCK(ErrorMessageLock);
02710     MesCall("GCDclean");
02711     MUNLOCK(ErrorMessageLock);
02712     Terminate(-1);
02713 }
02714 UnPack((UWORD *)r1,csize3,&csize2,&csize1);
02715 t2 = r1+ABS(csize3);
02716 for ( i = 0; i < csize2; i++ ) r2[i] = t2[i];
02717 r2 += csize2; *r2++ = 1;
02718 for ( i = 1; i < csize2; i++ ) *r2++ = 0;
02719 csize2 = INCLENG(csize2); *r2++ = csize2; *out2 = r2-out2;
02720 r1 += ABS(csize1); *r1++ = 1;
02721 for ( i = 1; i < ABS(csize1); i++ ) *r1++ = 0;
02722 csize1 = INCLENG(csize1); *r1++ = csize1; *out1 = r1-out1;
02723
02724 t1 = num; t2 = out1; i = *out1; NCOPY(t1,t2,i); *t1 = 0;
02725 if ( sign < 0 ) t1[-1] = -t1[-1];
02726 t1 = den; t2 = out2; i = *out2; NCOPY(t1,t2,i); *t1 = 0;
02727
02728 TermFree(out2,"GCDclean");
02729 TermFree(out1,"GCDclean");
02730 }
02731
02732 /*
02733     #] GCDclean :
02734     #[ PolyDiv :
02735
02736     Special stub function for polynomials that should fit inside a term.
02737     We make sure that the space is allocated by TermMalloc.
02738     This makes things much easier on the calling routines.
02739 */
02740
02741 WORD *PolyDiv(PHEAD WORD *a,WORD *b,char *text)
02742 {
02743     /*
02744         Probably the following would work now
02745     */
02746     DUMMYUSE(text);
02747     return(poly_div(BHEAD a,b,1));
02748     /*
02749         WORD *quo, *qq;
02750         WORD *x, *xx;
02751         LONG i;
02752         quo = poly_div(BHEAD a,b,1);
02753         x = TermMalloc(text);
02754         qq = quo; while ( *qq ) qq += *qq;
02755         i = (qq-quo+1);
02756         if ( i*sizeof(WORD) > (size_t)(AM.MaxTer) ) {
02757             DUMMYUSE(text);
02758             MLOCK(ErrorMessageLock);
02759             MesPrint("PolyDiv: Term too complex. Maybe increasing MaxTermSize can help");
02760             MUNLOCK(ErrorMessageLock);
02761             Terminate(-1);
02762         }
02763         xx = x; qq = quo;
02764         NCOPY(xx,qq,i)
02765         TermFree(quo,"poly_div");
02766         return(x);
02767     */
02768 }

```

```

02769
02770 /*
02771     #] PolyDiv :
02772     #] GCDfunction :
02773     #] DIVfunction :
02774
02775     Input: a div_ function that has two arguments inside a term.
02776     Action: Calculates [arg1/arg2] using polynomial techniques if needed.
02777     Output: The output result is combined with the remainder of the term
02778     and sent to Generator for further processing.
02779     Note that the output can be just a number or many terms.
02780     In case par == 0 the output is [arg1/arg2]
02781     In case par == 1 the output is [arg1%arg2]
02782     In case par == 2 the output is [inverse of arg1 modulus arg2]
02783     In case par == 3 the output is [arg1*arg2]
02784 */
02785
02786 WORD divrem[4] = { DIVFUNCTION, REMFUNCTION, INVERSEFUNCTION, MULFUNCTION };
02787
02788 int DIVfunction(PHEAD WORD *term,WORD level,int par)
02789 {
02790     GETBIDENTITY
02791     WORD *t, *tt, *r, *arg1 = 0, *arg2 = 0, *arg3 = 0, *termout;
02792     WORD *tstop, *tend, *r3, *rr, *rstop, tlength, rlength, newlength;
02793     WORD *proper1, *proper2, *proper3 = 0, numdol = -1;
02794     int numargs = 0, type1, type2, actionflag1, actionflag2;
02795     WORD startebuf = cbuf[AT.ebufnum].numrhs;
02796     int division = ( par <= 2 ); /* false for mul_ */
02797     if ( par < 0 || par > 3 ) {
02798         MLOCK(ErrorMessageLock);
02799         MesPrint("Internal error. Illegal parameter %d in DIVfunction.",par);
02800         MUNLOCK(ErrorMessageLock);
02801         Terminate(-1);
02802     }
02803     /*
02804     Find the function
02805     */
02806     tend = term + *term; tstop = tend - ABS(tend[-1]);
02807     t = term+1;
02808     while ( t < tstop ) {
02809         if ( *t != divrem[par] ) { t += t[1]; continue; }
02810         r = t + FUNHEAD;
02811         tt = t + t[1]; numargs = 0;
02812         while ( r < tt ) {
02813             if ( numargs == 0 ) { arg1 = r; }
02814             if ( numargs == 1 ) { arg2 = r; }
02815             if ( numargs == 2 && *r == -DOLLAREXPRESSION ) { numdol = r[1]; }
02816             numargs++;
02817             NEXTARG(r);
02818         }
02819         if ( numargs == 2 ) break;
02820         if ( division && numargs == 3 ) break;
02821         t = tt;
02822     }
02823     if ( t >= tstop ) {
02824         MLOCK(ErrorMessageLock);
02825         MesPrint("Internal error. Indicated div_ or rem_ function not encountered.");
02826         MUNLOCK(ErrorMessageLock);
02827         Terminate(-1);
02828     }
02829     /*
02830     We have two arguments in arg1 and arg2.
02831     */
02832     if ( division && *arg1 == -SNUMBER && arg1[1] == 0 ) {
02833         if ( *arg2 == -SNUMBER && arg2[1] == 0 ) {
02834             zerozero:;
02835             MLOCK(ErrorMessageLock);
02836             MesPrint("0/0 in either div_ or rem_ function.");
02837             MUNLOCK(ErrorMessageLock);
02838             Terminate(-1);
02839         }
02840         if ( numdol >= 0 ) PutTermInDollar(0,numdol);
02841         return(0);
02842     }
02843     if ( division && *arg2 == -SNUMBER && arg2[1] == 0 ) {
02844         divzero:;
02845         MLOCK(ErrorMessageLock);
02846         MesPrint("Division by zero in either div_ or rem_ function.");
02847         MUNLOCK(ErrorMessageLock);
02848         Terminate(-1);
02849     }
02850     if ( !division ) {
02851         if ( (*arg1 == -SNUMBER && arg1[1] == 0) ||
02852             (*arg2 == -SNUMBER && arg2[1] == 0) ) {
02853             return(0);
02854         }
02855     }

```

```

02856     if ( ( arg1 = ConvertArgument(BHEAD arg1, &type1) ) == 0 ) goto CalledFrom;
02857     if ( ( arg2 = ConvertArgument(BHEAD arg2, &type2) ) == 0 ) goto CalledFrom;
02858     if ( division && *arg1 == 0 ) {
02859         if ( *arg2 == 0 ) {
02860             M_free(arg2,"DIVfunction");
02861             M_free(arg1,"DIVfunction");
02862             goto zerozero;
02863         }
02864         M_free(arg2,"DIVfunction");
02865         M_free(arg1,"DIVfunction");
02866         if ( numdol >= 0 ) PutTermInDollar(0,numdol);
02867         return(0);
02868     }
02869     if ( division && *arg2 == 0 ) {
02870         M_free(arg2,"DIVfunction");
02871         M_free(arg1,"DIVfunction");
02872         goto divzero;
02873     }
02874     if ( !division && (*arg1 == 0 || *arg2 == 0) ) {
02875         M_free(arg2,"DIVfunction");
02876         M_free(arg1,"DIVfunction");
02877         return(0);
02878     }
02879     if ( ( proper1 = PutExtraSymbols(BHEAD arg1,startebuf,&actionflag1) ) == 0 ) goto CalledFrom;
02880     if ( ( proper2 = PutExtraSymbols(BHEAD arg2,startebuf,&actionflag2) ) == 0 ) goto CalledFrom;
02881     /*
02882     if ( type2 == 0 ) M_free(arg2,"DIVfunction");
02883     else {
02884         DOLLARS d = ((DOLLARS)arg2)-1;
02885         if ( d->factors ) M_free(d->factors,"Dollar factors");
02886         M_free(d,"Copy of dollar variable");
02887     }
02888     */
02889     M_free(arg2,"DIVfunction");
02890     /*
02891     if ( type1 == 0 ) M_free(arg1,"DIVfunction");
02892     else {
02893         DOLLARS d = ((DOLLARS)arg1)-1;
02894         if ( d->factors ) M_free(d->factors,"Dollar factors");
02895         M_free(d,"Copy of dollar variable");
02896     }
02897     */
02898     M_free(arg1,"DIVfunction");
02899     if ( par == 0 ) proper3 = poly_div(BHEAD proper1, proper2,0);
02900     else if ( par == 1 ) proper3 = poly_rem(BHEAD proper1, proper2,0);
02901     else if ( par == 2 ) proper3 = poly_inverse(BHEAD proper1, proper2);
02902     else if ( par == 3 ) proper3 = poly_mul(BHEAD proper1, proper2);
02903     if ( proper3 == 0 ) goto CalledFrom;
02904     if ( actionflag1 || actionflag2 ) {
02905         if ( ( arg3 = TakeExtraSymbols(BHEAD proper3,startebuf) ) == 0 ) goto CalledFrom;
02906         M_free(proper3,"DIVfunction");
02907     }
02908     else {
02909         arg3 = proper3;
02910     }
02911     M_free(proper2,"DIVfunction");
02912     M_free(proper1,"DIVfunction");
02913     cbuf[AT.ebufnum].numrhs = startebuf;
02914     if ( *arg3 ) {
02915         termout = AT.WorkPointer;
02916         tlength = tend[-1];
02917         tlength = REDLENG(tlength);
02918         r3 = arg3;
02919         while ( *r3 ) {
02920             tt = term + 1; rr = termout + 1;
02921             while ( tt < t ) *rr++ = *tt++;
02922             r = r3 + 1;
02923             r3 = r3 + *r3;
02924             rstop = r3 - ABS(r3[-1]);
02925             while ( r < rstop ) *rr++ = *r++;
02926             tt += t[1];
02927             while ( tt < tstop ) *rr++ = *ttt++;
02928             rlength = r3[-1];
02929             rlength = REDLENG(rlength);
02930             if ( MulRat(BHEAD (UWORD *)tstop,tlength,(UWORD *)rstop,rlength,
02931                 (UWORD *)rr,&newlength) < 0 ) goto CalledFrom;
02932             rlength = INCLENG(newlength);
02933             rr += ABS(rlength);
02934             rr[-1] = rlength;
02935             *termout = rr - termout;
02936             AT.WorkPointer = rr;
02937             if ( Generator(BHEAD termout,level) ) goto CalledFrom;
02938         }
02939         AT.WorkPointer = termout;
02940     }
02941     M_free(arg3,"DIVfunction");
02942     return(0);

```

```

02943 CalledFrom:
02944     MLOCK(ErrorMessageLock);
02945     MesCall("DIVfunction");
02946     MUNLOCK(ErrorMessageLock);
02947     SETERROR(-1)
02948 }
02949
02950 /*
02951 #] DIVfunction :
02952 #[ MULfunc :
02953
02954     Multiplies two polynomials and puts the results in TermMalloc space.
02955 */
02956
02957 WORD *MULfunc(PHEAD WORD *p1, WORD *p2)
02958 {
02959     WORD *prod, size1, size2, size3, *t, *tfill, *ps1, *ps2, sign1, sign2, error, *p3;
02960     UWORD *num1, *num2, *num3;
02961     int i;
02962     WORD oldsorttype = AR.SortType;
02963     COMPARE oldcompareroutine = (COMPARE) (AR.CompareRoutine);
02964     AR.SortType = SORTHIGHFIRST;
02965     AR.CompareRoutine = (COMPAREDUMMY) (&CompareSymbols);
02966     num3 = NumberMalloc("MULfunc");
02967     prod = TermMalloc("MULfunc");
02968     NewSort(BHEAD0);
02969     while ( *p1 ) {
02970         ps1 = p1+*p1; num1 = (UWORD *) (ps1 - ABS(ps1[-1])); size1 = ps1[-1];
02971         if ( size1 < 0 ) { sign1 = -1; size1 = -size1; }
02972         else sign1 = 1;
02973         size1 = (size1-1)/2;
02974         p3 = p2;
02975         while ( *p3 ) {
02976             ps2 = p3+*p3; num2 = (UWORD *) (ps2 - ABS(ps2[-1])); size2 = ps2[-1];
02977             if ( size2 < 0 ) { sign2 = -1; size2 = -size2; }
02978             else sign2 = 1;
02979             size2 = (size2-1)/2;
02980             if ( MulLong(num1, size1, num2, size2, num3, &size3) ) {
02981                 error = 1;
02982             }
02983             CalledFrom:
02984                 MLOCK(ErrorMessageLock);
02985                 MesPrint(" Error %d", error);
02986                 MesCall("MulFunc");
02987                 MUNLOCK(ErrorMessageLock);
02988                 Terminate(-1);
02989             }
02990             tfill = prod+1;
02991             t = p1+1; while ( t < (WORD *) num1 ) *tfill++ = *t++;
02992             t = p3+1; while ( t < (WORD *) num2 ) *tfill++ = *t++;
02993             t = (WORD *) num3;
02994             for ( i = 0; i < size3; i++ ) *tfill++ = *t++;
02995             *tfill++ = 1;
02996             for ( i = 1; i < size3; i++ ) *tfill++ = 0;
02997             *tfill++ = (2*size3+1)*sign1*sign2;
02998             prod[0] = tfill - prod;
02999             if ( SymbolNormalize(prod) ) { error = 2; goto CalledFrom; }
03000             if ( StoreTerm(BHEAD prod) ) { error = 3; goto CalledFrom; }
03001             p3 += *p3;
03002         }
03003         p1 += *p1;
03004     }
03005     NumberFree(num3, "MULfunc");
03006     EndSort(BHEAD prod, 1);
03007     AR.CompareRoutine = (COMPAREDUMMY) oldcompareroutine;
03008     AR.SortType = oldsorttype;
03009     return(prod);
03010 }
03011
03012 /*
03013 #] MULfunc :
03014 #[ ConvertArgument :
03015
03016     Converts an argument to a general notation in allocated space.
03017 */
03018 WORD *ConvertArgument(PHEAD WORD *arg, int *type)
03019 {
03020     WORD *output, *t, *r;
03021     int i, size;
03022     if ( *arg > 0 ) {
03023         output = (WORD *) Malloc1((*arg)*sizeof(WORD), "ConvertArgument");
03024         i = *arg - ARGHEAD; t = arg + ARGHEAD; r = output;
03025         NCOPY(r, t, i);
03026         *r = 0; *type = 0;
03027         return(output);
03028     }
03029     if ( *arg == -EXPRESSION ) {

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```

03030     *type = 0;
03031     return(CreateExpression(BHEAD arg[1]));
03032 }
03033 if ( *arg == -DOLLAREXPRESSION ) {
03034     DOLLARS d;
03035     *type = 1;
03036     d = DolToTerms(BHEAD arg[1]);
03037 /*
03038     The problem is that DolToTerms creates a copy of the dollar variable.
03039     If we just return d->where we create a memory leak. Hence we have to
03040     copy the contents of d->where to a new buffer
03041 */
03042     output = (WORD *)Malloc1((d->size+1)*sizeof(WORD),"Copy of dollar content");
03043     WCOPY(output,d->where,d->size+1);
03044     if ( d->factors ) { M_free(d->factors,"Dollar factors"); d->factors = 0; }
03045     M_free(d,"Copy of dollar variable");
03046     return(output);
03047 }
03048 #if ( FUNHEAD > 4 )
03049     size = FUNHEAD+5;
03050 #else
03051     size = 9;
03052 #endif
03053 output = (WORD *)Malloc1(size*sizeof(WORD),"ConvertArgument");
03054 switch(*arg) {
03055     case -SYMBOL:
03056         output[0] = 8; output[1] = SYMBOL; output[2] = 4; output[3] = arg[1];
03057         output[4] = 1; output[5] = 1; output[6] = 1; output[7] = 3;
03058         output[8] = 0;
03059         break;
03060     case -INDEX:
03061     case -VECTOR:
03062     case -MINVECTOR:
03063         output[0] = 7; output[1] = INDEX; output[2] = 3; output[3] = arg[1];
03064         output[4] = 1; output[5] = 1;
03065         if ( *arg == -MINVECTOR ) output[6] = -3;
03066         else output[6] = 3;
03067         output[7] = 0;
03068         break;
03069     case -SNUMBER:
03070         output[0] = 4;
03071         if ( arg[1] < 0 ) {
03072             output[1] = -arg[1]; output[2] = 1; output[3] = -3;
03073         }
03074         else {
03075             output[1] = arg[1]; output[2] = 1; output[3] = 3;
03076         }
03077         output[4] = 0;
03078         break;
03079     default:
03080         output[0] = FUNHEAD+4;
03081         output[1] = -*arg;
03082         output[2] = FUNHEAD;
03083         for ( i = 3; i <= FUNHEAD; i++ ) output[i] = 0;
03084         output[FUNHEAD+1] = 1;
03085         output[FUNHEAD+2] = 1;
03086         output[FUNHEAD+3] = 3;
03087         output[FUNHEAD+4] = 0;
03088         break;
03089 }
03090 *type = 0;
03091 return(output);
03092 }
03093
03094 /*
03095 #] ConvertArgument :
03096 #[ ExpandRat :
03097
03098 Expands the denominator of a PolyRatFun in the variable PolyFunVar.
03099 The output is a polyratfun with a single argument.
03100 In the case that there is a polyratfun with more than one argument
03101 or the dirtyflag is on, the argument(s) is/are normalized.
03102 The output overwrites the input.
03103 */
03104
03105 char *TheErrorMessage[] = {
03106     "PolyRatFun not of a type that FORM will expand: incorrect variable inside."
03107     ,"Division by zero in PolyRatFun encountered in ExpandRat."
03108     ,"Irregular code in PolyRatFun encountered in ExpandRat."
03109     ,"Called from ExpandRat."
03110     ,"Workspace overflow. Change parameter Workspace in setup file?"
03111 };
03112
03113 int ExpandRat(PHEAD WORD *fun)
03114 {
03115     WORD *r, *rr, *rrr, *tt, *tnext, *arg1, *arg2, *rmin = 0, *rmininv;
03116     WORD *rcoef, rsize, rcopy, *ow = AT.WorkPointer;

```



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03117     WORD *numerator, *denominator, *rnext;
03118     WORD *thecopy, *rc, ncoef, newcoef, *m, *mm, nco, *outarg = 0;
03119     UWORD co[2], co1[2], co2[2];
03120     WORD OldPolyFunPow = AR.PolyFunPow;
03121     int i, j, minpow = 0, eppow, first, error = 0, ipoly;
03122     if ( fun[1] == FUNHEAD ) { return(0); }
03123     tnext = fun + fun[1];
03124     if ( fun[1] == fun[FUNHEAD]+FUNHEAD ) { /* Single argument */
03125         if ( fun[2] == 0 ) { goto done; }
03126     /*
03127         We have to normalize the argument. This could make it shorter.
03128     */
03129     NormArg;;
03130         if ( outarg == 0 ) outarg = TermMalloc("ExpandRat")+ARGHEAD;
03131         AT.TrimPower = 1;
03132         NewSort(BHEAD0);
03133         r = fun+FUNHEAD+ARGHEAD;
03134         if ( AR.PolyFunExp == 2 ) { /* Find minimum power */
03135             WORD minpow2 = MAXPOWER, *rrm;
03136             rrm = r;
03137             while ( rrm < tnext ) {
03138                 if ( *rrm == 4 ) {
03139                     if ( minpow2 > 0 ) minpow2 = 0;
03140                 }
03141                 else if ( ABS(rrm[*rrm-1]) == (*rrm-1) ) {
03142                     if ( minpow2 > 0 ) minpow2 = 0;
03143                 }
03144                 else {
03145                     if ( rrm[1] == SYMBOL && rrm[2] == 4 && rrm[3] == AR.PolyFunVar ) {
03146                         if ( rrm[4] < minpow2 ) minpow2 = rrm[4];
03147                     }
03148                     else {
03149                         MesPrint("Illegal term in expanded polyratfun.");
03150                         goto onerror;
03151                     }
03152                 }
03153                 rrm += *rrm;
03154             }
03155             AR.PolyFunPow += minpow2;
03156         }
03157         while ( r < tnext ) {
03158             rr = r + *r;
03159             i = *r; rrr = outarg; NCOPY(rrr,r,i);
03160             Normalize(BHEAD outarg);
03161             if ( *outarg > 0 ) StoreTerm(BHEAD outarg);
03162         }
03163         r = fun+FUNHEAD+ARGHEAD;
03164         EndSort(BHEAD r,1);
03165         AT.TrimPower = 0;
03166         if ( *r == 0 ) {
03167             fun[FUNHEAD] = -SNUMBER; fun[FUNHEAD+1] = 0;
03168             fun[1] = FUNHEAD+2;
03169         }
03170         else {
03171             rr = fun+FUNHEAD;
03172             if ( ToFast(rr,rr) ) {
03173                 NEXTARG(rr); fun[1] = rr - fun;
03174             }
03175             else {
03176                 while ( *r ) r += *r;
03177                 *rr = r-rr; rr[1] = CLEANFLAG;
03178                 fun[1] = r - fun;
03179             }
03180         }
03181         fun[2] = CLEANFLAG;
03182         goto done;
03183     }
03184     /*
03185     First test whether we have only AR.PolyFunVar in the denominator
03186     */
03187     tt = fun + FUNHEAD;
03188     arg1 = arg2 = 0;
03189     if ( tt < tnext ) {
03190         arg1 = tt; NEXTARG(tt);
03191         if ( tt < tnext ) {
03192             arg2 = tt; NEXTARG(tt);
03193             if ( tt != tnext ) { arg1 = arg2 = 0; } /* more than two arguments */
03194         }
03195     }
03196     if ( arg2 == 0 ) {
03197         if ( *arg1 < 0 ) { fun[2] = CLEANFLAG; goto done; }
03198         if ( fun[2] == CLEANFLAG ) goto done;
03199         goto NormArg; /* Note: should not come here */
03200     }
03201     /*
03202     Produce the output argument in outarg
03203     */

```

```

03204     if ( outarg == 0 ) outarg = TermMalloc("ExpandRat")+ARGHEAD;
03205
03206     if ( *arg2 <= 0 ) {
03207 /*
03208         These cases are trivial.
03209         We try as much as possible to write the output directly into the
03210         function. We just have to be extremely careful not to overwrite
03211         relevant information before we are finished with it.
03212 */
03213         if ( *arg2 == -SYMBOL && arg2[1] == AR.PolyFunVar ) {
03214             rr = r = fun+FUNHEAD+ARGHEAD;
03215             if ( *arg1 < 0 ) {
03216                 if ( *arg1 == -SYMBOL ) {
03217                     if ( arg1[1] == AR.PolyFunVar ) {
03218                         *r++ = 4; *r++ = 1; *r++ = 1; *r++ = 3; *r = 0;
03219                     }
03220                     else {
03221                         *r++ = 10; *r++ = SYMBOL; *r++ = 6;
03222                         *r++ = arg1[1]; *r++ = 1;
03223                         *r++ = AR.PolyFunVar; *r++ = -1;
03224                         *r++ = 1; *r++ = 1; *r++ = 3; *r = 0;
03225                         Normalize(BHEAD rr);
03226                     }
03227                 }
03228                 else if ( *arg1 == -SNUMBER ) {
03229                     nco = arg1[1];
03230                     if ( nco == 0 ) { *r++ = 0; }
03231                     else {
03232                         *r++ = 8; *r++ = SYMBOL; *r++ = 4;
03233                         *r++ = AR.PolyFunVar; *r++ = -1;
03234                         *r++ = ABS(nco); *r++ = 1;
03235                         if ( nco < 0 ) *r++ = -3;
03236                         else *r++ = 3;
03237                         *r = 0;
03238                     }
03239                 }
03240                 else { error = 2; goto onerror; } /* should not happen! */
03241             }
03242             else { /* Multi-term numerator. */
03243                 m = arg1+ARGHEAD;
03244                 NewSort(BHEAD0); /* Technically maybe not needed */
03245                 if ( AR.PolyFunExp == 2 ) { /* Find minimum power */
03246                     WORD minpow2 = MAXPOWER, *rrm;
03247                     rrm = m;
03248                     while ( rrm < arg2 ) {
03249                         if ( *rrm == 4 ) {
03250                             if ( minpow2 > 0 ) minpow2 = 0;
03251                         }
03252                         else if ( ABS(rrm[*rrm-1]) == (*rrm-1) ) {
03253                             if ( minpow2 > 0 ) minpow2 = 0;
03254                         }
03255                         else {
03256                             if ( rrm[1] == SYMBOL && rrm[2] == 4 && rrm[3] == AR.PolyFunVar ) {
03257                                 if ( rrm[4] < minpow2 ) minpow2 = rrm[4];
03258                             }
03259                             else {
03260                                 MesPrint("Illegal term in expanded polyratfun.");
03261                                 goto onerror;
03262                             }
03263                         }
03264                         rrm += *rrm;
03265                     }
03266                     AR.PolyFunPow += minpow2-1;
03267                 }
03268                 while ( m < arg2 ) {
03269                     r = outarg;
03270                     rrr = r++; mm = m + *m;
03271                     *r++ = SYMBOL; *r++ = 4; *r++ = AR.PolyFunVar; *r++ = -1;
03272                     m++; while ( m < mm ) *r++ = *m++;
03273                     *rrr = r-rrr;
03274                     Normalize(BHEAD rrr);
03275                     StoreTerm(BHEAD rrr);
03276                 }
03277                 EndSort(BHEAD rr,1);
03278                 r = rr; while ( *r ) r += *r;
03279             }
03280             if ( *rr == 0 ) {
03281                 fun[FUNHEAD] = -SNUMBER; fun[FUNHEAD+1] = CLEANFLAG;
03282                 fun[1] = FUNHEAD+2;
03283             }
03284             else {
03285                 rr = fun+FUNHEAD;
03286                 *rr = r-rr;
03287                 rr[1] = CLEANFLAG;
03288                 if ( ToFast(rr,rr) ) {
03289                     NEXTARG(rr);
03290                     fun[1] = rr - fun;

```

```

03291         }
03292         else { fun[1] = r - fun; }
03293     }
03294     fun[2] = CLEANFLAG;
03295     goto done;
03296 }
03297 else if ( *arg2 == -SNUMBER ) {
03298     rr = r = outarg;
03299     if ( arg2[1] == 0 ) { error = 1; goto onerror; }
03300     if ( *arg1 == -SNUMBER ) { /* Things may not be normalized */
03301         if ( arg1[1] == 0 ) { *r++ = 0; }
03302         else {
03303             col[0] = ABS(arg1[1]); col[1] = 1;
03304             co2[0] = 1; co2[1] = ABS(arg2[1]);
03305             MulRat(BHEAD col,1,co2,1,co,&nco);
03306             *r++ = 4; *r++ = (WORD)(co[0]); *r++ = (WORD)(co[1]);
03307             if ( ( arg1[1] < 0 && arg2[1] > 0 ) ||
03308                 ( arg1[1] > 0 && arg2[1] < 0 ) ) *r++ = -3;
03309             else *r++ = 3;
03310             *r = 0;
03311         }
03312     }
03313     else if ( *arg1 == -SYMBOL ) {
03314         *r++ = 8; *r++ = SYMBOL; *r++ = 4;
03315         *r++ = arg1[1]; *r++ = 1;
03316         *r++ = 1; *r++ = ABS(arg2[1]);
03317         if ( arg2[1] < 0 ) *r++ = -3;
03318         else *r++ = 3;
03319         *r = 0;
03320     }
03321     else if ( *arg1 < 0 ) { error = 2; goto onerror; }
03322     else { /* Multi-term numerator. */
03323         m = arg1+ARGHEAD;
03324         NewSort(BHEAD0); /* Technically maybe not needed */
03325         if ( AR.PolyFunExp == 2 ) { /* Find minimum power */
03326             WORD minpow2 = MAXPOWER, *rrm;
03327             rrm = m;
03328             while ( rrm < arg2 ) {
03329                 if ( *rrm == 4 ) {
03330                     if ( minpow2 > 0 ) minpow2 = 0;
03331                 }
03332                 else if ( ABS(rrm[*rrm-1]) == (*rrm-1) ) {
03333                     if ( minpow2 > 0 ) minpow2 = 0;
03334                 }
03335                 else {
03336                     if ( rrm[1] == SYMBOL && rrm[2] == 4 && rrm[3] == AR.PolyFunVar ) {
03337                         if ( rrm[4] < minpow2 ) minpow2 = rrm[4];
03338                     }
03339                     else {
03340                         MesPrint("Illegal term in expanded polyratfun.");
03341                         goto onerror;
03342                     }
03343                 }
03344                 rrm += *rrm;
03345             }
03346             AR.PolyFunPow += minpow2;
03347         }
03348         while ( m < arg2 ) {
03349             r = rr;
03350             rrr = r++; mm = m + *m;
03351             *r++ = DENOMINATOR; *r++ = FUNHEAD + 2; *r++ = DIRTYFLAG;
03352             FILLFUN3(r);
03353             *r++ = arg2[0]; *r++ = arg2[1];
03354             m++; while ( m < mm ) *r++ = *m++;
03355             *rrr = r-rrr;
03356             if ( r < AT.WorkTop && r >= AT.WorkSpace )
03357                 AT.WorkPointer = r;
03358             Normalize(BHEAD rrr);
03359             if ( ABS(rrr[*rrr-1]) == *rrr-1 ) {
03360                 if ( AR.PolyFunPow >= 0 ) {
03361                     StoreTerm(BHEAD rrr);
03362                 }
03363             }
03364             else if ( rrr[1] == SYMBOL && rrr[2] == 4 &&
03365                 rrr[3] == AR.PolyFunVar && rrr[4] <= AR.PolyFunPow ) {
03366                 StoreTerm(BHEAD rrr);
03367             }
03368         }
03369         EndSort(BHEAD rr,1);
03370     }
03371     r = rr; while ( *r ) r += *r;
03372     i = r-rr;
03373     r = fun + FUNHEAD + ARGHEAD;
03374     NCOPY(r,rr,i);
03375     rr = fun + FUNHEAD;
03376     *rr = r - rr; rr[1] = CLEANFLAG;
03377     if ( ToFast(rr,rr) ) {

```

```

03378         NEXTARG(rr);
03379         fun[1] = rr - fun;
03380     }
03381     else { fun[1] = r - fun; }
03382     fun[2] = CLEANFLAG;
03383     goto done;
03384 }
03385 else { error = 0; goto onerror; }
03386 }
03387 else {
03388     r = arg2+ARGHEAD; /* The argument ends at tnext */
03389     first = 1;
03390     while ( r < tnext ) {
03391         rr = r + *r; rr -= ABS(rr[-1]);
03392         if ( r+1 == rr ) {
03393             if ( first ) { minpow = 0; first = 0; rmin = r; }
03394             else if ( minpow > 0 ) { minpow = 0; rmin = r; }
03395         }
03396         else if ( r[1] != SYMBOL || r[2] != 4 || r[3] != AR.PolyFunVar
03397             || r[4] > MAXPOWER ) { error = 0; goto onerror; }
03398         else if ( first ) { minpow = r[4]; first = 0; rmin = r; }
03399         else if ( r[4] < minpow ) { minpow = r[4]; rmin = r; }
03400         r += *r;
03401     }
03402     /*
03403     We have now:
03404     1: a numerator in arg1 which can contain several variables.
03405     2: a denominator in arg2 with at most only AR.PolyFunVar (ep).
03406     3: the minimum power in the denominator is minpow and the
03407        term with that minimum power is in rmin.
03408     Divide numerator and denominator by this minimum power.
03409     Determine the power range in the numerator.
03410     Call InvPoly.
03411     Multiply by the inverse in such a way that we never take more
03412     powers of ep than necessary.
03413     */
03414     /*
03415     One: put 1/rmin in AT.WorkPointer -> rmininv
03416     */
03417     AT.WorkPointer += AM.MaxTer/sizeof(WORD);
03418     if ( AT.WorkPointer + (AM.MaxTer/sizeof(WORD)) >= AT.WorkTop ) {
03419         error = 4; goto onerror;
03420     }
03421     rmininv = r = AT.WorkPointer;
03422     rr = rmin; i = *rmin; NCOPY(r,rr,i)
03423     if ( minpow != 0 ) { rmininv[4] = -rmininv[4]; }
03424     rsize = ABS(r[-1]);
03425     rcoef = r - rsize;
03426     rsize = (rsize-1)/2; rr = rcoef + rsize;
03427     for ( i = 0; i < rsize; i++ ) {
03428         rcopy = rcoef[i]; rcoef[i] = rr[i]; rr[i] = rcopy;
03429     }
03430     AT.WorkPointer = r;
03431     if ( *arg1 < 0 ) {
03432         ToGeneral(arg1,r,0);
03433         arg1 = r; r += *r; *r++ = 0; rcopy = 0;
03434         AT.WorkPointer = r;
03435     }
03436     else {
03437         r = arg1 + *arg1;
03438         rcopy = *r; *r++ = 0;
03439     }
03440     /*
03441     We can use MultiplyWithTerm.
03442     */
03443     AT.LeaveNegative = 1;
03444     numerator = MultiplyWithTerm(BHEAD arg1+ARGHEAD,rmininv,0);
03445     AT.LeaveNegative = 0;
03446     r[-1] = rcopy;
03447     r = numerator; while ( *r ) r += *r;
03448     AT.WorkPointer = r+1;
03449     rcopy = arg2[*arg2]; arg2[*arg2] = 0;
03450     denominator = MultiplyWithTerm(BHEAD arg2+ARGHEAD,rmininv,1);
03451     arg2[*arg2] = rcopy;
03452     r = denominator; while ( *r ) r += *r;
03453     AT.WorkPointer = r+1;
03454     /*
03455     Now find the minimum power of ep in the numerator.
03456     */
03457     r = numerator;
03458     first = 1;
03459     while ( *r ) {
03460         rr = r + *r; rr -= ABS(rr[-1]);
03461         if ( r+1 == rr ) {
03462             if ( first ) { minpow = 0; first = 0; }
03463             else if ( minpow > 0 ) { minpow = 0; }
03464         }

```

```

03465         else if ( r[1] != SYMBOL ) { error = 0; goto onerror; }
03466     else {
03467         for ( i = 3; i < r[2]; i += 2 ) {
03468             if ( r[i] == AR.PolyFunVar ) {
03469                 if ( first ) { minpow = r[i+1]; first = 0; }
03470                 else if ( r[i+1] < minpow ) minpow = r[i+1];
03471                 break;
03472             }
03473         }
03474         if ( i >= r[2] ) {
03475             if ( first ) { minpow = 0; first = 0; }
03476             else if ( minpow > 0 ) minpow = 0;
03477         }
03478     }
03479     r += *r;
03480 }
03481 /*
03482 We can invert the denominator.
03483 Note that the return value is an offset in AT.pWorkSpace.
03484 Hence there is no need to free memory afterwards.
03485 */
03486 if ( AR.PolyFunExp == 3 ) {
03487     ipoly = InvPoly(BHEAD denominator, AR.PolyFunPow-minpow, AR.PolyFunVar);
03488 }
03489 else {
03490     ipoly = InvPoly(BHEAD denominator, AR.PolyFunPow, AR.PolyFunVar);
03491 }
03492 /*
03493 Now we start the multiplying
03494 */
03495 NewSort(BHEAD0);
03496 r = numerator;
03497 while ( *r ) {
03498     /*
03499         1: Find power of ep.
03500     */
03501     rnext = r + *r;
03502     rrr = rnext - ABS(rnext[-1]);
03503     rr = r+1;
03504     eppow = 0;
03505     if ( rr < rrr ) {
03506         j = rr[1] - 2; rr += 2;
03507         while ( j > 0 ) {
03508             if ( *rr == AR.PolyFunVar ) { eppow = rr[1]; break; }
03509             j -= 2; rr += 2;
03510         }
03511     }
03512     /*
03513         2: Multiply by the proper terms in ipoly
03514     */
03515     for ( i = 0; i <= AR.PolyFunPow-eppow+minpow; i++ ) {
03516         if ( AT.pWorkSpace[ipoly+i] == 0 ) continue;
03517         /*
03518             Copy the term, add i to the power of ep and multiply coef.
03519         */
03520         rc = r;
03521         rr = thecopy = AT.WorkPointer;
03522         while ( rc < rrr ) *rr++ = *rc++;
03523         if ( i != 0 ) {
03524             *rr++ = SYMBOL; *rr++ = 4; *rr++ = AR.PolyFunVar; *rr++ = i;
03525         }
03526         ncoef = REDLENG(rnext[-1]);
03527         MulRat(BHEAD (UWORD *)rrr, ncoef,
03528             (UWORD *) (AT.pWorkSpace[ipoly+i]+1), AT.pWorkSpace[ipoly+i][0],
03529             (UWORD *)rr, &newcoef);
03530         ncoef = ABS(newcoef); rr += 2*ncoef;
03531         newcoef = INCLENG(newcoef);
03532         *rr++ = newcoef;
03533         *thecopy = rr - thecopy;
03534         AT.WorkPointer = rr;
03535         Normalize(BHEAD thecopy);
03536         if ( *thecopy > 0 ) StoreTerm(BHEAD thecopy);
03537         AT.WorkPointer = thecopy;
03538     }
03539     r = rnext;
03540 }
03541 /*
03542 Now we have all.
03543 */
03544 rr = fun + FUNHEAD; r = rr + ARGHEAD;
03545 EndSort(BHEAD r, 1);
03546 if ( *r == 0 ) {
03547     fun[1] = FUNHEAD+2; fun[2] = CLEANFLAG;
03548     fun[FUNHEAD] = -SNUMBER; fun[FUNHEAD+1] = 0;
03549 }
03550 else {
03551     while ( *r ) r += *r;

```

```

03552         rr[0] = r-rr; rr[1] = CLEANFLAG;
03553         if ( ToFast(rr,rr) ) { NEXTARG(rr); fun[1] = rr-fun; }
03554         else { fun[1] = r-fun; }
03555         fun[2] = CLEANFLAG;
03556     }
03557 }
03558 done:
03559     if ( outarg ) TermFree(outarg-ARGHEAD,"ExpandRat");
03560     AR.PolyFunPow = OldPolyFunPow;
03561     AT.WorkPointer = ow;
03562     AN.PolyNormFlag = 1;
03563     return(0);
03564 onerror:
03565     if ( outarg ) TermFree(outarg-ARGHEAD,"ExpandRat");
03566     AR.PolyFunPow = OldPolyFunPow;
03567     AT.WorkPointer = ow;
03568     MLOCK(ErrorMessageLock);
03569     MesPrint(TheErrorMessage[error]);
03570     MUNLOCK(ErrorMessageLock);
03571     Terminate(-1);
03572     return(-1);
03573 }
03574
03575 /*
03576 #] ExpandRat :
03577 #[ InvPoly :
03578
03579 The input polynomial is represented as a sequence of terms in ascending
03580 power. The first coefficient is 1. If we call this 1-a and
03581  $a = \sum_{j=1,n} x^j a(j)$ , and  $b = 1/(1-a)$  we can find the coefficients
03582 of b with the recursion
03583  $b(0) = 1, b(n) = \sum_{j=1,n} a(j) * b(n-j)$ 
03584 The variable is the symbol sym and we need maxpow powers in the answer.
03585 The answer is an array of pointers to the coefficients of the various
03586 powers as rational numbers in the notation signedsize,numerator,denominator
03587 We put these powers in the workspace and the answer is in AT.pWorkspace.
03588 Hence the return value is an offset in the pWorkspace.
03589 A zero pointer indicates that this coefficient is zero.
03590 */
03591
03592 int InvPoly(PHEAD WORD *inpoly, WORD maxpow, WORD sym)
03593 {
03594     int needed, inpointers, outpointers, maxinput = 0, i, j;
03595     WORD *t, *tt, *ttt, *w, *c, *cc, *ccc, lenc, lenc1, lenc2, rc, *c1, *c2;
03596     /*
03597     Step 0: allocate the space
03598     */
03599     needed = (maxpow+1)*2;
03600     WantAddPointers(needed);
03601     inpointers = AT.pWorkPointer;
03602     outpointers = AT.pWorkPointer+maxpow+1;
03603     for ( i = 0; i < needed; i++ ) AT.pWorkspace[inpointers+i] = 0;
03604     /*
03605     Step 1: determine the coefficients in inpoly
03606     often there is a maximum power that is much smaller than maxpow.
03607     keeping track of this can speed up things.
03608     */
03609     t = inpoly;
03610     w = AT.WorkPointer;
03611     while ( *t ) {
03612         if ( *t == 4 ) {
03613             if ( t[1] != 1 || t[2] != 1 || t[3] != 3 ) goto onerror;
03614             AT.pWorkspace[inpointers] = 0;
03615         }
03616         else if ( t[1] != SYMBOL || t[2] != 4 || t[3] != sym || t[4] < 0 ) goto onerror;
03617         else if ( t[4] > maxpow ) {} /* power outside useful range */
03618         else {
03619             if ( t[4] > maxinput ) maxinput = t[4];
03620             AT.pWorkspace[inpointers+t[4]] = w;
03621             tt = t + *t; rc = *--tt; /* we need - the coefficient! */
03622             rc = REDLENG(rc); *w++ = rc;
03623             ttt = t+5;
03624             while ( ttt < tt ) *w++ = *ttt++;
03625         }
03626         t += *t;
03627     }
03628     /*
03629     Step 2: compute the output. b(0) = 1.
03630     then the recursion starts.
03631     */
03632     AT.pWorkspace[outpointers] = w;
03633     *w++ = 1; *w++ = 1; *w++ = 1;
03634     c = TermMalloc("InvPoly");
03635     c1 = TermMalloc("InvPoly");
03636     c2 = TermMalloc("InvPoly");
03637     for ( j = 1; j <= maxpow; j++ ) {
03638     /*

```

```

03639         Start at c = a(j)*b(0) = a(j)
03640     */
03641     if ( ( cc = AT.pWorkspace[inpointers+j] ) != 0 ) {
03642         lenc = *cc++; /* reduced length */
03643         i = 2*ABS(lenc); ccc = c;
03644         NCOPY(ccc,cc,i);
03645     }
03646     else { lenc = 0; }
03647     for ( i = Min(j-1,maxinput); i > 0; i-- ) {
03648     /*
03649         c -> c + a(i)*b(j-i)
03650     */
03651         if ( AT.pWorkspace[inpointers+i] == 0
03652             || AT.pWorkspace[outpointers+j-i] == 0 ) {
03653             }
03654         else {
03655             if ( MulRat(BHEAD (UWORD
*) (AT.pWorkspace[inpointers+i]+1),AT.pWorkspace[inpointers+i][0],
03656                 (UWORD *) (AT.pWorkspace[outpointers+j-i]+1),AT.pWorkspace[outpointers+j-i][0],
03657                 (UWORD *)c1,&lenc1) ) goto calcerror;
03658             if ( lenc == 0 ) {
03659                 cc = c; c = c1; c1 = cc;
03660                 lenc = lenc1;
03661             }
03662             else {
03663                 if ( AddRat(BHEAD (UWORD *)c,lenc, (UWORD *)c1,lenc1, (UWORD *)c2,&lenc2) )
03664                     goto calcerror;
03665                 cc = c; c = c2; c2 = cc;
03666                 lenc = lenc2;
03667             }
03668         }
03669     }
03670     /*
03671     Copy c to the proper location
03672     */
03673     if ( lenc == 0 ) AT.pWorkspace[outpointers+j] = 0;
03674     else {
03675         AT.pWorkspace[outpointers+j] = w;
03676         *w++ = lenc;
03677         i = 2*ABS(lenc); ccc = c;
03678         NCOPY(w,ccc,i);
03679     }
03680     }
03681     AT.WorkPointer = w;
03682     TermFree(c2,"InvPoly");
03683     TermFree(c1,"InvPoly");
03684     TermFree(c,"InvPoly");
03685
03686     return(outpointers);
03687 onerror:
03688     MLOCK(ErrorMessageLock);
03689     MesPrint("Incorrect symbol field in InvPoly.");
03690     MUNLOCK(ErrorMessageLock);
03691     Terminate(-1);
03692     return(-1);
03693 calcerror:
03694     MLOCK(ErrorMessageLock);
03695     MesPrint("Called from InvPoly.");
03696     MUNLOCK(ErrorMessageLock);
03697     Terminate(-1);
03698     return(-1);
03699 }
03700
03701 /*
03702 #] InvPoly :
03703 */

```

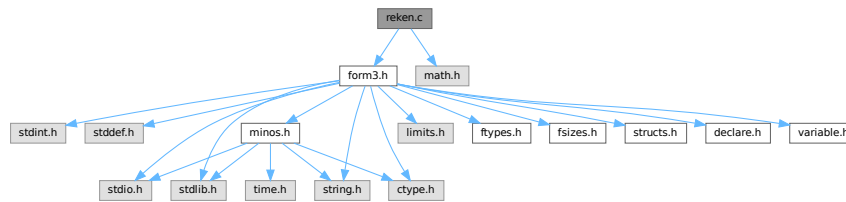
## 4.118 reken.c File Reference

```

#include "form3.h"
#include <math.h>

```

Include dependency graph for reken.c:



## Macros

- `#define GCDMAX 3`
- `#define NEWTRICK 1`
- `#define COPYLONG(x1, nx1, x2, nx2) { int i; for(i=0;i<ABS(nx2);i++)x1[i]=x2[i];nx1=nx2; }`
- `#define WARMUP 6`

## Functions

- void `Pack` (UWORD \*a, WORD \*na, UWORD \*b, WORD nb)
- void `UnPack` (UWORD \*a, WORD na, WORD \*denom, WORD \*numer)
- int `Mully` (PHEAD UWORD \*a, WORD \*na, UWORD \*b, WORD nb)
- int `Divvy` (PHEAD UWORD \*a, WORD \*na, UWORD \*b, WORD nb)
- int `AddRat` (PHEAD UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc)
- int `MulRat` (PHEAD UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc)
- int `DivRat` (PHEAD UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc)
- int `Simplify` (PHEAD UWORD \*a, WORD \*na, UWORD \*b, WORD \*nb)
- int `AccumGCD` (PHEAD UWORD \*a, WORD \*na, UWORD \*b, WORD nb)
- int `TakeRatRoot` (UWORD \*a, WORD \*n, WORD power)
- int `AddLong` (UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc)
- int `AddPLon` (UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc)
- void `SubPLon` (UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc)
- int `MulLong` (UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc)
- int `BigLong` (UWORD \*a, WORD na, UWORD \*b, WORD nb)
- int `DivLong` (UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc, UWORD \*d, WORD \*nd)
- int `RaisPow` (PHEAD UWORD \*a, WORD \*na, UWORD b)
- void `RaisPowCached` (PHEAD WORD x, WORD n, UWORD \*\*c, WORD \*nc)
- WORD `RaisPowMod` (WORD x, WORD n, WORD m)
- int `NormalModulus` (UWORD \*a, WORD \*na)
- int `MakeInverses` (void)
- int `GetModInverses` (WORD m1, WORD m2, WORD \*im1, WORD \*im2)
- int `GetLongModInverses` (PHEAD UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*ia, WORD \*nia, UWORD \*ib, WORD \*nib)
- int `Product` (UWORD \*a, WORD \*na, WORD b)
- UWORD `Quotient` (UWORD \*a, WORD \*na, WORD b)
- WORD `Remain10` (UWORD \*a, WORD \*na)
- WORD `Remain4` (UWORD \*a, WORD \*na)
- void `PrtLong` (UWORD \*a, WORD na, UBYTE \*s)
- int `GetLong` (UBYTE \*s, UWORD \*a, WORD \*na)
- int `GcdLong` (PHEAD UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc)



- int [GetBinom](#) (UWORD \*a, WORD \*na, WORD i1, WORD i2)
- int [LcmLong](#) (PHEAD UWORD \*a, WORD na, UWORD \*b, WORD nb, UWORD \*c, WORD \*nc)
- int [TakeLongRoot](#) (UWORD \*a, WORD \*n, WORD power)
- int [MakeRational](#) (WORD a, WORD m, WORD \*b, WORD \*c)
- int [MakeLongRational](#) (PHEAD UWORD \*a, WORD na, UWORD \*m, WORD nm, UWORD \*b, WORD \*nb)
- WORD [CompCoef](#) (WORD \*term1, WORD \*term2)
- int [Modulus](#) (WORD \*term)
- int [TakeModulus](#) (UWORD \*a, WORD \*na, UWORD \*cmodvec, WORD ncmmod, WORD par)
- int [TakeNormalModulus](#) (UWORD \*a, WORD \*na, UWORD \*c, WORD nc, WORD par)
- int [MakeModTable](#) (void)
- int [Factorial](#) (PHEAD WORD n, UWORD \*a, WORD \*na)
- int [Bernoulli](#) (WORD n, UWORD \*a, WORD \*na)
- WORD [NextPrime](#) (PHEAD WORD num)
- WORD [Moebius](#) (PHEAD WORD nn)
- void [iniwranf](#) (PHEAD0)
- UWORD [wranf](#) (PHEAD0)
- UWORD [iranf](#) (PHEAD UWORD imax)
- UBYTE \* [PreRandom](#) (UBYTE \*s)

### 4.118.1 Detailed Description

This file contains the numerical routines. The arithmetic in FORM is normally over the rational numbers. Hence there are routines for dealing with integers and with rational of 'arbitrary precision' (within limits) There are also routines for that calculus modulus an integer. In addition there are the routines for factorials and Bernoulli numbers. The random number function is currently only for internal purposes.

Definition in file [reken.c](#).

### 4.118.2 Macro Definition Documentation

#### 4.118.2.1 GCDMAX

```
#define GCDMAX 3
```

Definition at line 49 of file [reken.c](#).

#### 4.118.2.2 NEWTRICK

```
#define NEWTRICK 1
```

Definition at line 51 of file [reken.c](#).

#### 4.118.2.3 COPYLONG

```
#define COPYLONG(
    x1,
    nx1,
    x2,
    nx2 ) { int i; for(i=0;i<ABS(nx2);i++) x1[i]=x2[i];nx1=nx2; }
```

Definition at line 2902 of file [reken.c](#).

#### 4.118.2.4 WARMUP

```
#define WARMUP 6
```

Definition at line [3802](#) of file [reken.c](#).

### 4.118.3 Function Documentation

#### 4.118.3.1 Pack()

```
void Pack (
    UWORD * a,
    WORD * na,
    UWORD * b,
    WORD nb )
```

Definition at line [62](#) of file [reken.c](#).

#### 4.118.3.2 UnPack()

```
void UnPack (
    UWORD * a,
    WORD na,
    WORD * denom,
    WORD * numer )
```

Definition at line [100](#) of file [reken.c](#).

#### 4.118.3.3 Mully()

```
int Mully (
    PHEAD UWORD * a,
    WORD * na,
    UWORD * b,
    WORD nb )
```

Definition at line [126](#) of file [reken.c](#).

#### 4.118.3.4 Divvy()

```
int Divvy (
    PHEAD UWORD * a,
    WORD * na,
    UWORD * b,
    WORD nb )
```

Definition at line [172](#) of file [reken.c](#).

#### 4.118.3.5 AddRat()

```
int AddRat (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc )
```

Definition at line 210 of file [reken.c](#).

#### 4.118.3.6 MulRat()

```
int MulRat (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc )
```

Definition at line 378 of file [reken.c](#).

#### 4.118.3.7 DivRat()

```
int DivRat (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc )
```

Definition at line 500 of file [reken.c](#).

#### 4.118.3.8 Simplify()

```
int Simplify (
    PHEAD UWORD * a,
    WORD * na,
    UWORD * b,
    WORD * nb )
```

Definition at line 531 of file [reken.c](#).

#### 4.118.3.9 AccumGCD()

```
int AccumGCD (
    PHEAD UWORD * a,
    WORD * na,
    UWORD * b,
    WORD nb )
```

Definition at line 648 of file [reken.c](#).

#### 4.118.3.10 TakeRatRoot()

```
int TakeRatRoot (
    UWORD * a,
    WORD * n,
    WORD power )
```

Definition at line 681 of file [reken.c](#).

#### 4.118.3.11 AddLong()

```
int AddLong (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc )
```

Definition at line 706 of file [reken.c](#).

#### 4.118.3.12 AddPLon()

```
int AddPLon (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc )
```

Definition at line 751 of file [reken.c](#).

#### 4.118.3.13 SubPLon()

```
void SubPLon (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc )
```

Definition at line 804 of file [reken.c](#).

#### 4.118.3.14 MulLong()

```
int MulLong (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc )
```

Definition at line 842 of file [reken.c](#).

#### 4.118.3.15 BigLong()

```
int BigLong (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb )
```

Definition at line 945 of file [reken.c](#).

#### 4.118.3.16 DivLong()

```
int DivLong (
    UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc,
    UWORD * d,
    WORD * nd )
```

Definition at line 973 of file [reken.c](#).

#### 4.118.3.17 RaisPow()

```
int RaisPow (
    PHEAD UWORD * a,
    WORD * na,
    UWORD b )
```

Definition at line 1229 of file [reken.c](#).

#### 4.118.3.18 RaisPowCached()

```
void RaisPowCached (
    PHEAD WORD x,
    WORD n,
    UWORD ** c,
    WORD * nc )
```

Computes power  $x^n$  and caches the value

### 4.118.4 Description

Calculates the power  $x^n$  and stores the results for caching purposes. The pointer c (i.e., the pointer, and not what it points to) is overwritten. What it points to should not be overwritten in the calling function.

#### 4.118.5 Notes

- Caching is done in `AT.small_power[]`. This array is extended if necessary.

Definition at line 1297 of file [reken.c](#).

Referenced by [poly::divmod\\_heap\(\)](#), [poly::divmod\\_univar\(\)](#), and [poly::mul\\_heap\(\)](#).

##### 4.118.5.1 RaisPowMod()

```
WORD RaisPowMod (
    WORD x,
    WORD n,
    WORD m )
```

Definition at line 1384 of file [reken.c](#).

##### 4.118.5.2 NormalModulus()

```
int NormalModulus (
    UWORD * a,
    WORD * na )
```

Brings a modular representation in the range  $-p/2$  to  $+p/2$  The return value tells whether anything was done. Routine made in the general modulus revamp of July 2008 (JV).

Definition at line 1404 of file [reken.c](#).

Referenced by [AddCoef\(\)](#), and [MergePatches\(\)](#).

##### 4.118.5.3 MakeInverses()

```
int MakeInverses (
    void )
```

Makes a table of inverses in modular calculus The modulus is in `AC.cmod` and `AC.ncmod` One should notice that the table of inverses can only be made if the modulus fits inside a single FORM word. Otherwise the table lookup becomes too difficult and the table too long.

Definition at line 1441 of file [reken.c](#).

References [GetModInverses\(\)](#).

Here is the call graph for this function:



#### 4.118.5.4 GetModInverses()

```
int GetModInverses (
    WORD m1,
    WORD m2,
    WORD * im1,
    WORD * im2 )
```

Input m1 and m2, which are relative prime. determines  $a*m1+b*m2 = 1$  (and 1 is the gcd of m1 and m2) then  $a*m1 = 1 \bmod m2$  and hence  $im1 = a$ . and  $b*m2 = 1 \bmod m1$  and hence  $im2 = b$ . Set  $m1 = 0*m1+1*m2 = a1*m1+b1*m2$   $m2 = 1*m1+0*m2 = a2*m1+b2*m2$  If everything is OK, the return value is zero

Definition at line [1477](#) of file [reken.c](#).

Referenced by [poly::divmod\\_heap\(\)](#), [poly::divmod\\_univar\(\)](#), [MakeDollarMod\(\)](#), [MakeInverses\(\)](#), [MakeMod\(\)](#), [TakeContent\(\)](#), and [TakeSymbolContent\(\)](#).

#### 4.118.5.5 GetLongModInverses()

```
int GetLongModInverses (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * ia,
    WORD * nia,
    UWORD * ib,
    WORD * nib )
```

Definition at line [1509](#) of file [reken.c](#).

#### 4.118.5.6 Product()

```
int Product (
    UWORD * a,
    WORD * na,
    WORD b )
```

Definition at line [1586](#) of file [reken.c](#).

#### 4.118.5.7 Quotient()

```
UWORD Quotient (
    UWORD * a,
    WORD * na,
    WORD b )
```

Definition at line [1621](#) of file [reken.c](#).

#### 4.118.5.8 Remain10()

```
WORD Remain10 (
    UWORD * a,
    WORD * na )
```

Definition at line 1660 of file [reken.c](#).

#### 4.118.5.9 Remain4()

```
WORD Remain4 (
    UWORD * a,
    WORD * na )
```

Definition at line 1687 of file [reken.c](#).

#### 4.118.5.10 PrtLong()

```
void PrtLong (
    UWORD * a,
    WORD na,
    UBYTE * s )
```

Definition at line 1713 of file [reken.c](#).

#### 4.118.5.11 GetLong()

```
int GetLong (
    UBYTE * s,
    UWORD * a,
    WORD * na )
```

Definition at line 1775 of file [reken.c](#).

#### 4.118.5.12 GcdLong()

```
int GcdLong (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc )
```

Definition at line 2277 of file [reken.c](#).



#### 4.118.5.13 GetBinom()

```
int GetBinom (
    UWORD * a,
    WORD * na,
    WORD i1,
    WORD i2 )
```

Definition at line [2668](#) of file [reken.c](#).

#### 4.118.5.14 LcmLong()

```
int LcmLong (
    PHEAD UWORD * a,
    WORD na,
    UWORD * b,
    WORD nb,
    UWORD * c,
    WORD * nc )
```

Definition at line [2702](#) of file [reken.c](#).

#### 4.118.5.15 TakeLongRoot()

```
int TakeLongRoot (
    UWORD * a,
    WORD * n,
    WORD power )
```

Definition at line [2736](#) of file [reken.c](#).

#### 4.118.5.16 MakeRational()

```
int MakeRational (
    WORD a,
    WORD m,
    WORD * b,
    WORD * c )
```

Definition at line [2856](#) of file [reken.c](#).

#### 4.118.5.17 MakeLongRational()

```
int MakeLongRational (
    PHEAD UWORD * a,
    WORD na,
    UWORD * m,
    WORD nm,
    UWORD * b,
    WORD * nb )
```

Definition at line [2904](#) of file [reken.c](#).

**4.118.5.18 CompCoef()**

```
WORD CompCoef (
    WORD * term1,
    WORD * term2 )
```

Routine takes  $a_1 \bmod m_1$  and  $a_2 \bmod m_2$  and returns  $a \bmod m_1*m_2$  with  
 $a \bmod m_1 = a_1$  and  $a \bmod m_2 = a_2$

Chinese remainder:  $a \bmod (m_1*m_2) = q_1*m_1 + a_1$   $a \bmod (m_1*m_2) = q_2*m_2 + a_2$  Compute  $n_1$  such that  $(n_1*m_1) \bmod m_2$  is one  
 Compute  $n_2$  such that  $(n_2*m_2) \bmod m_1$  is one Then  $(a_1*n_2*m_2 + a_2*n_1*m_1) \bmod (m_1*m_2)$  is  $a \bmod (m_1*m_2)$

Definition at line 3048 of file [reken.c](#).

Referenced by [Compare1\(\)](#).

**4.118.5.19 Modulus()**

```
int Modulus (
    WORD * term )
```

Definition at line 3103 of file [reken.c](#).

**4.118.5.20 TakeModulus()**

```
int TakeModulus (
    UWORD * a,
    WORD * na,
    UWORD * cmodvec,
    WORD ncmod,
    WORD par )
```

Definition at line 3150 of file [reken.c](#).

**4.118.5.21 TakeNormalModulus()**

```
int TakeNormalModulus (
    UWORD * a,
    WORD * na,
    UWORD * c,
    WORD nc,
    WORD par )
```

Definition at line 3322 of file [reken.c](#).

**4.118.5.22 MakeModTable()**

```
int MakeModTable (
    void )
```

Definition at line 3364 of file [reken.c](#).

#### 4.118.5.23 Factorial()

```
int Factorial (
    PHEAD WORD n,
    UWORD * a,
    WORD * na )
```

Definition at line 3450 of file [reken.c](#).

#### 4.118.5.24 Bernoulli()

```
int Bernoulli (
    WORD n,
    UWORD * a,
    WORD * na )
```

Definition at line 3530 of file [reken.c](#).

#### 4.118.5.25 NextPrime()

```
WORD NextPrime (
    PHEAD WORD num )
```

Gives the next prime number in the list of prime numbers.

If the list isn't long enough we expand it. For ease in ParForm and because these lists shouldn't be very big we let each worker keep its own list.

The list is cut off at MAXPOWER, because we don't want to get into trouble that the power of a variable gets larger than the prime number.

Definition at line 3665 of file [reken.c](#).

#### 4.118.5.26 Moebius()

```
WORD Moebius (
    PHEAD WORD nn )
```

Definition at line 3729 of file [reken.c](#).

#### 4.118.5.27 iniwranf()

```
void iniwranf (
    PHEAD0 )
```

Definition at line 3820 of file [reken.c](#).

**4.118.5.28 wranf()**

```
UWORD wranf (
    PHEAD0 )
```

Definition at line [3869](#) of file [reken.c](#).

**4.118.5.29 iranf()**

```
UWORD iranf (
    PHEAD UWORD imax )
```

Definition at line [3888](#) of file [reken.c](#).

**4.118.5.30 PreRandom()**

```
UBYTE * PreRandom (
    UBYTE * s )
```

Definition at line [3913](#) of file [reken.c](#).

**4.119 reken.c**

[Go to the documentation of this file.](#)

```
00001
00011 /* #[] License : */
00012 /*
00013 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00014 *   When using this file you are requested to refer to the publication
00015 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00016 *   This is considered a matter of courtesy as the development was paid
00017 *   for by FOM the Dutch physics granting agency and we would like to
00018 *   be able to track its scientific use to convince FOM of its value
00019 *   for the community.
00020 *
00021 *   This file is part of FORM.
00022 *
00023 *   FORM is free software: you can redistribute it and/or modify it under the
00024 *   terms of the GNU General Public License as published by the Free Software
00025 *   Foundation, either version 3 of the License, or (at your option) any later
00026 *   version.
00027 *
00028 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00029 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00030 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00031 *   details.
00032 *
00033 *   You should have received a copy of the GNU General Public License along
00034 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00035 */
00036 /* #[] License : */
00037 /*
00038 *   #[] Includes : reken.c
00039 */
00040
00041 #include "form3.h"
00042 #include <math.h>
00043
00044 #ifdef WITHGMP
00045 #include <gmp.h>
00046 #define GMPSPREAD (GMP_LIMB_BITS/BITSINWORD)
00047 #endif
00048
00049 #define GCDMAX 3
00050
00051 #define NEWTRICK 1
```

```

00052 /*
00053  #] Includes :
00054  #[ RekenRational :
00055      #[ Pack :          void Pack(a,na,b,nb)
00056
00057      Packs the contents of the numerator a and the denominator b into
00058      one normalized fraction a.
00059  */
00060
00061 void Pack(UWORD *a, WORD *na, UWORD *b, WORD nb)
00062 {
00063     WORD c, sgn = 1, i;
00064     UWORD *to,*from;
00065     if ( (c = *na) == 0 ) {
00066         MLOCK(ErrorMessageLock);
00067         MesPrint("Caught a zero in Pack");
00068         MUNLOCK(ErrorMessageLock);
00069         return;
00070     }
00071     if ( nb == 0 ) {
00072         MLOCK(ErrorMessageLock);
00073         MesPrint("Division by zero in Pack");
00074         MUNLOCK(ErrorMessageLock);
00075         return;
00076     }
00077     if ( *na < 0 ) { sgn = -sgn; c = -c; }
00078     if ( nb < 0 ) { sgn = -sgn; nb = -nb; }
00079     *na = MaX(c,nb);
00080     to = a + c;
00081     i = *na - c;
00082     while ( --i >= 0 ) *to++ = 0;
00083     i = *na - nb;
00084     from = b;
00085     NCOPY(to,from,nb);
00086     while ( --i >= 0 ) *to++ = 0;
00087     if ( sgn < 0 ) *na = -*na;
00088 }
00089
00090
00091 /*
00092     #] Pack :
00093     #[ UnPack :          void UnPack(a,na,denom,number)
00094
00095     Determines the sizes of the numerator and the denominator in the
00096     normalized fraction a with length na.
00097  */
00098
00099 void UnPack(UWORD *a, WORD na, WORD *denom, WORD *number)
00100 {
00101     UWORD *pos;
00102     WORD i, sgn = na;
00103     if ( na < 0 ) { na = -na; }
00104     i = na;
00105     if ( i > 1 ) { /* Find the respective leading words */
00106         a += i;
00107         a--;
00108         pos = a + i;
00109         while ( !(*a) ) { i--; a--; }
00110         while ( !(*pos) ) { na--; pos--; }
00111     }
00112     *denom = na;
00113     if ( sgn < 0 ) i = -i;
00114     *number = i;
00115 }
00116
00117
00118 /*
00119     #] UnPack :
00120     #[ Mully :          WORD Mully(a,na,b,nb)
00121
00122     Multiplies the rational a by the Long b.
00123  */
00124
00125 int Mully(PHEAD UWORD *a, WORD *na, UWORD *b, WORD nb)
00126 {
00127     GETBIDENTITY
00128     UWORD *d, *e;
00129     WORD i, sgn = 1;
00130     WORD nd, ne, adenom, anumer;
00131     if ( !nb ) { *na = 0; return(0); }
00132     else if ( *b == 1 ) {
00133         if ( nb == 1 ) return(0);
00134         else if ( nb == -1 ) { *na = -*na; return(0); }
00135     }
00136     if ( *na < 0 ) { sgn = -sgn; *na = -*na; }
00137     if ( nb < 0 ) { sgn = -sgn; nb = -nb; }

```

```

00139     UnPack(a,*na,&adenom,&anumer);
00140     d = NumberMalloc("Mully"); e = NumberMalloc("Mully");
00141     for ( i = 0; i < nb; i++ ) { e[i] = *b++; }
00142     ne = nb;
00143     if ( Simplify(BHEAD a,*na,&adenom,e,&ne) ) goto MullyEr;
00144     if ( Mullong(a,anumer,e,ne,d,&nd) ) goto MullyEr;
00145     b = a+*na;
00146     for ( i = 0; i < *na; i++ ) { e[i] = *b++; }
00147     ne = adenom;
00148     *na = nd;
00149     b = d;
00150     *na = nd;
00151     for ( i = 0; i < *na; i++ ) { a[i] = *b++; }
00152     Pack(a,na,e,ne);
00153     if ( sgn < 0 ) *na = -*na;
00154     NumberFree(d,"Mully"); NumberFree(e,"Mully");
00155     return(0);
00156 MullyEr:
00157     MLOCK(ErrorMessageLock);
00158     MesCall("Mully");
00159     MUNLOCK(ErrorMessageLock);
00160     NumberFree(d,"Mully"); NumberFree(e,"Mully");
00161     SETERROR(-1)
00162 }
00163
00164 /*
00165     #] Mully :
00166     #[ Divvy :          WORD Divvy(a,na,b,nb)
00167
00168     Divides the rational a by the Long b.
00169 */
00170
00171 int Divvy(PHEAD UWORD *a, WORD *na, UWORD *b, WORD nb)
00172 {
00173     GETBIDENTITY
00174     UWORD *d,*e;
00175     WORD i, sgn = 1;
00176     WORD nd, ne, adenom, anumer;
00177     if ( !nb ) {
00178         MLOCK(ErrorMessageLock);
00179         MesPrint("Division by zero in Divvy");
00180         MUNLOCK(ErrorMessageLock);
00181         return(-1);
00182     }
00183     d = NumberMalloc("Divvy"); e = NumberMalloc("Divvy");
00184     if ( nb < 0 ) { sgn = -sgn; nb = -nb; }
00185     if ( *na < 0 ) { sgn = -sgn; *na = -*na; }
00186     UnPack(a,*na,&adenom,&anumer);
00187     for ( i = 0; i < nb; i++ ) { e[i] = *b++; }
00188     ne = nb;
00189     if ( Simplify(BHEAD a,&anumer,e,&ne) ) goto DivvyEr;
00190     if ( Mullong(a+*na,adenom,e,ne,d,&nd) ) goto DivvyEr;
00191     *na = anumer;
00192     Pack(a,na,d,nd);
00193     if ( sgn < 0 ) *na = -*na;
00194     NumberFree(d,"Divvy"); NumberFree(e,"Divvy");
00195     return(0);
00196 DivvyEr:
00197     MLOCK(ErrorMessageLock);
00198     MesCall("Divvy");
00199     MUNLOCK(ErrorMessageLock);
00200     NumberFree(d,"Divvy"); NumberFree(e,"Divvy");
00201     SETERROR(-1)
00202 }
00203
00204 /*
00205     #] Divvy :
00206     #[ AddRat :          WORD AddRat(a,na,b,nb,c,nc)
00207
00208 */
00209
00210 int AddRat(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
00211 {
00212     GETBIDENTITY
00213     UWORD *d, *e, *f, *g;
00214     WORD nd, ne, nf, ng, adenom, anumer, bdenom, bnumer;
00215     if ( !na ) {
00216         WORD i;
00217         *nc = nb;
00218         if ( nb < 0 ) nb = -nb;
00219         nb *= 2;
00220         for ( i = 0; i < nb; i++ ) *c++ = *b++;
00221         return(0);
00222     }
00223     else if ( !nb ) {
00224         WORD i;
00225         *nc = na;

```

```

00226     if ( na < 0 ) na = -na;
00227     na *= 2;
00228     for ( i = 0; i < na; i++ ) *c++ = *a++;
00229     return(0);
00230 }
00231 else if ( b[1] == 1 && a[1] == 1 ) {
00232     if ( na == 1 ) {
00233         if ( nb == 1 ) {
00234             *c = *a + *b;
00235             c[1] = 1;
00236             if ( *c < *a ) { c[2] = 1; c[3] = 0; *nc = 2; }
00237             else { *nc = 1; }
00238             return(0);
00239         }
00240         else if ( nb == -1 ) {
00241             if ( *b > *a ) {
00242                 *c = *b - *a; *nc = -1;
00243             }
00244             else if ( *b < *a ) {
00245                 *c = *a - *b; *nc = 1;
00246             }
00247             else *nc = 0;
00248             c[1] = 1;
00249             return(0);
00250         }
00251     }
00252     else if ( na == -1 ) {
00253         if ( nb == -1 ) {
00254             c[1] = 1;
00255             *c = *a + *b;
00256             if ( *c < *a ) { c[2] = 1; c[3] = 0; *nc = -2; }
00257             else { *nc = -1; }
00258             return(0);
00259         }
00260         else if ( nb == 1 ) {
00261             if ( *b > *a ) {
00262                 *c = *b - *a; *nc = 1;
00263             }
00264             else if ( *b < *a ) {
00265                 *c = *a - *b; *nc = -1;
00266             }
00267             else *nc = 0;
00268             c[1] = 1;
00269             return(0);
00270         }
00271     }
00272 }
00273 UnPack(a,na,&adenom,&anumer);
00274 UnPack(b,nb,&bdenom,&bnumber);
00275 if ( na < 0 ) na = -na;
00276 if ( nb < 0 ) nb = -nb;
00277 if ( na == 1 && nb == 1 ) {
00278     RLONG t1, t2, t3;
00279     t3 = ((RLONG)a[1])*((RLONG)b[1]);
00280     t1 = ((RLONG)a[0])*((RLONG)b[1]);
00281     t2 = ((RLONG)a[1])*((RLONG)b[0]);
00282     if ( ( anumer > 0 && bnumber > 0 ) || ( anumer < 0 && bnumber < 0 ) ) {
00283         if ( ( t1 = t1 + t2 ) < t2 ) {
00284             c[2] = 1;
00285             c[0] = (UWORD)t1;
00286             c[1] = (UWORD)(t1 » BITSINWORD);
00287             *nc = 3;
00288         }
00289         else {
00290             c[0] = (UWORD)t1;
00291             if ( ( c[1] = (UWORD)(t1 » BITSINWORD) ) != 0 ) *nc = 2;
00292             else *nc = 1;
00293         }
00294     }
00295     else {
00296         if ( t1 == t2 ) { *nc = 0; return(0); }
00297         if ( t1 > t2 ) {
00298             t1 -= t2;
00299         }
00300         else {
00301             t1 = t2 - t1;
00302             anumer = -anumer;
00303         }
00304         c[0] = (UWORD)t1;
00305         if ( ( c[1] = (UWORD)(t1 » BITSINWORD) ) != 0 ) *nc = 2;
00306         else *nc = 1;
00307     }
00308     if ( anumer < 0 ) *nc = -*nc;
00309     d = NumberMalloc("AddRat");
00310     d[0] = (UWORD)t3;
00311     if ( ( d[1] = (UWORD)(t3 » BITSINWORD) ) != 0 ) nd = 2;
00312     else nd = 1;

```

```

00313         if ( Simplify(BHEAD c,nc,d,&nd) ) goto AddRer1;
00314     }
00315 /*
00316     else if ( a[na] == 1 && b[nb] == 1 && adenom == 1 && bdenom == 1 ) {
00317         if ( AddLong(a,na,b,nb,c,&nc) ) goto AddRer2;
00318         i = ABS(nc); d = c + i; *d++ = 1;
00319         while ( --i > 0 ) *d++ = 0 ;
00320         return(0);
00321     }
00322 */
00323     else {
00324         d = NumberMalloc("AddRat"); e = NumberMalloc("AddRat");
00325         f = NumberMalloc("AddRat"); g = NumberMalloc("AddRat");
00326         if ( GcdLong(BHEAD a+na,adenom,b+nb,bdenom,d,&nd) ) goto AddRer;
00327         if ( *d == 1 && nd == 1 ) nd = 0;
00328         if ( nd ) {
00329             if ( DivLong(a+na,adenom,d,nd,e,&ne,c,nc) ) goto AddRer;
00330             if ( DivLong(b+nb,bdenom,d,nd,f,&nf,c,nc) ) goto AddRer;
00331             if ( MulLong(a,anumer,f,nf,c,nc) ) goto AddRer;
00332             if ( MulLong(b,bnumer,e,ne,g,&ng) ) goto AddRer;
00333         }
00334         else {
00335             if ( MulLong(a+na,adenom,b,bnumer,c,nc) ) goto AddRer;
00336             if ( MulLong(b+nb,bdenom,a,anumer,g,&ng) ) goto AddRer;
00337         }
00338         if ( AddLong(c,*nc,g,ng,c,nc) ) goto AddRer;
00339         if ( !*nc ) {
00340             NumberFree(g,"AddRat"); NumberFree(f,"AddRat");
00341             NumberFree(e,"AddRat"); NumberFree(d,"AddRat");
00342             return(0);
00343         }
00344         if ( nd ) {
00345             if ( Simplify(BHEAD c,nc,d,&nd) ) goto AddRer;
00346             if ( MulLong(e,ne,d,nd,g,&ng) ) goto AddRer;
00347             if ( MulLong(g,ng,f,nf,d,&nd) ) goto AddRer;
00348         }
00349         else {
00350             if ( MulLong(a+na,adenom,b+nb,bdenom,d,&nd) ) goto AddRer;
00351         }
00352         NumberFree(g,"AddRat"); NumberFree(f,"AddRat"); NumberFree(e,"AddRat");
00353     }
00354     Pack(c,nc,d,nd);
00355     NumberFree(d,"AddRat");
00356     return(0);
00357 AddRer:
00358     NumberFree(g,"AddRat"); NumberFree(f,"AddRat"); NumberFree(e,"AddRat");
00359 AddRer1:
00360     NumberFree(d,"AddRat");
00361 /* AddRer2: */
00362     MLOCK(ErrorMessageLock);
00363     MesCall("AddRat");
00364     MUNLOCK(ErrorMessageLock);
00365     SETERROR(-1)
00366 }
00367
00368 /*
00369     #] AddRat :
00370     #[ MulRat :          WORD MulRat(a,na,b,nb,c,nc)
00371
00372     Multiplies the rationals a and b. The Gcd of the individual
00373     pieces is divided out first to minimize the chances of spurious
00374     overflows.
00375 */
00376 */
00377
00378 int MulRat(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
00379 {
00380     WORD i;
00381     WORD sgn = 1;
00382     if ( *b == 1 && b[1] == 1 ) {
00383         if ( nb == 1 ) {
00384             *nc = na;
00385             i = ABS(na); i *= 2;
00386             while ( --i >= 0 ) *c++ = *a++;
00387             return(0);
00388         }
00389         else if ( nb == -1 ) {
00390             *nc = - na;
00391             i = ABS(na); i *= 2;
00392             while ( --i >= 0 ) *c++ = *a++;
00393             return(0);
00394         }
00395     }
00396     if ( *a == 1 && a[1] == 1 ) {
00397         if ( na == 1 ) {
00398             *nc = nb;
00399             i = ABS(nb); i *= 2;

```



```

00400         while ( --i >= 0 ) *c++ = *b++;
00401         return(0);
00402     }
00403     else if ( na == -1 ) {
00404         *nc = - nb;
00405         i = ABS(nb); i *= 2;
00406         while ( --i >= 0 ) *c++ = *b++;
00407         return(0);
00408     }
00409 }
00410 if ( na < 0 ) { na = -na; sgn = -sgn; }
00411 if ( nb < 0 ) { nb = -nb; sgn = -sgn; }
00412 if ( !na || !nb ) { *nc = 0; return(0); }
00413 if ( na != 1 || nb != 1 ) {
00414     GETBIDENTITY
00415     UWORD *xd,*xe, *xf,*xg;
00416     WORD dden, dnumr, eden, enumr;
00417     UnPack(a,na,&dden,&dnumr);
00418     UnPack(b,nb,&eden,&enumr);
00419     xd = NumberMalloc("MulRat"); xf = NumberMalloc("MulRat");
00420     for ( i = 0; i < dnumr; i++ ) xd[i] = a[i];
00421     a += na;
00422     for ( i = 0; i < dden; i++ ) xf[i] = a[i];
00423     xe = NumberMalloc("MulRat"); xg = NumberMalloc("MulRat");
00424     for ( i = 0; i < enumr; i++ ) xe[i] = b[i];
00425     b += nb;
00426     for ( i = 0; i < eden; i++ ) xg[i] = b[i];
00427     if ( Simplify(BHEAD xd,&dnumr,xg,&eden) ||
00428         Simplify(BHEAD xe,&enumr,xf,&dden) ||
00429         MulLong(xd,dnumr,xe,enumr,c,nc) ||
00430         MulLong(xf,dden,xg,eden,xd,&dnumr) ) {
00431         MLOCK(ErrorMessageLock);
00432         MesCall("MulRat");
00433         MUNLOCK(ErrorMessageLock);
00434         NumberFree(xd,"MulRat"); NumberFree(xe,"MulRat"); NumberFree(xf,"MulRat");
00435         NumberFree(xg,"MulRat");
00436         SETERROR(-1)
00437     }
00438     Pack(c,nc,xd,dnumr);
00439     NumberFree(xd,"MulRat"); NumberFree(xe,"MulRat"); NumberFree(xf,"MulRat");
00440     NumberFree(xg,"MulRat");
00441 }
00442 else {
00443     UWORD y;
00444     UWORD a0,a1,b0,b1;
00445     RLONG xx;
00446     y = a[0]; b1=b[1];
00447     do { a0 = y % b1; y = b1; } while ( ( b1 = a0 ) != 0 );
00448     if ( y != 1 ) {
00449         a0 = a[0] / y;
00450         b1 = b[1] / y;
00451     }
00452     else {
00453         a0 = a[0];
00454         b1 = b[1];
00455     }
00456     y=b[0]; a1=a[1];
00457     do { b0 = y % a1; y = a1; } while ( ( a1 = b0 ) != 0 );
00458     if ( y != 1 ) {
00459         a1 = a[1] / y;
00460         b0 = b[0] / y;
00461     }
00462     else {
00463         a1 = a[1];
00464         b0 = b[0];
00465     }
00466     xx = ((RLONG)a0)*b0;
00467     if ( xx & AWORDMASK ) {
00468         *nc = 2;
00469         c[0] = (UWORD)xx;
00470         c[1] = (UWORD)(xx >> BITSINWORD);
00471         xx = ((RLONG)a1)*b1;
00472         c[2] = (UWORD)xx;
00473         c[3] = (UWORD)(xx >> BITSINWORD);
00474     }
00475     else {
00476         c[0] = (UWORD)xx;
00477         xx = ((RLONG)a1)*b1;
00478         if ( xx & AWORDMASK ) {
00479             c[1] = 0;
00480             c[2] = (UWORD)xx;
00481             c[3] = (UWORD)(xx >> BITSINWORD);
00482             *nc = 2;
00483         }
00484         else {
00485             c[1] = (UWORD)xx;
00486             *nc = 1;
00487         }
00488     }
00489 }

```

```

00485     }
00486     }
00487 }
00488 if ( sgn < 0 ) *nc = -*nc;
00489 return(0);
00490 }
00491
00492 /*
00493     #] MulRat :
00494     #[ DivRat :      WORD DivRat(a,na,b,nb,c,nc)
00495
00496     Divides the rational a by the rational b.
00497
00498 */
00499
00500 int DivRat(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
00501 {
00502     GETBIDENTITY
00503     WORD i, j;
00504     int ret;
00505     UWORD *xd,*xe,xx;
00506     if ( !nb ) {
00507         MLOCK(ErrorMessageLock);
00508         MesPrint("Rational division by zero");
00509         MUNLOCK(ErrorMessageLock);
00510         return(-1);
00511     }
00512     j = i = (nb >= 0)? nb: -nb;
00513     xd = b; xe = b + i;
00514     do { xx = *xd; *xd++ = *xe; *xe++ = xx; } while ( --j > 0 );
00515     ret = MulRat(BHEAD a,na,b,nb,c,nc);
00516     xd = b; xe = b + i;
00517     do { xx = *xd; *xd++ = *xe; *xe++ = xx; } while ( --i > 0 );
00518     return(ret);
00519 }
00520
00521 /*
00522     #] DivRat :
00523     #[ Simplify :      WORD Simplify(a,na,b,nb)
00524
00525     Determines the greatest common denominator of a and b and
00526     divides both by it. A possible sign is put in a. This is
00527     the simplification of the fraction a/b.
00528
00529 */
00530
00531 int Simplify(PHEAD UWORD *a, WORD *na, UWORD *b, WORD *nb)
00532 {
00533     GETBIDENTITY
00534     UWORD *x1,*x2,*x3;
00535     UWORD *x4;
00536     WORD n1,n2,n3,n4,sgn = 1;
00537     WORD i;
00538     UWORD *Siscrat5, *Siscrat6, *Siscrat7, *Siscrat8;
00539     if ( *na < 0 ) { *na = -*na; sgn = -sgn; }
00540     if ( *nb < 0 ) { *nb = -*nb; sgn = -sgn; }
00541     Siscrat5 = NumberMalloc("Simplify"); Siscrat6 = NumberMalloc("Simplify");
00542     Siscrat7 = NumberMalloc("Simplify"); Siscrat8 = NumberMalloc("Simplify");
00543     x1 = Siscrat8; x2 = Siscrat7;
00544     if ( *nb == 1 ) {
00545         x3 = Siscrat6;
00546         if ( DivLong(a,*na,b,*nb,x1,&n1,x2,&n2) ) goto SimpErr;
00547         if ( !n2 ) {
00548             for ( i = 0; i < n1; i++ ) *a++ = *x1++;
00549             *na = n1;
00550             *b = 1;
00551         }
00552     } else {
00553         UWORD y1, y2, y3;
00554         y2 = *b;
00555         y3 = *x2;
00556         do { y1 = y2 % y3; y2 = y3; } while ( ( y3 = y1 ) != 0 );
00557         if ( ( *x2 = y2 ) != 1 ) {
00558             *b /= y2;
00559             if ( DivLong(a,*na,x2,(WORD)1,x1,&n1,x3,&n3) ) goto SimpErr;
00560             for ( i = 0; i < n1; i++ ) *a++ = *x1++;
00561             *na = n1;
00562         }
00563     }
00564 }
00565 #ifndef NEWTRICK
00566 else if ( *na >= GCDMAX && *nb >= GCDMAX ) {
00567     n1 = i = *na; x3 = a;
00568     NCOPY(x1,x3,i);
00569     x3 = b; n2 = i = *nb;
00570     NCOPY(x2,x3,i);
00571     x4 = Siscrat5;

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```

00572     x2 = Siscrat6;
00573     x3 = Siscrat7;
00574     if ( GcdLong(BHEAD Siscrat8,n1,Siscrat7,n2,x2,&n3) ) goto SimpErr;
00575     n2 = n3;
00576     if ( *x2 != 1 || n2 != 1 ) {
00577         DivLong(a,*na,x2,n2,x1,&n1,x4,&n4);
00578         *na = i = n1;
00579         NCOPY(a,x1,i);
00580         DivLong(b,*nb,x2,n2,x3,&n3,x4,&n4);
00581         *nb = i = n3;
00582         NCOPY(b,x3,i);
00583     }
00584 }
00585 #endif
00586 else {
00587     x4 = Siscrat5;
00588     n1 = i = *na; x3 = a;
00589     NCOPY(x1,x3,i);
00590     x3 = b; n2 = i = *nb;
00591     NCOPY(x2,x3,i);
00592     x1 = Siscrat8; x2 = Siscrat7; x3 = Siscrat6;
00593     for(;;){
00594         if ( DivLong(x1,n1,x2,n2,x4,&n4,x3,&n3) ) goto SimpErr;
00595         if ( !n3 ) break;
00596         if ( n2 == 1 ) {
00597             while ( ( *x1 = (*x2) % (*x3) ) != 0 ) { *x2 = *x3; *x3 = *x1; }
00598             *x2 = *x3;
00599             break;
00600         }
00601         if ( DivLong(x2,n2,x3,n3,x4,&n4,x1,&n1) ) goto SimpErr;
00602         if ( !n1 ) { x2 = x3; n2 = n3; x3 = Siscrat7; break; }
00603         if ( n3 == 1 ) {
00604             while ( ( *x2 = (*x3) % (*x1) ) != 0 ) { *x3 = *x1; *x1 = *x2; }
00605             *x2 = *x1;
00606             n2 = 1;
00607             break;
00608         }
00609         if ( DivLong(x3,n3,x1,n1,x4,&n4,x2,&n2) ) goto SimpErr;
00610         if ( !n2 ) { x2 = x1; n2 = n1; x1 = Siscrat7; break; }
00611         if ( n1 == 1 ) {
00612             while ( ( *x3 = (*x1) % (*x2) ) != 0 ) { *x1 = *x2; *x2 = *x3; }
00613             break;
00614         }
00615     }
00616     if ( *x2 != 1 || n2 != 1 ) {
00617         DivLong(a,*na,x2,n2,x1,&n1,x4,&n4);
00618         *na = i = n1;
00619         NCOPY(a,x1,i);
00620         DivLong(b,*nb,x2,n2,x3,&n3,x4,&n4);
00621         *nb = i = n3;
00622         NCOPY(b,x3,i);
00623     }
00624 }
00625 if ( sgn < 0 ) *na = -*na;
00626 NumberFree(Siscrat5,"Simplify"); NumberFree(Siscrat6,"Simplify");
00627 NumberFree(Siscrat7,"Simplify"); NumberFree(Siscrat8,"Simplify");
00628 return(0);
00629 SimpErr:
00630 MLOCK(ErrorMessageLock);
00631 MesCall("Simplify");
00632 MUNLOCK(ErrorMessageLock);
00633 NumberFree(Siscrat5,"Simplify"); NumberFree(Siscrat6,"Simplify");
00634 NumberFree(Siscrat7,"Simplify"); NumberFree(Siscrat8,"Simplify");
00635 SETERROR(-1)
00636 }
00637
00638 /*
00639 #] Simplify :
00640 #[ AccumGCD :      WORD AccumGCD(PHEAD a,na,b,nb)
00641
00642 Routine takes the rational GCD of the fractions in a and b and
00643 replaces a by the GCD of the two.
00644 The rational GCD is defined as the rational that consists of
00645 the GCD of the numerators divided by the GCD of the denominators
00646 */
00647
00648 int AccumGCD(PHEAD UWORD *a, WORD *na, UWORD *b, WORD nb)
00649 {
00650     GETBIDENTITY
00651     WORD nna,nnb,numa,numb,dena,denb,numc,denc;
00652     UWORD *GCDbuffer = NumberMalloc("AccumGCD");
00653     int i;
00654     nna = *na; if ( nna < 0 ) nna = -nna; nna = (nna-1)/2;
00655     nnb = nb; if ( nnb < 0 ) nnb = -nnb; nnb = (nnb-1)/2;
00656     UnPack(a,nna,&dena,&numa);
00657     UnPack(b,nnb,&denb,&numb);
00658     if ( GcdLong(BHEAD a,numa,b,numb,GCDbuffer,&numc) ) goto AccErr;

```

```

00659     numa = numc;
00660     for ( i = 0; i < numa; i++ ) a[i] = GCDbuffer[i];
00661     if ( GcdLong(BHEAD a+nna,dena,b+nnb,denb,GCDbuffer,&denc) ) goto AccErr;
00662     dena = denc;
00663     for ( i = 0; i < dena; i++ ) a[i+nna] = GCDbuffer[i];
00664     Pack(a,&numa,a+nna,dena);
00665     *na = INCLENG(numa);
00666     NumberFree(GCDbuffer,"AccumGCD");
00667     return(0);
00668 AccErr:
00669     MLOCK(ErrorMessageLock);
00670     MesCall("AccumGCD");
00671     MUNLOCK(ErrorMessageLock);
00672     NumberFree(GCDbuffer,"AccumGCD");
00673     SETERROR(-1)
00674 }
00675
00676 /*
00677     #] AccumGCD :
00678     #[ TakeRatRoot:
00679 */
00680
00681 int TakeRatRoot(WORD *a, WORD *n, WORD power)
00682 {
00683     WORD number,denom, nn;
00684     if ( ( power & 1 ) == 0 && *n < 0 ) return(1);
00685     if ( ABS(*n) == 1 && a[0] == 1 && a[1] == 1 ) return(0);
00686     nn = ABS(*n);
00687     UnPack(a,nn,&denom,&number);
00688     if ( TakeLongRoot(a+nn,&denom,power) ) return(1);
00689     if ( TakeLongRoot(a,&number,power) ) return(1);
00690     Pack(a,&number,a+nn,denom);
00691     if ( *n < 0 ) *n = -number;
00692     else *n = number;
00693     return(0);
00694 }
00695
00696 /*
00697     #] TakeRatRoot:
00698     #] RekenRational :
00699     #[ RekenLong :
00700     #[ AddLong :          WORD AddLong(a,na,b,nb,c,nc)
00701
00702     Long addition. Uses addition and subtraction of positive numbers.
00703
00704 */
00705
00706 int AddLong(WORD *a, WORD na, WORD *b, WORD nb, WORD *c, WORD *nc)
00707 {
00708     WORD sgn, res;
00709     if ( na < 0 ) {
00710         if ( nb < 0 ) {
00711             if ( AddPLon(a,-na,b,-nb,c,nc) ) return(-1);
00712             *nc = -*nc;
00713             return(0);
00714         }
00715         else {
00716             na = -na;
00717             sgn = -1;
00718         }
00719     }
00720     else {
00721         if ( nb < 0 ) {
00722             nb = -nb;
00723             sgn = 1;
00724         }
00725         else { return( AddPLon(a,na,b,nb,c,nc) ); }
00726     }
00727     if ( ( res = BigLong(a,na,b,nb) ) > 0 ) {
00728         SubPLon(a,na,b,nb,c,nc);
00729         if ( sgn < 0 ) *nc = -*nc;
00730     }
00731     else if ( res < 0 ) {
00732         SubPLon(b,nb,a,na,c,nc);
00733         if ( sgn > 0 ) *nc = -*nc;
00734     }
00735     else {
00736         *nc = 0;
00737         *c = 0;
00738     }
00739     return(0);
00740 }
00741
00742 /*
00743     #] AddLong :
00744     #[ AddPLon :          WORD AddPLon(a,na,b,nb,c,nc)
00745

```

```

00746     Adds two long integers a and b and puts the result in c.
00747     The length of a and b are na and nb. The length of c is returned in nc.
00748     c can be a or b.
00749 */
00750
00751 int AddPLon(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
00752 {
00753     UWORD carry = 0, e, nd = 0;
00754     while ( na && nb ) {
00755         e = *a;
00756         *c = e + *b + carry;
00757         if ( carry ) {
00758             if ( e < *c ) carry = 0;
00759         }
00760         else {
00761             if ( e > *c ) carry = 1;
00762         }
00763         a++; b++; c++; nd++; na--; nb--;
00764     }
00765     while ( na ) {
00766         if ( carry ) {
00767             *c = *a++ + carry;
00768             if ( *c++ ) carry = 0;
00769         }
00770         else *c++ = *a++;
00771         nd++; na--;
00772     }
00773     while ( nb ) {
00774         if ( carry ) {
00775             *c = *b++ + carry;
00776             if ( *c++ ) carry = 0;
00777         }
00778         else *c++ = *b++;
00779         nd++; nb--;
00780     }
00781     if ( carry ) {
00782         nd++;
00783         if ( nd > (UWORD)AM.MaxTal ) {
00784             MLOCK(ErrorMessageLock);
00785             MesPrint("Overflow in addition");
00786             MUNLOCK(ErrorMessageLock);
00787             return(-1);
00788         }
00789         *c++ = carry;
00790     }
00791     *nc = nd;
00792     return(0);
00793 }
00794
00795 /*
00796     #] AddPLon :
00797     #[ SubPLon :      void SubPLon(a,na,b,nb,c,nc)
00798
00799     Subtracts b from a. Assumes that a > b. Result in c.
00800     c can be a or b.
00801 */
00802
00803
00804 void SubPLon(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
00805 {
00806     UWORD borrow = 0, e, nd = 0;
00807     while ( nb ) {
00808         e = *a;
00809         if ( borrow ) {
00810             *c = e - *b - borrow;
00811             if ( *c < e ) borrow = 0;
00812         }
00813         else {
00814             *c = e - *b;
00815             if ( *c > e ) borrow = 1;
00816         }
00817         a++; b++; c++; na--; nb--; nd++;
00818     }
00819     while ( na ) {
00820         if ( borrow ) {
00821             if ( *a ) { *c++ = *a++ - 1; borrow = 0; }
00822             else { *c++ = (UWORD)(-1); a++; }
00823         }
00824         else *c++ = *a++;
00825         na--; nd++;
00826     }
00827     while ( nd && !*--c ) { nd--; }
00828     *nc = (WORD)nd;
00829 }
00830
00831 /*
00832     #] SubPLon :

```

```

00833      #[ MulLong :          WORD MulLong(a,na,b,nb,c,nc)
00834
00835      Does a Long multiplication. Assumes that WORD is half the size
00836      of a LONG to work out the scheme! The number of operations is
00837      the canonical na*nm multiplications.
00838      c should not overlap with a or b.
00839
00840      */
00841
00842      int MulLong(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
00843      {
00844          WORD sgn = 1;
00845          UWORD i, *ic, *ia;
00846          RLONG t, bb;
00847          if ( !na || !nb ) { *nc = 0; return(0); }
00848          if ( na < 0 ) { na = -na; sgn = -sgn; }
00849          if ( nb < 0 ) { nb = -nb; sgn = -sgn; }
00850          *nc = i = na + nb;
00851          if ( i > (UWORD)(AM.MaxTal+1) ) goto MulLov;
00852          ic = c;
00853      /*
00854      #[ GMP stuff :
00855      */
00856      #ifdef WITHGMP
00857          if (na > 3 && nb > 3) {
00858              /* mp_limb_t res; */
00859              UWORD *to, *from;
00860              int j;
00861              GETIDENTITY
00862              UWORD *DLscrat9 = NumberMalloc("MulLong"), *DLscratA = NumberMalloc("MulLong"), *DLscratB =
00863              NumberMalloc("MulLong");
00864              #if ( GMPSPREAD != 1 )
00865                  if ( na & 1 ) {
00866                      from = a; a = to = DLscrat9; j = na; NCOPY(to, from, j);
00867                      a[na++] = 0;
00868                      ++*nc;
00869                  } else
00870                  if ( (LONG)a & (sizeof(mp_limb_t)-1) ) {
00871                      from = a; a = to = DLscrat9; j = na; NCOPY(to, from, j);
00872                  }
00873              #if ( GMPSPREAD != 1 )
00874                  if ( nb & 1 ) {
00875                      from = b; b = to = DLscratA; j = nb; NCOPY(to, from, j);
00876                      b[nb++] = 0;
00877                      ++*nc;
00878                  } else
00879                  if ( (LONG)b & (sizeof(mp_limb_t)-1) ) {
00880                      from = b; b = to = DLscratA; j = nb; NCOPY(to, from, j);
00881                  }
00882                  if ( ( *nc > (WORD)i ) || ( (LONG)c & (LONG)(sizeof(mp_limb_t)-1) ) ) {
00883                      ic = DLscratB;
00884                  }
00885                  if ( na < nb ) {
00886                      /* res = */
00887                      mpn_mul((mp_ptr)ic, (mp_srcptr)b, nb/GMPSPREAD, (mp_srcptr)a, na/GMPSPREAD);
00888                  } else {
00889                      /* res = */
00890                      mpn_mul((mp_ptr)ic, (mp_srcptr)a, na/GMPSPREAD, (mp_srcptr)b, nb/GMPSPREAD);
00891                  }
00892                  while ( ic[i-1] == 0 ) i--;
00893                  *nc = i;
00894              /*
00895              if ( res == 0 ) *nc -= GMPSPREAD;
00896              else if ( res <= WORDMASK ) --*nc;
00897              */
00898              if ( ic != c ) {
00899                  j = *nc; NCOPY(c, ic, j);
00900              }
00901              if ( sgn < 0 ) *nc = -( *nc );
00902              NumberFree(DLscrat9, "MulLong"); NumberFree(DLscratA, "MulLong");
00903              NumberFree(DLscratB, "MulLong");
00904              return(0);
00905          }
00906      #endif
00907      /*
00908      #] GMP stuff :
00909      */
00910      do { *ic++ = 0; } while ( --i > 0 );
00911      do {
00912          ia = a;
00913          ic = c++;
00914          t = 0;
00915          i = na;

```

```

00918         bb = (RLONG) (*b++);
00919         do {
00920             t = (*ia++) * bb + t + *ic;
00921             *ic++ = (WORD)t;
00922             t >= BITSINWORD;          /* should actually be a swap */
00923         } while ( --i > 0 );
00924         if ( t ) *ic = (UWORD)t;
00925     } while ( --nb > 0 );
00926     if ( !*ic ) (*nc)--;
00927     if ( *nc > AM.MaxTal ) goto MulLov;
00928     if ( sgn < 0 ) *nc = -(*nc);
00929     return(0);
00930 MulLov:
00931     MLOCK(ErrorMessageLock);
00932     MesPrint("Overflow in Multiplication");
00933     MUNLOCK(ErrorMessageLock);
00934     return(-1);
00935 }
00936
00937 /*
00938     #] MulLong :
00939     #[ BigLong :          WORD BigLong(a,na,b,nb)
00940
00941     Returns > 0 if a > b, < 0 if b > a and 0 if a == b
00942
00943 */
00944
00945 int BigLong(UWORD *a, WORD na, UWORD *b, WORD nb)
00946 {
00947     a += na;
00948     b += nb;
00949     while ( na && !--a ) na--;
00950     while ( nb && !--b ) nb--;
00951     if ( nb < na ) return(1);
00952     if ( nb > na ) return(-1);
00953     while ( --na >= 0 ) {
00954         if ( *a > *b ) return(1);
00955         else if ( *b > *a ) return(-1);
00956         a--; b--;
00957     }
00958     return(0);
00959 }
00960
00961 /*
00962     #] BigLong :
00963     #[ DivLong :          WORD DivLong(a,na,b,nb,c,nc,d,nd)
00964
00965     This is the long division which knows a couple of exceptions.
00966     It uses therefore a recursive call for the renormalization.
00967     The quotient comes in c and the remainder in d.
00968     d may be overlapping with b. It may also be identical to a.
00969     c should not overlap with a, but it can overlap with b.
00970
00971 */
00972
00973 int DivLong(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c,
00974             WORD *nc, UWORD *d, WORD *nd)
00975 {
00976     WORD sgnA = 1, sgnB = 1, ne, nf, ng, nh;
00977     WORD i, ni;
00978     UWORD *w1, *w2;
00979     RLONG t, v;
00980     UWORD *e, *f, *ff, *g, norm, estim;
00981 #ifdef WITHGMP
00982     UWORD *DLscrat9, *DLscratA, *DLscratB, *DLscratC;
00983 #endif
00984     RLONG esthelp;
00985     if ( !nb ) {
00986         MLOCK(ErrorMessageLock);
00987         MesPrint("Division by zero");
00988         MUNLOCK(ErrorMessageLock);
00989         return(-1);
00990     }
00991     if ( !na ) { *nc = *nd = 0; return(0); }
00992     if ( na < 0 ) { sgnA = -sgnA; na = -na; }
00993     if ( nb < 0 ) { sgnB = -sgnB; nb = -nb; }
00994     if ( na < nb ) {
00995         for ( i = 0; i < na; i++ ) *d++ = *a++;
00996         *nd = na;
00997         *nc = 0;
00998     }
00999     else if ( nb == na && ( i = BigLong(b,nb,a,na) ) >= 0 ) {
01000         if ( i > 0 ) {
01001             for ( i = 0; i < na; i++ ) *d++ = *a++;
01002             *nd = na;
01003             *nc = 0;
01004         }

```

```

01005         else {
01006             *c = 1;
01007             *nc = 1;
01008             *nd = 0;
01009         }
01010     }
01011     else if ( nb == 1 ) {
01012         if ( *b == 1 ) {
01013             for ( i = 0; i < na; i++ ) *c++ = *a++;
01014             *nc = na;
01015             *nd = 0;
01016         }
01017         else {
01018             w1 = a+na;
01019             *nc = ni = na;
01020             *nd = 1;
01021             w2 = c+ni;
01022             v = (RLONG) (*b);
01023             t = (RLONG) (*--w1);
01024             while ( --ni >= 0 ) {
01025                 *--w2 = t / v;
01026                 t -= v * (*w2);
01027                 if ( ni ) {
01028                     t <= BITSINWORD;
01029                     t += *--w1;
01030                 }
01031             }
01032             if ( ( *d = (UWORD)t ) == 0 ) *nd = 0;
01033             if ( !*(c+na-1) ) (*nc)--;
01034         }
01035     }
01036     else {
01037         GETIDENTITY
01038     /*
01039         #! GMP stuff :
01040
01041         We start with copying a and b.
01042         Then we make space for c and d.
01043         Next we call mpn_tdiv_qr
01044         We adjust sizes and copy to c and d if needed.
01045         Finally the signs are settled.
01046     */
01047     #ifdef WITHGMP
01048         if ( na > 4 && nb > 3 ) {
01049             UWORD *ic, *id, *to, *from;
01050             int j = na - nb;
01051             DLscrat9 = NumberMalloc("DivLong"); DLscratA = NumberMalloc("DivLong");
01052             DLscratB = NumberMalloc("DivLong"); DLscratC = NumberMalloc("DivLong");
01053
01054             #if ( GMPSPREAD != 1 )
01055                 if ( na & 1 ) {
01056                     from = a; a = to = DLscrat9; i = na; NCOPY(to, from, i);
01057                     a[na++] = 0;
01058                 } else
01059             #endif
01060                 if ( (LONG)a & (sizeof(mp_limb_t)-1) ) {
01061                     from = a; a = to = DLscrat9; i = na; NCOPY(to, from, i);
01062                 }
01063
01064             #if ( GMPSPREAD != 1 )
01065                 if ( nb & 1 ) {
01066                     from = b; b = to = DLscratA; i = nb; NCOPY(to, from, i);
01067                     b[nb++] = 0;
01068                 } else
01069             #endif
01070                 if ( ( (LONG)b & (sizeof(mp_limb_t)-1) ) != 0 ) {
01071                     from = b; b = to = DLscratA; i = nb; NCOPY(to, from, i);
01072                 }
01073             #if ( GMPSPREAD != 1 )
01074             /*
01075                 Not recognizing this case was a nasty and extremely rare bug.
01076                 In the case that c == a and na is odd and nc == na and b
01077                 still needs to be used, the least significant UWORD of b got
01078                 overwritten by zero. (22-mar-2023)
01079             */
01080             ic = DLscratB; id = DLscratC;
01081         #else
01082             if ( ( (LONG)c & (sizeof(mp_limb_t)-1) ) != 0 ) ic = DLscratB;
01083             else ic = c;
01084
01085             if ( ( (LONG)d & (sizeof(mp_limb_t)-1) ) != 0 ) id = DLscratC;
01086             else id = d;
01087         #endif
01088         mpn_tdiv_qr((mp_limb_t *)ic, (mp_limb_t *)id, (mp_size_t)0,
01089             (const mp_limb_t *)a, (mp_size_t)(na/GMPSPREAD),
01090             (const mp_limb_t *)b, (mp_size_t)(nb/GMPSPREAD));
01091         while ( j >= 0 && ic[j] == 0 ) j--;

```



```

01092         j++; *nc = j;
01093         if ( c != ic ) { NCOPY(c,ic,j); }
01094         j = nb-1;
01095         while ( j >= 0 && id[j] == 0 ) j--;
01096         j++; *nd = j;
01097         if ( d != id ) { NCOPY(d,id,j); }
01098         if ( sgna < 0 ) { *nc = -(*nc); *nd = -(*nd); }
01099         if ( sgnb < 0 ) { *nc = -(*nc); }
01100         NumberFree(DLscrat9,"DivLong"); NumberFree(DLscratA,"DivLong");
01101         NumberFree(DLscratB,"DivLong"); NumberFree(DLscratC,"DivLong");
01102         return(0);
01103     }
01104 #endif
01105 /*
01106 #] GMP stuff :
01107 */
01108 /* Start with normalization operation */
01109
01110 e = NumberMalloc("DivLong"); f = NumberMalloc("DivLong"); g = NumberMalloc("DivLong");
01111 if ( b[nb-1] == (FULLMAX-1) ) norm = 1;
01112 else {
01113     norm = (UWORD) (((ULONG)FULLMAX) / (ULONG) ((b[nb-1]+1L)));
01114 }
01115 f[na] = 0;
01116 if ( MulLong(b,nb,&norm,1,e,&ne) ||
01117     MulLong(a,na,&norm,1,f,&nf) ) {
01118     NumberFree(e,"DivLong"); NumberFree(f,"DivLong"); NumberFree(g,"DivLong");
01119     return(-1);
01120 }
01121 if ( BigLong(f+nf-ne,ne,e,ne) >= 0 ) {
01122     SubPLon(f+nf-ne,ne,e,ne,f+nf-ne,&nh);
01123     w1 = c + (nf-ne);
01124     *nc = nf-ne+1;
01125 }
01126 else {
01127     nh = ne;
01128     *nc = nf-ne;
01129     w1 = 0;
01130 }
01131 w2 = c; i = *nc; do { *w2++ = 0; } while ( --i > 0 );
01132 nf = na;
01133 ni = nf-ne;
01134 esthelp = (RLONG) (e[ne-1]) + 1L;
01135 while ( nf >= ne ) {
01136     if ( (WORD)esthelp == 0 ) {
01137         estim = (WORD) ((( (RLONG) (f[nf])) «BITSINWORD) + f[nf-1] »BITSINWORD);
01138     }
01139     else {
01140         estim = (WORD) ((( (RLONG) (f[nf])) «BITSINWORD) + f[nf-1] ) / esthelp);
01141     }
01142     /* This estimate may be up to two too small */
01143     if ( estim ) {
01144         MulLong(e,ne,&estim,1,g,&ng);
01145         nh = ne + 1; if ( !f[ni+ne] ) nh--;
01146         SubPLon(f+ni,nh,g,ng,f+ni,&nh);
01147     }
01148     else {
01149         w2 = f+ni+ne; nh = ne+1;
01150         while ( ( nh > 0 ) && !*w2 ) { nh--; w2--; }
01151     }
01152     if ( BigLong(f+ni,nh,e,ne) >= 0 ) {
01153         estim++;
01154         SubPLon(f+ni,nh,e,ne,f+ni,&nh);
01155         if ( BigLong(f+ni,nh,e,ne) >= 0 ) {
01156             estim++;
01157             SubPLon(f+ni,nh,e,ne,f+ni,&nh);
01158             if ( BigLong(f+ni,nh,e,ne) >= 0 ) {
01159                 MLOCK(ErrorMessageLock);
01160                 MesPrint("Problems in DivLong");
01161                 AO.OutSkip = 3;
01162                 FiniLine();
01163                 i = na;
01164                 while ( --i >= 0 ) { TalToLine((UWORD) (*a++)); TokenToLine((UBYTE *) " "); }
01165                 FiniLine();
01166                 i = nb;
01167                 while ( --i >= 0 ) { TalToLine((UWORD) (*b++)); TokenToLine((UBYTE *) " "); }
01168                 AO.OutSkip = 0;
01169                 FiniLine();
01170                 MUNLOCK(ErrorMessageLock);
01171                 NumberFree(e,"DivLong"); NumberFree(f,"DivLong"); NumberFree(g,"DivLong");
01172                 return(-1);
01173             }
01174         }
01175     }
01176     c[ni] = estim;
01177     nf--;
01178     ni--;

```

```

01179     }
01180     if ( w1 ) *w1 = 1;
01181
01182     /* Finish with the renormalization operation */
01183
01184     if ( nh > 0 ) {
01185         if ( norm == 1 ) {
01186             *nd = i = nh; ff = f;
01187             NCOPY(d,ff,i);
01188         }
01189         else {
01190             w1 = f+nh;
01191             *nd = ni = nh;
01192             w2 = d+ni;
01193             v = norm;
01194             t = (RLONG) (*--w1);
01195             while ( --ni >= 0 ) {
01196                 *--w2 = t / v;
01197                 t -= v * (*w2);
01198                 if ( ni ) {
01199                     t <<= BITSINWORD;
01200                     t += *--w1;
01201                 }
01202             }
01203             if ( t ) {
01204                 MLOCK(ErrorMessageLock);
01205                 MesPrint("Error in DivLong");
01206                 MUNLOCK(ErrorMessageLock);
01207                 NumberFree(e,"DivLong"); NumberFree(f,"DivLong"); NumberFree(g,"DivLong");
01208                 return(-1);
01209             }
01210             if ( !(d+nh-1) ) (*nd)--;
01211         }
01212     }
01213     else { *nd = 0; }
01214     NumberFree(e,"DivLong"); NumberFree(f,"DivLong"); NumberFree(g,"DivLong");
01215 }
01216 if ( sgna < 0 ) { *nc = -(*nc); *nd = -(*nd); }
01217 if ( sgnb < 0 ) { *nc = -(*nc); }
01218 return(0);
01219 }
01220
01221 /*
01222     #] DivLong :
01223     #[ RaisPow :      WORD RaisPow(a,na,b)
01224
01225     Raises a to the power b. a is a Long integer and b >= 0.
01226     The method that is used works with a bitdecomposition of b.
01227 */
01228
01229 int RaisPow(PHEAD UWORD *a, WORD *na, UWORD b)
01230 {
01231     GETBIDENTITY
01232     WORD i, nu;
01233     UWORD *it, *iu, c;
01234     UWORD *is, *iss;
01235     WORD ns, nt, nmod;
01236     nmod = ABS(AN.ncmod);
01237     if ( !*na || ( ( *na == 1 ) && ( *a == 1 ) ) ) return(0);
01238     if ( !b ) { *na=1; *a=1; return(0); }
01239     is = NumberMalloc("RaisPow");
01240     it = NumberMalloc("RaisPow");
01241     for ( i = 0; i < ABS(*na); i++ ) is[i] = a[i];
01242     ns = *na;
01243     c = b;
01244     for ( i = 0; i < BITSINWORD; i++ ) {
01245         if ( !c ) break;
01246         c /= 2;
01247     }
01248     i--;
01249     c = 1 << i;
01250     while ( --i >= 0 ) {
01251         c /= 2;
01252         if (MulLong(is,ns,is,ns,it,&nt)) goto RaisOvl;
01253         if ( b & c ) {
01254             if ( MulLong(it,nt,a,*na,is,&ns) ) goto RaisOvl;
01255         }
01256         else {
01257             iu = is; is = it; it = iu;
01258             nu = ns; ns = nt; nt = nu;
01259         }
01260         if ( nmod != 0 ) {
01261             if ( DivLong(is,ns,(UWORD *)AC.cmod,nmod,it,&nt,is,&ns) ) goto RaisOvl;
01262         }
01263     }
01264     if ( ( nmod != 0 ) && ( ( AC.modmode & POSNEG ) != 0 ) ) {
01265         NormalModulus(is,&ns);

```

```

01266     }
01267     if ( ( *na = i = ns ) != 0 ) { iss = is; i=ABS(i); NCOPY(a,iss,i); }
01268     NumberFree(is,"RaisPow"); NumberFree(it,"RaisPow");
01269     return(0);
01270 RaisOvl:
01271     MLOCK(ErrorMessageLock);
01272     MesCall("RaisPow");
01273     MUNLOCK(ErrorMessageLock);
01274     NumberFree(is,"RaisPow"); NumberFree(it,"RaisPow");
01275     SETERROR(-1)
01276 }
01277
01278 /*
01279     #] RaisPow :
01280     #[ RaisPowCached :
01281 */
01282
01297 void RaisPowCached (PHEAD WORD x, WORD n, UWORD **c, WORD *nc) {
01298
01299     int i,j;
01300     WORD new_small_power_maxx, new_small_power_maxn, ID;
01301     WORD *new_small_power_n;
01302     UWORD **new_small_power;
01303
01304     /* check whether to extend the array */
01305     if (x>=AT.small_power_maxx || n>=AT.small_power_maxn) {
01306
01307         new_small_power_maxx = AT.small_power_maxx;
01308         if (x>=AT.small_power_maxx)
01309             new_small_power_maxx = MaX(2*AT.small_power_maxx, x+1);
01310
01311         new_small_power_maxn = AT.small_power_maxn;
01312         if (n>=AT.small_power_maxn)
01313             new_small_power_maxn = MaX(2*AT.small_power_maxn, n+1);
01314
01315         new_small_power_n = (WORD*)
01316         Malloc1(new_small_power_maxx*new_small_power_maxn*sizeof(WORD),"RaisPowCached");
01317         new_small_power = (UWORD **) Malloc1(new_small_power_maxx*new_small_power_maxn*sizeof(UWORD
01318         *),"RaisPowCached");
01319
01320         for (i=0; i<new_small_power_maxx * new_small_power_maxn; i++) {
01321             new_small_power_n[i] = 0;
01322             new_small_power [i] = NULL;
01323         }
01324
01325         for (i=0; i<AT.small_power_maxx; i++)
01326             for (j=0; j<AT.small_power_maxn; j++) {
01327                 new_small_power_n[i*new_small_power_maxn+j] =
01328                 AT.small_power_n[i*AT.small_power_maxn+j];
01329                 new_small_power [i*new_small_power_maxn+j] = AT.small_power
01330                 [i*AT.small_power_maxn+j];
01331             }
01332
01333         if (AT.small_power_n != NULL) {
01334             M_free(AT.small_power_n,"RaisPowCached");
01335             M_free(AT.small_power,"RaisPowCached");
01336         }
01337
01338         AT.small_power_maxx = new_small_power_maxx;
01339         AT.small_power_maxn = new_small_power_maxn;
01340         AT.small_power_n    = new_small_power_n;
01341         AT.small_power      = new_small_power;
01342     }
01343
01344     /* check whether the results is already calculated */
01345     ID = x * AT.small_power_maxn + n;
01346
01347     if (AT.small_power[ID] == NULL) {
01348         #ifdef OLDRAISPOWCACHED
01349             AT.small_power[ID] = NumberMalloc("RaisPowCached");
01350             AT.small_power_n[ID] = 1;
01351             AT.small_power[ID][0] = x;
01352             RaisPow(BHEAD AT.small_power[ID], &AT.small_power_n[ID], n);
01353         #else
01354             UWORD *c = NumberMalloc("RaisPowCached");
01355             WORD i, k = 1;
01356             c[0] = x;
01357             RaisPow(BHEAD c, &k, n);
01358         /*
01359             And now get the proper amount.
01360         */
01361         if ( AT.InNumMem < k ) { /* We should start a new buffer */
01362             AT.InNumMem = 5*AM.MaxTal;
01363             AT.NumMem = (UWORD *) Malloc1(AT.InNumMem*sizeof(UWORD), "RaisPowCached");
01364         /*
01365             MesPrint(" Got an extra %l UWORDS in RaisPowCached",AT.InNumMem);
01366         */

```

```

01363     }
01364     for ( i = 0; i < k; i++ ) AT.NumMem[i] = c[i];
01365     AT.small_power[ID] = AT.NumMem;
01366     AT.small_power_n[ID] = k;
01367     AT.NumMem += k;
01368     AT.InNumMem -= k;
01369     NumberFree(c, "RaisPowCached");
01370 #endif
01371 }
01372
01373 /* return the result */
01374 *c = AT.small_power[ID];
01375 *nc = AT.small_power_n[ID];
01376 }
01377
01378 /*
01379     #] RaisPowCached :
01380     #[ RaisPowMod :
01381
01382     Computes the power x^n mod m
01383 */
01384 WORD RaisPowMod (WORD x, WORD n, WORD m) {
01385     LONG y=1, z=x;
01386     while (n) {
01387         if (n&1) { y*=z; y%=m; }
01388         z*=z; z%=m;
01389         n /= 2;
01390     }
01391     return (WORD)y;
01392 }
01393
01394 /*
01395     #] RaisPowMod :
01396     #[ NormalModulus : int NormalModulus(UWORD *a,WORD *na)
01397 */
01400 int NormalModulus(UWORD *a,WORD *na)
01401 {
01402     WORD n;
01403     if ( AC.halfmod == 0 ) {
01404         LOCK(AC.halfmodlock);
01405         if ( AC.halfmod == 0 ) {
01406             UWORD two[1], remain[1];
01407             WORD dummy;
01408             two[0] = 2;
01409             AC.halfmod = (UWORD *)Malloc1((ABS(AC.ncmod))*sizeof(UWORD), "halfmod");
01410             DivLong((UWORD *)AC.cmod, (ABS(AC.ncmod)), two, 1
01411                 , (UWORD *)AC.halfmod, &(AC.nhalfmod), remain, &dummy);
01412         }
01413         UNLOCK(AC.halfmodlock);
01414     }
01415     n = ABS(*na);
01416     if ( BigLong(a,n,AC.halfmod,AC.nhalfmod) > 0 ) {
01417         SubPLon((UWORD *)AC.cmod, (ABS(AC.ncmod)), a, n, a, &n);
01418         if ( *na > 0 ) { *na = -n; }
01419         else { *na = n; }
01420         return(1);
01421     }
01422     return(0);
01423 }
01424
01425 /*
01426     #] NormalModulus :
01427     #[ MakeInverses :
01428 */
01429 int MakeInverses(void)
01430 {
01431     WORD n = AC.cmod[0], i, inv2;
01432     if ( AC.ncmod != 1 ) return(1);
01433     if ( AC.modinverses == 0 ) {
01434         LOCK(AC.halfmodlock);
01435         if ( AC.modinverses == 0 ) {
01436             AC.modinverses = (UWORD *)Malloc1(n*sizeof(UWORD), "modinverses");
01437             AC.modinverses[0] = 0;
01438             AC.modinverses[1] = 1;
01439             for ( i = 2; i < n; i++ ) {
01440                 if ( GetModInverses(i,n,
01441                     (WORD *)(&(AC.modinverses[i])),&inv2) ) {
01442                     SETERROR(-1)
01443                 }
01444             }
01445             UNLOCK(AC.halfmodlock);
01446         }
01447     }
01448     return(0);
01449 }
01450
01451 /*

```

```

01464         #] MakeInverses :
01465         #[ GetModInverses :
01466     */
01477 int GetModInverses(WORD m1, WORD m2, WORD *im1, WORD *im2)
01478 {
01479     WORD a1, a2, a3;
01480     WORD b1, b2, b3;
01481     WORD x = m1, y, c, d = m2;
01482     if ( x < 1 || d <= 1 ) goto somethingwrong;
01483     a1 = 0; a2 = 1;
01484     b1 = 1; b2 = 0;
01485     for(;;) {
01486         c = d/x; y = d%x; /* a good compiler makes this faster than y=d-c*x */
01487         if ( y == 0 ) break;
01488         a3 = a1-c*a2; a1 = a2; a2 = a3;
01489         b3 = b1-c*b2; b1 = b2; b2 = b3;
01490         d = x; x = y;
01491     }
01492     if ( x != 1 ) goto somethingwrong;
01493     if ( a2 < 0 ) a2 += m2;
01494     if ( b2 < 0 ) b2 += m1;
01495     if (im1!=NULL) *im1 = a2;
01496     if (im2!=NULL) *im2 = b2;
01497     return(0);
01498 somethingwrong:
01499     MLOCK(ErrorMessageLock);
01500     MesPrint("Error trying to determine inverses in GetModInverses");
01501     MUNLOCK(ErrorMessageLock);
01502     return(-1);
01503 }
01504 /*
01505     #] GetModInverses :
01506     #[ GetLongModInverses :
01507 */
01508
01509 int GetLongModInverses(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *ia, WORD *nia, UWORD *ib,
WORD *nib) {
01510
01511     UWORD *s, *t, *sa, *sb, *ta, *tb, *x, *y, *swap1;
01512     WORD ns, nt, nsa, nsb, nta, ntb, nx, ny, swap2;
01513
01514     s = NumberMalloc("GetLongModInverses");
01515     ns = na;
01516     WCOPY(s, a, ABS(ns));
01517
01518     t = NumberMalloc("GetLongModInverses");
01519     nt = nb;
01520     WCOPY(t, b, ABS(nt));
01521
01522     sa = NumberMalloc("GetLongModInverses");
01523     nsa = 1;
01524     sa[0] = 1;
01525
01526     sb = NumberMalloc("GetLongModInverses");
01527     nsb = 0;
01528
01529     ta = NumberMalloc("GetLongModInverses");
01530     nta = 0;
01531
01532     tb = NumberMalloc("GetLongModInverses");
01533     ntb = 1;
01534     tb[0] = 1;
01535
01536     x = NumberMalloc("GetLongModInverses");
01537     y = NumberMalloc("GetLongModInverses");
01538
01539     while (nt != 0) {
01540         DivLong(s, ns, t, nt, x, &nx, y, &ny);
01541         swap1=s; s=y; y=swap1;
01542         ns=ny;
01543         MulLong(x, nx, ta, nta, y, &ny);
01544         AddLong(sa, nsa, y, -ny, sa, &nsa);
01545         MulLong(x, nx, tb, ntb, y, &ny);
01546         AddLong(sb, nsb, y, -ny, sb, &nsb);
01547
01548         swap1=s; s=t; t=swap1;
01549         swap2=ns; ns=nt; nt=swap2;
01550         swap1=sa; sa=ta; ta=swap1;
01551         swap2=nsa; nsa=nta; nta=swap2;
01552         swap1=sb; sb=tb; tb=swap1;
01553         swap2=nsb; nsb=ntb; ntb=swap2;
01554     }
01555
01556     if (ia!=NULL) {
01557         *nia = nsa*ns;
01558         WCOPY(ia, sa, ABS(*nia));
01559     }

```

```

01560
01561     if (ib!=NULL) {
01562         *nib = nsb*ns;
01563         WCOPY(ib,sb,ABS(*nib));
01564     }
01565
01566     NumberFree(s,"GetLongModInverses");
01567     NumberFree(t,"GetLongModInverses");
01568     NumberFree(sa,"GetLongModInverses");
01569     NumberFree(sb,"GetLongModInverses");
01570     NumberFree(ta,"GetLongModInverses");
01571     NumberFree(tb,"GetLongModInverses");
01572     NumberFree(x,"GetLongModInverses");
01573     NumberFree(y,"GetLongModInverses");
01574
01575     return 0;
01576 }
01577
01578 /*
01579     #] GetLongModInverses :
01580     #[ Product :          WORD Product(a,na,b)
01581
01582     Multiplies the Long number in a with the WORD b.
01583
01584 */
01585
01586 int Product(UWORD *a, WORD *na, WORD b)
01587 {
01588     WORD i, sgn = 1;
01589     RLONG t, u;
01590     if ( *na < 0 ) { *na = -(*na); sgn = -sgn; }
01591     if ( b < 0 ) { b = -b; sgn = -sgn; }
01592     t = 0;
01593     u = (RLONG)b;
01594     for ( i = 0; i < *na; i++ ) {
01595         t += *a * u;
01596         *a++ = (UWORD)t;
01597         t >>= BITSINWORD;
01598     }
01599     if ( t > 0 ) {
01600         if ( ++(*na) > AM.MaxTal ) {
01601             MLOCK(ErrorMessageLock);
01602             MesPrint("Overflow in Product");
01603             MUNLOCK(ErrorMessageLock);
01604             return(-1);
01605         }
01606         *a = (UWORD)t;
01607     }
01608     if ( sgn < 0 ) *na = -(*na);
01609     return(0);
01610 }
01611
01612 /*
01613     #] Product :
01614     #[ Quotient :          UWORD Quotient(a,na,b)
01615
01616     Routine divides the long number a by b with the assumption that
01617     there is no remainder (like while computing binomials).
01618
01619 */
01620
01621 UWORD Quotient(UWORD *a, WORD *na, WORD b)
01622 {
01623     RLONG v, t;
01624     WORD i, j, sgn = 1;
01625     if ( ( i = *na ) < 0 ) { sgn = -1; i = -i; }
01626     if ( b < 0 ) { b = -b; sgn = -sgn; }
01627     if ( i == 1 ) {
01628         if ( ( *a /= (UWORD)b ) == 0 ) *na = 0;
01629         if ( sgn < 0 ) *na = -*na;
01630         return(0);
01631     }
01632     a += i;
01633     j = i;
01634     v = (RLONG)b;
01635     t = (RLONG)(--a);
01636     while ( --i >= 0 ) {
01637         *a = t / v;
01638         t -= v * (*a);
01639         if ( i ) {
01640             t <= BITSINWORD;
01641             t += --a;
01642         }
01643     }
01644     a += j - 1;
01645     if ( !*a ) j--;
01646     if ( sgn < 0 ) j = -j;

```

```

01647     *na = j;
01648     return(0);
01649 }
01650
01651 /*
01652     #] Quotient :
01653     #[ Remain10 :      WORD Remain10(a,na)
01654
01655     Routine divides a by 10 and gives the remainder as return value.
01656     The value of a will be the quotient! a must be positive.
01657 */
01658
01659 WORD Remain10(UWORD *a, WORD *na)
01660 {
01661     WORD i;
01662     RLONG t, u;
01663     UWORD *b;
01664     i = *na;
01665     t = 0;
01666     b = a + i - 1;
01667     while ( --i >= 0 ) {
01668         t += *b;
01669         *b-- = u = t / 10;
01670         t -= u * 10;
01671         if ( i > 0 ) t <= BITSINWORD;
01672     }
01673     if ( ( *na > 0 ) && !a[*na-1] ) (*na)--;
01674     return((WORD)t);
01675 }
01676
01677 /*
01678     #] Remain10 :
01679     #[ Remain4 :      WORD Remain4(a,na)
01680
01681     Routine divides a by 10000 and gives the remainder as return value.
01682     The value of a will be the quotient! a must be positive.
01683 */
01684
01685 WORD Remain4(UWORD *a, WORD *na)
01686 {
01687     WORD i;
01688     RLONG t, u;
01689     UWORD *b;
01690     i = *na;
01691     t = 0;
01692     b = a + i - 1;
01693     while ( --i >= 0 ) {
01694         t += *b;
01695         *b-- = u = t / 10000;
01696         t -= u * 10000;
01697         if ( i > 0 ) t <= BITSINWORD;
01698     }
01699     if ( ( *na > 0 ) && !a[*na-1] ) (*na)--;
01700     return((WORD)t);
01701 }
01702
01703 /*
01704     #] Remain4 :
01705     #[ PrtLong :      void PrtLong(a,na,s)
01706
01707     Puts the long number a in string s.
01708 */
01709
01710 void PrtLong(UWORD *a, WORD na, UBYTE *s)
01711 {
01712     GETIDENTITY
01713     WORD q, i;
01714     UBYTE *sa, *sb;
01715     UBYTE c;
01716     UWORD *bb, *b;
01717
01718     if ( na < 0 ) {
01719         *s++ = '-';
01720         na = -na;
01721     }
01722
01723     b = NumberMalloc("PrtLong");
01724     bb = b;
01725     i = na; while ( --i >= 0 ) *bb++ = *a++;
01726     a = b;
01727     if ( na > 2 ) {
01728         sa = s;
01729         do {
01730             q = Remain4(a,&na);

```

```

01734         *sa++ = (UBYTE) ('0' + (q%10));
01735         q /= 10;
01736         *sa++ = (UBYTE) ('0' + (q%10));
01737         q /= 10;
01738         *sa++ = (UBYTE) ('0' + (q%10));
01739         q /= 10;
01740         *sa++ = (UBYTE) ('0' + (q%10));
01741     } while ( na );
01742     while ( sa[-1] == '0' ) sa--;
01743     sb = s;
01744     s = sa;
01745     sa--;
01746     while ( sa > sb ) { c = *sa; *sa = *sb; *sb = c; sa--; sb++; }
01747 }
01748 else if ( na ) {
01749     sa = s;
01750     do {
01751         q = Remain10(a,&na);
01752         *sa++ = (UBYTE) ('0' + q);
01753     } while ( na );
01754     sb = s;
01755     s = sa;
01756     sa--;
01757     while ( sa > sb ) { c = *sa; *sa = *sb; *sb = c; sa--; sb++; }
01758 }
01759 else *s++ = '0';
01760 *s = '\0';
01761 NumberFree(b,"PrtLong");
01762 }
01763
01764 /*
01765     #] PrtLong :
01766     #[ GetLong :          WORD GetLong(s,a,na)
01767
01768     Reads a long number from a string.
01769     The string is zero terminated and contains only digits!
01770
01771     New algorithm: try to read 4 digits together before the result
01772     is accumulated.
01773 */
01774
01775 int GetLong(UBYTE *s, UWORD *a, WORD *na)
01776 {
01777     /*
01778         UWORD digit;
01779         *a = 0;
01780         *na = 0;
01781         while ( FG.cTable[*s] == 1 ) {
01782             digit = *s++ - '0';
01783             if ( *na && Product(a,na,(WORD)10) ) return(-1);
01784             if ( digit && AddLong(a,*na,&digit,(WORD)1,a,na) ) return(-1);
01785         }
01786         return(0);
01787     */
01788     UWORD digit, x = 0, y = 0;
01789     *a = 0;
01790     *na = 0;
01791     while ( FG.cTable[*s] == 1 ) {
01792         x = *s++ - '0';
01793         if ( FG.cTable[*s] != 1 ) { y = 10; break; }
01794         x = 10*x + *s++ - '0';
01795         if ( FG.cTable[*s] != 1 ) { y = 100; break; }
01796         x = 10*x + *s++ - '0';
01797         if ( FG.cTable[*s] != 1 ) { y = 1000; break; }
01798         x = 10*x + *s++ - '0';
01799         if ( *na && Product(a,na,(WORD)10000) ) return(-1);
01800         if ( ( digit = x ) != 0 && AddLong(a,*na,&digit,(WORD)1,a,na) )
01801             return(-1);
01802         y = 0;
01803     }
01804     if ( y ) {
01805         if ( *na && Product(a,na,(WORD)y) ) return(-1);
01806         if ( ( digit = x ) != 0 && AddLong(a,*na,&digit,(WORD)1,a,na) )
01807             return(-1);
01808     }
01809     return(0);
01810 }
01811
01812 /*
01813     #] GetLong :
01814     #[ GCD :          WORD GCD(a,na,b,nb,c,nc)
01815
01816     Algorithm to compute the GCD of two long numbers.
01817     See Knuth, sec 4.5.2 algorithm L.
01818
01819     We assume that both numbers are positive
01820

```



```

01821     NOTE!!!!. NumberMalloc gets called and it may not be freed
01822 */
01823
01824 #ifdef EXTRAGCD
01825
01826 #define Convert(ia,aa,naa) \
01827     if ( (LONG)ia < 0 ) { \
01828         ia = (ULONG)(~(LONG)ia); \
01829         aa[0] = ia; \
01830         if ( ( aa[1] = ia » BITSINWORD ) != 0 ) naa = -2; \
01831         else naa = -1; \
01832     } \
01833     else if ( ia == 0 ) { aa[0] = 0; naa = 0; } \
01834     else { \
01835         aa[0] = ia; \
01836         if ( ( aa[1] = ia » BITSINWORD ) != 0 ) naa = 2; \
01837         else naa = 1; \
01838     }
01839
01840 void GCD(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
01841 {
01842     int ja = 0, jb = 0, j;
01843     UWORD *r,*t;
01844     UWORD *x1, *x2, *x3;
01845     WORD nd,naa,nbb;
01846     ULONG ia,ib,ic,id,u,v,w,q,T;
01847     UWORD aa[2], bb[2];
01848 /*
01849     First eliminate easy powers of 2^...
01850 */
01851     while ( a[0] == 0 ) { na--; ja++; a++; }
01852     while ( b[0] == 0 ) { nb--; jb++; b++; }
01853     if ( ja > jb ) ja = jb;
01854     if ( ja > 0 ) {
01855         j = ja;
01856         do { *c++ = 0; } while ( --j > 0 );
01857     }
01858 /*
01859     Now arrange things such that a >= b
01860 */
01861     if ( na < nb ) {
01862         jb = na; na = nb; nb = jb;
01863     exch:
01864         r = a; a = b; b = r;
01865     }
01866     else if ( na == nb ) {
01867         r = a+na;
01868         t = b+nb;
01869         j = na;
01870         while ( --j >= 0 ) {
01871             if ( *--r > *--t ) break;
01872             if ( *r < *t ) goto exch;
01873         }
01874         if ( j < 0 ) {
01875     out:
01876             j = nb;
01877             NCOPY(c,b,j);
01878             *nc = nb+ja;
01879             return;
01880         }
01881     }
01882 /*
01883     {
01884         MLOCK(ErrorMessageLock);
01885         MesPrint("Ordered input, ja = %d", (WORD)ja);
01886         AO.OutSkip = 3;
01887         FiniLine();
01888         j = na; r = a;
01889         while ( --j >= 0 ) { TalToLine((UWORD)(*r++)); TokenToLine((UBYTE *) " "); }
01890         FiniLine();
01891         j = nb; r = b;
01892         while ( --j >= 0 ) { TalToLine((UWORD)(*r++)); TokenToLine((UBYTE *) " "); }
01893         AO.OutSkip = 0;
01894         FiniLine();
01895         MUNLOCK(ErrorMessageLock);
01896     }
01897 */
01898 /*
01899     We have now that A > B
01900     The loop recognizes the case that na-nb >= 1
01901     In that case we just have to divide!
01902 */
01903     r = x1 = NumberMalloc("GCD"); t = x2 = NumberMalloc("GCD"); x3 = NumberMalloc("GCD");
01904     j = na;
01905     NCOPY(r,a,j);
01906     j = nb;
01907     NCOPY(t,b,j);

```

```

01908
01909     for(;;) {
01910
01911         while ( na > nb ) {
01912 toobad:
01913             DivLong(x1,na,x2,nb,c,nc,x3,&nd);
01914             if ( nd == 0 ) { b = x2; goto out; }
01915             t = x1; x1 = x2; x2 = x3; x3 = t; na = nb; nb = nd;
01916             if ( na == 2 ) break;
01917         }
01918     /*
01919     Here we can use the shortcut.
01920 */
01921     if ( na == 2 ) {
01922         v = x1[0] + ( ((ULONG)x1[1]) << BITSINWORD );
01923         w = x2[0];
01924         if ( nb == 2 ) w += ((ULONG)x2[1]) << BITSINWORD;
01925 #ifdef EXTRAGCD2
01926         v = GCD2(v,w);
01927 #else
01928         do { u = v%w; v = w; w = u; } while ( w );
01929 #endif
01930         c[0] = (UWORD)v;
01931         if ( ( c[1] = (UWORD)(v >> BITSINWORD) ) != 0 ) *nc = 2+ja;
01932         else *nc = 1+ja;
01933         NumberFree(x1,"GCD"); NumberFree(x2,"GCD"); NumberFree(x3,"GCD");
01934         return;
01935     }
01936     if ( na == 1 ) {
01937         UWORD ui, uj;
01938         ui = x1[0]; uj = x2[0];
01939 #ifdef EXTRAGCD2
01940         ui = (UWORD)GCD2((ULONG)ui,(ULONG)uj);
01941 #else
01942         do { nd = ui%uj; ui = uj; uj = nd; } while ( nd );
01943 #endif
01944         c[0] = ui;
01945         *nc = 1 + ja;
01946         NumberFree(x1,"GCD"); NumberFree(x2,"GCD"); NumberFree(x3,"GCD");
01947         return;
01948     }
01949     ia = 1; ib = 0; ic = 0; id = 1;
01950     u = ( ((ULONG)x1[na-1]) << BITSINWORD ) + x1[na-2];
01951     v = ( ((ULONG)x2[nb-1]) << BITSINWORD ) + x2[nb-2];
01952
01953     while ( v+ic != 0 && v+id != 0 &&
01954             ( q = (u+ia)/(v+ic) ) == (u+ib)/(v+id) ) {
01955         T = ia-q*ic; ia = ic; ic = T;
01956         T = ib-q*id; ib = id; id = T;
01957         T = u - q*v; u = v; v = T;
01958     }
01959     if ( ib == 0 ) goto toobad;
01960     Convert(ia,aa,naa);
01961     Convert(ib,bb,nbb);
01962     MulLong(x1,na,aa,naa,x3,&nd);
01963     MulLong(x2,nb,bb,nbb,c,nc);
01964     AddLong(x3,nd,c,*nc,c,nc);
01965     Convert(ic,aa,naa);
01966     Convert(id,bb,nbb);
01967     MulLong(x1,na,aa,naa,x3,&nd);
01968     t = c; na = j = *nc; r = x1;
01969     NCOPY(r,t,j);
01970     MulLong(x2,nb,bb,nbb,c,nc);
01971     AddLong(x3,nd,c,*nc,x2,&nb);
01972     }
01973 }
01974
01975 #endif
01976
01977 /*
01978     #] GCD :
01979     #[ GcdLong :          WORD GcdLong(a,na,b,nb,c,nc)
01980
01981     Returns the Greatest Common Divider of a and b in c.
01982     If a and or b are zero an error message will be returned.
01983     The answer is always positive.
01984     In principle a and c can be the same.
01985 */
01986
01987 #ifndef NEWTRICK
01988 /*
01989     #[ Old Routine :
01990 */
01991
01992 int GcdLong(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
01993 {
01994     GETBIDENTITY

```

```

01995     if ( !na || !nb ) {
01996         if ( !na && !nb ) {
01997             MLOCK(ErrorMessageLock);
01998             MesPrint("Cannot take gcd");
01999             MUNLOCK(ErrorMessageLock);
02000             return(-1);
02001         }
02002     }
02003     if ( !na ) {
02004         *nc = abs(nb);
02005         NCOPY(c,b,*nc);
02006         *nc = abs(nb);
02007         return(0);
02008     }
02009     *nc = abs(na);
02010     NCOPY(c,a,*nc);
02011     *nc = abs(na);
02012     return(0);
02013 }
02014 if ( na < 0 ) na = -na;
02015 if ( nb < 0 ) nb = -nb;
02016 if ( na == 1 && nb == 1 ) {
02017     #ifdef EXTRAGCD2
02018         *c = (UWORD)GCD2((ULONG)*a,(ULONG)*b);
02019     #else
02020         UWORD x,y,z;
02021         x = *a;
02022         y = *b;
02023         do { z = x % y; x = y; } while ( ( y = z ) != 0 );
02024         *c = x;
02025     #endif
02026     *nc = 1;
02027 }
02028 else if ( na <= 2 && nb <= 2 ) {
02029     RLONG lx,ly,lz;
02030     if ( na == 2 ) { lx = ((RLONG)(a[1]))«BITSINWORD + *a; }
02031     else { lx = *a; }
02032     if ( nb == 2 ) { ly = ((RLONG)(b[1]))«BITSINWORD + *b; }
02033     else { ly = *b; }
02034     #ifdef LONGWORD
02035     #ifdef EXTRAGCD2
02036         lx = GCD2(lx,ly);
02037     #else
02038         do { lz = lx % ly; lx = ly; } while ( ( ly = lz ) != 0 );
02039     #endif
02040     #endif
02041     #else
02042         if ( lx < ly ) { lz = lx; lx = ly; ly = lz; }
02043         do {
02044             lz = lx % ly; lx = ly;
02045         } while ( ( ly = lz ) != 0 && ( lx & AWORDMASK ) != 0 );
02046         if ( ly ) {
02047             do { *c = ((UWORD)lx)%((UWORD)ly); lx = ly; } while ( ( ly = *c ) != 0 );
02048             *c = (UWORD)lx;
02049             *nc = 1;
02050         }
02051         else
02052     #endif
02053     {
02054         *c++ = (UWORD)lx;
02055         if ( ( *c = (UWORD)(lx » BITSINWORD) ) != 0 ) *nc = 2;
02056         else *nc = 1;
02057     }
02058 }
02059 else {
02060     #ifdef EXTRAGCD
02061         GCD(a,na,b,nb,c,nc);
02062     #else
02063     #ifdef NEWGCD
02064         UWORD *x3,*x1,*x2, *GLscrat7, *GLscrat8;
02065         WORD n1,n2,n3,n4;
02066         WORD i, j;
02067         x1 = c; x3 = a; n1 = i = na;
02068         NCOPY(x1,x3,i);
02069         GLscrat7 = NumberMalloc("GcdLong"); GLscrat8 = NumberMalloc("GcdLong");
02070         x2 = GLscrat8; x3 = b; n2 = i = nb;
02071         NCOPY(x2,x3,i);
02072         x1 = c; i = 0;
02073         while ( x1[0] == 0 ) { i += BITSINWORD; x1++; n1--; }
02074         while ( ( x1[0] & 1 ) == 0 ) { i++; SCHUIF(x1,n1) }
02075         x2 = GLscrat8; j = 0;
02076         while ( x2[0] == 0 ) { j += BITSINWORD; x2++; n2--; }
02077         while ( ( x2[0] & 1 ) == 0 ) { j++; SCHUIF(x2,n2) }
02078         if ( j > i ) j = i; /* powers of two in GCD */
02079         for(;;){
02080             if ( n1 > n2 ) {
02081 firstbig:

```

```

02082         SubPLon(x1,n1,x2,n2,x1,&n3);
02083         n1 = n3;
02084         if ( n1 == 0 ) {
02085             x1 = c;
02086             n1 = i = n2; NCOPY(x1,x2,i);
02087             break;
02088         }
02089         while ( ( x1[0] & 1 ) == 0 ) SCHUIF(x1,n1)
02090         if ( n1 == 1 ) {
02091             if ( DivLong(x2,n2,x1,n1,Glscrat7,&n3,x2,&n4) ) goto GcdErr;
02092             n2 = n4;
02093             if ( n2 == 0 ) {
02094                 i = n1; x2 = c; NCOPY(x2,x1,i);
02095                 break;
02096             }
02097 #ifdef EXTRAGCD2
02098             *c = (UWORD)GCD2((ULONG)x1[0],(ULONG)x2[0]);
02099 #else
02100             {
02101                 UWORD x,y,z;
02102                 x = x1[0];
02103                 y = x2[0];
02104                 do { z = x % y; x = y; } while ( ( y = z ) != 0 );
02105                 *c = x;
02106             }
02107 #endif
02108             n1 = 1;
02109             break;
02110         }
02111     }
02112     else if ( n1 < n2 ) {
02113 lastbig:
02114         SubPLon(x2,n2,x1,n1,x2,&n3);
02115         n2 = n3;
02116         if ( n2 == 0 ) {
02117             i = n1; x2 = c; NCOPY(x2,x1,i);
02118             break;
02119         }
02120         while ( ( x2[0] & 1 ) == 0 ) SCHUIF(x2,n2)
02121         if ( n2 == 1 ) {
02122             if ( DivLong(x1,n1,x2,n2,Glscrat7,&n3,x1,&n4) ) goto GcdErr;
02123             n1 = n4;
02124             if ( n1 == 0 ) {
02125                 x1 = c;
02126                 n1 = i = n2; NCOPY(x1,x2,i);
02127                 break;
02128             }
02129 #ifdef EXTRAGCD2
02130             *c = (UWORD)GCD2((ULONG)x2[0],(ULONG)x1[0]);
02131 #else
02132             {
02133                 UWORD x,y,z;
02134                 x = x2[0];
02135                 y = x1[0];
02136                 do { z = x % y; x = y; } while ( ( y = z ) != 0 );
02137                 *c = x;
02138             }
02139 #endif
02140             n1 = 1;
02141             break;
02142         }
02143     }
02144     else {
02145         for ( i = n1-1; i >= 0; i-- ) {
02146             if ( x1[i] > x2[i] ) goto firstbig;
02147             else if ( x1[i] < x2[i] ) goto lastbig;
02148         }
02149         i = n1; x2 = c; NCOPY(x2,x1,i);
02150         break;
02151     }
02152 }
02153 /*
02154 Now the GCD is in c but still needs j powers of 2.
02155 */
02156 x1 = c;
02157 while ( j >= BITSINWORD ) {
02158     for ( i = n1; i > 0; i-- ) x1[i] = x1[i-1];
02159     x1[0] = 0; n1++;
02160     j -= BITSINWORD;
02161 }
02162 if ( j > 0 ) {
02163     ULONG a1,a2 = 0;
02164     for ( i = 0; i < n1; i++ ) {
02165         a1 = x1[i]; a1 <= j;
02166         a2 += a1;
02167         x1[i] = a2;
02168         a2 >= BITSINWORD;

```

```

02169         }
02170         if ( a2 != 0 ) {
02171             x1[n1++] = a2;
02172         }
02173     }
02174     *nc = n1;
02175     NumberFree (GLscrat7,"GcdLong"); NumberFree (GLscrat8,"GcdLong");
02176 #else
02177     UWORD *x1,*x2,*x3,*x4,*c1,*c2;
02178     WORD n1,n2,n3,n4,i;
02179     x1 = c; x3 = a; n1 = i = na;
02180     NCOPY(x1,x3,i);
02181     x1 = c; c1 = x2 = NumberMalloc("GcdLong"); x3 = NumberMalloc("GcdLong"); x4 =
NumberMalloc("GcdLong");
02182     c2 = b; n2 = i = nb;
02183     NCOPY(c1,c2,i);
02184     for(;;){
02185         if ( DivLong(x1,n1,x2,n2,x4,&n4,x3,&n3) ) goto GcdErr;
02186         if ( !n3 ) { x1 = x2; n1 = n2; break; }
02187         if ( DivLong(x2,n2,x3,n3,x4,&n4,x1,&n1) ) goto GcdErr;
02188         if ( !n1 ) { x1 = x3; n1 = n3; break; }
02189         if ( DivLong(x3,n3,x1,n1,x4,&n4,x2,&n2) ) goto GcdErr;
02190         if ( !n2 ) {
02191             *nc = n1;
02192             NumberFree(x2,"GcdLong"); NumberFree(x3,"GcdLong"); NumberFree(x4,"GcdLong");
02193             return(0);
02194         }
02195     }
02196     *nc = i = n1;
02197     NCOPY(c,x1,i);
02198     NumberFree(x2,"GcdLong"); NumberFree(x3,"GcdLong"); NumberFree(x4,"GcdLong");
02199 #endif
02200 #endif
02201     }
02202     return(0);
02203 GcdErr:
02204     MLOCK(ErrorMessageLock);
02205     MesCall("GcdLong");
02206     MUNLOCK(ErrorMessageLock);
02207     SETERROR(-1)
02208 }
02209 /*
02210 #] Old Routine :
02211 */
02212 #else
02213 /*
02214     New routine for GcdLong that uses smart shortcuts.
02215     Algorithm by J. Vermaseren 15-nov-2006.
02216     It runs faster for very big numbers but only by a fixed factor.
02217     There is no improvement in the power behaviour.
02218     Improvement on the whole of hf9 (multiple zeta values at weight 9):
02219         Better than a factor 2 on a 32 bits architecture and 2.76 on a
02220         64 bits architecture.
02221     On hf10 (MZV's at weight 10), 64 bits architecture: factor 7.
02222
02223     If we have two long numbers (na,nb > GCDMAX) we will work in a
02224     truncated way. At the moment of writing (15-nov-2006) it isn't
02225     clear whether this algorithm is an invention or a reinvention.
02226     A short search on the web didn't show anything.
02227
02228     31-jul-2007:
02229     A better search shows that this is an adaptation of the Lehmer-Euclid
02230     algorithm, already described in Knuth. Here we can work without upper
02231     and lower limit because we are only interested in the GCD, not the
02232     extra numbers. Also it takes already some features of the double
02233     digit Lehmer-Euclid algorithm of Jeebelean it seems.
02234
02235     Maybe this can be programmed slightly better and we can get another
02236     few percent speed increase. Further improvements for the asymptotic
02237     case come from splitting the calculation as in Karatsuba and working
02238     with FFT divisions and multiplications etc. But this is when hundreds
02239     of words are involved at the least.
02240
02241     Algorithm
02242
02243     1: while ( na > nb || nb < GCDMAX ) {
02244         if ( nb == 0 ) { result in a }
02245         c = a % b;
02246         a = b;
02247         b = c;
02248     }
02249
02250     2: Make the truncated values in which a and b are the combinations
02251     of the top two words of a and b. The whole numbers are aa and bb now.
02252
02253     3: ma1 = 1; ma2 = 0; mb1 = 0; mb2 = 1;
02254     4: A = a; B = b; m = a/b; c = a - m*b;
02255         c = ma1*a+ma2*b-m*(mb1*a+mb2*b) = (ma1-m*mb1)*a+(ma2-m*mb2)*b

```

```

02255     mc1 = ma1-m*mb1; mc2 = ma2-m*mb2;
02256 5: a = b; ma1 = mb1; ma2 = mb2;
02257   b = c; mb1 = mc1; mb2 = mc2;
02258 6: if ( b != 0 && nb >= FULLMAX ) goto 4;
02259 7: Now construct the new quantities
02260     ma1*aa+ma2*bb and mb1*aa+mb2*bb
02261 8: goto 1;
02262
02263 The essence of the above algorithm is that we do the divisions only
02264 on relatively short numbers. Also usually there are many steps 4&5
02265 for each step 7. This eliminates many operations.
02266 The termination at FULLMAX is that we make errors by not considering
02267 the tail of the number. If we run b down all the way, the errors combine
02268 in such a way that the new numbers may be of the same order as the old
02269 numbers. By stopping halfway we don't get the error beyond halfway
02270 either. Unfortunately this means that a >= FULLMAX and hence na > nb
02271 which means that next we will have a complete division. But just once.
02272 Running the steps 4-6 till a < FULLMAX runs already into problems.
02273 It may be necessary to experiment a bit to obtain the optimum value
02274 of GCDMAX.
02275 */
02276
02277 int GcdLong(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
02278 {
02279     GETBIDENTITY
02280     UWORD x,y,z;
02281     UWORD *x1,*x2,*x3,*x4,*x5,*d;
02282     UWORD *GLscrat6, *GLscrat7, *GLscrat8, *GLscrat9, *GLscrat10;
02283     WORD n1,n2,n3,n4,n5,i;
02284     RLONG lx,ly,lz;
02285     LONG ma1, ma2, mb1, mb2, mc1, mc2, m;
02286     if ( !na || !nb ) {
02287         if ( !na && !nb ) {
02288             MLOCK(ErrorMessageLock);
02289             MesPrint("Cannot take gcd");
02290             MUNLOCK(ErrorMessageLock);
02291             return(-1);
02292         }
02293
02294         if ( !na ) {
02295             *nc = abs(nb);
02296             NCOPY(c,b,*nc);
02297             *nc = abs(nb);
02298             return(0);
02299         }
02300
02301         *nc = abs(na);
02302         NCOPY(c,a,*nc);
02303         *nc = abs(na);
02304         return(0);
02305     }
02306     if ( na < 0 ) na = -na;
02307     if ( nb < 0 ) nb = -nb;
02308 /*
02309     # [ GMP stuff :
02310 */
02311 #ifdef WITHGMP
02312     if ( na > 3 && nb > 3 ) {
02313         int ii;
02314         mp_limb_t *upa, *upb, *upc, xx;
02315         UWORD *uw, *u1, *u2;
02316         unsigned int tcounta, tcountb, tcountal, tcountbl;
02317         mp_size_t ana, anb, anc;
02318
02319         u1 = uw = NumberMalloc("GcdLong");
02320         upa = (mp_limb_t *)u1;
02321         ana = na; tcountal = 0;
02322         while ( a[0] == 0 ) { a++; ana--; tcountal++; }
02323         for ( ii = 0; ii < ana; ii++ ) { *uw++ = *a++; }
02324         if ( ( ana & 1 ) != 0 ) { *uw = 0; ana++; }
02325         ana /= 2;
02326
02327         u2 = uw = NumberMalloc("GcdLong");
02328         upb = (mp_limb_t *)u2;
02329         anb = nb; tcountbl = 0;
02330         while ( b[0] == 0 ) { b++; anb--; tcountbl++; }
02331         for ( ii = 0; ii < anb; ii++ ) { *uw++ = *b++; }
02332         if ( ( anb & 1 ) != 0 ) { *uw = 0; anb++; }
02333         anb /= 2;
02334
02335         xx = upa[0]; tcounta = 0;
02336         while ( ( xx & 15 ) == 0 ) { tcounta += 4; xx >>= 4; }
02337         while ( ( xx & 1 ) == 0 ) { tcounta += 1; xx >>= 1; }
02338         xx = upb[0]; tcountb = 0;
02339         while ( ( xx & 15 ) == 0 ) { tcountb += 4; xx >>= 4; }
02340         while ( ( xx & 1 ) == 0 ) { tcountb += 1; xx >>= 1; }
02341

```

```

02342     if ( tcounta ) {
02343         mpn_rshift(upa,upa,ana,tcounta);
02344         if ( upa[ana-1] == 0 ) ana--;
02345     }
02346     if ( tcountb ) {
02347         mpn_rshift(upb,upb,anb,tcountb);
02348         if ( upb[anb-1] == 0 ) anb--;
02349     }
02350
02351     upc = (mp_limb_t *) (NumberMalloc("GcdLong"));
02352     if ( ( ana > anb ) || ( ( ana == anb ) && ( upa[ana-1] >= upb[anb-1] ) ) ) {
02353         anc = mpn_gcd(upc,upa,ana,upb,anb);
02354     }
02355     else {
02356         anc = mpn_gcd(upc,upb,anb,upa,ana);
02357     }
02358
02359     tcounta = tcounta*BITSINWORD + tcounta;
02360     tcountb = tcountb*BITSINWORD + tcountb;
02361     if ( tcountb > tcounta ) tcountb = tcounta;
02362     tcounta = tcountb/BITSINWORD;
02363     tcountb = tcountb%BITSINWORD;
02364
02365     if ( tcountb ) {
02366         xx = mpn_lshift(upc,upc,anc,tcountb);
02367         if ( xx ) { upc[anc] = xx; anc++; }
02368     }
02369
02370     uw = (UWORD *)upc; anc *= 2;
02371     while ( uw[anc-1] == 0 ) anc--;
02372     for ( ii = 0; ii < (int)tcounta; ii++ ) *c++ = 0;
02373     for ( ii = 0; ii < anc; ii++ ) *c++ = *uw++;
02374     *nc = anc + tcounta;
02375     NumberFree(u1,"GcdLong"); NumberFree(u2,"GcdLong"); NumberFree((UWORD *) (upc),"GcdLong");
02376     return(0);
02377 }
02378 #endif
02379 /*
02380 #] GMP stuff :
02381 */
02382 /*
02383 #[ Easy cases :
02384 */
02385     if ( na == 1 && nb == 1 ) {
02386         x = *a;
02387         y = *b;
02388         do { z = x % y; x = y; } while ( ( y = z ) != 0 );
02389         *c = x;
02390         *nc = 1;
02391         return(0);
02392     }
02393     else if ( na <= 2 && nb <= 2 ) {
02394         if ( na == 2 ) { lx = ((RLONG) (a[1]))«BITSINWORD + *a; }
02395         else { lx = *a; }
02396         if ( nb == 2 ) { ly = ((RLONG) (b[1]))«BITSINWORD + *b; }
02397         else { ly = *b; }
02398         if ( lx < ly ) { lz = lx; lx = ly; ly = lz; }
02399     #if ( BITSINWORD == 16 )
02400         do {
02401             lz = lx % ly; lx = ly;
02402         } while ( ( ly = lz ) != 0 );
02403     #else
02404         do {
02405             lz = lx % ly; lx = ly;
02406         } while ( ( ly = lz ) != 0 && ( lx & AWORDMASK ) != 0 );
02407         if ( ly ) {
02408             x = (UWORD)lx; y = (UWORD)ly;
02409             do { *c = x % y; x = y; } while ( ( y = *c ) != 0 );
02410             *c = x;
02411             *nc = 1;
02412         }
02413         else
02414     #endif
02415     {
02416         *c++ = (UWORD)lx;
02417         if ( ( *c = (UWORD) (lx » BITSINWORD) ) != 0 ) *nc = 2;
02418         else *nc = 1;
02419     }
02420     return(0);
02421 }
02422 /*
02423 #] Easy cases :
02424 */
02425     GLscrat6 = NumberMalloc("GcdLong"); GLscrat7 = NumberMalloc("GcdLong");
02426     GLscrat8 = NumberMalloc("GcdLong");
02427     GLscrat9 = NumberMalloc("GcdLong"); GLscrat10 = NumberMalloc("GcdLong");
02428     restart;;

```

```

02429 /*
02430      #] Easy cases :
02431 */
02432     if ( na == 1 && nb == 1 ) {
02433         x = *a;
02434         y = *b;
02435         do { z = x % y; x = y; } while ( ( y = z ) != 0 );
02436         *c = x;
02437         *nc = 1;
02438     }
02439     else if ( na <= 2 && nb <= 2 ) {
02440         if ( na == 2 ) { lx = ((RLONG) (a[1]))«BITSINWORD) + *a; }
02441         else { lx = *a; }
02442         if ( nb == 2 ) { ly = ((RLONG) (b[1]))«BITSINWORD) + *b; }
02443         else { ly = *b; }
02444         if ( lx < ly ) { lz = lx; lx = ly; ly = lz; }
02445     #if ( BITSINWORD == 16 )
02446         do {
02447             lz = lx % ly; lx = ly;
02448         } while ( ( ly = lz ) != 0 );
02449     #else
02450         do {
02451             lz = lx % ly; lx = ly;
02452         } while ( ( ly = lz ) != 0 && ( lx & AWORDMASK ) != 0 );
02453         if ( ly ) {
02454             x = (UWORD)lx; y = (UWORD)ly;
02455             do { *c = x % y; x = y; } while ( ( y = *c ) != 0 );
02456             *c = x;
02457             *nc = 1;
02458         }
02459         else
02460     #endif
02461     {
02462         *c++ = (UWORD)lx;
02463         if ( ( *c = (UWORD)(lx » BITSINWORD) ) != 0 ) *nc = 2;
02464         else *nc = 1;
02465     }
02466 }
02467 /*
02468      #] Easy cases :
02469      #] Original code :
02470 */
02471     else if ( na < GCDMAX || nb < GCDMAX || na != nb ) {
02472         if ( na < nb ) {
02473             x2 = GLscrat8; x3 = a; n2 = i = na;
02474             NCOPY(x2,x3,i);
02475             x1 = c; x3 = b; n1 = i = nb;
02476             NCOPY(x1,x3,i);
02477         }
02478         else {
02479             x1 = c; x3 = a; n1 = i = na;
02480             NCOPY(x1,x3,i);
02481             x2 = GLscrat8; x3 = b; n2 = i = nb;
02482             NCOPY(x2,x3,i);
02483         }
02484         x1 = c; x2 = GLscrat8; x3 = GLscrat7; x4 = GLscrat6;
02485         for(;;){
02486             if ( DivLong(x1,n1,x2,n2,x4,&n4,x3,&n3) ) goto GcdErr;
02487             if ( !n3 ) { x1 = x2; n1 = n2; break; }
02488             if ( n2 <= 2 ) { a = x2; b = x3; na = n2; nb = n3; goto restart; }
02489             if ( n3 >= GCDMAX && n2 == n3 ) {
02490                 a = GLscrat9; b = GLscrat10; na = n2; nb = n3;
02491                 for ( i = 0; i < na; i++ ) a[i] = x2[i];
02492                 for ( i = 0; i < nb; i++ ) b[i] = x3[i];
02493                 goto newtrick;
02494             }
02495             if ( DivLong(x2,n2,x3,n3,x4,&n4,x1,&n1) ) goto GcdErr;
02496             if ( !n1 ) { x1 = x3; n1 = n3; break; }
02497             if ( n3 <= 2 ) { a = x3; b = x1; na = n3; nb = n1; goto restart; }
02498             if ( n1 >= GCDMAX && n1 == n3 ) {
02499                 a = GLscrat9; b = GLscrat10; na = n3; nb = n1;
02500                 for ( i = 0; i < na; i++ ) a[i] = x3[i];
02501                 for ( i = 0; i < nb; i++ ) b[i] = x1[i];
02502                 goto newtrick;
02503             }
02504             if ( DivLong(x3,n3,x1,n1,x4,&n4,x2,&n2) ) goto GcdErr;
02505             if ( !n2 ) { *nc = n1; goto normalend; }
02506             if ( n1 <= 2 ) { a = x1; b = x2; na = n1; nb = n2; goto restart; }
02507             if ( n2 >= GCDMAX && n2 == n1 ) {
02508                 a = GLscrat9; b = GLscrat10; na = n1; nb = n2;
02509                 for ( i = 0; i < na; i++ ) a[i] = x1[i];
02510                 for ( i = 0; i < nb; i++ ) b[i] = x2[i];
02511                 goto newtrick;
02512             }
02513         }
02514         *nc = i = n1;
02515         NCOPY(c,x1,i);

```



```

02516     }
02517     /*
02518     #] Original code :
02519     #[ New code :
02520     */
02521     else {
02522     /*
02523         This is the new algorithm starting at step 3.
02524
02525         3: ma1 = 1; ma2 = 0; mb1 = 0; mb2 = 1;
02526         4: A = a; B = b; m = a/b; c = a - m*b;
02527           c = ma1*a+ma2*b-m*(mb1*a+mb2*b) = (ma1-m*mb1)*a+(ma2-m*mb2)*b
02528           mc1 = ma1-m*mb1; mc2 = ma2-m*mb2;
02529         5: a = b; ma1 = mb1; ma2 = mb2;
02530           b = c; mb1 = mc1; mb2 = mc2;
02531         6: if ( b != 0 ) goto 4;
02532     */
02533     newtrick;;
02534         ma1 = 1; ma2 = 0; mb1 = 0; mb2 = 1;
02535         lx = ((RLONG)(a[na-1]))«BITSINWORD + a[na-2];
02536         ly = ((RLONG)(b[nb-1]))«BITSINWORD + b[nb-2];
02537         if ( ly > lx ) { lz = lx; lx = ly; ly = lz; d = a; a = b; b = d; }
02538         do {
02539             m = lx/ly;
02540             mc1 = ma1-m*mb1; mc2 = ma2-m*mb2;
02541             ma1 = mb1; ma2 = mb2; mb1 = mc1; mb2 = mc2;
02542             lz = lx - m*ly; lx = ly; ly = lz;
02543         } while ( ly >= FULLMAX );
02544     /*
02545         Next the construction of the two new numbers
02546
02547         7: Now construct the new quantities
02548           a = ma1*aa+ma2*bb and b = mb1*aa+mb2*bb
02549     */
02550         x1 = GLscrat6;
02551         x2 = GLscrat7;
02552         x3 = GLscrat8;
02553         x5 = GLscrat10;
02554         if ( ma1 < 0 ) {
02555             ma1 = -ma1;
02556             x1[0] = (UWORD)ma1;
02557             x1[1] = (UWORD)(ma1 » BITSINWORD);
02558             if ( x1[1] ) n1 = -2;
02559             else n1 = -1;
02560         }
02561         else {
02562             x1[0] = (UWORD)ma1;
02563             x1[1] = (UWORD)(ma1 » BITSINWORD);
02564             if ( x1[1] ) n1 = 2;
02565             else n1 = 1;
02566         }
02567         if ( MulLong(a,na,x1,n1,x2,&n2) ) goto GcdErr;
02568         if ( ma2 < 0 ) {
02569             ma2 = -ma2;
02570             x1[0] = (UWORD)ma2;
02571             x1[1] = (UWORD)(ma2 » BITSINWORD);
02572             if ( x1[1] ) n1 = -2;
02573             else n1 = -1;
02574         }
02575         else {
02576             x1[0] = (UWORD)ma2;
02577             x1[1] = (UWORD)(ma2 » BITSINWORD);
02578             if ( x1[1] ) n1 = 2;
02579             else n1 = 1;
02580         }
02581         if ( MulLong(b,nb,x1,n1,x3,&n3) ) goto GcdErr;
02582         if ( AddLong(x2,n2,x3,n3,c,&n4) ) goto GcdErr;
02583         if ( mb1 < 0 ) {
02584             mb1 = -mb1;
02585             x1[0] = (UWORD)mb1;
02586             x1[1] = (UWORD)(mb1 » BITSINWORD);
02587             if ( x1[1] ) n1 = -2;
02588             else n1 = -1;
02589         }
02590         else {
02591             x1[0] = (UWORD)mb1;
02592             x1[1] = (UWORD)(mb1 » BITSINWORD);
02593             if ( x1[1] ) n1 = 2;
02594             else n1 = 1;
02595         }
02596         if ( MulLong(a,na,x1,n1,x2,&n2) ) goto GcdErr;
02597         if ( mb2 < 0 ) {
02598             mb2 = -mb2;
02599             x1[0] = (UWORD)mb2;
02600             x1[1] = (UWORD)(mb2 » BITSINWORD);
02601             if ( x1[1] ) n1 = -2;
02602             else n1 = -1;

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```

02603     }
02604     else {
02605         x1[0] = (UWORD)mb2;
02606         x1[1] = (UWORD)(mb2 » BITSINWORD);
02607         if ( x1[1] ) n1 = 2;
02608         else      n1 = 1;
02609     }
02610     if ( MulLong(b,nb,x1,n1,x3,&n3) ) goto GcdErr;
02611     if ( AddLong(x2,n2,x3,n3,x5,&n5) ) goto GcdErr;
02612     a = c; na = n4; b = x5; nb = n5;
02613     if ( nb == 0 ) { *nc = n4; goto normalend; }
02614     x4 = GLscrat9;
02615     for ( i = 0; i < na; i++ ) x4[i] = a[i];
02616     a = x4;
02617     if ( na < 0 ) na = -na;
02618     if ( nb < 0 ) nb = -nb;
02619     /*
02620     The typical case now is that in a we have the last step to go
02621     to loose the leading word, while in b we have lost the leading word.
02622     We could go to DivLong now but we can also add an extra step that
02623     is less wasteful.
02624     In the case that the new leading word of b is extrememly short (like 1)
02625     we make a rather large error of course. In the worst case the whole
02626     will be intercepted by DivLong after all, but that is so rare that
02627     it shouldn't influence any timing in a measurable way.
02628     */
02629     if ( nb >= GCDMAX && na == nb+1 && b[nb-1] >= HALFMAX && b[nb-1] > a[na-1] ) {
02630         lx = (((RLONG)(a[na-1]))«BITSINWORD) + a[na-2];
02631         x1[0] = lx/b[nb-1]; n1 = 1;
02632         MulLong(b,nb,x1,n1,x2,&n2);
02633         n2 = -n2;
02634         AddLong(a,na,x2,n2,x4,&n4);
02635         if ( n4 == 0 ) {
02636             *nc = nb;
02637             for ( i = 0; i < nb; i++ ) c[i] = b[i];
02638             goto normalend;
02639         }
02640         if ( n4 < 0 ) n4 = -n4;
02641         a = b; na = nb; b = x4; nb = n4;
02642     }
02643     goto restart;
02644     /*
02645     #] New code :
02646     */
02647     }
02648 normalend:
02649     NumberFree(GLscrat6,"GcdLong"); NumberFree(GLscrat7,"GcdLong"); NumberFree(GLscrat8,"GcdLong");
02650     NumberFree(GLscrat9,"GcdLong"); NumberFree(GLscrat10,"GcdLong");
02651     return(0);
02652 GcdErr:
02653     MLOCK(ErrorMessageLock);
02654     MesCall("GcdLong");
02655     MUNLOCK(ErrorMessageLock);
02656     NumberFree(GLscrat6,"GcdLong"); NumberFree(GLscrat7,"GcdLong"); NumberFree(GLscrat8,"GcdLong");
02657     NumberFree(GLscrat9,"GcdLong"); NumberFree(GLscrat10,"GcdLong");
02658     SETERROR(-1)
02659 }
02660
02661 #endif
02662
02663 /*
02664     #] GcdLong :
02665     #[ GetBinom :      WORD GetBinom(a,na,i1,i2)
02666     */
02667
02668 int GetBinom(UWORD *a, WORD *na, WORD i1, WORD i2)
02669 {
02670     GETIDENTITY
02671     WORD j, k, l;
02672     UWORD *GBscrat3, *GBscrat4;
02673     if ( i1-i2 < i2 ) i2 = i1-i2;
02674     if ( i2 == 0 ) { *a = 1; *na = 1; return(0); }
02675     if ( i2 > i1 ) { *a = 0; *na = 0; return(0); }
02676     *a = i1; *na = 1;
02677     GBscrat3 = NumberMalloc("GetBinom"); GBscrat4 = NumberMalloc("GetBinom");
02678     for ( j = 2; j <= i2; j++ ) {
02679         GBscrat3[0] = i1+1-j;
02680         if ( MulLong(a,*na,GBscrat3,(WORD)1,GBscrat4,&k) ) goto CalledFrom;
02681         GBscrat3[0] = j;
02682         if ( DivLong(GBscrat4,k,GBscrat3,(WORD)1,a,na,GBscrat3,&l) ) goto CalledFrom;
02683     }
02684     NumberFree(GBscrat3,"GetBinom"); NumberFree(GBscrat4,"GetBinom");
02685     return(0);
02686 CalledFrom:
02687     MLOCK(ErrorMessageLock);
02688     MesCall("GetBinom");
02689     MUNLOCK(ErrorMessageLock);

```

```

02690     NumberFree(GBscrat3,"GetBinom"); NumberFree(GBscrat4,"GetBinom");
02691     SETERROR(-1)
02692 }
02693
02694 /*
02695     #] GetBinom :
02696     #[ LcmLong :      WORD LcmLong(a,na,b,nb)
02697
02698     Computes the LCM of the long numbers a and b and puts the result
02699     in c. c is allowed to be equal to a.
02700 */
02701
02702 int LcmLong(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
02703 {
02704     int error = 0;
02705     UWORD *d = NumberMalloc("LcmLong");
02706     UWORD *e = NumberMalloc("LcmLong");
02707     UWORD *f = NumberMalloc("LcmLong");
02708     WORD nd, ne, nf;
02709     GcdLong(BHEAD a, na, b, nb, d, &nd);
02710     DivLong(a,na,d,nd,e,&ne,f,&nf);
02711     if ( MulLong(b,nb,e,ne,c,nc) ) {
02712         MLOCK(ErrorMessageLock);
02713         MesCall("LcmLong");
02714         MUNLOCK(ErrorMessageLock);
02715         error = -1;
02716     }
02717     NumberFree(f,"LcmLong");
02718     NumberFree(e,"LcmLong");
02719     NumberFree(d,"LcmLong");
02720     return(error);
02721 }
02722
02723 /*
02724     #] LcmLong :
02725     #[ TakeLongRoot:   int TakeLongRoot(a,n,power)
02726
02727     Takes the 'power'-root of the long number in a.
02728     If the root could be taken the return value is zero.
02729     If the root could not be taken, the return value is 1.
02730     The root will be in a if it could be taken, otherwise there will be garbage
02731     Algorithm: (assume b is guess of root, b' better guess)
02732         b' = (a-(power-1)*b^power)/(n*b^(power-1))
02733     Note: power should be positive!
02734 */
02735
02736 int TakeLongRoot(UWORD *a, WORD *n, WORD power)
02737 {
02738     GETIDENTITY
02739     int numbits, guessbits, i, retval = 0;
02740     UWORD x, *b, *c, *d, *e;
02741     WORD na, nb, nc, nd, ne;
02742     if ( *n < 0 && ( power & 1 ) == 0 ) return(1);
02743     if ( power == 1 ) return(0);
02744     if ( *n < 0 ) { na = -*n; }
02745     else { na = *n; }
02746     if ( na == 1 ) {
02747         /* Special cases that are the most frequent */
02748         if ( a[0] == 1 ) return(0);
02749         if ( power < BITSINWORD && na == 1 && a[0] == (UWORD)(1<<power) ) {
02750             a[0] = 2; return(0);
02751         }
02752         if ( 2*power < BITSINWORD && na == 1 && a[0] == (UWORD)(1<<(2*power)) ) {
02753             a[0] = 4; return(0);
02754         }
02755     }
02756     /*
02757     Step 1: make a guess. We count the number of bits.
02758     numbits will be the 1+2log(a)
02759     */
02760     numbits = BITSINWORD*(na-1);
02761     x = a[na-1];
02762     while ( ( x >> 8 ) != 0 ) { numbits += 8; x >>= 8; }
02763     if ( ( x >> 4 ) != 0 ) { numbits += 4; x >>= 4; }
02764     if ( ( x >> 2 ) != 0 ) { numbits += 2; x >>= 2; }
02765     if ( ( x >> 1 ) != 0 ) numbits++;
02766     guessbits = numbits / power;
02767     if ( guessbits <= 0 ) return(1); /* root < 2 and 1 we did already */
02768     nb = guessbits/BITSINWORD;
02769     /*
02770     The recursion is:
02771     (b'-b) = (a/b^(power-1)-b)/n
02772             = (a/c-b)/n
02773             = (d-b)/n    (remainder of a/c is e)
02774             = c/n    (we reuse the scratch array c)
02775     Termination can be tricky. When a/c has no remainder and = b we have a root.
02776     When d = b but the remainder of a/c != 0, there is definitely no root.

```

```

02777 */
02778 b = NumberMalloc("TakeLongRoot"); c = NumberMalloc("TakeLongRoot");
02779 d = NumberMalloc("TakeLongRoot"); e = NumberMalloc("TakeLongRoot");
02780 for ( i = 0; i < nb; i++ ) { b[i] = 0; }
02781 b[nb] = 1 << (guessbits%BITSINWORD);
02782 nb++;
02783 for(;;) {
02784     nc = nb;
02785     for ( i = 0; i < nb; i++ ) c[i] = b[i];
02786     if ( RaisPow(BHEAD c,&nc,power-1) ) goto TLcall;
02787     if ( DivLong(a,na,c,nc,d,&nd,e,&ne) ) goto TLcall;
02788     nb = -nb;
02789     if ( AddLong(d,nd,b,nb,c,&nc) ) goto TLcall;
02790     nb = -nb;
02791     if ( nc == 0 ) {
02792         if ( ne == 0 ) break;
02793         retval = 1; break;
02794     /*
02795         else {
02796             NumberFree(b,"TakeLongRoot"); NumberFree(c,"TakeLongRoot");
02797             NumberFree(d,"TakeLongRoot"); NumberFree(e,"TakeLongRoot");
02798             return(1);
02799         }
02800     */
02801     }
02802     DivLong(c,nc,(UWORD *)(&power),1,d,&nd,e,&ne);
02803     if ( nd == 0 ) {
02804         retval = 1;
02805         break;
02806     /*
02807         NumberFree(b,"TakeLongRoot"); NumberFree(c,"TakeLongRoot");
02808         NumberFree(d,"TakeLongRoot"); NumberFree(e,"TakeLongRoot");
02809         return(1);
02810     */
02811     /*
02812         This code tries b+1 as a final possibility.
02813         We believe this is not needed
02814         UWORD one = 1;
02815         if ( AddLong(b,nb,&one,1,c,&nc) ) goto TLcall;
02816         if ( RaisPow(BHEAD c,&nc,power-1) ) goto TLcall;
02817         if ( DivLong(a,na,c,nc,d,&nd,e,&ne) ) goto TLcall;
02818         if ( ne != 0 ) return(1);
02819         nb = -nb;
02820         if ( SubLong(d,nd,b,nb,c,&nc) ) goto TLcall;
02821         nb = -nb;
02822         if ( nc != 0 ) {
02823             NumberFree(b,"TakeLongRoot"); NumberFree(c,"TakeLongRoot");
02824             NumberFree(d,"TakeLongRoot"); NumberFree(e,"TakeLongRoot");
02825             return(1);
02826         }
02827         break;
02828     */
02829     }
02830     if ( AddLong(b,nb,d,nd,b,&nb) ) goto TLcall;
02831 }
02832 for ( i = 0; i < nb; i++ ) a[i] = b[i];
02833 if ( *n < 0 ) *n = -nb;
02834 else *n = nb;
02835 NumberFree(b,"TakeLongRoot"); NumberFree(c,"TakeLongRoot");
02836 NumberFree(d,"TakeLongRoot"); NumberFree(e,"TakeLongRoot");
02837 return(retval);
02838 TLcall:
02839 MLOCK(ErrorMessageLock);
02840 MesCall("TakeLongRoot");
02841 MUNLOCK(ErrorMessageLock);
02842 NumberFree(b,"TakeLongRoot"); NumberFree(c,"TakeLongRoot");
02843 NumberFree(d,"TakeLongRoot"); NumberFree(e,"TakeLongRoot");
02844 Terminate(-1);
02845 return(-1);
02846 }
02847
02848 /*
02849     #] TakeLongRoot:
02850     #[ MakeRational:
02851
02852     Makes the integer a mod m into a traction b/c with |b|,|c| < sqrt(m)
02853     For the algorithm, see MakeLongRational.
02854 */
02855
02856 int MakeRational(WORD a,WORD m, WORD *b, WORD *c)
02857 {
02858     LONG x1,x2,x3,x4,y1,y2;
02859     if ( a < 0 ) { a = a+m; }
02860     if ( a <= 1 ) {
02861         if ( a > m/2 ) a = a-m;
02862         *b = a; *c = 1; return(0);
02863     }

```

```

02864     x1 = m; x2 = a;
02865     if ( x2*x2 >= m ) {
02866         y1 = x1/x2; y2 = x1%x2; x3 = 1; x4 = -y1; x1 = x2; x2 = y2;
02867         while ( x2*x2 >= m ) {
02868             y1 = x1/x2; y2 = x1%x2; x1 = x2; x2 = y2; y2 = x3-y1*x4; x3 = x4; x4 = y2;
02869         }
02870     }
02871     else x4 = 1;
02872     if ( x2 == 0 ) { return(1); }
02873     if ( x2 > m/2 ) *b = x2-m;
02874     else *b = x2;
02875     if ( x4 > m/2 ) { *c = x4-m; *c = -*c; *b = -*b; }
02876     else if ( x4 <= -m/2 ) { x4 += m; *c = x4; }
02877     else if ( x4 < 0 ) { x4 = -x4; *c = x4; *b = -*b; }
02878     else *c = x4;
02879     return(0);
02880 }
02881
02882 /*
02883     #] MakeRational:
02884     #[ MakeLongRational:
02885
02886     Converts the long number a mod m into the fraction b
02887     One of the properties of b is that num,den < sqrt(m)
02888     The algorithm: Start with:   m 0
02889                                a 1
02890     Make now c=m/a, c1=m/a      c c2=0-c1*1
02891     Make now d=a/c, d1=a/c      d d2=1-d1*c2
02892     Make now e=c/d, e1=c/d      e e2=1-e1*d2
02893     etc till in the first column we get a number < sqrt(m)
02894     We have then f,f2 and the fraction is f/f2.
02895     If at any moment we get a zero, m contained an unlucky prime.
02896
02897     Note that this can be made a lot faster when we make the same
02898     improvements as in the GCD routine. That is something for later.
02899 #ifndef WITHMAKERATIONAL
02900 */
02901
02902 #define COPYLONG(x1,nx1,x2,nx2) { int i; for(i=0;i<ABS(nx2);i++)x1[i]=x2[i];nx1=nx2; }
02903
02904 int MakeLongRational(PHEAD UWORD *a, WORD na, UWORD *m, WORD nm, UWORD *b, WORD *nb)
02905 {
02906     UWORD *root = NumberMalloc("MakeRational");
02907     UWORD *x1 = NumberMalloc("MakeRational");
02908     UWORD *x2 = NumberMalloc("MakeRational");
02909     UWORD *x3 = NumberMalloc("MakeRational");
02910     UWORD *x4 = NumberMalloc("MakeRational");
02911     UWORD *y1 = NumberMalloc("MakeRational");
02912     UWORD *y2 = NumberMalloc("MakeRational");
02913     WORD nroot,nx1,nx2,nx3,nx4,ny1,ny2 = 0,retval = 0;
02914     WORD sign = 1;
02915     /*
02916     Step 1: Take the square root of m
02917     */
02918     COPYLONG(root,nroot,m,nm)
02919     TakeLongRoot(root,&nroot,2);
02920     /*
02921     Step 2: Set the start values
02922     */
02923     if ( na < 0 ) { na = -na; sign = -sign; }
02924     COPYLONG(x1,nx1,m,nm)
02925     COPYLONG(x2,nx2,a,na)
02926     /*
02927     x3[0] = 0, nx3 = 0;
02928     x4[0] = 1, nx4 = 1;
02929     */
02930     /*
02931     The start operation needs some special attention because of the zero.
02932     */
02933     if ( BigLong(x2,nx2,root,nroot) <= 0 ) {
02934         x4[0] = 1, nx4 = 1;
02935         goto gottheanswer;
02936     }
02937     DivLong(x1,nx1,x2,nx2,y1,&ny1,y2,&ny2);
02938     if ( ny2 == 0 ) { retval = 1; goto cleanup; }
02939     COPYLONG(x1,nx1,x2,nx2)
02940     COPYLONG(x2,nx2,y2,ny2)
02941     x3[0] = 1; nx3 = 1;
02942     COPYLONG(x4,nx4,y1,ny1)
02943     nx4 = -nx4;
02944     /*
02945     Now the loop.
02946     */
02947     while ( BigLong(x2,nx2,root,nroot) > 0 ) {
02948         DivLong(x1,nx1,x2,nx2,y1,&ny1,y2,&ny2);
02949         if ( ny2 == 0 ) { retval = 1; goto cleanup; }
02950         COPYLONG(x1,nx1,x2,nx2)

```

```

02951     COPYLONG(x2,nx2,y2,ny2)
02952     MulLong(y1,ny1,x4,nx4,y2,&ny2);
02953     ny2 = -ny2;
02954     AddLong(x3,nx3,y2,ny2,y1,&ny1);
02955     COPYLONG(x3,nx3,x4,nx4)
02956     COPYLONG(x4,nx4,y1,ny1)
02957 }
02958 /*
02959     Now we have the answer. It is x2/x4. It has to be packed into b.
02960 */
02961 gottheanswer:
02962     if ( nx4 < 0 ) { sign = -sign; nx4 = -nx4; }
02963     COPYLONG(b,*nb,x2,nx2)
02964     Pack(b,nb,x4,nx4);
02965     if ( sign < 0 ) *nb = -*nb;
02966 cleanup:
02967     NumberFree(y2,"MakeRational");
02968     NumberFree(y1,"MakeRational");
02969     NumberFree(x4,"MakeRational");
02970     NumberFree(x3,"MakeRational");
02971     NumberFree(x2,"MakeRational");
02972     NumberFree(x1,"MakeRational");
02973     NumberFree(root,"MakeRational");
02974     return(retval);
02975 }
02976
02977 /*
02978 #endif
02979     #] MakeLongRational:
02980     #[ ChineseRemainder:
02981 */
02993 #ifdef WITHCHINESEREMAINDER
02994
02995 int ChineseRemainder(PHEAD MODNUM *a1, MODNUM *a2, MODNUM *a)
02996 {
02997     UWORD *inv1 = NumberMalloc("ChineseRemainder");
02998     UWORD *inv2 = NumberMalloc("ChineseRemainder");
02999     UWORD *fac1 = NumberMalloc("ChineseRemainder");
03000     UWORD *fac2 = NumberMalloc("ChineseRemainder");
03001     UWORD two[1];
03002     WORD ninv1, ninv2, nfac1, nfac2;
03003     if ( a1->na < 0 ) {
03004         AddLong(a1->a,a1->na,a1->m,a1->nm,a1->a,&(a1->na));
03005     }
03006     if ( a2->na < 0 ) {
03007         AddLong(a2->a,a2->na,a2->m,a2->nm,a2->a,&(a2->na));
03008     }
03009     MulLong(a1->m,a1->nm,a2->m,a2->nm,a->m,&(a->nm));
03010
03011     GetLongModInverses(BHEAD a1->m,a1->nm,a2->m,a2->nm,inv1,&ninv1,inv2,&ninv2);
03012     MulLong(inv1,ninv1,a1->m,a1->nm,fac1,&nfac1);
03013     MulLong(inv2,ninv2,a2->m,a2->nm,fac2,&nfac2);
03014
03015     MulLong(fac1,nfac1,a2->a,a2->na,inv1,&ninv1);
03016     MulLong(fac2,nfac2,a1->a,a1->na,inv2,&ninv2);
03017     AddLong(inv1,ninv1,inv2,ninv2,a->a,&(a->na));
03018
03019     two[0] = 2;
03020     MulLong(a->a,a->na,two,1,fac1,&nfac1);
03021     if ( BigLong(fac1,nfac1,a->m,a->nm) > 0 ) {
03022         a->nm = -a->nm;
03023         AddLong(a->a,a->na,a->m,a->nm,a->a,&(a->na));
03024         a->nm = -a->nm;
03025     }
03026     NumberFree(fac2,"ChineseRemainder");
03027     NumberFree(fac1,"ChineseRemainder");
03028     NumberFree(inv2,"ChineseRemainder");
03029     NumberFree(inv1,"ChineseRemainder");
03030     return(0);
03031 }
03032
03033 #endif
03034
03035 /*
03036     #] ChineseRemainder:
03037     #[ RekenLong :
03038     #[ RekenTerms :
03039     #[ CompCoef :      WORD CompCoef(term1,term2)
03040
03041     Compares the coefficients of term1 and term2 by subtracting them.
03042     This does more work than needed but this routine is only called
03043     when sorting functions and function arguments.
03044     (and comparing values
03045 */
03046 /* #define 64SAVE */
03047
03048 WORD CompCoef(WORD *term1, WORD *term2)

```

```

03049 {
03050     GETIDENTITY
03051     UWORD *c;
03052     WORD n1,n2,n3,*a;
03053     GETCOEF(term1,n1);
03054     GETCOEF(term2,n2);
03055     if ( term1[1] == 0 && n1 == 1 ) {
03056         if ( term2[1] == 0 && n2 == 1 ) return(0);
03057         if ( n2 < 0 ) return(1);
03058         return(-1);
03059     }
03060     else if ( term2[1] == 0 && n2 == 1 ) {
03061         if ( n1 < 0 ) return(-1);
03062         return(1);
03063     }
03064     if ( n1 > 0 ) {
03065         if ( n2 < 0 ) return(1);
03066     }
03067     else {
03068         if ( n2 > 0 ) return(-1);
03069         a = term1; term1 = term2; term2 = a;
03070         n3 = -n1; n1 = -n2; n2 = n3;
03071     }
03072     if ( term1[1] == 1 && term2[1] == 1 && n1 == 1 && n2 == 1 ) {
03073         if ( (UWORD)*term1 > (UWORD)*term2 ) return(1);
03074         else if ( (UWORD)*term1 < (UWORD)*term2 ) return(-1);
03075         else return(0);
03076     }
03077
03078     /*
03079     The next call should get dedicated code, as AddRat does more than
03080     strictly needed. Also more attention should be given to overflow.
03081     */
03082     c = NumberMalloc("CompCoef");
03083     if ( AddRat(BHEAD (UWORD *)term1,n1,(UWORD *)term2,-n2,c,&n3) ) {
03084         MLOCK(ErrorMessageLock);
03085         MesCall("CompCoef");
03086         MUNLOCK(ErrorMessageLock);
03087         NumberFree(c,"CompCoef");
03088         SETERROR(-1)
03089     }
03090     NumberFree(c,"CompCoef");
03091     return(n3);
03092 }
03093
03094 /*
03095     #] CompCoef :
03096     #[ Modulus :          WORD Modulus(term)
03097
03098     Routine takes the coefficient of term modulus b. The answer
03099     is in term again and the length of term is adjusted.
03100
03101     */
03102
03103 int Modulus(WORD *term)
03104 {
03105     WORD *t;
03106     WORD n1;
03107     t = term;
03108     GETCOEF(t,n1);
03109     if ( TakeModulus((UWORD *)t,&n1,AC.cmod,AC.ncmod,UNPACK) ) {
03110         MLOCK(ErrorMessageLock);
03111         MesCall("Modulus");
03112         MUNLOCK(ErrorMessageLock);
03113         SETERROR(-1)
03114     }
03115     if ( !n1 ) {
03116         *term = 0;
03117         return(0);
03118     }
03119     else if ( n1 > 0 ) {
03120         n1 *= 2;
03121         t += n1;          /* Note that n1 >= 0 */
03122         n1++;
03123     }
03124     else if ( n1 < 0 ) {
03125         n1 *= 2;
03126         t += -n1;
03127         n1--;
03128     }
03129     *t++ = n1;
03130     *term = WORDDIF(t,term);
03131     return(0);
03132 }
03133
03134 /*
03135     #] Modulus :

```

```

03136      # [ TakeModulus :      WORD TakeModulus(a,na,cmoivec,nmod,par)
03137
03138      Routine gets the rational number in a with reduced length na.
03139      It is called when AC.nmod != 0 and the number in AC.cmod is the
03140      number wrt which we need the modulus.
03141      The result is returned in a and na again.
03142
03143      If par == NOUNPACK we only do a single number, not a fraction.
03144      In addition we don't do fancy. We want a positive number and
03145      the input was supposed to be positive.
03146      We don't pack the result. The calling routine is responsible for that.
03147      This may not be a good idea. To be checked.
03148  */
03149
03150  int TakeModulus(UWORD *a, WORD *na, UWORD *cmoivec, WORD nmod, WORD par)
03151  {
03152      GETIDENTITY
03153      UWORD *c, *d, *e, *f, *g, *h;
03154      UWORD *x4,*x2;
03155      UWORD *x3,*x1,*x5,*x6,*x7,*x8;
03156      WORD y3,y1,y5,y6;
03157      WORD n1, i, y2, y4;
03158      WORD nh, tdenom, tnum, nmod;
03159      LONG x;
03160      if ( nmod == 0 ) return(0);          /* No modulus operation */
03161      nmod = ABS(nmod);
03162      n1 = *na;
03163      if ( ( par & UNPACK ) != 0 ) UnPack(a,n1,&tdenom,&tnum);
03164      else { tnum = n1; }
03165  /*
03166      We fish out the special case that the coefficient is short as well.
03167      There is no need to make lots of calls etc
03168  */
03169      if ( ( ( par & UNPACK ) == 0 ) && nmod == 1 && ( n1 == 1 || n1 == -1 ) ) {
03170          goto simplecase;
03171      }
03172      else if ( nmod == 1 && ( n1 == 1 || n1 == -1 ) ) {
03173          if ( a[1] != 1 ) {
03174              a[1] = a[1] % cmoivec[0];
03175              if ( a[1] == 0 ) {
03176                  MesPrint("Division by zero in short modulus arithmetic");
03177                  return(-1);
03178              }
03179              y1 = 0;
03180              if ( ( AC.modinverses != 0 ) && ( ( par & NOINVERSES ) == 0 ) ) {
03181                  y1 = AC.modinverses[a[1]];
03182              }
03183              else {
03184                  GetModInverses(a[1],cmoivec[0],&y1,&y2);
03185              }
03186              x = a[0];
03187              a[0] = (x*y1) % cmoivec[0];
03188              a[1] = 1;
03189          }
03190          else {
03191  simplecase:
03192              a[0] = a[0] % cmoivec[0];
03193          }
03194          if ( a[0] == 0 ) { *na = 0; return(0); }
03195          if ( ( AC.modmode & POSNEG ) != 0 ) {
03196              if ( a[0] > (UWORD)(cmoivec[0]/2) ) {
03197                  a[0] = cmoivec[0] - a[0];
03198                  *na = -*na;
03199              }
03200          }
03201          else if ( *na < 0 ) {
03202              *na = 1; a[0] = cmoivec[0] - a[0];
03203          }
03204          return(0);
03205      }
03206      c = NumberMalloc("TakeModulus"); d = NumberMalloc("TakeModulus"); e = NumberMalloc("TakeModulus");
03207      f = NumberMalloc("TakeModulus"); g = NumberMalloc("TakeModulus"); h = NumberMalloc("TakeModulus");
03208      n1 = ABS(n1);
03209      if ( DivLong(a,tnum,(UWORD *)cmoivec,nmod,
03210          c,&nh,a,&tnum) ) goto ModErr;
03211      if ( tnum == 0 ) { *na = 0; goto normalreturn; }
03212      if ( ( par & UNPACK ) == 0 ) {
03213          if ( ( AC.modmode & POSNEG ) != 0 ) {
03214              NormalModulus(a,&tnum);
03215          }
03216          else if ( tnum < 0 ) {
03217              SubPLon((UWORD *)cmoivec,nmod,a,-tnum,a,&tnum);
03218          }
03219          *na = tnum;
03220          goto normalreturn;
03221      }
03222      if ( tdenom == 1 && a[n1] == 1 ) {

```



```

03223     if ( ( AC.modmode & POSNEG ) != 0 ) {
03224         NormalModulus(a,&tnumer);
03225     }
03226     else if ( tnumer < 0 ) {
03227         SubPLon((UWORD *)cmodvec,nmod,a,-tnumer,a,&tnumer);
03228     }
03229     *na = tnumer;
03230     i = ABS(tnumer);
03231     a += i;
03232     *a++ = 1;
03233     while ( --i > 0 ) *a++ = 0;
03234     goto normalreturn;
03235 }
03236 if ( DivLong(a+n1,tdenom,(UWORD *)cmodvec,nmod,c,&nh,a+n1,&tdenom) ) goto ModErr;
03237 if ( !tdenom ) {
03238     MLOCK(ErrorMessageLock);
03239     MesPrint("Division by zero in modulus arithmetic");
03240     if ( AP.DebugFlag ) {
03241         AO.OutSkip = 3;
03242         FiniLine();
03243         i = *na;
03244         if ( i < 0 ) i = -i;
03245         while ( --i >= 0 ) { TalToLine((UWORD)(*a++)); TokenToLine((UBYTE *)" "); }
03246         i = *na;
03247         if ( i < 0 ) i = -i;
03248         while ( --i >= 0 ) { TalToLine((UWORD)(*a++)); TokenToLine((UBYTE *)" "); }
03249         TalToLine((UWORD)(*na));
03250         AO.OutSkip = 0;
03251         FiniLine();
03252     }
03253     MUNLOCK(ErrorMessageLock);
03254     NumberFree(c,"TakeModulus"); NumberFree(d,"TakeModulus"); NumberFree(e,"TakeModulus");
03255     NumberFree(f,"TakeModulus"); NumberFree(g,"TakeModulus"); NumberFree(h,"TakeModulus");
03256     return(-1);
03257 }
03258 if ( ( AC.modinverses != 0 ) && ( ( par & NOINVERSES ) == 0 )
03259 && ( tdenom == 1 || tdenom == -1 ) ) {
03260     *d = AC.modinverses[a[n1]]; y1 = 1; y2 = tdenom;
03261     if ( MulLong(a,tnumer,d,y1,c,&y3) ) goto ModErr;
03262     if ( DivLong(c,y3,(UWORD *)cmodvec,nmod,d,&y5,a,&tdenom) ) goto ModErr;
03263     if ( y2 < 0 ) tdenom = -tdenom;
03264 }
03265 else {
03266     x2 = (UWORD *)cmodvec; x1 = c; i = nmod; while ( --i >= 0 ) *x1++ = *x2++;
03267     x1 = c; x2 = a+n1; x3 = d; x4 = e; x5 = f; x6 = g;
03268     y1 = nmod; y2 = tdenom; y4 = 0; y5 = 1; *x5 = 1;
03269     for(;;) {
03270         if ( DivLong(x1,y1,x2,y2,h,&nh,x3,&y3) ) goto ModErr;
03271         if ( MulLong(x5,y5,h,nh,x6,&y6) ) goto ModErr;
03272         if ( AddLong(x4,y4,x6,-y6,x6,&y6) ) goto ModErr;
03273         if ( !y3 ) {
03274             if ( y2 != 1 || *x2 != 1 ) {
03275                 MLOCK(ErrorMessageLock);
03276                 MesPrint("Inverse in modulus arithmetic doesn't exist");
03277                 MesPrint("Denominator and modulus are not relative prime");
03278                 MUNLOCK(ErrorMessageLock);
03279                 goto ModErr;
03280             }
03281             break;
03282         }
03283         x7 = x1; x1 = x2; y1 = y2; x2 = x3; y2 = y3; x3 = x7;
03284         x8 = x4; x4 = x5; y4 = y5; x5 = x6; y5 = y6; x6 = x8;
03285     }
03286     if ( y5 < 0 && AddLong((UWORD *)cmodvec,nmod,x5,y5,x5,&y5) ) goto ModErr;
03287     if ( MulLong(a,tnumer,x5,y5,c,&y3) ) goto ModErr;
03288     if ( DivLong(c,y3,(UWORD *)cmodvec,nmod,d,&y5,a,&tdenom) ) goto ModErr;
03289 }
03290 if ( !tdenom ) { *na = 0; goto normalreturn; }
03291 if ( ( ( AC.modmode & POSNEG ) != 0 ) && ( ( par & FROMFUNCTION ) == 0 ) ) {
03292     NormalModulus(a,&tdenom);
03293 }
03294 else if ( tdenom < 0 ) {
03295     SubPLon((UWORD *)cmodvec,nmod,a,-tdenom,a,&tdenom);
03296 }
03297 *na = tdenom;
03298 i = ABS(tdenom);
03299 a += i;
03300 *a++ = 1;
03301 while ( --i > 0 ) *a++ = 0;
03302 normalreturn:
03303     NumberFree(c,"TakeModulus"); NumberFree(d,"TakeModulus"); NumberFree(e,"TakeModulus");
03304     NumberFree(f,"TakeModulus"); NumberFree(g,"TakeModulus"); NumberFree(h,"TakeModulus");
03305     return(0);
03306 ModErr:
03307     MLOCK(ErrorMessageLock);
03308     MesCall("TakeModulus");
03309     MUNLOCK(ErrorMessageLock);

```

```

03310     NumberFree(c,"TakeModulus"); NumberFree(d,"TakeModulus"); NumberFree(e,"TakeModulus");
03311     NumberFree(f,"TakeModulus"); NumberFree(g,"TakeModulus"); NumberFree(h,"TakeModulus");
03312     SETERROR(-1)
03313 }
03314
03315 /*
03316     #] TakeModulus :
03317     #[ TakeNormalModulus : WORD TakeNormalModulus(a,na,par)
03318
03319     added by Jan [01-09-2010]
03320 */
03321
03322 int TakeNormalModulus (UWORD *a, WORD *na, UWORD *c, WORD nc, WORD par)
03323 {
03324     WORD n;
03325     WORD nhalfc;
03326     UWORD *halfc;
03327
03328     GETIDENTITY;
03329
03330     /* determine c/2 by right shifting */
03331     halfc = NumberMalloc("TakeNormalModulus");
03332     nhalfc=nc;
03333     WCOPY(halfc,c,nc);
03334
03335     for (n=0; n<nhalfc; n++) {
03336         halfc[n] /= 2;
03337         if (n+1<nc) halfc[n] |= c[n+1] << (BITSINWORD-1);
03338     }
03339
03340     if (halfc[nhalfc-1]==0)
03341         nhalfc--;
03342
03343     /* takes care of the number never expanding, e.g., -1(mod 100) -> 99 -> -1 */
03344     if (BigLong(a,ABS(*na),halfc,nhalfc) > 0) {
03345
03346         TakeModulus(a,na,c,nc,par);
03347
03348         n = ABS(*na);
03349         if (BigLong(a,n,halfc,nhalfc) > 0) {
03350             SubPLon(c,nc,a,n,a,&n);
03351             *na = (*na > 0 ? -n : n);
03352         }
03353     }
03354
03355     NumberFree(halfc,"TakeNormalModulus");
03356     return(0);
03357 }
03358
03359 /*
03360     #] TakeNormalModulus :
03361     #[ MakeModTable : WORD MakeModTable()
03362 */
03363
03364 int MakeModTable(void)
03365 {
03366     LONG size, i, j, n;
03367     n = ABS(AC.ncmod);
03368     if ( AC.modpowers ) {
03369         M_free(AC.modpowers,"AC.modpowers");
03370         AC.modpowers = NULL;
03371     }
03372     if ( n > 2 ) {
03373         MLOCK(ErrorMessageLock);
03374         MesPrint("%No memory for modulus generator power table");
03375         MUNLOCK(ErrorMessageLock);
03376         Terminate(-1);
03377     }
03378     if ( n == 0 ) return(0);
03379     size = (LONG) (*AC.cmod);
03380     if ( n == 2 ) size += ((LONG)AC.cmod[1]<<BITSINWORD);
03381     AC.modpowers = (UWORD *)Malloca(size*n*sizeof(UWORD),"table for powers of modulus");
03382     if ( n == 1 ) {
03383         j = 1;
03384         for ( i = 0; i < size; i++ ) AC.modpowers[i] = 0;
03385         for ( i = 0; i < size; i++ ) {
03386             AC.modpowers[j] = (WORD)i;
03387             j *= *AC.powmod;
03388             j %= *AC.cmod;
03389         }
03390         for ( i = 2; i < size; i++ ) {
03391             if ( AC.modpowers[i] == 0 ) {
03392                 MLOCK(ErrorMessageLock);
03393                 MesPrint("&improper generator for this modulus");
03394                 MUNLOCK(ErrorMessageLock);
03395                 M_free(AC.modpowers,"AC.modpowers");
03396                 return(-1);

```

```

03397     }
03398     }
03399     AC.modpowers[1] = 0;
03400 }
03401 else {
03402     GETIDENTITY
03403     WORD nSkrat, n2;
03404     UWORD *MMskrat7 = NumberMalloc("MakeModTable"), *MMskratC = NumberMalloc("MakeModTable");
03405     *MMskratC = 1;
03406     nSkrat = 1;
03407     j = size * 2;
03408     for ( i = 0; i < j; i+=2 ) { AC.modpowers[i] = 0; AC.modpowers[i+1] = 0; }
03409     for ( i = 0; i < size; i++ ) {
03410         j = *MMskratC + (((LONG)MMskratC[1])«BITSINWORD);
03411         j *= 2;
03412         AC.modpowers[j] = (WORD)(i & WORDMASK);
03413         AC.modpowers[j+1] = (WORD)(i » BITSINWORD);
03414         Mullong((UWORD *)MMskratC,nSkrat,(UWORD *)AC.powmod,
03415             AC.npowmod,(UWORD *)MMskrat7,&n2);
03416         TakeModulus(MMskrat7,&n2,AC.cmod,AC.ncmod,NOUNPACK);
03417         *MMskratC = *MMskrat7; MMskratC[1] = MMskrat7[1]; nSkrat = n2;
03418     }
03419     NumberFree(MMskrat7,"MakeModTable"); NumberFree(MMskratC,"MakeModTable");
03420     j = size * 2;
03421     for ( i = 4; i < j; i+=2 ) {
03422         if ( AC.modpowers[i] == 0 && AC.modpowers[i+1] == 0 ) {
03423             MLOCK(ErrorMessageLock);
03424             MesPrint("&improper generator for this modulus");
03425             MUNLOCK(ErrorMessageLock);
03426             M_free(AC.modpowers,"AC.modpowers");
03427             return(-1);
03428         }
03429     }
03430     AC.modpowers[2] = AC.modpowers[3] = 0;
03431 }
03432 return(0);
03433 }
03434
03435 /*
03436     #] MakeModTable :
03437     #] RekenTerms :
03438     #[ Functions :
03439         #[ Factorial :      WORD Factorial(n,a,na)
03440
03441     Starts with only the value of fac_(0).
03442     Builds up what is needed and remembers it for the next time.
03443
03444     We have:
03445         AT.nfac: the number of the highest stored factorial
03446         AT.pfac: the array of locations in the array of stored factorials
03447         AT.factorials: the array with stored factorials
03448 */
03449
03450 int Factorial(PHEAD WORD n, UWORD *a, WORD *na)
03451 {
03452     GETBIDENTITY
03453     UWORD *b, *c;
03454     WORD nc;
03455     int i, j;
03456     LONG ii;
03457     if ( n > AT.nfac ) {
03458         if ( AT.factorials == 0 ) {
03459             AT.nfac = 0; AT.mfac = 50; AT.sfact = 400;
03460             AT.pfac = (LONG *)Mallocl1((AT.mfac+2)*sizeof(LONG),"factorials");
03461             AT.factorials = (UWORD *)Mallocl1(AT.sfact*sizeof(UWORD),"factorials");
03462             AT.factorials[0] = 1; AT.pfac[0] = 0; AT.pfac[1] = 1;
03463         }
03464         b = a;
03465         c = AT.factorials+AT.pfac[AT.nfac];
03466         nc = i = AT.pfac[AT.nfac+1] - AT.pfac[AT.nfac];
03467         while ( --i >= 0 ) *b++ = *c++;
03468         for ( j = AT.nfac+1; j <= n; j++ ) {
03469             Product(a,&nc,j);
03470             if ( nc > AM.MaxTal ) {
03471                 MLOCK(ErrorMessageLock);
03472                 MesPrint("Overflow in factorial. MaxTal = %d",AM.MaxTal);
03473                 MesPrint("Increase MaxTerm in %s",setupfilename);
03474                 MUNLOCK(ErrorMessageLock);
03475                 return(-1);
03476             }
03477             if ( j > AT.mfac ) { /* double the pfac buffer */
03478                 LONG *p;
03479                 p = (LONG *)Mallocl1((AT.mfac*2+2)*sizeof(LONG),"factorials");
03480                 i = AT.mfac;
03481                 for ( i = AT.mfac+1; i >= 0; i-- ) p[i] = AT.pfac[i];
03482                 M_free(AT.pfac,"factorial offsets"); AT.pfac = p; AT.mfac *= 2;
03483             }

```

```

03484         if ( AT.pfac[j] + nc >= AT.sfact ) { /* double the factorials buffer */
03485             UWORD *f;
03486             f = (UWORD *)Malloc1(AT.sfact*2*sizeof(UWORD),"factorials");
03487             ii = AT.sfact;
03488             c = AT.factorials; b = f;
03489             while ( --ii >= 0 ) *b++ = *c++;
03490             M_free(AT.factorials,"factorials");
03491             AT.factorials = f;
03492             AT.sfact *= 2;
03493         }
03494         b = a; c = AT.factorials + AT.pfac[j]; i = nc;
03495         while ( --i >= 0 ) *c++ = *b++;
03496         AT.pfac[j+1] = AT.pfac[j] + nc;
03497     }
03498     *na = nc;
03499     AT.nfac = n;
03500 }
03501 else if ( n == 0 ) {
03502     *a = 1; *na = 1;
03503 }
03504 else {
03505     *na = i = AT.pfac[n+1] - AT.pfac[n];
03506     b = AT.factorials + AT.pfac[n];
03507     while ( --i >= 0 ) *a++ = *b++;
03508 }
03509 return(0);
03510 }
03511
03512 /*
03513     #] Factorial :
03514     #[ Bernoulli :      WORD Bernoulli(n,a,na)
03515
03516     Starts with only the value of bernoulli_(0).
03517     Builds up what is needed and remembers it for the next time.
03518     b_0 = 1
03519     (n+1)*b_n = -b_{n-1}-sum_(i,1,n-1,b_i*b_{n-i})
03520     The n-1 plays only a role for b_2.
03521     We have hard coded b_0,b_1,b_2 and b_odd. After that:
03522     (2n+1)*b_{2n} = -sum_(i,1,n-1,b_{2i}*b_{2n-2i})
03523
03524     We have:
03525     AT.nBer: the number of the highest stored Bernoulli number
03526     AT.pBer: the array of locations in the array of stored Bernoulli numbers
03527     AT.bernoullis: the array with stored Bernoulli numbers
03528 */
03529
03530 int Bernoulli(WORD n, UWORD *a, WORD *na)
03531 {
03532     GETIDENTITY
03533     UWORD *b, *c, *scrib, *ntop, *ntopl;
03534     WORD i, i1, i2, nhalf, nqua, nscrib, nntop, nntopl, *oldworkpointer;
03535     UWORD twee = 2, twonplus1;
03536     int j;
03537     LONG ii;
03538     if ( n <= 1 ) {
03539         if ( n == 0 ) { a[0] = a[1] = 1; *na = 3; }
03540         else if ( n == 1 ) { a[0] = 1; a[1] = 2; *na = 3; }
03541         return(0);
03542     }
03543     if ( ( n & 1 ) != 0 ) { a[0] = a[1] = 0; *na = 0; return(0); }
03544     nhalf = n/2;
03545     if ( nhalf > AT.nBer ) {
03546         oldworkpointer = AT.WorkPointer;
03547         if ( AT.bernoullis == 0 ) {
03548             AT.nBer = 1; AT.mBer = 50; AT.sBer = 400;
03549             AT.pBer = (LONG *)Malloc1((AT.mBer+2)*sizeof(LONG),"bernoullis");
03550             AT.bernoullis = (UWORD *)Malloc1(AT.sBer*sizeof(UWORD),"bernoullis");
03551             AT.pBer[1] = 0; AT.pBer[2] = 3;
03552             AT.bernoullis[0] = 3; AT.bernoullis[1] = 1; AT.bernoullis[2] = 12;
03553             if ( nhalf == 1 ) {
03554                 a[0] = 1; a[1] = 12; *na = 3; return(0);
03555             }
03556         }
03557         while ( nhalf > AT.mBer ) {
03558             LONG *p;
03559             p = (LONG *)Malloc1((AT.mBer*2+1)*sizeof(LONG),"bernoullis");
03560             i = AT.mBer;
03561             for ( i = AT.mBer; i >= 0; i-- ) p[i] = AT.pBer[i];
03562             M_free(AT.pBer,"factorial pointers"); AT.pBer = p; AT.mBer *= 2;
03563         }
03564         for ( n = AT.nBer+1; n <= nhalf; n++ ) {
03565             scrib = (UWORD *) (AT.WorkPointer);
03566             nqua = n/2;
03567             if ( ( n & 1 ) == 1 ) {
03568                 nscrib = 0; ntop = scrib;
03569             }
03570             else {

```

```

03571         b = AT.bernoullis + AT.pBer[nqua];
03572         nscrib = *b++;
03573         i = (WORD)(REDLENG(nscrib));
03574         MulRat(BHEAD b,i,b,i,scrib,&nscrib);
03575         ntop = scrib + 2*nscrib;
03576         nqua--;
03577     }
03578     for ( j = 1; j <= nqua; j++ ) {
03579         b = AT.bernoullis + AT.pBer[j];
03580         c = AT.bernoullis + AT.pBer[n-j];
03581         i1 = (WORD)(*b); i2 = (WORD)(*c);
03582         i1 = REDLENG(i1);
03583         i2 = REDLENG(i2);
03584         MulRat(BHEAD b+1,i1,c+1,i2,ntop,&nntop);
03585         Mullt(BHEAD ntop,&nntop,&twee,1);
03586         if ( nscrib ) {
03587             i = (WORD)nntop; if ( i < 0 ) i = -i;
03588             ntop1 = ntop + 2*i;
03589             AddRat(BHEAD ntop,nntop,scrib,nscrib,ntop1,&nntop1);
03590         }
03591         else {
03592             ntop1 = ntop; nntop1 = nntop;
03593         }
03594         nscrib = i1 = (WORD)nntop1;
03595         if ( i1 < 0 ) i1 = -i1;
03596         i1 = 2*i1;
03597         for ( i = 0; i < i1; i++ ) scrib[i] = ntop1[i];
03598         ntop = scrib + i1;
03599     }
03600     twonplus1 = 2*n+1;
03601     Divvy(BHEAD scrib,&nscrib,&twonplus1,-1);
03602     i1 = INCLENG(nscrib);
03603     i2 = i1; if ( i2 < 0 ) i2 = -i2;
03604     i = (WORD)(AT.bernoullis[AT.pBer[n-1]]);
03605     if ( i < 0 ) i = -i;
03606     AT.pBer[n] = AT.pBer[n-1]+i;
03607     if ( AT.pBer[n] + i2 >= AT.sBer ) {
03608         UWORD *f;
03609         f = (UWORD *)Malloca(AT.sBer*2*sizeof(UWORD),"bernoullis");
03610         ii = AT.sBer;
03611         c = AT.bernoullis; b = f;
03612         while ( --ii >= 0 ) *b++ = *c++;
03613         M_free(AT.bernoullis,"bernoullis");
03614         AT.bernoullis = f;
03615         AT.sBer *= 2;
03616     }
03617     c = AT.bernoullis + AT.pBer[n]; b = scrib;
03618     *c++ = i1;
03619     for ( i = 1; i < i2; i++ ) *c++ = *b++;
03620 }
03621 AT.nBer = nhalf;
03622 AT.WorkPointer = oldworkpointer;
03623 }
03624 b = AT.bernoullis + AT.pBer[nhalf];
03625 *na = i = (WORD)(*b++);
03626 if ( i < 0 ) i = -i;
03627 i--;
03628 while ( --i >= 0 ) *a++ = *b++;
03629 return(0);
03630 }
03631
03632 /*
03633     #] Bernoulli :
03634     #[ NextPrime :
03635 */
03647 #if ( BITSINWORD == 32 )
03648
03649 void StartPrimeList(PHEAD0)
03650 {
03651     int i, j;
03652     AR.PrimeList[AR.numinprimelist++] = 3;
03653     for ( i = 5; i < 46340; i += 2 ) {
03654         for ( j = 0; j < AR.numinprimelist && AR.PrimeList[j]*AR.PrimeList[j] <= i; j++ ) {
03655             if ( i % AR.PrimeList[j] == 0 ) goto nexti;
03656         }
03657         AR.PrimeList[AR.numinprimelist++] = i;
03658     nexti:;
03659     }
03660     AR.notfirstprime = 1;
03661 }
03662 #endif
03663
03664
03665 WORD NextPrime(PHEAD WORD num)
03666 {
03667     int i, j;
03668     WORD *newpl;

```

```

03669     LONG newsize, x;
03670 #if ( BITSINWORD == 32 )
03671     if ( AR.notfirstprime == 0 ) StartPrimeList(BHEAD0);
03672 #endif
03673     if ( num > AT.inprimelist ) {
03674         while ( AT.inprimelist < num ) {
03675             if ( num >= AT.sizeprimelist ) {
03676                 if ( AT.sizeprimelist == 0 ) newsize = 32;
03677                 else newsize = 2*AT.sizeprimelist;
03678                 while ( num >= newsize ) newsize = newsize*2;
03679                 newpl = (WORD *)Malloc1(newsize*sizeof(WORD),"NextPrime");
03680                 for ( i = 0; i < AT.sizeprimelist; i++ ) {
03681                     newpl[i] = AT.primelist[i];
03682                 }
03683                 if ( AT.sizeprimelist > 0 ) {
03684                     M_free(AT.primelist,"NextPrime");
03685                 }
03686                 AT.sizeprimelist = newsize;
03687                 AT.primelist = newpl;
03688             }
03689             if ( AT.inprimelist < 0 ) { i = MAXPOSITIVE; }
03690             else { i = AT.primelist[AT.inprimelist]; }
03691             while ( i > MAXPOWER ) {
03692                 i -= 2; x = i;
03693 #if ( BITSINWORD == 32 )
03694                 for ( j = 0; j < AR.numinprimelist && AR.PrimeList[j]*(LONG)(AR.PrimeList[j]) <= x;
03695                     j++ ) {
03696                     if ( x % AR.PrimeList[j] == 0 ) goto nexti;
03697                 }
03698             #else
03699                 for ( j = 3; j*((LONG)j) <= x; j += 2 ) {
03700                     if ( x % j == 0 ) goto nexti;
03701                 }
03702             #endif
03703             AT.inprimelist++;
03704             AT.primelist[AT.inprimelist] = i;
03705             break;
03706         nexti:;
03707         }
03708         if ( i < MAXPOWER ) {
03709             MLOCK(ErrorMessageLock);
03710             MesPrint("There are not enough short prime numbers for this calculation");
03711             MesPrint("Try to use a computer with a %d-bits architecture",
03712                 (int)(BITSINWORD*4));
03713             MUNLOCK(ErrorMessageLock);
03714             Terminate(-1);
03715         }
03716     }
03717     return(AT.primelist[num]);
03718 }
03719
03720 /*
03721     #] NextPrime :
03722     #[ Moebius :
03723
03724     Returns the value of the Moebius function if n fits inside
03725     a WORD.
03726     The method we use is a bit like the sieve of Erathostenes.
03727 */
03728
03729 WORD Moebius(PHEAD WORD nn)
03730 {
03731     WORD i,n = nn, x;
03732     LONG newsize;
03733     SBYTE *newtable, mu;
03734 #if ( BITSINWORD == 32 )
03735     if ( AR.notfirstprime == 0 ) StartPrimeList(BHEAD0);
03736 #endif
03737 /*
03738     First we make sure that:
03739     a: the table is big enough.
03740     b: the number is not already in the table.
03741 */
03742     if ( nn >= AR.moebiustablesz ) {
03743         if ( AR.moebiustablesz <= 0 ) { newsize = (LONG)nn + 20; }
03744         else { newsize = (LONG)nn*2; }
03745         if ( newsize > MAXPOSITIVE ) newsize = MAXPOSITIVE;
03746         newtable = (SBYTE *)Malloc1(newsize*sizeof(SBYTE),"Moebius");
03747         for ( i = 0; i < AR.moebiustablesz; i++ ) newtable[i] = AR.moebiustable[i];
03748         for ( ; i < newsize; i++ ) newtable[i] = 2;
03749         if ( AR.moebiustablesz > 0 ) M_free(AR.moebiustable,"Moebius");
03750         AR.moebiustable = newtable;
03751         AR.moebiustablesz = newsize;
03752     }
03753     /* NOTE: nn == MAXPOSITIVE never fits in moebiustable. */
03754     if ( nn != MAXPOSITIVE && AR.moebiustable[nn] != 2 ) return((WORD)AR.moebiustable[nn]);

```

```

03755     mu = 1;
03756     if ( n == 1 ) goto putvalue;
03757     if ( n % 2 == 0 ) {
03758         n /= 2;
03759         if ( n % 2 == 0 ) { mu = 0; goto putvalue; }
03760         if ( AR.moebiustable[n] != 2 ) { mu = -AR.moebiustable[n]; goto putvalue; }
03761         mu = -mu;
03762         if ( n == 1 ) goto putvalue;
03763     }
03764     #if ( BITSINWORD == 32 )
03765     for ( i = 0; i < AR.numinprimelist; i++ ) {
03766         x = AR.PrimeList[i];
03767     #else
03768     for ( x = 3; x < MAXPOSITIVE; x += 2 ) {
03769     #endif
03770         if ( n % x == 0 ) {
03771             n /= x;
03772             if ( n % x == 0 ) { mu = 0; goto putvalue; }
03773             if ( AR.moebiustable[n] != 2 ) { mu = -AR.moebiustable[n]; goto putvalue; }
03774             mu = -mu;
03775             if ( n == 1 ) goto putvalue;
03776         }
03777         if ( n < x*x ) break; /* notice that x*x always fits inside a WORD */
03778     }
03779     mu = -mu;
03780     putvalue:
03781     if ( nn != MAXPOSITIVE ) AR.moebiustable[nn] = mu;
03782     return ( (WORD)mu );
03783 }
03784
03785 /*
03786     #] Moebius :
03787     #[ wranf :
03788
03789     A random number generator that generates random WORDs with a very
03790     long sequence. It is based on the Knuth generator.
03791
03792     We take some care that each thread can run its own, but each
03793     uses its own startup. Hence the seed includes the identity of
03794     the thread.
03795
03796     For NPAIR1, NPAIR2 we can use any pair from the table on page 28.
03797     Candidates are 24,55 (the example on the pages 171,172)
03798     or (33,97) or (38,89)
03799     These values are defined in fsizes.h and used in startup.c and threads.c
03800 */
03801
03802 #define WARMUP 6
03803
03804 static void wranfnew(PHEAD0)
03805 {
03806     int i;
03807     LONG j;
03808     for ( i = 0; i < AR.wranfnpair1; i++ ) {
03809         j = AR.wranfia[i] - AR.wranfia[i+(AR.wranfnpair2-AR.wranfnpair1)];
03810         if ( j < 0 ) j += (LONG)1 « (2*BITSINWORD-2);
03811         AR.wranfia[i] = j;
03812     }
03813     for ( i = AR.wranfnpair1; i < AR.wranfnpair2; i++ ) {
03814         j = AR.wranfia[i] - AR.wranfia[i-AR.wranfnpair1];
03815         if ( j < 0 ) j += (LONG)1 « (2*BITSINWORD-2);
03816         AR.wranfia[i] = j;
03817     }
03818 }
03819
03820 void iniwranf(PHEAD0)
03821 {
03822     int imax = AR.wranfnpair2-1;
03823     ULONG i, ii, seed = AR.wranfseed;
03824     LONG j, k;
03825     ULONG offset = 12345;
03826     #ifdef PARALLELCODE
03827     int id;
03828     #if defined(WITHPTHREADS)
03829     id = AT.identity;
03830     #elif defined(WITHMPI)
03831     id = PF.me;
03832     #endif
03833     seed += id;
03834     i = id + 1;
03835     if ( i > 1 ) {
03836         ULONG pow, accu;
03837         pow = offset; accu = 1;
03838         while ( i ) {
03839             if ( ( i & 1 ) != 0 ) accu *= pow;
03840             i /= 2; pow = pow*pow;
03841         }

```

```

03842     offset = accu;
03843 }
03844 #endif
03845 if ( seed < ((LONG)1«(BITSINWORD-1)) ) {
03846     j = ( (seed+31459L) « (BITSINWORD-2))+offset;
03847 }
03848 else if ( seed < ((LONG)1«(BITSINWORD+10-1)) ) {
03849     j = ( (seed+31459L) « (BITSINWORD-10-2))+offset;
03850 }
03851 else {
03852     j = ( (seed+31459L) « 1)+offset;
03853 }
03854 if ( ( seed & 1 ) == 1 ) seed++;
03855 j += seed;
03856 AR.wranfia[imax] = j;
03857 k = 1;
03858 for ( i = 0; i <= (ULONG)(imax); i++ ) {
03859     ii = (AR.wranfnpair1*i)%AR.wranfnpair2;
03860     AR.wranfia[ii] = k;
03861     k = ULongToLong((ULONG)j - (ULONG)k);
03862     if ( k < 0 ) k += (LONG)1 « (2*BITSINWORD-2);
03863     j = AR.wranfia[ii];
03864 }
03865 for ( i = 0; i < WARMUP; i++ ) wranfnew(BHEAD0);
03866 AR.wranfcall = 0;
03867 }
03868
03869 UWORD wranf(PHEAD0)
03870 {
03871     UWORD wval;
03872     if ( AR.wranfia == 0 ) {
03873         AR.wranfia = (ULONG *)Malloc1(AR.wranfnpair2*sizeof(ULONG),"wranf");
03874         iniwranf(BHEAD0);
03875     }
03876     if ( AR.wranfcall >= AR.wranfnpair2 ) {
03877         wranfnew(BHEAD0);
03878         AR.wranfcall = 0;
03879     }
03880     wval = (UWORD) (AR.wranfia[AR.wranfcall++])«(BITSINWORD-1));
03881     return(wval);
03882 }
03883
03884 /*
03885  Returns a random UWORD in the range (0,...,imax-1)
03886 */
03887
03888 UWORD iranf(PHEAD UWORD imax)
03889 {
03890     UWORD i;
03891     ULONG x, xmax;
03892     if (imax < 2) return 0;
03893     x = (LONG)1 « BITSINWORD;
03894     xmax = x - x%imax;
03895     while ( ( i = wranf(BHEAD0) ) >= xmax ) {}
03896     return(i%imax);
03897 }
03898
03899 /*
03900     #] wranf :
03901     #[ PreRandom :
03902
03903     The random number generator of the preprocessor.
03904     This one is completely different from the execution time generator
03905     random_(number). In the preprocessor we generate a floating point
03906     number in a string according to a distribution.
03907     Currently allowed are:
03908         RANDOM_(log,min,max)
03909         RANDOM_(lin,min,max)
03910     The return value is a string with the floating point number.
03911 */
03912
03913 UBYTE *PreRandom(UBYTE *s)
03914 {
03915     GETIDENTITY
03916     UBYTE *mode,*mins = 0,*maxs = 0, *outval;
03917     float num;
03918     double minval, maxval, value = 0;
03919     int linlog = -1;
03920     mode = s;
03921     while ( FG.cTable[*s] <= 1 ) s++;
03922     if ( *s == ',' ) { *s = 0; s++; }
03923     mins = s;
03924     while ( *s && *s != ',' ) s++;
03925     if ( *s == ',' ) { *s = 0; s++; }
03926     maxs = s;
03927     while ( *s && *s != ',' ) s++;
03928     if ( *s || *maxs == 0 || *mins == 0 ) {

```



```

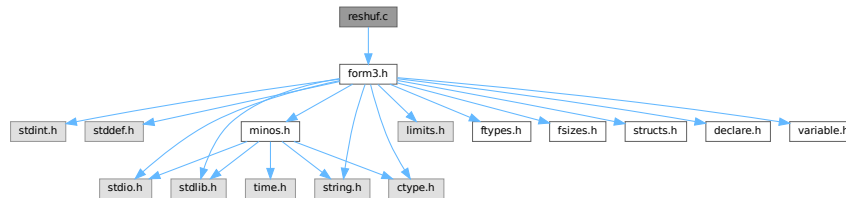
03929     MesPrint("@Illegal arguments in macro RANDOM_");
03930     Terminate(-1);
03931 }
03932 if ( StrICmp(mode, (UBYTE *)"lin") == 0 ) {
03933     linlog = 0;
03934 }
03935 else if ( StrICmp(mode, (UBYTE *)"log") == 0 ) {
03936     linlog = 1;
03937 }
03938 else {
03939     MesPrint("@Illegal mode argument in macro RANDOM_");
03940     Terminate(-1);
03941 }
03942
03943 sscanf((char *)mins, "%f", &num); minval = num;
03944 sscanf((char *)maxs, "%f", &num); maxval = num;
03945
03946 /*
03947  * Note on ParFORM: we should use the same random number on all the
03948  * processes in the compilation phase. The random number is generated
03949  * on the master and broadcast to the other processes.
03950  */
03951 {
03952     UWORD x;
03953     double xx;
03954 #ifdef WITHMPI
03955     x = 0;
03956     if ( PF.me == MASTER ) {
03957         x = wranf(BHEAD0);
03958     }
03959     x = (UWORD)PF_BroadcastNumber((LONG)x);
03960 #else
03961     x = wranf(BHEAD0);
03962 #endif
03963     xx = x/pow(2.0, (double) (BITSINWORD-1));
03964     if ( linlog == 0 ) {
03965         value = minval + (maxval-minval)*xx;
03966     }
03967     else if ( linlog == 1 ) {
03968         value = minval * pow(maxval/minval, xx);
03969     }
03970 }
03971
03972 outval = (UBYTE *)Malloca(64, "PreRandom");
03973 if ( ABS(value) < 0.00001 || ABS(value) > 1000000. ) {
03974     snprintf((char *)outval, 64, "%e", value);
03975 }
03976 else if ( ABS(value) < 0.0001 ) { snprintf((char *)outval, 64, "%10f", value); }
03977 else if ( ABS(value) < 0.001 ) { snprintf((char *)outval, 64, "%9f", value); }
03978 else if ( ABS(value) < 0.01 ) { snprintf((char *)outval, 64, "%8f", value); }
03979 else if ( ABS(value) < 0.1 ) { snprintf((char *)outval, 64, "%7f", value); }
03980 else if ( ABS(value) < 1. ) { snprintf((char *)outval, 64, "%6f", value); }
03981 else if ( ABS(value) < 10. ) { snprintf((char *)outval, 64, "%5f", value); }
03982 else if ( ABS(value) < 100. ) { snprintf((char *)outval, 64, "%4f", value); }
03983 else if ( ABS(value) < 1000. ) { snprintf((char *)outval, 64, "%3f", value); }
03984 else if ( ABS(value) < 10000. ) { snprintf((char *)outval, 64, "%2f", value); }
03985 else { snprintf((char *)outval, 64, "%1f", value); }
03986 return(outval);
03987 }
03988
03989 /*
03990     #] PreRandom :
03991     #] Functions :
03992 */

```

## 4.120 reshuf.c File Reference

```
#include "form3.h"
```

Include dependency graph for reshuf.c:



### Functions

- WORD [ReNumber](#) (PHEAD WORD \*term)
- void [FunLevel](#) (PHEAD WORD \*term)
- WORD [DetCurDum](#) (PHEAD WORD \*t)
- int [FullRenumber](#) (PHEAD WORD \*term, WORD par)
- void [MoveDummies](#) (PHEAD WORD \*term, WORD shift)
- void [AdjustRenumScratch](#) (PHEAD0)
- WORD [CountDo](#) (WORD \*term, WORD \*instruct)
- WORD [CountFun](#) (WORD \*term, WORD \*countfun)
- WORD [DimensionSubterm](#) (WORD \*subterm)
- WORD [DimensionTerm](#) (WORD \*term)
- WORD [DimensionExpression](#) (PHEAD WORD \*expr)
- int [MultDo](#) (PHEAD WORD \*term, WORD \*pattern)
- int [TryDo](#) (PHEAD WORD \*term, WORD \*pattern, WORD level)
- int [DoDistrib](#) (PHEAD WORD \*term, WORD level)
- int [EqualArg](#) (WORD \*parms, WORD num1, WORD num2)
- int [DoDelta3](#) (PHEAD WORD \*term, WORD level)
- int [TestPartitions](#) (WORD \*tfun, [PARTI](#) \*parti)
- int [DoPartitions](#) (PHEAD WORD \*term, WORD level)
- int [DoPermutations](#) (PHEAD WORD \*term, WORD level)
- int [DoShuffle](#) (WORD \*term, WORD level, WORD fun, WORD option)
- int [Shuffle](#) (WORD \*from1, WORD \*from2, WORD \*to)
- int [FinishShuffle](#) (WORD \*fini)
- int [DoStuffle](#) (WORD \*term, WORD level, WORD fun, WORD option)
- int [Stuffle](#) (WORD \*from1, WORD \*from2, WORD \*to)
- int [FinishStuffle](#) (WORD \*fini)
- WORD \* [StuffRootAdd](#) (WORD \*t1, WORD \*t2, WORD \*to)

### 4.120.1 Detailed Description

Mixed routines: Routines for relabelling dummy indices. The multiply command The distrib\_ function The tryreplace statement

Definition in file [reshuf.c](#).

## 4.120.2 Macro Definition Documentation

### 4.120.2.1 NEWCODE

```
#define NEWCODE
```

Definition at line 35 of file [reshuf.c](#).

## 4.120.3 Function Documentation

### 4.120.3.1 ReNumber()

```
WORD ReNumber (
    PHEAD WORD * term )
```

Definition at line 80 of file [reshuf.c](#).

### 4.120.3.2 FunLevel()

```
void FunLevel (
    PHEAD WORD * term )
```

Definition at line 130 of file [reshuf.c](#).

### 4.120.3.3 DetCurDum()

```
WORD DetCurDum (
    PHEAD WORD * t )
```

Definition at line 252 of file [reshuf.c](#).

### 4.120.3.4 FullRenumber()

```
int FullRenumber (
    PHEAD WORD * term,
    WORD par )
```

Definition at line 324 of file [reshuf.c](#).

### 4.120.3.5 MoveDummies()

```
void MoveDummies (
    PHEAD WORD * term,
    WORD shift )
```

Definition at line 436 of file [reshuf.c](#).

#### 4.120.3.6 AdjustRenumScratch()

```
void AdjustRenumScratch (
    PHEAD0 )
```

Definition at line 506 of file [reshuf.c](#).

#### 4.120.3.7 CountDo()

```
WORD CountDo (
    WORD * term,
    WORD * instruct )
```

Definition at line 552 of file [reshuf.c](#).

#### 4.120.3.8 CountFun()

```
WORD CountFun (
    WORD * term,
    WORD * countfun )
```

Definition at line 706 of file [reshuf.c](#).

#### 4.120.3.9 DimensionSubterm()

```
WORD DimensionSubterm (
    WORD * subterm )
```

Definition at line 867 of file [reshuf.c](#).

#### 4.120.3.10 DimensionTerm()

```
WORD DimensionTerm (
    WORD * term )
```

Definition at line 968 of file [reshuf.c](#).

#### 4.120.3.11 DimensionExpression()

```
WORD DimensionExpression (
    PHEAD WORD * expr )
```

Definition at line 1000 of file [reshuf.c](#).

#### 4.120.3.12 MultDo()

```
int MultDo (
    PHEAD WORD * term,
    WORD * pattern )
```

Definition at line [1044](#) of file [reshuf.c](#).

#### 4.120.3.13 TryDo()

```
int TryDo (
    PHEAD WORD * term,
    WORD * pattern,
    WORD level )
```

Definition at line [1072](#) of file [reshuf.c](#).

#### 4.120.3.14 DoDistrib()

```
int DoDistrib (
    PHEAD WORD * term,
    WORD level )
```

Definition at line [1119](#) of file [reshuf.c](#).

#### 4.120.3.15 EqualArg()

```
int EqualArg (
    WORD * parms,
    WORD num1,
    WORD num2 )
```

Definition at line [1379](#) of file [reshuf.c](#).

#### 4.120.3.16 DoDelta3()

```
int DoDelta3 (
    PHEAD WORD * term,
    WORD level )
```

Definition at line [1405](#) of file [reshuf.c](#).

#### 4.120.3.17 TestPartitions()

```
int TestPartitions (
    WORD * tfun,
    PARTI * parti )
```

Definition at line [1607](#) of file [reshuf.c](#).

#### 4.120.3.18 DoPartitions()

```
int DoPartitions (
    PHEAD WORD * term,
    WORD level )
```

Definition at line 1722 of file [reshuf.c](#).

#### 4.120.3.19 DoPermutations()

```
int DoPermutations (
    PHEAD WORD * term,
    WORD level )
```

Definition at line 2007 of file [reshuf.c](#).

#### 4.120.3.20 DoShuffle()

```
int DoShuffle (
    WORD * term,
    WORD level,
    WORD fun,
    WORD option )
```

Definition at line 2095 of file [reshuf.c](#).

#### 4.120.3.21 Shuffle()

```
int Shuffle (
    WORD * from1,
    WORD * from2,
    WORD * to )
```

Definition at line 2229 of file [reshuf.c](#).

#### 4.120.3.22 FinishShuffle()

```
int FinishShuffle (
    WORD * fini )
```

Definition at line 2421 of file [reshuf.c](#).

#### 4.120.3.23 DoStuffle()

```
int DoStuffle (
    WORD * term,
    WORD level,
    WORD fun,
    WORD option )
```

Definition at line 2487 of file [reshuf.c](#).

#### 4.120.3.24 Stuffle()

```
int Stuffle (
    WORD * from1,
    WORD * from2,
    WORD * to )
```

Definition at line 2702 of file [reshuf.c](#).

#### 4.120.3.25 FinishStuffle()

```
int FinishStuffle (
    WORD * fini )
```

Definition at line 2829 of file [reshuf.c](#).

#### 4.120.3.26 StuffRootAdd()

```
WORD * StuffRootAdd (
    WORD * t1,
    WORD * t2,
    WORD * to )
```

Definition at line 2876 of file [reshuf.c](#).

## 4.121 reshuf.c

[Go to the documentation of this file.](#)

```
00001
00009 /* #[] License : */
00010 /*
00011 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00012 * When using this file you are requested to refer to the publication
00013 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00014 * This is considered a matter of courtesy as the development was paid
00015 * for by FOM the Dutch physics granting agency and we would like to
00016 * be able to track its scientific use to convince FOM of its value
00017 * for the community.
00018 *
00019 * This file is part of FORM.
00020 *
00021 * FORM is free software: you can redistribute it and/or modify it under the
00022 * terms of the GNU General Public License as published by the Free Software
00023 * Foundation, either version 3 of the License, or (at your option) any later
00024 * version.
00025 *
00026 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00027 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00028 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00029 * details.
00030 *
00031 * You should have received a copy of the GNU General Public License along
00032 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00033 */
00034 /* #[] License : */
00035 #define NEWCODE
00036 /*
00037 #[] Includes : reshuf.c
00038 */
00039
00040 #include "form3.h"
00041
00042 /*
00043 #[] Includes :
```

```

00044      #[ Reshuf :
00045
00046      Routines to rearrange dummy indices, so that
00047      a: The notation becomes reasonably unique (the perfect job
00048         may consume very much time).
00049      b: The value of AR.CurDum is reset.
00050
00051      Also some routines are needed to aid in the reading of stored
00052      expressions. Also in those expressions there can be dummy
00053      indices, and there should be no conflict with the already
00054      existing dummies.
00055
00056      #[ ReNumber :
00057
00058      Reads the term, tests for dummies, and puts them in order.
00059      Note that this is kind of a first order approximation.
00060      There is quite some room to make this routine 'smart'
00061      First order:
00062          First index will be lowest, second will be next etc.
00063      Second order:
00064          Functions with more than one index and symmetry properties
00065          have some look ahead to see which index is the first to
00066          find its partner.
00067      Third order:
00068          Take the ordering of the functions into account.
00069      Fourth order:
00070          Try all permutations and see which one gives the 'minimal' term.
00071      Currently we use only the first order.
00072
00073      We need a scratch array for the numbers we find, and one for
00074      the addresses at which these numbers are.
00075      We can use the space for the Scrut arrays. There are 13 of those
00076      and each is AM.MaxTal UWORDS long.
00077
00078  */
00079
00080  WORD ReNumber(PHEAD WORD *term)
00081  {
00082      GETBIDENTITY
00083      WORD *d, *e, **p, **f;
00084      WORD n, i, j, old;
00085      AN.DumFound = AN.RenumScratch;
00086      AN.DumPlace = AN.PoinScratch;
00087      AN.DumFunPlace = AN.FunScratch;
00088      AN.NumFound = 0;
00089      FunLevel(BHEAD term);
00090      d = AN.RenumScratch;
00091      p = AN.PoinScratch;
00092      f = AN.FunScratch;
00093  /*
00094      Now the first level sorting.
00095  */
00096      i = AN.IndDum;
00097      n = AN.NumFound;
00098      while ( --n >= 0 ) {
00099          if ( *d > 0 ) {
00100              old = **p;
00101              **p = ++i;
00102              if ( *f ) **f |= DIRTYSYMFLAG;
00103              e = d;
00104              e++;
00105              for ( j = 1; j <= n; j++ ) {
00106                  if ( *e && *(p[j]) == old ) {
00107                      *(p[j]) = i;
00108                      if ( f[j] ) *(f[j]) |= DIRTYSYMFLAG;
00109                      *e = 0;
00110                  }
00111                  e++;
00112              }
00113          }
00114          p++;
00115          d++;
00116          f++;
00117      }
00118      return(i);
00119  }
00120
00121  /*
00122      #] ReNumber :
00123      #[ FunLevel :
00124
00125      Does one term in determining where the dummies are.
00126      Made to work recursively for functions.
00127
00128  */
00129
00130  void FunLevel(PHEAD WORD *term)

```



```

00131 {
00132     GETBIDENTITY
00133     WORD *t, *tstop, *r, *fun;
00134     WORD *m;
00135     t = r = term;
00136     r += *r;
00137     tstop = r - ABS(r[-1]);
00138     t++;
00139     if ( t < tstop ) do {
00140         r = t + t[1];
00141         switch ( *t ) {
00142             case SYMBOL:
00143             case DOTPRODUCT:
00144                 break;
00145             case VECTOR:
00146                 t += 3;
00147                 do {
00148                     if ( *t > AN.IndDum ) {
00149                         if ( AN.NumFound >= AN.MaxRenumScratch ) AdjustRenumScratch(BHEAD0);
00150                         AN.NumFound++;
00151                         *AN.DumFound++ = *t;
00152                         *AN.DumPlace++ = t;
00153                         *AN.DumFunPlace++ = 0;
00154                     }
00155                     t += 2;
00156                 } while ( t < r );
00157                 break;
00158             case SUBEXPRESSION:
00159 /*
00160             Still must hunt down the wildcards(?)
00161 */
00162                 break;
00163             case GAMMA:
00164                 t += FUNHEAD-2;
00165                 /* fall through */
00166             case DELTA:
00167             case INDEX:
00168                 t += 2;
00169                 while ( t < r ) {
00170                     if ( *t > AN.IndDum ) {
00171                         if ( AN.NumFound >= AN.MaxRenumScratch ) AdjustRenumScratch(BHEAD0);
00172                         AN.NumFound++;
00173                         *AN.DumFound++ = *t;
00174                         *AN.DumPlace++ = t;
00175                         *AN.DumFunPlace++ = 0;
00176                     }
00177                     t++;
00178                 }
00179                 break;
00180             case HAAKJE:
00181             case EXPRESSION:
00182             case SNUMBER:
00183             case LNUMBER:
00184                 break;
00185             default:
00186                 if ( *t < FUNCTION ) {
00187                     MLOCK(ErrorMessageLock);
00188                     MesPrint("Unexpected code in ReNumber");
00189                     MUNLOCK(ErrorMessageLock);
00190                     Terminate(-1);
00191                 }
00192                 fun = t+2;
00193                 if ( *t >= FUNCTION && functions[*t-FUNCTION].spec
00194                 >= TENSORFUNCTION ) {
00195                     t += FUNHEAD;
00196                     while ( t < r ) {
00197                         if ( *t > AN.IndDum ) {
00198                             if ( AN.NumFound >= AN.MaxRenumScratch ) AdjustRenumScratch(BHEAD0);
00199                             AN.NumFound++;
00200                             *AN.DumFound++ = *t;
00201                             *AN.DumPlace++ = t;
00202                             *AN.DumFunPlace++ = fun;
00203                         }
00204                         t++;
00205                     }
00206                     break;
00207                 }
00208                 t += FUNHEAD;
00209                 while ( t < r ) {
00210                     if ( *t > 0 ) {
00211
00212                         /* General function. Enter 'recursion'. */
00213
00214                         m = t + *t;
00215                         t += ARGHEAD;
00216                         while ( t < m ) {

```

```

00218             FunLevel(BHEAD t);
00219             t += *t;
00220         }
00221     }
00222     else {
00223         if ( *t == -INDEX ) {
00224             t++;
00225             if ( *t >= AN.IndDum ) {
00226                 if ( AN.NumFound >= AN.MaxRenumScratch ) AdjustRenumScratch(BHEAD0);
00227                 AN.NumFound++;
00228                 *AN.DumFound++ = *t;
00229                 *AN.DumPlace++ = t;
00230                 *AN.DumFunPlace++ = fun;
00231             }
00232             t++;
00233         }
00234         else if ( *t <= -FUNCTION ) t++;
00235         else t += 2;
00236     }
00237 }
00238 break;
00239 }
00240 t = r;
00241 } while ( t < tstop );
00242 }
00243
00244 /*
00245     #] FunLevel :
00246     #[ DetCurDum :
00247
00248     We look for indices in the range AM.IndDum to AM.IndDum+MAXDUMMIES.
00249     The maximum value is returned.
00250 */
00251
00252 WORD DetCurDum(PHEAD WORD *t)
00253 {
00254     GETBIDENTITY
00255     WORD maxval = AN.IndDum;
00256     WORD maxtop = AM.IndDum + WILDOFFSET;
00257     WORD *tstop, *m, *r, i;
00258     tstop = t + *t - 1;
00259     tstop -= ABS(*tstop);
00260     t++;
00261     while ( t < tstop ) {
00262         if ( *t == VECTOR ) {
00263             m = t + 3;
00264             t += t[1];
00265             while ( m < t ) {
00266                 if ( *m > maxval && *m < maxtop ) maxval = *m;
00267                 m += 2;
00268             }
00269         }
00270         else if ( *t == DELTA || *t == INDEX ) {
00271             m = t + 2;
00272         Singles:
00273             t += t[1];
00274             while ( m < t ) {
00275                 if ( *m > maxval && *m < maxtop ) maxval = *m;
00276                 m++;
00277             }
00278         }
00279         else if ( *t >= FUNCTION ) {
00280             if ( functions[*t-FUNCTION].spec >= TENSORFUNCTION ) {
00281                 m = t + FUNHEAD;
00282                 goto Singles;
00283             }
00284             r = t + FUNHEAD;
00285             t += t[1];
00286             while ( r < t ) { /* The arguments */
00287                 if ( *r < 0 ) {
00288                     if ( *r <= -FUNCTION ) r++;
00289                     else if ( *r == -INDEX ) {
00290                         if ( r[1] > maxval && r[1] < maxtop ) maxval = r[1];
00291                         r += 2;
00292                     }
00293                     else r += 2;
00294                 }
00295                 else {
00296                     m = r + ARGHEAD;
00297                     r += *r;
00298                     while ( m < r ) { /* Terms in the argument */
00299                         i = DetCurDum(BHEAD m);
00300                         if ( i > maxval && i < maxtop ) maxval = i;
00301                         m += *m;
00302                     }
00303                 }
00304             }

```

```

00305     }
00306     else {
00307         t += t[1];
00308     }
00309 }
00310 return(maxval);
00311 }
00312
00313 /*
00314     #] DetCurDum :
00315     #[ FullRenumber :
00316
00317     Does a full renumbering. May be slow if there are many indices.
00318     par = 1 Goes with a factorial!
00319     par = 0 All single exchanges only till there is no more improvement.
00320     Notice that there is a hole in the defense with respect to
00321     arguments inside functions inside functions.
00322 */
00323
00324 int FullRenumber(PHEAD WORD *term, WORD par)
00325 {
00326     GETBIDENTITY
00327     WORD *d, **p, **f, *w, *t, *best, *stac, *perm, a, *termtry;
00328     WORD n, i, j, k, ii;
00329     WORD *oldworkpointer = AT.WorkPointer;
00330     n = ReNumber(BHEAD term) - AM.IndDum;
00331     if ( n <= 1 ) return(0);
00332     Normalize(BHEAD term);
00333     if ( *term == 0 ) return(0);
00334     n = ReNumber(BHEAD term) - AM.IndDum;
00335     d = AN.RenumScratch;
00336     p = AN.PoinScratch;
00337     f = AN.FunScratch;
00338     if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
00339     k = AN.NumFound;
00340     best = w = AT.WorkPointer; t = term;
00341     for ( i = *term; i > 0; i-- ) *w++ = *t++;
00342     AT.WorkPointer = w;
00343     Normalize(BHEAD best);
00344     AT.WorkPointer = w = best + *best;
00345     stac = w+100;
00346     perm = stac + n + 1;
00347     termtry = perm + n + 1;
00348     for ( i = 1; i <= n; i++ ) perm[i] = i + AM.IndDum;
00349     for ( i = 1; i <= n; i++ ) stac[i] = i;
00350     for ( i = 0; i < k; i++ ) d[i] = *(p[i]) - AM.IndDum;
00351     if ( par == 0 ) { /* All single exchanges */
00352         for ( i = 1; i < n; i++ ) {
00353             for ( j = i+1; j <= n; j++ ) {
00354                 a = perm[j]; perm[j] = perm[i]; perm[i] = a;
00355                 for ( ii = 0; ii < k; ii++ ) {
00356                     *(p[ii]) = perm[d[ii]];
00357                     if ( f[ii] ) *(f[ii]) |= DIRTYSYMFLAG;
00358                 }
00359                 t = term; w = termtry;
00360                 for ( ii = 0; ii < *term; ii++ ) *w++ = *t++;
00361                 AT.WorkPointer = w;
00362                 if ( Normalize(BHEAD termtry) == 0 ) {
00363                     if ( *termtry == 0 ) goto Return0;
00364                     if ( ( ii = CompareTerms(BHEAD termtry,best,0) ) > 0 ) {
00365                         t = termtry; w = best;
00366                         for ( ii = 0; ii < *termtry; ii++ ) *w++ = *t++;
00367                         i = 0; break; /* restart from beginning */
00368                     }
00369                     else if ( ii == 0 && CompCoef(termtry,best) != 0 )
00370                         goto Return0;
00371                 }
00372                 /* if no success, set back */
00373                 a = perm[j]; perm[j] = perm[i]; perm[i] = a;
00374             }
00375         }
00376     }
00377     else if ( par == 1 ) { /* all permutations */
00378         j = n-1;
00379         for (;;) {
00380             if ( stac[j] == n ) {
00381                 a = perm[j]; perm[j] = perm[n]; perm[n] = a;
00382                 stac[j] = j;
00383                 j--;
00384                 if ( j <= 0 ) break;
00385                 continue;
00386             }
00387             if ( j != stac[j] ) {
00388                 a = perm[j]; perm[j] = perm[stac[j]]; perm[stac[j]] = a;
00389             }
00390             (stac[j])++;
00391             a = perm[j]; perm[j] = perm[stac[j]]; perm[stac[j]] = a;

```

```

00392         {
00393             for ( i = 0; i < k; i++ ) {
00394                 *(p[i]) = perm[d[i]];
00395                 if ( f[i] ) *(f[i]) |= DIRTSYMFFLAG;
00396             }
00397             t = term; w = termtry;
00398             for ( i = 0; i < *term; i++ ) *w++ = *t++;
00399             AT.WorkPointer = w;
00400             if ( Normalize(BHEAD termtry) == 0 ) {
00401                 if ( *termtry == 0 ) goto Return0;
00402                 if ( ( ii = CompareTerms(BHEAD termtry,best,0) ) > 0 ) {
00403                     t = termtry; w = best;
00404                     for ( i = 0; i < *termtry; i++ ) *w++ = *t++;
00405                 }
00406                 else if ( ii == 0 && CompCoef(termtry,best) != 0 )
00407                     goto Return0;
00408             }
00409         }
00410         if ( j < n-1 ) { j = n-1; }
00411     }
00412 }
00413 t = term; w = best;
00414 n = *best;
00415 for ( i = 0; i < n; i++ ) *t++ = *w++;
00416 AT.WorkPointer = oldworkpointer;
00417 return(0);
00418 Return0:
00419 *term = 0;
00420 AT.WorkPointer = oldworkpointer;
00421 return(0);
00422 }
00423
00424 /*
00425     #] FullRenumber :
00426     #[ MoveDummies :
00427
00428     Routine shifts the dummy indices by an amount 'shift'.
00429     It is an adaptation of DetCurDum.
00430     Needed for = ...*expression^power*...
00431     in which expression contains dummy indices.
00432     Note that this code should have been in ver1 already and has
00433     always been missing. Routine made 29-jan-2007.
00434 */
00435
00436 void MoveDummies(PHEAD WORD *term, WORD shift)
00437 {
00438     GETBIDENTITY
00439     WORD maxval = AN.IndDum;
00440     WORD maxtop = AM.IndDum + WILDOFFSET;
00441     WORD *tstop, *m, *r;
00442     tstop = term + *term - 1;
00443     tstop -= ABS(*tstop);
00444     term++;
00445     while ( term < tstop ) {
00446         if ( *term == VECTOR ) {
00447             m = term + 3;
00448             term += term[1];
00449             while ( m < term ) {
00450                 if ( *m > maxval && *m < maxtop ) *m += shift;
00451                 m += 2;
00452             }
00453         }
00454         else if ( *term == DELTA || *term == INDEX ) {
00455             m = term + 2;
00456         Singles:
00457             term += term[1];
00458             while ( m < term ) {
00459                 if ( *m > maxval && *m < maxtop ) *m += shift;
00460                 m++;
00461             }
00462         }
00463         else if ( *term >= FUNCTION ) {
00464             if ( functions[*term-FUNCTION].spec >= TENSORFUNCTION ) {
00465                 m = term + FUNHEAD;
00466                 goto Singles;
00467             }
00468             r = term + FUNHEAD;
00469             term += term[1];
00470             while ( r < term ) { /* The arguments */
00471                 if ( *r < 0 ) {
00472                     if ( *r <= -FUNCTION ) r++;
00473                     else if ( *r == -INDEX ) {
00474                         if ( r[1] > maxval && r[1] < maxtop ) r[1] += shift;
00475                         r += 2;
00476                     }
00477                     else r += 2;
00478                 }

```

```

00479         else {
00480             m = r + ARGHEAD;
00481             r += *r;
00482             while ( m < r ) { /* Terms in the argument */
00483                 MoveDummies(BHEAD m, shift);
00484                 m += *m;
00485             }
00486         }
00487     }
00488 }
00489 else {
00490     term += term[1];
00491 }
00492 }
00493 }
00494
00495 /*
00496     #] MoveDummies :
00497     #[ AdjustRenumScratch :
00498
00499     Extends the buffer for number of dummies that can be found in
00500     a term. Originally we had a fixed buffer at size 300 in the AN
00501     struct. Thomas Hahn ran out of that. Hence we have now made it
00502     a dynamical buffer.
00503     Note that the pointers used in FunLevel need adjustment as well.
00504 */
00505 void AdjustRenumScratch(PHEAD0)
00506 {
00507     GETBIDENTITY
00508     WORD newsize;
00509     int i;
00510     WORD **newpoin, *newnum;
00511     if ( AN.MaxRenumScratch == 0 ) newsize = 100;
00512     else newsize = AN.MaxRenumScratch*2;
00513     if ( newsize > MAXPOSITIVE/2 ) newsize = MAXPOSITIVE/2+1;
00514
00515     newpoin = (WORD **)Malloca1(newsize*sizeof(WORD *), "PoinScratch");
00516     for ( i = 0; i < AN.NumFound; i++ ) newpoin[i] = AN.PoinScratch[i];
00517     for ( ; i < newsize; i++ ) newpoin[i] = 0;
00518     if ( AN.PoinScratch ) M_free(AN.PoinScratch, "PoinScratch");
00519     AN.PoinScratch = newpoin;
00520     AN.DumPlace = newpoin + AN.NumFound;
00521
00522     newpoin = (WORD **)Malloca1(newsize*sizeof(WORD *), "FunScratch");
00523     for ( i = 0; i < AN.NumFound; i++ ) newpoin[i] = AN.FunScratch[i];
00524     for ( ; i < newsize; i++ ) newpoin[i] = 0;
00525     if ( AN.FunScratch ) M_free(AN.FunScratch, "FunScratch");
00526     AN.FunScratch = newpoin;
00527     AN.DumFunPlace = newpoin + AN.NumFound;
00528
00529     newnum = (WORD *)Malloca1(newsize*sizeof(WORD), "RenumScratch");
00530     for ( i = 0; i < AN.NumFound; i++ ) newnum[i] = AN.RenumScratch[i];
00531     for ( ; i < newsize; i++ ) newnum[i] = 0;
00532     if ( AN.RenumScratch ) M_free(AN.RenumScratch, "RenumScratch");
00533     AN.RenumScratch = newnum;
00534     AN.DumFound = newnum + AN.NumFound;
00535
00536     AN.MaxRenumScratch = newsize;
00537 }
00538
00539 /*
00540     #] AdjustRenumScratch :
00541     #] Reshuf :
00542     #[ Count :
00543     #[ CountDo :
00544
00545     This function executes the counting action in a count
00546     operation. The return value is the count of the term.
00547     Input is the term and a pointer to the instruction.
00548 */
00549 WORD CountDo(WORD *term, WORD *instruct)
00550 {
00551     WORD *m, *r, i, j, count = 0;
00552     WORD *stopper, *tstop, *r1 = 0, *r2 = 0;
00553     m = instruct;
00554     stopper = m + m[1];
00555     instruct += 3;
00556     tstop = term + *term; tstop -= ABS(tstop[-1]); term++;
00557     while ( term < tstop ) {
00558         switch ( *term ) {
00559             case SYMBOL:
00560                 i = term[1] - 2;
00561                 term += 2;
00562                 while ( i > 0 ) {

```

```

00566         m = instruct;
00567         while ( m < stopper ) {
00568             if ( *m == SYMBOL && m[2] == *term ) {
00569                 count += m[3] * term[1];
00570             }
00571             m += m[1];
00572         }
00573         term += 2;
00574         i -= 2;
00575     }
00576     break;
00577 case DOTPRODUCT:
00578     i = term[1] - 2;
00579     term += 2;
00580     while ( i > 0 ) {
00581         m = instruct;
00582         while ( m < stopper ) {
00583             if ( *m == DOTPRODUCT && ( ( m[2] == *term &&
00584                 m[3] == term[1]) || ( m[2] == term[1] &&
00585                 m[3] == *term ) ) ) {
00586                 count += m[4] * term[2];
00587                 break;
00588             }
00589             m += m[1];
00590         }
00591         m = instruct;
00592         while ( m < stopper ) {
00593             if ( *m == VECTOR && m[2] == *term &&
00594                 ( m[3] & DOTPBIT ) != 0 ) {
00595                 count += m[m[1]-1] * term[2];
00596             }
00597             m += m[1];
00598         }
00599         m = instruct;
00600         while ( m < stopper ) {
00601             if ( *m == VECTOR && m[2] == term[1] &&
00602                 ( m[3] & DOTPBIT ) != 0 ) {
00603                 count += m[m[1]-1] * term[2];
00604             }
00605             m += m[1];
00606         }
00607         term += 3;
00608         i -= 3;
00609     }
00610     break;
00611 case INDEX:
00612     j = 1;
00613     goto VectInd;
00614 case VECTOR:
00615     j = 2;
00616 VectInd:
00617     i = term[1] - 2;
00618     term += 2;
00619     while ( i > 0 ) {
00620         m = instruct;
00621         while ( m < stopper ) {
00622             if ( *m == VECTOR && m[2] == *term &&
00623                 ( m[3] & VECTBIT ) != 0 ) {
00624                 count += m[m[1]-1];
00625             }
00626             m += m[1];
00627         }
00628         term += j;
00629         i -= j;
00630     }
00631     break;
00632 default:
00633     if ( *term >= FUNCTION ) {
00634         i = *term;
00635         m = instruct;
00636         while ( m < stopper ) {
00637             if ( *m == FUNCTION && m[2] == i ) count += m[3];
00638             m += m[1];
00639         }
00640         if ( functions[i-FUNCTION].spec >= TENSORFUNCTION ) {
00641             i = term[1] - FUNHEAD;
00642             term += FUNHEAD;
00643             while ( i > 0 ) {
00644                 if ( *term < 0 ) {
00645                     m = instruct;
00646                     while ( m < stopper ) {
00647                         if ( *m == VECTOR && m[2] == *term &&
00648                             ( m[3] & FUNBIT ) != 0 ) {
00649                             count += m[m[1]-1];
00650                         }
00651                         m += m[1];
00652                     }

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```

00653             term++;
00654             i--;
00655         }
00656     }
00657     else {
00658         r = term + term[1];
00659         term += FUNHEAD;
00660         while ( term < r ) {
00661             if ( ( *term == -INDEX || *term == -VECTOR
00662                 || *term == -MINVECTOR ) && term[1] < MINSPEC ) {
00663                 m = instruct;
00664                 while ( m < stopper ) {
00665                     if ( *m == VECTOR && term[1] == m[2]
00666                         && ( m[3] & SETBIT ) != 0 ) {
00667                         r1 = SetElements + Sets[m[4]].first;
00668                         r2 = SetElements + Sets[m[4]].last;
00669                         while ( r1 < r2 ) {
00670                             if ( *r1 == i ) {
00671                                 count += m[m[1]-1];
00672                                 goto NextFF;
00673                             }
00674                             r1++;
00675                         }
00676                     }
00677                     m += m[1];
00678                 }
00679             NextFF:
00680                 term += 2;
00681             }
00682             else { NEXTARG(term) }
00683         }
00684     }
00685     break;
00686 }
00687 else {
00688     term += term[1];
00689 }
00690 break;
00691 }
00692 }
00693 return(count);
00694 }
00695
00696 /*
00697 #] CountDo :
00698 #[ CountFun :
00699
00700 This is the count function.
00701 The return value is the count of the term.
00702 Input is the term and a pointer to the count function.
00703
00704 */
00705
00706 WORD CountFun(WORD *term, WORD *countfun)
00707 {
00708     WORD *m, *r, i, j, count = 0, *instruct, *stopper, *tstop;
00709     m = countfun;
00710     stopper = m + m[1];
00711     instruct = countfun + FUNHEAD;
00712     tstop = term + *term; tstop -= ABS(tstop[-1]); term++;
00713     while ( term < tstop ) {
00714         switch ( *term ) {
00715             case SYMBOL:
00716                 i = term[1] - 2;
00717                 term += 2;
00718                 while ( i > 0 ) {
00719                     m = instruct;
00720                     while ( m < stopper ) {
00721                         if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
00722                         if ( *m == -SYMBOL && m[1] == *term
00723                             && m[2] == -SNUMBER && ( m + 2 ) < stopper ) {
00724                             count += m[3] * term[1]; m += 4;
00725                         }
00726                         else { NEXTARG(m) }
00727                     }
00728                     term += 2;
00729                     i -= 2;
00730                 }
00731                 break;
00732             case DOTPRODUCT:
00733                 i = term[1] - 2;
00734                 term += 2;
00735                 while ( i > 0 ) {
00736                     m = instruct;
00737                     while ( m < stopper ) {
00738                         if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
00739                         if ( *m == 9+ARGHEAD && m[ARGHEAD] == 9

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00740         && m[ARGHEAD+1] == DOTPRODUCT
00741         && m[ARGHEAD+9] == -SNUMBER && ( m + ARGHEAD+9 ) < stopper
00742         && ( ( m[ARGHEAD+3] == *term &&
00743         m[ARGHEAD+4] == term[1] ) ||
00744         ( m[ARGHEAD+3] == term[1] &&
00745         m[ARGHEAD+4] == *term ) ) {
00746             count += m[ARGHEAD+10] * term[2];
00747             m += ARGHEAD+11;
00748         }
00749         else { NEXTARG(m) }
00750     }
00751     m = instruct;
00752     while ( m < stopper ) {
00753         if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
00754         if ( ( *m == -VECTOR || *m == -MINVECTOR )
00755         && m[1] == *term &&
00756         m[2] == -SNUMBER && ( m+2 ) < stopper ) {
00757             count += m[3] * term[2]; m += 4;
00758         }
00759         NEXTARG(m)
00760     }
00761     m = instruct;
00762     while ( m < stopper ) {
00763         if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
00764         if ( ( *m == -VECTOR || *m == -MINVECTOR )
00765         && m[1] == term[1] &&
00766         m[2] == -SNUMBER && ( m+2 ) < stopper ) {
00767             count += m[3] * term[2];
00768             m += 4;
00769         }
00770         NEXTARG(m)
00771     }
00772     term += 3;
00773     i -= 3;
00774 }
00775 break;
00776 case INDEX:
00777     j = 1;
00778     goto VectInd;
00779 case VECTOR:
00780     j = 2;
00781 VectInd:
00782     i = term[1] - 2;
00783     term += 2;
00784     while ( i > 0 ) {
00785         m = instruct;
00786         while ( m < stopper ) {
00787             if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
00788             if ( ( *m == -VECTOR || *m == -MINVECTOR )
00789             && m[1] == *term &&
00790             m[2] == -SNUMBER && ( m+2 ) < stopper ) {
00791                 count += m[3]; m += 4;
00792             }
00793             NEXTARG(m)
00794         }
00795         term += j;
00796         i -= j;
00797     }
00798     break;
00799 default:
00800     if ( *term >= FUNCTION ) {
00801         i = *term;
00802         m = instruct;
00803         while ( m < stopper ) {
00804             if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
00805             if ( *m == -i && m[1] == -SNUMBER && ( m+1 ) < stopper ) {
00806                 count += m[2]; m += 3;
00807             }
00808             NEXTARG(m)
00809         }
00810         if ( functions[i-FUNCTION].spec >= TENSORFUNCTION ) {
00811             i = term[1] - FUNHEAD;
00812             term += FUNHEAD;
00813             while ( i > 0 ) {
00814                 if ( *term < 0 ) {
00815                     m = instruct;
00816                     while ( m < stopper ) {
00817                         if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
00818                         if ( ( *m == -VECTOR || *m == -INDEX
00819                         || *m == -MINVECTOR ) && m[1] == *term &&
00820                         m[2] == -SNUMBER && ( m+2 ) < stopper ) {
00821                             count += m[3]; m += 4;
00822                         }
00823                         else { NEXTARG(m) }
00824                     }
00825                 }
00826                 term++;
00827                 i--;

```



```

00827     }
00828     }
00829     else {
00830         r = term + term[1];
00831         term += FUNHEAD;
00832         while ( term < r ) {
00833             if ( ( *term == -INDEX || *term == -VECTOR
00834                 || *term == -MINVECTOR ) && term[1] < MINSPEC ) {
00835                 m = instruct;
00836                 while ( m < stopper ) {
00837                     if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
00838                     if ( *m == -VECTOR && m[1] == term[1]
00839                         && m[2] == -SNUMBER && (m+2) < stopper ) {
00840                         count += m[3];
00841                         m += 4;
00842                     }
00843                     else { NEXTARG(m) }
00844                 }
00845                 term += 2;
00846             }
00847             else { NEXTARG(term) }
00848         }
00849     }
00850     break;
00851 }
00852 else {
00853     term += term[1];
00854 }
00855 break;
00856 }
00857 }
00858 return(count);
00859 }
00860
00861 /*
00862     #] CountFun :
00863     #] Count :
00864     #[ DimensionSubterm :
00865 */
00866
00867 WORD DimensionSubterm(WORD *subterm)
00868 {
00869     WORD *r, *rstop, dim, i;
00870     LONG x = 0;
00871     rstop = subterm + subterm[1];
00872     if ( *subterm == SYMBOL ) {
00873         r = subterm + 2;
00874         while ( r < rstop ) {
00875             if ( *r <= NumSymbols && *r > -MAXPOWER ) {
00876                 dim = symbols[*r].dimension;
00877                 if ( dim == MAXPOSITIVE ) goto undefined;
00878                 x += dim * r[1];
00879                 if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
00880                 r += 2;
00881             }
00882             else if ( *r <= MAXVARIABLES ) {
00883 /*
00884                 Here we have an extra symbol. Store dimension in the compiler buffer
00885 */
00886                 i = MAXVARIABLES - *r;
00887                 dim = cbuf[AM.sbufnum].dimension[i];
00888                 if ( dim == MAXPOSITIVE ) goto undefined;
00889                 if ( dim == -MAXPOSITIVE ) goto outofrange;
00890                 x += dim * r[1];
00891                 if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
00892                 r += 2;
00893             }
00894             else { r += 2; }
00895         }
00896     }
00897     else if ( *subterm == DOTPRODUCT ) {
00898         r = subterm + 2;
00899         while ( r < rstop ) {
00900             dim = vectors[*r-AM.OffsetVector].dimension;
00901             if ( dim == MAXPOSITIVE ) goto undefined;
00902             x += dim * r[2];
00903             if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
00904             dim = vectors[r[1]-AM.OffsetVector].dimension;
00905             if ( dim == MAXPOSITIVE ) goto undefined;
00906             x += dim * r[2];
00907             if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
00908             r += 3;
00909         }
00910     }
00911     else if ( *subterm == VECTOR ) {
00912         r = subterm + 2;
00913         while ( r < rstop ) {

```

```

00914         dim = vectors[*r-AM.OffsetVector].dimension;
00915         if ( dim == MAXPOSITIVE ) goto undefined;
00916         x += dim;
00917         if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
00918         r += 2;
00919     }
00920 }
00921 else if ( *subterm == INDEX ) {
00922     r = subterm + 2;
00923     while ( r < rstop ) {
00924         if ( *r < 0 ) {
00925             dim = vectors[*r-AM.OffsetVector].dimension;
00926             if ( dim == MAXPOSITIVE ) goto undefined;
00927             x += dim;
00928             if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
00929         }
00930         r++;
00931     }
00932 }
00933 else if ( *subterm >= FUNCTION ) {
00934     dim = functions[*subterm-FUNCTION].dimension;
00935     if ( dim == MAXPOSITIVE ) goto undefined;
00936     x += dim;
00937     if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
00938     if ( functions[*subterm-FUNCTION].spec > 0 ) { /* tensor */
00939         r = subterm + FUNHEAD;
00940         while ( r < rstop ) {
00941             if ( *r < MINSPEC ) {
00942                 dim = vectors[*r-AM.OffsetVector].dimension;
00943                 if ( dim == MAXPOSITIVE ) goto undefined;
00944                 x += dim;
00945                 if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
00946             }
00947             r++;
00948         }
00949     }
00950 }
00951 return((WORD)x);
00952 undefined:
00953 return((WORD)MAXPOSITIVE);
00954 outofrange:
00955 return(-(WORD)MAXPOSITIVE);
00956 }
00957 /*
00958 #] DimensionSubterm :
00959 #[ DimensionTerm :
00960
00961 Returns the dimension of the given term.
00962 If there is any variable of which the dimension is not defined
00963 we return the code for undefined which is MAXPOSITIVE
00964 When the value is out of range we return -MAXPOSITIVE
00965 */
00966
00967 WORD DimensionTerm(WORD *term)
00968 {
00969     WORD *t, *tstop, dim;
00970     LONG x = 0;
00971     tstop = term + *term; tstop -= ABS(tstop[-1]);
00972     t = term+1;
00973     while ( t < tstop ) {
00974         dim = DimensionSubterm(t);
00975         if ( dim == MAXPOSITIVE ) goto undefined;
00976         if ( dim == -MAXPOSITIVE ) goto outofrange;
00977         x += dim;
00978         if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
00979         t += t[1];
00980     }
00981     return((WORD)x);
00982 undefined:
00983 return((WORD)MAXPOSITIVE);
00984 outofrange:
00985 return(-(WORD)MAXPOSITIVE);
00986 }
00987 /*
00988 #] DimensionTerm :
00989 #[ DimensionExpression :
00990
00991 Returns the dimension of the given expression.
00992 If there is any variable of which the dimension is not defined
00993 we return the code for undefined which is MAXPOSITIVE
00994 When the value is out of range we return -MAXPOSITIVE
00995 When the value is not consistent we return -MAXPOSITIVE.
00996 */
00997
00998 WORD DimensionExpression(PHEAD WORD *expr)

```

```

01001 {
01002     WORD dim, *term, *old, x = 0;
01003     int first = 1;
01004     term = expr;
01005     while ( *term ) {
01006         dim = DimensionTerm(term);
01007         if ( dim == MAXPOSITIVE ) goto undefined;
01008         if ( dim == -MAXPOSITIVE ) goto outofrange;
01009         if ( first ) { x = dim; }
01010         else if ( x != dim ) {
01011             old = AN.currentTerm;
01012             MLOCK(ErrorMessageLock);
01013             MesPrint("Dimension is not the same in the terms of the expression");
01014             term = expr;
01015             while ( *term ) {
01016                 AN.currentTerm = term;
01017                 MesPrint("  %T");
01018             }
01019             MUNLOCK(ErrorMessageLock);
01020             AN.currentTerm = old;
01021             return(-(WORD)MAXPOSITIVE);
01022         }
01023         term += *term;
01024     }
01025     return((WORD)x);
01026 undefined:
01027     return((WORD)MAXPOSITIVE);
01028 outofrange:
01029     old = AN.currentTerm;
01030     AN.currentTerm = term;
01031     MLOCK(ErrorMessageLock);
01032     MesPrint("Dimension out of range in %t in subexpression");
01033     MUNLOCK(ErrorMessageLock);
01034     AN.currentTerm = old;
01035     return(-(WORD)MAXPOSITIVE);
01036 }
01037
01038 /*
01039 #] DimensionExpression :
01040 #[ Multiply Term :
01041     #[ MultDo :
01042 */
01043
01044 int MultDo(PHEAD WORD *term, WORD *pattern)
01045 {
01046     GETBIDENTITY
01047     WORD *t, *r, i;
01048     t = term + *term;
01049     if ( pattern[2] > 0 ) {          /* Left multiply */
01050         i = *term - 1;
01051     }
01052     else {                          /* Right multiply */
01053         i = ABS(t[-1]);
01054     }
01055     *term += SUBEXPSIZE;
01056     r = t + SUBEXPSIZE;
01057     do { *--r = *--t; } while ( --i > 0 );
01058     r = pattern + 3;
01059     i = r[1];
01060     while ( --i >= 0 ) *t++ = *r++;
01061     AT.WorkPointer = term + *term;
01062     return(0);
01063 }
01064
01065 /*
01066 #] MultDo :
01067 #[ Multiply Term :
01068 #[ Try Term(s) :
01069     #[ TryDo :
01070 */
01071
01072 int TryDo(PHEAD WORD *term, WORD *pattern, WORD level)
01073 {
01074     GETBIDENTITY
01075     WORD *t, *r, *m, i, j;
01076     ReNumber(BHEAD term);
01077     Normalize(BHEAD term);
01078     m = r = term + *term;
01079     m++;
01080     i = pattern[2];
01081     t = pattern + 3;
01082     NCOPY(m,t,i)
01083     j = *term - 1;
01084     t = term + 1;
01085     NCOPY(m,t,j)
01086     *r = WORDDIFF(m,r);
01087     AT.WorkPointer = m;

```

```

01088     if ( ( j = Normalize(BHEAD r) ) == 0 || j == 1 ) {
01089         if ( *r == 0 ) return(0);
01090         ReNumber(BHEAD r); Normalize(BHEAD r);
01091         if ( *r == 0 ) return(0);
01092         if ( ( i = CompareTerms(BHEAD term,r,0) ) < 0 ) {
01093             *AN.RepPoint = 1;
01094             AR.expchanged = 1;
01095             return(Generator(BHEAD r,level));
01096         }
01097         if ( i == 0 && CompCoef(term,r) != 0 ) { return(0); }
01098     }
01099     AT.WorkPointer = r;
01100     return(Generator(BHEAD term,level));
01101 }
01102
01103 /*
01104     #] TryDo :
01105     #] Try Term(s) :
01106     #[ Distribute :
01107     #[ DoDistrib :
01108
01109     The routine that generates the terms ordered by a distrib_ function.
01110     The presence of a replaceable distrib_ function has been sensed
01111     in the routine TestSub and has been passed on to Generator.
01112     It is then Generator that calls this function in a way that is
01113     similar to calling the trace routines, except for that for the
01114     trace routines and the Levi-Civita tensors the arguments are put
01115     in temporary storage and here we leave them inside the term,
01116     because there is no knowing how long the field will be.
01117 */
01118
01119 int DoDistrib(PHEAD WORD *term, WORD level)
01120 {
01121     GETBIDENTITY
01122     WORD *t, *m, *r = 0, *stop, *tstop, *termout, *endhead, *starttail, *parms;
01123     WORD i, j, k, n, nn, ntype, fun1 = 0, fun2 = 0, typ1 = 0, typ2 = 0;
01124     WORD *arg, *oldwork, *mf, ktype = 0, atype = 0;
01125     WORD sgn, dirtyflag;
01126     AN.TeInFun = AR.TePos = 0;
01127     t = term;
01128     tstop = t + *t;
01129     stop = tstop - ABS(tstop[-1]);
01130     t++;
01131     while ( t < stop ) {
01132         r = t + t[1];
01133         if ( *t == DISTRIBUTION && t[FUNHEAD] == -SNUMBER
01134             && t[FUNHEAD+1] >= -2 && t[FUNHEAD+1] <= 2
01135             && t[FUNHEAD+2] == -SNUMBER
01136             && t[FUNHEAD+4] <= -FUNCTION
01137             && t[FUNHEAD+5] <= -FUNCTION ) {
01138             WORD *ttt = t+FUNHEAD+6, *tttstop = t+t[1];
01139             while ( ttt < tttstop ) {
01140                 if ( *ttt == -DOLLEXPRESSION ) break;
01141                 NEXTARG(ttt);
01142             }
01143             if ( ttt >= tttstop ) {
01144                 fun1 = -t[FUNHEAD+4];
01145                 fun2 = -t[FUNHEAD+5];
01146                 typ1 = functions[fun1-FUNCTION].spec;
01147                 typ2 = functions[fun2-FUNCTION].spec;
01148                 if ( typ1 > 0 || typ2 > 0 ) {
01149                     m = t + FUNHEAD+6;
01150                     r = t + t[1];
01151                     while ( m < r ) {
01152                         if ( *m != -INDEX && *m != -VECTOR && *m != -MINVECTOR )
01153                             break;
01154                         m += 2;
01155                     }
01156                     if ( m < r ) {
01157                         MLOCK(ErrorMessageLock);
01158                         MesPrint("Incompatible function types and arguments in distrib_");
01159                         MUNLOCK(ErrorMessageLock);
01160                         SETERROR(-1)
01161                     }
01162                 }
01163                 break;
01164             }
01165         }
01166         t = r;
01167     }
01168     dirtyflag = t[2];
01169     ntype = t[FUNHEAD+1];
01170     n = t[FUNHEAD+3];
01171 /*
01172     t points at the distrib_ function to be expanded.
01173     fun1,fun2 and typ1,typ2 are the two functions and their types.
01174     ntype indicates the action:

```

```

01175      0:  Make all possible divisions:  2^nargs
01176      1:  fun1 should get n arguments:  nargs! / ( n! (nargs-n)! )
01177      2:  fun2 should get n arguments:  nargs! / ( n! (nargs-n)! )
01178  The distinction between 1 and two is for noncommuting objects.
01179      3:  fun1 should get n arguments. Super symmetric option.
01180      4:  fun2 idem
01181  The super symmetric option involves:
01182      a:  arguments get sorted
01183      b:  identical arguments are seen as such. Hence not all their
01184          distributions are taken into account. It is as if after the
01185          distrib there is a symmetrize fun1; symmetrize fun2;
01186      c:  Hence if the occurrence of each argument is a,b,c,...
01187          and their occurrence in fun1 is a1,b1,c1,... and in fun2
01188          is a2,b2,c2,... then each term is generated (a1+a2)!/a1!/a2!
01189          (b1+b2)!/b1!/b2! (c1+c2)!/c1!/c2! ... times.
01190      d:  We have to make an array of occurrences and positions.
01191      e:  Then we sort the arguments indirectly.
01192      f:  Next we generate the argument lists in the same way as we
01193          generate powers of expressions with binomials. Hence we need
01194          a third array to keep track of the 'powers'
01195  */
01196      endhead = t;
01197      starttail = r;
01198      parms = m = t + FUNHEAD+6;
01199      i = 0;
01200      while ( m < r ) { /* Count arguments */
01201          i++;
01202          NEXTARG(m);
01203      }
01204      oldwork = AT.WorkPointer;
01205      arg = AT.WorkPointer + 1;
01206      arg[-1] = 0;
01207      termout = arg + i;
01208      switch ( ntype ) {
01209          case 0: ktype = 1; atype = n < 0 ? 1: 0; n = 0; break;
01210          case 1: ktype = 1; atype = 0; break;
01211          case 2: ktype = 0; atype = 0; break;
01212          case -1: ktype = 1; atype = 1; break;
01213          case -2: ktype = 0; atype = 1; break;
01214      }
01215      do {
01216  /*
01217          All distributions with n elements. We generate the array arg with
01218          all possible 1 and 0 patterns. 1 means in fun1 and 0 means in fun2.
01219  */
01220          if ( n > i ) return(0); /* 0 elements */
01221
01222          for ( j = 0; j < n; j++ ) arg[j] = 1;
01223          for ( j = n; j < i; j++ ) arg[j] = 0;
01224          for(;;) {
01225              sgn = 0;
01226              t = term;
01227              m = termout;
01228              while ( t < endhead ) *m++ = *t++;
01229              mf = m;
01230              *m++ = fun1;
01231              *m++ = FUNHEAD;
01232              *m++ = dirtyflag;
01233              #if FUNHEAD > 3
01234                  k = FUNHEAD -3;
01235                  while ( k-- > 0 ) *m++ = 0;
01236              #endif
01237              r = parms;
01238              for ( k = 0; k < i; k++ ) {
01239                  if ( arg[k] == ktype ) {
01240                      if ( *r <= -FUNCTION ) *m++ = *r++;
01241                      else if ( *r < 0 ) {
01242                          if ( typ1 > 0 ) {
01243                              if ( *r == -MINVECTOR ) sgn ^= 1;
01244                              r++;
01245                              *m++ = *r++;
01246                          }
01247                          else { *m++ = *r++; *m++ = *r++; }
01248                      }
01249                      else {
01250                          nn = *r;
01251                          NCOPY(m,r,nn);
01252                      }
01253                  }
01254                  else { NEXTARG(r) }
01255              }
01256              mf[1] = WORDDIFF(m,mf);
01257              mf = m;
01258              *m++ = fun2;
01259              *m++ = FUNHEAD;
01260              *m++ = dirtyflag;
01261              #if FUNHEAD > 3

```

```

01262         k = FUNHEAD -3;
01263         while ( k-- > 0 ) *m++ = 0;
01264     #endif
01265     r = parms;
01266     for ( k = 0; k < i; k++ ) {
01267         if ( arg[k] != ktype ) {
01268             if ( *r <= -FUNCTION ) *m++ = *r++;
01269             else if ( *r < 0 ) {
01270                 if ( typ2 > 0 ) {
01271                     if ( *r == -MINVECTOR ) sgn ^= 1;
01272                     r++;
01273                     *m++ = *r++;
01274                 }
01275                 else { *m++ = *r++; *m++ = *r++; }
01276             }
01277             else {
01278                 nn = *r;
01279                 NCOPY(m,r,nn);
01280             }
01281         }
01282         else { NEXTARG(r) }
01283     }
01284     mf[1] = WORDDIF(m,mf);
01285     #ifndef NUOVO
01286     if ( atype == 0 ) {
01287         WORD k1,k2;
01288         for ( k = 0; k < i-1; k++ ) {
01289             if ( arg[k] == 0 ) continue;
01290             k1 = 1; k2 = k;
01291             while ( k < i-1 && EqualArg(parms,k,k+1) ) { k++; k1++; }
01292             while ( k2 <= k && arg[k2] == 1 ) k2++;
01293             k2 = k-k2+1;
01294             /*
01295             Now we need k1!/(k2! (k1-k2)!)
01296             */
01297             if ( k2 != k1 && k2 != 0 ) {
01298                 if ( GetBinom((UWORD *)m+3,m+2,k1,k2) ) {
01299                     MLOCK(ErrorMessageLock);
01300                     MesCall("DoDistrib");
01301                     MUNLOCK(ErrorMessageLock);
01302                     SETERROR(-1)
01303                 }
01304                 m[1] = ( m[2] < 0 ? -m[2]: m[2] ) + 3;
01305                 *m = LNUMBER;
01306                 m += m[1];
01307             }
01308         }
01309     }
01310     #endif
01311     r = starttail;
01312     while ( r < tstop ) *m++ = *r++;
01313
01314     if ( atype ) { /* antisymmetric field */
01315         k = n;
01316         nn = 0;
01317         for ( j = 0; j < i && k > 0; j++ ) {
01318             if ( arg[j] == 1 ) k--;
01319             else nn += k;
01320         }
01321         sgn ^= nn & 1;
01322     }
01323
01324     if ( sgn ) m[-1] = -m[-1];
01325     *termout = WORDDIF(m,termout);
01326     AT.WorkPointer = m;
01327     if ( AT.WorkPointer > AT.WorkTop ) {
01328         MLOCK(ErrorMessageLock);
01329         MesWork();
01330         MUNLOCK(ErrorMessageLock);
01331         return(-1);
01332     }
01333     *AN.RepPoint = 1;
01334     AR.expchanged = 1;
01335     if ( Generator(BHEAD termout,level) ) Terminate(-1);
01336     #ifndef NUOVO
01337     {
01338         WORD k1;
01339         j = i - 1;
01340         k = 0;
01341         redok: while ( arg[j] == 1 && j >= 0 ) { j--; k++; }
01342         while ( arg[j] == 0 && j >= 0 ) j--;
01343         if ( j < 0 ) break;
01344         k1 = j;
01345         arg[j] = 0;
01346         while ( !atype && EqualArg(parms,j,j+1) ) {
01347             j++;
01348             if ( j >= i - k - 1 ) { j = k1; k++; goto redok; }

```

```

01349         arg[j] = 0;
01350     }
01351     while ( k >= 0 ) { j++; arg[j] = 1; k--; }
01352     j++;
01353     while ( j < i ) { arg[j] = 0; j++; }
01354 }
01355 #else
01356     j = i - 1;
01357     k = 0;
01358     while ( arg[j] == 1 && j >= 0 ) { j--; k++; }
01359     while ( arg[j] == 0 && j >= 0 ) j--;
01360     if ( j < 0 ) break;
01361     arg[j] = 0;
01362     while ( k >= 0 ) { j++; arg[j] = 1; k--; }
01363     j++;
01364     while ( j < i ) { arg[j] = 0; j++; }
01365 #endif
01366 }
01367 } while ( ntype == 0 && ++n <= i );
01368 AT.WorkPointer = oldwork;
01369 return(0);
01370 }
01371
01372 /*
01373     #] DoDistrib :
01374     #[ EqualArg :
01375
01376     Returns 1 if the arguments in the field are identical.
01377 */
01378
01379 int EqualArg(WORD *parms, WORD num1, WORD num2)
01380 {
01381     WORD *t1, *t2;
01382     WORD i;
01383     t1 = parms;
01384     while ( --num1 >= 0 ) { NEXTARG(t1); }
01385     t2 = parms;
01386     while ( --num2 >= 0 ) { NEXTARG(t2); }
01387     if ( *t1 != *t2 ) return(0);
01388     if ( *t1 < 0 ) {
01389         if ( *t1 <= -FUNCTION || t1[1] == t2[1] ) return(1);
01390         return(0);
01391     }
01392     i = *t1;
01393     while ( --i >= 0 ) {
01394         if ( *t1 != *t2 ) return(0);
01395         t1++; t2++;
01396     }
01397     return(1);
01398 }
01399
01400 /*
01401     #] EqualArg :
01402     #[ DoDelta3 :
01403 */
01404
01405 int DoDelta3(PHEAD WORD *term, WORD level)
01406 {
01407     GETBIDENTITY
01408     WORD *t, *m, *m1, *m2, *stopper, *tstop, *termout, *dels, *taken;
01409     WORD *ic, *jc, *factors;
01410     WORD num, num2, i, j, k, knum, a;
01411     AN.TeInFun = AR.TePos = 0;
01412     tstop = term + *term;
01413     stopper = tstop - ABS(tstop[-1]);
01414     t = term+1;
01415     while ( ( *t != DELTA3 || ((t[1]-FUNHEAD) & 1) != 0 ) && t < stopper )
01416         t += t[1];
01417     if ( t >= stopper ) {
01418         MLOCK(ErrorMessageLock);
01419         MesPrint("Internal error with dd_ function");
01420         MUNLOCK(ErrorMessageLock);
01421         Terminate(-1);
01422     }
01423     m1 = t; m2 = t + t[1];
01424     num = t[1] - FUNHEAD;
01425     if ( num == 0 ) {
01426         termout = t = AT.WorkPointer;
01427         m = term;
01428         while ( m < m1 ) *t++ = *m++;
01429         m = m2; while ( m < tstop ) *t++ = *m++;
01430         *termout = WORDDIF(t, termout);
01431         AT.WorkPointer = t;
01432         *AN.RepPoint = 1;
01433         AR.expchanged = 1;
01434         if ( Generator(BHEAD termout, level) ) {
01435             MLOCK(ErrorMessageLock);

```

```

01436         MesCall("Do dd_");
01437         MUNLOCK(ErrorMessageLock);
01438         SETERROR(-1)
01439     }
01440     AT.WorkPointer = termout;
01441     return(0);
01442 }
01443 t += FUNHEAD;
01444 /*
01445 Step 1: sort the arguments
01446 */
01447 for ( i = 1; i < num; i++ ) {
01448     if ( t[i] < t[i-1] ) {
01449         a = t[i]; t[i] = t[i-1]; t[i-1] = a;
01450         j = i - 1;
01451         while ( j > 0 ) {
01452             if ( t[j] >= t[j-1] ) break;
01453             a = t[j]; t[j] = t[j-1]; t[j-1] = a;
01454             j--;
01455         }
01456     }
01457 }
01458 /*
01459 Step 2: Order them by occurrence
01460 In 'taken' we have the array with the number of occurrences.
01461 in 'dels' is the type of object.
01462 */
01463 m = taken = AT.WorkPointer;
01464 for ( i = 0; i < num; i++ ) *m++ = 0;
01465 dels = m; knum = 0;
01466 for ( i = 0; i < num; knum++ ) {
01467     *m++ = t[i]; i++; taken[knum] = 1;
01468     while ( i < num ) {
01469         if ( t[i] != t[i-1] ) break;
01470         i++; (taken[knum])++;
01471     }
01472 }
01473 for ( i = 0; i < knum; i++ ) *m++ = taken[i];
01474 ic = m; num2 = num/2;
01475 jc = ic + num2;
01476 factors = jc + num2;
01477 termout = factors + num2;
01478 /*
01479 The recursion has num/2 steps
01480 */
01481 k = 0;
01482 while ( k >= 0 ) {
01483     if ( k >= num2 ) {
01484         t = termout; m = term;
01485         while ( m < m1 ) *t++ = *m++;
01486         *t++ = DELTA; *t++ = num+2;
01487         for ( i = 0; i < num2; i++ ) {
01488             *t++ = dels[ic[i]]; *t++ = dels[jc[i]];
01489         }
01490         for ( i = 0; i < num2; i++ ) {
01491             if ( ic[i] == jc[i] ) {
01492                 j = 1;
01493                 while ( i < num2-1 && ic[i] == ic[i+1] && ic[i] == jc[i+1] )
01494                     { i++; j++; }
01495                 for ( a = 1; a < j; a++ ) {
01496                     *t++ = SNUMBER; *t++ = 4; *t++ = 2*a+1; *t++ = 1;
01497                 }
01498                 for ( a = 0; a+1+i < num2; a++ ) {
01499                     if ( ic[a+i] != ic[a+i+1] ) break;
01500                 }
01501                 if ( a > 0 ) {
01502                     if ( GetBinom((UWORD *) (t+3), t+2, 2*j+a, a) ) {
01503                         MLOCK(ErrorMessageLock);
01504                         MesCall("Do dd_");
01505                         MUNLOCK(ErrorMessageLock);
01506                         SETERROR(-1)
01507                     }
01508                     t[1] = ( t[2] < 0 ? -t[2]: t[2] ) + 3;
01509                     *t = LNUMBER;
01510                     t += t[1];
01511                 }
01512             }
01513             else if ( factors[i] != 1 ) {
01514                 *t++ = SNUMBER; *t++ = 4; *t++ = factors[i]; *t++ = 1;
01515             }
01516         }
01517         for ( i = 0; i < num2-1; i++ ) {
01518             if ( ic[i] == jc[i] ) continue;
01519             j = 1;
01520             while ( i < num2-1 && jc[i] == jc[i+1] && ic[i] == ic[i+1] ) {
01521                 i++; j++;
01522             }

```



```

01523         for ( a = 0; a+i < num2-1; a++ ) {
01524             if ( ic[i+a] != ic[i+a+1] ) break;
01525         }
01526         if ( a > 0 ) {
01527             if ( GetBinom((UWORD *) (t+3),t+2,j+a,a) ) {
01528                 MLOCK(ErrorMessageLock);
01529                 MesCall("Do dd_");
01530                 MUNLOCK(ErrorMessageLock);
01531                 SETERROR(-1)
01532             }
01533             t[1] = ( t[2] < 0 ? -t[2]: t[2] ) + 3;
01534             *t = LNUMBER;
01535             t += t[1];
01536         }
01537     }
01538     m = m2;
01539     while ( m < tstop ) *t++ = *m++;
01540     *termout = WORDDIFF(t,termout);
01541     AT.WorkPointer = t;
01542     *AN.RepPoint = 1;
01543     AR.expchanged = 1;
01544     if ( Generator(BHEAD termout,level) ) {
01545         MLOCK(ErrorMessageLock);
01546         MesCall("Do dd_");
01547         MUNLOCK(ErrorMessageLock);
01548         SETERROR(-1)
01549     }
01550     k--;
01551     if ( k >= 0 ) goto nextj;
01552     else break;
01553 }
01554 for ( ic[k] = 0; ic[k] < knum; ic[k]++ ) {
01555     if ( taken[ic[k]] > 0 ) break;
01556 }
01557 if ( k > 0 && ic[k-1] == ic[k] ) jc[k] = jc[k-1];
01558 else jc[k] = ic[k];
01559 for ( ; jc[k] < knum; jc[k]++ ) {
01560     if ( taken[jc[k]] <= 0 ) continue;
01561     if ( ic[k] == jc[k] ) {
01562         if ( taken[jc[k]] <= 1 ) continue;
01563     /*
01564         factors[k] = taken[ic[k]];
01565         if ( ( factors[k] & 1 ) == 0 ) (factors[k])--;
01566     */
01567         taken[ic[k]] -= 2;
01568     }
01569     else {
01570         factors[k] = taken[jc[k]];
01571         (taken[ic[k]])--; (taken[jc[k]])--;
01572     }
01573     k++;
01574     goto nextk; /* This is the simulated recursion */
01575 nextj::
01576     (taken[ic[k]])++; (taken[jc[k]])++;
01577 }
01578 k--;
01579 if ( k >= 0 ) goto nextj;
01580 nextk::
01581 }
01582 AT.WorkPointer = taken;
01583 return(0);
01584 }
01585
01586 /*
01587 #] DoDelta3 :
01588 #[ TestPartitions :
01589
01590 Checks whether the function in tfun is a partitions_ function
01591 that can be expanded. If it can a number of relevant objects is
01592 inside the struct parti.
01593 This test is not entirely trivial because there are many restrictions
01594 w.r.t. the arguments.
01595 Syntax (still to be implemented)
01596 partitions_(number_of_partition_entries,[function,number,]^nope,arguments)
01597 [function,number,]: can be
01598     f,3     for a partition of 3 arguments
01599     f,0     for the remaining arguments (should be last)
01600     num1,f,num2 with num1 effectively a number of partitions but this
01601                counts as num1 entries.
01602     0,f,num2: all partitions have num2 arguments. No number of partition
01603                entries needed. If num2 does not divide the number of
01604                arguments there will be no action.
01605 */
01606
01607 int TestPartitions(WORD *tfun, PARTI *parti)
01608 {
01609     WORD *tnext = tfun + tfun[1];

```

```

01610     WORD *t, *tt;
01611     WORD argcount = 0, sum = 0, i, ipart, argremain;
01612     WORD tensorflag = 0;
01613     parti->psize = parti->nfun = parti->args = parti->nargs = 0;
01614     parti->numargs = parti->numpart = parti->where = 0;
01615     tt = t = tfun + FUNHEAD;
01616     while ( t < tnext ) { argcount++; NEXTARG(t); }
01617     if ( argcount < 1 ) goto No;
01618     t = tt;
01619     if ( *t != -SNUMBER ) goto No;
01620     if ( t[1] == 0 ) {
01621         t += 2;
01622         if ( *t <= -FUNCTION && t[1] == -SNUMBER && t[2] > 0 ) {
01623             if ( functions[-*t-FUNCTION].spec > 0 ) tensorflag = 1;
01624             if ( argcount-3 < 0 ) goto No;
01625             if ( ( argcount-3 ) % t[2] != 0 ) goto No;
01626         }
01627         else goto No;
01628         parti->numpart = (argcount-3)/t[2];
01629         parti->numargs = argcount - 3;
01630         parti->psize = (WORD *)Malloc1((parti->numpart*2+parti->numargs*2+2)
01631             *sizeof(WORD),"partitions");
01632         parti->nfun = parti->psize + parti->numpart;
01633         parti->args = parti->nfun + parti->numpart;
01634         parti->nargs = parti->args + parti->numargs;
01635         for ( i = 0; i < parti->numpart; i++ ) {
01636             parti->psize[i] = t[2];
01637             parti->nfun[i] = -t[0];
01638         }
01639         t += 3;
01640     }
01641     else if ( t[1] > 0 ) { /* Number of partitions */
01642 /*
01643     We can have sequences of function,number for one partition
01644     or number1,function,number2 for number1 partitions of size number2.
01645     The last partition can have number=0. It must be a single partition
01646     and it will take all remaining arguments.
01647     If any of the functions is a tensor, all arguments must be either
01648     vector or index.
01649 */
01650     parti->numpart = t[1]; t += 2;
01651     ipart = sum = 0; argremain = argcount - 1;
01652 /*
01653     At this point is seems better to make an allocation already that
01654     may be too big. The alternative is having to pass this code twice.
01655 */
01656     parti->psize = (WORD *)Malloc1((argcount*4+2)*sizeof(WORD),"partitions");
01657     parti->nfun = parti->psize+argcount;
01658     parti->args = parti->nfun+argcount;
01659     parti->nargs = parti->args+argcount;
01660     while ( ipart < parti->numpart ) {
01661         if ( *t <= -FUNCTION && t[1] == -SNUMBER && t[2] >= 0 ) {
01662             if ( functions[-*t-FUNCTION].spec > 0 ) tensorflag = 1;
01663             if ( t[2] == 0 ) {
01664                 if ( ipart+1 != parti->numpart ) goto WhatAPity;
01665                 argremain -= 2;
01666                 parti->nfun[ipart] = -*t;
01667                 parti->psize[ipart++] = argremain-sum;
01668                 ipart++;
01669                 sum = argremain;
01670             }
01671             else {
01672                 parti->nfun[ipart] = -*t;
01673                 parti->psize[ipart++] = t[2];
01674                 argremain -= 2;
01675                 sum += t[2];
01676             }
01677             t += 3;
01678         }
01679         else if ( *t == -SNUMBER && t[1] > 0 && ipart+t[1] <= parti->numpart
01680             && t[2] <= -FUNCTION && t[3] == -SNUMBER && t[4] > 0 ) {
01681             if ( functions[-t[2]-FUNCTION].spec > 0 ) tensorflag = 1;
01682             argremain -= 3;
01683             for ( i = 0; i < t[1]; i++ ) {
01684                 parti->nfun[ipart] = -t[2];
01685                 parti->psize[ipart++] = t[4];
01686                 sum += t[4];
01687             }
01688             if ( sum > argremain ) goto WhatAPity;
01689             t += 5;
01690         }
01691         else goto WhatAPity;
01692     }
01693     if ( sum != argremain ) goto WhatAPity;
01694     parti->numargs = argremain;
01695 }
01696 else goto No;

```

```

01697 /*
01698     Now load the offsets of the arguments and check if needed whether OK with tensor
01699 */
01700 for ( i = 0; i < parti->numargs; i++ ) {
01701     parti->args[i] = t - tfun;
01702     if ( tensorflag && ( *t != -VECTOR && *t != -INDEX ) ) goto WhatAPity;
01703     NEXTARG(t);
01704 }
01705 return(1);
01706 WhatAPity:
01707     M_free(parti->psize,"partitions");
01708     parti->psize = parti->nfun = parti->args = parti->nargs = 0;
01709     parti->numargs = parti->numpart = parti->where = 0;
01710 No:
01711     return(0);
01712 }
01713
01714 /*
01715     #] TestPartitions :
01716     #[ DoPartitions :
01717
01718     As we have only one AT.partitions we need to copy it locally
01719     if we keep needing it.
01720 */
01721
01722 int DoPartitions(PHEAD WORD *term, WORD level)
01723 {
01724     WORD x, i, j, im, *fun, ndiff, siz, tensorflag = 0;
01725     PARTI part = AT.partitions;
01726     WORD *array, **j3, **j3fill, **j3where;
01727     WORD a, pfill, *j2, *j2fill, j3size, ncoeff, ncoeffnum, nfac, ncoeff2, ncoeff3, n;
01728     UWORD *coeff, *coeffnum, *cfac, *coeff2, *coeff3, *c;
01729     /* Make AT.partitions ready for future use (if there is another function) */
01730     AT.partitions.psize = AT.partitions.nfun = AT.partitions.args = AT.partitions.nargs = 0;
01731     AT.partitions.numargs = AT.partitions.numpart = AT.partitions.where = 0;
01732 /*
01733     Start with bubble sorting the list of arguments. And the list of partitions.
01734 */
01735     fun = term + part.where;
01736     if ( functions[*fun-FUNCTION].spec > 0 ) tensorflag = 1;
01737     for ( i = 1; i < part.numargs; i++ ) {
01738         for ( j = i-1; j >= 0; j-- ) {
01739             if ( CompArg(fun+part.args[j+1],fun+part.args[j]) >= 0 ) break;
01740             x = part.args[j+1]; part.args[j+1] = part.args[j]; part.args[j] = x;
01741         }
01742     }
01743     for ( i = 1; i < part.numpart; i++ ) {
01744         for ( j = i-1; j >= 0; j-- ) {
01745             if ( part.psize[j+1] < part.psize[j] ) break;
01746             if ( part.psize[j+1] == part.psize[j] && part.nfun[j+1] <= part.nfun[j] ) break;
01747             x = part.psize[j+1]; part.psize[j+1] = part.psize[j]; part.psize[j] = x;
01748             x = part.nfun[j+1]; part.nfun[j+1] = part.nfun[j]; part.nfun[j] = x;
01749         }
01750     }
01751 /*
01752     Now we have the partitions sorted from high to low and the arguments
01753     have been sorted the regular way arguments are sorted in a symmetrize.
01754     The important thing is that identical arguments are adjacent.
01755     Assign the numbers (identical arguments have identical numbers).
01756 */
01757     ndiff = 1; part.nargs[0] = ndiff;
01758     for ( i = 1; i < part.numargs; i++ ) {
01759         if ( CompArg(fun+part.args[i],fun+part.args[i-1]) != 0 ) ndiff++;
01760         part.nargs[i] = ndiff;
01761     }
01762     part.nargs[part.numargs] = 0;
01763     coeffnum = NumberMalloc("partitionsn");
01764     coeff = NumberMalloc("partitions");
01765     coeff2 = NumberMalloc("partitions2");
01766     coeff3 = NumberMalloc("partitions3");
01767     cfac = NumberMalloc("partitions!");
01768     ncoeffnum = 1; coeffnum[0] = 1;
01769 /*
01770     The numerator of the coefficient will be n1!*n2!*...*n(ndiff)!
01771     We compute it only once.
01772 */
01773     j = 0;
01774     for ( i = 1; i <= ndiff; i++ ) {
01775         n = 0;
01776         while ( part.nargs[j] == i ) { n++; j++; }
01777         if ( n > 1 ) { /* 1! needs no attention */
01778             if ( Factorial(BHEAD n, cfac, &nfac) ) Terminate(-1);
01779             if ( MulLong(coeffnum,ncoeffnum,cfac,nfac,coeff2,&ncoeff2) ) Terminate(-1);
01780             c = coeffnum; coeffnum = coeff2; coeff2 = c;
01781             n = ncoeffnum; ncoeffnum = ncoeff2; ncoeff2 = n;
01782         }
01783     }

```

```

01784 /*
01785     Now comes the part where we have to make sure that
01786     a: we generate all partitions.
01787     b: we generate only different partitions.
01788     c: we get the proper combinatorics factor.
01789     Method:
01790     Suppose the largest partition needs n objects and there are m partitions.
01791     We allocate m arrays of n 'digits'. Make in the smaller partitions the
01792     appropriate leading digits zero.
01793     Divide the largest numbers (of the arguments) over the partitions as
01794     leftmost digits (after possible zeroes). The arrays, seen as numbers,
01795     should be such that each is less or equal to its left neighbour. Take the
01796     next largest numbers, etc. This generates unique partitions and all of
01797     them. Because we have a formula for the multiplicity, this should do it.
01798
01799     The general case. At a later stage we might put in a more economical
01800     version for special cases.
01801 */
01802     siz = part.psize[0];
01803     j3size = 2*(part.numpart+1)+2*(part.numargs+1);
01804     array = (WORD *)Malloc1((part.numpart+1)*siz*sizeof(WORD), "parts");
01805     j3 = (WORD **)Malloc1(j3size*sizeof(WORD *), "parts3");
01806     j2 = (WORD *)Malloc1((part.numpart+part.numargs+2)*sizeof(WORD), "parts2");
01807     j3fill = j3+(part.numpart+1);
01808     j3where = j3fill+(part.numpart+1);
01809     for ( i = 0; i < j3size; i++ ) j3[i] = 0;
01810     j2fill = j2+(part.numpart+1);
01811     for ( i = 0; i < part.numargs; i++ ) j2fill[i] = 0;
01812     for ( i = 0; i < part.numpart; i++ ) {
01813         j3[i] = array+i*siz;
01814         for ( j = 0; j < siz; j++ ) j3[i][j] = 0;
01815         j3fill[i] = j3[i]+(siz-part.psize[i]);
01816         j2[i] = part.psize[i]; /* Number of places still available */
01817     }
01818     j3[part.numpart] = array+part.numpart*siz;
01819     j2[part.numpart] = 0;
01820 /*
01821     Now comes a complicated two-level recursion in a and pfill.
01822 */
01823     a = part.numargs-1;
01824     pfill = 0;
01825 /*
01826     We start putting the last number in part.nargs in the first partition in array.
01827     For backtracking we need to know where we put this number. Hence j3where.
01828 */
01829     while ( a < part.numargs ) {
01830         while ( j2[pfill] <= 0 ) {
01831             pfill++;
01832             while ( pfill >= part.numpart ) { /* we have to pop */
01833                 a++;
01834                 if ( a >= part.numargs ) goto Done;
01835                 pfill = j2fill[a];
01836                 j2[pfill]++;
01837                 j3where[a][0] = 0;
01838                 j3fill[pfill]--;
01839                 pfill++;
01840             }
01841         }
01842         j3where[a] = j3fill[pfill];
01843         *(j3fill[pfill])++ = part.nargs[a];
01844         j2[pfill]--; j2fill[a] = pfill;
01845 /*
01846         Now test whether this is allowed.
01847 */
01848         if ( pfill > 0 && part.psize[pfill] == part.psize[pfill-1]
01849             && part.nfun[pfill] == part.nfun[pfill-1] ) { /* First check whether allowed */
01850             for ( im = 0; im < siz; im++ ) {
01851                 if ( j3[pfill-1][im] < j3[pfill][im] ) break;
01852                 if ( j3[pfill-1][im] > j3[pfill][im] ) im = siz;
01853             }
01854             if ( im < siz ) { /* not ordered. undo and raise pfill */
01855                 pfill = j2fill[a];
01856                 j2[pfill]++;
01857                 j3where[a][0] = 0;
01858                 j3fill[pfill]--;
01859                 pfill++;
01860                 continue; /* Note that j2[part.numpart] = 0 */
01861             }
01862         }
01863         a--;
01864         if ( a < 0 ) { /* Solution */
01865 /*
01866             #[ Solution :
01867
01868             Now we compose the output term. The input term contains
01869             three parts: head, partitions_, tail.
01870             partitions_ starts at term+part.where.

```

```

01871      We first put the function parts and worry about the coefficient later.
01872  */
01873      WORD *t, *to, *twhere = term+part.where, *t2, *tend = term+term, *termout;
01874      WORD num, jj, *targ, *tfun;
01875      t2 = twhere+twhere[1];
01876      to = termout = AT.WorkPointer;
01877      if ( termout + *term + part.numpart*FUNHEAD + AM.MaxTal >= AT.WorkTop ) {
01878          return(MesWork());
01879      }
01880      for ( i = 0; i < ncoeffnum; i++ ) coeff[i] = coeffnum[i];
01881      ncoeff = ncoeffnum;
01882      t = term; while ( t < twhere ) *to++ = *t++;
01883  /*
01884      Now the partitions
01885  */
01886      for ( i = 0; i < part.numpart; i++ ) {
01887          tfun = to;
01888          *to++ = part.nfun[i]; to++; FILLFUN(to);
01889          for ( j = 1; j <= part.psize[i]; j++ ) {
01890              num = j3[i][siz-j]; /* now we need an argument with this number */
01891              for ( jj = num-1; jj < part.numargs; jj++ ) {
01892                  if ( part.nargs[jj] == num ) break;
01893              }
01894              targ = part.args[jj]+twhere;
01895              if ( *targ < 0 ) {
01896                  if ( tensorflag ) targ++;
01897                  else if ( *targ > -FUNCTION ) *to++ = *targ++;
01898                  *to++ = *targ++;
01899              }
01900              else { jj = *targ; NCOPY(to,targ,jj); }
01901          }
01902          tfun[1] = to - tfun;
01903      }
01904  /*
01905      Now the denominators of the coefficient
01906      First identical functions/partitions
01907  */
01908      j = 1; n = 1;
01909      while ( j < part.numpart ) {
01910          for ( im = 0; im < siz; im++ ) {
01911              if ( part.nfun[j-1] != part.nfun[j] ) break;
01912              if ( j3[j-1][im] < j3[j][im] ) break;
01913              if ( j3[j-1][im] > j3[j][im] ) im = 2*siz+2;
01914          }
01915          if ( im == siz ) { n++; j++; continue; }
01916          if ( n > 1 ) {
01917              div1: if ( Factorial(BHEAD n, cfac, &nfac) ) Terminate(-1);
01918                  if ( DivLong(coeff,ncoeff,cfac,nfac,coeff2,&ncoeff2,coeff3,&ncoeff3) )
01919                      Terminate(-1);
01920                      c = coeff; coeff = coeff2; coeff2 = c;
01921                      n = ncoeff; ncoeff = ncoeff2; ncoeff2 = n;
01922                      }
01923                      n = 1; j++;
01924                      }
01925                      if ( n > 1 ) goto div1;
01926  /*
01927      Now identical elements inside the partitions
01928  */
01929      for ( i = 0; i < part.numpart; i++ ) {
01930          j = 0; while ( j3[i][j] == 0 ) j++;
01931          n = 1; j++;
01932          while ( j < siz ) {
01933              if ( j3[i][j-1] == j3[i][j] ) { n++; j++; }
01934              else {
01935                  if ( n > 1 ) {
01936                      div2: if ( Factorial(BHEAD n, cfac, &nfac) ) Terminate(-1);
01937                          if ( DivLong(coeff,ncoeff,cfac,nfac,coeff2,&ncoeff2,coeff3,&ncoeff3) )
01938                              Terminate(-1);
01939                              c = coeff; coeff = coeff2; coeff2 = c;
01940                              n = ncoeff; ncoeff = ncoeff2; ncoeff2 = n;
01941                              }
01942                              n = 1; j++;
01943                              }
01944                              if ( n > 1 ) goto div2;
01945  /*
01946      And put this inside the term. Normalize will take care of it.
01947  */
01948      if ( ncoeff != 1 || coeff[0] > 1 ) {
01949          if ( ncoeff == 1 && coeff[0] <= MAXPOSITIVE ) {
01950              *to++ = SNUMBER; *to++ = 4; *to++ = (WORD)(coeff[0]); *to++ = 1;
01951          }
01952          else {
01953              *to++ = LNUMBER; *to++ = ncoeff+3; *to++ = ncoeff;
01954              for ( i = 0; i < ncoeff; i++ ) *to++ = ((WORD *)coeff)[i];
01955          }

```

```

01956         }
01957  /*
01958         And the tail
01959  */
01960         while ( t2 < tend ) *to++ = *t2++;
01961         *termout = to-termout;
01962         AT.WorkPointer = to;
01963         if ( Generator(BHEAD termout,level) ) Terminate(-1);
01964         AT.WorkPointer = termout;
01965  /*
01966         #] Solution :
01967
01968         Now we can pop all a with the lowest value and one more.
01969  */
01970         a = 0;
01971         while ( part.nargs[a] == 1 ) {
01972             pfill = j2fill[a]; j2[pfill]++; j3where[a][0] = 0; j3fill[pfill]--; a++;
01973         }
01974         if ( a < part.numargs ) {
01975             pfill = j2fill[a]; j2[pfill]++; j3where[a][0] = 0; j3fill[pfill]--; a++;
01976         }
01977         a--;
01978         pfill++;
01979     }
01980     else if ( part.nargs[a] == part.nargs[a+1] ) {}
01981     else { pfill = 0; }
01982 }
01983 Done:
01984     M_free(j2,"parts2");
01985     M_free(j3,"parts3");
01986     M_free(array,"parts");
01987     NumberFree(cfac,"partitions!");
01988     NumberFree(coeff3,"partitions3");
01989     NumberFree(coeff2,"partitions2");
01990     NumberFree(coeff,"partitions");
01991     NumberFree(coeffnum,"partitionsn");
01992     M_free(part.psize,"partitions");
01993     part.psize = part.nfun = part.args = part.nargs = 0;
01994     part.numargs = part.numpart = part.where = 0;
01995     return(0);
01996 }
01997
01998 /*
01999     #] DoPartitions :
02000     #] Distribute :
02001     #[ DoPermutations :
02002
02003     Routine replaces the function perm_(f,args) by occurrences of f with
02004     all permutations of the args. This should always fit!
02005  */
02006
02007 int DoPermutations(PHEAD WORD *term, WORD level)
02008 {
02009     PERMP perm;
02010     WORD *oldworkpointer = AT.WorkPointer, *termout = AT.WorkPointer;
02011     WORD *t, *tstop, *tt, *ttstop, odd = 0;
02012     WORD *args[MAXMATCH], nargs, i, first, skip, *to, *from;
02013  /*
02014     Find function and count arguments. Check for odd/even
02015  */
02016     tstop = term+*term; tstop -= ABS(tstop[-1]);
02017     t = term+1;
02018     while ( t < tstop ) {
02019         if ( *t == PERMUTATIONS ) {
02020             if ( t[1] >= FUNHEAD+1 && t[FUNHEAD] <= -FUNCTION ) {
02021                 odd = 0; skip = 1;
02022             }
02023             else if ( t[1] >= FUNHEAD+3 && t[FUNHEAD] == -SNUMBER && t[FUNHEAD+2] <= -FUNCTION ) {
02024                 if ( t[FUNHEAD+1] % 2 == 1 ) odd = -1;
02025                 else odd = 0;
02026                 skip = 3;
02027             }
02028             else { t += t[1]; continue; }
02029             tt = t+FUNHEAD+skip; ttstop = t + t[1];
02030             nargs = 0;
02031             while ( tt < ttstop ) { NEXTARG(tt); nargs++; }
02032             tt = t+FUNHEAD+skip;
02033             if ( nargs > MAXMATCH ) {
02034                 MLOCK(ErrorMessageLock);
02035                 MesPrint("Too many arguments in function perm_. %d! is way too big", (WORD)MAXMATCH);
02036                 MUNLOCK(ErrorMessageLock);
02037                 SETERROR(-1)
02038             }
02039             i = 0;
02040             while ( tt < ttstop ) { args[i++] = tt; NEXTARG(tt); }
02041             perm.n = nargs;
02042             perm.sign = 0;

```

```

02043     perm.objects = args;
02044     first = 1;
02045     while ( (first = PermuteP(&perm,first) ) == 0 ) {
02046 /*
02047         Compose the output term
02048 */
02049         to = termout; from = term;
02050         while ( from < t ) *to++ = *from++;
02051         *to++ = -t[FUNHEAD+skip-1];
02052         *to++ = t[1] - skip;
02053         for ( i = 2; i < FUNHEAD; i++ ) *to++ = t[i];
02054         for ( i = 0; i < nargs; i++ ) {
02055             from = args[i];
02056             COPY1ARG(to,from);
02057         }
02058         from = t+t[1];
02059         tstop = term + *term;
02060         while ( from < tstop ) *to++ = *from++;
02061         if ( odd && ( ( perm.sign & 1 ) != 0 ) ) to[-1] = -to[-1];
02062         *termout = to - termout;
02063         AT.WorkPointer = to;
02064         if ( Generator(BHEAD termout,level) ) Terminate(-1);
02065         AT.WorkPointer = oldworkpointer;
02066     }
02067     return(0);
02068 }
02069 t += t[1];
02070 }
02071 return(0);
02072 }
02073
02074 /*
02075 #] DoPermutations :
02076 #[ DoShuffle :
02077
02078 Merges the arguments of all occurrences of function fun into a
02079 single occurrence of fun. The opposite of Distrib_
02080 Syntax:
02081     Shuffle[,once|all],fun;
02082     Shuffle[,once|all],$fun;
02083 The expansion of the dollar should give a single function.
02084 The dollar is indicated as usual with a negative value.
02085 option = 1 (once): generate identical results only once
02086 option = 0 (all): generate identical results with combinatorics (default)
02087 */
02088
02089 /*
02090 We use the Shuffle routine which has a large amount of combinatorics.
02091 It doesn't have grouped combinatorics as in (0,1,2)*(0,1,3) where the
02092 groups (0,1) also cause double terms.
02093 */
02094
02095 int DoShuffle(WORD *term, WORD level, WORD fun, WORD option)
02096 {
02097     GETIDENTITY
02098     SHvariables SHback, *SH = &(AN.SHvar);
02099     WORD *t1, *t2, *tstop, ncoef, n = fun, *to, *from;
02100     int i, error;
02101     LONG k;
02102     UWORD *newcombi;
02103
02104     if ( n < 0 ) {
02105         if ( ( n = DolToFunction(BHEAD -n) ) == 0 ) {
02106             MLOCK(ErrorMessageLock);
02107             MesPrint("$-variable in merge statement did not evaluate to a function.");
02108             MUNLOCK(ErrorMessageLock);
02109             return(1);
02110         }
02111     }
02112     if ( AT.WorkPointer + 3*(*term) + AM.MaxTal > AT.WorkTop ) {
02113         MLOCK(ErrorMessageLock);
02114         MesWork();
02115         MUNLOCK(ErrorMessageLock);
02116         return(-1);
02117     }
02118
02119     tstop = term + *term;
02120     ncoef = tstop[-1];
02121     tstop -= ABS(ncoef);
02122     t1 = term + 1;
02123     while ( t1 < tstop ) {
02124         if ( ( *t1 == n ) && ( t1+t1[1] < tstop ) && ( t1[1] > FUNHEAD ) ) {
02125             t2 = t1 + t1[1];
02126             if ( t2 >= tstop ) {
02127                 return(Generator(BHEAD term,level));
02128             }
02129             while ( t2 < tstop ) {

```

```

02130         if ( ( *t2 == n ) && ( t2[1] > FUNHEAD ) ) break;
02131         t2 += t2[1];
02132     }
02133     if ( t2 < tstop ) break;
02134 }
02135 t1 += t1[1];
02136 }
02137 if ( t1 >= tstop ) {
02138     return(Generator(BHEAD term,level));
02139 }
02140 *AN.RepPoint = 1;
02141 /*
02142 Now we have two occurrences of the function.
02143 Back up all relevant variables and load all the stuff that needs to be
02144 passed on.
02145 */
02146 SHback = AN.SHvar;
02147 SH->finishuf = &FinishShuffle;
02148 SH->do_uffle = &DoShuffle;
02149 SH->outterm = AT.WorkPointer;
02150 AT.WorkPointer += *term;
02151 SH->stop1 = t1 + t1[1];
02152 SH->stop2 = t2 + t2[1];
02153 SH->thefunction = n;
02154 SH->option = option;
02155 SH->level = level;
02156 SH->incoef = tstop;
02157 SH->nincoef = ncoef;
02158
02159 if ( AN.SHcombi == 0 || AN.SHcombisize == 0 ) {
02160     AN.SHcombisize = 200;
02161     AN.SHcombi = (UWORD *)Malloca(AN.SHcombisize*sizeof(UWORD), "AN.SHcombi");
02162     SH->combilast = 0;
02163     SHback.combilast = 0;
02164 }
02165 else {
02166     SH->combilast += AN.SHcombi[SH->combilast]+1;
02167     if ( SH->combilast >= AN.SHcombisize - 100 ) {
02168         newcombi = (UWORD *)Malloca(2*AN.SHcombisize*sizeof(UWORD), "AN.SHcombi");
02169         for ( k = 0; k < AN.SHcombisize; k++ ) newcombi[k] = AN.SHcombi[k];
02170         M_free(AN.SHcombi, "AN.SHcombi");
02171         AN.SHcombi = newcombi;
02172         AN.SHcombisize *= 2;
02173     }
02174 }
02175 AN.SHcombi[SH->combilast] = 1;
02176 AN.SHcombi[SH->combilast+1] = 1;
02177
02178 i = t1-term; to = SH->outterm; from = term;
02179 NCOPY(to,from,i)
02180 SH->outfun = to;
02181 for ( i = 0; i < FUNHEAD; i++ ) { *to++ = t1[i]; }
02182
02183 error = Shuffle(t1+FUNHEAD,t2+FUNHEAD,to);
02184
02185 AT.WorkPointer = SH->outterm;
02186 AN.SHvar = SHback;
02187 if ( error ) {
02188     MesCall("DoShuffle");
02189     return(-1);
02190 }
02191 return(0);
02192 }
02193
02194 /*
02195 #] DoShuffle :
02196 #[ Shuffle :
02197
02198 How to make shuffles:
02199
02200 We have two lists of arguments. We have to make a single
02201 shuffle of them. All combinations. Doubles should have as
02202 much as possible a combinatorics factor. Sometimes this is
02203 very difficult as in:
02204 (0,1,2)x(0,1,3) = -> (0,1) is a repeated pattern and the
02205 factor on that is difficult
02206 Simple way: (without combinatorics)
02207 repeat id f0(?c)*f(x1?,?a)*f(x2?,?b) =
02208         +f0(?c,x1)*f(?a)*f(x2,?b)
02209         +f0(?c,x2)*f(x1,?a)*f(?b);
02210
02211 Refinement:
02212 if ( x1 == x2 ) check how many more there are of the same.
02213 --> (n1,x) and (n2,x)
02214 id f0(?c)*f1((n1,x),?b)*f2((n2,x),?c) =
02215         +binom_(n1+n2,n1)*f0(?c,(n1+n2,x))*f1(?a)*f2(?b)
02216         +sum_(j,0,n1-1,binom_(n2+j,j)*f0(?c,(j+n2,x))
02217             *f1((n1-j),?a)*f2(?b))*force2

```



```

02217             +sum_(j,0,n2-1,binom_(n1+j,j)*f0(?c,(j+n1,x))
02218             *f1(?a)*f2((n2-j),?b))*force1
02219     The force operation can be executed directly
02220
02221     The next question is how to program this: recursively or linearly
02222     which would require simulation of a recursion. Recursive is clearest
02223     but we need to pass a number of arguments from the calling routine
02224     to the final routine. This is done with AN.SHvar.
02225
02226     We need space for the accumulation of the combinatoric factors.
02227 */
02228
02229 int Shuffle(WORD *from1, WORD *from2, WORD *to)
02230 {
02231     GETIDENTITY
02232     WORD *t, *fr, *next1, *next2, na, *fn1, *fn2, *tt;
02233     int i, n, n1, n2, j;
02234     LONG combilast;
02235     SHvariables *SH = &(AN.SHvar);
02236     if ( from1 == SH->stop1 && from2 == SH->stop2 ) {
02237         return(FiniShuffle(to));
02238     }
02239     else if ( from1 == SH->stop1 ) {
02240         i = SH->stop2 - from2; t = to; tt = from2; NCOPY(t,tt,i)
02241         return(FiniShuffle(t));
02242     }
02243     else if ( from2 == SH->stop2 ) {
02244         i = SH->stop1 - from1; t = to; tt = from1; NCOPY(t,tt,i)
02245         return(FiniShuffle(t));
02246     }
02247 /*
02248     Compare lead arguments
02249 */
02250     if ( AreArgsEqual(from1,from2) ) {
02251 /*
02252         First find out how many of each
02253 */
02254         next1 = from1; n1 = 1; NEXTARG(next1)
02255         while ( ( next1 < SH->stop1 ) && AreArgsEqual(from1,next1) ) {
02256             n1++; NEXTARG(next1)
02257         }
02258         next2 = from2; n2 = 1; NEXTARG(next2)
02259         while ( ( next2 < SH->stop2 ) && AreArgsEqual(from2,next2) ) {
02260             n2++; NEXTARG(next2)
02261         }
02262         combilast = SH->combilast;
02263 /*
02264         +binom_(n1+n2,n1)*f0(?c,(n1+n2,x))*f1(?a)*f2(?b)
02265 */
02266         t = to;
02267         n = n1 + n2;
02268         while ( --n >= 0 ) { fr = from1; CopyArg(t,fr) }
02269         if ( GetBinom((UWORD *) (t),&na,n1+n2,n1) ) goto shuffcall;
02270         if ( combilast + AN.SHcombi[combilast] + na + 2 >= AN.SHcombisize ) {
02271 /*
02272             We need more memory in this stack. Fortunately this is the
02273             only place where we have to do this, because the other factors
02274             are definitely smaller.
02275             Layout:  size, LongInteger, size, LongInteger, .....
02276             We start pointing at the last one.
02277 */
02278             UWORD *combi = (UWORD *)Malloca(2*AN.SHcombisize*2,"AN.SHcombi");
02279             LONG jj;
02280             for ( jj = 0; jj < AN.SHcombisize; jj++ ) combi[jj] = AN.SHcombi[jj];
02281             AN.SHcombisize *= 2;
02282             M_free(AN.SHcombi,"AN.SHcombi");
02283             AN.SHcombi = combi;
02284         }
02285         if ( MulLong((UWORD *) (AN.SHcombi+combilast+1),AN.SHcombi[combilast],
02286             (UWORD *) (t),na,
02287             (UWORD *) (AN.SHcombi+combilast+AN.SHcombi[combilast]+2),
02288             (WORD *) (AN.SHcombi+combilast+AN.SHcombi[combilast]+1)) ) goto shuffcall;
02289         SH->combilast = combilast + AN.SHcombi[combilast] + 1;
02290         if ( next1 >= SH->stop1 ) {
02291             fr = next2; i = SH->stop2 - fr;
02292             NCOPY(t,fr,i)
02293             if ( FiniShuffle(t) ) goto shuffcall;
02294         }
02295         else if ( next2 >= SH->stop2 ) {
02296             fr = next1; i = SH->stop1 - fr;
02297             NCOPY(t,fr,i)
02298             if ( FiniShuffle(t) ) goto shuffcall;
02299         }
02300         else {
02301             if ( Shuffle(next1,next2,t) ) goto shuffcall;
02302         }
02303         SH->combilast = combilast;

```

```

02304  /*
02305      +sum_(j,0,n1-1,binom_(n2+j,j)*f0(?c,(j+n2,x))
02306      *f1((n1-j),?a)*f2(?b))*force2
02307  */
02308      if ( next2 < SH->stop2 ) {
02309          t = to;
02310          n = n2;
02311          while ( --n >= 0 ) { fr = from1; CopyArg(t,fr) }
02312          for ( j = 0; j < n1; j++ ) {
02313              if ( GetBinom((UWORD *) (t), &na, n2+j, j) ) goto shuffcall;
02314              if ( Mullong((UWORD *) (AN.SHcombi+combilast+1), AN.SHcombi[combilast],
02315                          (UWORD *) (t), na,
02316                          (UWORD *) (AN.SHcombi+combilast+AN.SHcombi[combilast]+2),
02317                          (WORD *) (AN.SHcombi+combilast+AN.SHcombi[combilast]+1)) ) goto shuffcall;
02318              SH->combilast = combilast + AN.SHcombi[combilast] + 1;
02319              if ( j > 0 ) { fr = from1; CopyArg(t,fr) }
02320              fn2 = next2; tt = t;
02321              CopyArg(tt,fn2)
02322          }
02323          if ( fn2 >= SH->stop2 ) {
02324              n = n1-j;
02325              while ( --n >= 0 ) { fr = from1; CopyArg(tt,fr) }
02326              fr = next1; i = SH->stop1 - fr;
02327              NCOPY(tt,fr,i)
02328              if ( FiniShuffle(tt) ) goto shuffcall;
02329          }
02330          else {
02331              n = j; fn1 = from1; while ( --n >= 0 ) { NEXTARG(fn1) }
02332              if ( Shuffle(fn1,fn2,tt) ) goto shuffcall;
02333          }
02334          SH->combilast = combilast;
02335      }
02336  }
02337  /*
02338      +sum_(j,0,n2-1,binom_(n1+j,j)*f0(?c,(j+n1,x))
02339      *f1(?a)*f2((n2-j),?b))*force1
02340  */
02341      if ( next1 < SH->stop1 ) {
02342          t = to;
02343          n = n1;
02344          while ( --n >= 0 ) { fr = from1; CopyArg(t,fr) }
02345          for ( j = 0; j < n2; j++ ) {
02346              if ( GetBinom((UWORD *) (t), &na, n1+j, j) ) goto shuffcall;
02347              if ( Mullong((UWORD *) (AN.SHcombi+combilast+1), AN.SHcombi[combilast],
02348                          (UWORD *) (t), na,
02349                          (UWORD *) (AN.SHcombi+combilast+AN.SHcombi[combilast]+2),
02350                          (WORD *) (AN.SHcombi+combilast+AN.SHcombi[combilast]+1)) ) goto shuffcall;
02351              SH->combilast = combilast + AN.SHcombi[combilast] + 1;
02352              if ( j > 0 ) { fr = from1; CopyArg(t,fr) }
02353              fn1 = next1; tt = t;
02354              CopyArg(tt,fn1)
02355          }
02356          if ( fn1 >= SH->stop1 ) {
02357              n = n2-j;
02358              while ( --n >= 0 ) { fr = from1; CopyArg(tt,fr) }
02359              fr = next2; i = SH->stop2 - fr;
02360              NCOPY(tt,fr,i)
02361              if ( FiniShuffle(tt) ) goto shuffcall;
02362          }
02363          else {
02364              n = j; fn2 = from2; while ( --n >= 0 ) { NEXTARG(fn2) }
02365              if ( Shuffle(fn1,fn2,tt) ) goto shuffcall;
02366          }
02367          SH->combilast = combilast;
02368      }
02369  }
02370  }
02371  else {
02372  /*
02373      Argument from first list
02374  */
02375      t = to;
02376      fr = from1;
02377      CopyArg(t,fr)
02378      if ( fr >= SH->stop1 ) {
02379          fr = from2; i = SH->stop2 - fr;
02380          NCOPY(t,fr,i)
02381          if ( FiniShuffle(t) ) goto shuffcall;
02382      }
02383      else {
02384          if ( Shuffle(fr,from2,t) ) goto shuffcall;
02385      }
02386  /*
02387      Argument from second list
02388  */
02389      t = to;
02390      fr = from2;

```

```

02391     CopyArg(t,fr)
02392     if ( fr >= SH->stop2 ) {
02393         fr = from1; i = SH->stop1 - fr;
02394         NCOPY(t,fr,i)
02395         if ( FiniShuffle(t) ) goto shuffcall;
02396     }
02397     else {
02398         if ( Shuffle(from1,fr,t) ) goto shuffcall;
02399     }
02400 }
02401 return(0);
02402 shuffcall:
02403 MesCall("Shuffle");
02404 return(-1);
02405 }
02406
02407 /*
02408 #] Shuffle :
02409 #[ FinishShuffle :
02410
02411 The complications here are:
02412 1: We want to save space. We put the output term in 'out' straight
02413    on top of what we produced thusfar. We have to copy the early
02414    piece because once the term goes back to Generator, Normalize can
02415    change it in situ
02416 2: There can be other occurrence of the function between the two
02417    that we did. For shuffles that isn't likely, but we use this
02418    routine also for the stuffles and there it can happen.
02419 */
02420
02421 int FinishShuffle(WORD *fini)
02422 {
02423     GETIDENTITY
02424     WORD *t, *t1, *oldworkpointer = AT.WorkPointer, *tcoef, ntcoef, *out;
02425     int i;
02426     SHvariables *SH = &(AN.SHvar);
02427     SH->outfun[1] = fini - SH->outfun;
02428     if ( functions[SH->outfun[0]-FUNCTION].symmetric != 0 )
02429         SH->outfun[2] |= DIRTYSYMFLAG;
02430     out = fini; i = fini - SH->outterm; t = SH->outterm;
02431     NCOPY(fini,t,i)
02432     t = SH->stop1;
02433     t1 = t + t[1];
02434     while ( t1 < SH->stop2 ) { t = t1; t1 = t + t[1]; }
02435     t1 = SH->stop1;
02436     while ( t1 < t ) *fini++ = *t1++;
02437     t = SH->stop2;
02438     while ( t < SH->incoef ) *fini++ = *t++;
02439     tcoef = fini;
02440     ntcoef = SH->nincoef;
02441     i = ABS(ntcoef);
02442     NCOPY(fini,t,i);
02443     ntcoef = REDLENG(ntcoef);
02444     Mully(BHEAD (UWORD *)tcoef,&ntcoef,
02445         (UWORD *) (AN.SHcombi+SH->combilast+1),AN.SHcombi[SH->combilast]));
02446     ntcoef = INCLENG(ntcoef);
02447     fini = tcoef + ABS(ntcoef);
02448     if ( ( ( SH->option & 2 ) != 0 ) && ( ( SH->option & 256 ) != 0 ) ) ntcoef = -ntcoef;
02449     fini[-1] = ntcoef;
02450     i = *out = fini - out;
02451 /*
02452 Now check whether we have to do more
02453 */
02454 AT.WorkPointer = out + *out;
02455 if ( ( SH->option & 1 ) == 1 ) {
02456     if ( Generator(BHEAD out,SH->level) ) goto Finicall;
02457 }
02458 else {
02459     if ( DoStuffle(out,SH->level,SH->thefunction,SH->option) ) goto Finicall;
02460 }
02461 AT.WorkPointer = oldworkpointer;
02462 return(0);
02463 Finicall:
02464 AT.WorkPointer = oldworkpointer;
02465 MesCall("FinishShuffle");
02466 return(-1);
02467 }
02468
02469 /*
02470 #] FinishShuffle :
02471 #[ DoStuffle :
02472
02473 Stuffling is a variation of shuffling.
02474 In the stuffling we insist that the arguments are (short) integers. nonzero.
02475 The stuffle sum is  $x \text{ st } y = \text{sig}_-(x) * \text{sig}_-(y) * (\text{abs}(x) + \text{abs}(y))$ 
02476 The way we do this is:
02477 1: count the arguments in each function: n1, n2

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02478         2: take the minimum minval = min(n1,n2).
02479         3: for ( j = 0; j <= min; j++ ) take j elements in each of the lists.
02480         4: the j+1 groups of remaining arguments have to each be shuffled
02481         5: the j selected pairs have to be stuffle added.
02482         We can use many of the shuffle things.
02483         Considering the recursive nature of the generation we actually don't
02484         need to know n1, n2, minval.
02485     */
02486
02487 int DoStuffle(WORD *term, WORD level, WORD fun, WORD option)
02488 {
02489     GETIDENTITY
02490     SHvariables SHback, *SH = &(AN.SHvar);
02491     WORD *t1, *t2, *tstop, *t1stop, *t2stop, ncoef, n = fun, *to, *from;
02492     WORD *r1, *r2;
02493     int i, error;
02494     LONG k;
02495     UWORD *newcombi;
02496 #ifdef NEWCODE
02497     WORD *rr1, *rr2, i1, i2;
02498 #endif
02499     if ( n < 0 ) {
02500         if ( ( n = DolToFunction(BHEAD -n) ) == 0 ) {
02501             MLOCK(ErrorMessageLock);
02502             MesPrint("$-variable in merge statement did not evaluate to a function.");
02503             MUNLOCK(ErrorMessageLock);
02504             return(1);
02505         }
02506     }
02507     if ( AT.WorkPointer + 3*(term) + AM.MaxTal > AT.WorkTop ) {
02508         MLOCK(ErrorMessageLock);
02509         MesWork();
02510         MUNLOCK(ErrorMessageLock);
02511         return(-1);
02512     }
02513
02514     tstop = term + *term;
02515     ncoef = tstop[-1];
02516     tstop -= ABS(ncoef);
02517     t1 = term + 1;
02518 retry1:;
02519     while ( t1 < tstop ) {
02520         if ( ( *t1 == n ) && ( t1+t1[1] < tstop ) && ( t1[1] > FUNHEAD ) ) {
02521             t2 = t1 + t1[1];
02522             if ( t2 >= tstop ) {
02523                 return(Generator(BHEAD term,level));
02524             }
02525 retry2:;
02526             while ( t2 < tstop ) {
02527                 if ( ( *t2 == n ) && ( t2[1] > FUNHEAD ) ) break;
02528                 t2 += t2[1];
02529             }
02530             if ( t2 < tstop ) break;
02531         }
02532         t1 += t1[1];
02533     }
02534     if ( t1 >= tstop ) {
02535         return(Generator(BHEAD term,level));
02536     }
02537     /*
02538     Next we have to check that the arguments are of the correct type
02539     At the same time we can count them.
02540     */
02541 #ifndef NEWCODE
02542     t1stop = t1 + t1[1];
02543     r1 = t1 + FUNHEAD;
02544     while ( r1 < t1stop ) {
02545         if ( *r1 != -SNUMBER ) break;
02546         if ( r1[1] == 0 ) break;
02547         r1 += 2;
02548     }
02549     if ( r1 < t1stop ) { t1 = t2; goto retry1; }
02550     t2stop = t2 + t2[1];
02551     r2 = t2 + FUNHEAD;
02552     while ( r2 < t2stop ) {
02553         if ( *r2 != -SNUMBER ) break;
02554         if ( r2[1] == 0 ) break;
02555         r2 += 2;
02556     }
02557     if ( r2 < t2stop ) { t2 = t2 + t2[1]; goto retry2; }
02558 #else
02559     t1stop = t1 + t1[1];
02560     r1 = t1 + FUNHEAD;
02561     while ( r1 < t1stop ) {
02562         if ( *r1 == -SNUMBER ) {
02563             if ( r1[1] == 0 ) break;
02564             r1 += 2; continue;

```

```

02565     }
02566     else if ( *r1 == -SYMBOL ) {
02567         if ( ( symbols[r1[1]].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
02568             break;
02569         r1 += 2; continue;
02570     }
02571     if ( *r1 > 0 && *r1 == r1[ARGHEAD]+ARGHEAD ) {
02572         if ( ABS(r1[r1[0]-1]) == r1[0]-ARGHEAD-1 ) {}
02573         else if ( r1[ARGHEAD+1] == SYMBOL ) {
02574             rr1 = r1 + ARGHEAD + 3;
02575             i1 = rr1[-1]-2;
02576             while ( i1 > 0 ) {
02577                 if ( ( symbols[*rr1].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
02578                     break;
02579                 i1 -= 2; rr1 += 2;
02580             }
02581             if ( i1 > 0 ) break;
02582         }
02583         else break;
02584         rr1 = r1+*r1-1;
02585         i1 = (ABS(*rr1)-1)/2;
02586         while ( i1 > 1 ) {
02587             if ( rr1[-1] ) break;
02588             i1--; rr1--;
02589         }
02590         if ( i1 > 1 || rr1[-1] != 1 ) break;
02591         r1 += *r1;
02592     }
02593     else break;
02594 }
02595 if ( r1 < t1stop ) { t1 = t2; goto retry1; }
02596 t2stop = t2 + t2[1];
02597 r2 = t2 + FUNHEAD;
02598
02599 while ( r2 < t2stop ) {
02600     if ( *r2 == -SNUMBER ) {
02601         if ( r2[1] == 0 ) break;
02602         r2 += 2; continue;
02603     }
02604     else if ( *r2 == -SYMBOL ) {
02605         if ( ( symbols[r2[1]].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
02606             break;
02607         r2 += 2; continue;
02608     }
02609     if ( *r2 > 0 && *r2 == r2[ARGHEAD]+ARGHEAD ) {
02610         if ( ABS(r2[r2[0]-1]) == r2[0]-ARGHEAD-1 ) {}
02611         else if ( r2[ARGHEAD+1] == SYMBOL ) {
02612             rr2 = r2 + ARGHEAD + 3;
02613             i2 = rr2[-1]-2;
02614             while ( i2 > 0 ) {
02615                 if ( ( symbols[*rr2].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
02616                     break;
02617                 i2 -= 2; rr2 += 2;
02618             }
02619             if ( i2 > 0 ) break;
02620         }
02621         else break;
02622         rr2 = r2+*r2-1;
02623         i2 = (ABS(*rr2)-1)/2;
02624         while ( i2 > 1 ) {
02625             if ( rr2[-1] ) break;
02626             i2--; rr2--;
02627         }
02628         if ( i2 > 1 || rr2[-1] != 1 ) break;
02629         r2 += *r2;
02630     }
02631     else break;
02632 }
02633 if ( r2 < t2stop ) { t2 = t2 + t2[1]; goto retry2; }
02634 #endif
02635 /*
02636 OK, now we got two objects that can be used.
02637 */
02638 *AN.RepPoint = 1;
02639
02640 SHback = AN.SHvar;
02641 SH->finishuf = &FinishStuffle;
02642 SH->do_uffle = &DoStuffle;
02643 SH->outterm = AT.WorkPointer;
02644 AT.WorkPointer += *term;
02645 SH->ststop1 = t1 + t1[1];
02646 SH->ststop2 = t2 + t2[1];
02647 SH->thefunction = n;
02648 SH->option = option;
02649 SH->level = level;
02650 SH->incoef = tstop;
02651 SH->nincoef = ncoef;

```

```

02652     if ( AN.SHcombi == 0 || AN.SHcombisize == 0 ) {
02653         AN.SHcombisize = 200;
02654         AN.SHcombi = (UWORD *)Malloc1(AN.SHcombisize*sizeof(UWORD), "AN.SHcombi");
02655         SH->combilast = 0;
02656         SHback.combilast = 0;
02657     }
02658     else {
02659         SH->combilast += AN.SHcombi[SH->combilast]+1;
02660         if ( SH->combilast >= AN.SHcombisize - 100 ) {
02661             newcombi = (UWORD *)Malloc1(2*AN.SHcombisize*sizeof(UWORD), "AN.SHcombi");
02662             for ( k = 0; k < AN.SHcombisize; k++ ) newcombi[k] = AN.SHcombi[k];
02663             M_free(AN.SHcombi, "AN.SHcombi");
02664             AN.SHcombi = newcombi;
02665             AN.SHcombisize *= 2;
02666         }
02667     }
02668     AN.SHcombi[SH->combilast] = 1;
02669     AN.SHcombi[SH->combilast+1] = 1;
02670
02671     i = t1-term; to = SH->outterm; from = term;
02672     NCOPY(to, from, i)
02673     SH->outfun = to;
02674     for ( i = 0; i < FUNHEAD; i++ ) { *to++ = t1[i]; }
02675
02676     error = Stuffle(t1+FUNHEAD, t2+FUNHEAD, to);
02677
02678     AT.WorkPointer = SH->outterm;
02679     AN.SHvar = SHback;
02680     if ( error ) {
02681         MesCall("DoStuffle");
02682         return(-1);
02683     }
02684     return(0);
02685 }
02686
02687 /*
02688 #] DoStuffle :
02689 #[ Stuffle :
02690
02691 The way to generate the stuffles
02692 1: select an argument in the first list (for(j1=0;j1<last;j1++))
02693 2: select an argument in the second list (for(j2=0;j2<last;j2++))
02694 3: put values for SH->ststop1 and SH->ststop2 at these arguments.
02695 4: generate all shuffles of the arguments in front.
02696 5: Then put the stuffle sum of arg(j1) and arg(j2)
02697 6: Then continue calling Stuffle
02698 7: Once one gets exhausted, we can clean up the list and call FinishShuffle
02699 8: if ( ( SH->option & 2 ) != 0 ) the stuffle sum is negative.
02700 */
02701
02702 int Stuffle(WORD *from1, WORD *from2, WORD *to)
02703 {
02704     GETIDENTITY
02705     WORD *t, *tf, *next1, *next2, *st1, *st2, *savel, *save2;
02706     SHvariables *SH = &(AN.SHvar);
02707     int i, retval;
02708     /*
02709     First the special cases (exhausted list(s)):
02710     */
02711     savel = SH->stop1; save2 = SH->stop2;
02712     if ( from1 >= SH->ststop1 && from2 == SH->ststop2 ) {
02713         SH->stop1 = SH->ststop1;
02714         SH->stop2 = SH->ststop2;
02715         retval = FinishShuffle(to);
02716         SH->stop1 = savel; SH->stop2 = save2;
02717         return(retval);
02718     }
02719     else if ( from1 >= SH->ststop1 ) {
02720         i = SH->ststop2 - from2; t = to; tf = from2; NCOPY(t, tf, i)
02721         SH->stop1 = SH->ststop1;
02722         SH->stop2 = SH->ststop2;
02723         retval = FinishShuffle(t);
02724         SH->stop1 = savel; SH->stop2 = save2;
02725         return(retval);
02726     }
02727     else if ( from2 >= SH->ststop2 ) {
02728         i = SH->ststop1 - from1; t = to; tf = from1; NCOPY(t, tf, i)
02729         SH->stop1 = SH->ststop1;
02730         SH->stop2 = SH->ststop2;
02731         retval = FinishShuffle(t);
02732         SH->stop1 = savel; SH->stop2 = save2;
02733         return(retval);
02734     }
02735     /*
02736     Now the case that we have no stuffle sums.
02737     */
02738     SH->stop1 = SH->ststop1;

```

```

02739     SH->stop2 = SH->ststop2;
02740     SH->finishuf = &FinishShuffle;
02741     if ( Shuffle(from1,from2,to) ) goto stuffcall;
02742     SH->finishuf = &FinishStuffle;
02743     /*
02744     Now we have to select a pair, one from 1 and one from 2.
02745     */
02746     #ifndef NEWCODE
02747     st1 = from1; next1 = st1+2;          /* <----- */
02748     #else
02749     st1 = next1 = from1;
02750     NEXTARG(next1)
02751     #endif
02752     while ( next1 <= SH->ststop1 ) {
02753     #ifndef NEWCODE
02754     st2 = from2; next2 = st2+2;          /* <----- */
02755     #else
02756     next2 = st2 = from2;
02757     NEXTARG(next2)
02758     #endif
02759     while ( next2 <= SH->ststop2 ) {
02760         SH->stop1 = st1;
02761         SH->stop2 = st2;
02762         if ( st1 == from1 && st2 == from2 ) {
02763             t = to;
02764             #ifndef NEWCODE
02765             *t++ = -SNUMBER; *t++ = StuffAdd(st1[1],st2[1]);
02766             #else
02767             t = StuffRootAdd(st1,st2,t);
02768             #endif
02769             SH->option ^= 256;
02770             if ( Stuffle(next1,next2,t) ) goto stuffcall;
02771             SH->option ^= 256;
02772         }
02773         else if ( st1 == from1 ) {
02774             i = st2-from2;
02775             t = to; tf = from2; NCOPY(t,tf,i)
02776             #ifndef NEWCODE
02777             *t++ = -SNUMBER; *t++ = StuffAdd(st1[1],st2[1]);
02778             #else
02779             t = StuffRootAdd(st1,st2,t);
02780             #endif
02781             SH->option ^= 256;
02782             if ( Stuffle(next1,next2,t) ) goto stuffcall;
02783             SH->option ^= 256;
02784         }
02785         else if ( st2 == from2 ) {
02786             i = st1-from1;
02787             t = to; tf = from1; NCOPY(t,tf,i)
02788             #ifndef NEWCODE
02789             *t++ = -SNUMBER; *t++ = StuffAdd(st1[1],st2[1]);
02790             #else
02791             t = StuffRootAdd(st1,st2,t);
02792             #endif
02793             SH->option ^= 256;
02794             if ( Stuffle(next1,next2,t) ) goto stuffcall;
02795             SH->option ^= 256;
02796         }
02797         else {
02798             if ( Shuffle(from1,from2,to) ) goto stuffcall;
02799         }
02800     #ifndef NEWCODE
02801     st2 = next2; next2 += 2;          /* <----- */
02802     #else
02803     st2 = next2;
02804     NEXTARG(next2)
02805     #endif
02806     }
02807     #ifndef NEWCODE
02808     st1 = next1; next1 += 2;          /* <----- */
02809     #else
02810     st1 = next1;
02811     NEXTARG(next1)
02812     #endif
02813     }
02814     SH->stop1 = save1; SH->stop2 = save2;
02815     return(0);
02816 stuffcall:;
02817     MesCall("Stuffle");
02818     return(-1);
02819 }
02820
02821 /*
02822 #] Stuffle :
02823 #[ FinishStuffle :
02824
02825 The program only comes here from the Shuffle routine.

```

```

02826     It should add the stuffle sum and then call Stuffle again.
02827 */
02828
02829 int FinishStuffle(WORD *fini)
02830 {
02831     GETIDENTITY
02832     SHvariables *SH = &(AN.SHvar);
02833     #ifdef NEWCODE
02834     WORD *next1 = SH->stop1, *next2 = SH->stop2;
02835     fini = StuffRootAdd(next1,next2,fini);
02836     #else
02837     *fini++ = -SNUMBER; *fini++ = StuffAdd(SH->stop1[1],SH->stop2[1]);
02838     #endif
02839     SH->option ^= 256;
02840     #ifdef NEWCODE
02841     NEXTARG(next1)
02842     NEXTARG(next2)
02843     if ( Stuffle(next1,next2,fini) ) goto stuffcall;
02844     #else
02845     if ( Stuffle(SH->stop1+2,SH->stop2+2,fini) ) goto stuffcall;
02846     #endif
02847     SH->option ^= 256;
02848     return(0);
02849 stuffcall:;
02850     MesCall("FinishStuffle");
02851     return(-1);
02852 }
02853
02854 /*
02855     #] FinishStuffle :
02856     #[ StuffRootAdd :
02857
02858     Makes the stuffle sum of two arguments.
02859     The arguments can be of one of three types:
02860     1: -SNUMBER,num
02861     2: -SYMBOL,symbol
02862     3: Numerical (long) argument.
02863     4: Generic argument with (only) symbols that are roots of unity and
02864        a coefficient.
02865     We have excluded the case that both t1 and t2 are of type 1:
02866     The output should be written to 'to' and the new fill position should
02867     be the return value.
02868     'to' is inside the workspace.
02869
02870     The stuffle sum is sig_(t2)*t1+sig_(t1)*t2
02871     or sig_(t1)*sig_(t2)*(abs_(t1)+abs_(t2))
02872 */
02873
02874 #ifdef NEWCODE
02875 WORD *StuffRootAdd(WORD *t1, WORD *t2, WORD *to)
02876 {
02877     int type1, type2, type3, sgn, sgn1, sgn2, sgn3, pow, root, nosymbols, i;
02878     WORD *tt1, *tt2, it1, it2, *t3, *r, size1, size2, size3;
02879     WORD scratch[2];
02880     LONG x;
02881     if ( *t1 == -SNUMBER ) { type1 = 1; if ( t1[1] < 0 ) sgn1 = -1; else sgn1 = 1; }
02882     else if ( *t1 == -SYMBOL ) { type1 = 2; sgn1 = 1; }
02883     else if ( ABS(t1[*t1-1]) == *t1-ARGHEAD-1 ) {
02884         type1 = 3; if ( t1[*t1-1] < 0 ) sgn1 = -1; else sgn1 = 1; }
02885     else { type1 = 4; if ( t1[*t1-1] < 0 ) sgn1 = -1; else sgn1 = 1; }
02886     if ( *t2 == -SNUMBER ) { type2 = 1; if ( t2[1] < 0 ) sgn2 = -1; else sgn2 = 1; }
02887     else if ( *t2 == -SYMBOL ) { type2 = 2; sgn2 = 1; }
02888     else if ( ABS(t2[*t2-1]) == *t2-ARGHEAD-1 ) {
02889         type2 = 3; if ( t2[*t2-1] < 0 ) sgn2 = -1; else sgn2 = 1; }
02890     else { type2 = 4; if ( t2[*t2-1] < 0 ) sgn2 = -1; else sgn2 = 1; }
02891     if ( type1 > type2 ) {
02892         t3 = t1; t1 = t2; t2 = t3;
02893         type3 = type1; type1 = type2; type2 = type3;
02894         sgn3 = sgn1; sgn1 = sgn2; sgn2 = sgn3;
02895     }
02896     nosymbols = 1; sgn3 = 1;
02897     switch ( type1 ) {
02898     case 1:
02899         if ( type2 == 1 ) {
02900             x = sgn2 * t1[1];
02901             x += sgn1 * t2[1];
02902             if ( x > MAXPOSITIVE || x < -(MAXPOSITIVE+1) ) {
02903                 if ( x < 0 ) { sgn1 = -3; x = -x; }
02904                 else sgn1 = 3;
02905                 *to++ = ARGHEAD+4;
02906                 *to++ = 0;
02907                 FILLARG(to)
02908                 *to++ = 4; *to++ = (UWORD)x; *to++ = 1; *to++ = sgn1;
02909             }
02910         }
02911         else { *to++ = -SNUMBER; *to++ = (WORD)x; }
02912     }

```



```

02913     else if ( type2 == 2 ) {
02914         *to++ = ARGHEAD+8; *to++ = 0; FILLARG(to)
02915         *to++ = 8; *to++ = SYMBOL; *to++ = 4; *to++ = t2[1]; *to++ = 1;
02916         *to++ = ABS(t1[1])+1;
02917         *to++ = 1;
02918         *to++ = 3*sgn1;
02919     }
02920     else if ( type2 == 3 ) {
02921         tt1 = (WORD *)scratch; tt1[0] = ABS(t1[1]); size1 = 1;
02922         tt2 = t2+ARGHEAD+1; size2 = (ABS(t2[*t2-1])-1)/2;
02923         t3 = to;
02924         *to++ = 0; *to++ = 0; FILLARG(to) *to++ = 0;
02925         goto DoCoeffi;
02926     }
02927     else {
02928         /*
02929         t1 is (short) numeric, t2 has the symbol(s).
02930         */
02931         tt1 = (WORD *)scratch; tt1[0] = ABS(t1[1]); size1 = 1;
02932         tt2 = t2+ARGHEAD+1; tt2 += tt2[1]; size2 = (ABS(t2[*t2-1])-1)/2;
02933         t3 = to; i = tt2 - t2; r = t2;
02934         NCOPY(to,r,i)
02935         nosymbols = 0;
02936         goto DoCoeffi;
02937     }
02938     break;
02939 case 2:
02940     if ( type2 == 2 ) {
02941         if ( t1[1] == t2[1] ) {
02942             if ( ( symbols[t1[1]].maxpower == 4 )
02943                 && ( ( symbols[t1[1]].complex & VARTYPEMINUS ) == VARTYPEMINUS ) ) {
02944                 *to++ = -SNUMBER; *to++ = -2;
02945             }
02946             else if ( symbols[t1[1]].maxpower == 2 ) {
02947                 *to++ = -SNUMBER; *to++ = 2;
02948             }
02949             else {
02950                 *to++ = ARGHEAD+8; *to++ = 0; FILLARG(to)
02951                 *to++ = 8; *to++ = SYMBOL; *to++ = 4;
02952                 *to++ = t1[1]; *to++ = 2;
02953                 *to++ = 2; *to++ = 1; *to++ = 3;
02954             }
02955         }
02956         else {
02957             *to++ = ARGHEAD+10; *to++ = 0; FILLARG(to)
02958             *to++ = 10; *to++ = SYMBOL; *to++ = 6;
02959             if ( t1[1] < t2[1] ) {
02960                 *to++ = t1[1]; *to++ = 1; *to++ = t2[1]; *to++ = 1;
02961             }
02962             else {
02963                 *to++ = t2[1]; *to++ = 1; *to++ = t1[1]; *to++ = 1;
02964             }
02965             *to++ = 2; *to++ = 1; *to++ = 3;
02966         }
02967     }
02968     else if ( type2 == 3 ) {
02969         t3 = to;
02970         *to++ = 0; *to++ = 0; FILLARG(to) *to++ = 0;
02971         *to++ = SYMBOL; *to++ = 4; *to++ = t1[1]; *to++ = 1;
02972         tt1 = scratch; tt1[1] = 1; size1 = 1;
02973         tt2 = t2+ARGHEAD+1; size2 = (ABS(t2[*t2-1])-1)/2;
02974         nosymbols = 0;
02975         goto DoCoeffi;
02976     }
02977     else {
02978         tt1 = scratch; tt1[0] = 1; size1 = 1;
02979         t3 = to;
02980         *to++ = 0; *to++ = 0; FILLARG(to) *to++ = 0;
02981         *to++ = SYMBOL; *to++ = 0;
02982         tt2 = t2 + ARGHEAD+3; it2 = tt2[-1]-2;
02983         while ( it2 > 0 ) {
02984             if ( *tt2 == t1[1] ) {
02985                 pow = tt2[1]+1;
02986                 root = symbols[*tt2].maxpower;
02987                 if ( pow >= root ) pow -= root;
02988                 if ( ( symbols[*tt2].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
02989                     if ( ( root & 1 ) == 0 && pow >= root/2 ) {
02990                         pow -= root/2; sgn3 = -sgn3;
02991                     }
02992                 }
02993                 if ( pow != 0 ) {
02994                     *to++ = *tt2; *to++ = pow;
02995                 }
02996                 tt2 += 2; it2 -= 2;
02997                 break;
02998             }
02999             else if ( t1[1] < *tt2 ) {

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```

03000          *to++ = t1[1]; *to++ = 1; break;
03001      }
03002      else {
03003          *to++ = *tt2++; *to++ = *tt2++; it2 -= 2;
03004          if ( it2 <= 0 ) { *to++ = t1[1]; *to++ = 1; }
03005      }
03006  }
03007  while ( it2 > 0 ) { *to++ = *tt2++; *to++ = *tt2++; it2 -= 2; }
03008  if ( (to - t3) > ARGHEAD+3 ) {
03009      t3[ARGHEAD+2] = (to-t3)-ARGHEAD-1; /* size of the SYMBOL field */
03010      nosymbols = 0;
03011  }
03012  else {
03013      to = t3+ARGHEAD+1; /* no SYMBOL field */
03014  }
03015  size2 = (ABS(t2[*t2-1])-1)/2;
03016  goto DoCoeffi;
03017  }
03018  break;
03019  case 3:
03020      if ( type2 == 3 ) {
03021  /*
03022          Both are numeric
03023  */
03024          tt1 = t1+ARGHEAD+1; size1 = (ABS(t1[*t1-1])-1)/2;
03025          tt2 = t2+ARGHEAD+1; size2 = (ABS(t2[*t2-1])-1)/2;
03026          t3 = to;
03027          *to++ = 0; *to++ = 0; FILLARG(to) *to++ = 0;
03028          goto DoCoeffi;
03029      }
03030      else {
03031  /*
03032          t1 is (long) numeric, t2 has the symbol(s).
03033  */
03034          tt1 = t1+ARGHEAD+1; size1 = (ABS(t1[*t1-1])-1)/2;
03035          tt2 = t2+ARGHEAD+1; tt2 += tt2[1]; size2 = (ABS(t2[*t2-1])-1)/2;
03036          t3 = to; i = tt2 - t2; r = t2;
03037          NCOPY(to,r,i)
03038          nosymbols = 0;
03039          goto DoCoeffi;
03040      }
03041  break;
03042  case 4:
03043  /*
03044          Both have roots of unity
03045          1: Merge the lists and simplify if possible
03046  */
03047          tt1 = t1+ARGHEAD+3; it1 = tt1[-1]-2;
03048          tt2 = t2+ARGHEAD+3; it2 = tt2[-1]-2;
03049          t3 = to;
03050          *to++ = 0; *to++ = 0; FILLARG(to)
03051          *to++ = 0; *to++ = SYMBOL; *to++ = 0;
03052          while ( it1 > 0 && it2 > 0 ) {
03053              if ( *tt1 == *tt2 ) {
03054                  pow = tt1[1]+tt2[1];
03055                  root = symbols[*tt1].maxpower;
03056                  if ( pow >= root ) pow -= root;
03057                  if ( ( symbols[*tt1].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
03058                      if ( ( root & 1 ) == 0 && pow >= root/2 ) {
03059                          pow -= root/2; sgn3 = -sgn3;
03060                      }
03061                  }
03062                  if ( pow != 0 ) {
03063                      *to++ = *tt1; *to++ = pow;
03064                  }
03065                  tt1 += 2; tt2 += 2; it1 -= 2; it2 -= 2;
03066              }
03067              else if ( *tt1 < *tt2 ) {
03068                  *to++ = *tt1++; *to++ = *tt1++; it1 -= 2;
03069              }
03070              else {
03071                  *to++ = *tt2++; *to++ = *tt2++; it2 -= 2;
03072              }
03073          }
03074          while ( it1 > 0 ) { *to++ = *tt1++; *to++ = *tt1++; it1 -= 2; }
03075          while ( it2 > 0 ) { *to++ = *tt2++; *to++ = *tt2++; it2 -= 2; }
03076          if ( (to - t3) > ARGHEAD+3 ) {
03077              t3[ARGHEAD+2] = (to-t3)-ARGHEAD-1; /* size of the SYMBOL field */
03078              nosymbols = 0;
03079          }
03080          else {
03081              to = t3+ARGHEAD+1; /* no SYMBOL field */
03082          }
03083          size1 = (ABS(t1[*t1-1])-1)/2;
03084          size2 = (ABS(t2[*t2-1])-1)/2;
03085  /*
03086          Now tt1 and tt2 are pointing at their coefficients.

```

```

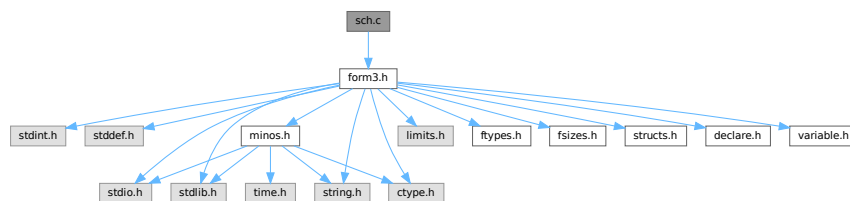
03087         sgn1 is the sign of 1, sgn2 is the sign of 2 and sgn3 is an extra
03088         overall sign.
03089     */
03090 DoCoeffi:
03091     if ( AddLong((UWORD *)tt1,size1,(UWORD *)tt2,size2,(UWORD *)to,&size3) ) {
03092         MLOCK(ErrorMessageLock);
03093         MesPrint("Called from StuffRootAdd");
03094         MUNLOCK(ErrorMessageLock);
03095         Terminate(-1);
03096     }
03097     sgn = sgn1*sgn2*sgn3;
03098     if ( nosymbols && size3 == 1 ) {
03099         if ( (UWORD)(to[0]) <= MAXPOSITIVE && sgn > 0 ) {
03100             sgn1 = to[0];
03101             to = t3; *to++ = -SNUMBER; *to++ = sgn1;
03102         }
03103         else if ( (UWORD)(to[0]) <= (MAXPOSITIVE+1) && sgn < 0 ) {
03104             sgn1 = to[0];
03105             to = t3; *to++ = -SNUMBER; *to++ = -sgn1;
03106         }
03107         else goto genericcoef;
03108     }
03109     else {
03110 genericcoef:
03111         to += size3;
03112         sgn = sgn*(2*size3+1);
03113         *to++ = 1;
03114         while ( size3 > 1 ) { *to++ = 0; size3--; }
03115         *to++ = sgn;
03116         t3[0] = to - t3;
03117         t3[ARGHEAD] = t3[0] - ARGHEAD;
03118     }
03119     break;
03120 }
03121 return(to);
03122 }
03123 #endif
03124
03125 /*
03126 #] StuffRootAdd :
03127 */

```

## 4.122 sch.c File Reference

#include "form3.h"

Include dependency graph for sch.c:



### Macros

- #define [va\\_dcl](#) int va\_alist;
- #define [va\\_start](#)(list) list = (UBYTE \*) &va\_alist
- #define [va\\_end](#)(list)
- #define [va\\_arg](#)(list, mode) (((mode \*) (list += sizeof(mode)))[-1])

### Typedefs

- typedef UBYTE \* [va\\_list](#)

## Functions

- `UBYTE *` [StrCopy](#) (`UBYTE *from`, `UBYTE *to`)
- `void` [AddToLine](#) (`UBYTE *s`)
- `void` [FiniLine](#) (`void`)
- `void` [IniLine](#) (`WORD extrablank`)
- `void` [LongToLine](#) (`UWORD *a`, `WORD na`)
- `void` [RatToLine](#) (`UWORD *a`, `WORD na`)
- `void` [TalToLine](#) (`UWORD x`)
- `void` [TokenToLine](#) (`UBYTE *s`)
- `UBYTE *` [CodeToLine](#) (`WORD number`, `UBYTE *Out`)
- `void` [MultiplyToLine](#) (`void`)
- `UBYTE *` [AddArrayIndex](#) (`WORD num`, `UBYTE *out`)
- `void` [PrtTerms](#) (`void`)
- `UBYTE *` [WrtPower](#) (`UBYTE *Out`, `WORD Power`)
- `void` [PrintTime](#) (`UBYTE *mess`)
- `void` [WriteLists](#) (`void`)
- `void` [WriteDictionary](#) (`DICTIONARY *dict`)
- `void` [WriteArgument](#) (`WORD *t`)
- `int` [WriteSubTerm](#) (`WORD *sterm`, `WORD first`)
- `int` [WriteInnerTerm](#) (`WORD *term`, `WORD first`)
- `int` [WriteTerm](#) (`WORD *term`, `WORD *lbrac`, `WORD first`, `WORD prtf`, `WORD br`)
- `int` [WriteExpression](#) (`WORD *terms`, `LONG ltot`)
- `int` [WriteAll](#) (`void`)
- `int` [WriteOne](#) (`UBYTE *name`, `int alreadyinline`, `int nosemi`, `WORD plus`)

### 4.122.1 Detailed Description

Contains the functions that deal with the writing of expressions/terms in a textual representation. (Dutch schrijven = to write)

Definition in file [sch.c](#).

### 4.122.2 Macro Definition Documentation

#### 4.122.2.1 `va_dcl`

```
#define va_dcl int va_alist;
```

Definition at line 48 of file [sch.c](#).

#### 4.122.2.2 `va_start`

```
#define va_start(  
    list ) list = (UBYTE *) &va_alist
```

Definition at line 49 of file [sch.c](#).

#### 4.122.2.3 va\_end

```
#define va_end(  
    list )
```

Definition at line 50 of file [sch.c](#).

#### 4.122.2.4 va\_arg

```
#define va_arg(  
    list,  
    mode ) (((mode *) (list += sizeof(mode))) [-1])
```

Definition at line 51 of file [sch.c](#).

### 4.122.3 Typedef Documentation

#### 4.122.3.1 va\_list

```
typedef UBYTE* va_list
```

Definition at line 47 of file [sch.c](#).

### 4.122.4 Function Documentation

#### 4.122.4.1 StrCopy()

```
UBYTE * StrCopy (  
    UBYTE * from,  
    UBYTE * to )
```

Definition at line 67 of file [sch.c](#).

#### 4.122.4.2 AddToLine()

```
void AddToLine (  
    UBYTE * s )
```

Definition at line 82 of file [sch.c](#).

#### 4.122.4.3 FiniLine()

```
void FiniLine (  
    void )
```

Definition at line 182 of file [sch.c](#).

#### 4.122.4.4 IniLine()

```
void IniLine (
    WORD extrablink )
```

Definition at line 267 of file [sch.c](#).

#### 4.122.4.5 LongToLine()

```
void LongToLine (
    UWORD * a,
    WORD na )
```

Definition at line 303 of file [sch.c](#).

#### 4.122.4.6 RatToLine()

```
void RatToLine (
    UWORD * a,
    WORD na )
```

Definition at line 337 of file [sch.c](#).

#### 4.122.4.7 TalToLine()

```
void TalToLine (
    UWORD x )
```

Definition at line 522 of file [sch.c](#).

#### 4.122.4.8 TokenToLine()

```
void TokenToLine (
    UBYTE * s )
```

Definition at line 551 of file [sch.c](#).

#### 4.122.4.9 CodeToLine()

```
UBYTE * CodeToLine (
    WORD number,
    UBYTE * Out )
```

Definition at line 658 of file [sch.c](#).

#### 4.122.4.10 MultiplyToLine()

```
void MultiplyToLine (
    void )
```

Definition at line 671 of file [sch.c](#).

#### 4.122.4.11 AddArrayIndex()

```
UBYTE * AddArrayIndex (
    WORD num,
    UBYTE * out )
```

Definition at line 696 of file [sch.c](#).

#### 4.122.4.12 PrtTerms()

```
void PrtTerms (
    void )
```

Definition at line 716 of file [sch.c](#).

#### 4.122.4.13 WrtPower()

```
UBYTE * WrtPower (
    UBYTE * Out,
    WORD Power )
```

Definition at line 738 of file [sch.c](#).

#### 4.122.4.14 PrintTime()

```
void PrintTime (
    UBYTE * mess )
```

Definition at line 784 of file [sch.c](#).

#### 4.122.4.15 WriteLists()

```
void WriteLists (
    void )
```

Definition at line 810 of file [sch.c](#).

#### 4.122.4.16 WriteDictionary()

```
void WriteDictionary (
    DICTIONARY * dict )
```

Definition at line 1307 of file [sch.c](#).

#### 4.122.4.17 WriteArgument()

```
void WriteArgument (
    WORD * t )
```

Definition at line 1424 of file [sch.c](#).

#### 4.122.4.18 WriteSubTerm()

```
int WriteSubTerm (
    WORD * sterm,
    WORD first )
```

Definition at line 1542 of file [sch.c](#).

#### 4.122.4.19 WriteInnerTerm()

```
int WriteInnerTerm (
    WORD * term,
    WORD first )
```

Definition at line 1951 of file [sch.c](#).

#### 4.122.4.20 WriteTerm()

```
int WriteTerm (
    WORD * term,
    WORD * lbrac,
    WORD first,
    WORD prtft,
    WORD br )
```

Definition at line 2148 of file [sch.c](#).

#### 4.122.4.21 WriteExpression()

```
int WriteExpression (
    WORD * terms,
    LONG ltot )
```

Definition at line 2470 of file [sch.c](#).

#### 4.122.4.22 WriteAll()

```
int WriteAll (
    void )
```

Definition at line 2501 of file [sch.c](#).



## 4.122.4.23 WriteOne()

```
int WriteOne (
    UBYTE * name,
    int alreadyinline,
    int nosemi,
    WORD plus )
```

Definition at line 2727 of file [sch.c](#).

## 4.123 sch.c

[Go to the documentation of this file.](#)

```
00001
00006 /* #[] License : */
00007 /*
00008  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00009  * When using this file you are requested to refer to the publication
00010  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011  * This is considered a matter of courtesy as the development was paid
00012  * for by FOM the Dutch physics granting agency and we would like to
00013  * be able to track its scientific use to convince FOM of its value
00014  * for the community.
00015  *
00016  * This file is part of FORM.
00017  *
00018  * FORM is free software: you can redistribute it and/or modify it under the
00019  * terms of the GNU General Public License as published by the Free Software
00020  * Foundation, either version 3 of the License, or (at your option) any later
00021  * version.
00022  *
00023  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024  * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026  * details.
00027  *
00028  * You should have received a copy of the GNU General Public License along
00029  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00030  */
00031 /* #[] License : */
00032 /*
00033  #[] Includes : sch.c
00034  */
00035
00036 #include "form3.h"
00037
00038 #ifdef ANSI
00039 #include <stdarg.h>
00040 #else
00041 #ifdef mBSD
00042 #include <varargs.h>
00043 #else
00044 #ifdef VMS
00045 #include <varargs.h>
00046 #else
00047 typedef UBYTE *va_list;
00048 #define va_dcl int va_alist;
00049 #define va_start(list) list = (UBYTE *) &va_alist
00050 #define va_end(list)
00051 #define va_arg(list,mode) (((mode *) (list += sizeof(mode)))[-1])
00052 #endif
00053 #endif
00054 #endif
00055
00056 static int startinline = 0;
00057 static char fcontchar = '&';
00058 static int noextralinefeed = 0;
00059 static int lowestlevel = 1;
00060
00061 /*
00062  #[] Includes :
00063  #[] schryf-Utilities :
00064  #[] StrCopy : UBYTE *StrCopy(from,to)
00065  */
00066
00067 UBYTE *StrCopy(UBYTE *from, UBYTE *to)
```

```

00068 {
00069     while( ( *to++ = *from++ ) != 0 );
00070     return(to-1);
00071 }
00072
00073 /*
00074     #] StrCopy :
00075     #[ AddToLine :          void AddToLine(s)
00076
00077     Puts the characters of s in the outputline. If the line becomes
00078     filled it is written.
00079
00080 */
00081
00082 void AddToLine(UBYTE *s)
00083 {
00084     UBYTE *Out;
00085     LONG num;
00086     int i;
00087     if ( AO.OutInBuffer ) { AddToDollarBuffer(s); return; }
00088     Out = AO.OutFill;
00089     while ( *s ) {
00090         if ( Out >= AO.OutStop ) {
00091             if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
00092                 *Out++ = fcontchar;
00093             }
00094             #ifdef WITHRETURN
00095                 *Out++ = CARRIAGERETURN;
00096             #endif
00097             *Out++ = LINEFEED;
00098             AO.PortFirst = 0;
00099             num = Out - AO.OutputLine;
00100
00101             if ( AC.LogHandle >= 0 ) {
00102                 if ( WriteFile(AC.LogHandle,AO.OutputLine+startinline
00103                     ,num-startinline) != (num-startinline) ) {
00104                     /*
00105                         We cannot write to an otherwise open log file.
00106                         The disk could be full of course.
00107                     */
00108                     #ifdef DEBUGGER
00109                         if ( BUG.logfileflag == 0 ) {
00110                             fprintf(stderr,"Panic: Cannot write to log file! Disk full?\n");
00111                             BUG.logfileflag = 1;
00112                         }
00113                         BUG.eflag = 1; BUG.printflag = 1;
00114                     #else
00115                         Terminate(-1);
00116                     #endif
00117                 }
00118             }
00119
00120             if ( ( AO.PrintType & PRINTLFILE ) == 0 ) {
00121                 #ifdef WITHRETURN
00122                     if ( num > 1 && AO.OutputLine[num-2] == CARRIAGERETURN ) {
00123                         AO.OutputLine[num-2] = LINEFEED;
00124                         num--;
00125                     }
00126                 #endif
00127                 if ( WriteFile(AM.StdOut,AO.OutputLine+startinline
00128                     ,num-startinline) != (num-startinline) ) {
00129                     #ifdef DEBUGGER
00130                         if ( BUG.stdoutflag == 0 ) {
00131                             fprintf(stderr,"Panic: Cannot write to standard output!\n");
00132                             BUG.stdoutflag = 1;
00133                         }
00134                         BUG.eflag = 1; BUG.printflag = 1;
00135                     #else
00136                         Terminate(-1);
00137                     #endif
00138                 }
00139             }
00140             /* thomasr 23/04/09: A continuation line has been started.
00141              * In Fortran90 we do not want a space after the initial
00142              * '&' character otherwise we might end up with something
00143              * like:
00144              * ... 2.&
00145              * & 0 ...
00146              */
00147             startinline = 0;
00148             for ( i = 0; i < AO.OutSkip; i++ ) AO.OutputLine[i] = ' ';
00149             Out = AO.OutputLine + AO.OutSkip;
00150             if ( ( AC.OutputMode == FORTRANMODE
00151                 || AC.OutputMode == PFORTRANMODE ) && AO.OutSkip == 7 ) {
00152                 /* thomasr 23/04/09: fix leading blank in fortran90 mode */
00153                 if(AC.IsFortran90 == ISFORTRAN90) {
00154                     Out[-1] = fcontchar;

```

```

00155         }
00156         else {
00157             Out[-2] = fcontchar;
00158             Out[-1] = ' ';
00159         }
00160     }
00161     if ( AO.IsBracket ) { *Out++ = ' ';
00162         if ( AC.OutputSpaces == NORMALFORMAT ) {
00163             *Out++ = ' '; *Out++ = ' '; }
00164     }
00165     *Out = '\0';
00166     if ( AC.OutputMode == FORTRANMODE
00167         || ( AC.OutputMode == CMODE && AO.FactorMode == 0 )
00168         || AC.OutputMode == PFORTRANMODE )
00169         AO.InFbrack++;
00170     }
00171     *Out++ = *s++;
00172 }
00173 *Out = '\0';
00174 AO.OutFill = Out;
00175 }
00176
00177 /*
00178     #] AddToLine :
00179     #[ FiniLine :      void FiniLine()
00180 */
00181
00182 void FiniLine(void)
00183 {
00184     UBYTE *Out;
00185     WORD i;
00186     LONG num;
00187     if ( AO.OutInBuffer ) return;
00188     Out = AO.OutFill;
00189     while ( Out > AO.OutputLine ) {
00190         if ( Out[-1] == ' ' ) Out--;
00191         else break;
00192     }
00193     i = (WORD)(Out-AO.OutputLine);
00194     if ( noextralinefeed == 0 ) {
00195         if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90
00196             && Out > AO.OutputLine ) {
00197             /*
00198                 *Out++ = fcontchar;
00199             */
00200         }
00201         #ifdef WITHRETURN
00202         *Out++ = CARRIAGERETURN;
00203         #endif
00204         *Out++ = LINEFEED;
00205         AO.FortFirst = 0;
00206     }
00207     num = Out - AO.OutputLine;
00208
00209     if ( AC.LogHandle >= 0 ) {
00210         if ( WriteFile(AC.LogHandle,AO.OutputLine+startinline,
00211             ,num-startinline) != (num-startinline) ) {
00212             #ifdef DEBUGGER
00213                 if ( BUG.logfileflag == 0 ) {
00214                     fprintf(stderr,"Panic: Cannot write to log file! Disk full?\n");
00215                     BUG.logfileflag = 1;
00216                 }
00217                 BUG.eflag = 1; BUG.printflag = 1;
00218             #else
00219                 Terminate(-1);
00220             #endif
00221         }
00222     }
00223
00224     if ( ( AO.PrintType & PRINTLFILE ) == 0 ) {
00225         #ifdef WITHRETURN
00226         if ( num > 1 && AO.OutputLine[num-2] == CARRIAGERETURN ) {
00227             AO.OutputLine[num-2] = LINEFEED;
00228             num--;
00229         }
00230         #endif
00231         if ( WriteFile(AM.Stdout,AO.OutputLine+startinline,
00232             ,num-startinline) != (num-startinline) ) {
00233             #ifdef DEBUGGER
00234                 if ( BUG.stdoutflag == 0 ) {
00235                     fprintf(stderr,"Panic: Cannot write to standard output!\n");
00236                     BUG.stdoutflag = 1;
00237                 }
00238                 BUG.eflag = 1; BUG.printflag = 1;
00239             #else
00240                 Terminate(-1);
00241             #endif

```

```

00242     }
00243 }
00244 startinline = 0;
00245 if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
00246     || ( AC.OutputMode == CMODE && AO.FactorMode == 0 ) ) AO.InFbrack++;
00247 Out = AO.OutputLine;
00248 AO.OutStop = Out + AC.LineLength;
00249 i = AO.OutSkip;
00250 while ( --i >= 0 ) *Out++ = ' ';
00251 if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
00252     && AO.OutSkip == 7 ) {
00253     Out[-2] = fcontchar;
00254     Out[-1] = ' ';
00255 }
00256 AO.OutFill = Out;
00257 }
00258
00259 /*
00260     #] FiniLine :
00261     #[ IniLine :          void IniLine(extrablank)
00262
00263     Initializes the output line for the type of output
00264 */
00265 void IniLine(WORD extrablank)
00266 {
00267     UBYTE *Out;
00270     Out = AO.OutputLine;
00271     AO.OutStop = Out + AC.LineLength;
00272     *Out++ = ' ';
00273     *Out++ = ' ';
00274     *Out++ = ' ';
00275     *Out++ = ' ';
00276     *Out++ = ' ';
00277     if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
00278         *Out++ = fcontchar;
00279         AO.OutSkip = 7;
00280     }
00281     else
00282         AO.OutSkip = 6;
00283     *Out++ = ' ';
00284     while ( extrablank > 0 ) {
00285         *Out++ = ' ';
00286         extrablank--;
00287     }
00288     AO.OutFill = Out;
00289 }
00290
00291 /*
00292     #] IniLine :
00293     #[ LongToLine :      void LongToLine(a,na)
00294
00295     Puts a Long integer in the output line. If it is only a single
00296     word long it is put in the line as a single token.
00297     The sign of a is ignored.
00298 */
00299 static UBYTE *LLscratch = 0;
00300
00301 void LongToLine(WORD *a, WORD na)
00302 {
00303     UBYTE *OutScratch;
00306     if ( LLscratch == 0 ) {
00307         LLscratch = (UBYTE *)Malloca(4*(AM.MaxTal*sizeof(WORD)+2)*sizeof(UBYTE), "LongToLine");
00308     }
00309     OutScratch = LLscratch;
00310     if ( na < 0 ) na = -na;
00311     if ( na > 1 ) {
00312         PrtLong(a, na, OutScratch);
00313         if ( AO.NoSpacesInNumbers || AC.OutputMode == REDUCEMODE ) {
00314             AO.BlockSpaces = 1;
00315             TokenToLine(OutScratch);
00316             AO.BlockSpaces = 0;
00317         }
00318         else {
00319             TokenToLine(OutScratch);
00320         }
00321     }
00322     else if ( !na ) TokenToLine((UBYTE *)"0");
00323     else TalToLine(*a);
00324 }
00325
00326 /*
00327     #] LongToLine :
00328     #[ RatToLine :      void RatToLine(a,na)

```

```

00329
00330     Puts a rational number in the output line. The sign is ignored.
00331
00332 */
00333
00334 static UBYTE *RLscratch = 0;
00335 static UWORD *RLscratE = 0;
00336
00337 void RatToLine(UWORD *a, WORD na)
00338 {
00339     GETIDENTITY
00340     WORD adenom, anumer;
00341     UWORD maxInt;
00342
00343     if ( AC.OutputMode == CMODE ) {
00344         // In C, integer literals over 2^32-1 are automatically promoted to longer types up to
00345         // unsigned long long, however FORTRAN has always given integers over 2^32-1 a floating suffix
00346         // in C mode. Retain that behaviour here for now. In the future, maxInt could be a user-
00347         // configurable parameter.
00348         maxInt = 4294967295;
00349     }
00350     else {
00351         // In Fortran modes, literals over 2^31-1 should be printed with a float suffix
00352         maxInt = 2147483647;
00353     }
00354
00355     if ( na < 0 ) na = -na;
00356
00357     if ( AC.OutNumberType == RATIONALMODE ) {
00358
00359         // Work out what is to be printed:
00360         WORD isOne = 0, isOneOver = 0, isIntegral = 0;
00361         WORD isLongNum = 0, isLongDen = 0;
00362         UnPack(a, na, &adenom, &anumer);
00363         if ( na == 1 && a[0] == 1 && a[1] == 1 ) { isOne = 1; isIntegral = 1; }
00364         else if ( adenom == 1 && a[na] == 1 ) { isIntegral = 1; }
00365         else if ( anumer == 1 && a[0] == 1 ) { isOneOver = 1; }
00366         if ( anumer > 1 || ( anumer == 1 && a[0] > maxInt ) ) { isLongNum = 1; };
00367         if ( adenom > 1 || ( adenom == 1 && a[na] > maxInt ) ) { isLongDen = 1; };
00368
00369         // Now sort out the float suffix for the numerator:
00370         UBYTE* suffNum = (UBYTE*)" ";
00371         UBYTE* suffDen = (UBYTE*)" ";
00372         if ( isLongNum || !isIntegral || AC.Fortran90Kind || ( AO.DoubleFlag & 4 ) == 4 ) {
00373             if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
00374                 if ( AC.Fortran90Kind ) { suffNum = AC.Fortran90Kind; }
00375                 else { suffNum = (UBYTE*)". "; }
00376             }
00377             else if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == CMODE ) {
00378                 if ( ( AO.DoubleFlag & 2 ) == 2 ) { suffNum = (UBYTE*)".Q0"; }
00379                 else if ( ( AO.DoubleFlag & 1 ) == 1 ) { suffNum = (UBYTE*)".D0"; }
00380                 else { suffNum = (UBYTE*)". "; }
00381             }
00382         }
00383         // The same again, for the denominator:
00384         if ( isLongDen || !isIntegral || AC.Fortran90Kind || ( AO.DoubleFlag & 4 ) == 4 ) {
00385             if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
00386                 if ( AC.Fortran90Kind ) { suffDen = AC.Fortran90Kind; }
00387                 else { suffDen = (UBYTE*)". "; }
00388             }
00389             else if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == CMODE ) {
00390                 if ( ( AO.DoubleFlag & 2 ) == 2 ) { suffDen = (UBYTE*)".Q0"; }
00391                 else if ( ( AO.DoubleFlag & 1 ) == 1 ) { suffDen = (UBYTE*)".D0"; }
00392                 else { suffDen = (UBYTE*)". "; }
00393             }
00394         }
00395         // In PFORTAN mode, rationals don't get a suffix if the integers are not long
00396         // Also the suffix is at least ".D0" (and never ".").
00397         if ( AC.OutputMode == PFORTANMODE ) {
00398             if ( isLongNum ) {
00399                 if ( ( AO.DoubleFlag & 2 ) == 2 ) { suffNum = (UBYTE*)".Q0"; }
00400                 else { suffNum = (UBYTE*)".D0"; }
00401             }
00402             if ( isLongDen ) {
00403                 if ( ( AO.DoubleFlag & 2 ) == 2 ) { suffDen = (UBYTE*)".Q0"; }
00404                 else { suffDen = (UBYTE*)".D0"; }
00405             }
00406         }
00407
00408         // Finally, print the number:
00409         // In PFORTAN we use
00410         //   one      if denom = numerator = 1
00411         //   integer   if denom = 1
00412         //   (one/integer) if numerator = 1
00413         //   ((one*integer)/integer) in the general case
00414         if ( AC.OutputMode == PFORTANMODE ) {
00415             if ( isOne ) {

```

```

00416         AddToLine((UBYTE *) "one");
00417     }
00418     else if ( isOneOver ) {
00419         AddToLine((UBYTE *) "(one/");
00420         LongToLine(a+na,adenom);
00421         AddToLine(suffDen);
00422         AddToLine((UBYTE*) ")");
00423     }
00424     else if ( isIntegral ) {
00425         LongToLine(a,anumer);
00426         AddToLine(suffNum);
00427     }
00428     else {
00429         // rational
00430         AddToLine((UBYTE *) "((one*");
00431         LongToLine(a,anumer);
00432         AddToLine(suffNum);
00433         AddToLine((UBYTE*) ")/");
00434         LongToLine(a+na,adenom);
00435         AddToLine(suffDen);
00436         AddToLine((UBYTE*) ")");
00437     }
00438 }
00439 // All other modes use the same printing code:
00440 else {
00441     // Numerator:
00442     LongToLine(a,anumer);
00443     AddToLine(suffNum);
00444     // Denominator, if it is not 1:
00445     if (!isIntegral) {
00446         AddToLine((UBYTE*) "/");
00447         LongToLine(a+na,adenom);
00448         AddToLine(suffDen);
00449     }
00450 }
00451 }
00452
00453 else {
00454 /*
00455     This is the float mode
00456 */
00457     UBYTE *OutScratch;
00458     WORD exponent = 0, i, ndig, newl;
00459     UWORD *c, *den, b = 10, dig[10];
00460     UBYTE *o, *out, cc;
00461 /*
00462     First we have to adjust the numerator and denominator
00463 */
00464     if ( RLscratch == 0 ) {
00465         RLscratch = (UBYTE *) Malloc1(4*(AM.MaxTal+2)*sizeof(UBYTE), "RatToLine");
00466         RLscratE = (UWORD *) Malloc1(2*(AM.MaxTal+2)*sizeof(UWORD), "RatToLine");
00467     }
00468     out = OutScratch = RLscratch;
00469     c = RLscratE; for ( i = 0; i < 2*na; i++ ) c[i] = a[i];
00470     UnPack(c, na, &adenom, &anumer);
00471     while ( BigLong(c, anumer, c+na, adenom) >= 0 ) {
00472         Divvy(BHEAD c, &na, &b, 1);
00473         UnPack(c, na, &adenom, &anumer);
00474         exponent++;
00475     }
00476     while ( BigLong(c, anumer, c+na, adenom) < 0 ) {
00477         Mully(BHEAD c, &na, &b, 1);
00478         UnPack(c, na, &adenom, &anumer);
00479         exponent--;
00480     }
00481 /*
00482     Now division will give a number between 1 and 9
00483 */
00484     den = c + na; i = 1;
00485     DivLong(c, anumer, den, adenom, dig, &ndig, c, &newl);
00486     *out++ = (UBYTE) (dig[0]+'0'); *out++ = '.';
00487     while ( newl && i < AC.OutNumberType ) {
00488         Pack(c, &newl, den, adenom);
00489         Mully(BHEAD c, &newl, &b, 1);
00490         na = newl;
00491         UnPack(c, na, &adenom, &anumer);
00492         den = c + na;
00493         DivLong(c, anumer, den, adenom, dig, &ndig, c, &newl);
00494         if ( ndig == 0 ) *out++ = '0';
00495         else *out++ = (UBYTE) (dig[0]+'0');
00496         i++;
00497     }
00498     *out++ = 'E';
00499     if ( exponent < 0 ) { exponent = -exponent; *out++ = '-'; }
00500     else { *out++ = '+'; }
00501     o = out;
00502     do {

```

```

00503         *out++ = (UBYTE)((exponent % 10)+'0');
00504         exponent /= 10;
00505     } while ( exponent );
00506     *out = 0; out--;
00507     while ( o < out ) { cc = *o; *o = *out; *out = cc; o++; out--; }
00508     TokenToLine(OutScratch);
00509 }
00510 }
00511
00512 /*
00513     #] RatToLine :
00514     #[ TalToLine :          void TalToLine(x)
00515
00516     Writes the unsigned number x to the output as a single token.
00517     Par indicates the number of leading blanks in the line.
00518     This parameter is needed here for the WriteLists routine.
00519
00520 */
00521
00522 void TalToLine(UWORD x)
00523 {
00524     UBYTE t[BITSINWORD/3+1];
00525     UBYTE *s;
00526     WORD i = 0, j;
00527     s = t;
00528     do { *s++ = (UBYTE)((x % 10)+'0'); i++; } while ( ( x /= 10 ) != 0 );
00529     *s-- = '\0';
00530     j = ( i - 1 ) >> 1;
00531     while ( j >= 0 ) {
00532         i = t[j]; t[j] = s[-j]; s[-j] = (UBYTE)i; j--;
00533     }
00534     TokenToLine(t);
00535 }
00536
00537 /*
00538     #] TalToLine :
00539     #[ TokenToLine :      void TokenToLine(s)
00540
00541     Puts s in the output buffer. If it doesn't fit the buffer is
00542     flushed first. This routine keeps tokens as one unit.
00543     Par indicates the number of leading blanks in the line.
00544     This parameter is needed here for the WriteLists routine.
00545
00546     Remark (27-oct-2007): i and j must be longer than WORD!
00547     It can happen that a number is so long that it has more than 2^15 or 2^31
00548     digits!
00549 */
00550
00551 void TokenToLine(UBYTE *s)
00552 {
00553     UBYTE *t, *Out;
00554     LONG num, i = 0, j;
00555     if ( AO.OutInBuffer ) { AddToDollarBuffer(s); return; }
00556     t = s; Out = AO.OutFill;
00557     while ( *t++ ) i++;
00558     while ( i > 0 ) {
00559         if ( ( Out + i ) >= AO.OutStop && ( ( i < ((AC.LineLength-AO.OutSkip)>1) )
00560             || ( (AO.OutStop-Out) < (i>2) ) ) ) {
00561             if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
00562                 *Out++ = fcontchar;
00563             }
00564             #ifdef WITHRETURN
00565             *Out++ = CARRIAGERETURN;
00566             #endif
00567             *Out++ = LINEFEED;
00568             AO.FortFirst = 0;
00569             num = Out - AO.OutputLine;
00570             if ( AC.LogHandle >= 0 ) {
00571                 if ( WriteFile(AC.LogHandle, AO.OutputLine+startinline,
00572                     num-startinline) != (num-startinline) ) {
00573                     #ifdef DEBUGGER
00574                     if ( BUG.logfileflag == 0 ) {
00575                         fprintf(stderr, "Panic: Cannot write to log file! Disk full?\n");
00576                         BUG.logfileflag = 1;
00577                     }
00578                     BUG.eflag = 1; BUG.printflag = 1;
00579                 #else
00580                 Terminate(-1);
00581                 #endif
00582             }
00583             }
00584             if ( ( AO.PrintType & PRINTLFILE ) == 0 ) {
00585                 #ifdef WITHRETURN
00586                 if ( num > 1 && AO.OutputLine[num-2] == CARRIAGERETURN ) {
00587                     AO.OutputLine[num-2] = LINEFEED;
00588                     num--;
00589                 }

```

```

00590 #endif
00591         if ( WriteFile(AM.Stdout,AO.OutputLine+startinline,
00592             num-startinline) != (num-startinline) ) {
00593 #ifdef DEBUGGER
00594             if ( BUG.stdoutflag == 0 ) {
00595                 fprintf(stderr,"Panic: Cannot write to standard output!\n");
00596                 BUG.stdoutflag = 1;
00597             }
00598             BUG.eflag = 1; BUG.printflag = 1;
00599 #else
00600             Terminate(-1);
00601 #endif
00602         }
00603     }
00604     startinline = 0;
00605     Out = AO.OutputLine;
00606     if ( AO.BlockSpaces == 0 ) {
00607         for ( j = 0; j < AO.OutSkip; j++ ) { *Out++ = ' '; }
00608         if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) ) {
00609             if ( AO.OutSkip == 7 ) {
00610                 Out[-2] = fcontchar;
00611                 Out[-1] = ' ';
00612             }
00613         }
00614     }
00615     /*
00616     Out = AO.OutputLine + AO.OutSkip;
00617     if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
00618         && AO.OutSkip == 7 ) {
00619         Out[-2] = fcontchar;
00620         Out[-1] = ' ';
00621     }
00622     else {
00623         for ( j = 0; j < AO.OutSkip; j++ ) { AO.OutputLine[j] = ' '; }
00624     }
00625     */
00626     if ( AO.IsBracket ) { *Out++ = ' '; *Out++ = ' '; *Out++ = ' '; }
00627     *Out = '\0';
00628     if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
00629         || ( AC.OutputMode == CMODE && AO.FactorMode == 0 ) ) AO.InFbrack++;
00630 }
00631 if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
00632     /* Very long numbers */
00633     if ( i > (WORD)(AO.OutStop-Out) ) j = (WORD)(AO.OutStop - Out);
00634     else j = i;
00635     i -= j;
00636     NCOPYB(Out,s,j);
00637 }
00638 else {
00639     if ( i > (WORD)(AO.OutStop-Out) ) j = (WORD)(AO.OutStop - Out - 1);
00640     else j = i;
00641     i -= j;
00642     NCOPYB(Out,s,j);
00643     if ( i > 0 ) *Out++ = '\\';
00644 }
00645 }
00646 *Out = '\0';
00647 AO.OutFill = Out;
00648 }
00649
00650 /*
00651     #] TokenToLine :
00652     #[ CodeToLine :         void CodeToLine(name,number,mode)
00653
00654     Writes a name and possibly its number to output as a single token.
00655
00656     */
00657
00658 UBYTE *CodeToLine(WORD number, UBYTE *Out)
00659 {
00660     Out = StrCopy((UBYTE *) "(", Out);
00661     Out = NumCopy(number, Out);
00662     Out = StrCopy((UBYTE *) ")", Out);
00663     return(Out);
00664 }
00665
00666 /*
00667     #] CodeToLine :
00668     #[ MultiplyToLine :
00669     */
00670
00671 void MultiplyToLine(void)
00672 {
00673     int i;
00674     if ( AO.CurrentDictionary > 0 && AO.CurDictSpecials > 0
00675         && AO.CurDictSpecials == DICT_DOSPECIALS ) {
00676         DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];

```



```

00677 /*
00678     Find the star:
00679 */
00680     for ( i = 0; i < dict->numelements; i++ ) {
00681         if ( dict->elements[i]->type != DICT_SPECIALCHARACTER ) continue;
00682         if ( (UBYTE)dict->elements[i]->lhs[0] == (UBYTE)('*') ) {
00683             TokenToLine((UBYTE *) (dict->elements[i]->rhs));
00684             return;
00685         }
00686     }
00687 }
00688 TokenToLine((UBYTE *) "*");
00689 }
00690
00691 /*
00692     #] MultiplyToLine :
00693     #[ AddArrayIndex :
00694 */
00695
00696 UBYTE *AddArrayIndex(WORD num, UBYTE *out)
00697 {
00698     if ( AC.OutputMode == CMODE ) {
00699         out = StrCopy((UBYTE *) "[", out);
00700         out = NumCopy(num, out);
00701         out = StrCopy((UBYTE *) "]", out);
00702     }
00703     else {
00704         out = StrCopy((UBYTE *) "(", out);
00705         out = NumCopy(num, out);
00706         out = StrCopy((UBYTE *) ")", out);
00707     }
00708     return(out);
00709 }
00710
00711 /*
00712     #] AddArrayIndex :
00713     #[ PrtTerms :          void PrtTerms()
00714 */
00715
00716 void PrtTerms(void)
00717 {
00718     UWORD a[2];
00719     WORD na;
00720     a[0] = (UWORD)AO.NumInBrack;
00721     a[1] = (UWORD)(AO.NumInBrack >> BITSINWORD);
00722     if ( a[1] ) na = 2;
00723     else na = 1;
00724     TokenToLine((UBYTE *) " ");
00725     LongToLine(a, na);
00726     if ( a[0] == 1 && na == 1 ) {
00727         TokenToLine((UBYTE *) " term");
00728     }
00729     else TokenToLine((UBYTE *) " terms");
00730     AO.NumInBrack = 0;
00731 }
00732
00733 /*
00734     #] PrtTerms :
00735     #[ WrtPower :
00736 */
00737
00738 UBYTE *WrtPower(UBYTE *Out, WORD Power)
00739 {
00740     if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
00741         || AC.OutputMode == REDUCEMODE ) {
00742         *Out++ = '*'; *Out++ = '*';
00743     }
00744     else if ( AC.OutputMode == CMODE ) *Out++ = ',';
00745     else {
00746         UBYTE *Out1 = IsExponentSign();
00747         if ( Out1 == 0 ) *Out++ = '^';
00748         else {
00749             while ( *Out1 ) *Out++ = *Out1++;
00750             *Out = 0;
00751         }
00752     }
00753     if ( Power >= 0 ) {
00754         if ( Power < 2*MAXPOWER )
00755             Out = NumCopy(Power, Out);
00756         else
00757             Out = StrCopy(FindSymbol((WORD)((LONG)Power-2*MAXPOWER)), Out);
00758     /*
00759         Out = StrCopy(VARNAME(symbols, (LONG)Power-2*MAXPOWER), Out); */
00759     if ( AC.OutputMode == CMODE ) *Out++ = ')';
00760     *Out = 0;
00761 }
00762 else {
00763     if ( ( AC.OutputMode >= FORTRANMODE || AC.OutputMode >= PFORTRANMODE

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00764         || AC.OutputMode >= REDUCEMODE ) && AC.OutputMode != CMODE )
00765         *Out++ = '(';
00766         *Out++ = '-';
00767         if ( Power > -2*MAXPOWER )
00768             Out = NumCopy(-Power,Out);
00769         else
00770             Out = StrCopy(FindSymbol( (WORD) ( (LONG)Power-2*MAXPOWER) ),Out);
00771         /* Out = StrCopy(VARNAME(symbols, (LONG) (-Power)-2*MAXPOWER),Out); */
00772         if ( AC.OutputMode >= FORTRANMODE || AC.OutputMode >= PFORTRANMODE
00773             || AC.OutputMode >= REDUCEMODE ) *Out++ = ')';
00774         *Out = 0;
00775     }
00776     return(Out);
00777 }
00778
00779 /*
00780     #] WrtPower :
00781     #[ PrintTime :
00782 */
00783
00784 void PrintTime(UBYTE *mess)
00785 {
00786     LONG millitime = TimeCPU(1);
00787     WORD timepart = (WORD)(millitime%1000);
00788     millitime /= 1000;
00789     timepart /= 10;
00790     MesPrint("At %s: Time = %7l.%2i sec",mess,millitime,timepart);
00791 }
00792
00793 /*
00794     #] PrintTime :
00795     #] schryf-Utilities :
00796     #[ schryf-Writes :
00797     #[ WriteLists :          void WriteLists()
00798
00799     Writes the namelists. If mode > 0 also the internal codes are given.
00800
00801 */
00802
00803 static UBYTE *symname[] = {
00804     (UBYTE *)"(cyclic)", (UBYTE *)"(reversecyclic)"
00805     , (UBYTE *)"(symmetric)", (UBYTE *)"(antisymmetric)" };
00806 static UBYTE *rsymname[] = {
00807     (UBYTE *)"(-cyclic)", (UBYTE *)"(-reversecyclic)"
00808     , (UBYTE *)"(-symmetric)", (UBYTE *)"(-antisymmetric)" };
00809
00810 void WriteLists(void)
00811 {
00812     GETIDENTITY
00813     WORD i, j, k, *skip;
00814     int first, startvalue;
00815     UBYTE *OutScr, *Out;
00816     EXPRESSIONS e;
00817     CBUF *C = cbuf+AC.cbunum;
00818     int olddict = AO.CurrentDictionary;
00819     skip = &AO.OutSkip;
00820     *skip = 0;
00821     AO.OutputLine = AO.OutFill = (UBYTE *)AT.WorkPointer;
00822     AO.CurrentDictionary = 0;
00823     FiniLine();
00824     OutScr = (UBYTE *)AT.WorkPointer + ( TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer) ) /2;
00825     if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
00826     else startvalue = FIRSTUSERSYMBOL;
00827     /*
00828         #[ Symbols :
00829     */
00830     if ( ( j = NumSymbols ) > startvalue ) {
00831         TokenToLine((UBYTE *)" Symbols");
00832         *skip = 3;
00833         FiniLine();
00834         for ( i = startvalue; i < j; i++ ) {
00835             if ( i >= BUILTINSYMBOLS && i < FIRSTUSERSYMBOL ) continue;
00836             Out = StrCopy(VARNAME(symbols,i),OutScr);
00837             if ( symbols[i].minpower > -MAXPOWER || symbols[i].maxpower < MAXPOWER ) {
00838                 Out = StrCopy((UBYTE *)"(",Out);
00839                 if ( symbols[i].minpower > -MAXPOWER )
00840                     Out = NumCopy(symbols[i].minpower,Out);
00841                 Out = StrCopy((UBYTE *)":",Out);
00842                 if ( symbols[i].maxpower < MAXPOWER )
00843                     Out = NumCopy(symbols[i].maxpower,Out);
00844                 Out = StrCopy((UBYTE *)")",Out);
00845             }
00846             if ( ( symbols[i].complex & VARTYPEIMAGINARY ) == VARTYPEIMAGINARY ) {
00847                 Out = StrCopy((UBYTE *)"#i",Out);
00848             }
00849             else if ( ( symbols[i].complex & VARTYPECOMPLEX ) == VARTYPECOMPLEX ) {
00850                 Out = StrCopy((UBYTE *)"#c",Out);
00851             }
00852         }
00853     }

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```

00851         }
00852         else if ( ( symbols[i].complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY ) {
00853             Out = StrCopy((UBYTE *)"#",Out);
00854             if ( ( symbols[i].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
00855                 Out = StrCopy((UBYTE *)"-",Out);
00856             }
00857             else {
00858                 Out = StrCopy((UBYTE *)"+",Out);
00859             }
00860             Out = NumCopy(symbols[i].maxpower,Out);
00861         }
00862         if ( AC.CodesFlag ) Out = CodeToLine(i,Out);
00863         if ( ( symbols[i].complex & VARTYPECOMPLEX ) == VARTYPECOMPLEX ) i++;
00864         StrCopy((UBYTE *)" ",Out);
00865         TokenToLine(OutScr);
00866     }
00867     *skip = 0;
00868     FiniLine();
00869 }
00870 /*
00871     #] Symbols :
00872     #[ Indices :
00873 */
00874 if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
00875 else startvalue = BUILTININDICES;
00876 if ( ( j = NumIndices ) > startvalue ) {
00877     TokenToLine((UBYTE *)" Indices");
00878     *skip = 3;
00879     FiniLine();
00880     for ( i = startvalue; i < j; i++ ) {
00881         Out = StrCopy(FindIndex(i+AM.OffsetIndex),OutScr);
00882         Out = StrCopy(VARNAME(indices,i),OutScr);
00883         if ( indices[i].dimension >= 0 ) {
00884             if ( indices[i].dimension != AC.lDefDim ) {
00885                 Out = StrCopy((UBYTE *)"=",Out);
00886                 Out = NumCopy(indices[i].dimension,Out);
00887             }
00888         }
00889         else if ( indices[i].dimension < 0 ) {
00890             Out = StrCopy((UBYTE *)"=",Out);
00891             Out = StrCopy(VARNAME(symbols,-indices[i].dimension),Out);
00892             if ( indices[i].nmin4 < -NMIN4SHIFT ) {
00893                 Out = StrCopy((UBYTE *)":",Out);
00894                 Out = StrCopy(VARNAME(symbols,-indices[i].nmin4-NMIN4SHIFT),Out);
00895             }
00896         }
00897         if ( AC.CodesFlag ) Out = CodeToLine(i+AM.OffsetIndex,Out);
00898         StrCopy((UBYTE *)" ",Out);
00899         TokenToLine(OutScr);
00900     }
00901     *skip = 0;
00902     FiniLine();
00903 }
00904 /*
00905     #] Indices :
00906     #[ Vectors :
00907 */
00908 if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
00909 else startvalue = BUILTINVECTORS;
00910 if ( ( j = NumVectors ) > startvalue ) {
00911     TokenToLine((UBYTE *)" Vectors");
00912     *skip = 3;
00913     FiniLine();
00914     for ( i = startvalue; i < j; i++ ) {
00915         Out = StrCopy(VARNAME(vectors,i),OutScr);
00916         if ( AC.CodesFlag ) Out = CodeToLine(i+AM.OffsetVector,Out);
00917         StrCopy((UBYTE *)" ",Out);
00918         TokenToLine(OutScr);
00919     }
00920     *skip = 0;
00921     FiniLine();
00922 }
00923 /*
00924     #] Vectors :
00925     #[ Functions :
00926 */
00927 if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
00928 else startvalue = AM.NumFixedFunctions;
00929 for ( k = 0; k < 2; k++ ) {
00930     first = 1;
00931     j = NumFunctions;
00932     for ( i = startvalue; i < j; i++ ) {
00933         if ( i > MAXBUILTINFUNCTION-FUNCTION
00934             && i < FIRSTUSERFUNCTION-FUNCTION ) continue;
00935         if ( ( k == 0 && functions[i].commute )
00936             || ( k != 0 && !functions[i].commute ) ) {
00937             if ( first ) {

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```

00938             TokenToLine((UBYTE *) (FG.FunNam[k]));
00939             *skip = 3;
00940             FiniLine();
00941             first = 0;
00942         }
00943         Out = StrCopy(VARNAME(functions,i),OutScr);
00944         if ( ( functions[i].complex & VARTYPEIMAGINARY ) == VARTYPEIMAGINARY ) {
00945             Out = StrCopy((UBYTE *)"#i",Out);
00946         }
00947         else if ( ( functions[i].complex & VARTYPECOMPLEX ) == VARTYPECOMPLEX ) {
00948             Out = StrCopy((UBYTE *)"#c",Out);
00949         }
00950         if ( functions[i].spec == VERTEXFUNCTION ) {
00951             Out = StrCopy((UBYTE *)"(Particle)",Out);
00952         }
00953         else if ( functions[i].spec >= TENSORFUNCTION ) {
00954             Out = StrCopy((UBYTE *)"(Tensor)",Out);
00955         }
00956         if ( functions[i].symmetric > 0 ) {
00957             if ( ( functions[i].symmetric & REVERSEORDER ) != 0 ) {
00958                 Out = StrCopy((UBYTE *) (rsymname[(functions[i].symmetric &
~REVERSEORDER)-1]),Out);
00959             }
00960             else {
00961                 Out = StrCopy((UBYTE *) (symname[functions[i].symmetric-1]),Out);
00962             }
00963         }
00964         if ( AC.CodesFlag ) Out = CodeToLine(i+FUNCTION,Out);
00965         if ( ( functions[i].complex & VARTYPECOMPLEX ) == VARTYPECOMPLEX ) i++;
00966         StrCopy((UBYTE *)" ",Out);
00967         TokenToLine(OutScr);
00968     }
00969 }
00970 *skip = 0;
00971 if ( first == 0 ) FiniLine();
00972 }
00973 /*
00974     #] Functions :
00975     #[ Sets :
00976 */
00977 if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
00978 else startvalue = AM.NumFixedSets;
00979 if ( ( j = AC.SetList.num ) > startvalue ) {
00980     WORD element, LastElement, type, number;
00981     TokenToLine((UBYTE *)" Sets");
00982     for ( i = startvalue; i < j; i++ ) {
00983         *skip = 3;
00984         FiniLine();
00985         if ( Sets[i].name < 0 ) {
00986             Out = StrCopy((UBYTE *)"{}",OutScr);
00987         }
00988         else {
00989             Out = StrCopy(VARNAME(Sets,i),OutScr);
00990         }
00991         if ( AC.CodesFlag ) Out = CodeToLine(i,Out);
00992         StrCopy((UBYTE *)":",Out);
00993         TokenToLine(OutScr);
00994         if ( i < AM.NumFixedSets ) {
00995             TokenToLine((UBYTE *)" ");
00996             TokenToLine((UBYTE *)fixedsets[i].description);
00997         }
00998         else if ( Sets[i].type == CRANGE ) {
00999             int iflag = 0;
01000             if ( Sets[i].first == 3*MAXPOWER ) {
01001             }
01002             else if ( Sets[i].first >= MAXPOWER ) {
01003                 TokenToLine((UBYTE *)"<=");
01004                 NumCopy(Sets[i].first-2*MAXPOWER,OutScr);
01005                 TokenToLine(OutScr);
01006                 iflag = 1;
01007             }
01008             else {
01009                 TokenToLine((UBYTE *)"<");
01010                 NumCopy(Sets[i].first,OutScr);
01011                 TokenToLine(OutScr);
01012                 iflag = 1;
01013             }
01014             if ( Sets[i].last == -3*MAXPOWER ) {
01015             }
01016             else if ( Sets[i].last <= -MAXPOWER ) {
01017                 if ( iflag ) TokenToLine((UBYTE *)",");
01018                 TokenToLine((UBYTE *)">=");
01019                 NumCopy(Sets[i].last+2*MAXPOWER,OutScr);
01020                 TokenToLine(OutScr);
01021             }
01022             else {
01023                 if ( iflag ) TokenToLine((UBYTE *)",");

```

```

01024         TokenToLine((UBYTE *)">");
01025         NumCopy(Sets[i].last, OutScr);
01026         TokenToLine(OutScr);
01027     }
01028 }
01029 else {
01030     element = Sets[i].first;
01031     LastElement = Sets[i].last;
01032     type = Sets[i].type;
01033     while ( element < LastElement ) {
01034         TokenToLine((UBYTE *)" ");
01035         number = SetElements[element++];
01036         switch ( type ) {
01037             case CSYMBOL:
01038                 if ( number < 0 ) {
01039                     StrCopy(VARNAME(symbols, -number), OutScr);
01040                     StrCopy((UBYTE *) "?", Out);
01041                     TokenToLine(OutScr);
01042                 }
01043                 else if ( number < MAXPOWER )
01044                     TokenToLine(VARNAME(symbols, number));
01045                 else {
01046                     NumCopy(number-2*MAXPOWER, OutScr);
01047                     TokenToLine(OutScr);
01048                 }
01049                 break;
01050             case CINDEX:
01051                 if ( number >= AM.IndDum ) {
01052                     Out = StrCopy((UBYTE *) "N", OutScr);
01053                     Out = NumCopy(number-(AM.IndDum), Out);
01054                     StrCopy((UBYTE *) "_?", Out);
01055                     TokenToLine(OutScr);
01056                 }
01057                 else if ( number >= AM.OffsetIndex + (WORD)WILDMASK ) {
01058                     Out = StrCopy(VARNAME(indices, number
01059                                     -AM.OffsetIndex-WILDMASK), OutScr);
01060                     StrCopy((UBYTE *) "?", Out);
01061                     TokenToLine(OutScr);
01062                 }
01063                 else if ( number >= AM.OffsetIndex ) {
01064                     TokenToLine(VARNAME(indices, number-AM.OffsetIndex));
01065                 }
01066                 else {
01067                     NumCopy(number, OutScr);
01068                     TokenToLine(OutScr);
01069                 }
01070                 break;
01071             case CVECTOR:
01072                 Out = OutScr;
01073                 if ( number < AM.OffsetVector ) {
01074                     number += WILDMASK;
01075                     Out = StrCopy((UBYTE *) "-", Out);
01076                 }
01077                 if ( number >= AM.OffsetVector + WILDOFFSET ) {
01078                     Out = StrCopy(VARNAME(vectors, number
01079                                     -AM.OffsetVector-WILDOFFSET), Out);
01080                     StrCopy((UBYTE *) "?", Out);
01081                 }
01082                 else {
01083                     Out = StrCopy(VARNAME(vectors, number-AM.OffsetVector), Out);
01084                 }
01085                 TokenToLine(OutScr);
01086                 break;
01087             case CFUNCTION:
01088                 if ( number >= FUNCTION + (WORD)WILDMASK ) {
01089                     Out = StrCopy(VARNAME(functions, number
01090                                     -FUNCTION-WILDMASK), OutScr);
01091                     StrCopy((UBYTE *) "?", Out);
01092                     TokenToLine(OutScr);
01093                 }
01094                 TokenToLine(VARNAME(functions, number-FUNCTION));
01095                 break;
01096             default:
01097                 NumCopy(number, OutScr);
01098                 TokenToLine(OutScr);
01099                 break;
01100         }
01101     }
01102 }
01103 }
01104 *skip = 0;
01105 FiniLine();
01106 }
01107 /*
01108 #] Sets :
01109 #[ Expressions :
01110 */

```

```

01111     if ( AS.ExecMode ) {
01112         e = Expressions;
01113         j = NumExpressions;
01114         first = 1;
01115         for ( i = 0; i < j; i++, e++ ) {
01116             if ( e->status >= 0 ) {
01117                 if ( first ) {
01118                     TokenToLine((UBYTE *) " Expressions");
01119                     *skip = 3;
01120                     FiniLine();
01121                     first = 0;
01122                 }
01123                 Out = StrCopy(AC.exprnames->namebuffer+e->name, OutScr);
01124                 Out = StrCopy((UBYTE *) (FG.ExprStat[e->status]), Out);
01125                 if ( AC.CodesFlag ) Out = CodeToLine(i, Out);
01126                 StrCopy((UBYTE *) " ", Out);
01127                 TokenToLine(OutScr);
01128             }
01129         }
01130         if ( !first ) {
01131             *skip = 0;
01132             FiniLine();
01133         }
01134     }
01135     e = Expressions;
01136     j = NumExpressions;
01137     first = 1;
01138     for ( i = 0; i < j; i++ ) {
01139         if ( e->printflag && ( e->status == LOCALEXPRESSION ||
01140             e->status == GLOBALEXPRESSION || e->status == UNHIDELEXPRESSION
01141             || e->status == UNHIDEGEXPRESSION ) ) {
01142             if ( first ) {
01143                 TokenToLine((UBYTE *) " Expressions to be printed");
01144                 *skip = 3;
01145                 FiniLine();
01146                 first = 0;
01147             }
01148             Out = StrCopy(AC.exprnames->namebuffer+e->name, OutScr);
01149             StrCopy((UBYTE *) " ", Out);
01150             TokenToLine(OutScr);
01151         }
01152         e++;
01153     }
01154     if ( !first ) {
01155         *skip = 0;
01156         FiniLine();
01157     }
01158     /*
01159         #] Expressions :
01160         #[ Dollars :
01161     */
01162
01163     if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
01164     else startvalue = BUILTINDOLLARS;
01165     if ( ( j = NumDollars ) > startvalue ) {
01166         TokenToLine((UBYTE *) " Dollar variables");
01167         *skip = 3;
01168         FiniLine();
01169         for ( i = startvalue; i < j; i++ ) {
01170             Out = StrCopy((UBYTE *) "$", OutScr);
01171             Out = StrCopy(DOLLARNAME(Dollars, i), Out);
01172             if ( AC.CodesFlag ) Out = CodeToLine(i, Out);
01173             StrCopy((UBYTE *) " ", Out);
01174             TokenToLine(OutScr);
01175         }
01176         *skip = 0;
01177         FiniLine();
01178     }
01179
01180     if ( ( j = NumPotModdollars ) > 0 ) {
01181         TokenToLine((UBYTE *) " Dollar variables to be modified");
01182         *skip = 3;
01183         FiniLine();
01184         for ( i = 0; i < j; i++ ) {
01185             Out = StrCopy((UBYTE *) "$", OutScr);
01186             Out = StrCopy(DOLLARNAME(Dollars, PotModdollars[i]), Out);
01187             for ( k = 0; k < NumModOptdollars; k++ )
01188                 if ( ModOptdollars[k].number == PotModdollars[i] ) break;
01189             if ( k < NumModOptdollars ) {
01190                 switch ( ModOptdollars[k].type ) {
01191                     case MODSUM:
01192                         Out = StrCopy((UBYTE *) "(sum)", Out);
01193                         break;
01194                     case MODMAX:
01195                         Out = StrCopy((UBYTE *) "(maximum)", Out);
01196                         break;
01197                     case MODMIN:

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01198         Out = StrCopy((UBYTE *)"(minimum)", Out);
01199         break;
01200     case MODLOCAL:
01201         Out = StrCopy((UBYTE *)"(local)", Out);
01202         break;
01203     default:
01204         Out = StrCopy((UBYTE *)"(?)", Out);
01205         break;
01206     }
01207 }
01208 StrCopy((UBYTE *)" ", Out);
01209 TokenToLine(OutScr);
01210 }
01211 *skip = 0;
01212 FiniLine();
01213 }
01214 /*
01215 #] Dollars :
01216 */
01217
01218 if ( AC.ncmod != 0 ) {
01219     TokenToLine((UBYTE *)"All arithmetic is modulus ");
01220     LongToLine((UWORD *)AC.cmod,ABS(AC.ncmod));
01221     if ( AC.ncmod > 0 ) TokenToLine((UBYTE *)" with powerreduction");
01222     else TokenToLine((UBYTE *)" without powerreduction");
01223     if ( ( AC.modmode & POSNEG ) != 0 ) TokenToLine((UBYTE *)" centered around 0");
01224     else TokenToLine((UBYTE *)" positive numbers only");
01225     FiniLine();
01226 }
01227 if ( AC.lDefDim != 4 ) {
01228     TokenToLine((UBYTE *)"The default dimension is ");
01229     if ( AC.lDefDim >= 0 ) {
01230         NumCopy(AC.lDefDim,OutScr);
01231         TokenToLine(OutScr);
01232     }
01233     else {
01234         TokenToLine(VARNAME(symbols,-AC.lDefDim));
01235         if ( AC.lDefDim4 != -NMIN4SHIFT ) {
01236             TokenToLine((UBYTE *)"");
01237             if ( AC.lDefDim4 >= -NMIN4SHIFT ) {
01238                 NumCopy(AC.lDefDim4,OutScr);
01239                 TokenToLine(OutScr);
01240             }
01241             else {
01242                 TokenToLine(VARNAME(symbols,-AC.lDefDim4-NMIN4SHIFT));
01243             }
01244         }
01245     }
01246     FiniLine();
01247 }
01248 if ( AC.lUnitTrace != 4 ) {
01249     TokenToLine((UBYTE *)"The trace of the unit matrix is ");
01250     if ( AC.lUnitTrace >= 0 ) {
01251         NumCopy(AC.lUnitTrace,OutScr);
01252         TokenToLine(OutScr);
01253     }
01254     else {
01255         TokenToLine(VARNAME(symbols,-AC.lUnitTrace));
01256     }
01257     FiniLine();
01258 }
01259 if ( AO.NumDictionaries > 0 ) {
01260     for ( i = 0; i < AO.NumDictionaries; i++ ) {
01261         WriteDictionary(AO.Dictionaries[i]);
01262     }
01263     if ( olddict > 0 )
01264         MesPrint("\nCurrently dictionary %s is active\n",
01265             AO.Dictionaries[olddict-1]->name);
01266     else
01267         MesPrint("\nCurrently there is no active dictionary\n");
01268 }
01269 if ( AC.CodesFlag ) {
01270     if ( C->numlhs > 0 ) {
01271         TokenToLine((UBYTE *)" Left Hand Sides:");
01272         AO.OutSkip = 3;
01273         for ( i = 1; i <= C->numlhs; i++ ) {
01274             FiniLine();
01275             skip = C->lhs[i];
01276             j = skip[1];
01277             while ( --j >= 0 ) { TalToLine((UWORD)(*skip++)); TokenToLine((UBYTE *)" "); }
01278         }
01279         AO.OutSkip = 0;
01280         FiniLine();
01281     }
01282     if ( C->numrhs > 0 ) {
01283         TokenToLine((UBYTE *)" Right Hand Sides:");
01284         AO.OutSkip = 3;

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```

01285         for ( i = 1; i <= C->numrhs; i++ ) {
01286             FiniLine();
01287             skip = C->rhs[i];
01288             while ( ( j = skip[0] ) != 0 ) {
01289                 while ( --j >= 0 ) { TalToLine((UWORD) (*skip++)); TokenToLine((UBYTE *) " "); }
01290             }
01291             FiniLine();
01292         }
01293         AO.OutSkip = 0;
01294         FiniLine();
01295     }
01296 }
01297 AO.CurrentDictionary = olddict;
01298 }
01299
01300 /*
01301     #] WriteLists :
01302     #[ WriteDictionary :
01303
01304     This routine is part of WriteLists and should be called from there.
01305 */
01306
01307 void WriteDictionary(DICTIONARY *dict)
01308 {
01309     GETIDENTITY
01310     int i, first;
01311     WORD *skip, na, *a, spec, *t, *tstop, j;
01312     UBYTE str[2], *OutScr, *Out;
01313     WORD oldoutputmode = AC.OutputMode, oldoutputsaces = AC.OutputSpaces;
01314     WORD oldoutskip = AO.OutSkip;
01315     AC.OutputMode = NORMALFORMAT;
01316     AC.OutputSpaces = NOSPACEFORMAT;
01317     MesPrint("===Contents of dictionary %s===", dict->name);
01318     skip = &AO.OutSkip;
01319     *skip = 3;
01320     AO.OutputLine = AO.OutFill = (UBYTE *) AT.WorkPointer;
01321     for ( j = 0; j < *skip; j++ ) *(AO.OutFill)++ = ' ';
01322
01323     OutScr = (UBYTE *) AT.WorkPointer + ( TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer) ) /2;
01324     for ( i = 0; i < dict->numelements; i++ ) {
01325         switch ( dict->elements[i]->type ) {
01326             case DICT_INTEGERNUMBER:
01327                 LongToLine((UWORD *) (dict->elements[i]->lhs), dict->elements[i]->size);
01328                 Out = OutScr; *Out = 0;
01329                 break;
01330             case DICT_RATIONALNUMBER:
01331                 a = dict->elements[i]->lhs;
01332                 na = a[a[0]-1]; na = (ABS(na)-1)/2;
01333                 RatToLine((UWORD *) (a+1), na);
01334                 Out = OutScr; *Out = 0;
01335                 break;
01336             case DICT_SYMBOL:
01337                 na = dict->elements[i]->lhs[0];
01338                 Out = StrCopy(VARNAME(symbols, na), OutScr);
01339                 break;
01340             case DICT_VECTOR:
01341                 na = dict->elements[i]->lhs[0]-AM.OffsetVector;
01342                 Out = StrCopy(VARNAME(vectors, na), OutScr);
01343                 break;
01344             case DICT_INDEX:
01345                 na = dict->elements[i]->lhs[0]-AM.OffsetIndex;
01346                 Out = StrCopy(VARNAME(indices, na), OutScr);
01347                 break;
01348             case DICT_FUNCTION:
01349                 na = dict->elements[i]->lhs[0]-FUNCTION;
01350                 Out = StrCopy(VARNAME(functions, na), OutScr);
01351                 break;
01352             case DICT_FUNCTION_WITH_ARGUMENTS:
01353                 t = dict->elements[i]->lhs;
01354                 na = *t-FUNCTION;
01355                 Out = StrCopy(VARNAME(functions, na), OutScr);
01356                 spec = functions[*t - FUNCTION].spec;
01357                 tstop = t + t[1];
01358                 first = 1;
01359                 if ( t[1] <= FUNHEAD ) {}
01360                 else if ( spec >= TENSORFUNCTION ) {
01361                     t += FUNHEAD; *Out++ = (UBYTE) '(';
01362                     while ( t < tstop ) {
01363                         if ( first == 0 ) *Out++ = (UBYTE) (',' );
01364                         else first = 0;
01365                         j = *t++;
01366                         if ( j >= 0 ) {
01367                             if ( j < AM.OffsetIndex ) { Out = NumCopy(j, Out); }
01368                             else if ( j < AM.IndDum ) {
01369                                 Out = StrCopy(VARNAME(indices, j-AM.OffsetIndex), Out);
01370                             }
01371                             else {

```



```

01372                                     MesPrint("Currently wildcards are not allowed in dictionary
elements");
01373                                     Terminate(-1);
01374                                     }
01375                                     }
01376                                     else {
01377                                         Out = StrCopy(VARNAME(vectors,j-AM.OffsetVector),Out);
01378                                     }
01379                                     }
01380                                     *Out++ = (UBYTE)'\''; *Out = 0;
01381                                     }
01382                                     else {
01383                                         t += FUNHEAD; *Out++ = (UBYTE)'\''; *Out = 0;
01384                                         TokenToLine(OutScr);
01385                                         while ( t < tstop ) {
01386                                             if ( !first ) TokenToLine((UBYTE *)",");
01387                                             WriteArgument(t);
01388                                             NEXTARG(t);
01389                                             first = 0;
01390                                         }
01391                                         Out = OutScr;
01392                                         *Out++ = (UBYTE)'\''; *Out = 0;
01393                                     }
01394                                     break;
01395                                     case DICT_SPECIALCHARACTER:
01396                                         str[0] = (UBYTE)(dict->elements[i]->lhs[0]);
01397                                         str[1] = 0;
01398                                         Out = StrCopy(str,OutScr);
01399                                         break;
01400                                     default:
01401                                         Out = OutScr; *Out = 0;
01402                                         break;
01403                                     }
01404                                     Out = StrCopy((UBYTE *)": \"",Out);
01405                                     Out = StrCopy((UBYTE *) (dict->elements[i]->rhs),Out);
01406                                     Out = StrCopy((UBYTE *) "\"",Out);
01407                                     TokenToLine(OutScr);
01408                                     FiniLine();
01409                                 }
01410                                 MesPrint("====End of dictionary %s====",dict->name);
01411                                 AC.OutputMode = oldoutputmode;
01412                                 AC.OutputSpaces = oldoutputspaces;
01413                                 AO.OutSkip = oldoutskip;
01414         }
01415     /*
01416     #] WriteDictionary :
01417     #[ WriteArgument :      void WriteArgument(WORD *t)
01418
01419     Write a single argument field. The general field goes to
01420     WriteExpression and the fast field is dealt with here.
01421     */
01422     void WriteArgument(WORD *t)
01423     {
01424         UBYTE buffer[180];
01425         UBYTE *Out;
01426         WORD i;
01427         int oldoutsidefun, oldlowestlevel = lowestlevel;
01428         lowestlevel = 0;
01429         if ( *t > 0 ) {
01430             oldoutsidefun = AC.outsidefun; AC.outsidefun = 0;
01431             WriteExpression(t+ARGHEAD, (LONG) (*t-ARGHEAD));
01432             AC.outsidefun = oldoutsidefun;
01433             goto CleanUp;
01434         }
01435         Out = buffer;
01436         if ( *t == -SNUMBER ) {
01437             NumCopy(t[1],Out);
01438         }
01439         else if ( *t == -SYMBOL ) {
01440             if ( t[1] >= MAXVARIABLES-cbuf[AM.sbufnum].numrhs ) {
01441                 Out = StrCopy(FindExtraSymbol(MAXVARIABLES-t[1]),Out);
01442             }
01443             else if ( AC.extrasymbols == 0 ) {
01444                 Out = NumCopy((MAXVARIABLES-t[1]),Out);
01445                 Out = StrCopy((UBYTE *) "_",Out);
01446             }
01447             else if ( AC.extrasymbols == 1 ) {
01448                 Out = AddArrayIndex((MAXVARIABLES-t[1]),Out);
01449             }
01450             else if ( AC.extrasymbols == 2 ) {
01451                 Out = NumCopy((MAXVARIABLES-t[1]),Out);
01452             }
01453         }
01454     }
01455     else if ( AC.extrasymbols == 2 ) {
01456         Out = NumCopy((MAXVARIABLES-t[1]),Out);
01457     }

```

```

01458 */
01459     }
01460     else {
01461         StrCopy(FindSymbol(t[1]),Out);
01462     /*      StrCopy(VARNAME(symbols,t[1]),Out); */
01463     }
01464 }
01465 else if ( *t == -VECTOR ) {
01466     if ( t[1] == FUNNYVEC ) { *Out++ = '?'; *Out = 0; }
01467     else
01468         StrCopy(FindVector(t[1]),Out);
01469 /*      StrCopy(VARNAME(vectors,t[1] - AM.OffsetVector),Out); */
01470 }
01471 else if ( *t == -MINVECTOR ) {
01472     *Out++ = '-';
01473     StrCopy(FindVector(t[1]),Out);
01474 /*      StrCopy(VARNAME(vectors,t[1] - AM.OffsetVector),Out); */
01475 }
01476 else if ( *t == -INDEX ) {
01477     if ( t[1] >= 0 ) {
01478         if ( t[1] < AM.OffsetIndex ) { NumCopy(t[1],Out); }
01479         else {
01480             i = t[1];
01481             if ( i >= AM.IndDum ) {
01482                 i -= AM.IndDum;
01483                 *Out++ = 'N';
01484                 Out = NumCopy(i,Out);
01485                 *Out++ = '-';
01486                 *Out++ = '?';
01487                 *Out = 0;
01488             }
01489             else {
01490                 i -= AM.OffsetIndex;
01491                 Out = StrCopy(FindIndex(i%WILDOFFSET+AM.OffsetIndex),Out);
01492 /*      Out = StrCopy(VARNAME(indices,i%WILDOFFSET),Out); */
01493                 if ( i >= WILDOFFSET ) { *Out++ = '?'; *Out = 0; }
01494             }
01495         }
01496     }
01497     else if ( t[1] == FUNNYVEC ) { *Out++ = '?'; *Out = 0; }
01498     else
01499         StrCopy(FindVector(t[1]),Out);
01500 /*      StrCopy(VARNAME(vectors,t[1] - AM.OffsetVector),Out); */
01501 }
01502 else if ( *t == -DOLLAREXPRESSION ) {
01503     DOLLARS d = Dollars + t[1];
01504     *Out++ = '$';
01505     StrCopy(AC.dollarnames->namebuffer+d->name,Out);
01506 }
01507 else if ( *t == -EXPRESSION ) {
01508     StrCopy(EXPRNAME(t[1]),Out);
01509 }
01510 else if ( *t == -SETSET ) {
01511     StrCopy(VARNAME(Sets,t[1]),Out);
01512 }
01513 else if ( *t <= -FUNCTION ) {
01514     StrCopy(FindFunction(-*t),Out);
01515 /*      StrCopy(VARNAME(functions,-*t-FUNCTION),Out); */
01516 }
01517 else {
01518     MesPrint("Illegal function argument while writing");
01519     goto CleanUp;
01520 }
01521 TokenToLine(buffer);
01522 CleanUp:
01523 lowestlevel = oldlowestlevel;
01524 return;
01525 }
01526
01527 /*
01528     #] WriteArgument :
01529     #[ WriteSubTerm :      WORD WriteSubTerm(sterm,first)
01530
01531     Writes a single subterm field to the output line.
01532     There is a recursion for functions.
01533
01534
01535 #define NUMSPECS 8
01536 UBYTE *specfunnames[NUMSPECS] = {
01537     (UBYTE *)"fac", (UBYTE *)"nargs", (UBYTE *)"binom"
01538     , (UBYTE *)"sign", (UBYTE *)"mod", (UBYTE *)"min", (UBYTE *)"max"
01539     , (UBYTE *)"invfac" };
01540 */
01541
01542 int WriteSubTerm(WORD *sterm, WORD first)
01543 {
01544     UBYTE buffer[80];

```

```

01545     UBYTE *Out, closepar[2] = { (UBYTE)'\'', 0};
01546     WORD *stopper, *t, *tt, i, j, po = 0;
01547     int oldoutsidefun;
01548     stopper = sterm + sterm[1];
01549     t = sterm + 2;
01550     switch ( *sterm ) {
01551     case SYMBOL :
01552         while ( t < stopper ) {
01553             if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
01554                 FiniLine();
01555                 if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
01556                 else IniLine(3);
01557                 if ( first ) TokenToLine((UBYTE *) " ");
01558             }
01559             if ( !first ) MultiplyToLine();
01560             if ( AC.OutputMode == CMODE && t[1] != 1 ) {
01561                 if ( AC.Cnumpows >= t[1] && t[1] > 0 ) {
01562                     po = t[1];
01563                     Out = StrCopy((UBYTE *) "POW",buffer);
01564                     Out = NumCopy(po,Out);
01565                     Out = StrCopy((UBYTE *) "(",Out);
01566                     TokenToLine(buffer);
01567                 }
01568                 else {
01569                     TokenToLine((UBYTE *) "pow(");
01570                 }
01571             }
01572             if ( *t < NumSymbols ) {
01573                 Out = StrCopy(FindSymbol(*t),buffer); t++;
01574                 /* Out = StrCopy(VARNAME(symbols,*t),buffer); t++; */
01575             }
01576             else {
01577                 /* see also routine PrintSubtermList.
01578
01579                 */
01580                 Out = StrCopy(FindExtraSymbol(MAXVARIABLES-*t),buffer);
01581                 /*
01582                 Out = StrCopy((UBYTE *) AC.extrasym,buffer);
01583                 if ( AC.extrasymbols == 0 ) {
01584                     Out = NumCopy((MAXVARIABLES-*t),Out);
01585                     Out = StrCopy((UBYTE *) "_",Out);
01586                 }
01587                 else if ( AC.extrasymbols == 1 ) {
01588                     Out = AddArrayIndex((MAXVARIABLES-*t),Out);
01589                 }
01590                 */
01591                 /*
01592                 else if ( AC.extrasymbols == 2 ) {
01593                     Out = NumCopy((MAXVARIABLES-*t),Out);
01594                 }
01595                 */
01596                 t++;
01597             }
01598             if ( AC.OutputMode == CMODE && po > 1
01599                 && AC.Cnumpows >= po ) {
01600                 Out = StrCopy((UBYTE *) ")",Out);
01601                 po = 0;
01602             }
01603             else if ( *t != 1 ) WrtPower(Out,*t);
01604             TokenToLine(buffer);
01605             t++;
01606             first = 0;
01607         }
01608         break;
01609     case VECTOR :
01610         while ( t < stopper ) {
01611             if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
01612                 FiniLine();
01613                 if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
01614                 else IniLine(3);
01615                 if ( first ) TokenToLine((UBYTE *) " ");
01616             }
01617             if ( !first ) MultiplyToLine();
01618
01619             Out = StrCopy(FindVector(*t),buffer);
01620             /* Out = StrCopy(VARNAME(vectors,*t - AM.OffsetVector),buffer); */
01621             t++;
01622             if ( AC.OutputMode == MATHEMATICAMODE ) *Out++ = '[';
01623             else *Out++ = '(';
01624             if ( *t >= AM.OffsetIndex ) {
01625                 i = *t++;
01626                 if ( i >= AM.IndDum ) {
01627                     i -= AM.IndDum;
01628                     *Out++ = 'N';
01629                     Out = NumCopy(i,Out);
01630                     *Out++ = '_';
01631                     *Out++ = '?';

```

```

01632         *Out = 0;
01633     }
01634     else
01635         Out = StrCopy(FindIndex(i),Out);
01636 /*         Out = StrCopy(VARNAME(indices,i - AM.OffsetIndex),Out); */
01637     }
01638     else if ( *t == FUNNYVEC ) { *Out++ = '?'; *Out = 0; }
01639     else {
01640         Out = NumCopy(*t++,Out);
01641     }
01642     if ( AC.OutputMode == MATHEMATICAMODE ) *Out++ = ']';
01643     else *Out++ = ')';
01644     *Out = 0;
01645     TokenToLine(buffer);
01646     first = 0;
01647 }
01648 break;
01649 case INDEX :
01650     while ( t < stopper ) {
01651         if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
01652             FiniLine();
01653             if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
01654             else IniLine(3);
01655             if ( first ) TokenToLine((UBYTE *) " ");
01656         }
01657         if ( !first ) MultiplyToLine();
01658         if ( *t >= 0 ) {
01659             if ( *t < AM.OffsetIndex ) {
01660                 TalToLine((UWORD)(*t++));
01661             }
01662             else {
01663                 i = *t++;
01664                 if ( i >= AM.IndDum ) {
01665                     i -= AM.IndDum;
01666                     Out = buffer;
01667                     *Out++ = 'N';
01668                     Out = NumCopy(i,Out);
01669                     *Out++ = '_';
01670                     *Out++ = '?';
01671                     *Out = 0;
01672                 }
01673                 else {
01674                     i -= AM.OffsetIndex;
01675                     Out = StrCopy(FindIndex(i%WILDOFFSET+AM.OffsetIndex),buffer);
01676 /*                     Out = StrCopy(VARNAME(indices,i%WILDOFFSET),buffer); */
01677                     if ( i >= WILDOFFSET ) { *Out++ = '?'; *Out = 0; }
01678                 }
01679                 TokenToLine(buffer);
01680             }
01681         }
01682         else {
01683             TokenToLine(FindVector(*t)); t++;
01684 /*             TokenToLine(VARNAME(vectors,*t - AM.OffsetVector)); t++; */
01685         }
01686         first = 0;
01687     }
01688     break;
01689 case DOLLAREXPRESSION:
01690     {
01691         DOLLARS d = Dollars + sterm[2];
01692         Out = StrCopy((UBYTE *) "$",buffer);
01693         Out = StrCopy(AC.dollarnames->namebuffer+d->name,Out);
01694         if ( sterm[3] != 1 ) WrtPower(Out,sterm[3]);
01695         TokenToLine(buffer);
01696     }
01697     first = 0;
01698     break;
01699 case DELTA :
01700     while ( t < stopper ) {
01701         if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
01702             FiniLine();
01703             if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
01704             else IniLine(3);
01705             if ( first ) TokenToLine((UBYTE *) " ");
01706         }
01707         if ( !first ) MultiplyToLine();
01708         Out = StrCopy((UBYTE *) "d_(",buffer);
01709         if ( *t >= AM.OffsetIndex ) {
01710             if ( *t < AM.IndDum ) {
01711                 Out = StrCopy(FindIndex(*t),Out);
01712 /*                 Out = StrCopy(VARNAME(indices,*t - AM.OffsetIndex),Out); */
01713                 t++;
01714             }
01715             else {
01716                 *Out++ = 'N';
01717                 Out = NumCopy( *t++ - AM.IndDum, Out);
01718                 *Out++ = '_';

```

```

01719         *Out++ = '?';
01720         *Out = 0;
01721     }
01722 }
01723 else if ( *t == FUNNYVEC ) { *Out++ = '?'; *Out = 0; }
01724 else {
01725     Out = NumCopy(*t++,Out);
01726 }
01727 *Out++ = ',';
01728 if ( *t >= AM.OffsetIndex ) {
01729     if ( *t < AM.IndDum ) {
01730         Out = StrCopy(FindIndex(*t),Out);
01731         Out = StrCopy(VARNAME(indices,*t - AM.OffsetIndex),Out); /*
01732         t++;
01733     }
01734     else {
01735         *Out++ = 'N';
01736         Out = NumCopy(*t++ - AM.IndDum,Out);
01737         *Out++ = '_';
01738         *Out++ = '?';
01739     }
01740 }
01741 else {
01742     Out = NumCopy(*t++,Out);
01743 }
01744 *Out++ = ')';
01745 *Out = 0;
01746 TokenToLine(buffer);
01747 first = 0;
01748 }
01749 break;
01750 case DOTPRODUCT :
01751     while ( t < stopper ) {
01752         if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
01753             FiniLine();
01754             if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
01755             else IniLine(3);
01756             if ( first ) TokenToLine((UBYTE *) " ");
01757         }
01758         if ( !first ) MultiplyToLine();
01759         if ( AC.OutputMode == CMODE && t[2] != 1 )
01760             TokenToLine((UBYTE *) "pow(");
01761         if ( AC.OutputMode == MATHEMATICAMODE )
01762             TokenToLine((UBYTE *) "(");
01763         Out = StrCopy(FindVector(*t),buffer);
01764         Out = StrCopy(VARNAME(vectors,*t - AM.OffsetVector),buffer); /*
01765         t++;
01766         if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
01767             || AC.OutputMode == CMODE )
01768             *Out++ = AO.FortDotChar;
01769         else *Out++ = '.';
01770         Out = StrCopy(FindVector(*t),Out);
01771         Out = StrCopy(VARNAME(vectors,*t - AM.OffsetVector),Out); /*
01772         if ( AC.OutputMode == MATHEMATICAMODE ) {
01773             *Out++ = ')';
01774             *Out = 0;
01775         }
01776         t++;
01777         if ( *t != 1 ) WrtPower(Out,*t);
01778         t++;
01779         TokenToLine(buffer);
01780         first = 0;
01781     }
01782     break;
01783 case EXPONENT :
01784     #if FUNHEAD != 2
01785         t += FUNHEAD - 2;
01786     #endif
01787     if ( !first ) MultiplyToLine();
01788     if ( AC.OutputMode == CMODE ) TokenToLine((UBYTE *) "pow(");
01789     else TokenToLine((UBYTE *) "(");
01790     WriteArgument(t);
01791     if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
01792         || AC.OutputMode == REDUCEMODE )
01793         TokenToLine((UBYTE *) ")*(");
01794     else if ( AC.OutputMode == CMODE ) TokenToLine((UBYTE *) ",");
01795     else {
01796         UBYTE *Out1 = IsExponentSign();
01797         if ( Out1 ) {
01798             TokenToLine((UBYTE *) " ");
01799             TokenToLine(Out1);
01800             TokenToLine((UBYTE *) "(");
01801         }
01802         else TokenToLine((UBYTE *) ")^(");
01803     }
01804     NEXTARG(t)
01805     WriteArgument(t);

```

```

01806         TokenToLine((UBYTE *)"");
01807         break;
01808     case DENOMINATOR :
01809 #if FUNHEAD != 2
01810         t += FUNHEAD - 2;
01811 #endif
01812         if ( first ) TokenToLine((UBYTE *)"1/(");
01813         else TokenToLine((UBYTE *)"/(");
01814         WriteArgument(t);
01815         TokenToLine((UBYTE *)"");
01816         break;
01817     case SUBEXPRESSION:
01818         if ( !first ) MultiplyToLine();
01819         TokenToLine((UBYTE *)"(");
01820         t = cbuf[sterm[4]].rhs[sterm[2]];
01821         tt = t;
01822         while ( *tt ) tt += *tt;
01823         oldoutsidfun = AC.outsidfun; AC.outsidfun = 0;
01824         if ( *t ) {
01825             WriteExpression(t, (LONG)(tt-t));
01826         }
01827         else {
01828             TokenToLine((UBYTE *)"0");
01829         }
01830         AC.outsidfun = oldoutsidfun;
01831         TokenToLine((UBYTE *)"");
01832         if ( sterms[3] != 1 ) {
01833             UBYTE *Out1 = IsExponentSign();
01834             if ( Out1 ) TokenToLine(Out1);
01835             else TokenToLine((UBYTE *)"^");
01836             Out = buffer;
01837             NumCopy(sterms[3], Out);
01838             TokenToLine(buffer);
01839         }
01840         break;
01841     default :
01842         if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
01843             FiniLine();
01844             if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
01845             else IniLine(3);
01846             if ( first ) TokenToLine((UBYTE *)" ");
01847         }
01848         if ( *sterm < FUNCTION ) {
01849             return(MesPrint("Illegal subterm while writing"));
01850         }
01851         if ( !first ) MultiplyToLine();
01852         first = 1;
01853         { UBYTE *tmp;
01854             if ( ( tmp = FindFunWithArgs(sterm) ) != 0 ) {
01855                 TokenToLine(tmp);
01856                 break;
01857             }
01858         }
01859         t += FUNHEAD-2;
01860
01861         if ( *sterm == GAMMA && t[-FUNHEAD+1] == FUNHEAD+1 ) {
01862             TokenToLine((UBYTE *)"gi_");
01863         }
01864         else {
01865             if ( *sterm != DUMFUN ) {
01866                 Out = StrCopy(FindFunction(*sterm), buffer);
01867                 Out = StrCopy(VARNAME(functions, *sterm - FUNCTION), buffer); /*
01868             }
01869             else { Out = buffer; *Out = 0; }
01870             if ( t >= stopper ) {
01871                 TokenToLine(buffer);
01872                 break;
01873             }
01874             if ( AC.OutputMode == MATHEMATICAMODE ) { *Out++ = '['; closepar[0] = (UBYTE)']'; }
01875             else { *Out++ = '('; }
01876             *Out = 0;
01877             TokenToLine(buffer);
01878         }
01879         i = functions[*sterm - FUNCTION].spec;
01880         if ( i >= TENSORFUNCTION ) {
01881             int curdict = AO.CurrentDictionary;
01882             if ( AO.CurrentDictionary && AO.CurDictNotInFunctions > 0 )
01883                 AO.CurrentDictionary = 0;
01884             t = sterms + FUNHEAD;
01885             while ( t < stopper ) {
01886                 if ( !first ) TokenToLine((UBYTE *)",");
01887                 else first = 0;
01888                 j = *t++;
01889                 if ( j >= 0 ) {
01890                     if ( j < AM.OffsetIndex ) TalToLine((UWORD)(j));
01891                     else if ( j < AM.IndDum ) {
01892                         i = j - AM.OffsetIndex;

```

```

01893                                     Out = StrCopy(FindIndex(i%WILDOFFSET+AM.OffsetIndex),buffer);
01894 /*                                     Out = StrCopy(VARNAME(indices,i%WILDOFFSET),buffer); */
01895                                     if ( i >= WILDOFFSET ) { *Out++ = '?'; *Out = 0; }
01896                                     TokenToLine(buffer);
01897                                     }
01898                                     else {
01899                                         Out = buffer;
01900                                         *Out++ = 'N';
01901                                         Out = NumCopy(j - AM.IndDum,Out);
01902                                         *Out++ = '_';
01903                                         *Out++ = '?';
01904                                         *Out = 0;
01905                                         TokenToLine(buffer);
01906                                     }
01907                                     }
01908                                     else if ( j == FUNNYVEC ) { TokenToLine((UBYTE *)"?"); }
01909                                     else if ( j > -WILDOFFSET ) {
01910                                         Out = buffer;
01911                                         Out = NumCopy((UWORD)(-j + 4),Out);
01912                                         *Out++ = '_';
01913                                         *Out = 0;
01914                                         TokenToLine(buffer);
01915                                     }
01916                                     else {
01917                                         TokenToLine(FindVector(j));
01918 /*                                     TokenToLine(VARNAME(vectors,j - AM.OffsetVector)); */
01919                                     }
01920                                     }
01921                                     AO.CurrentDictionary = curdict;
01922                                     }
01923                                     else {
01924                                         int curdict = AO.CurrentDictionary;
01925                                         if ( AO.CurrentDictionary && AO.CurDictNotInFunctions > 0 )
01926                                             AO.CurrentDictionary = 0;
01927                                         while ( t < stopper ) {
01928                                             if ( !first ) TokenToLine((UBYTE *)",");
01929                                             WriteArgument(t);
01930                                             NEXTARG(t);
01931                                             first = 0;
01932                                         }
01933                                         AO.CurrentDictionary = curdict;
01934                                     }
01935                                     TokenToLine(closepar);
01936                                     closepar[0] = (UBYTE)'\'';
01937                                     break;
01938                                     }
01939                                     return(0);
01940     }
01941
01942 /*
01943     #] WriteSubTerm :
01944     #[ WriteInnerTerm :      WORD WriteInnerTerm(term,first)
01945
01946     Writes the contents of term to the output.
01947     Only the part that is inside parentheses is written.
01948
01949 */
01950
01951 int WriteInnerTerm(WORD *term, WORD first)
01952 {
01953     WORD *t, *s, *s1, *s2, n, i, pow;
01954 #ifdef WITHFLOAT
01955     int FloatChars = 0;
01956     GETIDENTITY
01957 #endif
01958     t = term;
01959     s = t+1;
01960     GETCOEF(t,n);
01961     while ( s < t ) {
01962         if ( *s == HAAKJE ) break;
01963         s += s[1];
01964     }
01965     if ( s < t ) { s += s[1]; }
01966     else { s = term+1; }
01967
01968     if ( n < 0 || !first ) {
01969         if ( n > 0 ) { TOKENTOLINE(" + ","+") }
01970         else if ( n < 0 ) { n = -n; TOKENTOLINE(" - ","-") }
01971     }
01972     if ( AC.modpowers ) {
01973         if ( n == 1 && *t == 1 && t > s ) first = 1;
01974         else if ( ABS(AC.ncmod) == 1 ) {
01975             UBYTE *Out1 = IsExponentSign();
01976             LongToLine((UWORD *)AC.powmod,AC.npowmod);
01977             if ( Out1 ) TokenToLine(Out1);
01978             else TokenToLine((UBYTE *)"^");
01979             TalToLine(AC.modpowers[(LONG)((UWORD)*t)]);

```

```

01980         first = 0;
01981     }
01982     else {
01983         LONG jj;
01984         UBYTE *Out1 = IsExponentSign();
01985         LongToLine((UWORD *)AC.powmod,AC.npowmod);
01986         if ( Out1 ) TokenToLine(Out1);
01987         else TokenToLine((UBYTE *)"^");
01988         jj = (UWORD)*t;
01989         if ( n == 2 ) jj += ((LONG)t[1])«BITSINWORD;
01990         if ( AC.modpowers[jj+1] == 0 ) {
01991             TalToLine(AC.modpowers[jj]);
01992         }
01993         else {
01994             LongToLine(AC.modpowers+jj,2);
01995         }
01996         first = 0;
01997     }
01998 }
01999 else if ( n != 1 || *t != 1 || t[1] != 1 || t <= s ) {
02000     if ( lowestlevel && ( ( AO.PrintType & PRINTONEFUNCTION ) != 0 ) ) {
02001         FiniLine();
02002         if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
02003         else IniLine(3);
02004     }
02005     if ( AO.CurrentDictionary > 0 ) TransformRational((UWORD *)t,n);
02006     else RatToLine((UWORD *)t,n);
02007     first = 0;
02008 }
02009 #ifdef WITHFLOAT
02010 /*
02011     Check whether there is a 'proper' float_ function and no raw mode.
02012     If so, print as float.
02013     Raw mode is indicated as AO.FloatPrec < 0.
02014     AO.FloatPrec == 0 indicated as many digits as the precision allows.
02015 */
02016 else if ( AO.FloatPrec >= 0 && AT.aux_ != 0 ) {
02017     WORD *ss = s;
02018     while ( ss < t ) {
02019         if ( *ss == FLOATFUN ) {
02020             if ( ( FloatChars = PrintFloat(ss,AO.FloatPrec) ) != 0 ) {
02021                 TokenToLine(AO.floatspace);
02022                 first = 0;
02023             }
02024             break;
02025         }
02026         ss += ss[1];
02027     }
02028     if ( ss >= t ) first = 1;
02029 }
02030 #endif
02031 else first = 1;
02032 while ( s < t ) {
02033     if ( lowestlevel && ( (AO.PrintType & (PRINTONEFUNCTION | PRINTALL)) == PRINTONEFUNCTION ) ) {
02034         FiniLine();
02035         if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
02036         else IniLine(3);
02037     }
02038 }
02039 /*
02040     #[ NEWGAMMA :
02041 */
02042 #ifdef NEWGAMMA
02043     if ( *s == GAMMA ) { /* String them up */
02044         WORD *tt,*ss;
02045         ss = AT.WorkPointer;
02046         *ss++ = GAMMA;
02047         *ss++ = s[1];
02048         FILLFUN(ss)
02049         *ss++ = s[FUNHEAD];
02050         tt = s + FUNHEAD + 1;
02051         n = s[1] - FUNHEAD-1;
02052         do {
02053             while ( --n >= 0 ) *ss++ = *tt++;
02054             tt = s + s[1];
02055             while ( *tt == GAMMA && tt[FUNHEAD] == s[FUNHEAD] && tt < t ) {
02056                 s = tt;
02057                 tt += FUNHEAD + 1;
02058                 n = s[1] - FUNHEAD-1;
02059                 if ( n > 0 ) break;
02060             } while ( n > 0 );
02061             tt = AT.WorkPointer;
02062             AT.WorkPointer = ss;
02063             tt[1] = WORDDIF(ss,tt);
02064             if ( WriteSubTerm(tt,first) ) {
02065                 MesCall("WriteInnerTerm");

```



```

02067         SETERROR(-1)
02068     }
02069     AT.WorkPointer = tt;
02070 }
02071 else
02072 #endif
02073 /*
02074     #] NEWGAMMA :
02075 */
02076 #ifdef WITHFLOAT
02077     if ( *s == FLOATFUN && AO.FloatPrec >= 0 && AT.aux_ != 0 ) {
02078     }
02079     else
02080 #endif
02081     {
02082         if ( *s >= FUNCTION && AC.funpowers > 0
02083             && functions[*s-FUNCTION].spec == 0 && ( AC.funpowers == ALLFUNPOWERS ||
02084             ( AC.funpowers == COMFUNPOWERS && functions[*s-FUNCTION].commute == 0 ) ) ) {
02085             pow = 1;
02086             for(;;) {
02087                 s1 = s; s2 = s + s[1]; i = s[1];
02088                 if ( s2 < t ) {
02089                     while ( --i >= 0 && *s1 == *s2 ) { s1++; s2++; }
02090                     if ( i < 0 ) {
02091                         pow++; s = s+s[1];
02092                     }
02093                     else break;
02094                 }
02095                 else break;
02096             }
02097             if ( pow > 1 ) {
02098                 if ( AC.OutputMode == CMODE ) {
02099                     if ( !first ) MultiplyToLine();
02100                     TokenToLine((UBYTE *) "pow(");
02101                     first = 1;
02102                 }
02103                 if ( WriteSubTerm(s,first) ) {
02104                     MesCall("WriteInnerTerm");
02105                     SETERROR(-1)
02106                 }
02107                 if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
02108                     || AC.OutputMode == REDUCEMODE ) { TokenToLine((UBYTE *) "***"); }
02109                 else if ( AC.OutputMode == CMODE ) { TokenToLine((UBYTE *) ","); }
02110                 else {
02111                     UBYTE *Out1 = IsExponentSign();
02112                     if ( Out1 ) TokenToLine(Out1);
02113                     else TokenToLine((UBYTE *) "^");
02114                 }
02115                 TalToLine(pow);
02116                 if ( AC.OutputMode == CMODE ) TokenToLine((UBYTE *) ")");
02117             }
02118             else if ( WriteSubTerm(s,first) ) {
02119                 MesCall("WriteInnerTerm");
02120                 SETERROR(-1)
02121             }
02122         }
02123         else if ( WriteSubTerm(s,first) ) {
02124             MesCall("WriteInnerTerm");
02125             SETERROR(-1)
02126         }
02127     }
02128     first = 0;
02129     s += s[1];
02130 }
02131 return(0);
02132 }
02133
02134 /*
02135     #] WriteInnerTerm :
02136     #[ WriteTerm :          WORD WriteTerm(term,lbrac,first,prtf,br)
02137
02138     Writes a term to output. It tests the bracket information first.
02139     If there are no brackets or the bracket is the same all is passed
02140     to WriteInnerTerm. If there are brackets and the bracket is not
02141     the same as for the predecessor the old bracket is closed and
02142     a new one is opened.
02143     br indicates whether we are in a subexpression, barring zeroing
02144     AO.IsBracket
02145
02146 */
02147
02148 int WriteTerm(WORD *term, WORD *lbrac, WORD first, WORD prtf, WORD br)
02149 {
02150     WORD *t, *stopper, *b, n;
02151     int oldIsFortran90 = AC.IsFortran90, i;
02152     if ( *lbrac >= 0 ) {
02153         t = term + 1;

```

```

02154     stopper = (term + *term - 1);
02155     stopper -= ABS(*stopper) - 1;
02156     while ( t < stopper ) {
02157         if ( *t == HAAKJE ) {
02158             stopper = t;
02159             t = term+1;
02160             if ( *lbrac == ( n = WORDDIF(stopper,t) ) ) {
02161                 b = AO.bracket + 1;
02162                 t = term + 1;
02163                 while ( n > 0 && ( *b++ == *t++ ) ) { n--; }
02164                 if ( n <= 0 && ( ( AM.FortranCont <= 0 || AO.InFbrack < AM.FortranCont )
02165                     || ( lowestlevel == 0 ) ) ) {
02166                     /*
02167                         We continue inside a bracket.
02168                     */
02169                     AO.IsBracket = 1;
02170                     if ( ( prtft & PRINTCONTENTS ) != 0 ) {
02171                         AO.NumInBrack++;
02172                     }
02173                     else {
02174                         if ( WriteInnerTerm(term,0) ) goto WrtTmes;
02175                         if ( ( AO.PrintType & PRINTONETERM ) != 0 ) {
02176                             FiniLine();
02177                             TokenToLine((UBYTE *) "    ");
02178                         }
02179                     }
02180                     return(0);
02181                 }
02182                 t = term + 1;
02183                 n = WORDDIF(stopper,t);
02184             }
02185             /*
02186                 Close the bracket
02187             */
02188             if ( *lbrac ) {
02189                 if ( ( prtft & PRINTCONTENTS ) ) PrtTerms();
02190                 TOKENTOLINE(" )","")
02191                 if ( AC.OutputMode == CMODE && AO.FactorMode == 0 )
02192                     TokenToLine((UBYTE *) ";");
02193                 else if ( AO.FactorMode && ( n == 0 ) ) {
02194                     /*
02195                         This should not happen.
02196                     */
02197                     return(0);
02198                 }
02199                 AC.IsFortran90 = ISNOTFORTRAN90;
02200                 FiniLine();
02201                 AC.IsFortran90 = oldIsFortran90;
02202                 if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE
02203                     && AC.OutputSpaces == NORMALFORMAT
02204                     && AO.FactorMode == 0 ) FiniLine();
02205             }
02206             else {
02207                 if ( AC.OutputMode == CMODE && AO.FactorMode == 0 )
02208                     TokenToLine((UBYTE *) ";");
02209                 if ( AO.FortFirst == 0 ) {
02210                     if ( !first ) {
02211                         AC.IsFortran90 = ISNOTFORTRAN90;
02212                         FiniLine();
02213                         AC.IsFortran90 = oldIsFortran90;
02214                     }
02215                 }
02216             }
02217             if ( AO.FactorMode == 0 ) {
02218                 if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
02219                     && !first ) {
02220                     WORD oldmode = AC.OutputMode;
02221                     AC.OutputMode = 0;
02222                     IniLine(0);
02223                     AC.OutputMode = oldmode;
02224                     AO.OutSkip = 7;
02225                 }
02226                 if ( AO.FortFirst == 0 ) {
02227                     TokenToLine(AO.CurBufWrt);
02228                     TOKENTOLINE(" = ","=")
02229                     TokenToLine(AO.CurBufWrt);
02230                 }
02231                 else {
02232                     AO.FortFirst = 0;
02233                     TokenToLine(AO.CurBufWrt);
02234                     TOKENTOLINE(" = ","=")
02235                 }
02236             }
02237             else if ( AC.OutputMode == CMODE && !first ) {
02238                 IniLine(0);
02239                 if ( AO.FortFirst == 0 ) {
02240                     TokenToLine(AO.CurBufWrt);

```

```

02241         TOKENTOLINE(" += ", "+=")
02242     }
02243     else {
02244         AO.FortFirst = 0;
02245         TokenToLine(AO.CurBufWrt);
02246         TOKENTOLINE(" = ", "=")
02247     }
02248 }
02249 else if ( startinline == 0 ) {
02250     IniLine(0);
02251 }
02252 AO.InFbrack = 0;
02253 if ( ( *lbrac = n ) > 0 ) {
02254     b = AO.bracket;
02255     *b++ = n + 4;
02256     while ( --n >= 0 ) *b++ = *t++;
02257     *b++ = 1; *b++ = 1; *b = 3;
02258     AO.IsBracket = 0;
02259     if ( WriteInnerTerm(AO.bracket,0) ) {
02260         /* Error message */
02261         WORD i;
02262         t = term;
02263         AO.OutSkip = 3;
02264         FiniLine();
02265         i = *t;
02266         while ( --i >= 0 ) { TalToLine((UWORD)(t++));
02267             if ( AC.OutputSpaces == NORMALFORMAT )
02268                 TokenToLine((UBYTE *)" "); }
02269         AO.OutSkip = 0;
02270         FiniLine();
02271         MesCall("WriteTerm");
02272         SETERROR(-1)
02273     }
02274     TOKENTOLINE(" * ( ", "*(")
02275     AO.NumInBrack = 0;
02276     AO.IsBracket = 1;
02277     if ( ( prt & PRINTONETERM ) != 0 ) {
02278         first = 0;
02279         FiniLine();
02280         TokenToLine((UBYTE *)" ");
02281     }
02282     else first = 1;
02283 }
02284 else {
02285     AO.IsBracket = 0;
02286     first = 0;
02287 }
02288 }
02289 else {
02290     /*
02291     Here is the code that writes the glue between two factors.
02292     We should not forget factors that are zero!
02293     */
02294     if ( ( *lbrac = n ) > 0 ) {
02295         b = AO.bracket;
02296         *b++ = n + 4;
02297         while ( --n >= 0 ) *b++ = *t++;
02298         *b++ = 1; *b++ = 1; *b = 3;
02299         for ( i = AO.FactorNum+1; i < AO.bracket[4]; i++ ) {
02300             if ( first ) {
02301                 TOKENTOLINE(" ( 0 )", "(0)")
02302                 first = 0;
02303             }
02304             else {
02305                 TOKENTOLINE(" * ( 0 )", "* (0)")
02306             }
02307             FiniLine();
02308             IniLine(0);
02309         }
02310         AO.FactorNum = AO.bracket[4];
02311     }
02312     else {
02313         AO.NumInBrack = 0;
02314         return(0);
02315     }
02316     if ( first == 0 ) { TOKENTOLINE(" * ( ", "*(") }
02317     else { TOKENTOLINE(" ( ", "(") }
02318     AO.NumInBrack = 0;
02319     first = 1;
02320 }
02321 if ( ( prt & PRINTCONTENTS ) != 0 ) AO.NumInBrack++;
02322 else if ( WriteInnerTerm(term,first) ) goto WrtTmes;
02323 if ( ( AO.PrintType & PRINTONETERM ) != 0 ) {
02324     FiniLine();
02325     TokenToLine((UBYTE *)" ");
02326 }
02327 return(0);

```

```

02328         }
02329         else t += t[1];
02330     }
02331     if ( *lbrac > 0 ) {
02332         if ( ( prtfl & PRINTCONTENTS ) != 0 ) PrtTerms();
02333         TokenToLine((UBYTE *)" ");
02334         if ( AC.OutputMode == CMODE ) TokenToLine((UBYTE *)");");
02335         if ( AO.FortFirst == 0 ) {
02336             AC.IsFortran90 = ISNOTFORTRAN90;
02337             FiniLine();
02338             AC.IsFortran90 = oldIsFortran90;
02339         }
02340         if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE
02341             && AC.OutputSpaces == NORMALFORMAT ) FiniLine();
02342         if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
02343             && !first ) {
02344             WORD oldmode = AC.OutputMode;
02345             AC.OutputMode = 0;
02346             IniLine(0);
02347             AC.OutputMode = oldmode;
02348             AO.OutSkip = 7;
02349             if ( AO.FortFirst == 0 ) {
02350                 TokenToLine(AO.CurBufWrt);
02351                 TOKENTOLINE(" = ", "=")
02352                 TokenToLine(AO.CurBufWrt);
02353             }
02354             else {
02355                 AO.FortFirst = 0;
02356                 TokenToLine(AO.CurBufWrt);
02357                 TOKENTOLINE(" = ", "=")
02358             }
02359             /*
02360             TokenToLine(AO.CurBufWrt);
02361             TOKENTOLINE(" = ", "=")
02362             if ( AO.FortFirst == 0 )
02363                 TokenToLine(AO.CurBufWrt);
02364             else AO.FortFirst = 0;
02365             */
02366         }
02367         else if ( AC.OutputMode == CMODE && !first ) {
02368             IniLine(0);
02369             if ( AO.FortFirst == 0 ) {
02370                 TokenToLine(AO.CurBufWrt);
02371                 TOKENTOLINE(" += ", "+=")
02372             }
02373             else {
02374                 AO.FortFirst = 0;
02375                 TokenToLine(AO.CurBufWrt);
02376                 TOKENTOLINE(" = ", "=")
02377             }
02378             /*
02379             TokenToLine(AO.CurBufWrt);
02380             if ( AO.FortFirst == 0 ) { TOKENTOLINE(" += ", "+=") }
02381             else {
02382                 TOKENTOLINE(" = ", "=")
02383                 AO.FortFirst = 0;
02384             }
02385             */
02386         }
02387         else IniLine(0);
02388         *lbrac = 0;
02389         first = 1;
02390     }
02391 }
02392 if ( !br ) AO.IsBracket = 0;
02393 if ( ( AM.FortranCont > 0 && AO.InFbrack >= AM.FortranCont ) && lowestlevel ) {
02394     if ( AC.OutputMode == CMODE ) TokenToLine((UBYTE *)");");
02395     if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
02396         && !first ) {
02397         WORD oldmode = AC.OutputMode;
02398         if ( AO.FortFirst == 0 ) {
02399             AC.IsFortran90 = ISNOTFORTRAN90;
02400             FiniLine();
02401             AC.IsFortran90 = oldIsFortran90;
02402             AC.OutputMode = 0;
02403             IniLine(0);
02404             AC.OutputMode = oldmode;
02405             AO.OutSkip = 7;
02406             TokenToLine(AO.CurBufWrt);
02407             TOKENTOLINE(" = ", "=")
02408             TokenToLine(AO.CurBufWrt);
02409         }
02410         else {
02411             AO.FortFirst = 0;
02412             /*
02413             TokenToLine(AO.CurBufWrt);
02414             TOKENTOLINE(" = ", "=")

```

```

02415 */
02416     }
02417 /*
02418     TokenToLine(AO.CurBufWrt);
02419     TOKENTOLINE(" = ", "=")
02420     if ( AO.FortFirst == 0 )
02421         TokenToLine(AO.CurBufWrt);
02422     else AO.FortFirst = 0;
02423 */
02424 }
02425 else if ( AC.OutputMode == CMODE && !first ) {
02426     FiniLine();
02427     IniLine(0);
02428     if ( AO.FortFirst == 0 ) {
02429         TokenToLine(AO.CurBufWrt);
02430         TOKENTOLINE(" += ", "+=")
02431     }
02432     else {
02433         AO.FortFirst = 0;
02434         TokenToLine(AO.CurBufWrt);
02435         TOKENTOLINE(" = ", "=")
02436     }
02437 /*
02438     TokenToLine(AO.CurBufWrt);
02439     if ( AO.FortFirst == 0 ) { TOKENTOLINE(" += ", "+=") }
02440     else {
02441         TOKENTOLINE(" = ", "=")
02442         AO.FortFirst = 0;
02443     }
02444 */
02445 }
02446 else {
02447     FiniLine();
02448     IniLine(0);
02449 }
02450 AO.InFbrack = 0;
02451 }
02452 if ( WriteInnerTerm(term,first) ) goto WrtTmes;
02453 if ( ( AO.PrintType & PRINTONETERM ) != 0 ) {
02454     FiniLine();
02455     IniLine(0);
02456 }
02457 return(0);
02458 }
02459
02460 /*
02461     #] WriteTerm :
02462     #[ WriteExpression :      WORD WriteExpression(terms,ltot)
02463
02464     Writes a subexpression to output.
02465     The subexpression is in terms and contains ltot words.
02466     This is only used for function arguments.
02467 */
02468 */
02469
02470 int WriteExpression(WORD *terms, LONG ltot)
02471 {
02472     WORD *stopper;
02473     WORD first, btot;
02474     WORD OldIsBracket, OldPrintType = AO.PrintType;
02475     if ( !AC.outsidefun ) { AO.PrintType &= ~PRINTONETERM; first = 1; }
02476     else first = 0;
02477     stopper = terms + ltot;
02478     btot = -1;
02479     while ( terms < stopper ) {
02480         AO.IsBracket = OldIsBracket;
02481         if ( WriteTerm(terms,&btot,first,0,1) ) {
02482             MesCall("WriteExpression");
02483             SETERROR(-1)
02484         }
02485         first = 0;
02486         terms += *terms;
02487     }
02488 /* AO.IsBracket = 0; */
02489 AO.IsBracket = OldIsBracket;
02490 AO.PrintType = OldPrintType;
02491 return(0);
02492 }
02493
02494 /*
02495     #] WriteExpression :
02496     #[ WriteAll :          WORD WriteAll()
02497
02498     Writes all expressions that should be written
02499 */
02500
02501 int WriteAll(void)

```

```

02502 {
02503     GETIDENTITY
02504     WORD lbrac, first;
02505     WORD *t, *stopper, n, prtf;
02506     int oldIsFortran90 = AC.IsFortran90, i;
02507     POSITION pos;
02508     FILEHANDLE *f;
02509     EXPRESSIONS e;
02510     if ( AM.exitflag ) return(0);
02511 #ifdef WITHMPI
02512     if ( PF.me != MASTER ) {
02513         /*
02514          * For the slaves, we need to call Optimize() the same number of times
02515          * as the master. The first argument doesn't have any important role.
02516          */
02517         for ( n = 0; n < NumExpressions; n++ ) {
02518             e = &Expressions[n];
02519             if ( (!e->printflag) & PRINTON ) continue;
02520             switch ( e->status ) {
02521                 case LOCALEXPRESSION:
02522                 case GLOBALEXPRESSION:
02523                 case UNHIDELEXPRESSION:
02524                 case UNHIDEGEXPRESSION:
02525                     break;
02526                 default:
02527                     continue;
02528             }
02529             e->printflag = 0;
02530             PutPreVar(AM.oldnumextrasymbols, GetPreVar((UBYTE *)"EXTRASYNBOLS_", 0), 0, 1);
02531             if ( AO.OptimizationLevel > 0 ) {
02532                 if ( Optimize(0, 1) ) return(-1);
02533             }
02534         }
02535         return(0);
02536     }
02537 #endif
02538     SeekScratch(AR.outfile, &pos);
02539     if ( ResetScratch() ) {
02540         MesCall("WriteAll");
02541         SETERROR(-1)
02542     }
02543     AO.termbuf = AT.WorkPointer;
02544     AO.bracket = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer);
02545     AT.WorkPointer = (WORD *)(((UBYTE *) (AT.WorkPointer)) + AM.MaxTer*2);
02546     AO.OutFill = AO.OutputLine = (UBYTE *)AT.WorkPointer;
02547     AT.WorkPointer += 2*AC.LineLength;
02548     *(AR.CompressBuffer) = 0;
02549     first = 0;
02550     for ( n = 0; n < NumExpressions; n++ ) {
02551         if ( ( Expressions[n].printflag & PRINTON ) != 0 ) { first = 1; break; }
02552     }
02553     if ( !first ) goto EndWrite;
02554     AO.IsBracket = 0;
02555     AO.OutSkip = 3;
02556     AR.DeferFlag = 0;
02557     while ( GetTerm(BHEAD AO.termbuf) ) {
02558         t = AO.termbuf + 1;
02559         e = Expressions + AO.termbuf[3];
02560         n = e->status;
02561         if ( ( n == LOCALEXPRESSION || n == GLOBALEXPRESSION
02562             || n == UNHIDELEXPRESSION || n == UNHIDEGEXPRESSION ) &&
02563             ( ( prtf = e->printflag ) & PRINTON ) != 0 ) {
02564             e->printflag = 0;
02565             AO.NumInBrack = 0;
02566             PutPreVar(AM.oldnumextrasymbols,
02567                 GetPreVar((UBYTE *)"EXTRASYNBOLS_", 0), 0, 1);
02568             if ( ( prtf & PRINTLFILE ) != 0 ) {
02569                 if ( AC.LogHandle < 0 ) prtf &= ~PRINTLFILE;
02570             }
02571             AO.PrintType = prtf;
02572             /*
02573              if ( AC.OutputMode == VORTRANMODE ) {
02574                  UBYTE *oldOutFill = AO.OutFill, *oldOutputLine = AO.OutputLine;
02575                  AO.OutSkip = 6;
02576                  if ( Optimize(AO.termbuf[3], 1) ) goto AboWrite;
02577                  AO.OutSkip = 3;
02578                  AO.OutFill = oldOutFill; AO.OutputLine = oldOutputLine;
02579                  FiniLine();
02580                  continue;
02581              }
02582              else
02583              */
02584             if ( AO.OptimizationLevel > 0 ) {
02585                 UBYTE *oldOutFill = AO.OutFill, *oldOutputLine = AO.OutputLine;
02586                 AO.OutSkip = 6;
02587                 if ( Optimize(AO.termbuf[3], 1) ) goto AboWrite;
02588                 AO.OutSkip = 3;

```

```

02589         AO.OutFill = oldOutFill; AO.OutputLine = oldOutputLine;
02590         FiniLine();
02591         continue;
02592     }
02593     if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
02594         AO.OutSkip = 6;
02595     FiniLine();
02596     AO.CurBufWrt = EXPRNAME(AO.termbuf[3]);
02597     TokenToLine(AO.CurBufWrt);
02598     stopper = t + t[1];
02599     t += SUBEXPSIZE;
02600     if ( t < stopper ) {
02601         TokenToLine((UBYTE *) "(");
02602         first = 1;
02603         while ( t < stopper ) {
02604             n = *t;
02605             if ( !first ) TokenToLine((UBYTE *) ",");
02606             switch ( n ) {
02607                 case SYMTOSYM :
02608                     TokenToLine(FindSymbol(t[2]));
02609                     TokenToLine(VARNAME(symbols,t[2])); /*
02610                     break;
02611                 case VECTOVEC :
02612                     TokenToLine(FindVector(t[2]));
02613                     TokenToLine(VARNAME(vectors,t[2] - AM.OffsetVector)); /*
02614                     break;
02615                 case INDTOIND :
02616                     TokenToLine(FindIndex(t[2]));
02617                     TokenToLine(VARNAME(indices,t[2] - AM.OffsetIndex)); /*
02618                     break;
02619                 default :
02620                     TokenToLine(FindFunction(t[2]));
02621                     TokenToLine(VARNAME(functions,t[2] - FUNCTION)); /*
02622                     break;
02623             }
02624             t += t[1];
02625             first = 0;
02626         }
02627         TokenToLine((UBYTE *) ")");
02628     }
02629     TOKENTOLINE(" =", "=");
02630     if ( AC.OutputMode == MATHEMATICAMODE ) {
02631         TOKENTOLINE(" (","(");
02632     }
02633     lbrac = 0;
02634     AO.InFbrack = 0;
02635     if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
02636         AO.FortFirst = 1;
02637     else
02638         AO.FortFirst = 0;
02639     first = 1;
02640     if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
02641         AO.FactorMode = 1+e->numfactors;
02642         AO.FactorNum = 0; /* Which factor are we doing. For factors that are zero */
02643     }
02644     else {
02645         AO.FactorMode = 0;
02646     }
02647     while ( GetTerm(BHEAD AO.termbuf) ) {
02648         WORD *m;
02649         GETSTOP(AO.termbuf,m);
02650         if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
02651             && ( ( prtf & PRINTONETERM ) != 0 ) ) {}
02652         else {
02653             if ( first ) {
02654                 FiniLine();
02655                 IniLine(0);
02656             }
02657             if ( ( prtf & PRINTONETERM ) != 0 ) first = 0;
02658             if ( WriteTerm(AO.termbuf,&lbrac,first,prtf,0) )
02659                 goto AboWrite;
02660             first = 0;
02661         }
02662     }
02663     if ( AO.FactorMode ) {
02664         if ( first ) { AO.FactorNum = 1; TOKENTOLINE(" ( 0 )"," (0)") }
02665         else TOKENTOLINE(" )","");
02666         for ( i = AO.FactorNum+1; i <= e->numfactors; i++ ) {
02667             FiniLine();
02668             IniLine(0);
02669             TOKENTOLINE(" * ( 0 )","*(0)");
02670         }
02671         AO.FactorNum = e->numfactors;
02672         if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE )
02673             TokenToLine((UBYTE *) ",");
02674     }
02675     else if ( AO.FactorMode == 0 || first ) {

```

```

02676         if ( first ) { TOKENTOline(" 0","0" ) }
02677     else if ( lbrac ) {
02678         if ( ( prtf & PRINTCONTENTS ) != 0 ) PrtTerms();
02679         TOKENTOline(" ",")")
02680     }
02681     else if ( ( prtf & PRINTCONTENTS ) != 0 ) {
02682         TOKENTOline(" + 1 * ( ", "+1*(")
02683         PrtTerms();
02684         TOKENTOline(" )",")")
02685     }
02686     if ( AC.OutputMode == MATHEMATICAMODE ) {
02687         TokenToLine((UBYTE *)");");
02688     }
02689     if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE )
02690         TokenToLine((UBYTE *)";");
02691 }
02692 AO.OutSkip = 3;
02693 AC.IsFortran90 = ISNOTFORTRAN90;
02694 FiniLine();
02695 AC.IsFortran90 = oldIsFortran90;
02696 AO.FactorMode = 0;
02697 }
02698 else {
02699     do { } while ( GetTerm(BHEAD AO.termbuf) );
02700 }
02701 }
02702 if ( AC.OutputSpaces == NORMALFORMAT ) FiniLine();
02703 EndWrite:
02704 if ( AR.infile->handle >= 0 ) {
02705     SeekFile(AR.infile->handle, &(AR.infile->filesize), SEEK_SET);
02706 }
02707 AO.IsBracket = 0;
02708 AT.WorkPointer = AO.termbuf;
02709 SetScratch(AR.infile, &pos);
02710 f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
02711 return(0);
02712 AboWrite:
02713 SetScratch(AR.infile, &pos);
02714 f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
02715 MesCall("WriteAll");
02716 Terminate(-1);
02717 return(-1);
02718 }
02719
02720 /*
02721     #] WriteAll :
02722     #[ WriteOne :          WORD WriteOne(name,alreadyinline)
02723
02724     Writes one expression from the preprocessor
02725 */
02726
02727 int WriteOne(UBYTE *name, int alreadyinline, int nosemi, WORD plus)
02728 {
02729     GETIDENTITY
02730     WORD number;
02731     WORD lbrac, first;
02732     POSITION pos;
02733     FILEHANDLE *f;
02734     WORD prf;
02735
02736     if ( GetName(AC.exprnames, name, &number, NOAUTO) != CEXPRESSION ) {
02737         MesPrint("@%s is not an expression", name);
02738         return(-1);
02739     }
02740     switch ( Expressions[number].status ) {
02741         case HIDDENLEXPRESSION:
02742         case HIDDENGEXPRESSION:
02743         case HIDELEXPRESSION:
02744         case HIDEGEXPRESSION:
02745         case UNHIDELEXPRESSION:
02746         case UNHIDEGEXPRESSION:
02747     /*
02748         case DROPHLEXPRESSION:
02749         case DROPHGEXPRESSION:
02750     */
02751         AR.GetFile = 2;
02752         break;
02753         case LOCALEXPRESSION:
02754         case GLOBALEXPRESSION:
02755         case SKIPLEXPRESSION:
02756         case SKIPGEXPRESSION:
02757     /*
02758         case DROPLEXPRESSION:
02759         case DROPGEXPRESSION:
02760     */
02761         AR.GetFile = 0;
02762         break;

```



```

02763         default:
02764             MesPrint("@expressions %s is not active. It cannot be written",name);
02765             return(-1);
02766     }
02767     SeekScratch(AR.outfile,&pos);
02768
02769     f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
02770 /*
02771     if ( ResetScratch() ) {
02772         MesCall("WriteOne");
02773         SETERROR(-1)
02774     }
02775 */
02776     if ( AR.GetFile == 2 ) f = AR.hidefile;
02777     else f = AR.infile;
02778     prf = Expressions[number].printflag;
02779     if ( plus ) prf |= PRINTONETERM;
02780 /*
02781         Now position the file
02782 */
02783     if ( f->handle >= 0 ) {
02784         SetScratch(f,&(Expressions[number].onfile));
02785     }
02786     else {
02787         f->POfill = (WORD *) ((UBYTE *) (f->PObuffer)
02788             + BASEPOSITION(Expressions[number].onfile));
02789     }
02790     AO.termbuf = AT.WorkPointer;
02791     AO.bracket = (WORD *) ((UBYTE *) (AT.WorkPointer)) + AM.MaxTer;
02792     AT.WorkPointer = (WORD *) ((UBYTE *) (AT.WorkPointer)) + AM.MaxTer*2);
02793
02794     AO.OutFill = AO.OutputLine = (UBYTE *) AT.WorkPointer;
02795     AT.WorkPointer += 2*AC.LineLength;
02796     *(AR.CompressBuffer) = 0;
02797
02798     AO.IsBracket = 0;
02799     AO.OutSkip = 3;
02800     AR.DeferFlag = 0;
02801
02802     if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
02803         AO.OutSkip = 6;
02804     if ( GetTerm(BHEAD AO.termbuf) <= 0 ) {
02805         MesPrint("@ReadError in expression %s",name);
02806         goto AboWrite;
02807     }
02808 /*
02809     PutPreVar(AM.oldnumextrasymbols,
02810         GetPreVar((UBYTE *) "EXTRASYMBOLS_",0),0,1);
02811 */
02812 /*
02813     * Currently WriteOne() is called only from writeToChannel() with setting
02814     * AO.OptimizationLevel = 0, which means Optimize() is never called here.
02815     * So we don't need to think about how to ensure that the master and the
02816     * slaves call Optimize() at the same time. (TU 26 Jul 2013)
02817     */
02818     if ( AO.OptimizationLevel > 0 ) {
02819         AO.OutSkip = 6;
02820         if ( Optimize(AO.termbuf[3], 1) ) goto AboWrite;
02821         AO.OutSkip = 3;
02822         FiniLine();
02823     }
02824     else {
02825         lbrac = 0;
02826         AO.InFbrack = 0;
02827         AO.FortFirst = 0;
02828         first = 1;
02829         while ( GetTerm(BHEAD AO.termbuf) ) {
02830             WORD *m;
02831             GETSTOP(AO.termbuf,m);
02832             if ( first ) {
02833                 IniLine(0);
02834                 startinline = alreadyinline;
02835                 AO.OutFill = AO.OutputLine + startinline;
02836                 if ( WriteTerm(AO.termbuf,&lbrac,first,0,0) )
02837                     goto AboWrite;
02838                 first = 0;
02839             }
02840             else {
02841                 if ( ( prf & PRINTONETERM ) != 0 ) first = 1;
02842                 if ( first ) {
02843                     FiniLine();
02844                     IniLine(0);
02845                 }
02846                 first = 0;
02847                 if ( WriteTerm(AO.termbuf,&lbrac,first,0,0) )
02848                     goto AboWrite;
02849             }

```

```

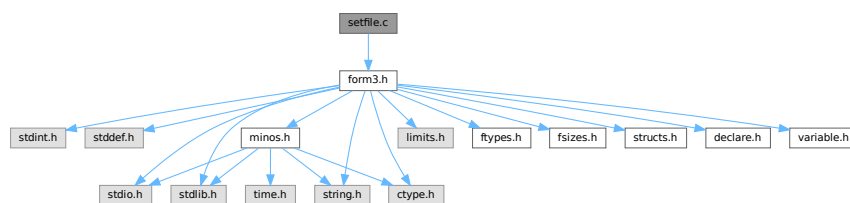
02850     }
02851     if ( first ) {
02852         IniLine(0);
02853         startinline = alreadyinline;
02854         AO.OutFill = AO.OutputLine + startinline;
02855         TOKENTOLINE(" 0","0");
02856     }
02857     else if ( lbrac ) {
02858         TOKENTOLINE(" )","");
02859     }
02860     if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE
02861         && nosemi == 0 ) TokenToLine((UBYTE *)");");
02862     AO.OutSkip = 3;
02863     if ( AC.OutputSpaces == NORMALFORMAT && nosemi == 0 ) {
02864         FiniLine();
02865     }
02866     else {
02867         noextralinefeed = 1;
02868         FiniLine();
02869         noextralinefeed = 0;
02870     }
02871 }
02872 AO.IsBracket = 0;
02873 AT.WorkPointer = AO.termbuf;
02874 SetScratch(f,&pos);
02875 f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
02876 AO.InFbrack = 0;
02877 return(0);
02878 AboWrite:
02879 SetScratch(AR.infile,&pos);
02880 f->POposition = pos;
02881 f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
02882 MesCall("WriteOne");
02883 Terminate(-1);
02884 return(-1);
02885 }
02886
02887 /*
02888     #] WriteOne :
02889     #] schryf-Writes :
02890 */

```

## 4.124 setfile.c File Reference

```
#include "form3.h"
```

Include dependency graph for setfile.c:



### Macros

- #define NUMERICALVALUE 0
- #define STRINGVALUE 1
- #define PATHVALUE 2
- #define ONOFFVALUE 3
- #define DEFINEVALUE 4
- #define SETBUFSIZE 257

## Functions

- int [DoSetups](#) (void)
- int [ProcessOption](#) (UBYTE \*s1, UBYTE \*s2, int filetype)
- [SETUPPARAMETERS](#) \* [GetSetupPar](#) (UBYTE \*s)
- int [RecalcSetups](#) (void)
- int [AllocSetups](#) (void)
- void [WriteSetup](#) (void)
- [SORTING](#) \* [AllocSort](#) (LONG inLargeSize, LONG inSmallSize, LONG inSmallEsize, LONG inTermsInSmall, int inMaxPatches, int inMaxFpatches, LONG inIOsize, int level)
- void [AllocSortFileName](#) ([SORTING](#) \*sort)
- [FILEHANDLE](#) \* [AllocFileHandle](#) (WORD par, char \*name)
- void [DeAllocFileHandle](#) ([FILEHANDLE](#) \*fh)
- int [MakeSetupAllocs](#) (void)
- int [TryFileSetups](#) (void)
- int [TryEnvironment](#) (void)

## Variables

- char [curdirp](#) [] = "."
- char [cursortdirp](#) [] = "."
- char [commentchar](#) [] = "\*"
- char [dotchar](#) [] = "\_"
- char [highfirst](#) [] = "highfirst"
- char [lowfirst](#) [] = "lowfirst"
- char [procedureextension](#) [] = "prc"
- [SETUPPARAMETERS](#) [setupparameters](#) []

### 4.124.1 Detailed Description

The routines that deal with the setup parameters.

Definition in file [setfile.c](#).

### 4.124.2 Macro Definition Documentation

#### 4.124.2.1 NUMERICALVALUE

```
#define NUMERICALVALUE 0
```

Definition at line 47 of file [setfile.c](#).

#### 4.124.2.2 STRINGVALUE

```
#define STRINGVALUE 1
```

Definition at line 48 of file [setfile.c](#).

#### 4.124.2.3 PATHVALUE

```
#define PATHVALUE 2
```

Definition at line 49 of file [setfile.c](#).

#### 4.124.2.4 ONOFFVALUE

```
#define ONOFFVALUE 3
```

Definition at line 50 of file [setfile.c](#).

#### 4.124.2.5 DEFINEVALUE

```
#define DEFINEVALUE 4
```

Definition at line 51 of file [setfile.c](#).

#### 4.124.2.6 SETBUFSIZE

```
#define SETBUFSIZE 257
```

Definition at line 1140 of file [setfile.c](#).

### 4.124.3 Function Documentation

#### 4.124.3.1 DoSetups()

```
int DoSetups (  
    void )
```

Definition at line 132 of file [setfile.c](#).

#### 4.124.3.2 ProcessOption()

```
int ProcessOption (  
    UBYTE * s1,  
    UBYTE * s2,  
    int filetype )
```

Definition at line 182 of file [setfile.c](#).

#### 4.124.3.3 GetSetupPar()

```
SETUPPARAMETERS * GetSetupPar (  
    UBYTE * s )
```

Definition at line 337 of file [setfile.c](#).

#### 4.124.3.4 RecalcSetups()

```
int RecalcSetups (
    void )
```

Definition at line 357 of file [setfile.c](#).

#### 4.124.3.5 AllocSetups()

```
int AllocSetups (
    void )
```

Definition at line 404 of file [setfile.c](#).

#### 4.124.3.6 WriteSetup()

```
void WriteSetup (
    void )
```

Definition at line 811 of file [setfile.c](#).

#### 4.124.3.7 AllocSort()

```
SORTING * AllocSort (
    LONG inLargeSize,
    LONG inSmallSize,
    LONG inSmallEsize,
    LONG inTermsInSmall,
    int inMaxPatches,
    int inMaxFpatches,
    LONG inIOSize,
    int level )
```

Definition at line 869 of file [setfile.c](#).

#### 4.124.3.8 AllocSortFileName()

```
void AllocSortFileName (
    SORTING * sort )
```

Definition at line 1019 of file [setfile.c](#).

#### 4.124.3.9 AllocFileHandle()

```
FILEHANDLE * AllocFileHandle (
    WORD par,
    char * name )
```

Definition at line 1044 of file [setfile.c](#).

#### 4.124.3.10 DeAllocFileHandle()

```
void DeAllocFileHandle (
    FILEHANDLE * fh )
```

Definition at line 1105 of file [setfile.c](#).

#### 4.124.3.11 MakeSetupAllocs()

```
int MakeSetupAllocs (
    void )
```

Definition at line 1122 of file [setfile.c](#).

#### 4.124.3.12 TryFileSetups()

```
int TryFileSetups (
    void )
```

Definition at line 1142 of file [setfile.c](#).

#### 4.124.3.13 TryEnvironment()

```
int TryEnvironment (
    void )
```

Definition at line 1228 of file [setfile.c](#).

### 4.124.4 Variable Documentation

#### 4.124.4.1 curdirp

```
char curdirp[] = "."
```

Definition at line 39 of file [setfile.c](#).

#### 4.124.4.2 cursortdirp

```
char cursortdirp[] = "."
```

Definition at line 40 of file [setfile.c](#).

#### 4.124.4.3 commentchar

```
char commentchar[] = "*"
```

Definition at line 41 of file [setfile.c](#).

#### 4.124.4.4 dotchar

```
char dotchar[] = "_"
```

Definition at line 42 of file [setfile.c](#).

#### 4.124.4.5 highfirst

```
char highfirst[] = "highfirst"
```

Definition at line 43 of file [setfile.c](#).

#### 4.124.4.6 lowfirst

```
char lowfirst[] = "lowfirst"
```

Definition at line 44 of file [setfile.c](#).

#### 4.124.4.7 procedureextension

```
char procedureextension[] = "prc"
```

Definition at line 45 of file [setfile.c](#).

#### 4.124.4.8 setupparameters

```
SETUPPARAMETERS setupparameters[]
```

Definition at line 53 of file [setfile.c](#).

## 4.125 setfile.c

[Go to the documentation of this file.](#)

```
00001
00005 /* # [ License : */
00006 /*
00007 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00008 * When using this file you are requested to refer to the publication
00009 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010 * This is considered a matter of courtesy as the development was paid
00011 * for by FOM the Dutch physics granting agency and we would like to
00012 * be able to track its scientific use to convince FOM of its value
00013 * for the community.
00014 *
00015 * This file is part of FORM.
00016 *
00017 * FORM is free software: you can redistribute it and/or modify it under the
00018 * terms of the GNU General Public License as published by the Free Software
00019 * Foundation, either version 3 of the License, or (at your option) any later
00020 * version.
00021 *
00022 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00023 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00024 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00025 * details.
00026 *
```

```

00027 *   You should have received a copy of the GNU General Public License along
00028 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
00029 */
00030 /* #] License : */
00031 /*
00032     #[ Includes :
00033
00034     Routines that deal with settings and the setup file
00035 */
00036
00037 #include "form3.h"
00038
00039 char curdirp[] = ".";
00040 char cursortdirp[] = ".";
00041 char commentchar[] = "*";
00042 char dotchar[] = "_";
00043 char highfirst[] = "highfirst";
00044 char lowfirst[] = "lowfirst";
00045 char procedureextension[] = "prc";
00046
00047 #define NUMERICALVALUE 0
00048 #define STRINGVALUE 1
00049 #define PATHVALUE 2
00050 #define ONOFFVALUE 3
00051 #define DEFINEVALUE 4
00052
00053 SETUPPARAMETERS setupparameters[] =
00054 {
00055     {(UBYTE *) "bracketindexsize", NUMERICALVALUE, 0, (LONG) MAXBRACKETBUFFERSIZE}
00056     , {(UBYTE *) "commentchar", STRINGVALUE, 0, (LONG) commentchar}
00057     , {(UBYTE *) "compresssize", NUMERICALVALUE, 0, (LONG) COMPRESSBUFFER}
00058     , {(UBYTE *) "constindex", NUMERICALVALUE, 0, (LONG) NUMFIXED}
00059     , {(UBYTE *) "continuationlines", NUMERICALVALUE, 0, (LONG) FORTRANCONTINUATIONLINES}
00060 #ifdef WITHFLOAT
00061     , {(UBYTE *) "defaultprecision", NUMERICALVALUE, 0, (LONG) DEFAULTPRECISION}
00062 #endif
00063     , {(UBYTE *) "define", DEFINEVALUE, 0, (LONG) 0}
00064     , {(UBYTE *) "dotchar", STRINGVALUE, 0, (LONG) dotchar}
00065     , {(UBYTE *) "factorizationcache", NUMERICALVALUE, 0, (LONG) FBUFFERSIZE}
00066     , {(UBYTE *) "filepatches", NUMERICALVALUE, 0, (LONG) MAXFPATCHES}
00067     , {(UBYTE *) "functionlevels", NUMERICALVALUE, 0, (LONG) MAXFLEVELS}
00068     , {(UBYTE *) "hidesize", NUMERICALVALUE, 0, (LONG) 0}
00069     , {(UBYTE *) "indir", PATHVALUE, 0, (LONG) curdirp}
00070     , {(UBYTE *) "indentspace", NUMERICALVALUE, 0, (LONG) INDENTSPACE}
00071     , {(UBYTE *) "insidefirst", ONOFFVALUE, 0, (LONG) 1}
00072     , {(UBYTE *) "jumpratio", NUMERICALVALUE, 0, (LONG) JUMPRATIO}
00073     , {(UBYTE *) "largepatches", NUMERICALVALUE, 0, (LONG) MAXPATCHES}
00074     , {(UBYTE *) "largesize", NUMERICALVALUE, 0, (LONG) LARGEBUFFER}
00075     , {(UBYTE *) "maxnumbersize", NUMERICALVALUE, 0, (LONG) 0}
00076 /* , {(UBYTE *) "maxnumbersize", NUMERICALVALUE, 0, (LONG) MAXNUMBERSIZE} */
00077     , {(UBYTE *) "maxtermsize", NUMERICALVALUE, 0, (LONG) MAXTER}
00078 #ifdef WITHFLOAT
00079     , {(UBYTE *) "maxweight", NUMERICALVALUE, 0, (LONG) MAXWEIGHT}
00080 #endif
00081     , {(UBYTE *) "maxwildcards", NUMERICALVALUE, 0, (LONG) MAXWILDC}
00082     , {(UBYTE *) "nospacesinnnumbers", ONOFFVALUE, 0, (LONG) 0}
00083     , {(UBYTE *) "numstorecaches", NUMERICALVALUE, 0, (LONG) NUMSTORECACHES}
00084     , {(UBYTE *) "nwritefinalstatistics", ONOFFVALUE, 0, (LONG) 0}
00085     , {(UBYTE *) "nwriteprocessstatistics", ONOFFVALUE, 0, (LONG) 0}
00086     , {(UBYTE *) "nwritestatistics", ONOFFVALUE, 0, (LONG) 0}
00087     , {(UBYTE *) "nwritethreadstatistics", ONOFFVALUE, 0, (LONG) 0}
00088     , {(UBYTE *) "oldfactarg", ONOFFVALUE, 0, (LONG) NEWFACTARG}
00089     , {(UBYTE *) "oldqcd", ONOFFVALUE, 0, (LONG) 1}
00090     , {(UBYTE *) "oldorder", ONOFFVALUE, 0, (LONG) 0}
00091     , {(UBYTE *) "oldparallelstatistics", ONOFFVALUE, 0, (LONG) 0}
00092     , {(UBYTE *) "parentheses", NUMERICALVALUE, 0, (LONG) MAXPARLEVEL}
00093     , {(UBYTE *) "path", PATHVALUE, 0, (LONG) curdirp}
00094     , {(UBYTE *) "procedureextension", STRINGVALUE, 0, (LONG) procedureextension}
00095     , {(UBYTE *) "processbucketsize", NUMERICALVALUE, 0, (LONG) DEFAULTPROCESSBUCKETSIZE}
00096     , {(UBYTE *) "resettimeonclear", ONOFFVALUE, 0, (LONG) 1}
00097     , {(UBYTE *) "scratchsize", NUMERICALVALUE, 0, (LONG) SCRATCHSIZE}
00098     , {(UBYTE *) "shmwinsize", NUMERICALVALUE, 0, (LONG) SHMWINSIZE}
00099     , {(UBYTE *) "sizestorecache", NUMERICALVALUE, 0, (LONG) SIZESTORECACHE}
00100     , {(UBYTE *) "smallextension", NUMERICALVALUE, 0, (LONG) SMALLOVERFLOW}
00101     , {(UBYTE *) "smallsize", NUMERICALVALUE, 0, (LONG) SMALLBUFFER}
00102     , {(UBYTE *) "sortiosize", NUMERICALVALUE, 0, (LONG) SORTIOSIZE}
00103     , {(UBYTE *) "sorttype", STRINGVALUE, 0, (LONG) lowfirst}
00104     , {(UBYTE *) "spectatorsize", NUMERICALVALUE, 0, (LONG) SPECTATORSIZE}
00105     , {(UBYTE *) "subfilepatches", NUMERICALVALUE, 0, (LONG) SMAXFPATCHES}
00106     , {(UBYTE *) "sublargepatches", NUMERICALVALUE, 0, (LONG) SMAXPATCHES}
00107     , {(UBYTE *) "sublargesize", NUMERICALVALUE, 0, (LONG) SLARGEBUFFER}
00108     , {(UBYTE *) "subsmallextension", NUMERICALVALUE, 0, (LONG) SSMAILOVERFLOW}
00109     , {(UBYTE *) "subsmallsize", NUMERICALVALUE, 0, (LONG) SSMAILLBUFFER}
00110     , {(UBYTE *) "subsortiosize", NUMERICALVALUE, 0, (LONG) SSORTIOSIZE}
00111     , {(UBYTE *) "subtermsinsmall", NUMERICALVALUE, 0, (LONG) STERMSSMALL}
00112     , {(UBYTE *) "tempdir", STRINGVALUE, 0, (LONG) curdirp}
00113     , {(UBYTE *) "tempsortdir", STRINGVALUE, 0, (LONG) cursortdirp}

```



```

00114     ,{(UBYTE *)"termsinsmall",          NUMERICALVALUE, 0, (LONG)TERMSSMALL}
00115     ,{(UBYTE *)"threadbucketsize",      NUMERICALVALUE, 0, (LONG)DEFAULTTHREADBUCKETSIZE}
00116     ,{(UBYTE *)"threadloadbalancing",    ONOFFVALUE, 0, (LONG)DEFAULTTHREADLOADBALANCING}
00117     ,{(UBYTE *)"threads",                NUMERICALVALUE, 0, (LONG)DEFAULTTHREADS}
00118     ,{(UBYTE *)"threadscratchoutsize",   NUMERICALVALUE, 0, (LONG)THREADSCRATCHOUTSIZE}
00119     ,{(UBYTE *)"threadscratchsize",      NUMERICALVALUE, 0, (LONG)THREADSCRATCHSIZE}
00120     ,{(UBYTE *)"threadsortfilesynch",    ONOFFVALUE, 0, (LONG)0}
00121     ,{(UBYTE *)"totalsize",              ONOFFVALUE, 0, (LONG)2}
00122     ,{(UBYTE *)"workspace",              NUMERICALVALUE, 0, (LONG)WORKBUFFER}
00123     ,{(UBYTE *)"wtimestats",             ONOFFVALUE, 0, (LONG)2}
00124 };
00125
00126 /*
00127     #] Includes :
00128     #[ Setups :
00129         #[ DoSetups :
00130 */
00131
00132 int DoSetups(void)
00133 {
00134     UBYTE *setbuffer, *s, *t, *u /*, c */;
00135     int errors = 0;
00136     setbuffer = LoadInputFile((UBYTE *)setupfilename,SETUPFILE);
00137     if ( setbuffer ) {
00138 /*
00139         The contents of the file are now in setbuffer.
00140         Each line is commentary or a single command.
00141         The buffer is terminated with a zero.
00142 */
00143         s = setbuffer;
00144         while ( *s ) {
00145             if ( *s == ' ' || *s == '\t' || *s == '*' || *s == '#' || *s == '\n' ) {
00146                 while ( *s && *s != '\n' ) s++;
00147             }
00148             else if ( tolower(*s) < 'a' || tolower(*s) > 'z' ) {
00149                 t = s;
00150                 while ( *s && *s != '\n' ) s++;
00151 /*
00152                 c = *s; *s = 0;
00153                 Error1("Setup file: Illegal statement: ",t);
00154                 errors++; *s = c;
00155 */
00156             }
00157             else {
00158                 t = s; /* name of the option */
00159                 while ( tolower(*s) >= 'a' && tolower(*s) <= 'z' ) s++;
00160                 *s++ = 0;
00161                 while ( *s == ' ' || *s == '\t' ) s++;
00162                 u = s; /* 'value' of the option */
00163                 while ( *s && *s != '\n' && *s != '\r' ) s++;
00164                 if ( *s ) *s++ = 0;
00165                 errors += ProcessOption(t,u,0);
00166             }
00167             while ( *s == '\n' || *s == '\r' ) s++;
00168         }
00169         M_free(setbuffer,"setup file buffer");
00170     }
00171     if ( errors ) return(1);
00172     else return(0);
00173 }
00174
00175 /*
00176     #] DoSetups :
00177     #[ ProcessOption :
00178 */
00179
00180 static char *proopl[3] = { "Setup file", "Setups in .frm file", "Setup in environment" };
00181
00182 int ProcessOption(UBYTE *s1, UBYTE *s2, int filetype)
00183 {
00184     SETUPPARAMETERS *sp;
00185     int n, giveback = 0, error = 0;
00186     UBYTE *s, *t, *s2ret;
00187     LONG x;
00188     sp = GetSetupPar(s1);
00189     if ( sp ) {
00190 /*
00191         We check now whether there are ` variables to be looked up in the
00192         environment. This is new (30-may-2008). This is only allowed in s2.
00193 */
00194 restart:;
00195         {
00196             UBYTE *s3,*s4,*s5,*s6, c, *start;
00197             int n1,n2,n3;
00198             s = s2;
00199             while ( *s ) {
00200                 if ( *s == '\\ ' ) s += 2;

```

```

00201         else if ( *s == ' ' ) {
00202             start = s; s++;
00203             while ( *s && *s != '\n' ) {
00204                 if ( *s == '\\ ' ) s++;
00205                 s++;
00206             }
00207             if ( *s == 0 ) {
00208                 MesPrint("%s: Illegal use of ` character for parameter %s"
00209                     ,proopl[filetype],s1);
00210                 return(1);
00211             }
00212             c = *s; *s = 0;
00213             s3 = (UBYTE *)getenv((char *) (start+1));
00214             if ( s3 == 0 ) {
00215                 MesPrint("%s: Cannot find environment variable %s for parameter %s"
00216                     ,proopl[filetype],start+1,s1);
00217                 return(1);
00218             }
00219             }
00220             *s = c; s++;
00221             n1 = start - s2; s4 = s3; n2 = 0;
00222             while ( *s4 ) {
00223                 if ( *s4 == '\\ ' ) { s4++; n2++; }
00224                 s4++; n2++;
00225             }
00226             s4 = s; n3 = 0;
00227             while ( *s4 ) {
00228                 if ( *s4 == '\\ ' ) { s4++; n3++; }
00229                 s4++; n3++;
00230             }
00231             s4 = (UBYTE *)Malloc1((n1+n2+n3+1)*sizeof(UBYTE),"environment in setup");
00232             s5 = s2; s6 = s4;
00233             while ( n1-- > 0 ) *s6++ = *s5++;
00234             s5 = s3;
00235             while ( n2-- > 0 ) *s6++ = *s5++;
00236             s5 = s;
00237             while ( n3-- > 0 ) *s6++ = *s5++;
00238             *s6 = 0;
00239             if ( giveback ) M_free(s2,"environment in setup");
00240             s2 = s4;
00241             giveback = 1;
00242             goto restart;
00243         }
00244         else s++;
00245     }
00246 }
00247 n = sp->type;
00248 s2ret = s2;
00249 switch ( n ) {
00250     case NUMERICALVALUE:
00251         ParseNumber(x,s2);
00252         if ( *s2 == 'K' ) { x = x * 1000; s2++; }
00253         else if ( *s2 == 'M' ) { x = x * 1000000; s2++; }
00254         else if ( *s2 == 'G' ) { x = x * 1000000000; s2++; }
00255         else if ( *s2 == 'T' ) { x = x * 1000000000000; s2++; }
00256         if ( *s2 && *s2 != ' ' && *s2 != '\t' ) {
00257             MesPrint("%s: Numerical value expected for parameter %s"
00258                 ,proopl[filetype],s1);
00259             error = 1; break;
00260         }
00261         sp->value = x;
00262         sp->flags = USEDFLAG;
00263         break;
00264     case STRINGVALUE:
00265         if ( StrICmp(s1,(UBYTE *)"tempsortdir") == 0 ) AM.havesortdir = 1;
00266         s = s2; t = s2;
00267         while ( *s ) {
00268             if ( *s == ' ' || *s == '\t' ) break;
00269             if ( *s == '\\ ' ) s++;
00270             *t++ = *s++;
00271         }
00272         *t = 0;
00273         if ( sp->flags == USEDFLAG && sp->value != 0 )
00274             M_free((void *) (sp->value),"Process option");
00275         sp->value = (LONG)strDupl(s2,"Process option");
00276         sp->flags = USEDFLAG;
00277         break;
00278     case PATHVALUE:
00279         if ( StrICmp(s1,(UBYTE *)"incdir") == 0 ) {
00280             AM.IncDir = 0;
00281         }
00282         else if ( StrICmp(s1,(UBYTE *)"path") == 0 ) {
00283             if ( AM.Path ) M_free(AM.Path,"path");
00284             AM.Path = 0;
00285         }
00286         else {
00287             MesPrint("Setups: %s not yet implemented",s1);

```

```

00288         error = 1;
00289         break;
00290     }
00291     if ( sp->flags == USEDFLAG && sp->value != 0 )
00292         M_free((void *) (sp->value), "Process option");
00293     sp->value = (LONG)strDup1(s2, "Process option");
00294     sp->flags = USEDFLAG;
00295     break;
00296 case ONOFFVALUE:
00297     if ( tolower(*s2) == 'o' && tolower(s2[1]) == 'n'
00298         && ( s2[2] == 0 || s2[2] == ' ' || s2[2] == '\t' ) )
00299         sp->value = 1;
00300     else if ( tolower(*s2) == 'o' && tolower(s2[1]) == 'f'
00301         && tolower(s2[2]) == 'f'
00302         && ( s2[3] == 0 || s2[3] == ' ' || s2[3] == '\t' ) )
00303         sp->value = 0;
00304     else {
00305         MesPrint("%s: Unrecognized option for parameter %s: %s"
00306             , proopl[filetype], s1, s2);
00307         error = 1; break;
00308     }
00309     sp->flags = USEDFLAG;
00310     break;
00311 case DEFINEVALUE:
00312 /*
00313     if ( sp->value ) M_free((UBYTE *) (sp->value), "Process option");
00314     sp->value = (LONG)strDup1(s2, "Process option");
00315 */
00316     if ( TheDefine(s2, 2) ) error = 1;
00317     break;
00318 default:
00319     Error1("Error in setupparameter table for:", s1);
00320     error = 1;
00321     break;
00322 }
00323 }
00324 else {
00325     MesPrint("%s: Keyword not recognized: %s", proopl[filetype], s1);
00326     error = 1;
00327 }
00328 if ( giveback ) M_free(s2ret, "environment in setup");
00329 return(error);
00330 }
00331
00332 /*
00333     #] ProcessOption :
00334     #[ GetSetupPar :
00335 */
00336
00337 SETUPPARAMETERS *GetSetupPar(UBYTE *s)
00338 {
00339     int hi, med, lo, i;
00340     lo = 0;
00341     hi = sizeof(setupparameters)/sizeof(SETUPPARAMETERS);
00342     do {
00343         med = ( hi + lo ) / 2;
00344         i = StrICmp(s, (UBYTE *) setupparameters[med].parameter);
00345         if ( i == 0 ) return(setupparameters+med);
00346         if ( i < 0 ) hi = med-1;
00347         else lo = med+1;
00348     } while ( hi >= lo );
00349     return(0);
00350 }
00351
00352 /*
00353     #] GetSetupPar :
00354     #[ RecalcSetups :
00355 */
00356
00357 int RecalcSetups(void)
00358 {
00359     SETUPPARAMETERS *sp, *spl;
00360
00361     spl = GetSetupPar((UBYTE *) "threads");
00362     if ( AM.totalnumberofthreads > 1 ) spl->value = AM.totalnumberofthreads - 1;
00363     else spl->value = 0;
00364 /*
00365     if ( spl->value > 0 ) AM.totalnumberofthreads = spl->value+1;
00366     if ( AM.totalnumberofthreads == 0 ) AM.totalnumberofthreads = 1;
00367 */
00368     sp = GetSetupPar((UBYTE *) "filepatches");
00369     if ( sp->value < AM.totalnumberofthreads-1 )
00370         sp->value = AM.totalnumberofthreads - 1;
00371
00372     sp = GetSetupPar((UBYTE *) "smallsize");
00373     spl = GetSetupPar((UBYTE *) "smallestextension");
00374     if ( 6*spl->value < 7*sp->value ) spl->value = (7*sp->value)/6;

```

```

00375     sp = GetSetupPar((UBYTE *)"termsinsmall");
00376     sp->value = ( sp->value + 15 ) & (-16L);
00377 #ifdef WITHPTHREADS
00378 {
00379     SETUPPARAMETERS *sp2;
00380     LONG totalsize, minimumsize;
00381     sp = GetSetupPar((UBYTE *)"largesize");
00382     totalsize = spl->value+sp->value;
00383     sp2 = GetSetupPar((UBYTE *)"maxtermsize");
00384     AM.MaxTer = sp2->value*sizeof(WORD);
00385     if ( AM.MaxTer < 200*(LONG)(sizeof(WORD)) ) AM.MaxTer = 200*(LONG)(sizeof(WORD));
00386     if ( AM.MaxTer > MAXPOSITIVE - 200*(LONG)(sizeof(WORD)) ) AM.MaxTer = MAXPOSITIVE -
200*(LONG)(sizeof(WORD));
00387     AM.MaxTer /= sizeof(WORD);
00388     AM.MaxTer *= sizeof(WORD);
00389     minimumsize = (AM.totalnumberofthreads-1)*(AM.MaxTer+
00390     NUMBEROFBLOCKSINSERT*MINIMUMNUMBEROFTERMS*AM.MaxTer);
00391     if ( totalsize < minimumsize ) {
00392         sp->value = minimumsize - spl->value;
00393     }
00394 }
00395 #endif
00396     return(0);
00397 }
00398
00399 /*
00400     #] RecalcSetups :
00401     #[ AllocSetups :
00402 */
00403
00404 int AllocSetups(void)
00405 {
00406     SETUPPARAMETERS *sp;
00407     LONG LargeSize, SmallSize, SmallEsize, TermsInSmall, IOSize;
00408     int MaxPatches, MaxFpatches, error = 0, i, size;
00409     UBYTE *s;
00410 #ifndef WITHPTHREADS
00411     int j;
00412 #endif
00413     sp = GetSetupPar((UBYTE *)"threads");
00414     if ( sp->value > 0 ) AM.totalnumberofthreads = sp->value+1;
00415
00416     AM.OutBuffer = (UBYTE *)Malloca1(AM.OutBufSize+1,"OutputBuffer");
00417     AP.PreAssignStack = (LONG *)Malloca1(AP.MaxPreAssignLevel*sizeof(LONG *),"PreAssignStack");
00418     for ( i = 0; i < AP.MaxPreAssignLevel; i++ ) AP.PreAssignStack[i] = 0;
00419     AC.iBuffer = (UBYTE *)Malloca1(AC.iBufferSize+1,"statement buffer");
00420     AC.iStop = AC.iBuffer + AC.iBufferSize-2;
00421     AP.preStart = (UBYTE *)Malloca1(AP.pSize,"instruction buffer");
00422     AP.preStop = AP.preStart + AP.pSize - 3;
00423     /* AP.PreIfStack is already allocated in StartPrepro(), but to be sure we
00424        "if" the freeing */
00425     if ( AP.PreIfStack ) M_free(AP.PreIfStack,"PreIfStack");
00426     AP.PreIfStack = (int *)Malloca1(AP.MaxPreIfLevel*sizeof(int),
00427        "Preprocessor if stack");
00428     AP.PreIfStack[0] = EXECUTINGIF;
00429     sp = GetSetupPar((UBYTE *)"insidefirst");
00430     AM.ginsidefirst = AC.minsidefirst = AC.insidefirst = sp->value;
00431 /*
00432     We need to consider eliminating this variable
00433 */
00434     sp = GetSetupPar((UBYTE *)"maxtermsize");
00435     AM.MaxTer = sp->value*sizeof(WORD);
00436     if ( AM.MaxTer < 200*(LONG)(sizeof(WORD)) ) AM.MaxTer = 200*(LONG)(sizeof(WORD));
00437     if ( AM.MaxTer > MAXPOSITIVE - 200*(LONG)(sizeof(WORD)) ) AM.MaxTer = MAXPOSITIVE -
200*(LONG)(sizeof(WORD));
00438     AM.MaxTer /= (LONG)sizeof(WORD);
00439     AM.MaxTer *= (LONG)sizeof(WORD);
00440 /*
00441     Allocate workspace.
00442 */
00443     sp = GetSetupPar((UBYTE *)"workspace");
00444     AM.WorkSize = sp->value;
00445 #ifdef WITHPTHREADS
00446 #else
00447     AT.WorkSpace = (WORD *)Malloca1(AM.WorkSize*sizeof(WORD), (char *) (sp->parameter));
00448     AT.WorkTop = AT.WorkSpace + AM.WorkSize;
00449     AT.WorkPointer = AT.WorkSpace;
00450 #endif
00451 /*
00452     Fixed indices
00453 */
00454     sp = GetSetupPar((UBYTE *)"constindex");
00455     if ( ( sp->value+100+5*WILDOFFSET ) > MAXPOSITIVE ) {
00456         MesPrint("Setting of %s in setupfile too large","constindex");
00457         AM.OffsetIndex = MAXPOSITIVE - 5*WILDOFFSET - 100;
00458         MesPrint("value corrected to maximum allowed: %d",AM.OffsetIndex);
00459     }

```

```

00460     else AM.OffsetIndex = sp->value + 1;
00461     AC.FixIndices = (WORD *)Malloc1((AM.OffsetIndex)*sizeof(WORD), (char *) (sp->parameter));
00462     AM.WilInd = AM.OffsetIndex + WILDOFFSET;
00463     AM.DumInd = AM.OffsetIndex + 2*WILDOFFSET;
00464     AM.IndDum = AM.DumInd + WILDOFFSET;
00465 #ifndef WITHPTHREADS
00466     AR.CurDum = AN.IndDum = AM.IndDum;
00467 #endif
00468     AM.mTraceDum = AM.IndDum + 2*WILDOFFSET;
00469
00470     sp = GetSetupPar((UBYTE *) "parentheses");
00471     AM.MaxParLevel = sp->value+1;
00472     AC.tokenarglevel = (WORD *)Malloc1((sp->value+1)*sizeof(WORD), (char *) (sp->parameter));
00473 /*
00474     Space during calculations
00475 */
00476     sp = GetSetupPar((UBYTE *) "maxnumbersize");
00477 /*
00478     size = ( sp->value + 11 ) & (-4);
00479     AM.MaxTal = size - 2;
00480     if ( AM.MaxTal > (AM.MaxTer/sizeof(WORD)-2)/2 )
00481         AM.MaxTal = (AM.MaxTer/sizeof(WORD)-2)/2;
00482     if ( AM.MaxTal < (AM.MaxTer/sizeof(WORD)-2)/4 )
00483         AM.MaxTal = (AM.MaxTer/sizeof(WORD)-2)/4;
00484 */
00485 /*
00486     There is too much confusion about MaxTal cq maxnumbersize.
00487     It seems better to fix it at its maximum value. This way we only worry
00488     about maxtermsize. This can be understood better by the 'innocent' user.
00489 */
00490     if ( sp->value == 0 ) {
00491         AM.MaxTal = (AM.MaxTer/sizeof(WORD)-2)/2;
00492     }
00493     else {
00494         size = ( sp->value + 11 ) & (-4);
00495         AM.MaxTal = size - 2;
00496         if ( (size_t)AM.MaxTal > (size_t)((AM.MaxTer/sizeof(WORD)-2)/2) )
00497             AM.MaxTal = (AM.MaxTer/sizeof(WORD)-2)/2;
00498     }
00499     AM.MaxTal &= -sizeof(WORD)*2;
00500
00501     sp->value = AM.MaxTal;
00502     AC.cmod = (UWORD *)Malloc1(AM.MaxTal*4*sizeof(UWORD), (char *) (sp->parameter));
00503     AM.gcmod = AC.cmod + AM.MaxTal;
00504     AC.powmod = AM.gcmod + AM.MaxTal;
00505     AM.gpowmod = AC.powmod + AM.MaxTal;
00506 /*
00507     The IO buffers for the input and output expressions.
00508     Fscr[2] will be assigned in a later stage for hiding expressions from
00509     the regular action. That will make the program faster.
00510 */
00511     sp = GetSetupPar((UBYTE *) "scratchsize");
00512     AM.ScratSize = sp->value/sizeof(WORD);
00513     if ( AM.ScratSize < 4*AM.MaxTer ) AM.ScratSize = 4*AM.MaxTer;
00514     AM.HideSize = AM.ScratSize;
00515     sp = GetSetupPar((UBYTE *) "hidesize");
00516     if ( sp->value > 0 ) {
00517         AM.HideSize = sp->value/sizeof(WORD);
00518         if ( AM.HideSize < 4*AM.MaxTer ) AM.HideSize = 4*AM.MaxTer;
00519     }
00520     sp = GetSetupPar((UBYTE *) "factorizationcache");
00521     AM.fbufferSize = sp->value;
00522 #ifdef WITHPTHREADS
00523     sp = GetSetupPar((UBYTE *) "threadscratchsize");
00524     AM.ThreadSkratSize = sp->value/sizeof(WORD);
00525     sp = GetSetupPar((UBYTE *) "threadscratchoutsize");
00526     AM.ThreadSkratOutSize = sp->value/sizeof(WORD);
00527 #endif
00528 #ifndef WITHPTHREADS
00529     for ( j = 0; j < 2; j++ ) {
00530         WORD *ScratchBuf;
00531         ScratchBuf = (WORD *)Malloc1(AM.ScratSize*sizeof(WORD), "scratchsize");
00532         AR.Fscr[j].POsize = AM.ScratSize * sizeof(WORD);
00533         AR.Fscr[j].POfull = AR.Fscr[j].POfill = AR.Fscr[j].PObuffer = ScratchBuf;
00534         AR.Fscr[j].POstop = AR.Fscr[j].PObuffer + AM.ScratSize;
00535         PUTZERO(AR.Fscr[j].POposition);
00536     }
00537     AR.Fscr[2].PObuffer = 0;
00538 #endif
00539     sp = GetSetupPar((UBYTE *) "threadbucketSize");
00540     AC.ThreadBucketSize = AM.gThreadBucketSize = AM.ggThreadBucketSize = sp->value;
00541     sp = GetSetupPar((UBYTE *) "threadloadbalancing");
00542     AC.ThreadBalancing = AM.gThreadBalancing = AM.ggThreadBalancing = sp->value;
00543     sp = GetSetupPar((UBYTE *) "threadsortfilesynch");
00544     AC.ThreadSortFileSynch = AM.gThreadSortFileSynch = AM.ggThreadSortFileSynch = sp->value;
00545 /*
00546     The size for shared memory window for onside MPI2 communications

```

```

00547 */
00548     sp = GetSetupPar((UBYTE *)"shmwinSize");
00549     AM.shmWinSize = sp->value/sizeof(WORD);
00550     if ( AM.shmWinSize < 4*AM.MaxTer ) AM.shmWinSize = 4*AM.MaxTer;
00551 /*
00552     The sort buffer
00553 */
00554     sp = GetSetupPar((UBYTE *)"smallSize");
00555     SmallSize = sp->value;
00556     sp = GetSetupPar((UBYTE *)"smallextension");
00557     SmallEsize = sp->value;
00558     sp = GetSetupPar((UBYTE *)"largesize");
00559     LargeSize = sp->value;
00560     sp = GetSetupPar((UBYTE *)"termsinsmall");
00561     TermsInSmall = sp->value;
00562     sp = GetSetupPar((UBYTE *)"largepatches");
00563     MaxPatches = sp->value;
00564     sp = GetSetupPar((UBYTE *)"filepatches");
00565     MaxFpatches = sp->value;
00566     sp = GetSetupPar((UBYTE *)"sortiosize");
00567     IOsize = sp->value;
00568     if ( IOsize < AM.MaxTer ) { IOsize = AM.MaxTer; sp->value = IOsize; }
00569 #ifndef WITHPTHREADS
00570 #ifdef WITHZLIB
00571     for ( j = 0; j < 2; j++ ) { AR.Fscr[j].ziosize = IOsize; }
00572 #endif
00573 #endif
00574     AM.S0 = 0;
00575     AM.S0 = AllocSort(LargeSize,SmallSize,SmallEsize,TermsInSmall
00576                     ,MaxPatches,MaxFpatches,IOsize,0);
00577     /* AM.S0->file.ziosize was already set to a (larger) value by AllocSort, here it is re-set. */
00578 #ifdef WITHZLIB
00579     AM.S0->file.ziosize = IOsize;
00580 #ifndef WITHPTHREADS
00581     AR.FoStage4[0].ziosize = IOsize;
00582     AR.FoStage4[1].ziosize = IOsize;
00583     AT.S0 = AM.S0;
00584 #endif
00585 #else
00586 #ifndef WITHPTHREADS
00587     AT.S0 = AM.S0;
00588 #endif
00589 #endif
00590 #ifndef WITHPTHREADS
00591     AR.FoStage4[0].POsize = ((IOsize+sizeof(WORD)-1)/sizeof(WORD))*sizeof(WORD);
00592     AR.FoStage4[1].POsize = ((IOsize+sizeof(WORD)-1)/sizeof(WORD))*sizeof(WORD);
00593 #endif
00594     sp = GetSetupPar((UBYTE *)"subsmallSize");
00595     AM.SSmallSize = sp->value;
00596     sp = GetSetupPar((UBYTE *)"subsmallextension");
00597     AM.SSmallEsize = sp->value;
00598     sp = GetSetupPar((UBYTE *)"sublargesize");
00599     AM.SLargeSize = sp->value;
00600     sp = GetSetupPar((UBYTE *)"subtermsinsmall");
00601     AM.STermsInSmall = sp->value;
00602     sp = GetSetupPar((UBYTE *)"sublargepatches");
00603     AM.SMaxPatches = sp->value;
00604     sp = GetSetupPar((UBYTE *)"subfilepatches");
00605     AM.SMaxFpatches = sp->value;
00606     sp = GetSetupPar((UBYTE *)"subsortiosize");
00607     AM.SIOsize = sp->value;
00608     /* As for IOsize, make sure SIOsize is at least as large as AM.MaxTer. */
00609     if ( AM.SIOsize < AM.MaxTer ) { AM.SIOsize = AM.MaxTer; sp->value = AM.SIOsize; }
00610     sp = GetSetupPar((UBYTE *)"spectatorsize");
00611     AM.SpectatorSize = sp->value;
00612 /*
00613     The next code is just for the moment (26-jan-1997) because we have
00614     the new parts combined with the old. Once the old parts are gone
00615     from the program, we can eliminate this code too.
00616 */
00617     sp = GetSetupPar((UBYTE *)"functionlevels");
00618     AM.maxFlevels = sp->value + 1;
00619 #ifdef WITHPTHREADS
00620 #else
00621     AT.Nest = (NESTING)Malloc1((LONG)sizeof(struct NeStInG)*AM.maxFlevels,"functionlevels");
00622     AT.NestStop = AT.Nest + AM.maxFlevels;
00623     AT.NestPoin = AT.Nest;
00624 #endif
00625
00626     sp = GetSetupPar((UBYTE *)"maxwildcards");
00627     AM.MaxWildcards = sp->value;
00628 #ifdef WITHPTHREADS
00629 #else
00630     AT.WildMask = (WORD *)Malloc1((LONG)AM.MaxWildcards*sizeof(WORD),"maxwildcards");
00631 #endif
00632
00633     sp = GetSetupPar((UBYTE *)"compresssize");

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```

00634     if ( sp->value < 2*AM.MaxTer ) sp->value = 2*AM.MaxTer;
00635     AM.CompressSize = sp->value;
00636 #ifndef WITHPTHREADS
00637     AR.CompressBuffer = (WORD *)Malloc1( (AM.CompressSize+10)*sizeof(WORD), "compresssize");
00638     AR.CompressPointer = AR.CompressBuffer;
00639     AR.ComprTop = AR.CompressBuffer + AM.CompressSize;
00640 #endif
00641     sp = GetSetupPar( (UBYTE *) "bracketindexsize");
00642     if ( sp->value < 20*AM.MaxTer ) sp->value = 20*AM.MaxTer;
00643     AM.MaxBracketBufferSize = sp->value/sizeof(WORD);
00644
00645     sp = GetSetupPar( (UBYTE *) "dotchar");
00646     AO.FortDotChar = ((UBYTE *) (sp->value))[0];
00647     sp = GetSetupPar( (UBYTE *) "commentchar");
00648     AP.cComChar = AP.ComChar = ((UBYTE *) (sp->value))[0];
00649     sp = GetSetupPar( (UBYTE *) "procedureextension");
00650 /*
00651     Check validity first.
00652 */
00653     s = (UBYTE *) (sp->value);
00654     if ( FG.cTable[*s] != 0 ) {
00655         MesPrint(" Illegal string for procedure extension %s", (UBYTE *) sp->value);
00656         error = -2;
00657     }
00658     else {
00659         s++;
00660         while ( *s ) {
00661             if ( *s == ' ' || *s == '\t' || *s == '\n' ) {
00662                 MesPrint(" Illegal string for procedure extension %s", (UBYTE *) sp->value);
00663                 error = -2;
00664                 break;
00665             }
00666             s++;
00667         }
00668     }
00669     AP.cprocedureExtension = strDup1( (UBYTE *) (sp->value), "procedureExtension");
00670     AP.procedureExtension = strDup1( AP.cprocedureExtension, "procedureExtension");
00671
00672     sp = GetSetupPar( (UBYTE *) "totalsize");
00673     if ( sp->value != 2 ) AM.PrintTotalSize = sp->value;
00674
00675     sp = GetSetupPar( (UBYTE *) "continuationlines");
00676     AM.FortranCont = sp->value;
00677     sp = GetSetupPar( (UBYTE *) "oldorder");
00678     AM.OldOrderFlag = sp->value;
00679     sp = GetSetupPar( (UBYTE *) "resettimeonclear");
00680     AM.resetTimeOnClear = sp->value;
00681     sp = GetSetupPar( (UBYTE *) "nospacesinnumbers");
00682     AO.NoSpacesInNumbers = AM.gNoSpacesInNumbers = AM.ggNoSpacesInNumbers = sp->value;
00683     sp = GetSetupPar( (UBYTE *) "indentspace");
00684     AO.IndentSpace = AM.gIndentSpace = AM.ggIndentSpace = sp->value;
00685     sp = GetSetupPar( (UBYTE *) "jumpratio");
00686     AM.jumpratio = sp->value;
00687     sp = GetSetupPar( (UBYTE *) "nwritestatistics");
00688     AC.StatsFlag = AM.gStatsFlag = AM.ggStatsFlag = 1-sp->value;
00689     sp = GetSetupPar( (UBYTE *) "nwritefinalstatistics");
00690     AC.FinalStats = AM.gFinalStats = AM.ggFinalStats = 1-sp->value;
00691     sp = GetSetupPar( (UBYTE *) "nwritethreadstatistics");
00692     AC.ThreadStats = AM.gThreadStats = AM.ggThreadStats = 1-sp->value;
00693     sp = GetSetupPar( (UBYTE *) "nwriteprocessstatistics");
00694     AC.ProcessStats = AM.gProcessStats = AM.ggProcessStats = 1-sp->value;
00695     sp = GetSetupPar( (UBYTE *) "oldparallelstatistics");
00696     AC.OldParallelStats = AM.gOldParallelStats = AM.ggOldParallelStats = sp->value;
00697     sp = GetSetupPar( (UBYTE *) "oldfactarg");
00698     AC.OldFactArgFlag = AM.gOldFactArgFlag = AM.ggOldFactArgFlag = sp->value;
00699     sp = GetSetupPar( (UBYTE *) "oldgcd");
00700     AC.OldGCDflag = AM.gOldGCDflag = AM.ggOldGCDflag = sp->value;
00701     sp = GetSetupPar( (UBYTE *) "wtimestats");
00702     if ( sp->value == 2 ) sp->value = AM.ggWTimeStatsFlag;
00703     AC.WTimeStatsFlag = AM.gWTimeStatsFlag = AM.ggWTimeStatsFlag = sp->value;
00704 #ifdef WITHFLOAT
00705     sp = GetSetupPar( (UBYTE *) "maxweight");
00706     AC.MaxWeight = AM.gMaxWeight = AM.ggMaxWeight = sp->value;
00707     sp = GetSetupPar( (UBYTE *) "defaultprecision");
00708     AC.DefaultPrecision = AM.gDefaultPrecision = AM.ggDefaultPrecision = sp->value;
00709 #endif
00710     sp = GetSetupPar( (UBYTE *) "sorttype");
00711     if ( StrICmp( (UBYTE *) "lowfirst", (UBYTE *) sp->value) == 0 ) {
00712         AC.lSortType = SORTLOWFIRST;
00713     }
00714     else if ( StrICmp( (UBYTE *) "highfirst", (UBYTE *) sp->value) == 0 ) {
00715         AC.lSortType = SORTHIGHFIRST;
00716     }
00717     else {
00718         MesPrint(" Illegal SortType specification: %s", (UBYTE *) sp->value);
00719         error = -2;
00720     }

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00721
00722     sp = GetSetupPar((UBYTE *)"processbucketsize");
00723     AM.hProcessBucketSize = AM.gProcessBucketSize =
00724     AC.ProcessBucketSize = AC.mProcessBucketSize = sp->value;
00725     /*
00726     The store caches (code installed 15-aug-2006 JV)
00727     */
00728     sp = GetSetupPar((UBYTE *)"numstorecaches");
00729     AM.NumStoreCaches = sp->value;
00730     sp = GetSetupPar((UBYTE *)"sizestorecache");
00731     AM.SizeStoreCache = sp->value;
00732     /* Make sure this is a multiple of sizeof(WORD). */
00733     AM.SizeStoreCache = ((AM.SizeStoreCache+sizeof(WORD)-1)/sizeof(WORD))*sizeof(WORD);
00734     #ifndef WITHPTHREADS
00735     /*
00736     Install the store caches (15-aug-2006 JV)
00737     Note that in the case of PTHREADS this is done in InitializeOneThread
00738     */
00739     AT.StoreCache = AT.StoreCacheAlloc = 0;
00740     if ( AM.NumStoreCaches > 0 ) {
00741         STORECACHE sa, sb;
00742         size = sizeof(struct StOrEcAcHe)+AM.SizeStoreCache;
00743         size = ((size-1)/sizeof(size_t)+1)*sizeof(size_t);
00744         AT.StoreCacheAlloc = (STORECACHE)Malloc1(size*AM.NumStoreCaches,"StoreCaches");
00745         AT.StoreCache = AT.StoreCacheAlloc;
00746         sa = AT.StoreCache;
00747         for ( j = 0; j < AM.NumStoreCaches; j++ ) {
00748             sb = (STORECACHE)(void *)((UBYTE *)sa+size);
00749             if ( j == AM.NumStoreCaches-1 ) {
00750                 sa->next = 0;
00751             }
00752             else {
00753                 sa->next = sb;
00754             }
00755             SETBASEPOSITION(sa->position,-1);
00756             SETBASEPOSITION(sa->toppos,-1);
00757             sa = sb;
00758         }
00759     }
00760     #endif
00761     /*
00762     And now some order sensitive things
00763     */
00764     if ( AM.Path == 0 ) {
00765         sp = GetSetupPar((UBYTE *)"path");
00766         AM.Path = strDup1((UBYTE *) (sp->value),"path");
00767     }
00768     if ( AM.IncDir == 0 ) {
00769         sp = GetSetupPar((UBYTE *)"incdir");
00770         AM.IncDir = strDup1((UBYTE *) (sp->value),"incdir");
00771     }
00772     /*
00773     if ( AM.TempDir == 0 ) {
00774         sp = GetSetupPar((UBYTE *)"tempdir");
00775         AM.TempDir = strDup1((UBYTE *) (sp->value),"tempdir");
00776     }
00777     */
00778     /*
00779     As part of the setup, we possibly need to work around a flint multi-threading
00780     bug which was fixed in version 3.1.0. This function should be called by the
00781     master only, before the worker threads start processing terms.
00782     */
00783     #ifndef WITHFLINT
00784     #if __FLINT_RELEASE < 30100
00785         flint_startup_init();
00786     #endif
00787     #endif
00788     return(error);
00789 }
00790
00791 /*
00792 #] AllocSetups :
00793 #[ WriteSetup :
00794
00795 The routine writes the values of the setup parameters.
00796 We should do this better. (JV, 21-may-2008)
00797 The way it should be done is:
00798     a: write the raw values.
00799     b: give readjusted values.
00800     c: give derived values.
00801 Because this is a difficult subject, it would be nice to have a LaTeX
00802 document that explains this all exactly. There should then be a
00803 mechanism to poke the values of the setup into the LaTeX document.
00804 probably the easiest way is to make a file with lots of \def definitions

```



```

00808     and have that included into the LaTeX file.
00809  */
00810
00811 void WriteSetup(void)
00812 {
00813     int n = sizeof(setupparameters)/sizeof(SETUPPARAMETERS);
00814     SETUPPARAMETERS *sp;
00815     MesPrint(" The setup parameters are:");
00816     for ( sp = setupparameters; n > 0; n--, sp++ ) {
00817         switch(sp->type){
00818             case NUMERICALVALUE:
00819                 MesPrint("    %s: %1", sp->parameter, sp->value);
00820                 break;
00821             case PATHVALUE:
00822                 if ( StrICmp(sp->parameter, (UBYTE *) "path") == 0 && AM.Path ) {
00823                     MesPrint("    %s: '%s'", sp->parameter, (UBYTE *) (AM.Path));
00824                     break;
00825                 }
00826                 if ( StrICmp(sp->parameter, (UBYTE *) "incdir") == 0 && AM.IncDir ) {
00827                     MesPrint("    %s: '%s'", sp->parameter, (UBYTE *) (AM.IncDir));
00828                     break;
00829                 }
00830                 /* fall through */
00831             case STRINGVALUE:
00832                 if ( StrICmp(sp->parameter, (UBYTE *) "tempdir") == 0 && AM.TempDir ) {
00833                     MesPrint("    %s: '%s'", sp->parameter, (UBYTE *) (AM.TempDir));
00834                 }
00835                 else if ( StrICmp(sp->parameter, (UBYTE *) "tempSortDir") == 0 && AM.TempSortDir ) {
00836                     MesPrint("    %s: '%s'", sp->parameter, (UBYTE *) (AM.TempSortDir));
00837                 }
00838                 else {
00839                     MesPrint("    %s: '%s'", sp->parameter, (UBYTE *) (sp->value));
00840                 }
00841                 break;
00842             case ONOFFVALUE:
00843                 if ( sp->value == 0 )
00844                     MesPrint("    %s: OFF", sp->parameter);
00845                 else if ( sp->value == 1 )
00846                     MesPrint("    %s: ON", sp->parameter);
00847                 break;
00848             case DEFINEVALUE:
00849                 /*
00850                  MesPrint("    %s: '%s'", sp->parameter, (UBYTE *) (sp->value));
00851                 */
00852                 break;
00853         }
00854     }
00855     AC.SetupFlag = 0;
00856 }
00857
00858 /*
00859     #] WriteSetup :
00860     #[ AllocSort :
00861
00862     Routine allocates a complete struct for sorting.
00863     To be used for the main allocation of the sort buffers, and
00864     in a later stage for the function and subroutine sort buffers.
00865     The level arg denotes a main buffer allocation (0) or a sub-buffer
00866     allocation (1), used only for printing warning messages for buffer
00867     size adjustments in DEBUGGING mode.
00868  */
00869 SORTING *AllocSort(LONG inLargeSize, LONG inSmallSize, LONG inSmallEsize, LONG inTermsInSmall,
00870                   int inMaxPatches, int inMaxFpatches, LONG inIOsize, int level)
00871 {
00872     DUMMYUSE(level); /* This is only used in DEBUGGING mode */
00873     LONG LargeSize = inLargeSize;
00874     LONG SmallSize = inSmallSize;
00875     LONG SmallEsize = inSmallEsize;
00876     LONG TermsInSmall = inTermsInSmall;
00877     int MaxPatches = inMaxPatches;
00878     int MaxFpatches = inMaxFpatches;
00879     LONG IOsize = inIOsize;
00880
00881     LONG longer, terms2insmall, sortsize, longerp;
00882     LONG IObuffersize = IOsize;
00883     LONG IOtry;
00884     SORTING *sort;
00885     int fname2Size = 0, j = 0;
00886     char *s;
00887     if ( AM.S0 != 0 ) {
00888         s = FG.fname2; fname2Size = 0;
00889         while ( *s ) { s++; fname2Size++; }
00890         fname2Size += 16;
00891     }
00892     if ( MaxFpatches < 4 ) MaxFpatches = 4;
00893     longer = MaxPatches > MaxFpatches ? MaxPatches : MaxFpatches;
00894     longerp = longer;

```

```

00895     while ( (1 < j) < longerp ) j++;
00896     longerp = (1 < j) + 1;
00897     longerp += sizeof(WORD*) - (longerp*sizeof(WORD *));
00898     longer++;
00899     longer += sizeof(WORD*) - (longerp*sizeof(WORD *));
00900     if ( SmallSize < 16*AM.MaxTer ) SmallSize = 16*AM.MaxTer+16;
00901     TermsInSmall = (TermsInSmall+15) & (-16L);
00902     terms2insmall = 2*TermsInSmall; /* Used to be just + 100 rather than *2 */
00903     if ( SmallSize < (SmallSize*3)/2 ) SmallSize = (SmallSize*3)/2;
00904     if ( LargeSize > 0 && LargeSize < 2*SmallSize ) LargeSize = 2*SmallSize;
00905     SmallEsize = (SmallEsize+15) & (-16L);
00906     if ( LargeSize < 0 ) LargeSize = 0;
00907     sortsize = sizeof(SORTING);
00908     sortsize = (sortsize+15)&(-16L);
00909     IObuffersize = (IObuffersize+sizeof(WORD)-1)/sizeof(WORD);
00910 /*
00911     The next statement fixes a bug. In the rare case that we have a
00912     problem here, we expand the size of the large buffer or the
00913     small extension
00914 */
00915     if ( (ULONG) ( LargeSize+SmallEsize ) < MaxFpatches*((IObuffersize
00916     +COMPINC)*sizeof(WORD)+2*AM.MaxTer) ) {
00917         if ( LargeSize == 0 )
00918             SmallSize = MaxFpatches*((IObuffersize+COMPINC)*sizeof(WORD)+2*AM.MaxTer);
00919         else
00920             LargeSize = MaxFpatches*((IObuffersize+COMPINC)*sizeof(WORD)+2*AM.MaxTer)
00921                 - SmallEsize;
00922     }
00923
00924     IOtry = ((LargeSize+SmallEsize)/MaxFpatches-2*AM.MaxTer)/sizeof(WORD)-COMPINC;
00925
00926     /* Here both IObuffersize and IOtry are in units of sizeof(WORD). */
00927     if ( IObuffersize < IOtry ) {
00928         IObuffersize = IOtry;
00929     }
00930
00931 #if DEBUGGING
00932     char *prefix;
00933     if ( level == 0 ) { prefix = ""; }
00934     else { prefix = "Sub"; }
00935     if ( LargeSize != inLargeSize ) { MesPrint("Warning: %sLargeSize adjusted: %l -> %l", prefix,
inLargeSize, LargeSize); }
00936     if ( SmallSize != inSmallSize ) { MesPrint("Warning: %sSmallSize adjusted: %l -> %l", prefix,
inSmallSize, SmallSize); }
00937     if ( SmallEsize != inSmallEsize ) { MesPrint("Warning: %sSmallEsize adjusted: %l -> %l", prefix,
inSmallEsize, SmallEsize); }
00938     if ( TermsInSmall != inTermsInSmall ) { MesPrint("Warning: %sTermsInSmall adjusted: %l -> %l",
prefix, inTermsInSmall, TermsInSmall); }
00939     if ( MaxPatches != inMaxPatches ) { MesPrint("Warning: MaxPatches adjusted: %d -> %d",
inMaxPatches, MaxPatches); }
00940     if ( MaxFpatches != inMaxFpatches ) { MesPrint("Warning: MaxFPatches adjusted: %d -> %d",
inMaxFpatches, MaxFpatches); }
00941     /* This one is always changed if the LargeSize has not been... */
00942     /* if ( IObuffersize != inIOsize/(LONG)sizeof(WORD) ) { MesPrint("Warning: IOsize adjusted: %l ->
%l", inIOsize/sizeof(WORD), IObuffersize); } */
00943 #endif
00944
00945     /* Allocate separate buffers for most struct members, for better testing and debugging with
valgrind. */
00946     sort = Malloc1(sizeof(*sort), "AllocSort: sorting struct");
00947
00948     sort->LargeSize = LargeSize/sizeof(WORD);
00949     sort->SmallSize = SmallSize/sizeof(WORD);
00950     sort->SmallEsize = SmallEsize/sizeof(WORD);
00951     sort->MaxPatches = MaxPatches;
00952     sort->MaxFpatches = MaxFpatches;
00953     sort->TermsInSmall = TermsInSmall;
00954     sort->Terms2InSmall = terms2insmall;
00955
00956     sort->sPointer = Malloc1(sizeof(*(sort->sPointer ))*terms2insmall, "AllocSort: sPointer");
00957     sort->Patches = Malloc1(sizeof(*(sort->Patches ))*longer, "AllocSort: Patches");
00958     sort->pStop = Malloc1(sizeof(*(sort->pStop ))*longer, "AllocSort: pStop");
00959     sort->poina = Malloc1(sizeof(*(sort->poina ))*longerp, "AllocSort: poina");
00960     sort->poin2a = Malloc1(sizeof(*(sort->poin2a ))*longerp, "AllocSort: poin2a");
00961
00962     sort->fPatches = Malloc1(sizeof(*(sort->fPatches ))*longer, "AllocSort: fPatches");
00963     sort->fPatchesStop = Malloc1(sizeof(*(sort->fPatchesStop))*longer, "AllocSort: fPatchesStop");
00964     sort->inPatches = Malloc1(sizeof(*(sort->inPatches ))*longer, "AllocSort: inPatches");
00965     sort->tree = Malloc1(sizeof(*(sort->tree ))*longerp, "AllocSort: tree");
00966     sort->used = Malloc1(sizeof(*(sort->used ))*longerp, "AllocSort: used");
00967
00968 #ifdef WITHZLIB
00969     sort->fpcompressed = Malloc1(sizeof(*(sort->fpcompressed ))*(longerp+2), "AllocSort:
fpcompressed");
00970     sort->fpincompressed = Malloc1(sizeof(*(sort->fpincompressed))*(longerp+2), "AllocSort:
fpincompressed");
00971     sort->zsparray = 0;

```

```

00972 #endif
00973
00974     sort->ktoi          = Malloc1(sizeof(*(sort->ktoi))*(longerp+2), "AllocSort: ktoi");
00975
00976     // The combined Large buffer and Small buffer (+ extension) are used.
00977     // They must be allocated together.
00978     sort->lBuffer       = Malloc1(sizeof(*(sort->lBuffer))*(sort->LargeSize+sort->SmallEsize),
"AllocSort: lBuffer+sBuffer");
00979     sort->lTop = sort->lBuffer+sort->LargeSize;
00980
00981     sort->sBuffer = sort->lTop;
00982     if ( sort->LargeSize == 0 ) { sort->lBuffer = 0; sort->lTop = 0; }
00983     sort->sTop = sort->sBuffer + sort->SmallSize;
00984     sort->sTop2 = sort->sBuffer + sort->SmallEsize;
00985     sort->sHalf = sort->sBuffer + (LONG)((sort->SmallSize+sort->SmallEsize)>>1);
00986
00987     sort->file.PObuffer = Malloc1(IObuffersize*sizeof(*(sort->file.PObuffer))+fname2Size+16,
"AllocSort: PObuffer");
00988     sort->file.POstop = sort->file.PObuffer+IObuffersize;
00989     sort->file.POsize = IObuffersize * sizeof(WORD);
00990     sort->file.POfill = sort->file.PObuffer;
00991     sort->file.active = 0;
00992     sort->file.handle = -1;
00993     PUTZERO(sort->file.POposition);
00994 #ifdef WITHPTHREADS
00995     sort->file.pthreadlock = dummylock;
00996 #endif
00997 #ifdef WITHZLIB
00998     sort->file.ziosize = IObuffersize*sizeof(WORD);
00999     sort->file.ziobuffer = 0;
01000 #endif
01001     if ( AM.S0 != 0 ) {
01002         sort->file.name = (char *) (sort->file.PObuffer + IObuffersize);
01003         AllocSortFileName(sort);
01004     }
01005     else sort->file.name = 0;
01006     sort->cBuffer = 0;
01007     sort->cBufferSize = 0;
01008     sort->f = 0;
01009     sort->PolyWise = 0;
01010
01011     return(sort);
01012 }
01013
01014 /*
01015     #] AllocSort :
01016     #[ AllocSortFileName :
01017 */
01018
01019 void AllocSortFileName(SORTING *sort)
01020 {
01021     GETIDENTITY
01022     char *s, *t;
01023     /*
01024         This is not the allocation before the tempfiles are determined.
01025         Hence we can use the name in FG.fname2 and modify the tail
01026     */
01027     s = FG.fname2; t = sort->file.name;
01028     while ( *s ) *t++ = *s++;
01029 #ifdef WITHPTHREADS
01030     t[-2] = 'F';
01031     snprintf(t-1,FG.fname2size-(t-1-sort->file.name),"%d.%d",identity,AN.filename);
01032 #else
01033     t[-2] = 'f';
01034     snprintf(t-1,FG.fname2size-(t-1-sort->file.name),"%d",AN.filename);
01035 #endif
01036     AN.filename++;
01037 }
01038
01039 /*
01040     #] AllocSortFileName :
01041     #[ AllocFileHandle :
01042 */
01043
01044 FILEHANDLE *AllocFileHandle(WORD par,char *name)
01045 {
01046     GETIDENTITY
01047     LONG allocation, Ssize;
01048     FILEHANDLE *fh;
01049     int i = 0;
01050     char *s, *t;
01051
01052     s = FG.fname2; i = 0;
01053     while ( *s ) { s++; i++; }
01054     if ( par == 0 ) { i += 16; Ssize = AM.SIOsize; }
01055     else { s = name; while ( *s ) { i++; s++; } i+= 2; Ssize = AM.SpectatorSize; }
01056

```

```

01057     allocation = sizeof(FILEHANDLE) + (Ssize+1)*sizeof(WORD) + i*sizeof(char)
01058                 +FG.fname2size+20;
01059     fh = (FILEHANDLE *)Malloc1(allocation,"FileHandle");
01060
01061     fh->PObuffer = (WORD *) (fh+1);
01062     fh->POstop = fh->PObuffer+Ssize;
01063     fh->POsize = Ssize * sizeof(WORD);
01064     fh->active = 0;
01065     fh->handle = -1;
01066     PUTZERO(fh->POposition);
01067 #ifdef WITHPTHREADS
01068     fh->pthreadlock = dummylock;
01069 #endif
01070     if ( par == 0 ) { /* sort file */
01071         if ( AM.S0 != 0 ) {
01072             fh->name = (char *) (fh->POstop + 1);
01073             s = FG.fname2; t = fh->name;
01074             while ( *s ) *t++ = *s++;
01075 #ifdef WITHPTHREADS
01076             t[-2] = 'F';
01077             snprintf(t-1,FG.fname2size-(t-1-fh->name), "%d-%d", identity, AN.filenum);
01078 #else
01079             t[-2] = 'f';
01080             snprintf(t-1,FG.fname2size-(t-1-fh->name), "%d", AN.filenum);
01081 #endif
01082             AN.filenum++;
01083         }
01084         else fh->name = 0;
01085     }
01086     else { /* Spectator file */
01087         fh->name = (char *) (fh->POstop + 1);
01088         s = FG.fname; t = fh->name;
01089         for ( i = 0; i < FG.fnamebase; i++ ) *t++ = *s++;
01090         s = name;
01091         while ( *s ) *t++ = *s++;
01092         *t = 0;
01093     }
01094     fh->POfill = fh->POfull = fh->PObuffer;
01095     return(fh);
01096 }
01097
01098 /*
01099     #] AllocFileHandle :
01100     #[ DeAllocFileHandle :
01101
01102     Made to repair deallocation of AN.filenum. 21-sep-2000
01103 */
01104
01105 void DeAllocFileHandle(FILEHANDLE *fh)
01106 {
01107     GETIDENTITY
01108     if ( fh->handle >= 0 ) {
01109         CloseFile(fh->handle);
01110         fh->handle = -1;
01111         remove(fh->name);
01112     }
01113     AN.filenum--; /* free namespace. was forgotten in first reading */
01114     M_free(fh,"Temporary FileHandle");
01115 }
01116
01117 /*
01118     #] DeAllocFileHandle :
01119     #[ MakeSetupAllocs :
01120 */
01121
01122 int MakeSetupAllocs(void)
01123 {
01124     if ( RecalcSetups() || AllocSetups() ) return(1);
01125     else return(0);
01126 }
01127
01128 /*
01129     #] MakeSetupAllocs :
01130     #[ TryFileSetups :
01131
01132     Routine looks in the input file for a start of the type
01133     [#-]
01134     #: setupparameter value
01135     It keeps looking until the first line that does not start with
01136     #-, #+ or #:
01137     Then it rewinds the input.
01138 */
01139
01140 #define SETBUFSIZE 257
01141
01142 int TryFileSetups(void)
01143 {

```

```

01144     LONG oldstreamposition;
01145     int oldstream;
01146     int error = 0, eqnum;
01147     int oldNoShowInput = AC.NoShowInput;
01148     UBYTE buff[SETBUFSIZE+1], *s, *t, *u, *settop, c;
01149     LONG linenum, prevline;
01150
01151     if ( AC.CurrentStream == 0 ) return(error);
01152     oldstream = AC.CurrentStream - AC.Streams;
01153     oldstreamposition = GetStreamPosition(AC.CurrentStream);
01154     linenum = AC.CurrentStream->linenum;
01155     prevline = AC.CurrentStream->prevline;
01156     eqnum = AC.CurrentStream->eqnum;
01157     AC.NoShowInput = 1;
01158     settop = buff + SETBUFSIZE;
01159     if ( AC.CurrentStream->type == INPUTSTREAM && oldstreamposition == 0 )
01160         AC.CurrentStream->fileposition = 0;
01161     for(;;) {
01162         c = GetInput();
01163         if ( c == '*' || c == '\n' ) {
01164             while ( c != '\n' && c != ENDOFINPUT ) c = GetInput();
01165             if ( c == ENDOFINPUT ) goto eoi;
01166             continue;
01167         }
01168         if ( c == ENDOFINPUT ) goto eoi;
01169         if ( c != '#' ) break;
01170         c = GetInput();
01171         if ( c == ENDOFINPUT ) goto eoi;
01172         if ( c != '-' && c != '+' && c != ':' ) break;
01173         if ( c != ':' ) {
01174             while ( c != '\n' && c != ENDOFINPUT ) c = GetInput();
01175             continue;
01176         }
01177         s = buff;
01178         while ( ( c = GetInput() ) == ' ' || c == '\t' || c == '\r' ) {}
01179         if ( c == ENDOFINPUT ) break;
01180         if ( c == LINEFEED ) continue;
01181         if ( c == 0 || c == ENDOFINPUT ) break;
01182         while ( c != LINEFEED ) {
01183             *s++ = c;
01184             c = GetInput();
01185             if ( c != LINEFEED && c != '\r' ) continue;
01186             if ( s >= settop ) {
01187                 while ( c != '\n' && c != ENDOFINPUT ) c = GetInput();
01188                 MesPrint("Setups in .frm file: Line too long. setup ignored");
01189                 error++; goto nextline;
01190             }
01191             *s++ = '\n';
01192             t = s = buff; /* name of the option */
01193             while ( tolower(*s) >= 'a' && tolower(*s) <= 'z' ) s++;
01194             if ( *s != '\n' ) {
01195                 *s++ = 0;
01196                 while ( *s == ' ' || *s == '\t' ) s++;
01197                 u = s; /* 'value' of the option */
01198                 while ( *s && *s != '\n' && *s != '\r' ) s++;
01199                 if ( *s ) *s++ = 0;
01200             }
01201             else {
01202                 /* The value is empty. */
01203                 u = s;
01204                 *s++ = 0;
01205             }
01206             error += ProcessOption(t,u,1);
01207         nextline:;
01208     }
01209     AC.NoShowInput = oldNoShowInput;
01210     AC.CurrentStream = AC.Streams + oldstream;
01211     PositionStream(AC.CurrentStream,oldstreamposition);
01212     AC.CurrentStream->linenum = linenum;
01213     AC.CurrentStream->prevline = prevline;
01214     AC.CurrentStream->eqnum = eqnum;
01215     ClearPushback();
01216     if ( AC.CurrentStream->type == INPUTSTREAM ) AC.CurrentStream->fileposition = -1;
01217     return(error);
01218 eoi:
01219     MesPrint("Input file without a program.");
01220     return(-1);
01221 }
01222
01223 /*
01224     #] TryFileSetups :
01225     #[ TryEnvironment :
01226 */
01227
01228 int TryEnvironment(void)
01229 {

```

```

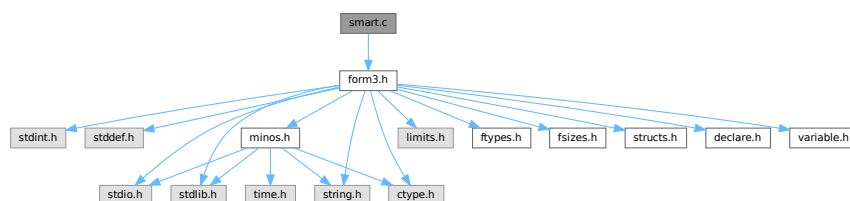
01230     char *s, *t, *u, varname[100];
01231     int i,imax = sizeof(setupparameters)/sizeof(SETUPPARAMETERS);
01232     int error = 0;
01233     varname[0] = 'F'; varname[1] = 'O'; varname[2] = 'R'; varname[3] = 'M';
01234     varname[4] = '_'; varname[5] = 0;
01235     for ( i = 0; i < imax; i++ ) {
01236         t = s = (char *) (setupparameters[i].parameter);
01237         u = varname+5;
01238         while ( *s ) { *u++ = (char) (toupper((unsigned char)*s)); s++; }
01239         *u = 0;
01240         s = (char *) (getenv(varname));
01241         if ( s ) {
01242             error += ProcessOption((UBYTE *)t, (UBYTE *)s,2);
01243         }
01244     }
01245     return(error);
01246 }
01247
01248 /*
01249     #] TryEnvironment :
01250     #] Setups :
01251 */

```

## 4.126 smart.c File Reference

#include "form3.h"

Include dependency graph for smart.c:



### Functions

- int [StudyPattern](#) (WORD \*lhs)
- int [MatchesPossible](#) (WORD \*pattern, WORD \*term)

#### 4.126.1 Detailed Description

The functions for smart pattern searches in combinations of functions. When many wildcards are involved and the functions are (anti)symmetric an exhaustive search for all possibilities may take very much time (like factorial in the number of wildcards) while a human can often see immediately that there cannot be a match. The routines here try to make FORM a bit smarter in this respect.

This is just the beginning. It still needs lots of work!

Definition in file [smart.c](#).

## 4.126.2 Function Documentation

### 4.126.2.1 StudyPattern()

```
int StudyPattern (
    WORD * lhs )
```

Definition at line 62 of file [smart.c](#).

### 4.126.2.2 MatchIsPossible()

```
int MatchIsPossible (
    WORD * pattern,
    WORD * term )
```

Definition at line 299 of file [smart.c](#).

## 4.127 smart.c

[Go to the documentation of this file.](#)

```
00001
00012 /* #[] License : */
00013 /*
00014 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00015 * When using this file you are requested to refer to the publication
00016 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00017 * This is considered a matter of courtesy as the development was paid
00018 * for by FOM the Dutch physics granting agency and we would like to
00019 * be able to track its scientific use to convince FOM of its value
00020 * for the community.
00021 *
00022 * This file is part of FORM.
00023 *
00024 * FORM is free software: you can redistribute it and/or modify it under the
00025 * terms of the GNU General Public License as published by the Free Software
00026 * Foundation, either version 3 of the License, or (at your option) any later
00027 * version.
00028 *
00029 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00030 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00031 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00032 * details.
00033 *
00034 * You should have received a copy of the GNU General Public License along
00035 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00036 */
00037 /* #[] License : */
00038 /*
00039 #[] Includes : function.c
00040 */
00041
00042 #include "form3.h"
00043
00044 /*
00045 #[] Includes :
00046 #[] StudyPattern :
00047
00048 Argument is a complete lhs of an id-statement
00049 Its last word is a zero (new convention(18-may-1997)) to indicate
00050 that no extra information is following. We can add there information
00051 about the pattern that will help during the actual pattern matching.
00052 Currently the WorkPointer points to just after this lhs.
00053
00054 Task of this routine: To study the functions, their symmetry properties
00055 and their wildcards to determine in which order the functions can
00056 be matched best. If the order should be different we can change it here.
00057
00058 Problem encountered 22-dec-2008 (JV): we don't take noncommuting
00059 functions into account.
```

```

00060 */
00061
00062 int StudyPattern(WORD *lhs)
00063 {
00064     GETIDENTITY
00065     WORD *fullproto, *pat, *p, *p1, *p2, *pstop, *info, f, nn;
00066     int numfun = 0, numsym = 0, allwilds = 0, i, j, k, nc;
00067     FUN_INFO *finf, *fmin, *f1, *f2, funscratch;
00068
00069     fullproto = lhs + IDHEAD;
00070     /* if ( *lhs == TYPEIF ) fullproto--; */
00071     pat = fullproto + fullproto[1];
00072     info = pat + *pat;
00073
00074     p = pat + 1;
00075     while ( p < info ) {
00076         if ( *p >= FUNCTION ) {
00077             numfun++;
00078             nn = *p - FUNCTION;
00079             if ( nn >= WILDOFFSET ) nn -= WILDOFFSET;
00080         /*
00081             We check here for cases that are not allowed like ?a inside
00082             symmetric functions or tensors.
00083         */
00084         if ( ( functions[nn].symmetric == SYMMETRIC ) ||
00085             ( functions[nn].symmetric == ANTISYMMETRIC ) ) {
00086             p2 = p+p[1]; p1 = p+FUNHEAD;
00087             if ( functions[nn].spec > 0 ) {
00088                 while ( p1 < p2 ) {
00089                     if ( *p1 == FUNNYWILD ) {
00090                         MesPrint("&Argument field wildcards are not allowed inside (anti)symmetric
functions or tensors");
00091                         return(1);
00092                     }
00093                     p1++;
00094                 }
00095             }
00096             else {
00097                 while ( p1 < p2 ) {
00098                     if ( *p1 == -ARGWILD ) {
00099                         MesPrint("&Argument field wildcards are not allowed inside (anti)symmetric
functions or tensors");
00100                         return(1);
00101                     }
00102                     NEXTARG(p1);
00103                 }
00104             }
00105         }
00106         p += p[1];
00107     }
00108     if ( numfun == 0 ) return(0);
00109     if ( ( lhs[2] & SUBMASK ) == SUBALL ) {
00110         p = pat + 1;
00111         while ( p < info ) {
00112             if ( *p == SYMBOL || *p == VECTOR || *p == DOTPRODUCT || *p == INDEX ) {
00113                 MesPrint("&id,all can have only functions and/or tensors in the lhs.");
00114                 return(1);
00115             }
00116             p += p[1];
00117         }
00118     }
00119 }
00120 /*
00121     We need now some room for the information about the functions
00122 */
00123 if ( numfun > AN.numfuninfo ) {
00124     if ( AN.FunInfo ) M_free(AN.FunInfo,"funinfo");
00125     AN.numfuninfo = numfun + 10;
00126     AN.FunInfo = (FUN_INFO *)Malloc1(AN.numfuninfo*sizeof(FUN_INFO),"funinfo");
00127 }
00128 /*
00129     Now collect the information. First the locations.
00130 */
00131 p = pat + 1; i = 0;
00132 while ( p < info ) {
00133     if ( *p >= FUNCTION ) AN.FunInfo[i++].location = p;
00134     p += p[1];
00135 }
00136 for ( i = 0, finf = AN.FunInfo; i < numfun; i++, finf++ ) {
00137     p = finf->location;
00138     pstop = p + p[1];
00139     f = *p;
00140     if ( f > FUNCTION+WILDOFFSET ) f -= WILDOFFSET;
00141     finf->numargs = finf->numfunnies = finf->numwildcards = 0;
00142     finf->symmet = functions[f-FUNCTION].symmetric;
00143     finf->tensor = functions[f-FUNCTION].spec;
00144     finf->commute = functions[f-FUNCTION].commute;

```



```

00145     if ( finf->tensor >= TENSORFUNCTION ) {
00146         p += FUNHEAD;
00147         while ( p < pstop ) {
00148             if ( *p == FUNNYWILD ) {
00149                 finf->numfunnies++; p+= 2; continue;
00150             }
00151             else if ( *p < 0 ) {
00152                 if ( *p >= AM.OffsetVector + WILDOFFSET && *p < MINSPEC ) {
00153                     finf->numwildcards++;
00154                 }
00155             }
00156             else {
00157                 if ( *p >= AM.OffsetIndex + WILDOFFSET &&
00158                     *p <= AM.OffsetIndex + 2*WILDOFFSET ) finf->numwildcards++;
00159             }
00160             finf->numargs++;
00161             p++;
00162         }
00163     }
00164     else {
00165         p += FUNHEAD;
00166         while ( p < pstop ) {
00167             if ( *p > 0 ) { finf->numargs++; p += *p; continue; }
00168             if ( *p <= -FUNCTION ) {
00169                 if ( *p <= -FUNCTION - WILDOFFSET ) finf->numwildcards++;
00170                 p++;
00171             }
00172             else if ( *p == -SYMBOL ) {
00173                 if ( p[1] >= 2*MAXPOWER ) finf->numwildcards++;
00174                 p += 2;
00175             }
00176             else if ( *p == -INDEX ) {
00177                 if ( p[1] >= AM.OffsetIndex + WILDOFFSET &&
00178                     p[1] <= AM.OffsetIndex + 2*WILDOFFSET ) finf->numwildcards++;
00179                 p += 2;
00180             }
00181             else if ( *p == -VECTOR || *p == -MINVECTOR ) {
00182                 if ( p[1] >= AM.OffsetVector + WILDOFFSET && p[1] < MINSPEC ) {
00183                     finf->numwildcards++;
00184                 }
00185                 p += 2;
00186             }
00187             else if ( *p == -ARGWILD ) {
00188                 finf->numfunnies++;
00189                 p += 2;
00190             }
00191             else { p += 2; }
00192             finf->numargs++;
00193         }
00194     }
00195     if ( finf->symmet ) {
00196         numsym++;
00197         allwilds += finf->numwildcards + finf->numfunnies;
00198     }
00199 }
00200 if ( numsym == 0 ) return(0);
00201 if ( allwilds == 0 ) return(0);
00202 /*
00203 We have the information in the array AN.FunInfo.
00204 We sort things and then write the sorted pattern.
00205 Of course we may not play with the order of the noncommuting functions.
00206 Of course we have to become even smarter in the future and look during
00207 the sorting which wildcards are assigned when.
00208 But for now this should do.
00209 */
00210 for ( nc = numfun-1; nc >= 0; nc-- ) { if ( AN.FunInfo[nc].commute ) break; }
00211
00212 finf = AN.FunInfo;
00213 for ( i = nc+2; i < numfun; i++ ) {
00214     fmin = finf; finf++;
00215     if ( ( finf->symmet < fmin->symmet ) || (
00216         ( finf->symmet == fmin->symmet ) &&
00217         ( ( finf->numwildcards+finf->numfunnies < fmin->numwildcards+fmin->numfunnies )
00218         || ( ( finf->numwildcards+finf->numfunnies == fmin->numwildcards+fmin->numfunnies )
00219         && ( finf->numwildcards < fmin->numfunnies ) ) ) ) ) {
00220         funscratch = AN.FunInfo[i];
00221         AN.FunInfo[i] = AN.FunInfo[i-1];
00222         AN.FunInfo[i-1] = funscratch;
00223         for ( j = i-1; j > nc && j > 0; j-- ) {
00224             f1 = AN.FunInfo[j];
00225             f2 = f1-1;
00226             if ( ( f1->symmet < f2->symmet ) || (
00227                 ( f1->symmet == f2->symmet ) &&
00228                 ( ( f1->numwildcards+f1->numfunnies < f2->numwildcards+f2->numfunnies )
00229                 || ( ( f1->numwildcards+f1->numfunnies == f2->numwildcards+f2->numfunnies )
00230                 && ( f1->numwildcards < f2->numfunnies ) ) ) ) ) {
00231                 funscratch = AN.FunInfo[j];

```

```

00232             AN.FunInfo[j] = AN.FunInfo[j-1];
00233             AN.FunInfo[j-1] = funscratch;
00234         }
00235         else break;
00236     }
00237 }
00238 }
00239 /*
00240 Now we rewrite the pattern. First into the space after it and then we
00241 copy it back. Be careful with the non-commutative functions. There the
00242 worst one should decide.
00243 */
00244 p = pat + 1;
00245 p2 = info;
00246 for ( i = 0; i < numfun; i++ ) {
00247     if ( i == nc ) {
00248         for ( k = 0; k <= nc; k++ ) {
00249             if ( AN.FunInfo[k].commute ) {
00250                 p1 = AN.FunInfo[k].location; j = p1[1];
00251                 NCOPY(p2,p1,j)
00252             }
00253         }
00254     }
00255     else if ( AN.FunInfo[i].commute == 0 ) {
00256         p1 = AN.FunInfo[i].location; j = p1[1];
00257         NCOPY(p2,p1,j)
00258     }
00259 }
00260 p = pat + 1; p1 = info;
00261 while ( p1 < p2 ) *p++ = *p1++;
00262 /*
00263 And now we place the relevant information in the info part
00264 */
00265 p2 = info+1;
00266 for ( i = 0; i < numfun; i++ ) {
00267     if ( i == nc ) {
00268         for ( k = 0; k <= nc; k++ ) {
00269             if ( AN.FunInfo[k].commute ) {
00270                 finf = AN.FunInfo + k;
00271                 *p2++ = finf->numargs;
00272                 *p2++ = finf->numwildcards;
00273                 *p2++ = finf->numfunnies;
00274                 *p2++ = finf->symmet;
00275             }
00276         }
00277     }
00278     else if ( AN.FunInfo[i].commute == 0 ) {
00279         finf = AN.FunInfo + i;
00280         *p2++ = finf->numargs;
00281         *p2++ = finf->numwildcards;
00282         *p2++ = finf->numfunnies;
00283         *p2++ = finf->symmet;
00284     }
00285 }
00286 *info = p2-info;
00287 lhs[1] = p2-lhs;
00288 return(0);
00289 }
00290
00291 /*
00292 #] StudyPattern :
00293 #[ MatchIsPossible :
00294
00295 We come here when there are functions and there is nontrivial
00296 symmetry related wildcarding.
00297 */
00298
00299 int MatchIsPossible(WORD *pattern, WORD *term)
00300 {
00301     GETIDENTITY
00302     WORD *info = pattern + *pattern;
00303     WORD *t, *tstop, *tt, *inf, *p;
00304     int numfun = 0, inpat, i, j, numargs;
00305     FUN_INFO *finf;
00306 /*
00307 We need a list of functions and their number of arguments
00308 */
00309 GETSTOP(term,tstop);
00310 t = term + 1;
00311 while ( t < tstop ) {
00312     if ( *t >= FUNCTION ) numfun++;
00313     t += t[1];
00314 }
00315 if ( numfun == 0 ) goto NotPossible;
00316 if ( *info == SETSET ) info += info[1];
00317 inpat = (*info-1)/4;
00318 if ( inpat > numfun ) goto NotPossible;

```

```

00319 /*
00320 We need now some room for the information about the functions
00321 */
00322 if ( numfun > AN.numfuninfo ) {
00323     if ( AN.FunInfo ) M_free(AN.FunInfo,"funinfo");
00324     AN.numfuninfo = numfun + 10;
00325     AN.FunInfo = (FUN_INFO *)Malloc1(AN.numfuninfo*sizeof(FUN_INFO),"funinfo");
00326 }
00327 t = term + 1; finf = AN.FunInfo;
00328 while ( t < tstop ) {
00329     if ( *t >= FUNCTION ) {
00330         finf->location = t;
00331         if ( functions[*t-FUNCTION].spec >= TENSORFUNCTION ) {
00332             numargs = t[1]-FUNHEAD;
00333             t += t[1];
00334         }
00335         else {
00336             numargs = 0;
00337             tt = t + t[1];
00338             t += FUNHEAD;
00339             while ( t < tt ) {
00340                 numargs++;
00341                 NEXTARG(t)
00342             }
00343         }
00344         finf->numargs = numargs;
00345         finf++;
00346     }
00347     else t += t[1];
00348 }
00349 /*
00350 Now we first find a partner for each function in the pattern
00351 with a fixed number of arguments
00352 */
00353 for ( i = 0, inf = info+1, p = pattern+1; i < inpat; i++, inf += 4, p+=p[1] ) {
00354     if ( inf[2] ) continue;
00355     if ( *p >= (FUNCTION+WILDOFFSET) ) continue;
00356     for ( j = 0, finf = AN.FunInfo; j < numfun; j++, finf++ ) {
00357         if ( *p == *(finf->location) && *inf == finf->numargs ) {
00358             finf->numargs = -finf->numargs-1;
00359             break;
00360         }
00361     }
00362     if ( j >= numfun ) goto NotPossible;
00363 }
00364 for ( i = 0, inf = info+1, p = pattern+1; i < inpat; i++, inf += 4, p+=p[1] ) {
00365     if ( inf[2] ) continue;
00366     if ( *p < (FUNCTION+WILDOFFSET) ) continue;
00367     for ( j = 0, finf = AN.FunInfo; j < numfun; j++, finf++ ) {
00368         if ( *inf == finf->numargs ) {
00369             finf->numargs = -finf->numargs-1;
00370             break;
00371         }
00372     }
00373     if ( j >= numfun ) goto NotPossible;
00374 }
00375 for ( i = 0, inf = info+1, p = pattern+1; i < inpat; i++, inf += 4, p+=p[1] ) {
00376     if ( inf[2] == 0 ) continue;
00377     if ( *p >= (FUNCTION+WILDOFFSET) ) continue;
00378     for ( j = 0, finf = AN.FunInfo; j < numfun; j++, finf++ ) {
00379         if ( *p == *(finf->location) && (*inf-inf[2]) <= finf->numargs ) {
00380             finf->numargs = -finf->numargs-1;
00381             break;
00382         }
00383     }
00384     if ( j >= numfun ) goto NotPossible;
00385 }
00386 for ( i = 0, inf = info+1, p = pattern+1; i < inpat; i++, inf += 4, p+=p[1] ) {
00387     if ( inf[2] == 0 ) continue;
00388     if ( *p < (FUNCTION+WILDOFFSET) ) continue;
00389     for ( j = 0, finf = AN.FunInfo; j < numfun; j++, finf++ ) {
00390         if ( (*inf-inf[2]) <= finf->numargs ) {
00391             finf->numargs = -finf->numargs-1;
00392             break;
00393         }
00394     }
00395     if ( j >= numfun ) goto NotPossible;
00396 }
00397 /*
00398 Thus far we have determined that for each function in the pattern
00399 there is a potential partner with enough arguments.
00400 For now the rest is up to the real pattern matcher.
00401
00402 To undo the disabling of the number of arguments we need this code
00403 for ( i = 0, finf = AN.FunInfo; i < numfun; i++, finf++ ) {
00404     if ( finf->numargs < 0 ) finf->numargs = -finf->numargs-1;
00405 }

```

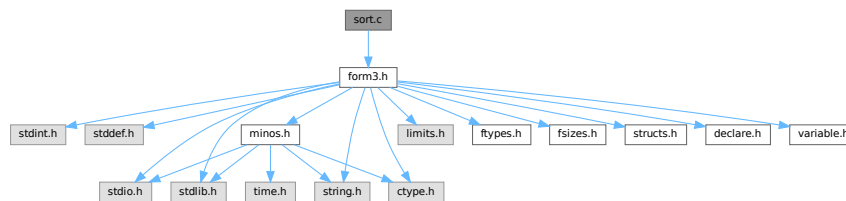
```

00406 */
00407     return(1);
00408 NotPossible:
00409 /*
00410     PrintTerm(pattern,"p");
00411     PrintTerm(term,"t");
00412 */
00413     return(0);
00414 }
00415
00416 /*
00417     #] MatchIsPossible :
00418 */

```

## 4.128 sort.c File Reference

#include "form3.h"  
 Include dependency graph for sort.c:



### Macros

- #define [NEWSPLITMERGE](#)
- #define [NEWSPLITMERGETIMSORT](#)
- #define [HUMANSTRLEN](#) 12
- #define [HUMANSUFFLEN](#) 4
- #define [INSLNGTH](#)(x) w[1] = FUNHEAD+ARGHEAD+x; w[FUNHEAD] = ARGHEAD+x;

### Functions

- void [HumanString](#) (char \*string, float input, const char suffix[HUMANSUFFLEN][4])
- void [WriteStats](#) ([POSITION](#) \*plspace, WORD par, WORD checkLogType)
- int [NewSort](#) (PHEAD0)
- LONG [EndSort](#) (PHEAD WORD \*buffer, int par)
- LONG [PutIn](#) ([FILEHANDLE](#) \*file, [POSITION](#) \*position, WORD \*buffer, WORD \*\*take, int npat)
- int [Sflush](#) ([FILEHANDLE](#) \*fi)
- WORD [PutOut](#) (PHEAD WORD \*term, [POSITION](#) \*position, [FILEHANDLE](#) \*fi, WORD ncomp)
- int [FlushOut](#) ([POSITION](#) \*position, [FILEHANDLE](#) \*fi, int compr)
- int [AddCoef](#) (PHEAD WORD \*\*ps1, WORD \*\*ps2)
- int [AddPoly](#) (PHEAD WORD \*\*ps1, WORD \*\*ps2)
- void [AddArgs](#) (PHEAD WORD \*s1, WORD \*s2, WORD \*m)
- WORD [Compare1](#) (PHEAD WORD \*term1, WORD \*term2, WORD level)
- WORD [CompareSymbols](#) (PHEAD WORD \*term1, WORD \*term2, WORD par)
- WORD [CompareHSymbols](#) (PHEAD WORD \*term1, WORD \*term2, WORD par)
- LONG [ComPress](#) (WORD \*\*ss, LONG \*n)
- LONG [SplitMerge](#) (PHEAD WORD \*\*Pointer, LONG number)

- void [GarbHand](#) (void)
- int [MergePatches](#) (WORD par)
- int [StoreTerm](#) (PHEAD WORD \*term)
- void [StageSort](#) (FILEHANDLE \*fout)
- int [SortWild](#) (WORD \*w, WORD nw)
- void [CleanUpSort](#) (int num)
- void [LowerSortLevel](#) (void)
- WORD \* [PolyRatFunSpecial](#) (PHEAD WORD \*t1, WORD \*t2)
- void [SimpleSplitMergeRec](#) (WORD \*array, WORD num, WORD \*auxarray)
- void [SimpleSplitMerge](#) (WORD \*array, WORD num)
- WORD [BinarySearch](#) (WORD \*array, WORD num, WORD x)

## Variables

- LONG [numcompares](#)
- char \* [totterms](#) [] = { " ", ">>", "-->" }
- const char [humanTermsSuffix](#) [HUMANSUFFLEN][4] = {"K ", "M ", "B ", "T "}
- const char [humanBytesSuffix](#) [HUMANSUFFLEN][4] = {"KiB", "MiB", "GiB", "TiB"}

## 4.128.1 Detailed Description

Contains the sort routines. We distinguish levels of sorting. The ground level is the sorting of terms in an expression. When a term has functions, the arguments can contain terms that need sorting, which this then done by raising the level. This can happen recursively. NewSort and EndSort automatically raise and lower the level. Because the ground level does some special things, sometimes we have to raise immediately to the second level skipping the ground level.

Special routines for the parallel sorting are in the file [threads.c](#). Also the sorting of terms in polynomials is special but most of that is controlled by changing the address of the compare routine. Other routines relevant for adding rational polynomials are in the file [polynito.c](#).

Definition in file [sort.c](#).

## 4.128.2 Macro Definition Documentation

### 4.128.2.1 NEWSPLITMERGE

```
#define NEWSPLITMERGE
```

Definition at line 53 of file [sort.c](#).

### 4.128.2.2 NEWSPLITMERGETIMSORT

```
#define NEWSPLITMERGETIMSORT
```

Definition at line 55 of file [sort.c](#).

#### 4.128.2.3 HUMANSTRLEN

```
#define HUMANSTRLEN 12
```

Definition at line 85 of file [sort.c](#).

#### 4.128.2.4 HUMANSUFFLEN

```
#define HUMANSUFFLEN 4
```

Definition at line 86 of file [sort.c](#).

#### 4.128.2.5 INSLENGTH

```
#define INSLENGTH(
    x ) w[1] = FUNHEAD+ARGHEAD+x; w[FUNHEAD] = ARGHEAD+x;
```

Definition at line 2318 of file [sort.c](#).

### 4.128.3 Function Documentation

#### 4.128.3.1 HumanString()

```
void HumanString (
    char * string,
    float input,
    const char suffix[HUMANSUFFLEN][4] )
```

Definition at line 89 of file [sort.c](#).

#### 4.128.3.2 WriteStats()

```
void WriteStats (
    POSITION * plspace,
    WORD par,
    WORD checkLogType )
```

Writes the statistics.

##### Parameters

|                     |                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <i>plspace</i>      | The size in bytes currently occupied                                                                                                     |
| <i>par</i>          | par = 0 = STATSSPLITMERGE after a splitmerge. par = 1 = STATSMERGETOFILE after merge to sortfile. par = 2 = STATSPOSTSORT after the sort |
| <i>checkLogType</i> | == CHECKLOGTYPE: The output should not go to the log file if AM.LogType is 1.                                                            |

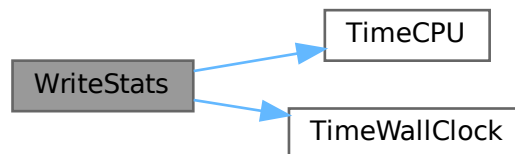
current expression is to be found in AR.CurExpr. terms are in S->TermsLeft. S->GenTerms.

Definition at line 121 of file [sort.c](#).

References [TimeCPU\(\)](#), and [TimeWallClock\(\)](#).

Referenced by [EndSort\(\)](#), [PF\\_Processor\(\)](#), and [StoreTerm\(\)](#).

Here is the call graph for this function:



#### 4.128.3.3 NewSort()

```
int NewSort (
    PHEAD0 )
```

Starts a new sort. At the lowest level this is a 'main sort' with the struct according to the parameters in S0. At higher levels this is a sort for functions, subroutines or dollars. We prepare the arrays and structs.

##### Returns

Regular convention (OK -> 0)

Definition at line 643 of file [sort.c](#).

Referenced by [generate\\_expression\(\)](#), [get\\_expression\(\)](#), [InFunction\(\)](#), [MakeDollarInteger\(\)](#), [MakeDollarMod\(\)](#), [PF\\_CollectModifiedDollars\(\)](#), [PF\\_Processor\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_sort\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [PolyFunMul\(\)](#), [Processor\(\)](#), [TakeArgContent\(\)](#), and [TestMatch\(\)](#).

#### 4.128.3.4 EndSort()

```
LONG EndSort (
    PHEAD WORD * buffer,
    int par )
```

Finishes a sort. At `AR.sLevel == 0` the output is to the regular output stream. When `AR.sLevel > 0`, the parameter `par` determines the actual output. The `AR.sLevel` will be popped. All ongoing stages are finished and if the sortfile is open it is closed. The statistics are printed when `AR.sLevel == 0` `par == 0` [Output](#) to the buffer. `par == 1` Sort for function arguments. The output will be copied into the buffer. It is assumed that this is in the `WorkSpace`. `par == 2` Sort for \$-variable. We return the address of the buffer that contains the output in `buffer` (treated like `WORD **`). We first catch the output in a file (unless we can intercept things after the small buffer has been sorted) Then we read from the file into a buffer. Only when `par == 0` data compression can be attempted at `AT.SS==AT.S0`.

## Parameters

|               |                               |
|---------------|-------------------------------|
| <i>buffer</i> | buffer for output when needed |
| <i>par</i>    | See above                     |

## Returns

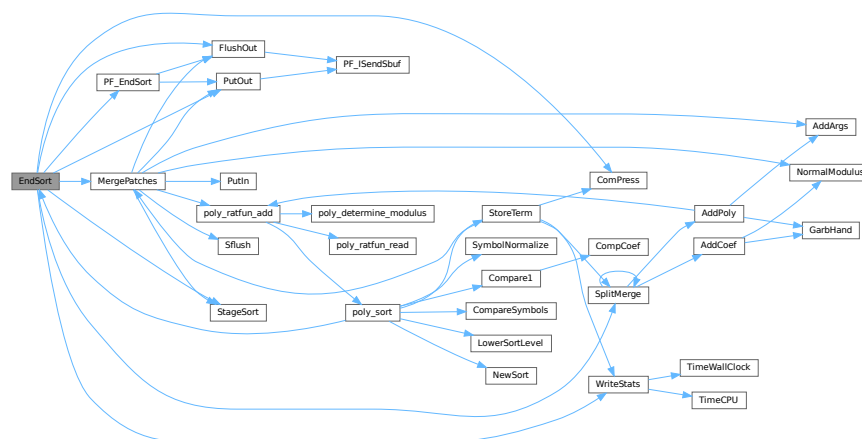
If negative: error. If positive: number of words in output.

Definition at line 733 of file [sort.c](#).

References [ComPress\(\)](#), [FlushOut\(\)](#), [MergePatches\(\)](#), [PF\\_EndSort\(\)](#), [PutOut\(\)](#), [SplitMerge\(\)](#), [StageSort\(\)](#), and [WriteStats\(\)](#).

Referenced by [generate\\_expression\(\)](#), [get\\_expression\(\)](#), [InFunction\(\)](#), [MakeDollarInteger\(\)](#), [MakeDollarMod\(\)](#), [PF\\_CollectModifiedDollars\(\)](#), [PF\\_Processor\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_sort\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [PolyFunMul\(\)](#), [Processor\(\)](#), [TakeArgContent\(\)](#), and [TestMatch\(\)](#).

Here is the call graph for this function:



## 4.128.3.5 PutIn()

```

LONG PutIn (
    FILEHANDLE * file,
    POSITION * position,
    WORD * buffer,
    WORD ** take,
    int npat )

```

Reads a new patch from position in file handle. It is put at buffer, anything after take is moved forward. This would be part of a term that hasn't been used yet. Because of this there should be some space before the start of the buffer

## Parameters

|                 |                                                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <i>file</i>     | The file system from which to read                                                                                                        |
| <i>position</i> | The position from which to read                                                                                                           |
| <i>buffer</i>   | The buffer into which to read                                                                                                             |
| <i>take</i>     | The unused tail should be moved before the buffer                                                                                         |
| <i>npat</i>     | The number of the patch. Is needed if the information was compressed with gzip, because each patch has its own independent gzip encoding. |



Definition at line 1304 of file [sort.c](#).

References [FiLe::handle](#).

Referenced by [MergePatches\(\)](#).

#### 4.128.3.6 Sflush()

```
int Sflush (
    FILEHANDLE * fi )
```

Puts the contents of a buffer to output Only to be used when there is a single patch in the large buffer.

##### Parameters

|           |                                                                    |
|-----------|--------------------------------------------------------------------|
| <i>fi</i> | The filesystem (or its cache) to which the patch should be written |
|-----------|--------------------------------------------------------------------|

Definition at line 1364 of file [sort.c](#).

References [FiLe::handle](#).

Referenced by [MergePatches\(\)](#).

#### 4.128.3.7 PutOut()

```
WORD PutOut (
    PHEAD WORD * term,
    POSITION * position,
    FILEHANDLE * fi,
    WORD ncomp )
```

Routine writes one term to file handle at position. It returns the new value of the position.

NOTE: For 'final output' we may have to index the brackets. See the struct BRACKETINDEX. We should maintain:  
 1: a list with brackets array with the brackets  
 2: a list of objects of type BRACKETINDEX. It contains array with either pointers or offsets to the list of brackets. starting positions in the file. The index may be tied to a maximum size. In that case we may have to prune the list occasionally.

##### Parameters

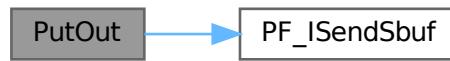
|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| <i>term</i>     | The term to be written                                    |
| <i>position</i> | The position in the file. Afterwards it is updated        |
| <i>fi</i>       | The file (or its cache) to which should be written        |
| <i>ncomp</i>    | Information about what type of compression should be used |

Definition at line 1450 of file [sort.c](#).

References [FiLe::handle](#), and [PF\\_ISendSbuf\(\)](#).

Referenced by [EndSort\(\)](#), [generate\\_expression\(\)](#), [MergePatches\(\)](#), [PF\\_EndSort\(\)](#), [PF\\_Processor\(\)](#), and [Processor\(\)](#).

Here is the call graph for this function:



#### 4.128.3.8 FlushOut()

```
int FlushOut (
    POSITION * position,
    FILEHANDLE * fi,
    int compr )
```

Completes output to an output file and writes the trailing zero.

##### Parameters

|                 |                                                          |
|-----------------|----------------------------------------------------------|
| <i>position</i> | The position in the file after writing                   |
| <i>fi</i>       | The file (or its cache)                                  |
| <i>compr</i>    | Indicates whether there should be compression with gzip. |

##### Returns

Regular conventions (OK -> 0).

Definition at line 1812 of file [sort.c](#).

References [FiLe::handle](#), [BrAcKeTiNfO::indexbuffer](#), and [PF\\_ISendSbuf\(\)](#).

Referenced by [EndSort\(\)](#), [MergePatches\(\)](#), [PF\\_EndSort\(\)](#), and [Processor\(\)](#).

Here is the call graph for this function:



#### 4.128.3.9 AddCoef()

```
int AddCoef (
    PHEAD WORD ** ps1,
    WORD ** ps2 )
```

Adds the coefficients of the terms \*ps1 and \*ps2. The problem comes when there is not enough space for a new longer coefficient. First a local solution is tried. If this is not successful we need to move terms around. The possibility of a garbage collection should not be ignored, as avoiding this costs very much extra space which is nearly wasted otherwise.

If the return value is zero the terms cancelled.

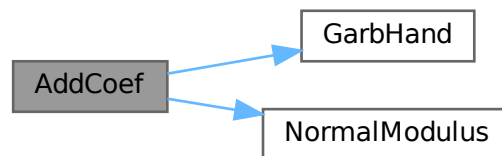
The resulting term is left in \*ps1.

Definition at line 2026 of file [sort.c](#).

References [GarbHand\(\)](#), and [NormalModulus\(\)](#).

Referenced by [SplitMerge\(\)](#).

Here is the call graph for this function:



#### 4.128.3.10 AddPoly()

```
int AddPoly (
    PHEAD WORD ** ps1,
    WORD ** ps2 )
```

Routine should be called when `S->PolyWise != 0`. It points then to the position of `AR.PolyFun` in both terms.

We add the contents of the arguments of the two polynomials. Special attention has to be given to special arguments. We have to reserve a space equal to the size of one term + the size of the argument of the other. The addition has to be done in this routine because not all objects are reentrant.

Newer addition (12-nov-2007). The `PolyFun` can have two arguments. In that case `S->PolyFlag` is 2 and we have to call the routine for adding rational polynomials. We have to be rather careful what happens with: The location of the output The order of the terms in the arguments At first we allow only univariate polynomials in the `PolyFun`. This restriction will be lifted a.s.a.p.

## Parameters

|            |                                              |
|------------|----------------------------------------------|
| <i>ps1</i> | A pointer to the position of the first term  |
| <i>ps2</i> | A pointer to the position of the second term |

## Returns

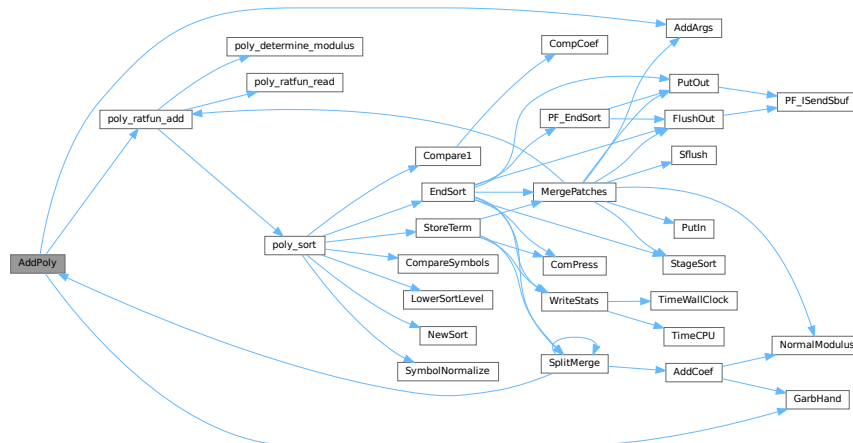
If zero the terms cancel. Otherwise the new term is in \*ps1.

Definition at line 2155 of file [sort.c](#).

References [AddArgs\(\)](#), [GarbHand\(\)](#), and [poly\\_ratfun\\_add\(\)](#).

Referenced by [SplitMerge\(\)](#).

Here is the call graph for this function:



## 4.128.3.11 AddArgs()

```

void AddArgs (
    PHEAD WORD * s1,
    WORD * s2,
    WORD * m )

```

Adds the arguments of two occurrences of the PolyFun.

## Parameters

|           |                                        |
|-----------|----------------------------------------|
| <i>s1</i> | Pointer to the first occurrence.       |
| <i>s2</i> | Pointer to the second occurrence.      |
| <i>m</i>  | Pointer to where the answer should be. |

Definition at line 2327 of file [sort.c](#).

Referenced by [AddPoly\(\)](#), and [MergePatches\(\)](#).

#### 4.128.3.12 Compare1()

```
WORD Compare1 (
    PHEAD WORD * term1,
    WORD * term2,
    WORD level )
```

Compares two terms. The answer is: 0 equal ( with exception of the coefficient if level == 0. ) >0 term1 comes first. <0 term2 comes first. Some special precautions may be needed to keep the CompCoef routine from generating overflows, although this is very unlikely in subterms. This routine should not return an error condition.

Originally this routine was called Compare. With the treatment of special polynomials with terms that contain only symbols and the need for extreme speed for the polynomial routines we made a special compare routine and now we store the address of the current compare routine in AR.CompareRoutine and have a macro Compare which makes all existing code work properly and we can just replace the routine on a thread by thread basis (each thread has its own AR struct).

##### Parameters

|              |                                                 |
|--------------|-------------------------------------------------|
| <i>term1</i> | First input term                                |
| <i>term2</i> | Second input term                               |
| <i>level</i> | The sorting level (may influence on the result) |

##### Returns

0 equal ( with exception of the coefficient if level == 0. ) >0 term1 comes first. <0 term2 comes first.

When there are floating point numbers active (float\_ = FLOATFUN) the presence of one or more float\_ functions is returned in AT.SortFloatMode: 0: no float\_ 1: float\_ in term1 only 2: float\_ in term2 only 3: float\_ in both terms

Definition at line 2620 of file [sort.c](#).

References [CompCoef\(\)](#).

Referenced by [InFunction\(\)](#), [poly\\_sort\(\)](#), and [StartVariables\(\)](#).

Here is the call graph for this function:



#### 4.128.3.13 CompareSymbols()

```
WORD CompareSymbols (
    PHEAD WORD * term1,
    WORD * term2,
    WORD par )
```

Compares the terms, based on the value of AN.polysortflag. If  $\text{term1} < \text{term2}$  the return value is -1 If  $\text{term1} > \text{term2}$  the return value is 1 If  $\text{term1} = \text{term2}$  the return value is 0 The coefficients may differ. The terms contain only a single subterm of type SYMBOL. If AN.polysortflag = 0 it is a 'regular' compare. If AN.polysortflag = 1 the sum of the powers is more important par is a dummy parameter to make the parameter field identical to that of Compare1 which is the regular compare routine in [sort.c](#)

Definition at line [3084](#) of file [sort.c](#).

Referenced by [InFunction\(\)](#), and [poly\\_sort\(\)](#).

#### 4.128.3.14 CompareHSymbols()

```
WORD CompareHSymbols (
    PHEAD WORD * term1,
    WORD * term2,
    WORD par )
```

Compares terms that can have only SYMBOL and HAAKJE subterms. If  $\text{term1} < \text{term2}$  the return value is -1 If  $\text{term1} > \text{term2}$  the return value is 1 If  $\text{term1} = \text{term2}$  the return value is 0 par is a dummy parameter to make the parameter field identical to that of Compare1 which is the regular compare routine in [sort.c](#)

Definition at line [3127](#) of file [sort.c](#).

#### 4.128.3.15 ComPress()

```
LONG ComPress (
    WORD ** ss,
    LONG * n )
```

Gets a list of pointers to terms and compresses the terms. In n it collects the number of terms and the return value of the function is the space that is occupied.

We have to pay some special attention to the compression of terms with a PolyFun. This PolyFun should occur only straight before the coefficient, so we can use the same trick as for the coefficient to sabotage compression of this object (Replace in the history the function pointer by zero. This is safe, because terms that would be identical otherwise would have been added).

##### Parameters

|           |                                              |
|-----------|----------------------------------------------|
| <i>ss</i> | Array of pointers to terms to be compressed. |
| <i>n</i>  | Number of pointers in ss.                    |

##### Returns

Total number of words needed for the compressed result.

Definition at line 3180 of file [sort.c](#).

Referenced by [EndSort\(\)](#), and [StoreTerm\(\)](#).

#### 4.128.3.16 SplitMerge()

```
LONG SplitMerge (
    PHEAD WORD ** Pointer,
    LONG number )
```

Algorithm by J.A.M.Vermaseren (31-7-1988)

Note that AN.SplitScratch and AN.InScratch are used also in GarbHand

Merge sort in memory. The input is an array of pointers. Sorting is done recursively by dividing the array in two equal parts and calling SplitMerge for each. When the parts are small enough we can do the compare and take the appropriate action. An addition is that we look for 'runs'. Sequences that are already ordered. This happens a lot when there is very little action in a module. This made FORM faster by a few percent.

##### Parameters

|                |                                                  |
|----------------|--------------------------------------------------|
| <i>Pointer</i> | The array of pointers to the terms to be sorted. |
| <i>number</i>  | The number of pointers in Pointer.               |

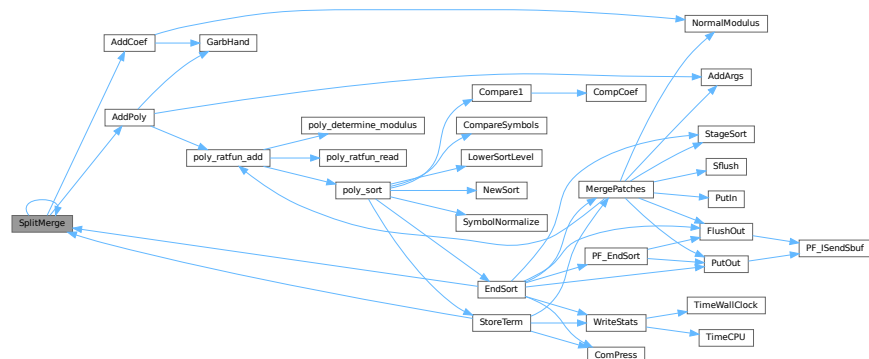
The terms are supposed to be sitting in the small buffer and there is supposed to be an extension to this buffer for when there are two terms that should be added and the result takes more space than each of the original terms. The notation guarantees that the result never needs more space than the sum of the spaces of the original terms.

Definition at line 3346 of file [sort.c](#).

References [AddCoef\(\)](#), [AddPoly\(\)](#), and [SplitMerge\(\)](#).

Referenced by [EndSort\(\)](#), [SplitMerge\(\)](#), and [StoreTerm\(\)](#).

Here is the call graph for this function:



**4.128.3.17 GarbHand()**

```
void GarbHand (
    void )
```

Garbage collection that takes place when the small extension is full and we need to place more terms there. When this is the case there are many holes in the small buffer and the whole can be compactified. The major complication is the buffer for SplitMerge. There are two options for temporary memory: 1: find some buffer that has enough space (maybe in the large buffer). 2: allocate a buffer. Give it back afterwards of course. If the small extension is properly dimensioned this routine should be called very rarely. Most of the time it will be called when the polyfun or polyratfun is active.

Definition at line 3626 of file [sort.c](#).

Referenced by [AddCoef\(\)](#), and [AddPoly\(\)](#).

**4.128.3.18 MergePatches()**

```
int MergePatches (
    WORD par )
```

The general merge routine. Can be used for the large buffer and the file merging. The array S->Patches tells where the patches start S->pStop tells where they end (has to be computed first). The end of a 'line to be merged' is indicated by a zero. If the end is reached without running into a zero or a term runs over the boundary of a patch it is a file merging operation and a new piece from the file is read in.

**Parameters**

|            |                                                                                                                                                                                    |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>par</i> | If <i>par</i> == 0 the sort is for file -> outputfile. If <i>par</i> == 1 the sort is for large buffer -> sortfile. If <i>par</i> == 2 the sort is for large buffer -> outputfile. |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Definition at line 3741 of file [sort.c](#).

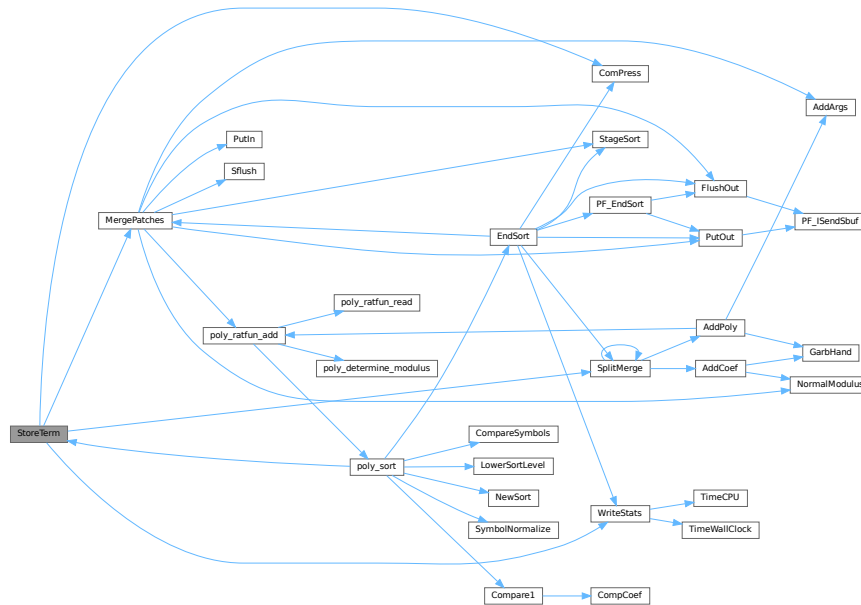
References [AddArgs\(\)](#), [FlushOut\(\)](#), [File::handle](#), [NormalModulus\(\)](#), [poly\\_ratfun\\_add\(\)](#), [PutIn\(\)](#), [PutOut\(\)](#), [Sflush\(\)](#), and [StageSort\(\)](#).

Referenced by [EndSort\(\)](#), and [StoreTerm\(\)](#).





Here is the call graph for this function:



#### 4.128.3.20 StageSort()

```
void StageSort (
    FILEHANDLE * fout )
```

Prepares a stage 4 or higher sort. Stage 4 sorts occur when the sort file contains more patches than can be merged in one pass.

Definition at line 4638 of file [sort.c](#).

References [FiLe::handle](#).

Referenced by [EndSort\(\)](#), and [MergePatches\(\)](#).

#### 4.128.3.21 SortWild()

```
int SortWild (
    WORD * w,
    WORD nw )
```

Sorts the wildcard entries in the parameter w. Double entries are removed. Full space taken is nw words. Routine serves for the reading of wildcards in the compiler. The entries come in the format: (type,4,number,0) in which the zero is reserved for the future replacement of 'number'.

##### Parameters

|           |                               |
|-----------|-------------------------------|
| <i>w</i>  | buffer with wildcard entries. |
| <i>nw</i> | number of wildcard entries.   |

## Returns

Normal conventions (OK -> 0)

Definition at line 4737 of file [sort.c](#).

### 4.128.3.22 CleanUpSort()

```
void CleanUpSort (
    int num )
```

Partially or completely frees function sort buffers.

Definition at line 4830 of file [sort.c](#).

### 4.128.3.23 LowerSortLevel()

```
void LowerSortLevel (
    void )
```

Lowers the level in the sort system.

Definition at line 4930 of file [sort.c](#).

Referenced by [generate\\_expression\(\)](#), [get\\_expression\(\)](#), [InFunction\(\)](#), [PF\\_CollectModifiedDollars\(\)](#), [PF\\_Processor\(\)](#), [poly\\_factorize\\_expression\(\)](#), [poly\\_sort\(\)](#), [poly\\_unfactorize\\_expression\(\)](#), [PolyFunMul\(\)](#), [Processor\(\)](#), and [TestMatch\(\)](#).

### 4.128.3.24 PolyRatFunSpecial()

```
WORD * PolyRatFunSpecial (
    PHEAD WORD * t1,
    WORD * t2 )
```

Definition at line 4947 of file [sort.c](#).

### 4.128.3.25 SimpleSplitMergeRec()

```
void SimpleSplitMergeRec (
    WORD * array,
    WORD num,
    WORD * auxarray )
```

Definition at line 5049 of file [sort.c](#).

### 4.128.3.26 SimpleSplitMerge()

```
void SimpleSplitMerge (
    WORD * array,
    WORD num )
```

Definition at line 5077 of file [sort.c](#).

#### 4.128.3.27 BinarySearch()

```
WORD BinarySearch (
    WORD * array,
    WORD num,
    WORD x )
```

Definition at line [5095](#) of file [sort.c](#).

### 4.128.4 Variable Documentation

#### 4.128.4.1 numcompares

```
LONG numcompares
```

Definition at line [75](#) of file [sort.c](#).

#### 4.128.4.2 toterms

```
char* toterms[] = { " ", " >>", "-->" }
```

Definition at line [83](#) of file [sort.c](#).

#### 4.128.4.3 humanTermsSuffix

```
const char humanTermsSuffix[HUMANSUFFLEN][4] = {"K ", "M ", "B ", "T "}
```

Definition at line [87](#) of file [sort.c](#).

#### 4.128.4.4 humanBytesSuffix

```
const char humanBytesSuffix[HUMANSUFFLEN][4] = {"KiB", "MiB", "GiB", "TiB"}
```

Definition at line [88](#) of file [sort.c](#).

## 4.129 sort.c

[Go to the documentation of this file.](#)

```

00001
00017 /* #[ License : */
00018 /*
00019 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00020 * When using this file you are requested to refer to the publication
00021 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00022 * This is considered a matter of courtesy as the development was paid
00023 * for by FOM the Dutch physics granting agency and we would like to
00024 * be able to track its scientific use to convince FOM of its value
00025 * for the community.
00026 *
00027 * This file is part of FORM.
00028 *
00029 * FORM is free software: you can redistribute it and/or modify it under the
00030 * terms of the GNU General Public License as published by the Free Software
00031 * Foundation, either version 3 of the License, or (at your option) any later
00032 * version.
00033 *
00034 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00035 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00036 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00037 * details.
00038 *
00039 * You should have received a copy of the GNU General Public License along
00040 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00041 */
00042 /* #[ License : */
00043 /*
00044 *#[ Includes : sort.c
00045
00046 Sort routines according to new conventions (25-jun-1997).
00047 This is more object oriented.
00048 The active sort is indicated by AT.SS which should agree with
00049 AN.FunSorts[AR.sLevel];
00050
00051 #define GZIPDEBUG
00052 */
00053 #define NEWSPLITMERGE
00054 /* Comment to turn off Timsort in SplitMerge for debugging: */
00055 #define NEWSPLITMERGETIMSORT
00056 /* During SplitMerge, print pointer array state on entry. Very spammy, for debugging. */
00057 /* #define SPLITMERGEDEBUG */
00058 /* Debug printing for GarbHand */
00059 /* #define TESTGARB */
00060
00061 #include "form3.h"
00062
00063 #ifdef WITHPTHREADS
00064 UBYTE THRbuf[100];
00065 #endif
00066
00067 #ifdef WITHSTATS
00068 extern LONG numwrites;
00069 extern LONG numreads;
00070 extern LONG numseeks;
00071 extern LONG nummallocs;
00072 extern LONG numfrees;
00073 #endif
00074
00075 LONG numcompares;
00076
00077 /*
00078 *#[ Includes :
00079 *#[ SortUtilities :
00080 *#[ WriteStats : void WriteStats(lspace,par,checkLogType)
00081 */
00082
00083 char *totterms[] = { " ", " ", " »", "-->" };
00084
00085 #define HUMANSTRLEN 12
00086 #define HUMANSUFFLEN 4
00087 const char humanTermsSuffix[HUMANSUFFLEN][4] = {"K ", "M ", "B ", "T "};
00088 const char humanBytesSuffix[HUMANSUFFLEN][4] = {"KiB", "MiB", "GiB", "TiB"};
00089 void HumanString(char* string, float input, const char suffix[HUMANSUFFLEN][4]) {
00090     int ind = -1;
00091     while (ind < 0 || (input >= 1000.0f && ind < HUMANSUFFLEN) ) {
00092         input /= 1000.0f;
00093         ind++;
00094     }
00095     if ( input <= 0.5f ) {
00096         snprintf(string, HUMANSTRLEN,
00097             " ( <1 %s)", suffix[ind]);

```

```

00098     }
00099     else {
00100         snprintf(string, HUMANSTRLEN,
00101             " (%3.f %s)", input, suffix[ind]);
00102     }
00103 }
00104
00121 void WriteStats(POSITION *plspace, WORD par, WORD checkLogType)
00122 {
00123     GETIDENTITY
00124     LONG millitime, y = 0x7FFFFFFFL >> 1;
00125     WORD timepart;
00126     SORTING *S;
00127     POSITION pp;
00128     int use_wtime;
00129     if ( AT.SS == AT.S0 && AC.StatsFlag ) {
00130 #ifdef WITHPTHREADS
00131         if ( AC.ThreadStats == 0 && identity > 0 ) return;
00132 #elif defined(WITHMPI)
00133         if ( AC.OldParallelStats ) return;
00134         if ( ! AC.ProcessStats && PF.me != MASTER ) return;
00135 #endif
00136         if ( Expressions == 0 ) return;
00137
00138         if ( par == STATSSPLITMERGE ) {
00139             if ( AC.ShortStatsMax == 0 ) return;
00140             AR.ShortSortCount++;
00141             if ( AR.ShortSortCount < AC.ShortStatsMax ) return;
00142         }
00143         AR.ShortSortCount = 0;
00144
00145         S = AT.SS;
00146
00147         char humanGenTermsText[HUMANSTRLEN] = "";
00148         char humanTermsLeftText[HUMANSTRLEN] = "";
00149         char humanBytesText[HUMANSTRLEN] = "";
00150         if ( AC.HumanStatsFlag ) {
00151             HumanString(humanGenTermsText, (float)(S->GenTerms), humanTermsSuffix);
00152             HumanString(humanTermsLeftText, (float)(S->TermsLeft), humanTermsSuffix);
00153             HumanString(humanBytesText, (float)(BASEPOSITION(*plspace)), humanBytesSuffix);
00154         }
00155
00156         MLOCK(ErrorMessageLock);
00157
00158         /* If the statistics should not go to the log file, temporarily hide the
00159          * LogHandle. This must be done within the ErrorMessageLock region to
00160          * avoid a data race between threads. */
00161         const WORD oldLogHandle = AC.LogHandle;
00162         if ( checkLogType && AM.LogType ) {
00163             AC.LogHandle = -1;
00164         }
00165
00166         if ( AC.ShortStats ) {}
00167         else {
00168 #ifdef WITHPTHREADS
00169             if ( identity > 0 ) {
00170                 MesPrint("                Thread %d reporting",identity);
00171             }
00172             else {
00173                 MesPrint("");
00174             }
00175 #elif defined(WITHMPI)
00176             if ( PF.me != MASTER ) {
00177                 MesPrint("                Process %d reporting",PF.me);
00178             }
00179             else {
00180                 MesPrint("");
00181             }
00182 #else
00183             MesPrint("");
00184 #endif
00185         }
00186         /*
00187          * We define WTimeStatsFlag as a flag to print the wall-clock time on
00188          * the *master*, not in workers. This can be confusing in thread
00189          * statistics when short statistics is used. Technically,
00190          * TimeWallClock() is not thread-safe in TFORM.
00191          */
00192         use_wtime = AC.WTimeStatsFlag;
00193 #if defined(WITHPTHREADS)
00194         if ( use_wtime && identity > 0 ) use_wtime = 0;
00195 #elif defined(WITHMPI)
00196         if ( use_wtime && PF.me != MASTER ) use_wtime = 0;
00197 #endif
00198         millitime = use_wtime ? TimeWallClock(1) * 10 : TimeCPU(1);
00199         timepart = (WORD)(millitime%1000);
00200         millitime /= 1000;

```

```

00201         timepart /= 10;
00202         if ( AC.ShortStats ) {
00203             #if defined(WITHPTHREADS) || defined(WITHMPI)
00204             #ifdef WITHPTHREADS
00205                 if ( identity > 0 ) {
00206                     #else
00207                     if ( PF.me != MASTER ) {
00208                         const int identity = PF.me;
00209                     #endif
00210                     if ( par == STATSSPLITMERGE || par == STATSPOSTSORT ) {
00211                         SETBASEPOSITION(pp,y);
00212                         if ( ISLESSPOS(*plspace,pp) ) {
00213                             MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%10p %s %s",identity,
00214                                 millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00215                                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00216                             /*
00217                             MesPrint("%d: %14s %17s %7l.%2is %8l>%10l%3s%10l:%10p",identity,
00218                                 EXPRNAME(AR.CurExpr),AC.Commercial,millitime,timepart,
00219                                 AN.ninterms,S->GenTerms,toterms[par],S->TermsLeft,plspace);
00220                             */
00221                         }
00222                     else {
00223                         y = 1000000000L;
00224                         SETBASEPOSITION(pp,y);
00225                         MULPOS(pp,100);
00226                         if ( ISLESSPOS(*plspace,pp) ) {
00227                             MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%11p %s %s",identity,
00228                                 millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00229                                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00230                         }
00231                     else {
00232                         MULPOS(pp,10);
00233                         if ( ISLESSPOS(*plspace,pp) ) {
00234                             MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%12p %s %s",identity,
00235                                 millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00236                                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00237                         }
00238                     else {
00239                         MULPOS(pp,10);
00240                         if ( ISLESSPOS(*plspace,pp) ) {
00241                             MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%13p %s %s",identity,
00242                                 millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00243                                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00244                         }
00245                     else {
00246                         MULPOS(pp,10);
00247                         if ( ISLESSPOS(*plspace,pp) ) {
00248                             MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%14p %s %s",identity,
00249                                 millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00250                                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00251                         }
00252                     else {
00253                         MULPOS(pp,10);
00254                         if ( ISLESSPOS(*plspace,pp) ) {
00255                             MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%15p %s %s",identity,
00256                                 millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00257                                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00258                         }
00259                     else {
00260                         MULPOS(pp,10);
00261                         if ( ISLESSPOS(*plspace,pp) ) {
00262                             MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%16p %s %s",identity,
00263                                 millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00264                                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00265                         }
00266                     else {
00267                         MULPOS(pp,10);
00268                         if ( ISLESSPOS(*plspace,pp) ) {
00269                             MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%17p %s %s",identity,
00270                                 millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00271                                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00272                         }
00273                     } } } } }
00274                 }
00275             }
00276         }
00277     else if ( par == STATSMERGETOFILE ) {
00278         SETBASEPOSITION(pp,y);
00279         if ( ISLESSPOS(*plspace,pp) ) {
00280             MesPrint("%d: %7l.%2is %10l:%10p",identity,millitime,timepart,
00281                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00282         }
00283     else {
00284         y = 1000000000L;
00285         SETBASEPOSITION(pp,y);
00286         MULPOS(pp,100);
00287         if ( ISLESSPOS(*plspace,pp) ) {

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```

00288         MesPrint("%d: %7l.%2is %10l:%11p",identity,millitime,timepart,
00289         S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00290     }
00291     else {
00292         MULPOS(pp,10);
00293         if ( ISLESSPOS(*plspace,pp) ) {
00294             MesPrint("%d: %7l.%2is %10l:%12p",identity,millitime,timepart,
00295             S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00296         }
00297         else {
00298             MULPOS(pp,10);
00299             if ( ISLESSPOS(*plspace,pp) ) {
00300                 MesPrint("%d: %7l.%2is %10l:%13p",identity,millitime,timepart,
00301                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00302             }
00303             else {
00304                 MULPOS(pp,10);
00305                 if ( ISLESSPOS(*plspace,pp) ) {
00306                     MesPrint("%d: %7l.%2is %10l:%14p",identity,millitime,timepart,
00307                     S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00308                 }
00309                 else {
00310                     MULPOS(pp,10);
00311                     if ( ISLESSPOS(*plspace,pp) ) {
00312                         MesPrint("%d: %7l.%2is %10l:%15p",identity,millitime,timepart,
00313                         S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00314                     }
00315                     else {
00316                         MULPOS(pp,10);
00317                         if ( ISLESSPOS(*plspace,pp) ) {
00318                             MesPrint("%d: %7l.%2is %10l:%16p",identity,millitime,timepart,
00319                             S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00320                         }
00321                         else {
00322                             MULPOS(pp,10);
00323                             if ( ISLESSPOS(*plspace,pp) ) {
00324                                 MesPrint("%d: %7l.%2is %10l:%17p",identity,millitime,timepart,
00325                                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00326                             }
00327                             } } } } }
00328     }
00329 } } else
00330 } } #endif
00331
00332 {
00333 if ( par == STATSSPLITMERGE || par == STATSPOSTSORT ) {
00334     SETBASEPOSITION(pp,y);
00335     if ( ISLESSPOS(*plspace,pp) ) {
00336         MesPrint("%7l.%2is %8l>%10l%3s%10l:%10p %s %s",
00337         millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00338         S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00339         /*
00340         MesPrint("%14s %17s %7l.%2is %8l>%10l%3s%10l:%10p",
00341         EXPRNAME(AR.CurExpr),AC.Commercial,millitime,timepart,
00342         AN.ninterms,S->GenTerms,toterms[par],S->TermsLeft,plspace);
00343         */
00344     }
00345     else {
00346         y = 1000000000L;
00347         SETBASEPOSITION(pp,y);
00348         MULPOS(pp,100);
00349         if ( ISLESSPOS(*plspace,pp) ) {
00350             MesPrint("%7l.%2is %8l>%10l%3s%10l:%11p %s %s",
00351             millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00352             S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00353         }
00354         else {
00355             MULPOS(pp,10);
00356             if ( ISLESSPOS(*plspace,pp) ) {
00357                 MesPrint("%7l.%2is %8l>%10l%3s%10l:%12p %s %s",
00358                 millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00359                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00360             }
00361             else {
00362                 MULPOS(pp,10);
00363                 if ( ISLESSPOS(*plspace,pp) ) {
00364                     MesPrint("%7l.%2is %8l>%10l%3s%10l:%13p %s %s",
00365                     millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00366                     S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00367                 }
00368                 else {
00369                     MULPOS(pp,10);
00370                     if ( ISLESSPOS(*plspace,pp) ) {
00371                         MesPrint("%7l.%2is %8l>%10l%3s%10l:%14p %s %s",
00372                         millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00373                         S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00374                     }

```



```

00375         else {
00376             MULPOS(pp,10);
00377             if ( ISLESSPOS(*plspace,pp) ) {
00378                 MesPrint("%7l.%2is %8l>%10l%3s%10l:%15p %s %s",
00379                     millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00380                     S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00381             }
00382         }
00383         else {
00384             MULPOS(pp,10);
00385             if ( ISLESSPOS(*plspace,pp) ) {
00386                 MesPrint("%7l.%2is %8l>%10l%3s%10l:%16p %s %s",
00387                     millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00388                     S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00389             }
00390         }
00391         else {
00392             MULPOS(pp,10);
00393             if ( ISLESSPOS(*plspace,pp) ) {
00394                 MesPrint("%7l.%2is %8l>%10l%3s%10l:%17p %s %s",
00395                     millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
00396                     S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00397             }
00398         }
00399     }
00400     else if ( par == STATSMERGETOFILE ) {
00401         SETBASEPOSITION(pp,y);
00402         if ( ISLESSPOS(*plspace,pp) ) {
00403             MesPrint("%7l.%2is %10l:%10p",millitime,timepart,
00404                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00405         }
00406         else {
00407             y = 1000000000L;
00408             SETBASEPOSITION(pp,y);
00409             MULPOS(pp,100);
00410             if ( ISLESSPOS(*plspace,pp) ) {
00411                 MesPrint("%7l.%2is %10l:%11p",millitime,timepart,
00412                     S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00413             }
00414             else {
00415                 MULPOS(pp,10);
00416                 if ( ISLESSPOS(*plspace,pp) ) {
00417                     MesPrint("%7l.%2is %10l:%12p",millitime,timepart,
00418                         S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00419                 }
00420                 else {
00421                     MULPOS(pp,10);
00422                     if ( ISLESSPOS(*plspace,pp) ) {
00423                         MesPrint("%7l.%2is %10l:%13p",millitime,timepart,
00424                             S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00425                     }
00426                     else {
00427                         MULPOS(pp,10);
00428                         if ( ISLESSPOS(*plspace,pp) ) {
00429                             MesPrint("%7l.%2is %10l:%14p",millitime,timepart,
00430                                 S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00431                         }
00432                         else {
00433                             MULPOS(pp,10);
00434                             if ( ISLESSPOS(*plspace,pp) ) {
00435                                 MesPrint("%7l.%2is %10l:%15p",millitime,timepart,
00436                                     S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00437                             }
00438                             else {
00439                                 MULPOS(pp,10);
00440                                 if ( ISLESSPOS(*plspace,pp) ) {
00441                                     MesPrint("%7l.%2is %10l:%16p",millitime,timepart,
00442                                         S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00443                                 }
00444                                 else {
00445                                     MULPOS(pp,10);
00446                                     if ( ISLESSPOS(*plspace,pp) ) {
00447                                         MesPrint("%7l.%2is %10l:%17p",millitime,timepart,
00448                                             S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
00449                                     }
00450                                 }
00451                             }
00452                         }
00453                     }
00454                 }
00455             }
00456         if ( par == STATSMERGETOFILE ) {
00457             if ( use_wtime ) {
00458                 MesPrint("WTime = %7l.%2i sec",millitime,timepart);
00459             }
00460             else {
00461                 MesPrint("Time = %7l.%2i sec",millitime,timepart);

```

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00462     }
00463     }
00464     else {
00465     #if ( BITSINLONG > 32 )
00466         if ( S->GenTerms >= 10000000000L ) {
00467             if ( use_wtime ) {
00468                 MesPrint("WTime = %7l.%2i sec    Generated terms = %16l%s",
00469                     millitime,timepart,S->GenTerms,humanGenTermsText);
00470             }
00471             else {
00472                 MesPrint("Time = %7l.%2i sec    Generated terms = %16l%s",
00473                     millitime,timepart,S->GenTerms,humanGenTermsText);
00474             }
00475         }
00476     else {
00477         if ( use_wtime ) {
00478             MesPrint("WTime = %7l.%2i sec    Generated terms = %10l%s",
00479                 millitime,timepart,S->GenTerms,humanGenTermsText);
00480         }
00481         else {
00482             MesPrint("Time = %7l.%2i sec    Generated terms = %10l%s",
00483                 millitime,timepart,S->GenTerms,humanGenTermsText);
00484         }
00485     }
00486 #else
00487     if ( use_wtime ) {
00488         MesPrint("WTime = %7l.%2i sec    Generated terms = %10l%s",
00489             millitime,timepart,S->GenTerms,humanGenTermsText);
00490     }
00491     else {
00492         MesPrint("Time = %7l.%2i sec    Generated terms = %10l%s",
00493             millitime,timepart,S->GenTerms,humanGenTermsText);
00494     }
00495 #endif
00496 }
00497 #if ( BITSINLONG > 32 )
00498     if ( par == STATSSPLITMERGE )
00499         if ( S->TermsLeft >= 10000000000L ) {
00500             MesPrint("%16s%8l Terms %s = %16l%s",EXPRNAME(AR.CurExpr),
00501                 AN.ninterms,FG.swmes[par],S->TermsLeft,humanTermsLeftText);
00502         }
00503         else {
00504             MesPrint("%16s%8l Terms %s = %10l%s",EXPRNAME(AR.CurExpr),
00505                 AN.ninterms,FG.swmes[par],S->TermsLeft,humanTermsLeftText);
00506         }
00507     else {
00508         if ( S->TermsLeft >= 10000000000L ) {
00509             #ifdef WITHPTHREADS
00510                 if ( identity > 0 && par == STATSPOSTSORT ) {
00511                     MesPrint("%16s          Terms in thread = %16l%s",
00512                         EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
00513                 }
00514                 else
00515                     #elif defined(WITHMPI)
00516                     if ( PF.me != MASTER && par == STATSPOSTSORT ) {
00517                         MesPrint("%16s          Terms in process= %16l%s",
00518                             EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
00519                     }
00520                     else
00521                         #endif
00522                         {
00523                             MesPrint("%16s          Terms %s = %16l%s",
00524                                 EXPRNAME(AR.CurExpr),FG.swmes[par],S->TermsLeft,humanTermsLeftText);
00525                         }
00526                 }
00527             else {
00528                 #ifdef WITHPTHREADS
00529                     if ( identity > 0 && par == STATSPOSTSORT ) {
00530                         MesPrint("%16s          Terms in thread = %10l%s",
00531                             EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
00532                     }
00533                     else
00534                         #elif defined(WITHMPI)
00535                         if ( PF.me != MASTER && par == STATSPOSTSORT ) {
00536                             MesPrint("%16s          Terms in process= %10l%s",
00537                                 EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
00538                         }
00539                         else
00540                             #endif
00541                             {
00542                                 MesPrint("%16s          Terms %s = %10l%s",
00543                                     EXPRNAME(AR.CurExpr),FG.swmes[par],S->TermsLeft,humanTermsLeftText);
00544                             }
00545                 }
00546             }
00547         }
00548     #else
00549         if ( par == STATSSPLITMERGE )

```

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00549         MesPrint("%16s%8l Terms %s = %10l%s",EXPRNAME(AR.CurExpr),
00550         AN.ninterms,FG.swmes[par],S->TermsLeft,humanTermsLeftText);
00551     else {
00552 #ifdef WITHPTHREADS
00553         if ( identity > 0 && par == STATSPOSTSORT ) {
00554             MesPrint("%16s          Terms in thread = %10l%s",
00555             EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
00556         }
00557     else
00558 #elif defined(WITHMPI)
00559         if ( PF.me != MASTER && par == STATSPOSTSORT ) {
00560             MesPrint("%16s          Terms in process= %10l%s",
00561             EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
00562         }
00563     else
00564 #endif
00565     {
00566         MesPrint("%16s          Terms %s = %10l%s",
00567         EXPRNAME(AR.CurExpr),FG.swmes[par],S->TermsLeft,humanTermsLeftText);
00568     }
00569 }
00570 #endif
00571 SETBASEPOSITION(pp,y);
00572 if ( ISLESSPOS(*plspace,pp) ) {
00573     MesPrint("%24s Bytes used      = %10p%s",AC.Commercial,plspace,humanBytesText);
00574 }
00575 else {
00576     y = 1000000000L;
00577     SETBASEPOSITION(pp,y);
00578     MULPOS(pp,100);
00579     if ( ISLESSPOS(*plspace,pp) ) {
00580         MesPrint("%24s Bytes used      =%11p%s",AC.Commercial,plspace,humanBytesText);
00581     }
00582     else {
00583         MULPOS(pp,10);
00584         if ( ISLESSPOS(*plspace,pp) ) {
00585             MesPrint("%24s Bytes used      =%12p%s",AC.Commercial,plspace,humanBytesText);
00586         }
00587         else {
00588             MULPOS(pp,10);
00589             if ( ISLESSPOS(*plspace,pp) ) {
00590                 MesPrint("%24s Bytes used      =%13p%s",AC.Commercial,plspace,humanBytesText);
00591             }
00592             else {
00593                 MULPOS(pp,10);
00594                 if ( ISLESSPOS(*plspace,pp) ) {
00595                     MesPrint("%24s Bytes used      =%14p%s",AC.Commercial,plspace,humanBytesText);
00596                 }
00597                 else {
00598                     MULPOS(pp,10);
00599                     if ( ISLESSPOS(*plspace,pp) ) {
00600                         MesPrint("%24s Bytes used      =%15p%s",AC.Commercial,plspace,humanBytesText);
00601                     }
00602                     else {
00603                         MULPOS(pp,10);
00604                         if ( ISLESSPOS(*plspace,pp) ) {
00605                             MesPrint("%24s Bytes used      =%16p%s",AC.Commercial,plspace,humanBytesText);
00606                         }
00607                         else {
00608                             MULPOS(pp,10);
00609                             if ( ISLESSPOS(*plspace,pp) ) {
00610                                 MesPrint("%24s Bytes used=%17p%s",AC.Commercial,plspace,humanBytesText);
00611                             }
00612                             } } } } }
00613     }
00614 } }
00615 #ifdef WITHSTATS
00616 MesPrint("Total number of writes: %l, reads: %l, seeks, %l"
00617         ,numwrites,numreads,numseeks);
00618 MesPrint("Total number of mallocs: %l, frees: %l"
00619         ,nummallocs,numfrees);
00620 #endif
00621 /* Put back the original LogHandle, it was changed if the statistics were
00622  * not printed in the log file. */
00623 AC.LogHandle = oldLogHandle;
00624
00625 MUNLOCK(ErrorMessageLock);
00626 }
00627 }
00628
00629 /*
00630     #] WriteStats :
00631     #[ NewSort :          WORD NewSort()
00632 */
00643 int NewSort(PHEAD0)
00644 {
00645     GETBIDENTITY

```

```

00646     SORTING *S, **newFS;
00647     int i, newsize;
00648     if ( AN.SoScratC == 0 )
00649         AN.SoScratC = (UWORD *)Malloc1(2*(AM.MaxTal+2)*sizeof(UWORD), "NewSort");
00650     AR.sLevel++;
00651     if ( AR.sLevel >= AN.NumFunSorts ) {
00652         if ( AN.NumFunSorts == 0 ) newsize = 100;
00653         else newsize = 2*AN.NumFunSorts;
00654         newFS = (SORTING **)Malloc1((newsize+1)*sizeof(SORTING *), "FunSort pointers");
00655         for ( i = 0; i < AN.NumFunSorts; i++ ) newFS[i] = AN.FunSorts[i];
00656         for ( ; i <= newsize; i++ ) newFS[i] = 0;
00657         if ( AN.FunSorts ) M_free(AN.FunSorts, "FunSort pointers");
00658         AN.FunSorts = newFS; AN.NumFunSorts = newsize;
00659     }
00660     if ( AR.sLevel == 0 ) {
00661         numcompares = 0;
00662
00663         AN.FunSorts[0] = AT.S0;
00664         if ( AR.PolyFun == 0 ) { AT.S0->PolyFlag = 0; }
00665         else if ( AR.PolyFunType == 1 ) { AT.S0->PolyFlag = 1; }
00666         else if ( AR.PolyFunType == 2 ) {
00667             if ( AR.PolyFunExp == 2
00668                 || AR.PolyFunExp == 3 ) AT.S0->PolyFlag = 1;
00669             else AT.S0->PolyFlag = 2;
00670         }
00671         AR.ShortSortCount = 0;
00672     }
00673     else {
00674         if ( AN.FunSorts[AR.sLevel] == 0 ) {
00675             AN.FunSorts[AR.sLevel] = AllocSort(
00676                 AM.SLargeSize, AM.SSmallSize, AM.SSmallEsize, AM.STermsInSmall
00677                 , AM.SMaxPatches, AM.SMaxFpatches, AM.SIOsize, 1);
00678         }
00679         AN.FunSorts[AR.sLevel]->PolyFlag = 0;
00680     }
00681     AT.SS = S = AN.FunSorts[AR.sLevel];
00682     S->sFill = S->sBuffer;
00683     S->lFill = S->lBuffer;
00684     S->lPatch = 0;
00685     S->fPatchN = 0;
00686     S->GenTerms = S->TermsLeft = S->GenSpace = S->SpaceLeft = 0;
00687     S->PoinFill = S->sPointer;
00688     *S->PoinFill = S->sFill;
00689     if ( AR.sLevel > 0 ) { S->PolyWise = 0; }
00690     PUTZERO(S->SizeInFile[0]); PUTZERO(S->SizeInFile[1]); PUTZERO(S->SizeInFile[2]);
00691     S->sTerms = 0;
00692     PUTZERO(S->file.POposition);
00693     S->stage4 = 0;
00694     if ( AR.sLevel > AN.MaxFunSorts ) AN.MaxFunSorts = AR.sLevel;
00695     /*
00696     The next variable is for the staged sort only.
00697     It should be treated differently
00698
00699     PUTZERO(AN.OldPosOut);
00700     */
00701     return(0);
00702 }
00703
00704 /*
00705 #] NewSort :
00706 #[ EndSort :          WORD EndSort(PHEAD buffer, par)
00707 */
00733 LONG EndSort(PHEAD WORD *buffer, int par)
00734 {
00735     GETBIDENTITY
00736     SORTING *S = AT.SS;
00737     WORD j, **ss, *to, *t;
00738     LONG sSpace, over, tover, spare, retval = 0;
00739     POSITION position, pp;
00740     off_t lSpace;
00741     FILEHANDLE *fout = 0, *oldoutfile = 0, *newout = 0;
00742
00743     if ( AM.exitflag && AR.sLevel == 0 ) return(0);
00744     #ifdef WITHMPI
00745     if( (retval = PF_EndSort()) > 0){
00746         oldoutfile = AR.outfile;
00747         retval = 0;
00748         goto RetRetval;
00749     }
00750     else if(retval < 0){
00751         retval = -1;
00752         goto RetRetval;
00753     }
00754     /* PF_EndSort returned 0: for S != AM.S0 and slaves still do the regular sort */
00755     #endif /* WITHMPI */
00756     oldoutfile = AR.outfile;

```

```

00757 /*      PolyFlag repair action
00758     if ( S == AT.S0 ) {
00759         if ( AR.PolyFun == 0 ) { S->PolyFlag = 0; }
00760         else if ( AR.PolyFunType == 1 ) { S->PolyFlag = 1; }
00761         else if ( AR.PolyFunType == 2 ) {
00762             if ( AR.PolyFunExp == 2
00763                 || AR.PolyFunExp == 3 ) S->PolyFlag = 1;
00764             else S->PolyFlag = 2;
00765         }
00766         S->PolyWise = 0;
00767     }
00768     else {
00769         S->PolyFlag = S->PolyWise = 0;
00770     }
00771 */
00772     S->PolyWise = 0;
00773     *(S->PoinFill) = 0;
00774 #ifdef SPLITTIME
00775     PrintTime((UBYTE *)"EndSort, before SplitMerge");
00776 #endif
00777     S->sPointer[SplitMerge(BHEAD S->sPointer,S->sTerms)] = 0;
00778 #ifdef SPLITTIME
00779     PrintTime((UBYTE *)"Endsort, after SplitMerge");
00780 #endif
00781     sSpace = 0;
00782     tover = over = S->sTerms;
00783     ss = S->sPointer;
00784     if ( over >= 0 ) {
00785         if ( S->lPatch > 0 || S->file.handle >= 0 ) {
00786             ss[over] = 0;
00787             sSpace = ComPress(ss,&sSpace);
00788             S->TermsLeft -= over - sSpace;
00789             if ( par == 1 ) { AR.outfile = newout = AllocFileHandle(0,(char *)0); }
00790         }
00791         else if ( S != AT.S0 ) {
00792             ss[over] = 0;
00793             if ( par == 2 ) {
00794                 sSpace = 3;
00795                 while ( ( t = *ss++ ) != 0 ) { sSpace += *t; }
00796                 if ( AN.tryterm > 0 && ( (sSpace+1)*sizeof(WORD) < (size_t)(AM.MaxTer) ) ) {
00797                     to = TermMalloc("$-sort space");
00798                 }
00799                 else {
00800                     LONG allocsp = sSpace+1;
00801                     if ( allocsp < MINALLOC ) allocsp = MINALLOC;
00802                     allocsp = (allocsp+7)/8*8;
00803                     to = (WORD *)Malloc1(allocsp*sizeof(WORD),"$-sort space");
00804                     if ( AN.tryterm > 0 ) AN.tryterm = 0;
00805                 }
00806                 *((WORD **)buffer) = to;
00807                 ss = S->sPointer;
00808                 while ( ( t = *ss++ ) != 0 ) {
00809                     j = *t; while ( --j >= 0 ) *to++ = *t++;
00810                 }
00811                 *to = 0;
00812                 retval = sSpace + 1;
00813             }
00814             else {
00815                 to = buffer;
00816                 sSpace = 0;
00817                 while ( ( t = *ss++ ) != 0 ) {
00818                     j = *t;
00819                     if ( ( sSpace += j ) > AM.MaxTer/((LONG)sizeof(WORD)) ) {
00820                         /* Too big! Get the total size for useful error message */
00821                         while ( ( t = *ss++ ) != 0 ) {
00822                             sSpace += *t;
00823                         }
00824                         MLOCK(ErrorMessageLock);
00825                         MesPrint("Sorted function argument too long (%d words). Increase MaxTermSize
00826 (%l words).", sSpace, AM.MaxTer/((LONG)sizeof(WORD)));
00827                         MUNLOCK(ErrorMessageLock);
00828                         retval = -1; goto RetRetVal;
00829                     }
00830                     while ( --j >= 0 ) *to++ = *t++;
00831                 }
00832                 *to = 0;
00833                 retval = to - buffer;
00834             }
00835             goto RetRetVal;
00836         }
00837         else {
00838             POSITION oldpos;
00839             if ( S == AT.S0 ) {
00840                 fout = AR.outfile;
00841                 *AR.CompressPointer = 0;
00842                 SeekScratch(AR.outfile,&position);

```

```

00843         else {
00844             fout = &(S->file);
00845             PUTZERO(position);
00846         }
00847         oldpos = position;
00848         S->TermsLeft = 0;
00849         /*
00850             Here we can go directly to the output.
00851         */
00852         #ifdef WITHZLIB
00853             { int oldgzipCompress = AR.gzipCompress;
00854               AR.gzipCompress = 0;
00855             #endif
00856             if ( tover > 0 ) {
00857                 ss = S->sPointer;
00858                 while ( ( t = *ss++ ) != 0 ) {
00859                     if ( *t ) S->TermsLeft++;
00860                 #ifdef WITHPTHREADS
00861                     if ( AS.MasterSort && ( fout == AR.outfile ) ) { PutToMaster(BHEAD t); }
00862                     else
00863                 #endif
00864                     if ( PutOut(BHEAD t,&position,fout,1) < 0 ) {
00865                         retval = -1; goto RetRetval;
00866                     }
00867                 }
00868             }
00869             #ifdef WITHPTHREADS
00870             if ( AS.MasterSort && ( fout == AR.outfile ) ) { PutToMaster(BHEAD 0); }
00871             else
00872         #endif
00873             if ( FlushOut(&position,fout,1) ) {
00874                 retval = -1; goto RetRetval;
00875             }
00876             #ifdef WITHZLIB
00877                 AR.gzipCompress = oldgzipCompress;
00878             }
00879             #endif
00880             #ifdef WITHPTHREADS
00881             if ( AS.MasterSort && ( fout == AR.outfile ) ) goto RetRetval;
00882             #endif
00883             #ifdef WITHMPI
00884             if ( PF.me != MASTER && PF.exprtodo < 0 ) goto RetRetval;
00885             #endif
00886             DIFPOS(oldpos,position,oldpos);
00887             S->SpaceLeft = BASEPOSITION(oldpos);
00888             WriteStats(&oldpos,STATSPOSTSORT,NOCHECKLOGTYPE);
00889             pp = oldpos;
00890             goto RetRetval;
00891         }
00892     }
00893     else if ( par == 1 && newout == 0 ) { AR.outfile = newout = AllocFileHandle(0,(char *)0); }
00894     sSpace++;
00895     lSpace = sSpace + (S->lFill - S->lBuffer) - (LONG)S->lPatch*(AM.MaxTer/sizeof(WORD));
00896     /* Note wrt MaxTer and lPatch: each patch starts with space for decompression */
00897     /* Not needed if only large buffer, but needed when using files (?) */
00898     SETBASEPOSITION(pp,lSpace);
00899     MULPOS(pp,sizeof(WORD));
00900     if ( S->file.handle >= 0 ) {
00901         ADD2POS(pp,S->fPatches[S->fPatchN]);
00902     }
00903     if ( S == AT.S0 ) {
00904         if ( S->lPatch > 0 || S->file.handle >= 0 ) {
00905             WriteStats(&pp,STATSSPLITMERGE,CHECKLOGTYPE);
00906         }
00907     }
00908     if ( par == 2 ) { AR.outfile = newout = AllocFileHandle(0,(char *)0); }
00909     if ( S->lPatch > 0 ) {
00910         if ( ( S->lPatch >= S->MaxPatches ) ||
00911             ( ( (WORD *)(((UBYTE *) (S->lFill + sSpace)) + 2*AM.MaxTer) ) >= S->lTop ) ) {
00912             /*
00913                 The large buffer is too full. Merge and write it
00914             */
00915             #ifdef GZIPDEBUG
00916                 MLOCK(ErrorMessageLock);
00917                 MesPrint("%w EndSort: lPatch = %d, MaxPatches = %d,lFill = %x, sSpace = %ld, MaxTer = %d,
00918 lTop = %x"
00919 ,S->lPatch,S->MaxPatches,S->lFill,sSpace,AM.MaxTer/sizeof(WORD),S->lTop);
00920                 MUNLOCK(ErrorMessageLock);
00921             #endif
00922             if ( MergePatches(1) ) {
00923                 MLOCK(ErrorMessageLock);
00924                 MesCall("EndSort");
00925                 MUNLOCK(ErrorMessageLock);
00926                 retval = -1; goto RetRetval;
00927             }
00928             S->lPatch = 0;

```

```

00929         pp = S->SizeInFile[1];
00930         MULPOS(pp, sizeof(WORD));
00931 #ifndef WITHPTHREADS
00932         if ( S == AT.S0 )
00933 #endif
00934         {
00935             POSITION pppp;
00936             SETBASEPOSITION(pppp, 0);
00937             SeekFile(S->file.handle, &pppp, SEEK_CUR);
00938             SeekFile(S->file.handle, &pp, SEEK_END);
00939             SeekFile(S->file.handle, &pppp, SEEK_SET);
00940             WriteStats(&pp, STATSMERGETOFILE, CHECKLOGTYPE);
00941             UpdateMaxSize();
00942         }
00943     }
00944     else {
00945         S->Patches[S->lPatch++] = S->lFill;
00946         to = (WORD *)(((UBYTE *) (S->lFill)) + AM.MaxTer);
00947         if ( tover > 0 ) {
00948             ss = S->sPointer;
00949             while ( ( t = *ss++ ) != 0 ) {
00950                 j = *t;
00951                 if ( j < 0 ) j = t[1] + 2;
00952                 while ( --j >= 0 ) *to++ = *t++;
00953             }
00954         }
00955         *to++ = 0;
00956         S->lFill = to;
00957         if ( S->file.handle < 0 ) {
00958             if ( MergePatches(2) ) {
00959                 MLOCK(ErrorMessageLock);
00960                 MesCall("EndSort");
00961                 MUNLOCK(ErrorMessageLock);
00962                 retval = -1; goto RetRetVal;
00963             }
00964             if ( S == AT.S0 ) {
00965                 pp = S->SizeInFile[2];
00966                 MULPOS(pp, sizeof(WORD));
00967 #ifdef WITHPTHREADS
00968                 if ( AS.MasterSort && ( fout == AR.outfile ) ) goto RetRetVal;
00969 #endif
00970                 WriteStats(&pp, STATSPOSTSORT, NOCHECKLOGTYPE);
00971                 UpdateMaxSize();
00972             }
00973             else {
00974                 if ( par == 2 && newout->handle >= 0 ) {
00975                     POSITION zeropos;
00976                     PUTZERO(zeropos);
00977 #ifdef ALLLOCK
00978                     LOCK(newout->pthreadlock);
00979 #endif
00980                     SeekFile(newout->handle, &zeropos, SEEK_SET);
00981                     to = (WORD *)Malloc1(BASEPOSITION(newout->filesize)+sizeof(WORD)*2
00982                                     , "$-buffer reading");
00983                     if ( AN.tryterm > 0 ) AN.tryterm = 0;
00984                     if ( ( retval = ReadFile(newout->handle, (UBYTE
00985 *)to, BASEPOSITION(newout->filesize)) ) !=
00986                         BASEPOSITION(newout->filesize) ) {
00987                         MLOCK(ErrorMessageLock);
00988                         MesPrint("Error reading information for $ variable");
00989                         MUNLOCK(ErrorMessageLock);
00990                         M_free(to, "$-buffer reading");
00991                         retval = -1;
00992                     }
00993                     else {
00994                         *((WORD **)buffer) = to;
00995                         retval /= sizeof(WORD);
00996                     }
00997 #ifdef ALLLOCK
00998                     UNLOCK(newout->pthreadlock);
00999 #endif
01000                 }
01001                 else if ( newout->handle >= 0 ) {
01002                     We land here if par == 1 (function arg sort) and PutOut has created a file.
01003                     This means that the term is larger than SIOsize, which we ensure is at least
01004                     as large as MaxTermSize in setfile.c. Thus we know already that the term won't
01005                     fit.
01006                     TooLarge:
01007                     MLOCK(ErrorMessageLock);
01008                     MesPrint("(1)Output should fit inside a single term. Increase MaxTermSize?");
01009                     MesCall("EndSort");
01010                     MUNLOCK(ErrorMessageLock);
01011                     retval = -1; goto RetRetVal;
01012                 }
01013             }

```

```

01014         t = newout->PObuffer;
01015         // We deal with the par == 2 case after RetRetval.
01016         if ( par != 2 ) {
01017             j = newout->POfill - t;
01018             to = buffer;
01019             if ( to >= AT.WorkSpace && to < AT.WorkTop && to+j > AT.WorkTop )
01020                 goto WorkspaceError;
01021             if ( j > AM.MaxTer ) {
01022                 MLOCK(ErrorMessageLock);
01023                 MesPrint("Encountered term of size: %d words.", j/(LONG)sizeof(WORD)
);
01024                 MUNLOCK(ErrorMessageLock);
01025                 goto TooLarge;
01026             }
01027             NCOPY(to,t,j);
01028             retval = to - buffer - 1;
01029         }
01030     }
01031 }
01032 goto RetRetval;
01033 }
01034 if ( MergePatches(1) ) { /* --> SortFile */
01035     MLOCK(ErrorMessageLock);
01036     MesCall("EndSort");
01037     MUNLOCK(ErrorMessageLock);
01038     retval = -1; goto RetRetval;
01039 }
01040 UpdateMaxSize();
01041 pp = S->SizeInFile[1];
01042 MULPOS(pp,sizeof(WORD));
01043 #ifndef WITHPTHREADS
01044     if ( S == AT.S0 )
01045 #endif
01046     {
01047         POSITION pppp;
01048         SETBASEPOSITION(pppp,0);
01049         SeekFile(S->file.handle,&pppp,SEEK_CUR);
01050         SeekFile(S->file.handle,&pp,SEEK_END);
01051         SeekFile(S->file.handle,&pppp,SEEK_SET);
01052         WriteStats(&pp,STATSMERGETOFILE,CHECKLOGTYPE);
01053     }
01054 #ifdef WITHEXRRXXX
01055     if ( S != AT.S0 ) {
01056         /*
01057             This is wrong! We have sorted to the sort file.
01058             Things are not sitting in the output yet.
01059         */
01060         if ( newout->handle >= 0 ) goto TooLarge;
01061         t = newout->PObuffer;
01062         j = newout->POfill - t;
01063         to = buffer;
01064         if ( to >= AT.WorkSpace && to < AT.WorkTop && to+j > AT.WorkTop )
01065             goto WorkspaceError;
01066         if ( j > AM.MaxTer ) goto TooLarge;
01067         NCOPY(to,t,j);
01068         goto RetRetval;
01069     }
01070 #endif
01071 }
01072 }
01073 if ( S->file.handle >= 0 ) {
01074     #ifdef GZIPDEBUG
01075         MLOCK(ErrorMessageLock);
01076         MesPrint("%w EndSort: fPatchN = %d, lPatch = %d, position = %12p"
, S->fPatchN, S->lPatch, &(S->fPatches[S->fPatchN]));
01078         MUNLOCK(ErrorMessageLock);
01079     #endif
01080     if ( S->lPatch <= 0 ) {
01081         StageSort(&(S->file));
01082         position = S->fPatches[S->fPatchN];
01083         ss = S->sPointer;
01084         if ( *ss ) {
01085             *AR.CompressPointer = 0;
01086         #ifdef WITHZLIB
01087             if ( S == AT.S0 && AR.NoCompress == 0 && AR.zipCompress > 0 )
01088                 S->fpcompressed[S->fPatchN] = 1;
01089             else
01090                 S->fpcompressed[S->fPatchN] = 0;
01091             SetupOutputGZIP(&(S->file));
01092         #endif
01093         while ( ( t = *ss++ ) != 0 ) {
01094             if ( PutOut(BHEAD t,&position,&(S->file),1) < 0 ) {
01095                 retval = -1; goto RetRetval;
01096             }
01097         }
01098         if ( FlushOut(&position,&(S->file),1) ) {
01099             retval = -1; goto RetRetval;

```



```

01100     }
01101     ++(S->fPatchN);
01102     S->fPatches[S->fPatchN] = position;
01103     UpdateMaxSize();
01104 #ifdef GZIPDEBUG
01105     MLOCK(ErrorMessageLock);
01106     MesPrint("%w EndSort+: fPatchN = %d, lPatch = %d, position = %12p"
01107             , S->fPatchN, S->lPatch, &(S->fPatches[S->fPatchN]));
01108     MUNLOCK(ErrorMessageLock);
01109 #endif
01110     }
01111     }
01112     AR.Stage4Name = 0;
01113 #ifdef WITHPTHREADS
01114     if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
01115         if ( S->file.handle >= 0 ) {
01116             SynchFile(S->file.handle);
01117         }
01118     }
01119 #endif
01120     UpdateMaxSize();
01121     if ( MergePatches(0) ) {
01122         MLOCK(ErrorMessageLock);
01123         MesCall("EndSort");
01124         MUNLOCK(ErrorMessageLock);
01125         retval = -1; goto RetRetVal;
01126     }
01127     S->stage4 = 0;
01128 #ifdef WITHPTHREADS
01129     if ( AS.MasterSort && ( fout == AR.outfile ) ) goto RetRetVal;
01130 #endif
01131     pp = S->SizeInFile[0];
01132     MULPOS(pp, sizeof(WORD));
01133     WriteStats(&pp, STATSPOSTSORT, NOCHECKLOGTYPE);
01134     UpdateMaxSize();
01135     }
01136 RetRetVal:
01137
01138 #ifdef WITHMPI
01139     /* NOTE: PF_EndSort has been changed such that it sets S->TermsLeft. (TU 30 Jun 2011) */
01140     if ( AR.sLevel == 0 && (PF.me == MASTER || PF.exprtodo >= 0) ) {
01141         Expressions[AR.CurExpr].counter = S->TermsLeft;
01142         Expressions[AR.CurExpr].size = pp;
01143     }
01144 #else
01145     if ( AR.sLevel == 0 ) {
01146         Expressions[AR.CurExpr].counter = S->TermsLeft;
01147         Expressions[AR.CurExpr].size = pp;
01148     } /*if ( AR.sLevel == 0 )*/
01149 #endif
01150     /*:[25nov2003 mt]*/
01151     if ( S->file.handle >= 0 && ( par != 1 ) && ( par != 2 ) ) {
01152         /* sortfile is still open */
01153         UpdateMaxSize();
01154 #ifdef WITHZLIB
01155         ClearSortGZIP(&(S->file));
01156 #endif
01157         CloseFile(S->file.handle);
01158         S->file.handle = -1;
01159         remove(S->file.name);
01160 #ifdef GZIPDEBUG
01161         MLOCK(ErrorMessageLock);
01162         MesPrint("%wEndSort: sortfile %s removed", S->file.name);
01163         MUNLOCK(ErrorMessageLock);
01164 #endif
01165     }
01166     AR.outfile = oldoutfile;
01167     AR.sLevel--;
01168     if ( AR.sLevel >= 0 ) AT.SS = AN.FunSorts[AR.sLevel];
01169     if ( par == 1 ) {
01170         if ( retval < 0 ) {
01171             UpdateMaxSize();
01172             if ( newout ) {
01173                 DeAllocFileHandle(newout);
01174                 newout = 0;
01175             }
01176         }
01177         else if ( newout ) {
01178             if ( newout->handle >= 0 ) {
01179                 MLOCK(ErrorMessageLock);
01180                 MesPrint("(2)Output should fit inside a single term. Increase MaxTermSize?");
01181                 MesCall("EndSort");
01182                 MUNLOCK(ErrorMessageLock);
01183                 Terminate(-1);
01184             }
01185             else if ( newout->POfill > newout->PObuffer ) {
01186                 /*

```

```

01187         Here we have to copy the contents of the 'file' into
01188         the buffer. We assume that this buffer lies in the Workspace.
01189         Hence
01190     */
01191     j = newout->POfill-newout->PObuffer;
01192     if ( buffer >= AT.WorkSpace && buffer < AT.WorkTop && buffer+j > AT.WorkTop )
01193         goto WorkspaceError;
01194     else {
01195         to = buffer; t = newout->PObuffer;
01196         while ( j-- > 0 ) *to++ = *t++;
01197     }
01198     UpdateMaxSize();
01199 }
01200 DeAllocFileHandle(newout);
01201 newout = 0;
01202 }
01203 }
01204 else if ( par == 2 ) {
01205     if ( newout ) {
01206         if ( retval == 0 ) {
01207             if ( newout->handle >= 0 ) {
01208 /*
01209             output resides at the moment in a file
01210             Find the size, make a buffer, copy into the buffer and clean up.
01211 */
01212             POSITION zeropos;
01213             PUTZERO(position);
01214 #ifdef ALLLOCK
01215             LOCK(newout->pthreadlock);
01216 #endif
01217             SeekFile(newout->handle,&position,SEEK_END);
01218             PUTZERO(zeropos);
01219             SeekFile(newout->handle,&zeropos,SEEK_SET);
01220             to = (WORD *)Malloca(BASEPOSITION(position)+sizeof(WORD)*3
01221             ,"$-buffer reading");
01222             if ( AN.tryterm > 0 ) AN.tryterm = 0;
01223             if ( ( retval = ReadFile(newout->handle,(UBYTE *)to,BASEPOSITION(position)) ) !=
01224             BASEPOSITION(position) ) {
01225                 MLOCK(ErrorMessageLock);
01226                 MesPrint("Error reading information for $ variable");
01227                 MUNLOCK(ErrorMessageLock);
01228                 M_free(to,"$-buffer reading");
01229                 retval = -1;
01230             }
01231             else {
01232                 *((WORD **)buffer) = to;
01233                 retval /= sizeof(WORD);
01234             }
01235 #ifdef ALLLOCK
01236             UNLOCK(newout->pthreadlock);
01237 #endif
01238         }
01239         else {
01240 /*
01241             output resides in the cache buffer and the file was never opened
01242 */
01243             LONG wsiz = newout->POfill - newout->PObuffer;
01244             if ( AN.tryterm > 0 && ( (wsiz+2)*sizeof(WORD) < (size_t)(AM.MaxTer) ) ) {
01245                 to = TermMalloc("$-sort space");
01246             }
01247             else {
01248                 LONG allocsp = wsiz+2;
01249                 if ( allocsp < MINALLOCC ) allocsp = MINALLOCC;
01250                 allocsp = ((allocsp+7)/8)*8;
01251                 to = (WORD *)Malloca(allocsp*sizeof(WORD),"$-buffer reading");
01252                 if ( AN.tryterm > 0 ) AN.tryterm = 0;
01253             }
01254             *((WORD **)buffer) = to; t = newout->PObuffer;
01255             retval = wsiz;
01256             NCOPY(to,t,wsiz);
01257         }
01258     }
01259     UpdateMaxSize();
01260     DeAllocFileHandle(newout);
01261     newout = 0;
01262 }
01263 }
01264 else {
01265     if ( newout ) {
01266         DeAllocFileHandle(newout);
01267         newout = 0;
01268     }
01269 }
01270 /*
01271 if ( AR.sLevel < 0 ) {
01272     MesPrint(" number of calls to compare was %1",numcompares);
01273 }

```

```

01274 */
01275     return(retval);
01276 WorkspaceError:
01277     MLOCK(ErrorMessageLock);
01278     MesWork();
01279     MesCall("EndSort");
01280     MUNLOCK(ErrorMessageLock);
01281     Terminate(-1);
01282     return(-1);
01283 }
01284
01285 /*
01286     #] EndSort :
01287     #[ PutIn :                LONG PutIn(handle,position,buffer,take,npat)
01288 */
01304 LONG PutIn(FILEHANDLE *file, POSITION *position, WORD *buffer, WORD **take, int npat)
01305 {
01306     LONG i, RetCode;
01307     WORD *from, *to;
01308     #ifndef WITHZLIB
01309         DUMMYUSE(npat);
01310     #endif
01311     from = buffer + ( file->POsize * sizeof(UBYTE) )/sizeof(WORD);
01312     i = from - *take;
01313     if ( i*((LONG)(sizeof(WORD))) > AM.MaxTer ) {
01314         MLOCK(ErrorMessageLock);
01315         MesPrint("Problems in PutIn");
01316         MUNLOCK(ErrorMessageLock);
01317         Terminate(-1);
01318     }
01319     to = buffer;
01320     while ( --i >= 0 ) *--to = *--from;
01321     *take = to;
01322     #ifdef WITHZLIB
01323         if ( ( RetCode = FillInputGZIP(file,position,(UBYTE *)buffer
01324             ,file->POsize,npat) ) < 0 ) {
01325             MLOCK(ErrorMessageLock);
01326             MesPrint("PutIn: We have RetCode = %x while reading %x bytes",
01327                 RetCode,file->POsize);
01328             MUNLOCK(ErrorMessageLock);
01329             Terminate(-1);
01330         }
01331     #else
01332         #ifdef ALLLOCK
01333             LOCK(file->pthreadlock);
01334         #endif
01335         SeekFile(file->handle,position,SEEK_SET);
01336         if ( ( RetCode = ReadFile(file->handle,(UBYTE *)buffer,file->POsize) ) < 0 ) {
01337             #ifdef ALLLOCK
01338                 UNLOCK(file->pthreadlock);
01339             #endif
01340             MLOCK(ErrorMessageLock);
01341             MesPrint("PutIn: We have RetCode = %x while reading %x bytes",
01342                 RetCode,file->POsize);
01343             MUNLOCK(ErrorMessageLock);
01344             Terminate(-1);
01345         }
01346         #ifdef ALLLOCK
01347             UNLOCK(file->pthreadlock);
01348         #endif
01349     #endif
01350     return(RetCode);
01351 }
01352
01353 /*
01354     #] PutIn :
01355     #[ Sflush :                WORD Sflush(file)
01356 */
01364 int Sflush(FILEHANDLE *fi)
01365 {
01366     LONG size, RetCode;
01367     #ifdef WITHZLIB
01368         GETIDENTITY
01369         int dobracketindex = 0;
01370         if ( AR.sLevel <= 0 && Expressions[AR.CurExpr].newbracketinfo
01371             && ( fi == AR.outfile || fi == AR.hidefile ) ) dobracketindex = 1;
01372     #endif
01373     if ( fi->handle < 0 ) {
01374         if ( ( RetCode = CreateFile(fi->name) ) >= 0 ) {
01375             #ifdef GZIPDEBUG
01376                 MLOCK(ErrorMessageLock);
01377                 MesPrint("%w Sflush created scratch file %s",fi->name);
01378                 MUNLOCK(ErrorMessageLock);
01379             #endif
01380             fi->handle = (WORD)RetCode;
01381             PUTZERO(fi->filesize);
01382             PUTZERO(fi->POposition);

```

```

01383     }
01384     else {
01385         MLOCK(ErrorMessageLock);
01386         MesPrint("Cannot create scratch file %s",fi->name);
01387         MUNLOCK(ErrorMessageLock);
01388         return(-1);
01389     }
01390 }
01391 #ifdef WITHZLIB
01392 if ( AT.SS == AT.S0 && !AR.NoCompress && AR.gzipCompress > 0
01393 && dobracketindex == 0 ) {
01394     if ( FlushOutputGZIP(fi) ) return(-1);
01395     fi->POfill = fi->PObuffer;
01396 }
01397 else
01398 #endif
01399 {
01400     #ifdef ALLLOCK
01401         LOCK(fi->pthreadlock);
01402     #endif
01403     size = (fi->POfill-fi->PObuffer)*sizeof(WORD);
01404     SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
01405     if ( WriteFile(fi->handle,(UBYTE *) (fi->PObuffer),size) != size ) {
01406         #ifdef ALLLOCK
01407             UNLOCK(fi->pthreadlock);
01408         #endif
01409         MLOCK(ErrorMessageLock);
01410         MesPrint("Write error while finishing sort. Disk full?");
01411         MUNLOCK(ErrorMessageLock);
01412         return(-1);
01413     }
01414     ADDPOS(fi->filesize,size);
01415     ADDPOS(fi->POposition,size);
01416     fi->POfill = fi->PObuffer;
01417     #ifdef ALLLOCK
01418         UNLOCK(fi->pthreadlock);
01419     #endif
01420 }
01421 return(0);
01422 }
01423
01424 /*
01425     #] Sflush :
01426     #[ PutOut :                WORD PutOut(term,position,file,ncomp)
01427 */
01450 WORD PutOut(PHEAD WORD *term, POSITION *position, FILEHANDLE *fi, WORD ncomp)
01451 {
01452     GETBIDENTITY
01453     WORD i, *p, ret, *r, *rr, j, k, first;
01454     int dobracketindex = 0;
01455     LONG RetCode;
01456
01457     if ( AT.SS != AT.S0 ) {
01458         /*
01459             For this case no compression should be used
01460         */
01461         if ( ( i = *term ) <= 0 ) return(0);
01462         ret = i;
01463         ADDPOS(*position,i*sizeof(WORD));
01464         p = fi->POfill;
01465         do {
01466             if ( p >= fi->POstop ) {
01467                 if ( fi->handle < 0 ) {
01468                     if ( ( RetCode = CreateFile(fi->name) ) >= 0 ) {
01469                         #ifdef GZIPDEBUG
01470                             MLOCK(ErrorMessageLock);
01471                             MesPrint("%w PutOut created sortfile %s",fi->name);
01472                             MUNLOCK(ErrorMessageLock);
01473                         #endif
01474                         fi->handle = (WORD)RetCode;
01475                         PUTZERO(fi->filesize);
01476                         PUTZERO(fi->POposition);
01477                     /*
01478                         Should not be here anymore?
01479                     */
01479                     #ifdef WITHZLIB
01480                         fi->ziobuffer = 0;
01481                     #endif
01482                 /*
01483                     }
01484                 else {
01485                     MLOCK(ErrorMessageLock);
01486                     MesPrint("Cannot create scratch file %s",fi->name);
01487                     MUNLOCK(ErrorMessageLock);
01488                     return(-1);
01489                 }
01490             }
01491         #ifdef ALLLOCK

```

```

01492                LOCK(fi->pthreadlock);
01493 #endif
01494                if ( fi == AR.hidefile ) {
01495                    LOCK(AS.inputslock);
01496                }
01497                SeekFile(fi->handle,&(fi->PPosition),SEEK_SET);
01498                if ( ( RetCode = WriteFile(fi->handle,(UBYTE *) (fi->PObuffer),fi->POsize) ) !=
fi->POsize ) {
01499                    if ( fi == AR.hidefile ) {
01500                        UNLOCK(AS.inputslock);
01501                    }
01502 #ifdef ALLLOCK
01503                    UNLOCK(fi->pthreadlock);
01504 #endif
01505                    MLOCK(ErrorMessageLock);
01506                    MesPrint("Write error during sort. Disk full?");
01507                    MesPrint("Attempt to write %l bytes on file %d at position %l5p",
01508                        fi->POsize,fi->handle,&(fi->PPosition));
01509                    MesPrint("RetCode = %l, Buffer address = %l",RetCode,(LONG) (fi->PObuffer));
01510                    MUNLOCK(ErrorMessageLock);
01511                    return(-1);
01512                }
01513                ADDPOS(fi->filesize,fi->POsize);
01514                p = fi->PObuffer;
01515                ADDPOS(fi->PPosition,fi->POsize);
01516                if ( fi == AR.hidefile ) {
01517                    UNLOCK(AS.inputslock);
01518                }
01519 #ifdef ALLLOCK
01520                UNLOCK(fi->pthreadlock);
01521 #endif
01522 #ifdef WITHPTHREADS
01523                if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
01524                    if ( fi->handle >= 0 ) SynchFile(fi->handle);
01525                }
01526 #endif
01527                }
01528                *p++ = *term++;
01529            } while ( --i > 0 );
01530            fi->POfull = fi->POfill = p;
01531            return(ret);
01532        }
01533        if ( ( AP.PreDebug & DUMPOUTTERMS ) == DUMPOUTTERMS ) {
01534            MLOCK(ErrorMessageLock);
01535 #ifdef WITHPTHREADS
01536            snprintf((char *) (THRbuf),100,"PutOut(%d)",AT.identity);
01537            PrintTerm(term,(char *) (THRbuf));
01538 #else
01539            PrintTerm(term,"PutOut");
01540 #endif
01541            MesPrint("ncomp = %d, AR.NoCompress = %d, AR.sLevel = %d",ncomp,AR.NoCompress,AR.sLevel);
01542            MesPrint("File %s, position %p",fi->name,position);
01543            MUNLOCK(ErrorMessageLock);
01544        }
01545
01546        if ( AR.sLevel <= 0 && Expressions[AR.CurExpr].newbracketinfo
01547            && ( fi == AR.outfile || fi == AR.hidefile ) ) dobracketindex = 1;
01548        r = rr = AR.CompressPointer;
01549        first = j = k = ret = 0;
01550        if ( ( i = *term ) != 0 ) {
01551            if ( i < 0 ) { /* Compressed term */
01552                i = term[1] + 2;
01553                if ( fi == AR.outfile || fi == AR.hidefile ) {
01554                    MLOCK(ErrorMessageLock);
01555                    MesPrint("Ran into precompressed term");
01556                    MUNLOCK(ErrorMessageLock);
01557                    Crash();
01558                    return(-1);
01559                }
01560            }
01561            else if ( !AR.NoCompress && ( ncomp > 0 ) && AR.sLevel <= 0 ) { /* Must compress */
01562                if ( dobracketindex ) {
01563                    PutBracketInIndex(BHEAD term,position);
01564                }
01565                j = *r++ - 1;
01566                p = term + 1;
01567                i--;
01568                if ( AR.PolyFun ) {
01569                    WORD *polystop, *sa;
01570                    sa = p + i;
01571                    sa -= ABS(sa[-1]);
01572                    polystop = p;
01573                    while ( polystop < sa && *polystop != AR.PolyFun ) {
01574                        polystop += polystop[1];
01575                    }
01576                    if ( polystop < sa ) {
01577                        if ( AR.PolyFunType == 2 ) polystop[2] &= ~MUSTCLEANPRF;

```

```

01578         while ( i > 0 && j > 0 && *p == *r && p < polystop ) {
01579             i--; j--; k--; p++; r++;
01580         }
01581     }
01582     else {
01583         while ( i > 0 && j > 0 && *p == *r && p < sa ) { i--; j--; k--; p++; r++; }
01584     }
01585 }
01586 #ifdef WITHFLOAT
01587     else if ( AC.DefaultPrecision ) {
01588         WORD *floatstop, *sa;
01589         sa = p + i;
01590         sa -= ABS(sa[-1]);
01591         floatstop = p;
01592         while ( floatstop < sa && *floatstop != FLOATFUN ) {
01593             floatstop += floatstop[1];
01594         }
01595         if ( floatstop < sa ) {
01596             while ( i > 0 && j > 0 && *p == *r && p < floatstop ) {
01597                 i--; j--; k--; p++; r++;
01598             }
01599         }
01600         else {
01601             while ( i > 0 && j > 0 && *p == *r && p < sa ) { i--; j--; k--; p++; r++; }
01602         }
01603     }
01604 #endif
01605     else {
01606         WORD *sa;
01607         sa = p + i;
01608         sa -= ABS(sa[-1]);
01609         while ( i > 0 && j > 0 && *p == *r && p < sa ) { i--; j--; k--; p++; r++; }
01610     }
01611     if ( k > -2 ) {
01612 nocompress:
01613         j = i = *term;
01614         k = 0;
01615         p = term;
01616         r = rr;
01617         NCOPY(r,p,j);
01618     }
01619     else {
01620         *rr = *term;
01621         term = p;
01622         j = i;
01623         NCOPY(r,p,j);
01624         j = i;
01625         i += 2;
01626         first = 2;
01627     }
01628 /* Sabotage getting into the coefficient next time */
01629 r[-(ABS(r[-1]))] = 0;
01630 if ( r >= AR.ComprTop ) {
01631     MLOCK(ErrorMessageLock);
01632     MesPrint("CompressSize of %10l is insufficient",AM.CompressSize);
01633     MUNLOCK(ErrorMessageLock);
01634     Crash();
01635     return(-1);
01636 }
01637 }
01638 else if ( !AR.NoCompress && ( ncomp < 0 ) && AR.sLevel <= 0 ) {
01639     /* No compress but put in compress buffer anyway */
01640     if ( dobracketindex ) {
01641         PutBracketInIndex(BHEAD term,position);
01642     }
01643     j = *r++ - 1;
01644     p = term + 1;
01645     i--;
01646     if ( AR.PolyFun ) {
01647         WORD *polystop, *sa;
01648         sa = p + i;
01649         sa -= ABS(sa[-1]);
01650         polystop = p;
01651         while ( polystop < sa && *polystop != AR.PolyFun ) {
01652             polystop += polystop[1];
01653         }
01654         if ( polystop < sa ) {
01655             if ( AR.PolyFunType == 2 ) polystop[2] &= ~MUSTCLEANPRF;
01656             while ( i > 0 && j > 0 && *p == *r && p < polystop ) {
01657                 i--; j--; k--; p++; r++;
01658             }
01659         }
01660         else {
01661             while ( i > 0 && j > 0 && *p == *r ) { i--; j--; k--; p++; r++; }
01662         }
01663     }
01664     else {

```

```

01665         while ( i > 0 && j > 0 && *p == *r ) { i--; j--; k--; p++; r++; }
01666     }
01667     goto nocompress;
01668 }
01669 else {
01670     if ( AR.PolyFunType == 2 ) {
01671         WORD *t, *tstop;
01672         tstop = term + *term;
01673         tstop -= ABS(tstop[-1]);
01674         t = term+1;
01675         while ( t < tstop ) {
01676             if ( *t == AR.PolyFun ) {
01677                 t[2] &= ~MUSTCLEANPRF;
01678             }
01679             t += t[1];
01680         }
01681     }
01682     if ( dobracketindex ) {
01683         PutBracketInIndex(BHEAD term,position);
01684     }
01685 }
01686 ret = i;
01687 ADDPOS(*position,i*sizeof(WORD));
01688 p = fi->POfill;
01689 do {
01690     if ( p >= fi->POstop ) {
01691 #ifdef WITHMPI /* [16mar1998 ar] */
01692         if ( PF.me != MASTER && AR.sLevel <= 0 && (fi == AR.outfile || fi == AR.hidefile) &&
PF.parallel && PF.exprtodo < 0 ) {
01693             PF_BUFFER *sbuf = PF.sbuf;
01694             sbuf->fill[sbuf->active] = fi->POstop;
01695             PF_ISendSbuf(MASTER,PF_BUFFER_MSGTAG);
01696             p = fi->PObuffer = fi->POfill = fi->POfull =
sbuf->buff[sbuf->active];
01697             fi->POstop = sbuf->stop[sbuf->active];
01698         }
01699     }
01700     else
01701 #endif /* WITHMPI [16mar1998 ar] */
01702     {
01703         if ( fi->handle < 0 ) {
01704             if ( ( RetCode = CreateFile(fi->name) ) >= 0 ) {
01705 #ifdef GZIPDEBUG
01706                 MLOCK(ErrorMessageLock);
01707                 MesPrint("%w PutOut created sortfile %s",fi->name);
01708                 MUNLOCK(ErrorMessageLock);
01709 #endif
01710                 fi->handle = (WORD)RetCode;
01711                 PUTZERO(fi->filesize);
01712                 PUTZERO(fi->POposition);
01713             /*
01714              Should not be here?
01715             */
01716             #ifdef WITHZLIB
01717                 fi->ziobuffer = 0;
01718             #endif
01719         }
01720         else {
01721             MLOCK(ErrorMessageLock);
01722             MesPrint("Cannot create scratch file %s",fi->name);
01723             MUNLOCK(ErrorMessageLock);
01724             return(-1);
01725         }
01726     }
01727 #ifdef WITHZLIB
01728     if ( !AR.NoCompress && ncomp > 0 && AR.gzipCompress > 0
&& dobracketindex == 0 && fi->zsp != 0 ) {
01729         fi->POfill = p;
01730         if ( PutOutputGZIP(fi) ) return(-1);
01731         p = fi->PObuffer;
01732     }
01733     else
01734 #endif
01735     {
01736         if ( fi == AR.hidefile ) {
01737             LOCK(AS.inputslock);
01738         }
01739         SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
01740         if ( ( RetCode = WriteFile(fi->handle,(UBYTE *) (fi->PObuffer),fi->POsize) ) !=
fi->POsize ) {
01741             if ( fi == AR.hidefile ) {
01742                 UNLOCK(AS.inputslock);
01743             }
01744             #ifdef ALLLOCK
01745                 UNLOCK(fi->pthreadlock);
01746             #endif

```

```

01750 #endif
01751         MLOCK(ErrorMessageLock);
01752         MesPrint("Write error during sort. Disk full?");
01753         MesPrint("Attempt to write %l bytes on file %d at position %15p",
01754                 fi->POsize, fi->handle, &(fi->POposition));
01755         MesPrint("RetCode = %l, Buffer address = %l", RetCode, (LONG) (fi->PObuffer));
01756         MUNLOCK(ErrorMessageLock);
01757         return(-1);
01758     }
01759     ADDPOS(fi->filesize, fi->POsize);
01760     p = fi->PObuffer;
01761     ADDPOS(fi->POposition, fi->POsize);
01762     if ( fi == AR.hidefile ) {
01763         UNLOCK(AS.inputslock);
01764     }
01765 #ifdef ALLLOCK
01766     UNLOCK(fi->pthreadlock);
01767 #endif
01768 #ifdef WITHPTHREADS
01769     if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
01770         if ( fi->handle >= 0 ) SynchFile(fi->handle);
01771     }
01772 #endif
01773     }
01774     }
01775     }
01776     if ( first ) {
01777         if ( first == 2 ) *p++ = k;
01778         else *p++ = j;
01779         first--;
01780     }
01781     else *p++ = *term++;
01782 /*
01783     if ( AP.DebugFlag ) {
01784         TalToLine((UWORD) (p[-1])); TokenToLine((UBYTE *) " ");
01785     }
01786 */
01787     } while ( --i > 0 );
01788     fi->POfull = fi->POfill = p;
01789 }
01790 /*
01791     if ( AP.DebugFlag ) {
01792         AO.OutSkip = 0;
01793         FiniLine();
01794     }
01795 */
01796     return(ret);
01797 }
01798
01799 /*
01800     #] PutOut :
01801     #[ FlushOut :                WORD FlushOut(position, file, compr)
01802 */
01812 int FlushOut(POSITION *position, FILEHANDLE *fi, int compr)
01813 {
01814     GETIDENTITY
01815     LONG size, RetCode;
01816     int dobracketindex = 0;
01817 #ifndef WITHZLIB
01818     DUMMYUSE(compr);
01819 #endif
01820     if ( AR.sLevel <= 0 && Expressions[AR.CurExpr].newbracketinfo
01821         && ( fi == AR.outfile || fi == AR.hidefile ) ) dobracketindex = 1;
01822 #ifndef WITHMPI /* [16mar1998 ar] */
01823     if ( PF.me != MASTER && AR.sLevel <= 0 && (fi == AR.outfile || fi == AR.hidefile) && PF.parallel
01824         && PF.exprtodo < 0 ) {
01825         PF_BUFFER *sbuf = PF.sbuf;
01826         if ( fi->POfill >= fi->POstop ) {
01827             sbuf->fill[sbuf->active] = fi->POstop;
01828             PF_ISendSbuf(MASTER, PF_BUFFER_MSGTAG);
01829             fi->POfull = fi->POfill = fi->PObuffer = sbuf->buff[sbuf->active];
01830             fi->POstop = sbuf->stop[sbuf->active];
01831         }
01832         * (fi->POfill)++ = 0;
01833         sbuf->fill[sbuf->active] = fi->POfill;
01834         PF_ISendSbuf(MASTER, PF_ENDBUFFER_MSGTAG);
01835         fi->PObuffer = fi->POfill = fi->POfull = sbuf->buff[sbuf->active];
01836         fi->POstop = sbuf->stop[sbuf->active];
01837         return(0);
01838     }
01839 #endif /* WITHMPI [16mar1998 ar] */
01840     if ( fi->POfill >= fi->POstop ) {
01841         if ( fi->handle < 0 ) {
01842             if ( ( RetCode = CreateFile(fi->name) ) >= 0 ) {
01843 #ifdef GZIPDEBUG
01844                 MLOCK(ErrorMessageLock);
01845                 MesPrint("%w FlushOut created scratch file %s", fi->name);

```



```

01845             MUNLOCK(ErrorMessageLock);
01846 #endif
01847             PUTZERO(fi->filesize);
01848             PUTZERO(fi->PPosition);
01849             fi->handle = (WORD)RetCode;
01850 /*
01851             Should not be here?
01852 #ifdef WITHZLIB
01853             fi->ziobuffer = 0;
01854 #endif
01855 */
01856         }
01857         else {
01858             MLOCK(ErrorMessageLock);
01859             MesPrint("Cannot create scratch file %s",fi->name);
01860             MUNLOCK(ErrorMessageLock);
01861             return(-1);
01862         }
01863     }
01864 #ifdef WITHZLIB
01865     if ( AT.SS == AT.S0 && !AR.NoCompress && AR.gzipCompress > 0
01866         && dobracketindex == 0 && ( compr > 0 ) && fi->zsp != 0 ) {
01867         if ( PutOutputGZIP(fi) ) return(-1);
01868         fi->POfill = fi->PObuffer;
01869     }
01870     else
01871 #endif
01872     {
01873 #ifdef ALLLOCK
01874         LOCK(fi->pthreadlock);
01875 #endif
01876         if ( fi == AR.hidefile ) {
01877             LOCK(AS.inputslock);
01878         }
01879         SeekFile(fi->handle,&(fi->PPosition),SEEK_SET);
01880         if ( ( RetCode = WriteFile(fi->handle,(UBYTE *) (fi->PObuffer),fi->POsize) ) != fi->POsize )
01881         {
01882 #ifdef ALLLOCK
01883             UNLOCK(fi->pthreadlock);
01884 #endif
01885             if ( fi == AR.hidefile ) {
01886                 UNLOCK(AS.inputslock);
01887             }
01888             MLOCK(ErrorMessageLock);
01889             MesPrint("Write error while sorting. Disk full?");
01890             MesPrint("Attempt to write %l bytes on file %d at position %l5p",
01891                 fi->POsize,fi->handle,&(fi->PPosition));
01892             MesPrint("RetCode = %l, Buffer address = %l",RetCode,(LONG) (fi->PObuffer));
01893             MUNLOCK(ErrorMessageLock);
01894             return(-1);
01895         }
01896         ADDPOS(fi->filesize,fi->POsize);
01897         fi->POfill = fi->PObuffer;
01898         ADDPOS(fi->PPosition,fi->POsize);
01899         if ( fi == AR.hidefile ) {
01900             UNLOCK(AS.inputslock);
01901         }
01902 #ifdef ALLLOCK
01903         UNLOCK(fi->pthreadlock);
01904 #endif
01905 #ifdef WITHPTHREADS
01906         if ( AS.MasterSort && AC.ThreadSortFileSynch && fi != AR.hidefile ) {
01907             if ( fi->handle >= 0 ) SynchFile(fi->handle);
01908         }
01909 #endif
01910     }
01911     *(fi->POfill)++ = 0;
01912     fi->POfull = fi->POfill;
01913 /*
01914     {
01915         UBYTE OutBuf[140];
01916         if ( AP.DebugFlag ) {
01917             AO.OutFill = AO.OutputLine = OutBuf;
01918             AO.OutSkip = 3;
01919             FiniLine();
01920             TokenToLine((UBYTE *) "End of expression written");
01921             FiniLine();
01922         }
01923     }
01924 */
01925     size = (fi->POfill-fi->PObuffer)*sizeof(WORD);
01926     if ( fi->handle >= 0 ) {
01927 #ifdef WITHZLIB
01928         if ( AT.SS == AT.S0 && !AR.NoCompress && AR.gzipCompress > 0
01929             && dobracketindex == 0 && ( compr > 0 ) && fi->zsp != 0 ) {
01930             if ( FlushOutputGZIP(fi) ) return(-1);

```

```

01931         fi->POfill = fi->PObuffer;
01932     }
01933     else
01934 #endif
01935     {
01936 #ifdef ALLLOCK
01937         LOCK(fi->pthreadlock);
01938 #endif
01939         if ( fi == AR.hidefile ) {
01940             LOCK(AS.inputslock);
01941         }
01942         SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
01943         /*
01944             MesPrint("FlushOut: writing %l bytes to position %l2p",size,&(fi->POposition));
01945         */
01946         if ( ( RetCode = WriteFile(fi->handle,(UBYTE *) (fi->PObuffer),size) ) != size ) {
01947 #ifdef ALLLOCK
01948             UNLOCK(fi->pthreadlock);
01949 #endif
01950             if ( fi == AR.hidefile ) {
01951                 UNLOCK(AS.inputslock);
01952             }
01953             MLOCK(ErrorMessageLock);
01954             MesPrint("Write error while finishing sorting. Disk full?");
01955             MesPrint("Attempt to write %l bytes on file %d at position %l5p",
01956                 size,fi->handle,&(fi->POposition));
01957             MesPrint("RetCode = %l, Buffer address = %l",RetCode,(LONG) (fi->PObuffer));
01958             MUNLOCK(ErrorMessageLock);
01959             return(-1);
01960         }
01961         ADDPOS(fi->filesize,size);
01962         ADDPOS(fi->POposition,size);
01963         fi->POfill = fi->PObuffer;
01964         if ( fi == AR.hidefile ) {
01965             UNLOCK(AS.inputslock);
01966         }
01967 #ifdef ALLLOCK
01968         UNLOCK(fi->pthreadlock);
01969 #endif
01970 #ifdef WITHPTHREADS
01971         if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
01972             if ( fi->handle >= 0 ) SynchFile(fi->handle);
01973         }
01974 #endif
01975     }
01976     }
01977     if ( dobracketindex ) {
01978         BRACKETINFO *b = Expressions[AR.CurExpr].newbracketinfo;
01979         if ( b->indexfill > 0 ) {
01980             DIFPOS(b->indexbuffer[b->indexfill-1].next,*position,Expressions[AR.CurExpr].onfile);
01981         }
01982     }
01983 #ifdef WITHZLIB
01984     if ( AT.SS == AT.S0 && !AR.NoCompress && AR.gzipCompress > 0
01985         && dobracketindex == 0 && ( compr > 0 ) && fi->zsp != 0 ) {
01986         PUTZERO(*position);
01987         if ( fi->handle >= 0 ) {
01988 #ifdef ALLLOCK
01989             LOCK(fi->pthreadlock);
01990 #endif
01991             SeekFile(fi->handle,position,SEEK_END);
01992 #ifdef ALLLOCK
01993             UNLOCK(fi->pthreadlock);
01994 #endif
01995         }
01996         else {
01997             ADDPOS(*position,((UBYTE *) fi->POfill-(UBYTE *) fi->PObuffer));
01998         }
01999     }
02000     else
02001 #endif
02002     {
02003         ADDPOS(*position,sizeof(WORD));
02004     }
02005     return(0);
02006 }
02007
02008 /*
02009     #] FlushOut :
02010     #[ AddCoef :          WORD AddCoef(pterm1,pterm2)
02011 */
02026 int AddCoef(PHEAD WORD **ps1, WORD **ps2)
02027 {
02028     GETBIDENTITY
02029     SORTING *S = AT.SS;
02030     WORD *s1, *s2;
02031     WORD l1, l2, i;

```

```

02032     WORD OutLen, *t, j;
02033     UWORD *OutCoef;
02034 #ifdef WITHFLOAT
02035     if ( AT.SortFloatMode ) return(AddWithFloat(BHEAD ps1,ps2));
02036 #endif
02037     OutCoef = AN.SoScratC;
02038     s1 = *ps1; s2 = *ps2;
02039     GETCOEF(s1,l1);
02040     GETCOEF(s2,l2);
02041     if ( AddRat(BHEAD (UWORD *)s1,l1,(UWORD *)s2,l2,OutCoef,&OutLen) ) {
02042         MLOCK(ErrorMessageLock);
02043         MesCall("AddCoef");
02044         MUNLOCK(ErrorMessageLock);
02045         Terminate(-1);
02046     }
02047     if ( AN.ncmod != 0 ) {
02048         if ( ( AC.modmode & POSNEG ) != 0 ) {
02049             NormalModulus(OutCoef,&OutLen);
02050 /*
02051         We had forgotten that this can also become smaller but the
02052         denominator isn't there. Correct in the other case
02053         17-may-2009 [JV]
02054 */
02055         j = ABS(OutLen); OutCoef[j] = 1;
02056         for ( i = 1; i < j; i++ ) OutCoef[j+i] = 0;
02057     }
02058     else if ( BigLong(OutCoef,OutLen,(UWORD *)AC.cmod,ABS(AN.ncmod)) >= 0 ) {
02059         SubPLon(OutCoef,OutLen,(UWORD *)AC.cmod,ABS(AN.ncmod),OutCoef,&OutLen);
02060         OutCoef[OutLen] = 1;
02061         for ( i = 1; i < OutLen; i++ ) OutCoef[OutLen+i] = 0;
02062     }
02063 }
02064 if ( !OutLen ) { *ps1 = *ps2 = 0; return(0); }
02065 OutLen *= 2;
02066 if ( OutLen < 0 ) i = - ( --OutLen );
02067 else i = ++OutLen;
02068 if ( l1 < 0 ) l1 = -l1;
02069 l1 *= 2; l1++;
02070 if ( i <= l1 ) { /* Fits in 1 */
02071     l1 -= i;
02072     **ps1 -= l1;
02073     s2 = (WORD *)OutCoef;
02074     while ( --i > 0 ) *s1++ = *s2++;
02075     *s1++ = OutLen;
02076     while ( --l1 >= 0 ) *s1++ = 0;
02077     goto RegEnd;
02078 }
02079 if ( l2 < 0 ) l2 = -l2;
02080 l2 *= 2; l2++;
02081 if ( i <= l2 ) { /* Fits in 2 */
02082     l2 -= i;
02083     **ps2 -= l2;
02084     s1 = (WORD *)OutCoef;
02085     while ( --i > 0 ) *s2++ = *s1++;
02086     *s2++ = OutLen;
02087     while ( --l2 >= 0 ) *s2++ = 0;
02088     *ps1 = *ps2;
02089     goto RegEnd;
02090 }
02091
02092 /* Doesn't fit. Make a new term. */
02093
02094 t = s1;
02095 s1 = *ps1;
02096 j = *s1++ + i - l1; /* Space needed */
02097 if ( ( S->sFill + j ) >= S->sTop2 ) {
02098     GarbHand();
02099     s1 = *ps1;
02100     t = s1 + *s1 - 1;
02101     j = *s1++ + i - l1; /* Space needed */
02102     l1 = *t;
02103     if ( l1 < 0 ) l1 = - l1;
02104     t -= l1-1;
02105 }
02106 s2 = S->sFill;
02107 *s2++ = j;
02108 while ( s1 < t ) *s2++ = *s1++;
02109 s1 = (WORD *)OutCoef;
02110 while ( --i > 0 ) *s2++ = *s1++;
02111 *s2++ = OutLen;
02112 *ps1 = S->sFill;
02113 S->sFill = s2;
02114 RegEnd:
02115 *ps2 = 0;
02116 if ( **ps1 > AM.MaxTer/((LONG)(sizeof(WORD))) ) {
02117     MLOCK(ErrorMessageLock);
02118     MesPrint("Term too complex after addition in sort. MaxTermSize = %10l",

```

```

02119         AM.MaxTer/sizeof(WORD));
02120         MUNLOCK(ErrorMessageLock);
02121         Terminate(-1);
02122     }
02123     return(1);
02124 }
02125
02126 /*
02127     #] AddCoef :
02128     #[ AddPoly :                WORD AddPoly(pterm1,pterm2)
02129 */
02155 int AddPoly(PHEAD WORD **ps1, WORD **ps2)
02156 {
02157     GETBIDENTITY
02158     SORTING *S = AT.SS;
02159     WORD i;
02160     WORD *s1, *s2, *m, *w, *t, oldpw = S->PolyWise;
02161     s1 = *ps1 + S->PolyWise;
02162     s2 = *ps2 + S->PolyWise;
02163     w = AT.WorkPointer;
02164 /*
02165     Add here the two arguments. Is a straight merge.
02166 */
02167     if ( S->PolyFlag == 2 && AR.PolyFunExp != 2 && AR.PolyFunExp != 3 ) {
02168         WORD **oldSplitScratch = AN.SplitScratch;
02169         LONG oldSplitScratchSize = AN.SplitScratchSize;
02170         LONG oldInScratch = AN.InScratch;
02171         WORD oldtype = AR.SortType;
02172         if ( (WORD *) ((UBYTE *)w + AM.MaxTer) >= AT.WorkTop ) {
02173             MLOCK(ErrorMessageLock);
02174             MesPrint("Program was adding polyratfun arguments");
02175             MesWork();
02176             MUNLOCK(ErrorMessageLock);
02177         }
02178         AR.SortType = SORTHIGHFIRST;
02179         S->PolyWise = 0;
02180         AN.SplitScratch = AN.SplitScratch1;
02181         AN.SplitScratchSize = AN.SplitScratchSize1;
02182         AN.InScratch = AN.InScratch1;
02183         poly_ratfun_add(BHEAD s1,s2);
02184         S->PolyWise = oldpw;
02185         AN.SplitScratch1 = AN.SplitScratch;
02186         AN.SplitScratchSize1 = AN.SplitScratchSize;
02187         AN.InScratch1 = AN.InScratch;
02188         AN.SplitScratch = oldSplitScratch;
02189         AN.SplitScratchSize = oldSplitScratchSize;
02190         AN.InScratch = oldInScratch;
02191         AT.WorkPointer = w;
02192         AR.SortType = oldtype;
02193         if ( w[1] <= FUNHEAD ||
02194             ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) ) {
02195             *ps1 = *ps2 = 0; return(0);
02196         }
02197     }
02198     else {
02199         if ( w + s1[1] + s2[1] + 12 + ARGHEAD >= AT.WorkTop ) {
02200             MLOCK(ErrorMessageLock);
02201             MesPrint("Program was adding polyfun arguments");
02202             MesWork();
02203             MUNLOCK(ErrorMessageLock);
02204         }
02205         AddArgs(BHEAD s1,s2,w);
02206     }
02207 /*
02208     Now we need to store the result in a convenient place.
02209 */
02210     if ( w[1] <= FUNHEAD ) { *ps1 = *ps2 = 0; return(0); }
02211     if ( w[1] <= s1[1] || w[1] <= s2[1] ) { /* Fits in place. */
02212         if ( w[1] > s1[1] ) {
02213             *ps1 = *ps2;
02214             s1 = s2;
02215         }
02216         t = s1 + s1[1];
02217         m = *ps1 + **ps1;
02218         i = w[1];
02219         NCOPY(s1,w,i);
02220         if ( s1 != t ) {
02221             while ( t < m ) *s1++ = *t++;
02222             **ps1 = WORDDIFF(s1,(*ps1));
02223         }
02224         *ps2 = 0;
02225     }
02226     else { /* Make new term */
02227 #ifdef TESTGARB
02228         s2 = *ps2;
02229 #endif
02230         *ps2 = 0;

```

```

02231         if ( (S->sFill + (**ps1 + w[1] - s1[1])) >= S->sTop2 ) {
02232 #ifdef TESTGARB
02233         MesPrint("-----Garbage collection-----");
02234 #endif
02235         AT.WorkPointer += w[1];
02236         GarbHand();
02237         AT.WorkPointer = w;
02238         s1 = *ps1;
02239         if ( (S->sFill + (**ps1 + w[1] - s1[1])) >= S->sTop2 ) {
02240 #ifdef TESTGARB
02241             UBYTE OutBuf[140];
02242             MLOCK(ErrorMessageLock);
02243             AO.OutFill = AO.OutputLine = OutBuf;
02244             AO.OutSkip = 3;
02245             FiniLine();
02246             i = *s2;
02247             while ( --i >= 0 ) {
02248                 TalToLine((UWORD) (*s2++)); TokenToLine((UBYTE *) " ");
02249             }
02250             FiniLine();
02251             AO.OutFill = AO.OutputLine = OutBuf;
02252             AO.OutSkip = 3;
02253             FiniLine();
02254             s2 = *ps1;
02255             i = *s2;
02256             while ( --i >= 0 ) {
02257                 TalToLine((UWORD) (*s2++)); TokenToLine((UBYTE *) " ");
02258             }
02259             FiniLine();
02260             AO.OutFill = AO.OutputLine = OutBuf;
02261             AO.OutSkip = 3;
02262             FiniLine();
02263             s2 = w;
02264             i = w[1];
02265             while ( --i >= 0 ) {
02266                 TalToLine((UWORD) (*s2++)); TokenToLine((UBYTE *) " ");
02267             }
02268             FiniLine();
02269             if ( AR.sLevel > 0 ) {
02270                 MesPrint("Please increase SubSmallExtension setup parameter.");
02271             }
02272             else {
02273                 MesPrint("Please increase SmallExtension setup parameter.");
02274             }
02275             MUNLOCK(ErrorMessageLock);
02276 #else
02277             MLOCK(ErrorMessageLock);
02278             if ( AR.sLevel > 0 ) {
02279                 MesPrint("Please increase SubSmallExtension setup parameter.");
02280             }
02281             else {
02282                 MesPrint("Please increase SmallExtension setup parameter.");
02283             }
02284             MUNLOCK(ErrorMessageLock);
02285 #endif
02286             Terminate(-1);
02287         }
02288     }
02289     t = *ps1;
02290     s2 = S->sFill;
02291     m = s2;
02292     i = S->PolyWise;
02293     NCOPY(s2,t,i);
02294     i = w[1];
02295     NCOPY(s2,w,i);
02296     t = t + t[1];
02297     w = *ps1 + **ps1;
02298     while ( t < w ) *s2++ = *t++;
02299     *m = WORDDIFF(s2,m);
02300     *ps1 = m;
02301     S->sFill = s2;
02302     if ( *m > AM.MaxTer/((LONG)sizeof(WORD)) ) {
02303         MLOCK(ErrorMessageLock);
02304         MesPrint("Term too complex after polynomial addition. MaxTermSize = %10l",
02305             AM.MaxTer/sizeof(WORD));
02306         MUNLOCK(ErrorMessageLock);
02307         Terminate(-1);
02308     }
02309 }
02310 return(1);
02311 }
02312
02313 /*
02314 #] AddPoly :
02315 #[ AddArgs : void AddArgs(arg1,arg2,to)
02316 */
02317

```

```

02318 #define INSLENGTH(x)  w[1] = FUNHEAD+ARGHEAD+x; w[FUNHEAD] = ARGHEAD+x;
02319
02327 void AddArgs(PHEAD WORD *s1, WORD *s2, WORD *m)
02328 {
02329     GETBIDENTITY
02330     WORD i1, i2;
02331     WORD *w = m, *mm, *t, *t1, *t2, *tstop1, *tstop2;
02332     WORD tempterm[8+FUNHEAD];
02333
02334     *m++ = AR.PolyFun; *m++ = 0; FILLFUN(m)
02335     *m++ = 0; *m++ = 0; FILLARG(m)
02336     if ( s1[FUNHEAD] < 0 || s2[FUNHEAD] < 0 ) {
02337         if ( s1[FUNHEAD] < 0 ) {
02338             if ( s2[FUNHEAD] < 0 ) { /* Both are special */
02339                 if ( s1[FUNHEAD] <= -FUNCTION ) {
02340                     if ( s2[FUNHEAD] == s1[FUNHEAD] ) {
02341                         *m++ = 4+FUNHEAD; *m++ = -s1[FUNHEAD]; *m++ = FUNHEAD;
02342                         FILLFUN(m)
02343                         *m++ = 2; *m++ = 1; *m++ = 3;
02344                         INSLENGTH(4+FUNHEAD)
02345                     }
02346                     else if ( s2[FUNHEAD] <= -FUNCTION ) {
02347                         i1 = functions[-FUNCTION-s1[FUNHEAD]].commute != 0;
02348                         i2 = functions[-FUNCTION-s2[FUNHEAD]].commute != 0;
02349                         if ( ( !i1 && i2 ) || ( i1 == i2 && i1 > i2 ) ) {
02350                             i1 = s2[FUNHEAD];
02351                             s2[FUNHEAD] = s1[FUNHEAD];
02352                             s1[FUNHEAD] = i1;
02353                         }
02354                         *m++ = 4+FUNHEAD; *m++ = -s1[FUNHEAD]; *m++ = FUNHEAD;
02355                         FILLFUN(m)
02356                         *m++ = 1; *m++ = 1; *m++ = 3;
02357                         *m++ = 4+FUNHEAD; *m++ = -s2[FUNHEAD]; *m++ = FUNHEAD;
02358                         FILLFUN(m)
02359                         *m++ = 1; *m++ = 1; *m++ = 3;
02360                         INSLENGTH(8+2*FUNHEAD)
02361                     }
02362                     else if ( s2[FUNHEAD] == -SYMBOL ) {
02363                         *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = s2[FUNHEAD+1]; *m++ = 1;
02364                         *m++ = 1; *m++ = 1; *m++ = 3;
02365                         *m++ = 4+FUNHEAD; *m++ = -s1[FUNHEAD]; *m++ = FUNHEAD;
02366                         FILLFUN(m)
02367                         *m++ = 1; *m++ = 1; *m++ = 3;
02368                         INSLENGTH(12+FUNHEAD)
02369                     }
02370                     else { /* number */
02371                         *m++ = 4;
02372                         *m++ = ABS(s2[FUNHEAD+1]); *m++ = 1; *m++ = s2[FUNHEAD+1] < 0 ? -3: 3;
02373                         *m++ = 4+FUNHEAD; *m++ = -s1[FUNHEAD]; *m++ = FUNHEAD;
02374                         FILLFUN(m)
02375                         *m++ = 1; *m++ = 1; *m++ = 3;
02376                         INSLENGTH(8+FUNHEAD)
02377                     }
02378                 }
02379             else if ( s1[FUNHEAD] == -SYMBOL ) {
02380                 if ( s2[FUNHEAD] == s1[FUNHEAD] ) {
02381                     if ( s1[FUNHEAD+1] == s2[FUNHEAD+1] ) {
02382                         *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = s1[FUNHEAD+1];
02383                         *m++ = 1; *m++ = 2; *m++ = 1; *m++ = 3;
02384                         INSLENGTH(8)
02385                     }
02386                     else {
02387                         if ( s1[FUNHEAD+1] > s2[FUNHEAD+1] )
02388                             { i1 = s1[FUNHEAD+1]; i2 = s1[FUNHEAD+1]; }
02389                         else { i1 = s1[FUNHEAD+1]; i2 = s2[FUNHEAD+1]; }
02390                         *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = i1;
02391                         *m++ = 1; *m++ = 1; *m++ = 1; *m++ = 3;
02392                         *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = i2;
02393                         *m++ = 1; *m++ = 1; *m++ = 1; *m++ = 3;
02394                         INSLENGTH(16)
02395                     }
02396                 }
02397             else if ( s2[FUNHEAD] <= -FUNCTION ) {
02398                 *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = s1[FUNHEAD+1]; *m++ = 1;
02399                 *m++ = 1; *m++ = 1; *m++ = 3;
02400                 *m++ = 4+FUNHEAD; *m++ = -s2[FUNHEAD]; *m++ = FUNHEAD;
02401                 FILLFUN(m)
02402                 *m++ = 1; *m++ = 1; *m++ = 3;
02403                 INSLENGTH(12+FUNHEAD)
02404             }
02405             else {
02406                 *m++ = 4;
02407                 *m++ = ABS(s2[FUNHEAD+1]); *m++ = 1; *m++ = s2[FUNHEAD+1] < 0 ? -3: 3;
02408                 *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = s1[FUNHEAD+1]; *m++ = 1;
02409                 *m++ = 1; *m++ = 1; *m++ = 3;
02410                 INSLENGTH(12)
02411             }

```

```

02412     }
02413     else { /* Must be -SNUMBER! */
02414         if ( s2[FUNHEAD] <= -FUNCTION ) {
02415             *m++ = 4;
02416             *m++ = ABS(s1[FUNHEAD+1]); *m++ = 1; *m++ = s1[FUNHEAD+1] < 0 ? -3: 3;
02417             *m++ = 4+FUNHEAD; *m++ = -s2[FUNHEAD]; *m++ = FUNHEAD;
02418             FILLFUN(m)
02419             *m++ = 1; *m++ = 1; *m++ = 3;
02420             INSLLENGTH(8+FUNHEAD)
02421         }
02422         else if ( s2[FUNHEAD] == -SYMBOL ) {
02423             *m++ = 4;
02424             *m++ = ABS(s1[FUNHEAD+1]); *m++ = 1; *m++ = s1[FUNHEAD+1] < 0 ? -3: 3;
02425             *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = s2[FUNHEAD+1]; *m++ = 1;
02426             *m++ = 1; *m++ = 1; *m++ = 3;
02427             INSLLENGTH(12)
02428         }
02429         else { /* Both are numbers. add. */
02430             LONG x1;
02431             x1 = (LONG)s1[FUNHEAD+1] + (LONG)s2[FUNHEAD+1];
02432             if ( x1 < 0 ) { i1 = (WORD)(-x1); i2 = -3; }
02433             else { i1 = (WORD)x1; i2 = 3; }
02434             if ( x1 && AN.ncmod != 0 ) {
02435                 m[0] = 4;
02436                 m[1] = i1;
02437                 m[2] = 1;
02438                 m[3] = i2;
02439                 if ( Modulus(m) ) Terminate(-1);
02440                 if ( *m == 0 ) w[1] = 0;
02441                 else {
02442                     if ( *m == 4 && ( m[1] & MAXPOSITIVE ) == m[1]
02443                         && m[3] == 3 ) {
02444                         i1 = m[1];
02445                         m -= ARGHEAD;
02446                         *m++ = -SNUMBER;
02447                         *m++ = i1;
02448                         INSLLENGTH(4)
02449                     }
02450                     else {
02451                         INSLLENGTH(*m)
02452                         m += *m;
02453                     }
02454                 }
02455             }
02456             else {
02457                 if ( x1 == 0 ) {
02458                     w[1] = FUNHEAD;
02459                 }
02460                 else if ( ( i1 & MAXPOSITIVE ) == i1 ) {
02461                     m -= ARGHEAD;
02462                     *m++ = -SNUMBER;
02463                     *m++ = (WORD)x1;
02464                     w[1] = FUNHEAD+2;
02465                 }
02466                 else {
02467                     *m++ = 4; *m++ = i1; *m++ = 1; *m++ = i2;
02468                     INSLLENGTH(4)
02469                 }
02470             }
02471         }
02472     }
02473 }
02474 else { /* Only s1 is special */
02475     slonly: /*
02476     */
02477     Compose a term in `tempterm'
02478     */
02479     t = tempterm;
02480     if ( s1[FUNHEAD] <= -FUNCTION ) {
02481         *t++ = 4+FUNHEAD; *t++ = -s1[FUNHEAD]; *t++ = FUNHEAD;
02482         FILLFUN(t)
02483         *t++ = 1; *t++ = 1; *t++ = 3;
02484     }
02485     else if ( s1[FUNHEAD] == -SYMBOL ) {
02486         *t++ = 8; *t++ = SYMBOL; *t++ = 4;
02487         *t++ = s1[FUNHEAD+1]; *t++ = 1;
02488         *t++ = 1; *t++ = 1; *t++ = 3;
02489     }
02490     else {
02491         *t++ = 4; *t++ = ABS(s1[FUNHEAD+1]);
02492         *t++ = 1; *t++ = s1[FUNHEAD+1] < 0 ? -3: 3;
02493     }
02494     tstop1 = t;
02495     s1 = tempterm;
02496     goto twogen;
02497 }
02498 }

```

```

02499         else {          /* Only s2 is special */
02500             t = s1;
02501             s1 = s2;
02502             s2 = t;
02503             goto slonly;
02504         }
02505     }
02506     else {
02507         int oldPolyFlag;
02508         tstop1 = s1 + s1[1];
02509         s1 += FUNHEAD+ARGHEAD;
02510 twogen:
02511         tstop2 = s2 + s2[1];
02512         s2 += FUNHEAD+ARGHEAD;
02513     /*
02514         Now we should merge the expressions in s1 and s2 into m.
02515     */
02516     oldPolyFlag = AT.SS->PolyFlag;
02517     AT.SS->PolyFlag = 0;
02518     while ( s1 < tstop1 && s2 < tstop2 ) {
02519         i1 = CompareTerms(BHEAD s1,s2,(WORD) (-1));
02520         if ( i1 > 0 ) {
02521             i2 = *s1;
02522             NCOPY(m,s1,i2);
02523         }
02524         else if ( i1 < 0 ) {
02525             i2 = *s2;
02526             NCOPY(m,s2,i2);
02527         }
02528         else { /* Coefficients should be added. */
02529             WORD i;
02530             t = s1+s1;
02531             i1 = t[-1];
02532             i2 = *s1 - ABS(i1);
02533             t2 = s2 + i2;
02534             s2 += *s2;
02535             mm = m;
02536             NCOPY(m,s1,i2);
02537             t1 = s1;
02538             s1 = t;
02539             i2 = s2[-1];
02540     /*
02541             t1,i1 is the first coefficient
02542             t2,i2 is the second coefficient
02543             It should be placed at m,i1
02544     */
02545             i1 = REDLENG(i1);
02546             i2 = REDLENG(i2);
02547             if ( AddRat(BHEAD (UWORD *)t1,i1,(UWORD *)t2,i2,(UWORD *)m,&i) ) {
02548                 MLOCK(ErrorMessageLock);
02549                 MesPrint("Addition of coefficients of PolyFun");
02550                 MUNLOCK(ErrorMessageLock);
02551                 Terminate(-1);
02552             }
02553             if ( i == 0 ) {
02554                 m = mm;
02555             }
02556             else {
02557                 i1 = INCLENG(i);
02558                 m += ABS(i1);
02559                 m[-1] = i1;
02560                 *mm = WORDDIFF(m,mm);
02561                 if ( AN.ncmod != 0 ) {
02562                     if ( Modulus(mm) ) Terminate(-1);
02563                     if ( !*mm ) m = mm;
02564                     else m = mm + *mm;
02565                 }
02566             }
02567         }
02568     }
02569     while ( s1 < tstop1 ) *m++ = *s1++;
02570     while ( s2 < tstop2 ) *m++ = *s2++;
02571     w[1] = WORDDIFF(m,w);
02572     w[FUNHEAD] = w[1] - FUNHEAD;
02573     if ( ToFast(w+FUNHEAD,w+FUNHEAD) ) {
02574         if ( w[FUNHEAD] <= -FUNCTION ) w[1] = FUNHEAD+1;
02575         else w[1] = FUNHEAD+2;
02576         if ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) w[1] = FUNHEAD;
02577     }
02578     /* AT.SS->PolyFlag = AR.PolyFunType; */
02579     AT.SS->PolyFlag = oldPolyFlag;
02580 }
02581 }
02582
02583 /*
02584 #] AddArgs :
02585 #[ Compare1 :          WORD Compare1(term1,term2,level)

```



```

02586 */
02620 WORD Compare1(PHEAD WORD *term1, WORD *term2, WORD level)
02621 {
02622     SORTING *S = AT.SS;
02623     WORD *stopper1, *stopper2, *t2;
02624     WORD *s1, *s2, *t1;
02625     WORD *stopex1, *stopex2;
02626     WORD c1, c2;
02627     WORD prevorder;
02628     WORD count = -1, localPoly, polyhit = -1;
02629
02630     if ( AR.sLevel == 0 ) {
02631         numcompares++;
02632     }
02633
02634     if ( S->PolyFlag ) {
02635         /*
02636             if ( S->PolyWise != 0 ) {
02637                 MLOCK(ErrorMessageLock);
02638                 MesPrint("S->PolyWise is not zero!!!!");
02639                 MUNLOCK(ErrorMessageLock);
02640             }
02641         */
02642         count = 0; localPoly = 1; S->PolyWise = polyhit = 0;
02643         S->PolyFlag = AR.PolyFunType;
02644         if ( AR.PolyFunType == 2 &&
02645             ( AR.PolyFunExp == 2 || AR.PolyFunExp == 3 ) ) S->PolyFlag = 1;
02646     }
02647     else { localPoly = 0; }
02648     #ifdef WITHFLOAT
02649     AT.SortFloatMode = 0;
02650     #endif
02651     prevorder = 0;
02652     GETSTOP(term1,s1);
02653     stopper1 = s1;
02654     GETSTOP(term2,stopper2);
02655     t1 = term1 + 1;
02656     t2 = term2 + 1;
02657     while ( t1 < stopper1 && t2 < stopper2 ) {
02658         if ( *t1 != *t2 ) {
02659             if ( *t1 == HAAKJE ) return(PREV(-1));
02660             if ( *t2 == HAAKJE ) return(PREV(1));
02661             if ( *t1 >= (FUNCTION-1) ) {
02662                 if ( *t2 < (FUNCTION-1) ) return(PREV(-1));
02663                 if ( *t1 < FUNCTION && *t2 < FUNCTION ) return(PREV(*t2-*t1));
02664                 if ( *t1 < FUNCTION ) return(PREV(1));
02665                 if ( *t2 < FUNCTION ) return(PREV(-1));
02666                 c1 = functions[*t1-FUNCTION].commute;
02667                 c2 = functions[*t2-FUNCTION].commute;
02668                 if ( !c1 ) {
02669                     if ( c2 ) return(PREV(1));
02670                     else return(PREV(*t2-*t1));
02671                 }
02672                 else {
02673                     if ( !c2 ) return(PREV(-1));
02674                     else return(PREV(*t2-*t1));
02675                 }
02676             }
02677             else return(PREV(*t2-*t1));
02678         }
02679         s1 = t1 + 2;
02680         s2 = t2 + 2;
02681         c1 = *t1;
02682         t1 += t1[1];
02683         t2 += t2[1];
02684         if ( localPoly && c1 < FUNCTION ) {
02685             polyhit = 1;
02686         }
02687         if ( c1 <= (FUNCTION-1)
02688             || ( c1 >= FUNCTION && functions[c1-FUNCTION].spec > 0 ) ) {
02689             if ( c1 == SYMBOL ) {
02690                 if ( *s1 == FACTORSYMBOL && *s2 == FACTORSYMBOL
02691                     && s1[-1] == 4 && s2[-1] == 4
02692                     && ( ( t1 < stopper1 && *t1 == HAAKJE )
02693                         || ( t1 == stopper1 && AT.fromindex ) ) ) {
02694                     /*
02695                         We have to be very careful with the criteria here, because
02696                         Compare1 is called both in the regular sorting and by the
02697                         routine that makes the bracket index. In the last case
02698                         there is no HAAKJE subterm.
02699                     */
02700                     if ( s1[1] != s2[1] ) return(s2[1]-s1[1]);
02701                     s1 += 2; s2 += 2;
02702                 }
02703                 else if ( AR.SortType >= SORTPOWERFIRST ) {
02704                     WORD i1 = 0, *r1;
02705                     r1 = s1;

```

```

02706         while ( s1 < t1 ) { i1 += s1[1]; s1 += 2; }
02707         s1 = r1; r1 = s2;
02708         while ( s2 < t2 ) { i1 -= s2[1]; s2 += 2; }
02709         s2 = r1;
02710         if ( i1 ) {
02711             if ( AR.SortType >= SORTANTIPOWER ) i1 = -i1;
02712             return(PREV(i1));
02713         }
02714     }
02715     while ( s1 < t1 ) {
02716         if ( s2 >= t2 ) {
02717             return(PREV(1)); /*
02718             if ( AR.SortType==SORTLOWFIRST ) {
02719                 return(PREV((s1[1]>0?-1:1)));
02720             }
02721             else {
02722                 return(PREV((s1[1]<0?-1:1)));
02723             }
02724         }
02725         if ( *s1 != *s2 ) {
02726             return(PREV(*s2-*s1)); /*
02727             if ( AR.SortType==SORTLOWFIRST ) {
02728                 if ( *s1 < *s2 ) {
02729                     return(PREV((s1[1]<0?1:-1)));
02730                 }
02731                 else {
02732                     return(PREV((s2[1]<0?-1:1)));
02733                 }
02734             }
02735             else {
02736                 if ( *s1 < *s2 ) {
02737                     return(PREV((s1[1]<0?-1:1)));
02738                 }
02739                 else {
02740                     return(PREV((s2[1]<0?1:-1)));
02741                 }
02742             }
02743         }
02744         s1++; s2++;
02745         if ( *s1 != *s2 ) return(
02746             PREV((AR.SortType==SORTLOWFIRST?*s2-*s1:*s1-*s2)));
02747         s1++; s2++;
02748     }
02749     if ( s2 < t2 ) {
02750         return(PREV(-1)); /*
02751         if ( AR.SortType==SORTLOWFIRST ) {
02752             return(PREV((s2[1]<0?-1:1)));
02753         }
02754         else {
02755             return(PREV((s2[1]<0?1:-1)));
02756         }
02757     }
02758 }
02759 else if ( c1 == DOTPRODUCT ) {
02760     if ( AR.SortType >= SORTPOWERFIRST ) {
02761         WORD i1 = 0, *r1;
02762         r1 = s1;
02763         while ( s1 < t1 ) { i1 += s1[2]; s1 += 3; }
02764         s1 = r1; r1 = s2;
02765         while ( s2 < t2 ) { i1 -= s2[2]; s2 += 3; }
02766         s2 = r1;
02767         if ( i1 ) {
02768             if ( AR.SortType >= SORTANTIPOWER ) i1 = -i1;
02769             return(PREV(i1));
02770         }
02771     }
02772     while ( s1 < t1 ) {
02773         if ( s2 >= t2 ) return(PREV(1));
02774         if ( *s1 != *s2 ) return(PREV(*s2-*s1));
02775         s1++; s2++;
02776         if ( *s1 != *s2 ) return(PREV(*s2-*s1));
02777         s1++; s2++;
02778         if ( *s1 != *s2 ) return(
02779             PREV((AR.SortType==SORTLOWFIRST?*s2-*s1:*s1-*s2)));
02780         s1++; s2++;
02781     }
02782     if ( s2 < t2 ) return(PREV(-1));
02783 }
02784 else {
02785     while ( s1 < t1 ) {
02786         if ( s2 >= t2 ) return(PREV(1));
02787         if ( *s1 != *s2 ) return(PREV(*s2-*s1));
02788         s1++; s2++;
02789     }
02790     if ( s2 < t2 ) return(PREV(-1));
02791 }
02792 }

```

```

02793         else {
02794 #if FUNHEAD != 2
02795     s1 += FUNHEAD-2;
02796     s2 += FUNHEAD-2;
02797 #endif
02798     if ( localPoly && c1 == AR.PolyFun ) {
02799         if ( count == 0 ) {
02800             if ( S->PolyFlag == 1 ) {
02801                 WORD i1, i2;
02802                 if ( *s1 > 0 ) i1 = *s1;
02803                 else if ( *s1 <= -FUNCTION ) i1 = 1;
02804                 else i1 = 2;
02805                 if ( *s2 > 0 ) i2 = *s2;
02806                 else if ( *s2 <= -FUNCTION ) i2 = 1;
02807                 else i2 = 2;
02808                 if ( s1+i1 == t1 && s2+i2 == t2 ) { /* This is the stuff */
02809 /*
02810             Test for scalar nature
02811 */
02812             if ( !polyhit ) {
02813                 WORD *u1, *u2, *ustop;
02814                 if ( *s1 < 0 ) {
02815                     if ( *s1 != -SNUMBER && *s1 != -SYMBOL && *s1 > -FUNCTION )
02816                         goto NoPoly;
02817                 }
02818                 else {
02819                     u1 = s1 + ARGHEAD;
02820                     while ( u1 < t1 ) {
02821                         u2 = u1 + *u1;
02822                         ustop = u2 - ABS(u2[-1]);
02823                         u1++;
02824                         while ( u1 < ustop ) {
02825                             if ( *u1 == INDEX ) goto NoPoly;
02826                             u1 += u1[1];
02827                         }
02828                         u1 = u2;
02829                     }
02830                 }
02831                 if ( *s2 < 0 ) {
02832                     if ( *s2 != -SNUMBER && *s2 != -SYMBOL && *s2 > -FUNCTION )
02833                         goto NoPoly;
02834                 }
02835                 else {
02836                     u1 = s2 + ARGHEAD;
02837                     while ( u1 < t2 ) {
02838                         u2 = u1 + *u1;
02839                         ustop = u2 - ABS(u2[-1]);
02840                         u1++;
02841                         while ( u1 < ustop ) {
02842                             if ( *u1 == INDEX ) goto NoPoly;
02843                             u1 += u1[1];
02844                         }
02845                         u1 = u2;
02846                     }
02847                 }
02848             }
02849             S->PolyWise = WORDDIFF(s1,term1);
02850             S->PolyWise -= FUNHEAD;
02851             count = 1;
02852             continue;
02853         }
02854         else {
02855 NoPoly:
02856             S->PolyWise = localPoly = 0;
02857         }
02858     }
02859     else if ( AR.PolyFunType == 2 ) {
02860         WORD i1, i2, i1a, i2a;
02861         if ( *s1 > 0 ) i1 = *s1;
02862         else if ( *s1 <= -FUNCTION ) i1 = 1;
02863         else i1 = 2;
02864         if ( *s2 > 0 ) i2 = *s2;
02865         else if ( *s2 <= -FUNCTION ) i2 = 1;
02866         else i2 = 2;
02867         if ( s1[i1] > 0 ) i1a = s1[i1];
02868         else if ( s1[i1] <= -FUNCTION ) i1a = 1;
02869         else i1a = 2;
02870         if ( s2[i2] > 0 ) i2a = s2[i2];
02871         else if ( s2[i2] <= -FUNCTION ) i2a = 1;
02872         else i2a = 2;
02873         if ( s1+i1+i1a == t1 && s2+i2+i2a == t2 ) { /* This is the stuff */
02874 /*
02875             Test for scalar nature
02876 */
02877             if ( !polyhit ) {
02878                 WORD *u1, *u2, *ustop;
02879                 if ( *s1 < 0 ) {

```

```

02880         if ( *s1 != -SNUMBER && *s1 != -SYMBOL && *s1 > -FUNCTION )
02881             goto NoPoly;
02882     }
02883     else {
02884         u1 = s1 + ARGHEAD;
02885         while ( u1 < s1+i1 ) {
02886             u2 = u1 + *u1;
02887             ustop = u2 - ABS(u2[-1]);
02888             u1++;
02889             while ( u1 < ustop ) {
02890                 if ( *u1 == INDEX ) goto NoPoly;
02891                 u1 += u1[1];
02892             }
02893             u1 = u2;
02894         }
02895     }
02896     if ( s1[i1] < 0 ) {
02897         if ( s1[i1] != -SNUMBER && s1[i1] != -SYMBOL && s1[i1] > -FUNCTION )
02898             goto NoPoly;
02899     }
02900     else {
02901         u1 = s1 +i1 + ARGHEAD;
02902         while ( u1 < t1 ) {
02903             u2 = u1 + *u1;
02904             ustop = u2 - ABS(u2[-1]);
02905             u1++;
02906             while ( u1 < ustop ) {
02907                 if ( *u1 == INDEX ) goto NoPoly;
02908                 u1 += u1[1];
02909             }
02910             u1 = u2;
02911         }
02912     }
02913     if ( *s2 < 0 ) {
02914         if ( *s2 != -SNUMBER && *s2 != -SYMBOL && *s2 > -FUNCTION )
02915             goto NoPoly;
02916     }
02917     else {
02918         u1 = s2 + ARGHEAD;
02919         while ( u1 < s2+i2 ) {
02920             u2 = u1 + *u1;
02921             ustop = u2 - ABS(u2[-1]);
02922             u1++;
02923             while ( u1 < ustop ) {
02924                 if ( *u1 == INDEX ) goto NoPoly;
02925                 u1 += u1[1];
02926             }
02927             u1 = u2;
02928         }
02929     }
02930     if ( s2[i2] < 0 ) {
02931         if ( s2[i2] != -SNUMBER && s2[i2] != -SYMBOL && s2[i2] > -FUNCTION )
02932             goto NoPoly;
02933     }
02934     else {
02935         u1 = s2 + i2 + ARGHEAD;
02936         while ( u1 < t2 ) {
02937             u2 = u1 + *u1;
02938             ustop = u2 - ABS(u2[-1]);
02939             u1++;
02940             while ( u1 < ustop ) {
02941                 if ( *u1 == INDEX ) goto NoPoly;
02942                 u1 += u1[1];
02943             }
02944             u1 = u2;
02945         }
02946     }
02947 }
02948 S->PolyWise = WORDDIFF(s1,term1);
02949 S->PolyWise -= FUNHEAD;
02950 count = 1;
02951 continue;
02952 }
02953 else {
02954     S->PolyWise = localPoly = 0;
02955 }
02956 }
02957 else {
02958     S->PolyWise = localPoly = 0;
02959 }
02960 }
02961 else {
02962     t1 = term1 + S->PolyWise;
02963     t2 = term2 + S->PolyWise;
02964     S->PolyWise = 0;
02965     localPoly = 0;
02966     continue;

```

```

02967     }
02968 }
02969 #ifdef WITHFLOAT
02970     if ( level == 0 && c1 == FLOATFUN && t1 == stopper1 && t2 == stopper2 && AT.aux_ != 0 ) {
02971 /*
02972         We have two FLOATFUN's. Test whether they are 'legal'
02973 */
02974         if ( TestFloat(s1-FUNHEAD) ) {
02975             if ( TestFloat(s2-FUNHEAD) ) { AT.SortFloatMode = 3; return(0); }
02976             else { return(1); }
02977         }
02978         else if ( TestFloat(s2-FUNHEAD) ) { return(-1); }
02979     }
02980 #endif
02981     while ( s1 < t1 ) {
02982 /*
02983         The next statement was added 9-nov-2001. It repaired a bad error
02984 */
02985         if ( s2 >= t2 ) return(PREV(-1));
02986 /*
02987         There is a little problem here with fast arguments
02988         We don't want to sacrifice speed, but we like to
02989         keep a rational ordering. This last one suffers in
02990         the solution that has been chosen here.
02991 */
02992         if ( AC.properorderflag ) {
02993             WORD oldpolyflag;
02994             oldpolyflag = S->PolyFlag;
02995             S->PolyFlag = 0;
02996             if ( ( c2 = -CompArg(s1,s2) ) != 0 ) {
02997                 S->PolyFlag = oldpolyflag; return(PREV(c2));
02998             }
02999             S->PolyFlag = oldpolyflag;
03000             NEXTARG(s1)
03001             NEXTARG(s2)
03002         }
03003         else {
03004             if ( *s1 > 0 ) {
03005                 if ( *s2 > 0 ) {
03006                     WORD oldpolyflag;
03007                     stopex1 = s1 + *s1;
03008                     if ( s2 >= t2 ) return(PREV(-1));
03009                     stopex2 = s2 + *s2;
03010                     s1 += ARGHEAD; s2 += ARGHEAD;
03011                     oldpolyflag = S->PolyFlag;
03012                     S->PolyFlag = 0;
03013                     while ( s1 < stopex1 ) {
03014                         if ( s2 >= stopex2 ) {
03015                             S->PolyFlag = oldpolyflag; return(PREV(-1));
03016                         }
03017                         if ( ( c2 = CompareTerms(BHEAD s1,s2,(WORD)1) ) != 0 ) {
03018                             S->PolyFlag = oldpolyflag; return(PREV(c2));
03019                         }
03020                         s1 += *s1;
03021                         s2 += *s2;
03022                     }
03023                     S->PolyFlag = oldpolyflag;
03024                     if ( s2 < stopex2 ) return(PREV(1));
03025                 }
03026                 else return(PREV(1));
03027             }
03028             else {
03029                 if ( *s2 > 0 ) return(PREV(-1));
03030                 if ( *s1 != *s2 ) { return(PREV(*s1-*s2)); }
03031                 if ( *s1 > -FUNCTION ) {
03032                     if ( *++s1 != *++s2 ) { return(PREV(*s2-*s1)); }
03033                 }
03034                 s1++; s2++;
03035             }
03036         }
03037     }
03038     if ( s2 < t2 ) return(PREV(1));
03039 }
03040 }
03041 #ifdef WITHFLOAT
03042     if ( level == 0 && t1 < stopper1 && *t1 == FLOATFUN && t1+t1[1] == stopper1
03043         && TestFloat(t1) && AT.aux_ != 0 ) {
03044         AT.SortFloatMode = 1; return(0);
03045     }
03046     else if ( level == 0 && t2 < stopper2 && *t2 == FLOATFUN && t2+t2[1] == stopper2
03047         && TestFloat(t2) && AT.aux_ != 0 ) {
03048         AT.SortFloatMode = 2; return(0);
03049     }
03050 #endif
03051 {
03052     if ( AR.SortType != SORTLOWFIRST ) {
03053         if ( t1 < stopper1 ) return(PREV(1));

```

```

03054         if ( t2 < stopper2 ) return(PREV(-1));
03055     }
03056     else {
03057         if ( t1 < stopper1 ) return(PREV(-1));
03058         if ( t2 < stopper2 ) return(PREV(1));
03059     }
03060 }
03061 if ( level == 3 ) return(CompCoef(term1,term2));
03062 if ( level >= 1 )
03063     return(CompCoef(term2,term1));
03064 return(0);
03065 }
03066
03067 /*
03068     #] Compare1 :
03069     #[ CompareSymbols :          WORD CompareSymbols(term1,term2,par)
03070 */
03084 WORD CompareSymbols(PHEAD WORD *term1, WORD *term2, WORD par)
03085 {
03086     int sum1, sum2;
03087     WORD *t1, *t2, *tt1, *tt2;
03088     int low, high;
03089     DUMMYUSE(par);
03090     if ( AR.SortType == SORTLOWFIRST ) { low = 1; high = -1; }
03091     else { low = -1; high = 1; }
03092     t1 = term1 + 1; tt1 = term1+*term1; tt1 -= ABS(tt1[-1]); t1 += 2;
03093     t2 = term2 + 1; tt2 = term2+*term2; tt2 -= ABS(tt2[-1]); t2 += 2;
03094     if ( AN.polysortflag > 0 ) {
03095         sum1 = 0; sum2 = 0;
03096         while ( t1 < tt1 ) { sum1 += t1[1]; t1 += 2; }
03097         while ( t2 < tt2 ) { sum2 += t2[1]; t2 += 2; }
03098         if ( sum1 < sum2 ) return(low);
03099         if ( sum1 > sum2 ) return(high);
03100         t1 = term1+3; t2 = term2 + 3;
03101     }
03102     while ( t1 < tt1 && t2 < tt2 ) {
03103         if ( *t1 > *t2 ) return(low);
03104         if ( *t1 < *t2 ) return(high);
03105         if ( t1[1] < t2[1] ) return(low);
03106         if ( t1[1] > t2[1] ) return(high);
03107         t1 += 2; t2 += 2;
03108     }
03109     if ( t1 < tt1 ) return(high);
03110     if ( t2 < tt2 ) return(low);
03111     return(0);
03112 }
03113
03114 /*
03115     #] CompareSymbols :
03116     #[ CompareHSymbols :          WORD CompareHSymbols(term1,term2,par)
03117 */
03127 WORD CompareHSymbols(PHEAD WORD *term1, WORD *term2, WORD par)
03128 {
03129     WORD *t1, *t2, *tt1, *tt2, *ttt1, *ttt2;
03130     DUMMYUSE(par);
03131     DUMMYUSE(AT.WorkPointer);
03132     t1 = term1 + 1; tt1 = term1+*term1; tt1 -= ABS(tt1[-1]); t1 += 2;
03133     t2 = term2 + 1; tt2 = term2+*term2; tt2 -= ABS(tt2[-1]); t2 += 2;
03134     while ( t1 < tt1 && t2 < tt2 ) {
03135         if ( *t1 != *t2 ) {
03136             if ( t1[0] < t2[0] ) return(-1);
03137             return(1);
03138         }
03139         else if ( *t1 == HAAKJE ) {
03140             t1 += 3; t2 += 3; continue;
03141         }
03142         ttt1 = t1+tt1[1]; ttt2 = t2+tt2[1];
03143         while ( t1 < ttt1 && t2 < ttt2 ) {
03144             if ( *t1 > *t2 ) return(-1);
03145             if ( *t1 < *t2 ) return(1);
03146             if ( t1[1] < t2[1] ) return(-1);
03147             if ( t1[1] > t2[1] ) return(1);
03148             t1 += 2; t2 += 2;
03149         }
03150         if ( t1 < ttt1 ) return(1);
03151         if ( t2 < ttt2 ) return(-1);
03152     }
03153     if ( t1 < tt1 ) return(1);
03154     if ( t2 < tt2 ) return(-1);
03155     return(0);
03156 }
03157
03158 /*
03159     #] CompareHSymbols :
03160     #[ ComPress :                  LONG ComPress(ss,n)
03161 */
03180 LONG ComPress(WORD **ss, LONG *n)

```

```

03181 {
03182     GETIDENTITY
03183     WORD *t, *s, j, k;
03184     LONG size = 0;
03185     int newsize, i;
03186 /*
03187     #[] debug :
03188
03189     WORD **sss = ss;
03190
03191     if ( AP.DebugFlag ) {
03192         UBYTE OutBuf[140];
03193         MLOCK(ErrorMessageLock);
03194         MesPrint("ComPress:");
03195         AO.OutFill = AO.OutputLine = OutBuf;
03196         AO.OutSkip = 3;
03197         FiniLine();
03198         ss = sss;
03199         while ( *ss ) {
03200             s = *ss++;
03201             j = *s;
03202             if ( j < 0 ) {
03203                 j = s[1] + 2;
03204             }
03205             while ( --j >= 0 ) {
03206                 TalToLine((UWORD) (*s++)); TokenToLine((UBYTE *) " ");
03207             }
03208             FiniLine();
03209         }
03210         AO.OutSkip = 0;
03211         FiniLine();
03212         MUNLOCK(ErrorMessageLock);
03213         ss = sss;
03214     }
03215
03216     #[] debug :
03217 */
03218     *n = 0;
03219     if ( AT.SS == AT.S0 && !AR.NoCompress ) {
03220         if ( AN.compressSize == 0 ) {
03221             if ( *ss ) { AN.compressSize = *ss + 64; }
03222             else { AN.compressSize = AM.MaxTer/sizeof(WORD) + 2; }
03223             AN.compressSpace = (WORD *) Malloc1(AN.compressSize*sizeof(WORD), "Compression");
03224         }
03225         AN.compressSpace[0] = 0;
03226         while ( *ss ) {
03227             k = 0;
03228             s = *ss;
03229             j = *s++;
03230             if ( j > AN.compressSize ) {
03231                 newsize = j + 64;
03232                 t = (WORD *) Malloc1(newsize*sizeof(WORD), "Compression");
03233                 t[0] = 0;
03234                 if ( AN.compressSpace ) {
03235                     for ( i = 0; i < *AN.compressSpace; i++ ) t[i] = AN.compressSpace[i];
03236                     M_free(AN.compressSpace, "Compression");
03237                 }
03238                 AN.compressSpace = t;
03239                 AN.compressSize = newsize;
03240             }
03241             t = AN.compressSpace;
03242             i = *t - 1;
03243             *t++ = j; j--;
03244             if ( AR.PolyFun ) {
03245                 WORD *polystop, *sa;
03246                 sa = s + j;
03247                 sa -= ABS(sa[-1]);
03248                 polystop = s;
03249                 while ( polystop < sa && *polystop != AR.PolyFun ) {
03250                     polystop += polystop[1];
03251                 }
03252                 while ( i > 0 && j > 0 && *s == *t && s < polystop ) {
03253                     i--; j--; s++; t++; k--;
03254                 }
03255             }
03256             else {
03257                 WORD *sa;
03258                 sa = s + j;
03259                 sa -= ABS(sa[-1]);
03260                 while ( i > 0 && j > 0 && *s == *t && s < sa ) { i--; j--; s++; t++; k--; }
03261             }
03262             if ( k < -1 ) {
03263                 s[-1] = j;
03264                 s[-2] = k;
03265                 *ss = s-2;
03266                 size += j + 2;
03267             }

```

```

03268         else {
03269             size += *AN.compressSpace;
03270             if ( k == -1 ) { t--; s--; j++; }
03271         }
03272         while ( --j >= 0 ) *t++ = *s++;
03273 /* Sabotage getting into the coefficient next time */
03274 t = AN.compressSpace + *AN.compressSpace;
03275 t[-(ABS(t[-1]))] = 0;
03276 ss++;
03277 (*n)++;
03278     }
03279 }
03280 else {
03281     while ( *ss ) {
03282         size += *(ss++);
03283         (*n)++;
03284     }
03285 }
03286 /*
03287     #] debug :
03288
03289     if ( AP.DebugFlag ) {
03290         UBYTE OutBuf[140];
03291         AO.OutFill = AO.OutputLine = OutBuf;
03292         AO.OutSkip = 3;
03293         FiniLine();
03294         ss = sss;
03295         while ( *ss ) {
03296             s = ss++;
03297             j = *s;
03298             if ( j < 0 ) {
03299                 j = s[1] + 2;
03300             }
03301             while ( --j >= 0 ) {
03302                 TalToLine((UWORD) (*s++)); TokenToLine((UBYTE *) " ");
03303             }
03304             FiniLine();
03305         }
03306         AO.OutSkip = 0;
03307         FiniLine();
03308     }
03309
03310     #] debug :
03311 */
03312     return(size);
03313 }
03314
03315 /*
03316     #] ComPress :
03317     #[ SplitMerge : void SplitMerge(Point,number)
03318 */
03344 #ifdef NEWSPLITMERGE
03345
03346 LONG SplitMerge(PHEAD WORD **Pointer, LONG number)
03347 {
03348     GETBIDENTITY
03349     SORTING *S = AT.SS;
03350     WORD **pp3, **pp1, **pp2, **pptop;
03351     LONG i, newleft, newright, split;
03352
03353 #ifdef SPLITMERGEDEBUG
03354     /* Print current array state on entry. */
03355     printf("%4ld: ", number);
03356     for (int ii = 0; ii < S->sTerms; ii++) {
03357         if ( (S->sPointer[ii]) ) {
03358             printf("%4d ", (unsigned) (S->sPointer[ii]-S->sBuffer));
03359         }
03360         else {
03361             printf(".... ");
03362         }
03363     }
03364     printf("\n");
03365     fflush(stdout);
03366 #endif
03367
03368     if ( number < 2 ) return(number);
03369     if ( number == 2 ) {
03370         pp1 = Pointer; pp2 = pp1 + 1;
03371         if ( ( i = CompareTerms(BHEAD *pp1,*pp2,(WORD)0) ) < 0 ) {
03372             pp3 = (WORD *) (*pp1); *pp1 = *pp2; *pp2 = (WORD *) pp3;
03373         }
03374         else if ( i == 0 ) {
03375             number--;
03376             if ( S->PolyWise ) { if ( AddPoly(BHEAD pp1,pp2) == 0 ) number = 0; }
03377             else { if ( AddCoef(BHEAD pp1,pp2) == 0 ) number = 0; }
03378         }
03379         return(number);

```



```

03380     }
03381     pptop = Pointer + number;
03382     split = number/2;
03383     newleft = SplitMerge(BHEAD Pointer,split);
03384     newright = SplitMerge(BHEAD Pointer+split,number-split);
03385     if ( newright == 0 ) return(newleft);
03386     /*
03387     We compare the last of the left with the first of the right
03388     If they are already in order, we will be done quickly.
03389     We may have to compactify the buffer because the recursion may
03390     have created holes. Also this compare may result in equal terms.
03391     Addition of 23-jul-1999. It makes things a bit faster.
03392     */
03393     if ( newleft > 0 && newright > 0 &&
03394         ( i = CompareTerms(BHEAD Pointer[newleft-1],Pointer[split],(WORD)0) ) >= 0 ) {
03395         pp2 = Pointer+split; pp1 = Pointer+newleft-1;
03396         if ( i == 0 ) {
03397             if ( S->PolyWise ) {
03398                 if ( AddPoly(BHEAD pp1,pp2) > 0 ) pp1++;
03399                 else newleft--;
03400             }
03401             else {
03402                 if ( AddCoef(BHEAD pp1,pp2) > 0 ) pp1++;
03403                 else newleft--;
03404             }
03405             *pp2++ = 0; newright--;
03406         }
03407         else pp1++;
03408         newleft += newright;
03409         if ( pp1 < pp2 ) {
03410             while ( --newright >= 0 ) *pp1++ = *pp2++;
03411             while ( pp1 < pptop ) *pp1++ = 0;
03412         }
03413         return(newleft);
03414     }
03415
03416     if ( split >= AN.SplitScratchSize ) {
03417         AN.SplitScratchSize = (split*3)/2+100;
03418         if ( AN.SplitScratchSize > S->Terms2InSmall/2 )
03419             AN.SplitScratchSize = S->Terms2InSmall/2;
03420         if ( AN.SplitScratch ) M_free(AN.SplitScratch,"AN.SplitScratch");
03421         AN.SplitScratch = (WORD **)Malloca1(AN.SplitScratchSize*sizeof(WORD *),"AN.SplitScratch");
03422     }
03423
03424     pp3 = AN.SplitScratch; pp1 = Pointer;
03425     /* Move rather than copy, so GarbHand can't double-count. */
03426     for ( i = 0; i < newleft; i++ ) { *pp3++ = *pp1; *pp1++ = 0; }
03427     AN.InScratch = newleft;
03428     pp1 = AN.SplitScratch; pp2 = Pointer + split; pp3 = Pointer;
03429
03430 #ifdef NEWSPLITMERGETIMSORT
03431     /*
03432     An improvement in the style of Timsort
03433     */
03434     while ( newleft > 8 ) {
03435         /* Check the middle of the LHS terms */
03436         LONG nnleft = newleft/2;
03437         if ( ( i = CompareTerms(BHEAD pp1[nnleft],*pp2,(WORD)0) ) < 0 ) {
03438             /* The terms are not in order. Break out and continue as normal. */
03439             break;
03440         }
03441         /* The terms merge or are in order. Copy pointers up to this point. */
03442         /* In the copy, zero the skipped pointers so GarbHand can't double-count. */
03443         for ( int iii = 0; iii < nnleft; iii++ ) {
03444             *pp3++ = *pp1;
03445             *pp1++ = 0;
03446         }
03447         newleft -= nnleft;
03448         if ( i == 0 ) {
03449             if ( S->PolyWise ) { i = AddPoly(BHEAD pp1,pp2); }
03450             else { i = AddCoef(BHEAD pp1,pp2); }
03451             if ( i == 0 ) {
03452                 /* The terms cancelled. The next term goes in *pp3. Don't move. */
03453             }
03454             else {
03455                 /* The terms added. Advance pp3. */
03456                 *pp3++ = *pp1;
03457             }
03458             /* We have taken a LHS (copy) and RHS term. */
03459             *pp2++ = 0;
03460             newright--;
03461             *pp1++ = 0;
03462             newleft--;
03463             break;
03464         }
03465     }
03466 #endif

```

```

03467
03468     while ( newleft > 0 && newright > 0 ) {
03469         if ( ( i = CompareTerms(BHEAD *pp1,*pp2,(WORD)0) ) < 0 ) {
03470             *pp3++ = *pp2;
03471             *pp2++ = 0;
03472             newright--;
03473         }
03474         else if ( i > 0 ) {
03475             *pp3++ = *pp1;
03476             *pp1++ = 0;
03477             newleft--;
03478         }
03479         else {
03480             if ( S->PolyWise ) { if ( AddPoly(BHEAD pp1,pp2) > 0 ) *pp3++ = *pp1; }
03481             else { if ( AddCoef(BHEAD pp1,pp2) > 0 ) *pp3++ = *pp1; }
03482             *pp1++ = 0; *pp2++ = 0; newleft--; newright--;
03483         }
03484     }
03485     for ( i = 0; i < newleft; i++ ) { *pp3++ = *pp1; *pp1++ = 0; }
03486     if ( pp3 == pp2 ) {
03487         pp3 += newright;
03488     } else {
03489         for ( i = 0; i < newright; i++ ) { *pp3++ = *pp2++; }
03490     }
03491     newleft = pp3 - Pointer;
03492     while ( pp3 < pptop ) *pp3++ = 0;
03493     AN.InScratch = 0;
03494     return(newleft);
03495 }
03496
03497 #else
03498
03499 LONG SplitMerge(PHEAD WORD **Pointer, LONG number)
03500 {
03501     GETBIDENTITY
03502     SORTING *S = AT.SS;
03503     WORD **pp3, **pp1, **pp2;
03504     LONG nleft, nright, i, newleft, newright;
03505     WORD **pptop;
03506
03507 #ifdef SPLITMERGEDEBUG
03508     /* Print current array state on entry. */
03509     printf("%4ld: ", number);
03510     for (int ii = 0; ii < S->sTerms; ii++) {
03511         if ( (S->sPointer)[ii] ) {
03512             printf("%4d ", (unsigned)(S->sPointer[ii]-S->sBuffer));
03513         }
03514         else {
03515             printf(".... ");
03516         }
03517     }
03518     printf("\n");
03519     fflush(stdout);
03520 #endif
03521
03522     if ( number < 2 ) return(number);
03523     if ( number == 2 ) {
03524         pp1 = Pointer; pp2 = pp1 + 1;
03525         if ( ( i = CompareTerms(BHEAD *pp1,*pp2,(WORD)0) ) < 0 ) {
03526             pp3 = (WORD **)(*pp1); *pp1 = *pp2; *pp2 = (WORD *)pp3;
03527         }
03528         else if ( i == 0 ) {
03529             number--;
03530             if ( S->PolyWise ) { if ( AddPoly(BHEAD pp1,pp2) == 0 ) { number = 0; } }
03531             else { if ( AddCoef(BHEAD pp1,pp2) == 0 ) { number = 0; } }
03532         }
03533         return(number);
03534     }
03535     pptop = Pointer + number;
03536     nleft = number » 1; nright = number - nleft;
03537     newleft = SplitMerge(BHEAD Pointer,nleft);
03538     newright = SplitMerge(BHEAD Pointer+nleft,nright);
03539     /*
03540     We compare the last of the left with the first of the right
03541     If they are already in order, we will be done quickly.
03542     We may have to compactify the buffer because the recursion may
03543     have created holes. Also this compare may result in equal terms.
03544     Addition of 23-jul-1999. It makes things a bit faster.
03545     */
03546     if ( newleft > 0 && newright > 0 &&
03547         ( i = CompareTerms(BHEAD Pointer[newleft-1],Pointer[nleft],(WORD)0) ) >= 0 ) {
03548         pp2 = Pointer+nleft; pp1 = Pointer+newleft-1;
03549         if ( i == 0 ) {
03550             if ( S->PolyWise ) {
03551                 if ( AddPoly(BHEAD pp1,pp2) > 0 ) pp1++;
03552                 else newleft--;
03553             }

```

```

03554         else {
03555             if ( AddCoef(BHEAD pp1,pp2) > 0 ) pp1++;
03556             else newleft--;
03557         }
03558         *pp2++ = 0; newright--;
03559     }
03560     else pp1++;
03561     newleft += newright;
03562     if ( pp1 < pp2 ) {
03563         while ( --newright >= 0 ) *pp1++ = *pp2++;
03564         while ( pp1 < pptop ) *pp1++ = 0;
03565     }
03566     return(newleft);
03567 }
03568 if ( nleft > AN.SplitScratchSize ) {
03569     AN.SplitScratchSize = (nleft*3)/2+100;
03570     if ( AN.SplitScratchSize > S->Terms2InSmall/2 )
03571         AN.SplitScratchSize = S->Terms2InSmall/2;
03572     if ( AN.SplitScratch ) M_free(AN.SplitScratch,"AN.SplitScratch");
03573     AN.SplitScratch = (WORD **)Malloca(AN.SplitScratchSize*sizeof(WORD *),"AN.SplitScratch");
03574 }
03575 pp3 = AN.SplitScratch; pp1 = Pointer; i = nleft;
03576 do { *pp3++ = *pp1; *pp1++ = 0; } while ( *pp1 && --i > 0 );
03577 if ( i > 0 ) { *pp3 = 0; i--; }
03578 AN.InScratch = nleft - i;
03579 pp1 = AN.SplitScratch; pp2 = Pointer + nleft; pp3 = Pointer;
03580 while ( nleft > 0 && nright > 0 && *pp1 && *pp2 ) {
03581     if ( ( i = CompareTerms(BHEAD *pp1,*pp2,(WORD)0) ) < 0 ) {
03582         *pp3++ = *pp2;
03583         *pp2++ = 0;
03584         nright--;
03585     }
03586     else if ( i > 0 ) {
03587         *pp3++ = *pp1;
03588         *pp1++ = 0;
03589         nleft--;
03590     }
03591     else {
03592         if ( S->PolyWise ) { if ( AddPoly(BHEAD pp1,pp2) > 0 ) *pp3++ = *pp1; }
03593         else { if ( AddCoef(BHEAD pp1,pp2) > 0 ) *pp3++ = *pp1; }
03594         *pp1++ = 0; *pp2++ = 0; nleft--; nright--;
03595     }
03596 }
03597 while ( --nleft >= 0 && *pp1 ) { *pp3++ = *pp1; *pp1++ = 0; }
03598 while ( --nright >= 0 && *pp2 ) { *pp3++ = *pp2++; }
03599 nleft = pp3 - Pointer;
03600 while ( pp3 < pptop ) *pp3++ = 0;
03601 AN.InScratch = 0;
03602 return(nleft);
03603 }
03604
03605 #endif
03606
03607 /*
03608     #] SplitMerge :
03609     #[ GarbHand :          void GarbHand()
03610 */
03626 void GarbHand(void)
03627 {
03628     GETIDENTITY
03629     SORTING *S = AT.SS;
03630     WORD **Point, *s2, *t, *garbuf, i;
03631     LONG k, total = 0;
03632     int tobereturned = 0;
03633 /*
03634     Compute the size needed. Put it in total.
03635 */
03636 #ifdef TESTGARB
03637     MLOCK(ErrorMessageLock);
03638     MesPrint("in: S->sFill = %l, S->sTop2 = %l",S->sFill,S->sTop2-S->sBuffer,S->sTop2-S->sBuffer);
03639 #endif
03640     Point = S->sPointer;
03641     k = S->sTerms;
03642     while ( --k >= 0 ) {
03643         if ( ( s2 = *Point++ ) != 0 ) { total += *s2; }
03644     }
03645     Point = AN.SplitScratch;
03646     k = AN.InScratch;
03647     while ( --k >= 0 ) {
03648         if ( ( s2 = *Point++ ) != 0 ) { total += *s2; }
03649     }
03650 #ifdef TESTGARB
03651     MesPrint("total = %l, nterms = %l",total,AN.InScratch);
03652     MUNLOCK(ErrorMessageLock);
03653 #endif
03654 /*
03655     Test now whether it fits. If so deal with the problem inside

```

```

03656     the memory at the tail of the large buffer.
03657 */
03658     if ( S->lBuffer != 0 && S->lFill + total <= S->lTop ) {
03659         garbuf = S->lFill;
03660     }
03661     else {
03662         garbuf = (WORD *)Malloc1(total*sizeof(WORD),"Garbage buffer");
03663         tobereturned = 1;
03664     }
03665     t = garbuf;
03666     Point = S->sPointer;
03667     k = S->sTerms;
03668     while ( --k >= 0 ) {
03669         if ( *Point ) {
03670             s2 = *Point++;
03671             i = *s2;
03672             NCOPY(t,s2,i);
03673         }
03674         else { Point++; }
03675     }
03676     Point = AN.SplitScratch;
03677     k = AN.InScratch;
03678     while ( --k >= 0 ) {
03679         if ( *Point ) {
03680             s2 = *Point++;
03681             i = *s2;
03682             NCOPY(t,s2,i);
03683         }
03684         else Point++;
03685     }
03686     s2 = S->sBuffer;
03687     t = garbuf;
03688     Point = S->sPointer;
03689     k = S->sTerms;
03690     while ( --k >= 0 ) {
03691         if ( *Point ) {
03692             *Point++ = s2;
03693             i = *t;
03694             NCOPY(s2,t,i);
03695         }
03696         else { Point++; }
03697     }
03698     Point = AN.SplitScratch;
03699     k = AN.InScratch;
03700     while ( --k >= 0 ) {
03701         if ( *Point ) {
03702             *Point++ = s2;
03703             i = *t;
03704             NCOPY(s2,t,i);
03705         }
03706         else Point++;
03707     }
03708     S->sFill = s2;
03709     #ifdef TESTGARB
03710     MLOCK(ErrorMessageLock);
03711     MesPrint("out: S->sFill = %l, S->sTop2 = %l",S->sFill-S->sBuffer,S->sTop2-S->sBuffer);
03712     if ( S->sFill >= S->sTop2 ) {
03713         MesPrint("We are in deep trouble");
03714     }
03715     MUNLOCK(ErrorMessageLock);
03716     #endif
03717     if ( tobereturned ) M_free(garbuf,"Garbage buffer");
03718     return;
03719 }
03720
03721 /*
03722     #] GarbHand :
03723     #[ MergePatches :          WORD MergePatches(par)
03724 */
03741 int MergePatches(WORD par)
03742 {
03743     GETIDENTITY
03744     SORTING *S = AT.SS;
03745     WORD **poin, **poin2, ul, k, i, im, *m1;
03746     WORD *p, lpat, mpat, level, ll, l2, rl, r2, r3, c;
03747     WORD *m2, *m3, r31, r33, ki, *rr;
03748     UWORD *coef;
03749     POSITION position;
03750     FILEHANDLE *fin, *fout;
03751     int fhandle;
03752 /*
03753     UBYTE *s;
03754 */
03755     #ifdef WITHZLIB
03756     POSITION position2;
03757     int oldgzipCompress = AR.gzipCompress;
03758     if ( par == 2 ) {

```

```

03759     AR.zipCompress = 0;
03760 }
03761 #endif
03762 fin = &S->file;
03763 fout = &(AR.FoStage4[0]);
03764 NewMerge:
03765     coef = AN.SoScratC;
03766     poin = S->poina; poin2 = S->poin2a;
03767     rr = AR.CompressPointer;
03768     *rr = 0;
03769 /*
03770     #[] Setup :
03771 */
03772     if ( par == 1 ) {
03773         fout = &(S->file);
03774         if ( fout->handle < 0 ) {
03775 FileMake:
03776             PUTZERO(AN.OldPosOut);
03777             if ( ( fhandle = CreateFile(fout->name) ) < 0 ) {
03778                 MLOCK(ErrorMessageLock);
03779                 MesPrint("Cannot create file %s",fout->name);
03780                 MUNLOCK(ErrorMessageLock);
03781                 goto ReturnError;
03782             }
03783 #ifdef GZIPDEBUG
03784             MLOCK(ErrorMessageLock);
03785             MesPrint("%w MergePatches created output file %s",fout->name);
03786             MUNLOCK(ErrorMessageLock);
03787 #endif
03788             fout->handle = fhandle;
03789             PUTZERO(fout->filesize);
03790             PUTZERO(fout->POposition);
03791 /*
03792             Should not be here?
03793 #ifdef WITHZLIB
03794             fout->ziobuffer = 0;
03795 #endif
03796 */
03797 #ifdef ALLLOCK
03798             LOCK(fout->pthreadlock);
03799 #endif
03800             SeekFile(fout->handle,&(fout->filesize),SEEK_SET);
03801 #ifdef ALLLOCK
03802             UNLOCK(fout->pthreadlock);
03803 #endif
03804             S->fPatchN = 0;
03805             PUTZERO(S->fPatches[0]);
03806             fout->POfill = fout->PObuffer;
03807             PUTZERO(fout->POposition);
03808         }
03809 ConMer:
03810         StageSort(fout);
03811 #ifdef WITHZLIB
03812         if ( S == AT.S0 && AR.NoCompress == 0 && AR.zipCompress > 0 )
03813             S->fpcompressed[S->fPatchN] = 1;
03814         else
03815             S->fpcompressed[S->fPatchN] = 0;
03816         SetupOutputGZIP(fout);
03817 #endif
03818     }
03819     else if ( par == 0 && S->stage4 > 0 ) {
03820 /*
03821         We will have to do our job more than once.
03822         Input is from S->file and output will go to AR.FoStage4.
03823         The file corresponding to this last one must be made now.
03824 */
03825         AR.Stage4Name ^= 1;
03826 /*
03827         s = (UBYTE *) (fout->name); while ( *s ) s++;
03828         if ( AR.Stage4Name ) s[-1] += 1;
03829         else
03830             s[-1] -= 1;
03831 */
03832         S->iPatches = S->fPatches;
03833         S->fPatches = S->inPatches;
03834         S->inPatches = S->iPatches;
03835         (S->inNum) = S->fPatchN;
03836         AN.OldPosIn = AN.OldPosOut;
03837 #ifdef WITHZLIB
03838         m1 = S->fpincompressed;
03839         S->fpincompressed = S->fpcompressed;
03840         S->fpcompressed = m1;
03841         for ( i = 0; i < S->inNum; i++ ) {
03842             S->fPatchesStop[i] = S->iPatches[i+1];
03843 #ifdef GZIPDEBUG
03844             MLOCK(ErrorMessageLock);
03845             MesPrint("%w fPatchesStop[%d] = %10p",i,&(S->fPatchesStop[i]));
03846             MUNLOCK(ErrorMessageLock);

```

```

03846 #endif
03847     }
03848 #endif
03849     S->stage4 = 0;
03850     goto FileMake;
03851 }
03852 else {
03853 #ifdef WITHZLIB
03854 /*
03855     The next statement is just for now
03856 */
03857     AR.gzipCompress = 0;
03858 #endif
03859     if ( par == 0 ) {
03860         S->iPatches = S->fPatches;
03861         S->inNum = S->fPatchN;
03862 #ifdef WITHZLIB
03863         m1 = S->fpincompressed;
03864         S->fpincompressed = S->fpcompressed;
03865         S->fpcompressed = m1;
03866         for ( i = 0; i < S->inNum; i++ ) {
03867             S->fPatchesStop[i] = S->fPatches[i+1];
03868 #ifdef GZIPDEBUG
03869             MLOCK(ErrorMessageLock);
03870             MesPrint("%w fPatchesStop[%d] = %10p", i, &(S->fPatchesStop[i]));
03871             MUNLOCK(ErrorMessageLock);
03872 #endif
03873         }
03874 #endif
03875     }
03876     fout = AR.outfile;
03877 }
03878 if ( par ) { /* Mark end of patches */
03879     S->Patches[S->lPatch] = S->lFill;
03880     for ( i = 0; i < S->lPatch; i++ ) {
03881         S->pStop[i] = S->Patches[i+1]-1;
03882         S->Patches[i] = (WORD *)(((UBYTE *) (S->Patches[i])) + AM.MaxTer);
03883     }
03884 }
03885 else { /* Load the patches */
03886     S->lPatch = (S->inNum);
03887 #ifdef WITHMPI
03888     if ( S->lPatch > 1 || ( (PF.exprtodo < 0) && (fout == AR.outfile || fout == AR.hidefile) ) ) {
03889 #else
03890     if ( S->lPatch > 1 ) {
03891 #endif
03892 #ifdef WITHZLIB
03893         SetupAllInputGZIP(S);
03894 #endif
03895         p = S->lBuffer;
03896         for ( i = 0; i < S->lPatch; i++ ) {
03897             p = (WORD *)(((UBYTE *)p)+2*AM.MaxTer+COMPINC*sizeof(WORD));
03898             S->Patches[i] = p;
03899             p = (WORD *)(((UBYTE *)p) + fin->POsize);
03900             S->pStop[i] = m2 = p;
03901 #ifdef WITHZLIB
03902             PutIn(fin, &(S->iPatches[i]), S->Patches[i], &m2, i);
03903 #else
03904             ADDPOS(S->iPatches[i], PutIn(fin, &(S->iPatches[i]), S->Patches[i], &m2, i));
03905 #endif
03906         }
03907     }
03908 }
03909 if ( fout->handle >= 0 ) {
03910     PUTZERO(position);
03911 #ifdef ALLLOCK
03912     LOCK(fout->pthreadlock);
03913 #endif
03914     SeekFile(fout->handle, &position, SEEK_END);
03915     ADDPOS(position, ((fout->POfill-fout->PObuffer)*sizeof(WORD)));
03916 #ifdef ALLLOCK
03917     UNLOCK(fout->pthreadlock);
03918 #endif
03919 }
03920 else {
03921     SETBASEPOSITION(position, (fout->POfill-fout->PObuffer)*sizeof(WORD));
03922 }
03923 /*
03924     #] Setup :
03925
03926     The old code had to be replaced because all output needs to go
03927     through PutOut. For this we have to go term by term and keep
03928     track of the compression.
03929 */
03930 if ( S->lPatch == 1 ) { /* Single patch --> direct copy. Very rare. */
03931     LONG length;
03932 }

```

```

03933         if ( fout->handle < 0 ) if ( Sflush(fout) ) goto PatCall;
03934         if ( par ) {           /* Memory to file */
03935 #ifdef WITHZLIB
03936 /*
03937         We fix here the problem that the thing needs to go through PutOut
03938 */
03939         m2 = m1 = *S->Patches; /* The m2 is to keep the compiler from complaining */
03940         while ( *m1 ) {
03941             if ( *m1 < 0 ) { /* Need to uncompress */
03942                 i = -( *m1++ ); m2 += i; im = *m1+i+1;
03943                 while ( i > 0 ) { *m1-- = *m2--; i--; }
03944                 *m1 = im;
03945             }
03946 #ifdef WITHPTHREADS
03947             /* Check here (and in the following) that we are at ground level
03948             (so S == AT.S0) to use PutToMaster. Control can reach here,
03949             with AS.MasterSort, but with S != AT.S0, when the SortBots are
03950             adding PolyRatFun and the sorting of their arguments requires
03951             large buffer patches to be merged. In this case, terms escape
03952             the PolyRatFun argument and end up at ground level, if we use
03953             PutToMaster. */
03954             if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
03955                 im = PutToMaster(BHEAD m1);
03956             }
03957             else
03958 #endif
03959                 if ( ( im = PutOut(BHEAD m1,&position,fout,1) ) < 0 ) goto ReturnError;
03960             ADDPOS(S->SizeInFile[par],im);
03961             m2 = m1;
03962             m1 += *m1;
03963         }
03964 #ifdef WITHPTHREADS
03965         if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
03966             PutToMaster(BHEAD 0);
03967         }
03968         else
03969 #endif
03970             if ( FlushOut(&position,fout,1) ) goto ReturnError;
03971             ADDPOS(S->SizeInFile[par],1);
03972 #else
03973 /* old code */
03974         length = (LONG) (*S->pStop)-(LONG) (*S->Patches)+sizeof(WORD);
03975         if ( WriteFile(fout->handle, (UBYTE *) (*S->Patches),length) != length )
03976             goto PatwCall;
03977         ADDPOS(position,length);
03978         ADDPOS(fout->POposition,length);
03979         ADDPOS(fout->filesize,length);
03980         ADDPOS(S->SizeInFile[par],length/sizeof(WORD));
03981 #endif
03982     }
03983     else {           /* File to file */
03984 #ifdef WITHZLIB
03985 /*
03986         Note: if we change FRONTSIZE we need to make the minimum value
03987         of SmallEsize in AllocSort correspondingly larger or smaller.
03988         Theoretically we could get close to 2*AM.MaxTer!
03989 */
03990         #define FRONTSIZE (2*AM.MaxTer)
03991         WORD *copybuf = (WORD *) (((UBYTE *) (S->sBuffer)) + FRONTSIZE);
03992         WORD *copytop;
03993         SetupAllInputGZIP(S);
03994         m1 = m2 = copybuf;
03995         position2 = S->iPatches[0];
03996         while ( ( length = FillInputGZIP(fin,&position2,
03997             (UBYTE *) copybuf,
03998             (S->SmallEsize*sizeof(WORD)-FRONTSIZE,0) ) > 0 ) {
03999             copytop = (WORD *) (((UBYTE *) copybuf)+length);
04000             while ( *m1 && ( ( *m1 > 0 && m1+m1 < copytop ) ||
04001                 ( *m1 < 0 && ( m1+1 < copytop ) && ( m1+m1[1]+1 < copytop ) ) ) )
04002 /*
04003         22-jun-2013 JV Extremely nasty bug that has been around for a while.
04004         What if the end is in the remaining part? We will loose terms!
04005         while ( *m1 && ( (WORD *) (((UBYTE *) (m1)) + AM.MaxTer ) < S->sTop2 ) )
04006 */
04007             {
04008                 if ( *m1 < 0 ) { /* Need to uncompress */
04009                     i = -( *m1++ ); m2 += i; im = *m1+i+1;
04010                     while ( i > 0 ) { *m1-- = *m2--; i--; }
04011                     *m1 = im;
04012                 }
04013 #ifdef WITHPTHREADS
04014                 if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
04015                     im = PutToMaster(BHEAD m1);
04016                 }
04017                 else
04018 #endif
04019                     if ( ( im = PutOut(BHEAD m1,&position,fout,1) ) < 0 ) goto ReturnError;

```

```

04020             ADDPOS(S->SizeInFile[par],im);
04021             m2 = m1;
04022             m1 += *m1;
04023         }
04024         if ( m1 < copytop && *m1 == 0 ) break;
04025     /*
04026         Now move the remaining part 'back'
04027     */
04028     m3 = copybuf;
04029     m1 = copytop;
04030     while ( m1 > m2 ) *--m3 = *--m1;
04031     m2 = m3;
04032     m1 = m2 + *m2;
04033 }
04034 if ( length < 0 ) {
04035     MLOCK(ErrorMessageLock);
04036     MesPrint("Readerror");
04037     goto PatCall2;
04038 }
04039 #ifdef WITHPTHREADS
04040     if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
04041         PutToMaster(BHEAD 0);
04042     }
04043     else
04044 #endif
04045     if ( FlushOut(&position,fout,1) ) goto ReturnError;
04046     ADDPOS(S->SizeInFile[par],1);
04047 #else
04048     /* old code */
04049     SeekFile(fin->handle,&(S->iPatches[0]),SEEK_SET); /* needed for stage4 */
04050     while ( ( length = ReadFile(fin->handle,
04051         (UBYTE *) (S->sBuffer),S->SmallSize*sizeof(WORD)) ) > 0 ) {
04052         if ( WriteFile(fout->handle,(UBYTE *) (S->sBuffer),length) != length )
04053             goto PatwCall;
04054         ADDPOS(position,length);
04055         ADDPOS(fout->PPosition,length);
04056         ADDPOS(fout->filesize,length);
04057         ADDPOS(S->SizeInFile[par],length/sizeof(WORD));
04058     }
04059     if ( length < 0 ) {
04060         MLOCK(ErrorMessageLock);
04061         MesPrint("Readerror");
04062         goto PatCall2;
04063     }
04064 #endif
04065 }
04066 goto EndOfAll;
04067 }
04068 else if ( S->lPatch > 0 ) {
04069     /* More than one patch. Construct the tree. */
04070
04071     lpat = 1;
04072     do { lpat *= 2; } while ( lpat < S->lPatch );
04073     mpat = ( lpat » 1 ) - 1;
04074     k = lpat - S->lPatch;
04075
04076     /* k is the number of empty places in the tree. they will
04077        be at the even positions from 2 to 2*k */
04078
04079     for ( i = 1; i < lpat; i++ ) {
04080         S->tree[i] = -1;
04081     }
04082     for ( i = 1; i <= k; i++ ) {
04083         im = ( i * 2 ) - 1;
04084         poin[im] = S->Patches[i-1];
04085         poin2[im] = poin[im] + *(poin[im]);
04086         S->used[i] = im;
04087         S->ktoi[im] = i-1;
04088         S->tree[mpat+i] = 0;
04089         poin[im-1] = poin2[im-1] = 0;
04090     }
04091     for ( i = (k*2)+1; i <= lpat; i++ ) {
04092         S->used[i-k] = i;
04093         S->ktoi[i] = i-k-1;
04094         poin[i] = S->Patches[i-k-1];
04095         poin2[i] = poin[i] + *(poin[i]);
04096     }
04097 }
04098 /*
04099     the array poin tells the position of the i-th element of the S->tree
04100     'S->used' is a stack with the S->tree elements that need to be entered
04101     into the S->tree. at the beginning this is S->lPatch. during the
04102     sort there will be only very few elements.
04103     poin2 is the next value of poin. it has to be determined
04104     before the comparisons as the position or the size of the
04105     term indicated by poin may change.
04106     S->ktoi translates a S->tree element back to its stream number.

```



```

04107
04108     start the sort
04109 */
04110     level = S->lPatch;
04111
04112     /* introduce one term */
04113 OneTerm:
04114     k = S->used[level];
04115     i = k + lpat - 1;
04116     if ( !*(poin[k]) ) {
04117         do { if ( !( i >= 1 ) ) goto EndOfMerge; } while ( !S->tree[i] );
04118         if ( S->tree[i] == -1 ) {
04119             S->tree[i] = 0;
04120             level--;
04121             goto OneTerm;
04122         }
04123         k = S->tree[i];
04124         S->used[level] = k;
04125         S->tree[i] = 0;
04126     }
04127 /*
04128     move terms down the tree
04129 */
04130     while ( i >= 1 ) {
04131         if ( S->tree[i] > 0 ) {
04132             if ( ( c = CompareTerms(BHEAD poin[S->tree[i]],poin[k],(WORD)0) ) > 0 ) {
04133                 /*
04134                     S->tree[i] is the smaller. Exchange and go on.
04135                 */
04136                 S->used[level] = S->tree[i];
04137                 S->tree[i] = k;
04138                 k = S->used[level];
04139             }
04140             else if ( !c ) { /* Terms are equal */
04141                 S->TermsLeft--;
04142             }
04143             /*
04144                 Here the terms are equal and their coefficients
04145                 have to be added.
04146             */
04147             l1 = *( m1 = poin[S->tree[i]] );
04148             l2 = *( m2 = poin[k] );
04149             if ( S->PolyWise ) { /* Here we work with PolyFun */
04150                 WORD *ttl, *w;
04151                 ttl = m1;
04152                 m1 += S->PolyWise;
04153                 m2 += S->PolyWise;
04154                 if ( S->PolyFlag == 2 ) {
04155                     w = poly_ratfun_add(BHEAD m1,m2);
04156                     if ( *ttl + w[l1] - m1[l1] > AM.MaxTer/((LONG)sizeof(WORD)) ) {
04157                         MLOCK(ErrorMessageLock);
04158                         MesPrint("Term too complex in PolyRatFun addition. MaxTermSize of %10l
04159 is too small",AM.MaxTer);
04160                         MUNLOCK(ErrorMessageLock);
04161                         Terminate(-1);
04162                     }
04163                     AT.WorkPointer = w;
04164                 }
04165                 else {
04166                     w = AT.WorkPointer;
04167                     if ( w + m1[l1] + m2[l1] > AT.WorkTop ) {
04168                         MLOCK(ErrorMessageLock);
04169                         MesPrint("A Workspace of %10l is too small",AM.WorkSize);
04170                         MUNLOCK(ErrorMessageLock);
04171                         Terminate(-1);
04172                     }
04173                     AddArgs(BHEAD m1,m2,w);
04174                 }
04175                 r1 = w[l1];
04176                 if ( r1 <= FUNHEAD
04177                     || ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) )
04178                     { goto cancelled; }
04179                 if ( r1 == m1[l1] ) {
04180                     NCOPY(m1,w,r1);
04181                 }
04182                 else if ( r1 < m1[l1] ) {
04183                     r2 = m1[l1] - r1;
04184                     m2 = w + r1;
04185                     m1 += m1[l1];
04186                     while ( --r1 >= 0 ) *--m1 = *--m2;
04187                     m2 = m1 - r2;
04188                     r1 = S->PolyWise;
04189                     while ( --r1 >= 0 ) *--m1 = *--m2;
04190                     *m1 -= r2;
04191                     poin[S->tree[i]] = m1;
04192                 }
04193                 else {
04194                     r2 = r1 - m1[l1];

```

```

04193             m2 = ttl - r2;
04194             r1 = S->PolyWise;
04195             m1 = ttl;
04196             *m1 += r2;
04197             poin[S->tree[i]] = m2;
04198             NCOPY(m2,m1,r1);
04199             r1 = w[l];
04200             NCOPY(m2,w,r1);
04201         }
04202     }
04203 #ifdef WITHFLOAT
04204     else if ( AT.SortFloatMode ) {
04205         WORD *term1, *term2;
04206         term1 = poin[S->tree[i]];
04207         term2 = poin[k];
04208         if ( MergeWithFloat(BHEAD &term1,&term2) == 0 )
04209             goto cancelled;
04210         poin[S->tree[i]] = term1;
04211     }
04212 #endif
04213     else {
04214         r1 = *( m1 += l1 - 1 );
04215         m1 -= ABS(r1) - 1;
04216         r1 = ( ( r1 > 0 ) ? (r1-1) : (r1+1) ) >> 1;
04217         r2 = *( m2 += l2 - 1 );
04218         m2 -= ABS(r2) - 1;
04219         r2 = ( ( r2 > 0 ) ? (r2-1) : (r2+1) ) >> 1;
04220
04221         if ( AddRat(BHEAD (UWORD *)m1,r1,(UWORD *)m2,r2,coef,&r3) ) {
04222             MLOCK(ErrorMessageLock);
04223             MesCall("MergePatches");
04224             MUNLOCK(ErrorMessageLock);
04225             SETERROR(-1)
04226         }
04227
04228         if ( AN.ncmod != 0 ) {
04229             if ( ( AC.modmode & POSNEG ) != 0 ) {
04230                 NormalModulus(coef,&r3);
04231             }
04232             else if ( BigLong(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod)) >= 0 ) {
04233                 WORD ii;
04234                 SubPLon(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod),coef,&r3);
04235                 coef[r3] = 1;
04236                 for ( ii = 1; ii < r3; ii++ ) coef[r3+ii] = 0;
04237             }
04238         }
04239         r3 *= 2;
04240         r33 = ( r3 > 0 ) ? ( r3 + 1 ) : ( r3 - 1 );
04241         if ( r3 < 0 ) r3 = -r3;
04242         if ( r1 < 0 ) r1 = -r1;
04243         r1 *= 2;
04244         r31 = r3 - r1;
04245         if ( !r3 ) { /* Terms cancel */
04246 cancelled:
04247             ul = S->used[level] = S->tree[i];
04248             S->tree[i] = -1;
04249             /*
04250             We skip to the next term in stream ul
04251             */
04252             im = *poin2[ul];
04253             if ( im < 0 ) {
04254                 r1 = poin2[ul][1] - im + 1;
04255                 m1 = poin2[ul] + 2;
04256                 m2 = poin[ul] - im + 1;
04257                 while ( ++im <= 0 ) *--m1 = *--m2;
04258                 *--m1 = r1;
04259                 poin2[ul] = m1;
04260                 im = r1;
04261             }
04262             poin[ul] = poin2[ul];
04263             ki = S->ktoi[ul];
04264             if ( !par && (poin[ul] + im + COMPINC) >= S->pStop[ki]
04265                 && im > 0 ) {
04266 #ifdef WITHZLIB
04267                 PutIn(fin,&(S->iPatches[ki]),S->Patches[ki],&(poin[ul]),ki);
04268 #else
04269                 ADDPOS(S->iPatches[ki],PutIn(fin,&(S->iPatches[ki]),
04270                     S->Patches[ki],&(poin[ul]),ki));
04271 #endif
04272                 poin2[ul] = poin[ul] + im;
04273             }
04274             else {
04275                 poin2[ul] += im;
04276             }
04277             S->used[++level] = k;
04278             S->TermsLeft--;
04279         }

```

```

04280         else if ( !r31 ) {           /* copy coef into term1 */
04281             goto CopCof2;
04282         }
04283         else if ( r31 < 0 ) {         /* copy coef into term1
04284                                     and adjust the length of term1 */
04285             goto CopCoef;
04286         }
04287         else {
04288             /*
04289              this is the dreaded calamity.
04290              is there enough space?
04291             */
04292             if( (poin[S->tree[i]]+l1+r31) >= poin2[S->tree[i]] ) {
04293                 /*
04294                  no space! now the special trick for which
04295                  we left 2*maxlng spaces open at the beginning
04296                  of each patch.
04297                 */
04298                 if ( (l1 + r31) > AM.MaxTer/((LONG)sizeof(WORD)) ) {
04299                     MLOCK(ErrorMessageLock);
04300                     MesPrint("Coefficient overflow during sort");
04301                     MUNLOCK(ErrorMessageLock);
04302                     goto ReturnError;
04303                 }
04304                 m2 = poin[S->tree[i]];
04305                 m3 = ( poin[S->tree[i]] -= r31 );
04306                 do { *m3++ = *m2++; } while ( m2 < m1 );
04307                 m1 = m3;
04308             }
04309             CopCoef:
04310                 *(poin[S->tree[i]]) += r31;
04311             CopCof2:
04312                 m2 = (WORD *)coef; im = r3;
04313                 NCOPY(m1,m2,im);
04314                 *m1 = r33;
04315             }
04316         }
04317         /*
04318         Now skip to the next term in stream k.
04319         */
04320     NextTerm:
04321         im = poin2[k][0];
04322         if ( im < 0 ) {
04323             r1 = poin2[k][1] - im + 1;
04324             m1 = poin2[k] + 2;
04325             m2 = poin[k] - im + 1;
04326             while ( ++im <= 0 ) *--m1 = *--m2;
04327             *--m1 = r1;
04328             poin2[k] = m1;
04329             im = r1;
04330         }
04331         poin[k] = poin2[k];
04332         ki = S->ktoi[k];
04333         if ( !par && ( (poin[k] + im + COMPINC) >= S->pStop[ki] )
04334             && im > 0 ) {
04335             #ifdef WITHZLIB
04336                 PutIn(fin,&(S->iPatches[ki]),S->Patches[ki],&(poin[k]),ki);
04337             #else
04338                 ADDPOS(S->iPatches[ki],PutIn(fin,&(S->iPatches[ki]),
04339                     S->Patches[ki],&(poin[k]),ki));
04340             #endif
04341             poin2[k] = poin[k] + im;
04342         }
04343         else {
04344             poin2[k] += im;
04345         }
04346         goto OneTerm;
04347     }
04348 }
04349 else if ( S->tree[i] < 0 ) {
04350     S->tree[i] = k;
04351     level--;
04352     goto OneTerm;
04353 }
04354 }
04355 /*
04356 found the smallest in the set. indicated by k.
04357 write to its destination.
04358 */
04359 #ifdef WITHPTHREADS
04360     if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
04361         im = PutToMaster(BHEAD poin[k]);
04362     }
04363     else
04364 #endif
04365     if ( ( im = PutOut(BHEAD poin[k],&position,fout,1) ) < 0 ) {
04366         MLOCK(ErrorMessageLock);

```

```

04367         MesPrint("Called from MergePatches with k = %d (stream %d)",k,S->ktoi[k]);
04368         MUNLOCK(ErrorMessageLock);
04369         goto ReturnError;
04370     }
04371     ADDPOS(S->SizeInFile[par],im);
04372     goto NextTerm;
04373 }
04374 else {
04375     goto NormalReturn;
04376 }
04377 EndOfMerge:
04378 #ifdef WITHPTHREADS
04379     if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
04380         PutToMaster(BHEAD 0);
04381     }
04382     else
04383 #endif
04384     if ( FlushOut(&position,fout,1) ) goto ReturnError;
04385     ADDPOS(S->SizeInFile[par],1);
04386 EndOfAll:
04387     if ( par == 1 ) { /* Set the fpatch pointers */
04388 #ifdef WITHZLIB
04389         SeekFile(fout->handle,&position,SEEK_CUR);
04390 #endif
04391         (S->fPatchN)++;
04392         S->fPatches[S->fPatchN] = position;
04393     }
04394     if ( par == 0 && fout != AR.outfile ) {
04395 /*
04396         Output went to sortfile. We have two possibilities:
04397         1: We are not finished with the current in-out cycle
04398           In that case we should pop to the next set of patches
04399         2: We finished a cycle and should clean up the in file
04400           Then we restart the sort.
04401 */
04402         (S->fPatchN)++;
04403         S->fPatches[S->fPatchN] = position;
04404         if ( ISNOTZEROPOS(AN.OldPosIn) ) { /* We are not done */
04405
04406             SeekFile(fin->handle,&(AN.OldPosIn),SEEK_SET);
04407 /*
04408             We don't need extra provisions for the zlib compression here.
04409             If part of an expression has been sorted, the whole has been so.
04410             This means that S->fpincompressed[] will remain the same
04411 */
04412             if ( (ULONG)ReadFile(fin->handle,(UBYTE *)(&(S->inNum)),(LONG)sizeof(WORD)) !=
04413                 sizeof(WORD)
04414             || (ULONG)ReadFile(fin->handle,(UBYTE *)(&AN.OldPosIn),(LONG)sizeof(POSITION)) !=
04415                 sizeof(POSITION)
04416             || (ULONG)ReadFile(fin->handle,(UBYTE *)S->iPatches,(LONG)((S->inNum)+1)
04417                 *sizeof(POSITION)) != ((S->inNum)+1)*sizeof(POSITION) ) {
04418                 MLOCK(ErrorMessageLock);
04419                 MesPrint("Read error fourth stage sorting");
04420                 MUNLOCK(ErrorMessageLock);
04421                 goto ReturnError;
04422             }
04423             *rr = 0;
04424 #ifdef WITHZLIB
04425             for ( i = 0; i < S->inNum; i++ ) {
04426                 S->fPatchesStop[i] = S->iPatches[i+1];
04427 #ifdef GZIPDEBUG
04428                 MLOCK(ErrorMessageLock);
04429                 MesPrint("%w fPatchesStop[%d] = %10p",i,&(S->fPatchesStop[i]));
04430                 MUNLOCK(ErrorMessageLock);
04431 #endif
04432             }
04433 #endif
04434             goto ConMer;
04435         }
04436     }
04437 /*
04438     if ( fin == &(AR.FoStage4[0]) ) {
04439         s = (UBYTE *) (fin->name); while ( *s ) s++;
04440         if ( AR.Stage4Name == 1 ) s[-1] -= 1;
04441         else s[-1] += 1;
04442     }
04443 */
04444 /* TruncateFile(fin->handle); */
04445     UpdateMaxSize();
04446 #ifdef WITHZLIB
04447     ClearSortGZIP(fin);
04448 #endif
04449     CloseFile(fin->handle);
04450     remove(fin->name); /* Gives disk space free again. */
04451 #ifdef GZIPDEBUG
04452     MLOCK(ErrorMessageLock);
04453     MesPrint("%w MergePatches removed in file %s",fin->name);

```

```

04454             MUNLOCK (ErrorMessageLock);
04455 #endif
04456 /*
04457             if ( fin == &(AR.FoStage4[0]) ) {
04458                 s = (UBYTE *) (fin->name); while ( *s ) s++;
04459                 if ( AR.Stage4Name == 1 ) s[-1] += 1;
04460                 else s[-1] -= 1;
04461             }
04462 */
04463             fin->handle = -1;
04464             { FILEHANDLE *ff = fin; fin = fout; fout = ff; }
04465             PUTZERO(S->SizeInFile[0]);
04466             goto NewMerge;
04467         }
04468     }
04469     if ( par == 0 ) {
04470 /*         TruncateFile(fin->handle); */
04471         UpdateMaxSize();
04472 #ifdef WITHZLIB
04473         ClearSortGZIP(fin);
04474 #endif
04475         CloseFile(fin->handle);
04476         remove(fin->name);
04477         fin->handle = -1;
04478 #ifdef GZIPDEBUG
04479         MLOCK(ErrorMessageLock);
04480         MesPrint("%w MergePatches removed in file %s",fin->name);
04481         MUNLOCK(ErrorMessageLock);
04482 #endif
04483     }
04484 NormalReturn:
04485 #ifdef WITHZLIB
04486     AR.gzipCompress = oldgzipCompress;
04487 #endif
04488     return(0);
04489 ReturnError:
04490 #ifdef WITHZLIB
04491     AR.gzipCompress = oldgzipCompress;
04492 #endif
04493     return(-1);
04494 #ifndef WITHZLIB
04495 PatwCall:
04496     MLOCK(ErrorMessageLock);
04497     MesPrint("Error while writing to file.");
04498     goto PatCall2;
04499 #endif
04500 PatCall:;
04501     MLOCK(ErrorMessageLock);
04502 PatCall2:;
04503     MesCall("MergePatches");
04504     MUNLOCK(ErrorMessageLock);
04505 #ifdef WITHZLIB
04506     AR.gzipCompress = oldgzipCompress;
04507 #endif
04508     SETERROR(-1)
04509 }
04510
04511 /*
04512         #] MergePatches :
04513         #[ StoreTerm :                WORD StoreTerm(term)
04514 */
04524 int StoreTerm(PHEAD WORD *term)
04525 {
04526     GETBIDENTITY
04527     SORTING *s = AT.SS;
04528     WORD **ss, *lfill, j, *t;
04529     POSITION pp;
04530     LONG lSpace, sSpace, RetCode, over, tover;
04531
04532     if ( ( ( AP.PreDebug & DUMPTOSORT ) == DUMPTOSORT ) && AR.sLevel == 0 ) {
04533 #ifdef WITHPTHREADS
04534         snprintf((char *) (THRbuf),100,"StoreTerm(%d)",AT.identity);
04535         PrintTerm(term, (char *) (THRbuf));
04536 #else
04537         PrintTerm(term, "StoreTerm");
04538 #endif
04539     }
04540     if ( AM.exitflag && AR.sLevel == 0 ) return(0);
04541     S->sFill = *(S->PoinFill);
04542     if ( S->sTerms >= S->TermsInSmall || ( S->sFill + *term ) >= S->sTop ) {
04543 /*         The small buffer is full. It has to be sorted and written.
04544 */
04545         tover = over = S->sTerms;
04546         ss = S->sPointer;
04547         ss[over] = 0;
04548 #ifdef SPLITTIME

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```

04550     PrintTime((UBYTE *) "Before SplitMerge");
04551 #endif
04552     ss[SplitMerge(BHEAD ss,over)] = 0;
04553 #ifdef SPLITTIME
04554     PrintTime((UBYTE *) "After SplitMerge");
04555 #endif
04556     sSpace = 0;
04557     if ( over > 0 ) {
04558         sSpace = ComPress(ss,&RetCode);
04559         S->TermsLeft -= over - RetCode;
04560     }
04561     sSpace++;
04562
04563     lSpace = sSpace + (S->lFill - S->lBuffer)
04564             - (AM.MaxTer/sizeof(WORD))*((LONG)S->lPatch);
04565     SETBASEPOSITION(pp,lSpace);
04566     MULPOS(pp,sizeof(WORD));
04567     if ( S->file.handle >= 0 ) {
04568         ADD2POS(pp,S->fPatches[S->fPatchN]);
04569     }
04570     if ( S == AT.S0 ) { /* Only statistics at ground level */
04571         WriteStats(&pp,STATSSPLITMERGE,CHECKLOGTYPE);
04572     }
04573     if ( ( S->lPatch >= S->MaxPatches ) ||
04574         ( ( (WORD *)(((UBYTE *) (S->lFill + sSpace)) + 2*AM.MaxTer ) ) >= S->lTop ) ) {
04575 /*
04576     The large buffer is too full. Merge and write it
04577 */
04578     if ( MergePatches(1) ) goto StoreCall;
04579 /*
04580     pp = S->SizeInFile[1];
04581     ADDPOS(pp,sSpace);
04582     MULPOS(pp,sizeof(WORD));
04583 */
04584     SETBASEPOSITION(pp,sSpace);
04585     MULPOS(pp,sizeof(WORD));
04586     ADD2POS(pp,S->fPatches[S->fPatchN]);
04587
04588     if ( S == AT.S0 ) { /* Only statistics at ground level */
04589         WriteStats(&pp,STATSMERGETOFILE,CHECKLOGTYPE);
04590     }
04591     S->lPatch = 0;
04592     S->lFill = S->lBuffer;
04593 }
04594 S->Patches[S->lPatch++] = S->lFill;
04595 lfill = (WORD *)(((UBYTE *) (S->lFill)) + AM.MaxTer);
04596 if ( tover > 0 ) {
04597     ss = S->sPointer;
04598     while ( ( t = *ss++ ) != 0 ) {
04599         j = *t;
04600         if ( j < 0 ) j = t[1] + 2;
04601         while ( --j >= 0 ){
04602             *lfill++ = *t++;
04603         }
04604     }
04605 }
04606 *lfill++ = 0;
04607 S->lFill = lfill;
04608 S->sTerms = 0;
04609 S->PoinFill = S->sPointer;
04610 *(S->PoinFill) = S->sFill = S->sBuffer;
04611 }
04612 j = *term;
04613 while ( --j >= 0 ) *S->sFill++ = *term++;
04614 S->sTerms++;
04615 S->GenTerms++;
04616 S->TermsLeft++;
04617 **S->PoinFill = S->sFill;
04618
04619     return(0);
04620
04621 StoreCall:
04622     MLOCK(ErrorMessageLock);
04623     MesCall("StoreTerm");
04624     MUNLOCK(ErrorMessageLock);
04625     SETERROR(-1)
04626 }
04627
04628 /*
04629     #] StoreTerm :
04630     #[ StageSort :                void StageSort(FILEHANDLE *fout)
04631 */
04632 void StageSort(FILEHANDLE *fout)
04633 {
04634     GETIDENTITY
04635     SORTING *S = AT.SS;
04636     if ( S->fPatchN >= S->MaxFpatches ) {

```

```

04643     POSITION position;
04644     if ( S != AT.SO ) {
04645 /*
04646         There are no proper provisions for stage 4 or higher sorts
04647         for function arguments and $ variables. The reason:
04648         The current code maps out the patches, based on the size of
04649         the buffers in the FoStage4 structs, while they are used
04650         inside the S->file struct that may have far smaller buffers.
04651         By itself that might still be repairable, but it goes completely
04652         wrong when during the sort polyRatFuns have to be added and they
04653         would go into stage4 (very rare but possible).
04654         The only really correct solution would be to put FoStage4 structs
04655         in all sort levels. Messy. (JV 8-oct-2018).
04656 */
04657         MLOCK(ErrorMessageLock);
04658         MesPrint("Currently Stage 4 sorts are not allowed for function arguments or $
variables.");
04659         MesPrint("Please increase correspondingsorting parameters (sub-) in the setup.");
04660         MUNLOCK(ErrorMessageLock);
04661         Terminate(-1);
04662     }
04663     PUTZERO(position);
04664     MLOCK(ErrorMessageLock);
04665 #ifdef WITHPTHREADS
04666     MesPrint("StageSort in thread %d",identity);
04667 #elif defined(WITHMPI)
04668     MesPrint("StageSort in process %d",PF.me);
04669 #else
04670     MesPrint("StageSort");
04671 #endif
04672     MUNLOCK(ErrorMessageLock);
04673     SeekFile(fout->handle,&position,SEEK_END);
04674 /*
04675     No extra compression data has to be written.
04676     S->fpincompressed should remain valid.
04677 */
04678     if ( (ULONG)WriteFile(fout->handle, (UBYTE *) (&(S->fPatchN)), (LONG)sizeof(WORD)) !=
sizeof(WORD)
|| (ULONG)WriteFile(fout->handle, (UBYTE *) (&(AN.OldPosOut)), (LONG)sizeof(POSITION)) !=
sizeof(POSITION)
|| (ULONG)WriteFile(fout->handle, (UBYTE *) (S->fPatches), (LONG) (S->fPatchN+1)
*sizeof(POSITION)) != (S->fPatchN+1)*sizeof(POSITION) ) {
04684         MLOCK(ErrorMessageLock);
04685         MesPrint("Write error while staging sort. Disk full?");
04686         MUNLOCK(ErrorMessageLock);
04687         Terminate(-1);
04688     }
04689     AN.OldPosOut = position;
04690     fout->filesize = position;
04691     ADDPOS(fout->filesize, (S->fPatchN+2)*sizeof(POSITION) + sizeof(WORD));
04692     fout->PPosition = fout->filesize;
04693     S->fPatches[0] = fout->filesize;
04694     S->fPatchN = 0;
04695
04696     if ( AR.FoStage4[0].PObuffer == 0 ) {
04697         AR.FoStage4[0].PObuffer = (WORD *)Malloc1(AR.FoStage4[0].POsize*sizeof(WORD)
,"Stage 4 buffer");
04698         AR.FoStage4[0].POfill = AR.FoStage4[0].PObuffer;
04699         AR.FoStage4[0].POstop = AR.FoStage4[0].PObuffer
+ AR.FoStage4[0].POsize/sizeof(WORD);
04702 #ifdef WITHPTHREADS
04703         AR.FoStage4[0].pthreadlock = dummylock;
04704 #endif
04705     }
04706     if ( AR.FoStage4[1].PObuffer == 0 ) {
04707         AR.FoStage4[1].PObuffer = (WORD *)Malloc1(AR.FoStage4[1].POsize*sizeof(WORD)
,"Stage 4 buffer");
04708         AR.FoStage4[1].POfill = AR.FoStage4[1].PObuffer;
04709         AR.FoStage4[1].POstop = AR.FoStage4[1].PObuffer
+ AR.FoStage4[1].POsize/sizeof(WORD);
04712 #ifdef WITHPTHREADS
04713         AR.FoStage4[1].pthreadlock = dummylock;
04714 #endif
04715     }
04716     S->stage4 = 1;
04717 }
04718 }
04719
04720 /*
04721     #] StageSort :
04722     #[ SortWild :           WORD SortWild(w,nw)
04723 */
04737 int SortWild(WORD *w, WORD nw)
04738 {
04739     GETIDENTITY
04740     WORD *v, *s, *m, k, i;
04741     WORD *pScreat, *stop, *sv;

```

```

04742     int error = 0;
04743     pScrat = AT.WorkPointer;
04744     if ( ( AT.WorkPointer + 8 * AM.MaxWildcards ) >= AT.WorkTop ) {
04745         MLOCK(ErrorMessageLock);
04746         MesWork();
04747         MUNLOCK(ErrorMessageLock);
04748         return(-1);
04749     }
04750     stop = w + nw;
04751     i = 0;
04752     while ( i < nw ) {
04753         m = w + i;
04754         v = m + m[1];
04755         while ( v < stop && (
04756             *v == FROMSET || *v == SETTONUM || *v == LOADDOLLAR ) ) v += v[1];
04757         while ( v < stop ) {
04758             if ( *v >= 0 ) {
04759                 if ( AM.Ordering[*v] < AM.Ordering[*m] ) {
04760                     m = v;
04761                 }
04762                 else if ( *v == *m ) {
04763                     if ( v[2] < m[2] ) {
04764                         m = v;
04765                     }
04766                     else if ( v[2] == m[2] ) {
04767                         s = m + m[1];
04768                         sv = v + v[1];
04769                         if ( s < stop && ( *s == FROMSET
04770                             || *s == SETTONUM || *s == LOADDOLLAR ) ) {
04771                             if ( sv < stop && ( *sv == FROMSET
04772                                 || *sv == SETTONUM || *sv == LOADDOLLAR ) ) {
04773                                 if ( s[2] != sv[2] ) {
04774                                     error = -1;
04775                                     MLOCK(ErrorMessageLock);
04776                                     MesPrint("&Wildcard set conflict");
04777                                     MUNLOCK(ErrorMessageLock);
04778                                 }
04779                             }
04780                             *v = -1;
04781                         }
04782                     }
04783                     else {
04784                         if ( sv < stop && ( *sv == FROMSET
04785                             || *sv == SETTONUM || *sv == LOADDOLLAR ) ) {
04786                             *m = -1;
04787                             m = v;
04788                         }
04789                     }
04790                     *v = -1;
04791                 }
04792             }
04793         }
04794     }
04795     v += v[1];
04796     while ( v < stop && ( *v == FROMSET
04797         || *v == SETTONUM || *v == LOADDOLLAR ) ) v += v[1];
04798 }
04799 s = pScrat;
04800 v = m;
04801 k = m[1];
04802 NCOPY(s,m,k);
04803 while ( m < stop && ( *m == FROMSET
04804     || *m == SETTONUM || *m == LOADDOLLAR ) ) {
04805     k = m[1];
04806     NCOPY(s,m,k);
04807 }
04808 *v = -1;
04809 pScrat = s;
04810 i = 0;
04811 while ( i < nw && ( w[i] < 0 || w[i] == FROMSET
04812     || w[i] == SETTONUM || w[i] == LOADDOLLAR ) ) i += w[i+1];
04813 }
04814 AC.NwildC = k = WORDDIFF(pScrat,AT.WorkPointer);
04815 s = AT.WorkPointer;
04816 m = w;
04817 NCOPY(m,s,k);
04818 AC.WildC = m;
04819 return(error);
04820 }
04821
04822 /*
04823     #] SortWild :
04824     #[ CleanupSort :                void CleanupSort(num)
04825 */
04830 void CleanupSort(int num)
04831 {
04832     GETIDENTITY

```



```

04833     SORTING *S;
04834     int minnum = num, i;
04835     if ( AN.FunSorts ) {
04836         if ( num == -1 ) {
04837             if ( AN.MaxFunSorts > 3 ) {
04838                 minnum = (AN.MaxFunSorts+4)/2;
04839             }
04840             else minnum = 4;
04841         }
04842         else if ( minnum == 0 ) minnum = 1;
04843         for ( i = minnum; i < AN.NumFunSorts; i++ ) {
04844             S = AN.FunSorts[i];
04845             if ( S ) {
04846                 if ( S->file.handle >= 0 ) {
04847                     /* TruncateFile(S->file.handle); */
04848                     UpdateMaxSize();
04849                     #ifdef WITHZLIB
04850                     ClearSortGZIP(&(S->file));
04851                     #endif
04852                     CloseFile(S->file.handle);
04853                     S->file.handle = -1;
04854                     remove(S->file.name);
04855                     #ifdef GZIPDEBUG
04856                     MLOCK(ErrorMessageLock);
04857                     MesPrint("%w CleanUpSort removed file %s",S->file.name);
04858                     MUNLOCK(ErrorMessageLock);
04859                     #endif
04860                 }
04861                 M_free(S->sPointer, "CleanUpSort: sPointer");
04862                 M_free(S->Patches, "CleanUpSort: Patches");
04863                 M_free(S->pStop, "CleanUpSort: pStop");
04864                 M_free(S->poina, "CleanUpSort: poina");
04865                 M_free(S->poin2a, "CleanUpSort: poin2a");
04866                 M_free(S->fPatches, "CleanUpSort: fPatches");
04867                 M_free(S->fPatchesStop, "CleanUpSort: fPatchesStop");
04868                 M_free(S->inPatches, "CleanUpSort: inPatches");
04869                 M_free(S->tree, "CleanUpSort: tree");
04870                 M_free(S->used, "CleanUpSort: used");
04871                 #ifdef WITHZLIB
04872                 M_free(S->fpcompressed, "CleanUpSort: fpcompressed");
04873                 M_free(S->fpincompressed, "CleanUpSort: fpincompressed");
04874                 #endif
04875                 M_free(S->ktoi, "CleanUpSort: ktoi");
04876                 M_free(S->lBuffer, "CleanUpSort: lBuffer+sBuffer");
04877                 M_free(S->file.PObuffer, "CleanUpSort: PObuffer");
04878                 M_free(S, "CleanUpSort: sorting struct");
04879             }
04880             AN.FunSorts[i] = 0;
04881         }
04882         AN.MaxFunSorts = minnum;
04883         if ( num == 0 ) {
04884             S = AN.FunSorts[0];
04885             if ( S ) {
04886                 if ( S->file.handle >= 0 ) {
04887                     /* TruncateFile(S->file.handle); */
04888                     UpdateMaxSize();
04889                     #ifdef WITHZLIB
04890                     ClearSortGZIP(&(S->file));
04891                     #endif
04892                     CloseFile(S->file.handle);
04893                     S->file.handle = -1;
04894                     remove(S->file.name);
04895                     #ifdef GZIPDEBUG
04896                     MLOCK(ErrorMessageLock);
04897                     MesPrint("%w CleanUpSort removed file %s",S->file.name);
04898                     MUNLOCK(ErrorMessageLock);
04899                     #endif
04900                 }
04901             }
04902         }
04903     }
04904     for ( i = 0; i < 2; i++ ) {
04905         if ( AR.FoStage4[i].handle >= 0 ) {
04906             UpdateMaxSize();
04907             #ifdef WITHZLIB
04908             ClearSortGZIP(&(AR.FoStage4[i]));
04909             #endif
04910             CloseFile(AR.FoStage4[i].handle);
04911             remove(AR.FoStage4[i].name);
04912             AR.FoStage4[i].handle = -1;
04913             #ifdef GZIPDEBUG
04914             MLOCK(ErrorMessageLock);
04915             MesPrint("%w CleanUpSort removed stage4 file %s",AR.FoStage4[i].name);
04916             MUNLOCK(ErrorMessageLock);
04917             #endif
04918         }
04919     }

```

```

04920 }
04921
04922 /*
04923     #] CleanUpSort :
04924     #[ LowerSortLevel :          void LowerSortLevel()
04925 */
04930 void LowerSortLevel(void)
04931 {
04932     GETIDENTITY
04933     if ( AR.sLevel >= 0 ) {
04934         AR.sLevel--;
04935         if ( AR.sLevel >= 0 ) AT.SS = AN.FunSorts[AR.sLevel];
04936     }
04937 }
04938
04939 /*
04940     #] LowerSortLevel :
04941     #[ PolyRatFunSpecial :
04942
04943     Keeps only the most divergent term in AR.PolyFunVar
04944     We assume that the terms are already in that notation.
04945 */
04946
04947 WORD *PolyRatFunSpecial(PHEAD WORD *t1, WORD *t2)
04948 {
04949     WORD *oldworkpointer = AT.WorkPointer, *t, *r;
04950     WORD expl, exp2;
04951     int i;
04952     t = t1+FUNHEAD;
04953     if ( *t == -SYMBOL ) {
04954         if ( t[1] != AR.PolyFunVar ) goto Illegal;
04955         expl = 1;
04956         if ( t[2] != -SNUMBER ) goto Illegal;
04957         t[3] = 1;
04958     }
04959     else if ( *t == -SNUMBER ) {
04960         t[1] = 1;
04961         t += 2;
04962         if ( *t == -SYMBOL ) {
04963             if ( t[1] != AR.PolyFunVar ) goto Illegal;
04964             expl = -1;
04965         }
04966         else if ( *t == -SNUMBER ) {
04967             t[1] = 1;
04968             expl = 0;
04969         }
04970         else if ( *t == ARGHEAD+8 && t[ARGHEAD] == 8 && t[ARGHEAD+1] == SYMBOL
04971             && t[ARGHEAD+3] == AR.PolyFunVar ) {
04972             t[ARGHEAD+5] = 1;
04973             t[ARGHEAD+6] = 1;
04974             t[ARGHEAD+7] = 3;
04975             expl = -t[ARGHEAD+4];
04976         }
04977         else goto Illegal;
04978     }
04979     else if ( *t == ARGHEAD+8 && t[ARGHEAD] == 8 && t[ARGHEAD+1] == SYMBOL
04980         && t[ARGHEAD+3] == AR.PolyFunVar ) {
04981         t[ARGHEAD+5] = 1;
04982         t[ARGHEAD+6] = 1;
04983         t[ARGHEAD+7] = 3;
04984         expl = t[ARGHEAD+4];
04985         t += *t;
04986         if ( *t != -SNUMBER ) goto Illegal;
04987         t[1] = 1;
04988     }
04989     else goto Illegal;
04990
04991     t = t2+FUNHEAD;
04992     if ( *t == -SYMBOL ) {
04993         if ( t[1] != AR.PolyFunVar ) goto Illegal;
04994         exp2 = 1;
04995         if ( t[2] != -SNUMBER ) goto Illegal;
04996         t[3] = 1;
04997     }
04998     else if ( *t == -SNUMBER ) {
04999         t[1] = 1;
05000         t += 2;
05001         if ( *t == -SYMBOL ) {
05002             if ( t[1] != AR.PolyFunVar ) goto Illegal;
05003             exp2 = -1;
05004         }
05005         else if ( *t == -SNUMBER ) {
05006             t[1] = 1;
05007             exp2 = 0;
05008         }
05009         else if ( *t == ARGHEAD+8 && t[ARGHEAD] == 8 && t[ARGHEAD+1] == SYMBOL
05010             && t[ARGHEAD+3] == AR.PolyFunVar ) {

```

```

05011         t[ARGHEAD+5] = 1;
05012         t[ARGHEAD+6] = 1;
05013         t[ARGHEAD+7] = 3;
05014         exp2 = -t[ARGHEAD+4];
05015     }
05016     else goto Illegal;
05017 }
05018 else if ( *t == ARGHEAD+8 && t[ARGHEAD] == 8 && t[ARGHEAD+1] == SYMBOL
05019         && t[ARGHEAD+3] == AR.PolyFunVar ) {
05020     t[ARGHEAD+5] = 1;
05021     t[ARGHEAD+6] = 1;
05022     t[ARGHEAD+7] = 3;
05023     exp2 = t[ARGHEAD+4];
05024     t += *t;
05025     if ( *t != -SNUMBER ) goto Illegal;
05026     t[1] = 1;
05027 }
05028 else goto Illegal;
05029
05030 if ( exp1 <= exp2 ) { i = t1[1]; r = t1; }
05031 else { i = t2[1]; r = t2; }
05032 t = oldworkpointer;
05033 NCOPY(t,r,i)
05034
05035 return(oldworkpointer);
05036 Illegal:
05037 MesPrint("Illegal occurrence of PolyRatFun with divergent option");
05038 Terminate(-1);
05039 return(0);
05040 }
05041
05042 /*
05043     #] PolyRatFunSpecial :
05044     #[ SimpleSplitMerge :
05045
05046     Sorts an array of WORDs. No adding of equal objects.
05047 */
05048
05049 void SimpleSplitMergeRec(WORD *array,WORD num,WORD *auxarray)
05050 {
05051     WORD n1,n2,i,j,k,*t1,*t2;
05052     if ( num < 2 ) return;
05053     if ( num == 2 ) {
05054         if ( array[0] > array[1] ) {
05055             EXCH(array[0],array[1])
05056         }
05057         return;
05058     }
05059     n1 = num/2;
05060     n2 = num - n1;
05061     SimpleSplitMergeRec(array,n1,auxarray);
05062     SimpleSplitMergeRec(array+n1,n2,auxarray);
05063     if ( array[n1-1] <= array[n1] ) return;
05064
05065     t1 = array; t2 = auxarray; i = n1; NCOPY(t2,t1,i);
05066     i = 0; j = n1; k = 0;
05067     while ( i < n1 && j < num ) {
05068         if ( auxarray[i] <= array[j] ) { array[k++] = auxarray[i++]; }
05069         else { array[k++] = array[j++]; }
05070     }
05071     while ( i < n1 ) array[k++] = auxarray[i++];
05072 /*
05073     Remember: remnants of j are still in place!
05074 */
05075 }
05076
05077 void SimpleSplitMerge(WORD *array,WORD num)
05078 {
05079     WORD *auxarray = Malloc1(sizeof(WORD)*num/2,"SimpleSplitMerge");
05080     SimpleSplitMergeRec(array,num,auxarray);
05081     M_free(auxarray,"SimpleSplitMerge");
05082 }
05083
05084 /*
05085     #] SimpleSplitMerge :
05086     #[ BinarySearch :
05087
05088     Searches in the sorted array with length num for the object x.
05089     If x is in the list, it returns the number of the array element
05090     that matched. If it is not in the list, it returns -1.
05091     If there are identical objects in the list, which one will
05092     match is quasi random.
05093 */
05094
05095 WORD BinarySearch(WORD *array,WORD num,WORD x)
05096 {
05097     WORD i, bot, top, med;

```

```

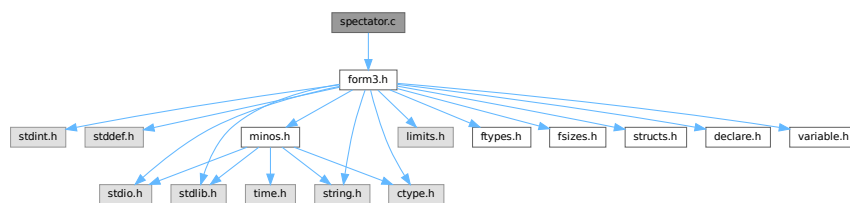
05098     if ( num < 8 ) {
05099         for ( i = 0; i < num; i++ ) if ( array[i] == x ) return(i);
05100         return(-1);
05101     }
05102     if ( array[0] > x || array[num-1] < x ) return(-1);
05103     bot = 0; top = num-1; med = (top+bot)/2;
05104     do {
05105         if ( array[med] == x ) return(med);
05106         if ( array[med] < x ) { bot = med+1; }
05107         else { top = med-1; }
05108         med = (top+bot)/2;
05109     } while ( med >= bot && med <= top );
05110     return(-1);
05111 }
05112
05113 /*
05114     #] BinarySearch :
05115     #] SortUtilities :
05116 */

```

## 4.130 spectator.c File Reference

#include "form3.h"

Include dependency graph for spectator.c:



### Functions

- int [CoCreateSpectator](#) (UBYTE \*inp)
- int [CoToSpectator](#) (UBYTE \*inp)
- int [CoRemoveSpectator](#) (UBYTE \*inp)
- int [CoEmptySpectator](#) (UBYTE \*inp)
- int [PutInSpectator](#) (WORD \*term, WORD specnum)
- void [FlushSpectators](#) (void)
- int [CoCopySpectator](#) (UBYTE \*inp)
- WORD [GetFromSpectator](#) (WORD \*term, WORD specnum)
- void [ClearSpectators](#) (WORD par)

### 4.130.1 Detailed Description

File contains the code for the spectator files and their control.

Definition in file [spectator.c](#).

## 4.130.2 Function Documentation

### 4.130.2.1 CoCreateSpectator()

```
int CoCreateSpectator (
    UBYTE * inp )
```

Definition at line 89 of file [spectator.c](#).

### 4.130.2.2 CoToSpectator()

```
int CoToSpectator (
    UBYTE * inp )
```

Definition at line 181 of file [spectator.c](#).

### 4.130.2.3 CoRemoveSpectator()

```
int CoRemoveSpectator (
    UBYTE * inp )
```

Definition at line 233 of file [spectator.c](#).

### 4.130.2.4 CoEmptySpectator()

```
int CoEmptySpectator (
    UBYTE * inp )
```

Definition at line 293 of file [spectator.c](#).

### 4.130.2.5 PutInSpectator()

```
int PutInSpectator (
    WORD * term,
    WORD specnum )
```

Definition at line 345 of file [spectator.c](#).

### 4.130.2.6 FlushSpectators()

```
void FlushSpectators (
    void )
```

Definition at line 402 of file [spectator.c](#).

#### 4.130.2.7 CoCopySpectator()

```
int CoCopySpectator (
    UBYTE * inp )
```

Definition at line 459 of file [spectator.c](#).

#### 4.130.2.8 GetFromSpectator()

```
WORD GetFromSpectator (
    WORD * term,
    WORD specnum )
```

Definition at line 601 of file [spectator.c](#).

#### 4.130.2.9 ClearSpectators()

```
void ClearSpectators (
    WORD par )
```

Definition at line 675 of file [spectator.c](#).

### 4.131 spectator.c

[Go to the documentation of this file.](#)

```
00001
00006 /* #[] License : */
00007 /*
00008 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 *   When using this file you are requested to refer to the publication
00010 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 *   This is considered a matter of courtesy as the development was paid
00012 *   for by FOM the Dutch physics granting agency and we would like to
00013 *   be able to track its scientific use to convince FOM of its value
00014 *   for the community.
00015 *
00016 *   This file is part of FORM.
00017 *
00018 *   FORM is free software: you can redistribute it and/or modify it under the
00019 *   terms of the GNU General Public License as published by the Free Software
00020 *   Foundation, either version 3 of the License, or (at your option) any later
00021 *   version.
00022 *
00023 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 *   details.
00027 *
00028 *   You should have received a copy of the GNU General Public License along
00029 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #[] License : */
00032 /*
00033 *   #[] includes :
00034 */
00035
00036 #include "form3.h"
00037
00038 /*
00039 *   #[] includes :
00040 *   #[] Commentary :
00041
00042 *   We use an array of SPECTATOR structs in AM.SpectatorFiles.
00043 *   When a spectator is removed this leaves a hole. This means that
00044 *   we cannot use AM.NumSpectatorFiles but always have to scan up to
```

```

00045     AM.SizeForSpectatorFiles which is the size of the array.
00046     An element is in use when it has a name. This is the name of the
00047     expression that is associated with it. There is also the number of
00048     the expression, but because the expressions are frequently renumbered
00049     at the end of a module, we always search for the spectators by name.
00050     The expression number is only valid in the current module.
00051     During execution we use the number of the spectator.
00052
00053     The FILEHANDLE struct is borrowed from the structs for the scratch
00054     files, but we do not keep copies for all workers as with the scratch
00055     files. This brings some limitations (but saves much space). Basically
00056     the reading can only be done by one master or worker. And because
00057     we use the buffer both for writing and for reading we cannot read and
00058     write in the same module.
00059
00060     Processor can see that an expression is to be filled with a spectator
00061     because we replace the compiler buffer number in the prototype by
00062     -specnum-1. Of course, after this filling has taken place we should
00063     make sure that in the next module there is a nonnegative number there.
00064     The input is then obtained from GetFromSpectator instead from GetTerm.
00065     This needed an extra argument in ThreadsProcessor. InParallelProcessor
00066     can figure it out by itself. ParFORM still needs to be treated for this.
00067
00068     The writing is to a single buffer. Hence it needs a lock. It is possible
00069     to give all workers their own buffers (at great memory cost) and merge
00070     the results when needed. That would be friendlier on ParFORM. We ALWAYS
00071     assume that the order of the terms in the spectator file is random.
00072
00073     In the first version there is no compression in the file. This could
00074     change in later versions because both the writing and the reading are
00075     purely sequential. Brackets are not possible.
00076
00077     Currently, ParFORM allows use of spectators only in the sequential
00078     mode. The parallelization is switched off in modules containing
00079     ToSpectator or CopySpectator. Workers never create or access to
00080     spectator files. Their file handles are always -1. We leave the
00081     parallelization of modules with spectators for future work.
00082
00083     #] Commentary :
00084     #[ CoCreateSpectator :
00085
00086     Syntax: CreateSpectator name_of_expr "filename";
00087 */
00088
00089 int CoCreateSpectator(UBYTE *inp)
00090 {
00091     UBYTE *p, *q, *filename, c, cc;
00092     WORD c1, c2, numexpr = 0, specnum, HadOne = 0;
00093     FILEHANDLE *fh;
00094     while ( *inp == ',' ) inp++;
00095     if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
00096         MesPrint("&Illegal name for expression");
00097         return(1);
00098     }
00099     c = *q; *q = 0;
00100     if ( GetVar(inp,&c1,&c2,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
00101         if ( c2 == CEXPRESSION &&
00102             Expressions[c1].status == DROPSPECTATOREXPRESSION ) {
00103             numexpr = c1;
00104             Expressions[numexpr].status = SPECTATOREXPRESSION;
00105             HadOne = 1;
00106         }
00107         else {
00108             MesPrint("&The name %s has been used already.",inp);
00109             *q = c;
00110             return(1);
00111         }
00112     }
00113     p = q+1;
00114     while ( *p == ',' ) p++;
00115     if ( *p != '"' ) goto Syntax;
00116     p++; filename = p;
00117     while ( *p && *p != '"' ) {
00118         if ( *p == '\\\\' ) p++;
00119         p++;
00120     }
00121     if ( *p != '"' ) goto Syntax;
00122     q = p+1;
00123     while ( *q && ( *q == ',' || *q == ' ' || *q == '\t' ) ) q++;
00124     if ( *q ) goto Syntax;
00125     cc = *p; *p = 0;
00126 /*
00127     Now we need to: create a struct for the spectator file.
00128 */
00129     if ( HadOne == 0 )
00130         numexpr = EntVar(CEXPRESSION,inp,SPECTATOREXPRESSION,0,0,0);
00131     fh = AllocFileHandle(1,(char *)filename);

```

```

00132 /*
00133      Make sure there is space in the AM.spectatorfiles array
00134 */
00135     if ( AM.NumSpectatorFiles >= AM.SizeForSpectatorFiles || AM.SpectatorFiles == 0 ) {
00136         int newsize, i;
00137         SPECTATOR *newspectators;
00138         if ( AM.SizeForSpectatorFiles == 0 ) {
00139             newsize = 10;
00140             AM.NumSpectatorFiles = AM.SizeForSpectatorFiles = 0;
00141         }
00142         else newsize = AM.SizeForSpectatorFiles*2;
00143         newspectators = (SPECTATOR *)Malloc1(newsize*sizeof(SPECTATOR),"Spectators");
00144         for ( i = 0; i < AM.NumSpectatorFiles; i++ )
00145             newspectators[i] = AM.SpectatorFiles[i];
00146         for ( ; i < newsize; i++ ) {
00147             newspectators[i].fh = 0;
00148             newspectators[i].name = 0;
00149             newspectators[i].exprnumber = -1;
00150             newspectators[i].flags = 0;
00151             PUTZERO(newspectators[i].position);
00152             PUTZERO(newspectators[i].readpos);
00153         }
00154         AM.SizeForSpectatorFiles = newsize;
00155         if ( AM.SpectatorFiles != 0 ) M_free(AM.SpectatorFiles,"Spectators");
00156         AM.SpectatorFiles = newspectators;
00157         specnum = AM.NumSpectatorFiles++;
00158     }
00159     else {
00160         for ( specnum = 0; specnum < AM.SizeForSpectatorFiles; specnum++ ) {
00161             if ( AM.SpectatorFiles[specnum].name == 0 ) break;
00162         }
00163         AM.NumSpectatorFiles++;
00164     }
00165     PUTZERO(AM.SpectatorFiles[specnum].position);
00166     AM.SpectatorFiles[specnum].name = (char *) (strDupl(inp,"Spectator expression name"));
00167     AM.SpectatorFiles[specnum].fh = fh;
00168     AM.SpectatorFiles[specnum].exprnumber = numexpr;
00169     *p = cc;
00170     return(0);
00171 Syntax:
00172     MesPrint("&Proper syntax is: CreateSpectator,exprname,\"filename\";");
00173     return(-1);
00174 }
00175
00176 /*
00177     #] CoCreateSpectator :
00178     #[ CoToSpectator :
00179 */
00180
00181 int CoToSpectator(UBYTE *inp)
00182 {
00183     UBYTE *q;
00184     WORD c1, numexpr;
00185     int i;
00186     while ( *inp == ',' ) inp++;
00187     if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
00188         MesPrint("&Illegal name for expression");
00189         return(1);
00190     }
00191     if ( *q != 0 ) goto Syntax;
00192     if ( GetVar(inp,&c1,&numexpr,ALLVARIABLES,NOAUTO) == NAMENOTFOUND ||
00193         c1 != CEXPRESSION ) {
00194         MesPrint("&%s is not a valid expression.",inp);
00195         return(1);
00196     }
00197     if ( Expressions[numexpr].status != SPECTATOREXPRESSION ) {
00198         MesPrint("&%s is not an active spectator.",inp);
00199         return(1);
00200     }
00201     for ( i = 0; i < AM.SizeForSpectatorFiles; i++ ) {
00202         if ( AM.SpectatorFiles[i].name != 0 ) {
00203             if ( StrCmp((UBYTE *) (AM.SpectatorFiles[i].name), (UBYTE *) (inp)) == 0 ) break;
00204         }
00205     }
00206     if ( i >= AM.SizeForSpectatorFiles ) {
00207         MesPrint("&Spectator %s not found.",inp);
00208         return(1);
00209     }
00210     if ( ( AM.SpectatorFiles[i].flags & READSPECTATORFLAG ) != 0 ) {
00211         MesPrint("&Spectator %s: It is not permitted to read from and write to the same spectator in
one module.",inp);
00212         return(1);
00213     }
00214     AM.SpectatorFiles[i].exprnumber = numexpr;
00215     Add3Com(TYPETOSPECTATOR,i);
00216 #ifndef WITHMPI
00217     /*

```



```

00218      * In ParFORM, ToSpectator has to be executed on the master.
00219      */
00220      AC.mparallelfldg |= NOPARALLEL_SPECTATOR;
00221 #endif
00222      return(0);
00223 Syntax:
00224      MesPrint("&Proper syntax is: ToSpectator,exprname;");
00225      return(-1);
00226 }
00227 /*
00228 #] CoToSpectator :
00229 #[ CoRemoveSpectator :
00231 */
00232
00233 int CoRemoveSpectator(UBYTE *inp)
00234 {
00235     UBYTE *q;
00236     WORD c1, numexpr;
00237     int i;
00238     SPECTATOR *sp;
00239     while ( *inp == ',' ) inp++;
00240     if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
00241         MesPrint("&Illegal name for expression");
00242         return(1);
00243     }
00244     if ( *q != 0 ) goto Syntax;
00245     if ( GetVar(inp,&c1,&numexpr,ALLVARIABLES,NOAUTO) == NAMENOTFOUND ||
00246         c1 != CEXPRESSION ) {
00247         MesPrint("&%s is not a valid expression.",inp);
00248         return(1);
00249     }
00250     if ( Expressions[numexpr].status != SPECTATOREXPRESSION ) {
00251         MesPrint("&%s is not a spectator.",inp);
00252         return(1);
00253     }
00254     for ( i = 0; i < AM.SizeForSpectatorFiles; i++ ) {
00255         if ( AM.SpectatorFiles[i].name != 0 ) {
00256             if ( StrCmp((UBYTE *) (AM.SpectatorFiles[i].name), (UBYTE *) (inp)) == 0 ) {
00257                 break;
00258             }
00259         }
00260     }
00261     if ( i >= AM.SizeForSpectatorFiles ) {
00262         MesPrint("&Spectator %s not found.",inp);
00263         return(1);
00264     }
00265     sp = AM.SpectatorFiles+i;
00266     Expressions[numexpr].status = DROPSPECTATOREXPRESSION;
00267     if ( sp->fh->handle != -1 ) {
00268         CloseFile(sp->fh->handle);
00269         sp->fh->handle = -1;
00270         remove(sp->fh->name);
00271     }
00272     M_free(sp->fh,"Temporary FileHandle");
00273     M_free(sp->name,"Spectator expression name");
00274
00275     PUTZERO(sp->position);
00276     PUTZERO(sp->readpos);
00277     sp->fh = 0;
00278     sp->name = 0;
00279     sp->exprnumber = -1;
00280     sp->flags = 0;
00281     AM.NumSpectatorFiles--;
00282     return(0);
00283 Syntax:
00284     MesPrint("&Proper syntax is: RemoveSpectator,exprname;");
00285     return(-1);
00286 }
00287 /*
00288 #] CoRemoveSpectator :
00289 #[ CoEmptySpectator :
00291 */
00292
00293 int CoEmptySpectator(UBYTE *inp)
00294 {
00295     UBYTE *q;
00296     WORD c1, numexpr;
00297     int i;
00298     SPECTATOR *sp;
00299     while ( *inp == ',' ) inp++;
00300     if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
00301         MesPrint("&Illegal name for expression");
00302         return(1);
00303     }
00304     if ( *q != 0 ) goto Syntax;

```

```

00305     if ( GetVar(inp,&c1,&numexpr,ALLVARIABLES,NOAUTO) == NAMENOTFOUND ||
00306         c1 != CEXPRESSION ) {
00307         MesPrint("%s is not a valid expression.",inp);
00308         return(1);
00309     }
00310     if ( Expressions[numexpr].status != SPECTATOREXPRESSION ) {
00311         MesPrint("%s is not a spectator.",inp);
00312         return(1);
00313     }
00314     for ( i = 0; i < AM.SizeForSpectatorFiles; i++ ) {
00315         if ( StrCmp((UBYTE *) (AM.SpectatorFiles[i].name), (UBYTE *) (inp)) == 0 ) break;
00316     }
00317     if ( i >= AM.SizeForSpectatorFiles ) {
00318         MesPrint("&Spectator %s not found.",inp);
00319         return(1);
00320     }
00321     sp = AM.SpectatorFiles+i;
00322     if ( sp->fh->handle != -1 ) {
00323         CloseFile(sp->fh->handle);
00324         sp->fh->handle = -1;
00325         remove(sp->fh->name);
00326     }
00327     sp->fh->POfill = sp->fh->POfull = sp->fh->PObuffer;
00328     PUTZERO(sp->position);
00329     PUTZERO(sp->readpos);
00330     return(0);
00331 Syntax:
00332 MesPrint("&Proper syntax is: EmptySpectator,exprname;");
00333     return(-1);
00334 }
00335
00336 /*
00337 #] CoEmptySpectator :
00338 #[ PutInSpectator :
00339
00340 We need to use locks! There is only one file.
00341 The code was copied (and modified) from PutOut.
00342 Here we use no compression.
00343 */
00344
00345 int PutInSpectator(WORD *term,WORD specnum)
00346 {
00347     GETBIDENTITY
00348     WORD i, *p, ret;
00349     LONG RetCode;
00350     SPECTATOR *sp = &(AM.SpectatorFiles[specnum]);
00351     FILEHANDLE *fi = sp->fh;
00352
00353     if ( ( i = *term ) <= 0 ) return(0);
00354     LOCK(fi->pthreadslck);
00355     ret = i;
00356     p = fi->POfill;
00357     do {
00358         if ( p >= fi->POstop ) {
00359             if ( fi->handle < 0 ) {
00360                 if ( ( RetCode = CreateFile(fi->name) ) >= 0 ) {
00361                     fi->handle = (WORD)RetCode;
00362                     PUTZERO(fi->filesize);
00363                     PUTZERO(fi->POposition);
00364                 }
00365                 else {
00366                     MLOCK(ErrorMessageLock);
00367                     MesPrint("Cannot create spectator file %s",fi->name);
00368                     MUNLOCK(ErrorMessageLock);
00369                     UNLOCK(fi->pthreadslck);
00370                     return(-1);
00371                 }
00372             }
00373             SeekFile(fi->handle,&(sp->position),SEEK_SET);
00374             if ( ( RetCode = WriteFile(fi->handle, (UBYTE *) (fi->PObuffer),fi->POsize) ) != fi->POsize
00375         ) {
00376                 MLOCK(ErrorMessageLock);
00377                 MesPrint("Error during spectator write. Disk full?");
00378                 MesPrint("Attempt to write %l bytes on file %d at position %l5p",
00379                     fi->POsize,fi->handle,&(fi->POposition));
00380                 MesPrint("RetCode = %l, Buffer address = %l",RetCode,(LONG) (fi->PObuffer));
00381                 MUNLOCK(ErrorMessageLock);
00382                 UNLOCK(fi->pthreadslck);
00383                 return(-1);
00384             }
00385             ADDPOS(fi->filesize,fi->POsize);
00386             p = fi->PObuffer;
00387             ADDPOS(sp->position,fi->POsize);
00388             fi->POposition = sp->position;
00389         }
00390         *p++ = *term++;
00391     } while ( --i > 0 );

```

```

00391     fi->POfull = fi->POfill = p;
00392     Expressions[AM.SpectatorFiles[specnum].exprnumber].counter++;
00393     UNLOCK(fi->pthreadlock);
00394     return(ret);
00395 }
00396
00397 /*
00398  #] PutInSpectator :
00399  #[ FlushSpectators :
00400  */
00401
00402 void FlushSpectators(void)
00403 {
00404     SPECTATOR *sp = AM.SpectatorFiles;
00405     FILEHANDLE *fh;
00406     LONG RetCode;
00407     int i;
00408     LONG size;
00409     if ( AM.NumSpectatorFiles <= 0 ) return;
00410     for ( i = 0; i < AM.SizeForSpectatorFiles; i++, sp++ ) {
00411         if ( sp->name == 0 ) continue;
00412         fh = sp->fh;
00413         if ( ( sp->flags & READSPECTATORFLAG ) != 0 ) { /* reset for writing */
00414             sp->flags &= ~READSPECTATORFLAG;
00415             fh->POfill = fh->PObuffer;
00416             if ( fh->handle >= 0 ) {
00417                 SeekFile(fh->handle, &(sp->position), SEEK_SET);
00418                 fh->POposition = sp->position;
00419             }
00420             continue;
00421         }
00422         if ( fh->POfill <= fh->PObuffer ) continue; /* is clean */
00423         if ( fh->handle < 0 ) { /* File needs to be created */
00424             if ( ( RetCode = CreateFile(fh->name) ) >= 0 ) {
00425                 PUTZERO(fh->filesize);
00426                 PUTZERO(fh->POposition);
00427                 fh->handle = (WORD)RetCode;
00428             }
00429             else {
00430                 MLOCK(ErrorMessageLock);
00431                 MesPrint("Cannot create spectator file %s", fh->name);
00432                 MUNLOCK(ErrorMessageLock);
00433                 Terminate(-1);
00434             }
00435             PUTZERO(sp->position);
00436         }
00437         SeekFile(fh->handle, &(sp->position), SEEK_SET);
00438         size = (fh->POfill - fh->PObuffer)*sizeof(WORD);
00439         if ( ( RetCode = WriteFile(fh->handle, (UBYTE *) (fh->PObuffer), size) ) != size ) {
00440             MLOCK(ErrorMessageLock);
00441             MesPrint("Write error synching spectator file. Disk full?");
00442             MesPrint("Attempt to write %l bytes on file %s at position %l5p",
00443                     size, fh->name, &(sp->position));
00444             MUNLOCK(ErrorMessageLock);
00445             Terminate(-1);
00446         }
00447         fh->POfill = fh->PObuffer;
00448         SeekFile(fh->handle, &(sp->position), SEEK_END);
00449         fh->POposition = sp->position;
00450     }
00451     return;
00452 }
00453
00454 /*
00455  #] FlushSpectators :
00456  #[ CoCopySpectator :
00457  */
00458
00459 int CoCopySpectator(UBYTE *inp)
00460 {
00461     GETIDENTITY
00462     UBYTE *q, c, *exprname, *p;
00463     WORD c1, c2, numexpr;
00464     int specnum, error = 0;
00465     SPECTATOR *sp;
00466     while ( *inp == ',' ) inp++;
00467     if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
00468         MesPrint("&Illegal name for expression");
00469         return(1);
00470     }
00471     if ( *q == 0 ) goto Syntax;
00472     c = *q; *q = 0;
00473     if ( GetVar(inp, &c1, &c2, ALLVARIABLES, NOAUTO) != NAMENOTFOUND ) {
00474         MesPrint("&%s is the name of an existing variable.", inp);
00475         return(1);
00476     }
00477     numexpr = EntVar(CEXPRESSION, inp, LOCALEXPRESSION, 0, 0, 0);

```

```

00478     p = q;
00479     exprname = inp;
00480     *q = c;
00481     while ( *q == ' ' || *q == ',' || *q == '\t' ) q++;
00482     if ( *q != '=' ) goto Syntax;
00483     q++;
00484     while ( *q == ' ' || *q == ',' || *q == '\t' ) q++;
00485     inp = q;
00486     if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
00487         MesPrint("%Illegal name for spectator expression");
00488         return(1);
00489     }
00490     if ( *q != 0 ) goto Syntax;
00491     if ( AM.NumSpectatorFiles <= 0 ) {
00492         MesPrint("&CopySpectator: There are no spectator expressions!");
00493         return(1);
00494     }
00495     sp = AM.SpectatorFiles;
00496     for ( specnum = 0; specnum < AM.SizeForSpectatorFiles; specnum++, sp++ ) {
00497         if ( sp->name != 0 ) {
00498             if ( StrCmp((UBYTE *) (sp->name), (UBYTE *) (inp)) == 0 ) break;
00499         }
00500     }
00501     if ( specnum >= AM.SizeForSpectatorFiles ) {
00502         MesPrint("&Spectator %s not found.",inp);
00503         return(1);
00504     }
00505     sp->flags |= READSPECTATORFLAG;
00506     PUTZERO(sp->fh->PPosition);
00507     PUTZERO(sp->readpos);
00508     sp->fh->POfill = sp->fh->PObuffer;
00509     if ( sp->fh->handle >= 0 ) {
00510         SeekFile(sp->fh->handle, &(sp->fh->PPosition), SEEK_SET);
00511     }
00512     /*
00513     Now we have:
00514     1: The name of the target expression: numexpr
00515     2: The spectator: sp (or specnum).
00516     Time for some action. We need:
00517     a: Write a prototype to create the expression
00518     b: Signal to Processor that this is a spectator.
00519     We do this by giving a negative compiler buffer number.
00520     */
00521     {
00522         WORD *OldWork, *w;
00523         POSITION pos;
00524         OldWork = w = AT.WorkPointer;
00525         *w++ = TYPEEXPRESSION;
00526         *w++ = 3+SUBEXPSIZE;
00527         *w++ = numexpr;
00528         AC.ProtoType = w;
00529         AR.CurExpr = numexpr;          /* Block expression numexpr */
00530         *w++ = SUBEXPRESSION;
00531         *w++ = SUBEXPSIZE;
00532         *w++ = numexpr;
00533         *w++ = 1;
00534         *w++ = -specnum-1;          /* Indicates "spectator" to Processor */
00535         FILLSUB(w)
00536         *w++ = 1;
00537         *w++ = 1;
00538         *w++ = 3;
00539         *w++ = 0;
00540         SeekScratch(AR.outfile, &pos);
00541         Expressions[numexpr].counter = 1;
00542         Expressions[numexpr].onfile = pos;
00543         Expressions[numexpr].whichbuffer = 0;
00544     #ifdef PARALLELCODE
00545         Expressions[numexpr].partodo = AC.inparallelflag;
00546     #endif
00547         OldWork[2] = w - OldWork - 3;
00548         AT.WorkPointer = w;
00549
00550         if ( PutOut(BHEAD OldWork+2, &pos, AR.outfile, 0) < 0 ) {
00551             c = *p; *p = 0;
00552             MesPrint("&Cannot create expression %s", exprname);
00553             *p = c;
00554             error = -1;
00555         }
00556         else {
00557             OldWork[2] = 4+SUBEXPSIZE;
00558             OldWork[4] = SUBEXPSIZE;
00559             OldWork[5] = numexpr;
00560             OldWork[SUBEXPSIZE+3] = 1;
00561             OldWork[SUBEXPSIZE+4] = 1;
00562             OldWork[SUBEXPSIZE+5] = 3;
00563             OldWork[SUBEXPSIZE+6] = 0;
00564             if ( PutOut(BHEAD OldWork+2, &pos, AR.outfile, 0) < 0

```

```

00565         || FlushOut (&pos,AR.outfile,0) ) {
00566             c = *p; *p = 0;
00567             MesPrint("&Cannot create expression %s",exprname);
00568             *p = c;
00569             error = -1;
00570         }
00571         AR.outfile->POfull = AR.outfile->POfill;
00572     }
00573     OldWork[2] = numexpr;
00574 /*
00575     Seems unnecessary (13-feb-2018)
00576
00577     AddNtoL(OldWork[1],OldWork);
00578 */
00579     AT.WorkPointer = OldWork;
00580     if ( AC.dumnumflag ) Add2Com(TYPEDETCURDUM)
00581 }
00582 #ifdef WITHMPI
00583 /*
00584     * In ParFORM, substitutions of spectators has to be done on the master.
00585     */
00586     AC.mparallelflag |= NOPARALLEL_SPECTATOR;
00587 #endif
00588     return(error);
00589 Syntax:
00590     MesPrint("&Proper syntax is: CopySpectator,exprname=spectatorname;");
00591     return(-1);
00592 }
00593
00594 /*
00595     #] CoCopySpectator :
00596     #[ GetFromSpectator :
00597
00598     Note that if we did things right, we do not need a lock for the reading.
00599 */
00600
00601 WORD GetFromSpectator(WORD *term,WORD specnum)
00602 {
00603     SPECTATOR *sp = &(AM.SpectatorFiles[specnum]);
00604     FILEHANDLE *fh = sp->fh;
00605     WORD i, size, *t = term;
00606     LONG InIn;
00607     if ( fh->handle < 0 ) {
00608         *term = 0;
00609         return(0);
00610     }
00611 /*
00612     sp->position marks the 'end' of the file: the point where writing should
00613     take place. sp->readpos marks from where to read.
00614     fh->POposition marks where the file is currently positioned.
00615     Note that when we read, we need to
00616 */
00617     if ( ISZEROPOS(sp->readpos) ) { /* we start reading. Fill buffer. */
00618         FillBuffer:
00619         SeekFile(fh->handle,&(sp->readpos),SEEK_SET);
00620         InIn = ReadFile(fh->handle,(UBYTE *) (fh->PObuffer),fh->POsize);
00621         if ( InIn < 0 || ( InIn & 1 ) ) {
00622             MLOCK(ErrorMessageLock);
00623             MesPrint("Error reading information for %s spectator",sp->name);
00624             MUNLOCK(ErrorMessageLock);
00625             Terminate(-1);
00626         }
00627         InIn /= sizeof(WORD);
00628         if ( InIn == 0 ) { *term = 0; return(0); }
00629         SeekFile(fh->handle,&(sp->readpos),SEEK_CUR);
00630         fh->POposition = sp->readpos;
00631         fh->POfull = fh->PObuffer+InIn;
00632         fh->POfill = fh->PObuffer;
00633     }
00634     if ( fh->POfill == fh->POfull ) { /* not even the size of the term! */
00635         if ( ISLESSPOS(sp->readpos,sp->position) ) goto FillBuffer;
00636         *term = 0;
00637         return(0);
00638     }
00639     size = *fh->POfill++; *t++ = size;
00640     for ( i = 1; i < size; i++ ) {
00641         if ( fh->POfill >= fh->POfull ) {
00642             SeekFile(fh->handle,&(sp->readpos),SEEK_SET);
00643             InIn = ReadFile(fh->handle,(UBYTE *) (fh->PObuffer),fh->POsize);
00644             if ( InIn < 0 || ( InIn & 1 ) ) {
00645                 MLOCK(ErrorMessageLock);
00646                 MesPrint("Error reading information for %s spectator",sp->name);
00647                 MUNLOCK(ErrorMessageLock);
00648                 Terminate(-1);
00649             }
00650             InIn /= sizeof(WORD);
00651             if ( InIn == 0 ) {

```

```

00652             MLOCK(ErrorMessageLock);
00653             MesPrint("Reading incomplete information for %s spectator",sp->name);
00654             MUNLOCK(ErrorMessageLock);
00655             Terminate(-1);
00656         }
00657         SeekFile(fh->handle,&(sp->readpos),SEEK_CUR);
00658         fh->POposition = sp->readpos;
00659         fh->POfull = fh->PObuffer+InIn;
00660         fh->POfill = fh->PObuffer;
00661     }
00662     *t++ = *fh->POfill++;
00663 }
00664 return(size);
00665 }
00666
00667 /*
00668 #] GetFromSpectator :
00669 #[ ClearSpectators :
00670
00671 Removes all spectators.
00672 In case of .store, the ones that are protected by .global stay.
00673 */
00674 void ClearSpectators(WORD par)
00675 {
00676     SPECTATOR *sp = AM.SpectatorFiles;
00677     WORD numexpr, c1;
00678     int i;
00679     if ( AM.NumSpectatorFiles > 0 ) {
00680         for ( i = 0; i < AM.SizeForSpectatorFiles; i++, sp++ ) {
00681             if ( sp->name == 0 ) continue;
00682             if ( ( sp->flags & GLOBALSPECTATORFLAG ) == 1 && par == STOREMODULE ) continue;
00683
00684             if ( GetVar((UBYTE *) (sp->name),&c1,&numexpr,ALLVARIABLES,NOAUTO) == NAMENOTFOUND ||
00685                 c1 != CEXPRESSION ) {
00686                 MesPrint("%s is not a valid expression.",sp->name);
00687                 continue;
00688             }
00689             Expressions[numexpr].status = DROPPEDEXPRESSION;
00690             if ( sp->fh->handle != -1 ) {
00691                 CloseFile(sp->fh->handle);
00692                 sp->fh->handle = -1;
00693                 remove(sp->fh->name);
00694             }
00695             M_free(sp->fh,"Temporary FileHandle");
00696             M_free(sp->name,"Spectator expression name");
00697             PUTZERO(sp->position);
00698             sp->fh = 0;
00699             sp->name = 0;
00700             sp->exprnumber = -1;
00701             sp->flags = 0;
00702             AM.NumSpectatorFiles--;
00703         }
00704     }
00705 }
00706 }
00707
00708 /*
00709 #] ClearSpectators :
00710 */

```

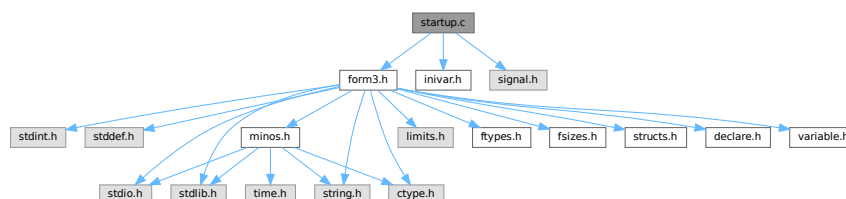
## 4.132 startup.c File Reference

```

#include "form3.h"
#include "inivar.h"
#include <signal.h>

```

Include dependency graph for startup.c:



## Macros

- #define [STRINGIFY](#)(x) STRINGIFY\_\_(x)
- #define [STRINGIFY\\_\\_](#)(x) #x
- #define [FORMNAME](#) "FORM"
- #define [VERSIONSTR\\_\\_](#) STRINGIFY(MAJORVERSION) "." STRINGIFY(MINORVERSION) "Beta"
- #define [VERSIONSTR](#) FORMNAME " " VERSIONSTR\_\_ " (" PRODUCTIONDATE ")"
- #define [TAKEPATH](#)(x) if(s[1]== '=' ){x=s+2;} else{x=\*argv++;argc--;}
- #define [DEFAULTFNAMELENGTH](#) 16

## Functions

- int [DoTail](#) (int argc, UBYTE \*\*argv)
- int [OpenInput](#) (void)
- void [ReserveTempFiles](#) (int par)
- void [StartVariables](#) (void)
- void [StartMore](#) (void)
- void [IniVars](#) (void)
- int [main](#) (int argc, char \*\*argv)
- void [CleanUp](#) (WORD par)
- void [TerminateImpl](#) (int errorcode, const char \*file, int line, const char \*function)
- void [PrintDeprecation](#) (const char \*feature, const char \*issue)
- void [PrintRunningTime](#) (void)
- LONG [GetRunningTime](#) (void)

## Variables

- UBYTE \* [emptystring](#) = (UBYTE \*)"."
- UBYTE \* [defaulttempfilename](#) = (UBYTE \*)"xformxxx.str"

### 4.132.1 Detailed Description

This file contains the main program. It also deals with the very early stages of the startup of FORM and the final stages when the program attempts some cleanup. Here is the routine that analyses the command tail.

Definition in file [startup.c](#).

### 4.132.2 Macro Definition Documentation

#### 4.132.2.1 STRINGIFY

```
#define STRINGIFY(  
    x ) STRINGIFY__(x)
```

Definition at line 57 of file [startup.c](#).

#### 4.132.2.2 STRINGIFY\_\_

```
#define STRINGIFY__(  
    x ) #x
```

Definition at line 58 of file [startup.c](#).

#### 4.132.2.3 FORMNAME

```
#define FORMNAME "FORM"
```

Definition at line 68 of file [startup.c](#).

#### 4.132.2.4 VERSIONSTR\_\_

```
#define VERSIONSTR__ STRINGIFY(MAJORVERSION) "." STRINGIFY(MINORVERSION) "Beta"
```

Definition at line 97 of file [startup.c](#).

#### 4.132.2.5 VERSIONSTR

```
#define VERSIONSTR FORMNAME " " VERSIONSTR__ " (" PRODUCTIONDATE ")"
```

Definition at line 101 of file [startup.c](#).

#### 4.132.2.6 TAKEPATH

```
#define TAKEPATH(  
    x ) if(s[1]== '=' ) {x=s+2;} else {x=*argv++;argc--;} 
```

Definition at line 235 of file [startup.c](#).

#### 4.132.2.7 DEFAULTFNAMELENGTH

```
#define DEFAULTFNAMELENGTH 16
```

Definition at line 664 of file [startup.c](#).

### 4.132.3 Function Documentation

#### 4.132.3.1 DoTail()

```
int DoTail (  
    int argc,  
    UBYTE ** argv )
```

Definition at line 237 of file [startup.c](#).



#### 4.132.3.2 OpenInput()

```
int OpenInput (
    void )
```

Definition at line 542 of file [startup.c](#).

#### 4.132.3.3 ReserveTempFiles()

```
void ReserveTempFiles (
    int par )
```

Definition at line 666 of file [startup.c](#).

#### 4.132.3.4 StartVariables()

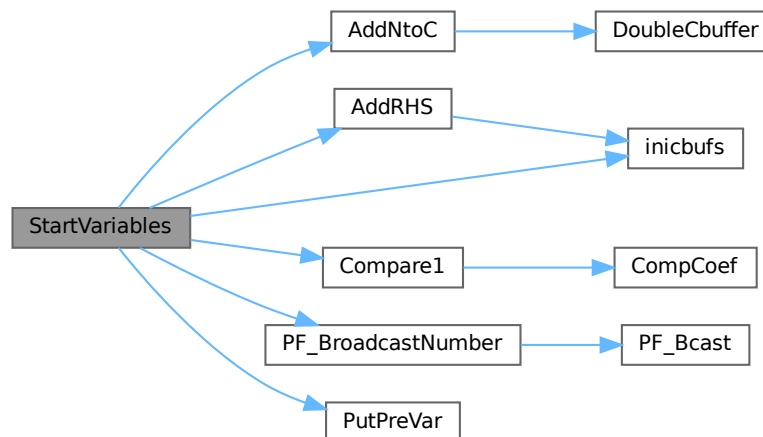
```
void StartVariables (
    void )
```

All functions (well, nearly all) are declared here.

Definition at line 905 of file [startup.c](#).

References [AddNtoC\(\)](#), [AddRHS\(\)](#), [Compare1\(\)](#), [inicbufs\(\)](#), [FuNcTiOn::name](#), [PF\\_BroadcastNumber\(\)](#), [PutPreVar\(\)](#), and [FuNcTiOn::symmetric](#).

Here is the call graph for this function:



#### 4.132.3.5 StartMore()

```
void StartMore (
    void )
```

Definition at line 1316 of file [startup.c](#).

#### 4.132.3.6 IniVars()

```
void IniVars (
    void )
```

Definition at line [1354](#) of file [startup.c](#).

#### 4.132.3.7 main()

```
int main (
    int argc,
    char ** argv )
```

Definition at line [1661](#) of file [startup.c](#).

#### 4.132.3.8 CleanUp()

```
void CleanUp (
    WORD par )
```

Definition at line [1760](#) of file [startup.c](#).

#### 4.132.3.9 TerminateImpl()

```
void TerminateImpl (
    int errorcode,
    const char * file,
    int line,
    const char * function )
```

Definition at line [1849](#) of file [startup.c](#).

#### 4.132.3.10 PrintDeprecation()

```
void PrintDeprecation (
    const char * feature,
    const char * issue )
```

Prints a deprecation warning for a given feature.

##### Parameters

|                |                                           |
|----------------|-------------------------------------------|
| <i>feature</i> | The name of the deprecated feature.       |
| <i>issue</i>   | The associated issue, e.g., "issues/700". |

Definition at line [2062](#) of file [startup.c](#).

#### 4.132.3.11 PrintRunningTime()

```
void PrintRunningTime (
    void )
```

Definition at line 2086 of file [startup.c](#).

#### 4.132.3.12 GetRunningTime()

```
LONG GetRunningTime (
    void )
```

Definition at line 2127 of file [startup.c](#).

### 4.132.4 Variable Documentation

#### 4.132.4.1 emptystring

```
UBYTE* emptystring = (UBYTE *)"."
```

Definition at line 660 of file [startup.c](#).

#### 4.132.4.2 defaulttempfilename

```
UBYTE* defaulttempfilename = (UBYTE *)"xformxxx.str"
```

Definition at line 661 of file [startup.c](#).

## 4.133 startup.c

[Go to the documentation of this file.](#)

```
00001
00008 /* #[] License : */
00009 /*
00010 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00011 * When using this file you are requested to refer to the publication
00012 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00013 * This is considered a matter of courtesy as the development was paid
00014 * for by FOM the Dutch physics granting agency and we would like to
00015 * be able to track its scientific use to convince FOM of its value
00016 * for the community.
00017 *
00018 * This file is part of FORM.
00019 *
00020 * FORM is free software: you can redistribute it and/or modify it under the
00021 * terms of the GNU General Public License as published by the Free Software
00022 * Foundation, either version 3 of the License, or (at your option) any later
00023 * version.
00024 *
00025 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00026 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00027 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00028 * details.
00029 *
00030 * You should have received a copy of the GNU General Public License along
00031 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00032 */
00033 /* #[] License : */
00034 /*
```

```

00035         #[] includes :
00036     */
00037
00038     #include "form3.h"
00039     #include "inivar.h"
00040
00041     #ifdef TRAP_SIGNALS
00042     #include "portsignals.h"
00043     #else
00044     #include <signal.h>
00045     #endif
00046     #ifdef ENABLE_BACKTRACE
00047     #include <execinfo.h>
00048     #ifdef LINUX
00049     #include <stdint.h>
00050     #include <inttypes.h>
00051     #endif
00052     #endif
00053
00054     /*
00055     * A macro for translating the contents of `x' into a string after expanding.
00056     */
00057     #define STRINGIFY(x)  STRINGIFY__(x)
00058     #define STRINGIFY__(x) #x
00059
00060     /*
00061     * FORMNAME = "FORM" or "TFORM" or "ParFORM".
00062     */
00063     #if defined(WITHPTHREADS)
00064     #define FORMNAME "TFORM"
00065     #elif defined(WITHMPI)
00066     #define FORMNAME "ParFORM"
00067     #else
00068     #define FORMNAME "FORM"
00069     #endif
00070
00071     /*
00072     * VERSIONSTR is the version information printed in the header line.
00073     */
00074     #ifdef HAVE_CONFIG_H
00075     /* We have also version.h. */
00076     #include "version.h"
00077     #ifndef REPO_VERSION
00078     #define REPO_VERSION STRINGIFY(REPO_MAJOR_VERSION) "." STRINGIFY(REPO_MINOR_VERSION)
00079     #endif
00080     #ifndef REPO_DATE
00081     /* The build date, instead of the repo date. */
00082     #define REPO_DATE __DATE__
00083     #endif
00084     #ifndef REPO_REVISION
00085     #define VERSIONSTR FORMNAME " " REPO_VERSION " (" REPO_DATE ", " REPO_REVISION ")"
00086     #else
00087     #define VERSIONSTR FORMNAME " " REPO_VERSION " (" REPO_DATE ")"
00088     #endif
00089     #define MAJORVERSION REPO_MAJOR_VERSION
00090     #define MINORVERSION REPO_MINOR_VERSION
00091     #else
00092     /*
00093     * Otherwise, form3.h defines MAJORVERSION, MINORVERSION and PRODUCTIONDATE,
00094     * possibly BETAVERSION.
00095     */
00096     #ifndef BETAVERSION
00097     #define VERSIONSTR__ STRINGIFY(MAJORVERSION) "." STRINGIFY(MINORVERSION) "Beta"
00098     #else
00099     #define VERSIONSTR__ STRINGIFY(MAJORVERSION) "." STRINGIFY(MINORVERSION)
00100     #endif
00101     #define VERSIONSTR FORMNAME " " VERSIONSTR__ " (" PRODUCTIONDATE ")"
00102     #endif
00103
00104     /*
00105     #[] includes :
00106     #[] PrintHeader :
00107     */
00108
00109     static void PrintHeader(int with_full_info)
00110     {
00111     #ifdef WITHMPI
00112     if ( PF.me == MASTER && !AM.silent ) {
00113     #else
00114     if ( !AM.silent ) {
00115     #endif
00116
00117         char buffer1[250], buffer2[80], *s = buffer1, *t = buffer2;
00118         WORD length, n;
00119         for ( n = 0; n < 250; n++ ) buffer1[n] = ' ';
00120         /*
00121         * NOTE: we expect that the compiler optimizes strlen("string literal")
00122         * to just a number.

```

```

00127         */
00128         if ( strlen(VERSIONSTR) <= 100 ) {
00129             strcpy(s,VERSIONSTR);
00130             s += strlen(VERSIONSTR);
00131             *s = 0;
00132         }
00133         else {
00134             /*
00135              * Truncate when it is too long.
00136              */
00137             strncpy(s,VERSIONSTR,97);
00138             s[97] = '.';
00139             s[98] = '.';
00140             s[99] = '.';
00141             s[100] = '\0';
00142             s += 100;
00143         }
00144     /*
00145      By now we omit the message about 32/64 bits. It should all be 64.
00146
00147      s += snprintf(s,250-(s-buffer1)," %d-bits", (WORD) (sizeof(WORD)*16));
00148      *s = 0;
00149  */
00150         if ( with_full_info ) {
00151             #if defined(WITHPTHREADS) || defined(WITHMPI)
00152             #if defined(WITHPTHREADS)
00153                 int nworkers = AM.totalnumberofthreads-1;
00154             #elif defined(WITHMPI)
00155                 int nworkers = PF.numtasks-1;
00156             #endif
00157             s += snprintf(s,250-(s-buffer1)," %d worker",nworkers);
00158             *s = 0;
00159             /*
00160              while ( *s ) s++; */
00161             if ( nworkers != 1 ) {
00162                 *s++ = 's';
00163                 *s = '\0';
00164             }
00165             #endif
00166             snprintf(t,80-(t-buffer2),"Run: %s",MakeDate());
00167             while ( *t ) t++;
00168
00169             /*
00170              * Align the date to the right, if it fits in a line.
00171              */
00172             length = (s-buffer1) + (t-buffer2);
00173             if ( length+2 <= AC.LineLength ) {
00174                 for ( n = AC.LineLength-length; n > 0; n-- ) *s++ = ' ';
00175                 *s = 0;
00176                 strcat(s,buffer2);
00177                 while ( *s ) s++;
00178             }
00179             else {
00180                 *s = 0;
00181                 strcat(s," ");
00182                 while ( *s ) s++;
00183                 *s = 0;
00184                 strcat(s,buffer2);
00185                 while ( *s ) s++;
00186             }
00187         }
00188     /*
00189      * If the header information doesn't fit in a line, we need to extend
00190      * the line length temporarily.
00191      */
00192     length = s-buffer1;
00193     if ( length <= AC.LineLength ) {
00194         MesPrint("%s",buffer1);
00195     }
00196     else {
00197         WORD oldLineLength = AC.LineLength;
00198         AC.LineLength = length;
00199         MesPrint("%s",buffer1);
00200         AC.LineLength = oldLineLength;
00201     }
00202 }
00203
00204 #ifdef WINDOWS
00205     PrintDeprecation("the native Windows version", "issues/623");
00206 #endif
00207 #if BITSINWORD == 16
00208     PrintDeprecation("the 32-bit version", "issues/624");
00209 #endif
00210 #ifdef WITHMPI
00211     PrintDeprecation("the MPI version (ParFORM)", "issues/625");
00212 #endif
00213     if ( AC.CheckpointFlag ) {

```

```

00214     PrintDeprecation("the checkpoint mechanism", "issues/626");
00215 }
00216 }
00217
00218 /*
00219     #] PrintHeader :
00220     #[ DoTail :
00221
00222     Routine reads the command tail and handles the commandline options.
00223     It sets the flags for later actions and stored pathnames for
00224     the setup file, include/prc/sub directories etc.
00225     Finally the name of the program is passed on.
00226     Note that we do not support interactive use yet. This will come
00227     to pass in the distant future when we can couple STedi to FORM.
00228     Routine made 23-feb-1993 by J.Vermaseren
00229 */
00230
00231 #ifdef WITHINTERACTION
00232 static UBYTE deflogname[] = "formsession.log";
00233 #endif
00234
00235 #define TAKEPATH(x) if(s[1]== '=' ){x=s+2;} else{x=*argv++;argc--;}
00236
00237 int DoTail(int argc, UBYTE **argv)
00238 {
00239     int errorflag = 0, onlyversion = 1;
00240     UBYTE *s, *t, *copy;
00241     int threadnum = 0;
00242     argc--; argv++;
00243     AM.ClearStore = 0;
00244     AM.TimeLimit = 0;
00245     AM.LogType = -1;
00246     AM.HoldFlag = AM.qError = AM.Interact = AM.FileOnlyFlag = 0;
00247     AM.InputFileName = AM.LogFileName = AM.IncDir = AM.TempDir = AM.TempSortDir =
00248     AM.SetupDir = AM.SetupFile = AM.Path = 0;
00249     AM.FromStdin = 0;
00250     /* Always use MultiRun, "-M" option is now ignored. */
00251     AM.MultiRun = 1;
00252     if ( argc < 1 ) {
00253         onlyversion = 0;
00254         goto printversion;
00255     }
00256     while ( argc >= 1 ) {
00257         s = *argv++; argc--;
00258         if ( *s == '-' || ( *s == '/' && ( argc > 0 || AM.Interact ) ) ) {
00259             s++;
00260             switch (*s) {
00261                 case 'c': /* Error checking only */
00262                     AM.qError = 1; break;
00263                 case 'C': /* Next arg is filename of log file */
00264                     TAKEPATH(AM.LogFileName) break;
00265                 case 'D':
00266                 case 'd': /* Next arg is define preprocessor var. */
00267                     t = copy = strDupl(*argv,"Dotail");
00268                     while ( *t && *t != '=' ) t++;
00269                     if ( *t == 0 ) {
00270                         if ( PutPreVar(copy,(UBYTE *) "1",0,0) < 0 ) return(-1);
00271                     }
00272                     else {
00273                         *t++ = 0;
00274                         if ( PutPreVar(copy,t,0,0) < 0 ) return(-1);
00275                         t[-1] = '=';
00276                     }
00277                     M_free(copy,"-d prevar");
00278                     argv++; argc--; break;
00279                 case 'f': /* Output only to regular log file */
00280                     AM.FileOnlyFlag = 1; AM.LogType = 0; break;
00281                 case 'F': /* Output only to log file. Further like L. */
00282                     AM.FileOnlyFlag = 1; AM.LogType = 1; break;
00283                 case 'h': /* For old systems: wait for key before exit */
00284                     AM.HoldFlag = 1; break;
00285                 case 'i':
00286                     if ( StrCmp(s, (UBYTE *) "ignore-deprecation") == 0 ) {
00287                         AM.IgnoreDeprecation = 1;
00288                         break;
00289                     }
00290             #ifdef WITHINTERACTION
00291                 /* Interactive session (not used yet) */
00292                 AM.Interact = 1;
00293                 break;
00294             #else
00295                 goto IllegalOption;
00296             #endif
00297                 case 'I': /* Next arg is dir for inc/prc/sub files */
00298                     TAKEPATH(AM.IncDir) break;
00299                 case 'l': /* Make regular log file */
00300                     if ( s[1] == 'l' ) AM.LogType = 1; /*compatibility! */

```

```

00301             else                AM.LogType = 0;
00302             break;
00303         case 'L': /* Make log file with only final statistics */
00304             AM.LogType = 1; break;
00305         case 'M': /* Multirun. Name of tempfiles will contain PID */
00306             /* This option is now ignored. We always use MultiRun. */
00307             /* AM.MultiRun = 1; */
00308             break;
00309         case 'm': /* Read number of threads */
00310         case 'w': /* Read number of workers */
00311             t = s++;
00312             threadnum = 0;
00313             while ( *s >= '0' && *s <= '9' )
00314                 threadnum = 10*threadnum + *s++ - '0';
00315             if ( *s ) {
00316 #ifdef WITHMPI
00317                 if ( PF.me == MASTER )
00318 #endif
00319                 printf("Illegal value for option m or w: %s\n",t);
00320                 errorflag++;
00321             }
00322             /* if ( threadnum == 1 ) threadnum = 0; */
00323             threadnum++;
00324             break;
00325         case 'W': /* Print the wall-clock time on the master. */
00326             AM.ggWTimeStatsFlag = 1;
00327             break;
00328         /*
00329         case 'n':
00330             Reserved for number of slaves without MPI
00331         */
00332         case 'p':
00333 #ifdef WITHEXTERNALCHANNEL
00334             /*There are two possibilities: -p|-pipe*/
00335             if(s[1]=='i'){
00336                 if( (s[2]=='p') && (s[3]=='e') && (s[4]=='\0') ) {
00337                     argc--;
00338                     /*Initialize pre-set external channels, see
00339                     the file extcmd.c:*/
00340                     if(initPresetExternalChannels(*argv++,AX.timeout)<1){
00341 #ifdef WITHMPI
00342                         if ( PF.me == MASTER )
00343 #endif
00344                         printf("Error initializing preset external channels\n");
00345                         errorflag++;
00346                     }
00347                     AX.timeout=-1; /*This indicates that preset channels
00348                     are initialized from cmdline*/
00349                 }else{
00350 #ifdef WITHMPI
00351                     if ( PF.me == MASTER )
00352 #endif
00353                     printf("Illegal option in call of FORM: %s\n",s);
00354                     errorflag++;
00355                 }
00356             }else
00357             #else
00358             if ( s[1] ) {
00359                 if ( ( s[1]=='i' ) && ( s[2] == 'p' ) && ( s[3] == 'e' )
00360                     && ( s[4] == '\0' ) ) {
00361 #ifdef WITHMPI
00362                     if ( PF.me == MASTER )
00363 #endif
00364                     printf("Illegal option: Pipes not supported on this system.\n");
00365                 }
00366                 else {
00367 #ifdef WITHMPI
00368                     if ( PF.me == MASTER )
00369 #endif
00370                     printf("Illegal option: %s\n",s);
00371                 }
00372                 errorflag++;
00373             }
00374             #endif
00375         #endif
00376         {
00377             /* Next arg is a path variable like in environment */
00378             TAKEPATH(AM.Path)
00379         }
00380         break;
00381         case 'q': /* Quiet option. Only output. Same as -si */
00382             AM.silent = 1; break;
00383         case 'R': /* recover from saved snapshot */
00384             AC.CheckpointFlag = -1;
00385             break;
00386         case 's': /* Next arg is dir with form.set to be used */
00387             if ( ( s[1] == 'o' ) && ( s[2] == 'r' ) && ( s[3] == 't' ) ) {

```

```

00388             if(s[4]== '=' ) {
00389                 AM.TempSortDir = s+5;
00390             }
00391             else {
00392                 AM.TempSortDir = *argv++;
00393                 argc--;
00394             }
00395         }
00396         else if ( s[1] == 'i' ) { /* compatibility: silent/quiet */
00397             AM.silent = 1;
00398         }
00399         else {
00400             TAKEPATH(AM.SetupDir)
00401         }
00402         break;
00403     case 'S': /* Next arg is setup file */
00404         TAKEPATH(AM.SetupFile) break;
00405     case 't': /* Next arg is directory for temp files */
00406         if ( s[1] == 's' ) {
00407             s++;
00408             AM.havesortdir = 1;
00409             TAKEPATH(AM.TempSortDir)
00410         }
00411         else {
00412             TAKEPATH(AM.TempDir)
00413         }
00414         break;
00415     case 'T': /* Print the total size used at end of job */
00416         AM.PrintTotalSize = 1; break;
00417     case 'v':
00418     printversion;;
00419     #ifdef WITHMPI
00420         if ( PF.me == MASTER )
00421     #endif
00422         PrintHeader(0);
00423         if ( onlyversion ) return(1);
00424         goto NoFile;
00425     case 'y': /* Preprocessor dumps output. No compilation. */
00426         AP.PreDebug = PREPROONLY; break;
00427     case 'z': /* The number following is a time limit in sec. */
00428         t = s++;
00429         AM.TimeLimit = 0;
00430         while ( *s >= '0' && *s <= '9' )
00431             AM.TimeLimit = 10*AM.TimeLimit + *s++ - '0';
00432         break;
00433     case 'Z': /* Removes the .str file on crash, no matter its contents */
00434         AM.ClearStore = 1; break;
00435     case '\0': /* "-" to use STDIN for the input. */
00436     #ifdef WITHMPI
00437         /* At the moment, ParFORM doesn't implement STDIN broadcasts. */
00438         if ( PF.me == MASTER )
00439             printf("Sorry, reading STDIN as input is currently not supported by
ParFORM\n");
00440     #endif
00441         errorflag++;
00442         AM.FromStdin = 1;
00443         AC.NoShowInput = 1; // disable input echoing by default
00444         break;
00445     default:
00446         if ( FG.cTable[*s] == 1 ) {
00447             AM.SkipClears = 0; t = s;
00448             while ( FG.cTable[*t] == 1 )
00449                 AM.SkipClears = 10*AM.SkipClears + *t++ - '0';
00450             if ( *t != 0 ) {
00451     #ifdef WITHMPI
00452                 if ( PF.me == MASTER )
00453     #endif
00454                 printf("Illegal numerical option in call of FORM: %s\n",s);
00455                 errorflag++;
00456             }
00457         }
00458         else {
00459     IllegalOption:
00460     #ifdef WITHMPI
00461         if ( PF.me == MASTER )
00462     #endif
00463             printf("Illegal option in call of FORM: %s\n",s);
00464             errorflag++;
00465         }
00466         break;
00467     }
00468 }
00469 else if ( argc == 0 && !AM.Interact ) AM.InputFileName = argv[-1];
00470 else {
00471     #ifdef WITHMPI
00472         if ( PF.me == MASTER )
00473     #endif

```



```

00474         printf("Illegal option in call of FORM: %s\n",s);
00475         errorflag++;
00476     }
00477 }
00478 AM.totalnumberofthreads = threadnum;
00479 if ( AM.InputFileName ) {
00480     if ( AM.FromStdin ) {
00481         printf("STDIN and the input filename cannot be specified simultaneously\n");
00482         errorflag++;
00483     }
00484     s = AM.InputFileName;
00485     while ( *s ) s++;
00486     if ( s < AM.InputFileName+4 ||
00487         s[-4] != '.' || s[-3] != 'f' || s[-2] != 'r' || s[-1] != 'm' ) {
00488         t = (UBYTE *)Malloc1((s-AM.InputFileName)+5,"adding .frm");
00489         s = AM.InputFileName;
00490         AM.InputFileName = t;
00491         while ( *s ) *t++ = *s++;
00492         *t++ = '.'; *t++ = 'f'; *t++ = 'r'; *t++ = 'm'; *t = 0;
00493     }
00494     if ( AM.LogType >= 0 && AM.LogFileName == 0 ) {
00495         AM.LogFileName = strDup1(AM.InputFileName,"name of logfile");
00496         s = AM.LogFileName;
00497         while ( *s ) s++;
00498         s[-3] = 'l'; s[-2] = 'o'; s[-1] = 'g';
00499     }
00500 }
00501 #ifdef WITHINTERACTION
00502     else if ( AM.Interact ) {
00503         if ( AM.LogType >= 0 ) {
00504             /*
00505              We may have to do better than just taking a name.
00506              It is not unique! This will be left for later.
00507             */
00508             AM.LogFileName = deflogname;
00509         }
00510     }
00511 #endif
00512     else if ( AM.FromStdin ) {
00513         /* Do nothing. */
00514     }
00515     else {
00516 NoFile:
00517 #ifdef WITHMPI
00518         if ( PF.me == MASTER )
00519 #endif
00520         printf("No filename specified in call of FORM\n");
00521         errorflag++;
00522     }
00523     if ( AM.Path == 0 ) AM.Path = (UBYTE *)getenv("FORMPATH");
00524     if ( AM.Path ) {
00525         /*
00526          * AM.Path is taken from argv or getenv. Reallocate it to avoid invalid
00527          * frees when AM.Path has to be changed.
00528          */
00529         AM.Path = strDup1(AM.Path,"DoTail Path");
00530     }
00531     return(errorflag);
00532 }
00533
00534 /*
00535     #] DoTail :
00536     #[ OpenInput :
00537
00538     Major task here after opening is to skip the proper number of
00539     .clear instructions if so desired without using interpretation
00540 */
00541
00542 int OpenInput(void)
00543 {
00544     int oldNoShowInput = AC.NoShowInput;
00545     UBYTE c;
00546     if ( !AM.Interact ) {
00547         if ( AM.FromStdin ) {
00548             if ( OpenStream(0,INPUTSTREAM,0,PREENOACTION) == 0 ) {
00549                 Error0("Cannot open STDIN");
00550                 return(-1);
00551             }
00552         }
00553         else {
00554             if ( OpenStream(AM.InputFileName,FILESTREAM,0,PREENOACTION) == 0 ) {
00555                 Error1("Cannot open file",AM.InputFileName);
00556                 return(-1);
00557             }
00558             if ( AC.CurrentStream->inbuffer <= 0 ) {
00559                 Error1("No input in file",AM.InputFileName);
00560                 return(-1);

```

```

00561     }
00562     }
00563     AC.NoShowInput = 1;
00564     while ( AM.SkipClears > 0 ) {
00565         c = GetInput();
00566         if ( c == ENDOFINPUT ) {
00567             Error0("Not enough .clear instructions in input file");
00568         }
00569         if ( c == '\\ ' ) {
00570             c = GetInput();
00571             if ( c == ENDOFINPUT )
00572                 Error0("Not enough .clear instructions in input file");
00573             continue;
00574         }
00575         if ( c == ' ' || c == '\t' ) continue;
00576         if ( c == '.' ) {
00577             c = GetInput();
00578             if ( tolower(c) == 'c' ) {
00579                 c = GetInput();
00580                 if ( tolower(c) == 'l' ) {
00581                     c = GetInput();
00582                     if ( tolower(c) == 'e' ) {
00583                         c = GetInput();
00584                         if ( tolower(c) == 'a' ) {
00585                             c = GetInput();
00586                             if ( tolower(c) == 'r' ) {
00587                                 c = GetInput();
00588                                 if ( FG.cTable[c] > 2 ) {
00589                                     AM.SkipClears--;
00590                                 }
00591                             }
00592                         }
00593                     }
00594                 }
00595             }
00596             while ( c != '\n' && c != '\r' && c != ENDOFINPUT ) {
00597                 c = GetInput();
00598                 if ( c == '\\ ' ) c = GetInput();
00599             }
00600         }
00601         else if ( c == '\n' || c == '\r' ) continue;
00602         else {
00603             while ( ( c = GetInput() ) != '\n' && c != '\r' ) {
00604                 if ( c == ENDOFINPUT ) {
00605                     Error0("Not enough .clear instructions in input file");
00606                 }
00607             }
00608         }
00609     }
00610     AC.NoShowInput = oldNoShowInput;
00611 }
00612 if ( AM.LogFileName ) {
00613 #ifdef WITHMPI
00614     if ( PF.me != MASTER ) {
00615         /*
00616          * Only the master writes to the log file. On slaves, we need
00617          * a dummy handle, without opening the file.
00618          */
00619         extern FILES **filelist; /* in tools.c */
00620         int i = CreateHandle();
00621         RWLOCKW(AM.handlelock);
00622         filelist[i] = (FILES *)123; /* Must be nonzero to prevent a reuse in CreateHandle. */
00623         UNRWLOCK(AM.handlelock);
00624         AC.LogHandle = i;
00625     }
00626     else
00627 #endif
00628     if ( AC.CheckpointFlag != -1 ) {
00629         if ( ( AC.LogHandle = CreateLogFile((char *) (AM.LogFileName)) ) < 0 ) {
00630             Error1("Cannot create logfile", AM.LogFileName);
00631             return(-1);
00632         }
00633     }
00634     else {
00635         if ( ( AC.LogHandle = OpenAddFile((char *) (AM.LogFileName)) ) < 0 ) {
00636             Error1("Cannot re-open logfile", AM.LogFileName);
00637             return(-1);
00638         }
00639     }
00640 }
00641 return(0);
00642 }
00643
00644 /*
00645     #] OpenInput :
00646     #[ ReserveTempFiles :
00647

```

```

00648         Order of preference:
00649         a: if there is a path in the commandtail, take that.
00650         b: if none, try in the form.set file.
00651         c: if still none, try in the environment for the variable FORMTMP
00652         d: if still none, try the current directory.
00653
00654         The parameter indicates action in the case of multithreaded running.
00655         par = 0 : We just run on a single processor. Keep everything normal.
00656         par = 1 : Multithreaded running startup phase 1.
00657         par = 2 : Multithreaded running startup phase 2.
00658     */
00659
00660     UBYTE *emptystring = (UBYTE *)".";
00661     UBYTE *defaulttmpfilename = (UBYTE *)"xformxxx.str";
00662     /* This is the length of the above default, but with 7 spaces for PID digits
00663        instead of the "xxx". (Previously FORM used 5 digits and this value was 14) */
00664     #define DEFAULTFNAMLENGTH 16
00665
00666     void ReserveTempFiles(int par)
00667     {
00668         GETIDENTITY
00669         SETUPPARAMETERS *sp;
00670         UBYTE *s, *t, *tenddir, *tenddir2, c;
00671         int i = 0;
00672         WORD j;
00673         if ( par == 0 || par == 1 ) {
00674             if ( AM.TempDir == 0 ) {
00675                 sp = GetSetupPar((UBYTE *)"tempdir");
00676                 if ( ( sp->flags & USEDFLAG ) != USEDFLAG ) {
00677                     AM.TempDir = (UBYTE *)getenv("FORMTMP");
00678                     if ( AM.TempDir == 0 ) AM.TempDir = emptystring;
00679                 }
00680                 else AM.TempDir = (UBYTE *) (sp->value);
00681             }
00682             if ( AM.TempSortDir == 0 ) {
00683                 if ( AM.havesortdir ) {
00684                     sp = GetSetupPar((UBYTE *)"tempsortdir");
00685                     AM.TempSortDir = (UBYTE *) (sp->value);
00686                 }
00687                 else {
00688                     AM.TempSortDir = (UBYTE *)getenv("FORMTMPSORT");
00689                     if ( AM.TempSortDir == 0 ) AM.TempSortDir = AM.TempDir;
00690                 }
00691             }
00692         }
00693         /* We have now in principle a path but we will use its first element only.
00694            Later that should become more complicated. Then we will use a path and
00695            when one device is full we can continue on the next one.
00696         */
00697         s = AM.TempDir; i = 200; /* Some extra for VMS */
00698         while ( *s && *s != PATHSEPARATOR ) { if ( *s == '\\') s++; s++; i++; }
00699         FG.fnamesize = sizeof(UBYTE)*(i+DEFAULTFNAMLENGTH);
00700         FG.fname = (char *)Malloc1(FG.fnamesize,"name for temporary files");
00701         s = AM.TempDir; t = (UBYTE *)FG.fname;
00702         while ( *s && *s != PATHSEPARATOR ) { if ( *s == '\\') s++; *t++ = *s++; }
00703         if ( (char *)t > FG.fname && t[-1] != SEPARATOR && t[-1] != ALTSEPARATOR )
00704             *t++ = SEPARATOR;
00705         *t = 0;
00706         tenddir = t;
00707         FG.fnamebase = t-(UBYTE *) (FG.fname);
00708
00709         s = AM.TempSortDir; i = 200; /* Some extra for VMS */
00710         while ( *s && *s != PATHSEPARATOR ) { if ( *s == '\\') s++; s++; i++; }
00711
00712         FG.fname2size = sizeof(UBYTE)*(i+DEFAULTFNAMLENGTH);
00713         FG.fname2 = (char *)Malloc1(FG.fname2size,"name for sort files");
00714         s = AM.TempSortDir; t = (UBYTE *)FG.fname2;
00715         while ( *s && *s != PATHSEPARATOR ) { if ( *s == '\\') s++; *t++ = *s++; }
00716         if ( (char *)t > FG.fname2 && t[-1] != SEPARATOR && t[-1] != ALTSEPARATOR )
00717             *t++ = SEPARATOR;
00718         *t = 0;
00719         tenddir2 = t;
00720         FG.fname2base = t-(UBYTE *) (FG.fname2);
00721
00722         t = tenddir;
00723         s = defaulttmpfilename;
00724         #ifdef WITHMPI
00725         {
00726             int iii;
00727             iii = sprintf((char*)t,FG.fnamesize-((char*)t)-FG.fname),"%d",PF.me);
00728             t+= iii;
00729             s+= iii; /* in case defaulttmpfilename is too short */
00730         }
00731         #endif
00732         while ( *s ) *t++ = *s++;
00733         *t = 0;
00734         /*

```

```

00735         There are problems when running many FORM jobs at the same time
00736         from make or minos. If they start up simultaneously, occasionally
00737         they can make the same .str file. We prevent this with first trying
00738         a file that contains the digits of the pid. If this file
00739         has already been taken we fall back on the old scheme.
00740         The whole is controlled with the -M (MultiRun) parameter in the
00741         command tail.
00742     */
00743     if ( AM.MultiRun ) {
00744         int num = ((int)GetPID())%10000000;
00745         t += 4;
00746         *t = 0;
00747         t[-1] = 'r';
00748         t[-2] = 't';
00749         t[-3] = 's';
00750         t[-4] = '.';
00751         t[-5] = (UBYTE)('0' + num%10);
00752         t[-6] = (UBYTE)('0' + (num/10)%10);
00753         t[-7] = (UBYTE)('0' + (num/100)%10);
00754         t[-8] = (UBYTE)('0' + (num/1000)%10);
00755         t[-9] = (UBYTE)('0' + (num/10000)%10);
00756         t[-10] = (UBYTE)('0' + (num/100000)%10);
00757         t[-11] = (UBYTE)('0' + num/1000000);
00758         if ( ( AC.StoreHandle = CreateFile((char *)FG.fname) ) < 0 ) {
00759             t[-5] = 'x'; t[-6] = 'x'; t[-7] = 'x'; t[-8] = 'x'; t[-9] = 'x'; t[-10] = 'x'; t[-11] =
00760             'x';
00761             goto classic;
00762         }
00763     else
00764     {
00765     classic;;
00766         for(;;) {
00767             if ( ( AC.StoreHandle = OpenFile((char *)FG.fname) ) < 0 ) {
00768                 if ( ( AC.StoreHandle = CreateFile((char *)FG.fname) ) >= 0 ) break;
00769             }
00770             else CloseFile(AC.StoreHandle);
00771             c = t[-5];
00772             if ( c == 'x' ) t[-5] = '0';
00773             else if ( c == '9' ) {
00774                 t[-5] = '0';
00775                 c = t[-6];
00776                 if ( c == 'x' ) t[-6] = '0';
00777                 else if ( c == '9' ) {
00778                     t[-6] = '0';
00779                     c = t[-7];
00780                     if ( c == 'x' ) t[-7] = '0';
00781                     else if ( c == '9' ) {
00782                         /*
00783                             Note that we tried 1111 names!
00784                         */
00785                         MesPrint("Name space for temp files exhausted");
00786                         t[-7] = 0;
00787                         MesPrint("Please remove files of the type %s or try a different directory"
00788                             ,FG.fname);
00789                         Terminate(-1);
00790                     }
00791                     else t[-7] = (UBYTE)(c+1);
00792                 }
00793                 else t[-6] = (UBYTE)(c+1);
00794             }
00795             else t[-5] = (UBYTE)(c+1);
00796         }
00797     }
00798     /*
00799     Now we should make sure that the tempsortdir cq tempsortfilename makes it
00800     into a similar construction.
00801     */
00802     s = tenddir; t = tenddir2; while ( *s ) *t++ = *s++;
00803     *t = 0;
00804
00805     /*
00806     Now we should assign a name to the main sort file and the two stage 4 files.
00807     */
00808     AM.S0->file.name = (char *)Malloc1(sizeof(char)*(i+DEFAULTFNAMELENGTH),"name for temporary
00809     files");
00809     s = (UBYTE *)AM.S0->file.name;
00810     t = (UBYTE *)FG.fname2;
00811     i = 1;
00812     while ( *t ) { *s++ = *t++; i++; }
00813     s[-2] = 'o'; *s = 0;
00814     }
00815     /*
00816     Try to create the sort file already, so we can Terminate earlier if this fails.
00817     */
00818     #ifdef WITHPTHREADS
00819     if ( par <= 1 ) {

```

```

00820 #endif
00821     if ( ( AM.S0->file.handle = CreateFile((char *)AM.S0->file.name) ) < 0 ) {
00822         MesPrint("Could not create sort file: %s", AM.S0->file.name);
00823         Terminate(-1);
00824     };
00825     /* Close and clean up the test file */
00826     CloseFile(AM.S0->file.handle);
00827     AM.S0->file.handle = -1;
00828     remove(AM.S0->file.name);
00829 #ifdef WITHPTHREADS
00830 }
00831 #endif
00832 /*
00833     With the stage4 and scratch file names we have to be a bit more careful.
00834     They are to be allocated after the threads are initialized when there
00835     are threads of course.
00836 */
00837     if ( par == 0 ) {
00838         s = (UBYTE *)((void *) (FG.fname2)); i = 0;
00839         while ( *s ) { s++; i++; }
00840         /* +1 for null terminator */
00841         s = (UBYTE *)Malloc1(sizeof(char)*(i+1),"name for stage4 file a");
00842         AR.FoStage4[1].name = (char *)s;
00843         t = (UBYTE *)FG.fname2;
00844         while ( *t ) *s++ = *t++;
00845         s[-2] = '4'; s[-1] = 'a'; *s = 0;
00846         s = (UBYTE *)((void *) (FG.fname)); i = 0;
00847         while ( *s ) { s++; i++; }
00848         /* +1 for null terminator */
00849         s = (UBYTE *)Malloc1(sizeof(char)*(i+1),"name for stage4 file b");
00850         AR.FoStage4[0].name = (char *)s;
00851         t = (UBYTE *)FG.fname;
00852         while ( *t ) *s++ = *t++;
00853         s[-2] = '4'; s[-1] = 'b'; *s = 0;
00854         for ( j = 0; j < 3; j++ ) {
00855             /* +1 for null terminator */
00856             s = (UBYTE *)Malloc1(sizeof(char)*(i+1),"name for scratch file");
00857             AR.Fscr[j].name = (char *)s;
00858             t = (UBYTE *)FG.fname;
00859             while ( *t ) *s++ = *t++;
00860             s[-2] = 'c'; s[-1] = (UBYTE)('0'+j); *s = 0;
00861         }
00862     }
00863 #ifdef WITHPTHREADS
00864     else if ( par == 2 ) {
00865         size_t tname;
00866         s = (UBYTE *)((void *) (FG.fname2)); i = 0;
00867         while ( *s ) { s++; i++; }
00868         /* +1 for null terminator, +10 for 32bit int, +1 for "." */
00869         tname = sizeof(char)*(i+12);
00870         s = (UBYTE *)Malloc1(tname,"name for stage4 file a");
00871         snprintf((char *)s,tname,"%s.%d",FG.fname2,AT.identity);
00872         s[i-2] = '4'; s[i-1] = 'a';
00873         AR.FoStage4[1].name = (char *)s;
00874         s = (UBYTE *)((void *) (FG.fname)); i = 0;
00875         while ( *s ) { s++; i++; }
00876         /* +1 for null terminator, +10 for 32bit int, +1 for "." */
00877         tname = sizeof(char)*(i+12);
00878         s = (UBYTE *)Malloc1(tname,"name for stage4 file b");
00879         snprintf((char *)s,tname,"%s.%d",FG.fname,AT.identity);
00880         s[i-2] = '4'; s[i-1] = 'b';
00881         AR.FoStage4[0].name = (char *)s;
00882         if ( AT.identity == 0 ) {
00883             for ( j = 0; j < 3; j++ ) {
00884                 /* +1 for null terminator */
00885                 s = (UBYTE *)Malloc1(sizeof(char)*(i+1),"name for scratch file");
00886                 AR.Fscr[j].name = (char *)s;
00887                 t = (UBYTE *)FG.fname;
00888                 while ( *t ) *s++ = *t++;
00889                 s[-2] = 'c'; s[-1] = (UBYTE)('0'+j); *s = 0;
00890             }
00891         }
00892     }
00893 #endif
00894 }
00895
00896 /*
00897     #] ReserveTempFiles :
00898     #[ StartVariables :
00899 */
00900
00901 #ifdef WITHPTHREADS
00902 ALLPRIVATES *DummyPointer = 0;
00903 #endif
00904
00905 void StartVariables(void)
00906 {

```

```

00907     int i, ii;
00908     PUTZERO(AM.zeropos);
00909     StartPrepro();
00910 /*
00911     The module counter:
00912 */
00913     AC.CModule=0;
00914 #ifdef WITHPTHREADS
00915 /*
00916     We need a value in AB because in the startup some routines may call AB[0].
00917 */
00918     AB = (ALLPRIVATES **) &DummyPointer;
00919 #endif
00920 /*
00921     separators used to delimit arguments in #call and #do, by default ',' and '|':
00922     Be sure, it is an empty set:
00923 */
00924     set_sub(AC.separators,AC.separators,AC.separators);
00925     set_set(',',AC.separators);
00926     set_set('|',AC.separators);
00927
00928     AM.BraceFactors[0] = 8;
00929     AM.BraceFactors[1] = SYMBOL;
00930     AM.BraceFactors[2] = 4;
00931     AM.BraceFactors[3] = FACTORSYMBOL;
00932     AM.BraceFactors[4] = 1;
00933     AM.BraceFactors[5] = 1;
00934     AM.BraceFactors[6] = 1;
00935     AM.BraceFactors[7] = 3;
00936
00937     AM.SkipClears = 0;
00938     AC.Cnumpows = 0;
00939     AC.OutputMode = 72;
00940     AC.OutputSpaces = NORMALFORMAT;
00941     AC.LineLength = 79;
00942     AM.gIsFortran90 = AC.IsFortran90 = ISNOTFORTRAN90;
00943     AM.gFortran90Kind = AC.Fortran90Kind = 0;
00944     AM.gCnumpows = 0;
00945     AC.exprfillwarning = 0;
00946     AM.gLineLength = 79;
00947     AM.OutBufSize = 80;
00948     AM.MaxStreamSize = MAXFILESTREAMSIZE;
00949     AP.MaxPreAssignLevel = 4;
00950     AC.iBufferSize = 512;
00951     AP.pSize = 128;
00952     AP.MaxPreIfLevel = 10;
00953     AP.cComChar = AP.ComChar = '*';
00954     AP.firstnamespace = 0;
00955     AP.lastnamespace = 0;
00956     AP.fullnamesize = 127;
00957     AP.fullname = (UBYTE *) Malloc1((AP.fullnamesize+1)*sizeof(UBYTE), "AP.fullname");
00958     AM.OffsetVector = -2*WILDOFFSET+MINSPEC;
00959     AC.cbufList.num = 0;
00960     AM.hparallelflag = AM.gparallelflag = 0;
00961     AC.parallelflag = AC.mparallelflag = PARALLELFLAG;
00962 #ifdef WITHMPI
00963     if ( PF.numtasks < 2 ) AM.hparallelflag |= NOPARALLEL_NPROC;
00964 #endif
00965     AC.tablefilling = 0;
00966     AC.models = 0;
00967     AC.nummodels = 0;
00968     AC.ModelLevel = 0;
00969     AC.modelspace = 0;
00970     AM.resetTimeOnClear = 1;
00971     AM.gnumextrasym = AM.ggnumextrasym = 0;
00972     AM.havesortdir = 0;
00973     AM.SpectatorFiles = 0;
00974     AM.NumSpectatorFiles = 0;
00975     AM.SizeForSpectatorFiles = 0;
00976 /*
00977     Information for the lists of variables. Part of error message and size:
00978 */
00979     AP.ProcList.message = "procedure";
00980     AP.ProcList.size = sizeof(PROCEDURE);
00981     AP.LoopList.message = "doloop";
00982     AP.LoopList.size = sizeof(DOLOOP);
00983     AP.PreVarList.message = "PreVariable";
00984     AP.PreVarList.size = sizeof(PREVAR);
00985     AC.SymbolList.message = "symbol";
00986     AC.SymbolList.size = sizeof(struct SyMbol);
00987     AC.IndexList.message = "index";
00988     AC.IndexList.size = sizeof(struct InDeX);
00989     AC.VectorList.message = "vector";
00990     AC.VectorList.size = sizeof(struct VeCtOr);
00991     AC.FunctionList.message = "function";
00992     AC.FunctionList.size = sizeof(struct FuNcTiOn);
00993     AC.SetList.message = "set";

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00994     AC.SetList.size = sizeof(struct SeTs);
00995     AC.SetElementList.message = "set element";
00996     AC.SetElementList.size = sizeof(WORD);
00997     AC.ExpressionList.message = "expression";
00998     AC.ExpressionList.size = sizeof(struct ExPrEsSiOn);
00999     AC.cbufList.message = "compiler buffer";
01000     AC.cbufList.size = sizeof(CBUF);
01001     AC.Channellist.message = "channel buffer";
01002     AC.Channellist.size = sizeof(CHANNEL);
01003     AP.DollarList.message = "$-variable";
01004     AP.DollarList.size = sizeof(struct DoLLArS);
01005     AC.DubiousList.message = "ambiguous variable";
01006     AC.DubiousList.size = sizeof(struct DuBiOuS);
01007     AC.TableBaseList.message = "list of tablebases";
01008     AC.TableBaseList.size = sizeof(DBASE);
01009     AC.TestValue = 0;
01010     AC.InnerTest = 0;
01011
01012     AC.AutoSymbolList.message = "autosymbol";
01013     AC.AutoSymbolList.size = sizeof(struct SyMbOl);
01014     AC.AutoIndexList.message = "autoindex";
01015     AC.AutoIndexList.size = sizeof(struct InDeX);
01016     AC.AutoVectorList.message = "autovector";
01017     AC.AutoVectorList.size = sizeof(struct VeCtOr);
01018     AC.AutoFunctionList.message = "autofunction";
01019     AC.AutoFunctionList.size = sizeof(struct FuNcTiOn);
01020     AC.PotModDolList.message = "potentially modified dollar";
01021     AC.PotModDolList.size = sizeof(WORD);
01022     AC.ModOptDolList.message = "moduleoptiondollar";
01023     AC.ModOptDolList.size = sizeof(MODOPTDOLLAR);
01024
01025     AO.FortDotChar = '_';
01026     AO.ErrorBlock = 0;
01027     AC.firstconstindex = 1;
01028     AO.Optimize.mctsconstant.fval = 1.0;
01029     AO.Optimize.horner = O_MCTS;
01030     AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
01031     AO.Optimize.method = O_GREEDY;
01032     AO.Optimize.mctstimelimit = 0;
01033     AO.Optimize.mctsnumexpand = 1000;
01034     AO.Optimize.mctsnumkeep = 10;
01035     AO.Optimize.mctsnumrepeat = 1;
01036     AO.Optimize.greedytimelimit = 0;
01037     AO.Optimize.greedyminnum = 10;
01038     AO.Optimize.greedymaxperc = 5;
01039     AO.Optimize.printstats = 0;
01040     AO.Optimize.debugflags = 0;
01041     AO.OptimizeResult.code = NULL;
01042     AO.inscheme = 0;
01043     AO.schemenum = 0;
01044     AO.wpos = 0;
01045     AO.wpoin = 0;
01046     AO.wlen = 0;
01047     AM.dollarzero = 0;
01048     AC.doloopstack = 0;
01049     AC.doloopstacksize = 0;
01050     AC.dolooplevel = 0;
01051 /*
01052     Set up the main name trees:
01053 */
01054     AC.varnames = MakeNameTree();
01055     AC.exprnames = MakeNameTree();
01056     AC.dollarname = MakeNameTree();
01057     AC.autonames = MakeNameTree();
01058     AC.activenames = &(AC.varnames);
01059     AP.preError = 0;
01060 /*
01061     Initialize the compiler:
01062 */
01063     inictable();
01064     AM.rbufnum = inicbufs(); /* Regular compiler buffer */
01065 #ifndef WITHPTHREADS
01066     AT.ebufnum = inicbufs(); /* Buffer for extras during execution */
01067     AT.fbufnum = inicbufs(); /* Buffer for caching in factorization */
01068     AT.allbufnum = inicbufs(); /* Buffer for id,all */
01069     AT.aebufnum = inicbufs(); /* Buffer for id,all */
01070     AN.tryterm = 0;
01071 #else
01072     AS.MasterSort = 0;
01073 #endif
01074     AM.dbufnum = inicbufs(); /* Buffer for dollar variables */
01075     AM.sbufnum = inicbufs(); /* Subterm buffer for polynomials and optimization */
01076     AC.ffbufnum = inicbufs(); /* Buffer number for user defined factorizations */
01077     AM.zbufnum = inicbufs(); /* For very special values */
01078     {
01079         CBUF *C = cbuf+AM.zbufnum;
01080         WORD one[5] = {4,1,1,3,0};

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01081     WORD zero = 0;
01082     AddrRHS (AM.zbufnum,1);
01083     AM.zerorhs = C->numrhs;
01084     AddNtoC (AM.zbufnum,1,&zero,17);
01085     AddrRHS (AM.zbufnum,1);
01086     AM.onerhs = C->numrhs;
01087     AddNtoC (AM.zbufnum,5,one,17);
01088 }
01089 AP.inside.inscbuf = inicbufs(); /* For the #inside instruction */
01090 /*
01091  Enter the built in objects
01092 */
01093 AC.Symbols = &(AC.SymbolList);
01094 AC.Indices = &(AC.IndexList);
01095 AC.Vectors = &(AC.VectorList);
01096 AC.Functions = &(AC.FunctionList);
01097 AC.vetofilling = 0;
01098
01099 AddDollar ((UBYTE *) "$", DOLUNDEFINED, 0, 0);
01100
01101 cbuf[AM.dbufnum].mnumlhs = cbuf[AM.dbufnum].numlhs;
01102 cbuf[AM.dbufnum].mnumrhs = cbuf[AM.dbufnum].numrhs;
01103
01104 AddSymbol ((UBYTE *) "i_", -MAXPOWER, MAXPOWER, VARTYPEIMAGINARY, 0);
01105 AddSymbol ((UBYTE *) "pi_", -MAXPOWER, MAXPOWER, VARTYPENONE, 0);
01106 /*
01107  coeff_ should have the number COEFFSYMBOL and den_ the number DENOMINATOR
01108  and the three should be in this order!
01109 */
01110 AddSymbol ((UBYTE *) "coeff_", -MAXPOWER, MAXPOWER, VARTYPENONE, 0);
01111 AddSymbol ((UBYTE *) "num_", -MAXPOWER, MAXPOWER, VARTYPENONE, 0);
01112 AddSymbol ((UBYTE *) "den_", -MAXPOWER, MAXPOWER, VARTYPENONE, 0);
01113 AddSymbol ((UBYTE *) "xarg_", -MAXPOWER, MAXPOWER, VARTYPENONE, 0);
01114 AddSymbol ((UBYTE *) "dimension_", -MAXPOWER, MAXPOWER, VARTYPENONE, 0);
01115 AddSymbol ((UBYTE *) "factor_", -MAXPOWER, MAXPOWER, VARTYPENONE, 0);
01116 AddSymbol ((UBYTE *) "sep_", -MAXPOWER, MAXPOWER, VARTYPENONE, 0);
01117 AddSymbol ((UBYTE *) "ee_", -MAXPOWER, MAXPOWER, VARTYPENONE, 0);
01118 AddSymbol ((UBYTE *) "em_", -MAXPOWER, MAXPOWER, VARTYPENONE, 0);
01119 i = BUILTINSYMBOLS; /* update this in ftypes.h when we add new symbols */
01120 /*
01121  Next we add a number of dummy symbols for ensuring that the user defined
01122  symbols start at a fixed given number FIRSTUSERSYMBOL
01123  We do want to give them unique names though that the user cannot access.
01124 */
01125 {
01126     char dumstr[20];
01127     for ( ; i < FIRSTUSERSYMBOL; i++ ) {
01128         snprintf (dumstr, 20, "%d:", i);
01129         AddSymbol ((UBYTE *) dumstr, -MAXPOWER, MAXPOWER, VARTYPENONE, 0);
01130     }
01131 }
01132
01133 AddIndex ((UBYTE *) "iarg_", 4, 0);
01134 AddVector ((UBYTE *) "parg_", VARTYPENONE, 0);
01135
01136 AM.NumFixedFunctions = sizeof(fixedfunctions)/sizeof(struct fixedfun);
01137 for ( i = 0; i < AM.NumFixedFunctions; i++ ) {
01138     ii = AddFunction ((UBYTE *) fixedfunctions[i].name
01139                     , fixedfunctions[i].commu
01140                     , fixedfunctions[i].tensor
01141                     , fixedfunctions[i].complx
01142                     , fixedfunctions[i].symmetric
01143                     , 0, -1, -1);
01144     if ( fixedfunctions[i].tensor == GAMMAFUNCTION )
01145         functions[ii].flags |= COULDCOMMUTE;
01146 }
01147 /*
01148  Next we add a number of dummy functions for ensuring that the user defined
01149  functions start at a fixed given number FIRSTUSERFUNCTION.
01150  We do want to give them unique names though that the user cannot access.
01151 */
01152 {
01153     char dumstr[20];
01154     for ( ; i < FIRSTUSERFUNCTION-FUNCTION; i++ ) {
01155         snprintf (dumstr, 20, ":%d::", i);
01156         AddFunction ((UBYTE *) dumstr, 0, 0, 0, 0, 0, -1, -1);
01157     }
01158 }
01159 AM.NumFixedSets = sizeof(fixedsets)/sizeof(struct fixedset);
01160 for ( i = 0; i < AM.NumFixedSets; i++ ) {
01161     ii = AddSet ((UBYTE *) fixedsets[i].name, fixedsets[i].dimension);
01162     Sets[ii].type = fixedsets[i].type;
01163 }
01164 AM.RepMax = MAXREPEAT;
01165 #ifndef WITHPTHREADS
01166 AT.RepCount = (int *) Malloc1 (LONG) ((AM.RepMax+3)*sizeof(int)), "repeat buffers";
01167 AN.RepPoint = AT.RepCount;

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01168     AT.RepTop = AT.RepCount + AM.RepMax;
01169     AN.polysortflag = 0;
01170     AN.subsubveto = 0;
01171 #endif
01172     AC.NumWildcardNames = 0;
01173     AC.WildcardBufferSize = 50;
01174     AC.WildcardNames = (UBYTE *)Malloc1((LONG)AC.WildcardBufferSize,"argument list names");
01175 #ifndef WITHPTHREADS
01176     AT.WildArgTaken = (WORD *)Malloc1((LONG)AC.WildcardBufferSize*sizeof(WORD)/2
01177                                     ,"argument list names");
01178     AT.WildcardBufferSize = AC.WildcardBufferSize;
01179     AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
01180     AT.nfac = AT.nBer = 0;
01181     AT.factorials = 0;
01182     AT.bernoullis = 0;
01183     AR.wranfia = 0;
01184     AR.wranfcall = 0;
01185     AR.wranfnpair1 = NPAIR1;
01186     AR.wranfnpair2 = NPAIR2;
01187     AR.wranfseed = 0;
01188 #endif
01189     AM.atstartup = 1;
01190     AM.oldnumextrasymbols = strDup1((UBYTE *)"OLDNUMEXTRASYMBOLS_", "oldnumextrasymbols");
01191     PutPreVar((UBYTE *)"VERSION_", (UBYTE *)STRINGIFY(MAJORVERSION), 0, 0);
01192     PutPreVar((UBYTE *)"SUBVERSION_", (UBYTE *)STRINGIFY(MINORVERSION), 0, 0);
01193     PutPreVar((UBYTE *)"DATE_", (UBYTE *)MakeDate(), 0, 0);
01194     PutPreVar((UBYTE *)"random_", (UBYTE *)"_____", (UBYTE *)"?a", 0);
01195     PutPreVar((UBYTE *)"optimminvar_", (UBYTE *)("0"), 0, 0);
01196     PutPreVar((UBYTE *)"optimmaxvar_", (UBYTE *)("0"), 0, 0);
01197     PutPreVar(AM.oldnumextrasymbols, (UBYTE *)("0"), 0, 0);
01198     PutPreVar((UBYTE *)"optimvalue_", (UBYTE *)("0"), 0, 0);
01199     PutPreVar((UBYTE *)"optimscheme_", (UBYTE *)("0"), 0, 0);
01200     PutPreVar((UBYTE *)"tolower_", (UBYTE *)("0"), (UBYTE *)("?a"), 0);
01201     PutPreVar((UBYTE *)"toupper_", (UBYTE *)("0"), (UBYTE *)("?a"), 0);
01202     PutPreVar((UBYTE *)"takeleft_", (UBYTE *)("0"), (UBYTE *)("?a"), 0);
01203     PutPreVar((UBYTE *)"takerright_", (UBYTE *)("0"), (UBYTE *)("?a"), 0);
01204     PutPreVar((UBYTE *)"keepleft_", (UBYTE *)("0"), (UBYTE *)("?a"), 0);
01205     PutPreVar((UBYTE *)"keepright_", (UBYTE *)("0"), (UBYTE *)("?a"), 0);
01206     PutPreVar((UBYTE *)"SYSTEMERROR_", (UBYTE *)("0"), 0, 0);
01207 /*
01208     Next are a few 'constants' for diagram generation
01209 */
01210     PutPreVar((UBYTE *)"ONEPI_", (UBYTE *)("1"), 0, 0);
01211     PutPreVar((UBYTE *)"WITHOUTINSERTIONS_", (UBYTE *)("2"), 0, 0);
01212     PutPreVar((UBYTE *)"NOTADPOLES_", (UBYTE *)("4"), 0, 0);
01213     PutPreVar((UBYTE *)"SYMMETRIZE_", (UBYTE *)("8"), 0, 0);
01214     PutPreVar((UBYTE *)"TOPOLOGIESONLY_", (UBYTE *)("16"), 0, 0);
01215     PutPreVar((UBYTE *)"NONODES_", (UBYTE *)("32"), 0, 0);
01216     PutPreVar((UBYTE *)"WITHEDGES_", (UBYTE *)("64"), 0, 0);
01217 /*
01218     Note that CHECKEXTERN is 128 */
01218     PutPreVar((UBYTE *)"WITHBLOCKS_", (UBYTE *)("256"), 0, 0);
01219     PutPreVar((UBYTE *)"WITHONEPISETS_", (UBYTE *)("512"), 0, 0);
01220     PutPreVar((UBYTE *)"NOSNAILS_", (UBYTE *)("1024"), 0, 0);
01221     PutPreVar((UBYTE *)"NOEXTSELF_", (UBYTE *)("2048"), 0, 0);
01222
01223     {
01224         char buf[41]; /* up to 128-bit */
01225         LONG pid;
01226 #ifndef WITHMPI
01227         pid = GetPID();
01228 #else
01229         pid = ( PF.me == MASTER ) ? GetPID() : (LONG)0;
01230         pid = PF_BroadcastNumber(pid);
01231 #endif
01232         LongCopy(pid, buf);
01233         PutPreVar((UBYTE *)"PID_", (UBYTE *)buf, 0, 0);
01234     }
01235     AM.atstartup = 0;
01236     AP.MaxPreTypes = 10;
01237     AP.NumPreTypes = 0;
01238     AP.PreTypes = (int *)Malloc1(sizeof(int)*(AP.MaxPreTypes+1), "preprocessor types");
01239     AP.inside.buffer = 0;
01240     AP.inside.size = 0;
01241
01242     AC.SortType = AC.lSortType = AM.gSortType = SORTLOWFIRST;
01243 #ifndef WITHPTHREADS
01244 #else
01245     AR.SortType = AC.SortType;
01246 #endif
01247     AC.LogHandle = -1;
01248     AC.SetList.numtemp = AC.SetList.num;
01249     AC.SetElementList.numtemp = AC.SetElementList.num;
01250
01251     GetName(AC.varnames, (UBYTE *)"exp_", &AM.expnum, NOAUTO);
01252     GetName(AC.varnames, (UBYTE *)"denom_", &AM.denomnum, NOAUTO);
01253     GetName(AC.varnames, (UBYTE *)"fac_", &AM.facnum, NOAUTO);
01254     GetName(AC.varnames, (UBYTE *)"invfac_", &AM.invfacnum, NOAUTO);

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01255     GetName(AC.varnames, (UBYTE *) "sum_", &AM.sumnum, NOAUTO);
01256     GetName(AC.varnames, (UBYTE *) "sump_", &AM.sumpnum, NOAUTO);
01257     GetName(AC.varnames, (UBYTE *) "term_", &AM.termfunnum, NOAUTO);
01258     GetName(AC.varnames, (UBYTE *) "match_", &AM.matchfunnum, NOAUTO);
01259     GetName(AC.varnames, (UBYTE *) "count_", &AM.countfunnum, NOAUTO);
01260     AM.termfunnum += FUNCTION;
01261     AM.matchfunnum += FUNCTION;
01262     AM.countfunnum += FUNCTION;
01263
01264     AC.ThreadStats = AM.gThreadStats = AM.ggThreadStats = 1;
01265     AC.FinalStats = AM.gFinalStats = AM.ggFinalStats = 1;
01266     AC.StatsFlag = AM.gStatsFlag = AM.ggStatsFlag = 1;
01267     AC.ThreadsFlag = AM.gThreadsFlag = AM.ggThreadsFlag = 1;
01268     AC.ThreadBalancing = AM.gThreadBalancing = AM.ggThreadBalancing = 1;
01269     AC.ThreadSortFileSynch = AM.gThreadSortFileSynch = AM.ggThreadSortFileSynch = 0;
01270     AC.ProcessStats = AM.gProcessStats = AM.ggProcessStats = 1;
01271     AC.OldParallelStats = AM.gOldParallelStats = AM.ggOldParallelStats = 0;
01272     AC.OldFactArgFlag = AM.gOldFactArgFlag = AM.ggOldFactArgFlag = NEWFACTARG;
01273     AC.OldGCDflag = AM.gOldGCDflag = AM.ggOldGCDflag = 1;
01274     AC.WTimeStatsFlag = AM.gWTimeStatsFlag = AM.ggWTimeStatsFlag = 0;
01275     AM.gcNumDollars = AP.DollarList.num;
01276     AC.SizeCommuteInSet = AM.gSizeCommuteInSet = 0;
01277 #ifdef WITHFLOAT
01278     AC.MaxWeight = AM.gMaxWeight = AM.ggMaxWeight = MAXWEIGHT;
01279     AC.DefaultPrecision = AM.gDefaultPrecision = AM.ggDefaultPrecision = DEFAULTPRECISION;
01280 #endif
01281     AC.CommuteInSet = 0;
01282
01283     AM.PrintTotalSize = 0;
01284
01285     AO.NoSpacesInNumbers = AM.gNoSpacesInNumbers = AM.ggNoSpacesInNumbers = 0;
01286     AO.IndentSpace = AM.gIndentSpace = AM.ggIndentSpace = INDENTSPACE;
01287     AO.BlockSpaces = 0;
01288     AO.OptimizationLevel = 0;
01289     PUTZERO(AS.MaxExprSize);
01290     PUTZERO(AC.StoreFileSize);
01291
01292 #ifdef WITHPTHREADS
01293     AC.inputnumbers = 0;
01294     AC.pfirstnum = 0;
01295     AC.numpfirstnum = AC.sizepfirstnum = 0;
01296 #endif
01297     AC.MemDebugFlag = 1;
01298
01299 #ifdef WITHEXTERNALCHANNEL
01300     AX.currentExternalChannel=0;
01301     AX.killSignal=SIGKILL;
01302     AX.killWholeGroup=1;
01303     AX.daemonize=1;
01304     AX.currentPrompt=0;
01305     AX.timeout=1000; /*One second to initialize preset channels*/
01306     AX.shellname=strDup1((UBYTE *) "/bin/sh -c", "external channel shellname");
01307     AX.stderrname=strDup1((UBYTE *) "/dev/null", "external channel stderrname");
01308 #endif
01309 }
01310
01311 /*
01312     #] StartVariables :
01313     #[ StartMore :
01314 */
01315
01316 void StartMore(void)
01317 {
01318 #ifdef WITHEXTERNALCHANNEL
01319     /*If env.variable "FORM_PIPES" is defined, we have to initialize
01320     corresponding pre-set external channels, see file extcmd.c.*/
01321     /*This line must be after all setup settings: in future, timeout
01322     could be changed at setup.*/
01323     if(AX.timeout>=0) /*if AX.timeout<0, this was done by cmdline option -pipe*/
01324         initPresetExternalChannels((UBYTE*)getenv("FORM_PIPES"), AX.timeout);
01325 #endif
01326
01327 #ifdef WITHMPI
01328 /*
01329     Define preprocessor variable PARALLELTASK_ as a process number, 0 is the master
01330     Define preprocessor variable NPARALLELTASKS_ as a total number of processes
01331 */
01332     {
01333         UBYTE buf[32];
01334         snprintf((char*)buf, 32, "%d", PF.me);
01335         PutPreVar((UBYTE *) "PARALLELTASK_", buf, 0, 0);
01336         snprintf((char*)buf, 32, "%d", PF.numtasks);
01337         PutPreVar((UBYTE *) "NPARALLELTASKS_", buf, 0, 0);
01338     }
01339 #else
01340     PutPreVar((UBYTE *) "PARALLELTASK_", (UBYTE *) "0", 0, 0);
01341     PutPreVar((UBYTE *) "NPARALLELTASKS_", (UBYTE *) "1", 0, 0);

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01342 #endif
01343
01344     PutPreVar((UBYTE *) "NAME_", AM.InputFileName ? AM.InputFileName : (UBYTE *) "STDIN", 0, 0);
01345 }
01346
01347 /*
01348     #] StartMore :
01349     #[ IniVars :
01350
01351     This routine initializes the parameters that may change during the run.
01352 */
01353
01354 void IniVars(void)
01355 {
01356     #ifdef WITHPTHREADS
01357         GETIDENTITY
01358     #else
01359         WORD *t;
01360     #endif
01361     WORD *fi, i, one = 1;
01362     CBUF *C = cbuf+AC.cbunum;
01363
01364     #ifdef WITHPTHREADS
01365         UBYTE buf[32];
01366         snprintf((char*)buf, 32, "%d", AM.totalnumberofthreads);
01367         PutPreVar((UBYTE *) "NTHREADS_", buf, 0, 1);
01368     #else
01369         PutPreVar((UBYTE *) "NTHREADS_", (UBYTE *) "1", 0, 1);
01370     #endif
01371
01372     AC.ShortStats = 0;
01373     AC.WarnFlag = 1;
01374     AR.SortType = AC.SortType = AC.lSortType = AM.gSortType;
01375     AC.OutputMode = 72;
01376     AC.OutputSpaces = NORMALFORMAT;
01377     AR.Eside = 0;
01378     AC.DumNum = 0;
01379     AC.ncmod = AM.gncmod = 0;
01380     AC.modmode = AM.gmodmode = 0;
01381     AC.npowmod = AM.gnpowmod = 0;
01382     AC.halfmod = 0; AC.nhalfmod = 0;
01383     AC.modinverses = 0;
01384     AC.lPolyFun = AM.gPolyFun = 0;
01385     AC.lPolyFunInv = AM.gPolyFunInv = 0;
01386     AC.lPolyFunType = AM.gPolyFunType = 0;
01387     AC.lPolyFunExp = AM.gPolyFunExp = 0;
01388     AC.lPolyFunVar = AM.gPolyFunVar = 0;
01389     AC.lPolyFunPow = AM.gPolyFunPow = 0;
01390     #ifdef WITHFLINT
01391         AC.FlintPolyFlag = 1;
01392     #else
01393         AC.FlintPolyFlag = 0;
01394     #endif
01395     AC.DirtPow = 0;
01396     AC.lDefDim = AM.gDefDim = 4;
01397     AC.lDefDim4 = AM.gDefDim4 = 0;
01398     AC.lUnitTrace = AM.gUnitTrace = 4;
01399     AC.NamesFlag = AM.gNamesFlag = 0;
01400     AC.CodesFlag = AM.gCodesFlag = 0;
01401     /* Printing a backtrace on crash is on by default for both normal and debug
01402        modes if FORM has been compiled with backtrace support. */
01403     #ifdef ENABLE_BACKTRACE
01404         AC.PrintBacktraceFlag = 1;
01405     #else
01406         AC.PrintBacktraceFlag = 0;
01407     #endif
01408     /* Human-readable statistics are off by default */
01409     AC.HumanStatsFlag = 0;
01410     AC.extrasymbols = AM.gextrasymbols = 0;
01411     AC.extrasym = (UBYTE *) Malloc1(2*sizeof(UBYTE), "extrasym");
01412     AM.gextrasym = (UBYTE *) Malloc1(2*sizeof(UBYTE), "extrasym");
01413     AM.ggextrasym = (UBYTE *) Malloc1(2*sizeof(UBYTE), "extrasym");
01414     AC.extrasym[0] = AM.gextrasym[0] = AM.ggextrasym[0] = 'Z';
01415     AC.extrasym[1] = AM.gextrasym[1] = AM.ggextrasym[1] = 0;
01416     AC.TokensWriteFlag = AM.gTokensWriteFlag = 0;
01417     AC.SetupFlag = 0;
01418     AC.LineLength = AM.gLineLength = 79;
01419     AC.NwildC = 0;
01420     AC.OutputMode = 0;
01421     AM.gOutputMode = 0;
01422     AC.OutputSpaces = NORMALFORMAT;
01423     AM.gOutputSpaces = NORMALFORMAT;
01424     AC.OutNumberType = RATIONALMODE;
01425     AM.gOutNumberType = RATIONALMODE;
01426     #ifdef WITHZLIB
01427         AR.gzipCompress = GZIPDEFAULT;
01428         AR.FoStage4[0].ziobuffer = 0;

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01429     AR.FoStage4[1].ziobuffer = 0;
01430 #ifdef WITHZSTD
01431     /* Zstd compression is on by default, if we have compiled with it */
01432     ZWRAP_useZSTDcompression(1);
01433 #endif
01434 #endif
01435     AR.BracketOn = 0;
01436     AC.bracketindexflag = 0;
01437     AT.bracketindexflag = 0;
01438     AT.bracketinfo = 0;
01439     AO.IsBracket = 0;
01440     AM.gfunpowers = AC.funpowers = COMFUNPOWERS;
01441     AC.parallelfldflag = AM.gparallelfldflag;
01442     AC.properorderflag = AM.gproperorderflag = PROPERORDERFLAG;
01443     AC.ProcessBucketSize = AC.mProcessBucketSize = AM.gProcessBucketSize;
01444     AC.ThreadBucketSize = AM.gThreadBucketSize;
01445     AC.ShortStatsMax = 0;
01446     AM.gShortStatsMax = 0;
01447     AM.ggShortStatsMax = 0;
01448
01449     GlobalSymbols      = NumSymbols;
01450     GlobalIndices      = NumIndices;
01451     GlobalVectors      = NumVectors;
01452     GlobalFunctions    = NumFunctions;
01453     GlobalSets         = NumSets;
01454     GlobalSetElements  = NumSetElements;
01455     AC.modpowers = (UWORD *)0;
01456
01457     i = AM.OffsetIndex;
01458     fi = AC.FixIndices;
01459     if ( i > 0 ) do { *fi++ = one; } while ( --i >= 0 );
01460     AR.sLevel = -1;
01461     AM.Ordering[0] = 5;
01462     AM.Ordering[1] = 6;
01463     AM.Ordering[2] = 7;
01464     AM.Ordering[3] = 0;
01465     AM.Ordering[4] = 1;
01466     AM.Ordering[5] = 2;
01467     AM.Ordering[6] = 3;
01468     AM.Ordering[7] = 4;
01469     for ( i = 8; i < 15; i++ ) AM.Ordering[i] = i;
01470     AM.gUniTrace[0] =
01471     AC.lUniTrace[0] = SNUMBER;
01472     AM.gUniTrace[1] =
01473     AC.lUniTrace[1] =
01474     AM.gUniTrace[2] =
01475     AC.lUniTrace[2] = 4;
01476     AM.gUniTrace[3] =
01477     AC.lUniTrace[3] = 1;
01478 #ifdef WITHPTHREADS
01479     AS.Balancing = 0;
01480 #else
01481     AT.MinVecArg[0] = 7+ARGHEAD;
01482     AT.MinVecArg[ARGHEAD] = 7;
01483     AT.MinVecArg[1+ARGHEAD] = INDEX;
01484     AT.MinVecArg[2+ARGHEAD] = 3;
01485     AT.MinVecArg[3+ARGHEAD] = 0;
01486     AT.MinVecArg[4+ARGHEAD] = 1;
01487     AT.MinVecArg[5+ARGHEAD] = 1;
01488     AT.MinVecArg[6+ARGHEAD] = -3;
01489     t = AT.FunArg;
01490     *t++ = 4+ARGHEAD+FUNHEAD;
01491     for ( i = 1; i < ARGHEAD; i++ ) *t++ = 0;
01492     *t++ = 4+FUNHEAD;
01493     *t++ = 0;
01494     *t++ = FUNHEAD;
01495     for ( i = 2; i < FUNHEAD; i++ ) *t++ = 0;
01496     *t++ = 1; *t++ = 1; *t++ = 3;
01497
01498 #ifdef WITHMPI
01499     AS.printflag = 0;
01500 #endif
01501
01502     AT.comsym[0] = 8;
01503     AT.comsym[1] = SYMBOL;
01504     AT.comsym[2] = 4;
01505     AT.comsym[3] = 0;
01506     AT.comsym[4] = 1;
01507     AT.comsym[5] = 1;
01508     AT.comsym[6] = 1;
01509     AT.comsym[7] = 3;
01510     AT.comnum[0] = 4;
01511     AT.comnum[1] = 1;
01512     AT.comnum[2] = 1;
01513     AT.comnum[3] = 3;
01514     AT.comfun[0] = FUNHEAD+4;
01515     AT.comfun[1] = FUNCTION;

```

```

01516     AT.comfun[2] = FUNHEAD;
01517     AT.comfun[3] = 0;
01518     #if FUNHEAD == 4
01519     AT.comfun[4] = 0;
01520     #endif
01521     AT.comfun[FUNHEAD+1] = 1;
01522     AT.comfun[FUNHEAD+2] = 1;
01523     AT.comfun[FUNHEAD+3] = 3;
01524     AT.comind[0] = 7;
01525     AT.comind[1] = INDEX;
01526     AT.comind[2] = 3;
01527     AT.comind[3] = 0;
01528     AT.comind[4] = 1;
01529     AT.comind[5] = 1;
01530     AT.comind[6] = 3;
01531     AT.locwildvalue[0] = SUBEXPRESSION;
01532     AT.locwildvalue[1] = SUBEXPSIZE;
01533     for ( i = 2; i < SUBEXPSIZE; i++ ) AT.locwildvalue[i] = 0;
01534     AT.mulpat[0] = TYPEMULT;
01535     AT.mulpat[1] = SUBEXPSIZE+3;
01536     AT.mulpat[2] = 0;
01537     AT.mulpat[3] = SUBEXPRESSION;
01538     AT.mulpat[4] = SUBEXPSIZE;
01539     AT.mulpat[5] = 0;
01540     AT.mulpat[6] = 1;
01541     for ( i = 7; i < SUBEXPSIZE+5; i++ ) AT.mulpat[i] = 0;
01542     AT.proexp[0] = SUBEXPSIZE+4;
01543     AT.proexp[1] = EXPRESSION;
01544     AT.proexp[2] = SUBEXPSIZE;
01545     AT.proexp[3] = -1;
01546     AT.proexp[4] = 1;
01547     for ( i = 5; i < SUBEXPSIZE+1; i++ ) AT.proexp[i] = 0;
01548     AT.proexp[SUBEXPSIZE+1] = 1;
01549     AT.proexp[SUBEXPSIZE+2] = 1;
01550     AT.proexp[SUBEXPSIZE+3] = 3;
01551     AT.proexp[SUBEXPSIZE+4] = 0;
01552     AT.dummysubexp[0] = SUBEXPRESSION;
01553     AT.dummysubexp[1] = SUBEXPSIZE+4;
01554     for ( i = 2; i < SUBEXPSIZE; i++ ) AT.dummysubexp[i] = 0;
01555     AT.dummysubexp[SUBEXPSIZE] = WILDDUMMY;
01556     AT.dummysubexp[SUBEXPSIZE+1] = 4;
01557     AT.dummysubexp[SUBEXPSIZE+2] = 0;
01558     AT.dummysubexp[SUBEXPSIZE+3] = 0;
01559
01560     AT.inprimelist = -1;
01561     AT.sizeprimelist = 0;
01562     AT.primelist = 0;
01563     AT.LeaveNegative = 0;
01564     AT.TrimPower = 0;
01565     AN.SplitScratch = 0;
01566     AN.SplitScratchSize = AN.InScratch = 0;
01567     AN.SplitScratch1 = 0;
01568     AN.SplitScratchSize1 = AN.InScratch1 = 0;
01569     AN.idfunctionflag = 0;
01570     #endif
01571     AO.OutputLine = AO.OutFill = BufferForOutput;
01572     AO.FactorMode = 0;
01573     C->Pointer = C->Buffer;
01574
01575     AP.PreOut = 0;
01576     AP.ComChar = AP.cComChar;
01577     AC.cbufnum = AM.rbufnum; /* Select the default compiler buffer */
01578     AC.HideLevel = 0;
01579     AP.PreAssignFlag = 0;
01580 }
01581
01582 /*
01583     #] IniVars :
01584     #[ Signal handlers :
01585 */
01586 /*[28apr2004 mt]:*/
01587 #ifdef TRAP_SIGNALS
01588
01589 static int exitInProgress = 0;
01590 static int trappedTerminate = 0;
01591
01592 /*INTSIGHANDLER : some systems require a signal handler to return an integer,
01593    so define the macro INTSIGHANDLER if compiler fails:*/
01594 #ifdef INTSIGHANDLER
01595 static int onErrSig(int i)
01596 #else
01597 static void onErrSig(int i)
01598 #endif
01599 {
01600     if (exitInProgress){
01601         signal(i,SIG_DFL);/* Use default behaviour*/
01602         raise (i);/*reproduce trapped signal*/

```

```

01603 #ifdef INTSIGHANDLER
01604     return(i);
01605 #else
01606     return;
01607 #endif
01608 }
01609     trappedTerminate = 1;
01610     /*[13jul2005 mt]*/ /*TerminateImpl(-1) on signal is here:*/
01611     Terminate(-1);
01612 }
01613
01614 #ifdef INTSIGHANDLER
01615 static void setNewSig(int i, int (*handler)(int))
01616 #else
01617 static void setNewSig(int i, void (*handler)(int))
01618 #endif
01619 {
01620     if(! (i<NSIG) )/* Invalid signal -- see comments in the file */
01621         return;
01622     if ( signal(i,SIG_IGN) !=SIG_IGN)
01623         /* if compiler fails here, try to define INTSIGHANDLER:*/
01624         signal(i,handler);
01625 }
01626
01627 void setSignalHandlers(void)
01628 {
01629     /* Reset various unrecoverable error signals:*/
01630     setNewSig(SIGSEGV,onErrSig);
01631     setNewSig(SIGFPE,onErrSig);
01632     setNewSig(SIGILL,onErrSig);
01633     setNewSig(SIGEMT,onErrSig);
01634     setNewSig(SIGSYS,onErrSig);
01635     setNewSig(SIGPIPE,onErrSig);
01636     setNewSig(SIGLOST,onErrSig);
01637     setNewSig(SIGXCPU,onErrSig);
01638     setNewSig(SIGXFSZ,onErrSig);
01639
01640     /* Reset interrupt signals:*/
01641     setNewSig(SIGTERM,onErrSig);
01642     setNewSig(SIGINT,onErrSig);
01643     setNewSig(SIGQUIT,onErrSig);
01644     setNewSig(SIGHUP,onErrSig);
01645     setNewSig(SIGALRM,onErrSig);
01646     setNewSig(SIGVTALRM,onErrSig);
01647     /* setNewSig(SIGPROF,onErrSig); */ /* Why did Tentukov forbid profilers?? */
01648 }
01649
01650 #endif
01651 /*:[28apr2004 mt]*/
01652 /*
01653     #] Signal handlers :
01654     #[ main :
01655 */
01656
01657 #ifdef WITHPTHREADS
01658 ALLPRIVATES *ABdummy[10];
01659 #endif
01660
01661 int main(int argc, char **argv)
01662 {
01663     int retval;
01664     bzero((void *)(&A),sizeof(A)); /* make sure A is initialized at zero */
01665     iniTools();
01666     #ifdef TRAP_SIGNALS
01667         setSignalHandlers();
01668     #endif
01669     #ifdef WINDOWS
01670         _setmode(_fileno(stdout),O_BINARY);
01671     #endif
01672
01673     #ifdef WITHPTHREADS
01674         AB = ABdummy;
01675         StartHandleLock();
01676         BeginIdentities();
01677     #else
01678         AM.SumTime = TimeCPU(0);
01679         TimeWallClock(0);
01680     #endif
01681
01682     #ifdef WITHMPI
01683         if ( PF_Init(&argc,&argv) ) exit(-1);
01684     #endif
01685
01686     StartFiles();
01687     StartVariables();
01688     #ifdef WITHMPI
01689     /*

```

```

01690      * Here MesPrint() is ready. We turn on AS.printflag to print possible
01691      * errors occurring on slaves in the initialization. With AS.printflag = -1
01692      * MesPrint() does not use the synchronized output. This may lead broken
01693      * texts in the output somewhat, but it is safer to implement in this way
01694      * for the situation in which some of MesPrint() calls use MLOCK()-MUNLOCK()
01695      * and some do not. In future if we set AS.printflag = 1 and modify the
01696      * source code such that all MesPrint() calls are sandwiched by MLOCK()-
01697      * MUNLOCK(), we need also to modify the code for the master to catch
01698      * messages corresponding to MUNLOCK() calls at some point.
01699      *
01700      * AS.printflag will be set to 0 in IniVars() to prevent slaves from
01701      * printing redundant errors in the preprocessor and compiler (e.g., syntax
01702      * errors).
01703      */
01704      AS.printflag = -1;
01705 #endif
01706
01707      if ( ( retval = DoTail(argc, (UBYTE **)argv) ) != 0 ) {
01708          if ( retval > 0 ) Terminate(0);
01709          else Terminate(-1);
01710      }
01711      if ( DoSetups() ) Terminate(-2);
01712 #ifdef WITHMPI
01713      /* It is messy if all errors in OpenInput() on slaves are printed. */
01714      AS.printflag = 0;
01715 #endif
01716      if ( OpenInput() ) Terminate(-3);
01717 #ifdef WITHMPI
01718      AS.printflag = -1;
01719 #endif
01720      if ( TryEnvironment() ) Terminate(-2);
01721      if ( TryFileSetups() ) Terminate(-2);
01722      if ( MakeSetupAllocs() ) Terminate(-2);
01723      StartMore();
01724      InitRecovery();
01725      CheckRecoveryFile();
01726      if ( AM.totalnumberofthreads == 0 ) AM.totalnumberofthreads = 1;
01727      AS.MultiThreaded = 0;
01728 #ifdef WITHPTHREADS
01729      if ( AM.totalnumberofthreads > 1 ) AS.MultiThreaded = 1;
01730      ReserveTempFiles(1);
01731      StartAllThreads(AM.totalnumberofthreads);
01732      IniFbufs();
01733 #else
01734      ReserveTempFiles(0);
01735      IniFbuffer(AT.fbufnum);
01736 #endif
01737      if ( !AM.FromStdin ) PrintHeader(1);
01738      IniVars();
01739      Globalize(1);
01740 #ifdef WITH_ALARM
01741      if ( AM.TimeLimit > 0 ) alarm(AM.TimeLimit);
01742 #endif
01743      TimeCPU(0);
01744      TimeChildren(0);
01745      TimeWallClock(0);
01746      PreProcessor();
01747      Terminate(0);
01748      return(0);
01749 }
01750 /*
01751      #] main :
01752      #[ CleanUp :
01753
01754      if par < 0 we have to keep the storage file.
01755      when par > 0 we ran into a .clear statement.
01756      In that case we keep the zero level input and the log file.
01757
01758 */
01759
01760 void CleanUp(WORD par)
01761 {
01762     GETIDENTITY
01763     int i;
01764
01765     if ( FG.fname ) {
01766 #ifdef WITHPTHREADS
01767         if ( B ) {
01768 #endif
01769             CleanUpSort(0);
01770             for ( i = 0; i < 3; i++ ) {
01771                 if ( AR.Fscr[i].handle >= 0 ) {
01772                     if ( AR.Fscr[i].name ) {
01773 /*
01774                         If there are more threads referring to the same file
01775                         only the one with the name is the owner of the file.
01776 */

```

```

01777             CloseFile(AR.Fscr[i].handle);
01778             remove(AR.Fscr[i].name);
01779         }
01780         AR.Fscr[i].handle = - 1;
01781         AR.Fscr[i].POfill = 0;
01782     }
01783 }
01784 #ifdef WITHPTHREADS
01785 }
01786 #endif
01787 if ( par > 0 ) {
01788 /*
01789     Close all input levels above the lowest?
01790 */
01791     }
01792     if ( AC.StoreHandle >= 0 && par <= 0 ) {
01793 #ifdef TRAP_SIGNALS
01794     if ( trappedTerminate ) { /* We don't throw .str if it has contents */
01795         POSITION pos;
01796         PUTZERO(pos);
01797         SeekFile(AC.StoreHandle,&pos,SEEK_END);
01798         if ( ISNOTZEROPOS(pos) ) {
01799             CloseFile(AC.StoreHandle);
01800             goto dontremove;
01801         }
01802     }
01803     CloseFile(AC.StoreHandle);
01804 #ifdef WITHPTHREADS
01805     if ( B )
01806 #endif
01807     if ( par >= 0 || AR.StoreData.Handle < 0 || AM.ClearStore ) {
01808         remove(FG.fname);
01809     }
01810 dontremove:;
01811 #else
01812     CloseFile(AC.StoreHandle);
01813 #ifdef WITHPTHREADS
01814     if ( B )
01815 #endif
01816     if ( par >= 0 || AR.StoreData.Handle < 0 || AM.ClearStore > 0 ) {
01817         remove(FG.fname);
01818     }
01819 #endif
01820     }
01821 }
01822 ClearSpectators(CLEARMODULE);
01823 /*
01824     Remove recovery file on exit if everything went well
01825 */
01826 if ( par == 0 ) {
01827     DeleteRecoveryFile();
01828 }
01829 /*
01830     Now the final message concerning the total time
01831 */
01832 if ( AC.LogHandle >= 0 && par <= 0 ) {
01833     WORD lh = AC.LogHandle;
01834     AC.LogHandle = -1;
01835 #ifdef WITHMPI
01836     if ( PF.me == MASTER ) /* Only the master opened the real file. */
01837 #endif
01838     CloseFile(lh);
01839 }
01840 }
01841
01842 /*
01843     #] CleanUp :
01844     #[ TerminateImpl :
01845 */
01846
01847 static int firstterminate = 1;
01848
01849 void TerminateImpl(int errorcode, const char* file, int line, const char* function)
01850 {
01851     if ( errorcode && firstterminate ) {
01852         firstterminate = 0;
01853
01854         MLOCK(ErrorMessageLock);
01855 #ifdef WITHPTHREADS
01856         MesPrint("Program terminating in thread %w at %");
01857 #elif defined(WITHMPI)
01858         MesPrint("Program terminating in process %w at %");
01859 #else
01860         MesPrint("Program terminating at %");
01861 #endif
01862         MesPrint("Terminate called from %s:%d (%s)", file, line, function);
01863     }

```



```

01864         if ( AC.PrintBacktraceFlag ) {
01865 #ifdef ENABLE_BACKTRACE
01866         void *stack[64];
01867         int stacksize, stop = 0;
01868         stacksize = backtrace(stack, sizeof(stack)/sizeof(stack[0]));
01869
01870         /* First check whether eu-addr2line is available */
01871         if ( !system("command -v eu-addr2line > /dev/null 2>&1") ) {
01872             MesPrint("Backtrace:");
01873             for (int i = 0; i < stacksize && !stop; i++) {
01874                 FILE *fp;
01875                 char cmd[512];
01876                 // Leave an initial space
01877                 cmd[0] = ' ';
01878                 MesPrint("###%2d:%", i);
01879                 snprintf(cmd+1, sizeof(cmd)-1, "eu-addr2line -s --pretty-print -f -i '%p'
--pid=%d\n", stack[i], getpid());
01880                 fp = popen(cmd+1, "r");
01881                 while ( fgets(cmd+1, sizeof(cmd)-1, fp) != NULL ) {
01882                     MesPrint("%s", cmd);
01883                     /* Don't show functions lower than "main" (or thread equivalent) */
01884                     if ( strstr(cmd, " main ") || strstr(cmd, " RunThread ") || strstr(cmd, "
RunSortBot ") ) {
01885                         stop = 1;
01886                     }
01887                 }
01888                 pclose(fp);
01889             }
01890         }
01891 #ifdef LINUX
01892         else if ( !system("command -v addr2line > /dev/null 2>&1") ) {
01893             /* Get the executable path. */
01894             char exe_path[PATH_MAX];
01895             {
01896                 ssize_t len = readlink("/proc/self/exe", exe_path, sizeof(exe_path) - 1);
01897                 if ( len != -1 ) {
01898                     exe_path[len] = '\0';
01899                 }
01900                 else {
01901                     goto backtrace_fallback;
01902                 }
01903             }
01904             /* Assume PIE binary and get the base address. */
01905             uintptr_t base_address = 0;
01906             {
01907                 char line[256];
01908                 FILE *maps = fopen("/proc/self/maps", "r");
01909                 if ( !maps ) {
01910                     goto backtrace_fallback;
01911                 }
01912                 /* See the format used by nommu_region_show() in fs/proc/nommu.c of the Linux
source. */
01913                 if ( fgets(line, sizeof(line), maps) ) {
01914                     sscanf(line, "%* SCNxPTR -", &base_address);
01915                 }
01916                 else {
01917                     fclose(maps);
01918                     goto backtrace_fallback;
01919                 }
01920                 fclose(maps);
01921             }
01922             char **strings;
01923             strings = backtrace_symbols(stack, stacksize);
01924             MesPrint("Backtrace:");
01925             for ( int i = 0; i < stacksize && !stop; i++ ) {
01926                 FILE *fp;
01927                 char cmd[PATH_MAX + 512];
01928                 // Leave an initial space
01929                 cmd[0] = ' ';
01930                 uintptr_t addr = (uintptr_t)stack[i] - base_address;
01931                 MesPrint("###%2d:%", i);
01932                 snprintf(cmd+1, sizeof(cmd)-1, "addr2line -e \"%s\" -i -p -s -f -C 0x% PRIxPTR,
exe_path, addr);
01933                 fp = popen(cmd+1, "r");
01934                 while ( fgets(cmd+1, sizeof(cmd)-1, fp) != NULL ) {
01935                     MesPrint("%s", cmd);
01936                     /* Don't show functions lower than "main" */
01937                     if ( strstr(cmd, " main ") || strstr(cmd, " RunThread ") || strstr(cmd, "
RunSortBot ") ) {
01938                         stop = 1;
01939                     }
01940                 }
01941                 pclose(fp);
01942             }
01943             free(strings);
01944         }
01945 #endif

```

```

01946         else {
01947             /* eu-addr2line not found */
01948 #ifdef LINUX
01949 backtrace_fallback: ;
01950 #endif
01951             char **strings;
01952             strings = backtrace_symbols(stack, stacksize);
01953             MesPrint("Backtrace:");
01954             for ( int i = 0; i < stacksize && !stop; i++ ) {
01955                 char *p = strings[i];
01956                 while ( *p && *p != ' (' ) p++;
01957                 MesPrint("%%2d: %s\n", i, p);
01958                 /* Don't show functions lower than "main" (or thread equivalent) */
01959                 if ( strstr(p, "(main)") || strstr(p, "(RunThread)") || strstr(p, "(RunSortBot)")
01960 ) {
01961                     stop = 1;
01962                 }
01963 #ifdef LINUX
01964                 MesPrint("Please install addr2line or eu-addr2line for readable stack information.");
01965 #else
01966                 MesPrint("Please install eu-addr2line for readable stack information.");
01967 #endif
01968                 free(strings);
01969             }
01970 #else
01971             MesPrint("FORM compiled without backtrace support.");
01972 #endif
01973     } /* if ( AC.PrintBacktraceFlag ) { */
01974
01975     MUNLOCK(ErrorMessageLock);
01976
01977     Crash();
01978 }
01979 #ifdef TRAP_SIGNALS
01980     exitInProgress=1;
01981 #endif
01982 #ifdef WITHEXTERNALCHANNEL
01983 /*
01984     This function can be called from the error handler, so it is better to
01985     clean up all started processes before any activity:
01986 */
01987     closeAllExternalChannels();
01988     AX.currentExternalChannel=0;
01989     /*[08may2006 mt]:*/
01990     AX.killSignal=SIGKILL;
01991     AX.killWholeGroup=1;
01992     AX.daemonize=1;
01993     /*:[08may2006 mt]*/
01994     if(AX.currentPrompt){
01995         M_free(AX.currentPrompt,"external channel prompt");
01996         AX.currentPrompt=0;
01997     }
01998     /*[08may2006 mt]:*/
01999     if(AX.shellname){
02000         M_free(AX.shellname,"external channel shellname");
02001         AX.shellname=0;
02002     }
02003     if(AX.stdername){
02004         M_free(AX.stdername,"external channel stderrname");
02005         AX.stdername=0;
02006     }
02007     /*:[08may2006 mt]*/
02008 #endif
02009 #ifdef WITHPTHREADS
02010     if ( !WhoAmI() && !errorcode ) {
02011         TerminateAllThreads();
02012     }
02013 #endif
02014     if ( AC.FinalStats ) {
02015         if ( AM.PrintTotalSize ) {
02016             MesPrint("Max. space for expressions: %19p bytes",&(AS.MaxExprSize));
02017         }
02018         PrintRunningTime();
02019     }
02020 #ifdef WITHMPI
02021     if ( AM.HoldFlag && PF.me == MASTER ) {
02022         WriteFile(AM.Stdout,(UBYTE *)("Hit any key "),12);
02023         PF_FlushStdOutBuffer();
02024         getchar();
02025     }
02026 #else
02027     if ( AM.HoldFlag ) {
02028         WriteFile(AM.Stdout,(UBYTE *)("Hit any key "),12);
02029         getchar();
02030     }
02031 #endif

```

```

02032 #ifdef WITHMPI
02033     PF_Terminate(errorcode);
02034 #endif
02035 /*
02036     We are about to terminate the program. If we are using flint, call the cleanup function.
02037     This keeps valgrind happy.
02038 */
02039 #ifdef WITHFLINT
02040     flint_final_cleanup_master();
02041 #endif
02042     CleanUp(errorcode);
02043     M_print();
02044 #ifdef VMS
02045     P_term(errorcode? 0: 1);
02046 #else
02047     P_term(errorcode);
02048 #endif
02049 }
02050
02051 /*
02052     #] TerminateImpl :
02053     #[ PrintDeprecation :
02054 */
02055
02062 void PrintDeprecation(const char *feature, const char *issue) {
02063 #ifdef WITHMPI
02064     if ( PF.me != MASTER ) return;
02065 #endif
02066     if ( AM.IgnoreDeprecation ) return;
02067
02068     UBYTE *e = (UBYTE *)getenv("FORM_IGNORE_DEPRECATION");
02069     if ( e && e[0] && StrCmp(e, (UBYTE *)"0") && StrICmp(e, (UBYTE *)"false") && StrICmp(e, (UBYTE
*)"no") ) return;
02070
02071     MesPrint("DeprecationWarning: We are considering deprecating %s.", feature);
02072     MesPrint("If you want this support to continue, leave a comment at:");
02073     MesPrint("");
02074     MesPrint("    https://github.com/vermaseren/form/%s", issue);
02075     MesPrint("");
02076     MesPrint("Otherwise, it will be discontinued in the future.");
02077     MesPrint("To suppress this warning, use the -ignore-deprecation command line option or");
02078     MesPrint("set the environment variable FORM_IGNORE_DEPRECATION=1.");
02079 }
02080
02081 /*
02082     #] PrintDeprecation :
02083     #[ PrintRunningTime :
02084 */
02085
02086 void PrintRunningTime(void)
02087 {
02088 #if (defined(WITHPTHREADS) && (defined(WITHPOSIXCLOCK) || defined(WINDOWS))) || defined(WITHMPI)
02089     LONG mastertime;
02090     LONG workertime;
02091     LONG wallclocktime;
02092     LONG totaltime;
02093 #if defined(WITHPTHREADS)
02094     if ( AB[0] != 0 ) {
02095         workertime = GetWorkerTimes();
02096     }
02097     else
02098         workertime = PF_GetSlaveTimes(); /* must be called on all processors */
02099 #endif
02100     mastertime = AM.SumTime + TimeCPU(1);
02101     wallclocktime = TimeWallClock(1);
02102     totaltime = mastertime+workertime;
02103     if ( !AM.silent ) {
02104         MesPrint("    %1.%2i sec + %1.%2i sec: %1.%2i sec out of %1.%2i sec",
02105             mastertime/1000, (WORD)((mastertime%1000)/10),
02106             workertime/1000, (WORD)((workertime%1000)/10),
02107             totaltime/1000, (WORD)((totaltime%1000)/10),
02108             wallclocktime/100, (WORD)(wallclocktime%100));
02109     }
02110 }
02111 #else
02112     LONG mastertime = AM.SumTime + TimeCPU(1);
02113     LONG wallclocktime = TimeWallClock(1);
02114     if ( !AM.silent ) {
02115         MesPrint("    %1.%2i sec out of %1.%2i sec",
02116             mastertime/1000, (WORD)((mastertime%1000)/10),
02117             wallclocktime/100, (WORD)(wallclocktime%100));
02118     }
02119 #endif
02120 }
02121
02122 /*
02123     #] PrintRunningTime :

```

```

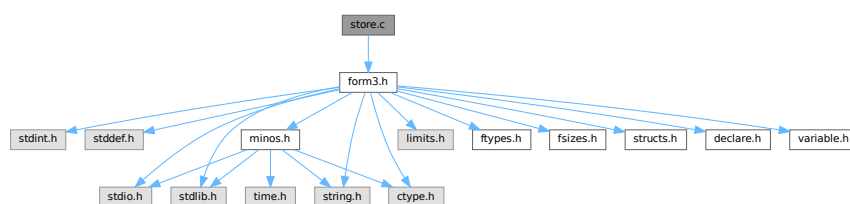
02124         #] GetRunningTime :
02125     */
02126
02127 LONG GetRunningTime(void)
02128 {
02129     #if defined(WITHPTHREADS) && (defined(WITHPOSIXCLOCK) || defined(WINDOWS))
02130         LONG mastertime;
02131         if ( AB[0] != 0 ) {
02132             /*
02133             #if ( defined(APPLE64) || defined(APPLE32) )
02134                 mastertime = AM.SumTime + TimeCPU(1);
02135                 return(mastertime);
02136             #else
02137             /*
02138                 LONG workertime = GetWorkerTimes();
02139                 mastertime = AM.SumTime + TimeCPU(1);
02140                 return(mastertime+workertime);
02141             /*
02142             #endif
02143             */
02144         }
02145         else {
02146             return(AM.SumTime + TimeCPU(1));
02147         }
02148     #elif defined(WITHMPI)
02149         LONG mastertime, t = 0;
02150         LONG workertime = PF_GetSlaveTimes(); /* must be called on all processors */
02151         if ( PF.me == MASTER ) {
02152             mastertime = AM.SumTime + TimeCPU(1);
02153             t = mastertime + workertime;
02154         }
02155         return PF_BroadcastNumber(t); /* must be called on all processors */
02156     #else
02157         return(AM.SumTime + TimeCPU(1));
02158     #endif
02159 }
02160
02161 /*
02162     #] GetRunningTime :
02163 */

```

## 4.134 store.c File Reference

#include "form3.h"

Include dependency graph for store.c:



### Macros

- #define [SAVEREVISION](#) 0x02

### Functions

- void [OpenTemp](#) (void)
- void [SeekScratch](#) (FILEHANDLE \*fi, POSITION \*pos)
- void [SetEndScratch](#) (FILEHANDLE \*f, POSITION \*position)

- void [SetEndHScratch](#) ([FILEHANDLE](#) \*f, [POSITION](#) \*position)
- void [SetScratch](#) ([FILEHANDLE](#) \*f, [POSITION](#) \*position)
- int [RevertScratch](#) (void)
- int [ResetScratch](#) (void)
- int [ReadFromScratch](#) ([FILEHANDLE](#) \*fi, [POSITION](#) \*pos, [UBYTE](#) \*buffer, [POSITION](#) \*length)
- int [AddToScratch](#) ([FILEHANDLE](#) \*fi, [POSITION](#) \*pos, [UBYTE](#) \*buffer, [POSITION](#) \*length, int withflush)
- int [CoSave](#) ([UBYTE](#) \*inp)
- int [CoLoad](#) ([UBYTE](#) \*inp)
- int [DeleteStore](#) ([WORD](#) par)
- int [PutInStore](#) ([INDEXENTRY](#) \*ind, [WORD](#) num)
- [WORD](#) [GetTerm](#) ([PHEAD](#) [WORD](#) \*term)
- [WORD](#) [GetOneTerm](#) ([PHEAD](#) [WORD](#) \*term, [FILEHANDLE](#) \*fi, [POSITION](#) \*pos, int par)
- [WORD](#) [GetMoreTerms](#) ([WORD](#) \*term)
- int [GetMoreFromMem](#) ([WORD](#) \*term, [WORD](#) \*\*tpoin)
- [WORD](#) [GetFromStore](#) ([WORD](#) \*to, [POSITION](#) \*position, [RENUMBER](#) renumber, [WORD](#) \*InCompState, [WORD](#) nexpr)
- void [DetVars](#) ([WORD](#) \*term, [WORD](#) par)
- int [ToStorage](#) ([EXPRESSIONS](#) e, [POSITION](#) \*length)
- [INDEXENTRY](#) \* [NextFileIndex](#) ([POSITION](#) \*indexpos)
- int [SetFileIndex](#) (void)
- int [VarStore](#) ([UBYTE](#) \*s, [WORD](#) n, [WORD](#) name, [WORD](#) namesize)
- int [TermRenumber](#) ([WORD](#) \*term, [RENUMBER](#) renumber, [WORD](#) nexpr)
- [WORD](#) [FindrNumber](#) ([WORD](#) n, [VARRENUM](#) \*v)
- [INDEXENTRY](#) \* [FindInIndex](#) ([WORD](#) expr, [FILEDATA](#) \*f, [WORD](#) par, [WORD](#) mode)
- [RENUMBER](#) [GetTable](#) ([WORD](#) expr, [POSITION](#) \*position, [WORD](#) mode)
- int [CopyExpression](#) ([FILEHANDLE](#) \*from, [FILEHANDLE](#) \*to)
- [LONG](#) [WriteStoreHeader](#) ([WORD](#) handle)
- int [ReadSaveHeader](#) (void)
- [LONG](#) [ReadSaveIndex](#) ([FILEINDEX](#) \*fileind)
- [LONG](#) [ReadSaveVariables](#) ([UBYTE](#) \*buffer, [UBYTE](#) \*top, [LONG](#) \*size, [LONG](#) \*outsize, [INDEXENTRY](#) \*ind, [LONG](#) \*stage)
- [UBYTE](#) \* [ReadSaveTerm32](#) ([UBYTE](#) \*bin, [UBYTE](#) \*binend, [UBYTE](#) \*\*bout, [UBYTE](#) \*boutend, [UBYTE](#) \*top, int terminbuf)
- [LONG](#) [ReadSaveExpression](#) ([UBYTE](#) \*buffer, [UBYTE](#) \*top, [LONG](#) \*size, [LONG](#) \*outsize)

### 4.134.1 Detailed Description

Contains all functions that deal with store-files and the system independent save-files.

Definition in file [store.c](#).

### 4.134.2 Macro Definition Documentation

#### 4.134.2.1 SAVEREVISION

```
#define SAVEREVISION 0x02
```

Definition at line 3953 of file [store.c](#).

### 4.134.3 Function Documentation

#### 4.134.3.1 OpenTemp()

```
void OpenTemp (
    void )
```

Definition at line 48 of file [store.c](#).

#### 4.134.3.2 SeekScratch()

```
void SeekScratch (
    FILEHANDLE * fi,
    POSITION * pos )
```

Definition at line 63 of file [store.c](#).

#### 4.134.3.3 SetEndScratch()

```
void SetEndScratch (
    FILEHANDLE * f,
    POSITION * position )
```

Definition at line 74 of file [store.c](#).

#### 4.134.3.4 SetEndHScratch()

```
void SetEndHScratch (
    FILEHANDLE * f,
    POSITION * position )
```

Definition at line 88 of file [store.c](#).

#### 4.134.3.5 SetScratch()

```
void SetScratch (
    FILEHANDLE * f,
    POSITION * position )
```

Definition at line 114 of file [store.c](#).

#### 4.134.3.6 RevertScratch()

```
int RevertScratch (
    void )
```

Definition at line 189 of file [store.c](#).

#### 4.134.3.7 ResetScratch()

```
int ResetScratch (
    void )
```

Definition at line 234 of file [store.c](#).

#### 4.134.3.8 ReadFromScratch()

```
int ReadFromScratch (
    FILEHANDLE * fi,
    POSITION * pos,
    UBYTE * buffer,
    POSITION * length )
```

Definition at line 272 of file [store.c](#).

#### 4.134.3.9 AddToScratch()

```
int AddToScratch (
    FILEHANDLE * fi,
    POSITION * pos,
    UBYTE * buffer,
    POSITION * length,
    int withflush )
```

Definition at line 299 of file [store.c](#).

#### 4.134.3.10 CoSave()

```
int CoSave (
    UBYTE * inp )
```

Definition at line 362 of file [store.c](#).

#### 4.134.3.11 CoLoad()

```
int CoLoad (
    UBYTE * inp )
```

Definition at line 572 of file [store.c](#).

#### 4.134.3.12 DeleteStore()

```
int DeleteStore (
    WORD par )
```

Definition at line 772 of file [store.c](#).

#### 4.134.3.13 PutInStore()

```
int PutInStore (
    INDEXENTRY * ind,
    WORD num )
```

Definition at line 873 of file [store.c](#).

#### 4.134.3.14 GetTerm()

```
WORD GetTerm (
    PHEAD WORD * term )
```

Definition at line 992 of file [store.c](#).

#### 4.134.3.15 GetOneTerm()

```
WORD GetOneTerm (
    PHEAD WORD * term,
    FILEHANDLE * fi,
    POSITION * pos,
    int par )
```

Definition at line 1267 of file [store.c](#).

#### 4.134.3.16 GetMoreTerms()

```
WORD GetMoreTerms (
    WORD * term )
```

Definition at line 1425 of file [store.c](#).

#### 4.134.3.17 GetMoreFromMem()

```
int GetMoreFromMem (
    WORD * term,
    WORD ** tpoint )
```

Definition at line 1521 of file [store.c](#).

#### 4.134.3.18 GetFromStore()

```
WORD GetFromStore (
    WORD * to,
    POSITION * position,
    RENUMBER renumber,
    WORD * InCompState,
    WORD nexpr )
```

Definition at line 1628 of file [store.c](#).



#### 4.134.3.19 DetVars()

```
void DetVars (
    WORD * term,
    WORD par )
```

Definition at line 1829 of file [store.c](#).

#### 4.134.3.20 ToStorage()

```
int ToStorage (
    EXPRESSIONS e,
    POSITION * length )
```

Definition at line 2016 of file [store.c](#).

#### 4.134.3.21 NextFileIndex()

```
INDEXENTRY * NextFileIndex (
    POSITION * indexpos )
```

Definition at line 2257 of file [store.c](#).

#### 4.134.3.22 SetFileIndex()

```
int SetFileIndex (
    void )
```

Reads the next file index and puts it into AR.StoreData.Index. TODO

##### Returns

= 0 everything okay, != 0 an error occurred

Definition at line 2320 of file [store.c](#).

References [WriteStoreHeader\(\)](#).

Here is the call graph for this function:



#### 4.134.3.23 VarStore()

```
int VarStore (
    UBYTE * s,
    WORD n,
    WORD name,
    WORD namesize )
```

Definition at line 2369 of file [store.c](#).

#### 4.134.3.24 TermRenumber()

```
int TermRenumber (
    WORD * term,
    RENUMBER renumber,
    WORD nexpr )
```

```
!! WORD *memterm=term; static LONG ctrap=0; !!!
```

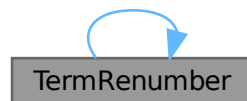
```
!! ctrap++; !!!
```

Definition at line 2427 of file [store.c](#).

References [ReNuMbEr::func](#), [ReNuMbEr::funnum](#), [ReNuMbEr::indi](#), [ReNuMbEr::indnum](#), [ReNuMbEr::symb](#), [ReNuMbEr::symnum](#), [TermRenumber\(\)](#), [ReNuMbEr::vecnum](#), and [ReNuMbEr::vect](#).

Referenced by [TermRenumber\(\)](#).

Here is the call graph for this function:



#### 4.134.3.25 FindrNumber()

```
WORD FindrNumber (
    WORD n,
    VARENUM * v )
```

Definition at line 2587 of file [store.c](#).

#### 4.134.3.26 FindInIndex()

```
INDEXENTRY * FindInIndex (
    WORD expr,
    FILEDATA * f,
    WORD par,
    WORD mode )
```

Definition at line 2664 of file [store.c](#).

#### 4.134.3.27 GetTable()

```
RENUMBER GetTable (
    WORD expr,
    POSITION * position,
    WORD mode )
```

Definition at line 2825 of file [store.c](#).

#### 4.134.3.28 CopyExpression()

```
int CopyExpression (
    FILEHANDLE * from,
    FILEHANDLE * to )
```

Definition at line 3307 of file [store.c](#).

#### 4.134.3.29 WriteStoreHeader()

```
LONG WriteStoreHeader (
    WORD handle )
```

Writes header with information about system architecture and FORM revision to an open store file.

Called by [SetFileIndex\(\)](#).

##### Parameters

|               |                                                     |
|---------------|-----------------------------------------------------|
| <i>handle</i> | specifies open file to which header will be written |
|---------------|-----------------------------------------------------|

##### Returns

= 0 everything okay, != 0 an error occurred

Definition at line 3964 of file [store.c](#).

References [STOREHEADER::endianness](#), [STOREHEADER::lenLONG](#), [STOREHEADER::lenPOINTER](#), [STOREHEADER::lenPOS](#), [STOREHEADER::lenWORD](#), [STOREHEADER::maxpower](#), [STOREHEADER::sFun](#), [STOREHEADER::sInd](#), [STOREHEADER::sSym](#), [STOREHEADER::sVec](#), and [STOREHEADER::wildoffset](#).

Referenced by [SetFileIndex\(\)](#).

**4.134.3.30 ReadSaveHeader()**

```
int ReadSaveHeader (
    void )
```

Reads the header in the save file and sets function pointers and flags according to the information found there. Must be called before any other ReadSave... function.

Currently works only for the exchange between 32bit and 64bit machines (WORD size must be 2 or 4 bytes)!

It is called by CoLoad().

**Returns**

= 0 everything okay, != 0 an error occurred

Definition at line [4057](#) of file [store.c](#).

**4.134.3.31 ReadSaveIndex()**

```
LONG ReadSaveIndex (
    FILEINDEX * fileind )
```

Reads a FILEINDEX from the open save file specified by AO.SaveData.Handle. Translations for adjusting endianness and data sizes are done if necessary.

Depends on the assumption that sizeof(FILEINDEX) is the same everywhere. If FILEINDEX or INDEXENTRY change, then this functions has to be adjusted.

Called by CoLoad() and FindInIndex().

**Parameters**

|                |                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------|
| <i>fileind</i> | contains the read FILEINDEX after successful return. must point to allocated, big enough memory. |
|----------------|--------------------------------------------------------------------------------------------------|

**Returns**

= 0 everything okay, != 0 an error occurred

Definition at line [4175](#) of file [store.c](#).

References [INFILEINDEX](#), [FiLeInDeX::next](#), [FiLeInDeX::number](#), and [FuNcTiOn::number](#).

**4.134.3.32 ReadSaveVariables()**

```
LONG ReadSaveVariables (
    UBYTE * buffer,
    UBYTE * top,
    LONG * size,
    LONG * outsize,
```

```
INDEXENTRY * ind,
LONG * stage )
```

Reads the variables from the open file specified by AO.SaveData.Handle. It reads the *\*size* bytes and writes them to the *\*buffer*. It is called by PutInStore().

If translation is necessary, the data might shrink or grow in size, then *\*size* is adjusted so that the reading and writing fits into the memory from the buffer to the top. The actual number of read bytes is returned in *\*size*, the number of written bytes is returned in *\*outsize*.

If the *\*size* is smaller than the actual size of the variables, this function will be called several times and needs to remember the current position in the variable structure. The parameter *stage* does this job. When [ReadSaveVariables\(\)](#) is called for the first time, this parameter should have the value -1.

The parameter *ind* is used to get the number of variables.

#### Parameters

|                |                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------|
| <i>buffer</i>  | read variables are written into this allocated memory                                                                      |
| <i>top</i>     | upper end of allocated memory                                                                                              |
| <i>size</i>    | number of bytes to read. might return a smaller number of read bytes if translation was necessary                          |
| <i>outsize</i> | if translation has be done, outsize contains the number of written bytes                                                   |
| <i>ind</i>     | pointer of INDEXENTRY for the current expression. read-only                                                                |
| <i>stage</i>   | should be -1 for the first call, will be increased by ReadSaveVariables to memorize the position in the variable structure |

#### Returns

= 0 everything okay, != 0 an error occurred

Definition at line [4361](#) of file [store.c](#).

References [InDeXeNtRy::nfunctions](#), [InDeXeNtRy::nindices](#), [InDeXeNtRy::nsymbols](#), [InDeXeNtRy::nvectors](#), and [FuNcTiOn::spec](#).

#### 4.134.3.33 ReadSaveTerm32()

```
UBYTE * ReadSaveTerm32 (
    UBYTE * bin,
    UBYTE * binend,
    UBYTE ** bout,
    UBYTE * boutend,
    UBYTE * top,
    int terminbuf )
```

Reads a single term from the given buffer at *bin* and write the translated term back to this buffer at *bout*.

[ReadSaveTerm32\(\)](#) is currently the only instantiation of a ReadSaveTerm-function. It only deals with data that already has the correct endianness and that is resized to 32bit words but without being renumbered or translated in any other way. It uses the compress buffer AR.CompressBuffer.

The function is reentrant in order to cope with nested function arguments. It is called by [ReadSaveExpression\(\)](#) and itself.

The *return value* indicates the position in the input buffer up to which the data has already been successfully processed. The parameter *bout* returns the corresponding position in the output buffer.

## Parameters

|                  |                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <i>bin</i>       | start of the input buffer                                                                                                       |
| <i>binend</i>    | end of the input buffer                                                                                                         |
| <i>bout</i>      | as input points to the beginning of the output buffer, as output points behind the already translated data in the output buffer |
| <i>boutend</i>   | end of already decompressed data in output buffer                                                                               |
| <i>top</i>       | end of output buffer                                                                                                            |
| <i>terminbuf</i> | flag whether decompressed data is already in the output buffer. used in recursive calls                                         |

## Returns

pointer to the next unprocessed data in the input buffer

Definition at line 4728 of file [store.c](#).

References [ReadSaveTerm32\(\)](#).

Referenced by [ReadSaveExpression\(\)](#), and [ReadSaveTerm32\(\)](#).

Here is the call graph for this function:



#### 4.134.3.34 ReadSaveExpression()

```

LONG ReadSaveExpression (
    UBYTE * buffer,
    UBYTE * top,
    LONG * size,
    LONG * outsize )
  
```

Reads an expression from the open file specified by `AO.SaveData.Handle`. The endianness flip and a resizing without renumbering is done in this function. Thereafter the buffer consists of chunks with a uniform maximal word size (32bit at the moment). The actual renumbering is then done by calling the function [ReadSaveTerm32\(\)](#). The result is returned in *buffer*.

If the translation at some point doesn't fit into the buffer anymore, the function returns and must be called again. In any case *size* returns the number of successfully read bytes, *outsize* returns the number of successfully written bytes, and the file will be positioned at the next byte after the successfully read data.

It is called by `PutInStore()`.

## Parameters

|                |                                                  |
|----------------|--------------------------------------------------|
| <i>buffer</i>  | output buffer, holds the (translated) expression |
| <i>top</i>     | end of buffer                                    |
| <i>size</i>    | number of read bytes                             |
| <i>outsize</i> | number of written bytes                          |

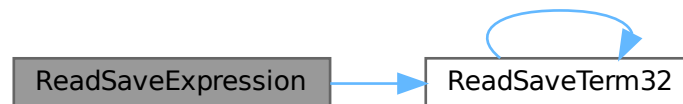
## Returns

= 0 everything okay, != 0 an error occurred

Definition at line 5138 of file [store.c](#).

References [ReadSaveTerm32\(\)](#).

Here is the call graph for this function:



## 4.135 store.c

[Go to the documentation of this file.](#)

```

00001
00006 /* #[ License : */
00007 /*
00008 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 *   When using this file you are requested to refer to the publication
00010 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 *   This is considered a matter of courtesy as the development was paid
00012 *   for by FOM the Dutch physics granting agency and we would like to
00013 *   be able to track its scientific use to convince FOM of its value
00014 *   for the community.
00015 *
00016 *   This file is part of FORM.
00017 *
00018 *   FORM is free software: you can redistribute it and/or modify it under the
00019 *   terms of the GNU General Public License as published by the Free Software
00020 *   Foundation, either version 3 of the License, or (at your option) any later
00021 *   version.
00022 *
00023 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 *   details.
00027 *
00028 *   You should have received a copy of the GNU General Public License along
00029 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #] License : */
00032 /*
00033 #define HIDEDEBUG
00034     #[ Includes : store.c
00035 */
00036

```

```

00037 #include "form3.h"
00038
00039 /*
00040 #] Includes :
00041 #[ StoreExpressions :
00042 #[ OpenTemp :
00043
00044     Opens the scratch files for the input -> output operations.
00045 */
00046
00047 void OpenTemp(void)
00048 {
00049     GETIDENTITY
00050     if ( AR.outfile->handle >= 0 ) {
00051         SeekFile(AR.outfile->handle, &(AR.outfile->filesize), SEEK_SET);
00052         AR.outfile->POposition = AR.outfile->filesize;
00053         AR.outfile->POfill = AR.outfile->PObuffer;
00054     }
00055 }
00056
00057 /*
00058 #] OpenTemp :
00059 #[ SeekScratch :
00060 */
00061
00062 void SeekScratch(FILEHANDLE *fi, POSITION *pos)
00063 {
00064     *pos = fi->POposition;
00065     ADDPOS(*pos, (TOLONG(fi->POfill)-TOLONG(fi->PObuffer)));
00066 }
00067
00068 /*
00069 #] SeekScratch :
00070 #[ SetEndScratch :
00071 */
00072
00073 void SetEndScratch(FILEHANDLE *f, POSITION *position)
00074 {
00075     if ( f->handle < 0 ) {
00076         SETBASEPOSITION(*position, (f->POfull-f->PObuffer)*sizeof(WORD));
00077     }
00078     else *position = f->filesize;
00079     SetScratch(f, position);
00080 }
00081
00082 /*
00083 #] SetEndScratch :
00084 #[ SetEndHScratch :
00085 */
00086
00087 void SetEndHScratch(FILEHANDLE *f, POSITION *position)
00088 {
00089     if ( f->handle < 0 ) {
00090         SETBASEPOSITION(*position, (f->POfull-f->PObuffer)*sizeof(WORD));
00091         f->POfill = f->POfull;
00092     }
00093     else {
00094         #ifdef HIDEDEBUG
00095             POSITION possize;
00096             PUTZERO(possize);
00097             SeekFile(f->handle, &possize, SEEK_END);
00098             MesPrint("SetEndHScratch: filesize(th) = %12p, filesize(ex) = %12p", &(f->filesize),
00099                     &(possize));
00100         #endif
00101         *position = f->filesize;
00102         f->POposition = f->filesize;
00103         f->POfill = f->POfull = f->PObuffer;
00104     }
00105     SetScratch(f, position);
00106 }
00107
00108 /*
00109 #] SetEndHScratch :
00110 #[ SetScratch :
00111 */
00112
00113 void SetScratch(FILEHANDLE *f, POSITION *position)
00114 {
00115     GETIDENTITY
00116     POSITION possize;
00117     LONG size, *whichInInBuf;
00118     if ( f == AR.hidefile ) whichInInBuf = &(AR.InHiBuf);
00119     else whichInInBuf = &(AR.InInBuf);
00120     #ifdef HIDEDEBUG
00121         if ( f == AR.hidefile ) MesPrint("In the hide file");
00122     else MesPrint("In the input file");
00123 
```



```

00124     MesPrint("SetScratch to position %15p",position);
00125     MesPrint("POposition = %15p, full = %1, fill = %1"
00126     , &(f->POposition), (f->POfull-f->PObuffer)*sizeof(WORD)
00127     , (f->POfill-f->PObuffer)*sizeof(WORD));
00128 #endif
00129     if ( ISLESSPOS(*position,f->POposition) ||
00130     ISGEPOSINC(*position,f->POposition, (f->POfull-f->PObuffer)*sizeof(WORD)) ) {
00131         if ( f->handle < 0 ) {
00132             if ( ISEQUALPOSINC(*position,f->POposition,
00133             (f->POfull-f->PObuffer)*sizeof(WORD)) ) goto endpos;
00134             MesPrint("Illegal position in SetScratch");
00135             Terminate(-1);
00136         }
00137         possize = *position;
00138         LOCK(AS.inputslock);
00139         SeekFile(f->handle,&possize,SEEK_SET);
00140         if ( ISNOTEQUALPOS(possize,*position) ) {
00141             UNLOCK(AS.inputslock);
00142             MesPrint("Cannot position file in SetScratch");
00143             Terminate(-1);
00144         }
00145 #ifdef HIIDEDEBUG
00146         MesPrint("SetScratch1(%w): position = %12p, size = %1, address =
00147         %x",position,f->POsize,f->PObuffer);
00148 #endif
00149         if ( ( size = ReadFile(f->handle, (UBYTE *) (f->PObuffer),f->POsize) ) < 0
00150         || ( size & 1 ) != 0 ) {
00151             UNLOCK(AS.inputslock);
00152             MesPrint("Read error in SetScratch");
00153             Terminate(-1);
00154         }
00155         UNLOCK(AS.inputslock);
00156         if ( size == 0 ) {
00157             f->PObuffer[0] = 0;
00158         }
00159         f->POfill = f->PObuffer;
00160         f->POposition = *position;
00161 #ifdef WORD2
00162         *whichInInBuf = size » 1;
00163 #else
00164         *whichInInBuf = size / TABLESIZE(WORD,UBYTE);
00165 #endif
00166         f->POfull = f->PObuffer + *whichInInBuf;
00167 #ifdef HIIDEDEBUG
00168         MesPrint("SetScratch2: size = %1, InInBuf = %1, fill = %1, full = %1"
00169         ,size,*whichInInBuf, (f->POfill-f->PObuffer)*sizeof(WORD)
00170         , (f->POfull-f->PObuffer)*sizeof(WORD));
00171 #endif
00172     } else {
00173     endpos:
00174         DIFPOS(possize,*position,f->POposition);
00175         f->POfill = (WORD *) (BASEPOSITION(possize)+(UBYTE *) (f->PObuffer));
00176         *whichInInBuf = f->POfull-f->POfill;
00177     }
00178 }
00179
00180 /*
00181     #] SetScratch :
00182     #[ RevertScratch :
00183
00184     Reverts the input/output directions. This way input comes
00185     always from AR.infile
00186 */
00187
00188 int RevertScratch(void)
00189 {
00190     GETIDENTITY
00191     FILEHANDLE *f;
00192     if ( AR.infile->handle >= 0 && AR.infile->handle != AR.outfile->handle ) {
00193         CloseFile(AR.infile->handle);
00194         AR.infile->handle = -1;
00195         remove(AR.infile->name);
00196     }
00197     f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
00198     AR.infile->POfull = AR.infile->POfill;
00199     AR.infile->POfill = AR.infile->PObuffer;
00200     if ( AR.infile->handle >= 0 ) {
00201         POSITION scrpos;
00202         PUTZERO(scrpos);
00203         SeekFile(AR.infile->handle,&scrpos,SEEK_SET);
00204         if ( ISNOTZEROPOS(scrpos) ) {
00205             return(MesPrint("Error with scratch output."));
00206         }
00207     }
00208     if ( ( AR.InInBuf = ReadFile(AR.infile->handle, (UBYTE *) (AR.infile->PObuffer)
00209     ,AR.infile->POsize) ) < 0 || AR.InInBuf & 1 ) {

```

```

00210         return(MesPrint("Error while reading from scratch file"));
00211     }
00212     else {
00213         AR.InInBuf /= TABLESIZE(WORD,UBYTE);
00214     }
00215     AR.infile->POfull = AR.infile->PObuffer + AR.InInBuf;
00216 }
00217 PUTZERO(AR.infile->POposition);
00218 AR.outfile->POfill = AR.outfile->POfull = AR.outfile->PObuffer;
00219 PUTZERO(AR.outfile->POposition);
00220 PUTZERO(AR.outfile->filesize);
00221 return(0);
00222 }
00223
00224 /*
00225     #] RevertScratch :
00226     #[ ResetScratch :
00227
00228     Resets the output scratch file to its beginning in such a way
00229     that the write routines can read it. The output buffers are
00230     left untouched as they may still be needed for extra declarations.
00231
00232 */
00233
00234 int ResetScratch(void)
00235 {
00236     GETIDENTITY
00237     FILEHANDLE *f;
00238     if ( AR.infile->handle >= 0 ) {
00239         CloseFile(AR.infile->handle); AR.infile->handle = -1;
00240         remove(AR.infile->name);
00241         PUTZERO(AR.infile->POposition);
00242         AR.infile->POfill = AR.infile->POfull = AR.infile->PObuffer;
00243     }
00244     if ( AR.outfile->handle >= 0 ) {
00245         POSITION scrpos;
00246         PUTZERO(scrpos);
00247         SeekFile(AR.outfile->handle,&scrpos,SEEK_SET);
00248         if ( ISNOTZEROPOS(scrpos) ) {
00249             return(MesPrint("Error with scratch output."));
00250         }
00251         if ( ( AR.InInBuf = ReadFile(AR.outfile->handle,(UBYTE *) (AR.outfile->PObuffer)
00252             ,AR.outfile->POsize) ) < 0 || AR.InInBuf & 1 ) {
00253             return(MesPrint("Error while reading from scratch file"));
00254         }
00255         else AR.InInBuf /= TABLESIZE(WORD,UBYTE);
00256         AR.outfile->POfull = AR.outfile->PObuffer + AR.InInBuf;
00257     }
00258     else AR.outfile->POfull = AR.outfile->POfill;
00259     AR.outfile->POfill = AR.outfile->PObuffer;
00260     PUTZERO(AR.outfile->POposition);
00261     f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
00262     return(0);
00263 }
00264
00265 /*
00266     #] ResetScratch :
00267     #[ ReadFromScratch :
00268
00269     Routine is used to copy files from scratch to hide.
00270 */
00271
00272 int ReadFromScratch(FILEHANDLE *fi, POSITION *pos, UBYTE *buffer, POSITION *length)
00273 {
00274     GETIDENTITY
00275     LONG l = BASEPOSITION(*length);
00276     if ( fi->handle < 0 ) {
00277         memcpy(buffer,fi->POfill,l);
00278     }
00279     else {
00280         SeekFile(fi->handle,pos,SEEK_SET);
00281         if ( ReadFile(fi->handle,buffer,l) != l ) {
00282             if ( fi == AR.hidefile )
00283                 MesPrint("Error reading from hide file.");
00284             else
00285                 MesPrint("Error reading from scratch file.");
00286             return(-1);
00287         }
00288     }
00289     return(0);
00290 }
00291
00292 /*
00293     #] ReadFromScratch :
00294     #[ AddToScratch :
00295
00296     Routine is used to copy files from scratch to hide.

```

```

00297 */
00298
00299 int AddToScratch(FILEHANDLE *fi, POSITION *pos, UBYTE *buffer, POSITION *length,
00300                 int withflush)
00301 {
00302     GETIDENTITY
00303     LONG l = BASEPOSITION(*length), avail;
00304     DUMMYUSE(pos)
00305     fi->POfill = fi->POfull;
00306     while ( fi->POfill+l/sizeof(WORD) > fi->POstop ) {
00307         avail = (fi->POstop-fi->POfill)*sizeof(WORD);
00308         if ( avail > 0 ) {
00309             memcpy(fi->POfill,buffer,avail);
00310             l -= avail; buffer += avail;
00311         }
00312         if ( fi->handle < 0 ) {
00313             if ( ( fi->handle = (WORD)CreateFile(fi->name) ) < 0 ) {
00314                 if ( fi == AR.hidefile )
00315                     MesPrint("Cannot create hide file %s",fi->name);
00316                 else
00317                     MesPrint("Cannot create scratch file %s",fi->name);
00318                 return(-1);
00319             }
00320             PUTZERO(fi->POposition);
00321         }
00322         SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
00323         if ( WriteFile(fi->handle,(UBYTE *)fi->PObuffer,fi->POsize) != fi->POsize )
00324             goto writeerror;
00325         ADDPOS(fi->POposition,fi->POsize);
00326         fi->POfill = fi->POfull = fi->PObuffer;
00327     }
00328     if ( l > 0 ) {
00329         memcpy(fi->POfill,buffer,l);
00330         fi->POfill += l/sizeof(WORD);
00331         fi->POfull = fi->POfill;
00332     }
00333     if ( withflush && fi->handle >= 0 && fi->POfill > fi->PObuffer ) { /* flush */
00334         l = (LONG)fi->POfill - (LONG)fi->PObuffer;
00335         SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
00336         if ( WriteFile(fi->handle,(UBYTE *)fi->PObuffer,l) != l ) goto writeerror;
00337         ADDPOS(fi->POposition,fi->POsize);
00338         fi->POfill = fi->POfull = fi->PObuffer;
00339     }
00340     if ( withflush && fi->handle >= 0 )
00341         SETBASEPOSITION(fi->filesize,TellFile(fi->handle));
00342     return(0);
00343 writeerror:
00344     if ( fi == AR.hidefile )
00345         MesPrint("Error writing to hide file. Disk full?");
00346     else
00347         MesPrint("Error writing to scratch file. Disk full?");
00348     return(-1);
00349 }
00350
00351 /*
00352     #] AddToScratch :
00353     #[ CoSave :
00354
00355     The syntax of the save statement is:
00356
00357     save filename
00358     save filename expr1 expr2
00359 */
00360
00361 int CoSave(UBYTE *inp)
00362 {
00363     GETIDENTITY
00364     UBYTE *p, c;
00365     WORD n = 0, i;
00366     WORD error = 0, type, number;
00367     WORD exprInStorageFlag = 0;
00368     LONG RetCode = 0, wSize;
00369     EXPRESSIONS e;
00370     INDEXENTRY *ind;
00371     INDEXENTRY *indold;
00372     WORD TMproto[SUBEXPSIZE];
00373     POSITION scrpos, scrposl, filesize;
00374     int ii, j = sizeof(FILEINDEX)/(sizeof(LONG));
00375     LONG *lo;
00376     while ( *inp == ',' ) inp++;
00377     p = inp;
00378
00379 #ifdef WITHMPI
00380     if ( PF.me != MASTER ) return(0);
00381 #endif
00382
00383

```

```

00384     if ( !*p ) return(MesPrint("No filename in save statement"));
00385     if ( FG.cTable[*p] > 1 && ( *p != '.' ) && ( *p != SEPARATOR ) && ( *p != ALTSEPARATOR ) )
00386         return(MesPrint("Illegal filename"));
00387     while ( ++p && *p != ',' ) {}
00388     c = *p;
00389     *p = 0;
00390     if ( !AP.preError ) {
00391         if ( ( RetCode = CreateFile((char *)inp) ) < 0 ) {
00392             return(MesPrint("Cannot open file %s",inp));
00393         }
00394     }
00395     AO.SaveData.Handle = (WORD)RetCode;
00396     PUTZERO(filesize);
00397
00398     e = Expressions;
00399     n = NumExpressions;
00400     if ( c ) { /* There follows a list of expressions */
00401         *p++ = c;
00402         inp = p;
00403         i = (WORD) (INFILEINDEX);
00404         if ( WriteStoreHeader(AO.SaveData.Handle) ) return(MesPrint("Error writing storage file
header"));
00405         /* PUTZERO(AO.SaveData.Index.number); */
00406         /* PUTZERO(AO.SaveData.Index.next); */
00407         lo = (LONG *)(&AO.SaveData.Index);
00408         for ( ii = 0; ii < j; ii++ ) *lo++ = 0;
00409         SETBASEPOSITION(AO.SaveData.Position, (LONG)sizeof(STOREHEADER));
00410         ind = AO.SaveData.Index.expression;
00411         if ( !AP.preError && WriteFile(AO.SaveData.Handle, (UBYTE *)(&(AO.SaveData.Index))
00412             , (LONG)sizeof(struct FileInDeX)) != (LONG)sizeof(struct FileInDeX) ) goto SavWrt;
00413         SeekFile(AO.SaveData.Handle, &(filesize), SEEK_END);
00414         /* ADDPOS(filesize, sizeof(struct FileInDeX)); */
00415
00416         do { /* Scan the list */
00417             if ( !FG.cTable[*p] || *p == '[' ) {
00418                 p = SkipAName(p);
00419                 if ( p == 0 ) return(-1);
00420             }
00421             c = *p; *p = 0;
00422             if ( GetVar(inp, &type, &number, CEXPRESSION, NOAUTO) != NAMENOTFOUND ) {
00423                 if ( e[number].status == STOREDEXPRESSION ) {
00424                     if ( StrLen(AC.exprnames->namebuffer+e[number].name) > MAXENAME ) {
00425                         char msg[100];
00426                         snprintf(msg, sizeof(msg), "saved expr name over %d char: %s", MAXENAME,
AC.exprnames->namebuffer+e[number].name);
00427                         Warning(msg);
00428                     }
00429                 }
00430                 /* Here we have to locate the stored expression, copy its index entry
00431                 possibly after making a new fileindex and then copy the whole
00432                 expression.
00433                 */
00434                 if ( AP.preError ) goto NextExpr;
00435                 TMproto[0] = EXPRESSION;
00436                 TMproto[1] = SUBEXPSIZE;
00437                 TMproto[2] = number;
00438                 TMproto[3] = 1;
00439                 { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
00440                 AT.TMaddr = TMproto;
00441                 if ( ( indold = FindInIndex(number, &AR.StoreData, 0, 0) ) != 0 ) {
00442                     if ( i <= 0 ) {
00443                         /*
00444                         AO.SaveData.Index.next = filesize;
00445                         */
00446                         SeekFile(AO.SaveData.Handle, &(AO.SaveData.Index.next), SEEK_END);
00447                         scrpos = AO.SaveData.Position;
00448                         SeekFile(AO.SaveData.Handle, &scrpos, SEEK_SET);
00449                         if ( ISNOTEQUALPOS(scrpos, AO.SaveData.Position) ) goto SavWrt;
00450                         if ( WriteFile(AO.SaveData.Handle, (UBYTE *)(&(AO.SaveData.Index))
00451                             , (LONG)sizeof(struct FileInDeX)) != (LONG)sizeof(struct FileInDeX) )
00452                             goto SavWrt;
00453                         i = (WORD) (INFILEINDEX);
00454                         AO.SaveData.Position = AO.SaveData.Index.next;
00455                         lo = (LONG *)(&AO.SaveData.Index);
00456                         for ( ii = 0; ii < j; ii++ ) *lo++ = 0;
00457                         ind = AO.SaveData.Index.expression;
00458                         scrpos = AO.SaveData.Position;
00459                         SeekFile(AO.SaveData.Handle, &scrpos, SEEK_SET);
00460                         if ( ISNOTEQUALPOS(scrpos, AO.SaveData.Position) ) goto SavWrt;
00461                         if ( WriteFile(AO.SaveData.Handle, (UBYTE *)(&(AO.SaveData.Index))
00462                             , (LONG)sizeof(struct FileInDeX)) != (LONG)sizeof(struct FileInDeX) )
00463                             goto SavWrt;
00464                         ADDPOS(filesize, sizeof(struct FileInDeX));
00465                     }
00466                     *ind = *indold;
00467                 }
00468             }
00469         } while ( *p );
00470     }

```

```

00468             ind->variables = SeekFile(AO.SaveData.Handle, &(AM.zeropos), SEEK_END);
00469  */
00470             ind->variables = filesize;
00471             ind->position = ind->variables;
00472             ADDPOS(ind->position, DIFBASE(indold->position, indold->variables));
00473             SeekFile(AR.StoreData.Handle, &(indold->variables), SEEK_SET);
00474             wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
00475             scrpos = ind->length;
00476             ADDPOS(scrpos, DIFBASE(ind->position, ind->variables));
00477             ADD2POS(filesize, scrpos);
00478             SETBASEPOSITION(scrpos1, wSize);
00479             do {
00480                 if ( ISLESSPOS(scrpos, scrpos1) ) wSize = BASEPOSITION(scrpos);
00481                 if ( ReadFile(AR.StoreData.Handle, (UBYTE *)AT.WorkPointer, wSize)
00482                     != wSize ) {
00483                     MesPrint("ReadError");
00484                     error = -1;
00485                     goto EndSave;
00486                 }
00487                 if ( WriteFile(AO.SaveData.Handle, (UBYTE *)AT.WorkPointer, wSize)
00488                     != wSize ) goto SavWrt;
00489                 ADDPOS(scrpos, -wSize);
00490             } while ( ISPOSPOS(scrpos) );
00491             ADDPOS(AO.SaveData.Index.number, 1);
00492             ind++;
00493         }
00494         else error = -1;
00495         i--;
00496     }
00497     else {
00498         MesPrint("%s is not a stored expression", inp);
00499         error = -1;
00500     }
00501 NextExpr;;
00502 }
00503 else {
00504     MesPrint("%s is not an expression", inp);
00505     error = -1;
00506 }
00507 *p = c;
00508 if ( c != ' ' && c ) {
00509     MesComp("Illegal character", inp, p);
00510     error = -1;
00511     goto EndSave;
00512 }
00513 if ( c ) c = *++p;
00514 inp = p;
00515 } while ( c );
00516 if ( !AP.preError ) {
00517     scrpos = AO.SaveData.Position;
00518     SeekFile(AO.SaveData.Handle, &scrpos, SEEK_SET);
00519     if ( ISNOTEQUALPOS(scrpos, AO.SaveData.Position) ) goto SavWrt;
00520 }
00521 if ( !AP.preError &&
00522     WriteFile(AO.SaveData.Handle, (UBYTE *)(&(AO.SaveData.Index))
00523         , (LONG)sizeof(struct FileInDeX) != (LONG)sizeof(struct FileInDeX) ) goto SavWrt;
00524 }
00525 else if ( !AP.preError ) { /* All stored expressions should be saved. Easy */
00526     /* Make sure there is at least one stored expression: */
00527     if ( n > 0 ) { do {
00528         if ( StrLen(AC.exprnames->namebuffer+e->name) > MAXENAME ) {
00529             char msg[100];
00530             snprintf(msg, sizeof(msg), "saved expr name over %d char: %s", MAXENAME,
00531                 AC.exprnames->namebuffer+e->name);
00532             Warning(msg);
00533         }
00534         if ( e->status == STOREDEXPRESSION ) exprInStorageFlag = 1;
00535         e++;
00536     } while ( --n > 0 ); }
00537     if ( exprInStorageFlag ) {
00538         wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
00539         PUTZERO(scrpos);
00540         SeekFile(AR.StoreData.Handle, &scrpos, SEEK_SET); /* Start at the beginning */
00541         scrpos = AR.StoreData.Fill; /* Number of bytes to be copied */
00542         SETBASEPOSITION(scrpos1, wSize);
00543         do {
00544             if ( ISLESSPOS(scrpos, scrpos1) ) wSize = BASEPOSITION(scrpos);
00545             if ( ReadFile(AR.StoreData.Handle, (UBYTE *)AT.WorkPointer, wSize) != wSize ) {
00546                 MesPrint("ReadError");
00547                 error = -1;
00548                 goto EndSave;
00549             }
00550             if ( WriteFile(AO.SaveData.Handle, (UBYTE *)AT.WorkPointer, wSize) != wSize )
00551                 goto SavWrt;
00552             ADDPOS(scrpos, -wSize);
00553         } while ( ISPOSPOS(scrpos) );

```

```

00554     }
00555 EndSave:
00556     if ( !AP.preError ) {
00557         CloseFile(AO.SaveData.Handle);
00558         AO.SaveData.Handle = -1;
00559     }
00560     return(error);
00561 SavWrt:
00562     MesPrint("WriteError");
00563     error = -1;
00564     goto EndSave;
00565 }
00566
00567 /*
00568     #] CoSave :
00569     #[ CoLoad :
00570 */
00571
00572 int CoLoad(UBYTE *inp)
00573 {
00574     GETIDENTITY
00575     INDEXENTRY *ind;
00576     LONG RetCode;
00577     UBYTE *p, c;
00578     WORD num, i, error = 0;
00579     WORD type, number, silentload = 0;
00580     WORD TMproto[SUBEXPSIZE];
00581     POSITION scrpos, firstposition;
00582     while ( *inp == ',' ) inp++;
00583     p = inp;
00584     if ( ( *p == ',' && p[1] == '-' ) || *p == '-' ) {
00585         if ( *p == ',' ) p++;
00586         p++;
00587         if ( *p == 's' || *p == 'S' ) {
00588             silentload = 1;
00589             while ( *p && ( *p != ',' && *p != '-' && *p != '+'
00590                 && *p != SEPARATOR && *p != ALTSEPARATOR && *p != '.' ) ) p++;
00591         }
00592         else if ( *p != ',' ) {
00593             return(MesPrint("Illegal option in Load statement"));
00594         }
00595         while ( *p == ',' ) p++;
00596     }
00597     inp = p;
00598     if ( !*p ) return(MesPrint("No filename in load statement"));
00599     if ( FG.cTable[*p] > 1 && ( *p != '.' ) && ( *p != SEPARATOR ) && ( *p != ALTSEPARATOR ) )
00600         return(MesPrint("Illegal filename"));
00601     while ( ++p && *p != ',' ) {}
00602     c = *p;
00603     *p = 0;
00604     if ( ( RetCode = OpenFile((char *)inp) ) < 0 ) {
00605         return(MesPrint("Cannot open file %s",inp));
00606     }
00607
00608     if ( SetFileIndex() ) {
00609         MesCall("CoLoad");
00610         SETERROR(-1)
00611     }
00612
00613     AO.SaveData.Handle = (WORD) (RetCode);
00614
00615 #ifndef SYSDEPENDENTSAVE
00616     if ( ReadFile(AO.SaveData.Handle, (UBYTE *) (&(AO.SaveData.Index)),
00617         (LONG)sizeof(struct FileInDeX) != (LONG)sizeof(struct FileInDeX) ) goto LoadRead;
00618 #else
00619     if ( ReadSaveHeader() ) goto LoadRead;
00620     TELLFILE(AO.SaveData.Handle,&firstposition);
00621     if ( ReadSaveIndex(&AO.SaveData.Index) ) goto LoadRead;
00622 #endif
00623     if ( c ) { /* There follows a list of expressions */
00624         *p++ = c;
00625         inp = p;
00626
00627         do { /* Scan the list */
00628             if ( !FG.cTable[*p] || *p == '[' ) {
00629                 p = SkipAName(p);
00630                 if ( p == 0 ) return(-1);
00631             }
00632             c = *p; *p = 0;
00633             if ( GetVar(inp,&type,&number,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
00634                 MesPrint("Conflicting name: %s",inp);
00635                 error = -1;
00636             }
00637             else {
00638                 if ( ( num = EntVar(CEXPRESSION,inp,STOREDEXPRESSION,0,0,0) ) >= 0 ) {
00639                     TMproto[0] = EXPRESSION;
00640                     TMproto[1] = SUBEXPSIZE;

```

```

00641         TMproto[2] = num;
00642         TMproto[3] = 1;
00643         { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
00644         AT.TMaddr = TMproto;
00645         SeekFile(AO.SaveData.Handle,&firstposition,SEEK_SET);
00646         AO.SaveData.Position = firstposition;
00647         if ( ReadSaveIndex(&AO.SaveData.Index) ) goto LoadRead;
00648         if ( ( ind = FindInIndex(num,&AO.SaveData,1,0) ) != 0 ) {
00649             if ( !error ) {
00650                 if ( PutInStore(ind,num) ) error = -1;
00651                 else if ( !AM.silent && silentload == 0 )
00652                     MesPrint(" %s loaded",ind->name);
00653             }
00654             /*
00655             !!! Added 1-feb-1998
00656             */
00657             Expressions[num].counter = -1;
00658         }
00659         else {
00660             MesPrint(" %s not found",inp);
00661             error = -1;
00662         }
00663     }
00664     else error = -1;
00665 }
00666 *p = c;
00667 if ( c != ',' && c ) {
00668     MesComp("Illegal character",inp,p);
00669     error = -1;
00670     goto EndLoad;
00671 }
00672 if ( c ) c = ++p;
00673 inp = p;
00674 } while ( c );
00675 scrpos = AR.StoreData.Position;
00676 SeekFile(AR.StoreData.Handle,&scrpos,SEEK_SET);
00677 if ( ISNOTEQUALPOS(scrpos,AR.StoreData.Position) ) goto LoadWrt;
00678 if ( WriteFile(AR.StoreData.Handle,(UBYTE *) (&(AR.StoreData.Index))
00679     ,(LONG)sizeof(struct FileInDeX)) != (LONG)sizeof(struct FileInDeX) ) goto LoadWrt;
00680 }
00681 else {
00682     /* All saved expressions should be stored. Easy */
00683     i = (WORD)BASEPOSITION(AO.SaveData.Index.number);
00684     ind = AO.SaveData.Index.expression;
00685 #ifdef SYSDEPENDENTSARE
00686     if ( i > 0 ) { do {
00687         if ( GetVar((UBYTE *) (ind->name),&type,&number,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
00688             MesPrint("Conflicting name: %s",ind->name);
00689             error = -1;
00690         }
00691     } else {
00692         if ( ( num = EntVar(CEXPRESSION,(UBYTE *) (ind->name),STOREDEXPRESSION,0,0,0) ) >= 0 )
00693         {
00694             if ( !error ) {
00695                 if ( PutInStore(ind,num) ) error = -1;
00696                 else if ( !AM.silent && silentload == 0 )
00697                     MesPrint(" %s loaded",ind->name);
00698             }
00699             else error = -1;
00700         }
00701         i--;
00702         if ( i == 0 && ISNOTZEROPOS(AO.SaveData.Index.next) ) {
00703             SeekFile(AO.SaveData.Handle,&(AO.SaveData.Index.next),SEEK_SET);
00704             if ( ReadFile(AO.SaveData.Handle,(UBYTE *) (&(AO.SaveData.Index)),
00705                 (LONG)sizeof(struct FileInDeX)) != (LONG)sizeof(struct FileInDeX) ) goto LoadRead;
00706             i = (WORD)BASEPOSITION(AO.SaveData.Index.number);
00707             ind = AO.SaveData.Index.expression;
00708         }
00709         else ind++;
00710     } while ( i > 0 ); }
00711 #else
00712     if ( i > 0 ) {
00713         do {
00714             if ( GetVar((UBYTE *) (ind->name),&type,&number,ALLVARIABLES,NOAUTO) != NAMENOTFOUND )
00715             {
00716                 MesPrint("Conflicting name: %s",ind->name);
00717                 error = -1;
00718             }
00719             else {
00720                 if ( ( num = EntVar(CEXPRESSION,(UBYTE *) (ind->name),STOREDEXPRESSION,0,0,0) ) >=
00721                 0 ) {
00722                     if ( !error ) {
00723                         if ( PutInStore(ind,num) ) error = -1;
00724                         else if ( !AM.silent && silentload == 0 )
00725                             MesPrint(" %s loaded",ind->name);
00726                     }
00727                 }
00728             }
00729             i--;
00730             if ( i == 0 && ISNOTZEROPOS(AO.SaveData.Index.next) ) {
00731                 SeekFile(AO.SaveData.Handle,&(AO.SaveData.Index.next),SEEK_SET);
00732                 if ( ReadFile(AO.SaveData.Handle,(UBYTE *) (&(AO.SaveData.Index)),
00733                     (LONG)sizeof(struct FileInDeX)) != (LONG)sizeof(struct FileInDeX) ) goto LoadRead;
00734                 i = (WORD)BASEPOSITION(AO.SaveData.Index.number);
00735                 ind = AO.SaveData.Index.expression;
00736             }
00737             else ind++;
00738         } while ( i > 0 ); }
00739 #endif
00740 }

```

```

00725         else error = -1;
00726     }
00727     i--;
00728     if ( i == 0 && (ISNOTZEROPOS(AO.SaveData.Index.next) || AO.bufferedInd) ) {
00729         SeekFile(AO.SaveData.Handle, &(AO.SaveData.Index.next), SEEK_SET);
00730         if ( ReadSaveIndex(&AO.SaveData.Index) ) goto LoadRead;
00731         i = (WORD)BASEPOSITION(AO.SaveData.Index.number);
00732         ind = AO.SaveData.Index.expression;
00733     }
00734     else ind++;
00735     } while ( i > 0 );
00736 }
00737 #endif
00738 }
00739 EndLoad:
00740 #ifndef SYSDEPENDENTSAVE
00741     if ( AO.powerFlag ) {
00742         MesPrint("WARNING: min-/maxpower had to be adjusted!");
00743     }
00744     if ( AO.resizeFlag ) {
00745         MesPrint("ERROR: could not downsize data!");
00746         return ( -2 );
00747     }
00748 #endif
00749     CloseFile(AO.SaveData.Handle);
00750     AO.SaveData.Handle = -1;
00751     SeekFile(AR.StoreData.Handle, &(AC.StoreFileSize), SEEK_END);
00752     return(error);
00753 LoadWrt:
00754     MesPrint("WriteError");
00755     error = -1;
00756     goto EndLoad;
00757 LoadRead:
00758     MesPrint("ReadError");
00759     error = -1;
00760     goto EndLoad;
00761 }
00762
00763 /*
00764     #] CoLoad :
00765     #[ DeleteStore :
00766
00767     Routine deletes the contents of the entire storage file.
00768     We close the file and recreate it.
00769     If par > 0 we have to remove the expressions from the namelists.
00770 */
00771
00772 int DeleteStore(WORD par)
00773 {
00774     GETIDENTITY
00775     char *s;
00776     WORD j, n = 0;
00777     EXPRESSIONS e_in, e_out;
00778     WORD DidClean = 0;
00779     if ( AR.StoreData.Handle >= 0 ) {
00780         if ( par > 0 ) {
00781             n = NumExpressions;
00782             j = 0;
00783             e_in = e_out = Expressions;
00784             if ( n > 0 ) { do {
00785                 if ( e_in->status == STOREDEXPRESSION ) {
00786                     NAMENODE *node = GetNode(AC.exprnames,
00787                         AC.exprnames->namebuffer+e_in->name);
00788                     node->type = CDELETE;
00789                     DidClean = 1;
00790                 }
00791             } else {
00792                 if ( e_out != e_in ) {
00793                     NAMENODE *node;
00794                     node = GetNode(AC.exprnames,
00795                         AC.exprnames->namebuffer+e_in->name);
00796                     node->number = (WORD)(e_out - Expressions);
00797                     e_out->onfile = e_in->onfile;
00798                     e_out->prototype = e_in->prototype;
00799                     e_out->printfFlag = 0;
00800                     e_out->status = e_in->status;
00801                     e_out->name = e_in->name;
00802                     e_out->inmem = e_in->inmem;
00803                     e_out->counter = e_in->counter;
00804                     e_out->numfactors = e_in->numfactors;
00805                     e_out->numdummies = e_in->numdummies;
00806                     e_out->compression = e_in->compression;
00807                     e_out->namesize = e_in->namesize;
00808                     e_out->whichbuffer = e_in->whichbuffer;
00809                     e_out->hidelevel = e_in->hidelevel;
00810                     e_out->node = e_in->node;
00811                     e_out->replace = e_in->replace;

```



```

00812             e_out->vflags = e_in->vflags;
00813             e_out->uflags = e_in->uflags;
00814 #ifdef PARALLELCODE
00815             e_out->partodo = e_in->partodo;
00816 #endif
00817         }
00818         e_out++;
00819         j++;
00820     }
00821     e_in++;
00822     } while ( --n > 0 ); }
00823     NumExpressions = j;
00824     if ( DidClean ) CompactifyTree(AC.exprnames,EXPRNAMES);
00825 }
00826 AR.StoreData.Handle = -1;
00827 CloseFile(AC.StoreHandle);
00828 AC.StoreHandle = -1;
00829 {
00830 /*
00831     Knock out the storage caches (25-apr-1990!)
00832 */
00833     STORECACHE st;
00834     st = (STORECACHE) (AT.StoreCache);
00835     while ( st ) {
00836         SETBASEPOSITION(st->position,-1);
00837         SETBASEPOSITION(st->toppos,-1);
00838         st = st->next;
00839     }
00840 #ifdef WITHPTHREADS
00841     for ( j = 1; j < AM.totalnumberofthreads; j++ ) {
00842         st = (STORECACHE) (AB[j]->T.StoreCache);
00843         while ( st ) {
00844             SETBASEPOSITION(st->position,-1);
00845             SETBASEPOSITION(st->toppos,-1);
00846             st = st->next;
00847         }
00848     }
00849 #endif
00850     }
00851     PUTZERO(AC.StoreFileSize);
00852     s = FG.fname; while ( *s ) s++;
00853 #ifdef VMS
00854     *s = ' '; s[1] = ' '; s[2] = 0;
00855     remove(FG.fname);
00856     *s = 0;
00857 #endif
00858     return(AC.StoreHandle = CreateFile(FG.fname));
00859 }
00860 else return(0);
00861 }
00862
00863 /*
00864     #] DeleteStore :
00865     #[ PutInStore :
00866
00867     Copies the expression indicated by ind from a load file to the
00868     internal storage file. A return value of zero indicates that
00869     everything is OK.
00870
00871 */
00872
00873 int PutInStore(INDEXENTRY *ind, WORD num)
00874 {
00875     GETIDENTITY
00876     INDEXENTRY *newind;
00877     LONG wSize;
00878 #ifndef SYSDEPENDENTSAVE
00879     LONG wSizeOut;
00880     LONG stage;
00881 #endif
00882     POSITION scrpos,scrpos1;
00883     newind = NextFileIndex(&(Expressions[num].onfile));
00884     *newind = *ind;
00885 #ifndef SYSDEPENDENTSAVE
00886     SETBASEPOSITION(newind->length, 0);
00887 #endif
00888     newind->variables = AR.StoreData.Fill;
00889     SeekFile(AR.StoreData.Handle,&(newind->variables),SEEK_SET);
00890     if ( ISNOTEQUALPOS(newind->variables,AR.StoreData.Fill) ) goto PutErrS;
00891     newind->position = newind->variables;
00892 #ifdef SYSDEPENDENTSAVE
00893     ADDPOS(newind->position,DIFBASE(ind->position,ind->variables));
00894 #endif
00895     /* set read position to ind->variables */
00896     scrpos = ind->variables;
00897     SeekFile(AO.SaveData.Handle,&scrpos,SEEK_SET);
00898     if ( ISNOTEQUALPOS(scrpos,ind->variables) ) goto PutErrS;

```

```

00899      /* set max size for read-in */
00900      wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
00901 #ifdef SYSDEPENDENTSAVE
00902     scrpos = ind->length;
00903     ADDPOS(scrpos,DIFBASE(ind->position,ind->variables));
00904     ADD2POS(AR.StoreData.Fill,scrpos);
00905 #endif
00906     SETBASEPOSITION(scrpos1,wSize);
00907 #ifdef SYSDEPENDENTSAVE
00908     /* prepare look-up table for tensor functions */
00909     if ( ind->nfunctions ) {
00910         AO.tensorList = (UBYTE *)Malloca1(MAXSAVEFUNCTION,"PutInStore");
00911     }
00912     SETBASEPOSITION(scrpos, DIFBASE(ind->position,ind->variables));
00913     /* copy variables first */
00914     stage = -1;
00915     do {
00916         wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
00917         if ( ISLESSPOS(scrpos,scrpos1) ) wSize = BASEPOSITION(scrpos);
00918         wSizeOut = wSize;
00919         if ( ReadSaveVariables(
00920             (UBYTE *)AT.WorkPointer, (UBYTE *)AT.WorkTop, &wSize, &wSizeOut, ind, &stage) ) {
00921             goto PutErrS;
00922         }
00923         if ( WriteFile(AR.StoreData.Handle, (UBYTE *)AT.WorkPointer, wSizeOut)
00924             != wSizeOut ) goto PutErrS;
00925         ADDPOS(scrpos,-wSize);
00926         ADDPOS(newind->position, wSizeOut);
00927         ADDPOS(AR.StoreData.Fill, wSizeOut);
00928     } while ( ISPOSPOS(scrpos) );
00929     /* then copy the expression itself */
00930     wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
00931     scrpos = ind->length;
00932 #endif
00933     do {
00934         wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
00935         if ( ISLESSPOS(scrpos,scrpos1) ) wSize = BASEPOSITION(scrpos);
00936 #ifdef SYSDEPENDENTSAVE
00937         if ( ReadFile(AO.SaveData.Handle, (UBYTE *)AT.WorkPointer,wSize)
00938             != wSize ) goto PutErrS;
00939         if ( WriteFile(AR.StoreData.Handle, (UBYTE *)AT.WorkPointer,wSize)
00940             != wSize ) goto PutErrS;
00941         ADDPOS(scrpos,-wSize);
00942 #else
00943         wSizeOut = wSize;
00944         if ( ReadSaveExpression((UBYTE *)AT.WorkPointer, (UBYTE *)AT.WorkTop, &wSize, &wSizeOut) ) {
00945             goto PutErrS;
00946         }
00947         if ( WriteFile(AR.StoreData.Handle, (UBYTE *)AT.WorkPointer, wSizeOut)
00948             != wSizeOut ) goto PutErrS;
00949         ADDPOS(scrpos,-wSize);
00950         ADDPOS(AR.StoreData.Fill, wSizeOut);
00951         ADDPOS(newind->length, wSizeOut);
00952     } while ( ISPOSPOS(scrpos) );
00953     /* free look-up table for tensor functions */
00954     if ( ind->nfunctions ) {
00955         M_free(AO.tensorList,"PutInStore");
00956     }
00957     scrpos = AR.StoreData.Position;
00958     SeekFile(AR.StoreData.Handle,&scrpos,SEEK_SET);
00959     if ( ISNOTEQUALPOS(scrpos,AR.StoreData.Position) ) goto PutErrS;
00960     if ( WriteFile(AR.StoreData.Handle, (UBYTE *)(&AR.StoreData.Index), (LONG)sizeof(FILEINDEX))
00961         == (LONG)sizeof(FILEINDEX) ) return(0);
00962 PutErrS:
00963     return(MesPrint("File error"));
00964 }
00965 /*
00966 #] PutInStore :
00967 #[ GetTerm :
00968
00969 Gets one term from input scratch stream.
00970 Puts it in 'term'.
00971 Returns the length of the term.
00972
00973 Used by Processor      (proces.c)
00974 WriteAll               (sch.c)
00975 WriteOne              (sch.c)
00976 GetMoreTerms          (store.c)
00977 ToStorage             (store.c)
00978 CoFillExpression      (comexpr.c)
00979 FactorInExpr          (factor.c)
00980 LoadOpti             (optim.c)
00981 PF_Processor          (parallel.c)

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```

00986             ThreadsProcessor (threads.c)
00987             In multi thread/processor mode all calls are done by the master.
00988             Note however that other routines, used by the threads, can use
00989             the same file. Hence we need to be careful about SeekFile and locks.
00990 */
00991
00992 WORD GetTerm(PHEAD WORD *term)
00993 {
00994     GETBIDENTITY
00995     WORD *inp, i, j = 0, len;
00996     LONG InIn, *whichInInBuf;
00997     WORD *r, *m, *mstop = 0, minsiz = 0, *bra = 0, *from;
00998     WORD first, *start = 0, testing = 0;
00999     FILEHANDLE *fi;
01000     AN.deferskipped = 0;
01001     if ( AR.GetFile == 2 ) {
01002         fi = AR.hidefile;
01003         whichInInBuf = &(AR.InHiBuf);
01004     }
01005     else {
01006         fi = AR.infile;
01007         whichInInBuf = &(AR.InInBuf);
01008     }
01009     InIn = *whichInInBuf;
01010     from = term;
01011     if ( AR.KeptInHold ) {
01012         r = AR.CompressBuffer;
01013         i = *r;
01014         AR.KeptInHold = 0;
01015         if ( i <= 0 ) { *term = 0; goto RegRet; }
01016         m = term;
01017         NCOPY(m,r,i);
01018         goto RegRet;
01019     }
01020     if ( AR.DeferFlag ) {
01021         m = AR.CompressBuffer;
01022         if ( *m > 0 ) {
01023             mstop = m + *m;
01024             mstop -= ABS(mstop[-1]);
01025             m++;
01026             while ( m < mstop ) {
01027                 if ( *m == HAAKJE ) {
01028                     testing = 1;
01029                     mstop = m + m[1];
01030                     bra = (WORD *)(((UBYTE *) (term)) + 2*AM.MaxTer);
01031                     m = AR.CompressBuffer+1;
01032                     r = bra;
01033                     while ( m < mstop ) *r++ = *m++;
01034                     mstop = r;
01035                     minsiz = WORDDIF(mstop,bra);
01036                     goto ReStart;
01037                 }
01038                 /* We have the bracket to be tested in bra till mstop
01039                 */
01040             }
01041             m += m[1];
01042         }
01043     }
01044     bra = (WORD *)(((UBYTE *) (term)) + 2*AM.MaxTer);
01045     mstop = bra+1;
01046     *bra = 0;
01047     minsiz = 1;
01048     testing = 1;
01049 }
01050 ReStart:
01051     first = 0;
01052     r = AR.CompressBuffer;
01053     if ( fi->handle >= 0 ) {
01054         if ( InIn <= 0 ) {
01055             ADDPOS(fi->POposition, (fi->POfull-fi->PObuffer)*sizeof(WORD));
01056             LOCK(AS.inputslock);
01057             SeekFile(fi->handle, &(fi->POposition), SEEK_SET);
01058             InIn = ReadFile(fi->handle, (UBYTE *) (fi->PObuffer), fi->POsize);
01059             UNLOCK(AS.inputslock);
01060             if ( ( InIn < 0 ) || ( InIn & 1 ) ) {
01061                 goto GTerr;
01062             }
01063 #ifdef WORD2
01064             InIn >= 1;
01065 #else
01066             InIn /= TABLESIZE(WORD,UBYTE);
01067 #endif
01068             *whichInInBuf = InIn;
01069             if ( !InIn ) { *r = 0; *from = 0; goto RegRet; }
01070             fi->POfill = fi->PObuffer;
01071             fi->POfull = fi->PObuffer + InIn;
01072         }

```

```

01073     inp = fi->POfill;
01074     if ( ( len = i = *inp ) == 0 ) {
01075         (*whichInInBuf)--;
01076         (fi->POfill)++;
01077         *r = 0;
01078         *from = 0;
01079         goto RegRet;
01080     }
01081     if ( i < 0 ) {
01082         InIn--;
01083         inp++;
01084         r++;
01085         start = term;
01086         *term++ = -i + 1;
01087         while ( ++i <= 0 ) *term++ = *r++;
01088         if ( InIn > 0 ) {
01089             i = *inp++;
01090             InIn--;
01091             *start += i;
01092             *(AR.CompressBuffer) = len = *start;
01093         }
01094         else {
01095             first = 1;
01096             goto NewIn;
01097         }
01098     }
01099     InIn -= i;
01100     if ( InIn < 0 ) {
01101         j = (WORD) (- InIn);
01102         i -= j;
01103     }
01104     else j = 0;
01105     while ( --i >= 0 ) {
01106         *r++ = *term++ = *inp++;
01107     }
01108     if ( j ) {
01109 NewIn:
01110         ADDPOS(fi->POposition, (fi->POfull-fi->PObuffer)*sizeof(WORD));
01111         LOCK(AS.inputslock);
01112         SeekFile(fi->handle, &(fi->POposition), SEEK_SET);
01113         InIn = ReadFile(fi->handle, (UBYTE *) (fi->PObuffer), fi->POsize);
01114         UNLOCK(AS.inputslock);
01115         if ( ( InIn <= 0 ) || ( InIn & 1 ) ) {
01116             goto GTerr;
01117         }
01118 #ifdef WORD2
01119         InIn >= 1;
01120 #else
01121         InIn /= TABLESIZE(WORD, UBYTE);
01122 #endif
01123         inp = fi->PObuffer;
01124         fi->POfull = inp + InIn;
01125
01126         if ( first ) {
01127             j = *inp++;
01128             InIn--;
01129             *start += j;
01130             *(AR.CompressBuffer) = len = *start;
01131         }
01132         InIn -= j;
01133         while ( --j >= 0 ) { *r++ = *term++ = *inp++; }
01134     }
01135     fi->POfill = inp;
01136     *whichInInBuf = InIn;
01137     AR.DefPosition = fi->POposition;
01138     ADDPOS(AR.DefPosition, ((UBYTE *) (fi->POfill)) - ((UBYTE *) (fi->PObuffer)));
01139 }
01140 else {
01141     inp = fi->POfill;
01142     if ( inp >= fi->POfull ) { *from = 0; goto RegRet; }
01143     len = j = *inp;
01144     if ( j < 0 ) {
01145         inp++;
01146         *term++ = *r++ = len = - j + 1 + *inp;
01147         while ( ++j <= 0 ) *term++ = *r++;
01148         j = *inp++;
01149     }
01150     else if ( !j ) j = 1;
01151     while ( --j >= 0 ) { *r++ = *term++ = *inp++; }
01152     fi->POfill = inp;
01153 /*%%%%ADDED 7-apr-2006 for Keep Brackets in bucket */
01154     SETBASEPOSITION(AR.DefPosition, ((UBYTE *) (fi->POfill)) - ((UBYTE *) (fi->PObuffer)));
01155     if ( inp > fi->POfull ) {
01156         goto GTerr;
01157     }
01158 }
01159 if ( r >= AR.ComprTop ) {

```

```

01160         MesPrint("CompressSize of %10l is insufficient",AM.CompressSize);
01161         Terminate(-1);
01162     }
01163     AR.CompressPointer = r; *r = 0;
01164     /*
01165         The next *from is a bug fix that made the program read in forbidden
01166         territory.
01167     */
01168     if ( testing && *from != 0 ) {
01169         WORD jj;
01170         r = from;
01171         jj = *r - 1 - ABS(*(r+*r-1));
01172         if ( jj < minsiz ) goto strip;
01173         r++;
01174         m = bra;
01175         while ( m < mstop ) {
01176             if ( *m != *r ) {
01177 strip:
01178                 r = from;
01179                 m = r + *r;
01180                 mstop = m - ABS(m[-1]);
01181                 r++;
01182                 while ( r < mstop ) {
01183                     if ( *r == HAAKJE ) {
01184                         *r++ = 1;
01185                         *r++ = 1;
01186                         *r++ = 3;
01187                         len = WORDDIF(r,from);
01188                         *from = len;
01189                         goto RegRet;
01190                     }
01191                     r += r[1];
01192                 }
01193                 goto RegRet;
01194             }
01195             m++;
01196             r++;
01197         }
01198         term = from;
01199         AN.deferskipped++;
01200         goto ReStart;
01201 RegRet:;
01202     /*
01203         #[ debug :
01204     */
01205     {
01206         UBYTE OutBuf[140];
01207     /*
01208         if ( AP.DebugFlag ) { */
01209         if ( ( AP.PreDebug & DUMPINTERMS ) == DUMPINTERMS ) {
01210             MLOCK(ErrorMessageLock);
01211             AO.OutFill = AO.OutputLine = OutBuf;
01212             AO.OutSkip = 3;
01213             FiniLine();
01214             r = from;
01215             i = *r;
01216             TokenToLine((UBYTE *)("Input: "));
01217             if ( i == 0 ) {
01218                 TokenToLine((UBYTE *)"zero");
01219             }
01220             else if ( i < 0 ) {
01221                 TokenToLine((UBYTE *)"negative!!");
01222             }
01223             else {
01224                 while ( --i >= 0 ) {
01225                     TalToLine((UWORD)(*r++)); TokenToLine((UBYTE *)" ");
01226                 }
01227             }
01228             FiniLine();
01229             MUNLOCK(ErrorMessageLock);
01230         }
01231     /*
01232         #[ debug :
01233     */
01234     return(*from);
01235 GTerr:
01236     MesPrint("Error while reading scratch file in GetTerm");
01237     Terminate(-1);
01238     return(-1);
01239 }
01240
01241 /*
01242     #[ GetTerm :
01243     #[ GetOneTerm :
01244
01245     Gets one term from stream AR.infile->handle.
01246     Puts it in 'term'.

```

```

01247     Returns the length of the term.
01248     Input is unbuffered.
01249     Compression via AR.CompressPointer
01250     par is actually in all calls a file handle
01251
01252     Routine is called from
01253         DoOnePow           Get one power of an expression
01254         Deferred          Get the contents of a bracket
01255         GetFirstBracket
01256         FindBracket
01257     We should do something about the lack of buffering.
01258     Maybe a buffer of a few times AM.MaxTer (MaxTermSize*sizeof(WORD)).
01259     Each thread will need its own buffer!
01260
01261     If par == 0 we use ReadPosFile which can fill the whole buffer.
01262     If par == 1 we use ReadFile and do actual read operations.
01263
01264     Note: we cannot use ReadPosFile when running in the master thread.
01265 */
01266
01267 WORD GetOneTerm(PHEAD WORD *term, FILEHANDLE *fi, POSITION *pos, int par)
01268 {
01269     GETBIDENTITY
01270     WORD i, *p;
01271     LONG j, siz;
01272     WORD *r, *rr = AR.CompressPointer;
01273     int error = 0;
01274     r = rr;
01275     if ( fi->handle >= 0 ) {
01276 #ifdef READONEBYONE
01277 #ifdef WITHPTHREADS
01278 /*
01279     This code needs some investigation.
01280     It may be that we should do this always.
01281     It may be that even for workers it is no good.
01282     We may have to make a variable like AM.ReadDirect with
01283     if ( AM.ReadDirect ) par = 1;
01284     and a user command like
01285     On ReadDirect;
01286 */
01287     if ( AT.identity > 0 ) par = 1;
01288 #endif
01289 #endif
01290 /*
01291     To be changed:
01292     1: check first whether the term lies completely inside the buffer
01293     2: if not a: use old strategy for AT.identity == 0 (master)
01294         b: for workers, position file and read buffer
01295 */
01296     if ( par == 0 ) {
01297         siz = ReadPosFile(BHEAD fi, (UBYTE *)term, 1L, pos);
01298     }
01299     else {
01300         LOCK(AS.inputslock);
01301         SeekFile(fi->handle, pos, SEEK_SET);
01302         siz = ReadFile(fi->handle, (UBYTE *)term, sizeof(WORD));
01303         UNLOCK(AS.inputslock);
01304         ADDPOS(*pos, siz);
01305     }
01306     if ( siz == sizeof(WORD) ) {
01307         p = term;
01308         j = i = *term++;
01309         if ( ( i > AM.MaxTer/((WORD)sizeof(WORD)) ) || ( -i >= AM.MaxTer/((WORD)sizeof(WORD)) ) )
01310         {
01311             error = 1;
01312             goto ErrGet;
01313         }
01314         r++;
01315         if ( i < 0 ) { /* Loading a compressed term */
01316             *p = -i + 1;
01317             while ( ++i <= 0 ) *term++ = *r++;
01318             if ( par == 0 ) {
01319                 siz = ReadPosFile(BHEAD fi, (UBYTE *)term, 1L, pos);
01320             }
01321             else {
01322                 LOCK(AS.inputslock);
01323                 SeekFile(fi->handle, pos, SEEK_SET);
01324                 siz = ReadFile(fi->handle, (UBYTE *)term, sizeof(WORD));
01325                 UNLOCK(AS.inputslock);
01326                 ADDPOS(*pos, sizeof(WORD));
01327             }
01328             if ( siz != sizeof(WORD) ) {
01329                 error = 2;
01330                 goto ErrGet;
01331             }
01332             *p += *term;
01333             j = *term;

```

```

01334         if ( ( j > AM.MaxTer/((WORD)sizeof(WORD)) ) || ( j <= 0 ) )
01335         {
01336             error = 3;
01337             goto ErrGet;
01338         }
01339         *rr = *p; /* Write the proper term size to *AR.CompressPointer */
01340     }
01341     else { /* Loading a regular term */
01342         if ( !j ) return(0);
01343         *rr = i; /* Write the proper term size to *AR.CompressPointer */
01344         j--;
01345     }
01346     i = (WORD)j;
01347     if ( par == 0 ) {
01348         siz = ReadPosFile(BHEAD fi, (UBYTE *)term, j, pos);
01349         j *= TABLESIZE(WORD, UBYTE);
01350     }
01351     else {
01352         j *= TABLESIZE(WORD, UBYTE);
01353         LOCK(AS.inputslock);
01354         SeekFile(fi->handle, pos, SEEK_SET);
01355         siz = ReadFile(fi->handle, (UBYTE *)term, j);
01356         UNLOCK(AS.inputslock);
01357         ADDPOS(*pos, j);
01358     }
01359     if ( siz != j ) {
01360         error = 4;
01361         goto ErrGet;
01362     }
01363     while ( --i >= 0 ) *r++ = *term++;
01364     if ( r >= AR.ComprTop ) {
01365         MLOCK(ErrorMessageLock);
01366         MesPrint("CompressSize of %10l is insufficient", AM.CompressSize);
01367         MUNLOCK(ErrorMessageLock);
01368         Terminate(-1);
01369     }
01370     AR.CompressPointer = r; *r = 0;
01371     return(*p);
01372 }
01373 error = 5;
01374 }
01375 else {
01376 /*
01377     Here the whole expression is in the buffer.
01378 */
01379     fi->POfill = (WORD *)((UBYTE *) (fi->PObuffer) + BASEPOSITION(*pos));
01380     p = fi->POfill;
01381     if ( p >= fi->POfull ) { *term = 0; return(0); }
01382     j = i = *p;
01383     if ( i < 0 ) {
01384         p++;
01385         j = *r++ = *term++ = -i + 1 + *p;
01386         while ( ++i <= 0 ) *term++ = *r++;
01387         i = *p++;
01388     }
01389     if ( i == 0 ) { i = 1; *r++ = 0; *term++ = 0; }
01390     else { while ( --i >= 0 ) { *r++ = *term++ = *p++; } }
01391     fi->POfill = p;
01392     SETBASEPOSITION(*pos, (UBYTE *) (fi->POfill) - (UBYTE *) (fi->PObuffer));
01393     if ( p <= fi->POfull ) {
01394         if ( r >= AR.ComprTop ) {
01395             MLOCK(ErrorMessageLock);
01396             MesPrint("CompressSize of %10l is insufficient", AM.CompressSize);
01397             MUNLOCK(ErrorMessageLock);
01398             Terminate(-1);
01399         }
01400         AR.CompressPointer = r; *r = 0;
01401         return((WORD)j);
01402     }
01403     error = 6;
01404 }
01405 ErrGet:
01406     MLOCK(ErrorMessageLock);
01407     MesPrint("Error while reading scratch file in GetOneTerm (%d)", error);
01408     MUNLOCK(ErrorMessageLock);
01409     Terminate(-1);
01410     return(-1);
01411 }
01412
01413 /*
01414     #] GetOneTerm :
01415     #[ GetMoreTerms :
01416     Routine collects more contents of brackets inside a function,
01417     indicated by the number in AC.CollectFun.
01418     The first term is in term already.
01419     We can keep calling GetTerm either till a bracket is finished
01420     or till it would make the term too long (> AM.MaxTer/2)

```

```

01421     In all cases this function makes that the routine GetTerm
01422     has a term in 'hold', so the AR.KeptInHold flag must be turned on.
01423 */
01424
01425 WORD GetMoreTerms(WORD *term)
01426 {
01427     GETIDENTITY
01428     WORD *t, *r, *m, *h, *tstop, i, inc, same;
01429     WORD extra;
01430     WORD retval = 0;
01431 /*
01432     We use 23% as a quasi-random default value.
01433 */
01434     extra = (AM.MaxTer/sizeof(WORD))*((LONG)100-AC.CollectPercentage)/100;
01435     if ( extra < 23 ) extra = 23;
01436 /*
01437     First find the bracket pointer
01438 */
01439     t = term + *term;
01440     tstop = t - ABS(t[-1]);
01441     h = term+1;
01442     while ( *h != HAAKJE && h < tstop ) h += h[1];
01443     if ( h >= tstop ) return(retval);
01444     inc = FUNHEAD+ARGHEAD+1-h[1];
01445     same = WORDDIF(h,term) + h[1] - 1;
01446     r = m = t + inc;
01447     tstop = h + h[1];
01448     while ( t > tstop ) *--r = *--t;
01449     r--;
01450     *r = WORDDIF(m,r);
01451     while ( GetTerm(BHEAD m) > 0 ) {
01452         r = m + 1;
01453         t = m + *m - 1;
01454         if ( same > ( i = ( *m - ABS(*t) -1 ) ) ) { /* Must fail */
01455             if ( AC.AltCollectFun && AS.CollectOverFlag == 2 ) AS.CollectOverFlag = 3;
01456             break;
01457         }
01458         t = term+1;
01459         i = same;
01460         while ( --i >= 0 ) {
01461             if ( *r != *t ) {
01462                 if ( AC.AltCollectFun && AS.CollectOverFlag == 2 ) AS.CollectOverFlag = 3;
01463                 goto FullTerm;
01464             }
01465             r++; t++;
01466         }
01467         if ( ( WORDDIF(m,term) + i + extra ) > (WORD)(AM.MaxTer/sizeof(WORD)) ) {
01468 /* 23 = 3 +20. The 20 is to have some extra for substitutions or whatever */
01469             if ( AS.CollectOverFlag == 0 && AC.AltCollectFun == 0 ) {
01470                 Warning("Bracket contents too long in Collect statement");
01471                 Warning("Contents spread over more than one term");
01472                 Warning("If possible: increase MaxTermSize in setfile");
01473                 AS.CollectOverFlag = 1;
01474             }
01475             else if ( AC.AltCollectFun ) {
01476                 AS.CollectOverFlag = 2;
01477             }
01478             break;
01479         }
01480         tstop = m + *m;
01481         *m -= same;
01482         m++;
01483         while ( r < tstop ) *m++ = *r++;
01484         retval++;
01485         if ( extra == 23 ) extra = ((AM.MaxTer/sizeof(WORD))/6);
01486     }
01487 FullTerm:
01488     h[1] = WORDDIF(m,h);
01489     if ( AS.CollectOverFlag > 1 ) {
01490         *h = AC.AltCollectFun;
01491         if ( AS.CollectOverFlag == 3 ) AS.CollectOverFlag = 1;
01492     }
01493     else *h = AC.CollectFun;
01494     h[2] |= DIRTYFLAG;
01495     h[FUNHEAD] = h[1] - FUNHEAD;
01496     h[FUNHEAD+1] = 0;
01497     if ( ToFast(h+FUNHEAD,h+FUNHEAD) ) {
01498         if ( h[FUNHEAD] <= -FUNCTION ) {
01499             h[1] = FUNHEAD+1;
01500             m = h + FUNHEAD+1;
01501         }
01502         else {
01503             h[1] = FUNHEAD+2;
01504             m = h + FUNHEAD+2;
01505         }
01506     }
01507     *m++ = 1;

```



```

01508     *m++ = 1;
01509     *m++ = 3;
01510     *term = WORDDIF(m,term);
01511     AR.KeptInHold = 1;
01512     return(retval);
01513 }
01514
01515 /*
01516     #] GetMoreTerms :
01517     #[ GetMoreFromMem :
01518
01519 */
01520
01521 int GetMoreFromMem(WORD *term, WORD **tpoin)
01522 {
01523     GETIDENTITY
01524     WORD *t, *r, *m, *h, *tstop, i, j, inc, same;
01525     LONG extra = 23;
01526 /*
01527     First find the bracket pointer
01528 */
01529     t = term + *term;
01530     tstop = t - ABS(t[-1]);
01531     h = term+1;
01532     while ( *h != HAAKJE && h < tstop ) h += h[1];
01533     if ( h >= tstop ) return(0);
01534     inc = FUNHEAD+ARGHEAD+1-h[1];
01535     same = WORDDIF(h,term) + h[1] - 1;
01536     r = m = t + inc;
01537     tstop = h + h[1];
01538     while ( t > tstop ) *--r = *--t;
01539     r--;
01540     *r = WORDDIF(m,r);
01541     while ( **tpoin ) {
01542         r = *tpoin; j = *r;
01543         for ( i = 0; i < j; i++ ) m[i] = *r++;
01544         *tpoin = r;
01545         r = m + 1;
01546         t = m + *m - 1;
01547         if ( same > ( i = ( *m - ABS(*t) - 1 ) ) ) { /* Must fail */
01548             if ( AC.AltCollectFun && AS.CollectOverFlag == 2 ) AS.CollectOverFlag = 3;
01549             break;
01550         }
01551         t = term+1;
01552         i = same;
01553         while ( --i >= 0 ) {
01554             if ( *r != *t ) {
01555                 if ( AC.AltCollectFun && AS.CollectOverFlag == 2 ) AS.CollectOverFlag = 3;
01556                 goto FullTerm;
01557             }
01558             r++; t++;
01559         }
01560         if ( ( WORDDIF(m,term) + i + extra ) > (LONG) (AM.MaxTer/(2*sizeof(WORD))) ) {
01561 /* 23 = 3 +20. The 20 is to have some extra for substitutions or whatever */
01562             if ( AS.CollectOverFlag == 0 && AC.AltCollectFun == 0 ) {
01563                 Warning("Bracket contents too long in Collect statement");
01564                 Warning("Contents spread over more than one term");
01565                 Warning("If possible: increase MaxTermSize in setfile");
01566                 AS.CollectOverFlag = 1;
01567             }
01568             else if ( AC.AltCollectFun ) {
01569                 AS.CollectOverFlag = 2;
01570             }
01571             break;
01572         }
01573         tstop = m + *m;
01574         *m -= same;
01575         m++;
01576         while ( r < tstop ) *m++ = *r++;
01577         if ( extra == 23 ) extra = ((AM.MaxTer/sizeof(WORD))/6);
01578     }
01579 FullTerm:
01580     h[1] = WORDDIF(m,h);
01581     if ( AS.CollectOverFlag > 1 ) {
01582         *h = AC.AltCollectFun;
01583         if ( AS.CollectOverFlag == 3 ) AS.CollectOverFlag = 1;
01584     }
01585     else *h = AC.CollectFun;
01586     h[2] |= DIRTYFLAG;
01587     h[FUNHEAD] = h[1] - FUNHEAD;
01588     h[FUNHEAD+1] = 0;
01589     if ( ToFast(h+FUNHEAD,h+FUNHEAD) ) {
01590         if ( h[FUNHEAD] <= -FUNCTION ) {
01591             h[1] = FUNHEAD+1;
01592             m = h + FUNHEAD+1;
01593         }
01594         else {

```

```

01595         h[1] = FUNHEAD+2;
01596         m = h + FUNHEAD+2;
01597     }
01598 }
01599 *m++ = 1;
01600 *m++ = 1;
01601 *m++ = 3;
01602 *term = WORDDIFF(m,term);
01603 AR.KeptInHold = 1;
01604 return(0);
01605 }
01606
01607 /*
01608     #] GetMoreFromMem :
01609     #[ GetFromStore :
01610
01611     Gets a single term from the storage file at position and puts
01612     it at 'to'.
01613     The value to be returned is the number of words read.
01614     Renumbering is done also.
01615     This is controlled by the renumber table, given in 'renumber'
01616
01617     This routine should work with a number of cache buffers. The
01618     exact number should be definable in form.set.
01619     The parameters are:
01620     AM.SizeStoreCache (4096)
01621     The numbers are the proposed default values.
01622
01623     The cache is a pure read cache.
01624 */
01625
01626 static int gfs = 0;
01627
01628 WORD GetFromStore(WORD *to, POSITION *position, RENUMBER renumber, WORD *InCompState, WORD nexpr)
01629 {
01630     GETIDENTITY
01631     LONG RetCode, num, first = 0;
01632     WORD *from, *m;
01633     struct StOrEcAcHe sc;
01634     STORECACHE s;
01635     STORECACHE snext, sold;
01636     WORD *r, *rr = AR.CompressPointer;
01637     r = rr;
01638     gfs++;
01639     sc.next = AT.StoreCache;
01640     sold = s = &sc;
01641     snext = s->next;
01642     while ( snext ) {
01643         sold = s;
01644         s = snext;
01645         snext = s->next;
01646         if ( BASEPOSITION(s->position) == -1 ) break;
01647         if ( ISLESSPOS(*position,s->toppos) &&
01648             ISGEPOS(*position,s->position) ) { /* Hit */
01649             if ( AT.StoreCache != s ) {
01650                 sold->next = s->next;
01651                 s->next = AT.StoreCache->next;
01652                 AT.StoreCache = s;
01653             }
01654             from = (WORD *) ((UBYTE *) (s->buffer)) + DIFBASE(*position,s->position));
01655             num = *from;
01656             if ( !num ) { return(*to = 0); }
01657             *InCompState = (WORD)num;
01658             m = to;
01659             if ( num < 0 ) {
01660                 from++;
01661                 ADDPOS(*position,sizeof(WORD));
01662                 *m++ = (WORD) (-num+1);
01663                 r++;
01664                 while ( ++num <= 0 ) *m++ = *r++;
01665                 if ( ISLESSPOS(*position,s->toppos) ) {
01666                     num = *from++;
01667                     *to += (WORD)num;
01668                     ADDPOS(*position,sizeof(WORD));
01669                     *InCompState = (WORD) (num + 2);
01670                 }
01671                 else {
01672                     first = 1;
01673                     goto InNew;
01674                 }
01675             }
01676             PastCon;;
01677             while ( num > 0 && ISLESSPOS(*position,s->toppos) ) {
01678                 *r++ = *m++ = *from++; ADDPOS(*position,sizeof(WORD)); num--;
01679             }
01680             if ( num > 0 ) {
01681 InNew:

```

```

01682         SETBASEPOSITION(s->position,-1);
01683         SETBASEPOSITION(s->toppos,-1);
01684         LOCK(AM.storefilelock);
01685         SeekFile(AR.StoreData.Handle,position,SEEK_SET);
01686         RetCode = ReadFile(AR.StoreData.Handle,(UBYTE *) (s->buffer),AM.SizeStoreCache);
01687         UNLOCK(AM.storefilelock);
01688         if ( RetCode < 0 ) goto PastErr;
01689         if ( !RetCode ) return( *to = 0 );
01690         s->position = *position;
01691         s->toppos = *position;
01692         ADDPOS(s->toppos,RetCode);
01693         from = s->buffer;
01694         if ( first ) {
01695             num = *from++;
01696             ADDPOS(*position,sizeof(WORD));
01697             *to += (WORD)num;
01698             /* This next line has always been commented, but uncommenting it
01699              * fixes a rare bug when loading certain save files. */
01700             first = 0;
01701             *InCompState = (WORD) (num + 2);
01702         }
01703         goto PastCon;
01704     }
01705     goto PastEnd;
01706 }
01707
01708 if ( AT.StoreCache ) { /* Fill the last buffer */
01709     s->position = *position;
01710     LOCK(AM.storefilelock);
01711     SeekFile(AR.StoreData.Handle,position,SEEK_SET);
01712     RetCode = ReadFile(AR.StoreData.Handle,(UBYTE *) (s->buffer),AM.SizeStoreCache);
01713     UNLOCK(AM.storefilelock);
01714     if ( RetCode < 0 ) goto PastErr;
01715     if ( !RetCode ) return( *to = 0 );
01716     s->toppos = *position;
01717     ADDPOS(s->toppos,RetCode);
01718     if ( AT.StoreCache != s ) {
01719         sold->next = s->next;
01720         s->next = AT.StoreCache->next;
01721         AT.StoreCache = s;
01722     }
01723     m = to;
01724     from = s->buffer;
01725     num = *from;
01726     if ( !num ) { return( *to = 0 ); }
01727     *InCompState = (WORD)num;
01728     if ( num < 0 ) {
01729         *m++ = (WORD) (-num+1);
01730         r++;
01731         from++;
01732         ADDPOS(*position,sizeof(WORD));
01733         while ( ++num <= 0 ) *m++ = *r++;
01734         num = *from++;
01735         *to += (WORD)num;
01736         ADDPOS(*position,sizeof(WORD));
01737         *InCompState = (WORD) (num+2);
01738     }
01739     goto PastCon;
01740 }
01741 /* No caching available */
01742 LOCK(AM.storefilelock);
01743 SeekFile(AR.StoreData.Handle,position,SEEK_SET);
01744 RetCode = ReadFile(AR.StoreData.Handle,(UBYTE *)to,(LONG) sizeof(WORD));
01745 SeekFile(AR.StoreData.Handle,position,SEEK_CUR);
01746 UNLOCK(AM.storefilelock);
01747 if ( RetCode != sizeof(WORD) ) {
01748     *to = 0;
01749     return( (WORD) RetCode );
01750 }
01751 if ( !*to ) return(0);
01752 m = to;
01753 if ( *to < 0 ) {
01754     num = *m++;
01755     *to = *r++ = (WORD) (-num + 1);
01756     while ( ++num <= 0 ) *m++ = *r++;
01757     LOCK(AM.storefilelock);
01758     SeekFile(AR.StoreData.Handle,position,SEEK_SET);
01759     RetCode = ReadFile(AR.StoreData.Handle,(UBYTE *)m,(LONG) sizeof(WORD));
01760     SeekFile(AR.StoreData.Handle,position,SEEK_CUR);
01761     UNLOCK(AM.storefilelock);
01762     if ( RetCode != sizeof(WORD) ) {
01763         MLOCK(ErrorMessageLock);
01764         MesPrint("@Error in compression of store file");
01765         MUNLOCK(ErrorMessageLock);
01766         return(-1);
01767     }
01768     num = *m;

```

```

01769         *to += (WORD)num;
01770         *InCompState = (WORD)(num + 2);
01771     }
01772     else {
01773         *InCompState = *to;
01774         num = *to - 1; m = to + 1; r = rr + 1;
01775     }
01776     first = num;
01777     num *= wsizeof(WORD);
01778     if ( num < 0 ) {
01779         MLOCK(ErrorMessageLock);
01780         MesPrint("@Error in stored expressions file at position %9p",position);
01781         MUNLOCK(ErrorMessageLock);
01782         return(-1);
01783     }
01784     LOCK(AM.storefilelock);
01785     SeekFile(AR.StoreData.Handle,position,SEEK_SET);
01786     RetCode = ReadFile(AR.StoreData.Handle,(UBYTE *)m,num);
01787     SeekFile(AR.StoreData.Handle,position,SEEK_CUR);
01788     UNLOCK(AM.storefilelock);
01789     if ( RetCode != num ) {
01790         MLOCK(ErrorMessageLock);
01791         MesPrint("@Error in stored expressions file at position %9p",position);
01792         MUNLOCK(ErrorMessageLock);
01793         return(-1);
01794     }
01795     NCOPY(r,m,first);
01796 PastEnd:
01797     *rr = *to;
01798     if ( r >= AR.ComprTop ) {
01799         MLOCK(ErrorMessageLock);
01800         MesPrint("CompressSize of %10l is insufficient",AM.CompressSize);
01801         MUNLOCK(ErrorMessageLock);
01802         Terminate(-1);
01803     }
01804     AR.CompressPointer = r; *r = 0;
01805     if ( !TermRenumber(to,renumber,nexpr) ) {
01806         MarkDirty(to,DIRTYSYMFLAG);
01807         if ( AR.CurDum > AM.IndDum && Expressions[nexpr].numdummies > 0 )
01808             MoveDummies(BHEAD to,AR.CurDum - AM.IndDum);
01809         return((WORD)*to);
01810     }
01811 PastErr:
01812     MLOCK(ErrorMessageLock);
01813     MesCall("GetFromStore");
01814     MUNLOCK(ErrorMessageLock);
01815     SETERROR(-1)
01816 }
01817
01818 /*
01819     #] GetFromStore :
01820     #[ DetVars :          void DetVars(term)
01821
01822     Determines which variables are used in term.
01823
01824     When par = 1 we are scanning a prototype expression which involves
01825     completely different rules.
01826 */
01827
01828 void DetVars(WORD *term, WORD par)
01829 {
01830     GETIDENTITY
01831     WORD *stopper;
01832     WORD *t, sym;
01833     WORD *sarg;
01834     stopper = term + *term - 1;
01835     stopper = stopper - ABS(*stopper) + 1;
01836     term++;
01837     if ( par ) { /* Prototype expression */
01838         WORD n;
01839         if ( ( n = NumSymbols ) > 0 ) {
01840             SYMBOLS tt;
01841             tt = symbols;
01842             do {
01843                 (tt++)->flags &= ~INUSE;
01844             } while ( --n > 0 );
01845         }
01846         if ( ( n = NumIndices ) > 0 ) {
01847             INDICES tt;
01848             tt = indices;
01849             do {
01850                 (tt++)->flags &= ~INUSE;
01851             } while ( --n > 0 );
01852         }
01853         if ( ( n = NumVectors ) > 0 ) {
01854             VECTORS tt;

```

```

01856         tt = vectors;
01857         do {
01858             (tt++)->flags &= ~INUSE;
01859         } while ( --n > 0 );
01860     }
01861     if ( ( n = NumFunctions ) > 0 ) {
01862         FUNCTIONS tt;
01863         tt = functions;
01864         do {
01865             (tt++)->flags &= ~INUSE;
01866         } while ( --n > 0 );
01867     }
01868     term += SUBEXPSIZE;
01869     while ( term < stopper ) {
01870         if ( *term == SYMTOSYM || *term == SYMTONUM ) {
01871             term += 2;
01872             AN.UsedSymbol[*term] = 1;
01873             symbols[*term].flags |= INUSE;
01874         }
01875         else if ( *term == VECTOVEC ) {
01876             term += 2;
01877             AN.UsedVector[*term-AM.OffsetVector] = 1;
01878             vectors[*term-AM.OffsetVector].flags |= INUSE;
01879         }
01880         else if ( *term == INDTOIND ) {
01881             term += 2;
01882             sym = indices[*term - AM.OffsetIndex].dimension;
01883             if ( sym < 0 ) AN.UsedSymbol[-sym] = 1;
01884             AN.UsedIndex[*term - AM.OffsetIndex] = 1;
01885             sym = indices[*term-AM.OffsetIndex].nmin4;
01886             if ( sym < -NMIN4SHIFT ) AN.UsedSymbol[-sym-NMIN4SHIFT] = 1;
01887             indices[*term-AM.OffsetIndex].flags |= INUSE;
01888         }
01889         else if ( *term == FUNTOFUN ) {
01890             term += 2;
01891             AN.UsedFunction[*term-FUNCTION] = 1;
01892             functions[*term-FUNCTION].flags |= INUSE;
01893         }
01894         term += 2;
01895     }
01896 }
01897 else {
01898     while ( term < stopper ) {
01899         t = term + term[1];
01900         if ( *term == SYMBOL ) {
01901             term += 2;
01902             do {
01903                 AN.UsedSymbol[*term] = 1;
01904                 term += 2;
01905             } while ( term < t );
01906         }
01907         else if ( *term == DOTPRODUCT ) {
01908             term += 2;
01909             do {
01910                 AN.UsedVector[( *term++ ) - AM.OffsetVector] = 1;
01911                 AN.UsedVector[( *term ) - AM.OffsetVector] = 1;
01912                 term += 2;
01913             } while ( term < t );
01914         }
01915         else if ( *term == VECTOR ) {
01916             term += 2;
01917             do {
01918                 AN.UsedVector[( *term++ ) - AM.OffsetVector] = 1;
01919                 if ( *term >= AM.OffsetIndex && *term < AM.DumInd ) {
01920                     sym = indices[*term - AM.OffsetIndex].dimension;
01921                     if ( sym < 0 ) AN.UsedSymbol[-sym] = 1;
01922                     AN.UsedIndex[*term - AM.OffsetIndex] = 1;
01923                     sym = indices[( *term++ ) - AM.OffsetIndex].nmin4;
01924                     if ( sym < -NMIN4SHIFT ) AN.UsedSymbol[-sym-NMIN4SHIFT] = 1;
01925                 }
01926                 else term++;
01927             } while ( term < t );
01928         }
01929         else if ( *term == INDEX || *term == LEVICIVITA || *term == GAMMA
01930             || *term == DELTA ) {
01931             /*
01932             Tensors:
01933                 term += 2;
01934             */
01935             if ( *term == INDEX || *term == DELTA ) term += 2;
01936             else {
01937                 Tensors:
01938                 term += FUNHEAD;
01939             }
01940             while ( term < t ) {
01941                 if ( *term >= AM.OffsetIndex && *term < AM.DumInd ) {
01942                     sym = indices[*term - AM.OffsetIndex].dimension;

```

```

01943         if ( sym < 0 ) AN.UsedSymbol[-sym] = 1;
01944         AN.UsedIndex[( *term ) - AM.OffsetIndex] = 1;
01945         sym = indices[ *term-AM.OffsetIndex ].nmin4;
01946         if ( sym < -NMIN4SHIFT ) AN.UsedSymbol[-sym-NMIN4SHIFT] = 1;
01947     }
01948     else if ( *term < (WILDOFFSET+AM.OffsetVector) )
01949         AN.UsedVector[( *term ) - AM.OffsetVector] = 1;
01950     term++;
01951 }
01952 }
01953 else if ( *term == HAAKJE ) term = t;
01954 else {
01955     if ( *term > MAXBUILTINFUNCTION )
01956         AN.UsedFunction[( *term )-FUNCTION] = 1;
01957     if ( *term >= FUNCTION && functions[ *term-FUNCTION ].spec
01958         >= TENSORFUNCTION && term[1] > FUNHEAD ) goto Tensors;
01959     term += FUNHEAD;          /* First argument */
01960     while ( term < t ) {
01961         sarg = term;
01962         NEXTARG(sarg)
01963         if ( *term > 0 ) {
01964             sarg = term + *term;    /* End of argument */
01965             term += ARGHEAD;        /* First term in argument */
01966             if ( term < sarg ) { do {
01967                 DetVars(term,par);
01968                 term += *term;
01969             } while ( term < sarg ); }
01970         }
01971     }
01972     else {
01973         if ( *term < -MAXBUILTINFUNCTION ) {
01974             AN.UsedFunction[- *term-FUNCTION] = 1;
01975         }
01976         else if ( *term == -SYMBOL ) {
01977             AN.UsedSymbol[term[1]] = 1;
01978         }
01979         else if ( *term == -INDEX ) {
01980             if ( term[1] < (WILDOFFSET+AM.OffsetVector) ) {
01981                 AN.UsedVector[term[1]-AM.OffsetVector] = 1;
01982             }
01983             else if ( term[1] >= AM.OffsetIndex && term[1] < AM.DumInd ) {
01984                 sym = indices[term[1] - AM.OffsetIndex].dimension;
01985                 if ( sym < 0 ) AN.UsedSymbol[-sym] = 1;
01986                 AN.UsedIndex[term[1] - AM.OffsetIndex] = 1;
01987                 sym = indices[term[1]-AM.OffsetIndex].nmin4;
01988                 if ( sym < -NMIN4SHIFT ) AN.UsedSymbol[-sym-NMIN4SHIFT] = 1;
01989             }
01990         }
01991         else if ( *term == -VECTOR || *term == -MINVECTOR ) {
01992             AN.UsedVector[term[1]-AM.OffsetVector] = 1;
01993         }
01994     }
01995     term = sarg;          /* Next argument */
01996 }
01997 term = t;
01998 }
01999 }
02000 }
02001
02002 /*
02003     #] DetVars :
02004     #[ ToStorage :
02005
02006     This routine takes an expression in the scratch buffer (indicated by e)
02007     and puts it in the storage file. The necessary actions are:
02008
02009     1: determine the list of the used variables.
02010     2: make an index entry.
02011     3: write the namelists.
02012     4: copy the 'length' bytes of the expression.
02013
02014 */
02015
02016 int ToStorage(EXPRESSIONS e, POSITION *length)
02017 {
02018     GETIDENTITY
02019     WORD *w, i, j;
02020     WORD *term;
02021     INDEXENTRY *indexent;
02022     LONG size;
02023     POSITION indexpos, scrpos;
02024     FILEHANDLE *f;
02025     if ( ( indexent = NextFileIndex(&indexpos) ) == 0 ) {
02026         MesCall("ToStorage");
02027         SETERROR(-1)
02028     }
02029     indexent->CompressSize = 0;    /* thus far no compression */

```

```

02030     f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
02031     if ( e->status == HIDDENEXPRESSION ) {
02032         AR.InHiBuf = 0; f = AR.hidefile; AR.GetFile = 2;
02033     }
02034     else {
02035         AR.InInBuf = 0; f = AR.infile;   AR.GetFile = 0;
02036     }
02037     if ( f->handle >= 0 ) {
02038         scrpos = e->onfile;
02039         SeekFile(f->handle,&scrpos,SEEK_SET);
02040         if ( ISNOTEQUALPOS(scrpos,e->onfile) ) {
02041             MesPrint("::Error in Scratch file");
02042             goto ErrReturn;
02043         }
02044         f->POposition = e->onfile;
02045         f->POfull = f->PObuffer;
02046         if ( e->status == HIDDENEXPRESSION ) AR.InHiBuf = 0;
02047         else AR.InInBuf = 0;
02048     }
02049     else {
02050         f->POfill = (WORD *) ((UBYTE *) (f->PObuffer)+BASEPOSITION(e->onfile));
02051     }
02052     w = AT.WorkPointer;
02053     AN.UsedSymbol   = w;   w += NumSymbols;
02054     AN.UsedVector   = w;   w += NumVectors;
02055     AN.UsedIndex    = w;   w += NumIndices;
02056     AN.UsedFunction = w;   w += NumFunctions;
02057     term = w;
02058     w = (WORD *) ((UBYTE *) (w)) + AM.MaxTer;
02059     if ( w > AT.WorkTop ) {
02060         MesWork();
02061         goto ErrReturn;
02062     }
02063     w = AN.UsedSymbol;
02064     i = NumSymbols + NumVectors + NumIndices + NumFunctions;
02065     do { *w++ = 0; } while ( --i > 0 );
02066     if ( GetTerm(BHEAD term) > 0 ) {
02067         DetVars(term,1);
02068         if ( GetTerm(BHEAD term) ) {
02069             do { DetVars(term,0); } while ( GetTerm(BHEAD term) > 0 );
02070         }
02071     }
02072     j = 0;
02073     w = AN.UsedSymbol;
02074     i = NumSymbols;
02075     while ( --i >= 0 ) { if ( *w++ ) j++; }
02076     indexent->nsymbols = j;
02077     /* size = j * sizeof(struct SyMbOl); */
02078     j = 0;
02079     w = AN.UsedIndex;
02080     i = NumIndices;
02081     while ( --i >= 0 ) { if ( *w++ ) j++; }
02082     indexent->nindices = j;
02083     /* size += j * sizeof(struct InDeX); */
02084     j = 0;
02085     w = AN.UsedVector;
02086     i = NumVectors;
02087     while ( --i >= 0 ) { if ( *w++ ) j++; }
02088     indexent->nvectors = j;
02089     /* size += j * sizeof(struct VeCtOr); */
02090     j = 0;
02091     w = AN.UsedFunction;
02092     i = NumFunctions;
02093     while ( --i >= 0 ) { if ( *w++ ) j++; }
02094     indexent->nfunctions = j;
02095     /* size += j * sizeof(struct FuNcTiOn); */
02096     indexent->length = *length;
02097     indexent->variables = AR.StoreData.Fill;
02098     /* indexent->position = AR.StoreData.Fill + size; */
02099     StrCopy(AC.exprnames->namebuffer+e->name,(UBYTE *) (indexent->name));
02100     SeekFile(AR.StoreData.Handle,&(AR.StoreData.Fill),SEEK_SET);
02101     AO.wlen = 100000;
02102     AO.wpos = (UBYTE *) Malloc1(AO.wlen,"AO.wpos buffer");
02103     AO.wpoin = AO.wpos;
02104     {
02105         SYMBOLS a;
02106         w = AN.UsedSymbol;
02107         a = symbols;
02108         j = 0;
02109         i = indexent->nsymbols;
02110         while ( --i >= 0 ) {
02111             while ( !*w ) { w++; a++; j++; }
02112             a->number = j;
02113             if ( VarStore((UBYTE *) a,(WORD) (sizeof(struct SyMbOl)),a->name,
02114                 a->namesize) ) goto ErrToSto;
02115             w++; j++; a++;
02116         }

```

```

02117     }
02118     {
02119         INDICES a;
02120         w = AN.UsedIndex;
02121         a = indices;
02122         j = 0;
02123         i = indexent->nindices;
02124         while ( --i >= 0 ) {
02125             while ( !*w ) { w++; a++; j++; }
02126             a->number = j;
02127             if ( VarStore((UBYTE *)a, (WORD) (sizeof(struct InDeX)), a->name,
02128                 a->namesize) ) goto ErrToSto;
02129             w++; j++; a++;
02130         }
02131     }
02132     {
02133         VECTORS a;
02134         w = AN.UsedVector;
02135         a = vectors;
02136         j = 0;
02137         i = indexent->nvectors;
02138         while ( --i >= 0 ) {
02139             while ( !*w ) { w++; a++; j++; }
02140             a->number = j;
02141             if ( VarStore((UBYTE *)a, (WORD) (sizeof(struct VeCtOr)), a->name,
02142                 a->namesize) ) goto ErrToSto;
02143             w++; j++; a++;
02144         }
02145     }
02146     {
02147         FUNCTIONS a;
02148         w = AN.UsedFunction;
02149         a = functions;
02150         j = 0;
02151         i = indexent->nfunctions;
02152         while ( --i >= 0 ) {
02153             while ( !*w ) { w++; a++; j++; }
02154             a->number = j;
02155             if ( VarStore((UBYTE *)a, (WORD) (sizeof(struct FuNcTiOn)), a->name,
02156                 a->namesize) ) goto ErrToSto;
02157             w++; a++; j++;
02158         }
02159     }
02160     if ( VarStore((UBYTE *)0L, (WORD)0, (WORD)0, (WORD)0) ) goto ErrToSto; /* Flush buffer */
02161     TELLFILE(AR.StoreData.Handle, &(indexent->position));
02162     indexent->size = (WORD) DIFBASE(indexent->position, indexent->variables);
02163 /*
02164     The following code was added when it became apparent (30-jan-2007)
02165     that we need provisions for extra space without upsetting existing
02166     .sav files. Here we can put as much as we want.
02167     Look in GetTable on how to recover numdummies.
02168     Forgetting numdummies has been in there from the beginning.
02169 */
02170     if ( WriteFile(AR.StoreData.Handle, (UBYTE *) (&(e->numdummies)), (LONG) sizeof(WORD)) !=
02171         sizeof(WORD) ) {
02172         MesPrint("Error while writing storage file");
02173         goto ErrReturn;
02174     }
02175     if ( WriteFile(AR.StoreData.Handle, (UBYTE *) (&(e->numfactors)), (LONG) sizeof(WORD)) !=
02176         sizeof(WORD) ) {
02177         MesPrint("Error while writing storage file");
02178         goto ErrReturn;
02179     }
02180     if ( WriteFile(AR.StoreData.Handle, (UBYTE *) (&(e->vflags)), (LONG) sizeof(WORD)) !=
02181         sizeof(WORD) ) {
02182         MesPrint("Error while writing storage file");
02183         goto ErrReturn;
02184     }
02185     if ( WriteFile(AR.StoreData.Handle, (UBYTE *) (&(e->uflags)), (LONG) sizeof(WORD)) !=
02186         sizeof(WORD) ) {
02187         MesPrint("Error while writing storage file");
02188         goto ErrReturn;
02189     }
02190     TELLFILE(AR.StoreData.Handle, &(indexent->position));
02191     if ( f->handle >= 0 ) {
02192         POSITION llength;
02193         llength = *length;
02194         SeekFile(f->handle, &(e->onfile), SEEK_SET);
02195         while ( ISPOSPOS(llength) ) {
02196             SETBASEPOSITION(scrpos, AO.wlen);
02197             if ( ISLESSPOS(llength, scrpos) ) size = BASEPOSITION(llength);
02198             else size = AO.wlen;
02199             if ( ReadFile(f->handle, AO.wpos, size) != size ) {
02200                 MesPrint("Error while reading scratch file");
02201                 goto ErrReturn;
02202             }
02203             if ( WriteFile(AR.StoreData.Handle, AO.wpos, size) != size ) {

```



```

02204         MesPrint("Error while writing storage file");
02205         goto ErrReturn;
02206     }
02207     ADDPOS(llength,-size);
02208 }
02209 }
02210 else {
02211     WORD *ppp;
02212     ppp = (WORD *) ((UBYTE *) (f->PObuffer) + BASEPOSITION(e->onfile));
02213     if ( WriteFile(AR.StoreData.Handle, (UBYTE *) ppp, BASEPOSITION(*length)) !=
02214         BASEPOSITION(*length) ) {
02215         MesPrint("Error while writing storage file");
02216         goto ErrReturn;
02217     }
02218 }
02219 ADD2POS(*length, indexent->position);
02220 e->onfile = indexpos;
02221 /*
02222     AR.StoreData.Fill = SeekFile(AR.StoreData.Handle, &(AM.zeropos), SEEK_END);
02223 */
02224     AR.StoreData.Fill = *length;
02225     SeekFile(AR.StoreData.Handle, &(AR.StoreData.Fill), SEEK_SET);
02226     scrpos = AR.StoreData.Position;
02227     ADDPOS(scrpos, sizeof(POSITION));
02228     SeekFile(AR.StoreData.Handle, &scrpos, SEEK_SET);
02229     if ( WriteFile(AR.StoreData.Handle, ((UBYTE *) &(AR.StoreData.Index.number))
02230         , (LONG) (sizeof(POSITION))) != sizeof(POSITION) ) goto ErrInSto;
02231     SeekFile(AR.StoreData.Handle, &indexpos, SEEK_SET);
02232     if ( WriteFile(AR.StoreData.Handle, (UBYTE *) indexent, (LONG) (sizeof(INDEXENTRY))) !=
02233         sizeof(INDEXENTRY) ) goto ErrInSto;
02234     FlushFile(AR.StoreData.Handle);
02235     SeekFile(AR.StoreData.Handle, &(AC.StoreFileSize), SEEK_END);
02236     f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
02237     if ( AO.wpos ) M_free(AO.wpos, "AO.wpos buffer");
02238     AO.wpos = AO.wpoin = 0;
02239     return(0);
02240 ErrToSto:
02241     MesPrint("---Error while storing namelists");
02242     goto ErrReturn;
02243 ErrInSto:
02244     MesPrint("Error in storage");
02245 ErrReturn:
02246     if ( AO.wpos ) M_free(AO.wpos, "AO.wpos buffer");
02247     AO.wpos = AO.wpoin = 0;
02248     f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
02249     return(-1);
02250 }
02251
02252 /*
02253     #] ToStorage :
02254     #[ NextFileIndex :
02255 */
02256
02257 INDEXENTRY *NextFileIndex(POSITION *indexpos)
02258 {
02259     GETIDENTITY
02260     INDEXENTRY *ind;
02261     int i, j = sizeof(FILEINDEX) / (sizeof(LONG));
02262     LONG *lo;
02263     if ( AR.StoreData.Handle <= 0 ) {
02264         if ( SetFileIndex() ) {
02265             MesCall("NextFileIndex");
02266             return(0);
02267         }
02268         SETBASEPOSITION(AR.StoreData.Index.number, 1);
02269 #ifdef SYSDEPENDENTSAVE
02270         SETBASEPOSITION(*indexpos, (2*sizeof(POSITION)));
02271 #else
02272         SETBASEPOSITION(*indexpos, (2*sizeof(POSITION)+sizeof(STOREHEADER)));
02273 #endif
02274         return(AR.StoreData.Index.expression);
02275     }
02276     while ( BASEPOSITION(AR.StoreData.Index.number) >= (LONG) (INFILEINDEX) ) {
02277         if ( ISNOTZEROPOS(AR.StoreData.Index.next) ) {
02278             SeekFile(AR.StoreData.Handle, &(AR.StoreData.Index.next), SEEK_SET);
02279             AR.StoreData.Position = AR.StoreData.Index.next;
02280             if ( ReadFile(AR.StoreData.Handle, (UBYTE
02281 *) (&AR.StoreData.Index), (LONG) (sizeof(FILEINDEX))) !=
02282                 (LONG) (sizeof(FILEINDEX)) ) goto ErrNextS;
02283         }
02284         else {
02285             PUTZERO(AR.StoreData.Index.number);
02286             SeekFile(AR.StoreData.Handle, &(AR.StoreData.Position), SEEK_SET);
02287             if ( WriteFile(AR.StoreData.Handle, (UBYTE
02288 *) (&(AR.StoreData.Fill)), (LONG) (sizeof(POSITION)))
02289                 != (LONG) (sizeof(POSITION)) ) goto ErrNextS;
02288             PUTZERO(AR.StoreData.Index.next);

```

```

02289         SeekFile(AR.StoreData.Handle, &(AR.StoreData.Fill), SEEK_SET);
02290         AR.StoreData.Position = AR.StoreData.Fill;
02291         lo = (LONG *)(&AR.StoreData.Index);
02292         for ( i = 0; i < j; i++ ) *lo++ = 0;
02293         if ( WriteFile(AR.StoreData.Handle, (UBYTE
*)(&AR.StoreData.Index), (LONG)(sizeof(FILEINDEX))) !=
02294             (LONG)(sizeof(FILEINDEX)) ) goto ErrNextS;
02295         ADDPOS(AR.StoreData.Fill, sizeof(FILEINDEX));
02296     }
02297 }
02298 *indexpos = AR.StoreData.Position;
02299 ADDPOS(*indexpos, (2*sizeof(POSITION)) +
02300     BASEPOSITION(AR.StoreData.Index.number) * sizeof(INDEXENTRY));
02301 ind = &AR.StoreData.Index.expression[BASEPOSITION(AR.StoreData.Index.number)];
02302 ADDPOS(AR.StoreData.Index.number, 1);
02303 return(ind);
02304 ErrNextS:
02305     MesPrint("Error in storage file");
02306     return(0);
02307 }
02308
02309 /*
02310     #] NextFileIndex :
02311     #[ SetFileIndex :
02312 */
02313
02320 int SetFileIndex(void)
02321 {
02322     GETIDENTITY
02323     int i, j = sizeof(FILEINDEX)/(sizeof(LONG));
02324     LONG *lo;
02325     if ( AR.StoreData.Handle < 0 ) {
02326         AR.StoreData.Handle = AC.StoreHandle;
02327         PUTZERO(AR.StoreData.Index.next);
02328         PUTZERO(AR.StoreData.Index.number);
02329 #ifdef SYSDEPENDENTSAVE
02330         SETBASEPOSITION(AR.StoreData.Fill, sizeof(FILEINDEX));
02331 #else
02332         if ( WriteStoreHeader(AR.StoreData.Handle) ) return(MesPrint("Error writing storage file
header"));
02333         SETBASEPOSITION(AR.StoreData.Fill, (LONG)sizeof(FILEINDEX)+(LONG)sizeof(STOREHEADER));
02334 #endif
02335         lo = (LONG *)(&AR.StoreData.Index);
02336         for ( i = 0; i < j; i++ ) *lo++ = 0;
02337         if ( WriteFile(AR.StoreData.Handle, (UBYTE *)(&AR.StoreData.Index), (LONG)(sizeof(FILEINDEX))) !=
02338             (LONG)(sizeof(FILEINDEX)) ) return(MesPrint("Error writing storage file"));
02339     }
02340     else {
02341         POSITION scrpos;
02342 #ifdef SYSDEPENDENTSAVE
02343         PUTZERO(scrpos);
02344 #else
02345         SETBASEPOSITION(scrpos, (LONG)(sizeof(STOREHEADER)));
02346 #endif
02347         SeekFile(AR.StoreData.Handle, &scrpos, SEEK_SET);
02348         if ( ReadFile(AR.StoreData.Handle, (UBYTE *)(&AR.StoreData.Index), (LONG)(sizeof(FILEINDEX))) !=
02349             (LONG)(sizeof(FILEINDEX)) ) return(MesPrint("Error reading storage file"));
02350     }
02351 #ifdef SYSDEPENDENTSAVE
02352     PUTZERO(AR.StoreData.Position);
02353 #else
02354     SETBASEPOSITION(AR.StoreData.Position, (LONG)(sizeof(STOREHEADER)));
02355 #endif
02356     return(0);
02357 }
02358
02359 /*
02360     #] SetFileIndex :
02361     #[ VarStore :
02362
02363     The n -= sizeof(WORD); makes that the real length comes in the
02364     padding space, provided there is padding space (it seems so).
02365     The reading of the information assumes this is the case and hence
02366     things work....
02367 */
02368
02369 int VarStore(UBYTE *s, WORD n, WORD name, WORD namesize)
02370 {
02371     GETIDENTITY
02372     UBYTE *t, *u;
02373     if ( s ) {
02374         n -= sizeof(WORD);
02375         t = (UBYTE *)AO.wpoin;
02376     /*
02377         u = (UBYTE *)AT.WorkTop;
02378     */

```

```

02379     u = AO.wpos+AO.wlen;
02380     while ( n > 0 && t < u ) { *t++ = *s++; n--; }
02381     while ( t >= u ) {
02382         if ( WriteFile(AR.StoreData.Handle,AO.wpos,AO.wlen) != AO.wlen ) return(-1);
02383         t = AO.wpos;
02384         while ( n > 0 && t < u ) { *t++ = *s++; n--; }
02385     }
02386     s = AC.varnames->namebuffer + name;
02387     n = namesize;
02388     n += sizeof(void *)-1; n &= -(sizeof(void *));
02389     *((WORD *)t) = n;
02390     t += sizeof(WORD);
02391     while ( n > 0 && t < u ) {
02392         if ( namesize > 0 ) { *t++ = *s++; namesize--; }
02393         else { *t++ = 0; }
02394         n--;
02395     }
02396     while ( t >= u ) {
02397         if ( WriteFile(AR.StoreData.Handle,AO.wpos,AO.wlen) != AO.wlen ) return(-1);
02398         t = AO.wpos;
02399         while ( n > 0 && t < u ) {
02400             if ( namesize > 0 ) { *t++ = *s++; namesize--; }
02401             else { *t++ = 0; }
02402             n--;
02403         }
02404     }
02405     AO.wpoin = t;
02406 }
02407 else {
02408     LONG size;
02409     size = AO.wpoin - AO.wpos;
02410     if ( WriteFile(AR.StoreData.Handle,AO.wpos,size) != size ) return(-1);
02411     AO.wpoin = AO.wpos;
02412 }
02413 return(0);
02414 }
02415
02416 /*
02417     #] VarStore :
02418     #[ TermRenummer :
02419
02420     rennumbers the variables inside term according to the information
02421     in struct renumber.
02422     The search is binary. This avoided having to read/write the
02423     expression twice when it was stored.
02424
02425 */
02426
02427 int TermRenummer(WORD *term, RENUMBER renumber, WORD nexpr)
02428 {
02429     WORD *stopper;
02430     WORD *t, *sarg, n;
02431     stopper = term + *term - 1;
02432     stopper = stopper - ABS(*stopper) + 1;
02433     term++;
02434     while ( term < stopper ) {
02435         if ( *term == SYMBOL ) {
02436             t = term + term[1];
02437             term += 2;
02438             do {
02439                 if ( ( n = FindrNumber(*term,&(renumber->symp)) ) < 0 ) goto ErrR;
02440                 *term = renumber->symnum[n];
02441                 term += 2;
02442             } while ( term < t );
02443         }
02444         else if ( *term == DOTPRODUCT ) {
02445             t = term + term[1];
02446             term += 2;
02447             do {
02448                 if ( ( n = FindrNumber(*term,&(renumber->vect)) ) < 0 ) goto ErrR;
02449                 *term++ = renumber->vecnum[n];
02450                 if ( ( n = FindrNumber(*term,&(renumber->vect)) ) < 0 ) goto ErrR;
02451                 *term = renumber->vecnum[n];
02452                 term += 2;
02453             } while ( term < t );
02454         }
02455         else if ( *term == VECTOR ) {
02456             t = term + term[1];
02457             term += 2;
02458             do {
02459                 if ( ( n = FindrNumber(*term,&(renumber->vect)) ) < 0 ) goto ErrR;
02460                 *term++ = renumber->vecnum[n];
02461                 if ( ( *term >= AM.OffsetIndex ) && ( *term < AM.IndDum ) ) {
02462                     if ( ( n = FindrNumber(*term,&(renumber->indi)) ) < 0 ) goto ErrR;
02463                     *term = renumber->indnum[n];
02464                     term += 2;
02465                 }
02466             } while ( term < t );
02467         }
02468     }
02469 }

```

```

02473         < 0 ) goto ErrR;
02474         *term++ = renumber->indnum[n];
02475     }
02476     else term++;
02477 } while ( term < t );
02478 }
02479 else if ( *term == INDEX || *term == LEVICIVITA || *term == GAMMA
02480 || *term == DELTA ) {
02481 Tensors:
02482     t = term + term[1];
02483     if ( *term == INDEX || *term == DELTA ) term += 2;
02484     else term += FUNHEAD;
02485 /*
02486     term += 2;
02487 */
02488     while ( term < t ) {
02489         if ( *term >= AM.OffsetIndex + WILDOFFSET ) {
02490 /*
02491     Still TOBEDONE
02492 */
02493         }
02494         else if ( ( *term >= AM.OffsetIndex ) && ( *term < AM.IndDum ) ) {
02495             if ( ( n = FindrNumber(*term,&(renumber->indi)) )
02496                 < 0 ) goto ErrR;
02497             *term = renumber->indnum[n];
02498         }
02499         else if ( *term < (WILDOFFSET+AM.OffsetVector) ) {
02500             if ( ( n = FindrNumber(*term,&(renumber->vect)) )
02501                 < 0 ) goto ErrR;
02502             *term = renumber->vecnum[n];
02503         }
02504         term++;
02505     }
02506 }
02507 else if ( *term == HAAKJE ) term += term[1];
02508 else {
02509     if ( *term > MAXBUILTINFUNCTION ) {
02510         if ( ( n = FindrNumber(*term,&(renumber->func)) )
02511             < 0 ) goto ErrR;
02512         *term = renumber->funnum[n];
02513     }
02514     if ( *term >= FUNCTION && functions[*term-FUNCTION].spec
02515         >= TENSORFUNCTION && term[1] > FUNHEAD ) goto Tensors;
02516     t = term + term[1]; /* General stopper */
02517     term += FUNHEAD; /* First argument */
02518     while ( term < t ) {
02519         sarg = term;
02520         NEXTARG(sarg)
02521         if ( *term > 0 ) {
02522 /*
02523     Problem here:
02524     Marking the argument as dirty attacks the heap
02525     very heavily and costs much computer time.
02526 */
02527         *term = 1;
02528         term += ARGHEAD-1;
02529         while ( term < sarg ) {
02530             if ( TermRenumber(term,renumber,nexpr) ) goto ErrR;
02531             term += *term;
02532         }
02533     }
02534     else {
02535         if ( *term <= -MAXBUILTINFUNCTION ) {
02536             if ( ( n = FindrNumber(-*term,&(renumber->func)) )
02537                 < 0 ) goto ErrR;
02538             *term = -renumber->funnum[n];
02539         }
02540         else if ( *term == -SYMBOL ) {
02541             term++;
02542             if ( ( n = FindrNumber(*term,
02543                 &(renumber->symp)) ) < 0 ) goto ErrR;
02544             *term = renumber->symnum[n];
02545         }
02546         else if ( *term == -INDEX ) {
02547             term++;
02548             if ( *term >= AM.OffsetIndex + WILDOFFSET ) {
02549 /*
02550     Still TOBEDONE
02551 */
02552             }
02553             else if ( ( *term >= AM.OffsetIndex ) && ( *term < AM.IndDum ) ) {
02554                 if ( ( n = FindrNumber(*term,&(renumber->indi)) )
02555                     < 0 ) goto ErrR;
02556                 *term = renumber->indnum[n];
02557             }
02558             else if ( *term < (WILDOFFSET+AM.OffsetVector) ) {
02559                 if ( ( n = FindrNumber(*term,&(renumber->vect)) )

```

```

02560             < 0 ) goto ErrR;
02561             *term = renumber->vecnum[n];
02562         }
02563     }
02564     else if ( *term == -VECTOR || *term == -MINVECTOR ) {
02565         term++;
02566         if ( ( n = FindrNumber(*term,&(renumber->vect)) )
02567             < 0 ) goto ErrR;
02568         *term = renumber->vecnum[n];
02569     }
02570 }
02571 term = sarg;          /* Next argument */
02572 }
02573 term = t;
02574 }
02575 }
02576 return(0);
02577 ErrR:
02578 MesCall("TermRenumber");
02579 SETERROR(-1)
02580 }
02581
02582 /*
02583     #] TermRenumber :
02584     #[ FindrNumber :
02585 */
02586
02587 WORD FindrNumber(WORD n, VARRENUM *v)
02588 {
02589     WORD *hi,*med,*lo;
02590     hi = v->hi;
02591     lo = v->lo;
02592     med = v->start;
02593     if ( *hi == 0 ) {
02594         if ( n != *hi ) {
02595             MesPrint("Serious problems coming up in FindrNumber");
02596             return(-1);
02597         }
02598         return(*hi);
02599     }
02600     while ( *med != n ) {
02601         if ( *med < n ) {
02602             if ( med == hi ) goto ErrFindr;
02603             lo = med;
02604             med = hi - ((WORDDIFF(hi,med))/2);
02605         }
02606         else {
02607             if ( med == lo ) goto ErrFindr;
02608             hi = med;
02609             med = lo + ((WORDDIFF(med,lo))/2);
02610         }
02611     }
02612     return(WORDDIFF(med,v->lo));
02613 ErrFindr:
02614 /*
02615     Reconstruction:
02616 */
02617     {
02618         int i;
02619         i = WORDDIFF(v->hi,v->lo);
02620         MesPrint("FindrNumber: n = %d, list has %d members",n,i);
02621         while ( i >= 0 ) {
02622             MesPrint("v->lo[%d] = %d",i,v->lo[i]); i--;
02623         }
02624         hi = v->hi;
02625         lo = v->lo;
02626         med = v->start;
02627         MesPrint("Start with %d,%d,%d",0,WORDDIFF(med,v->lo),WORDDIFF(hi,v->lo));
02628         while ( *med != n ) {
02629             if ( *med < n ) {
02630                 if ( med == hi ) goto ErrFindr2;
02631                 lo = med;
02632                 med = hi - ((WORDDIFF(hi,med))/2);
02633             }
02634             else {
02635                 if ( med == lo ) goto ErrFindr2;
02636                 hi = med;
02637                 med = ((WORDDIFF(med,lo))/2) + lo;
02638             }
02639             MesPrint("New: %d,%d,%d, *med =
02640 %d",WORDDIFF(lo,v->lo),WORDDIFF(med,v->lo),WORDDIFF(hi,v->lo),*med);
02641         }
02642         return(WORDDIFF(med,v->lo));
02643 ErrFindr2:
02644         return(MesPrint("Renumbering problems"));
02645     }

```

```

02646
02647 /*
02648     #] FindrNumber :
02649     #[ FindInIndex :
02650
02651     Finds an expression in the storage index if it exists.
02652     If found it returns a pointer to the index entry, otherwise zero.
02653     par = 0      Search by address (--> f == &AR.StoreData, called by GetTable, CoSave )
02654     par = 1      Search by name      (--> f == &AO.SaveData, called by CoLoad )
02655
02656     When comparing parameter fields the parameters of the expression
02657     to be searched are in AT.TMaddr. This includes the primary expression
02658     and a possible FROMBRAC information. The FROMBRAC is always last.
02659
02660     The parameter mode tells whether we should worry about arguments of
02661     a stored expression.
02662 */
02663
02664 INDEXENTRY *FindInIndex(WORD expr, FILEDATA *f, WORD par, WORD mode)
02665 {
02666     GETIDENTITY
02667     INDEXENTRY *ind;
02668     WORD i, hand, *m;
02669     WORD *start, *stop, *stop2, *m2, nomatch = 0;
02670     POSITION stindex, indexpos, scrpos;
02671     LONG number, num;
02672     stindex = f->Position;
02673     m = AT.TMaddr;
02674     stop = m + m[1];
02675     m += SUBEXPSIZE;
02676     start = m;
02677     while ( m < stop ) {
02678         if ( *m == FROMBRAC || *m == WILDCARDS ) break;
02679         m += m[1];
02680     }
02681     stop = m;
02682     if ( !par ) hand = AR.StoreData.Handle;
02683     else hand = AO.SaveData.Handle;
02684     for(;;) {
02685         if ( ( i = (WORD)BASEPOSITION(f->Index.number) ) != 0 ) {
02686             indexpos = f->Position;
02687             ADDPOS(indexpos, (2*sizeof(POSITION)));
02688             ind = f->Index.expression;
02689             do {
02690                 if ( ( !par && ISEQUALPOS(indexpos, Expressions[expr].onfile) )
02691                     || ( par && !StrCmp(EXPRNAME(expr), (UBYTE *) (ind->name)) ) ) {
02692                     nomatch = 1;
02693                 }
02694                 MesPrint("index: position: %8p",&(ind->position));
02695                 MesPrint("index: length: %8p",&(ind->length));
02696                 MesPrint("index: variables: %8p",&(ind->variables));
02697                 MesPrint("index: nsymbols: %d",ind->nsymbols);
02698                 MesPrint("index: nindices: %d",ind->nindices);
02699                 MesPrint("index: nvectors: %d",ind->nvectors);
02700                 MesPrint("index: nfunctions: %d",ind->nfunctions);
02701                 MesPrint("index: size: %d",ind->size);
02702             }
02703             if ( par ) return(ind);
02704             scrpos = ind->position;
02705             SeekFile(hand,&scrpos,SEEK_SET);
02706             if ( ISNOTEQUALPOS(scrpos,ind->position) ) goto ErrGt2;
02707             if ( ReadFile(hand, (UBYTE *)AT.WorkPointer, (LONG)sizeof(WORD)) !=
02708                 sizeof(WORD) || !*AT.WorkPointer ) goto ErrGt2;
02709             num = *AT.WorkPointer - 1;
02710             num *= wsizeof(WORD);
02711             if ( *AT.WorkPointer < 0 ||
02712                 ReadFile(hand, (UBYTE *) (AT.WorkPointer+1), num) != num ) goto ErrGt2;
02713             m = start; /* start of parameter field to be searched */
02714             m2 = AT.WorkPointer + 1;
02715             stop2 = m2 + m2[1];
02716             m2 += SUBEXPSIZE;
02717             while ( m < stop && m2 < stop2 ) {
02718                 if ( *m == SYMBOL ) {
02719                     if ( *m2 != SYMTOSYM ) break;
02720                     m2[3] = m[2];
02721                 }
02722                 else if ( *m == INDEX ) {
02723                     if ( m[2] >= 0 ) {
02724                         if ( *m2 != INDTOIND ) break;
02725                     }
02726                     else {
02727                         if ( *m2 != VECTOVEC ) break;
02728                     }
02729                     m2[3] = m[2];
02730                 }
02731                 else if ( *m >= FUNCTION ) {
02732                     if ( *m2 != FUNTOFUN ) break;
02733                 }

```

```

02733             m2[3] = *m;
02734         }
02735         else {}
02736         m += m[1];
02737         m2 += m2[1];
02738     }
02739     if ( ( m >= stop && m2 >= stop2 ) || mode == 0 ) {
02740         AT.WorkPointer = stop2;
02741
02742         return(ind);
02743     }
02744 }
02745 ind++;
02746 ADDPOS(indexpos, sizeof(INDEXENTRY));
02747 } while ( --i > 0 );
02748 }
02749 f->Position = f->Index.next;
02750 #ifndef SYSDEPENDENTSAVE
02751 if ( !ISNOTZEROPOS(f->Position) ) ADDPOS(f->Position, sizeof(STOREHEADER));
02752 number = sizeof(struct FileInDeX);
02753 #endif
02754 if ( ISEQUALPOS(f->Position, stindex) && !AO.bufferedInd ) goto ErrGetTab;
02755 if ( !par ) {
02756     SeekFile(AR.StoreData.Handle, &(f->Position), SEEK_SET);
02757     if ( ISNOTEQUALPOS(f->Position, AR.StoreData.Position) ) goto ErrGt2;
02758 #ifndef SYSDEPENDENTSAVE
02759     if ( ReadFile(f->Handle, (UBYTE *)(&(f->Index)), number) != number ) goto ErrGt2;
02760 #endif
02761 }
02762 else {
02763     SeekFile(AO.SaveData.Handle, &(f->Position), SEEK_SET);
02764     if ( ISNOTEQUALPOS(f->Position, AO.SaveData.Position) ) goto ErrGt2;
02765 #ifndef SYSDEPENDENTSAVE
02766     if ( ReadSaveIndex(&(f->Index)) ) goto ErrGt2;
02767 #endif
02768 }
02769 #ifndef SYSDEPENDENTSAVE
02770 number = sizeof(struct FileInDeX);
02771 if ( ReadFile(f->Handle, (UBYTE *)(&(f->Index)), number) !=
02772     number ) goto ErrGt2;
02773 #endif
02774 }
02775 ErrGetTab:
02776 if ( nomatch ) {
02777     MesPrint("Parameters of expression %s don't match."
02778         , EXPRNAME(expr));
02779 }
02780 else {
02781     MesPrint("Cannot find expression %s", EXPRNAME(expr));
02782 }
02783 return(0);
02784 ErrGt2:
02785 MesPrint("Readerror in IndexSearch");
02786 return(0);
02787 }
02788
02789 /*
02790 #] FindInIndex :
02791 #[ GetTable :
02792
02793 Locates stored files and constructs the renumbering tables.
02794 They are allocated in the Workspace.
02795 First the expression data are located. The Index is treated
02796 as a circularly linked buffer which is paged forwardly.
02797 If the indexentry is located (in ind) the two renumber tables
02798 have to be constructed.
02799 Finally the prototype has to be put in the proper buffer, so
02800 that wildcards can be passed. There should be a test with
02801 an already existing prototype that is constructed by the
02802 pattern matcher. This has not been put in yet.
02803
02804 There is a problem with the parallel processing.
02805 Feeding in the variables that were erased by a .store could in
02806 principle happen in different orders (ParFORM) or simultaneously
02807 (TFORM). The proper resolution is to have the compiler call GetTable
02808 when a stored expression is encountered.
02809
02810 This has been mended in development of TFORM by reading the
02811 symbol tables during compilation. See the call to GetTable
02812 in the CodeGenerator.
02813
02814 Next is the problem of FindInIndex which writes in AR.StoreData
02815 Copying this is expensive!
02816
02817 This Doesn't work well for TFORM yet!!!!!!!
02818 e[x1,x2] versus e[x2,x1] messes up.
02819 For the rest is the reloading during execution not thread safe.

```

```

02820
02821     The parameter mode tells whether we should worry about arguments
02822     of a stored expression.
02823 */
02824
02825 RENUMBER GetTable(WORD expr, POSITION *position, WORD mode)
02826 {
02827     GETIDENTITY
02828     WORD i, j;
02829     WORD *w;
02830     RENUMBER r;
02831     LONG num, nsize, xx;
02832     WORD jsym, jind, jvec, jfun;
02833     WORD k, type, error = 0, *oldw, *neww, *oldwork = AT.WorkPointer;
02834     struct SyMbOl SyM;
02835     struct InDeX InD;
02836     struct VeCtOr VeC;
02837     struct FuNcTiOn FuN;
02838     INDEXENTRY *ind;
02839 /*
02840     Prepare for FindInIndex to put the prototype in the Workspace.
02841     oldw will point at the "wildcards"
02842 */
02843 /*
02844     Bug fix. Look also in Generator.
02845 #ifndef WITHPTHREADS
02846
02847     if ( ( r = Expressions[expr].renum ) != 0 ) { }
02848     else {
02849         Expressions[expr].renum =
02850         r = (RENUMBER)Mallocl(sizeof(struct ReNuMbEr), "Renumber");
02851     }
02852 #else
02853     r = (RENUMBER)Mallocl(sizeof(struct ReNuMbEr), "Renumber");
02854 #endif
02855 */
02856     r = (RENUMBER)Mallocl(sizeof(struct ReNuMbEr), "Renumber");
02857
02858     oldw = AT.WorkPointer + 1 + SUBEXPSIZE;
02859 /*
02860     The prototype is loaded in the Workspace by the Index routine.
02861     After all it has to find an occurrence with the proper arguments.
02862     This sets the WorkPointer. Hence be careful now.
02863 */
02864     LOCK(AM.storefilelock);
02865     if ( ( ind = FindInIndex(expr, &AR.StoreData, 0, mode) ) == 0 ) {
02866         UNLOCK(AM.storefilelock);
02867         return(0);
02868     }
02869
02870     xx = ind->nsymbols+ind->nindices+ind->nvectors+ind->nfunctions;
02871     if ( xx == 0 ) {
02872         Expressions[expr].renumlists =
02873         w = AN.dummyrenumlist;
02874     }
02875     else {
02876 /*
02877 #ifndef WITHPTHREADS
02878     Expressions[expr].renumlists =
02879 #endif
02880 */
02881         w = (WORD *)Mallocl(sizeof(WORD)*(xx*2), "VarSpace");
02882     }
02883     r->symb.lo = w;
02884     r->symb.start = w + ind->nsymbols/2;
02885     w += ind->nsymbols;
02886     r->symb.hi = w - 1;
02887     r->symnum = w;
02888     w += ind->nsymbols;
02889
02890     r->indi.lo = w;
02891     r->indi.start = w + ind->nindices/2;
02892     w += ind->nindices;
02893     r->indi.hi = w - 1;
02894     r->indnum = w;
02895     w += ind->nindices;
02896
02897     r->vect.lo = w;
02898     r->vect.start = w + ind->nvectors/2;
02899     w += ind->nvectors;
02900     r->vect.hi = w - 1;
02901     r->vecnum = w;
02902     w += ind->nvectors;
02903
02904     r->func.lo = w;
02905     r->func.start = w + ind->nfunctions/2;
02906     w += ind->nfunctions;

```



```

02907     r->func.hi = w - 1;
02908     r->funnum = w;
02909     /* w += ind->nfunctions; */
02910
02911     SeekFile(AR.StoreData.Handle,&(ind->variables),SEEK_SET);
02912     *position = ind->position;
02913     jsym = ind->nsymbols;
02914     jvec = ind->nvectors;
02915     jind = ind->nindices;
02916     jfun = ind->nfunctions;
02917     /*
02918         # Symbols :
02919     */
02920     {
02921         SYMBOLS s = &SyM;
02922         w = r->symb.lo; j = jsym;
02923         for ( i = 0; i < j; i++ ) {
02924             if ( ReadFile(AR.StoreData.Handle, (UBYTE *)s, (LONG) (sizeof(struct SyMbOl)))
02925                 != sizeof(struct SyMbOl) ) goto ErrGt2;
02926             nsize = s->namesize; nsize += sizeof(void *)-1;
02927             nsize &= -sizeof(void *);
02928             if ( ReadFile(AR.StoreData.Handle, (UBYTE *) (AT.WorkPointer), nsize)
02929                 != nsize ) goto ErrGt2;
02930             *w = s->number;
02931             if ( ( s->flags & INUSE ) != 0 ) {
02932                 /* Find the replacement. It must exist! */
02933                 neww = oldw;
02934                 while ( *neww != SYMTOSYM || neww[2] != *w ) neww += neww[1];
02935                 k = neww[3];
02936             }
02937             else if ( GetVar((UBYTE *)AT.WorkPointer,&type,&k,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
02938                 if ( type != CSYMBOL ) {
02939                     MesPrint("Error: Conflicting types for %s", (AT.WorkPointer));
02940                     error = -1;
02941                 }
02942                 else {
02943                     if ( ( s->complex & (VARTYPEIMAGINARY|VARTYPECOMPLEX) ) !=
02944                         ( symbols[k].complex & (VARTYPEIMAGINARY|VARTYPECOMPLEX) ) ) {
02945                         MesPrint("Warning: Conflicting complexity for %s", AT.WorkPointer);
02946                         error = -1;
02947                     }
02948                     if ( ( s->complex & (VARTYPEROOTOFUNITY) ) !=
02949                         ( symbols[k].complex & (VARTYPEROOTOFUNITY) ) ) {
02950                         MesPrint("Warning: Conflicting root of unity properties for %s", AT.WorkPointer);
02951                         error = -1;
02952                     }
02953                     if ( ( s->complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY ) {
02954                         if ( s->maxpower != symbols[k].maxpower ) {
02955                             MesPrint("Warning: Conflicting n in n-th root of unity properties for
02956 %s", AT.WorkPointer);
02957                             error = -1;
02958                         }
02959                     }
02960                     else if ( ( s->minpower !=
02961                         symbols[k].minpower || s->maxpower !=
02962                         symbols[k].maxpower ) && AC.WarnFlag ) {
02963                         MesPrint("Warning: Conflicting power restrictions for %s", AT.WorkPointer);
02964                     }
02965                 }
02966             }
02967             else {
02968                 if ( ( k = EntVar(CSYMBOL, (UBYTE *) (AT.WorkPointer), s->complex, s->minpower,
02969                     s->maxpower, s->dimension) ) < 0 ) goto GetTcall;
02970             }
02971             *(w+j) = k;
02972             w++;
02973         }
02974     /*
02975         # Symbols :
02976         # Indices :
02977     */
02978     {
02979         INDICES s = &InD;
02980         w = r->indi.lo; j = jind;
02981         for ( i = 0; i < j; i++ ) {
02982             if ( ReadFile(AR.StoreData.Handle, (UBYTE *)s, (LONG) (sizeof(struct InDeX)))
02983                 != sizeof(struct InDeX) ) goto ErrGt2;
02984             nsize = s->namesize; nsize += sizeof(void *)-1;
02985             nsize &= -sizeof(void *);
02986             if ( ReadFile(AR.StoreData.Handle, (UBYTE *) (AT.WorkPointer), nsize)
02987                 != nsize ) goto ErrGt2;
02988             *w = s->number + AM.OffsetIndex;
02989             if ( s->dimension < 0 ) { /* Relabel the dimension */
02990                 s->dimension = -r->symnum[FindrNumber(-s->dimension,&(r->symb))];
02991                 if ( s->nmin4 < -NMIN4SHIFT ) { /* Relabel n-4 */
02992                     s->nmin4 = -r->symnum[FindrNumber(-s->nmin4-NMIN4SHIFT

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02993         , &(r->symbol))] -NMIN4SHIFT;
02994     }
02995 }
02996 if ( ( s->flags & INUSE ) != 0 ) {
02997     /* Find the replacement. It must exist! */
02998     neww = oldw;
02999     while ( *neww != INDTOIND || neww[2] != *w ) neww += neww[1];
03000     k = neww[3] - AM.OffsetIndex;
03001 }
03002 else if ( s->type == DUMMY ) {
03003     /*
03004     ----->          Here we may have to execute some renumbering
03005     */
03006 }
03007 else if ( GetVar((UBYTE *) (AT.WorkPointer), &type, &k, ALLVARIABLES, NOAUTO) != NAMENOTFOUND ) {
03008     if ( type != CINDEX ) {
03009         MesPrint("Error: Conflicting types for %s", (AT.WorkPointer));
03010         error = -1;
03011     }
03012     else {
03013         if ( s->type !=
03014             indices[k].type ) {
03015             MesPrint("Warning: %s is also a dummy index", (AT.WorkPointer));
03016             error = -1;
03017             goto GetTb3;
03018         }
03019         if ( s->dimension != indices[k].dimension ) {
03020             MesPrint("Warning: Conflicting dimensions for %s", (AT.WorkPointer));
03021             error = -1;
03022         }
03023     }
03024 }
03025 else {
03026 GetTb3:
03027     if ( ( k = EntVar(CINDEX, (UBYTE *) (AT.WorkPointer),
03028         s->dimension, 0, s->nmin4, 0) ) < 0 ) goto GetTcall;
03029 }
03030 * (w+j) = k + AM.OffsetIndex;
03031 w++;
03032 }
03033 }
03034 }
03035 /*
03036     #] Indices :
03037     #[ Vectors :
03038 */
03039 {
03040     VECTORS s = &Vec;
03041     w = r->vect.lo; j = jvec;
03042     for ( i = 0; i < j; i++ ) {
03043         if ( ReadFile(AR.StoreData.Handle, (UBYTE *) s, (LONG) (sizeof(struct VecCtOr)))
03044             != sizeof(struct VecCtOr) ) goto ErrGt2;
03045         nsize = s->namesize; nsize += sizeof(void *) - 1;
03046         nsize &= -sizeof(void *);
03047         if ( ReadFile(AR.StoreData.Handle, (UBYTE *) (AT.WorkPointer), nsize)
03048             != nsize ) goto ErrGt2;
03049         *w = s->number + AM.OffsetVector;
03050         if ( ( s->flags & INUSE ) != 0 ) {
03051             /* Find the replacement. It must exist! */
03052             neww = oldw;
03053             while ( *neww != VECTOVEC || neww[2] != *w ) neww += neww[1];
03054             k = neww[3] - AM.OffsetVector;
03055         }
03056         else if ( GetVar((UBYTE *) (AT.WorkPointer), &type, &k, ALLVARIABLES, NOAUTO) != NAMENOTFOUND ) {
03057             if ( type != CVECTOR ) {
03058                 MesPrint("Error: Conflicting types for %s", (AT.WorkPointer));
03059                 error = -1;
03060             }
03061             else {
03062                 if ( ( s->complex & (VARTYPEIMAGINARY|VARTYPECOMPLEX) ) !=
03063                     ( vectors[k].complex & (VARTYPEIMAGINARY|VARTYPECOMPLEX) ) ) {
03064                     MesPrint("Warning: Conflicting complexity for %s", (AT.WorkPointer));
03065                     error = -1;
03066                 }
03067             }
03068         }
03069         else {
03070             if ( ( k = EntVar(CVECTOR, (UBYTE *) (AT.WorkPointer),
03071                 s->complex, 0, 0, s->dimension) ) < 0 ) goto GetTcall;
03072         }
03073         * (w+j) = k + AM.OffsetVector;
03074         w++;
03075     }
03076 }
03077 /*
03078     #] Vectors :
03079     #[ Functions :

```

```

03080 */
03081 {
03082     FUNCTIONS s = &FuN;
03083     w = r->func.lo; j = jfun;
03084     for ( i = 0; i < j; i++ ) {
03085         if ( ReadFile(AR.StoreData.Handle, (UBYTE *)s, (LONG) (sizeof(struct FuNctiOn))
03086             != sizeof(struct FuNctiOn) ) goto ErrGt2;
03087         nsize = s->namesize; nsize += sizeof(void *)-1;
03088         nsize &= -sizeof(void *);
03089         if ( ReadFile(AR.StoreData.Handle, (UBYTE *) (AT.WorkPointer), nsize)
03090             != nsize ) goto ErrGt2;
03091         *w = s->number + FUNCTION;
03092         if ( ( s->flags & INUSE ) != 0 ) {
03093             /* Find the replacement. It must exist! */
03094             neww = oldw;
03095             while ( *neww != FUNTOFUN || neww[2] != *w ) neww += neww[1];
03096             k = neww[3] - FUNCTION;
03097         }
03098         else if ( GetVar((UBYTE *) (AT.WorkPointer), &type, &k, ALLVARIABLES, NOAUTO) != NAMENOTFOUND ) {
03099             if ( type != CFUNCTION ) {
03100                 MesPrint("Error: Conflicting types for %s", (AT.WorkPointer));
03101                 error = -1;
03102             }
03103             else {
03104                 if ( s->complex != functions[k].complex ) {
03105                     MesPrint("Warning: Conflicting complexity for %s", (AT.WorkPointer));
03106                     error = -1;
03107                 }
03108                 else if ( s->symmetric != functions[k].symmetric ) {
03109                     MesPrint("Warning: Conflicting symmetry properties for %s", (AT.WorkPointer));
03110                     error = -1;
03111                 }
03112                 else if ( ( s->maxnumargs != functions[k].maxnumargs )
03113                     || ( s->minnumargs != functions[k].minnumargs ) ) {
03114                     MesPrint("Warning: Conflicting argument restriction properties for
03115 %s", (AT.WorkPointer));
03116                     error = -1;
03117                 }
03118             }
03119             else {
03120                 if ( ( k = EntVar(CFUNCTION, (UBYTE *) (AT.WorkPointer),
03121                     s->complex, s->commute, s->spec, s->dimension) ) < 0 ) goto GetTcall;
03122                 functions[k].symmetric = s->symmetric;
03123                 functions[k].maxnumargs = s->maxnumargs;
03124                 functions[k].minnumargs = s->minnumargs;
03125             }
03126             * (w+j) = k + FUNCTION;
03127             w++;
03128         }
03129     }
03130     /*
03131         #] Functions :
03132
03133     Now we skip the prototype. This sets the start position at the first term
03134 */
03135     if ( error ) {
03136         UNLOCK(AM.storefilelock);
03137         AT.WorkPointer = oldwork;
03138         return(0);
03139     }
03140     {
03141         /*
03142             For clarity we look where we are.
03143             We want to know: is this position already known?
03144             Could we have inserted extra information here?
03145
03146             nummystery indicates extra words. We have currently in order
03147             (if they exist)
03148                 numdummies
03149                 numfactors
03150                 vflags
03151                 uflags
03152 */
03153         POSITION pos;
03154         int nummystery;
03155         TELLFILE(AR.StoreData.Handle, &pos);
03156         nummystery = DIFBASE(ind->position, pos);
03157         /*
03158             MesPrint("---> We are at position      %8p", &pos);
03159             MesPrint("---> The index says        at %8p", &(ind->position));
03160             MesPrint("---> There are %d mystery bytes", nummystery);
03161 */
03162         if ( nummystery > 0 ) {
03163             if ( ReadFile(AR.StoreData.Handle, (UBYTE *) AT.WorkPointer, (LONG) sizeof(WORD)) !=
03164                 sizeof(WORD) ) {

```

```

03166         UNLOCK(AM.storefilelock);
03167         AT.WorkPointer = oldwork;
03168         return(0);
03169     }
03170     Expressions[expr].numdummies = *AT.WorkPointer;
03171     /*
03172     MesPrint("--> numdummies = %d",Expressions[expr].numdummies);
03173     */
03174     nummystery -= sizeof(WORD);
03175 }
03176 else {
03177     Expressions[expr].numdummies = 0;
03178 }
03179 if ( nummystery > 0 ) {
03180     if ( ReadFile(AR.StoreData.Handle, (UBYTE *)AT.WorkPointer, (LONG)sizeof(WORD)) !=
03181         sizeof(WORD) ) {
03182         UNLOCK(AM.storefilelock);
03183         AT.WorkPointer = oldwork;
03184         return(0);
03185     }
03186     if ( ( AS.OldNumFactors == 0 ) || ( AS.NumOldNumFactors < NumExpressions ) ) {
03187         WORD *buffer;
03188         int capacity = 20;
03189         if (capacity < NumExpressions) capacity = NumExpressions * 2;
03190
03191         buffer = (WORD *)Malloc1(capacity * sizeof(WORD), "numfactors pointers");
03192         if (AS.OldNumFactors) {
03193             WCOPY(buffer, AS.OldNumFactors, AS.NumOldNumFactors);
03194             M_free(AS.OldNumFactors, "numfactors pointers");
03195         }
03196         AS.OldNumFactors = buffer;
03197
03198         buffer = (WORD *)Malloc1(capacity * sizeof(WORD), "vflags pointers");
03199         if (AS.Oldvflags) {
03200             WCOPY(buffer, AS.Oldvflags, AS.NumOldNumFactors);
03201             M_free(AS.Oldvflags, "vflags pointers");
03202         }
03203         AS.Oldvflags = buffer;
03204
03205         buffer = (WORD *)Malloc1(capacity * sizeof(WORD), "uflags pointers");
03206         if (AS.Olduflags) {
03207             WCOPY(buffer, AS.Olduflags, AS.NumOldNumFactors);
03208             M_free(AS.Olduflags, "uflags pointers");
03209         }
03210         AS.Oldvflags = buffer;
03211
03212         AS.NumOldNumFactors = capacity;
03213     }
03214
03215     AS.OldNumFactors[expr] =
03216     Expressions[expr].numfactors = *AT.WorkPointer;
03217     /*
03218     MesPrint("--> numfactors = %d",Expressions[expr].numfactors);
03219     */
03220     nummystery -= sizeof(WORD);
03221 }
03222 else {
03223     Expressions[expr].numfactors = 0;
03224 }
03225 if ( nummystery > 0 ) {
03226     if ( ReadFile(AR.StoreData.Handle, (UBYTE *)AT.WorkPointer, (LONG)sizeof(WORD)) !=
03227         sizeof(WORD) ) {
03228         UNLOCK(AM.storefilelock);
03229         AT.WorkPointer = oldwork;
03230         return(0);
03231     }
03232     AS.Oldvflags[expr] =
03233     Expressions[expr].vflags = *AT.WorkPointer;
03234     /*
03235     MesPrint("--> vflags = %d",Expressions[expr].vflags);
03236     */
03237     nummystery -= sizeof(WORD);
03238 }
03239 else {
03240     Expressions[expr].vflags = 0;
03241 }
03242 if ( nummystery > 0 ) {
03243     if ( ReadFile(AR.StoreData.Handle, (UBYTE *)AT.WorkPointer, (LONG)sizeof(WORD)) !=
03244         sizeof(WORD) ) {
03245         UNLOCK(AM.storefilelock);
03246         AT.WorkPointer = oldwork;
03247         return(0);
03248     }
03249     AS.Olduflags[expr] =
03250     Expressions[expr].uflags = *AT.WorkPointer;
03251     /*
03252     MesPrint("--> uflags = %d",Expressions[expr].uflags);

```

```

03253 */
03254         nummystery -= sizeof(WORD);
03255     }
03256     else {
03257         Expressions[expr].uflags = 0;
03258     }
03259 }
03260
03261 SeekFile(AR.StoreData.Handle,&(ind->position),SEEK_SET);
03262 if ( ReadFile(AR.StoreData.Handle,(UBYTE *)AT.WorkPointer,(LONG)sizeof(WORD)) !=
03263 sizeof(WORD) || !*AT.WorkPointer ) {
03264     UNLOCK(AM.storefilelock);
03265     AT.WorkPointer = oldwork;
03266     return(0);
03267 }
03268 num = *AT.WorkPointer - 1;
03269 num *= sizeof(WORD);
03270 if ( *AT.WorkPointer < 0 ||
03271 ReadFile(AR.StoreData.Handle,(UBYTE *) (AT.WorkPointer+1),num) != num ) {
03272     MesPrint("@Error in stored expressions file at position %10p",*position);
03273     UNLOCK(AM.storefilelock);
03274     AT.WorkPointer = oldwork;
03275     return(0);
03276 }
03277 UNLOCK(AM.storefilelock);
03278 ADDPOS(*position,num+sizeof(WORD));
03279 r->startposition = *position;
03280 AT.WorkPointer = oldwork;
03281 return(r);
03282 GetTcall:
03283 UNLOCK(AM.storefilelock);
03284 AT.WorkPointer = oldwork;
03285 MesCall("GetTable");
03286 return(0);
03287 ErrGt2:
03288 UNLOCK(AM.storefilelock);
03289 AT.WorkPointer = oldwork;
03290 MesPrint("Readererror in GetTable");
03291 return(0);
03292 }
03293
03294 /*
03295     #] GetTable :
03296     #[ CopyExpression :
03297
03298     Copies from one scratch buffer to another.
03299     We assume here that the complete 'from' scratch buffer is taken.
03300     We also assume that the 'from' buffer is positioned at the end of
03301     the expression.
03302
03303     The locks should be placed in the calling routine. We need basically
03304     AS.outputslock.
03305 */
03306
03307 int CopyExpression(FILEHANDLE *from, FILEHANDLE *to)
03308 {
03309     POSITION posfrom, poscopy;
03310     LONG fullsize,i;
03311     WORD *t1, *t2;
03312     int RetCode;
03313     SeekScratch(from,&posfrom);
03314     if ( from->handle < 0 ) { /* input is in memory */
03315         fullsize = (BASEPOSITION(posfrom))/sizeof(WORD);
03316         if ( ( to->POstop - to->POfull ) >= fullsize ) {
03317             /*
03318              Fits inside the buffer of the output. This will be fast.
03319             */
03320             t1 = from->PObuffer;
03321             t2 = to->POfull;
03322             NCOPY(t2,t1,fullsize)
03323             to->POfull = to->POfill = t2;
03324             goto WriteTrailer;
03325         }
03326         if ( to->handle < 0 ) { /* First open the file */
03327             if ( ( RetCode = CreateFile(to->name) ) >= 0 ) {
03328                 to->handle = (WORD)RetCode;
03329                 PUTZERO(to->filesize);
03330                 PUTZERO(to->POposition);
03331             }
03332             else {
03333                 MLOCK(ErrorMessageLock);
03334                 MesPrint("Cannot create scratch file %s",to->name);
03335                 MUNLOCK(ErrorMessageLock);
03336                 return(-1);
03337             }
03338         }
03339         t1 = from->PObuffer;

```

```

03340     while ( fullsize > 0 ) {
03341         i = to->PPostop - to->POfull;
03342         if ( i > fullsize ) i = fullsize;
03343         fullsize -= i;
03344         t2 = to->POfull;
03345         NCOPY(t2,t1,i)
03346         if ( fullsize > 0 ) {
03347             SeekFile(to->handle,&(to->POposition),SEEK_SET);
03348             if ( WriteFile(to->handle,((UBYTE *) (to->PObuffer)),to->POsize) != to->POsize ) {
03349                 MLOCK(ErrorMessageLock);
03350                 MesPrint("Error while writing to disk. Disk full?");
03351                 MUNLOCK(ErrorMessageLock);
03352                 return(-1);
03353             }
03354             ADDPOS(to->POposition,to->POsize);
03355             /* SeekFile(to->handle,&(to->POposition),SEEK_CUR); */
03356             to->filesize = to->POposition;
03357             to->POfill = to->POfull = to->PObuffer;
03358         }
03359         else {
03360             to->POfill = to->POfull = t2;
03361         }
03362     }
03363     goto WriteTrailer;
03364 }
03365 /*
03366 Now the input involves a file. This needs the use of the PObuffer of from.
03367 First make sure the tail of the buffer has been written
03368 */
03369 if ( ((UBYTE *) (from->POfill))-((UBYTE *) (from->PObuffer)) > 0 ) {
03370     if ( WriteFile(from->handle,((UBYTE *) (from->PObuffer)),((UBYTE *) (from->POfill))-((UBYTE
03371 *) (from->PObuffer)))
03372     != ((UBYTE *) (from->POfill))-((UBYTE *) (from->PObuffer)) ) {
03373         MLOCK(ErrorMessageLock);
03374         MesPrint("Error while writing to disk. Disk full?");
03375         MUNLOCK(ErrorMessageLock);
03376         return(-1);
03377     }
03378     SeekFile(from->handle,&(from->POposition),SEEK_CUR);
03379     posfrom = from->filesize = from->POposition;
03380     from->POfill = from->POfull = from->PObuffer;
03381 }
03382 /*
03383 Now copy the complete contents
03384 */
03385 PUTZERO(poscopy);
03386 SeekFile(from->handle,&poscopy,SEEK_SET);
03387 while ( ISLESSPOS(poscopy,posfrom) ) {
03388     fullsize = ReadFile(from->handle,((UBYTE *) (from->PObuffer)),from->POsize);
03389     if ( fullsize < 0 || ( fullsize % sizeof(WORD) ) != 0 ) {
03390         MLOCK(ErrorMessageLock);
03391         MesPrint("Error while reading from disk while copying expression.");
03392         MUNLOCK(ErrorMessageLock);
03393         return(-1);
03394     }
03395     fullsize /= sizeof(WORD);
03396     from->POfull = from->PObuffer + fullsize;
03397     t1 = from->PObuffer;
03398     if ( ( to->PPostop - to->POfull ) >= fullsize ) {
03399         /*
03400 Fits inside the buffer of the output. This will be fast.
03401 */
03402         t2 = to->POfull;
03403         NCOPY(t2,t1,fullsize)
03404         to->POfill = to->POfull = t2;
03405     }
03406     else {
03407         if ( to->handle < 0 ) { /* First open the file */
03408             if ( ( RetCode = CreateFile(to->name) ) >= 0 ) {
03409                 to->handle = (WORD)RetCode;
03410                 PUTZERO(to->POposition);
03411                 PUTZERO(to->filesize);
03412             }
03413             else {
03414                 MLOCK(ErrorMessageLock);
03415                 MesPrint("Cannot create scratch file %s",to->name);
03416                 MUNLOCK(ErrorMessageLock);
03417                 return(-1);
03418             }
03419         }
03420         while ( fullsize > 0 ) {
03421             i = to->PPostop - to->POfull;
03422             if ( i > fullsize ) i = fullsize;
03423             fullsize -= i;
03424             t2 = to->POfull;
03425             NCOPY(t2,t1,i)

```

```

03426         if ( fullsize > 0 ) {
03427             SeekFile(to->handle,&(to->PPosition),SEEK_SET);
03428             if ( WriteFile(to->handle,((UBYTE *) (to->Pbuffer)),to->Psize) != to->Psize ) {
03429                 MLOCK(ErrorMessageLock);
03430                 MesPrint("Error while writing to disk. Disk full?");
03431                 MUNLOCK(ErrorMessageLock);
03432                 return(-1);
03433             }
03434             ADDPOS(to->PPosition,to->Psize);
03435             /* SeekFile(to->handle,&(to->PPosition),SEEK_CUR); */
03436             to->filesize = to->Pposition;
03437             to->Pofill = to->Pofull = to->Pbuffer;
03438         }
03439         else {
03440             to->Pofill = to->Pofull = t2;
03441         }
03442     }
03443 }
03444 SeekFile(from->handle,&poscopy,SEEK_CUR);
03445 }
03446 WriteTrailer:
03447 if ( ( to->handle >= 0 ) && ( to->Pofill > to->Pbuffer ) ) {
03448     fullsize = (UBYTE *) (to->Pofill) - (UBYTE *) (to->Pbuffer);
03449     /*
03450     PUTZERO(to->Pposition);
03451     SeekFile(to->handle,&(to->Pposition),SEEK_END);
03452     */
03453     SeekFile(to->handle,&(to->filesize),SEEK_SET);
03454     if ( WriteFile(to->handle,((UBYTE *) (to->Pbuffer)),fullsize) != fullsize ) {
03455         MLOCK(ErrorMessageLock);
03456         MesPrint("Error while writing to disk. Disk full?");
03457         MUNLOCK(ErrorMessageLock);
03458         return(-1);
03459     }
03460     ADDPOS(to->filesize,fullsize);
03461     to->Pposition = to->filesize;
03462     to->Pofill = to->Pofull = to->Pbuffer;
03463 }
03464
03465 return(0);
03466 }
03467
03468 /*
03469     #] CopyExpression :
03470     #[ ExprStatus :
03471 */
03472
03473 #ifdef HIDEDEBUG
03474
03475 static UBYTE *statusexpr[] = {
03476     (UBYTE *) "LOCALEXPRESSION"
03477 , (UBYTE *) "SKIPLEXPRESSION"
03478 , (UBYTE *) "DROPLEXPRESSION"
03479 , (UBYTE *) "DROPEXPRESSION"
03480 , (UBYTE *) "GLOBALEXPRESSION"
03481 , (UBYTE *) "SKIPGEXPRESSION"
03482 , (UBYTE *) "DROPGEXPRESSION"
03483 , (UBYTE *) "UNKNOWN"
03484 , (UBYTE *) "STOREDEXPRESSION"
03485 , (UBYTE *) "HIDDENLEXPRESSION"
03486 , (UBYTE *) "HIDELEXPRESSION"
03487 , (UBYTE *) "DROPHLEXPRESSION"
03488 , (UBYTE *) "UNHIDELEXPRESSION"
03489 , (UBYTE *) "HIDDENGEXPRESSION"
03490 , (UBYTE *) "HIDEGEXPRESSION"
03491 , (UBYTE *) "DROPHGEXPRESSION"
03492 , (UBYTE *) "UNHIDEGEXPRESSION"
03493 , (UBYTE *) "INTOHIDELEXPRESSION"
03494 , (UBYTE *) "INTOHIDEGEXPRESSION"
03495 };
03496
03497 void ExprStatus(EXPRESSIONS e)
03498 {
03499     MesPrint("Expression %s(%d) has status %s(%d,%d). Buffer: %d, Position: %15p",
03500         AC.exprnames->namebuffer+e->name, (WORD) (e->Expressions),
03501         statusexpr[e->status],e->status,e->hidelevel,
03502         e->whichbuffer,&(e->onfile));
03503 }
03504
03505 #endif
03506
03507 /*
03508     #] ExprStatus :
03509     #] StoreExpressions :
03510     #[ System Independent Saved Expressions :
03511
03512     All functions concerned with the system independent reading of save-files

```

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03513     are here. They are called by the functions CoLoad, PutInStore,
03514     SetFileIndex, FindInIndex. In case no translation (endianness flip,
03515     resizing of words, renumbering) has to be done, they just do simple file
03516     reading. The function SaveFileHeader() for writing a header with
03517     information about the system architecture, FORM version, etc. is also
03518     located here.
03519
03520     #[] Flip :
03521 */
03522
03523 static void FlipN(UBYTE *p, int length)
03524 {
03525     UBYTE *q, buf;
03526     q = p + length;
03527     do {
03528         --q;
03529         buf = *p; *p = *q; *q = buf;
03530     } while ( ++p != q );
03531 }
03532
03533 static void Flip16(UBYTE *p)
03534 {
03535     uint16_t in = *((uint16_t *)p);
03536     uint16_t out = (uint16_t)( ((in) >> 8) & UINT16_C(0x00FF) | ((in) << 8) & UINT16_C(0xFF00) );
03537     *((uint16_t *)p) = out;
03538 }
03539
03540 static void Flip32(UBYTE *p)
03541 {
03542     uint32_t in = *((uint32_t *)p);
03543     uint32_t out =
03544         ( ((in) >> 24) & UINT32_C(0x000000FF) | ((in) >> 8) & UINT32_C(0x0000FF00) | \
03545         ((in) << 8) & UINT32_C(0x00FF0000) | ((in) << 24) & UINT32_C(0xFF000000) );
03546     *((uint32_t *)p) = out;
03547 }
03548
03549 #ifdef UINT64_C
03550 static void Flip64(UBYTE *p)
03551 {
03552     uint64_t in = *((uint64_t *)p);
03553     uint64_t out =
03554         ( ((in) >> 56) & UINT64_C(0x00000000000000FF) | ((in) >> 40) & UINT64_C(0x000000000000FF00) | \
03555         ((in) >> 24) & UINT64_C(0x0000000000FF0000) | ((in) >> 8) & UINT64_C(0x00000000FF000000) | \
03556         ((in) << 8) & UINT64_C(0x000000FF00000000) | ((in) << 24) & UINT64_C(0x0000FF0000000000) | \
03557         ((in) << 40) & UINT64_C(0x00FF000000000000) | ((in) << 56) & UINT64_C(0xFF00000000000000) );
03558     *((uint64_t *)p) = out;
03559 }
03560 #else
03561 static void Flip64(UBYTE *p) { FlipN(p, 8); }
03562 #endif /* UINT64_C */
03563
03564 static void Flip128(UBYTE *p) { FlipN(p, 16); }
03565
03566 /*
03567     #[] Flip :
03568     #[] Resize :
03569 */
03570
03571 static void ResizeDataBE(UBYTE *src, int slen, UBYTE *dst, int dlen)
03572 {
03573     if ( slen > dlen ) {
03574         src += slen - dlen;
03575         while ( dlen-- ) { *dst++ = *src++; }
03576     }
03577     else {
03578         int i = dlen - slen;
03579         while ( i-- ) { *dst++ = 0; }
03580         while ( slen-- ) { *dst++ = *src++; }
03581     }
03582 }
03583
03584 static void ResizeDataLE(UBYTE *src, int slen, UBYTE *dst, int dlen)
03585 {
03586     if ( slen > dlen ) {
03587         while ( dlen-- ) { *dst++ = *src++; }
03588     }
03589     else {
03590         int i = dlen - slen;
03591         while ( slen-- ) { *dst++ = *src++; }
03592         while ( i-- ) { *dst++ = 0; }
03593     }
03594 }
03595
03596
03597
03598
03599
03600
03601
03602
03603
03604
03605
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03608
03609
03610
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```

03643 static void Resize16t16(UBYTE *src, UBYTE *dst)
03644 {
03645     *((int16_t *)dst) = *((int16_t *)src);
03646 }
03647
03649 static void Resize16t32(UBYTE *src, UBYTE *dst)
03650 {
03651     int16_t in = *((int16_t *)src);
03652     int32_t out = (int32_t)in;
03653     *((int32_t *)dst) = out;
03654 }
03655
03657 #ifdef INT64_C
03658 static void Resize16t64(UBYTE *src, UBYTE *dst)
03659 {
03660     int16_t in = *((int16_t *)src);
03661     int64_t out = (int64_t)in;
03662     *((int64_t *)dst) = out;
03663 }
03664 #else
03665 static void Resize16t64(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 2, dst, 8); }
03666 #endif /* INT64_C */
03667
03669 static void Resize16t128(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 2, dst, 16); }
03670
03672 static void Resize32t32(UBYTE *src, UBYTE *dst)
03673 {
03674     *((int32_t *)dst) = *((int32_t *)src);
03675 }
03676
03678 #ifdef INT64_C
03679 static void Resize32t64(UBYTE *src, UBYTE *dst)
03680 {
03681     int32_t in = *((int32_t *)src);
03682     int64_t out = (int64_t)in;
03683     *((int64_t *)dst) = out;
03684 }
03685 #else
03686 static void Resize32t64(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 4, dst, 8); }
03687 #endif /* INT64_C */
03688
03690 static void Resize32t128(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 4, dst, 16); }
03691
03693 #ifdef INT64_C
03694 static void Resize64t64(UBYTE *src, UBYTE *dst)
03695 {
03696     *((int64_t *)dst) = *((int64_t *)src);
03697 }
03698 #else
03699 static void Resize64t64(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 8); }
03700 #endif /* INT64_C */
03701
03703 static void Resize64t128(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 16); }
03704
03706 static void Resize128t128(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 16); }
03707
03709 static void Resize32t16(UBYTE *src, UBYTE *dst)
03710 {
03711     int32_t in = *((int32_t *)src);
03712     int16_t out = (int16_t)in;
03713     if ( in > (1<15)-1 || in < -(1<15)+1 ) AO.resizeFlag |= 1;
03714     *((int16_t *)dst) = out;
03715 }
03716
03723 static void Resize32t16NC(UBYTE *src, UBYTE *dst)
03724 {
03725     int32_t in = *((int32_t *)src);
03726     int16_t out = (int16_t)in;
03727     *((int16_t *)dst) = out;
03728 }
03729
03730 #ifdef INT64_C
03732 static void Resize64t16(UBYTE *src, UBYTE *dst)
03733 {
03734     int64_t in = *((int64_t *)src);
03735     int16_t out = (int16_t)in;
03736     if ( in > (1<15)-1 || in < -(1<15)+1 ) AO.resizeFlag |= 1;
03737     *((int16_t *)dst) = out;
03738 }
03740 static void Resize64t16NC(UBYTE *src, UBYTE *dst)
03741 {
03742     int64_t in = *((int64_t *)src);
03743     int16_t out = (int16_t)in;
03744     *((int16_t *)dst) = out;
03745 }
03746 #else
03748 static void Resize64t16(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 2); }

```

```

03750 static void Resize64t16NC(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 2); }
03751 #endif /* INT64_C */
03752
03753 #ifdef INT64_C
03755 static void Resize64t32(UBYTE *src, UBYTE *dst)
03756 {
03757     int64_t in = *((int64_t *)src);
03758     int32_t out = (int32_t)in;
03759     if ( in > (INT64_C(1)«31)-1 || in < -(INT64_C(1)«31)+1 ) AO.resizeFlag |= 1;
03760     *((int32_t *)dst) = out;
03761 }
03763 static void Resize64t32NC(UBYTE *src, UBYTE *dst)
03764 {
03765     int64_t in = *((int64_t *)src);
03766     int32_t out = (int32_t)in;
03767     *((int32_t *)dst) = out;
03768 }
03769 #else
03771 static void Resize64t32(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 4); }
03773 static void Resize64t32NC(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 4); }
03774 #endif /* INT64_C */
03775
03777 static void Resize128t16(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 2); }
03778
03780 static void Resize128t16NC(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 2); }
03781
03783 static void Resize128t32(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 4); }
03784
03786 static void Resize128t32NC(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 4); }
03787
03789 static void Resize128t64(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 8); }
03790
03792 static void Resize128t64NC(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 8); }
03793
03794 /*
03795     #] Resize :
03796     #[ CheckPower and RenumVec :
03797 */
03798
03805 static void CheckPower32(UBYTE *p)
03806 {
03807     if ( *((int32_t *)p) < -MAXPOWER ) {
03808         AO.powerFlag |= 0x01;
03809         *((int32_t *)p) = -MAXPOWER;
03810     }
03811     p += sizeof(int32_t);
03812     if ( *((int32_t *)p) > MAXPOWER ) {
03813         AO.powerFlag |= 0x02;
03814         *((int32_t *)p) = MAXPOWER;
03815     }
03816 }
03817
03825 static void RenumVec32(UBYTE *p)
03826 {
03827     /* int32_t wilddoffset = *((int32_t *)AO.SaveHeader.wilddoffset); */
03828     void *dummy = (void *)AO.SaveHeader.wilddoffset; /* to remove a warning about strict-aliasing
rules in gcc */
03829     int32_t wilddoffset = *(int32_t *)dummy;
03830     int32_t in = *((int32_t *)p);
03831     in = in + 2*wilddoffset;
03832     in = in - 2*WILDDOFFSET;
03833     *((int32_t *)p) = in;
03834 }
03835
03836 /*
03837     #] CheckPower and RenumVec :
03838     #[ ResizeCoeff :
03839 */
03840
03854 static void ResizeCoeff32(UBYTE **bout, UBYTE *bend, UBYTE *top)
03855 {
03856     int i;
03857     int32_t sign;
03858     int32_t *in, *p;
03859     int32_t *out = (int32_t *)*bout;
03860     int32_t *end = (int32_t *)bend;
03861
03862     if ( sizeof(WORD) == 2 ) {
03863         /* 4 -> 2 */
03864         int32_t len = (end - 1 - out) / 2;
03865         int zeros = 2;
03866         p = out + len - 1;
03867
03868         if ( *p & 0xFFFF0000 ) --zeros;
03869         p += len;
03870         if ( *p & 0xFFFF0000 ) --zeros;
03871

```

```

03872     in = end - 1;
03873     sign = ( *in-- > 0 ) ? 1 : -1;
03874     p = out + 4*len;
03875     if ( zeros == 2 ) p -= 2;
03876     out = p--;
03877
03878     if ( zeros < 2 ) *p-- = *in » 16;
03879     *p-- = *in-- & 0x0000FFFF;
03880     for ( i = 1; i < len; ++i ) {
03881         *p-- = *in » 16;
03882         *p-- = *in-- & 0x0000FFFF;
03883     }
03884     if ( zeros < 2 ) *p-- = *in » 16;
03885     *p-- = *in-- & 0x0000FFFF;
03886     for ( i = 1; i < len; ++i ) {
03887         *p-- = *in » 16;
03888         *p-- = *in-- & 0x0000FFFF;
03889     }
03890
03891     *out = (out - p) * sign;
03892     *bout = (UBYTE *) (out+1);
03893
03894 }
03895 else {
03896     /* 2 -> 4 */
03897     int32_t len = (end - 1 - out) / 2;
03898     if ( len == 1 ) {
03899         *out = *(uint16_t *)out;
03900         ++out;
03901         *out = *(uint16_t *)out;
03902         ++out;
03903         ++out;
03904     }
03905     else {
03906         p = out;
03907         *out = *(uint16_t *)out;
03908         in = out + 1;
03909         for ( i = 1; i < len; ++i ) {
03910             /* shift */
03911             *out = (uint32_t) (*(uint16_t *)out)
03912                 + ((uint32_t) (*(uint16_t *)in) « 16);
03913             ++in;
03914             if ( ++i == len ) break;
03915             /* copy */
03916             ++out;
03917             *out = *(uint16_t *)in;
03918             ++in;
03919         }
03920         ++out;
03921         *out = *(uint16_t *)in;
03922         ++in;
03923         for ( i = 1; i < len; ++i ) {
03924             /* shift */
03925             *out = (uint32_t) (*(uint16_t *)out)
03926                 + ((uint32_t) (*(uint16_t *)in) « 16);
03927             ++in;
03928             if ( ++i == len ) break;
03929             /* copy */
03930             ++out;
03931             *out = *(uint16_t *)in;
03932             ++in;
03933         }
03934         ++out;
03935         if ( *in < 0 ) *out = -(out - p + 1);
03936         else *out = out - p + 1;
03937         ++out;
03938     }
03939
03940     if ( out > (int32_t *)top ) {
03941         MesPrint("Error in resizing coefficient!");
03942     }
03943
03944     *bout = (UBYTE *)out;
03945 }
03946 }
03947
03948 /*
03949     #] ResizeCoeff :
03950     #[ WriteStoreHeader :
03951 */
03952
03953 #define SAVEREVISION 0x02
03954
03955 LONG WriteStoreHeader(WORD handle)
03956 {
03957     /* template of the STOREHEADER */
03958     static STOREHEADER sh = {

```

```

03968     { 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF }, /* store header mark */
03969     0, 0, 0, 0, /* sizeof of WORD, LONG, POSITION, void* */
03970     { 0 }, /* endianness check number */
03971     0, 0, 0, 0, /* sizeof variable structs */
03972     { 0 }, /* maxpower */
03973     { 0 }, /* wilddoffset */
03974     SAVEREVISION, /* revision */
03975     { 0 } }; /* reserved */
03976 int endian, i;
03977
03978 /* if called for the first time ... */
03979 if ( sh.lenWORD == 0 ) {
03980     sh.lenWORD = sizeof(WORD);
03981     sh.lenLONG = sizeof(LONG);
03982     sh.lenPOS = sizeof(POSITION);
03983     sh.lenPOINTER = sizeof(void *);
03984
03985     endian = 1;
03986     for ( i = 1; i < (int)sizeof(int); ++i ) {
03987         endian <= 8;
03988         endian += i+1;
03989     }
03990     for ( i = 0; i < (int)sizeof(int); ++i ) sh.endianness[i] = ((char *)&endian)[i];
03991
03992     sh.sSym = sizeof(struct SyMb01);
03993     sh.sInd = sizeof(struct InDeX);
03994     sh.sVec = sizeof(struct VeCtOr);
03995     sh.sFun = sizeof(struct FuNcTiOn);
03996
03997 /*      *((WORD *)sh.maxpower) = MAXPOWER;
03998      *((WORD *)sh.wilddoffset) = WILDDOFFSET; */
03999     {
04000         WORD dumw[8];
04001         UBYTE *dummy;
04002         for ( i = 0; i < 8; i++ ) dumw[i] = 0;
04003         dummy = (UBYTE *)dumw;
04004         dumw[0] = (WORD)MAXPOWER;
04005         for ( i = 0; i < 16; i++ ) sh.maxpower[i] = dummy[i];
04006         dumw[0] = (WORD)WILDDOFFSET;
04007         for ( i = 0; i < 16; i++ ) sh.wilddoffset[i] = dummy[i];
04008     }
04009 }
04010
04011 return ( WriteFile(handle, (UBYTE *)&sh, (LONG)(sizeof(STOREHEADER)))
04012         != (LONG)(sizeof(STOREHEADER)) );
04013 }
04014
04015 /*
04016     #] WriteStoreHeader :
04017     #[ CompactifySizeof :
04018 */
04019
04027 static unsigned int CompactifySizeof(unsigned int size)
04028 {
04029     switch ( size ) {
04030         case 2: return 0;
04031         case 4: return 1;
04032         case 8: return 2;
04033         case 16: return 3;
04034         default: MesPrint("Error compactifying size.");
04035                 return 3;
04036     }
04037 }
04038
04039 /*
04040     #] CompactifySizeof :
04041     #[ ReadSaveHeader :
04042 */
04043
04057 int ReadSaveHeader(void)
04058 {
04059     /* Read-only tables of function pointers for conversions. */
04060     static void (*flipJumpTable[4])(UBYTE *) =
04061     { Flip16, Flip32, Flip64, Flip128 };
04062     static void (*resizeJumpTable[4][4])(UBYTE *, UBYTE *) = /* "own x saved"-sizes */
04063     { { Resizel6t16, Resizel32t16, Resizel64t16, Resizel128t16 },
04064       { Resizel6t32, Resizel32t32, Resizel64t32, Resizel128t32 },
04065       { Resizel6t64, Resizel32t64, Resizel64t64, Resizel128t64 },
04066       { Resizel6t128, Resizel32t128, Resizel64t128, Resizel128t128 } };
04067     static void (*resizencJumpTable[4][4])(UBYTE *, UBYTE *) = /* "own x saved"-sizes */
04068     { { Resizel6t16, Resizel32t16NC, Resizel64t16NC, Resizel128t16NC },
04069       { Resizel6t32, Resizel32t32, Resizel64t32NC, Resizel128t32NC },
04070       { Resizel6t64, Resizel32t64, Resizel64t64, Resizel128t64NC },
04071       { Resizel6t128, Resizel32t128, Resizel64t128, Resizel128t128 } };
04072
04073     int endian, i;
04074     WORD idxW = CompactifySizeof(sizeof(WORD));

```

```

04075     WORD idxL = CompactifySizeof(sizeof(LONG));
04076     WORD idxP = CompactifySizeof(sizeof(POSITION));
04077     WORD idxVP = CompactifySizeof(sizeof(void *));
04078
04079     AO.transFlag = 0;
04080     AO.powerFlag = 0;
04081     AO.resizeFlag = 0;
04082     AO.bufferedInd = 0;
04083
04084     if ( ReadFile(AO.SaveData.Handle, (UBYTE *)(&AO.SaveHeader),
04085                 (LONG)sizeof(STOREHEADER)) != (LONG)sizeof(STOREHEADER) )
04086         return(MesPrint("Error reading save file header"));
04087
04088     /* check whether save-file has no header. if yes then it is an old version
04089        of FORM -> go back to position 0 in file which then contains the first
04090        index and skip the rest. */
04091     for ( i = 0; i < 8; ++i ) {
04092         if ( AO.SaveHeader.headermark[i] != 0xFF ) {
04093             POSITION p;
04094             PUTZERO(p);
04095             SeekFile(AO.SaveData.Handle, &p, SEEK_SET);
04096             return ( 0 );
04097         }
04098     }
04099
04100     if ( AO.SaveHeader.revision != SAVEREVISION ) {
04101         return(MesPrint("Save file header from an old version. Cannot read this file."));
04102     }
04103
04104     endian = 1;
04105     for ( i = 1; i < (int)sizeof(int); ++i ) {
04106         endian <= 8;
04107         endian += i+1;
04108     }
04109     if ( ((char *)&endian)[0] < ((char *)&endian)[1] ) {
04110         /* this machine is big-endian */
04111         AO.ResizeData = ResizeDataBE;
04112     }
04113     else {
04114         /* this machine is little-endian */
04115         AO.ResizeData = ResizeDataLE;
04116     }
04117
04118     /* set AO.transFlag if ANY conversion has to be done later */
04119     if ( AO.SaveHeader.endianness[0] > AO.SaveHeader.endianness[1] ) {
04120         AO.transFlag = ( ((char *)&endian)[0] < ((char *)&endian)[1] );
04121     }
04122     else {
04123         AO.transFlag = ( ((char *)&endian)[0] > ((char *)&endian)[1] );
04124     }
04125     if ( (WORD)AO.SaveHeader.lenWORD != sizeof(WORD) ) AO.transFlag |= 0x02;
04126     if ( (WORD)AO.SaveHeader.lenLONG != sizeof(LONG) ) AO.transFlag |= 0x04;
04127     if ( (WORD)AO.SaveHeader.lenPOS != sizeof(POSITION) ) AO.transFlag |= 0x08;
04128     if ( (WORD)AO.SaveHeader.lenPOINTER != sizeof(void *) ) AO.transFlag |= 0x10;
04129
04130     AO.FlipWORD = flipJumpTable[idxW];
04131     AO.FlipLONG = flipJumpTable[idxL];
04132     AO.FlipPOS = flipJumpTable[idxP];
04133     AO.FlipPOINTER = flipJumpTable[idxVP];
04134
04135     /* Works only for machines where WORD is not greater than 32bit ! */
04136     AO.CheckPower = CheckPower32;
04137     AO.RenumberVec = RenumberVec32;
04138
04139     AO.ResizeWORD = resizeJumpTable[idxW][CompactifySizeof(AO.SaveHeader.lenWORD)];
04140     AO.ResizeNCWORD = resizeNCJumpTable[idxW][CompactifySizeof(AO.SaveHeader.lenWORD)];
04141     AO.ResizeLONG = resizeJumpTable[idxL][CompactifySizeof(AO.SaveHeader.lenLONG)];
04142     AO.ResizePOS = resizeJumpTable[idxP][CompactifySizeof(AO.SaveHeader.lenPOS)];
04143     AO.ResizePOINTER = resizeJumpTable[idxVP][CompactifySizeof(AO.SaveHeader.lenPOINTER)];
04144
04145     {
04146         WORD dumw[8];
04147         UBYTE *dummy;
04148         for ( i = 0; i < 8; i++ ) dumw[i] = 0;
04149         dummy = (UBYTE *)dumw;
04150         for ( i = 0; i < 16; i++ ) dummy[i] = AO.SaveHeader.maxpower[i];
04151         AO.mpower = dumw[0];
04152     }
04153
04154     return ( 0 );
04155 }
04156
04157 /*
04158     #] ReadSaveHeader :
04159     #[ ReadSaveIndex :
04160 */
04161

```

```

04175 LONG ReadSaveIndex(FILEINDEX *fileind)
04176 {
04177     /* do we need some translation for the FILEINDEX? */
04178     if ( AO.transFlag ) {
04179         /* if a translated FILEINDEX can hold less entries than the original
04180            FILEINDEX, then we need to buffer the extra entries in this static
04181            variable (can happen going from 32bit to 64bit */
04182         static FILEINDEX sbuffer;
04183
04184         FILEINDEX buffer;
04185         UBYTE *p, *q;
04186         int i;
04187
04188         /* shortcuts */
04189         int lenW = AO.SaveHeader.lenWORD;
04190         int lenL = AO.SaveHeader.lenLONG;
04191         int lenP = AO.SaveHeader.lenPOS;
04192
04193         /* if we have a buffered FILEINDEX then just return it */
04194         if ( AO.bufferedInd ) {
04195             *fileind = sbuffer;
04196             AO.bufferedInd = 0;
04197             return ( 0 );
04198         }
04199
04200         if ( ReadFile(AO.SaveData.Handle, (UBYTE *)fileind, sizeof(FILEINDEX))
04201             != sizeof(FILEINDEX) ) {
04202             return ( MesPrint("Error(1) reading stored expression.") );
04203         }
04204
04205         /* do we need to flip the endianness? */
04206         if ( AO.transFlag & 1 ) {
04207             LONG number;
04208             /* padding bytes */
04209             int padp = lenL - ((lenW*5+(MAXENAME + 1)) & (lenL-1));
04210             p = (UBYTE *)fileind;
04211             AO.FlipPOS(p); p += lenP;          /* next */
04212             AO.FlipPOS(p);                    /* number */
04213             AO.ResizePOS(p, (UBYTE *)&number);
04214             p += lenP;
04215             for ( i = 0; i < number; ++i ) {
04216                 AO.FlipPOS(p); p += lenP;      /* position */
04217                 AO.FlipPOS(p); p += lenP;      /* length */
04218                 AO.FlipPOS(p); p += lenP;      /* variables */
04219                 AO.FlipLONG(p); p += lenL;     /* CompressSize */
04220                 AO.FlipWORD(p); p += lenW;     /* nsymbols */
04221                 AO.FlipWORD(p); p += lenW;     /* nindices */
04222                 AO.FlipWORD(p); p += lenW;     /* nvectors */
04223                 AO.FlipWORD(p); p += lenW;     /* nfunctions */
04224                 AO.FlipWORD(p); p += lenW;     /* size */
04225                 p += padp;
04226             }
04227         }
04228
04229         /* do we need to resize data? */
04230         if ( AO.transFlag > 1 ) {
04231             LONG number, maxnumber;
04232             int n;
04233             /* padding bytes */
04234             int padp = lenL - ((lenW*5+(MAXENAME + 1)) & (lenL-1));
04235             int padq = sizeof(LONG) - ((sizeof(WORD)*5+(MAXENAME + 1)) & (sizeof(LONG)-1));
04236
04237             p = (UBYTE *)fileind; q = (UBYTE *)&buffer;
04238             AO.ResizePOS(p, q);                /* next */
04239             p += lenP; q += sizeof(POSITION);
04240             AO.ResizePOS(p, q);                /* number */
04241             p += lenP;
04242             number = BASEPOSITION(*((POSITION *)q));
04243             /* if FILEINDEX in file contains more entries than the FILEINDEX in
04244                memory can contain, then adjust the numbers and prepare for
04245                buffering */
04246             if ( number > (LONG)INFILEINDEX ) {
04247                 AO.bufferedInd = number-INFILEINDEX;
04248                 if ( AO.bufferedInd > (WORD)INFILEINDEX ) {
04249                     /* can happen when reading 32bit and writing >=128bit.
04250                        Fix: more than one static buffer for FILEINDEX */
04251                     return ( MesPrint("Too many index entries.") );
04252                 }
04253                 maxnumber = INFILEINDEX;
04254                 SETBASEPOSITION(*((POSITION *)q), INFILEINDEX);
04255             }
04256             else {
04257                 maxnumber = number;
04258             }
04259             q += sizeof(POSITION);
04260             /* read all INDEXENTRY that fit into the output buffer */
04261             for ( i = 0; i < maxnumber; ++i ) {

```

```

04262         AO.ResizePOS(p, q);                /* position */
04263         p += lenP; q += sizeof(POSITION);
04264         AO.ResizePOS(p, q);                /* length */
04265         p += lenP; q += sizeof(POSITION);
04266         AO.ResizePOS(p, q);                /* variables */
04267         p += lenP; q += sizeof(POSITION);
04268         AO.ResizeLONG(p, q);               /* CompressSize */
04269         p += lenL; q += sizeof(LONG);
04270         AO.ResizeWORD(p, q);               /* nsymbols */
04271         p += lenW; q += sizeof(WORD);
04272         AO.ResizeWORD(p, q);               /* nindices */
04273         p += lenW; q += sizeof(WORD);
04274         AO.ResizeWORD(p, q);               /* nvectors */
04275         p += lenW; q += sizeof(WORD);
04276         AO.ResizeWORD(p, q);               /* nfunctions */
04277         p += lenW; q += sizeof(WORD);
04278         AO.ResizeWORD(p, q);               /* size (unchanged!) */
04279         p += lenW; q += sizeof(WORD);
04280         n = MAXENAME + 1;
04281         NCOPYB(q, p, n)
04282         p += padp;
04283         q += padq;
04284     }
04285     /* read all the remaining INDEXENTRY and put them into the static buffer */
04286     if ( AO.bufferedInd ) {
04287         sbuffer.next = buffer.next;
04288         SETBASEPOSITION(sbuffer.number, AO.bufferedInd);
04289         q = (UBYTE *)&sbuffer + sizeof(POSITION) + sizeof(LONG);
04290         for ( i = maxnumber; i < number; ++i ) {
04291             AO.ResizePOS(p, q);                /* position */
04292             p += lenP; q += sizeof(POSITION);
04293             AO.ResizePOS(p, q);                /* length */
04294             p += lenP; q += sizeof(POSITION);
04295             AO.ResizePOS(p, q);                /* variables */
04296             p += lenP; q += sizeof(POSITION);
04297             AO.ResizeLONG(p, q);               /* CompressSize */
04298             p += lenL; q += sizeof(LONG);
04299             AO.ResizeWORD(p, q);               /* nsymbols */
04300             p += lenW; q += sizeof(WORD);
04301             AO.ResizeWORD(p, q);               /* nindices */
04302             p += lenW; q += sizeof(WORD);
04303             AO.ResizeWORD(p, q);               /* nvectors */
04304             p += lenW; q += sizeof(WORD);
04305             AO.ResizeWORD(p, q);               /* nfunctions */
04306             p += lenW; q += sizeof(WORD);
04307             AO.ResizeWORD(p, q);               /* size (unchanged!) */
04308             p += lenW; q += sizeof(WORD);
04309             n = MAXENAME + 1;
04310             NCOPYB(q, p, n)
04311             p += padp;
04312             q += padq;
04313         }
04314     }
04315     /* copy to output */
04316     p = (UBYTE *)fileind; q = (UBYTE *)&buffer n = sizeof(FILEINDEX);
04317     NCOPYB(p, q, n)
04318 }
04319 return ( 0 );
04320 } else {
04321     return ( ReadFile(AO.SaveData.Handle, (UBYTE *)fileind, sizeof(FILEINDEX))
04322             != sizeof(FILEINDEX) );
04323 }
04324 }
04325
04326 /*
04327     #] ReadSaveIndex :
04328     #[ ReadSaveVariables :
04329 */
04330
04331 LONG ReadSaveVariables(UBYTE *buffer, UBYTE *top, LONG *size, LONG *outsize, \
04332                       INDEXENTRY *ind, LONG *stage)
04333 {
04334     /* do we need some translation for the variables? */
04335     if ( AO.transFlag ) {
04336         /* counters for the number of already read symbols, indices, ... that
04337            need to remain valid between different calls to ReadSaveVariables().
04338            are initialized if stage == -1 */
04339         static WORD numReadSym;
04340         static WORD numReadInd;
04341         static WORD numReadVec;
04342         static WORD numReadFun;
04343
04344         POSITION pos;
04345         UBYTE *in, *out, *pp = 0, *end, *outbuf;
04346         LONG numread;
04347         WORD namelen, realnamelen;
04348         /* shortcuts */

```

```

04379     WORD lenW = AO.SaveHeader.lenWORD;
04380     WORD lenL = AO.SaveHeader.lenLONG;
04381     WORD lenP = AO.SaveHeader.lenPOINTER;
04382     WORD flip = AO.transFlag & 1;
04383
04384     /* remember file position in case we have to rewind */
04385     TELLFILE(AO.SaveData.Handle,&pos);
04386
04387     /* decide on the position of the in and out buffers.
04388        if the input is "bigger" than the output, we resize in-place, i.e.
04389        we immediately overwrite the source data by the translated data. in
04390        and out buffers start at the same place.
04391        if not, we read from the end of the given buffer and write at the
04392        beginning. */
04393     if ( (lenW > (WORD)sizeof(WORD))
04394     || ( (lenW == (WORD)sizeof(WORD))
04395         && ( (lenL > (WORD)sizeof(LONG))
04396             || ( (lenL == (WORD)sizeof(LONG)) && lenP > (WORD)sizeof(void *) )
04397         ) ) ) {
04398         in = out = buffer;
04399         end = buffer + *size;
04400     }
04401     else {
04402         /* data will grow roughly by sizeof(WORD)/lenW. the exact value is
04403            not important. if reading and writing areas start to overlap, the
04404            reading will already be near the end of the data and overwriting
04405            doesn't matter. */
04406         LONG newsize = (top - buffer) / (1 + sizeof(WORD)/lenW);
04407         end = top;
04408         out = buffer;
04409         in = end - newsize;
04410         if ( *size > newsize ) *size = newsize;
04411     }
04412 }
04413
04414 if ( ( numread = ReadFile(AO.SaveData.Handle, in, *size) ) != *size ) {
04415     return ( MesPrint("Error(2) reading stored expression.") );
04416 }
04417
04418 *size = 0;
04419 *outsize = 0;
04420
04421 /* first time in ReadSaveVariables(). initialize counters. */
04422 if ( *stage == -1 ) {
04423     numReadSym = 0;
04424     numReadInd = 0;
04425     numReadVec = 0;
04426     numReadFun = 0;
04427     ++*stage;
04428 }
04429
04430 while ( in < end ) {
04431     /* Symbols */
04432     if ( *stage == 0 ) {
04433         if ( ind->nsymbols <= numReadSym ) {
04434             ++*stage;
04435             continue;
04436         }
04437         if ( end - in < AO.SaveHeader.sSym ) {
04438             goto RSVEnd;
04439         }
04440         if ( flip ) {
04441             pp = in;
04442             AO.FlipLONG(pp); pp += lenL;
04443             while ( pp < in + AO.SaveHeader.sSym ) {
04444                 AO.FlipWORD(pp); pp += lenW;
04445             }
04446         }
04447         pp = in + AO.SaveHeader.sSym;
04448         AO.ResizeLONG(in, out); in += lenL; out += sizeof(LONG); /* name */
04449         AO.CheckPower(in);
04450         AO.ResizeWORD(in, out); in += lenW;
04451         if ( *((WORD *)out) == -AO.mpower ) *((WORD *)out) = -MAXPOWER;
04452         out += sizeof(WORD); /* minpower */
04453         AO.ResizeWORD(in, out); in += lenW;
04454         if ( *((WORD *)out) == AO.mpower ) *((WORD *)out) = MAXPOWER;
04455         out += sizeof(WORD); /* maxpower */
04456         AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* complex */
04457         AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* number */
04458         AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* flags */
04459         AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* node */
04460         AO.ResizeWORD(in, out); in += lenW; /* namesize */
04461         realnamelen = *((WORD *)out);
04462         realnamelen += sizeof(void *)-1; realnamelen &= -(sizeof(void *));
04463         out += sizeof(WORD);
04464         AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* dimension */
04465         while ( in < pp ) {

```



```

04466         AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD);
04467     }
04468     namelen = *((WORD *)out-1); /* cares for padding "bug" */
04469     if ( end - in < namelen ) {
04470         goto RSVEnd;
04471     }
04472     *((WORD *)out-1) = realnamelen;
04473     *size += AO.SaveHeader.sSym + namelen;
04474     *outsize += sizeof(struct SyMb01) + realnamelen;
04475     if ( realnamelen > namelen ) {
04476         int j = namelen;
04477         NCOPYB(out, in, j);
04478         out += realnamelen - namelen;
04479     }
04480     else {
04481         int j = realnamelen;
04482         NCOPYB(out, in, j);
04483         in += namelen - realnamelen;
04484     }
04485     ++numReadSym;
04486     continue;
04487 }
04488 /* Indices */
04489 if ( *stage == 1 ) {
04490     if ( ind->nindices <= numReadInd ) {
04491         ++*stage;
04492         continue;
04493     }
04494     if ( end - in < AO.SaveHeader.sInd ) {
04495         goto RSVEnd;
04496     }
04497     if ( flip ) {
04498         pp = in;
04499         AO.FlipLONG(pp); pp += lenL;
04500         while ( pp < in + AO.SaveHeader.sInd ) {
04501             AO.FlipWORD(pp); pp += lenW;
04502         }
04503     }
04504     pp = in + AO.SaveHeader.sInd;
04505     AO.ResizeLONG(in, out); in += lenL; out += sizeof(LONG); /* name */
04506     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* type */
04507     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* dimension */
04508     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* number */
04509     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* flags */
04510     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* nmin4 */
04511     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* node */
04512     AO.ResizeWORD(in, out); in += lenW; /* namesize */
04513     realnamelen = *((WORD *)out);
04514     realnamelen += sizeof(void *)-1; realnamelen &= -(sizeof(void *));
04515     out += sizeof(WORD);
04516     while ( in < pp ) {
04517         AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD);
04518     }
04519     namelen = *((WORD *)out-1); /* cares for padding "bug" */
04520     if ( end - in < namelen ) {
04521         goto RSVEnd;
04522     }
04523     *((WORD *)out-1) = realnamelen;
04524     *size += AO.SaveHeader.sInd + namelen;
04525     *outsize += sizeof(struct InDeX) + realnamelen;
04526     if ( realnamelen > namelen ) {
04527         int j = namelen;
04528         NCOPYB(out, in, j);
04529         out += realnamelen - namelen;
04530     }
04531     else {
04532         int j = realnamelen;
04533         NCOPYB(out, in, j);
04534         in += namelen - realnamelen;
04535     }
04536     ++numReadInd;
04537     continue;
04538 }
04539 /* Vectors */
04540 if ( *stage == 2 ) {
04541     if ( ind->nvectors <= numReadVec ) {
04542         ++*stage;
04543         continue;
04544     }
04545     if ( end - in < AO.SaveHeader.sVec ) {
04546         goto RSVEnd;
04547     }
04548     if ( flip ) {
04549         pp = in;
04550         AO.FlipLONG(pp); pp += lenL;
04551         while ( pp < in + AO.SaveHeader.sVec ) {
04552             AO.FlipWORD(pp); pp += lenW;

```

```

04553     }
04554 }
04555 pp = in + AO.SaveHeader.sVec;
04556 AO.ResizeLONG(in, out); in += lenL; out += sizeof(LONG); /* name */
04557 AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* complex */
04558 AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* number */
04559 AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* flags */
04560 AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* node */
04561 AO.ResizeWORD(in, out); in += lenW; /* namesize */
04562 realnamelen = *((WORD *)out);
04563 realnamelen += sizeof(void *)-1; realnamelen &= -(sizeof(void *));
04564 out += sizeof(WORD);
04565 AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* dimension */
04566 while ( in < pp ) {
04567     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD);
04568 }
04569 namelen = *((WORD *)out-1); /* cares for padding "bug" */
04570 if ( end - in < namelen ) {
04571     goto RSVEnd;
04572 }
04573 *((WORD *)out-1) = realnamelen;
04574 *size += AO.SaveHeader.sVec + namelen;
04575 *outsize += sizeof(struct VeCtOr) + realnamelen;
04576 if ( realnamelen > namelen ) {
04577     int j = namelen;
04578     NCOPYB(out, in, j)
04579     out += realnamelen - namelen;
04580 }
04581 else {
04582     int j = realnamelen;
04583     NCOPYB(out, in, j)
04584     in += namelen - realnamelen;
04585 }
04586 ++numReadVec;
04587 continue;
04588 }
04589 /* Functions */
04590 if ( *stage == 3 ) {
04591     if ( ind->nfunctions <= numReadFun ) {
04592         ++*stage;
04593         continue;
04594     }
04595     if ( end - in < AO.SaveHeader.sFun ) {
04596         goto RSVEnd;
04597     }
04598     if ( flip ) {
04599         pp = in;
04600         AO.FlipPOINTER(pp); pp += lenP;
04601         AO.FlipLONG(pp); pp += lenL;
04602         AO.FlipLONG(pp); pp += lenL;
04603         while ( pp < in + AO.SaveHeader.sFun ) {
04604             AO.FlipWORD(pp); pp += lenW;
04605         }
04606     }
04607     pp = in + AO.SaveHeader.sFun;
04608     outbuf = out;
04609     AO.ResizePOINTER(in, out); in += lenP; out += sizeof(void *); /* tabl */
04610     AO.ResizeLONG(in, out); in += lenL; out += sizeof(LONG); /* symminfo */
04611     AO.ResizeLONG(in, out); in += lenL; out += sizeof(LONG); /* name */
04612     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* commute */
04613     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* complex */
04614     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* number */
04615     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* flags */
04616     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* spec */
04617     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* symmetric */
04618     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* numargs */
04619     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* node */
04620     AO.ResizeWORD(in, out); in += lenW; /* namesize */
04621     realnamelen = *((WORD *)out);
04622     realnamelen += sizeof(void *)-1; realnamelen &= -(sizeof(void *));
04623     out += sizeof(WORD);
04624     AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* dimension */
04625     while ( in < pp ) {
04626         AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD);
04627     }
04628     namelen = *((WORD *)out-1); /* cares for padding "bug" */
04629     if ( end - in < namelen ) {
04630         goto RSVEnd;
04631     }
04632     *((WORD *)out-1) = realnamelen;
04633     *size += AO.SaveHeader.sFun + namelen;
04634     *outsize += sizeof(struct FuNcTiOn) + realnamelen;
04635     if ( realnamelen > namelen ) {
04636         int j = namelen;
04637         NCOPYB(out, in, j);
04638         out += realnamelen - namelen;
04639     }

```

```

04640         else {
04641             int j = realnamelen;
04642             NCOPYB(out, in, j);
04643             in += namelen - realnamelen;
04644         }
04645         ++numReadFun;
04646         /* we use the information whether a function is tensorial later in ReadSaveTerm */
04647         AO.tensorList[((FUNCTIONS)outbuf)->number+FUNCTION] =
04648             (UBYTE)((FUNCTIONS)outbuf)->spec == TENSORFUNCTION);
04649         continue;
04650     }
04651     /* handle numdummies */
04652     if ( end - in >= lenW ) {
04653         if ( flip ) AO.FlipWORD(in);
04654         AO.ResizeWORD(in, out);
04655         *size += lenW;
04656         *outsize += sizeof(WORD);
04657     }
04658     /* handle numfactors */
04659     if ( end - in >= lenW ) {
04660         if ( flip ) AO.FlipWORD(in);
04661         AO.ResizeWORD(in, out);
04662         *size += lenW;
04663         *outsize += sizeof(WORD);
04664     }
04665     /* handle vflags */
04666     if ( end - in >= lenW ) {
04667         if ( flip ) AO.FlipWORD(in);
04668         AO.ResizeWORD(in, out);
04669         *size += lenW;
04670         *outsize += sizeof(WORD);
04671     }
04672     /* handle uflags */
04673     if ( end - in >= lenW ) {
04674         if ( flip ) AO.FlipWORD(in);
04675         AO.ResizeWORD(in, out);
04676         *size += lenW;
04677         *outsize += sizeof(WORD);
04678     }
04679     return ( 0 );
04680 }
04681
04682 RSVEnd:
04683     /* we are here because the remaining buffer cannot hold the next
04684     struct. we position the file behind the last successfully translated
04685     struct and return. */
04686     ADDPOS(pos, *size);
04687     SeekFile(AO.SaveData.Handle, &pos, SEEK_SET);
04688     return ( 0 );
04689 } else {
04690     return ( ReadFile(AO.SaveData.Handle, buffer, *size) != *size );
04691 }
04692 }
04693
04694 /*
04695     #] ReadSaveVariables :
04696     #[ ReadSaveTerm :
04697 */
04698
04699 UBYTE *
04700 ReadSaveTerm32(UBYTE *bin, UBYTE *binend, UBYTE **bout, UBYTE *boutend, UBYTE *top, int terminbuf)
04701 {
04702     GETIDENTITY
04703     UBYTE *boutbuf;
04704     int32_t len, j, id;
04705     int32_t *r, *t, *coeff, *end, *newtermsize, *rend;
04706     int32_t *newsubterm;
04707     int32_t *in = (int32_t *)bin;
04708     int32_t *out = (int32_t *)*bout;
04709
04710     /* if called recursively the term is already decompressed in buffer.
04711     is this the case? */
04712     if ( terminbuf ) {
04713         /* don't do any decompression, just adjust the pointers */
04714         len = *out;
04715         end = out + len;
04716         r = in + 1;
04717         rend = (int32_t *)boutend;
04718         coeff = end - ABS(*end-1);
04719         newtermsize = (int32_t *)*bout;
04720         out = newtermsize + 1;
04721     }
04722     else {
04723         /* do decompression of necessary. always return if the space in the
04724         buffer is not sufficient */
04725         int32_t rbuf;

```

```

04755     r = (int32_t *)AR.CompressBuffer;
04756     rbuf = *r;
04757     len = j = *in;
04758     /* first copy from AR.CompressBuffer if necessary */
04759     if ( j < 0 ) {
04760         ++in;
04761         if ( (UBYTE *)in >= binend ) {
04762             return ( bin );
04763         }
04764         *out = len = -j + 1 + *in;
04765         end = out + *out;
04766         if ( (UBYTE *)end >= top ) {
04767             return ( bin );
04768         }
04769         ++out;
04770         *r++ = len;
04771         while ( ++j <= 0 ) {
04772             int32_t bb = *r++;
04773             *out++ = bb;
04774         }
04775         j = *in++;
04776     }
04777     else if ( j == 0 ) {
04778         /* care for padding words */
04779         while ( (UBYTE *)in < binend ) {
04780             *out++ = 0;
04781             if ( (UBYTE *)out > top ) {
04782                 return ( (UBYTE *)bin );
04783             }
04784             *r++ = 0;
04785             ++in;
04786         }
04787         *bout = (UBYTE *)out;
04788         return ( (UBYTE *)in );
04789     }
04790     else {
04791         end = out + len;
04792         if ( (UBYTE *)end >= top ) {
04793             return ( bin );
04794         }
04795     }
04796 }
04797 if ( (UBYTE *)in >= binend ) {
04798     *AR.CompressBuffer = rbuf;
04799     return ( bin );
04800 }
04801 if ( (UBYTE *)out >= top ) {
04802     return ( bin );
04803 }
04804 /* second copy from input buffer */
04805 while ( --j >= 0 ) {
04806     int32_t bb = *in++;
04807     *r++ = *out++ = bb;
04808 }
04809
04810 rend = r;
04811 r = (int32_t *)AR.CompressBuffer + 1;
04812 coeff = end - ABS(*end-1);
04813 newtermsize = (int32_t *)*bout;
04814 out = newtermsize + 1;
04815 }
04816
04817 /* iterate over subterms */
04818 while ( out < coeff ) {
04819
04820     id = *out++;
04821     ++r;
04822     t = out + *out - 1;
04823     newsubterm = out;
04824     ++out; ++r;
04825
04826     if ( id == SYMBOL ) {
04827         while ( out < t ) {
04828             ++out; ++r; /* symbol number */
04829             /* if exponent is too big, rewrite as exponent function */
04830             if ( ABS(*out) >= MAXPOWER ) {
04831                 int32_t *a, *b;
04832                 int32_t n;
04833                 int32_t num = *(out-1);
04834                 int32_t exp = *out;
04835                 coeff += 9;
04836                 end += 9;
04837                 t += 9;
04838                 if ( (UBYTE *)end > top ) return ( bin );
04839                 out -= 3;
04840                 *out++ = EXPONENT; /* id */
04841                 *out++ = 13; /* size */

```

```

04842         *out++ = 1;                /* dirtyflag */
04843         *out++ = -SYMBOL;          /* first short arg */
04844         *out++ = num;
04845         *out++ = 8;                /* second arg, size */
04846         *out++ = 0;                /* dirtyflag */
04847         *out++ = 6;                /* term size */
04848         *out++ = ABS(exp) & 0x0000FFFF;
04849         *out++ = ABS(exp) » 16;
04850         *out++ = 1;
04851         *out++ = 0;
04852         *out++ = ( exp < 0 ) ? -5 : 5;
04853         a = ++r;
04854         b = out;
04855         n = rend - r;
04856         NCOPYI32(b, a, n)
04857     }
04858     else {
04859         ++out; ++r;
04860     }
04861 }
04862 }
04863 else if ( id == DOTPRODUCT ) {
04864     while ( out < t ) {
04865         AO.RenumberVec((UBYTE *)out); /* vector 1 */
04866         ++out; ++r;
04867         AO.RenumberVec((UBYTE *)out); /* vector 2 */
04868         ++out; ++r;
04869         /* if exponent is too big, rewrite as exponent function */
04870         if ( ABS(*out) >= MAXPOWER ) {
04871             int32_t *a, *b;
04872             int32_t n;
04873             int32_t num1 = *(out-2);
04874             int32_t num2 = *(out-1);
04875             int32_t exp = *out;
04876             coeff += 17;
04877             end += 17;
04878             t += 17;
04879             if ( (UBYTE *)end > top ) return ( bin );
04880             out -= 4;
04881             *out++ = EXPONENT;        /* id */
04882             *out++ = 22;              /* size */
04883             *out++ = 1;               /* dirtyflag */
04884             *out++ = 11;              /* first arg, size */
04885             *out++ = 0;               /* dirtyflag */
04886             *out++ = 9;               /* term size */
04887             *out++ = DOTPRODUCT;      /* p1.p2 */
04888             *out++ = 5;               /* subterm size */
04889             *out++ = num1;            /* p1 */
04890             *out++ = num2;            /* p2 */
04891             *out++ = 1;               /* exponent */
04892             *out++ = 1;               /* coeff */
04893             *out++ = 1;
04894             *out++ = 3;
04895             *out++ = 8;               /* second arg, size */
04896             *out++ = 0;               /* dirtyflag */
04897             *out++ = 6;               /* term size */
04898             *out++ = ABS(exp) & 0x0000FFFF;
04899             *out++ = ABS(exp) » 16;
04900             *out++ = 1;
04901             *out++ = 0;
04902             *out++ = ( exp < 0 ) ? -5 : 5;
04903             a = ++r;
04904             b = out;
04905             n = rend - r;
04906             NCOPYI32(b, a, n)
04907         }
04908         else {
04909             ++out; ++r;
04910         }
04911     }
04912 }
04913 else if ( id == VECTOR ) {
04914     while ( out < t ) {
04915         AO.RenumberVec((UBYTE *)out); /* vector number */
04916         ++out; ++r;
04917         ++out; ++r; /* index, do nothing */
04918     }
04919 }
04920 else if ( id == INDEX ) {
04921     /* int32_t vectoroffset = -2 * *((int32_t *)AO.SaveHeader.wildoffset); */
04922     void *dummy = (void *)AO.SaveHeader.wildoffset; /* to remove a warning about
strict-aliasing rules in gcc */
04923     int32_t vectoroffset = -2 * *((int32_t *)dummy);
04924     while ( out < t ) {
04925         /* if there is a vector, renumber it */
04926         if ( *out < vectoroffset ) {
04927             AO.RenumberVec((UBYTE *)out);

```

```

04928         }
04929         ++out; ++r;
04930     }
04931 }
04932 else if ( id == SUBEXPRESSION ) {
04933     /* nothing to translate */
04934     while ( out < t ) {
04935         ++out; ++r;
04936     }
04937 }
04938 else if ( id == DELTA ) {
04939     /* nothing to translate */
04940     r += t - out;
04941     out = t;
04942 }
04943 else if ( id == HAAKJE ) {
04944     /* nothing to translate */
04945     r += t - out;
04946     out = t;
04947 }
04948 else if ( id == GAMMA || id == LEVICIVITA || (id >= FUNCTION && AO.tensorList[id]) ) {
04949     /* int32_t vectoroffset = -2 * *((int32_t *)AO.SaveHeader.wildoffset); */
04950     void *dummy = (void *)AO.SaveHeader.wildoffset; /* to remove a warning about
04951 strict-aliasing rules in gcc */
04952     int32_t vectoroffset = -2 * *((int32_t *)dummy);
04953     while ( out < t ) {
04954         /* if there is a vector as an argument, renumber it */
04955         if ( *out < vectoroffset ) {
04956             AO.RenumberVec((UBYTE *)out);
04957         }
04958         ++out; ++r;
04959     }
04960 }
04961 else if ( id >= FUNCTION ) {
04962     int32_t *argEnd;
04963     UBYTE *newbin;
04964     ++out; ++r; /* dirty flags */
04965     /* loop over arguments */
04966     while ( out < t ) {
04967         if ( *out < 0 ) {
04968             /* short notation arguments */
04969             switch ( -*out ) {
04970                 case SYMBOL:
04971                     ++out; ++r;
04972                     ++out; ++r;
04973                     break;
04974                 case SNUMBER:
04975                     argEnd = out+2;
04976                     ++out; ++r;
04977                     if ( sizeof(WORD) == 2 ) {
04978                         /* resize if needed */
04979                         if ( *out > (1<15)-1 || *out < -(1<15)+1 ) {
04980                             int32_t *a, *b;
04981                             int32_t n;
04982                             int32_t num = *out;
04983                             coeff += 6;
04984                             end += 6;
04985                             argEnd += 6;
04986                             t += 6;
04987                             if ( (UBYTE *)end > top ) return ( bin );
04988                             --out;
04989                             *out++ = 8; /* argument size */
04990                             *out++ = 0; /* dirtyflag */
04991                             *out++ = 6; /* term size */
04992                             *out++ = ABS(num) & 0x0000FFFF;
04993                             *out++ = ABS(num) » 16;
04994                             *out++ = 1;
04995                             *out++ = 0;
04996                             *out++ = ( num < 0 ) ? -5 : 5;
04997                             a = ++r;
04998                             b = out;
04999                             n = rend - r;
05000                             NCOPYI32(b, a, n)
05001                         }
05002                     }
05003                     else {
05004                         ++out; ++r;
05005                     }
05006                 }
05007             else {
05008                 ++out; ++r;
05009             }
05010             break;
05011         case VECTOR:
05012             ++out; ++r;
05013             AO.RenumberVec((UBYTE *)out);

```

```

05014             ++out; ++r;
05015             break;
05016         case INDEX:
05017             ++out; ++r;
05018             ++out; ++r;
05019             break;
05020         case MINVECTOR:
05021             ++out; ++r;
05022             AO.RenumberVec((UBYTE *)out);
05023             ++out; ++r;
05024             break;
05025         default:
05026             if ( -*out >= FUNCTION ) {
05027                 ++out; ++r;
05028                 break;
05029             } else {
05030                 MesPrint("short function code %d not implemented.", *out);
05031                 return ( (UBYTE *)in );
05032             }
05033     }
05034 }
05035 else {
05036     /* long arguments */
05037     int32_t *newargsize = out;
05038     argEnd = out + *out;
05039     ++out; ++r;
05040     ++out; ++r; /* dirty flags */
05041     while ( out < argEnd ) {
05042         int32_t *keepsizes = out + *out;
05043         int32_t lenbuf = *out;
05044         int32_t **ppp = &out; /* to avoid a compiler warning */
05045         /* recursion */
05046         newbin = ReadSaveTerm32((UBYTE *)r, binend, (UBYTE **)ppp, (UBYTE *)rend, top,
1);
05047         r += lenbuf;
05048         if ( newbin == (UBYTE *)r ) {
05049             return ( (UBYTE *)in );
05050         }
05051         /* if the term done by recursion has changed in size,
05052            we need to move the rest of the data accordingly */
05053         if ( out > keepsizes ) {
05054             int32_t *a, *b;
05055             int32_t n;
05056             int32_t extension = out - keepsizes;
05057             a = r;
05058             b = out;
05059             n = rend - r;
05060             NCOPYI32(b, a, n)
05061             coeff += extension;
05062             end += extension;
05063             argEnd += extension;
05064             t += extension;
05065         }
05066         else if ( out < keepsizes ) {
05067             int32_t *a, *b;
05068             int32_t n;
05069             int32_t extension = keepsizes - out;
05070             a = keepsizes;
05071             b = out;
05072             n = rend - r;
05073             NCOPYI32(b, a, n)
05074             coeff -= extension;
05075             end -= extension;
05076             argEnd -= extension;
05077             t -= extension;
05078         }
05079     }
05080     *newargsize = out - newargsize;
05081 }
05082 }
05083 }
05084 else {
05085     MesPrint("ID %d not recognized.", id);
05086     return ( (UBYTE *)in );
05087 }
05088
05089 *newsubterm = out - newsubterm + 1;
05090 }
05091
05092 if ( (UBYTE *)end >= top ) {
05093     return ( bin );
05094 }
05095
05096 /* do coefficient and adjust term size */
05097 boutbuf = *bout;
05098 *bout = (UBYTE *)out;
05099

```

```

05100     ResizeCoeff32(bout, (UBYTE *)end, top);
05101
05102     if ( *bout >= top ) {
05103         *bout = boutbuf;
05104         return ( bin );
05105     }
05106
05107     *newtermsize = (int32_t *)*bout - newtermsize;
05108
05109     return ( (UBYTE *)in );
05110 }
05111
05112 /*
05113     #] ReadSaveTerm :
05114     #[ ReadSaveExpression :
05115 */
05116
05138 LONG ReadSaveExpression(UBYTE *buffer, UBYTE *top, LONG *size, LONG *outsize)
05139 {
05140     if ( AO.transFlag ) {
05141         UBYTE *in, *end, *out, *outend, *p;
05142         POSITION pos;
05143         LONG half;
05144         WORD lenW = AO.SaveHeader.lenWORD;
05145
05146         /* remember the last file position in case an expression cannot be
05147            fully processed */
05148         TELLFILE(AO.SaveData.Handle,&pos);
05149
05150         /* adjust 'size' depending on whether the translated data is bigger or
05151            smaller */
05152         half = (top-buffer)/2;
05153         if ( *size > half ) *size = half;
05154         if ( lenW < (WORD)sizeof(WORD) ) {
05155             if ( *size * (LONG)sizeof(WORD)/lenW > half ) *size = half*lenW/(LONG)sizeof(WORD);
05156         }
05157         else {
05158             if ( *size > half ) *size = half;
05159         }
05160
05161         /* depending on the necessary resizing we position the input pointer
05162            either at the start of the buffer or in the middle. if the data will
05163            roughly remain the same size, we need only one processing step, so
05164            we put the 'in' at the middle and 'out' and the beginning. in the
05165            other cases we need two processing steps, so first we put 'in' at
05166            the beginning and write at the middle. the second step can then read
05167            from the middle and put its results at the beginning. */
05168         in = out = buffer;
05169         if ( lenW == sizeof(WORD) ) in += half;
05170         else out += half;
05171         end = in + *size;
05172         outend = out + *size;
05173
05174         if ( ReadFile(AO.SaveData.Handle, in, *size) != *size ) {
05175             return ( MesPrint("Error(3) reading stored expression.") );
05176         }
05177
05178         if ( AO.transFlag & 1 ) {
05179             p = in;
05180             end -= lenW;
05181             while ( p <= end ) {
05182                 AO.FlipWORD(p); p += lenW;
05183             }
05184             end += lenW;
05185         }
05186
05187         if ( lenW > (WORD)sizeof(WORD) ) {
05188             /* renumber first */
05189             do {
05190                 outend = out+*size;
05191                 if ( outend > top ) outend = top;
05192                 p = ReadSaveTerm32(in, end, &out, outend, top, 0);
05193                 if ( p == in ) break;
05194                 in = p;
05195             } while ( in <= end - lenW );
05196             /* then resize */
05197             *size = in - buffer;
05198             in = buffer + half;
05199             end = out;
05200             out = buffer;
05201
05202             while ( in < end ) {
05203                 /* resize without checking */
05204                 AO.ResizeNCWORD(in, out);
05205                 in += lenW; out += sizeof(WORD);
05206             }
05207         }
05208     }

```



```

05208         else {
05209             if ( lenW < (WORD)sizeof(WORD) ) {
05210                 /* resize first */
05211                 while ( in < end ) {
05212                     AO.ResizeWORD(in, out);
05213                     in += lenW; out += sizeof(WORD);
05214                 }
05215                 in = buffer + half;
05216                 end = out;
05217                 out = buffer;
05218             }
05219             /* then renumber */
05220             do {
05221                 p = ReadSaveTerm32(in, end, &out, buffer+half, buffer+half, 0);
05222                 if ( p == in ) break;
05223                 in = p;
05224             } while ( in <= end - sizeof(WORD) );
05225             *size = (in - buffer - half) * lenW / (ULONG)sizeof(WORD);
05226         }
05227         *outsize = out - buffer;
05228         ADDPOS(pos, *size);
05229         SeekFile(AO.SaveData.Handle, &pos, SEEK_SET);
05230
05231         return ( 0 );
05232     }
05233     else {
05234         return ( ReadFile(AO.SaveData.Handle, buffer, *size) != *size );
05235     }
05236 }
05237
05238 /*
05239     #] ReadSaveExpression :
05240     #] System Independent Saved Expressions :
05241 */

```

## 4.136 structs.h File Reference

This graph shows which files directly or indirectly include this file:



### Data Structures

- struct [PoSiTiOn](#)
- struct [STOREHEADER](#)
- struct [InDeXeNtRy](#)
- struct [FiLeInDeX](#)
- struct [FiLeDaTa](#)
- struct [VaRrEnUm](#)
- struct [ReNuMbEr](#)
- struct [LIST](#)
- struct [KEYWORD](#)
- struct [KEYWORDV](#)
- struct [NaMeNode](#)
- struct [NaMeTree](#)
- struct [tree](#)
- struct [MiNmAx](#)
- struct [BrAcKeTiNdEx](#)
- struct [BrAcKeTiNfO](#)
- struct [TaBlEs](#)
- struct [ExPrEsSiOn](#)
- struct [SyMbOl](#)

- struct [InDeX](#)
- struct [VeCtOr](#)
- struct [FuNcTiOn](#)
- struct [SeTs](#)
- struct [DuBiOuS](#)
- struct [FaCdOILaR](#)
- struct [DoLIaRS](#)
- struct [MoDoPtDoLIaRS](#)
- struct [fixedset](#)
- struct [TaBIeBAsEsUbInDeX](#)
- struct [TaBIeBAsE](#)
- struct [FUN\\_INFO](#)
- struct [PaRtlcLe](#)
- struct [VeRtEx](#)
- struct [MoDeL](#)
- struct [NaMeSpAcE](#)
- struct [FiLe](#)
- struct [StreaM](#)
- struct [SpecTatoR](#)
- struct [TrAcEs](#)
- struct [TrAcEn](#)
- struct [pReVaR](#)
- struct [INSIDEINFO](#)
- struct [PRELOAD](#)
- struct [PROCEDURE](#)
- struct [DoLoOp](#)
- struct [bit\\_field](#)
- struct [HANDLERS](#)
- struct [CbUf](#)
- struct [ChAnNeL](#)
- struct [SETUPPARAMETERS](#)
- struct [NeStInG](#)
- struct [StOrEcAcHe](#)
- struct [PeRmUtE](#)
- struct [PeRmUtEp](#)
- struct [DiStRiBuTe](#)
- struct [PaRtl](#)
- struct [sOrT](#)
- struct [POLYMOD](#)
- struct [SHvariables](#)
- struct [COST](#)
- struct [MODNUM](#)
- struct [OPTIMIZE](#)
- struct [OPTIMIZERESULT](#)
- struct [DICTIONARY\\_ELEMENT](#)
- struct [DICTIONARY](#)
- struct [SWITCHTABLE](#)
- struct [SWITCH](#)
- struct [TERMINFO](#)
- struct [M\\_const](#)
- struct [P\\_const](#)
- struct [C\\_const](#)
- struct [S\\_const](#)
- struct [R\\_const](#)
- struct [T\\_const](#)

- struct [N\\_const](#)
- struct [O\\_const](#)
- struct [X\\_const](#)
- struct [AllGlobals](#)
- struct [FixedGlobals](#)

## Macros

- #define [INFILEINDEX](#) ((512-2\*sizeof([POSITION](#)))/sizeof([INDEXENTRY](#)))
- #define [EMPTYININDEX](#) (512-2\*sizeof([POSITION](#))-[INFILEINDEX](#)\*sizeof([INDEXENTRY](#)))
- #define [PHEAD](#)
- #define [PHEAD0](#) void
- #define [BHEAD](#)
- #define [BHEAD0](#)

## Typedefs

- typedef struct [PoSiTiOn](#) **POSITION**
- typedef struct [InDeXeNtRy](#) **INDEXENTRY**
- typedef struct [FiLeInDeX](#) **FILEINDEX**
- typedef struct [FiLeDaTa](#) **FILEDATA**
- typedef struct [VaRrEnUm](#) **VARRENUM**
- typedef struct [ReNuMbEr](#) \* **RENUMBER**
- typedef struct [NaMeNode](#) **NAMENODE**
- typedef struct [NaMeTree](#) **NAMETREE**
- typedef struct [tree](#) **COMPTREE**
- typedef struct [MiNmAx](#) **MINMAX**
- typedef struct [BrAcKeTiNdEx](#) **BRACKETINDEX**
- typedef struct [BrAcKeTiNfO](#) **BRACKETINFO**
- typedef struct [TaBIes](#) \* **TABLES**
- typedef struct [ExPrEsSiOn](#) \* **EXPRESSIONS**
- typedef struct [SyMbOl](#) \* **SYMBOLS**
- typedef struct [InDeX](#) \* **INDICES**
- typedef struct [VeCtOr](#) \* **VECTORS**
- typedef struct [FuNcTiOn](#) \* **FUNCTIONS**
- typedef struct [SeTs](#) \* **SETS**
- typedef struct [DuBiOuS](#) \* **DUBIOUSV**
- typedef struct [FaCdOILaR](#) **FACDOLLAR**
- typedef struct [DoLIaR](#) \* **DOLLARS**
- typedef struct [MoDoPtDoLIaR](#) **MODOPTDOLLAR**
- typedef struct [fixedset](#) **FIXEDSET**
- typedef struct [TaBIEbAsEsUbInDeX](#) **TABLEBASESUBINDEX**
- typedef struct [TaBIEbAsE](#) **TABLEBASE**
- typedef struct [PaRtlcLe](#) **PARTICLE**
- typedef struct [VeRtEx](#) **VERTEX**
- typedef struct [MoDeL](#) **MODEL**
- typedef struct [NaMeSpAcE](#) **NAMESPACE**
- typedef struct [FiLe](#) **FILEHANDLE**
- typedef struct [StreaM](#) **STREAM**
- typedef struct [SpecTatoR](#) **SPECTATOR**
- typedef struct [TrAcEs](#) **TRACES**
- typedef struct [TrAcEn](#) \* **TRACEN**
- typedef struct [pReVaR](#) **PREVAR**

- typedef struct [DoLoOp](#) DOLOOP
- typedef struct [bit\\_field](#) set\_of\_char[32]
- typedef struct [bit\\_field](#) \* one\_byte
- typedef WORD(\* [FINISHUFFLE](#)) (WORD \*)
- typedef WORD(\* [DO\\_UFFLE](#)) (WORD \*, WORD, WORD, WORD)
- typedef WORD(\* [COMPAREDUMMY](#)) (WORD \*, WORD \*, WORD)
- typedef struct [CbUf](#) CBUF
- typedef struct [ChAnNeL](#) CHANNEL
- typedef struct [NeStInG](#) \* NESTING
- typedef struct [StOrEcAcHe](#) \* STORECACHE
- typedef struct [PeRmUtE](#) PERM
- typedef struct [PeRmUtEp](#) PERMP
- typedef struct [DiStRiBuTe](#) DISTRIBUTE
- typedef struct [PaRtl](#) PARTI
- typedef struct [sOrT](#) SORTING
- typedef struct [AllGlobals](#) ALLGLOBALS
- typedef WORD(\* [WCN](#)) ()
- typedef WORD(\* [WCN2](#)) ()
- typedef WORD(\* [COMPARE](#)) (PHEAD WORD \*, WORD \*, WORD)
- typedef struct [FixedGlobals](#) FIXEDGLOBALS

## Functions

- **STATIC\_ASSERT** (sizeof([STOREHEADER](#))==512)
- **STATIC\_ASSERT** (sizeof([FILEINDEX](#))==512)

### 4.136.1 Detailed Description

Contains definitions for global structs.

!!!CAUTION!!! Changes in this file will most likely have consequences for the recovery mechanism (see [checkpoint.c](#)). You need to care for the code in [checkpoint.c](#) as well and modify the code there accordingly!

The marker [D] is used in comments in this file to mark pointers to which dynamically allocated memory is assigned by a call to malloc() during runtime (in contrast to pointers that point into already allocated memory). This information is especially helpful if one needs to know which pointers need to be freed (cf. [checkpoint.c](#)).

Definition in file [structs.h](#).

### 4.136.2 Macro Definition Documentation

#### 4.136.2.1 INFILEINDEX

```
#define INFILEINDEX ((512-2*sizeof(POSITION))/sizeof(INDEXENTRY))
```

Maximum number of entries (struct [InDeXeNtRy](#)) in a file index (struct [FiLeInDeX](#)). Number is calculated such that the size of a file index is no more than 512 bytes.

Definition at line 119 of file [structs.h](#).

#### 4.136.2.2 EMPTYINDEX

```
#define EMPTYINDEX (512-2*sizeof(POSITION)-INFILEINDEX*sizeof(INDEXENTRY))
```

Number of empty filling bytes for a file index (struct [FiLeInDeX](#)). It is calculated such that the size of a file index is always 512 bytes.

Definition at line 124 of file [structs.h](#).

#### 4.136.2.3 PHEAD

```
#define PHEAD
```

Definition at line 2647 of file [structs.h](#).

#### 4.136.2.4 PHEAD0

```
#define PHEAD0 void
```

Definition at line 2648 of file [structs.h](#).

#### 4.136.2.5 BHEAD

```
#define BHEAD
```

Definition at line 2649 of file [structs.h](#).

#### 4.136.2.6 BHEAD0

```
#define BHEAD0
```

Definition at line 2650 of file [structs.h](#).

### 4.136.3 Typedef Documentation

#### 4.136.3.1 INDEXENTRY

```
typedef struct InDeXeNtRy INDEXENTRY
```

Defines the structure of an entry in a file index (see struct [FiLeInDeX](#)).

It represents one expression in the file.

#### 4.136.3.2 FILEINDEX

```
typedef struct FileInDeX FILEINDEX
```

Defines the structure of a file index in store-files and save-files.

It contains several entries (see struct [InDeXeNtRy](#)) up to a maximum of [INFILEINDEX](#).

The variable number has been made of type POSITION to avoid padding problems with some types of computers/↔ OS and keep system independence of the .sav files.

This struct is always 512 bytes long.

#### 4.136.3.3 VARRENUM

```
typedef struct VaRrEnUm VARRENUM
```

Contains the pointers to an array in which a binary search will be performed.

#### 4.136.3.4 RENUMBER

```
typedef struct ReNuMbEr * RENUMBER
```

Only symb.lo gets dynamically allocated. All other pointers points into this memory.

#### 4.136.3.5 NAMENODE

```
typedef struct NaMeNode NAMENODE
```

The names of variables are kept in an array. Elements of type [NAMENODE](#) define a tree (that is kept balanced) that make it easy and fast to look for variables. See also [NAMETREE](#).

#### 4.136.3.6 NAMETREE

```
typedef struct NaMeTree NAMETREE
```

A struct of type [NAMETREE](#) controls a complete (balanced) tree of names for the compiler. The compiler maintains several of such trees and the system has been set up in such a way that one could define more of them if we ever want to work with local name spaces.

#### 4.136.3.7 COMPTREE

```
typedef struct tree COMPTREE
```

The subexpressions in the compiler are kept track of in a (balanced) tree to reduce the need for subexpressions and hence save much space in large rhs expressions (like when we have xxxxxx occurrences of objects like  $f(x+1, x+1)$  in which each  $x+1$  becomes a subexpression. The struct that controls this tree is COMPTREE.

#### 4.136.3.8 TABLES

```
typedef struct TaBleS * TABLES
```

buffers, mm, flags, and prototype are always dynamically allocated, tablepointers only if needed (=0 if unallocated), boomlijst and argtail only for sparse tables.

Allocation is done for both the normal and the stub instance (spare), except for prototype and argtail which share memory.

#### 4.136.3.9 FUNCTIONS

```
typedef struct FuNcTiOn * FUNCTIONS
```

Contains all information about a function. Also used for tables. It is used in the [LIST](#) elements of AC.

#### 4.136.3.10 FILEHANDLE

```
typedef struct FiLe FILEHANDLE
```

The type FILEHANDLE is the struct that controls all relevant information of a file, whether it is open or not. The file may even not yet exist. There is a system of caches (PObuffer) and as long as the information to be written still fits inside the cache the file may never be created. There are variables that can store information about different types of files, like scratch files or sort files. Depending on what is available in the system we may also have information about gzip compression (currently sort file only) or locks (TFORM).

#### 4.136.3.11 STREAM

```
typedef struct StreaM STREAM
```

Input is read from 'streams' which are represented by objects of type STREAM. A stream can be a file, a do-loop, a procedure, the string value of a preprocessor variable ..... When a new stream is opened we have to keep information about where to fall back in the parent stream to allow this to happen even in the middle of reading names etc as would be the case with a`i`b

#### 4.136.3.12 TRACES

```
typedef struct TrAcEs TRACES
```

The struct TRACES keeps track of the progress during the expansion of a 4-dimensional trace. Each time a term gets generated the expansion tree continues in the next statement. When it returns it has to know where to continue. The 4-dimensional traces are more complicated than the n-dimensional traces (see TRACEN) because of the extra tricks that can be used. They are responsible for the shorter final expressions.

#### 4.136.3.13 TRACEN

```
typedef struct TrAcEn * TRACEN
```

The struct TRACEN keeps track of the progress during the expansion of a 4-dimensional trace. Each time a term gets generated the expansion tree continues in the next statement. When it returns it has to know where to continue.

#### 4.136.3.14 PREVAR

```
typedef struct pReVaR PREVAR
```

An element of the type PREVAR is needed for each preprocessor variable.

#### 4.136.3.15 DOLOOP

```
typedef struct DoLoOp DOLOOP
```

Each preprocessor do loop has a struct of type DOLOOP to keep track of all relevant parameters like where the beginning of the loop is, what the boundaries, increment and value of the loop parameter are, etc. Also we keep the whole loop inside a buffer of type PRELOAD

#### 4.136.3.16 set\_of\_char

```
typedef struct bit_field set_of_char[32]
```

Used in set\_in, set\_set, set\_del and set\_sub.

Definition at line 952 of file [structs.h](#).

#### 4.136.3.17 one\_byte

```
typedef struct bit_field* one_byte
```

Used in set\_in, set\_set, set\_del and set\_sub.

Definition at line 958 of file [structs.h](#).

#### 4.136.3.18 FINISHUFFLE

```
typedef WORD(* FINISHUFFLE) (WORD *)
```

Definition at line 978 of file [structs.h](#).

#### 4.136.3.19 DO\_UFFLE

```
typedef WORD(* DO_UFFLE) (WORD *, WORD, WORD, WORD)
```

Definition at line 979 of file [structs.h](#).

#### 4.136.3.20 COMPAREDUMMY

```
typedef WORD(* COMPAREDUMMY) (WORD *, WORD *, WORD)
```

Definition at line 984 of file [structs.h](#).



#### 4.136.3.21 CBUF

```
typedef struct CbUf CBUF
```

The CBUF struct is used by the compiler. It is a compiler buffer of which since version 3.0 there can be many.

#### 4.136.3.22 CHANNEL

```
typedef struct ChAnNeL CHANNEL
```

When we read input from text files we have to remember not only their handle but also their name. This is needed for error messages. Hence we call such a file a channel and reserve a struct of type `CHANNEL` to allow to lay this link.

#### 4.136.3.23 NESTING

```
typedef struct NeStInG * NESTING
```

The NESTING struct is used when we enter the argument of functions and there is the possibility that we have to change something there. Because functions can be nested we have to keep track of all levels of functions in case we have to move the outer layers to make room for a larger function argument.

#### 4.136.3.24 STORECACHE

```
typedef struct StOrEcAcHe * STORECACHE
```

The struct of type STORECACHE is used by a caching system for reading terms from stored expressions. Each thread should have its own system of caches.

#### 4.136.3.25 PERM

```
typedef struct PeRmUtE PERM
```

The struct PERM is used to generate all permutations when the pattern matcher has to try to match (anti)symmetric functions.

#### 4.136.3.26 PERMP

```
typedef struct PeRmUtEp PERMP
```

Like struct PERM but works with pointers.

#### 4.136.3.27 DISTRIBUTE

```
typedef struct DiStRiBuTe DISTRIBUTE
```

The struct DISTRIBUTE is used to help the pattern matcher when matching antisymmetric tensors.

#### 4.136.3.28 PARTI

```
typedef struct PaRtI PARTI
```

The struct PARTI is used to help determining whether a `partition_` function can be replaced.

#### 4.136.3.29 SORTING

```
typedef struct sOrT SORTING
```

The struct SORTING is used to control a sort operation. It includes a small and a large buffer and arrays for keeping track of various stages of the (merge) sorts. Each sort level has its own struct and different levels can have different sizes for its arrays. Also different threads have their own set of SORTING structs.

#### 4.136.3.30 ALLGLOBALS

```
typedef struct AllGlobals ALLGLOBALS
```

Without pthreads (FORM) the ALLGLOBALS struct has all the global variables

#### 4.136.3.31 WCN

```
typedef WORD(* WCN) ()
```

Definition at line 2657 of file [structs.h](#).

#### 4.136.3.32 WCN2

```
typedef WORD(* WCN2) ()
```

Definition at line 2658 of file [structs.h](#).

#### 4.136.3.33 COMPARE

```
typedef WORD(* COMPARE) (PHEAD WORD *, WORD *, WORD)
```

Definition at line 2661 of file [structs.h](#).

#### 4.136.3.34 FIXEDGLOBALS

```
typedef struct FixedGlobals FIXEDGLOBALS
```

The FIXEDGLOBALS struct is an anachronism. It started as the struct with global variables that needed initialization. It contains the elements `Operation` and `OperaFind` which define a very early way of automatically jumping to the proper operation. We find the results of it in parts of the file [opera.c](#). Later operations were treated differently in a more transparent way. We never changed the existing code. The most important part is currently the `cTable` which is used intensively in the compiler.

## 4.137 structs.h

[Go to the documentation of this file.](#)

```

00001
00017 /* #[ License : */
00018 /*
00019 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00020 *   When using this file you are requested to refer to the publication
00021 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00022 *   This is considered a matter of courtesy as the development was paid
00023 *   for by FOM the Dutch physics granting agency and we would like to
00024 *   be able to track its scientific use to convince FOM of its value
00025 *   for the community.
00026 *
00027 *   This file is part of FORM.
00028 *
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00040 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00041 */
00042 /* #] License : */
00043
00044 #ifndef __STRUCTS__
00045
00046 #define __STRUCTS__
00047 #ifdef _MSC_VER
00048 #include <wchar.h> /* off_t */
00049 #endif
00050
00051 /*
00052 *   #[ sav&store :
00053 */
00054
00059 typedef struct PosiTiOn {
00060     off_t pl;
00061 } POSITION;
00062
00063 /* Next are the index structs for stored and saved expressions */
00064
00075 typedef struct {
00076     UBYTE headermark[8];
00077     UBYTE lenWORD;
00078     UBYTE lenLONG;
00079     UBYTE lenPOS;
00080     UBYTE lenPOINTER;
00081     UBYTEendianness[16];
00082     UBYTE sSym;
00083     UBYTE sInd;
00084     UBYTE sVec;
00085     UBYTE sFun;
00086     UBYTE maxpower[16];
00087     UBYTE wilddoffset[16];
00088     UBYTE revision;
00089     UBYTE reserved[512-8-4-16-4-16-16-1];
00090 } STOREHEADER;
00091
00092
00093 STATIC_ASSERT(sizeof(STOREHEADER) == 512);
00094
00100 typedef struct InDeXeNtRy {
00101     POSITION position;
00102     POSITION length;
00103     POSITION variables;
00104     LONG CompressSize;
00105     WORD nsymbols;
00106     WORD nindices;
00107     WORD nvectors;
00108     WORD nfunctions;
00109     WORD size;
00110     SBYTE name[MAXENAME+1];
00111     PADPOSITION(0,1,0,5,MAXENAME+1);
00112 } INDEXENTRY;
00113
00119 #define INFILEINDEX ((512-2*sizeof(POSITION))/sizeof(INDEXENTRY))
00124 #define EMPTYINDEX (512-2*sizeof(POSITION)-INFILEINDEX*sizeof(INDEXENTRY))
00125
00138 typedef struct FiLeInDeX {

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```

00139     POSITION    next;
00140     POSITION    number;
00141     INDEXENTRY expression[INFILEINDEX];
00142     SBYTE      empty[EMPTYINDEX];
00143 } FILEINDEX;
00144
00145 STATIC_ASSERT(sizeof(FILEINDEX) == 512);
00146
00151 typedef struct FileDaTa {
00152     FILEINDEX Index;
00153     POSITION Fill;
00154     POSITION Position;
00155     WORD Handle;
00156     WORD dirtyflag;
00157     PADPOSITION(0,0,0,2,0);
00158 } FILEDATA;
00159
00166 typedef struct VaRrEnUm {
00167     WORD *start;
00168     WORD *lo;
00169     WORD *hi;
00170 } VARRENUM;
00171
00178 typedef struct ReNuMbEr {
00179     POSITION startposition;
00180     /* First stage renumbering */
00181     VARRENUM symb;
00182     VARRENUM indi;
00183     VARRENUM vect;
00184     VARRENUM func;
00185     /* Second stage renumbering */
00186     WORD *symnum;
00187     WORD *indnum;
00188     WORD *vecnum;
00189     WORD *funnum;
00190     PADPOSITION(4,0,0,0,sizeof(VARRENUM)*4);
00191 } *RENUMBER;
00192
00193 /*
00194     #] sav&store :
00195     #[ Variables :
00196 */
00197
00205 typedef struct {
00206     void *lijst;
00207     char *message;
00208     int num;
00209     int maxnum;
00210     int size;
00211     int numglobal;
00212     int numtemp;
00213     int numclear;
00214     PADPOINTER(0,6,0,0);
00215 } LIST;
00216
00222 typedef struct {
00223     char *name;
00224     TFUN func;
00225     int type;
00226     int flags;
00227 } KEYWORD;
00228
00234 typedef struct {
00235     char *name;
00236     int *var;
00237     int type;
00238     int flags;
00239 } KEYWORDV;
00240
00247 typedef struct NaMeNode {
00248     LONG name;
00249     WORD parent;
00250     WORD left;
00251     WORD right;
00252     WORD balance;
00253     WORD type;
00254     WORD number;
00255     PADLONG(0,6,0);
00256 } NAMENODE;
00257
00265 typedef struct NaMeTree {
00266     NAMENODE *namenode;
00267     UBYTE *namebuffer;
00271     LONG nodesize;
00272     LONG nodefill;
00273     LONG namesize;
00274     LONG namefill;

```

```

00275     LONG     oldnamefill;
00276     LONG     oldnodefill;
00277     LONG     globalnamefill;
00279     LONG     globalnodefill;
00280     LONG     clearnamefill;
00281     LONG     clearnodefill;
00282     WORD     headnode;
00283     PADPOINTER(10,0,1,0);
00284 } NAMETREE;
00285
00294 typedef struct tree {
00295     int parent;
00296     int left;
00297     int right;
00298     int value;
00299     int blnce;
00300     int usage;
00301 } COMPTREE;
00302
00307 typedef struct MiNmAx {
00308     WORD mini;
00309     WORD maxi;
00310     WORD size;
00311 } MINMAX;
00312
00317 typedef struct BrAcKeTiNdEx { /* For indexing brackets in local expressions */
00318     POSITION start; /* Place where bracket starts - start of expr */
00319     POSITION next; /* Place of next indexed bracket in expr */
00320     LONG bracket; /* Offset of position in bracketbuffer */
00321     LONG termsinbracket;
00322     PADPOSITION(0,2,0,0,0);
00323 } BRACKETINDEX;
00324
00329 typedef struct BrAcKeTiNfO {
00330     BRACKETINDEX *indexbuffer;
00331     WORD *bracketbuffer;
00332     LONG bracketbuffersize;
00333     LONG indexbuffersize;
00334     LONG bracketfill;
00335     LONG indexfill;
00336     WORD SortType;
00337     PADPOINTER(4,0,1,0);
00338 } BRACKETINFO;
00339
00350 typedef struct TaBlEs {
00351     WORD *tablepointers;
00352 #ifdef WITHPTHREADS
00353     WORD **prototype;
00354     WORD **pattern;
00355 #else
00356     WORD *prototype;
00357     WORD *pattern;
00358 #endif
00359     MINMAX *mm;
00360     WORD *flags;
00361     COMPTREE *boomlijst;
00362     UBYTE *argtail;
00364     struct TaBlEs *spare;
00365     WORD *buffers;
00366     LONG totind;
00367     LONG reserved;
00368     LONG defined;
00369     LONG mdefined;
00370     int prototypeSize;
00371     int numind;
00372     int bounds;
00373     int strict;
00374     int sparse;
00375     int numtree;
00376     int rootnum;
00377     int MaxTreeSize;
00378     WORD bufnum;
00379     WORD bufferssize;
00380     WORD buffersfill;
00381     WORD tablenum;
00382     WORD mode;
00383     WORD numdummies;
00384     PADPOINTER(4,8,6,0);
00385 } *TABLES;
00386
00391 typedef struct ExPrEsSiOn {
00392     POSITION onfile;
00393     POSITION prototype;
00394     POSITION size;
00395     RENUMBER renum; /* For Renumbering of global stored expressions */
00396     BRACKETINFO *bracketinfo;
00397     BRACKETINFO *newbracketinfo;

```

```

00398     WORD    *renumlists;
00400     WORD    *inmem;          /* If in memory like e.g. a polynomial */
00401     LONG    counter;
00402     LONG    name;
00403     WORD    hidelevel;
00404     WORD    vflags;          /* Various flags */
00405     WORD    printflag;
00406     WORD    status;
00407     WORD    replace;
00408     WORD    node;
00409     WORD    whichbuffer;
00410     WORD    namesize;
00411     WORD    compression;
00412     WORD    numdummies;
00413     WORD    numfactors;
00414     WORD    sizeprototype;
00415     WORD    uflags;          /* User flags */
00416 #ifdef PARALLELCODE
00417     WORD    partodo;          /* Whether to be done in parallel mode */
00418     PADPOSITION(5,2,0,14,0);
00419 #else
00420     PADPOSITION(5,2,0,13,0);
00421 #endif
00422 } *EXPRESSIONS;
00423
00428 typedef struct SyMbOl {      /* Don't change unless altering .sav too */
00429     LONG    name;            /* Location in names buffer */
00430     WORD    minpower;         /* Minimum power admissible */
00431     WORD    maxpower;         /* Maximum power admissible */
00432     WORD    complex;          /* Properties wrt complex conjugation */
00433     WORD    number;           /* Number when stored in file */
00434     WORD    flags;            /* Used to indicate usage when storing */
00435     WORD    node;
00436     WORD    namesize;
00437     WORD    dimension;        /* For dimensionality checks */
00438     PADLONG(0,8,0);
00439 } *SYMBOLS;
00440
00445 typedef struct InDeX {      /* Don't change unless altering .sav too */
00446     LONG    name;            /* Location in names buffer */
00447     WORD    type;             /* Regular or dummy */
00448     WORD    dimension;        /* Value of d_(n,n) or -number of symbol */
00449     WORD    number;           /* Number when stored in file */
00450     WORD    flags;            /* Used to indicate usage when storing */
00451     WORD    nmin4;            /* Used for n-4 if dimension < 0 */
00452     WORD    node;
00453     WORD    namesize;
00454     PADLONG(0,7,0);
00455 } *INDICES;
00456
00461 typedef struct VeCtOr {    /* Don't change unless altering .sav too */
00462     LONG    name;            /* Location in names buffer */
00463     WORD    complex;          /* Properties under complex conjugation */
00464     WORD    number;           /* Number when stored in file */
00465     WORD    flags;            /* Used to indicate usage when storing */
00466     WORD    node;
00467     WORD    namesize;
00468     WORD    dimension;        /* For dimensionality checks */
00469     PADLONG(0,6,0);
00470 } *VECTORS;
00471
00477 typedef struct FuNcTiOn {   /* Don't change unless altering .sav too */
00478     TABLES tabl;
00479     LONG    symminfo;
00480     LONG    name;
00481     WORD    commute;
00482     WORD    complex;
00483     WORD    number;
00484     WORD    flags;
00485     WORD    spec;
00486     WORD    symmetric;
00487     WORD    node;
00488     WORD    namesize;
00489     WORD    dimension;        /* For dimensionality checks */
00490     WORD    maxnumargs;
00491     WORD    minnumargs;
00492     PADPOINTER(2,0,11,0);
00493 } *FUNCTIONS;
00494
00499 typedef struct SeTs {
00500     LONG    name;            /* Location in names buffer */
00501     WORD    type;             /* Symbol, vector, index or function */
00502     WORD    first;            /* First element in setstore */
00503     WORD    last;             /* Last element in setstore (excluding) */
00504     WORD    node;
00505     WORD    namesize;
00506     WORD    dimension;        /* For dimensionality checks */

```

```

00507     WORD    flags;                /* Like: ordered */
00508     PADLONG(0,7,0);
00509 } *SETS;
00510
00515 typedef struct DuBiOuS {          /* Undeclared objects. Just for compiler. */
00516     LONG     name;                /* Location in names buffer */
00517     WORD     node;
00518     WORD     dummy;
00519     PADLONG(0,2,0);
00520 } *DUBIOUSV;
00521
00522 typedef struct FaCdOlLaR {
00523     WORD     *where;              /* A pointer(!) to the content */
00524     LONG     size;
00525     WORD     type;                /* Type can be DOLNUMBER or DOLTERMS */
00526     WORD     value;              /* in case it is a (short) number */
00527     PADPOINTER(1,0,2,0);
00528 } FACDOLLAR;
00529
00530 typedef struct DoLlArS {
00531     WORD     *where;              /* A pointer(!) to the object */
00532     FACDOLLAR *factors;           /* an array of factors. nfactors elements */
00533 #ifdef WITHPTHREADS
00534     pthread_mutex_t pthreadslockread;
00535     pthread_mutex_t pthreadslockwrite;
00536 #endif
00537     LONG     size;                /* The number of words */
00538     LONG     name;
00539     WORD     type;
00540     WORD     node;
00541     WORD     index;
00542     WORD     zero;
00543     WORD     numdummies;
00544     WORD     nfactors;
00545 #ifdef WITHPTHREADS
00546     PADPOINTER(2,0,6,sizeof(pthread_mutex_t)*2);
00547 #else
00548     PADPOINTER(2,0,6,0);
00549 #endif
00550 } *DOLLARS;
00551
00556 typedef struct MoDoPtDoLlArS {
00557 #ifdef WITHPTHREADS
00558     DOLLARS  dstruct;            /* If local dollar: list of DOLLARS for each thread */
00559 #endif
00560     WORD     number;
00561     WORD     type;
00562 #ifdef WITHPTHREADS
00563     PADPOINTER(0,0,2,0);
00564 #endif
00565 } MODOPTDOLLAR;
00566
00571 typedef struct fixedset {
00572     char *name;
00573     char *description;
00574     int type;
00575     int dimension;
00576 } FIXEDSET;
00577
00582 typedef struct TaBlEbAsEsUbInDeX {
00583     POSITION where;
00584     LONG size;
00585     PADPOSITION(0,1,0,0,0);
00586 } TABLEBASESUBINDEX;
00587
00592 typedef struct TaBlEbAsE {
00593     POSITION fillpoint;
00594     POSITION current;
00595     UBYTE *name;
00596     int *tablenumbers;            /* Number of each table */
00597     TABLEBASESUBINDEX *subindex; /* For each table */
00598     int numtables;
00599     PADPOSITION(3,0,1,0,0);
00600 } TABLEBASE;
00601
00608 typedef struct {
00609     WORD *location;
00610     int numargs;
00611     int numfunnies;
00612     int numwildcards;
00613     int symmet;
00614     int tensor;
00615     int commute;
00616     PADPOINTER(0,6,0,0);
00617 } FUN_INFO;
00618
00619 typedef struct PaRtIcLe {

```

```

00620     WORD number; /* Number of the function */
00621     WORD spin;    /* +/- dimension of SU(2) representation */
00622     WORD mass;    /* Number of symbol or 0 */
00623     WORD type;    /* -1: anti particle, 1: particle, 0: own antiparticle */
00624 } PARTICLE;
00625
00626 typedef struct VerTex {
00627     PARTICLE particles[MAXPARTICLES];
00628     WORD couplings[2*MAXCOUPLINGS];
00629     WORD nparticles;
00630     WORD ncouplings;
00631     WORD type;
00632     WORD error;
00633     WORD externonly;
00634     WORD spare;
00635 } VERTEX;
00636
00637 typedef struct MoDeL {
00638     VERTEX **vertices;
00639     WORD *couplings;
00640     UBYTE *name;
00641     void *grccmodel;
00642     WORD legcouple[MAXLEGS+2];
00643     WORD nparticles;
00644     WORD nvertices;
00645     WORD invertices;
00646     WORD sizevertices;
00647     WORD sizecouplings;
00648     WORD ncouplings;
00649     WORD error;
00650     WORD dummy;
00651 } MODEL;
00652
00653 /*
00654  * The struct NAMESPACE is used for the namespace info. It is a
00655  * doubly linked list of nested namespaces.
00656  */
00657
00658 typedef struct NaMeSpAcE {
00659     struct NaMeSpAcE *previous;
00660     struct NaMeSpAcE *next;
00661     NAMETREE *usenames;
00662     UBYTE *name;
00663 } NAMESPACE;
00664
00665
00666 /*
00667  #] Variables :
00668  #[ Files :
00669  */
00670
00682 typedef struct File {
00683     POSITION PPosition; /* File position */
00684     POSITION filesize; /* Because SEEK_END is unsafe on IBM */
00685     WORD *PObuffer; /* Address of the intermediate buffer */
00686     WORD *POstop; /* End of the buffer */
00687     WORD *POfill; /* Fill position of the buffer */
00688     WORD *POfull; /* Full buffer when only cached */
00689 #ifdef WITHPTHREADS
00690     WORD *wPObuffer; /* Address of the intermediate worker buffer */
00691     WORD *wPOstop; /* End of the worker buffer */
00692     WORD *wPOfill; /* Fill position of the worker buffer */
00693     WORD *wPOfull; /* Full buffer when only worker cached */
00694 #endif
00695     char *name; /* name of the file */
00696 #ifdef WITHZLIB
00697     z_stream zsp; /* The pointer to the stream struct for gzip */
00698     Bytef *ziobuffer; /* The output buffer for compression */
00699 #endif
00700     ULONG numblocks; /* Number of blocks in file */
00701     ULONG inbuffer; /* Block in the buffer */
00702     LONG PSize; /* size of the buffer */
00703 #ifdef WITHZLIB
00704     LONG ziosize; /* size of the zoutbuffer */
00705 #endif
00706 #ifdef WITHPTHREADS
00707     LONG wPSize; /* size of the worker buffer */
00708     pthread_mutex_t pthreadslock;
00709 #endif
00710     int handle;
00711     int active; /* File is open or closed. Not used. */
00712 #ifdef WITHPTHREADS
00713 #ifdef WITHZLIB
00714     PADPOSITION(11,5,2,0,sizeof(pthread_mutex_t));
00715 #else
00716     PADPOSITION(9,4,2,0,sizeof(pthread_mutex_t));
00717 #endif
00717 #endif

```



```

00718 #else
00719 #ifdef WITHZLIB
00720     PADPOSITION(7,4,2,0,0);
00721 #else
00722     PADPOSITION(5,3,2,0,0);
00723 #endif
00724 #endif
00725 } FILEHANDLE;
00726
00736 typedef struct Stream {
00737     off_t fileposition;
00738     off_t linenumber;
00739     off_t prevline;
00740     UBYTE *buffer;
00741     UBYTE *pointer;
00742     UBYTE *top;
00743     UBYTE *FoldName;
00744     UBYTE *name;
00745     UBYTE *pname;
00747     LONG buffersize;
00748     LONG bufferposition;
00749     LONG inbuffer;
00750     int previous;
00751     int handle;
00752     int type;
00753     int prevars;
00754     int previousNoShowInput;
00755     int eqnum;
00756     int afterwards;
00757     int olddelay;
00758     int oldnoshowinput;
00759     UBYTE isnextchar;
00760     UBYTE nextchar[2];
00761     UBYTE reserved;
00762     PADPOSITION(6,3,9,0,4);
00763 } STREAM;
00764
00765 typedef struct SpecTatoR {
00766     POSITION position; /* The place where we will be writing */
00767     POSITION readpos; /* The place from which we read */
00768     FILEHANDLE *fh;
00769     char *name; /* We identify the spectator by the name of the expression */
00770     WORD expnnumber; /* During running we use the number. */
00771     WORD flags; /* local, global? */
00772     PADPOSITION(2,0,0,2,0);
00773 } SPECTATOR;
00774
00775 /*
00776 #] Files :
00777 #[ Traces :
00778 */
00779
00789 typedef struct TrAcEs { /* For computing 4 dimensional traces */
00790     WORD *accu; /* NUMBER * 2 */
00791     WORD *accup;
00792     WORD *termp;
00793     WORD *perm; /* number */
00794     WORD *inlist; /* number */
00795     WORD *nt3; /* number/2 */
00796     WORD *nt4; /* number/2 */
00797     WORD *j3; /* number*2 */
00798     WORD *j4; /* number*2 */
00799     WORD *e3; /* number*2 */
00800     WORD *e4; /* number */
00801     WORD *eers; /* number/2 */
00802     WORD *mepf; /* number/2 */
00803     WORD *mdel; /* number/2 */
00804     WORD *pepf; /* number*2 */
00805     WORD *pdel; /* number*3/2 */
00806     WORD sgn;
00807     WORD stap;
00808     WORD stepl,kstep,mdum;
00809     WORD gamm,ad,a3,a4,lc3,lc4;
00810     WORD sign1,sign2,gamma5,num,level,factor,allsign;
00811     WORD finalstep;
00812     PADPOINTER(0,0,19,0);
00813 } TRACES;
00814
00822 typedef struct TrAcEn { /* For computing n dimensional traces */
00823     WORD *accu; /* NUMBER */
00824     WORD *accup;
00825     WORD *termp;
00826     WORD *perm; /* number */
00827     WORD *inlist; /* number */
00828     WORD sgn,num,level,factor,allsign;
00829     PADPOINTER(0,0,5,0);
00830 } *TRACEN;

```

```

00831
00832 /*
00833 #] Traces :
00834 #[ Preprocessor :
00835 */
00836
00841 typedef struct pReVaR {
00842     UBYTE *name;
00843     UBYTE *value;
00844     UBYTE *argnames;
00845     int nargs;
00846     int wildarg;
00847     PADPOINTER(0,2,0,0);
00848 } PREVAR;
00849
00854 typedef struct {
00855     WORD *buffer;
00856     int oldcompiletype;
00857     int oldparallelflag;
00858     int oldnumpotmoddollars;
00859     WORD size;
00860     WORD numdollars;
00861     WORD oldcbuf;
00862     WORD oldrbuf;
00863     WORD inscbuf;
00864     WORD oldcnumlhs;
00865     PADPOINTER(0,3,6,0);
00866 } INSIDEINFO;
00867
00873 typedef struct {
00874     UBYTE *buffer;
00875     LONG size;
00876     PADPOINTER(1,0,0,0);
00877 } PRELOAD;
00878
00883 typedef struct {
00884     PRELOAD p;
00885     UBYTE *name;
00886     int loadmode;
00887     PADPOINTER(0,1,0,0);
00888 } PROCEDURE;
00889
00897 typedef struct DoLoOp {
00898     PRELOAD p;
00899     UBYTE *name;
00900     UBYTE *vars; /* for {} or name of expression */
00901     UBYTE *contents;
00902     UBYTE *dollarname;
00903     LONG startlinenumber;
00904     LONG firstnum;
00905     LONG lastnum;
00906     LONG incnum;
00907     int type;
00908     int NoShowInput;
00909     int errorsinloop;
00910     int firstloopcall;
00911     WORD firstdollar; /* When >= 0 we have to get the value from a dollar */
00912     WORD lastdollar; /* When >= 0 we have to get the value from a dollar */
00913     WORD incdollar; /* When >= 0 we have to get the value from a dollar */
00914     WORD NumPreTypes;
00915     WORD PreIfLevel;
00916     WORD PreSwitchLevel;
00917     PADPOINTER(4,4,6,0);
00918 } DOLOOP;
00919
00927 struct bit_field { /* Assume 8 bits per byte */
00928     UBYTE bit_0 : 1;
00929     UBYTE bit_1 : 1;
00930     UBYTE bit_2 : 1;
00931     UBYTE bit_3 : 1;
00932     UBYTE bit_4 : 1;
00933     UBYTE bit_5 : 1;
00934     UBYTE bit_6 : 1;
00935     UBYTE bit_7 : 1;
00936 /*
00937     UINT bit_0 : 1;
00938     UINT bit_1 : 1;
00939     UINT bit_2 : 1;
00940     UINT bit_3 : 1;
00941     UINT bit_4 : 1;
00942     UINT bit_5 : 1;
00943     UINT bit_6 : 1;
00944     UINT bit_7 : 1;
00945 */
00946 };
00947
00952 typedef struct bit_field set_of_char[32];

```

```

00953
00958 typedef struct bit_field *one_byte;
00959
00964 typedef struct {
00965     WORD newlogonly;
00966     WORD newhandle;
00967     WORD oldhandle;
00968     WORD oldlogonly;
00969     WORD oldprinttype;
00970     WORD oldsilent;
00971 } HANDLERS;
00972
00973 /*
00974     #] Preprocessor :
00975     #[ Varia :
00976 */
00977
00978 typedef WORD (*FINISHUFFLE) (WORD *);
00979 typedef WORD (*DO_UFFLE) (WORD *,WORD,WORD,WORD);
00980
00981 #ifdef WITHPTHREADS
00982 typedef WORD (*COMPAREDUMMY) (void *,WORD *,WORD *,WORD);
00983 #else
00984 typedef WORD (*COMPAREDUMMY) (WORD *,WORD *,WORD);
00985 #endif
00986
00992 typedef struct CbUf {
00993     WORD *Buffer;
00994     WORD *Top;
00995     WORD *Pointer;
00996     WORD **lhs;
00997     WORD **rhs;
00998     LONG *CanCommu;
00999     LONG *NumTerms;
01000     WORD *numdum;
01001     WORD *dimension;
01002     COMPTREE *boomlijst;
01003     LONG BufferSize;
01004     int numlhs;
01005     int numrhs;
01006     int maxlhs;
01007     int maxrhs;
01008     int mnumlhs;
01009     int mnumrhs;
01010     int numtree;
01011     int rootnum;
01012     int MaxTreeSize;
01013     PADPOINTER(1,9,0,0);
01014 } CBUF;
01015
01023 typedef struct ChAnNeL {
01024     char *name;
01025     int handle;
01026     PADPOINTER(0,1,0,0);
01027 } CHANNEL;
01028
01038 typedef struct {
01039     UBYTE *parameter;
01040     int type;
01041     int flags;
01042     LONG value;
01043 } SETUPPARAMETERS;
01044
01053 typedef struct NeStInG {
01054     WORD *termsize;
01055     WORD *funsize;
01056     WORD *argsize;
01057 } *NESTING;
01058
01065 typedef struct StOrEcAcHe {
01066     POSITION position;
01067     POSITION toppos;
01068     struct StOrEcAcHe *next;
01069     WORD buffer[2];
01070     PADPOSITION(1,0,0,2,0);
01071 } *STORECACHE;
01072
01078 typedef struct PeRmUtE {
01079     WORD *objects;
01080     WORD sign;
01081     WORD n;
01082     WORD cycle[MAXMATCH];
01083     PADPOINTER(0,0,MAXMATCH+2,0);
01084 } PERM;
01085
01090 typedef struct PeRmUtEp {
01091     WORD **objects;

```

```

01092     WORD sign;
01093     WORD n;
01094     WORD cycle[MAXMATCH];
01095     PADPOINTER(0,0,MAXMATCH+2,0);
01096 } PERMP;
01097
01103 typedef struct DiStRiBuTe {
01104     WORD *obj1;
01105     WORD *obj2;
01106     WORD *out;
01107     WORD sign;
01108     WORD n1;
01109     WORD n2;
01110     WORD n;
01111     WORD cycle[MAXMATCH];
01112     PADPOINTER(0,0,(MAXMATCH+4),0);
01113 } DISTRIBUTE;
01114
01120 typedef struct PaRtI {
01121     WORD *psize; /* the sizes of the partitions */
01122     WORD *args; /* the offsets of the arguments to be partitioned */
01123     WORD *nargs; /* argument numbers (different number = different argument) */
01124     WORD *nfun; /* the functions into which the partitions go */
01125     WORD numargs; /* the number of arguments to be partitioned */
01126     WORD numpart; /* the number of partitions */
01127     WORD where; /* offset of the function in the term */
01128     PADPOINTER(0,0,3,0);
01129 } PARTI;
01130
01140 typedef struct sOrT {
01141     FILEHANDLE file; /* The own sort file */
01142     POSITION SizeInFile[3]; /* Sizes in the various files */
01143     WORD *lBuffer; /* The large buffer */
01144     WORD *lTop; /* End of the large buffer */
01145     WORD *lFill; /* The filling point of the large buffer */
01146     WORD *used; /* auxiliary during actual sort */
01147     WORD *sBuffer; /* The small buffer */
01148     WORD *sTop; /* End of the small buffer */
01149     WORD *sTop2; /* End of the extension of the small buffer */
01150     WORD *sHalf; /* Halfway point in the extension */
01151     WORD *sFill; /* Filling point in the small buffer */
01152     WORD **sPointer; /* Pointers to terms in the small buffer */
01153     WORD **PoinFill; /* Filling point for pointers to the sm.buf */
01154     WORD *cBuffer; /* Compress buffer (if it exists) */
01155     WORD **Patches; /* Positions of patches in large buffer */
01156     WORD **PStop; /* Ends of patches in the large buffer */
01157     WORD **poina; /* auxiliary during actual sort */
01158     WORD **poin2a; /* auxiliary during actual sort */
01159     WORD *ktoi; /* auxiliary during actual sort */
01160     WORD *tree; /* auxiliary during actual sort */
01161 #ifdef WITHZLIB
01162     WORD *fpccompressed; /* is output filepatch compressed? */
01163     WORD *fpincompressed; /* is input filepatch compressed? */
01164     z_stream zsparray; /* the array of zstreams for decompression */
01165 #endif
01166     POSITION *fPatches; /* Positions of output file patches */
01167     POSITION *inPatches; /* Positions of input file patches */
01168     POSITION *fPatchesStop; /* Positions of output file patches */
01169     POSITION *iPatches; /* Input file patches, Points to fPatches or inPatches */
01170     FILEHANDLE *f; /* The actual output file */
01171     FILEHANDLE **ff; /* Handles for a staged sort */
01172     LONG sTerms; /* Terms in small buffer */
01173     LONG LargeSize; /* Size of large buffer (in words) */
01174     LONG SmallSize; /* Size of small buffer (in words) */
01175     LONG SmallEsize; /* Size of small + extension (in words) */
01176     LONG TermsInSmall; /* Maximum number of terms in small buffer */
01177     LONG Terms2InSmall; /* with extension for polyfuns etc. */
01178     LONG GenTerms; /* Number of generated terms */
01179     LONG TermsLeft; /* Number of terms still in existence */
01180     LONG GenSpace; /* Amount of space of generated terms */
01181     LONG SpaceLeft; /* Space needed for still existing terms */
01182     LONG putinsize; /* Size of buffer in putin */
01183     LONG ninterms; /* Which input term ? */
01184     int MaxPatches; /* Maximum number of patches in large buffer */
01185     int MaxfPatches; /* Maximum number of patches in one filesort */
01186     int type; /* Main, function or sub(routine) */
01187     int lPatch; /* Number of patches in the large buffer */
01188     int fPatchN1; /* Number of patches in input file */
01189     int PolyWise; /* Is there a polyfun and if so, where? */
01190     int PolyFlag; /* */
01191     int cBufferSize; /* Size of the compress buffer */
01192     int maxtermsize; /* Keeps track for buffer allocations */
01193     int newmaxtermsize; /* Auxiliary for maxtermsize */
01194     int outputmode; /* Tells where the output is going */
01195     int stagelevel; /* In case we have a 'staged' sort */
01196     WORD fPatchN; /* Number of patches on file (output) */
01197     WORD inNum; /* Number of patches on file (input) */

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01198     WORD stage4;                /* Are we using stage4? */
01199 #ifdef WITHZLIB
01200     PADPOSITION(27,12,12,3,0);
01201 #else
01202     PADPOSITION(24,12,12,3,0);
01203 #endif
01204 } SORTING;
01205
01206 #ifdef WITHPTHREADS
01207 typedef struct SoRtBlOcK {
01214     pthread_mutex_t *MasterBlockLock;
01215     WORD **MasterStart;
01216     WORD **MasterFill;
01217     WORD **MasterStop;
01218     int MasterNumBlocks;
01219     int MasterBlock;
01220     int FillBlock;
01221     PADPOINTER(0,3,0,0);
01222 } SORTBLOCK;
01223 #endif
01224
01225 #ifdef DEBUGGER
01226 typedef struct DeBuGgInG {
01227     int eflag;
01228     int printflag;
01229     int logfileflag;
01230     int stdoutflag;
01231 } DEBUGSTR;
01232 #endif
01233
01234 #ifdef WITHPTHREADS
01235 typedef struct ThReAdBuCkEt {
01242     POSITION *deferbuffer;        /* For Keep Brackets: remember position */
01243     WORD *threadbuffer;         /* Here are the (primary) terms */
01244     WORD *compressbuffer;       /* For keep brackets we need the compressbuffer */
01245     LONG threadbuffersize;      /* Number of words in threadbuffer */
01246     LONG ddterms;               /* Number of primary+secondary terms represented */
01247     LONG firsttterm;            /* The number of the first term in the bucket */
01248     LONG firstbracket;          /* When doing complete brackets */
01249     LONG lastbracket;           /* When doing complete brackets */
01250     pthread_mutex_t lock;       /* For the load balancing phase */
01251     int free;                   /* Status of the bucket */
01252     int totnum;                 /* Total number of primary terms */
01253     int usenum;                 /* Which is the term being used at the moment */
01254     int busy;                   /* */
01255     int type;                   /* Doing brackets? */
01256     PADPOINTER(5,5,0,sizeof(pthread_mutex_t));
01257 } THREADBUCKET;
01258
01259 #endif
01260
01268 typedef struct {
01269     WORD *coefs;                /* The array of coefficients */
01270     WORD numsym;                /* The number of the symbol in the polynomial */
01271     WORD arraysize;             /* The size of the allocation of coefs */
01272     WORD polysize;              /* The maximum power in the polynomial */
01273     WORD modnum;                /* The prime number of the modulus */
01274 } POLYMOD;
01275
01276 typedef struct {
01277     WORD *outterm;              /* Used in DoShuffle/Merge/FinishShuffle system */
01278     WORD *outfun;
01279     WORD *incoef;
01280     WORD *stop1;
01281     WORD *stop2;
01282     WORD *ststop1;
01283     WORD *ststop2;
01284     FINISHUFFLE finishshuf;
01285     DO_UFFLE do_uffle;
01286     LONG combilast;
01287     WORD nincoef;
01288     WORD level;
01289     WORD thefunction;
01290     WORD option;
01291     PADPOINTER(1,0,4,0);
01292 } SHvariables;
01293
01294 typedef struct {
01295     LONG add;                   /* Used for computing calculational cost in optim.c */
01296     LONG mul;
01297     LONG div;
01298     LONG pow;
01299 } COST;
01300
01301 typedef struct {

```

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01302     UWORD    *a;        /* The number array */
01303     UWORD    *m;        /* The modulus array */
01304     WORD     na;        /* Size of the number */
01305     WORD     nm;        /* size of the number in the modulus array */
01306 } MODNUM;
01307
01308 /*
01309     Struct for optimizing outputs. If changed, do not forget to change
01310     the padding information in the AO struct.
01311 */
01312 typedef struct {
01313     union { /* we do this to allow padding */
01314         float fval;
01315         int ival[2]; /* This should be enough */
01316     } mctsconstant;
01317     int    horner;
01318     int    hornerdirection;
01319     int    method;
01320     int    mctstimelimit;
01321     int    mctsnumexpand;
01322     int    mctsnumkeep;
01323     int    mctsnumrepeat;
01324     int    greedytimelimit;
01325     int    greedyminnum;
01326     int    greedymaxperc;
01327     int    printstats;
01328     int    debugflags;
01329     int    schemeflags;
01330     int    mctsdecaymode;
01331     int    salter; /* Simulated annealing updates */
01332     union {
01333         float fval;
01334         int ival[2];
01335     } saMaxT; /* Maximum temperature of SA */
01336     union {
01337         float fval;
01338         int ival[2];
01339     } saMinT; /* Minimum temperature of SA */
01340     int    spare;
01341 } OPTIMIZE;
01342
01343 typedef struct {
01344     WORD    *code;
01345     UBYTE    *nameofexpr; /* It is easier to remember an expression by name */
01346     LONG    codesize; /* We need this for the checkpoints */
01347     WORD    exprnr; /* Problem here is: we renumber them in execute.c */
01348     WORD    minvar;
01349     WORD    maxvar;
01350
01351     PADPOSITION(2,1,0,3,0);
01352 } OPTIMIZERESULT;
01353
01354 typedef struct {
01355     WORD    *lhs; /* Object to be replaced */
01356     WORD    *rhs; /* Depending on the type it will be UBYTE* or WORD* */
01357     int    type;
01358     int    size; /* Size of the lhs */
01359 } DICTIONARY_ELEMENT;
01360
01361 typedef struct {
01362     DICTIONARY_ELEMENT **elements;
01363     UBYTE    *name;
01364     int    sizeelements;
01365     int    numelements;
01366     int    numbers; /* deal with numbers */
01367     int    variables; /* deal with single variables */
01368     int    characters; /* deal with special characters */
01369     int    funwith; /* deal with functions with arguments */
01370     int    gnumelements; /* if .global shrinks the dictionary */
01371     int    ranges;
01372 } DICTIONARY;
01373
01374 typedef struct {
01375     WORD    ncase;
01376     WORD    value;
01377     WORD    compbuffer;
01378 } SWITCHTABLE;
01379
01380 typedef struct {
01381     SWITCHTABLE *table;
01382     SWITCHTABLE defaultcase;
01383     SWITCHTABLE endswitch;
01384     WORD    typetable;
01385     WORD    maxcase;
01386     WORD    mincase;
01387     WORD    numcases;
01388     WORD    tablesizes;

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01389     WORD caseoffset;
01390     WORD iflevel;
01391     WORD whilelevel;
01392     WORD nestingsum;
01393     WORD padding;
01394 } SWITCH;
01395
01396 typedef struct {
01397     WORD *term;
01398     void *currentModel;
01399     void *currentMODEL;
01400     WORD *legcouple[MAXLEGS+1];
01401     LONG numtopo;
01402     LONG numdia;
01403     WORD diaoffset;
01404     WORD level;
01405     WORD externalset;
01406     WORD internalset;
01407     WORD numextern;
01408     WORD flags;
01409 } TERMINFO;
01410
01411 /*
01412 #] Varia :
01413 #[ A :
01414     #[ M : The M struct is for global settings at startup or .clear
01415 */
01416
01424 struct M_const {
01425     POSITION zeropos; /* (M) is zero */
01426     SORTING *S0;
01427     UWORD *gcmmod;
01428     UWORD *gpowmod;
01429     UBYTE *TempDir; /* (M) Path with where the temporary files go */
01430     UBYTE *TempSortDir; /* (M) Path with where the sort files go */
01431     UBYTE *IncDir; /* (M) Directory path for include files */
01432     UBYTE *InputFileName; /* (M) */
01433     UBYTE *LogFileName; /* (M) */
01434     UBYTE *OutBuffer; /* (M) Output buffer in pre.c */
01435     UBYTE *Path; /* (M) */
01436     UBYTE *SetupDir; /* (M) Directory with setup file */
01437     UBYTE *SetupFile; /* (M) Name of setup file */
01438     UBYTE *gFortran90Kind;
01439     UBYTE *gextrasymp;
01440     UBYTE *ggextrasymp;
01441     UBYTE *oldnumextrasymp;
01442     SPECTATOR *SpectatorFiles;
01443 #ifdef WITHPTHREADS
01444     pthread_rwlock_t handlelock; /* (M) */
01445     pthread_mutex_t storefilelock; /* (M) */
01446     pthread_mutex_t sbuflock; /* (M) Lock for writing in the AM.sbuffer */
01447     LONG ThreadScratSize; /* (M) Size of Fscr[0/2] buffers of the workers */
01448     LONG ThreadScratOutSize; /* (M) Size of Fscr[1] buffers of the workers */
01449 #endif
01450     LONG MaxTer; /* (M) Maximum term size. Fixed at setup. In Bytes!!! */
01451     LONG CompressSize; /* (M) Size of Compress buffer */
01452     LONG ScratSize; /* (M) Size of Fscr[] buffers */
01453     LONG HideSize; /* (M) Size of Fscr[2] buffer */
01454     LONG SizeStoreCache; /* (M) Size of the caches for reading global expr. */
01455     LONG MaxStreamSize; /* (M) Maximum buffer size in reading streams */
01456     LONG SIOsize; /* (M) Sort InputOutput buffer size */
01457     LONG SLargeSize; /* (M) */
01458     LONG SSmallEsize; /* (M) */
01459     LONG SSmallSize; /* (M) */
01460     LONG STermsInSmall; /* (M) */
01461     LONG MaxBracketBufferSize; /* (M) Max Size for B+ or AB+ per expression */
01462     LONG hProcessBucketSize; /* (M) */
01463     LONG gProcessBucketSize; /* (M) */
01464     LONG shmWinSize; /* (M) size for shared memory window used in communications */
01465     LONG OldChildTime; /* (M) Zero time. Needed in timer. */
01466     LONG OldSecTime; /* (M) Zero time for measuring wall clock time */
01467     LONG OldMilliTime; /* (M) Same, but milli seconds */
01468     LONG WorkSize; /* (M) Size of Workspace */
01469     LONG gThreadBucketSize; /* (C) */
01470     LONG ggThreadBucketSize; /* (C) */
01471     LONG SumTime; /* Used in .clear */
01472     LONG SpectatorSize; /* Size of the buffer in bytes */
01473     LONG TimeLimit; /* Limit in sec to the total real time */
01474 #ifdef WITHFLOAT
01475     LONG gDefaultPrecision; /* (M) Default precision in bits for float_ */
01476     LONG gMaxWeight; /* (M) Maximum weight for MZV or Euler */
01477     LONG ggDefaultPrecision; /* (M) Default precision in bits for float_ */
01478     LONG ggMaxWeight; /* (M) Maximum weight for MZV or Euler */
01479 #endif
01480     int FileOnlyFlag; /* (M) Writing only to file */
01481     int Interact; /* (M) Interactive mode flag */
01482     int MaxParLevel; /* (M) Maximum nesting of parentheses */

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01483     int      OutBufSize;          /* (M) size of OutBuffer */
01484     int      SMaxFpatches;        /* (M) */
01485     int      SMaxPatches;         /* (M) */
01486     int      StdOut;              /* (M) Regular output channel */
01487     int      ginsidefirst;        /* (M) Not used yet */
01488     int      gDefDim;             /* (M) */
01489     int      gDefDim4;            /* (M) */
01490     int      NumFixedSets;        /* (M) Number of the predefined sets */
01491     int      NumFixedFunctions;   /* (M) Number of built in functions */
01492     int      rbufnum;             /* (M) startup compiler buffer */
01493     int      dbufnum;             /* (M) dollar variables */
01494     int      sbufnum;             /* (M) subterm variables */
01495     int      zbufnum;             /* (M) special values */
01496     int      SkipClears;          /* (M) Number of .clear to skip at start */
01497     int      gTokensWriteFlag;   /* (M) */
01498     int      gfunpowers;          /* (M) */
01499     int      gStatsFlag;          /* (M) */
01500     int      gNamesFlag;         /* (M) */
01501     int      gCodesFlag;         /* (M) */
01502     int      gSortType;          /* (M) */
01503     int      gproperorderflag;    /* (M) */
01504     int      hparallelflag;      /* (M) */
01505     int      gparallelflag;      /* (M) */
01506     int      totalnumberofthreads; /* (M) */
01507     int      gSizeCommuteInSet;   /* (M) */
01508     int      gThreadStats;        /* (M) */
01509     int      ggThreadStats;       /* (M) */
01510     int      gFinalStats;         /* (M) */
01511     int      ggFinalStats;        /* (M) */
01512     int      gThreadsFlag;        /* (M) */
01513     int      ggThreadsFlag;       /* (M) */
01514     int      gThreadBalancing;    /* (M) */
01515     int      ggThreadBalancing;   /* (M) */
01516     int      gThreadSortFileSynch; /* (M) */
01517     int      ggThreadSortFileSynch; /* (M) */
01518     int      gProcessStats;       /* (M) */
01519     int      ggProcessStats;      /* (M) */
01520     int      gOldParallelStats;   /* (M) */
01521     int      ggOldParallelStats;  /* (M) */
01522     int      maxFlevels;          /* () maximum function levels */
01523     int      resetTimeOnClear;    /* (M) */
01524     int      gcNumDollars;        /* () number of dollars for .clear */
01525     int      MultiRun;            /* (M) */
01526     int      gNoSpacesInNumbers; /* For very long numbers */
01527     int      ggNoSpacesInNumbers; /* For very long numbers */
01528     int      gIsFortran90;        /* (M) */
01529     int      PrintTotalSize;       /* (M) */
01530     int      fbuffersize;          /* Size for the AT.fbufnum factorization caches */
01531     int      gOldFactArgFlag;      /* (M) */
01532     int      ggOldFactArgFlag;     /* (M) */
01533     int      gnumextrasym;         /* (M) */
01534     int      ggnumextrasym;        /* (M) */
01535     int      NumSpectatorFiles;    /* Elements used in AM.spectatorfiles; */
01536     int      SizeForSpectatorFiles; /* Size in AM.spectatorfiles; */
01537     int      gOldGCDflag;          /* (M) */
01538     int      ggOldGCDflag;         /* (M) */
01539     int      gWTimeStatsFlag;      /* (M) */
01540     int      ggWTimeStatsFlag;     /* (M) */
01541     int      jumpratio;            /* (M) */
01542     WORD     MaxTal;              /* (M) Maximum number of words in a number */
01543     WORD     IndDum;              /* (M) Basis value for dummy indices */
01544     WORD     DumInd;              /* (M) */
01545     WORD     WillInd;             /* (M) Offset for wildcard indices */
01546     WORD     gncmod;              /* (M) Global setting of modulus. size of gcmmod */
01547     WORD     gnpowmod;            /* (M) Global printing as powers. size gpowmod */
01548     WORD     gmodmode;            /* (M) Global mode for modulus */
01549     WORD     gUnitTrace;          /* (M) Global value of Tr[1] */
01550     WORD     gOutputMode;         /* (M) */
01551     WORD     gOutputSpaces;       /* (M) */
01552     WORD     gOutNumberType;      /* (M) */
01553     WORD     gCnumpows;           /* (M) */
01554     WORD     gUniTrace[4];        /* (M) */
01555     WORD     MaxWildcards;        /* (M) Maximum number of wildcards */
01556     WORD     mTraceDum;           /* (M) Position/Offset for generated dummies */
01557     WORD     OffsetIndex;         /* (M) */
01558     WORD     OffsetVector;        /* (M) */
01559     WORD     RepMax;              /* (M) Max repeat levels */
01560     WORD     LogType;             /* (M) Type of writing to log file */
01561     WORD     ggStatsFlag;         /* (M) */
01562     WORD     gLineLength;         /* (M) */
01563     WORD     qError;              /* (M) Only error checking {-c option} */
01564     WORD     FortranCont;         /* (M) Fortran Continuation character */
01565     WORD     HoldFlag;            /* (M) Exit on termination? */
01566     WORD     Ordering[15];        /* (M) Auxiliary for ordering wildcards */
01567     WORD     silent;              /* (M) Silent flag. Only results in output. */
01568     WORD     tracebackflag;       /* (M) For tracing errors */
01569     WORD     expnum;              /* (M) internal number of ^ function */

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01570     WORD    denomnum;          /* (M) internal number of / function */
01571     WORD    facnum;            /* (M) internal number of fac_ function */
01572     WORD    invfacnum;         /* (M) internal number of invfac_ function */
01573     WORD    sumnum;            /* (M) internal number of sum_ function */
01574     WORD    sumpnum;           /* (M) internal number of sump_ function */
01575     WORD    OldOrderFlag;      /* (M) Flag for allowing old statement order */
01576     WORD    termfunnum;        /* (M) internal number of term_ function */
01577     WORD    matchfunnum;       /* (M) internal number of match_ function */
01578     WORD    countfunnum;       /* (M) internal number of count_ function */
01579     WORD    gPolyFun;          /* (M) global value of PolyFun */
01580     WORD    gPolyFunInv;       /* (M) global value of Inverse of PolyFun */
01581     WORD    gPolyFunType;      /* (M) global value of PolyFun */
01582     WORD    gPolyFunExp;
01583     WORD    gPolyFunVar;
01584     WORD    gPolyFunPow;
01585     WORD    dollarzero;        /* (M) for dollars with zero value */
01586     WORD    atstartup;         /* To protect against DATE_ ending in \n */
01587     WORD    exitflag;          /* (R) For the exit statement */
01588     WORD    NumStoreCaches;     /* (I) Number of storage caches per processor */
01589     WORD    gIndentSpace;      /* For indentation in output */
01590     WORD    ggIndentSpace;
01591     WORD    gShortStatsMax;
01592     WORD    ggShortStatsMax;
01593     WORD    gextrasyms;
01594     WORD    ggextrasyms;
01595     WORD    zerorhs;
01596     WORD    onerhs;
01597     WORD    havesortdir;
01598     WORD    vectorzero;        /* p0_ */
01599     WORD    ClearStore;
01600     WORD    BracketFactors[8];
01601     BOOL    FromStdin;          /* read the input from STDIN */
01602     BOOL    IgnoreDeprecation; /* ignore deprecation warning */
01603 #ifdef WITHFLOAT
01604 #ifdef WITHPTHREADS
01605     PADPOSITION(17,30,62,84,(sizeof(pthread_rwlock_t)+sizeof(pthread_mutex_t)*2)+2);
01606 #else
01607     PADPOSITION(17,28,62,84,2);
01608 #endif
01609 #else
01610 #ifdef WITHPTHREADS
01611     PADPOSITION(17,30,62,84,(sizeof(pthread_rwlock_t)+sizeof(pthread_mutex_t)*2)+2);
01612 #else
01613     PADPOSITION(17,28,62,84,2);
01614 #endif
01615 #endif
01616 };
01617 /*
01618     #] M :
01619     #[ P : The P struct defines objects set by the preprocessor
01620 */
01621 struct P_const {
01622     LIST DollarList;            /* (R) Dollar variables. Contains pointers
                                to contents of the variables.*/
01623     LIST PreVarList;           /* (R) List of preprocessor variables
                                Points to contents. Can be changed */
01624     LIST LoopList;             /* (P) List of do loops */
01625     LIST ProcList;             /* (P) List of procedures */
01626     INSIDEINFO inside;         /* Information during #inside/#endside */
01627     NAMESPACE *firstnamespace; /* is zero when no namespace specified */
01628     NAMESPACE *lastnamespace; /* is zero when no namespace specified */
01629     UBYTE *fullname;           /* buffer for namespace expanded names */
01630     UBYTE **PreSwitchStrings; /* (P) The string in a switch */
01631     UBYTE *preStart;           /* (P) Preprocessor instruction buffer */
01632     UBYTE *preStop;            /* (P) end of preStart */
01633     UBYTE *preFill;            /* (P) Filling point in preStart */
01634     UBYTE *procedureExtension; /* (P) Extension for procedure files (prc) */
01635     UBYTE *cprocedureExtension; /* (P) Extension after .clear */
01636     LONG *PreAssignStack;      /* For nesting #name assignments */
01637     int *PreIfStack;           /* (P) Tracks nesting of #if */
01638     int *PreSwitchModes;       /* (P) Stack of switch status */
01639     int *PreTypes;            /* (P) stack of #call, #do etc nesting */
01640 #ifdef WITHPTHREADS
01641     pthread_mutex_t PreVarLock; /* (P) */
01642 #endif
01643     LONG StopWatchZero;        /* For `timer_' and #reset timer */
01644     LONG InOutBuf;             /* (P) Characters in the output buf in pre.c */
01645     LONG pSize;                /* (P) size of preStart */
01646     int PreAssignFlag;          /* (C) Indicates #assign -> catch dollar */
01647     int PreContinuation;        /* (C) Indicates whether the statement is new */
01648     int PreproFlag;            /* (P) Internal use to mark work on prepro instr. */
01649     int iBufError;             /* (P) Flag for errors with input buffer */
01650     int PreOut;                /* (P) Flag for #+ #- */
01651     int PreSwitchLevel;        /* (P) Nesting of #switch */
01652     int NumPreSwitchStrings;    /* (P) Size of PreSwitchStrings */
01653     int MaxPreTypes;           /* (P) Size of PreTypes */
01654     int NumPreTypes;           /* (P) Number of nesting objects in PreTypes */

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01664     int      MaxPreIfLevel;          /* (C) Maximum number of nested #if. Dynamic */
01665     int      PreIfLevel;              /* (C) Current position if PreIfStack */
01666     int      PreInsideLevel;          /* (C) #inside active? */
01667     int      DelayPrevar;              /* (P) Delaying prevar substitution */
01668     int      AllowDelay;               /* (P) Allow delayed prevar substitution */
01669     int      lhdollarerror;           /* (R) */
01670     int      eat;                      /* ( ) */
01671     int      gNumPre;                  /* (P) Number of preprocessor variables for .clear */
01672     int      PreDebug;                 /* (C) */
01673     int      OpenDictionary;
01674     int      PreAssignLevel;           /* For nesting # $name = ...; assignments */
01675     int      MaxPreAssignLevel;        /* For nesting # $name = ...; assignments */
01676     int      fullnamesize;             /* size of the fullname buffer */
01677     WORD     DebugFlag;                /* (P) For debugging purposes */
01678     WORD     preError;                 /* (P) Blocks certain types of execution */
01679     UBYTE     ComChar;                 /* (P) Commentary character */
01680     UBYTE     cComChar;                /* (P) Old commentary character for .clear */
01681     PADPOINTER(3,22,2,2);
01682 };
01683
01684 /*
01685     #] P :
01686     #[ C : The C struct defines objects changed by the compiler
01687 */
01688
01697 struct C_const {
01698     set_of_char separators;
01699     POSITION StoreFileSize;             /* ( ) Size of store file */
01700     NAMETREE *dollarnames;
01701     NAMETREE *exprnames;
01702     NAMETREE *varnames;
01703     LIST     ChannelList;
01704
01705     LIST     DubiousList;              /* Later also for write? */
01707     LIST     FunctionList;
01708     LIST     ExpressionList;
01709     LIST     IndexList;
01710     LIST     SetElementList;
01711     LIST     SetList;
01712     LIST     SymbolList;
01713     LIST     VectorList;
01714     LIST     PotModDollList;
01715     LIST     ModOptDollList;
01716     LIST     TableBaseList;
01717 /*
01718     Compile buffer variables
01719 */
01720     LIST     cbufList;
01721 /*
01722     Objects for auto declarations
01723 */
01724     LIST     AutoSymbolList;           /* (C) */
01725     LIST     AutoIndexList;            /* (C) */
01726     LIST     AutoVectorList;           /* (C) */
01727     LIST     AutoFunctionList;         /* (C) */
01728     NAMETREE *autonames;
01730     LIST     *Symbols;                 /* (C) Pointer for autodeclare. Which list is
01731                                     it searching. Later also for subroutines */
01732     LIST     *Indices;                 /* (C) id. */
01733     LIST     *Vectors;                 /* (C) id. */
01734     LIST     *Functions;               /* (C) id. */
01735     NAMETREE **activenames;
01737     STREAM    *Streams;
01738     STREAM    *CurrentStream;
01740     SWITCH    *SwitchArray;
01741     MODEL     **models;
01742     WORD       *SwitchHeap;
01743     LONG       *termstack;
01744     LONG       *termsortstack;
01745     UWORD      *cmmod;
01746     UWORD      *powmod;
01747     UWORD      *modpowers;
01748     UWORD      *halfmod;               /* (C) half the modulus when not zero */
01749     WORD       *ProtoType;             /* (C) The subexpression prototype {wildcards} */
01750     WORD       *WildC;                 /* (C) Filling point for wildcards. */
01751     LONG       *IfHeap;
01752     LONG       *IfCount;
01753     LONG       *IfStack;
01754     UBYTE      *iBuffer;
01755     UBYTE      *iPointer;
01756     UBYTE      *iStop;
01757     UBYTE      **LabelNames;
01758     WORD       *FixIndices;
01759     WORD       *termsumcheck;
01760     UBYTE      *WildcardNames;
01761     int         *Labels;
01762     SBYTE      *tokens;

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01763     SBYTE    *toptokens;
01764     SBYTE    *endoftokens;
01765     WORD     *tokenarglevel;
01766     UWORD    *modinverses;           /* Table for inverses of primes */
01767     UBYTE    *Fortran90Kind;        /* The kind of number in Fortran 90 as in _ki */
01768     WORD     **MultiBracketBuf;     /* Array of buffers for multi-level brackets */
01769     UBYTE    *extrasyms;            /* Array with the name for extra symbols in ToPolynomial */
01770     WORD     *doloopstack;          /* To keep track of begin and end of doloops */
01771     WORD     *doloopnest;          /* To keep track of nesting of doloops etc */
01772     char     *CheckpointRunAfter;
01774     char     *CheckpointRunBefore;
01776     WORD     *IfSumCheck;
01777     WORD     *CommuteInSet;         /* groups of noncommuting functions that can commute */
01778     UBYTE    *TestValue;           /* For debugging */
01779 #ifdef PARALLELCODE
01780     LONG     *inputnumbers;
01781     WORD     *pfirstnum;
01782 #endif
01783 #ifdef WITHPTHREADS
01784     pthread_mutex_t halfmodlock;   /* () Lock for adding buffer for halfmod */
01785 #endif
01786     LONG     argstack[MAXNEST];     /* (C) {contents} Stack for nesting of Argument */
01787     LONG     insidestack[MAXNEST];  /* (C) {contents} Stack for Argument or Inside. */
01788     LONG     inexprstack[MAXNEST]; /* (C) {contents} Stack for Argument or Inside. */
01789     LONG     iBufferSize;          /* (C) Size of the input buffer */
01790     LONG     TransEname;           /* (C) Used when a new definition overwrites
01791                                     an old expression. */
01792     LONG     ProcessBucketSize;    /* (C) */
01793     LONG     mProcessBucketSize;   /* (C) */
01794     LONG     CModule;              /* (C) Counter of current module */
01795     LONG     ThreadBucketSize;     /* (C) Roughly the maximum number of input terms */
01796     LONG     CheckpointStamp;
01797     LONG     CheckpointInterval;
01799 #ifdef WITHFLOAT
01800     LONG     DefaultPrecision;     /* (C) Default precision in bits for float_ */
01801     LONG     MaxWeight;            /* (C) Maximum weight for MZV or Euler */
01802     LONG     tDefaultPrecision;    /* (C) Default precision in bits for float_ */
01803     LONG     tMaxWeight;           /* (C) Maximum weight for MZV or Euler */
01804 #endif
01805     int      cbufnum;
01806     int      AutoDeclareFlag;
01808     int      NoShowInput;          /* (C) No listing of input as in .prc, #do */
01809     int      ShortStats;           /* (C) */
01810     int      compiletype;          /* (C) type of statement {DECLARATION etc} */
01811     int      firstconstindex;      /* (C) flag for giving first error message */
01812     int      insidefirst;          /* (C) Not used yet */
01813     int      minsidefirst;         /* (?) Not used yet */
01814     int      wildflag;             /* (C) Flag for marking use of wildcards */
01815     int      NumLabels;            /* (C) Number of labels {in Labels} */
01816     int      MaxLabels;            /* (C) Size of Labels array */
01817     int      lDefDim;              /* (C) */
01818     int      lDefDim4;             /* (C) */
01819     int      NumWildcardNames;     /* (C) Number of ?a variables */
01820     int      WildcardBufferSize;   /* (C) size of WildcardNames buffer */
01821     int      MaxIf;                /* (C) size of IfHeap, IfSumCheck, IfCount */
01822     int      NumStreams;           /* (C) */
01823     int      MaxNumStreams;        /* (C) */
01824     int      firstctypemessage;    /* (C) Flag for giving first st order error */
01825     int      tablecheck;           /* (C) For table checking */
01826     int      idoption;             /* (C) */
01827     int      BottomLevel;          /* (C) For propercount. Not used!!! */
01828     int      CompileLevel;         /* (C) Subexpression level */
01829     int      TokensWriteFlag;      /* (C) */
01830     int      UnsureDollarMode;     /* (C) ?Controls error messages undefined $'s */
01831     int      outsidefun;           /* (C) Used for writing Tables to file */
01832     int      funpowers;            /* (C) */
01833     int      WarnFlag;             /* (C) */
01834     int      StatsFlag;            /* (C) */
01835     int      NamesFlag;            /* (C) */
01836     int      CodesFlag;            /* (C) */
01837     int      SetupFlag;            /* (C) */
01838     int      SortType;             /* (C) */
01839     int      lSortType;            /* (C) */
01840     int      ThreadStats;          /* (C) */
01841     int      FinalStats;           /* (C) */
01842     int      OldParallelStats;     /* (C) */
01843     int      ThreadsFlag;
01844     int      ThreadBalancing;
01845     int      ThreadSortFileSynch;
01846     int      ProcessStats;         /* (C) */
01847     int      BracketNormalize;     /* (C) Indicates whether the bracket st is normalized */
01848     int      maxtermlevel;         /* (C) Size of termstack */
01849     int      dumnumflag;           /* (C) Where there dummy indices in tokenizer? */
01850     int      bracketindexflag;     /* (C) Are brackets going to be indexed? */
01851     int      parallelflag;         /* (C) parallel allowed? */
01852     int      mparallelflag;        /* (C) parallel allowed in this module? */
01853     int      inparallelflag;       /* (C) inparallel allowed? */

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01854     int    partodoflag;          /* (C) parallel allowed? */
01855     int    properorderflag;       /* (C) clean normalizing. */
01856     int    vetofilling;           /* (C) vetoes overwriting in tablebase stubs */
01857     int    tablefilling;          /* (C) to notify AddrHS we are filling a table */
01858     int    vetotablebasefill;     /* (C) For the load in tablebase */
01859     int    exprfillwarning;       /* (C) Warning has been printed for expressions in fill statements */
*/
01860     int    lhdollarflag;          /* (R) left hand dollar present */
01861     int    NoCompress;            /* (R) Controls native compression */
01862     int    IsFortran90;           /* Tells whether the Fortran is Fortran90 */
01863     int    MultiBracketLevels;    /* Number of elements in MultiBracketBuf */
01864     int    topolynomialflag;      /* To avoid ToPolynomial and FactArg together */
01865     int    ffbufnum;              /* Buffer number for user defined factorizations */
01866     int    OldFactArgFlag;
01867     int    MemDebugFlag;          /* Only used when MALLOCDDEBUG in tools.c */
01868     int    OldGCDflag;
01869     int    WTimeStatsFlag;
01870     int    SortReallocateFlag;    /* Controls reallocation of large+small buffer at module end.
01871                                     0 : Off
01872                                     1 : On, every module (set by On sortreallocate;)
01873                                     2 : On, single module (set by #sortreallocate) */
01874     int    PrintBacktraceFlag;    /* Print backtrace on terminate? */
01875     int    FlintPolyFlag;         /* Use Flint for polynomial arithmetic */
01876     int    HumanStatsFlag;        /* Print human-readable stats in the stats print? */
01877     int    doloopstacksize;
01878     int    dolooplevel;
01879     int    CheckpointFlag;
01883     int    SizeCommuteInSet;      /* Size of the CommuteInSet buffer */
01884 #ifdef PARALLELLECODE
01885     int    numpfirstnum;          /* For redefine */
01886     int    sizepfirstnum;         /* For redefine */
01887 #endif
01888     int    origin;                /* Determines whether .sort or ModuleOption */
01889     int    vectorlikeLHS;
01890     int    nummodels;
01891     int    modelspace;
01892     int    ModelLevel;
01893     WORD   argsumcheck[MAXNEST]; /* (C) Checking of nesting */
01894     WORD   insidesumcheck[MAXNEST]; /* (C) Checking of nesting */
01895     WORD   inexprsumcheck[MAXNEST]; /* (C) Checking of nesting */
01896     WORD   RepSumCheck[MAXREPEAT]; /* (C) Checks nesting of repeat, if, argument */
01897     WORD   lUnitTrace[4];         /* (C) */
01898     WORD   RepLevel;              /* (C) Tracks nesting of repeat. */
01899     WORD   arglevel;              /* (C) level of nested argument statements */
01900     WORD   insidelevel;           /* (C) level of nested inside statements */
01901     WORD   inexprlevel;           /* (C) level of nested inexpr statements */
01902     WORD   termlevel;             /* (C) level of nested term statements */
01903     WORD   MustTestTable;         /* (C) Indicates whether tables have been changed */
01904     WORD   DumNum;                /* (C) */
01905     WORD   ncmod;                 /* (C) Local setting of modulus. size of cmod */
01906     WORD   npowmod;               /* (C) Local printing as powers. size powmod */
01907     WORD   modmode;               /* (C) Mode for modulus calculus */
01908     WORD   nhalfmod;              /* relevant word size of halfmod when defined */
01909     WORD   DirtPow;               /* (C) Flag for changes in printing mod powers */
01910     WORD   lUnitTrace;            /* (C) Local value of Tr[1] */
01911     WORD   NwildC;                /* (C) Wildcard counter */
01912     WORD   ComDefer;              /* (C) defer brackets */
01913     WORD   CollectFun;            /* (C) Collect function number */
01914     WORD   AltCollectFun;         /* (C) Alternate Collect function number */
01915     WORD   OutputMode;            /* (C) */
01916     WORD   Cnumpows;
01917     WORD   OutputSpaces;          /* (C) */
01918     WORD   OutNumberType;         /* (C) Controls RATIONAL/FLOAT output */
01919     WORD   DidClean;              /* (C) Test whether nametree needs cleaning */
01920     WORD   IfLevel;               /* (C) */
01921     WORD   WhileLevel;            /* (C) */
01922     WORD   SwitchLevel;
01923     WORD   SwitchInArray;
01924     WORD   MaxSwitch;
01925     WORD   LogHandle;             /* (C) The Log File */
01926     WORD   LineLength;            /* (C) */
01927     WORD   StoreHandle;           /* (C) Handle of .str file */
01928     WORD   HideLevel;             /* (C) Hiding indicator */
01929     WORD   lPolyFun;              /* (C) local value of PolyFun */
01930     WORD   lPolyFunInv;           /* (C) local value of Inverse of PolyFun */
01931     WORD   lPolyFunType;          /* (C) local value of PolyFunType */
01932     WORD   lPolyFunExp;
01933     WORD   lPolyFunVar;
01934     WORD   lPolyFunPow;
01935     WORD   SymChangeFlag;         /* (C) */
01936     WORD   CollectPercentage;     /* (C) Collect function percentage */
01937     WORD   ShortStatsMax;         /* For On FewerStatistics 10; */
01938     WORD   extrasymbols;          /* Flag for the extra symbols output mode */
01939     WORD   PolyRatFunChanged;     /* Keeps track whether we changed in the compiler */
01940     WORD   ToBeInFactors;
01941     WORD   InnerTest;             /* For debugging */
01942 #ifdef WITHMPI

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01943     WORD    RhsExprInModuleFlag;    /* (C) Set by the compiler if RHS expressions exists. */
01944 #endif
01945     UBYTE    Commercial[COMMERCIALSIZE+2]; /* (C) Message to be printed in statistics */
01946     UBYTE    debugFlags[MAXFLAGS+2];    /* On/Off Flag number(s) */
01947 #ifdef WITHFLOAT
01948 #if defined(WITHPTHREADS)
01949     PADPOSITION(50,12+3*MAXNEST,77,48+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17+sizeof(pthread_mutex_t));
01950 #elif defined(WITHMPI)
01951     PADPOSITION(50,12+3*MAXNEST,77,49+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17);
01952 #else
01953     PADPOSITION(48,12+3*MAXNEST,75,48+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17);
01954 #endif
01955 #else
01956 #if defined(WITHPTHREADS)
01957     PADPOSITION(50,8+3*MAXNEST,77,48+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17+sizeof(pthread_mutex_t));
01958 #elif defined(WITHMPI)
01959     PADPOSITION(50,8+3*MAXNEST,77,49+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17);
01960 #else
01961     PADPOSITION(48,8+3*MAXNEST,75,48+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17);
01962 #endif
01963 #endif
01964 };
01965 /*
01966     #] C :
01967     #[ S : The S struct defines objects changed at the start of the run (Processor)
01968           Basically only set by the master.
01969 */
01977 struct S_const {
01978     POSITION MaxExprSize;    /* ( ) Maximum size of in/out/sort */
01979 #ifdef WITHPTHREADS
01980     pthread_mutex_t inputslock;
01981     pthread_mutex_t outputslock;
01982     pthread_mutex_t MaxExprSizeLock;
01983 #endif
01984     POSITION *OldOnFile;    /* (S) File positions of expressions */
01985     WORD *OldNumFactors;    /* ( ) NumFactors in (old) expression */
01986     WORD *Oldvflags;    /* ( ) vflags in (old) expression */
01987     WORD *Olduflags;    /* ( ) uflags in (old) expression */
01988 #ifdef WITHFLOAT
01989     void *delta_1;
01990 #endif
01991     int NumOldOnFile;    /* (S) Number of expressions in OldOnFile */
01992     int NumOldNumFactors;    /* (S) Number of expressions in OldNumFactors */
01993     int MultiThreaded;    /* (S) Are we running multi-threaded? */
01994 #ifdef WITHPTHREADS
01995     int MasterSort;    /* Final stage of sorting to the master */
01996 #endif
01997 #ifdef WITHMPI
01998     int printflag;    /* controls MesPrint() on each slave */
01999 #endif
02000     int Balancing;    /* For second stage loadbalancing */
02001     WORD ExecMode;    /* (S) */
02002
02003     WORD CollectOverFlag;    /* (R) Indicates overflow at Collect */
02004 #ifdef WITHPTHREADS
02005     WORD sLevel;    /* Copy of AR0.sLevel because it can get messy */
02006 #endif
02007 #if defined(WITHPTHREADS)
02008     PADPOSITION(3,0,5,3,sizeof(pthread_mutex_t)*3);
02009 #elif defined(WITHMPI)
02010     PADPOSITION(4,0,5,2,0);
02011 #else
02012     PADPOSITION(4,0,4,2,0);
02013 #endif
02014 };
02015 /*
02016     #] S :
02017     #[ R : The R struct defines objects changed at run time.
02018           They determine the environment that has to be transferred
02019           together with a term during multithreaded execution.
02020 */
02029 struct R_const {
02030     FILEDATA StoreData;    /* (O) */
02031     FILEHANDLE Fscr[3];    /* (R) Dollars etc play with it too */
02032     FILEHANDLE FoStage4[2];    /* (R) In Sort. Stage 4. */
02033     POSITION DefPosition;    /* (R) Deferred position of keep brackets. */
02034     FILEHANDLE *infile;    /* (R) Points alternately to Fscr[0] or Fscr[1] */
02035     FILEHANDLE *outfile;    /* (R) Points alternately to Fscr[1] or Fscr[0] */
02036     FILEHANDLE *hidefile;    /* (R) Points to Fscr[2] */
02037
02038     WORD *CompressBuffer;    /* (M) */
02039     WORD *ComprTop;    /* (M) */
02040     WORD *CompressPointer;    /* (R) */
02041     COMPAREDUMMY CompareRoutine;
02042     ULONG *wranfia;

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02043     SBYTE     *moebiustable;
02044
02045     LONG      OldTime;                /* (R) Zero time. Needed in timer. */
02046     LONG      InInBuf;                /* (R) Characters in input buffer. Scratch files. */
02047     LONG      InHiBuf;                /* (R) Characters in hide buffer. Scratch file. */
02048     LONG      pWorkSize;              /* (R) Size of pWorkSpace */
02049     LONG      lWorkSize;              /* (R) Size of lWorkSpace */
02050     LONG      posWorkSize;            /* (R) Size of posWorkSpace */
02051     ULONG     wranfseed;
02052     int        NoCompress;             /* (R) Controls native compression */
02053     int        gzipCompress;          /* (R) Controls gzip compression */
02054     int        Cnumlhs;               /* Local copy of cbuf[rbufnum].numlhs */
02055     int        outtohide;             /* Indicates that output is directly to hide */
02056 #ifdef WITHPTHREADS
02057     int        exprtodo;              /* The expression to do in parallel mode */
02058 #endif
02059     int        wranfcall;
02060     int        wranfnpair1;
02061     int        wranfnpair2;
02062 #if ( BITSINWORD == 32 )
02063     WORD       PrimeList[5000];
02064     WORD       numinprimelist;
02065     WORD       notfirstprime;
02066 #endif
02067     WORD       GetFile;               /* (R) Where to get the terms {like Hide} */
02068     WORD       KeptInHold;            /* (R) */
02069     WORD       BracketOn;             /* (R) Intensely used in poly_ */
02070     WORD       MaxBracket;            /* (R) Size of BrackBuf. Changed by poly_ */
02071     WORD       CurDum;               /* (R) Current maximum dummy number */
02072     WORD       DeferFlag;             /* (R) For deferred brackets */
02073     WORD       TePos;                /* (R) */
02074     WORD       sLevel;               /* (R) Sorting level */
02075     WORD       Stage4Name;           /* (R) Sorting only */
02076     WORD       GetOneFile;           /* (R) Getting from hide or regular */
02077     WORD       PolyFun;              /* (C) Number of the PolyFun function */
02078     WORD       PolyFunInv;           /* (C) Number of the Inverse of the PolyFun function */
02079     WORD       PolyFunType;          /* (I) value of PolyFunType */
02080     WORD       PolyFunExp;
02081     WORD       PolyFunVar;
02082     WORD       PolyFunPow;
02083     WORD       Eside;               /* (I) Tells which side of = sign */
02084     WORD       MaxDum;              /* Maximum dummy value in an expression */
02085     WORD       level;              /* Running level in Generator */
02086     WORD       expchanged;          /* (R) Info about expression */
02087     WORD       expflags;            /* (R) Info about expression */
02088     WORD       CurExpr;            /* (S) Number of current expression */
02089     WORD       SortType;            /* A copy of AC.SortType to play with */
02090     WORD       ShortSortCount;      /* For On FewerStatistics 10; */
02091     WORD       modeloptions;
02092     WORD       funoffset;
02093     WORD       moebiustablesize;
02094 #if ( BITSINWORD == 32 )
02095 #ifdef WITHPTHREADS
02096     PADPOSITION(8,7,8,5029,0);
02097 #else
02098     PADPOSITION(8,7,7,5029,0);
02099 #endif
02100 #else
02101 #ifdef WITHPTHREADS
02102     PADPOSITION(8,7,8,27,0);
02103 #else
02104     PADPOSITION(8,7,7,27,0);
02105 #endif
02106 #endif
02107 };
02108
02109 /*
02110     #] R :
02111     #[ T : These are variables that stay in each thread during multi threaded execution.
02112 */
02121 struct T_const {
02122 #ifdef WITHPTHREADS
02123     SORTBLOCK SB;
02124 #endif
02125     SORTING *S0;                    /* (-) The thread specific sort buffer */
02126     SORTING *SS;                   /* (R) Current sort buffer */
02127     NESTING Nest;                  /* (R) Nesting of function levels etc. */
02128     NESTING NestStop;              /* (R) */
02129     NESTING NestPoin;              /* (R) */
02130     WORD     *BrackBuf;            /* (R) Bracket buffer. Used by poly_ at runtime. */
02131     STORECACHE StoreCache;         /* (R) Cache for picking up stored expr. */
02132     STORECACHE StoreCacheAlloc;    /* (R) Permanent address of StoreCache to keep valgrind happy */
02133     WORD     **pWorkSpace;         /* (R) Workspace for pointers. Dynamic. */
02134     LONG     *lWorkSpace;          /* (R) Workspace for LONG. Dynamic. */
02135     POSITION  *posWorkSpace;        /* (R) Workspace for file positions */
02136     WORD     *WorkSpace;           /* (M) */
02137     WORD     *WorkTop;             /* (M) */

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02138     WORD    *WorkPointer;          /* (R) Pointer in the WorkSpace heap. */
02139     int     *RepCount;             /* (M) Buffer for repeat nesting */
02140     int     *RepTop;              /* (M) Top of RepCount buffer */
02141     WORD    *WildArgTaken;        /* (N) Stack for wildcard pattern matching */
02142     UWORD   *factorials;          /* (T) buffer of factorials. Dynamic. */
02143     WORD    *small_power_n;       /* length of the number */
02144     UWORD   **small_power;         /* the number*/
02145     UWORD   *bernoullis;          /* (T) The buffer with Bernoulli numbers. Dynamic. */
02146     WORD    *primelist;
02147     LONG    *pfac;                /* (T) array of positions of factorials. Dynamic. */
02148     LONG    *pBer;                /* (T) array of positions of Bernoulli's. Dynamic. */
02149     WORD    *TMaddr;              /* (R) buffer for TestSub */
02150     WORD    *WildMask;            /* (N) Wildcard info during pattern matching */
02151     WORD    *previousEfactor;     /* () Cache for factors in expressions */
02152     WORD    **TermMemHeap;        /* For TermMalloc. Set zero in Checkpoint */
02153     UWORD    **NumberMemHeap;     /* For NumberMalloc. Set zero in Checkpoint */
02154     UWORD    **CacheNumberMemHeap; /* For CacheNumberMalloc. Set zero in Checkpoint */
02155     BRACKETINFO *bracketinfo;
02156     WORD    **ListPoly;
02157     WORD    *ListSymbols;
02158     UWORD    *NumMem;
02159     WORD    *TopologiesTerm;
02160     WORD    *TopologiesStart;
02161 #ifdef WITHFLOAT
02162     int     *ind1l;
02163     int     *ind12;
02164     void     *mpf_tab1;
02165     void     *mpf_tab2;
02166     void     *aux_;
02167     void     *auxr_;
02168 #endif
02169     PARTI    partitions;
02170     LONG     sBer;                /* (T) Size of the Bernoullis buffer */
02171     LONG     pWorkPointer;        /* (R) Offset-pointer in pWorkSpace */
02172     LONG     lWorkPointer;        /* (R) Offset-pointer in lWorkSpace */
02173     LONG     posWorkPointer;      /* (R) Offset-pointer in posWorkSpace */
02174     LONG     InNumMem;
02175     int      sfact;               /* (T) size of the factorials buffer */
02176     int      mfac;                /* (T) size of the pfac array. */
02177     int      ebufnum;             /* (R) extra compiler buffer */
02178     int      fbufnum;             /* extra compiler buffer for factorization cache */
02179     int      allbufnum;           /* extra compiler buffer for id,all */
02180     int      aebufnum;           /* extra compiler buffer for id,all */
02181     int      idallflag;           /* indicates use of id,all buffers */
02182     int      idallnum;
02183     int      idallmaxnum;
02184     int      WildcardBufferSize; /* () local copy for updates */
02185 #ifdef WITHPTHREADS
02186     int      identity;            /* () When we work with B->T */
02187     int      LoadBalancing;      /* Needed for synchronization */
02188 #ifdef WITHSORTBOTS
02189     int      SortBotIn1;         /* Input stream 1 for a SortBot */
02190     int      SortBotIn2;         /* Input stream 2 for a SortBot */
02191 #endif
02192 #endif
02193     int      TermMemMax;         /* For TermMalloc. Set zero in Checkpoint */
02194     int      TermMemTop;         /* For TermMalloc. Set zero in Checkpoint */
02195     int      NumberMemMax;       /* For NumberMalloc. Set zero in Checkpoint */
02196     int      NumberMemTop;       /* For NumberMalloc. Set zero in Checkpoint */
02197     int      CacheNumberMemMax;  /* For CacheNumberMalloc. Set zero in Checkpoint */
02198     int      CacheNumberMemTop;  /* For CacheNumberMalloc. Set zero in Checkpoint */
02199     int      bracketindexflag;   /* Are brackets going to be indexed? */
02200     int      optintimes;         /* Number of the evaluation of the MCTS tree */
02201     int      ListSymbolsSize;
02202     int      NumListSymbols;
02203     int      numpoly;
02204     int      LeaveNegative;
02205     int      TrimPower;          /* Indicates trimming in polyratfun expansion */
02206     WORD     small_power_maxx;   /* size of the cache for small powers */
02207     WORD     small_power_maxn;   /* size of the cache for small powers */
02208     WORD     dummysubexp[SUBEXPSIZE+4]; /* () used in normal.c */
02209     WORD     comsym[8];          /* () Used in tools.c = {8,SYMBOL,4,0,1,1,1,3} */
02210     WORD     comnum[4];          /* () Used in tools.c = { 4,1,1,3 } */
02211     WORD     comfun[FUNHEAD+4]; /* () Used in tools.c = {7,FUNCTION,3,0,1,1,3} */
02212     /* or { 8,FUNCTION,4,0,0,1,1,3 } */
02213     WORD     comind[7];          /* () Used in tools.c = {7,INDEX,3,0,1,1,3} */
02214     WORD     MinVecArg[7+ARGHEAD]; /* (N) but should be more local */
02215     WORD     FunArg[4+ARGHEAD+FUNHEAD]; /* (N) but can be more local */
02216     WORD     locwildvalue[SUBEXPSIZE]; /* () Used in argument.c = {SUBEXPRESSION,SUBEXPSIZE,0,0,0} */
02217     WORD     mulpat[SUBEXPSIZE+5]; /* () Used in argument.c = {TYPEMULT, SUBEXPSIZE+3, 0, */
02218     /* SUBEXPRESSION, SUBEXPSIZE, 0, 1, 0, 0, 0 } */
02219     WORD     proexp[SUBEXPSIZE+5]; /* () Used in poly.c */
02220     WORD     TMout[40];          /* (R) Passing info */
02221     WORD     TMbuff;             /* (R) Communication between TestSub and Genera */
02222     WORD     Tmdolfac;           /* factor number for dollar */
02223     WORD     nfac;               /* (T) Number of highest stored factorial */
02224     WORD     nBer;               /* (T) Number of highest Bernoulli number. */

```

```

02225     WORD      mBer;                /* (T) Size of buffer pBer. */
02226     WORD      PolyAct;              /* (R) Used for putting the PolyFun at end. ini at 0 */
02227     WORD      RecFlag;              /* (R) Used in TestSub. ini at zero. */
02228     WORD      inprimelist;
02229     WORD      sizeprimelist;
02230     WORD      fromindex;            /* Tells the compare routine whether call from index */
02231     WORD      setinterntopo;         /* Set of internal momenta for topogen */
02232     WORD      setexterntopo;        /* Set of external momenta for topogen */
02233     WORD      TopologiesLevel;
02234     WORD      TopologiesOptions[2];
02235 #ifndef WITHFLOAT
02236     WORD      FloatPos;
02237     WORD      SortFloatMode;
02238 #endif
02239 #ifndef WITHFLOAT
02240 #ifndef WITHPTHREADS
02241     #ifndef WITHSORTBOTS
02242         PADPOINTER(5,27,107+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
02243     #else
02244         PADPOINTER(5,25,107+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
02245     #endif
02246 #else
02247         PADPOINTER(5,23,107+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
02248 #endif
02249 #else
02250 #ifndef WITHPTHREADS
02251     #ifndef WITHSORTBOTS
02252         PADPOINTER(5,27,105+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
02253     #else
02254         PADPOINTER(5,25,105+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
02255     #endif
02256 #else
02257         PADPOINTER(5,23,105+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
02258 #endif
02259 #endif
02260 };
02261 /*
02262     #] T :
02263     #[ N : The N struct contains variables used in running information
02264            that is inside blocks that should not be split, like pattern
02265            matching, traces etc. They are local for each thread.
02266            They don't need initializations.
02267 */
02276 struct N_const {
02277     POSITION OldPosIn;                /* (R) Used in sort. */
02278     POSITION OldPosOut;               /* (R) Used in sort */
02279     POSITION theposition;              /* ( ) Used in index.c */
02280     WORD *EndNest;                   /* (R) Nesting of function levels etc. */
02281     WORD *Frozen;                    /* (R) Bracket info */
02282     WORD *FullProto;                 /* (R) Prototype of a subexpression or table */
02283     WORD *cTerm;                     /* (R) Current term for coef_ and term_ */
02284     int *RepPoint;                   /* (R) Pointer in RepCount buffer. Tracks repeat */
02285     WORD *WildValue;                 /* (N) Wildcard info during pattern matching */
02286     WORD *WildStop;                  /* (N) Wildcard info during pattern matching */
02287     WORD *argaddress;                 /* (N) Used in pattern matching of arguments */
02288     WORD *RepFunList;                 /* (N) For pattern matching */
02289     WORD *patstop;                    /* (N) Used in pattern matching */
02290     WORD *terstop;                    /* (N) Used in pattern matching */
02291     WORD *terstart;                   /* (N) Used in pattern matching */
02292     WORD *terfirstcomm;               /* (N) Used in pattern matching */
02293     WORD *DumFound;                   /* (N) For renumbering indices {make local?} */
02294     WORD **DumPlace;                  /* (N) For renumbering indices {make local?} */
02295     WORD **DumFunPlace;               /* (N) For renumbering indices {make local?} */
02296     WORD *UsedSymbol;                 /* (N) When storing terms of a global expr. */
02297     WORD *UsedVector;                 /* (N) When storing terms of a global expr. */
02298     WORD *UsedIndex;                  /* (N) When storing terms of a global expr. */
02299     WORD *UsedFunction;               /* (N) When storing terms of a global expr. */
02300     WORD *MaskPointer;                 /* (N) For wildcard pattern matching */
02301     WORD *ForFindOnly;                 /* (N) For wildcard pattern matching */
02302     WORD *findTerm;                   /* (N) For wildcard pattern matching */
02303     WORD *findPattern;                 /* (N) For wildcard pattern matching */
02304 #ifndef WITHZLIB
02305     Bytef **ziobufnum;                /* ( ) Used in compress.c */
02306     Bytef *ziobuffers;                /* ( ) Used in compress.c */
02307 #endif
02308     WORD *dummyrenumlist;              /* ( ) Used in execute.c and store.c */
02309     int *funargs;                      /* ( ) Used in lus.c */
02310     WORD **funlocs;                   /* ( ) Used in lus.c */
02311     int *funinds;                      /* ( ) Used in lus.c */
02312     UWORD *NoScrat2;                  /* ( ) Used in normal.c */
02313     WORD *ReplaceScrat;                /* ( ) Used in normal.c */
02314     TRACES *tracestack;                /* ( ) used in opera.c */
02315     WORD *selecttermundo;              /* ( ) Used in pattern.c */
02316     WORD *patternbuffer;               /* ( ) Used in pattern.c */
02317     WORD *termbuffer;                  /* ( ) Used in pattern.c */
02318     WORD **PoinScrat;                  /* ( ) used in reshuf.c */
02319     WORD **FunScrat;                  /* ( ) used in reshuf.c */

```



```

02320     WORD    *RenumScratch;          /* () used in reshuf.c */
02321     FUN_INFO *FunInfo;              /* () Used in smart.c */
02322     WORD    **SplitScratch;         /* () Used in sort.c */
02323     WORD    **SplitScratch1;        /* () Used in sort.c */
02324     SORTING **FunSorts;             /* () Used in sort.c */
02325     UWORD    *SoScratC;             /* () Used in sort.c */
02326     WORD    *listinprint;           /* () Used in proces.c and message.c */
02327     WORD    *currentTerm;           /* () Used in proces.c and message.c */
02328     WORD    **arglist;              /* () Used in function.c */
02329     int      *tlistbuf;             /* () used in lus.c */
02330     #ifdef WHICHSUBEXPRESSION
02331     UWORD    *BinoScrat;            /* () Used in proces.c */
02332     #endif
02333     WORD    *compressSpace;         /* () Used in sort.c */
02334     #ifdef WITHPTHREADS
02335     THREADBUCKET *threadbuck;
02336     EXPRESSIONS expr;
02337     #endif
02338     UWORD    *SHcombi;
02339     WORD    *poly_vars;
02340     UWORD    *cmmod;                /* Local setting of modulus. Pointer to value. */
02341     SHvariables SHvar;
02342     LONG     deferskipped;           /* () Used in proces.c store.c and parallel.c */
02343     LONG     InScratch;             /* () Used in sort.c */
02344     LONG     SplitScratchSize;      /* () Used in sort.c */
02345     LONG     InScratch1;            /* () Used in sort.c */
02346     LONG     SplitScratchSize1;     /* () Used in sort.c */
02347     LONG     ninterms;              /* () Used in proces.c and sort.c */
02348     #ifdef WITHPTHREADS
02349     LONG     inputnumber;           /* () For use in redefine */
02350     LONG     lastinindex;
02351     #endif
02352     #ifdef WHICHSUBEXPRESSION
02353     LONG     last2;                 /* () Used in proces.c */
02354     LONG     last3;                 /* () Used in proces.c */
02355     #endif
02356     LONG     SHcombisize;
02357     int      NumTotWildArgs;         /* (N) Used in pattern matching */
02358     int      UseFindOnly;            /* (N) Controls pattern routines */
02359     int      UsedOtherFind;          /* (N) Controls pattern routines */
02360     int      ErrorInDollar;          /* (R) */
02361     int      numfargs;               /* () Used in lus.c */
02362     int      numflocs;               /* () Used in lus.c */
02363     int      nargs;                 /* () Used in lus.c */
02364     int      tohunt;                 /* () Used in lus.c */
02365     int      numoffuns;              /* () Used in lus.c */
02366     int      funisize;               /* () Used in lus.c */
02367     int      RSsize;                 /* () Used in normal.c */
02368     int      numtracesctack;         /* () used in opera.c */
02369     int      intracestack;           /* () used in opera.c */
02370     int      numfuninfo;             /* () Used in smart.c */
02371     int      NumFunSorts;            /* () Used in sort.c */
02372     int      MaxFunSorts;            /* () Used in sort.c */
02373     int      arglistsize;            /* () Used in function.c */
02374     int      tlistsize;              /* () used in lus.c */
02375     int      filenum;                /* () used in setfile.c */
02376     int      compressSize;           /* () Used in sort.c */
02377     int      polysortflag;
02378     int      nogroundlevel;          /* () Used to see whether pattern matching at groundlevel */
02379     int      subsubveto;             /* () Sabotage combining subexpressions in TestSub */
02380     WORD     MaxRenumScratch;        /* () used in reshuf.c */
02381     WORD     oldtype;                /* (N) WildCard info at pattern matching */
02382     WORD     oldvalue;               /* (N) WildCard info at pattern matching */
02383     WORD     NumWild;                /* (N) Used in Wildcard */
02384     WORD     RepFunNum;              /* (N) Used in pattern matching */
02385     WORD     DisorderFlag;           /* (N) Disorder option? Used in pattern matching */
02386     WORD     WildDirt;               /* (N) dirty in wildcard substitution. */
02387     WORD     NumFound;               /* (N) in reshuf only. Local? */
02388     WORD     WildReserve;            /* (N) Used in the wildcards */
02389     WORD     TeInFun;                /* (R) Passing type of action */
02390     WORD     TeSuOut;                /* (R) Passing info. Local? */
02391     WORD     WildArgs;               /* (R) */
02392     WORD     WildEat;                /* (R) */
02393     WORD     PolyNormFlag;           /* (R) For polynomial arithmetic */
02394     WORD     PolyFunTodo;            /* deals with expansions and multiplications */
02395     WORD     sizeselecttermundo;     /* () Used in pattern.c */
02396     WORD     patternbuffersize;      /* () Used in pattern.c */
02397     WORD     numlistinprint;         /* () Used in process.c */
02398     WORD     ncmod;                  /* () used as some type of flag to disable */
02399     WORD     ExpectedSign;
02400     WORD     SignCheck;
02401     WORD     IndDum;                 /* Active dummy indices */
02402     WORD     poly_num_vars;
02403     WORD     idfunctionflag;
02404     WORD     poly_vars_type;         /* type of allocation. For free. */
02405     WORD     tryterm;                /* For EndSort(...,2) */
02406     #ifdef WHICHSUBEXPRESSION

```

```

02407     WORD      nbino;                /* () Used in proces.c */
02408     WORD      last1;                /* () Used in proces.c */
02409 #endif
02410 #ifdef WITHPTHREADS
02411 #ifdef WHICHSUBEXPRESSION
02412 #ifdef WITHZLIB
02413     PADPOSITION(55,11,23,28,sizeof(SHvariables));
02414 #else
02415     PADPOSITION(53,11,23,28,sizeof(SHvariables));
02416 #endif
02417 #else
02418 #ifdef WITHZLIB
02419     PADPOSITION(54,9,23,26,sizeof(SHvariables));
02420 #else
02421     PADPOSITION(52,9,23,26,sizeof(SHvariables));
02422 #endif
02423 #endif
02424 #else
02425 #ifdef WHICHSUBEXPRESSION
02426 #ifdef WITHZLIB
02427     PADPOSITION(53,9,23,28,sizeof(SHvariables));
02428 #else
02429     PADPOSITION(51,9,23,28,sizeof(SHvariables));
02430 #endif
02431 #else
02432 #ifdef WITHZLIB
02433     PADPOSITION(52,7,23,26,sizeof(SHvariables));
02434 #else
02435     PADPOSITION(50,7,23,26,sizeof(SHvariables));
02436 #endif
02437 #endif
02438 #endif
02439 };
02440
02441 /*
02442     #] N :
02443     #[ O : The O struct concerns output variables
02444 */
02453 struct O_const {
02454     FILEDATA    SaveData;            /* (O) */
02455     STOREHEADER SaveHeader;          /* () System Independent save-Files */
02456     OPTIMIZERESULT OptimizeResult;
02457     UBYTE        *OutputLine;        /* (O) Sits also in debug statements */
02458     UBYTE        *OutStop;           /* (O) Top of OutputLine buffer */
02459     UBYTE        *OutFill;           /* (O) Filling point in OutputLine buffer */
02460     WORD         *bracket;           /* (O) For writing brackets */
02461     WORD         *termbuf;           /* (O) For writing terms */
02462     WORD         *tabstring;
02463     UBYTE        *wpos;              /* (O) Only when storing file {local?} */
02464     UBYTE        *wposin;            /* (O) Only when storing file {local?} */
02465     UBYTE        *DollarOutBuffer;    /* (O) Outputbuffer for Dollars */
02466     UBYTE        *CurBufWrt;        /* (O) Name of currently written expr. */
02467     void          (*FlipWORD)(UBYTE *); /* () Function pointers for translations. Initialized by
ReadSaveHeader() */
02468     void          (*FlipLONG)(UBYTE *);
02469     void          (*FlipPOS)(UBYTE *);
02470     void          (*FlipPOINTER)(UBYTE *);
02471     void          (*ResizeData)(UBYTE *,int,UBYTE *,int);
02472     void          (*ResizeWORD)(UBYTE *,UBYTE *);
02473     void          (*ResizeNCWORD)(UBYTE *,UBYTE *);
02474     void          (*ResizeLONG)(UBYTE *,UBYTE *);
02475     void          (*ResizePOS)(UBYTE *,UBYTE *);
02476     void          (*ResizePOINTER)(UBYTE *,UBYTE *);
02477     void          (*CheckPower)(UBYTE *);
02478     void          (*RenumberVec)(UBYTE *);
02479     DICTIONARY    **Dictionaries;
02480     UBYTE        *tensorList;        /* Dynamically allocated list with functions that are tensorial. */
02481     WORD         *inscheme;          /* for feeding a Horner scheme to Optimize */
02482 #ifdef WITHFLOAT
02483     UBYTE        *floatspace;
02484     LONG         floatsize;
02485 #endif
02486 /*----Leave NumInBrack as first non-pointer. This is used by the checkpoints---*/
02487     LONG         NumInBrack;          /* (O) For typing [] option in print */
02488     LONG         wlen;                /* (O) Used to store files. */
02489     LONG         DollarOutSizeBuffer; /* (O) Size of DollarOutBuffer */
02490     LONG         DollarInOutBuffer;   /* (O) Characters in DollarOutBuffer */
02491 #if defined(mBSD) && defined(MICROTIME)
02492     LONG         wrap;                /* (O) For statistics time. wrap around */
02493     LONG         wrapnum;             /* (O) For statistics time. wrap around */
02494 #endif
02495     OPTIMIZE      Optimize;
02496     int           OutInBuffer;        /* (O) Which routine does the writing */
02497     int           NoSpacesInNumbers; /* For very long numbers */
02498     int           BlockSpaces;        /* For very long numbers */
02499     int           CurrentDictionary;
02500     int           SizeDictionaries;

```

```

02501     int      NumDictionaries;
02502     int      CurDictNumbers;
02503     int      CurDictVariables;
02504     int      CurDictSpecials;
02505     int      CurDictFunWithArgs;
02506     int      CurDictNumberWarning;
02507     int      CurDictNotInFunctions;
02508     int      CurDictInDollars;
02509     int      gNumDictionaries;
02510 #ifdef WITHFLOAT
02511     int      FloatPrec;
02512 #endif
02513     WORD      schemenum;          /* for feeding a Horner scheme to Optimize */
02514     WORD      transFlag;          /* () >0 indicates that translations have to be done */
02515     WORD      powerFlag;          /* () >0 indicates that some exponents/powers had to be adjusted
*/
02516     WORD      mpower;             /* For maxpower adjustment to larger value */
02517     WORD      resizeFlag;         /* () >0 indicates that something went wrong when resizing words
*/
02518     WORD      bufferedInd;        /* () Contains extra INDEXENTRIES, see ReadSaveIndex() for an
explanation */
02519     WORD      OutSkip;            /* (0) How many chars to skip in output line */
02520     WORD      IsBracket;          /* (0) Controls brackets */
02521     WORD      InFbrack;          /* (0) For writing only */
02522     WORD      PrintType;          /* (0) */
02523     WORD      FortFirst;          /* (0) Only in sch.c */
02524     WORD      DoubleFlag;         /* (0) Output in double precision */
02525     WORD      IndentSpace;        /* For indentation in output */
02526     WORD      FactorMode;         /* When the output should be written as factors */
02527     WORD      FactorNum;          /* Number of factor currently treated */
02528     WORD      ErrorBlock;
02529     WORD      OptimizationLevel;  /* Level of optimization in the output */
02530     UBYTE     FortDotChar;        /* (0) */
02531 /*
02532     For the padding, please count also the number of int's in the OPTIMIZE struct.
02533 */
02534 #ifdef WITHFLOAT
02535 #if defined(mBSD) && defined(MICROTIME)
02536     PADPOSITION(25,7,36,18,1);
02537 #else
02538     PADPOSITION(25,5,36,18,1);
02539 #endif
02540 #else
02541 #if defined(mBSD) && defined(MICROTIME)
02542     PADPOSITION(25,6,36,17,1);
02543 #else
02544     PADPOSITION(25,4,36,17,1);
02545 #endif
02546 #endif
02547 };
02548 /*
02549     #] O :
02550     #[ X : The X struct contains variables that deal with the external channel
02551 */
02552 struct X_const {
02553     UBYTE *currentPrompt;
02554     UBYTE *shellname;             /* if !=NULL (default is "/bin/sh -c"), start in
the specified subshell*/
02555     UBYTE *stderrname;            /* If !=NULL (default if "/dev/null"), stderr is
redirected to the specified file*/
02556     int timeout;                  /* timeout to initialize preset channels.
If timeout<0, the preset channels are
already initialized*/
02557     int killSignal;               /* signal number, SIGKILL by default*/
02558     int killWholeGroup;           /* if 0, the signal is sent only to a process,
if !=0 (default) is sent to a whole process group*/
02559     int daemonize;                /* if !=0 (default), start in a daemon mode */
02560     int currentExternalChannel;
02561     PADPOINTER(0,5,0,0);
02562 };
02563 /*
02564     #] X :
02565     #[ Definitions :
02566
02567     Note: we changed the definition from C to Cc in version 5.
02568     The reason is that everywhere the pointer to the compiler
02569     buffer was called C as well. Somehow that gave no problems but
02570     who knows what the future might bring.
02571 */
02572 #ifdef WITHPTHREADS
02573 typedef struct AllGlobals {
02574     struct M_const M;
02575     struct C_const Cc;
02576     struct S_const S;
02577     struct O_const O;

```

```

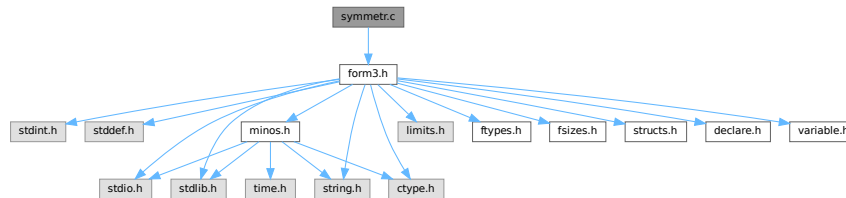
02597     struct P_const P;
02598     struct X_const X;
02599     PADPOSITION(0,0,0,0,sizeof(struct P_const)+sizeof(struct X_const));
02600 } ALLGLOBALS;
02601
02602 typedef struct AllPrivates {
02603     struct R_const R;
02604     struct N_const N;
02605     struct T_const T;
02606     PADPOSITION(0,0,0,0,sizeof(struct T_const));
02607 } ALLPRIVATES;
02608
02609 #else
02610 typedef struct AllGlobals {
02611     struct M_const M;
02612     struct C_const Cc;
02613     struct S_const S;
02614     struct R_const R;
02615     struct N_const N;
02616     struct O_const O;
02617     struct P_const P;
02618     struct T_const T;
02619     struct X_const X;
02620     PADPOSITION(0,0,0,0,sizeof(struct P_const)+sizeof(struct T_const)+sizeof(struct X_const));
02621 } ALLGLOBALS;
02622
02623 #endif
02624
02625 /*
02626     #] Definitions :
02627     #] A :
02628     #[ FG :
02629 */
02630
02631 #ifdef WITHPTHREADS
02632 #define PHEAD ALLPRIVATES *B,
02633 #define PHEAD0 ALLPRIVATES *B
02634 #define BHEAD B,
02635 #define BHEAD0 B
02636 #else
02637 #define PHEAD
02638 #define PHEAD0 void
02639 #define BHEAD
02640 #define BHEAD0
02641 #endif
02642
02643 #ifdef ANSI
02644 typedef WORD (*WCN) (PHEAD WORD *,WORD *,WORD,WORD);
02645 typedef WORD (*WCN2) (PHEAD WORD *,WORD *);
02646 #else
02647 typedef WORD (*WCN) ();
02648 typedef WORD (*WCN2) ();
02649 #endif
02650
02651 typedef WORD (*COMPARE) (PHEAD WORD *,WORD *,WORD);
02652
02653 typedef struct FixedGlobals {
02654     WCN      Operation[8];
02655     WCN2     OperaFind[6];
02656     char     *VarType[10];
02657     char     *ExprStat[21];
02658     char     *FunNam[2];
02659     char     *swmes[3];
02660     char     *fname;
02661     char     *fname2;
02662     UBYTE    *s_one;
02663     WORD     fnamebase;
02664     WORD     fname2base;
02665     WORD     fnamesize;
02666     WORD     fname2size;
02667     UINT     cTable[256];
02668 } FIXEDGLOBALS;
02669
02670 /*
02671     #] FG :
02672 */
02673 #endif

```

## 4.138 `symmetr.c` File Reference

```
#include "form3.h"
```

Include dependency graph for `symmetr.c`:



### Functions

- int [MatchE](#) (PHEAD WORD \*pattern, WORD \*fun, WORD \*inter, WORD par)
- int [Permute](#) ([PERM](#) \*perm, WORD first)
- int [PermuteP](#) ([PERMP](#) \*perm, WORD first)
- int [Distribute](#) ([DISTRIBUTE](#) \*d, WORD first)
- int [MatchCy](#) (PHEAD WORD \*pattern, WORD \*fun, WORD \*inter, WORD par)
- int [FunMatchCy](#) (PHEAD WORD \*pattern, WORD \*fun, WORD \*inter, WORD par)
- int [FunMatchSy](#) (PHEAD WORD \*pattern, WORD \*fun, WORD \*inter, WORD par)
- int [MatchArgument](#) (PHEAD WORD \*arg, WORD \*pat)

### 4.138.1 Detailed Description

The routines that deal with the pattern matching of functions with symmetric properties.

Definition in file [symmetr.c](#).

### 4.138.2 Function Documentation

#### 4.138.2.1 `MatchE()`

```
int MatchE (
    PHEAD WORD * pattern,
    WORD * fun,
    WORD * inter,
    WORD par )
```

Definition at line 56 of file [symmetr.c](#).

#### 4.138.2.2 `Permute()`

```
int Permute (
    PERM * perm,
    WORD first )
```

Definition at line 444 of file [symmetr.c](#).

#### 4.138.2.3 PermuteP()

```
int PermuteP (
    PERMP * perm,
    WORD first )
```

Definition at line 478 of file [symmetr.c](#).

#### 4.138.2.4 Distribute()

```
int Distribute (
    DISTRIBUTE * d,
    WORD first )
```

Definition at line 510 of file [symmetr.c](#).

#### 4.138.2.5 MatchCy()

```
int MatchCy (
    PHEAD WORD * pattern,
    WORD * fun,
    WORD * inter,
    WORD par )
```

Definition at line 571 of file [symmetr.c](#).

#### 4.138.2.6 FunMatchCy()

```
int FunMatchCy (
    PHEAD WORD * pattern,
    WORD * fun,
    WORD * inter,
    WORD par )
```

Definition at line 1067 of file [symmetr.c](#).

#### 4.138.2.7 FunMatchSy()

```
int FunMatchSy (
    PHEAD WORD * pattern,
    WORD * fun,
    WORD * inter,
    WORD par )
```

Definition at line 1525 of file [symmetr.c](#).

## 4.138.2.8 MatchArgument()

```
int MatchArgument (
    PHEAD WORD * arg,
    WORD * pat )
```

Definition at line 2052 of file [symmetr.c](#).

## 4.139 symmetr.c

[Go to the documentation of this file.](#)

```
00001
00006 /* #[] License : */
00007 /*
00008 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 * When using this file you are requested to refer to the publication
00010 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 * This is considered a matter of courtesy as the development was paid
00012 * for by FOM the Dutch physics granting agency and we would like to
00013 * be able to track its scientific use to convince FOM of its value
00014 * for the community.
00015 *
00016 * This file is part of FORM.
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00022 *
00023 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
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00025 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 * details.
00027 *
00028 * You should have received a copy of the GNU General Public License along
00029 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #[] License : */
00032 /*
00033 #[] Includes : function.c
00034 */
00035
00036 #include "form3.h"
00037
00038 /*
00039 #[] Includes :
00040 #[] MatchE : WORD MatchE(pattern,fun,inter,par)
00041
00042 Matches symmetric and antisymmetric tensors.
00043 Pattern and fun point at a tensor.
00044 Problem is the wildcarding and all its possible permutations.
00045 This routine loops over all of them and calls for each
00046 possible wildcarding the recursion in ScanFunctions.
00047 Note that this can be very costly.
00048
00049 Originally this routine did only Levi Civita tensors and hence
00050 it dealt only with commuting objects.
00051 Because of the backtracking we cannot fall back to the calling
00052 ScanFunctions routine and check the sequence of functions when
00053 non-commuting objects are involved.
00054 */
00055
00056 int MatchE(PHEAD WORD *pattern, WORD *fun, WORD *inter, WORD par)
00057 {
00058     GETBIDENTITY
00059     WORD *m, *t, *r, i;
00060     int retval;
00061     WORD *mstop, *tstop, j, newvalue, newfun;
00062     WORD fixvec[MAXMATCH], wvec[MAXMATCH], fixind[MAXMATCH], wcind[MAXMATCH];
00063     WORD tfixvec[MAXMATCH], tfixind[MAXMATCH];
00064     WORD vvc,vfix,ifix,iwc,tvfix,tfix,nv,ni;
00065     WORD sign = 0, *rstop, first1, first2, first3, funwild;
00066     WORD *OldWork, nwstore, oRepFunNum;
00067     PERM perm1,perm2;
00068     DISTRIBUTE distr;
00069     WORD *newpat, /* *newter, *instart, */ offset;
00070     /* instart = fun; */
```

```

00071     offset = WORDDIFF(fun,AN.terstart);
00072     if ( pattern[1] != fun[1] ) return(0);
00073     if ( *pattern >= FUNCTION+WILDOFFSET ) {
00074         if ( CheckWild(BHEAD *pattern-WILDOFFSET,FUNTOFUN,*fun,&newfun) ) return(0);
00075         funwild = 1;
00076     }
00077     else funwild = 0;
00078     mstop = pattern + pattern[1];
00079     tstop = fun + fun[1];
00080     m = pattern + FUNHEAD;
00081     t = fun + FUNHEAD;
00082     while ( m < mstop ) {
00083         if ( *m != *t ) break;
00084         m++; t++;
00085     }
00086     if ( m >= mstop ) {
00087         AN.RepFunList[AN.RepFunNum++] = offset;
00088         AN.RepFunList[AN.RepFunNum++] = 0;
00089         newpat = pattern + pattern[1];
00090         if ( funwild ) {
00091             m = AN.WildValue;
00092             t = OldWork = AT.WorkPointer;
00093             nwstore = i = (m[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
00094             r = AT.WildMask;
00095             if ( i > 0 ) {
00096                 do {
00097                     *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
00098                 } while ( --i > 0 );
00099             }
00100             if ( t >= AT.WorkTop ) {
00101                 MLOCK(ErrorMessageLock);
00102                 MesWork();
00103                 MUNLOCK(ErrorMessageLock);
00104                 return(-1);
00105             }
00106             AT.WorkPointer = t;
00107             AddWild(BHEAD *pattern-WILDOFFSET,FUNTOFUN,newfun);
00108             if ( newpat >= AN.patstop ) {
00109                 if ( AN.UseFindOnly == 0 ) {
00110                     if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
00111                         AN.UsedOtherFind = 1;
00112                         return(1);
00113                     }
00114                     retval = 0;
00115                 }
00116                 else return(1);
00117             }
00118             else {
00119                 /* newter = instart; */
00120                 retval = ScanFunctions(BHEAD newpat,inter,par);
00121             }
00122             if ( retval == 0 ) {
00123                 m = AN.WildValue;
00124                 t = OldWork; r = AT.WildMask; i = nwstore;
00125                 if ( i > 0 ) {
00126                     do {
00127                         *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *r++ = *t++;
00128                     } while ( --i > 0 );
00129                 }
00130             }
00131             AT.WorkPointer = OldWork;
00132             return(retval);
00133         }
00134         else {
00135             if ( newpat >= AN.patstop ) {
00136                 if ( AN.UseFindOnly == 0 ) {
00137                     if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
00138                         AN.UsedOtherFind = 1;
00139                         return(1);
00140                     }
00141                     else return(0);
00142                 }
00143                 else return(1);
00144             }
00145             /* newter = instart; */
00146             retval = ScanFunctions(BHEAD newpat,inter,par);
00147             return(retval);
00148         }
00149     }
00150     /* Now the recursion
00151     */
00152 }
00153 /*
00154 Strategy:
00155 1: match the fixed arguments
00156 2: match, permuting the wildcards if needed.
00157 3: keep track of sign.

```



```

00158 */
00159     vwc = 0;
00160     vfix = 0;
00161     ifix = 0;
00162     iwc = 0;
00163     r = pattern+FUNHEAD;
00164     while ( r < mstop ) {
00165         if ( *r < (AM.OffsetVector+WILDOFFSET) ) {
00166             fixvec[vfix++] = *r;          /* Fixed vectors */
00167             sign += vwc + ifix + iwc;
00168         }
00169         else if ( *r < MINSPEC ) {
00170             wcvec[wvc++] = *r;            /* Wildcard vectors */
00171             sign += ifix + iwc;
00172         }
00173         else if ( *r < (AM.OffsetIndex+WILDOFFSET) ) {
00174             fixind[ifix++] = *r;          /* Fixed indices */
00175             sign += iwc;
00176         }
00177         else if ( *r < (AM.OffsetIndex+(WILDOFFSET«1)) ) {
00178             wcind[iwc++] = *r;            /* Wildcard indices */
00179         }
00180         else {
00181             fixind[ifix++] = *r;          /* Generated indices ~ fixed */
00182             sign += iwc;
00183         }
00184         r++;
00185     }
00186     if ( iwc == 0 && vwc == 0 ) return(0);
00187     tvfix = tifix = 0;
00188     t = fun + FUNHEAD;
00189     m = fixvec;
00190     mstop = m + vfix;
00191     r = fixind;
00192     rstop = r + ifix;
00193     nv = 0; ni = 0;
00194     while ( t < tstop ) {
00195         if ( *t < 0 ) {
00196             nv++;
00197             if ( m < mstop && *t == *m ) {
00198                 m++;
00199             }
00200             else {
00201                 sign += WORDDIF(mstop,m);
00202                 tfixvec[tfix++] = *t;
00203             }
00204         }
00205         else {
00206             ni++;
00207             if ( r < rstop && *r == *t ) {
00208                 r++;
00209             }
00210             else {
00211                 sign += WORDDIF(rstop,r);
00212                 tfixind[tifix++] = *t;
00213             }
00214         }
00215         t++;
00216     }
00217     if ( m < mstop || r < rstop ) return(0);
00218     if ( tvfix < vwc || (tvfix+tifix) < (vwc+iwc) ) return(0);
00219     sign += ( nv - vfix - vwc ) & ni;
00220     /*
00221     Take now the wildcards that have an assignment already.
00222     See whether they match.
00223     */
00224     {
00225         WORD *wv, *wm, n;
00226         wm = AT.WildMask;
00227         wv = AN.WildValue;
00228         n = AN.NumWild;
00229         do {
00230             if ( *wm ) {
00231                 if ( *wv == VECTOVEC ) {
00232                     for ( ni = 0; ni < vwc; ni++ ) {
00233                         if ( wcvec[ni]-WILDOFFSET == wv[2] ) { /* Has been assigned */
00234                             sign += ni;
00235                             vwc--;
00236                             while ( ni < vwc ) {
00237                                 wcvec[ni] = wcvec[ni+1];
00238                                 ni++;
00239                             }
00240                             /* TryVect: */
00241                             for ( ni = 0; ni < tvfix; ni++ ) {
00242                                 if ( tfixvec[ni] == wv[3] ) {
00243                                     sign += ni;
00244                                     tvfix--;

```

```

00245             while ( ni < tvfix ) {
00246                 tfixvec[ni] = tfixvec[ni+1];
00247                 ni++;
00248             }
00249             goto NextWV;
00250         }
00251     }
00252     return(0);
00253 }
00254 }
00255 }
00256 else if ( *wv == INDTOIND ) {
00257     for ( ni = 0; ni < iwc; ni++ ) {
00258         if ( wcind[ni]-WILDOFFSET == wv[2] ) { /* Has been assigned */
00259             sign += ni;
00260             iwc--;
00261             while ( ni < iwc ) {
00262                 wcind[ni] = wcind[ni+1];
00263                 ni++;
00264             }
00265             for ( ni = 0; ni < tfix; ni++ ) {
00266                 if ( tfixind[ni] == wv[3] ) {
00267                     sign += ni;
00268                     tfix--;
00269                     while ( ni < tfix ) {
00270                         tfixind[ni] = tfixind[ni+1];
00271                         ni++;
00272                     }
00273                     goto NextWV;
00274                 }
00275             }
00276             /* goto TryVect; */
00277             return(0);
00278         }
00279     }
00280 }
00281 }
00282 else if ( *wv == VECTOSUB ) {
00283     for ( ni = 0; ni < vwc; ni++ ) {
00284         if ( wcvect[ni]-WILDOFFSET == wv[2] ) return(0);
00285     }
00286 }
00287 else if ( *wv == INDTOISUB ) {
00288     for ( ni = 0; ni < iwc; ni++ ) {
00289         if ( wcind[ni]-WILDOFFSET == wv[2] ) return(0);
00290     }
00291 }
00292 }
00293 NextWV:
00294     wm++;
00295     wv += wv[1];
00296     n--;
00297     if ( n > 0 ) {
00298         while ( n > 0 && ( *wv == FROMSET || *wv == SETTONUM
00299             || *wv == LOADDOLLAR ) ) { wv += wv[1]; wm++; n--; }
00300     /*
00301         Freak problem: doesn't test for n and ran into a remaining
00302         code equal to SETTONUM followed by a big number and then
00303         ran out of the memory.
00304
00305         while ( *wv == FROMSET || *wv == SETTONUM
00306             || ( *wv == LOADDOLLAR && n > 0 ) ) { wv += wv[1]; wm++; n--; }
00307     */
00308     }
00309     } while ( n > 0 );
00310 }
00311 /*
00312     Now there are only free wildcards left.
00313     Possibly the assigned values ate too many vectors.
00314     The rest has to be done the 'hard way' via permutations.
00315     This is too bad when there are 10 indices.
00316     This could cause 10! tries.
00317     We try to avoid the worst case by using a very special
00318     (somewhat slow) permutation routine that has as its worst
00319     cases some rather unlikely configurations, rather than some
00320     common ones (as would have been the case with the conventional
00321     permutation routine).
00322     assume:
00323         vvvvvvvvvvvviiiiiii (tvfix in tfixvec and tfix in tfixind)
00324         VVVVVVVVVVIIIIIIIII (vwc in wcvect and iwc in wcind)
00325     Note: all further assignments are possible at this point!
00326     Strategy:
00327         permute v
00328         permute i
00329         loop over the ordered distribution of the leftover v's
00330         through the i's.
00331     */

```

```

00332     if ( tvfix < vwc ) { return(0); }
00333     perm1.n = tvfix;
00334     perm1.sign = 0;
00335     perm1.objects = tfixvec;
00336     perm2.n = tfix;
00337     perm2.sign = 0;
00338     perm2.objects = tfixind;
00339     distr.n1 = tvfix - vwc;
00340     distr.n2 = tfix;
00341     distr.obj1 = tfixvec + vwc;
00342     distr.obj2 = tfixind;
00343     distr.out = fixvec;          /* For scratch */
00344     first1 = 1;
00345     /*
00346         Store the current Wildcard assignments
00347     */
00348     m = AN.WildValue;
00349     t = OldWork = AT.WorkPointer;
00350     nwstore = i = (m[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
00351     r = AT.WildMask;
00352     if ( i > 0 ) {
00353         do {
00354             *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
00355         } while ( --i > 0 );
00356     }
00357     if ( t >= AT.WorkTop ) {
00358         MLOCK(ErrorMessageLock);
00359         MesWork();
00360         MUNLOCK(ErrorMessageLock);
00361         return(-1);
00362     }
00363     AT.WorkPointer = t;
00364     while ( (first1 = Permute(&perm1,first1) ) == 0 ) {
00365         first2 = 1;
00366         while ( (first2 = Permute(&perm2,first2) ) == 0 ) {
00367             first3 = 1;
00368             while ( (first3 = Distribute(&distr,first3) ) == 0 ) {
00369                 /*
00370                     Make now the wildcard assignments
00371                 */
00372                 for ( i = 0; i < vwc; i++ ) {
00373                     j = wcvec[i] - WILDOFFSET;
00374                     if ( CheckWild(BHEAD j,VECTOVEC,tfixvec[i],&newvalue) )
00375                         goto NoCaseB;
00376                     AddWild(BHEAD j,VECTOVEC,newvalue);
00377                 }
00378                 for ( i = 0; i < iwc; i++ ) {
00379                     j = wcind[i] - WILDOFFSET;
00380                     if ( CheckWild(BHEAD j,INDTOIND,fixvec[i],&newvalue) )
00381                         goto NoCaseB;
00382                     AddWild(BHEAD j,INDTOIND,newvalue);
00383                 }
00384                 /*
00385                     Go into the recursion
00386                 */
00387                 oRepFunNum = AN.RepFunNum;
00388                 AN.RepFunList[AN.RepFunNum++] = offset;
00389                 AN.RepFunList[AN.RepFunNum++] =
00390                     ( perm1.sign + perm2.sign + distr.sign + sign ) & 1;
00391                 newpat = pattern + pattern[1];
00392                 if ( funwild ) AddWild(BHEAD *pattern-WILDOFFSET,FUNTOFUN,newfun);
00393                 if ( newpat >= AN.patstop ) {
00394                     if ( AN.UseFindOnly == 0 ) {
00395                         if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
00396                             AN.UsedOtherFind = 1;
00397                             return(1);
00398                         }
00399                     }
00400                     else return(1);
00401                 }
00402                 else {
00403                     /*
00404                         newter = instart; */
00405                     if ( ScanFunctions(BHEAD newpat,inter,par) ) { return(1); }
00406                 }
00407                 /*
00408                     Restore the old Wildcard assignments
00409                 */
00410                 AN.RepFunNum = oRepFunNum;
00411                 m = AN.WildValue;
00412                 t = OldWork; r = AT.WildMask; i = nwstore;
00413                 if ( i > 0 ) {
00414                     do {
00415                         *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
00416                     } while ( --i > 0 );
00417                 }
00418                 AT.WorkPointer = t;

```

```

00419     }
00420 }
00421 AT.WorkPointer = OldWork;
00422 return(0);
00423 }
00424
00425 /*
00426 #] MatchE :
00427 #[ Permute :                WORD Permute(perm,first)
00428
00429     Special permutation function.
00430     Works recursively.
00431     The aim is to cycle through in as fast a way as possible,
00432     to take care that each object hits the various positions
00433     already early in the game.
00434
00435     Start at two: -> cycle of two
00436     then three -> cycle of three
00437     etc;
00438     The innermost cycle is the longest. This is the opposite
00439     of the usual way of generating permutations and it is
00440     certainly not the fastest one. It allows for the fastest
00441     hit in the assignment of wildcards though.
00442 */
00443
00444 int Permute(PERM *perm, WORD first)
00445 {
00446     WORD *s, c, i, j;
00447     if ( first ) {
00448         perm->sign = ( perm->sign <= 1 ) ? 0: 1;
00449         for ( i = 0; i < perm->n; i++ ) perm->cycle[i] = 0;
00450         return(0);
00451     }
00452     i = perm->n;
00453     while ( --i > 0 ) {
00454         s = perm->objects;
00455         c = s[0];
00456         j = i;
00457         while ( --j >= 0 ) { *s = s[1]; s++; }
00458         *s = c;
00459         if ( ( i & 1 ) != 0 ) perm->sign ^= 1;
00460         if ( perm->cycle[i] < i ) {
00461             (perm->cycle[i])++;
00462             return(0);
00463         }
00464         else {
00465             perm->cycle[i] = 0;
00466         }
00467     }
00468     return(1);
00469 }
00470
00471 /*
00472 #] Permute :
00473 #[ PermuteP :                WORD PermuteP(perm,first)
00474
00475     Like Permute, but works on an array of pointers
00476 */
00477
00478 int PermuteP(PERMP *perm, WORD first)
00479 {
00480     WORD **s, *c, i, j;
00481     if ( first ) {
00482         perm->sign = ( perm->sign <= 1 ) ? 0: 1;
00483         for ( i = 0; i < perm->n; i++ ) perm->cycle[i] = 0;
00484         return(0);
00485     }
00486     i = perm->n;
00487     while ( --i > 0 ) {
00488         s = perm->objects;
00489         c = s[0];
00490         j = i;
00491         while ( --j >= 0 ) { *s = s[1]; s++; }
00492         *s = c;
00493         if ( ( i & 1 ) != 0 ) perm->sign ^= 1;
00494         if ( perm->cycle[i] < i ) {
00495             (perm->cycle[i])++;
00496             return(0);
00497         }
00498         else {
00499             perm->cycle[i] = 0;
00500         }
00501     }
00502     return(1);
00503 }
00504
00505 /*

```

```

00506     #] PermuteP :
00507     #[ Distribute :
00508     */
00509
00510 int Distribute(DISTRIBUTE *d, WORD first)
00511 {
00512     WORD *to, *from, *inc, *from2, i, j;
00513     if ( first ) {
00514         d->n = d->n1 + d->n2;
00515         to = d->out;
00516         from = d->obj2;
00517         for ( i = 0; i < d->n2; i++ ) {
00518             d->cycle[i] = 0;
00519             *to++ = *from++;
00520         }
00521         from = d->obj1;
00522         while ( i < d->n ) {
00523             d->cycle[i++] = 1;
00524             *to++ = *from++;
00525         }
00526         d->sign = 0;
00527         return(0);
00528     }
00529     if ( d->n1 == 0 || d->n2 == 0 ) return(1);
00530     j = 0;
00531     i = 0;
00532     inc = d->cycle;
00533     from = inc + d->n;
00534     while ( *inc ) { j++; inc++; }
00535     while ( inc < from && !*inc ) { i++; inc++; }
00536     if ( inc >= from ) return(1);
00537     d->sign ^= ((i&j)-j+1) & 1;
00538     *inc = 0;
00539     *--inc = 1;
00540     while ( --j >= 0 ) *--inc = 1;
00541     while ( --i > 0 ) *--inc = 0;
00542     to = d->out;
00543     from = d->obj1;
00544     from2 = d->obj2;
00545     for ( i = 0; i < d->n; i++ ) {
00546         if ( *inc++ ) {
00547             *to++ = *from++;
00548         }
00549         else {
00550             *to++ = *from2++;
00551         }
00552     }
00553     return(0);
00554 }
00555
00556 /*
00557     #] Distribute :
00558     #[ MatchCy :
00559
00560     Matching of (r)cyclic tensors.
00561     Parameters like in MatchE.
00562     The structure of the routine is much simpler, because the number
00563     of possibilities is much more limited.
00564     The major complication is the ?a-type wildcards
00565     We need a strategy for T(il?,?a,il?,?b). Which is the shorter
00566     match: ?a or ?b ? (if possible of course)
00567     This is also relevant in the case of the shortest match if there
00568     is more than one choice for il.
00569     */
00570
00571 int MatchCy(PHEAD WORD *pattern, WORD *fun, WORD *inter, WORD par)
00572 {
00573     GETBIDENTITY
00574     WORD *t, *tstop, *p, *pstop, *m, *r, *oldworkpointer = AT.WorkPointer;
00575     WORD *thewildcards, *multiplicity, *renum, wc, newvalue, oldwilval = 0;
00576     WORD *params, *lowlevel = 0;
00577     int argcount = 0, funnycount = 0, tcount = fun[1] - FUNHEAD;
00578     int type = 0, pnum, i, j, k, nwstore, iraise, itop, sumeat;
00579     CBUF *C = cbuf+AT.ebufnum;
00580     int ntw = 3*AN.NumTotWildArgs+1;
00581     LONG oldcpointer = C->Pointer - C->Buffer;
00582     WORD offset = fun-AN.terstart, *newpat;
00583
00584     if ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;
00585     pnum = pattern[0];
00586     nwstore = (AN.WildValue[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
00587     if ( pnum > FUNCTION + WILDOFFSET ) {
00588         pnum -= WILDOFFSET;
00589         if ( CheckWild(BHEAD pnum,FUNTOFUN,fun[0],&newvalue) ) return(0);
00590         oldwilval = 1;
00591         t = lowlevel = AT.WorkPointer;
00592         m = AN.WildValue;

```

```

00593     i = nwstore;
00594     r = AT.WildMask;
00595     if ( i > 0 ) {
00596         do {
00597             *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
00598         } while ( --i > 0 );
00599     }
00600     *t++ = C->numrhs;
00601     if ( t >= AT.WorkTop ) {
00602         MLOCK(ErrorMessageLock);
00603         MesWork();
00604         MUNLOCK(ErrorMessageLock);
00605         return(-1);
00606     }
00607     AT.WorkPointer = t;
00608     AddWild(BHEAD pnum,FUNTOFUN,newvalue);
00609 }
00610 if ( (functions[pnum-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;
00611
00612 /* First we have to make an inventory. Are there FUNNYWILD pointers? */
00613
00614 p = pattern + FUNHEAD;
00615 pstop = pattern + pattern[1];
00616 while ( p < pstop ) {
00617     if ( *p == FUNNYWILD ) { p += 2; funnycount++; }
00618     else { p++; argcount++; }
00619 }
00620 if ( argcount > tcount ) goto NoSuccess;
00621 if ( argcount < tcount && funnycount == 0 ) goto NoSuccess;
00622 if ( argcount == 0 && tcount == 0 && funnycount == 0 ) {
00623     AN.RepFunList[AN.RepFunNum++] = offset;
00624     AN.RepFunList[AN.RepFunNum++] = 0;
00625     newpat = pattern + pattern[1];
00626     if ( newpat >= AN.patstop ) {
00627         if ( AN.UseFindOnly == 0 ) {
00628             if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
00629                 AT.WorkPointer = oldworkpointer;
00630                 AN.UsedOtherFind = 1;
00631                 return(1);
00632             }
00633             j = 0;
00634         }
00635         else {
00636             AT.WorkPointer = oldworkpointer;
00637             return(1);
00638         }
00639     }
00640     else j = ScanFunctions(BHEAD newpat,inter,par);
00641     if ( j ) return(j);
00642     goto NoSuccess;
00643 }
00644 tstop = fun + fun[1];
00645
00646 /* Store the wildcard assignments */
00647
00648 params = AT.WorkPointer;
00649 thewildcards = t = params + tcount;
00650 t += ntw;
00651 if ( oldwilval ) lowlevel = oldworkpointer;
00652 else lowlevel = t;
00653 m = AN.WildValue;
00654 i = nwstore;
00655 if ( i > 0 ) {
00656     r = AT.WildMask;
00657     do {
00658         *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
00659     } while ( --i > 0 );
00660     *t++ = C->numrhs;
00661 }
00662 if ( t >= AT.WorkTop ) {
00663     MLOCK(ErrorMessageLock);
00664     MesWork();
00665     MUNLOCK(ErrorMessageLock);
00666     return(-1);
00667 }
00668 AT.WorkPointer = t;
00669 /*
00670 #[ Case 1: no funnies or all funnies must be empty. We just cycle through.
00671 */
00672 if ( argcount == tcount ) {
00673     if ( funnycount > 0 ) { /* Test all funnies first */
00674         p = pattern + FUNHEAD;
00675         t = fun + FUNHEAD;
00676         while ( p < pstop ) {
00677             if ( *p != FUNNYWILD ) { p++; continue; }
00678             AN.argaddress = t;
00679             if ( CheckWild(BHEAD p[1],ARGTOARG,0,t) ) goto nomatch;

```

```

00680         AddWild(BHEAD p[1],ARGTOARG,0);
00681         p += 2;
00682     }
00683     oldwilval = 1;
00684 }
00685 for ( k = 0; k <= type; k++ ) {
00686     if ( k == 0 ) {
00687         p = params; t = fun + FUNHEAD;
00688         while ( t < tstop ) *p++ = *t++;
00689     }
00690     else {
00691         p = params+tcount; t = fun + FUNHEAD;
00692         while ( t < tstop ) *--p = *t++;
00693     }
00694     for ( i = 0; i < tcount; i++ ) { /* The various cycles */
00695         p = pattern + FUNHEAD;
00696         wc = 0;
00697         for ( j = 0; j < tcount; j++, p++ ) { /* The arguments */
00698             while ( *p == FUNNYWILD ) p += 2;
00699             t = params + (i+j)%tcount;
00700             if ( *t == *p ) continue;
00701             if ( *p >= AM.OffsetIndex + WILDOFFSET
00702                 && *p < AM.OffsetIndex + 2*WILDOFFSET ) {
00703
00704                 /* Test wildcard index */
00705
00706                 wc = *p - WILDOFFSET;
00707                 if ( CheckWild(BHEAD wc,INDTOIND,*t,&newvalue) ) break;
00708                 AddWild(BHEAD wc,INDTOIND,newvalue);
00709             }
00710             else if ( *t < MINSPEC && p[j] < MINSPEC
00711                 && *p >= AM.OffsetVector + WILDOFFSET ) {
00712
00713                 /* Test wildcard vector */
00714
00715                 wc = *p - WILDOFFSET;
00716                 if ( CheckWild(BHEAD wc,VECTOVEC,*t,&newvalue) ) break;
00717                 AddWild(BHEAD wc,VECTOVEC,newvalue);
00718             }
00719             else break;
00720         }
00721         if ( j >= tcount ) { /* Match! */
00722
00723             /* Continue with other functions. Make sure of the funnies */
00724
00725             AN.RepFunList[AN.RepFunNum++] = offset;
00726             AN.RepFunList[AN.RepFunNum++] = 0;
00727
00728             if ( funnycount > 0 ) {
00729                 p = pattern + FUNHEAD;
00730                 t = fun + FUNHEAD;
00731                 while ( p < pstop ) {
00732                     if ( *p != FUNNYWILD ) { p++; continue; }
00733                     AN.argaddress = t;
00734                     AddWild(BHEAD p[1],ARGTOARG,0);
00735                     p += 2;
00736                 }
00737             }
00738             newpat = pattern + pattern[1];
00739             if ( newpat >= AN.patstop ) {
00740                 if ( AN.UseFindOnly == 0 ) {
00741                     if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
00742                         AT.WorkPointer = oldworkpointer;
00743                         AN.UsedOtherFind = 1;
00744                         return(1);
00745                     }
00746                     j = 0;
00747                 }
00748                 else {
00749                     AT.WorkPointer = oldworkpointer;
00750                     return(1);
00751                 }
00752             }
00753             else j = ScanFunctions(BHEAD newpat,inter,par);
00754             if ( j ) {
00755                 AT.WorkPointer = oldworkpointer;
00756                 return(j); /* Full match. Return our success */
00757             }
00758             AN.RepFunNum -= 2;
00759         }
00760
00761         /* No (deeper) match. -> reset wildcards and continue */
00762
00763         if ( wc && nwstore > 0 ) {
00764             j = nwstore;
00765             m = AN.WildValue;
00766             t = thewildcards + ntwa; r = AT.WildMask;

```

```

00767         if ( j > 0 ) {
00768             do {
00769                 *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
00770             } while ( --j > 0 );
00771         }
00772         C->numrhs = *t++;
00773         C->Pointer = C->Buffer + oldcpointer;
00774     }
00775 }
00776 }
00777 goto NoSuccess;
00778 }
00779 /*
00780 #] Case 1:
00781 #[ Case 2: One FUNNYWILD. Fix its length.
00782 */
00783 if ( funnycount == 1 ) {
00784     funnycount = tcount - argcount; /* Number of arguments to be eaten */
00785     for ( k = 0; k <= type; k++ ) {
00786         if ( k == 0 ) {
00787             p = params; t = fun + FUNHEAD;
00788             while ( t < tstop ) *p++ = *t++;
00789         }
00790         else {
00791             p = params+tcount; t = fun + FUNHEAD;
00792             while ( t < tstop ) *--p = *t++;
00793         }
00794         for ( i = 0; i < tcount; i++ ) { /* The various cycles */
00795             p = pattern + FUNHEAD;
00796             t = params;
00797             wc = 0;
00798             for ( j = 0; j < tcount; j++, p++, t++ ) { /* The arguments */
00799                 if ( *t == *p ) continue;
00800                 if ( *p == FUNNYWILD ) {
00801                     p++; wc = 1;
00802                     AN.argaddress = t;
00803                     if ( CheckWild(BHEAD *p,ARGTOARG|EATTENSOR,funnycount,t) ) break;
00804                     AddWild(BHEAD *p,ARGTOARG|EATTENSOR,funnycount);
00805                     j += funnycount-1; t += funnycount-1;
00806                 }
00807                 else if ( *p >= AM.OffsetIndex + WILDOFFSET
00808                     && *p < AM.OffsetIndex + 2*WILDOFFSET ) {
00809
00810                     /* Test wildcard index */
00811
00812                     wc = *p - WILDOFFSET;
00813                     if ( CheckWild(BHEAD wc,INDTOIND,*t,&newvalue) ) break;
00814                     AddWild(BHEAD wc,INDTOIND,newvalue);
00815                 }
00816                 else if ( *t < MINSPEC && *p < MINSPEC
00817                     && *p >= AM.OffsetVector + WILDOFFSET ) {
00818
00819                     /* Test wildcard vector */
00820
00821                     wc = *p - WILDOFFSET;
00822                     if ( CheckWild(BHEAD wc,VECTOVEC,*t,&newvalue) ) break;
00823                     AddWild(BHEAD wc,VECTOVEC,newvalue);
00824                 }
00825                 else break;
00826             }
00827             if ( j >= tcount ) { /* Match! */
00828
00829                 /* Continue with other functions. Make sure of the funnies */
00830
00831                 AN.RepFunList[AN.RepFunNum++] = offset;
00832                 AN.RepFunList[AN.RepFunNum++] = 0;
00833                 newpat = pattern + pattern[1];
00834                 if ( newpat >= AN.patstop ) {
00835                     if ( AN.UseFindOnly == 0 ) {
00836                         if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
00837                             AT.WorkPointer = oldworkpointer;
00838                             AN.UsedOtherFind = 1;
00839                             return(1);
00840                         }
00841                         j = 0;
00842                     }
00843                     else {
00844                         AT.WorkPointer = oldworkpointer;
00845                         return(1);
00846                     }
00847                 }
00848                 else j = ScanFunctions(BHEAD newpat,inter,par);
00849                 if ( j ) {
00850                     AT.WorkPointer = oldworkpointer;
00851                     return(j); /* Full match. Return our success */
00852                 }
00853                 AN.RepFunNum -= 2;

```



```

00854     }
00855
00856     /* No (deeper) match. -> reset wildcards and continue */
00857
00858     if ( wc ) {
00859         j = nwstore;
00860         m = AN.WildValue;
00861         t = thewildcards + ntw; r = AT.WildMask;
00862         if ( j > 0 ) {
00863             do {
00864                 *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
00865             } while ( --j > 0 );
00866         }
00867         C->numrhs = *t++;
00868         C->Pointer = C->Buffer + oldcpointer;
00869     }
00870     t = params;
00871     wc = *t;
00872     for ( j = 1; j < tcount; j++ ) { *t = t[1]; t++; }
00873     *t = wc;
00874 }
00875 }
00876 goto NoSuccess;
00877 }
00878 /*
00879 #] Case 2:
00880 #[ Case 3: More than one FUNNYWILD. Complicated.
00881 */
00882
00883 sumeat = tcount - argcount; /* Total number to be eaten by Funnies */
00884 /*
00885 In the first funnycount elements of 'thewildcards' we arrange
00886 for the summing over the various possibilities.
00887 The renumbering table is in thewildcards[2*funnycount]
00888 The multiplicity table is in thewildcards[funnycount]
00889 The number of arguments for each is in thewildcards[]
00890 */
00891 p = pattern+FUNHEAD;
00892 for ( i = funnycount; i < ntw; i++ ) thewildcards[i] = -1;
00893 multiplicity = thewildcards + funnycount;
00894 renum = multiplicity + funnycount;
00895 j = 0;
00896 while ( p < pstop ) {
00897     if ( *p != FUNNYWILD ) { p++; continue; }
00898     p++;
00899     if ( renum[*p] < 0 ) {
00900         renum[*p] = j;
00901         multiplicity[j] = 1;
00902         j++;
00903     }
00904     else multiplicity[renum[*p]]++;
00905     p++;
00906 }
00907 /*
00908 Strategy: First 'declared' has a tendency to be smaller
00909 */
00910 for ( i = 1; i < AN.NumTotWildArgs; i++ ) {
00911     if ( renum[i] < 0 ) continue;
00912     for ( j = i+1; j <= AN.NumTotWildArgs; j++ ) {
00913         if ( renum[j] < 0 ) continue;
00914         if ( renum[i] < renum[j] ) continue;
00915         k = multiplicity[renum[i]];
00916         multiplicity[renum[i]] = multiplicity[renum[j]];
00917         multiplicity[renum[j]] = k;
00918         k = renum[i]; renum[i] = renum[j]; renum[j] = k;
00919     }
00920 }
00921 for ( i = 0; i < funnycount; i++ ) thewildcards[i] = 0;
00922 iraise = funnycount-1;
00923 for ( ; ) {
00924     for ( i = 0, j = sumeat; i < iraise; i++ )
00925         j -= thewildcards[i]*multiplicity[i];
00926     if ( j < 0 || j % multiplicity[iraise] != 0 ) {
00927         if ( j > 0 ) {
00928             thewildcards[iraise-1]++;
00929             continue;
00930         }
00931         itop = iraise-1;
00932         while ( itop > 0 && j < 0 ) {
00933             j += thewildcards[itop]*multiplicity[itop];
00934             thewildcards[itop] = 0;
00935             itop--;
00936         }
00937         if ( itop <= 0 && j <= 0 ) break;
00938         thewildcards[itop]++;
00939         continue;
00940     }
}

```

```

00941     thewildcards[iraise] = j / multiplicity[iraise];
00942
00943     for ( k = 0; k <= type; k++ ) {
00944         if ( k == 0 ) {
00945             p = params; t = fun + FUNHEAD;
00946             while ( t < tstop ) *p++ = *t++;
00947         }
00948         else {
00949             p = params+tcount; t = fun + FUNHEAD;
00950             while ( t < tstop ) *--p = *t++;
00951         }
00952         for ( i = 0; i < tcount; i++ ) { /* The various cycles */
00953             p = pattern + FUNHEAD;
00954             t = params;
00955             wc = 0;
00956             for ( j = 0; j < tcount; j++, p++, t++ ) { /* The arguments */
00957                 if ( *t == *p ) continue;
00958                 if ( *p == FUNNYWILD ) {
00959                     p++; wc = thewildcards[renum[*p]];
00960                     AN.argaddress = t;
00961                     if ( CheckWild(BHEAD *p,ARGTOARG|EATTENSOR,wc,t) ) break;
00962                     AddWild(BHEAD *p,ARGTOARG|EATTENSOR,wc);
00963                     j += wc-1; t += wc-1; wc = 1;
00964                 }
00965                 else if ( *p >= AM.OffsetIndex + WILDOFFSET
00966                     && *p < AM.OffsetIndex + 2*WILDOFFSET ) {
00967
00968                     /* Test wildcard index */
00969
00970                     wc = *p - WILDOFFSET;
00971                     if ( CheckWild(BHEAD wc,INDTOIND,*t,&newvalue) ) break;
00972                     AddWild(BHEAD wc,INDTOIND,newvalue);
00973                 }
00974                 else if ( *t < MINSPEC && *p < MINSPEC
00975                     && *p >= AM.OffsetVector + WILDOFFSET ) {
00976
00977                     /* Test wildcard vector */
00978
00979                     wc = *p - WILDOFFSET;
00980                     if ( CheckWild(BHEAD wc,VECTOVEC,*t,&newvalue) ) break;
00981                     AddWild(BHEAD wc,VECTOVEC,newvalue);
00982                 }
00983                 else break;
00984             }
00985             if ( j >= tcount ) { /* Match! */
00986
00987                 /* Continue with other functions. Make sure of the funnies */
00988
00989                 AN.RepFunList[AN.RepFunNum++] = offset;
00990                 AN.RepFunList[AN.RepFunNum++] = 0;
00991                 newpat = pattern + pattern[1];
00992                 if ( newpat >= AN.patstop ) {
00993                     if ( AN.UseFindOnly == 0 ) {
00994                         if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
00995                             AT.WorkPointer = oldworkpointer;
00996                             AN.UsedOtherFind = 1;
00997                             return(1);
00998                         }
00999                         j = 0;
01000                     }
01001                     else {
01002                         AT.WorkPointer = oldworkpointer;
01003                         return(1);
01004                     }
01005                 }
01006                 else j = ScanFunctions(BHEAD newpat,inter,par);
01007                 if ( j ) {
01008                     AT.WorkPointer = oldworkpointer;
01009                     return(j); /* Full match. Return our success */
01010                 }
01011                 AN.RepFunNum -= 2;
01012             }
01013
01014             /* No (deeper) match. -> reset wildcards and continue */
01015
01016             if ( wc ) {
01017                 j = nwstore;
01018                 m = AN.WildValue;
01019                 t = thewildcards + ntwa; r = AT.WildMask;
01020                 if ( j > 0 ) {
01021                     do {
01022                         *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01023                     } while ( --j > 0 );
01024                 }
01025                 C->numrhs = *t++;
01026                 C->Pointer = C->Buffer + oldcpointer;
01027             }

```

```

01028         t = params;
01029         wc = *t;
01030         for ( j = 1; j < tcount; j++ ) { *t = t[1]; t++; }
01031         *t = wc;
01032     }
01033 }
01034 (thewildcards[iraise-1])++;
01035 }
01036 /*
01037 #] Case 3:
01038 */
01039 NoSuccess:
01040 if ( oldwilval > 0 ) {
01041 nomatch:
01042     j = nwstore;
01043     if ( j > 0 ) {
01044         m = AN.WildValue;
01045         t = lowlevel; r = AT.WildMask;
01046         if ( j > 0 ) {
01047             do {
01048                 *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01049             } while ( --j > 0 );
01050         }
01051         C->numrhs = *t++;
01052         C->Pointer = C->Buffer + oldcpointer;
01053     }
01054 }
01055 AT.WorkPointer = oldworkpointer;
01056 return(0);
01057 }
01058
01059 /*
01060 #] MatchCy :
01061 #[ FunMatchCy :
01062
01063     Matching of (r)cyclic functions.
01064     Like MatchCy, but now for general functions.
01065 */
01066
01067 int FunMatchCy(PHEAD WORD *pattern, WORD *fun, WORD *inter, WORD par)
01068 {
01069     GETBIDENTITY
01070     WORD *t, *tstop, *p, *pstop, *m, *r, *oldworkpointer = AT.WorkPointer;
01071     WORD **a, *thewildcards, *multiplicity, *renum, wc, wcc, oldwilval = 0;
01072     LONG ow = AT.pWorkPointer;
01073     WORD newvalue, *lowlevel = 0;
01074     int argcount = 0, funnycount = 0, tcount = 0;
01075     int type = 0, pn timer, i, j, k, nwstore, iraise, itop, sumeat;
01076     CBUF *C = cbuf+AT.ebufnum;
01077     int ntw = 3*AN.NumTotWildArgs+1;
01078     LONG oldcpointer = C->Pointer - C->Buffer;
01079     WORD offset = fun-AN.terstart, *newpat;
01080
01081     if ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;
01082     pn timer = pattern[0];
01083     nwstore = (AN.WildValue[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
01084     if ( pn timer > FUNCTION + WILDOFFSET ) {
01085         pn timer -= WILDOFFSET;
01086         if ( CheckWild(BHEAD pn timer,FUNTOFUN,fun[0],&newvalue) ) return(0);
01087         oldwilval = 1;
01088         t = lowlevel = oldworkpointer;
01089         m = AN.WildValue;
01090         i = nwstore;
01091         r = AT.WildMask;
01092         if ( i > 0 ) {
01093             do {
01094                 *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *r++ = *t++;
01095             } while ( --i > 0 );
01096         }
01097         *t++ = C->numrhs;
01098         if ( t >= AT.WorkTop ) {
01099             MLOCK(ErrorMessageLock);
01100             MesWork();
01101             MUNLOCK(ErrorMessageLock);
01102             return(-1);
01103         }
01104         AT.WorkPointer = t;
01105         AddWild(BHEAD pn timer,FUNTOFUN,newvalue);
01106     }
01107     if ( (functions[pn timer-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;
01108
01109     /* First we have to make an inventory. Are there -ARGWILD pointers? */
01110
01111     p = pattern + FUNHEAD;
01112     pstop = pattern + pattern[1];
01113     while ( p < pstop ) {
01114         if ( *p == -ARGWILD ) { p += 2; funnycount++; }

```

```

01115         else { NEXTARG(p); argcount++; }
01116     }
01117     t = fun + FUNHEAD;
01118     tstop = fun + fun[1];
01119     while ( t < tstop ) { NEXTARG(t); tcount++; }
01120
01121     if ( argcount > tcount ) return(0);
01122     if ( argcount < tcount && funnycount == 0 ) return(0);
01123     if ( argcount == 0 && tcount == 0 && funnycount == 0 ) {
01124         AN.RepFunList[AN.RepFunNum++] = offset;
01125         AN.RepFunList[AN.RepFunNum++] = 0;
01126         newpat = pattern + pattern[1];
01127         if ( newpat >= AN.patstop ) {
01128             if ( AN.UseFindOnly == 0 ) {
01129                 if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
01130                     AT.WorkPointer = oldworkpointer;
01131                     AN.UsedOtherFind = 1;
01132                     return(1);
01133                 }
01134                 j = 0;
01135             }
01136             else {
01137                 AT.WorkPointer = oldworkpointer;
01138                 return(1);
01139             }
01140         }
01141         else j = ScanFunctions(BHEAD newpat,inter,par);
01142         if ( j ) return(j);
01143         goto NoSuccess;
01144     }
01145
01146     /* Store the wildcard assignments */
01147
01148     WantAddPointers(tcount);
01149     AT.pWorkPointer += tcount;
01150     thewildcards = t = AT.WorkPointer;
01151     t += ntw;
01152     if ( oldwilval ) lowlevel = oldworkpointer;
01153     else lowlevel = t;
01154     m = AN.WildValue;
01155     i = nwstore;
01156     if ( i > 0 ) {
01157         r = AT.WildMask;
01158         do {
01159             *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
01160         } while ( --i > 0 );
01161         *t++ = C->numrhs;
01162     }
01163     if ( t >= AT.WorkTop ) {
01164         MLOCK(ErrorMessageLock);
01165         MesWork();
01166         MUNLOCK(ErrorMessageLock);
01167         return(-1);
01168     }
01169     AT.WorkPointer = t;
01170
01171     /*
01172     #[ Case 1: no funnies or all funnies must be empty. We just cycle through.
01173     */
01174     if ( argcount == tcount ) {
01175         if ( funnycount > 0 ) { /* Test all funnies first */
01176             p = pattern + FUNHEAD;
01177             t = fun + FUNHEAD;
01178             while ( p < pstop ) {
01179                 if ( *p != -ARGWILD ) { p++; continue; }
01180                 AN.argaddress = t;
01181                 if ( CheckWild(BHEAD p[1],ARGTOARG,0,t) ) goto nomatch;
01182                 AddWild(BHEAD p[1],ARGTOARG,0);
01183                 p += 2;
01184             }
01185             oldwilval = 1;
01186         }
01187         for ( k = 0; k <= type; k++ ) {
01188             if ( k == 0 ) {
01189                 a = AT.pWorkSpace+oww; t = fun + FUNHEAD;
01190                 while ( t < tstop ) { *a++ = t; NEXTARG(t); }
01191             }
01192             else {
01193                 a = AT.pWorkSpace+oww+tcount; t = fun + FUNHEAD;
01194                 while ( t < tstop ) { *--a = t; NEXTARG(t); }
01195             }
01196             for ( i = 0; i < tcount; i++ ) { /* The various cycles */
01197                 p = pattern + FUNHEAD;
01198                 wc = 0;
01199                 for ( j = 0; j < tcount; j++ ) { /* The arguments */
01200                     while ( *p == -ARGWILD ) p += 2;
01201                     t = AT.pWorkSpace[oww+((i+j)%tcount)];
01202                     if ( ( wcc = MatchArgument(BHEAD t,p) ) == 0 ) break;

```

```

01202         if ( wcc > 1 ) wc = 1;
01203         NEXTARG(p);
01204     }
01205     if ( j >= tcount ) { /* Match! */
01206
01207         /* Continue with other functions. Make sure of the funnies */
01208
01209         AN.RepFunList[AN.RepFunNum++] = offset;
01210         AN.RepFunList[AN.RepFunNum++] = 0;
01211
01212         if ( funnycount > 0 ) {
01213             p = pattern + FUNHEAD;
01214             t = fun + FUNHEAD;
01215             while ( p < pstop ) {
01216                 if ( *p != -ARGWILD ) { p++; continue; }
01217                 AN.argaddress = t;
01218                 AddWild(BHEAD p[1],ARGTOARG,0);
01219                 p += 2;
01220             }
01221         }
01222         newpat = pattern + pattern[1];
01223         if ( newpat >= AN.patstop ) {
01224             if ( AN.UseFindOnly == 0 ) {
01225                 if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
01226                     AT.WorkPointer = oldworkpointer;
01227                     AT.pWorkPointer = oww;
01228                     AN.UsedOtherFind = 1;
01229                     return(1);
01230                 }
01231                 j = 0;
01232             }
01233             else {
01234                 AT.WorkPointer = oldworkpointer;
01235                 AT.pWorkPointer = oww;
01236                 return(1);
01237             }
01238         }
01239         else j = ScanFunctions(BHEAD newpat,inter,par);
01240         if ( j ) {
01241             AT.WorkPointer = oldworkpointer;
01242             AT.pWorkPointer = oww;
01243             return(j); /* Full match. Return our success */
01244         }
01245         AN.RepFunNum -= 2;
01246     }
01247
01248     /* No (deeper) match. -> reset wildcards and continue */
01249
01250     if ( wc && nwstore > 0 ) {
01251         j = nwstore;
01252         m = AN.WildValue;
01253         t = thewildcards + ntwa; r = AT.WildMask;
01254         if ( j > 0 ) {
01255             do {
01256                 *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01257             } while ( --j > 0 );
01258         }
01259         C->numrhs = *t++;
01260         C->Pointer = C->Buffer + oldcpointer;
01261     }
01262 }
01263 }
01264 goto NoSuccess;
01265 }
01266 /*
01267 #] Case 1:
01268 #[ Case 2: One -ARGWILD. Fix its length.
01269 */
01270 if ( funnycount == 1 ) {
01271     funnycount = tcount - argcount; /* Number of arguments to be eaten */
01272     for ( k = 0; k <= type; k++ ) {
01273         if ( k == 0 ) {
01274             a = AT.pWorkSpace+oww; t = fun + FUNHEAD;
01275             while ( t < tstop ) { *a++ = t; NEXTARG(t); }
01276         }
01277         else {
01278             a = AT.pWorkSpace+oww+tcount; t = fun + FUNHEAD;
01279             while ( t < tstop ) { *--a = t; NEXTARG(t); }
01280         }
01281     }
01282     for ( i = 0; i < tcount; i++ ) { /* The various cycles */
01283         p = pattern + FUNHEAD;
01284         a = AT.pWorkSpace+oww;
01285         wc = 0;
01286         for ( j = 0; j < tcount; j++, a++ ) { /* The arguments */
01287             t = *a;
01288             if ( *p == -ARGWILD ) {
01289                 wc = 1;

```

```

01289         AN.argaddress = (WORD *)a;
01290         if ( CheckWild(BHEAD p[1],ARLTOARL,funnycount,(WORD *)a) ) break;
01291         AddWild(BHEAD p[1],ARLTOARL,funnycount);
01292         j += funnycount-1; a += funnycount-1;
01293     }
01294     else if ( MatchArgument(BHEAD t,p) == 0 ) break;
01295     NEXTARG(p);
01296 }
01297 if ( j >= tcount ) { /* Match! */
01298
01299     /* Continue with other functions. Make sure of the funnies */
01300
01301     AN.RepFunList[AN.RepFunNum++] = offset;
01302     AN.RepFunList[AN.RepFunNum++] = 0;
01303     newpat = pattern + pattern[1];
01304     if ( newpat >= AN.patstop ) {
01305         if ( AN.UseFindOnly == 0 ) {
01306             if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
01307                 AT.WorkPointer = oldworkpointer;
01308                 AT.pWorkPointer = ow;
01309                 AN.UsedOtherFind = 1;
01310                 return(1);
01311             }
01312             j = 0;
01313         }
01314         else {
01315             AT.WorkPointer = oldworkpointer;
01316             AT.pWorkPointer = ow;
01317             return(1);
01318         }
01319     }
01320     else j = ScanFunctions(BHEAD newpat,inter,par);
01321     if ( j ) {
01322         AT.WorkPointer = oldworkpointer;
01323         AT.pWorkPointer = ow;
01324         return(j); /* Full match. Return our success */
01325     }
01326     AN.RepFunNum -= 2;
01327 }
01328
01329 /* No (deeper) match. -> reset wildcards and continue */
01330
01331 if ( wc ) {
01332     j = nwstore;
01333     m = AN.WildValue;
01334     t = thewildcards + ntw; r = AT.WildMask;
01335     if ( j > 0 ) {
01336         do {
01337             *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01338         } while ( --j > 0 );
01339     }
01340     C->numrhs = *t++;
01341     C->Pointer = C->Buffer + oldcpointer;
01342 }
01343 a = AT.pWorkSpace+ow;
01344 t = *a;
01345 for ( j = 1; j < tcount; j++ ) { *a = a[1]; a++; }
01346 *a = t;
01347 }
01348 }
01349 goto NoSuccess;
01350 }
01351 /*
01352 #] Case 2:
01353 #[ Case 3: More than one -ARGWILD. Complicated.
01354 */
01355
01356 sumeat = tcount - argcount; /* Total number to be eaten by Funnies */
01357 /*
01358 In the first funnycount elements of 'thewildcards' we arrange
01359 for the summing over the various possibilities.
01360 The renumbering table is in thewildcards[2*funnycount]
01361 The multiplicity table is in thewildcards[funnycount]
01362 The number of arguments for each is in thewildcards[]
01363 */
01364 p = pattern+FUNHEAD;
01365 for ( i = funnycount; i < ntw; i++ ) thewildcards[i] = -1;
01366 multiplicity = thewildcards + funnycount;
01367 renum = multiplicity + funnycount;
01368 j = 0;
01369 while ( p < pstop ) {
01370     if ( *p != -ARGWILD ) { p++; continue; }
01371     p++;
01372     if ( renum[*p] < 0 ) {
01373         renum[*p] = j;
01374         multiplicity[j] = 1;
01375         j++;

```

```

01376     }
01377     else multiplicity[renum[*p]]++;
01378     p++;
01379 }
01380 /*
01381 Strategy: First 'declared' has a tendency to be smaller
01382 */
01383 for ( i = 1; i < AN.NumTotWildArgs; i++ ) {
01384     if ( renum[i] < 0 ) continue;
01385     for ( j = i+1; j <= AN.NumTotWildArgs; j++ ) {
01386         if ( renum[j] < 0 ) continue;
01387         if ( renum[i] < renum[j] ) continue;
01388         k = multiplicity[renum[i]];
01389         multiplicity[renum[i]] = multiplicity[renum[j]];
01390         multiplicity[renum[j]] = k;
01391         k = renum[i]; renum[i] = renum[j]; renum[j] = k;
01392     }
01393 }
01394 for ( i = 0; i < funnycount; i++ ) thewildcards[i] = 0;
01395 iraise = funnycount-1;
01396 for ( ;; ) {
01397     for ( i = 0, j = sumeat; i < iraise; i++ )
01398         j -= thewildcards[i]*multiplicity[i];
01399     if ( j < 0 || j % multiplicity[iraise] != 0 ) {
01400         if ( j > 0 ) {
01401             thewildcards[iraise-1]++;
01402             continue;
01403         }
01404         itop = iraise-1;
01405         while ( itop > 0 && j < 0 ) {
01406             j += thewildcards[itop]*multiplicity[itop];
01407             thewildcards[itop] = 0;
01408             itop--;
01409         }
01410         if ( itop <= 0 && j <= 0 ) break;
01411         thewildcards[itop]++;
01412         continue;
01413     }
01414     thewildcards[iraise] = j / multiplicity[iraise];
01415
01416     for ( k = 0; k <= type; k++ ) {
01417         if ( k == 0 ) {
01418             a = AT.pWorkSpace+oww; t = fun + FUNHEAD;
01419             while ( t < tstop ) { *a++ = t; NEXTARG(t); }
01420         }
01421         else {
01422             a = AT.pWorkSpace+oww+tcount; t = fun + FUNHEAD;
01423             while ( t < tstop ) { *--a = t; NEXTARG(t); }
01424         }
01425         for ( i = 0; i < tcount; i++ ) { /* The various cycles */
01426             p = pattern + FUNHEAD;
01427             a = AT.pWorkSpace+oww;
01428             wc = 0;
01429             for ( j = 0; j < tcount; j++, a++ ) { /* The arguments */
01430                 t = *a;
01431                 if ( *p == -ARGWILD ) {
01432                     wc = thewildcards[renum[p[1]]];
01433                     AN.argaddress = (WORD *)a;
01434                     if ( CheckWild(BHEAD p[1],ARLTOARL,wc,(WORD *)a) ) break;
01435                     AddWild(BHEAD p[1],ARLTOARL,wc);
01436                     j += wc-1; a += wc-1; wc = 1;
01437                 }
01438                 else if ( MatchArgument(BHEAD t,p) == 0 ) break;
01439                 NEXTARG(p);
01440             }
01441             if ( j >= tcount ) { /* Match! */
01442
01443                 /* Continue with other functions. Make sure of the funnies */
01444
01445                 AN.RepFunList[AN.RepFunNum++] = offset;
01446                 AN.RepFunList[AN.RepFunNum++] = 0;
01447                 newpat = pattern + pattern[1];
01448                 if ( newpat >= AN.patstop ) {
01449                     if ( AN.UseFindOnly == 0 ) {
01450                         if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
01451                             AT.WorkPointer = oldworkpointer;
01452                             AT.pWorkPointer = oww;
01453                             AN.UsedOtherFind = 1;
01454                             return(1);
01455                         }
01456                         j = 0;
01457                     }
01458                     else {
01459                         AT.WorkPointer = oldworkpointer;
01460                         AT.pWorkPointer = oww;
01461                         return(1);
01462                     }
01463                 }
01464             }
01465         }
01466     }

```

```

01463     }
01464     else j = ScanFunctions(BHEAD newpat,inter,par);
01465     if ( j ) {
01466         AT.WorkPointer = oldworkpointer;
01467         AT.pWorkPointer = oww;
01468         return(j); /* Full match. Return our success */
01469     }
01470     AN.RepFunNum -= 2;
01471 }
01472
01473 /* No (deeper) match. -> reset wildcards and continue */
01474
01475 if ( wc ) {
01476     j = nwstore;
01477     m = AN.WildValue;
01478     t = thewildcards + ntwa; r = AT.WildMask;
01479     if ( j > 0 ) {
01480         do {
01481             *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01482         } while ( --j > 0 );
01483     }
01484     C->numrhs = *t++;
01485     C->Pointer = C->Buffer + oldcpointer;
01486 }
01487 a = AT.pWorkSpace+oww;
01488 t = *a;
01489 for ( j = 1; j < tcount; j++ ) { *a = a[1]; a++; }
01490 *a = t;
01491 }
01492 }
01493 (thewildcards[iraise-1])++;
01494 }
01495 /*
01496 #] Case 3:
01497 */
01498 NoSuccess:
01499     if ( oldwilval > 0 ) {
01500 nomatch:;
01501         j = nwstore;
01502         m = AN.WildValue;
01503         t = lowlevel; r = AT.WildMask;
01504         if ( j > 0 ) {
01505             do {
01506                 *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01507             } while ( --j > 0 );
01508         }
01509         C->numrhs = *t++;
01510         C->Pointer = C->Buffer + oldcpointer;
01511     }
01512     AT.WorkPointer = oldworkpointer;
01513     AT.pWorkPointer = oww;
01514     return(0);
01515 }
01516
01517 /*
01518 #] FunMatchCy :
01519 #[ FunMatchSy :
01520
01521     Matching of (anti)symmetric functions.
01522     Like MatchE, but now for general functions.
01523 */
01524
01525 int FunMatchSy(PHEAD WORD *pattern, WORD *fun, WORD *inter, WORD par)
01526 {
01527     GETBIDENTITY
01528     WORD *t, *tstop, *p, *pstop, *m, *r, *oldworkpointer = AT.WorkPointer;
01529     WORD **a, *thewildcards, oldwilval = 0;
01530     WORD newvalue, *lowlevel = 0, num, assign;
01531     WORD *cycles;
01532     LONG oww = AT.pWorkPointer, lhpar, lhfunnies;
01533     int argcount = 0, funnycount = 0, tcount = 0, signs = 0, signfun = 0, signo;
01534     int type = 0, pnun, i, j, k, nwstore, iraise, cou2;
01535     CBUF *C = cbuf+AT.ebufnum;
01536     int ntwa = 3*AN.NumTotWildArgs+1;
01537     LONG oldcpointer = C->Pointer - C->Buffer;
01538     WORD offset = fun-AN.terstart, *newpat;
01539
01540     if ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;
01541     pnun = pattern[0];
01542     nwstore = (AN.WildValue[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
01543     if ( pnun > FUNCTION + WILDOFFSET ) {
01544         pnun -= WILDOFFSET;
01545         if ( CheckWild(BHEAD pnun,FUNTOFUN,fun[0],&newvalue) ) return(0);
01546         oldwilval = 1;
01547         t = lowlevel = oldworkpointer;
01548         m = AN.WildValue;
01549         i = nwstore;

```



```

01550     r = AT.WildMask;
01551     if ( i > 0 ) {
01552         do {
01553             *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
01554         } while ( --i > 0 );
01555     }
01556     *t++ = C->numrhs;
01557     if ( t >= AT.WorkTop ) {
01558         MLOCK(ErrorMessageLock);
01559         MesWork();
01560         MUNLOCK(ErrorMessageLock);
01561         return(-1);
01562     }
01563     AT.WorkPointer = t;
01564     AddWild(BHEAD pnun,FUNTOFUN,newvalue);
01565 }
01566 if ( (functions[pnum-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;
01567
01568 /* Try for a straight match. After all, both have been normalized */
01569
01570 if ( fun[1] == pattern[1] ) {
01571     i = fun[1]-FUNHEAD; p = pattern+FUNHEAD; t = fun + FUNHEAD;
01572     while ( --i >= 0 ) { if ( *p++ != *t++ ) break; }
01573     if ( i < 0 ) goto quicky;
01574 }
01575
01576 /* First we have to make an inventory. Are there -ARGWILD pointers? */
01577
01578 p = pattern + FUNHEAD;
01579 pstop = pattern + pattern[1];
01580 while ( p < pstop ) {
01581     if ( *p == -ARGWILD ) { p += 2; funnycount++; }
01582     else { NEXTARG(p); argcount++; }
01583 }
01584 t = fun + FUNHEAD;
01585 tstop = fun + fun[1];
01586 while ( t < tstop ) { NEXTARG(t); tcount++; }
01587
01588 if ( argcount > tcount ) return(0);
01589 if ( argcount < tcount && funnycount == 0 ) return(0);
01590 if ( argcount == 0 && tcount == 0 && funnycount == 0 ) {
01591 quicky:
01592     if ( AN.SignCheck && signs != AN.ExpectedSign ) goto NoSuccess;
01593     AN.RepFunList[AN.RepFunNum++] = offset;
01594     AN.RepFunList[AN.RepFunNum++] = signs;
01595     newpat = pattern + pattern[1];
01596     if ( newpat >= AN.patstop ) {
01597         if ( AN.UseFindOnly == 0 ) {
01598             if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
01599                 AT.WorkPointer = oldworkpointer;
01600                 AN.UsedOtherFind = 1;
01601                 return(1);
01602             }
01603             j = 0;
01604         }
01605         else {
01606             AT.WorkPointer = oldworkpointer;
01607             return(1);
01608         }
01609     }
01610     else j = ScanFunctions(BHEAD newpat,inter,par);
01611     if ( j ) {
01612         AT.WorkPointer = oldworkpointer;
01613         return(j);
01614     }
01615     goto NoSuccess;
01616 }
01617
01618 /* Store the wildcard assignments */
01619
01620 WantAddPointers(tcount+argcount+funnycount);
01621 AT.pWorkPointer += tcount+argcount+funnycount;
01622 thewildcards = t = AT.WorkPointer;
01623 t += ntw;
01624 if ( oldwilval ) lowlevel = oldworkpointer;
01625 else lowlevel = t;
01626 m = AN.WildValue;
01627 i = nwstore; assig = 0;
01628 if ( i > 0 ) {
01629     r = AT.WildMask;
01630     do {
01631         assig += *r;
01632         *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
01633     } while ( --i > 0 );
01634     *t++ = C->numrhs;
01635 }
01636 if ( t >= AT.WorkTop ) {

```

```

01637         MLOCK(ErrorMessageLock);
01638         MesWork();
01639         MUNLOCK(ErrorMessageLock);
01640         return(-1);
01641     }
01642     AT.WorkPointer = t;
01643
01644     /* Store pointers to the arguments */
01645
01646     t = fun + FUNHEAD; a = AT.pWorkSpace+oww;
01647     while ( t < tstop ) { *a++ = t; NEXTARG(t) }
01648     lhpars = a-AT.pWorkSpace;
01649     t = pattern + FUNHEAD;
01650     while ( t < pstop ) {
01651         if ( *t != -ARGWILD ) *a++ = t;
01652         NEXTARG(t)
01653     }
01654     lhfunnies = a-AT.pWorkSpace;
01655     t = pattern + FUNHEAD; cou2 = 0;
01656     while ( t < pstop ) {
01657         cou2++;
01658         if ( *t == -ARGWILD ) {
01659             *a++ = t;
01660         }
01661     }
01662     /* signfun: last ?a: tcount-argcount: number of arguments in ?a (assume one ?a)
01663        argcount+funnycount-cou2: arguments after ?a.
01664        Together tells whether moving ?a to end of list is even or odd
01665    */
01666     signfun = ((argcount+funnycount-cou2)*(tcount-argcount)) & 1;
01667     NEXTARG(t)
01668 }
01669 signs += signfun;
01670 if ( funnycount > 0 ) {
01671     if ( ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == SYMMETRIC )
01672         || ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC )
01673         || ( (functions[pnum-FUNCTION].symmetric & ~REVERSEORDER) == SYMMETRIC )
01674         || ( (functions[pnum-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC ) ) {
01675         AT.WorkPointer = oldworkpointer;
01676         AT.pWorkPointer = oww;
01677         MLOCK(ErrorMessageLock);
01678         MesPrint("Sorry: no argument field wildcards yet in (anti)symmetric functions");
01679         MUNLOCK(ErrorMessageLock);
01680         Terminate(-1);
01681     }
01682 }
01683 /*
01684 Sort the regular arguments by
01685 1: no wildcards, fast.
01686 2: wildcards that have been assigned.
01687 3: general arguments.
01688 4: wildcards without an assignment.
01689 */
01690 iraise = argcount;
01691 for ( i = 0; i < iraise; i++ ) {
01692     t = AT.pWorkSpace[i+lhpars];
01693     if ( *t > 0 ) { /* Category 3: general argument */
01694         continue;
01695     }
01696     else if ( *t <= -FUNCTION ) {
01697         if ( *t > -FUNCTION - WILDOFFSET ) goto cat1;
01698         type = FUNTOFUN; num = -*t - WILDOFFSET;
01699     }
01700     else if ( *t == -SYMBOL ) {
01701         if ( t[1] < 2*MAXPOWER ) goto cat1;
01702         type = SYMTOSYM; num = t[1] - 2*MAXPOWER;
01703     }
01704     else if ( *t == -INDEX ) {
01705         if ( t[1] < AM.OffsetIndex + WILDOFFSET ) goto cat1;
01706         type = INDTOIND; num = t[1] - WILDOFFSET;
01707     }
01708     else if ( *t == -VECTOR || *t == -MINVECTOR ) {
01709         if ( t[1] < AM.OffsetVector + WILDOFFSET ) goto cat1;
01710         type = VECTOVEC; num = t[1] - WILDOFFSET;
01711     }
01712     else goto cat1; /* Things like -SNUMBER etc. */
01713 }
01714 /*
01715 Now we have a wildcard and have to see whether it was assigned
01716 */
01717 m = AN.WildValue;
01718 j = nwstore;
01719 r = AT.WildMask;
01720 while ( --j >= 0 ) {
01721     if ( m[2] == num && *r ) {
01722         if ( type == *m ) break;
01723         if ( type == SYMTOSYM ) {
01724             if ( *m == SYMTONUM || *m == SYMTOSUB ) break;

```

```

01724         }
01725         else if ( type == INDTOIND ) {
01726             if ( *m == INDTOSUB ) break;
01727         }
01728         else if ( type == VECTOVEC ) {
01729             if ( *m == VECTOMIN || *m == VECTOSUB ) break;
01730         }
01731     }
01732     m += 4; r++;
01733 }
01734 if ( j < 0 ) { /* Category 4: Wildcard that was not assigned */
01735     a = AT.pWorkSpace+lhpar;
01736     iraise--;
01737     if ( iraise != i ) signs++;
01738     m = a[iraise];
01739     a[iraise] = a[i];
01740     a[i] = m; i--;
01741 }
01742 else { /* Category 2: Wildcard that was assigned */
01743     for ( j = 0; j < tcount; j++ ) {
01744         if ( MatchArgument(BHEAD AT.pWorkSpace[oww+j],t) ) {
01745             k = nwstore;
01746             r = AT.WildMask;
01747             num = 0;
01748             while ( --k >= 0 ) num += *r++;
01749             if ( num == assign ) { /* no wildcards were changed */
01750                 goto oneless;
01751             }
01752             break;
01753         }
01754     }
01755     if ( j >= tcount ) goto NoSuccess;
01756     j = nwstore;
01757     m = AN.WildValue;
01758     t = thewildcards + ntw; r = AT.WildMask;
01759     if ( j > 0 ) {
01760         do { /* undo assignment */
01761             *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01762         } while ( --j > 0 );
01763     }
01764     C->numrhs = *t++;
01765 }
01766 continue;
01767 cat1:
01768 for ( j = 0; j < tcount; j++ ) {
01769     m = AT.pWorkSpace[j+oww];
01770     if ( *t != *m ) continue;
01771     if ( *t < 0 ) {
01772         if ( *t <= -FUNCTION ) break;
01773         if ( t[1] == m[1] ) break;
01774     }
01775     else {
01776         k = *t; r = t;
01777         while ( --k >= 0 && *m++ == *r++ ) {}
01778         if ( k < 0 ) break;
01779     }
01780 }
01781 if ( j >= tcount ) goto NoSuccess; /* Even the fixed ones don't match */
01782 oneless:
01783 signs += j - i;
01784 /*
01785     The next statements replace the one that is commented out
01786 */
01787 tcount--;
01788 while ( j < tcount ) {
01789     AT.pWorkSpace[oww+j] = AT.pWorkSpace[oww+j+1]; j++;
01790 }
01791 /*
01792     AT.pWorkSpace[oww+j] = AT.pWorkSpace[oww+(-tcount)];
01793 */
01794 argcount--; j = i;
01795 while ( j < argcount ) {
01796     AT.pWorkSpace[lhpar+j] = AT.pWorkSpace[lhpar+j+1]; j++;
01797 }
01798 iraise--; i--;
01799 }
01800 /*
01801     Now we see whether there are any ARGWILD objects that have been
01802     assigned already. In that case the work simplifies considerably.
01803     Currently (12-nov-2001) only in (R)CYCLIC functions; hence we do not
01804     test the sign!
01805 */
01806 for ( i = 0; i < funnycount; i++ ) {
01807     k = AT.pWorkSpace[lhfunnies+i][1];
01808     m = AN.WildValue;
01809     j = nwstore;
01810     r = AT.WildMask;

```

```

01811     while ( --j >= 0 ) {
01812         if ( *m == ARGTOARG && m[2] == k ) break;
01813         m += 4; r++;
01814     }
01815     if ( *r == 0 ) continue; /* not assigned yet */
01816     m = cbuf[AT.ebufnum].rhs[m[3]];
01817     if ( *m > 0 ) { /* Tensor arguments */
01818         j = *m;
01819         if ( j > tcount - argcount ) goto NoSuccess;
01820         while ( --j >= 0 ) {
01821             m++;
01822             if ( *m < 0 ) type = -VECTOR;
01823             else if ( *m < AM.OffsetIndex ) type = -SNUMBER;
01824             else type = -INDEX;
01825             a = AT.pWorkSpace+oww;
01826             for ( k = 0; k < tcount; k++ ) {
01827                 if ( a[k][0] != type || a[k][1] != *m ) continue;
01828                 a[k] = a[--tcount];
01829                 goto nextjarg;
01830             }
01831             goto NoSuccess;
01832 nextjarg:;
01833         }
01834     }
01835     else {
01836         m++;
01837         while ( *m ) {
01838             for ( k = 0; k < tcount; k++ ) {
01839                 t = AT.pWorkSpace[oww+k];
01840                 if ( *t != *m ) continue;
01841                 r = m;
01842                 if ( *r < 0 ) {
01843                     if ( *r < -FUNCTION ) goto nextargw;
01844                     else if ( r[1] == t[1] ) goto nextargw;
01845                 }
01846                 else {
01847                     j = *r;
01848                     while ( --j >= 0 && *r++ == *t++ ) {}
01849                     if ( j < 0 ) goto nextargw;
01850                 }
01851             }
01852             goto NoSuccess;
01853 nextargw:;
01854             AT.pWorkSpace[oww+k] = AT.pWorkSpace[oww+(--tcount)];
01855             NEXTARG(m)
01856         }
01857         AT.pWorkSpace[lhfunnies+i] = AT.pWorkSpace[lhfunnies+(--funnycount)];
01858     }
01859     if ( tcount == 0 ) {
01860         if ( argcount > 0 ) goto NoSuccess;
01861         for ( i = 0; i < funnycount; i++ ) {
01862             AddWild(BHEAD AT.pWorkSpace[lhfunnies+i][1], ARGTOARG, 0);
01863         }
01864         goto quicky;
01865     }
01866 }
01867 /*
01868 We have now in lhpars first iraise elements with a dubious nature.
01869 Then argcount-iraise wildcards that have not been assigned.
01870 In lhfunnies we have funnycount ARGTOARG objects. ( (R)CyCLIC only )
01871
01872 First work our way through the 'dubious' objects
01873 We check whether assig changes.
01874 */
01875 for ( i = 0; i < iraise; i++ ) {
01876     for ( j = 0; j < tcount; j++ ) {
01877         if ( MatchArgument(BHEAD AT.pWorkSpace[oww+j], AT.pWorkSpace[lhpars+i]) ) {
01878             k = nwstore;
01879             r = AT.WildMask;
01880             num = 0;
01881             while ( --k >= 0 ) num += *r++;
01882             if ( num == assig ) { /* no wildcards were changed */
01883                 signs += j-i;
01884                 AT.pWorkSpace[oww+j] = AT.pWorkSpace[oww+(--tcount)];
01885                 if ( tcount > j ) signs += tcount-j-1;
01886                 argcount--;
01887                 a = AT.pWorkSpace + lhpars;
01888                 for ( j = i; j < argcount; j++ ) a[j] = a[j+1];
01889                 iraise--;
01890                 goto nextiraise;
01891             }
01892             else { /* We cannot use this yet */
01893                 j = nwstore;
01894                 m = AN.WildValue;
01895                 t = thewildcards + ntwa; r = AT.WildMask;
01896                 if ( j > 0 ) {
01897                     do { /* undo assignment */

```

```

01898             *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
01899         } while ( --j > 0 );
01900     }
01901     C->numrhs = *t++;
01902     C->Pointer = C->Buffer + oldcpointer;
01903     goto nexttiraise;
01904 }
01905 }
01906 }
01907 goto NoSuccess;
01908 nexttiraise:;
01909 }
01910 /*
01911     Now all leftover patterns have unassigned wildcards in them.
01912     From now on we are in potential factorial territory.
01913
01914     Strategy:
01915         1: cycle through the regular objects.
01916         2: save wildcard settings
01917         3: divide the ARGWILDS
01918         4: make permutations of leftover arguments
01919         5: try them all
01920 */
01921 cycles = AT.WorkPointer;
01922 for ( i = 0; i < tcount; i++ ) cycles[i] = tcount-i;
01923 AT.WorkPointer += tcount;
01924 signo = 0;
01925 /*MesPrint("<1> signs = %d",signs);*/
01926 for ( ;; ) {
01927     WORD oRepFunNum = AN.RepFunNum;
01928     for ( j = 0; j < argcount; j++ ) {
01929         if ( MatchArgument (BHEAD AT.pWorkSpace[oww+j],AT.pWorkSpace[lhpars+j]) == 0 ) {
01930             break;
01931         }
01932     }
01933     if ( j >= argcount ) {
01934 /*
01935         Thus far we have a match. Now the funnies
01936 */
01937         if ( funnycount ) {
01938             AT.WorkPointer = oldworkpointer;
01939             AT.pWorkPointer = oww;
01940             MLOCK(ErrorMessageLock);
01941             MesPrint("Sorry: no argument field wildcards yet in (anti)symmetric functions");
01942             MUNLOCK(ErrorMessageLock);
01943 /*
01944             Bugfix 31-oct-2001, reported by Kasper Peeters
01945             We returned here with value -1 but that is not caught.
01946             Extra note (12-nov-2001): the sign becomes a bit problematic
01947             if we have funnies. No more than one allowed in antisymmetric
01948             functions, or we have serious problems.
01949 */
01950             Terminate(-1);
01951         }
01952
01953         AN.RepFunList[AN.RepFunNum++] = offset;
01954         if ( ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC )
01955             || ( (functions[pnum-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC ) ) {
01956             AN.RepFunList[AN.RepFunNum++] = ( signs + signo ) & 1;
01957         }
01958         else {
01959             AN.RepFunList[AN.RepFunNum++] = 0;
01960         }
01961         newpat = pattern + pattern[1];
01962         if ( newpat >= AN.patstop ) {
01963             WORD countsgn, sgn = 0;
01964             for ( countsgn = oRepFunNum+1; countsgn < AN.RepFunNum; countsgn += 2 ) {
01965                 if ( AN.RepFunList[countsgn] ) sgn ^= 1;
01966             }
01967             if ( AN.SignCheck == 0 || sgn == AN.ExpectedSign ) {
01968                 AT.WorkPointer = oldworkpointer;
01969                 AT.pWorkPointer = oww;
01970                 return(1);
01971             }
01972             if ( AN.UseFindOnly == 0 ) {
01973                 if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
01974                     AT.WorkPointer = oldworkpointer;
01975                     AT.pWorkPointer = oww;
01976                     AN.UsedOtherFind = 1;
01977                     return(1);
01978                 }
01979             }
01980             j = 0;
01981         }
01982         else j = ScanFunctions(BHEAD newpat,inter,par);
01983         if ( j ) {
01984             WORD countsgn, sgn = 0;

```

```

01985         for ( countsgn = oRepFunNum+1; countsgn < AN.RepFunNum; countsgn += 2 ) {
01986             if ( AN.RepFunList[countsgn] ) sgn ^= 1;
01987         }
01988         if ( AN.SignCheck == 0 || sgn == AN.ExpectedSign ) {
01989             AT.WorkPointer = oldworkpointer;
01990             AT.pWorkPointer = ow;
01991             return(j);
01992         }
01993     }
01994     AN.RepFunNum = oRepFunNum;
01995     i = argcount - 1;
01996 }
01997 else i = j;
01998 j = nwstore;
01999 m = AN.WildValue;
02000 t = thewildcards + ntw; r = AT.WildMask;
02001 if ( j > 0 ) {
02002     do {
02003         *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
02004     } while ( --j > 0 );
02005 }
02006 C->numrhs = *t++;
02007 C->Pointer = C->Buffer + oldcpointer;
02008 /*
02009     On to the next cycle
02010 */
02011 a = AT.pWorkSpace + ow;
02012 for ( j = i+1, t = a[i]; j < tcount; j++ ) a[j-1] = a[j];
02013 a[tcount-1] = t; cycles[i]--;
02014 signo += tcount - i - 1;
02015 while ( cycles[i] <= 0 ) {
02016     cycles[i] = tcount - i;
02017     i--;
02018     if ( i < 0 ) goto NoSuccess;
02019 }
02020 /*
02021     MLOCK(ErrorMessageLock);
02022     MesPrint("Cycle i = %d",i);
02023     MUNLOCK(ErrorMessageLock);
02024 */
02025 for ( j = i+1, t = a[i]; j < tcount; j++ ) a[j-1] = a[j];
02026 a[tcount-1] = t; cycles[i]--;
02027 signo += tcount - i - 1;
02028 }
02029 NoSuccess:
02030 if ( oldwilval > 0 ) {
02031     j = nwstore;
02032     m = AN.WildValue;
02033     t = lowlevel; r = AT.WildMask;
02034     if ( j > 0 ) {
02035         do {
02036             *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
02037         } while ( --j > 0 );
02038     }
02039     C->numrhs = *t++;
02040     C->Pointer = C->Buffer + oldcpointer;
02041 }
02042 AT.WorkPointer = oldworkpointer;
02043 AT.pWorkPointer = ow;
02044 return(0);
02045 }
02046 /*
02047 #] FunMatchSy :
02048 #[ MatchArgument :
02049 */
02050 int MatchArgument(PHEAD WORD *arg, WORD *pat)
02051 {
02052     GETBIDENTITY
02053     WORD *m = pat, *t = arg, i, j, newvalue;
02054     WORD *argmstop = pat, *argtstop = arg;
02055     WORD *cto, *cfrom, *csav, ci;
02056     WORD oRepFunNum, *oRepFunList;
02057     WORD *oterstart, *oterstop, *opatstop;
02058     WORD wildargs, wildeat;
02059     WORD *mtrmstop, *ttrmstop, *msubstop, msizcoef;
02060     WORD *wildargtaken;
02061     int wc = 1;
02062     NEXTARG(argmstop);
02063     NEXTARG(argtstop);
02064     /*
02065     #[ Both fast :
02066     */
02067     if ( *m < 0 && *t < 0 ) {
02068         if ( *t <= -FUNCTION ) {

```

```

02072         if ( *t == *m ) {}
02073     else if ( *m <= -FUNCTION-WILDOFFSET
02074 && functions[-*t-FUNCTION].spec
02075 == functions[-*m-FUNCTION-WILDOFFSET].spec ) {
02076         i = -*m - WILDOFFSET; wc = 2;
02077         if ( CheckWild(BHEAD i,FUNTOFUN,-*t,&newvalue) ) {
02078             return(0);
02079         }
02080         AddWild(BHEAD i,FUNTOFUN,newvalue);
02081     }
02082     else if ( *m == -SYMBOL && m[1] >= 2*MAXPOWER ) {
02083         i = m[1] - 2*MAXPOWER;
02084         AN.argaddress = AT.FunArg;
02085         AT.FunArg[ARGHEAD+1] = -*t;
02086         if ( CheckWild(BHEAD i,SYMTOSUB,1,AN.argaddress) ) return(0);
02087         AddWild(BHEAD i,SYMTOSUB,0);
02088     }
02089     else return(0);
02090 }
02091 else if ( *t == *m ) {
02092     if ( t[1] == m[1] ) {}
02093     else if ( *t == -SYMBOL ) {
02094         j = SYMTOSYM;
02095 SymAll:         if ( ( i = m[1] - 2*MAXPOWER ) < 0 ) return(0);
02096         wc = 2;
02097         if ( CheckWild(BHEAD i,j,t[1],&newvalue) ) return(0);
02098         AddWild(BHEAD i,j,newvalue);
02099     }
02100     else if ( *t == -INDEX ) {
02101 IndAll:         i = m[1] - WILDOFFSET;
02102         if ( i < AM.OffsetIndex || i >= WILDOFFSET+AM.OffsetIndex )
02103             return(0);
02104         /* We kill the summed over indices here */
02105         wc = 2;
02106         if ( CheckWild(BHEAD i,INDTOIND,t[1],&newvalue) ) return(0);
02107         AddWild(BHEAD i,INDTOIND,newvalue);
02108     }
02109     else if ( *t == -VECTOR || *t == -MINVECTOR ) {
02110         i = m[1] - WILDOFFSET;
02111         if ( i < AM.OffsetVector ) return(0);
02112         wc = 2;
02113         if ( CheckWild(BHEAD i,VECTOVEC,t[1],&newvalue) ) return(0);
02114         AddWild(BHEAD i,VECTOVEC,newvalue);
02115     }
02116     else return(0);
02117 }
02118 else if ( *m == -INDEX && m[1] >= AM.OffsetIndex+WILDOFFSET
02119 && m[1] < AM.OffsetIndex+(WILDOFFSET<1) ) {
02120     if ( *t == -VECTOR ) goto IndAll;
02121     if ( *t == -SNUMBER && t[1] >= 0 && t[1] < AM.OffsetIndex ) goto IndAll;
02122     if ( *t == -MINVECTOR ) {
02123         i = m[1] - WILDOFFSET;
02124         AN.argaddress = AT.MinVecArg;
02125         AT.MinVecArg[ARGHEAD+3] = t[1];
02126         wc = 2;
02127         if ( CheckWild(BHEAD i,INDTOSUB,1,AN.argaddress) ) return(0);
02128         AddWild(BHEAD i,INDTOSUB,(WORD)0);
02129     }
02130     else return(0);
02131 }
02132 else if ( *m == -SYMBOL && m[1] >= 2*MAXPOWER && *t == -SNUMBER ) {
02133     j = SYMTONUM;
02134     goto SymAll;
02135 }
02136 else if ( *m == -VECTOR && *t == -MINVECTOR &&
02137 ( i = m[1] - WILDOFFSET ) >= AM.OffsetVector ) {
02138     wc = 2;
02139 /*
02140     AN.argaddress = AT.MinVecArg;
02141     AT.MinVecArg[ARGHEAD+3] = t[1];
02142     if ( CheckWild(BHEAD i,VECTOSUB,1,AN.argaddress) ) return(0);
02143     AddWild(BHEAD i,VECTOSUB,(WORD)0);
02144 */
02145     if ( CheckWild(BHEAD i,VECTOMIN,t[1],&newvalue) ) return(0);
02146     AddWild(BHEAD i,VECTOMIN,newvalue);
02147 }
02148 }
02149 else if ( *m == -MINVECTOR && *t == -VECTOR &&
02150 ( i = m[1] - WILDOFFSET ) >= AM.OffsetVector ) {
02151     wc = 2;
02152 /*
02153     AN.argaddress = AT.MinVecArg;
02154     AT.MinVecArg[ARGHEAD+3] = t[1];
02155     if ( CheckWild(BHEAD i,VECTOSUB,1,AN.argaddress) ) return(0);
02156     AddWild(BHEAD i,VECTOSUB,(WORD)0);
02157 */
02158     if ( CheckWild(BHEAD i,VECTOMIN,t[1],&newvalue) ) return(0);

```

```

02159         AddWild(BHEAD i, VECTOMIN, newvalue);
02160     }
02161     else return(0);
02162 }
02163 /*
02164 #] Both fast :
02165 #[ Fast arg :
02166 */
02167 else if ( *m > 0 && *t <= -FUNCTION ) {
02168     if ( ( m[ARGHEAD]+ARGHEAD == *m ) && m[*m-1] == 3
02169         && m[*m-2] == 1 && m[*m-3] == 1 && m[ARGHEAD+1] >= FUNCTION
02170         && m[ARGHEAD+2] == *m-ARGHEAD-4 ) { /* Check for f(?a) etc */
02171         WORD *mmmst, *mmm, mmmi;
02172         if ( m[ARGHEAD+1] >= FUNCTION+WILDOFFSET ) {
02173             mmmi = *m - WILDOFFSET;
02174             wc = 2;
02175             if ( CheckWild(BHEAD mmmi, FUNTOFUN, -*t, &newvalue) ) return(0);
02176             AddWild(BHEAD mmmi, FUNTOFUN, newvalue);
02177         }
02178         else if ( m[ARGHEAD+1] != -*t ) return(0);
02179     /*
02180         Only arguments allowed are ?a etc.
02181     */
02182     mmmst = m+*m-3;
02183     mmm = m + ARGHEAD + FUNHEAD + 1;
02184     while ( mmm < mmmst ) {
02185         if ( *mmm != -ARGWILD ) return(0);
02186         mmmi = 0;
02187         AN.argaddress = t; wc = 2;
02188         if ( CheckWild(BHEAD mmm[1], ARGTOARG, mmmi, t) ) return(0);
02189         AddWild(BHEAD mmm[1], ARGTOARG, mmmi);
02190         mmm += 2;
02191     }
02192 }
02193     else return(0);
02194 }
02195 /*
02196 #] Fast arg :
02197 #[ Fast pat :
02198 */
02199 else if ( *m < 0 && *t > 0 ) {
02200     if ( *m == -SYMBOL ) { /* SYMTOSUB */
02201         if ( m[1] < 2*MAXPOWER ) return(0);
02202         i = m[1] - 2*MAXPOWER;
02203         AN.argaddress = t; wc = 2;
02204         if ( CheckWild(BHEAD i, SYMTOSUB, 1, AN.argaddress) ) return(0);
02205         AddWild(BHEAD i, SYMTOSUB, 0);
02206     }
02207     else if ( *m == -VECTOR ) {
02208         if ( ( i = m[1] - WILDOFFSET ) < AM.OffsetVector ) return(0);
02209         AN.argaddress = t; wc = 2;
02210         if ( CheckWild(BHEAD i, VECTOSUB, 1, t) ) return(0);
02211         AddWild(BHEAD i, VECTOSUB, (WORD)0);
02212     }
02213     else if ( *m == -INDEX ) {
02214         if ( ( i = m[1] - WILDOFFSET ) < AM.OffsetIndex ) return(0);
02215         if ( i >= AM.OffsetIndex + WILDOFFSET ) return(0);
02216         AN.argaddress = t; wc = 2;
02217         if ( CheckWild(BHEAD i, INDOSUB, 1, AN.argaddress) ) return(0);
02218         AddWild(BHEAD i, INDOSUB, (WORD)0);
02219     }
02220     else return(0);
02221 }
02222 /*
02223 #] Fast pat :
02224 #[ Both general :
02225 */
02226 else if ( *m > 0 && *t > 0 ) {
02227     i = *m;
02228     do { if ( *m++ != *t++ ) break; } while ( --i > 0 );
02229     if ( i > 0 ) {
02230     /*
02231         Not an exact match here.
02232         We have to hope that the pattern contains a composite wildcard.
02233     */
02234         m = pat; t = arg;
02235         m += ARGHEAD; t += ARGHEAD; /* Point at (first?) term */
02236         mtrmstop = m + *m;
02237         ttrmstop = t + *t;
02238         if ( mtrmstop < argmstop ) return(0); /* More than one term */
02239         msizcoef = mtrmstop[-1];
02240         if ( msizcoef < 0 ) msizcoef = -msizcoef;
02241         msubstop = mtrmstop - msizcoef;
02242         m++;
02243         if ( m >= msubstop ) return(0); /* Only coefficient */
02244     /*
02245         Here we have a composite term. It can match provided it

```



```

02246 matches the entire argument. This argument must be a
02247 single term also and the coefficients should match
02248 (more or less).
02249 The matching takes:
02250 1: Match the functions etc. Nothing can be left.
02251 2: Match dotproducts and symbols. ONLY must match
02252 and nothing may be left.
02253 For safety it is best to take the term out and put it
02254 in workspace.
02255 */
02256 if ( argtstop > ttrmstop ) return(0);
02257 m--;
02258
02259 oterstart = AN.terstart;
02260 oterstop = AN.terstop;
02261 opatstop = AN.patstop;
02262 oRepFunList = AN.RepFunList;
02263 oRepFunNum = AN.RepFunNum;
02264 AN.RepFunNum = 0;
02265 wildargtaken = AT.WorkPointer;
02266 AN.RepFunList = wildargtaken + AN.NumTotWildArgs;
02267 AT.WorkPointer = (WORD *) ((UBYTE *) (AN.RepFunList)) + AM.MaxTer/2;
02268 csav = cto = AT.WorkPointer;
02269 cfrom = t;
02270 ci = *t;
02271 while ( --ci >= 0 ) *cto++ = *cfrom++;
02272 AT.WorkPointer = cto;
02273 ci = msizcoef;
02274 cfrom = mtrmstop;
02275 while ( --ci >= 0 ) {
02276     if ( *--cfrom != *--cto ) {
02277         AT.WorkPointer = wildargtaken;
02278         AN.RepFunList = oRepFunList;
02279         AN.RepFunNum = oRepFunNum;
02280         AN.terstart = oterstart;
02281         AN.terstop = oterstop;
02282         AN.patstop = opatstop;
02283         return(0);
02284     }
02285 }
02286 *m -= msizcoef;
02287 wildargs = AN.WildArgs;
02288 wildeat = AN.WildEat;
02289 for ( i = 0; i < wildargs; i++ ) wildargtaken[i] = AT.WildArgTaken[i];
02290 AN.ForFindOnly = 0; AN.UseFindOnly = 1;
02291 AN.nogroundlevel++;
02292 if ( FindRest(BHEAD csav,m) && ( AN.UsedOtherFind || FindOnly(BHEAD csav,m) ) ) { }
02293 else {
02294     *m += msizcoef;
02295     AT.WorkPointer = wildargtaken;
02296     AN.RepFunList = oRepFunList;
02297     AN.RepFunNum = oRepFunNum;
02298     AN.terstart = oterstart;
02299     AN.terstop = oterstop;
02300     AN.patstop = opatstop;
02301     AN.WildArgs = wildargs;
02302     AN.WildEat = wildeat;
02303     AN.nogroundlevel--;
02304     for ( i = 0; i < wildargs; i++ ) AT.WildArgTaken[i] = wildargtaken[i];
02305     return(0);
02306 }
02307 AN.nogroundlevel--;
02308 AN.WildArgs = wildargs;
02309 AN.WildEat = wildeat;
02310 for ( i = 0; i < wildargs; i++ ) AT.WildArgTaken[i] = wildargtaken[i];
02311 Substitute(BHEAD csav,m,1);
02312 cto = csav;
02313 cfrom = cto + *cto - msizcoef;
02314 cto++;
02315 *m += msizcoef;
02316 AT.WorkPointer = wildargtaken;
02317 AN.RepFunList = oRepFunList;
02318 AN.RepFunNum = oRepFunNum;
02319 AN.terstart = oterstart;
02320 AN.terstop = oterstop;
02321 AN.patstop = opatstop;
02322 if ( *cto != SUBEXPRESSION ) return(0);
02323 cto += cto[1];
02324 if ( cto < cfrom ) return(0);
02325 }
02326 }
02327 /*
02328 #] Both general :
02329 */
02330 else return(0);
02331 /*
02332 And now the success: (wc = 2 means that there was a wildcard involved)

```

```

02333 */
02334     return (wc);
02335 }
02336
02337 /*
02338     #] MatchArgument :
02339 */

```

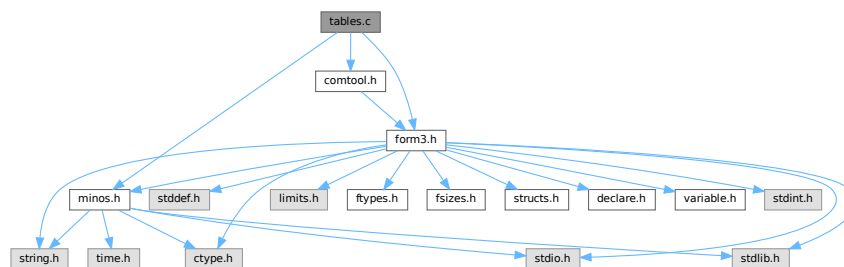
## 4.140 tables.c File Reference

```

#include "form3.h"
#include "minos.h"
#include "comtool.h"

```

Include dependency graph for tables.c:



## Functions

- void [ClearTableTree](#) (TABLES T)
- int [InsTableTree](#) (TABLES T, WORD \*tp)
- void [RedoTableTree](#) (TABLES T, int newsize)
- int [FindTableTree](#) (TABLES T, WORD \*tp, int inc)
- int [DoTableExpansion](#) (WORD \*term, WORD level)
- int [CoTableBase](#) (UBYTE \*s)
- int [FlipTable](#) (FUNCTIONS f, int type)
- int [SpareTable](#) (TABLES TT)
- DBASE \* [FindTB](#) (UBYTE \*name)
- int [CoTBcreate](#) (UBYTE \*s)
- int [CoTBopen](#) (UBYTE \*s)
- int [CoTBaddto](#) (UBYTE \*s)
- int [CoTBenter](#) (UBYTE \*s)
- int [CoTestUse](#) (UBYTE \*s)
- int [CheckTableDeclarations](#) (DBASE \*d)
- int [CoTBload](#) (UBYTE \*ss)
- int [TestUse](#) (WORD \*term, WORD level)
- int [CoTBaudit](#) (UBYTE \*s)
- int [CoTBon](#) (UBYTE \*s)
- int [CoTBoff](#) (UBYTE \*s)
- int [CoTBcleanup](#) (UBYTE \*s)
- int [CoTBreplace](#) (UBYTE \*s)
- int [CoTBuse](#) (UBYTE \*s)
- int [CoApply](#) (UBYTE \*s)
- int [CoTBhelp](#) (UBYTE \*s)

- void [ReWorkT](#) (WORD \*term, WORD \*funcs, WORD numfuncs)
- void [Apply](#) (WORD \*term, WORD level)
- int [ApplyExec](#) (WORD \*term, int maxtogo, WORD level)
- void [ApplyReset](#) (WORD level)
- void [TableReset](#) (void)
- int [ReleaseTB](#) (void)

## Variables

- char \* [helptb](#) []

## 4.140.1 Detailed Description

Contains all functions that deal with the table bases on the 'FORM level' The low level database routines are in [minos.c](#)

Definition in file [tables.c](#).

## 4.140.2 Function Documentation

### 4.140.2.1 ClearTableTree()

```
void ClearTableTree (  
    TABLES T )
```

Definition at line 72 of file [tables.c](#).

### 4.140.2.2 InsTableTree()

```
int InsTableTree (  
    TABLES T,  
    WORD * tp )
```

Definition at line 103 of file [tables.c](#).

### 4.140.2.3 RedoTableTree()

```
void RedoTableTree (  
    TABLES T,  
    int newsiz )
```

Definition at line 266 of file [tables.c](#).

### 4.140.2.4 FindTableTree()

```
int FindTableTree (  
    TABLES T,  
    WORD * tp,  
    int inc )
```

Definition at line 291 of file [tables.c](#).

#### 4.140.2.5 DoTableExpansion()

```
int DoTableExpansion (
    WORD * term,
    WORD level )
```

Definition at line 334 of file [tables.c](#).

#### 4.140.2.6 CoTableBase()

```
int CoTableBase (
    UBYTE * s )
```

Definition at line 627 of file [tables.c](#).

#### 4.140.2.7 FlipTable()

```
int FlipTable (
    FUNCTIONS f,
    int type )
```

Definition at line 671 of file [tables.c](#).

#### 4.140.2.8 SpareTable()

```
int SpareTable (
    TABLES TT )
```

Definition at line 694 of file [tables.c](#).

#### 4.140.2.9 FindTB()

```
DBASE * FindTB (
    UBYTE * name )
```

Definition at line 734 of file [tables.c](#).

#### 4.140.2.10 CoTBcreate()

```
int CoTBcreate (
    UBYTE * s )
```

Definition at line 756 of file [tables.c](#).

#### 4.140.2.11 CoTBopen()

```
int CoTBopen (
    UBYTE * s )
```

Definition at line 772 of file [tables.c](#).

#### 4.140.2.12 CoTBaddto()

```
int CoTBaddto (
    UBYTE * s )
```

Definition at line 801 of file [tables.c](#).

#### 4.140.2.13 CoTBenter()

```
int CoTBenter (
    UBYTE * s )
```

Definition at line 974 of file [tables.c](#).

#### 4.140.2.14 CoTestUse()

```
int CoTestUse (
    UBYTE * s )
```

Definition at line 1133 of file [tables.c](#).

#### 4.140.2.15 CheckTableDeclarations()

```
int CheckTableDeclarations (
    DBASE * d )
```

Definition at line 1178 of file [tables.c](#).

#### 4.140.2.16 CoTBload()

```
int CoTBload (
    UBYTE * ss )
```

Definition at line 1252 of file [tables.c](#).

#### 4.140.2.17 TestUse()

```
int TestUse (
    WORD * term,
    WORD level )
```

Definition at line 1414 of file [tables.c](#).

#### 4.140.2.18 CoTBaudit()

```
int CoTBaudit (
    UBYTE * s )
```

Definition at line 1474 of file [tables.c](#).

#### 4.140.2.19 CoTBon()

```
int CoTBon (
    UBYTE * s )
```

Definition at line [1514](#) of file [tables.c](#).

#### 4.140.2.20 CoTBoff()

```
int CoTBoff (
    UBYTE * s )
```

Definition at line [1545](#) of file [tables.c](#).

#### 4.140.2.21 CoTBcleanup()

```
int CoTBcleanup (
    UBYTE * s )
```

Definition at line [1576](#) of file [tables.c](#).

#### 4.140.2.22 CoTBreplace()

```
int CoTBreplace (
    UBYTE * s )
```

Definition at line [1588](#) of file [tables.c](#).

#### 4.140.2.23 CoTBuse()

```
int CoTBuse (
    UBYTE * s )
```

Definition at line [1603](#) of file [tables.c](#).

#### 4.140.2.24 CoApply()

```
int CoApply (
    UBYTE * s )
```

Definition at line [1797](#) of file [tables.c](#).

#### 4.140.2.25 CoTBhelp()

```
int CoTBhelp (
    UBYTE * s )
```

Definition at line [1884](#) of file [tables.c](#).

#### 4.140.2.26 ReWorkT()

```
void ReWorkT (
    WORD * term,
    WORD * funcs,
    WORD numfuncs )
```

Definition at line 1900 of file [tables.c](#).

#### 4.140.2.27 Apply()

```
void Apply (
    WORD * term,
    WORD level )
```

Definition at line 1981 of file [tables.c](#).

#### 4.140.2.28 ApplyExec()

```
int ApplyExec (
    WORD * term,
    int maxtogo,
    WORD level )
```

Definition at line 2039 of file [tables.c](#).

#### 4.140.2.29 ApplyReset()

```
void ApplyReset (
    WORD level )
```

Definition at line 2256 of file [tables.c](#).

#### 4.140.2.30 TableReset()

```
void TableReset (
    void )
```

Definition at line 2291 of file [tables.c](#).

#### 4.140.2.31 ReleaseTB()

```
int ReleaseTB (
    void )
```

Definition at line 2317 of file [tables.c](#).

## 4.140.3 Variable Documentation

### 4.140.3.1 helptb

char\* helptb[]

Definition at line 1852 of file [tables.c](#).

## 4.141 tables.c

[Go to the documentation of this file.](#)

```

00001
00006 /* #[ License : */
00007 /*
00008 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 *   When using this file you are requested to refer to the publication
00010 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 *   This is considered a matter of courtesy as the development was paid
00012 *   for by FOM the Dutch physics granting agency and we would like to
00013 *   be able to track its scientific use to convince FOM of its value
00014 *   for the community.
00015 *
00016 *   This file is part of FORM.
00017 *
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00020 *   Foundation, either version 3 of the License, or (at your option) any later
00021 *   version.
00022 *
00023 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 *   details.
00027 *
00028 *   You should have received a copy of the GNU General Public License along
00029 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #[ License : */
00032 /*
00033 *   #[ Includes :
00034 *
00035 *   File contains the routines for the tree structure of sparse tables
00036 *   We insert elements by
00037 *   InsTableTree(T,tp) with T the TABLES element and tp the pointer
00038 *   to the indices.
00039 *   We look for elements with
00040 *   FindTableTree(T,tp,inc) with T the TABLES element, tp the pointer to the
00041 *   indices or the function arguments and inc tells which of these options.
00042 *   The tree is cleared with ClearTableTree(T) and we rebuild the tree
00043 *   after a .store in which we lost a part of the table with
00044 *   RedoTableTree(T,newsize)
00045 *
00046 *   In T->tablepointers we have the lists of indices for each element.
00047 *   Additionally for each element there is an extension. There are
00048 *   TABLEEXTENSION WORDs reserved for that. The old system had two words
00049 *   One for the element in the rhs of the compile buffer and one for
00050 *   an additional rhs in case the original would be overwritten by a new
00051 *   definition, but the old was fixed by .global and hence it should be possible
00052 *   to restore it.
00053 *   New use (new = 24-sep-2001)
00054 *   rhs1,numCompBuffer1,rhs2,numCompBuffer2,usage
00055 *   Hence TABLEEXTENSION will be 5. Note that for 64 bits the use of the
00056 *   compiler buffer is overdoing it a bit, but it would be too complicated
00057 *   to try to give it special code.
00058 */
00059
00060 #include "form3.h"
00061 #include "minos.h"
00062 #include "comtool.h"
00063
00064 /* static UBYTE *sparse = (UBYTE *) "sparse"; */
00065 static UBYTE *tablebase = (UBYTE *) "tablebase";
00066
00067 /*
00068 *   #[ Includes :
00069 *   #[ ClearTableTree :
00070 */

```



```

00071
00072 void ClearTableTree(TABLES T)
00073 {
00074     COMPTREE *root;
00075     if ( T->boomlijst == 0 ) {
00076         T->MaxTreeSize = 125;
00077         T->boomlijst = (COMPTREE *)Malloc1(T->MaxTreeSize*sizeof(COMPTREE),
00078             "ClearTableTree");
00079     }
00080     root = T->boomlijst;
00081     T->numtree = 0;
00082     T->rootnum = 0;
00083     root->left = -1;
00084     root->right = -1;
00085     root->parent = -1;
00086     root->blnce = 0;
00087     root->value = -1;
00088     root->usage = 0;
00089 }
00090
00091 /*
00092 #] ClearTableTree :
00093 #[ InsTableTree :
00094
00095 int InsTableTree(TABLES T,WORD *,arglist)
00096     Searches for the element specified by the list of arguments.
00097     If found, it returns -(the offset in T->tablepointers)
00098     If not found, it will allocate a new element, balance the tree if
00099     necessary and return the number of the element in the boomlijst
00100     This number is always > 0, because we start from 1.
00101 */
00102
00103 int InsTableTree(TABLES T, WORD *tp)
00104 {
00105     COMPTREE *boomlijst, *q, *p, *s;
00106     WORD *v1, *v2, *v3, xstop;
00107     int ip, iq, is;
00108     if ( T->numtree + 1 >= T->MaxTreeSize ) {
00109         if ( T->MaxTreeSize == 0 ) ClearTableTree(T);
00110         else {
00111             is = T->MaxTreeSize * 2;
00112             s = (COMPTREE *)Malloc1(is*sizeof(COMPTREE),"InsTableTree");
00113             for ( ip = 0; ip < T->MaxTreeSize; ip++ ) { s[ip] = T->boomlijst[ip]; }
00114             if ( T->boomlijst ) M_free(T->boomlijst,"InsTableTree");
00115             T->boomlijst = s;
00116             T->MaxTreeSize = is;
00117         }
00118     }
00119     boomlijst = T->boomlijst;
00120     q = boomlijst + T->rootnum;
00121     if ( q->right == -1 ) { /* First element */
00122         T->numtree++;
00123         s = boomlijst+T->numtree;
00124         q->right = T->numtree;
00125         s->parent = T->rootnum;
00126         s->left = s->right = -1;
00127         s->blnce = 0;
00128         s->value = tp - T->tablepointers;
00129         s->usage = 0;
00130         return(T->numtree);
00131     }
00132     ip = q->right;
00133     if ( T->numind >= 0 ) xstop = T->numind;
00134     else xstop = *tp + 1;
00135     while ( ip >= 0 ) {
00136         p = boomlijst + ip;
00137         v1 = T->tablepointers + p->value;
00138         v2 = tp; v3 = tp + xstop;
00139         while ( *v1 == *v2 && v2 < v3 ) { v1++; v2++; }
00140         if ( v2 >= v3 ) return(-p->value);
00141         if ( *v1 > *v2 ) {
00142             iq = p->right;
00143             if ( iq >= 0 ) { ip = iq; }
00144             else {
00145                 T->numtree++;
00146                 is = T->numtree;
00147                 p->right = is;
00148                 s = boomlijst + is;
00149                 s->parent = ip; s->left = s->right = -1;
00150                 s->blnce = 0; s->value = tp - T->tablepointers;
00151                 s->usage = 0;
00152                 p->blnce++;
00153                 if ( p->blnce == 0 ) return(T->numtree);
00154                 goto balance;
00155             }
00156         }
00157         else if ( *v1 < *v2 ) {

```

```

00158         iq = p->left;
00159         if ( iq >= 0 ) { ip = iq; }
00160         else {
00161             T->numtree++;
00162             is = T->numtree;
00163             s = boomlijst+is;
00164             p->left = is;
00165             s->parent = ip; s->left = s->right = -1;
00166             s->blnce = 0; s->value = tp - T->tablepointers;
00167             s->usage = 0;
00168             p->blnce--;
00169             if ( p->blnce == 0 ) return(T->numtree);
00170             goto balance;
00171         }
00172     }
00173 }
00174 MesPrint("Serious problems in InsTableTree!\n");
00175 Terminate(-1);
00176 return(0);
00177 balance:;
00178 for (;;) {
00179     p = boomlijst + ip;
00180     iq = p->parent;
00181     if ( iq == T->rootnum ) break;
00182     q = boomlijst + iq;
00183     if ( ip == q->left ) q->blnce--;
00184     else q->blnce++;
00185     if ( q->blnce == 0 ) break;
00186     if ( q->blnce == -2 ) {
00187         if ( p->blnce == -1 ) { /* single rotation */
00188             q->left = p->right;
00189             p->right = iq;
00190             p->parent = q->parent;
00191             q->parent = ip;
00192             if ( boomlijst[p->parent].left == iq ) boomlijst[p->parent].left = ip;
00193             else boomlijst[p->parent].right = ip;
00194             if ( q->left >= 0 ) boomlijst[q->left].parent = iq;
00195             q->blnce = p->blnce = 0;
00196         }
00197         else { /* double rotation */
00198             s = boomlijst + is;
00199             q->left = s->right;
00200             p->right = s->left;
00201             s->right = iq;
00202             s->left = ip;
00203             if ( p->right >= 0 ) boomlijst[p->right].parent = ip;
00204             if ( q->left >= 0 ) boomlijst[q->left].parent = iq;
00205             s->parent = q->parent;
00206             q->parent = is;
00207             p->parent = is;
00208             if ( boomlijst[s->parent].left == iq )
00209                 boomlijst[s->parent].left = is;
00210             else boomlijst[s->parent].right = is;
00211             if ( s->blnce > 0 ) { q->blnce = s->blnce = 0; p->blnce = -1; }
00212             else if ( s->blnce < 0 ) { p->blnce = s->blnce = 0; q->blnce = 1; }
00213             else { p->blnce = s->blnce = q->blnce = 0; }
00214         }
00215         break;
00216     }
00217     else if ( q->blnce == 2 ) {
00218         if ( p->blnce == 1 ) { /* single rotation */
00219             q->right = p->left;
00220             p->left = iq;
00221             p->parent = q->parent;
00222             q->parent = ip;
00223             if ( boomlijst[p->parent].left == iq ) boomlijst[p->parent].left = ip;
00224             else boomlijst[p->parent].right = ip;
00225             if ( q->right >= 0 ) boomlijst[q->right].parent = iq;
00226             q->blnce = p->blnce = 0;
00227         }
00228         else { /* double rotation */
00229             s = boomlijst + is;
00230             q->right = s->left;
00231             p->left = s->right;
00232             s->left = iq;
00233             s->right = ip;
00234             if ( p->left >= 0 ) boomlijst[p->left].parent = ip;
00235             if ( q->right >= 0 ) boomlijst[q->right].parent = iq;
00236             s->parent = q->parent;
00237             q->parent = is;
00238             p->parent = is;
00239             if ( boomlijst[s->parent].left == iq ) boomlijst[s->parent].left = is;
00240             else boomlijst[s->parent].right = is;
00241             if ( s->blnce < 0 ) { q->blnce = s->blnce = 0; p->blnce = 1; }
00242             else if ( s->blnce > 0 ) { p->blnce = s->blnce = 0; q->blnce = -1; }
00243             else { p->blnce = s->blnce = q->blnce = 0; }
00244         }
00245     }
}

```

```

00245         break;
00246     }
00247     is = ip; ip = iq;
00248 }
00249 return(T->numtree);
00250 }
00251
00252 /*
00253 #] InsTableTree :
00254 #[ RedoTableTree :
00255
00256     To be used when a sparse table is trimmed due to a .store
00257     We rebuild the tree. In the future one could try to become faster
00258     at the cost of quite some complexity.
00259     We need to keep the first 'size' elements in the boomlijst.
00260     Kill all others and reconstruct the tree with the original ordering.
00261     This is very complicated! Because .store will either keep the whole
00262     table or remove the whole table we should not come here often.
00263     Hence we choose the slow solution for now.
00264 */
00265
00266 void RedoTableTree(TABLES T, int newsiz)
00267 {
00268     WORD *tp;
00269     int i;
00270     ClearTableTree(T);
00271     for ( i = 0, tp = T->tablepointers; i < newsiz; i++ ) {
00272         InsTableTree(T, tp);
00273         tp += ABS(T->numind)+TABLEEXTENSION;
00274     }
00275 }
00276
00277 /*
00278 #] RedoTableTree :
00279 #[ FindTableTree :
00280
00281     int FindTableTree(TABLES T, WORD *, arglist, int, inc)
00282         Searches for the element specified by the list of arguments.
00283         If found, it returns the offset in T->tablepointers
00284         If not found, it will return -1
00285         The list here is from the list of function arguments. Hence it
00286         has pairs of numbers -SNUMBER, index
00287         Actually inc says how many numbers there are and the above case is
00288         for inc = 2. For inc = 1 we have just a list of indices.
00289 */
00290
00291 int FindTableTree(TABLES T, WORD *tp, int inc)
00292 {
00293     COMPTREE *boomlijst = T->boomlijst, *q = boomlijst + T->rootnum, *p;
00294     WORD *v1, *v2, *v3, xstop;
00295     int ip, iq;
00296     if ( q->right == -1 ) return(-1);
00297     ip = q->right;
00298     if ( inc > 1 ) tp += inc-1;
00299     if ( T->numind >= 0 ) xstop = T->numind;
00300     else { /* We have to read the number of arguments first */
00301         if ( *tp <= 0 ) return(-1); /* Cannot be! */
00302         xstop = *tp+1;
00303     }
00304     while ( ip >= 0 ) {
00305         p = boomlijst + ip;
00306         v1 = T->tablepointers + p->value;
00307         v2 = tp; v3 = v1 + xstop;
00308         while ( *v1 == *v2 && v1 < v3 ) { v1++; v2 += inc; }
00309         if ( v1 == v3 ) {
00310             p->usage++;
00311             return(p->value);
00312         }
00313         if ( *v1 > *v2 ) {
00314             iq = p->right;
00315             if ( iq >= 0 ) { ip = iq; }
00316             else return(-1);
00317         }
00318         else if ( *v1 < *v2 ) {
00319             iq = p->left;
00320             if ( iq >= 0 ) { ip = iq; }
00321             else return(-1);
00322         }
00323     }
00324     MesPrint("Serious problems in FindTableTree\n");
00325     Terminate(-1);
00326     return(-1);
00327 }
00328
00329 /*
00330 #] FindTableTree :
00331 #[ DoTableExpansion :

```

```

00332 */
00333
00334 int DoTableExpansion(WORD *term, WORD level)
00335 {
00336     GETIDENTITY
00337     WORD *t, *tstop, *stopper, *termout, *m, *mm, *tp, *r, xx;
00338     WORD numsubexp, numbuf;
00339     TABLES T = 0;
00340     int i, j, num;
00341     AN.TeInFun = AR.TePos = 0;
00342     tstop = term + *term;
00343     stopper = tstop - ABS(tstop[-1]);
00344     t = term+1;
00345     while ( t < stopper ) {
00346         if ( *t != TABLEFUNCTION ) { t += t[1]; continue; }
00347         if ( t[FUNHEAD] > -FUNCTION ) { t += t[1]; continue; }
00348         T = functions[-t[FUNHEAD]-FUNCTION].tabl;
00349         if ( T == 0 ) { t += t[1]; continue; }
00350         if ( T->sparse ) T = T->sparse;
00351         if ( t[1] == FUNHEAD+2 && t[FUNHEAD+1] <= -FUNCTION ) break;
00352         if ( t[1] < FUNHEAD+1+2*ABS(T->numind) ) { t += t[1]; continue; }
00353         for ( i = 0; i < ABS(T->numind); i++ ) {
00354             if ( t[FUNHEAD+1+2*i] != -SYMBOL ) break;
00355         }
00356         if ( i >= ABS(T->numind) ) break;
00357         t += t[1];
00358     }
00359     if ( t >= stopper ) {
00360         MesPrint("Internal error: Missing table_ function");
00361         Terminate(-1);
00362     }
00363     /*
00364     Table in T. Now collect the numbers of the symbols;
00365     */
00366     termout = AT.WorkPointer;
00367     if ( T->sparse ) {
00368         for ( i = 0; i < T->totind; i++ ) {
00369             /*
00370             Loop over all table elements
00371             */
00372             m = termout + 1; mm = term + 1;
00373             while ( mm < t ) *m++ = *mm++;
00374             r = m;
00375             if ( t[1] == FUNHEAD+2 && t[FUNHEAD+1] <= -FUNCTION ) {
00376                 *m++ = -t[FUNHEAD+1];
00377                 tp = T->tablepointers + (ABS(T->numind)+TABLEEXTENSION)*i;
00378                 if ( T->numind < 0 ) {
00379                     xx = tp[0]+1;
00380                     *m++ = FUNHEAD+xx*2;
00381                     for ( j = 2; j < FUNHEAD; j++ ) *m++ = 0;
00382                     for ( j = 0; j < xx; j++ ) {
00383                         *m++ = -SNUMBER; *m++ = *tp++;
00384                     }
00385                 }
00386                 else {
00387                     *m++ = FUNHEAD+T->numind*2;
00388                     for ( j = 2; j < FUNHEAD; j++ ) *m++ = 0;
00389                     for ( j = 0; j < T->numind; j++ ) {
00390                         *m++ = -SNUMBER; *m++ = *tp++;
00391                     }
00392                 }
00393             }
00394             else if ( T->numind < 0 ) {
00395                 tp = T->tablepointers + (ABS(T->numind)+TABLEEXTENSION)*i;
00396                 xx = tp[0]+1;
00397                 *m++ = SYMBOL; *m++ = 2+xx*2; mm = t + FUNHEAD+1;
00398                 for ( j = 0; j < xx; j++, mm += 2, tp++ ) {
00399                     if ( *tp != 0 ) { *m++ = mm[1]; *m++ = *tp; }
00400                 }
00401                 r[1] = m-r;
00402                 if ( r[1] == 2 ) m = r;
00403             }
00404             else {
00405                 *m++ = SYMBOL; *m++ = 2+T->numind*2; mm = t + FUNHEAD+1;
00406                 tp = T->tablepointers + (T->numind+TABLEEXTENSION)*i;
00407                 for ( j = 0; j < T->numind; j++, mm += 2, tp++ ) {
00408                     if ( *tp != 0 ) { *m++ = mm[1]; *m++ = *tp; }
00409                 }
00410                 r[1] = m-r;
00411                 if ( r[1] == 2 ) m = r;
00412             }
00413             /*
00414             The next code replaces this old code
00415
00416             *m++ = SUBEXPRESSION;
00417             *m++ = SUBEXPSIZE;
00418             *m++ = *tp;

```

```

00419         *m++ = 1;
00420         *m++ = T->bufnum;
00421         FILLSUB(m);
00422         mm = t + t[1];
00423
00424         We had forgotten to take the parameters into account.
00425         Hence the subexpression prototype for wildcards was missed
00426         Now we slow things down a little bit, but we do not run
00427         any risks. There is still one problem. We have not checked
00428         that the prototype matches.
00429     */
00430     tp = T->tablepointers + (ABS(T->numind)+TABLEEXTENSION)*i
00431           +ABS(T->numind);
00432     numsubexp = tp[0]; numbuf = tp[1];
00433     r = m;
00434 #ifdef WITHPTHREADS
00435     tp = T->prototype[identity];
00436 #else
00437     tp = T->prototype;
00438 #endif
00439     for ( j = 0; j < tp[1]; j++ ) *m++ = tp[j];
00440     r[2] = numsubexp; r[4] = numbuf;
00441 /*
00442     r = m;
00443     tp = T->tablepointers + (ABS(T->numind)+TABLEEXTENSION)*i;
00444     *m++ = -t[FUNHEAD];
00445     if ( T->numind < 0 ) {
00446         xx = tp[0]+1;
00447         *m++ = t[1] - xx - T->numind - 1;
00448     }
00449     else {
00450         xx = T->numind;
00451         *m++ = t[1] - 1;
00452     }
00453     for ( j = 2; j < FUNHEAD; j++ ) *m++ = t[j];
00454     for ( j = 0; j < xx; j++ ) {
00455         *m++ = -SNUMBER; *m++ = *tp++;
00456     }
00457     tp = t + FUNHEAD + 1 + 2*T->numind;
00458     mm = t + t[1];
00459     while ( tp < mm ) *m++ = *tp++;
00460     r[1] = m-r;
00461 */
00462 /*
00463     From now on is old code
00464 */
00465     mm = t + t[1];
00466     while ( mm < tstop ) *m++ = *mm++;
00467     *termout = m - termout;
00468     AT.WorkPointer = m;
00469     if ( Generator(BHEAD termout,level) ) {
00470         MesCall("DoTableExpand");
00471         return(-1);
00472     }
00473     AT.WorkPointer = termout;
00474 }
00475 }
00476 else {
00477     for ( i = 0; i < T->totind; i++ ) {
00478 #if TABLEEXTENSION == 2
00479         if ( T->tablepointers[i] < 0 ) continue;
00480 #else
00481         if ( T->tablepointers[TABLEEXTENSION*i] < 0 ) continue;
00482 #endif
00483         m = termout + 1; mm = term + 1;
00484         while ( mm < t ) *m++ = *mm++;
00485         r = m;
00486         if ( t[1] == FUNHEAD+2 && t[FUNHEAD+1] <= -FUNCTION ) {
00487             *m++ = -t[FUNHEAD+1];
00488             *m++ = FUNHEAD+T->numind*2;
00489             for ( j = 2; j < FUNHEAD; j++ ) *m++ = 0;
00490             tp = T->tablepointers + (T->numind+TABLEEXTENSION)*i;
00491             for ( j = 0; j < T->numind; j++ ) {
00492                 if ( j > 0 ) {
00493                     num = T->mm[j].mini + ( i % T->mm[j-1].size ) / T->mm[j].size;
00494                 }
00495                 else {
00496                     num = T->mm[j].mini + i / T->mm[j].size;
00497                 }
00498                 *m++ = -SNUMBER; *m++ = num;
00499             }
00500         }
00501         else {
00502             *m++ = SYMBOL; *m++ = 2+T->numind*2; mm = t + FUNHEAD+1;
00503             for ( j = 0; j < T->numind; j++, mm += 2 ) {
00504                 if ( j > 0 ) {
00505                     num = T->mm[j].mini + ( i % T->mm[j-1].size ) / T->mm[j].size;

```

```

00506         }
00507         else {
00508             num = T->mm[j].mini + i / T->mm[j].size;
00509         }
00510         if ( num != 0 ) { *m++ = mm[1]; *m++ = num; }
00511     }
00512     r[1] = m-r;
00513     if ( r[1] == 2 ) m = r;
00514 }
00515 /*
00516     The next code replaces this old code
00517
00518     *m++ = SUBEXPRESSION;
00519     *m++ = SUBEXPSIZE;
00520     *m++ = *tp;
00521     *m++ = 1;
00522     *m++ = T->bufnum;
00523     FILLSUB(m);
00524     mm = t + t[1];
00525
00526     We had forgotten to take the parameters into account.
00527     Hence the subexpression prototype for wildcards was missed
00528     Now we slow things down a little bit, but we do not run
00529     any risks. There is still one problem. We have not checked
00530     that the prototype matches.
00531 */
00532     r = m;
00533     *m++ = -t[FUNHEAD];
00534     *m++ = t[1] - 1;
00535     for ( j = 2; j < FUNHEAD; j++ ) *m++ = t[j];
00536     for ( j = 0; j < T->numind; j++ ) {
00537         if ( j > 0 ) {
00538             num = T->mm[j].mini + ( i % T->mm[j-1].size ) / T->mm[j].size;
00539         }
00540         else {
00541             num = T->mm[j].mini + i / T->mm[j].size;
00542         }
00543         *m++ = -SNUMBER; *m++ = num;
00544     }
00545     tp = t + FUNHEAD + 1 + 2*T->numind;
00546     mm = t + t[1];
00547     while ( tp < mm ) *m++ = *tp++;
00548     r[1] = m - r;
00549 /*
00550     From now on is old code
00551 */
00552     while ( mm < tstop ) *m++ = *mm++;
00553     *termout = m - termout;
00554     AT.WorkPointer = m;
00555     if ( Generator(BHEAD termout, level) ) {
00556         MesCall("DoTableExpand");
00557         return(-1);
00558     }
00559 }
00560 }
00561 return(0);
00562 }
00563
00564 /*
00565 #] DoTableExpansion :
00566 #[ TableBase :
00567
00568 File with all the database related things.
00569 We have the routines for the generic database command
00570 TableBase,options;
00571 TB,options;
00572 Options are:
00573     Open "File.tbl";           Open for R/W
00574     Open "File.tbl", readonly; Open for R
00575     Create "File.tbl";         Create for write
00576     Load "File.tbl", tablename; Loads stubs of table
00577     Load "File.tbl";           Loads stubs of all tables
00578     Enter "File.tbl", tablename; Loads whole table
00579     Enter "File.tbl";          Loads all tables
00580     Audit "File.tbl", options; Print list of contents
00581     Replace "File.tbl", tablename; Saves a table (with overwrite)
00582     Replace "File.tbl", table element; Saves a table element
00583     Cleanup "File.tbl";        Makes tables contingent
00584     AddTo "File.tbl" tablename; Add if not yet there.
00585     AddTo "File.tbl" table element; Add if not yet there.
00586     Delete "File.tbl" tablename;
00587     Delete "File.tbl" table element;
00588
00589     On/Off substitute;
00590     On/Off compress "File.tbl";
00591     id tbl_(f?,?a) = f(?a);
00592     When a tbl_ is used, automatically the corresponding element is compiled

```

```

00593     at the start of the next module.
00594     if TB,On,substitute [tablename], use of table RHS (if loaded)
00595     if TB,Off,substitute [tablename], use of tbl_(table,...);
00596
00597
00598     Still needed: Something like OverLoad to allow loading parts of a table
00599     from more than one file. Date stamps needed? In that case we need a touch
00600     command as well.
00601
00602     If we put all our diagrams inside, we have to go outside the concept
00603     of tables.
00604
00605     #] TableBase :
00606     #[ CoTableBase :
00607
00608     To be followed by ,subkey
00609  */
00610 static KEYWORD tboptions[] = {
00611     {"addto",          (TFUN)CoTBaddto,      0,    PARTEST}
00612     ,{"audit",         (TFUN)CoTBaudit,      0,    PARTEST}
00613     ,{"cleanup",       (TFUN)CoTbcleanup,    0,    PARTEST}
00614     ,{"create",        (TFUN)CoTBcreate,     0,    PARTEST}
00615     ,{"enter",         (TFUN)CoTBenter,      0,    PARTEST}
00616     ,{"help",          (TFUN)CoTBhelp,       0,    PARTEST}
00617     ,{"load",          (TFUN)CoTBload,       0,    PARTEST}
00618     ,{"off",           (TFUN)CoTBoff,        0,    PARTEST}
00619     ,{"on",            (TFUN)CoTBon,         0,    PARTEST}
00620     ,{"open",          (TFUN)CoTBopen,       0,    PARTEST}
00621     ,{"replace",       (TFUN)CoTBreplace,    0,    PARTEST}
00622     ,{"use",           (TFUN)CoTBuse,        0,    PARTEST}
00623 };
00624
00625 static UBYTE *tablebasename = 0;
00626
00627 int CoTableBase(UBYTE *s)
00628 {
00629     UBYTE *option, c, *t;
00630     int i,optlistsize = sizeof(tboptions)/sizeof(KEYWORD), error = 0;
00631     while ( *s == ' ' ) s++;
00632     if ( *s != '"' ) {
00633         if ( ( tolower(*s) == 'h' ) && ( tolower(s[1]) == 'e' )
00634             && ( tolower(s[2]) == 'l' ) && ( tolower(s[3]) == 'p' )
00635             && ( FG.cTable[s[4]] > 1 ) ) {
00636             CoTBhelp(s);
00637             return(0);
00638         }
00639         proper::;
00640         MesPrint("&Proper syntax: TableBase \"filename\" options");
00641         return(1);
00642     }
00643     s++; tablebasename = s;
00644     while ( *s && *s != '"' ) s++;
00645     if ( *s != '"' ) goto proper;
00646     t = s; s++; *t = 0;
00647     while ( *s == ' ' || *s == '\\t' || *s == ',' ) s++;
00648     option = s;
00649     while ( FG.cTable[*s] == 0 ) s++;
00650     c = *s; *s = 0;
00651     for ( i = 0; i < optlistsize; i++ ) {
00652         if ( StrICmp(option, (UBYTE *) (tboptions[i].name)) == 0 ) {
00653             *s = c;
00654             while ( *s == ',' ) s++;
00655             error = (tboptions[i].func)(s);
00656             *t = '"';
00657             return(error);
00658         }
00659     }
00660     MesPrint("&Unrecognized option %s in TableBase statement",option);
00661     return(1);
00662 }
00663
00664 /*
00665     #] CoTableBase :
00666     #[ FlipTable :
00667
00668     Flips the table between use as 'stub' and regular use
00669  */
00670
00671 int FlipTable(FUNCTIONS f, int type)
00672 {
00673     TABLES T, TT;
00674     T = f->tabl;
00675     if ( ( TT = T->spare ) == 0 ) {
00676         MesPrint("Error: trying to change mode on a table that has no tablebase");
00677         return(-1);
00678     }
00679     if ( TT->mode == type ) f->tabl = TT;

```

```

00680     return(0);
00681 }
00682
00683 /*
00684 #] FlipTable :
00685 #[ SpareTable :
00686
00687     Creates a spare element for a table. This is used in the table bases.
00688     It is a (thus far) empty copy of the TT table.
00689     By using FlipTable we can switch between them and alter which version of
00690     a table we will be using. Note that this also causes some extra work in the
00691     ResetVariables and the Globalize routines.
00692 */
00693
00694 int SpareTable(TABLES TT)
00695 {
00696     TABLES T;
00697     T = (TABLES)Malloc1(sizeof(struct TableEs),"table");
00698     T->defined = T->mdefined = 0; T->sparse = TT->sparse; T->mm = 0; T->flags = 0;
00699     T->numtree = 0; T->rootnum = 0; T->MaxTreeSize = 0;
00700     T->boomlijst = 0;
00701     T->strict = TT->strict;
00702     T->bounds = TT->bounds;
00703     T->bufnum = inicbufs();
00704     T->argtail = TT->argtail;
00705     T->spare = TT;
00706     T->bufferssize = 8;
00707     T->buffers = (WORD *)Malloc1(sizeof(WORD)*T->bufferssize,"SpareTable buffers");
00708     T->buffersfill = 0;
00709     T->buffers[T->buffersfill++] = T->bufnum;
00710     T->mode = 0;
00711     T->numind = TT->numind;
00712     T->totind = 0;
00713     T->prototype = TT->prototype;
00714     T->pattern = TT->pattern;
00715     T->tablepointers = 0;
00716     T->reserved = 0;
00717     T->tablenum = 0;
00718     T->numdummies = 0;
00719     T->mm = (MINMAX *)Malloc1(ABS(T->numind)*sizeof(MINMAX),"table dimensions");
00720     T->flags = (WORD *)Malloc1(ABS(T->numind)*sizeof(WORD),"table flags");
00721     ClearTableTree(T);
00722     TT->spare = T;
00723     TT->mode = 1;
00724     return(0);
00725 }
00726
00727 /*
00728 #] SpareTable :
00729 #[ FindTB :
00730
00731     Looks for a tablebase with the given name in the active tablebases.
00732 */
00733
00734 DBASE *FindTB(UBYTE *name)
00735 {
00736     DBASE *d;
00737     int i;
00738     for ( i = 0; i < NumTableBases; i++ ) {
00739         d = tablebases+i;
00740         if ( d->name && ( StrCmp(name,(UBYTE *) (d->name)) == 0 ) ) { return(d); }
00741     }
00742     return(0);
00743 }
00744
00745 /*
00746 #] FindTB :
00747 #[ CoTBcreate :
00748
00749     Creates a new tablebase.
00750     Error is when there is already an active tablebase by this name.
00751     If a file with the given name exists already, but it does not correspond
00752     to an active table base, its contents will be lost.
00753     Note that tablebasename is a static variable, defined in CoTableBase
00754 */
00755
00756 int CoTBcreate(UBYTE *s)
00757 {
00758     DUMMYUSE(s);
00759     if ( FindTB(tablebasename) != 0 ) {
00760         MesPrint("&There is already an open TableBase with the name %s",tablebasename);
00761         return(-1);
00762     }
00763     NewDbase((char *)tablebasename,0);
00764     return(0);
00765 }
00766

```



```

00767 /*
00768 #] CoTBcreate :
00769 #[ CoTBopen :
00770 */
00771
00772 int CoTBopen(UBYTE *s)
00773 {
00774     DBASE *d;
00775     MLONG rw = 1;
00776
00777     SkipSpaces(&s);
00778
00779     if ( *s ) {
00780         if ( ConsumeOption(&s,"readonly") != 0 ) {
00781             rw = 0;
00782         } else {
00783             MesPrint("&Invalid option for TableBase open: %s, ignoring", s);
00784         }
00785     }
00786
00787     if ( ( d = FindTB(tablebasename) ) != 0 ) {
00788         MesPrint("&There is already an open TableBase with the name %s",tablebasename);
00789         return(-1);
00790     }
00791     d = GetDbase((char *)tablebasename, rw);
00792     if ( CheckTableDeclarations(d) ) return(-1);
00793     return(0);
00794 }
00795
00796 /*
00797 #] CoTBopen :
00798 #[ CoTBaddto :
00799 */
00800
00801 int CoTBaddto(UBYTE *s)
00802 {
00803     GETIDENTITY
00804     DBASE *d;
00805     UBYTE *tablename, c, *t, elementstring[ELEMENTSIZE+20], *ss, *es;
00806     WORD type, funnum, lbrac, first, num, *expr, *w;
00807     TABLES T = 0;
00808     MLONG basenumber;
00809     LONG x;
00810     int i, j, error = 0, sum;
00811     if ( ( d = FindTB(tablebasename) ) == 0 ) {
00812         MesPrint("&No open tablebase with the name %s",tablebasename);
00813         return(-1);
00814     }
00815
00816     if ( ( d->rwmode ) == 0 ) {
00817         MesPrint("&Tablebase with the name %s opened in read only mode",tablebasename);
00818         return(-1);
00819     }
00820     AO.DollarOutSizeBuffer = 32;
00821     AO.DollarOutBuffer = (UBYTE *)Malloc1(AO.DollarOutSizeBuffer,
00822         "TableOutBuffer");
00823
00824     /*
00825     Now loop through the names and start adding
00826     */
00827     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
00828     while ( *s ) {
00829         tablename = s;
00830         if ( ( s = SkipAName(s) ) == 0 ) goto tableabort;
00831         c = *s; *s = 0;
00832         if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
00833             || ( T = functions[funnum].tabl ) == 0 ) {
00834             MesPrint("&%s should be a previously declared table",tablename);
00835             *s = c; goto tableabort;
00836         }
00837         if ( T->sparse == 0 ) {
00838             MesPrint("&%s should be a sparse table",tablename);
00839             *s = c; goto tableabort;
00840         }
00841         basenumber = AddTableName(d,(char *)tablename,T);
00842         if ( T->sparse && ( T->mode == 1 ) ) T = T->sparse;
00843         if ( basenumber < 0 ) basenumber = -basenumber;
00844         else if ( basenumber == 0 ) { *s = c; goto tableabort; }
00845         *s = c;
00846         if ( *s == '(' ) { /* Addition of single element */
00847             s++; es = s;
00848             for ( i = 0, w = AT.WorkPointer; i < ABS(T->numind); i++ ) {
00849                 ParseSignedNumber(x,s);
00850                 if ( FG.cTable[s[-1]] != 1 || ( *s != ',' && *s != ')' ) ) {
00851                     MesPrint("&Table arguments in TableBase addto statement should be numbers");
00852                     return(1);
00853                 }
00854                 *w++ = x;
00855             }

```

```

00854         if ( *s == ' ' ) break;
00855         s++;
00856     }
00857     if ( *s != ' ' || i < ( ABS(T->numind) - 1 ) ) {
00858         MesPrint("&Incorrect number of table arguments in TableBase addto statement. Should be
%d"
00859             ,ABS(T->numind));
00860         error = 1;
00861     }
00862     c = *s; *s = 0;
00863     i = FindTableTree(T,AT.WorkPointer,1);
00864     if ( i < 0 ) {
00865         MesPrint("&Element %s has not been defined",es);
00866         error = 1;
00867         *s++ = c;
00868     }
00869     else if ( ExistsObject(d,basenumber,(char *)es) ) {}
00870     else {
00871         int dict = AO.CurrentDictionary;
00872         AO.CurrentDictionary = 0;
00873         sum = i + ABS(T->numind);
00874         /*
00875             See also commentary below
00876         */
00877         AO.DollarInOutBuffer = 1;
00878         AO.PrintType = 1;
00879         ss = AO.DollarOutBuffer;
00880         *ss = 0;
00881         AO.OutInBuffer = 1;
00882         #if ( TABLEEXTENSION == 2 )
00883             expr = cbuf[T->bufnum].rhs[T->tablepointers[sum]];
00884         #else
00885             expr = cbuf[T->tablepointers[sum+1]].rhs[T->tablepointers[sum]];
00886         #endif
00887         lbrac = 0; first = 0;
00888         while ( *expr ) {
00889             if ( WriteTerm(expr,&lbrac,first,PRINTON,0) ) {
00890                 error = 1; break;
00891             }
00892             expr += *expr;
00893         }
00894         AO.OutInBuffer = 0;
00895         AddObject(d,basenumber,(char *)es,(char *) (AO.DollarOutBuffer));
00896         *s++ = c;
00897         AO.CurrentDictionary = dict;
00898     }
00899 }
00900 else {
00901     /*
00902         Now we have to start looping through all defined elements of this table.
00903         We have to construct the arguments in text format.
00904     */
00905     for ( i = 0; i < T->totind; i++ ) {
00906         #if ( TABLEEXTENSION == 2 )
00907             if ( !T->sparse && T->tablepointers[i] < 0 ) continue;
00908         #else
00909             if ( !T->sparse && T->tablepointers[TABLEEXTENSION+i] < 0 ) continue;
00910         #endif
00911         sum = i * ( ABS(T->numind) + TABLEEXTENSION );
00912         t = elementstring;
00913         for ( j = 0; j < ABS(T->numind); j++, sum++ ) {
00914             if ( j > 0 ) *t++ = ',';
00915             num = T->tablepointers[sum];
00916             t = NumCopy(num,t);
00917             if ( ( t - elementstring ) >= ELEMENTSIZE ) {
00918                 MesPrint("&Table element specification takes more than %ld characters and cannot
be handled",
00919                     (MLONG) ELEMENTSIZE);
00920                 goto tableabort;
00921             }
00922         }
00923         if ( ExistsObject(d,basenumber,(char *)elementstring) ) { continue; }
00924         /*
00925             We have the number in basenumber and the element in elementstring.
00926             Now we need the rhs. We can use the code from WriteDollarToBuffer.
00927             Main complication: in the table compiler buffer there can be
00928             brackets. The dollars do not have those.....
00929         */
00930         AO.DollarInOutBuffer = 1;
00931         AO.PrintType = 1;
00932         ss = AO.DollarOutBuffer;
00933         *ss = 0;
00934         AO.OutInBuffer = 1;
00935         #if ( TABLEEXTENSION == 2 )
00936             expr = cbuf[T->bufnum].rhs[T->tablepointers[sum]];
00937         #else
00938             expr = cbuf[T->tablepointers[sum+1]].rhs[T->tablepointers[sum]];

```

```

00939 #endif
00940         lbrac = 0; first = 0;
00941         while ( *expr ) {
00942             if ( WriteTerm(expr,&lbrac,first,PRINTON,0) ) {
00943                 error = 1; break;
00944             }
00945             expr += *expr;
00946         }
00947         AO.OutInBuffer = 0;
00948         AddObject(d,basenumber, (char *)elementstring, (char *) (AO.DollarOutBuffer));
00949     }
00950 }
00951 while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
00952 }
00953 if ( WriteIniInfo(d) ) goto tableabort;
00954 M_free(AO.DollarOutBuffer,"DollarOutBuffer");
00955 AO.DollarOutBuffer = 0;
00956 AO.DollarOutSizeBuffer = 0;
00957 return(error);
00958 tableabort:;
00959 M_free(AO.DollarOutBuffer,"DollarOutBuffer");
00960 AO.DollarOutBuffer = 0;
00961 AO.DollarOutSizeBuffer = 0;
00962 AO.OutInBuffer = 0;
00963 return(1);
00964 }
00965
00966 /*
00967 #] CoTBaddto :
00968 #[ CoTBenter :
00969
00970 Loads the elements of the tables specified into memory and sends them
00971 one by one to the compiler as Fill statements.
00972 */
00973
00974 int CoTBenter(UBYTE *s)
00975 {
00976     DBASE *d;
00977     MLONG basenumber;
00978     UBYTE *arguments, *rhs, *buffer, *t, *u, c, *tablename;
00979     LONG size;
00980     int i, j, error = 0, error1 = 0, printall = 0;
00981     TABLES T = 0;
00982     WORD type, funnum;
00983     int dict = AO.CurrentDictionary;
00984     AO.CurrentDictionary = 0;
00985     if ( ( d = FindTB(tablebasename) ) == 0 ) {
00986         MesPrint("&No open tablebase with the name %s, check for existence of file or try readonly
mode when opening.",tablebasename);
00987         error = -1;
00988         goto Endofall;
00989     }
00990     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
00991     if ( *s == '!' ) { printall = 1; s++; }
00992     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
00993     if ( *s ) {
00994         while ( *s ) {
00995             tablename = s;
00996             if ( ( s = SkipAName(s) ) == 0 ) { error = 1; goto Endofall; }
00997             c = *s; *s = 0;
00998             if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
|| ( T = functions[funnum].tabl ) == 0 ) {
01000                 MesPrint("&%s should be a previously declared table",tablename);
01001                 basenumber = 0;
01002             }
01003             else if ( T->sparse == 0 ) {
01004                 MesPrint("&%s should be a sparse table",tablename);
01005                 basenumber = 0;
01006             }
01007             else { basenumber = GetTableName(d,(char *)tablename); }
01008             if ( T->sparse == 0 ) { SpareTable(T); }
01009             if ( basenumber > 0 ) {
01010                 for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
01011                     for ( j = 0; j < NUMOBJECTS; j++ ) {
01012                         if ( basenumber != d->iblocks[i]->objects[j].tablename )
01013                             continue;
01014                         arguments = (UBYTE *) (d->iblocks[i]->objects[j].element);
01015                         rhs = (UBYTE *) ReadObject(d,basenumber, (char *)arguments);
01016                         if ( printall ) {
01017                             if ( rhs ) {
01018                                 MesPrint("&%s(%s) = %s",tablename,arguments,rhs);
01019                             }
01020                             else {
01021                                 MesPrint("&%s(%s) = 0",tablename,arguments);
01022                             }
01023                         }
01024                     }
01025                 }
01026             }
01027         }
01028     }
01029     if ( rhs ) {

```

```

01025         u = rhs; while ( *u ) u++;
01026         size = u-rhs;
01027         u = arguments; while ( *u ) u++;
01028         size += u-arguments;
01029         u = tablename; while ( *u ) u++;
01030         size += u-tablename;
01031         size += 6;
01032         buffer = (UBYTE *)Malloc1(size,"TableBase copy");
01033         t = tablename; u = buffer;
01034         while ( *t ) *u++ = *t++;
01035         *u++ = '(';
01036         t = arguments;
01037         while ( *t ) *u++ = *t++;
01038         *u++ = ')'; *u++ = '=';
01039         t = rhs;
01040         while ( *t ) *u++ = *t++;
01041         if ( t == rhs ) *u++ = '0';
01042         *u++ = 0; *u = 0;
01043         M_free(rhs,"rhs in TBenter");
01044
01045         error1 = CoFill(buffer);
01046
01047         if ( error1 < 0 ) goto Endofall;
01048         if ( error1 != 0 ) error = error1;
01049         M_free(buffer,"TableBase copy");
01050     }
01051 }
01052 }
01053 }
01054 *s = c;
01055 while ( *s == ',' || *s == ' ' || *s == '\\t' ) s++;
01056 }
01057 }
01058 else {
01059     s = (UBYTE *) (d->tablenames); basenumber = 0;
01060     while ( *s ) {
01061         basenumber++;
01062         tablename = s; while ( *s ) s++; s++;
01063         while ( *s ) s++;
01064         s++;
01065         if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
01066             || ( T = functions[funnum].tabl ) == 0 ) {
01067             MesPrint("&%s should be a previously declared table",tablename);
01068         }
01069         else if ( T->sparse == 0 ) {
01070             MesPrint("&%s should be a sparse table",tablename);
01071         }
01072         if ( T->sparse == 0 ) { SpareTable(T); }
01073         for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
01074             for ( j = 0; j < NUMOBJECTS; j++ ) {
01075                 if ( d->iblocks[i]->objects[j].tablenumber == basenumber ) {
01076                     arguments = (UBYTE *) (d->iblocks[i]->objects[j].element);
01077                     rhs = (UBYTE *) ReadObject(d,basenumber,(char *)arguments);
01078                     if ( printall ) {
01079                         if ( rhs ) {
01080                             MesPrint("%s%s = %s",tablename,arguments,rhs);
01081                         }
01082                         else {
01083                             MesPrint("%s%s = 0",tablename,arguments);
01084                         }
01085                     }
01086                     if ( rhs ) {
01087                         u = rhs; while ( *u ) u++;
01088                         size = u-rhs;
01089                         u = arguments; while ( *u ) u++;
01090                         size += u-arguments;
01091                         u = tablename; while ( *u ) u++;
01092                         size += u-tablename;
01093                         size += 6;
01094                         buffer = (UBYTE *) Malloc1(size,"TableBase copy");
01095                         t = tablename; u = buffer;
01096                         while ( *t ) *u++ = *t++;
01097                         *u++ = '(';
01098                         t = arguments;
01099                         while ( *t ) *u++ = *t++;
01100                         *u++ = ')'; *u++ = '=';
01101                         t = rhs;
01102                         while ( *t ) *u++ = *t++;
01103                         if ( t == rhs ) *u++ = '0';
01104                         *u++ = 0; *u = 0;
01105                         M_free(rhs,"rhs in TBenter");
01106
01107                         error1 = CoFill(buffer);
01108
01109                         if ( error1 < 0 ) goto Endofall;
01110                         if ( error1 != 0 ) error = error1;
01111                         M_free(buffer,"TableBase copy");

```

```

01112     }
01113     }
01114     }
01115     }
01116     }
01117 }
01118 Endofall;;
01119 AO.CurrentDictionary = dict;
01120 return(error);
01121 }
01122
01123 /*
01124 #] CoTBenter :
01125 #[ CoTestUse :
01126
01127     Possibly to be followed by names of tables.
01128     We make an array of TABLES structs to be tested in AC.usedtables.
01129     Note: only sparse tables are allowed.
01130     No arguments means all tables.
01131 */
01132
01133 int CoTestUse(UBYTE *s)
01134 {
01135     GETIDENTITY
01136     UBYTE *tablename, c;
01137     WORD type, funnum, *w;
01138     TABLES T;
01139     int error = 0;
01140     w = AT.WorkPointer;
01141     *w++ = TYPETESTUSE; *w++ = 2;
01142     while ( *s ) {
01143         tablename = s;
01144         if ( ( s = SkipAName(s) ) == 0 ) return(1);
01145         c = *s; *s = 0;
01146         if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
01147             || ( T = functions[funnum].tabl ) == 0 ) {
01148             MesPrint("&%s should be a previously declared table",tablename);
01149             error = 1;
01150         }
01151         else if ( T->sparse == 0 ) {
01152             MesPrint("&%s should be a sparse table",tablename);
01153             error = 1;
01154         }
01155         *w++ = funnum + FUNCTION;
01156         *s = c;
01157         while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
01158     }
01159     AT.WorkPointer[1] = w - AT.WorkPointer;
01160 /*
01161     if ( AT.WorkPointer[1] > 2 ) {
01162         AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
01163     }
01164 */
01165     AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
01166     return(error);
01167 }
01168
01169 /*
01170 #] CoTestUse :
01171 #[ CheckTableDeclarations :
01172
01173     Checks that all tables in a tablebase have identical properties to
01174     possible previous declarations. If they have not been declared
01175     before, they are declared here.
01176 */
01177
01178 int CheckTableDeclarations(DBASE *d)
01179 {
01180     WORD type, funnum;
01181     UBYTE *s, *ss, *t, *command = 0;
01182     int k, error = 0, error1, i;
01183     TABLES T;
01184     LONG commandsize = 0;
01185
01186     s = (UBYTE *) (d->tablenames);
01187     for ( k = 0; k < d->topnumber; k++ ) {
01188         if ( GetVar(s,&type,&funnum,ANYTYPE,NOAUTO) == NAMENOTFOUND ) {
01189             /*
01190                 We have to declare the table
01191             */
01192             ss = s; i = 0; while ( *ss ) { ss++; i++; } /* name */
01193             ss++; while ( *ss ) { ss++; i++; } /* tail */
01194             if ( commandsize == 0 ) {
01195                 commandsize = i + 15;
01196                 if ( commandsize < 100 ) commandsize = 100;
01197             }
01198             if ( (i+11) > commandsize ) {

```

```

01199         if ( command ) { M_free(command,"table command"); command = 0; }
01200         commandsize = i+10;
01201     }
01202     if ( command == 0 ) {
01203         command = (UBYTE *)Malloc1(commandsize,"table command");
01204     }
01205     t = command; ss = tablebase; while ( *ss ) *t++ = *ss++;
01206     *t++ = ','; while ( *s ) *t++ = *s++;
01207     s++; while ( *s ) *t++ = *s++;
01208     *t++ = ')'; *t = 0; s++;
01209     error1 = DoTable(command,1);
01210     if ( error1 ) error = error1;
01211 }
01212 else if ( ( type != CFUNCTION )
01213         || ( T = functions[funnum].tabl ) == 0 )
01214         || ( T->sparse == 0 ) ) {
01215     MesPrint("%s has been declared previously, but not as a sparse table.",s);
01216     error = 1;
01217     while ( *s ) s++;
01218     s++;
01219     while ( *s ) s++;
01220     s++;
01221 }
01222 else {
01223 /*
01224     Test dimension and argtail. There should be an exact match.
01225     We are not going to rename arguments when reading the elements.
01226 */
01227     ss = s;
01228     while ( *s ) s++;
01229     s++;
01230     if ( StrCmp(s,T->argtail) ) {
01231         MesPrint("&Declaration of table %s in %s different from previous
declaration",ss,d->name);
01232         error = 1;
01233     }
01234     while ( *s ) s++;
01235     s++;
01236 }
01237 }
01238 if ( command ) { M_free(command,"table command"); }
01239 return(error);
01240 }
01241
01242 /*
01243 #] CheckTableDeclarations :
01244 #[ CoTbload :
01245
01246     Loads the table stubbs of the specified tables in the indicated
01247     tablebase. Syntax:
01248     TableBase "tablebasename.tbl" load [tablename(s)];
01249     If no tables are specified all tables are taken.
01250 */
01251
01252 int CoTbload(UBYTE *ss)
01253 {
01254     DBASE *d;
01255     UBYTE *s, *name, *t, *r, *command, *arguments, *tail;
01256     LONG commandsize;
01257     int num, cs, es, ns, ts, i, j, error = 0, error1;
01258     if ( ( d = FindTB(tablebasename) ) == 0 ) {
01259         MesPrint("&No open tablebase with the name %s",tablebasename);
01260         return(-1);
01261     }
01262     commandsize = 120;
01263     command = (UBYTE *)Malloc1(commandsize,"Fill command");
01264     AC.vetofilling = 1;
01265     if ( *ss ) {
01266         while ( *ss == ',' || *ss == ' ' || *ss == '\t' ) ss++;
01267         while ( *ss ) {
01268             name = ss; ss = SkipAName(ss); *ss = 0;
01269             s = (UBYTE *) (d->tablenames);
01270             num = 0; ns = 0;
01271             while ( *s ) {
01272                 num++;
01273                 if ( StrCmp(s,name) ) {
01274                     while ( *s ) s++;
01275                     s++;
01276                     while ( *s ) s++;
01277                     s++;
01278                     num++;
01279                     continue;
01280                 }
01281                 name = s; while ( *s ) s++; ns = s-name; s++;
01282                 tail = s; while ( *s ) s++; ts = s-tail; s++;
01283                 tail++; while ( FG.cTable[*tail] == 1 ) tail++;
01284 /*

```

```

01285         Go through all elements
01286     */
01287     for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
01288         for ( j = 0; j < NUMOBJECTS; j++ ) {
01289             if ( d->iblocks[i]->objects[j].tablename == num ) {
01290                 t = arguments = (UBYTE *) (d->iblocks[i]->objects[j].element);
01291                 while ( *t ) t++;
01292                 es = t - arguments;
01293                 cs = 2*es + 2*ns + ts + 10;
01294                 if ( cs > commandsize ) {
01295                     commandsize = 2*cs;
01296                     if ( command ) M_free(command,"Fill command");
01297                     command = (UBYTE *) Malloc1(commandsize,"Fill command");
01298                 }
01299                 r = command; t = name; while ( *t ) *r++ = *t++;
01300                 *r++ = '('; t = arguments; while ( *t ) *r++ = *t++;
01301                 *r++ = '='; *r++ = '='; *r++ = 't'; *r++ = 'b'; *r++ = 'l';
01302                 *r++ = '_'; *r++ = '('; t = name; while ( *t ) *r++ = *t++;
01303                 *r++ = ','; t = arguments; while ( *t ) *r++ = *t++;
01304                 t = tail; while ( *t ) {
01305                     if ( *t == '?' && r[-1] != ',' ) {
01306                         t++;
01307                         if ( FG.cTable[*t] == 0 || *t == '$' || *t == '[' ) {
01308                             t = SkipAName(t);
01309                             if ( *t == '[' ) {
01310                                 SKIPBRA1(t);
01311                             }
01312                         }
01313                     } else if ( *t == '{' ) {
01314                         SKIPBRA2(t);
01315                     }
01316                     else if ( *t ) { *r++ = *t++; continue; }
01317                 }
01318                 else *r++ = *t++;
01319             }
01320             *r++ = ')'; *r = 0;
01321         /*
01322         Still to do: replacemode or no replacemode?
01323     */
01324         AC.vetotablebasefill = 1;
01325         error1 = CoFill(command);
01326         AC.vetotablebasefill = 0;
01327         if ( error1 < 0 ) goto finishup;
01328         if ( error1 != 0 ) error = error1;
01329     }
01330 }
01331 }
01332 break;
01333 }
01334 while ( *ss == ',' || *ss == ' ' || *ss == '\t' ) ss++;
01335 }
01336 }
01337 else { /* do all of them */
01338     s = (UBYTE *) (d->tablenames);
01339     num = 0; ns = 0;
01340     while ( *s ) {
01341         num++;
01342         name = s; while ( *s ) s++; ns = s - name; s++;
01343         tail = s; while ( *s ) s++; ts = s - tail; s++;
01344         tail++; while ( FG.cTable[*tail] == 1 ) tail++;
01345     /*
01346     Go through all elements
01347     */
01348     for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
01349         for ( j = 0; j < NUMOBJECTS; j++ ) {
01350             if ( d->iblocks[i]->objects[j].tablename == num ) {
01351                 t = arguments = (UBYTE *) (d->iblocks[i]->objects[j].element);
01352                 while ( *t ) t++;
01353                 es = t - arguments;
01354                 cs = 2*es + 2*ns + ts + 10;
01355                 if ( cs > commandsize ) {
01356                     commandsize = 2*cs;
01357                     if ( command ) M_free(command,"Fill command");
01358                     command = (UBYTE *) Malloc1(commandsize,"Fill command");
01359                 }
01360                 r = command; t = name; while ( *t ) *r++ = *t++;
01361                 *r++ = '('; t = arguments; while ( *t ) *r++ = *t++;
01362                 *r++ = '='; *r++ = '='; *r++ = 't'; *r++ = 'b'; *r++ = 'l';
01363                 *r++ = '_'; *r++ = '('; t = name; while ( *t ) *r++ = *t++;
01364                 *r++ = ','; t = arguments; while ( *t ) *r++ = *t++;
01365                 t = tail; while ( *t ) {
01366                     if ( *t == '?' && r[-1] != ',' ) {
01367                         t++;
01368                         if ( FG.cTable[*t] == 0 || *t == '$' || *t == '[' ) {
01369                             t = SkipAName(t);
01370                             if ( *t == '[' ) {
01371                                 SKIPBRA1(t);

```

```

01372             }
01373         }
01374         else if ( *t == '{' ) {
01375             SKIPBRA2(t);
01376         }
01377         else if ( *t ) { *r++ = *t++; continue; }
01378     }
01379     else *r++ = *t++;
01380 }
01381 *r++ = ')'; *r = 0;
01382 /*
01383     Still to do: replacemode or no replacemode?
01384 */
01385     AC.vetotablebasefill = 1;
01386     error1 = CoFill(command);
01387     AC.vetotablebasefill = 0;
01388     if ( error1 < 0 ) goto finishup;
01389     if ( error1 != 0 ) error = error1;
01390 }
01391 }
01392 }
01393 }
01394 }
01395 finishup:;
01396     AC.vetofilling = 0;
01397     if ( command ) M_free(command,"Fill command");
01398     return(error);
01399 }
01400
01401 /*
01402     #] CoTBlod :
01403     #[ TestUse :
01404
01405     Look for tbl_(tablename,arguments)
01406     if tablename is encountered, check first whether the element is in
01407     use already. If not, check in the tables in AC.usedtables.
01408     If the element is not there, add it to AC.usedtables.
01409
01410
01411     We need the arguments of TestUse to see for which tables it is to be done
01412 */
01413
01414 int TestUse(WORD *term, WORD level)
01415 {
01416     WORD *tstop, *t, *m, *tstart, tabnum;
01417     WORD *funs, numfuns;
01418     int error = 0;
01419     TABLES T;
01420     LONG i;
01421     CBUF *C = cbuf+AM.rbufnum;
01422     int isp;
01423
01424     numfuns = C->lhs[level][1] - 2;
01425     funs = C->lhs[level] + 2;
01426     GETSTOP(term,tstop);
01427     t = term+1;
01428     while ( t < tstop ) {
01429         if ( *t != TABLESTUB ) { t += t[1]; continue; }
01430         tstart = t;
01431         m = t + FUNHEAD;
01432         t += t[1];
01433         if ( *m >= -FUNCTION ) continue;
01434         tabnum = -*m;
01435         if ( ( T = functions[tabnum-FUNCTION].tbl ) == 0 ) continue;
01436         if ( T->sparse == 0 ) continue;
01437     /*
01438         Check whether we have to test this one
01439     */
01440         if ( numfuns > 0 ) {
01441             for ( i = 0; i < numfuns; i++ ) {
01442                 if ( tabnum == funs[i] ) break;
01443             }
01444             if ( i >= numfuns && numfuns > 0 ) continue;
01445         }
01446     /*
01447         Test whether the element has been defined already.
01448         If not, mark it as used.
01449         Note: we only allow sparse tables (for now)
01450     */
01451         m++;
01452         for ( i = 0; i < ABS(T->numind); i++, m += 2 ) {
01453             if ( m >= t || *m != -SNUMBER ) break;
01454         }
01455         if ( ( i == ABS(T->numind) ) &&
01456             ( ( isp = FindTableTree(T,tstart+FUNHEAD+1,2) ) >= 0 ) ) {
01457             if ( ( T->tablepointers[isp+ABS(T->numind)+4] & ELEMENTLOADED ) == 0 ) {
01458                 T->tablepointers[isp+ABS(T->numind)+4] |= ELEMENTUSED;

```



```

01459         }
01460     }
01461     else {
01462         MesPrint("TestUse: Encountered a table element inside tbl_ that does not correspond to a
tablebase element");
01463         error = -1;
01464     }
01465 }
01466 return(error);
01467 }
01468
01469 /*
01470 #] TestUse :
01471 #[ CoTBaudit :
01472 */
01473
01474 int CoTBaudit(UBYTE *s)
01475 {
01476     DBASE *d;
01477     UBYTE *name, *tail;
01478     int i, j, error = 0, num;
01479
01480     if ( ( d = FindTB(tablebasename) ) == 0 ) {
01481         MesPrint("&No open tablebase with the name %s",tablebasename);
01482         return(-1);
01483     }
01484     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
01485     while ( *s ) {
01486         /*
01487             Get the options here
01488             They will mainly involve the sorting of the output.
01489         */
01490         s++;
01491     }
01492     s = (UBYTE *) (d->tablenames); num = 0;
01493     while ( *s ) {
01494         num++;
01495         name = s; while ( *s ) s++; s++;
01496         tail = s; while ( *s ) s++; s++;
01497         MesPrint("Table,sparse,%s%s",name,tail);
01498         for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
01499             for ( j = 0; j < NUMOBJECTS; j++ ) {
01500                 if ( d->iblocks[i]->objects[j].tablenumber == num ) {
01501                     MesPrint("    %s(%s)",name,d->iblocks[i]->objects[j].element);
01502                 }
01503             }
01504         }
01505     }
01506     return(error);
01507 }
01508
01509 /*
01510 #] CoTBaudit :
01511 #[ CoTBon :
01512 */
01513
01514 int CoTBon(UBYTE *s)
01515 {
01516     DBASE *d;
01517     UBYTE *ss, c;
01518     int error = 0;
01519     if ( ( d = FindTB(tablebasename) ) == 0 ) {
01520         MesPrint("&No open tablebase with the name %s",tablebasename);
01521         return(-1);
01522     }
01523     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
01524     while ( *s ) {
01525         ss = SkipAName(s);
01526         c = *ss; *ss = 0;
01527         if ( StrICmp(s,(UBYTE *) ("compress")) == 0 ) {
01528             d->mode &= ~NOCOMPRESS;
01529         }
01530         else {
01531             MesPrint("&subkey %s not defined in TableBase On statement");
01532             error = 1;
01533         }
01534         *ss = c; s = ss;
01535         while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
01536     }
01537     return(error);
01538 }
01539
01540 /*
01541 #] CoTBon :
01542 #[ CoTBoff :
01543 */
01544

```

```

01545 int CoTBoff(UBYTE *s)
01546 {
01547     DBASE *d;
01548     UBYTE *ss, c;
01549     int error = 0;
01550     if ( ( d = FindTB(tablebasename) ) == 0 ) {
01551         MesPrint("&No open tablebase with the name %s",tablebasename);
01552         return(-1);
01553     }
01554     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
01555     while ( *s ) {
01556         ss = SkipAName(s);
01557         c = *ss; *ss = 0;
01558         if ( StrICmp(s,(UBYTE *)("compress")) == 0 ) {
01559             d->mode |= NOCOMPRESS;
01560         }
01561         else {
01562             MesPrint("&subkey %s not defined in TableBase Off statement");
01563             error = 1;
01564         }
01565         *ss = c; s = ss;
01566         while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
01567     }
01568     return(error);
01569 }
01570
01571 /*
01572 #] CoTBoff :
01573 #[ CoTBcleanup :
01574 */
01575
01576 int CoTBcleanup(UBYTE *s)
01577 {
01578     DUMMYUSE(s);
01579     MesPrint("&TableBase Cleanup statement not yet implemented");
01580     return(1);
01581 }
01582
01583 /*
01584 #] CoTBcleanup :
01585 #[ CoTBreplace :
01586 */
01587
01588 int CoTBreplace(UBYTE *s)
01589 {
01590     DUMMYUSE(s);
01591     MesPrint("&TableBase Replace statement not yet implemented");
01592     return(1);
01593 }
01594
01595 /*
01596 #] CoTBreplace :
01597 #[ CoTBuse :
01598
01599 Here the actual table use as determined in TestUse causes the needed
01600 table elements to be loaded
01601 */
01602
01603 int CoTBuse(UBYTE *s)
01604 {
01605     GETIDENTITY
01606     DBASE *d;
01607     MLONG basenumber;
01608     UBYTE *arguments, *rhs, *buffer, *t, *u, c, *tablename, *p;
01609     LONG size, sum, x;
01610     int i, j, error = 0, error1 = 0, k;
01611     TABLES T = 0;
01612     WORD type, funnum, mode, *w;
01613     if ( ( d = FindTB(tablebasename) ) == 0 ) {
01614         MesPrint("&No open tablebase with the name %s",tablebasename);
01615         return(-1);
01616     }
01617     while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
01618     if ( *s ) {
01619         while ( *s ) {
01620             tablename = s;
01621             if ( ( s = SkipAName(s) ) == 0 ) return(1);
01622             c = *s; *s = 0;
01623             if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
01624                 || ( T = functions[funnum].tabl ) == 0 ) {
01625                 MesPrint("&%s should be a previously declared table",tablename);
01626                 basenumber = 0;
01627             }
01628             else if ( T->sparse == 0 ) {
01629                 MesPrint("&%s should be a sparse table",tablename);
01630                 basenumber = 0;
01631             }

```

```

01632         else { basenumber = GetTableName(d, (char *)tablename); }
01633 /*         if ( T->sparse == 0 ) { SpareTable(T); } */
01634         if ( basenumber > 0 ) {
01635             for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
01636                 for ( j = 0; j < NUMOBJECTS; j++ ) {
01637                     if ( d->iblocks[i]->objects[j].tablenumber != basenumber ) continue;
01638                     arguments = p = (UBYTE *) (d->iblocks[i]->objects[j].element);
01639 /*
01640                     Now translate the arguments and see whether we need
01641                     this one....
01642 */
01643                     for ( k = 0, w = AT.WorkPointer; k < ABS(T->numind); k++ ) {
01644                         ParseSignedNumber(x,p);
01645                         *w++ = x; p++;
01646                     }
01647                     sum = FindTableTree(T,AT.WorkPointer,1);
01648                     if ( sum < 0 ) {
01649                         MesPrint("Table %s in tablebase %s has not been loaded properly"
01650                                ,tablename,tablebasename);
01651                         error = 1;
01652                         continue;
01653                     }
01654                     sum += ABS(T->numind) + 4;
01655                     mode = T->tablepointers[sum];
01656                     if ( ( mode & ELEMENTLOADED ) == ELEMENTLOADED ) {
01657                         T->tablepointers[sum] &= ~ELEMENTUSED;
01658                         continue;
01659                     }
01660                     if ( ( mode & ELEMENTUSED ) == 0 ) continue;
01661 /*
01662                     We need this one!
01663 */
01664                     rhs = (UBYTE *)ReadijObject(d,i,j, (char *)arguments);
01665                     if ( rhs ) {
01666                         u = rhs; while ( *u ) u++;
01667                         size = u-rhs;
01668                         u = arguments; while ( *u ) u++;
01669                         size += u-arguments;
01670                         u = tablename; while ( *u ) u++;
01671                         size += u-tablename;
01672                         size += 6;
01673                         buffer = (UBYTE *)Malloca1(size,"TableBase copy");
01674                         t = tablename; u = buffer;
01675                         while ( *t ) *u++ = *t++;
01676                         *u++ = '(';
01677                         t = arguments;
01678                         while ( *t ) *u++ = *t++;
01679                         *u++ = ')'; *u++ = '=';
01680                         t = rhs;
01681                         while ( *t ) *u++ = *t++;
01682                         if ( t == rhs ) { *u++ = '0'; }
01683                         *u++ = 0; *u = 0;
01684                         M_free(rhs,"rhs in TBuse xxx");
01685
01686                         error1 = CoFill(buffer);
01687
01688                         if ( error1 < 0 ) { return(error); }
01689                         if ( error1 != 0 ) error = error1;
01690                         M_free(buffer,"TableBase copy");
01691                     }
01692                     T->tablepointers[sum] &= ~ELEMENTUSED;
01693                     T->tablepointers[sum] |= ELEMENTLOADED;
01694                 }
01695             }
01696         }
01697         *s = c;
01698         while ( *s == ',' || *s == ' ' || *s == '\\t' ) s++;
01699     }
01700 }
01701 else {
01702     s = (UBYTE *) (d->tablenames); basenumber = 0;
01703     while ( *s ) {
01704         basenumber++;
01705         tablename = s;
01706         while ( *s ) s++;
01707         s++;
01708         while ( *s ) s++;
01709         s++;
01710         if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
01711             || ( T = functions[funnum].tabl ) == 0 ) {
01712             MesPrint("%s should be a previously declared table",tablename);
01713         }
01714         else if ( T->sparse == 0 ) {
01715             MesPrint("%s should be a sparse table",tablename);
01716         }
01717         if ( T->sparse && T->mode == 0 ) {
01718             MesPrint("In table %s we have a problem with stubb orders in CoTBuse",tablename);

```

```

01719     error = -1;
01720 }
01721 /*      if ( T->spare == 0 ) { SpareTable(T); } */
01722 for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
01723     for ( j = 0; j < NUMOBJECTS; j++ ) {
01724         if ( d->iblocks[i]->objects[j].tablename == basenumber ) {
01725             arguments = p = (UBYTE *) (d->iblocks[i]->objects[j].element);
01726 /*
01727             Now translate the arguments and see whether we need
01728             this one....
01729 */
01730             for ( k = 0, w = AT.WorkPointer; k < ABS(T->numind); k++ ) {
01731                 ParseSignedNumber(x,p);
01732                 *w++ = x; p++;
01733             }
01734             sum = FindTableTree(T,AT.WorkPointer,1);
01735             if ( sum < 0 ) {
01736                 MesPrint("Table %s in tablebase %s has not been loaded properly"
01737                     ,tablename,tablebasename);
01738                 error = 1;
01739                 continue;
01740             }
01741             sum += ABS(T->numind) + 4;
01742             mode = T->tablepointers[sum];
01743             if ( ( mode & ELEMENTLOADED ) == ELEMENTLOADED ) {
01744                 T->tablepointers[sum] &= ~ELEMENTUSED;
01745                 continue;
01746             }
01747             if ( ( mode & ELEMENTUSED ) == 0 ) continue;
01748 /*
01749             We need this one!
01750 */
01751             rhs = (UBYTE *)ReadijObject(d,i,j,(char *)arguments);
01752             if ( rhs ) {
01753                 u = rhs; while ( *u ) u++;
01754                 size = u-rhs;
01755                 u = arguments; while ( *u ) u++;
01756                 size += u-arguments;
01757                 u = tablename; while ( *u ) u++;
01758                 size += u-tablename;
01759                 size += 6;
01760                 buffer = (UBYTE *)Malloca1(size,"TableBase copy");
01761                 t = tablename; u = buffer;
01762                 while ( *t ) *u++ = *t++;
01763                 *u++ = '(';
01764                 t = arguments;
01765                 while ( *t ) *u++ = *t++;
01766                 *u++ = ')'; *u++ = '=';
01767
01768                 t = rhs;
01769                 while ( *t ) *u++ = *t++;
01770                 if ( t == rhs ) { *u++ = '0'; }
01771                 *u++ = 0; *u = 0;
01772                 M_free(rhs,"rhs in TBase");
01773
01774                 error1 = CoFill(buffer);
01775
01776                 if ( error1 < 0 ) { return(error); }
01777                 if ( error1 != 0 ) error = error1;
01778                 M_free(buffer,"TableBase copy");
01779             }
01780             T->tablepointers[sum] &= ~ELEMENTUSED;
01781             T->tablepointers[sum] |= ELEMENTLOADED;
01782         }
01783     }
01784 }
01785 }
01786 }
01787 return(error);
01788 }
01789
01790 /*
01791 #] CoTBuse :
01792 #[ CoApply :
01793
01794 Possibly to be followed by names of tables.
01795 */
01796
01797 int CoApply(UBYTE *s)
01798 {
01799     GETIDENTITY
01800     UBYTE *tablename, c;
01801     WORD type, funnum, *w;
01802     TABLES T;
01803     LONG maxtogo = MAXPOSITIVE;
01804     int error = 0;
01805     w = AT.WorkPointer;

```

```

01806     if ( FG.cTable[*s] == 1 ) {
01807         maxtogo = 0;
01808         while ( FG.cTable[*s] == 1 ) {
01809             maxtogo = maxtogo*10 + (*s-'0');
01810             s++;
01811         }
01812         while ( *s == ',' ) s++;
01813         if ( maxtogo > MAXPOSITIVE || maxtogo < 0 ) maxtogo = MAXPOSITIVE;
01814     }
01815     *w++ = TYPEAPPLY; *w++ = 3; *w++ = maxtogo;
01816     while ( *s ) {
01817         tablename = s;
01818         if ( ( s = SkipAName(s) ) == 0 ) return(1);
01819         c = *s; *s = 0;
01820         if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
01821             || ( T = functions[funnum].tbl ) == 0 ) {
01822             MesPrint("%s should be a previously declared table",tablename);
01823             error = 1;
01824         }
01825         else if ( T->sparse == 0 ) {
01826             MesPrint("%s should be a sparse table",tablename);
01827             error = 1;
01828         }
01829         *w++ = funnum + FUNCTION;
01830         *s = c;
01831         while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
01832     }
01833     AT.WorkPointer[1] = w - AT.WorkPointer;
01834     /*
01835     if ( AT.WorkPointer[1] > 2 ) {
01836         AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
01837     }
01838     */
01839     AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
01840     /*
01841     AT.WorkPointer[0] = TYPEAPPLYRESET;
01842     AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
01843     */
01844     return(error);
01845 }
01846
01847 /*
01848 #] CoApply :
01849 #[ CoTBhelp :
01850 */
01851
01852 char *helptb[] = {
01853     "The TableBase statement is used as follows:"
01854     ,"TableBase \"file.tbl\" keyword subkey(s)"
01855     ,"    in which we have"
01856     ,"Keyword    Subkey(s)    Action"
01857     ,"open                Opens file.tbl for R/W"
01858     ,"create                Creates file.tbl for R/W. Old contents are lost"
01859     ,"load                Loads all stubs of all tables"
01860     ,"load    tablename(s) Loads all stubs the tables mentioned"
01861     ,"enter                Loads all stubs and rhs of all tables"
01862     ,"enter    tablename(s) Loads all stubs and rhs of the tables mentioned"
01863     ,"audit                Prints list of contents"
01864     /* ,"replace tablename    saves a table (with overwrite)" */
01865     /* ,"replace tableelement saves a table element (with overwrite)" */
01866     /* ,"cleanup                makes tables contingent" */
01867     ,"addto    tablename    adds all elements if not yet there"
01868     ,"addto    tableelement adds element if not yet there"
01869     /* ,"delete    tablename    removes table from tablebase" */
01870     /* ,"delete    tableelement removes element from tablebase" */
01871     ,"on    compress    elements are stored in gzip format (default)"
01872     ,"off    compress    elements are stored in uncompressed format"
01873     ,"use                compiles all needed elements"
01874     ,"use    tablename(s) compiles all needed elements of these tables"
01875     ,"    "
01876     ,"Related commands are:"
01877     ,"testuse                marks which tbl_ elements occur for all tables"
01878     ,"testuse    tablename(s) marks which tbl_ elements occur for given tables"
01879     ,"apply                replaces tbl_ if rhs available"
01880     ,"apply    tablename(s) replaces tbl_ for given tables if rhs available"
01881     ,"    "
01882     };
01883
01884 int CoTBhelp(UBYTE *s)
01885 {
01886     int i, ii = sizeof(helptb)/sizeof(char *);
01887     DUMMYUSE(s);
01888     for ( i = 0; i < ii; i++ ) MesPrint("%s",helptb[i]);
01889     return(0);
01890 }
01891
01892 /*

```

```

01893     #] CoTBhelp :
01894     #[ ReWorkT :
01895
01896     Replaces the STUBBS of the functions in the list.
01897     This gains one space. Hence we have to be very careful
01898 */
01899
01900 void ReWorkT(WORD *term, WORD *funs, WORD numfuns)
01901 {
01902     WORD *tstop, *tend, *m, *t, *tt, *mm, *mmm, *r, *rr;
01903     int i, j;
01904     tend = term + *term; tstop = tend - ABS(tend[-1]);
01905     m = t = term+1;
01906     while ( t < tstop ) {
01907         if ( *t == TABLESTUB ) {
01908             for ( i = 0; i < numfuns; i++ ) {
01909                 if ( -t[FUNHEAD] == funs[i] ) break;
01910             }
01911             if ( numfuns == 0 || i < numfuns ) { /* Hit */
01912                 i = t[1] - 1;
01913                 *m++ = -t[FUNHEAD]; *m++ = i; t += 2; i -= FUNHEAD;
01914                 if ( m < t ) { for ( j = 0; j < FUNHEAD-2; j++ ) *m++ = *t++; }
01915                 else { m += FUNHEAD-2; t += FUNHEAD-2; }
01916                 t++;
01917                 while ( i-- > 0 ) { *m++ = *t++; }
01918                 tt = t; mm = m;
01919                 if ( mm < tt ) {
01920                     while ( tt < tend ) *mm++ = *tt++;
01921                     *term = mm - term;
01922                     tend = term + *term; tstop = tend - ABS(tend[-1]);
01923                     t = m;
01924                 }
01925             }
01926             else { goto inc; }
01927         }
01928         else if ( *t >= FUNCTION ) {
01929             tt = t + t[1];
01930             mm = m;
01931             for ( j = 0; j < FUNHEAD; j++ ) {
01932                 if ( m == t ) { m++; t++; }
01933                 else *m++ = *t++;
01934             }
01935             while ( t < tt ) {
01936                 if ( *t <= -FUNCTION ) {
01937                     if ( m == t ) { m++; t++; }
01938                     else *m++ = *t++;
01939                 }
01940                 else if ( *t < 0 ) {
01941                     if ( m == t ) { m += 2; t += 2; }
01942                     else { *m++ = *t++; *m++ = *t++; }
01943                 }
01944                 else {
01945                     rr = t + *t; mmm = m;
01946                     for ( j = 0; j < ARGHEAD; j++ ) {
01947                         if ( m == t ) { m++; t++; }
01948                         else *m++ = *t++;
01949                     }
01950                     while ( t < rr ) {
01951                         r = t + *t;
01952                         ReWorkT(t, funs, numfuns);
01953                         j = *t;
01954                         if ( m == t ) { m += j; t += j; }
01955                         else { while ( j-- >= 0 ) *m++ = *t++; }
01956                         t = r;
01957                     }
01958                     *mmm = m - mmm;
01959                 }
01960             }
01961             mm[1] = m - mm;
01962             t = tt;
01963         }
01964         else {
01965             inc:
01966                 j = t[1];
01967                 if ( m < t ) { while ( j-- >= 0 ) *m++ = *t++; }
01968                 else { m += j; t += j; }
01969             }
01970         if ( m < t ) {
01971             while ( t < tend ) *m++ = *t++;
01972             *term = m - term;
01973         }
01974     }
01975
01976 /*
01977     #] ReWorkT :
01978     #[ Apply :
01979 */

```

```

01980
01981 void Apply(WORD *term, WORD level)
01982 {
01983     WORD *funcs, numfuncs;
01984     TABLES T;
01985     int i, j;
01986     CBUF *C = cbuf+AM.rbufnum;
01987     /*
01988     Point the tables in the proper direction
01989     */
01990     numfuncs = C->lhs[level][1] - 2;
01991     funcs = C->lhs[level] + 2;
01992     if ( numfuncs > 0 ) {
01993         for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
01994             if ( ( T = functions[i].tabl ) != 0 ) {
01995                 for ( j = 0; j < numfuncs; j++ ) {
01996                     if ( i == (funcs[j]-FUNCTION) && T->spare ) {
01997                         FlipTable(&(functions[i]),0);
01998                         break;
01999                     }
02000                 }
02001             }
02002         }
02003     }
02004     else {
02005         for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
02006             if ( ( T = functions[i].tabl ) != 0 ) {
02007                 if ( T->spare ) FlipTable(&(functions[i]),0);
02008             }
02009         }
02010     }
02011     /*
02012     Now the replacements everywhere of
02013     id tbl_(table,?a) = table(?a);
02014     Actually, this has to be done recursively.
02015     Note that we actually gain one space.
02016     */
02017     ReWorkT(term, funcs, numfuncs);
02018 }
02019
02020 /*
02021 #] Apply :
02022 #[ ApplyExec :
02023
02024 Replaces occurrences of tbl_(table,indices,pattern) by the proper
02025 rhs of table(indices,pattern). It does this up to maxtogo times
02026 in the given term. It starts with the occurrences inside the
02027 arguments of functions. If necessary it finishes at groundlevel.
02028 An infinite number of tries is indicated by maxtogo = 2^15-1 or 2^31-1.
02029 The occurrences are replaced by subexpressions. This allows TestSub
02030 to finish the job properly.
02031
02032 The main trick here is T = T->spare which turns to the proper rhs.
02033
02034 The return value is the number of substitutions that can still be made
02035 based on maxtogo. Hence, if the returnvalue is different from maxtogo
02036 there was a substitution.
02037 */
02038
02039 int ApplyExec(WORD *term, int maxtogo, WORD level)
02040 {
02041     GETIDENTITY
02042     WORD rhsnumber, *Tpattern, *funcs, numfuncs, funnum;
02043     WORD ii, *t, *tl, *w, *p, *m, *ml, *u, *r, tbufnum, csize, wilds;
02044     NESTING NN;
02045     int i, j, isp, stilltogo;
02046     CBUF *C;
02047     TABLES T;
02048     /*
02049     Startup. We need NestPoin for when we have to replace something deep down.
02050     */
02051     t = term;
02052     m = t + *t;
02053     csize = ABS(m[-1]);
02054     m -= csize;
02055     AT.NestPoin->termsize = t;
02056     if ( AT.NestPoin == AT.Nest ) AN.EndNest = t + *t;
02057     t++;
02058     /*
02059     First we look inside function arguments. Also when clean!
02060     */
02061     while ( t < m ) {
02062         if ( *t < FUNCTION ) { t += t[1]; continue; }
02063         if ( functions[*t-FUNCTION].spec > 0 ) { t += t[1]; continue; }
02064         AT.NestPoin->funsize = t;
02065         r = t + t[1];
02066         t += FUNHEAD;

```

```

02067     while ( t < r ) {
02068         if ( *t < 0 ) { NEXTARG(t); continue; }
02069         AT.NestPoin->argsize = t1 = t;
02070         u = t + *t;
02071         t += ARGHEAD;
02072         AT.NestPoin++;
02073         while ( t < u ) {
02074             /*
02075                 Now we loop over the terms inside a function argument
02076                 This defines a recursion and we have to call ApplyExec again.
02077                 The real problem is when we catch something and we have
02078                 to insert a subexpression pointer. This may use more or
02079                 less space and the whole term has to be readjusted.
02080                 This is why we have the NestPoin variables. They tell us
02081                 where the sizes of the term, the function and the arguments
02082                 are sitting, and also where the dirty flags are.
02083                 This readjusting is of course done in the groundlevel code.
02084                 Here we worry about the maxtogo count.
02085             */
02086             stilltogo = ApplyExec(t,maxtogo,level);
02087             if ( stilltogo != maxtogo ) {
02088                 if ( stilltogo <= 0 ) {
02089                     AT.NestPoin--;
02090                     return(stilltogo);
02091                 }
02092                 maxtogo = stilltogo;
02093                 u = t1 + *t1;
02094                 m = term + *term - csize;
02095             }
02096             t += *t;
02097         }
02098         AT.NestPoin--;
02099     }
02100 }
02101 /*
02102 Now we look at the ground level
02103 */
02104 C = cbuf+AM.rbufnum;
02105 t = term + 1;
02106 while ( t < m ) {
02107     if ( *t != TABLESTUB ) { t += t[1]; continue; }
02108     funnum = -t[FUNHEAD];
02109     if ( ( funnum < FUNCTION )
02110         || ( funnum >= FUNCTION+WILDOFFSET )
02111         || ( ( T = functions[funnum-FUNCTION].tab1 ) == 0 )
02112         || ( T->sparse == 0 )
02113         || ( T->spare == 0 ) ) { t += t[1]; continue; }
02114     numfuns = C->lhs[level][1] - 3;
02115     funs = C->lhs[level] + 3;
02116     if ( numfuns > 0 ) {
02117         for ( i = 0; i < numfuns; i++ ) {
02118             if ( funs[i] == funnum ) break;
02119         }
02120         if ( i >= numfuns ) { t += t[1]; continue; }
02121     }
02122     r = t + t[1];
02123     AT.NestPoin->funsize = t + 1;
02124     t1 = t;
02125     t += FUNHEAD + 1;
02126     /*
02127         Test whether the table catches
02128         Test 1: index arguments and range. isp will be the number
02129         of the element in the table.
02130     */
02131     T = T->sparse;
02132     #ifdef WITHPTHREADS
02133     Tpattern = T->pattern[identity];
02134     #else
02135     Tpattern = T->pattern;
02136     #endif
02137     p = Tpattern+FUNHEAD+1;
02138     for ( i = 0; i < ABS(T->numind); i++, t += 2 ) {
02139         if ( *t != -SNUMBER ) break;
02140     }
02141     if ( i < ABS(T->numind) ) { t = r; continue; }
02142     isp = FindTableTree(T,t1+FUNHEAD+1,2);
02143     if ( isp < 0 ) { t = r; continue; }
02144     rhsnumber = T->tablepointers[isp+ABS(T->numind)];
02145     #if ( TABLEEXTENSION == 2 )
02146     tbufnum = T->bunum;
02147     #else
02148     tbufnum = T->tablepointers[isp+ABS(T->numind)+1];
02149     #endif
02150     t = t1+FUNHEAD+2;
02151     ii = ABS(T->numind);
02152     while ( --ii >= 0 ) {
02153         *p = *t; t += 2; p += 2;

```



```

02154     }
02155     /*
02156         If there are more arguments we have to do some
02157         pattern matching. This should be easy. We adapted the
02158         pattern, so that the array indices match already.
02159     */
02160     #ifdef WITHPTHREADS
02161         AN.FullProto = T->prototype[identity];
02162     #else
02163         AN.FullProto = T->prototype;
02164     #endif
02165     AN.WildValue = AN.FullProto + SUBEXPSIZE;
02166     AN.WildStop = AN.FullProto+AN.FullProto[1];
02167     ClearWild(BHEAD0);
02168     AN.RepFunNum = 0;
02169     AN.RepFunList = AN.EndNest;
02170     AT.WorkPointer = (WORD *)(((UBYTE *) (AN.EndNest)) + AM.MaxTer/2);
02171     /*
02172         The RepFunList is after the term but not very relevant.
02173         We need because MatchFunction uses it
02174     */
02175     if ( AT.WorkPointer + t1[1] >= AT.WorkTop ) { MesWork(); }
02176     wilds = 0;
02177     w = AT.WorkPointer;
02178     *w++ = -t1[FUNHEAD];
02179     *w++ = t1[1] - 1;
02180     for ( i = 2; i < FUNHEAD; i++ ) *w++ = t1[i];
02181     t = t1 + FUNHEAD+1;
02182     while ( t < r ) *w++ = *t++;
02183     t = AT.WorkPointer;
02184     AT.WorkPointer = w;
02185     if ( MatchFunction(BHEAD Tpattern,t,&wilds) > 0 ) {
02186     /*
02187         Here we caught one. Now we should worry about:
02188         1: inserting the subexpression pointer with its wildcards
02189         2: NestPoin because we may not be at the lowest level
02190         The function starts at t1.
02191     */
02192     #ifdef WITHPTHREADS
02193         m1 = T->prototype[identity];
02194     #else
02195         m1 = T->prototype;
02196     #endif
02197     m1[2] = rhsnumber;
02198     m1[4] = tbufnum;
02199     t = t1;
02200     j = t[1];
02201     i = m1[1];
02202     if ( j > i ) {
02203         j = i - j;
02204         NCOPY(t,m1,i);
02205         m1 = AN.EndNest;
02206         while ( r < m1 ) *t++ = *r++;
02207         AN.EndNest = t;
02208         *term += j;
02209         NN = AT.NestPoin;
02210         while ( NN > AT.Nest ) {
02211             NN--;
02212             NN->termsize[0] += j;
02213             NN->funsize[1] += j;
02214             NN->argsize[0] += j;
02215             NN->funsize[2] |= DIRTYFLAG;
02216             NN->argsize[1] |= DIRTYFLAG;
02217         }
02218         m += j;
02219     }
02220     else if ( j < i ) {
02221         j = i-j;
02222         t = AN.EndNest;
02223         while ( t >= r ) { t[j] = *t; t--; }
02224         t = t1;
02225         NCOPY(t,m1,i);
02226         AN.EndNest += j;
02227         *term += j;
02228         NN = AT.NestPoin;
02229         while ( NN > AT.Nest ) {
02230             NN--;
02231             NN->termsize[0] += j;
02232             NN->funsize[1] += j;
02233             NN->argsize[0] += j;
02234             NN->funsize[2] |= DIRTYFLAG;
02235             NN->argsize[1] |= DIRTYFLAG;
02236         }
02237         m += j;
02238     }
02239     else {
02240         NCOPY(t,m1,j);

```

```

02241         }
02242         r = t1 + t1[1];
02243         maxtogo--;
02244         if ( maxtogo <= 0 ) return(maxtogo);
02245     }
02246     t = r;
02247 }
02248 return(maxtogo);
02249 }
02250
02251 /*
02252 #] ApplyExec :
02253 #[ ApplyReset :
02254 */
02255
02256 void ApplyReset(WORD level)
02257 {
02258     WORD *funcs, numfuncs;
02259     TABLES T;
02260     int i, j;
02261     CBUF *C = cbuf+AM.rbufnum;
02262
02263     numfuncs = C->lhs[level][1] - 2;
02264     funcs = C->lhs[level] + 2;
02265     if ( numfuncs > 0 ) {
02266         for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
02267             if ( ( T = functions[i].tabl ) != 0 ) {
02268                 for ( j = 0; j < numfuncs; j++ ) {
02269                     if ( i == (funcs[j]-FUNCTION) && T->spare ) {
02270                         FlipTable(&(functions[i]),1);
02271                         break;
02272                     }
02273                 }
02274             }
02275         }
02276     }
02277     else {
02278         for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
02279             if ( ( T = functions[i].tabl ) != 0 ) {
02280                 if ( T->spare ) FlipTable(&(functions[i]),1);
02281             }
02282         }
02283     }
02284 }
02285
02286 /*
02287 #] ApplyReset :
02288 #[ TableReset :
02289 */
02290
02291 void TableReset(void)
02292 {
02293     TABLES T;
02294     int i;
02295
02296     for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
02297         if ( ( T = functions[i].tabl ) != 0 && T->spare && T->mode == 0 ) {
02298             functions[i].tabl = T->spare;
02299         }
02300     }
02301 }
02302
02303 /*
02304 #] TableReset :
02305 #[ LoadTableElement :
02306 ?????
02307 int LoadTableElement(DBASE *d, TABLE *T, WORD num)
02308 {
02309 }
02310
02311 #] LoadTableElement :
02312 #[ ReleaseTB :
02313
02314 Releases all TableBases
02315 */
02316
02317 int ReleaseTB(void)
02318 {
02319     DBASE *d;
02320     int i;
02321     for ( i = NumTableBases - 1; i >= 0; i-- ) {
02322         d = tablebases+i;
02323         fclose(d->handle);
02324         FreeTableBase(d);
02325     }
02326     return(0);
02327 }

```

```

02328
02329 /*
02330      #] ReleaseTB :
02331 */

```

## 4.142 threads.c File Reference

### 4.142.1 Detailed Description

Routines for the interface of FORM with the pthreads library

This is the main part of the parallelization of TFORM. It is important to also look in the files [structs.h](#) and [variable.h](#) because the treatment of the A and B structs is essential (these structs are used by means of the macros AM, AP, AC, AS, AR, AT, AN, AO and AX). Also the definitions and use of the macros BHEAD and PHEAD should be looked up.

The sources are set up in such a way that if WITHPTHREADS isn't defined there is no trace of pthread parallelization. The reason is that TFORM is far more memory hungry than sequential FORM.

Special attention should also go to the locks. The proper use of the locks is essential and determines whether TFORM can work at all. We use the LOCK/UNLOCK macros which are empty in the case of sequential FORM. These locks are at many places in the source files when workers can interfere with each others data or with the data of the master.

Definition in file [threads.c](#).

## 4.143 threads.c

[Go to the documentation of this file.](#)

```

00001
00022 /* #] License : */
00023 /*
00024 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00025 *   When using this file you are requested to refer to the publication
00026 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00027 *   This is considered a matter of courtesy as the development was paid
00028 *   for by FOM the Dutch physics granting agency and we would like to
00029 *   be able to track its scientific use to convince FOM of its value
00030 *   for the community.
00031 *
00032 *   This file is part of FORM.
00033 *
00034 *   FORM is free software: you can redistribute it and/or modify it under the
00035 *   terms of the GNU General Public License as published by the Free Software
00036 *   Foundation, either version 3 of the License, or (at your option) any later
00037 *   version.
00038 *
00039 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00040 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00041 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00042 *   details.
00043 *
00044 *   You should have received a copy of the GNU General Public License along
00045 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00046 */
00047 /* #] License : */
00048
00049 #ifdef WITHPTHREADS
00050
00051 #define WHOLEBRACKETS
00052 /*
00053 *   #] Variables :
00054 *
00055 *   The sortbot additions are from 17-may-2007 and after. They constitute
00056 *   an attempt to make the final merge sorting faster for the master.
00057 *   This way the master has only one compare per term.

```

```

00058     It does add some complexity, but the final merge routine (MasterMerge)
00059     is much simpler for the sortbots. On the other hand the original merging is
00060     for a large part a copy of the MergePatches routine in sort.c and hence
00061     even though complex the bad part has been thoroughly debugged.
00062 */
00063
00064 #include "form3.h"
00065
00066 #ifdef WITH_ALARM
00067 // This is only required if we are blocking SIG_ALARM in the worker threads.
00068 #include <signal.h>
00069 #endif
00070
00071 #ifdef WITHFLOAT
00072 #include <gmp.h>
00073
00074 int PackFloat(WORD *,mpf_t);
00075 int UnpackFloat(mpf_t, WORD *);
00076 void RatToFloat(mpf_t result, UWORD *format, int ratsize);
00077 #endif
00078
00079 static int numberofthreads;
00080 static int numberofworkers;
00081 static int identityofthreads = 0;
00082 static int *listofavailables;
00083 static int topofavailables = 0;
00084 static pthread_key_t identitykey;
00085 static INILOCK(numberofthreadslock)
00086 static INILOCK(availabilitylock)
00087 static pthread_t *threadpointers = 0;
00088 static pthread_mutex_t *wakeuplocks;
00089 static pthread_mutex_t *wakeupmasterthreadlocks;
00090 static pthread_cond_t *wakeupconditions;
00091 static pthread_condattr_t *wakeupconditionattributes;
00092 static int *wakeup;
00093 static int *wakeupmasterthread;
00094 static INILOCK(wakeupmasterlock)
00095 static pthread_cond_t wakeupmasterconditions = PTHREAD_COND_INITIALIZER;
00096 static pthread_cond_t *wakeupmasterthreadconditions;
00097 static int wakeupmaster = 0;
00098 static int identityretval;
00099 /* static INILOCK(clearclocklock) */
00100 static LONG *timerinfo;
00101 static LONG *sumtimerinfo;
00102 static int numberclaimed;
00103
00104 static THREADBUCKET **threadbuckets, **freebuckets;
00105 static int numthreadbuckets;
00106 static int numberoffullbuckets;
00107
00108 /* static int numberbusy = 0; */
00109
00110 INILOCK(dummylock)
00111 INIRWLOCK(dummyrwlock)
00112 static pthread_cond_t dummywakeupcondition = PTHREAD_COND_INITIALIZER;
00113
00114 #ifdef WITHSORTBOTS
00115 static POSITION SortBotPosition;
00116 static int numberofsortbots;
00117 static INILOCK(wakeupsortbotlock)
00118 static pthread_cond_t wakeupsortbotconditions = PTHREAD_COND_INITIALIZER;
00119 static int topsortbotavailables = 0;
00120 static LONG numberoffterms;
00121 #endif
00122
00123 /*
00124     #] Variables :
00125     #[ Identity :
00126         #[ StartIdentity :
00127     */
00128 void StartIdentity(void)
00129 {
00130     pthread_key_create(&identitykey,FinishIdentity);
00131 }
00132
00133 /*
00134     #] StartIdentity :
00135     #[ FinishIdentity :
00136     */
00137 void FinishIdentity(void *keyp)
00138 {
00139     DUMMYUSE(keyp);
00140     free(keyp);
00141 }
00142
00143 /*
00144     #] FinishIdentity :

```

```

00154         #] SetIdentity :
00155     */
00160 int SetIdentity(int *identityretval)
00161 {
00162     /*
00163     #ifdef _MSC_VER
00164         printf("addr %d\n",&numberofthreadslock);
00165         printf("size %d\n",sizeof(numberofthreadslock));
00166     #endif
00167     */
00168     LOCK(numberofthreadslock);
00169     *identityretval = identityofthreads++;
00170     UNLOCK(numberofthreadslock);
00171     pthread_setspecific(identitykey, (void *)identityretval);
00172     return(*identityretval);
00173 }
00174
00175 /*
00176     #] SetIdentity :
00177     #] WhoAmI :
00178     */
00179
00190 int WhoAmI(void)
00191 {
00192     int *identity;
00193     /*
00194     First a fast exit for when there is at most one thread
00195     */
00196     if ( identityofthreads <= 1 ) return(0);
00197     /*
00198     Now the reading of the key.
00199     According to the book the statement should read:
00200
00201     pthread_getspecific(identitykey, (void **)(&identity));
00202
00203     but according to the information in pthread.h it is:
00204     */
00205     identity = (int *)pthread_getspecific(identitykey);
00206     return(*identity);
00207 }
00208
00209 /*
00210     #] WhoAmI :
00211     #] BeginIdentities :
00212     */
00218 void BeginIdentities(void)
00219 {
00220     StartIdentity();
00221     SetIdentity(&identityretval);
00222 }
00223
00224 /*
00225     #] BeginIdentities :
00226     #] Identity :
00227     #] StartHandleLock :
00228     */
00235 void StartHandleLock(void)
00236 {
00237     AM.handlelock = dummyrwlock;
00238 }
00239
00240 /*
00241     #] StartHandleLock :
00242     #] StartAllThreads :
00243     */
00261 int StartAllThreads(int number)
00262 {
00263     int identity, j, dummy, mul;
00264     ALLPRIVATES *B;
00265     pthread_t thethread;
00266     identity = WhoAmI();
00267
00268     #ifdef WITHSORTBOTS
00269     timerinfo = (LONG *)Malloc1(sizeof(LONG)*number*2,"timerinfo");
00270     sumtimerinfo = (LONG *)Malloc1(sizeof(LONG)*number*2,"sumtimerinfo");
00271     for ( j = 0; j < number*2; j++ ) { timerinfo[j] = 0; sumtimerinfo[j] = 0; }
00272     mul = 2;
00273     #else
00274     timerinfo = (LONG *)Malloc1(sizeof(LONG)*number,"timerinfo");
00275     sumtimerinfo = (LONG *)Malloc1(sizeof(LONG)*number,"sumtimerinfo");
00276     for ( j = 0; j < number; j++ ) { timerinfo[j] = 0; sumtimerinfo[j] = 0; }
00277     mul = 1;
00278     #endif
00279
00280     listofavailables = (int *)Malloc1(sizeof(int)*(number+1),"listofavailables");
00281     threadpointers = (pthread_t *)Malloc1(sizeof(pthread_t)*number*mul,"threadpointers");
00282     AB = (ALLPRIVATES **)Malloc1(sizeof(ALLPRIVATES *)*number*mul,"Private structs");

```

```

00283
00284     wakeup = (int *)Malloca(sizeof(int)*number*mul,"wakeup");
00285     wakeuplocks = (pthread_mutex_t *)Malloca(sizeof(pthread_mutex_t)*number*mul,"wakeuplocks");
00286     wakeupconditions = (pthread_cond_t
*)Malloca(sizeof(pthread_cond_t)*number*mul,"wakeupconditions");
00287     wakeupconditionattributes = (pthread_condattr_t *)
00288         Malloca(sizeof(pthread_condattr_t)*number*mul,"wakeupconditionattributes");
00289
00290     wakeupmasterthread = (int *)Malloca(sizeof(int)*number*mul,"wakeupmasterthread");
00291     wakeupmasterthreadlocks = (pthread_mutex_t
*)Malloca(sizeof(pthread_mutex_t)*number*mul,"wakeupmasterthreadlocks");
00292     wakeupmasterthreadconditions = (pthread_cond_t
*)Malloca(sizeof(pthread_cond_t)*number*mul,"wakeupmasterthread");
00293
00294     numberofthreads = number;
00295     numberofworkers = number - 1;
00296     threadpointers[identity] = pthread_self();
00297     topofavailables = 0;
00298
00299 #ifdef WITH_ALARM
00300     /* During thread creation, we block SIGALRM on the main thread. The created
00301        threads will inherit this. This is required for #timeout to work properly
00302        in TFORM: only the main thread should receive SIGALRM. */
00303     sigset_t sig_set;
00304     sigemptyset(&sig_set);
00305     sigaddset(&sig_set, SIGALRM);
00306     pthread_sigmask(SIG_BLOCK, &sig_set, NULL);
00307 #endif
00308
00309     for ( j = 1; j < number; j++ ) {
00310         if ( pthread_create(&thethread,NULL,RunThread,(void *)(&dummy)) )
00311             goto failure;
00312     }
00313     /*
00314     Now we initialize the master at the same time that the workers are doing so.
00315     */
00316     B = InitializeOneThread(identity);
00317     AR.infile = &(AR.Fscr[0]);
00318     AR.outfile = &(AR.Fscr[1]);
00319     AR.hidefile = &(AR.Fscr[2]);
00320     AM.sbuflock = dummylock;
00321     AS.inputslock = dummylock;
00322     AS.outputslock = dummylock;
00323     AS.MaxExprSizeLock = dummylock;
00324     AP.PreVarLock = dummylock;
00325     AC.halfmodlock = dummylock;
00326     MakeThreadBuckets(number,0);
00327     /*
00328     Now we wait for the workers to finish their startup.
00329     We don't want to initialize the sortbots yet and run the risk that
00330     some of them may end up with a lower number than one of the workers.
00331     */
00332     MasterWaitAll();
00333 #ifdef WITHSORTBOTS
00334     if ( numberofworkers > 2 ) {
00335         numberofsortbots = numberofworkers-2;
00336         for ( j = numberofworkers+1; j < 2*numberofworkers-1; j++ ) {
00337             if ( pthread_create(&thethread,NULL,RunSortBot,(void *)(&dummy)) )
00338                 goto failure;
00339         }
00340     }
00341     else {
00342         numberofsortbots = 0;
00343     }
00344     MasterWaitAllSortBots();
00345     DefineSortBotTree();
00346 #endif
00347     IniSortBlocks(number-1);
00348     AS.MasterSort = 0;
00349     AM.storefilelock = dummylock;
00350
00351 #ifdef WITH_ALARM
00352     /* Now we allow the main thread to receive SIGALRM again. */
00353     pthread_sigmask(SIG_UNBLOCK, &sig_set, NULL);
00354 #endif
00355
00356     /*
00357     MesPrint("AB = %x %x %x  %d",AB[0],AB[1],AB[2], identityofthreads);
00358     */
00359     return(0);
00360 failure:
00361     MesPrint("Cannot start %d threads",number);
00362     Terminate(-1);
00363     return(-1);
00364 }
00365
00366 /*

```

```

00367     #] StartAllThreads :
00368     #[ InitializeOneThread :
00369     */
00373     UBYTE *scratchname[] = { (UBYTE *)"scratchsize",
00374                             (UBYTE *)"scratchsize",
00375                             (UBYTE *)"hidesize" };
00402     ALLPRIVATES *InitializeOneThread(int identity)
00403     {
00404         WORD *t, *ScratchBuf;
00405         int i, j, bsize, *bp;
00406         LONG ScratchSize[3], IOSize;
00407         ALLPRIVATES *B;
00408         UBYTE *s;
00409
00410         wakeup[identity] = 0;
00411         wakeuplocks[identity] = dummylock;
00412         pthread_condattr_init(&(wakeupconditionattributes[identity]));
00413         pthread_condattr_setpshared(&(wakeupconditionattributes[identity]),PTHREAD_PROCESS_PRIVATE);
00414         wakeupconditions[identity] = dummywakeupcondition;
00415         pthread_cond_init(&(wakeupconditions[identity]),&(wakeupconditionattributes[identity]));
00416         wakeupmasterthread[identity] = 0;
00417         wakeupmasterthreadlocks[identity] = dummylock;
00418         wakeupmasterthreadconditions[identity] = dummywakeupcondition;
00419
00420         bsize = sizeof(ALLPRIVATES);
00421         bsize = (bsize+sizeof(int)-1)/sizeof(int);
00422         B = (ALLPRIVATES *)Malloc1(sizeof(int)*bsize,"B struct");
00423         for ( bp = (int *)B, j = 0; j < bsize; j++ ) *bp++ = 0;
00424
00425         AB[identity] = B;
00426     /*
00427         12-jun-2007 JV:
00428         For the timing one has to know a few things:
00429         The POSIX standard is that there is only a single process ID and that
00430         getrusage returns the time of all the threads together.
00431         Under Linux there are two methods though: The older LinuxThreads and NPTL.
00432         LinuxThreads gives each thread its own process id. This makes that we
00433         can time the threads with getrusage, and hence this was done. Under NPTL
00434         this has been 'corrected' and suddenly getrusage doesn't work anymore the
00435         way it used to. Now we need
00436         clock_gettime(CLOCK_THREAD_CPUTIME_ID,&timing)
00437         which is declared in <time.h> and we need -lrt extra in the link statement.
00438         (this is at least the case on blade02 at DESY-Zeuthen).
00439         See also the code in tools.c at the routine Timer.
00440         We may still have to include more stuff there.
00441     */
00442     if ( identity > 0 ) TimeCPU(0);
00443
00444     #ifdef WITHSORTBOTS
00445
00446     if ( identity > numberofworkers ) {
00447     /*
00448         Some workspace is needed when we have a PolyFun and we have to add
00449         two terms and the new result is going to be longer than the old result.
00450     */
00451         LONG length = AM.WorkSize*sizeof(WORD)/8+AM.MaxTer*2;
00452         AT.WorkSpace = (WORD *)Malloc1(length,"WorkSpace");
00453         AT.WorkTop = AT.WorkSpace + length/sizeof(WORD);
00454         AT.WorkPointer = AT.WorkSpace;
00455         AT.identity = identity;
00456     /*
00457         The SB struct gets treated in IniSortBlocks.
00458         The SortBotIn variables will be defined DefineSortBotTree.
00459     */
00460         if ( AN.ScratchC == 0 ) {
00461             AN.ScratchC = (UWORD *)Malloc1(2*(AM.MaxTal+2)*sizeof(UWORD),"Scratch in SortBot");
00462         }
00463         AT.SS = (SORTING *)Malloc1(sizeof(SORTING),"dummy sort buffer");
00464         AT.SS->PolyFlag = 0;
00465
00466         AT.comsym[0] = 8;
00467         AT.comsym[1] = SYMBOL;
00468         AT.comsym[2] = 4;
00469         AT.comsym[3] = 0;
00470         AT.comsym[4] = 1;
00471         AT.comsym[5] = 1;
00472         AT.comsym[6] = 1;
00473         AT.comsym[7] = 3;
00474         AT.comnum[0] = 4;
00475         AT.comnum[1] = 1;
00476         AT.comnum[2] = 1;
00477         AT.comnum[3] = 3;
00478         AT.comfun[0] = FUNHEAD+4;
00479         AT.comfun[1] = FUNCTION;
00480         AT.comfun[2] = FUNHEAD;
00481         AT.comfun[3] = 0;
00482     #if FUNHEAD > 3

```

```

00483         for ( i = 4; i <= FUNHEAD; i++ )
00484             AT.comfun[i] = 0;
00485     #endif
00486     AT.comfun[FUNHEAD+1] = 1;
00487     AT.comfun[FUNHEAD+2] = 1;
00488     AT.comfun[FUNHEAD+3] = 3;
00489     AT.comind[0] = 7;
00490     AT.comind[1] = INDEX;
00491     AT.comind[2] = 3;
00492     AT.comind[3] = 0;
00493     AT.comind[4] = 1;
00494     AT.comind[5] = 1;
00495     AT.comind[6] = 3;
00496
00497     AT.inprimelist = -1;
00498     AT.sizeprimelist = 0;
00499     AT.primelist = 0;
00500     AT.bracketinfo = 0;
00501
00502     AR.CompareRoutine = (COMPAREDUMMY) (&Compare1);
00503
00504     AR.sLevel = 0;
00505     AR.wranfia = 0;
00506     AR.wranfcall = 0;
00507     AR.wranfnpair1 = NPAIR1;
00508     AR.wranfnpair2 = NPAIR2;
00509     AN.NumFunSorts = 5;
00510     AN.MaxFunSorts = 5;
00511     AN.SplitScratch = 0;
00512     AN.SplitScratchSize = AN.InScratch = 0;
00513     AN.SplitScratch1 = 0;
00514     AN.SplitScratchSize1 = AN.InScratch1 = 0;
00515
00516     AN.FunSorts = (SORTING **)Malloc1((AN.NumFunSorts+1)*sizeof(SORTING *),"FunSort pointers");
00517     for ( i = 0; i <= AN.NumFunSorts; i++ ) AN.FunSorts[i] = 0;
00518     AN.FunSorts[0] = AT.S0 = AT.SS;
00519     AN.idfunctionflag = 0;
00520     AN.tryterm = 0;
00521
00522     return(B);
00523 }
00524 if ( identity == 0 && AN.SoScratC == 0 ) {
00525     AN.SoScratC = (UWORD *)Malloc1(2*(AM.MaxTal+2)*sizeof(UWORD),"Scratch in SortBot");
00526 }
00527 #endif
00528 AR.CurDum = AM.IndDum;
00529 for ( j = 0; j < 3; j++ ) {
00530     if ( identity == 0 ) {
00531         if ( j == 2 ) {
00532             ScratchSize[j] = AM.HideSize;
00533         }
00534         else {
00535             ScratchSize[j] = AM.ScratSize;
00536         }
00537         if ( ScratchSize[j] < 10*AM.MaxTer ) ScratchSize[j] = 10 * AM.MaxTer;
00538     }
00539     else {
00540         /*
00541             ScratchSize[j] = AM.ScratSize / (numberofthreads-1);
00542             ScratchSize[j] = ScratchSize[j] / 20;
00543             if ( ScratchSize[j] < 10*AM.MaxTer ) ScratchSize[j] = 10 * AM.MaxTer;
00544         */
00545         if ( j == 1 ) ScratchSize[j] = AM.ThreadScratOutSize;
00546         else ScratchSize[j] = AM.ThreadScratSize;
00547         if ( ScratchSize[j] < 4*AM.MaxTer ) ScratchSize[j] = 4 * AM.MaxTer;
00548         AR.Fscr[j].name = 0;
00549     }
00550     ScratchSize[j] = ( ScratchSize[j] + 255 ) / 256;
00551     ScratchSize[j] = ScratchSize[j] * 256;
00552     ScratchBuf = (WORD *)Malloc1(ScratchSize[j]*sizeof(WORD),(char *) (scratchname[j]));
00553     AR.Fscr[j].POsize = ScratchSize[j] * sizeof(WORD);
00554     AR.Fscr[j].POfull = AR.Fscr[j].POfill = AR.Fscr[j].PObuffer = ScratchBuf;
00555     AR.Fscr[j].POstop = AR.Fscr[j].PObuffer + ScratchSize[j];
00556     PUTZERO(AR.Fscr[j].POposition);
00557     AR.Fscr[j].pthreadlock = dummylock;
00558     AR.Fscr[j].wPOsize = AR.Fscr[j].POsize;
00559     AR.Fscr[j].wPObuffer = AR.Fscr[j].PObuffer;
00560     AR.Fscr[j].wPOfill = AR.Fscr[j].POfill;
00561     AR.Fscr[j].wPOfull = AR.Fscr[j].POfull;
00562     AR.Fscr[j].wPOstop = AR.Fscr[j].POstop;
00563 }
00564 AR.InInBuf = 0;
00565 AR.InHiBuf = 0;
00566 AR.Fscr[0].handle = -1;
00567 AR.Fscr[1].handle = -1;
00568 AR.Fscr[2].handle = -1;
00569 AR.FoStage4[0].handle = -1;

```



```

00570     AR.FoStage4[1].handle = -1;
00571     IOsize = AM.S0->file.POSize;
00572 #ifdef WITHZLIB
00573     AR.FoStage4[0].ziosize = IOsize;
00574     AR.FoStage4[1].ziosize = IOsize;
00575     AR.FoStage4[0].ziobuffer = 0;
00576     AR.FoStage4[1].ziobuffer = 0;
00577 #endif
00578     AR.FoStage4[0].POsize = ((IOsize+sizeof(WORD)-1)/sizeof(WORD))*sizeof(WORD);
00579     AR.FoStage4[1].POsize = ((IOsize+sizeof(WORD)-1)/sizeof(WORD))*sizeof(WORD);
00580
00581     AR.hidefile = &(AR.Fscr[2]);
00582     AR.StoreData.Handle = -1;
00583     AR.SortType = AC.SortType;
00584
00585     AN.IndDum = AM.IndDum;
00586
00587     if ( identity > 0 ) {
00588         s = (UBYTE *) (FG.fname); i = 0;
00589         while ( *s ) { s++; i++; }
00590         s = (UBYTE *) Malloc1(sizeof(char)*(i+12),"name for Fscr[0] file");
00591         snprintf((char *)s,i+12,"%s.%d",FG.fname,identity);
00592         s[i-3] = 's'; s[i-2] = 'c'; s[i-1] = '0';
00593         AR.Fscr[0].name = (char *)s;
00594         s = (UBYTE *) (FG.fname); i = 0;
00595         while ( *s ) { s++; i++; }
00596         s = (UBYTE *) Malloc1(sizeof(char)*(i+12),"name for Fscr[1] file");
00597         snprintf((char *)s,i+12,"%s.%d",FG.fname,identity);
00598         s[i-3] = 's'; s[i-2] = 'c'; s[i-1] = '1';
00599         AR.Fscr[1].name = (char *)s;
00600     }
00601
00602     AR.CompressBuffer = (WORD *) Malloc1((AM.CompressSize+10)*sizeof(WORD),"compresssize");
00603     AR.ComprTop = AR.CompressBuffer + AM.CompressSize;
00604     AR.CompareRoutine = (COMPAREDUMMY) (&Compare1);
00605 /*
00606     Here we make all allocations for the struct AT
00607     (which is AB[identity].T or B->T with B = AB+identity).
00608 */
00609     AT.WorkSpace = (WORD *) Malloc1(AM.WorkSize*sizeof(WORD),"WorkSpace");
00610     AT.WorkTop = AT.WorkSpace + AM.WorkSize;
00611     AT.WorkPointer = AT.WorkSpace;
00612
00613     AT.Nest = (NESTING) Malloc1((LONG) sizeof(struct NeStIng)*AM.maxFlevels,"functionlevels");
00614     AT.NestStop = AT.Nest + AM.maxFlevels;
00615     AT.NestPoin = AT.Nest;
00616
00617     AT.WildMask = (WORD *) Malloc1((LONG) AM.MaxWildcards*sizeof(WORD),"maxwildcards");
00618
00619     LOCK(availabilitylock);
00620     AT.ebufnum = inicbufs(); /* Buffer for extras during execution */
00621     AT.fbufnum = inicbufs(); /* Buffer for caching in factorization */
00622     AT.allbufnum = inicbufs(); /* Buffer for id,all */
00623     AT.aebufnum = inicbufs(); /* Buffer for id,all */
00624     UNLOCK(availabilitylock);
00625
00626     AT.RepCount = (int *) Malloc1((LONG) ((AM.RepMax+3)*sizeof(int)),"repeat buffers");
00627     AN.RepPoint = AT.RepCount;
00628     AN.polysortflag = 0;
00629     AN.subsubveto = 0;
00630     AN.tryterm = 0;
00631     AT.RepTop = AT.RepCount + AM.RepMax;
00632
00633     AT.WildArgTaken = (WORD *) Malloc1((LONG) AC.WildcardBufferSize*sizeof(WORD)/2
00634         , "argument list names");
00635     AT.WildcardBufferSize = AC.WildcardBufferSize;
00636     AT.previousEfactor = 0;
00637
00638     AT.identity = identity;
00639     AT.LoadBalancing = 0;
00640 /*
00641     Still to do: the SS stuff.
00642         the Fscr[3]
00643         the FoStage4[2]
00644 */
00645     if ( AT.WorkSpace == 0 ||
00646         AT.Nest == 0 ||
00647         AT.WildMask == 0 ||
00648         AT.RepCount == 0 ||
00649         AT.WildArgTaken == 0 ) goto OnError;
00650 /*
00651     And initializations
00652 */
00653     AT.comsym[0] = 8;
00654     AT.comsym[1] = SYMBOL;
00655     AT.comsym[2] = 4;
00656     AT.comsym[3] = 0;

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```

00657     AT.comsym[4] = 1;
00658     AT.comsym[5] = 1;
00659     AT.comsym[6] = 1;
00660     AT.comsym[7] = 3;
00661     AT.comnum[0] = 4;
00662     AT.comnum[1] = 1;
00663     AT.comnum[2] = 1;
00664     AT.comnum[3] = 3;
00665     AT.comfun[0] = FUNHEAD+4;
00666     AT.comfun[1] = FUNCTION;
00667     AT.comfun[2] = FUNHEAD;
00668     AT.comfun[3] = 0;
00669     #if FUNHEAD > 3
00670         for ( i = 4; i <= FUNHEAD; i++ )
00671             AT.comfun[i] = 0;
00672     #endif
00673     AT.comfun[FUNHEAD+1] = 1;
00674     AT.comfun[FUNHEAD+2] = 1;
00675     AT.comfun[FUNHEAD+3] = 3;
00676     AT.comind[0] = 7;
00677     AT.comind[1] = INDEX;
00678     AT.comind[2] = 3;
00679     AT.comind[3] = 0;
00680     AT.comind[4] = 1;
00681     AT.comind[5] = 1;
00682     AT.comind[6] = 3;
00683     AT.locwildvalue[0] = SUBEXPRESSION;
00684     AT.locwildvalue[1] = SUBEXPSIZE;
00685     for ( i = 2; i < SUBEXPSIZE; i++ ) AT.locwildvalue[i] = 0;
00686     AT.mulpat[0] = TYPMULT;
00687     AT.mulpat[1] = SUBEXPSIZE+3;
00688     AT.mulpat[2] = 0;
00689     AT.mulpat[3] = SUBEXPRESSION;
00690     AT.mulpat[4] = SUBEXPSIZE;
00691     AT.mulpat[5] = 0;
00692     AT.mulpat[6] = 1;
00693     for ( i = 7; i < SUBEXPSIZE+5; i++ ) AT.mulpat[i] = 0;
00694     AT.proexp[0] = SUBEXPSIZE+4;
00695     AT.proexp[1] = EXPRESSION;
00696     AT.proexp[2] = SUBEXPSIZE;
00697     AT.proexp[3] = -1;
00698     AT.proexp[4] = 1;
00699     for ( i = 5; i < SUBEXPSIZE+1; i++ ) AT.proexp[i] = 0;
00700     AT.proexp[SUBEXPSIZE+1] = 1;
00701     AT.proexp[SUBEXPSIZE+2] = 1;
00702     AT.proexp[SUBEXPSIZE+3] = 3;
00703     AT.proexp[SUBEXPSIZE+4] = 0;
00704     AT.dummysubexp[0] = SUBEXPRESSION;
00705     AT.dummysubexp[1] = SUBEXPSIZE+4;
00706     for ( i = 2; i < SUBEXPSIZE; i++ ) AT.dummysubexp[i] = 0;
00707     AT.dummysubexp[SUBEXPSIZE] = WILDDUMMY;
00708     AT.dummysubexp[SUBEXPSIZE+1] = 4;
00709     AT.dummysubexp[SUBEXPSIZE+2] = 0;
00710     AT.dummysubexp[SUBEXPSIZE+3] = 0;
00711
00712     AT.MinVecArg[0] = 7+ARGHEAD;
00713     AT.MinVecArg[ARGHEAD] = 7;
00714     AT.MinVecArg[1+ARGHEAD] = INDEX;
00715     AT.MinVecArg[2+ARGHEAD] = 3;
00716     AT.MinVecArg[3+ARGHEAD] = 0;
00717     AT.MinVecArg[4+ARGHEAD] = 1;
00718     AT.MinVecArg[5+ARGHEAD] = 1;
00719     AT.MinVecArg[6+ARGHEAD] = -3;
00720     t = AT.FunArg;
00721     *t++ = 4+ARGHEAD+FUNHEAD;
00722     for ( i = 1; i < ARGHEAD; i++ ) *t++ = 0;
00723     *t++ = 4+FUNHEAD;
00724     *t++ = 0;
00725     *t++ = FUNHEAD;
00726     for ( i = 2; i < FUNHEAD; i++ ) *t++ = 0;
00727     *t++ = 1; *t++ = 1; *t++ = 3;
00728
00729     AT.inprimelist = -1;
00730     AT.sizeprimelist = 0;
00731     AT.primelist = 0;
00732     AT.nfac = AT.nBer = 0;
00733     AT.factorials = 0;
00734     AT.bernoullis = 0;
00735     AR.wranfia = 0;
00736     AR.wranfcall = 0;
00737     AR.wranfnpair1 = NPAIR1;
00738     AR.wranfnpair2 = NPAIR2;
00739     AR.wranfseed = 0;
00740     AN.SplitScratch = 0;
00741     AN.SplitScratchSize = AN.InScratch = 0;
00742     AN.SplitScratch1 = 0;
00743     AN.SplitScratchSize1 = AN.InScratch1 = 0;

```

```

00744 /*
00745     Now the sort buffers. They depend on which thread. The master
00746     inherits the sortbuffer from AM.S0
00747 */
00748     if ( identity == 0 ) {
00749         AT.S0 = AM.S0;
00750     }
00751     else {
00752 /*
00753         For the moment we don't have special settings.
00754         They may become costly in virtual memory.
00755 */
00756         AT.S0 = AllocSort(AM.S0->LargeSize*sizeof(WORD)/numberofworkers
00757             ,AM.S0->SmallSize*sizeof(WORD)/numberofworkers
00758             ,AM.S0->SmallEsize*sizeof(WORD)/numberofworkers
00759             ,AM.S0->TermsInSmall
00760             ,AM.S0->MaxPatches
00761 /*
00762             ,AM.S0->MaxPatches/numberofworkers */
00763             ,AM.S0->MaxFpatches/numberofworkers
00764             ,AM.S0->file.POsize
00765             ,0);
00766     }
00767     AR.CompressPointer = AR.CompressBuffer;
00768 /*
00769     Install the store caches (15-aug-2006 JV)
00770 */
00771     AT.StoreCache = AT.StoreCacheAlloc = 0;
00772     if ( AM.NumStoreCaches > 0 ) {
00773         STORECACHE sa, sb;
00774         LONG size;
00775         size = sizeof(struct StOrEcAcHe)+AM.SizeStoreCache;
00776         size = ((size-1)/sizeof(size_t)+1)*sizeof(size_t);
00777         AT.StoreCacheAlloc = (STORECACHE)Malloc1(size*AM.NumStoreCaches,"StoreCaches");
00778         sa = AT.StoreCache = AT.StoreCacheAlloc;
00779         for ( i = 0; i < AM.NumStoreCaches; i++ ) {
00780             sb = (STORECACHE)(void *) ((UBYTE *)sa+size);
00781             if ( i == AM.NumStoreCaches-1 ) {
00782                 sa->next = 0;
00783             }
00784             else {
00785                 sa->next = sb;
00786             }
00787             SETBASEPOSITION(sa->position,-1);
00788             SETBASEPOSITION(sa->toppos,-1);
00789             sa = sb;
00790         }
00791     }
00792     ReserveTempFiles(2);
00793     return(B);
00794 OnError:;
00795     MLOCK(ErrorMessageLock);
00796     MesPrint("Error initializing thread %d",identity);
00797     MUNLOCK(ErrorMessageLock);
00798     Terminate(-1);
00799     return(B);
00800 }
00801
00802 /*
00803     #] InitializeOneThread :
00804     #[ FinalizeOneThread :
00805 */
00816 void FinalizeOneThread(int identity)
00817 {
00818     timerinfo[identity] = TimeCPU(1);
00819 }
00820
00821 /*
00822     #] FinalizeOneThread :
00823     #[ ClearAllThreads :
00824 */
00831 void ClearAllThreads(void)
00832 {
00833     int i;
00834     MasterWaitAll();
00835     for ( i = 1; i <= numberofworkers; i++ ) {
00836         WakeupThread(i,CLEARCLOCK);
00837     }
00838 #ifndef WITHSORTBOTS
00839     for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ ) {
00840         WakeupThread(i,CLEARCLOCK);
00841     }
00842 #endif
00843 }
00844
00845 /*
00846     #] ClearAllThreads :

```

```

00847     # [ TerminateAllThreads :
00848     /*
00855 void TerminateAllThreads(void)
00856 {
00857     int i;
00858     for ( i = 1; i <= numberofworkers; i++ ) {
00859         GetThread(i);
00860         WakeupThread(i, TERMINATETHREAD);
00861     }
00862 #ifdef WITHSORTBOTS
00863     for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ ) {
00864         WakeupThread(i, TERMINATETHREAD);
00865     }
00866 #endif
00867     for ( i = 1; i <= numberofworkers; i++ ) {
00868         pthread_join(threadpointers[i], NULL);
00869     }
00870 #ifdef WITHSORTBOTS
00871     for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ ) {
00872         pthread_join(threadpointers[i], NULL);
00873     }
00874 #endif
00875 }
00876
00877 /*
00878     #] TerminateAllThreads :
00879     # [ MakeThreadBuckets :
00880     /*
00906 int MakeThreadBuckets(int number, int par)
00907 {
00908     int i;
00909     LONG sizethreadbuckets;
00910     THREADBUCKET *thr;
00911     /*
00912     First we need a decent estimate. Not all terms should be maximal.
00913     Note that AM.MaxTer is in bytes!!!
00914     Maybe we should try to limit the size here a bit more effectively.
00915     This is a great consumer of memory.
00916     /*
00917     sizethreadbuckets = ( AC.ThreadBucketSize + 1 ) * AM.MaxTer + 2*sizeof(WORD);
00918     if ( AC.ThreadBucketSize >= 250 )     sizethreadbuckets /= 4;
00919     else if ( AC.ThreadBucketSize >= 90 )     sizethreadbuckets /= 3;
00920     else if ( AC.ThreadBucketSize >= 40 )     sizethreadbuckets /= 2;
00921     sizethreadbuckets /= sizeof(WORD);
00922
00923     if ( par == 0 ) {
00924         numthreadbuckets = 2*(number-1);
00925         threadbuckets = (THREADBUCKET **)Malloca1(numthreadbuckets*sizeof(THREADBUCKET
00926 *), "threadbuckets");
00927         freebuckets = (THREADBUCKET **)Malloca1(numthreadbuckets*sizeof(THREADBUCKET
00928 *), "threadbuckets");
00929     }
00930     if ( par > 0 ) {
00931         if ( sizethreadbuckets <= threadbuckets[0]->threadbuffersize ) return(0);
00932         for ( i = 0; i < numthreadbuckets; i++ ) {
00933             thr = threadbuckets[i];
00934             M_free(thr->deferbuffer, "deferbuffer");
00935         }
00936     }
00937     else {
00938         for ( i = 0; i < numthreadbuckets; i++ ) {
00939             threadbuckets[i] = (THREADBUCKET *)Malloca1(sizeof(THREADBUCKET), "threadbuckets");
00940             threadbuckets[i]->lock = dummylock;
00941         }
00942     }
00943     for ( i = 0; i < numthreadbuckets; i++ ) {
00944         thr = threadbuckets[i];
00945         thr->threadbuffersize = sizethreadbuckets;
00946         thr->free = BUCKETFREE;
00947         thr->deferbuffer = (POSITION *)Malloca1(2*sizethreadbuckets*sizeof(WORD)
00948             + (AC.ThreadBucketSize+1)*sizeof(POSITION), "deferbuffer");
00949         thr->threadbuffer = (WORD *) (thr->deferbuffer+AC.ThreadBucketSize+1);
00950         thr->compressbuffer = (WORD *) (thr->threadbuffer+sizethreadbuckets);
00951         thr->busy = BUCKETPREPARINGTERM;
00952         thr->usenum = thr->totnum = 0;
00953         thr->type = BUCKETDOINGTERMS;
00954     }
00955     return(0);
00956 }
00957
00958     #] MakeThreadBuckets :
00959     # [ GetTimerInfo :
00960     /*
00966 int GetTimerInfo(LONG** ti, LONG** sti)
00967 {

```

```

00968     *ti = timerinfo;
00969     *sti = sumtimerinfo;
00970 #ifdef WITHSORTBOTS
00971     return AM.totalnumberofthreads*2;
00972 #else
00973     return AM.totalnumberofthreads;
00974 #endif
00975 }
00976
00977 /*
00978     #] GetTimerInfo :
00979     #[ WriteTimerInfo :
00980 */
00981
00982 void WriteTimerInfo(LONG* ti, LONG* sti)
00983 {
00984     int i;
00985 #ifdef WITHSORTBOTS
00986     int max = AM.totalnumberofthreads*2;
00987 #else
00988     int max = AM.totalnumberofthreads;
00989 #endif
00990     for ( i=0; i<max; ++i ) {
00991         timerinfo[i] = ti[i];
00992         sumtimerinfo[i] = sti[i];
00993     }
00994 }
00995
01000 /*
01001     #] WriteTimerInfo :
01002     #[ GetWorkerTimes :
01003 */
01004
01005 LONG GetWorkerTimes(void)
01006 {
01007     LONG retval = 0;
01008     int i;
01009     for ( i = 1; i <= numberofworkers; i++ ) retval += timerinfo[i] + sumtimerinfo[i];
01010 #ifdef WITHSORTBOTS
01011     for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ )
01012         retval += timerinfo[i] + sumtimerinfo[i];
01013 #endif
01014     return(retval);
01015 }
01016
01021 /*
01022     #] GetWorkerTimes :
01023     #[ UpdateOneThread :
01024 */
01025
01026 int UpdateOneThread(int identity)
01027 {
01028     ALLPRIVATES *B = AB[identity], *B0 = AB[0];
01029     AR.GetFile = AR0.GetFile;
01030     AR.KeptInHold = AR0.KeptInHold;
01031     AR.CurExpr = AR0.CurExpr;
01032     AR.SortType = AC.SortType;
01033     if ( AT.WildcardBufferSize < AC.WildcardBufferSize ) {
01034         M_free(AT.WildArgTaken, "argument list names");
01035         AT.WildcardBufferSize = AC.WildcardBufferSize;
01036         AT.WildArgTaken = (WORD *)Malloc1((LONG)AC.WildcardBufferSize*sizeof(WORD)/2
01037             , "argument list names");
01038         if ( AT.WildArgTaken == 0 ) return(-1);
01039     }
01040     return(0);
01041 }
01042
01048 /*
01049     #] UpdateOneThread :
01050     #[ LoadOneThread :
01051 */
01052
01053 int LoadOneThread(int from, int identity, THREADBUCKET *thr, int par)
01054 {
01055     WORD *t1, *t2;
01056     ALLPRIVATES *B = AB[identity], *B0 = AB[from];
01057
01058     AR.DefPosition = AR0.DefPosition;
01059     AR.NoCompress = AR0.NoCompress;
01060     AR.gzipCompress = AR0.gzipCompress;
01061     AR.BraceOn = AR0.BraceOn;
01062     AR.CurDum = AR0.CurDum;
01063     AR.DeferFlag = AR0.DeferFlag;
01064     AR.TePos = 0;
01065     AR.sLevel = AR0.sLevel;
01066     AR.Stage4Name = AR0.Stage4Name;
01067     AR.GetOneFile = AR0.GetOneFile;
01068     AR.PolyFun = AR0.PolyFun;
01069     AR.PolyFunInv = AR0.PolyFunInv;
01070     AR.PolyFunType = AR0.PolyFunType;

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01083     AR.PolyFunExp = AR0.PolyFunExp;
01084     AR.PolyFunVar = AR0.PolyFunVar;
01085     AR.PolyFunPow = AR0.PolyFunPow;
01086     AR.Eside = AR0.Eside;
01087     AR.Cnumlhs = AR0.Cnumlhs;
01088 /*
01089     AR.MaxBracket = AR0.MaxBracket;
01090
01091     The compressbuffer contents are mainly relevant for keep brackets
01092     We should do this only if there is a keep brackets statement
01093     We may however still need the compressbuffer for expressions in the rhs.
01094 */
01095     if ( par >= 1 ) {
01096 /*
01097         We may not need this %%%% 7-apr-2006
01098 */
01099         t1 = AR.CompressBuffer; t2 = AR0.CompressBuffer;
01100         while ( t2 < AR0.CompressPointer ) *t1++ = *t2++;
01101         AR.CompressPointer = t1;
01102     }
01103     else {
01104         AR.CompressPointer = AR.CompressBuffer;
01105     }
01106     if ( AR.DeferFlag ) {
01107         if ( AR.infile->handle < 0 ) {
01108             AR.infile->POfill = AR0.infile->POfill;
01109         }
01110         else {
01111 /*
01112             We have to set the value of PPosition to something that will
01113             force a read in the first try.
01114 */
01115             AR.infile->POfull = AR.infile->POfill = AR.infile->PObuffer;
01116         }
01117     }
01118     if ( par == 0 ) {
01119         AN.threadbuck = thr;
01120         AN.ninterms = thr->firstterm;
01121     }
01122     else if ( par == 1 ) {
01123         WORD *tstop;
01124         t1 = thr->threadbuffer; tstop = t1 + *t1;
01125         t2 = AT.WorkPointer;
01126         while ( t1 < tstop ) *t2++ = *t1++;
01127         AN.ninterms = thr->firstterm;
01128     }
01129     AN.TeInFun = 0;
01130     AN.ncmod = AC.ncmod;
01131     AT.BrackBuf = AT0.BrackBuf;
01132     AT.bracketindexflag = AT0.bracketindexflag;
01133     AN.PolyFunTodo = 0;
01134 /*
01135     The relevant variables and the term are in their place.
01136     There is nothing more to do.
01137 */
01138     return(0);
01139 }
01140
01141 /*
01142 #] LoadOneThread :
01143 #[ BalanceRunThread :
01144 */
01145 int BalanceRunThread(PHEAD int identity, WORD *term, WORD level)
01146 {
01147     GETBIDENTITY
01148     ALLPRIVATES *BB;
01149     WORD *t, *tt;
01150     int i, *ti, *tti;
01151
01152     LoadOneThread(AT.identity,identity,0,2);
01153 /*
01154     Extra loading if needed. Quantities changed in Generator.
01155     Like the level that has to be passed.
01156 */
01157     BB = AB[identity];
01158     BB->R.level = level;
01159     BB->T.TMbuff = AT.TMbuff;
01160     ti = AT.RepCount; tti = BB->T.RepCount;
01161     i = AN.RepPoint - AT.RepCount;
01162     BB->N.RepPoint = BB->T.RepCount + i;
01163     for ( ; i >= 0; i-- ) tti[i] = ti[i];
01164
01165     t = term; i = *term;
01166     tt = BB->T.WorkSpace;
01167     NCOPY(tt,t,i);
01168     BB->T.WorkPointer = tt;

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01185
01186     WakeupThread(identity,HIGHERLEVELGENERATION);
01187
01188     return(0);
01189 }
01190
01191 /*
01192     #] BalanceRunThread :
01193     #[ SetWorkerFiles :
01194 */
01199 void SetWorkerFiles(void)
01200 {
01201     int id;
01202     ALLPRIVATES *B, *B0 = AB[0];
01203     for ( id = 1; id < AM.totalnumberofthreads; id++ ) {
01204         B = AB[id];
01205         AR.infile = &(AR.Fscr[0]);
01206         AR.outfile = &(AR.Fscr[1]);
01207         AR.hidefile = &(AR.Fscr[2]);
01208         AR.infile->handle = AR0.infile->handle;
01209         AR.hidefile->handle = AR0.hidefile->handle;
01210         if ( AR.infile->handle < 0 ) {
01211             AR.infile->PObuffer = AR0.infile->PObuffer;
01212             AR.infile->PStop = AR0.infile->PStop;
01213             AR.infile->POfill = AR0.infile->POfill;
01214             AR.infile->POfull = AR0.infile->POfull;
01215             AR.infile->POsize = AR0.infile->POsize;
01216             AR.InInBuf = AR0.InInBuf;
01217             AR.infile->POposition = AR0.infile->POposition;
01218             AR.infile->filesize = AR0.infile->filesize;
01219         }
01220         else {
01221             AR.infile->PObuffer = AR.infile->wPObuffer;
01222             AR.infile->PStop = AR.infile->wPStop;
01223             AR.infile->POfill = AR.infile->wPOfill;
01224             AR.infile->POfull = AR.infile->wPOfull;
01225             AR.infile->POsize = AR.infile->wPOsize;
01226             AR.InInBuf = 0;
01227             PUTZERO(AR.infile->POposition);
01228         }
01229     /*
01230         If there is some writing, it better happens to ones own outfile.
01231         Currently this is to be done only for InParallel.
01232         Merging of the outputs is then done by the CopyExpression routine.
01233     */
01234     {
01235         AR.outfile->PObuffer = AR.outfile->wPObuffer;
01236         AR.outfile->PStop = AR.outfile->wPStop;
01237         AR.outfile->POfill = AR.outfile->wPOfill;
01238         AR.outfile->POfull = AR.outfile->wPOfull;
01239         AR.outfile->POsize = AR.outfile->wPOsize;
01240         PUTZERO(AR.outfile->POposition);
01241     }
01242     if ( AR.hidefile->handle < 0 ) {
01243         AR.hidefile->PObuffer = AR0.hidefile->PObuffer;
01244         AR.hidefile->PStop = AR0.hidefile->PStop;
01245         AR.hidefile->POfill = AR0.hidefile->POfill;
01246         AR.hidefile->POfull = AR0.hidefile->POfull;
01247         AR.hidefile->POsize = AR0.hidefile->POsize;
01248         AR.InHiBuf = AR0.InHiBuf;
01249         AR.hidefile->POposition = AR0.hidefile->POposition;
01250         AR.hidefile->filesize = AR0.hidefile->filesize;
01251     }
01252     else {
01253         AR.hidefile->PObuffer = AR.hidefile->wPObuffer;
01254         AR.hidefile->PStop = AR.hidefile->wPStop;
01255         AR.hidefile->POfill = AR.hidefile->wPOfill;
01256         AR.hidefile->POfull = AR.hidefile->wPOfull;
01257         AR.hidefile->POsize = AR.hidefile->wPOsize;
01258         AR.InHiBuf = 0;
01259         PUTZERO(AR.hidefile->POposition);
01260     }
01261 }
01262 if ( AR0.StoreData.dirtyflag ) {
01263     for ( id = 1; id < AM.totalnumberofthreads; id++ ) {
01264         B = AB[id];
01265         AR.StoreData = AR0.StoreData;
01266     }
01267 }
01268 }
01269
01270 /*
01271     #] SetWorkerFiles :
01272     #[ RunThread :
01273 */
01281 void *RunThread(void *dummy)
01282 {

```

```

01283     WORD *term, *ttin, *tt, *ttco, *oldwork;
01284     int identity, wakeupsignal, identityretv, i, toberelased, errorcode;
01285     ALLPRIVATES *B;
01286     THREADBUCKET *thr;
01287     POSITION *ppdef;
01288     EXPRESSIONS e;
01289     DUMMYUSE(dummy);
01290     identity = SetIdentity(&identityretv);
01291     threadpointers[identity] = pthread_self();
01292     B = InitializeOneThread(identity);
01293     while ( ( wakeupsignal = ThreadWait(identity) ) > 0 ) {
01294         switch ( wakeupsignal ) {
01295             /*
01296              *
01297              */
01298             case STARTNEWEXPRESSION:
01299                 /*
01300                  *
01301                  * Set up the sort routines etc.
01302                  * Start with getting some buffers synchronized with the compiler
01303                  */
01304                 if ( UpdateOneThread(identity) ) {
01305                     MLOCK(ErrorMessageLock);
01306                     MesPrint("Update error in starting expression in thread %d in module
01307                             %d",identity,AC.CModule);
01308                     MUNLOCK(ErrorMessageLock);
01309                     Terminate(-1);
01310                 }
01311                 AR.DeferFlag = AC.ComDefer;
01312                 AR.sLevel = AS.sLevel;
01313                 AR.MaxDum = AM.IndDum;
01314                 AR.expchanged = AB[0]->R.expchanged;
01315                 AR.expflags = AB[0]->R.expflags;
01316                 AR.PolyFun = AB[0]->R.PolyFun;
01317                 AR.PolyFunInv = AB[0]->R.PolyFunInv;
01318                 AR.PolyFunType = AB[0]->R.PolyFunType;
01319                 AR.PolyFunExp = AB[0]->R.PolyFunExp;
01320                 AR.PolyFunVar = AB[0]->R.PolyFunVar;
01321                 AR.PolyFunPow = AB[0]->R.PolyFunPow;
01322             /*
01323              *
01324              * Now fire up the sort buffer.
01325              */
01326             NewSort (BHEAD0);
01327             break;
01328             /*
01329              *
01330              * #] STARTNEWEXPRESSION :
01331              * #[ LOWESTLEVELGENERATION :
01332              */
01333             case LOWESTLEVELGENERATION:
01334                 #ifndef INNERTEST
01335                 if ( AC.InnerTest ) {
01336                     if ( StrCmp(AC.TestValue, (UBYTE *)INNERTEST) == 0 ) {
01337                         MesPrint("Testing(Worker%d): value = %s",AT.identity,AC.TestValue);
01338                     }
01339                 }
01340                 #endif
01341                 e = Expressions + AR.CurExpr;
01342                 thr = AN.threadbuck;
01343                 ppdef = thr->deferbuffer;
01344                 ttin = thr->threadbuffer;
01345                 ttco = thr->compressbuffer;
01346                 term = AT.WorkPointer;
01347                 thr->usenum = 0;
01348                 toberelased = 0;
01349                 AN.inputnumber = thr->firstterm;
01350                 AN.ninterms = thr->firstterm;
01351                 do {
01352                     thr->usenum++; /* For if the master wants to steal the bucket */
01353                     tt = term; i = *ttin;
01354                     NCOPY(tt,ttin,i);
01355                     AT.WorkPointer = tt;
01356                     if ( AR.DeferFlag ) {
01357                         tt = AR.CompressBuffer; i = *ttco;
01358                         NCOPY(tt,ttco,i);
01359                         AR.CompressPointer = tt;
01360                         AR.DefPosition = ppdef[0]; ppdef++;
01361                     }
01362                 }
01363                 if ( thr->free == BUCKETTERMINATED ) {
01364                     /*
01365                      *
01366                      * The next statement allows the master to steal the bucket
01367                      * for load balancing purposes. We do still execute the current
01368                      * term, but afterwards we drop out.
01369                      * Once we have written the release code, we cannot use this
01370                      * bucket anymore. Hence the exit to the label bucketstolen.
01371                      */
01372                     if ( thr->usenum == thr->totnum ) {
01373                         thr->free = BUCKETCOMINGFREE;
01374                     }
01375                 }
01376             }
01377         }
01378     }

```



```

01369         else {
01370             thr->free = BUCKETRELEASED;
01371             tobereleased = 1;
01372         }
01373     }
01374     /*
01375         What if we want to steal and we set thr->free while
01376         the thread is inside the next code for a long time?
01377     if ( AT.LoadBalancing ) {
01378     */
01379         LOCK(thr->lock);
01380         thr->busy = BUCKETDOINGTERM;
01381         UNLOCK(thr->lock);
01382     /*
01383     }
01384     else {
01385         thr->busy = BUCKETDOINGTERM;
01386     }
01387     */
01388     AN.RepPoint = AT.RepCount + 1;
01389
01390     if ( ( e->vflags & ISFACTORIZED ) != 0 && term[1] == HAAKJE ) {
01391         StoreTerm(BHEAD term);
01392     }
01393     else {
01394         if ( AR.DeferFlag ) {
01395             AR.CurDum = AN.IndDum = Expressions[AR.CurExpr].numdummies + AM.IndDum;
01396         }
01397         else {
01398             AN.IndDum = AM.IndDum;
01399             AR.CurDum = ReNumber(BHEAD term);
01400         }
01401         if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMFLAG);
01402         if ( AN.ncmod ) {
01403             if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
01404             else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
01405         }
01406         else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
01407         if ( ( AP.PreDebug & THREADSDEBUG ) != 0 ) {
01408             MLOCK(ErrorMessageLock);
01409             MesPrint("Thread %w executing term:");
01410             PrintTerm(term,"LLG");
01411             MUNLOCK(ErrorMessageLock);
01412         }
01413         if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
01414             && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
01415             PolyFunClean(BHEAD term);
01416         }
01417         if ( Generator(BHEAD term,0) ) {
01418             LowerSortLevel();
01419             MLOCK(ErrorMessageLock);
01420             MesPrint("Error in processing one term in thread %d in module
01421 %d",identity,AC.CModule);
01422             MUNLOCK(ErrorMessageLock);
01423             Terminate(-1);
01424         }
01425         AN.ninterms++;
01426     /*
01427     if ( AT.LoadBalancing ) { */
01428         LOCK(thr->lock);
01429         thr->busy = BUCKETPREPARINGTERM;
01430         UNLOCK(thr->lock);
01431     /*
01432     }
01433     else {
01434         thr->busy = BUCKETPREPARINGTERM;
01435     }
01436     */
01437     if ( thr->free == BUCKETTERMINATED ) {
01438         if ( thr->usenum == thr->totnum ) {
01439             thr->free = BUCKETCOMINGFREE;
01440         }
01441         else {
01442             thr->free = BUCKETRELEASED;
01443             tobereleased = 1;
01444         }
01445     }
01446     if ( tobereleased ) goto bucketstolen;
01447     } while ( *ttin );
01448     thr->free = BUCKETCOMINGFREE;
01449     bucketstolen:;
01450     /*
01451     if ( AT.LoadBalancing ) { */
01452         LOCK(thr->lock);
01453         thr->busy = BUCKETTOBERELEASED;
01454         UNLOCK(thr->lock);
01455     }
01456     else {

```

```

01455         thr->busy = BUCKETTOBERELEASED;
01456     }
01457 */
01458     AT.WorkPointer = term;
01459     break;
01460 /*
01461     #] LOWESTLEVELGENERATION :
01462     #[ FINISHEXPRESSION :
01463 */
01464 #ifdef WITHSORTBOTS
01465     case CLAIMOUTPUT:
01466         LOCK(AT.SB.MasterBlockLock[1]);
01467         break;
01468 #endif
01469     case FINISHEXPRESSION:
01470 /*
01471         Finish the sort
01472
01473         Start with claiming the first block
01474         Once we have claimed it we can let the master know that
01475         everything is all right.
01476 */
01477         LOCK(AT.SB.MasterBlockLock[1]);
01478         ThreadClaimedBlock(identity);
01479 /*
01480         Entry for when we work with sortbots
01481 */
01482 #ifdef WITHSORTBOTS
01483         /* fall through */
01484         case FINISHEXPRESSION2:
01485 #endif
01486 /*
01487         Now we may need here an fsync on the sort file
01488 */
01489         if ( AC.ThreadSortFileSynch ) {
01490             if ( AT.S0->file.handle >= 0 ) {
01491                 SynchFile(AT.S0->file.handle);
01492             }
01493         }
01494         AT.SB.FillBlock = 1;
01495         AT.SB.MasterFill[1] = AT.SB.MasterStart[1];
01496         errorcode = EndSort(BHEAD AT.S0->sBuffer,0);
01497         UNLOCK(AT.SB.MasterBlockLock[AT.SB.FillBlock]);
01498         UpdateMaxSize();
01499         if ( errorcode ) {
01500             MLOCK(ErrorMessageLock);
01501             MesPrint("Error terminating sort in thread %d in module %d",identity,AC.CModule);
01502             MUNLOCK(ErrorMessageLock);
01503             Terminate(-1);
01504         }
01505         break;
01506 /*
01507     #] FINISHEXPRESSION :
01508     #[ CLEANUPEXPRESSION :
01509 */
01510     case CLEANUPEXPRESSION:
01511 /*
01512         Cleanup everything and wait for the next expression
01513 */
01514         if ( AR.outfile->handle >= 0 ) {
01515             CloseFile(AR.outfile->handle);
01516             AR.outfile->handle = -1;
01517             remove(AR.outfile->name);
01518             AR.outfile->POfill = AR.outfile->POfull = AR.outfile->PObuffer;
01519             PUTZERO(AR.outfile->POposition);
01520             PUTZERO(AR.outfile->filesize);
01521         }
01522         else {
01523             AR.outfile->POfill = AR.outfile->POfull = AR.outfile->PObuffer;
01524             PUTZERO(AR.outfile->POposition);
01525             PUTZERO(AR.outfile->filesize);
01526         }
01527     {
01528         CBUF *C = cbuf+AT.ebufnum;
01529         WORD **w, ii;
01530         if ( C->numrhs > 0 || C->numlhs > 0 ) {
01531             if ( C->rhs ) {
01532                 w = C->rhs; ii = C->numrhs;
01533                 do { *w++ = 0; } while ( --ii > 0 );
01534             }
01535             if ( C->lhs ) {
01536                 w = C->lhs; ii = C->numlhs;
01537                 do { *w++ = 0; } while ( --ii > 0 );
01538             }
01539             C->numlhs = C->numrhs = 0;
01540             ClearTree(AT.ebufnum);
01541             C->Pointer = C->Buffer;

```

```

01542         }
01543     }
01544     break;
01545 /*
01546     #] CLEANUPEXPRESSION :
01547     #[ HIGHERLEVELGENERATION :
01548 */
01549     case HIGHERLEVELGENERATION:
01550 /*
01551         When foliating halfway the tree.
01552         This should only be needed in a second level load balancing
01553 */
01554         term = AT.WorkSpace; AT.WorkPointer = term + *term;
01555         if ( Generator(BHEAD term,AR.level) ) {
01556             LowerSortLevel();
01557             MLOCK(ErrorMessageLock);
01558             MesPrint("Error in load balancing one term at level %d in thread %d in module
01559 %d",AR.level,AT.identity,AC.CModule);
01560             MUNLOCK(ErrorMessageLock);
01561             Terminate(-1);
01562         }
01563         AT.WorkPointer = term;
01564         break;
01565 /*
01566     #] HIGHERLEVELGENERATION :
01567     #[ STARTNEWMODULE :
01568 */
01569     case STARTNEWMODULE:
01570 /*
01571         For resetting variables.
01572 */
01573         SpecialCleanup(B);
01574         break;
01575 /*
01576     #] STARTNEWMODULE :
01577     #[ TERMINATETHREAD :
01578 */
01579     case TERMINATETHREAD:
01580         goto EndOfThread;
01581 /*
01582     #] TERMINATETHREAD :
01583     #[ DOONEEXPRESSION :
01584 */
01585         When a thread has to do a complete (not too big) expression.
01586         The number of the expression to be done is in AR.exprtodo.
01587         The code is mostly taken from Processor. The only difference
01588         is with what to do with the output.
01589         The output should go to the scratch buffer of the worker
01590         (which is free at the right moment). If this buffer is too
01591         small we have a problem. We could write to file or give the
01592         master what we have and from now on the master has to collect
01593         pieces until things are complete.
01594         Note: this assumes that the expressions don't keep their order.
01595         If they have to keep their order, don't use this feature.
01596 */
01597     case DOONEEXPRESSION: {
01598         POSITION position, outposition;
01599         FILEHANDLE *fi, *fout, *oldoutfile;
01600         LONG dd = 0;
01601         WORD oldBracketOn = AR.BracketOn;
01602         WORD *oldBrackBuf = AT.BrackBuf;
01603         WORD oldbracketindexflag = AT.bracketindexflag;
01604         WORD fromspectator = 0;
01605         e = Expressions + AR.exprtodo;
01606         i = AR.exprtodo;
01607         AR.CurExpr = i;
01608         AR.SortType = AC.SortType;
01609         AR.expchanged = 0;
01610         if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
01611             AR.BracketOn = 1;
01612             AT.BrackBuf = AM.BracketFactors;
01613             AT.bracketindexflag = 1;
01614         }
01615         position = AS.OldOnFile[i];
01616         if ( e->status == HIDDENLEXPRESSION || e->status == HIDDENEXPRESSION ) {
01617             AR.GetFile = 2; fi = AR.hidefile;
01618         }
01619         else {
01620             AR.GetFile = 0; fi = AR.infile;
01621         }
01622     }
01623 /*
01624     PUTZERO(fi->POposition);
01625     if ( fi->handle >= 0 ) {
01626         fi->POfill = fi->POfull = fi->PObuffer;
01627     }

```

```

01628 */
01629 SetScratch(fi,&position);
01630 term = oldwork = AT.WorkPointer;
01631 AR.CompressPointer = AR.CompressBuffer;
01632 AR.CompressPointer[0] = 0;
01633 AR.KeptInHold = 0;
01634 if ( GetTerm(BHEAD term) <= 0 ) {
01635     MLOCK(ErrorMessageLock);
01636     MesPrint("Expression %d has problems in scratchfile (t)",i);
01637     MUNLOCK(ErrorMessageLock);
01638     Terminate(-1);
01639 }
01640 if ( AT.bracketindexflag > 0 ) OpenBracketIndex(i);
01641 term[3] = i;
01642 if ( term[5] < 0 ) {
01643     fromspectator = -term[5];
01644     PUTZERO(AM.SpectatorFiles[fromspectator-1].readpos);
01645     term[5] = AC.cbufnum;
01646 }
01647 PUTZERO(outposition);
01648 fout = AR.outfile;
01649 fout->POfill = fout->POfull = fout->PObuffer;
01650 fout->POposition = outposition;
01651 if ( fout->handle >= 0 ) {
01652     fout->POposition = outposition;
01653 }
01654 /*
01655 The next statement is needed because we need the system
01656 to believe that the expression is at position zero for
01657 the moment. In this worker, with no memory of other expressions,
01658 it is. This is needed for when a bracket index is made
01659 because there e->onfile is an offset. Afterwards, when the
01660 expression is written to its final location in the masters
01661 output e->onfile will get its real value.
01662 */
01663 PUTZERO(e->onfile);
01664 if ( PutOut(BHEAD term,&outposition,fout,0) < 0 ) goto ProcErr;
01665
01666 AR.DeferFlag = AC.ComDefer;
01667
01668 AR.sLevel = AB[0]->R.sLevel;
01669 term = AT.WorkPointer;
01670 NewSort(BHEAD0);
01671 AR.MaxDum = AM.IndDum;
01672 AN.ninterms = 0;
01673 if ( fromspectator ) {
01674     while ( GetFromSpectator(term,fromspectator-1) ) {
01675         AT.WorkPointer = term + *term;
01676         AN.RepPoint = AT.RepCount + 1;
01677         AN.IndDum = AM.IndDum;
01678         AR.CurDum = ReNumber(BHEAD term);
01679         if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMFLAG);
01680         if ( AN.ncmod ) {
01681             if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
01682             else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
01683         }
01684         else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
01685         if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
01686             && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
01687             PolyFunClean(BHEAD term);
01688         }
01689         if ( Generator(BHEAD term,0) ) {
01690             LowerSortLevel(); goto ProcErr;
01691         }
01692     }
01693 }
01694 else {
01695     while ( GetTerm(BHEAD term) ) {
01696         SeekScratch(fi,&position);
01697         AN.ninterms++; dd = AN.deferskipped;
01698         if ( ( e->vflags & ISFACTORIZED ) != 0 && term[1] == HAAKJE ) {
01699             StoreTerm(BHEAD term);
01700         }
01701         else {
01702             if ( AC.CollectFun && *term <= (AM.MaxTer/(2*(LONG)sizeof(WORD))) ) {
01703                 if ( GetMoreTerms(term) < 0 ) {
01704                     LowerSortLevel(); goto ProcErr;
01705                 }
01706                 SeekScratch(fi,&position);
01707             }
01708             AT.WorkPointer = term + *term;
01709             AN.RepPoint = AT.RepCount + 1;
01710             if ( AR.DeferFlag ) {
01711                 AR.CurDum = AN.IndDum = Expressions[AR.exprtodo].numdummies;
01712             }
01713             else {
01714                 AN.IndDum = AM.IndDum;

```

```

01715         AR.CurDum = ReNumber(BHEAD term);
01716     }
01717     if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMLAG);
01718     if ( AN.ncmod ) {
01719         if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
01720         else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
01721     }
01722     else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
01723     if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
01724         && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
01725         PolyFunClean(BHEAD term);
01726     }
01727     if ( Generator(BHEAD term,0) ) {
01728         LowerSortLevel(); goto ProcErr;
01729     }
01730     AN.ninterms += dd;
01731     }
01732     SetScratch(fi,&position);
01733     if ( fi == AR.hidefile ) {
01734         AR.InHiBuf = (fi->POfull-fi->PObuffer)
01735             -DIFBASE(position,fi->POposition)/sizeof(WORD);
01736     }
01737     else {
01738         AR.InInBuf = (fi->POfull-fi->PObuffer)
01739             -DIFBASE(position,fi->POposition)/sizeof(WORD);
01740     }
01741     }
01742     }
01743     AN.ninterms += dd;
01744     if ( EndSort(BHEAD AT.S0->sBuffer,0) < 0 ) goto ProcErr;
01745     e->numdummies = AR.MaxDum - AM.IndDum;
01746     AR.BracketOn = oldBracketOn;
01747     AT.BrackBuf = oldBrackBuf;
01748     if ( ( e->vflags & TOBEFACTORED ) != 0 )
01749         poly_factorize_expression(e);
01750     else if ( ( ( e->vflags & TOBEUNFACTORED ) != 0 )
01751         && ( ( e->vflags & ISFACTORIZED ) != 0 ) )
01752         poly_unfactorize_expression(e);
01753     if ( AT.S0->TermsLeft ) e->vflags &= ~ISZERO;
01754     else e->vflags |= ISZERO;
01755     if ( AR.expchanged == 0 ) e->vflags |= ISUNMODIFIED;
01756     if ( AT.S0->TermsLeft ) AR.expflags |= ISZERO;
01757     if ( AR.expchanged ) AR.expflags |= ISUNMODIFIED;
01758     AR.GetFile = 0;
01759     AT.bracketindexflag = oldbracketindexflag;
01760     /*
01761     Now copy the whole thing from fout to AR0.outfile
01762     Do this in one go to keep the lock occupied as short as possible
01763     */
01764     SeekScratch(fout,&outposition);
01765     LOCK(AS.outputslock);
01766     oldoutfile = AB[0]->R.outfile;
01767     if ( e->status == INTOHIDELEXPRESSION || e->status == INTOHIDEGEXPRESSION ) {
01768         AB[0]->R.outfile = AB[0]->R.hidefile;
01769     }
01770     SeekScratch(AB[0]->R.outfile,&position);
01771     e->onfile = position;
01772     if ( CopyExpression(fout,AB[0]->R.outfile) < 0 ) {
01773         AB[0]->R.outfile = oldoutfile;
01774         UNLOCK(AS.outputslock);
01775         MLOCK(ErrorMessageLock);
01776         MesPrint("Error copying output of 'InParallel' expression to master. Thread:
01777         %d",identity);
01778         MUNLOCK(ErrorMessageLock);
01779         goto ProcErr;
01780     }
01781     AB[0]->R.outfile = oldoutfile;
01782     AB[0]->R.hidefile->POfull = AB[0]->R.hidefile->POfill;
01783     AB[0]->R.expflags = AR.expflags;
01784     UNLOCK(AS.outputslock);
01785     if ( fout->handle >= 0 ) { /* Now get rid of the file */
01786         CloseFile(fout->handle);
01787         fout->handle = -1;
01788         remove(fout->name);
01789         PUTZERO(fout->POposition);
01790         PUTZERO(fout->filesize);
01791         fout->POfill = fout->POfull = fout->PObuffer;
01792     }
01793     UpdateMaxSize();
01794     AT.WorkPointer = oldwork;
01795     } break;
01796     }
01797     } break;
01798     /*
01799     #] DOONEEXPRESSION :
01800     #[ DOBRACKETS :

```

```

01801
01802     In case we have a bracket index we can have the worker treat
01803     one or more of the entries in the bracket index.
01804     The advantage is that identical terms will meet each other
01805     sooner in the sorting and hence fewer compares will be needed.
01806     Also this way the master doesn't need to fill the buckets.
01807     The main problem is the load balancing which can become very
01808     bad when there is a long tail without things outside the bracket.
01809
01810     We get sent:
01811     1: The number of the first bracket to be done
01812     2: The number of the last bracket to be done
01813 */
01814     case DOBRACKETS: {
01815         BRACKETINFO *binfo;
01816         BRACKETINDEX *bi;
01817         FILEHANDLE *fi;
01818         POSITION stoppos, where;
01819         e = Expressions + AR.CurExpr;
01820         binfo = e->bracketinfo;
01821         thr = AN.threadbuck;
01822         bi = &(binfo->indexbuffer[thr->firstbracket]);
01823         if ( AR.GetFile == 2 ) fi = AR.hidefile;
01824         else fi = AR.infile;
01825         where = bi->start;
01826         ADD2POS(where, AS.OldOnFile[AR.CurExpr]);
01827         SetScratch(fi, &(where));
01828         stoppos = binfo->indexbuffer[thr->lastbracket].next;
01829         ADD2POS(stoppos, AS.OldOnFile[AR.CurExpr]);
01830         AN.ninterms = thr->firstterm;
01831     /*
01832         Now we have to put the 'value' of the bracket in the
01833         Compress buffer.
01834     */
01835         ttco = AR.CompressBuffer;
01836         tt = binfo->bracketbuffer + bi->bracket;
01837         i = *tt;
01838         NCOPY(ttco, tt, i)
01839         AR.CompressPointer = ttco;
01840         term = AT.WorkPointer;
01841         while ( GetTerm(BHEAD term) ) {
01842             SeekScratch(fi, &(where));
01843             AT.WorkPointer = term + *term;
01844             AN.IndDum = AM.IndDum;
01845             AR.CurDum = ReNumber(BHEAD term);
01846             if ( AC.SymChangeFlag ) MarkDirty(term, DIRTYSYMFLAG);
01847             if ( AN.ncmod ) {
01848                 if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term, DIRTYFLAG);
01849                 else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
01850             }
01851             else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
01852             if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
01853                 && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
01854                 PolyFunClean(BHEAD term);
01855             }
01856             if ( ( AP.PreDebug & THREADSDEBUG ) != 0 ) {
01857                 MLOCK(ErrorMessageLock);
01858                 MesPrint("Thread %w executing term:");
01859                 PrintTerm(term, "DoBrackets");
01860                 MUNLOCK(ErrorMessageLock);
01861             }
01862             AT.WorkPointer = term + *term;
01863             if ( Generator(BHEAD term, 0) ) {
01864                 LowerSortLevel();
01865                 MLOCK(ErrorMessageLock);
01866                 MesPrint("Error in processing one term in thread %d in module
01867 %d", identity, AC.CModule);
01867                 MUNLOCK(ErrorMessageLock);
01868                 Terminate(-1);
01869             }
01870             AN.ninterms++;
01871             SetScratch(fi, &(where));
01872             if ( ISGEPOS(where, stoppos) ) break;
01873         }
01874         AT.WorkPointer = term;
01875         thr->free = BUCKETCOMINGFREE;
01876         break;
01877     }
01878 /*
01879     #] DOBRACKETS :
01880     #[ CLEARCLOCK :
01881
01882     The program only comes here after a .clear
01883 */
01884     case CLEARCLOCK:
01885         LOCK(clearclocklock); /*
01886         sumtimerinfo[identity] += TimeCPU(1);

```

```

01887         timerinfo[identity] = TimeCPU(0);
01888     /*          UNLOCK(clearclocklock); */
01889         break;
01890     /*
01891         #] CLEARCLOCK :
01892         #[ MCTSEXPANDTREE :
01893     */
01894         case MCTSEXPANDTREE:
01895             AT.optimtimes = AB[0]->T.optimtimes;
01896             find_Horner_MCTS_expand_tree();
01897             break;
01898     /*
01899         #] MCTSEXPANDTREE :
01900         #[ OPTIMIZEEXPRESSION :
01901     */
01902         case OPTIMIZEEXPRESSION:
01903             optimize_expression_given_Horner();
01904             break;
01905     /*
01906         #] OPTIMIZEEXPRESSION :
01907     */
01908         default:
01909             MLOCK(ErrorMessageLock);
01910             MesPrint("Illegal wakeup signal %d for thread %d",wakeupsignal,identity);
01911             MUNLOCK(ErrorMessageLock);
01912             Terminate(-1);
01913             break;
01914     }
01915     /* we need the following update in case we are using checkpoints. then we
01916        need to readjust the clocks when recovering using this information */
01917     timerinfo[identity] = TimeCPU(1);
01918 }
01919 EndOfThread;;
01920 /*
01921     This is the end of the thread. We cleanup and exit.
01922     If we are using flint, call the per-thread cleanup function. This keep valgrind happy.
01923 */
01924 #ifdef WITHFLINT
01925     flint_final_cleanup_thread();
01926 #endif
01927     FinalizeOneThread(identity);
01928     return(0);
01929 ProcErr:
01930     Terminate(-1);
01931     return(0);
01932 }
01933
01934 /*
01935     #] RunThread :
01936     #[ RunSortBot :
01937 */
01938 #ifdef WITHSORTBOTS
01939 void *RunSortBot(void *dummy)
01940 {
01941     int identity, wakeupsignal, identityretv;
01942     ALLPRIVATES *B, *BB;
01943     DUMMYUSE(dummy);
01944     identity = SetIdentity(&identityretv);
01945     threadpointers[identity] = pthread_self();
01946     B = InitializeOneThread(identity);
01947     while ( ( wakeupsignal = SortBotWait(identity) ) > 0 ) {
01948         switch ( wakeupsignal ) {
01949             /*
01950                 #] INISORTBOT :
01951             */
01952             case INISORTBOT:
01953                 AR.CompressBuffer = AB[0]->R.CompressBuffer;
01954                 AR.ComprTop = AB[0]->R.ComprTop;
01955                 AR.CompressPointer = AB[0]->R.CompressPointer;
01956                 AR.CurExpr = AB[0]->R.CurExpr;
01957                 AR.PolyFun = AB[0]->R.PolyFun;
01958                 AR.PolyFunInv = AB[0]->R.PolyFunInv;
01959                 AR.PolyFunType = AB[0]->R.PolyFunType;
01960                 AR.PolyFunExp = AB[0]->R.PolyFunExp;
01961                 AR.PolyFunVar = AB[0]->R.PolyFunVar;
01962                 AR.PolyFunPow = AB[0]->R.PolyFunPow;
01963                 AR.SortType = AC.SortType;
01964                 if ( AR.PolyFun == 0 ) { AT.SS->PolyFlag = 0; }
01965                 else if ( AR.PolyFunType == 1 ) { AT.SS->PolyFlag = 1; }
01966                 else if ( AR.PolyFunType == 2 ) {
01967                     if ( AR.PolyFunExp == 2
01968                         || AR.PolyFunExp == 3 ) AT.SS->PolyFlag = 1;
01969                     else AT.SS->PolyFlag = 2;
01970                 }
01971                 AT.SS->PolyWise = 0;
01972                 AN.ncmod = AC.ncmod;

```

```

01981         LOCK(AT.SB.MasterBlockLock[1]);
01982         BB = AB[AT.SortBotIn1];
01983         LOCK(BB->T.SB.MasterBlockLock[BB->T.SB.MasterNumBlocks]);
01984         BB = AB[AT.SortBotIn2];
01985         LOCK(BB->T.SB.MasterBlockLock[BB->T.SB.MasterNumBlocks]);
01986         AT.SB.FillBlock = 1;
01987         AT.SB.MasterFill[1] = AT.SB.MasterStart[1];
01988         SETBASEPOSITION(AN.theposition,0);
01989         break;
01990 /*
01991         #] INISORTBOT :
01992         #[ RUNSORTBOT :
01993 */
01994         case RUNSORTBOT:
01995             SortBotMerge(B);
01996             break;
01997 /*
01998         #] RUNSORTBOT :
01999         #[ TERMINATETHREAD :
02000 */
02001         case TERMINATETHREAD:
02002             goto EndOfThread;
02003 /*
02004         #] TERMINATETHREAD :
02005         #[ CLEARCLOCK :
02006
02007         The program only comes here after a .clear
02008 */
02009         case CLEARCLOCK:
02010             LOCK(clearclocklock); */
02011             sumtimerinfo[identity] += TimeCPU(1);
02012             timerinfo[identity] = TimeCPU(0);
02013             UNLOCK(clearclocklock); */
02014             break;
02015 /*
02016         #] CLEARCLOCK :
02017 */
02018         default:
02019             MLOCK(ErrorMessageLock);
02020             MesPrint("Illegal wakeup signal %d for thread %d",wakeupsignal,identity);
02021             MUNLOCK(ErrorMessageLock);
02022             Terminate(-1);
02023             break;
02024     }
02025 }
02026 EndOfThread;;
02027 /*
02028     This is the end of the thread. We cleanup and exit.
02029     If we are using flint, call the per-thread cleanup function. This keep valgrind happy.
02030 */
02031 #ifdef WITHFLINT
02032     flint_final_cleanup_thread();
02033 #endif
02034     FinalizeOneThread(identity);
02035     return(0);
02036 }
02037
02038 #endif
02039
02040 /*
02041     #] RunSortBot :
02042     #[ IAmAvailable :
02043 */
02044 void IAmAvailable(int identity)
02045 {
02046     int top;
02047     LOCK(availabilitylock);
02048     top = topofavailables;
02049     listofavailables[topofavailables++] = identity;
02050     if ( top == 0 ) {
02051         UNLOCK(availabilitylock);
02052         LOCK(wakeupmasterlock);
02053         wakeupmaster = identity;
02054         pthread_cond_signal(&wakeupmasterconditions);
02055         UNLOCK(wakeupmasterlock);
02056     }
02057     else {
02058         UNLOCK(availabilitylock);
02059     }
02060 }
02061
02062 /*
02063     #] IAmAvailable :
02064     #[ GetAvailableThread :
02065 */
02066 int GetAvailableThread(void)
02067 {

```



```

02087     int retval = -1;
02088     LOCK(availabilitylock);
02089     if ( topofavailables > 0 ) retval = listofavailables[--topofavailables];
02090     UNLOCK(availabilitylock);
02091     if ( retval >= 0 ) {
02092 /*
02093         Make sure the thread is indeed waiting and not between
02094         saying that it is available and starting to wait.
02095 */
02096         LOCK(wakeuplocks[retval]);
02097         UNLOCK(wakeuplocks[retval]);
02098     }
02099     return(retval);
02100 }
02101
02102 /*
02103 #] GetAvailableThread :
02104 #[ ConditionalGetAvailableThread :
02105 */
02113 int ConditionalGetAvailableThread(void)
02114 {
02115     int retval = -1;
02116     if ( topofavailables > 0 ) {
02117         LOCK(availabilitylock);
02118         if ( topofavailables > 0 ) {
02119             retval = listofavailables[--topofavailables];
02120         }
02121         UNLOCK(availabilitylock);
02122         if ( retval >= 0 ) {
02123 /*
02124             Make sure the thread is indeed waiting and not between
02125             saying that it is available and starting to wait.
02126 */
02127             LOCK(wakeuplocks[retval]);
02128             UNLOCK(wakeuplocks[retval]);
02129         }
02130     }
02131     return(retval);
02132 }
02133
02134 /*
02135 #] ConditionalGetAvailableThread :
02136 #[ GetThread :
02137 */
02147 int GetThread(int identity)
02148 {
02149     int retval = -1, j;
02150     LOCK(availabilitylock);
02151     for ( j = 0; j < topofavailables; j++ ) {
02152         if ( identity == listofavailables[j] ) break;
02153     }
02154     if ( j < topofavailables ) {
02155         --topofavailables;
02156         for ( ; j < topofavailables; j++ ) {
02157             listofavailables[j] = listofavailables[j+1];
02158         }
02159         retval = identity;
02160     }
02161     UNLOCK(availabilitylock);
02162     return(retval);
02163 }
02164
02165 /*
02166 #] GetThread :
02167 #[ ThreadWait :
02168 */
02178 int ThreadWait(int identity)
02179 {
02180     int retval, top, j;
02181     LOCK(wakeuplocks[identity]);
02182     LOCK(availabilitylock);
02183     top = topofavailables;
02184     for ( j = topofavailables; j > 0; j-- )
02185         listofavailables[j] = listofavailables[j-1];
02186     listofavailables[0] = identity;
02187     topofavailables++;
02188     if ( top == 0 || topofavailables == numberofworkers ) {
02189         UNLOCK(availabilitylock);
02190         LOCK(wakeupmasterlock);
02191         wakeupmaster = identity;
02192         pthread_cond_signal(&wakeupmasterconditions);
02193         UNLOCK(wakeupmasterlock);
02194     }
02195     else {
02196         UNLOCK(availabilitylock);
02197     }
02198     while ( wakeup[identity] == 0 ) {

```

```

02199     pthread_cond_wait (&(wakeupconditions[identity]), &(wakeuplocks[identity]));
02200 }
02201     retval = wakeup[identity];
02202     wakeup[identity] = 0;
02203     UNLOCK(wakeuplocks[identity]);
02204     return(retval);
02205 }
02206
02207 /*
02208     #] ThreadWait :
02209     #[ SortBotWait :
02210 */
02211
02212 #ifndef WITHSORTBOTS
02222 int SortBotWait(int identity)
02223 {
02224     int retval;
02225     LOCK(wakeuplocks[identity]);
02226     LOCK(availabilitylock);
02227     topsortbotavailables++;
02228     if ( topsortbotavailables >= numberofsortbots ) {
02229         UNLOCK(availabilitylock);
02230         LOCK(wakeupsortbotlock);
02231         wakeupmaster = identity;
02232         pthread_cond_signal (&wakeupsortbotconditions);
02233         UNLOCK(wakeupsortbotlock);
02234     }
02235     else {
02236         UNLOCK(availabilitylock);
02237     }
02238     while ( wakeup[identity] == 0 ) {
02239         pthread_cond_wait (&(wakeupconditions[identity]), &(wakeuplocks[identity]));
02240     }
02241     retval = wakeup[identity];
02242     wakeup[identity] = 0;
02243     UNLOCK(wakeuplocks[identity]);
02244     return(retval);
02245 }
02246
02247 #endif
02248
02249 /*
02250     #] SortBotWait :
02251     #[ ThreadClaimedBlock :
02252 */
02263 int ThreadClaimedBlock(int identity)
02264 {
02265     LOCK(availabilitylock);
02266     numberclaimed++;
02267     if ( numberclaimed >= numberofworkers ) {
02268         UNLOCK(availabilitylock);
02269         LOCK(wakeupmasterlock);
02270         wakeupmaster = identity;
02271         pthread_cond_signal (&wakeupmasterconditions);
02272         UNLOCK(wakeupmasterlock);
02273     }
02274     else {
02275         UNLOCK(availabilitylock);
02276     }
02277     return(0);
02278 }
02279
02280 /*
02281     #] ThreadClaimedBlock :
02282     #[ MasterWait :
02283 */
02291 int MasterWait(void)
02292 {
02293     int retval;
02294     LOCK(wakeupmasterlock);
02295     while ( wakeupmaster == 0 ) {
02296         pthread_cond_wait (&wakeupmasterconditions, &wakeupmasterlock);
02297     }
02298     retval = wakeupmaster;
02299     wakeupmaster = 0;
02300     UNLOCK(wakeupmasterlock);
02301     return(retval);
02302 }
02303
02304 /*
02305     #] MasterWait :
02306     #[ MasterWaitThread :
02307 */
02314 int MasterWaitThread(int identity)
02315 {
02316     int retval;
02317     LOCK(wakeupmasterthreadlocks[identity]);

```

```

02318     while ( wakeupmasterthread[identity] == 0 ) {
02319         pthread_cond_wait (&(wakeupmasterthreadconditions[identity])
02320             , &(wakeupmasterthreadlocks[identity]));
02321     }
02322     retval = wakeupmasterthread[identity];
02323     wakeupmasterthread[identity] = 0;
02324     UNLOCK(wakeupmasterthreadlocks[identity]);
02325     return(retval);
02326 }
02327
02328 /*
02329  #] MasterWaitThread :
02330  #[ MasterWaitAll :
02331  */
02332 void MasterWaitAll(void)
02333 {
02334     LOCK(wakeupmasterlock);
02335     while ( topofavailables < numberofworkers ) {
02336         pthread_cond_wait (&wakeupmasterconditions, &wakeupmasterlock);
02337     }
02338     UNLOCK(wakeupmasterlock);
02339     return;
02340 }
02341
02342 /*
02343  #] MasterWaitAll :
02344  #[ MasterWaitAllSortBots :
02345  */
02346 #ifdef WITHSORTBOTS
02347 void MasterWaitAllSortBots(void)
02348 {
02349     LOCK(wakeupsortbotlock);
02350     while ( topsortbotavailables < numberofsortbots ) {
02351         pthread_cond_wait (&wakeupsortbotconditions, &wakeupsortbotlock);
02352     }
02353     UNLOCK(wakeupsortbotlock);
02354     return;
02355 }
02356 #endif
02357
02358 /*
02359  #] MasterWaitAllSortBots :
02360  #[ MasterWaitAllBlocks :
02361  */
02362 void MasterWaitAllBlocks(void)
02363 {
02364     LOCK(wakeupmasterlock);
02365     while ( numberclaimed < numberofworkers ) {
02366         pthread_cond_wait (&wakeupmasterconditions, &wakeupmasterlock);
02367     }
02368     UNLOCK(wakeupmasterlock);
02369     return;
02370 }
02371
02372 /*
02373  #] MasterWaitAllBlocks :
02374  #[ WakeupThread :
02375  */
02376 void WakeupThread(int identity, int signalnumber)
02377 {
02378     if ( signalnumber == 0 ) {
02379         MLOCK(ErrorMessageLock);
02380         MesPrint("Illegal wakeup signal for thread %d", identity);
02381         MUNLOCK(ErrorMessageLock);
02382         Terminate(-1);
02383     }
02384     LOCK(wakeuplocks[identity]);
02385     wakeup[identity] = signalnumber;
02386     pthread_cond_signal (&(wakeupconditions[identity]));
02387     UNLOCK(wakeuplocks[identity]);
02388 }
02389
02390 /*
02391  #] WakeupThread :
02392  #[ WakeupMasterFromThread :
02393  */
02394 void WakeupMasterFromThread(int identity, int signalnumber)
02395 {
02396     if ( signalnumber == 0 ) {
02397         MLOCK(ErrorMessageLock);
02398         MesPrint("Illegal wakeup signal for master %d", identity);
02399         MUNLOCK(ErrorMessageLock);
02400         Terminate(-1);
02401     }
02402 }

```

```

02438     LOCK(wakeupmasterthreadlocks[identity]);
02439     wakeupmasterthread[identity] = signalnumber;
02440     pthread_cond_signal(&(wakeupmasterthreadconditions[identity]));
02441     UNLOCK(wakeupmasterthreadlocks[identity]);
02442 }
02443
02444 /*
02445  #] WakeupMasterFromThread :
02446  #[ SendOneBucket :
02447 */
02453 int SendOneBucket(int type)
02454 {
02455     ALLPRIVATES *B0 = AB[0];
02456     THREADBUCKET *thr = 0;
02457     int j, k, id;
02458     for ( j = 0; j < numthreadbuckets; j++ ) {
02459         if ( threadbuckets[j]->free == BUCKETFILLED ) {
02460             thr = threadbuckets[j];
02461             for ( k = j+1; k < numthreadbuckets; k++ )
02462                 threadbuckets[k-1] = threadbuckets[k];
02463             threadbuckets[numthreadbuckets-1] = thr;
02464             break;
02465         }
02466     }
02467     AN0.ninterms++;
02468     while ( ( id = GetAvailableThread() ) < 0 ) { MasterWait(); }
02469 /*
02470     Prepare the thread. Give it the term and variables.
02471 */
02472     LoadOneThread(0,id,thr,0);
02473     thr->busy = BUCKETASSIGNED;
02474     thr->free = BUCKETINUSE;
02475     numberoffullbuckets--;
02476 /*
02477     And signal the thread to run.
02478     Form now on we may only interfere with this bucket
02479     1: after it has been marked BUCKETCOMINGFREE
02480     2: when thr->busy == BUCKETDOINGTERM and then only when protected by
02481        thr->lock. This would be for load balancing.
02482 */
02483     WakeupThread(id,type);
02484 /*     AN0.ninterms += thr->ddterms; */
02485     return(0);
02486 }
02487
02488 /*
02489  #] SendOneBucket :
02490  #[ InParallelProcessor :
02491 */
02511 int InParallelProcessor(void)
02512 {
02513     GETIDENTITY
02514     int i, id, retval = 0, num = 0;
02515     EXPRESSIONS e;
02516     if ( numberofworkers >= 2 ) {
02517         SetWorkerFiles();
02518         for ( i = 0; i < NumExpressions; i++ ) {
02519             e = Expressions+i;
02520             if ( e->partodo <= 0 ) continue;
02521             if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
02522                 || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
02523                 || e->status == INTOHIDELEXPRESSION || e->status == INTOHIDEGEXPRESSION ) {
02524             }
02525             else {
02526                 e->partodo = 0;
02527                 continue;
02528             }
02529             if ( e->counter == 0 ) { /* Expression with zero terms */
02530                 e->partodo = 0;
02531                 continue;
02532             }
02533 /*
02534             This expression should go to an idle worker
02535 */
02536             while ( ( id = GetAvailableThread() ) < 0 ) { MasterWait(); }
02537             LoadOneThread(0,id,0,-1);
02538             AB[id]->R.exprtodo = i;
02539             WakeupThread(id,DOONEEXPRESSION);
02540             num++;
02541         }
02542 /*
02543         Now we have to wait for all workers to finish
02544 */
02545         if ( num > 0 ) MasterWaitAll();
02546
02547         if ( AC.CollectFun ) AR.DeferFlag = 0;
02548     }

```

```

02549     else {
02550         for ( i = 0; i < NumExpressions; i++ ) {
02551             Expressions[i].partodo = 0;
02552         }
02553     }
02554     return(retval);
02555 }
02556
02557 /*
02558  #] InParallelProcessor :
02559  #[ ThreadsProcessor :
02560 */
02585 int ThreadsProcessor(EXPRESSIONS e, WORD LastExpression, WORD fromspectator)
02586 {
02587     ALLPRIVATES *B0 = AB[0], *B = B0;
02588     int id, oldgzipCompress, endofinput = 0, j, still, k, defcount = 0, bra = 0, first = 1;
02589     LONG dd = 0, ddd, thrbufsiz, thrbufsiz0, thrbufsiz2, numbucket = 0, numpasses;
02590     LONG num, i;
02591     WORD *oldworkpointer = AT0.WorkPointer, *tt, *ttco = 0, *tl = 0, ter, *tstop = 0, *t2;
02592     THREADBUCKET *thr = 0;
02593     FILEHANDLE *oldoutfile = AR0.outfile;
02594     GETTERM GetTermP = &GetTerm;
02595     POSITION eonfile = AS.OldOnFile[e-Expressions];
02596     numberoffullbuckets = 0;
02597 /*
02598     Start up all threads. The lock needs to be around the whole loop
02599     to keep processes from terminating quickly and putting themselves
02600     in the list of available threads again.
02601 */
02602     AM.tracebackflag = 1;
02603
02604     AS.sLevel = AR0.sLevel;
02605     LOCK(availabilitylock);
02606     toposofavailables = 0;
02607     for ( id = 1; id <= numberofworkers; id++ ) {
02608         WakeupThread(id, STARTNEWEXPRESSION);
02609     }
02610     UNLOCK(availabilitylock);
02611     NewSort(BHEAD0);
02612     AN0.ninterms = 1;
02613 /*
02614     Now for redefine
02615 */
02616     if ( AC.numpfirstnum > 0 ) {
02617         for ( j = 0; j < AC.numpfirstnum; j++ ) {
02618             AC.inputnumbers[j] = -1;
02619         }
02620     }
02621     MasterWaitAll();
02622 /*
02623     Determine a reasonable bucketsize.
02624     This is based on the value of AC.ThreadBucketSize and the number
02625     of terms. We want at least 5 buckets per worker at the moment.
02626     Some research should show whether this is reasonable.
02627
02628     The number of terms in the expression is in e->counter
02629 */
02630     thrbufsiz2 = thrbufsiz = AC.ThreadBucketSize-1;
02631     if ( ( e->counter / ( numberofworkers * 5 ) ) < thrbufsiz ) {
02632         thrbufsiz = e->counter / ( numberofworkers * 5 ) - 1;
02633         if ( thrbufsiz < 0 ) thrbufsiz = 0;
02634     }
02635     thrbufsiz0 = thrbufsiz;
02636     numpasses = 5; /* this is just for trying */
02637     thrbufsiz = thrbufsiz0 / ( 2 « numpasses );
02638 /*
02639     Mark all buckets as free and take the first.
02640 */
02641     for ( j = 0; j < numthreadbuckets; j++ )
02642         threadbuckets[j]->free = BUCKETFREE;
02643     thr = threadbuckets[0];
02644 /*
02645     #[ Whole brackets :
02646
02647     First we look whether we have to work with entire brackets
02648     This is the case when there is a non-NULL address in e->bracketinfo.
02649     Of course we shouldn't have interference from a collect or keep statement.
02650 */
02651 #ifdef WHOLEBRACKETS
02652     if ( e->bracketinfo && AC.CollectFun == 0 && AR0.DeferFlag == 0 ) {
02653         FILEHANDLE *curfile;
02654         int didone = 0;
02655         LONG num, n;
02656         AN0.expr = e;
02657         for ( n = 0; n < e->bracketinfo->indexfill; n++ ) {
02658             num = TreatIndexEntry(B0,n);
02659             if ( num > 0 ) {

```

```

02660             didone = 1;
02661  /*
02662             This bracket can be sent off.
02663             1: Look for an empty bucket
02664  */
02665  ReTry;;
02666             for ( j = 0; j < numthreadbuckets; j++ ) {
02667                 switch ( threadbuckets[j]->free ) {
02668                     case BUCKETFREE:
02669                         thr = threadbuckets[j];
02670                         goto Found1;
02671                     case BUCKETCOMINGFREE:
02672                         thr = threadbuckets[j];
02673                         thr->free = BUCKETFREE;
02674                         for ( k = j+1; k < numthreadbuckets; k++ )
02675                             threadbuckets[k-1] = threadbuckets[k];
02676                         threadbuckets[numthreadbuckets-1] = thr;
02677                         j--;
02678                         break;
02679                     default:
02680                         break;
02681                 }
02682             }
02683  Found1;;
02684             if ( j < numthreadbuckets ) {
02685  /*
02686                 Found an empty bucket. Fill it.
02687  */
02688                 thr->firstbracket = n;
02689                 thr->lastbracket = n + num - 1;
02690                 thr->type = BUCKETDOINGBRACKET;
02691                 thr->free = BUCKETFILLED;
02692                 thr->firstterm = AN0.ninterms;
02693                 for ( j = n; j < n+num; j++ ) {
02694                     AN0.ninterms += e->bracketinfo->indexbuffer[j].termsinbracket;
02695                 }
02696                 n += num-1;
02697                 numberoffullbuckets++;
02698                 if ( topofavailables > 0 ) {
02699                     SendOneBucket(DOBRACKETS);
02700                 }
02701             }
02702  /*
02703             All buckets are in use.
02704             Look/wait for an idle worker. Give it a bucket.
02705             After that, retry for a bucket
02706  */
02707             else {
02708                 while ( topofavailables <= 0 ) {
02709                     MasterWait();
02710                 }
02711                 SendOneBucket(DOBRACKETS);
02712                 goto ReTry;
02713             }
02714         }
02715     }
02716     if ( didone ) {
02717  /*
02718             And now put the input back in the original position.
02719  */
02720         switch ( e->status ) {
02721             case UNHIDELEXPRESSION:
02722             case UNHIDEGEXPRESSION:
02723             case DROPHLEXPRESSION:
02724             case DROPHGEXPRESSION:
02725             case HIDDENLEXPRESSION:
02726             case HIDDENGEXPRESSION:
02727                 curfile = AR0.hidefile;
02728                 break;
02729             default:
02730                 curfile = AR0.infile;
02731                 break;
02732         }
02733         SetScratch(curfile,&eonfile);
02734         GetTerm(B0,AT0.WorkPointer);
02735  /*
02736             Now we point the GetTerm that is used to the one that is selective
02737  */
02738         GetTermP = &GetTerm2;
02739  /*
02740             Next wait till there is a bucket available and initialize thr to it.
02741  */
02742         for(;;) {
02743             for ( j = 0; j < numthreadbuckets; j++ ) {
02744                 switch ( threadbuckets[j]->free ) {
02745                     case BUCKETFREE:
02746                         thr = threadbuckets[j];

```

```

02747         goto Found2;
02748     case BUCKETCOMINGFREE:
02749         thr = threadbuckets[j];
02750         thr->free = BUCKETFREE;
02751         for ( k = j+1; k < numthreadbuckets; k++ )
02752             threadbuckets[k-1] = threadbuckets[k];
02753         threadbuckets[numthreadbuckets-1] = thr;
02754         j--;
02755         break;
02756     default:
02757         break;
02758     }
02759     }
02760     while ( topofavailables <= 0 ) {
02761         MasterWait();
02762     }
02763     while ( topofavailables > 0 && numberoffullbuckets > 0 ) {
02764         SendOneBucket(DOBRACKETS);
02765     }
02766 }
02767 Found2:;
02768     while ( numberoffullbuckets > 0 ) {
02769         while ( topofavailables <= 0 ) {
02770             MasterWait();
02771         }
02772         while ( topofavailables > 0 && numberoffullbuckets > 0 ) {
02773             SendOneBucket(DOBRACKETS);
02774         }
02775     }
02776 /*
02777     Disable the 'warming up' with smaller buckets.
02778
02779     numpasses = 0;
02780     thrbufsiz = thrbufsiz0;
02781 */
02782     AN0.lastinindex = -1;
02783 }
02784 MasterWaitAll();
02785 }
02786 #endif
02787 /*
02788     #] Whole brackets :
02789
02790     Now the loop to start a bucket
02791 */
02792 for(;;) {
02793     if ( fromspectator ) {
02794         ter = GetFromSpectator(thr->threadbuffer,fromspectator-1);
02795         if ( ter == 0 ) fromspectator = 0;
02796     }
02797     else {
02798         ter = GetTermP(B0,thr->threadbuffer);
02799 /*
02800         At this point we could check whether the input term is
02801         just an expression that resides in a scratch file.
02802         If this is the case we should store the current input info
02803         (file and buffer content) and redirect the input.
02804         At the end we can go back to where we were.
02805         There are two possibilities: we are in the same scratchfile
02806         as the main input and the other expression is still in the
02807         input buffer, or we have to do some reading. The reading is
02808         of course done in GetTerm.
02809
02810         How to set this up can also be studied in TestSub (in file proces.c)
02811         where it checks for EXPRESSION.
02812 */
02813     }
02814     if ( ter < 0 ) break;
02815     if ( ter == 0 ) { endofinput = 1; goto Finalize; }
02816     dd = AN0.deferskipped;
02817     if ( AR0.DeferFlag ) {
02818         defcount = 0;
02819         thr->deferbuffer[defcount++] = AR0.DefPosition;
02820         ttco = thr->compressbuffer; t1 = AR0.CompressBuffer; j = *t1;
02821         NCOPY(ttco,t1,j);
02822     }
02823     else if ( first && ( AC.CollectFun == 0 ) ) { /* Brackets ? */
02824         first = 0;
02825         t1 = tstop = thr->threadbuffer;
02826         tstop += *tstop; tstop -= ABS(tstop[-1]);
02827         t1++;
02828         while ( t1 < tstop ) {
02829             if ( t1[0] == HAAKJE ) { bra = 1; break; }
02830             t1 += t1[1];
02831         }
02832         t1 = thr->threadbuffer;
02833     }

```

```

02834 /*
02835     Check whether we have a collect,function going. If so execute it.
02836 */
02837     if ( AC.CollectFun && *(thr->threadbuffer) < (AM.MaxTer/((LONG)sizeof(WORD))-10) ) {
02838         if ( ( dd = GetMoreTerms(thr->threadbuffer) ) < 0 ) {
02839             LowerSortLevel(); goto ProcErr;
02840         }
02841     }
02842 /*
02843     Check whether we have a priority task:
02844 */
02845     if ( topofavailables > 0 && numberoffullbuckets > 0 ) SendOneBucket(LOWESTLEVELGENERATION);
02846 /*
02847     Now put more terms in the bucket. Position tt after the first term
02848 */
02849     tt = thr->threadbuffer; tt += *tt;
02850     thr->totnum = 1;
02851     thr->usenum = 0;
02852 /*
02853     Next we worry about the 'slow startup' in which we make the initial
02854     buckets smaller, so that we get all threads busy as soon as possible.
02855 */
02856     if ( numpasses > 0 ) {
02857         numbucket++;
02858         if ( numbucket >= numberofworkers ) {
02859             numbucket = 0;
02860             numpasses--;
02861             if ( numpasses == 0 ) thrbufsiz = thrbufsiz0;
02862             else thrbufsiz = thrbufsiz0 / (2 << numpasses);
02863         }
02864         thrbufsiz2 = thrbufsiz + thrbufsiz/5; /* for completing brackets */
02865     }
02866 /*
02867     we have already 1+dd terms
02868 */
02869     while ( ( dd < thrbufsiz ) &&
02870         ( tt - thr->threadbuffer ) < ( thr->threadbuffersize - AM.MaxTer/((LONG)sizeof(WORD)) - 2
02871     ) ) {
02872 /*
02873     First check:
02874 */
02875     if ( topofavailables > 0 && numberoffullbuckets > 0 )
02876         SendOneBucket(LOWESTLEVELGENERATION);
02877 /*
02878     There is room in the bucket. Fill yet another term.
02879 */
02880     if ( GetTermP(B0,tt) == 0 ) { endofinput = 1; break; }
02881     dd++;
02882     thr->totnum++;
02883     dd += AN0.deferskipped;
02884     if ( AR0.DeferFlag ) {
02885         thr->deferbuffer[defcount++] = AR0.DefPosition;
02886         t1 = AR0.CompressBuffer; j = *t1;
02887         NCOPY(ttco,t1,j);
02888     }
02889     if ( AC.CollectFun && *tt < (AM.MaxTer/((LONG)sizeof(WORD))-10) ) {
02890         if ( ( ddd = GetMoreTerms(tt) ) < 0 ) {
02891             LowerSortLevel(); goto ProcErr;
02892         }
02893         dd += ddd;
02894     }
02895     t1 = tt;
02896     tt += *tt;
02897 }
02898 /*
02899     Check whether there are regular brackets and if we have no DeferFlag
02900     and no collect, we try to add more terms till we finish the current
02901     bracket. We should however not overdo it. Let us say: up to 20%
02902     more terms are allowed.
02903 */
02904     if ( bra ) {
02905         tstop = t1 + *t1; tstop -= ABS(tstop[-1]);
02906         t2 = t1+1;
02907         while ( t2 < tstop ) {
02908             if ( t2[0] == HAAKJE ) { break; }
02909             t2 += t2[1];
02910         }
02911         if ( t2[0] == HAAKJE ) {
02912             t2 += t2[1]; num = t2 - t1;
02913             while ( ( dd < thrbufsiz2 ) &&
02914                 ( tt - thr->threadbuffer ) < ( thr->threadbuffersize - AM.MaxTer - 2 ) ) {
02915 /*
02916     First check:
02917 */
02918             if ( topofavailables > 0 && numberoffullbuckets > 0 )
02919                 SendOneBucket(LOWESTLEVELGENERATION);
02920             /*

```



```

02918             There is room in the bucket. Fill yet another term.
02919  */
02920             if ( GetTermP(B0,tt) == 0 ) { endofinput = 1; break; }
02921  /*
02922             Same bracket?
02923  */
02924             tstop = tt + *tt; tstop -= ABS(tstop[-1]);
02925             if ( tstop-tt < num ) { /* Different: abort */
02926                 ARO.KeptInHold = 1;
02927                 break;
02928             }
02929             for ( i = 1; i < num; i++ ) {
02930                 if ( t1[i] != tt[i] ) break;
02931             }
02932             if ( i < num ) { /* Different: abort */
02933                 ARO.KeptInHold = 1;
02934                 break;
02935             }
02936  /*
02937             Same bracket. We need this term.
02938  */
02939             dd++;
02940             thr->totnum++;
02941             tt += *tt;
02942         }
02943     }
02944 }
02945 thr->ddterms = dd; /* total number of terms including keep brackets */
02946 thr->firstterm = AN0.ninterms;
02947 AN0.ninterms += dd;
02948 *tt = 0; /* mark end of bucket */
02949 thr->free = BUCKETFILLED;
02950 thr->type = BUCKETDOINGTERMS;
02951 numberoffullbuckets++;
02952 if ( topofavailables <= 0 && endofinput == 0 ) {
02953  /*
02954     Problem: topofavailables may already be > 0, but the
02955     thread has not yet gone into waiting. Can the signal get lost?
02956     How can we tell that a thread is waiting for a signal?
02957
02958     All threads are busy. Try to load up another bucket.
02959     In the future we could be more sophisticated.
02960     At the moment we load a complete bucket which could be
02961     1000 terms or even more.
02962     In principle it is better to keep a full bucket ready
02963     and check after each term we put in the next bucket. That
02964     way we don't waste time of the workers.
02965  */
02966     for ( j = 0; j < numthreadbuckets; j++ ) {
02967         switch ( threadbuckets[j]->free ) {
02968             case BUCKETFREE:
02969                 thr = threadbuckets[j];
02970                 if ( !endofinput ) goto NextBucket;
02971         /*
02972             If we are at the end of the input we mark
02973             the free buckets in a special way. That way
02974             we don't keep running into them.
02975         */
02976                 thr->free = BUCKETATEND;
02977                 break;
02978             case BUCKETCOMINGFREE:
02979                 thr = threadbuckets[j];
02980                 thr->free = BUCKETFREE;
02981         /*
02982             Bucket has just been finished.
02983             Put at the end of the list. We don't want
02984             an early bucket to wait to be treated last.
02985         */
02986                 for ( k = j+1; k < numthreadbuckets; k++ )
02987                     threadbuckets[k-1] = threadbuckets[k];
02988                 threadbuckets[numthreadbuckets-1] = thr;
02989                 j--; /* we have to redo the same number j. */
02990                 break;
02991             default:
02992                 break;
02993         }
02994     }
02995  /*
02996     We have no free bucket or we are at the end.
02997     The only thing we can do now is wait for a worker to come free,
02998     provided there are still buckets to send.
02999  */
03000 }
03001  /*
03002     Look for the next bucket to send. There is at least one full bucket!
03003  */
03004     for ( j = 0; j < numthreadbuckets; j++ ) {

```

```

03005         if ( threadbuckets[j]->free == BUCKETFILLED ) {
03006             thr = threadbuckets[j];
03007             for ( k = j+1; k < numthreadbuckets; k++ )
03008                 threadbuckets[k-1] = threadbuckets[k];
03009             threadbuckets[numthreadbuckets-1] = thr;
03010             break;
03011         }
03012     }
03013 /*
03014     Wait for a thread to become available
03015     The bucket we are going to use is in thr.
03016 */
03017 DoBucket::
03018     AN0.ninterms++;
03019     while ( ( id = GetAvailableThread() ) < 0 ) { MasterWait(); }
03020 /*
03021     Prepare the thread. Give it the term and variables.
03022 */
03023     LoadOneThread(0,id,thr,0);
03024     LOCK(thr->lock);
03025     thr->busy = BUCKETASSIGNED;
03026     UNLOCK(thr->lock);
03027     thr->free = BUCKETINUSE;
03028     numberoffullbuckets--;
03029 /*
03030     And signal the thread to run.
03031     Form now on we may only interfere with this bucket
03032     1: after it has been marked BUCKETCOMINGFREE
03033     2: when thr->busy == BUCKETDOINGTERM and then only when protected by
03034        thr->lock. This would be for load balancing.
03035 */
03036     WakeupThread(id,LOWESTLEVELGENERATION);
03037 /*
03038     AN0.ninterms += thr->ddterms; */
03039 /*
03040     Now look whether there is another bucket filled and a worker available
03041 */
03042     if ( topofavailables > 0 ) { /* there is a worker */
03043         for ( j = 0; j < numthreadbuckets; j++ ) {
03044             if ( threadbuckets[j]->free == BUCKETFILLED ) {
03045                 thr = threadbuckets[j];
03046                 for ( k = j+1; k < numthreadbuckets; k++ )
03047                     threadbuckets[k-1] = threadbuckets[k];
03048                 threadbuckets[numthreadbuckets-1] = thr;
03049                 goto DoBucket; /* and we found a bucket */
03050             }
03051         }
03052     } /*
03053         no bucket is loaded but there is a thread available
03054         find a bucket to load. If there is none (all are USED or ATEND)
03055         we jump out of the loop.
03056 */
03057     for ( j = 0; j < numthreadbuckets; j++ ) {
03058         switch ( threadbuckets[j]->free ) {
03059             case BUCKETFREE:
03060                 thr = threadbuckets[j];
03061                 if ( !endofinput ) goto NextBucket;
03062                 thr->free = BUCKETATEND;
03063                 break;
03064             case BUCKETCOMINGFREE:
03065                 thr = threadbuckets[j];
03066                 if ( endofinput ) {
03067                     thr->free = BUCKETATEND;
03068                 }
03069                 else {
03070                     thr->free = BUCKETFREE;
03071                     for ( k = j+1; k < numthreadbuckets; k++ )
03072                         threadbuckets[k-1] = threadbuckets[k];
03073                     threadbuckets[numthreadbuckets-1] = thr;
03074                     j--;
03075                 }
03076                 break;
03077             default:
03078                 break;
03079         }
03080     }
03081     if ( j >= numthreadbuckets ) break;
03082 }
03083 /*
03084     No worker available.
03085     Look for a bucket to load.
03086     Its number will be in "still"
03087 */
03088 Finalize::
03089     still = -1;
03090     for ( j = 0; j < numthreadbuckets; j++ ) {
03091         switch ( threadbuckets[j]->free ) {

```

```

03092         case BUCKETFREE:
03093             thr = threadbuckets[j];
03094             if ( !endofinput ) goto NextBucket;
03095             thr->free = BUCKETATEND;
03096             break;
03097         case BUCKETCOMINGFREE:
03098             thr = threadbuckets[j];
03099             if ( endofinput ) thr->free = BUCKETATEND;
03100             else {
03101                 thr->free = BUCKETFREE;
03102                 for ( k = j+1; k < numthreadbuckets; k++ )
03103                     threadbuckets[k-1] = threadbuckets[k];
03104                 threadbuckets[numthreadbuckets-1] = thr;
03105                 j--;
03106             }
03107             break;
03108         case BUCKETFILLED:
03109             if ( still < 0 ) still = j;
03110             break;
03111         default:
03112             break;
03113     }
03114 }
03115 if ( still < 0 ) {
03116 /*
03117     No buckets to be executed and no buckets FREE.
03118     We must be at the end. Break out of the loop.
03119 */
03120     break;
03121 }
03122 thr = threadbuckets[still];
03123 for ( k = still+1; k < numthreadbuckets; k++ )
03124     threadbuckets[k-1] = threadbuckets[k];
03125 threadbuckets[numthreadbuckets-1] = thr;
03126 goto DoBucket;
03127 }
03128 NextBucket;;
03129 }
03130 /*
03131     Now the stage one load balancing.
03132     If the load has been readjusted we have again filled buckets.
03133     In that case we jump back in the loop.
03134
03135     Tricky point: when do the workers see the new value of AT.LoadBalancing?
03136     It should activate the locks on thr->busy
03137 */
03138 if ( AC.ThreadBalancing ) {
03139     for ( id = 1; id <= numberofworkers; id++ ) {
03140         AB[id]->T.LoadBalancing = 1;
03141     }
03142     if ( LoadReadjusted() ) goto Finalize;
03143     for ( id = 1; id <= numberofworkers; id++ ) {
03144         AB[id]->T.LoadBalancing = 0;
03145     }
03146 }
03147 if ( AC.ThreadBalancing ) {
03148 /*
03149     The AS.Balancing flag should have Generator look for
03150     free workers and apply the "buro" method.
03151
03152     There is still a serious problem.
03153     When for instance a sum_, there may be space created in a local
03154     compiler buffer for a wildcard substitution or whatever.
03155     Compiler buffer execution scribble space.....
03156     This isn't copied along?
03157     Look up ebufnum. There are 12 places with AddRHS!
03158     Problem: one process allocates in ebuf. Then term is given to
03159     other process. It would like to use from this ebuf, but the sender
03160     finishes first and removes the ebuf (and/or overwrites it).
03161
03162     Other problem: local $ variables aren't copied along.
03163 */
03164     AS.Balancing = 0;
03165 }
03166 MasterWaitAll();
03167 AS.Balancing = 0;
03168 /*
03169     When we deal with the last expression we can now remove the input
03170     scratch file. This saves potentially much disk space (up to 1/3)
03171 */
03172 if ( LastExpression ) {
03173     UpdateMaxSize();
03174     if ( AR0.infile->handle >= 0 ) {
03175         CloseFile(AR0.infile->handle);
03176         AR0.infile->handle = -1;
03177         remove(AR0.infile->name);
03178         PUTZERO(AR0.infile->PPosition);

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03179         AR0.infile->POfill = AR0.infile->POfull = AR0.infile->PObuffer;
03180     }
03181 }
03182 /*
03183 We order the threads to finish in the MasterMerge routine
03184 It will start with waiting for all threads to finish.
03185 One could make an administration in which threads that have
03186 finished can start already with the final sort but
03187 1: The load balancing should not make this super urgent
03188 2: It would definitely not be very compatible with the second
03189    stage load balancing.
03190 */
03191 oldgzipCompress = AR0.gzipCompress;
03192 AR0.gzipCompress = 0;
03193 if ( AR0.outtohide ) AR0.outfile = AR0.hidefile;
03194 if ( MasterMerge() < 0 ) {
03195     if ( AR0.outtohide ) AR0.outfile = oldoutfile;
03196     AR0.gzipCompress = oldgzipCompress;
03197     goto ProcErr;
03198 }
03199 if ( AR0.outtohide ) AR0.outfile = oldoutfile;
03200 AR0.gzipCompress = oldgzipCompress;
03201 /*
03202 Now wait for all threads to be ready to give them the cleaning up signal.
03203 With the new MasterMerge routine we can do the cleanup already automatically
03204 avoiding having to send these signals.
03205 */
03206 MasterWaitAll();
03207 AR0.sLevel--;
03208 for ( id = 1; id < AM.totalnumberofthreads; id++ ) {
03209     if ( GetThread(id) > 0 ) WakeupThread(id,CLEANUPEXPRESSION);
03210 }
03211 e->numdummies = 0;
03212 for ( id = 1; id < AM.totalnumberofthreads; id++ ) {
03213     if ( AB[id]->R.MaxDum - AM.IndDum > e->numdummies )
03214         e->numdummies = AB[id]->R.MaxDum - AM.IndDum;
03215     AR0.expchanged |= AB[id]->R.expchanged;
03216 }
03217 /*
03218 And wait for all to be clean.
03219 */
03220 MasterWaitAll();
03221 AT0.WorkPointer = oldworkpointer;
03222 return(0);
03223 ProcErr:;
03224 return(-1);
03225 }
03226
03227 /*
03228 #] ThreadsProcessor :
03229 #[ LoadReadjusted :
03230 */
03249 int LoadReadjusted(void)
03250 {
03251     ALLPRIVATES *B0 = AB[0];
03252     THREADBUCKET *thr = 0, *thrtogo = 0;
03253     int numtogo, numfree, numbusy, n, nperbucket, extra, i, j, u, bus;
03254     LONG numinput;
03255     WORD *t1, *c1, *t2, *c2, *t3;
03256 /*
03257 Start with waiting for at least one free processor.
03258 We don't want the master competing for time when all are busy.
03259 */
03260 while ( topofavailables <= 0 ) MasterWait();
03261 /*
03262 Now look for the fullest bucket and make a list of free buckets
03263 The bad part is that most numbers can change at any moment.
03264 */
03265 restart:;
03266     numtogo = 0;
03267     numfree = 0;
03268     numbusy = 0;
03269     for ( j = 0; j < numthreadbuckets; j++ ) {
03270         thr = threadbuckets[j];
03271         if ( thr->free == BUCKETFREE || thr->free == BUCKETATEND
03272             || thr->free == BUCKETCOMINGFREE ) {
03273             freebuckets[numfree++] = thr;
03274         }
03275         else if ( thr->type != BUCKETDOINGTERMS ) {}
03276         else if ( thr->totnum > 1 ) { /* never steal from a bucket with one term */
03277             LOCK(thr->lock);
03278             bus = thr->busy;
03279             UNLOCK(thr->lock);
03280             if ( thr->free == BUCKETINUSE ) {
03281                 n = thr->totnum-thr->usenum;
03282                 if ( bus == BUCKETASSIGNED ) numbusy++;
03283                 else if ( ( bus != BUCKETASSIGNED )

```

```

03284         && ( n > numtogo ) ) {
03285             numtogo = n;
03286             thrtogo = thr;
03287         }
03288     }
03289     else if ( bus == BUCKETTOBERELEASED
03290         && thr->free == BUCKETRELEASED ) {
03291         freebuckets[numfree++] = thr;
03292         thr->free = BUCKETATEND;
03293         LOCK(thr->lock);
03294         thr->busy = BUCKETPREPARINGTERM;
03295         UNLOCK(thr->lock);
03296     }
03297 }
03298 }
03299 if ( numfree == 0 ) return(0); /* serious problem */
03300 if ( numtogo > 0 ) { /* provisionally there is something to be stolen */
03301     thr = thrtogo;
03302 /*
03303     If the number has changed there is good progress.
03304     Maybe there is another thread that needs assistance.
03305     We start all over.
03306 */
03307     if ( thr->totnum-thr->usenum < numtogo ) goto restart;
03308 /*
03309     If the thread is in the term loading phase
03310     (thr->busy == BUCKETPREPARINGTERM) we better stay away from it.
03311     We wait now for the thread to be busy, and don't allow it
03312     now to drop out of this state till we are done here.
03313     This all depends on whether AT.LoadBalancing == 1 is seen by
03314     the thread.
03315 */
03316     LOCK(thr->lock);
03317     if ( thr->busy != BUCKETDOINGTERM ) {
03318         UNLOCK(thr->lock);
03319         goto restart;
03320     }
03321     if ( thr->totnum-thr->usenum < numtogo ) {
03322         UNLOCK(thr->lock);
03323         goto restart;
03324     }
03325     thr->free = BUCKETTERMINATED;
03326 /*
03327     The above will signal the thread we want to terminate.
03328     Next all effort goes into making sure the landing is soft.
03329     Unfortunately we don't want to wait for a signal, because the thread
03330     may be working for a long time on a single term.
03331 */
03332     if ( thr->usenum == thr->totnum ) {
03333 /*
03334         Terminated in the mean time or by now working on the
03335         last term. Try again.
03336 */
03337         thr->free = BUCKETATEND;
03338         UNLOCK(thr->lock);
03339         goto restart;
03340     }
03341     goto intercepted;
03342 }
03343 /* This has always been commented. Indeed no lock is held here. */
03344 /* UNLOCK(thr->lock); */
03345 if ( numbusy > 0 ) {
03346     /* JD: this avoids large runtimes for tform tests under valgrind.
03347        What seems to happen is we return from here, goto Finalize, and
03348        end up in LoadReadjusted again without the threads having a
03349        chance to update their busy status. Then we end up here again.
03350        Sleep the thread for, say, lus to allow threads to acquire the lock. */
03351     struct timespec sleeptime;
03352     sleeptime.tv_sec = 0;
03353     sleeptime.tv_nsec = 1000L;
03354     nanosleep(&sleeptime, NULL);
03355     return(1); /* Wait a bit.... */
03356 }
03357 return(0);
03358 intercepted:
03359 /*
03360     We intercepted one successfully. Now it becomes interesting. Action:
03361     1: determine how many terms per free bucket.
03362     2: find the first untreated term.
03363     3: put the terms in the free buckets.
03364
03365     Remember: we still have the lock to avoid interference from the thread
03366     that is being robbed. We were holding it and then jumped here with
03367     goto intercepted.
03368 */
03369     numinput = thr->firstterm + thr->usenum;
03370     nperbucket = numtogo / numfree;

```

```

03371     extra = numtogo - nperbucket*numfree;
03372     if ( AR0.DeferFlag ) {
03373         t1 = thr->threadbuffer; c1 = thr->compressbuffer; u = thr->usenum;
03374         for ( n = 0; n < thr->usenum; n++ ) { t1 += *t1; c1 += *c1; }
03375         t3 = t1;
03376         if ( extra > 0 ) {
03377             for ( i = 0; i < extra; i++ ) {
03378                 thrtogo = freebuckets[i];
03379                 t2 = thrtogo->threadbuffer;
03380                 c2 = thrtogo->compressbuffer;
03381                 thrtogo->free = BUCKETFILLED;
03382                 thrtogo->type = BUCKETDOINGTERMS;
03383                 thrtogo->totnum = nperbucket+1;
03384                 thrtogo->ddterms = 0;
03385                 thrtogo->usenum = 0;
03386                 thrtogo->busy = BUCKETASSIGNED;
03387                 thrtogo->firstterm = numinput;
03388                 numinput += nperbucket+1;
03389                 for ( n = 0; n <= nperbucket; n++ ) {
03390                     j = *t1; NCOPY(t2,t1,j);
03391                     j = *c1; NCOPY(c2,c1,j);
03392                     thrtogo->deferbuffer[n] = thr->deferbuffer[u++];
03393                 }
03394                 *t2 = *c2 = 0;
03395             }
03396         }
03397         if ( nperbucket > 0 ) {
03398             for ( i = extra; i < numfree; i++ ) {
03399                 thrtogo = freebuckets[i];
03400                 t2 = thrtogo->threadbuffer;
03401                 c2 = thrtogo->compressbuffer;
03402                 thrtogo->free = BUCKETFILLED;
03403                 thrtogo->type = BUCKETDOINGTERMS;
03404                 thrtogo->totnum = nperbucket;
03405                 thrtogo->ddterms = 0;
03406                 thrtogo->usenum = 0;
03407                 thrtogo->busy = BUCKETASSIGNED;
03408                 thrtogo->firstterm = numinput;
03409                 numinput += nperbucket;
03410                 for ( n = 0; n < nperbucket; n++ ) {
03411                     j = *t1; NCOPY(t2,t1,j);
03412                     j = *c1; NCOPY(c2,c1,j);
03413                     thrtogo->deferbuffer[n] = thr->deferbuffer[u++];
03414                 }
03415                 *t2 = *c2 = 0;
03416             }
03417         }
03418     }
03419     else {
03420         t1 = thr->threadbuffer;
03421         for ( n = 0; n < thr->usenum; n++ ) { t1 += *t1; }
03422         t3 = t1;
03423         if ( extra > 0 ) {
03424             for ( i = 0; i < extra; i++ ) {
03425                 thrtogo = freebuckets[i];
03426                 t2 = thrtogo->threadbuffer;
03427                 thrtogo->free = BUCKETFILLED;
03428                 thrtogo->type = BUCKETDOINGTERMS;
03429                 thrtogo->totnum = nperbucket+1;
03430                 thrtogo->ddterms = 0;
03431                 thrtogo->usenum = 0;
03432                 thrtogo->busy = BUCKETASSIGNED;
03433                 thrtogo->firstterm = numinput;
03434                 numinput += nperbucket+1;
03435                 for ( n = 0; n <= nperbucket; n++ ) {
03436                     j = *t1; NCOPY(t2,t1,j);
03437                 }
03438                 *t2 = 0;
03439             }
03440         }
03441         if ( nperbucket > 0 ) {
03442             for ( i = extra; i < numfree; i++ ) {
03443                 thrtogo = freebuckets[i];
03444                 t2 = thrtogo->threadbuffer;
03445                 thrtogo->free = BUCKETFILLED;
03446                 thrtogo->type = BUCKETDOINGTERMS;
03447                 thrtogo->totnum = nperbucket;
03448                 thrtogo->ddterms = 0;
03449                 thrtogo->usenum = 0;
03450                 thrtogo->busy = BUCKETASSIGNED;
03451                 thrtogo->firstterm = numinput;
03452                 numinput += nperbucket;
03453                 for ( n = 0; n < nperbucket; n++ ) {
03454                     j = *t1; NCOPY(t2,t1,j);
03455                 }
03456                 *t2 = 0;
03457             }

```

```

03458     }
03459 }
03460 *t3 = 0; /* This is some form of extra insurance */
03461 if ( thr->free == BUCKETRELEASED && thr->busy == BUCKETTOBERELEASED ) {
03462     thr->free = BUCKETATEND; thr->busy = BUCKETPREPARINGTERM;
03463 }
03464 UNLOCK(thr->lock);
03465 return(1);
03466 }
03467 /*
03468 #] LoadReadjusted :
03469 #[ SortStrategy :
03471 */
03504 /*
03505 #] SortStrategy :
03506 #[ PutToMaster :
03507 */
03530 int PutToMaster(PHEAD WORD *term)
03531 {
03532     int i,j,nexti,ret = 0;
03533     WORD *t, *fill, *top, zero = 0;
03534     if ( term == 0 ) { /* Mark the end of the expression */
03535         t = &zero; j = 1;
03536     }
03537     else {
03538         t = term; ret = j = *term;
03539         if ( j == 0 ) { j = 1; } /* Just in case there is a spurious end */
03540     }
03541     i = AT.SB.FillBlock; /* The block we are working at */
03542     fill = AT.SB.MasterFill[i]; /* Where we are filling */
03543     top = AT.SB.MasterStop[i]; /* End of the block */
03544     while ( j > 0 ) {
03545         while ( j > 0 && fill < top ) {
03546             *fill++ = *t++; j--;
03547         }
03548         if ( j > 0 ) {
03549             /*
03550              We reached the end of the block.
03551              Get the next block and release this block.
03552              The order is important. This is why there should be at least
03553              4 blocks or deadlocks can occur.
03554             */
03555             nexti = i+1;
03556             if ( nexti > AT.SB.MasterNumBlocks ) {
03557                 nexti = 1;
03558             }
03559             LOCK(AT.SB.MasterBlockLock[nexti]);
03560             UNLOCK(AT.SB.MasterBlockLock[i]);
03561             AT.SB.MasterFill[i] = AT.SB.MasterStart[i];
03562             AT.SB.FillBlock = i = nexti;
03563             fill = AT.SB.MasterStart[i];
03564             top = AT.SB.MasterStop[i];
03565         }
03566     }
03567     AT.SB.MasterFill[i] = fill;
03568     return(ret);
03569 }
03570 /*
03571 #] PutToMaster :
03572 #[ SortBotOut :
03574 */
03575 #ifdef WITHSORTBOTS
03576 int
03577 SortBotOut(PHEAD WORD *term)
03578 {
03579     WORD im;
03580
03581     if ( AT.identity != 0 ) return(PutToMaster(BHEAD term));
03582
03583     if ( term == 0 ) {
03584         if ( FlushOut(&SortBotPosition,AR.outfile,1) ) return(-1);
03585         ADDPOS(AT.SS->SizeInFile[0],1);
03586         return(0);
03587     }
03588     else {
03589         numberofterms++;
03590         if ( ( im = PutOut(BHEAD term,&SortBotPosition,AR.outfile,1) ) < 0 ) {
03591             MLOCK(ErrorMessageLock);
03592             MesPrint("Called from MasterMerge/SortBotOut");
03593             MUNLOCK(ErrorMessageLock);
03594             return(-1);
03595         }
03596         ADDPOS(AT.SS->SizeInFile[0],im);
03597     }
03598 }

```

```

03607         return(im);
03608     }
03609 }
03610
03611 #endif
03612
03613 /*
03614  #] SortBotOut :
03615  #[ MasterMerge :
03616 */
03634 int MasterMerge(void)
03635 {
03636     ALLPRIVATES *B0 = AB[0], *B = 0;
03637     SORTING *S = AT0.SS;
03638     WORD **poin, **poin2, ul, k, i, im, *m1, j;
03639     WORD lpat, mpat, level, l1, l2, r1, r2, r3, c;
03640     WORD *m2, *m3, r31, r33, ki, *rr;
03641     UWORD *coef;
03642     POSITION position;
03643     FILEHANDLE *fin, *fout;
03644     #ifdef WITHSORTBOTS
03645     if ( numberofworkers > 2 ) return(SortBotMasterMerge());
03646     #endif
03647     fin = &S->file;
03648     if ( AR0.PolyFun == 0 ) { S->PolyFlag = 0; }
03649     else if ( AR0.PolyFunType == 1 ) { S->PolyFlag = 1; }
03650     else if ( AR0.PolyFunType == 2 ) {
03651         if ( AR0.PolyFunExp == 2
03652             || AR0.PolyFunExp == 3 ) S->PolyFlag = 1;
03653         else S->PolyFlag = 2;
03654     }
03655     S->TermsLeft = 0;
03656     coef = AN0.SoScratC;
03657     poin = S->poina; poin2 = S->poin2a;
03658     rr = AR0.CompressPointer;
03659     *rr = 0;
03660 /*
03661     #[ Setup :
03662 */
03663     S->inNum = numberofthreads;
03664     fout = AR0.outfile;
03665 /*
03666     Load the patches. The threads have to finish their sort first.
03667 */
03668     S->lPatch = S->inNum - 1;
03669 /*
03670     Claim all zero blocks. We need them anyway.
03671     In principle the workers should never get into these.
03672     We also claim all last blocks. This is a safety procedure that
03673     should prevent the workers from working their way around the clock
03674     before the master gets started again.
03675 */
03676     AS.MasterSort = 1;
03677     numberclaimed = 0;
03678     for ( i = 1; i <= S->lPatch; i++ ) {
03679         B = AB[i];
03680         LOCK(AT.SB.MasterBlockLock[0]);
03681         LOCK(AT.SB.MasterBlockLock[AT.SB.MasterNumBlocks]);
03682     }
03683 /*
03684     Now wake up the threads and have them start their final sorting.
03685     They should start with claiming their block and the master is
03686     not allowed to continue until that has been done.
03687     This waiting of the master will be done below in MasterWaitAllBlocks
03688 */
03689     for ( i = 0; i < S->lPatch; i++ ) {
03690         GetThread(i+1);
03691         WakeupThread(i+1,FINISHEXPRESSION);
03692     }
03693 /*
03694     Prepare the output file.
03695 */
03696     if ( fout->handle >= 0 ) {
03697         PUTZERO(position);
03698         SeekFile(fout->handle,&position,SEEK_END);
03699         ADDPOS(position,((fout->POfill-fout->PObuffer)*sizeof(WORD)));
03700     }
03701     else {
03702         SETBASEPOSITION(position,(fout->POfill-fout->PObuffer)*sizeof(WORD));
03703     }
03704 /*
03705     Wait for all threads to finish loading their first block.
03706 */
03707     MasterWaitAllBlocks();
03708 /*
03709     Claim all first blocks.
03710     We don't release the last blocks.

```



```

03711     The strategy is that we always keep the previous block.
03712     In principle it looks like it isn't needed for the last block but
03713     actually it is to keep the front from overrunning the tail when writing.
03714  */
03715     for ( i = 1; i <= S->lPatch; i++ ) {
03716         B = AB[i];
03717         LOCK(AT.SB.MasterBlockLock[1]);
03718         AT.SB.MasterBlock = 1;
03719     }
03720  /*
03721     #] Setup :
03722
03723     Now construct the tree:
03724  */
03725     lpat = 1;
03726     do { lpat <= 1; } while ( lpat < S->lPatch );
03727     mpat = ( lpat > 1 ) - 1;
03728     k = lpat - S->lPatch;
03729  /*
03730     k is the number of empty places in the tree. they will
03731     be at the even positions from 2 to 2*k
03732  */
03733     for ( i = 1; i < lpat; i++ ) { S->tree[i] = -1; }
03734     for ( i = 1; i <= k; i++ ) {
03735         im = ( i * 2 ) - 1;
03736         poin[im] = AB[i]->T.SB.MasterStart[AB[i]->T.SB.MasterBlock];
03737         poin2[im] = poin[im] + *(poin[im]);
03738         S->used[i] = im;
03739         S->ktoi[im] = i-1;
03740         S->tree[mpat+i] = 0;
03741         poin[im-1] = poin2[im-1] = 0;
03742     }
03743     for ( i = (k*2)+1; i <= lpat; i++ ) {
03744         S->used[i-k] = i;
03745         S->ktoi[i] = i-k-1;
03746         poin[i] = AB[i-k]->T.SB.MasterStart[AB[i-k]->T.SB.MasterBlock];
03747         poin2[i] = poin[i] + *(poin[i]);
03748     }
03749  /*
03750     the array poin tells the position of the i-th element of the S->tree
03751     'S->used' is a stack with the S->tree elements that need to be entered
03752     into the S->tree. at the beginning this is S->lPatch. during the
03753     sort there will be only very few elements.
03754     poin2 is the next value of poin. it has to be determined
03755     before the comparisons as the position or the size of the
03756     term indicated by poin may change.
03757     S->ktoi translates a S->tree element back to its stream number.
03758
03759     start the sort
03760  */
03761     level = S->lPatch;
03762  /*
03763     introduce one term
03764  */
03765     OneTerm:
03766     k = S->used[level];
03767     i = k + lpat - 1;
03768     if ( !(poin[k]) ) {
03769         do { if ( !( i >= 1 ) ) goto EndOfMerge; } while ( !S->tree[i] );
03770         if ( S->tree[i] == -1 ) {
03771             S->tree[i] = 0;
03772             level--;
03773             goto OneTerm;
03774         }
03775         k = S->tree[i];
03776         S->used[level] = k;
03777         S->tree[i] = 0;
03778     }
03779  /*
03780     move terms down the tree
03781  */
03782     while ( i >= 1 ) {
03783         if ( S->tree[i] > 0 ) {
03784             if ( ( c = CompareTerms(B0, poin[S->tree[i]],poin[k],(WORD)0) ) > 0 ) {
03785                 /*
03786                     S->tree[i] is the smaller. Exchange and go on.
03787                 */
03788                 S->used[level] = S->tree[i];
03789                 S->tree[i] = k;
03790                 k = S->used[level];
03791             }
03792             else if ( !c ) { /* Terms are equal */
03793                 /*
03794                     S->TermsLeft--;
03795                     Here the terms are equal and their coefficients
03796                     have to be added.
03797                 */

```

```

03798         l1 = *( m1 = poin[S->tree[i]] );
03799         l2 = *( m2 = poin[k] );
03800         if ( S->PolyWise ) { /* Here we work with PolyFun */
03801             WORD *ttl, *w;
03802             ttl = m1;
03803             m1 += S->PolyWise;
03804             m2 += S->PolyWise;
03805             if ( S->PolyFlag == 2 ) {
03806                 w = poly_ratfun_add(B0,m1,m2);
03807                 if ( *ttl + w[1] - m1[1] > AM.MaxTer/((LONG)sizeof(WORD)) ) {
03808                     MLOCK(ErrorMessageLock);
03809                     MesPrint("Term too complex in PolyRatFun addition. MaxTermSize of %10l is
too small",AM.MaxTer);
03810                     MUNLOCK(ErrorMessageLock);
03811                     Terminate(-1);
03812                 }
03813                 AT0.WorkPointer = w;
03814                 if ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 && w[1] > FUNHEAD ) {
03815                     goto cancelled;
03816                 }
03817             }
03818             else {
03819                 w = AT0.WorkPointer;
03820                 if ( w + m1[1] + m2[1] > AT0.WorkTop ) {
03821                     MLOCK(ErrorMessageLock);
03822                     MesPrint("MasterMerge: A Workspace of %10l is too small",AM.WorkSize);
03823                     MUNLOCK(ErrorMessageLock);
03824                     Terminate(-1);
03825                 }
03826                 AddArgs(B0,m1,m2,w);
03827             }
03828             r1 = w[1];
03829             if ( r1 <= FUNHEAD
03830                 || ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) )
03831                 { goto cancelled; }
03832             if ( r1 == m1[1] ) {
03833                 NCOPY(m1,w,r1);
03834             }
03835             else if ( r1 < m1[1] ) {
03836                 r2 = m1[1] - r1;
03837                 m2 = w + r1;
03838                 m1 += m1[1];
03839                 while ( --r1 >= 0 ) *--m1 = *--m2;
03840                 m2 = m1 - r2;
03841                 r1 = S->PolyWise;
03842                 while ( --r1 >= 0 ) *--m1 = *--m2;
03843                 *m1 -= r2;
03844                 poin[S->tree[i]] = m1;
03845             }
03846             else {
03847                 r2 = r1 - m1[1];
03848                 m2 = ttl - r2;
03849                 r1 = S->PolyWise;
03850                 m1 = ttl;
03851                 *m1 += r2;
03852                 poin[S->tree[i]] = m2;
03853                 NCOPY(m2,m1,r1);
03854                 r1 = w[1];
03855                 NCOPY(m2,w,r1);
03856             }
03857         }
03858         #ifdef WITHFLOAT
03859         else if ( AT.SortFloatMode ) {
03860             WORD *term1, *term2;
03861             term1 = poin[S->tree[i]];
03862             term2 = poin[k];
03863             if ( MergeWithFloat(B0, &term1,&term2) == 0 )
03864                 goto cancelled;
03865             poin[S->tree[i]] = term1;
03866         }
03867         #endif
03868         else {
03869             r1 = *( m1 += l1 - 1 );
03870             m1 -= ABS(r1) - 1;
03871             r1 = ( ( r1 > 0 ) ? (r1-1) : (r1+1) ) >> 1;
03872             r2 = *( m2 += l2 - 1 );
03873             m2 -= ABS(r2) - 1;
03874             r2 = ( ( r2 > 0 ) ? (r2-1) : (r2+1) ) >> 1;
03875
03876             if ( AddRat(B0, (UWORD *)m1,r1, (UWORD *)m2,r2,coef,&r3) ) {
03877                 MLOCK(ErrorMessageLock);
03878                 MesCall("MasterMerge");
03879                 MUNLOCK(ErrorMessageLock);
03880                 SETERROR(-1)
03881             }
03882
03883             if ( AN.ncmod != 0 ) {

```

```

03884         if ( ( AC.modmode & POSNEG ) != 0 ) {
03885             NormalModulus(coef,&r3);
03886         }
03887     else if ( BigLong(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod)) >= 0 ) {
03888         WORD ii;
03889         SubPLon(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod),coef,&r3);
03890         coef[r3] = 1;
03891         for ( ii = 1; ii < r3; ii++ ) coef[r3+ii] = 0;
03892     }
03893 }
03894 r3 *= 2;
03895 r33 = ( r3 > 0 ) ? ( r3 + 1 ) : ( r3 - 1 );
03896 if ( r3 < 0 ) r3 = -r3;
03897 if ( r1 < 0 ) r1 = -r1;
03898 r1 *= 2;
03899 r31 = r3 - r1;
03900 if ( !r3 ) {          /* Terms cancel */
03901 cancelled:
03902     ul = S->used[level] = S->tree[i];
03903     S->tree[i] = -1;
03904     /*
03905     We skip to the next term in stream ul
03906     */
03907     im = *poin2[ul];
03908     poin[ul] = poin2[ul];
03909     ki = S->ktoi[ul];
03910     if ( (poin[ul] + im + COMPINC) >=
03911         AB[ki+1]->T.SB.MasterStop[AB[ki+1]->T.SB.MasterBlock]
03912         && im > 0 ) {
03913     /*
03914         We made it to the end of the block. We have to
03915         release the previous block and claim the next.
03916     */
03917     B = AB[ki+1];
03918     i = AT.SB.MasterBlock;
03919     if ( i == 1 ) {
03920         UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterNumBlocks]);
03921     }
03922     else {
03923         UNLOCK(AT.SB.MasterBlockLock[i-1]);
03924     }
03925     if ( i == AT.SB.MasterNumBlocks ) {
03926     /*
03927         Move the remainder down into block 0
03928     */
03929     WORD *from, *to;
03930     to = AT.SB.MasterStart[1];
03931     from = AT.SB.MasterStop[i];
03932     while ( from > poin[ul] ) *--to = *--from;
03933     poin[ul] = to;
03934     i = 1;
03935     }
03936     else { i++; }
03937     LOCK(AT.SB.MasterBlockLock[i]);
03938     AT.SB.MasterBlock = i;
03939     poin2[ul] = poin[ul] + im;
03940 }
03941 else {
03942     poin2[ul] += im;
03943 }
03944 S->used[++level] = k;
03945 }
03946 else if ( !r31 ) {          /* copy coef into term1 */
03947     goto CopCof2;
03948 }
03949 else if ( r31 < 0 ) {          /* copy coef into term1
03950                                and adjust the length of term1 */
03951     goto CopCoef;
03952 }
03953 else {
03954     /*
03955     this is the dreaded calamity.
03956     is there enough space?
03957     */
03958     if( (poin[S->tree[i]]+11+r31) >= poin2[S->tree[i]] ) {
03959     /*
03960         no space! now the special trick for which
03961         we left 2*maxlng spaces open at the beginning
03962         of each patch.
03963     */
03964     if ( (11 + r31)*((LONG)sizeof(WORD)) >= AM.MaxTer ) {
03965         MLOCK(ErrorMessageLock);
03966         MesPrint("MasterMerge: Coefficient overflow during sort");
03967         MUNLOCK(ErrorMessageLock);
03968         goto ReturnError;
03969     }
03970     m2 = poin[S->tree[i]];

```

```

03971             m3 = ( poin[S->tree[i]] -= r31 );
03972             do { *m3++ = *m2++; } while ( m2 < m1 );
03973             m1 = m3;
03974         }
03975 CopCoef:
03976             *(poin[S->tree[i]]) += r31;
03977 CopCof2:
03978             m2 = (WORD *)coef; im = r3;
03979             NCOPY(m1,m2,im);
03980             *m1 = r33;
03981         }
03982     }
03983 /*
03984         Now skip to the next term in stream k.
03985 */
03986 NextTerm:
03987     im = poin2[k][0];
03988     poin[k] = poin2[k];
03989     ki = S->ktoi[k];
03990     if ( (poin[k] + im + COMPINC) >=
03991         AB[ki+1]->T.SB.MasterStop[AB[ki+1]->T.SB.MasterBlock]
03992         && im > 0 ) {
03993 /*
03994         We made it to the end of the block. We have to
03995         release the previous block and claim the next.
03996 */
03997         B = AB[ki+1];
03998         i = AT.SB.MasterBlock;
03999         if ( i == 1 ) {
04000             UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterNumBlocks]);
04001         }
04002         else {
04003             UNLOCK(AT.SB.MasterBlockLock[i-1]);
04004         }
04005         if ( i == AT.SB.MasterNumBlocks ) {
04006 /*
04007             Move the remainder down into block 0
04008 */
04009             WORD *from, *to;
04010             to = AT.SB.MasterStart[1];
04011             from = AT.SB.MasterStop[i];
04012             while ( from > poin[k] ) *--to = *--from;
04013             poin[k] = to;
04014             i = 1;
04015         }
04016         else { i++; }
04017         LOCK(AT.SB.MasterBlockLock[i]);
04018         AT.SB.MasterBlock = i;
04019         poin2[k] = poin[k] + im;
04020     }
04021     else {
04022         poin2[k] += im;
04023     }
04024     goto OneTerm;
04025 }
04026 }
04027 else if ( S->tree[i] < 0 ) {
04028     S->tree[i] = k;
04029     level--;
04030     goto OneTerm;
04031 }
04032 }
04033 /*
04034     found the smallest in the set. indicated by k.
04035     write to its destination.
04036 */
04037 S->TermsLeft++;
04038 if ( ( im = PutOut(B0,poin[k],&position,fout,1) ) < 0 ) {
04039     MLOCK(ErrorMessageLock);
04040     MesPrint("Called from MasterMerge with k = %d (stream %d)",k,S->ktoi[k]);
04041     MUNLOCK(ErrorMessageLock);
04042     goto ReturnError;
04043 }
04044 ADDPOS(S->SizeInFile[0],im);
04045 goto NextTerm;
04046 EndOfMerge:
04047 if ( FlushOut(&position,fout,1) ) goto ReturnError;
04048 ADDPOS(S->SizeInFile[0],1);
04049 CloseFile(fin->handle);
04050 remove(fin->name);
04051 fin->handle = -1;
04052 position = S->SizeInFile[0];
04053 MULPOS(position,sizeof(WORD));
04054 S->GenTerms = 0;
04055 for ( j = 1; j <= numberofworkers; j++ ) {
04056     S->GenTerms += AB[j]->T.SS->GenTerms;
04057 }

```

```

04058     WriteStats(&position, STATSPOSTSORT, NOCHECKLOGTYPE);
04059     Expressions[ARO.CurExpr].counter = S->TermsLeft;
04060     Expressions[ARO.CurExpr].size = position;
04061 /*
04062     Release all locks
04063 */
04064     for ( i = 1; i <= S->lPatch; i++ ) {
04065         B = AB[i];
04066         UNLOCK(AT.SB.MasterBlockLock[0]);
04067         if ( AT.SB.MasterBlock == 1 ) {
04068             UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterNumBlocks]);
04069         }
04070         else {
04071             UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterBlock-1]);
04072         }
04073         UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterBlock]);
04074     }
04075     AS.MasterSort = 0;
04076     return(0);
04077 ReturnError:
04078     for ( i = 1; i <= S->lPatch; i++ ) {
04079         B = AB[i];
04080         UNLOCK(AT.SB.MasterBlockLock[0]);
04081         if ( AT.SB.MasterBlock == 1 ) {
04082             UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterNumBlocks]);
04083         }
04084         else {
04085             UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterBlock-1]);
04086         }
04087         UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterBlock]);
04088     }
04089     AS.MasterSort = 0;
04090     return(-1);
04091 }
04092 /*
04093 #] MasterMerge :
04094 #[ SortBotMasterMerge :
04095 */
04096 /*
04097 #ifdef WITHSORTBOTS
04098
04115 int SortBotMasterMerge(void)
04116 {
04117     FILEHANDLE *fin, *fout;
04118     ALLPRIVATES *B = AB[0], *BB;
04119     POSITION position;
04120     SORTING *S = AT.SS;
04121     int i, j;
04122 /*
04123     Get the sortbots get to claim their writing blocks.
04124     We have to wait till all have been claimed because they also have to
04125     claim the last writing blocks of the workers to prevent the head of
04126     the circular buffer to overrun the tail.
04127
04128     Before waiting we can do some needed initializations.
04129     Also the master has to claim the last writing blocks of its input.
04130 */
04131     topsortbotavailables = 0;
04132     for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ ) {
04133         WakeupThread(i, INISORTBOT);
04134     }
04135
04136     AS.MasterSort = 1;
04137     fout = AR.outfile;
04138     numberofterms = 0;
04139     AR.CompressPointer[0] = 0;
04140     numberclaimed = 0;
04141     BB = AB[AT.SortBotIn1];
04142     LOCK(BB->T.SB.MasterBlockLock[BB->T.SB.MasterNumBlocks]);
04143     BB = AB[AT.SortBotIn2];
04144     LOCK(BB->T.SB.MasterBlockLock[BB->T.SB.MasterNumBlocks]);
04145
04146     MasterWaitAllSortBots();
04147 /*
04148     Now we can start up the workers. They will claim their writing blocks.
04149     Here the master will wait till all writing blocks have been claimed.
04150 */
04151     for ( i = 1; i <= numberofworkers; i++ ) {
04152         j = GetThread(i);
04153         WakeupThread(i, FINISHEXPRESSION);
04154     }
04155 /*
04156     Prepare the output file in the mean time.
04157 */
04158     if ( fout->handle >= 0 ) {
04159         PUTZERO(SortBotPosition);

```

```

04160         SeekFile(fout->handle, &SortBotPosition, SEEK_END);
04161         ADDPOS(SortBotPosition, ((fout->Pofill-fout->Pobuffer)*sizeof(WORD)));
04162     }
04163     else {
04164         SETBASEPOSITION(SortBotPosition, (fout->Pofill-fout->Pobuffer)*sizeof(WORD));
04165     }
04166     MasterWaitAllBlocks();
04167     /*
04168     Now we can start the sortbots after which the master goes in
04169     sortbot mode to do its part of the job (the very final merge and
04170     the writing to output file).
04171     */
04172     topsortbotavailables = 0;
04173     for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ ) {
04174         WakeupThread(i, RUNSORTBOT);
04175     }
04176     if ( SortBotMerge(BHEAD0) ) {
04177         MLOCK(ErrorMessageLock);
04178         MesPrint("Called from SortBotMasterMerge");
04179         MUNLOCK(ErrorMessageLock);
04180         AS.MasterSort = 0;
04181         return(-1);
04182     }
04183     /*
04184     And next the cleanup
04185     */
04186     if ( S->file.handle >= 0 )
04187     {
04188         fin = &S->file;
04189         CloseFile(fin->handle);
04190         remove(fin->name);
04191         fin->handle = -1;
04192     }
04193     position = S->SizeInFile[0];
04194     MULPOS(position, sizeof(WORD));
04195     S->GenTerms = 0;
04196     for ( j = 1; j <= numberofworkers; j++ ) {
04197         S->GenTerms += AB[j]->T.SS->GenTerms;
04198     }
04199     S->TermsLeft = numberofterms;
04200     WriteStats(&position, STATSPOSTSORT, NOCHECKLOGTYPE);
04201     Expressions[AR.CurExpr].counter = S->TermsLeft;
04202     Expressions[AR.CurExpr].size = position;
04203     AS.MasterSort = 0;
04204     /*
04205     The next statement is to prevent one of the sortbots not having
04206     completely cleaned up before the next module starts.
04207     If this statement is omitted every once in a while one of the sortbots
04208     is still running when the next expression starts and misses its
04209     initialization. The result is usually disastrous.
04210     */
04211     MasterWaitAllSortBots();
04212
04213     return(0);
04214 }
04215
04216 #endif
04217
04218 /*
04219 #] SortBotMasterMerge :
04220 #[ SortBotMerge :
04221 */
04222
04223 #ifdef WITHSORTBOTS
04224
04225 int SortBotMerge(PHEAD0)
04226 {
04227     GETBIDENTITY
04228     ALLPRIVATES *Bin1 = AB[AT.SortBotIn1], *Bin2 = AB[AT.SortBotIn2];
04229     WORD *term1, *term2, *next, *wp;
04230     int blin1, blin2; /* Current block numbers */
04231     int error = 0;
04232     WORD l1, l2, *m1, *m2, *w, r1, r2, r3, r33, r31, *ttl1, ii;
04233     WORD *to, *from, im, c;
04234     UWORD *coef;
04235     SORTING *S = AT.SS;
04236
04237     #ifndef WITHFLOAT
04238     WORD *fun1, *fun2, *fun3, *tstop1, *tstop2, size1, size2, size3, l3, jj, *m3;
04239     #endif
04240
04241     /*
04242     Set the pointers to the input terms and the output space
04243     */
04244     coef = AN.SoScratC;
04245     blin1 = 1;
04246     blin2 = 1;
04247     if ( AT.identity == 0 ) {
04248         wp = AT.WorkPointer;

```

```

04252     }
04253     else {
04254         wp = AT.WorkPointer = AT.WorkSpace;
04255     }
04256 /*
04257     Get the locks for reading the input
04258     This means that we can start once these locks have been cleared
04259     which means that there will be input.
04260 */
04261     LOCK(Bin1->T.SB.MasterBlockLock[blin1]);
04262     LOCK(Bin2->T.SB.MasterBlockLock[blin2]);
04263
04264     term1 = Bin1->T.SB.MasterStart[blin1];
04265     term2 = Bin2->T.SB.MasterStart[blin2];
04266     AT.SB.FillBlock = 1;
04267 /*
04268     Now the main loop. Keep going until one of the two hits the end.
04269 */
04270     while ( *term1 && *term2 ) {
04271         if ( ( c = CompareTerms(BHEAD term1,term2,(WORD)0) ) > 0 ) {
04272 /*
04273             #[ One is smallest :
04274 */
04275             if ( SortBotOut(BHEAD term1) < 0 ) {
04276                 MLOCK(ErrorMessageLock);
04277                 MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
04278                 MUNLOCK(ErrorMessageLock);
04279                 error = -1;
04280                 goto ReturnError;
04281             }
04282             im = *term1;
04283             next = term1 + im;
04284             if ( next >= Bin1->T.SB.MasterStop[blin1] || ( *next &&
04285                 next+*next+COMPINC > Bin1->T.SB.MasterStop[blin1] ) ) {
04286                 if ( blin1 == 1 ) {
04287                     UNLOCK(Bin1->T.SB.MasterBlockLock[Bin1->T.SB.MasterNumBlocks]);
04288                 }
04289                 else {
04290                     UNLOCK(Bin1->T.SB.MasterBlockLock[blin1-1]);
04291                 }
04292                 if ( blin1 == Bin1->T.SB.MasterNumBlocks ) {
04293 /*
04294                     Move the remainder down into block 0
04295 */
04296                     to = Bin1->T.SB.MasterStart[1];
04297                     from = Bin1->T.SB.MasterStop[Bin1->T.SB.MasterNumBlocks];
04298                     while ( from > next ) *--to = *--from;
04299                     next = to;
04300                     blin1 = 1;
04301                 }
04302                 else {
04303                     blin1++;
04304                 }
04305                 LOCK(Bin1->T.SB.MasterBlockLock[blin1]);
04306                 Bin1->T.SB.MasterBlock = blin1;
04307             }
04308             term1 = next;
04309 /*
04310             #[ One is smallest :
04311 */
04312         }
04313         else if ( c < 0 ) {
04314 /*
04315             #[ Two is smallest :
04316 */
04317             if ( SortBotOut(BHEAD term2) < 0 ) {
04318                 MLOCK(ErrorMessageLock);
04319                 MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
04320                 MUNLOCK(ErrorMessageLock);
04321                 error = -1;
04322                 goto ReturnError;
04323             }
04324 next2:
04325             im = *term2;
04326             next = term2 + im;
04327             if ( next >= Bin2->T.SB.MasterStop[blin2] || ( *next
04328                 && next+*next+COMPINC > Bin2->T.SB.MasterStop[blin2] ) ) {
04329                 if ( blin2 == 1 ) {
04330                     UNLOCK(Bin2->T.SB.MasterBlockLock[Bin2->T.SB.MasterNumBlocks]);
04331                 }
04332                 else {
04333                     UNLOCK(Bin2->T.SB.MasterBlockLock[blin2-1]);
04334                 }
04335                 if ( blin2 == Bin2->T.SB.MasterNumBlocks ) {
04336 /*
04337                     Move the remainder down into block 0
04338 */
04339                     to = Bin2->T.SB.MasterStart[1];

```

```

04339         from = Bin2->T.SB.MasterStop[Bin2->T.SB.MasterNumBlocks];
04340         while ( from > next ) --to = ---from;
04341         next = to;
04342         blin2 = 1;
04343     }
04344     else {
04345         blin2++;
04346     }
04347     LOCK(Bin2->T.SB.MasterBlockLock[blin2]);
04348     Bin2->T.SB.MasterBlock = blin2;
04349 }
04350 term2 = next;
04351 /*
04352     #] Two is smallest :
04353 */
04354 }
04355 else {
04356 /*
04357     #[ Equal :
04358 */
04359     l1 = *( m1 = term1 );
04360     l2 = *( m2 = term2 );
04361     if ( S->PolyWise ) { /* Here we work with PolyFun */
04362         ttl = m1;
04363         m1 += S->PolyWise;
04364         m2 += S->PolyWise;
04365         if ( S->PolyFlag == 2 ) {
04366             AT.WorkPointer = wp;
04367             w = poly_ratfun_add(BHEAD m1,m2);
04368             if ( *ttl + w[1] - m1[1] > AM.MaxTer/((LONG)sizeof(WORD)) ) {
04369                 MLOCK(ErrorMessageLock);
04370                 MesPrint("Term too complex in PolyRatFun addition. MaxTermSize of %10l is too
04371 small",AM.MaxTer);
04372                 MUNLOCK(ErrorMessageLock);
04373                 Terminate(-1);
04374             }
04375             AT.WorkPointer = wp;
04376             if ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 && w[1] > FUNHEAD ) {
04377                 goto cancelled;
04378             }
04379             else {
04380                 w = wp;
04381                 if ( w + m1[1] + m2[1] > AT.WorkTop ) {
04382                     MLOCK(ErrorMessageLock);
04383                     MesPrint("SortBotMerge(%d): A Maxtermsize of %10l is too small",
04384                         AT.identity,AM.MaxTer/sizeof(WORD));
04385                     MesPrint("m1[1] = %d, m2[1] = %d, Space =
04386 %1",m1[1],m2[1],(LONG)(AT.WorkTop-wp));
04387                     PrintTerm(term1,"term1");
04388                     PrintTerm(term2,"term2");
04389                     MesPrint("PolyWise = %d",S->PolyWise);
04390                     MUNLOCK(ErrorMessageLock);
04391                     Terminate(-1);
04392                 }
04393                 AddArgs(BHEAD m1,m2,w);
04394             }
04395             r1 = w[1];
04396             if ( r1 <= FUNHEAD
04397                 || ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) )
04398                 { goto cancelled; }
04399             if ( r1 == m1[1] ) {
04400                 NCOPY(m1,w,r1);
04401             }
04402             else if ( r1 < m1[1] ) {
04403                 r2 = m1[1] - r1;
04404                 m2 = w + r1;
04405                 m1 += m1[1];
04406                 while ( --r1 >= 0 ) ---m1 = ---m2;
04407                 m2 = m1 - r2;
04408                 r1 = S->PolyWise;
04409                 while ( --r1 >= 0 ) ---m1 = ---m2;
04410                 *m1 -= r2;
04411                 term1 = m1;
04412             }
04413             else {
04414                 r2 = r1 - m1[1];
04415                 m2 = ttl - r2;
04416                 r1 = S->PolyWise;
04417                 m1 = ttl;
04418                 *m1 += r2;
04419                 term1 = m2;
04420                 NCOPY(m2,m1,r1);
04421                 r1 = w[1];
04422                 NCOPY(m2,w,r1);
04423             }
04424         }
04425     }

```



```

04424 #ifdef WITHFLOAT
04425     else if ( AT.SortFloatMode ) {
04426 /*
04427     The terms are in m1/term1 and m2/term2 and their length in l1 and l2.
04428     We have to locate the floats which have already been
04429     verified in the compare routine. Once we have the new
04430     term we can jump to the code that writes away the
04431     result after adding the rationals.
04432 */
04433     tstop1 = m1+l1; size1 = tstop1[-1]; tstop1 -= ABS(size1);
04434     tstop2 = m2+l2; size2 = tstop2[-1]; tstop2 -= ABS(size2);
04435     if ( AT.SortFloatMode == 3 ) {
04436         fun1 = m1+l1;
04437         while ( fun1[0] != FLOATFUN && fun1+fun1[1] < tstop1 ) {
04438             fun1 += fun1[1];
04439         }
04440         if ( size1 < 0 ) fun1[FUNHEAD+3] = -fun1[FUNHEAD+3];
04441         UnpackFloat(aux1, fun1);
04442         fun2 = m2+l1;
04443         while ( fun2[0] != FLOATFUN && fun2+fun2[1] < tstop2 ) {
04444             fun2 += fun2[1];
04445         }
04446         if ( size2 < 0 ) fun2[FUNHEAD+3] = -fun2[FUNHEAD+3];
04447         UnpackFloat(aux2, fun2);
04448     }
04449     else if ( AT.SortFloatMode == 1 ) {
04450         fun1 = m1+l1;
04451         while ( fun1[0] != FLOATFUN && fun1+fun1[1] < tstop1 ) {
04452             fun1 += fun1[1];
04453         }
04454         if ( size1 < 0 ) fun1[FUNHEAD+3] = -fun1[FUNHEAD+3];
04455         UnpackFloat(aux1, fun1);
04456         fun2 = tstop2;
04457         RatToFloat(aux2, (UWORD *)fun2, size2);
04458     }
04459     else if ( AT.SortFloatMode == 2 ) {
04460         fun1 = tstop1;
04461         RatToFloat(aux1, (UWORD *)fun1, size1);
04462         fun2 = m2+l1;
04463         while ( fun2[0] != FLOATFUN && fun2+fun2[1] < tstop2 ) {
04464             fun2 += fun2[1];
04465         }
04466         if ( size2 < 0 ) fun2[FUNHEAD+3] = -fun2[FUNHEAD+3];
04467         UnpackFloat(aux2, fun2);
04468     }
04469     else {
04470         MLOCK(ErrorMessageLock);
04471         MesPrint("Illegal value %d for AT.SortFloatMode in
SortBotMerge.", AT.SortFloatMode);
04472         MUNLOCK(ErrorMessageLock);
04473         Terminate(-1);
04474         return(0);
04475     }
04476     mpf_add(aux3, aux1, aux2);
04477     size3 = mpf_sgn(aux3);
04478     if ( size3 == 0 ) { /* Cancelling! Rare! */
04479         goto cancelled;
04480     }
04481     else if ( size3 < 0 ) mpf_neg(aux3, aux3);
04482     fun3 = TermMalloc("MasterMerge");
04483     PackFloat(fun3, aux3);
04484     l3 = fun3[1]+(fun1-m1)+3; /* new size */
04485
04486     if ( l3 != l1 ) {
04487 /*
04488         Copy it all to wp
04489 */
04490         if ( (l3)*((LONG)sizeof(WORD)) >= AM.MaxTer ) {
04491             MLOCK(ErrorMessageLock);
04492             MesPrint("MasterMerge: Coefficient overflow during sort");
04493             MUNLOCK(ErrorMessageLock);
04494             goto ReturnError;
04495         }
04496         m3 = wp; m2 = term1;
04497         while ( m2 < fun1 ) *m3++ = *m2++;
04498         for ( jj = 0; jj < fun3[1]; jj++ ) *m3++ = fun3[jj];
04499         *m3++ = 1; *m3++ = 1;
04500         *m3++ = size3 < 0 ? -3: 3;
04501         *wp = m3-wp;
04502         TermFree(fun3, "MasterMerge");
04503         goto PutOutwp;
04504     }
04505     for ( jj = 0; jj < fun3[1]; jj++ ) fun1[jj] = fun3[jj];
04506     fun1 += fun3[1];
04507     *fun1++ = 1; *fun1++ = 1;
04508     *fun1++ = size3 < 0 ? -3: 3;
04509     *term1 = fun1-term1;

```

```

04510         TermFree(fun3,"MasterMerge");
04511     }
04512 #endif
04513     else {
04514         r1 = *( m1 += l1 - 1 );
04515         m1 -= ABS(r1) - 1;
04516         r1 = ( ( r1 > 0 ) ? (r1-1) : (r1+1) ) » 1;
04517         r2 = *( m2 += l2 - 1 );
04518         m2 -= ABS(r2) - 1;
04519         r2 = ( ( r2 > 0 ) ? (r2-1) : (r2+1) ) » 1;
04520
04521         if ( AddRat(BHEAD (UWORD *)m1,r1,(UWORD *)m2,r2,coef,&r3) ) {
04522             MLOCK(ErrorMessageLock);
04523             MesCall("SortBotMerge");
04524             MUNLOCK(ErrorMessageLock);
04525             SETERROR(-1)
04526         }
04527
04528         if ( AN.ncmod != 0 ) {
04529             if ( ( AC.modmode & POSNEG ) != 0 ) {
04530                 NormalModulus(coef,&r3);
04531             }
04532             else if ( BigLong(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod)) >= 0 ) {
04533                 SubPLon(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod),coef,&r3);
04534                 coef[r3] = 1;
04535                 for ( ii = 1; ii < r3; ii++ ) coef[r3+ii] = 0;
04536             }
04537         }
04538         if ( !r3 ) { goto cancelled; }
04539         r3 *= 2;
04540         r33 = ( r3 > 0 ) ? ( r3 + 1 ) : ( r3 - 1 );
04541         if ( r3 < 0 ) r3 = -r3;
04542         if ( r1 < 0 ) r1 = -r1;
04543         r1 *= 2;
04544         r31 = r3 - r1;
04545         if ( !r31 ) { /* copy coef into term1 */
04546             m2 = (WORD *)coef; im = r3;
04547             NCOPY(m1,m2,im);
04548             *m1 = r33;
04549         }
04550         else {
04551             to = wp; from = term1;
04552             while ( from < m1 ) *to++ = *from++;
04553             from = (WORD *)coef; im = r3;
04554             NCOPY(to,from,im);
04555             *to++ = r33;
04556             wp[0] = to - wp;
04557 PutOutwp:
04558             if ( SortBotOut(BHEAD wp) < 0 ) {
04559                 MLOCK(ErrorMessageLock);
04560                 MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
04561                 MUNLOCK(ErrorMessageLock);
04562                 error = -1;
04563                 goto ReturnError;
04564             }
04565             goto cancelled;
04566         }
04567     }
04568     if ( SortBotOut(BHEAD term1) < 0 ) {
04569         MLOCK(ErrorMessageLock);
04570         MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
04571         MUNLOCK(ErrorMessageLock);
04572         error = -1;
04573         goto ReturnError;
04574     }
04575 cancelled:; /* Now we need two new terms */
04576     im = *term1;
04577     next = term1 + im;
04578     if ( next >= Bin1->T.SB.MasterStop[blin1] || ( *next &&
04579 next+*next+COMPINC > Bin1->T.SB.MasterStop[blin1] ) ) {
04580         if ( blin1 == 1 ) {
04581             UNLOCK(Bin1->T.SB.MasterBlockLock[Bin1->T.SB.MasterNumBlocks]);
04582         }
04583         else {
04584             UNLOCK(Bin1->T.SB.MasterBlockLock[blin1-1]);
04585         }
04586         if ( blin1 == Bin1->T.SB.MasterNumBlocks ) {
04587             /*
04588             Move the remainder down into block 0
04589             */
04590             to = Bin1->T.SB.MasterStart[1];
04591             from = Bin1->T.SB.MasterStop[Bin1->T.SB.MasterNumBlocks];
04592             while ( from > next ) *--to = *--from;
04593             next = to;
04594             blin1 = 1;
04595         }
04596     }
    else {

```

```

04597         blin1++;
04598     }
04599     LOCK(Bin1->T.SB.MasterBlockLock[blin1]);
04600     Bin1->T.SB.MasterBlock = blin1;
04601 }
04602 term1 = next;
04603 goto next2;
04604 /*
04605     #] Equal :
04606 */
04607 }
04608 }
04609 /*
04610     Copy the tail
04611 */
04612 if ( *term1 ) {
04613 /*
04614     #] Tail in one :
04615 */
04616     while ( *term1 ) {
04617         if ( SortBotOut(BHEAD term1) < 0 ) {
04618             MLOCK(ErrorMessageLock);
04619             MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
04620             MUNLOCK(ErrorMessageLock);
04621             error = -1;
04622             goto ReturnError;
04623         }
04624         im = *term1;
04625         next = term1 + im;
04626         if ( next >= Bin1->T.SB.MasterStop[blin1] || ( *next &&
04627             next+*next+COMPINC > Bin1->T.SB.MasterStop[blin1] ) ) {
04628             if ( blin1 == 1 ) {
04629                 UNLOCK(Bin1->T.SB.MasterBlockLock[Bin1->T.SB.MasterNumBlocks]);
04630             }
04631             else {
04632                 UNLOCK(Bin1->T.SB.MasterBlockLock[blin1-1]);
04633             }
04634             if ( blin1 == Bin1->T.SB.MasterNumBlocks ) {
04635 /*
04636                 Move the remainder down into block 0
04637 */
04638                 to = Bin1->T.SB.MasterStart[1];
04639                 from = Bin1->T.SB.MasterStop[Bin1->T.SB.MasterNumBlocks];
04640                 while ( from > next ) *--to = *--from;
04641                 next = to;
04642                 blin1 = 1;
04643             }
04644             else {
04645                 blin1++;
04646             }
04647             LOCK(Bin1->T.SB.MasterBlockLock[blin1]);
04648             Bin1->T.SB.MasterBlock = blin1;
04649         }
04650         term1 = next;
04651     }
04652 /*
04653     #] Tail in one :
04654 */
04655 }
04656 else if ( *term2 ) {
04657 /*
04658     #] Tail in two :
04659 */
04660     while ( *term2 ) {
04661         if ( SortBotOut(BHEAD term2) < 0 ) {
04662             MLOCK(ErrorMessageLock);
04663             MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
04664             MUNLOCK(ErrorMessageLock);
04665             error = -1;
04666             goto ReturnError;
04667         }
04668         im = *term2;
04669         next = term2 + im;
04670         if ( next >= Bin2->T.SB.MasterStop[blin2] || ( *next
04671             && next+*next+COMPINC > Bin2->T.SB.MasterStop[blin2] ) ) {
04672             if ( blin2 == 1 ) {
04673                 UNLOCK(Bin2->T.SB.MasterBlockLock[Bin2->T.SB.MasterNumBlocks]);
04674             }
04675             else {
04676                 UNLOCK(Bin2->T.SB.MasterBlockLock[blin2-1]);
04677             }
04678             if ( blin2 == Bin2->T.SB.MasterNumBlocks ) {
04679 /*
04680                 Move the remainder down into block 0
04681 */
04682                 to = Bin2->T.SB.MasterStart[1];
04683                 from = Bin2->T.SB.MasterStop[Bin2->T.SB.MasterNumBlocks];

```

```

04684             while ( from > next ) *--to = *--from;
04685             next = to;
04686             blin2 = 1;
04687         }
04688         else {
04689             blin2++;
04690         }
04691         LOCK(Bin2->T.SB.MasterBlockLock[blin2]);
04692         Bin2->T.SB.MasterBlock = blin2;
04693     }
04694     term2 = next;
04695 }
04696 /*
04697     #] Tail in two :
04698 */
04699 }
04700 SortBotOut(BHEAD 0);
04701 ReturnError;;
04702 /*
04703     Release all locks
04704 */
04705 UNLOCK(Bin1->T.SB.MasterBlockLock[blin1]);
04706 if ( blin1 > 1 ) {
04707     UNLOCK(Bin1->T.SB.MasterBlockLock[blin1-1]);
04708 }
04709 else {
04710     UNLOCK(Bin1->T.SB.MasterBlockLock[Bin1->T.SB.MasterNumBlocks]);
04711 }
04712 UNLOCK(Bin2->T.SB.MasterBlockLock[blin2]);
04713 if ( blin2 > 1 ) {
04714     UNLOCK(Bin2->T.SB.MasterBlockLock[blin2-1]);
04715 }
04716 else {
04717     UNLOCK(Bin2->T.SB.MasterBlockLock[Bin2->T.SB.MasterNumBlocks]);
04718 }
04719 if ( AT.identity > 0 ) {
04720     UNLOCK(AT.SB.MasterBlockLock[AT.SB.FillBlock]);
04721 }
04722 /*
04723     And that was all folks
04724 */
04725 return(error);
04726 }
04727
04728 #endif
04729
04730 /*
04731     #] SortBotMerge :
04732     #[ IniSortBlocks :
04733 */
04734
04735 static int SortBlocksInitialized = 0;
04736
04737 int IniSortBlocks(int numworkers)
04738 {
04739     ALLPRIVATES *B;
04740     SORTING *S;
04741     LONG totalsize, workersize, blocksize, numberofterms;
04742     int maxter, id, j;
04743     int numberofblocks = NUMBEROFBLOCKSINSORT, numparts;
04744     WORD *w;
04745
04746     if ( SortBlocksInitialized ) return(0);
04747     SortBlocksInitialized = 1;
04748     if ( numworkers == 0 ) return(0);
04749
04750 #ifdef WITHSORTBOTS
04751     if ( numworkers > 2 ) {
04752         numparts = 2*numworkers - 2;
04753         numberofblocks = numberofblocks/2;
04754     }
04755     else {
04756         numparts = numworkers;
04757     }
04758 #else
04759     numparts = numworkers;
04760 #endif
04761
04762     S = AM.S0;
04763     totalsize = S->LargeSize + S->SmallSize;
04764     workersize = totalsize / numparts;
04765     maxter = AM.MaxTer/sizeof(WORD);
04766     blocksize = ( workersize - maxter )/numberofblocks;
04767     numberofterms = blocksize / maxter;
04768     if ( numberofterms < MINIMUMNUMBEROFTERMS ) {
04769         /*
04770             This should have been taken care of in RecalcSetups.
04771         */
04772     }

```

```

04777         MesPrint("We have a problem with the size of the blocks in IniSortBlocks");
04778         Terminate(-1);
04779     }
04780     /*
04781     Layout:  For each worker
04782               block 0: size is maxter WORDS
04783               numberofblocks blocks of size blocksize WORDS
04784     */
04785     w = S->lBuffer;
04786     if ( w == 0 ) w = S->sBuffer;
04787     for ( id = 1; id <= numparts; id++ ) {
04788         B = AB[id];
04789         AT.SB.MasterBlockLock = (pthread_mutex_t *)Malloc1(
04790             sizeof(pthread_mutex_t)*(numberofblocks+1), "MasterBlockLock");
04791         AT.SB.MasterStart = (WORD **)Malloc1(sizeof(WORD *)*(numberofblocks+1)*3, "MasterBlock");
04792         AT.SB.MasterFill = AT.SB.MasterStart + (numberofblocks+1);
04793         AT.SB.MasterStop = AT.SB.MasterFill + (numberofblocks+1);
04794         AT.SB.MasterNumBlocks = numberofblocks;
04795         AT.SB.MasterBlock = 0;
04796         AT.SB.FillBlock = 0;
04797         AT.SB.MasterFill[0] = AT.SB.MasterStart[0] = w;
04798         w += maxter;
04799         AT.SB.MasterStop[0] = w;
04800         AT.SB.MasterBlockLock[0] = dummylock;
04801         for ( j = 1; j <= numberofblocks; j++ ) {
04802             AT.SB.MasterFill[j] = AT.SB.MasterStart[j] = w;
04803             w += blocksize;
04804             AT.SB.MasterStop[j] = w;
04805             AT.SB.MasterBlockLock[j] = dummylock;
04806         }
04807     }
04808     if ( w > S->sTop2 ) {
04809         MesPrint("Counting problem in IniSortBlocks");
04810         Terminate(-1);
04811     }
04812     return(0);
04813 }
04814
04815 /*
04816 #] IniSortBlocks :
04817 #[ UpdateSortBlocks :
04818 */
04819
04824 int UpdateSortBlocks(int numworkers)
04825 {
04826     ALLPRIVATES *B;
04827     SORTING *S;
04828     LONG totalsize, workersize, blocksize, numberofterms;
04829     int maxter, id, j;
04830     int numberofblocks = NUMBEROFBLOCKSINSORT, numparts;
04831     WORD *w;
04832
04833     if ( numworkers == 0 ) return(0);
04834
04835     #ifdef WITHSORTBOTS
04836     if ( numworkers > 2 ) {
04837         numparts = 2*numworkers - 2;
04838         numberofblocks = numberofblocks/2;
04839     }
04840     else {
04841         numparts = numworkers;
04842     }
04843     #else
04844     numparts = numworkers;
04845     #endif
04846     S = AM.S0;
04847     totalsize = S->LargeSize + S->SmallEsize;
04848     workersize = totalsize / numparts;
04849     maxter = AM.MaxTer/sizeof(WORD);
04850     blocksize = ( workersize - maxter )/numberofblocks;
04851     numberofterms = blocksize / maxter;
04852     if ( numberofterms < MINIMUMNUMBEROFTERMS ) {
04853     /*
04854         This should have been taken care of in RecalcSetups.
04855     */
04856         MesPrint("We have a problem with the size of the blocks in UpdateSortBlocks");
04857         Terminate(-1);
04858     }
04859     /*
04860     Layout:  For each worker
04861               block 0: size is maxter WORDS
04862               numberofblocks blocks of size blocksize WORDS
04863     */
04864     w = S->lBuffer;
04865     if ( w == 0 ) w = S->sBuffer;
04866     for ( id = 1; id <= numparts; id++ ) {
04867         B = AB[id];

```

```

04868     AT.SB.MasterFill[0] = AT.SB.MasterStart[0] = w;
04869     w += maxter;
04870     AT.SB.MasterStop[0] = w;
04871     for ( j = 1; j <= numberofblocks; j++ ) {
04872         AT.SB.MasterFill[j] = AT.SB.MasterStart[j] = w;
04873         w += blocksize;
04874         AT.SB.MasterStop[j] = w;
04875     }
04876 }
04877 if ( w > S->sTop2 ) {
04878     MesPrint("Counting problem in UpdateSortBlocks");
04879     Terminate(-1);
04880 }
04881 return(0);
04882 }
04883
04884 /*
04885 #] UpdateSortBlocks :
04886 #[ DefineSortBotTree :
04887 */
04888
04889 #ifdef WITHSORTBOTS
04890 void DefineSortBotTree(void)
04891 {
04892     ALLPRIVATES *B;
04893     int n, i, from;
04894     if ( numberofworkers <= 2 ) return;
04895     n = numberofworkers*2-2;
04896     for ( i = numberofworkers+1, from = 1; i <= n; i++ ) {
04897         B = AB[i];
04898         AT.SortBotIn1 = from++;
04899         AT.SortBotIn2 = from++;
04900     }
04901     B = AB[0];
04902     AT.SortBotIn1 = from++;
04903     AT.SortBotIn2 = from++;
04904 }
04905 #endif
04906
04907 /*
04908 #] DefineSortBotTree :
04909 #[ GetTerm2 :
04910
04911 Routine does a GetTerm but only when a bracket index is involved and
04912 only from brackets that have been judged not suitable for treatment
04913 as complete brackets by a single worker. Whether or not a bracket should
04914 be treated by a single worker is decided by TreatIndexEntry
04915 */
04916 WORD GetTerm2(PHEAD WORD *term)
04917 {
04918     WORD *ttco, *tt, retval;
04919     LONG n,i;
04920     FILEHANDLE *fi;
04921     EXPRESSIONS e = AN.expr;
04922     BRACKETINFO *b = e->bracketinfo;
04923     BRACKETINDEX *bi = b->indexbuffer;
04924     POSITION where, eonfile = AS.OldOnFile[e-Expressions], bstart, bnext;
04925
04926     /*
04927     1: Get the current position.
04928     */
04929     switch ( e->status ) {
04930         case UNHIDELEXPRESSION:
04931         case UNHIDEGEXPRESSION:
04932         case DROPHLEXPRESSION:
04933         case DROPHGEXPRESSION:
04934         case HIDDENLEXPRESSION:
04935         case HIDDENGEXPRESSION:
04936             fi = AR.hidefile;
04937             break;
04938         default:
04939             fi = AR.infile;
04940             break;
04941     }
04942     if ( AR.KeptInHold ) {
04943         retval = GetTerm(BHEAD term);
04944         return(retval);
04945     }
04946     SeekScratch(fi,&where);
04947     if ( AN.lastinindex < 0 ) {
04948         /*
04949         We have to test whether we have to do the first bracket
04950         */
04951         if ( ( n = TreatIndexEntry(BHEAD 0) ) <= 0 ) {
04952             AN.lastinindex = n;

```

```

04960         where = bi[n].start;
04961         ADD2POS(where,eonfile);
04962         SetScratch(fi,&where);
04963 /*
04964         Put the bracket in the Compress buffer.
04965 */
04966         ttco = AR.CompressBuffer;
04967         tt = b->bracketbuffer + bi[0].bracket;
04968         i = *tt;
04969         NCOPY(ttco,tt,i)
04970         AR.CompressPointer = ttco;
04971         retval = GetTerm(BHEAD term);
04972         return(retval);
04973     }
04974     else AN.lastinindex = n-1;
04975 }
04976 /*
04977     2: Find the corresponding index number
04978     a: test whether it is in the current bracket
04979 */
04980     n = AN.lastinindex;
04981     bstart = bi[n].start;
04982     ADD2POS(bstart,eonfile);
04983     bnext = bi[n].next;
04984     ADD2POS(bnext,eonfile);
04985     if ( ISLESSPOS(bstart,where) && ISLESSPOS(where,bnext) ) {
04986         retval = GetTerm(BHEAD term);
04987         return(retval);
04988     }
04989     for ( n++ ; n < b->indexfill; n++ ) {
04990         i = TreatIndexEntry(BHEAD n);
04991         if ( i <= 0 ) {
04992 /*
04993         Put the bracket in the Compress buffer.
04994 */
04995         ttco = AR.CompressBuffer;
04996         tt = b->bracketbuffer + bi[n].bracket;
04997         i = *tt;
04998         NCOPY(ttco,tt,i)
04999         AR.CompressPointer = ttco;
05000         AN.lastinindex = n;
05001         where = bi[n].start;
05002         ADD2POS(where,eonfile);
05003         SetScratch(fi,&(where));
05004         retval = GetTerm(BHEAD term);
05005         return(retval);
05006     }
05007     else n += i - 1;
05008 }
05009 return(0);
05010 }
05011
05012 /*
05013 #] GetTerm2 :
05014 #[ TreatIndexEntry :
05015 */
05024 int TreatIndexEntry(PHEAD LONG n)
05025 {
05026     BRACKETINFO *b = AN.expr->bracketinfo;
05027     LONG numbra = b->indexfill - 1, i;
05028     LONG totterms;
05029     BRACKETINDEX *bi;
05030     POSITION pos1, average;
05031 /*
05032     1: number of the bracket should be such that there is one bucket
05033     for each worker remaining.
05034 */
05035     if ( ( numbra - n ) <= numberofworkers ) return(0);
05036 /*
05037     2: size of the bracket contents should be less than what remains in
05038     the expression divided by the number of workers.
05039 */
05040     bi = b->indexbuffer;
05041     DIFPOS(pos1,bi[numbra].next,bi[n].next); /* Size of what remains */
05042     BASEPOSITION(average) = DIVPOS(pos1,(3*numberofworkers));
05043     DIFPOS(pos1,bi[n].next,bi[n].start); /* Size of the current bracket */
05044     if ( ISLESSPOS(average,pos1) ) return(0);
05045 /*
05046     It passes.
05047     Now check whether we can do more brackets
05048 */
05049     totterms = bi->termsinbracket;
05050     if ( totterms > 2*AC.ThreadBucketSize ) return(1);
05051     for ( i = 1; i < numbra-n; i++ ) {
05052         DIFPOS(pos1,bi[n+i].next,bi[n].start); /* Size of the combined brackets */
05053         if ( ISLESSPOS(average,pos1) ) return(i);
05054         totterms += bi->termsinbracket;

```

```

05055         if ( totterms > 2*AC.ThreadBucketSize ) return(i+1);
05056     }
05057 /*
05058     We have a problem at the end of the system. Just do one.
05059 */
05060     return(1);
05061 }
05062
05063 /*
05064     #] TreatIndexEntry :
05065     #[ SetHideFiles :
05066 */
05067
05068 void SetHideFiles(void) {
05069     int i;
05070     ALLPRIVATES *B, *B0 = AB[0];
05071     for ( i = 1; i <= numberofworkers; i++ ) {
05072         B = AB[i];
05073         AR.hidefile->handle = AR0.hidefile->handle;
05074         if ( AR.hidefile->handle < 0 ) {
05075             AR.hidefile->PObuffer = AR0.hidefile->PObuffer;
05076             AR.hidefile->POstop = AR0.hidefile->POstop;
05077             AR.hidefile->POfill = AR0.hidefile->POfill;
05078             AR.hidefile->POfull = AR0.hidefile->POfull;
05079             AR.hidefile->POsize = AR0.hidefile->POsize;
05080             AR.hidefile->POposition = AR0.hidefile->POposition;
05081             AR.hidefile->filesize = AR0.hidefile->filesize;
05082         }
05083         else {
05084             AR.hidefile->PObuffer = AR.hidefile->wPObuffer;
05085             AR.hidefile->POstop = AR.hidefile->wPOstop;
05086             AR.hidefile->POfill = AR.hidefile->wPOfill;
05087             AR.hidefile->POfull = AR.hidefile->wPOfull;
05088             AR.hidefile->POsize = AR.hidefile->wPOsize;
05089             PUTZERO(AR.hidefile->POposition);
05090         }
05091     }
05092 }
05093
05094 /*
05095     #] SetHideFiles :
05096     #[ IniFbufs :
05097 */
05098
05099 void IniFbufs(void)
05100 {
05101     int i;
05102     for ( i = 0; i < AM.totalnumberofthreads; i++ ) {
05103         IniFbuffer(AB[i]->T.fbufnum);
05104     }
05105 }
05106
05107 /*
05108     #] IniFbufs :
05109     #[ SetMods :
05110 */
05111
05112 void SetMods(void)
05113 {
05114     ALLPRIVATES *B;
05115     int i, n, j;
05116     for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
05117         B = AB[j];
05118         AN.ncmod = AC.ncmod;
05119         if ( AN.cmod != 0 ) M_free(AN.cmod,"AN.cmod");
05120         n = ABS(AN.ncmod);
05121         AN.cmod = (UWORD *)Malloc1(sizeof(WORD)*n,"AN.cmod");
05122         for ( i = 0; i < n; i++ ) AN.cmod[i] = AC.cmod[i];
05123     }
05124 }
05125
05126 /*
05127     #] SetMods :
05128     #[ UnSetMods :
05129 */
05130
05131 void UnSetMods(void)
05132 {
05133     ALLPRIVATES *B;
05134     int j;
05135     for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
05136         B = AB[j];
05137         if ( AN.cmod != 0 ) M_free(AN.cmod,"AN.cmod");
05138         AN.cmod = 0;
05139     }
05140 }
05141

```



```

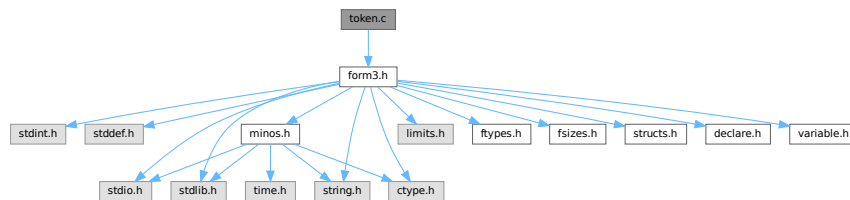
05142 /*
05143     #] UnSetMods :
05144     #[ find_Horner_MCTS_expand_tree_threaded :
05145 */
05146
05147 void find_Horner_MCTS_expand_tree_threaded(void) {
05148     int id;
05149     while (( id = GetAvailableThread() ) < 0)
05150         MasterWait();
05151     WakeupThread(id,MCTSEXPANDTREE);
05152 }
05153
05154 /*
05155     #] find_Horner_MCTS_expand_tree_threaded :
05156     #[ optimize_expression_given_Horner_threaded :
05157 */
05158
05159 extern void optimize_expression_given_Horner_threaded(void) {
05160     int id;
05161     while (( id = GetAvailableThread() ) < 0)
05162         MasterWait();
05163     WakeupThread(id,OPTIMIZEEXPRESSION);
05164 }
05165
05166 /*
05167     #] optimize_expression_given_Horner_threaded :
05168 */
05169
05170 #endif

```

## 4.144 token.c File Reference

#include "form3.h"

Include dependency graph for token.c:



### Macros

- #define [CHECKPOLY](#) {if(polyflag)MesPrint("&Illegal use of polynomial function"); polyflag = 0; }

### Functions

- int [tokenize](#) (UBYTE \*in, WORD leftright)
- void [WriteTokens](#) (SBYTE \*in)
- int [simp1token](#) (SBYTE \*s)
- int [simpwtoken](#) (SBYTE \*s)
- int [simp2token](#) (SBYTE \*s)
- int [simp3atoken](#) (SBYTE \*s, int mode)
- int [simp3btoken](#) (SBYTE \*s, int mode)
- int [simp4token](#) (SBYTE \*s)
- int [simp5token](#) (SBYTE \*s, int mode)
- int [simp6token](#) (SBYTE \*tokens, int mode)

## Variables

- `char * ttypes []`

### 4.144.1 Detailed Description

The tokenizer. This is a part of the compiler that does an intermediate type of translation. It does look up the names etc and can do a number of optimizations. The resulting output is a stream of bytes which can be processed by the code generator (in the file [compiler.c](#))

Definition in file [token.c](#).

### 4.144.2 Macro Definition Documentation

#### 4.144.2.1 CHECKPOLY

```
#define CHECKPOLY {if(polyflag)MesPrint("&Illegal use of polynomial function"); polyflag = 0;
}
```

Definition at line 56 of file [token.c](#).

### 4.144.3 Function Documentation

#### 4.144.3.1 tokenize()

```
int tokenize (
    UBYTE * in,
    WORD leftright )
```

Definition at line 58 of file [token.c](#).

#### 4.144.3.2 WriteTokens()

```
void WriteTokens (
    SBYTE * in )
```

Definition at line 657 of file [token.c](#).

#### 4.144.3.3 simp1token()

```
int simp1token (
    SBYTE * s )
```

Definition at line 705 of file [token.c](#).

#### 4.144.3.4 simpwtoken()

```
int simpwtoken (
    SBYTE * s )
```

Definition at line 794 of file [token.c](#).

#### 4.144.3.5 simp2token()

```
int simp2token (
    SBYTE * s )
```

Definition at line 948 of file [token.c](#).

#### 4.144.3.6 simp3atoken()

```
int simp3atoken (
    SBYTE * s,
    int mode )
```

Definition at line 1196 of file [token.c](#).

#### 4.144.3.7 simp3btoken()

```
int simp3btoken (
    SBYTE * s,
    int mode )
```

Definition at line 1437 of file [token.c](#).

#### 4.144.3.8 simp4token()

```
int simp4token (
    SBYTE * s )
```

Definition at line 1822 of file [token.c](#).

#### 4.144.3.9 simp5token()

```
int simp5token (
    SBYTE * s,
    int mode )
```

Definition at line 1982 of file [token.c](#).

#### 4.144.3.10 simp6token()

```
int simp6token (
    SBYTE * tokens,
    int mode )
```

Definition at line 2028 of file [token.c](#).

### 4.144.4 Variable Documentation

#### 4.144.4.1 ttypes

```
char* ttypes[]
```

##### Initial value:

```
= { "\n", "S", "I", "V", "F", "set", "E", "dotp", "#",
    "sub", "d_", "S", "dub", "(", ")", "?", "??", ".", "[", "]",
    ",", "(", ")", "*", "/", "^", "+", "-", "!", "end", "{", "}",
    "N_?", "conj", "()", "#d", "^d", "_", "snum" }
```

Definition at line 652 of file [token.c](#).

## 4.145 token.c

[Go to the documentation of this file.](#)

```
00001
00008 /* # [ License : */
00009 /*
00010 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00011 * When using this file you are requested to refer to the publication
00012 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00013 * This is considered a matter of courtesy as the development was paid
00014 * for by FOM the Dutch physics granting agency and we would like to
00015 * be able to track its scientific use to convince FOM of its value
00016 * for the community.
00017 *
00018 * This file is part of FORM.
00019 *
00020 * FORM is free software: you can redistribute it and/or modify it under the
00021 * terms of the GNU General Public License as published by the Free Software
00022 * Foundation, either version 3 of the License, or (at your option) any later
00023 * version.
00024 *
00025 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00026 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00027 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00028 * details.
00029 *
00030 * You should have received a copy of the GNU General Public License along
00031 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00032 */
00033 /* # [ License : */
00034 /*
00035 * # [ Includes :
00036 */
00037
00038 #include "form3.h"
00039
00040 /*
00041 * #] Includes :
00042 * # [ Compiler :
00043 * # [ tokenize :
00044
00045 * Takes the input in 'in' and translates it into tokens.
00046 * The tokens are put in the token buffer which starts at 'AC.tokens'
00047 * and runs till 'AC.toptokens'
00048 * We may assume that the various types of brackets match properly.
00049 * object = -1: after , or (
00050 * object = 0: name/variable/number etc is allowed
```

```

00051         object = 1: variable.
00052         object = 2: number
00053         object = 3: ) after subexpression
00054 */
00055
00056 #define CHECKPOLY {if(polyflag)MesPrint("&Illegal use of polynomial function"); polyflag = 0; }
00057
00058 int tokenize(UBYTE *in, WORD leftright)
00059 {
00060     int error = 0, object, funlevel = 0, bracelevel = 0, explevel = 0, numexp;
00061     int polyflag = 0;
00062     WORD number, type;
00063     UBYTE *s = in, c;
00064     SBYTE *out, *outtop, num[MAXNUMSIZE], *t;
00065     LONG i;
00066     if ( AC.tokens == 0 ) {
00067         SBYTE **ppp = &(AC.tokens); /* to avoid a compiler warning */
00068         SBYTE **pppp = &(AC.toptokens);
00069         DoubleBuffer((void **)ppp, (void **)pppp, sizeof(SBYTE), "start tokens");
00070     }
00071     out = AC.tokens;
00072     outtop = AC.toptokens - MAXNUMSIZE;
00073     AC.dumnumflag = 0;
00074     object = 0;
00075     while ( *in ) {
00076         if ( out > outtop ) {
00077             LONG oldsize = (LONG)(out - AC.tokens);
00078             SBYTE **ppp = &(AC.tokens); /* to avoid a compiler warning */
00079             SBYTE **pppp = &(AC.toptokens);
00080             DoubleBuffer((void **)ppp, (void **)pppp, sizeof(SBYTE), "expand tokens");
00081             out = AC.tokens + oldsize;
00082             outtop = AC.toptokens - MAXNUMSIZE;
00083         }
00084         switch ( FG.cTable[*in] ) {
00085             case 0: /* a-zA-Z */
00086                 CHECKPOLY
00087                 s = in++;
00088                 while ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1
00089                     || *in == '_' ) in++;
00090             dovariable: c = *in; *in = 0;
00091                 if ( object > 0 ) {
00092                     MesPrint("&Illegal position for %s",s);
00093                     if ( !error ) error = 1;
00094                 }
00095                 if ( out > AC.tokens && ( out[-1] == TWILDCARD || out[-1] == TNOT ) ) {
00096                     type = GetName(AC.varnames,s,&number,NOAUTO);
00097                 }
00098                 else {
00099                     type = GetName(AC.varnames,s,&number,WITHAUTO);
00100                 }
00101                 if ( type < 0 )
00102                     type = GetName(AC.exprnames,s,&number,NOAUTO);
00103                 switch ( type ) {
00104                     case CSYMBOL: *out++ = TSYMBOL; break;
00105                     case CINDEX:
00106                         if ( number >= (AM.IndDum-AM.OffsetIndex) ) {
00107                             if ( c != '?' ) {
00108                                 MesPrint("&Generated indices should be of the type Nnumber_?");
00109                                 error = 1;
00110                             }
00111                             else {
00112                                 *in++ = c; c = *in; *in = 0;
00113                                 AC.dumnumflag = 1;
00114                             }
00115                         }
00116                         *out++ = TINDEX;
00117                         break;
00118                     case CVECTOR: *out++ = TVECTOR; break;
00119                     case CFUNCTION:
00120 #ifdef WITHMPI
00121                         /*
00122                          * In the preprocessor, random functions in #svar=... and #inside
00123                          * may cause troubles, because the program flow on a slave may be
00124                          * different from those on others. We set AC.RhsExprInModuleFlag in order
00125                          * to make the change of $-variable be done on the master and thus keep the
00126                          * consistency among the master and all slave processes. The previous value
00127                          * of AC.RhsExprInModuleFlag will be restored after #svar=... and #inside.
00128                          */
00129                         if ( AP.PreAssignFlag || AP.PreInsideLevel ) {
00130                             switch ( number + FUNCTION ) {
00131                                 case RANDOMFUNCTION:
00132                                 case RANPERM:
00133                                     AC.RhsExprInModuleFlag = 1;
00134                             }
00135                         }
00136 #endif
00137                         *out++ = TFUNCTION;

```

```

00138             break;
00139         case CSET: *out++ = TSET; break;
00140         case CEXPRESSION: *out++ = TEXPRESSION;
00141             if ( leftright == LHSIDE ) {
00142                 if ( !error ) error = 1;
00143                 MesPrint("&Expression not allowed in LH-side of
substitution: %s",s);
00144             }
00145         /*[06nov2003 mt]:*/
00146         #ifdef WITHMPI
00147             else { /*RHSide*/
00148                 /* NOTE: We always set AC.RhsExprInModuleFlag regardless
of
00149                 * AP.PreAssignFlag or AP.PreInsideLevel because we
have to detect
00150                 * RHS expressions even in those cases. */
00151                 AC.RhsExprInModuleFlag = 1;
00152             }
00153             if ( !AP.PreAssignFlag && !AP.PreInsideLevel )
00154                 Expressions[number].vflags |= ISINRHS;
00155         #endif
00156         /*:[06nov2003 mt]*/
00157             if ( AC.exprfillwarning == 0 ) {
00158                 AC.exprfillwarning = 1;
00159             }
00160             break;
00161             case CDELTA: *out++ = TDELTA; *in = c;
00162                 object = 1; continue;
00163             case CDUBIOUS: *out++ = TDUBIOUS; break;
00164             default: *out++ = TDUBIOUS;
00165                 if ( !error ) error = 1;
00166                 MesPrint("&Undeclared variable %s",s);
00167                 number = AddDubious(s);
00168                 break;
00169             }
00170             object = 1;
00171         donumber: i = 0;
00172             do { num[i++] = (SBYTE)(number & 0x7F); number >>= 7; } while ( number );
00173             while ( --i >= 0 ) *out++ = num[i];
00174             *in = c;
00175             break;
00176             case 1: /* 0-9 */
00177             {
00178                 #ifdef WITHFLOAT
00179                     int spec;
00180                     UBYTE *in2;
00181                 #endif
00182                 CHECKPOLY
00183                 s = in;
00184                 while ( *s == '0' && FG.cTable[s[1]] == 1 ) s++;
00185                 in = s+1; i = 1;
00186                 while ( FG.cTable[*in] == 1 ) { in++; i++; }
00187                 if ( object > 0 ) {
00188                     c = *in; *in = 0;
00189                     MesPrint("&Illegal position for %s",s);
00190                     *in = c;
00191                     if ( !error ) error = 1;
00192                 }
00193                 if ( i == 1 && *in == '_' && ( *s == '5' || *s == '6'
|| *s == '7' ) ) {
00194                     in++; *out++ = TSGAMMA; *out++ = (SBYTE)(*s - '4');
00195                     object = 1;
00196                     break;
00197                 }
00198             }
00199             #ifdef WITHFLOAT
00200                 in2 = CheckFloat(in,&spec);
00201                 if ( in2 > in ) {
00202                     if ( spec == 1 ) {
00203                         *out++ = TNUMBER; *out++ = 0; s = in2;
00204                     }
00205                     else if ( spec == -1 ) {
00206                         MesPrint("&The floating point system has not been started: %s",in);
00207                         if ( !error ) error = 1;
00208                     }
00209                     else {
00210                         in = in2;
00211                     }
00212                 }
00213                 dofloat:
00214                     while ( out + (in-s) >= AC.toptokens ) {
00215                         LONG oldsize = (LONG)(out - AC.tokens);
00216                         SBYTE **ppp = &(AC.tokens); /* to avoid a compiler warning */
00217                         SBYTE **pppp = &(AC.toptokens);
00218                         DoubleBuffer((void **)ppp,(void **)pppp,sizeof(SBYTE),"more tokens");
00219                         out = AC.tokens + oldsize;
00220                         outtop = AC.toptokens - MAXNUMSIZE;
00221                     }
00222                     *out++ = TFLOAT;
00223                     while ( s < in ) *out++ = *s++;

```

```

00222     }
00223     }
00224     else
00225 #endif
00226     {
00227         *out++ = TNUMBER;
00228         if ( ( i & 1 ) != 0 ) *out++ = (SBYTE) (*s++ - '0');
00229         while ( out + (in-s)/2 >= AC.toptokens ) {
00230             LONG oldsize = (LONG) (out - AC.tokens);
00231             SBYTE **ppp = &(AC.tokens); /* to avoid a compiler warning */
00232             SBYTE **pppp = &(AC.toptokens);
00233             DoubleBuffer((void **)ppp, (void **)pppp, sizeof(SBYTE), "more tokens");
00234             out = AC.tokens + oldsize;
00235             outtop = AC.toptokens - MAXNUMSIZE;
00236         }
00237         while ( s < in ) { /* We store in base 100 */
00238             *out++ = (SBYTE) (( *s - '0' ) * 10 + ( s[1] - '0' ));
00239             s += 2;
00240         }
00241     }
00242     object = 2;
00243 }
00244 break;
00245 case 2: /* . $ _ ? # ' */
00246     CHECKPOLY
00247     if ( *in == '?' ) {
00248         if ( leftright == LHSIDE ) {
00249             if ( object == 1 ) { /* follows a name */
00250                 in++; *out++ = TWILDCARD;
00251                 if ( FG.cTable[in[0]] == 0 || in[0] == '[' || in[0] == '{' ) object = 0;
00252             }
00253             else if ( object == -1 ) { /* follows comma or ( */
00254                 in++; s = in;
00255                 while ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1 ) in++;
00256                 c = *in; *in = 0;
00257                 if ( FG.cTable[*s] != 0 ) {
00258                     MesPrint("&Illegal name for argument list variable %s", s);
00259                     error = 1;
00260                 }
00261                 else {
00262                     i = AddWildcardName((UBYTE *)s);
00263                     *in = c;
00264                     *out++ = TWILDARG;
00265                     *out++ = (SBYTE)i;
00266                 }
00267                 object = 1;
00268             }
00269             else {
00270                 MesPrint("&Illegal position for ?");
00271                 error = 1;
00272                 in++;
00273             }
00274         }
00275         else {
00276             if ( object != -1 ) goto IllPos;
00277             in++;
00278             if ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1 ) {
00279                 s = in;
00280                 while ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1 ) in++;
00281                 c = *in; *in = 0;
00282                 i = GetWildcardName((UBYTE *)s);
00283                 if ( i <= 0 ) {
00284                     MesPrint("&Undefined argument list variable %s", s);
00285                     error = 1;
00286                 }
00287                 *in = c;
00288                 *out++ = TWILDARG;
00289                 *out++ = (SBYTE)i;
00290             }
00291             else {
00292                 if ( AC.vectorlikeLHS == 0 ) {
00293                     MesPrint("&Generated index ? only allowed in vector substitution", s);
00294                     error = 1;
00295                 }
00296                 *out++ = TGENINDEX;
00297             }
00298             object = 1;
00299         }
00300     }
00301     else if ( *in == '.' ) {
00302         if ( object == 1 ) { /* follows a name */
00303             *out++ = TDOT;
00304             object = 0;
00305             in++;
00306         }
00307 #ifdef WITHFLOAT
00308         else if ( object == 0 || object == -1 ) {

```

```

00309 /*
00310                                     Test for floating point number
00311 */
00312 int spec;
00313 s = CheckFloat(in,&spec);
00314 if ( s > in ) {
00315     if ( spec == 1 ) {
00316         *out++ = TNUMBER; *out++ = 0;
00317         object = 2; in = s;
00318     }
00319     else if ( spec == -1 ) {
00320         MesPrint("The floating point system has not been started: %s",in);
00321         if ( !error ) error = 1;
00322     }
00323     else {
00324         UBYTE *a = s; s = in; in = a;
00325         goto dofloat;
00326     }
00327 }
00328     else goto IllPos;
00329 }
00330 #endif
00331     else goto IllPos;
00332 }
00333 else if ( *in == '$' ) { /* $ variable */
00334     in++;
00335     s = in;
00336     if ( FG.cTable[*in] == 0 ) {
00337         while ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1 ) in++;
00338         if ( *in == '_' && AP.PreAssignFlag == 2 ) in++;
00339         c = *in; *in = 0;
00340         if ( object > 0 ) {
00341             if ( object != 1 || leftright == RHSIDE ) {
00342                 MesPrint("&Illegal position for %s",s);
00343                 if ( !error ) error = 1;
00344             } /* else can be assignment in wildcard */
00345             else {
00346                 if ( ( number = GetDollar(s) ) < 0 ) {
00347                     number = AddDollar(s,0,0,0);
00348                 }
00349             }
00350         }
00351         else if ( ( number = GetDollar(s) ) < 0 ) {
00352             MesPrint("&Undefined variable %s",s);
00353             if ( !error ) error = 1;
00354             number = AddDollar(s,0,0,0);
00355         }
00356         *out++ = TDOLLAR;
00357         object = 1;
00358         if ( ( AC.exprfillwarning == 0 ) &&
00359             ( out > AC.tokens+1 ) && ( out[-2] != TWILDCARD ) ) {
00360             AC.exprfillwarning = 1;
00361         }
00362         goto donumber;
00363     }
00364     else {
00365         MesPrint("Illegal name for $ variable after %s",in);
00366         if ( !error ) error = 1;
00367     }
00368 }
00369 else if ( *in == '#' ) {
00370     in++;
00371     if ( object == 1 ) { /* follows a name */
00372         *out++ = TCONJUGATE;
00373     }
00374     else {
00375         MesPrint("&Illegal position for %#",in);
00376         error = 1;
00377     }
00378 }
00379     else goto IllPos;
00380     break;
00381 case 3: /* [ ] */
00382     CHECKPOLY
00383     if ( *in == '[' ) {
00384         if ( object == 1 ) { /* after name */
00385             t = out-1;
00386             if ( *t == RPARENTHESIS ) {
00387                 *out++ = LBRACE; *out++ = LPARENTHESIS;
00388                 bracelevel++; explevel = bracelevel;
00389             }
00390             else {
00391                 while ( *t >= 0 && t > AC.tokens ) t--;
00392                 if ( *t == TEXPRESSION ) {
00393                     *out++ = LBRACE; *out++ = LPARENTHESIS;
00394                     bracelevel++; explevel = bracelevel;
00395                 }

```



```

00396         else { *out++ = LBACE; bracelevel++; }
00397     }
00398     object = 0;
00399 }
00400 else { /* name. find matching */
00401     s = in;
00402     in = SkipAName(in);
00403     goto dovariable;
00404 }
00405 }
00406 else {
00407     if ( explevel > 0 && explevel == bracelevel ) {
00408         *out++ = RPARENTHESIS; explevel = 0;
00409     }
00410     *out++ = RBACE; object = 1; bracelevel--;
00411 }
00412 in++;
00413 break;
00414 case 4: /* ( ) = ; , */
00415     if ( *in == '(' ) {
00416         if ( funlevel >= AM.MaxParLevel ) {
00417             MesPrint("&More than %d levels of parentheses", AM.MaxParLevel);
00418             return(-1);
00419         }
00420         if ( object == 1 ) { /* After name -> function, vector */
00421             AC.tokenarglevel[funlevel++] = TYPEISFUN;
00422             *out++ = TFUNOPEN;
00423             if ( polyflag ) {
00424                 if ( in[1] != ')' && in[1] != ',' ) {
00425                     *out++ = TNUMBER; *out++ = (SBYTE) (polyflag);
00426                     *out++ = TCOMMA;
00427                     *out++ = LPARENTHESIS;
00428                 }
00429                 else {
00430                     *out++ = LPARENTHESIS;
00431                     *out++ = TNUMBER; *out++ = (SBYTE) (polyflag);
00432                 }
00433                 polyflag = 0;
00434             }
00435             else if ( in[1] != ')' && in[1] != ',' ) {
00436                 *out++ = LPARENTHESIS;
00437             }
00438         }
00439         else if ( object <= 0 ) {
00440             CHECKPOLY
00441             AC.tokenarglevel[funlevel++] = TYPEISSUB;
00442             *out++ = LPARENTHESIS;
00443         }
00444         else {
00445             polyflag = 0;
00446             AC.tokenarglevel[funlevel++] = TYPEISMYSTERY;
00447             MesPrint("&Illegal position for (: %s", in);
00448             if ( error >= 0 ) error = -1;
00449         }
00450         object = -1;
00451     }
00452     else if ( *in == ')' ) {
00453         funlevel--;
00454         if ( funlevel < 0 ) {
00455             /* if ( funflag == 0 ) { */
00456                 MesPrint("&There is an unmatched parenthesis");
00457                 if ( error >= 0 ) error = -1;
00458             /* } */
00459         }
00460         else if ( object <= 0
00461             && ( AC.tokenarglevel[funlevel] != TYPEISFUN
00462             || out[-1] != TFUNOPEN ) ) {
00463             MesPrint("&Illegal position for closing parenthesis.");
00464             if ( error >= 0 ) error = -1;
00465             if ( AC.tokenarglevel[funlevel] == TYPEISFUN ) object = 1;
00466             else object = 3;
00467         }
00468         else {
00469             if ( AC.tokenarglevel[funlevel] == TYPEISFUN ) {
00470                 if ( out[-1] == TFUNOPEN ) out--;
00471                 else {
00472                     if ( out[-1] != TCOMMA ) *out++ = RPARENTHESIS;
00473                     *out++ = TFUNCLOSE;
00474                 }
00475                 object = 1;
00476             }
00477             else if ( AC.tokenarglevel[funlevel] == TYPEISSUB ) {
00478                 *out++ = RPARENTHESIS;
00479                 object = 3;
00480             }
00481         }
00482     }

```

```

00483         else if ( *in == ',' ) {
00484             if ( /* object > 0 && */ funlevel > 0 &&
00485                 AC.tokenarglevel[funlevel-1] == TYPEISFUN ) {
00486                 if ( out[-1] != TFUNOPEN && out[-1] != TCOMMA )
00487                     *out++ = RPARENTHESIS;
00488                 else { *out++ = TNUMBER; *out++ = 0; }
00489                 *out++ = TCOMMA;
00490                 if ( in[1] != ',' && in[1] != ')' )
00491                     *out++ = LPARENTHESIS;
00492                 else if ( in[1] == ')' ) {
00493                     *out++ = TNUMBER; *out++ = 0;
00494                 }
00495             }
00496         /*
00497         else if ( object > 0 ) {
00498         }
00499         */
00500         else {
00501             MesPrint("&Illegal position for comma: %s",in);
00502             MesPrint("&Forgotten ; ?");
00503             if ( error >= 0 ) error = -1;
00504         }
00505         object = -1;
00506     }
00507     else goto IllPos;
00508     in++;
00509     break;
00510 case 5: /* + - * % / ^ : */
00511     CHECKPOLY
00512     if ( *in == ':' || *in == '%' ) goto IllPos;
00513     if ( *in == '*' || *in == '/' || *in == '^' ) {
00514         if ( object <= 0 ) {
00515             MesPrint("&Illegal position for operator: %s",in);
00516             if ( error >= 0 ) error = -1;
00517         }
00518         else if ( *in == '*' ) *out++ = TMULTIPLY;
00519         else if ( *in == '/' ) *out++ = TDIVIDE;
00520         else *out++ = TPOWER;
00521         in++;
00522     }
00523     else {
00524         i = 1;
00525         while ( *in == '+' || *in == '-' ) {
00526             if ( *in == '-' ) i = -i;
00527             in++;
00528         }
00529         if ( i == 1 ) {
00530             if ( out > AC.tokens && out[-1] != TFUNOPEN &&
00531                 out[-1] != LPARENTHESIS && out[-1] != TCOMMA
00532                 && out[-1] != LBRACE )
00533                 *out++ = TPLUS;
00534             }
00535             else *out++ = TMINUS;
00536         }
00537         object = 0;
00538         break;
00539 case 6: /* Whitespace */
00540     in++; break;
00541 case 7: /* { | } */
00542     CHECKPOLY
00543     if ( *in == '{' ) {
00544         if ( object > 0 ) {
00545             MesPrint("&Illegal position for %s",in);
00546             if ( !error ) error = 1;
00547         }
00548         s = in+1;
00549         SKIPBRA2(in)
00550         number = DoTempSet(s,in);
00551         in++;
00552         if ( number >= 0 ) {
00553             *out++ = TSET;
00554             i = 0;
00555             do { num[i++] = (SBYTE)(number & 0x7F); number >>= 7; } while ( number );
00556             while ( --i >= 0 ) *out++ = num[i];
00557         }
00558         else if ( error == 0 ) error = 1;
00559         object = 1;
00560     }
00561     else goto IllPos;
00562     break;
00563 case 8: /* ! & < > */
00564     CHECKPOLY
00565     if ( *in != '!' || leftright == RHSIDE
00566         || object != 1 || out[-1] != TWILDCARD ) goto IllPos;
00567     *out++ = TNOT;
00568     if ( FG.cTable[in[1]] == 0 || in[1] == '[' || in[1] == '{' ) object = 0;
00569     in++;

```

```

00570         break;
00571     default:
00572     IllPos:    MesPrint("&Illegal character at this position: %s",in);
00573         if ( error >= 0 ) error = -1;
00574         in++;
00575         polyflag = 0;
00576         break;
00577     }
00578 }
00579 *out++ = TENDOFIT;
00580 AC.endoftokens = out;
00581 if ( funlevel > 0 || bracelevel != 0 ) {
00582     if ( funlevel > 0 ) MesPrint("&Unmatched parentheses");
00583     if ( bracelevel != 0 ) MesPrint("&Unmatched braces");
00584     return(-1);
00585 }
00586 if ( AC.TokensWriteFlag ) WriteTokens(AC.tokens);
00587 /*
00588     Simplify fixed set elements
00589 */
00590 if ( error == 0 && simpltoken(AC.tokens) ) error = 1;
00591 /*
00592     Collect wildcards for the prototype. Simplify the leftover wildcards
00593 */
00594 if ( error == 0 && leftright == LHSIDE && simpwtoken(AC.tokens) )
00595     error = 1;
00596 /*
00597     Now prepare the set[n] objects in the RHS.
00598 */
00599 if ( error == 0 && leftright == RHSIDE && simp4token(AC.tokens) )
00600     error = 1;
00601 /*
00602     Simplify simple function arguments (and 1/fac_ and 1/invfac_)
00603 */
00604 if ( error == 0 && simp2token(AC.tokens) ) error = 1;
00605 /*
00606     Next we try to remove composite denominators or exponents and
00607     replace them by their internal functions. This may involve expanding
00608     the buffer. The return code of 3a is negative if there is an error
00609     and positive if indeed we need to do some work.
00610     simp3btoken does the work
00611 */
00612 numexp = 0;
00613 if ( error == 0 && ( numexp = simp3atoken(AC.tokens,leftright) ) < 0 )
00614     error = 1;
00615 if ( numexp > 0 ) {
00616     SBYTE *tt;
00617     out = AC.tokens;
00618     while ( *out != TENDOFIT ) out++;
00619     while ( out+numexp*9+20 > outtop ) {
00620         LONG oldsize = (LONG)(out - AC.tokens);
00621         SBYTE **pppp = &(AC.tokens); /* to avoid a compiler warning */
00622         SBYTE **pppp = &(AC.toptokens);
00623         DoubleBuffer((void **)pppp,(void **)pppp,sizeof(SBYTE),"out tokens");
00624         out = AC.tokens + oldsize;
00625         outtop = AC.toptokens - MAXNUMSIZE;
00626     }
00627     tt = out + numexp*9+20;
00628     while ( out >= AC.tokens ) { *tt-- = *out--; }
00629     while ( tt >= AC.tokens ) { *tt-- = EMPTY; }
00630     if ( error == 0 && simp3btoken(AC.tokens,leftright) ) error = 1;
00631     if ( error == 0 && simp2token(AC.tokens) ) error = 1;
00632 }
00633 /*
00634     In simp5token we test for special cases like sumvariables that are
00635     already wildcards, etc.
00636 */
00637 if ( error == 0 && simp5token(AC.tokens,leftright) ) error = 1;
00638 /*
00639     In simp6token we test for special cases like factorized expressions
00640     that occur in the RHS in an improper way.
00641 */
00642 if ( error == 0 && simp6token(AC.tokens,leftright) ) error = 1;
00643
00644 return(error);
00645 }
00646
00647 /*
00648     #] tokenize :
00649     #[ WriteTokens :
00650 */
00651
00652 char *ttypes[] = { "\n", "S", "I", "V", "F", "set", "E", "dotp", "#",
00653     "sub", "d_", "$", "dub", "(", ")", "?", "??", ".", "[", "]",
00654     ", ", "(", ")", "*", "/", "^", "+", "-", "!", "end", "{", "}",
00655     "N_?", "conj", "()", "#d", "^d", "_", "snum" };
00656

```

```

00657 void WriteTokens(SBYTE *in)
00658 {
00659     int numinline = 0, x, n = sizeof(ttypes)/sizeof(char *);
00660     char outbuf[81], *s, *out, c;
00661     out = outbuf;
00662     while ( *in != TENDOFIT ) {
00663         if ( *in < 0 ) {
00664             if ( *in >= -n ) {
00665                 s = ttypes[-*in];
00666                 while ( *s ) { *out++ = *s++; numinline++; }
00667             }
00668             else {
00669                 *out++ = '-'; x = -*in; numinline++;
00670                 goto writenumber;
00671             }
00672         }
00673         else {
00674             x = *in;
00675 writenumber:
00676             s = out;
00677             do {
00678                 *out++ = (char)(( x % 10 ) + '0');
00679                 numinline++;
00680                 x = x / 10;
00681             } while ( x );
00682             c = out[-1]; out[-1] = *s; *s = c;
00683         }
00684         if ( numinline > 70 ) {
00685             *out = 0;
00686             MesPrint("%s",outbuf);
00687             out = outbuf; numinline = 0;
00688         }
00689         else {
00690             *out++ = ' '; numinline++;
00691         }
00692         in++;
00693     }
00694     if ( numinline > 0 ) { *out = 0; MesPrint("%s",outbuf); }
00695 }
00696
00697 /*
00698     #] WriteTokens :
00699     #[ simpltoken :
00700
00701     Routine substitutes set elements if possible.
00702     This means sets with a fixed argument like setname[3].
00703 */
00704
00705 int simpltoken(SBYTE *s)
00706 {
00707     int error = 0, n, i, base;
00708     WORD numsub;
00709     SBYTE *fill = s, *start, *t, numtab[10];
00710     SETS set;
00711     while ( *s != TENDOFIT ) {
00712         if ( *s == RBRACE ) {
00713             start = fill-1;
00714             while ( *start != LBRACE ) start--;
00715             t = start - 1;
00716             while ( *t >= 0 ) t--;
00717             if ( *t == TSET && ( start[1] == TNUMBER || start[1] == TNUMBER1 ) ) {
00718                 base = start[1] == TNUMBER ? 100: 128;
00719                 start += 2;
00720                 numsub = *start++;
00721                 while ( *start >= 0 && start < fill )
00722                     { numsub = base*numsub + *start++; }
00723                 if ( start == fill ) {
00724                     start = t;
00725                     t++; n = *t++; while ( *t >= 0 ) { n = 128*n + *t++; }
00726                     set = Sets+n;
00727                     if ( ( set->type != CRANGE )
00728                         && ( numsub > 0 && numsub <= set->last-set->first ) ) {
00729                         fill = start;
00730                         n = SetElements[set->first+numsub-1];
00731                         switch (set->type) {
00732                             case CSYMBOL:
00733                                 if ( n > MAXPOWER ) {
00734                                     n -= 2*MAXPOWER;
00735                                     if ( n < 0 ) { n = -n; *fill++ = TMINUS; }
00736                                     *fill++ = TNUMBER1;
00737                                 }
00738                                 else *fill++ = TSYMBOL;
00739                                 break;
00740                             case CINDEX:
00741                                 if ( n < AM.OffsetIndex ) *fill++ = TNUMBER1;
00742                                 else {
00743                                     *fill++ = TINDEX;

```

```

00744             n -= AM.OffsetIndex;
00745         }
00746         break;
00747     case CVECTOR: *fill++ = TVECTOR;
00748         n -= AM.OffsetVector; break;
00749     case CFUNCTION: *fill++ = TFUNCTION;
00750         n -= FUNCTION; break;
00751     case CNUMBER: *fill++ = TNUMBER1; break;
00752     case CDUBIOUS: *fill++ = TDUBIOUS; n = 1; break;
00753     }
00754     i = 0;
00755     if ( n < 0 ) {
00756         MesPrint("Value of n = %d",n);
00757     }
00758     do { numtab[i++] = (SBYTE)(n & 0x7F); n >>= 7; } while ( n );
00759     while ( --i >= 0 ) *fill++ = numtab[i];
00760     }
00761     else {
00762         MesPrint("&Illegal element %d in set",numsub);
00763         error++;
00764     }
00765     s++; continue;
00766     }
00767     }
00768     *fill++ = *s++;
00769     }
00770     else *fill++ = *s++;
00771     }
00772     *fill++ = TENDOFIT;
00773     return(error);
00774 }
00775
00776 /*
00777     #] simpltoken :
00778     #[ simpwtoken :
00779
00780     Only to be called in the LHS.
00781     Hunts down the wildcards and writes them to the wildcardbuffer.
00782     Next it causes the ProtoType to be constructed.
00783     All wildcards are simplified into the trailing TWILDCARD,
00784     because the specifics are stored in the prototype.
00785     These specifics also include the transfer of wildcard values
00786     to $variables.
00787
00788     Types of wildcards:
00789     a?, a?set, a?!set, a?set[i], A?set1?set2, ?a
00790     After this we can strip the set information.
00791     We still need the ? because of the wildcarding offset in code generation
00792 */
00793
00794 int simpwtoken(SBYTE *s)
00795 {
00796     int error = 0, first = 1, notflag;
00797     WORD num, numto, numdollar, *w = AC.WildC, *wstart, *wtop;
00798     SBYTE *fill = s, *t, *v, *s0 = s;
00799     while ( *s != TENDOFIT ) {
00800         if ( *s == TWILDCARD ) {
00801             notflag = 0; t = fill;
00802             while ( t > s0 && t[-1] >= 0 ) t--;
00803             v = t; num = 0; *fill++ = *s++;
00804             while ( *v >= 0 ) num = 128*num + *v++;
00805             if ( t > s0 ) t--;
00806             AC.NwildC += 4;
00807             if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
00808             switch ( *t ) {
00809                 case TSYMBOL:
00810                 case TDUBIOUS:
00811                     *w++ = SYMTOSYM; *w++ = 4; *w++ = num; *w++ = num; break;
00812                 case TINDEX:
00813                     num += AM.OffsetIndex;
00814                     *w++ = INDTOIND; *w++ = 4; *w++ = num; *w++ = num; break;
00815                 case TVECTOR:
00816                     num += AM.OffsetVector;
00817                     *w++ = VECTOVEC; *w++ = 4; *w++ = num; *w++ = num; break;
00818                 case TFUNCTION:
00819                     num += FUNCTION;
00820                     *w++ = FUNTOFUN; *w++ = 4; *w++ = num; *w++ = num; break;
00821                 default:
00822                     MesPrint("&Illegal type of wildcard in LHS");
00823                     error = -1;
00824                     *w++ = SYMTOSYM; *w++ = 4; *w++ = num; *w++ = num; break;
00825             }
00826         }
00827     }
00828     /*
00829     Now the sets. The s pointer sits after the ?
00830     */
00831     wstart = w;

```

```

00831         if ( *s == TNOT && s[1] == TSET ) { notflag = 1; s++; }
00832     if ( *s == TSET ) {
00833         s++; num = 0; while ( *s >= 0 ) num = 128*num + *s++;
00834         if ( notflag == 0 && *s == TWILDCARD && s[1] == TSET ) {
00835             s += 2; numto = 0; while ( *s >= 0 ) numto = 128*numto + *s++;
00836             if ( num < AM.NumFixedSets || numto < AM.NumFixedSets
00837                 || Sets[num].type == CRANGE || Sets[numto].type == CRANGE ) {
00838                 MesPrint("&This type of set not allowed in this wildcard construction");
00839                 error = 1;
00840             }
00841             else {
00842                 AC.NwildC += 4;
00843                 if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
00844                 *w++ = FROMSET; *w++ = 4; *w++ = num; *w++ = numto;
00845                 wstart = w;
00846             }
00847         }
00848     else if ( notflag == 0 && *s == LBRACE && s[1] == TSYMBOL ) {
00849         if ( num < AM.NumFixedSets || Sets[num].type == CRANGE ) {
00850             MesPrint("&This type of set not allowed in this wildcard construction");
00851             error = 1;
00852         }
00853         v = s; s += 2;
00854         numto = 0; while ( *s >= 0 ) numto = 128*numto + *s++;
00855         if ( *s == TWILDCARD ) s++; /* most common mistake */
00856         if ( *s == RBRACE ) {
00857             s++;
00858             AC.NwildC += 8;
00859             if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
00860             *w++ = SETTONUM; *w++ = 4; *w++ = num; *w++ = numto;
00861             wstart = w;
00862             *w++ = SYMTOSYM; *w++ = 4; *w++ = numto; *w++ = 0;
00863         }
00864     else if ( *s == TDOLLAR ) {
00865         s++; numdollar = 0;
00866         while ( *s >= 0 ) numdollar = 128*numdollar + *s++;
00867         if ( *s == RBRACE ) {
00868             s++;
00869             AC.NwildC += 12;
00870             if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
00871             *w++ = SETTONUM; *w++ = 4; *w++ = num; *w++ = numto;
00872             wstart = w;
00873             *w++ = SYMTOSYM; *w++ = 4; *w++ = numto; *w++ = 0;
00874             *w++ = LOADDOLLAR; *w++ = 4; *w++ = numdollar;
00875             *w++ = numdollar;
00876         }
00877         else { s = v; goto singlewild; }
00878     }
00879     else { s = v; goto singlewild; }
00880 }
00881 singlewild:
00882     num += notflag * 2*WILDOFFSET;
00883     AC.NwildC += 4;
00884     if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
00885     *w++ = FROMSET; *w++ = 4; *w++ = num; *w++ = -WILDOFFSET;
00886     wstart = w;
00887 }
00888 }
00889 else if ( *s != TDOLLAR && *s != TENDOFIT && *s != RPARENTHESIS
00890 && *s != RBRACE && *s != TCOMMA && *s != TFUNCLOSE && *s != TMULTIPLY
00891 && *s != TPOWER && *s != TDIVIDE && *s != TPLUS && *s != TMINUS
00892 && *s != TPOWER1 && *s != TEMPTY && *s != TFUNOPEN && *s != TDOT ) {
00893     MesPrint("&Illegal type of wildcard in LHS");
00894     error = -1;
00895 }
00896 if ( *s == TDOLLAR ) {
00897     s++; numdollar = 0;
00898     while ( *s >= 0 ) numdollar = 128*numdollar + *s++;
00899     AC.NwildC += 4;
00900     if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
00901     wtop = w + 4;
00902     if ( wstart < w ) {
00903         while ( w > wstart ) { w[4] = w[0]; w--; }
00904     }
00905     *w++ = LOADDOLLAR; *w++ = 4; *w++ = numdollar; *w++ = numdollar;
00906     w = wtop;
00907 }
00908 }
00909 else if ( *s == TWILDARG ) {
00910     *fill++ = *s++;
00911     num = 0;
00912     while ( *s >= 0 ) { num = 128*num + *s; *fill++ = *s++; }
00913     AC.NwildC += 4;
00914     if ( AC.NwildC > 4*AM.MaxWildcards ) {
00915         firsterr:
00916         if ( first ) {
00917             MesPrint("&More than %d wildcards",AM.MaxWildcards);
00918             error = -1;

```

```

00918         first = 0;
00919     }
00920 }
00921 else { *w++ = ARGTOARG; *w++ = 4; *w++ = num; *w++ = -1; }
00922 if ( *s == TDOLLAR ) {
00923     s++; num = 0; while ( *s >= 0 ) num = 128*num + *s++;
00924     AC.NwildC += 4;
00925     if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
00926     *w++ = LOADDOLLAR; *w++ = 4; *w++ = num; *w++ = num;
00927 }
00928 }
00929 else *fill++ = *s++;
00930 }
00931 *fill++ = TENDOFIT;
00932 AC.WildC = w;
00933 return(error);
00934 }
00935
00936 /*
00937     #] simpwtoken :
00938     #[ simp2token :
00939
00940     Deals with function arguments.
00941     The tokenizer has given function arguments extra parentheses.
00942     We remove the double parentheses.
00943     Next we remove the parentheses around the simple arguments.
00944
00945     It also replaces /fac_() by *invfac_() and /invfac_() by *fac_()
00946 */
00947
00948 int simp2token(SBYTE *s)
00949 {
00950     SBYTE *to, *fill, *t, *v, *w, *s0 = s, *vv;
00951     int error = 0, n;
00952     /*
00953         Set substitutions
00954     */
00955     fill = to = s;
00956     while ( *s != TENDOFIT ) {
00957         if ( *s == LPARENTHESIS && s[1] == LPARENTHESIS ) {
00958             t = s+1; n = 0;
00959             while ( n >= 0 ) {
00960                 t++;
00961                 if ( *t == LPARENTHESIS ) n++;
00962                 else if ( *t == RPARENTHESIS ) n--;
00963             }
00964             if ( t[1] == RPARENTHESIS ) {
00965                 *t = EMPTY; s++;
00966             }
00967             *fill++ = *s++;
00968         }
00969         else if ( *s == EMPTY ) s++;
00970         else if ( *s == AM.facnum && ( fill > (s0+1) ) && fill[-2] == TDIVIDE
00971             && fill[-1] == TFUNCTION ) {
00972             fill[-2] = TMULTIPLY; *fill++ = (SBYTE)(AM.invfacnum); s++;
00973         }
00974         else if ( *s == AM.invfacnum && ( fill > (s0+1) ) && fill[-2] == TDIVIDE
00975             && fill[-1] == TFUNCTION ) {
00976             fill[-2] = TMULTIPLY; *fill++ = (SBYTE)(AM.facnum); s++;
00977         }
00978         else *fill++ = *s++;
00979     }
00980     *fill++ = TENDOFIT;
00981     /*
00982         Second round: try to locate 'simple' arguments and strip their brackets
00983
00984         We add (9-feb-2010) to the simple arguments integers of any size
00985     */
00986     fill = s = to;
00987     while ( *s != TENDOFIT ) {
00988         if ( *s == LPARENTHESIS ) {
00989             t = s; n = 0;
00990             while ( n >= 0 ) {
00991                 t++;
00992                 if ( *t == LPARENTHESIS ) n++;
00993                 else if ( *t == RPARENTHESIS ) n--;
00994             }
00995             if ( t[1] == TFUNCLOSE && s[1] != TWILDARG ) { /* Check for last argument in sum */
00996                 v = fill - 1; n = 0;
00997                 while ( n >= 0 && v >= to ) {
00998                     if ( *v == TFUNOPEN ) n--;
00999                     else if ( *v == TFUNCLOSE ) n++;
01000                     v--;
01001                 }
01002                 if ( v > to ) {
01003                     while ( *v >= 0 ) v--;
01004                     if ( *v == TFUNCTION ) { v++;

```

```

01005         n = 0; while ( *v >= 0 && v < fill ) n = 128*n + *v++;
01006         if ( n == AM.sumnum || n == AM.sumpnum ) {
01007             *fill++ = *s++; continue;
01008         }
01009         else if ( ( n == (FIRSTBRACKET-FUNCTION)
01010             || n == (TERMSINEXPR-FUNCTION)
01011             || n == (SIZEOFFUNCTION-FUNCTION)
01012             || n == (NUMFACTORS-FUNCTION)
01013             || n == (GCDFUNCTION-FUNCTION)
01014             || n == (DIVFUNCTION-FUNCTION)
01015             || n == (REMFUNCTON-FUNCTION)
01016             || n == (INVERSEFUNCTION-FUNCTION)
01017             || n == (MULFUNCTION-FUNCTION)
01018             || n == (FACTORIN-FUNCTION)
01019             || n == (FIRSTTERM-FUNCTION)
01020             || n == (CONTENTTERM-FUNCTION) )
01021             && fill[-1] == TFUNOPEN ) {
01022             v = s+1;
01023             if ( *v == TEXPRESSION ) {
01024                 v++;
01025                 n = 0; while ( *v >= 0 ) n = 128*n + *v++;
01026                 if ( v == t ) {
01027                     *t = EMPTY; s++;
01028                 }
01029             }
01030         }
01031     }
01032 }
01033 }
01034 if ( ( fill > to )
01035     && ( ( fill[-1] == TFUNOPEN || fill[-1] == TCOMMA )
01036         && ( t[1] == TFUNCLOSE || t[1] == TCOMMA ) ) ) {
01037     v = s + 1;
01038     switch ( *v ) {
01039     case TMINUS:
01040         v++;
01041         if ( *v == TVECTOR ) {
01042             w = v+1; while ( *w >= 0 ) w++;
01043             if ( w == t ) {
01044                 *t = EMPTY; s++;
01045             }
01046         }
01047     else {
01048         if ( *v == TNUMBER || *v == TNUMBER1 ) {
01049             if ( BITSINWORD == 16 ) { ULONG x; WORD base;
01050                 base = ( *v == TNUMBER ) ? 100: 128;
01051                 vv = v+1; x = 0; while ( *vv >= 0 ) { x = x*base + *vv++; }
01052                 if ( ( vv != t ) || ( ( vv - v ) > 4 ) || ( x > (MAXPOSITIVE+1) ) )
01053                     *fill++ = *s++;
01054                 else { *t = EMPTY; s++; break; }
01055             }
01056             else if ( BITSINWORD == 32 ) { ULONG x; WORD base;
01057                 base = ( *v == TNUMBER ) ? 100: 128;
01058                 vv = v+1; x = 0; while ( *vv >= 0 ) { x = x*base + *vv++; }
01059                 if ( ( vv != t ) || ( ( vv - v ) > 6 ) || ( x > (MAXPOSITIVE+1) ) )
01060                     *fill++ = *s++;
01061                 else { *t = EMPTY; s++; break; }
01062             }
01063             else {
01064                 if ( ( v+2 == t ) || ( v+3 == t && v[2] >= 0 ) )
01065                     { *t = EMPTY; s++; break; }
01066                 else *fill++ = *s++;
01067             }
01068         }
01069         else if ( *v == LPARENTHESIS && t[-1] == RPARENTHESIS ) {
01070             w = v; n = 0;
01071             while ( n >= 0 ) {
01072                 w++;
01073                 if ( *w == LPARENTHESIS ) n++;
01074                 else if ( *w == RPARENTHESIS ) n--;
01075             }
01076             if ( w == ( t-1 ) ) { *t = EMPTY; s++; }
01077             else *fill++ = *s++;
01078         }
01079         else *fill++ = *s++;
01080         break;
01081     }
01082     /* fall through */
01083     case TSETNUM:
01084         v++; while ( *v >= 0 ) v++;
01085         goto tcommon;
01086     case TSYMBOL:
01087         if ( ( v[1] == COEFFSYMBOL || v[1] == NUMERATORSYMBOL
01088             || v[1] == DENOMINATORSYMBOL ) && v[2] < 0 ) {
01089             *fill++ = *s++; break;
01090         }
01091     /* fall through */

```



```

01092         case TSET:
01093         case TVECTOR:
01094         case TINDEX:
01095         case TFUNCTION:
01096         case TDOLLAR:
01097         case TDUBIOUS:
01098         case TSGAMMA:
01099 tcommon:      v++; while ( *v >= 0 ) v++;
01100                 if ( v == t || ( v[0] == TWILDCARD && v+1 == t ) )
01101                 { *t = EMPTY; s++; }
01102                 else *fill++ = s++;
01103                 break;
01104         case TGENINDEX:
01105                 v++;
01106                 if ( v == t ) { *t = EMPTY; s++; }
01107                 else *fill++ = s++;
01108                 break;
01109         case TNUMBER:
01110         case TNUMBER1:
01111                 if ( BITSINWORD == 16 ) { ULONG x; WORD base;
01112                         base = ( *v == TNUMBER ) ? 100: 128;
01113                         vv = v+1; x = 0; while ( *vv >= 0 ) { x = x*base + *vv++; }
01114                         if ( ( vv != t ) || ( ( vv - v ) > 4 ) || ( x > MAXPOSITIVE ) )
01115                                 *fill++ = s++;
01116                         else { *t = EMPTY; s++; break; }
01117                 }
01118                 else if ( BITSINWORD == 32 ) { ULONG x; WORD base;
01119                         base = ( *v == TNUMBER ) ? 100: 128;
01120                         vv = v+1; x = 0; while ( *vv >= 0 ) { x = x*base + *vv++; }
01121                         if ( ( vv != t ) || ( ( vv - v ) > 6 ) || ( x > MAXPOSITIVE ) )
01122                                 *fill++ = s++;
01123                         else { *t = EMPTY; s++; break; }
01124                 }
01125                 else {
01126                         if ( ( v+2 == t ) || ( v+3 == t && v[2] >= 0 ) )
01127                                 { *t = EMPTY; s++; break; }
01128                         else *fill++ = s++;
01129                 }
01130                 break;
01131         case TWILDARG:
01132                 v++; while ( *v >= 0 ) v++;
01133                 if ( v == t ) { *t = EMPTY; s++; }
01134                 else *fill++ = s++;
01135                 break;
01136         case TEXPRESSON:
01137 /*
01138         First establish that there is only the expression
01139         in this argument.
01140 */
01141         vv = s+1;
01142         while ( vv < t ) {
01143                 if ( *vv != TEXPRESSON ) break;
01144                 vv++; while ( *vv >= 0 ) vv++;
01145         }
01146         if ( vv < t ) { *fill++ = s++; break; }
01147 /*
01148         Find the function
01149 */
01150         w = fill-1; n = 0;
01151         while ( n >= 0 && w >= to ) {
01152                 if ( *w == TFUNOPEN ) n--;
01153                 else if ( *w == TFUNCLOSE ) n++;
01154                 w--;
01155         }
01156         w--; while ( w > to && *w >= 0 ) w--;
01157         if ( *w != TFUNCTION ) { *fill++ = s++; break; }
01158         w++; n = 0;
01159         while ( *w >= 0 ) { n = 128*n + *w++; }
01160         if ( n == GCDFUNCTION-FUNCTION
01161             || n == DIVFUNCTION-FUNCTION
01162             || n == REMFUNCTION-FUNCTION
01163             || n == INVERSEFUNCTION-FUNCTION
01164             || n == MULFUNCTION-FUNCTION ) {
01165                 *t = EMPTY; s++;
01166         }
01167         else *fill++ = s++;
01168         break;
01169     default: *fill++ = s++; break;
01170 }
01171 }
01172     else *fill++ = s++;
01173 }
01174     else if ( *s == EMPTY ) s++;
01175     else *fill++ = s++;
01176 }
01177 *fill++ = TENDOFIT;
01178 return(error);

```

```

01179 }
01180
01181 /*
01182     #] simp2token :
01183     #[ simp3atoken :
01184
01185     We hunt for denominators and exponents that seem hidden.
01186     For the denominators we have to recognize:
01187         /fun /fun() /fun^power /fun()^power
01188         /set[n] /set[n]() /set[n]^power /set[n]()^power
01189         /symbol^power (power no number or symbol wildcard)
01190         /dotpr^power (id)
01191         /#^power (id)
01192         /() /()^power
01193         /vect /index /vect(anything) /vect(anything)^power
01194 */
01195
01196 int simp3atoken(SBYTE *s, int mode)
01197 {
01198     int error = 0, n, numexp = 0, denom, base, numprot, i;
01199     SBYTE *t, c;
01200     LONG num;
01201     WORD *prot;
01202     if ( mode == RHSIDE ) {
01203         prot = AC.ProtoType;
01204         numprot = prot[1] - SUBEXPSIZE;
01205         prot += SUBEXPSIZE;
01206     }
01207     else { prot = 0; numprot = 0; }
01208     while ( *s != TENDOFIT ) {
01209         denom = 1;
01210         if ( *s == TDIVIDE ) { denom = -1; s++; }
01211         c = *s;
01212         switch(c) {
01213             case TSYMBOL:
01214             case TNUMBER:
01215             case TNUMBER1:
01216                 s++; while ( *s >= 0 ) s++; /* skip the object */
01217                 if ( *s == TWILDCARD ) s++; /* and the possible wildcard */
01218             dosymbol:
01219                 if ( *s != TPOWER ) continue; /* No power -> done */
01220                 s++; /* Skip the power */
01221                 if ( *s == TMINUS ) s++; /* negative: no difference here */
01222                 if ( *s == TNUMBER || *s == TNUMBER1 ) {
01223                     base = *s == TNUMBER ? 100: 128; /* NUMBER = base 100 */
01224                     s++; /* Now we compose the power */
01225                     num = s++; /* If the number is way too large */
01226                     while ( *s >= 0 ) { /* it may look like not too big */
01227                         if ( num > MAXPOWER ) break; /* Hence... */
01228                         num = base*num + *s++;
01229                     }
01230                     while ( *s >= 0 ) s++; /* Finish the number if needed */
01231                     if ( *s == TPOWER ) goto doublepower;
01232                     if ( num <= MAXPOWER ) continue; /* Simple case */
01233                 }
01234                 else if ( *s == TSYMBOL && c != TNUMBER && c != TNUMBER1 ) {
01235                     s++; n = 0; while ( *s >= 0 ) { n = 128*n + *s++; }
01236                     if ( *s == TWILDCARD ) { s++;
01237                         if ( *s == TPOWER ) goto doublepower;
01238                         continue; }
01239                 /*
01240                 Now we have to test whether n happens to be a wildcard
01241                 */
01242                 if ( mode == RHSIDE ) {
01243                     n += 2*MAXPOWER;
01244                     for ( i = 0; i < numprot; i += 4 ) {
01245                         if ( prot[i+2] == n && prot[i] == SYMTOSYM ) break;
01246                     }
01247                     if ( i < numprot ) break;
01248                 }
01249                 if ( *s == TPOWER ) goto doublepower;
01250             }
01251             numexp++;
01252             break;
01253         case TINDEX:
01254             s++; while ( *s >= 0 ) s++;
01255             if ( *s == TWILDCARD ) s++;
01256         doindex:
01257             if ( denom < 0 || *s == TPOWER ) {
01258                 MesPrint("&Index to a power or in denominator is illegal");
01259                 error = 1;
01260             }
01261             break;
01262         case TVECTOR:
01263             s++; while ( *s >= 0 ) s++;
01264             if ( *s == TWILDCARD ) s++;
01265         dovector:

```

```

01266         if ( *s == TFUNOPEN ) {
01267             s++; n = 1;
01268             for(;;) {
01269                 if ( *s == TFUNOPEN ) {
01270                     n++;
01271                     MesPrint("&Illegal vector index");
01272                     error = 1;
01273                 }
01274                 else if ( *s == TFUNCLOSE ) {
01275                     n--;
01276                     if ( n <= 0 ) break;
01277                 }
01278                 s++;
01279             }
01280             s++;
01281         }
01282         else if ( *s == TDOT ) goto dodot;
01283         if ( denom < 0 || *s == TPOWER || *s == TPOWER1 ) numexp++;
01284         break;
01285     case TFUNCTION:
01286         s++; while ( *s >= 0 ) s++;
01287         if ( *s == TWILDCARD ) s++;
01288     dofunction:
01289         t = s;
01290         if ( *t == TFUNOPEN ) {
01291             t++; n = 1;
01292             for(;;) {
01293                 if ( *t == TFUNOPEN ) n++;
01294                 else if ( *t == TFUNCLOSE ) { if ( --n <= 0 ) break; }
01295                 t++;
01296             }
01297             t++; s++;
01298         }
01299         if ( denom < 0 || *t == TPOWER || *t == TPOWER1 ) numexp++;
01300         break;
01301     #ifdef WITHFLOAT
01302     case TFLOAT:
01303         s++; while ( *s >= 0 ) s++;
01304         if ( denom < 0 || *s == TPOWER || *s == TPOWER1 ) numexp++;
01305         break;
01306     #endif
01307     case TEXPRESSION:
01308         s++; while ( *s >= 0 ) s++;
01309         t = s;
01310         if ( *t == TFUNOPEN ) {
01311             t++; n = 1;
01312             for(;;) {
01313                 if ( *t == TFUNOPEN ) n++;
01314                 else if ( *t == TFUNCLOSE ) { if ( --n <= 0 ) break; }
01315                 t++;
01316             }
01317             t++;
01318         }
01319         if ( *t == LBRACE ) {
01320             t++; n = 1;
01321             for(;;) {
01322                 if ( *t == LBRACE ) n++;
01323                 else if ( *t == RBRACE ) { if ( --n <= 0 ) break; }
01324                 t++;
01325             }
01326             t++;
01327         }
01328         if ( denom < 0 || ( ( *t == TPOWER || *t == TPOWER1 )
01329             && t[1] == TMINUS ) ) numexp++;
01330         break;
01331     case TDOLLAR:
01332         s++; while ( *s >= 0 ) s++;
01333         if ( denom < 0 || ( ( *s == TPOWER || *s == TPOWER1 )
01334             && s[1] == TMINUS ) ) numexp++;
01335         break;
01336     case LPARENTHESIS:
01337         s++; n = 1; t = s;
01338         for(;;) {
01339             if ( *t == LPARENTHESIS ) n++;
01340             else if ( *t == RPARENTHESIS ) { if ( --n <= 0 ) break; }
01341             t++;
01342         }
01343         t++;
01344         if ( denom > 0 && ( *t == TPOWER || *t == TPOWER1 ) ) {
01345             if ( ( t[1] == TNUMBER || t[1] == TNUMBER1 ) && t[2] >= 0
01346                 && t[3] < 0 ) break;
01347             numexp++;
01348         }
01349         else if ( denom < 0 && ( *t == TPOWER || *t == TPOWER1 ) ) {
01350             if ( t[1] == TMINUS && ( t[2] == TNUMBER
01351                 || t[2] == TNUMBER1 ) && t[3] >= 0
01352                 && t[4] < 0 ) break;

```

```

01353         numexp++;
01354     }
01355     else if ( denom < 0 || ( ( *t == TPOWER || *t == TPOWER1 )
01356         && ( t[1] == TMINUS || t[1] == LPARENTHESIS ) ) ) numexp++;
01357     break;
01358 case TSET:
01359     s++; n = *s++; while ( *s >= 0 ) { n = 128*n + *s++; }
01360     n = Sets[n].type;
01361     switch ( n ) {
01362     case CSYMBOL: goto dosymbol;
01363     case CINDEX: goto doindex;
01364     case CVECTOR: goto dovector;
01365     case CFUNCTION: goto dofunction;
01366     case CNUMBER: goto dosymbol;
01367     case CMODEL:
01368         if ( denom < 0 || *s == TPOWER ) {
01369             MesPrint("&A model to a power or in denominator is illegal");
01370             error = 1;
01371         }
01372         break;
01373     case ANYTYPE:
01374         if ( denom < 0 || *s == TPOWER ) {
01375             MesPrint("&A set without type to a power or in denominator is illegal");
01376             error = 1;
01377         }
01378         break;
01379     default: error = 1; break;
01380     }
01381     break;
01382 case TDOT:
01383     s++;
01384     if ( *s == TVECTOR ) { s++; while ( *s >= 0 ) s++; }
01385     else if ( *s == TSET ) {
01386         s++; n = *s++; while ( *s >= 0 ) { n = 128*n + *s++; }
01387         if ( Sets[n].type != CVECTOR ) {
01388             MesPrint("&Set in dotproduct is not a set of vectors");
01389             error = 1;
01390         }
01391         if ( *s == LBRACE ) {
01392             s++; n = 1;
01393             for(;;) {
01394                 if ( *s == LBRACE ) n++;
01395                 else if ( *s == RBRACE ) { if ( --n <= 0 ) break; }
01396                 s++;
01397             }
01398             s++;
01399         }
01400         else {
01401             MesPrint("&Set without argument in dotproduct");
01402             error = 1;
01403         }
01404     }
01405     else if ( *s == TSETNUM ) {
01406         s++; n = *s++; while ( *s >= 0 ) { n = 128*n + *s++; }
01407         if ( *s != TVECTOR ) goto nodot;
01408         s++; n = *s++; while ( *s >= 0 ) { n = 128*n + *s++; }
01409         if ( Sets[n].type != CVECTOR ) {
01410             MesPrint("&Set in dotproduct is not a set of vectors");
01411             error = 1;
01412         }
01413     }
01414     else {
01415         MesPrint("&Illegal second element in dotproduct");
01416         error = 1;
01417         s++; while ( *s >= 0 ) s++;
01418     }
01419     goto dosymbol;
01420 default:
01421     s++; while ( *s >= 0 ) s++;
01422     break;
01423 }
01424 }
01425 if ( error ) return(-1);
01426 return(numexp);
01427 doublepower:
01428     MesPrint("&Dubious notation with object^power1^power2");
01429     return(-1);
01430 }
01431
01432 /*
01433     #] simp3atoken :
01434     #[ simp3btoken :
01435 */
01436
01437 int simp3btoken(SBYTE *s, int mode)
01438 {
01439     int error = 0, i, numprot, n, denom, base, inset = 0, dotp, sube = 0;

```

```

01440     SBYTE *t, c, *fill, *ff, *ss;
01441     LONG num;
01442     WORD *prot;
01443     if ( mode == RHSIDE ) {
01444         prot = AC.ProtoType;
01445         numprot = prot[1] - SUBEXPSIZE;
01446         prot += SUBEXPSIZE;
01447     }
01448     else { prot = 0; numprot = 0; }
01449     fill = s;
01450     while ( *s == EMPTY ) s++;
01451     while ( *s != TENDOFIT ) {
01452         if ( *s == EMPTY ) { s++; continue; }
01453         denom = 1;
01454         if ( *s == TDIVIDE ) { denom = -1; *fill++ = *s++; }
01455         ff = fill; ss = s; c = *s;
01456         if ( c == TSETNUM ) {
01457             *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
01458             c = *s;
01459         }
01460         dotp = 0;
01461         switch(c) {
01462             case TSYMBOL:
01463             case TNUMBER:
01464             case TNUMBER1:
01465                 *fill++ = *s++;
01466                 while ( *s >= 0 ) *fill++ = *s++;
01467                 if ( *s == TWILDCARD ) *fill++ = *s++;
01468 dosymbol:
01469                 t = s;
01470                 if ( *s != TPOWER ) continue;
01471                 *fill++ = *s++;
01472                 if ( *s == TMINUS ) *fill++ = *s++;
01473                 if ( *s == TPLUS ) s++;
01474                 if ( *s == TSETNUM ) {
01475                     *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
01476                     inset = 1;
01477                 }
01478                 else inset = 0;
01479                 if ( *s == TNUMBER || *s == TNUMBER1 ) {
01480                     base = *s == TNUMBER ? 100: 128;
01481                     *fill++ = *s++;
01482                     num = *s++; *fill++ = num;
01483                     while ( *s >= 0 ) {
01484                         if ( num > MAXPOWER ) break;
01485                         *fill++ = *s;
01486                         num = base*num + *s++;
01487                     }
01488                     while ( *s >= 0 ) *fill++ = *s++;
01489                     if ( num <= MAXPOWER ) continue;
01490                     goto putexpl;
01491                 }
01492                 else if ( *s == TSYMBOL && c != TNUMBER && c != TNUMBER1 ) {
01493                     *fill++ = *s++;
01494                     n = 0; while ( *s >= 0 ) { n = 128*n + *s; *fill++ = *s++; }
01495                     if ( *s == TWILDCARD ) { *fill++ = *s++;
01496                         if ( *s == TPOWER ) goto doublepower;
01497                     break; }
01498 /*
01499     Now we have to test whether n happens to be a wildcard
01500 */
01501                 if ( mode == RHSIDE && inset == 0 ) {
01502 /*
01503                     n += WILDOFFSET;*/
01504                     for ( i = 0; i < numprot; i += 4 ) {
01505                         if ( prot[i+2] == n && prot[i] == SYMTOSYM ) break;
01506                     }
01507                     if ( i < numprot ) break;
01508                 }
01509 putexpl:
01510                 fill = ff;
01511                 if ( denom < 0 ) fill[-1] = TMULTIPLY;
01512                 *fill++ = TFUNCTION; *fill++ = (SBYTE)(AM.expnum); *fill++ = TFUNOPEN;
01513                 if ( dotp ) *fill++ = LPARENTHESIS;
01514                 while ( ss < t ) *fill++ = *ss++;
01515                 if ( dotp ) *fill++ = RPARENTHESIS;
01516                 *fill++ = TCOMMA;
01517                 ss++; /* Skip TPOWER */
01518                 if ( *ss == TMINUS ) { denom = -denom; ss++; }
01519                 if ( denom < 0 ) {
01520                     *fill++ = LPARENTHESIS;
01521                     *fill++ = TMINUS;
01522                     while ( ss < s ) *fill++ = *ss++;
01523                     *fill++ = RPARENTHESIS;
01524                 }
01525                 else {
01526                     while ( ss < s ) *fill++ = *ss++;
01527                 }

```

```

01527         *fill++ = TFUNCLOSE;
01528         if ( *ss == TPOWER ) goto doublepower;
01529     }
01530     else { /* other objects can be composite */
01531         goto dofunpower;
01532     }
01533     break;
01534 case TINDEX:
01535     *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
01536     if ( *s == TWILDCARD ) *fill++ = *s++;
01537     break;
01538 case TVECTOR:
01539     *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
01540     if ( *s == TWILDCARD ) *fill++ = *s++;
01541 dovector:
01542     if ( *s == TFUNOPEN ) {
01543         while ( *s != TFUNCLOSE ) *fill++ = *s++;
01544         *fill++ = *s++;
01545     }
01546     else if ( *s == TDOT ) goto dodot;
01547     t = s;
01548     goto dofunpower;
01549 #ifdef WITHFLOAT
01550 case TFLOAT:
01551     *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
01552     t = s;
01553     sube = 0;
01554     goto dofunpower;
01555 #endif
01556 case TFUNCTION:
01557     *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
01558     if ( *s == TWILDCARD ) *fill++ = *s++;
01559 dofunction:
01560     t = s;
01561     if ( *t == TFUNOPEN ) {
01562         t++; n = 1;
01563         for(;;) {
01564             if ( *t == TFUNOPEN ) n++;
01565             else if ( *t == TFUNCLOSE ) { if ( --n <= 0 ) break; }
01566             t++;
01567         }
01568         t++; *fill++ = *s++;
01569     }
01570     sube = 0;
01571 dofunpower:
01572     if ( *t == TPOWER || *t == TPOWER1 ) {
01573         if ( sube ) {
01574             if ( ( t[1] == TNUMBER || t[1] == TNUMBER1 )
01575                 && denom > 0 ) {
01576                 if ( t[2] >= 0 && t[3] < 0 ) { sube = 0; break; }
01577             }
01578             else if ( t[1] == TMINUS && denom < 0 &&
01579                 ( t[2] == TNUMBER || t[2] == TNUMBER1 ) ) {
01580                 if ( t[2] >= 0 && t[3] < 0 ) { sube = 0; break; }
01581             }
01582             sube = 0;
01583         }
01584         fill = ff;
01585         *fill++ = TFUNCTION; *fill++ = (SBYTE) (AM.expnum); *fill++ = TFUNOPEN;
01586         *fill++ = LPARENTHESIS;
01587         while ( ss < t ) *fill++ = *ss++;
01588         t++;
01589         *fill++ = RPARENTHESIS; *fill++ = TCOMMA;
01590         if ( *t == TMINUS ) { t++; denom = -denom; }
01591         *fill++ = LPARENTHESIS;
01592         if ( denom < 0 ) *fill++ = TMINUS;
01593         if ( *t == LPARENTHESIS ) {
01594             *fill++ = *t++; n = 0;
01595             while ( n >= 0 ) {
01596                 if ( *t == LPARENTHESIS ) n++;
01597                 else if ( *t == RPARENTHESIS ) n--;
01598                 *fill++ = *t++;
01599             }
01600         }
01601         else if ( *t == TFUNCTION || *t == TDUBIOUS ) {
01602             *fill++ = *t++; while ( *t >= 0 ) *fill++ = *t++;
01603             if ( *t == TWILDCARD ) *fill++ = *t++;
01604             if ( *t == TFUNOPEN ) {
01605                 *fill++ = *t++; n = 0;
01606                 while ( n >= 0 ) {
01607                     if ( *t == TFUNOPEN ) n++;
01608                     else if ( *t == TFUNCLOSE ) n--;
01609                     *fill++ = *t++;
01610                 }
01611             }
01612         }
01613         else if ( *t == TSET ) {

```

```

01614         *fill++ = *t++; n = 0;
01615         while ( *t >= 0 ) { n = 128*n + *t; *fill++ = *t++; }
01616         if ( *t == LBRACE ) {
01617             if ( n < AM.NumFixedSets || Sets[n].type == CRANGE ) {
01618                 MesPrint("&This type of usage of sets is not allowed");
01619                 error = 1;
01620             }
01621             *fill++ = *t++; n = 0;
01622             while ( n >= 0 ) {
01623                 if ( *t == LBRACE ) n++;
01624                 else if ( *t == RBRACE ) n--;
01625                 *fill++ = *t++;
01626             }
01627         }
01628     }
01629     else if ( *t == TEXPRESSION ) {
01630         *fill++ = *t++; while ( *t >= 0 ) *fill++ = *t++;
01631         if ( *t == TFUNOPEN ) {
01632             *fill++ = *t++; n = 0;
01633             while ( n >= 0 ) {
01634                 if ( *t == TFUNOPEN ) n++;
01635                 else if ( *t == TFUNCLOSE ) n--;
01636                 *fill++ = *t++;
01637             }
01638         }
01639         if ( *t == LBRACE ) {
01640             *fill++ = *t++; n = 0;
01641             while ( n >= 0 ) {
01642                 if ( *t == LBRACE ) n++;
01643                 else if ( *t == RBRACE ) n--;
01644                 *fill++ = *t++;
01645             }
01646         }
01647     }
01648     else if ( *t == TVECTOR ) {
01649         *fill++ = *t++; while ( *t >= 0 ) *fill++ = *t++;
01650         if ( *t == TFUNOPEN ) {
01651             *fill++ = *t++; n = 0;
01652             while ( n >= 0 ) {
01653                 if ( *t == TFUNOPEN ) n++;
01654                 else if ( *t == TFUNCLOSE ) n--;
01655                 *fill++ = *t++;
01656             }
01657         }
01658     }
01659     else if ( *t == TDOT ) {
01660         *fill++ = *t++;
01661         if ( *t == TVECTOR || *t == TDUBIOUS ) {
01662             *fill++ = *t++; while ( *t >= 0 ) *fill++ = *t++;
01663         }
01664         else if ( *t == TSET ) {
01665             *fill++ = *t++; num = 0;
01666             while ( *t >= 0 ) { num = 128*num + *t; *fill++ = *t++; }
01667             if ( Sets[num].type != CVECTOR ) {
01668                 MesPrint("&Illegal set type in dotproduct");
01669                 error = 1;
01670             }
01671             if ( *t == LBRACE ) {
01672                 *fill++ = *t++; n = 0;
01673                 while ( n >= 0 ) {
01674                     if ( *t == LBRACE ) n++;
01675                     else if ( *t == RBRACE ) n--;
01676                     *fill++ = *t++;
01677                 }
01678             }
01679         }
01680         else if ( *t == TSETNUM ) {
01681             *fill++ = *t++;
01682             while ( *t >= 0 ) { *fill++ = *t++; }
01683             *fill++ = *t++;
01684             while ( *t >= 0 ) { *fill++ = *t++; }
01685         }
01686     }
01687     else {
01688         MesPrint("&Illegal second element in dotproduct");
01689         error = 1;
01690     }
01691 }
01692 else {
01693     *fill++ = *t++; while ( *t >= 0 ) *fill++ = *t++;
01694     if ( *t == TWILDCARD ) *fill++ = *t++;
01695 }
01696 *fill++ = RPARENTHESIS; *fill++ = TFUNCLOSE;
01697 if ( *t == TPOWER ) goto doublepower;
01698 while ( fill > ff ) *--t = *--fill;
01699 s = t;
01700 }
01701 else if ( denom < 0 ) {

```

```

01701         fill = ff; ff[-1] = TMULTIPLY;
01702         *fill++ = TFUNCTION; *fill++ = (SBYTE) (AM.denomnum);
01703         *fill++ = TFUNOPEN; *fill++ = LPARENTHESIS;
01704         while ( ss < t ) *fill++ = *ss++;
01705         *fill++ = RPARENTHESIS; *fill++ = TFUNCLOSE;
01706         while ( fill > ff ) *--t = *--fill;
01707         s = t; denom = 1; sube = 0;
01708         break;
01709     }
01710     sube = 0;
01711     break;
01712 case TEXPRESSION:
01713     *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
01714     t = s;
01715     if ( *t == TFUNOPEN ) {
01716         t++; n = 1;
01717         for(;;) {
01718             if ( *t == TFUNOPEN ) n++;
01719             else if ( *t == TFUNCLOSE ) { if ( --n <= 0 ) break; }
01720             t++;
01721         }
01722         t++;
01723     }
01724     if ( *t == LBRACE ) {
01725         t++; n = 1;
01726         for(;;) {
01727             if ( *t == LBRACE ) n++;
01728             else if ( *t == RBRACE ) { if ( --n <= 0 ) break; }
01729             t++;
01730         }
01731         t++;
01732     }
01733     if ( t > s || denom < 0 || ( ( *t == TPOWER || *t == TPOWER1 )
01734         && t[1] == TMINUS ) ) goto dofunpower;
01735     else goto dosymbol;
01736 case TDOLLAR:
01737     *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
01738     goto dosymbol;
01739 case LPARENTHESIS:
01740     *fill++ = *s++; n = 1; t = s;
01741     for(;;) {
01742         if ( *t == LPARENTHESIS ) n++;
01743         else if ( *t == RPARENTHESIS ) { if ( --n <= 0 ) break; }
01744         t++;
01745     }
01746     t++; sube = 1;
01747     goto dofunpower;
01748 case TSET:
01749     *fill++ = *s++; n = *s++; *fill++ = (SBYTE)n;
01750     while ( *s >= 0 ) { *fill++ = *s; n = 128*n + *s++; }
01751     n = Sets[n].type;
01752     switch ( n ) {
01753         case CSYMBOL: goto dosymbol;
01754         case CINDEX: break;
01755         case CVECTOR: goto dovector;
01756         case CFUNCTION: goto dofunction;
01757         case CNUMBER: goto dosymbol;
01758         case CMODEL: break;
01759         case ANYTYPE: break;
01760         default: error = 1; break;
01761     }
01762     break;
01763 case TDOT:
01764     *fill++ = *s++;
01765     if ( *s == TVECTOR ) {
01766         *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
01767     }
01768     else if ( *s == TSET ) {
01769         *fill++ = *s++; n = *s++; *fill++ = (SBYTE)n;
01770         while ( *s >= 0 ) { *fill++ = *s; n = 128*n + *s++; }
01771         if ( *s == LBRACE ) {
01772             if ( n < AM.NumFixedSets || Sets[n].type == CRANGE ) {
01773                 MesPrint("&This type of usage of sets is not allowed");
01774                 error = 1;
01775             }
01776             *fill++ = *s++; n = 1;
01777             for(;;) {
01778                 if ( *s == LBRACE ) n++;
01779                 else if ( *s == RBRACE ) { if ( --n <= 0 ) break; }
01780                 *fill++ = *s++;
01781             }
01782             *fill++ = *s++;
01783         }
01784         else {
01785             MesPrint("&Set without argument in dotproduct");
01786             error = 1;
01787         }
01788     }

```



```

01788         }
01789         else if ( *s == TSETNUM ) {
01790             *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
01791             if ( *s != TVECTOR ) goto nodot;
01792             *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
01793         }
01794         else {
01795     nodot:      MesPrint("&Illegal second element in dotproduct");
01796             error = 1;
01797             *fill++ = *s++;
01798             while ( *s >= 0 ) *fill++ = *s++;
01799         }
01800         dotp = 1;
01801         goto dosymbol;
01802     default:
01803         *fill++ = *s++;
01804         while ( *s >= 0 ) *fill++ = *s++;
01805         break;
01806     }
01807 }
01808 *fill = TENDOFIT;
01809 return(error);
01810 doublepower:;
01811 MesPrint("&Dubious notation with power of power");
01812 return(-1);
01813 }
01814
01815 /*
01816     #] simp3btoken :
01817     #[ simp4token :
01818
01819     Deal with the set[n] objects in the RHS.
01820 */
01821
01822 int simp4token(SBYTE *s)
01823 {
01824     int error = 0, n, nsym, settype;
01825     WORD i, *w, *wstop, level;
01826     SBYTE *const s0 = s;
01827     SBYTE *fill = s, *s1, *s2, *s3, type, slbuf[10];
01828     SBYTE *tbuf = s, *t, *t1;
01829
01830     while ( *s != TENDOFIT ) {
01831         if ( *s != TSET ) {
01832             if ( *s == TEMPTY ) s++;
01833             else *fill++ = *s++;
01834             continue;
01835         }
01836         if ( fill >= (s0+1) && fill[-1] == TWILDCARD ) { *fill++ = *s++; continue; }
01837         if ( fill >= (s0+2) && fill[-1] == TNOT && fill[-2] == TWILDCARD ) { *fill++ = *s++; continue; }
01838
01839         s1 = s++; n = 0; while ( *s >= 0 ) { n = 128*n + *s++; }
01840         i = Sets[n].type;
01841         if ( *s != LBRACE ) { while ( s1 < s ) *fill++ = *s1++; continue; }
01842         if ( n < AM.NumFixedSets || i == CRANGE ) {
01843             MesPrint("&It is not allowed to refer to individual elements of built in or ranged sets");
01844             error = 1;
01845         }
01846         s++;
01847         if ( *s != TSYMBOL && *s != TDOLLAR ) {
01848             MesPrint("&Set index in RHS is not a wildcard symbol or $-variable");
01849             error = 1;
01850             while ( s1 < s ) *fill++ = *s1++;
01851             continue;
01852         }
01853         settype = ( *s == TDOLLAR );
01854         s++; nsym = 0; s2 = s;
01855         while ( *s >= 0 ) nsym = 128*nsym + *s++;
01856         if ( *s != RBRACE ) {
01857             MesPrint("&Improper set argument in RHS");
01858             error = 1;
01859             while ( s1 < s ) *fill++ = *s1++;
01860             continue;
01861         }
01862         s++;
01863         /*
01864         Verify that nsym is a wildcard
01865         */
01866         if ( !settype ) {
01867             w = AC.ProtoType; wstop = w + w[1]; w += SUBEXPSIZE;
01868             while ( w < wstop ) {
01869                 if ( *w == SYMTOSYM && w[2] == nsym ) break;
01870                 w += w[1];
01871             }
01872             if ( w >= wstop ) {
01873                 It could still be a summation parameter!

```

```

01874 */
01875         t = fill - 1;
01876         while ( t >= tbuf ) {
01877             if ( *t == TFUNCLOSE ) {
01878                 level = 1; t--;
01879                 while ( t >= tbuf ) {
01880                     if ( *t == TFUNCLOSE ) level++;
01881                     else if ( *t == TFUNOPEN ) {
01882                         level--;
01883                         if ( level == 0 ) break;
01884                     }
01885                     t--;
01886                 }
01887             }
01888             else if ( *t == RBRACE ) {
01889                 level = 1; t--;
01890                 while ( t >= tbuf ) {
01891                     if ( *t == RBRACE ) level++;
01892                     else if ( *t == LBRACE ) {
01893                         level--;
01894                         if ( level == 0 ) break;
01895                     }
01896                     t--;
01897                 }
01898             }
01899             else if ( *t == RPARENTHESIS ) {
01900                 level = 1; t--;
01901                 while ( t >= tbuf ) {
01902                     if ( *t == RPARENTHESIS ) level++;
01903                     else if ( *t == LPARENTHESIS ) {
01904                         level--;
01905                         if ( level == 0 ) break;
01906                     }
01907                     t--;
01908                 }
01909             }
01910             else if ( *t == TFUNOPEN ) {
01911                 t1 = t-1;
01912                 while ( *t1 > 0 && t1 > tbuf ) t1--;
01913                 if ( *t1 == TFUNCTION ) {
01914                     t1++; level = 0;
01915                     while ( *t1 > 0 ) level = level*128+*t1++;
01916                     if ( level == (SUMF1-FUNCTION)
01917                         || level == (SUMF2-FUNCTION) ) {
01918                         t1 = t + 1;
01919                         if ( *t1 == LPARENTHESIS ) t1++;
01920                         if ( *t1 == TSYMBOL ) {
01921                             if ( ( t1[1] == COEFFSYMBOL
01922                                 || t1[1] == NUMERATORSYMBOL
01923                                 || t1[1] == DENOMINATORSYMBOL )
01924                                 && t1[2] < 0 ) {}
01925                             else {
01926                                 t1++; level = 0;
01927                                 while ( *t1 >= 0 && t1 < fill ) level = 128*level + *t1++;
01928                                 if ( level == nsym && t1 < fill ) {
01929                                     if ( t[1] == LPARENTHESIS
01930                                         && *t1 == RPARENTHESIS && t1[1] == TCOMMA ) break;
01931                                     if ( t[1] != LPARENTHESIS && *t1 == TCOMMA ) break;
01932                                 }
01933                             }
01934                         }
01935                     }
01936                 }
01937             }
01938             t--;
01939         }
01940         if ( t < tbuf ) {
01941             fill--;
01942             MesPrint("&Set index in RHS is not a wildcard symbol");
01943             error = 1;
01944             while ( s1 < s ) *fill++ = *s1++;
01945             continue;
01946         }
01947     }
01948 }
01949 /*
01950 Now replace by a set marker: TSETNUM,nsym,TYPE,setnumber
01951 */
01952 switch ( i ) {
01953     case CSYMBOL: type = TSYMBOL; break;
01954     case CINDEX: type = TINDEX; break;
01955     case CVECTOR: type = TVECTOR; break;
01956     case CFUNCTION: type = TFUNCTION; break;
01957     case CNUMBER: type = TNUMBER1; break;
01958     case CDUBIOUS: type = TDUBIOUS; break;
01959     default:
01960         MesPrint("&Unknown set type in simp4token");

```

```

01961         error = 1; type = CDUBIOUS; break;
01962     }
01963     s3 = slbuf; s1++;
01964     while ( *s1 >= 0 ) *s3++ = *s1++;
01965     *s3 = -1; s1 = slbuf;
01966     if ( settype ) *fill++ = TSETDOL;
01967     else           *fill++ = TSETNUM;
01968     while ( *s2 >= 0 ) *fill++ = *s2++;
01969     *fill++ = type; while ( *s1 >= 0 ) *fill++ = *s1++;
01970 }
01971 *fill++ = TENDOFIT;
01972 return(error);
01973 }
01974
01975 /*
01976     #] simp4token :
01977     #[ simp5token :
01978
01979     Making sure that first argument of sumfunction is not a wildcard already
01980 */
01981
01982 int simp5token(SBYTE *s, int mode)
01983 {
01984     int error = 0, n, type;
01985     WORD *w, *wstop;
01986     if ( mode == RHSIDE ) {
01987         while ( *s != TENDOFIT ) {
01988             if ( *s == TFUNCTION ) {
01989                 s++; n = 0; while ( *s >= 0 ) n = 128*n + *s++;
01990                 if ( n == AM.sumnum || n == AM.sumpnum ) {
01991                     if ( *s != TFUNOPEN ) continue;
01992                     s++;
01993                     if ( *s != TSYMBOL && *s != TINDEX ) continue;
01994                     type = *s++;
01995                     n = 0; while ( *s >= 0 ) n = 128*n + *s++;
01996                     if ( type == TINDEX ) n += AM.OffsetIndex;
01997                     if ( *s != TCOMMA ) continue;
01998                     w = AC.ProtoType;
01999                     wstop = w + w[1];
02000                     w += SUBEXPSIZE;
02001                     while ( w < wstop ) {
02002                         if ( w[2] == n ) {
02003                             if ( ( type == TSYMBOL && ( w[0] == SYMTOSYM
02004                                 || w[0] == SYMTONUM || w[0] == SYMTOSUB ) ) || (
02005                                 type == TINDEX && ( w[0] == INDTOIND
02006                                 || w[0] == INDTOSUB ) ) ) {
02007                                 error = 1;
02008                                 MesPrint("&Parameter of sum function is already a wildcard");
02009                             }
02010                         }
02011                         w += w[1];
02012                     }
02013                 }
02014             }
02015             else s++;
02016         }
02017     }
02018     return(error);
02019 }
02020
02021 /*
02022     #] simp5token :
02023     #[ simp6token :
02024
02025     Making sure that factorized expressions are used properly
02026 */
02027
02028 int simp6token(SBYTE *tokens, int mode)
02029 {
02030     /* EXPRESSIONS e = Expressions; */
02031     int error = 0, n;
02032     int level = 0, haveone = 0;
02033     SBYTE *s = tokens, *ss;
02034     LONG numterms;
02035     WORD funnum = 0;
02036     GETIDENTITY
02037     if ( mode == RHSIDE ) {
02038         while ( *s == TPLUS || *s == TMINUS ) s++;
02039         numterms = 1;
02040         while ( *s != TENDOFIT ) {
02041             if ( *s == LPARENTHESIS ) level++;
02042             else if ( *s == RPARENTHESIS ) level--;
02043             else if ( *s == TFUNOPEN ) level++;
02044             else if ( *s == TFUNCLOSE ) level--;
02045             else if ( ( *s == TPLUS || *s == TMINUS ) && level == 0 ) {
02046                 /*
02047                     Special exception: x^-1 etc.

```

```

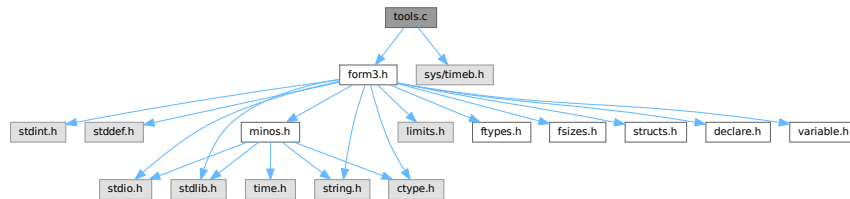
02048 */
02049         if ( s[-1] != TPOWER && s[-1] != TPLUS && s[-1] != TMINUS ) {
02050             numterms++;
02051         }
02052     }
02053     else if ( *s == TEXPRESSON ) {
02054         ss = s;
02055         s++; n = 0; while ( *s >= 0 ) n = 128*n + *s++;
02056
02057         if ( Expressions[n].status == STOREDEXPRESSION ) {
02058             POSITION position;
02059             /*
02060             #ifdef WITHPTHREADS
02061                 RENUMBER renumber;
02062             #endif
02063             */
02064             RENUMBER renumber;
02065
02066             WORD TMproto[SUBEXPSIZE];
02067             TMproto[0] = EXPRESSION;
02068             TMproto[1] = SUBEXPSIZE;
02069             TMproto[2] = n;
02070             TMproto[3] = 1;
02071             { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
02072             AT.TMaddr = TMproto;
02073             PUTZERO(position);
02074             /*
02075             if ( (
02076             #ifdef WITHPTHREADS
02077                 renumber =
02078             #endif
02079                 GetTable(n,&position,0) ) == 0 )
02080             */
02081             if ( ( renumber = GetTable(n,&position,0) ) == 0 )
02082             {
02083                 error = 1;
02084                 MesPrint("&Problems getting information about stored expression %s(4)"
02085                     ,EXPRNAME(n));
02086             }
02087             /*
02088             #ifdef WITHPTHREADS
02089             */
02090             if ( renumber->sympb.lo != AN.dummyrenumlist )
02091                 M_free(renumber->sympb.lo,"VarSpace");
02092             M_free(renumber,"Renumber");
02093             /*
02094             #endif
02095             */
02096         }
02097
02098         if ( ( ( AS.Oldvflags[n] & ISFACTORIZED ) != 0 ) && *s != LBRACE ) {
02099             if ( level == 0 ) {
02100                 haveone = 1;
02101             }
02102             else if ( error == 0 ) {
02103                 if ( ss[-1] != TFUNOPEN || funnum != NUMFACTORS-FUNCTION ) {
02104                     MesPrint("&Illegal use of factorized expression(s) in RHS");
02105                     error = 1;
02106                 }
02107             }
02108         }
02109         continue;
02110     }
02111     else if ( *s == TFUNCTION ) {
02112         s++; funnum = 0; while ( *s >= 0 ) funnum = 128*fnum + *s++;
02113         continue;
02114     }
02115     s++;
02116 }
02117 if ( haveone ) {
02118     if ( numterms > 1 ) {
02119         MesPrint("&Factorized expression in RHS in an expression of more than one term.");
02120         error = 1;
02121     }
02122     else if ( AC.ToBeInFactors == 0 ) {
02123         MesPrint("&Attempt to put a factorized expression inside an unfactorized
02124 expression.");
02125         error = 1;
02126     }
02127 }
02128 return(error);
02129 }
02130
02131 /*
02132     #] simp6token :
02133     #] Compiler :

```

02134 \*/

## 4.146 tools.c File Reference

```
#include "form3.h"
#include <sys/timeb.h>
Include dependency graph for tools.c:
```



### Macros

- #define [FILLVALUE](#) 0
- #define [COPYFILEBUFSIZE](#) 40960L
- #define [TERMMEMSTARTNUM](#) 16
- #define [TERMEXTRAWORDS](#) 10
- #define [NUMBERMEMSTARTNUM](#) 16
- #define [NUMBEREXTRAWORDS](#) 10L
- #define [DODOUBLE](#)(x)
- #define [DOEXPAND](#)(x)

### Functions

- UBYTE \* [LoadInputFile](#) (UBYTE \*filename, int type)
- UBYTE [ReadFromStream](#) (STREAM \*stream)
- UBYTE [GetFromStream](#) (STREAM \*stream)
- UBYTE [LookInStream](#) (STREAM \*stream)
- STREAM \* [OpenStream](#) (UBYTE \*name, int type, int prevarmode, int raiselow)
- int [LocateFile](#) (UBYTE \*\*name, int type)
- STREAM \* [CloseStream](#) (STREAM \*stream)
- STREAM \* [CreateStream](#) (UBYTE \*where)
- LONG [GetStreamPosition](#) (STREAM \*stream)
- void [PositionStream](#) (STREAM \*stream, LONG position)
- int [ReverseStatements](#) (STREAM \*stream)
- void [StartFiles](#) (void)
- int [OpenFile](#) (char \*name)
- int [OpenAddFile](#) (char \*name)
- int [ReOpenFile](#) (char \*name)
- int [CreateFile](#) (char \*name)
- int [CreateLogFile](#) (char \*name)
- void [CloseFile](#) (int handle)
- int [CopyFile](#) (char \*source, char \*dest)
- int [CreateHandle](#) (void)

- LONG [ReadFile](#) (int handle, UBYTE \*buffer, LONG size)
- LONG [ReadPosFile](#) (PHEAD [FILEHANDLE](#) \*fi, UBYTE \*buffer, LONG size, [POSITION](#) \*pos)
- LONG [WriteFileToFile](#) (int handle, UBYTE \*buffer, LONG size)
- void [SeekFile](#) (int handle, [POSITION](#) \*offset, int origin)
- LONG [TellFile](#) (int handle)
- void [TELLFILE](#) (int handle, [POSITION](#) \*position)
- void [FlushFile](#) (int handle)
- int [GetPosFile](#) (int handle, fpos\_t \*pospointer)
- int [SetPosFile](#) (int handle, fpos\_t \*pospointer)
- void [SynchFile](#) (int handle)
- void [TruncateFile](#) (int handle)
- int [GetChannel](#) (char \*name, int mode)
- int [GetAppendChannel](#) (char \*name)
- int [CloseChannel](#) (char \*name)
- void [UpdateMaxSize](#) (void)
- int [StrCmp](#) (UBYTE \*s1, UBYTE \*s2)
- int [StrlCmp](#) (UBYTE \*s1, UBYTE \*s2)
- int [StrHlCmp](#) (UBYTE \*s1, UBYTE \*s2)
- int [StrlCont](#) (UBYTE \*s1, UBYTE \*s2)
- int [CmpArray](#) (WORD \*t1, WORD \*t2, WORD n)
- int [ConWord](#) (UBYTE \*s1, UBYTE \*s2)
- int [StrLen](#) (UBYTE \*s)
- void [NumToStr](#) (UBYTE \*s, LONG x)
- void [WriteString](#) (int type, UBYTE \*str, int num)
- void [WriteUnfinString](#) (int type, UBYTE \*str, int num)
- UBYTE \* [AddToString](#) (UBYTE \*outstring, UBYTE \*extrastring, int par)
- UBYTE \* [strDup1](#) (UBYTE \*instring, char \*ifwrong)
- UBYTE \* [EndOfToken](#) (UBYTE \*s)
- UBYTE \* [ToToken](#) (UBYTE \*s)
- UBYTE \* [SkipField](#) (UBYTE \*s, int level)
- WORD [ReadSnum](#) (UBYTE \*\*p)
- UBYTE \* [NumCopy](#) (WORD y, UBYTE \*to)
- char \* [LongCopy](#) (LONG y, char \*to)
- char \* [LongLongCopy](#) (off\_t \*y, char \*to)
- UBYTE \* [MakeDate](#) (void)
- int [set\\_in](#) (UBYTE ch, [set\\_of\\_char](#) set)
- [one\\_byte](#) [set\\_set](#) (UBYTE ch, [set\\_of\\_char](#) set)
- [one\\_byte](#) [set\\_del](#) (UBYTE ch, [set\\_of\\_char](#) set)
- [one\\_byte](#) [set\\_sub](#) ([set\\_of\\_char](#) set, [set\\_of\\_char](#) set1, [set\\_of\\_char](#) set2)
- void [iniTools](#) (void)
- void \* [Malloc1](#) (LONG size, const char \*messageifwrong)
- void [M\\_free](#) (void \*x, const char \*where)
- void [M\\_check1](#) (void)
- void [M\\_print](#) (void)
- void [TermMallocAddMemory](#) (PHEAD0)
- void [NumberMallocAddMemory](#) (PHEAD0)
- void [CacheNumberMallocAddMemory](#) (PHEAD0)
- void \* [FromList](#) ([LIST](#) \*L)
- void \* [From0List](#) ([LIST](#) \*L)
- void \* [FromVarList](#) ([LIST](#) \*L)
- int [DoubleList](#) (void \*\*\*lijst, int \*oldsize, int objectsize, char \*nameoftype)
- int [DoubleLList](#) (void \*\*\*lijst, LONG \*oldsize, int objectsize, char \*nameoftype)
- void [DoubleBuffer](#) (void \*\*start, void \*\*stop, int size, char \*text)
- void [ExpandBuffer](#) (void \*\*buffer, LONG \*oldsize, int type)
- LONG [iexp](#) (LONG x, int p)

- void [ToGeneral](#) (WORD \*r, WORD \*m, WORD par)
- int [ToFast](#) (WORD \*r, WORD \*m)
- WORD [ToPolyFunGeneral](#) (PHEAD WORD \*term)
- int [IsLikeVector](#) (WORD \*arg)
- int [AreArgsEqual](#) (WORD \*arg1, WORD \*arg2)
- int [CompareArgs](#) (WORD \*arg1, WORD \*arg2)
- int [CompArg](#) (WORD \*s1, WORD \*s2)
- LONG [TimeWallClock](#) (WORD par)
- LONG [TimeChildren](#) (WORD par)
- LONG [TimeCPU](#) (WORD par)
- int [Crash](#) (void)
- int [TestTerm](#) (WORD \*term)
- int [DistrN](#) (int n, int \*cpl, int ncpl, int \*scratch)

## Variables

- FILES \*\* [filelist](#)
- int [numinfilelist](#) = 0
- int [filelistsize](#) = 0
- WRITEFILE [WriteFile](#) = &WriteFileToFile

## 4.146.1 Detailed Description

Low level routines for many types of task. There are routines for manipulating the input system (streams and files) routines for string manipulation, the memory allocation interface, and the clock. The last is the most sensitive to ports. In the past nearly every port to another OS or computer gave trouble. Nowadays it is slightly better but the poor POSIX compliance of LINUX again gave problems for the multithreaded version.

Definition in file [tools.c](#).

## 4.146.2 Macro Definition Documentation

### 4.146.2.1 FILLVALUE

```
#define FILLVALUE 0
```

Definition at line 49 of file [tools.c](#).

### 4.146.2.2 TERMMEMSTARTNUM

```
#define TERMMEMSTARTNUM 16
```

Provides memory for one term (or one small polynomial) This means that the memory is limited to a buffer of size AM.MaxTer plus a few extra words. In parallel versions, each worker has its own memory pool.

The way we use the memory is by: term = TermMalloc(BHEAD0); and later we free it by TermFree(BHEAD term);

Layout: We have a list of available pointers to buffers: AT.TermMemHeap Its size is AT.TermMemMax We take from the top (indicated by AT.TermMemTop). When we run out of buffers we assign new ones (doubling the amount) and we have to extend the AT.TermMemHeap array. Important: There is no checking that the returned memory is legal, ie is memory that was handed out earlier.

Definition at line 2466 of file [tools.c](#).

#### 4.146.2.3 TERMEXTRAWORDS

```
#define TERMEXTRAWORDS 10
```

Definition at line 2467 of file [tools.c](#).

#### 4.146.2.4 NUMBERMEMSTARTNUM

```
#define NUMBERMEMSTARTNUM 16
```

Provides memory for one Long number This means that the memory is limited to a buffer of size AM.MaxTal In parallel versions, each worker has its own memory pool.

The way we use the memory is by: num = NumberMalloc(BHEAD0); Number = AT.NumberMemHeap[num]; and later we free it by NumberFree(BHEAD num);

Layout: We have a list of available pointers to buffers: AT.NumberMemHeap Its size is AT.NumberMemMax We take from the top (indicated by AT.NumberMemTop). When we run out of buffers we assign new ones (doubling the amount) and we have to extend the AT.NumberMemHeap array. Important: There is no checking on the returned memory!!!!

Definition at line 2566 of file [tools.c](#).

#### 4.146.2.5 NUMBEREXTRAWORDS

```
#define NUMBEREXTRAWORDS 10L
```

Definition at line 2567 of file [tools.c](#).

#### 4.146.2.6 DODOUBLE

```
#define DODOUBLE(  
    x )
```

**Value:**

```
{ x *s, *t, *u; if ( *start ) { \
    oldsize = *(x **)stop - *(x **)start; newsize = 2*oldsize; \
    t = u = (x *)Mallc1(newsize*sizeof(x),text); s = *(x **)start; \
    for ( i = 0; i < oldsize; i++ ) { *t++ = *s++; } M_free(*start,"double"); } \
    else { newsize = 100; u = (x *)Mallc1(newsize*sizeof(x),text); } \
    *start = (void *)u; *stop = (void *) (u+newsize); }
```

Definition at line 2889 of file [tools.c](#).

#### 4.146.2.7 DOEXPAND

```
#define DOEXPAND(  
    x )
```

**Value:**

```
{ x *newbuffer, *t, *m; \
    t = newbuffer = (x *)Mallc1((newsize+2)*type,"ExpandBuffer"); \
    if ( *buffer ) { m = (x *)*buffer; i = *oldsize; \
        while ( --i >= 0 ) { *t++ = *m++; } M_free(*buffer,"ExpandBuffer"); \
    } *buffer = newbuffer; *oldsize = newsize; }
```

Definition at line 2915 of file [tools.c](#).



## 4.146.3 Function Documentation

### 4.146.3.1 LoadInputFile()

```
UBYTE * LoadInputFile (
    UBYTE * filename,
    int type )
```

Definition at line 112 of file [tools.c](#).

### 4.146.3.2 ReadFromStream()

```
UBYTE ReadFromStream (
    STREAM * stream )
```

Definition at line 151 of file [tools.c](#).

### 4.146.3.3 GetFromStream()

```
UBYTE GetFromStream (
    STREAM * stream )
```

Definition at line 286 of file [tools.c](#).

### 4.146.3.4 LookInStream()

```
UBYTE LookInStream (
    STREAM * stream )
```

Definition at line 309 of file [tools.c](#).

### 4.146.3.5 OpenStream()

```
STREAM * OpenStream (
    UBYTE * name,
    int type,
    int prevarmode,
    int raiselow )
```

Definition at line 321 of file [tools.c](#).

### 4.146.3.6 LocateFile()

```
int LocateFile (
    UBYTE ** name,
    int type )
```

Definition at line 576 of file [tools.c](#).

#### 4.146.3.7 CloseStream()

```
STREAM * CloseStream (
    STREAM * stream )
```

Definition at line 663 of file [tools.c](#).

#### 4.146.3.8 CreateStream()

```
STREAM * CreateStream (
    UBYTE * where )
```

Definition at line 782 of file [tools.c](#).

#### 4.146.3.9 GetStreamPosition()

```
LONG GetStreamPosition (
    STREAM * stream )
```

Definition at line 813 of file [tools.c](#).

#### 4.146.3.10 PositionStream()

```
void PositionStream (
    STREAM * stream,
    LONG position )
```

Definition at line 823 of file [tools.c](#).

#### 4.146.3.11 ReverseStatements()

```
int ReverseStatements (
    STREAM * stream )
```

Definition at line 855 of file [tools.c](#).

#### 4.146.3.12 StartFiles()

```
void StartFiles (
    void )
```

Definition at line 956 of file [tools.c](#).

#### 4.146.3.13 OpenFile()

```
int OpenFile (
    char * name )
```

Definition at line 983 of file [tools.c](#).

#### 4.146.3.14 OpenAddFile()

```
int OpenAddFile (
    char * name )
```

Definition at line 1002 of file [tools.c](#).

#### 4.146.3.15 ReOpenFile()

```
int ReOpenFile (
    char * name )
```

Definition at line 1023 of file [tools.c](#).

#### 4.146.3.16 CreateFile()

```
int CreateFile (
    char * name )
```

Definition at line 1043 of file [tools.c](#).

#### 4.146.3.17 CreateLogFile()

```
int CreateLogFile (
    char * name )
```

Definition at line 1060 of file [tools.c](#).

#### 4.146.3.18 CloseFile()

```
void CloseFile (
    int handle )
```

Definition at line 1078 of file [tools.c](#).

#### 4.146.3.19 CopyFile()

```
int CopyFile (
    char * source,
    char * dest )
```

Copies a file with name \*source to a file named \*dest. The involved files must not be open. Returns non-zero if an error occurred. Uses if possible the combined large and small sorting buffers as cache.

Definition at line 1101 of file [tools.c](#).

Referenced by [DoRecovery\(\)](#).

#### 4.146.3.20 CreateHandle()

```
int CreateHandle (
    void )
```

Definition at line 1162 of file [tools.c](#).

#### 4.146.3.21 ReadFile()

```
LONG ReadFile (
    int handle,
    UBYTE * buffer,
    LONG size )
```

Definition at line 1217 of file [tools.c](#).

#### 4.146.3.22 ReadPosFile()

```
LONG ReadPosFile (
    PHEAD FILEHANDLE * fi,
    UBYTE * buffer,
    LONG size,
    POSITION * pos )
```

Definition at line 1267 of file [tools.c](#).

#### 4.146.3.23 WriteFileToFile()

```
LONG WriteFileToFile (
    int handle,
    UBYTE * buffer,
    LONG size )
```

Definition at line 1333 of file [tools.c](#).

#### 4.146.3.24 SeekFile()

```
void SeekFile (
    int handle,
    POSITION * offset,
    int origin )
```

Definition at line 1371 of file [tools.c](#).

#### 4.146.3.25 TellFile()

```
LONG TellFile (
    int handle )
```

Definition at line 1396 of file [tools.c](#).

#### 4.146.3.26 TELLFILE()

```
void TELLFILE (
    int handle,
    POSITION * position )
```

Definition at line 1406 of file [tools.c](#).

#### 4.146.3.27 FlushFile()

```
void FlushFile (
    int handle )
```

Definition at line 1420 of file [tools.c](#).

#### 4.146.3.28 GetPosFile()

```
int GetPosFile (
    int handle,
    fpos_t * pospointer )
```

Definition at line 1434 of file [tools.c](#).

#### 4.146.3.29 SetPosFile()

```
int SetPosFile (
    int handle,
    fpos_t * pospointer )
```

Definition at line 1448 of file [tools.c](#).

#### 4.146.3.30 SynchFile()

```
void SynchFile (
    int handle )
```

Definition at line 1468 of file [tools.c](#).

#### 4.146.3.31 TruncateFile()

```
void TruncateFile (
    int handle )
```

Definition at line 1490 of file [tools.c](#).

**4.146.3.32 GetChannel()**

```
int GetChannel (
    char * name,
    int mode )
```

Definition at line [1510](#) of file [tools.c](#).

**4.146.3.33 GetAppendChannel()**

```
int GetAppendChannel (
    char * name )
```

Definition at line [1546](#) of file [tools.c](#).

**4.146.3.34 CloseChannel()**

```
int CloseChannel (
    char * name )
```

Definition at line [1576](#) of file [tools.c](#).

**4.146.3.35 UpdateMaxSize()**

```
void UpdateMaxSize (
    void )
```

Definition at line [1610](#) of file [tools.c](#).

**4.146.3.36 StrCmp()**

```
int StrCmp (
    UBYTE * s1,
    UBYTE * s2 )
```

Definition at line [1700](#) of file [tools.c](#).

**4.146.3.37 StrICmp()**

```
int StrICmp (
    UBYTE * s1,
    UBYTE * s2 )
```

Definition at line [1711](#) of file [tools.c](#).

**4.146.3.38 StrHICmp()**

```
int StrHICmp (
    UBYTE * s1,
    UBYTE * s2 )
```

Definition at line [1722](#) of file [tools.c](#).

**4.146.3.39 StrICont()**

```
int StrICont (
    UBYTE * s1,
    UBYTE * s2 )
```

Definition at line [1733](#) of file [tools.c](#).

**4.146.3.40 CmpArray()**

```
int CmpArray (
    WORD * t1,
    WORD * t2,
    WORD n )
```

Definition at line [1745](#) of file [tools.c](#).

**4.146.3.41 ConWord()**

```
int ConWord (
    UBYTE * s1,
    UBYTE * s2 )
```

Definition at line [1759](#) of file [tools.c](#).

**4.146.3.42 StrLen()**

```
int StrLen (
    UBYTE * s )
```

Definition at line [1771](#) of file [tools.c](#).

**4.146.3.43 NumToStr()**

```
void NumToStr (
    UBYTE * s,
    LONG x )
```

Definition at line [1783](#) of file [tools.c](#).

#### 4.146.3.44 WriteString()

```
void WriteString (
    int type,
    UBYTE * str,
    int num )
```

Definition at line [1807](#) of file [tools.c](#).

#### 4.146.3.45 WriteUnfinString()

```
void WriteUnfinString (
    int type,
    UBYTE * str,
    int num )
```

Definition at line [1841](#) of file [tools.c](#).

#### 4.146.3.46 AddToString()

```
UBYTE * AddToString (
    UBYTE * outstring,
    UBYTE * extrastring,
    int par )
```

Definition at line [1866](#) of file [tools.c](#).

#### 4.146.3.47 strDup1()

```
UBYTE * strDup1 (
    UBYTE * instring,
    char * ifwrong )
```

Definition at line [1904](#) of file [tools.c](#).

#### 4.146.3.48 EndOfToken()

```
UBYTE * EndOfToken (
    UBYTE * s )
```

Definition at line [1919](#) of file [tools.c](#).

#### 4.146.3.49 ToToken()

```
UBYTE * ToToken (
    UBYTE * s )
```

Definition at line [1931](#) of file [tools.c](#).



#### 4.146.3.50 SkipField()

```
UBYTE * SkipField (
    UBYTE * s,
    int level )
```

Definition at line 1946 of file [tools.c](#).

#### 4.146.3.51 ReadSnum()

```
WORD ReadSnum (
    UBYTE ** p )
```

Definition at line 1973 of file [tools.c](#).

#### 4.146.3.52 NumCopy()

```
UBYTE * NumCopy (
    WORD y,
    UBYTE * to )
```

Definition at line 1997 of file [tools.c](#).

#### 4.146.3.53 LongCopy()

```
char * LongCopy (
    LONG y,
    char * to )
```

Definition at line 2024 of file [tools.c](#).

#### 4.146.3.54 LongLongCopy()

```
char * LongLongCopy (
    off_t * y,
    char * to )
```

Definition at line 2051 of file [tools.c](#).

#### 4.146.3.55 MakeDate()

```
UBYTE * MakeDate (
    void )
```

Definition at line 2090 of file [tools.c](#).

#### 4.146.3.56 set\_in()

```
int set_in (
    UBYTE ch,
    set_of_char set )
```

Definition at line 2112 of file [tools.c](#).

#### 4.146.3.57 set\_set()

```
one_byte set_set (
    UBYTE ch,
    set_of_char set )
```

Definition at line 2132 of file [tools.c](#).

#### 4.146.3.58 set\_del()

```
one_byte set_del (
    UBYTE ch,
    set_of_char set )
```

Definition at line 2153 of file [tools.c](#).

#### 4.146.3.59 set\_sub()

```
one_byte set_sub (
    set_of_char set,
    set_of_char set1,
    set_of_char set2 )
```

Definition at line 2175 of file [tools.c](#).

#### 4.146.3.60 iniTools()

```
void iniTools (
    void )
```

Definition at line 2201 of file [tools.c](#).

#### 4.146.3.61 Malloc1()

```
void * Malloc1 (
    LONG size,
    const char * messageifwrong )
```

Definition at line 2222 of file [tools.c](#).

**4.146.3.62 M\_free()**

```
void M_free (
    void * x,
    const char * where )
```

Definition at line [2302](#) of file [tools.c](#).

**4.146.3.63 M\_check1()**

```
void M_check1 (
    void )
```

Definition at line [2435](#) of file [tools.c](#).

**4.146.3.64 M\_print()**

```
void M_print (
    void )
```

Definition at line [2436](#) of file [tools.c](#).

**4.146.3.65 TermMallocAddMemory()**

```
void TermMallocAddMemory (
    PHEAD0 )
```

Definition at line [2469](#) of file [tools.c](#).

**4.146.3.66 NumberMallocAddMemory()**

```
void NumberMallocAddMemory (
    PHEAD0 )
```

Definition at line [2573](#) of file [tools.c](#).

**4.146.3.67 CacheNumberMallocAddMemory()**

```
void CacheNumberMallocAddMemory (
    PHEAD0 )
```

Definition at line [2647](#) of file [tools.c](#).

**4.146.3.68 FromList()**

```
void * FromList (
    LIST * L )
```

Definition at line [2699](#) of file [tools.c](#).

#### 4.146.3.69 From0List()

```
void * From0List (
    LIST * L )
```

Definition at line 2725 of file [tools.c](#).

#### 4.146.3.70 FromVarList()

```
void * FromVarList (
    LIST * L )
```

Definition at line 2754 of file [tools.c](#).

#### 4.146.3.71 DoubleList()

```
int DoubleList (
    void *** lijst,
    int * oldsize,
    int objectsize,
    char * nameoftype )
```

Definition at line 2796 of file [tools.c](#).

#### 4.146.3.72 DoubleLList()

```
int DoubleLList (
    void *** lijst,
    LONG * oldsize,
    int objectsize,
    char * nameoftype )
```

Definition at line 2848 of file [tools.c](#).

#### 4.146.3.73 DoubleBuffer()

```
void DoubleBuffer (
    void ** start,
    void ** stop,
    int size,
    char * text )
```

Definition at line 2896 of file [tools.c](#).

#### 4.146.3.74 ExpandBuffer()

```
void ExpandBuffer (
    void ** buffer,
    LONG * oldsize,
    int type )
```

Definition at line 2921 of file [tools.c](#).

**4.146.3.75 iexp()**

```
LONG iexp (
    LONG x,
    int p )
```

Definition at line [2945](#) of file [tools.c](#).

**4.146.3.76 ToGeneral()**

```
void ToGeneral (
    WORD * r,
    WORD * m,
    WORD par )
```

Definition at line [2976](#) of file [tools.c](#).

**4.146.3.77 ToFast()**

```
int ToFast (
    WORD * r,
    WORD * m )
```

Definition at line [3020](#) of file [tools.c](#).

**4.146.3.78 ToPolyFunGeneral()**

```
WORD ToPolyFunGeneral (
    PHEAD WORD * term )
```

Definition at line [3073](#) of file [tools.c](#).

**4.146.3.79 IsLikeVector()**

```
int IsLikeVector (
    WORD * arg )
```

Definition at line [3148](#) of file [tools.c](#).

**4.146.3.80 AreArgsEqual()**

```
int AreArgsEqual (
    WORD * arg1,
    WORD * arg2 )
```

Definition at line [3176](#) of file [tools.c](#).

#### 4.146.3.81 CompareArgs()

```
int CompareArgs (
    WORD * arg1,
    WORD * arg2 )
```

Definition at line 3195 of file [tools.c](#).

#### 4.146.3.82 CompArg()

```
int CompArg (
    WORD * s1,
    WORD * s2 )
```

Definition at line 3226 of file [tools.c](#).

#### 4.146.3.83 TimeWallClock()

```
LONG TimeWallClock (
    WORD par )
```

Returns the wall-clock time.

##### Parameters

|            |                                                  |
|------------|--------------------------------------------------|
| <i>par</i> | If zero, the wall-clock time will be reset to 0. |
|------------|--------------------------------------------------|

##### Returns

The wall-clock time in centiseconds.

Definition at line 3396 of file [tools.c](#).

Referenced by [DoCheckpoint\(\)](#), [DoRecovery\(\)](#), [find\\_Horner\\_MCTS\(\)](#), [optimize\\_expression\\_given\\_Horner\(\)](#), [optimize\\_greedy\(\)](#), and [WriteStats\(\)](#).

#### 4.146.3.84 TimeChildren()

```
LONG TimeChildren (
    WORD par )
```

Definition at line 3452 of file [tools.c](#).

#### 4.146.3.85 TimeCPU()

```
LONG TimeCPU (
    WORD par )
```

Returns the CPU time.

## Parameters

|            |                                           |
|------------|-------------------------------------------|
| <i>par</i> | If zero, the CPU time will be reset to 0. |
|------------|-------------------------------------------|

## Returns

The CPU time in milliseconds.

Definition at line 3470 of file [tools.c](#).

Referenced by [PF\\_GetSlaveTimes\(\)](#), [PF\\_Processor\(\)](#), and [WriteStats\(\)](#).

#### 4.146.3.86 Crash()

```
int Crash (
    void )
```

Definition at line 3753 of file [tools.c](#).

#### 4.146.3.87 TestTerm()

```
int TestTerm (
    WORD * term )
```

Tests the consistency of the term. Returns 0 when the term is OK. Any nonzero value is trouble. In the current version the testing isn't 100% complete. For instance, we don't check the validity of the symbols nor do we check the range of their powers. Etc. This should be extended when the need is there.

## Parameters

|             |                       |
|-------------|-----------------------|
| <i>term</i> | the term to be tested |
|-------------|-----------------------|

Definition at line 3781 of file [tools.c](#).

References [TestTerm\(\)](#).

Referenced by [TestTerm\(\)](#).

Here is the call graph for this function:



#### 4.146.3.88 DistrN()

```
int DistrN (
    int n,
    int * cpl,
    int ncpl,
    int * scratch )
```

Definition at line [3967](#) of file [tools.c](#).

### 4.146.4 Variable Documentation

#### 4.146.4.1 filelist

```
FILES** filelist
```

Definition at line [65](#) of file [tools.c](#).

#### 4.146.4.2 numinfilelist

```
int numinfilelist = 0
```

Definition at line [66](#) of file [tools.c](#).

#### 4.146.4.3 filelistsize

```
int filelistsize = 0
```

Definition at line [67](#) of file [tools.c](#).

#### 4.146.4.4 WriteFile

```
WRITEFILE WriteFile = &WriteFileToFile
```

Definition at line [1357](#) of file [tools.c](#).



## 4.147 tools.c

[Go to the documentation of this file.](#)

```

00001
00011 /* #[ License : */
00012 /*
00013 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00014 *   When using this file you are requested to refer to the publication
00015 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00016 *   This is considered a matter of courtesy as the development was paid
00017 *   for by FOM the Dutch physics granting agency and we would like to
00018 *   be able to track its scientific use to convince FOM of its value
00019 *   for the community.
00020 *
00021 *   This file is part of FORM.
00022 *
00023 *   FORM is free software: you can redistribute it and/or modify it under the
00024 *   terms of the GNU General Public License as published by the Free Software
00025 *   Foundation, either version 3 of the License, or (at your option) any later
00026 *   version.
00027 *
00028 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00029 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00030 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00031 *   details.
00032 *
00033 *   You should have received a copy of the GNU General Public License along
00034 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00035 */
00036 /* #] License : */
00037 /*
00038 *   #[ Includes :
00039 *   Note: TERMMALLOCDEBUG tests part of the TermMalloc and NumberMalloc
00040 *   system. To work properly it needs MEMORYMACROS in declare.h
00041 *   not to be defined to make sure that all calls will be diverted
00042 *   to the routines here.
00043 #define TERMMALLOCDEBUG
00044 #define FILLVALUE 126
00045 #define MALLOCDEBUGOUTPUT
00046 #define MALLOCDEBUG 1
00047 */
00048 #ifndef FILLVALUE
00049     #define FILLVALUE 0
00050 #endif
00051
00052 /*
00053 *   The enhanced malloc debugger, see comments in the beginning of the
00054 *   file mallocprotect.h
00055 *   MALLOCPROTECT == -1 -- protect left side, used block is left-aligned.
00056 *   MALLOCPROTECT == 0 -- protect both sides, used block is left-aligned;
00057 *   MALLOCPROTECT == 1 -- protect both sides, used block is right-aligned;
00058 *   ATTENTION! The macro MALLOCPROTECT must be defined
00059 *   BEFORE #include mallocprotect.h
00060 #define MALLOCPROTECT 1
00061 */
00062
00063 #include "form3.h"
00064
00065 FILES **filelist;
00066 int numinfilelist = 0;
00067 int filelistsize = 0;
00068 #ifdef MALLOCDEBUG
00069 #define BANNER (4*sizeof(LONG))
00070 void *malloclist[60000];
00071 LONG mallocsizes[60000];
00072 char *mallocstrings[60000];
00073 int nummalloclist = 0;
00074 #endif
00075
00076 #ifdef GPP
00077 extern "C" getdtablesize();
00078 #endif
00079
00080 #ifdef WITHSTATS
00081 LONG numwrites = 0;
00082 LONG numreads = 0;
00083 LONG numseeks = 0;
00084 LONG nummallocs = 0;
00085 LONG numfrees = 0;
00086 #endif
00087
00088 #ifdef MALLOCPROTECT
00089 #ifdef TRAP_SIGNALS
00090 #error "MALLOCPROTECT":  undefine "TRAP_SIGNALS" in unix.h first!
00091 #endif

```

```

00092 #include "mallocprotect.h"
00093
00094 #ifdef M_alloc
00095 #undef M_alloc
00096 #endif
00097
00098 #define M_alloc mprotectMalloc
00099
00100 #endif
00101
00102 #ifdef TERMMALLOCDEBUG
00103 WORD **DebugHeap1, **DebugHeap2;
00104 #endif
00105
00106 /*
00107     #] Includes :
00108     #[ Streams :
00109         #[ LoadInputFile :
00110 */
00111
00112 UBYTE *LoadInputFile(UBYTE *filename, int type)
00113 {
00114     int handle;
00115     LONG filesize;
00116     UBYTE *buffer, *name = filename;
00117     POSITION scrpos;
00118     handle = LocateFile(&name,type);
00119     if ( handle < 0 ) return(0);
00120     PUTZERO(scrpos);
00121     SeekFile(handle,&scrpos,SEEK_END);
00122     TELLFILE(handle,&scrpos);
00123     filesize = BASEPOSITION(scrpos);
00124     PUTZERO(scrpos);
00125     SeekFile(handle,&scrpos,SEEK_SET);
00126     buffer = (UBYTE *)Malloca1(filesize+2,"LoadInputFile");
00127     if ( ReadFile(handle,buffer,filesize) != filesize ) {
00128         Error1("Read error for file ",name);
00129         M_free(buffer,"LoadInputFile");
00130         if ( name != filename ) M_free(name,"FromLoadInputFile");
00131         CloseFile(handle);
00132         return(0);
00133     }
00134     CloseFile(handle);
00135     if ( type == PROCEDUREFILE || type == SETUPFILE ) {
00136         buffer[filesize] = '\n';
00137         buffer[filesize+1] = 0;
00138     }
00139     else {
00140         buffer[filesize] = 0;
00141     }
00142     if ( name != filename ) M_free(name,"FromLoadInputFile");
00143     return(buffer);
00144 }
00145
00146 /*
00147     #] LoadInputFile :
00148     #[ ReadFromStream :
00149 */
00150
00151 UBYTE ReadFromStream(STREAM *stream)
00152 {
00153     UBYTE c;
00154     POSITION scrpos;
00155     #ifdef WITHPIPE
00156     if ( stream->type == PIPESTREAM ) {
00157     #ifndef WITHMPI
00158         FILE *f;
00159         int cc;
00160         RWLOCKR(AM.handlelock);
00161         f = (FILE *) (filelist[stream->handle]);
00162         UNRWLOCK(AM.handlelock);
00163         cc = getc(f);
00164         if ( cc == EOF ) return(ENDOFSTREAM);
00165         c = (UBYTE)cc;
00166     #else
00167         if ( stream->pointer >= stream->top ) {
00168             /* The master reads the pipe and broadcasts it to the slaves. */
00169             LONG len;
00170             if ( PF.me == MASTER ) {
00171                 FILE *f;
00172                 UBYTE *p, *end;
00173                 RWLOCKR(AM.handlelock);
00174                 f = (FILE *) filelist[stream->handle];
00175                 UNRWLOCK(AM.handlelock);
00176                 p = stream->buffer;
00177                 end = stream->buffer + stream->buffersize;
00178                 while ( p < end ) {

```

```

00179         int cc = getc(f);
00180         if ( cc == EOF ) {
00181             break;
00182         }
00183         *p++ = (UBYTE)cc;
00184     }
00185     len = p - stream->buffer;
00186     PF_BroadcastNumber(len);
00187 }
00188 else {
00189     len = PF_BroadcastNumber(0);
00190 }
00191 if ( len > 0 ) {
00192     PF_Bcast(stream->buffer, len);
00193 }
00194 stream->pointer = stream->buffer;
00195 stream->inbuffer = len;
00196 stream->top = stream->buffer + stream->inbuffer;
00197 if ( stream->pointer == stream->top ) return ENDOFSTREAM;
00198 }
00199 c = (UBYTE)*stream->pointer++;
00200 #endif
00201 if ( stream->eqnum == 1 ) { stream->eqnum = 0; stream->linenumber++; }
00202 if ( c == LINEFEED ) stream->eqnum = 1;
00203 return(c);
00204 }
00205 #endif
00206 /*[14apr2004 mt]:*/
00207 #ifdef WITHEXTERNALCHANNEL
00208     if ( stream->type == EXTERNALCHANNELSTREAM ) {
00209         int cc;
00210         cc = getcFromExtChannel();
00211         /*[18may20006 mt]:*/
00212         /*if ( cc == EOF ) return(ENDOFSTREAM);*/
00213         if ( cc < 0 ) {
00214             if( cc == EOF )
00215                 return(ENDOFSTREAM);
00216             else{
00217                 Error0("No current external channel");
00218                 Terminate(-1);
00219             }
00220         }/*if ( cc < 0 )*/
00221         /*:[18may20006 mt]:*/
00222         c = (UBYTE)cc;
00223         if ( stream->eqnum == 1 ) { stream->eqnum = 0; stream->linenumber++; }
00224         if ( c == LINEFEED ) stream->eqnum = 1;
00225         return(c);
00226     }
00227 #endif /*ifdef WITHEXTERNALCHANNEL*/
00228 /*:[14apr2004 mt]:*/
00229     if ( stream->type == INPUTSTREAM ) {
00230         if ( stream->pointer < stream->top ) {
00231             c = *stream->pointer++;
00232         }
00233         else {
00234             if ( ReadFile(stream->handle,&c,1) != 1 ) {
00235                 return(ENDOFSTREAM);
00236             }
00237             if ( stream->fileposition == 0 ) {
00238                 if ( !stream->buffer ) {
00239                     stream->buffer = (UBYTE *)Malloc1(stream->buffer, "input stream buffer");
00240                     stream->pointer = stream->top = stream->buffer;
00241                 }
00242             }
00243             else {
00244                 if ( stream->top - stream->buffer >= stream->buffer ) {
00245                     LONG oldsize = stream->buffer;
00246                     DoubleBuffer((void**)&stream->buffer, (void**)&stream->top, sizeof(UBYTE), "double input stream buffer");
00247                     stream->buffer = stream->buffer;
00248                     stream->pointer = stream->top = stream->buffer + oldsize;
00249                 }
00250             }
00251             *stream->pointer = c;
00252             stream->pointer = ++stream->top;
00253             stream->inbuffer++;
00254         }
00255     }
00256     if ( stream->eqnum == 1 ) { stream->eqnum = 0; stream->linenumber++; }
00257     if ( c == LINEFEED ) stream->eqnum = 1;
00258     return(c);
00259 }
00260 if ( stream->pointer >= stream->top ) {
00261     if ( stream->type != FILESTREAM ) return(ENDOFSTREAM);
00262     if ( stream->fileposition != stream->bufferposition+stream->inbuffer ) {
00263         stream->fileposition = stream->bufferposition+stream->inbuffer;
00264         SETBASEPOSITION(scrpos, stream->fileposition);

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00265         SeekFile(stream->handle, &scrpos, SEEK_SET);
00266     }
00267     stream->bufferposition = stream->fileposition;
00268     stream->inbuffer = ReadFile(stream->handle,
00269         stream->buffer, stream->buffersize);
00270     if ( stream->inbuffer <= 0 ) return(EOFSTREAM);
00271     stream->top = stream->buffer + stream->inbuffer;
00272     stream->pointer = stream->buffer;
00273     stream->fileposition = stream->bufferposition + stream->inbuffer;
00274 }
00275 if ( stream->eqnum == 1 ) { stream->eqnum = 0; stream->linenumber++; }
00276 c = *(stream->pointer)++;
00277 if ( c == LINEFEED ) stream->eqnum = 1;
00278 return(c);
00279 }
00280
00281 /*
00282     #] ReadFromStream :
00283     #[ GetFromStream :
00284 */
00285
00286 UBYTE GetFromStream(STREAM *stream)
00287 {
00288     UBYTE c1, c2;
00289     if ( stream->isnextchar > 0 ) {
00290         return(stream->nextchar[--stream->isnextchar]);
00291     }
00292     c1 = ReadFromStream(stream);
00293     if ( c1 == LINEFEED || c1 == CARRIAGERETURN ) {
00294         c2 = ReadFromStream(stream);
00295         if ( c2 == c1 || ( c2 != LINEFEED && c2 != CARRIAGERETURN ) ) {
00296             stream->isnextchar = 1;
00297             stream->nextchar[0] = c2;
00298         }
00299         return(LINEFEED);
00300     }
00301     else return(c1);
00302 }
00303
00304 /*
00305     #] GetFromStream :
00306     #[ LookInStream :
00307 */
00308
00309 UBYTE LookInStream(STREAM *stream)
00310 {
00311     UBYTE c = GetFromStream(stream);
00312     UngetFromStream(stream, c);
00313     return(c);
00314 }
00315
00316 /*
00317     #] LookInStream :
00318     #[ OpenStream :
00319 */
00320
00321 STREAM *OpenStream(UBYTE *name, int type, int prevarmode, int raiselow)
00322 {
00323     STREAM *stream;
00324     UBYTE *rhsofvariable, *s, *newname, c;
00325     POSITION scrpos;
00326     int handle, num;
00327     LONG filesize;
00328     switch ( type ) {
00329         case REVERSEFILESTREAM:
00330         case FILESTREAM:
00331     }
00332     /*
00333         Notice that FILESTREAM is only used for text files:
00334         The #include files and the main input file (.frm)
00335         Hence we do not worry about files longer than 2 Gbytes.
00336     */
00337     newname = name;
00338     handle = LocateFile(&newname, -1);
00339     if ( handle < 0 ) return(0);
00340     PUTZERO(scrpos);
00341     SeekFile(handle, &scrpos, SEEK_END);
00342     TELLFILE(handle, &scrpos);
00343     filesize = BASEPOSITION(scrpos);
00344     PUTZERO(scrpos);
00345     SeekFile(handle, &scrpos, SEEK_SET);
00346     if ( filesize > AM.MaxStreamSize && type == FILESTREAM )
00347         filesize = AM.MaxStreamSize;
00348     stream = CreateStream((UBYTE *) "filestream");
00349     /*
00350         The extra +1 in the Malloc1 is potentially needed in ReverseStatements!
00351     */
00352     stream->buffer = (UBYTE *) Malloc1(filesize+1, "name of input stream");

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00352     stream->inbuffer = ReadFile(handle,stream->buffer,filesize);
00353     if ( type == REVERSEFILESTREAM ) {
00354         if ( ReverseStatements(stream) ) {
00355             M_free(stream->buffer,"name of input stream");
00356             return(0);
00357         }
00358     }
00359     stream->top = stream->buffer + stream->inbuffer;
00360     stream->pointer = stream->buffer;
00361     stream->handle = handle;
00362     stream->bufferSize = fileSize;
00363     stream->fileposition = stream->inbuffer;
00364     if ( newname != name ) stream->name = newname;
00365     else if ( name ) stream->name = strdup(name,"name of input stream");
00366     else
00367         stream->name = 0;
00368     stream->prevline = stream->linenumber = 1;
00369     stream->eqnum = 0;
00370     break;
00371 case PREVARSTREAM:
00372     if ( ( rhsOfVariable = GetPreVar(name,WITHERROR) ) == 0 ) return(0);
00373     stream = CreateStream((UBYTE *)"var-stream");
00374     stream->buffer = stream->pointer = s = rhsOfVariable;
00375     while ( *s ) s++;
00376     stream->top = s;
00377     stream->inbuffer = s - stream->buffer;
00378     stream->name = AC.CurrentStream->name;
00379     stream->linenumber = AC.CurrentStream->linenumber;
00380     stream->prevline = AC.CurrentStream->prevline;
00381     stream->eqnum = AC.CurrentStream->eqnum;
00382     stream->pname = strdup(name,"stream->pname");
00383     stream->olddelay = AP.AllowDelay;
00384     s = stream->pname; while ( *s ) s++;
00385     while ( s[-1] == '+' || s[-1] == '-' ) s--;
00386     *s = 0;
00387     UnsetAllowDelay();
00388     break;
00389 case DOLLARSTREAM:
00390     if ( ( num = GetDollar(name) ) < 0 ) {
00391         WORD numfac = 0;
00392         /*
00393          Here we have to test first whether we have $x[1], $x[0]
00394          or just an undefined $x.
00395          */
00396         s = name; while ( *s && *s != '[' ) s++;
00397         if ( *s == 0 ) return(0);
00398         c = *s; *s = 0;
00399         if ( ( num = GetDollar(name) ) < 0 ) return(0);
00400         *s = c;
00401         s++;
00402         if ( *s == 0 || FG.cTable[*s] != 1 || *s == ']' ) {
00403             MesPrint("@Illegal factor number for dollar variable");
00404             return(0);
00405         }
00406         while ( *s && FG.cTable[*s] == 1 ) {
00407             numfac = 10*numfac+*s++-'0';
00408         }
00409         if ( *s != ']' || s[1] != 0 ) {
00410             MesPrint("@Illegal factor number for $ variable");
00411             return(0);
00412         }
00413         stream = CreateStream((UBYTE *)"dollar-stream");
00414         stream->buffer = stream->pointer = s = WriteDollarFactorToBuffer(num,numfac,1);
00415     }
00416     else {
00417         stream = CreateStream((UBYTE *)"dollar-stream");
00418         stream->buffer = stream->pointer = s = WriteDollarToBuffer(num,1);
00419     }
00420     while ( *s ) s++;
00421     stream->top = s;
00422     stream->inbuffer = s - stream->buffer;
00423     stream->name = AC.CurrentStream->name;
00424     stream->linenumber = AC.CurrentStream->linenumber;
00425     stream->prevline = AC.CurrentStream->prevline;
00426     stream->eqnum = AC.CurrentStream->eqnum;
00427     stream->pname = strdup(name,"stream->pname");
00428     s = stream->pname; while ( *s ) s++;
00429     while ( s[-1] == '+' || s[-1] == '-' ) s--;
00430     *s = 0;
00431     /* We 'stole' the buffer. Later we can free it. */
00432     AO.DollarOutSizeBuffer = 0;
00433     AO.DollarOutBuffer = 0;
00434     AO.DollarInOutBuffer = 0;
00435     break;
00436 case PREREADSTREAM:
00437 case PREREADSTREAM2:
00438 case PREREADSTREAM3:

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00439         case PRECALCSTREAM:
00440             stream = CreateStream((UBYTE *)"calculator");
00441             stream->buffer = stream->pointer = s = name;
00442             while ( *s ) s++;
00443             stream->top = s;
00444             stream->inbuffer = s - stream->buffer;
00445             stream->name = AC.CurrentStream->name;
00446             stream->linenumber = AC.CurrentStream->linenumber;
00447             stream->prevline = AC.CurrentStream->prevline;
00448             stream->eqnum = 0;
00449             break;
00450 #ifdef WITHPIPE
00451         case PIPESTREAM:
00452             stream = CreateStream((UBYTE *)"pipe");
00453 #ifndef WITHMPI
00454             {
00455                 FILE *f;
00456                 if ( ( f = popen((char *)name,"r") ) == 0 ) {
00457                     Error0("@Cannot create pipe");
00458                 }
00459                 stream->handle = CreateHandle();
00460                 RWLOCKW(AM.handlelock);
00461                 filelist[stream->handle] = (FILES *)f;
00462                 UNRWLOCK(AM.handlelock);
00463             }
00464             stream->buffer = stream->top = 0;
00465             stream->inbuffer = 0;
00466 #else
00467             {
00468                 /* Only the master opens the pipe. */
00469                 FILE *f;
00470                 if ( PF.me == MASTER ) {
00471                     f = popen((char *)name, "r");
00472                     PF_BroadcastNumber(f == 0);
00473                     if ( f == 0 ) Error0("@Cannot create pipe");
00474                 }
00475                 else {
00476                     if ( PF_BroadcastNumber(0) ) Error0("@Cannot create pipe");
00477                     f = (FILE *)123; /* dummy */
00478                 }
00479                 stream->handle = CreateHandle();
00480                 RWLOCKW(AM.handlelock);
00481                 filelist[stream->handle] = (FILES *)f;
00482                 UNRWLOCK(AM.handlelock);
00483             }
00484             /* stream->buffer as a send/receive buffer. */
00485             stream->buffer = (UBYTE *)Malloc1(stream->buffer, "pipe buffer");
00486             stream->inbuffer = 0;
00487             stream->top = stream->buffer;
00488             stream->pointer = stream->buffer;
00489 #endif
00490             stream->name = strDup1((UBYTE *)"pipe","pipe");
00491             stream->prevline = stream->linenumber = 1;
00492             stream->eqnum = 0;
00493             break;
00494 #endif
00495 #endif
00496 /*[14apr2004 mt]:*/
00497 #ifdef WITHEXTERNALCHANNEL
00498         case EXTERNALCHANNELSTREAM:
00499             /*Block*/
00500             int n, *tmpn;
00501             if( (n=getCurrentExternalChannel()) == 0 )
00502                 Error0("@No current external channel");
00503             stream = CreateStream((UBYTE *)"externalchannel");
00504             stream->handle = CreateHandle();
00505             tmpn = (int *)Malloc1(sizeof(int),"external channel handle");
00506             *tmpn = n;
00507             RWLOCKW(AM.handlelock);
00508             filelist[stream->handle] = (FILES *)tmpn;
00509             UNRWLOCK(AM.handlelock);
00510             /*Block*/
00511             stream->buffer = stream->top = 0;
00512             stream->inbuffer = 0;
00513             stream->name = strDup1((UBYTE *)"externalchannel","externalchannel");
00514             stream->prevline = stream->linenumber = 1;
00515             stream->eqnum = 0;
00516             break;
00517 #endif /*ifdef WITHEXTERNALCHANNEL*/
00518 /*:[14apr2004 mt]:*/
00519         case INPUTSTREAM:
00520             /*
00521              * Assume that "name" stores a file descriptor (UNIX) or a FILE
00522              * pointer (Windows). We don't close it automatically on closing
00523              * the INPUTSTREAM stream (e.g., for stdin).
00524              */
00525             stream = CreateStream((UBYTE *)"input stream");

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00526         stream->handle = CreateHandle();
00527     {
00528         FILES *f = (FILES *)Malloca(sizeof(int),"input stream handle");
00529         /* NOTE: in both cases name=0 indicates stdin. */
00530 #ifdef UNIX
00531         f->descriptor = (int)(ssize_t)name;
00532 #else
00533         f = name ? (FILES *)name : stdin;
00534 #endif
00535         RWLOCKW(AM.handlelock);
00536         filelist[stream->handle] = f;
00537         UNRWLOCK(AM.handlelock);
00538     }
00539     stream->buffer = stream->pointer = stream->top = 0;
00540     stream->inbuffer = 0;
00541     stream->name = strdup((UBYTE *) (name ? "INPUT" : "STDIN"),"input stream name");
00542     stream->prevline = stream->linenumber = 1;
00543     stream->eqnum = 0;
00544     /*
00545      * fileposition == -1: default
00546      * fileposition == 0: cache the input
00547      * See also: ReadFromStream, TryFileSetups
00548      */
00549     stream->fileposition = -1;
00550     break;
00551 default:
00552     return(0);
00553 }
00554 stream->bufferposition = 0;
00555 stream->isnextchar = 0;
00556 stream->type = type;
00557 stream->previousNoShowInput = AC.NoShowInput;
00558 stream->afterwards = raiselow;
00559 if ( AC.CurrentStream ) stream->previous = AC.CurrentStream - AC.Streams;
00560 else stream->previous = -1;
00561 stream->FoldName = 0;
00562 if ( prevarmode == 0 ) stream->prevvars = -1;
00563 else if ( prevarmode > 0 ) stream->prevvars = NumPre;
00564 else if ( prevarmode < 0 ) stream->prevvars = -prevarmode-1;
00565 AC.CurrentStream = stream;
00566 if ( type == PREREADSTREAM || type == PREREADSTREAM3 || type == PRECALCSTREAM
00567     || type == DOLLARSTREAM ) AC.NoShowInput = 1;
00568 return(stream);
00569 }
00570
00571 /*
00572     #] OpenStream :
00573     #[ LocateFile :
00574 */
00575
00576 int LocateFile(UBYTE **name, int type)
00577 {
00578     int handle, namesize, i;
00579     UBYTE *s, *to, *u1, *u2, *newname, *indir;
00580     handle = OpenFile((char *) (*name));
00581     if ( handle >= 0 ) return(handle);
00582     if ( type == SETUPFILE && AM.SetupFile ) {
00583         handle = OpenFile((char *) (AM.SetupFile));
00584         if ( handle >= 0 ) return(handle);
00585         MesPrint("Could not open setup file %s", (char *) (AM.SetupFile));
00586     }
00587     namesize = 4; s = *name;
00588     while ( *s ) { s++; namesize++; }
00589     if ( type == SETUPFILE ) indir = AM.SetupDir;
00590     else indir = AM.IncDir;
00591     if ( indir ) {
00592
00593         s = indir; i = 0;
00594         while ( *s ) { s++; i++; }
00595         newname = (UBYTE *)Malloca(namesize+i,"LocateFile");
00596         s = indir; to = newname;
00597         while ( *s ) *to++ = *s++;
00598         if ( to > newname && to[-1] != SEPARATOR ) *to++ = SEPARATOR;
00599         s = *name;
00600         while ( *s ) *to++ = *s++;
00601         *to = 0;
00602         handle = OpenFile((char *) newname);
00603         if ( handle >= 0 ) {
00604             *name = newname;
00605             return(handle);
00606         }
00607         M_free(newname,"LocateFile, incdir/file");
00608     }
00609     if ( type == SETUPFILE ) {
00610         handle = OpenFile(setupfilename);
00611         if ( handle >= 0 ) return(handle);
00612         s = (UBYTE *)getenv("FORMSETUP");

```

```

00613         if ( s ) {
00614             handle = OpenFile((char *)s);
00615             if ( handle >= 0 ) return(handle);
00616             MesPrint("Could not open setup file %s",s);
00617         }
00618     }
00619     if ( type != SETUPFILE && AM.Path ) {
00620         ul = AM.Path;
00621         while ( *ul ) {
00622             u2 = ul; i = 0;
00623 #ifndef WINDOWS
00624             while ( *ul && *ul != ';' ) {
00625                 ul++; i++;
00626             }
00627 #else
00628             while ( *ul && *ul != ':' ) {
00629                 if ( *ul == '\\\' ) ul++;
00630                 ul++; i++;
00631             }
00632 #endif
00633             newname = (UBYTE *)Malloc1(namesize+i,"LocateFile");
00634             s = u2; to = newname;
00635             while ( s < ul ) {
00636 #ifndef WINDOWS
00637                 if ( *s == '\\\' ) s++;
00638 #endif
00639                 *to++ = *s++;
00640             }
00641             if ( to > newname && to[-1] != SEPARATOR ) *to++ = SEPARATOR;
00642             s = *name;
00643             while ( *s ) *to++ = *s++;
00644             *to = 0;
00645             handle = OpenFile((char *)newname);
00646             if ( handle >= 0 ) {
00647                 *name = newname;
00648                 return(handle);
00649             }
00650             M_free(newname,"LocateFile Path/file");
00651             if ( *ul ) ul++;
00652         }
00653     }
00654     if ( type != SETUPFILE && type >= -1 ) Error1("LocateFile: Cannot find file",*name);
00655     return(-1);
00656 }
00657
00658 /*
00659     #] LocateFile :
00660     #[ CloseStream :
00661 */
00662
00663 STREAM *CloseStream(STREAM *stream)
00664 {
00665     int newstr = stream->previous, sgn;
00666     UBYTE *t, numbuf[24];
00667     LONG x;
00668     if ( stream->FoldName ) {
00669         M_free(stream->FoldName,"stream->FoldName");
00670         stream->FoldName = 0;
00671     }
00672     if ( stream->type == FILESTREAM || stream->type == REVERSEFILESTREAM ) {
00673         CloseFile(stream->handle);
00674         if ( stream->buffer != 0 ) M_free(stream->buffer,"name of input stream");
00675         stream->buffer = 0;
00676     }
00677 #ifdef WITHPIPE
00678     else if ( stream->type == PIPESTREAM ) {
00679         RWLOCKW(AM.handlelock);
00680 #ifdef WITHMPI
00681         if ( PF.me == MASTER )
00682 #endif
00683             pclose((FILE *) (filelist[stream->handle]));
00684             filelist[stream->handle] = 0;
00685             numinfilelist--;
00686             UNRWLOCK(AM.handlelock);
00687 #ifdef WITHMPI
00688         if ( stream->buffer != 0 ) {
00689             M_free(stream->buffer, "pipe buffer");
00690             stream->buffer = 0;
00691         }
00692 #endif
00693     }
00694 #endif
00695 /*[14apr2004 mt]:*/
00696 #ifdef WITHEXTERNALCHANNEL
00697     else if ( stream->type == EXTERNALCHANNELSTREAM ) {
00698         int *tmpn;
00699         RWLOCKW(AM.handlelock);

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```

00700         tmpn = (int *) (filelist[stream->handle]);
00701         filelist[stream->handle] = 0;
00702         numinfilelist--;
00703         UNRWLOCK(AM.handlelock);
00704         M_free(tmpn, "external channel handle");
00705     }
00706 #endif /*ifdef WITHEXTERNALCHANNEL*/
00707 /*:[14apr2004 mt]*/
00708     else if ( stream->type == INPUTSTREAM ) {
00709         FILES *f;
00710         RWLOCKW(AM.handlelock);
00711         f = filelist[stream->handle];
00712         filelist[stream->handle] = 0;
00713         numinfilelist--;
00714         UNRWLOCK(AM.handlelock);
00715         M_free(f, "input stream handle");
00716     }
00717     else if ( stream->type == PREVARSTREAM && (
00718 stream->afterwards == PRERAISEAFTER || stream->afterwards == PRELOWERAFTER ) ) {
00719         t = stream->buffer; x = 0; sgn = 1;
00720         while ( *t == '-' || *t == '+' ) {
00721             if ( *t == '-' ) sgn = -sgn;
00722             t++;
00723         }
00724         if ( FG.cTable[*t] == 1 ) {
00725             while ( *t && FG.cTable[*t] == 1 ) x = 10*x + *t++ - '0';
00726             if ( *t == 0 ) {
00727                 if ( stream->afterwards == PRERAISEAFTER ) x = sgn*x + 1;
00728                 else x = sgn*x - 1;
00729                 NumToStr(numbuf, x);
00730                 PutPreVar(stream->pname, numbuf, 0, 1);
00731             }
00732         }
00733     }
00734     else if ( stream->type == DOLLARSTREAM && (
00735 stream->afterwards == PRERAISEAFTER || stream->afterwards == PRELOWERAFTER ) ) {
00736         if ( stream->afterwards == PRERAISEAFTER ) x = 1;
00737         else x = -1;
00738         DollarRaiseLow(stream->pname, x);
00739     }
00740     else if ( stream->type == PRECALCSTREAM || stream->type == DOLLARSTREAM ) {
00741         if ( stream->buffer ) M_free(stream->buffer, "stream->buffer");
00742         stream->buffer = 0;
00743     }
00744     if ( stream->name && stream->type != PREVARSTREAM
00745 && stream->type != PREREADSTREAM && stream->type != PREREADSTREAM2 && stream->type !=
PREREADSTREAM3
00746 && stream->type != PRECALCSTREAM && stream->type != DOLLARSTREAM ) {
00747         M_free(stream->name, "stream->name");
00748     }
00749     stream->name = 0;
00750 /* if ( stream->type != FILESTREAM ) */
00751 AC.NoShowInput = stream->previousNoShowInput;
00752 stream->buffer = 0; /* To make sure we will not reuse it */
00753 stream->pointer = 0;
00754 /*
00755 Look whether we have to pop preprocessor variables.
00756 */
00757     if ( stream->prevars >= 0 ) {
00758         while ( NumPre > stream->prevars ) {
00759             NumPre--;
00760             M_free(PreVar[NumPre].name, "PreVar[NumPre].name");
00761             PreVar[NumPre].name = PreVar[NumPre].value = 0;
00762         }
00763     }
00764     if ( stream->type == PREVARSTREAM ) {
00765         AP.AllowDelay = stream->olddelay;
00766         ClearMacro(stream->pname);
00767         M_free(stream->pname, "stream->pname");
00768     }
00769     else if ( stream->type == DOLLARSTREAM ) {
00770         M_free(stream->pname, "stream->pname");
00771     }
00772     AC.NumStreams--;
00773     if ( newstr >= 0 ) return(AC.Streams + newstr);
00774     else return(0);
00775 }
00776 /*
00777 #] CloseStream :
00778 #[ CreateStream :
00779 */
00780
00781
00782 STREAM *CreateStream(UBYTE *where)
00783 {
00784     STREAM *newstreams;
00785     int numnewstreams, i;

```

```

00786     int offset;
00787     if ( AC.NumStreams >= AC.MaxNumStreams ) {
00788         if ( AC.MaxNumStreams == 0 ) numnewstreams = 10;
00789         else numnewstreams = 2*AC.MaxNumStreams;
00790         newstreams = (STREAM *)Malloc1(sizeof(STREAM)*(numnewstreams+1),"CreateStream");
00791         if ( AC.MaxNumStreams > 0 ) {
00792             offset = AC.CurrentStream - AC.Streams;
00793             for ( i = 0; i < AC.MaxNumStreams; i++ ) {
00794                 newstreams[i] = AC.Streams[i];
00795             }
00796             AC.CurrentStream = newstreams + offset;
00797         }
00798         else newstreams[0].previous = -1;
00799         AC.MaxNumStreams = numnewstreams;
00800         if ( AC.Streams ) M_free(AC.Streams,(char *)where);
00801         AC.Streams = newstreams;
00802     }
00803     newstreams = AC.Streams+AC.NumStreams++;
00804     newstreams->name = 0;
00805     return(newstreams);
00806 }
00807
00808 /*
00809     #] CreateStream :
00810     #[ GetStreamPosition :
00811 */
00812
00813 LONG GetStreamPosition(STREAM *stream)
00814 {
00815     return(stream->bufferposition + ((LONG)stream->pointer-(LONG)stream->buffer));
00816 }
00817
00818 /*
00819     #] GetStreamPosition :
00820     #[ PositionStream :
00821 */
00822
00823 void PositionStream(STREAM *stream, LONG position)
00824 {
00825     POSITION scrpos;
00826     if ( position >= stream->bufferposition
00827         && position < stream->bufferposition + stream->inbuffer ) {
00828         stream->pointer = stream->buffer + (position-stream->bufferposition);
00829     }
00830     else if ( stream->type == FILESTREAM ) {
00831         SETBASEPOSITION(scrpos,position);
00832         SeekFile(stream->handle,&scrpos,SEEK_SET);
00833         stream->inbuffer = ReadFile(stream->handle,stream->buffer,stream->buffersize);
00834         stream->pointer = stream->buffer;
00835         stream->top = stream->buffer + stream->inbuffer;
00836         stream->bufferposition = position;
00837         stream->fileposition = position + stream->inbuffer;
00838         stream->isnextchar = 0;
00839     }
00840     else {
00841         Error0("Illegal position for stream");
00842         Terminate(-1);
00843     }
00844 }
00845
00846 /*
00847     #] PositionStream :
00848     #[ ReverseStatements :
00849
00850     Reverses the order of the statements in the buffer.
00851     We allocate an extra buffer and copy a bit to and from.
00852     Note that there are some nasties that cannot be resolved.
00853 */
00854
00855 int ReverseStatements(STREAM *stream)
00856 {
00857     UBYTE *spare = (UBYTE *)Malloc1((stream->inbuffer+1)*sizeof(UBYTE),"Reverse copy");
00858     UBYTE *top = stream->buffer + stream->inbuffer, *in, *s, *ss, *out;
00859     out = spare+stream->inbuffer+1;
00860     in = stream->buffer;
00861     while ( in < top ) {
00862         s = in;
00863         if ( *s == AP.ComChar ) {
00864             toeol;
00865             for(;;) {
00866                 if ( s == top ) { *--out = '\n'; break; }
00867                 if ( *s == '\\\\' ) {
00868                     s++;
00869                     if ( s >= top ) { /* This is an error! */
00870                         MesPrint("@Irregular end of reverse include file.");
00871                         return(1);
00872                     }

```

```

00873     }
00874     else if ( *s == '\n' ) {
00875         s++; ss = s;
00876         while ( ss > in ) *--out = *--ss;
00877         in = s;
00878         if ( out[0] == AP.ComChar && ss+6 < s && out[3] == '#' ) {
00879             /*
00880                 For folds we have to exchange begin and end
00881             */
00882                 if ( out[4] == '[' ) out[4] = ']';
00883                 else if ( out[4] == ']' ) out[4] = '[';
00884             }
00885             break;
00886         }
00887         s++;
00888     }
00889     continue;
00890 }
00891 while ( s < top && ( *s == ' ' || *s == '\t' ) ) s++;
00892 if ( *s == '#' ) { /* preprocessor instruction */
00893     goto toeol; /* read to end of line */
00894 }
00895 if ( *s == '.' ) { /* end-of-module instruction */
00896     goto toeol; /* read to end of line */
00897 }
00898 /*
00899     Here we have a regular statement. In principle we scan to ; and its \n
00900     but there are special cases.
00901     1: ; inside a string (in print ".....;")
00902     2: multiple statements on one line.
00903     3: ; + commentary after some blanks.
00904     4: 'var' can cause problems.....
00905 */
00906 while ( s < top ) {
00907     if ( *s == ';' ) {
00908         s++;
00909         while ( s < top && ( *s == ' ' || *s == '\t' ) ) s++;
00910         while ( s < top && *s == '\n' ) s++;
00911         if ( s >= top && s[-1] != '\n' ) *s++ = '\n';
00912         ss = s;
00913         while ( ss > in ) *--out = *--ss;
00914         in = s;
00915         break;
00916     }
00917     else if ( *s == '"' ) {
00918         s++;
00919         while ( s < top ) {
00920             if ( *s == '"' ) break;
00921             if ( *s == '\\ ' ) { s++; }
00922             s++;
00923         }
00924         if ( s >= top ) goto irrend;
00925     }
00926     else if ( *s == '\\ ' ) {
00927         s++;
00928         if ( s >= top ) goto irrend;
00929     }
00930     s++;
00931 }
00932 if ( in < top ) { /* Like blank lines at the end */
00933     if ( s >= top && s[-1] != '\n' ) *s++ = '\n';
00934     ss = s;
00935     while ( ss > in ) *--out = *--ss;
00936     in = s;
00937 }
00938 }
00939 if ( out == spare ) stream->inbuffer++;
00940 if ( out > spare+1 ) {
00941     MesPrint("@Internal error in #reverseinclude instruction.");
00942     return(1);
00943 }
00944 memcpy((void *) (stream->buffer), (void *) out, (size_t) (stream->inbuffer*sizeof(UBYTE)));
00945 M_free(spare, "Reverse copy");
00946 return(0);
00947 }
00948
00949 /*
00950     #] ReverseStatements :
00951     #] Streams :
00952     #[ Files :
00953     #[ StartFiles :
00954 */
00955
00956 void StartFiles(void)
00957 {
00958     int i = CreateHandle();
00959     filelist[i] = Ustdout;

```

```

00960     AM.StdOut = i;
00961     AC.StoreHandle = -1;
00962     AC.LogHandle = -1;
00963 #ifndef WITHPTHREADS
00964     AR.Fscr[0].handle = -1;
00965     AR.Fscr[1].handle = -1;
00966     AR.Fscr[2].handle = -1;
00967     AR.FoStage4[0].handle = -1;
00968     AR.FoStage4[1].handle = -1;
00969     AR.infile = &(AR.Fscr[0]);
00970     AR.outfile = &(AR.Fscr[1]);
00971     AR.hidefile = &(AR.Fscr[2]);
00972     AR.StoreData.Handle = -1;
00973 #endif
00974     AC.Streams = 0;
00975     AC.MaxNumStreams = 0;
00976 }
00977
00978 /*
00979     #] StartFiles :
00980     #[ OpenFile :
00981 */
00982
00983 int OpenFile(char *name)
00984 {
00985     FILES *f;
00986     int i;
00987
00988     if ( ( f = Uopen(name,"rb") ) == 0 ) return(-1);
00989 /* Usetbuf(f,0); */
00990     i = CreateHandle();
00991     RWLOCKW(AM.handlelock);
00992     filelist[i] = f;
00993     UNRWLOCK(AM.handlelock);
00994     return(i);
00995 }
00996
00997 /*
00998     #] OpenFile :
00999     #[ OpenAddFile :
01000 */
01001
01002 int OpenAddFile(char *name)
01003 {
01004     FILES *f;
01005     int i;
01006     POSITION scrpos;
01007     if ( ( f = Uopen(name,"a+b") ) == 0 ) return(-1);
01008 /* Usetbuf(f,0); */
01009     i = CreateHandle();
01010     RWLOCKW(AM.handlelock);
01011     filelist[i] = f;
01012     UNRWLOCK(AM.handlelock);
01013     TELLFILE(i,&scrpos);
01014     SeekFile(i,&scrpos,SEEK_SET);
01015     return(i);
01016 }
01017
01018 /*
01019     #] OpenAddFile :
01020     #[ ReOpenFile :
01021 */
01022
01023 int ReOpenFile(char *name)
01024 {
01025     FILES *f;
01026     int i;
01027     POSITION scrpos;
01028     if ( ( f = Uopen(name,"r+b") ) == 0 ) return(-1);
01029     i = CreateHandle();
01030     RWLOCKW(AM.handlelock);
01031     filelist[i] = f;
01032     UNRWLOCK(AM.handlelock);
01033     TELLFILE(i,&scrpos);
01034     SeekFile(i,&scrpos,SEEK_SET);
01035     return(i);
01036 }
01037
01038 /*
01039     #] ReOpenFile :
01040     #[ CreateFile :
01041 */
01042
01043 int CreateFile(char *name)
01044 {
01045     FILES *f;
01046     int i;

```

```

01047     if ( ( f = Uopen(name,"w+b") ) == 0 ) return(-1);
01048     i = CreateHandle();
01049     RWLOCKW(AM.handlelock);
01050     filelist[i] = f;
01051     UNRWLOCK(AM.handlelock);
01052     return(i);
01053 }
01054
01055 /*
01056     #] CreateFile :
01057     #[ CreateLogFile :
01058 */
01059
01060 int CreateLogFile(char *name)
01061 {
01062     FILES *f;
01063     int i;
01064     if ( ( f = Uopen(name,"w+b") ) == 0 ) return(-1);
01065     Usetbuf(f,0);
01066     i = CreateHandle();
01067     RWLOCKW(AM.handlelock);
01068     filelist[i] = f;
01069     UNRWLOCK(AM.handlelock);
01070     return(i);
01071 }
01072
01073 /*
01074     #] CreateLogFile :
01075     #[ CloseFile :
01076 */
01077
01078 void CloseFile(int handle)
01079 {
01080     if ( handle >= 0 ) {
01081         FILES *f; /* we need this variable to be thread-safe */
01082         RWLOCKW(AM.handlelock);
01083         f = filelist[handle];
01084         filelist[handle] = 0;
01085         numinfilelist--;
01086         UNRWLOCK(AM.handlelock);
01087         Uclose(f);
01088     }
01089 }
01090
01091 /*
01092     #] CloseFile :
01093     #[ CopyFile :
01094 */
01095
01101 int CopyFile(char *source, char *dest)
01102 {
01103     #define COPYFILEBUFSIZE 40960L
01104     FILE *in, *out;
01105     size_t countin, countout, sumcount;
01106     char *buffer = NULL;
01107
01108     sumcount = (AM.S0->LargeSize+AM.S0->SmallEsize)*sizeof(WORD);
01109     if ( sumcount <= COPYFILEBUFSIZE ) {
01110         sumcount = COPYFILEBUFSIZE;
01111         buffer = (char*)Mallctl(sumcount, "file copy buffer");
01112     }
01113     else {
01114         buffer = (char *) (AM.S0->lBuffer);
01115     }
01116
01117     in = fopen(source, "rb");
01118     if ( in == NULL ) {
01119         perror("CopyFile: ");
01120         return(1);
01121     }
01122     out = fopen(dest, "wb");
01123     if ( out == NULL ) {
01124         perror("CopyFile: ");
01125         return(2);
01126     }
01127
01128     while ( !feof(in) ) {
01129         countin = fread(buffer, 1, sumcount, in);
01130         if ( countin != sumcount ) {
01131             if ( ferror(in) ) {
01132                 perror("CopyFile: ");
01133                 return(3);
01134             }
01135         }
01136         countout = fwrite(buffer, 1, countin, out);
01137         if ( countin != countout ) {
01138             perror("CopyFile: ");

```

```

01139         return(4);
01140     }
01141 }
01142
01143 fclose(in);
01144 fclose(out);
01145 if ( sumcount <= COPYFILEBUFSIZE ) {
01146     M_free(buffer, "file copy buffer");
01147 }
01148 return(0);
01149 }
01150
01151 /*
01152     #] CopyFile :
01153     #[ CreateHandle :
01154
01155     We need a lock here.
01156     Problem: the same lock is needed inside Malloc1 and M_free which
01157     is used in DoubleList when we use MALLOCDEBUG
01158
01159     Conclusion: MALLOCDEBUG will have to be a bit unsafe
01160 */
01161
01162 int CreateHandle(void)
01163 {
01164     int i, j;
01165     #ifndef MALLOCDEBUG
01166         RWLOCKW(AM.handlelock);
01167     #endif
01168     if ( filelistsize == 0 ) {
01169         filelistsize = 10;
01170         filelist = (FILES **)Malloc1(sizeof(FILES *)*filelistsize, "file handle");
01171         for ( j = 0; j < filelistsize; j++ ) filelist[j] = 0;
01172         numinfilelist = 1;
01173         i = 0;
01174     }
01175     else if ( numinfilelist >= filelistsize ) {
01176         void **fl = (void **)filelist;
01177         i = filelistsize;
01178         if ( DoubleList((void ***)&fl, &filelistsize, (int)sizeof(FILES *),
01179             "list of open files") != 0 ) Terminate(-1);
01180         filelist = (FILES **)fl;
01181         for ( j = i; j < filelistsize; j++ ) filelist[j] = 0;
01182         numinfilelist = i + 1;
01183     }
01184     else {
01185         i = filelistsize;
01186         for ( j = 0; j < filelistsize; j++ ) {
01187             if ( filelist[j] == 0 ) { i = j; break; }
01188         }
01189         numinfilelist++;
01190     }
01191     filelist[i] = (FILES *) (filelist); /* Just for now to not get into problems */
01192     /*
01193     The next code is not needed when we use open.
01194     It may be needed when we use fopen.
01195     fopen is used in minos.c without this central administration.
01196     */
01197     if ( numinfilelist > MAX_OPEN_FILES ) {
01198         #ifndef MALLOCDEBUG
01199             UNRWLOCK(AM.handlelock);
01200         #endif
01201         MesPrint("More than %d open files", MAX_OPEN_FILES);
01202         Error0("System limit. This limit is not due to FORM!");
01203     }
01204     else {
01205         #ifndef MALLOCDEBUG
01206             UNRWLOCK(AM.handlelock);
01207         #endif
01208     }
01209     return(i);
01210 }
01211
01212 /*
01213     #] CreateHandle :
01214     #[ ReadFile :
01215     */
01216
01217 LONG ReadFile(int handle, UBYTE *buffer, LONG size)
01218 {
01219     LONG inbuf = 0, r;
01220     FILES *f;
01221     char *b;
01222     b = (char *)buffer;
01223     for(;;) { /* Gotta do difficult because of VMS! */
01224         RWLOCKR(AM.handlelock);
01225         f = filelist[handle];

```

```

01226     UNRWLOCK(AM.handlelock);
01227 #ifdef WITHSTATS
01228     numreads++;
01229 #endif
01230     r = Uread(b,1,size,f);
01231     if ( r < 0 ) return(r);
01232     if ( r == 0 ) return(inbuf);
01233     inbuf += r;
01234     if ( r == size ) return(inbuf);
01235     if ( r > size ) return(-1);
01236     size -= r;
01237     b += r;
01238 }
01239 }
01240
01241 /*
01242     #] ReadFile :
01243     #[ ReadPosFile :
01244
01245     Gets words from a file(handle).
01246     First tries to get the information from the buffers.
01247     Reads a file at a position. Updates the position.
01248     Places a lock in the case of multithreading.
01249     Exists for multiple reading from the same file.
01250     size is the number of WORDs to read!!!!
01251
01252     We may need some strategy in the caching. This routine is used from
01253     GetOneTerm only. The problem is when it reads brackets and the
01254     brackets are read backwards. This is very uneconomical because
01255     each time it may read a large buffer.
01256     On the other hand, reading piece by piece in GetOneTerm takes
01257     much overhead as well.
01258     Two strategies come to mind:
01259     1: keep things as they are but limit the size of the buffers.
01260     2: have the position of 'pos' at about 1/3 of the buffer.
01261         this is of course guess work.
01262     Currently we have implemented the first method by creating the
01263     setup parameter threadscratchsize with the default value 100K.
01264     In the test program much bigger values gave a slower program.
01265 */
01266 LONG ReadPosFile(PHEAD FILEHANDLE *fi, UBYTE *buffer, LONG size, POSITION *pos)
01267 {
01268     GETBIDENTITY
01269     LONG i, retval = 0;
01270     WORD *b = (WORD *)buffer, *t;
01271
01272     if ( fi->handle < 0 ) {
01273         fi->POfill = (WORD *)((UBYTE *) (fi->PObuffer) + BASEPOSITION(*pos));
01274         t = fi->POfill;
01275         while ( size > 0 && fi->POfill < fi->POfull ) { *b++ = *t++; size--; }
01276     }
01277     else {
01278         if ( ISLESSPOS(*pos,fi->POposition) || ISGEPOSINC(*pos,fi->POposition,
01279             ((UBYTE *) (fi->POfull)-(UBYTE *) (fi->PObuffer))) ) {
01280             /*
01281                 The start is not inside the buffer. Fill the buffer.
01282             */
01283
01284             fi->POposition = *pos;
01285             LOCK(AS.inputslock);
01286             SeekFile(fi->handle,pos,SEEK_SET);
01287             retval = ReadFile(fi->handle,(UBYTE *) (fi->PObuffer),fi->POsize);
01288             UNLOCK(AS.inputslock);
01289             fi->POfull = fi->PObuffer+retval/sizeof(WORD);
01290             fi->POfill = fi->PObuffer;
01291             if ( fi != AR.hidefile ) AR.InInBuf = retval/sizeof(WORD);
01292             else AR.InHiBuf = retval/sizeof(WORD);
01293         }
01294         else {
01295             fi->POfill = (WORD *)((UBYTE *) (fi->PObuffer) + DIFBASE(*pos,fi->POposition));
01296         }
01297         if ( fi->POfill + size <= fi->POfull ) {
01298             t = fi->POfill;
01299             while ( size > 0 ) { *b++ = *t++; size--; }
01300         }
01301         else {
01302             for (;;) {
01303                 i = fi->POfull - fi->POfill; t = fi->POfill;
01304                 if ( i > size ) i = size;
01305                 size -= i;
01306                 while ( --i >= 0 ) *b++ = *t++;
01307                 if ( size == 0 ) break;
01308                 ADDPOS(fi->POposition,(UBYTE *) (fi->POfull)-(UBYTE *) (fi->PObuffer));
01309                 LOCK(AS.inputslock);
01310                 SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
01311                 retval = ReadFile(fi->handle,(UBYTE *) (fi->PObuffer),fi->POsize);
01312             }

```

```

01313         UNLOCK(AS.inputslock);
01314         fi->POfull = fi->PObuffer+retval/sizeof(WORD);
01315         fi->POfill = fi->PObuffer;
01316         if ( fi != AR.hidefile ) AR.InInBuf = retval/sizeof(WORD);
01317         else                     AR.InHiBuf = retval/sizeof(WORD);
01318         if ( retval == 0 ) { t = fi->POfill; break; }
01319     }
01320 }
01321 }
01322 retval = (UBYTE *)b - buffer;
01323 fi->POfill = t;
01324 ADDPOS(*pos,retval);
01325 return(retval);
01326 }
01327
01328 /*
01329     #] ReadPosFile :
01330     #[ WriteFile :
01331 */
01332
01333 LONG WriteFileToFile(int handle, UBYTE *buffer, LONG size)
01334 {
01335     FILES *f;
01336     LONG retval, totalwritten = 0, stilltowrite;
01337     RWLOCKR(AM.handlelock);
01338     f = filelist[handle];
01339     UNRWLOCK(AM.handlelock);
01340     while ( totalwritten < size ) {
01341         stilltowrite = size - totalwritten;
01342 #ifdef WITHSTATS
01343         numwrites++;
01344 #endif
01345         retval = Uwrite((char *)buffer+totalwritten,1,stilltowrite,f);
01346         if ( retval < 0 ) return(retval);
01347         if ( retval == 0 ) return(totalwritten);
01348         totalwritten += retval;
01349     }
01350 /*
01351 if ( handle == AC.LogHandle || handle == ERROROUT ) FlushFile(handle);
01352 */
01353     return(totalwritten);
01354 }
01355 #ifndef WITHMPI
01356 /*[17nov2005]:*/
01357 WRITEFILE WriteFile = &WriteFileToFile;
01358 /*
01359 LONG (*WriteFile)(int handle, UBYTE *buffer, LONG size) = &WriteFileToFile;
01360 */
01361 /*:[17nov2005]:*/
01362 #else
01363 WRITEFILE WriteFile = &PF_WriteFileToFile;
01364 #endif
01365
01366 /*
01367     #] WriteFile :
01368     #[ SeekFile :
01369 */
01370
01371 void SeekFile(int handle, POSITION *offset, int origin)
01372 {
01373     FILES *f;
01374     RWLOCKR(AM.handlelock);
01375     f = filelist[handle];
01376     UNRWLOCK(AM.handlelock);
01377 #ifdef WITHSTATS
01378     numseeks++;
01379 #endif
01380     if ( origin == SEEK_SET ) {
01381         Useek(f,BASEPOSITION(*offset),origin);
01382         SETBASEPOSITION(*offset,(Utell(f)));
01383         return;
01384     }
01385     else if ( origin == SEEK_END ) {
01386         Useek(f,0,origin);
01387     }
01388     SETBASEPOSITION(*offset,(Utell(f)));
01389 }
01390
01391 /*
01392     #] SeekFile :
01393     #[ TellFile :
01394 */
01395
01396 LONG TellFile(int handle)
01397 {
01398     POSITION pos;
01399     TELLFILE(handle,&pos);

```



```

01400 #ifdef WITHSTATS
01401     numseeks++;
01402 #endif
01403     return (BASEPOSITION(pos));
01404 }
01405
01406 void TELLFILE(int handle, POSITION *position)
01407 {
01408     FILES *f;
01409     RWLOCKR(AM.handlelock);
01410     f = filelist[handle];
01411     UNRWLOCK(AM.handlelock);
01412     SETBASEPOSITION(*position, (Utell(f)));
01413 }
01414
01415 /*
01416     #] TellFile :
01417     #[ FlushFile :
01418 */
01419
01420 void FlushFile(int handle)
01421 {
01422     FILES *f;
01423     RWLOCKR(AM.handlelock);
01424     f = filelist[handle];
01425     UNRWLOCK(AM.handlelock);
01426     Uflush(f);
01427 }
01428
01429 /*
01430     #] FlushFile :
01431     #[ GetPosFile :
01432 */
01433
01434 int GetPosFile(int handle, fpos_t *pospointer)
01435 {
01436     FILES *f;
01437     RWLOCKR(AM.handlelock);
01438     f = filelist[handle];
01439     UNRWLOCK(AM.handlelock);
01440     return (Ugetpos(f, pospointer));
01441 }
01442
01443 /*
01444     #] GetPosFile :
01445     #[ SetPosFile :
01446 */
01447
01448 int SetPosFile(int handle, fpos_t *pospointer)
01449 {
01450     FILES *f;
01451     RWLOCKR(AM.handlelock);
01452     f = filelist[handle];
01453     UNRWLOCK(AM.handlelock);
01454     return (Usetpos(f, (fpos_t *)pospointer));
01455 }
01456
01457 /*
01458     #] SetPosFile :
01459     #[ SynchFile :
01460
01461     It may be that when we use many sort files at the same time there
01462     is a big traffic jam in the cache. This routine is experimental,
01463     just to see whether this improves the situation.
01464     It could also be that the internal disk of the Quad opteron norma
01465     is very slow.
01466 */
01467
01468 void SynchFile(int handle)
01469 {
01470     FILES *f;
01471     if ( handle >= 0 ) {
01472         RWLOCKR(AM.handlelock);
01473         f = filelist[handle];
01474         UNRWLOCK(AM.handlelock);
01475         Usync(f);
01476     }
01477 }
01478
01479 /*
01480     #] SynchFile :
01481     #[ TruncateFile :
01482
01483     It may be that when we use many sort files at the same time there
01484     is a big traffic jam in the cache. This routine is experimental,
01485     just to see whether this improves the situation.
01486     It could also be that the internal disk of the Quad opteron norma

```

```

01487         is very slow.
01488     */
01489
01490 void TruncateFile(int handle)
01491 {
01492     FILES *f;
01493     if ( handle >= 0 ) {
01494         RWLOCKR(AM.handlelock);
01495         f = filelist[handle];
01496         UNRWLOCK(AM.handlelock);
01497         Utruncate(f);
01498     }
01499 }
01500
01501 /*
01502     #] TruncateFile :
01503     #[ GetChannel :
01504
01505     Checks whether we have this file already. If so, we return its
01506     handle. If not and mode == 0, we open the file first and add it
01507     to the buffers.
01508 */
01509
01510 int GetChannel(char *name,int mode)
01511 {
01512     CHANNEL *ch;
01513     int i;
01514     FILES *f;
01515     for ( i = 0; i < NumOutputChannels; i++ ) {
01516         if ( channels[i].name == 0 ) continue;
01517         if ( StrCmp((UBYTE *)name,(UBYTE *) (channels[i].name)) == 0 ) return(channels[i].handle);
01518     }
01519     if ( mode == 1 ) {
01520         MesPrint("&File %s in print statement is not open",name);
01521         MesPrint("    You should open it first with a #write or #append instruction");
01522         return(-1);
01523     }
01524     for ( i = 0; i < NumOutputChannels; i++ ) {
01525         if ( channels[i].name == 0 ) break;
01526     }
01527     if ( i < NumOutputChannels ) { ch = &(channels[i]); }
01528     else { ch = (CHANNEL *)FromList(&AC.Channellist); }
01529     ch->name = (char *)strDup1((UBYTE *)name,"name of channel");
01530     ch->handle = CreateFile(name);
01531     RWLOCKR(AM.handlelock);
01532     f = filelist[ch->handle];
01533     UNRWLOCK(AM.handlelock);
01534     Usetbuf(f,0); /* We turn the buffer off!!!!!!*/
01535     return(ch->handle);
01536 }
01537
01538 /*
01539     #] GetChannel :
01540     #[ GetAppendChannel :
01541
01542     Checks whether we have this file already. If so, we return its
01543     handle. If not, we open the file first and add it to the buffers.
01544 */
01545
01546 int GetAppendChannel(char *name)
01547 {
01548     CHANNEL *ch;
01549     int i;
01550     FILES *f;
01551     for ( i = 0; i < NumOutputChannels; i++ ) {
01552         if ( channels[i].name == 0 ) continue;
01553         if ( StrCmp((UBYTE *)name,(UBYTE *) (channels[i].name)) == 0 ) return(channels[i].handle);
01554     }
01555     for ( i = 0; i < NumOutputChannels; i++ ) {
01556         if ( channels[i].name == 0 ) break;
01557     }
01558     if ( i < NumOutputChannels ) { ch = &(channels[i]); }
01559     else { ch = (CHANNEL *)FromList(&AC.Channellist); }
01560     ch->name = (char *)strDup1((UBYTE *)name,"name of channel");
01561     ch->handle = OpenAddFile(name);
01562     RWLOCKR(AM.handlelock);
01563     f = filelist[ch->handle];
01564     UNRWLOCK(AM.handlelock);
01565     Usetbuf(f,0); /* We turn the buffer off!!!!!!*/
01566     return(ch->handle);
01567 }
01568
01569 /*
01570     #] GetAppendChannel :
01571     #[ CloseChannel :
01572
01573     Checks whether we have this file already. If so, we close it.

```

```

01574 */
01575
01576 int CloseChannel(char *name)
01577 {
01578     int i;
01579     for ( i = 0; i < NumOutputChannels; i++ ) {
01580         if ( channels[i].name == 0 ) continue;
01581         if ( channels[i].name[0] == 0 ) continue;
01582         if ( StrCmp((UBYTE *)name, (UBYTE *) (channels[i].name)) == 0 ) {
01583             CloseFile(channels[i].handle);
01584             M_free(channels[i].name, "CloseChannel");
01585             channels[i].name = 0;
01586             return(0);
01587         }
01588     }
01589     return(0);
01590 }
01591
01592 /*
01593     #] CloseChannel :
01594     #[ UpdateMaxSize :
01595
01596     Updates the maximum size of the combined input/output/hide scratch
01597     files, the sort files and the .str file.
01598     The result becomes only visible with either
01599         ON totalsize;
01600         #: totalsize ON;
01601     or the -T in the command tail.
01602
01603     To be called, whenever a file is closed/removed or truncated to zero.
01604
01605     We have no provisions yet for expressions that remain inside the
01606     small or large buffer during the sort. The space they use there is
01607     currently ignored.
01608 */
01609
01610 void UpdateMaxSize(void)
01611 {
01612     POSITION position, sumsize;
01613     int i;
01614     FILEHANDLE *scr;
01615 #ifdef WITHMPI
01616     /* Currently, it works only on the master. The sort files on the slaves
01617      * are ignored. (TU 11 Oct 2011) */
01618     if ( PF.me != MASTER ) return;
01619 #endif
01620     PUTZERO(sumsize);
01621     if ( AM.PrintTotalSize ) {
01622         /*
01623             First the three scratch files
01624         */
01625 #ifdef WITHPTHREADS
01626         scr = AB[0]->R.Fscr;
01627     #else
01628         scr = AR.Fscr;
01629     #endif
01630         for ( i = 0; i <=2; i++ ) {
01631             if ( scr[i].handle < 0 ) {
01632                 SETBASEPOSITION(position, (scr[i].POfull-scr[i].PObuffer)*sizeof(WORD));
01633             }
01634             else {
01635                 position = scr[i].filesize;
01636             }
01637             ADD2POS(sumsize, position);
01638         }
01639         /*
01640             Now the sort file(s)
01641         */
01642 #ifdef WITHPTHREADS
01643         {
01644             int j;
01645             ALLPRIVATES *B;
01646             for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
01647                 B = AB[j];
01648                 if ( AT.SS && AT.SS->file.handle >= 0 ) {
01649                     position = AT.SS->file.filesize;
01650                 }
01651                 MLOCK(ErrorMessageLock);
01652                 MesPrint("%d: %10p", j, &(AT.SS->file.filesize));
01653                 MUNLOCK(ErrorMessageLock);
01654                 /*
01655                     ADD2POS(sumsize, position);
01656                 */
01657                 if ( AR.FoStage4[0].handle >= 0 ) {
01658                     position = AR.FoStage4[0].filesize;
01659                     ADD2POS(sumsize, position);
01660                 }

```

```

01661     }
01662 }
01663 #else
01664 if ( AT.SS && AT.SS->file.handle >= 0 ) {
01665     position = AT.SS->file.filesize;
01666     ADD2POS(sumsize,position);
01667 }
01668 if ( AR.FoStage4[0].handle >= 0 ) {
01669     position = AR.FoStage4[0].filesize;
01670     ADD2POS(sumsize,position);
01671 }
01672 #endif
01673 /*
01674     And of course the str file.
01675 */
01676 ADD2POS(sumsize,AC.StoreFileSize);
01677 /*
01678     Finally the test whether it is bigger
01679 */
01680 if ( ISLESSPOS(AS.MaxExprSize,sumsize) ) {
01681 #ifdef WITHPTHREADS
01682     LOCK(AS.MaxExprSizeLock);
01683     if ( ISLESSPOS(AS.MaxExprSize,sumsize) ) AS.MaxExprSize = sumsize;
01684     UNLOCK(AS.MaxExprSizeLock);
01685 #else
01686     AS.MaxExprSize = sumsize;
01687 #endif
01688 }
01689 }
01690 return;
01691 }
01692
01693 /*
01694     #] UpdateMaxSize :
01695     #] Files :
01696     #[ Strings :
01697     #[ StrCmp :
01698 */
01699
01700 int StrCmp(UBYTE *s1, UBYTE *s2)
01701 {
01702     while ( *s1 && *s1 == *s2 ) { s1++; s2++; }
01703     return((int)*s1-(int)*s2);
01704 }
01705
01706 /*
01707     #] StrCmp :
01708     #[ StrICmp :
01709 */
01710
01711 int StrICmp(UBYTE *s1, UBYTE *s2)
01712 {
01713     while ( *s1 && tolower(*s1) == tolower(*s2) ) { s1++; s2++; }
01714     return((int)tolower(*s1)-(int)tolower(*s2));
01715 }
01716
01717 /*
01718     #] StrICmp :
01719     #[ StrHICmp :
01720 */
01721
01722 int StrHICmp(UBYTE *s1, UBYTE *s2)
01723 {
01724     while ( *s1 && tolower(*s1) == *s2 ) { s1++; s2++; }
01725     return((int)tolower(*s1)-(int)(*s2));
01726 }
01727
01728 /*
01729     #] StrHICmp :
01730     #[ StrICont :
01731 */
01732
01733 int StrICont(UBYTE *s1, UBYTE *s2)
01734 {
01735     while ( *s1 && tolower(*s1) == tolower(*s2) ) { s1++; s2++; }
01736     if ( *s1 == 0 ) return(0);
01737     return((int)tolower(*s1)-(int)tolower(*s2));
01738 }
01739
01740 /*
01741     #] StrICont :
01742     #[ CmpArray :
01743 */
01744
01745 int CmpArray(WORD *t1, WORD *t2, WORD n)
01746 {
01747     int i,x;

```

```

01748     for ( i = 0; i < n; i++ ) {
01749         if ( ( x = (int)(t1[i]-t2[i]) ) != 0 ) return(x);
01750     }
01751     return(0);
01752 }
01753
01754 /*
01755     #] CmpArray :
01756     #[ ConWord :
01757 */
01758
01759 int ConWord(UBYTE *s1, UBYTE *s2)
01760 {
01761     while ( *s1 && ( tolower(*s1) == tolower(*s2) ) ) { s1++; s2++; }
01762     if ( *s1 == 0 ) return(1);
01763     return(0);
01764 }
01765
01766 /*
01767     #] ConWord :
01768     #[ StrLen :
01769 */
01770
01771 int StrLen(UBYTE *s)
01772 {
01773     int i = 0;
01774     while ( *s ) { s++; i++; }
01775     return(i);
01776 }
01777
01778 /*
01779     #] StrLen :
01780     #[ NumToStr :
01781 */
01782
01783 void NumToStr(UBYTE *s, LONG x)
01784 {
01785     UBYTE *t, str[24];
01786     ULONG xx;
01787     t = str;
01788     if ( x < 0 ) { *s++ = '-'; xx = -x; }
01789     else xx = x;
01790     do {
01791         *t++ = xx % 10 + '0';
01792         xx /= 10;
01793     } while ( xx );
01794     while ( t > str ) *s++ = --t;
01795     *s = 0;
01796 }
01797
01798 /*
01799     #] NumToStr :
01800     #[ WriteString :
01801
01802     Writes a characterstring to the various outputs.
01803     The action may depend on the flags involved.
01804     The type of output is given by type, the string by str and the
01805     number of characters in it by num
01806 */
01807 void WriteString(int type, UBYTE *str, int num)
01808 {
01809     int error = 0;
01810
01811     if ( num > 0 && str[num-1] == 0 ) { num--; }
01812     else if ( num <= 0 || str[num-1] != LINEFEED ) {
01813         AddLineFeed(str,num);
01814     }
01815     /*[15apr2004 mt]:*/
01816     if(type == EXTERNALCHANNELOUT){
01817         if(WriteFile(0,str,num) != num) error = 1;
01818     }else
01819     /*:[15apr2004 mt]:*/
01820     if ( AM.silent == 0 || type == ERROROUT ) {
01821         if ( type == INPUTOUT ) {
01822             if ( !AM.FileOnlyFlag && WriteFile(AM.StdOut, (UBYTE *) "      ",4) != 4 ) error = 1;
01823             if ( AC.LogHandle >= 0 && WriteFile(AC.LogHandle, (UBYTE *) "      ",4) != 4 ) error = 1;
01824         }
01825         if ( !AM.FileOnlyFlag && WriteFile(AM.StdOut,str,num) != num ) error = 1;
01826         if ( AC.LogHandle >= 0 && WriteFile(AC.LogHandle,str,num) != num ) error = 1;
01827     }
01828     if ( error ) Terminate(-1);
01829 }
01830
01831 /*
01832     #] WriteString :
01833     #[ WriteUnfinString :
01834

```

```

01835         Writes a characterstring to the various outputs.
01836         The action may depend on the flags involved.
01837         The type of output is given by type, the string by str and the
01838         number of characters in it by num
01839     */
01840
01841 void WriteUnfinString(int type, UBYTE *str, int num)
01842 {
01843     int error = 0;
01844
01845     /*[15apr2004 mt]:*/
01846     if(type == EXTERNALCHANNELOUT){
01847         if(WriteFile(0,str,num) != num) error = 1;
01848     }else
01849     /*:[15apr2004 mt]*/
01850     if ( AM.silent == 0 || type == ERROROUT ) {
01851         if ( type == INPUTOUT ) {
01852             if ( !AM.FileOnlyFlag && WriteFile(AM.Stdout,(UBYTE *)"      ",4) != 4 ) error = 1;
01853             if ( AC.LogHandle >= 0 && WriteFile(AC.LogHandle,(UBYTE *)"      ",4) != 4 ) error = 1;
01854         }
01855         if ( !AM.FileOnlyFlag && WriteFile(AM.Stdout,str,num) != num ) error = 1;
01856         if ( AC.LogHandle >= 0 && WriteFile(AC.LogHandle,str,num) != num ) error = 1;
01857     }
01858     if ( error ) Terminate(-1);
01859 }
01860
01861 /*
01862     #] WriteUnfinString :
01863     #[ AddToString :
01864 */
01865
01866 UBYTE *AddToString(UBYTE *outstring, UBYTE *extrastring, int par)
01867 {
01868     UBYTE *s = extrastring, *t, *newstring;
01869     int n, nn;
01870     while ( *s ) { s++; }
01871     n = s-extrastring;
01872     if ( outstring == 0 ) {
01873         s = extrastring;
01874         t = outstring = (UBYTE *)Malloca1(n+1,"AddToString");
01875         NCOPY(t,s,n)
01876         *t++ = 0;
01877         return(outstring);
01878     }
01879     else {
01880         t = outstring;
01881         while ( *t ) t++;
01882         nn = t - outstring;
01883         t = newstring = (UBYTE *)Malloca1(n+nn+2,"AddToString");
01884         s = outstring;
01885         NCOPY(t,s,nn)
01886         if ( par == 1 ) *t++ = ',';
01887         s = extrastring;
01888         NCOPY(t,s,n)
01889         *t = 0;
01890         M_free(outstring,"AddToString");
01891         return(newstring);
01892     }
01893 }
01894
01895 /*
01896     #] AddToString :
01897     #[ strDupl :
01898
01899     string duplication with message passing for Malloca1, allowing
01900     this routine to give a more detailed error message if there
01901     is not enough memory.
01902 */
01903
01904 UBYTE *strDupl(UBYTE *instring, char *ifwrong)
01905 {
01906     UBYTE *s = instring, *to;
01907     while ( *s ) s++;
01908     to = s = (UBYTE *)Malloca1((s-instring)+1,ifwrong);
01909     while ( *instring ) *to++ = *instring++;
01910     *to = 0;
01911     return(s);
01912 }
01913
01914 /*
01915     #] strDupl :
01916     #[ EndOfToken :
01917 */
01918
01919 UBYTE *EndOfToken(UBYTE *s)
01920 {
01921     UBYTE c;

```

```

01922     while ( ( c = (BYTE)(FG.cTable[*s]) ) == 0 || c == 1 ) s++;
01923     return(s);
01924 }
01925
01926 /*
01927     #] EndOfToken :
01928     #[ ToToken :
01929 */
01930
01931 BYTE *ToToken(BYTE *s)
01932 {
01933     BYTE c;
01934     while ( *s && ( c = (BYTE)(FG.cTable[*s]) ) != 0 && c != 1 ) s++;
01935     return(s);
01936 }
01937
01938 /*
01939     #] ToToken :
01940     #[ SkipField :
01941
01942     Skips from s to the end of a declaration field.
01943     par is the number of parentheses that still has to be closed.
01944 */
01945
01946 BYTE *SkipField(BYTE *s, int level)
01947 {
01948     while ( *s ) {
01949         if ( *s == ',' && level == 0 ) return(s);
01950         if ( *s == '(' ) level++;
01951         else if ( *s == ')' ) { level--; if ( level < 0 ) level = 0; }
01952         else if ( *s == '[' ) {
01953             SKIPBRA1(s)
01954         }
01955         else if ( *s == '{' ) {
01956             SKIPBRA2(s)
01957         }
01958         s++;
01959     }
01960     return(s);
01961 }
01962
01963 /*
01964     #] SkipField :
01965     #[ ReadSnum :          WORD ReadSnum(p)
01966
01967     Reads a number that should fit in a word.
01968     The number should be unsigned and a negative return value
01969     indicates an irregularity.
01970
01971 */
01972
01973 WORD ReadSnum(BYTE **p)
01974 {
01975     LONG x = 0;
01976     BYTE *s;
01977     s = *p;
01978     if ( FG.cTable[*s] == 1 ) {
01979         do {
01980             x = ( x << 3 ) + ( x << 1 ) + ( *s++ - '0' );
01981             if ( x > MAXPOSITIVE ) return(-1);
01982         } while ( FG.cTable[*s] == 1 );
01983         *p = s;
01984         return((WORD)x);
01985     }
01986     else return(-1);
01987 }
01988
01989 /*
01990     #] ReadSnum :
01991     #[ NumCopy :
01992
01993     Adds the decimal representation of a number to a string.
01994
01995 */
01996
01997 BYTE *NumCopy(WORD y, BYTE *to)
01998 {
01999     BYTE *s;
02000     WORD i = 0, j;
02001     WORD x;
02002     if ( y < 0 ) {
02003         *to++ = '-';
02004     }
02005     x = WordAbs(y);
02006     s = to;
02007     do { *s++ = (BYTE)((x % 10) + '0'); i++; } while ( ( x /= 10 ) != 0 );
02008     *s-- = '\0';

```

```

02009     j = ( i - 1 ) >> 1;
02010     while ( j >= 0 ) {
02011         i = to[j]; to[j] = s[-j]; s[-j] = (UBYTE)i; j--;
02012     }
02013     return(s+1);
02014 }
02015
02016 /*
02017     #] NumCopy :
02018     #[ LongCopy :
02019
02020     Adds the decimal representation of a number to a string.
02021
02022 */
02023
02024 char *LongCopy(LONG y, char *to)
02025 {
02026     char *s;
02027     WORD i = 0, j;
02028     ULONG x;
02029     if ( y < 0 ) {
02030         *to++ = '-';
02031     }
02032     x = LongAbs(y);
02033     s = to;
02034     do { *s++ = (x % 10) + '0'; i++; } while ( ( x /= 10 ) != 0 );
02035     *s-- = '\0';
02036     j = ( i - 1 ) >> 1;
02037     while ( j >= 0 ) {
02038         i = to[j]; to[j] = s[-j]; s[-j] = (char)i; j--;
02039     }
02040     return(s+1);
02041 }
02042
02043 /*
02044     #] LongCopy :
02045     #[ LongLongCopy :
02046
02047     Adds the decimal representation of a number to a string.
02048     Bugfix feb 2003. y was not pointer!
02049 */
02050
02051 char *LongLongCopy(off_t *y, char *to)
02052 {
02053     /*
02054      * This code fails to print the maximum negative value on systems with two's
02055      * complement. To fix this, we need the unsigned version of off_t with the
02056      * same size, but unfortunately it is undefined. On the other hand, if a
02057      * system is configured with a 64-bit off_t, in practice one never reaches
02058      * 2^63 ~ 10^18 as of 2016. If one really reach such a big number, then it
02059      * would be the time to move on a 128-bit off_t.
02060      */
02061     off_t x = *y;
02062     char *s;
02063     WORD i = 0, j;
02064     if ( x < 0 ) { x = -x; *to++ = '-'; }
02065     s = to;
02066     do { *s++ = (x % 10) + '0'; i++; } while ( ( x /= 10 ) != 0 );
02067     *s-- = '\0';
02068     j = ( i - 1 ) >> 1;
02069     while ( j >= 0 ) {
02070         i = to[j]; to[j] = s[-j]; s[-j] = (char)i; j--;
02071     }
02072     return(s+1);
02073 }
02074
02075 /*
02076     #] LongLongCopy :
02077     #[ MakeDate :
02078
02079     Routine produces a string with the date and time of the run
02080 */
02081
02082 #ifdef ANSI
02083 #else
02084 #ifdef mBSD
02085 #else
02086 static char notime[] = "";
02087 #endif
02088 #endif
02089
02090 UBYTE *MakeDate(void)
02091 {
02092 #ifdef ANSI
02093     time_t tp;
02094     time(&tp);
02095     return((UBYTE *)ctime(&tp));

```



```

02096 #else
02097 #ifdef mBSD
02098     time_t tp;
02099     time(&tp);
02100     return((UBYTE *)ctime(&tp));
02101 #else
02102     return((UBYTE *)notime);
02103 #endif
02104 #endif
02105 }
02106
02107 /*
02108     #] MakeDate :
02109     #[ set_in :
02110         Returns 1 if ch is in set ; 0 if ch is not in set:
02111 */
02112 int set_in(UBYTE ch, set_of_char set)
02113 {
02114     set += ch/8;
02115     switch (ch % 8){
02116         case 0: return(set->bit_0);
02117         case 1: return(set->bit_1);
02118         case 2: return(set->bit_2);
02119         case 3: return(set->bit_3);
02120         case 4: return(set->bit_4);
02121         case 5: return(set->bit_5);
02122         case 6: return(set->bit_6);
02123         case 7: return(set->bit_7);
02124     }/*switch (ch % 8)*/
02125     return(-1);
02126 }/*set_in*/
02127
02128     #] set_in :
02129     #[ set_set :
02130         sets ch into set; returns *set:
02131 */
02132 one_byte set_set(UBYTE ch, set_of_char set)
02133 {
02134     one_byte tmp=(one_byte)set;
02135     set += ch/8;
02136     switch (ch % 8){
02137         case 0: set->bit_0=1;break;
02138         case 1: set->bit_1=1;break;
02139         case 2: set->bit_2=1;break;
02140         case 3: set->bit_3=1;break;
02141         case 4: set->bit_4=1;break;
02142         case 5: set->bit_5=1;break;
02143         case 6: set->bit_6=1;break;
02144         case 7: set->bit_7=1;break;
02145     }
02146     return(tmp);
02147 }/*set_set*/
02148
02149     #] set_set :
02150     #[ set_del :
02151         deletes ch from set; returns *set:
02152 */
02153 one_byte set_del(UBYTE ch, set_of_char set)
02154 {
02155     one_byte tmp=(one_byte)set;
02156     set += ch/8;
02157     switch (ch % 8){
02158         case 0: set->bit_0=0;break;
02159         case 1: set->bit_1=0;break;
02160         case 2: set->bit_2=0;break;
02161         case 3: set->bit_3=0;break;
02162         case 4: set->bit_4=0;break;
02163         case 5: set->bit_5=0;break;
02164         case 6: set->bit_6=0;break;
02165         case 7: set->bit_7=0;break;
02166     }
02167     return(tmp);
02168 }/*set_del*/
02169
02170     #] set_del :
02171     #[ set_sub :
02172         returns *set = set1\set2. This function may be used for initialising,
02173         set_sub(a,a,a) => now a is empty set :
02174 */
02175 one_byte set_sub(set_of_char set, set_of_char set1, set_of_char set2)
02176 {
02177     one_byte tmp=(one_byte)set;
02178     int i=0,j=0;
02179     while(j=0,i++<32)
02180     while(j<9)
02181         switch (j++){
02182             case 0: set->bit_0=(set1->bit_0&&(!set2->bit_0));break;

```

```

02183         case 1: set->bit_1=(set1->bit_1&&!set2->bit_1));break;
02184         case 2: set->bit_2=(set1->bit_2&&!set2->bit_2));break;
02185         case 3: set->bit_3=(set1->bit_3&&!set2->bit_3));break;
02186         case 4: set->bit_4=(set1->bit_4&&!set2->bit_4));break;
02187         case 5: set->bit_5=(set1->bit_5&&!set2->bit_5));break;
02188         case 6: set->bit_6=(set1->bit_6&&!set2->bit_6));break;
02189         case 7: set->bit_7=(set1->bit_7&&!set2->bit_7));break;
02190         case 8: set++;set1++;set2++;
02191     };
02192     return(tmp);
02193 }/*set_sub*/
02194 /*
02195     #] set_sub :
02196     #] Strings :
02197     #[ Mixed :
02198     #[ iniTools :
02199 */
02200
02201 void iniTools(void)
02202 {
02203     #ifdef MALLOCPROTECT
02204         if ( mprotectInit() ) exit(0);
02205     #endif
02206     return;
02207 }
02208
02209 /*
02210     #] iniTools :
02211     #[ Malloc1 :
02212
02213     Malloc routine with built in error checking,
02214     and a more detailed error message.
02215     Gives the user some idea of what is happening.
02216 */
02217
02218 #ifdef MALLOCDEBUG
02219 INILOCK(MallocLock)
02220 #endif
02221
02222 void *Malloc1(LONG size, const char *messageifwrong)
02223 {
02224     void *mem;
02225     #ifdef MALLOCDEBUG
02226         char *t, *u;
02227         int i;
02228         LOCK(MallocLock);
02229         /* MLOCK(ErrorMessageLock); */
02230         if ( size == 0 ) {
02231             MesPrint("%wAsking for 0 bytes in Malloc1");
02232         }
02233     #endif
02234     #ifdef WITHSTATS
02235         nummalloccs++;
02236     #endif
02237     if ( ( size & 7 ) != 0 ) { size = size - ( size&7 ) + 8; }
02238     #ifdef MALLOCDEBUG
02239         size += 2*BANNER;
02240     #endif
02241     mem = (void *)M_alloc(size);
02242     if ( mem == 0 ) {
02243         #ifndef MALLOCDEBUG
02244             MLOCK(ErrorMessageLock);
02245         #endif
02246         MesPrint("Attempted to allocate %l bytes.", size);
02247         Error1("No memory while allocating ",(UBYTE *)messageifwrong);
02248         #ifndef MALLOCDEBUG
02249             MUNLOCK(ErrorMessageLock);
02250         #else
02251             /* MUNLOCK(ErrorMessageLock); */
02252         #endif
02253         #ifdef MALLOCDEBUG
02254             UNLOCK(MallocLock);
02255         #endif
02256         Terminate(-1);
02257     }
02258     #ifdef MALLOCDEBUG
02259         malloccsizes[nummalloclist] = size;
02260         malloccstrings[nummalloclist] = (char *)messageifwrong;
02261         malloclist[nummalloclist++] = mem;
02262         if ( AC.MemDebugFlag && filelist ) MesPrint("%wMem1 at 0x%x: %l bytes.
02263 %s",mem,size,messageifwrong);
02264         {
02265             int i = nummalloclist-1;
02266             while ( --i >= 0 ) {
02267                 if ( (char *)mem < (((char *)malloclist[i]) + malloccsizes[i])
02268                     && (char *)malloclist[i] < ((char *)mem + size) ) {
02269                     if ( filelist ) MesPrint("This memory overlaps with the block at 0x%x"

```

```

02269             ,malloclist[i]);
02270         }
02271     }
02272 }
02273
02274 #ifdef MALLOCDDEBUGOUTPUT
02275     printf ("Malloc1: %s, allocated %li bytes at %.8lx\n",messageifwrong,size,(unsigned long)mem);
02276     fflush (stdout);
02277 #endif
02278
02279     t = (char *)mem;
02280     u = t + size;
02281     for ( i = 0; i < (int)BANNER; i++ ) { *t++ = FILLVALUE; *--u = FILLVALUE; }
02282     mem = (void *)t;
02283     M_check();
02284     /* MUNLOCK(ErrorMessageLock); */
02285     UNLOCK(MallocLock);
02286 #endif
02287 /*
02288     if ( size > 500000000L ) {
02289         MLOCK(ErrorMessageLock);
02290         MesPrint("Malloc1: %s, allocated %l bytes\n",messageifwrong,size);
02291         MUNLOCK(ErrorMessageLock);
02292     }
02293 */
02294     return(mem);
02295 }
02296
02297 /*
02298     #] Malloc1 :
02299     #[ M_free :
02300 */
02301
02302 void M_free(void *x, const char *where)
02303 {
02304     #ifdef MALLOCDDEBUG
02305         char *t = (char *)x;
02306         int i, j, k;
02307         LONG size = 0;
02308         x = (void *)(((char *)x)-BANNER);
02309         /* MLOCK(ErrorMessageLock); */
02310         if ( AC.MemDebugFlag ) MesPrint("%wFreeing 0x%x: %s",x,where);
02311         LOCK(MallocLock);
02312         for ( i = nummalloclist-1; i >= 0; i-- ) {
02313             if ( x == malloclist[i] ) {
02314                 size = mallocsizes[i];
02315                 for ( j = i+1; j < nummalloclist; j++ ) {
02316                     malloclist[j-1] = malloclist[j];
02317                     mallocsizes[j-1] = mallocsizes[j];
02318                     mallocstrings[j-1] = mallocstrings[j];
02319                 }
02320                 nummalloclist--;
02321                 break;
02322             }
02323         }
02324         if ( i < 0 ) {
02325             unsigned int xx = ((ULONG)x);
02326             printf("Error returning non-allocated address: 0x%x from %s\n"
02327                 ,xx,where);
02328             /* MUNLOCK(ErrorMessageLock); */
02329             UNLOCK(MallocLock);
02330             exit(-1);
02331         }
02332         else {
02333             for ( k = 0, j = 0; k < (int)BANNER; k++ ) {
02334                 if ( *--t != FILLVALUE ) j++;
02335             }
02336             if ( j ) {
02337                 LONG *tt = (LONG *)x;
02338                 MesPrint("%w!!!! Banner has been written in !!!!!: %x %x %x %x",
02339                     tt[0],tt[1],tt[2],tt[3]);
02340             }
02341             t += size;
02342             for ( k = 0, j = 0; k < (int)BANNER; k++ ) {
02343                 if ( *--t != FILLVALUE ) j++;
02344             }
02345             if ( j ) {
02346                 LONG *tt = (LONG *)x;
02347                 MesPrint("%w!!!! Tail has been written in !!!!!: %x %x %x %x",
02348                     tt[0],tt[1],tt[2],tt[3]);
02349             }
02350             M_check();
02351             /* MUNLOCK(ErrorMessageLock); */
02352             UNLOCK(MallocLock);
02353         }
02354     #else
02355         DUMMYUSE(where);

```

```

02356 #endif
02357 #ifdef WITHSTATS
02358     numfrees++;
02359 #endif
02360     if ( x ) {
02361 #ifdef MALLOCDEBUGOUTPUT
02362         printf ("M_free: %s, memory freed at %.8lx\n",where,(unsigned long)x);
02363         fflush(stdout);
02364 #endif
02365
02366 #ifdef MALLOCPROTECT
02367         mprotectFree((void *)x);
02368 #else
02369         free(x);
02370 #endif
02371     }
02372 }
02373
02374 /*
02375     #] M_free :
02376     #[ M_check :
02377 */
02378
02379 #ifdef MALLOCDEBUG
02380
02381 void M_check1() { MesPrint("Checking Malloc"); M_check(); }
02382
02383 void M_check()
02384 {
02385     int i,j,k,error = 0;
02386     char *t;
02387     LONG *tt;
02388     for ( i = 0; i < nummalloclist; i++ ) {
02389         t = (char *) (malloclist[i]);
02390         for ( k = 0, j = 0; k < (int)BANNER; k++ ) {
02391             if ( *t++ != FILLVALUE ) j++;
02392         }
02393         if ( j ) {
02394             tt = (LONG *) (malloclist[i]);
02395             MesPrint("%w!!!! Banner %d (%s) has been written in !!!!!: %x %x %x %x",
02396                 i,mallocstrings[i],tt[0],tt[1],tt[2],tt[3]);
02397             tt[0] = tt[1] = tt[2] = tt[3] = 0;
02398             error = 1;
02399         }
02400         t = (char *) (malloclist[i]) + malloccsizes[i];
02401         for ( k = 0, j = 0; k < (int)BANNER; k++ ) {
02402             if ( *--t != FILLVALUE ) j++;
02403         }
02404         if ( j ) {
02405             tt = (LONG *)t;
02406             MesPrint("%w!!!! Tail %d (%s) has been written in !!!!!: %x %x %x %x",
02407                 i,mallocstrings[i],tt[0],tt[1],tt[2],tt[3]);
02408             tt[0] = tt[1] = tt[2] = tt[3] = 0;
02409             error = 1;
02410         }
02411         if ( ( mallocstrings[i][0] == ' ' ) || ( mallocstrings[i][0] == '#' ) ) {
02412             MesPrint("%w!!!! Funny mallocstring");
02413             error = 1;
02414         }
02415     }
02416     if ( error ) {
02417         M_print();
02418         /* MUNLOCK(ErrorMessageLock); */
02419         UNLOCK(MallocLock);
02420         Terminate(-1);
02421     }
02422 }
02423
02424 void M_print()
02425 {
02426     int i;
02427     MesPrint("We have the following memory allocations left:");
02428     for ( i = 0; i < nummalloclist; i++ ) {
02429         MesPrint("0x%x: %l bytes. number %d: '%s'",malloclist[i],malloccsizes[i],i,mallocstrings[i]);
02430     }
02431 }
02432
02433 #else
02434
02435 void M_check1(void) {}
02436 void M_print(void) {}
02437
02438 #endif
02439
02440 /*
02441     #] M_check :
02442     #[ TermMalloc :

```

```

02443 */
02466 #define TERMMEMSTARTNUM 16
02467 #define TERMEXTRAWORDS 10
02468
02469 void TermMallocAddMemory(PHEAD0)
02470 {
02471     WORD *newbufs;
02472     int i, extra;
02473     if ( AT.TermMemMax == 0 ) extra = TERMMEMSTARTNUM;
02474     else extra = AT.TermMemMax;
02475     if ( AT.TermMemHeap ) M_free(AT.TermMemHeap,"TermMalloc");
02476     newbufs = (WORD *)Malloc1(extra*(AM.MaxTer+TERMEXTRAWORDS*sizeof(WORD)),"TermMalloc");
02477     AT.TermMemHeap = (WORD **)Malloc1((extra+AT.TermMemMax)*sizeof(WORD *),"TermMalloc");
02478     for ( i = 0; i < extra; i++ ) {
02479         AT.TermMemHeap[i] = newbufs + i*(AM.MaxTer/sizeof(WORD)+TERMEXTRAWORDS);
02480     }
02481 #ifdef TERMMALLOCDEBUG
02482     DebugHeap2 = (WORD **)Malloc1((extra+AT.TermMemMax)*sizeof(WORD *),"TermMalloc");
02483     for ( i = 0; i < AT.TermMemMax; i++ ) { DebugHeap2[i] = DebugHeap1[i]; }
02484     for ( i = 0; i < extra; i++ ) {
02485         DebugHeap2[i+AT.TermMemMax] = newbufs + i*(AM.MaxTer/sizeof(WORD)+TERMEXTRAWORDS);
02486     }
02487     if ( DebugHeap1 ) M_free(DebugHeap1,"TermMalloc");
02488     DebugHeap1 = DebugHeap2;
02489 #endif
02490     AT.TermMemTop = extra;
02491     AT.TermMemMax += extra;
02492 #ifdef TERMMALLOCDEBUG
02493     MesPrint("AT.TermMemMax is now %l",AT.TermMemMax);
02494 #endif
02495 }
02496
02497 #ifndef MEMORYMACROS
02498
02499 WORD *TermMalloc2(PHEAD char *text)
02500 {
02501     if ( AT.TermMemTop <= 0 ) TermMallocAddMemory(BHEAD0);
02502 #ifdef TERMMALLOCDEBUG
02503     MesPrint("TermMalloc: %s, %d",text,(AT.TermMemMax-AT.TermMemTop));
02504 #endif
02505 #ifdef MALLOCDEBUGOUTPUT
02506     MesPrint("TermMalloc: %s, %l/%l (%x)",text,AT.TermMemTop,AT.TermMemMax,AT.TermMemHeap[AT.TermMemTop-1]);
02507 #endif
02508     DUMMYUSE(text);
02509     return(AT.TermMemHeap[--AT.TermMemTop]);
02510 }
02511
02512 void TermFree2(PHEAD WORD *TermMem, char *text)
02513 {
02514 #ifdef TERMMALLOCDEBUG
02515     int i;
02516     for ( i = 0; i < AT.TermMemMax; i++ ) {
02517         if ( TermMem == DebugHeap1[i] ) break;
02518     }
02519     if ( i >= AT.TermMemMax ) {
02520         MesPrint(" ERROR: TermFree called with an address not given by TermMalloc.");
02521         Terminate(-1);
02522     }
02523 #endif
02524     DUMMYUSE(text);
02525     AT.TermMemHeap[AT.TermMemTop++] = TermMem;
02526 #ifdef TERMMALLOCDEBUG
02527     MesPrint("TermFree: %s, %d",text,(AT.TermMemMax-AT.TermMemTop));
02528 #endif
02529 #ifdef MALLOCDEBUGOUTPUT
02530     MesPrint("TermFree: %s, %l/%l (%x)",text,AT.TermMemTop,AT.TermMemMax,TermMem);
02531 #endif
02532 }
02533 #endif
02534
02535 /*
02536 #] TermMalloc :
02537 #[ NumberMalloc :
02538 */
02539 #define NUMBERMEMSTARTNUM 16
02540 #define NUMBEREXTRAWORDS 10L
02541
02542 #ifdef TERMMALLOCDEBUG
02543 UWORD **DebugHeap3, **DebugHeap4;

```

```

02571 #endif
02572
02573 void NumberMallocAddMemory(PHEAD0)
02574 {
02575     UWORD *newbufs;
02576     WORD extra;
02577     int i;
02578     if ( AT.NumberMemMax == 0 ) extra = NUMBERMEMSTARTNUM;
02579     else extra = AT.NumberMemMax;
02580     if ( AT.NumberMemHeap ) M_free(AT.NumberMemHeap, "NumberMalloc");
02581     newbufs = (UWORD *)Malloc1(extra*(AM.MaxTal+NUMBEREXTRAWORDS)*sizeof(UWORD), "NumberMalloc");
02582     AT.NumberMemHeap = (UWORD **)Malloc1((extra+AT.NumberMemMax)*sizeof(UWORD *), "NumberMalloc");
02583     for ( i = 0; i < extra; i++ ) {
02584         AT.NumberMemHeap[i] = newbufs + i*(LONG) (AM.MaxTal+NUMBEREXTRAWORDS);
02585     }
02586 #ifdef TERMMALLOCDEBUG
02587     DebugHeap4 = (UWORD **)Malloc1((extra+AT.NumberMemMax)*sizeof(WORD *), "NumberMalloc");
02588     for ( i = 0; i < AT.NumberMemMax; i++ ) { DebugHeap4[i] = DebugHeap3[i]; }
02589     for ( i = 0; i < extra; i++ ) {
02590         DebugHeap4[i+AT.NumberMemMax] = newbufs + i*(LONG) (AM.MaxTal+NUMBEREXTRAWORDS);
02591     }
02592     if ( DebugHeap3 ) M_free(DebugHeap3, "NumberMalloc");
02593     DebugHeap3 = DebugHeap4;
02594 #endif
02595     AT.NumberMemTop = extra;
02596     AT.NumberMemMax += extra;
02597     /*
02598     MesPrint("AT.NumberMemMax is now %l", AT.NumberMemMax);
02599     */
02600 }
02601
02602 #ifndef MEMORYMACROS
02603
02604 UWORD *NumberMalloc2(PHEAD char *text)
02605 {
02606     if ( AT.NumberMemTop <= 0 ) NumberMallocAddMemory(BHEAD text);
02607
02608 #ifdef MALLOCDEBUGOUTPUT
02609     if ( (AT.NumberMemMax-AT.NumberMemTop) > 10 )
02610         MesPrint("NumberMalloc: %s, %l/%l\n", text, AT.NumberMemTop, AT.NumberMemMax, AT.NumberMemHeap[AT.NumberMemTop-1]);
02611 #endif
02612     DUMMYUSE(text);
02613     return (AT.NumberMemHeap[--AT.NumberMemTop]);
02614 }
02615
02616 void NumberFree2(PHEAD UWORD *NumberMem, char *text)
02617 {
02618 #ifdef TERMMALLOCDEBUG
02619     int i;
02620     for ( i = 0; i < AT.NumberMemMax; i++ ) {
02621         if ( NumberMem == DebugHeap3[i] ) break;
02622     }
02623     if ( i >= AT.NumberMemMax ) {
02624         MesPrint(" ERROR: NumberFree called with an address not given by NumberMalloc.");
02625         Terminate(-1);
02626     }
02627 #endif
02628     DUMMYUSE(text);
02629     AT.NumberMemHeap[AT.NumberMemTop++] = NumberMem;
02630
02631 #ifdef MALLOCDEBUGOUTPUT
02632     if ( (AT.NumberMemMax-AT.NumberMemTop) > 10 )
02633         MesPrint("NumberFree: %s, %l/%l (%x)", text, AT.NumberMemTop, AT.NumberMemMax, NumberMem);
02634 #endif
02635 }
02636
02637 #endif
02638
02639 /*
02640 #] NumberMalloc :
02641 #[ CacheNumberMalloc :
02642
02643     Similar to NumberMalloc
02644 */
02645
02646 void CacheNumberMallocAddMemory(PHEAD0)
02647 {
02648     UWORD *newbufs;
02649     WORD extra;
02650     int i;
02651     if ( AT.CacheNumberMemMax == 0 ) extra = NUMBERMEMSTARTNUM;
02652     else extra = AT.CacheNumberMemMax;
02653     if ( AT.CacheNumberMemHeap ) M_free(AT.CacheNumberMemHeap, "NumberMalloc");
02654     newbufs = (UWORD *)Malloc1(extra*(AM.MaxTal+NUMBEREXTRAWORDS)*sizeof(UWORD), "CacheNumberMalloc");
02655     AT.CacheNumberMemHeap = (UWORD **)Malloc1((extra+AT.NumberMemMax)*sizeof(UWORD

```

```

    *), "CacheNumberMalloc");
02657     for ( i = 0; i < extra; i++ ) {
02658         AT.CacheNumberMemHeap[i] = newbufs + i*(LONG)(AM.MaxTal+NUMBEREXTRAWORDS);
02659     }
02660     AT.CacheNumberMemTop = extra;
02661     AT.CacheNumberMemMax += extra;
02662 }
02663
02664 #ifndef MEMORYMACROS
02665
02666 UWORD *CacheNumberMalloc2(PHEAD char *text)
02667 {
02668     if ( AT.CacheNumberMemTop <= 0 ) CacheNumberMallocAddMemory(BHEAD0);
02669
02670 #ifdef MALLOCDEBUGOUTPUT
02671     MesPrint("NumberMalloc: %s, %l/%l
02672             (%x)",text,AT.NumberMemTop,AT.NumberMemMax,AT.NumberMemHeap[AT.NumberMemTop-1]);
02673 #endif
02674     DUMMYUSE(text);
02675     return(AT.CacheNumberMemHeap[--AT.CacheNumberMemTop]);
02676 }
02677
02678 void CacheNumberFree2(PHEAD UWORD *NumberMem, char *text)
02679 {
02680     DUMMYUSE(text);
02681     AT.CacheNumberMemHeap[AT.CacheNumberMemTop++] = NumberMem;
02682
02683 #ifdef MALLOCDEBUGOUTPUT
02684     MesPrint("NumberFree: %s, %l/%l (%x)",text,AT.NumberMemTop,AT.NumberMemMax,NumberMem);
02685 #endif
02686 }
02687
02688 #endif
02689
02690 /*
02691     #] CacheNumberMalloc :
02692     #[ FromList :
02693
02694     Returns the next object in a list.
02695     If the list has been exhausted we double it (like a realloc)
02696     If the list has not been initialized yet we start with 12 elements.
02697 */
02698
02699 void *FromList(LIST *L)
02700 {
02701     void *newlist;
02702     int i, *old, *newL;
02703     if ( L->num >= L->maxnum || L->lijst == 0 ) {
02704         if ( L->maxnum == 0 ) L->maxnum = 12;
02705         else if ( L->lijst ) L->maxnum *= 2;
02706         newlist = Malloc1(L->maxnum * L->size,L->message);
02707         if ( L->lijst ) {
02708             i = ( L->num * L->size ) / sizeof(int);
02709             old = (int *)L->lijst; newL = (int *)newlist;
02710             while ( --i >= 0 ) *newL++ = *old++;
02711             if ( L->lijst ) M_free(L->lijst,"L->lijst FromList");
02712         }
02713         L->lijst = newlist;
02714     }
02715     return( ((char *) (L->lijst)) + L->size * (L->num)++ );
02716 }
02717
02718 /*
02719     #] FromList :
02720     #[ From0List :
02721
02722     Same as FromList, but we zero excess variables.
02723 */
02724
02725 void *From0List(LIST *L)
02726 {
02727     void *newlist;
02728     int i, *old, *newL;
02729     if ( L->num >= L->maxnum || L->lijst == 0 ) {
02730         if ( L->maxnum == 0 ) L->maxnum = 12;
02731         else if ( L->lijst ) L->maxnum *= 2;
02732         newlist = Malloc1(L->maxnum * L->size,L->message);
02733         i = ( L->num * L->size ) / sizeof(int);
02734         old = (int *) (L->lijst); newL = (int *) newlist;
02735         while ( --i >= 0 ) *newL++ = *old++;
02736         i = ( L->maxnum - L->num ) / sizeof(int);
02737         while ( --i >= 0 ) *newL++ = 0;
02738         if ( L->lijst ) M_free(L->lijst,"L->lijst From0List");
02739         L->lijst = newlist;
02740     }
02741     return( ((char *) (L->lijst)) + L->size * (L->num)++ );

```

```

02742 }
02743
02744 /*
02745     #] From0List :
02746     #[ FromVarList :
02747
02748     Returns the next object in a list of variables.
02749     If the list has been exhausted we double it (like a realloc)
02750     If the list has not been initialized yet we start with 12 elements.
02751     We allow at most MAXVARIABLES elements!
02752 */
02753
02754 void *FromVarList(LIST *L)
02755 {
02756     void *newlist;
02757     int i, *old, *newL;
02758     if ( L->num >= L->maxnum || L->lijst == 0 ) {
02759         if ( L->maxnum == 0 ) L->maxnum = 12;
02760         else if ( L->lijst ) {
02761             L->maxnum *= 2;
02762             if ( L == &(AP.DollarList) ) {
02763                 if ( L->maxnum > MAXDOLLARVARIABLES ) L->maxnum = MAXDOLLARVARIABLES;
02764                 if ( L->num >= MAXDOLLARVARIABLES ) {
02765                     MesPrint("!!!More than %1 objects in list of $-variables",
02766                             MAXDOLLARVARIABLES);
02767                     Terminate(-1);
02768                 }
02769             }
02770             else {
02771                 if ( L->maxnum > MAXVARIABLES ) L->maxnum = MAXVARIABLES;
02772                 if ( L->num >= MAXVARIABLES ) {
02773                     MesPrint("!!!More than %1 objects in list of variables",
02774                             MAXVARIABLES);
02775                     Terminate(-1);
02776                 }
02777             }
02778             newlist = Malloc1(L->maxnum * L->size, L->message);
02779             if ( L->lijst ) {
02780                 i = ( L->num * L->size ) / sizeof(int);
02781                 old = (int *) (L->lijst); newL = (int *) newlist;
02782                 while ( --i >= 0 ) *newL++ = *old++;
02783                 if ( L->lijst ) M_free(L->lijst, "L->lijst from VarList");
02784             }
02785             L->lijst = newlist;
02786         }
02787         return( ((char *) (L->lijst)) + L->size * ((L->num)++) );
02788     }
02789 }
02790
02791 /*
02792     #] FromVarList :
02793     #[ DoubleList :
02794 */
02795
02796 int DoubleList(void ***lijst, int *oldsize, int objectsize, char *nameoftype)
02797 {
02798     void **newlist;
02799     LONG i, newsize, fullsize;
02800     void **to, **from;
02801     static LONG maxlistsize = (LONG) (MAXPOSITIVE);
02802     if ( *lijst == 0 ) {
02803         if ( *oldsize > 0 ) newsize = *oldsize;
02804         else newsize = 100;
02805     }
02806     else newsize = *oldsize * 2;
02807     if ( newsize > maxlistsize ) {
02808         if ( *oldsize == maxlistsize ) {
02809             MesPrint("No memory for extra space in %s", nameoftype);
02810             return(-1);
02811         }
02812         newsize = maxlistsize;
02813     }
02814     fullsize = ( newsize * objectsize + sizeof(void *) - 1 ) & (~sizeof(void *));
02815     newlist = (void **) Malloc1(fullsize, nameoftype);
02816     if ( *lijst ) { /* Now some punning. DANGEROUS CODE in principle */
02817         to = newlist; from = *lijst; i = (*oldsize * objectsize) / sizeof(void *);
02818     }
02819     #ifdef MALLOCDEBUG
02820     if ( filelist ) MesPrint("    oldsize: %1, objectsize: %d, fullsize: %1"
02821                             , *oldsize, objectsize, fullsize);
02822     #endif
02823     /*
02824         while ( --i >= 0 ) *to++ = *from++;
02825     */
02826     if ( *lijst ) M_free(*lijst, "DoubleList");
02827     *lijst = newlist;
02828     *oldsize = newsize;

```



```

02829     return(0);
02830 /*
02831     int error;
02832     LONG lsize = *oldsize;
02833
02834     maxlistsize = (LONG)(MAXPOSITIVE);
02835     error = DoubleLList(lijst,&lsize,objectsize,nameoftype);
02836     *oldsize = lsize;
02837     maxlistsize = (LONG)(MAXLONG);
02838
02839     return(error);
02840 */
02841 }
02842
02843 /*
02844     #] DoubleList :
02845     #[ DoubleLList :
02846 */
02847
02848 int DoubleLList(void **lijst, LONG *oldsize, int objectsize, char *nameoftype)
02849 {
02850     void **newlist;
02851     LONG i, newsize, fullsize;
02852     void **to, **from;
02853     static LONG maxlistsize = (LONG)(MAXLONG);
02854     if ( *lijst == 0 ) {
02855         if ( *oldsize > 0 ) newsize = *oldsize;
02856         else newsize = 100;
02857     }
02858     else newsize = *oldsize * 2;
02859     if ( newsize > maxlistsize ) {
02860         if ( *oldsize == maxlistsize ) {
02861             MesPrint("No memory for extra space in %s",nameoftype);
02862             return(-1);
02863         }
02864         newsize = maxlistsize;
02865     }
02866     fullsize = ( newsize * objectsize + sizeof(void *)-1 ) & (~sizeof(void *));
02867     newlist = (void **)Malloca1(fullsize,nameoftype);
02868     if ( *lijst ) { /* Now some punning. DANGEROUS CODE in principle */
02869         to = newlist; from = *lijst; i = (*oldsize * objectsize)/sizeof(void *);
02870     }
02871     #ifdef MALLOCDEBUG
02872     if ( filelist ) MesPrint("    oldsize: %l, objectsize: %d, fullsize: %l"
02873         ,*oldsize,objectsize,fullsize);
02874     #endif
02875     /*
02876         while ( --i >= 0 ) *to++ = *from++;
02877     */
02878     if ( *lijst ) M_free(*lijst,"DoubleLList");
02879     *lijst = newlist;
02880     *oldsize = newsize;
02881     return(0);
02882 }
02883
02884 /*
02885     #] DoubleLList :
02886     #[ DoubleBuffer :
02887 */
02888
02889 #define DODOUBLE(x) { x *s, *t, *u; if ( *start ) { \
02890     oldsize = *(x **)stop - *(x **)start; newsize = 2*oldsize; \
02891     t = u = (x *)Malloca1(newsize*sizeof(x),text); s = *(x **)start; \
02892     for ( i = 0; i < oldsize; i++ ) {t++ = *s++;} M_free(*start,"double"); } \
02893     else { newsize = 100; u = (x *)Malloca1(newsize*sizeof(x),text); } \
02894     *start = (void *)u; *stop = (void *) (u+newsize); }
02895
02896 void DoubleBuffer(void **start, void **stop, int size, char *text)
02897 {
02898     LONG oldsize, newsize, i;
02899     if ( size == sizeof(char) ) DODOUBLE(char)
02900     else if ( size == sizeof(short) ) DODOUBLE(short)
02901     else if ( size == sizeof(int) ) DODOUBLE(int)
02902     else if ( size == sizeof(LONG) ) DODOUBLE(LONG)
02903     else if ( size % sizeof(int) == 0 ) DODOUBLE(int)
02904     else {
02905         MesPrint("---Cannot handle doubling buffers of size %d",size);
02906         Terminate(-1);
02907     }
02908 }
02909
02910 /*
02911     #] DoubleBuffer :
02912     #[ ExpandBuffer :
02913 */
02914
02915 #define DOEXPAND(x) { x *newbuffer, *t, *m; \

```

```

02916     t = newbuffer = (x *)Malloc1((newsize+2)*type,"ExpandBuffer");      \
02917     if ( *buffer ) { m = (x *)*buffer; i = *oldsize;                    \
02918         while ( --i >= 0 ) {t++ = *m++;} M_free(*buffer,"ExpandBuffer"); \
02919     } *buffer = newbuffer; *oldsize = newsize; }
02920
02921 void ExpandBuffer(void **buffer, LONG *oldsize, int type)
02922 {
02923     LONG newsize, i;
02924     if ( *oldsize <= 0 ) { newsize = 100; }
02925     else newsize = 2*( *oldsize );
02926     if ( type == sizeof(char) ) DOEXPAND(char)
02927     else if ( type == sizeof(short) ) DOEXPAND(short)
02928     else if ( type == sizeof(int) ) DOEXPAND(int)
02929     else if ( type == sizeof(LONG) ) DOEXPAND(LONG)
02930     else if ( type == sizeof(POSITION) ) DOEXPAND(POSITION)
02931     else {
02932         MesPrint("---Cannot handle expanding buffers with objects of size %d",type);
02933         Terminate(-1);
02934     }
02935 }
02936
02937 /*
02938     #] ExpandBuffer :
02939     #[ iexp :
02940
02941     Raises the long integer y to the power p.
02942     Returnvalue is long, regardless of overflow.
02943 */
02944
02945 LONG iexp(LONG x, int p)
02946 {
02947     int sign;
02948     ULONG y;
02949     ULONG ux;
02950     if ( x == 0 ) return(0);
02951     if ( p == 0 ) return(1);
02952     sign = x < 0 ? -1 : 1;
02953     if ( sign < 0 && ( p & 1 ) == 0 ) sign = 1;
02954     ux = LongAbs(x);
02955     if ( ux == 1 ) return(sign);
02956     if ( p < 0 ) return(0);
02957     y = 1;
02958     while ( p ) {
02959         if ( ( p & 1 ) != 0 ) y *= ux;
02960         p >>= 1;
02961         ux = ux*ux;
02962     }
02963     if ( sign < 0 ) y = -y;
02964     return ULONGToLong(y);
02965 }
02966
02967 /*
02968     #] iexp :
02969     #[ ToGeneral :
02970
02971     Convert a fast argument to a general argument
02972     Input in r, output in m.
02973     If par == 0 we need the argument header also.
02974 */
02975
02976 void ToGeneral(WORD *r, WORD *m, WORD par)
02977 {
02978     WORD *mm = m, j, k;
02979     if ( par ) m++;
02980     else { m[1] = 0; m += ARGHEAD + 1; }
02981     j = -*r++;
02982     k = 3;
02983     /* JV: Bugfix 1-feb-2016. Old code assumed FUNHEAD to be 2 */
02984     if ( j >= FUNCTION ) { *m++ = j; *m++ = FUNHEAD; FILLFUN(m) }
02985     else {
02986         switch ( j ) {
02987             case SYMBOL: *m++ = j; *m++ = 4; *m++ = *r++; *m++ = 1; break;
02988             case SNUMBER:
02989                 if ( *r > 0 ) { *m++ = *r; *m++ = 1; *m++ = 3; }
02990                 else if ( *r == 0 ) { m--; }
02991                 else { *m++ = -*r; *m++ = 1; *m++ = -3; }
02992                 goto MakeSize;
02993             case MINVECTOR:
02994                 k = -k;
02995                 /* fall through */
02996             case INDEX:
02997             case VECTOR:
02998                 *m++ = INDEX; *m++ = 3; *m++ = *r++;
02999                 break;
03000         }
03001     }
03002     *m++ = 1; *m++ = 1; *m++ = k;

```

```

03003 MakeSize:
03004     *mm = m-mm;
03005     if ( !par ) mm[ARGHEAD] = *mm-ARGHEAD;
03006 }
03007
03008 /*
03009     #] ToGeneral :
03010     #[ ToFast :
03011
03012     Checks whether an argument can be converted to fast notation
03013     If this can be done it does it.
03014     Important: m should be allowed to be equal to r!
03015     Return value is 1 if conversion took place.
03016     If there was conversion the answer is in m.
03017     If there was no conversion m hasn't been touched.
03018 */
03019
03020 int ToFast(WORD *r, WORD *m)
03021 {
03022     WORD i;
03023     if ( *r == ARGHEAD ) { *m++ = -SNUMBER; *m++ = 0; return(1); }
03024     if ( *r != r[ARGHEAD]+ARGHEAD ) return(0); /* > 1 term */
03025     r += ARGHEAD;
03026     if ( *r == 4 ) {
03027         if ( r[2] != 1 || r[1] <= 0 ) return(0);
03028         *m++ = -SNUMBER; *m = ( r[3] < 0 ) ? -r[1] : r[1]; return(1);
03029     }
03030     i = *r - 1;
03031     if ( r[i-1] != 1 || r[i-2] != 1 ) return(0);
03032     if ( r[i] != 3 ) {
03033         if ( r[i] == -3 && r[2] == *r-4 && r[2] == 3 && r[1] == INDEX
03034             && r[3] < MINSPEC ) {}
03035         else return(0);
03036     }
03037     else if ( r[2] != *r - 4 ) return(0);
03038     r++;
03039     if ( *r >= FUNCTION ) {
03040         if ( r[1] <= FUNHEAD ) { *m++ = -*r; return(1); }
03041     }
03042     else if ( *r == SYMBOL ) {
03043         if ( r[1] == 4 && r[3] == 1 )
03044             { *m++ = -SYMBOL; *m++ = r[2]; return(1); }
03045     }
03046     else if ( *r == INDEX ) {
03047         if ( r[1] == 3 ) {
03048             if ( r[2] >= MINSPEC ) {
03049                 if ( r[2] >= 0 && r[2] < AM.OffsetIndex ) *m++ = -SNUMBER;
03050                 else *m++ = -INDEX;
03051             }
03052             else {
03053                 if ( r[5] == -3 ) *m++ = -MINVECTOR;
03054                 else *m++ = -VECTOR;
03055             }
03056             *m++ = r[2];
03057             return(1);
03058         }
03059     }
03060     return(0);
03061 }
03062
03063 /*
03064     #] ToFast :
03065     #[ ToPolyFunGeneral :
03066
03067     Routine forces a polyratfun into general notation if needed.
03068     If no action was needed, the return value is zero.
03069     A positive return value indicates how many arguments were converted.
03070     The new term overwrite the old.
03071 */
03072
03073 WORD ToPolyFunGeneral(PHEAD WORD *term)
03074 {
03075     WORD *t = term+1, *tt, *to, *tol, *termout, *tstop, *tnext;
03076     WORD numarg, i, change = 0;
03077     tstop = term + *term; tstop -= ABS(tstop[-1]);
03078     termout = to = AT.WorkPointer;
03079     to++;
03080     while ( t < tstop ) { /* go through the subterms */
03081         if ( *t == AR.PolyFun ) {
03082             tt = t+FUNHEAD; tnext = t + t[1];
03083             numarg = 0;
03084             while ( tt < tnext ) { numarg++; NEXTARG(tt); }
03085             if ( numarg == 2 ) { /* this needs attention */
03086                 tt = t + FUNHEAD;
03087                 tol = to;
03088                 i = FUNHEAD; NCOPY(to,t,i);
03089                 while ( tt < tnext ) { /* Do the arguments */

```

```

03090         if ( *tt > 0 ) {
03091             i = *tt; NCOPY(to,tt,i);
03092         }
03093         else if ( *tt == -SYMBOL ) {
03094             tol[1] += 6+ARGHEAD; tol[2] |= MUSTCLEANPRF; change++;
03095             *to++ = 8+ARGHEAD; *to++ = 0; FILLARG(to);
03096             *to++ = 8; *to++ = SYMBOL; *to++ = 4; *to++ = tt[1];
03097             *to++ = 1; *to++ = 1; *to++ = 1; *to++ = 3;
03098             tt += 2;
03099         }
03100         else if ( *tt == -SNUMBER ) {
03101             if ( tt[1] > 0 ) {
03102                 tol[1] += 2+ARGHEAD; tol[2] |= MUSTCLEANPRF; change++;
03103                 *to++ = 4+ARGHEAD; *to++ = 0; FILLARG(to);
03104                 *to++ = 4; *to++ = tt[1]; *to++ = 1; *to++ = 3;
03105                 tt += 2;
03106             }
03107             else if ( tt[1] < 0 ) {
03108                 tol[1] += 2+ARGHEAD; tol[2] |= MUSTCLEANPRF; change++;
03109                 *to++ = 4+ARGHEAD; *to++ = 0; FILLARG(to);
03110                 *to++ = 4; *to++ = -tt[1]; *to++ = 1; *to++ = -3;
03111                 tt += 2;
03112             }
03113             else {
03114                 MLOCK(ErrorMessageLock);
03115                 MesPrint("Internal error: Zero in PolyRatFun");
03116                 MUNLOCK(ErrorMessageLock);
03117                 Terminate(-1);
03118             }
03119         }
03120     }
03121     t = tnext;
03122     continue;
03123 }
03124 }
03125 i = t[1]; NCOPY(to,t,i)
03126 }
03127 if ( change ) {
03128     tt = term + *term;
03129     while ( t < tt ) *to++ = *t++;
03130     *termout = to - termout;
03131     t = term; i = *termout; tt = termout;
03132     NCOPY(t,tt,i)
03133     AT.WorkPointer = term + *term;
03134 }
03135 return(change);
03136 }
03137
03138 /*
03139     #] ToPolyFunGeneral :
03140     #[ IsLikeVector :
03141
03142     Routine determines whether a function argument is like a vector.
03143     Returnvalue: 1: is vector or index
03144                 0: is not vector or index
03145                 -1: may be an index
03146 */
03147
03148 int IsLikeVector(WORD *arg)
03149 {
03150     WORD *sstop, *t, *tstop;
03151     if ( *arg < 0 ) {
03152         if ( *arg == -VECTOR || *arg == -INDEX ) return(1);
03153         if ( *arg == -SNUMBER && arg[1] >= 0 && arg[1] < AM.OffsetIndex )
03154             return(-1);
03155         return(0);
03156     }
03157     sstop = arg + *arg; arg += ARGHEAD;
03158     while ( arg < sstop ) {
03159         t = arg + *arg;
03160         tstop = t - ABS(t[-1]);
03161         arg++;
03162         while ( arg < tstop ) {
03163             if ( *arg == INDEX ) return(1);
03164             arg += arg[1];
03165         }
03166         arg = t;
03167     }
03168     return(0);
03169 }
03170
03171 /*
03172     #] IsLikeVector :
03173     #[ AreArgsEqual :
03174 */
03175
03176 int AreArgsEqual(WORD *arg1, WORD *arg2)

```

```

03177 {
03178     int i;
03179     if ( *arg2 != *arg1 ) return(0);
03180     if ( *arg1 > 0 ) {
03181         i = *arg1;
03182         while ( --i > 0 ) { if ( arg1[i] != arg2[i] ) return(0); }
03183         return(1);
03184     }
03185     else if ( *arg1 <= -FUNCTION ) return(1);
03186     else if ( arg1[1] == arg2[1] ) return(1);
03187     return(0);
03188 }
03189
03190 /*
03191     #] AreArgsEqual :
03192     #[ CompareArgs :
03193 */
03194
03195 int CompareArgs(WORD *arg1, WORD *arg2)
03196 {
03197     int i1,i2;
03198     if ( *arg1 > 0 ) {
03199         if ( *arg2 < 0 ) return(-1);
03200         i1 = *arg1-ARGHEAD; arg1 += ARGHEAD;
03201         i2 = *arg2-ARGHEAD; arg2 += ARGHEAD;
03202         while ( i1 > 0 && i2 > 0 ) {
03203             if ( *arg1 != *arg2 ) return((int) (*arg1)-(int) (*arg2));
03204             i1--; i2--; arg1++; arg2++;
03205         }
03206         return(i1-i2);
03207     }
03208     else if ( *arg2 > 0 ) return(1);
03209     else {
03210         if ( *arg1 != *arg2 ) {
03211             if ( *arg1 < *arg2 ) return(-1);
03212             else return(1);
03213         }
03214         if ( *arg1 <= -FUNCTION ) return(0);
03215         return((int) (arg1[1])-(int) (arg2[1]));
03216     }
03217 }
03218
03219 /*
03220     #] CompareArgs :
03221     #[ CompArg :
03222
03223     returns 1 if arg1 comes first, -1 if arg2 comes first, 0 if equal
03224 */
03225
03226 int CompArg(WORD *s1, WORD *s2)
03227 {
03228     GETIDENTITY
03229     WORD *st1, *st2, x[7];
03230     int k;
03231     if ( *s1 < 0 ) {
03232         if ( *s2 < 0 ) {
03233             if ( *s1 <= -FUNCTION && *s2 <= -FUNCTION ) {
03234                 if ( *s1 > *s2 ) return(-1);
03235                 if ( *s1 < *s2 ) return(1);
03236                 return(0);
03237             }
03238             if ( *s1 > *s2 ) return(1);
03239             if ( *s1 < *s2 ) return(-1);
03240             if ( *s1 <= -FUNCTION ) return(0);
03241             s1++; s2++;
03242             if ( *s1 > *s2 ) return(1);
03243             if ( *s1 < *s2 ) return(-1);
03244             return(0);
03245         }
03246         x[1] = AT.comsym[3];
03247         x[2] = AT.comnum[1];
03248         x[3] = AT.comnum[3];
03249         x[4] = AT.comind[3];
03250         x[5] = AT.comind[6];
03251         x[6] = AT.comfun[1];
03252         if ( *s1 == -SYMBOL ) {
03253             AT.comsym[3] = s1[1];
03254             st1 = AT.comsym+8; s1 = AT.comsym;
03255         }
03256         else if ( *s1 == -SNUMBER ) {
03257             if ( s1[1] < 0 ) {
03258                 AT.comnum[1] = -s1[1]; AT.comnum[3] = -3;
03259             }
03260             else {
03261                 AT.comnum[1] = s1[1]; AT.comnum[3] = 3;
03262             }
03263             st1 = AT.comnum+4;

```

```

03264         s1 = AT.comnum;
03265     }
03266     else if ( *s1 == -INDEX || *s1 == -VECTOR ) {
03267         AT.comind[3] = s1[1]; AT.comind[6] = 3;
03268         st1 = AT.comind+7; s1 = AT.comind;
03269     }
03270     else if ( *s1 == -MINVECTOR ) {
03271         AT.comind[3] = s1[1]; AT.comind[6] = -3;
03272         st1 = AT.comind+7; s1 = AT.comind;
03273     }
03274     else if ( *s1 <= -FUNCTION ) {
03275         AT.comfun[1] = -*s1;
03276         st1 = AT.comfun+FUNHEAD+4; s1 = AT.comfun;
03277     }
03278 /*
03279         Symmetrize during compilation of id statement when properorder
03280         needs this one. Code added 10-nov-2001
03281 */
03282     else if ( *s1 == -ARGWILD ) {
03283         return(-1);
03284     }
03285     else { goto argerror; }
03286     st2 = s2 + *s2; s2 += ARGHEAD;
03287     goto docompare;
03288 }
03289 else if ( *s2 < 0 ) {
03290     x[1] = AT.comsym[3];
03291     x[2] = AT.comnum[1];
03292     x[3] = AT.comnum[3];
03293     x[4] = AT.comind[3];
03294     x[5] = AT.comind[6];
03295     x[6] = AT.comfun[1];
03296     if ( *s2 == -SYMBOL ) {
03297         AT.comsym[3] = s2[1];
03298         st2 = AT.comsym+8; s2 = AT.comsym;
03299     }
03300     else if ( *s2 == -SNUMBER ) {
03301         if ( s2[1] < 0 ) {
03302             AT.comnum[1] = -s2[1]; AT.comnum[3] = -3;
03303             st2 = AT.comnum+4;
03304         }
03305         else if ( s2[1] == 0 ) {
03306             st2 = AT.comnum+4; s2 = st2;
03307         }
03308         else {
03309             AT.comnum[1] = s2[1]; AT.comnum[3] = 3;
03310             st2 = AT.comnum+4;
03311         }
03312         s2 = AT.comnum;
03313     }
03314     else if ( *s2 == -INDEX || *s2 == -VECTOR ) {
03315         AT.comind[3] = s2[1]; AT.comind[6] = 3;
03316         st2 = AT.comind+7; s2 = AT.comind;
03317     }
03318     else if ( *s2 == -MINVECTOR ) {
03319         AT.comind[3] = s2[1]; AT.comind[6] = -3;
03320         st2 = AT.comind+7; s2 = AT.comind;
03321     }
03322     else if ( *s2 <= -FUNCTION ) {
03323         AT.comfun[1] = -*s2;
03324         st2 = AT.comfun+FUNHEAD+4; s2 = AT.comfun;
03325     }
03326 /*
03327         Symmetrize during compilation of id statement when properorder
03328         needs this one. Code added 10-nov-2001
03329 */
03330     else if ( *s2 == -ARGWILD ) {
03331         return(1);
03332     }
03333     else { goto argerror; }
03334     st1 = s1 + *s1; s1 += ARGHEAD;
03335     goto docompare;
03336 }
03337 else {
03338     x[1] = AT.comsym[3];
03339     x[2] = AT.comnum[1];
03340     x[3] = AT.comnum[3];
03341     x[4] = AT.comind[3];
03342     x[5] = AT.comind[6];
03343     x[6] = AT.comfun[1];
03344     st1 = s1 + *s1; st2 = s2 + *s2;
03345     s1 += ARGHEAD; s2 += ARGHEAD;
03346 docompare:
03347     while ( s1 < st1 && s2 < st2 ) {
03348         if ( ( k = CompareTerms(BHEAD s1,s2,(WORD)2) ) != 0 ) {
03349             AT.comsym[3] = x[1];
03350             AT.comnum[1] = x[2];

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```

03351         AT.comnum[3] = x[3];
03352         AT.comind[3] = x[4];
03353         AT.comind[6] = x[5];
03354         AT.comfun[1] = x[6];
03355         return(-k);
03356     }
03357     s1 += *s1; s2 += *s2;
03358 }
03359 AT.comsym[3] = x[1];
03360 AT.comnum[1] = x[2];
03361 AT.comnum[3] = x[3];
03362 AT.comind[3] = x[4];
03363 AT.comind[6] = x[5];
03364 AT.comfun[1] = x[6];
03365 if ( s1 < st1 ) return(1);
03366 if ( s2 < st2 ) return(-1);
03367 }
03368 return(0);
03369
03370 argerror:
03371     MesPrint("Illegal type of short function argument in Normalize");
03372     Terminate(-1); return(0);
03373 }
03374
03375 /*
03376     #] CompArg :
03377     #[ TimeWallClock :
03378 */
03379
03380 #ifdef HAVE_CLOCK_GETTIME
03381 #include <time.h> /* for clock_gettime() */
03382 #else
03383 #ifdef HAVE_GETTIMEOFDAY
03384 #include <sys/time.h> /* for gettimeofday() */
03385 #else
03386 #include <sys/timeb.h> /* for ftime() */
03387 #endif
03388 #endif
03389
03396 LONG TimeWallClock(WORD par)
03397 {
03398     /*
03399      * NOTE: this function is not thread-safe. Operations on tp are not atomic.
03400      */
03401
03402 #ifdef HAVE_CLOCK_GETTIME
03403     struct timespec ts;
03404     clock_gettime(CLOCK_MONOTONIC, &ts);
03405
03406     if ( par ) {
03407         return(((LONG)(ts.tv_sec)-AM.OldSecTime)*100 +
03408             ((LONG)(ts.tv_nsec / 1000000)-AM.OldMilliTime)/10);
03409     }
03410     else {
03411         AM.OldSecTime = (LONG)(ts.tv_sec);
03412         AM.OldMilliTime = (LONG)(ts.tv_nsec / 1000000);
03413         return(0L);
03414     }
03415 #else
03416 #ifdef HAVE_GETTIMEOFDAY
03417     struct timeval t;
03418     LONG sec, msec;
03419     gettimeofday(&t, NULL);
03420     sec = (LONG)t.tv_sec;
03421     msec = (LONG)(t.tv_usec/1000);
03422     if ( par ) {
03423         return (sec-AM.OldSecTime)*100 + (msec-AM.OldMilliTime)/10;
03424     }
03425     else {
03426         AM.OldSecTime = sec;
03427         AM.OldMilliTime = msec;
03428         return(0L);
03429     }
03430 #else
03431     struct timeb tp;
03432     ftime(&tp);
03433
03434     if ( par ) {
03435         return(((LONG)(tp.time)-AM.OldSecTime)*100 +
03436             ((LONG)(tp.millitm)-AM.OldMilliTime)/10);
03437     }
03438     else {
03439         AM.OldSecTime = (LONG)(tp.time);
03440         AM.OldMilliTime = (LONG)(tp.millitm);
03441         return(0L);
03442     }
03443 #endif

```

```

03444 #endif
03445 }
03446
03447 /*
03448     #] TimeWallClock :
03449     #[ TimeChildren :
03450 */
03451
03452 LONG TimeChildren(WORD par)
03453 {
03454     if ( par ) return(Timer(1)-AM.OldChildTime);
03455     AM.OldChildTime = Timer(1);
03456     return(0L);
03457 }
03458
03459 /*
03460     #] TimeChildren :
03461     #[ TimeCPU :
03462 */
03463
03470 LONG TimeCPU(WORD par)
03471 {
03472     GETIDENTITY
03473     if ( par ) return(Timer(0)-AR.OldTime);
03474     AR.OldTime = Timer(0);
03475     return(0L);
03476 }
03477
03478 /*
03479     #] TimeCPU :
03480     #[ Timer :
03481 */
03482 #if defined(WINDOWS)
03483
03484 LONG Timer(int par)
03485 {
03486     #ifndef WITHPTHREADS
03487         static int initialized = 0;
03488         static HANDLE hProcess;
03489         FILETIME ftCreate, ftExit, ftKernel, ftUser;
03490         DUMMYUSE(par);
03491
03492         if ( !initialized ) {
03493             hProcess = OpenProcess(PROCESS_QUERY_INFORMATION, FALSE, GetCurrentProcessId());
03494         }
03495         if ( GetProcessTimes(hProcess, &ftCreate, &ftExit, &ftKernel, &ftUser) ) {
03496             PFILETIME pftKernel = &ftKernel; /* to avoid strict-aliasing rule warnings */
03497             PFILETIME pftUser = &ftUser;
03498             __int64 t = *(__int64 *)pftKernel + *(__int64 *)pftUser; /* in 100 nsec. */
03499             return (LONG)(t / 10000); /* in msec. */
03500         }
03501         return 0;
03502     #else
03503         LONG lResult = 0;
03504         HANDLE hThread;
03505         FILETIME ftCreate, ftExit, ftKernel, ftUser;
03506         DUMMYUSE(par);
03507
03508         hThread = OpenThread(THREAD_QUERY_INFORMATION, FALSE, GetCurrentThreadId());
03509         if ( hThread ) {
03510             if ( GetThreadTimes(hThread, &ftCreate, &ftExit, &ftKernel, &ftUser) ) {
03511                 PFILETIME pftKernel = &ftKernel; /* to avoid strict-aliasing rule warnings */
03512                 PFILETIME pftUser = &ftUser;
03513                 __int64 t = *(__int64 *)pftKernel + *(__int64 *)pftUser; /* in 100 nsec. */
03514                 lResult = (LONG)(t / 10000); /* in msec. */
03515             }
03516             CloseHandle(hThread);
03517         }
03518         return lResult;
03519     #endif
03520 }
03521
03522 #elif defined(UNIX)
03523 #include <sys/time.h>
03524 #include <sys/resource.h>
03525 #ifndef WITHPOSIXCLOCK
03526 #include <time.h>
03527 /*
03528     And include -lrt in the link statement (on blade02)
03529 */
03530 #endif
03531
03532 LONG Timer(int par)
03533 {
03534     #ifndef WITHPOSIXCLOCK
03535         /*
03536             Only to be used in combination with WITHPTHREADS

```



```

03537     This clock seems to be supported by the standard.
03538     The getrusage clock returns according to the standard only the combined
03539     time of the whole process. But in older versions of Linux LinuxThreads
03540     is used which gives a separate id to each thread and individual timings.
03541     In NPTL we get, according to the standard, one combined timing.
03542     To get individual timings we need to use
03543         clock_gettime(CLOCK_THREAD_CPUTIME_ID, &timing)
03544     with timing of the time
03545     struct timespec {
03546         time_t tv_sec;           Seconds.
03547         long tv_nsec;           Nanoseconds.
03548     };
03549
03550 */
03551     struct timespec t;
03552     if ( par == 0 ) {
03553         if ( clock_gettime(CLOCK_THREAD_CPUTIME_ID, &t) ) {
03554             MesPrint("Error in getting timing information");
03555         }
03556         return (LONG)t.tv_sec * 1000 + (LONG)t.tv_nsec / 1000000;
03557     }
03558     return(0);
03559 #else
03560     struct rusage rusage;
03561     if ( par == 1 ) {
03562         getrusage(RUSAGE_CHILDREN, &rusage);
03563         return(((LONG) (rusage.ru_utime.tv_sec) + (LONG) (rusage.ru_stime.tv_sec)) * 1000
03564             + (rusage.ru_utime.tv_usec / 1000 + rusage.ru_stime.tv_usec / 1000));
03565     }
03566     else {
03567         getrusage(RUSAGE_SELF, &rusage);
03568         return(((LONG) (rusage.ru_utime.tv_sec) + (LONG) (rusage.ru_stime.tv_sec)) * 1000
03569             + (rusage.ru_utime.tv_usec / 1000 + rusage.ru_stime.tv_usec / 1000));
03570     }
03571 #endif
03572 }
03573
03574 #elif defined(SUN)
03575 #define _TIME_T_
03576 #include <sys/time.h>
03577 #include <sys/resource.h>
03578
03579 LONG Timer(int par)
03580 {
03581     struct rusage rusage;
03582     if ( par == 1 ) {
03583         getrusage(RUSAGE_CHILDREN, &rusage);
03584         return(((LONG) (rusage.ru_utime.tv_sec) + (LONG) (rusage.ru_stime.tv_sec)) * 1000
03585             + (rusage.ru_utime.tv_usec / 1000 + rusage.ru_stime.tv_usec / 1000));
03586     }
03587     else {
03588         getrusage(RUSAGE_SELF, &rusage);
03589         return(((LONG) (rusage.ru_utime.tv_sec) + (LONG) (rusage.ru_stime.tv_sec)) * 1000
03590             + (rusage.ru_utime.tv_usec / 1000 + rusage.ru_stime.tv_usec / 1000));
03591     }
03592 }
03593
03594 #elif defined(RS6K)
03595 #include <sys/time.h>
03596 #include <sys/resource.h>
03597
03598 LONG Timer(int par)
03599 {
03600     struct rusage rusage;
03601     if ( par == 1 ) {
03602         getrusage(RUSAGE_CHILDREN, &rusage);
03603         return(((LONG) (rusage.ru_utime.tv_sec) + (LONG) (rusage.ru_stime.tv_sec)) * 1000
03604             + (rusage.ru_utime.tv_usec / 1000 + rusage.ru_stime.tv_usec / 1000));
03605     }
03606     else {
03607         getrusage(RUSAGE_SELF, &rusage);
03608         return(((LONG) (rusage.ru_utime.tv_sec) + (LONG) (rusage.ru_stime.tv_sec)) * 1000
03609             + (rusage.ru_utime.tv_usec / 1000 + rusage.ru_stime.tv_usec / 1000));
03610     }
03611 }
03612
03613 #elif defined(ANSI)
03614 LONG Timer(int par)
03615 {
03616     #ifdef ALPHA
03617         /* clock_t t, tikken = clock(); */
03618         /* MesPrint("ALPHA-clock = %l", (LONG)tikken); */
03619         /* t = tikken % CLOCKS_PER_SEC; */
03620         /* tikken /= CLOCKS_PER_SEC; */
03621         /* tikken *= 1000; */
03622         /* tikken += (t * 1000) / CLOCKS_PER_SEC; */
03623         /* return((LONG)tikken); */

```

```

03624 /* #define _TIME_T_ */
03625 #include <sys/time.h>
03626 #include <sys/resource.h>
03627 struct rusage rusage;
03628 if ( par == 1 ) {
03629     getrusage(RUSAGE_CHILDREN,&rusage);
03630     return(((LONG)(rusage.ru_utime.tv_sec)+(LONG)(rusage.ru_stime.tv_sec))*1000
03631         +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
03632 }
03633 else {
03634     getrusage(RUSAGE_SELF,&rusage);
03635     return(((LONG)(rusage.ru_utime.tv_sec)+(LONG)(rusage.ru_stime.tv_sec))*1000
03636         +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
03637 }
03638 #else
03639 #ifdef DEC_STATION
03640     clock_t tikken = clock();
03641     return((LONG)tikken/1000);
03642 #else
03643     clock_t t, tikken = clock();
03644     t = tikken % CLK_TCK;
03645     tikken /= CLK_TCK;
03646     tikken *= 1000;
03647     tikken += (t*1000)/CLK_TCK;
03648     return(tikken);
03649 #endif
03650 #endif
03651 }
03652 #elif defined(VMS)
03653
03654 #include <time.h>
03655 void times(tbuffer_t *buffer);
03656
03657 LONG
03658 Timer(int par)
03659 {
03660     tbuffer_t buffer;
03661     if ( par == 1 ) { return(0); }
03662     else {
03663         times(&buffer);
03664         return(buffer.proc_user_time * 10);
03665     }
03666 }
03667
03668 #elif defined(mBSD)
03669
03670 #ifdef MICROTIME
03671 /*
03672     There is only a CP time clock in microseconds here
03673     This can cause problems with AO.wrap around
03674 */
03675 #else
03676 #ifdef mBSD2
03677     #include <sys/types.h>
03678     #include <sys/times.h>
03679     #include <time.h>
03680     LONG pretime = 0;
03681 #else
03682     #define _TIME_T_
03683     #include <sys/time.h>
03684     #include <sys/resource.h>
03685 #endif
03686 #endif
03687
03688 LONG Timer(int par)
03689 {
03690     #ifdef MICROTIME
03691         LONG t;
03692         if ( par == 1 ) { return(0); }
03693         t = clock();
03694         if ( ( AO.wrapnum & 1 ) != 0 ) t ^= 0x80000000;
03695         if ( t < 0 ) {
03696             t ^= 0x80000000;
03697             wrapnum++;
03698             AO.wrap += 2147584;
03699         }
03700         return(AO.wrap+(t/1000));
03701     #else
03702     #ifdef mBSD2
03703         struct tms buffer;
03704         LONG ret;
03705         ULONG a1, a2, a3, a4;
03706         if ( par == 1 ) { return(0); }
03707         times(&buffer);
03708         a1 = (ULONG)buffer.tms_utime;
03709         a2 = a1 » 16;
03710         a3 = a1 & 0xFFFFL;

```

```

03711     a3 *= 1000;
03712     a2 = 1000*a2 + (a3 >> 16);
03713     a3 &= 0xFFFFL;
03714     a4 = a2/CLK_TCK;
03715     a2 %= CLK_TCK;
03716     a3 += a2 << 16;
03717     ret = (LONG)((a4 << 16) + a3 / CLK_TCK);
03718     /* ret = ((LONG)buffer.tms_utime * 1000)/CLK_TCK; */
03719     return(ret);
03720 #else
03721 #ifdef REALTIME
03722     struct timeval tp;
03723     struct timezone tzp;
03724     if ( par == 1 ) { return(0); }
03725     gettimeofday(&tp,&tzp); */
03726     return(tp.tv_sec*1000+tp.tv_usec/1000);
03727 #else
03728     struct rusage rusage;
03729     if ( par == 1 ) {
03730         getrusage(RUSAGE_CHILDREN,&rusage);
03731         return((rusage.ru_utime.tv_sec+rusage.ru_stime.tv_sec)*1000
03732             + (rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
03733     }
03734     else {
03735         getrusage(RUSAGE_SELF,&rusage);
03736         return((rusage.ru_utime.tv_sec+rusage.ru_stime.tv_sec)*1000
03737             + (rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
03738     }
03739 #endif
03740 #endif
03741 #endif
03742 }
03743
03744 #endif
03745
03746 /*
03747     #] Timer :
03748     #[ Crash :
03749
03750     Routine for debugging purposes
03751 */
03752
03753 int Crash(void)
03754 {
03755     int retval;
03756     #ifndef DEBUGGING
03757     int *zero = 0;
03758     retval = *zero;
03759     #else
03760     retval = 0;
03761     #endif
03762     return(retval);
03763 }
03764
03765 /*
03766     #] Crash :
03767     #[ TestTerm :
03768 */
03769
03781 int TestTerm(WORD *term)
03782 {
03783     int errorcode = 0, coeffsize;
03784     WORD *t, *tt, *tstop, *endterm, *targ, *targstop, *funstop, *argterm;
03785     endterm = term + *term;
03786     coeffsize = ABS(endterm[-1]);
03787     if ( coeffsize >= *term ) {
03788         MLOCK(ErrorMessageLock);
03789         MesPrint("TestTerm: Internal inconsistency in term. Coefficient too big.");
03790         MUNLOCK(ErrorMessageLock);
03791         errorcode = 1;
03792         goto finish;
03793     }
03794     if ( ( coeffsize < 3 ) || ( ( coeffsize & 1 ) != 1 ) ) {
03795         MLOCK(ErrorMessageLock);
03796         MesPrint("TestTerm: Internal inconsistency in term. Wrong size coefficient.");
03797         MUNLOCK(ErrorMessageLock);
03798         errorcode = 2;
03799         goto finish;
03800     }
03801     t = term+1;
03802     tstop = endterm - coeffsize;
03803     while ( t < tstop ) {
03804         switch ( *t ) {
03805             case SYMBOL:
03806             case DOTPRODUCT:
03807             case INDEX:
03808             case VECTOR:

```

```

03809         case DELTA:
03810         case HAAKJE:
03811             break;
03812         case SNUMBER:
03813         case LNUMBER:
03814             MLOCK(ErrorMessageLock);
03815             MesPrint("TestTerm: Internal inconsistency in term. L or S number");
03816             MUNLOCK(ErrorMessageLock);
03817             errorcode = 3;
03818             goto finish;
03819             break;
03820         case EXPRESSION:
03821         case SUBEXPRESSION:
03822         case DOLLAREXPRESSION:
03823     /*
03824         MLOCK(ErrorMessageLock);
03825         MesPrint("TestTerm: Internal inconsistency in term. Expression survives.");
03826         MUNLOCK(ErrorMessageLock);
03827         errorcode = 4;
03828         goto finish;
03829     */
03830         break;
03831         case SETSET:
03832         case MINVECTOR:
03833         case SETEXP:
03834         case ARGFIELD:
03835             MLOCK(ErrorMessageLock);
03836             MesPrint("TestTerm: Internal inconsistency in term. Illegal subterm.");
03837             MUNLOCK(ErrorMessageLock);
03838             errorcode = 5;
03839             goto finish;
03840             break;
03841         case ARGWILD:
03842             break;
03843         default:
03844             if ( *t <= 0 ) {
03845                 MLOCK(ErrorMessageLock);
03846                 MesPrint("TestTerm: Internal inconsistency in term. Illegal subterm number.");
03847                 MUNLOCK(ErrorMessageLock);
03848                 errorcode = 6;
03849                 goto finish;
03850             }
03851     /*
03852         This is a regular function.
03853     */
03854         if ( *t-FUNCTION >= NumFunctions ) {
03855             MLOCK(ErrorMessageLock);
03856             MesPrint("TestTerm: Internal inconsistency in term. Illegal function number");
03857             MUNLOCK(ErrorMessageLock);
03858             errorcode = 7;
03859             goto finish;
03860         }
03861         funstop = t + t[1];
03862         if ( funstop > tstop ) goto subtermsize;
03863         if ( t[2] != 0 ) {
03864             MLOCK(ErrorMessageLock);
03865             MesPrint("TestTerm: Internal inconsistency in term. Dirty flag nonzero.");
03866             MUNLOCK(ErrorMessageLock);
03867             errorcode = 8;
03868             goto finish;
03869         }
03870         targ = t + FUNHEAD;
03871         if ( targ > funstop ) {
03872             MLOCK(ErrorMessageLock);
03873             MesPrint("TestTerm: Internal inconsistency in term. Illegal function size.");
03874             MUNLOCK(ErrorMessageLock);
03875             errorcode = 9;
03876             goto finish;
03877         }
03878         if ( functions[*t-FUNCTION].spec >= TENSORFUNCTION ) {
03879         }
03880         else {
03881             while ( targ < funstop ) {
03882                 if ( *targ < 0 ) {
03883                     if ( *targ <= -(FUNCTION+NumFunctions) ) {
03884                         MLOCK(ErrorMessageLock);
03885                         MesPrint("TestTerm: Internal inconsistency in term. Illegal function
number in argument.");
03886                         MUNLOCK(ErrorMessageLock);
03887                         errorcode = 10;
03888                         goto finish;
03889                     }
03890                     if ( *targ <= -FUNCTION ) { targ++; }
03891                     else {
03892                         if ( ( *targ != -SYMBOL ) && ( *targ != -VECTOR )
&& ( *targ != -MINVECTOR )
&& ( *targ != -SNUMBER )

```

```

03895             && ( *targ != -ARGWILD )
03896             && ( *targ != -INDEX ) ) {
03897                 MLOCK(ErrorMessageLock);
03898                 MesPrint("TestTerm: Internal inconsistency in term. Illegal object in
argument.");
03899                 MUNLOCK(ErrorMessageLock);
03900                 errorcode = 11;
03901                 goto finish;
03902             }
03903             targ += 2;
03904         }
03905     }
03906     else if ( ( *targ < ARGHEAD ) || ( targ+*targ > funstop ) ) {
03907         MLOCK(ErrorMessageLock);
03908         MesPrint("TestTerm: Internal inconsistency in term. Illegal size of
argument.");
03909         MUNLOCK(ErrorMessageLock);
03910         errorcode = 12;
03911         goto finish;
03912     }
03913     else if ( targ[1] != 0 ) {
03914         MLOCK(ErrorMessageLock);
03915         MesPrint("TestTerm: Internal inconsistency in term. Dirty flag in argument.");
03916         MUNLOCK(ErrorMessageLock);
03917         errorcode = 13;
03918         goto finish;
03919     }
03920     else {
03921         targstop = targ + *targ;
03922         argterm = targ + ARGHEAD;
03923         while ( argterm < targstop ) {
03924             if ( ( *argterm < 4 ) || ( argterm + *argterm > targstop ) ) {
03925                 MLOCK(ErrorMessageLock);
03926                 MesPrint("TestTerm: Internal inconsistency in term. Illegal term size
in argument.");
03927                 MUNLOCK(ErrorMessageLock);
03928                 errorcode = 14;
03929                 goto finish;
03930             }
03931             if ( TestTerm(argterm) != 0 ) {
03932                 MLOCK(ErrorMessageLock);
03933                 MesPrint("TestTerm: Internal inconsistency in term. Called from
TestTerm.");
03934                 MUNLOCK(ErrorMessageLock);
03935                 errorcode = 15;
03936                 goto finish;
03937             }
03938             argterm += *argterm;
03939         }
03940         targ = targstop;
03941     }
03942 }
03943 }
03944 break;
03945 }
03946 tt = t + t[1];
03947 if ( tt > tstop ) {
03948 subtermsize:
03949     MLOCK(ErrorMessageLock);
03950     MesPrint("TestTerm: Internal inconsistency in term. Illegal subterm size.");
03951     MUNLOCK(ErrorMessageLock);
03952     errorcode = 100;
03953     goto finish;
03954 }
03955 t = tt;
03956 }
03957 return(errorcode);
03958 finish:
03959 return(errorcode);
03960 }
03961
03962 /*
03963     #] TestTerm :
03964     #[ DistrN :
03965 */
03966
03967 int DistrN(int n, int *cpl, int ncpl, int *scratch)
03968 {
03969     /*
03970     Divides n objects over ncpl bins (cpl), each time returning one
03971     of those distributions until there are no more after which the
03972     routine returns the value zero (otherwise one).
03973     The array scratch (size n) is kept for the intermediate information.
03974     The whole starts with scratch[0] == -2;
03975     */
03976     int i, j;
03977     if ( ncpl == 0 ) {

```

```

03978         if ( scratch[0] == -2 ) { scratch[0] = 0; return(1); }
03979         else return(0);
03980     }
03981     if ( scratch[0] == ncpl-1 ) {
03982         return(0);
03983     }
03984     else if ( scratch[0] == -2 ) {
03985         for ( i = 0; i < n; i++ ) scratch[i] = 0;
03986     }
03987     else {
03988         j = n-1;
03989         while ( j >= 0 ) {
03990             scratch[j]++;
03991             if ( scratch[j] < ncpl ) break;
03992             j--;
03993         }
03994         j++;
03995         while ( j < n ) { scratch[j] = scratch[j-1]; j++; }
03996     }
03997     for ( i = 0; i < ncpl; i++ ) cpl[i] = 0;
03998     for ( i = 0; i < n; i++ ) { cpl[scratch[i]]++; }
03999     return(1);
04000 }
04001
04002 /*
04003     #] DistrN :
04004     #] Mixed :
04005 */

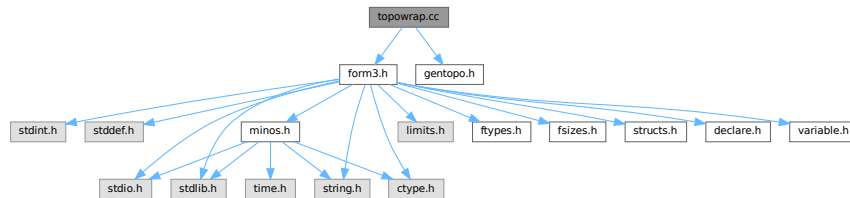
```

## 4.148 topowrap.cc File Reference

```
#include "form3.h"
```

```
#include "gentopo.h"
```

Include dependency graph for topowrap.cc:



### Data Structures

- struct [ToPoTyPe](#)

### Macros

- #define [MAXPOINTS](#) 120

### Typedefs

- typedef struct [ToPoTyPe](#) [TOPOTYPE](#)

### Functions

- int [GenerateVertices](#) ([TOPOTYPE](#) \*TopoInf, int pointsremaining, int level)
- int [GenerateTopologies](#) (PHEAD WORD nloops, WORD nlegs, WORD setvert, WORD setmax)
- void [toForm](#) ([T\\_EGraph](#) \*egraph)

### 4.148.1 Detailed Description

routines for conversion of topology and diagram output to FORM notation

Definition in file [topowrap.cc](#).

### 4.148.2 Macro Definition Documentation

#### 4.148.2.1 MAXPOINTS

```
#define MAXPOINTS 120
```

Definition at line 44 of file [topowrap.cc](#).

### 4.148.3 Function Documentation

#### 4.148.3.1 GenerateVertices()

```
int GenerateVertices (
    TOPOTYPE * TopoInf,
    int pointsremaining,
    int level )
```

Definition at line 72 of file [topowrap.cc](#).

#### 4.148.3.2 GenerateTopologies()

```
int GenerateTopologies (
    PHEAD WORD nloops,
    WORD nlegs,
    WORD setvert,
    WORD setmax )
```

Definition at line 123 of file [topowrap.cc](#).

#### 4.148.3.3 toForm()

```
void toForm (
    T_EGraph * egraph )
```

Definition at line 178 of file [topowrap.cc](#).

## 4.149 topowrap.cc

[Go to the documentation of this file.](#)

```

00001
00006 /* #[ License : */
00007 /*
00008 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 *   When using this file you are requested to refer to the publication
00010 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 *   This is considered a matter of courtesy as the development was paid
00012 *   for by FOM the Dutch physics granting agency and we would like to
00013 *   be able to track its scientific use to convince FOM of its value
00014 *   for the community.
00015 *
00016 *   This file is part of FORM.
00017 *
00018 *   FORM is free software: you can redistribute it and/or modify it under the
00019 *   terms of the GNU General Public License as published by the Free Software
00020 *   Foundation, either version 3 of the License, or (at your option) any later
00021 *   version.
00022 *
00023 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 *   details.
00027 *
00028 *   You should have received a copy of the GNU General Public License along
00029 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /*
00032     #[ License :
00033     #[ includes :
00034 */
00035
00036 extern "C" {
00037 #include "form3.h"
00038 }
00039
00040 #include "gentopo.h"
00041
00042 //using namespace std;
00043
00044 #define MAXPOINTS 120
00045
00046 typedef struct ToPoTyPe {
00047     int cldeg[MAXPOINTS], cnum[MAXPOINTS], clext[MAXPOINTS];
00048     int ncl, nloops, nlegs, npadding;
00049     WORD *vert;
00050     WORD *vertmax;
00051     WORD nvert;
00052     WORD sopi;
00053 } TOPOTYPE;
00054
00055 /*
00056     #[ includes :
00057     #[ GenerateVertices :
00058
00059     Routine is to be used recursively to work its way through a list
00060     of possible vertices. The array of vertices is in TopoInf->vert
00061     with TopoInf->nvert the number of possible vertices.
00062     Currently we allow in TopoInf->vert only vertices with 3 or more edges.
00063
00064     We work with a point system. Each n-point vertex contributes n-2 points.
00065     When all points are assigned, we can call mgraph->generate().
00066
00067     The number of vertices of a given number of edges is stored in
00068     TopoInf->cnum[..] but the loop that determines how many there are
00069     may be limited by the corresponding element in TopoInf->vertmax[level]
00070 */
00071
00072 int GenerateVertices(TOPOTYPE *TopoInf, int pointsremaining, int level)
00073 {
00074     int i, j;
00075
00076     for ( i = pointsremaining, j = 0; i >= 0; i -= TopoInf->vert[level]-2, j++ ) {
00077         if ( TopoInf->vertmax && TopoInf->vertmax[level] >= 0
00078             && j > TopoInf->vertmax[level] ) break;
00079         if ( i == 0 ) { // We got one!
00080             T_MGraph *mgraph;
00081             TopoInf->cldeg[TopoInf->ncl] = TopoInf->vert[level];
00082             TopoInf->cnum[TopoInf->ncl] = j;
00083             TopoInf->clext[TopoInf->ncl] = 0;
00084             TopoInf->ncl++;
00085
00086             mgraph = new T_MGraph(0, TopoInf->ncl, TopoInf->cldeg,
```



```

00087         TopoInf->clnum, TopoInf->clext, TopoInf->sopi);
00088
00089         mgraph->generate();
00090
00091         delete mgraph;
00092
00093         TopoInf->ncl--;
00094
00095         break;
00096     }
00097     if ( level < TopoInf->nvert-1 ) {
00098         if ( j > 0 ) {
00099             TopoInf->cldeg[TopoInf->ncl] = TopoInf->vert[level];
00100             TopoInf->clnum[TopoInf->ncl] = j;
00101             TopoInf->clext[TopoInf->ncl] = 0;
00102             TopoInf->ncl++;
00103         }
00104         if ( GenerateVertices(TopoInf,i,level+1) < 0 ) return(-1);
00105         if ( j > 0 ) { TopoInf->ncl--; }
00106     }
00107 }
00108 return(0);
00109 }
00110
00111 /*
00112  #] GenerateVertices :
00113  #[ GenerateTopologies :
00114
00115  Note that setmax, option1 and option2 are optional.
00116  Default values are -1,0,0
00117
00118  vert is a pointer to a set of numbers indicating the types of vertices
00119  that are allowed.
00120  nvert is the number of elements in vert.
00121 */
00122
00123 int GenerateTopologies(PHEAD WORD nloops, WORD nlegs, WORD setvert, WORD setmax)
00124 {
00125     TOPOTYPE TopoInf;
00126     int i, points, identical = 0;
00127     DUMMYUSE(AT.nfac);
00128
00129     if ( nlegs == -2 ) {
00130         nlegs = 2;
00131         identical = 1;
00132     }
00133     TopoInf.vert = &(SetElements[Sets[setvert].first]);
00134     TopoInf.nvert = Sets[setvert].last-Sets[setvert].first;
00135
00136     if ( setmax >= 0 ) TopoInf.vertmax = &(SetElements[Sets[setmax].first]);
00137     else TopoInf.vertmax = 0;
00138
00139     // point counting: an n-point vertex counts for n-2 points.
00140
00141     points = 2*nloops-2+nlegs;
00142     if ( points >= MAXPOINTS ) {
00143         MLOCK(ErrorMessageLock);
00144         MesPrint("GenerateTopologies: %d loops and %d legs considered excessive",nloops,nlegs);
00145         MUNLOCK(ErrorMessageLock);
00146         Terminate(-1);
00147     }
00148
00149     // First the external nodes.
00150
00151     for ( i = 0; i < nlegs; i++ ) {
00152         TopoInf.cldeg[i] = 1; TopoInf.clnum[i] = 1; TopoInf.clext[i] = 1;
00153     }
00154     if ( identical == 1 ) { /* Only propagator topologies..... */
00155         nlegs = 1;
00156         TopoInf.clnum[0] = 2;
00157     }
00158     TopoInf.ncl = nlegs;
00159     TopoInf.sopi = 1; // For now
00160     if ( GenerateVertices(&TopoInf,points,0) != 0 ) {
00161         MLOCK(ErrorMessageLock);
00162         MesPrint("Called from GenerateTopologies with %d loops and %d legs considered
00163 excessive",nloops,nlegs);
00164         MUNLOCK(ErrorMessageLock);
00165         Terminate(-1);
00166     }
00167     return(0);
00168 }
00169
00170 /*
00171  #] GenerateTopologies :
00172  #[ toForm :
00173 */

```

```

00173
00174 //=====
00175 // Output for FROM called by genTopo each time a new graph is
00176 // generated
00177 //
00178 void toForm(T_EGraph *egraph)
00179 {
00180     GETIDENTITY;
00181     //
00182     // skipext : boolean; whether to skip external node/line in the output.
00183     //
00184     // const int skipext = 1;
00185
00186     int n, lg, ed, i, fromset;
00187
00188     WORD *termout = AT.WorkPointer;
00189     WORD *t, *tt, *ttstop, *ttend, *ttail;
00190
00191     t = termout+1;
00192
00193     // First pick up part of the original term
00194
00195     tt = AT.TopologiesTerm + 1;
00196     i = AT.TopologiesStart - tt;
00197     NCOPY(t,tt,i)
00198     ttail = tt+tt[1];
00199
00200     // Now the vertices/nodes
00201     // Options are in AT.TopologiesOptions[]
00202
00203     for ( n = 0; n < egraph->nNodes; n++ ) {
00204         if ( ( AT.TopologiesOptions[1] & 1 ) == 1 && egraph->nodes[n].ext ) continue;
00205         tt = t;
00206         *t++ = NODEFUNCTION; t++; FILLFUN(t);
00207         *t++ = -SNUMBER; *t++ = n;
00208         for ( lg = 0; lg < egraph->nodes[n].deg; lg++ ) {
00209             ed = egraph->nodes[n].edges[lg];
00210             if ( ed >= 0 ) { *t++ = -VECTOR; }
00211             else { ed = -ed; *t++ = -MINVECTOR; }
00212
00213             // Now we need to pick up the proper set element.
00214
00215             fromset = egraph->edges[ed].ext ? AT.setexterntopo : AT.setinterntopo;
00216             *t++ = SetElements[Sets[fromset].first+egraph->edges[ed].momn-1];
00217         }
00218         tt[1] = t-tt;
00219     }
00220
00221     if ( ( AT.TopologiesOptions[0] & 1 ) == 1 ) {
00222         // Note that the edges count from 1.
00223         for ( n = 1; n <= egraph->nEdges; n++ ) {
00224             if ( ( AT.TopologiesOptions[1] & 1 ) == 1 && egraph->edges[n].ext ) continue;
00225             tt = t;
00226             *t++ = EDGE; t++; FILLFUN(t);
00227             *t++ = -SNUMBER; *t++ = egraph->edges[n].nodes[0];
00228             *t++ = -SNUMBER; *t++ = egraph->edges[n].nodes[1];
00229             *t++ = -VECTOR;
00230             fromset = egraph->edges[n].ext ? AT.setexterntopo : AT.setinterntopo;
00231             *t++ = SetElements[Sets[fromset].first+egraph->edges[n].momn-1];
00232             tt[1] = t-tt;
00233         }
00234     }
00235
00236     // Now the tail end
00237
00238     ttend = AT.TopologiesTerm; ttend = ttend+ttend[0];
00239     ttstop = ttend - ABS(ttend[-1]);
00240     i = ttstop - ttail;
00241     NCOPY(t,ttail,i)
00242
00243     // Finally the coefficient
00244     // The topological coefficient should be in egraph->wsum, egraph->nwsum
00245     // as a FORM Long number
00246
00247     #ifdef WITHFACTOR
00248     if ( egraph->nwsum == 1 && egraph->wsum[0] == 1 ) {
00249         while ( ttstop < ttend ) *t++ = *ttstop++;
00250     }
00251     else {
00252         WORD newsize;
00253         newsize = ttend[-1];
00254         newsize = REDLENG(newsize);
00255         if ( Divvy(BHEAD (UWORD *)ttstop,&newsize,egraph->wsum,egraph->nwsum) )
00256             goto OnError;
00257         newsize = INCLENG(newsize);
00258         i = ABS(newsize)-1;
00259         NCOPY(t,ttstop,i)

```

```

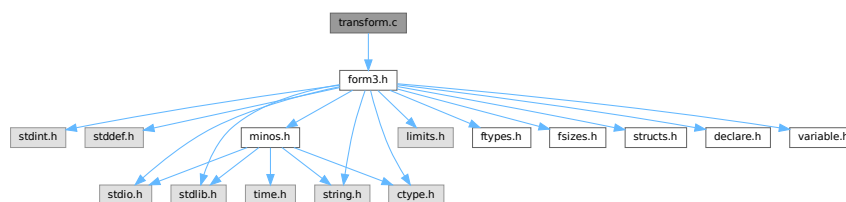
00260         *t++ = newsize;
00261     }
00262 #else
00263     while (ttstop < ttend ) *t++ = *ttstop++;
00264 #endif
00265     *termout = t - termout;
00266     AT.WorkPointer = t;
00267
00268     if ( Generator(BHEAD termout,AT.TopologiesLevel) < 0 ) {
00269 // OnError:
00271         MLOCK(ErrorMessageLock);
00272         MesPrint("Called from the topologies routine toForm");
00273         MUNLOCK(ErrorMessageLock);
00274         Terminate(-1);
00275     }
00276
00277     AT.WorkPointer = termout;
00278 }
00279
00280 /*
00281 #] toForm :
00282 */
00283

```

## 4.150 transform.c File Reference

```
#include "form3.h"
```

Include dependency graph for transform.c:



### Functions

- int [CoTransform](#) (UBYTE \*in)
- int [RunTransform](#) (PHEAD WORD \*term, WORD \*params)
- int [RunEncode](#) (PHEAD WORD \*fun, WORD \*args, WORD \*info)
- int [RunDecode](#) (PHEAD WORD \*fun, WORD \*args, WORD \*info)
- int [RunReplace](#) (PHEAD WORD \*fun, WORD \*args, WORD \*info)
- int [RunImplode](#) (WORD \*fun, WORD \*args)
- int [RunExplode](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunPermute](#) (PHEAD WORD \*fun, WORD \*args, WORD \*info)
- int [RunReverse](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunDedup](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunCycle](#) (PHEAD WORD \*fun, WORD \*args, WORD \*info)
- int [RunAddArg](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunMulArg](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunIsLyndon](#) (PHEAD WORD \*fun, WORD \*args, int par)
- WORD [RunToLyndon](#) (PHEAD WORD \*fun, WORD \*args, int par)
- int [RunDropArg](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunSelectArg](#) (PHEAD WORD \*fun, WORD \*args)
- int [RunZtoHArg](#) (PHEAD WORD \*fun, WORD \*args)

- int [RunHtoZArg](#) (PHEAD WORD \*fun, WORD \*args)
- int [TestArgNum](#) (int n, int totarg, WORD \*args)
- WORD [PutArgInScratch](#) (WORD \*arg, UWORD \*scrat)
- UBYTE \* [ReadRange](#) (UBYTE \*s, WORD \*out, int par)
- int [FindRange](#) (PHEAD WORD \*args, WORD \*arg1, WORD \*arg2, WORD totarg)

### 4.150.1 Detailed Description

Routines that deal with the transform statement and its varieties.

Definition in file [transform.c](#).

### 4.150.2 Function Documentation

#### 4.150.2.1 CoTransform()

```
int CoTransform (
    UBYTE * in )
```

Definition at line 87 of file [transform.c](#).

#### 4.150.2.2 RunTransform()

```
int RunTransform (
    PHEAD WORD * term,
    WORD * params )
```

Definition at line 732 of file [transform.c](#).

#### 4.150.2.3 RunEncode()

```
int RunEncode (
    PHEAD WORD * fun,
    WORD * args,
    WORD * info )
```

Definition at line 982 of file [transform.c](#).

#### 4.150.2.4 RunDecode()

```
int RunDecode (
    PHEAD WORD * fun,
    WORD * args,
    WORD * info )
```

Definition at line 1169 of file [transform.c](#).

#### 4.150.2.5 RunReplace()

```
int RunReplace (
    PHEAD WORD * fun,
    WORD * args,
    WORD * info )
```

Definition at line 1339 of file [transform.c](#).

#### 4.150.2.6 RunImplode()

```
int RunImplode (
    WORD * fun,
    WORD * args )
```

Definition at line 1817 of file [transform.c](#).

#### 4.150.2.7 RunExplode()

```
int RunExplode (
    PHEAD WORD * fun,
    WORD * args )
```

Definition at line 2018 of file [transform.c](#).

#### 4.150.2.8 RunPermute()

```
int RunPermute (
    PHEAD WORD * fun,
    WORD * args,
    WORD * info )
```

Definition at line 2132 of file [transform.c](#).

#### 4.150.2.9 RunReverse()

```
int RunReverse (
    PHEAD WORD * fun,
    WORD * args )
```

Definition at line 2359 of file [transform.c](#).

#### 4.150.2.10 RunDedup()

```
int RunDedup (
    PHEAD WORD * fun,
    WORD * args )
```

Definition at line 2446 of file [transform.c](#).

#### 4.150.2.11 RunCycle()

```
int RunCycle (
    PHEAD WORD * fun,
    WORD * args,
    WORD * info )
```

Definition at line [2535](#) of file [transform.c](#).

#### 4.150.2.12 RunAddArg()

```
int RunAddArg (
    PHEAD WORD * fun,
    WORD * args )
```

Definition at line [2687](#) of file [transform.c](#).

#### 4.150.2.13 RunMulArg()

```
int RunMulArg (
    PHEAD WORD * fun,
    WORD * args )
```

Definition at line [2776](#) of file [transform.c](#).

#### 4.150.2.14 RunIsLyndon()

```
int RunIsLyndon (
    PHEAD WORD * fun,
    WORD * args,
    int par )
```

Definition at line [2917](#) of file [transform.c](#).

#### 4.150.2.15 RunToLyndon()

```
WORD RunToLyndon (
    PHEAD WORD * fun,
    WORD * args,
    int par )
```

Definition at line [2995](#) of file [transform.c](#).

#### 4.150.2.16 RunDropArg()

```
int RunDropArg (
    PHEAD WORD * fun,
    WORD * args )
```

Definition at line [3107](#) of file [transform.c](#).

#### 4.150.2.17 RunSelectArg()

```
int RunSelectArg (
    PHEAD WORD * fun,
    WORD * args )
```

Definition at line 3133 of file [transform.c](#).

#### 4.150.2.18 RunZtoHArg()

```
int RunZtoHArg (
    PHEAD WORD * fun,
    WORD * args )
```

Definition at line 3161 of file [transform.c](#).

#### 4.150.2.19 RunHtoZArg()

```
int RunHtoZArg (
    PHEAD WORD * fun,
    WORD * args )
```

Definition at line 3217 of file [transform.c](#).

#### 4.150.2.20 TestArgNum()

```
int TestArgNum (
    int n,
    int totarg,
    WORD * args )
```

Definition at line 3283 of file [transform.c](#).

#### 4.150.2.21 PutArgInScratch()

```
WORD PutArgInScratch (
    WORD * arg,
    UWORD * scrat )
```

Definition at line 3352 of file [transform.c](#).

#### 4.150.2.22 ReadRange()

```
UBYTE * ReadRange (
    UBYTE * s,
    WORD * out,
    int par )
```

Definition at line 3388 of file [transform.c](#).

#### 4.150.2.23 FindRange()

```
int FindRange (
    PHEAD WORD * args,
    WORD * arg1,
    WORD * arg2,
    WORD totarg )
```

Definition at line 3548 of file [transform.c](#).

### 4.151 transform.c

[Go to the documentation of this file.](#)

```
00001
00005 /* #[] License : */
00006 /*
00007  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00008  * When using this file you are requested to refer to the publication
00009  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010  * This is considered a matter of courtesy as the development was paid
00011  * for by FOM the Dutch physics granting agency and we would like to
00012  * be able to track its scientific use to convince FOM of its value
00013  * for the community.
00014  *
00015  * This file is part of FORM.
00016  *
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00021  *
00022  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
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00025  * details.
00026  *
00027  * You should have received a copy of the GNU General Public License along
00028  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00029  */
00030 /* #[] License : */
00031 /*
00032  #[] Includes : transform.c
00033  */
00034
00035 #include "form3.h"
00036
00037 /*
00038  #[] Includes :
00039  #[] Transform :
00040  #[] Intro :
00041
00042  Here are the routines for the transform statement. This is a
00043  group of transformations on function arguments or groups of
00044  function arguments. The purpose of this command is that it
00045  avoids repetitive pattern matching.
00046  Syntax:
00047      Transform,SetOfFunctions,OneOrMoreTransformations;
00048  Each transformation is given by
00049      Replace(argfirst,arglast)=(,,)
00050      Encode(argfirst,arglast):base=#
00051      Decode(argfirst,arglast):base=#
00052      Implode(argfirst,arglast)
00053      Explode(argfirst,arglast)
00054      Permute(cycle)(cycle)(cycle)...(cycle)
00055      Reverse(argfirst,arglast)
00056      Dedup(argfirst,arglast)
00057      Cycle(argfirst,arglast)=+/-num
00058      IsLyndon(argfirst,arglast)=(yes,no)
00059      ToLyndon(argfirst,arglast)=(yes,no)
00060  In replace the extra information is
00061      a replace_() without the name of the replace_ function.
00062      This can be as in (0,1,1,0) or (xarg_,1-xarg_) to indicate
00063      a symbolic argument or (x,y,y,x) to exchange x and y, etc.
00064  In Encode and Decode argfirst is the most significant 'word' and
00065  arglast is the least significant 'word'.
00066  Note that we need to introduce the generic symbolic arguments xarg_,
```



```

00067     parg_, iarg_ and farg_.
00068     Examples:
00069         Transform,{H,E}
00070             ,Replace(1:'WEIGHT')=(0,1,1,0)
00071             ,Encode(1:'WEIGHT')=base(2);
00072         Transform,{H,E}
00073             ,Decode(1:'WEIGHT')=base(3)
00074             ,Replace(1:'WEIGHT')=(2,-1,1,0,0,1);
00075     Others that can be added:
00076         symmetrize?
00077
00078     6-may-2016: Changed MAXPOSITIVE2 into MAXPOSITIVE4. This makes room
00079         for the use of dollar variables as arguments.
00080
00081     #] Intro :
00082     #[ CoTransform :
00083 */
00084
00085 static WORD tranarray[10] = { SUBEXPRESSION, SUBEXPSIZE, 0, 1, 0, 0, 0, 0, 0, 0 };
00086
00087 int CoTransform(UBYTE *in)
00088 {
00089     GETIDENTITY
00090     UBYTE *s = in, c, *ss, *Tempbuf;
00091     WORD number, type, i, *work = AT.WorkPointer+2, *wp, range[2], one = 1;
00092     WORD numdol, *wstart;
00093     int error = 0, irhs;
00094     LONG x;
00095     while ( *in == ',' ) in++;
00096     wp = work + 1;
00097 /*
00098     #[ Sets :
00099
00100     First the set specification(s). No sets means all functions (dangerous!)
00101 */
00102     for(;;) {
00103         if ( *in == '{' ) {
00104             s = in+1;
00105             SKIPBRA2(in)
00106             number = DoTempSet(s,in);
00107             in++;
00108             if ( *in != ',' ) {
00109                 c = in[1]; in[1] = 0;
00110                 MesPrint("& %s: A set in a transform statement should be followed by a comma",s);
00111                 in[1] = c; in++;
00112                 if ( error == 0 ) error = 1;
00113             }
00114         }
00115         else if ( *in == '[' || FG.cTable[*in] == 0 ) {
00116             s = in;
00117             in = SkipAName(in);
00118             if ( *in != ',' ) break;
00119             c = *in; *in = 0;
00120             type = GetName(AC.varnames,s,&number,NOAUTO);
00121             if ( type == CFUNCTION ) { number += MAXVARIABLES + FUNCTION; }
00122             else if ( type != CSET ) {
00123                 MesPrint("& %s: A transform statement starts with sets of functions",s);
00124                 if ( error == 0 ) error = 1;
00125             }
00126             *in++ = c;
00127         }
00128         else {
00129             MesPrint("&Illegal syntax in Transform statement",s);
00130             if ( error == 0 ) error = 1;
00131             return(error);
00132         }
00133         if ( number >= 0 ) {
00134             if ( number < MAXVARIABLES ) {
00135 /*
00136                 Check that this is a set of functions
00137 */
00138                 if ( Sets[number].type != CFUNCTION ) {
00139                     MesPrint("&A set in a transform statement should be a set of functions");
00140                     if ( error == 0 ) error = 1;
00141                 }
00142             }
00143         }
00144         else if ( error == 0 ) error = 1;
00145 /*
00146         Now write the number to the right place
00147 */
00148         *wp++ = number;
00149         while ( *in == ',' ) in++;
00150     }
00151     *work = wp - work;
00152     work = wp; wp++;
00153 */

```

```

00154     #] Sets :
00155
00156     Now we should loop over the various transformations
00157 */
00158     while ( *s ) {
00159         in = s;
00160         if ( FG.cTable[*in] != 0 ) {
00161             MesPrint("&Illegal character in Transform statement");
00162             if ( error == 0 ) error = 1;
00163             return(error);
00164         }
00165         in = SkipAName(in);
00166         if ( *in == '>' || *in == '<' || *in == '+' || *in == '-' ) in++;
00167         ss = in;
00168         c = *ss; *ss = 0;
00169         if ( c != '(' ) {
00170             MesPrint("&Illegal syntax in specifying a transformation inside a Transform statement");
00171             if ( error == 0 ) error = 1;
00172             return(error);
00173         }
00174 /*
00175     #[ replace :
00176 */
00177     if ( StrICmp(s, (UBYTE *) "replace") == 0 ) {
00178 /*
00179         Subkeys: (,,) as in replace_(,,)
00180         The idea here is to read the subkeys as the argument
00181         of a replace_ function.
00182         We put the whole together as in the multiply statement (which
00183         could just be a replace_(...)) and compile it.
00184         Then we expand the tree with Generator and check the complete
00185         expression for legality.
00186 */
00187         type = REPLACEARG;
00188     doreplace:
00189         *ss = c;
00190         if ( ( in = ReadRange(in, range, 0) ) == 0 ) {
00191             if ( error == 0 ) error = 1;
00192             return(error);
00193         }
00194         in++;
00195 /*
00196         We have replace(#,#)=(...), and we want dum_(...) (DUMFUN)
00197         to send to the compiler. The pointer is after the '=';
00198 */
00199         s = in;
00200         if ( *s != '(' ) {
00201             MesPrint("&");
00202             if ( error == 0 ) error = 1;
00203             return(error);
00204         }
00205         SKIPBRA3(in);
00206         if ( *in != ')' ) {
00207             MesPrint("&");
00208             if ( error == 0 ) error = 1;
00209             return(error);
00210         }
00211         in++;
00212         if ( *in != ',' && *in != '\0' ) {
00213             MesPrint("&");
00214             if ( error == 0 ) error = 1;
00215             return(error);
00216         }
00217         i = in - s;
00218         ss = Tempbuf = (UBYTE *) Malloc1(i+5, "CoTransform/replace");
00219         *ss++ = 'd'; *ss++ = 'u'; *ss++ = 'm'; *ss++ = '_';
00220         NCOPY(ss, s, i)
00221         *ss++ = 0;
00222         AC.ProtoType = tranarray;
00223         tranarray[4] = AC.cbufnum;
00224         irhs = CompileAlgebra(Tempbuf, RHSIDE, AC.ProtoType);
00225         M_free(Tempbuf, "CoTransform/replace");
00226         if ( irhs < 0 ) {
00227             if ( error == 0 ) error = 1;
00228             return(error);
00229         }
00230         tranarray[2] = irhs;
00231 /*
00232         The result of the compilation goes through Generator during
00233         execution, because that takes care of $-variables.
00234         This is why we could not use replace_ and had to use dum_.
00235 */
00236         *wp++ = ARGGRANGE;
00237         *wp++ = range[0];
00238         *wp++ = range[1];
00239         *wp++ = type;
00240         *wp++ = SUBEXPSIZE+4;

```

```

00241         for ( i = 0; i < SUBEXPSIZE; i++ ) *wp++ = tranarray[i];
00242         *wp++ = 1;
00243         *wp++ = 1;
00244         *wp++ = 3;
00245         *work = wp-work;
00246         work = wp; *wp++ = 0;
00247         s = in;
00248     }
00249 /*
00250     #] replace :
00251     #[ encode/decode :
00252 */
00253     else if ( StrICmp(s,(UBYTE *)"decode" ) == 0 ) {
00254         type = DECODEARG;
00255         goto doencode;
00256     }
00257     else if ( StrICmp(s,(UBYTE *)"encode" ) == 0 ) {
00258         type = ENCODEARG;
00259 doencode:    *ss = c;
00260         if ( ( in = ReadRange(in,range,2) ) == 0 ) {
00261             if ( error == 0 ) error = 1;
00262             return(error);
00263         }
00264         in++;
00265         s = in; while ( FG.cTable[*in] == 0 ) in++;
00266         c = *in; *in = 0;
00267 /*
00268         Subkeys: base=# or base=$var
00269 */
00270         if ( StrICmp(s,(UBYTE *)"base" ) == 0 ) {
00271             *in = c;
00272             if ( *in != '=' ) {
00273                 MesPrint("&Illegal base specification in encode/decode transformation");
00274                 if ( error == 0 ) error = 1;
00275                 return(error);
00276             }
00277             in++;
00278             if ( *in == '$' ) {
00279                 in++; ss = in;
00280                 in = SkipAName(in);
00281                 c = *in; *in = 0;
00282                 if ( GetName(AC.dollarnames,ss,&numdol,NOAUTO) != CDOLLAR ) {
00283                     MesPrint("&%s is undefined",ss-1);
00284                     numdol = AddDollar(ss,DOLINDEX,&one,1);
00285                     return(1);
00286                 }
00287                 *in = c;
00288                 x = -numdol;
00289             }
00290             else {
00291                 x = 0;
00292                 while ( FG.cTable[*in] == 1 ) {
00293                     x = 10*x + *in++ - '0';
00294                     if ( x > MAXPOSITIVE4 ) {
00295 illsize:      MesPrint("&Illegal value for base in encode/decode transformation");
00296                     if ( error == 0 ) error = 1;
00297                     return(error);
00298                 }
00299             }
00300             if ( x <= 1 ) goto illsize;
00301         }
00302         if ( *in != ',' && *in != '\0' ) {
00303             MesPrint("&Illegal termination of transformation");
00304             if ( error == 0 ) error = 1;
00305             return(error);
00306         }
00307     }
00308     else {
00309         MesPrint("&Illegal option in encode/decode transformation");
00310         if ( error == 0 ) error = 1;
00311         return(error);
00312     }
00313 /*
00314     Now we can put the whole statement together
00315     We have the set(s) in work up to wp and the range in range.
00316     The base is in x and the type tells whether it is encode or decode.
00317 */
00318     *wp++ = ARGRANGE;
00319     *wp++ = range[0];
00320     *wp++ = range[1];
00321     *wp++ = type;
00322     *wp++ = 4;
00323     *wp++ = BASECODE;
00324     *wp++ = (WORD)x;
00325     *work = wp-work;
00326     work = wp; *wp++ = 0;
00327     s = in;

```

```

00328     }
00329  /*
00330  #] encode/decode :
00331  #[ implode :
00332  */
00333  else if ( StrICmp(s, (UBYTE *) "implode") == 0
00334           || StrICmp(s, (UBYTE *) "tosumnotation") == 0 ) {
00335  /*
00336           Subkeys: ?
00337  */
00338           type = IMplodeARG;
00339           *ss = c;
00340           if ( ( in = ReadRange(in, range, 1) ) == 0 ) {
00341               if ( error == 0 ) error = 1;
00342               return(error);
00343           }
00344           *wp++ = ARGRANGE;
00345           *wp++ = range[0];
00346           *wp++ = range[1];
00347           *wp++ = type;
00348           *work = wp-work;
00349           work = wp; *wp++ = 0;
00350           s = in;
00351     }
00352  /*
00353  #] implode :
00354  #[ explode :
00355  */
00356  else if ( StrICmp(s, (UBYTE *) "explode") == 0
00357           || StrICmp(s, (UBYTE *) "tointegralnotation") == 0 ) {
00358  /*
00359           Subkeys: ?
00360  */
00361           type = EXPLODEARG;
00362           *ss = c;
00363           if ( ( in = ReadRange(in, range, 1) ) == 0 ) {
00364               if ( error == 0 ) error = 1;
00365               return(error);
00366           }
00367           *wp++ = ARGRANGE;
00368           *wp++ = range[0];
00369           *wp++ = range[1];
00370           *wp++ = type;
00371           *work = wp-work;
00372           work = wp; *wp++ = 0;
00373           s = in;
00374     }
00375  /*
00376  #] explode :
00377  #[ permute :
00378  */
00379  else if ( StrICmp(s, (UBYTE *) "permute") == 0 ) {
00380           type = PERMUTEARG;
00381           *ss = c;
00382           *wp++ = ARGRANGE;
00383           *wp++ = 1;
00384           *wp++ = MAXPOSITIVE4;
00385           *wp++ = type;
00386  /*
00387           Now a sequence of cycles
00388  */
00389           do {
00390               wstart = wp; wp++;
00391               do {
00392                   in++;
00393                   if ( *in == '$' ) {
00394                       WORD number; UBYTE *t;
00395                       in++; t = in;
00396                       while ( FG.cTable[*in] < 2 ) in++;
00397                       c = *in; *in = 0;
00398                       if ( ( number = GetDollar(t) ) < 0 ) {
00399                           MesPrint("&Undefined variable %s", t);
00400                           if ( !error ) error = 1;
00401                           number = AddDollar(t, 0, 0, 0);
00402                       }
00403                       *in = c;
00404                       *wp++ = -number-1;
00405                   }
00406               } else {
00407                   x = 0;
00408                   while ( FG.cTable[*in] == 1 ) {
00409                       x = 10*x + *in++ - '0';
00410                       if ( x > MAXPOSITIVE4 ) {
00411                           MesPrint("&value in permute transformation too large");
00412                           if ( error == 0 ) error = 1;
00413                           return(error);
00414                       }

```

```

00415         }
00416         if ( x == 0 ) {
00417             MesPrint("&value 0 in permute transformation not allowed");
00418             if ( error == 0 ) error = 1;
00419             return(error);
00420         }
00421         *wp++ = (WORD)x-1;
00422     }
00423     } while ( *in == ',' );
00424     if ( *in != ')' ) {
00425         MesPrint("&Illegal syntax in permute transformation");
00426         if ( error == 0 ) error = 1;
00427         return(error);
00428     }
00429     in++;
00430     if ( *in != ',' && *in != '(' && *in != '\\0' ) {
00431         MesPrint("&Illegal ending in permute transformation");
00432         if ( error == 0 ) error = 1;
00433         return(error);
00434     }
00435     *wstart = wp-wstart;
00436     if ( *wstart == 1 ) wstart--;
00437     } while ( *in == '(' );
00438     *work = wp-work;
00439     work = wp; *wp++ = 0;
00440     s = in;
00441 }
00442 /*
00443 #] permute :
00444 #[ reverse :
00445 */
00446 else if ( StrICmp(s, (UBYTE *) "reverse") == 0 ) {
00447     type = REVERSEARG;
00448     *ss = c;
00449     if ( ( in = ReadRange(in, range, 1) ) == 0 ) {
00450         if ( error == 0 ) error = 1;
00451         return(error);
00452     }
00453     *wp++ = ARGRANGE;
00454     *wp++ = range[0];
00455     *wp++ = range[1];
00456     *wp++ = type;
00457     *work = wp-work;
00458     work = wp; *wp++ = 0;
00459     s = in;
00460 }
00461 /*
00462 #] reverse :
00463 #[ dedup :
00464 */
00465 else if ( StrICmp(s, (UBYTE *) "dedup") == 0 ) {
00466     type = DEDUPARG;
00467     *ss = c;
00468     if ( ( in = ReadRange(in, range, 1) ) == 0 ) {
00469         if ( error == 0 ) error = 1;
00470         return(error);
00471     }
00472     *wp++ = ARGRANGE;
00473     *wp++ = range[0];
00474     *wp++ = range[1];
00475     *wp++ = type;
00476     *work = wp-work;
00477     work = wp; *wp++ = 0;
00478     s = in;
00479 }
00480 /*
00481 #] dedup :
00482 #[ cycle :
00483 */
00484 else if ( StrICmp(s, (UBYTE *) "cycle") == 0 ) {
00485     type = CYCLEARG;
00486     *ss = c;
00487     if ( ( in = ReadRange(in, range, 0) ) == 0 ) {
00488         if ( error == 0 ) error = 1;
00489         return(error);
00490     }
00491     *wp++ = ARGRANGE;
00492     *wp++ = range[0];
00493     *wp++ = range[1];
00494     *wp++ = type;
00495 /*
00496 Now a sequence of cycles
00497 */
00498     in++;
00499     if ( *in == '+' ) {
00500     }
00501     else if ( *in == '-' ) {

```

```

00502         one = -1;
00503     }
00504     else {
00505         MesPrint("&Cycle in a Transform statement should be followed by +/-number/$");
00506         if ( error == 0 ) error = 1;
00507         return(error);
00508     }
00509     in++; x = 0;
00510     if ( *in == '$' ) {
00511         UBYTE *si = in;
00512         in++; si = in;
00513         while ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1 ) in++;
00514         c = *in; *in = 0;
00515         if ( ( x = GetDollar(si) ) < 0 ) {
00516             MesPrint("&Undefined $-variable in transform,cycle statement.");
00517             error = 1;
00518         }
00519         *in = c;
00520         if ( one < 0 ) x += MAXPOSITIVE4;
00521         x += MAXPOSITIVE2;
00522         *wp++ = x;
00523     }
00524     else {
00525         while ( FG.cTable[*in] == 1 ) {
00526             x = 10*x + *in++ - '0';
00527             if ( x > MAXPOSITIVE4 ) {
00528                 MesPrint("&Number in cycle in a Transform statement too big");
00529                 if ( error == 0 ) error = 1;
00530                 return(error);
00531             }
00532         }
00533         *wp++ = x*one;
00534     }
00535     *work = wp-work;
00536     work = wp; *wp++ = 0;
00537     s = in;
00538 }
00539 /*
00540 #] cycle :
00541 #[ islyndon/tolyndon :
00542 */
00543 else if ( StrICmp(s,(UBYTE *)"islyndon" ) == 0 ) {
00544     type = ISLYNDON;
00545     goto doreplace;
00546 }
00547 else if ( StrICmp(s,(UBYTE *)"islyndon<" ) == 0 ) {
00548     type = ISLYNDON;
00549     goto doreplace;
00550 }
00551 else if ( StrICmp(s,(UBYTE *)"islyndon-" ) == 0 ) {
00552     type = ISLYNDON;
00553     goto doreplace;
00554 }
00555 else if ( StrICmp(s,(UBYTE *)"islyndon>" ) == 0 ) {
00556     type = ISLYNDONR;
00557     goto doreplace;
00558 }
00559 else if ( StrICmp(s,(UBYTE *)"islyndon+" ) == 0 ) {
00560     type = ISLYNDONR;
00561     goto doreplace;
00562 }
00563 else if ( StrICmp(s,(UBYTE *)"tolyndon" ) == 0 ) {
00564     type = TOLYNDON;
00565     goto doreplace;
00566 }
00567 else if ( StrICmp(s,(UBYTE *)"tolyndon<" ) == 0 ) {
00568     type = TOLYNDON;
00569     goto doreplace;
00570 }
00571 else if ( StrICmp(s,(UBYTE *)"tolyndon-" ) == 0 ) {
00572     type = TOLYNDON;
00573     goto doreplace;
00574 }
00575 else if ( StrICmp(s,(UBYTE *)"tolyndon>" ) == 0 ) {
00576     type = TOLYNDONR;
00577     goto doreplace;
00578 }
00579 else if ( StrICmp(s,(UBYTE *)"tolyndon+" ) == 0 ) {
00580     type = TOLYNDONR;
00581     goto doreplace;
00582 }
00583 /*
00584 #] islyndon/tolyndon :
00585 #[ addarg :
00586 */
00587 else if ( StrICmp(s,(UBYTE *)"addargs" ) == 0 ) {
00588     type = ADDARG;

```

```

00589         *ss = c;
00590         if ( ( in = ReadRange(in,range,1) ) == 0 ) {
00591             if ( error == 0 ) error = 1;
00592             return(error);
00593         }
00594         *wp++ = ARGGRANGE;
00595         *wp++ = range[0];
00596         *wp++ = range[1];
00597         *wp++ = type;
00598         *work = wp-work;
00599         work = wp; *wp++ = 0;
00600         s = in;
00601     }
00602     /*
00603     #] addarg :
00604     #[ mularg :
00605     */
00606     else if ( ( StrICmp(s,(UBYTE *)"mulargs" ) == 0 )
00607         || ( StrICmp(s,(UBYTE *)"multiplyargs" ) == 0 ) ) {
00608         type = MULTIPLYARG;
00609         *ss = c;
00610         if ( ( in = ReadRange(in,range,1) ) == 0 ) {
00611             if ( error == 0 ) error = 1;
00612             return(error);
00613         }
00614         *wp++ = ARGGRANGE;
00615         *wp++ = range[0];
00616         *wp++ = range[1];
00617         *wp++ = type;
00618         *work = wp-work;
00619         work = wp; *wp++ = 0;
00620         s = in;
00621     }
00622     /*
00623     #] mularg :
00624     #[ droparg :
00625     */
00626     else if ( StrICmp(s,(UBYTE *)"dropargs" ) == 0 ) {
00627         type = DROPARG;
00628         *ss = c;
00629         if ( ( in = ReadRange(in,range,1) ) == 0 ) {
00630             if ( error == 0 ) error = 1;
00631             return(error);
00632         }
00633         *wp++ = ARGGRANGE;
00634         *wp++ = range[0];
00635         *wp++ = range[1];
00636         *wp++ = type;
00637         *work = wp-work;
00638         work = wp; *wp++ = 0;
00639         s = in;
00640     }
00641     /*
00642     #] droparg :
00643     #[ selectarg :
00644     */
00645     else if ( StrICmp(s,(UBYTE *)"selectargs" ) == 0 ) {
00646         type = SELECTARG;
00647         *ss = c;
00648         if ( ( in = ReadRange(in,range,1) ) == 0 ) {
00649             if ( error == 0 ) error = 1;
00650             return(error);
00651         }
00652         *wp++ = ARGGRANGE;
00653         *wp++ = range[0];
00654         *wp++ = range[1];
00655         *wp++ = type;
00656         *work = wp-work;
00657         work = wp; *wp++ = 0;
00658         s = in;
00659     }
00660     /*
00661     #] selectarg :
00662     #[ ZtoH :
00663     */
00664     else if ( StrICmp(s,(UBYTE *)"ztoh" ) == 0 ) {
00665         /*
00666         Subkeys: ?
00667         */
00668         type = ZTOHARG;
00669         *ss = c;
00670         if ( ( in = ReadRange(in,range,1) ) == 0 ) {
00671             if ( error == 0 ) error = 1;
00672             return(error);
00673         }
00674         *wp++ = ARGGRANGE;
00675         *wp++ = range[0];

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```

00676         *wp++ = range[1];
00677         *wp++ = type;
00678         *work = wp-work;
00679         work = wp; *wp++ = 0;
00680         s = in;
00681     }
00682     /*
00683     #] ZtoH :
00684     #[ HtoZ :
00685     */
00686     else if ( StrICmp(s, (UBYTE *) "htoz") == 0 ) {
00687     /*
00688         Subkeys: ?
00689     */
00690         type = HTOZARG;
00691         *ss = c;
00692         if ( ( in = ReadRange(in, range, 1) ) == 0 ) {
00693             if ( error == 0 ) error = 1;
00694             return(error);
00695         }
00696         *wp++ = ARGGRANGE;
00697         *wp++ = range[0];
00698         *wp++ = range[1];
00699         *wp++ = type;
00700         *work = wp-work;
00701         work = wp; *wp++ = 0;
00702         s = in;
00703     }
00704     /*
00705     #] HtoZ :
00706     */
00707     else {
00708         MesPrint("&Unknown transformation inside a Transform statement: %s", s);
00709         *ss = c;
00710         if ( error == 0 ) error = 1;
00711         return(error);
00712     }
00713     while ( *s == ',' ) s++;
00714 }
00715 AT.WorkPointer[0] = TYPETRANSFORM;
00716 AT.WorkPointer[1] = i = wp - AT.WorkPointer;
00717 AddNtoL(i, AT.WorkPointer);
00718 return(error);
00719 }
00720
00721 /*
00722     #] CoTransform :
00723     #[ RunTransform :
00724
00725     Executes the transform statement.
00726     This routine hunts down the functions and sends them to the various
00727     action routines.
00728     params: size, #set1, ..., #setn, transformations
00729
00730 */
00731
00732 int RunTransform(PHEAD WORD *term, WORD *params)
00733 {
00734     WORD *t, *tstop, *w, *m, *out, *in, *tt, retval;
00735     WORD *fun, *args, *info, *infoend, *onetransform, *funs, *endfun;
00736     WORD *thearg = 0, *iterm, *newterm, *nt, *oldwork = AT.WorkPointer, sign = 1;
00737     int i;
00738     out = tstop = term + *term;
00739     tstop -= ABS(tstop[-1]);
00740     in = term;
00741     t = term + 1;
00742     while ( t < tstop ) {
00743         endfun = onetransform = params + *params;
00744         funs = params + 1;
00745         if ( *t < FUNCTION ) {}
00746         else if ( funs == endfun ) { /* we do all functions */
00747             hit:;
00748             while ( in < t ) *out++ = *in++;
00749             tt = t + t[1]; fun = out;
00750             while ( in < tt ) *out++ = *in++;
00751             do {
00752                 args = onetransform + 1;
00753                 info = args; while ( *info <= MAXRANGEINDICATOR ) {
00754                     if ( *info == ALLARGS ) info++;
00755                     else if ( *info == NUMARG ) info += 2;
00756                     else if ( *info == ARGGRANGE ) info += 3;
00757                     else if ( *info == MAKEARGS ) info += 3;
00758                 }
00759                 switch ( *info ) {
00760                     case REPLACEARG:
00761                         if ( RunReplace(BHEAD fun, args, info) ) goto abo;
00762                         out = fun + fun[1];

```



```

00763         break;
00764     case ENCODEARG:
00765         if ( RunEncode(BHEAD fun,args,info) ) goto abo;
00766         out = fun + fun[1];
00767         break;
00768     case DECODEARG:
00769         if ( RunDecode(BHEAD fun,args,info) ) goto abo;
00770         out = fun + fun[1];
00771         break;
00772     case IMplodeARG:
00773         if ( RunImplode(fun,args) ) goto abo;
00774         out = fun + fun[1];
00775         break;
00776     case EXPLODEARG:
00777         if ( RunExplode(BHEAD fun,args) ) goto abo;
00778         out = fun + fun[1];
00779         break;
00780     case PERMUTEARG:
00781         if ( RunPermute(BHEAD fun,args,info) ) goto abo;
00782         out = fun + fun[1];
00783         break;
00784     case REVERSEARG:
00785         if ( RunReverse(BHEAD fun,args) ) goto abo;
00786         out = fun + fun[1];
00787         break;
00788     case DEDUPARG:
00789         if ( RunDedup(BHEAD fun,args) ) goto abo;
00790         out = fun + fun[1];
00791         break;
00792     case CYCLEARG:
00793         if ( RunCycle(BHEAD fun,args,info) ) goto abo;
00794         out = fun + fun[1];
00795         break;
00796     case ADDARG:
00797         if ( RunAddArg(BHEAD fun,args) ) goto abo;
00798         out = fun + fun[1];
00799         break;
00800     case MULTIPLYARG:
00801         if ( RunMulArg(BHEAD fun,args) ) goto abo;
00802         out = fun + fun[1];
00803         break;
00804     case ISLYNDON:
00805         if ( ( retval = RunIsLyndon(BHEAD fun,args,1) ) < -1 ) goto abo;
00806         goto returnvalues;
00807         break;
00808     case ISLYNDONR:
00809         if ( ( retval = RunIsLyndon(BHEAD fun,args,-1) ) < -1 ) goto abo;
00810         goto returnvalues;
00811         break;
00812     case TOLYNDON:
00813         if ( ( retval = RunToLyndon(BHEAD fun,args,1) ) < -1 ) goto abo;
00814         goto returnvalues;
00815         break;
00816     case TOLYNDONR:
00817         if ( ( retval = RunToLyndon(BHEAD fun,args,-1) ) < -1 ) goto abo;
00818         goto returnvalues;
00819         break;
00820         if ( retval == -1 ) break;
00821     /*
00822     Work out the yes/no stuff
00823     */
00824     AT.WorkPointer += 2*AM.MaxTer;
00825     if ( AT.WorkPointer > AT.WorkTop ) {
00826         MLOCK(ErrorMessageLock);
00827         MesWork();
00828         MUNLOCK(ErrorMessageLock);
00829         return(-1);
00830     }
00831     iterm = AT.WorkPointer;
00832     info++;
00833     for ( i = 0; i < *info; i++ ) iterm[i] = info[i];
00834     AT.WorkPointer = iterm + *iterm;
00835     AR.Eside = LHSIDEX;
00836     NewSort(BHEAD0);
00837     if ( Generator(BHEAD iterm,AR.Cnumlhs) ) {
00838         LowerSortLevel();
00839         AT.WorkPointer = oldwork;
00840         return(-1);
00841     }
00842     newterm = AT.WorkPointer;
00843     if ( EndSort(BHEAD newterm,1) < 0 ) {}
00844     if ( ( *newterm && *(newterm+*newterm) != 0 ) || *newterm == 0 ) {
00845         MLOCK(ErrorMessageLock);
00846         MesPrint("&yes/no information in islyndon/tolyndon does not evaluate into
a single term");
00847         MUNLOCK(ErrorMessageLock);
00848         return(-1);

```

```

00849     }
00850     AR.Eside = RHSIDE;
00851     i = *newterm; tt = iterm; nt = newterm;
00852     NCOPY(tt,nt,i);
00853     AT.WorkPointer = iterm + *iterm;
00854     info = iterm + 1;
00855     infoend = info+info[1];
00856     info += FUNHEAD;
00857
00858     if ( retval == 0 ) {
00859 /*
00860         Need second argument (=no)
00861 */
00862         if ( info >= infoend ) {
00863 abortlyndon;;
00864             MLOCK(ErrorMessageLock);
00865             MesPrint("There should be a yes and a no argument in
00866 islyndon/tolyndon");
00867             MUNLOCK(ErrorMessageLock);
00868             Terminate(-1);
00869         }
00870         NEXTARG(info)
00871         if ( info >= infoend ) goto abortlyndon;
00872         thearg = info;
00873     }
00874 /*
00875     Need first argument (=yes)
00876 */
00877     if ( info >= infoend ) goto abortlyndon;
00878     thearg = info;
00879     NEXTARG(info)
00880     if ( info >= infoend ) goto abortlyndon;
00881 }
00882 NEXTARG(info)
00883 if ( info < infoend ) goto abortlyndon;
00884 /*
00885     The argument in thearg needs to be copied
00886     We did not pull it through generator to guarantee
00887     that it is a single argument.
00888     The easiest way is to let the routine Normalize
00889     do the job and put everything in an exponent function
00890     with the power one.
00891 */
00892     if ( *thearg == -SNUMBER && thearg[1] == 0 ) {
00893         *term = 0; return(0);
00894     }
00895     if ( *thearg == -SNUMBER && thearg[1] == 1 ) { }
00896     else {
00897         fun = out;
00898         *out++ = EXPONENT; out++; *out++ = 1; FILLFUN3(out);
00899         COPY1ARG(out,thearg);
00900         *out++ = -SNUMBER; *out++ = 1;
00901         fun[1] = out-fun;
00902     }
00903     break;
00904 case DROPARG:
00905     if ( RunDropArg(BHEAD fun,args) ) goto abo;
00906     out = fun + fun[1];
00907     break;
00908 case SELECTARG:
00909     if ( RunSelectArg(BHEAD fun,args) ) goto abo;
00910     out = fun + fun[1];
00911     break;
00912 case ZTOHARG:
00913     {
00914         WORD s = RunZtoHArg(BHEAD fun,args);
00915         if ( s < 0 ) goto abo;
00916         if ( s == 1 ) sign = -sign;
00917         out = fun + fun[1];
00918     }
00919     break;
00920 case HTOZARG:
00921     {
00922         WORD s = RunHtoZArg(BHEAD fun,args);
00923         if ( s < 0 ) goto abo;
00924         if ( s == 1 ) sign = -sign;
00925         out = fun + fun[1];
00926     }
00927     break;
00928 default:
00929     MLOCK(ErrorMessageLock);
00930     MesPrint("Irregular code in execution of transform statement");
00931     MUNLOCK(ErrorMessageLock);
00932     Terminate(-1);
00933 }
00934 onetransform += *onetransform;

```

```

00935         } while ( *onetransform );
00936     }
00937     else {
00938         while ( funs < endfun ) { /* sum over sets */
00939             if ( *funs > MAXVARIABLES ) {
00940                 if ( *t == *funs-MAXVARIABLES ) goto hit;
00941             }
00942             else {
00943                 w = SetElements + Sets[*funs].first;
00944                 m = SetElements + Sets[*funs].last;
00945                 while ( w < m ) { /* sum over set elements */
00946                     if ( *w == *t ) goto hit;
00947                     w++;
00948                 }
00949             }
00950             funs++;
00951         }
00952     }
00953     t += t[1];
00954 }
00955 tt = term + *term; while ( in < tt ) *out++ = *in++;
00956 if ( sign == -1 ) out[-1] = -out[-1];
00957 *tt = i = out - tt;
00958 /*
00959 Now copy the whole thing back
00960 */
00961 NCOPY(term,tt,i)
00962 return(0);
00963 abo:
00964 MLOCK(ErrorMessageLock);
00965 MesCall("RunTransform");
00966 MUNLOCK(ErrorMessageLock);
00967 return(-1);
00968 }
00969
00970 /*
00971 #] RunTransform :
00972 #[ RunEncode :
00973
00974 The info is given by
00975     ENCODEARG,size,BASECODE,num
00976 and possibly more codes to follow.
00977 Only one range is allowed and for now, it should be fully numerical
00978 If the range is in reverse order, we need to either revert it
00979 first or work with an array of pointers.
00980 */
00981 int RunEncode(PHEAD WORD *fun, WORD *args, WORD *info)
00982 {
00983     WORD base, *f, *funstop, *fun1, *t, size1, size2, size3, *arg;
00984     int num, num1, num2, n, i, i1, i2;
00985     UWORD *scrat1, *scrat2, *scrat3;
00986     WORD *tt, *tstop, totarg, arg1, arg2;
00987     if ( functions[fun[0]-FUNCTION].spec != 0 ) return(0);
00988     if ( *args != ARG RANGE ) {
00989         MLOCK(ErrorMessageLock);
00990         MesPrint("Illegal range encountered in RunEncode");
00991         MUNLOCK(ErrorMessageLock);
00992         Terminate(-1);
00993     }
00994     tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
00995     while ( tt < tstop ) { totarg++; NEXTARG(tt); }
00996     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
00997     if ( arg1 > totarg || arg2 > totarg ) return(0);
00998
00999     if ( info[2] == BASECODE ) {
01000         base = info[3];
01001         if ( base <= 0 ) { /* is a dollar variable */
01002             i1 = -base;
01003             base = DolToNumber(BHEAD i1);
01004             if ( AN.ErrorInDollar || base < 2 ) {
01005                 MLOCK(ErrorMessageLock);
01006                 MesPrint("$%s does not have a number value > 1 in base/encode/transform statement in
module %1",
01007                     DOLLARNAME(Dollars,i1),AC.CModule);
01008                 MUNLOCK(ErrorMessageLock);
01009                 Terminate(-1);
01010             }
01011         }
01012     }
01013 /*
01014 Compute number of pointers needed and make sure there is space
01015 */
01016 if ( arg1 > arg2 ) { num1 = arg2; num2 = arg1; }
01017 else { num1 = arg1; num2 = arg2; }
01018 num = num2-num1+1;
01019 WantAddPointers(num);
01020 /*

```

```

01021         Collect the pointers in pWorkspace
01022     */
01023     n = 1; funstop = fun+fun[1]; f = fun+FUNHEAD;
01024     while ( n < num1 ) {
01025         if ( f >= funstop ) return(0);
01026         NEXTARG(f);
01027         n++;
01028     }
01029     fun1 = f; i = 0;
01030     while ( n <= num2 ) {
01031         if ( f >= funstop ) return(0);
01032         if ( *f != -SNUMBER ) {
01033             if ( *f < 0 ) return(0);
01034             t = f + *f - 1;
01035             i1 = ABS(*t);
01036             if ( (*f-i1) != (ARGHEAD+1) ) return(0); /* Not numerical */
01037             i1 = (i1-1)/2 - 1;
01038             t--;
01039             while ( i1 > 0 ) {
01040                 if ( *t != 0 ) return(0); /* Not an integer */
01041                 t--; i1--;
01042             }
01043         }
01044         AT.pWorkspace[AT.pWorkPointer+i] = f;
01045         i++;
01046         NEXTARG(f);
01047         n++;
01048     }
01049     /*
01050     f points now to after the arguments; fun1 at the first.
01051     Now check whether we need to revert the order
01052     */
01053     if ( arg1 > arg2 ) {
01054         i1 = 0; i2 = i-1;
01055         while ( i1 < i2 ) {
01056             t = AT.pWorkspace[AT.pWorkPointer+i1];
01057             AT.pWorkspace[AT.pWorkPointer+i1] = AT.pWorkspace[AT.pWorkPointer+i2];
01058             AT.pWorkspace[AT.pWorkPointer+i2] = t;
01059             i1++; i2--;
01060         }
01061     }
01062     /*
01063     Now we can put the thing together.
01064     x = arg1;
01065     x = base*x+arg2
01066     x = base*x+arg3 etc.
01067     We need three scratch arrays for long integers
01068         (see NumberMalloc in tools.c).
01069     */
01070     scrat1 = NumberMalloc("RunEncode");
01071     scrat2 = NumberMalloc("RunEncode");
01072     scrat3 = NumberMalloc("RunEncode");
01073     arg = AT.pWorkspace[AT.pWorkPointer];
01074     size1 = PutArgInScratch(arg,scrat1);
01075     i--;
01076     while ( i > 0 ) {
01077         if ( MulLong(scrat1,size1,(UWORD *)(&base),1,scrat2,&size2) ) {
01078             NumberFree(scrat3,"RunEncode");
01079             NumberFree(scrat2,"RunEncode");
01080             NumberFree(scrat1,"RunEncode");
01081             goto CalledFrom;
01082         }
01083         NEXTARG(arg);
01084         size3 = PutArgInScratch(arg,scrat3);
01085         if ( AddLong(scrat2,size2,scrat3,size3,scrat1,&size1) ) {
01086             NumberFree(scrat3,"RunEncode");
01087             NumberFree(scrat2,"RunEncode");
01088             NumberFree(scrat1,"RunEncode");
01089             goto CalledFrom;
01090         }
01091         i--;
01092     }
01093     /*
01094     Now put the output in place. There are two cases, one being much
01095     faster than the other. Hence we program both.
01096     Fast: it fits inside the old location.
01097     Slow: it does not.
01098     The total space is f-fun1
01099     */
01100     if ( size1 == 0 ) { /* Fits! */
01101         *fun1++ = -SNUMBER; *fun1++ = 0;
01102         while ( f < funstop ) *fun1++ = *f++;
01103         fun[1] = funstop-fun;
01104     }
01105     else if ( size1 == 1 && scrat1[0] <= MAXPOSITIVE ) { /* Fits! */
01106         *fun1++ = -SNUMBER; *fun1++ = scrat1[0];
01107         while ( f < funstop ) *fun1++ = *f++;

```

```

01108         fun[l] = funl-fun;
01109     }
01110     else if ( sizel == -1 && scratl[0] <= MAXPOSITIVE+1 ) { /* Fits! */
01111         *funl++ = -SNUMBER;
01112         if ( scratl[0] < MAXPOSITIVE ) *funl++ = scratl[0];
01113         else *funl++ = (WORD) (MAXPOSITIVE+1);
01114         while ( f < funstop ) *funl++ = *f++;
01115         fun[l] = funl-fun;
01116     }
01117     else if ( ABS(sizel)*2+2+ARGHEAD <= f-funl ) { /* Fits! */
01118         if ( sizel < 0 ) { size2 = sizel*2-1; sizel = -sizel; size3 = -size2; }
01119         else { size2 = 2*sizel+1; size3 = size2; }
01120         *funl++ = size3+ARGHEAD+1;
01121         *funl++ = 0; FILLARG(funl);
01122         *funl++ = size3+1;
01123         for ( i = 0; i < sizel; i++ ) *funl++ = scratl[i];
01124         *funl++ = 1;
01125         for ( i = 1; i < sizel; i++ ) *funl++ = 0;
01126         *funl++ = size2;
01127         while ( f < funstop ) *funl++ = *f++;
01128         fun[l] = funl-fun;
01129     }
01130     else { /* Does not fit */
01131         t = funstop;
01132         if ( sizel < 0 ) { size2 = sizel*2-1; sizel = -sizel; size3 = -size2; }
01133         else { size2 = 2*sizel+1; size3 = size2; }
01134         *t++ = size3+ARGHEAD+1;
01135         *t++ = 0; FILLARG(t);
01136         *t++ = size3+1;
01137         for ( i = 0; i < sizel; i++ ) *t++ = scratl[i];
01138         *t++ = 1;
01139         for ( i = 1; i < sizel; i++ ) *t++ = 0;
01140         *t++ = size2;
01141         while ( f < funstop ) *t++ = *f++;
01142         f = funstop;
01143         while ( f < t ) *funl++ = *f++;
01144         fun[l] = funl-fun;
01145     }
01146     NumberFree(scrat3,"RunEncode");
01147     NumberFree(scrat2,"RunEncode");
01148     NumberFree(scratl,"RunEncode");
01149 }
01150 else {
01151     MLOCK(ErrorMessageLock);
01152     MesPrint("Unimplemented type of encoding encountered in RunEncode");
01153     MUNLOCK(ErrorMessageLock);
01154     Terminate(-1);
01155 }
01156 return(0);
01157 CalledFrom:
01158 MLOCK(ErrorMessageLock);
01159 MesCall("RunEncode");
01160 MUNLOCK(ErrorMessageLock);
01161 return(-1);
01162 }
01163
01164 /*
01165     #] RunEncode :
01166     #[ RunDecode :
01167 */
01168
01169 int RunDecode(PHEAD WORD *fun, WORD *args, WORD *info)
01170 {
01171     WORD base, num, num1, num2, n, *f, *funstop, *funl, sizel, size2, size3, *t;
01172     WORD i1, i2, i, sig;
01173     UWORD *scratl, *scrat2, *scrat3;
01174     WORD *tt, *tstop, totarg, arg1, arg2;
01175     if ( functions[fun[0]-FUNCTION].spec != 0 ) return(0);
01176     if ( *args != ARGRANGE ) {
01177         MLOCK(ErrorMessageLock);
01178         MesPrint("Illegal range encountered in RunDecode");
01179         MUNLOCK(ErrorMessageLock);
01180         Terminate(-1);
01181     }
01182     tt = fun+FUNHEAD; tstop = fun+fun[l]; totarg = 0;
01183     while ( tt < tstop ) { totarg++; NEXTARG(tt); }
01184     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
01185     if ( arg1 > totarg && arg2 > totarg ) return(0);
01186     if ( info[2] == BASECODE ) {
01187         base = info[3];
01188         if ( base <= 0 ) { /* is a dollar variable */
01189             i1 = -base;
01190             base = DolToNumber(BHEAD i1);
01191             if ( AN.ErrorInDollar || base < 2 ) {
01192                 MLOCK(ErrorMessageLock);
01193                 MesPrint("%$s does not have a number value > 1 in base/decode/transform statement in
module %1",

```

```

01194             DOLLARNAME(Dollars,il),AC.CModule);
01195             MUNLOCK(ErrorMessageLock);
01196             Terminate(-1);
01197         }
01198     }
01199     /*
01200     Compute number of output arguments needed
01201     */
01202     if ( arg1 > arg2 ) { num1 = arg2; num2 = arg1; }
01203     else { num1 = arg1; num2 = arg2; }
01204     num = num2-num1+1;
01205     if ( num <= 1 ) return(0);
01206     /*
01207     Find argument num1
01208     */
01209     funstop = fun + fun[1];
01210     f = fun + FUNHEAD; n = 1;
01211     while ( f < funstop ) {
01212         if ( n == num1 ) break;
01213         NEXTARG(f); n++;
01214     }
01215     if ( f >= funstop ) return(0); /* not enough arguments */
01216     /*
01217     Check that f is integer
01218     */
01219     if ( *f == -SNUMBER ) {}
01220     else if ( *f < 0 ) return(0);
01221     else {
01222         t = f + *f - 1;
01223         i1 = ABS(*t);
01224         if ( (*f-i1) != (ARGHEAD+1) ) return(0); /* Not numerical */
01225         i1 = (i1-1)/2 - 1;
01226         t--;
01227         while ( i1 > 0 ) {
01228             if ( *t != 0 ) return(0); /* Not an integer */
01229             t--; i1--;
01230         }
01231     }
01232     fun1 = f;
01233     /*
01234     The argument that should be decoded is in fun1
01235     We have to copy it to scratch
01236     */
01237     scrat1 = NumberMalloc("RunEncode");
01238     scrat2 = NumberMalloc("RunEncode");
01239     scrat3 = NumberMalloc("RunEncode");
01240     size1 = PutArgInScratch(fun1,scrat1);
01241     if ( size1 < 0 ) { sig = -1; size1 = -size1; }
01242     else sig = 1;
01243     /*
01244     We can check first whether this number can be decoded
01245     */
01246     scrat2[0] = base; size2 = 1;
01247     if ( RaisPow(BHEAD scrat2,&size2,num) ) {
01248         NumberFree(scrat3,"RunEncode");
01249         NumberFree(scrat2,"RunEncode");
01250         NumberFree(scrat1,"RunEncode");
01251         goto CalledFrom;
01252     }
01253     if ( BigLong(scrat1,size1,scrat2,size2) >= 0 ) { /* Number too big */
01254         NumberFree(scrat3,"RunEncode");
01255         NumberFree(scrat2,"RunEncode");
01256         NumberFree(scrat1,"RunEncode");
01257         return(0);
01258     }
01259     /*
01260     We need num*2 spaces
01261     */
01262     if ( *fun1 > num*2 ) { /* shrink space */
01263         t = fun1 + 2*num; f = fun1 + *fun1;
01264         while ( f < funstop ) *t++ = *f++;
01265         fun[1] = t - fun;
01266     }
01267     else if ( *fun1 < num*2 ) { /* case includes -SNUMBER */
01268         if ( *fun1 < 0 ) { /* expand space from -SNUMBER */
01269             fun[1] += (num-1)*2;
01270             t = funstop + (num-1)*2;
01271         }
01272         else { /* expand space from general argument */
01273             fun[1] += 2*num - *fun1;
01274             t = funstop + 2*num - *fun1;
01275         }
01276         f = funstop;
01277         while ( f > fun1 ) *--t = *--f;
01278     }
01279     /*
01280     Now there is space for num -SNUMBER arguments filled from the top.

```

```

01281 */
01282     for ( i = num-1; i >= 0; i-- ) {
01283         DivLong(scrat1,size1,(UWORD *)(&base),1,scrat2,&size2,scrat3,&size3);
01284         funl[2*i] = -SNUMBER;
01285         if ( size3 == 0 ) funl[2*i+1] = 0;
01286         else funl[2*i+1] = (WORD)(scrat3[0])*sig;
01287         for ( i1 = 0; i1 < size2; i1++ ) scrat1[i1] = scrat2[i1];
01288         size1 = size2;
01289     }
01290     if ( size2 != 0 ) {
01291         MLOCK(ErrorMessageLock);
01292         MesPrint("RunDecode: number to be decoded is too big");
01293         MUNLOCK(ErrorMessageLock);
01294         NumberFree(scrat3,"RunEncode");
01295         NumberFree(scrat2,"RunEncode");
01296         NumberFree(scrat1,"RunEncode");
01297         goto CalledFrom;
01298     }
01299 /*
01300     Now check whether we should change the order of the arguments
01301 */
01302     if ( arg1 > arg2 ) {
01303         i1 = 1; i2 = 2*num-1;
01304         while ( i2 > i1 ) {
01305             i = funl[i1]; funl[i1] = funl[i2]; funl[i2] = i;
01306             i1 += 2; i2 -= 2;
01307         }
01308     }
01309     NumberFree(scrat3,"RunEncode");
01310     NumberFree(scrat2,"RunEncode");
01311     NumberFree(scrat1,"RunEncode");
01312 }
01313 else {
01314     MLOCK(ErrorMessageLock);
01315     MesPrint("Unimplemented type of encoding encountered in RunDecode");
01316     MUNLOCK(ErrorMessageLock);
01317     Terminate(-1);
01318 }
01319 return(0);
01320 CalledFrom:
01321     MLOCK(ErrorMessageLock);
01322     MesCall("RunDecode");
01323     MUNLOCK(ErrorMessageLock);
01324     return(-1);
01325 }
01326
01327 /*
01328     #] RunDecode :
01329     #[ RunReplace :
01330
01331     Gets the function, passes the arguments and looks whether they
01332     need to be treated. If so, the exact treatment is found in info.
01333     The info is given as if it is a function of type REPLACEMENT but
01334     its name is REPLACEARG (which is NOT a function).
01335     It is performed on the arguments.
01336     The output is at first written after fun and in the end overwrites fun.
01337 */
01338
01339 int RunReplace(PHEAD WORD *fun, WORD *args, WORD *info)
01340 {
01341     int n = 0, i, dirty = 0, totarg, nfix, nwild, ngeneral;
01342     WORD *t, *tt, *u, *tstop, *info1, *infoend, *oldwork = AT.WorkPointer;
01343     WORD *term, *newterm, *nt, *term1, *term2;
01344     WORD wild[4], mask, *term3, *term4, *oldmask = AT.WildMask;
01345     WORD n1, n2, doanyway;
01346     info++;
01347     t = fun; tstop = fun + fun[1]; u = tstop;
01348     for ( i = 0; i < FUNHEAD; i++ ) *u++ = *t++;
01349     tt = t;
01350     if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
01351         totarg = 0;
01352         while ( tt < tstop ) { totarg++; NEXTARG(tt); }
01353     }
01354     else {
01355         totarg = tstop - tt;
01356     }
01357 /*
01358     Now get the info through Generator to bring it to standard form.
01359     info points at a single term that should be sent to Generator.
01360
01361     We want to put the information in the WorkSpace but fun etc lies there
01362     already. This means that we have to move the WorkPointer quite high up.
01363 */
01364     AT.WorkPointer += 2*AM.MaxTer;
01365     if ( AT.WorkPointer > AT.WorkTop ) {
01366         MLOCK(ErrorMessageLock);
01367         MesWork();

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```

01368     MUNLOCK(ErrorMessageLock);
01369     return(-1);
01370 }
01371 term = AT.WorkPointer;
01372 for ( i = 0; i < *info; i++ ) term[i] = info[i];
01373 AT.WorkPointer = term + *term;
01374 AR.Eside = LHSIDEX;
01375 NewSort(BHEAD0);
01376 if ( Generator(BHEAD term,AR.Cnumlhs) ) {
01377     LowerSortLevel();
01378     AT.WorkPointer = oldwork;
01379     return(-1);
01380 }
01381 newterm = AT.WorkPointer;
01382 if ( EndSort(BHEAD newterm,1) < 0 ) {}
01383 if ( ( *newterm && *(newterm+*newterm) != 0 ) || *newterm == 0 ) {
01384     MLOCK(ErrorMessageLock);
01385     MesPrint("&information in replace transformation does not evaluate into a single term");
01386     MUNLOCK(ErrorMessageLock);
01387     return(-1);
01388 }
01389 AR.Eside = RHSIDE;
01390 i = *newterm; tt = term; nt = newterm;
01391 NCOPY(tt,nt,i);
01392 AT.WorkPointer = term + *term;
01393 info = term + 1;
01394
01395 term1 = term + *term;
01396 term2 = term1+1;
01397 *term2++ = REPLACEMENT;
01398 term2++; FILLFUN(term2)
01399 /*
01400 First we count the different types of objects
01401 */
01402 infoend = info + info[1];
01403 info1 = info + FUNHEAD;
01404 nfix = nwild = ngeneral = 0;
01405 while ( info1 < infoend ) {
01406     if ( *info1 == -SNUMBER ) {
01407         nfix++;
01408         info1 += 2; NEXTARG(info1)
01409     }
01410     else if ( *info1 <= -FUNCTION ) {
01411         if ( *info1 == -WILDARGFUN ) {
01412             nwild++;
01413             info1++; NEXTARG(info1)
01414         }
01415         else {
01416             *term2++ = *info1++; COPY1ARG(term2,info1)
01417             ngeneral++;
01418         }
01419     }
01420     else if ( *info1 == -INDEX ) {
01421         if ( info1[1] == WILDARGINDEX + AM.OffsetIndex ) {
01422             nwild++;
01423             info1 += 2; NEXTARG(info1)
01424         }
01425         else {
01426             *term2++ = *info1++; *term2++ = *info1++; COPY1ARG(term2,info1)
01427             ngeneral++;
01428         }
01429     }
01430     else if ( *info1 == -SYMBOL ) {
01431         if ( info1[1] == WILDARGSYMBOL ) {
01432             nwild++;
01433             info1 += 2; NEXTARG(info1)
01434         }
01435         else {
01436             *term2++ = *info1++; *term2++ = *info1++; COPY1ARG(term2,info1)
01437             ngeneral++;
01438         }
01439     }
01440     else if ( *info1 == -MINVECTOR || *info1 == -VECTOR ) {
01441         if ( info1[1] == WILDARGVECTOR + AM.OffsetVector ) {
01442             nwild++;
01443             info1 += 2; NEXTARG(info1)
01444         }
01445         else {
01446             *term2++ = *info1++; *term2++ = *info1++; COPY1ARG(term2,info1)
01447             ngeneral++;
01448         }
01449     }
01450     else {
01451         MLOCK(ErrorMessageLock);
01452         MesPrint("&irregular code found in replace transformation (RunReplace)");
01453         MUNLOCK(ErrorMessageLock);
01454         Terminate(-1);

```



```

01455     }
01456 }
01457 AT.WorkPointer = term2;
01458 *term1 = term2 - term1;
01459 term1[2] = *term1 - 1;
01460 /*
01461 And now stepping through the arguments
01462 */
01463 while ( t < tstop ) {
01464     n++; /* The number of the argument. Now check whether we need it */
01465     if ( TestArgNum(n,totarg,args) == 0 ) {
01466         if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
01467             if ( *t <= -FUNCTION ) { *u++ = *t++; }
01468             else if ( *t < 0 ) { *u++ = *t++; *u++ = *t++; }
01469             else { i = *t; NCOPY(u,t,i) }
01470         }
01471         else *u++ = *t++;
01472         continue;
01473     }
01474 /*
01475 Here we have in info effectively a replace_ function, but with
01476 additionally the possibility of integer arguments. We treat those first
01477 and for the rest we have to do some pattern matching.
01478 Note that the compilation routine should check that there is an
01479 even number of arguments in the replace function.
01480
01481 First we go for number -> something
01482 */
01483 doanyway = 0;
01484 if ( nfix > 0 ) {
01485     if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
01486         if ( *t == -SNUMBER ) {
01487             info1 = info + FUNHEAD;
01488             while ( info1 < infoend ) {
01489                 if ( *info1 == -SNUMBER ) {
01490                     if ( info1[1] == t[1] ) {
01491                         if ( info1[2] == -SNUMBER ) {
01492                             *u++ = -SNUMBER; *u++ = info1[3];
01493                             info1 += 4;
01494                         }
01495                         else {
01496                             info1 += 2;
01497                             if ( info1[0] <= -FUNCTION ) i = 1;
01498                             else if ( info1[0] < 0 ) i = 2;
01499                             else i = *info1;
01500                             NCOPY(u,info1,i)
01501                         }
01502                         t += 2; goto nextt;
01503                     }
01504                     info1 += 2;
01505                     NEXTARG(info1);
01506                 }
01507                 else {
01508                     NEXTARG(info1);
01509                     NEXTARG(info1);
01510                 }
01511             }
01512 /*
01513 Here we had no match in the style of 1->2. It could however
01514 be that xarg_ does something
01515 */
01516 doanyway = 1; n2 = t[1];
01517 }
01518 }
01519 else { /* Tensor */
01520     if ( *t < AM.OffsetIndex && *t >= 0 ) {
01521         info1 = info + FUNHEAD;
01522         while ( info1 < infoend ) {
01523             if ( ( *info1 == -SNUMBER ) && ( info1[1] == *t )
01524                 && ( ( info1[2] == -SNUMBER ) && ( info1[3] >= 0 )
01525                     && ( info1[3] < AM.OffsetIndex )
01526                     || ( info1[2] == -INDEX || info1[2] == -VECTOR
01527                         || info1[2] == -MINVECTOR ) ) ) {
01528                 *u++ = info1[3];
01529                 info1 += 4;
01530                 t++; goto nextt;
01531             }
01532             else {
01533                 NEXTARG(info1);
01534                 NEXTARG(info1);
01535             }
01536         }
01537     }
01538 }
01539 }
01540 else if ( *t == -SNUMBER ) {
01541     doanyway = 1; n2 = t[1];

```

```

01542     }
01543 /*
01544 First we try to catch those elements that have an exact match
01545 in the traditional replace_ part.
01546 This means that *t should be less than zero and match an entry
01547 in the replace_ function that we prepared.
01548 */
01549 if ( ngeneral > 0 ) {
01550     if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
01551         if ( *t < 0 ) {
01552             term3 = term1 + *term1;
01553             term4 = term1 + FUNHEAD;
01554             while ( term4 < term3 ) {
01555                 if ( *term4 == *t && ( *t <= -FUNCTION ||
01556                     ( t[1] == term4[1] ) ) ) break;
01557                 NEXTARG(term4)
01558             }
01559             if ( term4 < term3 ) goto dothisnow;
01560         }
01561     }
01562     else {
01563         term3 = term1 + *term1;
01564         term4 = term1 + FUNHEAD;
01565         while ( term4 < term3 ) {
01566             if ( ( term4[1] == *t ) &&
01567                 ( ( *term4 == -INDEX || *term4 == -VECTOR ||
01568                   ( *term4 == -SYMBOL && term4[1] < AM.OffsetIndex
01569                     && term4[1] >= 0 ) ) ) ) break;
01570             NEXTARG(term4)
01571         }
01572         if ( term4 < term3 ) goto dothisnow;
01573     }
01574 }
01575 /*
01576 First we eliminate the fixed arguments and make a 'new info'
01577 If there is anything left we can continue.
01578 Now we look for whole argument wildcards (arg_, parg_, iarg_ or farg_)
01579 */
01580 if ( nwild > 0 ) {
01581 /*
01582 If we have f(a)*replace_(xarg_,b(xarg_)) this gives f(b(a))
01583 In testing the wildcard we have CheckWild do the work.
01584 This means that we have to set up the special variables
01585 (AT.WildMask,AN.WildValue,AN.NumWild)
01586 */
01587 */
01588 wild[1] = 4;
01589 info1 = info + FUNHEAD;
01590 while ( info1 < infoend ) {
01591     if ( *info1 == -SYMBOL && info1[1] == WILDARGSYMBOL
01592         && ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) ) {
01593         wild[0] = SYMTOSUB;
01594         wild[2] = WILDARGSYMBOL;
01595         wild[3] = 0;
01596         AN.WildValue = wild;
01597         AT.WildMask = &mask;
01598         mask = 0;
01599         AN.NumWild = 1;
01600         if ( *t == -SYMBOL || ( *t > 0 && CheckWild(BHEAD WILDARGSYMBOL,SYMTOSUB,1,t) == 0
01601             || doanyway ) ) {
01602 /*
01603 We put the part in replace in a function and make
01604 a replace_(xarg_,(t argument)).
01605 */
01606 n1 = SYMBOL; n2 = WILDARGSYMBOL;
01607 info1 += 2;
01608 getthisone;;
01609 term3 = term2+1;
01610 if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
01611     *term3++ = DUMFUN; term3++; FILLFUN(term3)
01612     COPY1ARG(term3,info1)
01613 }
01614 else {
01615     *term3++ = fun[0]; term3++; FILLFUN(term3)
01616     *term3++ = *info1;
01617 }
01618 term2[2] = term3 - term2 - 1;
01619 tt = term3;
01620 *term3++ = REPLACEMENT;
01621 term3++; FILLFUN(term3)
01622 *term3++ = -n1;
01623 if ( n1 < FUNCTION ) *term3++ = n2;
01624 if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
01625     term4 = t;
01626     COPY1ARG(term3,term4)
01627 }

```

```

01628         else {
01629             *term3++ = *t;
01630         }
01631         tt[1] = term3 - tt;
01632         *term3++ = 1; *term3++ = 1; *term3++ = 3;
01633         *term2 = term3 - term2;
01634
01635         AT.WorkPointer = term3;
01636         NewSort(BHEAD0);
01637         if ( Generator(BHEAD term2,AR.Cnumlhs) ) {
01638             LowerSortLevel();
01639             AT.WorkPointer = oldwork;
01640             AT.WildMask = oldmask;
01641             return(-1);
01642         }
01643         term4 = AT.WorkPointer;
01644         if ( EndSort(BHEAD term4,1) < 0 ) {}
01645         if ( ( *term4 && *(term4+*term4) != 0 ) || *term4 == 0 ) {
01646             MLOCK(ErrorMessageLock);
01647             MesPrint("information in replace transformation does not evaluate into a
single term");
01648             MUNLOCK(ErrorMessageLock);
01649             return(-1);
01650         }
01651         /*
01652
01653         Now we can copy the new function argument to the output u
01654
01655         i = term4[2]-FUNHEAD;
01656         term3 = term4+FUNHEAD+1;
01657         NCOPY(u,term3,i)
01658         if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
01659             NEXTARG(t)
01660         }
01661         else t++;
01662         AT.WorkPointer = term2;
01663         goto nextt;
01664     }
01665     info1 += 2; NEXTARG(info1)
01666 }
01667 else if ( ( *info1 == -INDEX )
01668 && ( info[1] == WILDARGINDEX + AM.OffsetIndex ) ) {
01669     wild[0] = INDTOSSUB;
01670     wild[2] = WILDARGINDEX+AM.OffsetIndex;
01671     wild[3] = 0;
01672     AN.WildValue = wild;
01673     AT.WildMask = &mask;
01674     mask = 0;
01675     AN.NumWild = 1;
01676     if ( ( functions[fun[0]-FUNCTION].spec == TENSORFUNCTION )
01677 || ( *t == -INDEX || ( *t > 0 && CheckWild(BHEAD WILDARGINDEX,INDTOSSUB,1,t) == 0 )
) ) {
01678         /*
01679
01680         We put the part in replace in a function and make
01681         a replace_(xarg_,(t argument)).
01682
01683         n1 = INDEX; n2 = WILDARGINDEX+AM.OffsetIndex;
01684         info1 += 2;
01685         goto getthisone;
01686     }
01687     info1 += 2; NEXTARG(info1)
01688 }
01689 else if ( ( *info1 == -VECTOR )
01690 && ( info[1] == WILDARGVECTOR + AM.OffsetVector ) ) {
01691     wild[0] = VECTOSSUB;
01692     wild[2] = WILDARGVECTOR+AM.OffsetVector;
01693     wild[3] = 0;
01694     AN.WildValue = wild;
01695     AT.WildMask = &mask;
01696     mask = 0;
01697     AN.NumWild = 1;
01698     if ( functions[fun[0]-FUNCTION].spec == TENSORFUNCTION ) {
01699         if ( *t < MINSPEC ) {
01700             n1 = VECTOR; n2 = WILDARGVECTOR+AM.OffsetVector;
01701             info1 += 2;
01702             goto getthisone;
01703         }
01704     }
01705     else if ( *t == -VECTOR || *t == -MINVECTOR ||
( *t > 0 && CheckWild(BHEAD WILDARGVECTOR,VECTOSSUB,1,t) == 0 ) ) {
01706         /*
01707
01708         We put the part in replace in a function and make
01709         a replace_(xarg_,(t argument)).
01710
01711         n1 = VECTOR; n2 = WILDARGVECTOR+AM.OffsetVector;
01712         info1 += 2;
01713         goto getthisone;

```

```

01713         }
01714         info1 += 2; NEXTARG(info1)
01715     }
01716     else if ( *info1 == -WILDARGFUN ) {
01717         wild[0] = FUNTOFUN;
01718         wild[2] = WILDARGFUN;
01719         wild[3] = 0;
01720         AN.WildValue = wild;
01721         AT.WildMask = &mask;
01722         mask = 0;
01723         AN.NumWild = 1;
01724         if ( *t <= -FUNCTION || ( *t > 0 && CheckWild(BHEAD WILDARGFUN,FUNTOFUN,1,t) == 0
    ) ) {
01725     /*
01726         We put the part in replace in a function and make
01727         a replace_(xarg_,(t argument)).
01728     */
01729         n2 = n1 = -WILDARGFUN; /* n2 is to keep the compiler quiet */
01730         info1++;
01731         goto getthisone;
01732     }
01733     info1++; NEXTARG(info1)
01734 }
01735     else {
01736         NEXTARG(info1) NEXTARG(info1)
01737     }
01738 }
01739 }
01740 if ( ngeneral > 0 ) {
01741     /*
01742         They are all in a replace_ function.
01743         Compose the whole thing into a term with replace_()*dum_(arg)
01744         which will be given to Generator.
01745         If we have f(a(x))*replace_(x,b) this gives f(a(b))
01746     */
01747     dothisnow;
01748     term3 = term2; term4 = term1; i = *term1;
01749     NCOPY(term3,term4,i)
01750     term4 = term3;
01751     if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
01752         *term3++ = DUMFUN; term3++; FILLFUN(term3);
01753         tt = t;
01754         COPY1ARG(term3,tt)
01755     }
01756     else {
01757         *term3++ = fun[0]; term3++; FILLFUN(term3); *term3++ = *t;
01758     }
01759     term4[1] = term3-term4;
01760     *term3++ = 1; *term3++ = 1; *term3++ = 3;
01761     *term2 = term3-term2;
01762     AT.WorkPointer = term3;
01763     NewSort(BHEAD0);
01764     if ( Generator(BHEAD term2,AR.Cnumlhs) ) {
01765         LowerSortLevel();
01766         AT.WorkPointer = oldwork;
01767         AT.WildMask = oldmask;
01768         return(-1);
01769     }
01770     term4 = AT.WorkPointer;
01771     if ( EndSort(BHEAD term4,1) < 0 ) {}
01772     if ( ( *term4 && *(term4+*term4) != 0 ) || *term4 == 0 ) {
01773         MLOCK(ErrorMessageLock);
01774         MesPrint("&information in replace transformation does not evaluate into a single
term");
01775         MUNLOCK(ErrorMessageLock);
01776         return(-1);
01777     }
01778     /*
01779         Now we can copy the new function argument to the output u
01780     */
01781     i = term4[2]-FUNHEAD;
01782     term3 = term4+FUNHEAD+1;
01783     NCOPY(u,term3,i)
01784     NEXTARG(t)
01785     AT.WorkPointer = term2;
01786
01787     goto nextt;
01788 }
01789
01790 /*
01791     No catch. Copy the argument and continue.
01792 */
01793 if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
01794     if ( *t <= -FUNCTION ) { *u++ = *t++; }
01795     else if ( *t < 0 ) { *u++ = *t++; *u++ = *t++; }
01796     else { i = *t; NCOPY(u,t,i) }
01797 }

```

```

01798         else {
01799             *u++ = *t++;
01800         }
01801 nextt::;
01802     }
01803     i = u - tstop; tstop[1] = i; tstop[2] = dirty;
01804     t = fun; u = tstop; NCOPY(t,u,i)
01805     AT.WorkPointer = oldwork;
01806     AT.WildMask = oldmask;
01807     return(0);
01808 }
01809
01810 /*
01811     #] RunReplace :
01812     #[ RunImplode :
01813
01814     Note that we restrict ourselves to short integers and/or single symbols
01815 */
01816
01817 int RunImplode(WORD *fun, WORD *args)
01818 {
01819     GETIDENTITY
01820     WORD *tt, *tstop, totarg, arg1, arg2, num1, num2, il, n;
01821     WORD *f, *t, *ttt, *t4, *ff, *fff;
01822     WORD moveup, numzero, outspace;
01823     if ( functions[fun[0]-FUNCTION].spec != 0 ) return(0);
01824     if ( *args != ARGRANGE ) {
01825         MLOCK(ErrorMessageLock);
01826         MesPrint("Illegal range encountered in RunImplode");
01827         MUNLOCK(ErrorMessageLock);
01828         Terminate(-1);
01829     }
01830     tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
01831     while ( tt < tstop ) { totarg++; NEXTARG(tt); }
01832     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
01833 /*
01834     Get the proper range in forward direction and the number of arguments
01835 */
01836     if ( arg1 > arg2 ) { num1 = arg2; num2 = arg1; }
01837     else { num1 = arg1; num2 = arg2; }
01838     if ( num1 > totarg || num2 > totarg ) return(0);
01839 /*
01840     We need, for the most general case 4 spots for each:
01841         x,pow,coef,sign
01842     Hence we put these in the workspace above the term after tstop
01843 */
01844     n = 1; f = fun+FUNHEAD;
01845     while ( n < num1 ) {
01846         if ( f >= tstop ) return(0);
01847         NEXTARG(f);
01848         n++;
01849     }
01850     ff = f;
01851 /*
01852     We are now at the first argument to be done
01853     Go through the terms and test their validity.
01854     If one of them doesn't conform to the rules we don't do anything.
01855     The terms to be done are put in special notation after the function.
01856     Notation: numsymbol, power, |coef|, sign
01857     If numsymbol is negative there is no symbol.
01858     We do it this way because otherwise stepping backwards (as in range=(4,1))
01859     would be very difficult.
01860 */
01861     tt = tstop;
01862     while ( n <= num2 ) {
01863         if ( f >= tstop ) return(0);
01864         if ( *f == -SNUMBER ) { *tt++ = -1; *tt++ = 0;
01865             if ( f[1] < 0 ) { *tt++ = -f[1]; *tt++ = -1; }
01866             else { *tt++ = f[1]; *tt++ = 1; }
01867             f += 2;
01868         }
01869         else if ( *f == -SYMBOL ) { *tt++ = f[1]; *tt++ = 1; *tt++ = 1; *tt++ = 1; f += 2; }
01870         else if ( *f < 0 ) return(0);
01871         else {
01872             if ( *f != ( f[ARGHEAD]+ARGHEAD ) ) return(0); /* Not a single term */
01873             t = f + *f - 1;
01874             il = ABS(*t);
01875             if ( ( il > 3 ) || ( t[-1] != 1 ) ) return(0); /* Not an integer or too big */
01876             if ( (UWORD)(t[-2]) > MAXPOSITIVE4 ) return(0); /* number too big */
01877             if ( f[ARGHEAD] == il+1 ) { /* numerical which is fine */
01878                 *tt++ = -1; *tt++ = 0; *tt++ = t[-2];
01879                 if ( *t < 0 ) { *tt++ = -1; }
01880                 else { *tt++ = 1; }
01881             }
01882             else if ( ( f[ARGHEAD+1] != SYMBOL )
01883                 || ( f[ARGHEAD+2] != 4 )
01884                 || ( ( f[ARGHEAD+1]+f[ARGHEAD+2] ) < ( t-il ) ) ) return(0);

```

```

01885             /* not a single symbol with a coefficient */
01886         else {
01887             *tt++ = f[ARGHEAD+3];
01888             *tt++ = f[ARGHEAD+4];
01889             *tt++ = t[-2];
01890             if ( *t < 0 ) { *tt++ = -1; }
01891             else { *tt++ = 1; }
01892         }
01893         f += *f;
01894     }
01895     n++;
01896 }
01897 fff = f;
01898 /*
01899 At this point we can do the implosion.
01900 Requirement: no coefficient shall take more than one word.
01901 (a stricter requirement may be needed to keep the explosion contained)
01902 */
01903 if ( arg1 > arg2 ) {
01904 /*
01905     Work backward.
01906 */
01907     t = tt - 4; numzero = 0;
01908     while ( t >= tstop ) {
01909         if ( t[2] == 0 ) numzero++;
01910         else {
01911             if ( numzero > 0 ) {
01912                 t[2] += numzero;
01913                 t4 = t+4;
01914                 ttt = t4 + 4*numzero;
01915                 while ( ttt < tt ) *t4++ = *ttt++;
01916                 tt -= 4*numzero;
01917                 numzero = 0;
01918             }
01919         }
01920         t -= 4;
01921     }
01922 }
01923 else {
01924     t = tstop;
01925     numzero = 0; ttt = t;
01926     while ( t < tt ) {
01927         if ( t[2] == 0 ) numzero++;
01928         else {
01929             if ( numzero > 0 ) {
01930                 t[2] += numzero;
01931                 t4 = t;
01932                 while ( t4 < tt ) *ttt++ = *t4++;
01933                 tt -= 4*numzero;
01934                 t -= 4*numzero;
01935                 ttt = t + 4;
01936                 numzero = 0;
01937             }
01938             else {
01939                 ttt = t + 4;
01940             }
01941         }
01942         t += 4;
01943     }
01944 /*
01945     We may have numzero > 0 at the end. We leave them.
01946     Output space is currently from tstop to tt
01947 */
01948 }
01949 /*
01950 Now we compute the real output space needed
01951 */
01952 t = tstop; outspace = 0;
01953 while ( t < tt ) {
01954     if ( t[0] == -1 ) {
01955         if ( t[2] > MAXPOSITIVE4 ) { return(0); /* Number too big */ }
01956         outspace += 2;
01957     }
01958     else if ( t[1] == 1 && t[2] == 1 && t[3] == 1 ) { outspace += 2; }
01959     else { outspace += 8 + ARGHEAD; }
01960     t += 4;
01961 }
01962 if ( outspace < (fff-ff) ) {
01963     t = tstop;
01964     while ( t < tt ) {
01965         if ( t[0] == -1 ) { *ff++ = -SNUMBER; *ff++ = t[2]*t[3]; }
01966         else if ( t[1] == 1 && t[2] == 1 && t[3] == 1 ) {
01967             *ff++ = -SYMBOL; *ff++ = t[0];
01968         }
01969         else {
01970             *ff++ = 8+ARGHEAD; *ff++ = 0; FILLARG(ff);
01971             *ff++ = 8; *ff++ = SYMBOL; *ff++ = 4; *ff++ = t[0]; *ff++ = t[1];

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```

01972         *ff++ = t[2]; *ff++ = 1; *ff++ = t[3] > 0 ? 3: -3;
01973     }
01974     t += 4;
01975 }
01976 while ( fff < tstop ) *ff++ = *fff++;
01977 fun[1] = ff - fun;
01978 }
01979 else if ( outspace > (fff-ff) ) {
01980 /*
01981     Move the answer up by the required amount.
01982     Move the tail to its new location
01983     Move in things as for outspace == (fff-ff)
01984 */
01985     moveup = outspace-(fff-ff);
01986     ttt = tt + moveup;
01987     t = tt;
01988     while ( t > fff ) *--ttt = *--t;
01989     tt += moveup; tstop += moveup;
01990     fff += moveup;
01991     fun[1] += moveup;
01992     goto moveinto;
01993 }
01994 else {
01995 moveinto:
01996     t = tstop;
01997     while ( t < tt ) {
01998         if ( t[0] == -1 ) { *ff++ = -SNUMBER; *ff++ = t[2]*t[3]; }
01999         else if ( t[1] == 1 && t[2] == 1 && t[3] == 1 ) {
02000             *ff++ = -SYMBOL; *ff++ = t[0];
02001         }
02002         else {
02003             *ff++ = 8+ARGHEAD; *ff++ = 0; FILLARG(ff);
02004             *ff++ = 8; *ff++ = SYMBOL; *ff++ = 4; *ff++ = t[0]; *ff++ = t[1];
02005             *ff++ = t[2]; *ff++ = 1; *ff++ = t[3] > 0 ? 3: -3;
02006         }
02007         t += 4;
02008     }
02009 }
02010 return(0);
02011 }
02012
02013 /*
02014     #] RunImplode :
02015     #[ RunExplode :
02016 */
02017
02018 int RunExplode(PHEAD WORD *fun, WORD *args)
02019 {
02020     WORD arg1, arg2, num1, num2, *tt, *tstop, totarg, *tonew, *newfun;
02021     WORD *ff, *f;
02022     int reverse = 0, iarg, i, numzero;
02023     if ( functions[fun[0]-FUNCTION].spec != 0 ) return(0);
02024     if ( *args != ARGGRANGE ) {
02025         MLOCK(ErrorMessageLock);
02026         MesPrint("Illegal range encountered in RunExplode");
02027         MUNLOCK(ErrorMessageLock);
02028         Terminate(-1);
02029     }
02030     tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
02031     while ( tt < tstop ) { totarg++; NEXTARG(tt); }
02032     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
02033 /*
02034     Get the proper range in forward direction and the number of arguments
02035 */
02036     if ( arg1 > arg2 ) { num1 = arg2; num2 = arg1; reverse = 1; }
02037     else { num1 = arg1; num2 = arg2; }
02038     if ( num1 > totarg || num2 > totarg ) return(0);
02039     if ( tstop + AM.MaxTer > AT.WorkTop ) goto OverWork;
02040 /*
02041     We will make the new function after the old one in the workspace
02042     Find the first argument
02043 */
02044     tonew = newfun = tstop;
02045     ff = fun + FUNHEAD; iarg = 0;
02046     while ( ff < tstop ) {
02047         iarg++;
02048         if ( iarg == num1 ) {
02049             i = ff - fun; f = fun;
02050             NCOPY(tonew,f,i)
02051             break;
02052         }
02053         NEXTARG(ff)
02054     }
02055 /*
02056     We have reached the first argument to be done
02057 */
02058     while ( iarg <= num2 ) {

```

```

02059     if ( *ff == -SYMBOL || ( *ff == -SNUMBER && ff[1] == 0 ) )
02060     { *tonew++ = *ff++; *tonew++ = *ff++; }
02061     else if ( *ff == -SNUMBER ) {
02062         numzero = ABS(ff[1])-1;
02063         if ( reverse ) {
02064             *tonew++ = -SNUMBER; *tonew++ = ff[1] < 0 ? -1: 1;
02065             while ( numzero > 0 ) {
02066                 *tonew++ = -SNUMBER; *tonew++ = 0; numzero--;
02067             }
02068         }
02069         else {
02070             while ( numzero > 0 ) {
02071                 *tonew++ = -SNUMBER; *tonew++ = 0; numzero--;
02072             }
02073             *tonew++ = -SNUMBER; *tonew++ = ff[1] < 0 ? -1: 1;
02074         }
02075         ff += 2;
02076     }
02077     else if ( *ff < 0 ) { return(0); }
02078     else {
02079         if ( *ff != ARGHEAD+8 || ff[ARGHEAD] != 8
02080             || ff[ARGHEAD+1] != SYMBOL || ABS(ff[ARGHEAD+7]) != 3
02081             || ff[ARGHEAD+6] != 1 ) return(0);
02082         numzero = ff[ARGHEAD+5];
02083         if ( numzero >= MAXPOSITIVE4 ) return(0);
02084         numzero--;
02085         if ( reverse ) {
02086             if ( ff[ARGHEAD+7] > 0 ) { *tonew++ = -SNUMBER; *tonew++ = 1; }
02087             else {
02088                 *tonew++ = ARGHEAD+8; *tonew++ = 0; FILLARG(tonew)
02089                 *tonew++ = 8; *tonew++ = SYMBOL; *tonew++ = ff[ARGHEAD+3];
02090                 *tonew++ = ff[ARGHEAD+4]; *tonew++ = 1; *tonew++ = 1;
02091                 *tonew++ = -3;
02092             }
02093             while ( numzero > 0 ) {
02094                 *tonew++ = -SNUMBER; *tonew++ = 0; numzero--;
02095             }
02096         }
02097         else {
02098             while ( numzero > 0 ) {
02099                 *tonew++ = -SNUMBER; *tonew++ = 0; numzero--;
02100             }
02101             *tonew++ = ARGHEAD+8; *tonew++ = 0; FILLARG(tonew)
02102             *tonew++ = 8; *tonew++ = SYMBOL; *tonew++ = 4;
02103             *tonew++ = ff[ARGHEAD+3]; *tonew++ = ff[ARGHEAD+4];
02104             *tonew++ = 1; *tonew++ = 1;
02105             if ( ff[ARGHEAD+7] > 0 ) *tonew++ = 3;
02106             else *tonew++ = -3;
02107         }
02108         ff += *ff;
02109     }
02110     if ( tonew > AT.WorkTop ) goto OverWork;
02111     iarg++;
02112 }
02113 /*
02114 Copy the tail, settle the size and copy the whole thing back.
02115 */
02116 while ( ff < tstop ) *tonew++ = *ff++;
02117 i = newfun[1] = tonew-newfun;
02118 NCOPY(fun,newfun,i)
02119 return(0);
02120 OverWork::
02121 MLOCK(ErrorMessageLock);
02122 MesWork();
02123 MUNLOCK(ErrorMessageLock);
02124 return(-1);
02125 }
02126
02127 /*
02128 #] RunExplode :
02129 #[ RunPermute :
02130 */
02131
02132 int RunPermute(PHEAD WORD *fun, WORD *args, WORD *info)
02133 {
02134     WORD *tt, totarg, *tstop, arg1, arg2, n, num, i, *f, *f1, *f2, *infostop;
02135     WORD *in, *iw, withdollar;
02136     DOLLARS d;
02137     if ( *args != ARGRANGE ) {
02138         MLOCK(ErrorMessageLock);
02139         MesPrint("Illegal range encountered in RunPermute");
02140         MUNLOCK(ErrorMessageLock);
02141         Terminate(-1);
02142     }
02143     if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
02144         tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
02145         while ( tt < tstop ) { totarg++; NEXTARG(tt); }

```



```

02146     arg1 = 1; arg2 = totarg;
02147  /*
02148     We need to:
02149     1: get pointers to the arguments
02150     2: permute the pointers
02151     3: copy the arguments to safe territory in the new order
02152     4: copy this new order back in situ.
02153  */
02154     num = arg2-arg1+1;
02155     WantAddPointers(num); /* Guarantees the presence of enough pointers */
02156     f = fun+FUNHEAD; n = 1; i = 0;
02157     while ( n < arg1 ) { n++; NEXTARG(f) }
02158     f1 = f;
02159     while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; NEXTARG(f) }
02160  /*
02161     Now the permutations
02162  */
02163     info++;
02164     while ( *info ) {
02165         infostop = info + *info;
02166         info++;
02167         if ( *info > totarg ) return(0);
02168  /*
02169         Now we have a look whether there are dollar variables to be expanded
02170         We also shift out all values that are out of range.
02171  */
02172         withddollar = 0; in = info;
02173         while ( in < infostop ) {
02174             if ( *in < 0 ) { /* Dollar variable -(number+1) */
02175                 d = Dollars - *in - 1;
02176 #ifdef WITHPTHREADS
02177                 {
02178                     int nummodopt, dtype = -1, numdollar = -*in-1;
02179                     if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
02180                         for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
02181                             if ( numdollar == ModOptdollars[nummodopt].number ) break;
02182                         }
02183                         if ( nummodopt < NumModOptdollars ) {
02184                             dtype = ModOptdollars[nummodopt].type;
02185                             if ( dtype == MODLOCAL ) {
02186                                 d = ModOptdollars[nummodopt].dstruct+AT.identity;
02187                             }
02188                             else {
02189                                 LOCK(d->pthreadlockread);
02190                             }
02191                         }
02192                     }
02193                 }
02194 #endif
02195                 if ( ( d->type == DOLNUMBER || d->type == DOLTERMS )
02196                     && d->where[0] == 4 && d->where[4] == 0 ) {
02197                     if ( d->where[3] < 0 || d->where[2] != 1 || d->where[1] > totarg ) return(0);
02198                 }
02199                 else if ( d->type == DOLWILDARGS ) {
02200                     iw = d->where+1;
02201                     while ( *iw ) {
02202                         if ( *iw == -SNUMBER ) {
02203                             if ( iw[1] <= 0 || iw[1] > totarg ) return(0);
02204                         }
02205                         else goto IllType;
02206                         iw += 2;
02207                     }
02208                 }
02209                 else {
02210 IllType:
02211                     MLOCK(ErrorMessageLock);
02212                     MesPrint("Illegal type of $-variable in RunPermute");
02213                     MUNLOCK(ErrorMessageLock);
02214                     Terminate(-1);
02215                 }
02216                 withddollar++;
02217             }
02218             else if ( *in > totarg ) return(0);
02219             in++;
02220         }
02221         if ( withddollar ) { /* We need some space for a copy */
02222             WORD *incopy, *tocopy;
02223             incopy = TermMalloc("RunPermute");
02224             tocopy = incopy+1; in = info;
02225             while ( in < infostop ) {
02226                 if ( *in < 0 ) {
02227                     d = Dollars - *in - 1;
02228 #ifdef WITHPTHREADS
02229                     {
02230                         int nummodopt, dtype = -1, numdollar = -*in-1;
02231                         if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
02232                             for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {

```

```

02233         if ( numdollar == ModOptdollars[nummodopt].number ) break;
02234     }
02235     if ( nummodopt < NumModOptdollars ) {
02236         dtype = ModOptdollars[nummodopt].type;
02237         if ( dtype == MODLOCAL ) {
02238             d = ModOptdollars[nummodopt].dstruct+AT.identity;
02239         }
02240         else {
02241             LOCK(d->pthreadlockread);
02242         }
02243     }
02244 }
02245 }
02246 #endif
02247     if ( d->type == DOLNUMBER || d->type == DOLTERMS ) {
02248         *tocopy++ = d->where[1] - 1;
02249     }
02250     else if ( d->type == DOLWILDARGS ) {
02251         iw = d->where+1;
02252         while ( *iw ) {
02253             *tocopy++ = iw[1] - 1;
02254             iw += 2;
02255         }
02256     }
02257     in++;
02258 }
02259     else *tocopy++ = *in++;
02260 }
02261 *tocopy = 0;
02262 *incopy = tocopy - incopy;
02263 in = incopy+1;
02264 tt = AT.pWorkSpace[AT.pWorkPointer+*in];
02265 in++;
02266 while ( in < tocopy ) {
02267     if ( *in > totarg ) return(0);
02268     AT.pWorkSpace[AT.pWorkPointer+in[-1]] = AT.pWorkSpace[AT.pWorkPointer+*in];
02269     in++;
02270 }
02271 AT.pWorkSpace[AT.pWorkPointer+in[-1]] = tt;
02272 TermFree(incopy,"RunPermute");
02273 info = infostop;
02274 }
02275 else {
02276     tt = AT.pWorkSpace[AT.pWorkPointer+*info];
02277     info++;
02278     while ( info < infostop ) {
02279         if ( *info > totarg ) return(0);
02280         AT.pWorkSpace[AT.pWorkPointer+info[-1]] = AT.pWorkSpace[AT.pWorkPointer+*info];
02281         info++;
02282     }
02283     AT.pWorkSpace[AT.pWorkPointer+info[-1]] = tt;
02284 }
02285 }
02286 /*
02287 info++;
02288 while ( *info ) {
02289     infostop = info + *info;
02290     info++;
02291     if ( *info > totarg ) return(0);
02292     tt = AT.pWorkSpace[AT.pWorkPointer+*info];
02293     info++;
02294     while ( info < infostop ) {
02295         if ( *info > totarg ) return(0);
02296         AT.pWorkSpace[AT.pWorkPointer+info[-1]] = AT.pWorkSpace[AT.pWorkPointer+*info];
02297         info++;
02298     }
02299     AT.pWorkSpace[AT.pWorkPointer+info[-1]] = tt;
02300 }
02301 */
02302 /*
02303 And the final cleanup
02304 */
02305 if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
02306 f2 = tstop;
02307 for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; COPY1ARG(f2,f) }
02308 i = f2 - tstop;
02309 NCOPY(f1,tstop,i)
02310 }
02311 else { /* tensors */
02312     tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = tstop-tt;
02313     arg1 = 1; arg2 = totarg;
02314     num = arg2-arg1+1;
02315     WantAddPointers(num); /* Guarantees the presence of enough pointers */
02316     f = fun+FUNHEAD; n = 1; i = 0;
02317     while ( n < arg1 ) { n++; f++; }
02318     f1 = f;
02319     while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; f++; }

```

```

02320 /*
02321     Now the permutations
02322 */
02323     info++;
02324     while ( *info ) {
02325         infostop = info + *info;
02326         info++;
02327         if ( *info > totarg ) return(0);
02328         tt = AT.pWorkSpace[AT.pWorkPointer+*info];
02329         info++;
02330         while ( info < infostop ) {
02331             if ( *info > totarg ) return(0);
02332             AT.pWorkSpace[AT.pWorkPointer+info[-1]] = AT.pWorkSpace[AT.pWorkPointer+*info];
02333             info++;
02334         }
02335         AT.pWorkSpace[AT.pWorkPointer+info[-1]] = tt;
02336     }
02337 /*
02338     And the final cleanup
02339 */
02340     if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
02341     f2 = tstop;
02342     for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; *f2++ = *f++; }
02343     i = f2 - tstop;
02344     NCOPY(f1,tstop,i)
02345 }
02346 return(0);
02347 OverWork:;
02348 MLOCK(ErrorMessageLock);
02349 MesWork();
02350 MUNLOCK(ErrorMessageLock);
02351 return(-1);
02352 }
02353
02354 /*
02355     #] RunPermute :
02356     #[ RunReverse :
02357 */
02358
02359 int RunReverse(PHEAD WORD *fun, WORD *args)
02360 {
02361     WORD *tt, totarg, *tstop, arg1, arg2, n, num, i, *f, *f1, *f2, i1, i2;
02362     if ( *args != ARGRANGE ) {
02363         MLOCK(ErrorMessageLock);
02364         MesPrint("Illegal range encountered in RunReverse");
02365         MUNLOCK(ErrorMessageLock);
02366         Terminate(-1);
02367     }
02368     if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
02369         tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
02370         while ( tt < tstop ) { totarg++; NEXTARG(tt); }
02371         if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
02372 /*
02373     We need to:
02374     1: get pointers to the arguments
02375     2: reverse the order of the pointers
02376     3: copy the arguments to safe territory in the new order
02377     4: copy this new order back in situ.
02378 */
02379         if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
02380         if ( arg2 > totarg ) return(0);
02381
02382         num = arg2-arg1+1;
02383         WantAddPointers(num); /* Guarantees the presence of enough pointers */
02384         f = fun+FUNHEAD; n = 1; i = 0;
02385         while ( n < arg1 ) { n++; NEXTARG(f) }
02386         f1 = f;
02387         while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; NEXTARG(f) }
02388         i1 = i-1; i2 = 0;
02389         while ( i1 > i2 ) {
02390             tt = AT.pWorkSpace[AT.pWorkPointer+i1];
02391             AT.pWorkSpace[AT.pWorkPointer+i1] = AT.pWorkSpace[AT.pWorkPointer+i2];
02392             AT.pWorkSpace[AT.pWorkPointer+i2] = tt;
02393             i1--; i2++;
02394         }
02395         if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
02396         f2 = tstop;
02397         for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; COPY1ARG(f2,f) }
02398         i = f2 - tstop;
02399         NCOPY(f1,tstop,i)
02400     }
02401     else { /* Tensors */
02402         tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = tstop - tt;
02403         if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
02404 /*
02405     We need to:
02406     1: get pointers to the arguments

```

```

02407         2: reverse the order of the pointers
02408         3: copy the arguments to safe territory in the new order
02409         4: copy this new order back in situ.
02410 */
02411     if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
02412     if ( arg2 > totarg ) return(0);
02413
02414     num = arg2-arg1+1;
02415     WantAddPointers(num); /* Guarantees the presence of enough pointers */
02416     f = fun+FUNHEAD; n = 1; i = 0;
02417     while ( n < arg1 ) { n++; f++; }
02418     f1 = f;
02419     while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; f++; }
02420     i1 = i-1; i2 = 0;
02421     while ( i1 > i2 ) {
02422         tt = AT.pWorkSpace[AT.pWorkPointer+i1];
02423         AT.pWorkSpace[AT.pWorkPointer+i1] = AT.pWorkSpace[AT.pWorkPointer+i2];
02424         AT.pWorkSpace[AT.pWorkPointer+i2] = tt;
02425         i1--; i2++;
02426     }
02427     if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
02428     f2 = tstop;
02429     for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; *f2++ = *f++; }
02430     i = f2 - tstop;
02431     NCOPY(f1,tstop,i)
02432 }
02433 return(0);
02434 OverWork:;
02435 MLOCK(ErrorMessageLock);
02436 MesWork();
02437 MUNLOCK(ErrorMessageLock);
02438 return(-1);
02439 }
02440
02441 /*
02442     #] RunReverse :
02443     #[ RunDedup :
02444 */
02445
02446 int RunDedup(PHEAD WORD *fun, WORD *args)
02447 {
02448     WORD *tt, totarg, *tstop, arg1, arg2, n, i, j,k, *f, *f1, *f2, *fd, *fstart;
02449     if ( *args != ARGRANGE ) {
02450         MLOCK(ErrorMessageLock);
02451         MesPrint("Illegal range encountered in RunDedup");
02452         MUNLOCK(ErrorMessageLock);
02453         Terminate(-1);
02454     }
02455     if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
02456         tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
02457         while ( tt < tstop ) { totarg++; NEXTARG(tt); }
02458         if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
02459
02460         if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
02461         if ( arg2 > totarg ) return(0);
02462
02463         f = fun+FUNHEAD; n = 1;
02464         while ( n < arg1 ) { n++; NEXTARG(f) }
02465         f1 = f; // fast forward to first element in range
02466         i = 0; // new argument count
02467         fstart = f1;
02468
02469         for (; n <= arg2; n++ ) {
02470             f2 = fstart;
02471             for ( j = 0; j < i; j++ ) { // check all previous terms
02472                 fd = f2;
02473                 NEXTARG(fd)
02474                 for ( k = 0; k < fd-f2; k++ ) // byte comparison of args
02475                     if ( f2[k] != f[k] ) break;
02476
02477                 if ( k == fd-f2 ) break; // duplicate arg
02478                 f2 = fd;
02479             }
02480
02481             if ( j == i ) {
02482                 // unique factor, copy in situ
02483                 COPY1ARG(f1,f)
02484                 i++;
02485             } else {
02486                 NEXTARG(f)
02487             }
02488         }
02489
02490         // move the terms from after the range
02491         for ( j = n; j <= totarg; j++ ) {
02492             COPY1ARG(f1,f)
02493         }

```

```

02494     fun[1] = f1 - fun; // resize function
02495 }
02496 else { /* Tensors */
02497     tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = tstop - tt;
02498     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
02500     if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
02501     if ( arg2 > totarg ) return(0);
02503     f = fun+FUNHEAD;
02504     i = arg1; // new argument count
02505     n = i;
02507     for (; n <= arg2; n++) {
02508         for ( j = arg1; j < i; j++ ) { // check all previous terms
02509             if ( f[n-1] == f[j-1] ) break; // duplicate arg
02510         }
02511         if ( j == i ) {
02512             // unique factor, copy in situ
02513             f[i-1] = f[n-1];
02514             i++;
02515         }
02516     }
02517     // move the terms from after the range
02518     for (j = n; j <= totarg; j++, i++) {
02519         f[i-1] = f[j-1];
02520     }
02521     fun[1] = f + i - 1 - fun; // resize function
02522 }
02523 return(0);
02524 }
02525 /*
02526     #] RunDedup :
02527     #[ RunCycle :
02528 */
02529 int RunCycle(PHEAD WORD *fun, WORD *args, WORD *info)
02530 {
02531     WORD *tt, totarg, *tstop, arg1, arg2, n, num, i, j, *f, *f1, *f2, x, ncyc, cc;
02532     if ( *args != ARGGRANGE ) {
02533         MLOCK(ErrorMessageLock);
02534         MesPrint("Illegal range encountered in RunCycle");
02535         MUNLOCK(ErrorMessageLock);
02536         Terminate(-1);
02537     }
02538     ncyc = info[1];
02539     if ( ncyc >= MAXPOSITIVE2 ) { /* $ variable */
02540         ncyc -= MAXPOSITIVE2;
02541         if ( ncyc >= MAXPOSITIVE4 ) {
02542             ncyc -= MAXPOSITIVE4; /* -$ */
02543             cc = -1;
02544         }
02545         else cc = 1;
02546         ncyc = DolToNumber(BHEAD ncyc);
02547         if ( AN.ErrorInDollar ) {
02548             MesPrint(" Error in Dollar variable in transform,cycle()=$");
02549             return(-1);
02550         }
02551         if ( ncyc >= MAXPOSITIVE4 || ncyc <= -MAXPOSITIVE4 ) {
02552             MesPrint(" Illegal value from Dollar variable in transform,cycle()=$");
02553             return(-1);
02554         }
02555         ncyc *= cc;
02556     }
02557     if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
02558         tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
02559         while ( tt < tstop ) { totarg++; NEXTARG(tt); }
02560         if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
02561         if ( arg1 > arg2 ) { n = arg1; arg1 = arg2; arg2 = n; }
02562         if ( arg2 > totarg ) return(0);
02563     }
02564     /*
02565     We need to:
02566     1: get pointers to the arguments
02567     2: cycle the pointers
02568     3: copy the arguments to safe territory in the new order
02569     4: copy this new order back in situ.
02570     */
02571     num = arg2-arg1+1;
02572     WantAddPointers(num); /* Guarantees the presence of enough pointers */
02573     f = fun+FUNHEAD; n = 1; i = 0;
02574     while ( n < arg1 ) { n++; NEXTARG(f) }
02575     f1 = f;

```

```

02581     while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; NEXTARG(f) }
02582 /*
02583     Now the cycle(s). First minimize the number of cycles.
02584 */
02585     x = ncy;
02586     if ( x >= i ) {
02587         x %= i;
02588         if ( x > i/2 ) x -= i;
02589     }
02590     else if ( x <= -i ) {
02591         x = -((-x) % i);
02592         if ( x <= -i/2 ) x += i;
02593     }
02594     while ( x ) {
02595         if ( x > 0 ) {
02596             tt = AT.pWorkSpace[AT.pWorkPointer+i-1];
02597             for ( j = i-1; j > 0; j-- )
02598                 AT.pWorkSpace[AT.pWorkPointer+j] = AT.pWorkSpace[AT.pWorkPointer+j-1];
02599             AT.pWorkSpace[AT.pWorkPointer] = tt;
02600             x--;
02601         }
02602         else {
02603             tt = AT.pWorkSpace[AT.pWorkPointer];
02604             for ( j = 1; j < i; j++ )
02605                 AT.pWorkSpace[AT.pWorkPointer+j-1] = AT.pWorkSpace[AT.pWorkPointer+j];
02606             AT.pWorkSpace[AT.pWorkPointer+j-1] = tt;
02607             x++;
02608         }
02609     }
02610 /*
02611     And the final cleanup
02612 */
02613     if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
02614     f2 = tstop;
02615     for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; COPY1ARG(f2,f) }
02616     i = f2 - tstop;
02617     NCOPY(f1,tstop,i)
02618 }
02619 else { /* Tensors */
02620     tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = tstop - tt;
02621     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
02622     if ( arg1 > arg2 ) { n = arg1; arg1 = arg2; arg2 = n; }
02623     if ( arg2 > totarg ) return(0);
02624 /*
02625     We need to:
02626     1: get pointers to the arguments
02627     2: cycle the pointers
02628     3: copy the arguments to safe territory in the new order
02629     4: copy this new order back in situ.
02630 */
02631     num = arg2-arg1+1;
02632     WantAddPointers(num); /* Guarantees the presence of enough pointers */
02633     f = fun+FUNHEAD; n = 1; i = 0;
02634     while ( n < arg1 ) { n++; f++; }
02635     f1 = f;
02636     while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; f++; }
02637 /*
02638     Now the cycle(s). First minimize the number of cycles.
02639 */
02640     x = ncy;
02641     if ( x >= i ) {
02642         x %= i;
02643         if ( x > i/2 ) x -= i;
02644     }
02645     else if ( x <= -i ) {
02646         x = -((-x) % i);
02647         if ( x <= -i/2 ) x += i;
02648     }
02649     while ( x ) {
02650         if ( x > 0 ) {
02651             tt = AT.pWorkSpace[AT.pWorkPointer+i-1];
02652             for ( j = i-1; j > 0; j-- )
02653                 AT.pWorkSpace[AT.pWorkPointer+j] = AT.pWorkSpace[AT.pWorkPointer+j-1];
02654             AT.pWorkSpace[AT.pWorkPointer] = tt;
02655             x--;
02656         }
02657         else {
02658             tt = AT.pWorkSpace[AT.pWorkPointer];
02659             for ( j = 1; j < i; j++ )
02660                 AT.pWorkSpace[AT.pWorkPointer+j-1] = AT.pWorkSpace[AT.pWorkPointer+j];
02661             AT.pWorkSpace[AT.pWorkPointer+j-1] = tt;
02662             x++;
02663         }
02664     }
02665 /*
02666     And the final cleanup
02667 */

```

```

02668     if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
02669     f2 = tstop;
02670     for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; *f2++ = *f++; }
02671     i = f2 - tstop;
02672     NCOPY(f1,tstop,i)
02673 }
02674 return(0);
02675 OverWork:;
02676 MLOCK(ErrorMessageLock);
02677 MesWork();
02678 MUNLOCK(ErrorMessageLock);
02679 return(-1);
02680 }
02681
02682 /*
02683     #] RunCycle :
02684     #[ RunAddArg :
02685 */
02686
02687 int RunAddArg(PHEAD WORD *fun, WORD *args)
02688 {
02689     WORD *tt, totarg, *tstop, arg1, arg2, n, num, *f, *f1, *f2;
02690     WORD scribble[10*ARGHEAD];
02691     LONG space;
02692     if ( *args != ARGRANGE ) {
02693         MLOCK(ErrorMessageLock);
02694         MesPrint("Illegal range encountered in RunAddArg");
02695         MUNLOCK(ErrorMessageLock);
02696         Terminate(-1);
02697     }
02698     if ( functions[fun[0]-FUNCTION].spec == TENSORFUNCTION ) {
02699         MLOCK(ErrorMessageLock);
02700         MesPrint("Illegal attempt to add arguments of a tensor in AddArg");
02701         MUNLOCK(ErrorMessageLock);
02702         Terminate(-1);
02703     }
02704     tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
02705     while ( tt < tstop ) { totarg++; NEXTARG(tt); }
02706     /* ignore functions with no arguments */
02707     if ( totarg == 0 ) return(0);
02708     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
02709 /*
02710     We need to:
02711         1: establish that we actually need to add something
02712         2: start a sort
02713         3: if needed, convert arguments to long arguments
02714         4: send (terms in) argument to StoreTerm
02715         5: EndSort and copy the result back into the function
02716     Note that the function is in the workspace, above the term and no
02717     relevant information is trailing it.
02718 */
02719     if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
02720     if ( arg2 > totarg ) return(0);
02721     num = arg2-arg1+1;
02722     if ( num == 1 ) return(0);
02723     f = fun+FUNHEAD; n = 1;
02724     while ( n < arg1 ) { n++; NEXTARG(f) }
02725     f1 = f;
02726     NewSort(BHEAD0);
02727     while ( n <= arg2 ) {
02728         if ( *f > 0 ) {
02729             f2 = f + *f; f += ARGHEAD;
02730             while ( f < f2 ) { StoreTerm(BHEAD f); f += *f; }
02731         }
02732         else if ( *f == -SNUMBER && f[1] == 0 ) {
02733             f+= 2;
02734         }
02735         else {
02736             ToGeneral(f,scribble,1);
02737             StoreTerm(BHEAD scribble);
02738             NEXTARG(f);
02739         }
02740         n++;
02741     }
02742     if ( EndSort(BHEAD tstop+ARGHEAD,1) < 0 ) return(-1);
02743     num = 0;
02744     f2 = tstop+ARGHEAD;
02745     while ( *f2 ) { f2 += *f2; num++; }
02746     *tstop = f2-tstop;
02747     for ( n = 1; n < ARGHEAD; n++ ) tstop[n] = 0;
02748     if ( num == 1 && ToFast(tstop,tstop) == 1 ) {
02749         f2 = tstop; NEXTARG(f2);
02750     }
02751     if ( *tstop == ARGHEAD ) {
02752         *tstop = -SNUMBER; tstop[1] = 0;
02753         f2 = tstop+2;
02754     }

```

```

02755 /*
02756 Copy the trailing arguments after the new argument, then copy the whole back.
02757 */
02758 while ( f < tstop ) *f2++ = *f++;
02759 while ( f < f2 ) *f1++ = *f++;
02760 space = f1 - fun;
02761 if ( (space+8)*sizeof(WORD) > (UWORD)AM.MaxTer ) {
02762     MLOCK(ErrorMessageLock);
02763     MesWork();
02764     MUNLOCK(ErrorMessageLock);
02765     return(-1);
02766 }
02767 fun[1] = (WORD)space;
02768 return(0);
02769 }
02770
02771 /*
02772 #] RunAddArg :
02773 #[ RunMulArg :
02774 */
02775
02776 int RunMulArg(PHEAD WORD *fun, WORD *args)
02777 {
02778     WORD *t, totarg, *tstop, arg1, arg2, n, *f, nb, *m, i, *w;
02779     WORD *scratch, argbuf[20], argsize, *where, *newterm;
02780     LONG oldcpointer_pos;
02781     CBUF *C = cbuf + AT.ebufnum;
02782     if ( *args != ARGRANGE ) {
02783         MLOCK(ErrorMessageLock);
02784         MesPrint("Illegal range encountered in RunMulArg");
02785         MUNLOCK(ErrorMessageLock);
02786         Terminate(-1);
02787     }
02788     if ( functions[fun[0]-FUNCTION].spec == TENSORFUNCTION ) {
02789         MLOCK(ErrorMessageLock);
02790         MesPrint("Illegal attempt to multiply arguments of a tensor in MulArg");
02791         MUNLOCK(ErrorMessageLock);
02792         Terminate(-1);
02793     }
02794     t = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
02795     while ( t < tstop ) { totarg++; NEXTARG(t); }
02796     /* ignore functions with no arguments */
02797     if ( totarg == 0 ) return(0);
02798     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
02799     if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
02800     if ( arg1 > totarg ) return(0);
02801     if ( arg2 < 1 ) return(0);
02802     if ( arg1 < 1 ) arg1 = 1;
02803     if ( arg2 > totarg ) arg2 = totarg;
02804     if ( arg1 == arg2 ) return(0);
02805 /*
02806 Now we move the arguments to a compiler buffer
02807 Then we create a term in the workspace that is the product of
02808 subexpression pointers to the objects in the compiler buffer.
02809 Next we let Generator work out that term.
02810 Finally we pick up the results from EndSort and put it in the function.
02811 */
02812 f = fun+FUNHEAD; n = 1;
02813 while ( n < arg1 ) { n++; NEXTARG(f) }
02814 t = f;
02815 if ( fun >= AT.WorkSpace && fun < AT.WorkTop ) {
02816     if ( AT.WorkPointer < fun+fun[1] ) AT.WorkPointer = fun+fun[1];
02817 }
02818 scratch = AT.WorkPointer;
02819 w = scratch+1;
02820 oldcpointer_pos = C->Pointer-C->Buffer;
02821 nb = C->numrhs;
02822 while ( n <= arg2 ) {
02823     if ( *t > 0 ) {
02824         argsize = *t - ARGHEAD; where = t + ARGHEAD; t += *t;
02825     }
02826     else if ( *t <= -FUNCTION ) {
02827         argbuf[0] = FUNHEAD+4; argbuf[1] = -*t++; argbuf[2] = FUNHEAD;
02828         for ( i = 2; i < FUNHEAD; i++ ) argbuf[i+1] = 0;
02829         argbuf[FUNHEAD+1] = 1;
02830         argbuf[FUNHEAD+2] = 1;
02831         argbuf[FUNHEAD+3] = 3;
02832         argsize = argbuf[0];
02833         where = argbuf;
02834     }
02835     else if ( *t == -SYMBOL ) {
02836         argbuf[0] = 8; argbuf[1] = SYMBOL; argbuf[2] = 4;
02837         argbuf[3] = t[1]; argbuf[4] = 1;
02838         argbuf[5] = 1; argbuf[6] = 1; argbuf[7] = 3;
02839         argsize = 8; t += 2;
02840         where = argbuf;
02841     }

```



```

02842     else if ( *t == -VECTOR || *t == -MINVECTOR ) {
02843         argbuf[0] = 7; argbuf[1] = INDEX; argbuf[2] = 3;
02844         argbuf[3] = t[1];
02845         argbuf[4] = 1; argbuf[5] = 1;
02846         if ( *t == -MINVECTOR ) argbuf[6] = -3;
02847         else argbuf[6] = 3;
02848         argsize = 7; t += 2;
02849         where = argbuf;
02850     }
02851     else if ( *t == -INDEX ) {
02852         argbuf[0] = 7; argbuf[1] = INDEX; argbuf[2] = 3;
02853         argbuf[3] = t[1];
02854         argbuf[4] = 1; argbuf[5] = 1; argbuf[6] = 3;
02855         argsize = 7; t += 2;
02856         where = argbuf;
02857     }
02858     else if ( *t == -SNUMBER ) {
02859         if ( t[1] < 0 ) {
02860             argbuf[0] = 4; argbuf[1] = -t[1]; argbuf[2] = 1; argbuf[3] = -3;
02861         }
02862         else {
02863             argbuf[0] = 4; argbuf[1] = t[1]; argbuf[2] = 1; argbuf[3] = 3;
02864         }
02865         argsize = 4; t += 2;
02866         where = argbuf;
02867     }
02868     else {
02869         /* unreachable */
02870         return(1);
02871     }
02872 /*
02873     Now add the argbuf to AT.ebufnum
02874 */
02875     m = AddRHS(AT.ebufnum,1);
02876     while ( (m + argsize + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,17);
02877     for ( i = 0; i < argsize; i++ ) m[i] = where[i];
02878     m[i] = 0;
02879     C->Pointer = m + i + 1;
02880     n++;
02881     *w++ = SUBEXPRESSION; *w++ = SUBEXPSIZE; *w++ = C->numrhs; *w++ = 1;
02882     *w++ = AT.ebufnum; FILLSUB(w);
02883 }
02884 *w++ = 1; *w++ = 1; *w++ = 3;
02885 *scratch = w-scratch;
02886 AT.WorkPointer = w;
02887 NewSort(BHEAD0);
02888 Generator(BHEAD scratch,AR.Cnumlhs);
02889 newterm = AT.WorkPointer;
02890 EndSort(BHEAD newterm+ARGHEAD,1);
02891 C->Pointer = C->Buffer+oldcpointer_pos;
02892 C->numrhs = nb;
02893 w = newterm+ARGHEAD; while ( *w ) w += *w;
02894 *newterm = w-newterm; newterm[1] = 0;
02895 if ( ToFast(newterm,newterm) ) {
02896     if ( *newterm <= -FUNCTION ) w = newterm+1;
02897     else w = newterm+2;
02898 }
02899 while ( t < tstop ) *w++ = *t++;
02900 i = w - newterm;
02901 t = newterm; NCOPY(f,t,i);
02902 fun[1] = f-fun;
02903 AT.WorkPointer = scratch;
02904 if ( AT.WorkPointer > AT.WorkSpace && AT.WorkPointer < f ) AT.WorkPointer = f;
02905 return(0);
02906 }
02907 /*
02908 #] RunMulArg :
02909 #[ RunIsLyndon :
02910
02911     Determines whether the range constitutes a Lyndon word.
02912     The two cases of ordering are distinguished by the order of
02913     the numbers of the arguments in the range.
02914 */
02915
02916 int RunIsLyndon(PHEAD WORD *fun, WORD *args, int par)
02917 {
02918     WORD *tt, totarg, *tstop, arg1, arg2, arg, num, *f, n, i;
02919 /* WORD *f1; */
02920     WORD sign, i1, i2, retval;
02921     if ( fun[0] <= GAMMASEVEN && fun[0] >= GAMMA ) return(0);
02922     if ( *args != ARGRANGE ) {
02923         MLOCK(ErrorMessageLock);
02924         MesPrint("Illegal range encountered in RunIsLyndon");
02925         MUNLOCK(ErrorMessageLock);
02926         Terminate(-1);
02927     }
02928 }

```

```

02929     tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
02930     while ( tt < tstop ) { totarg++; NEXTARG(tt); }
02931     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
02932     if ( arg1 > totarg || arg2 > totarg ) return(-1);
02933     /*
02934     Now make a list of the relevant arguments.
02935     */
02936     if ( arg1 == arg2 ) return(1);
02937     if ( arg2 < arg1 ) { /* greater, rather than smaller */
02938         arg = arg1; arg1 = arg2; arg2 = arg; sign = 1;
02939     }
02940     else sign = 0;
02941     num = arg2-arg1+1;
02942     WantAddPointers(num); /* Guarantees the presence of enough pointers */
02943     f = fun+FUNHEAD; n = 1; i = 0;
02944     while ( n < arg1 ) { n++; NEXTARG(f) }
02945     /* f1 = f; */
02946     while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; NEXTARG(f) }
02947     /*
02948     If sign == 1 we should alter the order of the pointers first
02949     */
02950     if ( sign ) {
02951         i1 = i-1; i2 = 0;
02952         while ( i1 > i2 ) {
02953             tt = AT.pWorkSpace[AT.pWorkPointer+i1];
02954             AT.pWorkSpace[AT.pWorkPointer+i1] = AT.pWorkSpace[AT.pWorkPointer+i2];
02955             AT.pWorkSpace[AT.pWorkPointer+i2] = tt;
02956             i1--; i2++;
02957         }
02958     }
02959     /*
02960     The argument range is from f1 to f and the num pointers to the arguments
02961     are in AT.pWorkSpace[AT.pWorkPointer] to AT.pWorkSpace[AT.pWorkPointer+num-1]
02962     */
02963     for ( i1 = 1; i1 < num; i1++ ) {
02964         retval = par * CompArg(AT.pWorkSpace[AT.pWorkPointer+i1],
02965                               AT.pWorkSpace[AT.pWorkPointer]);
02966         if ( retval > 0 ) continue;
02967         if ( retval < 0 ) return(0);
02968         for ( i2 = 1; i2 < num; i2++ ) {
02969             retval = par * CompArg(AT.pWorkSpace[AT.pWorkPointer+(i1+i2)%num],
02970                                   AT.pWorkSpace[AT.pWorkPointer+i2]);
02971             if ( retval < 0 ) return(0);
02972             if ( retval > 0 ) goto nextil;
02973         }
02974     }
02975     /*
02976     If we come here the sequence is not unique.
02977     */
02978     return(0);
02979 nextil:;
02980     }
02981     return(1);
02982 }
02983 /*
02984 #] RunIsLyndon :
02985 #[ RunToLyndon :
02986
02987 Determines whether the range constitutes a Lyndon word.
02988 If not, we rotate it to a Lyndon word. If this is not possible
02989 we return the noLyndon condition.
02990 The two cases of ordering are distinguished by the order of
02991 the numbers of the arguments in the range.
02992 */
02993 WORD RunToLyndon(PHEAD WORD *fun, WORD *args, int par)
02994 {
02995     WORD *tt, totarg, *tstop, arg1, arg2, arg, num, *f, *f1, *f2, n, i;
02996     WORD sign, i1, i2, retval, unique;
02997     if ( fun[0] <= GAMMASEVEN && fun[0] >= GAMMA ) return(0);
02998     if ( *args != ARGRANGE ) {
02999         MLOCK(ErrorMessageLock);
03000         MesPrint("Illegal range encountered in RunToLyndon");
03001         MUNLOCK(ErrorMessageLock);
03002         Terminate(-1);
03003     }
03004     tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
03005     while ( tt < tstop ) { totarg++; NEXTARG(tt); }
03006     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
03007     if ( arg1 > totarg || arg2 > totarg ) return(-1);
03008     /*
03009     Now make a list of the relevant arguments.
03010     */
03011     if ( arg1 == arg2 ) return(1);
03012     if ( arg2 < arg1 ) { /* greater, rather than smaller */
03013         arg = arg1; arg1 = arg2; arg2 = arg; sign = 1;
03014     }
03015     else sign = 0;

```

```

03016     }
03017     else sign = 0;
03018
03019     num = arg2-arg1+1;
03020     WantAddPointers((2*num)); /* Guarantees the presence of enough pointers */
03021     f = fun+FUNHEAD; n = 1; i = 0;
03022     while ( n < arg1 ) { n++; NEXTARG(f) }
03023     f1 = f;
03024     while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; NEXTARG(f) }
03025 /*
03026     If sign == 1 we should alter the order of the pointers first
03027 */
03028     if ( sign ) {
03029         i1 = i-1; i2 = 0;
03030         while ( i1 > i2 ) {
03031             tt = AT.pWorkSpace[AT.pWorkPointer+i1];
03032             AT.pWorkSpace[AT.pWorkPointer+i1] = AT.pWorkSpace[AT.pWorkPointer+i2];
03033             AT.pWorkSpace[AT.pWorkPointer+i2] = tt;
03034             i1--; i2++;
03035         }
03036     }
03037 /*
03038     The argument range is from f1 to f and the num pointers to the arguments
03039     are in AT.pWorkSpace[AT.pWorkPointer] to AT.pWorkSpace[AT.pWorkPointer+num-1]
03040 */
03041     unique = 1;
03042     for ( i1 = 1; i1 < num; i1++ ) {
03043         retval = par * CompArg(AT.pWorkSpace[AT.pWorkPointer+i1],
03044                               AT.pWorkSpace[AT.pWorkPointer]);
03045         if ( retval > 0 ) continue;
03046         if ( retval < 0 ) {
03047             Rotate;
03048             /*
03049                 Rotate so that i1 becomes the zero element. Then start again.
03050             */
03051             for ( i2 = 0; i2 < num; i2++ ) {
03052                 AT.pWorkSpace[AT.pWorkPointer+num+i2] =
03053                     AT.pWorkSpace[AT.pWorkPointer+(i1+i2)%num];
03054             }
03055             for ( i2 = 0; i2 < num; i2++ ) {
03056                 AT.pWorkSpace[AT.pWorkPointer+i2] =
03057                     AT.pWorkSpace[AT.pWorkPointer+i2+num];
03058             }
03059             i1 = 0;
03060             goto nextil;
03061         }
03062         for ( i2 = 1; i2 < num; i2++ ) {
03063             retval = par * CompArg(AT.pWorkSpace[AT.pWorkPointer+(i1+i2)%num],
03064                                   AT.pWorkSpace[AT.pWorkPointer+i2]);
03065             if ( retval < 0 ) goto Rotate;
03066             if ( retval > 0 ) goto nextil;
03067         }
03068     /*
03069         If we come here the sequence is not unique.
03070     */
03071     unique = 0;
03072 nextil;
03073     }
03074     if ( sign ) {
03075         i1 = i-1; i2 = 0;
03076         while ( i1 > i2 ) {
03077             tt = AT.pWorkSpace[AT.pWorkPointer+i1];
03078             AT.pWorkSpace[AT.pWorkPointer+i1] = AT.pWorkSpace[AT.pWorkPointer+i2];
03079             AT.pWorkSpace[AT.pWorkPointer+i2] = tt;
03080             i1--; i2++;
03081         }
03082     }
03083 /*
03084     Now rewrite the arguments into the proper order
03085 */
03086     if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
03087     f2 = tstop;
03088     for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; COPY1ARG(f2,f) }
03089     i = f2 - tstop;
03090     NCOPY(f1,tstop,i)
03091 /*
03092     The return value indicates whether we have a Lyndon word
03093 */
03094     return(unique);
03095 OverWork;
03096     MLOCK(ErrorMessageLock);
03097     MesWork();
03098     MUNLOCK(ErrorMessageLock);
03099     return(-2);
03100 }
03101
03102 /*

```

```

03103         #] RunToLyndon :
03104         #[ RunDropArg :
03105     */
03106
03107 int RunDropArg(PHEAD WORD *fun, WORD *args)
03108 {
03109     WORD *t, *tstop, *f, totarg, arg1, arg2, n;
03110
03111     t = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
03112     while ( t < tstop ) { totarg++; NEXTARG(t); }
03113     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
03114     if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
03115     if ( arg1 > totarg ) return(0);
03116     if ( arg2 < 1 ) return(0);
03117     if ( arg1 < 1 ) arg1 = 1;
03118     if ( arg2 > totarg ) arg2 = totarg;
03119     f = fun+FUNHEAD; n = 1;
03120     while ( n < arg1 ) { n++; NEXTARG(f) }
03121     t = f;
03122     while ( n <= arg2 ) { n++; NEXTARG(t) }
03123     while ( t < tstop ) *f++ = *t++;
03124     fun[1] = f-fun;
03125     return(0);
03126 }
03127
03128 /*
03129     #] RunDropArg :
03130     #[ RunSelectArg :
03131 */
03132
03133 int RunSelectArg(PHEAD WORD *fun, WORD *args)
03134 {
03135     WORD *t, *tstop, *f, *tt, totarg, arg1, arg2, n;
03136
03137     t = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
03138     while ( t < tstop ) { totarg++; NEXTARG(t); }
03139     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
03140     if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
03141     if ( arg1 > totarg ) return(0);
03142     if ( arg2 < 1 ) return(0);
03143     if ( arg1 < 1 ) arg1 = 1;
03144     if ( arg2 > totarg ) arg2 = totarg;
03145     f = fun+FUNHEAD; n = 1; t = f;
03146     while ( n < arg1 ) { n++; NEXTARG(t) }
03147     while ( n <= arg2 ) {
03148         tt = t; NEXTARG(tt)
03149         while ( t < tt ) *f++ = *t++;
03150         n++;
03151     }
03152     fun[1] = f-fun;
03153     return(0);
03154 }
03155
03156 /*
03157     #] RunSelectArg :
03158     #[ RunZtoHArg :
03159 */
03160
03161 int RunZtoHArg(PHEAD WORD *fun, WORD *args)
03162 {
03163     WORD *tt, totarg, *tstop, arg1, arg2, n, i, *f, *f1;
03164     int sign = 0;
03165     WORD *t, *t1, *t2, *t3;
03166     if ( *args != ARGGRANGE ) {
03167         MLOCK(ErrorMessageLock);
03168         MesPrint("Illegal range encountered in RunZtoHArg.");
03169         MUNLOCK(ErrorMessageLock);
03170         Terminate(-1);
03171     }
03172     if ( functions[fun[0]-FUNCTION].spec != 0 ) {
03173         MLOCK(ErrorMessageLock);
03174         MesPrint("The ZtoH transformation can only be executed on regular functions with nonzero
integer arguments.");
03175         MUNLOCK(ErrorMessageLock);
03176         Terminate(-1);
03177     }
03178     tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
03179     while ( tt < tstop ) { totarg++; NEXTARG(tt); }
03180     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
03181 /*
03182     Check the arguments. Should be -SNUMBER x!=0
03183 */
03184     f = fun+FUNHEAD; n = 1;
03185     while ( n < arg1 ) { n++; NEXTARG(f) }
03186     f1 = f;
03187     for ( i = arg1; i <= arg2; i++, f += 2 ) {
03188         if ( *f != -SNUMBER || f[1] == 0 ) return(-1);

```

```

03189     }
03190     /*
03191     Now we need a copy.
03192     */
03193     t = f1; t1 = t2 = tt = TermMalloc("RunZtoHArg");
03194     while ( t < f ) { *t1++ = *t++; *t1++ = *t++; }
03195     t = f1;
03196     while ( t2 < t1 ) {
03197         t += 2;
03198         if ( t2[1] < 0 ) {
03199             t3 = t;
03200             while ( t3 < f ) { t3[1] = -t3[1]; t3 += 2; }
03201         }
03202         t2 += 2;
03203     }
03204     TermFree(tt,"RunZtoHArg");
03205     /*
03206     Now the overall sign.
03207     */
03208     while ( f1 < f ) { if ( f1[1] < 0 ) sign = 1-sign; f1 += 2; }
03209     return(sign);
03210 }
03211
03212 /*
03213     #] RunZtoHArg :
03214     #[ RunHtoZArg :
03215     */
03216
03217 int RunHtoZArg(PHEAD WORD *fun, WORD *args)
03218 {
03219     WORD *tt, totarg, *tstop, arg1, arg2, n, i, *f, *f1, *f2;
03220     int sign = 0;
03221     WORD *t, *t1, *t2;
03222     if ( *args != ARGRANGE ) {
03223         MLOCK(ErrorMessageLock);
03224         MesPrint("Illegal range encountered in RunZtoHArg.");
03225         MUNLOCK(ErrorMessageLock);
03226         Terminate(-1);
03227     }
03228     if ( functions[fun[0]-FUNCTION].spec != 0 ) {
03229         MLOCK(ErrorMessageLock);
03230         MesPrint("The HtoZ transformation can only be executed on regular functions with nonzero
integer arguments.");
03231         MUNLOCK(ErrorMessageLock);
03232         Terminate(-1);
03233     }
03234     tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
03235     while ( tt < tstop ) { totarg++; NEXTARG(tt); }
03236     if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
03237     /*
03238     Check the arguments. Should be -SNUMBER x!=0
03239     */
03240     f = fun+FUNHEAD; n = 1;
03241     while ( n < arg1 ) { n++; NEXTARG(f) }
03242     f2 = f1 = f;
03243     for ( i = arg1; i <= arg2; i++, f += 2 ) {
03244         if ( *f != -SNUMBER || f[1] == 0 ) return(-1);
03245     }
03246     /*
03247     First the overall sign.
03248     */
03249     while ( f2 < f ) { if ( f2[1] < 0 ) sign = 1-sign; f2 += 2; }
03250     /*
03251     Now we need a copy.
03252     */
03253     t = f1; t1 = tt = TermMalloc("RunHtoZArg");
03254     while ( t < f ) { *t1++ = *t++; *t1++ = *t++; }
03255     /*
03256     Now the transformation.
03257     */
03258     t = f1; t2 = tt + 2;
03259     while ( t2 < t1 ) {
03260         t += 2;
03261         if ( t2[-1] < 0 ) t[1] = -t[1];
03262         t2 += 2;
03263     }
03264     TermFree(tt,"RunHtoZArg");
03265     return(sign);
03266 }
03267
03268 /*
03269     #] RunHtoZArg :
03270     #[ TestArgNum :
03271
03272     Looks whether argument n is contained in any of the ranges
03273     specified in args. Args contains objects of the types
03274     ALLARGS

```

```

03275         NUMARG,num
03276         ARGRANGE,num1,num2
03277         The object MAKEARGS,num1,num2 is skipped
03278         Any other object terminates the range specifications.
03279
03280         Currently only ARGRANGE is used (10-may-2016)
03281 */
03282
03283 int TestArgNum(int n, int totarg, WORD *args)
03284 {
03285     GETIDENTITY
03286     WORD x1, x2;
03287     for(;;) {
03288         switch ( *args ) {
03289             case ALLARGS:
03290                 return(1);
03291             case NUMARG:
03292                 if ( n == args[1] ) return(1);
03293                 if ( args[1] >= MAXPOSITIVE4 ) {
03294                     x1 = args[1]-MAXPOSITIVE4;
03295                     if ( totarg-x1 == n ) return(1);
03296                 }
03297                 args += 2;
03298                 break;
03299             case ARGRANGE:
03300                 if ( args[1] >= MAXPOSITIVE2 ) {
03301                     x1 = args[1] - MAXPOSITIVE2;
03302                     if ( x1 > MAXPOSITIVE4 ) {
03303                         x1 = x1 - MAXPOSITIVE4;
03304                         x1 = DolToNumber(BHEAD x1);
03305                         x1 = totarg - x1;
03306                     }
03307                     else {
03308                         x1 = DolToNumber(BHEAD x1);
03309                     }
03310                 }
03311                 else if ( args[1] >= MAXPOSITIVE4 ) {
03312                     x1 = totarg-(args[1]-MAXPOSITIVE4);
03313                 }
03314                 else x1 = args[1];
03315                 if ( args[2] >= MAXPOSITIVE2 ) {
03316                     x2 = args[2] - MAXPOSITIVE2;
03317                     if ( x2 > MAXPOSITIVE4 ) {
03318                         x2 = x2 - MAXPOSITIVE4;
03319                         x2 = DolToNumber(BHEAD x2);
03320                         x2 = totarg - x2;
03321                     }
03322                     else {
03323                         x2 = DolToNumber(BHEAD x2);
03324                     }
03325                 }
03326                 else if ( args[2] >= MAXPOSITIVE4 ) {
03327                     x2 = totarg-(args[2]-MAXPOSITIVE4);
03328                 }
03329                 else x2 = args[2];
03330                 if ( x1 >= x2 ) {
03331                     if ( n >= x2 && n <= x1 ) return(1);
03332                 }
03333                 else {
03334                     if ( n >= x1 && n <= x2 ) return(1);
03335                 }
03336                 args += 3;
03337                 break;
03338             case MAKEARGS:
03339                 args += 3;
03340                 break;
03341             default:
03342                 return(0);
03343         }
03344     }
03345 }
03346
03347 /*
03348     #] TestArgNum :
03349     #[ PutArgInScratch :
03350 */
03351
03352 WORD PutArgInScratch(WORD *arg,UWORD *scrat)
03353 {
03354     WORD size, *t, i;
03355     if ( *arg == -SNUMBER ) {
03356         scrat[0] = ABS(arg[1]);
03357         if ( arg[1] < 0 ) size = -1;
03358         else size = 1;
03359     }
03360     else {
03361         t = arg+*arg-1;

```

```

03362         if ( *t < 0 ) { i = ((-*t)-1)/2; size = -i; }
03363         else           { i = ( *t -1)/2; size = i; }
03364         t = arg+ARGHEAD+1;
03365         NCOPY(scrat,t,i);
03366     }
03367     return(size);
03368 }
03369
03370 /*
03371     #] PutArgInScratch :
03372     #[ ReadRange :
03373
03374     Comes in at the bracket and leaves at the = sign
03375     Ranges can be:
03376         #1,#2 with # numbers. If the second is smaller than the
03377             first we work it backwards.
03378         first,#2 or #2,first
03379         #1,last or last,#1
03380         first,last or last,first
03381     First is represented by 1. Last is represented by MAXPOSITIVE4.
03382
03383     par = 0: we need the = after.
03384     par = 1: we need a , or '\0' after.
03385     par = 2: we need a :
03386 */
03387 UBYTE *ReadRange(UBYTE *s, WORD *out, int par)
03388 {
03389     UBYTE *in = s, *ss, c;
03390     LONG x1, x2;
03391
03392     SKIPBRA3(in)
03393     if ( par == 0 && in[1] != '=' ) {
03394         MesPrint("&A range in this type of transform statement should be followed by an = sign");
03395         return(0);
03396     }
03397     else if ( par == 1 && in[1] != ',' && in[1] != '\0' ) {
03398         MesPrint("&A range in this type of transform statement should be followed by a comma or
03399 end-of-statement");
03400         return(0);
03401     }
03402     else if ( par == 2 && in[1] != ':' ) {
03403         MesPrint("&A range in this type of transform statement should be followed by a :");
03404         return(0);
03405     }
03406     s++;
03407     if ( FG.cTable[*s] == 0 ) {
03408         ss = s; while ( FG.cTable[*s] == 0 ) s++;
03409         c = *s; *s = 0;
03410         if ( StrICmp(ss, (UBYTE *)"first") == 0 ) {
03411             *s = c;
03412             x1 = 1;
03413         }
03414         else if ( StrICmp(ss, (UBYTE *)"last") == 0 ) {
03415             *s = c;
03416             if ( c == '-' ) {
03417                 s++;
03418                 if ( *s == '$' ) {
03419                     s++; ss = s;
03420                     while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
03421                     c = *s; *s = 0;
03422                     if ( ( x1 = GetDollar(ss) ) < 0 ) goto Error;
03423                     *s = c;
03424                     x1 += MAXPOSITIVE2;
03425                 }
03426                 else {
03427                     x1 = 0;
03428                     while ( *s >= '0' && *s <= '9' ) {
03429                         x1 = 10*x1 + *s++ - '0';
03430                         if ( x1 >= MAXPOSITIVE4 ) {
03431                             MesPrint("&Fixed range indicator bigger than %1", (LONG)MAXPOSITIVE4);
03432                             return(0);
03433                         }
03434                     }
03435                     x1 += MAXPOSITIVE4;
03436                 }
03437             }
03438             else x1 = MAXPOSITIVE4;
03439         }
03440         else {
03441             MesPrint("&Illegal keyword inside range specification");
03442             return(0);
03443         }
03444     }
03445     else if ( FG.cTable[*s] == 1 ) {
03446         x1 = 0;
03447         while ( *s >= '0' && *s <= '9' ) {

```

```

03448         x1 = x1*10 + *s++ - '0';
03449         if ( x1 >= MAXPOSITIVE4 ) {
03450             MesPrint("&Fixed range indicator bigger than %1", (LONG)MAXPOSITIVE4);
03451             return(0);
03452         }
03453     }
03454 }
03455 else if ( *s == '$' ) {
03456     s++; ss = s;
03457     while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
03458     c = *s; *s = 0;
03459     if ( ( x1 = GetDollar(ss) ) < 0 ) goto Error;
03460     *s = c;
03461     x1 += MAXPOSITIVE2;
03462 }
03463 else {
03464     MesPrint("&Illegal character in range specification");
03465     return(0);
03466 }
03467 if ( *s != ',' ) {
03468     MesPrint("&A range is two indicators, separated by a comma or blank");
03469     return(0);
03470 }
03471 s++;
03472 if ( FG.cTable[*s] == 0 ) {
03473     ss = s; while ( FG.cTable[*s] == 0 ) s++;
03474     c = *s; *s = 0;
03475     if ( StrICmp(ss, (UBYTE *)"first") == 0 ) {
03476         *s = c;
03477         x2 = 1;
03478     }
03479     else if ( StrICmp(ss, (UBYTE *)"last") == 0 ) {
03480         *s = c;
03481         if ( c == '-' ) {
03482             s++;
03483             if ( *s == '$' ) {
03484                 s++; ss = s;
03485                 while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
03486                 c = *s; *s = 0;
03487                 if ( ( x2 = GetDollar(ss) ) < 0 ) goto Error;
03488                 *s = c;
03489                 x2 += MAXPOSITIVE2;
03490             }
03491             else {
03492                 x2 = 0;
03493                 while ( *s >= '0' && *s <= '9' ) {
03494                     x2 = 10*x2 + *s++ - '0';
03495                     if ( x2 >= MAXPOSITIVE4 ) {
03496                         MesPrint("&Fixed range indicator bigger than %1", (LONG)MAXPOSITIVE4);
03497                         return(0);
03498                     }
03499                 }
03500             }
03501             x2 += MAXPOSITIVE4;
03502         }
03503         else x2 = MAXPOSITIVE4;
03504     }
03505     else {
03506         MesPrint("&Illegal keyword inside range specification");
03507         return(0);
03508     }
03509 }
03510 else if ( FG.cTable[*s] == 1 ) {
03511     x2 = 0;
03512     while ( *s >= '0' && *s <= '9' ) {
03513         x2 = x2*10 + *s++ - '0';
03514         if ( x2 >= MAXPOSITIVE4 ) {
03515             MesPrint("&Fixed range indicator bigger than %1", (LONG)MAXPOSITIVE4);
03516             return(0);
03517         }
03518     }
03519 }
03520 else if ( *s == '$' ) {
03521     s++; ss = s;
03522     while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
03523     c = *s; *s = 0;
03524     if ( ( x2 = GetDollar(ss) ) < 0 ) goto Error;
03525     *s = c;
03526     x2 += MAXPOSITIVE2;
03527 }
03528 else {
03529     MesPrint("&Illegal character in range specification");
03530     return(0);
03531 }
03532 if ( s < in ) {
03533     MesPrint("&A range is two indicators, separated by a comma or blank between parentheses");
03534     return(0);

```



```

03535     }
03536     out[0] = x1; out[1] = x2;
03537     return(in+1);
03538 Error:
03539     MesPrint("&Undefined variable %s in range",ss);
03540     return(0);
03541 }
03542
03543 /*
03544     #] ReadRange :
03545     #[ FindRange :
03546 */
03547
03548 int FindRange(PHEAD WORD *args, WORD *arg1, WORD *arg2, WORD totarg)
03549 {
03550     WORD n[2], fromlast, i;
03551     for ( i = 0; i < 2; i++ ) {
03552         n[i] = args[i+1];
03553         fromlast = 0;
03554         if ( n[i] >= MAXPOSITIVE2 ) { /* This is a dollar variable */
03555             n[i] -= MAXPOSITIVE2;
03556             if ( n[i] >= MAXPOSITIVE4 ) {
03557                 fromlast = 1;
03558                 n[i] -= MAXPOSITIVE4; /* Now we have the number of the dollar variable */
03559             }
03560             n[i] = DolToNumber(BHEAD n[i]);
03561             if ( AN.ErrorInDollar ) {
03562                 MLOCK(ErrorMessageLock);
03563                 MesPrint("Illegal $ value in range while executing transform statement.");
03564                 MUNLOCK(ErrorMessageLock);
03565                 return(-1);
03566             }
03567             if ( fromlast ) n[i] = totarg-n[i];
03568         }
03569         else if ( n[i] >= MAXPOSITIVE4 ) { n[i] = totarg-(n[i]-MAXPOSITIVE4); }
03570         if ( n[i] <= 0 ) {
03571             MLOCK(ErrorMessageLock);
03572             MesPrint("Illegal non-positive value in range (%d) while executing transform statement.",
03573 i+1);
03574             MUNLOCK(ErrorMessageLock);
03575             return(-1);
03576         }
03577     }
03578     *arg1 = n[0];
03579     *arg2 = n[1];
03580     return(0);
03581 }
03582 /*
03583     #] FindRange :
03584     #] Transform :
03585 */

```

## 4.152 unix.h File Reference

### Macros

- #define [LINEFEED](#) '\n'
- #define [CARRIAGEReturn](#) 0x0D
- #define [WITHPIPE](#)
- #define [WITHSYSTEM](#)
- #define [WITHEXTERNALCHANNEL](#)
- #define [TRAPsignals](#)
- #define [P\\_term](#)(code) exit((int)(code<0?-code:code))
- #define [SEPARATOR](#) '/'
- #define [ALTSEPARATOR](#) '/'
- #define [PATHSEPARATOR](#) ':'
- #define [WITH\\_ENV](#)
- #define [WITH\\_ALARM](#)

### 4.152.1 Detailed Description

Settings for Unix-like systems.

Definition in file [unix.h](#).

### 4.152.2 Macro Definition Documentation

#### 4.152.2.1 LINEFEED

```
#define LINEFEED '\n'
```

Definition at line 33 of file [unix.h](#).

#### 4.152.2.2 CARRIAGEReturn

```
#define CARRIAGEReturn 0x0D
```

Definition at line 34 of file [unix.h](#).

#### 4.152.2.3 WITHPIPE

```
#define WITHPIPE
```

Definition at line 36 of file [unix.h](#).

#### 4.152.2.4 WITHSYSTEM

```
#define WITHSYSTEM
```

Definition at line 37 of file [unix.h](#).

#### 4.152.2.5 WITHEXTERNALCHANNEL

```
#define WITHEXTERNALCHANNEL
```

Definition at line 46 of file [unix.h](#).

#### 4.152.2.6 TRAPsignals

```
#define TRAPsignals
```

Definition at line 49 of file [unix.h](#).

#### 4.152.2.7 P\_term

```
#define P_term(  
    code )  exit((int) (code<0?-code:code))
```

Definition at line 52 of file [unix.h](#).

#### 4.152.2.8 SEPARATOR

```
#define SEPARATOR '/'
```

Definition at line 54 of file [unix.h](#).

#### 4.152.2.9 ALTSEPARATOR

```
#define ALTSEPARATOR '/'
```

Definition at line 55 of file [unix.h](#).

#### 4.152.2.10 PATHSEPARATOR

```
#define PATHSEPARATOR ':'
```

Definition at line 56 of file [unix.h](#).

#### 4.152.2.11 WITH\_ENV

```
#define WITH_ENV
```

Definition at line 57 of file [unix.h](#).

#### 4.152.2.12 WITH\_ALARM

```
#define WITH_ALARM
```

Definition at line 58 of file [unix.h](#).

## 4.153 unix.h

[Go to the documentation of this file.](#)

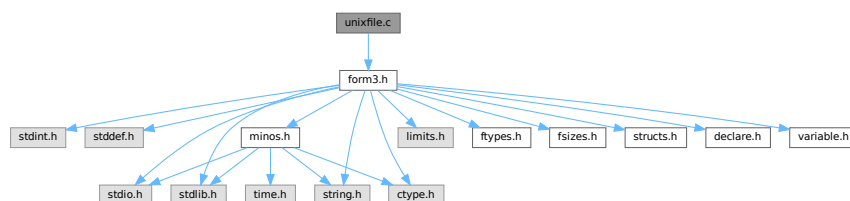
```

00001
00006 /* #[] License : */
00007 /*
00008 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009 *   When using this file you are requested to refer to the publication
00010 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011 *   This is considered a matter of courtesy as the development was paid
00012 *   for by FOM the Dutch physics granting agency and we would like to
00013 *   be able to track its scientific use to convince FOM of its value
00014 *   for the community.
00015 *
00016 *   This file is part of FORM.
00017 *
00018 *   FORM is free software: you can redistribute it and/or modify it under the
00019 *   terms of the GNU General Public License as published by the Free Software
00020 *   Foundation, either version 3 of the License, or (at your option) any later
00021 *   version.
00022 *
00023 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00026 *   details.
00027 *
00028 *   You should have received a copy of the GNU General Public License along
00029 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00030 */
00031 /* #[] License : */
00032
00033 #define LINEFEED '\n'
00034 #define CARRIAGERETURN 0x0D
00035
00036 #define WITHPIPE
00037 #define WITHSYSTEM
00038
00039 /*[13jul2005 mt]:*/
00040 /*With SAFESIGNAL defined, write() and read() syscalls are wrapped by
00041 the errno checkup*/
00042 /**define SAFESIGNAL*/
00043 /*:[13jul2005 mt]*/
00044
00045 /*[29apr2004 mt]:*/
00046 #define WITHEXTERNALCHANNEL
00047 /*
00048 */
00049 #define TRAP SIGNALS
00050 /*:[29apr2004 mt]*/
00051
00052 #define P_term(code)    exit((int)(code<0?-code:code))
00053
00054 #define SEPARATOR '/'
00055 #define ALTSEPARATOR '/'
00056 #define PATHSEPARATOR ':'
00057 #define WITH_ENV
00058 #define WITH_ALARM

```

## 4.154 unixfile.c File Reference

```
#include "form3.h"
Include dependency graph for unixfile.c:
```



## 4.154.1 Detailed Description

The interface to a fast variety of file routines in the unix system.

Definition in file [unixfile.c](#).

## 4.155 unixfile.c

[Go to the documentation of this file.](#)

```

00001
00005 /* #[ License : */
00006 /*
00007  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00008  *   When using this file you are requested to refer to the publication
00009  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010  *   This is considered a matter of courtesy as the development was paid
00011  *   for by FOM the Dutch physics granting agency and we would like to
00012  *   be able to track its scientific use to convince FOM of its value
00013  *   for the community.
00014  *
00015  *   This file is part of FORM.
00016  *
00017  *   FORM is free software: you can redistribute it and/or modify it under the
00018  *   terms of the GNU General Public License as published by the Free Software
00019  *   Foundation, either version 3 of the License, or (at your option) any later
00020  *   version.
00021  *
00022  *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00023  *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00024  *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00025  *   details.
00026  *
00027  *   You should have received a copy of the GNU General Public License along
00028  *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00029 */
00030 /* #[ License : */
00031 /*
00032  *   #[ Includes :
00033  *
00034  *   File with direct interface to the UNIX functions.
00035  *   This makes a big difference in speed!
00036 */
00037 #include "form3.h"
00038
00039
00040 /*[13jul2005 mt]:*/
00041 #ifdef SAFESIGNAL
00042 /*[15oct2004 mt]:*/
00043 /*To access errno variable and EINTR constant:*/
00044 #include <errno.h>
00045 /*:[15oct2004 mt]:*/
00046 #endif
00047 /*:[13jul2005 mt]:*/
00048
00049 #ifdef UFILES
00050
00051 FILES TheStdout = { 1 };
00052 FILES *Ustdout = &TheStdout;
00053
00054 #ifdef GPP
00055 extern "C" open();
00056 extern "C" close();
00057 extern "C" read();
00058 extern "C" write();
00059 extern "C" lseek();
00060 #endif
00061
00062 /*
00063  *   #[ Includes :
00064  *   #[ Uopen :
00065 */
00066
00067 FILES *Uopen(char *filename, char *mode)
00068 {
00069     FILES *f = (FILES *)Malloc1(sizeof(FILES), "Uopen");
00070     int flags = 0, rights = 0644;
00071     while ( *mode ) {
00072         if      ( *mode == 'r' ) { flags |= O_RDONLY; }

```

```

00073         else if ( *mode == 'w' ) { flags |= O_CREAT | O_TRUNC; }
00074         else if ( *mode == 'a' ) { flags |= O_APPEND; }
00075         else if ( *mode == 'b' ) { }
00076         else if ( *mode == '+' ) { flags |= O_RDWR; }
00077         mode++;
00078     }
00079     f->descriptor = open(filename,flags,rights);
00080     if ( f->descriptor >= 0 ) return(f);
00081     if ( ( flags & O_APPEND ) != 0 ) {
00082         flags |= O_CREAT;
00083         f->descriptor = open(filename,flags,rights);
00084         if ( f->descriptor >= 0 ) return(f);
00085     }
00086     M_free(f,"Uopen");
00087     return(0);
00088 }
00089
00090 /*
00091  #] Uopen :
00092  #[ Uclose :
00093  */
00094
00095 int Uclose(FILE *f)
00096 {
00097     int retval;
00098     if ( f ) {
00099         retval = close(f->descriptor);
00100         M_free(f,"Uclose");
00101         return(retval);
00102     }
00103     return(EOF);
00104 }
00105
00106 /*
00107  #] Uclose :
00108  #[ Uread :
00109  */
00110
00111 size_t Uread(char *ptr, size_t size, size_t nobj, FILE *f)
00112 {
00113     /*[13jul2005 mt]:*/
00114     #ifdef SAFESIGNAL
00115     /*[15oct2004 mt]:*/
00116     /*Operation read() can be interrupted by a signal. Note, this is rather unlikely,
00117     so we do not save size*nobj for future attempts*/
00118     size_t ret;
00119     /*If the syscall is interrupted by a signal before it
00120     succeeded in getting any progress, it must be repeated:*/
00121     while( ( ret=read(f->descriptor,ptr,size*nobj)<1)&&(errno == EINTR) );
00122     return(ret);
00123     #else
00124     #ifdef DEEPDEBUG
00125     {
00126         POSITION pos;
00127         SETBASEPOSITION(pos,lseek(f->descriptor,0L,SEEK_CUR));
00128         MesPrint("handle %d: reading %ld bytes from position %p\n",f->descriptor,size*nobj,pos);
00129     }
00130     #endif
00131     return(read(f->descriptor,ptr,size*nobj));
00132     #endif
00133 }
00134
00135 /*
00136  #] Uread :
00137  #[ Uwrite :
00138  */
00139
00140 size_t Uwrite(char *ptr, size_t size, size_t nobj, FILE *f)
00141 {
00142     /*[13jul2005 mt]:*/
00143     #ifdef SAFESIGNAL
00144     /*[15oct2004 mt]:*/
00145     /*Operation write() can be interrupted by a signal. */
00146     size_t ret;
00147     size_t thesize=size*nobj;
00148     /*If the syscall is interrupted by a signal before it
00149     succeeded in getting any progress, it must be repeated:*/
00150     while( ( ret=write(f->descriptor,ptr,thesize)<1 )&&(errno == EINTR) );
00151     return(ret);
00152     #else
00153     return(write(f->descriptor,ptr,thesize));
00154     #endif
00155 }

```

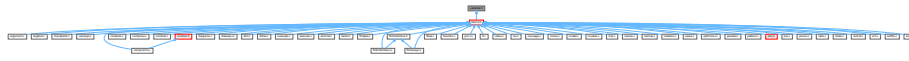
```

00160 #ifdef DEEPDEBUG
00161 {
00162     POSITION pos;
00163     SETBASEPOSITION(pos,lseek(f->descriptor,0L,SEEK_CUR));
00164     MesPrint("handle %d: writing %ld bytes to position %p\n",f->descriptor,size*nobj,pos);
00165 }
00166 #endif
00167     return(write(f->descriptor,ptr,size*nobj));
00168
00169 /*:[15oct2004 mt]*/
00170 #endif
00171 /*:[13jul2005 mt]*/
00172 }
00173
00174 /*
00175     #] Uwrite :
00176     #[ Useek :
00177 */
00178
00179 int Useek(FILE *f, off_t offset, int origin)
00180 {
00181     if ( f && ( lseek(f->descriptor,offset,origin) >= 0 ) ) return(0);
00182     return(-1);
00183 }
00184
00185 /*
00186     #] Useek :
00187     #[ Utell :
00188 */
00189
00190 off_t Utell(FILE *f)
00191 {
00192     if ( f ) return((off_t)lseek(f->descriptor,0L,SEEK_CUR));
00193     else return(-1);
00194 }
00195
00196 /*
00197     #] Utell :
00198     #[ Uflush :
00199 */
00200
00201 void Uflush(FILE *f) { DUMMYUSE(f); }
00202
00203 /*
00204     #] Uflush :
00205     #[ Ugetpos :
00206 */
00207
00208 int Ugetpos(FILE *f, fpos_t *ptr)
00209 {
00210     DUMMYUSE(f); DUMMYUSE(ptr);
00211     return(-1);
00212 }
00213
00214 /*
00215     #] Ugetpos :
00216     #[ Usetpos :
00217 */
00218
00219 int Usetpos(FILE *f,fpos_t *ptr)
00220 {
00221     DUMMYUSE(f); DUMMYUSE(ptr);
00222     return(-1);
00223 }
00224
00225 /*
00226     #] Usetpos :
00227     #[ Usetbuf :
00228 */
00229
00230 void Usetbuf(FILE *f, char *ptr) { DUMMYUSE(f); DUMMYUSE(ptr); }
00231
00232 /*
00233     #] Usetbuf :
00234 */
00235 #endif

```

## 4.156 variable.h File Reference

This graph shows which files directly or indirectly include this file:



### Macros

- #define [chartype](#) FG.cTable
- #define [Procedures](#) (([PROCEDURE](#) \*) (AP.ProcList.lijst))
- #define [NumProcedures](#) AP.ProcList.num
- #define [MaxProcedures](#) AP.ProcList.maxnum
- #define [DoLoops](#) (([DOLOOP](#) \*) (AP.LoopList.lijst))
- #define [NumDoLoops](#) AP.LoopList.num
- #define [MaxDoLoops](#) AP.LoopList.maxnum
- #define [PreVar](#) (([PREVAR](#) \*) (AP.PreVarList.lijst))
- #define [NumPre](#) AP.PreVarList.num
- #define [MaxNumPre](#) AP.PreVarList.maxnum
- #define [SetElements](#) (([WORD](#) \*) (AC.SetElementList.lijst))
- #define [Sets](#) (([SETS](#)) (AC.SetList.lijst))
- #define [functions](#) (([FUNCTIONS](#)) (AC.FunctionList.lijst))
- #define [indices](#) (([INDICES](#)) (AC.IndexList.lijst))
- #define [symbols](#) (([SYMBOLS](#)) (AC.SymbolList.lijst))
- #define [vectors](#) (([VECTORS](#)) (AC.VectorList.lijst))
- #define [tablebases](#) (([DBASE](#) \*) (AC.TableBaseList.lijst))
- #define [NumFunctions](#) AC.FunctionList.num
- #define [NumIndices](#) AC.IndexList.num
- #define [NumSymbols](#) AC.SymbolList.num
- #define [NumVectors](#) AC.VectorList.num
- #define [NumSets](#) AC.SetList.num
- #define [NumSetElements](#) AC.SetElementList.num
- #define [NumTableBases](#) AC.TableBaseList.num
- #define [GlobalFunctions](#) AC.FunctionList.numglobal
- #define [GlobalIndices](#) AC.IndexList.numglobal
- #define [GlobalSymbols](#) AC.SymbolList.numglobal
- #define [GlobalVectors](#) AC.VectorList.numglobal
- #define [GlobalSets](#) AC.SetList.numglobal
- #define [GlobalSetElements](#) AC.SetElementList.numglobal
- #define [cbuf](#) (([CBUF](#) \*) (AC.cbufList.lijst))
- #define [channels](#) (([CHANNEL](#) \*) (AC.ChannelList.lijst))
- #define [NumOutputChannels](#) AC.ChannelList.num
- #define [Dollars](#) (([DOLLARS](#)) (AP.DollarList.lijst))
- #define [NumDollars](#) AP.DollarList.num
- #define [Dubious](#) (([DUBIOUSV](#)) (AC.DubiousList.lijst))
- #define [NumDubious](#) AC.DubiousList.num
- #define [Expressions](#) (([EXPRESSIONS](#)) (AC.ExpressionList.lijst))
- #define [NumExpressions](#) AC.ExpressionList.num
- #define [autofunctions](#) (([FUNCTIONS](#)) (AC.AutoFunctionList.lijst))
- #define [autoindices](#) (([INDICES](#)) (AC.AutoIndexList.lijst))
- #define [autosymbols](#) (([SYMBOLS](#)) (AC.AutoSymbolList.lijst))
- #define [autovectors](#) (([VECTORS](#)) (AC.AutoVectorList.lijst))



- `#define xsymbol (cbuf[AM.sbufnum].rhs)`
- `#define numxsymbol (cbuf[AM.sbufnum].numrhs)`
- `#define PotModdollars ((WORD *) (AC.PotModDolList.lijst))`
- `#define NumPotModdollars AC.PotModDolList.num`
- `#define ModOptdollars ((MODOPTDOLLAR *) (AC.ModOptDolList.lijst))`
- `#define NumModOptdollars AC.ModOptDolList.num`
- `#define BUG A.bug;`
- `#define AC A.Cc`
- `#define AM A.M`
- `#define AN A.N`
- `#define AO A.O`
- `#define AP A.P`
- `#define AR A.R`
- `#define AS A.S`
- `#define AT A.T`
- `#define AX A.X`

## Variables

- `WRITEBUFTOEXTCHANNEL` `writeBufToExtChannel`
- `GETCFROMEXTCHANNEL` `getcFromExtChannel`
- `SETTERMINATORFOREXTCHANNEL` `setTerminatorForExternalChannel`
- `SETKILLMODEFOREXTCHANNEL` `setKillModeForExternalChannel`
- `WRITEFILE` `WriteFile`
- `ALLGLOBALS` `A`
- `FIXEDGLOBALS` `FG`
- `FIXEDSET` `fixedsets []`
- `char *` `setupfilename`

### 4.156.1 Detailed Description

Contains a number of defines to make the coding easier. Especially the defines for the use of the lists are very nice. And of course the AC for A.Cc and AT for either A.T or B->T are indispensable to keep FORM and TFORM in one set of sources.

Definition in file [variable.h](#).

### 4.156.2 Macro Definition Documentation

#### 4.156.2.1 chartype

```
#define chartype FG.cTable
```

Definition at line 77 of file [variable.h](#).

#### 4.156.2.2 Procedures

```
#define Procedures ((PROCEDURE *) (AP.ProcList.lijst))
```

Definition at line 79 of file [variable.h](#).

#### 4.156.2.3 NumProcedures

```
#define NumProcedures AP.ProcList.num
```

Definition at line 80 of file [variable.h](#).

#### 4.156.2.4 MaxProcedures

```
#define MaxProcedures AP.ProcList.maxnum
```

Definition at line 81 of file [variable.h](#).

#### 4.156.2.5 DoLoops

```
#define DoLoops ((DOLOOP *) (AP.LoopList.lijst))
```

Definition at line 82 of file [variable.h](#).

#### 4.156.2.6 NumDoLoops

```
#define NumDoLoops AP.LoopList.num
```

Definition at line 83 of file [variable.h](#).

#### 4.156.2.7 MaxDoLoops

```
#define MaxDoLoops AP.LoopList.maxnum
```

Definition at line 84 of file [variable.h](#).

#### 4.156.2.8 PreVar

```
#define PreVar ((PREVAR *) (AP.PreVarList.lijst))
```

Definition at line 85 of file [variable.h](#).

#### 4.156.2.9 NumPre

```
#define NumPre AP.PreVarList.num
```

Definition at line 86 of file [variable.h](#).

#### 4.156.2.10 MaxNumPre

```
#define MaxNumPre AP.PreVarList.maxnum
```

Definition at line 87 of file [variable.h](#).

#### 4.156.2.11 SetElements

```
#define SetElements ((WORD *) (AC.SetElementList.lijst))
```

Definition at line 88 of file [variable.h](#).

#### 4.156.2.12 Sets

```
#define Sets ((SETS) (AC.SetList.lijst))
```

Definition at line 89 of file [variable.h](#).

#### 4.156.2.13 functions

```
#define functions ((FUNCTIONS) (AC.FunctionList.lijst))
```

Definition at line 90 of file [variable.h](#).

#### 4.156.2.14 indices

```
#define indices ((INDICES) (AC.IndexList.lijst))
```

Definition at line 91 of file [variable.h](#).

#### 4.156.2.15 symbols

```
#define symbols ((SYMBOLS) (AC.SymbolList.lijst))
```

Definition at line 92 of file [variable.h](#).

#### 4.156.2.16 vectors

```
#define vectors ((VECTORS) (AC.VectorList.lijst))
```

Definition at line 93 of file [variable.h](#).

#### 4.156.2.17 tablebases

```
#define tablebases ((DBASE *) (AC.TableBaseList.lijst))
```

Definition at line 94 of file [variable.h](#).

#### 4.156.2.18 NumFunctions

```
#define NumFunctions AC.FunctionList.num
```

Definition at line 95 of file [variable.h](#).

#### 4.156.2.19 NumIndices

```
#define NumIndices AC.IndexList.num
```

Definition at line 96 of file [variable.h](#).

#### 4.156.2.20 NumSymbols

```
#define NumSymbols AC.SymbolList.num
```

Definition at line 97 of file [variable.h](#).

#### 4.156.2.21 NumVectors

```
#define NumVectors AC.VectorList.num
```

Definition at line 98 of file [variable.h](#).

#### 4.156.2.22 NumSets

```
#define NumSets AC.SetList.num
```

Definition at line 99 of file [variable.h](#).

#### 4.156.2.23 NumSetElements

```
#define NumSetElements AC.SetElementList.num
```

Definition at line 100 of file [variable.h](#).

#### 4.156.2.24 NumTableBases

```
#define NumTableBases AC.TableBaseList.num
```

Definition at line 101 of file [variable.h](#).

#### 4.156.2.25 GlobalFunctions

```
#define GlobalFunctions AC.FunctionList.numglobal
```

Definition at line 102 of file [variable.h](#).

#### 4.156.2.26 GlobalIndices

```
#define GlobalIndices AC.IndexList.numglobal
```

Definition at line 103 of file [variable.h](#).

#### 4.156.2.27 GlobalSymbols

```
#define GlobalSymbols AC.SymbolList.numglobal
```

Definition at line 104 of file [variable.h](#).

#### 4.156.2.28 GlobalVectors

```
#define GlobalVectors AC.VectorList.numglobal
```

Definition at line 105 of file [variable.h](#).

#### 4.156.2.29 GlobalSets

```
#define GlobalSets AC.SetList.numglobal
```

Definition at line 106 of file [variable.h](#).

#### 4.156.2.30 GlobalSetElements

```
#define GlobalSetElements AC.SetElementList.numglobal
```

Definition at line 107 of file [variable.h](#).

#### 4.156.2.31 cbuf

```
#define cbuf ((CBUF *) (AC.cbufList.lijt))
```

Definition at line 108 of file [variable.h](#).

#### 4.156.2.32 channels

```
#define channels ((CHANNEL *) (AC.ChannelList.lijt))
```

Definition at line 109 of file [variable.h](#).

#### 4.156.2.33 NumOutputChannels

```
#define NumOutputChannels AC.Channellist.num
```

Definition at line 110 of file [variable.h](#).

#### 4.156.2.34 Dollars

```
#define Dollars ((DOLLARS) (AP.DollarList.lijt))
```

Definition at line 111 of file [variable.h](#).

#### 4.156.2.35 NumDollars

```
#define NumDollars AP.DollarList.num
```

Definition at line 112 of file [variable.h](#).

#### 4.156.2.36 Dubious

```
#define Dubious ((DUBIOUSV) (AC.DubiousList.lijst))
```

Definition at line 113 of file [variable.h](#).

#### 4.156.2.37 NumDubious

```
#define NumDubious AC.DubiousList.num
```

Definition at line 114 of file [variable.h](#).

#### 4.156.2.38 Expressions

```
#define Expressions ((EXPRESSIONS) (AC.ExpressionList.lijst))
```

Definition at line 115 of file [variable.h](#).

#### 4.156.2.39 NumExpressions

```
#define NumExpressions AC.ExpressionList.num
```

Definition at line 116 of file [variable.h](#).

#### 4.156.2.40 autofunctions

```
#define autofunctions ((FUNCTIONS) (AC.AutoFunctionList.lijst))
```

Definition at line 117 of file [variable.h](#).

#### 4.156.2.41 autoindices

```
#define autoindices ((INDICES) (AC.AutoIndexList.lijst))
```

Definition at line 118 of file [variable.h](#).

#### 4.156.2.42 autosymbols

```
#define autosymbols ((SYMBOLS) (AC.AutoSymbolList.lijst))
```

Definition at line 119 of file [variable.h](#).

#### 4.156.2.43 autovectors

```
#define autovectors ((VECTORS) (AC.AutoVectorList.lijt))
```

Definition at line 120 of file [variable.h](#).

#### 4.156.2.44 xsymbol

```
#define xsymbol (cbuf[AM.sbufnum].rhs)
```

Definition at line 121 of file [variable.h](#).

#### 4.156.2.45 numxsymbol

```
#define numxsymbol (cbuf[AM.sbufnum].numrhs)
```

Definition at line 122 of file [variable.h](#).

#### 4.156.2.46 PotModdollars

```
#define PotModdollars ((WORD *) (AC.PotModDolList.lijt))
```

Definition at line 124 of file [variable.h](#).

#### 4.156.2.47 NumPotModdollars

```
#define NumPotModdollars AC.PotModDolList.num
```

Definition at line 125 of file [variable.h](#).

#### 4.156.2.48 ModOptdollars

```
#define ModOptdollars ((MODOPTDOLLAR *) (AC.ModOptDolList.lijt))
```

Definition at line 126 of file [variable.h](#).

#### 4.156.2.49 NumModOptdollars

```
#define NumModOptdollars AC.ModOptDolList.num
```

Definition at line 127 of file [variable.h](#).

#### 4.156.2.50 BUG

```
#define BUG A.bug;
```

Definition at line 129 of file [variable.h](#).

**4.156.2.51 AC**

```
#define AC A.Cc
```

Definition at line 145 of file [variable.h](#).

**4.156.2.52 AM**

```
#define AM A.M
```

Definition at line 146 of file [variable.h](#).

**4.156.2.53 AN**

```
#define AN A.N
```

Definition at line 147 of file [variable.h](#).

**4.156.2.54 AO**

```
#define AO A.O
```

Definition at line 148 of file [variable.h](#).

**4.156.2.55 AP**

```
#define AP A.P
```

Definition at line 149 of file [variable.h](#).

**4.156.2.56 AR**

```
#define AR A.R
```

Definition at line 150 of file [variable.h](#).

**4.156.2.57 AS**

```
#define AS A.S
```

Definition at line 151 of file [variable.h](#).

**4.156.2.58 AT**

```
#define AT A.T
```

Definition at line 152 of file [variable.h](#).



#### 4.156.2.59 AX

```
#define AX A.X
```

Definition at line 153 of file [variable.h](#).

### 4.156.3 Variable Documentation

#### 4.156.3.1 WriteFile

```
WRITEFILE WriteFile [extern]
```

Definition at line 1357 of file [tools.c](#).

#### 4.156.3.2 A

```
ALLGLOBALS A [extern]
```

Definition at line 133 of file [inivar.h](#).

#### 4.156.3.3 FG

```
FIXEDGLOBALS FG [extern]
```

Definition at line 34 of file [inivar.h](#).

#### 4.156.3.4 fixedsets

```
FIXEDSET fixedsets[] [extern]
```

Definition at line 258 of file [inivar.h](#).

#### 4.156.3.5 setupfilename

```
char* setupfilename [extern]
```

Definition at line 276 of file [inivar.h](#).

## 4.157 variable.h

[Go to the documentation of this file.](#)

```

00001 #ifndef __VARIABLE__
00002
00003 #define __VARIABLE__
00004
00013 /* #[] License : */
00014 /*
00015  * Copyright (C) 1984-2023 J.A.M. Vermaseren
00016  * When using this file you are requested to refer to the publication
00017  * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00018  * This is considered a matter of courtesy as the development was paid
00019  * for by FOM the Dutch physics granting agency and we would like to
00020  * be able to track its scientific use to convince FOM of its value
00021  * for the community.
00022  *
00023  * This file is part of FORM.
00024  *
00025  * FORM is free software: you can redistribute it and/or modify it under the
00026  * terms of the GNU General Public License as published by the Free Software
00027  * Foundation, either version 3 of the License, or (at your option) any later
00028  * version.
00029  *
00030  * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00031  * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00032  * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00033  * details.
00034  *
00035  * You should have received a copy of the GNU General Public License along
00036  * with FORM. If not, see <http://www.gnu.org/licenses/>.
00037  */
00038 /* #] License : */
00039
00040 /*See the file extcmd.c*/
00041
00042 #ifdef REMOVEDBY_MT
00043 extern int (*writeBufToExtChannel)(char *buf, size_t n);
00044 extern int (*getcFromExtChannel)();
00045 extern int (*setTerminatorForExternalChannel)(char *newterminator);
00046 #endif
00047 extern WRITEBUFTOEXTCHANNEL writeBufToExtChannel;
00048 extern GETCFROMEXTCHANNEL getcFromExtChannel;
00049 extern SETTERMINATORFOREXTCHANNEL setTerminatorForExternalChannel;
00050 extern SETKILLMODEFOREXTCHANNEL setKillModeForExternalChannel;
00051
00052 /*
00053 extern LONG (*WriteFile)(int handle, UBYTE *buffer, LONG number);
00054 */
00055 extern WRITEFILE WriteFile;
00056 /*:[17nov2005 mt]*/
00057
00058 extern ALLGLOBALS A;
00059 #ifdef WITHPTHREADS
00060 extern ALLPRIVATES **AB;
00061 #endif
00062
00063 extern FIXEDGLOBALS FG;
00064 extern FIXEDSET fixedsets[];
00065
00066 extern char *setupfilename;
00067
00068 EXTERNLOCK(ErrorMessageLock)
00069 EXTERNLOCK(FileReadLock)
00070 EXTERNLOCK(dummylock)
00071
00072 #ifdef VMS
00073 #include <stdio.h>
00074 extern FILE **FileStructs;
00075 #endif
00076
00077 #define chartype FG.cTable
00078
00079 #define Procedures ((PROCEDURE *) (AP.ProcList.lijst))
00080 #define NumProcedures AP.ProcList.num
00081 #define MaxProcedures AP.ProcList.maxnum
00082 #define DoLoops ((DOLOOP *) (AP.LoopList.lijst))
00083 #define NumDoLoops AP.LoopList.num
00084 #define MaxDoLoops AP.LoopList.maxnum
00085 #define PreVar ((PREVAR *) (AP.PreVarList.lijst))
00086 #define NumPre AP.PreVarList.num
00087 #define MaxNumPre AP.PreVarList.maxnum
00088 #define SetElements ((WORD *) (AC.SetElementList.lijst))
00089 #define Sets ((SETS) (AC.SetList.lijst))
00090 #define functions ((FUNCTIONS) (AC.FunctionList.lijst))

```

```

00091 #define indices ((INDICES) (AC.IndexList.lijst))
00092 #define symbols ((SYMBOLS) (AC.SymbolList.lijst))
00093 #define vectors ((VECTORS) (AC.VectorList.lijst))
00094 #define tablebases ((DBASE *) (AC.TableBaseList.lijst))
00095 #define NumFunctions AC.FunctionList.num
00096 #define NumIndices AC.IndexList.num
00097 #define NumSymbols AC.SymbolList.num
00098 #define NumVectors AC.VectorList.num
00099 #define NumSets AC.SetList.num
00100 #define NumSetElements AC.SetElementList.num
00101 #define NumTableBases AC.TableBaseList.num
00102 #define GlobalFunctions AC.FunctionList.numglobal
00103 #define GlobalIndices AC.IndexList.numglobal
00104 #define GlobalSymbols AC.SymbolList.numglobal
00105 #define GlobalVectors AC.VectorList.numglobal
00106 #define GlobalSets AC.SetList.numglobal
00107 #define GlobalSetElements AC.SetElementList.numglobal
00108 #define cbuf ((CBUF *) (AC.cbufList.lijst))
00109 #define channels ((CHANNEL *) (AC.ChannelList.lijst))
00110 #define NumOutputChannels AC.ChannelList.num
00111 #define Dollars ((DOLLARS) (AP.DollarList.lijst))
00112 #define NumDollars AP.DollarList.num
00113 #define Dubious ((DUBIOUSV) (AC.DubiousList.lijst))
00114 #define NumDubious AC.DubiousList.num
00115 #define Expressions ((EXPRESSIONS) (AC.ExpressionList.lijst))
00116 #define NumExpressions AC.ExpressionList.num
00117 #define autofunctions ((FUNCTIONS) (AC.AutoFunctionList.lijst))
00118 #define autoindices ((INDICES) (AC.AutoIndexList.lijst))
00119 #define autosymbols ((SYMBOLS) (AC.AutoSymbolList.lijst))
00120 #define autovectors ((VECTORS) (AC.AutoVectorList.lijst))
00121 #define xsymbol (cbuf[AM.sbufnum].rhs)
00122 #define numxsymbol (cbuf[AM.sbufnum].numrhs)
00123
00124 #define PotModdollars ((WORD *) (AC.PotModDolList.lijst))
00125 #define NumPotModdollars AC.PotModDolList.num
00126 #define ModOptdollars ((MODOPTDOLLAR *) (AC.ModOptDolList.lijst))
00127 #define NumModOptdollars AC.ModOptDolList.num
00128
00129 #define BUG A.bug;
00130
00131 #ifndef WITHPTHREADS
00132 #define AC A.Cc
00133 #define AM A.M
00134 #define AO A.O
00135 #define AP A.P
00136 #define AS A.S
00137 #define AX A.X
00138 #define AN B->N
00139 #define AR B->R
00140 #define AT B->T
00141 #define AN0 B0->N
00142 #define AR0 B0->R
00143 #define AT0 B0->T
00144 #else
00145 #define AC A.Cc
00146 #define AM A.M
00147 #define AN A.N
00148 #define AO A.O
00149 #define AP A.P
00150 #define AR A.R
00151 #define AS A.S
00152 #define AT A.T
00153 #define AX A.X
00154 #endif
00155
00156 #endif

```

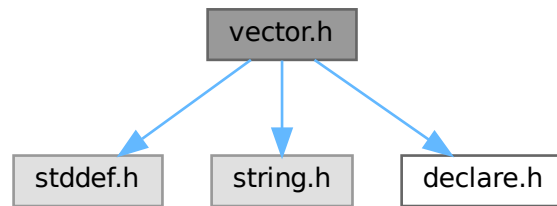
## 4.158 vector.h File Reference

```

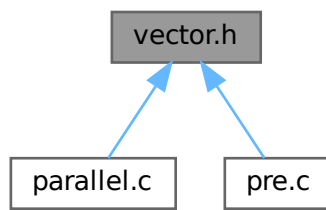
#include <stddef.h>
#include <string.h>
#include "declare.h"

```

Include dependency graph for vector.h:



This graph shows which files directly or indirectly include this file:



## Macros

- `#define VectorStruct(T)`
- `#define Vector(T, X) VectorStruct(T) X = { NULL, 0, 0 }`
- `#define DeclareVector(T, X) VectorStruct(T) X`
- `#define VectorInit(X)`
- `#define VectorFree(X)`
- `#define VectorPtr(X) ((X).ptr)`
- `#define VectorFront(X) ((X).ptr[0])`
- `#define VectorBack(X) ((X).ptr[(X).size - 1])`
- `#define VectorSize(X) ((X).size)`
- `#define VectorCapacity(X) ((X).capacity)`
- `#define VectorEmpty(X) ((X).size == 0)`
- `#define VectorClear(X) do { (X).size = 0; } while (0)`
- `#define VectorReserve(X, newcapacity)`
- `#define VectorPushBack(X, x)`
- `#define VectorPushBacks(X, src, n)`
- `#define VectorPopBack(X) do { (X).size --; } while (0)`
- `#define VectorInsert(X, index, x)`
- `#define VectorInserts(X, index, src, n)`
- `#define VectorErase(X, index)`
- `#define VectorErases(X, index, n)`

## 4.158.1 Detailed Description

An implementation of dynamic array.

Example:

```
size_t i;
Vector(int, vec);
VectorPushBack(vec, 1);
VectorPushBack(vec, 2);
VectorPushBack(vec, 3);
for ( i = 0; i < VectorSize(vec); i++ )
    printf("%d\n", VectorPtr(vec)[i]);
VectorFree(vec);
```

Definition in file [vector.h](#).

## 4.158.2 Macro Definition Documentation

### 4.158.2.1 VectorStruct

```
#define VectorStruct(
    T )
```

Value:

```
struct { \
    T *ptr; \
    size_t size; \
    size_t capacity; \
}
```

A struct for vector objects.

Parameters

|          |                       |
|----------|-----------------------|
| <i>T</i> | the type of elements. |
|----------|-----------------------|

Definition at line 65 of file [vector.h](#).

### 4.158.2.2 Vector

```
#define Vector(
    T,
    X ) VectorStruct(T) X = { NULL, 0, 0 }
```

Defines and initialises a vector X of the type T. The user must call [VectorFree\(X\)](#) after the use of X.

Parameters

|          |                           |
|----------|---------------------------|
| <i>T</i> | the type of elements.     |
| <i>X</i> | the name of vector object |

Definition at line 84 of file [vector.h](#).

#### 4.158.2.3 DeclareVector

```
#define DeclareVector(  
    T,  
    X )  VectorStruct (T) X
```

Declares a vector X of the type T. The user must call [VectorInit\(X\)](#) before the use of X.

##### Parameters

|          |                           |
|----------|---------------------------|
| <i>T</i> | the type of elements.     |
| <i>X</i> | the name of vector object |

Definition at line 99 of file [vector.h](#).

#### 4.158.2.4 VectorInit

```
#define VectorInit(  
    X )
```

##### Value:

```
do { \  
    (X).ptr = NULL; \  
    (X).size = 0; \  
    (X).capacity = 0; \  
} while (0)
```

Initialises a vector X of the type T. The user must call [VectorFree\(X\)](#) after the use of X.

##### Parameters

|          |                    |
|----------|--------------------|
| <i>X</i> | the vector object. |
|----------|--------------------|

Definition at line 113 of file [vector.h](#).

#### 4.158.2.5 VectorFree

```
#define VectorFree(  
    X )
```

##### Value:

```
do { \  
    M_free((X).ptr, "VectorFree:" #X); \  
    (X).ptr = NULL; \  
    (X).size = 0; \  
    (X).capacity = 0; \  
} while (0)
```

Frees the buffer allocated by the vector X.

##### Parameters

|          |                    |
|----------|--------------------|
| <i>X</i> | the vector object. |
|----------|--------------------|

Definition at line 130 of file [vector.h](#).

#### 4.158.2.6 VectorPtr

```
#define VectorPtr(  
    X )    ((X).ptr)
```

Returns the pointer to the buffer for the vector X. NULL when [VectorCapacity\(X\)](#) == 0.

##### Parameters

|   |                    |
|---|--------------------|
| X | the vector object. |
|---|--------------------|

##### Returns

the pointer to the allocated buffer for the vector.

Definition at line 150 of file [vector.h](#).

#### 4.158.2.7 VectorFront

```
#define VectorFront(  
    X )    ((X).ptr[0])
```

Returns the first element of the vector X. Undefined when [VectorSize\(X\)](#) == 0.

##### Parameters

|   |                    |
|---|--------------------|
| X | the vector object. |
|---|--------------------|

##### Returns

the first element of the vector.

Definition at line 165 of file [vector.h](#).

#### 4.158.2.8 VectorBack

```
#define VectorBack(  
    X )    ((X).ptr[(X).size - 1])
```

Returns the last element of the vector X. Undefined when [VectorSize\(X\)](#) == 0.

##### Parameters

|   |                    |
|---|--------------------|
| X | the vector object. |
|---|--------------------|

**Returns**

the last element of the vector.

Definition at line 180 of file [vector.h](#).

**4.158.2.9 VectorSize**

```
#define VectorSize(  
    X )    ((X).size)
```

Returns the size of the vector X.

**Parameters**

|          |                    |
|----------|--------------------|
| <i>X</i> | the vector object. |
|----------|--------------------|

**Returns**

the size of the vector.

Definition at line 194 of file [vector.h](#).

**4.158.2.10 VectorCapacity**

```
#define VectorCapacity(  
    X )    ((X).capacity)
```

Returns the capacity (the number of the already allocated elements) of the vector X.

**Parameters**

|          |                    |
|----------|--------------------|
| <i>X</i> | the vector object. |
|----------|--------------------|

**Returns**

the capacity of the vector.

Definition at line 208 of file [vector.h](#).

**4.158.2.11 VectorEmpty**

```
#define VectorEmpty(  
    X )    ((X).size == 0)
```

Returns true the size of the vector X is zero.



## Parameters

|          |                    |
|----------|--------------------|
| <i>X</i> | the vector object. |
|----------|--------------------|

## Returns

true if the vector has no elements, false otherwise.

Definition at line 222 of file [vector.h](#).

## 4.158.2.12 VectorClear

```
#define VectorClear(  
    X )    do { (X).size = 0; } while (0)
```

Sets the size of the vector *X* to zero.

## Parameters

|          |                    |
|----------|--------------------|
| <i>X</i> | the vector object. |
|----------|--------------------|

Definition at line 235 of file [vector.h](#).

## 4.158.2.13 VectorReserve

```
#define VectorReserve(  
    X,  
    newcapacity )
```

## Value:

```
do { \
    size_t v_tmp_newcapacity_ = (newcapacity); \
    if ( (X).capacity < v_tmp_newcapacity_ ) { \
        void *v_tmp_newptr_; \
        v_tmp_newcapacity_ = (v_tmp_newcapacity_ * 3) / 2; \
        if ( v_tmp_newcapacity_ < 4 ) v_tmp_newcapacity_ = 4; \
        v_tmp_newptr_ = Malloc1(sizeof((X).ptr[0]) * v_tmp_newcapacity_, "VectorReserve:" #X); \
        if ( (X).ptr != NULL ) { \
            memcpy(v_tmp_newptr_, (X).ptr, (X).size * sizeof((X).ptr[0])); \
            M_free((X).ptr, "VectorReserve:" #X); \
        } \
        (X).ptr = v_tmp_newptr_; \
        (X).capacity = v_tmp_newcapacity_; \
    } \
} while (0)
```

Requires that the capacity of the vector *X* is equal to or larger than *newcapacity*.

## Parameters

|                    |                              |
|--------------------|------------------------------|
| <i>X</i>           | the vector object.           |
| <i>newcapacity</i> | the capacity to be reserved. |

Definition at line 249 of file [vector.h](#).

**4.158.2.14 VectorPushBack**

```
#define VectorPushBack(
    X,
    x )
```

**Value:**

```
do { \
    VectorReserve((X), (X).size + 1); \
    (X).ptr[(X).size++] = (x); \
} while (0)
```

Adds an element *x* at the end of the vector *X*.

**Parameters**

|          |                          |
|----------|--------------------------|
| <i>X</i> | the vector object.       |
| <i>x</i> | the element to be added. |

Definition at line 277 of file [vector.h](#).

**4.158.2.15 VectorPushBacks**

```
#define VectorPushBacks(
    X,
    src,
    n )
```

**Value:**

```
do { \
    size_t v_tmp_n_ = (n); \
    VectorReserve((X), (X).size + v_tmp_n_); \
    memcpy((X).ptr + (X).size, (src), v_tmp_n_ * sizeof((X).ptr[0])); \
    (X).size += v_tmp_n_; \
} while (0)
```

Adds an *n* elements of *src* at the end of the vector *X*.

**Parameters**

|            |                                                                  |
|------------|------------------------------------------------------------------|
| <i>X</i>   | the vector object.                                               |
| <i>src</i> | the starting address of the buffer storing elements to be added. |
| <i>n</i>   | the number of elements to be added.                              |

Definition at line 295 of file [vector.h](#).

**4.158.2.16 VectorPopBack**

```
#define VectorPopBack(
    X ) do { (X).size --; } while (0)
```

Removes the last element of the vector *X*. [VectorSize\(X\)](#) must be  $> 0$ .

## Parameters

|          |                    |
|----------|--------------------|
| <i>X</i> | the vector object. |
|----------|--------------------|

Definition at line 314 of file [vector.h](#).

## 4.158.2.17 VectorInsert

```
#define VectorInsert(  
    X,  
    index,  
    x )
```

## Value:

```
do { \
    size_t v_tmp_index_ = (index); \
    VectorReserve((X), (X).size + 1); \
    memmove((X).ptr + v_tmp_index_ + 1, (X).ptr + v_tmp_index_, ((X).size - v_tmp_index_) * \
    sizeof((X).ptr[0])); \
    (X).ptr[v_tmp_index_] = (x); \
    (X).size++; \
} while (0)
```

Inserts an element *x* at the specified index of the vector *X*. The index must be  $0 \leq \text{index} < \text{VectorSize}(X)$ .

## Parameters

|              |                                                     |
|--------------|-----------------------------------------------------|
| <i>X</i>     | the vector object.                                  |
| <i>index</i> | the position at which the element will be inserted. |
| <i>x</i>     | the element to be inserted.                         |

Definition at line 330 of file [vector.h](#).

## 4.158.2.18 VectorInserts

```
#define VectorInserts(  
    X,  
    index,  
    src,  
    n )
```

## Value:

```
do { \
    size_t v_tmp_index_ = (index), v_tmp_n_ = (n); \
    VectorReserve((X), (X).size + v_tmp_n_); \
    memmove((X).ptr + v_tmp_index_ + v_tmp_n_, (X).ptr + v_tmp_index_, ((X).size - v_tmp_index_) * \
    sizeof((X).ptr[0])); \
    memcpy((X).ptr + v_tmp_index_, (src), v_tmp_n_ * sizeof((X).ptr[0])); \
    (X).size += v_tmp_n_; \
} while (0)
```

Inserts an *n* elements of *src* at the specified index of the vector *X*. The index must be  $0 \leq \text{index} < \text{VectorSize}(X)$ .

## Parameters

|              |                                                                     |
|--------------|---------------------------------------------------------------------|
| <i>X</i>     | the vector object.                                                  |
| <i>index</i> | the position at which the elements will be inserted.                |
| <i>src</i>   | the starting address of the buffer storing elements to be inserted. |
| <i>n</i>     | the number of elements to be inserted.                              |

Definition at line 353 of file [vector.h](#).

#### 4.158.2.19 VectorErase

```
#define VectorErase(  
    X,  
    index )
```

##### Value:

```
do { \
    size_t v_tmp_index_ = (index); \
    memmove((X).ptr + v_tmp_index_, (X).ptr + v_tmp_index_ + 1, ((X).size - v_tmp_index_ - 1) * \
    sizeof((X).ptr[0])); \
    (X).size--; \
} while (0)
```

Removes an element at the specified index of the vector X. The index must be  $0 \leq \text{index} < \text{VectorSize}(X)$ .

##### Parameters

|              |                                            |
|--------------|--------------------------------------------|
| <i>X</i>     | the vector object.                         |
| <i>index</i> | the position of the element to be removed. |

Definition at line 374 of file [vector.h](#).

#### 4.158.2.20 VectorErases

```
#define VectorErases(  
    X,  
    index,  
    n )
```

##### Value:

```
do { \
    size_t v_tmp_index_ = (index), v_tmp_n_ = (n); \
    memmove((X).ptr + v_tmp_index_, (X).ptr + v_tmp_index_ + v_tmp_n_, ((X).size - v_tmp_index_ - 1) * \
    sizeof((X).ptr[0])); \
    (X).size -= v_tmp_n_; \
} while (0)
```

Removes an n elements at the specified index of the vector X. The index must be  $0 \leq \text{index} < \text{VectorSize}(X) - n + 1$ .

##### Parameters

|              |                                                      |
|--------------|------------------------------------------------------|
| <i>X</i>     | the vector object.                                   |
| <i>index</i> | the starting position of the elements to be removed. |
| <i>n</i>     | the number of elements to be removed.                |

Definition at line 394 of file [vector.h](#).

## 4.159 vector.h

[Go to the documentation of this file.](#)

```

00001 #ifndef VECTOR_H_
00002 #define VECTOR_H_
00003
00020 /* # License : */
00021 /*
00022 * Copyright (C) 1984-2023 J.A.M. Vermaseren
00023 * When using this file you are requested to refer to the publication
00024 * J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00025 * This is considered a matter of courtesy as the development was paid
00026 * for by FOM the Dutch physics granting agency and we would like to
00027 * be able to track its scientific use to convince FOM of its value
00028 * for the community.
00029 *
00030 * This file is part of FORM.
00031 *
00032 * FORM is free software: you can redistribute it and/or modify it under the
00033 * terms of the GNU General Public License as published by the Free Software
00034 * Foundation, either version 3 of the License, or (at your option) any later
00035 * version.
00036 *
00037 * FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00038 * WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00039 * FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00040 * details.
00041 *
00042 * You should have received a copy of the GNU General Public License along
00043 * with FORM. If not, see <http://www.gnu.org/licenses/>.
00044 */
00045 /* # License : */
00046 /*
00047 # Includes :
00048 */
00049
00050 #include <stddef.h>
00051 #include <string.h>
00052 #include "declare.h"
00053
00054 /*
00055 # Includes :
00056 # Vector :
00057 # VectorStruct :
00058 */
00059
00065 #define VectorStruct(T) \
00066     struct { \
00067         T *ptr; \
00068         size_t size; \
00069         size_t capacity; \
00070     }
00071
00072 /*
00073 # VectorStruct :
00074 # Vector :
00075 */
00076
00084 #define Vector(T, X) \
00085     VectorStruct(T) X = { NULL, 0, 0 }
00086
00087 /*
00088 # Vector :
00089 # DeclareVector :
00090 */
00091
00099 #define DeclareVector(T, X) \
00100     VectorStruct(T) X
00101
00102 /*
00103 # DeclareVector :
00104 # VectorInit :
00105 */
00106
00113 #define VectorInit(X) \
00114     do { \
00115         (X).ptr = NULL; \
00116         (X).size = 0; \
00117         (X).capacity = 0; \
00118     } while (0)
00119
00120 /*
00121 # VectorInit :
00122 # VectorFree :
00123 */
00124
00130 #define VectorFree(X) \
00131     do { \
00132         M_free((X).ptr, "VectorFree:" #X); \
00133         (X).ptr = NULL; \

```

```

00134         (X).size = 0; \
00135         (X).capacity = 0; \
00136     } while (0)
00137
00138 /*
00139     #] VectorFree :
00140     #[ VectorPtr :
00141 */
00142
00150 #define VectorPtr(X) \
00151     ((X).ptr)
00152
00153 /*
00154     #] VectorPtr :
00155     #[ VectorFront :
00156 */
00157
00165 #define VectorFront(X) \
00166     ((X).ptr[0])
00167
00168 /*
00169     #] VectorFront :
00170     #[ VectorBack :
00171 */
00172
00180 #define VectorBack(X) \
00181     ((X).ptr[(X).size - 1])
00182
00183 /*
00184     #] VectorBack :
00185     #[ VectorSize :
00186 */
00187
00194 #define VectorSize(X) \
00195     ((X).size)
00196
00197 /*
00198     #] VectorSize :
00199     #[ VectorCapacity :
00200 */
00201
00208 #define VectorCapacity(X) \
00209     ((X).capacity)
00210
00211 /*
00212     #] VectorCapacity :
00213     #[ VectorEmpty :
00214 */
00215
00222 #define VectorEmpty(X) \
00223     ((X).size == 0)
00224
00225 /*
00226     #] VectorEmpty :
00227     #[ VectorClear :
00228 */
00229
00235 #define VectorClear(X) \
00236     do { (X).size = 0; } while (0)
00237
00238 /*
00239     #] VectorClear :
00240     #[ VectorReserve :
00241 */
00242
00249 #define VectorReserve(X, newcapacity) \
00250     do { \
00251         size_t v_tmp_newcapacity_ = (newcapacity); \
00252         if ( (X).capacity < v_tmp_newcapacity_ ) { \
00253             void *v_tmp_newptr_; \
00254             v_tmp_newcapacity_ = (v_tmp_newcapacity_ * 3) / 2; \
00255             if ( v_tmp_newcapacity_ < 4 ) v_tmp_newcapacity_ = 4; \
00256             v_tmp_newptr_ = Malloc1(sizeof((X).ptr[0]) * v_tmp_newcapacity_, "VectorReserve:" #X); \
00257             if ( (X).ptr != NULL ) { \
00258                 memcpy(v_tmp_newptr_, (X).ptr, (X).size * sizeof((X).ptr[0])); \
00259                 M_free((X).ptr, "VectorReserve:" #X); \
00260             } \
00261             (X).ptr = v_tmp_newptr_; \
00262             (X).capacity = v_tmp_newcapacity_; \
00263         } \
00264     } while (0)
00265
00266 /*
00267     #] VectorReserve :
00268     #[ VectorPushBack :
00269 */
00270

```

```

00277 #define VectorPushBack(X, x) \
00278     do { \
00279         VectorReserve((X), (X).size + 1); \
00280         (X).ptr[(X).size++] = (x); \
00281     } while (0)
00282
00283 /*
00284     #] VectorPushBack :
00285     #[ VectorPushBacks :
00286 */
00287
00295 #define VectorPushBacks(X, src, n) \
00296     do { \
00297         size_t v_tmp_n_ = (n); \
00298         VectorReserve((X), (X).size + v_tmp_n_); \
00299         memcpy((X).ptr + (X).size, (src), v_tmp_n_ * sizeof((X).ptr[0])); \
00300         (X).size += v_tmp_n_; \
00301     } while (0)
00302
00303 /*
00304     #] VectorPushBacks :
00305     #[ VectorPopBack :
00306 */
00307
00314 #define VectorPopBack(X) \
00315     do { (X).size --; } while (0)
00316
00317 /*
00318     #] VectorPopBack :
00319     #[ VectorInsert :
00320 */
00321
00330 #define VectorInsert(X, index, x) \
00331     do { \
00332         size_t v_tmp_index_ = (index); \
00333         VectorReserve((X), (X).size + 1); \
00334         memmove((X).ptr + v_tmp_index_ + 1, (X).ptr + v_tmp_index_, ((X).size - v_tmp_index_) * \
00335             sizeof((X).ptr[0])); \
00336         (X).ptr[v_tmp_index_] = (x); \
00337         (X).size++; \
00338     } while (0)
00339
00340 /*
00341     #] VectorInsert :
00342     #[ VectorInserts :
00343 */
00344
00353 #define VectorInserts(X, index, src, n) \
00354     do { \
00355         size_t v_tmp_index_ = (index), v_tmp_n_ = (n); \
00356         VectorReserve((X), (X).size + v_tmp_n_); \
00357         memmove((X).ptr + v_tmp_index_ + v_tmp_n_, (X).ptr + v_tmp_index_, ((X).size - v_tmp_index_) * \
00358             sizeof((X).ptr[0])); \
00359         memcpy((X).ptr + v_tmp_index_, (src), v_tmp_n_ * sizeof((X).ptr[0])); \
00360         (X).size += v_tmp_n_; \
00361     } while (0)
00362
00363 /*
00364     #] VectorInserts :
00365     #[ VectorErase :
00366 */
00367
00374 #define VectorErase(X, index) \
00375     do { \
00376         size_t v_tmp_index_ = (index); \
00377         memmove((X).ptr + v_tmp_index_, (X).ptr + v_tmp_index_ + 1, ((X).size - v_tmp_index_ - 1) * \
00378             sizeof((X).ptr[0])); \
00379         (X).size--; \
00380     } while (0)
00381
00382 /*
00383     #] VectorErase :
00384     #[ VectorErases :
00385 */
00394 #define VectorErases(X, index, n) \
00395     do { \
00396         size_t v_tmp_index_ = (index), v_tmp_n_ = (n); \
00397         memmove((X).ptr + v_tmp_index_, (X).ptr + v_tmp_index_ + v_tmp_n_, ((X).size - v_tmp_index_ - \
00398             1) * sizeof((X).ptr[0])); \
00399         (X).size -= v_tmp_n_; \
00400     } while (0)
00401
00402 /*
00403     #] VectorErases :
00404     #[ Vector :
00405 */

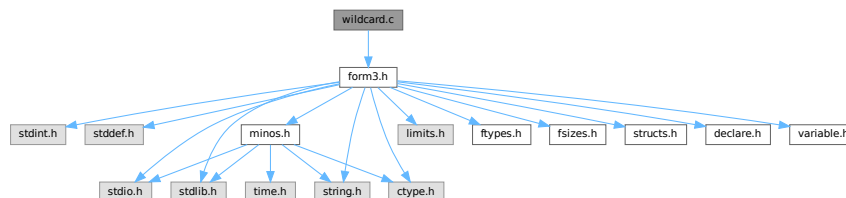
```

```
00405 #endif /* VECTOR_H_ */
```

## 4.160 wildcard.c File Reference

```
#include "form3.h"
```

Include dependency graph for wildcard.c:



### Macros

- #define [DEBUG\(x\)](#)

### Functions

- WORD [WildFill](#) (PHEAD WORD \*to, WORD \*from, WORD \*sub)
- int [ResolveSet](#) (PHEAD WORD \*from, WORD \*to, WORD \*subs)
- void [ClearWild](#) (PHEAD0)
- int [AddWild](#) (PHEAD WORD oldnumber, WORD type, WORD newnumber)
- int [CheckWild](#) (PHEAD WORD oldnumber, WORD type, WORD newnumber, WORD \*newval)
- int [DenToFunction](#) (WORD \*term, WORD numfun)

### 4.160.1 Detailed Description

Contains the functions that deal with the wildcards. During the pattern matching there are two steps: 1: check that a wildcard substitution is correct (if there was already an assignment for this variable, it is the same; it is part of the proper set; it is the proper type of variables, etc.) 2: make the assignment In addition we have to be able to clear assignments. During execution we have to make the actual replacements (WildFill)

Definition in file [wildcard.c](#).

### 4.160.2 Macro Definition Documentation

#### 4.160.2.1 DEBUG

```
#define DEBUG(
    x )
```

Definition at line 44 of file [wildcard.c](#).



## 4.160.3 Function Documentation

### 4.160.3.1 WildFill()

```
WORD WildFill (
    PHEAD WORD * to,
    WORD * from,
    WORD * sub )
```

Definition at line 65 of file [wildcard.c](#).

### 4.160.3.2 ResolveSet()

```
int ResolveSet (
    PHEAD WORD * from,
    WORD * to,
    WORD * subs )
```

Definition at line 1361 of file [wildcard.c](#).

### 4.160.3.3 ClearWild()

```
void ClearWild (
    PHEAD0 )
```

Definition at line 1518 of file [wildcard.c](#).

### 4.160.3.4 AddWild()

```
int AddWild (
    PHEAD WORD oldnumber,
    WORD type,
    WORD newnumber )
```

Definition at line 1544 of file [wildcard.c](#).

### 4.160.3.5 CheckWild()

```
int CheckWild (
    PHEAD WORD oldnumber,
    WORD type,
    WORD newnumber,
    WORD * newval )
```

Definition at line 1791 of file [wildcard.c](#).

### 4.160.3.6 DenToFunction()

```
int DenToFunction (
    WORD * term,
    WORD numFun )
```

Definition at line 2557 of file [wildcard.c](#).

## 4.161 wildcard.c

[Go to the documentation of this file.](#)

```
00001
00012 /* #[] License : */
00013 /*
00014 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00015 *   When using this file you are requested to refer to the publication
00016 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00017 *   This is considered a matter of courtesy as the development was paid
00018 *   for by FOM the Dutch physics granting agency and we would like to
00019 *   be able to track its scientific use to convince FOM of its value
00020 *   for the community.
00021 *
00022 *   This file is part of FORM.
00023 *
00024 *   FORM is free software: you can redistribute it and/or modify it under the
00025 *   terms of the GNU General Public License as published by the Free Software
00026 *   Foundation, either version 3 of the License, or (at your option) any later
00027 *   version.
00028 *
00029 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00030 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00031 *   FOR A PARTICULAR PURPOSE. See the GNU General Public License for more
00032 *   details.
00033 *
00034 *   You should have received a copy of the GNU General Public License along
00035 *   with FORM. If not, see <http://www.gnu.org/licenses/>.
00036 */
00037 /* #[] License : */
00038 /*
00039 *   #[] Includes : wildcard.c
00040 */
00041
00042 #include "form3.h"
00043
00044 #define DEBUG(x)
00045
00046 /*
00047 #define DEBUG(x) x
00048
00049 #[] Includes :
00050 #[] Wildcards :
00051 #[] WildFill :          WORD WildFill(to,from,sub)
00052
00053     Takes the term in from and puts it into to while
00054     making wildcard substitutions.
00055     The return value is the number of words put in to.
00056     The length as the first word of from is not copied.
00057
00058     There are two possible algorithms:
00059     1: For each element in `from': scan sub.
00060     2: For each wildcard in sub replace elements in term.
00061     The original algorithm used 1:
00062 */
00063
00064
00065 WORD WildFill(PHEAD WORD *to, WORD *from, WORD *sub)
00066 {
00067     GETBIDENTITY
00068     WORD i, j, *s, *t, *m, len, dflag, odirt, adirt;
00069     WORD *r, *u, *v, *w, *z, *zst, *zz, *subs, *accu, na, dirty = 0, *tstop;
00070     WORD *temp = 0, *uu, *oldcpointer, sgn;
00071     WORD subcount, setflag, *setlist = 0, si;
00072     accu = oldcpointer = AR.CompressPointer;
00073     t = sub;
00074     t += sub[1];
00075     s = sub + SUBEXPSIZE;
00076     i = 0;
```

```

00077     while ( s < t && *s != FROMBRAC ) {
00078         i++; s += s[1];
00079     }
00080     if ( !i ) {          /* No wildcards -> done quickly */
00081         j = i = *from;
00082         NCOPY(to,from,i);
00083         if ( dirty ) AN.WildDirt = dirty;
00084         return(j);
00085     }
00086     sgn = 0;
00087     subs = sub + SUBEXPSIZE;
00088     t = from;
00089     GETSTOP(t,r);
00090     t++;
00091     m = to + 1;
00092     if ( t < r ) do {
00093         uu = u = t + t[1];
00094         setflag = 0;
00095 ReSwitch:
00096         switch ( *t ) {
00097             case SYMBOL:
00098 /*
00099             #[ SYMBOLS :
00100 */
00101                 z = accu;
00102                 *m++ = *t++;
00103                 *m++ = *t++;
00104                 v = m;
00105                 while ( t < u ) {
00106                     *m = *t;
00107                     for ( si = 0; si < setflag; si += 2 ) {
00108                         if ( t == temp + setlist[si] ) goto sspow;
00109                     }
00110                     s = subs;
00111                     for ( j = 0; j < i; j++ ) {
00112                         if ( *t == s[2] ) {
00113                             if ( *s == SYMTOSYM ) {
00114                                 *m = s[3]; dirty = 1;
00115                                 break;
00116                             }
00117                             else if ( *s == SYMTONUM ) {
00118                                 dirty = 1;
00119                                 zst = z;
00120                                 *z++ = SNUMBER;
00121                                 *z++ = 4;
00122                                 *z++ = s[3];
00123                                 w = z;
00124                                 *z++ = *t;
00125                                 if ( ABS(*t) >= 2*MAXPOWER ) {
00126 DoPow:
00127                                     s = subs;
00128                                     for ( j = 0; j < i; j++ ) {
00129                                         if ( ( *s == SYMTONUM ) &&
00130                                             ( ABS(*t) - 2*MAXPOWER ) == s[2] ) {
00131                                             dirty = 1;
00132                                             *w = s[3];
00133                                             if ( *t < 0 ) *w = -*w;
00134                                             break;
00135                                         }
00136                                         if ( ( *s == SYMTOSYM ) &&
00137                                             ( ABS(*t) - 2*MAXPOWER ) == s[2] ) {
00138                                             dirty = 1;
00139                                             zz = z;
00140                                             while ( --zz >= zst ) {
00141                                                 zz[1+FUNHEAD+ARGHEAD] = *zz;
00142                                             }
00143                                             w += 1+FUNHEAD+ARGHEAD;
00144                                             *zst = EXPONENT;
00145                                             zst[2] = DIRTYFLAG;
00146                                             zst[FUNHEAD+ARGHEAD] = WORDDIF(z,zst)+4;
00147                                             zst[1+FUNHEAD] = 1;
00148                                             zst[FUNHEAD] = WORDDIF(z,zst)+4+ARGHEAD;
00149                                             z += FUNHEAD+ARGHEAD+1;
00150                                             *w = 1; /* exponent -> 1 */
00151                                             *z++ = 1;
00152                                             *z++ = 1;
00153                                             *z++ = 3;
00154                                             if ( *t > 0 ) {
00155                                                 *z++ = -SYMBOL;
00156                                                 *z++ = s[3];
00157                                             }
00158                                             else {
00159                                                 *z++ = ARGHEAD+8;
00160                                                 *z++ = 1;
00161                                                 *z++ = 8;
00162                                                 *z++ = SYMBOL;
00163                                                 *z++ = 4;
00164                                                 *z++ = s[3];

```

```

00164             *z++ = 1;
00165             *z++ = 1;
00166             *z++ = 1;
00167             *z++ = -3;
00168         }
00169         zst[1] = WORDDIF(z,zst);
00170         break;
00171     }
00172     if ( *s == SYMTOSUB &&
00173         ( ABS(*t) - 2*MAXPOWER ) == s[2] ) {
00174 MakeExp:
00175         dirty = 1;
00176         zz = z;
00177         while ( --zz >= zst ) {
00178             zz[1+FUNHEAD+ARGHEAD] = *zz;
00179         }
00180         w += 1+FUNHEAD+ARGHEAD;
00181         *zst = EXPONENT;
00182         zst[2] = DIRTYFLAG;
00183         zst[FUNHEAD+ARGHEAD] = WORDDIF(z,zst)+4;
00184         zst[1+FUNHEAD] = 1;
00185         zst[FUNHEAD] = WORDDIF(z,zst)+4+ARGHEAD;
00186         z += FUNHEAD+ARGHEAD+1;
00187         *w = 1; /* exponent -> 1 */
00188         *z++ = 1;
00189         *z++ = 1;
00190         *z++ = 3;
00191         *z++ = 4+SUBEXPSIZE+ARGHEAD;
00192         *z++ = 1;
00193         *z++ = 4+SUBEXPSIZE;
00194         *z++ = SUBEXPRESSION;
00195         *z++ = SUBEXPSIZE;
00196         *z++ = s[3];
00197         *z++ = 1;
00198         *z++ = AT.ebufnum;
00199         FILLSUB(z)
00200         *z++ = 1;
00201         *z++ = 1;
00202         *z++ = *t > 0 ? 3: -3;
00203         zst[1] = WORDDIF(z,zst);
00204         break;
00205     }
00206     s += s[1];
00207 }
00208 if ( !*w ) z = w - 3;
00209 t++;
00210 goto Seven;
00211 }
00212 else if ( *s == SYMTOSUB ) {
00213     dirty = 1;
00214     zst = z;
00215     *z++ = SUBEXPRESSION;
00216     *z++ = SUBEXPSIZE;
00217     *z++ = s[3];
00218     w = z;
00219     *z++ = *++t;
00220     *z++ = AT.ebufnum;
00221     FILLSUB(z)
00222     goto DoPow;
00223 }
00224 }
00225 s += s[1];
00226 }
00227 sspow:
00228 s = subs;
00229 **m = *++t;
00230 for ( si = 0; si < setflag; si += 2 ) {
00231     if ( t == temp + setlist[si] ) {
00232         t++; m++;
00233         goto Seven;
00234     }
00235 }
00236 for ( j = 0; j < i; j++ ) {
00237     if ( ( ABS(*t) - 2*MAXPOWER ) == s[2] ) {
00238         if ( *s == SYMTONUM ) {
00239             dirty = 1;
00240             *m = s[3];
00241             if ( *t < 0 ) *m = -*m;
00242             break;
00243         }
00244         else if ( *s == SYMTOSYM ) {
00245             dirty = 1;
00246             *z++ = EXPONENT;
00247             if ( *t < 0 ) *z++ = FUNHEAD+ARGHEAD+10;
00248             else *z++ = 4+FUNHEAD;
00249             *z++ = 0;
00250             FILLFUN3(z)

```

```

00251             *z++ = -SYMBOL;
00252             *z++ = m[-1];
00253             if ( *t < 0 ) {
00254                 *z++ = ARGHEAD+8;
00255                 *z++ = 0;
00256                 *z++ = 8;
00257                 *z++ = SYMBOL;
00258                 *z++ = 4;
00259                 *z++ = s[3];
00260                 *z++ = 1;
00261                 *z++ = 1;
00262                 *z++ = 1;
00263                 *z = -3;
00264             }
00265             else {
00266                 *z++ = -SYMBOL;
00267                 *z++ = s[3];
00268             }
00269             m -= 2;
00270             break;
00271         }
00272         else if ( *s == SYMTOSUB ) {
00273             zst = z;
00274             *z++ = SYMBOL;
00275             *z++ = 4;
00276             *z++ = *-m;
00277             w = z;
00278             *z++ = *t;
00279             goto MakeExp;
00280         }
00281     }
00282     s += s[1];
00283 }
00284 t++;
00285 if ( *m ) m++;
00286 else m--;
00287 Seven;;
00288 }
00289 j = WORDDIF(m,v);
00290 if ( !j ) m -= 2;
00291 else v[-1] = j + 2;
00292 s = accu;
00293 while ( s < z ) *m++ = *s++;
00294 break;
00295 /*
00296 #] SYMBOLS :
00297 */
00298 case DOTPRODUCT:
00299 /*
00300 #[ DOTPRODUCTS :
00301 */
00302     *m++ = *t++;
00303     *m++ = *t++;
00304     v = m;
00305     z = accu;
00306     while ( t < u ) {
00307         *m = *t;
00308         subcount = 0;
00309         /* Process the first vector of the DOTPRODUCT */
00310         for ( si = 0; si < setflag; si += 2 ) {
00311             if ( t == temp + setlist[si] ) goto ss2;
00312         }
00313         s = subs;
00314         for ( j = 0; j < i; j++ ) {
00315             if ( *t == s[2] ) {
00316                 if ( *s == VECTOVEC ) {
00317                     *m = s[3]; dirty = 1; break;
00318                 }
00319                 if ( *s == VECTOMIN ) {
00320                     *m = s[3];
00321                     dirty = 1;
00322                     if ( ( ABS(t[2]) - 2*MAXPOWER ) < 0 ) {
00323                         /* The power is a number */
00324                         sgn += t[2];
00325                     }
00326                     else {
00327                         /* The power is a wildcard. Put a -, resolve later. */
00328                         sgn++;
00329                     }
00330                     break;
00331                 }
00332                 if ( *s == VECTOSUB ) {
00333                     *m = s[3]; dirty = 1; subcount = 1; break;
00334                 }
00335             }
00336             s += s[1];
00337         }

```

```

00338 ss2:
00339     ***m = ***t;
00340     s = subs;
00341     /* Process the second vector of the DOTPRODUCT */
00342     for ( si = 0; si < setflag; si += 2 ) {
00343         if ( t == temp + setlist[si] ) goto ss3;
00344     }
00345     for ( j = 0; j < i; j++ ) {
00346         if ( *t == s[2] ) {
00347             if ( *s == VECTOVEC ) {
00348                 *m = s[3]; dirty = 1; break;
00349             }
00350             if ( *s == VECTOMIN ) {
00351                 *m = s[3];
00352                 dirty = 1;
00353                 if ( ( ABS(t[1]) - 2*MAXPOWER ) < 0 ) {
00354                     /* The power is a number */
00355                     sgn += t[1];
00356                 }
00357                 else {
00358                     /* The power is a wildcard. Put a -, resolve later. */
00359                     sgn++;
00360                 }
00361                 break;
00362             }
00363             if ( *s == VECTOSUB ) {
00364                 *m = s[3]; dirty = 1; subcount += 2; break;
00365             }
00366         }
00367         s += s[1];
00368     }
00369 ss3:
00370     ***m = ***t;
00371     /* Process the power */
00372     if ( ( ABS(*t) - 2*MAXPOWER ) < 0 ) goto RegPow;
00373     s = subs;
00374     for ( j = 0; j < i; j++ ) {
00375         if ( ( ABS(*t) - 2*MAXPOWER ) == s[2] ) {
00376             if ( *s == SYMTONUM ) {
00377                 *m = s[3];
00378                 /* Since the power is a wildcard, sgn is 0,1,2 depending whether
00379                    there were 0,1,2 VECTOMIN in the DOTPRODUCT. Multiply by the
00380                    power, which is positive currently. */
00381                 sgn *= *m;
00382                 /* Now flip the sign of the power, if the wildcard came with a - */
00383                 if ( *t < 0 ) *m = -*m;
00384                 dirty = 1;
00385                 break;
00386             }
00387             if ( *s <= SYMTOSUB ) {
00388
00389                 Here we put together a power function with the proper
00390                 arguments. Note that a p?.q? resolves to a single power.
00391
00392                 m -= 2;
00393                 *z++ = EXPONENT;
00394                 w = z;
00395                 if ( subcount == 0 ) {
00396                     *z++ = 17+FUNHEAD+2*ARGHEAD;
00397                     *z++ = DIRTYFLAG;
00398                     FILLFUN3(z)
00399                     *z++ = 9+ARGHEAD;
00400                     *z++ = 0;
00401                     FILLARG(z)
00402                     *z++ = 9;
00403                     *z++ = DOTPRODUCT;
00404                     *z++ = 5;
00405                     *z++ = *m;
00406                     *z++ = m[1];
00407                     *z++ = 1;
00408                     *z++ = 1;
00409                     *z++ = 1;
00410                     *z++ = 3;
00411                     if ( *s == SYMTOSYM ) {
00412                         *z++ = 8+ARGHEAD;
00413                         *z++ = 0;
00414                         FILLARG(z)
00415                         *z++ = 8;
00416                         *z++ = SYMBOL;
00417                         *z++ = 4;
00418                         *z++ = s[3];
00419                         *z++ = 1;
00420                     }
00421                     else {
00422                         *z++ = 4+SUBEXPSIZE+ARGHEAD;
00423                         *z++ = 1;
00424                         FILLARG(z)
00425                         *z++ = 4+SUBEXPSIZE;

```

```

00425         *z++ = SUBEXPRESSION;
00426         *z++ = SUBEXPSIZE;
00427         *z++ = s[3];
00428         *z++ = 1;
00429         *z++ = AT.ebufnum;
00430         FILLSUB(z)
00431     }
00432     *z++ = 1; *z++ = 1;
00433     *z++ = ( s[2] > 0 ) ? 3: -3;
00434 }
00435 else if ( subcount == 3 ) {
00436     *z++ = 20+2*SUBEXPSIZE+FUNHEAD+2*ARGHEAD;
00437     *z++ = DIRTYFLAG;
00438     FILLFUN3(z)
00439     *z++ = 12+2*SUBEXPSIZE+ARGHEAD;
00440     *z++ = 1;
00441     *z++ = 12+2*SUBEXPSIZE;
00442     *z++ = SUBEXPRESSION;
00443     *z++ = 4+SUBEXPSIZE;
00444     *z++ = *m + 1;
00445     *z++ = 1;
00446     *z++ = AT.ebufnum;
00447     FILLSUB(z)
00448     *z++ = INDTOIND;
00449     *z++ = 4;
00450     *z++ = FUNNYVEC;
00451     *z++ = ++AR.CurDum;
00452
00453     *z++ = SUBEXPRESSION;
00454     *z++ = 4+SUBEXPSIZE;
00455     *z++ = m[1] + 1;
00456     *z++ = 1;
00457     *z++ = AT.ebufnum;
00458     FILLSUB(z)
00459     *z++ = INDTOIND;
00460     *z++ = 4;
00461     *z++ = FUNNYVEC;
00462     *z++ = AR.CurDum;
00463     *z++ = 1; *z++ = 1; *z++ = 3;
00464 }
00465 else {
00466     if ( subcount == 2 ) {
00467         j = *m; *m = m[1]; m[1] = j;
00468     }
00469     *z++ = 16+SUBEXPSIZE+FUNHEAD+2*ARGHEAD;
00470     *z++ = DIRTYFLAG;
00471     FILLFUN3(z)
00472     *z++ = 8+SUBEXPSIZE+ARGHEAD;
00473     *z++ = 1;
00474     *z++ = 8+SUBEXPSIZE;
00475     *z++ = SUBEXPRESSION;
00476     *z++ = 4+SUBEXPSIZE;
00477     *z++ = *m + 1;
00478     *z++ = 1;
00479     *z++ = AT.ebufnum;
00480     FILLSUB(z)
00481     *z++ = INDTOIND;
00482     *z++ = 4;
00483     *z++ = FUNNYVEC;
00484     *z++ = m[1];
00485     *z++ = 1; *z++ = 1; *z++ = 3;
00486 }
00487 if ( *s == SYMTOSYM ) {
00488     if ( s[2] > 0 ) {
00489         *z++ = -SYMBOL;
00490         *z++ = s[3];
00491         t++;
00492         *w = z-w+1;
00493         goto NextDot;
00494     }
00495     *z++ = 8+ARGHEAD;
00496     *z++ = 0;
00497     *z++ = 8;
00498     *z++ = SYMBOL;
00499     *z++ = 4;
00500     *z++ = s[3];
00501     *z++ = 1;
00502 }
00503 else {
00504     *z++ = 4+SUBEXPSIZE+ARGHEAD;
00505     *z++ = 1;
00506     *z++ = 4+SUBEXPSIZE;
00507     *z++ = SUBEXPRESSION;
00508     *z++ = SUBEXPSIZE;
00509     *z++ = s[3];
00510     *z++ = 1;
00511     *z++ = AT.ebufnum;

```

```

00512             FILLSUB(z)
00513         }
00514         *z++ = 1; *z++ = 1;
00515         *z++ = ( s[2] > 0 ) ? 3: -3;
00516         t++;
00517         *w = z-w+1;
00518         goto NextDot;
00519     }
00520 }
00521 s += s[1];
00522 }
00523 RegPow: if ( *m ) m++;
00524 else { m -= 2; subcount = 0; }
00525 t++;
00526 if ( subcount ) {
00527     m -= 3;
00528     if ( subcount == 3 ) {
00529         if ( m[2] < 0 ) {
00530             j = (-m[2]) * (2*SUBEXPSIZE+8);
00531             *z++ = DENOMINATOR;
00532             *z++ = j + 8 + FUNHEAD + ARGHEAD;
00533             *z++ = DIRTYFLAG;
00534             FILLFUN3(z)
00535             *z++ = j + 8 + ARGHEAD;
00536             *z++ = 1;
00537             *z++ = j + 8;
00538             while ( m[2] < 0 ) {
00539                 (m[2])++;
00540                 *z++ = SUBEXPRESSION;
00541                 *z++ = 4+SUBEXPSIZE;
00542                 *z++ = *m + 1;
00543                 *z++ = 1;
00544                 *z++ = AT.ebufnum;
00545                 FILLSUB(z)
00546                 *z++ = INDTOIND;
00547                 *z++ = 4;
00548                 *z++ = FUNNYVEC;
00549                 *z++ = ++AR.CurDum;
00550                 *z++ = SUBEXPRESSION;
00551                 *z++ = 8+SUBEXPSIZE;
00552                 *z++ = m[1] + 1;
00553                 *z++ = 1;
00554                 *z++ = AT.ebufnum;
00555                 FILLSUB(z)
00556                 *z++ = INDTOIND;
00557                 *z++ = 4;
00558                 *z++ = FUNNYVEC;
00559                 *z++ = AR.CurDum;
00560                 *z++ = SYMTOSYM; /* Needed to avoid */
00561                 *z++ = 4; /* problems with */
00562                 *z++ = 1000; /* conversion to */
00563                 *z++ = 1000; /* square of subexp*/
00564             }
00565             *z++ = 1; *z++ = 1; *z++ = 3;
00566         }
00567         else {
00568             while ( m[2] > 0 ) {
00569                 (m[2])--;
00570                 *z++ = SUBEXPRESSION;
00571                 *z++ = 4+SUBEXPSIZE;
00572                 *z++ = *m + 1;
00573                 *z++ = 1;
00574                 *z++ = AT.ebufnum;
00575                 FILLSUB(z)
00576                 *z++ = INDTOIND;
00577                 *z++ = 4;
00578                 *z++ = FUNNYVEC;
00579                 *z++ = ++AR.CurDum;
00580                 *z++ = SUBEXPRESSION;
00581                 *z++ = 4+SUBEXPSIZE;
00582                 *z++ = m[1] + 1;
00583                 *z++ = 1;
00584                 *z++ = AT.ebufnum;
00585                 FILLSUB(z)
00586                 *z++ = INDTOIND;
00587                 *z++ = 4;
00588                 *z++ = FUNNYVEC;
00589                 *z++ = AR.CurDum;
00590             }
00591         }
00592     }
00593     else {
00594         if ( subcount == 2 ) {
00595             j = *m; *m = m[1]; m[1] = j;
00596         }
00597         if ( m[2] < 0 ) {
00598             *z++ = DENOMINATOR;

```



```

00599             *z++ = 8+SUBEXPSIZE+FUNHEAD+ARGHEAD;
00600             *z++ = DIRTYFLAG;
00601             FILLFUN3(z)
00602             *z++ = 8+SUBEXPSIZE+ARGHEAD;
00603             *z++ = 1;
00604             *z++ = 8+SUBEXPSIZE;
00605         }
00606         *z++ = SUBEXPRESSION;
00607         *z++ = 4+SUBEXPSIZE;
00608         *z++ = *m + 1;
00609         *z++ = ABS(m[2]);
00610         *z++ = AT.ebufnum;
00611         FILLSUB(z)
00612         *z++ = INDTOIND;
00613         *z++ = 4;
00614         *z++ = FUNNYVEC;
00615         *z++ = m[1];
00616         if ( m[2] < 0 ) {
00617             *z++ = 1; *z++ = 1; *z++ = 3;
00618         }
00619     }
00620 }
00621 NextDot:;
00622     }
00623     if ( m <= v ) m = v - 2;
00624     else v[-1] = WORDDIF(m,v) + 2;
00625     if ( z > accu ) {
00626         j = WORDDIF(z,accu);
00627         z = accu;
00628         NCOPY(m,z,j);
00629     }
00630     break;
00631 /*
00632 #] DOTPRODUCTS :
00633 */
00634 case SETSET:
00635 /*
00636 #[ SETS :
00637 */
00638     temp = accu + ((AR.ComprTop - accu)»1)&(-2));
00639     if ( ResolveSet(BHEAD t,temp,sub) ) {
00640         Terminate(-1);
00641     }
00642     setlist = t + 2 + t[3];
00643     setflag = t[1] - 2 - t[3]; /* Number of elements * 2 */
00644     t = temp; u = t + t[1];
00645     goto ReSwitch;
00646 /*
00647 #] SETS :
00648 */
00649 case VECTOR:
00650 /*
00651 #[ VECTORS :
00652 */
00653     *m++ = *t++;
00654     *m++ = *t++;
00655     v = m;
00656     z = accu;
00657     while ( t < u ) {
00658         *m = *t;
00659         for ( si = 0; si < setflag; si += 2 ) {
00660             if ( t == temp + setlist[si] ) goto ss4;
00661         }
00662         s = subs;
00663         for ( j = 0; j < i; j++ ) {
00664             if ( *t == s[2] ) {
00665                 if ( *s == INDTOIND || *s == VECTOVEC ) {
00666                     *m = s[3]; dirty = 1; break;
00667                 }
00668                 if ( *s == VECTOMIN ) {
00669                     *m = s[3]; dirty = 1; sgn++; break;
00670                 }
00671                 else if ( *s == VECTOSUB ) {
00672                     *z++ = SUBEXPRESSION;
00673                     *z++ = 4+SUBEXPSIZE;
00674                     *z++ = s[3]+1;
00675                     *z++ = 1;
00676                     *z++ = AT.ebufnum;
00677                     FILLSUB(z)
00678                     *z++ = VECTOVEC;
00679                     *z++ = 4;
00680                     *z++ = FUNNYVEC;
00681                     *z++ = *++t;
00682                     m--;
00683                     s = subs;
00684                     for ( j = 0; j < i; j++ ) {
00685                         if ( z[-1] == s[2] ) {

```

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00686                                     if ( *s == INDTOIND || *s == VECTOVEC ) {
00687                                         z[-1] = s[3];
00688                                         break;
00689                                     }
00690                                     if ( *s == INDTOSUB || *s == VECTOSUB ) {
00691                                         z[-1] = ++AR.CurDum;
00692                                         *z++ = SUBEXPRESSION;
00693                                         *z++ = 4+SUBEXPSIZE;
00694                                         *z++ = s[3]+1;
00695                                         *z++ = 1;
00696                                         *z++ = AT.ebufnum;
00697                                         FILLSUB(z)
00698                                         if ( *s == INDTOSUB ) *z++ = INDTOIND;
00699                                         else *z++ = VECTOSUB;
00700                                         *z++ = 4;
00701                                         *z++ = FUNNYVEC;
00702                                         *z++ = AR.CurDum;
00703                                         break;
00704                                     }
00705                                     }
00706                                     s += s[1];
00707                                 }
00708                                 dirty = 1;
00709                                 break;
00710                             }
00711                             else if ( *s == INDTOSUB ) {
00712                                 *z++ = SUBEXPRESSION;
00713                                 *z++ = 4+SUBEXPSIZE;
00714                                 *z++ = s[3]+1;
00715                                 *z++ = 1;
00716                                 *z++ = AT.ebufnum;
00717                                 FILLSUB(z)
00718                                 *z++ = INDTOIND;
00719                                 *z++ = 4;
00720                                 *z++ = FUNNYVEC;
00721                                 m -= 2;
00722                                 *z++ = m[1];
00723                                 dirty = 1;
00724                                 t++;
00725                                 break;
00726                             }
00727                         }
00728                         s += s[1];
00729                     }
00730                     m++; t++;
00731                 }
00732                 if ( m <= v ) m = v-2;
00733                 else v[-1] = WORDDIF(m,v)+2;
00734                 if ( z > accu ) {
00735                     j = WORDDIF(z,accu); z = accu;
00736                     NCOPY(m,z,j);
00737                 }
00738                 break;
00739             /*
00740             #] VECTORS :
00741             */
00742             case INDEX:
00743             /*
00744             #[ INDEX :
00745             */
00746                 *m++ = *t++;
00747                 *m++ = *t++;
00748                 v = m;
00749                 z = accu;
00750                 while ( t < u ) {
00751                     *m = *t;
00752                     for ( si = 0; si < setflag; si += 2 ) {
00753                         if ( t == temp + setlist[si] ) goto ss5;
00754                     }
00755                     s = subs;
00756                     for ( j = 0; j < i; j++ ) {
00757                         if ( *t == s[2] ) {
00758                             if ( *s == INDTOIND || *s == VECTOVEC )
00759                                 { *m = s[3]; dirty = 1; break; }
00760                             if ( *s == VECTOMIN )
00761                                 { *m = s[3]; dirty = 1; sgn++; break; }
00762                             else if ( *s == VECTOSUB || *s == INDTOSUB ) {
00763                                 *z++ = SUBEXPRESSION;
00764                                 *z++ = SUBEXPSIZE;
00765                                 *z++ = s[3];
00766                                 *z++ = 1;
00767                                 *z++ = AT.ebufnum;
00768                                 FILLSUB(z)
00769                                 m--;
00770                                 dirty = 1;
00771                                 break;
00772                             }

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```

00773         }
00774         s += s[1];
00775     }
00776 ss5:      m++; t++;
00777     }
00778     if ( m <= v ) m = v-2;
00779     else v[-1] = WORDDIF(m,v)+2;
00780     if ( z > accu ) {
00781         j = WORDDIF(z,accu); z = accu;
00782         NCOPY(m,z,j);
00783     }
00784     break;
00785 /*
00786     #] INDEX :
00787 */
00788     case DELTA:
00789     case LEVICIVITA:
00790     case GAMMA:
00791 /*
00792     #[ SPECIALS :
00793 */
00794     v = m;
00795     *m++ = *t++;
00796     *m++ = *t++;
00797     #if FUNHEAD > 2
00798         if ( t[-2] != DELTA ) *m++ = *t++;
00799     #endif
00800 Tensors:
00801     COPYFUN3(m,t)
00802     z = accu;
00803     while ( t < u ) {
00804         *m = *t;
00805         for ( si = 0; si < setflag; si += 2 ) {
00806             if ( t == temp + setlist[si] ) goto ss6;
00807         }
00808         s = subs;
00809         if ( *m == FUNNYWILD ) {
00810             CBUF *C = cbuf+AT.ebufnum;
00811             t++;
00812             for ( j = 0; j < i; j++ ) {
00813                 if ( *s == ARGTOARG && *t == s[2] ) {
00814                     v[2] |= DIRTYFLAG;
00815                     if ( s[3] < 0 ) { /* empty */
00816                         t++; break;
00817                     }
00818                     w = C->rhs[s[3]];
00819                     DEBUG(MesPrint("Thread %w(a): s[3] = %d, w=(%d,%d,%d,%d)",s[3],w[0],w[1],w[2],w[3]));
00820                     j = *w++;
00821                     if ( j > 0 ) {
00822                         NCOPY(m,w,j);
00823                     }
00824                     else {
00825                         while ( *w ) {
00826                             if ( *w == -INDEX || *w == -VECTOR
00827                                 || *w == -MINVECTOR
00828                                 || ( *w == -SNUMBER && w[1] >= 0
00829                                     && w[1] < AM.OffsetIndex ) ) {
00830                                 if ( *w == -MINVECTOR ) sgn++;
00831                                 w++;
00832                                 *m++ = *w++;
00833                             }
00834                             else {
00835                                 MLOCK(ErrorMessageLock);
00836                                 DEBUG(MesPrint("Thread %w(aa): *w = %d",*w));
00837                                 MesPrint("Illegal substitution of argument field in
00838 tensor");
00839                                 MUNLOCK(ErrorMessageLock);
00840                                 SETERROR(-1)
00841                             }
00842                         }
00843                         t++;
00844                         break;
00845                     }
00846                     s += s[1];
00847                 }
00848             }
00849             else {
00850                 for ( j = 0; j < i; j++ ) {
00851                     if ( *t == s[2] ) {
00852                         if ( *s == INDTOIND || *s == VECTOVEC )
00853                             { *m = s[3]; dirty = 1; break; }
00854                         if ( *s == VECTOMIN )
00855                             { *m = s[3]; dirty = 1; sgn++; break; }
00856                         else if ( *s == VECTOSUB || *s == INDTOSUB ) {
00857                             *m = ++AR.CurDum;
00858                             *z++ = SUBEXPRESSION;

```

```

00859                                     *z++ = 4+SUBEXPSIZE;
00860                                     *z++ = s[3]+1;
00861                                     *z++ = 1;
00862                                     *z++ = AT.ebufnum;
00863                                     FILLSUB(z)
00864                                     *z++ = INDTOIND;
00865                                     *z++ = 4;
00866                                     *z++ = FUNNYVEC;
00867                                     *z++ = AR.CurDum;
00868                                     dirty = 1;
00869                                     break;
00870                                 }
00871                             }
00872                             s += s[1];
00873                         }
00874                         if ( j < i && *v != DELTA ) v[2] |= DIRTYFLAG;
00875 ss6:                             m++; t++;
00876                     }
00877                 }
00878                 v[1] = WORDDIF(m,v);
00879                 if ( z > accu ) {
00880                     j = WORDDIF(z,accu); z = accu;
00881                     NCOPY(m,z,j);
00882                 }
00883                 break;
00884 /*
00885 #] SPECIALS :
00886 */
00887 case SUBEXPRESSION:
00888 /*
00889 #[ SUBEXPRESSION :
00890 */
00891     dirty = 1;
00892     tstop = t + t[1];
00893     *m++ = *t++;
00894     *m++ = *t++;
00895     *m++ = *t++;
00896     *m++ = *t++;
00897     if ( t[-1] >= 2*MAXPOWER || t[-1] <= -2*MAXPOWER ) {
00898         s = subs;
00899         for ( j = 0; j < i; j++ ) {
00900             if ( *s == SYMTONUM &&
00901                 ( ABS(t[-1]) - 2*MAXPOWER ) == s[2] ) {
00902                 m[-1] = s[3];
00903                 if ( t[-1] < 0 ) m[-1] = -m[-1];
00904                 break;
00905             }
00906             s += s[1];
00907         }
00908     }
00909     *m++ = *t++;
00910     COPYSUB(m,t)
00911     while ( t < tstop ) {
00912         for ( si = 0; si < setflag; si += 2 ) {
00913             if ( t == temp + setlist[si] - 2 ) goto ss7;
00914         }
00915         s = subs;
00916         for ( j = 0; j < i; j++ ) {
00917             if ( s[2] == t[2] ) {
00918                 if ( ( *s <= SYMTOSUB && *t <= SYMTOSUB )
00919                     || ( *s == *t && *s < FROMBRAC )
00920                     || ( *s == VECTOVEC && ( *t == VECTOSUB || *t == VECTOMIN ) )
00921                     || ( *s == VECTOSUB && ( *t == VECTOVEC || *t == VECTOMIN ) )
00922                     || ( *s == VECTOMIN && ( *t == VECTOSUB || *t == VECTOVEC ) )
00923                     || ( *s == INDTOIND && *t == INDTOIND )
00924                     || ( *s == INDTOIND && *t == INDTOIND ) ) {
00925                     WORD *vv = m;
00926 /*
00927                     *t = *s; Wrong!!! Overwrites compiler buffer */
00928                     j = t[1];
00929                     NCOPY(m,t,j);
00930                     vv[0] = s[0];
00931                     vv[3] = s[3];
00932                     goto sr7;
00933                 }
00934             }
00935             s += s[1];
00936 ss7:             j = t[1];
00937             NCOPY(m,t,j);
00938 sr7:;
00939         }
00940         break;
00941 /*
00942 #] SUBEXPRESSION :
00943 */
00944 case EXPRESSION:
00945 /*

```

```

00946      #[] EXPRESSION :
00947      /*
00948          dirty = 1;
00949          tstop = t + t[1];
00950          v = m;
00951          *m++ = *t++;
00952          *m++ = *t++;
00953          *m++ = *t++;
00954          *m++ = *t++;
00955          s = subs;
00956          for ( j = 0; j < i; j++ ) {
00957              if ( ( ABS(t[-1]) - 2*MAXPOWER ) == s[2] ) {
00958                  if ( *s == SYMTONUM ) {
00959                      m[-1] = s[3];
00960                      if ( t[-1] < 0 ) m[-1] = -m[-1];
00961                      break;
00962                  }
00963                  else if ( *s <= SYMTOSUB ) {
00964                      MLOCK(ErrorMessageLock);
00965                      MesPrint("Wildcard power of expression should be a number");
00966                      MUNLOCK(ErrorMessageLock);
00967                      SETERROR(-1)
00968                  }
00969              }
00970              s += s[1];
00971          }
00972          *m++ = *t++;
00973          COPYSUB(m,t)
00974          while ( t < tstop && *t != WILDCARDS ) {
00975              j = t[1];
00976              NCOPY(m,t,j);
00977          }
00978          if ( t < tstop && *t == WILDCARDS ) {
00979              *m++ = *t;
00980              s = sub;
00981              j = s[1];
00982              *m++ = j+2;
00983              NCOPY(m,s,j);
00984              t += t[1];
00985          }
00986          if ( t < tstop && *t == FROMBRAC ) {
00987              w = m;
00988              *m++ = *t;
00989              *m++ = t[1];
00990              if ( WildFill(BHEAD m,t+2,sub) < 0 ) {
00991                  MLOCK(ErrorMessageLock);
00992                  MesCall("WildFill");
00993                  MUNLOCK(ErrorMessageLock);
00994                  SETERROR(-1)
00995              }
00996              m += *m;
00997              w[1] = m - w;
00998              t += t[1];
00999          }
01000          while ( t < tstop ) {
01001              j = t[1];
01002              NCOPY(m,t,j);
01003          }
01004          v[1] = m-v;
01005          break;
01006      /*
01007      #[] EXPRESSION :
01008      /*
01009      default:
01010      /*
01011      #[] FUNCTIONS :
01012      /*
01013          if ( *t >= FUNCTION ) {
01014              dflag = 0;
01015              na = 0;
01016              *m = *t;
01017              for ( si = 0; si < setflag; si += 2 ) {
01018                  if ( t == temp + setlist[si] ) {
01019                      dflag = DIRTYFLAG; goto ss8;
01020                  }
01021              }
01022              s = subs;
01023              for ( j = 0; j < i; j++ ) {
01024                  if ( *s == FUNTOFUN && *t == s[2] )
01025                      { *m = s[3]; dirty = 1; dflag = DIRTYFLAG; break; }
01026                  s += s[1];
01027              }
01028          ss8:
01029              v = m;
01030              if ( *t >= FUNCTION && functions[*t-FUNCTION].spec
01031              >= TENSORFUNCTION ) {
01032                  if ( *m < FUNCTION || functions[*m-FUNCTION].spec

```

```

01033             MLOCK(ErrorMessageLock);
01034             MesPrint("Illegal wildcarding of regular function to tensorfunction");
01035             MUNLOCK(ErrorMessageLock);
01036             SETERROR(-1)
01037         }
01038         m++; t++;
01039         *m++ = *t++;
01040         *m++ = *t++ | dflag;
01041         goto Tensors;
01042     }
01043     m++; t++;
01044     *m++ = *t++;
01045     *m++ = *t++ | dflag;
01046     COPYFUN3(m,t)
01047     z = accu;
01048     while ( t < u ) {          /* do an argument */
01049         if ( *t < 0 ) {
01050             /*
01051              *# Simple arguments :
01052              */
01053             CBUF *C = cbuf+AT.ebufnum;
01054             for ( si = 0; si < setflag; si += 2 ) {
01055                 if ( *t <= -FUNCTION ) {
01056                     if ( t == temp + setlist[si] ) {
01057                         v[2] |= DIRTYFLAG; goto ss10; }
01058                     }
01059                     else {
01060                         if ( t == temp + setlist[si]-1 ) {
01061                             v[2] |= DIRTYFLAG; goto ss9; }
01062                         }
01063                     }
01064                 if ( *t == -ARGWILD ) {
01065                     s = subs;
01066                     for ( j = 0; j < i; j++ ) {
01067                         if ( *s == ARGTOARG && s[2] == t[1] ) break;
01068                         s += s[1];
01069                     }
01070                     v[2] |= DIRTYFLAG;
01071                     w = C->rhs[s[3]];
01072                     DEBUG(MesPrint("Thread %w(b): s[3] = %d, w=(%d,%d,%d,%d)",s[3],w[0],w[1],w[2],w[3]));
01073                     if ( *w == 0 ) {
01074                         w++;
01075                         while ( *w ) {
01076                             if ( *w > 0 ) j = *w;
01077                             else if ( *w <= -FUNCTION ) j = 1;
01078                             else j = 2;
01079                             NCOPY(m,w,j);
01080                         }
01081                     }
01082                     else {
01083                         j = *w++;
01084                         while ( --j >= 0 ) {
01085                             if ( *w < MINSPEC ) *m++ = -VECTOR;
01086                             else if ( *w >= 0 && *w < AM.OffsetIndex )
01087                                 *m++ = -SNUMBER;
01088                             else *m++ = -INDEX;
01089                             *m++ = *w++;
01090                         }
01091                     }
01092                     t += 2;
01093                     dirty = 1;
01094                     if ( ( *v == NUMARGSFUN || *v == NUMTERMSFUN )
01095                         && t >= u && m == v + FUNHEAD ) {
01096                         m = v;
01097                         *m++ = SNUMBER; *m++ = 3; *m++ = 0;
01098                         break;
01099                     }
01100                 }
01101                 else if ( *t <= -FUNCTION ) {
01102                     *m = *t;
01103                     s = subs;
01104                     for ( j = 0; j < i; j++ ) {
01105                         if ( -*t == s[2] ) {
01106                             if ( *s == FUNTOFUN )
01107                                 { *m = -s[3]; dirty = 1; v[2] |= DIRTYFLAG; break; }
01108                             }
01109                         s += s[1];
01110                     }
01111                     m++; t++;
01112                 }
01113                 else if ( *t == -SYMBOL ) {
01114                     *m++ = *t++;
01115                     *m = *t;
01116                     s = subs;
01117                     for ( j = 0; j < i; j++ ) {
01118                         if ( *t == s[2] && *s <= SYMTOSUB ) {
01119                             dirty = 1; v[2] |= DIRTYFLAG;

```

```

01120         if ( AR.PolyFunType == 2 && v[0] == AR.PolyFun )
01121             v[2] |= MUSTCLEANPRF;
01122         if ( *s == SYMTOSYM ) *m = s[3];
01123     else if ( *s == SYMTONUM ) {
01124         m[-1] = -SNUMBER;
01125         *m = s[3];
01126     }
01127     else if ( *s == SYMTOSUB ) {
01128         ToSub:
01129         m--;
01130         w = C->rhs[s[3]];
01131         DEBUG(MesPrint("Thread %w(c): s[3] = %d, w=(%d,%d,%d,%d)",s[3],w[0],w[1],w[2],w[3]));
01132         s = m;
01133         m += 2;
01134         while ( *w ) {
01135             j = *w;
01136             NCOPY(m,w,j);
01137         }
01138         *s = WORDDIF(m,s);
01139         s[1] = 0;
01140         *m = 0;
01141         if ( t[-1] == -MINVECTOR ) {
01142             w = s+2;
01143             while ( *w ) {
01144                 w += *w;
01145                 w[-1] = -w[-1];
01146             }
01147             if ( ToFast(s,s) ) {
01148                 if ( *s <= -FUNCTION ) m = s;
01149                 else m = s + 1;
01150             }
01151             else m--;
01152         }
01153         break;
01154     }
01155     s += s[1];
01156 }
01157 m++; t++;
01158 }
01159 else if ( *t == -INDEX ) {
01160     *m++ = *t++;
01161     *m = *t;
01162     s = subs;
01163     for ( j = 0; j < i; j++ ) {
01164         if ( *t == s[2] ) {
01165             if ( *s == INDTOIND || *s == VECTOVEC ) {
01166                 *m = s[3];
01167                 if ( *m < MINSPEC ) m[-1] = -VECTOR;
01168                 else if ( *m >= 0 && *m < AM.OffsetIndex )
01169                     m[-1] = -SNUMBER;
01170                 else m[-1] = -INDEX;
01171             }
01172             else if ( *s == VECTOSUB || *s == INDTOSUB ) {
01173                 m[-1] = -INDEX;
01174                 *m = ++AR.CurDum;
01175                 *z++ = SUBEXPRESSION;
01176                 *z++ = 4+SUBEXPSIZE;
01177                 *z++ = s[3]+1;
01178                 *z++ = 1;
01179                 *z++ = AT.ebufnum;
01180                 FILLSUB(z);
01181                 *z++ = INDTOIND;
01182                 *z++ = 4;
01183                 *z++ = FUNNYVEC;
01184                 *z++ = AR.CurDum;
01185             }
01186             v[2] |= DIRTYFLAG; dirty = 1;
01187             break;
01188         }
01189         s += s[1];
01190     }
01191     m++; t++;
01192 }
01193 else if ( *t == -VECTOR || *t == -MINVECTOR ) {
01194     *m++ = *t++;
01195     *m = *t;
01196     s = subs;
01197     for ( j = 0; j < i; j++ ) {
01198         if ( *t == s[2] ) {
01199             if ( *s == VECTOVEC ) *m = s[3];
01200             else if ( *s == VECTOMIN ) {
01201                 *m = s[3];
01202                 if ( t[-1] == -VECTOR )
01203                     m[-1] = -MINVECTOR;
01204                 else
01205                     m[-1] = -VECTOR;
01206             }

```

```

01207         else if ( *s == VECTOSUB ) goto ToSub;
01208         dirty = 1; v[2] |= DIRTYFLAG;
01209         break;
01210     }
01211     s += s[1];
01212 }
01213 m++; t++;
01214 }
01215 else if ( *t == -SNUMBER ) {
01216     *m++ = *t++;
01217     *m = *t;
01218     s = subs;
01219     for ( j = 0; j < i; j++ ) {
01220         if ( *t == s[2] && *s >= NUMTONUM && *s <= NUMTOSUB ) {
01221             dirty = 1; v[2] |= DIRTYFLAG;
01222             if ( *s == NUMTONUM ) *m = s[3];
01223             else if ( *s == NUMTOSYM ) {
01224                 m[-1] = -SYMBOL;
01225                 *m = s[3];
01226             }
01227             else if ( *s == NUMTOIND ) {
01228                 m[-1] = -INDEX;
01229                 *m = s[3];
01230             }
01231             else if ( *s == NUMTOSUB ) goto ToSub;
01232             break;
01233         }
01234         s += s[1];
01235     }
01236     m++; t++;
01237 }
01238 else {
01239     *m++ = *t++;
01240     *m++ = *t++;
01241 }
01242 na = WORDDIF(z,accu);
01243 /*
01244 #] Simple arguments :
01245 */
01246 }
01247 else {
01248     w = m;
01249     zz = t;
01250     NEXTARG(zz)
01251     odirt = AN.WildDirt; AN.WildDirt = 0;
01252     AR.CompressPointer = accu + na;
01253     for ( j = 0; j < ARGHEAD; j++ ) *m++ = *t++;
01254     j = 0;
01255     adirt = 0;
01256     while ( t < zz ) { /* do a term */
01257         if ( ( len = WildFill(BHEAD m,t,sub) ) < 0 ) {
01258             MLOCK(ErrorMessageLock);
01259             MesCall("WildFill");
01260             MUNLOCK(ErrorMessageLock);
01261             SETERROR(-1)
01262         }
01263         if ( AN.WildDirt ) {
01264             adirt = AN.WildDirt;
01265             AN.WildDirt = 0;
01266         }
01267         m += len;
01268         t += *t;
01269     }
01270     *w = WORDDIF(m,w); /* Fill parameter length */
01271     if ( adirt ) {
01272         dirty = w[1] = 1; v[2] |= DIRTYFLAG;
01273         if ( AR.PolyFunType == 2 && v[0] == AR.PolyFun )
01274             v[2] |= MUSTCLEANPRF;
01275         AN.WildDirt = adirt;
01276     }
01277     else {
01278         AN.WildDirt = odirt;
01279     }
01280     if ( ToFast(w,w) ) {
01281         if ( *w <= -FUNCTION ) {
01282             if ( *w == NUMARGSFUN || *w == NUMTERMSFUN ) {
01283                 *w = -SNUMBER; w[1] = 0; m = w + 2;
01284             }
01285             else m = w+1;
01286         }
01287         else m = w+2;
01288     }
01289     AR.CompressPointer = oldcpointer;
01290 }
01291 }
01292 v[1] = WORDDIF(m,v); /* Fill function length */
01293 s = accu;

```



```

01294         NCOPY(m,s,na);
01295  /*
01296         Now some code to speed up a few special cases
01297  */
01298         if ( v[0] == EXPONENT ) {
01299             if ( v[1] == FUNHEAD+4 && v[FUNHEAD] == -SYMBOL &&
01300                 v[FUNHEAD+2] == -SNUMBER && v[FUNHEAD+3] < MAXPOWER
01301                 && v[FUNHEAD+3] > -MAXPOWER ) {
01302                 v[0] = SYMBOL;
01303                 v[1] = 4;
01304                 v[2] = v[FUNHEAD+1];
01305                 v[3] = v[FUNHEAD+3];
01306                 m = v+4;
01307             }
01308             else if ( v[1] == FUNHEAD+ARGHEAD+11
01309                     && v[FUNHEAD] == ARGHEAD+9
01310                     && v[FUNHEAD+ARGHEAD] == 9
01311                     && v[FUNHEAD+ARGHEAD+1] == DOTPRODUCT
01312                     && v[FUNHEAD+ARGHEAD+8] == 3
01313                     && v[FUNHEAD+ARGHEAD+7] == 1
01314                     && v[FUNHEAD+ARGHEAD+6] == 1
01315                     && v[FUNHEAD+ARGHEAD+5] == 1
01316                     && v[FUNHEAD+ARGHEAD+9] == -SNUMBER
01317                     && v[FUNHEAD+ARGHEAD+10] < MAXPOWER
01318                     && v[FUNHEAD+ARGHEAD+10] > -MAXPOWER ) {
01319                 v[0] = DOTPRODUCT;
01320                 v[1] = 5;
01321                 v[2] = v[FUNHEAD+ARGHEAD+3];
01322                 v[3] = v[FUNHEAD+ARGHEAD+4];
01323                 v[4] = v[FUNHEAD+ARGHEAD+10];
01324                 m = v+5;
01325             }
01326         }
01327     }
01328     else { while ( t < u ) *m++ = *t++; }
01329  /*
01330         #] FUNCTIONS :
01331  */
01332     }
01333     t = uu;
01334     } while ( t < r );
01335     t = from; /* Copy coefficient */
01336     t += *t;
01337     if ( r < t ) do { *m++ = *r++; } while ( r < t );
01338     if ( ( sgn & 1 ) != 0 ) m[-1] = -m[-1];
01339     *to = WORDDIF(m,to);
01340     if ( dirty ) AN.WildDirt = dirty;
01341     return(*to);
01342 }
01343
01344  /*
01345     #] WildFill :
01346     #[ ResolveSet :          WORD ResolveSet(from,to,subs)
01347
01348     The set syntax is:
01349     SET,length,subterm,where,whichmember[,where,whichmember]
01350
01351     setlength is 2*n+1 with n the number of set substitutions.
01352     length = setlength + subtermlength + 2
01353
01354     At 'where' is the number of the set and 'whichmember' is the
01355     number of the element. This is still a symbol/dollar and we
01356     have to find the substitution in the wildcards.
01357     The output is the subterm in which the setelements have been
01358     substituted. This is ready for further wildcard substitutions.
01359  */
01360
01361 int ResolveSet(PHEAD WORD *from, WORD *to, WORD *subs)
01362 {
01363     GETBIDENTITY
01364     WORD *m, *s, *w, j, i, ii, i3, flag, num;
01365     DOLLARS d = 0;
01366     #ifdef WITHPTHREADS
01367     int nummodopt, dtype = -1;
01368     #endif
01369     m = to; /* pointer in output */
01370     s = from + 2;
01371     w = s + s[1];
01372     while ( s < w ) *m++ = *s++;
01373     j = (from[1] - WORDDIF(w,from) ) » 1;
01374     m = subs + subs[1];
01375     subs += SUBEXPSIZE;
01376     s = subs;
01377     i = 0;
01378     while ( s < m ) { i++; s += s[1]; }
01379     m = to;
01380     if ( *m >= FUNCTION && functions[*m-FUNCTION].spec

```

```

01381     >= TENSORFUNCTION ) flag = 0;
01382     else flag = 1;
01383     while ( --j >= 0 ) {
01384         if ( w[1] >= 0 ) {
01385             s = subs;
01386             for ( ii = 0; ii < i; ii++ ) {
01387                 if ( *s == SYMTONUM && s[2] == w[1] ) { num = s[3]; goto GotOne; }
01388                 s += s[1];
01389             }
01390             MLOCK(ErrorMessageLock);
01391             MesPrint(" Unresolved setelement during substitution");
01392             MUNLOCK(ErrorMessageLock);
01393             return(-1);
01394         }
01395         else { /* Dollar ! */
01396             d = Dollars - w[1];
01397 #ifndef WITHPTHREADS
01398             if ( AS.MultiThreaded ) {
01399                 for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
01400                     if ( -w[1] == ModOptdollars[nummodopt].number ) break;
01401                 }
01402                 if ( nummodopt < NumModOptdollars ) {
01403                     dtype = ModOptdollars[nummodopt].type;
01404                     if ( dtype == MODLOCAL ) {
01405                         d = ModOptdollars[nummodopt].dstruct+AT.identity;
01406                     }
01407                     else {
01408                         LOCK(d->pthreadslackread);
01409                     }
01410                 }
01411             }
01412 #endif
01413             if ( d->type == DOLNUMBER || d->type == DOLTERMS ) {
01414                 if ( d->where[0] == 4 && d->where[3] == 3 && d->where[2] == 1
01415                     && d->where[1] > 0 && d->where[4] == 0 ) {
01416                     num = d->where[1]; goto GotOne;
01417                 }
01418             }
01419             else if ( d->type == DOLINDEX ) {
01420                 if ( d->index > 0 && d->index < AM.OffsetIndex ) {
01421                     num = d->index; goto GotOne;
01422                 }
01423             }
01424             else if ( d->type == DOLARGUMENT ) {
01425                 if ( d->where[0] == -SNUMBER && d->where[1] > 0 ) {
01426                     num = d->where[1]; goto GotOne;
01427                 }
01428             }
01429             else if ( d->type == DOLWILDARGS ) {
01430                 if ( d->where[0] == 1 &&
01431                     d->where[1] > 0 && d->where[1] < AM.OffsetIndex ) {
01432                     num = d->where[1]; goto GotOne;
01433                 }
01434                 if ( d->where[0] == 0 && d->where[1] < 0 && d->where[3] == 0 ) {
01435                     if ( ( d->where[1] == -SNUMBER && d->where[2] > 0 )
01436                         || ( d->where[1] == -INDEX && d->where[2] > 0
01437                             && d->where[2] < AM.OffsetIndex ) ) {
01438                         num = d->where[2]; goto GotOne;
01439                     }
01440                 }
01441             }
01442 #ifndef WITHPTHREADS
01443             if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslackread); }
01444 #endif
01445             MLOCK(ErrorMessageLock);
01446             MesPrint("Unusable type of variable %s in set substitution",
01447                 AC.dollarnames->namebuffer+d->name);
01448             MUNLOCK(ErrorMessageLock);
01449             return(-1);
01450         }
01451     GotOne:;
01452 #ifndef WITHPTHREADS
01453     if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslackread); }
01454 #endif
01455     ii = m[*w];
01456     if ( ii >= 2*MAXPOWER ) i3 = ii - 2*MAXPOWER;
01457     else if ( ii <= -2*MAXPOWER ) i3 = -ii - 2*MAXPOWER;
01458     else i3 = ( ii >= 0 ) ? ii: -ii - 1;
01459
01460     if ( num > ( Sets[i3].last - Sets[i3].first ) || num <= 0 ) {
01461         MLOCK(ErrorMessageLock);
01462         MesPrint("Array bound check during set substitution");
01463         MesPrint("    value is %d",num);
01464         MUNLOCK(ErrorMessageLock);
01465         return(-1);
01466     }
01467     m[*w] = (SetElements+Sets[i3].first)[num-1];

```

```

01468         if ( Sets[i3].type == CSYMBOL && m[*w] > MAXPOWER ) {
01469             if ( ii >= 2*MAXPOWER ) m[*w] -= 2*MAXPOWER;
01470             else if ( ii <= -2*MAXPOWER ) m[*w] = -(m[*w] - 2*MAXPOWER);
01471             else {
01472                 m[*w] -= MAXPOWER;
01473                 if ( m[*w] < MAXPOWER ) m[*w] -= 2*MAXPOWER;
01474                 if ( flag ) MakeDirty(m,m+*w,1);
01475             }
01476         }
01477         else if ( Sets[i3].type == CSYMBOL ) {
01478             if ( ii >= 2*MAXPOWER ) m[*w] += 2*MAXPOWER;
01479             else if ( ii <= -2*MAXPOWER ) m[*w] = -m[*w] - 2*MAXPOWER;
01480             else if ( ii < 0 ) m[*w] = - m[*w];
01481         }
01482         else if ( ii < 0 ) m[*w] = - m[*w];
01483         w += 2;
01484     }
01485     m = to;
01486     if ( *m >= FUNCTION && functions[*m-FUNCTION].spec
01487     >= TENSORFUNCTION ) {
01488         w = from + 2 + from[3];
01489         if ( *w == 0 ) { /* We had function -> tensor */
01490             m = from + 2 + FUNHEAD; s = to + FUNHEAD;
01491             while ( m < w ) {
01492                 if ( *m == -INDEX || *m == -VECTOR ) {}
01493                 else if ( *m == -ARGWILD ) { *s++ = FUNNYWILD; }
01494                 else {
01495                     MLOCK(ErrorMessageLock);
01496                     MesPrint("Illegal argument in tensor after set substitution");
01497                     MUNLOCK(ErrorMessageLock);
01498                     SETERROR(-1)
01499                 }
01500                 *s++ = m[1];
01501                 m += 2;
01502             }
01503             to[1] = WORDDIFF(s,to);
01504         }
01505     }
01506     return(0);
01507 }
01508
01509 /*
01510     #] ResolveSet :
01511     #[ ClearWild :          void ClearWild()
01512
01513     Clears the current wildcard settings and makes them ready for
01514     CheckWild and AddWild.
01515
01516 */
01517 void ClearWild(PHEAD0)
01518 {
01519     GETBIDENTITY
01520     WORD n, nn, *w;
01521     n = (AN.WildValue[-SUBEXPSIZE+1]-SUBEXPSIZE)/4; /* Number of wildcards */
01522     AN.NumWild = nn = n;
01523     if ( n > 0 ) {
01524         w = AT.WildMask;
01525         do { *w++ = 0; } while ( --n > 0 );
01526         w = AN.WildValue;
01527         do {
01528             if ( *w == SYMTONUM ) *w = SYMTOSYM;
01529             w += w[1];
01530         } while ( --nn > 0 );
01531     }
01532 }
01533
01534
01535 /*
01536     #] ClearWild :
01537     #[ AddWild :          WORD AddWild(oldnumber,type,newnumber)
01538
01539     Adds a wildcard assignment.
01540     Extra parameter in AN.argaddress;
01541
01542 */
01543 int AddWild(PHEAD WORD oldnumber, WORD type, WORD newnumber)
01544 {
01545     GETBIDENTITY
01546     WORD *w, *m, n, k, i = -1;
01547     CBUF *C = cbuf+AT.ebufnum;
01548     WORD eattensor = type & EATTENSOR;
01549     type = type & ~EATTENSOR;
01550     DEBUG(WORD *mm;)
01551     AN.WildReserve = 0;
01552     m = AT.WildMask;
01553     w = AN.WildValue;

```

```

01555     n = AN.NumWild;
01556     if ( n <= 0 ) { return(-1); }
01557     if ( type <= SYMTOSUB ) {
01558         do {
01559             if ( w[2] == oldnumber && *w <= SYMTOSUB ) {
01560                 if ( n > 1 && w[4] == SETTONUM ) i = w[7];
01561                 *w = type;
01562                 if ( *m != 2 ) *m = 1;
01563                 if ( type != SYMTOSUB ) {
01564                     if ( type == SYMTONUM ) AN.MaskPointer = m;
01565                     w[3] = newnumber;
01566                     goto FlipOn;
01567                 }
01568                 m = AddRHS(AT.ebufnum,1);
01569                 w[3] = C->numrhs;
01570                 w = AN.argaddress;
01571             DEBUG(mm = m;);
01572             n = *w - ARGHEAD;
01573             w += ARGHEAD;
01574             while ( (m + n + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,4);
01575             while ( --n >= 0 ) *m++ = *w++;
01576             *m++ = 0;
01577             C->rhs[C->numrhs+1] = m;
01578             DEBUG(MesPrint("Thread %w(d): m=(%d,%d,%d,%d) (%d)",mm[0],mm[1],mm[2],mm[3],C->numrhs);)
01579             C->Pointer = m;
01580             goto FlipOn;
01581         }
01582         m++; w += w[1];
01583     } while ( --n > 0 );
01584 }
01585 else if ( type == ARGTOARG ) {
01586     do {
01587         if ( w[2] == oldnumber && *w == ARGTOARG ) {
01588             *m = 1;
01589             m = AddRHS(AT.ebufnum,1);
01590             w[3] = C->numrhs;
01591             w = AN.argaddress;
01592             DEBUG(mm=m;);
01593             if ( eattensor ) {
01594                 n = newnumber;
01595                 *m++ = n;
01596                 w = AN.argaddress;
01597             }
01598             else {
01599                 while ( --newnumber >= 0 ) { NEXTARG(w) }
01600                 n = WORDDIF(w,AN.argaddress);
01601                 w = AN.argaddress;
01602                 *m++ = 0;
01603             }
01604             while ( (m + n + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,5);
01605             DEBUG(if ( mm != m-1 ) MesPrint("Thread %w(e): Alarm!"); mm = m-1;);
01606             while ( --n >= 0 ) *m++ = *w++;
01607             *m++ = 0;
01608             C->rhs[C->numrhs+1] = m;
01609             C->Pointer = m;
01610             DEBUG(MesPrint("Thread %w(e): w=(%d,%d,%d,%d) (%d)",mm[0],mm[1],mm[2],mm[3],C->numrhs);)
01611             return(0);
01612         }
01613         m++; w += w[1];
01614     } while ( --n > 0 );
01615 }
01616 else if ( type == ARLTOARL ) {
01617     do {
01618         if ( w[2] == oldnumber && *w == ARGTOARG ) {
01619             WORD **a;
01620             *m = 1;
01621             m = AddRHS(AT.ebufnum,1);
01622             w[3] = C->numrhs;
01623             DEBUG(mm=m;);
01624             a = (WORD **)(AN.argaddress); n = 0; k = newnumber;
01625             while ( --newnumber >= 0 ) {
01626                 w = *a++;
01627                 if ( *w > 0 ) n += *w;
01628                 else if ( *w <= -FUNCTION ) n++;
01629                 else n += 2;
01630             }
01631             *m++ = 0;
01632             while ( (m + n + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,6);
01633             DEBUG(if ( mm != m-1 ) MesPrint("Thread %w(f): Alarm!"); mm = m-1;);
01634             a = (WORD **)(AN.argaddress);
01635             while ( --k >= 0 ) {
01636                 w = *a++;
01637                 if ( *w > 0 ) { n = *w; NCOPY(m,w,n); }
01638                 else if ( *w <= -FUNCTION ) *m++ = *w++;
01639                 else { *m++ = *w++; *m++ = *w++; }
01640             }
01641             *m++ = 0;

```

```

01642             C->rhs[C->numrhs+1] = m;
01643 DEBUG(MesPrint("Thread %w(f): w=(%d,%d,%d,%d) (%d) ",mm[0],mm[1],mm[2],mm[3],C->numrhs);)
01644             C->Pointer = m;
01645             return(0);
01646         }
01647         m++; w += w[1];
01648     } while ( --n > 0 );
01649 }
01650 else if ( type == VECTOSUB || type == INDTOSUB ) {
01651     WORD *ss, *sstop, *tt, *ttstop, j, *v1, *v2 = 0;
01652     do {
01653         if ( w[2] == oldnumber && ( *w == type ||
01654             ( type == VECTOSUB && ( *w == VECTOVEC || *w == VECTOMIN ) )
01655             || ( type == INDTOIND && *w == INDTOIND ) ) ) {
01656             if ( n > 1 && w[4] == SETTONUM ) i = w[7];
01657             *w = type;
01658             *m = 1;
01659             m = AddRHS(AT.ebufnum,1);
01660             w[3] = C->numrhs;
01661             w = AN.argaddress;
01662             n = *w - ARGHEAD;
01663             w += ARGHEAD;
01664             while ( (m + n + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,7);
01665             while ( --n >= 0 ) *m++ = *w++;
01666             *m++ = 0;
01667             C->rhs[C->numrhs+1] = m;
01668             C->Pointer = m;
01669             m = AddRHS(AT.ebufnum,1);
01670             w = AN.argaddress;
01671             n = *w - ARGHEAD;
01672             w += ARGHEAD;
01673             while ( (m + n + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,8);
01674             sstop = w + n;
01675             while ( w < sstop ) { /* Run over terms */
01676                 tt = w + *w; ttstop = tt - ABS(tt[-1]);
01677                 ss = m; m++; w++;
01678                 while ( w < ttstop ) { /* Subterms */
01679                     if ( *w != INDEX ) {
01680                         j = w[1];
01681                         NCOPY(m,w,j);
01682                     }
01683                     else {
01684                         v1 = m;
01685                         *m++ = *w++;
01686                         *m++ = j = *w++;
01687                         j -= 2;
01688                         while ( --j >= 0 ) {
01689                             if ( *w >= MINSPEC ) *m++ = *w++;
01690                             else v2 = w++;
01691                         }
01692                         j = WORDDIFF(m,v1);
01693                         if ( j != v1[1] ) {
01694                             if ( j <= 2 ) m -= 2;
01695                             else v1[1] = j;
01696                             *m++ = VECTOR;
01697                             *m++ = 4;
01698                             *m++ = *v2;
01699                             *m++ = FUNNYVEC;
01700                         }
01701                     }
01702                 }
01703                 while ( w < tt ) *m++ = *w++;
01704                 *ss = WORDDIFF(m,ss);
01705             }
01706             *m++ = 0;
01707             C->rhs[C->numrhs+1] = m;
01708             C->Pointer = m;
01709             if ( m > C->Top ) {
01710                 MLOCK(ErrorMessageLock);
01711                 MesPrint("Internal problems with extra compiler buffer");
01712                 MUNLOCK(ErrorMessageLock);
01713                 Terminate(-1);
01714             }
01715             goto FlipOn;
01716         }
01717         m++; w += w[1];
01718     } while ( --n > 0 );
01719 }
01720 else {
01721     do {
01722         if ( w[2] == oldnumber && ( *w == type || ( type == VECTOVEC
01723             && ( *w == VECTOMIN || *w == VECTOSUB ) ) || ( type == VECTOMIN
01724             && ( *w == VECTOVEC || *w == VECTOSUB ) )
01725             || ( type == INDTOIND && *w == INDTOIND ) ) ) {
01726             if ( n > 1 && w[4] == SETTONUM ) i = w[7];
01727             *w = type;
01728             w[3] = newnumber;

```

```

01729             *m = 1;
01730             goto FlipOn;
01731         }
01732         m++; w += w[1];
01733     } while ( --n > 0 );
01734 }
01735 MLOCK(ErrorMessageLock);
01736 MesPrint("Bug in AddWild.");
01737 MUNLOCK(ErrorMessageLock);
01738 return(-1);
01739 FlipOn:
01740     if ( i >= 0 ) {
01741         m = AT.WildMask;
01742         w = AN.WildValue;
01743         n = AN.NumWild;
01744         while ( --n >= 0 ) {
01745             if ( w[2] == i && *w == SYMTONUM ) {
01746                 *m = 2;
01747                 return(0);
01748             }
01749             m++; w += w[1];
01750         }
01751         MLOCK(ErrorMessageLock);
01752         MesPrint(" Bug in AddWild with passing set[i]");
01753         MUNLOCK(ErrorMessageLock);
01754     /*
01755         For the moment we want to crash here. That is easier with debugging.
01756     */
01757     #ifdef WITHPTHREADS
01758         { WORD *s = 0;
01759             *s++ = 1;
01760         }
01761     #endif
01762     Terminate(-1);
01763 }
01764 return(0);
01765 }
01766
01767 /*
01768     #] AddWild :
01769     #[ CheckWild :          WORD CheckWild(oldnumber,type,newnumber,newval)
01770
01771     Tests whether a wildcard assignment is allowed.
01772     A return value of zero means that it is allowed (nihil obstat).
01773     If the variable has been assigned already its existing
01774     assignment is returned in AN.oldvalue and AN.oldtype, which are
01775     global variables.
01776
01777     Note the special problem with name?set[i]. Here we have to pass
01778     an extra assignment. This cannot be done via globals as we
01779     call CheckWild sometimes twice before calling AddWild.
01780     Trick: Check the assignment of the number and if OK put it
01781     in place, but don't alter the used flag (if needed).
01782     Then AddWild can alter the used flag but the value is there.
01783     As long as this trick is 'hanging' we turn on the flag:
01784     'AN.WildReserve' which is either turned off by AddWild or by
01785     a failing call to CheckWild.
01786
01787     With ARGTOARG the tensors give the number of arguments
01788     or-ed with EATTENSOR which is at least 8192.
01789 */
01790
01791 int CheckWild(PHEAD WORD oldnumber, WORD type, WORD newnumber, WORD *newval)
01792 {
01793     GETBIDENTITY
01794     WORD *w, *m, *s, n, old2, inset;
01795     WORD n2, oldval, dirty, i, j;
01796     int notflag = 0, retblock = 0;
01797     CBUF *c = cbuf+AT.ebufnum;
01798     WORD eattensor = type & EATTENSOR;
01799     type = type & ~EATTENSOR;
01800     m = AT.WildMask;
01801     w = AN.WildValue;
01802     n = AN.NumWild;
01803     if ( n <= 0 ) { AN.oldtype = -1; AN.WildReserve = 0; return(-1); }
01804     switch ( type ) {
01805         case SYMTONUM :
01806             *newval = newnumber;
01807             do {
01808                 if ( w[2] == oldnumber && *w <= SYMTOSUB ) {
01809                     old2 = *w;
01810                     if ( !*m ) goto TestSet;
01811                     AN.MaskPointer = m;
01812                     if ( *w == SYMTONUM && w[3] == newnumber ) {
01813                         return(0);
01814                     }
01815                     AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;

```

```

01816         }
01817         m++; w += w[1];
01818     } while ( --n > 0 );
01819     break;
01820 case SYMTOSYM :
01821     *newval = newnumber;
01822     do {
01823         if ( w[2] == oldnumber && *w <= SYMTOSUB ) {
01824             old2 = *w;
01825             if ( *w == SYMTOSYM ) {
01826                 if ( !*m ) goto TestSet;
01827                 if ( newnumber >= 0 && (w+4) < AN.WildStop
01828                     && ( w[4] == FROMSET || w[4] == SETTONUM )
01829                     && w[7] >= 0 ) goto TestSet;
01830                 if ( w[3] == newnumber ) return(0);
01831             }
01832             else {
01833                 if ( !*m ) goto TestSet;
01834             }
01835             goto NoM;
01836         }
01837         m++; w += w[1];
01838     } while ( --n > 0 );
01839     break;
01840 case SYMTOSUB :
01841     /*
01842     Now newval contains the pointer to the argument.
01843     */
01844     {
01845     /*
01846     Search for vector or index nature. If so: reject.
01847     */
01848     WORD *ss, *sstop, *tt, *ttstop;
01849     ss = newval;
01850     sstop = ss + *ss;
01851     ss += ARGHEAD;
01852     while ( ss < sstop ) {
01853         tt = ss + *ss;
01854         ttstop = tt - ABS(tt[-1]);
01855         ss++;
01856         while ( ss < ttstop ) {
01857             if ( *ss == INDEX ) goto NoMatch;
01858             ss += ss[1];
01859         }
01860         ss = tt;
01861     }
01862     }
01863     do {
01864         if ( w[2] == oldnumber && *w <= SYMTOSUB ) {
01865             old2 = *w;
01866             if ( *w == SYMTONUM || *w == SYMTOSYM ) {
01867                 if ( !*m ) {
01868                     s = w + w[1];
01869                     if ( s >= AN.WildStop || *s != SETTONUM )
01870                         goto TestSet;
01871                 }
01872             }
01873             else if ( *w == SYMTOSUB ) {
01874                 if ( !*m ) {
01875                     s = w + w[1];
01876                     if ( s >= AN.WildStop || *s != SETTONUM )
01877                         goto TestSet;
01878                 }
01879                 n = *newval - 2;
01880                 newval += 2;
01881                 m = C->rhs[w[3]];
01882                 if ( (C->rhs[w[3]+1] - m - 1) == n ) {
01883                     while ( n > 0 ) {
01884                         if ( *m != *newval ) {
01885                             m++; newval++; break;
01886                         }
01887                         m++; newval++;
01888                         n--;
01889                     }
01890                     if ( n <= 0 ) return(0);
01891                 }
01892             }
01893             AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;
01894         }
01895         m++; w += w[1];
01896     } while ( --n > 0 );
01897     break;
01898 case ARGTOARG :
01899     do {
01900         if ( w[2] == oldnumber && *w == ARGTOARG ) {
01901             if ( !*m ) return(0); /* nihil obstat */
01902             m = C->rhs[w[3]];

```

```

01903         if ( eattensor ) {
01904             n = newnumber;
01905             if ( *m != 0 ) {
01906                 if ( n == *m ) {
01907                     m++;
01908                     while ( --n >= 0 ) {
01909                         if ( *m != *newval ) {
01910                             m++; newval++; break;
01911                         }
01912                         m++; newval++;
01913                     }
01914                     if ( n < 0 ) return(0);
01915                 }
01916             }
01917             else {
01918                 m++;
01919                 while ( --n >= 0 ) {
01920                     if ( *newval != m[1] || ( *m != -INDEX
01921                         && *m != -VECTOR && *m != -SNUMBER ) ) break;
01922                     m += 2;
01923                     newval++;
01924                 }
01925                 if ( n < 0 && *m == 0 ) return(0);
01926             }
01927         }
01928         else {
01929             i = newnumber;
01930             if ( *m != 0 ) { /* Tensor field */
01931                 if ( *m == i ) {
01932                     m++;
01933                     while ( --i >= 0 ) {
01934                         if ( *m != newval[1]
01935                             || ( *newval != -VECTOR
01936                                 && *newval != -INDEX
01937                                 && *newval != -SNUMBER ) ) break;
01938                         newval += 2;
01939                         m++;
01940                     }
01941                     if ( i < 0 ) return(0);
01942                 }
01943             }
01944             else {
01945                 m++;
01946                 s = newval;
01947                 while ( --i >= 0 ) { NEXTARG(s) }
01948                 n = WORDDIF(s,newval);
01949                 while ( --n >= 0 ) {
01950                     if ( *m != *newval ) {
01951                         m++; newval++; break;
01952                     }
01953                     m++; newval++;
01954                 }
01955                 if ( n < 0 && *m == 0 ) return(0);
01956             }
01957         }
01958         AN.oldtype = *w; AN.oldvalue = w[3]; goto NoMatch;
01959     }
01960     m++; w += w[1];
01961 } while ( --n > 0 );
01962 break;
01963 case ARLTOARL :
01964     do {
01965         if ( w[2] == oldnumber && *w == ARGTOARG ) {
01966             WORD **a;
01967             if ( !*m ) return(0); /* nihil obstat */
01968             m = C->rhs[w[3]];
01969             i = newnumber;
01970             a = (WORD **)newval;
01971             if ( *m != 0 ) { /* Tensor field */
01972                 if ( *m == i ) {
01973                     m++;
01974                     while ( --i >= 0 ) {
01975                         s = *a++;
01976                         if ( *m != s[1]
01977                             || ( *s != -VECTOR
01978                                 && *s != -INDEX
01979                                 && *s != -SNUMBER ) ) break;
01980                         m++;
01981                     }
01982                     if ( i < 0 ) return(0);
01983                 }
01984             }
01985             else {
01986                 m++;
01987                 while ( --i >= 0 ) {
01988                     s = *a++;
01989                     if ( *s > 0 ) {

```



```

01990             n = *s;
01991             while ( --n >= 0 ) {
01992                 if ( *s != *m ) {
01993                     s++; m++; break;
01994                 }
01995                 s++; m++;
01996             }
01997             if ( n >= 0 ) break;
01998         }
01999         else if ( *s <= -FUNCTION ) {
02000             if ( *s != *m ) {
02001                 s++; m++; break;
02002             }
02003             s++; m++;
02004         }
02005         else {
02006             if ( *s != *m ) {
02007                 s++; m++; break;
02008             }
02009             s++; m++;
02010             if ( *s != *m ) {
02011                 s++; m++; break;
02012             }
02013             s++; m++;
02014         }
02015     }
02016     if ( i < 0 && *m == 0 ) return(0);
02017 }
02018 AN.oldtype = *w; AN.oldvalue = w[3]; goto NoMatch;
02019 }
02020 m++; w += w[1];
02021 } while ( --n > 0 );
02022 break;
02023 case VECTOSUB :
02024 case INDTOSUB :
02025 /*
02026     Now newval contains the pointer to the argument(s).
02027 */
02028 {
02029 /*
02030     Search for vector or index nature. If not so: reject.
02031 */
02032 WORD *ss, *sstop, *tt, *ttstop, count, jt;
02033 ss = newval;
02034 sstop = ss + *ss;
02035 ss += ARGHEAD;
02036 while ( ss < sstop ) {
02037     tt = ss + *ss;
02038     ttstop = tt - ABS(tt[-1]);
02039     ss++;
02040     count = 0;
02041     while ( ss < ttstop ) {
02042         if ( *ss == INDEX ) {
02043             jt = ss[1] - 2; ss += 2;
02044             while ( --jt >= 0 ) {
02045                 if ( *ss < MINSPEC ) count++;
02046                 ss++;
02047             }
02048         }
02049         else ss += ss[1];
02050     }
02051     if ( count != 1 ) goto NoMatch;
02052     ss = tt;
02053 }
02054 }
02055 do {
02056     if ( w[2] == oldnumber ) {
02057         old2 = *w;
02058         if ( ( type == VECTOSUB && ( *w == VECTOVEC || *w == VECTOMIN ) )
02059             || ( type == INDTOSUB && *w == INDTOIND ) ) {
02060             if ( !*m ) goto TestSet;
02061             AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;
02062         }
02063         else if ( *w == type ) {
02064             if ( !*m ) goto TestSet;
02065             if ( type != INDTOIND && type != INDTOSUB ) { /* Prevent double index */
02066                 n = *newval - 2;
02067                 newval += 2;
02068                 m = C->rhs[w[3]];
02069                 if ( (C->rhs[w[3]+1] - m - 1) == n ) {
02070                     while ( n > 0 ) {
02071                         if ( *m != *newval ) {
02072                             m++; newval++; break;
02073                         }
02074                         m++; newval++;
02075                         n--;
02076                     }

```

```

02077             if ( n <= 0 ) return(0);
02078         }
02079     }
02080     AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;
02081 }
02082 }
02083 m++; w += w[1];
02084 } while ( --n > 0 );
02085 break;
02086 default :
02087     *newval = newnumber;
02088     do {
02089         if ( w[2] == oldnumber ) {
02090             if ( *w == type ) {
02091                 old2 = *w;
02092                 if ( !*m ) goto TestSet;
02093                 if ( newnumber >= 0 && (w+4) < AN.WildStop &&
02094                     ( w[4] == FROMSET || w[4] == SETTONUM )
02095                     && w[7] >= 0 ) goto TestSet;
02096                 if ( newnumber < 0 && *w == VECTOVEC
02097                     && (w+4) < AN.WildStop && ( w[4] == FROMSET
02098                     || w[4] == SETTONUM ) && w[7] >= 0 ) goto TestSet;
02099             /*
02100              The next statement kills multiple indices -> vector
02101             */
02102             if ( *w == INDTOIND && w[3] < 0 ) goto NoMatch;
02103             if ( w[3] == newnumber ) {
02104                 if ( *w != FUNTOFUN || newnumber < FUNCTION
02105                     || functions[newnumber-FUNCTION].spec ==
02106                     functions[oldnumber-FUNCTION].spec )
02107                     return(0);
02108             }
02109             AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;
02110         }
02111         else if ( ( type == VECTOVEC &&
02112             ( *w == VECTOSUB || *w == VECTOMIN ) )
02113             || ( type == INDTOIND && *w == INDTOSUB ) ) {
02114             if ( *m ) goto NoMatch;
02115             old2 = *w;
02116             goto TestSet;
02117         }
02118         else if ( type == VECTOMIN &&
02119             ( *w == VECTOSUB || *w == VECTOVEC ) ) {
02120             if ( *m ) goto NoMatch;
02121             old2 = *w;
02122             goto TestSet;
02123         }
02124     }
02125     m++; w += w[1];
02126     if ( n > 1 && ( *w == FROMSET
02127         || *w == SETTONUM ) ) { n--; m++; w += w[1]; }
02128 } while ( --n > 0 );
02129 break;
02130 }
02131 AN.oldtype = -1;
02132 AN.oldvalue = -1;
02133 AN.WildReserve = 0;
02134 MLOCK(ErrorMessageLock);
02135 MesPrint("Inconsistency in Wildcard prototype.");
02136 MUNLOCK(ErrorMessageLock);
02137 return(-1);
02138 NoMatch:
02139     AN.WildReserve = 0;
02140     return(1+retblock);
02141 /*
02142     Here we test the compatibility with a set specification.
02143 */
02144 TestSet:
02145     dirty = *m;
02146     oldval = w[3];
02147     w += w[1];
02148     if ( w < AN.WildStop && ( *w == FROMSET || *w == SETTONUM ) ) {
02149         WORD k;
02150         s = w;
02151         j = w[2]; n2 = w[3];
02152     /*
02153         if SETTONUM:  x?j[n2]
02154         if FROMSET:   x?j?n2    or x?j and n2 = -WOLDOFFSET.
02155     */
02156     if ( j > WILDOFFSET ) {
02157         j -= 2*WILDOFFSET;
02158         notflag = 1;
02159     /*
02160         ??????
02161     */
02162     AN.oldtype = -1;
02163     AN.oldvalue = -1;

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```

02164     }
02165     if ( j < AM.NumFixedSets ) { /* special set */
02166         retblock = 1;
02167         switch ( j ) {
02168             case POS_:
02169                 if ( type != SYMTONUM ||
02170                     newnumber <= 0 ) goto NoMnot;
02171                 break;
02172             case POS0_:
02173                 if ( type != SYMTONUM ||
02174                     newnumber < 0 ) goto NoMnot;
02175                 break;
02176             case NEG_:
02177                 if ( type != SYMTONUM ||
02178                     newnumber >= 0 ) goto NoMnot;
02179                 break;
02180             case NEG0_:
02181                 if ( type != SYMTONUM ||
02182                     newnumber > 0 ) goto NoMnot;
02183                 break;
02184             case EVEN_:
02185                 if ( type != SYMTONUM ||
02186                     ( newnumber & 1 ) != 0 ) goto NoMnot;
02187                 break;
02188             case ODD_:
02189                 if ( type != SYMTONUM ||
02190                     ( newnumber & 1 ) == 0 ) goto NoMnot;
02191                 break;
02192             case Z_:
02193                 if ( type != SYMTONUM ) goto NoMnot;
02194                 break;
02195             case SYMBOL_:
02196                 if ( type != SYMTOSYM ) goto NoMnot;
02197                 break;
02198             case FIXED_:
02199                 if ( type != INDTOIND ||
02200                     newnumber >= AM.OffsetIndex ||
02201                     newnumber < 0 ) goto NoMnot;
02202                 break;
02203             case INDEX_:
02204                 if ( type != INDTOIND ||
02205                     newnumber < 0 ) goto NoMnot;
02206                 break;
02207             case Q_:
02208                 if ( type == SYMTONUM ) break;
02209                 if ( type == SYMTOSUB ) {
02210                     WORD *ss, *sstop;
02211                     ss = newval;
02212                     sstop = ss + *ss;
02213                     ss += ARGHEAD;
02214                     if ( ss >= sstop ) break;
02215                     if ( ss + *ss < sstop ) goto NoMnot;
02216                     if ( ABS(sstop[-1]) == ss[0]-1 ) break;
02217                 }
02218                 goto NoMnot;
02219             case DUMMYINDEX_:
02220                 if ( type != INDTOIND ||
02221                     newnumber < AM.IndDum || newnumber >= AM.IndDum+MAXDUMMIES ) goto NoMnot;
02222                 break;
02223             case VECTOR_:
02224                 if ( type != VECTOVEC ) goto NoMnot;
02225                 break;
02226             default:
02227                 goto NoMnot;
02228         }
02229     Mnot:
02230         if ( notflag ) goto NoM;
02231         return(0);
02232     NoMnot:
02233         if ( !notflag ) goto NoM;
02234         return(0);
02235     }
02236     else if ( Sets[j].type == CRANGE ) {
02237         if ( ( type == SYMTONUM )
02238             || ( type == INDTOIND && ( newnumber > 0
02239                 && newnumber <= AM.OffsetIndex ) ) ) {
02240             if ( Sets[j].first < MAXPOWER ) {
02241                 if ( newnumber >= Sets[j].first ) goto NoMnot;
02242             }
02243             else if ( Sets[j].first < 3*MAXPOWER ) {
02244                 if ( newnumber+2*MAXPOWER > Sets[j].first ) goto NoMnot;
02245             }
02246             if ( Sets[j].last > -MAXPOWER ) {
02247                 if ( newnumber <= Sets[j].last ) goto NoMnot;
02248             }
02249             else if ( Sets[j].last > -3*MAXPOWER ) {
02250                 if ( newnumber-2*MAXPOWER < Sets[j].last ) goto NoMnot;

```

```

02251         }
02252         goto Mnot;
02253     }
02254     goto NoMnot;
02255 }
02256 /*
02257 Now we have to determine which set element
02258 */
02259 w = SetElements + Sets[j].first;
02260 m = SetElements + Sets[j].last;
02261 if ( w == m ) {
02262 /*
02263 The set is empty! This is not a match unless we have !{}, in
02264 which case it is.
02265 */
02266 if ( notflag ) return 0;
02267 AN.oldtype = old2;
02268 AN.oldvalue = oldval;
02269 goto NoMatch;
02270 }
02271
02272 if ( ( Sets[j].flags & ORDEREDSET ) == ORDEREDSET ) {
02273 /*
02274 We search first and ask questions later
02275 */
02276 i = BinarySearch(w, Sets[j].last - Sets[j].first, newnumber);
02277 if ( i < 0 ) { /* no matter what, it is not in the set. */
02278     goto NoMnot;
02279 }
02280 else {
02281 /*
02282 We can set the proper parameters now to make only the
02283 checks for the given set element.
02284 After that we jump into the appropriate loop.
02285 */
02286 w = m = SetElements + i;
02287 i++;
02288 if ( Sets[j].type == -1 || Sets[j].type == CNUMBER ) {
02289     goto insideloop1;
02290 }
02291 else {
02292     goto insideloop2;
02293 }
02294 }
02295 }
02296 i = 1;
02297 if ( Sets[j].type == -1 || Sets[j].type == CNUMBER ) {
02298     do {
02299         insideloop1:
02300         if ( notflag ) {
02301             switch ( type ) {
02302                 case SYMTOSYM:
02303                     if ( Sets[j].type == CNUMBER ) {}
02304                     else {
02305                         if ( *w == newnumber ) goto NoMatch;
02306                     }
02307                     break;
02308                 case SYMTONUM:
02309                 case INDTOIND:
02310                     if ( *w == newnumber ) goto NoMatch;
02311                     break;
02312                 default:
02313                     break;
02314             }
02315         }
02316         else if ( type != SYMTONUM && type != INDTOIND
02317             && type != SYMTOSYM ) goto NoMatch;
02318         else if ( type == SYMTOSYM && Sets[j].type == CNUMBER ) goto NoMatch;
02319         else if ( *w == newnumber ) {
02320             if ( *s == SETTONUM ) {
02321                 if ( n2 == oldnumber && type
02322                     <= SYMTOSUB ) goto NoMatch;
02323                 m = AT.WildMask;
02324                 w = AN.WildValue;
02325                 n = AN.NumWild;
02326                 while ( --n >= 0 ) {
02327                     if ( w[2] == n2 && *w <= SYMTOSUB ) {
02328                         if ( !*m ) {
02329                             *w = SYMTONUM;
02330                             w[3] = i;
02331                             AN.WildReserve = 1;
02332                             return(0);
02333                         }
02334                         if ( *w != SYMTONUM )
02335                             goto NoMatch;
02336                         if ( w[3] == i ) return(0);
02337                         i = w[3];

```

```

02338             j = (SetElements + Sets[j].first)[i];
02339             if ( j == n2 ) return(0);
02340             goto NoMatch;
02341         }
02342         m++; w += w[1];
02343     }
02344 }
02345 else if ( n2 >= 0 ) {
02346     *newval = *(w - Sets[j].first + Sets[n2].first);
02347     if ( *newval > MAXPOWER ) *newval -= 2*MAXPOWER;
02348     if ( dirty && *newval != oldval ) {
02349         *newval = oldval; goto NoMatch;
02350     }
02351 }
02352 return(0);
02353 }
02354 i++;
02355 } while ( ++w < m );
02356 }
02357 else {
02358     do {
02359         insideloop2:
02360         inset = *w;
02361         if ( notflag ) {
02362             switch ( type ) {
02363             case SYMTONUM:
02364             case SYMTOSYM:
02365                 if ( ( type == SYMTOSYM && *w == newnumber )
02366                     || ( type == SYMTONUM && *w-2*MAXPOWER == newnumber ) ) {
02367                     goto NoMatch;
02368                 }
02369                 /* fall through */
02370             case SYMTOSUB:
02371                 if ( *w < 0 ) {
02372                     WORD *mm = AT.WildMask, *mmm, *part;
02373                     WORD *ww = AN.WildValue;
02374                     WORD nn = AN.NumWild;
02375                     k = -*w;
02376                     while ( --nn >= 0 ) {
02377                         if ( *mm && ww[2] == k && ww[0] == type ) {
02378                             if ( type != SYMTOSUB ) {
02379                                 if ( ww[3] == newnumber ) goto NoMatch;
02380                             }
02381                             else {
02382                                 mmm = C->rhs[ww[3]];
02383                                 nn = *newval-2;
02384                                 part = newval+2;
02385                                 if ( (C->rhs[ww[3]+1]-mmm-1) == nn ) {
02386                                     while ( --nn >= 0 ) {
02387                                         if ( *mmm != *part ) {
02388                                             mmm++; part++; break;
02389                                         }
02390                                         mmm++; part++;
02391                                     }
02392                                     if ( nn < 0 ) goto NoMatch;
02393                                 }
02394                             }
02395                             break;
02396                         }
02397                         mm++; ww += ww[1];
02398                     }
02399                 }
02400                 break;
02401             case VECTOMIN:
02402                 if ( type == VECTOMIN ) {
02403                     if ( inset >= AM.OffsetVector ) { i++; continue; }
02404                     inset += WILDMASK;
02405                 }
02406                 /* fall through */
02407             case VECTOVEC:
02408                 if ( inset == newnumber ) goto NoMatch;
02409                 /* fall through */
02410             case VECTOSUB:
02411                 if ( inset - WILDOFFSET >= AM.OffsetVector ) {
02412                     WORD *mm = AT.WildMask, *mmm, *part;
02413                     WORD *ww = AN.WildValue;
02414                     WORD nn = AN.NumWild;
02415                     k = inset - WILDOFFSET;
02416                     while ( --nn >= 0 ) {
02417                         if ( *mm && ww[2] == k && ww[0] == type ) {
02418                             if ( type == VECTOVEC ) {
02419                                 if ( ww[3] == newnumber ) goto NoMatch;
02420                             }
02421                             else {
02422                                 mmm = C->rhs[ww[3]];
02423                                 nn = *newval-2;
02424                                 part = newval+2;

```

```

02425         if ( (C->rhs[ww[3]+1]-mmm-1) == nn ) {
02426             while ( --nn >= 0 ) {
02427                 if ( *mmm != *part ) {
02428                     mmm++; part++; break;
02429                 }
02430                 mmm++; part++;
02431             }
02432             if ( nn < 0 ) goto NoMatch;
02433         }
02434     }
02435     break;
02436 }
02437 mm++; ww += ww[1];
02438 }
02439 }
02440 break;
02441 case INDTOIND:
02442     if ( *w == newnumber ) goto NoMatch;
02443     /* fall through */
02444 case INTOSUB:
02445     if ( *w - (WORD)WILDMASK >= AM.OffsetIndex ) {
02446         WORD *mm = AT.WildMask, *mmm, *part;
02447         WORD *ww = AN.WildValue;
02448         WORD nn = AN.NumWild;
02449         k = *w - WILDMASK;
02450         while ( --nn >= 0 ) {
02451             if ( *mm && ww[2] == k && ww[0] == type ) {
02452                 if ( type == INDTOIND ) {
02453                     if ( ww[3] == newnumber ) goto NoMatch;
02454                 }
02455                 else {
02456                     mmm = C->rhs[ww[3]];
02457                     nn = *newval-2;
02458                     part = newval+2;
02459                     if ( (C->rhs[ww[3]+1]-mmm-1) == nn ) {
02460                         while ( --nn >= 0 ) {
02461                             if ( *mmm != *part ) {
02462                                 mmm++; part++; break;
02463                             }
02464                             mmm++; part++;
02465                         }
02466                         if ( nn < 0 ) goto NoMatch;
02467                     }
02468                 }
02469                 break;
02470             }
02471             mm++; ww += ww[1];
02472         }
02473     }
02474     break;
02475 case FUNTOFUN:
02476     if ( *w == newnumber ) goto NoMatch;
02477     if ( ( type == FUNTOFUN &&
02478           ( k = *w - WILDMASK ) > FUNCTION ) ) {
02479         WORD *mm = AT.WildMask;
02480         WORD *ww = AN.WildValue;
02481         WORD nn = AN.NumWild;
02482         while ( --nn >= 0 ) {
02483             if ( *mm && ww[2] == k && ww[0] == type ) {
02484                 if ( ww[3] == newnumber ) goto NoMatch;
02485                 break;
02486             }
02487             mm++; ww += ww[1];
02488         }
02489     }
02490 default:
02491     break;
02492 }
02493 }
02494 else {
02495     if ( type == VECTOMIN ) {
02496         if ( inset >= AM.OffsetVector ) { i++; continue; }
02497         inset += WILDMASK;
02498     }
02499     if ( ( inset == newnumber && type != SYMTONUM ) ||
02500         ( type == SYMTONUM && inset-2*MAXPOWER == newnumber ) ) {
02501         if ( *s == SETTONUM ) {
02502             if ( n2 == oldnumber && type
02503                 <= SYMTOSUB ) goto NoMatch;
02504             m = AT.WildMask;
02505             w = AN.WildValue;
02506             n = AN.NumWild;
02507             while ( --n >= 0 ) {
02508                 if ( w[2] == n2 && *w <= SYMTOSUB ) {
02509                     if ( !*m ) {
02510                         *w = SYMTONUM;
02511                         w[3] = i;

```

```

02512             AN.WildReserve = 1;
02513             return(0);
02514         }
02515         if ( *w != SYMTONUM )
02516             goto NoMatch;
02517         if ( w[3] == i ) return(0);
02518         i = w[3];
02519         j = (SetElements + Sets[j].first)[i];
02520         if ( j == n2 ) return(0);
02521         goto NoMatch;
02522     }
02523     m++; w += w[1];
02524 }
02525 }
02526 else if ( n2 >= 0 ) {
02527     *newval = *(w - Sets[j].first + Sets[n2].first);
02528     if ( *newval > MAXPOWER ) *newval -= 2*MAXPOWER;
02529     if ( dirty && *newval != oldval ) {
02530         *newval = oldval; goto NoMatch;
02531     }
02532 }
02533 return(0);
02534 }
02535 }
02536 i++;
02537 } while ( ++w < m );
02538 }
02539 if ( notflag ) return(0);
02540 AN.oldtype = old2; AN.oldvalue = oldval; goto NoMatch;
02541 }
02542 else { return(0); }
02543
02544 NoM:
02545     AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;
02546 }
02547
02548 /*
02549     #] CheckWild :
02550     #] Wildcards :
02551     #[ DenToFunction :
02552
02553     Renames the denominator function into a function with the given number.
02554     For the syntax see     Denominators,function;
02555 */
02556
02557 int DenToFunction(WORD *term, WORD numfun)
02558 {
02559     int action = 0;
02560     WORD *t, *tstop, *tnext, *arg, *argstop, *targ;
02561     t = term+1;
02562     tstop = term + *term; tstop -= ABS(tstop[-1]);
02563     while ( t < tstop ) {
02564         if ( *t == DENOMINATOR ) {
02565             *t = numfun; t[2] |= DIRTYFLAG; action = 1;
02566         }
02567         tnext = t + t[1];
02568         if ( *t >= FUNCTION && functions[*t-FUNCTION].spec <= 0 ) {
02569             arg = t + FUNHEAD;
02570             while ( arg < tnext ) {
02571                 if ( *arg > 0 ) {
02572                     targ = arg + ARGHEAD; argstop = arg + *arg;
02573                     while ( targ < argstop ) {
02574                         if ( DenToFunction(targ,numfun) ) {
02575                             arg[1] |= DIRTYFLAG; t[2] |= DIRTYFLAG; action = 1;
02576                         }
02577                         targ += *targ;
02578                     }
02579                     arg = argstop;
02580                 }
02581                 else if ( *arg <= -FUNCTION ) arg++;
02582                 else arg += 2;
02583             }
02584         }
02585         t = tnext;
02586     }
02587     return(action);
02588 }
02589
02590 /*
02591     #] DenToFunction :
02592 */

```





# Index

- `_CHECK`
    - `parallel.c`, [2064](#)
  - `__CHECK`
    - `parallel.c`, [2065](#)
  - `~AEdge`
    - `AEdge`, [8](#)
  - `~ANode`
    - `ANode`, [12](#)
  - `~AStack`
    - `AStack`, [27](#)
  - `~Assign`
    - `Assign`, [17](#)
  - `~ECand`
    - `ECand`, [94](#)
  - `~EEdge`
    - `EEdge`, [96](#)
  - `~EGraph`
    - `EGraph`, [104](#)
  - `~ENode`
    - `ENode`, [115](#)
  - `~EStack`
    - `EStack`, [119](#)
  - `~Interaction`
    - `Interaction`, [160](#)
  - `~MCBlock`
    - `MCBlock`, [193](#)
  - `~MCBridge`
    - `MCBridge`, [195](#)
  - `~MCOpi`
    - `MCOpi`, [203](#)
  - `~MConn`
    - `MConn`, [197](#)
  - `~MGraph`
    - `MGraph`, [207](#)
  - `~MNodeClass`
    - `MNodeClass`, [221](#)
  - `~MOrbits`
    - `MOrbits`, [237](#)
  - `~Model`
    - `Model`, [226](#)
  - `~NCand`
    - `NCand`, [266](#)
  - `~NStack`
    - `NStack`, [275](#)
  - `~Options`
    - `Options`, [297](#)
  - `~Output`
    - `Output`, [304](#)
  - `~PNodeClass`
    - `PNodeClass`, [335](#)
  - `~Particle`
    - `Particle`, [323](#)
  - `~Process`
    - `Process`, [364](#)
  - `~SGroup`
    - `SGroup`, [386](#)
  - `~SProcess`
    - `SProcess`, [404](#)
  - `~T_EGraph`
    - `T_EGraph`, [442](#)
  - `~T_MGraph`
    - `T_MGraph`, [447](#)
  - `~T_MNodeClass`
    - `T_MNodeClass`, [453](#)
  - `~fmpz`
    - `flint::fmpz`, [138](#)
  - `~mpoly`
    - `flint::mpoly`, [240](#)
  - `~mpoly_ctx`
    - `flint::mpoly_ctx`, [241](#)
  - `~mpoly_factor`
    - `flint::mpoly_factor`, [242](#)
  - `~poly`
    - `flint::poly`, [338](#)
    - `poly`, [341](#)
  - `~poly_factor`
    - `flint::poly_factor`, [357](#)
- `A`
  - `inivar.h`, [1682](#)
  - `variable.h`, [3207](#)
- `a`
  - `MODNUM`, [235](#)
- `a3`
  - `TrAcEs`, [477](#)
- `a4`
  - `TrAcEs`, [477](#)
- `ABS`
  - `declare.h`, [800](#)
- `ABSFUNCTON`
  - `ftypes.h`, [1395](#)
- `AC`
  - `variable.h`, [3205](#)
- `accu`
  - `TrAcEn`, [471](#)
  - `TrAcEs`, [474](#)
- `AccumGCD`
  - `declare.h`, [824](#)
  - `reken.c`, [2553](#)

- accup
  - TrAcEn, [471](#)
  - TrAcEs, [474](#)
- acode
  - Particle, [326](#)
  - PInput, [333](#)
- ACOSFUNCTION
  - ftypes.h, [1399](#)
- ACOSHFUNTION
  - ftypes.h, [1400](#)
- active
  - FiLe, [130](#)
  - PF\_BUFFER, [330](#)
- activenames
  - C\_const, [46](#)
- ad
  - TrAcEs, [476](#)
- add
  - COST, [76](#)
  - Fraction, [140](#), [141](#)
  - poly, [349](#)
- Add2Com
  - declare.h, [814](#)
- Add2ComStrings
  - compcomm.c, [617](#)
  - declare.h, [928](#)
- ADD2POS
  - declare.h, [812](#)
- Add3Com
  - declare.h, [814](#)
- Add4Com
  - declare.h, [814](#)
- Add5Com
  - declare.h, [814](#)
- add\_factor
  - factorized\_poly, [126](#)
- ADDARG
  - ftypes.h, [1447](#)
- AddArgs
  - declare.h, [824](#)
  - sort.c, [2722](#)
- AddArrayIndex
  - declare.h, [822](#)
  - sch.c, [2653](#)
- addArtic
  - MConn, [198](#)
- addBlock
  - MConn, [198](#)
- addBlockSelf
  - MConn, [199](#)
- addBridge
  - MConn, [198](#)
- AddCoef
  - declare.h, [824](#)
  - sort.c, [2720](#)
- AddComString
  - compcomm.c, [617](#)
  - declare.h, [957](#)
- AddDictionary
  - declare.h, [1021](#)
  - dict.c, [1076](#)
- AddDollar
  - declare.h, [892](#)
  - names.c, [1832](#)
- AddDubious
  - declare.h, [893](#)
  - names.c, [1832](#)
- addEdge
  - Assign, [18](#)
- AddExpression
  - declare.h, [892](#)
  - names.c, [1833](#)
- addFLine
  - EGraph, [108](#)
- AddFloats
  - float.c, [1288](#)
- AddFunction
  - declare.h, [893](#)
  - names.c, [1829](#)
- addGroup
  - SGroup, [387](#)
- AddIndex
  - declare.h, [893](#)
  - names.c, [1828](#)
- addInteraction
  - Model, [227](#)
- addInteractionEnd
  - Model, [227](#)
- AddLHS
  - comtool.c, [764](#)
  - declare.h, [925](#)
- AddLineFeed
  - declare.h, [801](#)
- AddLong
  - declare.h, [825](#)
  - reken.c, [2554](#)
- AddName
  - declare.h, [889](#)
  - names.c, [1824](#)
- AddNtoC
  - comtool.c, [766](#)
  - declare.h, [927](#)
- AddNtoL
  - comtool.c, [765](#)
  - declare.h, [926](#)
- AddObject
  - declare.h, [987](#)
  - minos.c, [1733](#)
- addOPic
  - MConn, [198](#)
- addParticle
  - Model, [226](#)
- addParticleEnd
  - Model, [226](#)
- AddPLon
  - declare.h, [825](#)

- reken.c, 2554
- AddPoly
  - declare.h, 825
  - sort.c, 2721
- ADDPOS
  - declare.h, 810
- AddPotModdollar
  - declare.h, 968
  - dollar.c, 1101
- AddRat
  - declare.h, 827
  - reken.c, 2552
- AddRatToFloat
  - float.c, 1288
- AddRHS
  - comtool.c, 765
  - declare.h, 926
- AddSet
  - declare.h, 894
  - names.c, 1831
- AddSymbol
  - declare.h, 892
  - names.c, 1828
- AddTableName
  - declare.h, 987
  - minos.c, 1732
- AddToCB
  - declare.h, 805
- AddToCom
  - compcomm.c, 617
  - declare.h, 928
- AddToDictionary
  - declare.h, 1021
  - dict.c, 1076
- AddToDollarBuffer
  - declare.h, 962
  - dollar.c, 1094
- AddToIndex
  - declare.h, 985
  - minos.c, 1733
- AddToLine
  - declare.h, 827
  - sch.c, 2651
- AddToPreTypes
  - declare.h, 924
  - pre.c, 2329
- AddToScratch
  - declare.h, 1025
  - store.c, 2829
- AddToString
  - declare.h, 916
  - tools.c, 3086
- AddVector
  - declare.h, 893
  - names.c, 1829
- AddWild
  - declare.h, 827
  - wildcard.c, 3223
- AddWildcardName
  - declare.h, 924
  - names.c, 1827
- AddWithFloat
  - float.c, 1294
- adjMat
  - MGraph, 212
  - T\_MGraph, 449
- AdjustRenumScratch
  - declare.h, 847
  - reshuf.c, 2609
- aebufnum
  - T\_const, 434
- AEdge, 7
  - ~AEdge, 8
  - AEdge, 8
  - cand, 8
  - DUMMYPADDING, 8
  - nlegs, 8
  - nodes, 8
  - ptcl, 8
- aedges
  - ANode, 13
- aelegs
  - ANode, 13
- afterwards
  - Stream, 419
- agraph
  - AStack, 28
  - SProcess, 407
- agrcount
  - Process, 365
  - SProcess, 409
- ald
  - EGraph, 109
- alfatable1
  - compiler.c, 726
- all\_variables
  - poly, 345
- ALLARGS
  - ftypes.h, 1445
- allAssigned
  - Assign, 18
- allbufnum
  - T\_const, 434
- ALLDIRTY
  - ftypes.h, 1420
- ALLFUNCTIONS
  - ftypes.h, 1404
- ALLFUNPOWERS
  - ftypes.h, 1440
- ALLGLOBALS
  - structs.h, 2904
- AllGlobals, 9
  - Cc, 10
  - M, 10
  - N, 10
  - O, 11

- P, [11](#)
- R, [10](#)
- S, [10](#)
- T, [11](#)
- X, [11](#)
- ALLINTEGERDOUBLE
  - ftypes.h, [1387](#)
- AllLoops
  - declare.h, [1030](#)
  - lus.c, [1687](#)
- ALLMZVFUNCTIONS
  - ftypes.h, [1404](#)
- AllocFileHandle
  - declare.h, [973](#)
  - setfile.c, [2691](#)
- AllocSetups
  - declare.h, [903](#)
  - setfile.c, [2691](#)
- AllocSort
  - declare.h, [904](#)
  - setfile.c, [2691](#)
- AllocSortFileName
  - declare.h, [904](#)
  - setfile.c, [2691](#)
- AllOnePI
  - lus.c, [1689](#)
- AllowDelay
  - P\_const, [315](#)
- allPart
  - Model, [229](#)
- allParticles
  - Model, [228](#)
- AllPaths
  - declare.h, [1031](#)
  - lus.c, [1688](#)
- allsign
  - TrAcEn, [472](#)
  - TrAcEs, [478](#)
- ALLVARIABLES
  - ftypes.h, [1376](#)
- ALSODOLLARS
  - ftypes.h, [1443](#)
- ALSOFUNARGS
  - ftypes.h, [1443](#)
- ALSOPOWERS
  - ftypes.h, [1443](#)
- ALSOREVERSE
  - ftypes.h, [1387](#)
- AltCollectFun
  - C\_const, [66](#)
- ALTSEPARATOR
  - fwin.h, [1496](#)
  - unix.h, [3193](#)
- AM
  - variable.h, [3206](#)
- AN
  - variable.h, [3206](#)
- aname
  - Particle, [325](#)
  - PInput, [332](#)
- ANDCOND
  - ftypes.h, [1437](#)
- ANNOUNCE
  - checkpoint.c, [541](#)
- ANode, [11](#)
  - ~ANode, [12](#)
  - aedges, [13](#)
  - aelegs, [13](#)
  - ANode, [12](#)
  - anodes, [13](#)
  - cand, [13](#)
  - deg, [13](#)
  - newleg, [12](#)
  - nlegs, [13](#)
- anodes
  - ANode, [13](#)
- antiParticle
  - Model, [228](#)
- ANTISYMMETRIC
  - ftypes.h, [1432](#)
- ANYTYPE
  - ftypes.h, [1377](#)
- AO
  - variable.h, [3206](#)
- AP
  - variable.h, [3206](#)
- aparticle
  - Particle, [324](#)
- Apply
  - declare.h, [838](#)
  - tables.c, [2965](#)
- ApplyExec
  - declare.h, [839](#)
  - tables.c, [2965](#)
- ApplyReset
  - declare.h, [839](#)
  - tables.c, [2965](#)
- AR
  - variable.h, [3206](#)
- AreArgsEqual
  - declare.h, [901](#)
  - tools.c, [3091](#)
- arg1
  - optimization, [291](#)
- arg2
  - optimization, [291](#)
- argaddress
  - N\_const, [246](#)
- argag
  - Options, [302](#)
- ARGBUFFER, [14](#)
  - buffer, [14](#)
  - dollar, [14](#)
  - dummy, [15](#)
  - size, [14](#)
  - type, [15](#)

- ArgDotproductMerge
  - argument.c, [492](#)
  - declare.h, [932](#)
- argemg
  - Options, [302](#)
- ArgFactorize
  - argument.c, [491](#)
  - declare.h, [929](#)
- ARGFIELD
  - ftypes.h, [1391](#)
- ARGHEAD
  - ftypes.h, [1420](#)
- arglevel
  - C\_const, [64](#)
- arglist
  - N\_const, [251](#)
- arglistsize
  - N\_const, [254](#)
- argmg
  - Options, [302](#)
- argnames
  - pReVaR, [360](#)
- ARRANGE
  - ftypes.h, [1445](#)
- args
  - PaRtl, [320](#)
- argsize
  - NeStInG, [269](#)
- argstack
  - C\_const, [52](#)
- argsumcheck
  - C\_const, [63](#)
- ArgSymbolMerge
  - argument.c, [492](#)
  - declare.h, [932](#)
- argtail
  - TaBIEs, [461](#)
- ARGTOARG
  - ftypes.h, [1417](#)
- argument.c, [489](#), [495](#)
  - ArgDotproductMerge, [492](#)
  - ArgFactorize, [491](#)
  - ArgSymbolMerge, [492](#)
  - ArgumentExplode, [490](#)
  - ArgumentImplode, [490](#)
  - CleanupArgCache, [491](#)
  - execarg, [490](#)
  - execterm, [490](#)
  - FindArg, [491](#)
  - InsertArg, [491](#)
  - MakeInteger, [493](#)
  - MakeMod, [493](#)
  - NEWORDER, [490](#)
  - SortWeights, [494](#)
  - TakeArgContent, [492](#)
- argument\_to\_poly
  - poly, [347](#)
- ArgumentExplode
  - argument.c, [490](#)
  - declare.h, [982](#)
- ArgumentImplode
  - argument.c, [490](#)
  - declare.h, [982](#)
- ARGWILD
  - ftypes.h, [1390](#)
- ARLTOARL
  - ftypes.h, [1418](#)
- arraysize
  - POLYMOD, [358](#)
- articals
  - MConn, [199](#)
- AS
  - variable.h, [3206](#)
- ASINFUNCTION
  - ftypes.h, [1399](#)
- ASINHFUNCTION
  - ftypes.h, [1400](#)
- Assign, [15](#)
  - ~Assign, [17](#)
  - addEdge, [18](#)
  - allAssigned, [18](#)
  - Assign, [17](#)
  - assignAllVertices, [17](#)
  - assignVertex, [20](#)
  - assignPLeg, [21](#)
  - assignVertex, [18](#)
  - astack, [24](#)
  - candPart, [20](#)
  - candPartClassify, [21](#)
  - checkAG, [17](#)
  - checkOrderCpl, [21](#)
  - checkpoint0, [25](#)
  - cmpNodes, [22](#)
  - cmpPermGraph, [22](#)
  - connect, [19](#)
  - cpleft, [25](#)
  - detEdge, [21](#)
  - edges, [25](#)
  - edgeSym, [22](#)
  - egraph, [23](#)
  - fillEGraph, [19](#)
  - fromMGraph, [18](#)
  - getLegParticle, [19](#)
  - isIsomorphic, [22](#)
  - isOrdLegs, [22](#)
  - isOrdPLeg, [21](#)
  - legEdgeParticle, [19](#)
  - legPart, [20](#)
  - mgraph, [23](#)
  - model, [23](#)
  - nAGraphs, [25](#)
  - nAOPI, [25](#)
  - nEdges, [24](#)
  - nETotal, [24](#)
  - nExtern, [24](#)
  - nNodes, [24](#)

- nodes, 25
- opt, 24
- orbits, 24
- pnclass, 24
- prCand, 17
- proc, 23
- reordLeg, 19
- restoreCouple, 23
- saveCouple, 22
- selectLeg, 18
- selectVertex, 18
- selectVertexSimp, 18
- selUnAssLeg, 20
- selUnAssVertex, 20
- selUnAssVertexSimp, 20
- sproc, 23
- subCouple, 23
- updateCandNode, 21
- wAGraphs, 25
- wAOPI, 25
- assign
  - SProcess, 405
- assignAllVertices
  - Assign, 17
- AssignDollar
  - declare.h, 962
  - dollar.c, 1094
- assigned
  - EGraph, 110
- assignIVertex
  - Assign, 20
- assignPLeg
  - Assign, 21
- assignVertex
  - Assign, 18
- AStack, 26
  - ~AStack, 27
  - agraph, 28
  - AStack, 27
  - checkPoint, 27
  - eSize, 29
  - eStack, 28
  - eStackP, 29
  - nSize, 28
  - nStack, 28
  - nStackP, 28
  - prStack, 27
  - pushEdge, 28
  - pushNode, 28
  - restore, 27
  - setAGraph, 27
- astack
  - Assign, 24
  - Process, 366
  - SProcess, 406
- AT
  - variable.h, 3206
- ATAN2FUNCTION
  - ftypes.h, 1399
- ATANFUNCTION
  - ftypes.h, 1399
- ATANHFUNCTION
  - ftypes.h, 1400
- ATENDOFMODULE
  - ftypes.h, 1375
- atstartup
  - M\_const, 190
- AutoDeclareFlag
  - C\_const, 54
- AutoFunctionList
  - C\_const, 45
- autofunctions
  - variable.h, 3204
- AutoIndexList
  - C\_const, 45
- autoindices
  - variable.h, 3204
- AUTONAMES
  - ftypes.h, 1444
- autonames
  - C\_const, 45
- AutoSymbolList
  - C\_const, 45
- autosymbols
  - variable.h, 3204
- AutoVectorList
  - C\_const, 45
- autovectors
  - variable.h, 3204
- AWORDMASK
  - form3.h, 1329
- AX
  - variable.h, 3206
- BACKINOUT
  - declare.h, 806
- balance
  - NaMeNode, 261
- Balancing
  - S\_const, 382
- BASECODE
  - ftypes.h, 1448
- BASEPOSITION
  - declare.h, 812
- bconn
  - EGraph, 113
- begin
  - Options, 300
- beginProc
  - Options, 300
- beginSubProc
  - Options, 300
- Bernoulli
  - declare.h, 844
  - reken.c, 2561
- BERNOULLIFUNCTION
  - ftypes.h, 1396

- bernoullis
    - T\_const, [430](#)
  - BETAVERSION
    - form3.h, [1326](#)
  - BHEAD
    - structs.h, [2899](#)
  - BHEAD0
    - structs.h, [2899](#)
  - biconnE
    - EGraph, [106](#)
  - biconnME
    - MGraph, [209](#)
  - bicount
    - EGraph, [113](#)
    - MGraph, [215](#)
  - bidef
    - EGraph, [113](#)
    - MGraph, [215](#)
  - BigLong
    - declare.h, [828](#)
    - reken.c, [2554](#)
  - biinitE
    - EGraph, [106](#)
  - bilow
    - EGraph, [113](#)
    - MGraph, [215](#)
  - BinarySearch
    - declare.h, [974](#)
    - sort.c, [2729](#)
  - BinomGen
    - declare.h, [828](#)
    - ratio.c, [2501](#)
  - BINOMIAL
    - ftypes.h, [1394](#)
  - bisearchE
    - EGraph, [106](#)
  - bisearchME
    - MGraph, [209](#)
  - bit\_0
    - bit\_field, [30](#)
  - bit\_1
    - bit\_field, [30](#)
  - bit\_2
    - bit\_field, [30](#)
  - bit\_3
    - bit\_field, [30](#)
  - bit\_4
    - bit\_field, [30](#)
  - bit\_5
    - bit\_field, [30](#)
  - bit\_6
    - bit\_field, [30](#)
  - bit\_7
    - bit\_field, [30](#)
  - bit\_field, [29](#)
    - bit\_0, [30](#)
    - bit\_1, [30](#)
    - bit\_2, [30](#)
  - bit\_3, [30](#)
  - bit\_4, [30](#)
  - bit\_5, [30](#)
  - bit\_6, [30](#)
  - bit\_7, [30](#)
- BITSINLONG
    - form3.h, [1328](#)
  - BITSINWORD
    - form3.h, [1327](#)
  - blksp
    - MConn, [200](#)
  - blkstk
    - MConn, [200](#)
  - blkstkptr
    - MConn, [202](#)
  - blnce
    - tree, [479](#)
  - BLOCK
    - ftypes.h, [1403](#)
  - block
    - MGraph, [215](#)
  - blocks
    - MConn, [199](#)
  - BlockSpaces
    - O\_const, [282](#)
  - BOOL
    - form3.h, [1335](#)
  - boomlijst
    - CbUf, [73](#)
    - TaBIEs, [461](#)
  - BottomLevel
    - C\_const, [56](#)
  - bounds
    - TaBIEs, [463](#)
  - BrackBuf
    - T\_const, [429](#)
  - bracket
    - BrAcKeTiNdEx, [32](#)
    - O\_const, [279](#)
  - bracketbuffer
    - BrAcKeTiNfO, [33](#)
  - bracketbuffersize
    - BrAcKeTiNfO, [33](#)
  - BRACKETCURRENTEXPR
    - execute.c, [1157](#)
  - BracketFactors
    - M\_const, [192](#)
  - bracketfill
    - BrAcKeTiNfO, [33](#)
  - BrAcKeTiNdEx, [31](#)
    - bracket, [32](#)
    - next, [32](#)
    - start, [32](#)
    - termsinbracket, [32](#)
  - bracketindexflag
    - C\_const, [59](#)
    - T\_const, [435](#)
  - BrAcKeTiNfO, [32](#)

- bracketbuffer, 33
- bracketbuffersize, 33
- bracketfill, 33
- indexbuffer, 33
- indexbuffersize, 33
- indexfill, 34
- SortType, 34
- BracketInfo, 34
  - BracketInfo, 35
  - dummy, 35
  - num\_terms, 35
  - operator<, 35
  - p, 36
  - pattern, 35
- bracketinfo
  - ExPrEsSiOn, 122
  - T\_const, 432
- BracketNormalize
  - C\_const, 59
  - declare.h, 858
  - normal.c, 1875
- BracketOn
  - R\_const, 374
- BRACKETOTHEREXPR
  - execute.c, 1157
- bridges
  - MConn, 199
- buff
  - PF\_BUFFER, 329
- Buffer
  - CbUf, 71
- buffer
  - ARGBUFFER, 14
  - INSIDEINFO, 158
  - longMultiStruct, 167
  - PRELOAD, 359
  - StOrEcAcHe, 412
  - StreaM, 417
- bufferedInd
  - O\_const, 285
- BufferForOutput
  - inivar.h, 1682
- buffernum
  - SuBbUf, 420
- bufferposition
  - StreaM, 418
- buffers
  - TaBIes, 462
- bufferfill
  - TaBIes, 465
- BufferSize
  - CbUf, 73
- bufferize
  - StreaM, 418
- bufferssize
  - TaBIes, 464
- bufIPstruct, 36
  - e, 37
  - i, 37
- bufnum
  - TaBIes, 464
- bufpos
  - longMultiStruct, 167
- BUG
  - variable.h, 3205
- bugtool.c, 535, 536
  - ExprStatus, 536
- build\_Horner\_tree
  - optimize.cc, 1993
- buildTree
  - optimize.cc, 1997
- BUILTINDOLLARS
  - ftypes.h, 1406
- BUILTININDICES
  - ftypes.h, 1406
- BUILTINSYMBOLS
  - ftypes.h, 1405
- BUILTINVECTORS
  - ftypes.h, 1406
- bzero
  - form3.h, 1334
- c1PI
  - MGraph, 212
  - T\_MGraph, 449
- c1PINoTadBlock
  - MGraph, 213
- C\_const, 37
  - activenames, 46
  - AltCollectFun, 66
  - arglevel, 64
  - argstack, 52
  - argsumcheck, 63
  - AutoDeclareFlag, 54
  - AutoFunctionList, 45
  - AutoIndexList, 45
  - autonames, 45
  - AutoSymbolList, 45
  - AutoVectorList, 45
  - BottomLevel, 56
  - bracketindexflag, 59
  - BracketNormalize, 59
  - cbufList, 44
  - cbufnum, 54
  - ChannelList, 42
  - CheckpointFlag, 62
  - CheckpointInterval, 53
  - CheckpointRunAfter, 51
  - CheckpointRunBefore, 51
  - CheckpointStamp, 53
  - cmod, 47
  - CModule, 53
  - Cnumpows, 66
  - CodesFlag, 57
  - CollectFun, 66
  - CollectPercentage, 69
  - ComDefer, 66



Commercial, 70  
CommutInSet, 52  
CompileLevel, 56  
compiletype, 54  
CurrentStream, 46  
debugFlags, 70  
DidClean, 67  
DirtPow, 65  
dollarnames, 42  
dolooplevel, 62  
doloopnest, 51  
doloopstack, 51  
doloopstacksize, 62  
DubiousList, 43  
DumNum, 65  
dumnumflag, 59  
endoftokens, 50  
ExpressionList, 43  
exprfillwarning, 60  
exprnames, 42  
extrasym, 51  
extrasymbols, 69  
ffbufnum, 61  
FinalStats, 58  
firstconstindex, 54  
firstctypemessage, 56  
FixIndices, 49  
FlintPolyFlag, 62  
Fortran90Kind, 51  
FunctionList, 43  
Functions, 46  
funpowers, 57  
halfmod, 48  
HideLevel, 68  
HumanStatsFlag, 62  
iBuffer, 48  
iBufferSize, 52  
idoption, 56  
IfCount, 48  
IfHeap, 48  
IfLevel, 67  
IfStack, 48  
IfSumCheck, 52  
IndexList, 43  
Indices, 45  
inexprlevel, 64  
inexprstack, 52  
inexprsumcheck, 64  
InnerTest, 69  
inparallelflag, 60  
insidefirst, 54  
insidelevel, 64  
insidestack, 52  
insidesumcheck, 64  
iPointer, 49  
IsFortran90, 61  
iStop, 49  
LabelNames, 49  
Labels, 50  
IDefDim, 55  
IDefDim4, 55  
Ihdollarflag, 60  
LineLength, 68  
LogHandle, 67  
IPolyFun, 68  
IPolyFunExp, 68  
IPolyFunInv, 68  
IPolyFunPow, 69  
IPolyFunType, 68  
IPolyFunVar, 68  
ISortType, 58  
IUniTrace, 64  
IUnitTrace, 66  
MaxIf, 56  
MaxLabels, 55  
MaxNumStreams, 56  
MaxSwitch, 67  
maxtermlevel, 59  
MemDebugFlag, 61  
minsidefirst, 55  
ModelLevel, 63  
models, 46  
modelspace, 63  
modinverses, 50  
modmode, 65  
ModOptDolList, 44  
modpowers, 47  
mparallelflag, 59  
mProcessBucketSize, 53  
MultiBracketBuf, 51  
MultiBracketLevels, 61  
MustTestTable, 65  
NamesFlag, 57  
ncmod, 65  
nhalfmod, 65  
NoCompress, 61  
NoShowInput, 54  
npowmod, 65  
NumLabels, 55  
nummodels, 63  
NumStreams, 56  
NumWildcardNames, 55  
NwildC, 66  
OldFactArgFlag, 61  
OldGCDflag, 61  
OldParallelStats, 58  
origin, 63  
OutNumberType, 67  
OutputMode, 66  
OutputSpaces, 66  
outsidefun, 57  
parallelflag, 59  
partodoflag, 60  
PolyRatFunChanged, 69  
PotModDolList, 44  
powmod, 47

PrintBacktraceFlag, 62  
 ProcessBucketSize, 53  
 ProcessStats, 59  
 properorderflag, 60  
 ProtoType, 48  
 RepLevel, 64  
 RepSumCheck, 64  
 separators, 42  
 SetElementList, 43  
 SetList, 43  
 SetupFlag, 58  
 ShortStats, 54  
 ShortStatsMax, 69  
 SizeCommutelnSet, 63  
 SortReallocateFlag, 62  
 SortType, 58  
 StatsFlag, 57  
 StoreFileSize, 42  
 StoreHandle, 68  
 Streams, 46  
 SwitchArray, 46  
 SwitchHeap, 47  
 SwitchlnArray, 67  
 SwitchLevel, 67  
 SymbolList, 44  
 Symbols, 45  
 SymChangeFlag, 69  
 TableBaseList, 44  
 tablecheck, 56  
 tablefilling, 60  
 termlevel, 65  
 termsortstack, 47  
 termstack, 47  
 termsumcheck, 49  
 TestValue, 52  
 ThreadBalancing, 58  
 ThreadBucketSize, 53  
 ThreadsFlag, 58  
 ThreadSortFileSynch, 59  
 ThreadStats, 58  
 ToBeInFactors, 69  
 tokenarglevel, 50  
 tokens, 50  
 TokensWriteFlag, 57  
 topolynomialflag, 61  
 toptokens, 50  
 TransEname, 53  
 UnsureDollarMode, 57  
 varnames, 42  
 vectorlikeLHS, 63  
 VectorList, 44  
 Vectors, 46  
 vetofilling, 60  
 vetotablebasefill, 60  
 WarnFlag, 57  
 WhileLevel, 67  
 WildC, 48  
 WildcardBufferSize, 55  
 WildcardNames, 49  
 wildflag, 55  
 WTimeStatsFlag, 62  
 cache\_buffer  
     checkpoint.c, 545  
 cache\_fill  
     checkpoint.c, 545  
 CACHE\_SIZE  
     checkpoint.c, 538  
 CACHED\_SNAPSHOT  
     checkpoint.c, 538  
 CacheNumberFree  
     declare.h, 816  
 CacheNumberMalloc  
     declare.h, 816  
 CacheNumberMallocAddMemory  
     declare.h, 1000  
     tools.c, 3089  
 CacheNumberMemHeap  
     T\_const, 432  
 CacheNumberMemMax  
     T\_const, 435  
 CacheNumberMemTop  
     T\_const, 435  
 calcHash  
     node, 273  
 CalculateEuler  
     float.c, 1291  
 CalculateMZV  
     float.c, 1291  
 CalculateMZVhalf  
     float.c, 1290  
 CanCommu  
     CbUf, 72  
 cand  
     AEdge, 8  
     ANode, 13  
 candPart  
     Assign, 20  
 candPartClassify  
     Assign, 21  
 CARRIAGERETURN  
     fwin.h, 1496  
     unix.h, 3192  
 caseoffset  
     SWITCH, 422  
 CatchDollar  
     declare.h, 962  
     dollar.c, 1094  
 CBUF  
     structs.h, 2902  
 CbUf, 70  
     boomlijst, 73  
     Buffer, 71  
     BufferSize, 73  
     CanCommu, 72  
     dimension, 73  
     lhs, 72

- maxlhs, [73](#)
- maxrhs, [74](#)
- MaxTreeSize, [74](#)
- mnumlhs, [74](#)
- mnumrhs, [74](#)
- numdum, [72](#)
- numlhs, [73](#)
- numrhs, [73](#)
- NumTerms, [72](#)
- numtree, [74](#)
- Pointer, [71](#)
- rhs, [72](#)
- rootnum, [74](#)
- Top, [71](#)
- cbuf
  - variable.h, [3203](#)
- cBuffer
  - sOrT, [395](#)
- cBufferSize
  - sOrT, [399](#)
- cbufList
  - C\_const, [44](#)
- cbufnum
  - C\_const, [54](#)
- Cc
  - AllGlobals, [10](#)
- cComChar
  - P\_const, [316](#)
- cdeg
  - PGInput, [331](#)
- CDELETE
  - ftypes.h, [1377](#)
- CDELTA
  - ftypes.h, [1378](#)
- cDiag
  - MGraph, [212](#)
- CDOLLAR
  - ftypes.h, [1378](#)
- CDOTPRODUCT
  - ftypes.h, [1378](#)
- CDOUBLEDOT
  - ftypes.h, [1379](#)
- CDUBIOUS
  - ftypes.h, [1379](#)
- CEXPRESSION
  - ftypes.h, [1378](#)
- CFD
  - minos.c, [1729](#)
- CFUN
  - ftypes.h, [1456](#)
- CFUNCTION
  - ftypes.h, [1378](#)
- cgen
  - SGroup, [388](#)
- ChainIn
  - declare.h, [982](#)
  - function.c, [1472](#)
- ChainOut
  - declare.h, [982](#)
  - function.c, [1472](#)
- CHANNEL
  - structs.h, [2903](#)
- ChAnNeL, [75](#)
  - handle, [75](#)
  - name, [75](#)
- ChannelList
  - C\_const, [42](#)
- channels
  - variable.h, [3203](#)
- characters
  - DICTIONARY, [83](#)
- CharOut
  - declare.h, [905](#)
  - pre.c, [2317](#)
- chartype
  - variable.h, [3199](#)
- CHECK
  - parallel.c, [2064](#)
- check\_memory
  - poly, [343](#)
- checkAG
  - Assign, [17](#)
- CHECKEXTERN
  - ftypes.h, [1455](#)
- CheckFloat
  - float.c, [1293](#)
- CHECKLOGTYPE
  - ftypes.h, [1389](#)
- checkOrderCpl
  - Assign, [21](#)
- checkPoint
  - AStack, [27](#)
- checkpoint.c, [537](#), [545](#)
  - ANNOUNCE, [541](#)
  - cache\_buffer, [545](#)
  - cache\_fill, [545](#)
  - CACHE\_SIZE, [538](#)
  - CACHED\_SNAPSHOT, [538](#)
  - CheckRecoveryFile, [542](#)
  - DeleteRecoveryFile, [542](#)
  - DoCheckpoint, [544](#)
  - DoRecovery, [543](#)
  - flush\_cache, [543](#)
  - fwrite\_cached, [543](#)
  - InitRecovery, [542](#)
  - R\_COPY\_B, [539](#)
  - R\_COPY\_LIST, [540](#)
  - R\_COPY\_NAMETREE, [541](#)
  - R\_COPY\_S, [540](#)
  - R\_FREE, [538](#)
  - R\_FREE\_NAMETREE, [539](#)
  - R\_FREE\_STREAM, [539](#)
  - R\_SET, [539](#)
  - RecoveryFilename, [542](#)
  - S\_FLUSH\_B, [540](#)
  - S\_WRITE\_B, [539](#)

- S\_WRITE\_DOLLAR, [541](#)
  - S\_WRITE\_LIST, [540](#)
  - S\_WRITE\_NAMETREE, [541](#)
  - S\_WRITE\_S, [540](#)
- checkpoint0
  - Assign, [25](#)
- CheckpointFlag
  - C\_const, [62](#)
- CheckpointInterval
  - C\_const, [53](#)
- CheckpointRunAfter
  - C\_const, [51](#)
- CheckpointRunBefore
  - C\_const, [51](#)
- CheckpointStamp
  - C\_const, [53](#)
- CHECKPOLY
  - token.c, [3048](#)
- CheckPower
  - O\_const, [281](#)
- CheckRecoveryFile
  - checkpoint.c, [542](#)
  - declare.h, [997](#)
- CheckTableDeclarations
  - declare.h, [838](#)
  - tables.c, [2963](#)
- CheckWild
  - declare.h, [828](#)
  - wildcard.c, [3223](#)
- childs
  - tree\_node, [481](#)
- CHISHOLM
  - ftypes.h, [1387](#)
- Chisholm
  - declare.h, [828](#)
  - opera.c, [1957](#)
- chkMomConsv
  - EGraph, [107](#)
- chkOrd
  - T\_MNodeClass, [455](#)
- CINDEX
  - ftypes.h, [1377](#)
- cl2mcl
  - PNodeClass, [337](#)
- cl2nd
  - PNodeClass, [337](#)
  - SProcess, [407](#)
- clCmp
  - MNodeClass, [221](#)
  - T\_MNodeClass, [454](#)
- cldeg
  - NCInput, [268](#)
  - ToPoTyPe, [469](#)
- CleanDollarFactors
  - declare.h, [966](#)
  - dollar.c, [1099](#)
- CleanExpr
  - declare.h, [828](#)
- execute.c, [1157](#)
- CLEANFLAG
  - ftypes.h, [1419](#)
- CleanUp
  - declare.h, [829](#)
  - startup.c, [2800](#)
- cleanup
  - flintinterface.h, [1270](#)
- cleanup\_master
  - flintinterface.h, [1270](#)
- CleanupArgCache
  - argument.c, [491](#)
  - declare.h, [931](#)
- CleanUpSort
  - declare.h, [973](#)
  - sort.c, [2729](#)
- CleanupTerm
  - declare.h, [977](#)
  - execute.c, [1159](#)
- CLEAR
  - ftypes.h, [1438](#)
- ClearBracketIndex
  - declare.h, [980](#)
  - index.c, [1672](#)
- clearcbuf
  - comtool.c, [764](#)
  - declare.h, [972](#)
- clearGroup
  - SGroup, [387](#)
- ClearMacro
  - declare.h, [907](#)
  - pre.c, [2321](#)
- CLEARMODULE
  - ftypes.h, [1369](#)
- ClearMZVTables
  - float.c, [1293](#)
- clearnamefill
  - NaMeTree, [265](#)
- clearnodefill
  - NaMeTree, [265](#)
- ClearOptimize
  - declare.h, [970](#)
  - optimize.cc, [2006](#)
- ClearPushback
  - declare.h, [904](#)
  - pre.c, [2317](#)
- ClearSpectators
  - declare.h, [1024](#)
  - spectator.c, [2788](#)
- ClearStore
  - M\_const, [192](#)
- ClearTableTree
  - declare.h, [971](#)
  - tables.c, [2961](#)
- ClearTree
  - comtool.c, [767](#)
  - declare.h, [962](#)
- ClearWild

- declare.h, [829](#)
- wildcard.c, [3223](#)
- ClearWildcardNames
  - declare.h, [924](#)
  - names.c, [1827](#)
- clext
  - ToPoTyPe, [469](#)
- clist
  - Interaction, [161](#)
  - MGraph, [211](#)
  - MNodeClass, [223](#)
  - Process, [366](#)
  - SProcess, [408](#)
  - T\_MGraph, [448](#)
  - T\_MNodeClass, [455](#)
- clmat
  - MNodeClass, [223](#)
  - T\_MNodeClass, [456](#)
- clnum
  - NCInput, [268](#)
  - ToPoTyPe, [469](#)
- clord
  - MNodeClass, [223](#)
- closeAllExternalChannels
  - declare.h, [990](#)
  - extcmd.c, [1194](#)
- CloseChannel
  - declare.h, [898](#)
  - tools.c, [3084](#)
- closeExternalChannel
  - declare.h, [990](#)
  - extcmd.c, [1193](#)
- CloseFile
  - declare.h, [895](#)
  - tools.c, [3081](#)
- CloseStream
  - declare.h, [888](#)
  - tools.c, [3079](#)
- clss
  - MNode, [219](#)
  - T\_MNode, [452](#)
- cltyp
  - NCInput, [268](#)
- cmaxd
  - NCInput, [268](#)
  - ToPoTyPe, [469](#)
- cmaxdeg
  - MNode, [219](#)
  - MNodeClass, [223](#)
  - Particle, [326](#)
  - PNodeClass, [336](#)
- cmind
  - NCInput, [268](#)
  - ToPoTyPe, [469](#)
- cminddeg
  - MNode, [219](#)
  - MNodeClass, [223](#)
  - Particle, [326](#)
- PNodeClass, [336](#)
- cmod
  - C\_const, [47](#)
  - N\_const, [251](#)
- CMODE
  - ftypes.h, [1386](#)
- CMODEL
  - ftypes.h, [1379](#)
- CModule
  - C\_const, [53](#)
- cmp
  - node, [272](#)
- CmpArray
  - declare.h, [902](#)
  - tools.c, [3085](#)
- cmpArray
  - T\_MNodeClass, [455](#)
- cmpMNCArray
  - MNodeClass, [222](#)
- cmpMom
  - EGraph, [107](#)
- cmpNodes
  - Assign, [22](#)
- cmpPermGraph
  - Assign, [22](#)
- cnamlist
  - MInput, [217](#)
- cnlist
  - Model, [229](#)
- cNoTadBlock
  - MGraph, [213](#)
- cNoTadpole
  - MGraph, [213](#)
- cnum
  - PGInput, [331](#)
- CNUMBER
  - ftypes.h, [1378](#)
- Cnumlhs
  - R\_const, [373](#)
- Cnumpows
  - C\_const, [66](#)
- CoAntiBracket
  - compcomm.c, [624](#)
  - declare.h, [933](#)
- CoAntiPutInside
  - compcomm.c, [631](#)
  - declare.h, [936](#)
- CoAntiSymmetrize
  - compcomm.c, [620](#)
  - declare.h, [934](#)
- CoApply
  - declare.h, [937](#)
  - tables.c, [2964](#)
- CoArgExplode
  - compcomm.c, [627](#)
  - declare.h, [934](#)
- CoArgImplode
  - compcomm.c, [627](#)

- declare.h, 934
- CoArgToExtraSymbol
  - compcomm.c, 628
  - declare.h, 1007
- CoArgument
  - compcomm.c, 618
  - declare.h, 934
- CoAssign
  - comexpr.c, 585
  - declare.h, 960
- CoAuto
  - declare.h, 954
  - names.c, 1832
- CoBracket
  - compcomm.c, 624
  - declare.h, 935
- CoBreak
  - compcomm.c, 631
  - declare.h, 954
- CoCanonicalize
  - declare.h, 835
  - diagrams.c, 1052
- CoCase
  - compcomm.c, 631
  - declare.h, 954
- CoCFunction
  - declare.h, 936
  - names.c, 1830
- CoChainin
  - compcomm.c, 622
  - declare.h, 953
- CoChainout
  - compcomm.c, 622
  - declare.h, 953
- CoChisholm
  - compcomm.c, 621
  - declare.h, 953
- CoClearTable
  - compcomm.c, 628
  - declare.h, 953
- CoClearUserFlag
  - compcomm.c, 632
  - declare.h, 1029
- CoCollect
  - compcomm.c, 612
  - declare.h, 936
- CoCommutelnSet
  - declare.h, 894
  - names.c, 1829
- CoCompress
  - compcomm.c, 612
  - declare.h, 936
- CoContract
  - compcomm.c, 617
  - declare.h, 936
- CoCopySpectator
  - declare.h, 1024
  - spectator.c, 2787
- CoCreateAll
  - compcomm.c, 632
  - declare.h, 1030
- CoCreateAllLoops
  - compcomm.c, 632
  - declare.h, 1029
- CoCreateAllPaths
  - compcomm.c, 632
  - declare.h, 1029
- CoCreateSpectator
  - declare.h, 1023
  - spectator.c, 2787
- CoCTable
  - declare.h, 952
  - names.c, 1831
- CoCTensor
  - declare.h, 936
  - names.c, 1830
- CoCycleSymmetrize
  - compcomm.c, 620
  - declare.h, 937
- code
  - OPTIMIZERESULT, 295
- CoDeallocateTable
  - comexpr.c, 585
  - declare.h, 966
- CodeFactors
  - compiler.c, 726
  - declare.h, 961
- CoDefault
  - compcomm.c, 631
  - declare.h, 955
- CodeGenerator
  - compiler.c, 725
  - declare.h, 960
- CoDelete
  - compcomm.c, 613
  - declare.h, 937
- CoDenominators
  - compcomm.c, 628
  - declare.h, 937
- CodesFlag
  - C\_const, 57
- codesize
  - OPTIMIZERESULT, 295
- CodeToLine
  - declare.h, 822
  - sch.c, 2652
- CoDimension
  - declare.h, 937
  - names.c, 1829
- CoDiscard
  - compcomm.c, 617
  - declare.h, 937
- CoDisorder
  - comexpr.c, 583
  - declare.h, 938
- CoDo

- compcomm.c, [629](#)
- declare.h, [1007](#)
- CoDrop
  - compcomm.c, [615](#)
  - declare.h, [938](#)
- CoDropCoefficient
  - compcomm.c, [628](#)
  - declare.h, [938](#)
- CoDropSymbols
  - compcomm.c, [628](#)
  - declare.h, [938](#)
- coeff
  - optimization, [291](#)
- COEFFI
  - ftypes.h, [1435](#)
- coefficient
  - poly, [346](#)
- coefficient\_list\_divmod
  - poly, [348](#)
- coefficients\_modulo
  - poly, [346](#)
- COEFFICIENTSONLY
  - ftypes.h, [1443](#)
- COEFFSYMBOL
  - ftypes.h, [1404](#)
- coefs
  - POLYMOD, [358](#)
- CoElse
  - compcomm.c, [624](#)
  - declare.h, [938](#)
- CoElseif
  - compcomm.c, [625](#)
  - declare.h, [938](#)
- CoEmptySpectator
  - declare.h, [1024](#)
  - spectator.c, [2787](#)
- CoEndArgument
  - compcomm.c, [618](#)
  - declare.h, [938](#)
- CoEndDo
  - compcomm.c, [629](#)
  - declare.h, [1007](#)
- CoEndIf
  - compcomm.c, [625](#)
  - declare.h, [939](#)
- CoEndInExpression
  - compcomm.c, [623](#)
  - declare.h, [935](#)
- CoEndInside
  - compcomm.c, [618](#)
  - declare.h, [939](#)
- CoEndModel
  - declare.h, [1029](#)
- CoEndRepeat
  - compcomm.c, [623](#)
  - declare.h, [939](#)
- CoEndSwitch
  - compcomm.c, [631](#)
- declare.h, [955](#)
- CoEndTerm
  - compcomm.c, [626](#)
  - declare.h, [939](#)
- CoEndWhile
  - compcomm.c, [625](#)
  - declare.h, [939](#)
- CoEvaluate
  - float.c, [1295](#)
- CoExit
  - compcomm.c, [622](#)
  - declare.h, [939](#)
- CoExtraSymbols
  - compcomm.c, [629](#)
  - declare.h, [1007](#)
- CoFactArg
  - compcomm.c, [619](#)
  - declare.h, [939](#)
- CoFactDollar
  - compcomm.c, [629](#)
  - declare.h, [940](#)
- CoFactorize
  - compcomm.c, [630](#)
  - declare.h, [940](#)
- CoFill
  - comexpr.c, [584](#)
  - declare.h, [940](#)
- CoFillExpression
  - comexpr.c, [584](#)
  - declare.h, [941](#)
- CoFindLoop
  - compcomm.c, [625](#)
  - declare.h, [964](#)
- CoFixIndex
  - compcomm.c, [614](#)
  - declare.h, [941](#)
- CoFlags
  - compcomm.c, [613](#)
  - declare.h, [946](#)
- CoFormat
  - compcomm.c, [612](#)
  - declare.h, [941](#)
- CoFromPolynomial
  - compcomm.c, [628](#)
  - declare.h, [1006](#)
- CoFunction
  - declare.h, [894](#)
  - names.c, [1829](#)
- CoFunPowers
  - compcomm.c, [626](#)
  - declare.h, [964](#)
- CoGlobal
  - comexpr.c, [582](#)
  - declare.h, [941](#)
- CoGlobalFactorized
  - comexpr.c, [582](#)
  - declare.h, [941](#)
- CoGoTo

- compcomm.c, [617](#)
- declare.h, [941](#)
- CoHide
  - compcomm.c, [616](#)
  - declare.h, [949](#)
- Cold
  - comexpr.c, [583](#)
  - declare.h, [941](#)
- ColdExpression
  - comexpr.c, [584](#)
  - declare.h, [960](#)
- ColdNew
  - comexpr.c, [583](#)
  - declare.h, [942](#)
- ColdOld
  - comexpr.c, [582](#)
  - declare.h, [942](#)
- Colf
  - compcomm.c, [624](#)
  - declare.h, [942](#)
- ColfMatch
  - comexpr.c, [583](#)
  - declare.h, [942](#)
- ColfNoMatch
  - comexpr.c, [583](#)
  - declare.h, [942](#)
- ColIndex
  - declare.h, [942](#)
  - names.c, [1828](#)
- ColnExpression
  - compcomm.c, [623](#)
  - declare.h, [935](#)
- ColnParallel
  - compcomm.c, [622](#)
  - declare.h, [935](#)
- ColInside
  - compcomm.c, [618](#)
  - declare.h, [934](#)
- ColInsideFirst
  - compcomm.c, [613](#)
  - declare.h, [942](#)
- ColIntoHide
  - compcomm.c, [616](#)
  - declare.h, [949](#)
- CoKeep
  - compcomm.c, [613](#)
  - declare.h, [943](#)
- CoLabel
  - compcomm.c, [618](#)
  - declare.h, [943](#)
- CollectFun
  - C\_const, [66](#)
- CollectOverFlag
  - S\_const, [382](#)
- CollectPercentage
  - C\_const, [69](#)
- CoLoad
  - declare.h, [943](#)
- store.c, [2829](#)
- CoLocal
  - comexpr.c, [582](#)
  - declare.h, [943](#)
- CoLocalFactorized
  - comexpr.c, [582](#)
  - declare.h, [943](#)
- CoMakeInteger
  - compcomm.c, [619](#)
  - declare.h, [946](#)
- CoMany
  - comexpr.c, [583](#)
  - declare.h, [943](#)
- combilast
  - SHvariables, [391](#)
- ComChar
  - P\_const, [316](#)
- ComDefer
  - C\_const, [66](#)
- CoMerge
  - compcomm.c, [627](#)
  - declare.h, [943](#)
- CoMetric
  - compcomm.c, [614](#)
  - declare.h, [944](#)
- comexpr.c, [581](#), [585](#)
  - CoAssign, [585](#)
  - CoDeallocateTable, [585](#)
  - CoDisorder, [583](#)
  - CoFill, [584](#)
  - CoFillExpression, [584](#)
  - CoGlobal, [582](#)
  - CoGlobalFactorized, [582](#)
  - Cold, [583](#)
  - ColdExpression, [584](#)
  - ColdNew, [583](#)
  - ColdOld, [582](#)
  - ColfMatch, [583](#)
  - ColfNoMatch, [583](#)
  - CoLocal, [582](#)
  - CoLocalFactorized, [582](#)
  - CoMany, [583](#)
  - CoMulti, [583](#)
  - CoMultiply, [584](#)
  - CoOnce, [584](#)
  - CoOnly, [584](#)
  - CoPrintTable, [585](#)
  - CoSelect, [584](#)
  - DoExpr, [582](#)
- comfun
  - T\_const, [437](#)
- COMFUNPOWERS
  - ftypes.h, [1440](#)
- comind
  - T\_const, [437](#)
- commentchar
  - setfile.c, [2692](#)
- Commercial



- C\_const, [70](#)
- COMMERCIALSIZE
  - fsizes.h, [1345](#)
- Commute
  - declare.h, [829](#)
  - normal.c, [1873](#)
- commute
  - FUN\_INFO, [143](#)
  - FuNcTiOn, [145](#)
- CommutelnSet
  - C\_const, [52](#)
- comnum
  - T\_const, [437](#)
- CoModel
  - declare.h, [1028](#)
- CoModOption
  - declare.h, [944](#)
  - module.c, [1763](#)
- CoModuleOption
  - declare.h, [944](#)
  - module.c, [1762](#)
- CoModulus
  - compcomm.c, [623](#)
  - declare.h, [944](#)
- CompactifyTree
  - declare.h, [891](#)
  - names.c, [1826](#)
- COMPARE
  - structs.h, [2904](#)
- Compare1
  - declare.h, [830](#)
  - sort.c, [2723](#)
- compare\_degree\_vector
  - poly, [349](#)
- CompareArgs
  - declare.h, [901](#)
  - tools.c, [3091](#)
- COMPAREDUMMY
  - structs.h, [2902](#)
- CompareFunctions
  - declare.h, [829](#)
  - normal.c, [1873](#)
- CompareHSymbols
  - declare.h, [991](#)
  - sort.c, [2724](#)
- CompareRoutine
  - R\_const, [372](#)
- CompareSymbols
  - declare.h, [991](#)
  - sort.c, [2723](#)
- CompareTerms
  - declare.h, [820](#)
- CompArg
  - declare.h, [829](#)
  - tools.c, [3092](#)
- compbuffer
  - SWITCHTABLE, [424](#)
- CompCoef
  - declare.h, [830](#)
  - reken.c, [2559](#)
- compcomm.c, [609](#), [633](#)
  - Add2ComStrings, [617](#)
  - AddComString, [617](#)
  - AddToCom, [617](#)
  - CoAntiBracket, [624](#)
  - CoAntiPutInside, [631](#)
  - CoAntiSymmetrize, [620](#)
  - CoArgExplode, [627](#)
  - CoArgImplode, [627](#)
  - CoArgToExtraSymbol, [628](#)
  - CoArgument, [618](#)
  - CoBracket, [624](#)
  - CoBreak, [631](#)
  - CoCase, [631](#)
  - CoChainin, [622](#)
  - CoChainout, [622](#)
  - CoChisholm, [621](#)
  - CoClearTable, [628](#)
  - CoClearUserFlag, [632](#)
  - CoCollect, [612](#)
  - CoCompress, [612](#)
  - CoContract, [617](#)
  - CoCreateAll, [632](#)
  - CoCreateAllLoops, [632](#)
  - CoCreateAllPaths, [632](#)
  - CoCycleSymmetrize, [620](#)
  - CoDefault, [631](#)
  - CoDelete, [613](#)
  - CoDenominators, [628](#)
  - CoDiscard, [617](#)
  - CoDo, [629](#)
  - CoDrop, [615](#)
  - CoDropCoefficient, [628](#)
  - CoDropSymbols, [628](#)
  - CoElse, [624](#)
  - CoElseif, [625](#)
  - CoEndArgument, [618](#)
  - CoEndDo, [629](#)
  - CoEndIf, [625](#)
  - CoEndInExpression, [623](#)
  - CoEndInside, [618](#)
  - CoEndRepeat, [623](#)
  - CoEndSwitch, [631](#)
  - CoEndTerm, [626](#)
  - CoEndWhile, [625](#)
  - CoExit, [622](#)
  - CoExtraSymbols, [629](#)
  - CoFactArg, [619](#)
  - CoFactDollar, [629](#)
  - CoFactorize, [630](#)
  - CoFindLoop, [625](#)
  - CoFixIndex, [614](#)
  - CoFlags, [613](#)
  - CoFormat, [612](#)
  - CoFromPolynomial, [628](#)
  - CoFunPowers, [626](#)

- CoGoTo, 617
- CoHide, 616
- Colf, 624
- ColnExpression, 623
- ColnParallel, 622
- ColnInside, 618
- ColnInsideFirst, 613
- ColnIntoHide, 616
- CoKeep, 613
- CoLabel, 618
- CoMakeInteger, 619
- CoMerge, 627
- CoMetric, 614
- CoModulus, 623
- CoMultiBracket, 624
- CoNFactorize, 630
- CoNoDrop, 615
- CoNoHide, 616
- CoNoIntoHide, 616
- CoNormalize, 618
- CoNoSkip, 616
- CoNotInParallel, 622
- CoNoUnHide, 616
- CoNPrint, 614
- CoNUnFactorize, 630
- CoNWrite, 620
- CoOff, 613
- CoOn, 613
- CoOptimizeOption, 630
- CoPolyFun, 626
- CoPolyRatFun, 626
- CoPopHide, 615
- CoPrint, 614
- CoPrintB, 614
- CoProcessBucket, 627
- CoProperCount, 613
- CoPushHide, 614
- CoPutInside, 630
- CoRatio, 620
- CoRCycleSymmetrize, 620
- CoRedefine, 620
- CoRenumber, 621
- CoRepeat, 623
- CoReplaceLoop, 625
- CoSetExitFlag, 623
- CoSetUserFlag, 632
- CoSkip, 615
- CoSort, 626
- CoSplitArg, 619
- CoSplitFirstArg, 619
- CoSplitLastArg, 619
- CoStuff, 627
- CoSum, 621
- CoSwitch, 631
- CoSymmetrize, 619
- CoTerm, 626
- CoThreadBucket, 627
- CoToPolynomial, 628
- CoToTensor, 621
- CoToVector, 621
- CoTrace4, 621
- CoTraceN, 621
- CoTryReplace, 623
- CoUnFactorize, 630
- CoUnHide, 616
- CoUnitTrace, 626
- CountComp, 624
- CoWhile, 625
- CoWrite, 620
- DoArgPlode, 627
- DoArgument, 618
- DoBrackets, 624
- DoChain, 622
- DoFactorize, 630
- DoFindLoop, 625
- DoInParallel, 622
- DoPrint, 614
- DoPutInside, 631
- DoSymmetrize, 619
- GetDoParam, 629
- GetIfDollarFactor, 629
- SetExpr, 615
- SetExprCases, 615
- setonoff, 612
- CompGroup
  - declare.h, 830
  - function.c, 1471
- CompileAlgebra
  - compiler.c, 725
  - declare.h, 957
- CompileLevel
  - C\_const, 56
- compiler.c, 722, 727
  - alfatable1, 726
  - CodeFactors, 726
  - CodeGenerator, 725
  - CompileAlgebra, 725
  - CompileStatement, 725
  - CompileSubExpressions, 725
  - CompleteTerm, 726
  - findcommand, 724
  - GenerateFactors, 726
  - inictable, 724
  - insubexpbuffers, 727
  - IsIdStatement, 725
  - IsRHS, 724
  - OPTION0, 723
  - OPTION1, 723
  - OPTION2, 723
  - ParenthesesTest, 724
  - REDUCESUBEXPBUFFERS, 724
  - SkipAName, 724
  - subexpbuffers, 726
  - TestTables, 725
  - topsubexpbuffers, 726
- COMPILERBUFFER

- fsizes.h, 1346
- CompileStatement
  - compiler.c, 725
  - declare.h, 920
- CompileSubExpressions
  - compiler.c, 725
  - declare.h, 960
- compiletype
  - C\_const, 54
- COMPINC
  - fsizes.h, 1347
- CompleteTerm
  - compiler.c, 726
  - declare.h, 961
- complex
  - FuNcTiOn, 145
  - SyMbOl, 425
  - VeCtOr, 484
- compMat
  - MGraph, 208
- ComposeTableNames
  - declare.h, 987
  - minos.c, 1732
- ComPress
  - declare.h, 923
  - sort.c, 2724
- compress.c, 753, 754
- COMPRESSBUFFER
  - fsizes.h, 1345
- CompressBuffer
  - R\_const, 371
- compression
  - ExPrEsSiOn, 123
- CompressPointer
  - R\_const, 372
- CompressSize
  - InDeXeNtRy, 154
  - M\_const, 174
- compressSize
  - N\_const, 255
- compressSpace
  - N\_const, 251
- ComprTop
  - R\_const, 372
- COMPTREE
  - structs.h, 2900
- comsym
  - T\_const, 436
- comtool.c, 762, 768
  - AddLHS, 764
  - AddNtoC, 766
  - AddNtoL, 765
  - AddRHS, 765
  - clearcbuf, 764
  - ClearTree, 767
  - DoubleCbuffer, 764
  - FindTree, 767
  - finishcbuf, 763
  - inicbufs, 763
  - IniFbuffer, 767
  - InsTree, 767
  - numcommute, 768
  - RedoTree, 767
- comtool.h, 775, 776
- CoMulti
  - comexpr.c, 583
  - declare.h, 944
- CoMultiBracket
  - compcomm.c, 624
  - declare.h, 936
- CoMultiply
  - comexpr.c, 584
  - declare.h, 944
- CoNFactorize
  - compcomm.c, 630
  - declare.h, 940
- CoNFunction
  - declare.h, 945
  - names.c, 1830
- conid
  - EEdge, 99
- CONJUGATION
  - ftypes.h, 1396
- connComp
  - EGraph, 105
- connect
  - Assign, 19
- connectClass
  - MGraph, 209
- connectLeg
  - MGraph, 209
- connectNode
  - MGraph, 209
- connVisit
  - EGraph, 106
- CoNoDrop
  - compcomm.c, 615
  - declare.h, 945
- CoNoHide
  - compcomm.c, 616
  - declare.h, 950
- CoNoIntoHide
  - compcomm.c, 616
  - declare.h, 950
- CoNormalize
  - compcomm.c, 618
  - declare.h, 945
- CoNoSkip
  - compcomm.c, 616
  - declare.h, 945
- CoNotInParallel
  - compcomm.c, 622
  - declare.h, 935
- CoNoUnHide
  - compcomm.c, 616
  - declare.h, 950

CoNPrint  
   compcomm.c, 614  
   declare.h, 945  
 ConstructName  
   declare.h, 1027  
   pre.c, 2333  
 CoNTable  
   declare.h, 952  
   names.c, 1831  
 CoNTensor  
   declare.h, 945  
   names.c, 1830  
 ContentMerge  
   declare.h, 978  
   execute.c, 1160  
 contents  
   DoLoOp, 91  
 CONTENTTERM  
   ftypes.h, 1401  
 CONTRACT  
   ftypes.h, 1415  
 CoNUnFactorize  
   compcomm.c, 630  
   declare.h, 940  
 ConvertArgument  
   declare.h, 1012  
   ratio.c, 2506  
 convertblock  
   declare.h, 984  
   minos.c, 1730  
 ConvertFromPoly  
   declare.h, 1005  
   notation.c, 1939  
 convertiniinfo  
   declare.h, 984  
   minos.c, 1730  
 convertnamesblock  
   declare.h, 984  
   minos.c, 1730  
 ConvertToPoly  
   declare.h, 1005  
   notation.c, 1938  
 ConWord  
   declare.h, 902  
   tools.c, 3085  
 CoNWrite  
   compcomm.c, 620  
   declare.h, 945  
 CoOff  
   compcomm.c, 613  
   declare.h, 946  
 CoOn  
   compcomm.c, 613  
   declare.h, 946  
 CoOnce  
   comexpr.c, 584  
   declare.h, 946  
 CoOnly  
   comexpr.c, 584  
   declare.h, 946  
 CoOptimizeOption  
   compcomm.c, 630  
   declare.h, 946  
 CoParticle  
   declare.h, 1028  
 CoPolyFun  
   compcomm.c, 626  
   declare.h, 947  
 CoPolyRatFun  
   compcomm.c, 626  
   declare.h, 947  
 CoPopHide  
   compcomm.c, 615  
   declare.h, 949  
 CoPrint  
   compcomm.c, 614  
   declare.h, 947  
 CoPrintB  
   compcomm.c, 614  
   declare.h, 947  
 CoPrintTable  
   comexpr.c, 585  
   declare.h, 966  
 CoProcessBucket  
   compcomm.c, 627  
   declare.h, 949  
 CoProperCount  
   compcomm.c, 613  
   declare.h, 947  
 CoPushHide  
   compcomm.c, 614  
   declare.h, 949  
 CoPutInside  
   compcomm.c, 630  
   declare.h, 935  
 copy  
   EEdge, 97  
   EGraph, 104  
   ENode, 116  
   MNodeClass, 221  
   T\_MNodeClass, 454  
 COPY1ARG  
   declare.h, 808  
 COPYARG  
   declare.h, 806  
 CopyArg  
   declare.h, 806  
 CopyExpression  
   declare.h, 988  
   store.c, 2833  
 CopyFile  
   declare.h, 896  
   tools.c, 3081  
 COPYFUN  
   declare.h, 807  
 COPYFUN3

- declare.h, [807](#)
- COPYLONG
  - reken.c, [2551](#)
- COPYSUB
  - declare.h, [807](#)
- CopyTree
  - declare.h, [891](#)
  - names.c, [1827](#)
- CoRatio
  - compcomm.c, [620](#)
  - declare.h, [948](#)
- CoRCycleSymmetrize
  - compcomm.c, [620](#)
  - declare.h, [947](#)
- CoRedefine
  - compcomm.c, [620](#)
  - declare.h, [948](#)
- CoRemoveSpectator
  - declare.h, [1024](#)
  - spectator.c, [2787](#)
- CoRenumber
  - compcomm.c, [621](#)
  - declare.h, [948](#)
- CoRepeat
  - compcomm.c, [623](#)
  - declare.h, [948](#)
- CoReplaceLoop
  - compcomm.c, [625](#)
  - declare.h, [964](#)
- CoSave
  - declare.h, [948](#)
  - store.c, [2829](#)
- CoSelect
  - comexpr.c, [584](#)
  - declare.h, [948](#)
- CoSet
  - declare.h, [948](#)
  - names.c, [1831](#)
- CoSetExitFlag
  - compcomm.c, [623](#)
  - declare.h, [949](#)
- CoSetUserFlag
  - compcomm.c, [632](#)
  - declare.h, [1029](#)
- COSFUNCTION
  - ftypes.h, [1399](#)
- COSHFUNCTION
  - ftypes.h, [1399](#)
- CoSkip
  - compcomm.c, [615](#)
  - declare.h, [949](#)
- CoSort
  - compcomm.c, [626](#)
  - declare.h, [950](#)
- CoSplitArg
  - compcomm.c, [619](#)
  - declare.h, [950](#)
- CoSplitFirstArg
  - compcomm.c, [619](#)
  - declare.h, [950](#)
- CoSplitLastArg
  - compcomm.c, [619](#)
  - declare.h, [951](#)
- COST, [75](#)
  - add, [76](#)
  - div, [76](#)
  - mul, [76](#)
  - pow, [76](#)
- CoStuffle
  - compcomm.c, [627](#)
  - declare.h, [944](#)
- CoSum
  - compcomm.c, [621](#)
  - declare.h, [951](#)
- CoSwitch
  - compcomm.c, [631](#)
  - declare.h, [954](#)
- CoSymbol
  - declare.h, [951](#)
  - names.c, [1828](#)
- CoSymmetrize
  - compcomm.c, [619](#)
  - declare.h, [951](#)
- CoTable
  - declare.h, [951](#)
  - names.c, [1830](#)
- CoTableBase
  - declare.h, [937](#)
  - tables.c, [2962](#)
- CoTBaddto
  - declare.h, [955](#)
  - tables.c, [2962](#)
- CoTBaudit
  - declare.h, [955](#)
  - tables.c, [2963](#)
- CoTBCleanup
  - declare.h, [955](#)
  - tables.c, [2964](#)
- CoTBcreate
  - declare.h, [955](#)
  - tables.c, [2962](#)
- CoTBenter
  - declare.h, [955](#)
  - tables.c, [2963](#)
- CoTBhelp
  - declare.h, [956](#)
  - tables.c, [2964](#)
- CoTBload
  - declare.h, [956](#)
  - tables.c, [2963](#)
- CoTBoff
  - declare.h, [956](#)
  - tables.c, [2964](#)
- CoTBon
  - declare.h, [956](#)
  - tables.c, [2963](#)

- CoTBopen
  - declare.h, [956](#)
  - tables.c, [2962](#)
- CoTBreplace
  - declare.h, [956](#)
  - tables.c, [2964](#)
- CoTBus
  - declare.h, [956](#)
  - tables.c, [2964](#)
- CoTerm
  - compcomm.c, [626](#)
  - declare.h, [951](#)
- CoTestUse
  - declare.h, [957](#)
  - tables.c, [2963](#)
- CoThreadBucket
  - compcomm.c, [627](#)
  - declare.h, [957](#)
- CoToFloat
  - float.c, [1293](#)
- CoToPolynomial
  - compcomm.c, [628](#)
  - declare.h, [1006](#)
- CoToRat
  - float.c, [1294](#)
- CoToSpectator
  - declare.h, [1024](#)
  - spectator.c, [2787](#)
- CoToTensor
  - compcomm.c, [621](#)
  - declare.h, [952](#)
- CoToVector
  - compcomm.c, [621](#)
  - declare.h, [952](#)
- CoTrace4
  - compcomm.c, [621](#)
  - declare.h, [952](#)
- CoTraceN
  - compcomm.c, [621](#)
  - declare.h, [952](#)
- CoTransform
  - declare.h, [953](#)
  - transform.c, [3146](#)
- CoTryReplace
  - compcomm.c, [623](#)
  - declare.h, [953](#)
- COULDCOMMUTE
  - ftypes.h, [1451](#)
- CoUnFactorize
  - compcomm.c, [630](#)
  - declare.h, [940](#)
- CoUnHide
  - compcomm.c, [616](#)
  - declare.h, [950](#)
- CoUnitTrace
  - compcomm.c, [626](#)
  - declare.h, [947](#)
- count
  - PNodeClass, [336](#)
- count\_operators
  - optimize.cc, [1990](#), [1991](#)
- count\_operators\_cse
  - optimize.cc, [1996](#)
- count\_operators\_cse\_topdown
  - optimize.cc, [1997](#)
- CountComp
  - compcomm.c, [624](#)
  - declare.h, [933](#)
- CountDo
  - declare.h, [831](#)
  - reshuf.c, [2610](#)
- counter
  - ExPrEsSiOn, [122](#)
- CountFun
  - declare.h, [831](#)
  - reshuf.c, [2610](#)
- COUNTFUNCTION
  - ftypes.h, [1396](#)
- countfunnum
  - M\_const, [189](#)
- CountTerms1
  - declare.h, [983](#)
  - execute.c, [1160](#)
- coupl
  - FGInput, [127](#)
- couple
  - PNodeClass, [336](#)
- couplings
  - MoDeL, [233](#)
  - VeRtEx, [485](#)
- CoVector
  - declare.h, [954](#)
  - names.c, [1829](#)
- CoVertex
  - declare.h, [1029](#)
- CoWhile
  - compcomm.c, [625](#)
  - declare.h, [954](#)
- CoWrite
  - compcomm.c, [620](#)
  - declare.h, [954](#)
- cple
  - NCInput, [269](#)
  - PGInput, [331](#)
- cplgcp
  - Model, [231](#)
- cplglg
  - Model, [231](#)
- cplgnvl
  - Model, [231](#)
- cplgvl
  - Model, [231](#)
- cpilleft
  - Assign, [25](#)
- cprocedureExtension
  - P\_const, [312](#)

- CRANGE
  - ftypes.h, [1379](#)
- Crash
  - declare.h, [983](#)
  - tools.c, [3093](#)
- CreateExpression
  - declare.h, [1011](#)
  - ratio.c, [2504](#)
- CreateFile
  - declare.h, [895](#)
  - tools.c, [3081](#)
- CreateHandle
  - declare.h, [896](#)
  - tools.c, [3081](#)
- CreateLogFile
  - declare.h, [895](#)
  - tools.c, [3081](#)
- CreateStream
  - declare.h, [928](#)
  - tools.c, [3080](#)
- CreateVertex
  - declare.h, [1028](#)
- csav
  - SGroup, [388](#)
- CSEEq, [76](#)
  - operator(), [77](#)
- CSEHash, [77](#)
  - operator(), [77](#)
- CSET
  - ftypes.h, [1378](#)
- CSUBEXP
  - ftypes.h, [1378](#)
- csum
  - Interaction, [161](#)
- CSYMBOL
  - ftypes.h, [1377](#)
- cTable
  - FixedGlobals, [137](#)
- CTD
  - minos.c, [1729](#)
- cTerm
  - N\_const, [246](#)
- ctloop
  - MCOpi, [204](#)
- ctotal
  - Process, [365](#)
- ctyp
  - PGInput, [331](#)
- CurBufWrt
  - O\_const, [280](#)
- curcl
  - MGraph, [212](#)
  - T\_MGraph, [450](#)
- CurDictFunWithArgs
  - O\_const, [283](#)
- CurDictInDollars
  - O\_const, [284](#)
- CurDictNotInFunctions
  - O\_const, [284](#)
- CurDictNumbers
  - O\_const, [283](#)
- CurDictNumberWarning
  - O\_const, [283](#)
- CurDictSpecials
  - O\_const, [283](#)
- CurDictVariables
  - O\_const, [283](#)
- curdirp
  - setfile.c, [2692](#)
- CurDum
  - R\_const, [374](#)
- CurExpr
  - R\_const, [377](#)
- current
  - TaBlEbAsE, [457](#)
- CURRENTBRACKET
  - execute.c, [1157](#)
- CurrentDictionary
  - O\_const, [283](#)
- currentExternalChannel
  - X\_const, [488](#)
- currentMODEL
  - TERMINFO, [466](#)
- currentModel
  - TERMINFO, [466](#)
- currentPrompt
  - X\_const, [487](#)
- CurrentStream
  - C\_const, [46](#)
- currentTerm
  - N\_const, [251](#)
- cursortdirp
  - setfile.c, [2692](#)
- cut
  - EEdge, [99](#)
- cvallist
  - lInput, [150](#)
- CVECTOR
  - ftypes.h, [1377](#)
- CVECTOR1
  - ftypes.h, [1379](#)
- cycle
  - DiStRiBuTe, [86](#)
  - PeRmUtE, [327](#)
  - PeRmUtEp, [328](#)
- CYCLE1
  - declare.h, [805](#)
- CYCLEARG
  - ftypes.h, [1446](#)
- CYCLESYMMETRIC
  - ftypes.h, [1432](#)
- d
  - flint::fmpz, [139](#)
  - flint::mpoly, [240](#)
  - flint::mpoly\_ctx, [241](#)
  - flint::mpoly\_factor, [242](#)

- flint::poly, 338
- flint::poly\_factor, 357
- DAEMON
  - pre.c, 2316
- daemonize
  - X\_const, 488
- data
  - node, 273
- date
  - objects, 287
- dbase, 78
  - fullname, 80
  - handle, 79
  - iblocks, 79
  - info, 78
  - mode, 78
  - name, 79
  - nblocks, 79
  - rwmode, 79
  - tablenamefill, 79
  - tablenames, 80
  - tablenamessize, 79
  - topnumber, 79
- DBGOUT
  - parallel.c, 2065
- DBGOUT\_NINTERMS
  - parallel.c, 2065
- dbufnum
  - M\_const, 179
- DeAllocFileHandle
  - declare.h, 973
  - setfile.c, 2691
- DEBUG
  - wildcard.c, 3222
- DebugFlag
  - P\_const, 316
- debugFlags
  - C\_const, 70
- debugflags
  - OPTIMIZE, 294
- DECLARATION
  - ftypes.h, 1375
- declare.h, 777, 1032
  - ABS, 800
  - AccumGCD, 824
  - Add2Com, 814
  - Add2ComStrings, 928
  - ADD2POS, 812
  - Add3Com, 814
  - Add4Com, 814
  - Add5Com, 814
  - AddArgs, 824
  - AddArrayIndex, 822
  - AddCoef, 824
  - AddComString, 957
  - AddDictionary, 1021
  - AddDollar, 892
  - AddDubious, 893
  - AddExpression, 892
  - AddFunction, 893
  - AddIndex, 893
  - AddLHS, 925
  - AddLineFeed, 801
  - AddLong, 825
  - AddName, 889
  - AddNtoC, 927
  - AddNtoL, 926
  - AddObject, 987
  - AddPLon, 825
  - AddPoly, 825
  - ADDPOS, 810
  - AddPotModdollar, 968
  - AddRat, 827
  - AddRHS, 926
  - AddSet, 894
  - AddSymbol, 892
  - AddTableName, 987
  - AddToCB, 805
  - AddToCom, 928
  - AddToDictionary, 1021
  - AddToDollarBuffer, 962
  - AddToIndex, 985
  - AddToLine, 827
  - AddToPreTypes, 924
  - AddToScratch, 1025
  - AddToString, 916
  - AddVector, 893
  - AddWild, 827
  - AddWildcardName, 924
  - AdjustRenumScratch, 847
  - AllLoops, 1030
  - AllocFileHandle, 973
  - AllocSetups, 903
  - AllocSort, 904
  - AllocSortFileName, 904
  - AllPaths, 1031
  - Apply, 838
  - ApplyExec, 839
  - ApplyReset, 839
  - AreArgsEqual, 901
  - ArgDotproductMerge, 932
  - ArgFactorize, 929
  - ArgSymbolMerge, 932
  - ArgumentExplode, 982
  - ArgumentImplode, 982
  - AssignDollar, 962
  - BACKINOUT, 806
  - BASEPOSITION, 812
  - Bernoulli, 844
  - BigLong, 828
  - BinarySearch, 974
  - BinomGen, 828
  - BracketNormalize, 858
  - CacheNumberFree, 816
  - CacheNumberMalloc, 816
  - CacheNumberMallocAddMemory, 1000



CatchDollar, [962](#)  
ChainIn, [982](#)  
ChainOut, [982](#)  
CharOut, [905](#)  
CheckRecoveryFile, [997](#)  
CheckTableDeclarations, [838](#)  
CheckWild, [828](#)  
Chisholm, [828](#)  
CleanDollarFactors, [966](#)  
CleanExpr, [828](#)  
CleanUp, [829](#)  
CleanupArgCache, [931](#)  
CleanupSort, [973](#)  
CleanupTerm, [977](#)  
ClearBracketIndex, [980](#)  
clearcbuf, [972](#)  
ClearMacro, [907](#)  
ClearOptimize, [970](#)  
ClearPushback, [904](#)  
ClearSpectators, [1024](#)  
ClearTableTree, [971](#)  
ClearTree, [962](#)  
ClearWild, [829](#)  
ClearWildcardNames, [924](#)  
closeAllExternalChannels, [990](#)  
CloseChannel, [898](#)  
closeExternalChannel, [990](#)  
CloseFile, [895](#)  
CloseStream, [888](#)  
CmpArray, [902](#)  
CoAntiBracket, [933](#)  
CoAntiPutInside, [936](#)  
CoAntiSymmetrize, [934](#)  
CoApply, [937](#)  
CoArgExplode, [934](#)  
CoArgImplode, [934](#)  
CoArgToExtraSymbol, [1007](#)  
CoArgument, [934](#)  
CoAssign, [960](#)  
CoAuto, [954](#)  
CoBracket, [935](#)  
CoBreak, [954](#)  
CoCanonicalize, [835](#)  
CoCase, [954](#)  
CoCFunction, [936](#)  
CoChainin, [953](#)  
CoChainout, [953](#)  
CoChisholm, [953](#)  
CoClearTable, [953](#)  
CoClearUserFlag, [1029](#)  
CoCollect, [936](#)  
CoCommutelnSet, [894](#)  
CoCompress, [936](#)  
CoContract, [936](#)  
CoCopySpectator, [1024](#)  
CoCreateAll, [1030](#)  
CoCreateAllLoops, [1029](#)  
CoCreateAllPaths, [1029](#)  
CoCreateSpectator, [1023](#)  
CoCTable, [952](#)  
CoCTensor, [936](#)  
CoCycleSymmetrize, [937](#)  
CoDeallocateTable, [966](#)  
CodeFactors, [961](#)  
CoDefault, [955](#)  
CodeGenerator, [960](#)  
CoDelete, [937](#)  
CoDenominators, [937](#)  
CodeToLine, [822](#)  
CoDimension, [937](#)  
CoDiscard, [937](#)  
CoDisorder, [938](#)  
CoDo, [1007](#)  
CoDrop, [938](#)  
CoDropCoefficient, [938](#)  
CoDropSymbols, [938](#)  
CoElse, [938](#)  
CoElseif, [938](#)  
CoEmptySpectator, [1024](#)  
CoEndArgument, [938](#)  
CoEndDo, [1007](#)  
CoEndIf, [939](#)  
CoEndInExpression, [935](#)  
CoEndInside, [939](#)  
CoEndModel, [1029](#)  
CoEndRepeat, [939](#)  
CoEndSwitch, [955](#)  
CoEndTerm, [939](#)  
CoEndWhile, [939](#)  
CoExit, [939](#)  
CoExtraSymbols, [1007](#)  
CoFactArg, [939](#)  
CoFactDollar, [940](#)  
CoFactorize, [940](#)  
CoFill, [940](#)  
CoFillExpression, [941](#)  
CoFindLoop, [964](#)  
CoFixIndex, [941](#)  
CoFlags, [946](#)  
CoFormat, [941](#)  
CoFromPolynomial, [1006](#)  
CoFunction, [894](#)  
CoFunPowers, [964](#)  
CoGlobal, [941](#)  
CoGlobalFactorized, [941](#)  
CoGoTo, [941](#)  
CoHide, [949](#)  
Cold, [941](#)  
ColdExpression, [960](#)  
ColdNew, [942](#)  
ColdOld, [942](#)  
Colf, [942](#)  
ColfMatch, [942](#)  
ColfNoMatch, [942](#)  
ColIndex, [942](#)  
ColnExpression, [935](#)

- CoInParallel, [935](#)
- CoInside, [934](#)
- CoInsideFirst, [942](#)
- CoIntoHide, [949](#)
- CoKeep, [943](#)
- CoLabel, [943](#)
- CoLoad, [943](#)
- CoLocal, [943](#)
- CoLocalFactorized, [943](#)
- CoMakeInteger, [946](#)
- CoMany, [943](#)
- CoMerge, [943](#)
- CoMetric, [944](#)
- Commute, [829](#)
- CoModel, [1028](#)
- CoModOption, [944](#)
- CoModuleOption, [944](#)
- CoModulus, [944](#)
- CompactifyTree, [891](#)
- Compare1, [830](#)
- CompareArgs, [901](#)
- CompareFunctions, [829](#)
- CompareHSymbols, [991](#)
- CompareSymbols, [991](#)
- CompareTerms, [820](#)
- CompArg, [829](#)
- CompCoef, [830](#)
- CompGroup, [830](#)
- CompileAlgebra, [957](#)
- CompileStatement, [920](#)
- CompileSubExpressions, [960](#)
- CompleteTerm, [961](#)
- ComposeTableNames, [987](#)
- ComPress, [923](#)
- CoMulti, [944](#)
- CoMultiBracket, [936](#)
- CoMultiply, [944](#)
- CoNFactorize, [940](#)
- CoNFunction, [945](#)
- CoNoDrop, [945](#)
- CoNoHide, [950](#)
- CoNoIntoHide, [950](#)
- CoNormalize, [945](#)
- CoNoSkip, [945](#)
- CoNotInParallel, [935](#)
- CoNoUnHide, [950](#)
- CoNPrint, [945](#)
- ConstructName, [1027](#)
- CoNTable, [952](#)
- CoNTensor, [945](#)
- ContentMerge, [978](#)
- CoNUnFactorize, [940](#)
- ConvertArgument, [1012](#)
- convertblock, [984](#)
- ConvertFromPoly, [1005](#)
- convertiniinfo, [984](#)
- convertnamesblock, [984](#)
- ConvertToPoly, [1005](#)
- ConWord, [902](#)
- CoNWrite, [945](#)
- CoOff, [946](#)
- CoOn, [946](#)
- CoOnce, [946](#)
- CoOnly, [946](#)
- CoOptimizeOption, [946](#)
- CoParticle, [1028](#)
- CoPolyFun, [947](#)
- CoPolyRatFun, [947](#)
- CoPopHide, [949](#)
- CoPrint, [947](#)
- CoPrintB, [947](#)
- CoPrintTable, [966](#)
- CoProcessBucket, [949](#)
- CoProperCount, [947](#)
- CoPushHide, [949](#)
- CoPutInside, [935](#)
- COPY1ARG, [808](#)
- COPYARG, [806](#)
- CopyArg, [806](#)
- CopyExpression, [988](#)
- CopyFile, [896](#)
- COPYFUN, [807](#)
- COPYFUN3, [807](#)
- COPYSUB, [807](#)
- CopyTree, [891](#)
- CoRatio, [948](#)
- CoRCycleSymmetrize, [947](#)
- CoRedefine, [948](#)
- CoRemoveSpectator, [1024](#)
- CoRenumber, [948](#)
- CoRepeat, [948](#)
- CoReplaceLoop, [964](#)
- CoSave, [948](#)
- CoSelect, [948](#)
- CoSet, [948](#)
- CoSetExitFlag, [949](#)
- CoSetUserFlag, [1029](#)
- CoSkip, [949](#)
- CoSort, [950](#)
- CoSplitArg, [950](#)
- CoSplitFirstArg, [950](#)
- CoSplitLastArg, [951](#)
- CoStuffle, [944](#)
- CoSum, [951](#)
- CoSwitch, [954](#)
- CoSymbol, [951](#)
- CoSymmetrize, [951](#)
- CoTable, [951](#)
- CoTableBase, [937](#)
- CoTBaddto, [955](#)
- CoTBaudit, [955](#)
- CoTBCleanup, [955](#)
- CoTBcreate, [955](#)
- CoTBenter, [955](#)
- CoTBhelp, [956](#)
- CoTBload, [956](#)

CoTBoff, [956](#)  
CoTBon, [956](#)  
CoTBopen, [956](#)  
CoTBreplace, [956](#)  
CoTBuse, [956](#)  
CoTerm, [951](#)  
CoTestUse, [957](#)  
CoThreadBucket, [957](#)  
CoToPolynomial, [1006](#)  
CoToSpectator, [1024](#)  
CoToTensor, [952](#)  
CoToVector, [952](#)  
CoTrace4, [952](#)  
CoTraceN, [952](#)  
CoTransform, [953](#)  
CoTryReplace, [953](#)  
CoUnFactorize, [940](#)  
CoUnHide, [950](#)  
CoUnitTrace, [947](#)  
CountComp, [933](#)  
CountDo, [831](#)  
CountFun, [831](#)  
CountTerms1, [983](#)  
CoVector, [954](#)  
CoVertex, [1029](#)  
CoWhile, [954](#)  
CoWrite, [954](#)  
Crash, [983](#)  
CreateExpression, [1011](#)  
CreateFile, [895](#)  
CreateHandle, [896](#)  
CreateLogFile, [895](#)  
CreateStream, [928](#)  
CreateVertex, [1028](#)  
CYCLE1, [805](#)  
DeAllocFileHandle, [973](#)  
Deferred, [832](#)  
defineChannel, [990](#)  
DeleteObject, [986](#)  
DeleteRecoveryFile, [998](#)  
DeleteStore, [833](#)  
DenToFunction, [983](#)  
DetCommu, [829](#)  
DetCurDum, [833](#)  
DetVars, [833](#)  
DictFromBytes, [1023](#)  
DictToBytes, [1023](#)  
DIFBASE, [811](#)  
DIFPOS, [811](#)  
DimensionExpression, [832](#)  
DimensionSubterm, [831](#)  
DimensionTerm, [831](#)  
Distribute, [833](#)  
DistrN, [1027](#)  
DIVfunction, [1011](#)  
DivLong, [834](#)  
DIVPOS, [811](#)  
DivRat, [834](#)  
Divvy, [834](#)  
DoArgPlode, [934](#)  
DoArgument, [929](#)  
DoBrackets, [933](#)  
DoBreakDo, [914](#)  
DoCall, [913](#)  
DoCanonicalize, [835](#)  
DoChain, [953](#)  
DoCheckpoint, [999](#)  
DoClearOptimize, [910](#)  
DoClearUserFlag, [1028](#)  
DoCommentChar, [916](#)  
DoContinueDo, [914](#)  
DoDebug, [914](#)  
DoDefine, [906](#)  
DoDelta, [834](#)  
DoDelta3, [834](#)  
DoDimension, [893](#)  
DoDistrib, [836](#)  
DoDo, [914](#)  
DoElements, [894](#)  
DoElse, [913](#)  
DoElseif, [913](#)  
DoEnddo, [914](#)  
DoEndif, [913](#)  
DoEndInside, [915](#)  
DoEndNamespace, [1027](#)  
DoEndprocedure, [914](#)  
DoEndSwitch, [1026](#)  
DoesCommu, [829](#)  
DoExecStatement, [920](#)  
DoExecute, [885](#)  
DoExpr, [960](#)  
DoExternal, [908](#)  
DoFactDollar, [909](#)  
DoFactorize, [940](#)  
DoFindLoop, [964](#)  
DoFromExternal, [908](#)  
Dolf, [913](#)  
Dolfdef, [912](#)  
Dolfndef, [913](#)  
DolfStatement, [840](#)  
Dolfydef, [912](#)  
DoInclude, [908](#)  
DoInParallel, [935](#)  
DoinParallel, [981](#)  
DoInside, [914](#)  
DollarFactorize, [966](#)  
DOLLARNAME, [809](#)  
DollarRaiseLow, [893](#)  
DolToFunction, [965](#)  
DolToIndex, [966](#)  
DolToLong, [966](#)  
DolToNumber, [965](#)  
DolToSymbol, [965](#)  
DolToTensor, [965](#)  
DolToTerms, [974](#)  
DolToVector, [965](#)

- DoMessage, [910](#)
- DoModDollar, [981](#)
- DoModLocal, [981](#)
- DoModMax, [981](#)
- DoModMin, [981](#)
- DoModSum, [981](#)
- DoNamespace, [1027](#)
- DoNoParallel, [980](#)
- DonotinParallel, [982](#)
- DoOnePow, [840](#)
- DoOptimize, [910](#)
- DoParallel, [980](#)
- DoPartitions, [835](#)
- DoPermutations, [837](#)
- DoPipe, [912](#)
- DoPipeStatement, [920](#)
- DoPolyfun, [920](#)
- DoPolyratfun, [920](#)
- DoPrcExtension, [916](#)
- DoPreAdd, [1020](#)
- DoPreAddSeparator, [989](#)
- DoPreAppend, [910](#)
- DoPreAppendPath, [1025](#)
- DoPreAssign, [911](#)
- DoPreBreak, [911](#)
- DoPreCase, [911](#)
- DoPreClose, [915](#)
- DoPreCloseDictionary, [1020](#)
- DoPreCreate, [911](#)
- DoPreDefault, [911](#)
- DoPreEndSwitch, [911](#)
- DoPreExchange, [912](#)
- DoPreOpenDictionary, [1020](#)
- DoPreOut, [910](#)
- DoPrePrependPath, [1025](#)
- DoPrePrintTimes, [915](#)
- DoPreRemove, [915](#)
- DoPreReset, [916](#)
- DoPreRmSeparator, [989](#)
- DoPreShow, [912](#)
- DoPreSortReallocate, [915](#)
- DoPreSwitch, [911](#)
- DoPreUseDictionary, [1020](#)
- DoPreWrite, [915](#)
- DoPrint, [928](#)
- DoProcedure, [915](#)
- DoProcessBucket, [981](#)
- DoPrompt, [909](#)
- DoPutInside, [933](#)
- DoRecovery, [998](#)
- DoRedefine, [906](#)
- DoReverseInclude, [908](#)
- DoRevert, [841](#)
- DoRmExternal, [909](#)
- DoSetExternal, [909](#)
- DoSetExternalAttr, [909](#)
- DoSetRandom, [909](#)
- DoSetups, [889](#)
- DoSetUserFlag, [1028](#)
- DoShattering, [836](#)
- DoShtuffle, [820](#)
- DoShuffle, [837](#)
- DoSkipExtraSymbols, [910](#)
- DoStuffle, [837](#)
- DoSumF1, [841](#)
- DoSumF2, [841](#)
- DoSwitch, [1026](#)
- DoSymmetrize, [929](#)
- DoSystem, [912](#)
- DoTable, [951](#)
- DoTableExpansion, [836](#)
- DoTail, [886](#)
- DoTempSet, [894](#)
- DoTerminate, [913](#)
- DoTheta, [841](#)
- DoTimeOutAfter, [910](#)
- DoToExternal, [908](#)
- DoTopologyCanonicalize, [836](#)
- DoubleBuffer, [900](#)
- DoubleCbuffer, [925](#)
- DoubleIfBuffers, [928](#)
- DoubleList, [900](#)
- DoubleLList, [900](#)
- DoubleSwitchBuffers, [1026](#)
- DoUndefine, [908](#)
- DoUse, [1027](#)
- DropCoefficient, [858](#)
- DropSymbols, [859](#)
- DUMMYUSE, [810](#)
- DumpNode, [891](#)
- DumpTree, [891](#)
- EmptyTable, [952](#)
- EndOfToken, [919](#)
- EndSort, [842](#)
- EntVar, [843](#)
- EpfCon, [843](#)
- EpfFind, [843](#)
- EpfGen, [843](#)
- EqualArg, [843](#)
- Error0, [887](#)
- Error1, [887](#)
- Error2, [887](#)
- EvalDoLoopArg, [974](#)
- EvalPrelf, [918](#)
- EXCH, [801](#)
- ExchangeDollars, [976](#)
- ExchangeExpressions, [976](#)
- EXCHINOUT, [805](#)
- EXCHN, [801](#)
- execarg, [884](#)
- ExecInside, [934](#)
- ExecModule, [919](#)
- ExecStore, [919](#)
- execterm, [884](#)
- ExistsObject, [986](#)
- ExpandBuffer, [900](#)

- ExpandRat, [1012](#)
- ExpandTripleDots, [922](#)
- EXPRNAME, [809](#)
- ExprStatus, [1000](#)
- EXTERNLOCK, [818](#)
- EXTERNRWLOCK, [818](#)
- ExtraSymbol, [858](#)
- ExtraSymFun, [1008](#)
- Factorial, [843](#)
- FactorIn, [844](#)
- FactorInExpr, [844](#)
- FILLARG, [806](#)
- FILLEXP, [808](#)
- FILLFUN, [807](#)
- FILLFUN3, [807](#)
- FILLSUB, [807](#)
- FindAll, [844](#)
- FindArg, [931](#)
- FindBracket, [980](#)
- FindCase, [1026](#)
- findcommand, [898](#)
- FindDictionary, [1021](#)
- FindExtraSymbol, [1023](#)
- FindFunction, [1023](#)
- FindFunWithArgs, [1023](#)
- FindIndex, [1023](#)
- FindInIndex, [822](#)
- FindInKeyWord, [906](#)
- FindKeyWord, [906](#)
- FindLocalSubterm, [1005](#)
- FindLus, [964](#)
- FindMulti, [844](#)
- FindOnce, [844](#)
- FindOnly, [845](#)
- FindRange, [1002](#)
- FindRest, [845](#)
- FindrNumber, [845](#)
- FindSubexpression, [1006](#)
- FindSubterm, [1005](#)
- FindSymbol, [1022](#)
- FindTableNumber, [988](#)
- FindTableTree, [972](#)
- FindTB, [838](#)
- FindTree, [961](#)
- FindVar, [839](#)
- FindVector, [1022](#)
- FinilLine, [845](#)
- finishcbuf, [972](#)
- FinishShuffle, [837](#)
- FinishStuffle, [838](#)
- FinishShuffle, [820](#)
- FinilTerm, [845](#)
- FlipTable, [982](#)
- FLUSHCONSOLE, [804](#)
- FlushFile, [897](#)
- FlushOut, [846](#)
- FlushSpectators, [1025](#)
- FORM\_INLINE, [815](#)
- FreeNameTree, [892](#)
- FreeTableBase, [987](#)
- FromOList, [899](#)
- FromList, [899](#)
- FromPolyRatFun, [1009](#)
- FromVarList, [900](#)
- FullCleanUp, [919](#)
- FullRenummer, [963](#)
- FullSymmetrize, [875](#)
- FunLevel, [846](#)
- FunMatchCy, [854](#)
- FunMatchSy, [855](#)
- GarbHand, [847](#)
- GCDclean, [1009](#)
- GCDfunction, [1008](#)
- GCDfunction3, [1008](#)
- GcdLong, [847](#)
- GCDterms, [1009](#)
- GenDiagrams, [835](#)
- GenerateFactors, [961](#)
- GenerateTopologies, [836](#)
- Generator, [847](#)
- GenLoops, [1030](#)
- GenPaths, [1031](#)
- GenTopologies, [835](#)
- GetAppendChannel, [898](#)
- GetAutoName, [890](#)
- GETBIDENTITY, [819](#)
- GetBinom, [849](#)
- GETCFROMEXTCHANNEL, [820](#)
- GetChannel, [898](#)
- GetChar, [904](#)
- GETCOEF, [800](#)
- GetContent, [977](#)
- getCurrentExternalChannel, [990](#)
- GetDbase, [985](#)
- GetDollar, [920](#)
- GetDollarNumber, [909](#)
- GetDoNum, [968](#)
- GetDoParam, [1007](#)
- GetFirstBracket, [977](#)
- GetFirstTerm, [977](#)
- GetFromSpectator, [1024](#)
- GetFromStore, [850](#)
- GetFromStream, [887](#)
- GetFunction, [890](#)
- GETIDENTITY, [819](#)
- GetIfDollarFactor, [1007](#)
- GetIfDollarNum, [839](#)
- GetInput, [904](#)
- GetLabel, [960](#)
- GetLastExprName, [890](#)
- GetLong, [850](#)
- GetLongModInverses, [868](#)
- GetModInverses, [867](#)
- GetMoreFromMem, [850](#)
- GetMoreTerms, [850](#)
- GetName, [889](#)

GetNode, 889  
GetNumber, 890  
GetOName, 891  
GetOneTerm, 850  
GetPosFile, 897  
GetPreVar, 902  
GetRunningTime, 864  
GetSetupPar, 903  
GETSTOP, 800  
GetStreamPosition, 925  
GetTable, 851  
GetTableName, 988  
GETTERM, 821  
GetTerm, 851  
GetVar, 890  
GetWildcardName, 924  
Globalize, 924  
Glue, 851  
HighWarning, 885  
HowMany, 983  
iexp, 900  
INCLENG, 800  
Include, 908  
InFunction, 851  
inicbufs, 898  
inictable, 898  
IniFbuffer, 1008  
IniLine, 852  
INILOCK, 818  
IniModule, 905  
INIRWLOCK, 818  
IniSpecialModule, 905  
iniTools, 1000  
initPresetExternalChannels, 989  
InitRecovery, 997  
IniVars, 852  
iniwranf, 992  
InsertArg, 931  
InsertTerm, 852  
InsideDollar, 974  
InsTableTree, 972  
InsTree, 961  
InvPoly, 1012  
iranf, 992  
ISEQUALPOS, 812  
ISEQUALPOSINC, 811  
IsExponentSign, 1021  
ISGEPOS, 813  
ISGEPOSINC, 811  
IsIdStatement, 957  
ISLESSPOS, 813  
IsLikeVector, 901  
ISMINPOS, 812  
IsMultipleOf, 979  
IsMultiplySign, 1022  
ISNEGPOS, 814  
ISNOTEQUALPOS, 813  
ISNOTZEROPOS, 813  
ISPOSPOS, 813  
IsRHS, 957  
IsSetMember, 979  
ISZEROPOS, 813  
LcmLong, 847  
LinkTree, 891  
LoadInputFile, 904  
LoadInstruction, 906  
LoadModel, 1029  
LoadStatement, 906  
LocalConvertToPoly, 1005  
LocateBase, 872  
LocateFile, 888  
LOCK, 818  
LongCopy, 921  
LongLongCopy, 921  
LongToLine, 853  
LookInStream, 887  
LoopOutput, 1030  
LowerSortLevel, 973  
Lus, 963  
M\_alloc, 820  
M\_check1, 979  
M\_free, 923  
M\_print, 979  
MakeDate, 899  
MakeDirty, 853  
MakeDollarInteger, 967  
MakeDollarMod, 967  
MakeDubious, 890  
MakeGlobal, 919  
MakeInteger, 930  
MakeInverses, 867  
MakeLongRational, 971  
MakeMod, 930  
MakeModTable, 854  
MakeNameTree, 892  
MakeRational, 971  
MakeSetupAllocs, 976  
Malloc1, 886  
MarkDirty, 853  
MarkPolyRatFunDirty, 817  
MatchArgument, 855  
MatchCy, 854  
MatchE, 854  
MatchFunction, 855  
MatchIsPossible, 965  
MaX, 799  
MEMORYMACROS, 815  
MergeDotproductLists, 1011  
MergePatches, 855  
MergeSymbolLists, 1011  
MesCall, 921  
MesCerr, 856  
MesComp, 856  
MesNesting, 817  
MessPreNesting, 924  
MesWork, 921

MiN, 799  
MLOCK, 819  
ModuleInstruction, 905  
Modulus, 856  
Moebius, 992  
MoveDummies, 857  
MULfunc, 1011  
MulLong, 857  
Mully, 857  
MULPOS, 811  
MulRat, 857  
MultDo, 857  
MultiplyToLine, 1022  
MultiplyWithTerm, 1010  
MUNLOCK, 819  
NameConflict, 895  
NCOPY, 802  
NCOPYB, 803  
NCOPYI, 803  
NCOPYI32, 803  
NeedNumber, 803  
NestingChecksum, 817  
NewDbase, 987  
NewSort, 858  
NEXTARG, 808  
NextFileIndex, 823  
NextPrime, 992  
Normalize, 858  
NormalModulus, 867  
NormPolyTerm, 1004  
NOTSTARTPOS, 812  
NumberFree, 816  
NumberMalloc, 816  
NumberMallocAddMemory, 999  
numcommute, 963  
NumCopy, 921  
NumToStr, 917  
OpenAddFile, 895  
OpenBracketIndex, 980  
OpenDbase, 986  
openExternalChannel, 989  
OpenFile, 895  
OpenInput, 886  
OpenStream, 888  
OpenTemp, 859  
Optimize, 969  
optimize\_print\_code, 1020  
Pack, 859  
ParenthesesTest, 958  
ParseNumber, 802  
ParseSign, 802  
ParseSignedNumber, 802  
PasteFile, 859  
PasteTerm, 823  
PathOutput, 1031  
Permute, 860  
PermuteP, 860  
poly\_div, 1013  
poly\_factorize\_argument, 1016  
poly\_factorize\_dollar, 1017  
poly\_factorize\_expression, 1017  
poly\_free\_poly\_vars, 1019  
poly\_gcd, 1012  
poly\_inverse, 1014  
poly\_mul, 1014  
poly\_ratfun\_add, 1014  
poly\_ratfun\_normalize, 1015  
poly\_rem, 1013  
poly\_unfactorize\_expression, 1018  
PolyDiv, 1009  
PolyFunClean, 854  
PolyFunDirty, 854  
PolyFunMul, 860  
PolyRatFunSpecial, 973  
POPPREASSIGNLEVEL, 817  
PopPreVars, 905  
PopVariables, 861  
PositionStream, 888  
pParseObject, 918  
PreCalc, 917  
PreCmp, 917  
PreEq, 917  
PreEval, 917  
PrelfDollarEval, 978  
PrelfEval, 918  
PreLoad, 918  
PrepPoly, 861  
PreProcessor, 899  
PreProInstruction, 905  
PreRandom, 993  
PreSkip, 918  
PREV, 809  
PrintDeprecation, 864  
PrintExtraSymbol, 1006  
PrintRunningTime, 864  
PrintSeq, 922  
PrintSubTerm, 922  
PrintSubtermList, 1006  
PrintTerm, 922  
PrintTermC, 922  
PrintTime, 980  
PrintWords, 922  
ProcessOption, 888  
Processor, 862  
Product, 863  
PrtLong, 863  
PrtTerms, 863  
PruneExtraSymbols, 1008  
PUSHPREASSIGNLEVEL, 817  
PutArgInScratch, 1002  
PutBracket, 864  
PutBracketInIndex, 980  
PutExtraSymbols, 1010  
PutIn, 864  
PutInside, 859  
PutInSpectator, 1024

- PutInStore, [865](#)
- PutInVflags, [877](#)
- PutOut, [865](#)
- PutPreVar, [886](#)
- PutTableNames, [987](#)
- PutTermInDollar, [963](#)
- PUTZERO, [812](#)
- Quotient, [866](#)
- RaisPow, [866](#)
- RaisPowCached, [866](#)
- RaisPowMod, [867](#)
- RatioFind, [868](#)
- RatioGen, [868](#)
- RatToLine, [868](#)
- ReadFile, [896](#)
- ReadFromScratch, [1025](#)
- ReadFromStream, [887](#)
- ReadijObject, [986](#)
- ReadIndex, [984](#)
- ReadIniInfo, [985](#)
- ReadObject, [986](#)
- ReadParticle, [1028](#)
- ReadPolyRatFun, [1008](#)
- ReadPosFile, [896](#)
- ReadRange, [1002](#)
- ReadSaveExpression, [994](#)
- ReadSaveHeader, [993](#)
- ReadSaveIndex, [993](#)
- ReadSaveTerm32, [995](#)
- ReadSaveVariables, [996](#)
- ReadSnum, [869](#)
- RecalcSetups, [903](#)
- RecoveryFilename, [998](#)
- REDLENG, [800](#)
- RedoTableTree, [972](#)
- RedoTree, [961](#)
- ReleaseTB, [991](#)
- Remain10, [869](#)
- Remain4, [869](#)
- RemoveDictionary, [1021](#)
- RemoveDollars, [983](#)
- ReNumber, [869](#)
- ReOpenFile, [895](#)
- ReplaceDollar, [892](#)
- ReserveTempFiles, [921](#)
- ResetScratch, [869](#)
- ResetVariables, [924](#)
- ResolveSet, [869](#)
- ReverseStatements, [888](#)
- RevertScratch, [870](#)
- ReWorkT, [839](#)
- RunAddArg, [1003](#)
- RunCycle, [1003](#)
- RunDecode, [1001](#)
- RunDedup, [1004](#)
- RunDropArg, [1004](#)
- RunEncode, [1001](#)
- RunExplode, [1002](#)
- RunHtoZArg, [1004](#)
- RunImplode, [1001](#)
- RunIsLyndon, [1003](#)
- RunMulArg, [1003](#)
- RunPermute, [1002](#)
- RunReplace, [1001](#)
- RunReverse, [1003](#)
- RunSelectArg, [1004](#)
- RunToLyndon, [1003](#)
- RunTransform, [1001](#)
- RunZtoHArg, [1004](#)
- RWLOCKR, [819](#)
- RWLOCKW, [819](#)
- ScanFunctions, [870](#)
- SeekFile, [897](#)
- SeekScratch, [870](#)
- selectExternalChannel, [990](#)
- set\_del, [989](#)
- set\_in, [988](#)
- set\_set, [988](#)
- set\_sub, [989](#)
- SETBASELENGTH, [810](#)
- SETBASEPOSITION, [810](#)
- SetDictionaryOptions, [1021](#)
- SetEndHScratch, [870](#)
- SetEndScratch, [870](#)
- SETERROR, [810](#)
- SetExpr, [928](#)
- SetExprCases, [975](#)
- SetFileIndex, [870](#)
- SETKILLMODEFOREXTERNALCHANNEL, [821](#)
- SetMods, [884](#)
- setonoff, [928](#)
- SetPosFile, [897](#)
- SetScratch, [885](#)
- SetSpecialMode, [919](#)
- SETSTARTPOS, [812](#)
- SETTERMINATORFOREXTERNALCHANNEL, [820](#)
- Sflush, [871](#)
- SGN, [800](#)
- ShrinkDictionary, [1022](#)
- Shuffle, [837](#)
- simp1token, [958](#)
- simp2token, [958](#)
- simp3atoken, [958](#)
- simp3btoken, [959](#)
- simp4token, [959](#)
- simp5token, [959](#)
- simp6token, [959](#)
- SimpleSplitMerge, [974](#)
- SimpleSplitMergeRec, [974](#)
- Simplify, [871](#)
- simpwtoken, [958](#)
- SizeOfDollar, [978](#)
- SizeOfExpression, [978](#)
- SkipAName, [959](#)
- SKIPBLANKS, [804](#)



SKIPBRA1, 804  
SKIPBRA2, 804  
SKIPBRA3, 804  
SKIPBRA4, 805  
SKIPBRA5, 805  
SkipField, 901  
SkipName, 1027  
SortTheList, 964  
SortWeights, 932  
SortWild, 871  
SpareTable, 885  
SpecialCleanup, 884  
SplitMerge, 872  
StageSort, 923  
StartFiles, 899  
StartLoops, 1030  
StartPrepro, 912  
StartVariables, 821  
StoreTerm, 873  
str\_dup, 984  
StrCmp, 901  
StrCopy, 823  
strDup1, 885  
StrHICmp, 902  
StrICmp, 901  
StrICont, 902  
StrLen, 902  
StudyPattern, 965  
StuffAdd, 801  
Stuffle, 837  
StuffRootAdd, 838  
SubPLon, 874  
SubsInAll, 975  
Substitute, 874  
SwitchSplitMerge, 1026  
SwitchSplitMergeRec, 1026  
SymbolNormalize, 991  
SymFind, 874  
SymGen, 874  
Symmetrize, 875  
SynchFile, 897  
TableReset, 839  
TABLESIZE, 809  
TakeArgContent, 929  
TakeContent, 1010  
TakeDollarContent, 966  
TakeExtraSymbols, 1010  
TakeIDfunction, 976  
TakeLongRoot, 971  
TakeModulus, 875  
TakeNormalModulus, 875  
TakeRatRoot, 971  
TakeSymbolContent, 1009  
TalToLine, 875  
TELLFILE, 821  
TellFile, 897  
TenVec, 876  
TenVecFind, 876  
TermAssign, 963  
TermFree, 816  
Terminate, 810  
TerminatImpl, 889  
TermMalloc, 816  
TermMallocAddMemory, 1000  
TermRenumbr, 876  
TermsInBracket, 983  
TermsInDollar, 978  
TermsInExpression, 978  
TestArgNum, 1002  
TestDoLoop, 1012  
TestDrop, 876  
TestEndDoLoop, 1012  
TestFunFlag, 991  
TestMatch, 877  
TestName, 894  
TestPartitions, 835  
TestSelect, 975  
TestSub, 878  
TestTables, 959  
TestTerm, 1000  
TestUse, 838  
TheDefine, 907  
TheUndefine, 907  
TimeChildren, 878  
TimeCPU, 878  
TimeWallClock, 879  
ToFast, 903  
ToGeneral, 903  
tokenize, 958  
TOKENTOLINE, 801  
TokenToLine, 879  
TOLONG, 814  
ToPolyFunGeneral, 903  
ToStorage, 879  
ToToken, 920  
Trace4, 879  
Trace4Gen, 879  
Trace4no, 880  
TraceFind, 880  
TraceN, 880  
TraceNgen, 880  
TraceNno, 880  
Traces, 880  
TransferBuffer, 976  
TransformRational, 1022  
TranslateExpression, 978  
TreatPolyRatFun, 993  
Trick, 881  
TruncateFile, 898  
TryDo, 881  
TryEnvironment, 988  
TryFileSetups, 976  
TryRecover, 802  
TwoExprCompare, 979  
UngetChar, 802  
UngetFromStream, 801

- UNLOCK, [818](#)
- UnPack, [881](#)
- UNRWLOCK, [819](#)
- UnsetAllowDelay, [905](#)
- UnSetDictionary, [1021](#)
- UnSetMods, [884](#)
- UpdateMaxSize, [1006](#)
- UpdatePositions, [979](#)
- VARNAME, [809](#)
- VarStore, [881](#)
- WantAddLongs, [815](#)
- WantAddPointers, [815](#)
- WantAddPositions, [815](#)
- Warning, [885](#)
- WCOPY, [803](#)
- WildDollars, [963](#)
- WildFill, [881](#)
- WORDDIF, [809](#)
- wranf, [992](#)
- WriteAll, [882](#)
- WriteArgument, [882](#)
- WRITEBUFTOEXTCHANNEL, [820](#)
- WriteDictionary, [1022](#)
- WriteDollarFactorToBuffer, [962](#)
- WriteDollarToBuffer, [962](#)
- WriteExpression, [882](#)
- WRITEFILE, [821](#)
- WriteFileToFile, [896](#)
- WriteIndex, [985](#)
- WriteIndexBlock, [984](#)
- WriteIniInfo, [985](#)
- WriteInnerTerm, [882](#)
- WriteLists, [882](#)
- WriteNamesBlock, [985](#)
- WriteObject, [986](#)
- WriteOne, [882](#)
- WriteSetup, [883](#)
- WriteStats, [883](#)
- WriteStoreHeader, [997](#)
- WriteString, [916](#)
- WriteSubTerm, [883](#)
- WriteTerm, [884](#)
- writeToChannel, [990](#)
- WriteTokens, [958](#)
- WriteUnfinString, [916](#)
- WrtPower, [823](#)
- wsizeof, [809](#)
- ZEROARG, [806](#)
- ZeroFillRange, [808](#)
- DeclareVector
  - vector.h, [3211](#)
- DECODEARG
  - ftypes.h, [1446](#)
- DEDUPARG
  - ftypes.h, [1447](#)
- defaultcase
  - SWITCH, [421](#)
- DEFAULTFNAMELENGTH
  - startup.c, [2798](#)
- DEFAULTPROCESSBUCKETSIZE
  - fsizes.h, [1348](#)
- defaulttempfilename
  - startup.c, [2801](#)
- DEFAULTTHREADBUCKETSIZE
  - fsizes.h, [1349](#)
- DEFAULTTHREADLOADBALANCING
  - fsizes.h, [1349](#)
- DEFAULTTHREADS
  - fsizes.h, [1348](#)
- defaulttv
  - OptDef, [289](#)
- DeferFlag
  - R\_const, [375](#)
- Deferred
  - declare.h, [832](#)
  - proces.c, [2432](#)
- deferskipped
  - N\_const, [252](#)
- defineChannel
  - declare.h, [990](#)
  - pre.c, [2331](#)
- defined
  - TaBIEs, [462](#)
- DEFINEVALUE
  - setfile.c, [2690](#)
- DEFINITION
  - ftypes.h, [1375](#)
- defpart
  - MInput, [217](#)
  - Model, [231](#)
- DefPosition
  - R\_const, [371](#)
- deg
  - ANode, [13](#)
  - ENode, [117](#)
  - MNode, [219](#)
  - NCand, [267](#)
  - NStack, [275](#)
  - PNodeClass, [336](#)
  - T\_ENode, [445](#)
  - T\_MNode, [452](#)
- degree
  - poly, [345](#)
- degree\_vector
  - poly, [349](#)
- DelayPrevar
  - P\_const, [315](#)
- deleted
  - EEdge, [98](#)
- DeleteObject
  - declare.h, [986](#)
  - minos.c, [1734](#)
- DeleteRecoveryFile
  - checkpoint.c, [542](#)
  - declare.h, [998](#)
- DeleteStore

- declare.h, [833](#)
  - store.c, [2829](#)
- delGroup
  - SGroup, [387](#)
- DELTA
  - ftypes.h, [1391](#)
- DELTA2
  - ftypes.h, [1395](#)
- DELTA3
  - ftypes.h, [1393](#)
- deltaEuler
  - float.c, [1290](#)
- deltaEulerC
  - float.c, [1290](#)
- deltaMZV
  - float.c, [1290](#)
- DELTAP
  - ftypes.h, [1396](#)
- den
  - Fraction, [142](#)
- DENOMINATOR
  - ftypes.h, [1392](#)
- DENOMINATORSYMBOL
  - ftypes.h, [1404](#)
- denomnum
  - M\_const, [188](#)
- DENSETABLE
  - ftypes.h, [1454](#)
- DenToFunction
  - declare.h, [983](#)
  - wildcard.c, [3223](#)
- derivative
  - poly, [346](#)
- description
  - fixedset, [137](#)
- det
  - ECand, [95](#)
  - EStack, [119](#)
- DetCommu
  - declare.h, [829](#)
  - normal.c, [1874](#)
- DetCurDum
  - declare.h, [833](#)
  - reshuf.c, [2609](#)
- detEdge
  - Assign, [21](#)
- DetVars
  - declare.h, [833](#)
  - store.c, [2830](#)
- DGraph, [80](#)
  - edges, [81](#)
  - nedges, [81](#)
  - nnodes, [81](#)
  - nodes, [81](#)
- DIAGRAMS
  - ftypes.h, [1402](#)
- diagrams.c, [1052](#), [1053](#)
  - CoCanonicalize, [1052](#)
  - DoCanonicalize, [1052](#)
  - DoShattering, [1053](#)
  - DoTopologyCanonicalize, [1052](#)
- diaoffset
  - TERMINFO, [467](#)
- diawrap.cc, [1061](#)
- dict.c, [1074](#), [1079](#)
  - AddDictionary, [1076](#)
  - AddToDictionary, [1076](#)
  - DictFromBytes, [1078](#)
  - DictToBytes, [1078](#)
  - DoPreAdd, [1078](#)
  - DoPreCloseDictionary, [1078](#)
  - DoPreOpenDictionary, [1077](#)
  - DoPreUseDictionary, [1078](#)
  - FindDictionary, [1076](#)
  - FindExtraSymbol, [1076](#)
  - FindFunction, [1076](#)
  - FindFunWithArgs, [1076](#)
  - FindIndex, [1076](#)
  - FindSymbol, [1075](#)
  - FindVector, [1075](#)
  - IsExponentSign, [1075](#)
  - IsMultiplySign, [1075](#)
  - RemoveDictionary, [1077](#)
  - SetDictionaryOptions, [1077](#)
  - ShrinkDictionary, [1077](#)
  - TransformRational, [1075](#)
  - UnSetDictionary, [1077](#)
  - UseDictionary, [1077](#)
- DICT\_ALLNUMBERS
  - ftypes.h, [1452](#)
- DICT\_DOFUNWITHARGS
  - ftypes.h, [1452](#)
- DICT\_DOSPECIALS
  - ftypes.h, [1452](#)
- DICT\_DOVARIABLES
  - ftypes.h, [1452](#)
- DICT\_FUNCTION
  - ftypes.h, [1453](#)
- DICT\_FUNCTION\_WITH\_ARGUMENTS
  - ftypes.h, [1454](#)
- DICT\_INDEX
  - ftypes.h, [1453](#)
- DICT\_INDOLLARS
  - ftypes.h, [1453](#)
- DICT\_INTEGERNUMBER
  - ftypes.h, [1453](#)
- DICT\_INTEGERONLY
  - ftypes.h, [1451](#)
- DICT\_NOFUNWITHARGS
  - ftypes.h, [1452](#)
- DICT\_NONUMBERS
  - ftypes.h, [1451](#)
- DICT\_NOSPECIALS
  - ftypes.h, [1452](#)
- DICT\_NOTINDOLLARS
  - ftypes.h, [1453](#)

- DICT\_NOVARIABLES
  - ftypes.h, [1452](#)
- DICT\_RANGE
  - ftypes.h, [1454](#)
- DICT\_RATIONALNUMBER
  - ftypes.h, [1453](#)
- DICT\_RATIONALONLY
  - ftypes.h, [1452](#)
- DICT\_SPECIALCHARACTER
  - ftypes.h, [1454](#)
- DICT\_SYMBOL
  - ftypes.h, [1453](#)
- DICT\_VECTOR
  - ftypes.h, [1453](#)
- DictFromBytes
  - declare.h, [1023](#)
  - dict.c, [1078](#)
- Dictionaries
  - O\_const, [281](#)
- DICTIONARY, [81](#)
  - characters, [83](#)
  - elements, [82](#)
  - funwith, [83](#)
  - gnumelements, [83](#)
  - name, [82](#)
  - numbers, [82](#)
  - numelements, [82](#)
  - ranges, [83](#)
  - sizeelements, [82](#)
  - variables, [82](#)
- DICTIONARY\_ELEMENT, [83](#)
  - lhs, [84](#)
  - rhs, [84](#)
  - size, [84](#)
  - type, [84](#)
- DictToBytes
  - declare.h, [1023](#)
  - dict.c, [1078](#)
- DidClean
  - C\_const, [67](#)
- DIFBASE
  - declare.h, [811](#)
- DIFPOS
  - declare.h, [811](#)
- dimension
  - CbUf, [73](#)
  - fixedset, [138](#)
  - FuNcTiOn, [147](#)
  - InDeX, [151](#)
  - SeTs, [384](#)
  - SyMbOl, [425](#)
  - VeCtOr, [484](#)
- DimensionExpression
  - declare.h, [832](#)
  - reshuf.c, [2610](#)
- DimensionSubterm
  - declare.h, [831](#)
  - reshuf.c, [2610](#)
- DIMENSIONSYMBOL
  - ftypes.h, [1405](#)
- DimensionTerm
  - declare.h, [831](#)
  - reshuf.c, [2610](#)
- dir
  - EEdge, [100](#)
- dirEdge
  - EGraph, [106](#)
- DirtPow
  - C\_const, [65](#)
- DIRTYFLAG
  - ftypes.h, [1419](#)
- dirtyflag
  - FiLeDaTa, [132](#)
- DIRTYSYMFLAG
  - ftypes.h, [1419](#)
- discardDisc
  - MGraph, [214](#)
  - T\_MGraph, [450](#)
- discardIso
  - MGraph, [214](#)
  - T\_MGraph, [451](#)
- discardOrd
  - T\_MGraph, [450](#)
- discardRefine
  - MGraph, [214](#)
  - T\_MGraph, [450](#)
- DisOrderFlag
  - N\_const, [256](#)
- DISTRIBUTE
  - structs.h, [2903](#)
- DiStRiBuTe, [84](#)
  - cycle, [86](#)
  - n, [85](#)
  - n1, [85](#)
  - n2, [85](#)
  - obj1, [85](#)
  - obj2, [85](#)
  - out, [85](#)
  - sign, [85](#)
- Distribute
  - declare.h, [833](#)
  - symmetr.c, [2932](#)
- DISTRIBUTION
  - ftypes.h, [1393](#)
- DistrN
  - declare.h, [1027](#)
  - tools.c, [3093](#)
- div
  - COST, [76](#)
  - poly, [350](#)
- DivFloats
  - float.c, [1288](#)
- DIVFUNCTION
  - ftypes.h, [1397](#)
- DIVfunction
  - declare.h, [1011](#)

- ratio.c, 2506
- divides
  - poly, 351
- DivLong
  - declare.h, 834
  - reken.c, 2555
- DivLong\_fix\_args
  - optimize.cc, 1993
- divmod
  - poly, 351
- divmod\_heap
  - poly, 354
- divmod\_mpoly
  - flintinterface.h, 1270
- divmod\_one\_term
  - poly, 353
- divmod\_poly
  - flintinterface.h, 1271
- divmod\_univar
  - poly, 353
- DIVPOS
  - declare.h, 811
- DivRat
  - declare.h, 834
  - reken.c, 2553
- divrem
  - ratio.c, 2507
- Divvy
  - declare.h, 834
  - reken.c, 2552
- DNode, 86
  - extloop, 86
  - intrct, 86
- do\_optimization
  - optimize.cc, 1999
- DO\_UFFLE
  - structs.h, 2902
- do\_uffle
  - SHvariables, 391
- DOALL
  - ftypes.h, 1451
- DoArgPlode
  - compcomm.c, 627
  - declare.h, 934
- DoArgument
  - compcomm.c, 618
  - declare.h, 929
- DoBrackets
  - compcomm.c, 624
  - declare.h, 933
- DoBreakDo
  - declare.h, 914
  - pre.c, 2323
- DoCall
  - declare.h, 913
  - pre.c, 2322
- DoCanonicalize
  - declare.h, 835
- diagrams.c, 1052
- DoChain
  - compcomm.c, 622
  - declare.h, 953
- DoCheckpoint
  - checkpoint.c, 544
  - declare.h, 999
- DoClearOptimize
  - declare.h, 910
  - pre.c, 2332
- DoClearUserFlag
  - declare.h, 1028
  - pre.c, 2334
- DoCommentChar
  - declare.h, 916
  - pre.c, 2321
- DoContinueDo
  - declare.h, 914
  - pre.c, 2323
- DoDebug
  - declare.h, 914
  - pre.c, 2322
- DoDefine
  - declare.h, 906
  - pre.c, 2321
- DoDelta
  - declare.h, 834
  - normal.c, 1873
- DoDelta3
  - declare.h, 834
  - reshuf.c, 2611
- DoDimension
  - declare.h, 893
  - names.c, 1828
- DoDistrib
  - declare.h, 836
  - reshuf.c, 2611
- DoDo
  - declare.h, 914
  - pre.c, 2323
- DODOUBLE
  - tools.c, 3078
- DoElements
  - declare.h, 894
  - names.c, 1831
- DoElse
  - declare.h, 913
  - pre.c, 2323
- DoElseif
  - declare.h, 913
  - pre.c, 2323
- DoEnddo
  - declare.h, 914
  - pre.c, 2323
- DoEndif
  - declare.h, 913
  - pre.c, 2324
- DoEndInside

- declare.h, [915](#)
- pre.c, [2325](#)
- DoEndNamespace
  - declare.h, [1027](#)
  - pre.c, [2333](#)
- DoEndprocedure
  - declare.h, [914](#)
  - pre.c, [2324](#)
- DoEndSwitch
  - declare.h, [1026](#)
  - if.c, [1656](#)
- DoesCommu
  - declare.h, [829](#)
  - normal.c, [1874](#)
- DOESNOTCOMMUTE
  - ftypes.h, [1451](#)
- DoExecStatement
  - declare.h, [920](#)
  - module.c, [1765](#)
- DoExecute
  - declare.h, [885](#)
  - execute.c, [1158](#)
- DOEXPAND
  - tools.c, [3078](#)
- DoExpr
  - comexpr.c, [582](#)
  - declare.h, [960](#)
- DoExternal
  - declare.h, [908](#)
  - pre.c, [2330](#)
- DoFactDollar
  - declare.h, [909](#)
  - pre.c, [2331](#)
- DoFactorize
  - compcomm.c, [630](#)
  - declare.h, [940](#)
- DoFindLoop
  - compcomm.c, [625](#)
  - declare.h, [964](#)
- DoFromExternal
  - declare.h, [908](#)
  - pre.c, [2330](#)
- Dolf
  - declare.h, [913](#)
  - pre.c, [2324](#)
- Dolfdef
  - declare.h, [912](#)
  - pre.c, [2324](#)
- Dolfndef
  - declare.h, [913](#)
  - pre.c, [2324](#)
- DolfStatement
  - declare.h, [840](#)
  - if.c, [1655](#)
- Dolfydef
  - declare.h, [912](#)
  - pre.c, [2324](#)
- DoInclude
  - declare.h, [908](#)
  - pre.c, [2322](#)
- DoInParallel
  - compcomm.c, [622](#)
  - declare.h, [935](#)
- DoinParallel
  - declare.h, [981](#)
  - module.c, [1765](#)
- DoInside
  - declare.h, [914](#)
  - pre.c, [2324](#)
- DOLARGUMENT
  - ftypes.h, [1439](#)
- DOLINDEX
  - ftypes.h, [1440](#)
- dollar
  - ARGBUFFER, [14](#)
- dollar.c, [1092](#), [1102](#)
  - AddPotModdollar, [1101](#)
  - AddToDollarBuffer, [1094](#)
  - AssignDollar, [1094](#)
  - CatchDollar, [1094](#)
  - CleanDollarFactors, [1099](#)
  - DollarFactorize, [1099](#)
  - DollarRaiseLow, [1097](#)
  - DoToFunction, [1095](#)
  - DoToIndex, [1095](#)
  - DoToLong, [1096](#)
  - DoToNumber, [1095](#)
  - DoToSymbol, [1095](#)
  - DoToTensor, [1095](#)
  - DoToTerms, [1096](#)
  - DoToVector, [1095](#)
  - EvalDoLoopArg, [1098](#)
  - ExchangeDollars, [1096](#)
  - ExecInside, [1096](#)
  - GetDoNum, [1101](#)
  - InsideDollar, [1096](#)
  - IsMultipleOf, [1097](#)
  - IsSetMember, [1097](#)
  - MakeDollarInteger, [1099](#)
  - MakeDollarMod, [1100](#)
  - PrelfDollarEval, [1097](#)
  - PutTermInDollar, [1094](#)
  - SizeOfDollar, [1096](#)
  - STEP2, [1093](#)
  - TakeDollarContent, [1099](#)
  - TermAssign, [1094](#)
  - TermsInDollar, [1096](#)
  - TestDoLoop, [1098](#)
  - TestEndDoLoop, [1098](#)
  - TranslateExpression, [1097](#)
  - TwoExprCompare, [1097](#)
  - WildDollars, [1095](#)
  - WriteDollarFactorToBuffer, [1094](#)
  - WriteDollarToBuffer, [1094](#)
- dollar\_buf, [87](#)
- DOLLAREXP2

- ftypes.h, [1390](#)
- DOLLAREXPRESSION
  - ftypes.h, [1390](#)
- DollarFactorize
  - declare.h, [966](#)
  - dollar.c, [1099](#)
- DOLLARFLAG
  - ftypes.h, [1434](#)
- DollarInOutBuffer
  - O\_const, [282](#)
- DollarList
  - P\_const, [311](#)
- DOLLARNAME
  - declare.h, [809](#)
- dollarname
  - DoLoOp, [91](#)
- DOLLARNAMES
  - ftypes.h, [1444](#)
- dollarname
  - C\_const, [42](#)
- DollarOutBuffer
  - O\_const, [279](#)
- DollarOutSizeBuffer
  - O\_const, [282](#)
- DollarRaiseLow
  - declare.h, [893](#)
  - dollar.c, [1097](#)
- DoLIaRS, [87](#)
  - factors, [88](#)
  - index, [88](#)
  - name, [88](#)
  - nfactors, [89](#)
  - node, [88](#)
  - numdummies, [89](#)
  - size, [88](#)
  - type, [88](#)
  - where, [88](#)
  - zero, [88](#)
- Dollars
  - variable.h, [3203](#)
- DOLLARSTREAM
  - ftypes.h, [1367](#)
- dollarzero
  - M\_const, [190](#)
- DOLNUMBER
  - ftypes.h, [1439](#)
- DOLOOP
  - structs.h, [2902](#)
- DoLoOp, [89](#)
  - contents, [91](#)
  - dollarname, [91](#)
  - errorsinloop, [92](#)
  - firstdollar, [92](#)
  - firstloopcall, [92](#)
  - firstnum, [91](#)
  - incdollar, [92](#)
  - incnum, [91](#)
  - lastdollar, [92](#)
  - lastnum, [91](#)
  - name, [90](#)
  - NoShowInput, [92](#)
  - NumPreTypes, [92](#)
  - p, [90](#)
  - PrelfLevel, [92](#)
  - PreSwitchLevel, [93](#)
  - startlinenumber, [91](#)
  - type, [91](#)
  - vars, [90](#)
- dolooplevel
  - C\_const, [62](#)
- doloopnest
  - C\_const, [51](#)
- DoLoops
  - variable.h, [3200](#)
- doloopstack
  - C\_const, [51](#)
- doloopstacksize
  - C\_const, [62](#)
- DOLSUBTERM
  - ftypes.h, [1439](#)
- DOLTERMS
  - ftypes.h, [1439](#)
- DolToFunction
  - declare.h, [965](#)
  - dollar.c, [1095](#)
- DolToIndex
  - declare.h, [966](#)
  - dollar.c, [1095](#)
- DolToLong
  - declare.h, [966](#)
  - dollar.c, [1096](#)
- DolToNumber
  - declare.h, [965](#)
  - dollar.c, [1095](#)
- DolToSymbol
  - declare.h, [965](#)
  - dollar.c, [1095](#)
- DolToTensor
  - declare.h, [965](#)
  - dollar.c, [1095](#)
- DolToTerms
  - declare.h, [974](#)
  - dollar.c, [1096](#)
- DolToVector
  - declare.h, [965](#)
  - dollar.c, [1095](#)
- DOLUNDEFINED
  - ftypes.h, [1439](#)
- DOLWILDARGS
  - ftypes.h, [1439](#)
- DOLZERO
  - ftypes.h, [1440](#)
- DoMessage
  - declare.h, [910](#)
  - pre.c, [2325](#)
- DoModDollar

- declare.h, [981](#)
- module.c, [1765](#)
- DoModLocal
  - declare.h, [981](#)
  - module.c, [1764](#)
- DoModMax
  - declare.h, [981](#)
  - module.c, [1764](#)
- DoModMin
  - declare.h, [981](#)
  - module.c, [1764](#)
- DoModSum
  - declare.h, [981](#)
  - module.c, [1764](#)
- DoNamespace
  - declare.h, [1027](#)
  - pre.c, [2333](#)
- DONE
  - proces.c, [2424](#)
- DoNoParallel
  - declare.h, [980](#)
  - module.c, [1764](#)
- DonotinParallel
  - declare.h, [982](#)
  - module.c, [1765](#)
- DoOnePow
  - declare.h, [840](#)
  - proces.c, [2431](#)
- DoOptimize
  - declare.h, [910](#)
  - pre.c, [2332](#)
- DoParallel
  - declare.h, [980](#)
  - module.c, [1764](#)
- DoPartitions
  - declare.h, [835](#)
  - reshuf.c, [2611](#)
- DoPermutations
  - declare.h, [837](#)
  - reshuf.c, [2612](#)
- DoPipe
  - declare.h, [912](#)
  - pre.c, [2325](#)
- DoPipeStatement
  - declare.h, [920](#)
  - module.c, [1765](#)
- DoPolyfun
  - declare.h, [920](#)
  - module.c, [1763](#)
- DoPolyratfun
  - declare.h, [920](#)
  - module.c, [1763](#)
- DoPrcExtension
  - declare.h, [916](#)
  - pre.c, [2325](#)
- DoPreAdd
  - declare.h, [1020](#)
  - dict.c, [1078](#)
- DoPreAddSeparator
  - declare.h, [989](#)
  - pre.c, [2329](#)
- DoPreAppend
  - declare.h, [910](#)
  - pre.c, [2326](#)
- DoPreAppendPath
  - declare.h, [1025](#)
  - pre.c, [2332](#)
- DoPreAssign
  - declare.h, [911](#)
  - pre.c, [2321](#)
- DoPreBreak
  - declare.h, [911](#)
  - pre.c, [2326](#)
- DoPreCase
  - declare.h, [911](#)
  - pre.c, [2327](#)
- DoPreClose
  - declare.h, [915](#)
  - pre.c, [2326](#)
- DoPreCloseDictionary
  - declare.h, [1020](#)
  - dict.c, [1078](#)
- DoPreCreate
  - declare.h, [911](#)
  - pre.c, [2326](#)
- DoPreDefault
  - declare.h, [911](#)
  - pre.c, [2327](#)
- DoPreEndSwitch
  - declare.h, [911](#)
  - pre.c, [2327](#)
- DoPreExchange
  - declare.h, [912](#)
  - pre.c, [2322](#)
- DoPreOpenDictionary
  - declare.h, [1020](#)
  - dict.c, [1077](#)
- DoPreOut
  - declare.h, [910](#)
  - pre.c, [2325](#)
- DoPrePrependPath
  - declare.h, [1025](#)
  - pre.c, [2332](#)
- DoPrePrintTimes
  - declare.h, [915](#)
  - pre.c, [2325](#)
- DoPreRemove
  - declare.h, [915](#)
  - pre.c, [2326](#)
- DoPreReset
  - declare.h, [916](#)
  - pre.c, [2332](#)
- DoPreRmSeparator
  - declare.h, [989](#)
  - pre.c, [2330](#)
- DoPreShow



- declare.h, [912](#)
- pre.c, [2327](#)
- DoPreSortReallocate
  - declare.h, [915](#)
  - pre.c, [2325](#)
- DoPreSwitch
  - declare.h, [911](#)
  - pre.c, [2327](#)
- DoPreUseDictionary
  - declare.h, [1020](#)
  - dict.c, [1078](#)
- DoPreWrite
  - declare.h, [915](#)
  - pre.c, [2326](#)
- DoPrint
  - compcomm.c, [614](#)
  - declare.h, [928](#)
- DoProcedure
  - declare.h, [915](#)
  - pre.c, [2326](#)
- DoProcessBucket
  - declare.h, [981](#)
  - module.c, [1764](#)
- DoPrompt
  - declare.h, [909](#)
  - pre.c, [2330](#)
- DoPutInside
  - compcomm.c, [631](#)
  - declare.h, [933](#)
- DoRecovery
  - checkpoint.c, [543](#)
  - declare.h, [998](#)
- DoRedefine
  - declare.h, [906](#)
  - pre.c, [2321](#)
- DoReverseInclude
  - declare.h, [908](#)
  - pre.c, [2322](#)
- DoRevert
  - declare.h, [841](#)
  - normal.c, [1874](#)
- DoRmExternal
  - declare.h, [909](#)
  - pre.c, [2330](#)
- DoSetExternal
  - declare.h, [909](#)
  - pre.c, [2330](#)
- DoSetExternalAttr
  - declare.h, [909](#)
  - pre.c, [2330](#)
- DoSetRandom
  - declare.h, [909](#)
  - pre.c, [2331](#)
- DoSetups
  - declare.h, [889](#)
  - setfile.c, [2690](#)
- DoSetUserFlag
  - declare.h, [1028](#)
- pre.c, [2334](#)
- DoShattering
  - declare.h, [836](#)
  - diagrams.c, [1053](#)
- DoShtuffle
  - declare.h, [820](#)
- DoShuffle
  - declare.h, [837](#)
  - reshuf.c, [2612](#)
- DoSkipExtraSymbols
  - declare.h, [910](#)
  - pre.c, [2332](#)
- DoStuffle
  - declare.h, [837](#)
  - reshuf.c, [2612](#)
- DoSumF1
  - declare.h, [841](#)
  - ratio.c, [2502](#)
- DoSumF2
  - declare.h, [841](#)
  - ratio.c, [2502](#)
- DoSwitch
  - declare.h, [1026](#)
  - if.c, [1656](#)
- DoSymmetrize
  - compcomm.c, [619](#)
  - declare.h, [929](#)
- DoSystem
  - declare.h, [912](#)
  - pre.c, [2327](#)
- DoTable
  - declare.h, [951](#)
  - names.c, [1830](#)
- DoTableExpansion
  - declare.h, [836](#)
  - tables.c, [2961](#)
- DoTail
  - declare.h, [886](#)
  - startup.c, [2798](#)
- dotchar
  - setfile.c, [2692](#)
- DoTempSet
  - declare.h, [894](#)
  - names.c, [1832](#)
- DoTerminate
  - declare.h, [913](#)
  - pre.c, [2323](#)
- DoTheta
  - declare.h, [841](#)
  - normal.c, [1873](#)
- DoTimeOutAfter
  - declare.h, [910](#)
  - pre.c, [2333](#)
- DoToExternal
  - declare.h, [908](#)
  - pre.c, [2331](#)
- DoTopologyCanonicalize
  - declare.h, [836](#)

- diagrams.c, [1052](#)
- DOTPBIT
  - ftypes.h, [1438](#)
- DOTPRODUCT
  - ftypes.h, [1389](#)
- DoubleBuffer
  - declare.h, [900](#)
  - tools.c, [3090](#)
- DoubleCbuffer
  - comtool.c, [764](#)
  - declare.h, [925](#)
- DoubleFlag
  - O\_const, [285](#)
- DOUBLEFORTRANMODE
  - ftypes.h, [1386](#)
- DoubleIfBuffers
  - declare.h, [928](#)
  - if.c, [1656](#)
- DoubleList
  - declare.h, [900](#)
  - tools.c, [3090](#)
- DoubleLList
  - declare.h, [900](#)
  - tools.c, [3090](#)
- DOUBLEPRECISIONFLAG
  - ftypes.h, [1386](#)
- DoubleSwitchBuffers
  - declare.h, [1026](#)
  - if.c, [1656](#)
- DoubleTable
  - float.c, [1289](#)
- DoUndefine
  - declare.h, [908](#)
  - pre.c, [2322](#)
- DoUse
  - declare.h, [1027](#)
  - pre.c, [2333](#)
- DROP
  - ftypes.h, [1421](#)
- DROPARG
  - ftypes.h, [1447](#)
- DropCoefficient
  - declare.h, [858](#)
  - normal.c, [1874](#)
- DROPGEXPRESSION
  - ftypes.h, [1422](#)
- DROPHGEXPRESSION
  - ftypes.h, [1423](#)
- DROPHLEXPRESSION
  - ftypes.h, [1423](#)
- DROPLEXPRESSION
  - ftypes.h, [1422](#)
- DROPEDEXPRESSION
  - ftypes.h, [1422](#)
- DROPSPECTATOREXPRESSION
  - ftypes.h, [1424](#)
- DropSymbols
  - declare.h, [859](#)
- normal.c, [1874](#)
- DuBiOuS, [93](#)
  - dummy, [93](#)
  - name, [93](#)
  - node, [93](#)
- Dubious
  - variable.h, [3204](#)
- DubiousList
  - C\_const, [43](#)
- DumFound
  - N\_const, [247](#)
- DUMFUN
  - ftypes.h, [1393](#)
- DumFunPlace
  - N\_const, [247](#)
- DumInd
  - M\_const, [185](#)
- DUMMY
  - ftypes.h, [1437](#)
- dummy
  - ARGBUFFER, [15](#)
  - BracketInfo, [35](#)
  - DuBiOuS, [93](#)
  - MoDeL, [234](#)
- DUMMYBUFFER
  - ftypes.h, [1445](#)
- DUMMYFLAG
  - ftypes.h, [1429](#)
- DUMMYFUN
  - ftypes.h, [1393](#)
- DUMMYINDEX\_
  - ftypes.h, [1427](#)
- DUMMYPADDING
  - AEdge, [8](#)
  - EGraph, [112](#)
  - ENode, [118](#)
  - MCBlock, [194](#)
  - MConn, [202](#)
  - Model, [231](#)
  - OptDef, [289](#)
  - Options, [302](#)
  - SProcess, [409](#)
- DUMMYPADDING1
  - EGraph, [114](#)
  - MGraph, [215](#)
- DUMMYPADDING2
  - MGraph, [216](#)
- dummyrenumlist
  - N\_const, [248](#)
- dummysubexp
  - T\_const, [436](#)
- DUMMYTEN
  - ftypes.h, [1393](#)
- DUMMYUSE
  - declare.h, [810](#)
- DumNum
  - C\_const, [65](#)
- dumnumflag

- C\_const, 59
- DUMPINTERMS
  - ftypes.h, 1372
- DumPlace
  - N\_const, 247
- DumpNode
  - declare.h, 891
  - names.c, 1826
- DUMPOUTTERMS
  - ftypes.h, 1372
- DUMPTOCOMPILER
  - ftypes.h, 1372
- DUMPTOPARALLEL
  - ftypes.h, 1373
- DUMPTOSORT
  - ftypes.h, 1372
- DumpTree
  - declare.h, 891
  - names.c, 1826
- e
  - bufIPstruct, 37
- e3
  - TrAcEs, 475
- e4
  - TrAcEs, 475
- eat
  - P\_const, 315
- EATTENSOR
  - ftypes.h, 1419
- ebufnum
  - T\_const, 433
- ECand, 94
  - ~ECand, 94
  - det, 95
  - ECand, 94
  - nplist, 95
  - plist, 95
  - prECand, 95
- eclass
  - SGroup, 388
- econn
  - EGraph, 109
- EDGE
  - ftypes.h, 1403
- edgen
  - EStack, 119
- edges
  - Assign, 25
  - DGraph, 81
  - EGraph, 109
  - ENode, 117
  - MCBlock, 194
  - T\_EGraph, 444
  - T\_ENode, 445
- edgeSym
  - Assign, 22
- edtype
  - EEdge, 99
- EEdge, 95
  - ~EEdge, 96
  - conid, 99
  - copy, 97
  - cut, 99
  - deleted, 98
  - dir, 100
  - edtype, 99
  - EEdge, 96
  - egraph, 98
  - emom, 99
  - ext, 98
  - extMom, 99
  - id, 98
  - lmom, 99
  - nlegs, 98
  - nodes, 98
  - opicom, 99
  - print, 97
  - ptcl, 98
  - setEMom, 97
  - setId, 97
  - setLMom, 97
  - setType, 97
  - visited, 99
- eers
  - TrAcEs, 475
- EESYMBOL
  - ftypes.h, 1405
- EFLine, 100
  - EFLine, 100
  - elist, 101
  - fkind, 101
  - ftype, 101
  - nlist, 101
  - print, 101
- EGraph, 102
  - ~EGraph, 104
  - addFLine, 108
  - ald, 109
  - assigned, 110
  - bconn, 113
  - biconnE, 106
  - bicount, 113
  - bidef, 113
  - biinitE, 106
  - bilow, 113
  - bisearchE, 106
  - chkMomConsv, 107
  - cmpMom, 107
  - connComp, 105
  - connVisit, 106
  - copy, 104
  - dirEdge, 106
  - DUMMYPADDING, 112
  - DUMMYPADDING1, 114
  - econn, 109
  - edges, 109

- EGraph, [104](#)
- endSetExtLoop, [105](#)
- esym, [110](#)
- extMom, [113](#)
- extMomConsv, [107](#)
- extperm, [110](#)
- findRoot, [106](#)
- flines, [113](#)
- fltrace, [108](#)
- fromDGraph, [104](#)
- fromMGraph, [104](#)
- fsign, [110](#)
- getFLines, [107](#)
- groupLMom, [107](#)
- gSubld, [110](#)
- isExternal, [105](#)
- isFermion, [105](#)
- isOptE, [104](#)
- legParticle, [108](#)
- loopm, [113](#)
- maxdeg, [112](#)
- mgraph, [109](#)
- mld, [109](#)
- model, [108](#)
- multp, [110](#)
- nadj2ptv, [112](#)
- nEdges, [111](#)
- nExtern, [111](#)
- nflines, [114](#)
- nLoops, [112](#)
- nNodes, [111](#)
- nodes, [109](#)
- nopi2p, [112](#)
- nopicomp, [112](#)
- nsym, [110](#)
- nsym1, [110](#)
- opi2plp, [112](#)
- opiCount, [113](#)
- opt, [108](#)
- pld, [111](#)
- prFLines, [107](#)
- print, [104](#)
- proc, [108](#)
- sEdges, [111](#)
- setExtern, [105](#)
- setExtLoop, [105](#)
- sld, [109](#)
- sLoops, [111](#)
- sMaxdeg, [111](#)
- sNodes, [111](#)
- sproc, [109](#)
- totalc, [112](#)
- egraph
  - Assign, [23](#)
  - EEdge, [98](#)
  - ENode, [117](#)
  - MGraph, [212](#)
  - SProcess, [407](#)
  - T\_MGraph, [450](#)
- elem
  - SGroup, [388](#)
- element
  - objects, [288](#)
- ELEMENTLOADED
  - ftypes.h, [1443](#)
- elements
  - DICTIONARY, [82](#)
- ELEMENTUSED
  - ftypes.h, [1442](#)
- elist
  - EFLine, [101](#)
- emom
  - EEdge, [99](#)
- empty
  - FiLeInDeX, [134](#)
- EMPTYINDEX
  - ftypes.h, [1429](#)
- EMPTYININDEX
  - structs.h, [2898](#)
- emptystring
  - startup.c, [2801](#)
- EmptyTable
  - declare.h, [952](#)
  - names.c, [1831](#)
- EMSYMBOL
  - ftypes.h, [1405](#)
- ENCODEARG
  - ftypes.h, [1445](#)
- END
  - ftypes.h, [1437](#)
- end
  - Options, [300](#)
- endAGraph
  - SProcess, [405](#)
- endianness
  - STOREHEADER, [414](#)
- endmg
  - Options, [302](#)
- endMGraph
  - SProcess, [405](#)
- ENDMODULE
  - ftypes.h, [1369](#)
- EndNest
  - N\_const, [245](#)
- ENDOFINPUT
  - ftypes.h, [1368](#)
- ENDOFSTREAM
  - ftypes.h, [1367](#)
- EndOfToken
  - declare.h, [919](#)
  - tools.c, [3086](#)
- endoftokens
  - C\_const, [50](#)
- endProc
  - Options, [300](#)
- endSetExtern

- T\_EGraph, 443
- endSetExtLoop
  - EGraph, 105
- EndSort
  - declare.h, 842
  - sort.c, 2717
- endSubProc
  - Options, 300
- endswitch
  - SWITCH, 422
- EndTable
  - float.c, 1290
- ENode, 114
  - ~ENode, 115
  - copy, 116
  - deg, 117
  - DUMMYPADDING, 118
  - edges, 117
  - egraph, 117
  - ENode, 115
  - extloop, 117
  - id, 117
  - initAss, 116
  - intrct, 117
  - klow, 118
  - maxdeg, 117
  - ndtype, 117
  - print, 116
  - setExtern, 116
  - setId, 116
  - setType, 116
  - visited, 118
- entriesinindex
  - iniinfo, 156
- EntVar
  - declare.h, 843
  - names.c, 1826
- EpfCon
  - declare.h, 843
  - opera.c, 1955
- EpfFind
  - declare.h, 843
  - opera.c, 1955
- EpfGen
  - declare.h, 843
  - opera.c, 1956
- eqnidxs
  - optimization, 291
- eqnum
  - StreaM, 419
- EQUAL
  - ftypes.h, 1437
- EqualArg
  - declare.h, 843
  - reshuf.c, 2611
- error
  - MoDeL, 234
  - VerRtEx, 486
- Error0
  - declare.h, 887
  - message.c, 1714
- Error1
  - declare.h, 887
  - message.c, 1714
- Error2
  - declare.h, 887
  - message.c, 1714
- ErrorBlock
  - O\_const, 286
- ErrorInDollar
  - N\_const, 253
- ERROROUT
  - ftypes.h, 1373
- errorsinloop
  - DoLoOp, 92
- Eside
  - R\_const, 376
- eSize
  - AStack, 29
- EStack, 118
  - ~EStack, 119
  - det, 119
  - edgen, 119
  - EStack, 119
  - nplist, 119
  - plist, 119
  - print, 119
- eStack
  - AStack, 28
- eStackP
  - AStack, 29
- esym
  - EGraph, 110
  - MGraph, 212
  - T\_EGraph, 444
  - T\_MGraph, 449
- EvalDoLoopArg
  - declare.h, 974
  - dollar.c, 1098
- EvalPrelf
  - declare.h, 918
  - pre.c, 2328
- evaluate.c, 1147
- EvaluateEuler
  - float.c, 1294
- EVEN\_
  - ftypes.h, 1427
- EXCH
  - declare.h, 801
- ExchangeDollars
  - declare.h, 976
  - dollar.c, 1096
- ExchangeExpressions
  - declare.h, 976
  - execute.c, 1158
- EXCHINOUT

- declare.h, [805](#)
- EXCHN
  - declare.h, [801](#)
- execarg
  - argument.c, [490](#)
  - declare.h, [884](#)
- ExecInside
  - declare.h, [934](#)
  - dollar.c, [1096](#)
- ExecMode
  - S\_const, [382](#)
- ExecModule
  - declare.h, [919](#)
  - module.c, [1763](#)
- ExecStore
  - declare.h, [919](#)
  - module.c, [1763](#)
- execterm
  - argument.c, [490](#)
  - declare.h, [884](#)
- execute.c, [1156](#), [1161](#)
  - BRACKETCURRENTEXPR, [1157](#)
  - BRACKETOTHEREXPR, [1157](#)
  - CleanExpr, [1157](#)
  - CleanupTerm, [1159](#)
  - ContentMerge, [1160](#)
  - CountTerms1, [1160](#)
  - CURRENTBRACKET, [1157](#)
  - DoExecute, [1158](#)
  - ExchangeExpressions, [1158](#)
  - GetContent, [1159](#)
  - GetFirstBracket, [1159](#)
  - GetFirstTerm, [1159](#)
  - MakeGlobal, [1157](#)
  - NOBRACKETACTIVE, [1157](#)
  - PopVariables, [1157](#)
  - PutBracket, [1158](#)
  - PutInVflags, [1158](#)
  - SetMods, [1158](#)
  - SizeOfExpression, [1160](#)
  - SpecialCleanup, [1158](#)
  - TermsInBracket, [1160](#)
  - TermsInExpression, [1160](#)
  - TestDrop, [1157](#)
  - UnSetMods, [1158](#)
  - UpdatePositions, [1160](#)
- EXECUTINGIF
  - ftypes.h, [1371](#)
- EXECUTINGPRESWITCH
  - ftypes.h, [1372](#)
- ExistsObject
  - declare.h, [986](#)
  - minos.c, [1734](#)
- exitflag
  - M\_const, [191](#)
- expand\_memory
  - poly, [343](#)
- ExpandBuffer
  - declare.h, [900](#)
  - tools.c, [3090](#)
- ExpandEuler
  - float.c, [1291](#)
- ExpandMZV
  - float.c, [1291](#)
- ExpandRat
  - declare.h, [1012](#)
  - ratio.c, [2506](#)
- ExpandTripleDots
  - declare.h, [922](#)
  - pre.c, [2319](#)
- expchanged
  - R\_const, [376](#)
- ExpectedSign
  - N\_const, [258](#)
- expflags
  - R\_const, [376](#)
- EXPLODEARG
  - ftypes.h, [1446](#)
- expnum
  - M\_const, [188](#)
- EXPONENT
  - ftypes.h, [1391](#)
- exprbufsize
  - ParallelVars, [319](#)
- EXPRESSION
  - ftypes.h, [1389](#)
- ExPrEsSiOn, [120](#)
  - bracketinfo, [122](#)
  - compression, [123](#)
  - counter, [122](#)
  - hidelevel, [122](#)
  - inmem, [122](#)
  - name, [122](#)
  - namesize, [123](#)
  - newbracketinfo, [122](#)
  - node, [123](#)
  - numdummies, [124](#)
  - numfactors, [124](#)
  - onfile, [121](#)
  - printf, [123](#)
  - prototype, [121](#)
  - renum, [121](#)
  - renumlists, [122](#)
  - replace, [123](#)
  - size, [121](#)
  - sizeprototype, [124](#)
  - status, [123](#)
  - uflags, [124](#)
  - vflags, [123](#)
  - whichbuffer, [123](#)
- expression
  - FiLeInDeX, [134](#)
- ExpressionList
  - C\_const, [43](#)
- EXPRESSIONNUMBER
  - ftypes.h, [1449](#)

- EXPRESSIONONLY
  - ftypes.h, 1377
- Expressions
  - variable.h, 3204
- exprfillwarning
  - C\_const, 60
- EXPRHEAD
  - ftypes.h, 1420
- EXPRNAME
  - declare.h, 809
- EXPRNAMES
  - ftypes.h, 1444
- exprnames
  - C\_const, 42
- exprnr
  - OPTIMIZERESULT, 296
- exprnumber
  - SpecTatoR, 402
- EXPRSOUT
  - ftypes.h, 1373
- ExprStat
  - FixedGlobals, 135
- ExprStatus
  - bugtool.c, 536
  - declare.h, 1000
- exprtodo
  - ParallelVars, 319
- EXPTOEXP
  - ftypes.h, 1418
- ext
  - EEdge, 98
  - T\_EEdge, 440
  - T\_ENode, 445
  - T\_MNode, 452
- extcmd.c, 1192, 1194
  - closeAllExternalChannels, 1194
  - closeExternalChannel, 1193
  - getCurrentExternalChannel, 1193
  - initPresetExternalChannels, 1193
  - openExternalChannel, 1193
  - selectExternalChannel, 1193
- EXTERNALCHANNELOUT
  - ftypes.h, 1373
- EXTERNALCHANNELSTREAM
  - ftypes.h, 1367
- externalset
  - TERMINFO, 467
- EXTERNLOCK
  - declare.h, 818
- externonly
  - VerRtEx, 486
- EXTERNRWLOCK
  - declare.h, 818
- EXTEUCLIDEAN
  - ftypes.h, 1401
- extloop
  - DNode, 86
  - ENode, 117
  - MNode, 219
- extMom
  - EEdge, 99
  - EGraph, 113
- extMomConsv
  - EGraph, 107
- extonly
  - Particle, 326
  - PInput, 333
- extperm
  - EGraph, 110
  - SProcess, 409
- EXTRAPARAMETER
  - ftypes.h, 1438
- extrasym
  - C\_const, 51
- EXTRASymbol
  - ftypes.h, 1449
- ExtraSymbol
  - declare.h, 858
  - normal.c, 1873
- extrasymbols
  - C\_const, 69
- EXTRASymFUN
  - ftypes.h, 1400
- ExtraSymFun
  - declare.h, 1008
  - notation.c, 1940
- extself
  - MGraph, 214
- f
  - sOrT, 397
- FaCdOILaR, 124
  - size, 125
  - type, 125
  - value, 125
  - where, 125
- facnum
  - M\_const, 189
- FACTARGFLAG
  - ftypes.h, 1448
- factor
  - factorized\_poly, 126
  - TrAcEn, 472
  - TrAcEs, 478
- factor.c, 1213, 1214
  - FactorIn, 1214
  - FactorInExpr, 1214
- FACTORIAL
  - ftypes.h, 1394
- Factorial
  - declare.h, 843
  - reken.c, 2560
- factorials
  - T\_const, 430
- FACTORIN
  - ftypes.h, 1398
- FactorIn

- declare.h, [844](#)
- factor.c, [1214](#)
- FactorInExpr
  - declare.h, [844](#)
  - factor.c, [1214](#)
- factorize\_mpoly
  - flintinterface.h, [1271](#)
- factorize\_poly
  - flintinterface.h, [1271](#)
- factorized\_poly, [125](#)
  - add\_factor, [126](#)
  - factor, [126](#)
  - power, [126](#)
  - tostring, [126](#)
- FactorMode
  - O\_const, [286](#)
- FactorNum
  - O\_const, [286](#)
- factors
  - DoLIaRS, [88](#)
- FACTORSYMBOL
  - ftypes.h, [1405](#)
- FALSE\_EXPR
  - pre.c, [2317](#)
- FATAL\_SIG\_ERROR
  - portsignals.h, [2309](#)
- FBUFFERSIZE
  - fsizes.h, [1350](#)
- fbuffersize
  - M\_const, [183](#)
- fbufnum
  - T\_const, [434](#)
- ff
  - sOrT, [397](#)
- ffbufnum
  - C\_const, [61](#)
- FG
  - inivar.h, [1682](#)
  - variable.h, [3207](#)
- FGInput, [126](#)
  - coupl, [127](#)
  - finalc, [127](#)
  - finaln, [127](#)
  - initlc, [127](#)
  - initln, [127](#)
  - nfinal, [127](#)
  - ninitl, [127](#)
- fh
  - SpecTatoR, [402](#)
- FiLe, [128](#)
  - active, [130](#)
  - filesize, [129](#)
  - handle, [130](#)
  - inbuffer, [130](#)
  - name, [129](#)
  - numblocks, [130](#)
  - PObuffer, [129](#)
  - POfill, [129](#)
  - POfull, [129](#)
  - POposition, [129](#)
  - POsize, [130](#)
  - POstop, [129](#)
- file
  - sOrT, [394](#)
- FiLeDaTa, [131](#)
  - dirtyflag, [132](#)
  - Fill, [132](#)
  - Handle, [132](#)
  - Index, [132](#)
  - Position, [132](#)
- FILEHANDLE
  - structs.h, [2901](#)
- FILEINDEX
  - structs.h, [2899](#)
- FiLeInDeX, [133](#)
  - empty, [134](#)
  - expression, [134](#)
  - next, [134](#)
  - number, [134](#)
- filelist
  - tools.c, [3094](#)
- filelistsize
  - tools.c, [3094](#)
- filenum
  - N\_const, [255](#)
- FileOnlyFlag
  - M\_const, [177](#)
- fileposition
  - StreamM, [416](#)
- FILES
  - form3.h, [1331](#)
- filesize
  - FiLe, [129](#)
- FILESTREAM
  - ftypes.h, [1366](#)
- Fill
  - FiLeDaTa, [132](#)
- fill
  - PF\_BUFFER, [329](#)
- FILLARG
  - declare.h, [806](#)
- fillEGraph
  - Assign, [19](#)
- FILLEXPRESS
  - declare.h, [808](#)
- FILLFUN
  - declare.h, [807](#)
- FILLFUN3
  - declare.h, [807](#)
- fillpoint
  - TabIEbAsE, [457](#)
- FILLSUB
  - declare.h, [807](#)
- FILLVALUE
  - tools.c, [3077](#)
- finalc



- FGInput, 127
- finaln
  - FGInput, 127
- finalPart
  - Process, 365
- FinalStats
  - C\_const, 58
- finalstep
  - TrAcEs, 478
- find\_Horner\_MCTS
  - optimize.cc, 1997
- find\_Horner\_MCTS\_expand\_tree
  - optimize.cc, 1997
- find\_optimizations
  - optimize.cc, 1999
- FindAll
  - declare.h, 844
  - pattern.c, 2173
- FindArg
  - argument.c, 491
  - declare.h, 931
- FindBracket
  - declare.h, 980
  - index.c, 1672
- FindCase
  - declare.h, 1026
  - if.c, 1656
- findcommand
  - compiler.c, 724
  - declare.h, 898
- FindDictionary
  - declare.h, 1021
  - dict.c, 1076
- FindExtraSymbol
  - declare.h, 1023
  - dict.c, 1076
- FindFunction
  - declare.h, 1023
  - dict.c, 1076
- FindFunWithArgs
  - declare.h, 1023
  - dict.c, 1076
- FindIndex
  - declare.h, 1023
  - dict.c, 1076
- FindInIndex
  - declare.h, 822
  - store.c, 2832
- FindInKeyWord
  - declare.h, 906
  - pre.c, 2320
- findInteractionCode
  - Model, 227
- findInteractionName
  - Model, 227
- FindKeyWord
  - declare.h, 906
  - pre.c, 2320
- FindLocalSubterm
  - declare.h, 1005
  - notation.c, 1939
- FINDLOOP
  - ftypes.h, 1440
- FindLus
  - declare.h, 964
  - lus.c, 1687
- findMClass
  - Model, 228
- FindMulti
  - declare.h, 844
  - findpat.c, 1225
- FindOnce
  - declare.h, 844
  - findpat.c, 1225
- FindOnly
  - declare.h, 845
  - findpat.c, 1225
- findParticleCode
  - Model, 227
- findParticleName
  - Model, 227
- findpat.c, 1224, 1226
  - FindMulti, 1225
  - FindOnce, 1225
  - FindOnly, 1225
  - FindRest, 1225
- findPattern
  - N\_const, 248
- FindRange
  - declare.h, 1002
  - transform.c, 3149
- FindRest
  - declare.h, 845
  - findpat.c, 1225
- FindrNumber
  - declare.h, 845
  - store.c, 2832
- findRoot
  - EGraph, 106
- FindSubexpression
  - declare.h, 1006
  - notation.c, 1940
- FindSubterm
  - declare.h, 1005
  - notation.c, 1939
- FindSymbol
  - declare.h, 1022
  - dict.c, 1075
- FindTableNumber
  - declare.h, 988
  - minos.c, 1733
- FindTableTree
  - declare.h, 972
  - tables.c, 2961
- FindTB
  - declare.h, 838

- tables.c, [2962](#)
- findTerm
  - N\_const, [248](#)
- FindTree
  - comtool.c, [767](#)
  - declare.h, [961](#)
- FindVar
  - declare.h, [839](#)
  - if.c, [1655](#)
- FindVector
  - declare.h, [1022](#)
  - dict.c, [1075](#)
- FinilLine
  - declare.h, [845](#)
  - sch.c, [2651](#)
- FINISH
  - ftypes.h, [1441](#)
- finishcbuf
  - comtool.c, [763](#)
  - declare.h, [972](#)
- finished
  - tree\_node, [482](#)
- FinishShuffle
  - declare.h, [837](#)
  - reshuf.c, [2612](#)
- FinishStuffle
  - declare.h, [838](#)
  - reshuf.c, [2613](#)
- finishuf
  - SHvariables, [391](#)
- FINISHUFFLE
  - structs.h, [2902](#)
- FinlShuffle
  - declare.h, [820](#)
- FinlTerm
  - declare.h, [845](#)
  - proces.c, [2429](#)
- first
  - SeTs, [383](#)
- first\_variable
  - poly, [345](#)
- FIRSTBRACKET
  - ftypes.h, [1397](#)
- firstconstindex
  - C\_const, [54](#)
- firstctypemessage
  - C\_const, [56](#)
- firstdollar
  - DoLoOp, [92](#)
- firstindexblock
  - iniinfo, [156](#)
- firstloopcall
  - DoLoOp, [92](#)
- FIRSTMODULE
  - ftypes.h, [1368](#)
- firstnameblock
  - iniinfo, [157](#)
- firstnamespace
  - P\_const, [311](#)
- firstnum
  - DoLoOp, [91](#)
- FIRSTTERM
  - ftypes.h, [1401](#)
- FIRSTUSERFUNCTION
  - ftypes.h, [1404](#)
- FIRSTUSERSYMBOL
  - ftypes.h, [1405](#)
- fix\_sign\_fmpz\_mpoly\_ratfun
  - flintinterface.h, [1276](#)
- fix\_sign\_fmpz\_poly\_ratfun
  - flintinterface.h, [1276](#)
- fixarg
  - optimize.cc, [1992](#)
- FIXED\_
  - ftypes.h, [1427](#)
- FIXEDGLOBALS
  - structs.h, [2904](#)
- FixedGlobals, [134](#)
  - cTable, [137](#)
  - ExprStat, [135](#)
  - fname, [136](#)
  - fname2, [136](#)
  - fname2base, [136](#)
  - fname2size, [137](#)
  - fnamebase, [136](#)
  - framesize, [136](#)
  - FunNam, [136](#)
  - OperaFind, [135](#)
  - Operation, [135](#)
  - s\_one, [136](#)
  - swmes, [136](#)
  - VarType, [135](#)
- fixedset, [137](#)
  - description, [137](#)
  - dimension, [138](#)
  - name, [137](#)
  - type, [138](#)
- fixedsets
  - inivar.h, [1682](#)
  - variable.h, [3207](#)
- FixIndices
  - C\_const, [49](#)
- fkind
  - EFLine, [101](#)
- flags
  - FuNcTiOn, [146](#)
  - InDeX, [151](#)
  - indexblock, [152](#)
  - KEYWORD, [163](#)
  - KEYWORDV, [164](#)
  - SeTs, [384](#)
  - SETUPPARAMETERS, [385](#)
  - SpecTatoR, [402](#)
  - SyMbOl, [425](#)
  - TaBlEs, [461](#)
  - TERMINFO, [467](#)

- VeCtOr, [484](#)
- flg0
  - MNodeClass, [224](#)
- flg1
  - MNodeClass, [224](#)
- flg2
  - MNodeClass, [224](#)
- flines
  - EGraph, [113](#)
- flint::fmpz, [138](#)
  - ~fmpz, [138](#)
  - d, [139](#)
  - fmpz, [138](#)
  - print, [139](#)
- flint::mpoly, [239](#)
  - ~mpoly, [240](#)
  - d, [240](#)
  - mpoly, [240](#)
  - print, [240](#)
- flint::mpoly\_ctx, [240](#)
  - ~mpoly\_ctx, [241](#)
  - d, [241](#)
  - mpoly\_ctx, [241](#)
- flint::mpoly\_factor, [241](#)
  - ~mpoly\_factor, [242](#)
  - d, [242](#)
  - mpoly\_factor, [242](#)
- flint::poly, [337](#)
  - ~poly, [338](#)
  - d, [338](#)
  - poly, [338](#)
  - print, [338](#)
- flint::poly\_factor, [356](#)
  - ~poly\_factor, [357](#)
  - d, [357](#)
  - poly\_factor, [357](#)
- flint\_div
  - flintwrap.cc, [1279](#)
- flint\_factorize\_argument
  - flintwrap.cc, [1280](#)
- flint\_factorize\_dollar
  - flintwrap.cc, [1280](#)
- flint\_final\_cleanup\_master
  - flintwrap.cc, [1279](#)
- flint\_final\_cleanup\_thread
  - flintwrap.cc, [1279](#)
- flint\_gcd
  - flintwrap.cc, [1280](#)
- flint\_inverse
  - flintwrap.cc, [1280](#)
- flint\_mul
  - flintwrap.cc, [1280](#)
- flint\_ratfun\_add
  - flintwrap.cc, [1280](#)
- flint\_ratfun\_normalize
  - flintwrap.cc, [1281](#)
- flint\_rem
  - flintwrap.cc, [1281](#)
- flint\_startup\_init
  - flintwrap.cc, [1281](#)
- flintinterface.cc, [1242](#), [1243](#)
  - IFW, [1242](#)
- flintinterface.h, [1268](#), [1277](#)
  - cleanup, [1270](#)
  - cleanup\_master, [1270](#)
  - divmod\_mpoly, [1270](#)
  - divmod\_poly, [1271](#)
  - factorize\_mpoly, [1271](#)
  - factorize\_poly, [1271](#)
  - fix\_sign\_fmpz\_mpoly\_ratfun, [1276](#)
  - fix\_sign\_fmpz\_poly\_ratfun, [1276](#)
  - fmpz\_get\_form, [1272](#)
  - fmpz\_set\_form, [1272](#)
  - form\_sort, [1271](#)
  - from\_argument\_mpoly, [1271](#)
  - from\_argument\_poly, [1272](#)
  - gcd\_mpoly, [1272](#)
  - gcd\_poly, [1272](#)
  - get\_variables, [1273](#)
  - inverse\_poly, [1273](#)
  - mul\_mpoly, [1273](#)
  - mul\_poly, [1273](#)
  - ratfun\_add\_mpoly, [1273](#)
  - ratfun\_add\_poly, [1274](#)
  - ratfun\_normalize\_mpoly, [1274](#)
  - ratfun\_normalize\_poly, [1274](#)
  - ratfun\_read\_mpoly, [1274](#)
  - ratfun\_read\_poly, [1274](#)
  - simplify\_fmpz, [1276](#)
  - simplify\_fmpz\_poly, [1276](#)
  - startup\_init, [1275](#)
  - to\_argument\_mpoly, [1275](#)
  - to\_argument\_poly, [1275](#)
  - var\_map\_t, [1270](#)
- FlintPolyFlag
  - C\_const, [62](#)
- flintwrap.cc, [1279](#), [1281](#)
  - flint\_div, [1279](#)
  - flint\_factorize\_argument, [1280](#)
  - flint\_factorize\_dollar, [1280](#)
  - flint\_final\_cleanup\_master, [1279](#)
  - flint\_final\_cleanup\_thread, [1279](#)
  - flint\_gcd, [1280](#)
  - flint\_inverse, [1280](#)
  - flint\_mul, [1280](#)
  - flint\_ratfun\_add, [1280](#)
  - flint\_ratfun\_normalize, [1281](#)
  - flint\_rem, [1281](#)
  - flint\_startup\_init, [1281](#)
- FlipLONG
  - O\_const, [280](#)
- FlipPOINTER
  - O\_const, [280](#)
- FlipPOS
  - O\_const, [280](#)
- FlipTable

- declare.h, [982](#)
- tables.c, [2962](#)
- FlipWORD
  - O\_const, [280](#)
- flist
  - MNodeClass, [223](#)
  - MOrbits, [239](#)
  - T\_MNodeClass, [456](#)
- float.c, [1285](#), [1295](#)
  - AddFloats, [1288](#)
  - AddRatToFloat, [1288](#)
  - AddWithFloat, [1294](#)
  - CalculateEuler, [1291](#)
  - CalculateMZV, [1291](#)
  - CalculateMZVhalf, [1290](#)
  - CheckFloat, [1293](#)
  - ClearMZVTables, [1293](#)
  - CoEvaluate, [1295](#)
  - CoToFloat, [1293](#)
  - CoToRat, [1294](#)
  - deltaEuler, [1290](#)
  - deltaEulerC, [1290](#)
  - deltaMZV, [1290](#)
  - DivFloats, [1288](#)
  - DoubleTable, [1289](#)
  - EndTable, [1290](#)
  - EvaluateEuler, [1294](#)
  - ExpandEuler, [1291](#)
  - ExpandMZV, [1291](#)
  - FloatFunToRat, [1292](#)
  - FloatToInteger, [1287](#)
  - FloatToRat, [1288](#)
  - Form\_mpf\_clear, [1287](#)
  - Form\_mpf\_empty, [1292](#)
  - Form\_mpf\_init, [1287](#)
  - Form\_mpf\_set\_prec\_raw, [1287](#)
  - FormtoZ, [1287](#)
  - GMPSPREAD, [1286](#)
  - IntegerToFloat, [1287](#)
  - MergeWithFloat, [1294](#)
  - MulFloats, [1288](#)
  - MulRatToFloat, [1289](#)
  - PackFloat, [1291](#)
  - PrintFloat, [1293](#)
  - RatToFloat, [1292](#)
  - ReadFloat, [1292](#)
  - SetFloatPrecision, [1293](#)
  - SetupMPFTables, [1293](#)
  - SetupMZVTables, [1293](#)
  - SimpleDelta, [1289](#)
  - SimpleDeltaC, [1289](#)
  - SingleTable, [1289](#)
  - TestFloat, [1292](#)
  - ToFloat, [1294](#)
  - ToRat, [1294](#)
  - UnpackFloat, [1291](#)
  - WITHCUTOFF, [1286](#)
  - ZeroTable, [1292](#)
  - ZtoForm, [1287](#)
- FloatFunToRat
  - float.c, [1292](#)
- FLOATMODE
  - ftypes.h, [1430](#)
- FloatToInteger
  - float.c, [1287](#)
- FloatToRat
  - float.c, [1288](#)
- fltrace
  - EGraph, [108](#)
- flush\_cache
  - checkpoint.c, [543](#)
- FLUSHCONSOLE
  - declare.h, [804](#)
- FlushFile
  - declare.h, [897](#)
  - tools.c, [3083](#)
- FlushOut
  - declare.h, [846](#)
  - sort.c, [2720](#)
- FlushSpectators
  - declare.h, [1025](#)
  - spectator.c, [2787](#)
- fmpz
  - flint::fmpz, [138](#)
- fmpz\_get\_form
  - flintinterface.h, [1272](#)
- fmpz\_set\_form
  - flintinterface.h, [1272](#)
- fname
  - FixedGlobals, [136](#)
- fname2
  - FixedGlobals, [136](#)
- fname2base
  - FixedGlobals, [136](#)
- fname2size
  - FixedGlobals, [137](#)
- fnamebase
  - FixedGlobals, [136](#)
- fnamesize
  - FixedGlobals, [136](#)
- FoldName
  - Stream, [417](#)
- forallignment
  - T\_MNodeClass, [456](#)
- ForFindOnly
  - N\_const, [248](#)
- form3.h, [1324](#), [1336](#)
  - AWORDMASK, [1329](#)
  - BETAVERSION, [1326](#)
  - BITSINLONG, [1328](#)
  - BITSINWORD, [1327](#)
  - BOOL, [1335](#)
  - bzero, [1334](#)
  - FILES, [1331](#)
  - form\_alignof, [1330](#)
  - FULLMAX, [1329](#)

- GetPID, [1334](#)
- inline, [1327](#)
- LONG, [1334](#)
- LONG\_MAX\_VALUE, [1328](#)
- LONG\_MIN\_VALUE, [1328](#)
- MAJORVERSION, [1326](#)
- MAX\_OPEN\_FILES, [1334](#)
- MAXLONG, [1329](#)
- MAXPOSITIVE, [1329](#)
- MAXPOSITIVE2, [1329](#)
- MAXPOSITIVE4, [1329](#)
- MINORVERSION, [1326](#)
- MLONG, [1335](#)
- NDEBUG, [1327](#)
- PADDUMMY, [1330](#)
- PADINT, [1331](#)
- PADLONG, [1330](#)
- PADPOINTER, [1330](#)
- PADPOSITION, [1330](#)
- PADWORD, [1331](#)
- PRODUCTIONDATE, [1326](#)
- RLONG, [1335](#)
- SBYTE, [1335](#)
- SPECMASK, [1328](#)
- STATIC\_ASSERT, [1327](#)
- STATIC\_ASSERT\_\_1, [1327](#)
- STATIC\_ASSERT\_\_2, [1327](#)
- STATIC\_ASSERT\_\_3, [1327](#)
- TOPBITONLY, [1328](#)
- TOPLONGBITONLY, [1328](#)
- UBYTE, [1335](#)
- Uclose, [1332](#)
- Uflush, [1332](#)
- Ugetpos, [1333](#)
- UINT, [1335](#)
- ULONG, [1334](#)
- Uopen, [1331](#)
- Uread, [1332](#)
- Useek, [1332](#)
- Usetbuf, [1332](#)
- Usetpos, [1333](#)
- Ustdout, [1333](#)
- Usync, [1333](#)
- Utell, [1333](#)
- Utruncate, [1333](#)
- UWORD, [1334](#)
- Uwrite, [1332](#)
- WILDMASK, [1329](#)
- WITHSORTBOTS, [1331](#)
- WORD, [1334](#)
- WORD\_MAX\_VALUE, [1328](#)
- WORD\_MIN\_VALUE, [1328](#)
- WORDMASK, [1329](#)
- form\_alignof
  - form3.h, [1330](#)
- FORM\_INLINE
  - declare.h, [815](#)
- Form\_mpf\_clear
  - float.c, [1287](#)
- Form\_mpf\_empty
  - float.c, [1292](#)
- Form\_mpf\_init
  - float.c, [1287](#)
- Form\_mpf\_set\_prec\_raw
  - float.c, [1287](#)
- form\_sort
  - flintinterface.h, [1271](#)
- FORMNAME
  - startup.c, [2798](#)
- FormtoZ
  - float.c, [1287](#)
- FortDotChar
  - O\_const, [286](#)
- FortFirst
  - O\_const, [285](#)
- Fortran90Kind
  - C\_const, [51](#)
- FortranCont
  - M\_const, [188](#)
- FORTRANCONTINUATIONLINES
  - fsizes.h, [1345](#)
- FORTRANMODE
  - ftypes.h, [1385](#)
- FoStage4
  - R\_const, [371](#)
- fPatches
  - sOrT, [396](#)
- fPatchesStop
  - sOrT, [396](#)
- fPatchN
  - sOrT, [400](#)
- fPatchN1
  - sOrT, [399](#)
- Fraction, [139](#)
  - add, [140](#), [141](#)
  - den, [142](#)
  - Fraction, [140](#)
  - gcd, [141](#)
  - isEq, [141](#)
  - normal, [141](#)
  - num, [142](#)
  - print, [140](#)
  - ratio, [142](#)
  - setValue, [140](#)
  - sub, [141](#)
- freeIg
  - MNode, [220](#)
  - T\_MNode, [452](#)
- FreeNameTree
  - declare.h, [892](#)
  - names.c, [1827](#)
- FreeTableBase
  - declare.h, [987](#)
  - minos.c, [1732](#)
- from
  - PF\_BUFFER, [330](#)

- From0List
  - declare.h, [899](#)
  - tools.c, [3089](#)
- from\_argument\_mpoly
  - flintinterface.h, [1271](#)
- from\_argument\_poly
  - flintinterface.h, [1272](#)
- from\_coefficient\_list
  - poly, [348](#)
- FROMBRAC
  - ftypes.h, [1418](#)
- fromDGraph
  - EGraph, [104](#)
- FROMDISK
  - minos.h, [1753](#)
- FROMFUNCTION
  - ftypes.h, [1444](#)
- fromindex
  - T\_const, [439](#)
- FromList
  - declare.h, [899](#)
  - tools.c, [3089](#)
- fromMGraph
  - Assign, [18](#)
  - EGraph, [104](#)
- FROMMODULEOPTION
  - ftypes.h, [1449](#)
- fromPerm
  - MOrbits, [238](#)
- FROMPOINTINSTRUCTION
  - ftypes.h, [1449](#)
- FromPolyRatFun
  - declare.h, [1009](#)
  - ratio.c, [2505](#)
- FROMSET
  - ftypes.h, [1418](#)
- FromStdin
  - M\_const, [192](#)
- FromVarList
  - declare.h, [900](#)
  - tools.c, [3090](#)
- Frozen
  - N\_const, [245](#)
- Fscr
  - R\_const, [371](#)
- fsign
  - EGraph, [110](#)
- fsizes.h, [1341](#), [1351](#)
  - COMMERCIALSIZE, [1345](#)
  - COMPILERBUFFER, [1346](#)
  - COMPINC, [1347](#)
  - COMPRESSBUFFER, [1345](#)
  - DEFAULTPROCESSBUCKETSIZE, [1348](#)
  - DEFAULTTHREADBUCKETSIZE, [1349](#)
  - DEFAULTTHREADLOADBALANCING, [1349](#)
  - DEFAULTTHREADS, [1348](#)
  - FBUFSIZE, [1350](#)
  - FORTRANCONTINUATIONLINES, [1345](#)
  - GZIPDEFAULT, [1348](#)
  - INDENTSPACE, [1349](#)
  - INITNAMESIZE, [1344](#)
  - INITNODESIZE, [1344](#)
  - JUMPRATIO, [1350](#)
  - MAXBRACKETBUFFERSIZE, [1348](#)
  - MAXCOUPLINGS, [1351](#)
  - MAXENAME, [1343](#)
  - MAXFLAGS, [1345](#)
  - MAXFLEVELS, [1347](#)
  - MAXFPATCHES, [1346](#)
  - MAXIF, [1344](#)
  - MAXLABELS, [1345](#)
  - MAXLEGS, [1351](#)
  - MAXLEVELS, [1345](#)
  - MAXLHS, [1345](#)
  - MAXLINELENGTH, [1350](#)
  - MAXLOOPS, [1345](#)
  - MAXMATCH, [1344](#)
  - MAXMULTIBRACKETLEVELS, [1350](#)
  - MAXNEST, [1344](#)
  - MAXNUMBERSIZE, [1343](#)
  - MAXNUMSIZE, [1348](#)
  - MAXPARLEVEL, [1343](#)
  - MAXPARTICLES, [1350](#)
  - MAXPATCHES, [1346](#)
  - MAXPRENAMESIZE, [1343](#)
  - MAXREPEAT, [1343](#)
  - MAXSAVEFUNCTION, [1343](#)
  - MAXWILDC, [1346](#)
  - MINALLOC, [1350](#)
  - MULTIINDENTSPACE, [1349](#)
  - NORMSIZE, [1343](#)
  - NPAIR1, [1350](#)
  - NPAIR2, [1350](#)
  - NUMFACS, [1344](#)
  - NUMFIXED, [1344](#)
  - NUMOPTIONS, [1351](#)
  - NUMSTORECACHES, [1349](#)
  - NUMTABLEENTRIES, [1346](#)
  - SFHSIZE, [1348](#)
  - SHMWINSIZE, [1348](#)
  - SIZEFACS, [1344](#)
  - SIZESTORECACHE, [1349](#)
  - SLARGEBUFFER, [1347](#)
  - SMAFPATCHES, [1347](#)
  - SMAXPATCHES, [1347](#)
  - SORTIOSIZE, [1346](#)
  - SPECTATORSIZE, [1347](#)
  - SSMALLBUFFER, [1346](#)
  - SSMALLOVERFLOW, [1346](#)
  - SSORTIOSIZE, [1347](#)
  - STERMSSMALL, [1347](#)
  - TABLEEXTENSION, [1348](#)
  - THREADSCRATCHOUTSIZE, [1349](#)
  - THREADSCRATCHSIZE, [1349](#)
- ftype
  - EFLine, [101](#)

- ftypes.h, [1354](#), [1457](#)
  - ABSFUNCTION, [1395](#)
  - ACOSFUNCTION, [1399](#)
  - ACOSHFUNCTION, [1400](#)
  - ADDARG, [1447](#)
  - ALLARGS, [1445](#)
  - ALLDIRTY, [1420](#)
  - ALLFUNCTIONS, [1404](#)
  - ALLFUNPOWERS, [1440](#)
  - ALLINTEGERDOUBLE, [1387](#)
  - ALLMZVFUNCTIONS, [1404](#)
  - ALLVARIABLES, [1376](#)
  - ALSODOLLARS, [1443](#)
  - ALSOFUNARGS, [1443](#)
  - ALSOPOWERS, [1443](#)
  - ALSOREVERSE, [1387](#)
  - ANDCOND, [1437](#)
  - ANTISYMMETRIC, [1432](#)
  - ANYTYPE, [1377](#)
  - ARGFIELD, [1391](#)
  - ARGHEAD, [1420](#)
  - ARGRANGE, [1445](#)
  - ARGTOARG, [1417](#)
  - ARGWILD, [1390](#)
  - ARLTOARL, [1418](#)
  - ASINFUNCTION, [1399](#)
  - ASINHFUNCTION, [1400](#)
  - ATAN2FUNCTION, [1399](#)
  - ATANFUNCTION, [1399](#)
  - ATANHFUNCTION, [1400](#)
  - ATENDOFMODULE, [1375](#)
  - AUTONAMES, [1444](#)
  - BASECODE, [1448](#)
  - BERNOULLIFUNCTION, [1396](#)
  - BINOMIAL, [1394](#)
  - BLOCK, [1403](#)
  - BUILTINDOLLARS, [1406](#)
  - BUILTININDICES, [1406](#)
  - BUILTINSYMBOLS, [1405](#)
  - BUILTINVECTORS, [1406](#)
  - CDELETE, [1377](#)
  - CDELTA, [1378](#)
  - CDOLLAR, [1378](#)
  - CDOTPRODUCT, [1378](#)
  - CDOUBLEDOT, [1379](#)
  - CDUBIOUS, [1379](#)
  - CEXPRESSION, [1378](#)
  - CFUN, [1456](#)
  - CFUNCTION, [1378](#)
  - CHECKEXTERN, [1455](#)
  - CHECKLOGTYPE, [1389](#)
  - CHISHOLM, [1387](#)
  - CINDEX, [1377](#)
  - CLEANFLAG, [1419](#)
  - CLEAR, [1438](#)
  - CLEARMODULE, [1369](#)
  - CMODE, [1386](#)
  - CMODEL, [1379](#)
  - CNUMBER, [1378](#)
  - COEFFI, [1435](#)
  - COEFFICIENTSONLY, [1443](#)
  - COEFFSYMBOL, [1404](#)
  - COMFUNPOWERS, [1440](#)
  - CONJUGATION, [1396](#)
  - CONTENTTERM, [1401](#)
  - CONTRACT, [1415](#)
  - COSFUNCTION, [1399](#)
  - COSHFUNCTION, [1399](#)
  - COULDCOMMUTE, [1451](#)
  - COUNTFUNCTION, [1396](#)
  - CRANGE, [1379](#)
  - CSET, [1378](#)
  - CSUBEXP, [1378](#)
  - CSYMBOL, [1377](#)
  - CVECTOR, [1377](#)
  - CVECTOR1, [1379](#)
  - CYCLEARG, [1446](#)
  - CYCLESYMMETRIC, [1432](#)
  - DECLARATION, [1375](#)
  - DECODEARG, [1446](#)
  - DEDUPARG, [1447](#)
  - DEFINITION, [1375](#)
  - DELTA, [1391](#)
  - DELTA2, [1395](#)
  - DELTA3, [1393](#)
  - DELTAP, [1396](#)
  - DENOMINATOR, [1392](#)
  - DENOMINATORSYMBOL, [1404](#)
  - DENSETABLE, [1454](#)
  - DIAGRAMS, [1402](#)
  - DICT\_ALLNUMBERS, [1452](#)
  - DICT\_DOFUNWITHARGS, [1452](#)
  - DICT\_DOSPECIALS, [1452](#)
  - DICT\_DOVARIABLES, [1452](#)
  - DICT\_FUNCTION, [1453](#)
  - DICT\_FUNCTION\_WITH\_ARGUMENTS, [1454](#)
  - DICT\_INDEX, [1453](#)
  - DICT\_INDOLLARS, [1453](#)
  - DICT\_INTEGERNUMBER, [1453](#)
  - DICT\_INTEGERONLY, [1451](#)
  - DICT\_NOFUNWITHARGS, [1452](#)
  - DICT\_NONUMBERS, [1451](#)
  - DICT\_NOSPECIALS, [1452](#)
  - DICT\_NOTINDOLLARS, [1453](#)
  - DICT\_NOVARIABLES, [1452](#)
  - DICT\_RANGE, [1454](#)
  - DICT\_RATIONALNUMBER, [1453](#)
  - DICT\_RATIONALONLY, [1452](#)
  - DICT\_SPECIALCHARACTER, [1454](#)
  - DICT\_SYMBOL, [1453](#)
  - DICT\_VECTOR, [1453](#)
  - DIMENSIONSYPMBOL, [1405](#)
  - DIRTYFLAG, [1419](#)
  - DIRTYSYMFLAG, [1419](#)
  - DISTRIBUTION, [1393](#)
  - DIVFUNCTION, [1397](#)

- DOALL, [1451](#)
- DOESNOTCOMMUTE, [1451](#)
- DOLARGUMENT, [1439](#)
- DOLINDEX, [1440](#)
- DOLLAREXP2, [1390](#)
- DOLLAREXPRESSION, [1390](#)
- DOLLARFLAG, [1434](#)
- DOLLARNAMES, [1444](#)
- DOLLARSTREAM, [1367](#)
- DOLNUMBER, [1439](#)
- DOLSUBTERM, [1439](#)
- DOLTERMS, [1439](#)
- DOLUNDEFINED, [1439](#)
- DOLWILDARGS, [1439](#)
- DOLZERO, [1440](#)
- DOTPBIT, [1438](#)
- DOTPRODUCT, [1389](#)
- DOUBLEFORTRANMODE, [1386](#)
- DOUBLEPRECISIONFLAG, [1386](#)
- DROP, [1421](#)
- DROPARG, [1447](#)
- DROPGEXPRESSION, [1422](#)
- DROPHGEXPRESSION, [1423](#)
- DROPHLEXPRESSION, [1423](#)
- DROPLEXPRESSION, [1422](#)
- DROPPEDEXPRESSION, [1422](#)
- DROPSPECTATOREXPRESSION, [1424](#)
- DUMFUN, [1393](#)
- DUMMY, [1437](#)
- DUMMYBUFFER, [1445](#)
- DUMMYFLAG, [1429](#)
- DUMMYFUN, [1393](#)
- DUMMYINDEX\_, [1427](#)
- DUMMYTEN, [1393](#)
- DUMPINTERMS, [1372](#)
- DUMPOUTTERMS, [1372](#)
- DUMPTOCOMPILER, [1372](#)
- DUMPTOPARALLEL, [1373](#)
- DUMPTOSORT, [1372](#)
- EATTENSOR, [1419](#)
- EDGE, [1403](#)
- EESYMBOL, [1405](#)
- ELEMENTLOADED, [1443](#)
- ELEMENTUSED, [1442](#)
- EMPTYINDEX, [1429](#)
- EMSYMBOL, [1405](#)
- ENCODEARG, [1445](#)
- END, [1437](#)
- ENDMODULE, [1369](#)
- ENDOFINPUT, [1368](#)
- ENDOFSTREAM, [1367](#)
- EQUAL, [1437](#)
- ERROROUT, [1373](#)
- EVEN\_, [1427](#)
- EXECUTINGIF, [1371](#)
- EXECUTINGPRESWITCH, [1372](#)
- EXPLODEARG, [1446](#)
- EXPONENT, [1391](#)
- EXPRESSION, [1389](#)
- EXPRESSIONNUMBER, [1449](#)
- EXPRESSIONONLY, [1377](#)
- EXPRHEAD, [1420](#)
- EXPRNAMES, [1444](#)
- EXPRSOUT, [1373](#)
- EXPTOEXP, [1418](#)
- EXTERNALCHANNELOUT, [1373](#)
- EXTERNALCHANNELSTREAM, [1367](#)
- EXTEUCLIDEAN, [1401](#)
- EXTRAPARAMETER, [1438](#)
- EXTRASymbol, [1449](#)
- EXTRASymFUN, [1400](#)
- FACTARGFLAG, [1448](#)
- FACTORIAL, [1394](#)
- FACTORIN, [1398](#)
- FACTORSYMBOL, [1405](#)
- FILESTREAM, [1366](#)
- FINDLOOP, [1440](#)
- FINISH, [1441](#)
- FIRSTBRACKET, [1397](#)
- FIRSTMODULE, [1368](#)
- FIRSTTERM, [1401](#)
- FIRSTUSERFUNCTION, [1404](#)
- FIRSTUSERSYMBOL, [1405](#)
- FIXED\_, [1427](#)
- FLOATMODE, [1430](#)
- FORTTRANMODE, [1385](#)
- FROMBRAC, [1418](#)
- FROMFUNCTION, [1444](#)
- FROMMODULEOPTION, [1449](#)
- FROMPOINTINSTRUCTION, [1449](#)
- FROMSET, [1418](#)
- FUNBIT, [1438](#)
- FUNCTION, [1391](#)
- FUNCTIONONLY, [1377](#)
- FUNCTIONSORT, [1429](#)
- FUNHEAD, [1420](#)
- FUNNYDOLLAR, [1429](#)
- FUNNYVEC, [1428](#)
- FUNNYWILD, [1428](#)
- FUNTOFUN, [1417](#)
- GAMMA, [1392](#)
- GAMMA1, [1428](#)
- GAMMA5, [1428](#)
- GAMMA6, [1428](#)
- GAMMA7, [1428](#)
- GAMMAFIVE, [1392](#)
- GAMMAFUNCTION, [1426](#)
- GAMMAI, [1392](#)
- GAMMASEVEN, [1392](#)
- GAMMASIX, [1392](#)
- GCDFUNCTION, [1397](#)
- GENCOMMUTE, [1438](#)
- GENERALFUNCTION, [1426](#)
- GENNONCOMMUTE, [1439](#)
- GFIVE, [1434](#)
- GIDENT, [1434](#)



GLOBAL, [1438](#)  
GLOBALEXPRESSION, [1422](#)  
GLOBALMODULE, [1368](#)  
GLOBALSPECTATORFLAG, [1454](#)  
GMINUS, [1434](#)  
GPLUS, [1434](#)  
GREATER, [1436](#)  
GREATEREQUAL, [1436](#)  
HAAKJE, [1391](#)  
HAAKJE0, [1390](#)  
HEADERFILE, [1368](#)  
HIDDENGEXPRESSION, [1422](#)  
HIDDENLEXPRESSION, [1422](#)  
HIDE, [1421](#)  
HIDEGEXPRESSION, [1423](#)  
HIDELEXPRESSION, [1423](#)  
HTOZARG, [1448](#)  
IDFUNCTION, [1402](#)  
IDHEAD, [1434](#)  
IFDOLLAR, [1435](#)  
IFDOLLAREXTRA, [1435](#)  
IFEXPRESSION, [1435](#)  
IFFLOATNUMBER, [1436](#)  
IFISFACTORIZED, [1436](#)  
IFOCCURS, [1436](#)  
IFUSERFLAG, [1436](#)  
IMPLODEARG, [1446](#)  
INCEXPRESSION, [1423](#)  
INDEX, [1389](#)  
INDEX\_, [1427](#)  
INDEXONLY, [1376](#)  
INDTOIND, [1417](#)  
INDTOSUB, [1417](#)  
INPUTOUT, [1373](#)  
INPUTSTREAM, [1367](#)  
INTFUNCTION, [1395](#)  
INTOHIDE, [1421](#)  
INTOHIDEGEXPRESSION, [1424](#)  
INTOHIDELEXPRESSION, [1423](#)  
INUSE, [1451](#)  
INVERSEFACTORIAL, [1394](#)  
INVERSEFUNCTION, [1401](#)  
INVERSETABLE, [1443](#)  
ISCOMPRESSED, [1430](#)  
ISFACTORIZED, [1431](#)  
ISFORTRAN90, [1387](#)  
ISINRHS, [1430](#)  
ISLYNDON, [1446](#)  
ISLYNDONR, [1446](#)  
ISNOTFORTRAN90, [1387](#)  
ISUNMODIFIED, [1430](#)  
ISYMBOL, [1404](#)  
ISZERO, [1430](#)  
KEEPZERO, [1431](#)  
LBRACE, [1381](#)  
LESS, [1436](#)  
LESSEQUAL, [1436](#)  
LEVICIVITA, [1394](#)  
LHSIDE, [1385](#)  
LHSIDEX, [1385](#)  
LI2FUNCTION, [1400](#)  
LINFUNCTION, [1400](#)  
LISTEDLOOP, [1374](#)  
LNFUNCTION, [1398](#)  
LNUMBER, [1391](#)  
LOADDOLLAR, [1418](#)  
LOCALEXPRESSION, [1421](#)  
LONGNUMBER, [1435](#)  
LOOKINGFORELSE, [1371](#)  
LOOKINGFORENDIF, [1371](#)  
LPARENTHESIS, [1381](#)  
LRPARENTHESSES, [1383](#)  
MAINSORT, [1429](#)  
MAKEARGS, [1445](#)  
MAKERATIONAL, [1401](#)  
MAPLEMODE, [1385](#)  
MATCH, [1435](#)  
MATCHFUNCTION, [1396](#)  
MATHEMATICAMODE, [1385](#)  
MAXBUILTINFUNCTION, [1403](#)  
MAXFUNCTION, [1395](#)  
MAXPOWEROF, [1397](#)  
MAXRANGEINDICATOR, [1445](#)  
MINFUNCTION, [1395](#)  
MINPOWEROF, [1398](#)  
MINSPEC, [1429](#)  
MINVECTOR, [1390](#)  
MIXED, [1376](#)  
MIXED2, [1375](#)  
MOD2FUNCTION, [1394](#)  
MODFUNCTION, [1394](#)  
MODLOCAL, [1442](#)  
MODMAX, [1442](#)  
MODMIN, [1442](#)  
MODNONE, [1442](#)  
MODSUM, [1442](#)  
MODULEINSTACK, [1371](#)  
MOEBIUS, [1402](#)  
MULFUNCTION, [1402](#)  
MULTIPLEOF, [1435](#)  
MULTIPLYARG, [1447](#)  
MUSTCLEANPRF, [1420](#)  
NAMENOTFOUND, [1439](#)  
NEG0\_, [1426](#)  
NEG\_, [1426](#)  
NEWFACTARG, [1448](#)  
NEWLYDEFINEDEXPRESSION, [1425](#)  
NEWSTATEMENT, [1371](#)  
NMIN4SHIFT, [1389](#)  
NOAUTO, [1376](#)  
NOCHECKLOGTYPE, [1389](#)  
NODEFUNCTION, [1403](#)  
NODOUBLEMASK, [1386](#)  
NOEXTSELF, [1456](#)  
NOFUNPOWERS, [1440](#)  
NOINDEX, [1429](#)

- NOINVERSES, 1443
- NOLYNDON, 1448
- NONODES, 1455
- NOPARALLEL\_CONVPOLY, 1369
- NOPARALLEL\_DOLLAR, 1369
- NOPARALLEL\_NPROC, 1370
- NOPARALLEL\_RHS, 1369
- NOPARALLEL\_SPECTATOR, 1370
- NOPARALLEL\_TBLDOLLAR, 1370
- NOPARALLEL\_USER, 1370
- NOQUADMASK, 1387
- NORMALFORMAT, 1387
- NORMALIZEFLAG, 1434
- NOSNAILS, 1456
- NOSPACEFORMAT, 1387
- NOTADPOLES, 1455
- NOTEQUAL, 1437
- NOTRICK, 1388
- NOUNPACK, 1444
- NUMARG, 1445
- NUMARGSFUN, 1394
- NUMERATORSYMBOL, 1404
- NUMERICLOOP, 1374
- NUMFACTORS, 1401
- NUMSPECSETS, 1430
- NUMTERMSFUN, 1397
- NUMTOIND, 1419
- NUMTONUM, 1419
- NUMTOSUB, 1419
- NUMTOSYM, 1419
- O\_BACKWARD, 1450
- O\_CSE, 1449
- O\_CSEGREEDY, 1449
- O\_FORWARD, 1450
- O\_FORWARDANDBACKWARD, 1450
- O\_FORWARDORBACKWARD, 1450
- O\_GREEDY, 1450
- O\_MCTS, 1450
- O\_NONE, 1449
- O\_OCCURRENCE, 1450
- O\_SIMULATED\_ANNEALING, 1450
- ODD\_, 1427
- OLDFACTARG, 1448
- OLDSTATEMENT, 1371
- ONEEXPRESSION, 1374
- ONEPARTICLEIRREDUCIBLE, 1455
- ONEPI, 1403
- ONLYFUNCTIONS, 1451
- OPTHEAD, 1451
- ORCOND, 1437
- ORDEREDSET, 1454
- PARALLELFLAG, 1370
- PARTEST, 1376
- PARTITIONS, 1402
- PATTERNFUNCTION, 1396
- PERMUTATIONS, 1402
- PERMUTEARG, 1446
- PFORTRANMODE, 1386
- PHI, 1403
- PIPESTREAM, 1366
- PISYMBOL, 1404
- POLYADD, 1441
- POLYDIV, 1441
- POLYFUN, 1369
- POLYGCD, 1441
- POLYINTFAC, 1442
- POLYMUL, 1441
- POLYNORM, 1442
- POLYREM, 1441
- POLYSUB, 1441
- POS0\_, 1426
- POS\_, 1426
- POSITIVEONLY, 1444
- POSNEG, 1443
- PRECALCSTREAM, 1367
- PRELOWERAFTER, 1370
- PRENOACTION, 1370
- PREPROONLY, 1372
- PRERAISEAFTER, 1370
- PREREADSTREAM, 1366
- PREREADSTREAM2, 1367
- PREREADSTREAM3, 1367
- PRETYPEDO, 1374
- PRETYPEIF, 1374
- PRETYPEINSIDE, 1375
- PRETYPENONE, 1374
- PRETYPEPROCEDURE, 1374
- PRETYPESWITCH, 1374
- PREVARSTREAM, 1366
- PRIMENUMBER, 1401
- PRINTALL, 1425
- PRINTCONTENT, 1425
- PRINTCONTENTS, 1424
- PRINTLFILE, 1425
- PRINTOFF, 1424
- PRINTON, 1424
- PRINTONEFUNCTION, 1425
- PRINTONETERM, 1425
- PROCEDUREFILE, 1368
- PROPERORDERFLAG, 1440
- PUTFIRST, 1402
- PVFUNV, 1456
- PVFUNWP, 1456
- Q\_, 1427
- QUADRUPLEFORTRANMODE, 1386
- QUADRUPLEPRECISIONFLAG, 1386
- RANDOMFUNCTION, 1400
- RANPERM, 1401
- RATIO, 1416
- RATIONALMODE, 1430
- RBRACE, 1381
- RCYCLESYMMETRIC, 1432
- READSPECTATORFLAG, 1454
- REDEFINEDEXPRESSION, 1425
- REDUCEMODE, 1385
- REGULAR, 1441

REGULAREXPRESSION, 1425  
REGULARSYMBOL, 1449  
REMFUNCTION, 1397  
REPLACEARG, 1445  
REPLACELOOP, 1440  
REPLACEMENT, 1393  
REVERSEARG, 1446  
REVERSEFILESTREAM, 1367  
REVERSEFUNCTION, 1393  
REVERSEORDER, 1432  
RHSIDE, 1385  
ROOTFUNCTION, 1396  
RPARENTHESIS, 1381  
SEARCHINGPRECASE, 1372  
SEARCHINGPREENDSWITCH, 1372  
SELECTARG, 1447  
SEPARATESYMBOL, 1405  
SETBIT, 1438  
SETEXP, 1390  
SETFUNCTION, 1392  
SETNUMBER, 1418  
SETONLY, 1377  
SETSET, 1390  
SETTONUM, 1418  
SETUPFILE, 1368  
SIGFUNCTION, 1395  
SIGNFUN, 1394  
SINFUNCTION, 1398  
SINHFUNCTION, 1399  
SIZEOFFUNCTION, 1403  
SKIP, 1421  
SKIPGEXPRESSION, 1422  
SKIPLEXPRESSION, 1421  
SKIPUNHIDEGEXPRESSION, 1424  
SKIPUNHIDELEXPRESSION, 1424  
SNUMBER, 1391  
SORT, 1437  
SORTANTIPOWER, 1388  
SORTHIGHFIRST, 1388  
SORTLOWFIRST, 1388  
SORTMODULE, 1369  
SORTPOWERFIRST, 1388  
SPARSETABLE, 1454  
SPECIFICATION, 1375  
SPECTATOREXPRESSION, 1424  
SQRTFUNCTION, 1398  
STATEMENT, 1375  
STATSMERGETOFILE, 1388  
STATSOUT, 1373  
STATSPOSTSORT, 1388  
STATSSPLITMERGE, 1388  
STORE, 1437  
STOREDEXPRESSION, 1422  
STOREMODULE, 1369  
SUBAFTER, 1433  
SUBAFTERNOT, 1434  
SUBALL, 1433  
SUBDISORDER, 1433  
SUBEXPR, 1435  
SUBEXPRESSION, 1390  
SUBEXPSIZE, 1420  
SUBMANY, 1433  
SUBMASK, 1433  
SUBMULTI, 1432  
SUBONCE, 1432  
SUBONLY, 1433  
SUBROUTINEFILE, 1368  
SUBSELECT, 1433  
SUBSORT, 1430  
SUBTERMUSED1, 1420  
SUBTERMUSED2, 1420  
SUBVECTOR, 1433  
SUMF1, 1392  
SUMF2, 1393  
SUMMEDIND, 1428  
SUMNUM1, 1416  
SUMNUM2, 1416  
SYMBOL, 1389  
SYMBOL\_, 1427  
SYMBOLONLY, 1376  
SYMMETRIC, 1432  
SYMMETRIZE, 1416  
SYMTONUM, 1416  
SYMTOSUB, 1417  
SYMTOSYM, 1416  
TABLEBASEFILE, 1368  
TABLEFUNCTION, 1397  
TABLESTUB, 1398  
TAKETRACE, 1415  
TANFUNCTION, 1399  
TANHFUNCTION, 1400  
TCOMMA, 1382  
TCONJUGATE, 1383  
TDELTA, 1380  
TDIVIDE, 1382  
TDOLLAR, 1380  
TDOT, 1381  
TDOTPRODUCT, 1380  
TDUBIOUS, 1381  
EMPTY, 1384  
TENDOFIT, 1383  
TENSORFUNCTION, 1426  
TENVEC, 1416  
TERMFUNCTION, 1396  
TERMSINBRACKET, 1398  
TERMSINEXPR, 1397  
TEXPRESSION, 1380  
TFLOAT, 1384  
TFUN, 1457  
TFUN1, 1457  
TFUNCLOSE, 1382  
TFUNCTION, 1380  
TFUNOPEN, 1382  
TGENINDEX, 1383  
THETA, 1395  
THETA2, 1395

THREADSDEBUG, 1373  
TINDEX, 1379  
TMINUS, 1382  
TMPPOLYFUN, 1391  
TMULTIPLY, 1382  
TNOT, 1383  
TNUMBER, 1380  
TNUMBER1, 1383  
TOBEFACTORED, 1431  
TOBEUNFACTORED, 1431  
TOLYNDON, 1447  
TOLYNDONR, 1447  
TOOUTPUT, 1375  
TOPO, 1403  
TOPOLOGIES, 1402  
TOPOLOGIESONLY, 1455  
TOPOLYNOMIALFLAG, 1448  
TPLUS, 1382  
TPOWER, 1382  
TPOWER1, 1384  
TSET, 1380  
TSETCLOSE, 1383  
TSETDOL, 1384  
TSETNUM, 1384  
TSETOPEN, 1383  
TSGAMMA, 1384  
TSUBEXP, 1380  
TSYMBOL, 1379  
TVECTOR, 1379  
TWILDARG, 1381  
TWILDCARD, 1381  
TYPEADJUSTBOUNDS, 1411  
TYPEALLLOOPS, 1415  
TYPEALLPATHS, 1415  
TYPEAPPLY, 1412  
TYPEAPPLYRESET, 1412  
TYPEARG, 1408  
TYPEARGEXPLODE, 1413  
TYPEARGHEADSIZE, 1421  
TYPEARGIMPLODE, 1413  
TYPEARGTOEXTRASymbol, 1414  
TYPEASSIGN, 1410  
TYPECANONICALIZE, 1414  
TYPECHAININ, 1412  
TYPECHAINOUT, 1412  
TYPECHISHOLM, 1408  
TYPECLEARUSERFLAG, 1415  
TYPECOUNT, 1407  
TYPEDENOMINATORS, 1413  
TYPEDETCURDUM, 1411  
TYPEDISCARD, 1407  
TYPEDOLOOP, 1414  
TYPEDROPCOEFFICIENT, 1413  
TYPEDROPSYMBOLS, 1414  
TYPEELIF, 1408  
TYPEELSE, 1408  
TYPEENDDOLOOP, 1414  
TYPEENDIF, 1408  
TYPEENDREPEAT, 1407  
TYPEENDSWITCH, 1415  
TYPEEXIT, 1409  
TYPEEXPRESSION, 1406  
TYPEFACTARG, 1410  
TYPEFACTARG2, 1410  
TYPEFACTOR, 1413  
TYPEFINDLOOP, 1410  
TYPEFPRINT, 1409  
TYPEFROMPOLYNOMIAL, 1414  
TYPEGOTO, 1407  
TYPEIDNEW, 1406  
TYPEIDOLD, 1406  
TYPEIF, 1407  
TYPEINEXPRESSION, 1411  
TYPEINSIDE, 1411  
TYPEISFUN, 1384  
TYPEISMYSTERY, 1385  
TYPEISSUB, 1384  
TYPEMERGE, 1412  
TPEMULT, 1407  
TYPENORM, 1408  
TYPENORM2, 1409  
TYPENORM3, 1409  
TYPENORM4, 1412  
TYPEOPERATION, 1407  
TYPEPRINT, 1409  
TYPEPUTINSIDE, 1414  
TYPEREDEFPRE, 1409  
TYPERENUMBER, 1410  
TYPEREPEAT, 1407  
TYPEREVERSE, 1408  
TYPESETEXIT, 1409  
TYPESETUSERFLAG, 1415  
TYPESORT, 1411  
TYPESPLITARG, 1409  
TYPESPLITARG2, 1410  
TYPESPLITFIRSTARG, 1411  
TYPESPLITLASTARG, 1412  
TYPESTUFFLE, 1413  
TYPESUM, 1408  
TYPESUMFIX, 1410  
TYPESWITCH, 1415  
TYPETERM, 1411  
TYPETESTUSE, 1412  
TYPETOPOLYNOMIAL, 1413  
TYPETOSPECTATOR, 1414  
TYPETRANSFORM, 1413  
TYPETRY, 1410  
TYPEUNRAVEL, 1411  
UNHIDE, 1421  
UNHIDEGEXPRESSION, 1423  
UNHIDELEXPRESSION, 1423  
UNPACK, 1444  
USEDFLAG, 1429  
VARNAMES, 1444  
VARTYPECOMPLEX, 1431  
VARTYPEIMAGINARY, 1431

- VARTYPEMINUS, [1432](#)
- VARTYPENONE, [1431](#)
- VARTYPEROOTOFUNITY, [1431](#)
- VECTBIT, [1438](#)
- VECTOMIN, [1417](#)
- VECTOR, [1389](#)
- VECTOR\_, [1428](#)
- VECTORONLY, [1376](#)
- VECTOSUB, [1417](#)
- VECTOVEC, [1417](#)
- VERTEXFUNCTION, [1426](#)
- VORTTRANMODE, [1386](#)
- WILDARGFUN, [1398](#)
- WILDARGINDEX, [1406](#)
- WILDARGSYMBOL, [1405](#)
- WILDARGVECTOR, [1406](#)
- WILDCARDS, [1418](#)
- WILDDUMMY, [1416](#)
- WITHAUTO, [1376](#)
- WITHBLOCKS, [1456](#)
- WITHEDGES, [1455](#)
- WITHERROR, [1366](#)
- WITHINSERTIONS, [1455](#)
- WITHONEPI, [1456](#)
- WITHOUTERROR, [1366](#)
- WITHOUTSEMICOLON, [1371](#)
- WITHSEMICOLON, [1371](#)
- WITHSYMMETRIZE, [1455](#)
- WRITEOUT, [1373](#)
- YESLYNDON, [1448](#)
- Z\_, [1427](#)
- ZTOHARG, [1447](#)
- full
  - PF\_BUFFER, [329](#)
- FullCleanUp
  - declare.h, [919](#)
  - module.c, [1763](#)
- FULLMAX
  - form3.h, [1329](#)
- fullName
  - dbase, [80](#)
  - P\_const, [311](#)
- fullnamesize
  - P\_const, [316](#)
- FullProto
  - N\_const, [245](#)
- FullRenumbr
  - declare.h, [963](#)
  - reshuf.c, [2609](#)
- FullSymmetrize
  - declare.h, [875](#)
  - function.c, [1471](#)
- FUN\_INFO, [142](#)
  - commute, [143](#)
  - location, [143](#)
  - numargs, [143](#)
  - numfunnies, [143](#)
  - numwildcards, [143](#)
  - symmet, [143](#)
  - tensor, [143](#)
- FunArg
  - T\_const, [437](#)
- funargs
  - N\_const, [248](#)
- FUNBIT
  - ftypes.h, [1438](#)
- func
  - KEYWORD, [163](#)
  - ReNuMbEr, [379](#)
- FUNCTION
  - ftypes.h, [1391](#)
- FuNcTiOn, [144](#)
  - commute, [145](#)
  - complex, [145](#)
  - dimension, [147](#)
  - flags, [146](#)
  - maxnumargs, [147](#)
  - minnumargs, [147](#)
  - name, [145](#)
  - namesize, [146](#)
  - node, [146](#)
  - number, [145](#)
  - spec, [146](#)
  - symmetric, [146](#)
  - symminfo, [145](#)
  - tabl, [145](#)
- function.c, [1470](#), [1473](#)
  - ChainIn, [1472](#)
  - ChainOut, [1472](#)
  - CompGroup, [1471](#)
  - FullSymmetrize, [1471](#)
  - MakeDirty, [1470](#)
  - MarkDirty, [1470](#)
  - MatchFunction, [1472](#)
  - PolyFunClean, [1471](#)
  - PolyFunDirty, [1471](#)
  - ScanFunctions, [1472](#)
  - SymFind, [1472](#)
  - SymGen, [1471](#)
  - Symmetrize, [1471](#)
- FunctionList
  - C\_const, [43](#)
- FUNCTIONONLY
  - ftypes.h, [1377](#)
- FUNCTIONS
  - structs.h, [2901](#)
- Functions
  - C\_const, [46](#)
- functions
  - variable.h, [3201](#)
- FUNCTIONSORT
  - ftypes.h, [1429](#)
- FUNHEAD
  - ftypes.h, [1420](#)
- funinds
  - N\_const, [249](#)

- FunInfo
  - N\_const, [250](#)
- funisize
  - N\_const, [254](#)
- FunLevel
  - declare.h, [846](#)
  - reshuf.c, [2609](#)
- funlocs
  - N\_const, [248](#)
- FunMatchCy
  - declare.h, [854](#)
  - symmetr.c, [2932](#)
- FunMatchSy
  - declare.h, [855](#)
  - symmetr.c, [2932](#)
- FunNam
  - FixedGlobals, [136](#)
- funnum
  - ReNuMbEr, [380](#)
- FUNNYDOLLAR
  - ftypes.h, [1429](#)
- FUNNYVEC
  - ftypes.h, [1428](#)
- FUNNYWILD
  - ftypes.h, [1428](#)
- funoffset
  - R\_const, [377](#)
- funpowers
  - C\_const, [57](#)
- FunScratch
  - N\_const, [250](#)
- funsizes
  - NeStInG, [269](#)
- FunSorts
  - N\_const, [250](#)
- FUNTOFUN
  - ftypes.h, [1417](#)
- funwith
  - DICTIONARY, [83](#)
- fval
  - OPTIMIZE, [292](#)
- fwin.h, [1495](#), [1497](#)
  - ALTSEPARATOR, [1496](#)
  - CARRIAGERETURN, [1496](#)
  - LINEFEED, [1496](#)
  - P\_term, [1496](#)
  - PATHSEPARATOR, [1496](#)
  - SEPARATOR, [1496](#)
  - WITH\_ENV, [1497](#)
  - WITHRETURN, [1496](#)
  - WITHSYSTEM, [1496](#)
- fwrite\_cached
  - checkpoint.c, [543](#)
- gamm
  - TrAcEs, [476](#)
- GAMMA
  - ftypes.h, [1392](#)
- GAMMA1
  - ftypes.h, [1428](#)
- GAMMA5
  - ftypes.h, [1428](#)
- gamma5
  - TrAcEs, [477](#)
- GAMMA6
  - ftypes.h, [1428](#)
- GAMMA7
  - ftypes.h, [1428](#)
- GAMMAFIVE
  - ftypes.h, [1392](#)
- GAMMAFUNCTION
  - ftypes.h, [1426](#)
- GAMMAI
  - ftypes.h, [1392](#)
- GAMMASEVEN
  - ftypes.h, [1392](#)
- GAMMASIX
  - ftypes.h, [1392](#)
- GarbHand
  - declare.h, [847](#)
  - sort.c, [2725](#)
- gcd
  - Fraction, [141](#)
- gcd\_heuristic\_failed, [147](#)
- gcd\_heuristic\_possible
  - polygcd.cc, [2256](#)
- gcd\_linear\_helper
  - polygcd.cc, [2257](#)
- gcd\_mpoly
  - flintinterface.h, [1272](#)
- gcd\_poly
  - flintinterface.h, [1272](#)
- GCDclean
  - declare.h, [1009](#)
  - ratio.c, [2505](#)
- GCDFUNCTION
  - ftypes.h, [1397](#)
- GCDfunction
  - declare.h, [1008](#)
  - ratio.c, [2502](#)
- GCDfunction3
  - declare.h, [1008](#)
  - ratio.c, [2502](#)
- GcdLong
  - declare.h, [847](#)
  - reken.c, [2558](#)
- GcdLong\_fix\_args
  - optimize.cc, [1993](#)
- GCDMAX
  - reken.c, [2551](#)
- GCDterms
  - declare.h, [1009](#)
  - ratio.c, [2504](#)
- gcmmod
  - M\_const, [172](#)
- gcNumDollars
  - M\_const, [183](#)

- gCnumpows
  - M\_const, [186](#)
- gCodesFlag
  - M\_const, [180](#)
- gDefDim
  - M\_const, [178](#)
- gDefDim4
  - M\_const, [178](#)
- GENCOMMUTE
  - ftypes.h, [1438](#)
- GenDiagrams
  - declare.h, [835](#)
- GENERALFUNCTION
  - ftypes.h, [1426](#)
- generate
  - MGraph, [207](#)
  - SProcess, [404](#)
  - T\_MGraph, [447](#)
- generate\_expression
  - optimize.cc, [2003](#)
- generate\_instructions
  - optimize.cc, [1995](#)
- generate\_output
  - optimize.cc, [2002](#)
- GenerateFactors
  - compiler.c, [726](#)
  - declare.h, [961](#)
- GenerateTopologies
  - declare.h, [836](#)
  - topowrap.cc, [3141](#)
- GenerateVertices
  - topowrap.cc, [3141](#)
- Generator
  - declare.h, [847](#)
  - proces.c, [2429](#)
- GenLoops
  - declare.h, [1030](#)
  - lus.c, [1688](#)
- genNext
  - SGroup, [387](#)
- GENNONCOMMUTE
  - ftypes.h, [1439](#)
- GenPaths
  - declare.h, [1031](#)
  - lus.c, [1688](#)
- GenSpace
  - sOrT, [398](#)
- GenTerms
  - sOrT, [398](#)
- gentopo.cc, [1497](#)
- gentopo.h, [1516](#)
- GenTopologies
  - declare.h, [835](#)
- get\_brackets
  - optimize.cc, [1990](#)
- get\_expression
  - optimize.cc, [1989](#)
- get\_variables
  - flintinterface.h, [1273](#)
  - poly, [347](#)
- GetAppendChannel
  - declare.h, [898](#)
  - tools.c, [3084](#)
- GetAutoName
  - declare.h, [890](#)
  - names.c, [1825](#)
- GETBIDENTITY
  - declare.h, [819](#)
- GetBinom
  - declare.h, [849](#)
  - reken.c, [2558](#)
- GETCFROMEXTCHANNEL
  - declare.h, [820](#)
- GetChannel
  - declare.h, [898](#)
  - tools.c, [3083](#)
- GetChar
  - declare.h, [904](#)
  - pre.c, [2317](#)
- GETCOEF
  - declare.h, [800](#)
- GetContent
  - declare.h, [977](#)
  - execute.c, [1159](#)
- getCurrentExternalChannel
  - declare.h, [990](#)
  - extcmd.c, [1193](#)
- GetDbase
  - declare.h, [985](#)
  - minos.c, [1731](#)
- getDef
  - Options, [300](#)
- GetDollar
  - declare.h, [920](#)
  - names.c, [1826](#)
- GetDollarNumber
  - declare.h, [909](#)
  - pre.c, [2331](#)
- GetDolNum
  - declare.h, [968](#)
  - dollar.c, [1101](#)
- GetDoParam
  - compcomm.c, [629](#)
  - declare.h, [1007](#)
- GetFile
  - R\_const, [374](#)
- GetFirstBracket
  - declare.h, [977](#)
  - execute.c, [1159](#)
- GetFirstTerm
  - declare.h, [977](#)
  - execute.c, [1159](#)
- getFLines
  - EGraph, [107](#)
- GetFromSpectator
  - declare.h, [1024](#)

- spectator.c, 2788
- GetFromStore
  - declare.h, 850
  - store.c, 2830
- GetFromStream
  - declare.h, 887
  - tools.c, 3079
- GetFunction
  - declare.h, 890
  - names.c, 1825
- GETIDENTITY
  - declare.h, 819
- GetIfDollarFactor
  - compcomm.c, 629
  - declare.h, 1007
- GetIfDollarNum
  - declare.h, 839
  - if.c, 1655
- GetInput
  - declare.h, 904
  - pre.c, 2317
- GetLabel
  - declare.h, 960
  - names.c, 1833
- GetLastExprName
  - declare.h, 890
  - names.c, 1825
- getLegParticle
  - Assign, 19
- GetLong
  - declare.h, 850
  - reken.c, 2558
- GetLongModInverses
  - declare.h, 868
  - reken.c, 2557
- GetModInverses
  - declare.h, 867
  - reken.c, 2556
- GetMoreFromMem
  - declare.h, 850
  - store.c, 2830
- GetMoreTerms
  - declare.h, 850
  - store.c, 2830
- GetName
  - declare.h, 889
  - names.c, 1824
- GetNode
  - declare.h, 889
  - names.c, 1824
- GetNumber
  - declare.h, 890
  - names.c, 1825
- GetOName
  - declare.h, 891
  - names.c, 1825
- GetOneFile
  - R\_const, 375
- GetOneTerm
  - declare.h, 850
  - store.c, 2830
- GetPID
  - form3.h, 1334
- GetPosFile
  - declare.h, 897
  - tools.c, 3083
- getpower
  - optimize.cc, 1991
- GetPreVar
  - declare.h, 902
  - pre.c, 2318
- GetRunningTime
  - declare.h, 864
  - startup.c, 2801
- GetSetupPar
  - declare.h, 903
  - setfile.c, 2690
- GETSTOP
  - declare.h, 800
- GetStreamPosition
  - declare.h, 925
  - tools.c, 3080
- GetTable
  - declare.h, 851
  - store.c, 2833
- GetTableName
  - declare.h, 988
  - minos.c, 1732
- GETTERM
  - declare.h, 821
- GetTerm
  - declare.h, 851
  - store.c, 2830
- getValue
  - Options, 298
- GetVar
  - declare.h, 890
  - names.c, 1825
- GetWildcardName
  - declare.h, 924
  - names.c, 1828
- gextrasym
  - M\_const, 174
- gextrasymbols
  - M\_const, 191
- gFinalStats
  - M\_const, 181
- GFIVE
  - ftypes.h, 1434
- gFortran90Kind
  - M\_const, 174
- gfunpowers
  - M\_const, 179
- ggextrasym
  - M\_const, 174
- ggextrasymbols



- M\_const, [192](#)
- ggFinalStats
  - M\_const, [181](#)
- ggIndentSpace
  - M\_const, [191](#)
- ggNoSpacesInNumbers
  - M\_const, [183](#)
- ggnumextrasym
  - M\_const, [184](#)
- ggOldFactArgFlag
  - M\_const, [184](#)
- ggOldGCDflag
  - M\_const, [184](#)
- ggOldParallelStats
  - M\_const, [182](#)
- ggProcessStats
  - M\_const, [182](#)
- ggShortStatsMax
  - M\_const, [191](#)
- ggStatsFlag
  - M\_const, [187](#)
- ggThreadBalancing
  - M\_const, [182](#)
- ggThreadBucketSize
  - M\_const, [177](#)
- ggThreadsFlag
  - M\_const, [181](#)
- ggThreadSortFileSynch
  - M\_const, [182](#)
- ggThreadStats
  - M\_const, [181](#)
- ggWTimeStatsFlag
  - M\_const, [185](#)
- gld
  - T\_EGraph, [443](#)
- GIDENT
  - ftypes.h, [1434](#)
- gIndentSpace
  - M\_const, [191](#)
- ginsidefirst
  - M\_const, [178](#)
- glsFortran90
  - M\_const, [183](#)
- gLineLength
  - M\_const, [187](#)
- GLOBAL
  - ftypes.h, [1438](#)
- GLOBALEXPRESSION
  - ftypes.h, [1422](#)
- GlobalFunctions
  - variable.h, [3202](#)
- GlobalIndices
  - variable.h, [3202](#)
- Globalize
  - declare.h, [924](#)
  - names.c, [1833](#)
- GLOBALMODULE
  - ftypes.h, [1368](#)
- globalnamefill
  - NaMeTree, [265](#)
- globalnodefill
  - NaMeTree, [265](#)
- GlobalSetElements
  - variable.h, [3203](#)
- GlobalSets
  - variable.h, [3203](#)
- GLOBALSPECTATORFLAG
  - ftypes.h, [1454](#)
- GlobalSymbols
  - variable.h, [3202](#)
- GlobalVectors
  - variable.h, [3203](#)
- Glue
  - declare.h, [851](#)
  - ratio.c, [2502](#)
- GMINUS
  - ftypes.h, [1434](#)
- gmodmode
  - M\_const, [186](#)
- GMPSPREAD
  - float.c, [1286](#)
- gNamesFlag
  - M\_const, [180](#)
- gncmod
  - M\_const, [185](#)
- gNoSpacesInNumbers
  - M\_const, [183](#)
- gnpowmod
  - M\_const, [186](#)
- GNUC\_PREREQ
  - parallel.h, [2139](#)
- gNumDictionaries
  - O\_const, [284](#)
- gnumelements
  - DICTIONARY, [83](#)
- gnumextrasym
  - M\_const, [184](#)
- gNumPre
  - P\_const, [315](#)
- gOldFactArgFlag
  - M\_const, [184](#)
- gOldGCDflag
  - M\_const, [184](#)
- gOldParallelStats
  - M\_const, [182](#)
- gOutNumberType
  - M\_const, [186](#)
- gOutputMode
  - M\_const, [186](#)
- gOutputSpaces
  - M\_const, [186](#)
- gparallelflag
  - M\_const, [180](#)
- GPLUS
  - ftypes.h, [1434](#)
- gPolyFun

- M\_const, 190
- gPolyFunExp
  - M\_const, 190
- gPolyFunInv
  - M\_const, 190
- gPolyFunPow
  - M\_const, 190
- gPolyFunType
  - M\_const, 190
- gPolyFunVar
  - M\_const, 190
- gpowmod
  - M\_const, 172
- gProcessBucketSize
  - M\_const, 176
- gProcessStats
  - M\_const, 182
- gproperorderflag
  - M\_const, 180
- grcc.cc, 1518
- grcc.h, 1636
- grccmodel
  - MoDeL, 233
- grccparam.h, 1652
- GREATER
  - ftypes.h, 1436
- GREATEREQUAL
  - ftypes.h, 1436
- greedymaxperc
  - OPTIMIZE, 294
- greedyminnum
  - OPTIMIZE, 293
- greedytimelimit
  - OPTIMIZE, 293
- group
  - MGraph, 210
- groupLMom
  - EGraph, 107
- gShortStatsMax
  - M\_const, 191
- gSizeCommutelnSet
  - M\_const, 181
- gSortType
  - M\_const, 180
- gStatsFlag
  - M\_const, 180
- gSubId
  - EGraph, 110
- gThreadBalancing
  - M\_const, 181
- gThreadBucketSize
  - M\_const, 177
- gThreadsFlag
  - M\_const, 181
- gThreadSortFileSynch
  - M\_const, 182
- gThreadStats
  - M\_const, 181
- gTokensWriteFlag
  - M\_const, 179
- gUniTrace
  - M\_const, 186
- gUnitTrace
  - M\_const, 186
- gWTimeStatsFlag
  - M\_const, 185
- gzipCompress
  - R\_const, 373
- GZIPDEFAULT
  - fsizes.h, 1348
- HAAKJE
  - ftypes.h, 1391
- HAAKJE0
  - ftypes.h, 1390
- halfmod
  - C\_const, 48
- Handle
  - FiLeDaTa, 132
- handle
  - ChAnNeL, 75
  - dbase, 79
  - FiLe, 130
  - StreaM, 418
- HANDLERS, 147
  - newhandle, 148
  - newlogonly, 148
  - oldhandle, 148
  - oldlogonly, 148
  - oldprinttype, 148
  - oldsilent, 148
- hash
  - node, 273
- hash\_range
  - optimize.cc, 1995
- havesortdir
  - M\_const, 192
- HEADERFILE
  - ftypes.h, 1368
- headermark
  - STOREHEADER, 413
- headnode
  - NaMeTree, 266
- helptb
  - tables.c, 2966
- hi
  - VaRrEnUm, 483
- HIDDENEXPRESSION
  - ftypes.h, 1422
- HIDDENLEXPRESSION
  - ftypes.h, 1422
- HIDE
  - ftypes.h, 1421
- hidefile
  - R\_const, 371
- HIDEGEXPRESSION
  - ftypes.h, 1423

- HideLevel
  - C\_const, 68
- hidelevel
  - ExPrEsSiOn, 122
- HIDELEXPRESSION
  - ftypes.h, 1423
- HideSize
  - M\_const, 175
- highfirst
  - setfile.c, 2693
- HighWarning
  - declare.h, 885
  - message.c, 1715
- HoldFlag
  - M\_const, 188
- horner
  - OPTIMIZE, 292
- Horner\_tree
  - optimize.cc, 1994
- hornerdirection
  - OPTIMIZE, 293
- HowMany
  - declare.h, 983
  - if.c, 1656
- hparallelflag
  - M\_const, 180
- hProcessBucketSize
  - M\_const, 176
- HTOZARG
  - ftypes.h, 1448
- humanBytesSuffix
  - sort.c, 2730
- HumanStatsFlag
  - C\_const, 62
- HumanString
  - sort.c, 2716
- HUMANSTRLEN
  - sort.c, 2715
- HUMANSUFFLEN
  - sort.c, 2716
- humanTermsSuffix
  - sort.c, 2730
- i
  - bufIPstruct, 37
- iblocks
  - dbase, 79
- iBufError
  - P\_const, 314
- iBuffer
  - C\_const, 48
- iBufferSize
  - C\_const, 52
- icode
  - lInput, 149
  - Interaction, 162
- id
  - EEdge, 98
  - ENode, 117
  - Interaction, 161
  - MNode, 219
  - Particle, 325
  - Process, 365
  - SProcess, 408
  - T\_MNode, 452
- idallflag
  - T\_const, 434
- idallmaxnum
  - T\_const, 434
- idallnum
  - T\_const, 434
- IDFUNCTION
  - ftypes.h, 1402
- idfunctionflag
  - N\_const, 258
- IDHEAD
  - ftypes.h, 1434
- idoption
  - C\_const, 56
- iexp
  - declare.h, 900
  - tools.c, 3090
- if.c, 1655, 1657
  - DoEndSwitch, 1656
  - DolfStatement, 1655
  - DoSwitch, 1656
  - DoubleIfBuffers, 1656
  - DoubleSwitchBuffers, 1656
  - FindCase, 1656
  - FindVar, 1655
  - GetIfDollarNum, 1655
  - HowMany, 1656
  - SwitchSplitMerge, 1657
  - SwitchSplitMergeRec, 1657
- IfCount
  - C\_const, 48
- IFDOLLAR
  - ftypes.h, 1435
- IFDOLLAREXTRA
  - ftypes.h, 1435
- IFEXPRESSION
  - ftypes.h, 1435
- IFFLOATNUMBER
  - ftypes.h, 1436
- IfHeap
  - C\_const, 48
- IFISFACTORIZED
  - ftypes.h, 1436
- IfLevel
  - C\_const, 67
- iflevel
  - SWITCH, 422
- IFOCCURS
  - ftypes.h, 1436
- IfStack
  - C\_const, 48
- IfSumCheck

- C\_const, [52](#)
- IFUSERFLAG
  - ftypes.h, [1436](#)
- IFW
  - flintinterface.cc, [1242](#)
- IgnoreDeprecation
  - M\_const, [193](#)
- lInput, [149](#)
  - cvallist, [150](#)
  - icode, [149](#)
  - name, [149](#)
  - nplistn, [149](#)
  - plistc, [150](#)
  - plistn, [149](#)
- ilist
  - NCand, [267](#)
  - NStack, [276](#)
- IMPLODEARG
  - ftypes.h, [1446](#)
- improve
  - optimization, [291](#)
- INANDOUT
  - minos.h, [1754](#)
- inbuffer
  - File, [130](#)
  - Stream, [418](#)
- IncDir
  - M\_const, [173](#)
- incdollar
  - DoLoOp, [92](#)
- INCEXPRESSION
  - ftypes.h, [1423](#)
- INCLENG
  - declare.h, [800](#)
- Include
  - declare.h, [908](#)
  - pre.c, [2322](#)
- incMat
  - MNodeClass, [222](#)
  - T\_MNodeClass, [455](#)
- incnum
  - DoLoOp, [91](#)
- incoef
  - SHvariables, [390](#)
- IndDum
  - M\_const, [185](#)
  - N\_const, [258](#)
- INDENTSPACE
  - fsizes.h, [1349](#)
- IndentSpace
  - O\_const, [285](#)
- INDEX
  - ftypes.h, [1389](#)
- InDeX, [150](#)
  - dimension, [151](#)
  - flags, [151](#)
  - name, [150](#)
  - namesize, [151](#)
  - nmin4, [151](#)
  - node, [151](#)
  - number, [151](#)
  - type, [150](#)
- Index
  - FiLeDaTa, [132](#)
- index
  - DoLIaRS, [88](#)
  - PF\_BUFFER, [330](#)
- index.c, [1671](#), [1673](#)
  - ClearBracketIndex, [1672](#)
  - FindBracket, [1672](#)
  - OpenBracketIndex, [1672](#)
  - PutBracketInIndex, [1672](#)
  - PutInside, [1672](#)
- INDEX\_
  - ftypes.h, [1427](#)
- indexblock, [152](#)
  - flags, [152](#)
  - objects, [152](#)
  - position, [152](#)
  - previousblock, [152](#)
- indexbuffer
  - BrAcKeTiNfO, [33](#)
- indexbuffersize
  - BrAcKeTiNfO, [33](#)
- INDEXENTRY
  - structs.h, [2899](#)
- InDeXeNtRy, [153](#)
  - CompressSize, [154](#)
  - length, [154](#)
  - name, [155](#)
  - nfunctions, [155](#)
  - nindices, [154](#)
  - nsymbols, [154](#)
  - nvectors, [155](#)
  - position, [154](#)
  - size, [155](#)
  - variables, [154](#)
- indexfill
  - BrAcKeTiNfO, [34](#)
- IndexList
  - C\_const, [43](#)
- INDEXONLY
  - ftypes.h, [1376](#)
- indi
  - ReNuMbEr, [379](#)
- Indices
  - C\_const, [45](#)
- indices
  - parallel.h, [2140](#)
  - variable.h, [3201](#)
- indnum
  - ReNuMbEr, [379](#)
- INDTOIND
  - ftypes.h, [1417](#)
- INDTOSUB
  - ftypes.h, [1417](#)

- inexprlevel
  - C\_const, [64](#)
- inexprstack
  - C\_const, [52](#)
- inexprsumcheck
  - C\_const, [64](#)
- InFbrack
  - O\_const, [285](#)
- infile
  - R\_const, [371](#)
- INFILEINDEX
  - structs.h, [2898](#)
- info
  - dbase, [78](#)
- InFunction
  - declare.h, [851](#)
  - proces.c, [2426](#)
- InHiBuf
  - R\_const, [372](#)
- inicbufs
  - comtool.c, [763](#)
  - declare.h, [898](#)
- inictable
  - compiler.c, [724](#)
  - declare.h, [898](#)
- IniFbuffer
  - comtool.c, [767](#)
  - declare.h, [1008](#)
- iniinfo, [156](#)
  - entriesinindex, [156](#)
  - firstindexblock, [156](#)
  - firstnameblock, [157](#)
  - lastindexblock, [156](#)
  - lastnameblock, [157](#)
  - numberofindexblocks, [156](#)
  - numberofnamesblocks, [157](#)
  - numeroftables, [156](#)
- IniLine
  - declare.h, [852](#)
  - sch.c, [2651](#)
- INILOCK
  - declare.h, [818](#)
- IniModule
  - declare.h, [905](#)
  - pre.c, [2319](#)
- InInBuf
  - R\_const, [372](#)
- INIRWLOCK
  - declare.h, [818](#)
- IniSpecialModule
  - declare.h, [905](#)
  - pre.c, [2319](#)
- init
  - MCBlock, [194](#)
  - MConn, [198](#)
  - MCOpi, [203](#)
  - MGraph, [207](#)
  - MNodeClass, [221](#)
  - T\_EGraph, [442](#)
  - T\_MNodeClass, [454](#)
- initAss
  - ENode, [116](#)
- initlc
  - FGInput, [127](#)
- initln
  - FGInput, [127](#)
- initlPart
  - Process, [365](#)
- INITNAMESIZE
  - fsizes.h, [1344](#)
- INITNODESIZE
  - fsizes.h, [1344](#)
- iniTools
  - declare.h, [1000](#)
  - tools.c, [3088](#)
- initPerm
  - MOrbits, [238](#)
- initPresetExternalChannels
  - declare.h, [989](#)
  - extcmd.c, [1193](#)
- InitRecovery
  - checkpoint.c, [542](#)
  - declare.h, [997](#)
- inivar.h, [1681](#), [1683](#)
  - A, [1682](#)
  - BufferForOutput, [1682](#)
  - FG, [1682](#)
  - fixedsets, [1682](#)
  - setupfilename, [1682](#)
- IniVars
  - declare.h, [852](#)
  - startup.c, [2799](#)
- iniwranf
  - declare.h, [992](#)
  - reken.c, [2561](#)
- inline
  - form3.h, [1327](#)
- inlist
  - TrAcEn, [472](#)
  - TrAcEs, [474](#)
- inmem
  - ExPrEsSiOn, [122](#)
- InnerTest
  - C\_const, [69](#)
- inNum
  - sOrT, [400](#)
- InNumMem
  - T\_const, [433](#)
- InOutBuf
  - P\_const, [313](#)
- inparallelflag
  - C\_const, [60](#)
- inPatches
  - sOrT, [396](#)
- inprimelist
  - T\_const, [439](#)

- InputFileName
  - M\_const, 173
- INPUTONLY
  - minos.h, 1754
- INPUTOUT
  - ftypes.h, 1373
- INPUTSTREAM
  - ftypes.h, 1367
- inscbuf
  - INSIDEINFO, 159
- inscheme
  - O\_const, 281
- InScratch
  - N\_const, 252
- InScratch1
  - N\_const, 252
- InsertArg
  - argument.c, 491
  - declare.h, 931
- InsertTerm
  - declare.h, 852
  - proces.c, 2427
- inside
  - P\_const, 311
- InsideDollar
  - declare.h, 974
  - dollar.c, 1096
- insidefirst
  - C\_const, 54
- INSIDEINFO, 157
  - buffer, 158
  - inscbuf, 159
  - numdollars, 158
  - oldcbuf, 158
  - oldcnumlhs, 159
  - oldcompiletype, 158
  - oldnumpotmoddollars, 158
  - oldparallelflag, 158
  - oldrbuf, 159
  - size, 158
- insidelevel
  - C\_const, 64
- insidestack
  - C\_const, 52
- insidesumcheck
  - C\_const, 64
- INSLNGTH
  - sort.c, 2716
- InsTableTree
  - declare.h, 972
  - tables.c, 2961
- InsTree
  - comtool.c, 767
  - declare.h, 961
- insubexpbuffers
  - compiler.c, 727
- integer\_lcoeff
  - poly, 345
- IntegerToFloat
  - float.c, 1287
- Interact
  - M\_const, 177
- Interaction, 159
  - ~Interaction, 160
  - clist, 161
  - csum, 161
  - icode, 162
  - id, 161
  - Interaction, 160
  - loop, 162
  - mdl, 161
  - name, 161
  - nclist, 162
  - nlegs, 162
  - nplist, 162
  - nslist, 162
  - plist, 161
  - prInteraction, 161
  - slist, 161
- interacts
  - Model, 230
- internalset
  - TERMINFO, 467
- INTFUNCTION
  - ftypes.h, 1395
- INTOHide
  - ftypes.h, 1421
- INTOHideGEXPRESSION
  - ftypes.h, 1424
- INTOHideLEXPRESSION
  - ftypes.h, 1423
- intPart
  - Model, 230
- intracestack
  - N\_const, 254
- intrct
  - DNode, 86
  - ENode, 117
- INUSE
  - ftypes.h, 1451
- inverse\_poly
  - flintinterface.h, 1273
- INVERSEFACTORIAL
  - ftypes.h, 1394
- INVERSEFUNCTION
  - ftypes.h, 1401
- INVERSETABLE
  - ftypes.h, 1443
- invertices
  - MoDeL, 234
- invfacnum
  - M\_const, 189
- InvPoly
  - declare.h, 1012
  - ratio.c, 2506
- iPatches

- sOrT, [397](#)
- iPointer
  - C\_const, [49](#)
- iranf
  - declare.h, [992](#)
  - reken.c, [2562](#)
- is\_dense\_univariate
  - poly, [344](#)
- is\_integer
  - poly, [344](#)
- is\_monomial
  - poly, [344](#)
- is\_one
  - poly, [344](#)
- is\_zero
  - poly, [344](#)
- IsBracket
  - O\_const, [285](#)
- ISCOMPRESSED
  - ftypes.h, [1430](#)
- isConnected
  - MGraph, [208](#)
- isEq
  - Fraction, [141](#)
- ISEQUALPOS
  - declare.h, [812](#)
- ISEQUALPOSINC
  - declare.h, [811](#)
- IsExponentSign
  - declare.h, [1021](#)
  - dict.c, [1075](#)
- isExternal
  - EGraph, [105](#)
  - MGraph, [207](#)
- ISFACTORIZED
  - ftypes.h, [1431](#)
- isFermion
  - EGraph, [105](#)
- ISFORTRAN90
  - ftypes.h, [1387](#)
- IsFortran90
  - C\_const, [61](#)
- ISGEPOS
  - declare.h, [813](#)
- ISGEPOSINC
  - declare.h, [811](#)
- IsIdStatement
  - compiler.c, [725](#)
  - declare.h, [957](#)
- ISINRHS
  - ftypes.h, [1430](#)
- isIsomorphic
  - Assign, [22](#)
  - MGraph, [208](#)
- ISLESSPOS
  - declare.h, [813](#)
- IsLikeVector
  - declare.h, [901](#)
- tools.c, [3091](#)
- ISLYNDON
  - ftypes.h, [1446](#)
- ISLYNDONR
  - ftypes.h, [1446](#)
- ISMINPOS
  - declare.h, [812](#)
- IsMultipleOf
  - declare.h, [979](#)
  - dollar.c, [1097](#)
- IsMultiplySign
  - declare.h, [1022](#)
  - dict.c, [1075](#)
- ISNEGPOS
  - declare.h, [814](#)
- isNeutral
  - Particle, [324](#)
- isnextchar
  - StreaM, [419](#)
- ISNOTEQUALPOS
  - declare.h, [813](#)
- ISNOTFORTRAN90
  - ftypes.h, [1387](#)
- ISNOTZEROPOS
  - declare.h, [813](#)
- isOptE
  - EGraph, [104](#)
- isOptM
  - MGraph, [209](#)
- isOrdLegs
  - Assign, [22](#)
- isOrdPLeg
  - Assign, [21](#)
- ISPOSPOS
  - declare.h, [813](#)
- IsRHS
  - compiler.c, [724](#)
  - declare.h, [957](#)
- IsSetMember
  - declare.h, [979](#)
  - dollar.c, [1097](#)
- iStop
  - C\_const, [49](#)
- ISUNMODIFIED
  - ftypes.h, [1430](#)
- ISYMBOL
  - ftypes.h, [1404](#)
- ISZERO
  - ftypes.h, [1430](#)
- ISZEROPOS
  - declare.h, [813](#)
- ival
  - OPTIMIZE, [292](#)
- j3
  - TrAcEs, [475](#)
- j4
  - TrAcEs, [475](#)
- JUMPRATIO

- fsizes.h, [1350](#)
- jumpratio
  - M\_const, [185](#)
- KEEPZERO
  - ftypes.h, [1431](#)
- KeptInHold
  - R\_const, [374](#)
- KEYWORD, [163](#)
  - flags, [163](#)
  - func, [163](#)
  - name, [163](#)
  - type, [163](#)
- KEYWORDV, [164](#)
  - flags, [164](#)
  - name, [164](#)
  - type, [164](#)
  - var, [164](#)
- KILL
  - pre.c, [2316](#)
- KILLALL
  - pre.c, [2316](#)
- killSignal
  - X\_const, [487](#)
- killWholeGroup
  - X\_const, [488](#)
- klow
  - ENode, [118](#)
- kstep
  - TrAcEs, [476](#)
- ktoi
  - sOrT, [396](#)
- I
  - node, [273](#)
- LabelNames
  - C\_const, [49](#)
- Labels
  - C\_const, [50](#)
- LargeSize
  - sOrT, [397](#)
- last
  - SeTs, [383](#)
- last\_monomial
  - poly, [349](#)
- last\_monomial\_index
  - poly, [349](#)
- lastdollar
  - DoLoOp, [92](#)
- lastindexblock
  - iniinfo, [156](#)
- lastLen
  - longMultiStruct, [168](#)
- lastnameblock
  - iniinfo, [157](#)
- lastnamespace
  - P\_const, [311](#)
- lastnum
  - DoLoOp, [91](#)
- LBRACE
  - ftypes.h, [1381](#)
- lBuffer
  - sOrT, [394](#)
- lc3
  - TrAcEs, [477](#)
- lc4
  - TrAcEs, [477](#)
- LcmLong
  - declare.h, [847](#)
  - reken.c, [2559](#)
- lcoeff\_multivar
  - poly, [346](#)
- lcoeff\_univar
  - poly, [346](#)
- lDefDim
  - C\_const, [55](#)
- lDefDim4
  - C\_const, [55](#)
- LeaveNegative
  - T\_const, [436](#)
- left
  - NaMeNode, [260](#)
  - NoDe, [270](#)
  - tree, [479](#)
- legcouple
  - MoDeL, [233](#)
  - TERMINFO, [466](#)
- legEdgeParticle
  - Assign, [19](#)
- legPart
  - Assign, [20](#)
- legParticle
  - EGraph, [108](#)
- length
  - InDeXeNtRy, [154](#)
- lenLONG
  - STOREHEADER, [413](#)
- lenPOINTER
  - STOREHEADER, [413](#)
- lenPOS
  - STOREHEADER, [413](#)
- lenWORD
  - STOREHEADER, [413](#)
- LESS
  - ftypes.h, [1436](#)
- LESSEQUAL
  - ftypes.h, [1436](#)
- level
  - R\_const, [376](#)
  - SHvariables, [391](#)
  - TERMINFO, [467](#)
  - TrAcEn, [472](#)
  - TrAcEs, [478](#)
- LEVICIVITA
  - ftypes.h, [1394](#)
- IFill
  - sOrT, [394](#)



- lhdollarerror
  - P\_const, 315
- lhdollarflag
  - C\_const, 60
- lhs
  - CbUf, 72
  - DICTIONARY\_ELEMENT, 84
- LHSIDE
  - ftypes.h, 1385
- LHSIDEX
  - ftypes.h, 1385
- LI2FUNCTION
  - ftypes.h, 1400
- lijst
  - LIST, 165
- LINEFEED
  - fwin.h, 1496
  - unix.h, 3192
- LineLength
  - C\_const, 68
- linenumber
  - StreaM, 416
- LINFUNCTION
  - ftypes.h, 1400
- LinkTree
  - declare.h, 891
  - names.c, 1827
- LIST, 165
  - lijst, 165
  - maxnum, 165
  - message, 165
  - num, 165
  - numclear, 166
  - numglobal, 166
  - numtemp, 166
  - size, 166
- LISTEDLOOP
  - ftypes.h, 1374
- listinprint
  - N\_const, 250
- ListPoly
  - T\_const, 432
- ListSymbols
  - T\_const, 432
- ListSymbolsSize
  - T\_const, 435
- lloser
  - NoDe, 271
- lmom
  - EEdge, 99
- LNFUNCTION
  - ftypes.h, 1398
- LNUMBER
  - ftypes.h, 1391
- lo
  - VaRrEnUm, 483
- LOADDOLLAR
  - ftypes.h, 1418
- LoadInputFile
  - declare.h, 904
  - tools.c, 3079
- LoadInstruction
  - declare.h, 906
  - pre.c, 2319
- loadmode
  - PROCEDURE, 362
- LoadModel
  - declare.h, 1029
- LoadStatement
  - declare.h, 906
  - pre.c, 2319
- LocalConvertToPoly
  - declare.h, 1005
  - notation.c, 1939
- LOCALEXPRESSION
  - ftypes.h, 1421
- LocateBase
  - declare.h, 872
  - minos.c, 1730
- LocateFile
  - declare.h, 888
  - tools.c, 3079
- location
  - FUN\_INFO, 143
- LOCK
  - declare.h, 818
- locwildvalue
  - T\_const, 437
- log
  - ParallelVars, 319
- LogFileName
  - M\_const, 173
- LogHandle
  - C\_const, 67
- LogType
  - M\_const, 187
- LONG
  - form3.h, 1334
- LONG\_MAX\_VALUE
  - form3.h, 1328
- LONG\_MIN\_VALUE
  - form3.h, 1328
- LongCopy
  - declare.h, 921
  - tools.c, 3087
- LongLongCopy
  - declare.h, 921
  - tools.c, 3087
- longMultiStruct, 167
  - buffer, 167
  - bufpos, 167
  - lastLen, 168
  - next, 168
  - nPacks, 167
  - packpos, 167
- LONGNUMBER

- ftypes.h, 1435
- LongToLine
  - declare.h, 853
  - sch.c, 2652
- LOOKINGFORELSE
  - ftypes.h, 1371
- LOOKINGFORENDIF
  - ftypes.h, 1371
- LookInStream
  - declare.h, 887
  - tools.c, 3079
- loop
  - Interaction, 162
  - MCBlock, 194
  - MCOpI, 204
  - Process, 366
  - SProcess, 408
- LoopList
  - P\_const, 311
- loopm
  - EGraph, 113
- LoopOutput
  - declare.h, 1030
  - lus.c, 1688
- LowerSortLevel
  - declare.h, 973
  - sort.c, 2729
- lowfirst
  - setfile.c, 2693
- LPARENTHESIS
  - ftypes.h, 1381
- IPatch
  - sOrT, 399
- IPolyFun
  - C\_const, 68
- IPolyFunExp
  - C\_const, 68
- IPolyFunInv
  - C\_const, 68
- IPolyFunPow
  - C\_const, 69
- IPolyFunType
  - C\_const, 68
- IPolyFunVar
  - C\_const, 68
- LRPARENTHESSES
  - ftypes.h, 1383
- ISortType
  - C\_const, 58
- lsrc
  - NoDe, 271
- ITop
  - sOrT, 394
- IUniTrace
  - C\_const, 64
- IUnitTrace
  - C\_const, 66
- Lus
  - declare.h, 963
  - lus.c, 1687
- lus.c, 1686, 1690
  - AllLoops, 1687
  - AllOnePI, 1689
  - AllPaths, 1688
  - FindLus, 1687
  - GenLoops, 1688
  - GenPaths, 1688
  - LoopOutput, 1688
  - Lus, 1687
  - OutputOnePI, 1689
  - PathOutput, 1689
  - RemoveBridges, 1689
  - SortTheList, 1687
  - StartLoops, 1687
  - TakeOneLine, 1689
- IWorkPointer
  - T\_const, 433
- IWorkSize
  - R\_const, 373
- IWorkSpace
  - T\_const, 429
- M
  - AllGlobals, 10
- m
  - MODNUM, 235
- M\_alloc
  - declare.h, 820
- M\_check1
  - declare.h, 979
  - tools.c, 3089
- M\_const, 168
  - atstartup, 190
  - BracketFactors, 192
  - ClearStore, 192
  - CompressSize, 174
  - countfunnum, 189
  - dbufnum, 179
  - denomnum, 188
  - dollarzero, 190
  - DumInd, 185
  - exitflag, 191
  - expnum, 188
  - facnum, 189
  - fbufferize, 183
  - FileOnlyFlag, 177
  - FortranCont, 188
  - FromStdin, 192
  - gcmmod, 172
  - gcNumDollars, 183
  - gCnumpows, 186
  - gCodesFlag, 180
  - gDefDim, 178
  - gDefDim4, 178
  - gextrasym, 174
  - gextrasymbols, 191
  - gFinalStats, 181

gFortran90Kind, 174  
gfunpowers, 179  
ggextrasym, 174  
ggextrasymbols, 192  
ggFinalStats, 181  
ggIndentSpace, 191  
ggNoSpacesInNumbers, 183  
ggnumextrasym, 184  
ggOldFactArgFlag, 184  
ggOldGCDflag, 184  
ggOldParallelStats, 182  
ggProcessStats, 182  
ggShortStatsMax, 191  
ggStatsFlag, 187  
ggThreadBalancing, 182  
ggThreadBucketSize, 177  
ggThreadsFlag, 181  
ggThreadSortFileSynch, 182  
ggThreadStats, 181  
ggWTimeStatsFlag, 185  
gIndentSpace, 191  
ginsidefirst, 178  
gIsFortran90, 183  
gLineLength, 187  
gmodmode, 186  
gNamesFlag, 180  
gncmod, 185  
gNoSpacesInNumbers, 183  
gnpowmod, 186  
gnumextrasym, 184  
gOldFactArgFlag, 184  
gOldGCDflag, 184  
gOldParallelStats, 182  
gOutNumberType, 186  
gOutputMode, 186  
gOutputSpaces, 186  
gparallelflag, 180  
gPolyFun, 190  
gPolyFunExp, 190  
gPolyFunInv, 190  
gPolyFunPow, 190  
gPolyFunType, 190  
gPolyFunVar, 190  
gpowmod, 172  
gProcessBucketSize, 176  
gProcessStats, 182  
gproperorderflag, 180  
gShortStatsMax, 191  
gSizeCommutelnSet, 181  
gSortType, 180  
gStatsFlag, 180  
gThreadBalancing, 181  
gThreadBucketSize, 177  
gThreadsFlag, 181  
gThreadSortFileSynch, 182  
gThreadStats, 181  
gTokensWriteFlag, 179  
gUnitTrace, 186  
gWTimeStatsFlag, 185  
havesortdir, 192  
HideSize, 175  
HoldFlag, 188  
hparallelflag, 180  
hProcessBucketSize, 176  
IgnoreDeprecation, 193  
IncDir, 173  
IndDum, 185  
InputFileName, 173  
Interact, 177  
invfacnum, 189  
jumpratio, 185  
LogFileName, 173  
LogType, 187  
matchfunnum, 189  
MaxBracketBufferSize, 176  
maxFlevels, 182  
MaxParLevel, 177  
MaxStreamSize, 175  
MaxTal, 185  
MaxTer, 174  
MaxWildcards, 187  
mTraceDum, 187  
MultiRun, 183  
NumFixedFunctions, 179  
NumFixedSets, 178  
NumSpectatorFiles, 184  
NumStoreCaches, 191  
OffsetIndex, 187  
OffsetVector, 187  
OldChildTime, 176  
OldMilliTime, 176  
oldnumextrasymbols, 174  
OldOrderFlag, 189  
OldSecTime, 176  
onerhs, 192  
Ordering, 188  
OutBuffer, 173  
OutBufSize, 178  
Path, 173  
PrintTotalSize, 183  
qError, 188  
rbufnum, 179  
RepMax, 187  
resetTimeOnClear, 183  
S0, 172  
sbufnum, 179  
ScratSize, 174  
SetupDir, 173  
SetupFile, 173  
shmWinSize, 176  
silent, 188  
SIOsize, 175  
SizeForSpectatorFiles, 184  
SizeStoreCache, 175  
SkipClears, 179

- SLargeSize, [175](#)
- SMaxFpatches, [178](#)
- SMaxPatches, [178](#)
- SpectatorFiles, [174](#)
- SpectatorSize, [177](#)
- SSmallEsize, [175](#)
- SSmallSize, [175](#)
- StdOut, [178](#)
- STermsInSmall, [175](#)
- sumnum, [189](#)
- sumpnum, [189](#)
- SumTime, [177](#)
- TempDir, [172](#)
- TempSortDir, [173](#)
- termfunnum, [189](#)
- TimeLimit, [177](#)
- totalnumberofthreads, [180](#)
- tracebackflag, [188](#)
- vectorzero, [192](#)
- Willnd, [185](#)
- WorkSize, [176](#)
- zbufnum, [179](#)
- zeropos, [172](#)
- zerorhs, [192](#)
- M\_free
  - declare.h, [923](#)
  - tools.c, [3088](#)
- M\_print
  - declare.h, [979](#)
  - tools.c, [3089](#)
- main
  - startup.c, [2800](#)
- MAINSORT
  - ftypes.h, [1429](#)
- MAJORVERSION
  - form3.h, [1326](#)
- MAKEARGS
  - ftypes.h, [1445](#)
- MakeDate
  - declare.h, [899](#)
  - tools.c, [3087](#)
- MakeDirty
  - declare.h, [853](#)
  - function.c, [1470](#)
- MakeDollarInteger
  - declare.h, [967](#)
  - dollar.c, [1099](#)
- MakeDollarMod
  - declare.h, [967](#)
  - dollar.c, [1100](#)
- MakeDubious
  - declare.h, [890](#)
  - names.c, [1832](#)
- MakeGlobal
  - declare.h, [919](#)
  - execute.c, [1157](#)
- MakeInteger
  - argument.c, [493](#)
  - declare.h, [930](#)
- MakeInverses
  - declare.h, [867](#)
  - reken.c, [2556](#)
- MakeLongRational
  - declare.h, [971](#)
  - reken.c, [2559](#)
- MakeMod
  - argument.c, [493](#)
  - declare.h, [930](#)
- MakeModTable
  - declare.h, [854](#)
  - reken.c, [2560](#)
- MakeNameTree
  - declare.h, [892](#)
  - names.c, [1827](#)
- MAKERATIONAL
  - ftypes.h, [1401](#)
- MakeRational
  - declare.h, [971](#)
  - reken.c, [2559](#)
- MakeSetupAllocs
  - declare.h, [976](#)
  - setfile.c, [2692](#)
- Malloc1
  - declare.h, [886](#)
  - tools.c, [3088](#)
- mallocprotect.h, [1708](#)
- MAPLEMODE
  - ftypes.h, [1385](#)
- MarkDirty
  - declare.h, [853](#)
  - function.c, [1470](#)
- MarkPolyRatFunDirty
  - declare.h, [817](#)
- MaskPointer
  - N\_const, [248](#)
- mass
  - PaRtlcLe, [322](#)
- MASTER
  - parallel.h, [2137](#)
- MATCH
  - ftypes.h, [1435](#)
- match
  - SProcess, [405](#)
- MatchArgument
  - declare.h, [855](#)
  - symmetr.c, [2932](#)
- MatchCy
  - declare.h, [854](#)
  - symmetr.c, [2932](#)
- MatchE
  - declare.h, [854](#)
  - symmetr.c, [2931](#)
- MATCHFUNCTION
  - ftypes.h, [1396](#)
- MatchFunction
  - declare.h, [855](#)

- function.c, [1472](#)
- matchfunnum
  - M\_const, [189](#)
- MatchIsPossible
  - declare.h, [965](#)
  - smart.c, [2709](#)
- MATHEMATICAMODE
  - ftypes.h, [1385](#)
- MaX
  - declare.h, [799](#)
- MAX\_OPEN\_FILES
  - form3.h, [1334](#)
- MaxBracket
  - R\_const, [374](#)
- MAXBRACKETBUFFERSIZE
  - fsizes.h, [1348](#)
- MaxBracketBufferSize
  - M\_const, [176](#)
- MAXBUILTINFUNCTION
  - ftypes.h, [1403](#)
- maxcase
  - SWITCH, [422](#)
- MAXCOUPLINGS
  - fsizes.h, [1351](#)
- maxcpl
  - Model, [230](#)
- maxdeg
  - EGraph, [112](#)
  - ENode, [117](#)
  - MGraph, [212](#)
  - MNodeClass, [224](#)
  - T\_EGraph, [444](#)
  - T\_MGraph, [448](#)
  - T\_MNodeClass, [456](#)
- MaxDoLoops
  - variable.h, [3200](#)
- MaxDum
  - R\_const, [376](#)
- MAXENAME
  - fsizes.h, [1343](#)
- MaxExprSize
  - S\_const, [381](#)
- MAXFLAGS
  - fsizes.h, [1345](#)
- MAXFLEVELS
  - fsizes.h, [1347](#)
- maxFlevels
  - M\_const, [182](#)
- MAXFPATCHES
  - fsizes.h, [1346](#)
- MaxFpatches
  - sOrT, [399](#)
- MAXFUNCTION
  - ftypes.h, [1395](#)
- MaxFunSorts
  - N\_const, [254](#)
- maxi
  - MiNmAx, [216](#)
- MAXIF
  - fsizes.h, [1344](#)
- MaxIf
  - C\_const, [56](#)
- MAXLABELS
  - fsizes.h, [1345](#)
- MaxLabels
  - C\_const, [55](#)
- MAXLEGS
  - fsizes.h, [1351](#)
- MAXLEVELS
  - fsizes.h, [1345](#)
- MAXLHS
  - fsizes.h, [1345](#)
- maxlhs
  - CbUf, [73](#)
- MAXLINELENGTH
  - fsizes.h, [1350](#)
- MAXLONG
  - form3.h, [1329](#)
- maxloop
  - Model, [230](#)
- MAXLOOPS
  - fsizes.h, [1345](#)
- MAXMATCH
  - fsizes.h, [1344](#)
- MAXMULTIBRACKETLEVELS
  - fsizes.h, [1350](#)
- MAXNEST
  - fsizes.h, [1344](#)
- maxnlegs
  - Model, [230](#)
  - Process, [366](#)
- maxnum
  - LIST, [165](#)
- maxnumargs
  - FuNcTiOn, [147](#)
- MAXNUMBEROFNONCOMTERMS
  - normal.c, [1873](#)
- MAXNUMBERSIZE
  - fsizes.h, [1343](#)
- MaxNumPre
  - variable.h, [3200](#)
- MAXNUMSIZE
  - fsizes.h, [1348](#)
- MaxNumStreams
  - C\_const, [56](#)
- MAXPARLEVEL
  - fsizes.h, [1343](#)
- MaxParLevel
  - M\_const, [177](#)
- MAXPARTICLES
  - fsizes.h, [1350](#)
- MAXPATCHES
  - fsizes.h, [1346](#)
- MaxPatches
  - sOrT, [398](#)
- MAXPOINTS

- topowrap.cc, [3141](#)
- MAXPOSITIVE
  - form3.h, [1329](#)
- MAXPOSITIVE2
  - form3.h, [1329](#)
- MAXPOSITIVE4
  - form3.h, [1329](#)
- maxpower
  - STOREHEADER, [415](#)
  - SyMbOI, [424](#)
- MAXPOWEROF
  - ftypes.h, [1397](#)
- MaxPreAssignLevel
  - P\_const, [316](#)
- MaxPrelfLevel
  - P\_const, [314](#)
- MAXPRENAMESIZE
  - fsizes.h, [1343](#)
- MaxPreTypes
  - P\_const, [314](#)
- MaxProcedures
  - variable.h, [3200](#)
- MAXRANGEINDICATOR
  - ftypes.h, [1445](#)
- MaxRenumScratch
  - N\_const, [255](#)
- MAXREPEAT
  - fsizes.h, [1343](#)
- maxrhs
  - CbUf, [74](#)
- MAXSAVEFUNCTION
  - fsizes.h, [1343](#)
- MaxStreamSize
  - M\_const, [175](#)
- MaxSwitch
  - C\_const, [67](#)
- MaxTal
  - M\_const, [185](#)
- MaxTer
  - M\_const, [174](#)
- maxtermlevel
  - C\_const, [59](#)
- maxtermsize
  - sOrT, [399](#)
- MaxTreeSize
  - CbUf, [74](#)
  - TaBIEs, [464](#)
- maxvar
  - OPTIMIZERESULT, [296](#)
- MAXWILDC
  - fsizes.h, [1346](#)
- MaxWildcards
  - M\_const, [187](#)
- mBer
  - T\_const, [438](#)
- MCBlock, [193](#)
  - ~MCBlock, [193](#)
  - DUMMYPPADDING, [194](#)
- edges, [194](#)
- init, [194](#)
- loop, [194](#)
- MCBlock, [193](#)
- nartps, [194](#)
- nmedges, [194](#)
- MCBridge, [195](#)
  - ~MCBridge, [195](#)
  - MCBridge, [195](#)
  - next, [195](#)
  - nodes, [195](#)
- MCLEANFLAG
  - minos.h, [1753](#)
- MConn, [196](#)
  - ~MConn, [197](#)
  - addArtic, [198](#)
  - addBlock, [198](#)
  - addBlockSelf, [199](#)
  - addBridge, [198](#)
  - addOPic, [198](#)
  - articuls, [199](#)
  - blksp, [200](#)
  - blkstk, [200](#)
  - blkstkptr, [202](#)
  - blocks, [199](#)
  - bridges, [199](#)
  - DUMMYPPADDING, [202](#)
  - init, [198](#)
  - MConn, [197](#)
  - na1blocks, [201](#)
  - narticuls, [201](#)
  - nblksp, [202](#)
  - nblocks, [201](#)
  - nbridges, [201](#)
  - nctopic, [200](#)
  - ne0bridges, [201](#)
  - ne1bridges, [201](#)
  - neblocks, [201](#)
  - nlpopic, [200](#)
  - nopic, [200](#)
  - nopisp, [202](#)
  - nselfloops, [201](#)
  - opics, [199](#)
  - opisp, [199](#)
  - opistk, [200](#)
  - opistkptr, [202](#)
  - print, [199](#)
  - pushEdge, [198](#)
  - pushNode, [198](#)
  - sedges, [200](#)
  - snodes, [200](#)
- mconn
  - MGraph, [215](#)
- MCOpi, [202](#)
  - ~MCOpi, [203](#)
  - ctloop, [204](#)
  - init, [203](#)
  - loop, [204](#)

- MCOp, [203](#)
- nedges, [204](#)
- next, [204](#)
- nlegs, [204](#)
- nnodes, [203](#)
- nodes, [203](#)
- mcts\_best\_schemes
  - optimize.cc, [2009](#)
- mcts\_expr\_score
  - optimize.cc, [2008](#)
- mcts\_factorized
  - optimize.cc, [2008](#)
- mcts\_root
  - optimize.cc, [2008](#)
- mcts\_separated
  - optimize.cc, [2008](#)
- mcts\_vars
  - optimize.cc, [2008](#)
- mctsdecaymode
  - OPTIMIZE, [294](#)
- mctsnumexpand
  - OPTIMIZE, [293](#)
- mctsnumkeep
  - OPTIMIZE, [293](#)
- mctsnumrepeat
  - OPTIMIZE, [293](#)
- mctstimelimit
  - OPTIMIZE, [293](#)
- mdefined
  - TaBIEs, [463](#)
- mdel
  - TrAcEs, [475](#)
- MDIRTYFLAG
  - minos.h, [1753](#)
- mdl
  - Interaction, [161](#)
  - Particle, [325](#)
- mdum
  - TrAcEs, [476](#)
- me
  - ParallelVars, [318](#)
- mean
  - OptDef, [289](#)
- MemDebugFlag
  - C\_const, [61](#)
- MEMORYMACROS
  - declare.h, [815](#)
- mepf
  - TrAcEs, [475](#)
- merge\_operators
  - optimize.cc, [1998](#)
- MergeDotproductLists
  - declare.h, [1011](#)
  - ratio.c, [2504](#)
- MergePatches
  - declare.h, [855](#)
  - sort.c, [2726](#)
- MergeSymbolLists
  - declare.h, [1011](#)
  - ratio.c, [2504](#)
- MergeWithFloat
  - float.c, [1294](#)
- MesCall
  - declare.h, [921](#)
  - message.c, [1715](#)
- MesCerr
  - declare.h, [856](#)
  - message.c, [1715](#)
- MesComp
  - declare.h, [856](#)
  - message.c, [1715](#)
- MesNesting
  - declare.h, [817](#)
- MesPrint
  - message.c, [1714](#)
- message
  - LIST, [165](#)
- message.c, [1713](#), [1716](#)
  - Error0, [1714](#)
  - Error1, [1714](#)
  - Error2, [1714](#)
  - HighWarning, [1715](#)
  - MesCall, [1715](#)
  - MesCerr, [1715](#)
  - MesComp, [1715](#)
  - MesPrint, [1714](#)
  - MesWork, [1714](#)
  - PrintSeq, [1716](#)
  - PrintSubTerm, [1716](#)
  - PrintTerm, [1715](#)
  - PrintTermC, [1715](#)
  - PrintWords, [1716](#)
  - Warning, [1714](#)
- MessPreNesting
  - declare.h, [924](#)
  - pre.c, [2329](#)
- MesWork
  - declare.h, [921](#)
  - message.c, [1714](#)
- method
  - OPTIMIZE, [293](#)
- mfac
  - T\_const, [433](#)
- MGraph, [205](#)
  - ~MGraph, [207](#)
  - adjMat, [212](#)
  - biconnME, [209](#)
  - bicount, [215](#)
  - bidef, [215](#)
  - bilow, [215](#)
  - bisearchME, [209](#)
  - block, [215](#)
  - c1PI, [212](#)
  - c1PINoTadBlock, [213](#)
  - cDiag, [212](#)
  - clist, [211](#)

- cNoTadBlock, [213](#)
- cNoTadpole, [213](#)
- compMat, [208](#)
- connectClass, [209](#)
- connectLeg, [209](#)
- connectNode, [209](#)
- curl, [212](#)
- discardDisc, [214](#)
- discardIso, [214](#)
- discardRefine, [214](#)
- DUMMYPADDING1, [215](#)
- DUMMYPADDING2, [216](#)
- egraph, [212](#)
- esym, [212](#)
- extself, [214](#)
- generate, [207](#)
- group, [210](#)
- init, [207](#)
- isConnected, [208](#)
- isExternal, [207](#)
- isIsomorphic, [208](#)
- isOptM, [209](#)
- maxdeg, [212](#)
- mconn, [215](#)
- MGraph, [206](#)
- mId, [210](#)
- mindeg, [211](#)
- modmat, [215](#)
- nCallRefine, [213](#)
- nClasses, [211](#)
- nEdges, [211](#)
- newGraph, [210](#)
- nExtern, [211](#)
- ngconn, [213](#)
- ngen, [213](#)
- nLoops, [211](#)
- nNodes, [211](#)
- nodes, [210](#)
- nsym, [212](#)
- opi, [214](#)
- opiloop, [214](#)
- opt, [210](#)
- orbits, [210](#)
- permMat, [208](#)
- pId, [211](#)
- print, [207](#)
- printAdjMat, [207](#)
- refineClass, [208](#)
- selfloop, [214](#)
- tadblock, [215](#)
- tadpole, [214](#)
- visit, [208](#)
- wscon, [213](#)
- wsopi, [213](#)
- mgraph
  - Assign, [23](#)
  - EGraph, [109](#)
  - SProcess, [407](#)
- mgrcount
  - Process, [365](#)
  - SProcess, [409](#)
- mId
  - EGraph, [109](#)
  - MGraph, [210](#)
- MiN
  - declare.h, [799](#)
- MINALLOC
  - fsizes.h, [1350](#)
- mincase
  - SWITCH, [422](#)
- mindeg
  - MGraph, [211](#)
  - T\_MGraph, [448](#)
- MINFUNCTION
  - ftypes.h, [1395](#)
- mini
  - MiNmAx, [216](#)
- MiNmAx, [216](#)
  - maxi, [216](#)
  - mini, [216](#)
  - size, [216](#)
- minnumargs
  - FuNcTiOn, [147](#)
- MINORVERSION
  - form3.h, [1326](#)
- minos.c, [1727](#), [1735](#)
  - AddObject, [1733](#)
  - AddTableName, [1732](#)
  - AddToIndex, [1733](#)
  - CFD, [1729](#)
  - ComposeTableNames, [1732](#)
  - convertblock, [1730](#)
  - convertiniinfo, [1730](#)
  - convertnamesblock, [1730](#)
  - CTD, [1729](#)
  - DeleteObject, [1734](#)
  - ExistsObject, [1734](#)
  - FindTableNumber, [1733](#)
  - FreeTableBase, [1732](#)
  - GetDbase, [1731](#)
  - GetTableName, [1732](#)
  - LocateBase, [1730](#)
  - minosread, [1729](#)
  - minoswrite, [1729](#)
  - NewDbase, [1732](#)
  - OpenDbase, [1732](#)
  - PutTableNames, [1733](#)
  - ReadijObject, [1734](#)
  - ReadIndex, [1730](#)
  - ReadIniInfo, [1731](#)
  - ReadObject, [1733](#)
  - str\_dup, [1730](#)
  - withoutflush, [1734](#)
  - WriteIndex, [1731](#)
  - WriteIndexBlock, [1731](#)
  - WriteIniInfo, [1731](#)



- WriteNamesBlock, [1731](#)
- WriteObject, [1733](#)
- minos.h, [1752](#), [1755](#)
  - FROMDISK, [1753](#)
  - INANDOUT, [1754](#)
  - INPUTONLY, [1754](#)
  - MCLEANFLAG, [1753](#)
  - MDIRTYFLAG, [1753](#)
  - minosread, [1754](#)
  - minoswrite, [1754](#)
  - NOCOMPRESS, [1754](#)
  - OUTPUTONLY, [1754](#)
  - TODISK, [1753](#)
  - withoutflush, [1755](#)
- minosread
  - minos.c, [1729](#)
  - minos.h, [1754](#)
- minoswrite
  - minos.c, [1729](#)
  - minos.h, [1754](#)
- minpower
  - SyMbOI, [424](#)
- MINPOWEROF
  - ftypes.h, [1398](#)
- MInput, [217](#)
  - cnamlist, [217](#)
  - defpart, [217](#)
  - name, [217](#)
  - ncouple, [217](#)
- minsidefirst
  - C\_const, [55](#)
- MINSPEC
  - ftypes.h, [1429](#)
- minvar
  - OPTIMIZERESULT, [296](#)
- MinVecArg
  - T\_const, [437](#)
- MINVECTOR
  - ftypes.h, [1390](#)
- MIXED
  - ftypes.h, [1376](#)
- MIXED2
  - ftypes.h, [1375](#)
- mkCIMat
  - MNodeClass, [222](#)
  - T\_MNodeClass, [454](#)
- mkFlist
  - MNodeClass, [222](#)
  - T\_MNodeClass, [454](#)
- mkNdCI
  - MNodeClass, [222](#)
  - T\_MNodeClass, [454](#)
- mkSlaveInfile
  - ParallelVars, [319](#)
- mkSPProcess
  - Process, [364](#)
- MLOCK
  - declare.h, [819](#)
- MLONG
  - form3.h, [1335](#)
- mm
  - TabIEs, [461](#)
- MNode, [218](#)
  - clss, [219](#)
  - cmaxdeg, [219](#)
  - cmindeg, [219](#)
  - deg, [219](#)
  - extloop, [219](#)
  - freelg, [220](#)
  - id, [219](#)
  - MNode, [218](#)
  - visited, [220](#)
- MNodeClass, [220](#)
  - ~MNodeClass, [221](#)
  - clCmp, [221](#)
  - clist, [223](#)
  - clmat, [223](#)
  - clord, [223](#)
  - cmaxdeg, [223](#)
  - cmindeg, [223](#)
  - cmpMNCArray, [222](#)
  - copy, [221](#)
  - flg0, [224](#)
  - flg1, [224](#)
  - flg2, [224](#)
  - flist, [223](#)
  - incMat, [222](#)
  - init, [221](#)
  - maxdeg, [224](#)
  - mkCIMat, [222](#)
  - mkFlist, [222](#)
  - mkNdCI, [222](#)
  - MNodeClass, [221](#)
  - nClasses, [224](#)
  - ndcl, [223](#)
  - nNodes, [224](#)
  - printMat, [221](#)
  - reorder, [222](#)
- mnumlhs
  - CbUf, [74](#)
- mnumrhs
  - CbUf, [74](#)
- mod
  - poly, [351](#)
- MOD2FUNCTION
  - ftypes.h, [1394](#)
- mode
  - dbase, [78](#)
  - TabIEs, [465](#)
- MoDeL, [232](#)
  - couplings, [233](#)
  - dummy, [234](#)
  - error, [234](#)
  - grccmodel, [233](#)
  - invertices, [234](#)
  - legcouple, [233](#)

- name, [233](#)
- ncouplings, [234](#)
- nparticles, [233](#)
- nvertices, [233](#)
- sizecouplings, [234](#)
- sizevertices, [234](#)
- vertices, [233](#)
- Model, [225](#)
  - ~Model, [226](#)
  - addInteraction, [227](#)
  - addInteractionEnd, [227](#)
  - addParticle, [226](#)
  - addParticleEnd, [226](#)
  - allPart, [229](#)
  - allParticles, [228](#)
  - antiParticle, [228](#)
  - cnlist, [229](#)
  - cplgcp, [231](#)
  - cplglg, [231](#)
  - cplgnvl, [231](#)
  - cplgvl, [231](#)
  - defpart, [231](#)
  - DUMMYPADDING, [231](#)
  - findInteractionCode, [227](#)
  - findInteractionName, [227](#)
  - findMClass, [228](#)
  - findParticleCode, [227](#)
  - findParticleName, [227](#)
  - interacts, [230](#)
  - intPart, [230](#)
  - maxcpl, [230](#)
  - maxloop, [230](#)
  - maxnlegs, [230](#)
  - Model, [226](#)
  - nallPart, [229](#)
  - name, [229](#)
  - ncouple, [229](#)
  - ncplgcp, [231](#)
  - nInteracts, [230](#)
  - nintPart, [230](#)
  - normalParticle, [228](#)
  - nParticles, [229](#)
  - particleCode, [228](#)
  - particleName, [227](#)
  - particles, [229](#)
  - pdef, [229](#)
  - prModel, [226](#)
  - prParticleArray, [228](#)
  - skipFLine, [231](#)
  - vdef, [230](#)
- model
  - Assign, [23](#)
  - EGraph, [108](#)
  - Options, [301](#)
  - Output, [307](#)
  - Process, [364](#)
  - SProcess, [406](#)
- model.c, [1756](#)
- ModelLevel
  - C\_const, [63](#)
- modeloptions
  - R\_const, [377](#)
- models
  - C\_const, [46](#)
- modelspace
  - C\_const, [63](#)
- MODFUNCTION
  - ftypes.h, [1394](#)
- modinverses
  - C\_const, [50](#)
- MODLOCAL
  - ftypes.h, [1442](#)
- modmat
  - MGraph, [215](#)
- MODMAX
  - ftypes.h, [1442](#)
- MODMIN
  - ftypes.h, [1442](#)
- modmode
  - C\_const, [65](#)
- modn
  - poly, [356](#)
- MODNONE
  - ftypes.h, [1442](#)
- MODNUM, [235](#)
  - a, [235](#)
  - m, [235](#)
  - na, [235](#)
  - nm, [235](#)
- modnum
  - POLYMOD, [358](#)
- MoDoPtDoLIaRS, [236](#)
  - number, [236](#)
  - type, [236](#)
- ModOptdollars
  - variable.h, [3205](#)
- ModOptDolList
  - C\_const, [44](#)
- modp
  - poly, [356](#)
- modpowers
  - C\_const, [47](#)
- MODSUM
  - ftypes.h, [1442](#)
- module.c, [1762](#), [1766](#)
  - CoModOption, [1763](#)
  - CoModuleOption, [1762](#)
  - DoExecStatement, [1765](#)
  - DoinParallel, [1765](#)
  - DoModDollar, [1765](#)
  - DoModLocal, [1764](#)
  - DoModMax, [1764](#)
  - DoModMin, [1764](#)
  - DoModSum, [1764](#)
  - DoNoParallel, [1764](#)
  - DonotinParallel, [1765](#)

- DoParallel, 1764
- DoPipeStatement, 1765
- DoPolyfun, 1763
- DoPolyratfun, 1763
- DoProcessBucket, 1764
- ExecModule, 1763
- ExecStore, 1763
- FullCleanUp, 1763
- ModuleInstruction, 1762
- SetSpecialMode, 1763
- MODULEINSTACK
  - ftypes.h, 1371
- ModuleInstruction
  - declare.h, 905
  - module.c, 1762
- Modulus
  - declare.h, 856
  - reken.c, 2560
- MOEBIUS
  - ftypes.h, 1402
- Moebius
  - declare.h, 992
  - reken.c, 2561
- moebiustable
  - R\_const, 372
- moebiustablesizesize
  - R\_const, 377
- momc
  - T\_EEdge, 440
- momn
  - T\_EEdge, 440
- monomial\_compare
  - poly, 349
- monomial\_larger, 236
  - monomial\_larger, 236
  - operator(), 237
- MOrbits, 237
  - ~MOrbits, 237
  - flist, 239
  - fromPerm, 238
  - initPerm, 238
  - MOrbits, 237
  - nd2or, 239
  - nNodes, 238
  - nOrbits, 238
  - or2nd, 239
  - print, 238
  - toOrbits, 238
- MoveDummies
  - declare.h, 857
  - reshuf.c, 2609
- mparallelflag
  - C\_const, 59
- mpi.c, 1775, 1791
  - MPI\_ERRCODE\_CHECK, 1776
  - PF\_Bcast, 1780
  - PF\_Broadcast, 1786
  - PF\_IRecvRbuf, 1779
  - PF\_ISendSbuf, 1778
  - PF\_LibInit, 1777
  - PF\_LibTerminate, 1778
  - PF\_LongMultiBroadcast, 1790
  - PF\_LongMultiPackImpl, 1788
  - PF\_LongMultiUnpackImpl, 1789
  - PF\_LongSinglePack, 1786
  - PF\_LongSingleReceive, 1788
  - PF\_LongSingleSend, 1787
  - PF\_LongSingleUnpack, 1787
  - PF\_maxDollarChunkSize, 1791
  - PF\_Pack, 1783
  - PF\_PACKSIZE, 1776
  - PF\_PackString, 1784
  - PF\_PrepareLongMultiPack, 1788
  - PF\_PrepareLongSinglePack, 1786
  - PF\_PreparePack, 1782
  - PF\_PrintPackBuf, 1782
  - PF\_Probe, 1778
  - PF\_RawProbe, 1782
  - PF\_RawRecv, 1781
  - PF\_RawSend, 1781
  - PF\_RealTime, 1777
  - PF\_Receive, 1785
  - PF\_RecvWbuf, 1779
  - PF\_Send, 1785
  - PF\_Unpack, 1783
  - PF\_UnpackString, 1784
  - PF\_WaitRbuf, 1780
- MPI\_Barrier
  - mpidbg.h, 1813
- MPI\_Bcast
  - mpidbg.h, 1813
- MPI\_Bsend
  - mpidbg.h, 1810
- MPI\_Cancel
  - mpidbg.h, 1813
- MPI\_ERRCODE\_CHECK
  - mpi.c, 1776
- MPI\_lbsend
  - mpidbg.h, 1811
- MPI\_lprobe
  - mpidbg.h, 1813
- MPI\_irecv
  - mpidbg.h, 1811
- MPI\_irsend
  - mpidbg.h, 1811
- MPI\_lsend
  - mpidbg.h, 1811
- MPI\_issend
  - mpidbg.h, 1811
- MPI\_Probe
  - mpidbg.h, 1813
- MPI\_Recv
  - mpidbg.h, 1810
- MPI\_Rsend
  - mpidbg.h, 1811
- MPI\_Send

- mpidbg.h, [1810](#)
- MPI\_Ssend
  - mpidbg.h, [1810](#)
- MPI\_Test
  - mpidbg.h, [1812](#)
- MPI\_Test\_cancelled
  - mpidbg.h, [1813](#)
- MPI\_Testall
  - mpidbg.h, [1812](#)
- MPI\_Testany
  - mpidbg.h, [1812](#)
- MPI\_Testsome
  - mpidbg.h, [1812](#)
- MPI\_Wait
  - mpidbg.h, [1811](#)
- MPI\_Waitall
  - mpidbg.h, [1812](#)
- MPI\_Waitany
  - mpidbg.h, [1812](#)
- MPI\_Waitsome
  - mpidbg.h, [1812](#)
- mpidbg.h, [1809](#), [1814](#)
  - MPI\_Barrier, [1813](#)
  - MPI\_Bcast, [1813](#)
  - MPI\_Bsend, [1810](#)
  - MPI\_Cancel, [1813](#)
  - MPI\_Ibsend, [1811](#)
  - MPI\_Iprobe, [1813](#)
  - MPI\_Irecv, [1811](#)
  - MPI\_Irsend, [1811](#)
  - MPI\_Isend, [1811](#)
  - MPI\_Issend, [1811](#)
  - MPI\_Probe, [1813](#)
  - MPI\_Recv, [1810](#)
  - MPI\_Rsend, [1811](#)
  - MPI\_Send, [1810](#)
  - MPI\_Ssend, [1810](#)
  - MPI\_Test, [1812](#)
  - MPI\_Test\_cancelled, [1813](#)
  - MPI\_Testall, [1812](#)
  - MPI\_Testany, [1812](#)
  - MPI\_Testsome, [1812](#)
  - MPI\_Wait, [1811](#)
  - MPI\_Waitall, [1812](#)
  - MPI\_Waitany, [1812](#)
  - MPI\_Waitsome, [1812](#)
  - MPIDBG\_EXTARG, [1810](#)
  - MPIDBG\_Out, [1810](#)
  - MPIDBG\_RANK, [1810](#)
- MPIDBG\_EXTARG
  - mpidbg.h, [1810](#)
- MPIDBG\_Out
  - mpidbg.h, [1810](#)
- MPIDBG\_RANK
  - mpidbg.h, [1810](#)
- mpoly
  - flint::mpoly, [240](#)
- mpoly\_ctx
- flint::mpoly\_ctx, [241](#)
- mpoly\_factor
  - flint::mpoly\_factor, [242](#)
- mpower
  - O\_const, [284](#)
- mProcessBucketSize
  - C\_const, [53](#)
- mTraceDum
  - M\_const, [187](#)
- mul
  - COST, [76](#)
  - poly, [350](#)
- mul\_heap
  - poly, [352](#)
- mul\_mpoly
  - flintinterface.h, [1273](#)
- mul\_one\_term
  - poly, [352](#)
- mul\_poly
  - flintinterface.h, [1273](#)
- mul\_univar
  - poly, [352](#)
- MulFloats
  - float.c, [1288](#)
- MULfunc
  - declare.h, [1011](#)
  - ratio.c, [2506](#)
- MULFUNCTION
  - ftypes.h, [1402](#)
- MulLong
  - declare.h, [857](#)
  - reken.c, [2554](#)
- Mully
  - declare.h, [857](#)
  - reken.c, [2552](#)
- mulpat
  - T\_const, [437](#)
- MULPOS
  - declare.h, [811](#)
- MulRat
  - declare.h, [857](#)
  - reken.c, [2553](#)
- MulRatToFloat
  - float.c, [1289](#)
- MultDo
  - declare.h, [857](#)
  - reshuf.c, [2610](#)
- MultiBracketBuf
  - C\_const, [51](#)
- MultiBracketLevels
  - C\_const, [61](#)
- MULTIINDENTSPACE
  - fsizes.h, [1349](#)
- MULTIPLEOF
  - ftypes.h, [1435](#)
- MULTIPLYARG
  - ftypes.h, [1447](#)
- MultiplyToLine

- declare.h, [1022](#)
- sch.c, [2652](#)
- MultiplyWithTerm
  - declare.h, [1010](#)
  - ratio.c, [2503](#)
- MultiRun
  - M\_const, [183](#)
- MultiThreaded
  - S\_const, [382](#)
- multp
  - EGraph, [110](#)
- MUNLOCK
  - declare.h, [819](#)
- MUSTCLEANPRF
  - ftypes.h, [1420](#)
- MustTestTable
  - C\_const, [65](#)
- my\_random\_shuffle
  - optimize.cc, [1989](#)
- mytime.cc, [1822](#)
- mytime.h, [1822](#)
- N
  - AllGlobals, [10](#)
- n
  - DiStRiBuTe, [85](#)
  - PeRmUtE, [327](#)
  - PeRmUtEp, [328](#)
- n1
  - DiStRiBuTe, [85](#)
- n1PI
  - T\_MGraph, [449](#)
- n2
  - DiStRiBuTe, [85](#)
- N\_const, [242](#)
  - argaddress, [246](#)
  - arglist, [251](#)
  - arglistsize, [254](#)
  - cmod, [251](#)
  - compressSize, [255](#)
  - compressSpace, [251](#)
  - cTerm, [246](#)
  - currentTerm, [251](#)
  - deferskipped, [252](#)
  - DisOrderFlag, [256](#)
  - DumFound, [247](#)
  - DumFunPlace, [247](#)
  - dummyrenumlist, [248](#)
  - DumPlace, [247](#)
  - EndNest, [245](#)
  - ErrorInDollar, [253](#)
  - ExpectedSign, [258](#)
  - filenum, [255](#)
  - findPattern, [248](#)
  - findTerm, [248](#)
  - ForFindOnly, [248](#)
  - Frozen, [245](#)
  - FullProto, [245](#)
  - funargs, [248](#)
  - funinds, [249](#)
  - FunInfo, [250](#)
  - funisize, [254](#)
  - funlocs, [248](#)
  - FunScratch, [250](#)
  - FunSorts, [250](#)
  - idfunctionflag, [258](#)
  - IndDum, [258](#)
  - InScratch, [252](#)
  - InScratch1, [252](#)
  - intracestack, [254](#)
  - listinprint, [250](#)
  - MaskPointer, [248](#)
  - MaxFunSorts, [254](#)
  - MaxRenumScratch, [255](#)
  - nargs, [253](#)
  - ncmod, [258](#)
  - ninterms, [252](#)
  - nogroundlevel, [255](#)
  - NoScrat2, [249](#)
  - numfargs, [253](#)
  - numflocs, [253](#)
  - NumFound, [256](#)
  - numfuninfo, [254](#)
  - NumFunSorts, [254](#)
  - numlistinprint, [257](#)
  - numoffuns, [253](#)
  - NumTotWildArgs, [252](#)
  - numtracesctack, [254](#)
  - NumWild, [256](#)
  - OldPosIn, [245](#)
  - OldPosOut, [245](#)
  - oldtype, [255](#)
  - oldvalue, [256](#)
  - patstop, [246](#)
  - patternbuffer, [249](#)
  - patternbuffersize, [257](#)
  - PoinScratch, [249](#)
  - poly\_num\_vars, [258](#)
  - poly\_vars, [251](#)
  - poly\_vars\_type, [258](#)
  - PolyFunTodo, [257](#)
  - PolyNormFlag, [257](#)
  - polysortflag, [255](#)
  - RenumScratch, [250](#)
  - RepFunList, [246](#)
  - RepFunNum, [256](#)
  - ReplaceScrat, [249](#)
  - RepPoint, [246](#)
  - RSSize, [254](#)
  - selecttermundo, [249](#)
  - SHcombi, [251](#)
  - SHcombisize, [252](#)
  - SHvar, [251](#)
  - SignCheck, [258](#)
  - sizeselecttermundo, [257](#)
  - SoScratC, [250](#)
  - SplitScratch, [250](#)

- SplitScratch1, [250](#)
- SplitScratchSize, [252](#)
- SplitScratchSize1, [252](#)
- subsubveto, [255](#)
- TeInFun, [256](#)
- terfirstcomm, [247](#)
- termbuffer, [249](#)
- terstart, [247](#)
- terstop, [246](#)
- TeSuOut, [257](#)
- theposition, [245](#)
- tlisbuf, [251](#)
- tlisize, [255](#)
- tohunt, [253](#)
- tracestack, [249](#)
- tryterm, [259](#)
- UsedFunction, [248](#)
- UsedIndex, [247](#)
- UsedOtherFind, [253](#)
- UsedSymbol, [247](#)
- UsedVector, [247](#)
- UseFindOnly, [253](#)
- WildArgs, [257](#)
- WildDirt, [256](#)
- WildEat, [257](#)
- WildReserve, [256](#)
- WildStop, [246](#)
- WildValue, [246](#)
- na
  - MODNUM, [235](#)
- na1blocks
  - MConn, [201](#)
- nadj2ptv
  - EGraph, [112](#)
- nAGraphs
  - Assign, [25](#)
  - Process, [368](#)
  - SProcess, [410](#)
- nallPart
  - Model, [229](#)
- name
  - ChAnNeL, [75](#)
  - dbase, [79](#)
  - DICTIONARY, [82](#)
  - DoLIaRS, [88](#)
  - DoLoOp, [90](#)
  - DuBiOuS, [93](#)
  - ExPrEsSiOn, [122](#)
  - FiLe, [129](#)
  - fixedset, [137](#)
  - FuNcTiOn, [145](#)
  - IInput, [149](#)
  - InDeX, [150](#)
  - InDeXeNtRy, [155](#)
  - Interaction, [161](#)
  - KEYWORD, [163](#)
  - KEYWORDV, [164](#)
  - MInput, [217](#)
  - MoDeL, [233](#)
  - Model, [229](#)
  - NaMeNode, [260](#)
  - NaMeSpAcE, [263](#)
  - OptDef, [289](#)
  - Particle, [325](#)
  - PInput, [332](#)
  - pReVaR, [360](#)
  - PROCEDURE, [362](#)
  - SeTs, [383](#)
  - SpecTatoR, [402](#)
  - StreaM, [417](#)
  - SyMbOl, [424](#)
  - TaBIeBaSE, [458](#)
  - VeCtOr, [484](#)
- nameblock, [259](#)
  - names, [259](#)
  - position, [259](#)
  - previousblock, [259](#)
- namebuffer
  - NaMeTree, [264](#)
- NameConflict
  - declare.h, [895](#)
  - names.c, [1833](#)
- namefill
  - NaMeTree, [264](#)
- NAMENODE
  - structs.h, [2900](#)
- NaMeNode, [260](#)
  - balance, [261](#)
  - left, [260](#)
  - name, [260](#)
  - number, [261](#)
  - parent, [260](#)
  - right, [261](#)
  - type, [261](#)
- namenode
  - NaMeTree, [264](#)
- NAMENOTFOUND
  - ftypes.h, [1439](#)
- nameofexpr
  - OPTIMIZERESULT, [295](#)
- names
  - nameblock, [259](#)
- names.c, [1823](#), [1834](#)
  - AddDollar, [1832](#)
  - AddDubious, [1832](#)
  - AddExpression, [1833](#)
  - AddFunction, [1829](#)
  - AddIndex, [1828](#)
  - AddName, [1824](#)
  - AddSet, [1831](#)
  - AddSymbol, [1828](#)
  - AddVector, [1829](#)
  - AddWildcardName, [1827](#)
  - ClearWildcardNames, [1827](#)
  - CoAuto, [1832](#)
  - CoCFunction, [1830](#)

- CoCommutInSet, [1829](#)
- CoCTable, [1831](#)
- CoCTensor, [1830](#)
- CoDimension, [1829](#)
- CoFunction, [1829](#)
- CoIndex, [1828](#)
- CompactifyTree, [1826](#)
- CoNFunction, [1830](#)
- CoNTable, [1831](#)
- CoNTensor, [1830](#)
- CopyTree, [1827](#)
- CoSet, [1831](#)
- CoSymbol, [1828](#)
- CoTable, [1830](#)
- CoVector, [1829](#)
- DoDimension, [1828](#)
- DoElements, [1831](#)
- DoTable, [1830](#)
- DoTempSet, [1832](#)
- DumpNode, [1826](#)
- DumpTree, [1826](#)
- EmptyTable, [1831](#)
- EntVar, [1826](#)
- FreeNameTree, [1827](#)
- GetAutoName, [1825](#)
- GetDollar, [1826](#)
- GetFunction, [1825](#)
- GetLabel, [1833](#)
- GetLastExprName, [1825](#)
- GetName, [1824](#)
- GetNode, [1824](#)
- GetNumber, [1825](#)
- GetOName, [1825](#)
- GetVar, [1825](#)
- GetWildcardName, [1828](#)
- Globalize, [1833](#)
- LinkTree, [1827](#)
- MakeDubious, [1832](#)
- MakeNameTree, [1827](#)
- NameConflict, [1833](#)
- RemoveDollars, [1833](#)
- ReplaceDollar, [1832](#)
- ResetVariables, [1833](#)
- TestName, [1834](#)
- NamesFlag
  - C\_const, [57](#)
- namesize
  - ExPrEsSiOn, [123](#)
  - FuNcTiOn, [146](#)
  - InDeX, [151](#)
  - NaMeTree, [264](#)
  - SeTs, [384](#)
  - SyMbOl, [425](#)
  - VeCtOr, [484](#)
- NaMeSpAcE, [262](#)
  - name, [263](#)
  - next, [262](#)
  - previous, [262](#)
  - usenames, [262](#)
- NAMETREE
  - structs.h, [2900](#)
- NaMeTree, [263](#)
  - clearnamefill, [265](#)
  - clearnodefill, [265](#)
  - globalnamefill, [265](#)
  - globalnodefill, [265](#)
  - headnode, [266](#)
  - namebuffer, [264](#)
  - namefill, [264](#)
  - namenode, [264](#)
  - namesize, [264](#)
  - nodefill, [264](#)
  - nodesize, [264](#)
  - oldnamefill, [265](#)
  - oldnodefill, [265](#)
- nAOPI
  - Assign, [25](#)
  - Process, [368](#)
  - SProcess, [410](#)
- nargs
  - N\_const, [253](#)
  - PaRtl, [320](#)
  - pReVaR, [360](#)
- narticuls
  - MConn, [201](#)
- nartps
  - MBlock, [194](#)
- nBer
  - T\_const, [438](#)
- nblksp
  - MConn, [202](#)
- nblocks
  - dbase, [79](#)
  - MConn, [201](#)
- nBridges
  - T\_MGraph, [449](#)
- nbridges
  - MConn, [201](#)
- nCallRefine
  - MGraph, [213](#)
  - T\_MGraph, [450](#)
- NCand, [266](#)
  - ~NCand, [266](#)
  - deg, [267](#)
  - ilist, [267](#)
  - NCand, [266](#)
  - nilist, [267](#)
  - prNCand, [267](#)
  - st, [267](#)
- ncase
  - SWITCHTABLE, [423](#)
- NCInput, [268](#)
  - cldeg, [268](#)
  - clnum, [268](#)
  - cltyp, [268](#)
  - cmaxd, [268](#)

- cmind, [268](#)
  - cple, [269](#)
  - ptcl, [268](#)
- ncl
  - ToPoTyPe, [470](#)
- nclass
  - PNodeClass, [337](#)
  - SProcess, [407](#)
- nClasses
  - MGraph, [211](#)
  - MNodeClass, [224](#)
  - T\_MGraph, [448](#)
  - T\_MNodeClass, [455](#)
- nclist
  - Interaction, [162](#)
- ncmod
  - C\_const, [65](#)
  - N\_const, [258](#)
- NCOPY
  - declare.h, [802](#)
- NCOPYB
  - declare.h, [803](#)
- NCOPYI
  - declare.h, [803](#)
- NCOPYI32
  - declare.h, [803](#)
- ncouple
  - MInput, [217](#)
  - Model, [229](#)
  - SProcess, [408](#)
- ncouplings
  - MoDeL, [234](#)
  - VeRtEx, [486](#)
- ncplgcp
  - Model, [231](#)
- nctopic
  - MConn, [200](#)
- nd2cl
  - PNodeClass, [337](#)
  - SProcess, [407](#)
- nd2or
  - MOrbits, [239](#)
- ndcl
  - MNodeClass, [223](#)
  - T\_MNodeClass, [456](#)
- NDEBUG
  - form3.h, [1327](#)
- ndiag
  - T\_MGraph, [448](#)
- ndtype
  - ENode, [117](#)
- ne0bridges
  - MConn, [201](#)
- ne1bridges
  - MConn, [201](#)
- neblocks
  - MConn, [201](#)
- neclass
  - SGroup, [388](#)
- nEdges
  - Assign, [24](#)
  - EGraph, [111](#)
  - MGraph, [211](#)
  - SProcess, [408](#)
  - T\_EGraph, [443](#)
  - T\_MGraph, [447](#)
- nedges
  - DGraph, [81](#)
  - MCOpI, [204](#)
- NeedNumber
  - declare.h, [803](#)
- NEG0\_
  - ftypes.h, [1426](#)
- NEG\_
  - ftypes.h, [1426](#)
- nElem
  - SGroup, [387](#)
- nelem
  - SGroup, [388](#)
- Nest
  - T\_const, [428](#)
- NESTING
  - structs.h, [2903](#)
- NeStInG, [269](#)
  - argsize, [269](#)
  - funsize, [269](#)
  - termsize, [269](#)
- NestingChecksum
  - declare.h, [817](#)
- nestingsum
  - SWITCH, [423](#)
- NestPoin
  - T\_const, [428](#)
- NestStop
  - T\_const, [428](#)
- nETotal
  - Assign, [24](#)
- neutral
  - Particle, [325](#)
- newAGraph
  - Options, [301](#)
- newbracketinfo
  - ExPrEsSiOn, [122](#)
- NEWCODE
  - reshuf.c, [2609](#)
- NewDbase
  - declare.h, [987](#)
  - minos.c, [1732](#)
- NEWFACTARG
  - ftypes.h, [1448](#)
- newGraph
  - MGraph, [210](#)
- newGroup
  - SGroup, [386](#)
- newhandle
  - HANDLERS, [148](#)



- newleg
  - ANode, [12](#)
- newlogonly
  - HANDLERS, [148](#)
- NEWLYDEFINEDEXPRESSION
  - ftypes.h, [1425](#)
- newmaxtermsize
  - sOrT, [400](#)
- newMGraph
  - Options, [301](#)
- NEWORDER
  - argument.c, [490](#)
- NewSort
  - declare.h, [858](#)
  - sort.c, [2717](#)
- NEWSPLITMERGE
  - sort.c, [2715](#)
- NEWSPLITMERGETIMSORT
  - sort.c, [2715](#)
- NEWSTATEMENT
  - ftypes.h, [1371](#)
- NEWTRICK
  - reken.c, [2551](#)
- next
  - BrAcKeTiNdEx, [32](#)
  - FiLeInDeX, [134](#)
  - longMultiStruct, [168](#)
  - MCBridge, [195](#)
  - MCOpI, [204](#)
  - NaMeSpAcE, [262](#)
  - StOrEcAcHe, [412](#)
- NEXTARG
  - declare.h, [808](#)
- nextchar
  - StreaM, [419](#)
- nextElem
  - SGroup, [387](#)
- nExtern
  - Assign, [24](#)
  - EGraph, [111](#)
  - MGraph, [211](#)
  - Process, [366](#)
  - SProcess, [408](#)
  - T\_EGraph, [444](#)
- NextFileIndex
  - declare.h, [823](#)
  - store.c, [2831](#)
- NextPrime
  - declare.h, [992](#)
  - reken.c, [2561](#)
- nfac
  - T\_const, [438](#)
- nfactors
  - DoLIARs, [89](#)
- nfal
  - FGInput, [127](#)
  - Process, [365](#)
  - SProcess, [407](#)
- nflines
  - EGraph, [114](#)
- nfun
  - PaRtl, [320](#)
- nfunctions
  - InDeXeNtRy, [155](#)
- ngconn
  - MGraph, [213](#)
  - T\_MGraph, [450](#)
- ngen
  - MGraph, [213](#)
  - T\_MGraph, [450](#)
- ngraphs
  - Process, [367](#)
- nhalfmod
  - C\_const, [65](#)
- nilist
  - NCand, [267](#)
  - NStack, [276](#)
- nincoef
  - SHvariables, [391](#)
- nindices
  - InDeXeNtRy, [154](#)
- ninitl
  - FGInput, [127](#)
  - Process, [365](#)
  - SProcess, [407](#)
- nInteracts
  - Model, [230](#)
- ninterms
  - N\_const, [252](#)
  - sOrT, [398](#)
- nintPart
  - Model, [230](#)
- nlegs
  - AEdge, [8](#)
  - ANode, [13](#)
  - EEdge, [98](#)
  - Interaction, [162](#)
  - MCOpI, [204](#)
  - ToPoTyPe, [470](#)
- nlist
  - EFLine, [101](#)
- nLoops
  - EGraph, [112](#)
  - MGraph, [211](#)
  - T\_MGraph, [448](#)
- nloops
  - ToPoTyPe, [470](#)
- nlpopic
  - MConn, [200](#)
- nm
  - MODNUM, [235](#)
- nmedges
  - MCBlock, [194](#)
- nMGraphs
  - Process, [367](#)
  - SProcess, [409](#)

- nmin4
  - InDeX, [151](#)
- NMIN4SHIFT
  - ftypes.h, [1389](#)
- nMOPI
  - Process, [367](#)
  - SProcess, [409](#)
- nNodes
  - Assign, [24](#)
  - EGraph, [111](#)
  - MGraph, [211](#)
  - MNodeClass, [224](#)
  - MOrbits, [238](#)
  - SProcess, [408](#)
  - T\_EGraph, [443](#)
  - T\_MGraph, [447](#)
  - T\_MNodeClass, [455](#)
- nnodes
  - DGraph, [81](#)
  - MCOpI, [203](#)
  - PNodeClass, [337](#)
  - SGroup, [388](#)
- NOAUTO
  - ftypes.h, [1376](#)
- NOBRACKETACTIVE
  - execute.c, [1157](#)
- NOCHECKLOGTYPE
  - ftypes.h, [1389](#)
- NOCOMPRESS
  - minos.h, [1754](#)
- NoCompress
  - C\_const, [61](#)
  - R\_const, [373](#)
- NODE
  - parallel.c, [2066](#)
- NoDe, [270](#)
  - left, [270](#)
  - lloser, [271](#)
  - lsrc, [271](#)
  - rght, [270](#)
  - rloser, [271](#)
  - rsrc, [271](#)
- node, [271](#)
  - calcHash, [273](#)
  - cmp, [272](#)
  - data, [273](#)
  - DoLIArS, [88](#)
  - DuBiOuS, [93](#)
  - ExPrEsSiOn, [123](#)
  - FuNcTiOn, [146](#)
  - hash, [273](#)
  - InDeX, [151](#)
  - l, [273](#)
  - node, [272](#)
  - operator(), [272](#)
  - r, [273](#)
  - SeTs, [384](#)
  - sign, [273](#)
  - SyMbOl, [425](#)
  - VeCtOr, [484](#)
- NodeEq, [274](#)
  - operator(), [274](#)
- nodefill
  - NaMeTree, [264](#)
- NODEFUNCTION
  - ftypes.h, [1403](#)
- NodeHash, [274](#)
  - operator(), [274](#)
- noden
  - NStack, [275](#)
- nodes
  - AEdge, [8](#)
  - Assign, [25](#)
  - DGraph, [81](#)
  - EEdge, [98](#)
  - EGraph, [109](#)
  - MCBridge, [195](#)
  - MCOpI, [203](#)
  - MGraph, [210](#)
  - T\_EEdge, [440](#)
  - T\_EGraph, [444](#)
  - T\_MGraph, [448](#)
- nodesize
  - NaMeTree, [264](#)
- NODOUBLEMASK
  - ftypes.h, [1386](#)
- NOEXTSELF
  - ftypes.h, [1456](#)
- NOFUNPOWERS
  - ftypes.h, [1440](#)
- nogroundlevel
  - N\_const, [255](#)
- NOINDEX
  - ftypes.h, [1429](#)
- NOINVERSES
  - ftypes.h, [1443](#)
- NOLYNDON
  - ftypes.h, [1448](#)
- NONODES
  - ftypes.h, [1455](#)
- NOPARALLEL\_CONVPOLY
  - ftypes.h, [1369](#)
- NOPARALLEL\_DOLLAR
  - ftypes.h, [1369](#)
- NOPARALLEL\_NPROC
  - ftypes.h, [1370](#)
- NOPARALLEL\_RHS
  - ftypes.h, [1369](#)
- NOPARALLEL\_SPECTATOR
  - ftypes.h, [1370](#)
- NOPARALLEL\_TBLDOLLAR
  - ftypes.h, [1370](#)
- NOPARALLEL\_USER
  - ftypes.h, [1370](#)
- nopi
  - Process, [367](#)

- nopi2p
  - EGraph, [112](#)
- nopic
  - MConn, [200](#)
- nopicomp
  - EGraph, [112](#)
- nopisp
  - MConn, [202](#)
- NOQUADMASK
  - ftypes.h, [1387](#)
- nOrbits
  - MOrbits, [238](#)
- normal
  - Fraction, [141](#)
- normal.c, [1872](#), [1875](#)
  - BracketNormalize, [1875](#)
  - Commute, [1873](#)
  - CompareFunctions, [1873](#)
  - DetCommu, [1874](#)
  - DoDelta, [1873](#)
  - DoesCommu, [1874](#)
  - DoRevert, [1874](#)
  - DoTheta, [1873](#)
  - DropCoefficient, [1874](#)
  - DropSymbols, [1874](#)
  - ExtraSymbol, [1873](#)
  - MAXNUMBEROFNONCOMTERMS, [1873](#)
  - Normalize, [1873](#)
  - SymbolNormalize, [1874](#)
  - TestFunFlag, [1875](#)
  - TreatPolyRatFun, [1874](#)
- NORMALFORMAT
  - ftypes.h, [1387](#)
- Normalize
  - declare.h, [858](#)
  - normal.c, [1873](#)
- normalize
  - poly, [348](#)
- NORMALIZEFLAG
  - ftypes.h, [1434](#)
- NormalModulus
  - declare.h, [867](#)
  - reken.c, [2556](#)
- normalParticle
  - Model, [228](#)
- NormPolyTerm
  - declare.h, [1004](#)
  - notation.c, [1938](#)
- NORMSIZE
  - fsizes.h, [1343](#)
- NoScrat2
  - N\_const, [249](#)
- NOSHELL
  - pre.c, [2317](#)
- NoShowInput
  - C\_const, [54](#)
  - DoLoOp, [92](#)
- NOSNAILS
  - ftypes.h, [1456](#)
- NOSPACEFORMAT
  - ftypes.h, [1387](#)
- NoSpacesInNumbers
  - O\_const, [282](#)
- NOTADPOLES
  - ftypes.h, [1455](#)
- notation.c, [1938](#), [1941](#)
  - ConvertFromPoly, [1939](#)
  - ConvertToPoly, [1938](#)
  - ExtraSymFun, [1940](#)
  - FindLocalSubterm, [1939](#)
  - FindSubexpression, [1940](#)
  - FindSubterm, [1939](#)
  - LocalConvertToPoly, [1939](#)
  - NormPolyTerm, [1938](#)
  - PrintExtraSymbol, [1940](#)
  - PrintSubtermList, [1940](#)
  - PruneExtraSymbols, [1940](#)
- NOTEQUAL
  - ftypes.h, [1437](#)
- NOTRICK
  - ftypes.h, [1388](#)
- NOTSTARTPOS
  - declare.h, [812](#)
- NOUNPACK
  - ftypes.h, [1444](#)
- nPacks
  - longMultiStruct, [167](#)
- npadding
  - ToPoTyPe, [470](#)
- NPAIR1
  - fsizes.h, [1350](#)
- NPAIR2
  - fsizes.h, [1350](#)
- nParticles
  - Model, [229](#)
- nparticles
  - MoDeL, [233](#)
  - VerRtEx, [486](#)
- nplist
  - ECand, [95](#)
  - EStack, [119](#)
  - Interaction, [162](#)
- nplistn
  - lInput, [149](#)
- npowmod
  - C\_const, [65](#)
- nselfloops
  - MConn, [201](#)
- NSIG
  - portsignals.h, [2309](#)
- nSize
  - AStack, [28](#)
- nslist
  - Interaction, [162](#)
- NStack, [275](#)
  - ~NStack, [275](#)

- deg, [275](#)
- ilist, [276](#)
- nilist, [276](#)
- noden, [275](#)
- NStack, [275](#)
- print, [275](#)
- st, [276](#)
- nStack
  - AStack, [28](#)
- nStackP
  - AStack, [28](#)
- nSubproc
  - Process, [366](#)
- nsym
  - EGraph, [110](#)
  - MGraph, [212](#)
  - T\_EGraph, [444](#)
  - T\_MGraph, [449](#)
- nsym1
  - EGraph, [110](#)
- nsymbols
  - InDeXeNtRy, [154](#)
- nt3
  - TrAcEs, [474](#)
- nt4
  - TrAcEs, [474](#)
- num
  - Fraction, [142](#)
  - LIST, [165](#)
  - TrAcEn, [472](#)
  - TrAcEs, [477](#)
- num\_terms
  - BracketInfo, [35](#)
- num\_visits
  - tree\_node, [482](#)
- NUMARG
  - ftypes.h, [1445](#)
- numargs
  - FUN\_INFO, [143](#)
  - PaRtl, [320](#)
- NUMARGSFUN
  - ftypes.h, [1394](#)
- number
  - FiLeInDeX, [134](#)
  - FuNcTiOn, [145](#)
  - InDeX, [151](#)
  - MoDoPtDoLIaRS, [236](#)
  - NaMeNode, [261](#)
  - PaRtlcLe, [321](#)
  - SyMbOl, [425](#)
  - VeCtOr, [484](#)
- number\_of\_terms
  - poly, [345](#)
- NUMBEREXTRAWORDS
  - tools.c, [3078](#)
- NumberFree
  - declare.h, [816](#)
- NumberMalloc
  - declare.h, [816](#)
  - NumberMallocAddMemory
    - declare.h, [999](#)
    - tools.c, [3089](#)
  - NumberMemHeap
    - T\_const, [431](#)
  - NumberMemMax
    - T\_const, [435](#)
  - NUMBERMEMSTARTNUM
    - tools.c, [3078](#)
  - NumberMemTop
    - T\_const, [435](#)
  - numberofindexblocks
    - iniinfo, [156](#)
  - numberofnamesblocks
    - iniinfo, [157](#)
  - numberoftables
    - iniinfo, [156](#)
  - numbers
    - DICTIONARY, [82](#)
  - numblocks
    - FiLe, [130](#)
  - numbufs
    - PF\_BUFFER, [330](#)
  - numcases
    - SWITCH, [422](#)
  - numclear
    - LIST, [166](#)
  - numcommute
    - comtool.c, [768](#)
    - declare.h, [963](#)
  - numcompares
    - sort.c, [2730](#)
  - NumCopy
    - declare.h, [921](#)
    - tools.c, [3087](#)
  - numdia
    - TERMINFO, [467](#)
  - NumDictionaries
    - O\_const, [283](#)
  - NumDollars
    - variable.h, [3203](#)
  - numdollars
    - INSIDEINFO, [158](#)
  - NumDoLoops
    - variable.h, [3200](#)
  - NumDubious
    - variable.h, [3204](#)
  - numdum
    - CbUf, [72](#)
  - numdummies
    - DoLIaRS, [89](#)
    - ExPrEsSiOn, [124](#)
    - TaBIes, [465](#)
  - numelements
    - DICTIONARY, [82](#)
  - NUMERATORSYMBOL
    - ftypes.h, [1404](#)

NUMERICALLOOP  
  ftypes.h, [1374](#)  
NUMERICALVALUE  
  setfile.c, [2689](#)  
NumExpressions  
  variable.h, [3204](#)  
numextern  
  TERMINFO, [467](#)  
NUMFACS  
  fsizes.h, [1344](#)  
NUMFACTORS  
  ftypes.h, [1401](#)  
numfactors  
  ExPrEsSiOn, [124](#)  
numfargs  
  N\_const, [253](#)  
NUMFIXED  
  fsizes.h, [1344](#)  
NumFixedFunctions  
  M\_const, [179](#)  
NumFixedSets  
  M\_const, [178](#)  
numflocs  
  N\_const, [253](#)  
NumFound  
  N\_const, [256](#)  
NumFunctions  
  variable.h, [3201](#)  
numfuninfo  
  N\_const, [254](#)  
numfunnies  
  FUN\_INFO, [143](#)  
NumFunSorts  
  N\_const, [254](#)  
numglobal  
  LIST, [166](#)  
NumInBrack  
  O\_const, [282](#)  
numind  
  TaBIes, [463](#)  
NumIndices  
  variable.h, [3201](#)  
numinfilelist  
  tools.c, [3094](#)  
NumLabels  
  C\_const, [55](#)  
numlhs  
  CbUf, [73](#)  
numlistinprint  
  N\_const, [257](#)  
NumListSymbols  
  T\_const, [436](#)  
NumMem  
  T\_const, [432](#)  
nummodels  
  C\_const, [63](#)  
NumModOptdollars  
  variable.h, [3205](#)  
numoffuns  
  N\_const, [253](#)  
NumOldNumFactors  
  S\_const, [382](#)  
NumOldOnFile  
  S\_const, [382](#)  
NUMOPTIONS  
  fsizes.h, [1351](#)  
NumOutputChannels  
  variable.h, [3203](#)  
numpart  
  PaRtl, [321](#)  
numpoly  
  T\_const, [436](#)  
NumPotModdollars  
  variable.h, [3205](#)  
NumPre  
  variable.h, [3200](#)  
NumPreSwitchStrings  
  P\_const, [314](#)  
NumPreTypes  
  DoLoOp, [92](#)  
  P\_const, [314](#)  
NumProcedures  
  variable.h, [3199](#)  
numrbufs  
  ParallelVars, [319](#)  
numrhs  
  CbUf, [73](#)  
numsbufs  
  ParallelVars, [319](#)  
NumSetElements  
  variable.h, [3202](#)  
NumSets  
  variable.h, [3202](#)  
NUMSPECSETS  
  ftypes.h, [1430](#)  
NumSpectatorFiles  
  M\_const, [184](#)  
NUMSTORECACHES  
  fsizes.h, [1349](#)  
NumStoreCaches  
  M\_const, [191](#)  
NumStreams  
  C\_const, [56](#)  
numsym  
  POLYMOD, [358](#)  
NumSymbols  
  variable.h, [3202](#)  
NumTableBases  
  variable.h, [3202](#)  
NUMTABLEENTRIES  
  fsizes.h, [1346](#)  
numtables  
  TaBIeBaSE, [458](#)  
numtasks  
  ParallelVars, [318](#)  
numtemp

- LIST, 166
- NumTerms
  - CbUf, 72
- NUMTERMSFUN
  - ftypes.h, 1397
- NUMTOIND
  - ftypes.h, 1419
- NUMTONUM
  - ftypes.h, 1419
- numtopo
  - TERMINFO, 466
- NumToStr
  - declare.h, 917
  - tools.c, 3085
- NUMTOSUB
  - ftypes.h, 1419
- NUMTOSYM
  - ftypes.h, 1419
- NumTotWildArgs
  - N\_const, 252
- numtracesctack
  - N\_const, 254
- numtree
  - CbUf, 74
  - TaBIEs, 464
- NumVectors
  - variable.h, 3202
- NumWild
  - N\_const, 256
- NumWildcardNames
  - C\_const, 55
- numwildcards
  - FUN\_INFO, 143
- numxsymbol
  - variable.h, 3205
- nvectors
  - InDeXeNtRy, 155
- nvert
  - SProcess, 408
  - ToPoTyPe, 470
- nvertices
  - MoDeL, 233
- NwildC
  - C\_const, 66
- O
  - AllGlobals, 11
- O\_BACKWARD
  - ftypes.h, 1450
- O\_const, 276
  - BlockSpaces, 282
  - bracket, 279
  - bufferedInd, 285
  - CheckPower, 281
  - CurBufWrt, 280
  - CurDictFunWithArgs, 283
  - CurDictInDollars, 284
  - CurDictNotInFunctions, 284
  - CurDictNumbers, 283
  - CurDictNumberWarning, 283
  - CurDictSpecials, 283
  - CurDictVariables, 283
  - CurrentDictionary, 283
  - Dictionaries, 281
  - DollarInOutBuffer, 282
  - DollarOutBuffer, 279
  - DollarOutSizeBuffer, 282
  - DoubleFlag, 285
  - ErrorBlock, 286
  - FactorMode, 286
  - FactorNum, 286
  - FlipLONG, 280
  - FlipPOINTER, 280
  - FlipPOS, 280
  - FlipWORD, 280
  - FortDotChar, 286
  - FortFirst, 285
  - gNumDictionaries, 284
  - IndentSpace, 285
  - InFbrack, 285
  - inscheme, 281
  - IsBracket, 285
  - mpower, 284
  - NoSpacesInNumbers, 282
  - NumDictionaries, 283
  - NumInBrack, 282
  - OptimizationLevel, 286
  - Optimize, 282
  - OptimizeResult, 278
  - OutFill, 279
  - OutInBuffer, 282
  - OutputLine, 278
  - OutSkip, 285
  - OutStop, 279
  - powerFlag, 284
  - PrintType, 285
  - ReNUMBERVec, 281
  - ResizeData, 280
  - resizeFlag, 284
  - ResizeLONG, 281
  - ResizeNCWORD, 280
  - ResizePOINTER, 281
  - ResizePOS, 281
  - ResizeWORD, 280
  - SaveData, 278
  - SaveHeader, 278
  - schemenum, 284
  - SizeDictionaries, 283
  - tabstring, 279
  - tensorList, 281
  - termbuf, 279
  - transFlag, 284
  - wlen, 282
  - wpoin, 279
  - wpos, 279
- O\_CSE
  - ftypes.h, 1449

- O\_CSEGREEDY
  - ftypes.h, 1449
- O\_FORWARD
  - ftypes.h, 1450
- O\_FORWARDANDBACKWARD
  - ftypes.h, 1450
- O\_FORWARDORBACKWARD
  - ftypes.h, 1450
- O\_GREEDY
  - ftypes.h, 1450
- O\_MCTS
  - ftypes.h, 1450
- O\_NONE
  - ftypes.h, 1449
- O\_OCCURRENCE
  - ftypes.h, 1450
- O\_SIMULATED\_ANNEALING
  - ftypes.h, 1450
- obj1
  - DiStRiBuTe, 85
- obj2
  - DiStRiBuTe, 85
- objects, 287
  - date, 287
  - element, 288
  - indexblock, 152
  - PeRmUtE, 327
  - PeRmUtEp, 328
  - position, 287
  - size, 287
  - spare1, 288
  - spare2, 288
  - spare3, 288
  - tablenumber, 287
  - uncompressed, 287
- occurrence\_order
  - optimize.cc, 1991
- ODD\_
  - ftypes.h, 1427
- OffsetIndex
  - M\_const, 187
- OffsetVector
  - M\_const, 187
- oldcbuf
  - INSIDEINFO, 158
- OldChildTime
  - M\_const, 176
- oldcnumlhs
  - INSIDEINFO, 159
- oldcompiletype
  - INSIDEINFO, 158
- olddelay
  - StreaM, 419
- OLDFACTARG
  - ftypes.h, 1448
- OldFactArgFlag
  - C\_const, 61
- OldGCDflag
  - C\_const, 61
- oldhandle
  - HANDLERS, 148
- oldlogonly
  - HANDLERS, 148
- OldMilliTime
  - M\_const, 176
- oldnamefill
  - NaMeTree, 265
- oldnodefill
  - NaMeTree, 265
- oldnoshowinput
  - StreaM, 419
- oldnumextrasymbols
  - M\_const, 174
- OldNumFactors
  - S\_const, 381
- oldnumpotmoddollars
  - INSIDEINFO, 158
- OldOnFile
  - S\_const, 381
- OldOrderFlag
  - M\_const, 189
- oldparalleflag
  - INSIDEINFO, 158
- OldParallelStats
  - C\_const, 58
- OldPosIn
  - N\_const, 245
- OldPosOut
  - N\_const, 245
- oldprnttype
  - HANDLERS, 148
- oldrbuf
  - INSIDEINFO, 159
- OldSecTime
  - M\_const, 176
- oldsilent
  - HANDLERS, 148
- OLDSTATEMENT
  - ftypes.h, 1371
- OldTime
  - R\_const, 372
- oldtype
  - N\_const, 255
- Olduflags
  - S\_const, 382
- oldvalue
  - N\_const, 256
- Oldvflags
  - S\_const, 381
- OMPI\_SKIP\_MPICXX
  - parallel.h, 2140
- one\_byte
  - structs.h, 2902
- ONEEXPRESSION
  - ftypes.h, 1374
- ONEPARTICLEIRREDUCIBLE

- ftypes.h, [1455](#)
- ONEPI
  - ftypes.h, [1403](#)
- onerhs
  - M\_const, [192](#)
- onfile
  - ExPrEsSiOn, [121](#)
- ONLYFUNCTIONS
  - ftypes.h, [1451](#)
- ONOFFVALUE
  - setfile.c, [2690](#)
- OpenAddFile
  - declare.h, [895](#)
  - tools.c, [3080](#)
- OpenBracketIndex
  - declare.h, [980](#)
  - index.c, [1672](#)
- OpenDbase
  - declare.h, [986](#)
  - minos.c, [1732](#)
- OpenDictionary
  - P\_const, [315](#)
- openExternalChannel
  - declare.h, [989](#)
  - extcmd.c, [1193](#)
- OpenFile
  - declare.h, [895](#)
  - tools.c, [3080](#)
- OpenInput
  - declare.h, [886](#)
  - startup.c, [2798](#)
- OpenStream
  - declare.h, [888](#)
  - tools.c, [3079](#)
- OpenTemp
  - declare.h, [859](#)
  - store.c, [2828](#)
- OPER\_ADD
  - optimize.cc, [2007](#)
- OPER\_COMMA
  - optimize.cc, [2007](#)
- OPER\_MUL
  - optimize.cc, [2007](#)
- opera.c, [1955](#), [1958](#)
  - Chisholm, [1957](#)
  - EpFCon, [1955](#)
  - EpFFind, [1955](#)
  - EpFGen, [1956](#)
  - TenVec, [1958](#)
  - TenVecFind, [1958](#)
  - Trace4, [1956](#)
  - Trace4Gen, [1956](#)
  - Trace4no, [1956](#)
  - TraceFind, [1957](#)
  - TraceN, [1957](#)
  - TraceNgen, [1957](#)
  - TraceNno, [1957](#)
  - Traces, [1957](#)
  - Trick, [1956](#)
- OperaFind
  - FixedGlobals, [135](#)
- Operation
  - FixedGlobals, [135](#)
- operator!=
  - poly, [343](#)
- operator<
  - BracketInfo, [35](#)
  - optimization, [290](#)
- operator<<
  - polyfact.cc, [2238](#)
- operator()
  - CSEEq, [77](#)
  - CSEHash, [77](#)
  - monomial\_larger, [237](#)
  - node, [272](#)
  - NodeEq, [274](#)
  - NodeHash, [274](#)
- operator+
  - poly, [342](#)
- operator+=
  - poly, [341](#)
- operator-
  - poly, [342](#)
- operator=
  - poly, [341](#)
- operator/
  - poly, [342](#)
- operator/=
  - poly, [341](#)
- operator=
  - poly, [343](#)
  - tree\_node, [481](#)
- operator==
  - poly, [342](#)
- operator%
  - poly, [342](#)
- operator%=
  - poly, [342](#)
- operator[]
  - poly, [343](#)
- operator\*
  - poly, [342](#)
- operator\*=
  - poly, [341](#)
- opi
  - MGraph, [214](#)
  - T\_EGraph, [444](#)
- opi2plp
  - EGraph, [112](#)
- opicomp
  - EEdge, [99](#)
- opiCount
  - EGraph, [113](#)
- opics
  - MConn, [199](#)
- opiloop



- MGraph, 214
- opisp
  - MConn, 199
- opistk
  - MConn, 200
- opistkptr
  - MConn, 202
- opt
  - Assign, 24
  - EGraph, 108
  - MGraph, 210
  - Output, 307
  - Process, 364
  - SProcess, 406
  - ToPoTyPe, 469
- OptDef, 288
  - defaultv, 289
  - DUMMYPADDING, 289
  - mean, 289
  - name, 289
- OPTHEAD
  - ftypes.h, 1451
- optimization, 289
  - arg1, 291
  - arg2, 291
  - coeff, 291
  - eqnidxs, 291
  - improve, 291
  - operator<, 290
  - type, 291
- OptimizationLevel
  - O\_const, 286
- OPTIMIZE, 292
  - debugflags, 294
  - fval, 292
  - greedymaxperc, 294
  - greedyminnum, 293
  - greedytimelimit, 293
  - horner, 292
  - hornerdirection, 293
  - ival, 292
  - mctsdecaymode, 294
  - mctsnumexpand, 293
  - mctsnumkeep, 293
  - mctsnumrepeat, 293
  - mctstimelimit, 293
  - method, 293
  - printstats, 294
  - salter, 294
  - schemeflags, 294
  - spare, 294
- Optimize
  - declare.h, 969
  - O\_const, 282
  - optimize.cc, 2004
- optimize.cc, 1987, 2009
  - build\_Horner\_tree, 1993
  - buildTree, 1997
  - ClearOptimize, 2006
  - count\_operators, 1990, 1991
  - count\_operators\_cse, 1996
  - count\_operators\_cse\_topdown, 1997
  - DivLong\_fix\_args, 1993
  - do\_optimization, 1999
  - find\_Horner\_MCTS, 1997
  - find\_Horner\_MCTS\_expand\_tree, 1997
  - find\_optimizations, 1999
  - fixarg, 1992
  - GcdLong\_fix\_args, 1993
  - generate\_expression, 2003
  - generate\_instructions, 1995
  - generate\_output, 2002
  - get\_brackets, 1990
  - get\_expression, 1989
  - getpower, 1991
  - hash\_range, 1995
  - Horner\_tree, 1994
  - mcts\_best\_schemes, 2009
  - mcts\_expr\_score, 2008
  - mcts\_factorized, 2008
  - mcts\_root, 2008
  - mcts\_separated, 2008
  - mcts\_vars, 2008
  - merge\_operators, 1998
  - my\_random\_shuffle, 1989
  - occurrence\_order, 1991
  - OPER\_ADD, 2007
  - OPER\_COMMA, 2007
  - OPER\_MUL, 2007
  - Optimize, 2004
  - optimize\_best\_Horner\_schemes, 2007
  - optimize\_best\_instr, 2008
  - optimize\_best\_num\_oper, 2008
  - optimize\_best\_vars, 2008
  - optimize\_expr, 2007
  - optimize\_expression\_given\_Horner, 2002
  - optimize\_greedy, 2000
  - optimize\_num\_vars, 2007
  - optimize\_print\_code, 2004
  - partial\_factorize, 2000
  - print\_instr, 1989
  - print\_tree, 1995
  - recycle\_variables, 2001
  - simulated\_annealing, 1997
  - term\_compare, 1994
- optimize\_best\_Horner\_schemes
  - optimize.cc, 2007
- optimize\_best\_instr
  - optimize.cc, 2008
- optimize\_best\_num\_oper
  - optimize.cc, 2008
- optimize\_best\_vars
  - optimize.cc, 2008
- optimize\_expr
  - optimize.cc, 2007
- optimize\_expression\_given\_Horner

- optimize.cc, [2002](#)
- optimize\_greedy
  - optimize.cc, [2000](#)
- optimize\_num\_vars
  - optimize.cc, [2007](#)
- optimize\_print\_code
  - declare.h, [1020](#)
  - optimize.cc, [2004](#)
- OPTIMIZERESULT, [295](#)
  - code, [295](#)
  - codesize, [295](#)
  - exprnr, [296](#)
  - maxvar, [296](#)
  - minvar, [296](#)
  - nameofexpr, [295](#)
- OptimizeResult
  - O\_const, [278](#)
- optentimes
  - T\_const, [435](#)
- option
  - SHvariables, [392](#)
- OPTION0
  - compiler.c, [723](#)
- OPTION1
  - compiler.c, [723](#)
- OPTION2
  - compiler.c, [723](#)
- Options, [296](#)
  - ~Options, [297](#)
  - argag, [302](#)
  - argemg, [302](#)
  - argmg, [302](#)
  - begin, [300](#)
  - beginProc, [300](#)
  - beginSubProc, [300](#)
  - DUMMYPADDING, [302](#)
  - end, [300](#)
  - endmg, [302](#)
  - endProc, [300](#)
  - endSubProc, [300](#)
  - getDef, [300](#)
  - getValue, [298](#)
  - model, [301](#)
  - newAGraph, [301](#)
  - newMGraph, [301](#)
  - Options, [297](#)
  - out, [301](#)
  - outag, [302](#)
  - outmg, [302](#)
  - outModel, [301](#)
  - print, [298](#)
  - printLevel, [299](#)
  - printModel, [299](#)
  - proc, [301](#)
  - setDefaultValue, [298](#)
  - setEndMG, [299](#)
  - setErExit, [299](#)
  - setOutAG, [299](#)
  - setOutMG, [299](#)
  - setOutputF, [298](#)
  - setOutputP, [298](#)
  - setValue, [298](#)
  - sproc, [301](#)
  - time0, [303](#)
  - time1, [303](#)
  - values, [302](#)
- or2nd
  - MOrbits, [239](#)
- orbits
  - Assign, [24](#)
  - MGraph, [210](#)
- ORCOND
  - ftypes.h, [1437](#)
- ORDEREDSET
  - ftypes.h, [1454](#)
- Ordering
  - M\_const, [188](#)
- origin
  - C\_const, [63](#)
- out
  - DiStRiBuTe, [85](#)
  - Options, [301](#)
- outag
  - Options, [302](#)
- outBeginF
  - Output, [305](#)
- outBeginP
  - Output, [305](#)
- OutBuffer
  - M\_const, [173](#)
- OutBufSize
  - M\_const, [178](#)
- outEGraphF
  - Output, [307](#)
- outEGraphP
  - Output, [307](#)
- outEndF
  - Output, [305](#)
- outEndP
  - Output, [305](#)
- outfile
  - R\_const, [371](#)
- OutFill
  - O\_const, [279](#)
- outfun
  - SHvariables, [390](#)
- outgrf
  - Output, [308](#)
- outgrfp
  - Output, [308](#)
- outgrp
  - Output, [308](#)
- outgrpp
  - Output, [308](#)
- OutInBuffer
  - O\_const, [282](#)

- outmg
  - Options, [302](#)
- outModel
  - Options, [301](#)
- outModelF
  - Output, [307](#)
- outModelP
  - Output, [307](#)
- OutNumberType
  - C\_const, [67](#)
- outproc
  - Output, [308](#)
- outProcBegin0
  - Output, [306](#)
- outProcBeginF
  - Output, [306](#)
- outProcBeginP
  - Output, [306](#)
- outProcEndF
  - Output, [306](#)
- outProcEndP
  - Output, [306](#)
- Output, [303](#)
  - ~Output, [304](#)
  - model, [307](#)
  - opt, [307](#)
  - outBeginF, [305](#)
  - outBeginP, [305](#)
  - outEGraphF, [307](#)
  - outEGraphP, [307](#)
  - outEndF, [305](#)
  - outEndP, [305](#)
  - outgrf, [308](#)
  - outgrfp, [308](#)
  - outgrp, [308](#)
  - outgrpp, [308](#)
  - outModelF, [307](#)
  - outModelP, [307](#)
  - outproc, [308](#)
  - outProcBegin0, [306](#)
  - outProcBeginF, [306](#)
  - outProcBeginP, [306](#)
  - outProcEndF, [306](#)
  - outProcEndP, [306](#)
  - Output, [304](#)
  - outSProcBeginF, [306](#)
  - outSProcBeginP, [306](#)
  - proc, [307](#)
  - procl, [308](#)
  - setOutgrf, [305](#)
  - setOutgrp, [305](#)
  - sproc, [308](#)
- OutputLine
  - O\_const, [278](#)
- OutputMode
  - C\_const, [66](#)
- outputmode
  - sOrT, [400](#)
- OutputOnePI
  - lus.c, [1689](#)
- OUTPUTONLY
  - minos.h, [1754](#)
- OutputSpaces
  - C\_const, [66](#)
- outsidefun
  - C\_const, [57](#)
- OutSkip
  - O\_const, [285](#)
- outSProcBeginF
  - Output, [306](#)
- outSProcBeginP
  - Output, [306](#)
- OutStop
  - O\_const, [279](#)
- outterm
  - SHvariables, [390](#)
- outtohide
  - R\_const, [373](#)
- P
  - AllGlobals, [11](#)
- p
  - BracketInfo, [36](#)
  - DoLoOp, [90](#)
  - PROCEDURE, [362](#)
- p1
  - PoSiTiOn, [359](#)
- P\_const, [309](#)
  - AllowDelay, [315](#)
  - cComChar, [316](#)
  - ComChar, [316](#)
  - cprocedureExtension, [312](#)
  - DebugFlag, [316](#)
  - DelayPrevar, [315](#)
  - DollarList, [311](#)
  - eat, [315](#)
  - firstnamespace, [311](#)
  - fullname, [311](#)
  - fullnamesize, [316](#)
  - gNumPre, [315](#)
  - iBufError, [314](#)
  - InOutBuf, [313](#)
  - inside, [311](#)
  - lastnamespace, [311](#)
  - lhdollarerror, [315](#)
  - LoopList, [311](#)
  - MaxPreAssignLevel, [316](#)
  - MaxPrelfLevel, [314](#)
  - MaxPreTypes, [314](#)
  - NumPreSwitchStrings, [314](#)
  - NumPreTypes, [314](#)
  - OpenDictionary, [315](#)
  - PreAssignFlag, [313](#)
  - PreAssignLevel, [316](#)
  - PreAssignStack, [312](#)
  - PreContinuation, [313](#)
  - PreDebug, [315](#)

- preError, 316
- preFill, 312
- PreffLevel, 314
- PreffStack, 312
- PreInsideLevel, 315
- PreOut, 314
- PreproFlag, 313
- preStart, 312
- preStop, 312
- PreSwitchLevel, 314
- PreSwitchModes, 313
- PreSwitchStrings, 312
- PreTypes, 313
- PreVarList, 311
- procedureExtension, 312
- ProcList, 311
- pSize, 313
- StopWatchZero, 313
- P\_term
  - fwin.h, 1496
  - unix.h, 3192
- Pack
  - declare.h, 859
  - reken.c, 2552
- PACK\_LONG
  - parallel.c, 2064
- PackFloat
  - float.c, 1291
- packpos
  - longMultiStruct, 167
- padding
  - SWITCH, 423
  - T\_EEdge, 441
- PADDUMMY
  - form3.h, 1330
- PADINT
  - form3.h, 1331
- PADLONG
  - form3.h, 1330
- PADPOINTER
  - form3.h, 1330
- PADPOSITION
  - form3.h, 1330
- PADWORD
  - form3.h, 1331
- parallel
  - ParallelVars, 318
- parallel.c, 2061, 2087
  - \_CHECK, 2064
  - \_\_CHECK, 2065
  - CHECK, 2064
  - DBGOUT, 2065
  - DBGOUT\_NINTERMS, 2065
  - NODE, 2066
  - PACK\_LONG, 2064
  - PF, 2086
  - PF\_BroadcastBuffer, 2076
  - PF\_BroadcastCBuf, 2081
  - PF\_BroadcastExpFlags, 2081
  - PF\_BroadcastExpr, 2082
  - PF\_BroadcastModifiedDollars, 2079
  - PF\_BroadcastNumber, 2075
  - PF\_BroadcastPreDollar, 2077
  - PF\_BroadcastRedefinedPreVars, 2080
  - PF\_BroadcastRHS, 2082
  - PF\_BroadcastString, 2076
  - PF\_CollectModifiedDollars, 2078
  - PF\_Deferred, 2071
  - PF\_EndSort, 2070
  - PF\_FlushStdOutBuffer, 2086
  - PF\_FreeErrorMessageBuffers, 2086
  - PF\_GetSlaveTimes, 2075
  - PF\_Init, 2073
  - PF\_InParallelProcessor, 2083
  - PF\_IRecvRbuf, 2068
  - PF\_LibInit, 2066
  - PF\_LibTerminate, 2067
  - PF\_MLock, 2085
  - PF\_MUnlock, 2085
  - PF\_Probe, 2067
  - PF\_Processor, 2072
  - PF\_RawProbe, 2070
  - PF\_RawRecv, 2069
  - PF\_RawSend, 2069
  - PF\_RealTime, 2066
  - PF\_RecvFile, 2084
  - PF\_RecvWbuf, 2067
  - PF\_RestoreInsideInfo, 2081
  - PF\_SendFile, 2083
  - PF\_SNDFILEBUFSIZE, 2065
  - PF\_STATS\_SIZE, 2065
  - PF\_StoreInsideInfo, 2080
  - PF\_Terminate, 2074
  - PF\_WaitRbuf, 2068
  - PF\_WriteFileToFile, 2085
  - PRINTFBUF, 2063
  - recvBuffer, 2066
  - SWAP, 2063
  - UNPACK\_LONG, 2064
- parallel.h, 2135, 2168
  - GNUPREREQ, 2139
  - indices, 2140
  - MASTER, 2137
  - OMPI\_SKIP\_MPICXX, 2140
  - PF, 2167
  - PF\_ANY\_MSGTAG, 2140
  - PF\_ANY\_SOURCE, 2140
  - PF\_Bcast, 2141
  - PF\_Broadcast, 2146
  - PF\_BroadcastBuffer, 2157
  - PF\_BroadcastCBuf, 2161
  - PF\_BroadcastExpFlags, 2162
  - PF\_BroadcastExpr, 2163
  - PF\_BroadcastModifiedDollars, 2160
  - PF\_BroadcastNumber, 2156
  - PF\_BroadcastPreDollar, 2158

- PF\_BroadcastRedefinedPreVars, [2161](#)
- PF\_BroadcastRHS, [2164](#)
- PF\_BroadcastString, [2157](#)
- PF\_BUFFER\_MSGTAG, [2138](#)
- PF\_BYTE, [2140](#)
- PF\_CollectModifiedDollars, [2159](#)
- PF\_COMM, [2140](#)
- PF\_DATA\_MSGTAG, [2138](#)
- PF\_Deferred, [2152](#)
- PF\_DOLLAR\_MSGTAG, [2138](#)
- PF\_EMPTY\_MSGTAG, [2139](#)
- PF\_ENDBUFFER\_MSGTAG, [2138](#)
- PF\_EndSort, [2151](#)
- PF\_ENDSORT\_MSGTAG, [2138](#)
- PF\_FlushStdOutBuffer, [2167](#)
- PF\_GetSlaveTimes, [2156](#)
- PF\_Init, [2154](#)
- PF\_InParallelProcessor, [2164](#)
- PF\_INT, [2140](#)
- PF\_ISendSbuf, [2141](#)
- PF\_LOG\_MSGTAG, [2139](#)
- PF\_LongMultiBroadcast, [2151](#)
- PF\_LongMultiPack, [2141](#)
- PF\_LongMultiPackImpl, [2149](#)
- PF\_LongMultiUnpack, [2141](#)
- PF\_LongMultiUnpackImpl, [2150](#)
- PF\_LongSinglePack, [2147](#)
- PF\_LongSingleReceive, [2148](#)
- PF\_LongSingleSend, [2148](#)
- PF\_LongSingleUnpack, [2147](#)
- PF\_maxDollarChunkSize, [2167](#)
- PF\_MISC\_MSGTAG, [2139](#)
- PF\_MLock, [2166](#)
- PF\_MUnlock, [2166](#)
- PF\_OPT\_COLLECT\_MSGTAG, [2139](#)
- PF\_OPT\_HORNER\_MSGTAG, [2139](#)
- PF\_OPT\_MCTS\_MSGTAG, [2139](#)
- PF\_Pack, [2143](#)
- PF\_PackString, [2144](#)
- PF\_PrepareLongMultiPack, [2149](#)
- PF\_PrepareLongSinglePack, [2146](#)
- PF\_PreparePack, [2143](#)
- PF\_Processor, [2153](#)
- PF\_RawRecv, [2142](#)
- PF\_RawSend, [2142](#)
- PF\_READY\_MSGTAG, [2138](#)
- PF\_Receive, [2146](#)
- PF\_RecvFile, [2165](#)
- PF\_RESET, [2137](#)
- PF\_RestoreInsideInfo, [2163](#)
- PF\_Send, [2145](#)
- PF\_SendFile, [2164](#)
- PF\_STDOUT\_MSGTAG, [2139](#)
- PF\_StoreInsideInfo, [2163](#)
- PF\_TERM\_MSGTAG, [2138](#)
- PF\_Terminate, [2155](#)
- PF\_TIME, [2137](#)
- PF\_Unpack, [2144](#)
- PF\_UnpackString, [2145](#)
- PF\_WriteFileToFile, [2166](#)
- PARALLELFLAG
  - ftypes.h, [1370](#)
- parallelflag
  - C\_const, [59](#)
- ParallelVars, [317](#)
  - exprbufsize, [319](#)
  - exprtodo, [319](#)
  - log, [319](#)
  - me, [318](#)
  - mkSlaveInfile, [319](#)
  - numrbufs, [319](#)
  - numsbufs, [319](#)
  - numtasks, [318](#)
  - parallel, [318](#)
  - rbufs, [318](#)
  - rhsInParallel, [318](#)
  - sbuf, [318](#)
  - slavebuf, [318](#)
- parameter
  - SETUPPARAMETERS, [385](#)
- parent
  - NaMeNode, [260](#)
  - tree, [479](#)
- ParenthesesTest
  - compiler.c, [724](#)
  - declare.h, [958](#)
- ParseNumber
  - declare.h, [802](#)
- ParseSign
  - declare.h, [802](#)
- ParseSignedNumber
  - declare.h, [802](#)
- PARTEST
  - ftypes.h, [1376](#)
- PARTI
  - structs.h, [2903](#)
- PaRtl, [320](#)
  - args, [320](#)
  - nargs, [320](#)
  - nfun, [320](#)
  - numargs, [320](#)
  - numpart, [321](#)
  - psize, [320](#)
  - where, [321](#)
- partial\_factorize
  - optimize.cc, [2000](#)
- PaRtlcLe, [321](#)
  - mass, [322](#)
  - number, [321](#)
  - spin, [321](#)
  - type, [322](#)
- Particle, [322](#)
  - ~Particle, [323](#)
  - acode, [326](#)
  - aname, [325](#)
  - aparticle, [324](#)

- cmaxdeg, [326](#)
  - cmindeg, [326](#)
  - extonly, [326](#)
  - id, [325](#)
  - isNeutral, [324](#)
  - mdl, [325](#)
  - name, [325](#)
  - neutral, [325](#)
  - Particle, [323](#)
  - particleCode, [324](#)
  - particleName, [324](#)
  - pcode, [325](#)
  - prParticle, [324](#)
  - pType, [325](#)
  - typeGName, [324](#)
  - typeName, [324](#)
- particle
  - PNodeClass, [336](#)
- particleCode
  - Model, [228](#)
  - Particle, [324](#)
- particleName
  - Model, [227](#)
  - Particle, [324](#)
- particles
  - Model, [229](#)
  - VerTex, [485](#)
- PARTITIONS
  - ftypes.h, [1402](#)
- partitions
  - T\_const, [432](#)
- partodoflag
  - C\_const, [60](#)
- PasteFile
  - declare.h, [859](#)
  - proces.c, [2428](#)
- PasteTerm
  - declare.h, [823](#)
  - proces.c, [2428](#)
- Patches
  - sOrT, [395](#)
- Path
  - M\_const, [173](#)
- PathOutput
  - declare.h, [1031](#)
  - lus.c, [1689](#)
- PATHSEPARATOR
  - fwin.h, [1496](#)
  - unix.h, [3193](#)
- PATHVALUE
  - setfile.c, [2689](#)
- patstop
  - N\_const, [246](#)
- pattern
  - BracketInfo, [35](#)
  - TaBIEs, [461](#)
- pattern.c, [2171](#), [2174](#)
  - FindAll, [2173](#)
  - PutInBuffers, [2172](#)
  - SubsInAll, [2173](#)
  - Substitute, [2173](#)
  - TakeIDfunction, [2174](#)
  - TestMatch, [2172](#)
  - TestSelect, [2173](#)
  - TransferBuffer, [2174](#)
- patternbuffer
  - N\_const, [249](#)
- patternbuffersize
  - N\_const, [257](#)
- PATTERNFUNCTION
  - ftypes.h, [1396](#)
- pBer
  - T\_const, [431](#)
- pcode
  - Particle, [325](#)
  - PGInput, [332](#)
  - PInput, [333](#)
- pdef
  - Model, [229](#)
- pdel
  - TrAcEs, [476](#)
- pepf
  - TrAcEs, [475](#)
- PERM
  - structs.h, [2903](#)
- perm
  - TrAcEn, [472](#)
  - TrAcEs, [474](#)
- permg
  - SGroup, [389](#)
- permMat
  - MGraph, [208](#)
- PERMP
  - structs.h, [2903](#)
- perms
  - SGroup, [389](#)
- PERMUTATIONS
  - ftypes.h, [1402](#)
- PeRmUtE, [326](#)
  - cycle, [327](#)
  - n, [327](#)
  - objects, [327](#)
  - sign, [327](#)
- Permute
  - declare.h, [860](#)
  - symmetr.c, [2931](#)
- PERMUTEARG
  - ftypes.h, [1446](#)
- PeRmUtEp, [327](#)
  - cycle, [328](#)
  - n, [328](#)
  - objects, [328](#)
  - sign, [328](#)
- PermuteP
  - declare.h, [860](#)
  - symmetr.c, [2931](#)

- PF
  - [parallel.c](#), 2086
  - [parallel.h](#), 2167
- PF\_ANY\_MSGTAG
  - [parallel.h](#), 2140
- PF\_ANY\_SOURCE
  - [parallel.h](#), 2140
- PF\_Bcast
  - [mpi.c](#), 1780
  - [parallel.h](#), 2141
- PF\_Broadcast
  - [mpi.c](#), 1786
  - [parallel.h](#), 2146
- PF\_BroadcastBuffer
  - [parallel.c](#), 2076
  - [parallel.h](#), 2157
- PF\_BroadcastCBuf
  - [parallel.c](#), 2081
  - [parallel.h](#), 2161
- PF\_BroadcastExpFlags
  - [parallel.c](#), 2081
  - [parallel.h](#), 2162
- PF\_BroadcastExpr
  - [parallel.c](#), 2082
  - [parallel.h](#), 2163
- PF\_BroadcastModifiedDollars
  - [parallel.c](#), 2079
  - [parallel.h](#), 2160
- PF\_BroadcastNumber
  - [parallel.c](#), 2075
  - [parallel.h](#), 2156
- PF\_BroadcastPreDollar
  - [parallel.c](#), 2077
  - [parallel.h](#), 2158
- PF\_BroadcastRedefinedPreVars
  - [parallel.c](#), 2080
  - [parallel.h](#), 2161
- PF\_BroadcastRHS
  - [parallel.c](#), 2082
  - [parallel.h](#), 2164
- PF\_BroadcastString
  - [parallel.c](#), 2076
  - [parallel.h](#), 2157
- PF\_BUFFER, 328
  - [active](#), 330
  - [buff](#), 329
  - [fill](#), 329
  - [from](#), 330
  - [full](#), 329
  - [index](#), 330
  - [numbufs](#), 330
  - [request](#), 330
  - [retstat](#), 330
  - [status](#), 329
  - [stop](#), 329
  - [tag](#), 330
  - [type](#), 330
- PF\_BUFFER\_MSGTAG
  - [parallel.h](#), 2138
- PF\_BYTE
  - [parallel.h](#), 2140
- PF\_CollectModifiedDollars
  - [parallel.c](#), 2078
  - [parallel.h](#), 2159
- PF\_COMM
  - [parallel.h](#), 2140
- PF\_DATA\_MSGTAG
  - [parallel.h](#), 2138
- PF\_Deferred
  - [parallel.c](#), 2071
  - [parallel.h](#), 2152
- PF\_DOLLAR\_MSGTAG
  - [parallel.h](#), 2138
- PF\_EMPTY\_MSGTAG
  - [parallel.h](#), 2139
- PF\_ENDBUFFER\_MSGTAG
  - [parallel.h](#), 2138
- PF\_EndSort
  - [parallel.c](#), 2070
  - [parallel.h](#), 2151
- PF\_ENDSORT\_MSGTAG
  - [parallel.h](#), 2138
- PF\_FlushStdOutBuffer
  - [parallel.c](#), 2086
  - [parallel.h](#), 2167
- PF\_FreeErrorMessageBuffers
  - [parallel.c](#), 2086
- PF\_GetSlaveTimes
  - [parallel.c](#), 2075
  - [parallel.h](#), 2156
- PF\_Init
  - [parallel.c](#), 2073
  - [parallel.h](#), 2154
- PF\_InParallelProcessor
  - [parallel.c](#), 2083
  - [parallel.h](#), 2164
- PF\_INT
  - [parallel.h](#), 2140
- PF\_IRecvRbuf
  - [mpi.c](#), 1779
  - [parallel.c](#), 2068
- PF\_ISendSbuf
  - [mpi.c](#), 1778
  - [parallel.h](#), 2141
- PF\_LibInit
  - [mpi.c](#), 1777
  - [parallel.c](#), 2066
- PF\_LibTerminate
  - [mpi.c](#), 1778
  - [parallel.c](#), 2067
- PF\_LOG\_MSGTAG
  - [parallel.h](#), 2139
- PF\_LongMultiBroadcast
  - [mpi.c](#), 1790
  - [parallel.h](#), 2151
- PF\_LongMultiPack

parallel.h, [2141](#)  
PF\_LongMultiPackImpl  
  mpi.c, [1788](#)  
  parallel.h, [2149](#)  
PF\_LongMultiUnpack  
  parallel.h, [2141](#)  
PF\_LongMultiUnpackImpl  
  mpi.c, [1789](#)  
  parallel.h, [2150](#)  
PF\_LongSinglePack  
  mpi.c, [1786](#)  
  parallel.h, [2147](#)  
PF\_LongSingleReceive  
  mpi.c, [1788](#)  
  parallel.h, [2148](#)  
PF\_LongSingleSend  
  mpi.c, [1787](#)  
  parallel.h, [2148](#)  
PF\_LongSingleUnpack  
  mpi.c, [1787](#)  
  parallel.h, [2147](#)  
PF\_maxDollarChunkSize  
  mpi.c, [1791](#)  
  parallel.h, [2167](#)  
PF\_MISC\_MSGTAG  
  parallel.h, [2139](#)  
PF\_MLock  
  parallel.c, [2085](#)  
  parallel.h, [2166](#)  
PF\_MUnlock  
  parallel.c, [2085](#)  
  parallel.h, [2166](#)  
PF\_OPT\_COLLECT\_MSGTAG  
  parallel.h, [2139](#)  
PF\_OPT\_HORNER\_MSGTAG  
  parallel.h, [2139](#)  
PF\_OPT\_MCTS\_MSGTAG  
  parallel.h, [2139](#)  
PF\_Pack  
  mpi.c, [1783](#)  
  parallel.h, [2143](#)  
PF\_PACKSIZE  
  mpi.c, [1776](#)  
PF\_PackString  
  mpi.c, [1784](#)  
  parallel.h, [2144](#)  
PF\_PrepareLongMultiPack  
  mpi.c, [1788](#)  
  parallel.h, [2149](#)  
PF\_PrepareLongSinglePack  
  mpi.c, [1786](#)  
  parallel.h, [2146](#)  
PF\_PreparePack  
  mpi.c, [1782](#)  
  parallel.h, [2143](#)  
PF\_PrintPackBuf  
  mpi.c, [1782](#)  
PF\_Probe  
  mpi.c, [1778](#)  
  parallel.c, [2067](#)  
PF\_Processor  
  parallel.c, [2072](#)  
  parallel.h, [2153](#)  
PF\_RawProbe  
  mpi.c, [1782](#)  
  parallel.c, [2070](#)  
PF\_RawRecv  
  mpi.c, [1781](#)  
  parallel.c, [2069](#)  
  parallel.h, [2142](#)  
PF\_RawSend  
  mpi.c, [1781](#)  
  parallel.c, [2069](#)  
  parallel.h, [2142](#)  
PF\_READY\_MSGTAG  
  parallel.h, [2138](#)  
PF\_RealTime  
  mpi.c, [1777](#)  
  parallel.c, [2066](#)  
PF\_Receive  
  mpi.c, [1785](#)  
  parallel.h, [2146](#)  
PF\_RecvFile  
  parallel.c, [2084](#)  
  parallel.h, [2165](#)  
PF\_RecvWbuf  
  mpi.c, [1779](#)  
  parallel.c, [2067](#)  
PF\_RESET  
  parallel.h, [2137](#)  
PF\_RestoreInsideInfo  
  parallel.c, [2081](#)  
  parallel.h, [2163](#)  
PF\_Send  
  mpi.c, [1785](#)  
  parallel.h, [2145](#)  
PF\_SendFile  
  parallel.c, [2083](#)  
  parallel.h, [2164](#)  
PF\_SNDFILEBUFSIZE  
  parallel.c, [2065](#)  
PF\_STATS\_SIZE  
  parallel.c, [2065](#)  
PF\_STDOUT\_MSGTAG  
  parallel.h, [2139](#)  
PF\_StoreInsideInfo  
  parallel.c, [2080](#)  
  parallel.h, [2163](#)  
PF\_TERM\_MSGTAG  
  parallel.h, [2138](#)  
PF\_Terminate  
  parallel.c, [2074](#)  
  parallel.h, [2155](#)  
PF\_TIME  
  parallel.h, [2137](#)  
PF\_Unpack



- mpi.c, [1783](#)
  - parallel.h, [2144](#)
- PF\_UnpackString
  - mpi.c, [1784](#)
  - parallel.h, [2145](#)
- PF\_WaitRbuf
  - mpi.c, [1780](#)
  - parallel.c, [2068](#)
- PF\_WriteFileToFile
  - parallel.c, [2085](#)
  - parallel.h, [2166](#)
- pfac
  - T\_const, [431](#)
- PFORTRANMODE
  - ftypes.h, [1386](#)
- PGInput, [331](#)
  - cdeg, [331](#)
  - cnum, [331](#)
  - cple, [331](#)
  - ctyp, [331](#)
  - pcode, [332](#)
  - pname, [332](#)
- pgq
  - SGroup, [389](#)
- pgr
  - SGroup, [389](#)
- PHEAD
  - structs.h, [2899](#)
- PHEAD0
  - structs.h, [2899](#)
- PHI
  - ftypes.h, [1403](#)
- pId
  - EGraph, [111](#)
  - MGraph, [211](#)
  - T\_EGraph, [443](#)
  - T\_MGraph, [447](#)
- PInput, [332](#)
  - acode, [333](#)
  - aname, [332](#)
  - extonly, [333](#)
  - name, [332](#)
  - pcode, [333](#)
  - ptypec, [333](#)
  - ptypen, [333](#)
- PIPESTREAM
  - ftypes.h, [1366](#)
- PISYMBOL
  - ftypes.h, [1404](#)
- plist
  - ECand, [95](#)
  - EStack, [119](#)
  - Interaction, [161](#)
- plisc
  - lInput, [150](#)
- plisn
  - lInput, [149](#)
- pname
  - PGInput, [332](#)
  - Stream, [418](#)
- pnclass
  - Assign, [24](#)
  - SProcess, [406](#)
- PNodeClass, [334](#)
  - ~PNodeClass, [335](#)
  - cl2mcl, [337](#)
  - cl2nd, [337](#)
  - cmaxdeg, [336](#)
  - cmindeg, [336](#)
  - count, [336](#)
  - couple, [336](#)
  - deg, [336](#)
  - nclass, [337](#)
  - nd2cl, [337](#)
  - nnodes, [337](#)
  - particle, [336](#)
  - PNodeClass, [335](#)
  - prElem, [335](#)
  - prPNodeClass, [335](#)
  - sproc, [336](#)
  - type, [336](#)
- PObuffer
  - FiLe, [129](#)
- POfill
  - FiLe, [129](#)
- POfull
  - FiLe, [129](#)
- poin2a
  - sOrT, [396](#)
- poina
  - sOrT, [396](#)
- PoinFill
  - sOrT, [395](#)
- PoinScratch
  - N\_const, [249](#)
- Pointer
  - CbUf, [71](#)
- pointer
  - Stream, [417](#)
- poly, [339](#)
  - ~poly, [341](#)
  - add, [349](#)
  - all\_variables, [345](#)
  - argument\_to\_poly, [347](#)
  - check\_memory, [343](#)
  - coefficient, [346](#)
  - coefficient\_list\_divmod, [348](#)
  - coefficients\_modulo, [346](#)
  - compare\_degree\_vector, [349](#)
  - degree, [345](#)
  - degree\_vector, [349](#)
  - derivative, [346](#)
  - div, [350](#)
  - divides, [351](#)
  - divmod, [351](#)
  - divmod\_heap, [354](#)

divmod\_one\_term, 353  
divmod\_univar, 353  
expand\_memory, 343  
first\_variable, 345  
flint::poly, 338  
from\_coefficient\_list, 348  
get\_variables, 347  
integer\_lcoeff, 345  
is\_dense\_univariate, 344  
is\_integer, 344  
is\_monomial, 344  
is\_one, 344  
is\_zero, 344  
last\_monomial, 349  
last\_monomial\_index, 349  
lcoeff\_multivar, 346  
lcoeff\_univar, 346  
mod, 351  
modn, 356  
modp, 356  
monomial\_compare, 349  
mul, 350  
mul\_heap, 352  
mul\_one\_term, 352  
mul\_univar, 352  
normalize, 348  
number\_of\_terms, 345  
operator!=, 343  
operator+, 342  
operator+=, 341  
operator-, 342  
operator-=, 341  
operator/, 342  
operator/=, 341  
operator=, 343  
operator==, 342  
operator%, 342  
operator%=: 342  
operator[], 343  
operator\*, 342  
operator\*=: 341  
poly, 340, 341  
poly\_to\_argument, 347  
poly\_to\_argument\_with\_den, 347  
pop\_heap, 355  
push\_heap, 355  
setmod, 346  
sign, 345  
simple\_poly, 346, 347  
size\_of\_form\_notation, 348  
size\_of\_form\_notation\_with\_den, 348  
size\_of\_terms, 356  
sub, 350  
terms, 356  
termscopy, 343  
to\_coefficient\_list, 348  
to\_string, 349  
total\_degree, 345  
poly.cc, 2201  
poly.h, 2234  
poly\_determine\_modulus  
  polywrap.cc, 2276  
poly\_div  
  declare.h, 1013  
  polywrap.cc, 2277  
poly\_divmod  
  polywrap.cc, 2277  
poly\_factor  
  flint::poly\_factor, 357  
poly\_factorize  
  polywrap.cc, 2281  
poly\_factorize\_argument  
  declare.h, 1016  
  polywrap.cc, 2282  
poly\_factorize\_dollar  
  declare.h, 1017  
  polywrap.cc, 2283  
poly\_factorize\_expression  
  declare.h, 1017  
  polywrap.cc, 2283  
poly\_fix\_minus\_signs  
  polywrap.cc, 2281  
poly\_free\_poly\_vars  
  declare.h, 1019  
  polywrap.cc, 2286  
poly\_gcd  
  declare.h, 1012  
  polywrap.cc, 2276  
poly\_inverse  
  declare.h, 1014  
  polywrap.cc, 2285  
poly\_mul  
  declare.h, 1014  
  polywrap.cc, 2286  
poly\_num\_vars  
  N\_const, 258  
poly\_ratfun\_add  
  declare.h, 1014  
  polywrap.cc, 2279  
poly\_ratfun\_normalize  
  declare.h, 1015  
  polywrap.cc, 2280  
poly\_ratfun\_read  
  polywrap.cc, 2277  
poly\_rem  
  declare.h, 1013  
  polywrap.cc, 2277  
poly\_sort  
  polywrap.cc, 2278  
poly\_to\_argument  
  poly, 347  
poly\_to\_argument\_with\_den  
  poly, 347  
poly\_unfactorize\_expression  
  declare.h, 1018  
  polywrap.cc, 2284

- poly\_vars
  - N\_const, [251](#)
- poly\_vars\_type
  - N\_const, [258](#)
- PolyAct
  - T\_const, [438](#)
- POLYADD
  - ftypes.h, [1441](#)
- POLYDIV
  - ftypes.h, [1441](#)
- PolyDiv
  - declare.h, [1009](#)
  - ratio.c, [2505](#)
- polyfact.cc, [2237](#), [2239](#)
  - operator<<, [2238](#)
- polyfact.h, [2254](#)
- PolyFlag
  - sOrT, [399](#)
- POLYFUN
  - ftypes.h, [1369](#)
- PolyFun
  - R\_const, [375](#)
- PolyFunClean
  - declare.h, [854](#)
  - function.c, [1471](#)
- PolyFunDirty
  - declare.h, [854](#)
  - function.c, [1471](#)
- PolyFunExp
  - R\_const, [376](#)
- PolyFunInv
  - R\_const, [375](#)
- PolyFunMul
  - declare.h, [860](#)
  - proces.c, [2434](#)
- PolyFunPow
  - R\_const, [376](#)
- PolyFunTodo
  - N\_const, [257](#)
- PolyFunType
  - R\_const, [375](#)
- PolyFunVar
  - R\_const, [376](#)
- POLYGCD
  - ftypes.h, [1441](#)
- polygcd.cc, [2256](#), [2257](#)
  - gcd\_heuristic\_possible, [2256](#)
  - gcd\_linear\_helper, [2257](#)
- polygcd.h, [2273](#)
- POLYINTFAC
  - ftypes.h, [1442](#)
- POLYMOD, [357](#)
  - arraysize, [358](#)
  - coefs, [358](#)
  - modnum, [358](#)
  - numsym, [358](#)
  - polysize, [358](#)
- POLYMUL
  - ftypes.h, [1441](#)
- POLYNORM
  - ftypes.h, [1442](#)
- PolyNormFlag
  - N\_const, [257](#)
- PolyRatFunChanged
  - C\_const, [69](#)
- PolyRatFunSpecial
  - declare.h, [973](#)
  - sort.c, [2729](#)
- POLYREM
  - ftypes.h, [1441](#)
- polysize
  - POLYMOD, [358](#)
- polysortflag
  - N\_const, [255](#)
- POLYSUB
  - ftypes.h, [1441](#)
- PolyWise
  - sOrT, [399](#)
- polywrap.cc, [2275](#), [2286](#)
  - poly\_determine\_modulus, [2276](#)
  - poly\_div, [2277](#)
  - poly\_divmod, [2277](#)
  - poly\_factorize, [2281](#)
  - poly\_factorize\_argument, [2282](#)
  - poly\_factorize\_dollar, [2283](#)
  - poly\_factorize\_expression, [2283](#)
  - poly\_fix\_minus\_signs, [2281](#)
  - poly\_free\_poly\_vars, [2286](#)
  - poly\_gcd, [2276](#)
  - poly\_inverse, [2285](#)
  - poly\_mul, [2286](#)
  - poly\_ratfun\_add, [2279](#)
  - poly\_ratfun\_normalize, [2280](#)
  - poly\_ratfun\_read, [2277](#)
  - poly\_rem, [2277](#)
  - poly\_sort, [2278](#)
  - poly\_unfactorize\_expression, [2284](#)
- POLYWRAP\_DENOMPOWER\_INCREASE\_FACTOR,  
[2286](#)
- POLYWRAP\_DENOMPOWER\_INCREASE\_FACTOR
  - polywrap.cc, [2286](#)
- pop\_heap
  - poly, [355](#)
- POposition
  - FiLe, [129](#)
- POPPREASSIGNLEVEL
  - declare.h, [817](#)
- PopPreVars
  - declare.h, [905](#)
  - pre.c, [2318](#)
- PopVariables
  - declare.h, [861](#)
  - execute.c, [1157](#)
- portsignals.h, [2308](#), [2312](#)
  - FATAL\_SIG\_ERROR, [2309](#)
  - NSIG, [2309](#)

- SIGALRM, [2311](#)
- SIGEMT, [2310](#)
- SIGFPE, [2309](#)
- SIGHUP, [2311](#)
- SIGILL, [2310](#)
- SIGINT, [2311](#)
- SIGLOST, [2310](#)
- SIGPIPE, [2310](#)
- SIGPROF, [2311](#)
- SIGQUIT, [2311](#)
- SIGSEGV, [2309](#)
- SIGSYS, [2310](#)
- SIGTERM, [2310](#)
- SIGVTALRM, [2311](#)
- SIGXCPU, [2310](#)
- SIGXFSZ, [2310](#)
- POS0\_
  - ftypes.h, [1426](#)
- POS\_
  - ftypes.h, [1426](#)
- PoSiTiOn, [358](#)
  - p1, [359](#)
- Position
  - FiLeDaTa, [132](#)
- position
  - indexblock, [152](#)
  - InDeXeNtRy, [154](#)
  - nameblock, [259](#)
  - objects, [287](#)
  - SpecTatoR, [401](#)
  - StOrEcAcHe, [411](#)
- PositionStream
  - declare.h, [888](#)
  - tools.c, [3080](#)
- POSITIVEONLY
  - ftypes.h, [1444](#)
- POsize
  - FiLe, [130](#)
- POSNEG
  - ftypes.h, [1443](#)
- POstop
  - FiLe, [129](#)
- posWorkPointer
  - T\_const, [433](#)
- posWorkSize
  - R\_const, [373](#)
- posWorkSpace
  - T\_const, [429](#)
- PotModdollars
  - variable.h, [3205](#)
- PotModDoLlist
  - C\_const, [44](#)
- pow
  - COST, [76](#)
- power
  - factorized\_poly, [126](#)
- powerFlag
  - O\_const, [284](#)
- powmod
  - C\_const, [47](#)
- pParseObject
  - declare.h, [918](#)
  - pre.c, [2329](#)
- prCand
  - Assign, [17](#)
- pre.c, [2313](#), [2334](#)
  - AddToPreTypes, [2329](#)
  - CharOut, [2317](#)
  - ClearMacro, [2321](#)
  - ClearPushback, [2317](#)
  - ConstructName, [2333](#)
  - DAEMON, [2316](#)
  - defineChannel, [2331](#)
  - DoBreakDo, [2323](#)
  - DoCall, [2322](#)
  - DoClearOptimize, [2332](#)
  - DoClearUserFlag, [2334](#)
  - DoCommentChar, [2321](#)
  - DoContinueDo, [2323](#)
  - DoDebug, [2322](#)
  - DoDefine, [2321](#)
  - DoDo, [2323](#)
  - DoElse, [2323](#)
  - DoElseif, [2323](#)
  - DoEnddo, [2323](#)
  - DoEndif, [2324](#)
  - DoEndInside, [2325](#)
  - DoEndNamespace, [2333](#)
  - DoEndprocedure, [2324](#)
  - DoExternal, [2330](#)
  - DoFactDollar, [2331](#)
  - DoFromExternal, [2330](#)
  - Dof, [2324](#)
  - Dofdef, [2324](#)
  - Dofndef, [2324](#)
  - Dofydef, [2324](#)
  - DoInclude, [2322](#)
  - DoInside, [2324](#)
  - DoMessage, [2325](#)
  - DoNamespace, [2333](#)
  - DoOptimize, [2332](#)
  - DoPipe, [2325](#)
  - DoPrcExtension, [2325](#)
  - DoPreAddSeparator, [2329](#)
  - DoPreAppend, [2326](#)
  - DoPreAppendPath, [2332](#)
  - DoPreAssign, [2321](#)
  - DoPreBreak, [2326](#)
  - DoPreCase, [2327](#)
  - DoPreClose, [2326](#)
  - DoPreCreate, [2326](#)
  - DoPreDefault, [2327](#)
  - DoPreEndSwitch, [2327](#)
  - DoPreExchange, [2322](#)
  - DoPreOut, [2325](#)
  - DoPrePrependPath, [2332](#)

- DoPrePrintTimes, [2325](#)
- DoPreRemove, [2326](#)
- DoPreReset, [2332](#)
- DoPreRmSeparator, [2330](#)
- DoPreShow, [2327](#)
- DoPreSortReallocate, [2325](#)
- DoPreSwitch, [2327](#)
- DoPreWrite, [2326](#)
- DoProcedure, [2326](#)
- DoPrompt, [2330](#)
- DoRedefine, [2321](#)
- DoReverseInclude, [2322](#)
- DoRmExternal, [2330](#)
- DoSetExternal, [2330](#)
- DoSetExternalAttr, [2330](#)
- DoSetRandom, [2331](#)
- DoSetUserFlag, [2334](#)
- DoSkipExtraSymbols, [2332](#)
- DoSystem, [2327](#)
- DoTerminate, [2323](#)
- DoTimeOutAfter, [2333](#)
- DoToExternal, [2331](#)
- DoUndefine, [2322](#)
- DoUse, [2333](#)
- EvalPrelf, [2328](#)
- ExpandTripleDots, [2319](#)
- FALSE\_EXPR, [2317](#)
- FindInKeyWord, [2320](#)
- FindKeyWord, [2320](#)
- GetChar, [2317](#)
- GetDollarNumber, [2331](#)
- GetInput, [2317](#)
- GetPreVar, [2318](#)
- Include, [2322](#)
- IniModule, [2319](#)
- IniSpecialModule, [2319](#)
- KILL, [2316](#)
- KILLALL, [2316](#)
- LoadInstruction, [2319](#)
- LoadStatement, [2319](#)
- MessPreNesting, [2329](#)
- NOSHELL, [2317](#)
- PopPreVars, [2318](#)
- pParseObject, [2329](#)
- PreCalc, [2329](#)
- PreCmp, [2328](#)
- PreEq, [2328](#)
- PreEval, [2329](#)
- PrelfEval, [2328](#)
- PreLoad, [2327](#)
- PreProcessor, [2319](#)
- PreProInstruction, [2319](#)
- PreSkip, [2328](#)
- PutPreVar, [2318](#)
- SHELL, [2316](#)
- SKIPBUFSIZE, [2316](#)
- SkipName, [2333](#)
- StartPrepro, [2328](#)
- STDERR, [2316](#)
- STRINGIFY, [2315](#)
- STRINGIFY\_\_, [2315](#)
- TERMINAL, [2317](#)
- TheDefine, [2320](#)
- TheUndefine, [2321](#)
- TRUE\_EXPR, [2316](#)
- UnsetAllowDelay, [2318](#)
- UserFlags, [2334](#)
- writeToChannel, [2331](#)
- PreAssignFlag
  - P\_const, [313](#)
- PreAssignLevel
  - P\_const, [316](#)
- PreAssignStack
  - P\_const, [312](#)
- PreCalc
  - declare.h, [917](#)
  - pre.c, [2329](#)
- PRECALCSTREAM
  - ftypes.h, [1367](#)
- prECand
  - ECand, [95](#)
- PreCmp
  - declare.h, [917](#)
  - pre.c, [2328](#)
- PreContinuation
  - P\_const, [313](#)
- PreDebug
  - P\_const, [315](#)
- PreEq
  - declare.h, [917](#)
  - pre.c, [2328](#)
- preError
  - P\_const, [316](#)
- PreEval
  - declare.h, [917](#)
  - pre.c, [2329](#)
- preFill
  - P\_const, [312](#)
- PrelfDollarEval
  - declare.h, [978](#)
  - dollar.c, [1097](#)
- PrelfEval
  - declare.h, [918](#)
  - pre.c, [2328](#)
- PrelfLevel
  - DoLoOp, [92](#)
  - P\_const, [314](#)
- PrelfStack
  - P\_const, [312](#)
- PreInsideLevel
  - P\_const, [315](#)
- prElem
  - PNodeClass, [335](#)
- PRELOAD, [359](#)
  - buffer, [359](#)
  - size, [359](#)

- PreLoad
  - declare.h, [918](#)
  - pre.c, [2327](#)
- PRELOWERAFTER
  - ftypes.h, [1370](#)
- PRENOACTION
  - ftypes.h, [1370](#)
- PreOut
  - P\_const, [314](#)
- PrepPoly
  - declare.h, [861](#)
  - proces.c, [2433](#)
- PreProcessor
  - declare.h, [899](#)
  - pre.c, [2319](#)
- PreproFlag
  - P\_const, [313](#)
- PreProInstruction
  - declare.h, [905](#)
  - pre.c, [2319](#)
- PREPROONLY
  - ftypes.h, [1372](#)
- PRERAISEAFTER
  - ftypes.h, [1370](#)
- PreRandom
  - declare.h, [993](#)
  - reken.c, [2562](#)
- PREREADESTREAM
  - ftypes.h, [1366](#)
- PREREADESTREAM2
  - ftypes.h, [1367](#)
- PREREADESTREAM3
  - ftypes.h, [1367](#)
- PreSkip
  - declare.h, [918](#)
  - pre.c, [2328](#)
- preStart
  - P\_const, [312](#)
- preStop
  - P\_const, [312](#)
- PreSwitchLevel
  - DoLoOp, [93](#)
  - P\_const, [314](#)
- PreSwitchModes
  - P\_const, [313](#)
- PreSwitchStrings
  - P\_const, [312](#)
- PRETYPEDO
  - ftypes.h, [1374](#)
- PRETYPEIF
  - ftypes.h, [1374](#)
- PRETYPEINSIDE
  - ftypes.h, [1375](#)
- PRETYPENONE
  - ftypes.h, [1374](#)
- PRETYPEPROCEDURE
  - ftypes.h, [1374](#)
- PreTypes
  - P\_const, [313](#)
- PRETYPESWITCH
  - ftypes.h, [1374](#)
- PREV
  - declare.h, [809](#)
- PREVAR
  - structs.h, [2901](#)
- PreVar
  - variable.h, [3200](#)
- pReVaR, [360](#)
  - argnames, [360](#)
  - name, [360](#)
  - nargs, [360](#)
  - value, [360](#)
  - wildarg, [361](#)
- PreVarList
  - P\_const, [311](#)
- prevars
  - Stream, [419](#)
- PREVARSTREAM
  - ftypes.h, [1366](#)
- previous
  - NaMeSpAcE, [262](#)
  - Stream, [418](#)
- previousblock
  - indexblock, [152](#)
  - nameblock, [259](#)
- previousEfactor
  - T\_const, [431](#)
- previousNoShowInput
  - Stream, [419](#)
- prevline
  - Stream, [417](#)
- prFLines
  - EGraph, [107](#)
- primelist
  - T\_const, [431](#)
- PRIMENUMBER
  - ftypes.h, [1401](#)
- print
  - EEdge, [97](#)
  - EFLine, [101](#)
  - EGraph, [104](#)
  - ENode, [116](#)
  - EStack, [119](#)
  - flint::fmpz, [139](#)
  - flint::mpoly, [240](#)
  - flint::poly, [338](#)
  - Fraction, [140](#)
  - MConn, [199](#)
  - MGraph, [207](#)
  - MOrbits, [238](#)
  - NStack, [275](#)
  - Options, [298](#)
  - SGroup, [386](#)
  - T\_EGraph, [442](#)
- print\_instr
  - optimize.cc, [1989](#)

- print\_tree
  - optimize.cc, 1995
- printAdjMat
  - MGraph, 207
- PRINTALL
  - ftypes.h, 1425
- PrintBacktraceFlag
  - C\_const, 62
- PRINTCONTENT
  - ftypes.h, 1425
- PRINTCONTENTS
  - ftypes.h, 1424
- PrintDeprecation
  - declare.h, 864
  - startup.c, 2800
- prInteraction
  - Interaction, 161
- PrintExtraSymbol
  - declare.h, 1006
  - notation.c, 1940
- PRINTFBUF
  - parallel.c, 2063
- printflag
  - ExPrEsSiOn, 123
- PrintFloat
  - float.c, 1293
- printLevel
  - Options, 299
- PRINTLFILE
  - ftypes.h, 1425
- printMat
  - MNodeClass, 221
  - T\_MNodeClass, 454
- printModel
  - Options, 299
- PRINTOFF
  - ftypes.h, 1424
- PRINTON
  - ftypes.h, 1424
- PRINTONEFUNCTION
  - ftypes.h, 1425
- PRINTONETERM
  - ftypes.h, 1425
- PrintRunningTime
  - declare.h, 864
  - startup.c, 2800
- printscratch
  - proces.c, 2435
- PrintSeq
  - declare.h, 922
  - message.c, 1716
- printstats
  - OPTIMIZE, 294
- PrintSubTerm
  - declare.h, 922
  - message.c, 1716
- PrintSubtermList
  - declare.h, 1006
- notation.c, 1940
- PrintTerm
  - declare.h, 922
  - message.c, 1715
- PrintTermC
  - declare.h, 922
  - message.c, 1715
- PrintTime
  - declare.h, 980
  - sch.c, 2653
- PrintTotalSize
  - M\_const, 183
- PrintType
  - O\_const, 285
- PrintWords
  - declare.h, 922
  - message.c, 1716
- prModel
  - Model, 226
- prNCand
  - NCand, 267
- proc
  - Assign, 23
  - EGraph, 108
  - Options, 301
  - Output, 307
  - SProcess, 406
- PROCEDURE, 361
  - loadmode, 362
  - name, 362
  - p, 362
- procedureExtension
  - P\_const, 312
- procedureextension
  - setfile.c, 2693
- PROCEDUREFILE
  - ftypes.h, 1368
- Procedures
  - variable.h, 3199
- proces.c, 2423, 2435
  - Deferred, 2432
  - DONE, 2424
  - DoOnePow, 2431
  - FiniTerm, 2429
  - Generator, 2429
  - InFunction, 2426
  - InsertTerm, 2427
  - PasteFile, 2428
  - PasteTerm, 2428
  - PolyFunMul, 2434
  - PrepPoly, 2433
  - printscratch, 2435
  - Processor, 2425
  - TestSub, 2426
- Process, 362
  - ~Process, 364
  - agrcount, 365
  - astack, 366

- clist, [366](#)
- ctotal, [365](#)
- finalPart, [365](#)
- id, [365](#)
- initlPart, [365](#)
- loop, [366](#)
- maxnlegs, [366](#)
- mgrcount, [365](#)
- mkSProcess, [364](#)
- model, [364](#)
- nAGraphs, [368](#)
- nAOPI, [368](#)
- nExtern, [366](#)
- nfinal, [365](#)
- ngraphs, [367](#)
- ninitl, [365](#)
- nMGraphs, [367](#)
- nMOPI, [367](#)
- nopi, [367](#)
- nSubproc, [366](#)
- opt, [364](#)
- Process, [364](#)
- prProcess, [364](#)
- sec, [368](#)
- sproc, [366](#)
- sptbl, [366](#)
- wAGraphs, [368](#)
- wAOPI, [368](#)
- wgraphs, [367](#)
- wMGraphs, [367](#)
- wMOPI, [367](#)
- wopi, [367](#)
- ProcessBucketSize
  - C\_const, [53](#)
- ProcessOption
  - declare.h, [888](#)
  - setfile.c, [2690](#)
- Processor
  - declare.h, [862](#)
  - proces.c, [2425](#)
- ProcessStats
  - C\_const, [59](#)
- proclid
  - Output, [308](#)
- ProcList
  - P\_const, [311](#)
- Product
  - declare.h, [863](#)
  - reken.c, [2557](#)
- PRODUCTIONDATE
  - form3.h, [1326](#)
- proexp
  - T\_const, [437](#)
- PROPERORDERFLAG
  - ftypes.h, [1440](#)
- properorderflag
  - C\_const, [60](#)
- ProtoType
  - C\_const, [48](#)
- prototype
  - ExPrEsSiOn, [121](#)
  - TaBlEs, [460](#)
- prototypeSize
  - TaBlEs, [463](#)
- prParticle
  - Particle, [324](#)
- prParticleArray
  - Model, [228](#)
- prPNodeClass
  - PNodeClass, [335](#)
- prProcess
  - Process, [364](#)
- prSProcess
  - SProcess, [404](#)
- prStack
  - AStack, [27](#)
- PrtLong
  - declare.h, [863](#)
  - reken.c, [2558](#)
- PrtTerms
  - declare.h, [863](#)
  - sch.c, [2653](#)
- PruneExtraSymbols
  - declare.h, [1008](#)
  - notation.c, [1940](#)
- pSize
  - P\_const, [313](#)
- psize
  - PaRtl, [320](#)
- psq
  - SGroup, [389](#)
- psr
  - SGroup, [389](#)
- pStop
  - sOrT, [396](#)
- ptcl
  - AEdge, [8](#)
  - EEdge, [98](#)
  - NCInput, [268](#)
- pType
  - Particle, [325](#)
- ptypec
  - PInput, [333](#)
- ptypen
  - PInput, [333](#)
- push\_heap
  - poly, [355](#)
- pushEdge
  - AStack, [28](#)
  - MConn, [198](#)
- pushNode
  - AStack, [28](#)
  - MConn, [198](#)
- PUSHPREASSIGNLEVEL
  - declare.h, [817](#)
- PutArgInScratch



- declare.h, [1002](#)
- transform.c, [3149](#)
- PutBracket
  - declare.h, [864](#)
  - execute.c, [1158](#)
- PutBracketInIndex
  - declare.h, [980](#)
  - index.c, [1672](#)
- PutExtraSymbols
  - declare.h, [1010](#)
  - ratio.c, [2503](#)
- PUTFIRST
  - ftypes.h, [1402](#)
- PutIn
  - declare.h, [864](#)
  - sort.c, [2718](#)
- PutInBuffers
  - pattern.c, [2172](#)
- PutInside
  - declare.h, [859](#)
  - index.c, [1672](#)
- putinsize
  - sOrT, [398](#)
- PutInSpectator
  - declare.h, [1024](#)
  - spectator.c, [2787](#)
- PutInStore
  - declare.h, [865](#)
  - store.c, [2829](#)
- PutInVflags
  - declare.h, [877](#)
  - execute.c, [1158](#)
- PutOut
  - declare.h, [865](#)
  - sort.c, [2719](#)
- PutPreVar
  - declare.h, [886](#)
  - pre.c, [2318](#)
- PutTableNames
  - declare.h, [987](#)
  - minos.c, [1733](#)
- PutTermInDollar
  - declare.h, [963](#)
  - dollar.c, [1094](#)
- PUTZERO
  - declare.h, [812](#)
- PVFUNV
  - ftypes.h, [1456](#)
- PVFUNWP
  - ftypes.h, [1456](#)
- pWorkPointer
  - T\_const, [433](#)
- pWorkSize
  - R\_const, [373](#)
- pWorkSpace
  - T\_const, [429](#)
- Q\_
  - ftypes.h, [1427](#)
- qError
  - M\_const, [188](#)
- QUADRUPLEFORTRANMODE
  - ftypes.h, [1386](#)
- QUADRUPLEPRECISIONFLAG
  - ftypes.h, [1386](#)
- Quotient
  - declare.h, [866](#)
  - reken.c, [2557](#)
- R
  - AllGlobals, [10](#)
- r
  - node, [273](#)
- R\_const, [369](#)
  - BracketOn, [374](#)
  - Cnumlhs, [373](#)
  - CompareRoutine, [372](#)
  - CompressBuffer, [371](#)
  - CompressPointer, [372](#)
  - ComprTop, [372](#)
  - CurDum, [374](#)
  - CurExpr, [377](#)
  - DeferFlag, [375](#)
  - DefPosition, [371](#)
  - Esides, [376](#)
  - expchanged, [376](#)
  - expflags, [376](#)
  - FoStage4, [371](#)
  - Fscr, [371](#)
  - funoffset, [377](#)
  - GetFile, [374](#)
  - GetOneFile, [375](#)
  - gzipCompress, [373](#)
  - hidefile, [371](#)
  - infile, [371](#)
  - InHiBuf, [372](#)
  - InInBuf, [372](#)
  - KeptInHold, [374](#)
  - level, [376](#)
  - IWorkSize, [373](#)
  - MaxBracket, [374](#)
  - MaxDum, [376](#)
  - modeloptions, [377](#)
  - moebiustable, [372](#)
  - moebiustablesizes, [377](#)
  - NoCompress, [373](#)
  - OldTime, [372](#)
  - outfile, [371](#)
  - outtohide, [373](#)
  - PolyFun, [375](#)
  - PolyFunExp, [376](#)
  - PolyFunInv, [375](#)
  - PolyFunPow, [376](#)
  - PolyFunType, [375](#)
  - PolyFunVar, [376](#)
  - posWorkSize, [373](#)
  - pWorkSize, [373](#)
  - ShortSortCount, [377](#)

- sLevel, 375
- SortType, 377
- Stage4Name, 375
- StoreData, 371
- TePos, 375
- wranfcall, 374
- wranfia, 372
- wranfnpair1, 374
- wranfnpair2, 374
- wranfseed, 373
- R\_COPY\_B
  - checkpoint.c, 539
- R\_COPY\_LIST
  - checkpoint.c, 540
- R\_COPY\_NAMETREE
  - checkpoint.c, 541
- R\_COPY\_S
  - checkpoint.c, 540
- R\_FREE
  - checkpoint.c, 538
- R\_FREE\_NAMETREE
  - checkpoint.c, 539
- R\_FREE\_STREAM
  - checkpoint.c, 539
- R\_SET
  - checkpoint.c, 539
- RaisPow
  - declare.h, 866
  - reken.c, 2555
- RaisPowCached
  - declare.h, 866
  - reken.c, 2555
- RaisPowMod
  - declare.h, 867
  - reken.c, 2556
- RANDOMFUNCTION
  - ftypes.h, 1400
- ranges
  - DICTIONARY, 83
- RANPERM
  - ftypes.h, 1401
- ratfun\_add\_mpoly
  - flintinterface.h, 1273
- ratfun\_add\_poly
  - flintinterface.h, 1274
- ratfun\_normalize\_mpoly
  - flintinterface.h, 1274
- ratfun\_normalize\_poly
  - flintinterface.h, 1274
- ratfun\_read\_mpoly
  - flintinterface.h, 1274
- ratfun\_read\_poly
  - flintinterface.h, 1274
- RATIO
  - ftypes.h, 1416
- ratio
  - Fraction, 142
- ratio.c, 2500, 2507
- BinomGen, 2501
- ConvertArgument, 2506
- CreateExpression, 2504
- DIVfunction, 2506
- divrem, 2507
- DoSumF1, 2502
- DoSumF2, 2502
- ExpandRat, 2506
- FromPolyRatFun, 2505
- GCDclean, 2505
- GCDfunction, 2502
- GCDfunction3, 2502
- GCDterms, 2504
- Glue, 2502
- InvPoly, 2506
- MergeDotproductLists, 2504
- MergeSymbolLists, 2504
- MULfunc, 2506
- MultiplyWithTerm, 2503
- PolyDiv, 2505
- PutExtraSymbols, 2503
- RatioFind, 2501
- RatioGen, 2501
- ReadPolyRatFun, 2504
- TakeContent, 2503
- TakeExtraSymbols, 2503
- TakeSymbolContent, 2505
- TheErrorMessage, 2507
- RatioFind
  - declare.h, 868
  - ratio.c, 2501
- RatioGen
  - declare.h, 868
  - ratio.c, 2501
- RATIONALMODE
  - ftypes.h, 1430
- RatToFloat
  - float.c, 1292
- RatToLine
  - declare.h, 868
  - sch.c, 2652
- RBRACE
  - ftypes.h, 1381
- rbufnum
  - M\_const, 179
- rbufs
  - ParallelVars, 318
- RCYCLESYMMETRIC
  - ftypes.h, 1432
- ReadFile
  - declare.h, 896
  - tools.c, 3082
- ReadFloat
  - float.c, 1292
- ReadFromScratch
  - declare.h, 1025
  - store.c, 2829
- ReadFromStream

- declare.h, [887](#)
- tools.c, [3079](#)
- ReadObjObject
  - declare.h, [986](#)
  - minos.c, [1734](#)
- ReadIndex
  - declare.h, [984](#)
  - minos.c, [1730](#)
- ReadIniInfo
  - declare.h, [985](#)
  - minos.c, [1731](#)
- ReadObject
  - declare.h, [986](#)
  - minos.c, [1733](#)
- ReadParticle
  - declare.h, [1028](#)
- ReadPolyRatFun
  - declare.h, [1008](#)
  - ratio.c, [2504](#)
- readpos
  - SpecTatoR, [401](#)
- ReadPosFile
  - declare.h, [896](#)
  - tools.c, [3082](#)
- ReadRange
  - declare.h, [1002](#)
  - transform.c, [3149](#)
- ReadSaveExpression
  - declare.h, [994](#)
  - store.c, [2836](#)
- ReadSaveHeader
  - declare.h, [993](#)
  - store.c, [2833](#)
- ReadSaveIndex
  - declare.h, [993](#)
  - store.c, [2834](#)
- ReadSaveTerm32
  - declare.h, [995](#)
  - store.c, [2835](#)
- ReadSaveVariables
  - declare.h, [996](#)
  - store.c, [2834](#)
- ReadSnum
  - declare.h, [869](#)
  - tools.c, [3087](#)
- READSPECTATORFLAG
  - ftypes.h, [1454](#)
- RecalcSetups
  - declare.h, [903](#)
  - setfile.c, [2690](#)
- RecFlag
  - T\_const, [438](#)
- RecoveryFilename
  - checkpoint.c, [542](#)
  - declare.h, [998](#)
- recvBuffer
  - parallel.c, [2066](#)
- recycle\_variables
  - optimize.cc, [2001](#)
- REDEFINEDEXPRESSION
  - ftypes.h, [1425](#)
- REDLENG
  - declare.h, [800](#)
- RedoTableTree
  - declare.h, [972](#)
  - tables.c, [2961](#)
- RedoTree
  - comtool.c, [767](#)
  - declare.h, [961](#)
- REDUCEMODE
  - ftypes.h, [1385](#)
- REDUCESUBEXPBUFFERS
  - compiler.c, [724](#)
- refineClass
  - MGraph, [208](#)
- REGULAR
  - ftypes.h, [1441](#)
- REGULAREXPRESSION
  - ftypes.h, [1425](#)
- REGULARSYMBOL
  - ftypes.h, [1449](#)
- reken.c, [2549](#), [2562](#)
  - AccumGCD, [2553](#)
  - AddLong, [2554](#)
  - AddPLon, [2554](#)
  - AddRat, [2552](#)
  - Bernoulli, [2561](#)
  - BigLong, [2554](#)
  - CompCoef, [2559](#)
  - COPYLONG, [2551](#)
  - DivLong, [2555](#)
  - DivRat, [2553](#)
  - Divvy, [2552](#)
  - Factorial, [2560](#)
  - GcdLong, [2558](#)
  - GCDMAX, [2551](#)
  - GetBinom, [2558](#)
  - GetLong, [2558](#)
  - GetLongModInverses, [2557](#)
  - GetModInverses, [2556](#)
  - iniwranf, [2561](#)
  - iranf, [2562](#)
  - LcmLong, [2559](#)
  - MakeInverses, [2556](#)
  - MakeLongRational, [2559](#)
  - MakeModTable, [2560](#)
  - MakeRational, [2559](#)
  - Modulus, [2560](#)
  - Moebius, [2561](#)
  - MulLong, [2554](#)
  - Mully, [2552](#)
  - MulRat, [2553](#)
  - NEWTRICK, [2551](#)
  - NextPrime, [2561](#)
  - NormalModulus, [2556](#)
  - Pack, [2552](#)

- PreRandom, [2562](#)
- Product, [2557](#)
- PrtLong, [2558](#)
- Quotient, [2557](#)
- RaisPow, [2555](#)
- RaisPowCached, [2555](#)
- RaisPowMod, [2556](#)
- Remain10, [2557](#)
- Remain4, [2558](#)
- Simplify, [2553](#)
- SubPLon, [2554](#)
- TakeLongRoot, [2559](#)
- TakeModulus, [2560](#)
- TakeNormalModulus, [2560](#)
- TakeRatRoot, [2553](#)
- UnPack, [2552](#)
- WARMUP, [2551](#)
- wranf, [2561](#)
- ReleaseTB
  - declare.h, [991](#)
  - tables.c, [2965](#)
- Remain10
  - declare.h, [869](#)
  - reken.c, [2557](#)
- Remain4
  - declare.h, [869](#)
  - reken.c, [2558](#)
- REMFUNCTION
  - ftypes.h, [1397](#)
- RemoveBridges
  - lus.c, [1689](#)
- RemoveDictionary
  - declare.h, [1021](#)
  - dict.c, [1077](#)
- RemoveDollars
  - declare.h, [983](#)
  - names.c, [1833](#)
- renum
  - ExPrEsSiOn, [121](#)
- RENUMBER
  - structs.h, [2900](#)
- ReNuMbEr, [378](#)
  - func, [379](#)
  - funnum, [380](#)
  - indi, [379](#)
  - indnum, [379](#)
  - startposition, [378](#)
  - symb, [378](#)
  - symnum, [379](#)
  - vecnum, [380](#)
  - vect, [379](#)
- ReNumber
  - declare.h, [869](#)
  - reshuf.c, [2609](#)
- ReNumberVec
  - O\_const, [281](#)
- renumlists
  - ExPrEsSiOn, [122](#)
- RenumScratch
  - N\_const, [250](#)
- ReOpenFile
  - declare.h, [895](#)
  - tools.c, [3081](#)
- reorder
  - MNodeClass, [222](#)
- reordLeg
  - Assign, [19](#)
- RepCount
  - T\_const, [430](#)
- RepFunList
  - N\_const, [246](#)
- RepFunNum
  - N\_const, [256](#)
- replace
  - ExPrEsSiOn, [123](#)
- REPLACEARG
  - ftypes.h, [1445](#)
- ReplaceDollar
  - declare.h, [892](#)
  - names.c, [1832](#)
- REPLACELOOP
  - ftypes.h, [1440](#)
- REPLACEMENT
  - ftypes.h, [1393](#)
- ReplaceScrat
  - N\_const, [249](#)
- RepLevel
  - C\_const, [64](#)
- RepMax
  - M\_const, [187](#)
- RepPoint
  - N\_const, [246](#)
- RepSumCheck
  - C\_const, [64](#)
- RepTop
  - T\_const, [430](#)
- request
  - PF\_BUFFER, [330](#)
- reserved
  - STOREHEADER, [415](#)
  - StreaM, [420](#)
  - TabIEs, [462](#)
- ReserveTempFiles
  - declare.h, [921](#)
  - startup.c, [2799](#)
- ResetScratch
  - declare.h, [869](#)
  - store.c, [2828](#)
- resetTimeOnClear
  - M\_const, [183](#)
- ResetVariables
  - declare.h, [924](#)
  - names.c, [1833](#)
- reshuf.c, [2608](#), [2613](#)
  - AdjustRenumScratch, [2609](#)
  - CountDo, [2610](#)

- CountFun, [2610](#)
- DetCurDum, [2609](#)
- DimensionExpression, [2610](#)
- DimensionSubterm, [2610](#)
- DimensionTerm, [2610](#)
- DoDelta3, [2611](#)
- DoDistrib, [2611](#)
- DoPartitions, [2611](#)
- DoPermutations, [2612](#)
- DoShuffle, [2612](#)
- DoStuff, [2612](#)
- EqualArg, [2611](#)
- FinishShuffle, [2612](#)
- FinishStuff, [2613](#)
- FullRenum, [2609](#)
- FunLevel, [2609](#)
- MoveDummies, [2609](#)
- MultDo, [2610](#)
- NEWCODE, [2609](#)
- ReNumber, [2609](#)
- Shuffle, [2612](#)
- Stuff, [2612](#)
- StuffRootAdd, [2613](#)
- TestPartitions, [2611](#)
- TryDo, [2611](#)
- ResizeData
  - O\_const, [280](#)
- resizeFlag
  - O\_const, [284](#)
- ResizeLONG
  - O\_const, [281](#)
- ResizeNCWORD
  - O\_const, [280](#)
- ResizePOINTER
  - O\_const, [281](#)
- ResizePOS
  - O\_const, [281](#)
- ResizeWORD
  - O\_const, [280](#)
- ResolveSet
  - declare.h, [869](#)
  - wildcard.c, [3223](#)
- restore
  - AStack, [27](#)
- restoreCouple
  - Assign, [23](#)
- resultAGraph
  - SProcess, [406](#)
- resultMGraph
  - SProcess, [405](#)
- retstat
  - PF\_BUFFER, [330](#)
- REVERSEARG
  - ftypes.h, [1446](#)
- REVERSEFILESTREAM
  - ftypes.h, [1367](#)
- REVERSEFUNCTION
  - ftypes.h, [1393](#)
- REVERSEORDER
  - ftypes.h, [1432](#)
- ReverseStatements
  - declare.h, [888](#)
  - tools.c, [3080](#)
- RevertScratch
  - declare.h, [870](#)
  - store.c, [2828](#)
- revision
  - STOREHEADER, [415](#)
- ReWorkT
  - declare.h, [839](#)
  - tables.c, [2964](#)
- right
  - NoDe, [270](#)
- rhs
  - CbUf, [72](#)
  - DICTIONARY\_ELEMENT, [84](#)
- RHSIDE
  - ftypes.h, [1385](#)
- rhsInParallel
  - ParallelVars, [318](#)
- right
  - NaMeNode, [261](#)
  - tree, [479](#)
- RLONG
  - form3.h, [1335](#)
- rloser
  - NoDe, [271](#)
- ROOTFUNCTION
  - ftypes.h, [1396](#)
- rootnum
  - CbUf, [74](#)
  - TaBIEs, [464](#)
- RPARENTHESIS
  - ftypes.h, [1381](#)
- rsrc
  - NoDe, [271](#)
- RSsize
  - N\_const, [254](#)
- RunAddArg
  - declare.h, [1003](#)
  - transform.c, [3148](#)
- RunCycle
  - declare.h, [1003](#)
  - transform.c, [3147](#)
- RunDecode
  - declare.h, [1001](#)
  - transform.c, [3146](#)
- RunDedup
  - declare.h, [1004](#)
  - transform.c, [3147](#)
- RunDropArg
  - declare.h, [1004](#)
  - transform.c, [3148](#)
- RunEncode
  - declare.h, [1001](#)
  - transform.c, [3146](#)

- RunExplode
  - declare.h, [1002](#)
  - transform.c, [3147](#)
- RunHtoZArg
  - declare.h, [1004](#)
  - transform.c, [3149](#)
- RunImplode
  - declare.h, [1001](#)
  - transform.c, [3147](#)
- RunIsLyndon
  - declare.h, [1003](#)
  - transform.c, [3148](#)
- RunMulArg
  - declare.h, [1003](#)
  - transform.c, [3148](#)
- RunPermute
  - declare.h, [1002](#)
  - transform.c, [3147](#)
- RunReplace
  - declare.h, [1001](#)
  - transform.c, [3146](#)
- RunReverse
  - declare.h, [1003](#)
  - transform.c, [3147](#)
- RunSelectArg
  - declare.h, [1004](#)
  - transform.c, [3148](#)
- RunToLyndon
  - declare.h, [1003](#)
  - transform.c, [3148](#)
- RunTransform
  - declare.h, [1001](#)
  - transform.c, [3146](#)
- RunZtoHArg
  - declare.h, [1004](#)
  - transform.c, [3149](#)
- RWLOCKR
  - declare.h, [819](#)
- RWLOCKW
  - declare.h, [819](#)
- rwmode
  - dbase, [79](#)
- S
  - AllGlobals, [10](#)
- S0
  - M\_const, [172](#)
  - T\_const, [428](#)
- S\_const, [380](#)
  - Balancing, [382](#)
  - CollectOverFlag, [382](#)
  - ExecMode, [382](#)
  - MaxExprSize, [381](#)
  - MultiThreaded, [382](#)
  - NumOldNumFactors, [382](#)
  - NumOldOnFile, [382](#)
  - OldNumFactors, [381](#)
  - OldOnFile, [381](#)
  - Olduflags, [382](#)
  - Oldvflags, [381](#)
- S\_FLUSH\_B
  - checkpoint.c, [540](#)
- s\_one
  - FixedGlobals, [136](#)
- S\_WRITE\_B
  - checkpoint.c, [539](#)
- S\_WRITE\_DOLLAR
  - checkpoint.c, [541](#)
- S\_WRITE\_LIST
  - checkpoint.c, [540](#)
- S\_WRITE\_NAMETREE
  - checkpoint.c, [541](#)
- S\_WRITE\_S
  - checkpoint.c, [540](#)
- salter
  - OPTIMIZE, [294](#)
- saveCouple
  - Assign, [22](#)
- SaveData
  - O\_const, [278](#)
- SaveHeader
  - O\_const, [278](#)
- SAVEREVISION
  - store.c, [2827](#)
- sBer
  - T\_const, [433](#)
- sbuf
  - ParallelVars, [318](#)
- sBuffer
  - sOrT, [394](#)
- sbufnum
  - M\_const, [179](#)
- SBYTE
  - form3.h, [1335](#)
- ScanFunctions
  - declare.h, [870](#)
  - function.c, [1472](#)
- sch.c, [2649](#), [2655](#)
  - AddArrayIndex, [2653](#)
  - AddToLine, [2651](#)
  - CodeToLine, [2652](#)
  - FinilLine, [2651](#)
  - IniLine, [2651](#)
  - LongToLine, [2652](#)
  - MultiplyToLine, [2652](#)
  - PrintTime, [2653](#)
  - PrtTerms, [2653](#)
  - RatToLine, [2652](#)
  - StrCopy, [2651](#)
  - TalToLine, [2652](#)
  - TokenToLine, [2652](#)
  - va\_arg, [2651](#)
  - va\_dcl, [2650](#)
  - va\_end, [2650](#)
  - va\_list, [2651](#)
  - va\_start, [2650](#)
  - WriteAll, [2654](#)

- WriteArgument, [2653](#)
- WriteDictionary, [2653](#)
- WriteExpression, [2654](#)
- WriteInnerTerm, [2654](#)
- WriteLists, [2653](#)
- WriteOne, [2654](#)
- WriteSubTerm, [2654](#)
- WriteTerm, [2654](#)
- WrtPower, [2653](#)
- schemeFlags
  - OPTIMIZE, [294](#)
- schemenum
  - O\_const, [284](#)
- ScratSize
  - M\_const, [174](#)
- SEARCHINGPRECASE
  - ftypes.h, [1372](#)
- SEARCHINGPREENDSWITCH
  - ftypes.h, [1372](#)
- sec
  - Process, [368](#)
- sEdges
  - EGraph, [111](#)
- sedges
  - MConn, [200](#)
- SeekFile
  - declare.h, [897](#)
  - tools.c, [3082](#)
- SeekScratch
  - declare.h, [870](#)
  - store.c, [2828](#)
- SELECTARG
  - ftypes.h, [1447](#)
- selectExternalChannel
  - declare.h, [990](#)
  - extcmd.c, [1193](#)
- selectLeg
  - Assign, [18](#)
- selecttermundo
  - N\_const, [249](#)
- selectVertex
  - Assign, [18](#)
- selectVertexSimp
  - Assign, [18](#)
- selfloop
  - MGraph, [214](#)
- selOPI
  - T\_MGraph, [448](#)
- selUnAssLeg
  - Assign, [20](#)
- selUnAssVertex
  - Assign, [20](#)
- selUnAssVertexSimp
  - Assign, [20](#)
- SEPARATESYMBOL
  - ftypes.h, [1405](#)
- SEPARATOR
  - fwin.h, [1496](#)
- unix.h, [3193](#)
- separators
  - C\_const, [42](#)
- set\_del
  - declare.h, [989](#)
  - tools.c, [3088](#)
- set\_in
  - declare.h, [988](#)
  - tools.c, [3087](#)
- set\_of\_char
  - structs.h, [2902](#)
- set\_set
  - declare.h, [988](#)
  - tools.c, [3088](#)
- set\_sub
  - declare.h, [989](#)
  - tools.c, [3088](#)
- setAGraph
  - AStack, [27](#)
- SETBASELENGTH
  - declare.h, [810](#)
- SETBASEPOSITION
  - declare.h, [810](#)
- SETBIT
  - ftypes.h, [1438](#)
- SETBUFSIZE
  - setfile.c, [2690](#)
- setDefaultValue
  - Options, [298](#)
- SetDictionaryOptions
  - declare.h, [1021](#)
  - dict.c, [1077](#)
- SetElementList
  - C\_const, [43](#)
- SetElements
  - variable.h, [3200](#)
- setEMom
  - EEdge, [97](#)
- SetEndHScratch
  - declare.h, [870](#)
  - store.c, [2828](#)
- setEndMG
  - Options, [299](#)
- SetEndScratch
  - declare.h, [870](#)
  - store.c, [2828](#)
- setErExit
  - Options, [299](#)
- SETERROR
  - declare.h, [810](#)
- SETEXP
  - ftypes.h, [1390](#)
- SetExpr
  - compcomm.c, [615](#)
  - declare.h, [928](#)
- SetExprCases
  - compcomm.c, [615](#)
  - declare.h, [975](#)

- setExtern
  - EGraph, 105
  - ENode, 116
  - T\_EGraph, 443
- setexterntopo
  - T\_const, 439
- setExtLoop
  - EGraph, 105
- setfile.c, 2688, 2693
  - AllocFileHandle, 2691
  - AllocSetups, 2691
  - AllocSort, 2691
  - AllocSortFileName, 2691
  - commentchar, 2692
  - curdirp, 2692
  - cursortdirp, 2692
  - DeAllocFileHandle, 2691
  - DEFINEVALUE, 2690
  - DoSetups, 2690
  - dotchar, 2692
  - GetSetupPar, 2690
  - highfirst, 2693
  - lowfirst, 2693
  - MakeSetupAllocs, 2692
  - NUMERICALVALUE, 2689
  - ONOFFVALUE, 2690
  - PATHVALUE, 2689
  - procedureextension, 2693
  - ProcessOption, 2690
  - RecalcSetups, 2690
  - SETBUFSIZE, 2690
  - setupparameters, 2693
  - STRINGVALUE, 2689
  - TryEnvironment, 2692
  - TryFileSetups, 2692
  - WriteSetup, 2691
- SetFileIndex
  - declare.h, 870
  - store.c, 2831
- SetFloatPrecision
  - float.c, 1293
- SETFUNCTION
  - ftypes.h, 1392
- setId
  - EEdge, 97
  - ENode, 116
- setinterntopo
  - T\_const, 439
- SETKILLMODEFOREXTERNALCHANNEL
  - declare.h, 821
- SetList
  - C\_const, 43
- setLMom
  - EEdge, 97
- setmod
  - poly, 346
- SetModes
  - declare.h, 884
- execute.c, 1158
- SETNUMBER
  - ftypes.h, 1418
- SETONLY
  - ftypes.h, 1377
- setonoff
  - compcomm.c, 612
  - declare.h, 928
- setOutAG
  - Options, 299
- setOutgrf
  - Output, 305
- setOutgrp
  - Output, 305
- setOutMG
  - Options, 299
- setOutputF
  - Options, 298
- setOutputP
  - Options, 298
- SetPosFile
  - declare.h, 897
  - tools.c, 3083
- SeTs, 383
  - dimension, 384
  - first, 383
  - flags, 384
  - last, 383
  - name, 383
  - namesize, 384
  - node, 384
  - type, 383
- Sets
  - variable.h, 3201
- SetScratch
  - declare.h, 885
  - store.c, 2828
- SETSET
  - ftypes.h, 1390
- SetSpecialMode
  - declare.h, 919
  - module.c, 1763
- SETSTARTPOS
  - declare.h, 812
- SETTERMINATORFOREXTERNALCHANNEL
  - declare.h, 820
- SETTONUM
  - ftypes.h, 1418
- setType
  - EEdge, 97
  - ENode, 116
- SetupDir
  - M\_const, 173
- SETUPFILE
  - ftypes.h, 1368
- SetupFile
  - M\_const, 173
- setupfilename



- inivar.h, [1682](#)
- variable.h, [3207](#)
- SetupFlag
  - C\_const, [58](#)
- SetupMPFTables
  - float.c, [1293](#)
- SetupMZVTables
  - float.c, [1293](#)
- SETUPPARAMETERS, [384](#)
  - flags, [385](#)
  - parameter, [385](#)
  - type, [385](#)
  - value, [385](#)
- setupparameters
  - setfile.c, [2693](#)
- setValue
  - Fraction, [140](#)
  - Options, [298](#)
- sfact
  - T\_const, [433](#)
- SFHSIZE
  - fsizes.h, [1348](#)
- sFill
  - sOrT, [395](#)
- Sflush
  - declare.h, [871](#)
  - sort.c, [2719](#)
- sFun
  - STOREHEADER, [414](#)
- SGN
  - declare.h, [800](#)
- sgn
  - TrAcEn, [472](#)
  - TrAcEs, [476](#)
- SGroup, [385](#)
  - ~SGroup, [386](#)
  - addGroup, [387](#)
  - cgen, [388](#)
  - clearGroup, [387](#)
  - csav, [388](#)
  - delGroup, [387](#)
  - eclass, [388](#)
  - elem, [388](#)
  - genNext, [387](#)
  - neclass, [388](#)
  - nElem, [387](#)
  - nelem, [388](#)
  - newGroup, [386](#)
  - nextElem, [387](#)
  - nnodes, [388](#)
  - permg, [389](#)
  - perms, [389](#)
  - pgq, [389](#)
  - pgr, [389](#)
  - print, [386](#)
  - psq, [389](#)
  - psr, [389](#)
  - SGroup, [386](#)
  - size, [388](#)
- sHalf
  - sOrT, [395](#)
- SHcombi
  - N\_const, [251](#)
- SHcombisize
  - N\_const, [252](#)
- SHELL
  - pre.c, [2316](#)
- shellname
  - X\_const, [487](#)
- SHMWINSIZE
  - fsizes.h, [1348](#)
- shmWinSize
  - M\_const, [176](#)
- ShortSortCount
  - R\_const, [377](#)
- ShortStats
  - C\_const, [54](#)
- ShortStatsMax
  - C\_const, [69](#)
- ShrinkDictionary
  - declare.h, [1022](#)
  - dict.c, [1077](#)
- Shuffle
  - declare.h, [837](#)
  - reshuf.c, [2612](#)
- SHvar
  - N\_const, [251](#)
- SHvariables, [390](#)
  - combilast, [391](#)
  - do\_uffle, [391](#)
  - finishuf, [391](#)
  - incoef, [390](#)
  - level, [391](#)
  - nincoef, [391](#)
  - option, [392](#)
  - outfun, [390](#)
  - outterm, [390](#)
  - stop1, [390](#)
  - stop2, [390](#)
  - ststop1, [391](#)
  - ststop2, [391](#)
  - thefunction, [391](#)
- sld
  - EGraph, [109](#)
- SIGALRM
  - portsignals.h, [2311](#)
- SIGEMT
  - portsignals.h, [2310](#)
- SIGFPE
  - portsignals.h, [2309](#)
- SIGFUNCTION
  - ftypes.h, [1395](#)
- SIGHUP
  - portsignals.h, [2311](#)
- SIGILL
  - portsignals.h, [2310](#)

- SIGINT
  - portsignals.h, 2311
- SIGLOST
  - portsignals.h, 2310
- sign
  - DiStRiBuTe, 85
  - node, 273
  - PeRmUtE, 327
  - PeRmUtEp, 328
  - poly, 345
- sign1
  - TrAcEs, 477
- sign2
  - TrAcEs, 477
- SignCheck
  - N\_const, 258
- SIGNFUN
  - ftypes.h, 1394
- SIGPIPE
  - portsignals.h, 2310
- SIGPROF
  - portsignals.h, 2311
- SIGQUIT
  - portsignals.h, 2311
- SIGSEGV
  - portsignals.h, 2309
- SIGSYS
  - portsignals.h, 2310
- SIGTERM
  - portsignals.h, 2310
- SIGVTALRM
  - portsignals.h, 2311
- SIGXCPU
  - portsignals.h, 2310
- SIGXFSZ
  - portsignals.h, 2310
- silent
  - M\_const, 188
- simp1token
  - declare.h, 958
  - token.c, 3048
- simp2token
  - declare.h, 958
  - token.c, 3049
- simp3atoken
  - declare.h, 958
  - token.c, 3049
- simp3btoken
  - declare.h, 959
  - token.c, 3049
- simp4token
  - declare.h, 959
  - token.c, 3049
- simp5token
  - declare.h, 959
  - token.c, 3049
- simp6token
  - declare.h, 959
- token.c, 3049
- simple\_poly
  - poly, 346, 347
- SimpleDelta
  - float.c, 1289
- SimpleDeltaC
  - float.c, 1289
- SimpleSplitMerge
  - declare.h, 974
  - sort.c, 2729
- SimpleSplitMergeRec
  - declare.h, 974
  - sort.c, 2729
- Simplify
  - declare.h, 871
  - reken.c, 2553
- simplify\_fmpz
  - flintinterface.h, 1276
- simplify\_fmpz\_poly
  - flintinterface.h, 1276
- simpwtoken
  - declare.h, 958
  - token.c, 3048
- simulated\_annealing
  - optimize.cc, 1997
- sInd
  - STOREHEADER, 414
- SINFUNCTION
  - ftypes.h, 1398
- SingleTable
  - float.c, 1289
- SINHFUNCTION
  - ftypes.h, 1399
- SIOsize
  - M\_const, 175
- size
  - ARGBUFFER, 14
  - DICTIONARY\_ELEMENT, 84
  - DoLIaRS, 88
  - ExPrEsSiOn, 121
  - FaCdOILaR, 125
  - InDeXeNtRy, 155
  - INSIDEINFO, 158
  - LIST, 166
  - MiNmAx, 216
  - objects, 287
  - PRELOAD, 359
  - SGroup, 388
  - TaBlEbAsEsUbInDeX, 459
- size\_of\_form\_notation
  - poly, 348
- size\_of\_form\_notation\_with\_den
  - poly, 348
- size\_of\_terms
  - poly, 356
- SizeCommutelnSet
  - C\_const, 63
- sizecouplings

- MoDeL, [234](#)
- SizeDictionaries
  - O\_const, [283](#)
- sizeelements
  - DICTIONARY, [82](#)
- SIZEFACS
  - fsizes.h, [1344](#)
- SizeForSpectatorFiles
  - M\_const, [184](#)
- SizeInFile
  - sOrT, [394](#)
- SizeOfDollar
  - declare.h, [978](#)
  - dollar.c, [1096](#)
- SizeOfExpression
  - declare.h, [978](#)
  - execute.c, [1160](#)
- SIZEOFFUNCTION
  - ftypes.h, [1403](#)
- sizeprimelist
  - T\_const, [439](#)
- sizeprototype
  - ExPrEsSiOn, [124](#)
- sizeselecttermundo
  - N\_const, [257](#)
- SIZESTORECACHE
  - fsizes.h, [1349](#)
- SizeStoreCache
  - M\_const, [175](#)
- sizevertices
  - MoDeL, [234](#)
- SKIP
  - ftypes.h, [1421](#)
- SkipAName
  - compiler.c, [724](#)
  - declare.h, [959](#)
- SKIPBLANKS
  - declare.h, [804](#)
- SKIPBRA1
  - declare.h, [804](#)
- SKIPBRA2
  - declare.h, [804](#)
- SKIPBRA3
  - declare.h, [804](#)
- SKIPBRA4
  - declare.h, [805](#)
- SKIPBRA5
  - declare.h, [805](#)
- SKIPBUFSIZE
  - pre.c, [2316](#)
- SkipClears
  - M\_const, [179](#)
- SkipField
  - declare.h, [901](#)
  - tools.c, [3086](#)
- skipFLine
  - Model, [231](#)
- SKIPGEXPRESSION
  - ftypes.h, [1422](#)
- SKIPLEXPRESSION
  - ftypes.h, [1421](#)
- SkipName
  - declare.h, [1027](#)
  - pre.c, [2333](#)
- SKIPUNHIDEGEXPRESSION
  - ftypes.h, [1424](#)
- SKIPUNHIDELEXPRESSION
  - ftypes.h, [1424](#)
- SLARGEBUFFER
  - fsizes.h, [1347](#)
- SLargeSize
  - M\_const, [175](#)
- slavebuf
  - ParallelVars, [318](#)
- sLevel
  - R\_const, [375](#)
- slist
  - Interaction, [161](#)
- sLoops
  - EGraph, [111](#)
- small\_power
  - T\_const, [430](#)
- small\_power\_maxn
  - T\_const, [436](#)
- small\_power\_maxx
  - T\_const, [436](#)
- small\_power\_n
  - T\_const, [430](#)
- SmallEsize
  - sOrT, [397](#)
- SmallSize
  - sOrT, [397](#)
- smart.c, [2708](#), [2709](#)
  - MatchIsPossible, [2709](#)
  - StudyPattern, [2709](#)
- sMaxdeg
  - EGraph, [111](#)
- SMAFXPATCHES
  - fsizes.h, [1347](#)
- SMaxFpatches
  - M\_const, [178](#)
- SMAXPATCHES
  - fsizes.h, [1347](#)
- SMaxPatches
  - M\_const, [178](#)
- sNodes
  - EGraph, [111](#)
- snodes
  - MConn, [200](#)
- SNUMBER
  - ftypes.h, [1391](#)
- sopi
  - ToPoTyPe, [470](#)
- SORT
  - ftypes.h, [1437](#)
- sOrT, [392](#)

- cBuffer, [395](#)
- cBufferSize, [399](#)
- f, [397](#)
- ff, [397](#)
- file, [394](#)
- fPatches, [396](#)
- fPatchesStop, [396](#)
- fPatchN, [400](#)
- fPatchN1, [399](#)
- GenSpace, [398](#)
- GenTerms, [398](#)
- inNum, [400](#)
- inPatches, [396](#)
- iPatches, [397](#)
- ktoi, [396](#)
- LargeSize, [397](#)
- lBuffer, [394](#)
- lFill, [394](#)
- lPatch, [399](#)
- lTop, [394](#)
- MaxFpatches, [399](#)
- MaxPatches, [398](#)
- maxtermsize, [399](#)
- newmaxtermsize, [400](#)
- ninterms, [398](#)
- outputmode, [400](#)
- Patches, [395](#)
- poin2a, [396](#)
- poina, [396](#)
- PoinFill, [395](#)
- PolyFlag, [399](#)
- PolyWise, [399](#)
- pStop, [396](#)
- putinsize, [398](#)
- sBuffer, [394](#)
- sFill, [395](#)
- sHalf, [395](#)
- SizeInFile, [394](#)
- SmallEsize, [397](#)
- SmallSize, [397](#)
- SpaceLeft, [398](#)
- sPointer, [395](#)
- stage4, [400](#)
- stagelevel, [400](#)
- sTerms, [397](#)
- sTop, [395](#)
- sTop2, [395](#)
- Terms2InSmall, [398](#)
- TermsInSmall, [397](#)
- TermsLeft, [398](#)
- tree, [396](#)
- type, [399](#)
- used, [394](#)
- sort.c, [2714](#), [2731](#)
  - AddArgs, [2722](#)
  - AddCoef, [2720](#)
  - AddPoly, [2721](#)
  - BinarySearch, [2729](#)
  - CleanUpSort, [2729](#)
  - Compare1, [2723](#)
  - CompareHSymbols, [2724](#)
  - CompareSymbols, [2723](#)
  - ComPress, [2724](#)
  - EndSort, [2717](#)
  - FlushOut, [2720](#)
  - GarbHand, [2725](#)
  - humanBytesSuffix, [2730](#)
  - HumanString, [2716](#)
  - HUMANSTRLEN, [2715](#)
  - HUMANSUFFLEN, [2716](#)
  - humanTermsSuffix, [2730](#)
  - INSLNGTH, [2716](#)
  - LowerSortLevel, [2729](#)
  - MergePatches, [2726](#)
  - NewSort, [2717](#)
  - NEWSPLITMERGE, [2715](#)
  - NEWSPLITMERGETIMSORT, [2715](#)
  - numcompares, [2730](#)
  - PolyRatFunSpecial, [2729](#)
  - PutIn, [2718](#)
  - PutOut, [2719](#)
  - Sflush, [2719](#)
  - SimpleSplitMerge, [2729](#)
  - SimpleSplitMergeRec, [2729](#)
  - SortWild, [2728](#)
  - SplitMerge, [2725](#)
  - StageSort, [2728](#)
  - StoreTerm, [2727](#)
  - totterms, [2730](#)
  - WriteStats, [2716](#)
- SORTANTIPOWER
  - ftypes.h, [1388](#)
- SORTHIGHFIRST
  - ftypes.h, [1388](#)
- SORTING
  - structs.h, [2904](#)
- SORTIOSIZE
  - fsizes.h, [1346](#)
- SORTLOWFIRST
  - ftypes.h, [1388](#)
- SORTMODULE
  - ftypes.h, [1369](#)
- SORTPOWERFIRST
  - ftypes.h, [1388](#)
- SortReallocateFlag
  - C\_const, [62](#)
- SortTheList
  - declare.h, [964](#)
  - lus.c, [1687](#)
- SortType
  - BrAcKeTiNfO, [34](#)
  - C\_const, [58](#)
  - R\_const, [377](#)
- SortWeights
  - argument.c, [494](#)
  - declare.h, [932](#)

- SortWild
  - declare.h, [871](#)
  - sort.c, [2728](#)
- SoScratC
  - N\_const, [250](#)
- SpaceLeft
  - sOrT, [398](#)
- spare
  - OPTIMIZE, [294](#)
  - TaBIEs, [462](#)
  - VeRtEx, [486](#)
- spare1
  - objects, [288](#)
- spare2
  - objects, [288](#)
- spare3
  - objects, [288](#)
- SpareTable
  - declare.h, [885](#)
  - tables.c, [2962](#)
- sparse
  - TaBIEs, [463](#)
- SPARSETABLE
  - ftypes.h, [1454](#)
- spec
  - FuNcTiOn, [146](#)
- SpecialCleanup
  - declare.h, [884](#)
  - execute.c, [1158](#)
- SPECIFICATION
  - ftypes.h, [1375](#)
- SPECMASK
  - form3.h, [1328](#)
- SpecTatoR, [401](#)
  - exprnumber, [402](#)
  - fh, [402](#)
  - flags, [402](#)
  - name, [402](#)
  - position, [401](#)
  - readpos, [401](#)
- spectator.c, [2786](#), [2788](#)
  - ClearSpectators, [2788](#)
  - CoCopySpectator, [2787](#)
  - CoCreateSpectator, [2787](#)
  - CoEmptySpectator, [2787](#)
  - CoRemoveSpectator, [2787](#)
  - CoToSpectator, [2787](#)
  - FlushSpectators, [2787](#)
  - GetFromSpectator, [2788](#)
  - PutInSpectator, [2787](#)
- SPECTATOREXPRESSON
  - ftypes.h, [1424](#)
- SpectatorFiles
  - M\_const, [174](#)
- SPECTATORSIZE
  - fsizes.h, [1347](#)
- SpectatorSize
  - M\_const, [177](#)
- spin
  - PaRtlcLe, [321](#)
- SplitMerge
  - declare.h, [872](#)
  - sort.c, [2725](#)
- SplitScratch
  - N\_const, [250](#)
- SplitScratch1
  - N\_const, [250](#)
- SplitScratchSize
  - N\_const, [252](#)
- SplitScratchSize1
  - N\_const, [252](#)
- sPointer
  - sOrT, [395](#)
- sproc
  - Assign, [23](#)
  - EGraph, [109](#)
  - Options, [301](#)
  - Output, [308](#)
  - PNodeClass, [336](#)
  - Process, [366](#)
- SProcess, [402](#)
  - ~SProcess, [404](#)
  - agraph, [407](#)
  - agrcount, [409](#)
  - assign, [405](#)
  - astack, [406](#)
  - cl2nd, [407](#)
  - clist, [408](#)
  - DUMMYPADDING, [409](#)
  - egraph, [407](#)
  - endAGraph, [405](#)
  - endMGraph, [405](#)
  - extperm, [409](#)
  - generate, [404](#)
  - id, [408](#)
  - loop, [408](#)
  - match, [405](#)
  - mgraph, [407](#)
  - mgrcount, [409](#)
  - model, [406](#)
  - nAGraphs, [410](#)
  - nAOPI, [410](#)
  - nclass, [407](#)
  - ncouple, [408](#)
  - nd2cl, [407](#)
  - nEdges, [408](#)
  - nExtern, [408](#)
  - nfinal, [407](#)
  - ninitl, [407](#)
  - nMGraphs, [409](#)
  - nMOPI, [409](#)
  - nNodes, [408](#)
  - nvert, [408](#)
  - opt, [406](#)
  - pnclass, [406](#)
  - proc, [406](#)

- prSProcess, [404](#)
- resultAGraph, [406](#)
- resultMGraph, [405](#)
- SProcess, [404](#)
- tCouple, [409](#)
- toMNodeClass, [405](#)
- wAGraphs, [410](#)
- wAOPI, [410](#)
- wMGraphs, [409](#)
- wMOPI, [410](#)
- sptbl
  - Process, [366](#)
- SQRTFUNCTION
  - ftypes.h, [1398](#)
- SS
  - T\_const, [428](#)
- SSMALLBUFFER
  - fsizes.h, [1346](#)
- SSmallEsize
  - M\_const, [175](#)
- SSMALLOVERFLOW
  - fsizes.h, [1346](#)
- SSmallSize
  - M\_const, [175](#)
- SSORTIOSIZE
  - fsizes.h, [1347](#)
- sSym
  - STOREHEADER, [414](#)
- st
  - NCand, [267](#)
  - NStack, [276](#)
- stage4
  - sOrT, [400](#)
- Stage4Name
  - R\_const, [375](#)
- stagelevel
  - sOrT, [400](#)
- StageSort
  - declare.h, [923](#)
  - sort.c, [2728](#)
- stap
  - TrAcEs, [476](#)
- start
  - BrAcKeTiNdEx, [32](#)
  - VaRrEnUm, [483](#)
- StartFiles
  - declare.h, [899](#)
  - tools.c, [3080](#)
- startlinenumber
  - DoLoOp, [91](#)
- StartLoops
  - declare.h, [1030](#)
  - lus.c, [1687](#)
- StartMore
  - startup.c, [2799](#)
- startposition
  - ReNuMbEr, [378](#)
- StartPrepro
  - declare.h, [912](#)
  - pre.c, [2328](#)
- startup.c, [2796](#), [2801](#)
  - CleanUp, [2800](#)
  - DEFAULTFNAMELENGTH, [2798](#)
  - defaulttempfilename, [2801](#)
  - DoTail, [2798](#)
  - emptystring, [2801](#)
  - FORMNAME, [2798](#)
  - GetRunningTime, [2801](#)
  - IniVars, [2799](#)
  - main, [2800](#)
  - OpenInput, [2798](#)
  - PrintDeprecation, [2800](#)
  - PrintRunningTime, [2800](#)
  - ReserveTempFiles, [2799](#)
  - StartMore, [2799](#)
  - StartVariables, [2799](#)
  - STRINGIFY, [2797](#)
  - STRINGIFY\_\_, [2797](#)
  - TAKEPATH, [2798](#)
  - TerminatImpl, [2800](#)
  - VERSIONSTR, [2798](#)
  - VERSIONSTR\_\_, [2798](#)
- startup\_init
  - flintinterface.h, [1275](#)
- StartVariables
  - declare.h, [821](#)
  - startup.c, [2799](#)
- STATEMENT
  - ftypes.h, [1375](#)
- STATIC\_ASSERT
  - form3.h, [1327](#)
- STATIC\_ASSERT\_\_1
  - form3.h, [1327](#)
- STATIC\_ASSERT\_\_2
  - form3.h, [1327](#)
- STATIC\_ASSERT\_\_3
  - form3.h, [1327](#)
- StatsFlag
  - C\_const, [57](#)
- STATSMERGETOFILE
  - ftypes.h, [1388](#)
- STATSOUT
  - ftypes.h, [1373](#)
- STATSPOSTSORT
  - ftypes.h, [1388](#)
- STATSSPLITMERGE
  - ftypes.h, [1388](#)
- status
  - ExPrEsSiOn, [123](#)
  - PF\_BUFFER, [329](#)
- STDERR
  - pre.c, [2316](#)
- stderrname
  - X\_const, [487](#)
- StdOut
  - M\_const, [178](#)

- step1
  - TrAcEs, 476
- STEP2
  - dollar.c, 1093
- sTerms
  - sOrT, 397
- STermsInSmall
  - M\_const, 175
- STERMSSMALL
  - fsizes.h, 1347
- sTop
  - sOrT, 395
- stop
  - PF\_BUFFER, 329
- stop1
  - SHvariables, 390
- sTop2
  - sOrT, 395
- stop2
  - SHvariables, 390
- StopWatchZero
  - P\_const, 313
- STORE
  - ftypes.h, 1437
- store.c, 2826, 2837
  - AddToScratch, 2829
  - CoLoad, 2829
  - CopyExpression, 2833
  - CoSave, 2829
  - DeleteStore, 2829
  - DetVars, 2830
  - FindInIndex, 2832
  - FindrNumber, 2832
  - GetFromStore, 2830
  - GetMoreFromMem, 2830
  - GetMoreTerms, 2830
  - GetOneTerm, 2830
  - GetTable, 2833
  - GetTerm, 2830
  - NextFileIndex, 2831
  - OpenTemp, 2828
  - PutInStore, 2829
  - ReadFromScratch, 2829
  - ReadSaveExpression, 2836
  - ReadSaveHeader, 2833
  - ReadSaveIndex, 2834
  - ReadSaveTerm32, 2835
  - ReadSaveVariables, 2834
  - ResetScratch, 2828
  - RevertScratch, 2828
  - SAVEREVISION, 2827
  - SeekScratch, 2828
  - SetEndHScratch, 2828
  - SetEndScratch, 2828
  - SetFileIndex, 2831
  - SetScratch, 2828
  - TermRenummer, 2832
  - ToStorage, 2831
  - VarStore, 2831
  - WriteStoreHeader, 2833
- STORECACHE
  - structs.h, 2903
- StOrEcAcHe, 411
  - buffer, 412
  - next, 412
  - position, 411
  - toppos, 411
- StoreCache
  - T\_const, 429
- StoreCacheAlloc
  - T\_const, 429
- StoreData
  - R\_const, 371
- STOREEXPRESSION
  - ftypes.h, 1422
- StoreFileSize
  - C\_const, 42
- StoreHandle
  - C\_const, 68
- STOREHEADER, 412
  - endianness, 414
  - headermark, 413
  - lenLONG, 413
  - lenPOINTER, 413
  - lenPOS, 413
  - lenWORD, 413
  - maxpower, 415
  - reserved, 415
  - revision, 415
  - sFun, 414
  - sInd, 414
  - sSym, 414
  - sVec, 414
  - wildoffset, 415
- STOREMODULE
  - ftypes.h, 1369
- StoreTerm
  - declare.h, 873
  - sort.c, 2727
- str\_dup
  - declare.h, 984
  - minos.c, 1730
- StrCmp
  - declare.h, 901
  - tools.c, 3084
- StrCopy
  - declare.h, 823
  - sch.c, 2651
- strDup1
  - declare.h, 885
  - tools.c, 3086
- STREAM
  - structs.h, 2901
- StreaM, 416
  - afterwards, 419
  - buffer, 417

- bufferposition, 418
- bufferize, 418
- eqnum, 419
- fileposition, 416
- FoldName, 417
- handle, 418
- inbuffer, 418
- isnextchar, 419
- linenumber, 416
- name, 417
- nextchar, 419
- olddelay, 419
- oldnoshowinput, 419
- pname, 418
- pointer, 417
- prevars, 419
- previous, 418
- previousNoShowInput, 419
- prevline, 417
- reserved, 420
- top, 417
- type, 418
- Streams
  - C\_const, 46
- StrHICmp
  - declare.h, 902
  - tools.c, 3084
- StrICmp
  - declare.h, 901
  - tools.c, 3084
- StrICont
  - declare.h, 902
  - tools.c, 3085
- strict
  - TaBIEs, 463
- STRINGIFY
  - pre.c, 2315
  - startup.c, 2797
- STRINGIFY\_\_
  - pre.c, 2315
  - startup.c, 2797
- STRINGVALUE
  - setfile.c, 2689
- StrLen
  - declare.h, 902
  - tools.c, 3085
- structs.h, 2895, 2905
  - ALLGLOBALS, 2904
  - BHEAD, 2899
  - BHEAD0, 2899
  - CBUF, 2902
  - CHANNEL, 2903
  - COMPARE, 2904
  - COMPAREDUMMY, 2902
  - COMPTREE, 2900
  - DISTRIBUTE, 2903
  - DO\_UFFLE, 2902
  - DOLOOP, 2902
  - EMPTYININDEX, 2898
  - FILEHANDLE, 2901
  - FILEINDEX, 2899
  - FINISHUFFLE, 2902
  - FIXEDGLOBALS, 2904
  - FUNCTIONS, 2901
  - INDEXENTRY, 2899
  - INFILEINDEX, 2898
  - NAMENODE, 2900
  - NAMETREE, 2900
  - NESTING, 2903
  - one\_byte, 2902
  - PARTI, 2903
  - PERM, 2903
  - PERMP, 2903
  - PHEAD, 2899
  - PHEAD0, 2899
  - PREVAR, 2901
  - RENUMBER, 2900
  - set\_of\_char, 2902
  - SORTING, 2904
  - STORECACHE, 2903
  - STREAM, 2901
  - TABLES, 2900
  - TRACEN, 2901
  - TRACES, 2901
  - VARRENUM, 2900
  - WCN, 2904
  - WCN2, 2904
- ststop1
  - SHvariables, 391
- ststop2
  - SHvariables, 391
- StudyPattern
  - declare.h, 965
  - smart.c, 2709
- StuffAdd
  - declare.h, 801
- Stuffle
  - declare.h, 837
  - reshuf.c, 2612
- StuffRootAdd
  - declare.h, 838
  - reshuf.c, 2613
- sub
  - Fraction, 141
  - poly, 350
- SUBAFTER
  - ftypes.h, 1433
- SUBAFTERNOT
  - ftypes.h, 1434
- SUBALL
  - ftypes.h, 1433
- SuBbUf, 420
  - buffernum, 420
  - subexpnum, 420
- subCouple
  - Assign, 23



- SUBDISORDER
  - ftypes.h, [1433](#)
- subexpbuffers
  - compiler.c, [726](#)
- subexpnum
  - SuBbUf, [420](#)
- SUBEXPR
  - ftypes.h, [1435](#)
- SUBEXPRESSION
  - ftypes.h, [1390](#)
- SUBEXPSIZE
  - ftypes.h, [1420](#)
- subindex
  - TaBIEbAsE, [458](#)
- SUBMANY
  - ftypes.h, [1433](#)
- SUBMASK
  - ftypes.h, [1433](#)
- SUBMULTI
  - ftypes.h, [1432](#)
- SUBONCE
  - ftypes.h, [1432](#)
- SUBONLY
  - ftypes.h, [1433](#)
- SubPLon
  - declare.h, [874](#)
  - reken.c, [2554](#)
- SUBROUTINEFILE
  - ftypes.h, [1368](#)
- SUBSELECT
  - ftypes.h, [1433](#)
- SubsInAll
  - declare.h, [975](#)
  - pattern.c, [2173](#)
- SUBSORT
  - ftypes.h, [1430](#)
- Substitute
  - declare.h, [874](#)
  - pattern.c, [2173](#)
- subsubveto
  - N\_const, [255](#)
- SUBTERMUSED1
  - ftypes.h, [1420](#)
- SUBTERMUSED2
  - ftypes.h, [1420](#)
- SUBVECTOR
  - ftypes.h, [1433](#)
- sum\_results
  - tree\_node, [481](#)
- SUMF1
  - ftypes.h, [1392](#)
- SUMF2
  - ftypes.h, [1393](#)
- SUMMEDIND
  - ftypes.h, [1428](#)
- sumnum
  - M\_const, [189](#)
- SUMNUM1
  - ftypes.h, [1416](#)
- SUMNUM2
  - ftypes.h, [1416](#)
- sumpnum
  - M\_const, [189](#)
- SumTime
  - M\_const, [177](#)
- sVec
  - STOREHEADER, [414](#)
- SWAP
  - parallel.c, [2063](#)
- SWITCH, [421](#)
  - caseoffset, [422](#)
  - defaultcase, [421](#)
  - endswitch, [422](#)
  - iflevel, [422](#)
  - maxcase, [422](#)
  - mincase, [422](#)
  - nestingsum, [423](#)
  - numcases, [422](#)
  - padding, [423](#)
  - table, [421](#)
  - tablesize, [422](#)
  - typetable, [422](#)
  - whilelevel, [423](#)
- SwitchArray
  - C\_const, [46](#)
- SwitchHeap
  - C\_const, [47](#)
- SwitchInArray
  - C\_const, [67](#)
- SwitchLevel
  - C\_const, [67](#)
- SwitchSplitMerge
  - declare.h, [1026](#)
  - if.c, [1657](#)
- SwitchSplitMergeRec
  - declare.h, [1026](#)
  - if.c, [1657](#)
- SWITCHTABLE, [423](#)
  - compbuffer, [424](#)
  - ncase, [423](#)
  - value, [423](#)
- swmes
  - FixedGlobals, [136](#)
- symb
  - ReNuMbEr, [378](#)
- SYMBOL
  - ftypes.h, [1389](#)
- SyMbOI, [424](#)
  - complex, [425](#)
  - dimension, [425](#)
  - flags, [425](#)
  - maxpower, [424](#)
  - minpower, [424](#)
  - name, [424](#)
  - namesize, [425](#)
  - node, [425](#)

- number, 425
- SYMBOL\_
  - ftypes.h, 1427
- SymbolList
  - C\_const, 44
- SymbolNormalize
  - declare.h, 991
  - normal.c, 1874
- SYMBOLONLY
  - ftypes.h, 1376
- Symbols
  - C\_const, 45
- symbols
  - variable.h, 3201
- SymChangeFlag
  - C\_const, 69
- SymFind
  - declare.h, 874
  - function.c, 1472
- SymGen
  - declare.h, 874
  - function.c, 1471
- symmet
  - FUN\_INFO, 143
- symmetr.c, 2931, 2933
  - Distribute, 2932
  - FunMatchCy, 2932
  - FunMatchSy, 2932
  - MatchArgument, 2932
  - MatchCy, 2932
  - MatchE, 2931
  - Permute, 2931
  - PermuteP, 2931
- SYMMETRIC
  - ftypes.h, 1432
- symmetric
  - FuNcTiOn, 146
- SYMMETRIZE
  - ftypes.h, 1416
- Symmetrize
  - declare.h, 875
  - function.c, 1471
- symminfo
  - FuNcTiOn, 145
- symnum
  - ReNuMbEr, 379
- SYMTONUM
  - ftypes.h, 1416
- SYMTOSUB
  - ftypes.h, 1417
- SYMTOSYM
  - ftypes.h, 1416
- SynchFile
  - declare.h, 897
  - tools.c, 3083
- T
  - AllGlobals, 11
- T\_const, 426
- aebufnum, 434
- allbufnum, 434
- bernoullis, 430
- BrackBuf, 429
- bracketindexflag, 435
- bracketinfo, 432
- CacheNumberMemHeap, 432
- CacheNumberMemMax, 435
- CacheNumberMemTop, 435
- comfun, 437
- comind, 437
- comnum, 437
- comsym, 436
- dummysubexp, 436
- ebufnum, 433
- factorials, 430
- fbufnum, 434
- fromindex, 439
- FunArg, 437
- idallflag, 434
- idallmaxnum, 434
- idallnum, 434
- InNumMem, 433
- inprimelist, 439
- LeaveNegative, 436
- ListPoly, 432
- ListSymbols, 432
- ListSymbolsSize, 435
- locwildvalue, 437
- IWorkPointer, 433
- IWorkspace, 429
- mBer, 438
- mfac, 433
- MinVecArg, 437
- mulpat, 437
- nBer, 438
- Nest, 428
- NestPoin, 428
- NestStop, 428
- nfac, 438
- NumberMemHeap, 431
- NumberMemMax, 435
- NumberMemTop, 435
- NumListSymbols, 436
- NumMem, 432
- numpoly, 436
- optentimes, 435
- partitions, 432
- pBer, 431
- pfac, 431
- PolyAct, 438
- posWorkPointer, 433
- posWorkspace, 429
- previousEfactor, 431
- primelist, 431
- proexp, 437
- pWorkPointer, 433
- pWorkspace, 429

- RecFlag, 438
- RepCount, 430
- RepTop, 430
- S0, 428
- sBer, 433
- setexterntopo, 439
- setinterntopo, 439
- sfact, 433
- sizeprimelist, 439
- small\_power, 430
- small\_power\_maxn, 436
- small\_power\_maxx, 436
- small\_power\_n, 430
- SS, 428
- StoreCache, 429
- StoreCacheAlloc, 429
- TermMemHeap, 431
- TermMemMax, 434
- TermMemTop, 435
- TMaddr, 431
- TMbuff, 438
- TMdolfac, 438
- TMout, 438
- TopologiesLevel, 439
- TopologiesOptions, 439
- TopologiesStart, 432
- TopologiesTerm, 432
- TrimPower, 436
- WildArgTaken, 430
- WildcardBufferSize, 434
- WildMask, 431
- WorkPointer, 430
- WorkSpace, 429
- WorkTop, 429
- T\_EEdge, 440
  - ext, 440
  - momc, 440
  - momn, 440
  - nodes, 440
  - padding, 441
- T\_EGraph, 441
  - ~T\_EGraph, 442
  - edges, 444
  - endSetExtern, 443
  - esym, 444
  - gld, 443
  - init, 442
  - maxdeg, 444
  - nEdges, 443
  - nExtern, 444
  - nNodes, 443
  - nodes, 444
  - nsym, 444
  - opi, 444
  - pId, 443
  - print, 442
  - setExtern, 443
  - T\_EGraph, 442
- T\_ENode, 445
  - deg, 445
  - edges, 445
  - ext, 445
- T\_MGraph, 446
  - ~T\_MGraph, 447
  - adjMat, 449
  - c1PI, 449
  - clist, 448
  - curcl, 450
  - discardDisc, 450
  - discardIso, 451
  - discardOrd, 450
  - discardRefine, 450
  - egraph, 450
  - esym, 449
  - generate, 447
  - maxdeg, 448
  - mindeg, 448
  - n1PI, 449
  - nBridges, 449
  - nCallRefine, 450
  - nClasses, 448
  - ndiag, 448
  - nEdges, 447
  - ngconn, 450
  - ngen, 450
  - nLoops, 448
  - nNodes, 447
  - nodes, 448
  - nsym, 449
  - pId, 447
  - selOPI, 448
  - T\_MGraph, 447
  - wsopi, 449
  - wsum, 449
- T\_MNode, 451
  - clss, 452
  - deg, 452
  - ext, 452
  - freelg, 452
  - id, 452
  - T\_MNode, 451
  - visited, 452
- T\_MNodeClass, 453
  - ~T\_MNodeClass, 453
  - chkOrd, 455
  - clCmp, 454
  - clist, 455
  - clmat, 456
  - cmpArray, 455
  - copy, 454
  - flist, 456
  - foralignment, 456
  - incMat, 455
  - init, 454
  - maxdeg, 456
  - mkCIMat, 454

- mkFlist, [454](#)
- mkNdCl, [454](#)
- nClasses, [455](#)
- ndcl, [456](#)
- nNodes, [455](#)
- printMat, [454](#)
- T\_MNodeClass, [453](#)
- tabl
  - FuNcTiOn, [145](#)
- table
  - SWITCH, [421](#)
- TaBIEbAsE, [457](#)
  - current, [457](#)
  - fillpoint, [457](#)
  - name, [458](#)
  - numtables, [458](#)
  - subindex, [458](#)
  - tablenumbers, [458](#)
- TABLEBASEFILE
  - ftypes.h, [1368](#)
- TableBaseList
  - C\_const, [44](#)
- tablebases
  - variable.h, [3201](#)
- TaBIEbAsEsUbInDeX, [458](#)
  - size, [459](#)
  - where, [459](#)
- tablecheck
  - C\_const, [56](#)
- TABLEEXTENSION
  - fsizes.h, [1348](#)
- tablefilling
  - C\_const, [60](#)
- TABLEFUNCTION
  - ftypes.h, [1397](#)
- tablenamefill
  - dbase, [79](#)
- tablenames
  - dbase, [80](#)
- tablenamessize
  - dbase, [79](#)
- tablenum
  - TaBIEs, [465](#)
- tablenumber
  - objects, [287](#)
- tablenumbers
  - TaBIEbAsE, [458](#)
- tablepointers
  - TaBIEs, [460](#)
- TableReset
  - declare.h, [839](#)
  - tables.c, [2965](#)
- TABLES
  - structs.h, [2900](#)
- TaBIEs, [459](#)
  - argtail, [461](#)
  - boomlijst, [461](#)
  - bounds, [463](#)
  - buffers, [462](#)
  - buffersfill, [465](#)
  - bufferssize, [464](#)
  - bunum, [464](#)
  - defined, [462](#)
  - flags, [461](#)
  - MaxTreeSize, [464](#)
  - mdefined, [463](#)
  - mm, [461](#)
  - mode, [465](#)
  - numdummies, [465](#)
  - numind, [463](#)
  - numtree, [464](#)
  - pattern, [461](#)
  - prototype, [460](#)
  - prototypeSize, [463](#)
  - reserved, [462](#)
  - rootnum, [464](#)
  - spare, [462](#)
  - sparse, [463](#)
  - strict, [463](#)
  - tablenum, [465](#)
  - tablepointers, [460](#)
  - totind, [462](#)
- tables.c, [2960](#), [2966](#)
  - Apply, [2965](#)
  - ApplyExec, [2965](#)
  - ApplyReset, [2965](#)
  - CheckTableDeclarations, [2963](#)
  - ClearTableTree, [2961](#)
  - CoApply, [2964](#)
  - CoTableBase, [2962](#)
  - CoTBaddto, [2962](#)
  - CoTBaudit, [2963](#)
  - CoTBCleanup, [2964](#)
  - CoTBcreate, [2962](#)
  - CoTBenter, [2963](#)
  - CoTBhelp, [2964](#)
  - CoTBload, [2963](#)
  - CoTBoff, [2964](#)
  - CoTBon, [2963](#)
  - CoTBopen, [2962](#)
  - CoTBreplace, [2964](#)
  - CoTBuse, [2964](#)
  - CoTestUse, [2963](#)
  - DoTableExpansion, [2961](#)
  - FindTableTree, [2961](#)
  - FindTB, [2962](#)
  - FlipTable, [2962](#)
  - helpTB, [2966](#)
  - InsTableTree, [2961](#)
  - RedoTableTree, [2961](#)
  - ReleaseTB, [2965](#)
  - ReWorkT, [2964](#)
  - SpareTable, [2962](#)
  - TableReset, [2965](#)
  - TestUse, [2963](#)
- TABLESIZE

- declare.h, [809](#)
- tablesize
  - SWITCH, [422](#)
- TABLESTUB
  - ftypes.h, [1398](#)
- tabstring
  - O\_const, [279](#)
- tadblock
  - MGraph, [215](#)
- tadpole
  - MGraph, [214](#)
- tag
  - PF\_BUFFER, [330](#)
- TakeArgContent
  - argument.c, [492](#)
  - declare.h, [929](#)
- TakeContent
  - declare.h, [1010](#)
  - ratio.c, [2503](#)
- TakeDollarContent
  - declare.h, [966](#)
  - dollar.c, [1099](#)
- TakeExtraSymbols
  - declare.h, [1010](#)
  - ratio.c, [2503](#)
- TakeIDfunction
  - declare.h, [976](#)
  - pattern.c, [2174](#)
- TakeLongRoot
  - declare.h, [971](#)
  - reken.c, [2559](#)
- TakeModulus
  - declare.h, [875](#)
  - reken.c, [2560](#)
- TakeNormalModulus
  - declare.h, [875](#)
  - reken.c, [2560](#)
- TakeOneLine
  - lus.c, [1689](#)
- TAKEPATH
  - startup.c, [2798](#)
- TakeRatRoot
  - declare.h, [971](#)
  - reken.c, [2553](#)
- TakeSymbolContent
  - declare.h, [1009](#)
  - ratio.c, [2505](#)
- TAKETRACE
  - ftypes.h, [1415](#)
- TalToLine
  - declare.h, [875](#)
  - sch.c, [2652](#)
- TANFUNCTION
  - ftypes.h, [1399](#)
- TANHFUNCTION
  - ftypes.h, [1400](#)
- TCOMMA
  - ftypes.h, [1382](#)
- TCONJUGATE
  - ftypes.h, [1383](#)
- tCouple
  - SProcess, [409](#)
- TDELTA
  - ftypes.h, [1380](#)
- TDIVIDE
  - ftypes.h, [1382](#)
- TDOLLAR
  - ftypes.h, [1380](#)
- TDOT
  - ftypes.h, [1381](#)
- TDOTPRODUCT
  - ftypes.h, [1380](#)
- TDUBIOUS
  - ftypes.h, [1381](#)
- TeInFun
  - N\_const, [256](#)
- TELLFILE
  - declare.h, [821](#)
  - tools.c, [3082](#)
- TellFile
  - declare.h, [897](#)
  - tools.c, [3082](#)
- TempDir
  - M\_const, [172](#)
- TempSortDir
  - M\_const, [173](#)
- EMPTY
  - ftypes.h, [1384](#)
- TENDOFIT
  - ftypes.h, [1383](#)
- tensor
  - FUN\_INFO, [143](#)
- TENSORFUNCTION
  - ftypes.h, [1426](#)
- tensorList
  - O\_const, [281](#)
- TENVEC
  - ftypes.h, [1416](#)
- TenVec
  - declare.h, [876](#)
  - opera.c, [1958](#)
- TenVecFind
  - declare.h, [876](#)
  - opera.c, [1958](#)
- TePos
  - R\_const, [375](#)
- terfirstcomm
  - N\_const, [247](#)
- term
  - TERMINFO, [466](#)
- term\_compare
  - optimize.cc, [1994](#)
- TermAssign
  - declare.h, [963](#)
  - dollar.c, [1094](#)
- termbuf

- O\_const, [279](#)
- termbuffer
  - N\_const, [249](#)
- TERMEXTRAWORDS
  - tools.c, [3077](#)
- TermFree
  - declare.h, [816](#)
- TERMFUNCTION
  - ftypes.h, [1396](#)
- termfunnum
  - M\_const, [189](#)
- TERMINAL
  - pre.c, [2317](#)
- Terminate
  - declare.h, [810](#)
- TerminatImpl
  - declare.h, [889](#)
  - startup.c, [2800](#)
- TERMINFO, [466](#)
  - currentMODEL, [466](#)
  - currentModel, [466](#)
  - diaoffset, [467](#)
  - externalset, [467](#)
  - flags, [467](#)
  - internalset, [467](#)
  - legcouple, [466](#)
  - level, [467](#)
  - numdia, [467](#)
  - numextern, [467](#)
  - numtopo, [466](#)
  - term, [466](#)
- termlevel
  - C\_const, [65](#)
- TermMalloc
  - declare.h, [816](#)
- TermMallocAddMemory
  - declare.h, [1000](#)
  - tools.c, [3089](#)
- TermMemHeap
  - T\_const, [431](#)
- TermMemMax
  - T\_const, [434](#)
- TERMMEMSTARTNUM
  - tools.c, [3077](#)
- TermMemTop
  - T\_const, [435](#)
- termp
  - TrAcEn, [471](#)
  - TrAcEs, [474](#)
- TermRenumber
  - declare.h, [876](#)
  - store.c, [2832](#)
- terms
  - poly, [356](#)
- Terms2InSmall
  - sOrT, [398](#)
- termscopy
  - poly, [343](#)
- TERMSINBRACKET
  - ftypes.h, [1398](#)
- TermsInBracket
  - declare.h, [983](#)
  - execute.c, [1160](#)
- termsinbracket
  - BrAcKeTiNdEx, [32](#)
- TermsInDollar
  - declare.h, [978](#)
  - dollar.c, [1096](#)
- TERMSINEXPR
  - ftypes.h, [1397](#)
- TermsInExpression
  - declare.h, [978](#)
  - execute.c, [1160](#)
- TermsInSmall
  - sOrT, [397](#)
- termsize
  - NeStInG, [269](#)
- TermsLeft
  - sOrT, [398](#)
- termstortstack
  - C\_const, [47](#)
- termstack
  - C\_const, [47](#)
- termsumcheck
  - C\_const, [49](#)
- terstart
  - N\_const, [247](#)
- terstop
  - N\_const, [246](#)
- TestArgNum
  - declare.h, [1002](#)
  - transform.c, [3149](#)
- TestDoLoop
  - declare.h, [1012](#)
  - dollar.c, [1098](#)
- TestDrop
  - declare.h, [876](#)
  - execute.c, [1157](#)
- TestEndDoLoop
  - declare.h, [1012](#)
  - dollar.c, [1098](#)
- TestFloat
  - float.c, [1292](#)
- TestFunFlag
  - declare.h, [991](#)
  - normal.c, [1875](#)
- TestMatch
  - declare.h, [877](#)
  - pattern.c, [2172](#)
- TestName
  - declare.h, [894](#)
  - names.c, [1834](#)
- TestPartitions
  - declare.h, [835](#)
  - reshuf.c, [2611](#)
- TestSelect

- declare.h, [975](#)
- pattern.c, [2173](#)
- TestSub
  - declare.h, [878](#)
  - proces.c, [2426](#)
- TestTables
  - compiler.c, [725](#)
  - declare.h, [959](#)
- TestTerm
  - declare.h, [1000](#)
  - tools.c, [3093](#)
- TestUse
  - declare.h, [838](#)
  - tables.c, [2963](#)
- TestValue
  - C\_const, [52](#)
- TeSuOut
  - N\_const, [257](#)
- TEXPRESSION
  - ftypes.h, [1380](#)
- TFLOAT
  - ftypes.h, [1384](#)
- TFUN
  - ftypes.h, [1457](#)
- TFUN1
  - ftypes.h, [1457](#)
- TFUNCLOSE
  - ftypes.h, [1382](#)
- TFUNCTION
  - ftypes.h, [1380](#)
- TFUNOPEN
  - ftypes.h, [1382](#)
- TGENINDEX
  - ftypes.h, [1383](#)
- TheDefine
  - declare.h, [907](#)
  - pre.c, [2320](#)
- TheErrorMessage
  - ratio.c, [2507](#)
- thefunction
  - SHvariables, [391](#)
- theposition
  - N\_const, [245](#)
- THETA
  - ftypes.h, [1395](#)
- THETA2
  - ftypes.h, [1395](#)
- TheUndefine
  - declare.h, [907](#)
  - pre.c, [2321](#)
- ThreadBalancing
  - C\_const, [58](#)
- ThreadBucketSize
  - C\_const, [53](#)
- threads.c, [2993](#)
- THREADSCRATCHOUTSIZE
  - fsizes.h, [1349](#)
- THREADSCRATCHSIZE
  - fsizes.h, [1349](#)
- THREADSDEBUG
  - ftypes.h, [1373](#)
- ThreadsFlag
  - C\_const, [58](#)
- ThreadSortFileSynch
  - C\_const, [59](#)
- ThreadStats
  - C\_const, [58](#)
- time0
  - Options, [303](#)
- time1
  - Options, [303](#)
- TimeChildren
  - declare.h, [878](#)
  - tools.c, [3092](#)
- TimeCPU
  - declare.h, [878](#)
  - tools.c, [3092](#)
- TimeLimit
  - M\_const, [177](#)
- timeout
  - X\_const, [487](#)
- TimeWallClock
  - declare.h, [879](#)
  - tools.c, [3092](#)
- TINDEX
  - ftypes.h, [1379](#)
- tlistbuf
  - N\_const, [251](#)
- tlistsize
  - N\_const, [255](#)
- TMaddr
  - T\_const, [431](#)
- TMbuff
  - T\_const, [438](#)
- TMdolfac
  - T\_const, [438](#)
- TMINUS
  - ftypes.h, [1382](#)
- TMout
  - T\_const, [438](#)
- TMPPOLYFUN
  - ftypes.h, [1391](#)
- TMULTIPLY
  - ftypes.h, [1382](#)
- TNOT
  - ftypes.h, [1383](#)
- TNUMBER
  - ftypes.h, [1380](#)
- TNUMBER1
  - ftypes.h, [1383](#)
- to\_argument\_mpoly
  - flintinterface.h, [1275](#)
- to\_argument\_poly
  - flintinterface.h, [1275](#)
- to\_coefficient\_list
  - poly, [348](#)

- to\_string
  - poly, [349](#)
- TOBEFACTORED
  - ftypes.h, [1431](#)
- ToBelnFactors
  - C\_const, [69](#)
- TOBEUNFACTORED
  - ftypes.h, [1431](#)
- TODISK
  - minos.h, [1753](#)
- ToFast
  - declare.h, [903](#)
  - tools.c, [3091](#)
- ToFloat
  - float.c, [1294](#)
- toForm
  - topowrap.cc, [3141](#)
- ToGeneral
  - declare.h, [903](#)
  - tools.c, [3091](#)
- tohunt
  - N\_const, [253](#)
- token.c, [3047](#), [3050](#)
  - CHECKPOLY, [3048](#)
  - simp1token, [3048](#)
  - simp2token, [3049](#)
  - simp3atoken, [3049](#)
  - simp3btoken, [3049](#)
  - simp4token, [3049](#)
  - simp5token, [3049](#)
  - simp6token, [3049](#)
  - simpwtoken, [3048](#)
  - tokenize, [3048](#)
  - ttypes, [3050](#)
  - WriteTokens, [3048](#)
- tokenarglevel
  - C\_const, [50](#)
- tokenize
  - declare.h, [958](#)
  - token.c, [3048](#)
- tokens
  - C\_const, [50](#)
- TokensWriteFlag
  - C\_const, [57](#)
- TOKENTOLINE
  - declare.h, [801](#)
- TokenToLine
  - declare.h, [879](#)
  - sch.c, [2652](#)
- TOLONG
  - declare.h, [814](#)
- TOLYNDON
  - ftypes.h, [1447](#)
- TOLYNDONR
  - ftypes.h, [1447](#)
- toMNodeClass
  - SProcess, [405](#)
- tools.c, [3075](#), [3095](#)
  - AddToString, [3086](#)
  - AreArgsEqual, [3091](#)
  - CacheNumberMallocAddMemory, [3089](#)
  - CloseChannel, [3084](#)
  - CloseFile, [3081](#)
  - CloseStream, [3079](#)
  - CmpArray, [3085](#)
  - CompareArgs, [3091](#)
  - CompArg, [3092](#)
  - ConWord, [3085](#)
  - CopyFile, [3081](#)
  - Crash, [3093](#)
  - CreateFile, [3081](#)
  - CreateHandle, [3081](#)
  - CreateLogFile, [3081](#)
  - CreateStream, [3080](#)
  - DistrN, [3093](#)
  - DODOUBLE, [3078](#)
  - DOEXPAND, [3078](#)
  - DoubleBuffer, [3090](#)
  - DoubleList, [3090](#)
  - DoubleLList, [3090](#)
  - EndOfToken, [3086](#)
  - ExpandBuffer, [3090](#)
  - filelist, [3094](#)
  - filelistsize, [3094](#)
  - FILLVALUE, [3077](#)
  - FlushFile, [3083](#)
  - FromOList, [3089](#)
  - FromList, [3089](#)
  - FromVarList, [3090](#)
  - GetAppendChannel, [3084](#)
  - GetChannel, [3083](#)
  - GetFromStream, [3079](#)
  - GetPosFile, [3083](#)
  - GetStreamPosition, [3080](#)
  - iexp, [3090](#)
  - iniTools, [3088](#)
  - IsLikeVector, [3091](#)
  - LoadInputFile, [3079](#)
  - LocateFile, [3079](#)
  - LongCopy, [3087](#)
  - LongLongCopy, [3087](#)
  - LookInStream, [3079](#)
  - M\_check1, [3089](#)
  - M\_free, [3088](#)
  - M\_print, [3089](#)
  - MakeDate, [3087](#)
  - Malloc1, [3088](#)
  - NUMBEREXTRAWORDS, [3078](#)
  - NumberMallocAddMemory, [3089](#)
  - NUMBERMEMSTARTNUM, [3078](#)
  - NumCopy, [3087](#)
  - numinfilelist, [3094](#)
  - NumToStr, [3085](#)
  - OpenAddFile, [3080](#)
  - OpenFile, [3080](#)
  - OpenStream, [3079](#)



- PositionStream, [3080](#)
- ReadFile, [3082](#)
- ReadFromStream, [3079](#)
- ReadPosFile, [3082](#)
- ReadSnum, [3087](#)
- ReOpenFile, [3081](#)
- ReverseStatements, [3080](#)
- SeekFile, [3082](#)
- set\_del, [3088](#)
- set\_in, [3087](#)
- set\_set, [3088](#)
- set\_sub, [3088](#)
- SetPosFile, [3083](#)
- SkipField, [3086](#)
- StartFiles, [3080](#)
- StrCmp, [3084](#)
- strDup1, [3086](#)
- StrHICmp, [3084](#)
- StrICmp, [3084](#)
- StrICont, [3085](#)
- StrLen, [3085](#)
- SynchFile, [3083](#)
- TELLFILE, [3082](#)
- TellFile, [3082](#)
- TERMEXTRAWORDS, [3077](#)
- TermMallocAddMemory, [3089](#)
- TERMMEMSTARTNUM, [3077](#)
- TestTerm, [3093](#)
- TimeChildren, [3092](#)
- TimeCPU, [3092](#)
- TimeWallClock, [3092](#)
- ToFast, [3091](#)
- ToGeneral, [3091](#)
- ToPolyFunGeneral, [3091](#)
- ToToken, [3086](#)
- TruncateFile, [3083](#)
- UpdateMaxSize, [3084](#)
- WriteFile, [3094](#)
- WriteFileToFile, [3082](#)
- WriteString, [3085](#)
- WriteUnfinString, [3086](#)
- toOrbits
  - MOrbits, [238](#)
- TOOUTPUT
  - ftypes.h, [1375](#)
- Top
  - CbUf, [71](#)
- top
  - StreaM, [417](#)
- TOPBITONLY
  - form3.h, [1328](#)
- TOPLONGBITONLY
  - form3.h, [1328](#)
- topnumber
  - dbase, [79](#)
- TOPO
  - ftypes.h, [1403](#)
- TOPOLOGIES
  - ftypes.h, [1402](#)
- TopologiesLevel
  - T\_const, [439](#)
- TOPOLOGIESONLY
  - ftypes.h, [1455](#)
- TopologiesOptions
  - T\_const, [439](#)
- TopologiesStart
  - T\_const, [432](#)
- TopologiesTerm
  - T\_const, [432](#)
- ToPolyFunGeneral
  - declare.h, [903](#)
  - tools.c, [3091](#)
- TOPOLYNOMIALFLAG
  - ftypes.h, [1448](#)
- topolynomialflag
  - C\_const, [61](#)
- ToPoTyPe, [468](#)
  - cldeg, [469](#)
  - clext, [469](#)
  - clnum, [469](#)
  - cmaxd, [469](#)
  - cmind, [469](#)
  - ncl, [470](#)
  - nlegs, [470](#)
  - nloops, [470](#)
  - npadding, [470](#)
  - nvert, [470](#)
  - opt, [469](#)
  - sopi, [470](#)
  - vert, [469](#)
  - vertmax, [469](#)
- topowrap.cc, [3140](#), [3142](#)
  - GenerateTopologies, [3141](#)
  - GenerateVertices, [3141](#)
  - MAXPOINTS, [3141](#)
  - toForm, [3141](#)
- toppos
  - StOrEcAcHe, [411](#)
- topsubexpbuffers
  - compiler.c, [726](#)
- toptokens
  - C\_const, [50](#)
- ToRat
  - float.c, [1294](#)
- ToStorage
  - declare.h, [879](#)
  - store.c, [2831](#)
- tostring
  - factorized\_poly, [126](#)
- total\_degree
  - poly, [345](#)
- totalc
  - EGraph, [112](#)
- totalnumberofthreads
  - M\_const, [180](#)
- totermis

- sort.c, [2730](#)
- totind
  - TaBIEs, [462](#)
- ToToken
  - declare.h, [920](#)
  - tools.c, [3086](#)
- TPLUS
  - ftypes.h, [1382](#)
- TPOWER
  - ftypes.h, [1382](#)
- TPOWER1
  - ftypes.h, [1384](#)
- Trace4
  - declare.h, [879](#)
  - opera.c, [1956](#)
- Trace4Gen
  - declare.h, [879](#)
  - opera.c, [1956](#)
- Trace4no
  - declare.h, [880](#)
  - opera.c, [1956](#)
- tracebackflag
  - M\_const, [188](#)
- TraceFind
  - declare.h, [880](#)
  - opera.c, [1957](#)
- TRACEN
  - structs.h, [2901](#)
- TrAcEn, [471](#)
  - accu, [471](#)
  - accup, [471](#)
  - allsign, [472](#)
  - factor, [472](#)
  - inlist, [472](#)
  - level, [472](#)
  - num, [472](#)
  - perm, [472](#)
  - sgn, [472](#)
  - termp, [471](#)
- TraceN
  - declare.h, [880](#)
  - opera.c, [1957](#)
- TraceNgen
  - declare.h, [880](#)
  - opera.c, [1957](#)
- TraceNno
  - declare.h, [880](#)
  - opera.c, [1957](#)
- TRACES
  - structs.h, [2901](#)
- TrAcEs, [473](#)
  - a3, [477](#)
  - a4, [477](#)
  - accu, [474](#)
  - accup, [474](#)
  - ad, [476](#)
  - allsign, [478](#)
  - e3, [475](#)
  - e4, [475](#)
  - eers, [475](#)
  - factor, [478](#)
  - finalstep, [478](#)
  - gamm, [476](#)
  - gamma5, [477](#)
  - inlist, [474](#)
  - j3, [475](#)
  - j4, [475](#)
  - kstep, [476](#)
  - lc3, [477](#)
  - lc4, [477](#)
  - level, [478](#)
  - mdel, [475](#)
  - mdum, [476](#)
  - mepf, [475](#)
  - nt3, [474](#)
  - nt4, [474](#)
  - num, [477](#)
  - pdel, [476](#)
  - pepf, [475](#)
  - perm, [474](#)
  - sgn, [476](#)
  - sign1, [477](#)
  - sign2, [477](#)
  - stap, [476](#)
  - step1, [476](#)
  - termp, [474](#)
- Traces
  - declare.h, [880](#)
  - opera.c, [1957](#)
- tracestack
  - N\_const, [249](#)
- TransEname
  - C\_const, [53](#)
- TransferBuffer
  - declare.h, [976](#)
  - pattern.c, [2174](#)
- transFlag
  - O\_const, [284](#)
- transform.c, [3145](#), [3150](#)
  - CoTransform, [3146](#)
  - FindRange, [3149](#)
  - PutArgInScratch, [3149](#)
  - ReadRange, [3149](#)
  - RunAddArg, [3148](#)
  - RunCycle, [3147](#)
  - RunDecode, [3146](#)
  - RunDedup, [3147](#)
  - RunDropArg, [3148](#)
  - RunEncode, [3146](#)
  - RunExplode, [3147](#)
  - RunHtoZArg, [3149](#)
  - RunImplode, [3147](#)
  - RunIsLyndon, [3148](#)
  - RunMulArg, [3148](#)
  - RunPermute, [3147](#)
  - RunReplace, [3146](#)

- RunReverse, [3147](#)
- RunSelectArg, [3148](#)
- RunToLyndon, [3148](#)
- RunTransform, [3146](#)
- RunZtoHArg, [3149](#)
- TestArgNum, [3149](#)
- TransformRational
  - declare.h, [1022](#)
  - dict.c, [1075](#)
- TranslateExpression
  - declare.h, [978](#)
  - dollar.c, [1097](#)
- TRAP SIGNALS
  - unix.h, [3192](#)
- TreatPolyRatFun
  - declare.h, [993](#)
  - normal.c, [1874](#)
- tree, [478](#)
  - blnce, [479](#)
  - left, [479](#)
  - parent, [479](#)
  - right, [479](#)
  - sOrT, [396](#)
  - usage, [480](#)
  - value, [479](#)
- tree\_node, [480](#)
  - childs, [481](#)
  - finished, [482](#)
  - num\_visits, [482](#)
  - operator=, [481](#)
  - sum\_results, [481](#)
  - tree\_node, [481](#)
  - var, [482](#)
- Trick
  - declare.h, [881](#)
  - opera.c, [1956](#)
- TrimPower
  - T\_const, [436](#)
- TRUE\_EXPR
  - pre.c, [2316](#)
- TruncateFile
  - declare.h, [898](#)
  - tools.c, [3083](#)
- TryDo
  - declare.h, [881](#)
  - reshuf.c, [2611](#)
- TryEnvironment
  - declare.h, [988](#)
  - setfile.c, [2692](#)
- TryFileSetups
  - declare.h, [976](#)
  - setfile.c, [2692](#)
- TryRecover
  - declare.h, [802](#)
- tryterm
  - N\_const, [259](#)
- TSET
  - ftypes.h, [1380](#)
- TSETCLOSE
  - ftypes.h, [1383](#)
- TSETDOL
  - ftypes.h, [1384](#)
- TSETNUM
  - ftypes.h, [1384](#)
- TSETOPEN
  - ftypes.h, [1383](#)
- TSGAMMA
  - ftypes.h, [1384](#)
- TSUBEXP
  - ftypes.h, [1380](#)
- TSYMBOL
  - ftypes.h, [1379](#)
- ttypes
  - token.c, [3050](#)
- TVECTOR
  - ftypes.h, [1379](#)
- TWILDARG
  - ftypes.h, [1381](#)
- TWILDCARD
  - ftypes.h, [1381](#)
- TwoExprCompare
  - declare.h, [979](#)
  - dollar.c, [1097](#)
- type
  - ARGBUFFER, [15](#)
  - DICTIONARY\_ELEMENT, [84](#)
  - DoLIaRS, [88](#)
  - DoLoOp, [91](#)
  - FaCdOILaR, [125](#)
  - fixedset, [138](#)
  - InDeX, [150](#)
  - KEYWORD, [163](#)
  - KEYWORDV, [164](#)
  - MoDoPtDoLIaRS, [236](#)
  - NaMeNode, [261](#)
  - optimization, [291](#)
  - PaRtlcLe, [322](#)
  - PF\_BUFFER, [330](#)
  - PNodeClass, [336](#)
  - SeTs, [383](#)
  - SETUPPARAMETERS, [385](#)
  - sOrT, [399](#)
  - StreaM, [418](#)
  - VeRtEx, [486](#)
- TYPEADJUSTBOUNDS
  - ftypes.h, [1411](#)
- TYPEALLLOOPS
  - ftypes.h, [1415](#)
- TYPEALLPATHS
  - ftypes.h, [1415](#)
- TYPEAPPLY
  - ftypes.h, [1412](#)
- TYPEAPPLYRESET
  - ftypes.h, [1412](#)
- TYPEARG
  - ftypes.h, [1408](#)

- TYPEARGEXPLODE
  - ftypes.h, [1413](#)
- TYPEARGHEADSIZE
  - ftypes.h, [1421](#)
- TYPEARGIMPLODE
  - ftypes.h, [1413](#)
- TYPEARGTOEXTRASymbol
  - ftypes.h, [1414](#)
- TYPEASSIGN
  - ftypes.h, [1410](#)
- TYPECANONICALIZE
  - ftypes.h, [1414](#)
- TYPECHAININ
  - ftypes.h, [1412](#)
- TYPECHAINOUT
  - ftypes.h, [1412](#)
- TYPECHISHOLM
  - ftypes.h, [1408](#)
- TYPECLEARUSERFLAG
  - ftypes.h, [1415](#)
- TYPECOUNT
  - ftypes.h, [1407](#)
- TYPEDENOMINATORS
  - ftypes.h, [1413](#)
- TYPEDETCURDUM
  - ftypes.h, [1411](#)
- TYPEDISCARD
  - ftypes.h, [1407](#)
- TYPEDOLOOP
  - ftypes.h, [1414](#)
- TYPEDROPCOEFFICIENT
  - ftypes.h, [1413](#)
- TYPEDROPSYMBOLS
  - ftypes.h, [1414](#)
- TYPEELIF
  - ftypes.h, [1408](#)
- TYPEELSE
  - ftypes.h, [1408](#)
- TYPEENDDOLOOP
  - ftypes.h, [1414](#)
- TYPEENDIF
  - ftypes.h, [1408](#)
- TYPEENDREPEAT
  - ftypes.h, [1407](#)
- TYPEENDSWITCH
  - ftypes.h, [1415](#)
- TYPEEXIT
  - ftypes.h, [1409](#)
- TYPEEXPRESSION
  - ftypes.h, [1406](#)
- TYPEFACTARG
  - ftypes.h, [1410](#)
- TYPEFACTARG2
  - ftypes.h, [1410](#)
- TYPEFACTOR
  - ftypes.h, [1413](#)
- TYPEFINDLOOP
  - ftypes.h, [1410](#)
- TYPEFPRINT
  - ftypes.h, [1409](#)
- TYPEFROMPOLYNOMIAL
  - ftypes.h, [1414](#)
- typeGName
  - Particle, [324](#)
- TYPEGOTO
  - ftypes.h, [1407](#)
- TYPEIDNEW
  - ftypes.h, [1406](#)
- TYPEIDOLD
  - ftypes.h, [1406](#)
- TYPEIF
  - ftypes.h, [1407](#)
- TYPEINEXPRESSION
  - ftypes.h, [1411](#)
- TYPEINSIDE
  - ftypes.h, [1411](#)
- TYPEISFUN
  - ftypes.h, [1384](#)
- TYPEISMYSTERY
  - ftypes.h, [1385](#)
- TYPEISSUB
  - ftypes.h, [1384](#)
- TYPEMERGE
  - ftypes.h, [1412](#)
- TYPEMULT
  - ftypes.h, [1407](#)
- typeName
  - Particle, [324](#)
- TYPENORM
  - ftypes.h, [1408](#)
- TYPENORM2
  - ftypes.h, [1409](#)
- TYPENORM3
  - ftypes.h, [1409](#)
- TYPENORM4
  - ftypes.h, [1412](#)
- TYPEOPERATION
  - ftypes.h, [1407](#)
- TYPEPRINT
  - ftypes.h, [1409](#)
- TYPEPUTINSIDE
  - ftypes.h, [1414](#)
- TYPEREDEFPRE
  - ftypes.h, [1409](#)
- TYPERENUMBER
  - ftypes.h, [1410](#)
- TYPEREPEAT
  - ftypes.h, [1407](#)
- TYPEREVERSE
  - ftypes.h, [1408](#)
- TYPESETEXIT
  - ftypes.h, [1409](#)
- TYPESETUSERFLAG
  - ftypes.h, [1415](#)
- TYPESORT
  - ftypes.h, [1411](#)

- TYPESPLITARG
  - ftypes.h, [1409](#)
- TYPESPLITARG2
  - ftypes.h, [1410](#)
- TYPESPLITFIRSTARG
  - ftypes.h, [1411](#)
- TYPESPLITLASTARG
  - ftypes.h, [1412](#)
- TYPESTUFFLE
  - ftypes.h, [1413](#)
- TYPESUM
  - ftypes.h, [1408](#)
- TYPESUMFIX
  - ftypes.h, [1410](#)
- TYPESWITCH
  - ftypes.h, [1415](#)
- typetable
  - SWITCH, [422](#)
- TYPETERM
  - ftypes.h, [1411](#)
- TYPETESTUSE
  - ftypes.h, [1412](#)
- TYPETOPOLYNOMIAL
  - ftypes.h, [1413](#)
- TYPETOSPECTATOR
  - ftypes.h, [1414](#)
- TYPETRANSFORM
  - ftypes.h, [1413](#)
- TYPETRY
  - ftypes.h, [1410](#)
- TYPEUNRAVEL
  - ftypes.h, [1411](#)
- UBYTE
  - form3.h, [1335](#)
- Uclose
  - form3.h, [1332](#)
- uflags
  - ExPrEsSiOn, [124](#)
- Uflush
  - form3.h, [1332](#)
- Ugetpos
  - form3.h, [1333](#)
- UINT
  - form3.h, [1335](#)
- ULONG
  - form3.h, [1334](#)
- uncompressed
  - objects, [287](#)
- UngetChar
  - declare.h, [802](#)
- UngetFromStream
  - declare.h, [801](#)
- UNHIDE
  - ftypes.h, [1421](#)
- UNHIDEGEXPRESSION
  - ftypes.h, [1423](#)
- UNHIDELEXPRESSON
  - ftypes.h, [1423](#)
- unix.h, [3191](#), [3194](#)
  - ALTSEPARATOR, [3193](#)
  - CARRIAGEReturn, [3192](#)
  - LINEFEED, [3192](#)
  - P\_term, [3192](#)
  - PATHSEPARATOR, [3193](#)
  - SEPARATOR, [3193](#)
  - TRAP SIGNALS, [3192](#)
  - WITH\_ALARM, [3193](#)
  - WITH\_ENV, [3193](#)
  - WITHEXTERNALCHANNEL, [3192](#)
  - WITHPIPE, [3192](#)
  - WITHSYSTEM, [3192](#)
- unixfile.c, [3194](#), [3195](#)
- UNLOCK
  - declare.h, [818](#)
- UNPACK
  - ftypes.h, [1444](#)
- UnPack
  - declare.h, [881](#)
  - reken.c, [2552](#)
- UNPACK\_LONG
  - parallel.c, [2064](#)
- UnpackFloat
  - float.c, [1291](#)
- UNRWLOCK
  - declare.h, [819](#)
- UnsetAllowDelay
  - declare.h, [905](#)
  - pre.c, [2318](#)
- UnSetDictionary
  - declare.h, [1021](#)
  - dict.c, [1077](#)
- UnSetMods
  - declare.h, [884](#)
  - execute.c, [1158](#)
- UnsureDollarMode
  - C\_const, [57](#)
- Uopen
  - form3.h, [1331](#)
- updateCandNode
  - Assign, [21](#)
- UpdateMaxSize
  - declare.h, [1006](#)
  - tools.c, [3084](#)
- UpdatePositions
  - declare.h, [979](#)
  - execute.c, [1160](#)
- Uread
  - form3.h, [1332](#)
- usage
  - tree, [480](#)
- used
  - sOrT, [394](#)
- USEDFLAG
  - ftypes.h, [1429](#)
- UsedFunction
  - N\_const, [248](#)

- UseDictionary
  - dict.c, [1077](#)
- UsedIndex
  - N\_const, [247](#)
- UsedOtherFind
  - N\_const, [253](#)
- UsedSymbol
  - N\_const, [247](#)
- UsedVector
  - N\_const, [247](#)
- Useseek
  - form3.h, [1332](#)
- UseFindOnly
  - N\_const, [253](#)
- usenames
  - NaMeSpAcE, [262](#)
- UserFlags
  - pre.c, [2334](#)
- Usetbuf
  - form3.h, [1332](#)
- Usetpos
  - form3.h, [1333](#)
- Ustdout
  - form3.h, [1333](#)
- Usync
  - form3.h, [1333](#)
- Utell
  - form3.h, [1333](#)
- Utruncate
  - form3.h, [1333](#)
- UWORD
  - form3.h, [1334](#)
- Uwrite
  - form3.h, [1332](#)
- va\_arg
  - sch.c, [2651](#)
- va\_dcl
  - sch.c, [2650](#)
- va\_end
  - sch.c, [2650](#)
- va\_list
  - sch.c, [2651](#)
- va\_start
  - sch.c, [2650](#)
- value
  - FaCdOILaR, [125](#)
  - pReVaR, [360](#)
  - SETUPPARAMETERS, [385](#)
  - SWITCHTABLE, [423](#)
  - tree, [479](#)
- values
  - Options, [302](#)
- var
  - KEYWORDV, [164](#)
  - tree\_node, [482](#)
- var\_map\_t
  - flintinterface.h, [1270](#)
- variable.h, [3198](#), [3208](#)
- A, [3207](#)
- AC, [3205](#)
- AM, [3206](#)
- AN, [3206](#)
- AO, [3206](#)
- AP, [3206](#)
- AR, [3206](#)
- AS, [3206](#)
- AT, [3206](#)
- autofunctions, [3204](#)
- autoindices, [3204](#)
- autosymbols, [3204](#)
- autovectors, [3204](#)
- AX, [3206](#)
- BUG, [3205](#)
- cbuf, [3203](#)
- channels, [3203](#)
- chartype, [3199](#)
- Dollars, [3203](#)
- DoLoops, [3200](#)
- Dubious, [3204](#)
- Expressions, [3204](#)
- FG, [3207](#)
- fixedsets, [3207](#)
- functions, [3201](#)
- GlobalFunctions, [3202](#)
- GlobalIndices, [3202](#)
- GlobalSetElements, [3203](#)
- GlobalSets, [3203](#)
- GlobalSymbols, [3202](#)
- GlobalVectors, [3203](#)
- indices, [3201](#)
- MaxDoLoops, [3200](#)
- MaxNumPre, [3200](#)
- MaxProcedures, [3200](#)
- ModOptdollars, [3205](#)
- NumDollars, [3203](#)
- NumDoLoops, [3200](#)
- NumDubious, [3204](#)
- NumExpressions, [3204](#)
- NumFunctions, [3201](#)
- NumIndices, [3201](#)
- NumModOptdollars, [3205](#)
- NumOutputChannels, [3203](#)
- NumPotModdollars, [3205](#)
- NumPre, [3200](#)
- NumProcedures, [3199](#)
- NumSetElements, [3202](#)
- NumSets, [3202](#)
- NumSymbols, [3202](#)
- NumTableBases, [3202](#)
- NumVectors, [3202](#)
- numxsymbol, [3205](#)
- PotModdollars, [3205](#)
- PreVar, [3200](#)
- Procedures, [3199](#)
- SetElements, [3200](#)
- Sets, [3201](#)

- setupfilename, [3207](#)
- symbols, [3201](#)
- tablebases, [3201](#)
- vectors, [3201](#)
- WriteFile, [3207](#)
- xsymbol, [3205](#)
- variables
  - DICTIONARY, [82](#)
  - InDeXeNtRy, [154](#)
- VARNAME
  - declare.h, [809](#)
- VARNAMES
  - ftypes.h, [1444](#)
- varnames
  - C\_const, [42](#)
- VARRENUM
  - structs.h, [2900](#)
- VaRrEnUm, [482](#)
  - hi, [483](#)
  - lo, [483](#)
  - start, [483](#)
- vars
  - DoLoOp, [90](#)
- VarStore
  - declare.h, [881](#)
  - store.c, [2831](#)
- VarType
  - FixedGlobals, [135](#)
- VARTYPECOMPLEX
  - ftypes.h, [1431](#)
- VARTYPEIMAGINARY
  - ftypes.h, [1431](#)
- VARTYPEMINUS
  - ftypes.h, [1432](#)
- VARTYPENONE
  - ftypes.h, [1431](#)
- VARTPEROOTOFUNITY
  - ftypes.h, [1431](#)
- vdef
  - Model, [230](#)
- vecnum
  - ReNuMbEr, [380](#)
- vect
  - ReNuMbEr, [379](#)
- VECTBIT
  - ftypes.h, [1438](#)
- VECTOMIN
  - ftypes.h, [1417](#)
- VECTOR
  - ftypes.h, [1389](#)
- VeCtOr, [483](#)
  - complex, [484](#)
  - dimension, [484](#)
  - flags, [484](#)
  - name, [484](#)
  - namesize, [484](#)
  - node, [484](#)
  - number, [484](#)
- Vector
  - vector.h, [3211](#)
- vector.h, [3209](#), [3218](#)
  - DeclareVector, [3211](#)
  - Vector, [3211](#)
  - VectorBack, [3213](#)
  - VectorCapacity, [3214](#)
  - VectorClear, [3215](#)
  - VectorEmpty, [3214](#)
  - VectorErase, [3218](#)
  - VectorErases, [3218](#)
  - VectorFree, [3212](#)
  - VectorFront, [3213](#)
  - VectorInit, [3212](#)
  - VectorInsert, [3217](#)
  - VectorInserts, [3217](#)
  - VectorPopBack, [3216](#)
  - VectorPtr, [3213](#)
  - VectorPushBack, [3215](#)
  - VectorPushBacks, [3216](#)
  - VectorReserve, [3215](#)
  - VectorSize, [3214](#)
  - VectorStruct, [3211](#)
- VECTOR\_
  - ftypes.h, [1428](#)
- VectorBack
  - vector.h, [3213](#)
- VectorCapacity
  - vector.h, [3214](#)
- VectorClear
  - vector.h, [3215](#)
- VectorEmpty
  - vector.h, [3214](#)
- VectorErase
  - vector.h, [3218](#)
- VectorErases
  - vector.h, [3218](#)
- VectorFree
  - vector.h, [3212](#)
- VectorFront
  - vector.h, [3213](#)
- VectorInit
  - vector.h, [3212](#)
- VectorInsert
  - vector.h, [3217](#)
- VectorInserts
  - vector.h, [3217](#)
- vectorlikeLHS
  - C\_const, [63](#)
- VectorList
  - C\_const, [44](#)
- VECTORONLY
  - ftypes.h, [1376](#)
- VectorPopBack
  - vector.h, [3216](#)
- VectorPtr
  - vector.h, [3213](#)
- VectorPushBack

- vector.h, 3215
- VectorPushBacks
  - vector.h, 3216
- VectorReserve
  - vector.h, 3215
- Vectors
  - C\_const, 46
- vectors
  - variable.h, 3201
- VectorSize
  - vector.h, 3214
- VectorStruct
  - vector.h, 3211
- vectorzero
  - M\_const, 192
- VECTOSUB
  - ftypes.h, 1417
- VECTOVEC
  - ftypes.h, 1417
- VERSIONSTR
  - startup.c, 2798
- VERSIONSTR\_\_
  - startup.c, 2798
- vert
  - ToPoTyPe, 469
- VeRtEx, 485
  - couplings, 485
  - error, 486
  - externonly, 486
  - ncouplings, 486
  - nparticles, 486
  - particles, 485
  - spare, 486
  - type, 486
- VERTEXFUNCTION
  - ftypes.h, 1426
- vertices
  - MoDeL, 233
- vertmax
  - ToPoTyPe, 469
- vetofilling
  - C\_const, 60
- vetotablebasefill
  - C\_const, 60
- vflags
  - ExPrEsSiOn, 123
- visit
  - MGraph, 208
- visited
  - EEdge, 99
  - ENode, 118
  - MNode, 220
  - T\_MNode, 452
- VORTTRANMODE
  - ftypes.h, 1386
- wAGraphs
  - Assign, 25
  - Process, 368
- SProcess, 410
- WantAddLongs
  - declare.h, 815
- WantAddPointers
  - declare.h, 815
- WantAddPositions
  - declare.h, 815
- wAOPI
  - Assign, 25
  - Process, 368
  - SProcess, 410
- WARMUP
  - reken.c, 2551
- WarnFlag
  - C\_const, 57
- Warning
  - declare.h, 885
  - message.c, 1714
- WCN
  - structs.h, 2904
- WCN2
  - structs.h, 2904
- WCOPY
  - declare.h, 803
- wgraphs
  - Process, 367
- where
  - DoLIARs, 88
  - FaCdOILaR, 125
  - PaRtl, 321
  - TaBlEbAsEsUbInDeX, 459
- whichbuffer
  - ExPrEsSiOn, 123
- WhileLevel
  - C\_const, 67
- whilelevel
  - SWITCH, 423
- wildarg
  - pReVaR, 361
- WILDARGFUN
  - ftypes.h, 1398
- WILDARGINDEX
  - ftypes.h, 1406
- WildArgs
  - N\_const, 257
- WILDARGSYMBOL
  - ftypes.h, 1405
- WildArgTaken
  - T\_const, 430
- WILDARGVECTOR
  - ftypes.h, 1406
- WildC
  - C\_const, 48
- wildcard.c, 3222, 3224
  - AddWild, 3223
  - CheckWild, 3223
  - ClearWild, 3223
  - DEBUG, 3222



- DenToFunction, [3223](#)
- ResolveSet, [3223](#)
- WildFill, [3223](#)
- WildcardBufferSize
  - C\_const, [55](#)
  - T\_const, [434](#)
- WildcardNames
  - C\_const, [49](#)
- WILDCARDS
  - ftypes.h, [1418](#)
- WildDirt
  - N\_const, [256](#)
- WildDollars
  - declare.h, [963](#)
  - dollar.c, [1095](#)
- WILDDUMMY
  - ftypes.h, [1416](#)
- WildEat
  - N\_const, [257](#)
- WildFill
  - declare.h, [881](#)
  - wildcard.c, [3223](#)
- wildflag
  - C\_const, [55](#)
- WILDMASK
  - form3.h, [1329](#)
- WildMask
  - T\_const, [431](#)
- wildoffset
  - STOREHEADER, [415](#)
- WildReserve
  - N\_const, [256](#)
- WildStop
  - N\_const, [246](#)
- WildValue
  - N\_const, [246](#)
- Willnd
  - M\_const, [185](#)
- WITH\_ALARM
  - unix.h, [3193](#)
- WITH\_ENV
  - fwin.h, [1497](#)
  - unix.h, [3193](#)
- WITHAUTO
  - ftypes.h, [1376](#)
- WITHBLOCKS
  - ftypes.h, [1456](#)
- WITHCUTOFF
  - float.c, [1286](#)
- WITHEDGES
  - ftypes.h, [1455](#)
- WITHERROR
  - ftypes.h, [1366](#)
- WITHEXTERNALCHANNEL
  - unix.h, [3192](#)
- WITHINSERTIONS
  - ftypes.h, [1455](#)
- WITHONEPI
  - ftypes.h, [1456](#)
- WITHOUTERROR
  - ftypes.h, [1366](#)
- withoutflush
  - minos.c, [1734](#)
  - minos.h, [1755](#)
- WITHOUTSEMICOLON
  - ftypes.h, [1371](#)
- WITHPIPE
  - unix.h, [3192](#)
- WITHRETURN
  - fwin.h, [1496](#)
- WITHSEMICOLON
  - ftypes.h, [1371](#)
- WITHSORTBOTS
  - form3.h, [1331](#)
- WITHSYMMETRIZE
  - ftypes.h, [1455](#)
- WITHSYSTEM
  - fwin.h, [1496](#)
  - unix.h, [3192](#)
- wlen
  - O\_const, [282](#)
- wMGraphs
  - Process, [367](#)
  - SProcess, [409](#)
- wMOPI
  - Process, [367](#)
  - SProcess, [410](#)
- wopi
  - Process, [367](#)
- WORD
  - form3.h, [1334](#)
- WORD\_MAX\_VALUE
  - form3.h, [1328](#)
- WORD\_MIN\_VALUE
  - form3.h, [1328](#)
- WORDDIF
  - declare.h, [809](#)
- WORDMASK
  - form3.h, [1329](#)
- WorkPointer
  - T\_const, [430](#)
- WorkSize
  - M\_const, [176](#)
- WorkSpace
  - T\_const, [429](#)
- WorkTop
  - T\_const, [429](#)
- wpoin
  - O\_const, [279](#)
- wpos
  - O\_const, [279](#)
- wranf
  - declare.h, [992](#)
  - reken.c, [2561](#)
- wranfcall
  - R\_const, [374](#)

- wranfia
  - R\_const, [372](#)
- wranfnpair1
  - R\_const, [374](#)
- wranfnpair2
  - R\_const, [374](#)
- wranfseed
  - R\_const, [373](#)
- WriteAll
  - declare.h, [882](#)
  - sch.c, [2654](#)
- WriteArgument
  - declare.h, [882](#)
  - sch.c, [2653](#)
- WRITEBUFTOEXTCHANNEL
  - declare.h, [820](#)
- WriteDictionary
  - declare.h, [1022](#)
  - sch.c, [2653](#)
- WriteDollarFactorToBuffer
  - declare.h, [962](#)
  - dollar.c, [1094](#)
- WriteDollarToBuffer
  - declare.h, [962](#)
  - dollar.c, [1094](#)
- WriteExpression
  - declare.h, [882](#)
  - sch.c, [2654](#)
- WRITEFILE
  - declare.h, [821](#)
- WriteFile
  - tools.c, [3094](#)
  - variable.h, [3207](#)
- WriteFileToFile
  - declare.h, [896](#)
  - tools.c, [3082](#)
- WriteIndex
  - declare.h, [985](#)
  - minos.c, [1731](#)
- WriteIndexBlock
  - declare.h, [984](#)
  - minos.c, [1731](#)
- WriteIniInfo
  - declare.h, [985](#)
  - minos.c, [1731](#)
- WriteInnerTerm
  - declare.h, [882](#)
  - sch.c, [2654](#)
- WriteLists
  - declare.h, [882](#)
  - sch.c, [2653](#)
- WriteNamesBlock
  - declare.h, [985](#)
  - minos.c, [1731](#)
- WriteObject
  - declare.h, [986](#)
  - minos.c, [1733](#)
- WriteOne
  - declare.h, [882](#)
  - sch.c, [2654](#)
- WRITEOUT
  - ftypes.h, [1373](#)
- WriteSetup
  - declare.h, [883](#)
  - setfile.c, [2691](#)
- WriteStats
  - declare.h, [883](#)
  - sort.c, [2716](#)
- WriteStoreHeader
  - declare.h, [997](#)
  - store.c, [2833](#)
- WriteString
  - declare.h, [916](#)
  - tools.c, [3085](#)
- WriteSubTerm
  - declare.h, [883](#)
  - sch.c, [2654](#)
- WriteTerm
  - declare.h, [884](#)
  - sch.c, [2654](#)
- writelnToChannel
  - declare.h, [990](#)
  - pre.c, [2331](#)
- WriteTokens
  - declare.h, [958](#)
  - token.c, [3048](#)
- WriteUnfinString
  - declare.h, [916](#)
  - tools.c, [3086](#)
- WrtPower
  - declare.h, [823](#)
  - sch.c, [2653](#)
- wskon
  - MGraph, [213](#)
- wsizeof
  - declare.h, [809](#)
- wsopi
  - MGraph, [213](#)
  - T\_MGraph, [449](#)
- wsum
  - T\_MGraph, [449](#)
- WTimeStatsFlag
  - C\_const, [62](#)
- X
  - AllGlobals, [11](#)
- X\_const, [487](#)
  - currentExternalChannel, [488](#)
  - currentPrompt, [487](#)
  - daemonize, [488](#)
  - killSignal, [487](#)
  - killWholeGroup, [488](#)
  - shellname, [487](#)
  - stderrname, [487](#)
  - timeout, [487](#)
- xsymbol
  - variable.h, [3205](#)

YESLYNDON  
    ftypes.h, [1448](#)

Z\_  
    ftypes.h, [1427](#)

zbufnum  
    M\_const, [179](#)

zero  
    DoLIaRS, [88](#)

ZEROARG  
    declare.h, [806](#)

ZeroFillRange  
    declare.h, [808](#)

zeropos  
    M\_const, [172](#)

zerorhs  
    M\_const, [192](#)

ZeroTable  
    float.c, [1292](#)

ZtoForm  
    float.c, [1287](#)

ZTOHARG  
    ftypes.h, [1447](#)